

Isabelle/STARK: A Formalization of zk-STARK in Isabelle/HOL

Diego Marmsoler

August 6, 2026

Department of Computer Science, University of Exeter, Exeter, UK
`d.marmsoler@exeter.ac.uk`

Contents

1	Core Infrastructure and Protocol	1
1.1	Finite Non-negative Real Numbers	1
1.1.1	Type definition	1
1.1.2	Basic operations	2
1.1.3	Algebra and order instances	4
1.1.4	Representation simp rules	4
1.1.5	Conversions from natural numbers	5
1.1.6	Arithmetic simp rules	6
1.1.7	Finite sums	7
1.2	Probability Distributions	8
1.2.1	Finite Set Aggregation	9
1.2.2	Probability Maps	11
1.2.3	Probability distributions	19
1.2.4	Distribution Items	21
1.2.5	Mapper	33
1.2.6	Relator	33
1.2.7	BNF Setup	33
1.2.8	Code Generator Setup	48
1.3	Finite Map setup	51
1.4	Hashing	52
1.4.1	Weakest Precondition Calculus	59
1.4.2	Code Generator Setup	59
1.5	Channels	60
1.5.1	Boolean Interpretation	125
1.5.2	GF5 Interpretation	127
1.6	The Field GF(5)	128

1.7	Merkle Trees	133
1.7.1	Weakest Precondition Calculus	134
1.7.2	Boolean interpretation	134
1.7.3	GF5 interpretation	136
1.7.4	Domain-separated protocol Merkle interface	136
1.7.5	Sanity Check	136
1.8	Probabilistic State Monad	151
1.8.1	Monad Operations	164
1.8.2	Weakest Precondition Calculus	165
1.8.3	Code generator	177
1.9	Stark Protocol	181
1.9.1	Preliminaries	181
1.9.2	Stark Specifications	185
1.9.3	Prover	199
1.9.4	Verifier	218
2	Completeness	221
2.1	Core Completeness Infrastructure	222
2.2	Algebraic Completeness Facts	283
2.3	Honest Transcript and Hash Replay	292
2.4	Verifier Replay	364
2.5	Honest Query Checks	456
2.6	Honest FRI Checks	518
2.7	Final Completeness Proof	584
2.8	Completeness Entry Point	649
3	Executable Example	649
3.1	Square Sequence Example	650
3.1.1	Polynomial Constraints	650
3.1.2	Interpreting the stark locale	651
3.1.3	Interpretation of prover locale	656
3.1.4	Interpretation of verifier locale	656
3.1.5	Creating and Verifying Proof	657
4	Soundness	657
4.1	Core Soundness Infrastructure	657
4.1.1	Proof structure for soundness	661
4.2	FRI Interfaces	1027
4.2.1	Algebraic FRI Soundness Interface	1027
4.2.2	Global FRI table bindings	1080
4.3	Oracle Budgets and Security Experiment	1083
4.4	Staged Composition Accounting	1692
4.5	Partial Merkle Openings and Algebraic Reductions	1871
4.6	Query and Partial-Opening Reductions	2124

4.7	Relevant Drift and Reachable FRI Evidence	3557
4.8	FRI Cover and Verifier-Tied Reductions	3834
4.9	FRI Replay and Active Bounds	4416
4.10	Concrete FRI Query Bounds and Public Endpoint	5167

1 Core Infrastructure and Protocol

```

theory NNReal
  imports
    Complex-Main
    HOL-Library.FSet
    HOL-Library.Conditional-Parametricity
begin

```

1.1 Finite Non-negative Real Numbers

HOL-Probability already provides the extended non-negative real numbers (usually called *ennreal*). That type is ideal for measure theory: non-negative integrals and measures may naturally take the value infinity.

The probabilistic state monad in this development only uses finite, discrete probability weights. For that purpose it is useful to have a smaller type of finite non-negative real numbers, implemented as a subtype of *real*. This avoids proof obligations about infinity while keeping non-negativity as a type invariant.

1.1.1 Type definition

```

typedef nnreal = {x :: real.  $0 \leq x$ }
  morphisms nn2real real2nn
  by auto

setup-lifting type-definition-nnreal

lemma nn2real-nonneg[simp]:  $0 \leq \text{nn2real } x$ 
  using nn2real[of x] by simp

lemma real2nn-nn2real[simp]:  $\text{real2nn } (\text{nn2real } x) = x$ 
  by (rule nn2real-inverse)

lemma nn2real-real2nn[simp]:
  assumes  $0 \leq x$ 
  shows  $\text{nn2real } (\text{real2nn } x) = x$ 
  using assms by (auto intro: real2nn-inverse)

```

1.1.2 Basic operations

instantiation *nnreal* :: *zero*
begin

lift-definition *zero-nnreal* :: *nnreal* **is** *0* **by** *simp*

instance ..

end

instantiation *nnreal* :: *one*
begin

lift-definition *one-nnreal* :: *nnreal* **is** *1* **by** *simp*

instance ..

end

instantiation *nnreal* :: *plus*
begin

lift-definition *plus-nnreal* :: *nnreal* $\Rightarrow$ *nnreal* $\Rightarrow$ *nnreal* **is** $(+)$ **by** *simp*

instance ..

end

instantiation *nnreal* :: *times*
begin

lift-definition *times-nnreal* :: *nnreal* $\Rightarrow$ *nnreal* $\Rightarrow$ *nnreal* **is** $(*)$ **by** *simp*

instance ..

end

instantiation *nnreal* :: *ord*
begin

lift-definition *less-eq-nnreal* :: *nnreal* $\Rightarrow$ *nnreal* $\Rightarrow$ *bool* **is** $(\leq)$.

lift-definition *less-nnreal* :: *nnreal* $\Rightarrow$ *nnreal* $\Rightarrow$ *bool* **is** $(<)$.

instance ..

end

instantiation *nnreal* :: *minus*


```

begin

lift-definition minus-nnreal :: nnreal  $\Rightarrow$  nnreal  $\Rightarrow$  nnreal
  is  $\lambda x y. \max 0 (x - y)$ 
  by simp

instance ..

end

instantiation nnreal :: divide
begin

lift-definition divide-nnreal :: nnreal  $\Rightarrow$  nnreal  $\Rightarrow$  nnreal is (/) by simp

instance ..

end

instantiation nnreal :: inverse
begin

lift-definition inverse-nnreal :: nnreal  $\Rightarrow$  nnreal is inverse by simp

instance ..

end

instantiation nnreal :: equal
begin

lift-definition equal-nnreal :: nnreal  $\Rightarrow$  nnreal  $\Rightarrow$  bool is (=) .

instance by (intro-classes, transfer, auto)

end

```

1.1.3 Algebra and order instances

```

instantiation nnreal :: linordered-semidom
begin

instance
  by (intro-classes; transfer, (simp|linarith|algebra))
  (metis max.absorb2 max.order-iff mult-nonneg-nonpos nle-le right-diff-distrib' zero-le-mult-iff)

end

instantiation nnreal :: comm-monoid-diff

```

begin

instance

by *intro-classes (transfer; auto simp: max-def)*

end

1.1.4 Representation simp rules

lemma *nn2real-eq-iff[simp]: nn2real x = nn2real y $\longleftrightarrow$ x = y*

by *transfer simp*

lemma *nn2real-le-iff[simp]: nn2real x $\leq$ nn2real y $\longleftrightarrow$ x $\leq$ y*

by *transfer simp*

lemma *nn2real-less-iff[simp]: nn2real x < nn2real y $\longleftrightarrow$ x < y*

by *transfer simp*

lemma *nn2real-0[simp]: nn2real 0 = 0*

by *transfer simp*

lemma *nn2real-1[simp]: nn2real 1 = 1*

by *transfer simp*

lemma *nn2real-add[simp]: nn2real (x + y) = nn2real x + nn2real y*

by *transfer simp*

lemma *nn2real-mult[simp]: nn2real (x * y) = nn2real x * nn2real y*

by *transfer simp*

lemma *nn2real-minus[simp]:*

nn2real (x - y) = max 0 (nn2real x - nn2real y)

by *transfer simp*

lemma *nn2real-inverse[simp]: nn2real (inverse x) = inverse (nn2real x)*

by *transfer simp*

lemma *nn2real-divide[simp]: nn2real (x / y) = nn2real x / nn2real y*

by *transfer simp*

lemma *nn2real-numeral[simp]: nn2real (numeral n) = numeral n*

by *(induct n) (simp-all only: numeral.simps nn2real-1 nn2real-add)*

lemma *nn2real-eq-transfer-aux[transfer-rule]:*

Transfer.Rel (rel-fun (=) (rel-fun pcr-nnreal (=))) ($\lambda x y. \text{nn2real } x = y$) (=)

by *(smt (verit) Rel-def cr-nnreal-def nn2real-eq-iff nnreal.pcr-cr-eq
rel-funI)*

lemma *real2nn-0[simp]: real2nn 0 = 0*

```

apply transfer-start
  apply transfer-step
    apply transfer-step
      apply transfer-step
        apply (rule nn2real-eq-transfer-aux)
          apply transfer-end
            by simp

```

```

lemma real2nn-1[simp]: real2nn 1 = 1
  apply transfer-start
    apply transfer-step
      apply transfer-step
        apply transfer-step
          apply (rule nn2real-eq-transfer-aux)
            apply transfer-end
              by simp

```

```

lemma real2nn-add[simp]:
  assumes  $0 \leq x$   $0 \leq y$ 
  shows real2nn (x + y) = real2nn x + real2nn y
  by (metis assms(1,2) nn2real-add nn2real-real2nn real2nn-nn2real)

```

```

lemma real2nn-mult[simp]:
  assumes  $0 \leq x$   $0 \leq y$ 
  shows real2nn (x * y) = real2nn x * real2nn y
  by (simp add: assms(1,2) times-nnreal-def)

```

1.1.5 Conversions from natural numbers

lift-definition *nnreal* :: *nat* $\Rightarrow$ *nnreal* **is of-nat by** *linarith*

parametric-constant *of-nat-transfer[transfer-rule]*: *of-nat-def*

```

lemma nn2real-nnreal[simp]: nn2real (nnreal n) = real n
  by transfer simp

```

```

lemma real2nn-eq-transfer-aux[transfer-rule]:
  Transfer.Rel (rel-fun pcr-nnreal (rel-fun (=) (==))) ( $\lambda y x. x = \text{real2nn } y$ ) (=)
  by (smt (verit) Rel-abs Rel-def cr-nnreal-def nnreal.pcr-cr-eq
    real2nn-nn2real)

```

```

lemma nnreal-of-nat[simp]: nnreal n = of-nat n
  by transfer simp

```

```

lemma nnreal-leq-simp[simp]: nnreal x  $\leq$  nnreal y  $\longleftrightarrow x \leq y$ 
  by transfer simp

```

```

lemma nnreal-less-simp[simp]: nnreal x < nnreal y  $\longleftrightarrow x < y$ 
  by transfer simp

```

lemma *nnreal-0[simp]*: $\text{nnreal } 0 = 0$
by *transfer simp*

lemma *nnreal-Suc[simp]*: $\text{nnreal } (\text{Suc } n) = \text{nnreal } n + 1$
by *transfer simp*

lemma *nnreal-eq-0-iff[simp]*: $\text{nnreal } n = 0 \longleftrightarrow n = 0$
by *transfer simp*

1.1.6 Arithmetic simp rules

lemma *zero-least[simp]*: $0 \leq (x::\text{nnreal})$
by *transfer simp*

lemma *zero-less-iff-neq-zero[simp]*: $0 < (x::\text{nnreal}) \longleftrightarrow x \neq 0$
using *zero-least[of x]* **by** (*auto simp: le-less*)

lemma *nnreal-add-eq-0-iff[simp]*:
 $(x + y = 0) \longleftrightarrow x = 0 \wedge y = (0::\text{nnreal})$
by *transfer auto*

lemma *nnreal-mult-eq-0-iff[simp]*:
 $(x * y = 0) \longleftrightarrow x = 0 \vee y = (0::\text{nnreal})$
by *transfer auto*

lemma *div-one-nnreal[simp]*: $(p::\text{nnreal}) / 1 = p$
by *transfer simp*

lemma *divide-self*: $(a::\text{nnreal}) \neq 0 \implies a / a = 1$
by *transfer simp*

lemma *times-div-1*:
assumes $x \neq 0$
shows $x * (1 / x) = (1::\text{nnreal})$
using *assms* **by** *transfer simp*

lemma *zero-div[simp]*: $0 / (q::\text{nnreal}) = 0$
by *transfer simp*

lemma *divide-eq-0-nnreal[simp]*:
 $((a::\text{nnreal}) / q = 0) \longleftrightarrow a = 0 \vee q = 0$
by *transfer auto*

lemma *add-divide-nnreal*:
 $((a::\text{nnreal}) + b) / q = a / q + b / q$
by *transfer (simp add: add-divide-distrib)*

lemma *minus-nnreal-code[code]*:

$nn2real\ (x - y) = (if\ x \geq y\ then\ nn2real\ x - nn2real\ y\ else\ 0)$
by *transfer auto*

1.1.7 Finite sums

lemma *fsum-fset-sum*:

$fsum\ f\ A = (\sum_{x \in fset\ A} f\ x)$
unfolding *fsum-def* **apply** *transfer*
by (*metis fsum.F.rep-eq fsum-def*)

lemma *fsum-divide-nnreal*:

$fsum\ (\lambda x. f\ x / q)\ A = fsum\ f\ A / (q::nnreal)$

proof (*induct A*)

case *empty*

show *?case*

by (*simp add: fsum-fset-sum*)

next

case (*insert x A*)

then show *?case*

by (*simp add: fsum-fset-sum add-divide-nnreal*)

qed

lemma *sum-divide-nnreal*:

assumes *finite A*

shows $(\sum_{x \in A} f\ x / (q::nnreal)) = (\sum_{x \in A} f\ x) / q$

using *assms*

proof *induct*

case *empty*

show *?case* **by** *simp*

next

case (*insert x A*)

then show *?case*

by (*simp add: add-divide-nnreal*)

qed

lemma *fsum-mult-left-nnreal*:

$fsum\ (\lambda x. (c::nnreal) * f\ x)\ A = c * fsum\ f\ A$

unfolding *fsum-fset-sum*

by (*simp add: sum-distrib-left*)

lemma *sum-constant-nnreal*:

assumes *finite S*

shows $(\sum_{x \in S} c) = nnreal\ (card\ S) * (c::nnreal)$

using *assms*

proof *induction*

case *empty*

show *?case* **by** *simp*

next

case (*insert x S*)

```

    then show ?case
      by (simp add: algebra-simps)
qed

lemma sum-1-nnreal:
  assumes finite S
  shows  $(\sum_{x \in S}. (1::nnreal)) = nnreal (card S)$ 
  using assms
proof induction
  case empty
  show ?case by simp
next
  case (insert x S)
  then show ?case by (simp add: Groups.add-ac(2))
qed

end

```

```

theory Prob-Dist
  imports
    NNReal
    HOL-Library.Monad-Syntax
    HOL-Library.FSet
begin

```

1.2 Probability Distributions

This is a formalization of a datatype for finite probability distributions.

The mathematical presentation uses finitely supported probability maps. This gives a direct specification of distributions and `bind`: a distribution is a partial map whose domain is finite and whose weights sum to one, and `bind` computes the probability of a result by summing over the finite support of the first computation.

The most direct code equation for `bind` is not executable for arbitrary result and state types, however. The map operation `dom` needs to enumerate the support of a map, and Isabelle’s generated code can only do that for types in the `enum` class. Thus a purely map-based implementation of `bind` would impose an unwanted executability restriction on the types used by the monad.

To avoid this, the theory keeps the map-based definitions as the abstract specification, but provides alternative executable equations in terms of finite sets. The finite-set representation explicitly stores the support and weights, so generated code can iterate over those stored elements without requiring the underlying types to be enumerable. This makes the executable version suitable for arbitrary result and state types.

A previous list-based version has the same basic idea, but converting finite sets to lists in a canonical way requires an ordering on the element type, so that approach introduces a *linorder* constraint. The finite-set implementation used here is therefore more general: it

avoids both the *enum* constraint of map-domain enumeration and the *linorder* constraint of canonical list conversion.

The theory also proofs monadic laws for probability distributions and registers the type as a BNF.

1.2.1 Finite Set Aggregation

Finite set sums lemma *fsum-ffilter-neutral*:

assumes $\bigwedge x. x \in A \implies \neg P x \implies f x = 0$
 shows $fsum\ f\ (ffilter\ P\ A) = fsum\ f\ A$
 unfolding *fsum-fset-sum*
 apply (rule *sum.mono-neutral-left*)
 using *assms* by *auto*

lemma *fsum-ffilter-if*:

$fsum\ f\ (ffilter\ P\ A) = fsum\ (\lambda x. \text{if } P\ x \text{ then } f\ x \text{ else } 0)\ A$

proof –

have $fsum\ f\ (ffilter\ P\ A) = (\sum_{x \in \{x \in fset\ A. P\ x\}}. f\ x)$
 apply (simp add: *fsum-fset-sum*)
 by (metis *Collect-conj-eq Int-commute Collect-mem-eq*)
 also have $\dots = (\sum_{x \in fset\ A. \text{if } P\ x \text{ then } f\ x \text{ else } 0})$
 by (rule *sum.inter-filter*) simp
 also have $\dots = fsum\ (\lambda x. \text{if } P\ x \text{ then } f\ x \text{ else } 0)\ A$
 by (simp add: *fsum-fset-sum*)
 finally show ?thesis .

qed

Summing pairs by key definition *sum-fset-x* :: $'a \Rightarrow ('a \times 'w :: comm-monoid-add)\ fset \Rightarrow 'w$ where

$sum-fset-x\ x\ xs = fsum\ snd\ (ffilter\ (\lambda p. fst\ p = x)\ xs)$

lemma *sum-fset-x-notin-fst*:

assumes $x \notin fimage\ fst\ A$
 shows $sum-fset-x\ x\ A = 0$

proof –

have $ffilter\ (\lambda p. fst\ p = x)\ A = \{\}\}$
 using *assms*
 by (rule-tac *fset-eqI*) *auto*
 then show ?thesis
 unfolding *sum-fset-x-def* by (simp add: *fsum-fset-sum*)

qed

lemma *sum-fset-x-unique*:

assumes *uniq*: *inj-on* *fst* (*fset* *A*)
 assumes *mem*: $(x, p) \in A$
 shows $sum-fset-x\ x\ A = p$

proof –

have $ffilter\ (\lambda q. fst\ q = x)\ A = \{(x, p)\}$

```

    using uniq mem
    apply (rule-tac fset-eqI) apply (auto simp: inj-on-def) by fastforce
then show ?thesis
    unfolding sum-fset-x-def
    by (simp add: fsum-fset-sum)
qed

```

Squashing pairs by key definition *squish* :: (*'a* × *'w*::comm-monoid-add) fset ⇒ (*'a* × *'w*) fset where

$$\text{squish } xs = \text{fimage } (\lambda x. (x, \text{sum-fset-}x \ x \ xs)) (\text{fimage } \text{fst } xs)$$

lemma *squish-fst-unique*:
 assumes $(x, p) \in \text{squish } A$ $(x, q) \in \text{squish } A$
 shows $p = q$
 using assms unfolding *squish-def* by auto

lemma *squish-inj-on-fst*:
 inj-on fst (fset (squish A))
 unfolding *squish-def* inj-on-def
 by auto

lemma *squish-id-if-inj-on-fst*:
 assumes *uniq*: inj-on fst (fset A)
 shows *squish* A = A
 proof (rule fset-eqI)
 fix $z :: 'a \times 'b$
 obtain $x \ p$ where $z = (x, p)$
 by (cases z)
 show $(z \in \text{squish } A) = (z \in A)$
 proof
 assume $z \in \text{squish } A$
 then obtain q where $(x, q) \in A$ and $p = \text{sum-fset-}x \ x \ A$
 unfolding *squish-def* by auto
 then have $\text{sum-fset-}x \ x \ A = q$
 using *uniq* by (simp add: sum-fset-x-unique)
 then show $z \in A$
 using $q \ p$ unfolding *z* by simp
 next
 assume $zA: z \in A$
 then have $\text{sum-fset-}x \ x \ A = p$
 using *uniq* unfolding *z* by (simp add: sum-fset-x-unique)
 then show $z \in \text{squish } A$
 using zA unfolding *z squish-def* by force
 qed
 qed

lemma *fsum-squish*:
 $\text{fsum snd } (\text{squish } A) = \text{fsum snd } A$
 proof –


```

have inj: inj-on ( $\lambda x. (x, \text{sum-fset-}x\ x\ A)$ ) ( $\text{fst } \text{'fset } A$ )
  by (auto intro!: inj-onI)

have fsum snd (squish A) =
  ( $\sum x \in \text{fst } \text{'fset } A. \text{sum-fset-}x\ x\ A$ )
  unfolding squish-def
  apply (simp add: fsum-fset-sum fimage.rep-eq sum.reindex[OF inj] o-def)
  by (smt (verit) image-cong image-image inj split-pairs sum.reindex-cong)
also have ... =
  ( $\sum x \in \text{fst } \text{'fset } A. \sum p \in \{p \in \text{fset } A. \text{fst } p = x\}. \text{snd } p$ )
  unfolding sum-fset-x-def
  apply (intro sum.cong refl) apply (simp add: fsum-fset-sum Int-def)
  by (metis (lifting) Collect-cong)
also have ... = ( $\sum p \in \text{fset } A. \text{snd } p$ )
  by (rule sum.group) auto
also have ... = fsum snd A
  by (simp add: fsum-fset-sum)
finally show ?thesis .
qed

```

1.2.2 Probability Maps

Summing probability maps **definition** $\text{sum-map} :: ('a \rightarrow \text{nnreal}) \Rightarrow \text{nnreal}$ **where**
 $\text{sum-map } f \equiv (\sum i \in \text{dom } f. \text{the } (f\ i))$

type-synonym $\text{prob} = \text{nnreal}$

Probability weights are represented by *nnreal*: finite, non-negative real numbers. Keeping non-negativity as a type invariant avoids carrying explicit $0 \leq p$ assumptions through sums, bind, map, expectation, and normalization proofs.

Plain *real* would admit negative weights, so every constructor and intermediate probability calculation would need extra invariants and proof obligations to rule them out. Strictly positive reals are also not a good fit: the development needs a zero weight for absent probabilities, empty sums, pruning zero-mass outcomes, and intermediate aggregation before the stored support is normalized.

The separate invariant on the distribution type below that excludes zero weights from the domain is therefore intentional. The weight type supports zero where computations need it, while distributions store only the non-zero support.

abbreviation some-map

where

$\text{some-map} \equiv (\text{let } x = \text{SOME } x :: 'a. \text{True in } (\text{Map.empty}(x := \text{Some } 1)))$

Dirac delta distribution **definition** $\text{delta-map} :: 'a \Rightarrow 'a \rightarrow \text{prob}$

where

$\text{delta-map } x = \text{Map.empty}(x := \text{Some } 1)$

Normalization **definition** $\text{norm-fset} :: ('a \times \text{prob}) \text{fset} \Rightarrow ('a \times \text{prob}) \text{fset}$

where

```

norm-fset xs =
  (if fsum snd xs = 0
   then {|(undefined, 1)|}
   else
    (let
      xs' = squish xs;
      q = fsum snd xs'
      in ffilter (λp. snd p ≠ 0) (fimage (λ(x, p). (x, p / q)) xs'))

```

Maps from finite sets **definition** $\text{map-of-val} :: ('a \times 'b) \text{fset} \Rightarrow 'a \Rightarrow 'b$ **where**

```

map-of-val A x =
  snd (fthe-elem (ffilter (λp. fst p = x) A))

```

definition $\text{map-of} :: ('a \times 'b) \text{fset} \Rightarrow 'a \rightarrow 'b$ **where**

```

map-of A x =
  (if x |∈| fimage fst A then Some (map-of-val A x) else None)

```

lemma dom-map-of-subset :

```

dom (map-of A) ⊆ fset (fimage fst A)
unfolding map-of-def
by (auto split:if-split-asm)

```

lemma finite-dom-map-of [simp]:

```

finite (dom (map-of A))
using dom-map-of-subset[of A]
by (meson finite-fset finite-subset)

```

lemma map-of-singleton :

```

map-of {|(x, y)|} = Map.empty(x := Some y)

```

proof

```

fix z
show map-of {|(x, y)|} z = (Map.empty(x := Some y)) z
proof (cases z = x)
  case True
    have filt: ffilter (λp. fst p = x) {|(x, y)|} = {|(x, y)|}
      by (rule fset-eqI) auto
    show ?thesis
      using True
      unfolding map-of-def map-of-val-def
      by (simp add: filt)
  next
    case False
    then show ?thesis
      unfolding map-of-def
      by simp
qed
qed

```

```

lemma map-of-val-unique-key:
  assumes uniq:  $\bigwedge y z. (x, y) \in A \implies (x, z) \in A \implies y = z$ 
  assumes mem:  $(x, y) \in A$ 
  shows map-of-val  $A\ x = y$ 
proof -
  have filt: ffilter  $(\lambda p. \text{fst } p = x)\ A = \{|(x, y)|\}$ 
  proof (rule fset-eqI)
    fix  $p$ 
    show  $(p \in \text{ffilter } (\lambda p. \text{fst } p = x)\ A) = (p \in \{|(x, y)|\})$ 
    proof
      assume  $p: p \in \text{ffilter } (\lambda p. \text{fst } p = x)\ A$ 
      then obtain  $z$  where p-eq:  $p = (x, z)$  and pA:  $(x, z) \in A$ 
      by (cases p) auto
      have  $z = y$ 
      using uniq[OF pA mem] .
      then show  $p \in \{|(x, y)|\}$ 
      using p-eq by simp
    next
    assume  $p \in \{|(x, y)|\}$ 
    then show  $p \in \text{ffilter } (\lambda p. \text{fst } p = x)\ A$ 
    using mem by auto
  qed
qed

  show ?thesis
  unfolding map-of-val-def
  by (simp add: filt)
qed

lemma map-of-nonzero-on-dom:
  assumes uniq:  $\bigwedge x y z. (x, y) \in A \implies (x, z) \in A \implies y = z$ 
  assumes nonzero:  $\bigwedge x p. (x, p) \in A \implies p \neq 0$ 
  assumes x:  $x \in \text{dom } (\text{map-of } A)$ 
  shows the  $(\text{map-of } A\ x) \neq 0$ 
proof -
  from  $x$  have xin:  $x \in \text{fimage fst } A$ 
  unfolding map-of-def by (auto split: if-splits)
  then obtain  $p$  where xp:  $(x, p) \in A$ 
  by auto
  have map-of-val  $A\ x = p$ 
  by (rule map-of-val-unique-key[OF uniq xp])
  then have map-of  $A\ x = \text{Some } p$ 
  using xin unfolding map-of-def by simp
  then show ?thesis
  using nonzero[OF xp] by simp
qed

definition map-of-fset ::  $('a \times \text{prob}) \text{fset} \Rightarrow 'a \multimap \text{prob}$  where
  map-of-fset  $A =$ 

```

```

    (let q = fsum snd A in
      if q = 0 then Map.empty(undefined := Some 1)
      else map-of (norm-fset A))

lemma finite-dom-map-of-fset: finite (dom (map-of-fset A))
  unfolding map-of-fset-def
  by (simp add: Let-def)

lemma norm-fset-fst-unique:
  assumes (x, p) |∈| norm-fset A (x, q) |∈| norm-fset A
  shows p = q
proof (cases fsum snd A = 0)
  case True
  then show ?thesis
    using assms unfolding norm-fset-def by auto
next
  case False
  from assms False obtain r s where
    r: (x, r) |∈| squish A p = r / fsum snd (squish A)
    and s: (x, s) |∈| squish A q = s / fsum snd (squish A)
    unfolding norm-fset-def Let-def by auto
  then have r = s
    using squish-fst-unique by fast
  then show ?thesis
    using r s by simp
qed

lemma norm-fset-nonzero-snd:
  assumes (x, p) |∈| norm-fset A
  shows p ≠ 0
  using assms
  unfolding norm-fset-def
  by (auto simp: Let-def split: if-splits prod.splits)

lemma fsum-norm-fset:
  fsum snd (norm-fset A) = 1
proof (cases fsum snd A = 0)
  case True
  then show ?thesis
    unfolding norm-fset-def by (simp add: fsum-fset-sum)
next
  case False
  define S where S = squish A
  define q where q = fsum snd S

  have q0: q ≠ 0
    using False fsum-squish unfolding S-def q-def apply simp
    by (simp add: fsum-squish)

```

```

have norm:
  norm-fset A =
    ffilter ( $\lambda p. \text{snd } p \neq 0$ ) (fimage ( $\lambda p. (\text{fst } p, \text{snd } p / q)$ ) S)
  using False unfolding norm-fset-def S-def q-def
  by (simp add: Let-def case-prod-unfold)

have inj-norm: inj-on ( $\lambda p. (\text{fst } p, \text{snd } p / q)$ ) (fset S)
proof (rule inj-onI)
  fix a b
  assume aS:  $a \in \text{fset } S$  and bS:  $b \in \text{fset } S$ 
  and eq:  $(\text{fst } a, \text{snd } a / q) = (\text{fst } b, \text{snd } b / q)$ 
  obtain x p where a:  $a = (x, p)$  by (cases a)
  obtain y r where b:  $b = (y, r)$  by (cases b)
  from eq have xy:  $x = y$ 
  by (simp add: a b)
  have p = r
  using squish-fst-unique[of x p A r] aS bS xy
  unfolding a b S-def by simp
  then show  $a = b$ 
  using xy by (simp add: a b)
qed

have fsum snd (norm-fset A) =
  fsum snd (ffilter ( $\lambda p. \text{snd } p \neq 0$ ) (fimage ( $\lambda p. (\text{fst } p, \text{snd } p / q)$ ) S))
  by (simp add: norm)
also have ... = fsum snd (fimage ( $\lambda p. (\text{fst } p, \text{snd } p / q)$ ) S)
  by (rule fsum-ffilter-neutral) auto
also have ... = fsum ( $\lambda p. \text{snd } p / q$ ) S
  using inj-norm
  by (simp add: fsum-fset-sum fimage.rep-eq sum.reindex o-def)
also have ... =  $q / q$ 
  by (simp add: fsum-divide-nnreal q-def)
also have ... = 1
  by (rule divide-self[OF q0])
finally show ?thesis .
qed

lemma norm-fset-singleton-1:
  norm-fset  $\{|(x, 1)|\} = \{|(x, 1::\text{nnreal})|\}$ 
proof –
  let ?A =  $\{|(x, 1::\text{nnreal})|\}$ 

  have total: fsum snd ?A = 1
  by (simp add: fsum-fset-sum)

  have sx: sum-fset-x x ?A = 1
  unfolding sum-fset-x-def
  by (simp add: fsum-fset-sum)

```

```

have squish: squish ?A = ?A
proof (rule fset-eqI)
  fix z
  show (z |∈| squish ?A) = (z |∈| ?A)
    unfolding squish-def
    using sx by auto
qed

have img:
  fimage (λ(x, p). (x, p / fsum snd ?A)) (squish ?A) = ?A
proof (rule fset-eqI)
  fix z
  show (z |∈| fimage (λ(x, p). (x, p / fsum snd ?A)) (squish ?A)) =
    (z |∈| ?A)
    using total squish by auto
qed

show ?thesis
  unfolding norm-fset-def
  using total squish img
  apply (simp add: Let-def) by fastforce
qed

lemma sum-map-of:
  assumes uniq:  $\bigwedge x y z. (x, y) |∈| A \implies (x, z) |∈| A \implies y = z$ 
  shows sum-map (map-of A) = fsum snd A
proof -
  have dom: dom (map-of A) = fset (fimage fst A)
    unfolding map-of-def apply auto
    apply (metis option.discI)
    by force
  have inj: inj-on fst (fset A)
  proof (rule inj-onI)
    fix p q
    assume pA:  $p \in \text{fset } A$  and qA:  $q \in \text{fset } A$  and fst:  $\text{fst } p = \text{fst } q$ 
    obtain x y where p:  $p = (x, y)$  by (cases p)
    obtain z where q:  $q = (x, z)$ 
    using fst p by (cases q) auto
    have y = z
      using uniq[of x y z] pA qA unfolding p q by simp
    then show p = q
      unfolding p q by simp
  qed
  have val:  $p |∈| A \implies \text{the } (\text{map-of } A \text{ } (\text{fst } p)) = \text{snd } p$  for p
  proof -
    assume pA:  $p |∈| A$ 
    have filt:  $\text{ffilter } (\lambda q. \text{fst } q = \text{fst } p) A = \{|p|\}$ 

```

```

proof (rule fset-eqI)
  fix q
  show ( $q \in \text{ffilter } (\lambda q. \text{fst } q = \text{fst } p) A$ ) = ( $q \in \{|p|\}$ )
  proof
    assume  $qf: q \in \text{ffilter } (\lambda q. \text{fst } q = \text{fst } p) A$ 
    then have  $qA: q \in A$  and  $fq: \text{fst } q = \text{fst } p$  by auto
    obtain  $x y$  where  $p: p = (x, y)$  by (cases p)
    obtain  $z$  where  $q: q = (x, z)$ 
      using  $fq p$  by (cases q) auto
    have  $z = y$ 
      using  $\text{uniq}[of\ x\ z\ y]\ qA\ pA$  unfolding  $p\ q$  by simp
    then show  $q \in \{|p|\}$ 
      unfolding  $p\ q$  by simp
  next
    assume  $q \in \{|p|\}$ 
    then show  $q \in \text{ffilter } (\lambda q. \text{fst } q = \text{fst } p) A$ 
      using  $pA$  by auto
  qed
qed
then show ?thesis
  using  $pA$  unfolding map-of-def map-of-val-def by auto
qed

have  $\text{sum-map } (\text{map-of } A) =$ 
  ( $\sum x \in \text{fset } (\text{fimage } \text{fst } A). \text{the } (\text{map-of } A\ x)$ )
  unfolding sum-map-def dom by simp
also have ... = ( $\sum p \in \text{fset } A. \text{the } (\text{map-of } A\ (\text{fst } p))$ )
  using inj by (simp add: fimage.rep-eq sum.reindex)
also have ... = ( $\sum p \in \text{fset } A. \text{snd } p$ )
  using val by (intro sum.cong) auto
also have ... =  $\text{fsum } \text{snd } A$ 
  by (simp add: fsum-fset-sum)
finally show ?thesis .
qed

lemma sum-map-1:  $\text{sum-map } (\text{map-of-fset } A) = 1$ 
proof (cases fsum snd A = 0)
  case True
    then show ?thesis
      unfolding map-of-fset-def sum-map-def by (simp add: Let-def)
  next
    case False
      have  $\text{sum-map } (\text{map-of-fset } A) = \text{sum-map } (\text{map-of } (\text{norm-fset } A))$ 
        using False unfolding map-of-fset-def by (simp add: Let-def)
      also have ... =  $\text{fsum } \text{snd } (\text{norm-fset } A)$ 
        by (rule sum-map-of) (rule norm-fset-fst-unique)
      also have ... = 1
        by (rule fsum-norm-fset)
      finally show ?thesis .

```

qed

lemma *map-of-fset-nonzero-on-dom*:

assumes $x \in \text{dom } (\text{map-of-fset } A)$

shows *the* $(\text{map-of-fset } A \ x) \neq 0$

proof (*cases fsum snd A = 0*)

case *True*

then show *?thesis*

using *assms* **unfolding** *map-of-fset-def* **by** (*simp add: Let-def*)

next

case *False*

then have *map-eq*: $\text{map-of-fset } A = \text{map-of } (\text{norm-fset } A)$

unfolding *map-of-fset-def* **by** (*simp add: Let-def*)

show *?thesis*

unfolding *map-eq*

proof (*rule map-of-nonzero-on-dom*)

show $\bigwedge x \ y \ z. (x, y) \in \text{norm-fset } A \implies (x, z) \in \text{norm-fset } A \implies y = z$

by (*rule norm-fset-fst-unique*)

show $\bigwedge x \ p. (x, p) \in \text{norm-fset } A \implies p \neq 0$

by (*rule norm-fset-nonzero-snd*)

show $x \in \text{dom } (\text{map-of } (\text{norm-fset } A))$

using *assms map-eq* **by** *simp*

qed

qed

lemma *delta-map-map-of-fset-singleton*:

$\text{delta-map } x = \text{map-of-fset } \{|(x, 1)|\}$

proof –

let *?A* = $\{|(x, 1::\text{nnreal})|\}$

have *total*: $\text{fsum snd } ?A = 1$

by (*simp add: fsum-fset-sum*)

have *norm*: $\text{norm-fset } ?A = ?A$

by (*rule norm-fset-singleton-1*)

have $\text{map-of-fset } ?A = \text{map-of } ?A$

unfolding *map-of-fset-def*

using *total norm*

by (*simp add: Let-def*)

also have $\dots = \text{Map.empty}(x := \text{Some } 1)$

by (*rule map-of-singleton*)

finally show *?thesis*

unfolding *delta-map-def* **by** *simp*

qed

abbreviation *option-default* **where** *option-default a* $\equiv \text{case-option } (0::\text{nnreal}) \text{ id } a$

lemma *option-default-if-zero*[simp]:
 $\text{option-default } (if\ p = 0\ then\ None\ else\ Some\ p) = p$
by (cases $p = (0::nnreal)$) simp-all

Raw items **definition** *raw-items* :: $('a \rightarrow prob) \Rightarrow ('a \times prob)$ set **where**
 $\text{raw-items } f = \text{image } (\lambda x. (x, \text{the } (f\ x)))\ (\text{dom } f)$

1.2.3 Probability distributions

typedef $'a\ dist =$
 $\{f. \text{finite } (\text{dom } f) \wedge \text{sum-map } f = 1 \wedge (\forall x \in \text{dom } f. \text{the } (f\ x) \neq 0)\}$
 $:: ('a \rightarrow prob)$ set
morphisms *dist* *abs-dist*
proof
fix $x::'a$
show $\text{some-map } \in$
 $\{f. \text{finite } (\text{dom } f) \wedge \text{sum-map } f = 1 \wedge (\forall x \in \text{dom } f. \text{the } (f\ x) \neq 0)\}$
unfolding *sum-map-def* **by** *simp*
qed

setup-lifting *type-definition-dist*

lemma *norm-fset-inj-on-fst*:
 $\text{inj-on } \text{fst } (fset\ (\text{norm-fset } A))$
proof (cases $fsum\ \text{snd } A = 0$)
case *True*
then show *?thesis*
unfolding *norm-fset-def* **by** (simp add: *inj-on-def*)
next
case *False*
then show *?thesis*
using *squish-inj-on-fst*[of *A*]
unfolding *inj-on-def*
unfolding *norm-fset-def* **apply** (auto simp: *Let-def* *split*: *prod.splits*)
by *fastforce*
qed

lift-definition *dist-of-fset* :: $('a \times prob) \text{ fset} \Rightarrow 'a\ dist$
is *map-of-fset*
by (auto simp add: *finite-dom-map-of-fset* *sum-map-1* *map-of-fset-nonzero-on-dom*)

Delta Distributions **definition** *delta-dist* :: $'a \Rightarrow 'a\ dist$
where $\text{delta-dist } x = \text{dist-of-fset } \{|(x, 1)|\}$

lemma *dist-delta-dist*:
 $\text{dist } (\text{delta-dist } x) = \text{delta-map } x$
unfolding *delta-dist-def*
by (simp add: *dist-of-fset.rep-eq* *delta-map-map-of-fset-singleton*)

Uniform Distributions **definition** *dist-uniform* :: 'a fset $\Rightarrow$ 'a dist
where *dist-uniform* *xs* = *dist-of-fset* (($\lambda x. (x, 1::prob)$) |[†] *xs*)

lemma *fsum-snd-uniform-ones*:

fsum snd (($\lambda x. (x, 1::nnreal)$) |[†] *xs*) = *nnreal* (*fcard xs*)

proof –

have *inj*: *inj-on* ($\lambda x. (x, 1::nnreal)$) (*fset xs*)

by (*auto intro!*: *inj-onI*)

have *fsum snd* (($\lambda x. (x, 1::nnreal)$) |[†] *xs*) =
 $(\sum_{x \in \text{fset } xs} (1::nnreal))$

by (*simp add*: *fsum-fset-sum fimage.rep-eq sum.reindex[OF inj] o-def*)

also have ... = *nnreal* (*card* (*fset xs*))

by (*rule sum-1-nnreal simp*)

also have ... = *nnreal* (*fcard xs*)

by (*metis fcard.rep-eq*)

finally show *?thesis* .

qed

lemma *norm-fset-uniform-ones*:

assumes *xs* $\neq \{\}\{\}$

shows

norm-fset (($\lambda x. (x, 1::nnreal)$) |[†] *xs*) =
 $((\lambda x. (x, 1 / \text{nnreal } (\text{fcard } xs))) |[†] \text{fset } xs)$

proof –

let *?A* = (($\lambda x. (x, 1::nnreal)$) |[†] *xs*)

let *?B* = (($\lambda x. (x, 1 / \text{nnreal } (\text{fcard } xs))$) |[†] *xs*)

have *total*: *fsum snd* *?A* = *nnreal* (*fcard xs*)

by (*rule fsum-snd-uniform-ones*)

have *nonzero*: *fsum snd* *?A* $\neq 0$

using *assms total* **by** *simp*

have *inj*: *inj-on fst* (*fset* *?A*)

by (*auto simp*: *fimage.rep-eq inj-on-def*)

have *squish-A*: *squish* *?A* = *?A*

by (*rule squish-id-if-inj-on-fst[OF inj]*)

have *img*:

fimage ($\lambda(x, p). (x, p / \text{fsum snd } ?A)$) *?A* = *?B*

using *total*

by (*rule-tac fset-eqI*) (*auto simp*: *fimage.rep-eq*)

have *filt*:

ffilter ($\lambda p. \text{snd } p \neq 0$) *?B* = *?B*

using *assms*

by (*rule-tac fset-eqI*) *auto*

```

show ?thesis
  unfolding norm-fset-def
  using nonzero squish-A img filt
  by (simp add: Let-def)
qed

```

Die definition $die :: nat \Rightarrow nat \text{ dist}$
 where $die\ n = dist\text{-uniform}\ (fset\text{-of-list}\ ([1..<Suc\ n]))$

Coin definition $coin :: 'a \Rightarrow 'a \Rightarrow 'a \text{ dist}$
 where $coin\ f\ x\ y = dist\text{-of-fset}\ \{|(x,f), (y,1-f)|\}$

1.2.4 Distribution Items

```

context
  includes fset.lifting
begin

```

lift-definition $items :: 'a \text{ dist} \Rightarrow ('a \times prob) \text{ fset}$
is raw-items unfolding raw-items-def by simp

end

lemma map-of-val-unique:

```

  assumes uniq: inj-on fst (fset A)
  assumes mem:  $(x, y) \in A$ 
  shows map-of-val A x = y

```

proof –

```

  have filt: ffilter  $(\lambda p. fst\ p = x)$  A =  $\{|(x, y)|\}$ 

```

proof (rule fset-eqI)

fix p

```

  show  $(p \in \text{filter } (\lambda p. fst\ p = x)\ A) = (p \in \{|(x, y)|\})$ 

```

proof

```

  assume  $p: p \in \text{ffilter } (\lambda p. fst\ p = x)\ A$ 

```

```

  then have  $pA: p \in A$  and  $fst\text{-}p: fst\ p = x$ 

```

```

    by auto

```

```

  have  $p = (x, y)$ 

```

```

    using uniq  $pA$  mem  $fst\text{-}p$ 

```

```

    unfolding inj-on-def

```

```

    by fastforce

```

```

  then show  $p \in \{|(x, y)|\}$ 

```

```

    by simp

```

next

```

  assume  $p \in \{|(x, y)|\}$ 

```

```

  then show  $p \in \text{ffilter } (\lambda p. fst\ p = x)\ A$ 

```

```

    using mem by auto

```

qed

qed

```

show ?thesis
  unfolding map-of-val-def
  by (simp add: filt)
qed

lemma raw-items-map-of-unique:
  assumes uniq: inj-on fst (fset A)
  shows raw-items (map-of A) = fset A
proof
  show raw-items (map-of A)  $\subseteq$  fset A
  proof
    fix z
    assume z:  $z \in \text{raw-items (map-of A)}$ 
    then obtain x where
      xdom:  $x \in \text{dom (map-of A)}$ 
      and z-eq:  $z = (x, \text{the (map-of A x)})$ 
      unfolding raw-items-def by auto

    from xdom have xin:  $x \in \text{fimage fst A}$ 
      unfolding map-of-def by (auto split: if-splits)

    then obtain y where xy:  $(x, y) \in A$ 
      by auto

    have the (map-of A x) = y
      using xin map-of-val-unique[OF uniq xy]
      unfolding map-of-def by simp

    then show  $z \in \text{fset A}$ 
      using z-eq xy by simp
  qed
next
  show fset A  $\subseteq$  raw-items (map-of A)
  proof
    fix z
    assume zA:  $z \in \text{fset A}$ 
    obtain x y where z-eq:  $z = (x, y)$ 
      by (cases z)

    have xy:  $(x, y) \in A$ 
      using zA z-eq by simp

    have xin:  $x \in \text{fimage fst A}$ 
      using xy apply auto by force

    have map-x:  $\text{map-of A x} = \text{Some y}$ 
      using xin map-of-val-unique[OF uniq xy]
      unfolding map-of-def by simp
  
```

```

then have  $x \in \text{dom } (\text{map-of } A)$ 
  by auto

moreover have  $z = (x, \text{the } (\text{map-of } A \ x))$ 
  using z-eq map-x by simp

ultimately show  $z \in \text{raw-items } (\text{map-of } A)$ 
  unfolding raw-items-def by auto
qed
qed

lemma raw-items-map-of-fset:
   $\text{raw-items } (\text{map-of-fset } xs) = \text{fset } (\text{norm-fset } xs)$ 
proof (cases fsum snd xs = 0)
  case True
  then show ?thesis
    unfolding raw-items-def map-of-fset-def norm-fset-def
    by (simp add: Let-def)
next
  case False
  have  $\text{raw-items } (\text{map-of-fset } xs) = \text{raw-items } (\text{map-of } (\text{norm-fset } xs))$ 
    using False
    unfolding map-of-fset-def
    by (simp add: Let-def)
  also have  $\dots = \text{fset } (\text{norm-fset } xs)$ 
    by (rule raw-items-map-of-unique) (rule norm-fset-inj-on-fst)
  finally show ?thesis .
qed

lemma fset-eq-transfer-aux[transfer-rule]:
  Transfer.Rel
  (rel-fun (pcr-fset (rel-prod (=) (=))))
  (rel-fun (rel-fset (rel-prod (=) (=)))) (=))
  ( $\lambda a \ b. a = \text{fset } b$ ) (=)
  unfolding rel-fun-def
  by (simp add: Rel-def cr-fset-def fset.pcr-cr-eq fset.rel-eq fset-cong prod.rel-eq)

definition sum-fset-by ::
   $'a \Rightarrow ('i \Rightarrow 'a) \Rightarrow ('i \Rightarrow \text{prob}) \Rightarrow 'i \text{ fset} \Rightarrow \text{prob}$ 
where
  sum-fset-by x key weight I =
    fsum weight (ffilter ( $\lambda i. \text{key } i = x$ ) I)

definition map-of-fset-by ::
   $('i \Rightarrow 'a) \Rightarrow ('i \Rightarrow \text{prob}) \Rightarrow 'i \text{ fset} \Rightarrow 'a \multimap \text{prob}$ 
where
  map-of-fset-by key weight I =
    (let q = fsum weight I in

```

if $q = 0$ then $\text{Map.empty}(\text{undefined} := \text{Some } 1)$
 else $(\lambda x.$
 let $p = \text{sum-fset-by } x \text{ key weight } I / q$
 in if $p = 0$ then None else $\text{Some } p)$)

lemma *fsum-by-total*:

fsum $(\lambda x. \text{sum-fset-by } x \text{ key weight } I) (\text{fimage key } I) = \text{fsum weight } I$

proof –

have *fsum* $(\lambda x. \text{sum-fset-by } x \text{ key weight } I) (\text{fimage key } I) =$
 $(\sum_{x \in \text{key}} ' \text{fset } I. \sum_{i \in \{i \in \text{fset } I. \text{key } i = x\}}. \text{weight } i)$

unfolding *sum-fset-by-def*

apply (*simp add: fsum-fset-sum fimage.rep-eq*)

by (*metis Int-commute Collect-conj-eq Collect-mem-eq*)

also have ... $= (\sum_{i \in \text{fset } I}. \text{weight } i)$

apply (*rule sum.group*) **by** *auto*

also have ... $= \text{fsum weight } I$

by (*simp add: fsum-fset-sum*)

finally show *?thesis* .

qed

lemma *finite-dom-map-of-fset-by[simp]*:

finite $(\text{dom } (\text{map-of-fset-by key weight } I))$

proof –

have $\text{dom } (\text{map-of-fset-by key weight } I) \subseteq \text{fset } (\text{fimage key } I) \cup \{\text{undefined}\}$

unfolding *map-of-fset-by-def sum-fset-by-def*

apply (*auto simp: Let-def fsum-fset-sum*)

by (*metis fmember-filter[of - $\lambda uub. \text{key } uub = - I$] filter-fset[of $\lambda uub. \text{key } uub = - I$]
 image-iff[of - $\text{key fset } I$] option.discI[of None]
 sum.not-neutral-contains-not-neutral[of $\text{weight fset } (\text{ffilter } (\lambda uub. \text{key } uub = -) I)$])*)

then show *?thesis*

by (*simp add: finite-subset*)

qed

lemma *sum-fset-by-notin*:

assumes $x \notin \text{fset } (\text{fimage key } I)$

shows $\text{sum-fset-by } x \text{ key weight } I = 0$

using *assms*

unfolding *sum-fset-by-def*

apply (*simp add: fsum-fset-sum fimage.rep-eq*)

by (*smt (verit) Int-emptyI image-iff mem-Collect-eq nnreal-0 sum.empty*)

lemma *dom-map-of-fset-by*:

assumes $q: \text{fsum weight } I \neq 0$

shows

$\text{dom } (\text{map-of-fset-by key weight } I) =$
 $\{x \in \text{fset } (\text{fimage key } I). \text{sum-fset-by } x \text{ key weight } I \neq 0\}$

proof

show $\text{dom } (\text{map-of-fset-by key weight } I)$

$\subseteq \{x \in \text{fset } (\text{fimage key } I). \text{sum-fset-by } x \text{ key weight } I \neq 0\}$

```

proof
  fix  $x$ 
  assume  $x: x \in \text{dom } (\text{map-of-fset-by key weight } I)$ 
  then have  $\text{nz}: \text{sum-fset-by } x \text{ key weight } I \neq 0$ 
    using  $q$ 
    unfolding  $\text{map-of-fset-by-def}$ 
    by (auto simp: Let-def split: if-splits)

  moreover have  $x \in \text{fset } (\text{fimage key } I)$ 
    using  $\text{nz sum-fset-by-notin}[of x key I weight]$ 
    by blast

  ultimately show  $x \in \{x \in \text{fset } (\text{fimage key } I). \text{sum-fset-by } x \text{ key weight } I \neq 0\}$ 
    by simp
qed
next
  show  $\{x \in \text{fset } (\text{fimage key } I). \text{sum-fset-by } x \text{ key weight } I \neq 0\}$ 
     $\subseteq \text{dom } (\text{map-of-fset-by key weight } I)$ 
    using  $q$ 
    unfolding  $\text{map-of-fset-by-def}$ 
    by (auto simp: Let-def)
qed

lemma  $\text{sum-map-map-of-fset-by}[simp]:$ 
   $\text{sum-map } (\text{map-of-fset-by key weight } I) = 1$ 
proof (cases fsum weight I = 0)
  case True
    then show ?thesis
      unfolding  $\text{map-of-fset-by-def sum-map-def}$ 
      by simp
  next
    case False
    let  $?q = \text{fsum weight } I$ 
    let  $?S = \{x \in \text{fset } (\text{fimage key } I). \text{sum-fset-by } x \text{ key weight } I \neq 0\}$ 

    have  $\text{dom}: \text{dom } (\text{map-of-fset-by key weight } I) = ?S$ 
      using False by (rule dom-map-of-fset-by)

    have val:
       $\bigwedge x. x \in ?S \implies$ 
         $\text{the } (\text{map-of-fset-by key weight } I x) =$ 
         $\text{sum-fset-by } x \text{ key weight } I / ?q$ 
      using False
      unfolding  $\text{map-of-fset-by-def}$ 
      by (auto simp: Let-def)

    have  $\text{sum-map } (\text{map-of-fset-by key weight } I) =$ 
       $(\sum x \in ?S. \text{the } (\text{map-of-fset-by key weight } I x))$ 
      unfolding  $\text{sum-map-def dom}$  by simp

```

```

also have ... =
  (∑ x∈?S. sum-fset-by x key weight I / ?q)
by (intro sum.cong refl val)
also have ... =
  (∑ x∈fset (fimage key I). sum-fset-by x key weight I / ?q)
apply (rule sum.mono-neutral-left)
  apply simp
  apply auto
done
also have ... =
  (∑ x∈fset (fimage key I). sum-fset-by x key weight I) / ?q
by (rule sum-divide-nnreal) simp
also have ... = ?q / ?q
  using fsum-by-total[of key weight I]
  by (simp add: fsum-fset-sum)
also have ... = 1
  by (rule divide-self[OF False])
finally show ?thesis .
qed

```

```

definition squish-by ::
  ('i ⇒ 'a) ⇒ ('i ⇒ prob) ⇒ 'i fset ⇒ ('a × prob) fset
where
  squish-by key weight I =
    fimage (λx. (x, sum-fset-by x key weight I)) (fimage key I)

```

```

definition dist-of-fset-by ::
  ('i ⇒ 'a) ⇒ ('i ⇒ prob) ⇒ 'i fset ⇒ 'a dist
where
  dist-of-fset-by key weight I =
    dist-of-fset (squish-by key weight I)

```

```

lemma fimage-fst-squish-by[simp]:
  fimage fst (squish-by key weight I) = fimage key I
unfolding squish-by-def
by (rule fset-eqI) auto

```

```

lemma squish-by-inj-on-fst:
  inj-on fst (fset (squish-by key weight I))
unfolding squish-by-def inj-on-def
by auto

```

```

lemma fsum-snd-squish-by:
  fsum snd (squish-by key weight I) = fsum weight I
proof -
  have inj:
    inj-on (λx. (x, sum-fset-by x key weight I)) (fset (fimage key I))
  by (auto intro!: inj-onI)

```



```

have fsum snd (squish-by key weight I) =
  fsum ( $\lambda x$ . sum-fset-by x key weight I) (fimage key I)
  unfolding squish-by-def
  apply (simp add: fsum-fset-sum fimage.rep-eq sum.reindex[OF inj] o-def)
  by (smt (verit) fset.set-map image-cong image-image inj sndI sum.reindex-cong)
also have ... = fsum weight I
  by (rule fsum-by-total)
finally show ?thesis .
qed

lemma sum-fset-x-squish-by:
  sum-fset-x x (squish-by key weight I) = sum-fset-by x key weight I
proof (cases x | $\in$ | fimage key I)
  case True
  then have (x, sum-fset-by x key weight I) | $\in$ | squish-by key weight I
    unfolding squish-by-def by auto
  then show ?thesis
    using squish-by-inj-on-fst
    by (metis squish-by-inj-on-fst  $\langle$ (x, sum-fset-by x key weight I) | $\in$ | squish-by key weight I $\rangle$ 
      sum-fset-x-unique)
  next
  case False
  then show ?thesis
    using sum-fset-x-notin-fst[of x squish-by key weight I]
    sum-fset-by-notin[of x key I weight]
    by simp
qed

lemma norm-fset-squish-by-1:
  assumes total: fsum weight I = 1
  shows
    norm-fset (squish-by key weight I) =
      ffilter ( $\lambda p$ . snd p  $\neq$  0) (squish-by key weight I)
proof -
  let ?A = squish-by key weight I

  have total-A: fsum snd ?A = 1
    using total by (simp add: fsum-snd-squish-by)

  have squish-A: squish ?A = ?A
    by (rule squish-id-if-inj-on-fst) (rule squish-by-inj-on-fst)

  have div-A: fimage ( $\lambda(x, p)$ . (x, p / fsum snd ?A)) ?A = ?A
    using total-A
    by (rule-tac fset-eqI) auto

  show ?thesis
    unfolding norm-fset-def

```

```

    by (simp add: total-A squish-A div-A Let-def)
qed

lemma map-of-filtered-unique:
  assumes uniq: inj-on fst (fset A)
  shows
    map-of (ffilter (λp. snd p ≠ 0) A) x =
      (if sum-fset-x x A = 0 then None else Some (sum-fset-x x A))
proof (cases x |∈| fimage fst A)
  case False
  then have sx: sum-fset-x x A = 0
    by (rule sum-fset-x-notin-fst)
  moreover have x |∉| fimage fst (ffilter (λp. snd p ≠ 0) A)
    using False by auto
  ultimately show ?thesis
    unfolding map-of-def by simp
next
  case True
  then obtain p where pA: (x, p) |∈| A
    by auto
  have sx: sum-fset-x x A = p
    using uniq pA by (rule sum-fset-x-unique)

  show ?thesis
proof (cases p = 0)
  case True
  have x |∉| fimage fst (ffilter (λp. snd p ≠ 0) A)
    using uniq pA True
    apply (auto simp: inj-on-def) by fastforce
  then show ?thesis
    using sx True unfolding map-of-def by simp
next
  case False
  have filt:
    ffilter (λq. fst q = x) (ffilter (λp. snd p ≠ 0) A) = {(x, p)}
    using uniq pA False
    apply (rule-tac fset-eqI) apply (auto simp: inj-on-def) by fastforce
  have xin: x |∈| fimage fst (ffilter (λp. snd p ≠ 0) A)
    using pA False apply auto by force
  have val:
    map-of-val (ffilter (λp. snd p ≠ 0) A) x = p
    unfolding map-of-val-def
    by (simp add: filt)
  show ?thesis
    using sx False xin val
    unfolding map-of-def by simp
qed
qed

```

lemma *map-of-fset-squish-by-1*:
assumes *total*: *fsum weight I = 1*
shows *map-of-fset (squish-by key weight I) = map-of-fset-by key weight I*
proof
fix *x*
let *?A = squish-by key weight I*

have *total-A*: *fsum snd ?A = 1*
using *total* **by** (*simp add: fsum-snd-squish-by*)

have *map-of-fset ?A x =*
map-of (ffilter (λp. snd p ≠ 0) ?A) x
unfolding *map-of-fset-def*
using *total-A norm-fset-squish-by-1 [OF total]*
apply *simp by (rule arg-cong2)*
also have ... =
(if sum-fset-x x ?A = 0 then None else Some (sum-fset-x x ?A))
by (*rule map-of-filtered-unique*) (*rule squish-by-inj-on-fst*)
also have ... =
(if sum-fset-by x key weight I = 0 then None else Some (sum-fset-by x key weight I))
by (*simp add: sum-fset-x-squish-by*)
also have ... = *map-of-fset-by key weight I x*
using *total*
unfolding *map-of-fset-by-def*
by *simp*
finally show *map-of-fset ?A x = map-of-fset-by key weight I x .*
qed

lemma *dist-of-fset-by-rep-eq-1*:
assumes *total*: *fsum weight I = 1*
shows *dist (dist-of-fset-by key weight I) = map-of-fset-by key weight I*
unfolding *dist-of-fset-by-def dist-of-fset.rep-eq*
using *map-of-fset-squish-by-1 [OF total]* .

lemma *sum-fset-by-id*:
sum-fset-by x id weight I = (if x |∈| I then weight x else 0)
unfolding *sum-fset-by-def fsum-fset-sum*
by (*simp add: ffilter.rep-eq*)

lemma *dist-of-fset-by-id-apply*:
assumes *total*: *fsum weight I = 1*
shows *dist (dist-of-fset-by id weight I) x =*
(let p = (if x |∈| I then weight x else 0) in
if p = 0 then None else Some p)
using *dist-of-fset-by-rep-eq-1 [OF total, of id] total*
by (*simp add: map-of-fset-by-def sum-fset-by-id*)

lemma *sum-map-dist[simp]*:

sum-map (*dist d*) = 1
by *transfer simp*

lemma *finite-dom-dist*[*simp*]:
finite (*dom* (*dist d*))
by *transfer simp*

lemma *dist-nonzero-on-dom*[*simp*]:
assumes $x \in \text{dom } (\text{dist } d)$
shows $\text{the } (\text{dist } d \ x) \neq 0$
using *assms* **by** *transfer simp*

lemma *fset-items*:
fset (*items d*) = $(\lambda x. (x, \text{the } (\text{dist } d \ x))) \text{ ` dom } (\text{dist } d)$
apply *transfer* **apply** (*simp add: raw-items-def*)
by (*metis fset.rep-transfer Rel-def*)

lemma *fsum-items*:
fsum *h* (*items d*) = $(\sum x \in \text{dom } (\text{dist } d). h \ (x, \text{the } (\text{dist } d \ x)))$
proof –
have *inj*: *inj-on* $(\lambda x. (x, \text{the } (\text{dist } d \ x))) \ (\text{dom } (\text{dist } d))$
by (*auto intro!: inj-onI*)
show *?thesis*
unfolding *fsum-fset-sum fset-items*
by (*subst sum.reindex[OF inj]*) *simp*
qed

lemma *fsum-items-key*:
fsum $(\lambda xp. \text{if fst } xp = x \text{ then snd } xp \text{ else } 0)$ (*items d*) =
option-default (*dist d x*)
unfolding *fsum-items*
apply *simp*
using *id-apply* **by** *fastforce*

lemma *fsum-snd-items*[*simp*]:
fsum *snd* (*items d*) = 1
unfolding *fsum-items sum-map-def* **apply** *simp*
by (*metis sum-map-def sum-map-dist*)

lemma *fsum-fbind-fimage-Pair*:
fsum *h* (*fbind* *A* $(\lambda a. \text{fimage } (\lambda b. (a, b)) \ (B \ a))$) =
fsum $(\lambda a. \text{fsum } (\lambda b. h \ (a, b)) \ (B \ a)) \ A$
proof –
have *set-eq*:
fset (*fbind* *A* $(\lambda a. \text{fimage } (\lambda b. (a, b)) \ (B \ a))$) =
 $(\text{SIGMA } a:\text{fset } A. \text{fset } (B \ a))$
by (*auto simp: fbind.rep-eq fimage.rep-eq Set.bind-def*)
have *sum h* $(\text{SIGMA } a:\text{fset } A. \text{fset } (B \ a)) =$

```

    (∑ a∈fset A. ∑ b∈fset (B a). h (a, b))
  by (subst sum.Sigma) auto
then show ?thesis
  unfolding fsum-fset-sum
  by (simp add: set-eq)
qed

lemma fsum-fbind-fimage-Pair-right:
  fsum h (fbind A (λa. fimage (λb. (b, a)) (B a))) =
    fsum (λa. fsum (λb. h (b, a)) (B a)) A
proof -
  have set-eq:
    fset (fbind A (λa. fimage (λb. (b, a)) (B a))) =
      (⋃ a∈fset A. (λb. (b, a)) ' fset (B a))
  by (auto simp: fbind.rep-eq fimage.rep-eq Set.bind-def)
  have sum h (⋃ a∈fset A. (λb. (b, a)) ' fset (B a)) =
    (∑ a∈fset A. sum h ((λb. (b, a)) ' fset (B a)))
  by (subst sum.UNION-disjoint) auto
  also have ... = (∑ a∈fset A. ∑ b∈fset (B a). h (b, a))
  by (intro sum.cong refl, subst sum.reindex) (auto simp: inj-on-def)
  finally have sum-eq:
    sum h (⋃ a∈fset A. (λb. (b, a)) ' fset (B a)) =
      (∑ a∈fset A. ∑ b∈fset (B a). h (b, a)) .
  then show ?thesis
    unfolding fsum-fset-sum
    by (simp add: set-eq)
qed

```

```

lemma sum-fset-by-bind-items:
  sum-fset-by x
    (λ(ms, ks). fst ks)
    (λ(ms, ks). snd ms * snd ks)
    (fbind (items d) (λms.
      case ms of ((a, s'), p) ⇒
        fimage (λks. (ms, ks)) (items (k a s')))) =
    (∑ i∈dom (dist d).
      the (dist d i) * option-default (dist (k (fst i) (snd i)) x))
  unfolding sum-fset-by-def
  apply (simp add: fsum-ffilter-if fsum-fbind-fimage-Pair case-prod-unfold)
  unfolding fsum-items
  apply (intro sum.cong refl)
  apply (simp add: fsum-mult-left-nnreal fsum-items-key)
  by (simp add: dom-def option.case-eq-if)

```

```

lemma fsum-items-mult-left:
  fsum (λx. c * snd x) (items d) = c
proof -
  have fsum (λx. c * snd x) (items d) =
    c * fsum snd (items d)

```

```

    by (rule fsum-mult-left-nnreal)
  also have ... = c
    by simp
  finally show ?thesis .
qed

lemma fsum-bind-items-total:
  fsum
    (λ(ms, ks). snd ms * snd ks)
    (fbind (items d) (λms.
      case ms of ((a, s^), p) ⇒
        fimage (λks. (ms, ks)) (items (k a s^)))) = 1
proof -
  have fsum
    (λ(ms, ks). snd ms * snd ks)
    (fbind (items d) (λms.
      case ms of ((a, s^), p) ⇒
        fimage (λks. (ms, ks)) (items (k a s^)))) =
    fsum
      (λms. fsum (λks. snd ms * snd ks)
        (items (k (fst (fst ms)) (snd (fst ms)))))
      (items d)
    by (simp add: fsum-fbind-fimage-Pair case-prod-unfold)

  also have ... = fsum snd (items d)
    by (intro fsum.cong refl) (simp add: fsum-items-mult-left)

  also have ... = 1
    by simp

  finally show ?thesis .
qed

```

1.2.5 Mapper

```

definition dist-map:: ('a ⇒ 'b) ⇒ 'a dist ⇒ 'b dist
where
  dist-map f a = dist-of-fset-by (f ∘ fst) snd (items a)

```

```

definition set-dist :: 'a dist ⇒ 'a set
where
  set-dist d = dom (dist d)

```

```

lemma set-dist-dist-of-fset-by-id-subset:
  assumes total: fsum weight I = 1
  shows set-dist (dist-of-fset-by id weight I) ⊆ fset I
  using dist-of-fset-by-id-apply[OF total]
  unfolding set-dist-def
  by (auto simp: Let-def split: if-splits)

```

1.2.6 Relator

definition $rel-dist :: ('a \Rightarrow 'b \Rightarrow bool) \Rightarrow 'a \ dist \Rightarrow 'b \ dist \Rightarrow bool$

where

$rel-dist \ R \ p \ q \longleftrightarrow$
 $(\exists r :: ('a \times 'b) \ dist.$
 $set-dist \ r \subseteq \{(x, y). \ R \ x \ y\} \wedge$
 $dist-map \ fst \ r = p \wedge$
 $dist-map \ snd \ r = q)$

lemma $nnreal-mult-div-cancel$:

fixes $a \ b :: nnreal$
assumes $b \neq 0$
shows $a * b / b = a$
using $assms$ **by** $transfer \ simp$

1.2.7 BNF Setup

definition $glue-items ::$

$('a \times 'b) \ dist \Rightarrow ('b \times 'c) \ dist \Rightarrow$
 $((('a \times 'b) \times prob) \times (('b \times 'c) \times prob)) \ fset$

where

$glue-items \ p \ q =$
 $fbind \ (items \ p)$
 $(\lambda ab. \ fimage \ (\lambda bc. \ (ab, bc))$
 $(ffilter \ (\lambda bc. \ snd \ (fst \ ab) = fst \ (fst \ bc)) \ (items \ q)))$

definition $glue-weight ::$

$('a \times 'b) \ dist \Rightarrow$
 $((('a \times 'b) \times prob) \times (('b \times 'c) \times prob)) \Rightarrow prob$

where

$glue-weight \ p \ x =$
 $snd \ (fst \ x) * snd \ (snd \ x) /$
 $the \ (dist \ (dist-map \ snd \ p) \ (snd \ (fst \ (fst \ x))))$

definition $glue-key ::$

$((('a \times 'b) \times prob) \times (('b \times 'c) \times prob)) \Rightarrow 'a \times 'c$

where

$glue-key \ x = (fst \ (fst \ (fst \ x)), snd \ (fst \ (snd \ x)))$

definition $glue-source ::$

$('a \times 'b) \ dist \Rightarrow ('b \times 'c) \ dist \Rightarrow$
 $((('a \times 'b) \times prob) \times (('b \times 'c) \times prob)) \ dist$

where

$glue-source \ p \ q = dist-of-fset-by \ id \ (glue-weight \ p) \ (glue-items \ p \ q)$

definition $glue-dist :: ('a \times 'b) \ dist \Rightarrow ('b \times 'c) \ dist \Rightarrow ('a \times 'c) \ dist$

where

$glue-dist \ p \ q = dist-map \ glue-key \ (glue-source \ p \ q)$

lemma *dist-map-rep-eq*:

dist (dist-map f d) = map-of-fset-by (f ∘ fst) snd (items d)

unfolding *dist-map-def*

by (*rule dist-of-fset-by-rep-eq-1*) *simp*

lemma *sum-fset-by-items*:

sum-fset-by x (f ∘ fst) snd (items d) =

(∑ y ∈ {y ∈ dom (dist d). f y = x}. the (dist d y))

proof –

let *?g* = $\lambda y. (y, \text{the } (dist\ d\ y))$

let *?S* = $\{y \in dom\ (dist\ d). f\ y = x\}$

have *inj*: *inj-on* *?g* $\{y \in dom\ (dist\ d). f\ y = x\}$

by (*auto intro!*: *inj-onI*)

have *filt*:

$\{i \in ?g\ ' dom\ (dist\ d). (f \circ fst)\ i = x\} =$

$?g\ '\{y \in dom\ (dist\ d). f\ y = x\}$

by *auto*

have *set-eq*:

$\{i. (f \circ fst)\ i = x\} \cap fset\ (items\ d) = ?g\ '\ ?S$

by (*auto simp*: *fset-items*)

have *sum snd* $(\{i. (f \circ fst)\ i = x\} \cap fset\ (items\ d)) =$

sum snd $(?g\ '\ ?S)$

using *set-eq* **by** *simp*

also have ... = $(\sum y \in ?S. snd\ (?g\ y))$

by (*simp add*: *sum.reindex[OF inj]*)

also have ... = $(\sum y \in ?S. the\ (dist\ d\ y))$

by *simp*

finally have *sum-eq*:

sum snd $(\{i. (f \circ fst)\ i = x\} \cap fset\ (items\ d)) =$

$(\sum y \in ?S. the\ (dist\ d\ y))$.

have *set-eq'*:

$\{i. i \in items\ d \wedge (f \circ fst)\ i = x\} = ?g\ '\ ?S$

by (*auto simp*: *fset-items*)

have *sum-eq'*:

$(\sum i \mid i \in items\ d \wedge (f \circ fst)\ i = x. snd\ i) =$

$(\sum y \in ?S. the\ (dist\ d\ y))$

using *set-eq'* **by** (*simp add*: *sum.reindex[OF inj]*)

show *?thesis*

unfolding *sum-fset-by-def fsum-fset-sum*

using *sum-eq* **by** (*simp add*: *ffilter.rep-eq Int-def*)

qed

lemma *dist-map-apply*:

dist (dist-map f d) x =

(let p = (∑ y ∈ {y ∈ dom (dist d). f y = x}. the (dist d y))

in if p = 0 then None else Some p)

proof –

have *total*: *fsum snd* $(items\ d) \neq 0$

by *simp*


```

show ?thesis
  unfolding dist-map-rep-eq map-of-fset-by-def sum-fset-by-items
  using total by (simp add: Let-def)
qed

lemma fsum-items-filter-dist-map:
  fsum snd (ffilter (λi. f (fst i) = x) (items d)) =
    (∑ y ∈ {y ∈ dom (dist d). f y = x}. the (dist d y))
  using sum-fset-by-items[of x f d]
  by (simp add: sum-fset-by-def comp-def)

lemma fsum-items-filter-dist-map-on-dom:
  assumes x ∈ dom (dist (dist-map f d))
  shows fsum snd (ffilter (λi. f (fst i) = x) (items d)) =
    the (dist (dist-map f d) x)
proof -
  let ?p = (∑ y ∈ {y ∈ dom (dist d). f y = x}. the (dist d y))
  have dist (dist-map f d) x = Some ?p
    using assms by (simp add: dist-map-apply Let-def domIff split: if-splits)
  then show ?thesis
    by (simp add: fsum-items-filter-dist-map)
qed

lemma dist-map-dist-of-fset-by-id-apply:
  assumes total: fsum weight I = 1
  shows dist (dist-map f (dist-of-fset-by id weight I)) x =
    (let p = fsum weight (ffilter (λi. f i = x) I)
     in if p = 0 then None else Some p)
proof -
  let ?d = dist-of-fset-by id weight I
  let ?D = {y ∈ fset I. weight y ≠ 0}
  let ?S = {y ∈ dom (dist ?d). f y = x}
  let ?T = {y ∈ fset I. f y = x}
  have T-eq: ?T = {i. f i = x} ∩ fset I
    by auto
  have dom-eq: dom (dist ?d) = ?D
    using dist-of-fset-by-id-apply[OF total]
    by (auto simp: Let-def split: if-splits)
  have sum-eq:
    (∑ y ∈ ?S. the (dist ?d y)) = (∑ y ∈ ?T. weight y)
  proof -
    have (∑ y ∈ ?S. the (dist ?d y)) = (∑ y ∈ {y ∈ ?T. weight y ≠ 0}. weight y)
      using dist-of-fset-by-id-apply[OF total]
      by (intro sum.cong) (auto simp: dom-eq Let-def split: if-splits)
    also have ... = (∑ y ∈ ?T. weight y)
      by (rule sum.mono-neutral-left) auto
    finally show ?thesis .
  qed
qed
show ?thesis

```

unfolding *dist-map-apply fsum-fset-sum*
using *sum-eq* **by** (*simp add: ffilter.rep-eq Let-def T-eq*)
qed

lemma *set-dist-dist-map:*

set-dist (dist-map f d) = f ‘ set-dist d

proof –

have *sum-nonzero:*

$(\sum y \in \{y \in \text{dom } (dist\ d). f\ y = f\ x\}. the\ (dist\ d\ y)) \neq 0$

if *x*: $x \in \text{dom } (dist\ d)$ **for** *x*

proof –

let *?S* = $\{y \in \text{dom } (dist\ d). f\ y = f\ x\}$

have *fin*: *finite ?S*

by *simp*

have *xS*: $x \in ?S$

using *x* **by** *simp*

moreover **have** $0 < the\ (dist\ d\ x)$

using *x* **by** *simp*

ultimately **show** *?thesis*

using *sum-pos2[OF fin xS, of $\lambda y. the\ (dist\ d\ y)$]*

by *auto*

qed

show *?thesis*

proof

show *set-dist (dist-map f d) $\subseteq$ f ‘ set-dist d*

proof

fix *z*

assume *z*: $z \in \text{set-dist } (dist\text{-map } f\ d)$

let *?S* = $\{y \in \text{dom } (dist\ d). f\ y = z\}$

have *fin*: *finite ?S*

by *simp*

have *p-ne*: $(\sum y \in ?S. the\ (dist\ d\ y)) \neq 0$

using *z* **unfolding** *set-dist-def dist-map-apply*

by (*auto simp: Let-def split: if-splits*)

have $\exists y. y \in ?S$

proof (*rule ccontr*)

assume $\neg (\exists y. y \in ?S)$

then **have** *?S* = $\{\}$

by *auto*

then **show** *False*

using *p-ne* **by** (*metis sum.empty*)

qed

then **obtain** *y* **where** $y \in ?S$

by *blast*

then **show** $z \in f\ ' \text{set-dist } d$

unfolding *set-dist-def* **by** *auto*

qed

next

```

show  $f \text{ ' } \text{set-dist } d \subseteq \text{set-dist } (\text{dist-map } f \text{ } d)$ 
proof
  fix  $z$ 
  assume  $z \in f \text{ ' } \text{set-dist } d$ 
  then obtain  $x$  where  $x: x \in \text{dom } (\text{dist } d)$  and  $z: z = f \text{ } x$ 
    unfolding set-dist-def by auto
  have  $(\sum_{y \in \{y \in \text{dom } (\text{dist } d). f \text{ } y = z\}. \text{ the } (\text{dist } d \text{ } y))} \neq 0$ 
    using sum-nonzero[OF x] z by simp
  then show  $z \in \text{set-dist } (\text{dist-map } f \text{ } d)$ 
    unfolding set-dist-def dist-map-apply
    by (simp add: Let-def domIff)
qed
qed
qed

lemma fsum-glue-weight-inner:
  fixes  $p :: ('a \times 'b) \text{ dist}$  and  $q :: ('b \times 'c) \text{ dist}$ 
  assumes mid:  $\text{dist-map snd } p = \text{dist-map fst } q$ 
  and  $ab: ab \in \text{items } p$ 
  shows  $fsum (\lambda bc. \text{glue-weight } p (ab, bc))$ 
     $(\text{ffilter } (\lambda bc. \text{snd } (\text{fst } ab) = \text{fst } (\text{fst } bc)) (\text{items } q)) = \text{snd } ab$ 
proof -
  let  $?b = \text{snd } (\text{fst } ab)$ 
  let  $?den = \text{the } (\text{dist } (\text{dist-map snd } p) \text{ } ?b)$ 
  have fst-ab-dom:  $\text{fst } ab \in \text{dom } (\text{dist } p)$ 
    using  $ab$  by (auto simp: fset-items)
  have b-dom:  $?b \in \text{dom } (\text{dist } (\text{dist-map snd } p))$ 
    using fst-ab-dom set-dist-dist-map[of snd p]
    unfolding set-dist-def by auto
  then have den-ne:  $?den \neq 0$ 
    by simp
  have sum-q:
     $fsum \text{ snd } (\text{ffilter } (\lambda bc. \text{fst } (\text{fst } bc) = ?b) (\text{items } q)) = ?den$ 
    using fsum-items-filter-dist-map-on-dom[of ?b fst q] mid b-dom
    by simp
  have  $fsum (\lambda bc. \text{glue-weight } p (ab, bc))$ 
     $(\text{ffilter } (\lambda bc. \text{snd } (\text{fst } ab) = \text{fst } (\text{fst } bc)) (\text{items } q)) =$ 
     $fsum (\lambda bc. \text{snd } ab * \text{snd } bc / ?den)$ 
     $(\text{ffilter } (\lambda bc. \text{fst } (\text{fst } bc) = ?b) (\text{items } q))$ 
    unfolding glue-weight-def by (simp add: eq-commute)
  also have ... =
     $\text{snd } ab * fsum \text{ snd } (\text{ffilter } (\lambda bc. \text{fst } (\text{fst } bc) = ?b) (\text{items } q)) / ?den$ 
    by (simp add: fsum-divide-nnreal fsum-mult-left-nnreal)
  also have ... =  $\text{snd } ab$ 
    using sum-q den-ne by (simp add: nnreal-mult-div-cancel)
  finally show thesis .
qed

```

lemma *fsum-glue-weight-inner-left*:

```

fixes  $p :: ('a \times 'b) \text{ dist}$  and  $q :: ('b \times 'c) \text{ dist}$ 
assumes  $\text{mid}: \text{dist-map snd } p = \text{dist-map fst } q$ 
and  $\text{bc}: \text{bc} \in | \text{items } q$ 
shows  $\text{fsum } (\lambda ab. \text{glue-weight } p \text{ } (ab, bc))$ 
 $(\text{ffilter } (\lambda ab. \text{snd } (\text{fst } ab) = \text{fst } (\text{fst } bc)) (\text{items } p)) = \text{snd } bc$ 
proof –
  let  $?b = \text{fst } (\text{fst } bc)$ 
  let  $?den = \text{the } (\text{dist } (\text{dist-map snd } p) ?b)$ 
  have  $\text{fst-bc-dom}: \text{fst } bc \in \text{dom } (\text{dist } q)$ 
  using  $\text{bc}$  by  $(\text{auto simp: fset-items})$ 
  have  $\text{b-dom-q}: ?b \in \text{dom } (\text{dist } (\text{dist-map fst } q))$ 
  using  $\text{fst-bc-dom set-dist-dist-map[of fst } q]$ 
  unfolding  $\text{set-dist-def}$  by  $\text{auto}$ 
  then have  $\text{b-dom-p}: ?b \in \text{dom } (\text{dist } (\text{dist-map snd } p))$ 
  using  $\text{mid}$  by  $\text{simp}$ 
  then have  $\text{den-ne}: ?den \neq 0$ 
  by  $\text{simp}$ 
  have  $\text{sum-p}:$ 
 $\text{fsum snd } (\text{ffilter } (\lambda ab. \text{snd } (\text{fst } ab) = ?b) (\text{items } p)) = ?den$ 
  using  $\text{fsum-items-filter-dist-map-on-dom[of ?b snd } p]$   $\text{b-dom-p}$ 
  by  $\text{simp}$ 
  have  $\text{fsum } (\lambda ab. \text{glue-weight } p \text{ } (ab, bc))$ 
 $(\text{ffilter } (\lambda ab. \text{snd } (\text{fst } ab) = \text{fst } (\text{fst } bc)) (\text{items } p)) =$ 
 $\text{fsum } (\lambda ab. \text{snd } bc * \text{snd } ab / ?den)$ 
 $(\text{ffilter } (\lambda ab. \text{snd } (\text{fst } ab) = ?b) (\text{items } p))$ 
  unfolding  $\text{glue-weight-def}$  by  $(\text{simp add: mult.commute})$ 
  also have  $\dots =$ 
 $\text{snd } bc * \text{fsum snd } (\text{ffilter } (\lambda ab. \text{snd } (\text{fst } ab) = ?b) (\text{items } p)) / ?den$ 
  by  $(\text{simp add: fsum-divide-nnreal fsum-mult-left-nnreal})$ 
  also have  $\dots = \text{snd } bc$ 
  using  $\text{sum-p den-ne}$  by  $(\text{simp add: nnreal-mult-div-cancel})$ 
  finally show  $?thesis$  .
qed

```

```

lemma  $\text{glue-items-commute}:$ 
 $\text{glue-items } p \text{ } q =$ 
 $\text{fbind } (\text{items } q)$ 
 $(\lambda bc. \text{fimage } (\lambda ab. (ab, bc))$ 
 $(\text{ffilter } (\lambda ab. \text{snd } (\text{fst } ab) = \text{fst } (\text{fst } bc)) (\text{items } p)))$ 
unfolding  $\text{glue-items-def}$ 
by  $(\text{rule fset-eqI}) (\text{auto simp: fbind.rep-eq fimage.rep-eq Set.bind-def})$ 

```

```

lemma  $\text{fsum-glue-weight}:$ 
fixes  $p :: ('a \times 'b) \text{ dist}$  and  $q :: ('b \times 'c) \text{ dist}$ 
assumes  $\text{mid}: \text{dist-map snd } p = \text{dist-map fst } q$ 
shows  $\text{fsum } (\text{glue-weight } p) (\text{glue-items } p \text{ } q) = 1$ 
proof –
  have  $\text{fsum } (\text{glue-weight } p) (\text{glue-items } p \text{ } q) =$ 
 $\text{fsum } (\lambda ab. \text{fsum } (\lambda bc. \text{glue-weight } p \text{ } (ab, bc)))$ 

```

```

    (ffilter (λbc. snd (fst ab) = fst (fst bc)) (items q))) (items p)
  unfolding glue-items-def
  by (simp add: fsum-fbind-fimage-Pair)
also have ... = fsum snd (items p)
  by (intro fsum.cong refl) (simp add: fsum-glue-weight-inner[OF mid])
also have ... = 1
  by simp
finally show ?thesis .
qed

lemma fsum-glue-weight-fst:
  fixes p :: ('a × 'b) dist and q :: ('b × 'c) dist
  assumes mid: dist-map snd p = dist-map fst q
  shows fsum (glue-weight p)
    (ffilter (λi. fst (glue-key i) = a) (glue-items p q)) =
    fsum snd (ffilter (λab. fst (fst ab) = a) (items p))
proof -
  have fsum (glue-weight p)
    (ffilter (λi. fst (glue-key i) = a) (glue-items p q)) =
    fsum (λab. fsum (λbc.
      if fst (fst ab) = a then glue-weight p (ab, bc) else 0)
      (ffilter (λbc. snd (fst ab) = fst (fst bc)) (items q))) (items p)
  unfolding glue-items-def
  by (simp add: fsum-ffilter-if fsum-fbind-fimage-Pair glue-key-def)
also have ... = fsum (λab. if fst (fst ab) = a then snd ab else 0) (items p)
proof (intro fsum.cong refl)
  fix ab
  assume ab: ab |∈| items p
  show fsum (λbc. if fst (fst ab) = a then glue-weight p (ab, bc) else 0)
    (ffilter (λbc. snd (fst ab) = fst (fst bc)) (items q)) =
    (if fst (fst ab) = a then snd ab else 0)
  proof (cases fst (fst ab) = a)
    case True
    then show ?thesis
      using fsum-glue-weight-inner[OF mid ab] by simp
  next
    case False
    then show ?thesis
      by (simp add: fsum-fset-sum)
  qed
qed
also have ... = fsum snd (ffilter (λab. fst (fst ab) = a) (items p))
  by (simp add: fsum-ffilter-if)
finally show ?thesis .
qed

lemma fsum-glue-weight-snd:
  fixes p :: ('a × 'b) dist and q :: ('b × 'c) dist
  assumes mid: dist-map snd p = dist-map fst q

```

```

shows  $fsum\ (glue-weight\ p)$ 
  ( $ffilter\ (\lambda i. snd\ (glue-key\ i) = c)\ (glue-items\ p\ q)$ ) =
   $fsum\ snd\ (ffilter\ (\lambda bc. snd\ (fst\ bc) = c)\ (items\ q))$ 
proof –
  have  $fsum\ (glue-weight\ p)$ 
    ( $ffilter\ (\lambda i. snd\ (glue-key\ i) = c)\ (glue-items\ p\ q)$ ) =
     $fsum\ (\lambda bc. fsum\ (\lambda ab.$ 
       $if\ snd\ (fst\ bc) = c\ then\ glue-weight\ p\ (ab,\ bc)\ else\ 0)$ 
      ( $ffilter\ (\lambda ab. snd\ (fst\ ab) = fst\ (fst\ bc))\ (items\ p))$ ) ( $items\ q$ )
    unfolding  $glue-items-commute$ 
    by ( $simp\ add: fsum-ffilter-if\ fsum-fbind-fimage-Pair-right\ glue-key-def$ )
  also have ... =  $fsum\ (\lambda bc. if\ snd\ (fst\ bc) = c\ then\ snd\ bc\ else\ 0)\ (items\ q)$ 
proof ( $intro\ fsum.cong\ refl$ )
  fix  $bc$ 
  assume  $bc: bc \in items\ q$ 
  show  $fsum\ (\lambda ab. if\ snd\ (fst\ bc) = c\ then\ glue-weight\ p\ (ab,\ bc)\ else\ 0)$ 
    ( $ffilter\ (\lambda ab. snd\ (fst\ ab) = fst\ (fst\ bc))\ (items\ p)$ ) =
    ( $if\ snd\ (fst\ bc) = c\ then\ snd\ bc\ else\ 0$ )
  proof ( $cases\ snd\ (fst\ bc) = c$ )
  case  $True$ 
  then show ?thesis
    using  $fsum-glue-weight-inner-left[OF\ mid\ bc]$  by  $simp$ 
  next
  case  $False$ 
  then show ?thesis
    by ( $simp\ add: fsum-fset-sum$ )
  qed
qed
also have ... =  $fsum\ snd\ (ffilter\ (\lambda bc. snd\ (fst\ bc) = c)\ (items\ q))$ 
  by ( $simp\ add: fsum-ffilter-if$ )
finally show ?thesis .
qed

```

```

lemma  $dist-map-id$ :
   $dist-map\ id = id$ 
proof ( $rule\ ext, rule\ dist-inject[THEN\ iffD1], rule\ ext$ )
  fix  $d::'a\ dist$  and  $x::'a$ 
  show  $dist.dist\ (dist-map\ id\ d)\ x = dist.dist\ (id\ d)\ x$ 
proof ( $cases\ x \in dom\ (dist\ d)$ )
  case  $True$ 
  have  $S: \{y \in dom\ (dist\ d). id\ y = x\} = \{x\}$ 
    using  $True$  by  $auto$ 
  have  $D: dist\ d\ x = Some\ (the\ (dist\ d\ x))$ 
    using  $True$  by  $auto$ 
  have  $NZ: the\ (dist\ d\ x) \neq 0$ 
    using  $True$  by  $simp$ 
  have  $dist\ (dist-map\ id\ d)\ x =$ 
    ( $let\ p = (\sum_{y \in \{y \in dom\ (dist\ d). id\ y = x\}}. the\ (dist\ d\ y))$ 
       $in\ if\ p = 0\ then\ None\ else\ Some\ p$ )

```

```

    by (rule dist-map-apply)
  also have ... = Some (the (dist d x))
    using S NZ by simp
  also have ... = dist d x
    using D by simp
  finally show ?thesis by simp
next
case False
have S: {y ∈ dom (dist d). id y = x} = {}
  using False by auto
have S': {y. y = x ∧ y ∈ dom (dist d)} = {}
  using False by auto
have p0: (∑ y | y = x ∧ y ∈ dom (dist d). the (dist d y)) = 0
  by (simp add: S')
have D: dist d x = None
  using False by auto
have dist (dist-map id d) x =
  (let p = (∑ y ∈ {y ∈ dom (dist d). id y = x}. the (dist d y))
   in if p = 0 then None else Some p)
  by (rule dist-map-apply)
also have ... = None
  using S S' p0 by simp
also have ... = dist d x
  using D by simp
finally show ?thesis by simp
qed
qed

lemma dist-map-delta-dist[simp]:
  dist-map f (delta-dist x) = delta-dist (f x)
proof (rule dist-inject[THEN iffD1], rule ext)
  fix y
  show dist (dist-map f (delta-dist x)) y = dist (delta-dist (f x)) y
proof (cases f x = y)
  case True
  have {z ∈ dom (dist (delta-dist x)). f z = y} = {x}
    using True by (auto simp: dist-delta-dist delta-map-def)
  then show ?thesis
    using True by (simp add: dist-map-apply dist-delta-dist delta-map-def)
next
case False
have {z ∈ dom (dist (delta-dist x)). f z = y} = {}
  using False by (auto simp: dist-delta-dist delta-map-def)
then show ?thesis
  using False by (simp add: dist-map-apply dist-delta-dist delta-map-def)
qed
qed

lemma dist-map-cong:

```

```

assumes  $\bigwedge x. x \in \text{set-dist } d \implies f\ x = g\ x$ 
shows  $\text{dist-map } f\ d = \text{dist-map } g\ d$ 
proof (rule dist-inject[THEN iffD1], rule ext)
  fix  $y$ 
  have  $\{x \in \text{dom } (\text{dist } d). f\ x = y\} = \{x \in \text{dom } (\text{dist } d). g\ x = y\}$ 
    using assms unfolding set-dist-def by (auto simp: domIff)
  then show  $\text{dist } (\text{dist-map } f\ d)\ y = \text{dist } (\text{dist-map } g\ d)\ y$ 
    by (simp add: dist-map-apply)
qed

```

```

lemma sum-partition-by-image:
  fixes  $w :: 'a \Rightarrow \text{nnreal}$ 
  assumes fin: finite M
  shows
     $(\sum y \in \{y \in f\ ` M. g\ y = x\}. \sum z \in \{z \in M. f\ z = y\}. w\ z) =$ 
     $(\sum z \in \{z \in M. g\ (f\ z) = x\}. w\ z)$ 
proof -
  let  $?Y = \{y \in f\ ` M. g\ y = x\}$ 
  have finY: finite ?Y
    using fin by simp
  have finM: finite M
    by (rule fin)
  have filter-sum:
     $(\sum z \in \{z \in M. f\ z = y\}. w\ z) =$ 
     $(\sum z \in M. \text{if } f\ z = y \text{ then } w\ z \text{ else } 0) \text{ for } y$ 
proof -
  have  $(\sum z \in M. \text{if } f\ z = y \text{ then } w\ z \text{ else } 0) =$ 
     $(\sum z \in \{z \in M. f\ z = y\}. w\ z)$ 
    using fin
    by (intro sum.mono-neutral-cong-right) auto
  then show ?thesis by simp
qed

```

```

have  $(\sum y \in ?Y. \sum z \in \{z \in M. f\ z = y\}. w\ z) =$ 
   $(\sum y \in ?Y. \sum z \in M. \text{if } f\ z = y \text{ then } w\ z \text{ else } 0)$ 
  by (rule sum.cong) (simp-all add: filter-sum)
also have ... =
   $(\sum z \in M. \sum y \in ?Y. \text{if } f\ z = y \text{ then } w\ z \text{ else } 0)$ 
  by (rule sum.swap)
also have ... =
   $(\sum z \in M. \text{if } g\ (f\ z) = x \text{ then } w\ z \text{ else } 0)$ 
proof (intro sum.cong refl)
  fix  $z$ 
  assume  $z: z \in M$ 
  show  $(\sum y \in ?Y. \text{if } f\ z = y \text{ then } w\ z \text{ else } 0) =$ 
     $(\text{if } g\ (f\ z) = x \text{ then } w\ z \text{ else } 0)$ 
  proof (cases  $g\ (f\ z) = x$ )
    case True
    then have  $fz: f\ z \in ?Y$ 

```



```

    using z by simp
  have  $(\sum y \in ?Y. \text{if } f z = y \text{ then } w z \text{ else } 0) =$ 
     $(\sum y \in ?Y. \text{if } y = f z \text{ then } w z \text{ else } 0)$ 
    by (intro sum.cong refl) simp
  also have ... = w z
    using finY fz by (simp)
  finally show ?thesis
    using True by simp
next
case False
then have  $\bigwedge y. y \in ?Y \implies f z \neq y$ 
  by auto
then show ?thesis
  using False by simp
qed
qed
also have ... =  $(\sum z \in \{z \in M. g (f z) = x\}. w z)$ 
  using finM by (rule sum.inter-filter[symmetric])
finally show ?thesis .
qed

lemma dist-map-comp:
  dist-map (g ∘ f) = dist-map g ∘ dist-map f
proof (rule ext)
  fix d
  show dist-map (g ∘ f) d = (dist-map g ∘ dist-map f) d
  proof (rule dist-inject[THEN iffD1], rule ext)
    fix x
    let ?M = dom (dist d)
    let ?lhs =  $(\sum y \in \{y \in \text{dom } (dist (dist-map f d)). g y = x\}. the (dist (dist-map f d) y))$ 
    let ?rhs =  $(\sum z \in \{z \in ?M. (g \circ f) z = x\}. the (dist d z))$ 

    have dom-map:  $\text{dom } (dist (dist-map f d)) = f^{-1} ?M$ 
      using set-dist-dist-map[of f d]
      unfolding set-dist-def by simp

    have lhs-sum:
      ?lhs =
         $(\sum y \in \{y \in f^{-1} ?M. g y = x\}. \sum z \in \{z \in ?M. f z = y\}. the (dist d z))$ 
    proof -
      have Aeq:
         $\{y \in \text{dom } (dist (dist-map f d)). g y = x\} =$ 
         $\{y \in f^{-1} ?M. g y = x\}$ 
        by (simp add: dom-map)
      show ?thesis
        unfolding Aeq
    proof (intro sum.cong refl)

```

```

fix  $y$ 
assume  $y \in \{y \in f^{-1} ?M. g\ y = x\}$ 
then have  $y\text{-dom}: y \in \text{dom} (\text{dist} (\text{dist-map } f\ d))$ 
  by (simp add: dom-map)
have  $nz:$ 
   $(\sum z \in \{z \in ?M. f\ z = y\}. \text{the } (\text{dist } d\ z)) \neq 0$ 
  using  $y\text{-dom}$  unfolding dist-map-apply
  by (auto simp: Let-def split: if-splits)
have  $\text{dist} (\text{dist-map } f\ d)\ y =$ 
   $\text{Some } (\sum z \in \{z \in ?M. f\ z = y\}. \text{the } (\text{dist } d\ z))$ 
  unfolding dist-map-apply using  $nz$  by simp
then show  $\text{the } (\text{dist} (\text{dist-map } f\ d)\ y) =$ 
   $(\sum z \in \{z \in ?M. f\ z = y\}. \text{the } (\text{dist } d\ z))$ 
  by simp
qed
qed

have part:
   $(\sum y \in \{y \in f^{-1} ?M. g\ y = x\}. \sum z \in \{z \in ?M. f\ z = y\}. \text{the } (\text{dist } d\ z)) =$ 
   $(\sum z \in \{z \in ?M. g\ (f\ z) = x\}. \text{the } (\text{dist } d\ z))$ 
  by (rule sum-partition-by-image) simp
have  $\text{sums-eq}: ?lhs = ?rhs$ 
  using lhs-sum part by (simp add: comp-def)

have left:
   $\text{dist} (\text{dist-map } (\lambda z. g\ (f\ z))\ d)\ x =$ 
   $(\text{let } p = ?rhs \text{ in if } p = 0 \text{ then None else Some } p)$ 
  by (simp add: dist-map-apply)
have right:
   $\text{dist} (\text{dist-map } g\ (\text{dist-map } f\ d))\ x =$ 
   $(\text{let } p = ?lhs \text{ in if } p = 0 \text{ then None else Some } p)$ 
  by (simp add: dist-map-apply)

show  $\text{dist} (\text{dist-map } (g \circ f)\ d)\ x =$ 
   $\text{dist} ((\text{dist-map } g \circ \text{dist-map } f)\ d)\ x$ 
  unfolding comp-def
  using left right sums-eq by simp
qed
qed

lemma dist-map-fst-glue-dist:
  fixes  $p :: ('a \times 'b)\ \text{dist}$  and  $q :: ('b \times 'c)\ \text{dist}$ 
  assumes  $\text{mid}: \text{dist-map } \text{snd } p = \text{dist-map } \text{fst } q$ 
  shows  $\text{dist-map } \text{fst } (\text{glue-dist } p\ q) = \text{dist-map } \text{fst } p$ 
proof (rule dist-inject[THEN iffD1], rule ext)
  fix  $a$ 
  have  $\text{total}: \text{fsum } (\text{glue-weight } p)\ (\text{glue-items } p\ q) = 1$ 
  by (rule fsum-glue-weight[OF mid])

```

```

have comp-fst:
  dist-map fst (glue-dist p q) = dist-map (fst ∘ glue-key) (glue-source p q)
  unfolding glue-dist-def
  by (metis comp-apply dist-map-comp)
have left:
  dist (dist-map fst (glue-dist p q)) a =
    (let s = fsum (glue-weight p)
     (ffilter (λi. fst (glue-key i) = a) (glue-items p q))
     in if s = 0 then None else Some s)
  unfolding comp-fst glue-source-def
  using dist-map-dist-of-fset-by-id-apply[OF total, of fst ∘ glue-key a]
  by (simp add: comp-def)
have right:
  dist (dist-map fst p) a =
    (let s = fsum snd (ffilter (λab. fst (fst ab) = a) (items p))
     in if s = 0 then None else Some s)
  by (simp add: dist-map-apply fsum-items-filter-dist-map[symmetric])
show dist (dist-map fst (glue-dist p q)) a = dist (dist-map fst p) a
  using left right fsum-glue-weight-fst[OF mid, of a] by simp
qed

```

```

lemma dist-map-snd-glue-dist:
  fixes p :: ('a × 'b) dist and q :: ('b × 'c) dist
  assumes mid: dist-map snd p = dist-map fst q
  shows dist-map snd (glue-dist p q) = dist-map snd q
proof (rule dist-inject[THEN iffD1], rule ext)
  fix c
  have total: fsum (glue-weight p) (glue-items p q) = 1
  by (rule fsum-glue-weight[OF mid])
  have comp-snd:
    dist-map snd (glue-dist p q) = dist-map (snd ∘ glue-key) (glue-source p q)
    unfolding glue-dist-def
    by (metis comp-apply dist-map-comp)
  have left:
    dist (dist-map snd (glue-dist p q)) c =
      (let s = fsum (glue-weight p)
       (ffilter (λi. snd (glue-key i) = c) (glue-items p q))
       in if s = 0 then None else Some s)
    unfolding comp-snd glue-source-def
    using dist-map-dist-of-fset-by-id-apply[OF total, of snd ∘ glue-key c]
    by (simp add: comp-def)
  have right:
    dist (dist-map snd q) c =
      (let s = fsum snd (ffilter (λbc. snd (fst bc) = c) (items q))
       in if s = 0 then None else Some s)
    by (simp add: dist-map-apply fsum-items-filter-dist-map[symmetric])
  show dist (dist-map snd (glue-dist p q)) c = dist (dist-map snd q) c
  using left right fsum-glue-weight-snd[OF mid, of c] by simp
qed

```

lemma *set-dist-glue-dist-subset*:

fixes $p :: ('a \times 'b) \text{ dist}$ **and** $q :: ('b \times 'c) \text{ dist}$

assumes $\text{mid}: \text{dist-map snd } p = \text{dist-map fst } q$

and $Rp: \text{set-dist } p \subseteq \{(a, b). R \ a \ b\}$

and $Sq: \text{set-dist } q \subseteq \{(b, c). S \ b \ c\}$

shows $\text{set-dist } (\text{glue-dist } p \ q) \subseteq \{(a, c). (R \ OO \ S) \ a \ c\}$

proof

fix ac

assume $ac: ac \in \text{set-dist } (\text{glue-dist } p \ q)$

let $?src = \text{glue-source } p \ q$

have $\text{total}: \text{fsum } (\text{glue-weight } p) (\text{glue-items } p \ q) = 1$

by $(\text{rule fsum-glue-weight}[OF \ \text{mid}])$

obtain x **where** $x\text{-src}: x \in \text{set-dist } ?src$ **and** $ac\text{-}x: ac = \text{glue-key } x$

using $ac \text{ set-dist-dist-map}[of \ \text{glue-key } ?src]$

unfolding glue-dist-def **by** auto

have $x\text{-}I: x \in \text{fset } (\text{glue-items } p \ q)$

using $x\text{-src} \text{ set-dist-dist-of-fset-by-id-subset}[OF \ \text{total}]$

unfolding glue-source-def **by** auto

then obtain $ab \ bc$ **where** $x: x = (ab, bc)$

and $ab: ab \in \text{items } p$

and $bc: bc \in \text{items } q$

and $\text{match}: \text{snd } (\text{fst } ab) = \text{fst } (\text{fst } bc)$

unfolding glue-items-def

by $(\text{auto simp: fbind.rep-eq fimage.rep-eq Set.bind-def})$

have $ab\text{-supp}: \text{fst } ab \in \text{set-dist } p$

using $ab \text{ unfolding set-dist-def by (auto simp: fset-items)}$

have $bc\text{-supp}: \text{fst } bc \in \text{set-dist } q$

using $bc \text{ unfolding set-dist-def by (auto simp: fset-items)}$

show $ac \in \{(a, c). (R \ OO \ S) \ a \ c\}$

using $Rp \ Sq \ ab\text{-supp} \ bc\text{-supp} \ \text{match} \ ac\text{-}x$

unfolding $x \ \text{glue-key-def}$

by $(\text{cases } \text{fst } ab; \text{cases } \text{fst } bc; \text{auto})$

qed

lemma *rel-dist-OO*:

$(\text{rel-dist } R \ OO \ \text{rel-dist } S) \leq \text{rel-dist } (R \ OO \ S)$

proof

fix $x \ z$

assume $(\text{rel-dist } R \ OO \ \text{rel-dist } S) \ x \ z$

then obtain y **where** $xy: \text{rel-dist } R \ x \ y$ **and** $yz: \text{rel-dist } S \ y \ z$

by auto

obtain p **where** $pR: \text{set-dist } p \subseteq \{(a, b). R \ a \ b\}$

and $p\text{-fst}: \text{dist-map fst } p = x$

and $p\text{-snd}: \text{dist-map snd } p = y$

using $xy \text{ unfolding rel-dist-def by auto}$

obtain q **where** $qS: \text{set-dist } q \subseteq \{(b, c). S \ b \ c\}$

and $q\text{-fst}: \text{dist-map fst } q = y$

and $q\text{-snd}: \text{dist-map snd } q = z$

```

    using yz unfolding rel-dist-def by auto
  have mid: dist-map snd p = dist-map fst q
    using p-snd q-fst by simp
  have rel-dist (R OO S) x z
    unfolding rel-dist-def
  proof (intro exI conjI)
    show set-dist (glue-dist p q)  $\subseteq$   $\{(a, c). (R \text{ OO } S) a c\}$ 
      by (rule set-dist-glue-dist-subset[OF mid pR qS])
    show dist-map fst (glue-dist p q) = x
      using dist-map-fst-glue-dist[OF mid] p-fst by simp
    show dist-map snd (glue-dist p q) = z
      using dist-map-snd-glue-dist[OF mid] q-snd by simp
  qed
  then show rel-dist (R OO S) x z .
qed

bnf dist: 'a dist
  map: dist-map
  sets: set-dist
  bd: natLeq
  wits: delta-dist
  rel: rel-dist
proof -
  show dist-map id = id
    by (rule dist-map-id)
next
  show  $\bigwedge f g. \text{dist-map } (g \circ f) = \text{dist-map } g \circ \text{dist-map } f$ 
    by (rule dist-map-comp)
next
  show  $\bigwedge x f g. (\bigwedge z. z \in \text{set-dist } x \implies f z = g z) \implies \text{dist-map } f x = \text{dist-map } g x$ 
    by (rule dist-map-cong)
next
  show  $\bigwedge f. \text{set-dist} \circ \text{dist-map } f = (\cdot) f \circ \text{set-dist}$ 
    by (simp add: fun-eq-iff set-dist-dist-map)
next
  show card-order natLeq
    by (rule natLeq-card-order)
next
  show BNF-Cardinal-Arithmetic.cinfinite natLeq
    by (rule natLeq-cinfinite)
next
  show regularCard natLeq
    by (rule regularCard-natLeq)
next
  show  $\bigwedge x. \text{ordLess2 } (\text{card-of } (\text{set-dist } x)) \text{ natLeq}$ 
    unfolding set-dist-def
    by (simp add: finite-iff-ordLess-natLeq[symmetric])
next
  show  $\bigwedge R S. \text{rel-dist } R \text{ OO rel-dist } S \leq \text{rel-dist } (R \text{ OO } S)$ 

```

```

    by (rule rel-dist-OO)
next
show  $\bigwedge R. \text{rel-dist } R =$ 
  ( $\lambda x y. \exists z. \text{set-dist } z \subseteq \{(x, y). R \ x \ y\} \wedge$ 
     $\text{dist-map fst } z = x \wedge \text{dist-map snd } z = y$ )
  by (auto simp: fun-eq-iff rel-dist-def)
next
show  $\bigwedge b z. b \in \text{set-dist } (\text{delta-dist } z) \implies b = z$ 
  unfolding set-dist-def by (simp add: dist-delta-dist delta-map-def)
qed

```

1.2.8 Code Generator Setup

We implement distributions using finite sets of key-value pairs. Note that we use a code datatype here since we want to use a different datatype for implementation as the raw datatype which was used to define the type.

code-datatype *dist-of-fset*

```

lemma delta-dist-code[code]:
   $\text{delta-dist } x = \text{dist-of-fset } \{|(x, 1)|\}$ 
  unfolding delta-dist-def by simp

```

```

declare [[code drop: squish]]

```

```

lemma squish-code[code]:
   $\text{squish } xs = \text{fbind } (\text{fimage fst } xs) (\lambda x. \{|(x, \text{sum-fset-} x \ x \ xs)|\})$ 
  unfolding squish-def
  by (rule fset-eqI) (auto simp: fbind.rep-eq fimage.rep-eq Set.bind-def)

```

```

lemma dist-code[code]:  $\text{dist } (\text{dist-of-fset } A) = (\text{let } q = \text{fsum snd } A \text{ in}$ 
   $\text{if } q = 0 \text{ then } \text{Map.empty}(\text{undefined} := \text{Some } 1)$ 
   $\text{else } \text{map-of } (\text{norm-fset } A))$ 
  by (simp add: dist-of-fset.rep-eq map-of-fset-def)

```

```

lemma dist-uniform-code[code]:
   $\text{dist-uniform } xs = \text{dist-of-fset } ((\lambda x. (x, 1::\text{prob})) \mid^q xs)$ 
  unfolding dist-uniform-def by simp

```

```

lemma items-code[code]:
   $\text{items } (\text{dist-of-fset } xs) = \text{norm-fset } xs$ 
apply transfer-start
  apply transfer-step
  apply transfer-step
  apply transfer-step
  apply (rule fset-eq-transfer-aux)
  apply transfer-step
  apply transfer-end
  by (rule raw-items-map-of-fset)

```

declare $[[code\ drop:\ squish-by]]$

lemma *squish-by-code* $[code]:$

squish-by key weight I =

fbind (fimage key I) ($\lambda x. \{|(x, sum-fset-by\ x\ key\ weight\ I)|\}$)

unfolding *squish-by-def*

by (*rule fset-eqI*) (*auto simp: fbind.rep-eq fimage.rep-eq Set.bind-def*)

Examples **thm** *eq-transfer*

instantiation *dist::(type) equal*

begin

lift-definition *equal-dist::'a dist $\Rightarrow$ 'a dist $\Rightarrow$ bool*

is (=) .

instance **by** (*intro-classes, transfer, simp*)

end

lemma *eq-fset-set-transfer:*

assumes *bi-unique A*

shows *Transfer.Rel (rel-fun (pcr-fset A) (rel-fun (pcr-fset A) (=))) (=) (=)*

using *assms FSet.fset.bi-unique eq-transfer*

unfolding *Transfer.Rel-def*

by *blast*

lemma *eq-iff-raw-items-eq:*

fixes *a b*

assumes *finite (dom a) $\wedge$ sum-map a = 1 $\wedge$ ($\forall x \in dom\ a. the\ (a\ x) \neq 0$)*

and *finite (dom b) $\wedge$ sum-map b = 1 $\wedge$ ($\forall x \in dom\ b. the\ (b\ x) \neq 0$)*

shows *(a = b) = (raw-items a = raw-items b)*

proof

assume *a = b*

then show *raw-items a = raw-items b*

by *simp*

next

assume *raw-eq: raw-items a = raw-items b*

show *a = b*

proof

fix *x*

show *a x = b x*

proof (*cases x $\in$ dom a*)

case *True*

then have *(x, the (a x)) $\in$ raw-items a*

unfolding *raw-items-def* **by** *blast*

then have *(x, the (a x)) $\in$ raw-items b*

using *raw-eq* **by** *simp*

then have *x $\in$ dom b the (a x) = the (b x)*

unfolding *raw-items-def* **by** *auto*

then show *?thesis*

using *True* **by** (*cases a x; cases b x*) *auto*

next

```

case False
show ?thesis
proof (cases  $x \in \text{dom } b$ )
  case True
  then have  $(x, \text{the } (b \ x)) \in \text{raw-items } b$ 
    unfolding raw-items-def by blast
  then have  $(x, \text{the } (b \ x)) \in \text{raw-items } a$ 
    using raw-eq by simp
  then have  $x \in \text{dom } a$ 
    unfolding raw-items-def by auto
  then show ?thesis
    using False by simp
next
  case False
  then show ?thesis
    using  $\langle x \notin \text{dom } a \rangle$  by (simp add: domIff)
qed
qed
qed
qed

lemma equal-dist-code[code]:
  equal-class.equal  $a \ b = (\text{items } a = \text{items } b)$ 
apply transfer-start
  apply transfer-step
  apply transfer-step
  apply (rule eq-fset-set-transfer)
  apply transfer-step
  apply transfer-step
  apply transfer-step
  apply transfer-step
  apply transfer-step
  apply transfer-end
by (rule eq-iff-raw-items-eq)

lemma
  dist-of-fset  $\{|(True, 1), (False, 1), (False, 1)|\}$ 
  = dist-of-fset  $\{|(True, 0.5), (False, 0.5)|\}$ 
by eval

lemma
  dist-of-fset  $\{|(1::nat, 1), (2, 1), (2, 1)|\}$ 
  = dist-of-fset  $\{|(1::nat, 1), (2, 1), (2, 1)|\}$ 
by eval

lemma dist-uniform  $\{|1, 2, 3::nat|\} = \text{dist-of-fset } \{|(1, 1), (2, 1), (3, 1)|\}$ 
by eval

lemma

```



```

    delta-dist (True) = dist-of-fset {|(True, 1)|}
  by eval

lemma coin 0.5 (1::nat) 0 = dist-of-fset {|(1, (1 / 2)), (0, (1 / 2))|}
  by eval

lemma die 6 = dist-of-fset {|(1, 1), (2, 1), (3, 1), (4, 1), (5, 1), (6, 1)|}
  by eval

end

theory Hash-Monad
  imports Main Prob-Monad HOL-Library.Finite-Map HOL.Enum
begin



### 1.3 Finite Map setup

definition fmap-of-fun :: ('a::finite  $\Rightarrow$  'b option)  $\Rightarrow$  ('a, 'b) fmap where
  fmap-of-fun f = Abs-fmap f
lemma fmlookup-fmap-of-fun[simp]:
  fmlookup (fmap-of-fun f) = f
  unfolding fmap-of-fun-def
  by (simp add: Abs-fmap-inverse)
lemma fmap-of-fun-fmlookup[simp]:
  fmap-of-fun (fmlookup m) = m
  by (rule fmap-ext) simp

instantiation fmap :: (enum, enum) enum
begin
definition enum-fmap :: ('a, 'b) fmap list where
  enum-fmap = map fmap-of-fun (enum-class.enum :: ('a  $\Rightarrow$  'b option) list)
definition enum-all-fmap :: (('a, 'b) fmap  $\Rightarrow$  bool)  $\Rightarrow$  bool where
  enum-all-fmap P =
    enum-class.enum-all ( $\lambda f :: 'a \Rightarrow 'b \text{ option. } P \text{ (fmap-of-fun f)}$ )
definition enum-ex-fmap :: (('a, 'b) fmap  $\Rightarrow$  bool)  $\Rightarrow$  bool where
  enum-ex-fmap P =
    enum-class.enum-ex ( $\lambda f :: 'a \Rightarrow 'b \text{ option. } P \text{ (fmap-of-fun f)}$ )
instance
proof standard
  show UNIV = set (enum-class.enum :: ('a, 'b) fmap list)
    unfolding enum-fmap-def
  proof (rule subset-antisym)
    show UNIV  $\subseteq$  set (map fmap-of-fun
      (enum-class.enum :: ('a  $\Rightarrow$  'b option) list))
    by (metis UNIV-I enum-UNIV fmap-of-fun-fmlookup image-eqI set-map subsetI)
    show set (map fmap-of-fun
      (enum-class.enum :: ('a  $\Rightarrow$  'b option) list))  $\subseteq$  UNIV
    by simp

```

```

qed
show distinct (enum-class.enum :: ('a, 'b) fmap list)
proof -
  have inj-on fmap-of-fun (set (enum-class.enum :: ('a ⇒ 'b option) list))
    by (rule inj-onI) (metis fmlookup-fmap-of-fun)
  then show ?thesis
    by (simp add: enum-fmap-def enum-distinct distinct-map)
qed
show enum-class.enum-all P = Ball UNIV P for P :: ('a, 'b) fmap ⇒ bool
  apply (auto simp: enum-all-fmap-def enum-all-UNIV)
  by (metis fmap-of-fun-fmlookup)
show enum-class.enum-ex P = Bex UNIV P for P :: ('a, 'b) fmap ⇒ bool
  apply (auto simp: enum-ex-fmap-def enum-ex-UNIV)
  by (metis fmap-of-fun-fmlookup)
qed
end

```

1.4 Hashing

Protocols should not encode semantically different random-oracle queries as the same raw value before hashing. The STARK protocol uses field-valued hash outputs and messages, but Merkle leaves, Merkle internal nodes, and Fiat-Shamir challenge queries need distinct input domains. The datatype below provides that shared domain-separated input vocabulary.

```

datatype 'a protocol-hash-input =
  MerkleLeaf 'a
| MerkleNode 'a 'a
| FiatShamirChallenge 'a
| TraceFriChallenge nat 'a
| CompositionFriChallenge nat 'a
| AlphaChallenge nat 'a
| QueryIndexChallenge nat 'a

record ('a, 'b) hash =
  HashMap :: ('a, 'b) fmap

instantiation hash-ext :: (enum, enum, enum) enum
begin
definition enum-hash-ext :: ('a, 'b, 'c) hash-ext list where
  enum-hash-ext =
    map (λ(m :: ('a, 'b) fmap, z :: 'c). (hash-ext m z :: ('a, 'b, 'c) hash-ext))
      (List.product
        (enum-class.enum :: ('a, 'b) fmap list)
        (enum-class.enum :: 'c list))
definition enum-all-hash-ext :: (('a, 'b, 'c) hash-ext ⇒ bool) ⇒ bool where
  enum-all-hash-ext P =
    enum-class.enum-all (λm :: ('a, 'b) fmap.
      enum-class.enum-all (λz :: 'c. P (hash-ext m z :: ('a, 'b, 'c) hash-ext)))

```

```

definition enum-ex-hash-ext :: (('a, 'b, 'c) hash-ext  $\Rightarrow$  bool)  $\Rightarrow$  bool where
  enum-ex-hash-ext P =
    enum-class.enum-ex ( $\lambda m :: ('a, 'b)$  fmap.
      enum-class.enum-ex ( $\lambda z :: 'c.$  P (hash-ext m z :: ('a, 'b, 'c) hash-ext)))
instance
proof standard
  show UNIV = set (enum-class.enum :: ('a, 'b, 'c) hash-ext list)
    unfolding enum-hash-ext-def
  proof (rule subset-antisym)
    show UNIV  $\subseteq$ 
      set (map ( $\lambda(m :: ('a, 'b)$  fmap, z :: 'c).
        (hash-ext m z :: ('a, 'b, 'c) hash-ext))
        (List.product
          (enum-class.enum :: ('a, 'b) fmap list)
          (enum-class.enum :: 'c list)))
      by (auto, case-tac x, auto simp: enum-UNIV)
    show set (map ( $\lambda(m :: ('a, 'b)$  fmap, z :: 'c).
      (hash-ext m z :: ('a, 'b, 'c) hash-ext))
      (List.product
        (enum-class.enum :: ('a, 'b) fmap list)
        (enum-class.enum :: 'c list)))  $\subseteq$  UNIV
      by simp
  qed
show distinct (enum-class.enum :: ('a, 'b, 'c) hash-ext list)
  by (auto simp: enum-hash-ext-def enum-distinct distinct-map inj-on-def
    intro!: distinct-product)
show enum-class.enum-all P = Ball UNIV P for P :: ('a, 'b, 'c) hash-ext  $\Rightarrow$  bool
  unfolding enum-all-hash-ext-def
  by (auto simp: enum-all-UNIV; metis hash.ext-surjective)
show enum-class.enum-ex P = Bex UNIV P for P :: ('a, 'b, 'c) hash-ext  $\Rightarrow$  bool
  unfolding enum-ex-hash-ext-def
  by (auto simp: enum-ex-UNIV; metis hash.ext-surjective)
qed
end

```

type-synonym ('a, 'b, 'c, 'd) hash-monad = ('c, ('a, 'b, 'd) hash-scheme) state-monad

definition hash-default :: 'a::finite dist
where hash-default = dist-uniform (fUNIV :: 'a fset)

definition option-default-dist :: 'a::finite option $\Rightarrow$ 'a dist
where option-default-dist a = case-option hash-default delta-dist a

definition hash-dist :: 'a $\Rightarrow$ ('a, 'b::finite, 'c) hash-scheme $\Rightarrow$ 'b dist
where
 hash-dist a s = option-default-dist (fmlookup (HashMap s) a)

definition modify-HashMap :: 'a $\Rightarrow$ 'b $\Rightarrow$ ('a, 'b, unit, 'd) hash-monad **where**
 modify-HashMap x a = modify ($\lambda s.$ s[HashMap := fmupd x a (HashMap s)])

definition *apply-hash* :: 'b $\Rightarrow$ ('b, 'a, 'a::finite, 'c) hash-monad **where**
apply-hash x = lift (hash-dist x)

definition *hash* :: 'b $\Rightarrow$ ('b, 'a, 'a::finite, 'c) hash-monad
where
hash x $\equiv$
do {
a $\leftarrow$ *apply-hash* x;
modify-HashMap x a;
return a
}

instantiation *fmap* :: (type, type) ord
begin

definition *less-eq* a b $\equiv$ ($\forall x. \text{fmlookup } a \ x = \text{None} \vee \text{fmlookup } a \ x = \text{fmlookup } b \ x$)

definition *less* a b $\equiv$ *less-eq* a b $\wedge$ ($\exists x. \text{fmlookup } a \ x = \text{None} \wedge \text{fmlookup } b \ x \neq \text{None}$)

instance ..

end

instantiation *hash-ext* :: (type, type, type) ord
begin

definition *less-eq* a b $\equiv$ (HashMap a) $\leq$ (HashMap b)

definition *less* a b $\equiv$ (HashMap a) $<$ (HashMap b)

instance ..

end

definition *less-eq-hash* :: ('a, 'b) hash $\Rightarrow$ ('a, 'b) hash $\Rightarrow$ bool
where *less-eq-hash* (a::('a, 'b) hash) (b::('a, 'b) hash) $\equiv$ (HashMap a) $\leq$ (HashMap b)

definition *less-hash* :: ('a, 'b) hash $\Rightarrow$ ('a, 'b) hash $\Rightarrow$ bool
where *less-hash* (a::('a, 'b) hash) (b::('a, 'b) hash) $\equiv$ (HashMap a) $<$ (HashMap b)

lemma *dist-expect-eq-1*:

assumes $\bigwedge x. x \in \text{dom } (\text{dist } d) \implies Q \ x = 1$

shows *dist-expect* d Q = 1

proof –

have *dist-expect* d Q = ($\sum x \in \text{dom } (\text{dist } d). \text{the } (\text{dist } d \ x)$)

unfolding *dist-expect-def*

by (*intro sum.cong refl*) (*simp add: assms*)

also have ... = *sum-map* (dist d)

```

    unfolding sum-map-def by simp
  also have ... = 1
    by simp
  finally show ?thesis .
qed

lemma hash-expect-known:
  assumes fmllookup (HashMap s) x = Some y
  shows
    dist-expect (lift-dist (hash-dist x) s)
      ( $\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 0 \mid \text{Some } (z, u) \Rightarrow \text{if } y = z \text{ then } 1 \text{ else } 0$ ) = 1
  proof -
    have lift-dist (hash-dist x) s = delta-dist (Some (y, s))
      using assms
    by (simp add: lift-dist-def hash-dist-def option-default-dist-def)
  then show ?thesis
    by simp
qed

lemma hash-expect-known-extension:
  assumes fmllookup (HashMap s0) x = Some y
    and s0  $\leq$  t
  shows
    dist-expect (lift-dist (hash-dist x) t)
      ( $\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 0 \mid \text{Some } (z, u) \Rightarrow \text{if } y = z \text{ then } 1 \text{ else } 0$ ) = 1
  proof -
    have fmllookup (HashMap t) x = Some y
      using assms
    unfolding less-eq-hash-ext-def less-eq-fmap-def
    by (metis option.distinct(1))
  then show ?thesis
    by (rule hash-expect-known)
qed

lemma hash-expect-after-update-extension:
  assumes s0  $\setminus$  (HashMap := fmupd x y (HashMap s0))  $\leq$  t
  shows
    dist-expect (lift-dist (hash-dist x) t)
      ( $\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 0 \mid \text{Some } (z, u) \Rightarrow \text{if } y = z \text{ then } 1 \text{ else } 0$ ) = 1
  using assms
  by (intro hash-expect-known-extension) simp-all

lemma hash-expect-after-update-extension-dist-map:
  assumes s0  $\setminus$  (HashMap := fmupd x y (HashMap s0))  $\leq$  t
  shows
    dist-expect (dist-map ( $\lambda z. \text{Some } (z, t)$ ) (hash-dist x t))
      ( $\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 0 \mid \text{Some } (z, u) \Rightarrow \text{if } y = z \text{ then } 1 \text{ else } 0$ ) = 1
  using hash-expect-after-update-extension[OF assms]
  by (simp add: lift-dist-def)

```

lemma *hash-dist-expect-after-update-extension*:
assumes $s0 \llbracket \text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s0) \rrbracket \leq t$
shows $\text{dist-expect } (\text{hash-dist } x \ t) \ (\lambda z. \text{if } y = z \text{ then } 1 \text{ else } 0) = 1$
using *hash-expect-after-update-extension*[*OF assms*]
by (*simp add: lift-dist-def dist-expect-map*)

lemma *None-notin-dist-map-Some*[*simp*]:
 $\text{None} \notin \text{dom } (\text{dist } (\text{dist-map } (\lambda x. \text{Some } (x, s)) \ d))$
using *set-dist-dist-map*[*of* $\lambda x. \text{Some } (x, s) \ d$]
unfolding *set-dist-def*
by *auto*

lemma *hash-update-extends*:
assumes $y \in \text{dom } (\text{dist } (\text{hash-dist } x \ s))$
shows $s \leq s \llbracket \text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s) \rrbracket$
unfolding *less-eq-hash-ext-def less-eq-fmap-def*
proof (*intro allI*)
fix z
show $\text{fmlookup } (\text{HashMap } s) \ z = \text{None} \vee$
 $\text{fmlookup } (\text{HashMap } s) \ z =$
 $\text{fmlookup } (\text{HashMap } (s \llbracket \text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s) \rrbracket)) \ z$
proof (*cases fmlookup (HashMap s) x*)
case *None*
then show *?thesis*
by (*cases z = x simp-all*)
next
case (*Some old*)
then have $y = \text{old}$
using *assms*
unfolding *hash-dist-def option-default-dist-def*
by (*simp add: dist-delta-dist delta-map-def*)
with Some show *?thesis*
by (*cases z = x simp-all*)
qed
qed

lemma *dist-expect-then-hash-after-extension*:
assumes $\bigwedge r \ t.$
 $\text{Some } (r, t) \in \text{dom } (\text{dist } d) \implies$
 $s0 \llbracket \text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s0) \rrbracket \leq t$
shows
 $\text{dist-expect } d$
 $(\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 1 \mid \text{Some } (r, t) \Rightarrow$
 $\text{dist-expect } (\text{lift-dist } (\text{hash-dist } x) \ t)$
 $(\lambda z. \text{case } z \text{ of } \text{None} \Rightarrow 0 \mid \text{Some } (z, u) \Rightarrow \text{if } y = z \text{ then } 1 \text{ else } 0)) = 1$
proof (*rule dist-expect-eq-1*)
fix z
assume $z: z \in \text{dom } (\text{dist } d)$

```

show (case z of None  $\Rightarrow$  1 | Some (r, t)  $\Rightarrow$ 
  dist-expect (lift-dist (hash-dist x) t)
  ( $\lambda z$ . case z of None  $\Rightarrow$  0 | Some (z, u)  $\Rightarrow$  if y = z then 1 else 0)) = 1
proof (cases z)
  case None
  then show ?thesis by simp
next
  case (Some rt)
  then obtain r t where rt: z = Some (r, t)
  by (cases rt) simp
  then show ?thesis
  using assms[of r t] z
  by (simp add: hash-expect-after-update-extension)
qed
qed

```

lemma dist-expect-then-hash-after-extension-dist-map:

```

assumes  $\bigwedge r$  t.
  Some (r, t)  $\in$  dom (dist d)  $\implies$ 
  s0(|HashMap := fmupd x y (HashMap s0)|)  $\leq$  t
shows
  dist-expect d
  ( $\lambda r$ . case r of None  $\Rightarrow$  1 | Some (r, t)  $\Rightarrow$ 
    dist-expect (dist-map ( $\lambda z$ . Some (z, t)) (hash-dist x t))
    ( $\lambda z$ . case z of None  $\Rightarrow$  0 | Some (z, u)  $\Rightarrow$  if y = z then 1 else 0)) = 1
proof –
  show ?thesis
proof (rule dist-expect-eq-1)
  fix z
  assume z: z  $\in$  dom (dist d)
  show (case z of None  $\Rightarrow$  1 | Some (r, t)  $\Rightarrow$ 
    dist-expect (dist-map ( $\lambda z$ . Some (z, t)) (hash-dist x t))
    ( $\lambda z$ . case z of None  $\Rightarrow$  0 | Some (z, u)  $\Rightarrow$  if y = z then 1 else 0)) = 1
  proof (cases z)
    case None
    then show ?thesis by simp
  next
    case (Some rt)
    then obtain r t where rt: z = Some (r, t)
    by (cases rt) simp
    then show ?thesis
    proof –
      have dist-expect (lift-dist (hash-dist x) t)
        ( $\lambda z$ . case z of None  $\Rightarrow$  0 | Some (z, u)  $\Rightarrow$  if y = z then 1 else 0) = 1
      using assms[of r t] z rt
      by (simp add: hash-expect-after-update-extension)
      then show ?thesis
      using rt by (simp add: lift-dist-def)
    qed
  qed

```

qed
 qed
 qed

lemma *send-intermediate-send-same*:

assumes *intermediate-no-fail*:

$\bigwedge a s'. \text{None} \notin$
 $\text{dom} (\text{dist} (\text{execute} (\text{between } a)$
 $(s' \backslash \text{HashMap} := \text{fmupd } x \ a \ (\text{HashMap } s^{\wedge}))))$

assumes *intermediate-extends*:

$\bigwedge a s' r t.$
 $\text{Some } (r, t) \in$
 $\text{dom} (\text{dist} (\text{execute} (\text{between } a)$
 $(s' \backslash \text{HashMap} := \text{fmupd } x \ a \ (\text{HashMap } s^{\wedge})))) \implies$
 $s' \backslash \text{HashMap} := \text{fmupd } x \ a \ (\text{HashMap } s^{\wedge}) \leq t$

shows

wp-event

(do {
 $a \leftarrow \text{hash } x;$
 $- \leftarrow \text{between } a;$
 $b \leftarrow \text{hash } x;$
 $\text{return } (a = b)$
 })
 $(\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow \text{False} \mid \text{Some } (ok, -) \Rightarrow ok)$
 $s = 1$

unfolding *wp-event-def hash-def apply-hash-def modify-HashMap-def*

apply (*simp add: wpsimps*)

apply (*rule dist-expect-eq-1*)

apply (*simp add: wp-def*)

apply (*rule dist-expect-eq-1*)

apply (*case-tac xaa*)

apply (*simp add: intermediate-no-fail*)

apply (*clarsimp simp: wpsimps split: prod.splits*)

apply (*rule hash-dist-expect-after-update-extension*)

by (*rule-tac a=xa and s'=s and r=x1 and t=x2 in intermediate-extends*) (*simp add: domIff*)

1.4.1 Weakest Precondition Calculus

lemma *wp-hash[wpsimps]*:

$\text{wp} (\text{hash } x) \ Q \ s =$
 $\text{dist-expect} (\text{hash-dist } x \ s)$
 $(\lambda xa. \ Q \ (\text{Some } (xa, s \backslash \text{HashMap} := \text{fmupd } x \ xa \ (\text{HashMap } s))))$
by (*simp add: hash-def wpsimps apply-hash-def modify-HashMap-def*)

lemma *wp-hashI[wp]*:

assumes $P \leq$
 $\text{dist-expect} (\text{hash-dist } x \ s)$
 $(\lambda xa. \ Q \ (\text{Some } (xa, s \backslash \text{HashMap} := \text{fmupd } x \ xa \ (\text{HashMap } s))))$
shows $P \leq \text{wp} (\text{hash } x) \ Q \ s$


```
using assms by (simp add:wpsimps)
```

1.4.2 Code Generator Setup

hash-default does not execute since it is using *fUNIV*. Thus, we provide an alternative implementation for enumerable datatypes which can be used for execution.

```
Enums definition hash-default-enum :: 'a::enum dist
  where hash-default-enum = hash-default
```

```
declare hash-default-enum-def[symmetric,code]
```

```
lemma hash-default-enum-code[code]:
  hash-default-enum = dist-uniform (fset-of-list (Enum.enum :: 'a::enum list))
proof -
  have Abs-fset (UNIV :: 'a set) = fset-of-list (Enum.enum :: 'a list)
    by (rule fset-inject[THEN iffD1])
    (simp add: Abs-fset-inverse fset-of-list.rep-eq enum-UNIV)
  then have (fUNIV :: 'a fset) = fset-of-list (Enum.enum :: 'a list)
    by (simp add:FSet.top-fset.abs-eq)
  then show ?thesis
    unfolding hash-default-enum-def hash-default-def by simp
qed
```

```
value fmupd (1::nat) (2::nat) fmempty
```

```
value execute (hash True) (HashMap = (fmempty::(bool, bool) fmap))
```

```
value execute (apply-hash True) (HashMap = (fmempty::(bool, bool) fmap))
```

```
value execute (modify-HashMap True True) (HashMap = (fmempty::(bool, bool) fmap))
```

```
value execute (put True) False
```

```
value execute (
  do {
    a ← hash True;
    x ← hash False;
    b ← hash True;
    return (a = b)
  }) (HashMap = (fmempty::(bool, bool) fmap))
```

```
end
```

```
theory Channel-Core
imports Hash-Monad Merkle-Tree
begin
```

1.5 Channels

lemma *no-failure-bindI*:

assumes *m-no-fail*: $\text{None} \notin \text{dom} (\text{dist} (\text{execute } m \ s))$

and *k-no-fail*:

$\bigwedge x \ t. \text{Some } (x, t) \in \text{set-dist} (\text{execute } m \ s) \implies$

$\text{None} \notin \text{dom} (\text{dist} (\text{execute } (k \ x) \ t))$

shows $\text{None} \notin \text{dom} (\text{dist} (\text{execute } (m \gg k) \ s))$

proof

assume $\text{None} \in \text{dom} (\text{dist} (\text{execute } (m \gg k) \ s))$

then have *none*:

$\text{None} \in \text{dom}$

$(\text{map-bind } (\text{dist} \circ \text{execute } m))$

$(\text{bind-cont-map } (\text{map-fun id dist} \circ \text{execute} \circ k)) \ s)$

unfolding *sm-bind.rep-eq dist-bind.rep-eq*

by (*simp add: o-def map-fun-def*)

then have

$\text{None} \in$

$(\bigcup_{i \in \text{dom} (\text{dist} (\text{execute } m \ s))}.$

$\text{dom} (\text{bind-cont-map } (\text{map-fun id dist} \circ \text{execute} \circ k) \ i))$

using *dom-map-bind[of dist o execute m*

bind-cont-map (map-fun id dist o execute o k) s]

by (*simp add: o-def*)

then obtain *i* **where**

i-dom: $i \in \text{dom} (\text{dist} (\text{execute } m \ s))$

and *none-i*:

$\text{None} \in \text{dom} (\text{bind-cont-map } (\text{map-fun id dist} \circ \text{execute} \circ k) \ i)$

by *auto*

show *False*

proof (*cases i*)

case *None*

then show *?thesis*

using *i-dom m-no-fail* **by** *simp*

next

case (*Some xt*)

then obtain *x t* **where** *xt*: $i = \text{Some } (x, t)$

by (*cases xt*) *auto*

have *i-set*: $\text{Some } (x, t) \in \text{set-dist} (\text{execute } m \ s)$

using *i-dom* **unfolding** *xt set-dist-def* **.**

have $\text{None} \notin \text{dom} (\text{dist} (\text{execute } (k \ x) \ t))$

by (*rule k-no-fail[OF i-set]*)

moreover have

$\text{dom} (\text{bind-cont-map } (\text{map-fun id dist} \circ \text{execute} \circ k) (\text{Some } (x, t))) =$

$\text{dom} (\text{dist} (\text{execute } (k \ x) \ t))$

unfolding *bind-cont-map-def* **by** *simp*

ultimately show *?thesis*

using *none-i* **unfolding** *xt* **by** *simp*

qed

qed

lemma *no-failure-return*[simp]:
None $\notin$ *dom* (*dist* (*execute* (*return* *x*) *s*))
unfolding *return.rep-eq* *dist-return-def* *dist-delta-dist* *delta-map-def*
by *simp*

lemma *no-failure-get*[simp]:
None $\notin$ *dom* (*dist* (*execute* *get* *s*))
unfolding *get.rep-eq* *dist-get-def* *dist-delta-dist* *delta-map-def*
by *simp*

lemma *no-failure-put*[simp]:
None $\notin$ *dom* (*dist* (*execute* (*put* *s'*) *s*))
unfolding *put.rep-eq* *dist-put-def* *dist-delta-dist* *delta-map-def*
by *simp*

lemma *no-failure-lift*[simp]:
None $\notin$ *dom* (*dist* (*execute* (*lift* *d*) *s*))
unfolding *lift.rep-eq* *lift-dist-def* *set-dist-def*
using *set-dist-dist-map*[of $\lambda a. \text{Some } (a, s) \ d \ s$]
unfolding *set-dist-def* **by** *auto*

lemma *no-failure-modify*[simp]:
None $\notin$ *dom* (*dist* (*execute* (*modify* *f*) *s*))
unfolding *modify-def*
by (*rule* *no-failure-bindI*) *simp-all*

lemma *throw-no-Some-set-dist*:
Some *x* $\notin$ *set-dist* (*execute* (*throw* :: (*'a*, *'s*) *state-monad*) *s*)
unfolding *set-dist-def* *throw.rep-eq* *dist-throw-def* *dist-delta-dist*
delta-map-def
by *simp*

record *'a channel* = (*'a*, *'a*) *hash* +
State :: *'a*
Transcript :: *'a list*

type-synonym (*'a*, *'b*, *'c*) *c-monad* = (*'b*, (*'a*, *'c*) *channel-scheme*) *state-monad*

definition *receive-random-field-element* :: (*'a*, *'a::finite*, *'b*) *c-monad*
where
receive-random-field-element $\equiv$
do {
s $\leftarrow$ *get*;
r $\leftarrow$ *hash* (*State* *s*);
return *r*
}

record *'a protocol-channel* = (*'a protocol-hash-input*, *'a*) *hash* +
PState :: *'a*

PTranscript :: 'a list
PTraceFriCounter :: nat
PCompositionFriCounter :: nat
PAlphaCounter :: nat
PQueryCounter :: nat

type-synonym ('a, 'b, 'c) *protocol-c-monad* =
('b, ('a, 'c) *protocol-channel-scheme*) *state-monad*

definition *protocol-receive-tagged-random-field-element*
:: ('a $\Rightarrow$ 'a *protocol-hash-input*) $\Rightarrow$
('a, 'a::finite, 'b) *protocol-c-monad*

where

protocol-receive-tagged-random-field-element tag $\equiv$
do {
 s $\leftarrow$ *get*;
 r $\leftarrow$ *hash* (*tag* (*PState s*));
 return *r*
}

definition *protocol-receive-random-field-element*
:: ('a, 'a::finite, 'b) *protocol-c-monad*

where

protocol-receive-random-field-element $\equiv$
protocol-receive-tagged-random-field-element FiatShamirChallenge

definition *protocol-receive-counted-tagged-random-field-element*

:: (((('a, 'b) *protocol-channel-scheme*) $\Rightarrow$ nat) $\Rightarrow$
(((('a, 'b) *protocol-channel-scheme*) $\Rightarrow$ ('a, 'b) *protocol-channel-scheme*) $\Rightarrow$
(nat $\Rightarrow$ 'a $\Rightarrow$ 'a *protocol-hash-input*) $\Rightarrow$
('a, 'a::finite, 'b) *protocol-c-monad*

where

protocol-receive-counted-tagged-random-field-element counter bump tag $\equiv$
do {
 s $\leftarrow$ *get*;
 r $\leftarrow$ *hash* (*tag* (*counter s*) (*PState s*));
 modify bump;
 return *r*
}

definition *protocol-send*

:: ('a $\Rightarrow$ 'a $\Rightarrow$ 'a) $\Rightarrow$ 'a $\Rightarrow$
('a, unit, 'b) *protocol-c-monad*

where

protocol-send absorb x $\equiv$
modify
($\lambda s.$ s[
 PState := *absorb* (*PState s*) *x*,
 PTranscript := *x* # *PTranscript s*])

definition *protocol-read*

$:: ('a \Rightarrow 'a \Rightarrow 'a) \Rightarrow ('a, 'a, 'b) \text{ protocol-c-monad}$

where

protocol-read absorb $\equiv$

```
do {
  s ← get;
  assert (PTranscript s ≠ []);
  modify
    (λs. s⟦
      PState := absorb (PState s) (hd (PTranscript s)),
      PTranscript := tl (PTranscript s)⟧);
  return (hd (PTranscript s))
}
```

lemma *protocol-hash-no-failure*:

$\text{None} \notin \text{dom} (\text{dist} (\text{execute} (\text{hash } x) s))$

unfolding *hash-def apply-hash-def modify-HashMap-def*

by (*intro no-failure-bindI*) *simp-all*

lemma *protocol-receive-tagged-random-field-element-no-failure*:

$\text{None} \notin$

$\text{dom} (\text{dist} (\text{execute} (\text{protocol-receive-tagged-random-field-element tag}) s))$

unfolding *protocol-receive-tagged-random-field-element-def*

by (*intro no-failure-bindI*) (*simp-all add: protocol-hash-no-failure*)

lemma *protocol-receive-random-field-element-no-failure*:

$\text{None} \notin$

$\text{dom} (\text{dist} (\text{execute } \text{protocol-receive-random-field-element } s))$

unfolding *protocol-receive-random-field-element-def*

by (*rule protocol-receive-tagged-random-field-element-no-failure*)

lemma *protocol-receive-counted-tagged-random-field-element-no-failure*:

$\text{None} \notin$

$\text{dom} (\text{dist} (\text{execute} (\text{protocol-receive-counted-tagged-random-field-element counter bump tag}) s))$

unfolding *protocol-receive-counted-tagged-random-field-element-def*

by (*intro no-failure-bindI*) (*simp-all add: protocol-hash-no-failure*)

lemma *protocol-send-outcome*:

assumes $\text{Some } ((), t) \in \text{set-dist} (\text{execute} (\text{protocol-send absorb } x) s)$

shows

```
t = s⟦
  PState := absorb (PState s) x,
  PTranscript := x # PTranscript s⟧
```

using *assms*

unfolding *protocol-send-def modify-def*

by (auto elim!: protocol-merkle.set-dist-bindE)

lemma *protocol-read-cons-outcome:*

assumes *Some (y, t) ∈ set-dist (execute (protocol-read absorb) (s(PTranscript := x # xs)))*

shows

$y = x \wedge$
 $t = s()$
 $PState := absorb (PState s) x,$
 $PTranscript := xs()$

using *assms*

unfolding *protocol-read-def modify-def assert-def*

by (auto elim!: protocol-merkle.set-dist-bindE)

lemma *protocol-hash-channel-preserves:*

assumes *Some (h, t) ∈ set-dist (execute (hash x) s)*

shows $PState t = PState s$

and $PTranscript t = PTranscript s$
and $PTraceFriCounter t = PTraceFriCounter s$
and $PCompositionFriCounter t = PCompositionFriCounter s$
and $PAlphaCounter t = PAlphaCounter s$
and $PQueryCounter t = PQueryCounter s$

proof –

from *assms* **obtain** *a s1* **where**

a: Some (a, s1) ∈ set-dist (execute (apply-hash x) s)
and *rest: Some (h, t) ∈ set-dist (execute (modify-HashMap x a >>= (λ-. return a)) s1)*

proof –

have *Some (h, t) ∈ set-dist (execute (apply-hash x >>= (λa. modify-HashMap x a >>= (λ-. return a))) s)*
using *assms* **unfolding** *hash-def* .

then show *?thesis*

by (rule protocol-merkle.set-dist-bindE) (rule that)

qed

have *a-img: Some (a, s1) ∈ (λa. Some (a, s)) ‘ set-dist (hash-dist x s)*

using *a*

unfolding *apply-hash-def lift.rep-eq lift-dist-def*

by (simp add: set-dist-dist-map)

then have *s1: s1 = s*

by *auto*

from *rest* **obtain** *u s2* **where**

mod: Some (u, s2) ∈ set-dist (execute (modify-HashMap x a) s1)
and *ret: Some (h, t) ∈ set-dist (execute (return a) s2)*
by (elim protocol-merkle.set-dist-bindE)

from *mod* **have** *s2: s2 = s1(HashMap := fupd x a (HashMap s1))*

unfolding *modify-HashMap-def modify-def*

by (auto elim!: protocol-merkle.set-dist-bindE)

```

from ret have t: t = s2
  by auto
show PState t = PState s
  unfolding t s2 s1 by simp
show PTranscript t = PTranscript s
  unfolding t s2 s1 by simp
show PTraceFriCounter t = PTraceFriCounter s
  unfolding t s2 s1 by simp
show PCompositionFriCounter t = PCompositionFriCounter s
  unfolding t s2 s1 by simp
show PAlphaCounter t = PAlphaCounter s
  unfolding t s2 s1 by simp
show PQueryCounter t = PQueryCounter s
  unfolding t s2 s1 by simp
qed

lemma protocol-hash-preserves-other-lookup:
  assumes outcome: Some (h, t) ∈ set-dist (execute (hash x) s)
  and neq: y ≠ x
  shows fmlookup (HashMap t) y = fmlookup (HashMap s) y
proof -
  from outcome obtain a s1 where
    a: Some (a, s1) ∈ set-dist (execute (apply-hash x) s)
    and rest: Some (h, t) ∈
      set-dist (execute (modify-HashMap x a >>= (λ-. return a)) s1)
  proof -
    have Some (h, t) ∈
      set-dist (execute
        (apply-hash x >>= (λa. modify-HashMap x a >>= (λ-. return a))) s)
    using outcome unfolding hash-def .
    then show ?thesis
      by (rule protocol-merkle.set-dist-bindE) (rule that)
  qed
have s1: s1 = s
  using a
  unfolding apply-hash-def lift.rep-eq lift-dist-def
  by (auto simp: set-dist-dist-map)
from rest obtain u s2 where
  mod: Some (u, s2) ∈ set-dist (execute (modify-HashMap x a) s1)
  and ret: Some (h, t) ∈ set-dist (execute (return a) s2)
  by (elim protocol-merkle.set-dist-bindE)
have s2: s2 = s1 (HashMap := fmupd x a (HashMap s1))
  using mod unfolding modify-HashMap-def modify-def
  by (auto elim!: protocol-merkle.set-dist-bindE)
have t: t = s2
  using ret by auto
show ?thesis
  unfolding t s2 s1 using neq by simp
qed

```

lemma *protocol-receive-random-field-element-outcome:*

assumes *outcome:*

Some (r, t) ∈
set-dist (execute protocol-receive-random-field-element s)

shows

s ≤ t ∧
PState t = PState s ∧
PTranscript t = PTranscript s ∧
fmlookup (HashMap t) (FiatShamirChallenge (PState s)) = Some r

proof –

from *outcome* **obtain** *s0* **where**

get-s: Some (s0, s0) ∈ set-dist (execute get s)

and *rest:*

Some (r, t) ∈
set-dist
(execute (hash (FiatShamirChallenge (PState s0))) ≫= return) s0)

unfolding *protocol-receive-random-field-element-def*

protocol-receive-tagged-random-field-element-def

by (*auto elim!:* *protocol-merkle.set-dist-bindE*)

have *s0: s0 = s*

using *get-s* **by** *simp*

have *hash-out:*

Some (r, t) ∈
set-dist (execute (hash (FiatShamirChallenge (PState s)))) s)

using *rest* **unfolding** *s0* **by** *simp*

show *?thesis*

using *protocol-merkle.hash-outcome[OF hash-out]*

protocol-hash-channel-preserves[OF hash-out]

by *simp*

qed

lemma *protocol-receive-tagged-random-field-element-outcome:*

assumes *outcome:*

Some (r, t) ∈
set-dist
(execute (protocol-receive-tagged-random-field-element tag) s)

shows

s ≤ t ∧
PState t = PState s ∧
PTranscript t = PTranscript s ∧
fmlookup (HashMap t) (tag (PState s)) = Some r

proof –

from *outcome* **obtain** *s0* **where**

get-s: Some (s0, s0) ∈ set-dist (execute get s)

and *rest:*

Some (r, t) ∈
set-dist (execute (hash (tag (PState s0))) ≫= return) s0)

unfolding *protocol-receive-tagged-random-field-element-def*

by (auto elim!: protocol-merkle.set-dist-bindE)
 have $s0: s0 = s$
 using get-s by simp
 have hash-out:
 Some $(r, t) \in \text{set-dist } (\text{execute } (\text{hash } (\text{tag } (PState\ s))))\ s$
 using rest unfolding $s0$ by simp
 show ?thesis
 using protocol-merkle.hash-outcome[OF hash-out]
 protocol-hash-channel-preserves[OF hash-out]
 by simp
 qed

lemma protocol-receive-tagged-random-field-element-preserves-other-lookup:

assumes outcome:
 Some $(r, t) \in$
 set-dist
 (execute (protocol-receive-tagged-random-field-element tag) s)
 and neq: $y \neq \text{tag } (PState\ s)$
 shows $\text{fmlookup } (HashMap\ t)\ y = \text{fmlookup } (HashMap\ s)\ y$
 proof –
 from outcome obtain $s0$ where
 get-s: Some $(s0, s0) \in \text{set-dist } (\text{execute } \text{get } s)$
 and rest:
 Some $(r, t) \in$
 set-dist (execute (hash (tag (PState $s0$))) $\gg=$ return) $s0$)
 unfolding protocol-receive-tagged-random-field-element-def
 by (auto elim!: protocol-merkle.set-dist-bindE)
 have $s0: s0 = s$
 using get-s by simp
 from rest obtain $h\ u$ where
 hash-out: Some $(h, u) \in \text{set-dist } (\text{execute } (\text{hash } (\text{tag } (PState\ s))))\ s$
 and ret: Some $(r, t) \in \text{set-dist } (\text{execute } (\text{return } h)\ u)$
 unfolding $s0$ by (elim protocol-merkle.set-dist-bindE)
 have $t: t = u$ and $r: r = h$
 using ret by auto
 show ?thesis
 unfolding t
 by (rule protocol-hash-preserves-other-lookup[OF hash-out neq])
 qed

lemma protocol-receive-counted-tagged-random-field-element-outcome:

assumes outcome:
 Some $(r, t) \in$
 set-dist
 (execute
 (protocol-receive-counted-tagged-random-field-element counter bump tag) s)
 and bump-hash: $\bigwedge u. HashMap\ (\text{bump } u) = HashMap\ u$
 and bump-state: $\bigwedge u. PState\ (\text{bump } u) = PState\ u$
 and bump-transcript: $\bigwedge u. PTranscript\ (\text{bump } u) = PTranscript\ u$

shows
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)\ (tag\ (counter\ s)\ (PState\ s)) = Some\ r$
proof –
from *outcome* **obtain** $s0\ h\ u\ v$ **where**
 $get\text{-}s: Some\ (s0,\ s0) \in set\text{-}dist\ (execute\ get\ s)$
and *hash-out*:
 $Some\ (h,\ u) \in$
 $set\text{-}dist\ (execute\ (hash\ (tag\ (counter\ s0)\ (PState\ s0))))\ s0)$
and *mod*:
 $Some\ ((),\ v) \in set\text{-}dist\ (execute\ (modify\ bump)\ u)$
and *ret*: $Some\ (r,\ t) \in set\text{-}dist\ (execute\ (return\ h)\ v)$
unfolding *protocol-receive-counted-tagged-random-field-element-def*
by (*auto elim!*: *protocol-merkle.set-dist-bindE*)
have $s0: s0 = s$
using *get-s* **by** *simp*
have $v: v = bump\ u$
using *mod* **unfolding** *modify-def*
by (*auto elim!*: *protocol-merkle.set-dist-bindE*)
have $t: t = v$ **and** $r: r = h$
using *ret* **by** *auto*
have $ext\text{-}su: s \leq u$
using *protocol-merkle.hash-outcome*[*OF hash-out*[*unfolded s0*]]
by *simp*
have *lookup-u*:
 $fmlookup\ (HashMap\ u)\ (tag\ (counter\ s)\ (PState\ s)) = Some\ h$
using *protocol-merkle.hash-outcome*[*OF hash-out*[*unfolded s0*]]
by *simp*
have *channel-u*:
 $PState\ u = PState\ s \wedge PTranscript\ u = PTranscript\ s$
using *protocol-hash-channel-preserves*[*OF hash-out*[*unfolded s0*]]
by *simp*
have *hash-t*: $HashMap\ t = HashMap\ u$
unfolding $t\ v$ **by** (*simp add*: *bump-hash*)
show *?thesis*
using *ext-su lookup-u channel-u*
unfolding $t\ v\ r$ *less-eq-hash-ext-def*
by (*simp add*: *bump-hash bump-state bump-transcript*)
qed

lemma *protocol-receive-counted-tagged-random-field-element-preserves-other-lookup*:
assumes *outcome*:
 $Some\ (r,\ t) \in$
 $set\text{-}dist$
 $(execute$
 $(protocol\text{-}receive\text{-}counted\text{-}tagged\text{-}random\text{-}field\text{-}element\ counter\ bump\ tag)\ s)$
and *neq*: $y \neq tag\ (counter\ s)\ (PState\ s)$

and bump-hash: $\bigwedge u. \text{HashMap } (\text{bump } u) = \text{HashMap } u$
shows $\text{fmlookup } (\text{HashMap } t) \ y = \text{fmlookup } (\text{HashMap } s) \ y$
proof –
from *outcome* **obtain** $s0 \ h \ u \ v$ **where**
get-s: $\text{Some } (s0, s0) \in \text{set-dist } (\text{execute } \text{get } s)$
and *hash-out:*
 $\text{Some } (h, u) \in$
 $\text{set-dist } (\text{execute } (\text{hash } (\text{tag } (\text{counter } s0) \ (P\text{State } s0)))) \ s0)$
and *mod:*
 $\text{Some } ((), v) \in \text{set-dist } (\text{execute } (\text{modify } \text{bump}) \ u)$
and *ret:* $\text{Some } (r, t) \in \text{set-dist } (\text{execute } (\text{return } h) \ v)$
unfolding *protocol-receive-counted-tagged-random-field-element-def*
by (*auto elim!:* *protocol-merkle.set-dist-bindE*)
have $s0: s0 = s$
using *get-s* **by** *simp*
have $v: v = \text{bump } u$
using *mod* **unfolding** *modify-def*
by (*auto elim!:* *protocol-merkle.set-dist-bindE*)
have $t: t = v$
using *ret* **by** *auto*
have *hash-pres:*
 $\text{fmlookup } (\text{HashMap } u) \ y = \text{fmlookup } (\text{HashMap } s) \ y$
by (*rule protocol-hash-preserves-other-lookup*[*OF hash-out*[*unfolded s0*] *neq*])
show *?thesis*
unfolding $t \ v$ **using** *hash-pres* **by** (*simp add:* *bump-hash*)
qed

lemma *protocol-merkle-create-no-failure:*
 $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{protocol-merkle.create } xs) \ s))$
proof (*induction xs arbitrary: s rule: protocol-merkle.create.induct*)
case 1
then show *?case* **by** *simp*
next
case (2 x)
then show *?case*
by (*simp add:* *no-failure-bindI protocol-hash-no-failure*)
next
case (3 $v \ vb \ vc$)
then show *?case*
by (*simp add:* *Let-def no-failure-bindI protocol-hash-no-failure*)
qed

lemma *protocol-merkle-create-preserves-channel:*
assumes *outcome:*
 $\text{Some } (r, t) \in \text{set-dist } (\text{execute } (\text{protocol-merkle.create } xs) \ s)$
shows $P\text{State } t = P\text{State } s$
and $P\text{Transcript } t = P\text{Transcript } s$
and $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s$
and $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s$

```

and  $P\text{AlphaCounter } t = P\text{AlphaCounter } s$ 
and  $P\text{QueryCounter } t = P\text{QueryCounter } s$ 
proof –
  have both:
     $P\text{State } t = P\text{State } s \wedge$ 
     $P\text{Transcript } t = P\text{Transcript } s \wedge$ 
     $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s \wedge$ 
     $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s \wedge$ 
     $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$ 
     $P\text{QueryCounter } t = P\text{QueryCounter } s$ 
  if outcome:
     $\text{Some } (r, t) \in \text{set-dist } (\text{execute } (\text{protocol-merkle.create } xs) s)$ 
  for  $xs \ r \ s \ t$ 
  using outcome
proof (induction xs arbitrary: r s t rule: protocol-merkle.create.induct)
  case 1
  then show ?case by simp
next
  case (2  $x$ )
  from 2.prem obtain  $h \ u$  where
     $h: \text{Some } (h, u) \in \text{set-dist } (\text{execute } (\text{hash } x) s)$ 
  and ret:
     $\text{Some } (r, t) \in$ 
     $\text{set-dist } (\text{execute } (\text{return } \langle \rangle, h, \langle \rangle)) u$ 
  by (auto elim!: protocol-merkle.set-dist-bindE)
  have  $P\text{State } u = P\text{State } s \wedge$ 
     $P\text{Transcript } u = P\text{Transcript } s \wedge$ 
     $P\text{TraceFriCounter } u = P\text{TraceFriCounter } s \wedge$ 
     $P\text{CompositionFriCounter } u = P\text{CompositionFriCounter } s \wedge$ 
     $P\text{AlphaCounter } u = P\text{AlphaCounter } s \wedge$ 
     $P\text{QueryCounter } u = P\text{QueryCounter } s$ 
  using protocol-hash-channel-preserves[OF h] by simp
  then show ?case
  using ret by simp
next
  case (3  $v \ vb \ vc$ )
  from 3.prem obtain  $l \ s1 \ rn \ s2 \ val \ s3$  where
    l:
       $\text{Some } (l, s1) \in$ 
       $\text{set-dist } (\text{execute } (\text{protocol-merkle.create } (v \# \text{take } (\text{length } vc \ \text{div } 2) \ (vb \# vc)))) s)$ 
  and rn:
       $\text{Some } (rn, s2) \in$ 
       $\text{set-dist } (\text{execute } (\text{protocol-merkle.create } (\text{drop } (\text{length } vc \ \text{div } 2) \ (vb \# vc)))) s1)$ 
  and val:
       $\text{Some } (val, s3) \in$ 

```

```

      set-dist (execute (hash (MerkleNode (value l) (value rn))) s2)
and ret:
  Some (r, t) ∈
    set-dist (execute (return ⟨l, val, rn⟩) s3)
by (auto simp: Let-def elim!: protocol-merkle.set-dist-bindE)
have s1:
  PState s1 = PState s ∧
  PTranscript s1 = PTranscript s ∧
  PTraceFriCounter s1 = PTraceFriCounter s ∧
  PCompositionFriCounter s1 = PCompositionFriCounter s ∧
  PAlphaCounter s1 = PAlphaCounter s ∧
  PQueryCounter s1 = PQueryCounter s
using 3.IH(1)[of length (v # vb # vc) div 2 l s1 s] l
by simp
have s2:
  PState s2 = PState s1 ∧
  PTranscript s2 = PTranscript s1 ∧
  PTraceFriCounter s2 = PTraceFriCounter s1 ∧
  PCompositionFriCounter s2 = PCompositionFriCounter s1 ∧
  PAlphaCounter s2 = PAlphaCounter s1 ∧
  PQueryCounter s2 = PQueryCounter s1
using 3.IH(2)[of length (v # vb # vc) div 2 rn s2 s1] rn
by simp
have s3:
  PState s3 = PState s2 ∧
  PTranscript s3 = PTranscript s2 ∧
  PTraceFriCounter s3 = PTraceFriCounter s2 ∧
  PCompositionFriCounter s3 = PCompositionFriCounter s2 ∧
  PAlphaCounter s3 = PAlphaCounter s2 ∧
  PQueryCounter s3 = PQueryCounter s2
using protocol-hash-channel-preserves[OF val] by simp
show ?case
  using s1 s2 s3 ret by simp
qed
show PState t = PState s
  using both[OF assms] by simp
show PTranscript t = PTranscript s
  using both[OF assms] by simp
show PTraceFriCounter t = PTraceFriCounter s
  using both[OF assms] by simp
show PCompositionFriCounter t = PCompositionFriCounter s
  using both[OF assms] by simp
show PAlphaCounter t = PAlphaCounter s
  using both[OF assms] by simp
show PQueryCounter t = PQueryCounter s
  using both[OF assms] by simp
qed

```

lemma protocol-merkle-create-preserves-non-merkle-lookup:

```

assumes outcome:
  Some (r, t) ∈ set-dist (execute (protocol-merkle.create xs) s)
  and not-leaf:  $\bigwedge x. x \in \text{set } xs \implies y \neq x$ 
  and not-node:  $\bigwedge x z. y \neq \text{MerkleNode } x z$ 
shows fmlookup (HashMap t) y = fmlookup (HashMap s) y
using outcome not-leaf
proof (induction xs arbitrary: r s t rule: protocol-merkle.create.induct)
  case 1
  then show ?case
    by simp
next
  case (2 x)
  from 2.prem(1) obtain h u where
    h: Some (h, u) ∈ set-dist (execute (hash x) s)
    and ret:
      Some (r, t) ∈
        set-dist (execute (return ⟨⟨⟩, h, ⟨⟩⟩) u)
    by (auto elim!: protocol-merkle.set-dist-bindE)
  have h-pres: fmlookup (HashMap u) y = fmlookup (HashMap s) y
    by (rule protocol-hash-preserves-other-lookup[OF h])
    (rule 2.prem(2), simp)
  show ?case
    using h-pres ret by simp
next
  case (3 v vb vc)
  let ?left = v # take (length vc div 2) (vb # vc)
  let ?right = drop (length vc div 2) (vb # vc)
  have left-eq:
    ?left = take (length (v # vb # vc) div 2) (v # vb # vc)
    by simp
  have right-eq:
    ?right = drop (length (v # vb # vc) div 2) (v # vb # vc)
    by simp
  from 3.prem(1) obtain l s1 rn s2 val s3 where
    l:
      Some (l, s1) ∈
        set-dist (execute (protocol-merkle.create ?left) s)
    and rn:
      Some (rn, s2) ∈
        set-dist (execute (protocol-merkle.create ?right) s1)
    and val:
      Some (val, s3) ∈
        set-dist (execute (hash (MerkleNode (value l) (value rn))) s2)
    and ret:
      Some (r, t) ∈
        set-dist (execute (return ⟨l, val, rn⟩) s3)
    by (auto simp: Let-def elim!: protocol-merkle.set-dist-bindE)
  have left-not-leaf:  $\bigwedge x. x \in \text{set } ?\text{left} \implies y \neq x$ 
proof –

```

```

fix x
assume mem:  $x \in \text{set } ?\text{left}$ 
show  $y \neq x$ 
proof (cases  $x = v$ )
  case True
  then show ?thesis
    using 3.premis(2)[of x] by simp
next
  case False
  then have  $x \in \text{set } (\text{take } (\text{length } vc \text{ div } 2) (vb \# vc))$ 
    using mem by simp
  then have  $x \in \text{set } (vb \# vc)$ 
    by (rule in-set-takeD)
  then show ?thesis
    using 3.premis(2)[of x] by simp
qed
qed
have right-not-leaf:  $\bigwedge x. x \in \text{set } ?\text{right} \implies y \neq x$ 
proof -
  fix x
  assume mem:  $x \in \text{set } ?\text{right}$ 
  then have  $x \in \text{set } (vb \# vc)$ 
    by (rule in-set-dropD)
  then show  $y \neq x$ 
    using 3.premis(2)[of x] by simp
qed
have l':
  Some (l, s1)  $\in$ 
    set-dist (execute
      (protocol-merkle.create
        (take (length (v # vb # vc) div 2) (v # vb # vc))) s)
  using l by (simp only: left-eq[symmetric])
have rn':
  Some (rn, s2)  $\in$ 
    set-dist (execute
      (protocol-merkle.create
        (drop (length (v # vb # vc) div 2) (v # vb # vc))) s1)
  using rn by (simp only: right-eq[symmetric])
have left-not-leaf':
 $\bigwedge x. x \in \text{set } (\text{take } (\text{length } (v \# vb \# vc) \text{ div } 2) (v \# vb \# vc)) \implies$ 
 $y \neq x$ 
proof -
  fix x
  assume mem:  $x \in \text{set } (\text{take } (\text{length } (v \# vb \# vc) \text{ div } 2) (v \# vb \# vc))$ 
  have  $x \in \text{set } ?\text{left}$ 
    using mem by (simp only: left-eq)
  then show  $y \neq x$ 
    by (rule left-not-leaf)
qed

```

```

have right-not-leaf':
   $\bigwedge x. x \in \text{set } (\text{drop } (\text{length } (v \# vb \# vc) \text{ div } 2) (v \# vb \# vc)) \implies$ 
     $y \neq x$ 
proof –
  fix x
  assume mem:  $x \in \text{set } (\text{drop } (\text{length } (v \# vb \# vc) \text{ div } 2) (v \# vb \# vc))$ 
  have  $x \in \text{set } ?\text{right}$ 
    using mem by (simp only: right-eq)
  then show  $y \neq x$ 
    by (rule right-not-leaf)
qed
have left-pres:  $\text{fmlookup } (\text{HashMap } s1) y = \text{fmlookup } (\text{HashMap } s) y$ 
  by (rule 3.IH(1))
    [of length (v # vb # vc) div 2 l s1 s,
      OF refl l' left-not-leaf']
have right-pres:  $\text{fmlookup } (\text{HashMap } s2) y = \text{fmlookup } (\text{HashMap } s1) y$ 
  by (rule 3.IH(2))
    [of length (v # vb # vc) div 2 rn s2 s1,
      OF refl rn' right-not-leaf']
have hash-pres:  $\text{fmlookup } (\text{HashMap } s3) y = \text{fmlookup } (\text{HashMap } s2) y$ 
  by (rule protocol-hash-preserves-other-lookup[OF val])
    (rule not-node)
show ?case
  using left-pres right-pres hash-pres ret by simp
qed

lemma protocol-create-preserves-non-merkle-lookup:
  assumes outcome:  $\text{Some } (r, t) \in \text{set-dist } (\text{execute } (\text{protocol-create } xs) s)$ 
  and not-leaf:  $\bigwedge x. y \neq \text{MerkleLeaf } x$ 
  and not-node:  $\bigwedge x z. y \neq \text{MerkleNode } x z$ 
  shows  $\text{fmlookup } (\text{HashMap } t) y = \text{fmlookup } (\text{HashMap } s) y$ 
proof –
  have outcome':
     $\text{Some } (r, t) \in$ 
       $\text{set-dist } (\text{execute } (\text{protocol-merkle.create } (\text{map MerkleLeaf } xs)) s)$ 
    using outcome unfolding protocol-create-def .
  show ?thesis
    by (rule protocol-merkle-create-preserves-non-merkle-lookup[OF outcome'])
      (use not-leaf not-node in auto)
qed

locale protocol-channel-compat =
  fixes concat ::  $'a::\text{finite} \Rightarrow 'a \Rightarrow 'a$ 
begin

definition create
  ::  $'a \text{ list} \Rightarrow ('a \text{ protocol-hash-input}, 'a, 'a \text{ tree}, 'b) \text{ hash-monad}$ 
  where create xs = protocol-create xs

```


definition *check-authentication-path*

$:: \text{nat} \Rightarrow \text{nat} \Rightarrow 'a \Rightarrow 'a \text{ list} \Rightarrow$
 $('a \text{ protocol-hash-input}, 'a, 'a, 'b) \text{ hash-monad}$

where

check-authentication-path len idx leaf path =
protocol-check-authentication-path len idx leaf path

definition *created-tree*

$:: 'a \text{ list} \Rightarrow 'a \text{ tree} \Rightarrow$
 $('a \text{ protocol-hash-input}, 'a, 'b) \text{ hash-scheme} \Rightarrow \text{bool}$
where *created-tree xs r s* $\longleftrightarrow$ *protocol-created-tree xs r s*

definition *receive-random-field-element*

$:: ('a, 'a, 'b) \text{ protocol-c-monad}$
where
receive-random-field-element = *protocol-receive-random-field-element*

definition *receive-tagged-random-field-element*

$:: ('a \Rightarrow 'a \text{ protocol-hash-input}) \Rightarrow$
 $('a, 'a, 'b) \text{ protocol-c-monad}$
where
receive-tagged-random-field-element tag =
protocol-receive-tagged-random-field-element tag

definition *receive-trace-fri-challenge*

$:: ('a, 'a, 'b) \text{ protocol-c-monad}$
where
receive-trace-fri-challenge =
protocol-receive-counted-tagged-random-field-element
PTraceFriCounter
 $(\lambda s. s \Downarrow PTraceFriCounter := Suc (PTraceFriCounter s))$
TraceFriChallenge

definition *receive-composition-fri-challenge*

$:: ('a, 'a, 'b) \text{ protocol-c-monad}$
where
receive-composition-fri-challenge =
protocol-receive-counted-tagged-random-field-element
PCompositionFriCounter
 $(\lambda s. s \Downarrow PCompositionFriCounter := Suc (PCompositionFriCounter s))$
CompositionFriChallenge

definition *receive-alpha-challenge*

$:: ('a, 'a, 'b) \text{ protocol-c-monad}$
where
receive-alpha-challenge =
protocol-receive-counted-tagged-random-field-element
PAlphaCounter
 $(\lambda s. s \Downarrow PAlphaCounter := Suc (PAlphaCounter s))$

AlphaChallenge

definition *receive-query-index-challenge*

$:: ('a, 'a, 'b) \text{ protocol-c-monad}$

where

receive-query-index-challenge =
protocol-receive-counted-tagged-random-field-element
PQueryCounter
 $(\lambda s. s(PQueryCounter := Suc (PQueryCounter s)))$
QueryIndexChallenge

definition *send* $:: 'a \Rightarrow ('a, unit, 'b) \text{ protocol-c-monad}$

where *send* $x = \text{protocol-send concat } x$

definition *read* $:: ('a, 'a, 'b) \text{ protocol-c-monad}$

where *read* = *protocol-read concat*

lemmas *set-dist-bindE* = *protocol-merkle.set-dist-bindE*

lemmas *hash-outcome* = *protocol-merkle.hash-outcome*

lemmas *hash-ext-refl* = *protocol-merkle.hash-ext-refl*

lemmas *hash-ext-trans* = *protocol-merkle.hash-ext-trans*

lemmas *hash-extension-lookup* = *protocol-merkle.hash-extension-lookup*

lemma *hash-no-failure*:

$None \notin \text{dom } (\text{dist } (\text{execute } (\text{hash } x) s))$
by (*rule protocol-hash-no-failure*)

lemmas *hash-channel-preserves* = *protocol-hash-channel-preserves*

lemma *send-no-failure*:

$None \notin \text{dom } (\text{dist } (\text{execute } (\text{send } x) s))$
unfolding *send-def protocol-send-def*
by *simp*

lemma *create-no-failure*:

$None \notin \text{dom } (\text{dist } (\text{execute } (\text{create } xs) s))$
unfolding *create-def protocol-create-def*
by (*rule protocol-merkle-create-no-failure*)

lemma *created-tree-mono*:

assumes *created-tree xs r s0*
and $s0 \leq s$
shows *created-tree xs r s*
using *assms*
unfolding *created-tree-def*
by (*rule protocol-created-tree-mono*)

lemma *create-outcome*:

assumes $\text{Some } (r, t) \in \text{set-dist } (\text{execute } (\text{create } xs) s)$

shows $s \leq t \wedge \text{created-tree } xs \ r \ t$
using *protocol-create-outcome*[*OF* *assms*[*unfolded create-def*]]
unfolding *created-tree-def* **by** *simp*

lemma *create-get-authentication-path-len*:
assumes *Some* $(r, t) \in \text{set-dist } (\text{execute } (\text{create } xs) \ s)$
and $0 < \text{lgth}$
and $\text{lgth} \leq \text{length } xs$
shows $\text{length } (\text{get-authentication-path } \text{lgth } \text{idx } r) = \text{floor-log } \text{lgth}$
using *protocol-merkle.create-get-authentication-path-len*
 $[\text{OF } \text{assms}(1)[\text{unfolded create-def protocol-create-def}] \ \text{assms}(2)]$
 $\text{assms}(3)$
by *simp*

lemma *create-preserves-channel*:
assumes *Some* $(r, t) \in \text{set-dist } (\text{execute } (\text{create } xs) \ s)$
shows $P\text{State } t = P\text{State } s$
and $P\text{Transcript } t = P\text{Transcript } s$
and $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s$
and $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s$
and $P\text{AlphaCounter } t = P\text{AlphaCounter } s$
and $P\text{QueryCounter } t = P\text{QueryCounter } s$
using *protocol-merkle.create-preserves-channel*
 $[\text{OF } \text{assms}[\text{unfolded create-def protocol-create-def}]]$
by *simp-all*

lemma *create-preserves-non-merkle-lookup*:
assumes *outcome*: *Some* $(r, t) \in \text{set-dist } (\text{execute } (\text{create } xs) \ s)$
and *not-leaf*: $\bigwedge x. y \neq \text{MerkleLeaf } x$
and *not-node*: $\bigwedge x \ z. y \neq \text{MerkleNode } x \ z$
shows $\text{fmlookup } (\text{HashMap } t) \ y = \text{fmlookup } (\text{HashMap } s) \ y$
using *protocol-create-preserves-non-merkle-lookup*
 $[\text{OF } \text{outcome}[\text{unfolded create-def}] \ \text{not-leaf not-node}]$
by *simp*

lemma *create-preserves-challenge-lookups*:
assumes *outcome*: *Some* $(r, t) \in \text{set-dist } (\text{execute } (\text{create } xs) \ s)$
shows
 $\text{fmlookup } (\text{HashMap } t) \ (\text{AlphaChallenge } i \ x) =$
 $\text{fmlookup } (\text{HashMap } s) \ (\text{AlphaChallenge } i \ x)$
 $\text{fmlookup } (\text{HashMap } t) \ (\text{QueryIndexChallenge } i \ x) =$
 $\text{fmlookup } (\text{HashMap } s) \ (\text{QueryIndexChallenge } i \ x)$
 $\text{fmlookup } (\text{HashMap } t) \ (\text{TraceFriChallenge } i \ x) =$
 $\text{fmlookup } (\text{HashMap } s) \ (\text{TraceFriChallenge } i \ x)$
 $\text{fmlookup } (\text{HashMap } t) \ (\text{CompositionFriChallenge } i \ x) =$
 $\text{fmlookup } (\text{HashMap } s) \ (\text{CompositionFriChallenge } i \ x)$

proof –

show $\text{fmlookup } (\text{HashMap } t) \ (\text{AlphaChallenge } i \ x) =$
 $\text{fmlookup } (\text{HashMap } s) \ (\text{AlphaChallenge } i \ x)$

by (rule create-preserves-non-merkle-lookup[OF outcome]) simp-all
 show fmlookup (HashMap t) (QueryIndexChallenge i x) =
 fmlookup (HashMap s) (QueryIndexChallenge i x)
 by (rule create-preserves-non-merkle-lookup[OF outcome]) simp-all
 show fmlookup (HashMap t) (TraceFriChallenge i x) =
 fmlookup (HashMap s) (TraceFriChallenge i x)
 by (rule create-preserves-non-merkle-lookup[OF outcome]) simp-all
 show fmlookup (HashMap t) (CompositionFriChallenge i x) =
 fmlookup (HashMap s) (CompositionFriChallenge i x)
 by (rule create-preserves-non-merkle-lookup[OF outcome]) simp-all
 qed

lemma check-created-tree-outcome:

assumes created: created-tree xs r s0
 and length-xs: length xs = 2^n
 and i-bound: $i < \text{length } xs$
 and v: $v = xs[i]$
 and ext: $s0 \leq s$
 and check:
 Some (h, t) $\in$ set-dist (execute
 (check-authentication-path (length xs) i v
 (get-authentication-path (length xs) i r)) s)
 shows $h = \text{value } r \wedge s \leq t$
 using protocol-check-created-tree-outcome
 [OF created[unfolded created-tree-def] length-xs i-bound v ext
 check[unfolded check-authentication-path-def]]
 by simp

lemma receive-random-field-element-no-failure:

None $\notin$ dom (dist (execute receive-random-field-element s))
 unfolding receive-random-field-element-def
 by (rule protocol-receive-random-field-element-no-failure)

lemma receive-tagged-random-field-element-no-failure:

None $\notin$
 dom (dist (execute (receive-tagged-random-field-element tag) s))
 unfolding receive-tagged-random-field-element-def
 by (rule protocol-receive-tagged-random-field-element-no-failure)

lemma receive-trace-fri-challenge-no-failure:

None $\notin$ dom (dist (execute receive-trace-fri-challenge s))
 unfolding receive-trace-fri-challenge-def
 by (rule protocol-receive-counted-tagged-random-field-element-no-failure)

lemma receive-composition-fri-challenge-no-failure:

None $\notin$ dom (dist (execute receive-composition-fri-challenge s))
 unfolding receive-composition-fri-challenge-def
 by (rule protocol-receive-counted-tagged-random-field-element-no-failure)

lemma *receive-alpha-challenge-no-failure:*
None $\notin$ *dom* (*dist* (*execute* *receive-alpha-challenge* *s*))
unfolding *receive-alpha-challenge-def*
by (*rule* *protocol-receive-counted-tagged-random-field-element-no-failure*)

lemma *receive-query-index-challenge-no-failure:*
None $\notin$ *dom* (*dist* (*execute* *receive-query-index-challenge* *s*))
unfolding *receive-query-index-challenge-def*
by (*rule* *protocol-receive-counted-tagged-random-field-element-no-failure*)

lemma *receive-random-field-element-outcome:*
assumes *Some* (*r*, *t*) $\in$
set-dist (*execute* *receive-random-field-element* *s*)
shows
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)\ (FiatShamirChallenge\ (PState\ s)) = Some\ r$
using *protocol-receive-random-field-element-outcome*
 $[OF\ assms[unfolding\ receive-random-field-element-def]]$
by *simp*

lemma *receive-tagged-random-field-element-outcome:*
assumes *Some* (*r*, *t*) $\in$
set-dist (*execute* (*receive-tagged-random-field-element* *tag*) *s*)
shows
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)\ (tag\ (PState\ s)) = Some\ r$
using *protocol-receive-tagged-random-field-element-outcome*
 $[OF\ assms[unfolding\ receive-tagged-random-field-element-def]]$
by *simp*

lemma *receive-tagged-random-field-element-preserves-other-lookup:*
assumes *Some* (*r*, *t*) $\in$
set-dist (*execute* (*receive-tagged-random-field-element* *tag*) *s*)
and $y \neq tag\ (PState\ s)$
shows $fmlookup\ (HashMap\ t)\ y = fmlookup\ (HashMap\ s)\ y$
using *protocol-receive-tagged-random-field-element-preserves-other-lookup*
 $[OF\ assms(1)[unfolding\ receive-tagged-random-field-element-def]\ assms(2)]$
by *simp*

lemma *receive-trace-fri-challenge-outcome:*
assumes *Some* (*r*, *t*) $\in$ *set-dist* (*execute* *receive-trace-fri-challenge* *s*)
shows
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$

$fmlookup (HashMap\ t)$
 $(TraceFriChallenge\ (PTraceFriCounter\ s)\ (PState\ s)) = Some\ r$
using *protocol-receive-counted-tagged-random-field-element-outcome*
 $[OF\ assms[unfolding\ receive-trace-fri-challenge-def]]$
by *simp*

lemma *receive-composition-fri-challenge-outcome:*

assumes $Some\ (r, t) \in set-dist\ (execute\ receive-composition-fri-challenge\ s)$
shows
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)$
 $(CompositionFriChallenge\ (PCompositionFriCounter\ s)\ (PState\ s)) =$
 $Some\ r$
using *protocol-receive-counted-tagged-random-field-element-outcome*
 $[OF\ assms[unfolding\ receive-composition-fri-challenge-def]]$
by *simp*

lemma *receive-alpha-challenge-outcome:*

assumes $Some\ (r, t) \in set-dist\ (execute\ receive-alpha-challenge\ s)$
shows
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)$
 $(AlphaChallenge\ (PAlphaCounter\ s)\ (PState\ s)) = Some\ r$
using *protocol-receive-counted-tagged-random-field-element-outcome*
 $[OF\ assms[unfolding\ receive-alpha-challenge-def]]$
by *simp*

lemma *receive-query-index-challenge-outcome:*

assumes $Some\ (r, t) \in set-dist\ (execute\ receive-query-index-challenge\ s)$
shows
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)$
 $(QueryIndexChallenge\ (PQueryCounter\ s)\ (PState\ s)) = Some\ r$
using *protocol-receive-counted-tagged-random-field-element-outcome*
 $[OF\ assms[unfolding\ receive-query-index-challenge-def]]$
by *simp*

lemma *receive-trace-fri-challenge-preserves-other-lookup:*

assumes $Some\ (r, t) \in set-dist\ (execute\ receive-trace-fri-challenge\ s)$
and $y \neq TraceFriChallenge\ (PTraceFriCounter\ s)\ (PState\ s)$
shows $fmlookup\ (HashMap\ t)\ y = fmlookup\ (HashMap\ s)\ y$
using *protocol-receive-counted-tagged-random-field-element-preserves-other-lookup*

[OF assms(1)[unfolded receive-trace-fri-challenge-def] assms(2)]
 by simp

lemma *receive-composition-fri-challenge-preserves-other-lookup:*

assumes *Some* $(r, t) \in$
 set-dist (*execute* *receive-composition-fri-challenge* s)
and $y \neq$ *CompositionFriChallenge* (*PCompositionFriCounter* s) (*PState* s)
shows *fmlookup* (*HashMap* t) $y =$ *fmlookup* (*HashMap* s) y
using *protocol-receive-counted-tagged-random-field-element-preserves-other-lookup*
 [OF assms(1)[unfolded receive-composition-fri-challenge-def] assms(2)]
 by simp

lemma *receive-alpha-challenge-preserves-other-lookup:*

assumes *Some* $(r, t) \in$ *set-dist* (*execute* *receive-alpha-challenge* s)
and $y \neq$ *AlphaChallenge* (*PAlphaCounter* s) (*PState* s)
shows *fmlookup* (*HashMap* t) $y =$ *fmlookup* (*HashMap* s) y
using *protocol-receive-counted-tagged-random-field-element-preserves-other-lookup*
 [OF assms(1)[unfolded receive-alpha-challenge-def] assms(2)]
 by simp

lemma *receive-query-index-challenge-preserves-other-lookup:*

assumes *Some* $(r, t) \in$ *set-dist* (*execute* *receive-query-index-challenge* s)
and $y \neq$ *QueryIndexChallenge* (*PQueryCounter* s) (*PState* s)
shows *fmlookup* (*HashMap* t) $y =$ *fmlookup* (*HashMap* s) y
using *protocol-receive-counted-tagged-random-field-element-preserves-other-lookup*
 [OF assms(1)[unfolded receive-query-index-challenge-def] assms(2)]
 by simp

lemma *receive-trace-fri-challenge-counter-outcome:*

assumes *Some* $(r, t) \in$ *set-dist* (*execute* *receive-trace-fri-challenge* s)
shows

$PTraceFriCounter\ t = Suc\ (PTraceFriCounter\ s) \wedge$
 $PCompositionFriCounter\ t = PCompositionFriCounter\ s \wedge$
 $PAlphaCounter\ t = PAlphaCounter\ s \wedge$
 $PQueryCounter\ t = PQueryCounter\ s$

proof –

from *assms* **obtain** $s0\ h\ u\ v$ **where**

get-s: *Some* $(s0, s0) \in$ *set-dist* (*execute* *get* s)

and *hash-out*:

Some $(h, u) \in$
set-dist (*execute*
 (*hash* (*TraceFriChallenge* (*PTraceFriCounter* $s0$) (*PState* $s0$))) $s0$)

and *mod*:

Some $((), v) \in$
set-dist (*execute*
 (*modify* $(\lambda s. s(PTraceFriCounter := Suc\ (PTraceFriCounter\ s)))$) u)

and *ret*: *Some* $(r, t) \in$ *set-dist* (*execute* (*return* h) v)

unfolding *receive-trace-fri-challenge-def*

protocol-receive-counted-tagged-random-field-element-def

```

    by (auto elim!: protocol-merkle.set-dist-bindE)
  have s0: s0 = s
    using get-s by simp
  have v: v = u(|PTraceFriCounter := Suc (PTraceFriCounter u)|)
    using mod unfolding modify-def
    by (auto elim!: protocol-merkle.set-dist-bindE)
  have t: t = v
    using ret by simp
  have counters-u:
    PTraceFriCounter u = PTraceFriCounter s ∧
    PCompositionFriCounter u = PCompositionFriCounter s ∧
    PAlphaCounter u = PAlphaCounter s ∧
    PQueryCounter u = PQueryCounter s
    using protocol-hash-channel-preserves[OF hash-out[unfolded s0]]
    by simp
  show ?thesis
    unfolding t v using counters-u by simp
qed

lemma receive-composition-fri-challenge-counter-outcome:
  assumes Some (r, t) ∈ set-dist (execute receive-composition-fri-challenge s)
  shows
    PTraceFriCounter t = PTraceFriCounter s ∧
    PCompositionFriCounter t = Suc (PCompositionFriCounter s) ∧
    PAlphaCounter t = PAlphaCounter s ∧
    PQueryCounter t = PQueryCounter s
proof -
  from assms obtain s0 h u v where
    get-s: Some (s0, s0) ∈ set-dist (execute get s)
    and hash-out:
      Some (h, u) ∈
        set-dist (execute
          (hash (CompositionFriChallenge
            (PCompositionFriCounter s0) (PState s0))) s0)
    and mod:
      Some ((), v) ∈
        set-dist (execute
          (modify (λs. s(|
            PCompositionFriCounter := Suc (PCompositionFriCounter s)|)) u)
    and ret: Some (r, t) ∈ set-dist (execute (return h) v)
  unfolding receive-composition-fri-challenge-def
    protocol-receive-counted-tagged-random-field-element-def
  by (auto elim!: protocol-merkle.set-dist-bindE)
  have s0: s0 = s
    using get-s by simp
  have v: v = u(|
    PCompositionFriCounter := Suc (PCompositionFriCounter u)|)
    using mod unfolding modify-def
    by (auto elim!: protocol-merkle.set-dist-bindE)

```



```

have  $t$ :  $t = v$ 
  using ret by simp
have counters-u:
   $PTraceFriCounter\ u = PTraceFriCounter\ s \wedge$ 
   $PCompositionFriCounter\ u = PCompositionFriCounter\ s \wedge$ 
   $PAlphaCounter\ u = PAlphaCounter\ s \wedge$ 
   $PQueryCounter\ u = PQueryCounter\ s$ 
  using protocol-hash-channel-preserves[OF hash-out[unfolded s0]]
  by simp
show ?thesis
  unfolding  $t\ v$  using counters-u by simp
qed

lemma receive-alpha-challenge-counter-outcome:
assumes  $Some\ (r, t) \in set-dist\ (execute\ receive-alpha-challenge\ s)$ 
shows
   $PTraceFriCounter\ t = PTraceFriCounter\ s \wedge$ 
   $PCompositionFriCounter\ t = PCompositionFriCounter\ s \wedge$ 
   $PAlphaCounter\ t = Suc\ (PAlphaCounter\ s) \wedge$ 
   $PQueryCounter\ t = PQueryCounter\ s$ 
proof –
from assms obtain  $s0\ h\ u\ v$  where
  get-s:  $Some\ (s0, s0) \in set-dist\ (execute\ get\ s)$ 
  and hash-out:
     $Some\ (h, u) \in$ 
     $set-dist\ (execute$ 
       $(hash\ (AlphaChallenge\ (PAlphaCounter\ s0)\ (PState\ s0)))\ s0)$ 
  and mod:
     $Some\ ((), v) \in$ 
     $set-dist\ (execute$ 
       $(modify\ (\lambda s. s(PAlphaCounter := Suc\ (PAlphaCounter\ s))))\ u)$ 
  and ret:  $Some\ (r, t) \in set-dist\ (execute\ (return\ h)\ v)$ 
  unfolding receive-alpha-challenge-def
    protocol-receive-counted-tagged-random-field-element-def
  by (auto elim!: protocol-merkle.set-dist-bindE)
have  $s0$ :  $s0 = s$ 
  using get-s by simp
have  $v$ :  $v = u(PAlphaCounter := Suc\ (PAlphaCounter\ u))$ 
  using mod unfolding modify-def
  by (auto elim!: protocol-merkle.set-dist-bindE)
have  $t$ :  $t = v$ 
  using ret by simp
have counters-u:
   $PTraceFriCounter\ u = PTraceFriCounter\ s \wedge$ 
   $PCompositionFriCounter\ u = PCompositionFriCounter\ s \wedge$ 
   $PAlphaCounter\ u = PAlphaCounter\ s \wedge$ 
   $PQueryCounter\ u = PQueryCounter\ s$ 
  using protocol-hash-channel-preserves[OF hash-out[unfolded s0]]
  by simp

```

show *?thesis*
unfolding t v **using** *counters-u* **by** *simp*
qed

lemma *receive-random-field-element-counter-outcome:*
assumes *Some* $(r, t) \in \text{set-dist}$ (*execute receive-random-field-element* s)
shows
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s \wedge$
 $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = P\text{QueryCounter } s$

proof –
from *assms* **obtain** $s0$ **where**
 $\text{get-}s$: *Some* $(s0, s0) \in \text{set-dist}$ (*execute get* s)
and *hash-out*:
 $\text{Some } (r, t) \in$
 $\text{set-dist } (\text{execute } (\text{hash } (\text{FiatShamirChallenge } (P\text{State } s0))) \gg= \text{return}) s0)$
unfolding *receive-random-field-element-def*
protocol-receive-random-field-element-def
protocol-receive-tagged-random-field-element-def
by (*auto elim!*: *protocol-merkle.set-dist-bindE*)
have $s0$: $s0 = s$
using *get-s* **by** *simp*
from *hash-out* **obtain** h u **where**
hash-u:
 $\text{Some } (h, u) \in$
 $\text{set-dist } (\text{execute } (\text{hash } (\text{FiatShamirChallenge } (P\text{State } s)))) s)$
and *ret*: *Some* $(r, t) \in \text{set-dist}$ (*execute (return* $h)$ u)
unfolding $s0$ **by** (*auto elim!*: *protocol-merkle.set-dist-bindE*)
have t : $t = u$
using *ret* **by** *simp*
show *?thesis*
using *protocol-hash-channel-preserves*[*OF hash-u*]
unfolding t **by** *simp*
qed

lemma *receive-query-index-challenge-counter-outcome:*
assumes *Some* $(r, t) \in \text{set-dist}$ (*execute receive-query-index-challenge* s)
shows
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s \wedge$
 $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = \text{Suc } (P\text{QueryCounter } s)$

proof –
from *assms* **obtain** $s0$ h u v **where**
 $\text{get-}s$: *Some* $(s0, s0) \in \text{set-dist}$ (*execute get* s)
and *hash-out*:
 $\text{Some } (h, u) \in$
 $\text{set-dist } (\text{execute } (\text{hash } (\text{FiatShamirChallenge } (P\text{State } s0)))) s)$

```

    (hash (QueryIndexChallenge (PQueryCounter s0) (PState s0))) s0)
  and mod:
    Some ((), v) ∈
      set-dist (execute
        (modify (λs. s(PQueryCounter := Suc (PQueryCounter s)))) u)
  and ret: Some (r, t) ∈ set-dist (execute (return h) v)
  unfolding receive-query-index-challenge-def
    protocol-receive-counted-tagged-random-field-element-def
  by (auto elim!: protocol-merkle.set-dist-bindE)
have s0: s0 = s
  using get-s by simp
have v: v = u(PQueryCounter := Suc (PQueryCounter u))
  using mod unfolding modify-def
  by (auto elim!: protocol-merkle.set-dist-bindE)
have t: t = v
  using ret by simp
have counters-u:
  PTraceFriCounter u = PTraceFriCounter s ∧
  PCompositionFriCounter u = PCompositionFriCounter s ∧
  PAlphaCounter u = PAlphaCounter s ∧
  PQueryCounter u = PQueryCounter s
  using protocol-hash-channel-preserves[OF hash-out[unfolded s0]]
  by simp
show ?thesis
  unfolding t v using counters-u by simp
qed

```

lemma *bind-preserves-lookup*:

```

fixes m :: ('x, ('a, 'b) protocol-channel-scheme) state-monad
  and k :: 'x ⇒ ('y, ('a, 'b) protocol-channel-scheme) state-monad
assumes outcome: Some (y, t) ∈ set-dist (execute (m ≫ k) s)
  and m-pres:
    ∧x u. Some (x, u) ∈ set-dist (execute m s) ⇒
      fmlookup (HashMap u) key = fmlookup (HashMap s) key
  and k-pres:
    ∧x u y t. Some (x, u) ∈ set-dist (execute m s) ⇒
      Some (y, t) ∈ set-dist (execute (k x) u) ⇒
        fmlookup (HashMap t) key = fmlookup (HashMap u) key
shows fmlookup (HashMap t) key = fmlookup (HashMap s) key
proof -
  from outcome obtain x u where
    m-out: Some (x, u) ∈ set-dist (execute m s)
  and k-out: Some (y, t) ∈ set-dist (execute (k x) u)
  by (elim set-dist-bindE)
have fmlookup (HashMap t) key = fmlookup (HashMap u) key
  by (rule k-pres[OF m-out k-out])
also have ... = fmlookup (HashMap s) key
  by (rule m-pres[OF m-out])
finally show ?thesis .

```

qed

lemma *mmap-preserves-lookup*:

fixes $ms :: ('x, ('a, 'b) \text{ protocol-channel-scheme}) \text{ state-monad list}$
assumes *step*:

$\bigwedge m \ x \ s \ t. m \in \text{set } ms \implies$
 $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{fmlookup } (\text{HashMap } t) \ \text{key} = \text{fmlookup } (\text{HashMap } s) \ \text{key}$
and *outcome*: $\text{Some } (xs, t) \in \text{set-dist } (\text{execute } (\text{mmap } ms) \ s)$
shows $\text{fmlookup } (\text{HashMap } t) \ \text{key} = \text{fmlookup } (\text{HashMap } s) \ \text{key}$
using *step outcome*

proof (*induction ms arbitrary: xs s t*)

case *Nil*

then show *?case*

by *simp*

next

case (*Cons m ms*)

from *Cons.prem1(2)* **obtain** $x \ u \ ys$ **where**

m-out: $\text{Some } (x, u) \in \text{set-dist } (\text{execute } m \ s)$
and *tail-out*: $\text{Some } (ys, t) \in \text{set-dist } (\text{execute } (\text{mmap } ms) \ u)$
and *xs-eq*: $xs = x \ \# \ ys$
by (*auto elim!: set-dist-bindE*)

have *head-pres*: $\text{fmlookup } (\text{HashMap } u) \ \text{key} = \text{fmlookup } (\text{HashMap } s) \ \text{key}$
by (*rule Cons.prem1(1)*) (*use m-out in simp-all*)

have *step-tail*:

$\bigwedge m \ x \ s \ t. m \in \text{set } ms \implies$
 $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{fmlookup } (\text{HashMap } t) \ \text{key} = \text{fmlookup } (\text{HashMap } s) \ \text{key}$

proof –

fix $m \ x \ s \ t$

assume *mem*: $m \in \text{set } ms$

and *out*: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s)$

show $\text{fmlookup } (\text{HashMap } t) \ \text{key} = \text{fmlookup } (\text{HashMap } s) \ \text{key}$
by (*rule Cons.prem1(1)*) (*use mem out in simp-all*)

qed

have *tail-pres*: $\text{fmlookup } (\text{HashMap } t) \ \text{key} = \text{fmlookup } (\text{HashMap } u) \ \text{key}$
by (*rule Cons.IH[OF step-tail tail-out]*)

show *?case*

using *head-pres tail-pres* **by** *simp*

qed

lemma *mfold2-preserves-lookup*:

fixes $f :: 'x \Rightarrow (\text{unit}, ('a, 'b) \text{ protocol-channel-scheme}) \text{ state-monad}$
assumes *step*:

$\bigwedge x \ s \ t. x \in \text{set } xs \implies$
 $\text{Some } ((), t) \in \text{set-dist } (\text{execute } (f \ x) \ s) \implies$
 $\text{fmlookup } (\text{HashMap } t) \ \text{key} = \text{fmlookup } (\text{HashMap } s) \ \text{key}$
and *outcome*: $\text{Some } ((), t) \in \text{set-dist } (\text{execute } (\text{mfold2 } f \ xs) \ s)$
shows $\text{fmlookup } (\text{HashMap } t) \ \text{key} = \text{fmlookup } (\text{HashMap } s) \ \text{key}$

```

    using step outcome
  proof (induction xs arbitrary: s t)
    case Nil
    then show ?case
      by simp
  next
    case (Cons x xs)
  from Cons.prem1(2) obtain u where
    head-out: Some ((), u) ∈ set-dist (execute (f x) s)
    and tail-out: Some ((), t) ∈ set-dist (execute (mfold2 f xs) u)
    by (auto elim!: set-dist-bindE)
  have head-pres: fmllookup (HashMap u) key = fmllookup (HashMap s) key
    by (rule Cons.prem1(1)) (use head-out in simp-all)
  have step-tail:
     $\bigwedge x s t. x \in \text{set } xs \implies$ 
     $\text{Some } ((), t) \in \text{set-dist } (\text{execute } (f x) s) \implies$ 
     $\text{fmllookup } (\text{HashMap } t) \text{ key} = \text{fmllookup } (\text{HashMap } s) \text{ key}$ 
  proof -
    fix x s t
    assume mem: x ∈ set xs
    and out: Some ((), t) ∈ set-dist (execute (f x) s)
    show fmllookup (HashMap t) key = fmllookup (HashMap s) key
      by (rule Cons.prem1(1)) (use mem out in simp-all)
    qed
  have tail-pres: fmllookup (HashMap t) key = fmllookup (HashMap u) key
    by (rule Cons.IH[OF step-tail tail-out])
  show ?case
    using head-pres tail-pres by simp
  qed

lemma mfold-preserves-lookup:
  fixes ms :: ('x ⇒ ('a, 'b) protocol-channel-scheme) state-monad list
  assumes step:
     $\bigwedge m a x s t. m \in \text{set } ms \implies$ 
     $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (m a) s) \implies$ 
     $\text{fmllookup } (\text{HashMap } t) \text{ key} = \text{fmllookup } (\text{HashMap } s) \text{ key}$ 
    and outcome: Some (x, t) ∈ set-dist (execute (mfold a ms) s)
  shows fmllookup (HashMap t) key = fmllookup (HashMap s) key
    using step outcome
  proof (induction ms arbitrary: a x s t)
    case Nil
    then show ?case
      by simp
  next
    case (Cons m ms)
  from Cons.prem1(2) obtain y u where
    head-out: Some (y, u) ∈ set-dist (execute (m a) s)
    and tail-out: Some (x, t) ∈ set-dist (execute (mfold y ms) u)
    by (auto elim!: set-dist-bindE)

```

have *head-pres*: $\text{fmlookup } (\text{HashMap } u) \text{ key} = \text{fmlookup } (\text{HashMap } s) \text{ key}$
by (rule *Cons.prem*(1)) (use *head-out* **in** *simp-all*)
have *step-tail*:
 $\bigwedge m \ a \ x \ s \ t. m \in \text{set } ms \implies$
 $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (m \ a) \ s) \implies$
 $\text{fmlookup } (\text{HashMap } t) \text{ key} = \text{fmlookup } (\text{HashMap } s) \text{ key}$
proof –
fix $m \ a \ x \ s \ t$
assume *mem*: $m \in \text{set } ms$
and *out*: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (m \ a) \ s)$
show $\text{fmlookup } (\text{HashMap } t) \text{ key} = \text{fmlookup } (\text{HashMap } s) \text{ key}$
by (rule *Cons.prem*(1)) (use *mem out* **in** *simp-all*)
qed
have *tail-pres*: $\text{fmlookup } (\text{HashMap } t) \text{ key} = \text{fmlookup } (\text{HashMap } u) \text{ key}$
by (rule *Cons.IH*[*OF step-tail tail-out*])
show ?*case*
using *head-pres tail-pres* **by** *simp*
qed

lemma *ntimes-preserves-lookup*:
fixes $m :: ('x, ('a, 'b) \text{ protocol-channel-scheme}) \text{ state-monad}$
assumes *step*:
 $\bigwedge x \ s \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{fmlookup } (\text{HashMap } t) \text{ key} = \text{fmlookup } (\text{HashMap } s) \text{ key}$
and *outcome*: $\text{Some } (xs, t) \in \text{set-dist } (\text{execute } (\text{ntimes } m \ n) \ s)$
shows $\text{fmlookup } (\text{HashMap } t) \text{ key} = \text{fmlookup } (\text{HashMap } s) \text{ key}$
using *step outcome*
proof (induction n arbitrary: $xs \ s \ t$)
case 0
then show ?*case*
by *simp*
next
case (*Suc* n)
from *Suc.prem*(2) **obtain** $x \ u \ ys$ **where**
 $\text{head-out}: \text{Some } (x, u) \in \text{set-dist } (\text{execute } m \ s)$
and $\text{tail-out}: \text{Some } (ys, t) \in \text{set-dist } (\text{execute } (\text{ntimes } m \ n) \ u)$
and $xs\text{-eq}: xs = x \ \# \ ys$
by (auto *elim!*: *set-dist-bindE*)
have *head-pres*: $\text{fmlookup } (\text{HashMap } u) \text{ key} = \text{fmlookup } (\text{HashMap } s) \text{ key}$
by (rule *Suc.prem*(1)[*OF head-out*])
have *tail-pres*: $\text{fmlookup } (\text{HashMap } t) \text{ key} = \text{fmlookup } (\text{HashMap } u) \text{ key}$
by (rule *Suc.IH*[*OF Suc.prem*(1) *tail-out*])
show ?*case*
using *head-pres tail-pres* **by** *simp*
qed

lemma *receive-random-field-element-known-outcome*:
assumes *lookup*:
 $\text{fmlookup } (\text{HashMap } s) (\text{FiatShamirChallenge } (PState \ s)) = \text{Some } r$

and *outcome*:
Some $(r', t) \in \text{set-dist } (\text{execute receive-random-field-element } s)$
shows
 $r' = r \wedge \text{PState } t = \text{PState } s \wedge$
 $\text{PTranscript } t = \text{PTranscript } s \wedge s \leq t$
proof –
have *out*:
 $s \leq t \wedge$
 $\text{PState } t = \text{PState } s \wedge$
 $\text{PTranscript } t = \text{PTranscript } s \wedge$
 $\text{fmlookup } (\text{HashMap } t) (\text{FiatShamirChallenge } (\text{PState } s)) = \text{Some } r'$
by (rule *receive-random-field-element-outcome*[*OF outcome*])
have *old*:
 $\text{fmlookup } (\text{HashMap } t) (\text{FiatShamirChallenge } (\text{PState } s)) = \text{Some } r$
using *lookup out*
by (*meson protocol-merkle.hash-extension-lookup*)
show ?*thesis*
using *out old* **by** *auto*
qed

lemma *receive-tagged-random-field-element-known-outcome*:
assumes *lookup*:
 $\text{fmlookup } (\text{HashMap } s) (\text{tag } (\text{PState } s)) = \text{Some } r$
and *outcome*:
Some $(r', t) \in$
 $\text{set-dist } (\text{execute } (\text{receive-tagged-random-field-element } \text{tag}) s)$
shows
 $r' = r \wedge \text{PState } t = \text{PState } s \wedge$
 $\text{PTranscript } t = \text{PTranscript } s \wedge s \leq t$
proof –
have *out*:
 $s \leq t \wedge$
 $\text{PState } t = \text{PState } s \wedge$
 $\text{PTranscript } t = \text{PTranscript } s \wedge$
 $\text{fmlookup } (\text{HashMap } t) (\text{tag } (\text{PState } s)) = \text{Some } r'$
by (rule *receive-tagged-random-field-element-outcome*[*OF outcome*])
have *old*:
 $\text{fmlookup } (\text{HashMap } t) (\text{tag } (\text{PState } s)) = \text{Some } r$
using *lookup out*
by (*meson protocol-merkle.hash-extension-lookup*)
show ?*thesis*
using *out old* **by** *auto*
qed

lemma *receive-trace-fri-challenge-known-outcome*:
assumes *fmlookup* $(\text{HashMap } s)$
 $(\text{TraceFriChallenge } (\text{PTraceFriCounter } s) (\text{PState } s)) = \text{Some } r$
and *Some* $(r', t) \in \text{set-dist } (\text{execute receive-trace-fri-challenge } s)$
shows

$r' = r \wedge PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge s \leq t$
proof –
have *out*:
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)$
 $(TraceFriChallenge\ (PTraceFriCounter\ s)\ (PState\ s)) = Some\ r'$
by (*rule receive-trace-fri-challenge-outcome*[*OF assms*(2)])
have *old*:
 $fmlookup\ (HashMap\ t)$
 $(TraceFriChallenge\ (PTraceFriCounter\ s)\ (PState\ s)) = Some\ r$
using *assms*(1) *out*
by (*meson protocol-merkle.hash-extension-lookup*)
show ?thesis
using *out old* **by** *auto*
qed

lemma *receive-composition-fri-challenge-known-outcome*:
assumes $fmlookup\ (HashMap\ s)$
 $(CompositionFriChallenge\ (PCompositionFriCounter\ s)\ (PState\ s)) = Some\ r$
and $Some\ (r', t) \in set-dist\ (execute\ receive-composition-fri-challenge\ s)$
shows
 $r' = r \wedge PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge s \leq t$
proof –
have *out*:
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)$
 $(CompositionFriChallenge\ (PCompositionFriCounter\ s)\ (PState\ s)) =$
 $Some\ r'$
by (*rule receive-composition-fri-challenge-outcome*[*OF assms*(2)])
have *old*:
 $fmlookup\ (HashMap\ t)$
 $(CompositionFriChallenge\ (PCompositionFriCounter\ s)\ (PState\ s)) =$
 $Some\ r$
using *assms*(1) *out*
by (*meson protocol-merkle.hash-extension-lookup*)
show ?thesis
using *out old* **by** *auto*
qed

lemma *receive-alpha-challenge-known-outcome*:
assumes $fmlookup\ (HashMap\ s)$
 $(AlphaChallenge\ (PAlphaCounter\ s)\ (PState\ s)) = Some\ r$
and $Some\ (r', t) \in set-dist\ (execute\ receive-alpha-challenge\ s)$

shows
 $r' = r \wedge PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge s \leq t$
proof –
have *out*:
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)$
 $(AlphaChallenge\ (PAlphaCounter\ s)\ (PState\ s)) = Some\ r'$
by (rule *receive-alpha-challenge-outcome*[*OF* *assms*(2)])
have *old*:
 $fmlookup\ (HashMap\ t)$
 $(AlphaChallenge\ (PAlphaCounter\ s)\ (PState\ s)) = Some\ r$
using *assms*(1) *out*
by (meson *protocol-merkle.hash-extension-lookup*)
show ?*thesis*
using *out old* **by** *auto*
qed

lemma *receive-query-index-challenge-known-outcome*:
assumes $fmlookup\ (HashMap\ s)$
 $(QueryIndexChallenge\ (PQueryCounter\ s)\ (PState\ s)) = Some\ r$
and $Some\ (r', t) \in set-dist\ (execute\ receive-query-index-challenge\ s)$
shows
 $r' = r \wedge PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge s \leq t$
proof –
have *out*:
 $s \leq t \wedge$
 $PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)$
 $(QueryIndexChallenge\ (PQueryCounter\ s)\ (PState\ s)) = Some\ r'$
by (rule *receive-query-index-challenge-outcome*[*OF* *assms*(2)])
have *old*:
 $fmlookup\ (HashMap\ t)$
 $(QueryIndexChallenge\ (PQueryCounter\ s)\ (PState\ s)) = Some\ r$
using *assms*(1) *out*
by (meson *protocol-merkle.hash-extension-lookup*)
show ?*thesis*
using *out old* **by** *auto*
qed

lemma *read-no-failure*:
assumes $PTranscript\ s \neq []$
shows $None \notin dom\ (dist\ (execute\ read\ s))$
using *assms*
unfolding *read-def protocol-read-def assert-def*

by (intro no-failure-bindI) simp-all

lemma read-hash-extends:

assumes $\text{Some } (x, t) \in \text{set-dist } (\text{execute read } s)$
 shows $s \leq t$

proof (cases PTranscript s)

case Nil

from assms obtain s0 u where

get-s: $\text{Some } (s0, s0) \in \text{set-dist } (\text{execute get } s)$

and assert-u:

$\text{Some } ((), u) \in$

$\text{set-dist } (\text{execute } (\text{assert } (\text{PTranscript } s0 \neq [])) s0)$

unfolding read-def protocol-read-def

by (auto elim!: set-dist-bindE)

have s0 = s

using get-s by simp

with Nil assert-u have False

unfolding assert-def

using throw-no-Some-set-dist by force

then show ?thesis by simp

next

case (Cons a list)

then show ?thesis

using assms

unfolding read-def protocol-read-def assert-def

less-eq-hash-ext-def less-eq-fmap-def modify-def

by (auto elim!: set-dist-bindE)

qed

lemma receive-random-field-element-extends:

assumes $\text{Some } (r, t) \in$

$\text{set-dist } (\text{execute receive-random-field-element } s)$

shows $s \leq t$

using receive-random-field-element-outcome[OF assms] by simp

lemma receive-tagged-random-field-element-extends:

assumes $\text{Some } (r, t) \in$

$\text{set-dist } (\text{execute } (\text{receive-tagged-random-field-element tag}) s)$

shows $s \leq t$

using receive-tagged-random-field-element-outcome[OF assms] by simp

lemma receive-trace-fri-challenge-extends:

assumes $\text{Some } (r, t) \in \text{set-dist } (\text{execute receive-trace-fri-challenge } s)$

shows $s \leq t$

using receive-trace-fri-challenge-outcome[OF assms] by simp

lemma receive-composition-fri-challenge-extends:

assumes $\text{Some } (r, t) \in$

$\text{set-dist } (\text{execute receive-composition-fri-challenge } s)$

shows $s \leq t$
using *receive-composition-fri-challenge-outcome*[*OF assms*] **by** *simp*

lemma *receive-alpha-challenge-extends*:
assumes *Some* $(r, t) \in \text{set-dist (execute receive-alpha-challenge } s)$
shows $s \leq t$
using *receive-alpha-challenge-outcome*[*OF assms*] **by** *simp*

lemma *receive-query-index-challenge-extends*:
assumes *Some* $(r, t) \in \text{set-dist (execute receive-query-index-challenge } s)$
shows $s \leq t$
using *receive-query-index-challenge-outcome*[*OF assms*] **by** *simp*

lemma *send-outcome*:
assumes *Some* $((), t) \in \text{set-dist (execute (send } x) s)$
shows
 $t = s \langle$
 $\quad PState := \text{concat } (PState\ s)\ x,$
 $\quad PTranscript := x \# PTranscript\ s \rangle$
using *protocol-send-outcome*[*OF assms*[*unfolded send-def*]] .

lemma *read-cons-outcome*:
assumes *Some* $(y, t) \in$
 $\quad \text{set-dist (execute read (s \langle PTranscript := x \# xs \rangle))}$
shows
 $y = x \wedge$
 $t = s \langle$
 $\quad PState := \text{concat } (PState\ s)\ x,$
 $\quad PTranscript := xs \rangle$
using *protocol-read-cons-outcome*[*OF assms*[*unfolded read-def*]] .

lemma *read-nonempty-outcome*:
assumes *out*: *Some* $(y, t) \in \text{set-dist (execute read } s)$
and *tr*: $PTranscript\ s = x \# xs$
shows
 $y = x \wedge$
 $t = s \langle PState := \text{concat } (PState\ s)\ x, PTranscript := xs \rangle$
proof –
have *s-eq*: $s \langle PTranscript := x \# xs \rangle = s$
using *tr* **by** (*cases* *s*) *simp*
have *out'*: *Some* $(y, t) \in$
 $\quad \text{set-dist (execute read (s \langle PTranscript := x \# xs \rangle))}$
using *out* **unfolding** *s-eq* .
show *?thesis*
using *read-cons-outcome*[*OF out'*] *tr* **by** *simp*
qed

lemma *random-then-read-cons-outcome*:
assumes *lookup*: *fmllookup* (*HashMap* *s*) (*FiatShamirChallenge* (*PState* *s*)) = *Some* *x*

and *outcome*:
Some $((r, y), t) \in$
set-dist (*execute*
 (*do* {
 $r \leftarrow \text{receive-random-field-element};$
 $y \leftarrow \text{read};$
 $\text{return } (r, y)$
 }) ($s(P\text{Transcript} := x \# xs)$))
shows
 $r = x \wedge$
 $y = x \wedge$
 $P\text{State } t = \text{concat } (P\text{State } s) \ x \wedge$
 $P\text{Transcript } t = xs$
proof –
let $?s = s(P\text{Transcript} := x \# xs)$
have *outcome'*:
Some $((r, y), t) \in$
set-dist (*execute*
 (*receive-random-field-element* $\gg$
 ($\lambda r. \text{read} \gg (\lambda y. \text{return } (r, y))$)) $?s$)
using *outcome* **by** *simp*
from *outcome'* **obtain** *r-state* **where**
rand:
Some $(r, r\text{-state}) \in \text{set-dist } (\text{execute } \text{receive-random-field-element } ?s)$
and *read-y*:
Some $(y, t) \in \text{set-dist } (\text{execute } \text{read } r\text{-state})$
by (*auto elim!*: *set-dist-bindE*)
have *lookup'*: $\text{fmlookup } (\text{HashMap } ?s) (\text{FiatShamirChallenge } (P\text{State } ?s)) = \text{Some } x$
using *lookup* **by** *simp*
have *rand-res*:
 $r = x \wedge P\text{State } r\text{-state} = P\text{State } s \wedge P\text{Transcript } r\text{-state} = x \# xs$
using *receive-random-field-element-known-outcome*[*OF* *lookup'* *rand*] **by** *simp*
have *r-state-update*: $r\text{-state}(P\text{Transcript} := x \# xs) = r\text{-state}$
using *rand-res* **by** *simp*
have *read-y'*:
Some $(y, t) \in \text{set-dist } (\text{execute } \text{read } (r\text{-state}(P\text{Transcript} := x \# xs)))$
using *read-y* **unfolding** *r-state-update* .
have *read-res*:
 $y = x \wedge P\text{State } t = \text{concat } (P\text{State } s) \ x \wedge P\text{Transcript } t = xs$
using *read-cons-outcome*[*OF* *read-y'*] *rand-res* **by** *simp*
show *?thesis*
using *rand-res* *read-res* **by** *simp*
qed

lemma *random-then-read-assert-no-failure*:
assumes *lookup*:
 $\text{fmlookup } (\text{HashMap } s) (\text{FiatShamirChallenge } (P\text{State } s)) = \text{Some } x$
shows
 $\text{None} \notin \text{dom } (\text{dist } (\text{execute }$

```

  (do {
    r ← receive-random-field-element;
    y ← read;
    assert (r = y);
    return y
  }) (s(PTranscript := x # xs)))
proof -
  let ?s = s(PTranscript := x # xs)
  let ?prefix =
    do {
      r ← receive-random-field-element;
      y ← read;
      return (r, y)
    }
  have prefix-no-failure:
    None ∉ dom (dist (execute ?prefix ?s))
  proof (rule no-failure-bindI)
  show None ∉ dom (dist (execute receive-random-field-element ?s))
    by (rule receive-random-field-element-no-failure)
  fix r t
  assume rand: Some (r, t) ∈ set-dist (execute receive-random-field-element ?s)
  have lookup-s:
    fmlookup (HashMap ?s) (FiatShamirChallenge (PState ?s)) = Some x
    using lookup by simp
  have tr-t: PTranscript t = x # xs
    using receive-random-field-element-known-outcome[OF lookup-s rand] by simp
  show None ∉ dom (dist (execute (read ≫ (λy. return (r, y))) t))
  proof (rule no-failure-bindI)
  show None ∉ dom (dist (execute read t))
    by (rule read-no-failure) (simp add: tr-t)
  fix y u
  assume Some (y, u) ∈ set-dist (execute read t)
  show None ∉ dom (dist (execute (return (r, y)) u))
    by simp
  qed
qed
have full:
  do {
    r ← receive-random-field-element;
    y ← read;
    assert (r = y);
    return y
  } =
  (?prefix ≫ (λ(r, y). assert (r = y) ≫ (λ-. return y)))
  by (simp add: sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full
proof (rule no-failure-bindI[OF prefix-no-failure])
  fix ry t

```

```

assume prefix:
  Some (ry, t) ∈ set-dist (execute ?prefix ?s)
obtain r y where ry: ry = (r, y)
  by (cases ry)
have replay: r = y
  using random-then-read-cons-outcome[OF lookup prefix[unfolded ry]] by simp
show None ∉ dom
  (dist (execute ((case ry of (r, y) ⇒ assert (r = y) ≫ (λ-. return y))) t))
  unfolding ry replay assert-def by simp
qed
qed

```

lemma *read-then-random-cons-outcome*:

```

assumes lookup:
  fmlookup (HashMap s)
  (FiatShamirChallenge (concat (PState s) x)) = Some r
and outcome:
  Some ((y, r'), t) ∈
    set-dist (execute
      (do {
        y ← read;
        r ← receive-random-field-element;
        return (y, r)
      }) (s[PTranscript := x # xs]))

```

shows

```

y = x ∧
r' = r ∧
PState t = concat (PState s) x ∧
PTranscript t = xs

```

proof –

```

let ?s = s[PTranscript := x # xs]
have outcome':
  Some ((y, r'), t) ∈
    set-dist (execute
      (read ≫
        (λy. receive-random-field-element ≫ (λr. return (y, r)))) ?s)
  using outcome by simp
from outcome' obtain read-state where
  read-y: Some (y, read-state) ∈ set-dist (execute read ?s)
  and rand:
    Some (r', t) ∈ set-dist (execute receive-random-field-element read-state)
  by (auto elim!: set-dist-bindE)
have read-res:
  y = x ∧
  read-state = s[PState := concat (PState s) x, PTranscript := xs]
  using read-cons-outcome[OF read-y] by simp
have rand-res:
  r' = r ∧
  PState t = concat (PState s) x ∧

```

```

    PTranscript t = xs
    using receive-random-field-element-known-outcome[OF - rand] lookup read-res by simp
  show ?thesis
    using read-res rand-res by simp
qed

```

```

lemma mfold2-send-outcome:
  assumes Some ((), t) ∈ set-dist (execute (mfold2 send xs) s)
  shows
    t = s⟦
      PState := foldl concat (PState s) xs,
      PTranscript := rev xs @ PTranscript s⟧
  using assms
proof (induction xs arbitrary: s t)
  case Nil
  then show ?case by simp
next
  case (Cons x xs)
  from Cons.prem1 obtain s1 where
    send-x: Some ((), s1) ∈ set-dist (execute (send x) s)
    and rest: Some ((), t) ∈ set-dist (execute (mfold2 send xs) s1)
    by (auto elim!: set-dist-bindE)
  have s1:
    s1 = s⟦PState := concat (PState s) x,
      PTranscript := x # PTranscript s⟧
  using send-outcome[OF send-x] .
  show ?case
    using Cons.IH[OF rest] unfolding s1 by simp
qed

```

```

lemma ntimes-read-outcome:
  assumes Some (ys, t) ∈
    set-dist (execute (ntimes read (length xs)) (s⟦PTranscript := xs⟧))
  shows
    ys = xs ∧
    t = s⟦PState := foldl concat (PState s) xs, PTranscript := []⟧
  using assms
proof (induction xs arbitrary: s ys t)
  case Nil
  then show ?case by simp
next
  case (Cons x xs)
  have prem1':
    Some (ys, t) ∈
      set-dist (execute
        (read ≫ (λy. ntimes read (length xs) ≫ (λys. return (y # ys))))
        (s⟦PTranscript := x # xs⟧))
  using Cons.prem1 by simp
  from prem1' obtain y s1 where

```

```

  read-x: Some (y, s1) ∈ set-dist (execute read (s(PTranscript := x # xs)))
  and rest1:
    Some (ys, t) ∈
      set-dist (execute (ntimes read (length xs) >>= (λys. return (y # ys))) s1)
  by (rule set-dist-bindE) blast
  from rest1 obtain ys' s2 where
    reads: Some (ys', s2) ∈ set-dist (execute (ntimes read (length xs)) s1)
    and ret: Some (ys, t) ∈ set-dist (execute (return (y # ys')) s2)
    by (auto elim!: set-dist-bindE)
  have read-res:
    y = x ∧ s1 = s(PState := concat (PState s) x, PTranscript := xs)
    using read-cons-outcome[OF read-x] .
  let ?s0 = s(PState := concat (PState s) x)
  have s1: s1 = ?s0(PTranscript := xs)
    using read-res by simp
  have tail:
    ys' = xs ∧
    s2 = ?s0(PState := foldl concat (PState ?s0) xs, PTranscript := [])
    using Cons.IH[OF reads[unfolded s1]] .
  have ys-t: ys = y # ys' ∧ t = s2
    using ret by simp
  show ?case
    using read-res tail ys-t by simp
qed

```

theorem *honest-transcript-replay-mfold2-ntimes:*

```

  assumes init: PTranscript s = []
  and sent: Some ((), sent) ∈ set-dist (execute (mfold2 send xs) s)
  and replay:
    Some (ys, replayed) ∈
      set-dist (execute
        (ntimes read (length xs))
        (sent(PState := PState s, PTranscript := rev (PTranscript sent))))
  shows ys = xs ∧ replayed = sent(PTranscript := [])

```

proof –

```

  have sent-eq:
    sent =
      s(
        PState := foldl concat (PState s) xs,
        PTranscript := rev xs
      )
    using mfold2-send-outcome[OF sent] init by simp
  let ?replay-start = sent(PState := PState s)
  have start-eq:
    sent(PState := PState s, PTranscript := rev (PTranscript sent)) =
      ?replay-start(PTranscript := xs)
    using sent-eq by simp
  have replay-out:
    ys = xs ∧

```



```

    replayed =
      ?replay-start(
        PState := foldl concat (PState ?replay-start) xs,
        PTranscript := []
      )
    using ntimes-read-outcome[OF replay[unfolded start-eq]] .
  then show ?thesis
    using sent-eq by simp
qed

lemma ntimes-read-prefix-outcome:
  assumes Some (ys, t) ∈
    set-dist (execute (ntimes read (length xs)) (s(PTranscript := xs @ rest)))
  shows
    ys = xs ∧
    t = s(PState := foldl concat (PState s) xs, PTranscript := rest)
  using assms
proof (induction xs arbitrary: s ys t)
  case Nil
  then show ?case by simp
next
  case (Cons x xs)
  have prems':
    Some (ys, t) ∈
      set-dist (execute
        (read ≫ (λy. ntimes read (length xs) ≫ (λys. return (y # ys))))
        (s(PTranscript := x # xs @ rest)))
  using Cons.prems by simp
  from prems' obtain y s1 where
    read-x: Some (y, s1) ∈ set-dist (execute read (s(PTranscript := x # xs @ rest)))
  and rest1:
    Some (ys, t) ∈
      set-dist (execute (ntimes read (length xs) ≫ (λys. return (y # ys))) s1)
  by (rule set-dist-bindE) blast
  from rest1 obtain ys' s2 where
    reads: Some (ys', s2) ∈ set-dist (execute (ntimes read (length xs)) s1)
  and ret: Some (ys, t) ∈ set-dist (execute (return (y # ys')) s2)
  by (auto elim!: set-dist-bindE)
  have read-res:
    y = x ∧ s1 = s(PState := concat (PState s) x, PTranscript := xs @ rest)
  using read-cons-outcome[OF read-x] .
  let ?s0 = s(PState := concat (PState s) x)
  have s1: s1 = ?s0(PTranscript := xs @ rest)
  using read-res by simp
  have tail:
    ys' = xs ∧
    s2 = ?s0(PState := foldl concat (PState ?s0) xs, PTranscript := rest)
  using Cons.IH[OF reads[unfolded s1]] .
  have ys-t: ys = y # ys' ∧ t = s2

```

```

    using ret by simp
  show ?case
    using read-res tail ys-t by simp
qed

lemma ntimes-read-no-failure:
  assumes  $n \leq \text{length } (P\text{Transcript } s)$ 
  shows  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{ntimes read } n) s))$ 
  using assms
proof (induction n arbitrary: s)
  case 0
  then show ?case by simp
next
  case (Suc n)
  have nonempty:  $P\text{Transcript } s \neq []$ 
  using Suc.prem by (cases  $P\text{Transcript } s$ ) auto
  have nf-read:  $\text{None} \notin \text{dom } (\text{dist } (\text{execute read } s))$ 
  by (rule read-no-failure[OF nonempty])
  have nf-cont:
     $\bigwedge x t. \text{Some } (x, t) \in \text{set-dist } (\text{execute read } s) \implies$ 
     $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{ntimes read } n \gg (\lambda xs. \text{return } (x \# xs))) t))$ 
  proof -
    fix x t
    assume out:  $\text{Some } (x, t) \in \text{set-dist } (\text{execute read } s)$ 
    obtain y ys where tr:  $P\text{Transcript } s = y \# ys$ 
    using nonempty by (cases  $P\text{Transcript } s$ ) auto
    have t-eq:
       $t = s(P\text{State} := \text{concat } (P\text{State } s) y, P\text{Transcript} := ys)$ 
    using read-nonempty-outcome[OF out tr] by simp
    have  $n \leq \text{length } ys$ 
    using Suc.prem tr by simp
    then have nf-tail:  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{ntimes read } n) t))$ 
    using Suc.IH[of t] unfolding t-eq by simp
    show  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{ntimes read } n \gg (\lambda xs. \text{return } (x \# xs))) t))$ 
    by (rule no-failure-bindI[OF nf-tail]) simp
  qed
  show ?case
    unfolding ntimes.simps
    by (rule no-failure-bindI[OF nf-read nf-cont])
qed

lemma ntimes-read-hash-extends:
  assumes  $\text{Some } (xs, t) \in \text{set-dist } (\text{execute } (\text{ntimes read } n) s)$ 
  shows  $s \leq t$ 
  using assms

```

```

proof (induction n arbitrary: xs s t)
  case 0
  then show ?case by (simp add: hash-ext-refl)
next
  case (Suc n)
  from Suc.premis obtain x ys u where
    read: Some (x, u) ∈ set-dist (execute read s)
    and reads: Some (ys, t) ∈ set-dist (execute (ntimes read n) u)
    by (auto elim!: set-dist-bindE)
  have s ≤ u
    by (rule read-hash-extends[OF read])
  moreover have u ≤ t
    by (rule Suc.IH[OF reads])
  ultimately show ?case
    by (rule hash-ext-trans)
qed

lemma check-authentication-path-no-failure:
  None ∉ dom (dist (execute (check-authentication-path len i v path) s))
  unfolding check-authentication-path-def protocol-check-authentication-path-def
proof (induction path arbitrary: len i v s)
  case Nil
  then show ?case
    by (simp add: protocol-hash-no-failure)
next
  case (Cons a path)
  show ?case
  proof (cases i < len div 2)
    case True
    then show ?thesis
      by (simp add: no-failure-bindI protocol-hash-no-failure Cons.IH)
  next
  case False
  then show ?thesis
    by (simp add: no-failure-bindI protocol-hash-no-failure Cons.IH)
  qed
qed

lemma check-authentication-path-hash-extends:
  assumes outcome:
    Some (h, t) ∈
      set-dist (execute (check-authentication-path len i v path) s)
  shows s ≤ t
  using outcome
  unfolding check-authentication-path-def protocol-check-authentication-path-def
proof (induction path arbitrary: len i v s h t)
  case Nil
  then have hash-step:
    Some (h, t) ∈ set-dist (execute (hash (MerkleLeaf v)) s)

```

```

    by simp
  show ?case
    by (rule hash-outcome(1)[OF hash-step])
next
case (Cons a path)
show ?case
proof (cases i < len div 2)
  case True
  from Cons.premis obtain x u where
    rec:
      Some (x, u) ∈
        set-dist (execute
          (protocol-merkle.check-authentication-path
            (len div 2) i (MerkleLeaf v) path) s)
  and hash:
      Some (h, t) ∈ set-dist (execute (hash (MerkleNode x a)) u)
  using True by (auto elim!: set-dist-bindE)
  have s ≤ u
    by (rule Cons.IH[OF rec])
  moreover have u ≤ t
    using hash-outcome(1)[OF hash] .
  ultimately show ?thesis
    by (rule hash-ext-trans)
next
case False
  from Cons.premis obtain x u where
    rec:
      Some (x, u) ∈
        set-dist (execute
          (protocol-merkle.check-authentication-path
            (len div 2) (i - len div 2) (MerkleLeaf v) path) s)
  and hash:
      Some (h, t) ∈ set-dist (execute (hash (MerkleNode a x)) u)
  using False by (auto elim!: set-dist-bindE)
  have s ≤ u
    by (rule Cons.IH[OF rec])
  moreover have u ≤ t
    using hash-outcome(1)[OF hash] .
  ultimately show ?thesis
    by (rule hash-ext-trans)
qed
qed

```

lemma *check-authentication-path-preserves-channel:*
assumes $\text{Some } (h, t) \in$
 $\text{set-dist } (\text{execute } (\text{check-authentication-path } \text{len } i \text{ v path}) \text{ s})$
shows $P\text{State } t = P\text{State } s$
and $P\text{Transcript } t = P\text{Transcript } s$
and $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s$

```

and  $PCompositionFriCounter\ t = PCompositionFriCounter\ s$ 
and  $PAlphaCounter\ t = PAlphaCounter\ s$ 
and  $PQueryCounter\ t = PQueryCounter\ s$ 
proof –
have both:
   $PState\ t = PState\ s \wedge$ 
   $PTranscript\ t = PTranscript\ s \wedge$ 
   $PTraceFriCounter\ t = PTraceFriCounter\ s \wedge$ 
   $PCompositionFriCounter\ t = PCompositionFriCounter\ s \wedge$ 
   $PAlphaCounter\ t = PAlphaCounter\ s \wedge$ 
   $PQueryCounter\ t = PQueryCounter\ s$ 
if outcome:
   $Some\ (h, t) \in$ 
     $set-dist\ (execute\ (check-authentication-path\ len\ i\ v\ path)\ s)$ 
for  $len\ i\ v\ s\ h\ t$ 
using outcome
unfolding check-authentication-path-def protocol-check-authentication-path-def
proof (induction path arbitrary: len i v s h t)
case Nil
have hash-step:
   $Some\ (h, t) \in set-dist\ (execute\ (hash\ (MerkleLeaf\ v))\ s)$ 
using Nil.prems by simp
show ?case
using hash-channel-preserves(1)[OF hash-step]
  hash-channel-preserves(2)[OF hash-step]
  hash-channel-preserves(3)[OF hash-step]
  hash-channel-preserves(4)[OF hash-step]
  hash-channel-preserves(5)[OF hash-step]
  hash-channel-preserves(6)[OF hash-step]
by simp
next
case (Cons a path)
show ?case
proof (cases i < len div 2)
case True
from Cons.prems obtain  $x\ u$  where
  rec:
     $Some\ (x, u) \in$ 
       $set-dist\ (execute$ 
         $(protocol-merkle.check-authentication-path$ 
           $(len\ div\ 2)\ i\ (MerkleLeaf\ v)\ path)\ s)$ 
and hash:
     $Some\ (h, t) \in set-dist\ (execute\ (hash\ (MerkleNode\ x\ a))\ u)$ 
using True by (auto elim!:: set-dist-bindE)
have rec-pres:
   $PState\ u = PState\ s \wedge$ 
   $PTranscript\ u = PTranscript\ s \wedge$ 
   $PTraceFriCounter\ u = PTraceFriCounter\ s \wedge$ 
   $PCompositionFriCounter\ u = PCompositionFriCounter\ s \wedge$ 

```

```

    PAlphaCounter u = PAlphaCounter s ∧
    PQueryCounter u = PQueryCounter s
using Cons.IH[OF rec] .
have hash-pres:
  PState t = PState u ∧
  PTranscript t = PTranscript u ∧
  PTraceFriCounter t = PTraceFriCounter u ∧
  PCompositionFriCounter t = PCompositionFriCounter u ∧
  PAlphaCounter t = PAlphaCounter u ∧
  PQueryCounter t = PQueryCounter u
using hash-channel-preserves(1)[OF hash]
  hash-channel-preserves(2)[OF hash]
  hash-channel-preserves(3)[OF hash]
  hash-channel-preserves(4)[OF hash]
  hash-channel-preserves(5)[OF hash]
  hash-channel-preserves(6)[OF hash] by simp
show ?thesis
  using rec-pres hash-pres by simp
next
case False
from Cons.premis obtain x u where
  rec:
    Some (x, u) ∈
      set-dist (execute
        (protocol-merkle.check-authentication-path
          (len div 2) (i - len div 2) (MerkleLeaf v) path) s)
and hash:
    Some (h, t) ∈ set-dist (execute (hash (MerkleNode a x)) u)
using False by (auto elim!: set-dist-bindE)
have rec-pres:
  PState u = PState s ∧
  PTranscript u = PTranscript s ∧
  PTraceFriCounter u = PTraceFriCounter s ∧
  PCompositionFriCounter u = PCompositionFriCounter s ∧
  PAlphaCounter u = PAlphaCounter s ∧
  PQueryCounter u = PQueryCounter s
using Cons.IH[OF rec] .
have hash-pres:
  PState t = PState u ∧
  PTranscript t = PTranscript u ∧
  PTraceFriCounter t = PTraceFriCounter u ∧
  PCompositionFriCounter t = PCompositionFriCounter u ∧
  PAlphaCounter t = PAlphaCounter u ∧
  PQueryCounter t = PQueryCounter u
using hash-channel-preserves(1)[OF hash]
  hash-channel-preserves(2)[OF hash]
  hash-channel-preserves(3)[OF hash]
  hash-channel-preserves(4)[OF hash]
  hash-channel-preserves(5)[OF hash]

```

```

    hash-channel-preserves(6)[OF hash] by simp
  show ?thesis
    using rec-pres hash-pres by simp
qed
qed
show PState t = PState s
  using both[OF assms] by simp
show PTranscript t = PTranscript s
  using both[OF assms] by simp
show PTraceFriCounter t = PTraceFriCounter s
  using both[OF assms] by simp
show PCompositionFriCounter t = PCompositionFriCounter s
  using both[OF assms] by simp
show PAlphaCounter t = PAlphaCounter s
  using both[OF assms] by simp
show PQueryCounter t = PQueryCounter s
  using both[OF assms] by simp
qed

lemma check-authentication-path-preserves-non-merkle-lookup:
  assumes outcome: Some (h, t) ∈
    set-dist (execute (check-authentication-path len i v path) s)
    and not-leaf:  $\bigwedge x. y \neq \text{MerkleLeaf } x$ 
    and not-node:  $\bigwedge x z. y \neq \text{MerkleNode } x z$ 
  shows fmllookup (HashMap t) y = fmllookup (HashMap s) y
  using outcome
  unfolding check-authentication-path-def protocol-check-authentication-path-def
proof (induction path arbitrary: len i v s h t)
  case Nil
  have hash-step:
    Some (h, t) ∈ set-dist (execute (hash (MerkleLeaf v)) s)
  using Nil.premis by simp
  show ?case
    by (rule protocol-hash-preserves-other-lookup[OF hash-step])
      (rule not-leaf)
next
  case (Cons a path)
  show ?case
  proof (cases i < len div 2)
    case True
    from Cons.premis obtain x u where
      rec:
        Some (x, u) ∈
          set-dist (execute
            (protocol-merkle.check-authentication-path
              (len div 2) i (MerkleLeaf v) path) s)
    and hash-step:
      Some (h, t) ∈ set-dist (execute (hash (MerkleNode x a)) u)
    using True by (auto elim!: set-dist-bindE)

```

```

have rec-pres:
  fmlookup (HashMap u) y = fmlookup (HashMap s) y
  by (rule Cons.IH[OF rec])
have hash-pres:
  fmlookup (HashMap t) y = fmlookup (HashMap u) y
  by (rule protocol-hash-preserves-other-lookup[OF hash-step])
    (rule not-node)
show ?thesis
  using hash-pres rec-pres by simp
next
case False
from Cons.premis obtain x u where
  rec:
    Some (x, u) ∈
      set-dist (execute
        (protocol-merkle.check-authentication-path
          (len div 2) (i - len div 2) (MerkleLeaf v) path) s)
  and hash-step:
    Some (h, t) ∈ set-dist (execute (hash (MerkleNode a x)) u)
  using False by (auto elim!: set-dist-bindE)
have rec-pres:
  fmlookup (HashMap u) y = fmlookup (HashMap s) y
  by (rule Cons.IH[OF rec])
have hash-pres:
  fmlookup (HashMap t) y = fmlookup (HashMap u) y
  by (rule protocol-hash-preserves-other-lookup[OF hash-step])
    (rule not-node)
show ?thesis
  using hash-pres rec-pres by simp
qed
qed

```

lemma check-authentication-path-preserves-challenge-lookups:

assumes outcome: Some (h, t) ∈
 set-dist (execute (check-authentication-path len i v path) s)

shows

fmlookup (HashMap t) (AlphaChallenge j x) =
 fmlookup (HashMap s) (AlphaChallenge j x)
 fmlookup (HashMap t) (QueryIndexChallenge j x) =
 fmlookup (HashMap s) (QueryIndexChallenge j x)
 fmlookup (HashMap t) (TraceFriChallenge j x) =
 fmlookup (HashMap s) (TraceFriChallenge j x)
 fmlookup (HashMap t) (CompositionFriChallenge j x) =
 fmlookup (HashMap s) (CompositionFriChallenge j x)

proof –

show fmlookup (HashMap t) (AlphaChallenge j x) =
 fmlookup (HashMap s) (AlphaChallenge j x)
by (rule check-authentication-path-preserves-non-merkle-lookup[
 OF outcome]) simp-all


```

show fmlookup (HashMap t) (QueryIndexChallenge j x) =
  fmlookup (HashMap s) (QueryIndexChallenge j x)
by (rule check-authentication-path-preserves-non-merkle-lookup[
  OF outcome]) simp-all
show fmlookup (HashMap t) (TraceFriChallenge j x) =
  fmlookup (HashMap s) (TraceFriChallenge j x)
by (rule check-authentication-path-preserves-non-merkle-lookup[
  OF outcome]) simp-all
show fmlookup (HashMap t) (CompositionFriChallenge j x) =
  fmlookup (HashMap s) (CompositionFriChallenge j x)
by (rule check-authentication-path-preserves-non-merkle-lookup[
  OF outcome]) simp-all
qed

lemma honest-query-decommitment-no-failure:
assumes created: created-tree xs tree s0
and len-pow: length xs = 2 ^ n
and idx: i < length xs
and leaf: v = xs ! i
and ext: s0 ≤ s
and path-len: length (get-authentication-path (length xs) i tree) = floor-log (length xs)
and root: fr = value tree
shows
  None ∉ dom (dist (execute
    (do {
      qh ← read;
      qh-path ← ntimes read (floor-log (length xs));
      ap ← check-authentication-path (length xs) i qh qh-path;
      assert (ap = fr);
      return qh
    }) (s(PTranscript := v # get-authentication-path (length xs) i tree @ rest))))
proof -
let ?path = get-authentication-path (length xs) i tree
let ?s = s(PTranscript := v # ?path @ rest)
let ?prefix =
  do {
    qh ← read;
    qh-path ← ntimes read (floor-log (length xs));
    ap ← check-authentication-path (length xs) i qh qh-path;
    return (qh, qh-path, ap)
  }
have nf-read: None ∉ dom (dist (execute read ?s))
by (rule read-no-failure) simp
have nf-after-read:
  ∧ qh s1. Some (qh, s1) ∈ set-dist (execute read ?s) ⇒
    None ∉ dom
    (dist (execute
      (ntimes read (floor-log (length xs))) ≫=
      (λqh-path. check-authentication-path (length xs) i qh qh-path ≫=

```

```

      (λap. return (qh, qh-path, ap)))) s1))
proof -
  fix qh s1
  assume read-gh: Some (qh, s1) ∈ set-dist (execute read ?s)
  have read-gh-res:
    s1 = s[PState := concat (PState s) v, PTranscript := ?path @ rest]
    using read-cons-outcome[OF read-gh] by simp
  have enough-path: floor-log (length xs) ≤ length (PTranscript s1)
    using path-len unfolding read-gh-res by simp
  have nf-reads:
    None ∉ dom (dist (execute (ntimes read (floor-log (length xs))) s1))
    by (rule ntimes-read-no-failure[OF enough-path])
  have nf-after-reads:
    ∧ qh-path s2. Some (qh-path, s2) ∈
      set-dist (execute (ntimes read (floor-log (length xs))) s1) ⇒
      None ∉ dom
        (dist (execute
          (check-authentication-path (length xs) i qh qh-path ≫=
            (λap. return (qh, qh-path, ap))) s2))
proof -
  fix qh-path s2
  assume Some (qh-path, s2) ∈
    set-dist (execute (ntimes read (floor-log (length xs))) s1)
  have nf-check:
    None ∉ dom
      (dist (execute (check-authentication-path (length xs) i qh qh-path) s2))
    by (rule check-authentication-path-no-failure)
  show None ∉ dom
    (dist (execute
      (check-authentication-path (length xs) i qh qh-path ≫=
        (λap. return (qh, qh-path, ap))) s2))
    by (rule no-failure-bindI[OF nf-check]) simp
qed
show None ∉ dom
  (dist (execute
    (ntimes read (floor-log (length xs)) ≫=
      (λqh-path. check-authentication-path (length xs) i qh qh-path ≫=
        (λap. return (qh, qh-path, ap)))) s1))
  by (rule no-failure-bindI[OF nf-reads nf-after-reads])
qed
have prefix-nf:
  None ∉ dom (dist (execute ?prefix ?s))
  by (rule no-failure-bindI[OF nf-read nf-after-read])
have prefix-out:
  ∧ res t. Some (res, t) ∈ set-dist (execute ?prefix ?s) ⇒
    case res of (qh, qh-path, ap) ⇒ ap = fr
proof -
  fix res t
  assume out: Some (res, t) ∈ set-dist (execute ?prefix ?s)

```

```

obtain qh qh-path ap where res-eq: res = (qh, qh-path, ap)
  by (cases res) auto
from out[unfolded res-eq] obtain s1 s2 where
  read-qh: Some (qh, s1) ∈ set-dist (execute read ?s)
  and reads:
    Some (qh-path, s2) ∈
      set-dist (execute (ntimes read (floor-log (length xs))) s1)
  and check:
    Some (ap, t) ∈
      set-dist (execute (check-authentication-path (length xs) i qh qh-path) s2)
  by (auto elim!: set-dist-bindE)
have qh-eq: qh = v
  using read-cons-outcome[OF read-qh] by simp
have s1-eq:
  s1 = s(PState := concat (PState s) v, PTranscript := ?path @ rest)
  using read-cons-outcome[OF read-qh] by simp
have reads-out:
  qh-path = ?path
  using ntimes-read-prefix-outcome[OF reads[unfolded path-len[symmetric] s1-eq]]
  by simp
have s-start: s ≤ ?s
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have start-s1: ?s ≤ s1
  by (rule read-hash-extends[OF read-qh])
have s1-s2: s1 ≤ s2
  by (rule ntimes-read-hash-extends[OF reads])
have s0-s2: s0 ≤ s2
  using ext s-start start-s1 s1-s2 by (meson hash-ext-trans)
have ap = value tree
  using check-created-tree-outcome
    [OF created len-pow idx leaf s0-s2 check[unfolded qh-eq reads-out]]
  by simp
then show case res of (qh, qh-path, ap) ⇒ ap = fr
  unfolding res-eq root by simp
qed
have full:
  do {
    qh ← read;
    qh-path ← ntimes read (floor-log (length xs));
    ap ← check-authentication-path (length xs) i qh qh-path;
    assert (ap = fr);
    return qh
  } =
  (?prefix ≫ (λ(qh, qh-path, ap). assert (ap = fr) ≫ (λ-. return qh)))
  by (simp add: sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full
proof (rule no-failure-bindI[OF prefix-nf])
fix res t

```

```

assume out: Some (res, t) ∈ set-dist (execute ?prefix ?s)
obtain qh qh-path ap where res-eq: res = (qh, qh-path, ap)
  by (cases res) auto
have ap-eq: ap = fr
  using prefix-out[OF out] unfolding res-eq by simp
show None ∉ dom
  (dist (execute
    ((case res of (qh, qh-path, ap) ⇒
      assert (ap = fr) >>= (λ-. return qh))) t))
  unfolding res-eq ap-eq assert-def by simp
qed
qed

end

end

```

```

theory Channel-Transcript
  imports Channel-Core
begin

```

```

locale channel = merkle-tree +
  constrains concat :: 'a::finite ⇒ 'a ⇒ 'a
begin

```

```

definition send:: 'a ⇒ ('a, unit, 'b) c-monad
where
  send x ≡
    modify (λs. s⌊State:=concat (State s) x, Transcript:=x # (Transcript s)⌋)

```

```

definition send2:: 'a ⇒ ('a, unit, 'b) c-monad
where
  send2 x ≡
    modify (λs. s⌊Transcript:=x # (Transcript s)⌋)

```

The idea is that read recovers the state of write

```

definition read :: ('a, 'a, 'b) c-monad
where
  read ≡
    do {
      s ← get;
      assert (Transcript s ≠ []);
      modify (λs. s⌊State:=concat (State s) (hd (Transcript s)), Transcript:=tl (Transcript s)⌋);
      return (hd (Transcript s))
    }

```

```

lemma send-outcome:
  assumes Some ((), t) ∈ set-dist (execute (send x) s)

```

shows $t = s \langle \text{State} := \text{concat} (\text{State } s) x, \text{Transcript} := x \# \text{Transcript } s \rangle$
using *assms*
unfolding *send-def modify-def*
by (*auto elim! set-dist-bindE*)

lemma *read-cons-outcome:*

assumes *Some* $(y, t) \in \text{set-dist} (\text{execute read } (s \langle \text{Transcript} := x \# xs \rangle))$
shows
 $y = x \wedge$
 $t = s \langle \text{State} := \text{concat} (\text{State } s) x, \text{Transcript} := xs \rangle$
using *assms*
unfolding *read-def modify-def assert-def*
by (*auto elim! set-dist-bindE*)

lemma *read-nonempty-outcome:*

assumes *out: Some* $(y, t) \in \text{set-dist} (\text{execute read } s)$
and *tr: Transcript* $s = x \# xs$
shows
 $y = x \wedge$
 $t = s \langle \text{State} := \text{concat} (\text{State } s) x, \text{Transcript} := xs \rangle$

proof –

have *s-eq:* $s \langle \text{Transcript} := x \# xs \rangle = s$
using *tr* **by** (*cases s simp*)
have *out': Some* $(y, t) \in$
 $\text{set-dist} (\text{execute read } (s \langle \text{Transcript} := x \# xs \rangle))$
using *out unfolding s-eq* .
show *?thesis*
using *read-cons-outcome[OF out'] tr* **by** *simp*

qed

lemma *read-no-failure:*

assumes *Transcript* $s \neq []$
shows *None* $\notin \text{dom} (\text{dist} (\text{execute read } s))$
using *assms*
unfolding *read-def assert-def*
by (*intro no-failure-bindI simp-all*)

lemma *hash-no-failure:*

$\text{None} \notin \text{dom} (\text{dist} (\text{execute} (\text{hash } x) s))$
unfolding *hash-def apply-hash-def modify-HashMap-def*
by (*intro no-failure-bindI simp-all*)

lemma *receive-random-field-element-no-failure:*

$\text{None} \notin \text{dom} (\text{dist} (\text{execute receive-random-field-element } s))$
unfolding *receive-random-field-element-def*
by (*intro no-failure-bindI (simp-all add: hash-no-failure)*)

lemma *receive-random-field-element-extends:*

assumes *Some* $(r, t) \in \text{set-dist} (\text{execute receive-random-field-element } s)$

shows $s \leq t$
proof –
 from *assms* **obtain** $s0$ **where**
 get-s: $\text{Some } (s0, s0) \in \text{set-dist } (\text{execute } \text{get } s)$
 and *rest*:
 $\text{Some } (r, t) \in$
 $\text{set-dist } (\text{execute } (\text{hash } (\text{State } s0) \ggg \text{return}) s0)$
 unfolding *receive-random-field-element-def*
 by (*auto elim!*: *set-dist-bindE*)
 have $s0$: $s0 = s$
 using *get-s* **by** *simp*
 have *hash-out*:
 $\text{Some } (r, t) \in \text{set-dist } (\text{execute } (\text{hash } (\text{State } s)) s)$
 using *rest* **unfolding** $s0$ **by** *simp*
 show *?thesis*
 using *hash-outcome*(1)[*OF hash-out*] .
qed

lemma *hash-channel-preserves*:
 assumes $\text{Some } (h, t) \in \text{set-dist } (\text{execute } (\text{hash } x) s)$
 shows $\text{State } t = \text{State } s$
 and $\text{Transcript } t = \text{Transcript } s$
proof –
 from *assms* **obtain** a $s1$ **where**
 a : $\text{Some } (a, s1) \in \text{set-dist } (\text{execute } (\text{apply-hash } x) s)$
 and *rest*: $\text{Some } (h, t) \in$
 $\text{set-dist } (\text{execute } (\text{modify-HashMap } x a \ggg (\lambda-. \text{return } a)) s1)$
proof –
 have $\text{Some } (h, t) \in$
 $\text{set-dist } (\text{execute } (\text{apply-hash } x \ggg (\lambda a. \text{modify-HashMap } x a \ggg (\lambda-. \text{return } a)))) s)$
 using *assms* **unfolding** *hash-def* .
 then show *?thesis*
 by (*rule set-dist-bindE*) (*rule that*)
qed
 have *a-img*: $\text{Some } (a, s1) \in (\lambda a. \text{Some } (a, s)) \text{ ' set-dist } (\text{hash-dist } x s)$
 using a
 unfolding *apply-hash-def lift.rep-eq lift-dist-def*
 by (*simp add*: *set-dist-dist-map*)
 then have $s1$: $s1 = s$
 by *auto*
 from *rest* **obtain** u $s2$ **where**
 mod : $\text{Some } (u, s2) \in \text{set-dist } (\text{execute } (\text{modify-HashMap } x a) s1)$
 and ret : $\text{Some } (h, t) \in \text{set-dist } (\text{execute } (\text{return } a) s2)$
 by (*elim set-dist-bindE*)
 from mod **have** $s2$: $s2 = s1 \langle \text{HashMap} := \text{fmupd } x a (\text{HashMap } s1) \rangle$
 unfolding *modify-HashMap-def modify-def*
 by (*auto elim!*: *set-dist-bindE*)
 from ret **have** t : $t = s2$

```

    by auto
  show State t = State s
    unfolding t s2 s1 by simp
  show Transcript t = Transcript s
    unfolding t s2 s1 by simp
qed

lemma receive-random-field-element-known-outcome:
  assumes lookup: fmlookup (HashMap s) (State s) = Some r
    and outcome: Some (r', t) ∈ set-dist (execute receive-random-field-element s)
  shows
    r' = r ∧ State t = State s ∧ Transcript t = Transcript s ∧ s ≤ t
proof -
  from outcome obtain s0 where
    get-s: Some (s0, s0) ∈ set-dist (execute get s)
  and rest:
    Some (r', t) ∈
      set-dist (execute (hash (State s0) >> return) s0)
  unfolding receive-random-field-element-def
  by (auto elim!: set-dist-bindE)
  have s0: s0 = s
    using get-s by simp
  have hash-out:
    Some (r', t) ∈ set-dist (execute (hash (State s)) s)
    using rest unfolding s0 by simp
  have ext: s ≤ t
    using hash-outcome(1)[OF hash-out] .
  have lookup-r': fmlookup (HashMap t) (State s) = Some r'
    using hash-outcome(2)[OF hash-out] .
  have lookup-r: fmlookup (HashMap t) (State s) = Some r
    using hash-extension-lookup[OF lookup ext] .
  moreover have State t = State s ∧ Transcript t = Transcript s
    using hash-channel-preserves[OF hash-out] by simp
  ultimately show ?thesis
    using ext lookup-r' by auto
qed

lemma random-then-read-cons-outcome:
  assumes lookup: fmlookup (HashMap s) (State s) = Some x
    and outcome:
      Some ((r, y), t) ∈
        set-dist (execute
          (do {
            r ← receive-random-field-element;
            y ← read;
            return (r, y)
          }) (s⟦Transcript := x # xs⟧)))
  shows
    r = x ∧

```

```

    y = x ∧
    State t = concat (State s) x ∧
    Transcript t = xs
proof –
  let ?s = s(Transcript := x # xs)
  have outcome':
    Some ((r, y), t) ∈
      set-dist (execute
        (receive-random-field-element >>=
          (λr. read >>= (λy. return (r, y))))) ?s)
  using outcome by simp
  from outcome' obtain r-state where
    rand:
      Some (r, r-state) ∈ set-dist (execute receive-random-field-element ?s)
  and read-y:
    Some (y, t) ∈ set-dist (execute read r-state)
  by (auto elim!: set-dist-bindE)
  have lookup': fmlookup (HashMap ?s) (State ?s) = Some x
  using lookup by simp
  have rand-res:
    r = x ∧ State r-state = State s ∧ Transcript r-state = x # xs
  using receive-random-field-element-known-outcome[OF lookup' rand] by simp
  have r-state-update: r-state(Transcript := x # xs) = r-state
  using rand-res by simp
  have read-y':
    Some (y, t) ∈ set-dist (execute read (r-state(Transcript := x # xs)))
  using read-y unfolding r-state-update .
  have read-res:
    y = x ∧ State t = concat (State s) x ∧ Transcript t = xs
  using read-cons-outcome[OF read-y'] rand-res by simp
  show ?thesis
  using rand-res read-res by simp
qed

```

```

lemma random-then-read-assert-no-failure:
  assumes lookup: fmlookup (HashMap s) (State s) = Some x
  shows
    None ∉ dom (dist (execute
      (do {
        r ← receive-random-field-element;
        y ← read;
        assert (r = y);
        return y
      }) (s(Transcript := x # xs))))
proof –
  let ?s = s(Transcript := x # xs)
  let ?prefix =
    do {
      r ← receive-random-field-element;

```



```

    y ← read;
    return (r, y)
  }
have prefix-no-failure:
  None ∉ dom (dist (execute ?prefix ?s))
proof (rule no-failure-bindI)
  show None ∉ dom (dist (execute receive-random-field-element ?s))
    by (rule receive-random-field-element-no-failure)
  fix r t
  assume rand: Some (r, t) ∈ set-dist (execute receive-random-field-element ?s)
  have lookup-s: fmllookup (HashMap ?s) (State ?s) = Some x
    using lookup by simp
  have tr-t: Transcript t = x # xs
    using receive-random-field-element-known-outcome[OF lookup-s rand] by simp
  show None ∉ dom (dist (execute (read ≫ (λy. return (r, y))) t))
  proof (rule no-failure-bindI)
    show None ∉ dom (dist (execute read t))
      by (rule read-no-failure) (simp add: tr-t)
    fix y u
    assume Some (y, u) ∈ set-dist (execute read t)
    show None ∉ dom (dist (execute (return (r, y)) u))
      by simp
    qed
  qed
have full:
  do {
    r ← receive-random-field-element;
    y ← read;
    assert (r = y);
    return y
  } =
  (?prefix ≫ (λ(r, y). assert (r = y) ≫ (λ-. return y)))
  by (simp add: sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full
proof (rule no-failure-bindI[OF prefix-no-failure])
  fix ry t
  assume prefix:
    Some (ry, t) ∈ set-dist (execute ?prefix ?s)
  obtain r y where ry: ry = (r, y)
    by (cases ry)
  have replay: r = y
    using random-then-read-cons-outcome[OF lookup prefix[unfolded ry]] by simp
  show None ∉ dom
    (dist (execute ((case ry of (r, y) ⇒ assert (r = y) ≫ (λ-. return y))) t))
    unfolding ry replay assert-def by simp
  qed
qed

```

lemma *read-then-random-cons-outcome*:

assumes *lookup*:

fmlookup (*HashMap s*) (*concat (State s) x*) = *Some r*

and *outcome*:

Some ((y, r'), t) ∈
set-dist (execute
(do {
y ← read;
r ← receive-random-field-element;
return (y, r)
}) (s⟦Transcript := x # xs⟧))

shows

y = x ∧
r' = r ∧
State t = concat (State s) x ∧
Transcript t = xs

proof –

let *?s = s⟦Transcript := x # xs⟧*

have *outcome'*:

Some ((y, r'), t) ∈
set-dist (execute
(read ≫=
(λy. receive-random-field-element ≫= (λr. return (y, r)))) ?s)

using *outcome* **by** *simp*

from *outcome'* **obtain** *read-state* **where**

read-y: Some (y, read-state) ∈ set-dist (execute read ?s)

and *rand*:

Some (r', t) ∈ set-dist (execute receive-random-field-element read-state)

by (*auto elim! set-dist-bindE*)

have *read-res*:

y = x ∧
read-state = s⟦State := concat (State s) x, Transcript := xs⟧

using *read-cons-outcome*[*OF read-y*] **by** *simp*

have *rand-res*:

r' = r ∧
State t = concat (State s) x ∧
Transcript t = xs

using *receive-random-field-element-known-outcome*[*OF - rand*] *lookup read-res* **by** *simp*

show *?thesis*

using *read-res rand-res* **by** *simp*

qed

lemma *mfold2-send-outcome*:

assumes *Some ((), t) ∈ set-dist (execute (mfold2 send xs) s)*

shows

t =
s⟦
State := foldl concat (State s) xs,
Transcript := rev xs @ Transcript s

```

     $\rangle$ 
    using assms
proof (induction xs arbitrary: s t)
    case Nil
    then show ?case by simp
next
    case (Cons x xs)
    from Cons.prems obtain s1 where
      send-x: Some ( $\langle \rangle$ , s1)  $\in$  set-dist (execute (send x) s)
      and rest: Some ( $\langle \rangle$ , t)  $\in$  set-dist (execute (mfold2 send xs) s1)
      by (auto elim!: set-dist-bindE)
    have s1:
      s1 = s(State := concat (State s) x, Transcript := x # Transcript s)
      using send-outcome[OF send-x] .
    show ?case
      using Cons.IH[OF rest] unfolding s1 by simp
qed

lemma ntimes-read-outcome:
  assumes Some (ys, t)  $\in$  set-dist (execute (ntimes read (length xs)) (s(Transcript := xs)))
  shows
    ys = xs  $\wedge$ 
    t = s(State := foldl concat (State s) xs, Transcript :=  $\langle \rangle$ )
  using assms
proof (induction xs arbitrary: s ys t)
  case Nil
  then show ?case by simp
next
  case (Cons x xs)
  have prems':
    Some (ys, t)  $\in$ 
      set-dist (execute
        (read  $\gg$  ( $\lambda y$ . ntimes read (length xs)  $\gg$  ( $\lambda ys$ . return (y # ys))))
        (s(Transcript := x # xs)))
    using Cons.prems by simp
  from prems' obtain y s1 where
    read-x: Some (y, s1)  $\in$  set-dist (execute read (s(Transcript := x # xs)))
    and rest1:
      Some (ys, t)  $\in$ 
        set-dist (execute (ntimes read (length xs)  $\gg$  ( $\lambda ys$ . return (y # ys))) s1)
      by (rule set-dist-bindE) blast
  from rest1 obtain ys' s2 where
    reads: Some (ys', s2)  $\in$  set-dist (execute (ntimes read (length xs)) s1)
    and ret: Some (ys, t)  $\in$  set-dist (execute (return (y # ys')) s2)
    by (auto elim!: set-dist-bindE)
  have read-res:
    y = x  $\wedge$  s1 = s(State := concat (State s) x, Transcript := xs)
    using read-cons-outcome[OF read-x] .
  let ?s0 = s(State := concat (State s) x)

```

```

have  $s1$ :  $s1 = ?s0 \langle \text{Transcript} := xs \rangle$ 
  using read-res by simp
have tail:
   $ys' = xs \wedge$ 
   $s2 = ?s0 \langle \text{State} := \text{foldl concat } (\text{State } ?s0) \ xs, \ \text{Transcript} := [] \rangle$ 
  using Cons.IH[OF reads[unfolded  $s1$ ]] .
have  $ys-t$ :  $ys = y \# ys' \wedge t = s2$ 
  using ret by simp
show ?case
  using read-res tail ys-t by simp
qed

lemma ntimes-read-prefix-outcome:
assumes Some ( $ys, t$ )  $\in$ 
  set-dist (execute (ntimes read (length xs)) ( $s \langle \text{Transcript} := xs @ \text{rest} \rangle$ ))
shows
   $ys = xs \wedge$ 
   $t = s \langle \text{State} := \text{foldl concat } (\text{State } s) \ xs, \ \text{Transcript} := \text{rest} \rangle$ 
using assms
proof (induction xs arbitrary: s ys t)
case Nil
  then show ?case by simp
next
case (Cons x xs)
have prems':
  Some ( $ys, t$ )  $\in$ 
  set-dist (execute
    (read  $\gg (\lambda y. \text{ntimes read } (\text{length } xs) \gg (\lambda ys. \text{return } (y \# ys)))$ )
    ( $s \langle \text{Transcript} := x \# xs @ \text{rest} \rangle$ ))
  using Cons.prems by simp
from prems' obtain  $y \ s1$  where
  read-x: Some ( $y, s1$ )  $\in$  set-dist (execute read ( $s \langle \text{Transcript} := x \# xs @ \text{rest} \rangle$ ))
  and rest1:
    Some ( $ys, t$ )  $\in$ 
    set-dist (execute (ntimes read (length xs)  $\gg (\lambda ys. \text{return } (y \# ys))$ )  $s1$ )
  by (rule set-dist-bindE) blast
from rest1 obtain  $ys' \ s2$  where
  reads: Some ( $ys', s2$ )  $\in$  set-dist (execute (ntimes read (length xs))  $s1$ )
  and ret: Some ( $ys, t$ )  $\in$  set-dist (execute (return ( $y \# ys'$ ))  $s2$ )
  by (auto elim!: set-dist-bindE)
have read-res:
   $y = x \wedge s1 = s \langle \text{State} := \text{concat } (\text{State } s) \ x, \ \text{Transcript} := xs @ \text{rest} \rangle$ 
  using read-cons-outcome[OF read-x] .
let  $?s0 = s \langle \text{State} := \text{concat } (\text{State } s) \ x \rangle$ 
have  $s1$ :  $s1 = ?s0 \langle \text{Transcript} := xs @ \text{rest} \rangle$ 
  using read-res by simp
have tail:
   $ys' = xs \wedge$ 
   $s2 = ?s0 \langle \text{State} := \text{foldl concat } (\text{State } ?s0) \ xs, \ \text{Transcript} := \text{rest} \rangle$ 

```

```

    using Cons.IH[OF reads[unfolded s1]] .
  have ys-t: ys = y # ys'  $\wedge$  t = s2
    using ret by simp
  show ?case
    using read-res tail ys-t by simp
qed

theorem honest-transcript-replay-mfold2-ntimes:
  assumes init: Transcript s = []
    and sent: Some ((), sent)  $\in$  set-dist (execute (mfold2 send xs) s)
    and replay:
      Some (ys, replayed)  $\in$ 
        set-dist (execute
          (ntimes read (length xs))
          (sent(State := State s, Transcript := rev (Transcript sent))))
  shows ys = xs  $\wedge$  replayed = sent(Transcript := [])
proof -
  have sent-eq:
    sent =
      s(
        State := foldl concat (State s) xs,
        Transcript := rev xs
      )
  using mfold2-send-outcome[OF sent] init by simp
  let ?replay-start = sent(State := State s)
  have start-eq:
    sent(State := State s, Transcript := rev (Transcript sent)) =
      ?replay-start(Transcript := xs)
  using sent-eq by simp
  have replay-out:
    ys = xs  $\wedge$ 
    replayed =
      ?replay-start(
        State := foldl concat (State ?replay-start) xs,
        Transcript := []
      )
  using ntimes-read-outcome[OF replay[unfolded start-eq]] .
  then show ?thesis
    using sent-eq by simp
qed

lemma ntimes-read-no-failure:
  assumes n  $\leq$  length (Transcript s)
  shows None  $\notin$  dom (dist (execute (ntimes read n) s))
  using assms
proof (induction n arbitrary: s)
  case 0
  then show ?case by simp
next

```

```

case (Suc n)
have nonempty: Transcript s ≠ []
  using Suc.prem by (cases Transcript s) auto
have nf-read: None ∉ dom (dist (execute read s))
  by (rule read-no-failure[OF nonempty])
have nf-cont:
   $\bigwedge x t. \text{Some } (x, t) \in \text{set-dist } (\text{execute read } s) \implies$ 
    None ∉ dom
    (dist (execute
      (ntimes read n  $\gg$  ( $\lambda xs. \text{return } (x \# xs)$ )) t))
proof -
  fix x t
  assume out: Some (x, t) ∈ set-dist (execute read s)
  obtain y ys where tr: Transcript s = y # ys
  using nonempty by (cases Transcript s) auto
  have t-eq: t = s[State := concat (State s) y, Transcript := ys]
  using read-nonempty-outcome[OF out tr] by simp
  have n ≤ length ys
  using Suc.prem tr by simp
  then have nf-tail: None ∉ dom (dist (execute (ntimes read n) t))
  using Suc.IH[of t] unfolding t-eq by simp
  show None ∉ dom
    (dist (execute
      (ntimes read n  $\gg$  ( $\lambda xs. \text{return } (x \# xs)$ )) t))
  by (rule no-failure-bindI[OF nf-tail]) simp
qed
show ?case
  by (simp add: no-failure-bindI[OF nf-read nf-cont])
qed

lemma check-authentication-path-no-failure:
  None ∉ dom (dist (execute (check-authentication-path len i v path) s))
proof (induction path arbitrary: len i v s)
  case Nil
  then show ?case
    by (simp add: hash-no-failure)
next
  case (Cons a path)
  show ?case
  proof (cases i < len div 2)
    case True
    then show ?thesis
      by (simp add: no-failure-bindI hash-no-failure Cons.IH)
    next
    case False
    then show ?thesis
      by (simp add: no-failure-bindI hash-no-failure Cons.IH)
  qed
qed

```

lemma *check-authentication-path-preserves-channel*:

assumes $\text{Some } (h, t) \in \text{set-dist } (\text{execute } (\text{check-authentication-path } \text{len } i \text{ } v \text{ path}) \text{ } s)$

shows $\text{State } t = \text{State } s$
and $\text{Transcript } t = \text{Transcript } s$

proof –

have *both*:

$\text{State } t = \text{State } s \wedge \text{Transcript } t = \text{Transcript } s$

if *outcome*:

$\text{Some } (h, t) \in \text{set-dist } (\text{execute } (\text{check-authentication-path } \text{len } i \text{ } v \text{ path}) \text{ } s)$

for $\text{len } i \text{ } v \text{ } s \text{ } h \text{ } t$

using *outcome*

proof (*induction path arbitrary: len i v s h t*)

case *Nil*

have *hash-step*: $\text{Some } (h, t) \in \text{set-dist } (\text{execute } (\text{hash } v) \text{ } s)$

using *Nil.prem*s **by** *simp*

show *?case*

using *hash-channel-preserves(1)[OF hash-step]*
hash-channel-preserves(2)[OF hash-step]

by *simp*

next

case (*Cons a path*)

show *?case*

proof (*cases i < len div 2*)

case *True*

from *Cons.prem*s **obtain** $x \text{ } s1$ **where**

rest:

$\text{Some } (x, s1) \in$
 $\text{set-dist } (\text{execute } (\text{check-authentication-path } (\text{len div } 2) \text{ } i \text{ } v \text{ path}) \text{ } s)$

and *hash-step*: $\text{Some } (h, t) \in \text{set-dist } (\text{execute } (\text{hash } (\text{concat } x \text{ } a)) \text{ } s1)$

using *True* **by** (*auto elim!:: set-dist-bindE*)

have *rec*: $\text{State } s1 = \text{State } s \wedge \text{Transcript } s1 = \text{Transcript } s$

using *Cons.IH*[*OF rest*] .

have *hash-pres*: $\text{State } t = \text{State } s1 \wedge \text{Transcript } t = \text{Transcript } s1$

using *hash-channel-preserves(1)[OF hash-step]*
hash-channel-preserves(2)[OF hash-step]

by *simp*

show *?thesis*

using *rec hash-pres* **by** *simp*

next

case *False*

from *Cons.prem*s **obtain** $x \text{ } s1$ **where**

rest:

$\text{Some } (x, s1) \in$
 $\text{set-dist } (\text{execute } (\text{check-authentication-path } (\text{len div } 2) \text{ } (i - \text{len div } 2) \text{ } v \text{ path}) \text{ } s)$

and *hash-step*: $\text{Some } (h, t) \in \text{set-dist } (\text{execute } (\text{hash } (\text{concat } a \text{ } x)) \text{ } s1)$

using *False* **by** (*auto elim!:: set-dist-bindE*)

have *rec*: $\text{State } s1 = \text{State } s \wedge \text{Transcript } s1 = \text{Transcript } s$

using *Cons.IH*[*OF rest*] .

```

have hash-pres: State  $t = \text{State } s1 \wedge \text{Transcript } t = \text{Transcript } s1$ 
  using hash-channel-preserves(1)[OF hash-step]
    hash-channel-preserves(2)[OF hash-step]
  by simp
show ?thesis
  using rec hash-pres by simp
qed
qed
show State  $t = \text{State } s$ 
  using both[OF assms] by simp
show Transcript  $t = \text{Transcript } s$ 
  using both[OF assms] by simp
qed

lemma honest-query-decommitment-no-failure:
  assumes created: created-tree  $xs$  tree  $s0$ 
    and len-pow: length  $xs = 2^n$ 
    and idx:  $i < \text{length } xs$ 
    and leaf:  $v = xs[i]$ 
    and ext:  $s0 \leq s$ 
    and path-len: length (get-authentication-path (length  $xs$ )  $i$  tree) = floor-log (length  $xs$ )
    and root:  $fr = \text{value tree}$ 
  shows
    None  $\notin \text{dom (dist (execute$ 
      (do {
         $qh \leftarrow \text{read};$ 
         $qh\text{-path} \leftarrow \text{ntimes read (floor-log (length } xs));$ 
         $ap \leftarrow \text{check-authentication-path (length } xs) i qh qh\text{-path};$ 
         $\text{assert (} ap = fr \text{);}$ 
         $\text{return } qh$ 
      }) (s[Transcript :=  $v \# \text{get-authentication-path (length } xs) i \text{ tree @ rest}]$ )))
  proof -
    let ?path = get-authentication-path (length  $xs$ )  $i$  tree
    let ?s = s[Transcript :=  $v \# ?path @ \text{rest}$ ]
    let ?prefix =
      do {
         $qh \leftarrow \text{read};$ 
         $qh\text{-path} \leftarrow \text{ntimes read (floor-log (length } xs));$ 
         $ap \leftarrow \text{check-authentication-path (length } xs) i qh qh\text{-path};$ 
         $\text{return (} qh, qh\text{-path}, ap \text{)}$ 
      }
    have nf-read: None  $\notin \text{dom (dist (execute read ?s))}$ 
    by (rule read-no-failure) simp
    have nf-after-read:
       $\wedge qh s1. \text{Some (} qh, s1 \text{)} \in \text{set-dist (execute read ?s)} \implies$ 
      None  $\notin \text{dom}$ 
      (dist (execute
        (ntimes read (floor-log (length  $xs$ )))  $\gg$ 
        ( $\lambda qh\text{-path. check-authentication-path (length } xs) i qh qh\text{-path} \gg$ 

```



```

      (λap. return (qh, qh-path, ap)))) s1))
proof -
  fix qh s1
  assume read-gh: Some (qh, s1) ∈ set-dist (execute read ?s)
  have read-gh-res:
    s1 = s[State := concat (State s) v, Transcript := ?path @ rest]
    using read-cons-outcome[OF read-gh] by simp
  have enough-path: floor-log (length xs) ≤ length (Transcript s1)
    using path-len unfolding read-gh-res by simp
  have nf-reads:
    None ∉ dom (dist (execute (ntimes read (floor-log (length xs))) s1))
    by (rule ntimes-read-no-failure[OF enough-path])
  have nf-after-reads:
    ∧ qh-path s2. Some (qh-path, s2) ∈
      set-dist (execute (ntimes read (floor-log (length xs))) s1) ⇒
      None ∉ dom
        (dist (execute
          (check-authentication-path (length xs) i qh qh-path ≫=
            (λap. return (qh, qh-path, ap))) s2))
proof -
  fix qh-path s2
  assume Some (qh-path, s2) ∈
    set-dist (execute (ntimes read (floor-log (length xs))) s1)
  have nf-check:
    None ∉ dom
      (dist (execute (check-authentication-path (length xs) i qh qh-path) s2))
    by (rule check-authentication-path-no-failure)
  show None ∉ dom
    (dist (execute
      (check-authentication-path (length xs) i qh qh-path ≫=
        (λap. return (qh, qh-path, ap))) s2))
  proof (rule no-failure-bindI[OF nf-check])
    fix ap t
    assume Some (ap, t) ∈
      set-dist (execute (check-authentication-path (length xs) i qh qh-path) s2)
    show None ∉ dom (dist (execute (return (qh, qh-path, ap)) t))
      by simp
  qed
qed
qed
show None ∉ dom
  (dist (execute
    (ntimes read (floor-log (length xs)) ≫=
      (λqh-path. check-authentication-path (length xs) i qh qh-path ≫=
        (λap. return (qh, qh-path, ap)))) s1))
  using no-failure-bindI[OF nf-reads nf-after-reads] .
qed
have prefix-no-failure: None ∉ dom (dist (execute ?prefix ?s))
  using no-failure-bindI[OF nf-read nf-after-read] .
have full:

```

```

do {
   $qh \leftarrow \text{read};$ 
   $qh\text{-path} \leftarrow \text{ntimes read (floor-log (length xs))};$ 
   $ap \leftarrow \text{check-authentication-path (length xs) } i \text{ } qh \text{ } qh\text{-path};$ 
   $\text{assert } (ap = fr);$ 
   $\text{return } qh$ 
} =
( $?prefix \gg= (\lambda(qh, qh\text{-path}, ap). \text{assert } (ap = fr) \gg= (\lambda-. \text{return } qh))$ )
by (simp add: sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full
proof (rule no-failure-bindI[OF prefix-no-failure])
  fix out t
  assume out:
    Some (out, t)  $\in$  set-dist (execute ?prefix ?s)
  obtain qh qh-path ap where out-eq: out = (qh, qh-path, ap)
  by (cases out) auto
  from out[unfolded out-eq] obtain s1 s2 where
    read-leaf: Some (qh, s1)  $\in$  set-dist (execute read ?s)
  and read-path:
    Some (qh-path, s2)  $\in$ 
      set-dist (execute (ntimes read (floor-log (length xs))) s1)
  and check:
    Some (ap, t)  $\in$ 
      set-dist (execute
        (check-authentication-path (length xs) i qh qh-path) s2)
  by (auto elim!: set-dist-bindE)
  have read-leaf-res:
     $qh = v \wedge$ 
     $s1 = s \parallel \text{State} := \text{concat (State } s) \text{ } v, \text{ Transcript} := ?path @ \text{rest})$ 
  using read-cons-outcome[OF read-leaf] by simp
  let ?s-leaf =  $s \parallel \text{State} := \text{concat (State } s) \text{ } v$ 
  have s1-eq:  $s1 = ?s\text{-leaf} \parallel \text{Transcript} := ?path @ \text{rest}$ 
  using read-leaf-res by simp
  have path-replay:
     $qh\text{-path} = ?path \wedge$ 
     $s2 = ?s\text{-leaf} \parallel$ 
       $\text{State} := \text{foldl concat (State } ?s\text{-leaf) } ?path,$ 
       $\text{Transcript} := \text{rest})$ 
  using ntimes-read-prefix-outcome[OF read-path[unfolded path-len[symmetric] s1-eq]]
  by simp
  have s0-s2:  $s0 \leq s2$ 
  using ext read-leaf-res path-replay
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  have check-honest:
    Some (ap, t)  $\in$ 
      set-dist (execute
        (check-authentication-path (length xs) i v ?path) s2)
  using check read-leaf-res path-replay by simp

```

```

have ap-eq: ap = value tree
  using check-created-tree-outcome[
    OF created len-pow idx leaf s0-s2 check-honest]
  by simp
then show None  $\notin$  dom
  (dist (execute
    ((case out of (qh, qh-path, ap)  $\Rightarrow$  assert (ap = fr)  $\gg$  ( $\lambda$ -. return qh))) t))
  unfolding out-eq root assert-def by simp
qed
qed

end

end

```

```

theory Channel-Instances
  imports Channel-Transcript
begin

```

1.5.1 Boolean Interpretation

```

global-interpretation test-channel:
  channel concat-bool
  defines test-send = test-channel.send
    and test-send2 = test-channel.send2
    and test-read = test-channel.read
  .

```

```

value execute (receive-random-field-element) ( $\langle$ HashMap = (fmempty::(bool, bool) fmap), State
= True, Transcript=[])

```

```

value execute (test-send True) ( $\langle$ HashMap = (fmempty::(bool, bool) fmap), State = True, Transcript=[])

```

The idea is that if we process a transcript then we recover the state and get the same random element. To this end we need to execute the monad on the final *HashMap*, the initial state and the reversed transcript. Each read then modifies the state in the same way as a send which leads to the same result for the hash value used to create the random value.

```

value
  execute (
    do {
      test-send True;
      r1  $\leftarrow$  receive-random-field-element;
      test-send False;
      r2  $\leftarrow$  receive-random-field-element;
      return r1
    }
  ) ( $\langle$ HashMap = (fmempty::(bool, bool) fmap), State = True, Transcript=[])

```

lemma

```

execute (
  do {
    test-read;
    r1 ← receive-random-field-element;
    test-read;
    r2 ← receive-random-field-element;
    return (r1 ∧ ¬ r2)
  }) (HashMap = fmap-of-list [(True, True), (False, False)], State = True, Transcript = rev
[False, True])
=
dist-of-fset ({|(Some (True, (HashMap = fmap-of-list [(True, True), (False, False)], State =
False, Transcript = [])), 1)|})
by eval

```

lemma

```

execute (
  do {
    test-read;
    r1 ← receive-random-field-element;
    test-read;
    r2 ← receive-random-field-element;
    return (r1 ∧ r2)
  }) (HashMap = fmap-of-list [(True, True), (False, True)], State = True, Transcript = rev
[False, True])
=
dist-of-fset ({|(Some (True, (HashMap = fmap-of-list [(True, True), (False, True)], State =
False, Transcript = [])), 1)|})
by eval

```

lemma

```

execute (
  do {
    test-read;
    r1 ← receive-random-field-element;
    test-read;
    r2 ← receive-random-field-element;
    return (¬ r1 ∧ r2)
  }) (HashMap = fmap-of-list [(True, False), (False, True)], State = True, Transcript = rev
[False, True])
=
dist-of-fset ({|(Some (True, (HashMap = fmap-of-list [(True, False), (False, True)], State =
False, Transcript = [])), 1)|})
by eval

```

lemma

```

execute (
  do {

```

```

    test-read;
    r1 ← receive-random-field-element;
    test-read;
    r2 ← receive-random-field-element;
    return (¬ r1 ∧ ¬ r2)
  }) (HashMap = fmap-of-list [(True, False), (False, False)], State = True, Transcript = rev
[False, True])
=
dist-of-fset ({|(Some (True, (HashMap = fmap-of-list [(True, False), (False, False)], State =
False, Transcript = [])), 1)|})
by eval

```

1.5.2 GF5 Interpretation

global-interpretation *gf5-channel*:

```

channel concat-gf5
defines gf5-send = gf5-channel.send
and gf5-send2 = gf5-channel.send2
and gf5-read = gf5-channel.read
.

```

lemma *execute (receive-random-field-element)* (HashMap = (fmempty::(gf5, gf5) fmap), State = 0, Transcript=[])

```

=
dist-of-fset (
  {|(Some (4, (HashMap = fmap-of-list [(0, 4)], State = 0, Transcript = [])), (1 / 5)),
   (Some (3, (HashMap = fmap-of-list [(0, 3)], State = 0, Transcript = [])), (1 / 5)),
   (Some (2, (HashMap = fmap-of-list [(0, 2)], State = 0, Transcript = [])), (1 / 5)),
   (Some (1, (HashMap = fmap-of-list [(0, 1)], State = 0, Transcript = [])), (1 / 5)),
   (Some (0, (HashMap = fmap-of-list [(0, 0)], State = 0, Transcript = [])), (1 / 5))|})
by eval

```

lemma *execute (gf5-send 1)* (HashMap = (fmempty::(gf5, gf5) fmap), State = 0, Transcript=[])

```

=
dist-of-fset ({|(Some (), (HashMap = fmap-of-list [], State = 1, Transcript = [1])), 1)|})
by eval

```

end

theory *Channel*

imports *Channel-Instances*

begin

Compatibility umbrella for the channel development. The imported theories define the protocol state, transcript operations, Fiat-Shamir oracle calls, and concrete field instances used by the STARK prover and verifier.

end

```

theory Galois-Field-5
  imports HOL-Number-Theory.Number-Theory
           HOL-Library.Cardinality
           Berlekamp-Zassenhaus.Finite-Field
begin

```

1.6 The Field GF(5)

This theory provides a small executable finite field instance used by the square-sequence example. The instance is intentionally small so that example proof generation and verification remain feasible inside Isabelle.

```

typedef gf5 = {0 .. 4 :: int}
  by auto

```

```

setup-lifting type-definition-gf5

```

```

lift-definition gf5-0:: gf5
is 0 by simp

```

```

lift-definition gf5-1:: gf5
is 1 by simp

```

```

lift-definition gf5-2:: gf5
is 2 by simp

```

```

lift-definition gf5-3:: gf5
is 3 by simp

```

```

lift-definition gf5-4:: gf5
is 4 by simp

```

```

lemma [simp]: gf5-0 ≠ gf5-1
  by transfer simp

```

```

lemma [simp]: gf5-0 ≠ gf5-2
  by transfer simp

```

```

lemma [simp]: gf5-0 ≠ gf5-3
  by transfer simp

```

```

lemma [simp]: gf5-0 ≠ gf5-4
  by transfer simp

```

```

lemma [simp]: gf5-1 ≠ gf5-2
  by transfer simp

```

```

lemma [simp]: gf5-1 ≠ gf5-3
  by transfer simp

```

```

lemma [simp]: gf5-1 ≠ gf5-4
  by transfer simp

lemma [simp]: gf5-2 ≠ gf5-3
  by transfer simp

lemma [simp]: gf5-2 ≠ gf5-4
  by transfer simp

lemma [simp]: gf5-3 ≠ gf5-4
  by transfer simp

instantiation gf5 :: field
begin
lift-definition zero-gf5 :: gf5 is 0 by auto
lift-definition one-gf5 :: gf5 is 1 by auto
lift-definition uminus-gf5 :: gf5 ⇒ gf5 is λx. (5 - x) mod 5 by auto
lift-definition plus-gf5 :: gf5 ⇒ gf5 ⇒ gf5 is λx y. (x + y) mod 5 by auto
definition minus-gf5 :: gf5 ⇒ gf5 ⇒ gf5 where minus-gf5 x y = (x + (uminus y))
lift-definition times-gf5 :: gf5 ⇒ gf5 ⇒ gf5 is λx y. (x * y) mod 5 by auto
lift-definition inverse-gf5 :: gf5 ⇒ gf5 is λx. if x = 2 then 3 else if x = 3 then 2 else x by
  auto
definition divide-gf5 :: gf5 ⇒ gf5 ⇒ gf5 where divide-gf5 x y = (x * inverse y)

instance
  apply standard
    apply (transfer, simp add: mod-mult-left-eq mod-mult-right-eq mult.assoc)
    apply (transfer, simp add: mult.commute)
    apply (transfer, simp)
    apply (transfer, presburger)
    apply (transfer, presburger)
    apply (transfer, simp)
    apply (transfer, presburger)
    apply (simp add: minus-gf5-def)
    apply (transfer, simp add: distrib-right mod-add-eq mod-mult-left-eq)
    apply (transfer, simp)
    apply (transfer, (case-tac a=1; auto), (case-tac a=4; auto))
    apply (simp add: divide-gf5-def)
  by (transfer, presburger)
end

lift-definition to-nat::gf5 ⇒ nat
is nat .

lift-definition of-nat::nat ⇒ gf5
is λx::nat. int (x mod 5) by auto

lemma of-nat-to-nat[simp]: of-nat (to-nat x) = x

```

```

apply transfer by fastforce

lemma gf5-cases[cases type:gf5]:
  fixes x::gf5
  obtains
     $x = gf5-0 \mid x = gf5-1 \mid x = gf5-2 \mid x = gf5-3 \mid x = gf5-4$ 
  apply transfer
  by fastforce

lemma gf5-Univ: (UNIV::gf5 set) = {gf5-0, gf5-1, gf5-2, gf5-3, gf5-4}
  using gf5-cases by blast

instantiation gf5 :: enum
begin

definition enum-gf5 :: gf5 list
  where enum-gf5 = [gf5-0, gf5-1, gf5-2, gf5-3, gf5-4]

definition enum-all-gf5 :: (gf5  $\Rightarrow$  bool)  $\Rightarrow$  bool
  where enum-all-gf5 P  $\equiv$  P gf5-0  $\wedge$  P gf5-1  $\wedge$  P gf5-2  $\wedge$  P gf5-3  $\wedge$  P gf5-4

definition enum-ex-gf5 :: (gf5  $\Rightarrow$  bool)  $\Rightarrow$  bool
  where enum-ex-gf5 P  $\equiv$  P gf5-0  $\vee$  P gf5-1  $\vee$  P gf5-2  $\vee$  P gf5-3  $\vee$  P gf5-4

instance
  apply (standard)
  by (auto simp add: gf5-Univ enum-gf5-def enum-all-gf5-def enum-ex-gf5-def)
end

instance gf5 :: finite-field
proof
  interpret type-definition Rep-gf5 Abs-gf5 {0 .. 4 :: int}
  by (rule type-definition-gf5)
  show finite (UNIV :: gf5 set)
  by simp
qed

instantiation gf5 :: equal
begin
lift-definition equal-gf5 :: gf5  $\Rightarrow$  gf5  $\Rightarrow$  bool
is (=) parametric eq-transfer .

instance
  by (standard) (transfer, simp)
end

lemma card-5-iff:  $\text{card } S = 5 \longleftrightarrow (\exists v\ w\ x\ y\ z. S = \{v, w, x, y, z\}$ 
   $\wedge v \neq w \wedge v \neq x \wedge v \neq y \wedge v \neq z$ 
   $\wedge w \neq x \wedge w \neq y \wedge w \neq z$ 

```



```

 $\wedge x \neq y \wedge x \neq z$ 
 $\wedge y \neq z$ )
by (fastforce simp: card-Suc-eq numeral-eq-Suc)

lemma gf5-size[simp]: CARD (gf5) = 5
proof -
  have ( $\exists v\ w\ x\ y\ z. \{gf5-0, gf5-1, gf5-2, gf5-3, gf5-4\} = \{v, w, x, y, z\} \wedge v \neq w \wedge v \neq x \wedge v \neq y \wedge v \neq z \wedge w \neq x \wedge w \neq y \wedge w \neq z \wedge x \neq y \wedge x \neq z \wedge y \neq z$ )
  by (metis gf5-0.rep-eq gf5-1.rep-eq gf5-2.rep-eq gf5-3.rep-eq gf5-4.rep-eq numeral-eq-iff numeral-eq-one-iff semiring-norm(85) verit-eq-simplify(12,14,8) zero-neq-numeral)
  then show ?thesis using card-5-iff gf5-Univ by fastforce
qed

```

Proth numbers: https://en.wikipedia.org/wiki/Proth_prime, they are useful for FTTs

```

class proth = finite +
  assumes proth-card:  $\exists k\ n. CARD('a) = k * 2^n + 1$ 

class proth-field = finite-field + proth + prime-card

instantiation gf5::proth-field
begin
instance proof
  show normalization-semidom-class.prime CARD(gf5) by simp
next
  show  $\exists k\ n. CARD(gf5) = k * 2^n + 1$ 
  apply (simp)
  by (metis power-even-eq power2-eq-square numeral-Bit0-eq-double)
qed
end

```

```

lemma gen-2-gf5:
  fixes x::gf5
  assumes  $x \neq gf5-0$ 
  shows  $\exists i. gf5-2^i = x$ 
proof (cases x)
  case 1
  with assms show ?thesis by satx
next
  case 2
  then show ?thesis
    by (metis gf5-1-def one-gf5-def power-0)
next
  case 3
  then show ?thesis
    by (meson power-Suc0-right)
next
  case 4

```

```

moreover have  $gf5-2 * gf5-2 * gf5-2 = gf5-3$ 
  by transfer simp
moreover have  $((gf5-2::gf5)^3) = gf5-2 * gf5-2 * gf5-2$ 
  by (rule power3-eq-cube)
ultimately show ?thesis by metis
next
case 5
moreover have  $gf5-2 * gf5-2 = gf5-4$ 
  by transfer simp
moreover have  $((gf5-2::gf5)^2) = gf5-2 * gf5-2$ 
  by (rule power2-eq-square)
ultimately show ?thesis by metis
qed

lemma to-nat (0::gf5) = 0 by eval
lemma to-nat (1::gf5) = 1 by eval
lemma to-nat (2::gf5) = 2 by eval
lemma to-nat (7::gf5) = 2 by eval
lemma  $((of-nat\ 0)::gf5) = 0$  by eval
lemma  $((of-nat\ 1)::gf5) = 1$  by eval
lemma  $((of-nat\ 2)::gf5) = 2$  by eval
lemma  $((of-nat\ 3)::gf5) = 3$  by eval

lemma gf5-0: 0 = gf5-0 apply transfer by presburger

end

```

```

theory Merkle-Tree
imports
  HOL-Library.Tree
  HOL-Library.Discrete-Functions
  Galois-Field-5
  Hash-Monad
begin

```

1.7 Merkle Trees

This theory defines Merkle tree creation, authentication paths, and checking in the probabilistic hash monad. The STARK prover uses these commitments for trace, composition, and FRI layers, while the verifier checks sampled openings against the recorded roots.

```

fun get-authentication-path::nat ⇒ nat ⇒ 'a tree ⇒ 'a list
where
  get-authentication-path lgth idx (Node l v r) =
    (if lgth = 1 then [] else
      (if idx < lgth div 2
        then (value r) # (get-authentication-path (lgth div 2) idx l)
        else (value l) # (get-authentication-path (lgth div 2) (idx - (lgth div 2)) r)))

```

```

locale merkle-tree =
  fixes concat :: 'a::finite  $\Rightarrow$  'a  $\Rightarrow$  'b
begin

fun create::'b list  $\Rightarrow$  ('b, 'a, 'a tree, 'c) hash-monad
where
  create [] = return  $\langle \rangle$ 
| create [x] =
  do {
    h  $\leftarrow$  hash x;
    return  $\langle \langle \rangle, h, \langle \rangle \rangle$ 
  }
| create xs =
  do {
    let i = length xs div 2;
    ln  $\leftarrow$  create (take i xs);
    rn  $\leftarrow$  create (drop i xs);
    val  $\leftarrow$  hash (concat (value ln) (value rn));
    return  $\langle ln, val, rn \rangle$ 
  }

fun check-authentication-path:: nat  $\Rightarrow$  nat  $\Rightarrow$  'b  $\Rightarrow$  'a list  $\Rightarrow$  ('b, 'a, 'a, 'c) hash-monad
where
  check-authentication-path lgth idx leaf-data [] = hash leaf-data
| check-authentication-path lgth idx leaf-data (a#ap) =
  (if idx < lgth div 2
   then
    do {
      x  $\leftarrow$  check-authentication-path (lgth div 2) idx leaf-data ap;
      hash (concat x a)
    }
   else
    do {
      x  $\leftarrow$  check-authentication-path (lgth div 2) (idx - (lgth div 2)) leaf-data ap;
      hash (concat a x)
    })
end

```

1.7.1 Weakest Precondition Calculus

```

context merkle-tree
begin

```

```

lemma wp-check-authentication-path[wp]:
  assumes ap = []  $\Longrightarrow$  P  $\leq$  wp (hash leaf-data) Q s
  assumes  $\bigwedge a$  list.
    ap = a # list  $\Longrightarrow$ 
    idx < lgth div 2  $\Longrightarrow$ 

```

```

    P ≤ wp (check-authentication-path (lgth div 2) idx leaf-data list ≫= (λx. hash (concat x
a))) Q s
  assumes ∧ a list.
    ap = a # list ⇒
    ¬ idx < lgth div 2 ⇒
    P ≤ wp (check-authentication-path (lgth div 2) (idx - lgth div 2) leaf-data list ≫=
      (λx. hash (concat a x)))
      Q s
  shows P ≤ wp (check-authentication-path lgth idx leaf-data ap) Q s
  using assms by (cases ap) (auto)

lemma wp-create[wp]:
  assumes xs = [] ⇒ P ≤ wp (return ⟨⟩) Q s
  assumes ∧ x. xs = [x] ⇒ P ≤ wp (hash x ≫= (λh. return ⟨⟨⟩, h, ⟨⟩⟩)) Q s
  assumes ∧ v vb vc.
    xs = v # vb # vc ⇒
    P ≤ wp (local.create (v # take (length vc div 2) (vb # vc)) ≫=
      (λln. local.create (drop (length vc div 2) (vb # vc)) ≫=
        (λrn. hash (concat (value ln) (value rn)) ≫= (λval. return ⟨ln, val, rn⟩))))
      Q s
  shows P ≤ wp (create xs) Q s
  using assms by (cases xs rule:create.cases) (auto)

end

```

1.7.2 Boolean interpretation

definition *concat-bool*:: *bool* ⇒ *bool* ⇒ *bool*
where
concat-bool = (∧)

global-interpretation *test-merkle*:
merkle-tree concat-bool
defines *test-create* = *test-merkle.create*
and *test-check-authentication-path* = *test-merkle.check-authentication-path*
 .

lemma
execute (test-create [True, True, True, True]) (HashMap = (fmempty::(bool, bool) fmap)) =
dist-of-fset (
 { |(Some (⟨⟨⟨⟨⟩, False, ⟨⟩), True, ⟨⟨⟩, False, ⟨⟩⟩), False, ⟨⟨⟨⟩, False, ⟨⟩⟩, True, ⟨⟨⟩, False, ⟨⟩⟩⟩,|
 (HashMap = fmap-of-list [(True, False), (False, True)]))|},
 (1 / 4)),
 (Some (⟨⟨⟨⟨⟩, False, ⟨⟩), False, ⟨⟨⟩, False, ⟨⟩⟩), False, ⟨⟨⟨⟩, False, ⟨⟩⟩, False, ⟨⟨⟩, False, ⟨⟩⟩⟩,|
 (HashMap = fmap-of-list [(True, False), (False, False)]))|},
 (1 / 4)),
 (Some (⟨⟨⟨⟨⟩, True, ⟨⟩), True, ⟨⟨⟩, True, ⟨⟩⟩), True, ⟨⟨⟨⟩, True, ⟨⟩⟩, True, ⟨⟨⟩, True, ⟨⟩⟩⟩,|
 (HashMap = fmap-of-list [(True, True)]))|}, (1 / 2))|})
by eval

lemma

```

execute (test-create [True, False, True, False]) (HashMap = (fmempty::(bool, bool) fmap)) =
dist-of-fset (
  { |(Some (⟨⟨⟨, False, ⟨⟩), True, ⟨⟩, True, ⟨⟩⟩), False, ⟨⟨⟨, False, ⟨⟩, True, ⟨⟩, True, ⟨⟩⟩⟩,
    (HashMap = fmap-of-list [(True, False), (False, True)])),
    (1 / 4)),
    (Some (⟨⟨⟨, False, ⟨⟩, False, ⟨⟩, False, ⟨⟩⟩), False, ⟨⟨⟨, False, ⟨⟩, False, ⟨⟩, False, ⟨⟩⟩⟩,
    (HashMap = fmap-of-list [(True, False), (False, False)])),
    (1 / 4)),
    (Some (⟨⟨⟨, True, ⟨⟩, True, ⟨⟩, True, ⟨⟩⟩), True, ⟨⟨⟨, True, ⟨⟩, True, ⟨⟩, True, ⟨⟩⟩⟩,
    (HashMap = fmap-of-list [(True, True), (False, True)])), (1 / 4)),
    (Some (⟨⟨⟨, True, ⟨⟩, False, ⟨⟩, False, ⟨⟩⟩), False, ⟨⟨⟨, True, ⟨⟩, False, ⟨⟩, False, ⟨⟩⟩⟩,
    (HashMap = fmap-of-list [(True, True), (False, False)])),
    (1 / 4))|}
by eval

```

lemma *execute*

```

(do {
  let x = [True, True, True, True];
  r ← test-create x;
  let ap = get-authentication-path (length x) 2 r;
  r2 ← test-check-authentication-path (length x) 2 True ap;
  return (r2 = value r)
}) (HashMap = (fmempty::(bool, bool) fmap))
=
dist-of-fset (
  { |(Some (True, (HashMap = fmap-of-list [(True, True)])), (1 / 2)), (Some (True, (HashMap
= fmap-of-list [(True, False), (False, False)])), (1 / 4)),
    (Some (True, (HashMap = fmap-of-list [(True, False), (False, True)])), (1 / 4))|}
by eval

```

1.7.3 GF5 interpretation**definition** *concat-gf5*:: gf5 ⇒ gf5 ⇒ gf5**where***concat-gf5* = (+)**global-interpretation** *merkle-gf5*:*merkle-tree concat-gf5***defines** *gf5-create* = *merkle-gf5.create***and** *gf5-check-authentication-path* = *merkle-gf5.check-authentication-path*

.

value *execute* (*gf5-create* [1, 2, 3]) (HashMap = (fmempty::(gf5, gf5) fmap))

1.7.4 Domain-separated protocol Merkle interface

The generic Merkle locale already separates hash outputs from hash inputs: leaves have a separate input type, hash outputs have the field type, and internal nodes are encoded by the locale parameter. Instantiating that parameter with the constructor `MerkleNode` and wrapping leaves with `MerkleLeaf` gives the STARK protocol a domain-separated Merkle API without duplicating the existing Merkle proofs.

global-interpretation *protocol-merkle*:

```
merkle-tree (λx y. MerkleNode x y)
defines protocol-merkle-create-raw = protocol-merkle.create
and protocol-merkle-check-authentication-path-raw =
  protocol-merkle.check-authentication-path
.
```

definition *protocol-create*

```
:: 'a::finite list ⇒
  ('a protocol-hash-input, 'a, 'a tree, 'c) hash-monad
where
  protocol-create xs =
    protocol-merkle.create (map MerkleLeaf xs)
```

definition *protocol-check-authentication-path*

```
:: nat ⇒ nat ⇒ 'a::finite ⇒ 'a list ⇒
  ('a protocol-hash-input, 'a, 'a, 'c) hash-monad
where
  protocol-check-authentication-path len idx leaf path =
    protocol-merkle.check-authentication-path len idx (MerkleLeaf leaf) path
```

1.7.5 Sanity Check

context *merkle-tree*

begin

The `created-tree` predicate is a proof-only invariant connecting the result of `create` to the final hash-map state. It is needed because `create` samples hashes probabilistically, while `authentication-path` checking later recomputes hashes. To prove the recomputation succeeds with probability 1, the proof needs to know those sampled hashes are already recorded in the hash map, so future calls to `hash` replay the same values instead of sampling fresh ones. So the `created-tree` predicate is not part of the Merkle API. It is a ghost invariant used to bridge "this tree was produced by `create`" and "checking its authentication path will replay the same root hash."

fun *created-tree*::'b list ⇒ 'a tree ⇒ ('b, 'a, 'c) hash-scheme ⇒ bool

where

```
  created-tree [] t s ⟷ t = ⟨⟩
| created-tree [x] t s ⟷
  (∃ h. t = ⟨⟨⟩, h, ⟨⟩⟩ ∧ fmllookup (HashMap s) x = Some h)
| created-tree xs t s ⟷
  (let i = length xs div 2 in
```

$$\begin{aligned} \exists l h r. t = \langle l, h, r \rangle \wedge \\ \text{created-tree } (\text{take } i \text{ xs}) \text{ l s} \wedge \\ \text{created-tree } (\text{drop } i \text{ xs}) \text{ r s} \wedge \\ \text{fmlookup } (\text{HashMap } s) (\text{concat } (\text{value } l) (\text{value } r)) = \text{Some } h \end{aligned}$$

declare *created-tree.simps*[*simp del*]

lemma *set-dist-bindE*:

assumes *Some* (*y*, *t*) $\in$ *set-dist* (*execute* (*m* $\gg$ *f*) *s*)
obtains *x u* **where** *Some* (*x*, *u*) $\in$ *set-dist* (*execute* *m s*)
and *Some* (*y*, *t*) $\in$ *set-dist* (*execute* (*f x*) *u*)

proof –

let *?K* = *bind-cont-map* (*map-fun id dist* $\circ$ *execute* $\circ$ *f*)

have *bind-dom*:

Some (*y*, *t*) $\in$ *dom* (*map-bind* (*dist* $\circ$ *execute m*) *?K s*)

using *assms*

unfolding *set-dist-def sm-bind.rep-eq dist-bind.rep-eq*

by (*simp add: o-def map-fun-def*)

then have *union*:

Some (*y*, *t*) $\in$

$(\bigcup_{i \in \text{dom } (\text{dist } (\text{execute } m \text{ s}))} \text{dom } (?K i))$

using *dom-map-bind[of dist $\circ$ execute m ?K s]*

by *auto*

then obtain *i* **where** *i-dom*: *i* $\in$ *dom* (*dist* (*execute m s*))

and *y-dom*: *Some* (*y*, *t*) $\in$ *dom* (*?K i*)

by *auto*

obtain *x u* **where** *i*: *i* = *Some* (*x*, *u*)

proof (*cases i*)

case *None*

then have *False*

using *y-dom*

unfolding *bind-cont-map-def delta-map-def*

by *simp*

then show *?thesis ..*

next

case (*Some p*)

then obtain *a b* **where** *i* = *Some* (*a*, *b*)

by (*cases p*) *auto*

then show *?thesis*

by (*rule that*)

qed

have *Some* (*x*, *u*) $\in$ *set-dist* (*execute m s*)

using *i-dom i* **unfolding** *set-dist-def* **by** *simp*

moreover have *Some* (*y*, *t*) $\in$ *set-dist* (*execute* (*f x*) *u*)

using *y-dom i*

unfolding *set-dist-def bind-cont-map-def*

by (*simp add: o-def map-fun-def*)

ultimately show *?thesis*

by (*rule that*)

qed

lemma *set-dist-return-iff*[simp]:

Some (y, t) ∈ set-dist (execute (return x) s) ⟷ y = x ∧ t = s

unfolding *set-dist-def return.rep-eq dist-return-def*

by (*simp add: dist-delta-dist delta-map-def*)

lemma *set-dist-get-iff*[simp]:

Some (y, t) ∈ set-dist (execute get s) ⟷ y = s ∧ t = s

unfolding *set-dist-def get.rep-eq dist-get-def*

by (*simp add: dist-delta-dist delta-map-def*)

lemma *set-dist-put-iff*[simp]:

Some (y, t) ∈ set-dist (execute (put s') s) ⟷ y = () ∧ t = s'

unfolding *set-dist-def put.rep-eq dist-put-def*

by (*simp add: dist-delta-dist delta-map-def*)

lemma *hash-outcome*:

assumes *Some (h, t) ∈ set-dist (execute (hash x) s)*

shows *s ≤ t*

and *fmlookup (HashMap t) x = Some h*

proof –

from *assms* **obtain** *a s1* **where**

a: Some (a, s1) ∈ set-dist (execute (apply-hash x) s)

and *rest: Some (h, t) ∈*

set-dist (execute (modify-HashMap x a >>= (λ-. return a)) s1)

proof –

have *Some (h, t) ∈*

set-dist (execute

(apply-hash x >>= (λa. modify-HashMap x a >>= (λ-. return a))) s)

using *assms* **unfolding** *hash-def* .

then show *?thesis*

by (*rule set-dist-bindE*) (*rule that*)

qed

have *a-img: Some (a, s1) ∈ (λa. Some (a, s)) ‘ set-dist (hash-dist x s)*

using *a*

unfolding *apply-hash-def lift.rep-eq lift-dist-def*

by (*simp add: set-dist-dist-map*)

then have *s1: s1 = s*

by *auto*

from *a-img* **have** *a-dom: a ∈ dom (dist (hash-dist x s))*

unfolding *set-dist-def* **by** *auto*

from *rest* **obtain** *u s2* **where**

mod: Some (u, s2) ∈ set-dist (execute (modify-HashMap x a) s1)

and *ret: Some (h, t) ∈ set-dist (execute (return a) s2)*

by (*elim set-dist-bindE*)

from *mod* **have** *s2: s2 = s1 (HashMap := fmupd x a (HashMap s1))*

unfolding *modify-HashMap-def modify-def*

by (*auto elim!: set-dist-bindE*)


```

from ret have h: h = a and t: t = s2
  by auto
show s ≤ t
  using a-dom unfolding s1 s2 h t by (rule hash-update-extends)
show fmlookup (HashMap t) x = Some h
  using a-dom unfolding s1 s2 h t hash-dist-def option-default-dist-def
  by (cases fmlookup (HashMap s) x) (auto simp: dist-delta-dist delta-map-def)
qed

```

```

lemma hash-extension-lookup:
  assumes fmlookup (HashMap s0) x = Some y
  and s0 ≤ s
  shows fmlookup (HashMap s) x = Some y
  using assms
  unfolding less-eq-hash-ext-def less-eq-fmap-def
  by (metis option.distinct(1))

```

```

lemma hash-ext-trans:
  assumes s0 ≤ s1 and s1 ≤ s2
  shows s0 ≤ (s2::('b, 'a, 'c) hash-scheme)
  using assms
  unfolding less-eq-hash-ext-def less-eq-fmap-def
  by (metis order.trans)

```

```

lemma hash-ext-refl:
  s ≤ (s::('b, 'a, 'c) hash-scheme)
  unfolding less-eq-hash-ext-def less-eq-fmap-def
  by simp

```

```

lemma created-tree-mono:
  assumes created-tree xs t s0
  and s0 ≤ s
  shows created-tree xs t s
  using assms
  by (induction xs t s0 arbitrary: s rule: created-tree.induct)
  (auto simp: Let-def created-tree.simps dest: hash-extension-lookup)

```

```

lemma create-outcome:
  assumes Some (r, t) ∈ set-dist (execute (create xs) s)
  shows s ≤ t ∧ created-tree xs r t
  using assms
proof (induction xs arbitrary: r s t rule: measure-induct-rule[of length])
  case (less xs)
  show ?case
  proof (cases xs rule: create.cases)
  case 1
  with less.prems show ?thesis
  by (auto simp: created-tree.simps intro: hash-ext-refl)
next

```

```

case (2 x)
from less.premis[unfolded 2] obtain h s1 where
  h: Some (h, s1) ∈ set-dist (execute (hash x) s)
  and ret: Some (r, t) ∈
    set-dist (execute (return ⟨⟨⟩, h, ⟨⟩⟩) s1)
  by (auto elim!: set-dist-bindE)
then show ?thesis
  using hash-outcome unfolding 2 by (auto simp: created-tree.simps)
next
case (3 v vb vc)
let ?i = length xs div 2
have len-ge: 2 ≤ length xs
  unfolding 3 by simp
have take-less: length (take ?i xs) < length xs
  using len-ge by simp
have drop-less: length (drop ?i xs) < length xs
  using len-ge by simp
have take-eq: take ?i xs = v # take (length vc div 2) (vb # vc)
  unfolding 3 by simp
have drop-eq: drop ?i xs = drop (length vc div 2) (vb # vc)
  unfolding 3 by simp
have split-create: Some (r, t) ∈ set-dist (execute
  (create (take ?i xs) >>=
    (λl. create (drop ?i xs) >>=
      (λrn. hash (concat (value l) (value rn)) >>=
        (λval. return ⟨l, val, rn⟩)))) s)
  using less.premis unfolding 3 take-eq drop-eq by simp
from split-create obtain l s1 where
  l: Some (l, s1) ∈ set-dist (execute (create (take ?i xs)) s)
  and rest1: Some (r, t) ∈
    set-dist (execute
      (create (drop ?i xs) >>=
        (λrn. hash (concat (value l) (value rn)) >>=
          (λval. return ⟨l, val, rn⟩)))) s1)
  by (rule set-dist-bindE) (rule that)
from rest1 obtain rn s2 where
  rn: Some (rn, s2) ∈ set-dist (execute (create (drop ?i xs)) s1)
  and rest2: Some (r, t) ∈
    set-dist (execute
      (hash (concat (value l) (value rn)) >>=
        (λval. return ⟨l, val, rn⟩))) s2)
  by (rule set-dist-bindE) (rule that)
from rest2 obtain val s3 where
  val: Some (val, s3) ∈ set-dist (execute (hash (concat (value l) (value rn)))) s2)
  and ret: Some (r, t) ∈ set-dist (execute (return ⟨l, val, rn⟩) s3)
  by (auto elim!: set-dist-bindE)
have left-out: s ≤ s1 ∧ created-tree (take ?i xs) l s1
  by (rule less.IH[OF take-less l])
have right-out: s1 ≤ s2 ∧ created-tree (drop ?i xs) rn s2

```

```

    by (rule less.IH[OF drop-less rn])
  have s2-s3:  $s2 \leq s3$  and root-lookup:
    fmlookup (HashMap s3) (concat (value l) (value rn)) = Some val
    using hash-outcome[OF val] by auto
  from ret have r:  $r = \langle l, val, rn \rangle$  and t:  $t = s3$ 
    by auto
  have s1-s3:  $s1 \leq s3$ 
    using right-out s2-s3 by (auto intro: hash-ext-trans)
  have st3:  $s \leq s3$ 
    using left-out s1-s3 by (auto intro: hash-ext-trans)
  have created-tree (take ?i xs) l s3
    using left-out s1-s3 by (auto intro: created-tree-mono)
  moreover have created-tree (drop ?i xs) rn s3
    using right-out s2-s3 by (auto intro: created-tree-mono)
  ultimately have created-tree xs r t
    using root-lookup unfolding r t 3 by (auto simp: created-tree.simps Let-def)
  with st3 show ?thesis
    unfolding t by simp
qed
qed

lemma hash-replay:
  assumes fmlookup (HashMap s0) x = Some y
    and  $s0 \leq s$ 
  shows wp-event (hash x) ( $\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow \text{False} \mid \text{Some } (z, -) \Rightarrow z = y$ )  $s = 1$ 
proof -
  have flip:
    ( $\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 0 \mid \text{Some } (z, u) \Rightarrow \text{if } z = y \text{ then } 1 \text{ else } 0$ ) =
    ( $\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 0 \mid \text{Some } (z, u) \Rightarrow \text{if } y = z \text{ then } 1 \text{ else } 0$ )
    by (rule ext) (auto split: option.splits prod.splits)
  have dist-expect (lift-dist (hash-dist x) s)
    ( $\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 0 \mid \text{Some } (z, u) \Rightarrow \text{if } z = y \text{ then } 1 \text{ else } 0$ ) = 1
    using hash-expect-known-extension[OF assms]
    unfolding flip .
  then show ?thesis
    unfolding wp-event-def wp-hash lift-dist-def
    by (simp add: dist-expect-map)
qed

lemma singleton-index:
  assumes  $xs = [x]$ 
    and  $i < \text{length } xs$ 
  shows  $i = 0$  and  $xs ! i = x$ 
  using assms by (cases i; auto)+

lemma check-created-tree-outcome:
  assumes created-tree xs r s0
    and  $\text{length } xs = 2^n$ 
    and  $i < \text{length } xs$ 

```

```

and  $v = xs ! i$ 
and  $s0 \leq s$ 
and  $\text{Some } (h, t) \in \text{set-dist } (\text{execute } (\text{check-authentication-path } (\text{length } xs) i v$ 
   $(\text{get-authentication-path } (\text{length } xs) i r)) s)$ 
shows  $h = \text{value } r \wedge s \leq t$ 
using assms
proof (induction n arbitrary: xs r i v s0 s h t)
  case 0
  have  $\text{len1: length } xs = 1$ 
  proof -
    have  $\text{length } xs = (2::\text{nat}) \wedge 0$ 
    using  $0.\text{prems}(2)$  .
    also have  $\dots = 1$ 
    by (rule power-0)
    finally show ?thesis .
  qed
obtain  $x$  where  $xs: xs = [x]$ 
proof (cases xs)
  case Nil
  then show ?thesis
  using  $\text{len1}$  by simp
next
  case (Cons y ys)
  then have  $\text{Suc } (\text{length } ys) = \text{Suc } 0$ 
  using  $\text{len1}$  by (simp only: length-Cons)
  then have  $\text{length } ys = 0$ 
  by simp
  then have  $ys = []$ 
  by (simp only: length-0-conv)
  then show ?thesis
  using Cons that by simp
qed
then obtain  $val$  where  $r: r = \langle \rangle, val, \langle \rangle$ 
  and  $\text{lookup: fmlookup } (\text{HashMap } s0) x = \text{Some } val$ 
  using  $0.\text{prems}(1)$  by (auto simp: created-tree.simps)
have  $i0: i = 0$ 
  using  $\text{singleton-index}(1)[OF xs 0.\text{prems}(3)]$  by assumption
have  $v\text{-}x: v = x$ 
  using  $0.\text{prems}(4)$   $\text{singleton-index}(2)[OF xs 0.\text{prems}(3)]$  by simp
have  $\text{outcome-hash: Some } (h, t) \in \text{set-dist } (\text{execute } (\text{hash } x) s)$ 
  using  $0.\text{prems}(6)$  unfolding  $xs r v\text{-}x$  by simp
have  $s\text{-}t: s \leq t$ 
  by (rule hash-outcome(1)[OF outcome-hash])
have  $s0\text{-}t: s0 \leq t$ 
  using  $0.\text{prems}(5)$   $s\text{-}t$  by (meson hash-ext-trans)
have  $\text{fmlookup } (\text{HashMap } t) x = \text{Some } val$ 
  by (rule hash-extension-lookup[OF lookup s0-t])
moreover have  $\text{fmlookup } (\text{HashMap } t) x = \text{Some } h$ 

```

```

    by (rule hash-outcome(2)[OF outcome-hash])
ultimately show  $h = \text{value } r \wedge s \leq t$ 
  unfolding  $r$  using  $s-t$  by auto
next
case (Suc  $n$ )
let  $?m = 2 \wedge n$ 
let  $?lxs = \text{take } ?m \text{ } xs$ 
let  $?rxs = \text{drop } ?m \text{ } xs$ 
have  $\text{len: length } xs = 2 * ?m$ 
  using Suc.prem(2) by simp
then have  $\text{mid: length } xs \text{ div } 2 = ?m$ 
  by simp
obtain  $a \ b \ cs$  where  $xs\text{-cons: } xs = a \# b \# cs$ 
  using  $\text{len}$  by (cases  $xs$ ; cases  $tl \ xs$ ) auto
have  $m\text{-eq: } 2 \wedge n = \text{Suc } (\text{length } cs \text{ div } 2)$ 
  using  $\text{mid}$  unfolding  $xs\text{-cons}$  by simp
have  $\text{take-mid-eq: } ?lxs = a \# \text{take } (\text{length } cs \text{ div } 2) \ (b \# cs)$ 
  unfolding  $xs\text{-cons } m\text{-eq}$  by simp
have  $\text{drop-mid-eq: } ?rxs = \text{drop } (\text{length } cs \text{ div } 2) \ (b \# cs)$ 
  unfolding  $xs\text{-cons } m\text{-eq}$  by simp
from Suc.prem(1) obtain  $l \ \text{root} \ rr$  where
   $r: r = \langle l, \text{root}, rr \rangle$ 
  and  $l\text{-created': created-tree } (a \# \text{take } (\text{length } cs \text{ div } 2) \ (b \# cs)) \ l \ s0$ 
  and  $r\text{-created': created-tree } (\text{drop } (\text{length } cs \text{ div } 2) \ (b \# cs)) \ rr \ s0$ 
  and  $\text{lookup: fmllookup } (\text{HashMap } s0) \ (\text{concat } (\text{value } l) \ (\text{value } rr)) = \text{Some } \text{root}$ 
  unfolding  $xs\text{-cons}$ 
  by (auto simp: created-tree.simps Let-def mid intro: that)
have  $l\text{-created: created-tree } ?lxs \ l \ s0$ 
  using  $l\text{-created'}$  unfolding  $\text{take-mid-eq}$  .
have  $r\text{-created: created-tree } ?rxs \ rr \ s0$ 
  using  $r\text{-created'}$  unfolding  $\text{drop-mid-eq}$  .
have  $\text{len-l: length } ?lxs = 2 \wedge n$ 
  using  $\text{len}$  by simp
have  $\text{len-r: length } ?rxs = 2 \wedge n$ 
  using  $\text{len}$  by simp
show  $h = \text{value } r \wedge s \leq t$ 
proof (cases  $i < ?m$ )
case True
have  $v\text{-l: } v = ?lxs \ ! \ i$ 
  using Suc.prem(3,4) True  $\text{len}$  by (simp add: nth-take)
have  $i\text{-l: } i < \text{length } ?lxs$ 
  using True  $\text{len-l}$  by simp
have  $\text{split-check: Some } (h, t) \in \text{set-dist } (\text{execute } (\text{check-authentication-path } (\text{length } ?lxs) \ i \ v) (\text{get-authentication-path } (\text{length } ?lxs) \ i \ l) \gg (\lambda x. \text{hash } (\text{concat } x \ (\text{value } rr)))) \ s)$ 
  using Suc.prem(6) True unfolding  $r$ 
  by (simp add: mid  $\text{len-l}$   $\text{len}$ )
from  $\text{split-check}$  obtain  $x \ s1$  where

```

```

    check-l: Some (x, s1) ∈ set-dist (execute
      (check-authentication-path (length ?lxs) i v
        (get-authentication-path (length ?lxs) i l)) s)
    and hash-root: Some (h, t) ∈ set-dist (execute (hash (concat x (value rr))) s1)
    by (rule set-dist-bindE) (rule that)
  have left-out: x = value l ∧ s ≤ s1
    by (rule Suc.IH[OF l-created len-l i-l v-l Suc.prem5] check-l)
  have x: x = value l
    using left-out by simp
  have s-s1: s ≤ s1
    using left-out by simp
  have s1-t: s1 ≤ t
    using hash-outcome(1)[OF hash-root] .
  have s0-t: s0 ≤ t
    using Suc.prem5 s-s1 s1-t by (meson hash-ext-trans)
  have fmlookup (HashMap t) (concat (value l) (value rr)) = Some root
    by (rule hash-extension-lookup[OF lookup s0-t])
  moreover have fmlookup (HashMap t) (concat (value l) (value rr)) = Some h
    using hash-outcome(2)[OF hash-root] unfolding x .
  ultimately show h = value r ∧ s ≤ t
    unfolding r using s-s1 s1-t by (auto intro: hash-ext-trans)
next
case False
have i-ge: ?m ≤ i
  using False by simp
have i-r: i - ?m < length ?rxs
  using Suc.prem3 len i-ge by simp
have v-r: v = ?rxs ! (i - ?m)
  using Suc.prem3,4 i-ge len
  by (simp add: nth-drop)
have split-check: Some (h, t) ∈ set-dist (execute
  (check-authentication-path (length ?rxs) (i - ?m) v
    (get-authentication-path (length ?rxs) (i - ?m) rr)) s)
  (λx. hash (concat (value l) x))) s)
  using Suc.prem6 False unfolding r
  by (simp add: mid len-r len)
from split-check obtain x s1 where
  check-r: Some (x, s1) ∈ set-dist (execute
    (check-authentication-path (length ?rxs) (i - ?m) v
      (get-authentication-path (length ?rxs) (i - ?m) rr)) s)
    and hash-root: Some (h, t) ∈ set-dist (execute (hash (concat (value l) x)) s1)
    by (rule set-dist-bindE) (rule that)
  have right-out: x = value rr ∧ s ≤ s1
    by (rule Suc.IH[OF r-created len-r i-r v-r Suc.prem5] check-r)
  have x: x = value rr
    using right-out by simp
  have s-s1: s ≤ s1
    using right-out by simp
  have s1-t: s1 ≤ t

```

```

    using hash-outcome(1)[OF hash-root] .
  have s0-t:  $s0 \leq t$ 
    using Suc.prem(5) s-s1 s1-t by (meson hash-ext-trans)
  have fmllookup (HashMap t) (concat (value l) (value rr)) = Some root
    by (rule hash-extension-lookup[OF lookup s0-t])
  moreover have fmllookup (HashMap t) (concat (value l) (value rr)) = Some h
    using hash-outcome(2)[OF hash-root] unfolding x .
  ultimately show  $h = \text{value } r \wedge s \leq t$ 
    unfolding r using s-s1 s1-t by (auto intro: hash-ext-trans)
qed
qed

lemma check-created-tree:
  assumes created-tree xs r s0
    and length xs =  $2^n$ 
    and  $i < \text{length } xs$ 
    and  $v = xs ! i$ 
    and  $s0 \leq s$ 
    and  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{check-authentication-path } (\text{length } xs) i v) (\text{get-authentication-path } (\text{length } xs) i r)) s))$ 
  shows wp-event
    (check-authentication-path (length xs) i v
      (get-authentication-path (length xs) i r))
    ( $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } (r', -) \Rightarrow r' = \text{value } r) s = 1$ 
  unfolding wp-event-def wp-def
proof (rule dist-expect-eq-1)
  fix p
  assume p:  $p \in \text{dom } (\text{dist } (\text{execute } (\text{check-authentication-path } (\text{length } xs) i v) (\text{get-authentication-path } (\text{length } xs) i r)) s))$ 
  obtain a b where p-eq:  $p = \text{Some } (a, b)$ 
    using p asms(6)
  by (cases p) auto
  have support:  $\text{Some } (a, b) \in \text{set-dist } (\text{execute } (\text{check-authentication-path } (\text{length } xs) i v) (\text{get-authentication-path } (\text{length } xs) i r)) s$ 
    using p unfolding p-eq set-dist-def .
  then have a-eq:  $a = \text{value } r$ 
    using check-created-tree-outcome[OF asms(1-5) support] by simp
  have goal: (if (case p of None  $\Rightarrow$  False  $\mid$  Some  $(r', -) \Rightarrow r' = \text{value } r$ ) then 1 else 0) = 1
    using a-eq by (simp add: p-eq)
  from goal show (if (case p of None  $\Rightarrow$  False  $\mid$  Some  $(r', -) \Rightarrow r' = \text{value } r$ ) then 1 else 0) = 1 .
qed

proposition create-check:
  assumes length x =  $2^n$ 
    and  $i < \text{length } x$ 

```

```

and  $v = x ! i$ 
and  $\text{None} \notin \text{dom} (\text{dist} (\text{execute}$ 
  (do {
     $r \leftarrow \text{create } x;$ 
     $\text{let } ap = \text{get-authentication-path } (\text{length } x) \text{ } i \text{ } r;$ 
     $r2 \leftarrow \text{check-authentication-path } (\text{length } x) \text{ } i \text{ } v \text{ } ap;$ 
     $\text{return } (r2 = \text{value } r)$ 
  }) ( $\text{HashMap} = \text{fmempty}$ )))
shows  $1 \leq \text{wp-event}$ 
  (do {
     $r \leftarrow \text{create } x;$ 
     $\text{let } ap = \text{get-authentication-path } (\text{length } x) \text{ } i \text{ } r;$ 
     $r2 \leftarrow \text{check-authentication-path } (\text{length } x) \text{ } i \text{ } v \text{ } ap;$ 
     $\text{return } (r2 = \text{value } r)$ 
  }) ( $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } (r, -) \Rightarrow r$ ) ( $\text{HashMap} = \text{fmempty}$ )
proof -
  let  $?s0 = (\text{HashMap} = \text{fmempty})$ 
  let  $?m = \text{do } \{$ 
     $r \leftarrow \text{create } x;$ 
     $\text{let } ap = \text{get-authentication-path } (\text{length } x) \text{ } i \text{ } r;$ 
     $r2 \leftarrow \text{check-authentication-path } (\text{length } x) \text{ } i \text{ } v \text{ } ap;$ 
     $\text{return } (r2 = \text{value } r)$ 
   $\}$ 
  have  $\text{wp-event } ?m (\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } (r, -) \Rightarrow r) ?s0 = 1$ 
  unfolding  $\text{wp-event-def wp-def}$ 
  proof (rule  $\text{dist-expect-eq-1}$ )
  fix  $p$ 
  assume  $p: p \in \text{dom} (\text{dist} (\text{execute } ?m ?s0))$ 
  obtain  $ok \text{ } sf$  where  $p\text{-eq}: p = \text{Some } (ok, sf)$ 
  using  $p \text{ } \text{assms}(4)$ 
  by (cases  $p$ ) auto
  have  $p\text{-set}: \text{Some } (ok, sf) \in \text{set-dist } (\text{execute } ?m ?s0)$ 
  using  $p$  unfolding  $p\text{-eq set-dist-def}$  .
  from  $p\text{-set}$  obtain  $r \text{ } s1$  where
     $\text{create-r}: \text{Some } (r, s1) \in \text{set-dist } (\text{execute } (\text{create } x) ?s0)$ 
    and  $\text{rest}: \text{Some } (ok, sf) \in \text{set-dist } (\text{execute}$ 
      ( $\text{let } ap = \text{get-authentication-path } (\text{length } x) \text{ } i \text{ } r \text{ in}$ 
         $\text{check-authentication-path } (\text{length } x) \text{ } i \text{ } v \text{ } ap \gg=$ 
        ( $\lambda r2. \text{return } (r2 = \text{value } r)$ ))  $s1$ )
      by (rule  $\text{set-dist-bindE}$ ) (rule that)
    from  $\text{rest}$  obtain  $r2 \text{ } s2$  where
       $\text{check-r}: \text{Some } (r2, s2) \in \text{set-dist } (\text{execute}$ 
        ( $\text{check-authentication-path } (\text{length } x) \text{ } i \text{ } v$ 
          ( $\text{get-authentication-path } (\text{length } x) \text{ } i \text{ } r$ ))  $s1$ )
        and  $\text{ret}: \text{Some } (ok, sf) \in \text{set-dist } (\text{execute } (\text{return } (r2 = \text{value } r)) \text{ } s2)$ 
        by (auto simp: Let-def elim!: set-dist-bindE)
      have  $\text{created}: \text{created-tree } x \text{ } r \text{ } s1$ 
      using  $\text{create-outcome}[OF \text{ create-r}]$  by simp
      have  $r2 = \text{value } r$ 

```



```

    using check-created-tree-outcome[
      OF created assms(1-3) hash-ext-refl check-r] by simp
  with ret have ok: ok
    by auto
  have goal: (if (case p of None  $\Rightarrow$  False | Some (r, -)  $\Rightarrow$  r) then 1 else 0) = 1
    using ok by (simp add: p-eq)
  from goal show (if (case p of None  $\Rightarrow$  False | Some (r, -)  $\Rightarrow$  r) then 1 else 0) = 1 .
qed
then show ?thesis
  by simp
qed
end

```

definition *protocol-created-tree*

```

:: 'a::finite list  $\Rightarrow$  'a tree  $\Rightarrow$ 
  ('a protocol-hash-input, 'a, 'c) hash-scheme  $\Rightarrow$  bool
where
  protocol-created-tree xs r s  $\longleftrightarrow$ 
    protocol-merkle.created-tree (map MerkleLeaf xs) r s

```

lemma *protocol-created-tree-mono*:

```

assumes protocol-created-tree xs r s0
  and s0  $\leq$  s
shows protocol-created-tree xs r s
using assms
unfolding protocol-created-tree-def
by (rule protocol-merkle.created-tree-mono)

```

lemma *protocol-create-outcome*:

```

assumes Some (r, t)  $\in$  set-dist (execute (protocol-create xs) s)
shows s  $\leq$  t  $\wedge$  protocol-created-tree xs r t
proof -
  have raw:
    Some (r, t)  $\in$ 
      set-dist (execute (protocol-merkle.create (map MerkleLeaf xs)) s)
  using assms unfolding protocol-create-def .
show ?thesis
  using protocol-merkle.create-outcome[OF raw]
  unfolding protocol-created-tree-def by simp
qed

```

lemma *protocol-check-created-tree-outcome*:

```

assumes created: protocol-created-tree xs r s0
  and length-xs: length xs =  $2^n$ 
  and i-bound:  $i <$  length xs
  and v: v = xs ! i
  and ext: s0  $\leq$  s
  and check:

```



```

then show ?thesis
  unfolding r by (simp add: floor-log-Suc-zero)
next
case (Cons y zs)
let ?i = length xs div 2
have xs-eq: xs = x # y # zs
  using xs-Cons ⟨ys = y # zs⟩ by simp
have len-ge: 2 ≤ length xs
  using xs-eq by simp
have take-eq: take ?i xs = x # take (length zs div 2) (y # zs)
  unfolding xs-eq by simp
have drop-eq: drop ?i xs = drop (length zs div 2) (y # zs)
  unfolding xs-eq by simp
have split-create: Some (r, t) ∈ set-dist (execute
  (create (take ?i xs) >>=
    (λl. create (drop ?i xs) >>=
      (λrn. hash (concat (value l) (value rn)) >>=
        (λval. return ⟨l, val, rn⟩)))) s)
  using less.premis(1) unfolding xs-eq take-eq drop-eq by simp
from split-create obtain l s1 where
  l: Some (l, s1) ∈ set-dist (execute (create (take ?i xs)) s)
  and rest1: Some (r, t) ∈
    set-dist (execute
      (create (drop ?i xs) >>=
        (λrn. hash (concat (value l) (value rn)) >>=
          (λval. return ⟨l, val, rn⟩)))) s1)
  by (rule set-dist-bindE) (rule that)
from rest1 obtain rr s2 where
  rr: Some (rr, s2) ∈ set-dist (execute (create (drop ?i xs)) s1)
  and rest2: Some (r, t) ∈
    set-dist (execute
      (hash (concat (value l) (value rr)) >>=
        (λval. return ⟨l, val, rr⟩)) s2)
  by (rule set-dist-bindE) (rule that)
from rest2 obtain val s3 where
  val: Some (val, s3) ∈ set-dist (execute (hash (concat (value l) (value rr))) s2)
  and ret: Some (r, t) ∈ set-dist (execute (return ⟨l, val, rr⟩) s3)
  by (auto elim!: set-dist-bindE)
have r: r = ⟨l, val, rr⟩
  using ret by simp
show ?thesis
proof (cases lgth = 1)
case True
  then show ?thesis
    unfolding r by (simp add: floor-log-Suc-zero)
next
case False
  have lgth-ge: 2 ≤ lgth
    using False less.premis by simp

```

```

have half-pos:  $0 < \text{lgth div } 2$ 
  using lgth-ge by simp
have take-less:  $\text{length } (\text{take } ?i \text{ } xs) < \text{length } xs$ 
  using len-ge by simp
have drop-less:  $\text{length } (\text{drop } ?i \text{ } xs) < \text{length } xs$ 
  using len-ge by simp
have half-le-i:  $\text{lgth div } 2 \leq ?i$ 
  using less.premis(3) by simp
have half-le-take:  $\text{lgth div } 2 \leq \text{length } (\text{take } ?i \text{ } xs)$ 
  using half-le-i by simp
have half-le-drop:  $\text{lgth div } 2 \leq \text{length } (\text{drop } ?i \text{ } xs)$ 
  using half-le-i len-ge by simp
have left-len:
   $\text{length } (\text{get-authentication-path } (\text{lgth div } 2) \text{ } idx \text{ } l) = \text{floor-log } (\text{lgth div } 2)$ 
  by (rule less.IH[OF take-less l half-pos half-le-take])
have right-len:
   $\text{length } (\text{get-authentication-path } (\text{lgth div } 2) \text{ } (idx - \text{lgth div } 2) \text{ } rr) =$ 
   $\text{floor-log } (\text{lgth div } 2)$ 
  by (rule less.IH[OF drop-less rr half-pos half-le-drop])
have floor-eq:  $\text{floor-log } \text{lgth} = \text{Suc } (\text{floor-log } (\text{lgth div } 2))$ 
  using floor-log-rec[OF lgth-ge] .
show ?thesis
proof (cases  $idx < \text{lgth div } 2$ )
  case True
  have  $\text{length } (\text{get-authentication-path } \text{lgth } idx \text{ } r) =$ 
     $\text{Suc } (\text{floor-log } (\text{lgth div } 2))$ 
  using False True left-len unfolding r by simp
  then show ?thesis
    by (simp only: floor-eq)
  next
  case False
  have  $\text{length } (\text{get-authentication-path } \text{lgth } idx \text{ } r) =$ 
     $\text{Suc } (\text{floor-log } (\text{lgth div } 2))$ 
  using  $\langle \text{lgth} \neq 1 \rangle$  False right-len unfolding r by simp
  then show ?thesis
    by (simp only: floor-eq)
qed
qed
qed
qed
qed

lemma get-authentication-path-len:
  assumes  $\text{execute } (\text{create } xs) \text{ } s = d$ 
  and  $x \in \text{dom } (\text{dist } d)$ 
  and  $x = \text{Some } (r, h)$ 
  and  $xs \neq []$ 
  shows  $\text{length } (\text{get-authentication-path } (\text{length } xs) \text{ } idx \text{ } r) = \text{floor-log } (\text{length } xs)$ 
proof -

```

```

have support: Some ( $r, h$ )  $\in$  set-dist (execute (create xs)  $s$ )
  using assms(1-3) unfolding set-dist-def by simp
show ?thesis
  by (rule create-get-authentication-path-len[OF support])
    (use assms(4) in simp-all)
qed

end

end

```

```

theory Prob-Monad
  imports
    Prob-Dist
begin

```

1.8 Probabilistic State Monad

This theory defines the finite probabilistic state monad used throughout the formalization. It includes monadic combinators, failure, assertions, and a weakest-precondition calculus for probabilistic events.

```

typedef ('a, 's) state-monad = UNIV::('s  $\Rightarrow$  (('a  $\times$  's) option) dist) set
```

morphisms *execute create*

by *simp*

```

lemma execute-create: execute (create f) =  $f$  using create-inverse by simp

```

```

setup-lifting type-definition-state-monad

```

```

Bind definition map-bind :: ('s  $\Rightarrow$  ('a  $\rightarrow$  prob))  $\Rightarrow$  ('a  $\Rightarrow$  ('b  $\rightarrow$  prob))  $\Rightarrow$  ('s  $\Rightarrow$  ('b  $\rightarrow$  prob))
where

```

```

  map-bind m k s =
    ( $\lambda b.$ 
      let  $p =$ 
        ( $\sum_{i \in \text{dom } (m s)}$ 
          (the ( $m s i$ ))
            * (option-default ( $k i b$ )))
        in if  $p = 0$  then None else Some p)

```

```

lemma map-bind-nonzero-on-dom:

```

```

  assumes  $x \in \text{dom } (map-bind m k s)$ 

```

```

  shows the (map-bind m k s x)  $\neq 0$ 

```

```

proof –

```

```

  show ?thesis

```

```

    using assms

```

```

    unfolding map-bind-def

```

```

    by (auto simp: Let-def split: if-splits)

```

qed

lemma *option-default-map-bind*:

option-default (*map-bind* *m k s x*) =
 ($\sum i \in \text{dom } (m \ s).$
 the (*m s i*) * *option-default* (*k i x*))
unfolding *map-bind-def* **by** (*simp add: Let-def*)

lemma *map-bind-eq-from-option-default*:

assumes
 option-default (*map-bind* *m k s x*) =
 option-default (*map-bind* *m' k' s' x*)
shows *map-bind* *m k s x* = *map-bind* *m' k' s' x*

proof –

show *?thesis*
using *assms*
unfolding *map-bind-def*
by (*simp add: Let-def*)

qed

lemma *map-bind-cong*:

assumes *m-eq*: *m s* = *m' s*
assumes *k-eq*: $\bigwedge i. i \in \text{dom } (m \ s) \implies k \ i = k' \ i$
shows *map-bind* *m k s* = *map-bind* *m' k' s*

proof

fix *x*
have *dom-eq*: *dom* (*m' s*) = *dom* (*m s*)
 using *m-eq* **by** *simp*
show *map-bind* *m k s x* = *map-bind* *m' k' s x*
 unfolding *map-bind-def Let-def*
 using *m-eq dom-eq k-eq*
 by (*intro if-cong refl arg-cong[where f=Some] sum.cong*) *auto*

qed

lemma *map-bind-cong-state*:

assumes *m-eq*: *m s* = *m' s'*
assumes *k-eq*: $\bigwedge i. i \in \text{dom } (m \ s) \implies k \ i = k' \ i$
shows *map-bind* *m k s* = *map-bind* *m' k' s'*

proof

fix *x*
have *dom-eq*: *dom* (*m' s'*) = *dom* (*m s*)
 using *m-eq* **by** *simp*
show *map-bind* *m k s x* = *map-bind* *m' k' s' x*
 unfolding *map-bind-def Let-def*
 using *m-eq dom-eq k-eq*
 by (*intro if-cong refl arg-cong[where f=Some] sum.cong*) *auto*

qed

lemma *dom-map-bind*:

```

dom (map-bind m k s) =
  {x ∈ (⋃ i ∈ dom (m s). dom (k i)).
   (∑ i ∈ dom (m s).
    the (m s i) * option-default (k i x)) ≠ 0}
proof -
  let ?M = dom (m s)
  let ?K = λi. k i
  let ?D = ⋃ i ∈ ?M. dom (?K i)
  let ?p = λx. ∑ i ∈ ?M. the (m s i) * option-default (?K i x)

  have outside-zero: x ∉ ?D ⟹ ?p x = 0 for x
  proof -
    assume x: x ∉ ?D
    have ∧i. i ∈ ?M ⟹ option-default (?K i x) = 0
      using x by (simp add: domIff)
    then show ?p x = 0
      by simp
  qed

  show ?thesis
  proof
    show dom (map-bind m k s) ⊆ {x ∈ ?D. ?p x ≠ 0}
    proof
      fix x
      assume x-dom: x ∈ dom (map-bind m k s)

      have px-ne: ?p x ≠ 0
        using x-dom
        unfolding map-bind-def
        by (fastforce simp: Let-def split: if-splits)

      moreover have x ∈ ?D
        using px-ne outside-zero by blast

      ultimately show x ∈ {x ∈ ?D. ?p x ≠ 0}
        by simp
    qed
  next
    show {x ∈ ?D. ?p x ≠ 0} ⊆ dom (map-bind m k s)
    proof
      fix x
      assume x: x ∈ {x ∈ ?D. ?p x ≠ 0}

      show x ∈ dom (map-bind m k s)
        using x
        unfolding map-bind-def
        apply (simp add: Let-def) by auto
    qed
  qed

```

qed

lemma *sum-option-default-superset*:
assumes *fin-D*: *finite D*
assumes *dom-subset*: $\text{dom } f \subseteq D$
shows $(\sum x \in D. \text{option-default } (f x)) = \text{sum-map } f$
proof –
have $(\sum x \in D. \text{option-default } (f x)) =$
 $(\sum x \in D \cap \text{dom } f. \text{option-default } (f x))$
using *fin-D*
apply (*intro sum.mono-neutral-right*)
apply *auto*
by (*metis not-None-eq option.case-eq-if*)
also have $\dots = (\sum x \in \text{dom } f. \text{option-default } (f x))$
using *dom-subset* **by** (*simp add: inf.absorb2*)
also have $\dots = (\sum x \in \text{dom } f. \text{the } (f x))$
by (*intro sum.cong*) *auto*
finally show *?thesis*
unfolding *sum-map-def* .
qed

lemma *sum-option-default-mult-superset*:
assumes *fin-D*: *finite D*
assumes *dom-subset*: $\text{dom } f \subseteq D$
shows $(\sum x \in D. \text{option-default } (f x) * g x) =$
 $(\sum x \in \text{dom } f. \text{the } (f x) * g x)$
proof –
have $(\sum x \in D. \text{option-default } (f x) * g x) =$
 $(\sum x \in \text{dom } f. \text{option-default } (f x) * g x)$
using *fin-D dom-subset*
by (*intro sum.mono-neutral-right*) (*auto simp: dom-def*)
also have $\dots = (\sum x \in \text{dom } f. \text{the } (f x) * g x)$
by (*intro sum.cong*) *auto*
finally show *?thesis* .
qed

lemma *the-map-bind-on-dom*:
assumes $x \in \text{dom } (\text{map-bind } m k s)$
shows
 $\text{the } (\text{map-bind } m k s x) =$
 $(\sum i \in \text{dom } (m s). \text{the } (m s i) * \text{option-default } (k i x))$
proof –
show *?thesis*
using *assms*
unfolding *map-bind-def*
by (*fastforce simp: Let-def split: if-splits*)
qed


```

lemma finite-dom-bind:
  assumes fin-m: finite (dom (m s))
  assumes fin-k:  $\forall i \in \text{dom } (m\ s). \text{finite } (\text{dom } (k\ i))$ 
  shows finite (dom (map-bind m k s))
proof -
  let ?D =  $\bigcup i \in \text{dom } (m\ s). \text{dom } (k\ i)$ 

  have fin-D: finite ?D
    using fin-m fin-k by auto

  have finite
    {x  $\in$  ?D.
     ( $\sum i \in \text{dom } (m\ s).$ 
      the (m s i) * option-default (k i x)  $\neq$  0)}
    using fin-D fin-m by simp

  then show ?thesis
    by (simp add: dom-map-bind)
qed

lemma sum-map-bind-1:
  assumes fin-m: finite (dom (m s))
  assumes fin-k:  $\forall i \in \text{dom } (m\ s). \text{finite } (\text{dom } (k\ i))$ 
  assumes sum-m: sum-map (m s) = 1
  assumes sum-k:  $\forall i \in \text{dom } (m\ s). \text{sum-map } (k\ i) = 1$ 
  shows sum-map (map-bind m k s) = 1
proof -
  let ?M = dom (m s)
  let ?K =  $\lambda i. k\ i$ 
  let ?D =  $\bigcup i \in ?M. \text{dom } (?K\ i)$ 
  let ?p =  $\lambda x. \sum i \in ?M. \text{the } (m\ s\ i) * \text{option-default } (?K\ i\ x)$ 

  have fin-D: finite ?D
    using fin-m fin-k by auto

  have sum-map (map-bind m k s) =
    ( $\sum x \in \text{dom } (\text{map-bind } m\ k\ s). \text{the } (\text{map-bind } m\ k\ s\ x)$ )
    unfolding sum-map-def by simp

  also have ... = ( $\sum x \in \text{dom } (\text{map-bind } m\ k\ s). ?p\ x$ )
    by (intro sum.cong) (simp-all add: the-map-bind-on-dom)

  also have ... = ( $\sum x \in \{x \in ?D. ?p\ x \neq 0\}. ?p\ x$ )
    by (simp add: dom-map-bind)

  also have ... = ( $\sum x \in ?D. ?p\ x$ )
    using fin-D
    apply (intro sum.mono-neutral-cong-left) using assms(1) by auto

```

```

also have ... = (∑ i ∈ ?M. ∑ x ∈ ?D.
  the (m s i) * option-default (?K i x))
by (rule sum.swap)

also have ... = (∑ i ∈ ?M.
  the (m s i) * (∑ x ∈ ?D. option-default (?K i x)))
by (simp add: sum-distrib-left)

also have ... = (∑ i ∈ ?M. the (m s i) * sum-map (?K i))
proof (intro sum.cong refl)
  fix i
  assume i: i ∈ ?M
  have dom (?K i) ⊆ ?D
    using i by auto
  then show
    the (m s i) * (∑ x ∈ ?D. option-default (?K i x)) =
    the (m s i) * sum-map (?K i)
    using sum-option-default-superset[OF fin-D, of ?K i]
    by simp
qed

also have ... = (∑ i ∈ ?M. the (m s i) * 1)
  using sum-k by simp

also have ... = sum-map (m s)
  unfolding sum-map-def by simp

also have ... = 1
  using sum-m .

finally show ?thesis .
qed

lemma map-bind-delta-left:
  fixes k :: 'a ⇒ 'b → prob
  assumes clean: ⋀x. x ∈ dom (k a) ⇒ the (k a x) ≠ 0
  shows map-bind (λ-. delta-map a) k s = k a
proof
  fix x :: 'b
  show map-bind (λ-. delta-map a) k s x = k a x
proof (cases k a x)
  case None
  then show ?thesis
    unfolding map-bind-def delta-map-def
    by (simp add: Let-def)
next
  case (Some p)
  then have p ≠ 0
    using clean[of x] by auto

```

```

    then show ?thesis
      unfolding map-bind-def delta-map-def
      using Some by (simp add: Let-def)
  qed
qed

lemma map-bind-delta-left-state:
  fixes k :: 'a  $\Rightarrow$  'b  $\rightarrow$  prob
  assumes clean:  $\bigwedge x. x \in \text{dom } (k \text{ (f s)}) \implies \text{the } (k \text{ (f s) } x) \neq 0$ 
  shows map-bind ( $\lambda t. \text{delta-map } (f t)$ ) k s = k (f s)
proof
  fix x :: 'b
  show map-bind ( $\lambda t. \text{delta-map } (f t)$ ) k s x = k (f s) x
  proof (cases k (f s) x)
    case None
    then show ?thesis
      unfolding map-bind-def delta-map-def
      by (simp add: Let-def)
    next
    case (Some p)
    then have p  $\neq 0$ 
      using clean[of x] by auto
    then show ?thesis
      unfolding map-bind-def delta-map-def
      using Some by (simp add: Let-def)
  qed
qed

lemma map-bind-delta-right:
  fixes m :: 's  $\Rightarrow$  'a  $\rightarrow$  prob
  assumes fin: finite (dom (m s))
  assumes clean:  $\bigwedge x. x \in \text{dom } (m s) \implies \text{the } (m s x) \neq 0$ 
  shows map-bind m ( $\lambda a. \text{delta-map } a$ ) s = m s
proof
  fix x :: 'a
  have sum-delta:
    ( $\sum_{i \in \text{dom } (m s)} \text{the } (m s i) * \text{option-default } (\text{delta-map } i x) =$ 
 $\text{if } x \in \text{dom } (m s) \text{ then the } (m s x) \text{ else } 0$ )
  proof -
    have ( $\sum_{i \in \text{dom } (m s)} \text{the } (m s i) * \text{option-default } (\text{delta-map } i x) =$ 
 $(\sum_{i \in \text{dom } (m s)} \text{if } i = x \text{ then the } (m s i) \text{ else } 0)$ )
      by (intro sum.cong refl) (simp add: delta-map-def)
    also have ... = (if x  $\in$  dom (m s) then the (m s x) else 0)
      using fin by (simp add: sum.delta)
    finally show ?thesis .
  qed

```

```

qed
show map-bind m (λa. delta-map a) s x = m s x
proof (cases x ∈ dom (m s))
  case True
  then have m s x = Some (the (m s x))
    by auto
  moreover have the (m s x) ≠ 0
    using True by (rule clean)
  ultimately show ?thesis
    unfolding map-bind-def
    using sum-delta True by (simp add: Let-def)
next
case False
then have m s x = None
  by auto
then show ?thesis
  unfolding map-bind-def
  using sum-delta False by (simp add: Let-def)
qed
qed

```

lemma *map-bind-assoc*:

```

fixes m :: 's ⇒ 'a → prob
and k :: 'a ⇒ 'b → prob
and h :: 'b ⇒ 'c → prob
assumes fin-m: finite (dom (m s))
assumes fin-k: ∧i. i ∈ dom (m s) ⇒ finite (dom (k i))
shows
  map-bind (map-bind m k) h s =
    map-bind m (λa. map-bind (λ-. k a) h ()) s
proof
  fix x :: 'c

```

```

let ?M = dom (m s)
let ?K = λi. k i
let ?D = ⋃ i ∈ ?M. dom (?K i)
let ?H = λj. option-default (h j x)
let ?p = λj. ∑ i ∈ ?M. the (m s i) * option-default (?K i j)

```

```

have fin-D: finite ?D
  using fin-m fin-k by auto

```

have *left-sum*:

```

(∑ j ∈ dom (map-bind m k s).
  the (map-bind m k s j) * ?H j) =
  (∑ j ∈ ?D. ?p j * ?H j)

```

proof –

```

have (∑ j ∈ dom (map-bind m k s).
  the (map-bind m k s j) * ?H j) =

```

```

  (∑ j ∈ {j ∈ ?D. ?p j ≠ 0}.
    the (map-bind m k s j) * ?H j)
  by (simp add: dom-map-bind)
also have ... =
  (∑ j ∈ {j ∈ ?D. ?p j ≠ 0}. ?p j * ?H j)
  by (intro sum.cong refl) (simp add: dom-map-bind the-map-bind-on-dom)
also have ... = (∑ j ∈ ?D. ?p j * ?H j)
  using fin-D
  by (intro sum.mono-neutral-left) auto
finally show ?thesis .
qed

```

```

have swap-sum:
  (∑ j ∈ ?D. ?p j * ?H j) =
  (∑ i ∈ ?M.
    the (m s i) *
    (∑ j ∈ dom (?K i).
      the (?K i j) * ?H j))
proof -
  have (∑ j ∈ ?D. ?p j * ?H j) =
    (∑ j ∈ ?D. ∑ i ∈ ?M.
      the (m s i) * option-default (?K i j) * ?H j)
    by (simp add: sum-distrib-left algebra-simps)
  also have ... =
    (∑ i ∈ ?M. ∑ j ∈ ?D.
      the (m s i) * option-default (?K i j) * ?H j)
    by (rule sum.swap)
  also have ... =
    (∑ i ∈ ?M.
      the (m s i) *
      (∑ j ∈ ?D. option-default (?K i j) * ?H j))
    by (simp add: sum-distrib-left algebra-simps)
  also have ... =
    (∑ i ∈ ?M.
      the (m s i) *
      (∑ j ∈ dom (?K i). the (?K i j) * ?H j))
  proof (intro sum.cong refl)
    fix i
    assume i: i ∈ ?M
    have dom (?K i) ⊆ ?D
      using i by auto
    then show
      the (m s i) *
      (∑ j ∈ ?D. option-default (?K i j) * ?H j) =
      the (m s i) *
      (∑ j ∈ dom (?K i). the (?K i j) * ?H j)
      using sum-option-default-mult-superset[OF fin-D, of ?K i ?H]
      by simp
  qed
qed

```

finally show *?thesis* .
qed

have *sums-eq*:

$(\sum j \in \text{dom } (\text{map-bind } m \ k \ s).$
 $\text{the } (\text{map-bind } m \ k \ s \ j) * ?H \ j) =$
 $(\sum i \in ?M.$
 $\text{the } (m \ s \ i) *$
 $\text{option-default } (\text{map-bind } (\lambda-. \ k \ i) \ h \ () \ x))$

proof –

have $(\sum j \in \text{dom } (\text{map-bind } m \ k \ s).$
 $\text{the } (\text{map-bind } m \ k \ s \ j) * ?H \ j) =$
 $(\sum i \in ?M.$
 $\text{the } (m \ s \ i) *$
 $(\sum j \in \text{dom } (?K \ i). \text{the } (?K \ i \ j) * ?H \ j))$
using *left-sum swap-sum* **by** *simp*
also have ... =
 $(\sum i \in ?M.$
 $\text{the } (m \ s \ i) *$
 $\text{option-default } (\text{map-bind } (\lambda-. \ k \ i) \ h \ () \ x))$
by (*intro sum.cong refl*) (*simp add: option-default-map-bind*)
finally show *?thesis* .

qed

show

$\text{map-bind } (\text{map-bind } m \ k) \ h \ s \ x =$
 $\text{map-bind } m \ (\lambda a. \text{map-bind } (\lambda-. \ k \ a) \ h \ ()) \ s \ x$

proof (*rule map-bind-eq-from-option-default*)

show

$\text{option-default } (\text{map-bind } (\text{map-bind } m \ k) \ h \ s \ x) =$
 $\text{option-default } (\text{map-bind } m \ (\lambda a. \text{map-bind } (\lambda-. \ k \ a) \ h \ ()) \ s \ x)$
using *sums-eq*
by (*simp add: option-default-map-bind*)

qed

qed

definition *bind-cont-map* ::

$('a \Rightarrow 's \Rightarrow ((b \times 's) \text{ option} \rightarrow \text{prob})) \Rightarrow$
 $('a \times 's) \text{ option} \Rightarrow ((b \times 's) \text{ option} \rightarrow \text{prob})$

where

bind-cont-map $k \ r =$
 $(\text{case } r \text{ of}$
 $\text{None} \Rightarrow \text{delta-map } \text{None}$
 $| \text{Some } (a, s') \Rightarrow k \ a \ s')$

lemma *bind-cont-map-return*:

bind-cont-map $(\lambda a \ s. \text{delta-map } (\text{Some } (a, s))) \ r = \text{delta-map } r$
unfolding *bind-cont-map-def*
by (*cases r*) *auto*

```

lemma finite-dom-bind-cont-mapI:
  assumes  $\bigwedge a s. \text{finite } (\text{dom } (k \ a \ s))$ 
  shows  $\text{finite } (\text{dom } (\text{bind-cont-map } k \ r))$ 
  using assms
  unfolding bind-cont-map-def
  by (cases r) (auto simp: delta-map-def split: prod.splits)

lemma sum-map-bind-cont-mapI:
  assumes  $\bigwedge a s. \text{sum-map } (k \ a \ s) = 1$ 
  shows  $\text{sum-map } (\text{bind-cont-map } k \ r) = 1$ 
  using assms
  unfolding bind-cont-map-def sum-map-def
  by (cases r) (auto simp: delta-map-def split: prod.splits)

lift-definition dist-bind ::
   $('s \Rightarrow (('a \times 's) \text{ option}) \text{ dist}) \Rightarrow$ 
   $('a \Rightarrow ('s \Rightarrow (('b \times 's) \text{ option}) \text{ dist})) \Rightarrow$ 
   $('s \Rightarrow (('b \times 's) \text{ option}) \text{ dist})$ 
is  $\lambda m k. \text{map-bind } m \ (\text{bind-cont-map } k)$ 
proof -
  fix  $m :: 's \Rightarrow (('a \times 's) \text{ option} \rightarrow \text{prob})$ 
  and  $k :: 'a \Rightarrow 's \Rightarrow (('b \times 's) \text{ option} \rightarrow \text{prob})$ 
  and  $s :: 's$ 
  assume  $m$ :
     $\bigwedge s. \text{finite } (\text{dom } (m \ s)) \wedge \text{sum-map } (m \ s) = 1 \wedge$ 
     $(\forall x \in \text{dom } (m \ s). \text{the } (m \ s \ x) \neq 0)$ 
  assume  $k$ :
     $\bigwedge a s. \text{finite } (\text{dom } (k \ a \ s)) \wedge \text{sum-map } (k \ a \ s) = 1 \wedge$ 
     $(\forall x \in \text{dom } (k \ a \ s). \text{the } (k \ a \ s \ x) \neq 0)$ 
  show  $\text{finite } (\text{dom } (\text{map-bind } m \ (\text{bind-cont-map } k) \ s)) \wedge$ 
     $\text{sum-map } (\text{map-bind } m \ (\text{bind-cont-map } k) \ s) = 1 \wedge$ 
     $(\forall x \in \text{dom } (\text{map-bind } m \ (\text{bind-cont-map } k) \ s).$ 
     $\text{the } (\text{map-bind } m \ (\text{bind-cont-map } k) \ s \ x) \neq 0)$ 
  proof (intro conjI ballI)
    show  $\text{finite } (\text{dom } (\text{map-bind } m \ (\text{bind-cont-map } k) \ s))$ 
    by (rule finite-dom-bind) (use m k in
       $\langle \text{auto intro!: finite-dom-bind-cont-mapI simp: bind-cont-map-def delta-map-def split: option.splits prod.splits} \rangle$ )
    show  $\text{sum-map } (\text{map-bind } m \ (\text{bind-cont-map } k) \ s) = 1$ 
    by (rule sum-map-bind-1) (use m k in
       $\langle \text{auto intro!: sum-map-bind-cont-mapI simp: bind-cont-map-def delta-map-def sum-map-def split: option.splits prod.splits} \rangle$ )
    fix  $x$ 
    assume  $x \in \text{dom } (\text{map-bind } m \ (\text{bind-cont-map } k) \ s)$ 
    then show  $\text{the } (\text{map-bind } m \ (\text{bind-cont-map } k) \ s \ x) \neq 0$ 
    by (rule map-bind-nonzero-on-dom)
  qed
qed

```

lift-definition

$sm\text{-}bind :: ('a, 's) \text{ state-monad} \Rightarrow ('a \Rightarrow ('b, 's) \text{ state-monad}) \Rightarrow ('b, 's) \text{ state-monad}$
(infixl >>= 60)
is $dist\text{-}bind$.

Return definition $dist\text{-}return :: 'a \Rightarrow ('s \Rightarrow (('a \times 's) \text{ option}) \text{ dist})$
where $dist\text{-}return\ a = (\lambda s. \text{ delta-dist } (Some\ (a, s)))$

lift-definition $return :: 'a \Rightarrow ('a, 's) \text{ state-monad}$
is $dist\text{-}return$.

definition $dist\text{-}throw :: 's \Rightarrow (('a \times 's) \text{ option}) \text{ dist}$
where $dist\text{-}throw = (\lambda -. \text{ delta-dist } None)$

lift-definition $throw :: ('a, 's) \text{ state-monad}$
is $dist\text{-}throw$.

adhoc-overloading $Monad\text{-}Syntax.bind \rightleftharpoons sm\text{-}bind$

Monad laws lemma $dist\text{-}bind\text{-}return\text{-}left$:

$dist\text{-}bind\ (dist\text{-}return\ a)\ k = k\ a$

proof

fix s
show $dist\text{-}bind\ (dist\text{-}return\ a)\ k\ s = k\ a\ s$
apply $(rule\ dist\text{-}inject[THEN\ iffD1])$
unfolding $dist\text{-}bind.rep\text{-}eq\ dist\text{-}return\text{-}def$
apply $(simp\ add: o\text{-}def\ map\text{-}fun\text{-}def\ dist\text{-}delta\text{-}dist)$
apply $(subst\ map\text{-}bind\text{-}delta\text{-}left\text{-}state[where\ f=\lambda t. Some\ (a, t)])$
by $(auto\ simp: bind\text{-}cont\text{-}map\text{-}def\ split: option.splits\ prod.splits)$

qed

lemma $sm\text{-}bind\text{-}return\text{-}left[simp]$:

$sm\text{-}bind\ (return\ a)\ k = k\ a$
by $transfer\ (rule\ dist\text{-}bind\text{-}return\text{-}left)$

lemma $dist\text{-}bind\text{-}return\text{-}right$:

$dist\text{-}bind\ m\ (\lambda a. dist\text{-}return\ a) = m$

proof

fix s
have $cont$:
 $bind\text{-}cont\text{-}map\ (\lambda x\ xa. \text{ delta-map } (Some\ (x, xa))) = (\lambda r. \text{ delta-map } r)$
by $(rule\ ext)\ (simp\ add: bind\text{-}cont\text{-}map\text{-}return)$
show $dist\text{-}bind\ m\ (\lambda a. dist\text{-}return\ a)\ s = m\ s$
apply $(rule\ dist\text{-}inject[THEN\ iffD1])$
unfolding $dist\text{-}bind.rep\text{-}eq\ dist\text{-}return\text{-}def$
apply $(simp\ add: o\text{-}def\ map\text{-}fun\text{-}def\ dist\text{-}delta\text{-}dist)$
apply $(subst\ cont)$
by $(simp\ add: map\text{-}bind\text{-}delta\text{-}right)$

qed

lemma *sm-bind-return-right*[simp]:
 $sm\text{-}bind\ m\ return = m$
by *transfer* (*rule dist-bind-return-right*)

lemma *dist-bind-assoc*:
 $dist\text{-}bind\ (dist\text{-}bind\ m\ k)\ h =$
 $dist\text{-}bind\ m\ (\lambda a. dist\text{-}bind\ (k\ a)\ h)$

proof
fix *s*
let *?M* = $\lambda s. dist\ (m\ s)$
let *?K* = $bind\text{-}cont\text{-}map\ (\lambda a\ t. dist\ (k\ a\ t))$
let *?H* = $bind\text{-}cont\text{-}map\ (\lambda a\ t. dist\ (h\ a\ t))$
let *?R* = $bind\text{-}cont\text{-}map\ (\lambda a\ t. map\text{-}bind\ (\lambda u. dist\ (k\ a\ u))\ ?H\ t)$
have *assoc*:
 $map\text{-}bind\ (map\text{-}bind\ ?M\ ?K)\ ?H\ s =$
 $map\text{-}bind\ ?M\ (\lambda r. map\text{-}bind\ (\lambda -. ?K\ r)\ ?H\ ())\ s$
by (*rule map-bind-assoc*) (*auto intro!*: *finite-dom-bind-cont-mapI*)
have *cont*:
 $map\text{-}bind\ (\lambda -. ?K\ r)\ ?H\ () = ?R\ r$ **for** *r*
proof (*cases r*)
case *None*
then show *?thesis*
apply (*simp add: bind-cont-map-def*)
apply (*subst map-bind-delta-left*[**where** *a=None*])
apply (*auto simp: bind-cont-map-def delta-map-def split: option.splits prod.splits*)
done
next
case (*Some as*)
obtain *a t* **where** *r*: $r = Some\ (a, t)$
using *Some* **by** (*cases as*) *simp*
show *?thesis*
unfolding *r bind-cont-map-def*
apply (*simp add: map-fun-def*)
by (*rule map-bind-cong-state*) *simp-all*
qed
show
 $dist\text{-}bind\ (dist\text{-}bind\ m\ k)\ h\ s =$
 $dist\text{-}bind\ m\ (\lambda a. dist\text{-}bind\ (k\ a)\ h)\ s$
apply (*rule dist-inject*[*THEN iffD1*])
unfolding *dist-bind.rep-eq*
using *assoc*
apply (*simp add: o-def map-fun-def dist-bind.rep-eq*)
apply (*rule map-bind-cong*)
apply *simp*
using *cont*
by *simp*
qed

lemma *sm-bind-assoc*:

sm-bind (*sm-bind* *m* *k*) *h* = *sm-bind* *m* ($\lambda a.$ *sm-bind* (*k* *a*) *h*)
by *transfer* (*rule dist-bind-assoc*)

1.8.1 Monad Operations

Basic Operators **definition** *dist-get* :: *'s* $\Rightarrow$ (((*'s* $\times$ *'s*) *option*) *dist*)
where

dist-get = ($\lambda s.$ *delta-dist* (*Some* (*s,s*)))

lift-definition *get* :: (*'s*, *'s*) *state-monad*
is *dist-get* .

definition *dist-put* :: *'s* $\Rightarrow$ *'s* $\Rightarrow$ ((*unit* $\times$ *'s*) *option*) *dist* **where**
dist-put *s* = ($\lambda -.$ *delta-dist* (*Some* (*()*,*s*)))

lift-definition *put* :: *'s* $\Rightarrow$ (*unit*, *'s*) *state-monad*
is *dist-put* .

definition *lift-dist* :: (*'s* $\Rightarrow$ *'a dist*) $\Rightarrow$ *'s* $\Rightarrow$ (((*'a* $\times$ *'s*) *option*) *dist*)
where
lift-dist *d* = ($\lambda s.$ *dist-map* ($\lambda a.$ *Some* (*a*, *s*)) (*d* *s*))

lift-definition *lift* :: (*'s* $\Rightarrow$ *'a dist*) $\Rightarrow$ (*'a*, *'s*) *state-monad*
is *lift-dist* .

definition *modify* :: (*'s* $\Rightarrow$ *'s*) $\Rightarrow$ (*unit*, *'s*) *state-monad* **where**
modify *f* = *get* $\gg=$ ($\lambda s::'s.$ *put* (*f* *s*))

definition *assert* :: *bool* $\Rightarrow$ (*unit*, *'s*) *state-monad* **where**
assert *b* = (*if* *b* *then return* *()* *else throw*)

primrec *ntimes* :: (*'a*, *'s*) *state-monad* $\Rightarrow$ *nat* $\Rightarrow$ (*'a list*, *'s*) *state-monad*
where
ntimes *m* 0 = *return* []
| *ntimes* *m* (*Suc* *n*) =
 do {
 x $\leftarrow$ *m*;
 xs $\leftarrow$ *ntimes* *m* *n*;
 return (*x* # *xs*)
 }

primrec *mfold2* :: (*'a* $\Rightarrow$ (*unit*, *'s*) *state-monad*) $\Rightarrow$ *'a list* $\Rightarrow$ (*unit*, *'s*) *state-monad*
where
mfold2 *m* [] = *return* *()*
| *mfold2* *m* (*x* # *xs*) =
 do {
 m *x*;

```

    mfold2 m xs
  }

```

```

primrec mmap :: ('a,'s) state-monad list  $\Rightarrow$  ('a list, 's) state-monad
where
  mmap [] = return []
| mmap (m # ms) =
  do {
    x  $\leftarrow$  m;
    xs  $\leftarrow$  mmap ms;
    return (x # xs)
  }

```

```

primrec mfold :: 'a  $\Rightarrow$  ('a  $\Rightarrow$  ('a,'s) state-monad) list  $\Rightarrow$  ('a, 's) state-monad
where
  mfold a [] = return a
| mfold a (m # ms) =
  do {
    x  $\leftarrow$  m a;
    mfold x ms
  }

```

Probabilistic Choice **definition** die :: nat $\Rightarrow$ (nat, 's) state-monad
where die n = create (λs . dist-uniform (fset-of-list (map (λx . Some (x,s)) [1.. $\text{Suc } n$])))

definition coin :: prob $\Rightarrow$ 'a $\Rightarrow$ 'a $\Rightarrow$ ('a, 's) state-monad
where coin f x y = create (λs . dist-of-fset {|(Some (x,s),f),(Some (y,s),1-f)|})

1.8.2 Weakest Precondition Calculus

definition dist-expect :: 'a dist $\Rightarrow$ ('a $\Rightarrow$ prob) $\Rightarrow$ prob **where**
 dist-expect d Q = ($\sum x \in \text{dom } (dist \ d)$. the (dist d x) * Q x)

lemma dist-expect-items:
 dist-expect d Q = fsum ($\lambda(x, p)$. p * Q x) (items d)
unfolding dist-expect-def fsum-items
by simp

lemma dist-expect-return[simp]:
 dist-expect (delta-dist x) Q = Q x
unfolding dist-expect-def
by (simp add: dist-delta-dist delta-map-def)

lemma dist-expect-0[simp]:
 dist-expect d (λ -. 0) = 0
unfolding dist-expect-def **by** simp

lemma dist-expect-indicator:
 dist-expect d (λx . if x = a then 1 else 0) =

```

    option-default (dist d a)
proof -
  have dist-expect d (λx. if x = a then 1 else 0) =
    (∑ x∈dom (dist d). if x = a then the (dist d x) else 0)
  unfolding dist-expect-def
  by (intro sum.cong refl) simp
  also have ... = (if a ∈ dom (dist d) then the (dist d a) else 0)
  by (simp add: sum.delta)
  also have ... = option-default (dist d a)
  by (cases dist d a) auto
  finally show ?thesis .
qed

```

lemma *dist-expect-None*:

```

dist-expect d (λr. case r of None ⇒ 1 | Some - ⇒ 0) =
  option-default (dist d None)
proof -
  have eq: (λr. case r of None ⇒ 1 | Some - ⇒ 0) =
    (λr. if r = None then 1 else 0)
  by (rule ext) (simp split: option.splits)
  have dist-expect d (λr. case r of None ⇒ 1 | Some - ⇒ 0) =
    dist-expect d (λr. if r = None then 1 else 0)
  by (simp add: eq)
  also have ... = option-default (dist d None)
  by (rule dist-expect-indicator)
  finally show ?thesis .
qed

```

lemma *dist-expect-map*:

```

dist-expect (dist-map f d) Q = dist-expect d (λx. Q (f x))
proof -
  let ?M = dom (dist d)
  have dom-map: dom (dist (dist-map f d)) = f ‘ ?M
  using set-dist-dist-map[of f d]
  unfolding set-dist-def by simp

  have dist-expect (dist-map f d) Q =
    (∑ x∈f ‘ ?M. the (dist (dist-map f d) x) * Q x)
  unfolding dist-expect-def dom-map by simp
  also have ... =
    (∑ x∈f ‘ ?M.
      (∑ y∈{y ∈ ?M. f y = x}. the (dist d y)) * Q x)
proof (intro sum.cong refl)
  fix x
  assume x: x ∈ f ‘ ?M
  then have x-dom: x ∈ dom (dist (dist-map f d))
  using dom-map by simp
  have the (dist (dist-map f d) x) =
    fsum snd (ffilter (λi. f (fst i) = x) (items d))

```

```

    using fsum-items-filter-dist-map-on-dom[OF x-dom]
    by simp
  also have ... = (∑ y∈{y ∈ ?M. f y = x}. the (dist d y))
    by (rule fsum-items-filter-dist-map)
  finally show
    the (dist (dist-map f d) x) * Q x =
      (∑ y∈{y ∈ ?M. f y = x}. the (dist d y)) * Q x
    by simp
qed
also have ... =
  (∑ x∈f ' ?M.
    (∑ y∈{y ∈ ?M. f y = x}. the (dist d y) * Q (f y)))
proof (intro sum.cong refl)
  fix x
  show
    (∑ y∈{y ∈ ?M. f y = x}. the (dist d y)) * Q x =
      (∑ y∈{y ∈ ?M. f y = x}. the (dist d y) * Q (f y))
    by (subst sum-distrib-right) (intro sum.cong refl, auto)
qed
also have ... = (∑ y∈?M. the (dist d y) * Q (f y))
  by (rule sum.group) auto
also have ... = dist-expect d (λx. Q (f x))
  unfolding dist-expect-def by simp
finally show ?thesis .
qed

lemma fimage-fst-norm-fset-subset:
  assumes fsum snd A ≠ 0
  shows fset (fimage fst (norm-fset A)) ⊆ fset (fimage fst A)
proof
  fix x
  assume x ∈ fset (fimage fst (norm-fset A))
  then obtain p where xp: (x, p) |∈| norm-fset A
    by auto
  then show x ∈ fset (fimage fst A)
    using assms
    unfolding norm-fset-def squish-def
    by (force simp: Let-def split: prod.splits)
qed

lemma dom-dist-of-fset-subset:
  assumes fsum snd A ≠ 0
  shows dom (dist (dist-of-fset A)) ⊆ fset (fimage fst A)
proof -
  have dist-eq: dist (dist-of-fset A) = map-of (norm-fset A)
    using assms
    unfolding dist-of-fset.rep-eq map-of-fset-def
    by (simp add: Let-def)
  have dom (dist (dist-of-fset A)) ⊆ fset (fimage fst (norm-fset A))

```

```

    unfolding dist-eq
    by (rule dom-map-of-subset)
  also have ...  $\subseteq$  fset (fimage fst A)
    by (rule fimage-fst-norm-fset-subset[OF assms])
  finally show ?thesis .
qed

lemma dom-dist-uniform-subset:
  assumes  $xs \neq \{\}\}$ 
  shows dom (dist (dist-uniform xs))  $\subseteq$  fset xs
proof -
  let ?A = (( $\lambda x. (x, 1::prob)$ )  $\mid$   $\mid$  xs)
  have total: fsum snd ?A  $\neq 0$ 
    using assms by (simp add: fsum-snd-uniform-ones)
  have dom (dist (dist-uniform xs))  $\subseteq$  fset (fimage fst ?A)
    unfolding dist-uniform-def
    by (rule dom-dist-of-fset-subset[OF total])
  also have ...  $\subseteq$  fset xs
    by auto
  finally show ?thesis .
qed

lemma fsum-snd-coin-nonzero:
  fixes p :: prob
  shows fsum snd  $\{|(a, p), (b, 1 - p)|\} \neq 0$ 
proof -
  have one-minus-zero: ( $1::prob$ ) - 0 = 1
    by transfer simp
  have  $p \neq 0 \vee 1 - p \neq 0$ 
  proof (cases p = 0)
    case True
    have  $1 - p = 1$ 
      using True one-minus-zero by simp
    then show ?thesis
      by simp
  next
    case False
    then show ?thesis
      by simp
  qed
  then show ?thesis
  proof
    assume p:  $p \neq 0$ 
    have  $(a, p) \in \{|(a, p), (b, 1 - p)|\}$ 
      by simp
    then show ?thesis
      using p
      unfolding fsum-fset-sum
      by (auto simp: sum-nonneg-eq-0-iff)
  qed

```

```

next
  assume  $q: 1 - p \neq 0$ 
  have  $(b, 1 - p) \in \text{fset } \{|(a, p), (b, 1 - p)|\}$ 
    by simp
  then show ?thesis
    using  $q$ 
    unfolding fsum-fset-sum
    by (auto simp: sum-nonneg-eq-0-iff)
qed
qed

lemma dist-expect-bind:
  dist-expect (dist-bind  $m$   $k$   $s$ )  $Q$  =
    dist-expect ( $m$   $s$ )
    ( $\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow Q \text{ None} \mid \text{Some } (x, s') \Rightarrow \text{dist-expect } (k \ x \ s') \ Q$ )
proof -
  let ?dm = dist  $\circ$   $m$ 
  let ?dk = map-fun id dist  $\circ$   $k$ 
  let ?M = dom (?dm  $s$ )
  let ?K = bind-cont-map ?dk
  let ?D =  $\bigcup_{i \in ?M} \text{dom } (?K \ i)$ 
  let ?B = map-bind ?dm ?K  $s$ 

  have fin-D: finite ?D
    by (auto simp: bind-cont-map-def delta-map-def split: option.splits prod.splits)

  have dom-B-subset: dom ?B  $\subseteq$  ?D
    by (auto simp: dom-map-bind)

  have dist-expect (dist-bind  $m$   $k$   $s$ )  $Q$  =
    ( $\sum_{x \in \text{dom } ?B} \text{the } (?B \ x) * Q \ x$ )
    unfolding dist-expect-def dist-bind.rep-eq by simp
  also have ... = ( $\sum_{x \in ?D} \text{option-default } (?B \ x) * Q \ x$ )
    using sum-option-default-mult-superset[OF fin-D dom-B-subset, of  $Q$ ]
    by simp
  also have ... =
    ( $\sum_{x \in ?D} \text{the } (dist \ (m \ s) \ i) * \text{option-default } (?K \ i \ x) * Q \ x$ )
    by (intro sum.cong refl) (simp add: option-default-map-bind)
  also have ... =
    ( $\sum_{x \in ?D} \sum_{i \in ?M} Q \ x * (\text{the } (dist \ (m \ s) \ i) * \text{option-default } (?K \ i \ x))$ )
proof (intro sum.cong refl)
  fix  $x$ 
  assume  $x \in ?D$ 
  have
    ( $\sum_{i \in ?M} \text{the } (dist \ (m \ s) \ i) * \text{option-default } (?K \ i \ x) * Q \ x =$ 
      ( $\sum_{i \in ?M} (\text{the } (dist \ (m \ s) \ i) * \text{option-default } (?K \ i \ x)) * Q \ x$ )
    by (simp add: sum-distrib-right)

```

```

also have ... =
  (∑ i ∈ ?M. Q x * (the (dist (m s) i) * option-default (?K i x)))
  by (intro sum.cong refl) (simp add: algebra-simps)
finally show
  (∑ i ∈ ?M. the (dist (m s) i) * option-default (?K i x)) * Q x =
  (∑ i ∈ ?M. Q x * (the (dist (m s) i) * option-default (?K i x))) .
qed
also have ... =
  (∑ i ∈ ?M. ∑ x ∈ ?D.
    Q x * (the (dist (m s) i) * option-default (?K i x)))
  by (rule sum.swap)
also have ... =
  (∑ i ∈ ?M.
    the (dist (m s) i) *
    (∑ x ∈ ?D. option-default (?K i x) * Q x))
  by (simp add: sum-distrib-left algebra-simps)
also have ... =
  (∑ i ∈ ?M.
    the (dist (m s) i) *
    (∑ x ∈ dom (?K i). the (?K i x) * Q x))
proof (intro sum.cong refl)
  fix i
  assume i: i ∈ ?M
  have dom (?K i) ⊆ ?D
    using i by auto
  then show
    the (dist (m s) i) *
    (∑ x ∈ ?D. option-default (?K i x) * Q x) =
    the (dist (m s) i) *
    (∑ x ∈ dom (?K i). the (?K i x) * Q x)
    using sum-option-default-mult-superset[OF fin-D, of ?K i Q]
    by simp
qed
also have ... =
  dist-expect (m s)
  (λr. case r of None ⇒ Q None | Some (x, s') ⇒ dist-expect (k x s') Q)
unfolding dist-expect-def bind-cont-map-def
apply (rule sum.cong)
apply simp
apply (rename-tac i)
apply (case-tac i)
apply (auto simp: case-prod-unfold dist-delta-dist delta-map-def split: prod.splits)
done
finally show ?thesis .
qed

definition wp :: ('a, 's) state-monad ⇒ (('a × 's) option ⇒ prob) ⇒ 's ⇒ prob where
  wp m Q s =
    dist-expect (execute m s)

```


$(\lambda r. Q\ r)$

lemma *wp-items*:

wp m Q s =
*fsum ($\lambda(r, p). p * Q\ r$) (items (execute m s))*
unfolding *wp-def dist-expect-items*
by (*simp add: case-prod-unfold*)

definition *wp-event* :: $('a, 's)$ *state-monad* $\Rightarrow ((a \times 's)$ *option* $\Rightarrow$ *bool*) $\Rightarrow 's \Rightarrow$ *prob* **where**
wp-event m P s = wp m ($\lambda x. \text{if } P\ x \text{ then } 1 \text{ else } 0$) s

definition *outcome-prob* ::

$('a, 's)$ *state-monad* $\Rightarrow 's \Rightarrow 'a \Rightarrow 's \Rightarrow$ *prob* **where**
outcome-prob m s x s' = option-default (dist (execute m s) (Some (x, s')))

definition *failure-prob* :: $('a, 's)$ *state-monad* $\Rightarrow 's \Rightarrow$ *prob* **where**

failure-prob m s = option-default (dist (execute m s) None)

definition *success-prob* :: $('a, 's)$ *state-monad* $\Rightarrow 's \Rightarrow$ *prob* **where**

success-prob m s = wp m ($\lambda -. 1$) s

definition *wp-error* :: $('a, 's)$ *state-monad* $\Rightarrow 's \Rightarrow$ *prob* **where**

wp-error m s =
wp m ($\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 1 \mid \text{Some } - \Rightarrow 0$) s

definition *wp-success* :: $('a, 's)$ *state-monad* $\Rightarrow ('a \Rightarrow 's \Rightarrow$ *prob*) $\Rightarrow 's \Rightarrow$ *prob* **where**

wp-success m Q s =
wp m ($\lambda r. \text{case } r \text{ of } \text{None} \Rightarrow 0 \mid \text{Some } (x, s') \Rightarrow Q\ x\ s')$ s

named-theorems *wpsimps*

named-theorems *wp*

lemma *wp-return[wpsimps]*:

wp (return x) Q s = Q (Some (x, s))
unfolding *wp-def return.rep-eq dist-return-def*
by *simp*

lemma *wp-bind[wpsimps]*:

wp (m $\gg=$ k) Q s =
wp m ($\lambda r. \text{case } r \text{ of}$
None $\Rightarrow Q\ \text{None}$
$\mid \text{Some } (x, s') \Rightarrow wp\ (k\ x)\ Q\ s')$ s
unfolding *wp-def sm-bind.rep-eq*
apply (*simp add: dist-expect-bind*)
apply (*rule arg-cong[where f= $\lambda F. \text{dist-expect (execute m s) F}$]*)
by (*rule ext*) (*auto split: option.splits prod.splits*)

lemma *wp-get[wpsimps]*:

wp get Q s = Q (Some (s, s))

unfolding *wp-def get.rep-eq dist-get-def* **by** *simp*

lemma *wp-put[wsimps]*:

wp (put s') Q s = Q (Some ((), s'))

unfolding *wp-def put.rep-eq dist-put-def* **by** *simp*

lemma *wp-lift[wsimps]*:

wp (lift d) Q s = dist-expect (d s) ($\lambda x. Q (Some (x, s))$)

unfolding *wp-def lift.rep-eq lift-dist-def*

by (*simp add: dist-expect-map*)

lemma *wp-modify[wsimps]*:

wp (modify f) Q s = Q (Some ((), f s))

unfolding *modify-def* **by** (*simp add: wsimps*)

lemma *wp-throw[wsimps]*:

wp throw Q s = Q None

unfolding *wp-def throw.rep-eq dist-throw-def* **by** *simp*

lemma *wp-create[wsimps]*:

wp (create f) Q s = dist-expect (f s) Q

unfolding *wp-def execute-create* **by** *simp*

lemma *wp-die[wsimps]*:

wp (die n) Q s =

dist-expect

*(dist-uniform (fset-of-list (map ($\lambda x. Some (x, s)$) [1..*Suc* n]))) Q*

unfolding *die-def* **by** (*simp add: wsimps*)

lemma *wp-coin[wsimps]*:

wp (coin p x y) Q s =

dist-expect (dist-of-fset {|(Some (x, s), p), (Some (y, s), 1 - p)|}) Q

unfolding *coin-def* **by** (*simp add: wsimps*)

lemma *wp-returnI[wp]*:

assumes $P \leq Q (Some (x, s))$

shows $P \leq wp (return x) Q s$

using *assms* **by** (*simp add: wsimps*)

lemma *wp-bindI[wp]*:

assumes $P \leq wp m (\lambda r. case r of$

None $\Rightarrow$ Q None

| Some (x, s') $\Rightarrow wp (k x) Q s'$) s

shows $P \leq wp (m \gg k) Q s$

using *assms* **by** (*simp add: wsimps*)

lemma *wp-getI[wp]*:

assumes $P \leq Q (Some (s, s))$

shows $P \leq wp get Q s$

```

using assms by (simp add: wpsimps)

lemma wp-putI[wp]:
  assumes  $P \leq Q \text{ (Some } ((), s^{\wedge}) \text{)}$ 
  shows  $P \leq wp \text{ (put } s^{\wedge}) Q s$ 
  using assms by (simp add: wpsimps)

lemma wp-modifyI[wp]:
  assumes  $P \leq Q \text{ (Some } ((), f s) \text{)}$ 
  shows  $P \leq wp \text{ (modify } f) Q s$ 
  using assms by (simp add: wpsimps)

lemma wp-liftI[wp]:
  assumes  $P \leq dist\text{-}expect \text{ (d s) } (\lambda x. Q \text{ (Some } (x, s)))$ 
  shows  $P \leq wp \text{ (lift d) } Q s$ 
  using assms by (simp add: wpsimps)

lemma wp-throwI[wp]:
  assumes  $P \leq Q \text{ None}$ 
  shows  $P \leq wp \text{ throw } Q s$ 
  using assms by (simp add: wpsimps)

lemma wp-createI[wp]:
  assumes  $P \leq dist\text{-}expect \text{ (f s) } Q$ 
  shows  $P \leq wp \text{ (create } f) Q s$ 
  using assms by (simp add: wpsimps)

lemma wp-dieI[wp]:
  assumes  $P \leq$ 
     $dist\text{-}expect$ 
     $(dist\text{-}uniform \text{ (fset-of-list (map } (\lambda x. Some \text{ (x, s)) } [1..<Suc \text{ n}]))) Q$ 
  shows  $P \leq wp \text{ (die n) } Q s$ 
  using assms by (simp add: wpsimps)

lemma wp-coinI[wp]:
  assumes  $P \leq dist\text{-}expect \text{ (dist-of-fset } \{|(Some \text{ (x, s), } p), (Some \text{ (y, s), } 1 - p)|\}) Q$ 
  shows  $P \leq wp \text{ (coin p x y) } Q s$ 
  using assms by (simp add: wpsimps)

lemma wp-error-eq-failure-prob:
  wp-error m s = failure-prob m s
  unfolding wp-error-def wp-def failure-prob-def
  by (simp add: dist-expect-None)

lemma wp-error-return[wpsimps]:
  wp-error (return x) s = 0
  unfolding wp-error-def by (simp add: wpsimps)

lemma wp-error-bind[wpsimps]:

```

```

wp-error (m  $\gg$  k) s =
  wp m ( $\lambda r$ . case r of
    None  $\Rightarrow$  1
    | Some (x, s')  $\Rightarrow$  wp-error (k x) s') s
unfolding wp-error-def
apply (simp add: wpsimps)
apply (rule arg-cong[where f= $\lambda F$ . wp m F s])
by (rule ext) (auto split: option.splits prod.splits)

lemma wp-error-get[wpsimps]:
  wp-error get s = 0
unfolding wp-error-def by (simp add: wpsimps)

lemma wp-error-put[wpsimps]:
  wp-error (put s') s = 0
unfolding wp-error-def by (simp add: wpsimps)

lemma wp-error-modify[wpsimps]:
  wp-error (modify f) s = 0
unfolding wp-error-def by (simp add: wpsimps)

lemma wp-error-lift[wpsimps]:
  wp-error (lift d) s = 0
unfolding wp-error-def by (simp add: wpsimps)

lemma wp-error-throw[wpsimps]:
  wp-error throw s = 1
unfolding wp-error-def by (simp add: wpsimps)

lemma wp-error-create[wpsimps]:
  wp-error (create f) s = option-default (dist (f s) None)
by (simp add: wp-error-eq-failure-prob failure-prob-def execute-create)

lemma wp-error-coin[wpsimps]:
  wp-error (coin p x y) s = 0
proof –
  let ?A = {|(Some (x, s), p), (Some (y, s), 1 - p)|}
  have total: fsum snd ?A  $\neq$  0
    by (rule fsum-snd-coin-nonzero)
  have None  $\notin$  dom (dist (dist-of-fset ?A))
    using dom-dist-of-fset-subset[OF total]
    by auto
  then show ?thesis
    unfolding wp-error-def
    by (simp add: wpsimps dist-expect-None domIff)
qed

lemma wp-error-die-pos[wpsimps]:
  assumes 0 < n

```

```

  shows wp-error (die n) s = 0
proof -
  let ?xs = fset-of-list (map (λx. Some (x, s)) [1..Suc n])
  have xs-nonempty: ?xs ≠ {}
    using assms by auto
  have none-xs: None ∉ fset ?xs
    by (auto simp: fset-of-list.rep-eq)
  have supp: dom (dist (dist-uniform ?xs)) ⊆ fset ?xs
    by (rule dom-dist-uniform-subset[OF xs-nonempty])
  have None ∉ dom (dist (dist-uniform ?xs))
    using supp none-xs by blast
  then show ?thesis
    unfolding wp-error-def
    by (simp add: wpsimps dist-expect-None domIff)
qed

```

```

lemma wp-success-return[wpsimps]:
  wp-success (return x) Q s = Q x s
  unfolding wp-success-def by (simp add: wpsimps)

```

```

lemma wp-success-bind[wpsimps]:
  wp-success (m >>= k) Q s =
    wp-success m (λx s'. wp-success (k x) Q s') s
  unfolding wp-success-def
  apply (simp add: wpsimps)
  apply (rule arg-cong[where f=λF. wp m F s])
  by (rule ext) (auto split: option.splits prod.splits)

```

```

lemma wp-success-get[wpsimps]:
  wp-success get Q s = Q s s
  unfolding wp-success-def by (simp add: wpsimps)

```

```

lemma wp-success-put[wpsimps]:
  wp-success (put s') Q s = Q () s'
  unfolding wp-success-def by (simp add: wpsimps)

```

```

lemma wp-success-modify[wpsimps]:
  wp-success (modify f) Q s = Q () (f s)
  unfolding wp-success-def by (simp add: wpsimps)

```

```

lemma wp-success-lift[wpsimps]:
  wp-success (lift d) Q s = dist-expect (d s) (λx. Q x s)
  unfolding wp-success-def by (simp add: wpsimps)

```

```

lemma wp-success-throw[wpsimps]:
  wp-success throw Q s = 0
  unfolding wp-success-def by (simp add: wpsimps)

```

```

lemma wp-error-return-check:

```

$wp\text{-}error\ (return\ x)\ s = 0$
by (*simp add: wpsimps*)

lemma *wp-error-throw-check*:
 $wp\text{-}error\ throw\ s = 1$
by (*simp add: wpsimps*)

lemma *wp-throw-error-post-check*:
 $wp\ throw\ (\lambda r. case\ r\ of\ None \Rightarrow 1\ |\ Some\ - \Rightarrow 0)\ s = 1$
by (*simp add: wpsimps*)

lemma *wp-error-bind-check*:
 $wp\text{-}error\ (m \ggg k)\ s =$
 $wp\ m\ (\lambda r. case\ r\ of\ None \Rightarrow 1\ |\ Some\ (x, s') \Rightarrow wp\text{-}error\ (k\ x)\ s')\ s$
by (*simp add: wpsimps*)

lemma *failure-prob-throw[simp]*:
 $failure\text{-}prob\ throw\ s = 1$
unfolding *failure-prob-def throw.rep-eq dist-throw-def dist-delta-dist delta-map-def*
by *simp*

lemma *failure-prob-return[simp]*:
 $failure\text{-}prob\ (return\ x)\ s = 0$
unfolding *failure-prob-def return.rep-eq dist-return-def dist-delta-dist delta-map-def*
by *simp*

lemma *success-prob-return[simp]*:
 $success\text{-}prob\ (return\ x)\ s = 1$
unfolding *success-prob-def* **by** (*simp add: wpsimps*)

lemma *outcome-prob-notin-dom*:
assumes $Some\ (x, s') \notin dom\ (dist\ (execute\ m\ s))$
shows $outcome\text{-}prob\ m\ s\ x\ s' = 0$
using *assms* **unfolding** *outcome-prob-def* **by** (*simp add: domIff*)

lemma *all-outcomes-above-thresholdI*:
assumes $\bigwedge x\ s'. outcome\text{-}prob\ m\ s\ x\ s' > t \implies P\ x\ s'$
shows $\forall x\ s'. outcome\text{-}prob\ m\ s\ x\ s' > t \longrightarrow P\ x\ s'$
using *assms* **by** *blast*

lemma *all-outcomes-above-threshold-domI*:
assumes
 $\bigwedge x\ s'. Some\ (x, s') \in dom\ (dist\ (execute\ m\ s)) \implies$
 $outcome\text{-}prob\ m\ s\ x\ s' > t \implies P\ x\ s'$
shows $\forall x\ s'. outcome\text{-}prob\ m\ s\ x\ s' > t \longrightarrow P\ x\ s'$
proof (*intro allI impI*)
fix $x\ s'$
assume *gt*: $outcome\text{-}prob\ m\ s\ x\ s' > t$
show $P\ x\ s'$

```

proof (cases Some (x, s') ∈ dom (dist (execute m s)))
  case True
  then show ?thesis
    using assms gt by blast
next
  case False
  then have outcome-prob m s x s' = 0
    by (rule outcome-prob-notin-dom)
  then have False
    using gt zero-least[of t] by order
  then show ?thesis
    by simp
qed
qed

```

Examples lemma return-outcomes-above-zero-example:

```

∀ y t. outcome-prob (return x) s y t > 0 ⟶ y = x ∧ t = s
unfolding outcome-prob-def return.rep-eq dist-return-def dist-delta-dist delta-map-def
by (simp add: domIff)

```

lemma return-threshold-example:

```

∀ y t. outcome-prob (return x) s y t > 0 ⟶ y = x
using return-outcomes-above-zero-example[of x s] by blast

```

1.8.3 Code generator

lemmas

```

sm-bind-def[code]
execute-create[code]

```

lemma sum-fset-by-bind-items-generic:

```

sum-fset-by x
  (λ(ms, ks). fst ks)
  (λ(ms, ks). snd ms * snd ks)
  (fbind (items d) (λms.
    fimage (λks. (ms, ks)) (items (k (fst ms)))))) =
  (∑ i ∈ dom (dist d).
    the (dist d i) * option-default (dist (k i) x))
unfolding sum-fset-by-def
apply (simp add: fsum-ffilter-if fsum-fbind-fimage-Pair case-prod-unfold)
unfolding fsum-items
apply (intro sum.cong refl)
apply (simp add: fsum-mult-left-nnreal fsum-items-key)
by (simp add: dom-def option.case-eq-if)

```

lemma fsum-bind-items-total-generic:

```

fsum
  (λ(ms, ks). snd ms * snd ks)
  (fbind (items d) (λms.

```

```

      fimage (λks. (ms, ks)) (items (k (fst ms)))))) = 1
proof -
  have fsum
    (λ(ms, ks). snd ms * snd ks)
    (fbind (items d) (λms.
      fimage (λks. (ms, ks)) (items (k (fst ms)))))) =
    fsum
      (λms. fsum (λks. snd ms * snd ks) (items (k (fst ms))))
      (items d)
  by (simp add: fsum-fbind-fimage-Pair case-prod-unfold)

  also have ... = fsum snd (items d)
  by (intro fsum.cong refl) (simp add: fsum-items-mult-left)

  also have ... = 1
  by simp

  finally show ?thesis .
qed

lemma dist-bind-code[code]:
  fixes m :: 's ⇒ (('a × 's) option) dist
  and k :: 'a ⇒ 's ⇒ (('b × 's) option) dist
  shows
    dist-bind m k s =
      dist-of-fset-by
        (λ(ms, ks). fst ks)
        (λ(ms, ks). snd ms * snd ks)
        (fbind (items (m s)) (λms.
          fimage (λks. (ms, ks))
            (items (case fst ms of None ⇒ delta-dist None | Some (x, s') ⇒ k x s')))))
proof (rule dist-inject[THEN iffD1])
  let ?I =
    fbind (items (m s)) (λms.
      fimage (λks. (ms, ks))
        (items (case fst ms of None ⇒ delta-dist None | Some (x, s') ⇒ k x s'))))
  let ?key = λ(ms, ks). fst ks
  let ?weight = λ(ms, ks). snd ms * snd ks
  let ?next =
    (λr :: ('a × 's) option.
      case r of None ⇒ delta-dist (None :: ('b × 's) option)
      | Some (x, s') ⇒ k x s')
  have next-eq:
    bind-cont-map (map-fun id dist ∘ k) i = dist (?next i) for i
  by (cases i) (auto simp: bind-cont-map-def dist-delta-dist o-def map-fun-def split: prod.splits)

  have next-eq-raw:
    bind-cont-map (λx xa. dist (k x xa)) i = dist (?next i) for i

```



```

by (cases i) (auto simp: bind-cont-map-def dist-delta-dist split: prod.splits)

have total: fsum ?weight ?I = 1
  by (rule fsum-bind-items-total-generic)

show dist (dist-bind m k s) =
  dist (dist-of-fset-by ?key ?weight ?I)
proof
  fix x :: ('b × 's) option

  have by-sum:
    sum-fset-by x ?key ?weight ?I =
      (∑ i∈dom (dist (m s)).
        the (dist (m s) i) *
        option-default (dist (?next i) x))
    by (rule sum-fset-by-bind-items-generic[of x m s ?next])

  have rhs:
    dist (dist-of-fset-by ?key ?weight ?I) x =
      map-of-fset-by ?key ?weight ?I x
    using dist-of-fset-by-rep-eq-1[OF total]
    by metis

  have dist (dist-bind m k s) x =
    (let p =
      (∑ i∈dom (dist (m s)).
        the (dist (m s) i) *
        option-default (dist (?next i) x))
    in if p = 0 then None else Some p)
    unfolding dist-bind.rep-eq map-bind-def
    by (simp add: Let-def o-def map-fun-def next-eq next-eq-raw split: option.splits prod.splits)
  also have ... = map-of-fset-by ?key ?weight ?I x
    using total by-sum
    unfolding map-of-fset-by-def
    by (simp add: Let-def)
  also have ... = dist (dist-of-fset-by ?key ?weight ?I) x
    using rhs by simp
  finally show
    dist (dist-bind m k s) x =
      dist (dist-of-fset-by ?key ?weight ?I) x .

qed
qed

```

Some basic examples `fun mymap1::bool ⇒ ((bool × bool) option → prob)`
where
`mymap1 True = Map.map-of [(Some (True,True), 0.5),(Some (False,True), 0.5)]`
`| mymap1 False = Map.map-of [(Some (False,True), 0.5),(Some (False,False), 0.5)]`

```

fun mymap2::(bool × bool) option ⇒ ((bool × bool) option → prob)
where
  mymap2 None = delta-map None
| mymap2 (Some (True, True)) = Map.map-of [(Some (True,False), 0.5),(Some (False,False),
0.5)]
| mymap2 (Some (False, True)) = Map.map-of [(Some (False,False), 0.5),(Some (False,True),
0.5)]
| mymap2 (Some (True, False)) = Map.map-of [(Some (True,False), 0.5),(Some (False,False),
0.5)]
| mymap2 (Some (False, False)) = Map.map-of [(Some (False,False), 0.5),(Some (True,False),
0.5)]

value map (map-bind mymap1 mymap2 True) [Some (True,False),Some (False,False),Some
(True,True),Some (False,True),None]

value (execute ((return 1)::(nat,nat) state-monad) 0)

value (execute (
  do {
    x ← return (1::nat);
    return (2::nat);
    return x
  } ::(nat,bool) state-monad) False)

value (execute (
  do {
    number ← die 6;
    if number = 6
      then coin 0.5 1 0
      else coin 0.1 1 0
  } ::(nat,bool) state-monad) False)

value execute (coin 0.5 True False) True

value wp-event (coin 0.5 True False) (λa. ∃ b. a = Some (True, b)) True

value execute (
  do {
    number ← coin 0.5 True False;
    if number
      then coin 0.2 True False
      else coin 0.7 True False
  }) False

value wp-event (
  do {
    number ← coin 0.5 True False;
    if number
      then coin 0.5 True False

```

```

    else coin 0.5 True False
  }) (λa. ∃ b. a = Some (b, False)) False

```

end

```

theory Stark
imports
  Main
  Merkle-Tree
  Channel
  Polynomial-Interpolation.Lagrange-Interpolation

```

begin

1.9 Stark Protocol

This theory defines the executable prover and verifier monads for the STARK protocol formalized in this entry. It also fixes the common domain parameters, transcript operations, Merkle commitments, query sampler, and FRI folding routines used by the completeness and soundness developments.

The formalization follows the structure of the StarkWare STARK tutorial and the algebraic framework described in the STARK literature. The field class uses Proth fields because their multiplicative groups provide the power-of-two subgroups used for FFT-style evaluation domains.

1.9.1 Preliminaries

definition *ceil-log* :: *nat* ⇒ *nat* **where**
ceil-log *n* = (if *n* ≤ 1 then 0 else Suc (floor-log (*n* − 1)))

Filter list by predicate over length There exists *nths* in List which does what we want but it cannot be executed. Thus, we introduce a new function with a corresponding code equation which makes it executable.

definition *nths-pred*
where
nths-pred *xs* *P* ≡ *nths* *xs* (Collect *P*)

lemma *nths-pred-nths*[*code*]:
nths-pred *xs* *P* = map fst (filter (λ*p*. *P* (snd *p*)) (zip *xs* ([0..*length* *xs*])))
unfolding *nths-pred-def* *nths-def* **by** *simp*

lemma *nths-pred-alt-def*[*simp*]:
nths-pred *xs* *P* = *nths* *xs* (Collect *P*)
unfolding *nths-pred-def* **..**

```

lemma nths-pred ([5,6,3,4]::nat list) even = [5,3] by eval
lemma nths-pred ([5,6,3]::nat list) even = [5,3] by eval
lemma nths-pred ([5,6,3]::nat list) odd = [6] by eval
lemma nths-pred ([6,3]::nat list) even = [6] by eval
lemma nths-pred ([6,3]::nat list) odd = [3] by eval

```

```

lemma nths-pred-length-leq:
  length (nths-pred xs P) ≤ length xs
  unfolding nths-pred-def nths-def
  by (metis length-filter-le length-map map-fst-zip map-nth)

```

Many properties can be more easily proven if we use *zip xs (rev [0..<length xs])*. Thus, we introduce the following lemmas which can be used to convert the lemmas back to *nths-pred*.

```

lemma filter-zip-lcomp:
  assumes xs ≠ []
  shows
    filter (λp. P (snd p)) (zip xs [0..<length xs]) = [(xs!i,i). i ← filter P [0..<length xs]]
  using assms
proof (induction xs rule: rev-induct)
  case Nil
  then show ?case by simp
next
  case (snoc x xs)
  then show ?case
  proof (cases P (length xs))
  case True
  then have
    filter (λp. P (snd p)) (zip (xs @ [x]) [0..<length (xs @ [x])])
    = (filter (λp. P (snd p)) (zip (xs) [0..<(length xs)]) @ [(x, (length xs))]
    using snoc.prems by auto
  also from snoc.IH
  have ... = map (λi. (xs ! i, i)) (filter P [0..<length xs]) @ [(x, (length xs))]
    by fastforce
  also have ... = map (λi. ((xs @ [x]) ! i, i)) (filter P [0..<length (xs @ [x])])
    using snoc.prems True by (auto simp add: nth-append-left)
  finally show ?thesis .
next
  case False
  then have
    filter (λp. P (snd p)) (zip (xs @ [x]) [0..<length (xs @ [x])])
    = (filter (λp. P (snd p)) (zip (xs) [0..<(length xs)])
    using snoc.prems by auto
  also from snoc have ... = map (λi. (xs ! i, i)) (filter P [0..<length xs]) by fastforce
  also have ... = map (λi. ((xs @ [x]) ! i, i)) (filter P [0..<length ((xs @ [x])])
    using snoc.prems False by (auto simp add: nth-append-left)
  finally show ?thesis .
qed
qed

```

```

lemma filter-zip-rev-lcomp:
  assumes  $xs \neq []$ 
  shows
    filter ( $\lambda p. P (snd\ p)$ ) (zip  $xs$  (rev [0.. $length\ xs$ ]))
    = [( $xs!(length\ xs - (Suc\ i)), i$ ) .  $i \leftarrow filter\ P\ (rev\ [0.. $length\ xs$ ])$ ]
  using assms
proof (induction  $xs$ )
  case Nil
  then show ?case by simp
next
  case (Cons  $x\ xs$ )
  then show ?case
proof (cases  $P\ (length\ xs)$ )
  case True
  then have
    filter ( $\lambda p. P (snd\ p)$ ) (zip ( $x \# xs$ ) (rev [0.. $length\ (x \# xs)$ ]))
    = ( $x, (length\ xs)$ ) # (filter ( $\lambda p. P (snd\ p)$ ) (zip  $xs$  (rev [0.. $length\ xs$ ])))
    using Cons.prems by auto
  also from Cons.IH have
    ... = ( $x, (length\ xs)$ ) # map ( $\lambda i. (xs!\ (length\ xs - (Suc\ i)), i)$ ) (filter  $P$  (rev [0.. $length\ xs$ ]))
  by fastforce
  also have
    ... = map ( $\lambda i. ((x \# xs)!\ (length\ (x \# xs) - Suc\ i), i)$ ) (filter  $P$  (rev [0.. $length\ (x \# xs)$ ]))
  using Cons.prems True by simp
  finally show ?thesis .
next
  case False
  then have
    filter ( $\lambda p. P (snd\ p)$ ) (zip ( $x \# xs$ ) (rev [0.. $length\ (x \# xs)$ ]))
    = filter ( $\lambda p. P (snd\ p)$ ) (zip  $xs$  (rev [0.. $length\ xs$ ]))
    using Cons.prems by auto
  also from Cons.IH
  have ... = map ( $\lambda i. (xs!\ (length\ xs - (Suc\ i)), i)$ ) (filter  $P$  (rev [0.. $length\ xs$ ]))
  by fastforce
  also have ... = map ( $\lambda i. ((x \# xs)!\ (length\ (x \# xs) - Suc\ i), i)$ ) (filter  $P$  (rev [0.. $length\ (x \# xs)$ ]))
  using Cons.prems False by simp
  finally show ?thesis .
qed
qed

```

```

lemma length-filter-zip-length-rev:
  length (filter ( $\lambda p. P (snd\ p)$ ) (zip  $xs$  ([0.. $length\ xs$ ])))
  = length (filter ( $\lambda p. P (snd\ p)$ ) (zip  $xs$  (rev [0.. $length\ xs$ ])))
proof (cases  $xs$ )
  case Nil

```

```

    then show ?thesis by simp
next
case (Cons a list)
then have length (map (λi. (xs ! i, i)) (filter P [0..zip-lcomp[of xs P] filter-zip-rev-lcomp[of xs P]
    by fastforce
qed

```

The following two lemmas are example of what can be easily proven if we use *rev* $[0.. $\text{length } xs]$$

```

lemma nth-pred-even-length-eq:
  length (nth-pred xs even) = (Suc (length xs)) div 2
proof -
  have length (map fst (filter (λp. even (snd p)) (zip xs (rev [0..zip-length-rev[of even xs] by (simp add: nth-def)
qed

```

```

lemma nth-pred-odd-length-eq:
  length (nth-pred xs odd) = (length xs) div 2
proof -
  have length (map fst (filter (λp. odd (snd p)) (zip xs (rev [0..zip-length-rev[of odd xs] by (simp add: nth-def)
qed

```

```

lemma nth-pred-even-length-less:
  assumes length xs > 1
  shows length (nth-pred xs even) < length xs
  using assms nth-pred-even-length-eq[of xs]
  by linarith

```

```

lemma nth-pred-odd-length-less:
  assumes length xs > 1
  shows length (nth-pred xs odd) < length xs
  using assms nth-pred-odd-length-eq[of xs]
  by linarith

```

Basic polynomial definitions definition *XP*

where

$$XP\ x \equiv \text{monom } x\ 1$$

definition *CP*

where

$$CP\ x \equiv \text{monom } x\ 0$$

definition *prod*

where

$prod\ xs \equiv fold\ (\lambda s. (*)\ (XP\ 1 - CP\ s))\ xs\ 1$

1.9.2 Stark Specifications

definition *degrees*

where $degrees\ clength\ spec = insert\ 0\ \{d * (clength - 1) - length\ rs \mid a\ rs\ d. (a, rs, d) \in set\ spec\}$

definition *exact-order* :: $'a::monoid-mult \Rightarrow nat \Rightarrow bool$

where

$exact-order\ a\ n \longleftrightarrow$

$0 < n \wedge a \wedge^n = 1 \wedge (\forall m. 0 < m \wedge m < n \longrightarrow a \wedge^m \neq 1)$

locale *stark* =

protocol-channel-compatible +

constrains $\omega::'f::proth-field$

and $shift::'f$

and $g::'f$

and $h::'f$

and $vals::'f\ list$

and $concat::'f \Rightarrow 'f \Rightarrow 'f$

fixes $\omega::'f::proth-field$ — This is assumed to be a generator of $'f$

and $shift::'f$ — This shifts H to the evaluation-domain coset.

and $scale::nat$ — This is the scaling or blow-up factor

and $clength::nat$

— The length of the computation, it does need to be a power of two or else needs to be padded to the next power of 2

and $powers::nat$

— The number of powers of the generator required for the spec $[0..<powers]$. This is basically the size of the list which is passed to the first element of *spec*

and $to-nat::'f \Rightarrow nat$

and $of-nat::nat \Rightarrow 'f$

— Functions to transfer from $'f$ to the field and back

and $size::nat$

— The size of $'f$

and $spec::('f\ poly\ list \Rightarrow 'f\ poly) \times nat\ list \times nat)\ list$

— The specification consists of a list of three elements: A constraint polynomial, i.e., a function which creates a polynomial from a list of polynomials (the input here is assumed to be the trace polynomial scaled by powers of the generator); A list of roots for a given constraint polynomial; The degree of the constraint polynomial (actually this could be computed but for now we give it explicitly)

and $rounds::nat$

assumes $size-card: size = CARD('f)$ — This is just there since CARD cannot be computed

and $clength-pow: \exists k. clength = 2 \wedge^k$

— We assume the trace has correct length. In reality this is not required since we can just add padding.

and $scale-pow: \exists k. scale = 2 \wedge^k$ — This needs to be a power of two to make definition of h

well-defined

and *omega-order*: *exact-order omega (size - 1)*
 — *omega* is a primitive generator of the multiplicative group.
and *eval-domain-dvd*: *clength * scale dvd size - 1*
and *eval-domain-nontrivial*: *1 < clength * scale*
and *of-nat-to-nat*: $\forall x. \text{of-nat} (\text{to-nat } x) = x$
and *to-nat-range*:

range to-nat = {0..<size}

— Exact-uniformity assumption for the current query sampler, which samples a field element and maps it through *to-nat* and modulo reduction. A direct sampler over the query sample space, or a rejection sampler with a proved distribution, would remove this divisibility condition from the protocol locale.

and *powers-le-clength*: *powers ≤ clength*

and *powers-pos*: *0 < powers*

and *query-sample-space-size-dvd*:

*clength * scale - Max (set [0..<powers]) * scale dvd size*

— Exact-uniformity assumption for modulo reduction into the current query sample space.

It is specific to the current field-element encoding based sampler.

and *shift-nonzero*: *shift ≠ 0*

and *two-nonzero*: *(2::'f) ≠ 0*

and *spec-degree-fits*:

$\bigwedge c \text{ roots } d.$

$(c, \text{roots}, d) \in \text{set spec} \implies$

$d * (\text{clength} - 1) - \text{length roots} < \text{clength} * \text{scale}$

and *constraint-degree-wellformed-raw*:

$\bigwedge f c \text{ roots } d.$

$\text{degree } f < \text{clength} \implies$

$(c, \text{roots}, d) \in \text{set spec} \implies$

degree

$(c \text{ (map}$

$(\lambda p. f \circ_p XP ((\text{omega} \wedge ((\text{size} - 1) \text{ div } \text{clength})) \wedge p))$

$[0..<\text{powers}]))$

$\leq d * (\text{clength} - 1)$

and *spec-roots-in-range*:

$\bigwedge c \text{ roots } d \text{ } r.$

$(c, \text{roots}, d) \in \text{set spec} \implies$

$r \in \text{set roots} \implies$

$r < \text{clength}$

and *spec-roots-distinct*:

$\bigwedge c \text{ roots } d.$

$(c, \text{roots}, d) \in \text{set spec} \implies$

distinct roots

and *shift-notin-H-raw*:

$\text{shift} \notin$

$\text{set (map ((\wedge) (\text{omega} \wedge ((\text{size} - 1) \text{ div } (\text{clength} * \text{scale}))))$

$[0..<\text{scale} * \text{clength}]$

and *to-nat-of-nat-bounded*: $\bigwedge n. n \leq \text{Max} (\text{degrees clength spec}) \implies \text{to-nat} (\text{of-nat } n) = n$

begin


```

Lemmas lemma scale-pos:  $0 < scale$ 
  using scale-pow
  by auto

lemma clength-pos:  $0 < clength$ 
  using clength-pow by auto

lemma eval-domain-size-pos:  $0 < clength * scale$ 
  using eval-domain-nontrivial by linarith

lemma of-nat-to-nat-field:
  of-nat (to-nat x) = x
  using of-nat-to-nat by simp

lemma to-nat-inj:
  inj to-nat
proof (rule injI)
  fix x y
  assume eq: to-nat x = to-nat y
  have of-nat (to-nat x) = of-nat (to-nat y)
    using eq by simp
  then show x = y
    using of-nat-to-nat-field by simp
qed

lemma finite-range-to-nat[simp]:
  finite (range to-nat)
  by simp

lemma card-range-to-nat:
  card (range to-nat) = size
proof -
  have range-eq: range to-nat = to-nat ' (UNIV :: 'f set)
    by blast
  have inj-univ: inj-on to-nat (UNIV :: 'f set)
    using to-nat-inj by (simp add: inj-on-def inj-def)
  have card-img:
    card (to-nat ' (UNIV :: 'f set)) = card (UNIV :: 'f set)
    by (rule card-image[OF inj-univ])
  have card-univ: card (UNIV :: 'f set) = size
    using size-card by simp
  show ?thesis
    using range-eq card-img card-univ by simp
qed

lemma eval-domain-length-power:  $\exists n. clength * scale = 2 \wedge n$ 
proof -
  obtain k l where clength =  $2 \wedge k$  and scale =  $2 \wedge l$ 
    using clength-pow scale-pow by blast

```

```

then have clength * scale = 2 ^ (k + l)
  by (simp add: power-add)
then show ?thesis
  by blast
qed

```

```

lemma omega-power-size-minus-one:
  omega ^ (size - 1) = 1
using omega-order unfolding exact-order-def by simp

```

```

lemma omega-nonzero-derived:
  omega ≠ 0
proof
  assume omega-zero: omega = 0
  have size-pos: 0 < size - 1
    using omega-order unfolding exact-order-def by simp
  then obtain n where size-eq: size - 1 = Suc n
    by (cases size - 1) auto
  have (0::'f) = 1
    using omega-power-size-minus-one omega-zero size-eq by simp
  then show False
    by simp
qed

```

Basic definitions G is a subgroup of $'f^x$ (the multiplicative group of $'f$). It is generated by generator g and contains exactly as many elements as the length of the (padded) execution trace.

```

definition g
where
  g ≡ omega ^ ((size - 1) div clength)

```

```

definition trace-powers-of :: 'f poly ⇒ 'f poly list
where trace-powers-of f = map (λp. f ∘p XP (g ^ p)) [0..<powers]

```

```

lemma trace-powers-of-length[simp]:
  length (trace-powers-of f) = powers
unfolding trace-powers-of-def by simp

```

```

lemma constraint-degree-wellformed:
  — This locale fact records a syntactic degree side condition for spec. A stronger syntactic
  degree-analysis interface could derive it in a later refinement of the specification layer.
  assumes degree f < clength
  and (c, roots, d) ∈ set spec
  shows degree (c (trace-powers-of f)) ≤ d * (clength - 1)
  using constraint-degree-wellformed-raw[OF assms]
  unfolding trace-powers-of-def g-def .

```

```

definition G

```

where

$G = \text{map } ((\wedge) g) [0..<clength]$

H is again subgroup of f^x . Its size is the size of G scaled by the scaling factor $scale$. It is generated by generator h .

definition h

where

$h \equiv \text{omega}^\wedge((size - 1) \text{ div } (clength * scale))$

definition H

where

$H = \text{map } ((\wedge) h) [0..<scale * clength]$ — Since both scale and clength are powers of two their product is a power of two as well

lemma *shift-notin-H*:

$shift \notin \text{set } H$

using *shift-notin-H-raw*

unfolding $H\text{-def } h\text{-def}$.

We generate a Coset(<https://en.wikipedia.org/wiki/Coset>) of H for our evaluation domain by multiplying each element with a nonzero shift.

definition *eval-domain*

where

$\text{eval-domain} = \text{map } ((* \text{ shift}) H$

This function creates the indices which need to be evaluated for the de-commitment phase. Note that the verifier needs to obtain values of f for all the powers in powers to compute the constraint polynomial. Also, in the flat representation a power actually means that the index is multiplied by the scale factor.

definition *powers-scaled*

where

$\text{powers-scaled } idx \equiv \text{map } (\lambda x. idx + (x * scale)) [0 ..< powers]$

definition *index*

where

$\text{index } x = x \text{ mod } ((clength * scale) - ((\text{Max } (\text{set } [0 ..< powers])) * scale))$

definition *g-map*

where

$g\text{-map } rs = (\text{map } ((\wedge) g) rs)$

lemma *domain-alignment*:

$g = h^\wedge scale$

proof —

let $?N = clength * scale$

obtain q **where** $q\text{-def: } size - 1 = ?N * q$

using *eval-domain-dvd* **unfolding** *dvd-def* **by** *blast*

have *div-mult*:

```

  (size - 1) div clength = ((size - 1) div (clength * scale)) * scale
  using q-def scale-pos clength-pos
  by (simp add: mult.assoc)
  have h ^ scale = omega ^ (((size - 1) div (clength * scale)) * scale)
  unfolding h-def by (simp add: power-mult)
  also have ... = omega ^ ((size - 1) div clength)
  using div-mult by simp
  finally show ?thesis
  unfolding g-def by simp
qed

```

```

lemma g-exact-order:
  exact-order g clength
proof -
  let ?N = clength
  let ?M = size - 1
  have dvd-clength: clength dvd ?M
  using eval-domain-dvd by (meson dvd-mult-left dvd-trans)
  obtain q where q-def: ?M = ?N * q
  using dvd-clength unfolding dvd-def by blast
  have N-pos: 0 < ?N
  using clength-pos .
  have q-pos: 0 < q
  using q-def omega-order unfolding exact-order-def by (cases q) auto
  have gN: g ^ ?N = 1
  proof -
    have g ^ ?N = omega ^ (((size - 1) div ?N) * ?N)
    unfolding g-def by (simp add: power-mult)
    also have ... = omega ^ (size - 1)
    using q-def N-pos by (simp add: mult.commute)
    also have ... = 1
    by (rule omega-power-size-minus-one)
    finally show ?thesis .
  qed

```

```

qed
have no-smaller:  $\bigwedge m. 0 < m \wedge m < ?N \implies g ^ m \neq 1$ 
proof
  fix m
  assume m-bounds: 0 < m  $\wedge$  m < ?N
  assume gm: g ^ m = 1
  have exp-lt: q * m < ?M
  using m-bounds q-def q-pos by (simp add: mult-less-mono2)
  have exp-pos: 0 < q * m
  using m-bounds q-pos by simp
  have g-eq: g = omega ^ q
  using q-def N-pos unfolding g-def by simp
  have omega ^ (q * m) = 1
  using gm g-eq by (simp add: power-mult)
  then show False
  using omega-order exp-pos exp-lt unfolding exact-order-def by blast

```

```

qed
show ?thesis
  unfolding exact-order-def using N-pos gN no-smaller by blast
qed

lemma g-nonzero:
  g ≠ 0
  unfolding g-def using omega-nonzero-derived by simp

lemma g-power-inj-on-range:
  assumes i-bound: i < clength
    and j-bound: j < clength
    and eq: g ^ i = g ^ j
  shows i = j
proof (rule ccontr)
  assume neg: i ≠ j
  consider i < j | j < i
    using neg by linarith
  then show False
proof cases
  case 1
  have diff-pos: 0 < j - i
    using 1 by simp
  have diff-lt: j - i < clength
    using j-bound by simp
  have gi-nonzero: g ^ i ≠ 0
    using g-nonzero by simp
  have j-decomp: j = i + (j - i)
    using 1 by simp
  have mult-eq: g ^ j = g ^ i * g ^ (j - i)
    by (subst j-decomp) (simp add: power-add)
  have cancel-eq: g ^ i * 1 = g ^ i * g ^ (j - i)
    using eq mult-eq by simp
  have g ^ i = 0 ∨ 1 = g ^ (j - i)
    using cancel-eq by (simp only: mult-cancel-left)
  then have g ^ (j - i) = 1
    using g-nonzero by auto
  then show False
    using g-exact-order diff-pos diff-lt unfolding exact-order-def by blast
next
  case 2
  have diff-pos: 0 < i - j
    using 2 by simp
  have diff-lt: i - j < clength
    using i-bound by simp
  have gj-nonzero: g ^ j ≠ 0
    using g-nonzero by simp
  have i-decomp: i = j + (i - j)
    using 2 by simp

```

```

have mult-eq:  $g^i = g^j * g^{(i-j)}$ 
  by (subst i-decomp) (simp add: power-add)
have cancel-eq:  $g^j * 1 = g^j * g^{(i-j)}$ 
  using eq mult-eq by simp
have  $g^j = 0 \vee 1 = g^{(i-j)}$ 
  using cancel-eq by (simp only: mult-cancel-left)
then have  $g^{(i-j)} = 1$ 
  using g-nonzero by auto
then show False
  using g-exact-order diff-pos diff-lt unfolding exact-order-def by blast
qed
qed

```

lemma *h-order*:

$h^{(clength * scale)} = 1$

proof –

let $?N = clength * scale$

obtain q **where** $q\text{-def}$: $size - 1 = ?N * q$

using eval-domain-dvd **unfolding** dvd-def **by** blast

have $h^{?N} = \omega^{((size - 1) \text{ div } ?N) * ?N}$

unfolding h-def **by** (simp add: power-mult)

also have $\dots = \omega^{(size - 1)}$

using q-def eval-domain-size-pos **by** (simp add: mult.commute)

also have $\dots = 1$

by (rule omega-power-size-minus-one)

finally show ?thesis .

qed

lemma *h-exact-order*:

$exact\text{-}order\ h\ (clength * scale)$

proof –

let $?N = clength * scale$

let $?M = size - 1$

obtain q **where** $q\text{-def}$: $?M = ?N * q$

using eval-domain-dvd **unfolding** dvd-def **by** blast

have $N\text{-pos}$: $0 < ?N$

using eval-domain-size-pos .

have $q\text{-pos}$: $0 < q$

using q-def omega-order **unfolding** exact-order-def **by** (cases q) auto

have hN : $h^{?N} = 1$

by (rule h-order)

have no-smaller: $\bigwedge m. 0 < m \wedge m < ?N \implies h^m \neq 1$

proof

fix m

assume $m\text{-bounds}$: $0 < m \wedge m < ?N$

assume hm : $h^m = 1$

have $exp\text{-lt}$: $q * m < ?M$

using m-bounds q-def q-pos **by** (simp add: mult-less-mono2)

have $exp\text{-pos}$: $0 < q * m$

```

    using m-bounds q-pos by simp
  have h-eq:  $h = \omega^q$ 
    using q-def N-pos unfolding h-def by simp
  have  $\omega^{q * m} = 1$ 
    using hm h-eq by (simp add: power-mult)
  then show False
    using omega-order exp-pos exp-lt unfolding exact-order-def by blast
qed
show ?thesis
  unfolding exact-order-def using N-pos hN no-smaller by blast
qed

```

lemma *square-eq-one-imp-eq-one-or-neg-one:*

```

  fixes  $x :: 'f$ 
  assumes  $x * x = 1$ 
  shows  $x = 1 \vee x = -1$ 
proof -
  have  $(x - 1) * (x + 1) = 0$ 
    using assms by (simp add: algebra-simps)
  then have  $x - 1 = 0 \vee x + 1 = 0$ 
    by (simp add: mult-eq-0-iff)
  then show ?thesis
  proof
    assume  $x - 1 = 0$ 
    then show ?thesis by simp
  next
    assume  $x + 1 = 0$ 
    then have  $x = -1$ 
      by (simp add: add-eq-0-iff2)
    then show ?thesis by simp
  qed
qed

```

lemma *h-half-order:*

```

   $h^{((clength * scale) \text{ div } 2)} = -1$ 
proof -
  let  $?N = clength * scale$ 
  obtain  $k \ l$  where clength-eq:  $clength = 2^k$  and scale-eq:  $scale = 2^l$ 
    using clength-pow scale-pow by blast
  have N-power:  $?N = 2^{(k + l)}$ 
    using clength-eq scale-eq by (simp add: power-add)
  have N-gt-one:  $1 < ?N$ 
    using eval-domain-nontrivial .
  then have exp-pos:  $0 < ?N \text{ div } 2$ 
    by simp
  have exp-lt:  $?N \text{ div } 2 < ?N$ 
    using N-gt-one by simp
  have h-exp-ne-one:  $h^{(?N \text{ div } 2)} \neq 1$ 
    using h-exact-order exp-pos exp-lt unfolding exact-order-def by blast

```

```

have square-one: (h ^ (?N div 2)) * (h ^ (?N div 2)) = 1
proof -
  have even-N: even ?N
    using N-power N-gt-one by (cases k + l) simp-all
  have (h ^ (?N div 2)) * (h ^ (?N div 2)) = h ^ (?N div 2 + ?N div 2)
    by (simp add: power-add)
  also have ... = h ^ ?N
  proof -
    have ?N div 2 + ?N div 2 = ?N
      using even-two-times-div-two[OF even-N] by presburger
    then show ?thesis
      by simp
  qed
  also have ... = 1
    by (rule h-order)
  finally show ?thesis .
qed
show ?thesis
  using square-eq-one-imp-eq-one-or-neg-one[OF square-one] h-exp-ne-one by blast
qed

lemma h-power-mod-domain:
  h ^ (k mod (clength * scale)) = h ^ k
proof -
  let ?N = clength * scale
  have order: h ^ ?N = 1
    using h-order .
  have k-eq: k = (k div ?N) * ?N + k mod ?N
    using div-mult-mod-eq[of k ?N] by (simp add: mult.commute)
  have period: h ^ ((k div ?N) * ?N) = 1
  proof -
    have h ^ ((k div ?N) * ?N) = h ^ (?N * (k div ?N))
      by (simp add: mult.commute)
    also have ... = (h ^ ?N) ^ (k div ?N)
      by (simp only: power-mult)
    also have ... = 1
      using order by simp
    finally show ?thesis .
  qed
  have h ^ k = h ^ ((k div ?N) * ?N + k mod ?N)
    using k-eq by simp
  also have ... = h ^ ((k div ?N) * ?N) * h ^ (k mod ?N)
    by (simp add: power-add)
  also have ... = h ^ (k mod ?N)
    using period by simp
  finally show ?thesis
    by simp
qed

```


lemma *h-power-in-H*:

$h \wedge k \in \text{set } H$

proof –

let $?N = \text{clength} * \text{scale}$

have *mod-bound*: $k \bmod ?N < \text{scale} * \text{clength}$

using *eval-domain-size-pos*

by (*simp add: mult.commute*)

have $h \wedge (k \bmod ?N) \in \text{set } H$

unfolding *H-def* using *mod-bound* by *simp*

then show *?thesis*

using *h-power-mod-domain*[*of k*] by *simp*

qed

lemma *eval-coset-disjoint-from-H*:

assumes *idx-bound*: $i < \text{clength} * \text{scale}$

shows $h \wedge i * \text{shift} \notin \text{set } H$

proof

let $?N = \text{clength} * \text{scale}$

assume *in-H*: $h \wedge i * \text{shift} \in \text{set } H$

then obtain *j* where *j-bound*: $j < ?N$

and *eq*: $h \wedge i * \text{shift} = h \wedge j$

unfolding *H-def* by (*auto simp: mult.commute*)

have *h-nonzero*: $h \neq 0$

unfolding *h-def* using *omega-nonzero-derived* by *simp*

have *h-i-nonzero*: $h \wedge i \neq 0$

using *h-nonzero* by *simp*

have *inverse-eq*: $\text{inverse } (h \wedge i) = h \wedge (?N - i)$

proof (*rule inverse-unique*)

have $h \wedge i * h \wedge (?N - i) = h \wedge (i + (?N - i))$

by (*simp add: power-add*)

also have $\dots = h \wedge ?N$

using *idx-bound* by *simp*

also have $\dots = 1$

using *h-order* .

finally show $h \wedge i * h \wedge (?N - i) = 1$.

qed

have *shift-eq*: $\text{shift} = h \wedge j * \text{inverse } (h \wedge i)$

proof –

have *eq'*: $\text{shift} * h \wedge i = h \wedge j$

using *eq* by (*simp only: mult.commute*)

have $\text{shift} = \text{shift} * (h \wedge i * \text{inverse } (h \wedge i))$

using *h-i-nonzero* by *simp*

also have $\dots = \text{shift} * h \wedge i * \text{inverse } (h \wedge i)$

by (*simp add: mult.assoc*)

also have $\dots = h \wedge j * \text{inverse } (h \wedge i)$

using *eq'* by *simp*

finally show *?thesis* .

qed

have $\text{shift} = h \wedge (j + (?N - i))$

```

    using shift-eq inverse-eq by (simp add: power-add)
  then have shift  $\in$  set H
    using h-power-in-H[of j + (?N - i)] by simp
  then show False
    using shift-notin-H by contradiction
qed

lemma clength-lt-size:
  clength < size
proof -
  have size-minus-pos:  $0 < \text{size} - 1$ 
    using omega-order unfolding exact-order-def by simp
  have size-minus-ge:  $\text{clength} * \text{scale} \leq \text{size} - 1$ 
    using eval-domain-dvd eval-domain-size-pos size-minus-pos
    by (meson dvd-imp-le)
  have  $\text{clength} \leq \text{clength} * \text{scale}$ 
    using scale-pos by simp
  also have  $\dots \leq \text{size} - 1$ 
    using size-minus-ge .
  finally show ?thesis
    using size-minus-pos by linarith
qed

definition maxDegree
  where maxDegree = Max (degrees clength spec)

lemma maxDegree-code[code]:
maxDegree = foldl ( $\lambda x \ (-,rs,d). \text{max } x \ (d * (\text{clength} - 1) - \text{length } rs)$ ) 0 spec
proof -
  let ?deg =  $\lambda(-, rs, d). d * (\text{clength} - 1) - \text{length } rs$ 
  have Max-insert-max:
     $\text{finite } A \implies \text{Max } (\text{insert } (\text{max } x \ y) \ A) = \text{Max } (\text{insert } x \ (\text{insert } y \ A))$ 
    for A :: nat set and x y
    by (cases A = {}) (auto)
  have deg-set:  $\{d * (\text{clength} - 1) - \text{length } rs \mid a \ rs \ d. (a,rs,d) \in \text{set spec}\} =$ 
    ?deg ' set spec
    by force
  have aux: foldl ( $\lambda x \ s. \text{max } x \ (\text{?deg } s)$ ) n spec = Max (insert n (?deg ' set spec))
    for n
  proof (induction spec arbitrary: n)
    case Nil
    then show ?case by simp
  next
    case (Cons s spec)
    then show ?case
      by (simp add: Max-insert-max)
  qed
  from aux[of 0] show ?thesis
    unfolding maxDegree-def using deg-set by (simp add: degrees-def split-beta')

```

qed

lemma *maxDegree-less-eval-domain*:

maxDegree < *clength* * *scale*

proof –

have *finite-degrees*: *finite* (*degrees clength spec*)

proof –

have $\{d * (clength - 1) - length\ roots \mid c\ roots\ d.\ (c, roots, d) \in set\ spec\} =$
 $(\lambda(c, roots, d). d * (clength - 1) - length\ roots)\ 'set\ spec$

by *force*

then show *?thesis*

unfolding *degrees-def* **by** *simp*

qed

have *nonempty-degrees*: *degrees clength spec* $\neq \{\}$

unfolding *degrees-def* **by** *simp*

have *all-lt*: $\bigwedge x. x \in degrees\ clength\ spec \implies x < clength * scale$

unfolding *degrees-def*

using *eval-domain-size-pos spec-degree-fits*

by *force*

then show *?thesis*

unfolding *maxDegree-def*

using *finite-degrees nonempty-degrees*

by (*simp add: Max-less-iff*)

qed

lemma *g-map-distinct*:

assumes *spec-entry*: $(c, roots, d) \in set\ spec$

shows *distinct* (*g-map roots*)

proof –

have *roots-distinct*: *distinct roots*

by (*rule spec-roots-distinct[OF spec-entry]*)

have *roots-in-range*: $\bigwedge r. r \in set\ roots \implies r < clength$

by (*rule spec-roots-in-range[OF spec-entry]*)

have *inj*: *inj-on* $((\bigwedge) g)$ (*set roots*)

using *roots-in-range* **by** (*auto intro!: inj-onI dest!: g-power-inj-on-range*)

show *?thesis*

unfolding *g-map-def*

using *roots-distinct inj* **by** (*simp add: distinct-map*)

qed

lemma *g-powers-in-H*:

assumes *r-bound*: $r < clength$

shows $g \wedge r \in set\ H$

proof –

have *idx-bound*: $scale * r < scale * clength$

using *r-bound scale-pos* **by** *simp*

have $g \wedge r = h \wedge (scale * r)$

using *domain-alignment* **by** (*simp add: power-mult*)

then show *?thesis*

unfolding H -def using idx -bound by simp
qed

lemma g -map-subset- H :
 assumes $spec$ -entry: $(c, roots, d) \in set\ spec$
 shows $set\ (g\text{-map}\ roots) \subseteq set\ H$
 unfolding g -map-def
 using $spec$ -roots-in-range[OF $spec$ -entry] g -powers-in- H
 by auto

lemma $query$ -domain-disjoint:
 assumes $spec$ -entry: $(c, roots, d) \in set\ spec$
 and idx -bound: $i < clength * scale$
 shows $h \wedge i * shift \notin set\ (g\text{-map}\ roots)$
 using $eval$ -coset-disjoint-from- H [OF idx -bound] g -map-subset- H [OF $spec$ -entry]
 by auto

Composition polynomial The composition polynomial is a random linear combination of the constraint polynomials divided by their roots. Note that in reality we don't compute the product polynomial as here. Instead we use a trick to optimize the computation (described here under Succinctness: <https://medium.com/starkware/arithmeticization-ii-403c3b3f4355>) Also note that it needs to be parametrized by f since the prover uses it with the real f whereas the verifier uses another f . Maybe we need different versions for prover and verifier later on.

definition cp
 where
 $cp\ as\ fs \equiv fold\ (\lambda(a, (p, rs, -)). (+) ((CP\ a) * ((p\ fs)\ div\ (prod\ (g\text{-map}\ rs))))\ (zip\ as\ spec)\ 0$
 end

1.9.3 Prover

Only the prover knows the actual execution trace.

locale $prover = stark +$
constrains $omega :: 'f :: proth$ -field
 and $shift :: 'f$
 and $concat :: 'f \Rightarrow 'f \Rightarrow 'f$
 and $spec :: (('f\ poly\ list \Rightarrow 'f\ poly) \times nat\ list \times nat)\ list$
fixes $vals :: 'f\ list$ — This is the actual computation trace

assumes $trace$ -length: $clength = length\ vals$

begin

Trace Polynomial The interpolated polynomial from the trace over G . We do it using Lagrange interpolation. In practice it is usually done using IFFT (inverse FFT – Fast

Fourier) but could not find it in the AFP.

definition *f*

where

f $\equiv$ *lagrange-interpolation-poly* (*zip G vals*)

Now we evaluate the trace polynomial on the larger domain, creating a Reed-Solomon error correction code.

definition *f-eval*

where

f-eval $\equiv$ *map* (*poly f*) *eval-domain*

Similar as the trace polynomial we evaluate the composition polynomial on the larger domain.

definition *f-powers*

where

f-powers $\equiv$ *map* ($\lambda p. f \circ_p XP (g \hat{p})$) [*0* ..< *powers*]

definition *cp-eval*

where

cp-eval as $\equiv$ *map* (*poly* (*cp as f-powers*)) *eval-domain*

FRI Commitments **definition** *next-fri-domain* :: '*f list* $\Rightarrow$ '*f list*

where

next-fri-domain fd $\equiv$ *map* ($\lambda x. x * x$) (*take* ((*length fd*) *div* (2::nat)) *fd*)

definition *next-fri-polynomial* :: '*f poly* $\Rightarrow$ '*f* $\Rightarrow$ '*f poly*

where

next-fri-polynomial p b $\equiv$

let

c = *coeffs p*;

odd-coefficients = *nths-pred c odd*;

even-coefficients = *nths-pred c even*;

odd = *CP b * poly-of-list odd-coefficients*;

even = *poly-of-list even-coefficients*

in odd + even

lemma *degree-Poly-less*:

assumes *degree p* > 0

and *length xs* < *length* (*coeffs p*)

shows *degree* (*Poly xs*) < *degree p*

proof –

from *assms*(2) **have** *length* (*strip-while* (*HOL.eq* 0) *xs*) < *length* (*coeffs p*)

by (*meson dual-order.strict-trans1 length-strip-while-le verit-comp-simplify1* (3))

then have *degree* (*Poly* (*strip-while* (*HOL.eq* 0) *xs*)) < *degree p*

using *length-coeffs*[*of* (*Poly* (*nths-pred* (*coeffs p*) *even*))]

by (*metis Poly-coeffs add-less-cancel-right assms*(1) *coeffs-Poly degree-0 length-coeffs not-gr-zero*)

```

then show  $\text{degree } (\text{Poly } xs) < \text{degree } p$ 
  by (metis Poly-coeffs coeffs-Poly)
qed

lemma degree-less-degree:
  assumes  $\text{degree } p > 0$ 
    and next-fri-polynomial  $p \ b = p'$ 
  shows  $\text{degree } p' < \text{degree } p$ 
proof –
  have *:  $\text{length } (\text{coeffs } p) > 1$  using assms length-coeffs
    by (simp add: degree-eq-length-coeffs)

  define odd-coefficients where  $\text{odd-coefficients} \equiv \text{nths-pred } (\text{coeffs } p) \ \text{odd}$ 
  define oddp where  $\text{oddp} \equiv \text{CP } b * \text{Poly } \text{odd-coefficients}$ 
  define even-coefficients where  $\text{even-coefficients} \equiv \text{nths-pred } (\text{coeffs } p) \ \text{even}$ 
  define evenp where  $\text{evenp} \equiv \text{Poly } \text{even-coefficients}$ 

  from * have  $\text{length } \text{odd-coefficients} < \text{length } (\text{coeffs } p)$ 
    using nths-pred-odd-length-less odd-coefficients-def by simp
  then have  $\text{degree } (\text{Poly } (\text{odd-coefficients})) < \text{degree } p$ 
    using assms(1) degree-Poly-less by metis

  moreover have  $\text{degree } (\text{CP } b) = 0$  unfolding CP-def by (simp add: monom-0)
  ultimately have  $\text{degree } \text{oddp} < \text{degree } p$ 
    unfolding oddp-def using degree-mult-le[of CP b (Poly odd-coefficients)] by simp

  moreover from * have  $\text{length } \text{even-coefficients} < \text{length } (\text{coeffs } p)$ 
    using nths-pred-even-length-less even-coefficients-def by simp
  then have  $\text{degree } (\text{Poly } (\text{even-coefficients})) < \text{degree } p$ 
    using assms(1) degree-Poly-less by metis
  then have  $\text{degree } \text{evenp} < \text{degree } p$  unfolding evenp-def by satx

  moreover have  $p' = \text{oddp} + \text{evenp}$ 
    using assms
    unfolding oddp-def evenp-def odd-coefficients-def even-coefficients-def next-fri-polynomial-def
    by (auto simp add: Let-def)
  ultimately show ?thesis by (simp add: degree-add-less)
qed

definition next-fri-layer::  $'f \ \text{poly} \Rightarrow 'f \ \text{list} \Rightarrow 'f \Rightarrow 'f \ \text{poly} \times 'f \ \text{list} \times 'f \ \text{list}$ 
where
  next-fri-layer  $p \ d \ b \equiv$ 
    let
       $\text{next-poly} = \text{next-fri-polynomial } p \ b;$ 
       $\text{next-domain} = \text{next-fri-domain } d;$ 
       $\text{next-layer} = \text{map } (\text{poly } \text{next-poly}) \ \text{next-domain}$ 
    in (next-poly, next-domain, next-layer)

primrec fri-commit-with ::

```

```

('f, 'f, unit) protocol-c-monad
⇒ nat
⇒ 'f poly list
⇒ 'f list list
⇒ 'f list list
⇒ 'f tree list
⇒ ('f, 'f poly list × 'f list list × 'f list list × 'f tree list, unit) protocol-c-monad
where
  fri-commit-with receive-challenge 0 ps ds ls ms =
    return (ps, ds, ls, ms)
| fri-commit-with receive-challenge (Suc n) ps ds ls ms =
  do {
    send (value (last ms));
    b ← receive-challenge;
    let (next-poly, next-domain, next-layer) =
      next-fri-layer (last ps) (last ds) b;
    m ← create next-layer;
    fri-commit-with receive-challenge n
      (ps @ [next-poly])
      (ds @ [next-domain])
      (ls @ [next-layer])
      (ms @ [m])
  }

```

```

primrec fri-commit ::
  nat
  ⇒ 'f poly list
  ⇒ 'f list list
  ⇒ 'f list list
  ⇒ 'f tree list
  ⇒ ('f, 'f poly list × 'f list list × 'f list list × 'f tree list, unit) protocol-c-monad
where
  fri-commit 0 ps ds ls ms = return (ps, ds, ls, ms)
| fri-commit (Suc n) ps ds ls ms =
  do {
    send (value (last ms));
    b ← receive-random-field-element;
    let (next-poly, next-domain, next-layer) = next-fri-layer (last ps) (last ds) b;
    m ← create next-layer;
    fri-commit n
      (ps @ [next-poly])
      (ds @ [next-domain])
      (ls @ [next-layer])
      (ms @ [m])
  }

```

definition trace-fri-commit

where

trace-fri-commit =

fri-commit-with receive-trace-fri-challenge

definition *composition-fri-commit*

where

composition-fri-commit =
fri-commit-with receive-composition-fri-challenge

lemma *fri-commit-with-receive-random-field-element*:

fri-commit-with receive-random-field-element n ps ds ls ms =
fri-commit n ps ds ls ms

by (*induction n arbitrary: ps ds ls ms*) *simp-all*

lemma *fri-commit-lengths*:

assumes *Some ((ps', ds', ls', ms'), t) ∈*
set-dist (execute (fri-commit n ps ds ls ms) s)

shows

length ps' = length ps + n ∧
length ds' = length ds + n ∧
length ls' = length ls + n ∧
length ms' = length ms + n

using *assms*

proof (*induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t*)

case 0

then show *?case* **by** *simp*

next

case (*Suc n*)

from *Suc.prem*s **obtain** *s1 b s2 next-poly next-domain next-layer m s3* **where**

send-root: Some ((), s1) ∈ set-dist (execute (send (value (last ms)))) s)

and *rand: Some (b, s2) ∈ set-dist (execute receive-random-field-element s1)*

and *next-layer-eq: next-fri-layer (last ps) (last ds) b = (next-poly, next-domain, next-layer)*

and *create-m: Some (m, s3) ∈ set-dist (execute (create next-layer) s2)*

and *rest: Some ((ps', ds', ls', ms'), t) ∈*

set-dist (execute
(fri-commit n
(ps @ [next-poly])
(ds @ [next-domain])
(ls @ [next-layer])
(ms @ [m])) s3)

by (*auto simp: Let-def split: prod.splits elim!: set-dist-bindE*)

show *?case*

using *Suc.IH[OF rest]* **by** *simp*

qed

lemma *fri-commit-extends*:

assumes *Some ((ps', ds', ls', ms'), t) ∈*
set-dist (execute (fri-commit n ps ds ls ms) s)

shows *s ≤ t*

using *assms*

proof (*induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t*)


```

case 0
then show ?case
  using hash-ext-refl by simp
next
case (Suc n)
from Suc.premis obtain s1 b s2 next-poly next-domain next-layer m s3 where
  send-root: Some ((), s1) ∈ set-dist (execute (send (value (last ms)))) s)
  and rand: Some (b, s2) ∈ set-dist (execute receive-random-field-element s1)
  and next-layer-eq: next-fri-layer (last ps) (last ds) b = (next-poly, next-domain, next-layer)
  and create-m: Some (m, s3) ∈ set-dist (execute (create next-layer) s2)
  and rest: Some ((ps', ds', ls', ms'), t) ∈
    set-dist (execute
      (fri-commit n
        (ps @ [next-poly])
        (ds @ [next-domain])
        (ls @ [next-layer])
        (ms @ [m])) s3)
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have s-s1: s ≤ s1
  using send-outcome[OF send-root]
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s1-s2: s1 ≤ s2
  using receive-random-field-element-extends[OF rand] .
have s2-s3: s2 ≤ s3
  using create-outcome[OF create-m] by simp
have s3-t: s3 ≤ t
  using Suc.IH[OF rest] .
show ?case
  using s-s1 s1-s2 s2-s3 s3-t by (meson hash-ext-trans)
qed

```

lemma fri-commit-preserves-prefix:

```

assumes outcome:
  Some ((ps', ds', ls', ms'), t) ∈
    set-dist (execute (fri-commit n ps ds ls ms) s)
shows
  take (length ps) ps' = ps ∧
  take (length ds) ds' = ds ∧
  take (length ls) ls' = ls ∧
  take (length ms) ms' = ms
using outcome
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t)
case 0
  then show ?case by simp
next
case (Suc n)
from Suc.premis obtain s1 b s2 next-poly next-domain next-layer m s3 where
  send-root: Some ((), s1) ∈ set-dist (execute (send (value (last ms)))) s)
  and rand: Some (b, s2) ∈ set-dist (execute receive-random-field-element s1)

```

and *next-layer-eq*: *next-fri-layer* (last *ps*) (last *ds*) *b* = (*next-poly*, *next-domain*, *next-layer*)
and *create-m*: *Some* (*m*, *s3*) ∈ *set-dist* (*execute* (*create next-layer*) *s2*)
and *rest*: *Some* ((*ps'*, *ds'*, *ls'*, *ms'*), *t*) ∈
set-dist (*execute*
(*fri-commit* *n*
(*ps* @ [*next-poly*])
(*ds* @ [*next-domain*])
(*ls* @ [*next-layer*])
(*ms* @ [*m*])) *s3*)
by (*auto simp*: *Let-def split: prod.splits elim!: set-dist-bindE*)
have *ih*: *take* (*length* (*ps* @ [*next-poly*])) *ps'* = *ps* @ [*next-poly*] ∧
take (*length* (*ds* @ [*next-domain*])) *ds'* = *ds* @ [*next-domain*] ∧
take (*length* (*ls* @ [*next-layer*])) *ls'* = *ls* @ [*next-layer*] ∧
take (*length* (*ms* @ [*m*])) *ms'* = *ms* @ [*m*]
using *Suc.IH[OF rest]* .
have *take* (*length ps*) *ps'* = *ps*
proof –
have *take* (*length ps*) *ps'* =
take (*length ps*) (*take* (*length* (*ps* @ [*next-poly*])) *ps'*)
by (*simp add: take-take*)
also have ... = *ps*
using *ih* **by** *simp*
finally show *?thesis* .
qed
moreover have *take* (*length ds*) *ds'* = *ds*
proof –
have *take* (*length ds*) *ds'* =
take (*length ds*) (*take* (*length* (*ds* @ [*next-domain*])) *ds'*)
by (*simp add: take-take*)
also have ... = *ds*
using *ih* **by** *simp*
finally show *?thesis* .
qed
moreover have *take* (*length ls*) *ls'* = *ls*
proof –
have *take* (*length ls*) *ls'* =
take (*length ls*) (*take* (*length* (*ls* @ [*next-layer*])) *ls'*)
by (*simp add: take-take*)
also have ... = *ls*
using *ih* **by** *simp*
finally show *?thesis* .
qed
moreover have *take* (*length ms*) *ms'* = *ms*
proof –
have *take* (*length ms*) *ms'* =
take (*length ms*) (*take* (*length* (*ms* @ [*m*])) *ms'*)
by (*simp add: take-take*)
also have ... = *ms*
using *ih* **by** *simp*

```

    finally show ?thesis .
qed
ultimately show ?case
  by simp
qed

```

lemma *fri-commit-layers-are-evaluations:*

assumes *init:*

$\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$

and *outcome:*

$\text{Some } ((ps', ds', ls', ms'), t) \in$
 $\text{set-dist } (\text{execute } (\text{fri-commit } n \text{ } ps \text{ } ds \text{ } ls \text{ } ms) \text{ } s)$

and *lens:* $\text{length } ps = \text{length } ds \text{ } \text{length } ds = \text{length } ls$

shows

$\bigwedge i. i < \text{length } ps' \implies i < \text{length } ds' \implies i < \text{length } ls' \implies$
 $ls' ! i = \text{map } (\text{poly } (ps' ! i)) (ds' ! i)$

using *init outcome lens*

proof (*induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t i*)

case 0

then show ?case **by** *simp*

next

case (*Suc n*)

from *Suc.prem5* **obtain** *s1 b s2 next-poly next-domain next-layer m s3* **where**

send-root: $\text{Some } ((), s1) \in \text{set-dist } (\text{execute } (\text{send } (\text{value } (\text{last } ms)))) \text{ } s)$

and *rand:* $\text{Some } (b, s2) \in \text{set-dist } (\text{execute } \text{receive-random-field-element } s1)$

and *next-layer-eq:* $\text{next-fri-layer } (\text{last } ps) (\text{last } ds) \text{ } b = (\text{next-poly}, \text{next-domain}, \text{next-layer})$

and *create-m:* $\text{Some } (m, s3) \in \text{set-dist } (\text{execute } (\text{create next-layer}) \text{ } s2)$

and *rest:* $\text{Some } ((ps', ds', ls', ms'), t) \in$

$\text{set-dist } (\text{execute}$
 $(\text{fri-commit } n$
 $(ps @ [\text{next-poly}])$
 $(ds @ [\text{next-domain}])$
 $(ls @ [\text{next-layer}])$
 $(ms @ [m])) \text{ } s3)$

by (*auto simp: Let-def split: prod.splits elim!: set-dist-bindE*)

have *next-eval:* $\text{next-layer} = \text{map } (\text{poly next-poly}) \text{ next-domain}$

using *next-layer-eq* **unfolding** *next-fri-layer-def* **by** (*auto simp: Let-def split: prod.splits*)

have *init':*

$\bigwedge i. i < \text{length } (ps @ [\text{next-poly}]) \implies$
 $i < \text{length } (ds @ [\text{next-domain}]) \implies$
 $i < \text{length } (ls @ [\text{next-layer}]) \implies$
 $(ls @ [\text{next-layer}]) ! i =$
 $\text{map } (\text{poly } ((ps @ [\text{next-poly}]) ! i)) ((ds @ [\text{next-domain}]) ! i)$

proof –

fix *i*

assume *i-ps:* $i < \text{length } (ps @ [\text{next-poly}])$

and *i-ds:* $i < \text{length } (ds @ [\text{next-domain}])$

and *i-ls:* $i < \text{length } (ls @ [\text{next-layer}])$

```

show (ls @ [next-layer]) ! i =
  map (poly ((ps @ [next-poly]) ! i)) ((ds @ [next-domain]) ! i)
proof (cases i < length ps)
  case True
  then show ?thesis
    using Suc.premis(4)[of i] Suc.premis(6,7)
    by (simp add: nth-append)
  next
  case False
  then have i-eq: i = length ps
    using i-ps by simp
  then show ?thesis
    using Suc.premis(6,7) next-eval by (simp add: nth-append)
  qed
qed
show ?case
  using Suc.IH[OF - - init' rest] Suc.premis(1-3,6,7) by simp
qed

```

lemma fri-commit-created-trees:

```

assumes init:
   $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$ 
    created-tree (ls ! i) (ms ! i) (s::f protocol-channel)
and outcome:
  Some ((ps', ds', ls', ms'), t)  $\in$ 
    set-dist (execute (fri-commit n ps ds ls ms) s)
and lens: length ls = length ms
shows  $\bigwedge i. i < \text{length } ls' \implies i < \text{length } ms' \implies$ 
  created-tree (ls' ! i) (ms' ! i) t
using init outcome lens
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t i)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.premis(4) obtain s1 b s2 next-poly next-domain next-layer m s3 where
    send-root: Some ((), s1)  $\in$  set-dist (execute (send (value (last ms))) s)
    and rand: Some (b, s2)  $\in$  set-dist (execute receive-random-field-element s1)
    and next-layer-eq: next-fri-layer (last ps) (last ds) b = (next-poly, next-domain, next-layer)
    and create-m: Some (m, s3)  $\in$  set-dist (execute (create next-layer) s2)
    and rest: Some ((ps', ds', ls', ms'), t)  $\in$ 
      set-dist (execute
        (fri-commit n
          (ps @ [next-poly])
          (ds @ [next-domain])
          (ls @ [next-layer])
          (ms @ [m])) s3)
    by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
  have s-s1: s  $\leq$  s1

```

```

using send-outcome[OF send-root]
unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s1-s2:  $s1 \leq s2$ 
  using receive-random-field-element-extends[OF rand] .
have s2-s3:  $s2 \leq s3$ 
  using create-outcome[OF create-m] by simp
have s-s3:  $s \leq s3$ 
  using s-s1 s1-s2 s2-s3 by (meson hash-ext-trans)
have created-m: created-tree next-layer m s3
  using create-outcome[OF create-m] by simp
have init':
   $\bigwedge i. i < \text{length } (ls @ [next-layer]) \implies$ 
     $i < \text{length } (ms @ [m]) \implies$ 
    created-tree ((ls @ [next-layer]) ! i) ((ms @ [m]) ! i) s3
proof -
  fix i
  assume i-ls:  $i < \text{length } (ls @ [next-layer])$ 
  and i-ms:  $i < \text{length } (ms @ [m])$ 
  show created-tree ((ls @ [next-layer]) ! i) ((ms @ [m]) ! i) s3
  proof (cases  $i < \text{length } ls$ )
    case True
    then have created-tree (ls ! i) (ms ! i) s
      using Suc.premis(3)[of i] Suc.premis(5) by simp
    then show ?thesis
      using True Suc.premis(5) s-s3
      by (simp add: nth-append created-tree-mono)
    next
    case False
    then have  $i = \text{length } ls$ 
      using i-ls by simp
    then show ?thesis
      using Suc.premis(5) created-m by (simp add: nth-append)
  qed
qed
show ?case
  using Suc.IH[OF Suc.premis(1,2) init' rest] Suc.premis(5) by simp
qed

```

lemma fri-commit-with-lengths:

```

assumes Some ((ps', ds', ls', ms'), t) ∈
  set-dist (execute (fri-commit-with receive-challenge n ps ds ls ms) s)
shows
  length ps' = length ps + n ∧
  length ds' = length ds + n ∧
  length ls' = length ls + n ∧
  length ms' = length ms + n
using assms
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t)
  case 0

```

then show ?case by simp
 next
 case (Suc n)
 from Suc.premis obtain s1 b s2 next-poly next-domain next-layer m s3 where
 send-root: Some ((), s1) ∈ set-dist (execute (send (value (last ms)))) s
 and rand: Some (b, s2) ∈ set-dist (execute receive-challenge s1)
 and next-layer-eq:
 next-fri-layer (last ps) (last ds) b =
 (next-poly, next-domain, next-layer)
 and create-m: Some (m, s3) ∈ set-dist (execute (create next-layer) s2)
 and rest: Some ((ps', ds', ls', ms'), t) ∈
 set-dist (execute
 (fri-commit-with receive-challenge n
 (ps @ [next-poly])
 (ds @ [next-domain])
 (ls @ [next-layer])
 (ms @ [m])) s3)
 by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
 show ?case
 using Suc.IH[OF rest] by simp
 qed

lemma fri-commit-with-extends:

assumes receiver-extends:

$\bigwedge b s t. \text{Some } (b, t) \in \text{set-dist } (\text{execute receive-challenge } s) \implies$
 $s \leq t$

and outcome:

Some ((ps', ds', ls', ms'), t) ∈
 set-dist (execute (fri-commit-with receive-challenge n ps ds ls ms) s)

shows $s \leq t$

using outcome

proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t)

case 0

then show ?case

using hash-ext-refl by simp

next

case (Suc n)

from Suc.premis obtain s1 b s2 next-poly next-domain next-layer m s3 where

send-root: Some ((), s1) ∈ set-dist (execute (send (value (last ms)))) s

and rand: Some (b, s2) ∈ set-dist (execute receive-challenge s1)

and next-layer-eq:

next-fri-layer (last ps) (last ds) b =
 (next-poly, next-domain, next-layer)

and create-m: Some (m, s3) ∈ set-dist (execute (create next-layer) s2)

and rest: Some ((ps', ds', ls', ms'), t) ∈

set-dist (execute
 (fri-commit-with receive-challenge n
 (ps @ [next-poly])
 (ds @ [next-domain])

```

      (ls @ [next-layer])
      (ms @ [m])) s3)
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have s-s1: s ≤ s1
  using send-outcome[OF send-root]
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s1-s2: s1 ≤ s2
  using receiver-extends[OF rand] .
have s2-s3: s2 ≤ s3
  using create-outcome[OF create-m] by simp
have s3-t: s3 ≤ t
  using Suc.IH[OF rest] .
show ?case
  using s-s1 s1-s2 s2-s3 s3-t by (meson hash-ext-trans)
qed

lemma fri-commit-with-preserves-prefix:
  assumes outcome:
    Some ((ps', ds', ls', ms'), t) ∈
      set-dist (execute (fri-commit-with receive-challenge n ps ds ls ms) s)
  shows
    take (length ps) ps' = ps ∧
    take (length ds) ds' = ds ∧
    take (length ls) ls' = ls ∧
    take (length ms) ms' = ms
  using outcome
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.premis obtain s1 b s2 next-poly next-domain next-layer m s3 where
    send-root: Some ((), s1) ∈ set-dist (execute (send (value (last ms)))) s)
    and rand: Some (b, s2) ∈ set-dist (execute receive-challenge s1)
    and next-layer-eq:
      next-fri-layer (last ps) (last ds) b =
        (next-poly, next-domain, next-layer)
    and create-m: Some (m, s3) ∈ set-dist (execute (create next-layer) s2)
    and rest: Some ((ps', ds', ls', ms'), t) ∈
      set-dist (execute
        (fri-commit-with receive-challenge n
          (ps @ [next-poly])
          (ds @ [next-domain])
          (ls @ [next-layer])
          (ms @ [m])) s3)
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have ih: take (length (ps @ [next-poly])) ps' = ps @ [next-poly] ∧
  take (length (ds @ [next-domain])) ds' = ds @ [next-domain] ∧
  take (length (ls @ [next-layer])) ls' = ls @ [next-layer] ∧

```

```

    take (length (ms @ [m])) ms' = ms @ [m]
  using Suc.IH[OF rest] .
have take (length ps) ps' = ps
proof -
  have take (length ps) ps' =
    take (length ps) (take (length (ps @ [next-poly])) ps')
  by (simp add: take-take)
  also have ... = ps
  using ih by simp
  finally show ?thesis .
qed
moreover have take (length ds) ds' = ds
proof -
  have take (length ds) ds' =
    take (length ds) (take (length (ds @ [next-domain])) ds')
  by (simp add: take-take)
  also have ... = ds
  using ih by simp
  finally show ?thesis .
qed
moreover have take (length ls) ls' = ls
proof -
  have take (length ls) ls' =
    take (length ls) (take (length (ls @ [next-layer])) ls')
  by (simp add: take-take)
  also have ... = ls
  using ih by simp
  finally show ?thesis .
qed
moreover have take (length ms) ms' = ms
proof -
  have take (length ms) ms' =
    take (length ms) (take (length (ms @ [m])) ms')
  by (simp add: take-take)
  also have ... = ms
  using ih by simp
  finally show ?thesis .
qed
ultimately show ?case
by simp
qed

lemma fri-commit-with-layers-are-evaluations:
  assumes init:
     $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
     $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
  and outcome:
    Some ((ps', ds', ls', ms'), t)  $\in$ 
    set-dist (execute (fri-commit-with receive-challenge n ps ds ls ms) s)

```



```

    and lens: length ps = length ds length ds = length ls
  shows
     $\bigwedge i. i < \text{length } ps' \implies i < \text{length } ds' \implies i < \text{length } ls' \implies$ 
     $ls' ! i = \text{map } (\text{poly } (ps' ! i)) (ds' ! i)$ 
  using init outcome lens
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t i)
  case 0
  then show ?case by simp
next
case (Suc n)
from Suc.prem5 obtain s1 b s2 next-poly next-domain next-layer m s3 where
  send-root: Some ((), s1)  $\in$  set-dist (execute (send (value (last ms)))) s)
  and rand: Some (b, s2)  $\in$  set-dist (execute receive-challenge s1)
  and next-layer-eq:
    next-fri-layer (last ps) (last ds) b =
      (next-poly, next-domain, next-layer)
  and create-m: Some (m, s3)  $\in$  set-dist (execute (create next-layer) s2)
  and rest: Some ((ps', ds', ls', ms'), t)  $\in$ 
    set-dist (execute
      (fri-commit-with receive-challenge n
        (ps @ [next-poly])
        (ds @ [next-domain])
        (ls @ [next-layer])
        (ms @ [m])) s3)
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have next-eval: next-layer = map (poly next-poly) next-domain
  using next-layer-eq unfolding next-fri-layer-def
  by (auto simp: Let-def split: prod.splits)
have init':
   $\bigwedge i. i < \text{length } (ps @ [next-poly]) \implies$ 
   $i < \text{length } (ds @ [next-domain]) \implies$ 
   $i < \text{length } (ls @ [next-layer]) \implies$ 
   $(ls @ [next-layer]) ! i =$ 
   $\text{map } (\text{poly } ((ps @ [next-poly]) ! i)) ((ds @ [next-domain]) ! i)$ 
proof -
  fix i
  assume i-ps:  $i < \text{length } (ps @ [next-poly])$ 
  and i-ds:  $i < \text{length } (ds @ [next-domain])$ 
  and i-ls:  $i < \text{length } (ls @ [next-layer])$ 
  show (ls @ [next-layer]) ! i =
    map (poly ((ps @ [next-poly]) ! i)) ((ds @ [next-domain]) ! i)
proof (cases  $i < \text{length } ps$ )
  case True
  then show ?thesis
    using Suc.prem5(4)[of i] Suc.prem5(6,7)
    by (simp add: nth-append)
next
case False
  then have i-eq:  $i = \text{length } ps$ 

```

```

    using i-ps by simp
  then show ?thesis
    using Suc.premis(6,7) next-eval by (simp add: nth-append)
qed
qed
show ?case
  using Suc.IH[OF - - init' rest] Suc.premis(1-3,6,7) by simp
qed

lemma fri-commit-with-created-trees:
  assumes receiver-extends:
     $\bigwedge b\ s\ t. \text{Some } (b, t) \in \text{set-dist } (\text{execute receive-challenge } s) \implies s \leq t$ 
  and init:
     $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies \text{created-tree } (ls ! i) (ms ! i) (s::f \text{ protocol-channel})$ 
  and outcome:
     $\text{Some } ((ps', ds', ls', ms'), t) \in \text{set-dist } (\text{execute } (\text{fri-commit-with receive-challenge } n\ ps\ ds\ ls\ ms)\ s)$ 
  and lens:  $\text{length } ls = \text{length } ms$ 
  shows  $\bigwedge i. i < \text{length } ls' \implies i < \text{length } ms' \implies \text{created-tree } (ls' ! i) (ms' ! i) t$ 
  using init outcome lens
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t i)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.premis(4) obtain s1 b s2 next-poly next-domain next-layer m s3 where
    send-root:  $\text{Some } ((), s1) \in \text{set-dist } (\text{execute } (\text{send } (\text{value } (\text{last } ms))))\ s$ 
    and rand:  $\text{Some } (b, s2) \in \text{set-dist } (\text{execute receive-challenge } s1)$ 
    and next-layer-eq:
       $\text{next-fri-layer } (\text{last } ps) (\text{last } ds)\ b = (\text{next-poly}, \text{next-domain}, \text{next-layer})$ 
    and create-m:  $\text{Some } (m, s3) \in \text{set-dist } (\text{execute } (\text{create next-layer})\ s2)$ 
    and rest:  $\text{Some } ((ps', ds', ls', ms'), t) \in \text{set-dist } (\text{execute } (\text{fri-commit-with receive-challenge } n\ (ps @ [\text{next-poly}])\ (ds @ [\text{next-domain}])\ (ls @ [\text{next-layer}])\ (ms @ [m]))\ s3)$ 
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
  have s-s1:  $s \leq s1$ 
  using send-outcome[OF send-root]
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  have s1-s2:  $s1 \leq s2$ 
  using receiver-extends[OF rand] .
  have s2-s3:  $s2 \leq s3$ 

```

```

using create-outcome[OF create-m] by simp
have s-s3:  $s \leq s3$ 
using s-s1 s1-s2 s2-s3 by (meson hash-ext-trans)
have created-m: created-tree next-layer m s3
using create-outcome[OF create-m] by simp
have init':
 $\bigwedge i. i < \text{length } (ls @ [next-layer]) \implies$ 
 $i < \text{length } (ms @ [m]) \implies$ 
 $\text{created-tree } ((ls @ [next-layer]) ! i) ((ms @ [m]) ! i) s3$ 
proof –
  fix i
  assume i-ls:  $i < \text{length } (ls @ [next-layer])$ 
  and i-ms:  $i < \text{length } (ms @ [m])$ 
  show created-tree  $((ls @ [next-layer]) ! i) ((ms @ [m]) ! i) s3$ 
  proof (cases  $i < \text{length } ls$ )
    case True
    then have created-tree  $(ls ! i) (ms ! i) s$ 
    using Suc.premis(3)[of i] Suc.premis(5) by simp
    then show ?thesis
    using True Suc.premis(5) s-s3
    by (simp add: nth-append created-tree-mono)
  next
  case False
  then have  $i = \text{length } ls$ 
  using i-ls by simp
  then show ?thesis
  using Suc.premis(5) created-m by (simp add: nth-append)
  qed
qed
show ?case
using Suc.IH[OF Suc.premis(1,2) init' rest] Suc.premis(5) by simp
qed

```

lemma trace-fri-commit-extends:

```

assumes Some  $((ps', ds', ls', ms'), t) \in$ 
 $\text{set-dist } (\text{execute } (\text{trace-fri-commit } n \text{ } ps \text{ } ds \text{ } ls \text{ } ms) \text{ } s)$ 
shows  $s \leq t$ 
unfolding trace-fri-commit-def
by (rule fri-commit-with-extends
  [OF receive-trace-fri-challenge-extends assms[unfolded trace-fri-commit-def]])

```

lemma trace-fri-commit-lengths:

```

assumes Some  $((ps', ds', ls', ms'), t) \in$ 
 $\text{set-dist } (\text{execute } (\text{trace-fri-commit } n \text{ } ps \text{ } ds \text{ } ls \text{ } ms) \text{ } s)$ 
shows
 $\text{length } ps' = \text{length } ps + n \wedge$ 
 $\text{length } ds' = \text{length } ds + n \wedge$ 
 $\text{length } ls' = \text{length } ls + n \wedge$ 
 $\text{length } ms' = \text{length } ms + n$ 

```

by (rule fri-commit-with-lengths[OF assms[unfolded trace-fri-commit-def]])

lemma trace-fri-commit-preserves-prefix:

assumes Some $((ps', ds', ls', ms'), t) \in$
 $set-dist (execute (trace-fri-commit n ps ds ls ms) s)$

shows

take (length ps) $ps' = ps \wedge$
take (length ds) $ds' = ds \wedge$
take (length ls) $ls' = ls \wedge$
take (length ms) $ms' = ms$

by (rule fri-commit-with-preserves-prefix
[*OF* assms[unfolded trace-fri-commit-def]])

lemma trace-fri-commit-layers-are-evaluations:

assumes *init*:

$\bigwedge i. i < length ps \implies i < length ds \implies i < length ls \implies$
 $ls ! i = map (poly (ps ! i)) (ds ! i)$

and *outcome*:

Some $((ps', ds', ls', ms'), t) \in$
 $set-dist (execute (trace-fri-commit n ps ds ls ms) s)$

and *lens*: $length ps = length ds \wedge length ds = length ls$

shows

$\bigwedge i. i < length ps' \implies i < length ds' \implies i < length ls' \implies$
 $ls' ! i = map (poly (ps' ! i)) (ds' ! i)$

by (rule fri-commit-with-layers-are-evaluations
[*OF* *init* *outcome*[unfolded trace-fri-commit-def] *lens*])

lemma trace-fri-commit-created-trees:

assumes *init*:

$\bigwedge i. i < length ls \implies i < length ms \implies$
 $created-tree (ls ! i) (ms ! i) (s::f protocol-channel)$

and *outcome*:

Some $((ps', ds', ls', ms'), t) \in$
 $set-dist (execute (trace-fri-commit n ps ds ls ms) s)$

and *lens*: $length ls = length ms$

shows $\bigwedge i. i < length ls' \implies i < length ms' \implies$
 $created-tree (ls' ! i) (ms' ! i) t$

by (rule fri-commit-with-created-trees
[*OF* *receive-trace-fri-challenge-extends* *init*
outcome[unfolded trace-fri-commit-def] *lens*])

lemma composition-fri-commit-extends:

assumes Some $((ps', ds', ls', ms'), t) \in$

$set-dist (execute (composition-fri-commit n ps ds ls ms) s)$

shows $s \leq t$

unfolding composition-fri-commit-def

by (rule fri-commit-with-extends
[*OF* *receive-composition-fri-challenge-extends*
assms[unfolded composition-fri-commit-def]])

lemma *composition-fri-commit-lengths:*

assumes *Some* $((ps', ds', ls', ms'), t) \in$
 $set-dist (execute (composition-fri-commit\ n\ ps\ ds\ ls\ ms)\ s)$
shows
 $length\ ps' = length\ ps + n \wedge$
 $length\ ds' = length\ ds + n \wedge$
 $length\ ls' = length\ ls + n \wedge$
 $length\ ms' = length\ ms + n$
by *(rule fri-commit-with-lengths*
 $[OF\ assms[unfolding\ composition-fri-commit-def]])$

lemma *composition-fri-commit-preserves-prefix:*

assumes *Some* $((ps', ds', ls', ms'), t) \in$
 $set-dist (execute (composition-fri-commit\ n\ ps\ ds\ ls\ ms)\ s)$
shows
 $take\ (length\ ps)\ ps' = ps \wedge$
 $take\ (length\ ds)\ ds' = ds \wedge$
 $take\ (length\ ls)\ ls' = ls \wedge$
 $take\ (length\ ms)\ ms' = ms$
by *(rule fri-commit-with-preserves-prefix*
 $[OF\ assms[unfolding\ composition-fri-commit-def]])$

lemma *composition-fri-commit-layers-are-evaluations:*

assumes *init:*
 $\bigwedge i. i < length\ ps \implies i < length\ ds \implies i < length\ ls \implies$
 $ls\ !\ i = map\ (poly\ (ps\ !\ i))\ (ds\ !\ i)$
and *outcome:*
 $Some\ ((ps', ds', ls', ms'), t) \in$
 $set-dist (execute (composition-fri-commit\ n\ ps\ ds\ ls\ ms)\ s)$
and *lens:* $length\ ps = length\ ds\ length\ ds = length\ ls$
shows
 $\bigwedge i. i < length\ ps' \implies i < length\ ds' \implies i < length\ ls' \implies$
 $ls'\ !\ i = map\ (poly\ (ps'\ !\ i))\ (ds'\ !\ i)$
by *(rule fri-commit-with-layers-are-evaluations*
 $[OF\ init\ outcome[unfolding\ composition-fri-commit-def]\ lens])$

lemma *composition-fri-commit-created-trees:*

assumes *init:*
 $\bigwedge i. i < length\ ls \implies i < length\ ms \implies$
 $created-tree\ (ls\ !\ i)\ (ms\ !\ i)\ (s::'f\ protocol-channel)$
and *outcome:*
 $Some\ ((ps', ds', ls', ms'), t) \in$
 $set-dist (execute (composition-fri-commit\ n\ ps\ ds\ ls\ ms)\ s)$
and *lens:* $length\ ls = length\ ms$
shows $\bigwedge i. i < length\ ls' \implies i < length\ ms' \implies$
 $created-tree\ (ls'\ !\ i)\ (ms'\ !\ i)\ t$
by *(rule fri-commit-with-created-trees*
 $[OF\ receive-composition-fri-challenge-extends\ init$

outcome[unfolded composition-fri-commit-def] lens])

Decommitments *definition decommit-on-fri-layers*

where

```
decommit-on-fri-layers fs =
  map
    (λ(l, m) idx. do {
      let len = length l;
      let idx' = idx mod len;
      let sidx = (idx' + (len div 2)) mod len;
      send (l ! idx');
      mfold2 send (get-authentication-path len idx' m);
      send (l ! sidx);
      mfold2 send (get-authentication-path len sidx m);
      return idx'
    }) fs
```

definition *decommit-on-query*

where

```
decommit-on-query idx f-merkle ≡
  map
    (λi. do {
      send (f-eval ! i);
      mfold2 send (get-authentication-path (length (f-eval)) i f-merkle)
    })
    (powers-scaled idx)
```

Stark Proof The following is a STARK proof for vals.

definition *prover-monad*

where

```
prover-monad ≡ (do {
  f-merkle ← create f-eval;
  send (value f-merkle); — Send the root of a Merkle tree which contains the evaluations over
the larger domain.
  let f-nrounds = ceil-log clength;
  (f-ps, f-ds, f-ls, f-ms) ←
    trace-fri-commit f-nrounds [f] [eval-domain] [f-eval] [f-merkle];
  send (hd (last f-ls)); — Send the final constant for the trace FRI proof.
  as ← mmap (replicate (length spec) (
    do {
      a ← receive-alpha-challenge;
      send a;
      return a
    }
  )); — Send the random factors (alphas) for the composition polynomial.
  let cp' = cp as f-powers;
  send (of-nat (degree cp')); — Send the degree of the composition polynomial.
  cp-merkle ← create (cp-eval as);
  let nrounds = ceil-log (degree cp' + 1);
```

```

(ps, ds, ls, ms) ←
  composition-fri-commit nrounds [cp'] [eval-domain] [cp-eval as] [cp-merkle];
  — Again, we create a Merkle tree which contains the evaluations over the larger domain.
  send (hd (last ls)); — Send the final constant before the verifier samples the query index.
  ntimes (do {
    idx ← receive-query-index-challenge; — Check the validity for a random index. This should
    be repeated multiple times.
    let idx' = index (to-nat idx); — Make sure index is not out of bounds
    mmap (decommit-on-query idx' f-merkle);
    mfold idx' (decommit-on-fri-layers (butlast (zip f-ls f-ms))); — Check that the trace com-
    mitment is low-degree.
    mfold idx' (decommit-on-fri-layers (butlast (zip ls ms))); — We don't de-commit on the
    constant polynomial
    return ()
  }) rounds
})

```

definition *sproof*

where

```

sproof ≡
  execute prover-monad
  (|HashMap = fmempty, PState = 0, PTranscript = [],
   PTraceFriCounter = 0, PCompositionFriCounter = 0,
   PAlphaCounter = 0, PQueryCounter = 0|)

```

end

1.9.4 Verifier

locale *verifier* = *stark* +

```

  constrains omega::'f::proth-field
  and shift::'f
  and concat :: 'f ⇒ 'f ⇒ 'f
  and spec :: (('f poly list ⇒ 'f poly) × nat list × nat) list
  fixes spec2:: (('f list ⇒ 'f) × nat list) list
  assumes specs-aligned:
    list-all2
    (λ(c, roots, d) (c2, roots2)).
    roots2 = roots ∧
    (∀ qs x.
      powers ≤ length qs →
      c2 (map (λq. poly q x) qs) = poly (c qs) x)
    spec spec2

```

begin

lemma *spec2-length*:

length spec2 = length spec
using *list-all2-lengthD[OF specs-aligned]* **by** *simp*

lemma *specs-agree:*
assumes *pair:*
 $((c, \text{roots}, d), (c2, \text{roots2})) \in \text{set } (\text{zip spec spec2})$
and *qs-len:* $\text{powers} \leq \text{length qs}$
shows
 $\text{roots2} = \text{roots} \wedge$
 $c2 (\text{map } (\lambda q. \text{poly } q x) \text{ qs}) = \text{poly } (c \text{ qs}) x$
proof –
obtain *i* **where** *i-bound:* $i < \text{length spec } i < \text{length spec2}$
and *spec-i:* $\text{spec } ! i = (c, \text{roots}, d)$
and *spec2-i:* $\text{spec2 } ! i = (c2, \text{roots2})$
using *pair* **by** (*auto simp: set-zip*)
have *aligned:*
 $(\lambda(c, \text{roots}, d) (c2, \text{roots2}).$
 $\text{roots2} = \text{roots} \wedge$
 $(\forall \text{qs } x.$
 $\text{powers} \leq \text{length qs} \longrightarrow$
 $c2 (\text{map } (\lambda q. \text{poly } q x) \text{ qs}) = \text{poly } (c \text{ qs}) x))$
 $(\text{spec } ! i) (\text{spec2 } ! i)$
 by (*rule list-all2-nthD[OF specs-aligned i-bound(1)]*)
show *?thesis*
using *aligned spec-i spec2-i qs-len* **by** *simp*
qed

definition *cp-eval*

where

$\text{cp-eval as fs } x \equiv \text{fold } (\lambda(a, (p, \text{rs})). (+) (a * ((p \text{ fs}) \text{ div poly } (\text{prod } (g\text{-map rs})) x))) (\text{zip as spec2}) 0$

Validate Committments **definition** *receive-fri-commits*

where

receive-fri-commits =
do {
 $r \leftarrow \text{read};$
 $b \leftarrow \text{receive-random-field-element};$
 return (b, r)
}

definition *receive-fri-commits-with*

where

receive-fri-commits-with receive-challenge =
do {
 $r \leftarrow \text{read};$
 $b \leftarrow \text{receive-challenge};$
 return (b, r)
}

definition *receive-trace-fri-commits*

where

receive-trace-fri-commits =
receive-fri-commits-with receive-trace-fri-challenge

definition *receive-composition-fri-commits*

where

receive-composition-fri-commits =
receive-fri-commits-with receive-composition-fri-challenge

lemma *receive-fri-commits-with-receive-random-field-element*:

receive-fri-commits-with receive-random-field-element = *receive-fri-commits*

unfolding *receive-fri-commits-with-def receive-fri-commits-def* **by** *simp*

lemma *receive-fri-commits-with-no-failure*:

assumes *receiver-nf*:

$\bigwedge s. \text{None} \notin \text{dom} (\text{dist} (\text{execute } \text{receive-challenge } s))$

and *tr*: $P\text{Transcript } s \neq []$

shows

$\text{None} \notin \text{dom} (\text{dist} (\text{execute} (\text{receive-fri-commits-with } \text{receive-challenge}) s))$

using *tr*

unfolding *receive-fri-commits-with-def*

by (*intro no-failure-bindI*)

(*simp-all add: read-no-failure receiver-nf*)

lemma *receive-trace-fri-commits-no-failure*:

assumes $P\text{Transcript } s \neq []$

shows $\text{None} \notin \text{dom} (\text{dist} (\text{execute } \text{receive-trace-fri-commits } s))$

unfolding *receive-trace-fri-commits-def*

by (*rule receive-fri-commits-with-no-failure*)

(*rule receive-trace-fri-challenge-no-failure, rule assms*)

lemma *receive-composition-fri-commits-no-failure*:

assumes $P\text{Transcript } s \neq []$

shows $\text{None} \notin \text{dom} (\text{dist} (\text{execute } \text{receive-composition-fri-commits } s))$

unfolding *receive-composition-fri-commits-def*

by (*rule receive-fri-commits-with-no-failure*)

(*rule receive-composition-fri-challenge-no-failure, rule assms*)

Validate Decommitments **definition** *validate*

where

validate lgth idx leaf-data l rv $\equiv$ *check-authentication-path lgth idx leaf-data l = rv*

definition *check-decommit-on-query*

where

check-decommit-on-query fr idx $\equiv$

map

```

( $\lambda i$ . do {
   $qh \leftarrow \text{read}$ ;
  let  $\text{len} = \text{scale} * \text{clength}$ ; —  $\text{scale} * \text{clength}$  is length of  $f\text{-eval}$ 
   $qh\text{-path} \leftarrow \text{ntimes read (floor-log len)}$ ;
   $ap \leftarrow \text{check-authentication-path len } i \text{ } qh \text{ } qh\text{-path}$ ;
  assert ( $ap = fr$ );
  return  $qh$ 
})
(powers-scaled  $idx$ )

```

definition *receive-query-commits*

where

```

receive-query-commits  $fl =$ 
  map ( $\lambda(b,f) (i,x,\text{len},\text{pow})$ ).
  do {
     $xp \leftarrow \text{read}$ ;
     $xp' \leftarrow \text{ntimes read (floor-log len)}$ ;
     $xn \leftarrow \text{read}$ ;
     $xn' \leftarrow \text{ntimes read (floor-log len)}$ ;
    assert ( $xp = x$ );
     $ap \leftarrow \text{check-authentication-path len } i \text{ } xp \text{ } xp'$ ;
    assert ( $ap = f$ );
    let  $\text{sidx} = (i + \text{len div } 2) \bmod \text{len}$ ;
     $ap \leftarrow \text{check-authentication-path len sidx } xn \text{ } xn'$ ;
    assert ( $ap = f$ );
    let  $gp = (xp + xn) \text{ div } 2$ ;
    let  $hp = (xp - xn) \text{ div } (2 * ((h \hat{=} i) * \text{shift})^{\text{pow}})$ ;
    let  $x = gp + b * hp$ ;
    return ( $i \bmod (\text{len div } 2), x, \text{len div } 2, \text{pow} + \text{pow}$ )
  })  $fl$ 

```

definition *verify-monad*

where

```

verify-monad  $\equiv$  (do {
   $fr \leftarrow \text{read}$ ;
   $f\text{-fl} \leftarrow \text{ntimes receive-trace-fri-commits (ceil-log clength)}$ ;
   $f\text{-final} \leftarrow \text{read}$ ;
   $as \leftarrow \text{mmap (replicate (length spec) ($ 
    do {
       $a0 \leftarrow \text{receive-alpha-challenge}$ ;
      let  $a0' = a0$ ;
       $a1 \leftarrow \text{read}$ ;
      let  $a1' = a1$ ;
      assert ( $a0' = a1'$ );
      return  $a1'$ 
    })
  );
   $dg \leftarrow \text{read}$ ;
  assert ( $\text{to-nat } dg \leq \text{maxDegree}$ ); — Degree check
   $fl \leftarrow \text{ntimes receive-composition-fri-commits (ceil-log (to-nat } dg + 1))$ ;

```

```

final ← read;
ntimes (do {
  idx ← receive-query-index-challenge;
  let idx' = index (to-nat idx); — Make sure index is not out of bounds
  fv ← mmap (check-decommit-on-query fr idx');

  (f-i, f-x, f-len, f-pow) ← mfold (idx', hd fv, clength * scale, 1) (receive-query-commits f-fl);
  assert (f-x = f-final);

  (i,x,len,pow) ← mfold (idx', cp-eval as fv (h^idx' * shift), clength * scale, 1) (receive-query-commits fl);

  assert (x = final) — Check that the final FRI value is the committed constant polynomial
}) rounds
})

```

definition *verify*

where

verify p ≡ *execute verify-monad (p(PState := 0))*

end

end

2 Completeness

theory *Completeness-Core*

imports *Stark-Core.Stark*

begin

2.1 Core Completeness Infrastructure

Core verification locale, honest-trace predicates, and WP/no-failure infrastructure.

locale *verification* =

p: prover concat omega shift scale clength powers to-nat of-nat size spec rounds vals +

v: verifier concat omega shift scale clength powers to-nat of-nat size spec rounds spec2

for *concat* :: '*f*::proth-field ⇒ '*f* ⇒ '*f*

and *omega* :: '*f*

and *shift* :: '*f*

and *scale* :: nat

and *clength* :: nat

and *powers* :: nat

and *of-nat*:: nat ⇒ '*f*

and *to-nat*:: '*f* ⇒ nat

and *size*:: nat

and *spec*:: (('f poly list ⇒ '*f* poly) × nat list × nat) list

```

and rounds :: nat
and spec2:: (('f list  $\Rightarrow$  'f)  $\times$  nat list) list
and vals :: 'f list
+
assumes constraint-degree-bound:
   $\bigwedge c$  roots  $d$ .
     $(c, \text{roots}, d) \in \text{set spec} \implies$ 
     $\text{degree } (c \text{ p.f-powers}) \leq d * (\text{clength} - 1)$ 
begin

fun root-check
where
  root-check p [] = True
| root-check p (r#rs) = (poly p r = 0  $\wedge$  root-check p rs)

definition exec
where exec  $\equiv$  do {
  fr  $\leftarrow$  p.prover-monad;
  modify ( $\lambda c$ . c(|PState := 0, PTranscript:=rev (PTranscript c),
    PTraceFriCounter := 0, PCompositionFriCounter := 0,
    PAlphaCounter := 0, PQueryCounter := 0|));
  v.verify-monad
}

definition init-state
where init-state  $\equiv$ 
  (|HashMap = fmempty, PState = 0, PTranscript = [],
    PTraceFriCounter = 0, PCompositionFriCounter = 0,
    PAlphaCounter = 0, PQueryCounter = 0|)

definition verifier-replay-state
where verifier-replay-state s  $\equiv$ 
  s(|PState := 0, PTranscript := rev (PTranscript s),
    PTraceFriCounter := 0, PCompositionFriCounter := 0,
    PAlphaCounter := 0, PQueryCounter := 0|)

definition constraint-roots-vanish
where
  constraint-roots-vanish c roots  $\longleftrightarrow$ 
     $(\forall r \in \text{set } (p.g\text{-map roots}). \text{poly } (c \text{ p.f-powers}) r = 0)$ 

definition declared-degree-sound
where
  declared-degree-sound c roots d  $\longleftrightarrow$ 
     $\text{degree } (c \text{ p.f-powers}) \leq d * (\text{clength} - 1)$ 

definition roots-distinct
where
  roots-distinct roots  $\longleftrightarrow$  distinct (p.g-map roots)

```

definition *honest-trace-algebra*

where

honest-trace-algebra $\longleftrightarrow$
 $(\forall (c, \text{roots}, d) \in \text{set spec.}$
 $\text{constraint-roots-vanish } c \text{ roots})$

definition *honest-trace-valid*

where

honest-trace-valid $\longleftrightarrow$ *honest-trace-algebra*

definition *no-failure*

where *no-failure* $m \ s \longleftrightarrow \text{None} \notin \text{dom} (\text{dist} (\text{execute } m \ s))$

definition *transcript-extends*

where *transcript-extends* $t \ s \longleftrightarrow (\exists xs. \text{PTranscript } t = xs @ \text{PTranscript } s)$

lemma *declared-degree-sound-from-constraint-degree-bound:*

assumes $(c, \text{roots}, d) \in \text{set spec}$
shows *declared-degree-sound* $c \ \text{roots} \ d$
using *constraint-degree-bound* [OF *assms*]
unfolding *declared-degree-sound-def* .

lemma *roots-distinct-from-spec-roots-distinct:*

assumes $(c, \text{roots}, d) \in \text{set spec}$
shows *roots-distinct* roots
using *p.g-map-distinct* [OF *assms*]
unfolding *roots-distinct-def* .

lemma *maxDegree-less-eval-domain:*

$p.\text{maxDegree} < \text{clength} * \text{scale}$
by (rule *p.maxDegree-less-eval-domain*)

lemma *wp-error-assert* [wpsimps]:

wp-error $(\text{assert } b) \ s = (\text{if } b \text{ then } 0 \text{ else } 1)$
unfolding *assert-def* **by** (*simp add: wpsimps*)

lemma *no-failure-assert-bindI:*

assumes b
and $\text{None} \notin \text{dom} (\text{dist} (\text{execute } (k \ ()) \ s))$
shows $\text{None} \notin \text{dom} (\text{dist} (\text{execute } (\text{assert } b \gg k) \ s))$
using *assms* **unfolding** *assert-def* **by** *simp*

lemma *no-failure-bind-returnI:*

assumes $\text{None} \notin \text{dom} (\text{dist} (\text{execute } m \ s))$
shows $\text{None} \notin \text{dom} (\text{dist} (\text{execute } (m \gg (\lambda x. \text{return } (f \ x))) \ s))$
by (rule *no-failure-bindI* [OF *assms*]) *simp*

lemma *no-failure-mfoldI:*

```

assumes step:
   $\bigwedge m\ a\ s. m \in \text{set } ms \implies$ 
     $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (m\ a)\ s))$ 
shows  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{mfold } a\ ms)\ s))$ 
using step
proof (induction ms arbitrary: a s)
  case Nil
    then show ?case by simp
next
  case (Cons m ms)
    show ?case
      unfolding mfold.simps
    proof (rule no-failure-bindI)
      show  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (m\ a)\ s))$ 
      using Cons.prems by simp
    next
      fix x t
      assume  $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (m\ a)\ s)$ 
      show  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{mfold } x\ ms)\ t))$ 
      by (rule Cons.IH) (use Cons.prems in simp)
    qed
  qed

lemma no-failure-mfold-invariantI:
  assumes init:  $I\ a\ s$ 
  and step-nf:
     $\bigwedge m\ a\ s. m \in \text{set } ms \implies I\ a\ s \implies$ 
       $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (m\ a)\ s))$ 
  and step-inv:
     $\bigwedge m\ a\ s\ x\ t. m \in \text{set } ms \implies I\ a\ s \implies$ 
       $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (m\ a)\ s) \implies I\ x\ t$ 
  shows  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{mfold } a\ ms)\ s))$ 
  using init step-nf step-inv
proof (induction ms arbitrary: a s)
  case Nil
    then show ?case by simp
next
  case (Cons m ms)
    show ?case
      unfolding mfold.simps
    proof (rule no-failure-bindI)
      show  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (m\ a)\ s))$ 
      using Cons.prems by simp
    next
      fix x t
      assume outcome:  $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (m\ a)\ s)$ 
      have inv-xt:  $I\ x\ t$ 
      using Cons.prems(1) Cons.prems(3)[of m a s x t] outcome by simp
      show  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{mfold } x\ ms)\ t))$ 

```

```

proof (rule Cons.IH[OF inv-xt])
  fix  $m' a' s'$ 
  assume  $m' \in \text{set } ms$  and  $I a' s'$ 
  then show  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (m' a') s'))$ 
    using Cons.prem by simp
next
  fix  $m' a' s' x' t'$ 
  assume  $m'\text{-in}: m' \in \text{set } ms$  and  $\text{inv}: I a' s'$ 
    and  $\text{out}: \text{Some } (x', t') \in \text{set-dist } (\text{execute } (m' a') s')$ 
  show  $I x' t'$ 
    using Cons.prem(3)[of  $m' a' s' x' t'$ ]  $m'\text{-in}$   $\text{inv}$   $\text{out}$  by simp
qed
qed
qed

```

lemma no-failure-mfold-map-index-fromI:

```

assumes  $\text{init}: I k a s$ 
and  $\text{step-nf}$ :
   $\bigwedge j x a s. k \leq j \implies j < k + \text{length } xs \implies$ 
     $x = xs ! (j - k) \implies I j a s \implies$ 
     $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (F x a) s))$ 
and  $\text{step-inv}$ :
   $\bigwedge j x a s y t. k \leq j \implies j < k + \text{length } xs \implies$ 
     $x = xs ! (j - k) \implies I j a s \implies$ 
     $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (F x a) s) \implies$ 
     $I (\text{Suc } j) y t$ 
shows  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{mfold } a (\text{map } F xs)) s))$ 
using  $\text{init}$   $\text{step-nf}$   $\text{step-inv}$ 
proof (induction  $xs$  arbitrary:  $k a s$ )
  case Nil
  then show ?case by simp
next
  case (Cons  $x xs$ )
  show ?case
    unfolding  $\text{mfold.simps list.map}$ 
  proof (rule no-failure-bindI)
    show  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (F x a) s))$ 
      by (rule Cons.prem(2)[of  $k x a s$ ]) (use Cons.prem(1) in simp-all)
  next
    fix  $y t$ 
    assume  $\text{step-out}: \text{Some } (y, t) \in \text{set-dist } (\text{execute } (F x a) s)$ 
    have  $\text{inv-next}: I (\text{Suc } k) y t$ 
      by (rule Cons.prem(3)[of  $k x a s y t$ ])
      (use Cons.prem(1)  $\text{step-out}$  in simp-all)
    show  $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{mfold } y (\text{map } F xs)) t))$ 
  proof (rule Cons.IH[OF inv-next])
    fix  $j x' a' s'$ 
    assume  $ge: \text{Suc } k \leq j$ 
    and  $lt: j < \text{Suc } k + \text{length } xs$ 

```

```

    and x'-eq: x' = xs ! (j - Suc k)
    and inv: I j a' s'
  have j-gt: k < j
    using ge by simp
  have j-lt: j < k + length (x # xs)
    using lt by simp
  have nth-eq: x' = (x # xs) ! (j - k)
    using x'-eq j-gt by simp
  show None ∉ dom (dist (execute (F x' a') s'))
    by (rule Cons.prem2(2)[OF - j-lt nth-eq inv]) (use ge in simp)
next
  fix j x' a' s' y' t'
  assume ge: Suc k ≤ j
    and lt: j < Suc k + length xs
    and x'-eq: x' = xs ! (j - Suc k)
    and inv: I j a' s'
    and out: Some (y', t') ∈ set-dist (execute (F x' a') s')
  have j-gt: k < j
    using ge by simp
  have j-lt: j < k + length (x # xs)
    using lt by simp
  have nth-eq: x' = (x # xs) ! (j - k)
    using x'-eq j-gt by simp
  show I (Suc j) y' t'
    by (rule Cons.prem3(3)[OF - j-lt nth-eq inv out]) (use ge in simp)
qed
qed
qed

lemma no-failure-mfold-map-indexI:
  assumes init: I 0 a s
  and step-nf:
     $\bigwedge j x a s. j < \text{length } xs \implies$ 
     $x = xs ! j \implies I j a s \implies$ 
    None ∉ dom (dist (execute (F x a) s))
  and step-inv:
     $\bigwedge j x a s y t. j < \text{length } xs \implies$ 
     $x = xs ! j \implies I j a s \implies$ 
    Some (y, t) ∈ set-dist (execute (F x a) s)  $\implies$ 
    I (Suc j) y t
  shows None ∉ dom (dist (execute (mfold a (map F xs)) s))
  by (rule no-failure-mfold-map-index-fromI[
    where I=I and k=0 and xs=xs and F=F, OF init])
    (use step-nf step-inv in simp-all)

```

```

lemma no-failure-mmapI:
  assumes step:  $\bigwedge m s. m \in \text{set } ms \implies \text{None} \notin \text{dom } (\text{dist } (\text{execute } m \ s))$ 
  shows None ∉ dom (dist (execute (mmap ms) s))
  using step

```



```

proof (induction ms arbitrary: s)
  case Nil
  then show ?case by simp
next
  case (Cons m ms)
  show ?case
    unfolding mmap.simps
  proof (rule no-failure-bindI)
    show None  $\notin$  dom (dist (execute m s))
    using Cons.prems by simp
  next
  fix x t
  assume Some (x, t)  $\in$  set-dist (execute m s)
  show None  $\notin$  dom (dist (execute (mmap ms  $\gg$  ( $\lambda xs.$  return (x  $\#$  xs))) t))
    by (rule no-failure-bind-returnI)
      (rule Cons.IH, use Cons.prems in simp)
  qed
qed

```

```

lemma no-failure-mfold2I:
  assumes step:  $\bigwedge x s. x \in \text{set } xs \implies \text{None} \notin \text{dom} (\text{dist} (\text{execute } (f x) s))$ 
  shows None  $\notin$  dom (dist (execute (mfold2 f xs s))
  using step
proof (induction xs arbitrary: s)
  case Nil
  then show ?case by simp
next
  case (Cons x xs)
  show ?case
    unfolding mfold2.simps
  proof (rule no-failure-bindI)
    show None  $\notin$  dom (dist (execute (f x s))
    using Cons.prems by simp
  next
  fix y t
  assume Some (y, t)  $\in$  set-dist (execute (f x s))
  show None  $\notin$  dom (dist (execute (mfold2 f xs t))
    by (rule Cons.IH) (use Cons.prems in simp)
  qed
qed

```

```

lemma no-failure-ntimesI:
  assumes step:  $\bigwedge s. \text{None} \notin \text{dom} (\text{dist} (\text{execute } m s))$ 
  shows None  $\notin$  dom (dist (execute (ntimes m n s))
  using assms
proof (induction n arbitrary: s)
  case 0
  then show ?case by simp
next

```

```

case (Suc n)
show ?case
  unfolding ntimes.simps
proof (rule no-failure-bindI[OF Suc.prem])
  fix x t
  assume Some (x, t) ∈ set-dist (execute m s)
  show None ∉ dom (dist (execute (ntimes m n ≫ (λxs. return (x # xs))) t))
    by (rule no-failure-bind-returnI) (rule Suc.IH[OF Suc.prem])
qed
qed

lemma no-failure-ntimes-invariantI:
  assumes init: I s
  and step-nf:  $\bigwedge s. I s \implies \text{None} \notin \text{dom} (\text{dist} (\text{execute } m \ s))$ 
  and step-inv:
     $\bigwedge x \ s \ t. I s \implies \text{Some } (x, t) \in \text{set-dist} (\text{execute } m \ s) \implies I t$ 
  shows None ∉ dom (dist (execute (ntimes m n) s))
  using init
proof (induction n arbitrary: s)
  case 0
  then show ?case by simp
next
  case (Suc n)
  show ?case
    unfolding ntimes.simps
  proof (rule no-failure-bindI[OF step-nf[OF Suc.prem]])
  fix x t
  assume step: Some (x, t) ∈ set-dist (execute m s)
  have I t
    by (rule step-inv[OF Suc.prem step])
  then show None ∉ dom (dist (execute (ntimes m n ≫ (λxs. return (x # xs))) t))
    by (rule no-failure-bind-returnI[OF Suc.IH])
  qed
qed

lemma ntimes-hash-extends:
  fixes m :: ('a, ('f, 'b) protocol-channel-scheme) state-monad
  assumes step-ext:
     $\bigwedge x \ (s :: ('f, 'b) \text{ protocol-channel-scheme}) \ t.$ 
     $\text{Some } (x, t) \in \text{set-dist} (\text{execute } m \ s) \implies s \leq t$ 
  and outcome: Some (xs, t) ∈ set-dist (execute (ntimes m n) s)
  shows s ≤ t
  using outcome
proof (induction n arbitrary: xs s t)
  case 0
  then show ?case
    using p.hash-ext-refl[of s] by simp
next
  case (Suc n)

```

```

have expanded:
  Some (xs, t) ∈
    set-dist (execute
      (m  $\gg$  ( $\lambda x.$  ntimes m n  $\gg$  ( $\lambda ys.$  return (x # ys)))) s)
  using Suc.prems by simp
show ?case
proof (rule p.set-dist-bindE[OF expanded])
  fix x u
  assume step: Some (x, u) ∈ set-dist (execute m s)
  and rest:
    Some (xs, t) ∈
      set-dist (execute (ntimes m n  $\gg$  ( $\lambda ys.$  return (x # ys)))) u)
  show ?thesis
proof (rule p.set-dist-bindE[OF rest])
  fix ys v
  assume tail-v: Some (ys, v) ∈ set-dist (execute (ntimes m n) u)
  and ret: Some (xs, t) ∈ set-dist (execute (return (x # ys)) v)
  have tail: Some (ys, t) ∈ set-dist (execute (ntimes m n) u)
  using tail-v ret by simp
  have s ≤ u
  by (rule step-ext[OF step])
  moreover have u ≤ t
  by (rule Suc.IH[OF tail])
  ultimately show ?thesis
  unfolding less-eq-hash-ext-def less-eq-fmap-def
  by (metis option.distinct(1))
qed
qed
qed

lemma ntimes-transcript-extends:
  assumes step-ext:
     $\bigwedge x s t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m s) \implies \text{transcript-extends } t s$ 
  and outcome: Some (xs, t) ∈ set-dist (execute (ntimes m n) s)
  shows transcript-extends t s
  using outcome
proof (induction n arbitrary: xs s t)
  case 0
  then show ?case
    unfolding transcript-extends-def by simp
next
  case (Suc n)
  have expanded:
    Some (xs, t) ∈
      set-dist (execute
        (m  $\gg$  ( $\lambda x.$  ntimes m n  $\gg$  ( $\lambda ys.$  return (x # ys)))) s)
    using Suc.prems by simp
  show ?case
proof (rule p.set-dist-bindE[OF expanded])

```

```

fix  $x\ u$ 
assume step:  $\text{Some } (x, u) \in \text{set-dist } (\text{execute } m\ s)$ 
and rest:
   $\text{Some } (xs, t) \in$ 
     $\text{set-dist } (\text{execute } (ntimes\ m\ n \gg (\lambda ys. \text{return } (x \# ys))))\ u)$ 
show ?thesis
proof (rule p.set-dist-bindE[OF rest])
  fix  $ys\ v$ 
  assume tail-v:  $\text{Some } (ys, v) \in \text{set-dist } (\text{execute } (ntimes\ m\ n)\ u)$ 
  and ret:  $\text{Some } (xs, t) \in \text{set-dist } (\text{execute } (\text{return } (x \# ys))\ v)$ 
  have tail:  $\text{Some } (ys, t) \in \text{set-dist } (\text{execute } (ntimes\ m\ n)\ u)$ 
  using tail-v ret by simp
  have transcript-extends  $u\ s$ 
  by (rule step-ext[OF step])
  moreover have transcript-extends  $t\ u$ 
  by (rule Suc.IH[OF tail])
  ultimately show ?thesis
  unfolding transcript-extends-def by auto
qed
qed
qed

lemma ntimes-Suc-outcomeE:
assumes outcome:  $\text{Some } (xs, t) \in \text{set-dist } (\text{execute } (ntimes\ m\ (\text{Suc } n))\ s)$ 
obtains  $x\ ys\ u$  where
   $\text{Some } (x, u) \in \text{set-dist } (\text{execute } m\ s)$ 
   $\text{Some } (ys, t) \in \text{set-dist } (\text{execute } (ntimes\ m\ n)\ u)$ 
   $xs = x \# ys$ 
proof –
have expanded:
   $\text{Some } (xs, t) \in$ 
     $\text{set-dist } (\text{execute } (m \gg (\lambda x. ntimes\ m\ n \gg (\lambda ys. \text{return } (x \# ys))))\ s)$ 
using outcome by simp
show ?thesis
proof (rule p.set-dist-bindE[OF expanded])
  fix  $x\ u$ 
  assume step:  $\text{Some } (x, u) \in \text{set-dist } (\text{execute } m\ s)$ 
  and rest:
     $\text{Some } (xs, t) \in$ 
       $\text{set-dist } (\text{execute } (ntimes\ m\ n \gg (\lambda ys. \text{return } (x \# ys))))\ u)$ 
show ?thesis
proof (rule p.set-dist-bindE[OF rest])
  fix  $ys\ v$ 
  assume tail-v:  $\text{Some } (ys, v) \in \text{set-dist } (\text{execute } (ntimes\ m\ n)\ u)$ 
  and ret:  $\text{Some } (xs, t) \in \text{set-dist } (\text{execute } (\text{return } (x \# ys))\ v)$ 
  have tail:  $\text{Some } (ys, t) \in \text{set-dist } (\text{execute } (ntimes\ m\ n)\ u)$ 
  using tail-v ret by simp
  have xs-eq:  $xs = x \# ys$ 
  using ret by simp

```

```

    show ?thesis
    by (rule that[OF step tail xs-eq])
qed
qed
qed

```

definition *prover-query-round*

where

```

prover-query-round f-merkle f-ls f-ms ls ms ≡
do {
  idx ← p.receive-query-index-challenge;
  mmap (p.decommit-on-query (p.index (to-nat idx)) f-merkle);
  mfold (p.index (to-nat idx)) (p.decommit-on-fri-layers (butlast (zip f-ls f-ms)));
  mfold (p.index (to-nat idx)) (p.decommit-on-fri-layers (butlast (zip ls ms)));
  return ()
}

```

definition *verifier-query-round*

where

```

verifier-query-round fr f-fl f-final as fl final ≡
do {
  idx ← p.receive-query-index-challenge;
  let idx' = p.index (to-nat idx);
  fv ← mmap (v.check-decommit-on-query fr idx');
  (f-i, f-x, f-len, f-pw) ←
    mfold
      (idx', hd fv, clength * scale, 1)
      (v.receive-query-commits f-fl);
  assert (f-x = f-final);
  (i, x, len, pw) ←
    mfold
      (idx', v.cp-eval as fv (p.h ^ idx' * shift), clength * scale, 1)
      (v.receive-query-commits fl);
  assert (x = final)
}

```

lemma *send-no-failure:*

```

None ∉ dom (dist (execute (p.send x) s))
by (rule p.send-no-failure)

```

lemma *create-no-failure:*

```

None ∉ dom (dist (execute (p.create xs) s))
by (rule p.create-no-failure)

```

lemma *fri-commit-no-failure:*

```

None ∉ dom (dist (execute (p.fri-commit n ps ds ls ms) s))

```

proof (*induction n arbitrary: ps ds ls ms s*)

case 0

then show ?case **by** simp

```

next
  case (Suc n)
  show ?case
    unfolding p.fri-commit.simps
  proof (rule no-failure-bindI[OF send-no-failure])
    fix u s1
    assume Some (u, s1) ∈ set-dist (execute (p.send (value (last ms))) s)
    show None ∉ dom (dist (execute
      (p.receive-random-field-element >>=
        (λb. let (next-poly, next-domain, next-layer) =
          p.next-fri-layer (last ps) (last ds) b
          in p.create next-layer >>=
            (λm. p.fri-commit n
              (ps @ [next-poly]) (ds @ [next-domain])
              (ls @ [next-layer]) (ms @ [m])))) s1)))
    proof (rule no-failure-bindI[OF p.receive-random-field-element-no-failure])
      fix b s2
      assume Some (b, s2) ∈ set-dist (execute p.receive-random-field-element s1)
      obtain next-poly next-domain next-layer where next-layer-eq:
        p.next-fri-layer (last ps) (last ds) b =
          (next-poly, next-domain, next-layer)
      by (cases p.next-fri-layer (last ps) (last ds) b)
      show None ∉ dom (dist (execute
        (let (next-poly, next-domain, next-layer) =
          p.next-fri-layer (last ps) (last ds) b
          in p.create next-layer >>=
            (λm. p.fri-commit n
              (ps @ [next-poly]) (ds @ [next-domain])
              (ls @ [next-layer]) (ms @ [m])))) s2)))
      unfolding next-layer-eq split Let-def
    proof (rule no-failure-bindI[OF create-no-failure])
      fix m s3
      assume Some (m, s3) ∈ set-dist (execute (p.create next-layer) s2)
      show None ∉ dom (dist (execute
        (p.fri-commit n (ps @ [next-poly]) (ds @ [next-domain])
          (ls @ [next-layer]) (ms @ [m])) s3))
      by (rule Suc.IH)
    qed
  qed
qed
qed
qed

```

lemma *fri-commit-with-no-failure:*
assumes *receiver-no-failure:*
 $\bigwedge s. \text{None} \notin \text{dom} (\text{dist} (\text{execute} \text{receive-challenge } s))$
shows
 $\text{None} \notin \text{dom} (\text{dist} (\text{execute} (p.\text{fri-commit-with receive-challenge } n \text{ ps ds ls ms } s)))$
proof (*induction n arbitrary: ps ds ls ms s*)

```

case 0
then show ?case by simp
next
case (Suc n)
show ?case
  unfolding p.fri-commit-with.simps
proof (rule no-failure-bindI[OF send-no-failure])
  fix u s1
  assume Some (u, s1) ∈ set-dist (execute (p.send (value (last ms)))) s)
  show None ∉ dom (dist (execute
    (receive-challenge >>=
      (λb. let (next-poly, next-domain, next-layer) =
        p.next-fri-layer (last ps) (last ds) b
        in p.create next-layer >>=
          (λm. p.fri-commit-with receive-challenge n
            (ps @ [next-poly]) (ds @ [next-domain])
            (ls @ [next-layer]) (ms @ [m])))) s1)))
  proof (rule no-failure-bindI[OF receiver-no-failure])
  fix b s2
  assume Some (b, s2) ∈ set-dist (execute receive-challenge s1)
  obtain next-poly next-domain next-layer where next-layer-eq:
    p.next-fri-layer (last ps) (last ds) b =
      (next-poly, next-domain, next-layer)
  by (cases p.next-fri-layer (last ps) (last ds) b)
  show None ∉ dom (dist (execute
    (let (next-poly, next-domain, next-layer) =
      p.next-fri-layer (last ps) (last ds) b
      in p.create next-layer >>=
        (λm. p.fri-commit-with receive-challenge n
          (ps @ [next-poly]) (ds @ [next-domain])
          (ls @ [next-layer]) (ms @ [m])))) s2)))
  unfolding next-layer-eq split Let-def
  proof (rule no-failure-bindI[OF create-no-failure])
  fix m s3
  assume Some (m, s3) ∈ set-dist (execute (p.create next-layer) s2)
  show None ∉ dom (dist (execute
    (p.fri-commit-with receive-challenge n
      (ps @ [next-poly]) (ds @ [next-domain])
      (ls @ [next-layer]) (ms @ [m])) s3)))
  by (rule Suc.IH)
  qed
qed
qed
qed

```

lemma trace-fri-commit-no-failure:

```

  None ∉ dom (dist (execute (p.trace-fri-commit n ps ds ls ms) s))
  unfolding p.trace-fri-commit-def
  by (rule fri-commit-with-no-failure)

```

(rule *p.receive-trace-fri-challenge-no-failure*)

lemma *composition-fri-commit-no-failure*:

$\text{None} \notin \text{dom} (\text{dist} (\text{execute} (p.\text{composition-fri-commit } n \text{ } ps \text{ } ds \text{ } ls \text{ } ms) s))$

unfolding *p.composition-fri-commit-def*

by (rule *fri-commit-with-no-failure*)

(rule *p.receive-composition-fri-challenge-no-failure*)

lemma *honest-alpha-send-round-no-failure*:

$\text{None} \notin \text{dom} (\text{dist} (\text{execute}$

(do {

$a \leftarrow p.\text{receive-alpha-challenge};$

$p.\text{send } a;$

return a

}) s))

by (intro *no-failure-bindI*)

(simp-all add: *p.receive-alpha-challenge-no-failure send-no-failure*)

lemma *prover-decommit-on-query-step-no-failure*:

$\text{None} \notin \text{dom} (\text{dist} (\text{execute}$

(do {

$p.\text{send } x;$

$\text{mfold2 } p.\text{send path}$

}) s))

proof (rule *no-failure-bindI*[*OF send-no-failure*])

fix $u \text{ } t$

assume $\text{Some} (u, t) \in \text{set-dist} (\text{execute} (p.\text{send } x) s)$

show $\text{None} \notin \text{dom} (\text{dist} (\text{execute} (\text{mfold2 } p.\text{send path}) t))$

by (rule *no-failure-mfold2I*) (simp add: *send-no-failure*)

qed

lemma *prover-decommit-on-query-no-failure*:

$\text{None} \notin \text{dom} (\text{dist} (\text{execute} (\text{mmap} (p.\text{decommit-on-query } idx \text{ } f\text{-merkle})) s))$

unfolding *p.decommit-on-query-def*

by (rule *no-failure-mmapI*)

(auto intro!: *prover-decommit-on-query-step-no-failure*)

lemma *prover-decommit-on-fri-layer-no-failure*:

$\text{None} \notin \text{dom} (\text{dist} (\text{execute}$

(do {

let $len = \text{length } l;$

let $idx' = idx \bmod len;$

let $sidx = (idx' + (len \text{ div } 2)) \bmod len;$

$p.\text{send } (l ! idx');$

$\text{mfold2 } p.\text{send } (\text{get-authentication-path } len \text{ } idx' \text{ } m);$

$p.\text{send } (l ! sidx);$

$\text{mfold2 } p.\text{send } (\text{get-authentication-path } len \text{ } sidx \text{ } m);$

return idx'

}) s))

unfolding *Let-def*
proof (rule no-failure-bindI[OF send-no-failure])
fix u s1
assume Some (u, s1) ∈ set-dist (execute (p.send (l ! (idx mod length l))) s)
show None ∉ dom (dist (execute
(mfold2 p.send (get-authentication-path (length l) (idx mod length l) m) ≫
(λ-. p.send (l ! ((idx mod length l + length l div 2) mod length l)) ≫
(λ-. mfold2 p.send
(get-authentication-path (length l)
((idx mod length l + length l div 2) mod length l) m) ≫
(λ-. return (idx mod length l)))) s1))
proof (rule no-failure-bindI)
show None ∉ dom (dist (execute
(mfold2 p.send (get-authentication-path (length l) (idx mod length l) m) s1))
by (rule no-failure-mfold2I) (simp add: send-no-failure)
next
fix u2 s2
assume Some (u2, s2) ∈ set-dist (execute
(mfold2 p.send (get-authentication-path (length l) (idx mod length l) m) s1)
show None ∉ dom (dist (execute
(p.send (l ! ((idx mod length l + length l div 2) mod length l)) ≫
(λ-. mfold2 p.send
(get-authentication-path (length l)
((idx mod length l + length l div 2) mod length l) m) ≫
(λ-. return (idx mod length l)))) s2))
proof (rule no-failure-bindI[OF send-no-failure])
fix u3 s3
assume Some (u3, s3) ∈ set-dist (execute
(p.send (l ! ((idx mod length l + length l div 2) mod length l)) s2)
show None ∉ dom (dist (execute
(mfold2 p.send
(get-authentication-path (length l)
((idx mod length l + length l div 2) mod length l) m) ≫
(λ-. return (idx mod length l)))) s3))
by (rule no-failure-bind-returnI)
(rule no-failure-mfold2I, simp add: send-no-failure)
qed
qed
qed

lemma prover-decommit-on-fri-layers-no-failure:
None ∉ dom (dist (execute (mfold idx (p.decommit-on-fri-layers fs)) s))
unfolding p.decommit-on-fri-layers-def
by (rule no-failure-mfoldI)
(auto intro!: prover-decommit-on-fri-layer-no-failure)

lemma prover-query-round-no-failure:
None ∉ dom (dist (execute (prover-query-round f-merkle f-ls f-ms ls ms) s))
unfolding prover-query-round-def

```

proof (rule no-failure-bindI[OF p.receive-query-index-challenge-no-failure])
  fix idx s1
  assume Some (idx, s1) ∈ set-dist (execute p.receive-query-index-challenge s)
  show None ∉ dom (dist (execute
    (mmap (p.decommit-on-query (p.index (to-nat idx)) f-merkle) >>=
      (λ-. mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip f-ls f-ms))) >>=
        (λ-. mfold (p.index (to-nat idx))
          (p.decommit-on-fri-layers (butlast (zip ls ms))) >>=
          (λ-. return ()))))) s1))
proof (rule no-failure-bindI[OF prover-decommit-on-query-no-failure])
  fix u2 s2
  assume Some (u2, s2) ∈ set-dist (execute
    (mmap (p.decommit-on-query (p.index (to-nat idx)) f-merkle)) s1)
  show None ∉ dom (dist (execute
    (mfold (p.index (to-nat idx))
      (p.decommit-on-fri-layers (butlast (zip f-ls f-ms))) >>=
      (λ-. mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip ls ms))) >>=
        (λ-. return ()))) s2))
proof (rule no-failure-bindI[OF prover-decommit-on-fri-layers-no-failure])
  fix u3 s3
  assume Some (u3, s3) ∈ set-dist (execute
    (mfold (p.index (to-nat idx))
      (p.decommit-on-fri-layers (butlast (zip f-ls f-ms)))) s2)
  show None ∉ dom (dist (execute
    (mfold (p.index (to-nat idx))
      (p.decommit-on-fri-layers (butlast (zip ls ms))) >>=
      (λ-. return ()))) s3))
  by (rule no-failure-bind-returnI)
    (rule prover-decommit-on-fri-layers-no-failure)
  qed
qed
qed

```

lemma prover-after-fri-layers-tail-no-failure:

```

  None ∉ dom (dist (execute
    (do {
      p.send (hd (last ls));
      ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds
    } s))
proof (rule no-failure-bindI[OF send-no-failure])
  fix u s1
  assume Some (u, s1) ∈ set-dist (execute (p.send (hd (last ls))) s)
  show None ∉ dom (dist (execute
    (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds) s1))
  by (rule no-failure-ntimesI) (rule prover-query-round-no-failure)
qed

```

lemma *prover-after-fri-commit-no-failure*:
 $\text{None} \notin \text{dom} (\text{dist} (\text{execute}$
 $((\text{case } \text{fri-data} \text{ of } (ps, ds, ls, ms) \Rightarrow$
 $\text{do } \{$
 $\text{p.send } (\text{hd } (\text{last } ls));$
 $\text{ntimes } (\text{prover-query-round } f\text{-merkle } f\text{-ls } f\text{-ms } ls \text{ ms}) \text{ rounds}$
 $\}))) \text{ s}))$
proof –
obtain $ps \ ds \ ls \ ms$ **where** $\text{fri-data-eq}: \text{fri-data} = (ps, ds, ls, ms)$
by $(\text{cases } \text{fri-data})$
show $?thesis$
unfolding fri-data-eq case-prod-unfold
by $(\text{rule } \text{prover-after-fri-layers-tail-no-failure})$
qed

lemma *prover-after-fri-commit-named-no-failure*:
 $\text{None} \notin \text{dom} (\text{dist} (\text{execute}$
 $((\text{case } \text{fri-data} \text{ of } (ps, ds, ls, ms) \Rightarrow$
 $\text{do } \{$
 $\text{p.send } (\text{hd } (\text{last } ls));$
 $\text{ntimes } (\text{prover-query-round } f\text{-merkle } f\text{-ls } f\text{-ms } ls \text{ ms}) \text{ rounds}$
 $\}))) \text{ s}))$
proof –
obtain $ps \ ds \ ls \ ms$ **where** $\text{fri-data-eq}: \text{fri-data} = (ps, ds, ls, ms)$
by $(\text{cases } \text{fri-data})$
show $?thesis$
unfolding fri-data-eq
apply $(\text{simp add: case-prod-unfold})$
apply $(\text{rule no-failure-bindI}[OF \text{ send-no-failure}])$
apply $(\text{rule no-failure-ntimesI})$
apply $(\text{rule prover-query-round-no-failure})$
done
qed

lemma *prover-after-alphas-no-failure*:
 $\text{None} \notin \text{dom} (\text{dist} (\text{execute}$
 $(\text{do } \{$
 $\text{p.send } (\text{of-nat } (\text{degree } (\text{p.cp as p.f-powers})));$
 $\text{cp-merkle} \leftarrow \text{p.create } (\text{p.cp-eval as});$
 $(ps, ds, ls, ms) \leftarrow$
 $\text{p.composition-fri-commit } (\text{ceil-log } (\text{degree } (\text{p.cp as p.f-powers}) + 1))$
 $\text{[p.cp as p.f-powers] [p.eval-domain] [p.cp-eval as] [cp-merkle];}$
 $\text{p.send } (\text{hd } (\text{last } ls));$
 $\text{ntimes } (\text{prover-query-round } f\text{-merkle } f\text{-ls } f\text{-ms } ls \text{ ms}) \text{ rounds}$
 $\}) \text{ s}))$
unfolding Let-def
by $(\text{intro no-failure-bindI no-failure-bind-returnI})$
 $(\text{simp-all add: create-no-failure composition-fri-commit-no-failure}$
 $\text{send-no-failure})$

*p.receive-random-field-element-no-failure prover-decommit-on-query-no-failure
p.receive-alpha-challenge-no-failure p.receive-query-index-challenge-no-failure
prover-decommit-on-fri-layers-no-failure prover-after-fri-commit-no-failure
prover-after-fri-commit-named-no-failure)*

lemma *prover-after-trace-fri-commit-no-failure:*

None $\notin$ *dom* (*dist* (*execute*
((*case trace-fri-data of* (*f-ps*, *f-ds*, *f-ls*, *f-ms*) $\Rightarrow$
do {
p.send (*hd* (*last f-ls*));
as $\leftarrow$ *mmap* (*replicate* (*length spec*)
(*do* {
a $\leftarrow$ *p.receive-alpha-challenge*;
p.send a;
return a
}));
p.send (*of-nat* (*degree* (*p.cp as p.f-powers*)));
cp-merkle $\leftarrow$ *p.create* (*p.cp-eval as*);
(*ps*, *ds*, *ls*, *ms*) $\leftarrow$
p.composition-fri-commit (*ceil-log* (*degree* (*p.cp as p.f-powers*) + 1))
[*p.cp as p.f-powers*] [*p.eval-domain*] [*p.cp-eval as*] [*cp-merkle*];
p.send (*hd* (*last ls*));
ntimes (*prover-query-round f-merkle f-ls f-ms ls ms*) *rounds*
})) *s*))

proof —

obtain *f-ps f-ds f-ls f-ms* **where** *trace-fri-data-eq:*
trace-fri-data = (f-ps, f-ds, f-ls, f-ms)
by (*cases trace-fri-data*)
show *?thesis*
unfolding *trace-fri-data-eq case-prod-unfold Let-def*
by (*intro no-failure-bindI no-failure-mmapI*)
(*simp-all add: send-no-failure honest-alpha-send-round-no-failure
prover-after-alphas-no-failure create-no-failure
composition-fri-commit-no-failure
p.receive-alpha-challenge-no-failure p.receive-query-index-challenge-no-failure
no-failure-ntimesI prover-query-round-no-failure split: prod.splits*)

qed

lemma *prover-monad-no-failure:*

None $\notin$ *dom* (*dist* (*execute p.prover-monad init-state*))
unfolding *p.prover-monad-def Let-def*
apply (*rule no-failure-bindI*)
apply (*rule create-no-failure*)
apply (*rule no-failure-bindI*)
apply (*rule send-no-failure*)
apply (*rule no-failure-bindI*)
apply (*rule trace-fri-commit-no-failure*)
apply (*simp only: prover-query-round-def[symmetric]*)
apply (*rule prover-after-trace-fri-commit-no-failure*)

done

lemma *mfold-map-index-outcome-invariant-fromI:*

assumes *init*: $I\ k\ a\ s$

and *step-inv*:

$\bigwedge j\ x\ a\ s\ y\ t. k \leq j \implies j < k + \text{length}\ xs \implies$
 $x = xs ! (j - k) \implies I\ j\ a\ s \implies$
 $\text{Some}\ (y, t) \in \text{set-dist}\ (\text{execute}\ (F\ x\ a)\ s) \implies$
 $I\ (\text{Suc}\ j)\ y\ t$

and *outcome*:

$\text{Some}\ (y, t) \in \text{set-dist}\ (\text{execute}\ (\text{mfold}\ a\ (\text{map}\ F\ xs))\ s)$

shows $I\ (k + \text{length}\ xs)\ y\ t$

using *init step-inv outcome*

proof (*induction xs arbitrary: k a s y t*)

case *Nil*

then show ?case **by** *simp*

next

case (*Cons x xs*)

from *Cons.prem1* **obtain** *y0 t0* **where**

head: $\text{Some}\ (y0, t0) \in \text{set-dist}\ (\text{execute}\ (F\ x\ a)\ s)$

and *tail*:

$\text{Some}\ (y, t) \in \text{set-dist}\ (\text{execute}\ (\text{mfold}\ y0\ (\text{map}\ F\ xs))\ t0)$

unfolding *mfold.simps* **by** (*auto elim!: p.set-dist-bindE*)

have *inv-next*: $I\ (\text{Suc}\ k)\ y0\ t0$

by (*rule Cons.prem2*) [*of k x a s y0 t0*]

(*use Cons.prem1*) *head* **in** *simp-all*)

have *tail-inv*: $I\ (\text{Suc}\ k + \text{length}\ xs)\ y\ t$

proof (*rule Cons.IH*) [*OF inv-next - tail*]

fix *j x' a' s' y' t'*

assume *ge*: $\text{Suc}\ k \leq j$

and *lt*: $j < \text{Suc}\ k + \text{length}\ xs$

and *x'-eq*: $x' = xs ! (j - \text{Suc}\ k)$

and *inv*: $I\ j\ a'\ s'$

and *out*: $\text{Some}\ (y', t') \in \text{set-dist}\ (\text{execute}\ (F\ x'\ a')\ s')$

have *j-gt*: $k < j$

using *ge* **by** *simp*

have *j-lt*: $j < k + \text{length}\ (x \# xs)$

using *lt* **by** *simp*

have *nth-eq*: $x' = (x \# xs) ! (j - k)$

using *x'-eq j-gt* **by** *simp*

show $I\ (\text{Suc}\ j)\ y'\ t'$

by (*rule Cons.prem2*) [*OF - j-lt nth-eq inv out*] (*use ge* **in** *simp*)

qed

then show ?case **by** *simp*

qed

lemma *mfold-map-index-outcome-invariantI:*

assumes *init*: $I\ 0\ a\ s$

and *step-inv*:

$\bigwedge j \ x \ a \ s \ y \ t. \ j < \text{length } xs \implies$
 $x = xs ! j \implies I \ j \ a \ s \implies$
 $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (F \ x \ a) \ s) \implies$
 $I \ (\text{Suc } j) \ y \ t$
and *outcome*:
 $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (\text{mfold } a \ (\text{map } F \ xs)) \ s)$
shows $I \ (\text{length } xs) \ y \ t$
using *mfold-map-index-outcome-invariant-fromI* [
where $I=I$ **and** $k=0$ **and** $xs=xs$ **and** $F=F$, *OF init - outcome*]
step-inv
by *simp*

lemma *set-dist-bindI*:

assumes $\text{Some } (x, u) \in \text{set-dist } (\text{execute } m \ s)$
and $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (f \ x) \ u)$
shows $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (m \gg f) \ s)$
proof –
let $?K = \text{bind-cont-map } (\text{map-fun id dist } \circ \text{execute } \circ f)$
have $x\text{-dom}$: $\text{Some } (x, u) \in \text{dom } (\text{dist } (\text{execute } m \ s))$
using *assms(1)* **unfolding** *set-dist-def* .
have $y\text{-dom}'$: $\text{Some } (y, t) \in \text{dom } (\text{dist } (\text{execute } (f \ x) \ u))$
using *assms(2)* **unfolding** *set-dist-def* .
have $y\text{-dom}$: $\text{Some } (y, t) \in \text{dom } (?K \ (\text{Some } (x, u)))$
using *assms(2)*
unfolding *set-dist-def bind-cont-map-def*
by (*simp add: o-def map-fun-def*)
have $x\text{-pos}$: $0 < \text{the } (\text{dist } (\text{execute } m \ s) \ (\text{Some } (x, u)))$
using $x\text{-dom}$ **by** *simp*
have $y\text{-pos}$: $0 < \text{the } (\text{dist } (\text{execute } (f \ x) \ u) \ (\text{Some } (y, t)))$
using $y\text{-dom}'$ **by** *simp*
have $y\text{-opt-pos}$:
 $0 < \text{option-default } (?K \ (\text{Some } (x, u)) \ (\text{Some } (y, t)))$
using $y\text{-dom}' \ y\text{-pos}$
by (*cases dist (execute (f x) u) (Some (y, t))*)
(auto simp: bind-cont-map-def o-def map-fun-def domIff)
have term-pos :
 $0 <$
 $\text{the } (\text{dist } (\text{execute } m \ s) \ (\text{Some } (x, u))) * \text{option-default } (?K \ (\text{Some } (x, u)) \ (\text{Some } (y, t)))$
using $x\text{-pos } y\text{-opt-pos}$ **by** *simp*
have sum-nonzero :
 $(\sum i \in \text{dom } (\text{dist } (\text{execute } m \ s)). \text{the } (\text{dist } (\text{execute } m \ s) \ i) * \text{option-default } (?K \ i \ (\text{Some } (y, t)))) \neq 0$

proof –

have $0 <$
 $(\sum i \in \text{dom } (\text{dist } (\text{execute } m \ s)). \text{the } (\text{dist } (\text{execute } m \ s) \ i) * \text{option-default } (?K \ i \ (\text{Some } (y, t))))$

```

using sum-pos2[OF finite-dom-dist x-dom, of
  λi. the (dist (execute m s) i) *
  option-default (?K i (Some (y, t)))]
  term-pos
by auto
then show ?thesis by simp
qed
have Some (y, t) ∈
  {z ∈ (⋃ i ∈ dom (dist (execute m s)). dom (?K i)).
  (∑ i ∈ dom (dist (execute m s)).
  the (dist (execute m s) i) * option-default (?K i z)) ≠ 0}
using x-dom y-dom sum-nonzero by auto
then show ?thesis
  unfolding set-dist-def sm-bind.rep-eq dist-bind.rep-eq
  by (simp add: o-def map-fun-def dom-map-bind)
qed

lemma set-dist-execute-throw-empty[simp]:
  Some x ∉ set-dist (execute throw s)
  unfolding throw.rep-eq dist-throw-def dist-delta-dist delta-map-def set-dist-def
  by simp

lemma assert-outcomeD:
  assumes outcome: Some ((), t) ∈ set-dist (execute (assert b) s)
  shows b t = s
proof –
  show b
  proof (cases b)
    case True
    then show ?thesis .
  next
    case False
    then show ?thesis
    using outcome by (simp add: assert-def)
  qed
then show t = s
  using outcome by (simp add: assert-def)
qed

lemma assert-hash-extends:
  assumes Some ((), t) ∈
    set-dist (execute (assert b) (s::('f, 'a) protocol-channel-scheme))
  shows s ≤ t
  using assert-outcomeD(2)[OF assms] p.hash-ext-refl[of s] by simp

lemma modify-verifier-replay-state-outcome:
  assumes Some (u, t) ∈
    set-dist (execute
      (modify (λc. c[|PState := 0, PTranscript := rev (PTranscript c),

```

```

      PTraceFriCounter := 0, PCompositionFriCounter := 0,
      PAlphaCounter := 0, PQueryCounter := 0))) s)
shows u = ()  $\wedge$  t = verifier-replay-state s
using assms
unfolding verifier-replay-state-def modify-def
by (auto elim!: p.set-dist-bindE)

lemma transcript-extends-refl[simp]:
  transcript-extends s s
unfolding transcript-extends-def by simp

lemma transcript-extends-trans:
  assumes transcript-extends u s
    and transcript-extends t u
shows transcript-extends t s
using assms unfolding transcript-extends-def by auto

lemma read-hash-extends:
  assumes Some (x, t)  $\in$  set-dist (execute p.read s)
shows s  $\leq$  t
by (rule p.read-hash-extends[OF assms])

lemma ntimes-read-hash-extends:
  assumes Some (xs, t)  $\in$  set-dist (execute (ntimes p.read n) s)
shows s  $\leq$  t
using assms
proof (induction n arbitrary: xs s t)
  case 0
  then show ?case
    by (simp add: p.hash-ext-refl)
next
  case (Suc n)
  from Suc.premis obtain x ys u where
    read: Some (x, u)  $\in$  set-dist (execute p.read s)
    and reads:
      Some (ys, t)  $\in$  set-dist (execute (ntimes p.read n) u)
    by (auto elim!: p.set-dist-bindE)
  have s  $\leq$  u
    using read by (rule read-hash-extends)
  moreover have u  $\leq$  t
    using reads by (rule Suc.IH)
  ultimately show ?case
    by (rule p.hash-ext-trans)
qed

lemma check-authentication-path-hash-extends:
  assumes outcome:
    Some (h, t)  $\in$ 
      set-dist (execute (p.check-authentication-path len i v path) s)

```


shows $s \leq t$
by (rule *p.check-authentication-path-hash-extends*[*OF outcome*])

lemma *honest-trace-valid-raw-algebra*:

assumes *honest-trace-valid*

shows

$\forall (c, \text{roots}, d) \in \text{set spec.}$
 $(\forall r \in \text{set } (p.g\text{-map roots}). \text{poly } (c \text{ p.f-powers}) r = 0) \wedge$
 $\text{degree } (c \text{ p.f-powers}) \leq d * (\text{clength} - 1) \wedge$
 $\text{distinct } (p.g\text{-map roots})$

using *assms*

unfolding *honest-trace-valid-def honest-trace-algebra-def*

constraint-roots-vanish-def

using *constraint-degree-bound p.g-map-distinct*

by *force*

lemma *honest-trace-valid-algebra*:

assumes *honest-trace-valid*

shows *honest-trace-algebra*

using *assms unfolding honest-trace-valid-def by simp*

lemma *straight-line-honest-transcript-replay*:

assumes $P\text{Transcript } s = []$

and $\text{Some } ((), \text{sent}) \in \text{set-dist } (\text{execute } (\text{mfold2 } p.\text{send } xs) s)$

and $\text{Some } (ys, \text{replayed}) \in$

$\text{set-dist } (\text{execute}$

$(\text{ntimes } p.\text{read } (\text{length } xs))$

$(\text{sent}(|P\text{State} := P\text{State } s, P\text{Transcript} := \text{rev } (P\text{Transcript } \text{sent})|)))$

shows $ys = xs \wedge \text{replayed} = \text{sent}(|P\text{Transcript} := []|)$

using *p.honest-transcript-replay-mfold2-ntimes*[*OF assms*] .

lemma *honest-alpha-round-replay-no-failure*:

assumes $\text{lookup: fmllookup } (\text{HashMap } s) (\text{AlphaChallenge } (P\text{AlphaCounter } s) (P\text{State } s)) =$

Some a

shows

$\text{None} \notin \text{dom } (\text{dist } (\text{execute}$

$(\text{do } \{$

$a0 \leftarrow p.\text{receive-alpha-challenge};$

$\text{let } a0' = a0;$

$a1 \leftarrow p.\text{read};$

$\text{let } a1' = a1;$

$\text{assert } (a0' = a1');$

$\text{return } a1'$

$\}) (s(|P\text{Transcript} := a \# \text{rest}|)))$

proof –

let $?start = s(|P\text{Transcript} := a \# \text{rest}|)$

let $?prefix =$

$\text{do } \{$

$a0 \leftarrow p.\text{receive-alpha-challenge};$

```

    a1 ← p.read;
    return (a0, a1)
  }
have lookup-start:
  fmllookup (HashMap ?start) (AlphaChallenge (PAlphaCounter ?start) (PState ?start)) = Some
a
  using lookup by simp
have prefix-no-failure:
  None ∉ dom (dist (execute ?prefix ?start))
proof (rule no-failure-bindI)
  show None ∉ dom (dist (execute p.receive-alpha-challenge ?start))
  by (rule p.receive-alpha-challenge-no-failure)
next
fix a0 t
assume rand: Some (a0, t) ∈ set-dist (execute p.receive-alpha-challenge ?start)
have tr-t: PTranscript t = a # rest
  using p.receive-alpha-challenge-known-outcome[OF lookup-start rand] by simp
show None ∉ dom (dist (execute (p.read ≫ (λa1. return (a0, a1))) t))
proof (rule no-failure-bindI)
  show None ∉ dom (dist (execute p.read t))
  by (rule p.read-no-failure) (simp add: tr-t)
next
fix a1 u
assume Some (a1, u) ∈ set-dist (execute p.read t)
show None ∉ dom (dist (execute (return (a0, a1)) u))
  by simp
qed
qed
have full:
  (do {
    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }) =
  (?prefix ≫ (λ(a0, a1). assert (a0 = a1) ≫ (λ-. return a1)))
  by (simp add: Let-def sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full
proof (rule no-failure-bindI[OF prefix-no-failure])
  fix aa t
  assume prefix: Some (aa, t) ∈ set-dist (execute ?prefix ?start)
  obtain a0 a1 where aa-eq: aa = (a0, a1)
  by (cases aa)
  from prefix[unfolded aa-eq] obtain r-state where
  rand:
    Some (a0, r-state) ∈ set-dist (execute p.receive-alpha-challenge ?start)

```

```

and read:
  Some (a1, t) ∈ set-dist (execute p.read r-state)
by (auto elim!: p.set-dist-bindE)
have rand-res:
  a0 = a ∧ PState r-state = PState s ∧ PTranscript r-state = a # rest
  using p.receive-alpha-challenge-known-outcome[OF lookup-start rand] by simp
have read-start:
  r-state = r-state(PTranscript := a # rest)
  using rand-res by simp
have read':
  Some (a1, t) ∈ set-dist (execute p.read (r-state(PTranscript := a # rest)))
  by (subst read-start[symmetric]) (rule read)
have a1 = a
  using p.read-cons-outcome[OF read'] rand-res by simp
then show None ∉ dom
  (dist (execute ((case aa of (a0, a1) ⇒ assert (a0 = a1) >>> (λ-. return a1))) t))
  using rand-res unfolding aa-eq assert-def by simp
qed
qed

lemma honest-alpha-round-replay-outcome:
  assumes lookup: fmlookup (HashMap s) (AlphaChallenge (PAlphaCounter s) (PState s)) =
  Some a
  and outcome:
    Some (y, t) ∈ set-dist (execute
      (do {
        a0 ← p.receive-alpha-challenge;
        let a0' = a0;
        a1 ← p.read;
        let a1' = a1;
        assert (a0' = a1');
        return a1'
      }) (s(PTranscript := a # rest)))
  shows
    y = a ∧
    PState t = concat (PState s) a ∧
    PTranscript t = rest ∧
    PTraceFriCounter t = PTraceFriCounter s ∧
    PCompositionFriCounter t = PCompositionFriCounter s ∧
    PAlphaCounter t = Suc (PAlphaCounter s) ∧
    PQueryCounter t = PQueryCounter s ∧
    s ≤ t
proof -
  let ?start = s(PTranscript := a # rest)
  from outcome obtain a0 r-state read-state where
    rand:
      Some (a0, r-state) ∈ set-dist (execute p.receive-alpha-challenge ?start)
  and read:
    Some (y, read-state) ∈ set-dist (execute p.read r-state)

```

```

and assert-ok: Some ( $()$ ,  $t$ )  $\in$  set-dist (execute (assert ( $a0 = y$ )) read-state)
by (auto simp: Let-def elim!: p.set-dist-bindE)
have lookup-start: fmlookup (HashMap ?start) (AlphaChallenge (PAlphaCounter ?start) (PState
?start)) = Some a
  using lookup by simp
have rand-known:
   $a0 = a \wedge$ 
   $PState\ r\text{-}state = PState\ s \wedge$ 
   $PTranscript\ r\text{-}state = a \# rest \wedge$ 
   $?start \leq r\text{-}state$ 
  using p.receive-alpha-challenge-known-outcome[OF lookup-start rand] by simp
have read-start:
   $r\text{-}state = r\text{-}state(PTranscript := a \# rest)$ 
  using rand-known by simp
have read':
   $Some\ (y, read\text{-}state) \in$ 
   $set\text{-}dist\ (execute\ p.read\ (r\text{-}state(PTranscript := a \# rest)))$ 
  by (subst read-start[symmetric]) (rule read)
have read-res:
   $y = a \wedge$ 
   $read\text{-}state =$ 
   $r\text{-}state(PState := concat\ (PState\ r\text{-}state)\ a,\ PTranscript := rest)$ 
  using p.read-cons-outcome[OF read'] by simp
have t-eq:  $t = read\text{-}state$ 
  using assert-ok rand-known read-res
  unfolding assert-def by simp
have alpha-counter:
   $PAlphaCounter\ t = Suc\ (PAlphaCounter\ s)$ 
  using p.receive-alpha-challenge-counter-outcome[OF rand] rand-known read-res
  unfolding t-eq by simp
have other-counters:
   $PTraceFriCounter\ t = PTraceFriCounter\ s \wedge$ 
   $PCompositionFriCounter\ t = PCompositionFriCounter\ s \wedge$ 
   $PQueryCounter\ t = PQueryCounter\ s$ 
  using p.receive-alpha-challenge-counter-outcome[OF rand] rand-known read-res
  unfolding t-eq by simp
have s-t:  $s \leq t$ 
proof –
  have s-start:  $s \leq ?start$ 
    unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  have start-r:  $?start \leq r\text{-}state$ 
    using rand-known by simp
  have r-read:  $r\text{-}state \leq read\text{-}state$ 
    using read-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  show ?thesis
    using s-start start-r r-read unfolding t-eq
    by (meson p.hash-ext-trans)
qed
show ?thesis

```

using *rand-known read-res t-eq alpha-counter other-counters s-t* **by** *simp*
qed

lemma *honest-alpha-round-replay-parts-outcome:*

assumes *lookup: fmlookup (HashMap s) (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a*

and *rand:*

Some (a0, r-state) ∈

set-dist (execute p.receive-alpha-challenge (s(PTranscript := a # rest)))

and *read:*

Some (y, read-state) ∈ set-dist (execute p.read r-state)

and *assert-ok:*

Some ((), t) ∈ set-dist (execute (assert (a0 = y)) read-state)

shows

y = a ∧

PState t = concat (PState s) a ∧

PTranscript t = rest ∧

PTraceFriCounter t = PTraceFriCounter s ∧

PCompositionFriCounter t = PCompositionFriCounter s ∧

PAlphaCounter t = Suc (PAlphaCounter s) ∧

PQueryCounter t = PQueryCounter s ∧

s ≤ t

proof –

let *?start = s(PTranscript := a # rest)*

have *lookup-start: fmlookup (HashMap ?start) (AlphaChallenge (PAlphaCounter ?start) (PState ?start)) = Some a*

using *lookup* **by** *simp*

have *rand-known:*

a0 = a ∧

PState r-state = PState s ∧

PTranscript r-state = a # rest ∧

?start ≤ r-state

using *p.receive-alpha-challenge-known-outcome[OF lookup-start rand]* **by** *simp*

have *read-start:*

r-state = r-state(PTranscript := a # rest)

using *rand-known* **by** *simp*

have *read':*

Some (y, read-state) ∈

set-dist (execute p.read (r-state(PTranscript := a # rest)))

by (*subst read-start[symmetric]*) (*rule read*)

have *read-res:*

y = a ∧

read-state =

r-state(PState := concat (PState r-state) a, PTranscript := rest)

using *p.read-cons-outcome[OF read']* **by** *simp*

have *t-eq: t = read-state*

using *assert-ok rand-known read-res*

unfolding *assert-def* **by** *simp*

have *alpha-counter:*

```

PAlphaCounter t = Suc (PAlphaCounter s)
using p.receive-alpha-challenge-counter-outcome[OF rand] rand-known read-res
unfolding t-eq by simp
have other-counters:
  PTraceFriCounter t = PTraceFriCounter s ∧
  PCompositionFriCounter t = PCompositionFriCounter s ∧
  PQueryCounter t = PQueryCounter s
using p.receive-alpha-challenge-counter-outcome[OF rand] rand-known read-res
unfolding t-eq by simp
have s-t: s ≤ t
proof –
  have s-start: s ≤ ?start
    unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  have start-r: ?start ≤ r-state
    using rand-known by simp
  have r-read: r-state ≤ read-state
    using read-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  show ?thesis
    using s-start start-r r-read unfolding t-eq
    by (meson p.hash-ext-trans)
qed
show ?thesis
  using rand-known read-res t-eq alpha-counter other-counters s-t by simp
qed

lemma honest-receive-fri-commit-replay-outcome:
  assumes lookup: fmlookup (HashMap s) (FiatShamirChallenge (concat (PState s) fri-root)) =
Some challenge
  and outcome:
    Some ((b, r), t) ∈
      set-dist (execute v.receive-fri-commits (s⟦PTranscript := fri-root # rest⟧))
  shows
    r = fri-root ∧
    b = challenge ∧
    PState t = concat (PState s) fri-root ∧
    PTranscript t = rest ∧
    s ≤ t
proof –
  from outcome obtain u where
    read-root: Some (r, u) ∈ set-dist (execute p.read (s⟦PTranscript := fri-root # rest⟧))
    and rand: Some (b, t) ∈ set-dist (execute p.receive-random-field-element u)
    unfolding v.receive-fri-commits-def
    by (auto elim!: p.set-dist-bindE)
  have read-res:
    r = fri-root ∧
    u = s⟦PState := concat (PState s) fri-root, PTranscript := rest⟧
    using p.read-cons-outcome[OF read-root] by simp
  have lookup-u: fmlookup (HashMap u) (FiatShamirChallenge (PState u)) = Some challenge
    using lookup read-res by simp

```

```

have rand-res:
   $b = \text{challenge} \wedge P\text{State } t = \text{concat } (P\text{State } s) \text{ fri-root} \wedge P\text{Transcript } t = \text{rest} \wedge u \leq t$ 
  using  $p.\text{receive-random-field-element-known-outcome}[OF \text{ lookup-}u \text{ rand}]$  rand-res by simp
have s-u:  $s \leq u$ 
  using read-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s-t:  $s \leq t$ 
  using s-u rand-res by (meson  $p.\text{hash-ext-trans}$ )
show ?thesis
  using read-res rand-res s-t by simp
qed

lemma honest-receive-fri-commit-no-failure:
  assumes  $P\text{Transcript } s \neq []$ 
  shows
     $\text{None} \notin \text{dom } (\text{dist } (\text{execute } v.\text{receive-fri-commits } s))$ 
  using assms
  unfolding  $v.\text{receive-fri-commits-def}$ 
  by (intro no-failure-bindI) (simp-all add:  $p.\text{read-no-failure } p.\text{receive-random-field-element-no-failure}$ )

lemma receive-fri-commit-preserves-query-counter:
  assumes outcome:
     $\text{Some } (br, t) \in \text{set-dist } (\text{execute } v.\text{receive-fri-commits } s)$ 
  shows  $P\text{QueryCounter } t = P\text{QueryCounter } s$ 
proof –
  from outcome obtain  $r u b$  where
    read-root:  $\text{Some } (r, u) \in \text{set-dist } (\text{execute } p.\text{read } s)$ 
    and rand:  $\text{Some } (b, t) \in \text{set-dist } (\text{execute } p.\text{receive-random-field-element } u)$ 
    unfolding  $v.\text{receive-fri-commits-def}$ 
    by (auto elim!:  $p.\text{set-dist-bindE}$ )
  have q-read:  $P\text{QueryCounter } u = P\text{QueryCounter } s$ 
  proof (cases  $P\text{Transcript } s$ )
    case Nil
    then have False
    using read-root
    unfolding  $p.\text{read-def } \text{protocol-read-def } \text{assert-def}$ 
    by (auto elim!:  $p.\text{set-dist-bindE}$ )
    then show ?thesis by simp
  next
    case (Cons  $x xs$ )
    then show ?thesis
    using  $p.\text{read-nonempty-outcome}[OF \text{ read-root Cons}]$  by simp
  qed
  have q-rand:  $P\text{QueryCounter } t = P\text{QueryCounter } u$ 
  using  $p.\text{receive-random-field-element-counter-outcome}[OF \text{ rand}]$  by simp
  show ?thesis
  using q-read q-rand by simp
qed

lemma receive-fri-commits-preserves-query-counter:

```

```

assumes outcome:
  Some (brs, t) ∈ set-dist (execute (ntimes v.receive-fri-commits n) s)
shows PQueryCounter t = PQueryCounter s
using outcome
proof (induction n arbitrary: brs s t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.prems obtain br u brs' where
    head: Some (br, u) ∈ set-dist (execute v.receive-fri-commits s)
    and tail:
      Some (brs', t) ∈
        set-dist (execute (ntimes v.receive-fri-commits n) u)
    by (auto elim!: p.set-dist-bindE)
  have q-head: PQueryCounter u = PQueryCounter s
    by (rule receive-fri-commit-preserves-query-counter[OF head])
  have q-tail: PQueryCounter t = PQueryCounter u
    by (rule Suc.IH[OF tail])
  show ?case
    using q-head q-tail by simp
qed

lemma honest-receive-fri-commits-list-outcome:
  assumes lookups:
     $\forall i < \text{length roots.}$ 
    fmllookup (HashMap s) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) roots)))
  =
    Some (challenges ! i)
    and len: length challenges = length roots
    and outcome:
      Some (brs, t) ∈
        set-dist (execute (ntimes v.receive-fri-commits (length roots))
          (s(PTranscript := roots @ rest))))
  shows
    brs = zip challenges roots ∧
    PState t = foldl concat (PState s) roots ∧
    PTranscript t = rest ∧
    s ≤ t
  using lookups len outcome
proof (induction roots arbitrary: challenges brs s t rest)
  case Nil
  have brs-eq: brs = []
    using Nil.prems( $\mathcal{I}$ ) by simp
  have t-eq: t = s(PTranscript := rest)
    using Nil.prems( $\mathcal{I}$ ) by simp
  have s-t: s ≤ t
    unfolding t-eq less-eq-hash-ext-def less-eq-fmap-def by simp
  show ?case

```



```

    using brs-eq t-eq s-t Nil.premis(2) by simp
next
case (Cons root roots)
from Cons.premis(2) obtain challenge challenges' where challenges-eq:
  challenges = challenge # challenges'
by (cases challenges) auto
from Cons.premis(3) obtain br u brs' where
  head:
    Some (br, u) ∈
      set-dist (execute v.receive-fri-commits
        (s(PTranscript := root # (roots @ rest))))))
  and tail:
    Some (brs', t) ∈
      set-dist (execute (ntimes v.receive-fri-commits (length roots)) u)
  and brs-eq: br = br # brs'
by (auto elim!: p.set-dist-bindE)
have lookup-root:
  fmlookup (HashMap s) (FiatShamirChallenge (concat (PState s) root)) = Some challenge
  using spec[OF Cons.premis(1), of 0] challenges-eq by simp
obtain b r where br-eq: br = (b, r)
  by (cases br)
have head-out:
  r = root ∧
  b = challenge ∧
  PState u = concat (PState s) root ∧
  PTranscript u = roots @ rest ∧
  s ≤ u
  using honest-receive-fri-commit-replay-outcome[
    OF lookup-root head[unfolded br-eq]]
  unfolding br-eq by simp
have tail-lookups:
  ∀ i < length roots.
    fmlookup (HashMap u) (FiatShamirChallenge (foldl concat (PState u) (take (Suc i) roots)))
=
  Some (challenges' ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have old-lookup:
    fmlookup (HashMap s)
      (FiatShamirChallenge (foldl concat (PState s) (take (Suc (Suc i)) (root # roots)))) =
    Some ((challenge # challenges') ! Suc i)
  using spec[OF Cons.premis(1), of Suc i] i-bound challenges-eq by simp
  have key-eq:
    foldl concat (PState s) (take (Suc (Suc i)) (root # roots)) =
    foldl concat (PState u) (take (Suc i) roots)
  using head-out by simp
  show fmlookup (HashMap u) (FiatShamirChallenge (foldl concat (PState u) (take (Suc i)
roots))) =

```

```

    Some (challenges' ! i)
    using p.hash-extension-lookup[OF old-lookup, of u] head-out key-eq i-bound
    by simp
qed
have u-update: u(PTranscript := roots @ rest) = u
    using head-out by simp
have tail':
    Some (brs', t) ∈
        set-dist (execute (ntimes v.receive-fri-commits (length roots))
            (u(PTranscript := roots @ rest)))
    using tail unfolding u-update .
have tail-len: length challenges' = length roots
    using Cons.prem(2) challenges-eq by simp
have tail-out:
    brs' = zip challenges' roots ∧
    PState t = foldl concat (PState u) roots ∧
    PTranscript t = rest ∧
    u ≤ t
    using Cons.IH[OF tail-lookups tail-len tail] .
have s-t: s ≤ t
    using head-out tail-out by (meson p.hash-ext-trans)
show ?case
    using brs-eq br-eq head-out tail-out s-t challenges-eq by simp
qed

lemma honest-receive-fri-commits-list-no-failure:
    assumes lookups:
        ∀ i < length roots.
            fmlookup (HashMap s) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) roots)))
    =
        Some (challenges ! i)
    and len: length challenges = length roots
    shows
        None ∉ dom (dist (execute
            (ntimes v.receive-fri-commits (length roots))
            (s(PTranscript := roots @ rest))))
    using lookups len
proof (induction roots arbitrary: challenges s rest)
    case Nil
    then show ?case by simp
next
    case (Cons root roots)
    from Cons.prem(2) obtain challenge challenges' where challenges-eq:
        challenges = challenge # challenges'
    by (cases challenges) auto
    have lookup-root:
        fmlookup (HashMap s) (FiatShamirChallenge (concat (PState s) root)) = Some challenge
    using spec[OF Cons.prem(1), of 0] challenges-eq by simp
    have head-nf:

```

```

None ∉ dom (dist (execute v.receive-fri-commits
  (s(PTranscript := root # (roots @ rest))))))
by (rule honest-receive-fri-commit-no-failure) simp
have case-eq:
  ?case =
    (None ∉ dom (dist (execute
      (v.receive-fri-commits >>=
        (λbr. ntimes v.receive-fri-commits (length roots) >>=
          (λbrs. return (br # brs))))
      (s(PTranscript := root # (roots @ rest))))))
    by simp
show ?case
  unfolding case-eq
proof (rule no-failure-bindI[OF head-nf])
  fix br u
  assume head:
    Some (br, u) ∈
      set-dist (execute v.receive-fri-commits
        (s(PTranscript := root # roots @ rest)))
  obtain b r where br-eq: br = (b, r)
  by (cases br)
  have head-out:
    r = root ∧
    b = challenge ∧
    PState u = concat (PState s) root ∧
    PTranscript u = roots @ rest ∧
    s ≤ u
  using honest-receive-fri-commit-replay-outcome[
    OF lookup-root head[unfolded br-eq]]
  unfolding br-eq by simp
  have tail-lookups:
    ∀ i < length roots.
      fmlookup (HashMap u) (FiatShamirChallenge (foldl concat (PState u) (take (Suc i) roots)))
=
    Some (challenges' ! i)
  proof (intro allI impI)
    fix i
    assume i-bound: i < length roots
    have old-lookup:
      fmlookup (HashMap s)
        (FiatShamirChallenge (foldl concat (PState s) (take (Suc (Suc i)) (root # roots)))) =
        Some ((challenge # challenges') ! Suc i)
    using spec[OF Cons.premis(1), of Suc i] i-bound challenges-eq by simp
    have key-eq:
      foldl concat (PState s) (take (Suc (Suc i)) (root # roots)) =
      foldl concat (PState u) (take (Suc i) roots)
    using head-out by simp
    show fmlookup (HashMap u) (FiatShamirChallenge (foldl concat (PState u) (take (Suc i)
roots))) =

```

```

    Some (challenges' ! i)
    using p.hash-extension-lookup[OF old-lookup, of u] head-out key-eq i-bound
    by simp
qed
have tail-len: length challenges' = length roots
  using Cons.premis(2) challenges-eq by simp
have tail-nf-start:
  None  $\notin$  dom (dist (execute
    (ntimes v.receive-fri-commits (length roots))
    (u(PTranscript := roots @ rest))))
  using Cons.IH[OF tail-lookups tail-len, of rest] .
have u-update: u(PTranscript := roots @ rest) = u
  using head-out by simp
have tail-nf:
  None  $\notin$  dom (dist (execute
    (ntimes v.receive-fri-commits (length roots)) u))
  using tail-nf-start unfolding u-update .
show None  $\notin$  dom (dist (execute
  (ntimes v.receive-fri-commits (length roots)  $\gg$ 
    ( $\lambda$ brs. return (br # brs))) u))
  by (rule no-failure-bindI[OF tail-nf]) simp
qed
qed

lemma honest-receive-fri-commit-with-no-failure:
  assumes receiver-nf:
     $\bigwedge s. \text{None} \notin \text{dom} (\text{dist} (\text{execute} \text{receive-challenge } s))$ 
    and tr: PTranscript s  $\neq$  []
  shows
    None  $\notin$  dom (dist
      (execute (v.receive-fri-commits-with receive-challenge) s))
  using tr
  unfolding v.receive-fri-commits-with-def
  by (intro no-failure-bindI)
  (simp-all add: p.read-no-failure receiver-nf)

lemma honest-receive-trace-fri-commit-replay-outcome:
  assumes lookup:
    fmlookup (HashMap s)
    (TraceFriChallenge (PTraceFriCounter s)
      (concat (PState s) fri-root)) = Some challenge
  and outcome:
    Some ((b, r), t)  $\in$ 
    set-dist (execute
      v.receive-trace-fri-commits
      (s(PTranscript := fri-root # rest)))
  shows
    r = fri-root  $\wedge$ 
    b = challenge  $\wedge$ 

```

$PState\ t = concat\ (PState\ s)\ fri_root \wedge$
 $PTranscript\ t = rest \wedge$
 $PTraceFriCounter\ t = Suc\ (PTraceFriCounter\ s) \wedge$
 $s \leq t$
proof –
from *outcome* **obtain** *u* **where**
read-root:
Some $(r, u) \in$
set-dist $(execute\ p.read$
 $(s(PTranscript := fri_root \# rest)))$
and *rand*:
Some $(b, t) \in set-dist\ (execute\ p.receive-trace-fri-challenge\ u)$
unfolding *v.receive-trace-fri-commits-def v.receive-fri-commits-with-def*
by $(auto\ elim!:\ p.set-dist-bindE)$
have *read-res*:
 $r = fri_root \wedge$
 $u = s(PState := concat\ (PState\ s)\ fri_root,\ PTranscript := rest)$
using *p.read-cons-outcome* $[OF\ read-root]$ **by** *simp*
have *lookup-u*:
 $fmlookup\ (HashMap\ u)\ (TraceFriChallenge\ (PTraceFriCounter\ u)\ (PState\ u)) = Some\ chal-$
lenge
using *lookup read-res* **by** *simp*
have *rand-res*:
 $b = challenge \wedge PState\ t = concat\ (PState\ s)\ fri_root \wedge$
 $PTranscript\ t = rest \wedge u \leq t$
using *p.receive-trace-fri-challenge-known-outcome* $[OF\ lookup-u\ rand]$
read-res **by** *simp*
have *rand-counter*:
 $PTraceFriCounter\ t = Suc\ (PTraceFriCounter\ s)$
using *p.receive-trace-fri-challenge-counter-outcome* $[OF\ rand]$ *read-res*
by *simp*
have *s-u*: $s \leq u$
using *read-res* **unfolding** *less-eq-hash-ext-def less-eq-fmap-def* **by** *simp*
have *s-t*: $s \leq t$
using *s-u rand-res* **by** $(meson\ p.hash-ext-trans)$
show *?thesis*
using *read-res rand-res rand-counter s-t* **by** *simp*
qed

lemma *honest-receive-composition-fri-commit-replay-outcome*:

assumes *lookup*:
 $fmlookup\ (HashMap\ s)$
 $(CompositionFriChallenge\ (PCompositionFriCounter\ s)$
 $(concat\ (PState\ s)\ fri_root))) =$
 $Some\ challenge$
and *outcome*:
 $Some\ ((b, r), t) \in$
 $set-dist\ (execute$
 $v.receive-composition-fri-commits$

```

      (s(PTranscript := fri-root # rest)))
shows
  r = fri-root ∧
  b = challenge ∧
  PState t = concat (PState s) fri-root ∧
  PTranscript t = rest ∧
  PCompositionFriCounter t = Suc (PCompositionFriCounter s) ∧
  s ≤ t
proof -
  from outcome obtain u where
    read-root:
      Some (r, u) ∈
        set-dist (execute p.read
          (s(PTranscript := fri-root # rest)))
    and rand:
      Some (b, t) ∈
        set-dist (execute p.receive-composition-fri-challenge u)
  unfolding v.receive-composition-fri-commits-def
    v.receive-fri-commits-with-def
  by (auto elim!: p.set-dist-bindE)
  have read-res:
    r = fri-root ∧
    u = s(PState := concat (PState s) fri-root, PTranscript := rest)
  using p.read-cons-outcome[OF read-root] by simp
  have lookup-u:
    fmlookup (HashMap u) (CompositionFriChallenge (PCompositionFriCounter u) (PState u))
  =
    Some challenge
  using lookup read-res by simp
  have rand-res:
    b = challenge ∧ PState t = concat (PState s) fri-root ∧
    PTranscript t = rest ∧ u ≤ t
  using p.receive-composition-fri-challenge-known-outcome[OF lookup-u rand]
    read-res by simp
  have rand-counter:
    PCompositionFriCounter t = Suc (PCompositionFriCounter s)
  using p.receive-composition-fri-challenge-counter-outcome[OF rand] read-res
    by simp
  have s-u: s ≤ u
  using read-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  have s-t: s ≤ t
  using s-u rand-res by (meson p.hash-ext-trans)
  show ?thesis
  using read-res rand-res rand-counter s-t by simp
qed

lemma read-preserves-counters:
  assumes outcome: Some (r, t) ∈ set-dist (execute p.read s)
  shows

```

```

    PTraceFriCounter t = PTraceFriCounter s ∧
    PCompositionFriCounter t = PCompositionFriCounter s ∧
    PAlphaCounter t = PAlphaCounter s ∧
    PQueryCounter t = PQueryCounter s
proof (cases PTranscript s)
  case Nil
  then have False
  using outcome
  unfolding p.read-def protocol-read-def assert-def
  by (auto elim!: p.set-dist-bindE)
  then show ?thesis by simp
next
  case (Cons x xs)
  then show ?thesis
  using p.read-nonempty-outcome[OF outcome Cons] by simp
qed

lemma ntimes-read-preserves-query-counter:
  assumes outcome: Some (xs, t) ∈ set-dist (execute (ntimes p.read n) s)
  shows PQueryCounter t = PQueryCounter s
  using outcome
proof (induction n arbitrary: xs s t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.premis obtain x u xs' where
    head: Some (x, u) ∈ set-dist (execute p.read s)
    and tail: Some (xs', t) ∈ set-dist (execute (ntimes p.read n) u)
    by (auto elim!: p.set-dist-bindE)
  have q-head: PQueryCounter u = PQueryCounter s
  using read-preserves-counters[OF head] by simp
  have q-tail: PQueryCounter t = PQueryCounter u
  by (rule Suc.IH[OF tail])
  show ?case
  using q-head q-tail by simp
qed

lemma receive-trace-fri-commit-preserves-other-counters:
  assumes outcome:
    Some (br, t) ∈ set-dist (execute v.receive-trace-fri-commits s)
  shows
    PCompositionFriCounter t = PCompositionFriCounter s ∧
    PAlphaCounter t = PAlphaCounter s ∧
    PQueryCounter t = PQueryCounter s
proof –
  from outcome obtain r u b where
    read-root: Some (r, u) ∈ set-dist (execute p.read s)
    and rand: Some (b, t) ∈ set-dist (execute p.receive-trace-fri-challenge u)

```

unfolding $v.receive\text{-}trace\text{-}fri\text{-}commits\text{-}def$ $v.receive\text{-}fri\text{-}commits\text{-}with\text{-}def$
by $(auto\ elim!:\ p.set\text{-}dist\text{-}bindE)$
have $read\text{-}res$:
 $PCompositionFriCounter\ u = PCompositionFriCounter\ s \wedge$
 $PAlphaCounter\ u = PAlphaCounter\ s \wedge$
 $PQueryCounter\ u = PQueryCounter\ s$
using $read\text{-}preserves\text{-}counters[OF\ read\text{-}root]$ **by** $simp$
show $?thesis$
using $p.receive\text{-}trace\text{-}fri\text{-}challenge\text{-}counter\text{-}outcome[OF\ rand]$ $read\text{-}res$
by $simp$
qed

lemma $receive\text{-}composition\text{-}fri\text{-}commit\text{-}preserves\text{-}other\text{-}counters$:

assumes $outcome$:

$Some\ (br,\ t) \in set\text{-}dist\ (execute\ v.receive\text{-}composition\text{-}fri\text{-}commits\ s)$

shows

$PTraceFriCounter\ t = PTraceFriCounter\ s \wedge$

$PAlphaCounter\ t = PAlphaCounter\ s \wedge$

$PQueryCounter\ t = PQueryCounter\ s$

proof –

from $outcome$ **obtain** $r\ u\ b$ **where**

$read\text{-}root$: $Some\ (r,\ u) \in set\text{-}dist\ (execute\ p.read\ s)$

and $rand$:

$Some\ (b,\ t) \in set\text{-}dist\ (execute\ p.receive\text{-}composition\text{-}fri\text{-}challenge\ u)$

unfolding $v.receive\text{-}composition\text{-}fri\text{-}commits\text{-}def$

$v.receive\text{-}fri\text{-}commits\text{-}with\text{-}def$

by $(auto\ elim!:\ p.set\text{-}dist\text{-}bindE)$

have $read\text{-}res$:

$PTraceFriCounter\ u = PTraceFriCounter\ s \wedge$

$PAlphaCounter\ u = PAlphaCounter\ s \wedge$

$PQueryCounter\ u = PQueryCounter\ s$

using $read\text{-}preserves\text{-}counters[OF\ read\text{-}root]$ **by** $simp$

show $?thesis$

using $p.receive\text{-}composition\text{-}fri\text{-}challenge\text{-}counter\text{-}outcome[OF\ rand]$

$read\text{-}res$

by $simp$

qed

lemma $receive\text{-}trace\text{-}fri\text{-}commits\text{-}preserves\text{-}other\text{-}counters$:

assumes $outcome$:

$Some\ (brs,\ t) \in$

$set\text{-}dist\ (execute\ (ntimes\ v.receive\text{-}trace\text{-}fri\text{-}commits\ n)\ s)$

shows

$PCompositionFriCounter\ t = PCompositionFriCounter\ s \wedge$

$PAlphaCounter\ t = PAlphaCounter\ s \wedge$

$PQueryCounter\ t = PQueryCounter\ s$

using $outcome$

proof $(induction\ n\ arbitrary:\ brs\ s\ t)$

case 0


```

    then show ?case by simp
next
case (Suc n)
from Suc.premis obtain br u brs' where
  head: Some (br, u) ∈ set-dist (execute v.receive-trace-fri-commits s)
  and tail:
    Some (brs', t) ∈
      set-dist (execute (ntimes v.receive-trace-fri-commits n) u)
  by (auto elim!: p.set-dist-bindE)
have head-pres:
  PCompositionFriCounter u = PCompositionFriCounter s ∧
  PAlphaCounter u = PAlphaCounter s ∧
  PQueryCounter u = PQueryCounter s
  using receive-trace-fri-commit-preserves-other-counters[OF head] .
have tail-pres:
  PCompositionFriCounter t = PCompositionFriCounter u ∧
  PAlphaCounter t = PAlphaCounter u ∧
  PQueryCounter t = PQueryCounter u
  using Suc.IH[OF tail] .
show ?case
  using head-pres tail-pres by simp
qed

```

```

lemma receive-composition-fri-commits-preserves-other-counters:
  assumes outcome:
    Some (brs, t) ∈
      set-dist (execute (ntimes v.receive-composition-fri-commits n) s)
  shows
    PTraceFriCounter t = PTraceFriCounter s ∧
    PAlphaCounter t = PAlphaCounter s ∧
    PQueryCounter t = PQueryCounter s
  using outcome
proof (induction n arbitrary: brs s t)
  case 0
  then show ?case by simp
next
case (Suc n)
from Suc.premis obtain br u brs' where
  head:
    Some (br, u) ∈
      set-dist (execute v.receive-composition-fri-commits s)
  and tail:
    Some (brs', t) ∈
      set-dist (execute (ntimes v.receive-composition-fri-commits n) u)
  by (auto elim!: p.set-dist-bindE)
have head-pres:
  PTraceFriCounter u = PTraceFriCounter s ∧
  PAlphaCounter u = PAlphaCounter s ∧
  PQueryCounter u = PQueryCounter s

```

using *receive-composition-fri-commit-preserves-other-counters*[*OF head*] .
have *tail-pres*:
 $PTraceFriCounter\ t = PTraceFriCounter\ u \wedge$
 $PAlphaCounter\ t = PAlphaCounter\ u \wedge$
 $PQueryCounter\ t = PQueryCounter\ u$
using *Suc.IH*[*OF tail*] .
show *?case*
using *head-pres tail-pres by simp*
qed

lemma *honest-receive-fri-commits-list-outcome-with*:
fixes *counter* :: (*'f*, *'a*) *protocol-channel-scheme* $\Rightarrow$ *nat*
and *tag* :: *nat* $\Rightarrow$ *'f* $\Rightarrow$ *'f protocol-hash-input*
assumes *step*:
 $\bigwedge s\ root\ rest\ challenge\ b\ r\ t.$
 $fmlookup\ (HashMap\ s)\ (tag\ (counter\ s)\ (concat\ (PState\ s)\ root)) =$
 $Some\ challenge \Rightarrow$
 $Some\ ((b,\ r),\ t) \in$
 $set-dist\ (execute\ receive-commits$
 $(s(PTranscript := root \# rest))) \Rightarrow$
 $r = root \wedge$
 $b = challenge \wedge$
 $PState\ t = concat\ (PState\ s)\ root \wedge$
 $PTranscript\ t = rest \wedge$
 $counter\ t = Suc\ (counter\ s) \wedge$
 $s \leq t$
and *counter-update*:
 $\bigwedge s\ tr.\ counter\ (s(PTranscript := tr)) = counter\ s$
and *lookups*:
 $\forall i < length\ roots.$
 $fmlookup\ (HashMap\ s)$
 $(tag\ (counter\ s + i)$
 $(foldl\ concat\ (PState\ s)\ (take\ (Suc\ i)\ roots))) =$
 $Some\ (challenges\ !\ i)$
and *len*: $length\ challenges = length\ roots$
and *outcome*:
 $Some\ (brs,\ t) \in$
 $set-dist\ (execute\ (ntimes\ receive-commits\ (length\ roots))$
 $(s(PTranscript := roots @ rest)))$
shows
 $brs = zip\ challenges\ roots \wedge$
 $PState\ t = foldl\ concat\ (PState\ s)\ roots \wedge$
 $PTranscript\ t = rest \wedge$
 $counter\ t = counter\ s + length\ roots \wedge$
 $s \leq t$
using *lookups len outcome*
proof (*induction roots arbitrary: challenges brs s t rest*)
case *Nil*
have *brs-eq*: $brs = []$

```

    using Nil.premis(3) by simp
  have t-eq:  $t = s(PTranscript := rest)$ 
    using Nil.premis(3) by simp
  have s-t:  $s \leq t$ 
    unfolding t-eq less-eq-hash-ext-def less-eq-fmap-def by simp
  have counter-t:  $counter\ t = counter\ s$ 
    using t-eq counter-update by simp
  show ?case
    using brs-eq t-eq s-t counter-t Nil.premis(2) by simp
next
case (Cons root roots)
from Cons.premis(2) obtain challenge challenges' where challenges-eq:
  challenges = challenge # challenges'
  by (cases challenges) auto
from Cons.premis(3) obtain br u brs' where
  head:
    Some (br, u) ∈
      set-dist (execute receive-commits
        ( $s(PTranscript := root \# (roots \text{ @ } rest)))$ )
  and tail:
    Some (brs', t) ∈
      set-dist (execute (ntimes receive-commits (length roots)) u)
  and brs-eq:  $brs = br \# brs'$ 
  by (auto elim!: p.set-dist-bindE)
have lookup-root:
  fmllookup (HashMap s)
    (tag (counter s) (concat (PState s) root)) = Some challenge
  using spec[OF Cons.premis(1), of 0] challenges-eq by simp
obtain b r where br-eq:  $br = (b, r)$ 
  by (cases br)
have head-out:
   $r = root \wedge$ 
   $b = challenge \wedge$ 
   $PState\ u = concat\ (PState\ s)\ root \wedge$ 
   $PTranscript\ u = roots \text{ @ } rest \wedge$ 
   $counter\ u = Suc\ (counter\ s) \wedge$ 
   $s \leq u$ 
  using step[OF lookup-root head[unfolded br-eq]]
  unfolding br-eq by simp
have tail-lookups:
   $\forall i < length\ roots.$ 
  fmllookup (HashMap u)
    (tag (counter u + i)
      (foldl concat (PState u) (take (Suc i) roots))) =
    Some (challenges' ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < length\ roots$ 
  have old-lookup:

```

```

fmlookup (HashMap s)
  (tag (counter s + Suc i)
    (foldl concat (PState s) (take (Suc (Suc i)) (root # roots)))) =
  Some ((challenge # challenges') ! Suc i)
using spec[OF Cons.prem1, of Suc i] i-bound challenges-eq by simp
have key-eq:
  foldl concat (PState s) (take (Suc (Suc i)) (root # roots)) =
  foldl concat (PState u) (take (Suc i) roots)
using head-out by simp
show fmlookup (HashMap u)
  (tag (counter u + i)
    (foldl concat (PState u) (take (Suc i) roots))) =
  Some (challenges' ! i)
using p.hash-extension-lookup[OF old-lookup, of u] head-out key-eq i-bound
by simp
qed
have u-update: u(PTranscript := roots @ rest) = u
using head-out by simp
have tail':
  Some (brs', t) ∈
    set-dist (execute (ntimes receive-commits (length roots))
      (u(PTranscript := roots @ rest)))
using tail unfolding u-update .
have tail-len: length challenges' = length roots
using Cons.prem2 challenges-eq by simp
have tail-out:
  brs' = zip challenges' roots ∧
  PState t = foldl concat (PState u) roots ∧
  PTranscript t = rest ∧
  counter t = counter u + length roots ∧
  u ≤ t
using Cons.IH[OF tail-lookups tail-len tail] .
have s-t: s ≤ t
using head-out tail-out by (meson p.hash-ext-trans)
show ?case
using brs-eq br-eq head-out tail-out s-t challenges-eq by simp
qed

```

lemma honest-receive-fri-commits-list-no-failure-with:
fixes counter :: ('f, 'a) protocol-channel-scheme ⇒ nat
and tag :: nat ⇒ 'f ⇒ 'f protocol-hash-input
assumes step:

```

  ∧ s root rest challenge b r t.
  fmlookup (HashMap s) (tag (counter s) (concat (PState s) root)) =
    Some challenge ⇒
  Some ((b, r), t) ∈
    set-dist (execute receive-commits
      (s(PTranscript := root # rest))) ⇒
  r = root ∧

```

```

    b = challenge ∧
    PState t = concat (PState s) root ∧
    PTranscript t = rest ∧
    counter t = Suc (counter s) ∧
    s ≤ t
  and step-nf:
    ∧ s. PTranscript s ≠ [] ⇒
      None ∉ dom (dist (execute receive-commits s))
  and lookups:
    ∀ i < length roots.
      fmllookup (HashMap s)
        (tag (counter s + i)
          (foldl concat (PState s) (take (Suc i) roots))) =
        Some (challenges ! i)
  and len: length challenges = length roots
shows
  None ∉ dom (dist (execute
    (ntimes receive-commits (length roots))
    (s(PTranscript := roots @ rest))))
using lookups len
proof (induction roots arbitrary: challenges s rest)
  case Nil
  then show ?case by simp
next
  case (Cons root roots)
  from Cons.prem1(2) obtain challenge challenges' where challenges-eq:
    challenges = challenge # challenges'
  by (cases challenges) auto
  have lookup-root:
    fmllookup (HashMap s)
      (tag (counter s) (concat (PState s) root)) = Some challenge
  using spec[OF Cons.prem1(1), of 0] challenges-eq by simp
  have head-nf:
    None ∉ dom (dist (execute receive-commits
      (s(PTranscript := root # (roots @ rest)))))
  by (rule step-nf) simp
  have case-eq:
    ?case =
      (None ∉ dom (dist (execute
        (receive-commits >>=
          (λ br. ntimes receive-commits (length roots) >>=
            (λ brs. return (br # brs)))))
        (s(PTranscript := root # (roots @ rest)))))
  by simp
  show ?case
  unfolding case-eq
proof (rule no-failure-bindI[OF head-nf])
  fix br u
  assume head:

```

```

    Some (br, u) ∈
      set-dist (execute receive-commits
        (s(PTranscript := root # roots @ rest)))
obtain b r where br-eq: br = (b, r)
    by (cases br)
have head-out:
  r = root ∧
  b = challenge ∧
  PState u = concat (PState s) root ∧
  PTranscript u = roots @ rest ∧
  counter u = Suc (counter s) ∧
  s ≤ u
using step[OF lookup-root head[unfolded br-eq]]
unfolding br-eq by simp
have tail-lookups:
  ∀ i < length roots.
    fmllookup (HashMap u)
      (tag (counter u + i)
        (foldl concat (PState u) (take (Suc i) roots))) =
      Some (challenges' ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have old-lookup:
    fmllookup (HashMap s)
      (tag (counter s + Suc i)
        (foldl concat (PState s) (take (Suc (Suc i)) (root # roots)))) =
    Some ((challenge # challenges') ! Suc i)
    using spec[OF Cons.prem1, of Suc i] i-bound challenges-eq by simp
  have key-eq:
    foldl concat (PState s) (take (Suc (Suc i)) (root # roots)) =
    foldl concat (PState u) (take (Suc i) roots)
    using head-out by simp
  show fmllookup (HashMap u)
    (tag (counter u + i)
      (foldl concat (PState u) (take (Suc i) roots))) =
    Some (challenges' ! i)
    using p.hash-extension-lookup[OF old-lookup, of u] head-out key-eq i-bound
    by simp
qed
have tail-len: length challenges' = length roots
  using Cons.prem2 challenges-eq by simp
have tail-nf-start:
  None ∉ dom (dist (execute
    (ntimes receive-commits (length roots))
    (u(PTranscript := roots @ rest))))
  using Cons.IH[OF tail-lookups tail-len, of rest] .
have u-update: u(PTranscript := roots @ rest) = u
  using head-out by simp

```

```

have tail-nf:
  None  $\notin$  dom (dist (execute
    (ntimes receive-commits (length roots)) u))
  using tail-nf-start unfolding u-update .
show None  $\notin$  dom (dist (execute
  (ntimes receive-commits (length roots)  $\gg$ 
    ( $\lambda$ brs. return (br # brs))) u))
  by (rule no-failure-bindI[OF tail-nf]) simp
qed
qed

lemma honest-receive-trace-fri-commits-list-outcome:
assumes lookups:
   $\forall i < \text{length roots.}$ 
  fmlookup (HashMap s)
    (TraceFriChallenge (PTraceFriCounter s + i)
      (foldl concat (PState s) (take (Suc i) roots))) =
    Some (challenges ! i)
and len: length challenges = length roots
and outcome:
  Some (brs, t)  $\in$ 
    set-dist (execute (ntimes v.receive-trace-fri-commits (length roots))
      (s(PTranscript := roots @ rest)))
shows
  brs = zip challenges roots  $\wedge$ 
  PState t = foldl concat (PState s) roots  $\wedge$ 
  PTranscript t = rest  $\wedge$ 
  s  $\leq$  t
proof -
have counter-update:
   $\bigwedge s \text{ tr. } P\text{TraceFriCounter } (s(P\text{Transcript} := \text{tr})) = P\text{TraceFriCounter } s$ 
by simp
have res:
  brs = zip challenges roots  $\wedge$ 
  PState t = foldl concat (PState s) roots  $\wedge$ 
  PTranscript t = rest  $\wedge$ 
  PTraceFriCounter t = PTraceFriCounter s + length roots  $\wedge$ 
  s  $\leq$  t
by (rule honest-receive-fri-commits-list-outcome-with
  [where counter=PTraceFriCounter and tag=TraceFriChallenge,
    OF honest-receive-trace-fri-commit-replay-outcome counter-update lookups len outcome])
then show ?thesis by simp
qed

```

```

lemma honest-receive-composition-fri-commits-list-outcome:
assumes lookups:
   $\forall i < \text{length roots.}$ 
  fmlookup (HashMap s)
    (CompositionFriChallenge

```

```

    (PCompositionFriCounter s + i)
    (foldl concat (PState s) (take (Suc i) roots))) =
    Some (challenges ! i)
  and len: length challenges = length roots
  and outcome:
    Some (brs, t) ∈
    set-dist (execute
      (ntimes v.receive-composition-fri-commits (length roots))
      (s(PTranscript := roots @ rest)))
  shows
    brs = zip challenges roots ∧
    PState t = foldl concat (PState s) roots ∧
    PTranscript t = rest ∧
    s ≤ t
  proof -
    have counter-update:
      ∧s tr. PCompositionFriCounter
        (s(PTranscript := tr)) = PCompositionFriCounter s
    by simp
    have res:
      brs = zip challenges roots ∧
      PState t = foldl concat (PState s) roots ∧
      PTranscript t = rest ∧
      PCompositionFriCounter t = PCompositionFriCounter s + length roots ∧
      s ≤ t
    by (rule honest-receive-fri-commits-list-outcome-with
      [where counter=PCompositionFriCounter and tag=CompositionFriChallenge,
        OF honest-receive-composition-fri-commit-replay-outcome
        counter-update lookups len outcome])
    then show ?thesis by simp
  qed

  lemma honest-receive-trace-fri-commits-list-no-failure:
    assumes lookups:
      ∀ i < length roots.
        fmlookup (HashMap s)
          (TraceFriChallenge (PTraceFriCounter s + i)
            (foldl concat (PState s) (take (Suc i) roots))) =
            Some (challenges ! i)
    and len: length challenges = length roots
    shows
      None ∉ dom (dist (execute
        (ntimes v.receive-trace-fri-commits (length roots))
        (s(PTranscript := roots @ rest))))
    using lookups len
  proof (induction roots arbitrary: challenges s rest)
    case Nil
    then show ?case by simp
  next

```



```

case (Cons root roots)
from Cons.prem(2) obtain challenge challenges' where challenges-eq:
  challenges = challenge # challenges'
by (cases challenges) auto
have lookup-root:
  fmlookup (HashMap s)
    (TraceFriChallenge (PTraceFriCounter s) (concat (PState s) root)) =
    Some challenge
using spec[OF Cons.prem(1), of 0] challenges-eq by simp
have head-nf:
  None  $\notin$  dom (dist (execute v.receive-trace-fri-commits
    (s(PTranscript := root # (roots @ rest))))))
by (rule v.receive-trace-fri-commits-no-failure) simp
have case-eq:
  ?case =
    (None  $\notin$  dom (dist (execute
      (v.receive-trace-fri-commits  $\gg$ 
        ( $\lambda$ br. ntimes v.receive-trace-fri-commits (length roots)  $\gg$ 
          ( $\lambda$ brs. return (br # brs))))
      (s(PTranscript := root # (roots @ rest))))))
by simp
show ?case
  unfolding case-eq
proof (rule no-failure-bindI[OF head-nf])
  fix br u
  assume head:
    Some (br, u)  $\in$ 
      set-dist (execute v.receive-trace-fri-commits
        (s(PTranscript := root # roots @ rest)))
  obtain b r where br-eq: br = (b, r)
  by (cases br)
  have head-out:
    r = root  $\wedge$ 
    b = challenge  $\wedge$ 
    PState u = concat (PState s) root  $\wedge$ 
    PTranscript u = roots @ rest  $\wedge$ 
    PTraceFriCounter u = Suc (PTraceFriCounter s)  $\wedge$ 
    s  $\leq$  u
  using honest-receive-trace-fri-commit-replay-outcome[
    OF lookup-root head[unfolded br-eq]]
  unfolding br-eq by simp
  have tail-lookups:
     $\forall i < \text{length roots.}$ 
    fmlookup (HashMap u)
      (TraceFriChallenge
        (PTraceFriCounter u + i)
        (foldl concat (PState u) (take (Suc i) roots))) =
      Some (challenges' ! i)
  proof (intro allI impI)

```

```

fix  $i$ 
assume  $i$ -bound:  $i < \text{length roots}$ 
have  $\text{old-lookup}$ :
   $\text{fmlookup (HashMap } s)$ 
    ( $\text{TraceFriChallenge}$ 
      ( $\text{PTraceFriCounter } s + \text{Suc } i$ )
      ( $\text{foldl concat (PState } s) (\text{take (Suc (Suc } i)) (\text{root \# roots})) =$ 
         $\text{Some } ((\text{challenge \# challenges}') ! \text{Suc } i)$ 
      using  $\text{spec}[OF \text{ Cons.premis}(1), \text{ of Suc } i]$   $i$ -bound  $\text{challenges-eq}$ 
      by  $\text{simp}$ 
    have  $\text{key-eq}$ :
       $\text{foldl concat (PState } s) (\text{take (Suc (Suc } i)) (\text{root \# roots})) =$ 
         $\text{foldl concat (PState } u) (\text{take (Suc } i) \text{ roots})$ 
      using  $\text{head-out}$  by  $\text{simp}$ 
    show  $\text{fmlookup (HashMap } u)$ 
      ( $\text{TraceFriChallenge}$ 
        ( $\text{PTraceFriCounter } u + i$ )
        ( $\text{foldl concat (PState } u) (\text{take (Suc } i) \text{ roots})) =$ 
           $\text{Some } (\text{challenges}' ! i)$ 
        using  $p.\text{hash-extension-lookup}[OF \text{ old-lookup}, \text{ of } u]$   $\text{head-out key-eq}$ 
         $i$ -bound
        by  $\text{simp}$ 
      qed
    have  $\text{tail-len}$ :  $\text{length challenges}' = \text{length roots}$ 
      using  $\text{Cons.premis}(2)$   $\text{challenges-eq}$  by  $\text{simp}$ 
    have  $\text{tail-nf-start}$ :
       $\text{None} \notin \text{dom } (\text{dist } (\text{execute}$ 
        ( $\text{ntimes } v.\text{receive-trace-fri-commits } (\text{length roots}))$ 
        ( $u(P\text{Transcript} := \text{roots} @ \text{rest}))$ 
      using  $\text{Cons.IH}[OF \text{ tail-lookups tail-len}, \text{ of rest}]$  .
    have  $u\text{-update}$ :  $u(P\text{Transcript} := \text{roots} @ \text{rest}) = u$ 
      using  $\text{head-out}$  by  $\text{simp}$ 
    have  $\text{tail-nf}$ :
       $\text{None} \notin \text{dom } (\text{dist } (\text{execute}$ 
        ( $\text{ntimes } v.\text{receive-trace-fri-commits } (\text{length roots})) u)$ 
      using  $\text{tail-nf-start unfolding } u\text{-update}$  .
    show  $\text{None} \notin \text{dom } (\text{dist } (\text{execute}$ 
      ( $\text{ntimes } v.\text{receive-trace-fri-commits } (\text{length roots}) \gg=$ 
        ( $\lambda \text{brs. return } (\text{br} \# \text{brs})) u)$ 
      by ( $\text{rule no-failure-bindI}[OF \text{ tail-nf}]$ )  $\text{simp}$ 
    qed
  qed

lemma  $\text{honest-receive-composition-fri-commits-list-no-failure}$ :
assumes  $\text{lookups}$ :
   $\forall i < \text{length roots.}$ 
   $\text{fmlookup (HashMap } s)$ 
    ( $\text{CompositionFriChallenge}$ 
      ( $\text{PCompositionFriCounter } s + i$ )

```

```

    (foldl concat (PState s) (take (Suc i) roots))) =
    Some (challenges ! i)
  and len: length challenges = length roots
shows
  None  $\notin$  dom (dist (execute
    (ntimes v.receive-composition-fri-commits (length roots))
    (s(PTranscript := roots @ rest))))
  using lookups len
proof (induction roots arbitrary: challenges s rest)
  case Nil
  then show ?case by simp
next
  case (Cons root roots)
  from Cons.premis(2) obtain challenge challenges' where challenges-eq:
    challenges = challenge # challenges'
  by (cases challenges) auto
  have lookup-root:
    fmlookup (HashMap s)
      (CompositionFriChallenge (PCompositionFriCounter s)
        (concat (PState s) root)) =
      Some challenge
  using spec[OF Cons.premis(1), of 0] challenges-eq by simp
  have head-nf:
    None  $\notin$  dom (dist (execute v.receive-composition-fri-commits
      (s(PTranscript := root # (roots @ rest)))))
  by (rule v.receive-composition-fri-commits-no-failure) simp
  have case-eq:
    ?case =
    (None  $\notin$  dom (dist (execute
      (v.receive-composition-fri-commits  $\gg$ 
        ( $\lambda$ br. ntimes v.receive-composition-fri-commits (length roots)  $\gg$ 
          ( $\lambda$ brs. return (br # brs)))))
      (s(PTranscript := root # (roots @ rest)))))
  by simp
  show ?case
  unfolding case-eq
proof (rule no-failure-bindI[OF head-nf])
  fix br u
  assume head:
    Some (br, u)  $\in$ 
    set-dist (execute v.receive-composition-fri-commits
      (s(PTranscript := root # roots @ rest)))
  obtain b r where br-eq: br = (b, r)
  by (cases br)
  have head-out:
    r = root  $\wedge$ 
    b = challenge  $\wedge$ 
    PState u = concat (PState s) root  $\wedge$ 
    PTranscript u = roots @ rest  $\wedge$ 

```

```

  PCompositionFriCounter u = Suc (PCompositionFriCounter s) ∧
  s ≤ u
using honest-receive-composition-fri-commit-replay-outcome[
  OF lookup-root head[unfolded br-eq]]
unfolding br-eq by simp
have tail-lookups:
  ∀ i < length roots.
    fmlookup (HashMap u)
      (CompositionFriChallenge
        (PCompositionFriCounter u + i)
        (foldl concat (PState u) (take (Suc i) roots))) =
      Some (challenges' ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have old-lookup:
    fmlookup (HashMap s)
      (CompositionFriChallenge
        (PCompositionFriCounter s + Suc i)
        (foldl concat (PState s) (take (Suc (Suc i)) (root # roots)))) =
    Some ((challenge # challenges') ! Suc i)
  using spec[OF Cons.prem1(1), of Suc i] i-bound challenges-eq
  by simp
  have key-eq:
    foldl concat (PState s) (take (Suc (Suc i)) (root # roots)) =
    foldl concat (PState u) (take (Suc i) roots)
  using head-out by simp
  show fmlookup (HashMap u)
    (CompositionFriChallenge
      (PCompositionFriCounter u + i)
      (foldl concat (PState u) (take (Suc i) roots))) =
    Some (challenges' ! i)
  using p.hash-extension-lookup[OF old-lookup, of u] head-out key-eq
  i-bound
  by simp
qed
have tail-len: length challenges' = length roots
  using Cons.prem2(2) challenges-eq by simp
have tail-nf-start:
  None ∉ dom (dist (execute
    (ntimes v.receive-composition-fri-commits (length roots))
    (u(PTranscript := roots @ rest))))
  using Cons.IH[OF tail-lookups tail-len, of rest] .
have u-update: u(PTranscript := roots @ rest) = u
  using head-out by simp
have tail-nf:
  None ∉ dom (dist (execute
    (ntimes v.receive-composition-fri-commits (length roots)) u))
  using tail-nf-start unfolding u-update .

```

show $\text{None} \notin \text{dom} (\text{dist} (\text{execute}$
 $(\text{ntimes } v.\text{receive-composition-fri-commits} (\text{length roots}) \gg$
 $(\lambda \text{brs. return } (\text{br} \# \text{brs}))) \text{ u}))$
by $(\text{rule no-failure-bindI}[\text{OF tail-nf}]) \text{ simp}$
qed
qed

lemma *honest-fri-commit-lengths*:
assumes $\text{Some} ((ps', ds', ls', ms'), t) \in$
 $\text{set-dist} (\text{execute} (p.\text{fri-commit } n \text{ ps ds ls ms}) s)$
shows
 $\text{length } ps' = \text{length } ps + n \wedge$
 $\text{length } ds' = \text{length } ds + n \wedge$
 $\text{length } ls' = \text{length } ls + n \wedge$
 $\text{length } ms' = \text{length } ms + n$
using $p.\text{fri-commit-lengths}[\text{OF assms}]$.

lemma *honest-fri-commit-extends*:
assumes $\text{Some} ((ps', ds', ls', ms'), t) \in$
 $\text{set-dist} (\text{execute} (p.\text{fri-commit } n \text{ ps ds ls ms}) s)$
shows $s \leq t$
using $p.\text{fri-commit-extends}[\text{OF assms}]$.

lemma *honest-fri-commit-layers-are-evaluations*:
assumes *init*:
 $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$
and *outcome*:
 $\text{Some} ((ps', ds', ls', ms'), t) \in$
 $\text{set-dist} (\text{execute} (p.\text{fri-commit } n \text{ ps ds ls ms}) s)$
and *lens*: $\text{length } ps = \text{length } ds \text{ length } ds = \text{length } ls$
and *bounds*: $i < \text{length } ps' \ i < \text{length } ds' \ i < \text{length } ls'$
shows $ls' ! i = \text{map } (\text{poly } (ps' ! i)) (ds' ! i)$
using $p.\text{fri-commit-layers-are-evaluations}[\text{OF init outcome lens bounds}]$.

lemma *honest-fri-commit-created-trees*:
assumes *init*:
 $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies p.\text{created-tree } (ls ! i) (ms ! i) s$
and *outcome*:
 $\text{Some} ((ps', ds', ls', ms'), t) \in$
 $\text{set-dist} (\text{execute} (p.\text{fri-commit } n \text{ ps ds ls ms}) s)$
and *lens*: $\text{length } ls = \text{length } ms$
and *bounds*: $i < \text{length } ls' \ i < \text{length } ms'$
shows $p.\text{created-tree } (ls' ! i) (ms' ! i) t$
using $p.\text{fri-commit-created-trees}[\text{OF init outcome lens bounds}]$.

lemma *honest-prover-monad-outcome*:
assumes *outcome*:
 $\text{Some } (final, t) \in \text{set-dist} (\text{execute } p.\text{prover-monad init-state})$

obtains $f\text{-merkle } s1 \ s2 \ f\text{-nrounds } f\text{-ps } f\text{-ds } f\text{-ls } f\text{-ms } s\text{-trace-fri}$
 $s\text{-trace-final } as \ s3 \ cp' \ s4 \ cp\text{-merkle } s5 \ nrounds \ ps \ ds \ ls \ ms \ s6 \ s\text{-final}$
where
 $Some \ (f\text{-merkle}, s1) \in$
 $set\text{-dist} \ (execute \ (p.create \ p.f\text{-eval}) \ init\text{-state})$
 $p.created\text{-tree} \ p.f\text{-eval} \ f\text{-merkle} \ s1$
 $Some \ ((), s2) \in set\text{-dist} \ (execute \ (p.send \ (value \ f\text{-merkle})) \ s1)$
 $f\text{-nrounds} = \text{ceil-log } clength$
 $Some \ ((f\text{-ps}, f\text{-ds}, f\text{-ls}, f\text{-ms}), s\text{-trace-fri}) \in$
 $set\text{-dist} \ (execute$
 $(p.trace\text{-fri-commit} \ f\text{-nrounds} \ [p.f] \ [p.eval\text{-domain}] \ [p.f\text{-eval}]$
 $[f\text{-merkle}]) \ s2)$
 $length \ f\text{-ps} = Suc \ f\text{-nrounds}$
 $length \ f\text{-ds} = Suc \ f\text{-nrounds}$
 $length \ f\text{-ls} = Suc \ f\text{-nrounds}$
 $length \ f\text{-ms} = Suc \ f\text{-nrounds}$
 $\bigwedge i. i < length \ f\text{-ps} \implies i < length \ f\text{-ds} \implies i < length \ f\text{-ls} \implies$
 $f\text{-ls} ! i = map \ (poly \ (f\text{-ps} ! i)) \ (f\text{-ds} ! i)$
 $\bigwedge i. i < length \ f\text{-ls} \implies i < length \ f\text{-ms} \implies$
 $p.created\text{-tree} \ (f\text{-ls} ! i) \ (f\text{-ms} ! i) \ s\text{-trace-fri}$
 $Some \ ((), s\text{-trace-final}) \in set\text{-dist} \ (execute \ (p.send \ (hd \ (last \ f\text{-ls}))) \ s\text{-trace-fri})$
 $Some \ (as, s3) \in$
 $set\text{-dist} \ (execute$
 $(mmap \ (replicate \ (length \ spec)$
 $(do \ \{$
 $a \leftarrow p.receive\text{-alpha-challenge};$
 $p.send \ a;$
 $return \ a$
 $\}))) \ s\text{-trace-final})$
 $cp' = p.cp \ as \ p.f\text{-powers}$
 $Some \ ((), s4) \in set\text{-dist} \ (execute \ (p.send \ (of\text{-nat} \ (degree \ cp'))) \ s3)$
 $Some \ (cp\text{-merkle}, s5) \in set\text{-dist} \ (execute \ (p.create \ (p.cp\text{-eval} \ as)) \ s4)$
 $p.created\text{-tree} \ (p.cp\text{-eval} \ as) \ cp\text{-merkle} \ s5$
 $nrounds = \text{ceil-log} \ (degree \ cp' + 1)$
 $Some \ ((ps, ds, ls, ms), s6) \in$
 $set\text{-dist} \ (execute$
 $(p.composition\text{-fri-commit} \ nrounds \ [cp'] \ [p.eval\text{-domain}]$
 $[p.cp\text{-eval} \ as] \ [cp\text{-merkle}]) \ s5)$
 $length \ ps = Suc \ nrounds$
 $length \ ds = Suc \ nrounds$
 $length \ ls = Suc \ nrounds$
 $length \ ms = Suc \ nrounds$
 $\bigwedge i. i < length \ ps \implies i < length \ ds \implies i < length \ ls \implies$
 $ls ! i = map \ (poly \ (ps ! i)) \ (ds ! i)$
 $\bigwedge i. i < length \ ls \implies i < length \ ms \implies$
 $p.created\text{-tree} \ (ls ! i) \ (ms ! i) \ s6$
 $Some \ ((), s\text{-final}) \in set\text{-dist} \ (execute \ (p.send \ (hd \ (last \ ls))) \ s6)$
 $Some \ (final, t) \in$
 $set\text{-dist} \ (execute$

```

      (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds) s-final)
proof –
from outcome obtain f-merkle s1 s2 f-ps f-ds f-ls f-ms s-trace-fri
      s-trace-final as s3 s4 cp-merkle s5 ps ds ls ms s6 s-final where
      create-f:
        Some (f-merkle, s1) ∈
          set-dist (execute (p.create p.f-eval) init-state)
and send-f:
        Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and trace-fri:
        Some ((f-ps, f-ds, f-ls, f-ms), s-trace-fri) ∈
          set-dist (execute
            (p.trace-fri-commit (ceil-log clength)
              [p.f] [p.eval-domain] [p.f-eval] [f-merkle]) s2)
and trace-final-send:
        Some ((), s-trace-final) ∈
          set-dist (execute (p.send (hd (last f-ls))) s-trace-fri)
and alphas:
        Some (as, s3) ∈
          set-dist (execute
            (mmap (replicate (length spec)
              (do {
                a ← p.receive-alpha-challenge;
                p.send a;
                return a
              }))) s-trace-final)
and send-degree:
        Some ((), s4) ∈
          set-dist (execute (p.send (of-nat (degree (p.cp as p.f-powers)))) s3)
and create-cp:
        Some (cp-merkle, s5) ∈
          set-dist (execute (p.create (p.cp-eval as)) s4)
and fri:
        Some ((ps, ds, ls, ms), s6) ∈
          set-dist (execute
            (p.composition-fri-commit
              (ceil-log (degree (p.cp as p.f-powers) + 1))
              [p.cp as p.f-powers] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and final-send:
        Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and round-tail:
        Some (final, t) ∈
          set-dist (execute
            (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds) s-final)
unfolding p.prover-monad-def
by (auto simp: Let-def init-state-def prover-query-round-def
      split: prod.splits elim!: p.set-dist-bindE)
let ?cp' = p.cp as p.f-powers
let ?nrounds = ceil-log (degree ?cp' + 1)

```

```

have created-f: p.created-tree p.f-eval f-merkle s1
  using p.create-outcome[OF create-f] by simp
let ?f-nrounds = ceil-log clength
have trace-fri-lengths:
  length f-ps = Suc ?f-nrounds
  length f-ds = Suc ?f-nrounds
  length f-ls = Suc ?f-nrounds
  length f-ms = Suc ?f-nrounds
  using p.trace-fri-commit-lengths[OF trace-fri] by simp-all
have trace-fri-layers:
   $\bigwedge i. i < \text{length } f\text{-ps} \implies i < \text{length } f\text{-ds} \implies i < \text{length } f\text{-ls} \implies$ 
     $f\text{-ls} ! i = \text{map } (\text{poly } (f\text{-ps} ! i)) (f\text{-ds} ! i)$ 
proof (rule p.trace-fri-commit-layers-are-evaluations[OF - trace-fri])
  fix i
  assume  $i < \text{length } [p.f]$   $i < \text{length } [p.\text{eval-domain}]$   $i < \text{length } [p.f\text{-eval}]$ 
  then show  $[p.f\text{-eval}] ! i =$ 
     $\text{map } (\text{poly } ([p.f] ! i)) ([p.\text{eval-domain}] ! i)$ 
  unfolding p.f-eval-def by simp
qed simp-all
have trace-fri-created:
   $\bigwedge i. i < \text{length } f\text{-ls} \implies i < \text{length } f\text{-ms} \implies$ 
     $p.\text{created-tree } (f\text{-ls} ! i) (f\text{-ms} ! i) \text{ s-trace-fri}$ 
proof (rule p.trace-fri-commit-created-trees[OF - trace-fri])
  fix i
  assume  $i < \text{length } [p.f\text{-eval}]$   $i < \text{length } [f\text{-merkle}]$ 
  then show  $p.\text{created-tree } ([p.f\text{-eval}] ! i) ([f\text{-merkle}] ! i) \text{ s2}$ 
  proof –
    have  $s1 \leq s2$ 
    using p.send-outcome[OF send-f]
    unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
    then have  $p.\text{created-tree } p.f\text{-eval } f\text{-merkle } s2$ 
    using p.created-tree-mono[OF created-f] by blast
    then show ?thesis
    using  $\langle i < \text{length } [p.f\text{-eval}] \rangle \langle i < \text{length } [f\text{-merkle}] \rangle$  by simp
  qed
qed simp-all
have created-cp:  $p.\text{created-tree } (p.\text{cp-eval } as) \text{ cp-merkle } s5$ 
  using p.create-outcome[OF create-cp] by simp
have fri-lengths:
  length ps = Suc ?nrounds
  length ds = Suc ?nrounds
  length ls = Suc ?nrounds
  length ms = Suc ?nrounds
  using p.composition-fri-commit-lengths[OF fri] by simp-all
have fri-layers:
   $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
     $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
proof (rule p.composition-fri-commit-layers-are-evaluations[OF - fri])
  fix i

```



```

assume  $i < \text{length } [?cp]$   $i < \text{length } [p.\text{eval-domain}]$   $i < \text{length } [p.\text{cp-eval as}]$ 
then show  $[p.\text{cp-eval as}] ! i =$ 
   $\text{map } (\text{poly } ([?cp] ! i)) ([p.\text{eval-domain}] ! i)$ 
  unfolding  $p.\text{cp-eval-def}$  by  $\text{simp}$ 
qed  $\text{simp-all}$ 
have  $\text{fri-created}:$ 
   $\wedge i. i < \text{length } ls \implies i < \text{length } ms \implies$ 
   $p.\text{created-tree } (ls ! i) (ms ! i) s6$ 
proof ( $\text{rule } p.\text{composition-fri-commit-created-trees}[OF - \text{fri}]$ )
  fix  $i$ 
  assume  $i < \text{length } [p.\text{cp-eval as}]$   $i < \text{length } [cp\text{-merkle}]$ 
  then show  $p.\text{created-tree } ([p.\text{cp-eval as}] ! i) ([cp\text{-merkle}] ! i) s5$ 
  using  $\text{created-cp}$  by  $\text{simp}$ 
qed  $\text{simp-all}$ 
show  $?thesis$ 
by ( $\text{rule that}$ 
   $OF \text{ create-f created-f send-f refl trace-fri trace-fri-lengths}(1)$ 
   $\text{trace-fri-lengths}(2) \text{ trace-fri-lengths}(3) \text{ trace-fri-lengths}(4)$ 
   $\text{trace-fri-layers trace-fri-created trace-final-send alphas refl send-degree create-cp created-cp}$ 
   $\text{refl fri fri-lengths}(1) \text{ fri-lengths}(2) \text{ fri-lengths}(3) \text{ fri-lengths}(4)$ 
   $\text{fri-layers fri-created final-send round-tail})$ )
qed

lemma  $\text{honest-verifier-monad-outcome}:$ 
assumes  $\text{outcome}:$ 
   $\text{Some } (\text{result}, t) \in \text{set-dist } (\text{execute } v.\text{verify-monad } s)$ 
obtains  $\text{fr } s1 \text{ f-fl } s\text{-fri } f\text{-final } s\text{-fri-final as } s2 \text{ dg } s3 \text{ s4 fl } s5 \text{ final } s6$ 
where
   $\text{Some } (\text{fr}, s1) \in \text{set-dist } (\text{execute } p.\text{read } s)$ 
   $\text{Some } (f\text{-fl}, s\text{-fri}) \in$ 
   $\text{set-dist } (\text{execute } (\text{ntimes } v.\text{receive-trace-fri-commits } (\text{ceil-log clength})) s1)$ 
   $\text{Some } (f\text{-final}, s\text{-fri-final}) \in \text{set-dist } (\text{execute } p.\text{read } s\text{-fri})$ 
   $\text{Some } (\text{as}, s2) \in$ 
   $\text{set-dist } (\text{execute } (\text{mmap } (\text{replicate } (\text{length spec})$ 
   $(\text{do } \{$ 
   $a0 \leftarrow p.\text{receive-alpha-challenge};$ 
   $\text{let } a0' = a0;$ 
   $a1 \leftarrow p.\text{read};$ 
   $\text{let } a1' = a1;$ 
   $\text{assert } (a0' = a1');$ 
   $\text{return } a1'$ 
   $\}))) s\text{-fri-final})$ 
   $\text{Some } (\text{dg}, s3) \in \text{set-dist } (\text{execute } p.\text{read } s2)$ 
   $\text{Some } ((), s4) \in$ 
   $\text{set-dist } (\text{execute } (\text{assert } (\text{to-nat dg} \leq p.\text{maxDegree})) s3)$ 
   $\text{to-nat dg} \leq p.\text{maxDegree}$ 
   $s4 = s3$ 

```

Some (fl, s5) ∈
set-dist (execute
(ntimes v.receive-composition-fri-commits
(ceil-log (to-nat dg + 1))) s4)
Some (final, s6) ∈ set-dist (execute p.read s5)
Some (result, t) ∈
set-dist (execute
(ntimes (verifier-query-round fr f-fl f-final as fl final) rounds) s6)
proof –
from *outcome* **obtain** *fr s1 f-fl s-fri f-final s-fri-final as s2 dg s3 s4 fl s5 final s6* **where**
read-fr: Some (fr, s1) ∈ set-dist (execute p.read s)
and *trace-fri-reads:*
Some (f-fl, s-fri) ∈
set-dist (execute
(ntimes v.receive-trace-fri-commits (ceil-log clength)) s1)
and *read-trace-final:*
Some (f-final, s-fri-final) ∈ set-dist (execute p.read s-fri)
and *alphas:*
Some (as, s2) ∈
set-dist (execute
(mmap (replicate (length spec)
(do {
a0 ← p.receive-alpha-challenge;
let a0' = a0;
a1 ← p.read;
let a1' = a1;
assert (a0' = a1');
return a1'
}})) s-fri-final)
and *read-dg: Some (dg, s3) ∈ set-dist (execute p.read s2)*
and *degree-assert:*
Some ((), s4) ∈
set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s3)
and *fri-reads:*
Some (fl, s5) ∈
set-dist (execute
(ntimes v.receive-composition-fri-commits
(ceil-log (to-nat dg + 1))) s4)
and *read-final: Some (final, s6) ∈ set-dist (execute p.read s5)*
and *round-tail:*
Some (result, t) ∈
set-dist (execute
(ntimes (verifier-query-round fr f-fl f-final as fl final) rounds) s6)
unfolding *v.verify-monad-def*
by (*auto simp: Let-def verifier-query-round-def*
split: prod.splits elim!: p.set-dist-bindE)
have *degree-ok: to-nat dg ≤ p.maxDegree*
using *degree-assert*
by (*cases to-nat dg ≤ p.maxDegree*) (*auto simp: assert-def*)

```

have  $s4\text{-eq}$ :  $s4 = s3$ 
  using degree-assert
  by (cases to-nat  $dg \leq p.\text{maxDegree}$ ) (auto simp: assert-def)
show ?thesis
  by (rule that[
    OF read-fr trace-fri-reads read-trace-final alphas read-dg degree-assert degree-ok s4-eq
    fri-reads
    read-final round-tail])
qed

lemma exec-success-splits:
assumes outcome:
  Some (result, t) ∈ set-dist (execute exec init-state)
obtains prover-result prover-state replay-state
where
  Some (prover-result, prover-state) ∈
    set-dist (execute p.prover-monad init-state)
  Some ((), replay-state) ∈
    set-dist (execute
      (modify (λc. c(PState := 0, PTranscript := rev (PTranscript c),
        PTraceFriCounter := 0, PCompositionFriCounter := 0,
        PAlphaCounter := 0, PQueryCounter := 0))
      prover-state)
    replay-state = verifier-replay-state prover-state
  Some (result, t) ∈ set-dist (execute v.verify-monad replay-state)
proof –
from outcome obtain prover-result prover-state replay-unit replay-state where
  prover:
    Some (prover-result, prover-state) ∈
      set-dist (execute p.prover-monad init-state)
  and reset:
    Some (replay-unit, replay-state) ∈
      set-dist (execute
        (modify (λc. c(PState := 0, PTranscript := rev (PTranscript c),
          PTraceFriCounter := 0, PCompositionFriCounter := 0,
          PAlphaCounter := 0, PQueryCounter := 0))
        prover-state)
      and verifier:
        Some (result, t) ∈ set-dist (execute v.verify-monad replay-state)
    unfolding exec-def
    by (auto elim!: p.set-dist-bindE)
  have reset-res:
    replay-unit = () ∧ replay-state = verifier-replay-state prover-state
    using modify-verifier-replay-state-outcome[OF reset] .
  show ?thesis
    by (rule that[OF prover - - verifier])
      (use reset reset-res in simp-all)
qed

```

lemma *exec-success-prover-outcome:*

assumes *outcome:*

Some (result, t) ∈ set-dist (execute exec init-state)

obtains *prover-result prover-state replay-state*

f-merkle s1 s2 f-nrounds f-ps f-ds f-ls f-ms s-trace-fri

s-trace-final as s3 cp' s4 cp-merkle s5 nrounds ps ds ls ms s6 s-final

where

Some (prover-result, prover-state) ∈

set-dist (execute p.prover-monad init-state)

replay-state = verifier-replay-state prover-state

Some (f-merkle, s1) ∈

set-dist (execute (p.create p.f-eval) init-state)

p.created-tree p.f-eval f-merkle s1

Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)

f-nrounds = ceil-log clength

Some ((f-ps, f-ds, f-ls, f-ms), s-trace-fri) ∈

set-dist (execute

(p.trace-fri-commit f-nrounds [p.f] [p.eval-domain] [p.f-eval]
[f-merkle]) s2)

length f-ps = Suc f-nrounds

length f-ds = Suc f-nrounds

length f-ls = Suc f-nrounds

length f-ms = Suc f-nrounds

∧ i. i < length f-ps ⇒ i < length f-ds ⇒ i < length f-ls ⇒

f-ls ! i = map (poly (f-ps ! i)) (f-ds ! i)

∧ i. i < length f-ls ⇒ i < length f-ms ⇒

p.created-tree (f-ls ! i) (f-ms ! i) s-trace-fri

Some ((), s-trace-final) ∈ set-dist (execute (p.send (hd (last f-ls))) s-trace-fri)

Some (as, s3) ∈

set-dist (execute

(mmap (replicate (length spec)

(do {

a ← p.receive-alpha-challenge;

p.send a;

return a

}})) s-trace-final)

cp' = p.cp as p.f-powers

Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)

Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)

p.created-tree (p.cp-eval as) cp-merkle s5

nrounds = ceil-log (degree cp' + 1)

Some ((ps, ds, ls, ms), s6) ∈

set-dist (execute

(p.composition-fri-commit nrounds [cp'] [p.eval-domain]
[p.cp-eval as] [cp-merkle]) s5)

length ps = Suc nrounds

length ds = Suc nrounds

length ls = Suc nrounds

length ms = Suc nrounds

$\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$
 $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$
 $p.\text{created-tree } (ls ! i) (ms ! i) s6$
 $\text{Some } ((), s\text{-final}) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{hd } (\text{last } ls)))) s6)$
 $\text{Some } (\text{prover-result}, \text{prover-state}) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{ntimes } (\text{prover-query-round } f\text{-merkle } f\text{-ls } f\text{-ms } ls \text{ ms}) \text{ rounds}) s\text{-final})$

proof –

from *exec-success-splits*[*OF* *outcome*] **obtain** *prover-result* *prover-state* *replay-state* **where**

prover:

$\text{Some } (\text{prover-result}, \text{prover-state}) \in$
 $\text{set-dist } (\text{execute } p.\text{prover-monad } \text{init-state})$

and *replay*: *replay-state* = *verifier-replay-state* *prover-state*

by *blast*

show *?thesis*

proof (*rule honest-prover-monad-outcome*[*OF* *prover*])

fix *f-merkle* *s1* *s2* *f-nrounds* *f-ps* *f-ds* *f-ls* *f-ms* *s-trace-fri*
s-trace-final *as* *s3* *cp'* *s4* *cp-merkle* *s5* *nrounds* *ps* *ds* *ls* *ms* *s6* *s-final*

assume *create-f*:

$\text{Some } (f\text{-merkle}, s1) \in$
 $\text{set-dist } (\text{execute } (p.\text{create } p.f\text{-eval}) \text{init-state})$

and *created-f*: $p.\text{created-tree } p.f\text{-eval } f\text{-merkle } s1$

and *send-f*: $\text{Some } ((), s2) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{value } f\text{-merkle})) s1)$

and *f-nrounds-def*: $f\text{-nrounds} = \text{ceil-log } \text{clength}$

and *trace-fri*:

$\text{Some } ((f\text{-ps}, f\text{-ds}, f\text{-ls}, f\text{-ms}), s\text{-trace-fri}) \in$
 $\text{set-dist } (\text{execute}$
 $(p.\text{trace-fri-commit } f\text{-nrounds } [p.f] [p.\text{eval-domain}] [p.f\text{-eval}]$
 $[f\text{-merkle}]) s2)$

and *trace-len-ps*: $\text{length } f\text{-ps} = \text{Suc } f\text{-nrounds}$

and *trace-len-ds*: $\text{length } f\text{-ds} = \text{Suc } f\text{-nrounds}$

and *trace-len-ls*: $\text{length } f\text{-ls} = \text{Suc } f\text{-nrounds}$

and *trace-len-ms*: $\text{length } f\text{-ms} = \text{Suc } f\text{-nrounds}$

and *trace-layers*:

$\bigwedge i. i < \text{length } f\text{-ps} \implies i < \text{length } f\text{-ds} \implies i < \text{length } f\text{-ls} \implies$
 $f\text{-ls} ! i = \text{map } (\text{poly } (f\text{-ps} ! i)) (f\text{-ds} ! i)$

and *trace-trees*:

$\bigwedge i. i < \text{length } f\text{-ls} \implies i < \text{length } f\text{-ms} \implies$
 $p.\text{created-tree } (f\text{-ls} ! i) (f\text{-ms} ! i) s\text{-trace-fri}$

and *trace-final-send*:

$\text{Some } ((), s\text{-trace-final}) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{hd } (\text{last } f\text{-ls})))) s\text{-trace-fri})$

and *alphas*:

$\text{Some } (as, s3) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{mmap } (\text{replicate } (\text{length } \text{spec})$
 $(\text{do } \{$
 $a \leftarrow p.\text{receive-alpha-challenge};$
 $p.\text{send } a;$

```

      return a
    }))) s-trace-final)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some (((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp')))) s3)
and create-cp:
  Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and created-cp: p.created-tree (p.cp-eval as) cp-merkle s5
and nrounds-def: nrounds = ceil-log (degree cp' + 1)
and fri:
  Some ((ps, ds, ls, ms), s6) ∈
    set-dist (execute
      (p.composition-fri-commit nrounds [cp'] [p.eval-domain]
        [p.cp-eval as] [cp-merkle]) s5)
and len-ps: length ps = Suc nrounds
and len-ds: length ds = Suc nrounds
and len-ls: length ls = Suc nrounds
and len-ms: length ms = Suc nrounds
and layers:
   $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
   $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
and trees:
   $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$ 
   $p.\text{created-tree } (ls ! i) (ms ! i) s6$ 
and final-send:
  Some (((), s-final) ∈ set-dist (execute (p.send (hd (last ls)))) s6)
and round-tail:
  Some (prover-result, prover-state) ∈
    set-dist (execute
      (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds) s-final)
show ?thesis
using prover replay create-f created-f send-f f-nrounds-def trace-fri
  trace-len-ps trace-len-ds trace-len-ls trace-len-ms trace-layers
  trace-trees trace-final-send alphas cp'-def send-degree
  create-cp created-cp nrounds-def fri len-ps len-ds len-ls len-ms
  layers trees final-send round-tail
by (rule that)
qed
qed

```

lemma *exec-success-verifier-replay-outcome:*
assumes *outcome:*
 Some (result, t) ∈ set-dist (execute exec init-state)
obtains prover-result prover-state replay-state
 fr s1 f-fl s-fri f-final s-fri-final as s2 dg s3 s4 fl s5 final s6
where
 Some (prover-result, prover-state) ∈
 set-dist (execute p.prover-monad init-state)
 replay-state = verifier-replay-state prover-state

Some (*fr*, *s1*) ∈ *set-dist* (*execute p.read replay-state*)
Some (*f-fl*, *s-fri*) ∈
set-dist (*execute*
(ntimes v.receive-trace-fri-commits (ceil-log clength)) s1)
Some (*f-final*, *s-fri-final*) ∈ *set-dist* (*execute p.read s-fri*)
Some (*as*, *s2*) ∈
set-dist (*execute*
(mmap (replicate (length spec)
(do {
a0 ← p.receive-alpha-challenge;
let a0' = a0;
a1 ← p.read;
let a1' = a1;
assert (a0' = a1');
return a1'
}})) s-fri-final)
Some (*dg*, *s3*) ∈ *set-dist* (*execute p.read s2*)
Some (*()*, *s4*) ∈
set-dist (*execute (assert (to-nat dg ≤ p.maxDegree)) s3*)
to-nat dg ≤ p.maxDegree
s4 = s3
Some (*fl*, *s5*) ∈
set-dist (*execute*
(ntimes v.receive-composition-fri-commits
(ceil-log (to-nat dg + 1))) s4)
Some (*final*, *s6*) ∈ *set-dist* (*execute p.read s5*)
Some (*result*, *t*) ∈
set-dist (*execute*
(ntimes (verifier-query-round fr f-fl f-final as fl final) rounds) s6)

proof –

from *exec-success-splits*[*OF outcome*] **obtain** *prover-result prover-state replay-state* **where**
prover:

Some (*prover-result*, *prover-state*) ∈
set-dist (*execute p.prover-monad init-state*)

and *replay*: *replay-state = verifier-replay-state prover-state*

and *verifier:*

Some (*result*, *t*) ∈ *set-dist* (*execute v.verify-monad replay-state*)

by *blast*

show *?thesis*

proof (*rule honest-verifier-monad-outcome*[*OF verifier*])

fix *fr s1 f-fl s-fri f-final s-fri-final as s2 dg s3 s4 fl s5 final s6*

assume *read-fr*: *Some* (*fr*, *s1*) ∈ *set-dist* (*execute p.read replay-state*)

and *trace-fri-reads*:

Some (*f-fl*, *s-fri*) ∈
set-dist (*execute*
(ntimes v.receive-trace-fri-commits (ceil-log clength)) s1)

and *read-trace-final*:

Some (*f-final*, *s-fri-final*) ∈ *set-dist* (*execute p.read s-fri*)

and *alphas*:

```

    Some (as, s2) ∈
      set-dist (execute
        (mmap (replicate (length spec)
          (do {
            a0 ← p.receive-alpha-challenge;
            let a0' = a0;
            a1 ← p.read;
            let a1' = a1;
            assert (a0' = a1');
            return a1'
          }))) s-fri-final)
  and read-dg: Some (dg, s3) ∈ set-dist (execute p.read s2)
  and degree-assert:
    Some ((), s4) ∈
      set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s3)
  and degree-ok: to-nat dg ≤ p.maxDegree
  and s4-eq: s4 = s3
  and fri-reads:
    Some (fl, s5) ∈
      set-dist (execute
        (ntimes v.receive-composition-fri-commits
          (ceil-log (to-nat dg + 1))) s4)
  and read-final: Some (final, s6) ∈ set-dist (execute p.read s5)
  and round-tail:
    Some (result, t) ∈
      set-dist (execute
        (ntimes (verifier-query-round fr f-fl f-final as fl final) rounds) s6)
  show ?thesis
  by (rule that[
    OF prover replay read-fr trace-fri-reads read-trace-final alphas read-dg degree-assert
    degree-ok s4-eq
    fri-reads read-final round-tail])
  qed
qed

end

end

```

```

theory Completeness-Algebra
  imports Completeness-Core
begin

```

2.2 Algebraic Completeness Facts

Deterministic polynomial and degree facts used by honest completeness.

```

context verification
begin

```


lemma *degree-message-decodes*:
assumes $\text{degree } q \leq p.\text{maxDegree}$
shows $\text{to-nat } (\text{of-nat } (\text{degree } q)) = \text{degree } q$
using *assms* **by** (*simp add: p.maxDegree-def degrees-def p.to-nat-of-nat-bounded*)

lemma *degree-assert-hol*:
assumes $\text{degree } q \leq p.\text{maxDegree}$
shows $\text{to-nat } (\text{of-nat } (\text{degree } q)) \leq p.\text{maxDegree}$
using *assms degree-message-decodes[OF assms]* **by** *simp*

lemma *root-factor-pCons*:
fixes $r :: 'f$
shows $XP\ 1 - CP\ r = pCons\ (-\ r)\ 1$
proof $-$
have $xp: XP\ (1::'f) = pCons\ 0\ 1$
unfolding *XP-def* **by** (*simp add: monom-0 monom-Suc*)
have $cp: CP\ r = pCons\ r\ 0$
unfolding *CP-def* **by** (*simp add: monom-0*)
show *?thesis*
unfolding $xp\ cp$
by (*simp only: diff-pCons pCons-eq-iff*)
(simp only: diff-conv-add-uminus add-0-left diff-0-right minus-zero add-0-right)
qed

lemma *poly-root-factor*:
fixes $r\ x :: 'f$
shows $\text{poly } (XP\ 1 - CP\ r)\ x = x - r$
unfolding *root-factor-pCons* **by** (*simp add: algebra-simps*)

lemma *root-factor-nonzero*:
fixes $r :: 'f$
shows $XP\ 1 - CP\ r \neq 0$
unfolding *root-factor-pCons* **by** *simp*

lemma *root-factor-degree*:
fixes $r :: 'f$
shows $\text{degree } (XP\ 1 - CP\ r) = 1$
unfolding *root-factor-pCons* **by** *simp*

lemma *prod-fold-acc*:
fixes $\text{roots} :: 'f\ \text{list}$
shows $\text{fold } (\lambda s\ acc. (XP\ 1 - CP\ s) * acc)\ \text{roots}\ acc =$
 $\text{fold } (\lambda s\ acc. (XP\ 1 - CP\ s) * acc)\ \text{roots}\ (1 :: 'f\ \text{poly}) * acc$
proof (*induction roots arbitrary: acc*)
case *Nil*
then show *?case* **by** *simp*
next
case (*Cons r roots*)

```

let ?F = λs acc. (XP 1 - CP s) * acc
let ?fac = XP 1 - CP r
have lhs:
  fold ?F (r # roots) acc =
    fold ?F roots 1 * (?fac * acc)
  using Cons.IH[of ?fac * acc] by simp
have fold ?F (r # roots) acc =
  (fold ?F roots 1 * ?fac) * acc
  using lhs by (simp only: mult.assoc)
also have ... = fold ?F roots ?fac * acc
  using Cons.IH[of ?fac, symmetric] by simp
also have ... = fold ?F (r # roots) 1 * acc
  by simp
finally show ?case .
qed

lemma prod-Cons:
  fixes r :: 'f and roots :: 'f list
  shows prod (r # roots) = (XP 1 - CP r) * (prod roots :: 'f poly)
proof -
  let ?F = λs acc. (XP 1 - CP s) * acc
  have prod (r # roots) = fold ?F roots ((XP 1 - CP r) * 1)
    unfolding prod-def by simp
  also have ... = fold ?F roots 1 * ((XP 1 - CP r) * 1)
    by (rule prod-fold-acc)
  also have ... = (prod roots :: 'f poly) * (XP 1 - CP r)
    unfolding prod-def by simp
  also have ... = (XP 1 - CP r) * (prod roots :: 'f poly)
    by (rule mult.commute)
  finally show ?thesis .
qed

lemma prod-roots-nonzero:
  fixes roots :: 'f list
  shows prod roots ≠ (0 :: 'f poly)
proof (induction roots)
  case Nil
  then show ?case
    unfolding prod-def by simp
next
  case (Cons r roots)
  then show ?case
    unfolding prod-Cons
    using root-factor-nonzero[of r]
    by simp
qed

lemma degree-prod-roots:
  fixes roots :: 'f list

```

```

  shows degree (prod roots :: 'f poly) = length roots
proof (induction roots)
  case Nil
  then show ?case
    unfolding prod-def by simp
next
  case (Cons r roots)
  have degree (prod (r # roots) :: 'f poly) =
    degree ((XP 1 - CP r) * prod roots :: 'f poly)
    unfolding prod-Cons by simp
  also have ... = degree (XP 1 - CP r :: 'f poly) + degree (prod roots :: 'f poly)
    by (rule degree-mult-eq)
    (use root-factor-nonzero[of r] prod-roots-nonzero[of roots] in simp-all)
  also have ... = Suc (length roots)
    using Cons.IH root-factor-degree[of r] by simp
  finally show ?case
    by simp
qed

lemma prod-roots-dvd:
  fixes roots :: 'f list
  assumes vanish:  $\bigwedge r. r \in \text{set roots} \implies \text{poly } q \ r = 0$ 
    and distinct: distinct roots
  shows prod roots dvd q
  using assms
proof (induction roots arbitrary: q)
  case Nil
  then show ?case
    unfolding prod-def by simp
next
  case (Cons r roots)
  have r-root: poly q r = 0
    using Cons.prem1 by simp
  have fac-dvd: (XP 1 - CP r) dvd q
  proof -
    have pCons (- r) 1 dvd q
    proof -
      have [:- r, 1:] dvd q
        using r-root by (simp add: poly-eq-0-iff-dvd)
      moreover have [:- r, 1:] = pCons (- r) 1
        by simp
      ultimately show ?thesis
        by simp
    qed
  qed
  then show ?thesis
    unfolding root-factor-pCons .
qed
then obtain q' where q-def:  $q = (XP \ 1 - CP \ r) * q'$ 
  unfolding dvd-def by auto

```

```

have roots-distinct: distinct roots
  using Cons.premis(2) by simp
have vanish-q':  $\bigwedge x. x \in \text{set roots} \implies \text{poly } q' x = 0$ 
proof -
  fix x
  assume x-in:  $x \in \text{set roots}$ 
  have x-ne:  $x \neq r$ 
    using Cons.premis(2) x-in by auto
  have 0 = poly q x
    using Cons.premis(1) x-in by simp
  also have ... = poly (XP 1 - CP r) x * poly q' x
    unfolding q-def by simp
  finally have zero-eq:  $0 = \text{poly } (XP 1 - CP r) x * \text{poly } q' x$  .
  then have prod-zero:  $\text{poly } (XP 1 - CP r) x * \text{poly } q' x = 0$ 
    by simp
  moreover have poly (XP 1 - CP r) x  $\neq 0$ 
  proof -
    have poly (XP 1 - CP r) x = x - r
      by (rule poly-root-factor)
    then show ?thesis
      using x-ne by simp
  qed
  ultimately show poly q' x = 0
    by (metis mult-eq-0-iff)
  qed
have prod-tail-dvd: prod roots dvd q'
  by (rule Cons.IH[OF vanish-q' roots-distinct])
then obtain k where q'-def:  $q' = \text{prod roots} * k$ 
  unfolding dvd-def by auto
have q = ((XP 1 - CP r) * prod roots) * k
  unfolding q-def q'-def by (simp add: mult.assoc)
also have ... = prod (r # roots) * k
  using prod-Cons[of r roots] by simp
finally have q = prod (r # roots) * k .
then show ?case
  unfolding dvd-def by blast
qed

lemma prod-eval-nonzero:
  fixes roots :: 'f list
  assumes x-not-root:  $x \notin \text{set roots}$ 
  shows poly (prod roots :: 'f poly) x  $\neq 0$ 
  using x-not-root
proof (induction roots)
  case Nil
  then show ?case
    unfolding prod-def by simp
next
  case (Cons r roots)

```

```

have  $x \neq r$ 
  using Cons.prems by simp
moreover have poly (prod roots :: 'f poly)  $x \neq 0$ 
  using Cons by simp
ultimately show ?case
  unfolding prod-Cons
  using poly-root-factor[of r x] by simp
qed

lemma h-power-mod-domain:
   $p.h \wedge (k \bmod (\text{clength} * \text{scale})) = p.h \wedge k$ 
  by (rule p.h-power-mod-domain)

lemma h-power-in-H:
   $p.h \wedge k \in \text{set } p.H$ 
  by (rule p.h-power-in-H)

lemma eval-coset-disjoint-from-H:
  assumes idx-bound:  $i < \text{clength} * \text{scale}$ 
  shows  $p.h \wedge i * \text{shift} \notin \text{set } p.H$ 
  by (rule p.eval-coset-disjoint-from-H[OF idx-bound])

lemma g-powers-in-H:
  assumes r-bound:  $r < \text{clength}$ 
  shows  $p.g \wedge r \in \text{set } p.H$ 
  by (rule p.g-powers-in-H[OF r-bound])

lemma g-map-subset-H:
  assumes spec-entry:  $(c, \text{roots}, d) \in \text{set spec}$ 
  shows  $\text{set } (p.g\text{-map roots}) \subseteq \text{set } p.H$ 
  by (rule p.g-map-subset-H[OF spec-entry])

lemma query-domain-disjoint:
  assumes spec-entry:  $(c, \text{roots}, d) \in \text{set spec}$ 
  and idx-bound:  $i < \text{clength} * \text{scale}$ 
  shows  $p.h \wedge i * \text{shift} \notin \text{set } (p.g\text{-map roots})$ 
  by (rule p.query-domain-disjoint[OF spec-entry idx-bound])

lemma honest-trace-algebra-prod-roots-dvd:
  assumes hta: honest-trace-algebra
  and spec-entry:  $(c, \text{roots}, d) \in \text{set spec}$ 
  shows prod (p.g-map roots) dvd c p.f-powers
proof -
  have vanish:
     $\bigwedge r. r \in \text{set } (p.g\text{-map roots}) \implies \text{poly } (c \text{ p.f-powers}) r = 0$ 
  using hta spec-entry
  unfolding honest-trace-algebra-def constraint-roots-vanish-def
  by force
  have distinct: distinct (p.g-map roots)

```

```

    by (rule p.g-map-distinct[OF spec-entry])
  show ?thesis
    by (rule prod-roots-dvd[OF vanish distinct])
qed

lemma poly-div-eval-dvd:
  fixes q r :: 'f poly
  assumes dvd: r dvd q
    and denom: poly r x  $\neq$  0
  shows poly (q div r) x = poly q x div poly r x
proof -
  have q-eq: q div r * r = q
    using dvd by simp
  then have poly (q div r) x * poly r x = poly q x
    by (metis poly-mult)
  then show ?thesis
    using denom by (simp add: field-simps)
qed

lemma honest-trace-algebra-quotient-eval:
  assumes hta: honest-trace-algebra
    and spec-entry: (c, roots, d)  $\in$  set spec
    and idx-bound: idx < clength * scale
  shows
    poly ((c p.f-powers) div prod (p.g-map roots)) (p.h ^ idx * shift) =
      poly (c p.f-powers) (p.h ^ idx * shift) div
        poly (prod (p.g-map roots)) (p.h ^ idx * shift)
proof -
  have dvd: prod (p.g-map roots) dvd c p.f-powers
    by (rule honest-trace-algebra-prod-roots-dvd[OF hta spec-entry])
  have denom: poly (prod (p.g-map roots)) (p.h ^ idx * shift)  $\neq$  0
    by (rule prod-eval-nonzero)
    (use query-domain-disjoint[OF spec-entry idx-bound] in simp)
  show ?thesis
    by (rule poly-div-eval-dvd[OF dvd denom])
qed

lemma degree-div-dvd-le:
  fixes q r :: 'f poly
  assumes dvd: r dvd q
    and r-nonzero: r  $\neq$  0
  shows degree (q div r)  $\leq$  degree q - degree r
proof (cases q div r = 0)
  case True
    then show ?thesis by simp
next
  case False
    have q-eq: q div r * r = q
      using dvd by simp

```

```

have degree q = degree (q div r * r)
  using q-eq by simp
also have ... = degree (q div r) + degree r
  by (rule degree-mult-eq[OF False r-nonzero])
finally show ?thesis
  by simp
qed

lemma honest-trace-algebra-quotient-degree:
  assumes honest-trace-algebra
  and (c, roots, d) ∈ set spec
  shows
    degree ((c p.f-powers) div (prod (p.g-map roots))) ≤
      d * (clength - 1) - length roots
proof -
  let ?q = c p.f-powers
  let ?roots = p.g-map roots
  have vanish:  $\bigwedge r. r \in \text{set } ?\text{roots} \implies \text{poly } ?q \ r = 0$ 
    using assms
    unfolding honest-trace-algebra-def constraint-roots-vanish-def
    by force
  have deg: degree ?q ≤ d * (clength - 1)
    by (rule constraint-degree-bound[OF assms(2)])
  have distinct: distinct ?roots
    by (rule p.g-map-distinct[OF assms(2)])
  have dvd: prod ?roots dvd ?q
    by (rule prod-roots-dvd[OF vanish distinct])
  have qdiv-le:
    degree (?q div prod ?roots) ≤ degree ?q - degree (prod ?roots)
    by (rule degree-div-dvd-le[OF dvd prod-roots-nonzero])
  have roots-len: length ?roots = length roots
    unfolding p.g-map-def by simp
  have prod-deg: degree (prod ?roots) = length roots
    using degree-prod-roots[of ?roots] roots-len by simp
  have degree (?q div prod ?roots) ≤ degree ?q - length roots
    using qdiv-le prod-deg by simp
  also have ... ≤ d * (clength - 1) - length roots
    using deg by simp
  finally show ?thesis .
qed

```

```

lemma maxDegree-ge-spec:
  assumes (c, roots, d) ∈ set spec
  shows d * (clength - 1) - length roots ≤ p.maxDegree
proof -
  let ?S = {d * (clength - 1) - length rs | a rs d. (a,rs,d) ∈ set spec}
  have finite-S: finite ?S
  proof -
    have ?S = (λ(-, rs, d). d * (clength - 1) - length rs) ‘ set spec

```

```

    by force
    then show ?thesis by simp
qed
have elem:  $d * (\text{clength} - 1) - \text{length roots} \in ?S$ 
    using assms by force
have  $d * (\text{clength} - 1) - \text{length roots} \leq \text{Max} (\text{insert } 0 ?S)$ 
    by (rule Max-ge) (use finite-S elem in auto)
then show ?thesis
    unfolding p.maxDegree-def degrees-def .
qed

lemma composition-term-degree:
  assumes hta: honest-trace-algebra
    and spec-entry:  $(c, \text{roots}, d) \in \text{set spec}$ 
  shows
     $\text{degree} (CP\ a * ((c\ p.f\text{-powers}) \text{ div } (\text{prod } (p.g\text{-map roots})))) \leq p.\text{maxDegree}$ 
proof -
  have qdeg:
     $\text{degree} ((c\ p.f\text{-powers}) \text{ div } (\text{prod } (p.g\text{-map roots}))) \leq$ 
     $d * (\text{clength} - 1) - \text{length roots}$ 
    using hta spec-entry by (rule honest-trace-algebra-quotient-degree)
  have  $\text{degree} (CP\ a * ((c\ p.f\text{-powers}) \text{ div } (\text{prod } (p.g\text{-map roots})))) \leq$ 
     $\text{degree} (CP\ a) + \text{degree} ((c\ p.f\text{-powers}) \text{ div } (\text{prod } (p.g\text{-map roots})))$ 
    by (rule degree-mult-le)
  also have  $\dots \leq d * (\text{clength} - 1) - \text{length roots}$ 
    using qdeg unfolding CP-def by (simp add: monom-0)
  also have  $\dots \leq p.\text{maxDegree}$ 
    using spec-entry by (rule maxDegree-ge-spec)
  finally show ?thesis .
qed

lemma cp-fold-degree-aux:
  assumes terms:
     $\bigwedge a\ c\ \text{roots}\ d.$ 
     $(a, (c, \text{roots}, d)) \in \text{set xs} \implies$ 
     $\text{degree} (CP\ a * ((c\ p.f\text{-powers}) \text{ div } (\text{prod } (p.g\text{-map roots})))) \leq p.\text{maxDegree}$ 
  and acc:  $\text{degree acc} \leq p.\text{maxDegree}$ 
  shows
     $\text{degree}$ 
     $(\text{fold}$ 
       $(\lambda(a, (c, \text{roots}, -)).$ 
         $(+) (CP\ a * ((c\ p.f\text{-powers}) \text{ div } (\text{prod } (p.g\text{-map roots}))))$ 
       $\text{xs acc}) \leq p.\text{maxDegree}$ 
    using terms acc
proof (induction xs arbitrary: acc)
  case Nil
  then show ?case by simp
next
  case (Cons x xs)

```



```

obtain  $a\ c\ roots\ d$  where  $x = (a, (c, roots, d))$ 
  by (cases  $x$ ) (auto split: prod.splits)
have term-deg:
   $degree\ (CP\ a * ((c\ p.f-powers)\ div\ (prod\ (p.g-map\ roots)))) \leq p.maxDegree$ 
  using Cons.premis(1) [of a c roots d] unfolding  $x$  by simp
have  $acc'$ :
   $degree$ 
     $((\lambda(a, (c, roots, -)).$ 
       $(+) (CP\ a * ((c\ p.f-powers)\ div\ (prod\ (p.g-map\ roots))))$ 
       $x\ acc) \leq p.maxDegree$ 
  using term-deg Cons.premis(2)
  unfolding  $x$ 
  by (auto intro: degree-add-le)
show ?case
  using Cons.IH[OF - acc'] Cons.premis(1)
  by auto
qed

lemma composition-degree-from-honest-trace-algebra:
  assumes honest-trace-algebra
  shows  $degree\ (p.cp\ as\ p.f-powers) \leq p.maxDegree$ 
  unfolding p.cp-def
proof (rule cp-fold-degree-aux)
  fix  $a\ c\ roots\ d$ 
  assume  $(a, (c, roots, d)) \in set\ (zip\ as\ spec)$ 
  then have spec-entry:  $(c, roots, d) \in set\ spec$ 
  by (auto dest!: set-zip-rightD)
  show  $degree\ (CP\ a * ((c\ p.f-powers)\ div\ (prod\ (p.g-map\ roots)))) \leq p.maxDegree$ 
  by (rule composition-term-degree[OF assms spec-entry])
qed simp

lemma composition-degree-from-honest-trace-valid:
  assumes honest-trace-valid
  shows  $degree\ (p.cp\ as\ p.f-powers) \leq p.maxDegree$ 
  using assms
  unfolding honest-trace-valid-def
  by (intro composition-degree-from-honest-trace-algebra) simp

lemma honest-degree-message-assert-hol:
  assumes honest-trace-valid
  shows  $to-nat\ (of-nat\ (degree\ (p.cp\ as\ p.f-powers))) \leq p.maxDegree$ 
  by (rule degree-assert-hol[OF composition-degree-from-honest-trace-valid[OF assms]])

lemma honest-degree-message-rounds:
  assumes honest-trace-valid
  shows  $ceil-log\ (to-nat\ (of-nat\ (degree\ (p.cp\ as\ p.f-powers)))) + 1 =$ 
   $ceil-log\ (degree\ (p.cp\ as\ p.f-powers) + 1)$ 
  using degree-message-decodes[OF composition-degree-from-honest-trace-valid[OF assms]]
  by simp

```

end

end

```
theory Completeness-Transcript
  imports Completeness-Algebra
begin
```

2.3 Honest Transcript and Hash Replay

Transcript-extension and hash-extension lemmas for honest prover execution.

```
context verification
begin
```

```
lemma transcript-send-outcome:
  assumes Some  $((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x) s)$ 
  shows  $t = s(PState := \text{concat } (PState s) x, PTranscript := x \# PTranscript s)$ 
  using  $p.\text{send-outcome}[OF \text{ assms}]$  .
```

```
lemma send-transcript-extends:
  assumes Some  $((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x) s)$ 
  shows  $\text{transcript-extends } t s$ 
  using  $\text{transcript-send-outcome}[OF \text{ assms}]$ 
  unfolding  $\text{transcript-extends-def}$  by auto
```

```
lemma send-hash-extends:
  assumes Some  $((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x) s)$ 
  shows  $s \leq t$ 
  using  $p.\text{send-outcome}[OF \text{ assms}]$ 
  unfolding  $\text{less-eq-hash-ext-def less-eq-fmap-def}$  by simp
```

```
lemma transcript-read-cons-outcome:
  assumes Some  $(y, t) \in \text{set-dist } (\text{execute } p.\text{read } (s(PTranscript := x \# xs)))$ 
  shows
     $y = x \wedge$ 
     $t = s(PState := \text{concat } (PState s) x, PTranscript := xs)$ 
  using  $p.\text{read-cons-outcome}[OF \text{ assms}]$  .
```

```
lemma create-preserves-transcript:
  assumes  $\text{outcome: Some } (r, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs) s)$ 
  shows  $PTranscript t = PTranscript s$ 
  by (rule  $p.\text{create-preserves-channel}(2)[OF \text{ outcome}]$ )
```

```
lemma create-preserves-state:
  assumes  $\text{outcome: Some } (r, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs) s)$ 
  shows  $PState t = PState s$ 
  by (rule  $p.\text{create-preserves-channel}(1)[OF \text{ outcome}]$ )
```

lemma *create-transcript-extends*:
assumes *Some* $(r, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs) s)$
shows *transcript-extends* $t s$
using *create-preserves-transcript*[*OF assms*]
unfolding *transcript-extends-def* **by** *auto*

lemma *create-hash-extends*:
assumes *Some* $(r, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs) s)$
shows $s \leq t$
using *p.create-outcome*[*OF assms*] **by** *simp*

lemma *receive-random-field-element-preserves-transcript*:
assumes *Some* $(a, t) \in \text{set-dist } (\text{execute } p.\text{receive-random-field-element } s)$
shows $P\text{Transcript } t = P\text{Transcript } s$
using *p.receive-random-field-element-outcome*[*OF assms*] **by** *simp*

lemma *receive-random-field-element-transcript-extends*:
assumes *Some* $(a, t) \in \text{set-dist } (\text{execute } p.\text{receive-random-field-element } s)$
shows *transcript-extends* $t s$
using *receive-random-field-element-preserves-transcript*[*OF assms*]
unfolding *transcript-extends-def* **by** *auto*

lemma *receive-alpha-challenge-preserves-transcript*:
assumes *Some* $(a, t) \in \text{set-dist } (\text{execute } p.\text{receive-alpha-challenge } s)$
shows $P\text{Transcript } t = P\text{Transcript } s$
using *p.receive-alpha-challenge-outcome*[*OF assms*] **by** *simp*

lemma *receive-alpha-challenge-transcript-extends*:
assumes *Some* $(a, t) \in \text{set-dist } (\text{execute } p.\text{receive-alpha-challenge } s)$
shows *transcript-extends* $t s$
using *receive-alpha-challenge-preserves-transcript*[*OF assms*]
unfolding *transcript-extends-def* **by** *auto*

lemma *receive-query-index-challenge-preserves-transcript*:
assumes *Some* $(a, t) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s)$
shows $P\text{Transcript } t = P\text{Transcript } s$
using *p.receive-query-index-challenge-outcome*[*OF assms*] **by** *simp*

lemma *receive-query-index-challenge-transcript-extends*:
assumes *Some* $(a, t) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s)$
shows *transcript-extends* $t s$
using *receive-query-index-challenge-preserves-transcript*[*OF assms*]
unfolding *transcript-extends-def* **by** *auto*

lemma *receive-random-field-element-outcome*:
assumes *outcome*: *Some* $(a, t) \in \text{set-dist } (\text{execute } p.\text{receive-random-field-element } s)$
shows
 $s \leq t \wedge$

$PState\ t = PState\ s \wedge$
 $PTranscript\ t = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ t)\ (FiatShamirChallenge\ (PState\ s)) = Some\ a$
by (rule *p.receive-random-field-element-outcome*[*OF outcome*])

lemma *bind-transcript-extends*:

assumes *outcome*: $Some\ (y, t) \in set-dist\ (execute\ (m \gg k)\ s)$
and *m-ext*: $\bigwedge x\ u. Some\ (x, u) \in set-dist\ (execute\ m\ s) \implies$
 $transcript-extends\ u\ s$
and *k-ext*: $\bigwedge x\ u\ y\ t. Some\ (x, u) \in set-dist\ (execute\ m\ s) \implies$
 $Some\ (y, t) \in set-dist\ (execute\ (k\ x)\ u) \implies$
 $transcript-extends\ t\ u$

shows *transcript-extends* *t s*

proof –

from *outcome* **obtain** *x u* **where**

$mx: Some\ (x, u) \in set-dist\ (execute\ m\ s)$
and *ky*: $Some\ (y, t) \in set-dist\ (execute\ (k\ x)\ u)$
by (auto elim!: *p.set-dist-bindE*)

show *?thesis*

using *m-ext*[*OF mx*] *k-ext*[*OF mx ky*] **by** (rule *transcript-extends-trans*)

qed

lemma *mmap-transcript-extends*:

assumes *outcome*: $Some\ (xs, t) \in set-dist\ (execute\ (mmap\ ms)\ s)$
and *step*: $\bigwedge m\ x\ u\ v. m \in set\ ms \implies$
 $Some\ (x, v) \in set-dist\ (execute\ m\ u) \implies$
 $transcript-extends\ v\ u$

shows *transcript-extends* *t s*

using *outcome step*

proof (induction *ms* arbitrary: *xs s t*)

case *Nil*

then show *?case* **by** *simp*

next

case (*Cons m ms*)

from *Cons.prem*s(1) **obtain** *x u xs'* **where**

$head: Some\ (x, u) \in set-dist\ (execute\ m\ s)$
and *tail*: $Some\ (xs', t) \in set-dist\ (execute\ (mmap\ ms)\ u)$
by (auto elim!: *p.set-dist-bindE*)

have *m-in*: $m \in set\ (m \# ms)$

by *simp*

have *head-ext*: *transcript-extends* *u s*

using *Cons.prem*s(2)[*OF m-in head*] .

have *tail-step*:

$\bigwedge m'\ x'\ u'\ v'. m' \in set\ ms \implies$
 $Some\ (x', v') \in set-dist\ (execute\ m'\ u') \implies$
 $transcript-extends\ v'\ u'$

using *Cons.prem*s(2) **by** *auto*

have *tail-ext*: *transcript-extends* *t u*

by (rule *Cons.IH*[*OF tail tail-step*])

```

show ?case
  using head-ext tail-ext by (rule transcript-extends-trans)
qed

lemma mfold-transcript-extends:
  assumes outcome: Some (x, t) ∈ set-dist (execute (mfold a ms) s)
  and step:  $\bigwedge m\ a0\ y\ u\ v. m \in \text{set } ms \implies$ 
    Some (y, v) ∈ set-dist (execute (m a0) u)  $\implies$ 
    transcript-extends v u
  shows transcript-extends t s
  using outcome step
proof (induction ms arbitrary: a x s t)
  case Nil
  then show ?case by simp
next
  case (Cons m ms)
  from Cons.prem1 obtain y u where
    head: Some (y, u) ∈ set-dist (execute (m a) s)
    and tail: Some (x, t) ∈ set-dist (execute (mfold y ms) u)
    by (auto elim!: p.set-dist-bindE)
  have m-in: m ∈ set (m # ms)
    by simp
  have head-ext: transcript-extends u s
    using Cons.prem2[OF m-in head] .
  have tail-step:
     $\bigwedge m'\ a0\ y'\ u'\ v'. m' \in \text{set } ms \implies$ 
    Some (y', v') ∈ set-dist (execute (m' a0) u')  $\implies$ 
    transcript-extends v' u'
  proof -
    fix m' a0 y' u' v'
    assume m'-in-tail: m' ∈ set ms
    and out: Some (y', v') ∈ set-dist (execute (m' a0) u')
    have m'-in: m' ∈ set (m # ms)
      using m'-in-tail by simp
    show transcript-extends v' u'
      using Cons.prem2[OF m'-in out] .
  qed
  have tail-ext: transcript-extends t u
    by (rule Cons.IH[OF tail tail-step])
  show ?case
    using head-ext tail-ext by (rule transcript-extends-trans)
qed

```

```

lemma mmap-hash-extends:
  fixes s t :: ('f, 'more) protocol-channel-scheme
  assumes outcome: Some (xs, t) ∈ set-dist (execute (mmap ms) s)
  and step:  $\bigwedge m\ x\ u\ v. m \in \text{set } ms \implies$ 
    Some (x, v) ∈ set-dist (execute m u)  $\implies$ 
    u ≤ v

```

```

shows  $s \leq t$ 
using outcome step
proof (induction ms arbitrary: xs s t)
case Nil
have t-eq:  $t = s$ 
using Nil.premis(1) by simp
show ?case
unfolding t-eq less-eq-hash-ext-def less-eq-fmap-def by simp
next
case (Cons m ms)
from Cons.premis(1) obtain x u xs' where
head: Some (x, u)  $\in$  set-dist (execute m s)
and tail: Some (xs', t)  $\in$  set-dist (execute (mmap ms) u)
by (auto elim!: p.set-dist-bindE)
have m-in:  $m \in \text{set } (m \# ms)$ 
by simp
have head-ext:  $s \leq u$ 
using Cons.premis(2)[OF m-in head] .
have tail-step:
 $\bigwedge m' x' u' v'. m' \in \text{set } ms \implies$ 
Some (x', v')  $\in$  set-dist (execute m' u')  $\implies$ 
 $u' \leq v'$ 
proof -
fix m' x' u' v'
assume m'-in-tail:  $m' \in \text{set } ms$ 
and out: Some (x', v')  $\in$  set-dist (execute m' u')
have m'-in:  $m' \in \text{set } (m \# ms)$ 
using m'-in-tail by simp
show  $u' \leq v'$ 
using Cons.premis(2)[OF m'-in out] .
qed
have tail-ext:  $u \leq t$ 
by (rule Cons.IH[OF tail tail-step])
show ?case
using head-ext tail-ext by (rule p.hash-ext-trans)
qed

```

```

lemma mfold-hash-extends:
fixes s t :: ('f, 'more) protocol-channel-scheme
assumes outcome: Some (x, t)  $\in$  set-dist (execute (mfold a ms) s)
and step:  $\bigwedge m a0 y u v. m \in \text{set } ms \implies$ 
Some (y, v)  $\in$  set-dist (execute (m a0) u)  $\implies$ 
 $u \leq v$ 
shows  $s \leq t$ 
using outcome step
proof (induction ms arbitrary: a x s t)
case Nil
have t-eq:  $t = s$ 
using Nil.premis(1) by simp

```

```

show ?case
  unfolding t-eq less-eq-hash-ext-def less-eq-fmap-def by simp
next
case (Cons m ms)
from Cons.prem1 obtain y u where
  head: Some (y, u) ∈ set-dist (execute (m a) s)
  and tail: Some (x, t) ∈ set-dist (execute (mfold y ms) u)
  by (auto elim!: p.set-dist-bindE)
have m-in: m ∈ set (m # ms)
  by simp
have head-ext: s ≤ u
  using Cons.prem2[OF m-in head] .
have tail-step:
   $\bigwedge m' a0 y' u' v'. m' \in \text{set } ms \implies$ 
   $\text{Some } (y', v') \in \text{set-dist } (\text{execute } (m' a0) u') \implies$ 
   $u' \leq v'$ 
proof -
  fix m' a0 y' u' v'
  assume m'-in-tail: m' ∈ set ms
  and out: Some (y', v') ∈ set-dist (execute (m' a0) u')
  have m'-in: m' ∈ set (m # ms)
    using m'-in-tail by simp
  show u' ≤ v'
    using Cons.prem2[OF m'-in out] .
qed
have tail-ext: u ≤ t
  by (rule Cons.IH[OF tail tail-step])
show ?case
  using head-ext tail-ext by (rule p.hash-ext-trans)
qed

```

```

lemma mmap-preserves-query-counter:
  fixes s t :: ('f, 'more) protocol-channel-scheme
  assumes outcome: Some (xs, t) ∈ set-dist (execute (mmap ms) s)
  and step:  $\bigwedge m x u v. m \in \text{set } ms \implies$ 
     $\text{Some } (x, v) \in \text{set-dist } (\text{execute } m u) \implies$ 
    PQueryCounter v = PQueryCounter u
  shows PQueryCounter t = PQueryCounter s
  using outcome step
proof (induction ms arbitrary: xs s t)
  case Nil
  then show ?case by simp
next
case (Cons m ms)
from Cons.prem1 obtain x u xs' where
  head: Some (x, u) ∈ set-dist (execute m s)
  and tail: Some (xs', t) ∈ set-dist (execute (mmap ms) u)
  by (auto elim!: p.set-dist-bindE)
have m-in: m ∈ set (m # ms)

```

```

    by simp
  have q-head: PQueryCounter u = PQueryCounter s
    by (rule Cons.prem(2)[OF m-in head])
  have tail-step:
     $\bigwedge m' x' u' v'. m' \in \text{set } ms \implies$ 
     $\text{Some } (x', v') \in \text{set-dist } (\text{execute } m' u') \implies$ 
     $PQueryCounter v' = PQueryCounter u'$ 
  proof -
    fix m' x' u' v'
    assume m'-in-tail: m'  $\in$  set ms
    and out: Some (x', v')  $\in$  set-dist (execute m' u')
    have m'-in: m'  $\in$  set (m # ms)
    using m'-in-tail by simp
    show PQueryCounter v' = PQueryCounter u'
      by (rule Cons.prem(2)[OF m'-in out])
  qed
  have q-tail: PQueryCounter t = PQueryCounter u
    by (rule Cons.IH[OF tail tail-step])
  show ?case
    using q-head q-tail by simp
qed

lemma mfold-preserves-query-counter:
  fixes s t :: ('f, 'more) protocol-channel-scheme
  assumes outcome: Some (x, t)  $\in$  set-dist (execute (mfold a ms) s)
  and step:  $\bigwedge m a0 y u v. m \in \text{set } ms \implies$ 
     $\text{Some } (y, v) \in \text{set-dist } (\text{execute } (m a0) u) \implies$ 
     $PQueryCounter v = PQueryCounter u$ 
  shows PQueryCounter t = PQueryCounter s
  using outcome step
proof (induction ms arbitrary: a x s t)
  case Nil
  then show ?case by simp
next
  case (Cons m ms)
  from Cons.prem(1) obtain y u where
    head: Some (y, u)  $\in$  set-dist (execute (m a) s)
    and tail: Some (x, t)  $\in$  set-dist (execute (mfold y ms) u)
    by (auto elim!: p.set-dist-bindE)
  have m-in: m  $\in$  set (m # ms)
    by simp
  have q-head: PQueryCounter u = PQueryCounter s
    by (rule Cons.prem(2)[OF m-in head])
  have tail-step:
     $\bigwedge m' a0 y' u' v'. m' \in \text{set } ms \implies$ 
     $\text{Some } (y', v') \in \text{set-dist } (\text{execute } (m' a0) u') \implies$ 
     $PQueryCounter v' = PQueryCounter u'$ 
  proof -
    fix m' a0 y' u' v'

```



```

assume  $m'$ -in-tail:  $m' \in \text{set } ms$ 
and out:  $\text{Some } (y', v') \in \text{set-dist } (\text{execute } (m' \text{ a0}) \ u')$ 
have  $m'$ -in:  $m' \in \text{set } (m \# ms)$ 
using  $m'$ -in-tail by simp
show  $P\text{QueryCounter } v' = P\text{QueryCounter } u'$ 
by (rule Cons.premss(2)[OF  $m'$ -in out])
qed
have  $q$ -tail:  $P\text{QueryCounter } t = P\text{QueryCounter } u$ 
by (rule Cons.IH[OF tail tail-step])
show ?case
using  $q$ -head  $q$ -tail by simp
qed

lemma mfold2-send-transcript-extends:
assumes  $\text{Some } ((), t) \in \text{set-dist } (\text{execute } (\text{mfold2 } p.\text{send } xs) \ s)$ 
shows transcript-extends  $t \ s$ 
using  $p.\text{mfold2-send-outcome}$ [OF assms]
unfolding transcript-extends-def by auto

lemma mfold2-send-hash-extends:
assumes  $\text{Some } ((), t) \in \text{set-dist } (\text{execute } (\text{mfold2 } p.\text{send } xs) \ s)$ 
shows  $s \leq t$ 
using  $p.\text{mfold2-send-outcome}$ [OF assms]
unfolding less-eq-hash-ext-def less-eq-fmap-def by simp

lemma mfold2-send-preserves-query-counter:
assumes  $\text{Some } ((), t) \in \text{set-dist } (\text{execute } (\text{mfold2 } p.\text{send } xs) \ s)$ 
shows  $P\text{QueryCounter } t = P\text{QueryCounter } s$ 
using  $p.\text{mfold2-send-outcome}$ [OF assms] by simp

lemma fri-commit-transcript-extends:
assumes outcome:
   $\text{Some } ((ps', ds', ls', ms'), t) \in$ 
   $\text{set-dist } (\text{execute } (p.\text{fri-commit } n \ ps \ ds \ ls \ ms) \ s)$ 
shows transcript-extends  $t \ s$ 
using outcome
proof (induction  $n$  arbitrary:  $ps \ ds \ ls \ ms \ s \ ps' \ ds' \ ls' \ ms' \ t$ )
case 0
then show ?case by simp
next
case (Suc  $n$ )
from Suc.premss obtain  $s1 \ b \ s2 \ \text{next-poly} \ \text{next-domain} \ \text{next-layer} \ m \ s3$  where
  send-root:  $\text{Some } ((), s1) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{value } (\text{last } ms)))) \ s)$ 
and rand:  $\text{Some } (b, s2) \in \text{set-dist } (\text{execute } p.\text{receive-random-field-element } s1)$ 
and next-layer-eq:  $p.\text{next-fri-layer } (\text{last } ps) \ (\text{last } ds) \ b =$ 
   $(\text{next-poly}, \text{next-domain}, \text{next-layer})$ 
and create-m:  $\text{Some } (m, s3) \in \text{set-dist } (\text{execute } (p.\text{create next-layer}) \ s2)$ 
and rest:  $\text{Some } ((ps', ds', ls', ms'), t) \in$ 
   $\text{set-dist } (\text{execute } (p.\text{fri-commit } n \ ps \ ds \ ls \ ms) \ s)$ 

```

```

    (p.fri-commit n
      (ps @ [next-poly])
      (ds @ [next-domain])
      (ls @ [next-layer])
      (ms @ [m])) s3)
  by (auto simp: Let-def split: prod.splits elim!: p.set-dist-bindE)
have s-s1: transcript-extends s1 s
  using send-transcript-extends[OF send-root] .
have s1-s2: transcript-extends s2 s1
  using receive-random-field-element-transcript-extends[OF rand] .
have s2-s3: transcript-extends s3 s2
  using create-transcript-extends[OF create-m] .
have s3-t: transcript-extends t s3
  using Suc.IH[OF rest] .
show ?case
  using s-s1 s1-s2 s2-s3 s3-t by (meson transcript-extends-trans)
qed

lemma fri-commit-replay-data:
  assumes outcome:
    Some ((ps', ds', ls', ms'), t) ∈
      set-dist (execute (p.fri-commit n ps ds ls ms) s)
  shows ∃ roots challenges.
    length roots = n ∧
    length challenges = n ∧
    PTranscript t = rev roots @ PTranscript s ∧
    PState t = foldl concat (PState s) roots ∧
    s ≤ t ∧
    (∀ i < length roots.
      fmllookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) roots)))
    =
      Some (challenges ! i))
  using outcome
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t)
  case 0
  then show ?case
    by (auto simp: p.hash-ext-refl)
next
  case (Suc n)
  from Suc.premis obtain s1 b s2 next-poly next-domain next-layer m s3 where
    send-root: Some ((), s1) ∈ set-dist (execute (p.send (value (last ms))) s)
    and rand: Some (b, s2) ∈ set-dist (execute p.receive-random-field-element s1)
    and next-layer-eq: p.next-fri-layer (last ps) (last ds) b =
      (next-poly, next-domain, next-layer)
    and create-m: Some (m, s3) ∈ set-dist (execute (p.create next-layer) s2)
    and rest:
      Some ((ps', ds', ls', ms'), t) ∈
        set-dist (execute
          (p.fri-commit n

```

```

      (ps @ [next-poly])
      (ds @ [next-domain])
      (ls @ [next-layer])
      (ms @ [m])) s3)
  by (auto simp: Let-def split: prod.splits elim!: p.set-dist-bindE)
let ?root = value (last ms)
have send-res:
  s1 = s(PState := concat (PState s) ?root, PTranscript := ?root # PTranscript s)
  using p.send-outcome[OF send-root] .
have rand-data:
  s1 ≤ s2 ∧
  PState s2 = PState s1 ∧
  PTranscript s2 = PTranscript s1 ∧
  fmlookup (HashMap s2) (FiatShamirChallenge (PState s1)) = Some b
  using receive-random-field-element-outcome[OF rand] .
have create-state: PState s3 = PState s2
  using create-preserves-state[OF create-m] .
have create-tr: PTranscript s3 = PTranscript s2
  using create-preserves-transcript[OF create-m] .
have s2-s3: s2 ≤ s3
  using create-hash-extends[OF create-m] .
from Suc.IH[OF rest] obtain roots challenges where
  roots-len: length roots = n
  and challenges-len: length challenges = n
  and tr-t: PTranscript t = rev roots @ PTranscript s3
  and st-t: PState t = foldl concat (PState s3) roots
  and s3-t: s3 ≤ t
  and lookups-tail:
    ∀ i < length roots.
      fmlookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s3) (take (Suc i)
roots))) =
        Some (challenges ! i)
  by blast
have s-s1: s ≤ s1
  using send-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s-t: s ≤ t
  using s-s1 rand-data s2-s3 s3-t by (meson p.hash-ext-trans)
have s2-t: s2 ≤ t
  using s2-s3 s3-t by (rule p.hash-ext-trans)
have lookup-head:
  fmlookup (HashMap t) (FiatShamirChallenge (concat (PState s) ?root)) = Some b
proof -
  have lookup-s2: fmlookup (HashMap s2) (FiatShamirChallenge (PState s1)) = Some b
    using rand-data by simp
  have fmlookup (HashMap t) (FiatShamirChallenge (PState s1)) = Some b
    using p.hash-extension-lookup[OF lookup-s2 s2-t] .
  then show ?thesis
    using send-res by simp
qed

```

```

have tr-final:
  PTranscript t = rev (?root # roots) @ PTranscript s
  using tr-t create-tr rand-data send-res by simp
have st-final:
  PState t = foldl concat (PState s) (?root # roots)
  using st-t create-state rand-data send-res by simp
have lookups:
   $\forall i < \text{length } (?root \# roots).$ 
    fmllookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) (?root
# roots)))) =
      Some ((b # challenges) ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{length } (?root \# roots)$ 
  show fmllookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i)
(?root # roots)))) =
    Some ((b # challenges) ! i)
proof (cases i)
  case 0
  then show ?thesis
    using lookup-head by simp
next
  case (Suc j)
  then have j-bound:  $j < \text{length } roots$ 
    using i-bound by simp
  have key-eq:
    foldl concat (PState s) (take (Suc (Suc j)) (?root # roots)) =
      foldl concat (PState s3) (take (Suc j) roots)
    using send-res rand-data create-state by simp
  show ?thesis
    using lookups-tail j-bound key-eq Suc by simp
qed
qed
show ?case
proof (rule exI[where  $x = ?root \# roots$ ])
  show  $\exists \text{ challenges.}$ 
     $\text{length } (?root \# roots) = \text{Suc } n \wedge$ 
     $\text{length } \text{challenges} = \text{Suc } n \wedge$ 
     $P\text{Transcript } t = \text{rev } (?root \# roots) @ P\text{Transcript } s \wedge$ 
     $P\text{State } t = \text{foldl } \text{concat } (P\text{State } s) (?root \# roots) \wedge$ 
     $s \leq t \wedge$ 
     $(\forall i < \text{length } (?root \# roots).$ 
      fmllookup (HashMap t)
        (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) (?root # roots)))) =
        Some (challenges ! i))
proof (rule exI[where  $x = b \# challenges$ ])
  show  $\text{length } (?root \# roots) = \text{Suc } n \wedge$ 
     $\text{length } (b \# challenges) = \text{Suc } n \wedge$ 
     $P\text{Transcript } t = \text{rev } (?root \# roots) @ P\text{Transcript } s \wedge$ 

```

```

    PState t = foldl concat (PState s) (?root # roots) ∧
    s ≤ t ∧
    (∀ i < length (?root # roots).
      fmllookup (HashMap t)
        (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) (?root # roots)))) =
        Some ((b # challenges) ! i))
    using roots-len challenges-len tr-final st-final s-t lookups by simp
  qed
qed
qed

lemma fri-commit-replay-state:
  assumes outcome:
    Some ((ps', ds', ls', ms'), t) ∈
      set-dist (execute (p.fri-commit n ps ds ls ms) s)
  shows ∃ roots.
    length roots = n ∧
    PTranscript t = rev roots @ PTranscript s ∧
    PState t = foldl concat (PState s) roots ∧
    s ≤ t
  using outcome
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t)
  case 0
  then show ?case
    by (auto simp: p.hash-ext-refl)
next
  case (Suc n)
  from Suc.premis obtain s1 b s2 next-poly next-domain next-layer m s3 where
    send-root: Some ((), s1) ∈ set-dist (execute (p.send (value (last ms))) s)
    and rand: Some (b, s2) ∈ set-dist (execute p.receive-random-field-element s1)
    and next-layer-eq: p.next-fri-layer (last ps) (last ds) b =
      (next-poly, next-domain, next-layer)
    and create-m: Some (m, s3) ∈ set-dist (execute (p.create next-layer) s2)
    and rest:
      Some ((ps', ds', ls', ms'), t) ∈
        set-dist (execute
          (p.fri-commit n
            (ps @ [next-poly])
            (ds @ [next-domain])
            (ls @ [next-layer])
            (ms @ [m])) s3)
    by (auto simp: Let-def split: prod.splits elim!: p.set-dist-bindE)
  let ?root = value (last ms)
  have send-res:
    s1 = s(PState := concat (PState s) ?root, PTranscript := ?root # PTranscript s)
    using p.send-outcome[OF send-root] .
  have rand-data:
    s1 ≤ s2 ∧ PState s2 = PState s1 ∧ PTranscript s2 = PTranscript s1
    using receive-random-field-element-outcome[OF rand] by simp

```

```

have create-state: PState s3 = PState s2
  using create-preserves-state[OF create-m] .
have create-tr: PTranscript s3 = PTranscript s2
  using create-preserves-transcript[OF create-m] .
have s-s1: s ≤ s1
  using send-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s2-s3: s2 ≤ s3
  using create-hash-extends[OF create-m] .
from Suc.IH[OF rest] obtain roots where
  roots-len: length roots = n
  and tr-t: PTranscript t = rev roots @ PTranscript s3
  and st-t: PState t = foldl concat (PState s3) roots
  and s3-t: s3 ≤ t
  by blast
have tr-final: PTranscript t = rev (?root # roots) @ PTranscript s
  using tr-t create-tr rand-data send-res by simp
have st-final: PState t = foldl concat (PState s) (?root # roots)
  using st-t create-state rand-data send-res by simp
have s-t: s ≤ t
  using s-s1 rand-data s2-s3 s3-t by (meson p.hash-ext-trans)
show ?case
  by (intro exI[where x=?root # roots])
    (use roots-len tr-final st-final s-t in simp)
qed

```

lemma *fri-commit-successor-layers*:

```

assumes outcome:
  Some ((ps', ds', ls', ms'), t) ∈
    set-dist (execute (p.fri-commit n ps ds ls ms) s)
and lens:
  length ps = length ds
  length ds = length ls
  length ls = length ms
and nonempty: ps ≠ []
shows ∃ challenges.
  length challenges = n ∧
  (∀ j < n.
    p.next-fri-layer
      (ps' ! (length ps - 1 + j))
      (ds' ! (length ps - 1 + j))
      (challenges ! j) =
        (ps' ! (length ps + j),
         ds' ! (length ps + j),
         ls' ! (length ps + j)))
  using outcome lens nonempty
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t)
  case 0
  then show ?case by auto
next

```

```

case (Suc n)
from Suc.prem(1) obtain s1 b s2 next-poly next-domain next-layer m s3 where
  send-root:
    Some (((), s1) ∈ set-dist (execute (p.send (value (last ms))) s)
  and rand:
    Some (b, s2) ∈ set-dist (execute p.receive-random-field-element s1)
  and next-layer-eq:
    p.next-fri-layer (last ps) (last ds) b =
      (next-poly, next-domain, next-layer)
  and create-m:
    Some (m, s3) ∈ set-dist (execute (p.create next-layer) s2)
  and rest:
    Some ((ps', ds', ls', ms'), t) ∈
      set-dist (execute
        (p.fri-commit n
          (ps @ [next-poly])
          (ds @ [next-domain])
          (ls @ [next-layer])
          (ms @ [m])) s3)
    by (auto simp: Let-def split: prod.splits elim!: p.set-dist-bindE)
have lens-tail:
  length (ps @ [next-poly]) = length (ds @ [next-domain])
  length (ds @ [next-domain]) = length (ls @ [next-layer])
  length (ls @ [next-layer]) = length (ms @ [m])
  using Suc.prem(2-4) by simp-all
have nonempty-tail: ps @ [next-poly] ≠ []
  by simp
from Suc.IH[OF rest lens-tail nonempty-tail] obtain challenges where
  challenges-len: length challenges = n
  and tail-layers:
    ∀ j < n.
      p.next-fri-layer
        (ps' ! (length (ps @ [next-poly]) - 1 + j))
        (ds' ! (length (ps @ [next-poly]) - 1 + j))
        (challenges ! j) =
          (ps' ! (length (ps @ [next-poly]) + j),
           ds' ! (length (ps @ [next-poly]) + j),
           ls' ! (length (ps @ [next-poly]) + j))
  by blast
have prefix:
  take (length (ps @ [next-poly])) ps' = ps @ [next-poly] ∧
  take (length (ds @ [next-domain])) ds' = ds @ [next-domain] ∧
  take (length (ls @ [next-layer])) ls' = ls @ [next-layer]
  using p.fri-commit-preserves-prefix[OF rest] by simp
have head-ps: ps' ! (length ps - 1) = last ps
proof -
  obtain q where len-ps: length ps = Suc q
  using Suc.prem(5) by (cases ps) auto
  have idx-lt: length ps - 1 < length (ps @ [next-poly])

```

```

    using Suc.premis(5) by simp
  have ps'-nth: ps' ! (length ps - 1) =
    (ps @ [next-poly]) ! (length ps - 1)
  proof -
    have (take (length (ps @ [next-poly])) ps') ! (length ps - 1) =
      (ps @ [next-poly]) ! (length ps - 1)
      using conjunct1[OF prefix] by simp
    then show ?thesis
      using idx-lt by simp
  qed
  have append-nth:
    (ps @ [next-poly]) ! (length ps - 1) = ps ! (length ps - 1)
    using len-ps by (simp add: nth-append)
  have last-ps: last ps = ps ! (length ps - 1)
    using Suc.premis(5) by (simp add: last-conv-nth)
  show ?thesis
    using ps'-nth append-nth last-ps by simp
  qed
  have head-ds: ds' ! (length ps - 1) = last ds
  proof -
    have ds-ne: ds ≠ []
      using Suc.premis(2,5) by auto
    obtain q where len-ds: length ds = Suc q
      using ds-ne by (cases ds) auto
    have idx-lt: length ps - 1 < length (ds @ [next-domain])
      using Suc.premis(2,5) by simp
    have ds'-nth: ds' ! (length ps - 1) =
      (ds @ [next-domain]) ! (length ps - 1)
    proof -
      have (take (length (ds @ [next-domain])) ds') ! (length ps - 1) =
        (ds @ [next-domain]) ! (length ps - 1)
        using conjunct1[OF conjunct2[OF prefix]] by simp
      then show ?thesis
        using idx-lt by simp
    qed
    have append-nth:
      (ds @ [next-domain]) ! (length ps - 1) = ds ! (length ds - 1)
      using Suc.premis(2) len-ds by (simp add: nth-append)
    have last-ds: last ds = ds ! (length ds - 1)
      using ds-ne by (simp add: last-conv-nth)
    show ?thesis
      using ds'-nth append-nth last-ds by simp
  qed
  have head-next-ps: ps' ! length ps = next-poly
  proof -
    have pref: take (Suc (length ps)) ps' = ps @ [next-poly]
      using conjunct1[OF prefix] by simp
    have ps' ! length ps = (take (Suc (length ps)) ps') ! length ps
      by simp
  qed

```



```

    also have ... = next-poly
      using pref by simp
    finally show ?thesis .
  qed
  have head-next-ds: ds' ! length ps = next-domain
  proof -
    have pref: take (Suc (length ds)) ds' = ds @ [next-domain]
      using conjunct1[OF conjunct2[OF prefix]] Suc.premis(2) by simp
    have ds' ! length ps = (take (Suc (length ds)) ds') ! length ps
      using Suc.premis(2) by simp
    also have ... = next-domain
      using pref Suc.premis(2) by simp
    finally show ?thesis .
  qed
  have head-next-ls: ls' ! length ps = next-layer
  proof -
    have pref: take (Suc (length ls)) ls' = ls @ [next-layer]
      using conjunct2[OF conjunct2[OF prefix]] Suc.premis(2,3) by simp
    have ls' ! length ps = (take (Suc (length ls)) ls') ! length ps
      using Suc.premis(2,3) by simp
    also have ... = next-layer
      using pref Suc.premis(2,3) by simp
    finally show ?thesis .
  qed
  have layers:
     $\forall j < \text{Suc } n.$ 
     $p.\text{next-fri-layer}$ 
     $(ps' ! (\text{length } ps - 1 + j))$ 
     $(ds' ! (\text{length } ps - 1 + j))$ 
     $((b \# \text{challenges}) ! j) =$ 
     $(ps' ! (\text{length } ps + j),$ 
     $ds' ! (\text{length } ps + j),$ 
     $ls' ! (\text{length } ps + j))$ 
  proof (intro allI impI)
    fix j
    assume j-bound:  $j < \text{Suc } n$ 
    show  $p.\text{next-fri-layer}$ 
       $(ps' ! (\text{length } ps - 1 + j))$ 
       $(ds' ! (\text{length } ps - 1 + j))$ 
       $((b \# \text{challenges}) ! j) =$ 
       $(ps' ! (\text{length } ps + j),$ 
       $ds' ! (\text{length } ps + j),$ 
       $ls' ! (\text{length } ps + j))$ 
    proof (cases j)
      case 0
      then show ?thesis
        using next-layer-eq head-ps head-ds head-next-ps head-next-ds head-next-ls
        by simp
    next
  
```

```

case (Suc k)
then have k-bound:  $k < n$ 
  using j-bound by simp
have idx1:
   $\text{length } (ps @ [next-poly]) - 1 + k = \text{length } ps + k$ 
  by simp
have idx2:
   $\text{length } (ps @ [next-poly]) + k = \text{length } ps + \text{Suc } k$ 
  by simp
have idx-current:
   $\text{length } ps - 1 + \text{Suc } k = \text{length } ps + k$ 
proof -
  obtain q where  $\text{length } ps = \text{Suc } q$ 
  using Suc.prems(5) by (cases ps) auto
  then show ?thesis by simp
qed
have idx-left:
   $\text{Suc } (\text{length } ps - 1 + k) = \text{length } ps + k$ 
proof -
  obtain q where  $\text{length } ps = \text{Suc } q$ 
  using Suc.prems(5) by (cases ps) auto
  then show ?thesis by simp
qed
have tail-k:
  p.next-fri-layer
  (ps' ! ( $\text{length } ps + k$ ))
  (ds' ! ( $\text{length } ps + k$ ))
  (challenges ! k) =
  (ps' !  $\text{Suc } (\text{length } ps + k)$ ),
  (ds' !  $\text{Suc } (\text{length } ps + k)$ ),
  (ls' !  $\text{Suc } (\text{length } ps + k)$ )
  using tail-layers k-bound by simp
show ?thesis
  unfolding Suc
  apply (subst idx-current)
  apply (subst idx-current)
  using tail-k
  apply simp
  done
qed
qed
show ?case
  by (intro exI[where  $x=b \# \text{challenges}$ ])
  (use challenges-len layers in simp)
qed

```

lemma *fri-commit-replay-successor-data*:

assumes *outcome*:

Some ((*ps'*, *ds'*, *ls'*, *ms'*), *t*) ∈

```

    set-dist (execute (p.fri-commit n ps ds ls ms) s)
  and lens:
    length ps = length ds
    length ds = length ls
    length ls = length ms
  and nonempty: ps ≠ []
shows ∃ roots challenges.
  length roots = n ∧
  length challenges = n ∧
  PTranscript t = rev roots @ PTranscript s ∧
  PState t = foldl concat (PState s) roots ∧
  s ≤ t ∧
  (∀ i < length roots.
    fmllookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) roots)))
  =
    Some (challenges ! i)) ∧
  (∀ j < n.
    roots ! j = value (ms' ! (length ms - 1 + j))) ∧
  (∀ j < n.
    p.next-fri-layer
      (ps' ! (length ps - 1 + j))
      (ds' ! (length ps - 1 + j))
      (challenges ! j) =
        (ps' ! (length ps + j),
         ds' ! (length ps + j),
         ls' ! (length ps + j)))
  using outcome lens nonempty
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t)
  case 0
  then show ?case
  by (auto simp: p.hash-ext-refl)
next
  case (Suc n)
  from Suc.premis(1) obtain s1 b s2 next-poly next-domain next-layer m s3 where
    send-root:
      Some ((), s1) ∈ set-dist (execute (p.send (value (last ms))) s)
  and rand:
      Some (b, s2) ∈ set-dist (execute p.receive-random-field-element s1)
  and next-layer-eq:
      p.next-fri-layer (last ps) (last ds) b =
        (next-poly, next-domain, next-layer)
  and create-m:
      Some (m, s3) ∈ set-dist (execute (p.create next-layer) s2)
  and rest:
      Some ((ps', ds', ls', ms'), t) ∈
        set-dist (execute
          (p.fri-commit n
            (ps @ [next-poly])
            (ds @ [next-domain])

```

```

      (ls @ [next-layer])
      (ms @ [m])) s3)
  by (auto simp: Let-def split: prod.splits elim!: p.set-dist-bindE)
let ?root = value (last ms)
have lens-tail:
  length (ps @ [next-poly]) = length (ds @ [next-domain])
  length (ds @ [next-domain]) = length (ls @ [next-layer])
  length (ls @ [next-layer]) = length (ms @ [m])
  using Suc.premis(2-4) by simp-all
have nonempty-tail: ps @ [next-poly] ≠ []
  by simp
from Suc.IH[OF rest lens-tail nonempty-tail] obtain roots challenges where
  roots-len: length roots = n
  and challenges-len: length challenges = n
  and tr-t: PTranscript t = rev roots @ PTranscript s3
  and st-t: PState t = foldl concat (PState s3) roots
  and s3-t: s3 ≤ t
  and lookups-tail:
    ∀ i < length roots.
      fmllookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s3) (take (Suc i)
roots))) =
        Some (challenges ! i)
  and roots-tail:
    ∀ j < n.
      roots ! j =
        value (ms' ! (length (ms @ [m]) - 1 + j))
  and tail-layers:
    ∀ j < n.
      p.next-fri-layer
      (ps' ! (length (ps @ [next-poly]) - 1 + j))
      (ds' ! (length (ps @ [next-poly]) - 1 + j))
      (challenges ! j) =
      (ps' ! (length (ps @ [next-poly]) + j),
       ds' ! (length (ps @ [next-poly]) + j),
       ls' ! (length (ps @ [next-poly]) + j))
  by blast
have send-res:
  s1 = s(PState := concat (PState s) ?root, PTranscript := ?root # PTranscript s)
  using p.send-outcome[OF send-root] .
have rand-data:
  s1 ≤ s2 ∧
  PState s2 = PState s1 ∧
  PTranscript s2 = PTranscript s1 ∧
  fmllookup (HashMap s2) (FiatShamirChallenge (PState s1)) = Some b
  using receive-random-field-element-outcome[OF rand] .
have create-state: PState s3 = PState s2
  using create-preserves-state[OF create-m] .
have create-tr: PTranscript s3 = PTranscript s2
  using create-preserves-transcript[OF create-m] .

```

```

have s2-s3:  $s2 \leq s3$ 
  using create-hash-extends[OF create-m] .
have s-s1:  $s \leq s1$ 
  using send-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s-t:  $s \leq t$ 
  using s-s1 rand-data s2-s3 s3-t by (meson p.hash-ext-trans)
have s2-t:  $s2 \leq t$ 
  using s2-s3 s3-t by (rule p.hash-ext-trans)
have lookup-head:
  fmllookup (HashMap t) (FiatShamirChallenge (concat (PState s) ?root)) = Some b
proof -
  have lookup-s2: fmllookup (HashMap s2) (FiatShamirChallenge (PState s1)) = Some b
    using rand-data by simp
  have fmllookup (HashMap t) (FiatShamirChallenge (PState s1)) = Some b
    using p.hash-extension-lookup[OF lookup-s2 s2-t] .
  then show ?thesis
    using send-res by simp
qed
have tr-final:
  PTranscript t = rev (?root # roots) @ PTranscript s
  using tr-t create-tr rand-data send-res by simp
have st-final:
  PState t = foldl concat (PState s) (?root # roots)
  using st-t create-state rand-data send-res by simp
have lookups:
   $\forall i < \text{length } (?root \# \text{roots}).$ 
  fmllookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) (?root
# roots)))) =
    Some ((b # challenges) ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{length } (?root \# \text{roots})$ 
  show fmllookup (HashMap t)
    (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) (?root # roots)))) =
    Some ((b # challenges) ! i)
  proof (cases i)
    case 0
    then show ?thesis
      using lookup-head by simp
  next
    case (Suc j)
    then have j-bound:  $j < \text{length roots}$ 
      using i-bound by simp
    have key-eq:
      foldl concat (PState s) (take (Suc (Suc j)) (?root # roots)) =
      foldl concat (PState s3) (take (Suc j) roots)
      using send-res rand-data create-state by simp
    show ?thesis
      using lookups-tail j-bound key-eq Suc by simp
  end
end

```

```

qed
qed
have prefix:
  take (length (ps @ [next-poly])) ps' = ps @ [next-poly] ∧
  take (length (ds @ [next-domain])) ds' = ds @ [next-domain] ∧
  take (length (ls @ [next-layer])) ls' = ls @ [next-layer] ∧
  take (length (ms @ [m])) ms' = ms @ [m]
  using p.fri-commit-preserves-prefix[OF rest] by simp
have head-ms: ms' ! (length ms - 1) = last ms
proof -
  have ms-ne: ms ≠ []
    using Suc.premis(2-5) by auto
  obtain q where len-ms: length ms = Suc q
    using ms-ne by (cases ms) auto
  have idx-lt: length ms - 1 < length (ms @ [m])
    using ms-ne by simp
  have ms'-nth: ms' ! (length ms - 1) =
    (ms @ [m]) ! (length ms - 1)
  proof -
    have (take (length (ms @ [m])) ms') ! (length ms - 1) =
      (ms @ [m]) ! (length ms - 1)
      using conjunct2[OF conjunct2[OF conjunct2[OF prefix]]] by simp
    then show ?thesis
      using idx-lt by simp
  qed
  have append-nth:
    (ms @ [m]) ! (length ms - 1) = ms ! (length ms - 1)
    using len-ms by (simp add: nth-append)
  have last-ms: last ms = ms ! (length ms - 1)
    using ms-ne by (simp add: last-conv-nth)
  show ?thesis
    using ms'-nth append-nth last-ms by simp
qed
have head-ps: ps' ! (length ps - 1) = last ps
proof -
  obtain q where len-ps: length ps = Suc q
    using Suc.premis(5) by (cases ps) auto
  have idx-lt: length ps - 1 < length (ps @ [next-poly])
    using Suc.premis(5) by simp
  have ps'-nth: ps' ! (length ps - 1) =
    (ps @ [next-poly]) ! (length ps - 1)
  proof -
    have (take (length (ps @ [next-poly])) ps') ! (length ps - 1) =
      (ps @ [next-poly]) ! (length ps - 1)
      using conjunct1[OF prefix] by simp
    then show ?thesis
      using idx-lt by simp
  qed
  have append-nth:

```

```

    (ps @ [next-poly]) ! (length ps - 1) = ps ! (length ps - 1)
    using len-ps by (simp add: nth-append)
  have last-ps: last ps = ps ! (length ps - 1)
    using Suc.premis(5) by (simp add: last-conv-nth)
  show ?thesis
    using ps'-nth append-nth last-ps by simp
qed
have head-ds: ds' ! (length ps - 1) = last ds
proof -
  have ds-ne: ds ≠ []
    using Suc.premis(2,5) by auto
  obtain q where len-ds: length ds = Suc q
    using ds-ne by (cases ds) auto
  have idx-lt: length ps - 1 < length (ds @ [next-domain])
    using Suc.premis(2,5) by simp
  have ds'-nth: ds' ! (length ps - 1) =
    (ds @ [next-domain]) ! (length ps - 1)
  proof -
    have (take (length (ds @ [next-domain])) ds') ! (length ps - 1) =
      (ds @ [next-domain]) ! (length ps - 1)
      using conjunct1[OF conjunct2[OF prefix]] by simp
    then show ?thesis
      using idx-lt by simp
  qed
  have append-nth:
    (ds @ [next-domain]) ! (length ps - 1) = ds ! (length ds - 1)
    using Suc.premis(2) len-ds by (simp add: nth-append)
  have last-ds: last ds = ds ! (length ds - 1)
    using ds-ne by (simp add: last-conv-nth)
  show ?thesis
    using ds'-nth append-nth last-ds by simp
qed
have head-next-ps: ps' ! length ps = next-poly
proof -
  have pref: take (Suc (length ps)) ps' = ps @ [next-poly]
    using conjunct1[OF prefix] by simp
  have ps' ! length ps = (take (Suc (length ps)) ps') ! length ps
    by simp
  also have ... = next-poly
    using pref by simp
  finally show ?thesis .
qed
have head-next-ds: ds' ! length ps = next-domain
proof -
  have pref: take (Suc (length ds)) ds' = ds @ [next-domain]
    using conjunct1[OF conjunct2[OF prefix]] Suc.premis(2) by simp
  have ds' ! length ps = (take (Suc (length ds)) ds') ! length ps
    using Suc.premis(2) by simp
  also have ... = next-domain

```

```

    using pref Suc.premis(2) by simp
  finally show ?thesis .
qed
have head-next-ls: ls' ! length ps = next-layer
proof -
  have pref: take (Suc (length ls)) ls' = ls @ [next-layer]
    using conjunct2[OF conjunct2[OF prefix]] Suc.premis(2,3) by simp
  have ls' ! length ps = (take (Suc (length ls)) ls') ! length ps
    using Suc.premis(2,3) by simp
  also have ... = next-layer
    using pref Suc.premis(2,3) by simp
  finally show ?thesis .
qed
have layers:
   $\forall j < \text{Suc } n.$ 
  p.next-fri-layer
  (ps' ! (length ps - 1 + j))
  (ds' ! (length ps - 1 + j))
  ((b # challenges) ! j) =
  (ps' ! (length ps + j),
   ds' ! (length ps + j),
   ls' ! (length ps + j))
proof (intro allI impI)
  fix j
  assume j-bound: j < Suc n
  show p.next-fri-layer
    (ps' ! (length ps - 1 + j))
    (ds' ! (length ps - 1 + j))
    ((b # challenges) ! j) =
    (ps' ! (length ps + j),
     ds' ! (length ps + j),
     ls' ! (length ps + j))
  proof (cases j)
    case 0
    then show ?thesis
      using next-layer-eq head-ps head-ds head-next-ps head-next-ds head-next-ls
      by simp
  next
    case (Suc k)
    then have k-bound: k < n
      using j-bound by simp
    have idx-current:
      length ps - 1 + Suc k = length ps + k
    proof -
      obtain q where length ps = Suc q
        using Suc.premis(5) by (cases ps) auto
      then show ?thesis by simp
    qed
    have tail-k:

```



```

    p.next-fri-layer
    (ps' ! (length ps + k))
    (ds' ! (length ps + k))
    (challenges ! k) =
      (ps' ! Suc (length ps + k),
       ds' ! Suc (length ps + k),
       ls' ! Suc (length ps + k))
  using tail-layers k-bound by simp
show ?thesis
  unfolding Suc
  apply (subst idx-current)
  apply (subst idx-current)
  using tail-k
  apply simp
done
qed
qed
have roots-aligned:
   $\forall j < \text{Suc } n.$ 
  ( $\text{?root} \# \text{roots}$ ) ! j = value (ms' ! (length ms - 1 + j))
proof (intro allI impI)
  fix j
  assume j-bound: j < Suc n
  show ( $\text{?root} \# \text{roots}$ ) ! j = value (ms' ! (length ms - 1 + j))
  proof (cases j)
    case 0
    then show ?thesis
      using head-ms by simp
  next
    case (Suc k)
    then have k-bound: k < n
      using j-bound by simp
    have idx-current:
      length ms - 1 + Suc k = length ms + k
    proof -
      obtain q where length ms = Suc q
      using Suc.prem5(2-5) by (cases ms) auto
      then show ?thesis by simp
    qed
    have tail-k:
      roots ! k = value (ms' ! (length ms + k))
      using roots-tail k-bound by simp
  show ?thesis
    unfolding Suc
    apply (subst idx-current)
    using tail-k
    apply simp
  done
qed

```

```

qed
show ?case
  by (intro exI[where  $x=?root \# roots$ ] exI[where  $x=b \# challenges$ ])
    (use roots-len challenges-len tr-final st-final s-t lookups roots-aligned layers in simp)
qed

lemma fri-commit-initial-replay-successor-data:
assumes outcome:
  Some ((ps, ds, ls, ms), t) ∈
    set-dist (execute
      (p.fri-commit n [cp] [d] [l] [m]) s)
shows ∃ roots challenges.
  length roots = n ∧
  length challenges = n ∧
  PTranscript t = rev roots @ PTranscript s ∧
  PState t = foldl concat (PState s) roots ∧
  s ≤ t ∧
  (∀ i < length roots.
    fmlookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) roots)))
  =
    Some (challenges ! i)) ∧
  (∀ j < n. roots ! j = value (ms ! j)) ∧
  (∀ j < n.
    p.next-fri-layer
      (ps ! j)
      (ds ! j)
      (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j))
proof –
have data:
  ∃ roots challenges.
  length roots = n ∧
  length challenges = n ∧
  PTranscript t = rev roots @ PTranscript s ∧
  PState t = foldl concat (PState s) roots ∧
  s ≤ t ∧
  (∀ i < length roots.
    fmlookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) roots)))
  =
    Some (challenges ! i)) ∧
  (∀ j < n. roots ! j = value (ms ! (length [m] - 1 + j))) ∧
  (∀ j < n.
    p.next-fri-layer
      (ps ! (length [cp] - 1 + j))
      (ds ! (length [cp] - 1 + j))
      (challenges ! j) =
      (ps ! (length [cp] + j),
       ds ! (length [cp] + j),
       ls ! (length [cp] + j)))

```

```

    by (rule fri-commit-replay-successor-data[OF outcome]) simp-all
  from data obtain roots challenges where
    len-roots: length roots = n
    and len-challenges: length challenges = n
    and tr: PTranscript t = rev roots @ PTranscript s
    and st: PState t = foldl concat (PState s) roots
    and ext: s ≤ t
    and lookups:
      ∀ i < length roots.
        fmlookup (HashMap t) (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) roots)))
  =
    Some (challenges ! i)
  and roots-aligned:
    ∀ j < n. roots ! j = value (ms ! (length [m] - 1 + j))
  and layers:
    ∀ j < n.
      p.next-fri-layer
        (ps ! (length [cp] - 1 + j))
        (ds ! (length [cp] - 1 + j))
        (challenges ! j) =
        (ps ! (length [cp] + j),
         ds ! (length [cp] + j),
         ls ! (length [cp] + j))
  by blast
  have roots-simple:
    ∀ j < n. roots ! j = value (ms ! j)
  using roots-aligned by simp
  have layers-simple:
    ∀ j < n.
      p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j)
  using layers by simp
  show ?thesis
  by (intro exI[where x=roots] exI[where x=challenges])
    (use len-roots len-challenges tr st ext lookups roots-simple layers-simple in simp)
qed

```

lemma *fri-commit-with-replay-successor-data*:

assumes *send-counter*:

$$\bigwedge x s t. \text{Some } ((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x) s) \implies \text{counter } t = \text{counter } s$$

and *receiver-data*:

$$\bigwedge b s t. \text{Some } (b, t) \in \text{set-dist } (\text{execute } \text{receive-challenge } s) \implies s \leq t \wedge P\text{State } t = P\text{State } s \wedge P\text{Transcript } t = P\text{Transcript } s \wedge \text{counter } t = \text{Suc } (\text{counter } s) \wedge \text{fmlookup } (\text{HashMap } t) (\text{tag } (\text{counter } s) (P\text{State } s)) = \text{Some } b$$

and *create-counter*:

$\bigwedge xs\ m\ s\ t. \text{Some } (m, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs) s) \implies$
 $\text{counter } t = \text{counter } s$
and *outcome*:
 $\text{Some } ((ps', ds', ls', ms'), t) \in$
 $\text{set-dist } (\text{execute } (p.\text{fri-commit-with receive-challenge } n\ ps\ ds\ ls\ ms) s)$
and *lens*:
 $\text{length } ps = \text{length } ds$
 $\text{length } ds = \text{length } ls$
 $\text{length } ls = \text{length } ms$
and *nonempty*: $ps \neq []$
shows $\exists \text{roots challenges.}$
 $\text{length roots} = n \wedge$
 $\text{length challenges} = n \wedge$
 $P\text{Transcript } t = \text{rev roots } @ P\text{Transcript } s \wedge$
 $P\text{State } t = \text{foldl concat } (P\text{State } s) \text{ roots } \wedge$
 $s \leq t \wedge$
 $\text{counter } t = \text{counter } s + n \wedge$
 $(\forall i < \text{length roots.}$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{tag } (\text{counter } s + i)$
 $(\text{foldl concat } (P\text{State } s) (\text{take } (\text{Suc } i) \text{ roots}))) =$
 $\text{Some } (\text{challenges } ! i)) \wedge$
 $(\forall j < n.$
 $\text{roots } ! j = \text{value } (ms' ! (\text{length } ms - 1 + j))) \wedge$
 $(\forall j < n.$
 $p.\text{next-fri-layer}$
 $(ps' ! (\text{length } ps - 1 + j))$
 $(ds' ! (\text{length } ps - 1 + j))$
 $(\text{challenges } ! j) =$
 $(ps' ! (\text{length } ps + j),$
 $ds' ! (\text{length } ps + j),$
 $ls' ! (\text{length } ps + j)))$
using *outcome lens nonempty*
proof (*induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t*)
case 0
then show ?case
by (*auto simp: p.hash-ext-refl*)
next
case (Suc n)
from *Suc.prem1* **obtain** *s1 b s2 next-poly next-domain next-layer m s3* **where**
send-root:
 $\text{Some } ((), s1) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{value } (\text{last } ms))) s)$
and *rand*:
 $\text{Some } (b, s2) \in \text{set-dist } (\text{execute receive-challenge } s1)$
and *next-layer-eq*:
 $p.\text{next-fri-layer } (\text{last } ps) (\text{last } ds) b =$
 $(\text{next-poly}, \text{next-domain}, \text{next-layer})$
and *create-m*:
 $\text{Some } (m, s3) \in \text{set-dist } (\text{execute } (p.\text{create next-layer}) s2)$

```

and rest:
  Some ((ps', ds', ls', ms'), t) ∈
    set-dist (execute
      (p.fri-commit-with receive-challenge n
        (ps @ [next-poly])
        (ds @ [next-domain])
        (ls @ [next-layer])
        (ms @ [m])) s3)
    by (auto simp: Let-def split: prod.splits elim!: p.set-dist-bindE)
let ?root = value (last ms)
have lens-tail:
  length (ps @ [next-poly]) = length (ds @ [next-domain])
  length (ds @ [next-domain]) = length (ls @ [next-layer])
  length (ls @ [next-layer]) = length (ms @ [m])
  using Suc.premis(2-4) by simp-all
have nonempty-tail: ps @ [next-poly] ≠ []
  by simp
from Suc.IH[OF rest lens-tail nonempty-tail] obtain roots challenges where
  roots-len: length roots = n
  and challenges-len: length challenges = n
  and tr-t: PTranscript t = rev roots @ PTranscript s3
  and st-t: PState t = foldl concat (PState s3) roots
  and s3-t: s3 ≤ t
  and counter-t: counter t = counter s3 + n
  and lookups-tail:
    ∀ i < length roots.
      fmlookup (HashMap t)
        (tag (counter s3 + i)
          (foldl concat (PState s3) (take (Suc i) roots))) =
          Some (challenges ! i)
  and roots-tail:
    ∀ j < n.
      roots ! j =
        value (ms' ! (length (ms @ [m]) - 1 + j))
  and tail-layers:
    ∀ j < n.
      p.next-fri-layer
        (ps' ! (length (ps @ [next-poly]) - 1 + j))
        (ds' ! (length (ps @ [next-poly]) - 1 + j))
        (challenges ! j) =
        (ps' ! (length (ps @ [next-poly]) + j),
         ds' ! (length (ps @ [next-poly]) + j),
         ls' ! (length (ps @ [next-poly]) + j))
  by blast
have send-res:
  s1 = s(PState := concat (PState s) ?root,
    PTranscript := ?root # PTranscript s)
  using p.send-outcome[OF send-root] .
have rand-data:

```

```

s1 ≤ s2 ∧
PState s2 = PState s1 ∧
PTranscript s2 = PTranscript s1 ∧
counter s2 = Suc (counter s1) ∧
fmlookup (HashMap s2) (tag (counter s1) (PState s1)) = Some b
using receiver-data[OF rand] .
have create-state: PState s3 = PState s2
using create-preserves-state[OF create-m] .
have create-tr: PTranscript s3 = PTranscript s2
using create-preserves-transcript[OF create-m] .
have counter-s1: counter s1 = counter s
using send-counter[OF send-root] .
have counter-s2: counter s2 = Suc (counter s1)
using rand-data by simp
have counter-s3: counter s3 = counter s2
using create-counter[OF create-m] .
have counter-s3-s: counter s3 = Suc (counter s)
using counter-s1 counter-s2 counter-s3 by simp
have counter-final: counter t = counter s + Suc n
using counter-t counter-s3-s by simp
have s2-s3: s2 ≤ s3
using create-hash-extends[OF create-m] .
have s-s1: s ≤ s1
using send-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s-t: s ≤ t
using s-s1 rand-data s2-s3 s3-t by (meson p.hash-ext-trans)
have s2-t: s2 ≤ t
using s2-s3 s3-t by (rule p.hash-ext-trans)
have lookup-head:
fmlookup (HashMap t)
(tag (counter s) (concat (PState s) ?root)) = Some b
proof -
have lookup-s2:
fmlookup (HashMap s2) (tag (counter s1) (PState s1)) = Some b
using rand-data by simp
have fmlookup (HashMap t) (tag (counter s1) (PState s1)) = Some b
using p.hash-extension-lookup[OF lookup-s2 s2-t] .
then show ?thesis
using send-res counter-s1 by simp
qed
have tr-final:
PTranscript t = rev (?root # roots) @ PTranscript s
using tr-t create-tr rand-data send-res by simp
have st-final:
PState t = foldl concat (PState s) (?root # roots)
using st-t create-state rand-data send-res by simp
have lookups:
∀ i < length (?root # roots).
fmlookup (HashMap t)

```

```

      (tag (counter s + i)
        (foldl concat (PState s) (take (Suc i) (?root # roots)))) =
      Some ((b # challenges) ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length (?root # roots)
  show fmlookup (HashMap t)
    (tag (counter s + i)
      (foldl concat (PState s) (take (Suc i) (?root # roots)))) =
    Some ((b # challenges) ! i)
proof (cases i)
  case 0
  then show ?thesis
    using lookup-head by simp
next
  case (Suc j)
  then have j-bound: j < length roots
    using i-bound by simp
  have key-eq:
    foldl concat (PState s) (take (Suc (Suc j)) (?root # roots)) =
    foldl concat (PState s3) (take (Suc j) roots)
    using send-res rand-data create-state by simp
  show ?thesis
    using lookups-tail j-bound key-eq Suc counter-s3-s by simp
qed
qed
have prefix:
  take (length (ps @ [next-poly])) ps' = ps @ [next-poly] ∧
  take (length (ds @ [next-domain])) ds' = ds @ [next-domain] ∧
  take (length (ls @ [next-layer])) ls' = ls @ [next-layer] ∧
  take (length (ms @ [m])) ms' = ms @ [m]
  using p.fri-commit-with-preserves-prefix[OF rest] by simp
have head-ms: ms' ! (length ms - 1) = last ms
proof -
  have ms-ne: ms ≠ []
    using Suc.premis(2-5) by auto
  obtain q where len-ms: length ms = Suc q
    using ms-ne by (cases ms) auto
  have idx-lt: length ms - 1 < length (ms @ [m])
    using ms-ne by simp
  have ms'-nth: ms' ! (length ms - 1) =
    (ms @ [m]) ! (length ms - 1)
  proof -
    have (take (length (ms @ [m])) ms') ! (length ms - 1) =
      (ms @ [m]) ! (length ms - 1)
      using conjunct2[OF conjunct2[OF conjunct2[OF prefix]]] by simp
    then show ?thesis
      using idx-lt by simp
  qed

```

```

have append-nth:
  (ms @ [m]) ! (length ms - 1) = ms ! (length ms - 1)
  using len-ms by (simp add: nth-append)
have last-ms: last ms = ms ! (length ms - 1)
  using ms-ne by (simp add: last-conv-nth)
show ?thesis
  using ms'-nth append-nth last-ms by simp
qed
have head-ps: ps' ! (length ps - 1) = last ps
proof -
  obtain q where len-ps: length ps = Suc q
  using Suc.premis(5) by (cases ps) auto
  have idx-lt: length ps - 1 < length (ps @ [next-poly])
  using Suc.premis(5) by simp
  have ps'-nth: ps' ! (length ps - 1) =
    (ps @ [next-poly]) ! (length ps - 1)
  proof -
    have (take (length (ps @ [next-poly])) ps') ! (length ps - 1) =
      (ps @ [next-poly]) ! (length ps - 1)
    using conjunct1[OF prefix] by simp
    then show ?thesis
      using idx-lt by simp
  qed
  have append-nth:
    (ps @ [next-poly]) ! (length ps - 1) = ps ! (length ps - 1)
    using len-ps by (simp add: nth-append)
  have last-ps: last ps = ps ! (length ps - 1)
    using Suc.premis(5) by (simp add: last-conv-nth)
  show ?thesis
    using ps'-nth append-nth last-ps by simp
qed
have head-ds: ds' ! (length ps - 1) = last ds
proof -
  have ds-ne: ds ≠ []
  using Suc.premis(2,5) by auto
  obtain q where len-ds: length ds = Suc q
  using ds-ne by (cases ds) auto
  have idx-lt: length ps - 1 < length (ds @ [next-domain])
  using Suc.premis(2,5) by simp
  have ds'-nth: ds' ! (length ps - 1) =
    (ds @ [next-domain]) ! (length ps - 1)
  proof -
    have (take (length (ds @ [next-domain])) ds') ! (length ps - 1) =
      (ds @ [next-domain]) ! (length ps - 1)
    using conjunct1[OF conjunct2[OF prefix]] by simp
    then show ?thesis
      using idx-lt by simp
  qed
  have append-nth:

```



```

    (ds @ [next-domain]) ! (length ps - 1) = ds ! (length ds - 1)
    using Suc.prem(2) len-ds by (simp add: nth-append)
have last-ds: last ds = ds ! (length ds - 1)
    using ds-ne by (simp add: last-conv-nth)
show ?thesis
    using ds'-nth append-nth last-ds by simp
qed
have head-next-ps: ps' ! length ps = next-poly
proof -
    have pref: take (Suc (length ps)) ps' = ps @ [next-poly]
        using conjunct1[OF prefix] by simp
    have ps' ! length ps = (take (Suc (length ps)) ps') ! length ps
        by simp
    also have ... = next-poly
        using pref by simp
    finally show ?thesis .
qed
have head-next-ds: ds' ! length ps = next-domain
proof -
    have pref: take (Suc (length ds)) ds' = ds @ [next-domain]
        using conjunct1[OF conjunct2[OF prefix]] Suc.prem(2) by simp
    have ds' ! length ps = (take (Suc (length ds)) ds') ! length ps
        using Suc.prem(2) by simp
    also have ... = next-domain
        using pref Suc.prem(2) by simp
    finally show ?thesis .
qed
have head-next-ls: ls' ! length ps = next-layer
proof -
    have pref: take (Suc (length ls)) ls' = ls @ [next-layer]
        using conjunct2[OF conjunct2[OF prefix]] Suc.prem(2,3) by simp
    have ls' ! length ps = (take (Suc (length ls)) ls') ! length ps
        using Suc.prem(2,3) by simp
    also have ... = next-layer
        using pref Suc.prem(2,3) by simp
    finally show ?thesis .
qed
have layers:
  ∀ j < Suc n.
    p.next-fri-layer
      (ps' ! (length ps - 1 + j))
      (ds' ! (length ps - 1 + j))
      ((b # challenges) ! j) =
      (ps' ! (length ps + j),
       ds' ! (length ps + j),
       ls' ! (length ps + j))
proof (intro allI impI)
  fix j
  assume j-bound: j < Suc n

```

```

show  $p.\text{next-fri-layer}$ 
  ( $ps' ! (\text{length } ps - 1 + j)$ )
  ( $ds' ! (\text{length } ps - 1 + j)$ )
  ( $(b \# \text{challenges}) ! j =$ 
    ( $ps' ! (\text{length } ps + j)$ ,
     ( $ds' ! (\text{length } ps + j)$ ,
     ( $ls' ! (\text{length } ps + j)$ ))
proof (cases  $j$ )
  case 0
  then show ?thesis
    using next-layer-eq head-ps head-ds head-next-ps head-next-ds head-next-ls
    by simp
next
  case (Suc  $k$ )
  then have  $k\text{-bound}$ :  $k < n$ 
    using  $j\text{-bound}$  by simp
  have  $\text{idx-current}$ :
     $\text{length } ps - 1 + \text{Suc } k = \text{length } ps + k$ 
  proof –
    obtain  $q$  where  $\text{length } ps = \text{Suc } q$ 
    using Suc.prem5 by (cases  $ps$ ) auto
    then show ?thesis by simp
  qed
  have  $\text{tail-k}$ :
     $p.\text{next-fri-layer}$ 
    ( $ps' ! (\text{length } ps + k)$ )
    ( $ds' ! (\text{length } ps + k)$ )
    ( $\text{challenges} ! k =$ 
      ( $ps' ! \text{Suc } (\text{length } ps + k)$ ,
       ( $ds' ! \text{Suc } (\text{length } ps + k)$ ,
       ( $ls' ! \text{Suc } (\text{length } ps + k)$ ))
    using tail-layers  $k\text{-bound}$  by simp
  show ?thesis
    unfolding Suc
    apply (subst  $\text{idx-current}$ )
    apply (subst  $\text{idx-current}$ )
    using  $\text{tail-k}$ 
    apply simp
    done
  qed
qed
have  $\text{roots-aligned}$ :
   $\forall j < \text{Suc } n.$ 
  ( $(\text{?root} \# \text{roots}) ! j = \text{value } (ms' ! (\text{length } ms - 1 + j))$ )
proof (intro allI impI)
  fix  $j$ 
  assume  $j\text{-bound}$ :  $j < \text{Suc } n$ 
  show ( $(\text{?root} \# \text{roots}) ! j = \text{value } (ms' ! (\text{length } ms - 1 + j))$ )
  proof (cases  $j$ )

```

```

case 0
then show ?thesis
  using head-ms by simp
next
case (Suc k)
then have k-bound:  $k < n$ 
  using j-bound by simp
have idx-current:
   $\text{length } ms - 1 + \text{Suc } k = \text{length } ms + k$ 
proof -
  obtain q where  $\text{length } ms = \text{Suc } q$ 
  using Suc.premis(2-5) by (cases ms) auto
  then show ?thesis by simp
qed
have tail-k:
   $\text{roots} ! k = \text{value } (ms' ! (\text{length } ms + k))$ 
  using roots-tail k-bound by simp
show ?thesis
  unfolding Suc
  apply (subst idx-current)
  using tail-k
  apply simp
  done
qed
qed
show ?case
proof (rule exI[where  $x = ?\text{root} \# \text{roots}$ ])
show  $\exists$  challenges.
   $\text{length } (? \text{root} \# \text{roots}) = \text{Suc } n \wedge$ 
   $\text{length } \text{challenges} = \text{Suc } n \wedge$ 
   $P\text{Transcript } t = \text{rev } (? \text{root} \# \text{roots}) @ P\text{Transcript } s \wedge$ 
   $P\text{State } t = \text{foldl concat } (P\text{State } s) (? \text{root} \# \text{roots}) \wedge$ 
   $s \leq t \wedge$ 
   $\text{counter } t = \text{counter } s + \text{Suc } n \wedge$ 
   $(\forall i < \text{length } (? \text{root} \# \text{roots}).$ 
     $\text{fmlookup } (\text{HashMap } t)$ 
     $(\text{tag } (\text{counter } s + i)$ 
       $(\text{foldl concat } (P\text{State } s) (\text{take } (\text{Suc } i) (? \text{root} \# \text{roots})))) =$ 
       $\text{Some } (\text{challenges} ! i)) \wedge$ 
   $(\forall j < \text{Suc } n.$ 
     $(? \text{root} \# \text{roots}) ! j = \text{value } (ms' ! (\text{length } ms - 1 + j))) \wedge$ 
   $(\forall j < \text{Suc } n.$ 
     $p.\text{next-fri-layer}$ 
     $(ps' ! (\text{length } ps - 1 + j))$ 
     $(ds' ! (\text{length } ps - 1 + j))$ 
     $(\text{challenges} ! j) =$ 
     $(ps' ! (\text{length } ps + j),$ 
       $ds' ! (\text{length } ps + j),$ 
       $ls' ! (\text{length } ps + j)))$ 

```

```

proof (rule exI[where  $x=b \# challenges$ ])
  show  $length \ (?\text{root} \# roots) = \text{Suc } n \wedge$ 
     $length \ (b \# challenges) = \text{Suc } n \wedge$ 
     $P\text{Transcript } t = \text{rev } (?\text{root} \# roots) @ P\text{Transcript } s \wedge$ 
     $P\text{State } t = \text{foldl concat } (P\text{State } s) \ (?\text{root} \# roots) \wedge$ 
     $s \leq t \wedge$ 
     $counter \ t = counter \ s + \text{Suc } n \wedge$ 
     $(\forall i < length \ (?\text{root} \# roots).$ 
       $\text{fmlookup } (\text{HashMap } t)$ 
         $(\text{tag } (counter \ s + i)$ 
           $(\text{foldl concat } (P\text{State } s) \ (\text{take } (\text{Suc } i) \ (?\text{root} \# roots)))) =$ 
           $\text{Some } ((b \# challenges) ! i)) \wedge$ 
     $(\forall j < \text{Suc } n.$ 
       $(?\text{root} \# roots) ! j = \text{value } (ms' ! (length \ ms - 1 + j))) \wedge$ 
     $(\forall j < \text{Suc } n.$ 
       $p.\text{next-fri-layer}$ 
         $(ps' ! (length \ ps - 1 + j))$ 
         $(ds' ! (length \ ps - 1 + j))$ 
         $((b \# challenges) ! j) =$ 
         $(ps' ! (length \ ps + j),$ 
           $ds' ! (length \ ps + j),$ 
           $ls' ! (length \ ps + j)))$ 
      using  $roots\text{-len } challenges\text{-len } tr\text{-final } st\text{-final } s\text{-t } counter\text{-final}$ 
         $lookups \ roots\text{-aligned } layers$ 
      by simp
    qed
  qed
qed

lemma fri-commit-with-preserves-counter:
  fixes  $counter :: 'f \text{ protocol-channel} \Rightarrow nat$ 
  assumes send-counter:
     $\bigwedge x \ s \ t. \text{Some } ((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x) \ s) \implies$ 
     $counter \ t = counter \ s$ 
  and receiver-counter:
     $\bigwedge b \ s \ t. \text{Some } (b, t) \in \text{set-dist } (\text{execute } \text{receive-challenge } s) \implies$ 
     $counter \ t = counter \ s$ 
  and create-counter:
     $\bigwedge xs \ m \ s \ t. \text{Some } (m, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs) \ s) \implies$ 
     $counter \ t = counter \ s$ 
  and outcome:
     $\text{Some } ((ps', ds', ls', ms'), t) \in$ 
     $\text{set-dist } (\text{execute } (p.\text{fri-commit-with receive-challenge } n \ ps \ ds \ ls \ ms) \ s)$ 
  shows  $counter \ t = counter \ s$ 
  using outcome
proof (induction n arbitrary: ps ds ls ms s ps' ds' ls' ms' t)
  case 0
  then show ?case by simp

```

```

next
case (Suc n)
from Suc.premis obtain s1 b s2 next-poly next-domain next-layer m s3 where
  send-root: Some ((), s1) ∈ set-dist (execute (p.send (value (last ms))) s)
  and rand: Some (b, s2) ∈ set-dist (execute receive-challenge s1)
  and next-layer-eq:
    p.next-fri-layer (last ps) (last ds) b =
      (next-poly, next-domain, next-layer)
  and create-m:
    Some (m, s3) ∈ set-dist (execute (p.create next-layer) s2)
  and rest:
    Some ((ps', ds', ls', ms'), t) ∈
      set-dist (execute
        (p.fri-commit-with receive-challenge n
          (ps @ [next-poly])
          (ds @ [next-domain])
          (ls @ [next-layer])
          (ms @ [m])) s3)
    by (auto simp: Let-def split: prod.splits elim!: p.set-dist-bindE)
have c-s1: counter s1 = counter s
  by (rule send-counter[OF send-root])
have c-s2: counter s2 = counter s1
  by (rule receiver-counter[OF rand])
have c-s3: counter s3 = counter s2
  by (rule create-counter[OF create-m])
have c-t: counter t = counter s3
  by (rule Suc.IH[OF rest])
show ?case
  using c-s1 c-s2 c-s3 c-t by simp
qed

```

lemma trace-fri-commit-preserves-alpha-counter:

```

assumes outcome:
  Some ((ps, ds, ls, ms), t) ∈
    set-dist (execute (p.trace-fri-commit n ps0 ds0 ls0 ms0) s)
shows PAlphaCounter t = PAlphaCounter s

```

proof –

```

have send-counter:
   $\bigwedge x s t. \text{Some } ((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x) s) \implies$ 
    PAlphaCounter t = PAlphaCounter s
  using p.send-outcome by fastforce
have receiver-counter:
   $\bigwedge b s t. \text{Some } (b, t) \in \text{set-dist } (\text{execute } p.\text{receive-trace-fri-challenge } s) \implies$ 
    PAlphaCounter t = PAlphaCounter s
  using p.receive-trace-fri-challenge-counter-outcome by fastforce
have create-counter:
   $\bigwedge xs m s t. \text{Some } (m, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs) s) \implies$ 
    PAlphaCounter t = PAlphaCounter s
  by (rule p.create-preserves-channel(5))

```

show *?thesis*
unfolding *p.trace-fri-commit-def*
by (rule *fri-commit-with-preserves-counter* [
 where *counter=PAAlphaCounter* **and** *receive-challenge=p.receive-trace-fri-challenge*,
 OF send-counter receiver-counter create-counter
 outcome[unfolded p.trace-fri-commit-def]])
qed

lemma *trace-fri-commit-preserves-composition-counter*:
assumes *outcome*:
 Some ((ps, ds, ls, ms), t) ∈
 set-dist (execute (p.trace-fri-commit n ps0 ds0 ls0 ms0) s)
shows *PCompositionFriCounter t = PCompositionFriCounter s*
proof –
have *send-counter*:
 $\bigwedge x\ s\ t. \text{Some } ((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x)\ s) \implies$
 PCompositionFriCounter t = PCompositionFriCounter s
using *p.send-outcome* **by** *fastforce*
have *receiver-counter*:
 $\bigwedge b\ s\ t. \text{Some } (b, t) \in \text{set-dist } (\text{execute } p.\text{receive-trace-fri-challenge } s) \implies$
 PCompositionFriCounter t = PCompositionFriCounter s
using *p.receive-trace-fri-challenge-counter-outcome* **by** *fastforce*
have *create-counter*:
 $\bigwedge x\ m\ s\ t. \text{Some } (m, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs)\ s) \implies$
 PCompositionFriCounter t = PCompositionFriCounter s
by (rule *p.create-preserves-channel(4)*)
show *?thesis*
unfolding *p.trace-fri-commit-def*
by (rule *fri-commit-with-preserves-counter* [
 where *counter=PCompositionFriCounter* **and** *receive-challenge=p.receive-trace-fri-challenge*,
 OF send-counter receiver-counter create-counter
 outcome[unfolded p.trace-fri-commit-def]])
qed

lemma *fri-commit-preserves-query-counter*:
assumes *outcome*:
 Some ((ps, ds, ls, ms), t) ∈
 set-dist (execute (p.fri-commit n ps0 ds0 ls0 ms0) s)
shows *PQueryCounter t = PQueryCounter s*
proof –
have *send-counter*:
 $\bigwedge x\ s\ t. \text{Some } ((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x)\ s) \implies$
 PQueryCounter t = PQueryCounter s
using *p.send-outcome* **by** *fastforce*
have *receiver-counter*:
 $\bigwedge b\ s\ t. \text{Some } (b, t) \in \text{set-dist } (\text{execute } p.\text{receive-random-field-element } s) \implies$
 PQueryCounter t = PQueryCounter s
using *p.receive-random-field-element-counter-outcome* **by** *fastforce*
have *create-counter*:

$\bigwedge xs\ m\ s\ t. \text{Some } (m, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs)\ s) \implies$
 $P\text{QueryCounter } t = P\text{QueryCounter } s$
by (rule *p.create-preserves-channel*(6))
have *outcome'*:
 $\text{Some } ((ps, ds, ls, ms), t) \in$
 $\text{set-dist } (\text{execute } (p.\text{fri-commit-with } p.\text{receive-random-field-element } n\ ps0\ ds0\ ls0\ ms0)\ s)$
using *outcome* *p.fri-commit-with-receive-random-field-element* **by** *simp*
show *?thesis*
by (rule *fri-commit-with-preserves-counter*[
where *counter*=*PQueryCounter* **and** *receive-challenge*=*p.receive-random-field-element*,
OF *send-counter* *receiver-counter* *create-counter* *outcome'*])
qed

lemma *trace-fri-commit-preserves-query-counter*:

assumes *outcome*:
 $\text{Some } ((ps, ds, ls, ms), t) \in$
 $\text{set-dist } (\text{execute } (p.\text{trace-fri-commit } n\ ps0\ ds0\ ls0\ ms0)\ s)$
shows $P\text{QueryCounter } t = P\text{QueryCounter } s$
proof –
have *send-counter*:
 $\bigwedge x\ s\ t. \text{Some } ((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x)\ s) \implies$
 $P\text{QueryCounter } t = P\text{QueryCounter } s$
using *p.send-outcome* **by** *fastforce*
have *receiver-counter*:
 $\bigwedge b\ s\ t. \text{Some } (b, t) \in \text{set-dist } (\text{execute } p.\text{receive-trace-fri-challenge } s) \implies$
 $P\text{QueryCounter } t = P\text{QueryCounter } s$
using *p.receive-trace-fri-challenge-counter-outcome* **by** *fastforce*
have *create-counter*:
 $\bigwedge xs\ m\ s\ t. \text{Some } (m, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs)\ s) \implies$
 $P\text{QueryCounter } t = P\text{QueryCounter } s$
by (rule *p.create-preserves-channel*(6))
show *?thesis*
unfolding *p.trace-fri-commit-def*
by (rule *fri-commit-with-preserves-counter*[
where *counter*=*PQueryCounter* **and** *receive-challenge*=*p.receive-trace-fri-challenge*,
OF *send-counter* *receiver-counter* *create-counter*
outcome[*unfolded* *p.trace-fri-commit-def*]])
qed

lemma *composition-fri-commit-preserves-query-counter*:

assumes *outcome*:
 $\text{Some } ((ps, ds, ls, ms), t) \in$
 $\text{set-dist } (\text{execute } (p.\text{composition-fri-commit } n\ ps0\ ds0\ ls0\ ms0)\ s)$
shows $P\text{QueryCounter } t = P\text{QueryCounter } s$
proof –
have *send-counter*:
 $\bigwedge x\ s\ t. \text{Some } ((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x)\ s) \implies$
 $P\text{QueryCounter } t = P\text{QueryCounter } s$

```

using p.send-outcome by fastforce
have receiver-counter:
   $\bigwedge b \ s \ t. \text{Some } (b, t) \in \text{set-dist } (\text{execute } p.\text{receive-composition-fri-challenge } s) \implies$ 
     $P\text{QueryCounter } t = P\text{QueryCounter } s$ 
  using p.receive-composition-fri-challenge-counter-outcome by fastforce
have create-counter:
   $\bigwedge x \ s \ m \ s \ t. \text{Some } (m, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs) \ s) \implies$ 
     $P\text{QueryCounter } t = P\text{QueryCounter } s$ 
  by (rule p.create-preserves-channel(6))
show ?thesis
unfolding p.composition-fri-commit-def
by (rule fri-commit-with-preserves-counter[
  where counter=PQueryCounter and receive-challenge=p.receive-composition-fri-challenge,
  OF send-counter receiver-counter create-counter
  outcome[unfolded p.composition-fri-commit-def]])
qed

lemma trace-fri-commit-initial-replay-successor-data:
assumes outcome:
   $\text{Some } ((ps, ds, ls, ms), t) \in$ 
     $\text{set-dist } (\text{execute } (p.\text{trace-fri-commit } n \ [cp] \ [d] \ [l] \ [m]) \ s)$ 
shows  $\exists \text{roots challenges.}$ 
   $\text{length roots} = n \wedge$ 
   $\text{length challenges} = n \wedge$ 
   $P\text{Transcript } t = \text{rev roots} \ @ \ P\text{Transcript } s \wedge$ 
   $P\text{State } t = \text{foldl concat } (P\text{State } s) \ \text{roots} \wedge$ 
   $s \leq t \wedge$ 
   $(\forall i < \text{length roots.}$ 
     $\text{fmlookup } (\text{HashMap } t)$ 
     $(\text{TraceFriChallenge } (P\text{TraceFriCounter } s + i)$ 
       $(\text{foldl concat } (P\text{State } s) \ (\text{take } (\text{Suc } i) \ \text{roots}))) =$ 
       $\text{Some } (\text{challenges } ! \ i)) \wedge$ 
   $(\forall j < n. \ \text{roots } ! \ j = \text{value } (ms \ ! \ j)) \wedge$ 
   $(\forall j < n.$ 
     $p.\text{next-fri-layer}$ 
     $(ps \ ! \ j)$ 
     $(ds \ ! \ j)$ 
     $(\text{challenges } ! \ j) =$ 
     $(ps \ ! \ \text{Suc } j, ds \ ! \ \text{Suc } j, ls \ ! \ \text{Suc } j))$ 
proof –
have send-counter:
   $\bigwedge x \ s \ t. \text{Some } ((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x) \ s) \implies$ 
     $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s$ 
proof –
fix  $x \ s \ t$ 
assume send:  $\text{Some } ((), t) \in \text{set-dist } (\text{execute } (p.\text{send } x) \ s)$ 
show  $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s$ 
  using p.send-outcome[OF send] by simp

```


qed
have *receiver*:
 $\bigwedge b \ s \ t. \text{Some } (b, t) \in$
 $\text{set-dist } (\text{execute } p.\text{receive-trace-fri-challenge } s) \implies$
 $s \leq t \wedge$
 $P\text{State } t = P\text{State } s \wedge$
 $P\text{Transcript } t = P\text{Transcript } s \wedge$
 $P\text{TraceFriCounter } t = \text{Suc } (P\text{TraceFriCounter } s) \wedge$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{TraceFriChallenge } (P\text{TraceFriCounter } s) (P\text{State } s)) = \text{Some } b$
proof –
fix $b \ s \ t$
assume *rand*:
 $\text{Some } (b, t) \in \text{set-dist } (\text{execute } p.\text{receive-trace-fri-challenge } s)$
show $s \leq t \wedge$
 $P\text{State } t = P\text{State } s \wedge$
 $P\text{Transcript } t = P\text{Transcript } s \wedge$
 $P\text{TraceFriCounter } t = \text{Suc } (P\text{TraceFriCounter } s) \wedge$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{TraceFriChallenge } (P\text{TraceFriCounter } s) (P\text{State } s)) = \text{Some } b$
using $p.\text{receive-trace-fri-challenge-outcome}[OF \ \text{rand}]$
 $p.\text{receive-trace-fri-challenge-counter-outcome}[OF \ \text{rand}]$
by *simp*
qed
have *create-counter*:
 $\bigwedge xs \ m \ s \ t. \text{Some } (m, t) \in \text{set-dist } (\text{execute } (p.\text{create } xs) \ s) \implies$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s$
by $(\text{rule } p.\text{create-preserves-channel}(3))$
have *data*:
 $\exists \text{roots challenges.}$
 $\text{length roots} = n \wedge$
 $\text{length challenges} = n \wedge$
 $P\text{Transcript } t = \text{rev roots } @ \ P\text{Transcript } s \wedge$
 $P\text{State } t = \text{foldl concat } (P\text{State } s) \ \text{roots} \wedge$
 $s \leq t \wedge$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s + n \wedge$
 $(\forall i < \text{length roots.}$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{TraceFriChallenge } (P\text{TraceFriCounter } s + i)$
 $(\text{foldl concat } (P\text{State } s) (\text{take } (\text{Suc } i) \ \text{roots}))) =$
 $\text{Some } (\text{challenges } ! \ i)) \wedge$
 $(\forall j < n. \text{roots } ! \ j = \text{value } (ms \ ! \ (\text{length } [m] - 1 + j))) \wedge$
 $(\forall j < n.$
 $p.\text{next-fri-layer}$
 $(ps \ ! \ (\text{length } [cp] - 1 + j))$
 $(ds \ ! \ (\text{length } [cp] - 1 + j))$
 $(\text{challenges } ! \ j) =$
 $(ps \ ! \ (\text{length } [cp] + j),$
 $ds \ ! \ (\text{length } [cp] + j),$

```

      ls ! (length [cp] + j)))
unfolding p.trace-fri-commit-def
by (rule fri-commit-with-replay-successor-data
    [where counter=PTraceFriCounter and tag=TraceFriChallenge,
      OF send-counter receiver create-counter
      outcome[unfolded p.trace-fri-commit-def]]) simp-all
then show ?thesis by auto
qed

```

lemma composition-fri-commit-initial-replay-successor-data:

```

assumes outcome:
  Some ((ps, ds, ls, ms), t) ∈
    set-dist (execute
      (p.composition-fri-commit n [cp] [d] [l] [m]) s)
shows ∃ roots challenges.
  length roots = n ∧
  length challenges = n ∧
  PTranscript t = rev roots @ PTranscript s ∧
  PState t = foldl concat (PState s) roots ∧
  s ≤ t ∧
  (∀ i < length roots.
    fmlookup (HashMap t)
      (CompositionFriChallenge (PCompositionFriCounter s + i)
        (foldl concat (PState s) (take (Suc i) roots))) =
      Some (challenges ! i)) ∧
  (∀ j < n. roots ! j = value (ms ! j)) ∧
  (∀ j < n.
    p.next-fri-layer
      (ps ! j)
      (ds ! j)
      (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j))

```

proof –

```

have send-counter:
  ∧ x s t. Some ((), t) ∈ set-dist (execute (p.send x) s) ⇒
    PCompositionFriCounter t = PCompositionFriCounter s

```

proof –

```

fix x s t
assume send: Some ((), t) ∈ set-dist (execute (p.send x) s)
show PCompositionFriCounter t = PCompositionFriCounter s
using p.send-outcome[OF send] by simp

```

qed

have receiver:

```

  ∧ b s t. Some (b, t) ∈
    set-dist (execute p.receive-composition-fri-challenge s) ⇒
    s ≤ t ∧
    PState t = PState s ∧
    PTranscript t = PTranscript s ∧
    PCompositionFriCounter t = Suc (PCompositionFriCounter s) ∧

```

```

    fmlookup (HashMap t)
    (CompositionFriChallenge (PCompositionFriCounter s) (PState s)) = Some b
proof –
  fix b s t
  assume rand:
    Some (b, t) ∈
      set-dist (execute p.receive-composition-fri-challenge s)
  show s ≤ t ∧
    PState t = PState s ∧
    PTranscript t = PTranscript s ∧
    PCompositionFriCounter t = Suc (PCompositionFriCounter s) ∧
    fmlookup (HashMap t)
      (CompositionFriChallenge (PCompositionFriCounter s) (PState s)) = Some b
  using p.receive-composition-fri-challenge-outcome[OF rand]
    p.receive-composition-fri-challenge-counter-outcome[OF rand]
  by simp
qed
have create-counter:
  ∧xs m s t. Some (m, t) ∈ set-dist (execute (p.create xs) s) ⇒
    PCompositionFriCounter t = PCompositionFriCounter s
  by (rule p.create-preserves-channel(4))
have data:
  ∃ roots challenges.
    length roots = n ∧
    length challenges = n ∧
    PTranscript t = rev roots @ PTranscript s ∧
    PState t = foldl concat (PState s) roots ∧
    s ≤ t ∧
    PCompositionFriCounter t = PCompositionFriCounter s + n ∧
    (∀ i < length roots.
      fmlookup (HashMap t)
        (CompositionFriChallenge (PCompositionFriCounter s + i)
          (foldl concat (PState s) (take (Suc i) roots))) =
        Some (challenges ! i)) ∧
    (∀ j < n. roots ! j = value (ms ! (length [m] - 1 + j))) ∧
    (∀ j < n.
      p.next-fri-layer
        (ps ! (length [cp] - 1 + j))
        (ds ! (length [cp] - 1 + j))
        (challenges ! j) =
        (ps ! (length [cp] + j),
         ds ! (length [cp] + j),
         ls ! (length [cp] + j)))
  unfolding p.composition-fri-commit-def
  by (rule fri-commit-with-replay-successor-data
    [where counter=PCompositionFriCounter and tag=CompositionFriChallenge,
      OF send-counter receiver create-counter
      outcome[unfolded p.composition-fri-commit-def]]) simp-all
then show ?thesis by auto

```

qed

lemma *fri-commit-initial-heads*:

assumes *outcome*:

Some $((ps, ds, ls, ms), t) \in$
set-dist (*execute*
 $(p.fri-commit\ n\ [cp]\ [d]\ [l]\ [m])\ s)$

shows

$ps ! 0 = cp$
 $ds ! 0 = d$
 $ls ! 0 = l$
 $ms ! 0 = m$

proof –

have *lens*:

$length\ ps = Suc\ n$
 $length\ ds = Suc\ n$
 $length\ ls = Suc\ n$
 $length\ ms = Suc\ n$

using *p.fri-commit-lengths*[*OF outcome*] **by** *simp-all*

have *prefixes*:

$take\ (length\ [cp])\ ps = [cp]$
 $take\ (length\ [d])\ ds = [d]$
 $take\ (length\ [l])\ ls = [l]$
 $take\ (length\ [m])\ ms = [m]$

using *p.fri-commit-preserves-prefix*[*OF outcome*] **by** *simp-all*

show $ps ! 0 = cp$

using *lens prefixes* **by** (*cases ps*) *auto*

show $ds ! 0 = d$

using *lens prefixes* **by** (*cases ds*) *auto*

show $ls ! 0 = l$

using *lens prefixes* **by** (*cases ls*) *auto*

show $ms ! 0 = m$

using *lens prefixes* **by** (*cases ms*) *auto*

qed

lemma *trace-fri-commit-initial-heads*:

assumes *outcome*:

Some $((ps, ds, ls, ms), t) \in$
set-dist (*execute*
 $(p.trace-fri-commit\ n\ [cp]\ [d]\ [l]\ [m])\ s)$

shows

$ps ! 0 = cp$
 $ds ! 0 = d$
 $ls ! 0 = l$
 $ms ! 0 = m$

proof –

have *lens*:

$length\ ps = Suc\ n$
 $length\ ds = Suc\ n$

```

length ls = Suc n
length ms = Suc n
using p.trace-fri-commit-lengths[OF outcome] by simp-all
have prefixes:
  take (length [cp]) ps = [cp]
  take (length [d]) ds = [d]
  take (length [l]) ls = [l]
  take (length [m]) ms = [m]
  using p.trace-fri-commit-preserves-prefix[OF outcome] by simp-all
show ps ! 0 = cp
  using lens prefixes by (cases ps) auto
show ds ! 0 = d
  using lens prefixes by (cases ds) auto
show ls ! 0 = l
  using lens prefixes by (cases ls) auto
show ms ! 0 = m
  using lens prefixes by (cases ms) auto
qed

lemma composition-fri-commit-initial-heads:
assumes outcome:
  Some ((ps, ds, ls, ms), t) ∈
    set-dist (execute
      (p.composition-fri-commit n [cp] [d] [l] [m]) s)
shows
  ps ! 0 = cp
  ds ! 0 = d
  ls ! 0 = l
  ms ! 0 = m
proof –
have lens:
  length ps = Suc n
  length ds = Suc n
  length ls = Suc n
  length ms = Suc n
  using p.composition-fri-commit-lengths[OF outcome] by simp-all
have prefixes:
  take (length [cp]) ps = [cp]
  take (length [d]) ds = [d]
  take (length [l]) ls = [l]
  take (length [m]) ms = [m]
  using p.composition-fri-commit-preserves-prefix[OF outcome] by simp-all
show ps ! 0 = cp
  using lens prefixes by (cases ps) auto
show ds ! 0 = d
  using lens prefixes by (cases ds) auto
show ls ! 0 = l
  using lens prefixes by (cases ls) auto
show ms ! 0 = m

```

using *lens prefixes* by (cases *ms*) auto
qed

definition *fri-commit-outcome* where

fri-commit-outcome *n cp d l m ps ds ls ms s t* $\longleftrightarrow$
Some ((*ps*, *ds*, *ls*, *ms*), *t*) $\in$
set-dist (*execute* (*p.fri-commit* *n [cp] [d] [l] [m]*) *s*) $\vee$
Some ((*ps*, *ds*, *ls*, *ms*), *t*) $\in$
set-dist (*execute* (*p.trace-fri-commit* *n [cp] [d] [l] [m]*) *s*) $\vee$
Some ((*ps*, *ds*, *ls*, *ms*), *t*) $\in$
set-dist (*execute* (*p.composition-fri-commit* *n [cp] [d] [l] [m]*) *s*)

lemma *fri-commit-outcomeI*:

assumes *Some* ((*ps*, *ds*, *ls*, *ms*), *t*) $\in$
set-dist (*execute* (*p.fri-commit* *n [cp] [d] [l] [m]*) *s*)
shows *fri-commit-outcome* *n cp d l m ps ds ls ms s t*
using *assms* **unfolding** *fri-commit-outcome-def* **by** *blast*

lemma *trace-fri-commit-outcomeI*:

assumes *Some* ((*ps*, *ds*, *ls*, *ms*), *t*) $\in$
set-dist (*execute* (*p.trace-fri-commit* *n [cp] [d] [l] [m]*) *s*)
shows *fri-commit-outcome* *n cp d l m ps ds ls ms s t*
using *assms* **unfolding** *fri-commit-outcome-def* **by** *blast*

lemma *composition-fri-commit-outcomeI*:

assumes *Some* ((*ps*, *ds*, *ls*, *ms*), *t*) $\in$
set-dist (*execute* (*p.composition-fri-commit* *n [cp] [d] [l] [m]*) *s*)
shows *fri-commit-outcome* *n cp d l m ps ds ls ms s t*
using *assms* **unfolding** *fri-commit-outcome-def* **by** *blast*

lemma *fri-commit-outcome-initial-heads*:

assumes *fri-commit-outcome* *n cp d l m ps ds ls ms s t*
shows
ps ! 0 = *cp*
ds ! 0 = *d*
ls ! 0 = *l*
ms ! 0 = *m*
using *assms*
unfolding *fri-commit-outcome-def*
by (*auto dest: fri-commit-initial-heads trace-fri-commit-initial-heads*
composition-fri-commit-initial-heads)

lemma *decommit-on-query-step-transcript-extends*:

assumes *step-in*: *m* $\in$ *set* (*p.decommit-on-query idx f-merkle*)
and *outcome*: *Some* (*x*, *t*) $\in$ *set-dist* (*execute m s*)
shows *transcript-extends t s*

proof —

from *step-in* **obtain** *i* **where**

m-eq:

```

    m = do {
      p.send (p.f-eval ! i);
      mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)
    }
  unfolding p.decommit-on-query-def by auto
from outcome[unfolded m-eq] obtain u where
  send-leaf: Some ((), u) ∈ set-dist (execute (p.send (p.f-eval ! i)) s)
and send-path:
  Some (x, t) ∈
    set-dist (execute
      (mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)) u)
by (auto elim!: p.set-dist-bindE)
have s-u: transcript-extends u s
  using send-transcript-extends[OF send-leaf] .
have x-unit: x = ()
  by (cases x) simp
have path-out:
  Some ((), t) ∈
    set-dist (execute
      (mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)) u)
  using send-path x-unit by simp
have u-t: transcript-extends t u
  by (rule mfold2-send-transcript-extends[OF path-out])
show ?thesis
  using s-u u-t by (rule transcript-extends-trans)
qed

```

lemma *decommit-on-query-step-hash-extends:*

```

  assumes step-in: m ∈ set (p.decommit-on-query idx f-merkle)
  and outcome: Some (x, t) ∈ set-dist (execute m s)
  shows s ≤ t
proof -
  from step-in obtain i where
    m-eq:
      m = do {
        p.send (p.f-eval ! i);
        mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)
      }
  unfolding p.decommit-on-query-def by auto
from outcome[unfolded m-eq] obtain u where
  send-leaf: Some ((), u) ∈ set-dist (execute (p.send (p.f-eval ! i)) s)
and send-path:
  Some (x, t) ∈
    set-dist (execute
      (mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)) u)
  by (auto elim!: p.set-dist-bindE)
have s-u: s ≤ u
  using send-hash-extends[OF send-leaf] .
have x-unit: x = ()

```

by (cases x) simp
 have path-out:
 Some $((), t) \in$
 set-dist (execute
 (mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)) u)
 using send-path x-unit by simp
 have u-t: $u \leq t$
 using mfold2-send-hash-extends[OF path-out] .
 show ?thesis
 using s-u u-t by (rule p.hash-ext-trans)
 qed

lemma decommit-on-query-step-preserves-query-counter:

assumes step-in: $m \in \text{set } (p.\text{decommit-on-query } \text{idx } f\text{-merkle})$
 and outcome: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s)$
 shows $P\text{QueryCounter } t = P\text{QueryCounter } s$

proof –

from step-in obtain i where

m-eq:
 $m = \text{do } \{$
 $p.\text{send } (p.f\text{-eval } ! \ i);$
 $\text{mfold2 } p.\text{send } (\text{get-authentication-path } (\text{length } p.f\text{-eval}) \ i \ f\text{-merkle})$
 $\}$

unfolding p.decommit-on-query-def by auto

from outcome[unfolded m-eq] obtain u where

send-leaf: $\text{Some } ((), u) \in \text{set-dist } (\text{execute } (p.\text{send } (p.f\text{-eval } ! \ i)) \ s)$

and send-path:

Some $(x, t) \in$
 set-dist (execute
 (mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)) u)
 by (auto elim!: p.set-dist-bindE)

have x-unit: $x = ()$

by (cases x) simp

have path-out:

Some $((), t) \in$
 set-dist (execute
 (mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)) u)
 using send-path x-unit by simp

have q-u: $P\text{QueryCounter } u = P\text{QueryCounter } s$

using p.send-outcome[OF send-leaf] by simp

have q-t: $P\text{QueryCounter } t = P\text{QueryCounter } u$

by (rule mfold2-send-preserves-query-counter[OF path-out])

show ?thesis

using q-u q-t by simp

qed

lemma decommit-on-fri-layers-step-transcript-extends:

assumes step-in: $m \in \text{set } (p.\text{decommit-on-fri-layers } fs)$

and outcome: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (m \ \text{idx}) \ s)$

shows *transcript-extends* t s

proof –

from *step-in* **obtain** l *tree* **where**

pair-in: $(l, \text{tree}) \in \text{set } fs$

and *m-eq*:

$m = (\lambda idx. \text{do } \{$

$\text{let } len = \text{length } l;$

$\text{let } idx' = idx \bmod len;$

$\text{let } sidx = (idx' + (len \text{ div } 2)) \bmod len;$

$p.\text{send } (l ! idx');$

$\text{mfold2 } p.\text{send } (\text{get-authentication-path } len \ idx' \ \text{tree});$

$p.\text{send } (l ! sidx);$

$\text{mfold2 } p.\text{send } (\text{get-authentication-path } len \ sidx \ \text{tree});$

$\text{return } idx'$

$\})$

unfolding $p.\text{decommit-on-fri-layers-def}$ **by** *auto*

let $?len = \text{length } l$

let $?idx' = idx \bmod ?len$

let $?sidx = (?idx' + (?len \text{ div } 2)) \bmod ?len$

from *outcome*[*unfolded m-eq Let-def*] **obtain** $u1 \ u2 \ u3 \ u4$ **where**

send-xp: $\text{Some } ((), u1) \in \text{set-dist } (\text{execute } (p.\text{send } (l ! ?idx')) \ s)$

and *path-xp*:

$\text{Some } ((), u2) \in$

$\text{set-dist } (\text{execute } (\text{mfold2 } p.\text{send } (\text{get-authentication-path } ?len \ ?idx' \ \text{tree})) \ u1)$

and *send-xn*: $\text{Some } ((), u3) \in \text{set-dist } (\text{execute } (p.\text{send } (l ! ?sidx)) \ u2)$

and *path-xn*:

$\text{Some } ((), u4) \in$

$\text{set-dist } (\text{execute } (\text{mfold2 } p.\text{send } (\text{get-authentication-path } ?len \ ?sidx \ \text{tree})) \ u3)$

and *ret*: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (\text{return } ?idx') \ u4)$

by (*auto elim!*: $p.\text{set-dist-bindE}$)

have $s-u1$: *transcript-extends* $u1 \ s$

using *send-transcript-extends*[*OF send-xp*] .

have $u1-u2$: *transcript-extends* $u2 \ u1$

using *mfold2-send-transcript-extends*[*OF path-xp*] .

have $u2-u3$: *transcript-extends* $u3 \ u2$

using *send-transcript-extends*[*OF send-xn*] .

have $u3-u4$: *transcript-extends* $u4 \ u3$

using *mfold2-send-transcript-extends*[*OF path-xn*] .

have $u4-t$: *transcript-extends* $t \ u4$

using *ret* **by** *simp*

show *?thesis*

using $s-u1 \ u1-u2 \ u2-u3 \ u3-u4 \ u4-t$ **by** (*meson transcript-extends-trans*)

qed

lemma *decommit-on-fri-layers-step-hash-extends*:

assumes *step-in*: $m \in \text{set } (p.\text{decommit-on-fri-layers } fs)$

and *outcome*: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (m \ idx) \ s)$

shows $s \leq t$

proof –

from *step-in* **obtain** l *tree* **where**
pair-in: $(l, \text{tree}) \in \text{set fs}$
and *m-eq*:
 $m = (\lambda \text{idx}. \text{do } \{$
 $\text{let } \text{len} = \text{length } l;$
 $\text{let } \text{idx}' = \text{idx} \bmod \text{len};$
 $\text{let } \text{sid}x = (\text{idx}' + (\text{len} \text{div } 2)) \bmod \text{len};$
 $\text{p.send } (l ! \text{idx}')$
 $\text{mfold2 } \text{p.send } (\text{get-authentication-path } \text{len } \text{idx}' \text{ tree});$
 $\text{p.send } (l ! \text{sid}x);$
 $\text{mfold2 } \text{p.send } (\text{get-authentication-path } \text{len } \text{sid}x \text{ tree});$
 $\text{return } \text{idx}'$
 $\})$
unfolding *p.decommit-on-fri-layers-def* **by** *auto*
let $?len = \text{length } l$
let $?idx' = \text{idx} \bmod ?len$
let $?sidx = (?idx' + (?len \text{div } 2)) \bmod ?len$
from *outcome[unfolded m-eq Let-def]* **obtain** $u1 \ u2 \ u3 \ u4$ **where**
send-xp: $\text{Some } ((), u1) \in \text{set-dist } (\text{execute } (\text{p.send } (l ! ?idx')) \ s)$
and *path-xp*:
 $\text{Some } ((), u2) \in$
 $\text{set-dist } (\text{execute } (\text{mfold2 } \text{p.send } (\text{get-authentication-path } ?len ?idx' \text{ tree})) \ u1)$
and *send-xn*: $\text{Some } ((), u3) \in \text{set-dist } (\text{execute } (\text{p.send } (l ! ?sidx)) \ u2)$
and *path-xn*:
 $\text{Some } ((), u4) \in$
 $\text{set-dist } (\text{execute } (\text{mfold2 } \text{p.send } (\text{get-authentication-path } ?len ?sidx \text{ tree})) \ u3)$
and *ret*: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (\text{return } ?idx') \ u4)$
by (*auto elim!*: *p.set-dist-bindE*)
have $s\text{-}u1: s \leq u1$
using *send-hash-extends*[*OF send-xp*] .
have $u1\text{-}u2: u1 \leq u2$
using *mfold2-send-hash-extends*[*OF path-xp*] .
have $u2\text{-}u3: u2 \leq u3$
using *send-hash-extends*[*OF send-xn*] .
have $u3\text{-}u4: u3 \leq u4$
using *mfold2-send-hash-extends*[*OF path-xn*] .
have $u4\text{-}t: u4 \leq t$
using *ret* **unfolding** *less-eq-hash-ext-def less-eq-fmap-def* **by** *simp*
show *?thesis*
using $s\text{-}u1 \ u1\text{-}u2 \ u2\text{-}u3 \ u3\text{-}u4 \ u4\text{-}t$ **by** (*meson p.hash-ext-trans*)
qed

lemma *decommit-on-fri-layers-step-preserves-query-counter*:

assumes *step-in*: $m \in \text{set } (p.\text{decommit-on-fri-layers } \text{fs})$
and *outcome*: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } (m \ \text{idx}) \ s)$
shows $P\text{QueryCounter } t = P\text{QueryCounter } s$

proof –

from *step-in* **obtain** l *tree* **where**

m-eq:

```

    m = (λidx. do {
      let len = length l;
      let idx' = idx mod len;
      let sidx = (idx' + (len div 2)) mod len;
      p.send (l ! idx');
      mfold2 p.send (get-authentication-path len idx' tree);
      p.send (l ! sidx);
      mfold2 p.send (get-authentication-path len sidx tree);
      return idx'
    })
  unfolding p.decommit-on-fri-layers-def by auto
let ?len = length l
let ?idx' = idx mod ?len
let ?sidx = (?idx' + (?len div 2)) mod ?len
from outcome[unfolded m-eq Let-def] obtain u1 u2 u3 u4 where
  send-xp: Some ((), u1) ∈ set-dist (execute (p.send (l ! ?idx')) s)
and path-xp:
  Some ((), u2) ∈
    set-dist (execute (mfold2 p.send (get-authentication-path ?len ?idx' tree)) u1)
and send-xn: Some ((), u3) ∈ set-dist (execute (p.send (l ! ?sidx)) u2)
and path-xn:
  Some ((), u4) ∈
    set-dist (execute (mfold2 p.send (get-authentication-path ?len ?sidx tree)) u3)
and ret: Some (x, t) ∈ set-dist (execute (return ?idx') u4)
by (auto elim!: p.set-dist-bindE)
have q-u1: PQueryCounter u1 = PQueryCounter s
  using p.send-outcome[OF send-xp] by simp
have q-u2: PQueryCounter u2 = PQueryCounter u1
  by (rule mfold2-send-preserves-query-counter[OF path-xp])
have q-u3: PQueryCounter u3 = PQueryCounter u2
  using p.send-outcome[OF send-xn] by simp
have q-u4: PQueryCounter u4 = PQueryCounter u3
  by (rule mfold2-send-preserves-query-counter[OF path-xn])
have t-eq: t = u4
  using ret by simp
show ?thesis
  using q-u1 q-u2 q-u3 q-u4 unfolding t-eq by simp
qed

lemma prover-query-round-hash-extends:
  assumes outcome:
    Some (x, t) ∈ set-dist (execute (prover-query-round f-merkle f-ls f-ms ls ms) s)
  shows s ≤ t
proof -
  from outcome obtain idx s1 qouts s2 f-fri-idx s3 fri-idx where
    rand: Some (idx, s1) ∈ set-dist (execute p.receive-query-index-challenge s)
  and query:
    Some (qouts, s2) ∈
      set-dist (execute (mmap (p.decommit-on-query (p.index (to-nat idx)) f-merkle)) s1)

```

```

and trace-fri:
  Some (f-fri-idx, s3) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip f-ls f-ms)))) s2)
and fri:
  Some (fri-idx, t) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip ls ms)))) s3)
unfolding prover-query-round-def
by (auto elim!: p.set-dist-bindE)
have s-s1: s ≤ s1
  using p.receive-query-index-challenge-extends[OF rand] .
have s1-s2: s1 ≤ s2
  by (rule mmap-hash-extends[OF query])
    (rule decommit-on-query-step-hash-extends)
have s2-s3: s2 ≤ s3
  by (rule mfold-hash-extends[OF trace-fri])
    (rule decommit-on-fri-layers-step-hash-extends)
have s3-t: s3 ≤ t
  by (rule mfold-hash-extends[OF fri])
    (rule decommit-on-fri-layers-step-hash-extends)
show ?thesis
  using s-s1 s1-s2 s2-s3 s3-t by (meson p.hash-ext-trans)
qed

lemma prover-query-round-query-counter:
assumes outcome:
  Some (x, t) ∈ set-dist (execute (prover-query-round f-merkle f-ls f-ms ls ms) s)
shows PQueryCounter t = Suc (PQueryCounter s)
proof –
from outcome obtain idx s1 qouts s2 f-fri-idx s3 fri-idx where
  rand: Some (idx, s1) ∈ set-dist (execute p.receive-query-index-challenge s)
and query:
  Some (qouts, s2) ∈
    set-dist (execute (mmap (p.decommit-on-query (p.index (to-nat idx)) f-merkle)) s1)
and trace-fri:
  Some (f-fri-idx, s3) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip f-ls f-ms)))) s2)
and fri:
  Some (fri-idx, t) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip ls ms)))) s3)
unfolding prover-query-round-def
by (auto elim!: p.set-dist-bindE)

```

```

have q-s1: PQueryCounter s1 = Suc (PQueryCounter s)
  using p.receive-query-index-challenge-counter-outcome[OF rand] by simp
have q-s2: PQueryCounter s2 = PQueryCounter s1
  by (rule mmap-preserves-query-counter[OF query])
  (rule decommit-on-query-step-preserves-query-counter)
have q-s3: PQueryCounter s3 = PQueryCounter s2
  by (rule mfold-preserves-query-counter[OF trace-fri])
  (rule decommit-on-fri-layers-step-preserves-query-counter)
have q-t: PQueryCounter t = PQueryCounter s3
  by (rule mfold-preserves-query-counter[OF fri])
  (rule decommit-on-fri-layers-step-preserves-query-counter)
show ?thesis
  using q-s1 q-s2 q-s3 q-t by simp
qed

lemma prover-query-round-transcript-extends:
  assumes outcome:
    Some (x, t) ∈ set-dist (execute (prover-query-round f-merkle f-ls f-ms ls ms) s)
  shows transcript-extends t s
proof -
  from outcome obtain idx s1 qouts s2 f-fri-idx s3 fri-idx where
    rand: Some (idx, s1) ∈ set-dist (execute p.receive-query-index-challenge s)
  and query:
    Some (qouts, s2) ∈
      set-dist (execute (mmap (p.decommit-on-query (p.index (to-nat idx)) f-merkle)) s1)
  and trace-fri:
    Some (f-fri-idx, s3) ∈
      set-dist (execute
        (mfold (p.index (to-nat idx))
          (p.decommit-on-fri-layers (butlast (zip f-ls f-ms)))) s2)
  and fri:
    Some (fri-idx, t) ∈
      set-dist (execute
        (mfold (p.index (to-nat idx))
          (p.decommit-on-fri-layers (butlast (zip ls ms)))) s3)
  unfolding prover-query-round-def
  by (auto elim!: p.set-dist-bindE)
have s-s1: transcript-extends s1 s
  using receive-query-index-challenge-transcript-extends[OF rand] .
have s1-s2: transcript-extends s2 s1
  by (rule mmap-transcript-extends[OF query])
  (rule decommit-on-query-step-transcript-extends)
have s2-s3: transcript-extends s3 s2
  by (rule mfold-transcript-extends[OF trace-fri])
  (rule decommit-on-fri-layers-step-transcript-extends)
have s3-t: transcript-extends t s3
  by (rule mfold-transcript-extends[OF fri])
  (rule decommit-on-fri-layers-step-transcript-extends)
show ?thesis

```

using $s\text{-}s1\ s1\text{-}s2\ s2\text{-}s3\ s3\text{-}t$ **by** (*meson transcript-extends-trans*)
qed

lemma *prover-query-round-outcomeE*:
assumes *outcome*:
Some $(x, t) \in \text{set-dist } (\text{execute } (\text{prover-query-round } f\text{-merkle } f\text{-ls } f\text{-ms } ls\ ms) s)$
obtains $idx\ s1\ qouts\ s2\ f\text{-fri-idx}\ s3\ fri\text{-idx}$
where
Some $(idx, s1) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s)$
Some $(qouts, s2) \in$
set-dist (*execute*
(mmap (*p.decommit-on-query* (*p.index* (*to-nat idx*)) *f-merkle*)) *s1*)
Some $(f\text{-fri-idx}, s3) \in$
set-dist (*execute*
(mfold (*p.index* (*to-nat idx*))
(p.decommit-on-fri-layers (*butlast* (*zip f-ls f-ms*)))) *s2*)
Some $(fri\text{-idx}, t) \in$
set-dist (*execute*
(mfold (*p.index* (*to-nat idx*))
(p.decommit-on-fri-layers (*butlast* (*zip ls ms*)))) *s3*)
 $x = ()$

proof –
from *outcome* **obtain** $idx\ s1\ qouts\ s2\ f\text{-fri-idx}\ s3\ fri\text{-idx}\ s4$ **where**
rand: *Some* $(idx, s1) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s)$
and *query*:
Some $(qouts, s2) \in$
set-dist (*execute*
(mmap (*p.decommit-on-query* (*p.index* (*to-nat idx*)) *f-merkle*)) *s1*)
and *trace-fri*:
Some $(f\text{-fri-idx}, s3) \in$
set-dist (*execute*
(mfold (*p.index* (*to-nat idx*))
(p.decommit-on-fri-layers (*butlast* (*zip f-ls f-ms*)))) *s2*)
and *fri*:
Some $(fri\text{-idx}, s4) \in$
set-dist (*execute*
(mfold (*p.index* (*to-nat idx*))
(p.decommit-on-fri-layers (*butlast* (*zip ls ms*)))) *s3*)
and *ret*: *Some* $(x, t) \in \text{set-dist } (\text{execute } (\text{return } ()) s4)$
unfolding *prover-query-round-def*
by (*auto elim!*: *p.set-dist-bindE*)
have *x-unit*: $x = ()$
using *ret* **by** *simp*
have *t-eq*: $t = s4$
using *ret* **by** *simp*
show *?thesis*
by (*rule that*[*OF rand query trace-fri - x-unit*])
(use fri t-eq in simp)

qed

lemma *random-send-round-transcript:*

assumes *outcome:*

Some $(a, t) \in$
set-dist (*execute*
 (*do* {
 $a \leftarrow p.receive\text{-}alpha\text{-}challenge$;
 $p.send\ a$;
 $return\ a$
 }) *s*)

shows $PTranscript\ t = a \# PTranscript\ s$

proof –

from *outcome* **obtain** $u\ v$ **where**

rand: *Some* $(a, u) \in set\text{-}dist\ (execute\ p.receive\text{-}alpha\text{-}challenge\ s)$

and *send-a*: *Some* $((), v) \in set\text{-}dist\ (execute\ (p.send\ a)\ u)$

and *ret*: *Some* $(a, t) \in set\text{-}dist\ (execute\ (return\ a)\ v)$

by (*auto elim!*: *p.set-dist-bindE*)

have $t\text{-}v$: $t = v$

using *ret* **by** *simp*

have $tr\text{-}u$: $PTranscript\ u = PTranscript\ s$

using *receive-alpha-challenge-preserves-transcript*[*OF rand*] .

have $v = u \Downarrow PState := concat\ (PState\ u)\ a, PTranscript := a \# PTranscript\ u)$

using *p.send-outcome*[*OF send-a*] .

then show *?thesis*

using $t\text{-}v\ tr\text{-}u$ **by** *simp*

qed

lemma *random-send-round-parts-outcome:*

assumes *rand*: *Some* $(a, u) \in set\text{-}dist\ (execute\ p.receive\text{-}alpha\text{-}challenge\ s)$

and *send-a*: *Some* $((), t) \in set\text{-}dist\ (execute\ (p.send\ a)\ u)$

shows

$s \leq t \wedge$

$PState\ t = concat\ (PState\ s)\ a \wedge$

$PTranscript\ t = a \# PTranscript\ s \wedge$

$PTraceFriCounter\ t = PTraceFriCounter\ s \wedge$

$PCompositionFriCounter\ t = PCompositionFriCounter\ s \wedge$

$PAlphaCounter\ t = Suc\ (PAlphaCounter\ s) \wedge$

$PQueryCounter\ t = PQueryCounter\ s \wedge$

$fmlookup\ (HashMap\ t)\ (AlphaChallenge\ (PAlphaCounter\ s)\ (PState\ s)) = Some\ a$

proof –

have *rand-out*:

$s \leq u \wedge$

$PState\ u = PState\ s \wedge$

$PTranscript\ u = PTranscript\ s \wedge$

$fmlookup\ (HashMap\ u)\ (AlphaChallenge\ (PAlphaCounter\ s)\ (PState\ s)) = Some\ a$

by (*rule* *p.receive-alpha-challenge-outcome*[*OF rand*])

have $s\text{-}u$: $s \leq u$

using *rand-out* **by** *simp*

have $lookup\text{-}u$: $fmlookup\ (HashMap\ u)\ (AlphaChallenge\ (PAlphaCounter\ s)\ (PState\ s)) = Some$

a
using *rand-out* **by** *simp*
have *state-u*: $PState\ u = PState\ s$
using *rand-out* **by** *simp*
have *tr-u*: $PTranscript\ u = PTranscript\ s$
using *rand-out* **by** *simp*
have *alpha-counter-u*: $PAlphaCounter\ u = Suc\ (PAlphaCounter\ s)$
using *p.receive-alpha-challenge-counter-outcome*[*OF rand*] **by** *simp*
have *other-counters-u*:
 $PTraceFriCounter\ u = PTraceFriCounter\ s \wedge$
 $PCompositionFriCounter\ u = PCompositionFriCounter\ s \wedge$
 $PQueryCounter\ u = PQueryCounter\ s$
using *p.receive-alpha-challenge-counter-outcome*[*OF rand*] **by** *simp*
have *t-eq*: $t = u(PState := concat\ (PState\ u)\ a,\ PTranscript := a \# PTranscript\ u)$
using *p.send-outcome*[*OF send-a*] .
have *u-t*: $u \leq t$
unfolding *t-eq* *less-eq-hash-ext-def* *less-eq-fmap-def* **by** *simp*
have *s-t*: $s \leq t$
using *s-u* *u-t* **by** (rule *p.hash-ext-trans*)
have *lookup-t*: $fmllookup\ (HashMap\ t)\ (AlphaChallenge\ (PAlphaCounter\ s)\ (PState\ s)) = Some$
 a
using *p.hash-extension-lookup*[*OF lookup-u u-t*] .
show *?thesis*
using *s-t* *state-u* *tr-u* *alpha-counter-u* *other-counters-u* *lookup-t*
unfolding *t-eq* **by** *simp*
qed

lemma *random-send-round-outcome*:

assumes *outcome*:

$Some\ (a,\ t) \in$
 $set-dist\ (execute$
 $(do\ \{$
 $a \leftarrow p.receive-alpha-challenge;$
 $p.send\ a;$
 $return\ a$
 $\})\ s)$

shows

$s \leq t \wedge$
 $PState\ t = concat\ (PState\ s)\ a \wedge$
 $PTranscript\ t = a \# PTranscript\ s \wedge$
 $PTraceFriCounter\ t = PTraceFriCounter\ s \wedge$
 $PCompositionFriCounter\ t = PCompositionFriCounter\ s \wedge$
 $PAlphaCounter\ t = Suc\ (PAlphaCounter\ s) \wedge$
 $PQueryCounter\ t = PQueryCounter\ s \wedge$
 $fmllookup\ (HashMap\ t)\ (AlphaChallenge\ (PAlphaCounter\ s)\ (PState\ s)) = Some\ a$

proof –

from *outcome* **obtain** $u\ v$ **where**

rand: $Some\ (a,\ u) \in set-dist\ (execute\ p.receive-alpha-challenge\ s)$

and *send-a*: $Some\ ((),\ v) \in set-dist\ (execute\ (p.send\ a)\ u)$


```

    and ret: Some (a, t) ∈ set-dist (execute (return a) v)
    by (auto elim!: p.set-dist-bindE)
  have t-v: t = v
    using ret by simp
  show ?thesis
    using random-send-round-parts-outcome[OF rand send-a] unfolding t-v .
qed

```

lemma *alpha-mmap-transcript*:

```

  assumes outcome:
    Some (as, t) ∈
      set-dist (execute
        (mmap (replicate n
          (do {
            a ← p.receive-alpha-challenge;
            p.send a;
            return a
          }))) s)
  shows length as = n ∧ PTranscript t = rev as @ PTranscript s
  using outcome
proof (induction n arbitrary: as s t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  let ?round =
    do {
      a ← p.receive-alpha-challenge;
      p.send a;
      return a
    }
  from Suc.prems obtain a u0 u as' where
    rand: Some (a, u0) ∈ set-dist (execute p.receive-alpha-challenge s)
    and send-a: Some ((), u) ∈ set-dist (execute (p.send a) u0)
    and rest:
      Some (as', t) ∈
        set-dist (execute (mmap (replicate n ?round)) u)
    and as-eq: as = a # as'
    by (auto elim!: p.set-dist-bindE)
  have tr-u0: PTranscript u0 = PTranscript s
    using receive-alpha-challenge-preserves-transcript[OF rand] .
  have tr-u: PTranscript u = a # PTranscript s
    using p.send-outcome[OF send-a] tr-u0 by simp
  have ih: length as' = n ∧ PTranscript t = rev as' @ PTranscript u
    using Suc.IH[OF rest] .
  show ?case
    using as-eq tr-u ih by simp
qed

```

```

lemma alpha-mmap-outcome:
  assumes outcome:
    Some (as, t) ∈
      set-dist (execute
        (mmap (replicate n
          (do {
            a ← p.receive-alpha-challenge;
            p.send a;
            return a
          }))) s)
  shows
    length as = n ∧
    s ≤ t ∧
    PState t = foldl concat (PState s) as ∧
    PTranscript t = rev as @ PTranscript s ∧
    PTraceFriCounter t = PTraceFriCounter s ∧
    PCompositionFriCounter t = PCompositionFriCounter s ∧
    PAlphaCounter t = PAlphaCounter s + length as ∧
    PQueryCounter t = PQueryCounter s ∧
    (∀ i < length as.
      fmlookup (HashMap t)
        (AlphaChallenge (PAlphaCounter s + i)
          (foldl concat (PState s) (take i as))) = Some (as ! i))
    using outcome
  proof (induction n arbitrary: as s t)
    case 0
    then show ?case
      unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  next
    case (Suc n)
    let ?round =
      do {
        a ← p.receive-alpha-challenge;
        p.send a;
        return a
      }
    from Suc.prems obtain a u0 u as' where
      rand: Some (a, u0) ∈ set-dist (execute p.receive-alpha-challenge s)
      and send-a: Some ((), u) ∈ set-dist (execute (p.send a) u0)
      and tail:
        Some (as', t) ∈
          set-dist (execute (mmap (replicate n ?round)) u)
      and as-eq: as = a # as'
      by (auto elim!: p.set-dist-bindE)
    have head-data:
      s ≤ u ∧
      PState u = concat (PState s) a ∧
      PTranscript u = a # PTranscript s ∧
      PTraceFriCounter u = PTraceFriCounter s ∧

```

```

PCompositionFriCounter u = PCompositionFriCounter s ∧
PAlphaCounter u = Suc (PAlphaCounter s) ∧
PQueryCounter u = PQueryCounter s ∧
fmlookup (HashMap u) (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a
using random-send-round-parts-outcome[OF rand send-a] .
have tail-data:
  length as' = n ∧
  u ≤ t ∧
  PState t = foldl concat (PState u) as' ∧
  PTranscript t = rev as' @ PTranscript u ∧
  PTraceFriCounter t = PTraceFriCounter u ∧
  PCompositionFriCounter t = PCompositionFriCounter u ∧
  PAlphaCounter t = PAlphaCounter u + length as' ∧
  PQueryCounter t = PQueryCounter u ∧
  (∀ i < length as'.
    fmlookup (HashMap t)
      (AlphaChallenge (PAlphaCounter u + i)
        (foldl concat (PState u) (take i as'))) = Some (as' ! i))
    using Suc.IH[OF tail] .
have s-t: s ≤ t
  using head-data tail-data by (meson p.hash-ext-trans)
have lookup-head:
  fmlookup (HashMap t) (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a
  using head-data tail-data by (meson p.hash-extension-lookup)
have lookup-all:
  ∀ i < length (a # as').
    fmlookup (HashMap t)
      (AlphaChallenge (PAlphaCounter s + i)
        (foldl concat (PState s) (take i (a # as')))) =
      Some ((a # as') ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length (a # as')
  show fmlookup (HashMap t)
    (AlphaChallenge (PAlphaCounter s + i)
      (foldl concat (PState s) (take i (a # as')))) =
    Some ((a # as') ! i)
  proof (cases i)
    case 0
    then show ?thesis
      using lookup-head by simp
  next
    case (Suc j)
    then have j-bound: j < length as'
      using i-bound by simp
    have fold-eq:
      foldl concat (PState s) (take (Suc j) (a # as')) =
        foldl concat (PState u) (take j as')
      using head-data by simp

```

```

    have alpha-u: PAlphaCounter u = Suc (PAlphaCounter s)
      using head-data by simp
    show ?thesis
      using tail-data j-bound fold-eq Suc alpha-u by simp
  qed
  qed
  show ?case
    using as-eq head-data tail-data s-t lookup-all by simp
  qed

lemma honest-alpha-mmap-replay-no-failure:
  assumes lookups:
     $\forall i < \text{length } as.$ 
    fmlookup (HashMap s)
      (AlphaChallenge (PAlphaCounter s + i)
        (foldl concat (PState s) (take i as))) = Some (as ! i)
  shows
    None  $\notin$  dom (dist (execute
      (mmap (replicate (length as)
        (do {
          a0  $\leftarrow$  p.receive-alpha-challenge;
          let a0' = a0;
          a1  $\leftarrow$  p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        }))))
      (s(PTranscript := as @ rest))))
    using lookups
  proof (induction as arbitrary: s rest)
    case Nil
      then show ?case by simp
    next
      case (Cons a as)
        let ?round =
          do {
            a0  $\leftarrow$  p.receive-alpha-challenge;
            let a0' = a0;
            a1  $\leftarrow$  p.read;
            let a1' = a1;
            assert (a0' = a1');
            return a1'
          }
        have lookup-a: fmlookup (HashMap s) (AlphaChallenge (PAlphaCounter s) (PState s)) = Some
a
          using spec[OF Cons.premis, of 0] by simp
        have head-nf:
          None  $\notin$  dom (dist (execute ?round
            (s(PTranscript := a # (as @ rest)))))

```

```

using honest-alpha-round-replay-no-failure[OF lookup-a, of as @ rest]
by simp
have full-eq:
  mmap (replicate (length (a # as)) ?round) =
    (?round  $\gg$  (λx. mmap (replicate (length as) ?round)  $\gg$  (λxs. return (x # xs))))
by simp
have start-eq:
  s(PTranscript := (a # as) @ rest) =
    s(PTranscript := a # (as @ rest))
by simp
show ?case
  unfolding full-eq start-eq
proof (rule no-failure-bindI[OF head-nf])
  fix y t
  assume head:
    Some (y, t) ∈
      set-dist (execute ?round (s(PTranscript := a # as @ rest)))
  have head-out:
    y = a ∧
    PState t = concat (PState s) a ∧
    PTranscript t = as @ rest ∧
    PAlphaCounter t = Suc (PAlphaCounter s) ∧
    s ≤ t
    using honest-alpha-round-replay-outcome[OF lookup-a head] by simp
  have tail-lookups:
    ∀ i < length as.
      fmllookup (HashMap t)
        (AlphaChallenge (PAlphaCounter t + i)
          (foldl concat (PState t) (take i as))) = Some (as ! i)
  proof (intro allI impI)
    fix i
    assume i-bound: i < length as
    have old-lookup:
      fmllookup (HashMap s)
        (AlphaChallenge (PAlphaCounter s + Suc i)
          (foldl concat (PState s) (take (Suc i) (a # as)))) =
        Some ((a # as) ! Suc i)
      using spec[OF Cons.premis, of Suc i] i-bound by simp
    have key-eq:
      foldl concat (PState s) (take (Suc i) (a # as)) =
        foldl concat (PState t) (take i as)
      using head-out by simp
    show fmllookup (HashMap t)
      (AlphaChallenge (PAlphaCounter t + i)
        (foldl concat (PState t) (take i as))) =
        Some (as ! i)
      using p.hash-extension-lookup[OF old-lookup, of t] head-out key-eq i-bound
      by simp
  qed

```

```

have tail-nf-start:
  None  $\notin$  dom (dist (execute
    (mmap (replicate (length as) ?round))
    (t(PTranscript := as @ rest))))
  using Cons.IH[OF tail-lookups, of rest] .
have t-update: t(PTranscript := as @ rest) = t
  using head-out by simp
have tail-nf:
  None  $\notin$  dom (dist (execute
    (mmap (replicate (length as) ?round)) t))
  using tail-nf-start unfolding t-update .
show None  $\notin$  dom (dist (execute
  (mmap (replicate (length as) ?round)  $\gg$  (λxs. return (y # xs))) t))
  by (rule no-failure-bindI[OF tail-nf] simp)
qed
qed

lemma honest-alpha-mmap-replay-outcome:
assumes lookups:
  ∀ i < length as.
    fmlookup (HashMap s)
      (AlphaChallenge (PAlphaCounter s + i)
        (foldl concat (PState s) (take i as))) = Some (as ! i)
and outcome:
  Some (ys, t) ∈ set-dist (execute
    (mmap (replicate (length as)
      (do {
        a0 ← p.receive-alpha-challenge;
        let a0' = a0;
        a1 ← p.read;
        let a1' = a1;
        assert (a0' = a1');
        return a1'
      }))))
    (s(PTranscript := as @ rest)))
shows
  ys = as ∧
  PState t = foldl concat (PState s) as ∧
  PTranscript t = rest ∧
  PTraceFriCounter t = PTraceFriCounter s ∧
  PCompositionFriCounter t = PCompositionFriCounter s ∧
  PAlphaCounter t = PAlphaCounter s + length as ∧
  PQueryCounter t = PQueryCounter s ∧
  s ≤ t
using lookups outcome
proof (induction as arbitrary: ys s t rest)
case Nil
have t-eq: t = s(PTranscript := rest)
  using Nil.prem(2) by simp

```

```

have ys-eq: ys = []
  using Nil.prems(2) by simp
have s-t: s ≤ t
  unfolding t-eq less-eq-hash-ext-def less-eq-fmap-def by simp
show ?case
  using ys-eq t-eq s-t by simp
next
case (Cons a as)
let ?round =
  do {
    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }
from Cons.prems(2) obtain a0 r-state y read-state u ys' where
  rand:
    Some (a0, r-state) ∈
      set-dist (execute p.receive-alpha-challenge
        (s(PTranscript := a # (as @ rest))))
  and read:
    Some (y, read-state) ∈ set-dist (execute p.read r-state)
  and assert-ok:
    Some ((), u) ∈ set-dist (execute (assert (a0 = y)) read-state)
  and tail:
    Some (ys', t) ∈
      set-dist (execute (mmap (replicate (length as) ?round)) u)
  and ys-eq: ys = y # ys'
  by (auto elim!: p.set-dist-bindE)
have lookup-a: fmlookup (HashMap s) (AlphaChallenge (PAlphaCounter s) (PState s)) = Some
a
  using spec[OF Cons.prems(1), of 0] by simp
have head-out:
  y = a ∧
    PState u = concat (PState s) a ∧
    PTranscript u = as @ rest ∧
    PTraceFriCounter u = PTraceFriCounter s ∧
    PCompositionFriCounter u = PCompositionFriCounter s ∧
    PAlphaCounter u = Suc (PAlphaCounter s) ∧
    PQueryCounter u = PQueryCounter s ∧
    s ≤ u
  using honest-alpha-round-replay-parts-outcome[
    OF lookup-a rand read assert-ok] .
have tail-lookups:
  ∀ i < length as.
    fmlookup (HashMap u)
      (AlphaChallenge (PAlphaCounter u + i)

```

```

      (foldl concat (PState u) (take i as))) = Some (as ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length as
  have old-lookup:
    fmlookup (HashMap s)
      (AlphaChallenge (PAlphaCounter s + Suc i)
        (foldl concat (PState s) (take (Suc i) (a # as)))) =
      Some ((a # as) ! Suc i)
  using spec[OF Cons.prem1, of Suc i] i-bound by simp
  have key-eq:
    foldl concat (PState s) (take (Suc i) (a # as)) =
    foldl concat (PState u) (take i as)
  using head-out by simp
  show fmlookup (HashMap u)
    (AlphaChallenge (PAlphaCounter u + i)
      (foldl concat (PState u) (take i as))) =
    Some (as ! i)
  using p.hash-extension-lookup[OF old-lookup, of u] head-out key-eq i-bound
  by simp
qed
have u-update: u(PTranscript := as @ rest) = u
  using head-out by simp
have tail':
  Some (ys', t) ∈
    set-dist (execute (mmap (replicate (length as) ?round))
      (u(PTranscript := as @ rest)))
  using tail unfolding u-update .
have tail-out:
  ys' = as ∧
  PState t = foldl concat (PState u) as ∧
  PTranscript t = rest ∧
  PTraceFriCounter t = PTraceFriCounter u ∧
  PCompositionFriCounter t = PCompositionFriCounter u ∧
  PAlphaCounter t = PAlphaCounter u + length as ∧
  PQueryCounter t = PQueryCounter u ∧
  u ≤ t
  using Cons.IH[OF tail-lookups tail] .
have s-t: s ≤ t
  using head-out tail-out by (meson p.hash-ext-trans)
show ?case
  using ys-eq head-out tail-out s-t by simp
qed

```

lemma honest-alpha-replay-round-preserves-query-counter:

```

assumes outcome:
  Some (y, t) ∈ set-dist (execute
    (do {
      a0 ← p.receive-alpha-challenge;

```



```

    let  $a0'$  =  $a0$ ;
     $a1 \leftarrow p.read$ ;
    let  $a1'$  =  $a1$ ;
    assert ( $a0' = a1'$ );
    return  $a1'$ 
  })  $s$ )
shows  $PQueryCounter\ t = PQueryCounter\ s$ 
proof –
from outcome obtain  $a0\ r\text{-}state\ read\text{-}state$  where
  rand:
    Some ( $a0, r\text{-}state$ )  $\in$  set-dist (execute  $p.receive\text{-}alpha\text{-}challenge\ s$ )
  and read:
    Some ( $y, read\text{-}state$ )  $\in$  set-dist (execute  $p.read\ r\text{-}state$ )
  and assert-ok: Some ( $()$ ,  $t$ )  $\in$  set-dist (execute (assert ( $a0 = y$ )) read-state)
  by (auto simp: Let-def elim!:  $p.set\text{-}dist\text{-}bindE$ )
have  $q\text{-}rand$ :  $PQueryCounter\ r\text{-}state = PQueryCounter\ s$ 
  using  $p.receive\text{-}alpha\text{-}challenge\text{-}counter\text{-}outcome[OF\ rand]$  by simp
have  $q\text{-}read$ :  $PQueryCounter\ read\text{-}state = PQueryCounter\ r\text{-}state$ 
  using  $read\text{-}preserves\text{-}counters[OF\ read]$  by simp
have  $t\text{-}eq$ :  $t = read\text{-}state$ 
  using  $assert\text{-}outcomeD(2)[OF\ assert\text{-}ok]$  .
show ?thesis
  using  $q\text{-}rand\ q\text{-}read$  unfolding  $t\text{-}eq$  by simp
qed

```

lemma *honest-alpha-mmap-replay-preserves-query-counter*:

```

assumes outcome:
  Some ( $ys, t$ )  $\in$  set-dist (execute
    (mmap (replicate  $n$ 
      (do {
         $a0 \leftarrow p.receive\text{-}alpha\text{-}challenge$ ;
        let  $a0'$  =  $a0$ ;
         $a1 \leftarrow p.read$ ;
        let  $a1'$  =  $a1$ ;
        assert ( $a0' = a1'$ );
        return  $a1'$ 
      })))  $s$ )
shows  $PQueryCounter\ t = PQueryCounter\ s$ 
using outcome
proof (induction  $n$  arbitrary:  $ys\ s\ t$ )
  case 0
  then show ?case by simp
next
  case (Suc  $n$ )
  let ?round =
    do {
       $a0 \leftarrow p.receive\text{-}alpha\text{-}challenge$ ;
      let  $a0'$  =  $a0$ ;
       $a1 \leftarrow p.read$ ;

```

```

    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }
from Suc.premis obtain a0 r-state y read-state u ys' where
  rand:
    Some (a0, r-state) ∈ set-dist (execute p.receive-alpha-challenge s)
and read:
    Some (y, read-state) ∈ set-dist (execute p.read r-state)
and assert-ok:
    Some ((), u) ∈ set-dist (execute (assert (a0 = y)) read-state)
and tail: Some (ys', t) ∈ set-dist (execute (mmap (replicate n ?round)) u)
by (auto elim!: p.set-dist-bindE)
have q-rand: PQueryCounter r-state = PQueryCounter s
  using p.receive-alpha-challenge-counter-outcome[OF rand] by simp
have q-read: PQueryCounter read-state = PQueryCounter r-state
  using read-preserves-counters[OF read] by simp
have u-eq: u = read-state
  using assert-outcomeD(2)[OF assert-ok] .
have q-head: PQueryCounter u = PQueryCounter s
  using q-rand q-read unfolding u-eq by simp
have q-tail: PQueryCounter t = PQueryCounter u
  by (rule Suc.IH[OF tail])
show ?case
  using q-head q-tail by simp
qed

```

lemma prover-root-alpha-degree-transcript:

```

assumes create-f:
  Some (f-merkle, s1) ∈
    set-dist (execute (p.create p.f-eval) init-state)
and send-f:
  Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and alphas:
  Some (as, s3) ∈
    set-dist (execute
      (mmap (replicate (length spec)
        (do {
          a ← p.receive-alpha-challenge;
          p.send a;
          return a
        }))) s2)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
shows
  length as = length spec ∧
  PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
proof –

```

```

have tr-s1: PTranscript s1 = []
  using create-preserves-transcript[OF create-f]
  unfolding init-state-def by simp
have tr-s2: PTranscript s2 = [value f-merkle]
  using p.send-outcome[OF send-f] tr-s1 by simp
have alpha-tr:
  length as = length spec  $\wedge$  PTranscript s3 = rev as @ PTranscript s2
  using alpha-mmap-transcript[OF alphas] .
have tr-s4: PTranscript s4 = of-nat (degree cp^) # PTranscript s3
  using p.send-outcome[OF send-degree] by simp
show ?thesis
  using alpha-tr tr-s2 tr-s4 by simp
qed

lemma prover-later-phases-transcript-extends:
  assumes create-cp:
    Some (cp-merkle, s5)  $\in$  set-dist (execute (p.create (p.cp-eval as)) s4)
  and fri:
    Some ((ps, ds, ls, ms), s6)  $\in$ 
      set-dist (execute
        (p.fri-commit nrounds [cp^] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
  and final-send:
    Some ((), s-final)  $\in$  set-dist (execute (p.send (hd (last ls))) s6)
  and random-idx:
    Some (idx, s7)  $\in$  set-dist (execute p.receive-query-index-challenge s-final)
  and query-decommit:
    Some (qouts, s8)  $\in$ 
      set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
  and fri-decommit:
    Some (fri-idx, s9)  $\in$ 
      set-dist (execute
        (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
  and prover-state-eq: prover-state = s9
  shows transcript-extends prover-state s4
proof -
  have s4-s5: transcript-extends s5 s4
    using create-transcript-extends[OF create-cp] .
  have s5-s6: transcript-extends s6 s5
    using fri-commit-transcript-extends[OF fri] .
  have s6-final: transcript-extends s-final s6
    using send-transcript-extends[OF final-send] .
  have final-s7: transcript-extends s7 s-final
    using receive-query-index-challenge-transcript-extends[OF random-idx] .
  have s7-s8: transcript-extends s8 s7
    by (rule mmap-transcript-extends[OF query-decommit])
      (rule decommit-on-query-step-transcript-extends)
  have s8-s9: transcript-extends s9 s8
    by (rule mfold-transcript-extends[OF fri-decommit])
      (rule decommit-on-fri-layers-step-transcript-extends)

```

```

have s9-prover: transcript-extends prover-state s9
  using prover-state-eq by simp
show ?thesis
  using s4-s5 s5-s6 s6-final final-s7 s7-s8 s8-s9 s9-prover
  by (meson transcript-extends-trans)
qed

lemma prover-later-phases-hash-extends:
  assumes send-degree:
    Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp^))) s3)
  and create-cp:
    Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
  and fri:
    Some ((ps, ds, ls, ms), s6) ∈
      set-dist (execute
        (p.fri-commit nrounds [cp^] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
  and final-send:
    Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
  and random-idx:
    Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
  and query-decommit:
    Some (qouts, s8) ∈
      set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
  and fri-decommit:
    Some (fri-idx, s9) ∈
      set-dist (execute
        (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
  and prover-state-eq: prover-state = s9
  shows s3 ≤ prover-state
proof -
  have s3-s4: s3 ≤ s4
    using send-hash-extends[OF send-degree] .
  have s4-s5: s4 ≤ s5
    using create-hash-extends[OF create-cp] .
  have s5-s6: s5 ≤ s6
    using honest-fri-commit-extends[OF fri] .
  have s6-final: s6 ≤ s-final
    using send-hash-extends[OF final-send] .
  have final-s7: s-final ≤ s7
    using p.receive-query-index-challenge-extends[OF random-idx] .
  have s7-s8: s7 ≤ s8
    by (rule mmap-hash-extends[OF query-decommit])
      (rule decommit-on-query-step-hash-extends)
  have s8-s9: s8 ≤ s9
    by (rule mfold-hash-extends[OF fri-decommit])
      (rule decommit-on-fri-layers-step-hash-extends)
  have s9-prover: s9 ≤ prover-state
    using prover-state-eq p.hash-ext-refl by simp
  show ?thesis

```

```

using s3-s4 s4-s5 s5-s6 s6-final final-s7 s7-s8 s8-s9 s9-prover
by (meson p.hash-ext-trans)
qed

lemma prover-after-fri-transcript-extends:
  assumes final-send:
    Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
  and random-idx:
    Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
  and query-decommit:
    Some (qouts, s8) ∈
      set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
  and fri-decommit:
    Some (fri-idx, s9) ∈
      set-dist (execute
        (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
  and prover-state-eq: prover-state = s9
  shows transcript-extends prover-state s6
proof -
  have s6-final: transcript-extends s-final s6
    using send-transcript-extends[OF final-send] .
  have final-s7: transcript-extends s7 s-final
    using receive-query-index-challenge-transcript-extends[OF random-idx] .
  have s7-s8: transcript-extends s8 s7
    by (rule mmap-transcript-extends[OF query-decommit])
      (rule decommit-on-query-step-transcript-extends)
  have s8-s9: transcript-extends s9 s8
    by (rule mfold-transcript-extends[OF fri-decommit])
      (rule decommit-on-fri-layers-step-transcript-extends)
  have s9-prover: transcript-extends prover-state s9
    using prover-state-eq by simp
  show ?thesis
    using s6-final final-s7 s7-s8 s8-s9 s9-prover by (meson transcript-extends-trans)
qed

```

```

lemma prover-after-fri-hash-extends:
  assumes final-send:
    Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
  and random-idx:
    Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
  and query-decommit:
    Some (qouts, s8) ∈
      set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
  and fri-decommit:
    Some (fri-idx, s9) ∈
      set-dist (execute
        (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
  and prover-state-eq: prover-state = s9
  shows s6 ≤ prover-state

```

proof –
 have $s6\text{-final}$: $s6 \leq s\text{-final}$
 using send-hash-extends [OF final-send] .
 have $\text{final-}s7$: $s\text{-final} \leq s7$
 using $p.\text{receive-query-index-challenge-extends}$ [OF random-idx] .
 have $s7\text{-}s8$: $s7 \leq s8$
 by ($\text{rule mmap-hash-extends}$ [OF query-decommit])
 ($\text{rule decommit-on-query-step-hash-extends}$)
 have $s8\text{-}s9$: $s8 \leq s9$
 by ($\text{rule mfold-hash-extends}$ [OF fri-decommit])
 ($\text{rule decommit-on-fri-layers-step-hash-extends}$)
 have $s9\text{-prover}$: $s9 \leq \text{prover-state}$
 using $\text{prover-state-eq } p.\text{hash-ext-refl}$ **by** simp
show $?thesis$
 using $s6\text{-final } \text{final-}s7 \ s7\text{-}s8 \ s8\text{-}s9 \ s9\text{-prover}$ **by** ($\text{meson } p.\text{hash-ext-trans}$)
qed

lemma $\text{prover-after-final-hash-extends}$:
assumes random-idx :
 $\text{Some } (idx, s7) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s\text{-final})$
and query-decommit :
 $\text{Some } (qouts, s8) \in$
 $\text{set-dist } (\text{execute } (\text{mmap } (p.\text{decommit-on-query } idx' \ f\text{-merkle})) \ s7)$
and fri-decommit :
 $\text{Some } (fri\text{-idx}, s9) \in$
 $\text{set-dist } (\text{execute } (\text{mfold } idx' \ (p.\text{decommit-on-fri-layers } (\text{butlast } (\text{zip } ls \ ms)))) \ s8)$
and prover-state-eq : $\text{prover-state} = s9$
shows $s\text{-final} \leq \text{prover-state}$

proof –
 have $\text{final-}s7$: $s\text{-final} \leq s7$
 using $p.\text{receive-query-index-challenge-extends}$ [OF random-idx] .
 have $s7\text{-}s8$: $s7 \leq s8$
 by ($\text{rule mmap-hash-extends}$ [OF query-decommit])
 ($\text{rule decommit-on-query-step-hash-extends}$)
 have $s8\text{-}s9$: $s8 \leq s9$
 by ($\text{rule mfold-hash-extends}$ [OF fri-decommit])
 ($\text{rule decommit-on-fri-layers-step-hash-extends}$)
 have $s9\text{-prover}$: $s9 \leq \text{prover-state}$
 using $\text{prover-state-eq } p.\text{hash-ext-refl}$ **by** simp
show $?thesis$
 using $\text{final-}s7 \ s7\text{-}s8 \ s8\text{-}s9 \ s9\text{-prover}$ **by** ($\text{meson } p.\text{hash-ext-trans}$)
qed

lemma $\text{prover-query-random-lookup-in-replay-state}$:
assumes random-idx :
 $\text{Some } (idx, s7) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s\text{-final})$
and query-decommit :
 $\text{Some } (qouts, s8) \in$

```

    set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
  and fri-decommit:
    Some (fri-idx, s9) ∈
      set-dist (execute
        (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
  and prover-state-eq: prover-state = s9
  and replay: replay-state = verifier-replay-state prover-state
  shows fmllookup (HashMap replay-state) (QueryIndexChallenge (PQueryCounter s-final) (PState
s-final)) = Some idx
proof -
  have lookup-s7: fmllookup (HashMap s7) (QueryIndexChallenge (PQueryCounter s-final) (PState
s-final)) = Some idx
    using p.receive-query-index-challenge-outcome[OF random-idx] by simp
  have s7-prover: s7 ≤ prover-state
proof -
  have s7-s8: s7 ≤ s8
    by (rule mmap-hash-extends[OF query-decommit])
      (rule decommit-on-query-step-hash-extends)
  have s8-s9: s8 ≤ s9
    by (rule mfold-hash-extends[OF fri-decommit])
      (rule decommit-on-fri-layers-step-hash-extends)
  have s9-prover: s9 ≤ prover-state
    using prover-state-eq p.hash-ext-refl by simp
  show ?thesis
    using s7-s8 s8-s9 s9-prover by (meson p.hash-ext-trans)
qed
have lookup-prover:
  fmllookup (HashMap prover-state) (QueryIndexChallenge (PQueryCounter s-final) (PState
s-final)) = Some idx
  using p.hash-extension-lookup[OF lookup-s7 s7-prover] .
  show ?thesis
    using lookup-prover replay
    unfolding verifier-replay-state-def by simp
qed

```

lemma prover-final-query-counter-in-replay-state:

```

  assumes create-f:
    Some (f-merkle, s1) ∈
      set-dist (execute (p.create p.f-eval) init-state)
  and send-f:
    Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
  and alphas:
    Some (as, s3) ∈
      set-dist (execute
        (mmap (replicate (length spec)
          (do {
            a ← p.receive-alpha-challenge;
            p.send a;
            return a
          }

```

```

      }))) s2)
and send-degree:
  Some ( $()$ ,  $s4$ )  $\in$  set-dist (execute ( $p$ .send degree-msg)  $s3$ )
and create-cp:
  Some ( $cp$ -merkle,  $s5$ )  $\in$  set-dist (execute ( $p$ .create cp-eval)  $s4$ )
and fri:
  Some ( $(ps, ds, ls, ms)$ ,  $s6$ )  $\in$ 
    set-dist (execute ( $p$ .fri-commit  $n$   $ps0$   $ds0$   $ls0$   $ms0$ )  $s5$ )
and final-send:
  Some ( $()$ ,  $s$ -final)  $\in$  set-dist (execute ( $p$ .send final-msg)  $s6$ )
and replay: replay-state = verifier-replay-state prover-state
shows PQueryCounter  $s$ -final = PQueryCounter replay-state
proof –
  have  $q$ -s1-zero: PQueryCounter  $s1$  = 0
    using  $p$ .create-preserves-channel(6)[OF create-f]
    unfolding init-state-def by simp
  have  $q$ -s2-zero: PQueryCounter  $s2$  = 0
    using  $p$ .send-outcome[OF send-f]  $q$ -s1-zero by simp
  have  $q$ -s3-zero: PQueryCounter  $s3$  = 0
    using  $\alpha$ -mmap-outcome[OF alphas]  $q$ -s2-zero by simp
  have  $q$ -s4-zero: PQueryCounter  $s4$  = 0
    using  $p$ .send-outcome[OF send-degree]  $q$ -s3-zero by simp
  have  $q$ -s5-zero: PQueryCounter  $s5$  = 0
    using  $p$ .create-preserves-channel(6)[OF create-cp]  $q$ -s4-zero by simp
  have  $q$ -s6-zero: PQueryCounter  $s6$  = 0
    using fri-commit-preserves-query-counter[OF fri]  $q$ -s5-zero by simp
  have  $q$ -final-zero: PQueryCounter  $s$ -final = 0
    using  $p$ .send-outcome[OF final-send]  $q$ -s6-zero by simp
  have  $q$ -replay-zero: PQueryCounter replay-state = 0
    using replay unfolding verifier-replay-state-def by simp
  show ?thesis
    using  $q$ -final-zero  $q$ -replay-zero by simp
qed

```

```

lemma prover-alpha-lookups-in-replay-state:
assumes create-f:
  Some ( $f$ -merkle,  $s1$ )  $\in$ 
    set-dist (execute ( $p$ .create p.f-eval) init-state)
and send-f:
  Some ( $()$ ,  $s2$ )  $\in$  set-dist (execute ( $p$ .send (value f-merkle))  $s1$ )
and alphas:
  Some ( $as$ ,  $s3$ )  $\in$ 
    set-dist (execute
      (mmap (replicate (length spec)
        (do {
           $a \leftarrow p$ .receive-alpha-challenge;
           $p$ .send  $a$ ;
          return  $a$ 
        })))  $s2$ )

```



```

and later-hash:  $s3 \leq \text{prover-state}$ 
and replay:  $\text{replay-state} = \text{verifier-replay-state prover-state}$ 
shows
   $\forall i < \text{length as.}$ 
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i)
        (foldl concat (concat (PState replay-state) (value f-merkle)) (take i as))) =
        Some (as ! i)
proof –
  have st-s1: PState s1 = 0
    using create-preserves-state[OF create-f]
    unfolding init-state-def by simp
  have send-res:
    s2 = s1 (
      PState := concat (PState s1) (value f-merkle),
      PTranscript := value f-merkle # PTranscript s1)
    using p.send-outcome[OF send-f] .
  have st-s2: PState s2 = concat 0 (value f-merkle)
    using st-s1 send-res by simp
  have alpha-s1-zero: PAlphaCounter s1 = 0
    using p.create-preserves-channel(5)[OF create-f]
    unfolding init-state-def by simp
  have alpha-s2-zero: PAlphaCounter s2 = 0
    using p.send-outcome[OF send-f] alpha-s1-zero by simp
  have alpha-data:
    length as = length spec  $\wedge$ 
    s2  $\leq$  s3  $\wedge$ 
    PState s3 = foldl concat (PState s2) as  $\wedge$ 
    PTranscript s3 = rev as @ PTranscript s2  $\wedge$ 
    ( $\forall i < \text{length as.}$ 
      fmlookup (HashMap s3)
        (AlphaChallenge (PAlphaCounter s2 + i)
          (foldl concat (PState s2) (take i as))) = Some (as ! i))
    using alpha-mmap-outcome[OF alphas] by simp
  show ?thesis
  proof (intro allI impI)
    fix i
    assume i-bound:  $i < \text{length as}$ 
    have lookup-s3:
      fmlookup (HashMap s3)
        (AlphaChallenge (PAlphaCounter s2 + i)
          (foldl concat (PState s2) (take i as))) = Some (as ! i)
      using alpha-data i-bound by simp
    have lookup-final:
      fmlookup (HashMap prover-state)
        (AlphaChallenge (PAlphaCounter s2 + i)
          (foldl concat (PState s2) (take i as))) =
        Some (as ! i)
      using p.hash-extension-lookup[OF lookup-s3 later-hash] .

```

```

show fmlookup (HashMap replay-state)
  (AlphaChallenge (PAlphaCounter replay-state + i)
    (foldl concat (concat (PState replay-state) (value f-merkle)) (take i as))) =
  Some (as ! i)
using lookup-final st-s2 replay alpha-s2-zero
unfolding verifier-replay-state-def by simp
qed
qed

end

end

```

```

theory Completeness-Replay
imports Completeness-Transcript
begin

```

2.4 Verifier Replay

Verifier replay of honest root, alpha, degree, and FRI-prefix transcript phases.

```

context verification
begin

```

```

lemma verifier-replay-root-alpha-degree-prefix:
assumes prefix:
  PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
and later: transcript-extends prover-state s4
and replay: replay-state = verifier-replay-state prover-state
shows  $\exists$  rest.
  PTranscript replay-state = value f-merkle # as @ [of-nat (degree cp')] @ rest
proof –
from later obtain xs where
  tr-final: PTranscript prover-state = xs @ PTranscript s4
unfolding transcript-extends-def by auto
have PTranscript replay-state =
  rev (xs @ (of-nat (degree cp') # rev as @ [value f-merkle]))
using replay tr-final prefix
unfolding verifier-replay-state-def by simp
also have ... = value f-merkle # as @ [of-nat (degree cp')] @ rev xs
by simp
finally show ?thesis by blast
qed

```

```

lemma verifier-root-read-from-replay-prefix:
assumes prefix:
  PTranscript replay-state = value f-merkle # as @ [of-nat (degree cp')] @ rest
and read-fr:
  Some (fr, s1)  $\in$  set-dist (execute p.read replay-state)

```

shows
 $fr = \text{value } f\text{-merkle} \wedge$
 $P\text{Transcript } s1 = as @ [of\text{-nat } (degree\ cp')] @ rest$
proof –
have *start-eq*:
 $replay\text{-state} =$
 $replay\text{-state}(\text{PTranscript} := \text{value } f\text{-merkle} \# (as @ [of\text{-nat } (degree\ cp')] @ rest))$
using *prefix* **by** *simp*
have *read-start*:
 $Some\ (fr, s1) \in$
 $set\text{-dist}\ (execute\ p.read$
 $(replay\text{-state}(\text{PTranscript} := \text{value } f\text{-merkle} \# (as @ [of\text{-nat } (degree\ cp')] @ rest))))$
by (*subst start-eq[symmetric]*) (*rule read-fr*)
have *replay*:
 $fr = \text{value } f\text{-merkle} \wedge$
 $s1 = replay\text{-state}(\text{PState} := \text{concat}\ (\text{PState}\ replay\text{-state})\ (\text{value } f\text{-merkle}),$
 $\text{PTranscript} := as @ [of\text{-nat } (degree\ cp')] @ rest)$
by (*rule p.read-cons-outcome[OF read-start]*)
then show *?thesis* **by** *simp*
qed

lemma *honest-root-alpha-degree-prefix-no-failure*:
assumes *htv*: *honest-trace-valid*
and *len-as*: $length\ as = length\ spec$
and *prefix*:
 $P\text{Transcript } replay\text{-state} =$
 $\text{value } f\text{-merkle} \# as @ [of\text{-nat } (degree\ (p.cp\ as\ p.f\text{-powers}))] @ rest$
and *lookups*:
 $\forall i < length\ as.$
 $fmllookup\ (HashMap\ replay\text{-state})$
 $(AlphaChallenge\ (PAlphaCounter\ replay\text{-state} + i)$
 $(foldl\ concat\ (concat\ (\text{PState}\ replay\text{-state})\ (\text{value } f\text{-merkle}))\ (take\ i\ as))) =$
 $Some\ (as\ !\ i)$

shows
 $None \notin dom\ (dist\ (execute$
 $(do\ \{$
 $fr \leftarrow p.read;$
 $as' \leftarrow mmap\ (replicate\ (length\ spec)$
 $(do\ \{$
 $a0 \leftarrow p.receive\text{-alpha}\text{-challenge};$
 $let\ a0' = a0;$
 $a1 \leftarrow p.read;$
 $let\ a1' = a1;$
 $assert\ (a0' = a1');$
 $return\ a1'$
 $\}));$
 $dg \leftarrow p.read;$

```

    assert (to-nat dg ≤ p.maxDegree);
    return (fr, as', dg)
  }) replay-state))
proof –
  let ?dg = of-nat (degree (p.cp as p.f-powers))
  let ?round =
    do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }
  show ?thesis
proof (rule no-failure-bindI)
  show None ∉ dom (dist (execute p.read replay-state))
    by (rule p.read-no-failure) (use prefix in simp)
next
  fix fr s1
  assume read-fr: Some (fr, s1) ∈ set-dist (execute p.read replay-state)
  have read-start:
    Some (fr, s1) ∈
      set-dist (execute p.read
        (replay-state(PTranscript := value f-merkle # (as @ [?dg] @ rest))))
  proof –
    have start-eq:
      replay-state =
        replay-state(PTranscript := value f-merkle # (as @ [?dg] @ rest))
    using prefix by simp
  show ?thesis
    by (subst start-eq[symmetric]) (rule read-fr)
qed
have read-res:
  fr = value f-merkle ∧
  s1 =
    replay-state(
      PState := concat (PState replay-state) (value f-merkle),
      PTranscript := as @ [?dg] @ rest)
  using p.read-cons-outcome[OF read-start] by simp
have alpha-lookups:
  ∀ i < length as.
    fmlookup (HashMap s1)
      (AlphaChallenge (PAlphaCounter s1 + i)
        (foldl concat (PState s1) (take i as))) = Some (as ! i)
  using lookups read-res by simp
have alpha-nf:
  None ∉ dom (dist (execute
    (mmap (replicate (length spec) ?round)) s1))

```

```

proof –
  have s1-update:  $s1 \langle P\text{Transcript} := as @ [?dg] @ rest \rangle = s1$ 
    using read-res by simp
  have nf:
     $None \notin \text{dom} (\text{dist} (\text{execute}$ 
       $(\text{mmap} (\text{replicate} (\text{length } as) ?round))$ 
       $(s1 \langle P\text{Transcript} := as @ [?dg] @ rest \rangle)))$ 
    using honest-alpha-mmap-replay-no-failure[OF alpha-lookups, of ?dg # rest]
    by simp
  show ?thesis
    using nf unfolding len-as[symmetric] s1-update .
qed
show  $None \notin \text{dom} (\text{dist} (\text{execute}$ 
   $(\text{mmap} (\text{replicate} (\text{length } spec) ?round) \gg=$ 
   $(\lambda as'. p.\text{read} \gg=$ 
   $(\lambda dg. \text{assert} (\text{to-nat } dg \leq p.\text{maxDegree}) \gg=$ 
   $(\lambda -. \text{return} (fr, as', dg))))))$ 
   $s1))$ 
proof (rule no-failure-bindI[OF alpha-nf])
  fix as' s2
  assume alpha-out:
     $Some (as', s2) \in$ 
     $\text{set-dist} (\text{execute} (\text{mmap} (\text{replicate} (\text{length } spec) ?round)) s1)$ 
  have s1-update:  $s1 \langle P\text{Transcript} := as @ [?dg] @ rest \rangle = s1$ 
    using read-res by simp
  have alpha-out':
     $Some (as', s2) \in$ 
     $\text{set-dist} (\text{execute} (\text{mmap} (\text{replicate} (\text{length } as) ?round))$ 
     $(s1 \langle P\text{Transcript} := as @ [?dg] @ rest \rangle)))$ 
    using alpha-out unfolding len-as[symmetric] s1-update .
  have alpha-res:
     $as' = as \wedge$ 
     $P\text{State } s2 = \text{foldl concat} (P\text{State } s1) as \wedge$ 
     $P\text{Transcript } s2 = ?dg \# rest \wedge$ 
     $s1 \leq s2$ 
    using honest-alpha-mmap-replay-outcome[OF alpha-lookups alpha-out'] by simp
  show  $None \notin \text{dom} (\text{dist} (\text{execute}$ 
     $(p.\text{read} \gg=$ 
     $(\lambda dg. \text{assert} (\text{to-nat } dg \leq p.\text{maxDegree}) \gg=$ 
     $(\lambda -. \text{return} (fr, as', dg)))) s2))$ 
  proof (rule no-failure-bindI)
    show  $None \notin \text{dom} (\text{dist} (\text{execute } p.\text{read } s2))$ 
    by (rule p.read-no-failure) (use alpha-res in simp)
  next
    fix dg s3
    assume read-dg:  $Some (dg, s3) \in \text{set-dist} (\text{execute } p.\text{read } s2)$ 
    have read-dg-start:
       $Some (dg, s3) \in$ 
       $\text{set-dist} (\text{execute } p.\text{read} (s2 \langle P\text{Transcript} := ?dg \# rest \rangle)))$ 

```

```

proof -
  have s2-update:  $s2 = s2(PTranscript := ?dg \# rest)$ 
    using alpha-res by simp
  show ?thesis
    by (subst s2-update[symmetric]) (rule read-dg)
qed
have dg-res:
   $dg = ?dg \wedge$ 
   $s3 = s2(PState := concat (PState s2) ?dg, PTranscript := rest)$ 
  using p.read-cons-outcome[OF read-dg-start] by simp
have degree-ok:  $to\_nat\ dg \leq p.maxDegree$ 
  using honest-degree-message-assert-hol[OF htv, of as] dg-res by simp
show  $None \notin dom\ (dist\ (execute$ 
  ( $assert\ (to\_nat\ dg \leq p.maxDegree) \gg (\lambda-. return\ (fr, as', dg)))\ s3))$ 
  using degree-ok by (simp add: assert-def)
qed
qed
qed
qed

lemma honest-root-alpha-degree-prefix-outcome:
assumes htv: honest-trace-valid
and len-as:  $length\ as = length\ spec$ 
and prefix:
   $PTranscript\ replay\_state =$ 
   $value\ f\_merkle \# as @ [of\_nat\ (degree\ (p.cp\ as\ p.f\_powers))] @ rest$ 
and lookups:
   $\forall i < length\ as.$ 
   $fmllookup\ (HashMap\ replay\_state)$ 
  ( $AlphaChallenge\ (PAlphaCounter\ replay\_state + i)$ 
  ( $foldl\ concat\ (concat\ (PState\ replay\_state)\ (value\ f\_merkle))\ (take\ i\ as))) =$ 
   $Some\ (as\ !\ i)$ 
and outcome:
   $Some\ ((fr, as', dg), t) \in set\_dist\ (execute$ 
  ( $do\ \{$ 
     $fr \leftarrow p.read;$ 
     $as' \leftarrow mmap\ (replicate\ (length\ spec)$ 
      ( $do\ \{$ 
         $a0 \leftarrow p.receive\_alpha\_challenge;$ 
         $let\ a0' = a0;$ 
         $a1 \leftarrow p.read;$ 
         $let\ a1' = a1;$ 
         $assert\ (a0' = a1');$ 
         $return\ a1'$ 
       $\}));$ 
     $dg \leftarrow p.read;$ 
     $assert\ (to\_nat\ dg \leq p.maxDegree);$ 
     $return\ (fr, as', dg)$ 
   $\})\ replay\_state)$ 

```

```

shows
  fr = value f-merkle ∧
  as' = as ∧
  dg = of-nat (degree (p.cp as p.f-powers)) ∧
  PState t =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers))) ∧
  PTranscript t = rest ∧
  replay-state ≤ t
proof -
  let ?dg = of-nat (degree (p.cp as p.f-powers))
  let ?round =
    do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }
  from outcome obtain s1 s2 s3 where
    read-fr: Some (fr, s1) ∈ set-dist (execute p.read replay-state)
    and alphas:
      Some (as', s2) ∈
        set-dist (execute (mmap (replicate (length spec) ?round)) s1)
    and read-dg: Some (dg, s3) ∈ set-dist (execute p.read s2)
    and degree-assert:
      Some ((), t) ∈ set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s3)
    by (auto elim!: p.set-dist-bindE)
  have read-start:
    Some (fr, s1) ∈
      set-dist (execute p.read
        (replay-state(PTranscript := value f-merkle # (as @ [?dg] @ rest))))
  proof -
    have start-eq:
      replay-state =
        replay-state(PTranscript := value f-merkle # (as @ [?dg] @ rest))
    using prefix by simp
    show ?thesis
      by (subst start-eq[symmetric]) (rule read-fr)
  qed
  have read-res:
    fr = value f-merkle ∧
    s1 =
      replay-state(
        PState := concat (PState replay-state) (value f-merkle),
        PTranscript := as @ [?dg] @ rest)
    using p.read-cons-outcome[OF read-start] by simp
  have alpha-lookups:

```

```

   $\forall i < \text{length } as.$ 
   $\text{fmlookup } (\text{HashMap } s1)$ 
     $(\text{AlphaChallenge } (\text{PAlphaCounter } s1 + i)$ 
       $(\text{foldl concat } (\text{PState } s1) (\text{take } i \text{ as}))) = \text{Some } (as ! i)$ 
     $\text{using lookups read-res by simp}$ 
  have  $s1\text{-update}$ :  $s1(P\text{Transcript} := as @ [?dg] @ rest) = s1$ 
     $\text{using read-res by simp}$ 
  have  $\alpha\text{-out}$ :
     $\text{Some } (as', s2) \in$ 
       $\text{set-dist } (\text{execute } (\text{mmap } (\text{replicate } (\text{length } as) ?round)))$ 
       $(s1(P\text{Transcript} := as @ [?dg] @ rest))$ 
     $\text{using alphas unfolding len-as[symmetric] s1-update .}$ 
  have  $\alpha\text{-res}$ :
     $as' = as \wedge$ 
     $\text{PState } s2 = \text{foldl concat } (\text{PState } s1) as \wedge$ 
     $\text{PTranscript } s2 = ?dg \# rest \wedge$ 
     $s1 \leq s2$ 
     $\text{using honest-alpha-mmap-replay-outcome[OF alpha-lookups alpha-out] by simp}$ 
  have  $\text{read-dg-start}$ :
     $\text{Some } (dg, s3) \in$ 
       $\text{set-dist } (\text{execute } p.\text{read } (s2(P\text{Transcript} := ?dg \# rest)))$ 
  proof -
    have  $s2\text{-update}$ :  $s2 = s2(P\text{Transcript} := ?dg \# rest)$ 
       $\text{using alpha-res by simp}$ 
    show  $?thesis$ 
       $\text{by (subst s2-update[symmetric]) (rule read-dg)}$ 
  qed
  have  $dg\text{-res}$ :
     $dg = ?dg \wedge$ 
     $s3 = s2(P\text{State} := \text{concat } (\text{PState } s2) ?dg, P\text{Transcript} := rest)$ 
     $\text{using p.read-cons-outcome[OF read-dg-start] by simp}$ 
  have  $\text{degree-ok}$ :  $\text{to-nat } dg \leq p.\text{maxDegree}$ 
     $\text{using honest-degree-message-assert-hol[OF htv, of as] dg-res by simp}$ 
  have  $t\text{-eq}$ :  $t = s3$ 
     $\text{using degree-assert degree-ok by (simp add: assert-def)}$ 
  have  $\text{replay-s1}$ :  $\text{replay-state} \leq s1$ 
     $\text{using read-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp}$ 
  have  $s2\text{-s3}$ :  $s2 \leq s3$ 
     $\text{using dg-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp}$ 
  have  $\text{replay-t}$ :  $\text{replay-state} \leq t$ 
     $\text{using replay-s1 alpha-res s2-s3 unfolding t-eq by (meson p.hash-ext-trans)}$ 
  show  $?thesis$ 
     $\text{using read-res alpha-res dg-res t-eq replay-t by simp}$ 
  qed

```

```

lemma  $\text{honest-root-alpha-degree-fri-prefix-no-failure}$ :
  assumes  $htv$ :  $\text{honest-trace-valid}$ 
  and  $\text{len-as}$ :  $\text{length } as = \text{length spec}$ 
  and  $\text{len-challenges}$ :  $\text{length challenges} = \text{length roots}$ 

```



```

and prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @ roots @ rest
and alpha-lookups:
  ∀ i < length as.
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i)
        (foldl concat (concat (PState replay-state) (value f-merkle)) (take i as))) =
        Some (as ! i)
and fri-lookups:
  ∀ i < length roots.
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          (of-nat (degree (p.cp as p.f-powers))))
        (take (Suc i) roots))) =
        Some (challenges ! i)
shows
  None ∉ dom (dist (execute
    (do {
      fr ← p.read;
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        })));
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      fl ← ntimes v.receive-fri-commits (length roots);
      return (fr, as', dg, fl)
    }) replay-state))
proof –
  let ?dg = of-nat (degree (p.cp as p.f-powers))
  let ?round =
    do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }
  let ?prefix =
    do {
      fr ← p.read;

```

```

    as' ← mmap (replicate (length spec) ?round);
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    return (fr, as', dg)
  }
let ?full =
  do {
    fr ← p.read;
    as' ← mmap (replicate (length spec) ?round);
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    fl ← ntimes v.receive-fri-commits (length roots);
    return (fr, as', dg, fl)
  }
have prefix-nf:
  None ∉ dom (dist (execute ?prefix replay-state))
proof -
  have prefix-root-alpha-degree:
    PTranscript replay-state =
      value f-merkle # as @ [?dg] @ (roots @ rest)
  using prefix by simp
  show ?thesis
    by (rule honest-root-alpha-degree-prefix-no-failure[
      OF htv len-as prefix-root-alpha-degree alpha-lookups])
qed
have full-eq:
  ?full =
    (?prefix ≫=
      (λout.
        ntimes v.receive-fri-commits (length roots) ≫=
          (λfl. return (case out of (fr, as', dg) ⇒ (fr, as', dg, fl)))))
  by (simp add: sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full-eq
proof (rule no-failure-bindI[OF prefix-nf])
  fix out s-prefix
  assume prefix-out:
    Some (out, s-prefix) ∈ set-dist (execute ?prefix replay-state)
  obtain fr as' dg where out-eq: out = (fr, as', dg)
  by (cases out)
  have prefix-out':
    Some ((fr, as', dg), s-prefix) ∈ set-dist (execute ?prefix replay-state)
  using prefix-out unfolding out-eq .
  have prefix-res:
    fr = value f-merkle ∧
    as' = as ∧
    dg = ?dg ∧
    PState s-prefix =
      concat (foldl concat (concat (PState replay-state) (value f-merkle)) as) ?dg ∧

```

```

    PTranscript s-prefix = roots @ rest ∧
    replay-state ≤ s-prefix
proof –
  have prefix-root-alpha-degree:
    PTranscript replay-state =
      value f-merkle # as @ [?dg] @ (roots @ rest)
    using prefix by simp
  show ?thesis
    by (rule honest-root-alpha-degree-prefix-outcome[
      OF htv len-as prefix-root-alpha-degree alpha-lookups prefix-out])
qed
have fri-lookups-prefix:
  ∀ i < length roots.
    fmlookup (HashMap s-prefix)
      (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-replay:
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          ?dg)
        (take (Suc i) roots))) =
      Some (challenges ! i)
    using fri-lookups i-bound by simp
  show fmlookup (HashMap s-prefix)
    (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
    Some (challenges ! i)
    using p.hash-extension-lookup[OF lookup-replay, of s-prefix] prefix-res
    by simp
qed
have fri-nf-start:
  None ∉ dom (dist (execute
    (ntimes v.receive-fri-commits (length roots))
    (s-prefix(PTranscript := roots @ rest))))
  using honest-receive-fri-commits-list-no-failure[
    OF fri-lookups-prefix len-challenges, of rest] .
have s-prefix-update: s-prefix(PTranscript := roots @ rest) = s-prefix
  using prefix-res by simp
have fri-nf:
  None ∉ dom (dist (execute
    (ntimes v.receive-fri-commits (length roots)) s-prefix))
  using fri-nf-start unfolding s-prefix-update .
show None ∉ dom (dist (execute
  (ntimes v.receive-fri-commits (length roots)) >>=
  (λfl. return (case out of (fr, as', dg) ⇒ (fr, as', dg, fl))))
  s-prefix))

```

```

    by (rule no-failure-bind-returnI[OF fri-nf])
qed
qed

lemma honest-root-alpha-degree-fri-final-prefix-no-failure:
  assumes htv: honest-trace-valid
  and len-as: length as = length spec
  and len-challenges: length challenges = length roots
  and prefix:
    PTranscript replay-state =
      value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
        roots @ [final-msg] @ rest
  and alpha-lookups:
    ∀ i < length as.
      fmlookup (HashMap replay-state)
        (AlphaChallenge (PAlphaCounter replay-state + i)
          (foldl concat (concat (PState replay-state) (value f-merkle)) (take i as))) =
          Some (as ! i)
  and fri-lookups:
    ∀ i < length roots.
      fmlookup (HashMap replay-state)
        (FiatShamirChallenge (foldl concat
          (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
            (of-nat (degree (p.cp as p.f-powers))))
          (take (Suc i) roots))) =
          Some (challenges ! i)
  shows
    None ∉ dom (dist (execute
      (do {
        fr ← p.read;
        as' ← mmap (replicate (length spec)
          (do {
            a0 ← p.receive-alpha-challenge;
            let a0' = a0;
            a1 ← p.read;
            let a1' = a1;
            assert (a0' = a1');
            return a1'
          })));
        dg ← p.read;
        assert (to-nat dg ≤ p.maxDegree);
        fl ← ntimes v.receive-fri-commits (length roots);
        final ← p.read;
        return (fr, as', dg, fl, final)
      }) replay-state))
  proof -
    let ?dg = of-nat (degree (p.cp as p.f-powers))
    let ?round =
      do {

```

```

    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }
let ?prefix =
do {
  fr ← p.read;
  as' ← mmap (replicate (length spec) ?round);
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  return (fr, as', dg)
}
let ?rad-fri =
do {
  fr ← p.read;
  as' ← mmap (replicate (length spec) ?round);
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits (length roots);
  return (fr, as', dg, fl)
}
have prefix-nf:
  None ∉ dom (dist (execute ?rad-fri replay-state))
proof -
  have prefix': PTranscript replay-state =
    value f-merkle # as @ [?dg] @ roots @ ([final-msg] @ rest)
  using prefix by simp
  show ?thesis
  by (rule honest-root-alpha-degree-fri-prefix-no-failure[
    OF htv len-as len-challenges prefix' alpha-lookups fri-lookups])
qed
have full-eq:
  (do {
    fr ← p.read;
    as' ← mmap (replicate (length spec) ?round);
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    fl ← ntimes v.receive-fri-commits (length roots);
    final ← p.read;
    return (fr, as', dg, fl, final)
  }) =
  (?rad-fri ≫=
    (λout. p.read ≫=
      (λfinal. return (case out of (fr, as', dg, fl) ⇒
        (fr, as', dg, fl, final))))))
  by (simp add: sm-bind-assoc split: prod.splits)

```

```

show ?thesis
  unfolding full-eq
proof (rule no-failure-bindI[OF prefix-nf])
  fix out s-fri
  assume prefix-out:
    Some (out, s-fri) ∈ set-dist (execute ?rad-fri replay-state)
  obtain fr as' dg fl where out-eq: out = (fr, as', dg, fl)
  by (cases out) auto
from prefix-out[unfolded out-eq] obtain s1 s2 s3 s-prefix where
  read-fr: Some (fr, s1) ∈ set-dist (execute p.read replay-state)
  and alpha-out:
    Some (as', s2) ∈
      set-dist (execute (mmap (replicate (length spec) ?round)) s1)
  and read-dg: Some (dg, s3) ∈ set-dist (execute p.read s2)
  and degree-assert:
    Some ((), s-prefix) ∈
      set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s3)
  and fri-out:
    Some (fl, s-fri) ∈
      set-dist (execute (ntimes v.receive-fri-commits (length roots)) s-prefix)
  by (auto elim!: p.set-dist-bindE split: prod.splits)
have prefix3-out:
  Some ((fr, as', dg), s-prefix) ∈ set-dist (execute ?prefix replay-state)
  apply (rule set-dist-bindI[OF read-fr])
  apply (rule set-dist-bindI[OF alpha-out])
  apply (rule set-dist-bindI[OF read-dg])
  apply (rule set-dist-bindI[OF degree-assert])
  by simp
have prefix3:
  PTranscript replay-state =
    value f-merkle # as @ [?dg] @ (roots @ [final-msg] @ rest)
  using prefix by simp
have prefix-res:
  fr = value f-merkle ∧
  as' = as ∧
  dg = ?dg ∧
  PState s-prefix =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as) ?dg ∧
  PTranscript s-prefix = roots @ [final-msg] @ rest ∧
  replay-state ≤ s-prefix
  by (rule honest-root-alpha-degree-prefix-outcome[
    OF htv len-as prefix3 alpha-lookups prefix3-out])
have fri-lookups-prefix:
  ∀ i < length roots.
    fmlookup (HashMap s-prefix)
      (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i

```

```

assume i-bound:  $i < \text{length roots}$ 
have lookup-replay:
  fmlookup (HashMap replay-state)
    (FiatShamirChallenge (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        ?dg)
      (take (Suc i) roots))) =
    Some (challenges ! i)
using fri-lookups i-bound by simp
show fmlookup (HashMap s-prefix)
  (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
  Some (challenges ! i)
using p.hash-extension-lookup[OF lookup-replay, of s-prefix] prefix-res
by simp
qed
have fri-out':
  Some (fl, s-fri)  $\in$ 
    set-dist (execute (ntimes v.receive-fri-commits (length roots))
      (s-prefix(PTranscript := roots @ ([final-msg] @ rest))))
proof –
  have s-prefix(PTranscript := roots @ ([final-msg] @ rest)) = s-prefix
    using prefix-res by (cases s-prefix) simp
  then show ?thesis
    using fri-out by simp
qed
have tr-fri: PTranscript s-fri = final-msg # rest
  using honest-receive-fri-commits-list-outcome[
    OF fri-lookups-prefix len-challenges fri-out']
  by simp
show None  $\notin \text{dom}$  (dist (execute
  (p.read  $\gg$ 
    ( $\lambda \text{final. return (case out of (fr, as', dg, fl) \Rightarrow$ 
      (fr, as', dg, fl, final)))) s-fri))
  unfolding out-eq
  by (rule no-failure-bind-returnI)
    (rule p.read-no-failure, simp add: tr-fri)
qed
qed

lemma honest-root-alpha-degree-fri-final-prefix-outcome:
assumes htv: honest-trace-valid
and len-as: length as = length spec
and len-challenges: length challenges = length roots
and prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
    roots @ [final-msg] @ rest
and alpha-lookups:
   $\forall i < \text{length as.}$ 

```

```

    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i)
        (foldl concat (concat (PState replay-state) (value f-merkle)) (take i as))) =
      Some (as ! i)
  and fri-lookups:
    ∀ i < length roots.
      fmlookup (HashMap replay-state)
        (FiatShamirChallenge (foldl concat
          (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
            (of-nat (degree (p.cp as p.f-powers))))
          (take (Suc i) roots))) =
          Some (challenges ! i)
  and outcome:
    Some ((fr, as', dg, fl, final), t) ∈ set-dist (execute
      (do {
        fr ← p.read;
        as' ← mmap (replicate (length spec)
          (do {
            a0 ← p.receive-alpha-challenge;
            let a0' = a0;
            a1 ← p.read;
            let a1' = a1;
            assert (a0' = a1');
            return a1'
          })));
        dg ← p.read;
        assert (to-nat dg ≤ p.maxDegree);
        fl ← ntimes v.receive-fri-commits (length roots);
        final ← p.read;
        return (fr, as', dg, fl, final)
      }) replay-state)
  shows
    fr = value f-merkle ∧
    as' = as ∧
    dg = of-nat (degree (p.cp as p.f-powers)) ∧
    fl = zip challenges roots ∧
    final = final-msg ∧
    PState t =
      concat
        (foldl concat
          (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
            (of-nat (degree (p.cp as p.f-powers))))
          roots)
        final-msg ∧
    PTranscript t = rest ∧
    replay-state ≤ t
  proof -
    let ?dg = of-nat (degree (p.cp as p.f-powers))
    let ?round =

```



```

do {
  a0 ← p.receive-alpha-challenge;
  let a0' = a0;
  a1 ← p.read;
  let a1' = a1;
  assert (a0' = a1');
  return a1'
}
let ?rad =
do {
  fr ← p.read;
  as' ← mmap (replicate (length spec) ?round);
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  return (fr, as', dg)
}
from outcome obtain s1 s2 s3 s-prefix s-fri where
  read-fr: Some (fr, s1) ∈ set-dist (execute p.read replay-state)
  and alpha-out:
    Some (as', s2) ∈
      set-dist (execute (mmap (replicate (length spec) ?round)) s1)
  and read-dg: Some (dg, s3) ∈ set-dist (execute p.read s2)
  and degree-assert:
    Some ((), s-prefix) ∈
      set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s3)
  and fri-out:
    Some (fl, s-fri) ∈
      set-dist (execute (ntimes v.receive-fri-commits (length roots)) s-prefix)
  and read-final:
    Some (final, t) ∈ set-dist (execute p.read s-fri)
  by (auto elim!: p.set-dist-bindE)
have rad-out:
  Some ((fr, as', dg), s-prefix) ∈ set-dist (execute ?rad replay-state)
  apply (rule set-dist-bindI[OF read-fr])
  apply (rule set-dist-bindI[OF alpha-out])
  apply (rule set-dist-bindI[OF read-dg])
  apply (rule set-dist-bindI[OF degree-assert])
  by simp
have rad-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [?dg] @ (roots @ [final-msg] @ rest)
  using prefix by simp
have rad-res:
  fr = value f-merkle ∧
  as' = as ∧
  dg = ?dg ∧
  PState s-prefix =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as) ?dg ∧
  PTranscript s-prefix = roots @ [final-msg] @ rest ∧

```

```

    replay-state ≤ s-prefix
  by (rule honest-root-alpha-degree-prefix-outcome[
    OF htv len-as rad-prefix alpha-lookups rad-out])
have fri-lookups-prefix:
  ∀ i < length roots.
    fmlookup (HashMap s-prefix)
      (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-replay:
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          ?dg)
        (take (Suc i) roots))) =
      Some (challenges ! i)
  using fri-lookups i-bound by simp
show fmlookup (HashMap s-prefix)
  (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
  Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-replay, of s-prefix] rad-res
  by simp
qed
have s-prefix-update:
  s-prefix(PTranscript := roots @ [final-msg] @ rest) = s-prefix
  using rad-res by simp
have fri-out':
  Some (fl, s-fri) ∈
    set-dist (execute (ntimes v.receive-fri-commits (length roots))
      (s-prefix(PTranscript := roots @ ([final-msg] @ rest))))
  using fri-out unfolding s-prefix-update by simp
have fri-res:
  fl = zip challenges roots ∧
  PState s-fri = foldl concat (PState s-prefix) roots ∧
  PTranscript s-fri = [final-msg] @ rest ∧
  s-prefix ≤ s-fri
  using honest-receive-fri-commits-list-outcome[
    OF fri-lookups-prefix len-challenges fri-out'] .
have read-final':
  Some (final, t) ∈
    set-dist (execute p.read (s-fri(PTranscript := final-msg # rest)))
proof -
  have update: s-fri = s-fri(PTranscript := final-msg # rest)
    using fri-res by simp
  show ?thesis
    by (subst update[symmetric]) (rule read-final)
qed

```

```

have final-res:
  final = final-msg ∧
    t = s-fri(PState := concat (PState s-fri) final-msg, PTranscript := rest)
  using p.read-cons-outcome[OF read-final] by simp
have replay-t: replay-state ≤ t
proof −
  have s-fri-t: s-fri ≤ t
    using final-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  show ?thesis
    using rad-res fri-res s-fri-t by (meson p.hash-ext-trans)
qed
show ?thesis
  using rad-res fri-res final-res replay-t by simp
qed

lemma honest-root-alpha-degree-fri-final-dynamic-prefix-outcome:
assumes htv: honest-trace-valid
and len-as: length as = length spec
and len-challenges: length challenges = length roots
and roots-rounds:
  length roots = ceil-log (degree (p.cp as p.f-powers) + 1)
and prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
      roots @ [final-msg] @ rest
and alpha-lookups:
  ∀ i < length as.
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i)
        (foldl concat (concat (PState replay-state) (value f-merkle)) (take i as))) =
        Some (as ! i)
and fri-lookups:
  ∀ i < length roots.
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          (of-nat (degree (p.cp as p.f-powers))))
        (take (Suc i) roots))) =
        Some (challenges ! i)
and outcome:
  Some ((fr, as', dg, fl, final), t) ∈ set-dist (execute
    (do {
      fr ← p.read;
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;

```

```

    assert (a0' = a1');
    return a1'
  ));
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
  final ← p.read;
  return (fr, as', dg, fl, final)
}) replay-state)
shows
fr = value f-merkle ∧
as' = as ∧
dg = of-nat (degree (p.cp as p.f-powers)) ∧
fl = zip challenges roots ∧
final = final-msg ∧
PState t =
  concat
    (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        (of-nat (degree (p.cp as p.f-powers)))))
      roots)
    final-msg ∧
PTranscript t = rest ∧
replay-state ≤ t
proof –
let ?dg = of-nat (degree (p.cp as p.f-powers))
let ?round =
  do {
    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }
let ?rad =
  do {
    fr ← p.read;
    as' ← mmap (replicate (length spec) ?round);
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    return (fr, as', dg)
  }
from outcome obtain s1 s2 s3 s-prefix s-fri where
  read-fr: Some (fr, s1) ∈ set-dist (execute p.read replay-state)
and alpha-out:
  Some (as', s2) ∈
    set-dist (execute (mmap (replicate (length spec) ?round)) s1)
and read-dg: Some (dg, s3) ∈ set-dist (execute p.read s2)

```

```

and degree-assert:
  Some ((l), s-prefix) ∈
    set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s3)
and fri-out:
  Some (fl, s-fri) ∈
    set-dist (execute (ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1)))) s-prefix)
and read-final:
  Some (final, t) ∈ set-dist (execute p.read s-fri)
by (auto elim!: p.set-dist-bindE)
have rad-out:
  Some ((fr, as', dg), s-prefix) ∈ set-dist (execute ?rad replay-state)
apply (rule set-dist-bindI[OF read-fr])
apply (rule set-dist-bindI[OF alpha-out])
apply (rule set-dist-bindI[OF read-dg])
apply (rule set-dist-bindI[OF degree-assert])
by simp
have rad-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [?dg] @ (roots @ [final-msg] @ rest)
using prefix by simp
have rad-res:
  fr = value f-merkle ∧
  as' = as ∧
  dg = ?dg ∧
  PState s-prefix =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as) ?dg ∧
  PTranscript s-prefix = roots @ [final-msg] @ rest ∧
  replay-state ≤ s-prefix
by (rule honest-root-alpha-degree-prefix-outcome[
  OF htv len-as rad-prefix alpha-lookups rad-out])
have rounds-eq: ceil-log (to-nat dg + 1) = length roots
using honest-degree-message-rounds[OF htv, of as] rad-res roots-rounds by simp
have fri-lookups-prefix:
  ∀ i < length roots.
    fmlookup (HashMap s-prefix)
      (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
fix i
assume i-bound: i < length roots
have lookup-replay:
  fmlookup (HashMap replay-state)
    (FiatShamirChallenge (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        ?dg)
      (take (Suc i) roots))) =
    Some (challenges ! i)
using fri-lookups i-bound by simp
show fmlookup (HashMap s-prefix)

```

```

    (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
    Some (challenges ! i)
    using p.hash-extension-lookup[OF lookup-replay, of s-prefix] rad-res
    by simp
qed
have s-prefix-update:
  s-prefix(PTranscript := roots @ [final-msg] @ rest) = s-prefix
  using rad-res by simp
have fri-out':
  Some (fl, s-fri) ∈
    set-dist (execute (ntimes v.receive-fri-commits (length roots))
      (s-prefix(PTranscript := roots @ ([final-msg] @ rest))))
  using fri-out rounds-eq unfolding s-prefix-update by simp
have fri-res:
  fl = zip challenges roots ∧
  PState s-fri = foldl concat (PState s-prefix) roots ∧
  PTranscript s-fri = [final-msg] @ rest ∧
  s-prefix ≤ s-fri
  using honest-receive-fri-commits-list-outcome[
    OF fri-lookups-prefix len-challenges fri-out'] .
have read-final':
  Some (final, t) ∈
    set-dist (execute p.read (s-fri(PTranscript := final-msg # rest)))
proof -
  have update: s-fri = s-fri(PTranscript := final-msg # rest)
    using fri-res by simp
  show ?thesis
    by (subst update[symmetric]) (rule read-final)
qed
have final-res:
  final = final-msg ∧
  t = s-fri(PState := concat (PState s-fri) final-msg, PTranscript := rest)
  using p.read-cons-outcome[OF read-final'] by simp
have replay-t: replay-state ≤ t
proof -
  have s-fri-t: s-fri ≤ t
    using final-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  show ?thesis
    using rad-res fri-res s-fri-t by (meson p.hash-ext-trans)
qed
show ?thesis
  using rad-res fri-res final-res replay-t by simp
qed

```

lemma honest-root-alpha-degree-fri-final-prefix-preserves-query-counter:

assumes outcome:

Some ((fr, as', dg, fl, final), t) ∈ set-dist (execute
 (do {
 fr ← p.read;

```

    as' ← mmap (replicate (length spec)
      (do {
        a0 ← p.receive-alpha-challenge;
        let a0' = a0;
        a1 ← p.read;
        let a1' = a1;
        assert (a0' = a1');
        return a1'
      }));
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    fl ← ntimes v.receive-fri-commits n;
    final ← p.read;
    return (fr, as', dg, fl, final)
  }) s)
shows PQueryCounter t = PQueryCounter s
proof –
  let ?round =
    do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }
from outcome obtain s-read s-alpha s-dg s-assert s-fri where
  read-fr: Some (fr, s-read) ∈ set-dist (execute p.read s)
and alpha-out:
  Some (as', s-alpha) ∈
    set-dist (execute (mmap (replicate (length spec) ?round)) s-read)
and read-dg: Some (dg, s-dg) ∈ set-dist (execute p.read s-alpha)
and assert-dg:
  Some ((), s-assert) ∈
    set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s-dg)
and fri-out:
  Some (fl, s-fri) ∈
    set-dist (execute (ntimes v.receive-fri-commits n) s-assert)
and read-final: Some (final, t) ∈ set-dist (execute p.read s-fri)
by (auto elim!: p.set-dist-bindE)
have q-read: PQueryCounter s-read = PQueryCounter s
  using read-preserves-counters[OF read-fr] by simp
have q-alpha: PQueryCounter s-alpha = PQueryCounter s-read
  by (rule honest-alpha-mmap-replay-preserves-query-counter[OF alpha-out])
have q-dg: PQueryCounter s-dg = PQueryCounter s-alpha
  using read-preserves-counters[OF read-dg] by simp
have s-assert-eq: s-assert = s-dg
  using assert-outcomeD(2)[OF assert-dg] .
have q-fri: PQueryCounter s-fri = PQueryCounter s-assert

```

```

    by (rule receive-fri-commits-preserves-query-counter[OF fri-out])
  have q-final: PQueryCounter t = PQueryCounter s-fri
    using read-preserves-counters[OF read-final] by simp
  show ?thesis
    using q-read q-alpha q-dg q-fri q-final unfolding s-assert-eq by simp
qed

```

lemma *honest-root-alpha-degree-fri-final-dynamic-prefix-preserves-query-counter:*

assumes *outcome:*

```

  Some ((fr, as', dg, fl, final), t) ∈ set-dist (execute
    (do {
      fr ← p.read;
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        })));
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
      final ← p.read;
      return (fr, as', dg, fl, final)
    }) s)

```

shows $PQueryCounter\ t = PQueryCounter\ s$

proof –

let *?round* =

```

  do {
    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }

```

from *outcome* **obtain** *s-read s-alpha s-dg s-assert s-fri* **where**

read-fr: $Some\ (fr,\ s-read) \in set-dist\ (execute\ p.read\ s)$

and *alpha-out:*

$Some\ (as',\ s-alpha) \in set-dist\ (execute\ (mmap\ (replicate\ (length\ spec)\ ?round))\ s-read)$

and *read-dg:* $Some\ (dg,\ s-dg) \in set-dist\ (execute\ p.read\ s-alpha)$

and *assert-dg:*

$Some\ ((),\ s-assert) \in set-dist\ (execute\ (assert\ (to-nat\ dg \leq p.maxDegree))\ s-dg)$

and *fri-out:*

$Some\ (fl,\ s-fri) \in$


```

    set-dist (execute (ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1))) s-assert)
  and read-final: Some (final, t) ∈ set-dist (execute p.read s-fri)
  by (auto elim!: p.set-dist-bindE)
  have q-read: PQueryCounter s-read = PQueryCounter s
    using read-preserves-counters[OF read-fr] by simp
  have q-alpha: PQueryCounter s-alpha = PQueryCounter s-read
    by (rule honest-alpha-mmap-replay-preserves-query-counter[OF alpha-out])
  have q-dg: PQueryCounter s-dg = PQueryCounter s-alpha
    using read-preserves-counters[OF read-dg] by simp
  have s-assert-eq: s-assert = s-dg
    using assert-outcomeD(2)[OF assert-dg] .
  have q-fri: PQueryCounter s-fri = PQueryCounter s-assert
    by (rule receive-fri-commits-preserves-query-counter[OF fri-out])
  have q-final: PQueryCounter t = PQueryCounter s-fri
    using read-preserves-counters[OF read-final] by simp
  show ?thesis
    using q-read q-alpha q-dg q-fri q-final unfolding s-assert-eq by simp
qed

```

lemma honest-root-alpha-degree-fri-final-dynamic-prefix-no-failure:

```

  assumes htv: honest-trace-valid
  and len-as: length as = length spec
  and len-challenges: length challenges = length roots
  and roots-rounds:
    length roots = ceil-log (degree (p.cp as p.f-powers) + 1)
  and prefix:
    PTranscript replay-state =
      value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
        roots @ [final-msg] @ rest
  and alpha-lookups:
    ∀ i < length as.
      fmlookup (HashMap replay-state)
        (AlphaChallenge (PAlphaCounter replay-state + i)
          (foldl concat (concat (PState replay-state) (value f-merkle)) (take i as))) =
        Some (as ! i)
  and fri-lookups:
    ∀ i < length roots.
      fmlookup (HashMap replay-state)
        (FiatShamirChallenge (foldl concat
          (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
            (of-nat (degree (p.cp as p.f-powers))))
          (take (Suc i) roots))) =
        Some (challenges ! i)

```

shows

```

  None ∉ dom (dist (execute
    (do {
      fr ← p.read;
      as' ← mmap (replicate (length spec)
        (do {

```

```

    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }));
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
  final ← p.read;
  return (fr, as', dg, fl, final)
}) replay-state))
proof –
let ?dg = of-nat (degree (p.cp as p.f-powers))
let ?round =
do {
  a0 ← p.receive-alpha-challenge;
  let a0' = a0;
  a1 ← p.read;
  let a1' = a1;
  assert (a0' = a1');
  return a1'
}
let ?rad =
do {
  fr ← p.read;
  as' ← mmap (replicate (length spec) ?round);
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  return (fr, as', dg)
}
have rad-prefix:
PTranscript replay-state =
  value f-merkle # as @ [?dg] @ (roots @ [final-msg] @ rest)
using prefix by simp
have rad-nf:
None ∉ dom (dist (execute ?rad replay-state))
by (rule honest-root-alpha-degree-prefix-no-failure[
  OF htv len-as rad-prefix alpha-lookups])
have full-eq:
(do {
  fr ← p.read;
  as' ← mmap (replicate (length spec) ?round);
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
  final ← p.read;
  return (fr, as', dg, fl, final)

```

```

    }) =
    (?rad >>=
      (λout. case out of (fr, as', dg) ⇒
        ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1)) >>=
          (λfl. p.read >>=
            (λfinal. return (fr, as', dg, fl, final))))))
  by (simp add: sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full-eq
proof (rule no-failure-bindI[OF rad-nf])
  fix out s-prefix
  assume rad-out: Some (out, s-prefix) ∈ set-dist (execute ?rad replay-state)
  obtain fr as' dg where out-eq: out = (fr, as', dg)
  by (cases out) auto
  have rad-out':
    Some ((fr, as', dg), s-prefix) ∈ set-dist (execute ?rad replay-state)
  using rad-out unfolding out-eq .
  have rad-res:
    fr = value f-merkle ∧
    as' = as ∧
    dg = ?dg ∧
    PState s-prefix =
      concat (foldl concat (concat (PState replay-state) (value f-merkle)) as) ?dg ∧
    PTranscript s-prefix = roots @ [final-msg] @ rest ∧
    replay-state ≤ s-prefix
  by (rule honest-root-alpha-degree-prefix-outcome[
    OF htv len-as rad-prefix alpha-lookups rad-out'])
  have rounds-eq: ceil-log (to-nat dg + 1) = length roots
  using honest-degree-message-rounds[OF htv, of as] rad-res roots-rounds by simp
  have fri-lookups-prefix:
    ∀ i < length roots.
      fmlookup (HashMap s-prefix)
        (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
        Some (challenges ! i)
  proof (intro allI impI)
    fix i
    assume i-bound: i < length roots
    have lookup-replay:
      fmlookup (HashMap replay-state)
        (FiatShamirChallenge (foldl concat
          (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
            ?dg)
          (take (Suc i) roots))) =
        Some (challenges ! i)
      using fri-lookups i-bound by simp
    show fmlookup (HashMap s-prefix)
      (FiatShamirChallenge (foldl concat (PState s-prefix) (take (Suc i) roots))) =
      Some (challenges ! i)
    using p.hash-extension-lookup[OF lookup-replay, of s-prefix] rad-res

```

```

    by simp
qed
have s-prefix-update:
  s-prefix( $PTranscript := roots @ [final-msg] @ rest$ ) = s-prefix
  using rad-res by simp
have fri-nf-start:
  None  $\notin$  dom (dist (execute
    (ntimes v.receive-fri-commits (length roots))
    (s-prefix( $PTranscript := roots @ ([final-msg] @ rest)$ ))))
  using honest-receive-fri-commits-list-no-failure[
    OF fri-lookups-prefix len-challenges, of  $[final-msg] @ rest$ ] .
have fri-nf:
  None  $\notin$  dom (dist (execute
    (ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1))) s-prefix))
  using fri-nf-start unfolding rounds-eq s-prefix-update by simp
have cont-nf:
  None  $\notin$  dom (dist (execute
    (ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1))  $\gg$ 
    ( $\lambda fl. p.read \gg (\lambda final. return (fr, as', dg, fl, final))$ )) s-prefix))
proof (rule no-failure-bindI[OF fri-nf])
  fix fl t
  assume fri-out:
    Some (fl, t)  $\in$ 
    set-dist (execute
      (ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1))) s-prefix)
  have fri-out-start:
    Some (fl, t)  $\in$ 
    set-dist (execute
      (ntimes v.receive-fri-commits (length roots))
      (s-prefix( $PTranscript := roots @ ([final-msg] @ rest)$ )))
    using fri-out unfolding rounds-eq s-prefix-update by simp
  have tr-t:  $PTranscript t = final-msg \# rest$ 
    using honest-receive-fri-commits-list-outcome[
      OF fri-lookups-prefix len-challenges fri-out-start]
    by simp
  show None  $\notin$  dom
    (dist (execute (p.read  $\gg$ 
      ( $\lambda final. return (fr, as', dg, fl, final)$ )) t))
    by (rule no-failure-bind-returnI)
      (rule p.read-no-failure, simp add: tr-t)
qed
show None  $\notin$  dom (dist (execute
  ((case out of (fr, as', dg)  $\Rightarrow$ 
    ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1))  $\gg$ 
    ( $\lambda fl. p.read \gg (\lambda final. return (fr, as', dg, fl, final))$ )) s-prefix))
  unfolding out-eq
  using cont-nf by simp
qed
qed

```

lemma *honest-alpha-degree-fri-final-dynamic-prefix-outcome:*
assumes *htv: honest-trace-valid*
and *len-as: length as = length spec*
and *len-challenges: length challenges = length roots*
and *roots-rounds:*
 $\text{length roots} = \text{ceil-log } (\text{degree } (p.\text{cp as } p.\text{f-powers}) + 1)$
and *prefix:*
 $P\text{Transcript replay-state} =$
 $\text{as } @ [\text{of-nat } (\text{degree } (p.\text{cp as } p.\text{f-powers}))] @ \text{roots } @ [\text{final-msg}] @ \text{rest}$
and *alpha-lookups:*
 $\forall i < \text{length as.}$
 $\text{fmlookup } (\text{HashMap replay-state})$
 $(\text{AlphaChallenge } (P\text{AlphaCounter replay-state} + i)$
 $(\text{foldl concat } (P\text{State replay-state}) (\text{take } i \text{ as}))) =$
 $\text{Some } (\text{as } ! i)$
and *fri-lookups:*
 $\forall i < \text{length roots.}$
 $\text{fmlookup } (\text{HashMap replay-state})$
 $(\text{CompositionFriChallenge } (P\text{CompositionFriCounter replay-state} + i) (\text{foldl concat}$
 $(\text{concat } (\text{foldl concat } (P\text{State replay-state}) \text{as})$
 $(\text{of-nat } (\text{degree } (p.\text{cp as } p.\text{f-powers}))))$
 $(\text{take } (\text{Suc } i) \text{ roots}))) =$
 $\text{Some } (\text{challenges } ! i)$
and *outcome:*
 $\text{Some } ((\text{as}', \text{dg}, \text{fl}, \text{final}), t) \in \text{set-dist } (\text{execute}$
 $(\text{do } \{$
 $\text{as}' \leftarrow \text{mmap } (\text{replicate } (\text{length spec})$
 $(\text{do } \{$
 $\text{a0} \leftarrow p.\text{receive-alpha-challenge};$
 $\text{let } \text{a0}' = \text{a0};$
 $\text{a1} \leftarrow p.\text{read};$
 $\text{let } \text{a1}' = \text{a1};$
 $\text{assert } (\text{a0}' = \text{a1}');$
 $\text{return } \text{a1}'$
 $\}));$
 $\text{dg} \leftarrow p.\text{read};$
 $\text{assert } (\text{to-nat } \text{dg} \leq p.\text{maxDegree});$
 $\text{fl} \leftarrow \text{ntimes } v.\text{receive-composition-fri-commits } (\text{ceil-log } (\text{to-nat } \text{dg} + 1));$
 $\text{final} \leftarrow p.\text{read};$
 $\text{return } (\text{as}', \text{dg}, \text{fl}, \text{final})$
 $\}) \text{ replay-state})$
shows
 $\text{as}' = \text{as} \wedge$
 $\text{dg} = \text{of-nat } (\text{degree } (p.\text{cp as } p.\text{f-powers})) \wedge$
 $\text{fl} = \text{zip challenges roots} \wedge$
 $\text{final} = \text{final-msg} \wedge$
 $P\text{State } t =$
 concat

```

    (foldl concat
      (concat (foldl concat (PState replay-state) as)
        (of-nat (degree (p.cp as p.f-powers))))
      roots)
    final-msg ∧
    PTranscript t = rest ∧
    replay-state ≤ t
proof –
  let ?dg = of-nat (degree (p.cp as p.f-powers))
  let ?round =
    do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }
  let ?ad =
    do {
      as' ← mmap (replicate (length spec) ?round);
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      return (as', dg)
    }
  from outcome obtain s-alpha s-dg s-prefix s-fri where
    alpha-out:
      Some (as', s-alpha) ∈
        set-dist (execute (mmap (replicate (length spec) ?round)) replay-state)
    and read-dg: Some (dg, s-dg) ∈ set-dist (execute p.read s-alpha)
    and degree-assert:
      Some ((), s-prefix) ∈
        set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s-dg)
    and fri-out:
      Some (fl, s-fri) ∈
        set-dist (execute
          (ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1)))
          s-prefix)
    and read-final:
      Some (final, t) ∈ set-dist (execute p.read s-fri)
    by (auto elim!: p.set-dist-bindE)
  have alpha-start:
    Some (as', s-alpha) ∈
      set-dist (execute (mmap (replicate (length as) ?round))
        (replay-state[PTranscript := as @ [?dg] @ roots @ [final-msg] @ rest]))
  proof –
    have alpha-out-as:
      Some (as', s-alpha) ∈
        set-dist (execute (mmap (replicate (length as) ?round)) replay-state)

```

```

    using alpha-out len-as by simp
  have replay-update:
    replay-state =
      replay-state(PTranscript := as @ [?dg] @ roots @ [final-msg] @ rest)
    using prefix by (simp add: append-assoc)
  show ?thesis
    by (subst replay-update[symmetric]) (rule alpha-out-as)
qed
have alpha-res:
  as' = as ∧
  PState s-alpha = foldl concat (PState replay-state) as ∧
  PTranscript s-alpha = ?dg # roots @ [final-msg] @ rest ∧
  PCompositionFriCounter s-alpha = PCompositionFriCounter replay-state ∧
  replay-state ≤ s-alpha
  using honest-alpha-mmap-replay-outcome[OF alpha-lookups alpha-start] by simp
have read-dg':
  Some (dg, s-dg) ∈
    set-dist (execute p.read (s-alpha(PTranscript := ?dg # roots @ [final-msg] @ rest)))
proof -
  have s-alpha-update:
    s-alpha = s-alpha(PTranscript := ?dg # roots @ [final-msg] @ rest)
    using alpha-res by simp
  then show ?thesis
    by (subst s-alpha-update[symmetric]) (rule read-dg)
qed
have dg-res:
  dg = ?dg ∧
  s-dg = s-alpha(PState := concat (PState s-alpha) ?dg,
    PTranscript := roots @ [final-msg] @ rest)
  using p.read-cons-outcome[OF read-dg'] by simp
have degree-ok: to-nat dg ≤ p.maxDegree
  using honest-degree-message-assert-hol[OF htv, of as] dg-res by simp
have prefix-eq: s-prefix = s-dg
  using degree-assert degree-ok by (simp add: assert-def)
have rounds-eq: ceil-log (to-nat dg + 1) = length roots
  using honest-degree-message-rounds[OF htv, of as] dg-res roots-rounds by simp
have fri-lookups-prefix:
  ∀ i < length roots.
    fmlookup (HashMap s-prefix)
      (CompositionFriChallenge
        (PCompositionFriCounter s-prefix + i)
        (foldl concat (PState s-prefix) (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-replay:
    fmlookup (HashMap replay-state)
      (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat

```

```

    (concat (foldl concat (PState replay-state) as) ?dg)
    (take (Suc i) roots))) =
    Some (challenges ! i)
  using fri-lookups i-bound by simp
have lookup-alpha:
  fmlookup (HashMap s-alpha)
    (CompositionFriChallenge (PCompositionFriCounter s-alpha + i) (foldl concat
      (concat (foldl concat (PState replay-state) as) ?dg)
      (take (Suc i) roots))) =
    Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-replay, of s-alpha] alpha-res by simp
show fmlookup (HashMap s-prefix)
  (CompositionFriChallenge
    (PCompositionFriCounter s-prefix + i)
    (foldl concat (PState s-prefix) (take (Suc i) roots))) =
    Some (challenges ! i)
  using lookup-alpha alpha-res dg-res prefix-eq by simp
qed
have fri-out':
  Some (fl, s-fri) ∈
  set-dist (execute (ntimes v.receive-composition-fri-commits (length roots))
    (s-prefix(PTranscript := roots @ ([final-msg] @ rest))))
  using fri-out rounds-eq alpha-res dg-res prefix-eq by simp
have fri-res:
  fl = zip challenges roots ∧
  PState s-fri = foldl concat (PState s-prefix) roots ∧
  PTranscript s-fri = [final-msg] @ rest ∧
  s-prefix ≤ s-fri
  using honest-receive-composition-fri-commits-list-outcome[
    OF fri-lookups-prefix len-challenges fri-out'] .
have read-final':
  Some (final, t) ∈
  set-dist (execute p.read (s-fri(PTranscript := final-msg # rest)))
proof -
  have s-fri-update: s-fri = s-fri(PTranscript := final-msg # rest)
    using fri-res by simp
  then show ?thesis
    by (subst s-fri-update[symmetric]) (rule read-final)
qed
have final-res:
  final = final-msg ∧
  t = s-fri(PState := concat (PState s-fri) final-msg, PTranscript := rest)
  using p.read-cons-outcome[OF read-final'] by simp
have replay-t: replay-state ≤ t
proof -
  have s-alpha-s-dg: s-alpha ≤ s-dg
    using dg-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  have s-fri-t: s-fri ≤ t
    using final-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp

```



```

show ?thesis
  using alpha-res s-alpha-s-dg prefix-eq fri-res s-fri-t by (meson p.hash-ext-trans)
qed
show ?thesis
  using alpha-res dg-res prefix-eq fri-res final-res replay-t by simp
qed

lemma honest-alpha-degree-fri-final-dynamic-prefix-no-failure:
assumes htv: honest-trace-valid
and len-as: length as = length spec
and len-challenges: length challenges = length roots
and roots-rounds:
  length roots = ceil-log (degree (p.cp as p.f-powers) + 1)
and prefix:
  PTranscript replay-state =
    as @ [of-nat (degree (p.cp as p.f-powers))] @ roots @ [final-msg] @ rest
and alpha-lookups:
  ∀ i < length as.
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i)
        (foldl concat (PState replay-state) (take i as))) =
      Some (as ! i)
and fri-lookups:
  ∀ i < length roots.
    fmlookup (HashMap replay-state)
      (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat
        (concat (foldl concat (PState replay-state) as)
          (of-nat (degree (p.cp as p.f-powers))))
        (take (Suc i) roots))) =
      Some (challenges ! i)
shows
  None ∉ dom (dist (execute
    (do {
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        })));
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      fl ← ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1));
      final ← p.read;
      return (as', dg, fl, final)
    }) replay-state))
proof -

```

```

let ?dg = of-nat (degree (p.cp as p.f-powers))
let ?round =
  do {
    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }
let ?ad =
  do {
    as' ← mmap (replicate (length spec) ?round);
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    return (as', dg)
  }
have alpha-nf-start:
  None ∉ dom (dist (execute
    (mmap (replicate (length as) ?round))
    (replay-state(PTranscript := as @ [?dg] @ roots @ [final-msg] @ rest))))
  using honest-alpha-mmap-replay-no-failure[OF alpha-lookups, of ?dg # roots @ [final-msg]
@ rest]
  by (simp add: append-assoc)
have alpha-nf:
  None ∉ dom (dist (execute
    (mmap (replicate (length spec) ?round)) replay-state))
proof –
  have replay-update:
    replay-state =
      replay-state(PTranscript := as @ [?dg] @ roots @ [final-msg] @ rest)
  using prefix by (simp add: append-assoc)
  have alpha-nf-as:
    None ∉ dom (dist (execute
      (mmap (replicate (length as) ?round)) replay-state))
    by (subst replay-update) (rule alpha-nf-start)
  show ?thesis
    using alpha-nf-as len-as by simp
qed
show ?thesis
proof (rule no-failure-bindI[OF alpha-nf])
  fix as' s-alpha
  assume alpha-out:
    Some (as', s-alpha) ∈
      set-dist (execute (mmap (replicate (length spec) ?round)) replay-state)
  have alpha-start:
    Some (as', s-alpha) ∈
      set-dist (execute (mmap (replicate (length as) ?round))
        (replay-state(PTranscript := as @ [?dg] @ roots @ [final-msg] @ rest)))

```

```

proof –
  have alpha-out-as:
    Some (as', s-alpha) ∈
      set-dist (execute (mmap (replicate (length as) ?round)) replay-state)
    using alpha-out len-as by simp
  have replay-update:
    replay-state =
      replay-state(PTranscript := as @ [?dg] @ roots @ [final-msg] @ rest)
    using prefix by (simp add: append-assoc)
  show ?thesis
    by (subst replay-update[symmetric]) (rule alpha-out-as)
qed
have alpha-res:
  as' = as ∧
    PState s-alpha = foldl concat (PState replay-state) as ∧
    PTranscript s-alpha = ?dg # roots @ [final-msg] @ rest ∧
    PCompositionFriCounter s-alpha = PCompositionFriCounter replay-state ∧
    replay-state ≤ s-alpha
  using honest-alpha-mmap-replay-outcome[OF alpha-lookups alpha-start] by simp
show None ∉ dom (dist (execute
  (p.read >>=
    (λdg. assert (to-nat dg ≤ p.maxDegree) >>=
      (λ-. ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1)) >>=
        (λfl. p.read >>= (λfinal. return (as', dg, fl, final)))))) s-alpha))
proof (rule no-failure-bindI)
  show None ∉ dom (dist (execute p.read s-alpha))
    by (rule p.read-no-failure) (use alpha-res in simp)
next
  fix dg s-dg
  assume read-dg: Some (dg, s-dg) ∈ set-dist (execute p.read s-alpha)
  have read-dg':
    Some (dg, s-dg) ∈
      set-dist (execute p.read (s-alpha(PTranscript := ?dg # roots @ [final-msg] @ rest)))
  proof –
    have s-alpha-update:
      s-alpha = s-alpha(PTranscript := ?dg # roots @ [final-msg] @ rest)
      using alpha-res by simp
    then show ?thesis
      by (subst s-alpha-update[symmetric]) (rule read-dg)
  qed
have dg-res:
  dg = ?dg ∧
    s-dg = s-alpha(PState := concat (PState s-alpha) ?dg,
      PTranscript := roots @ [final-msg] @ rest)
  using p.read-cons-outcome[OF read-dg'] by simp
have degree-ok: to-nat dg ≤ p.maxDegree
  using honest-degree-message-assert-hol[OF htv, of as] dg-res by simp
have rounds-eq: ceil-log (to-nat dg + 1) = length roots
  using honest-degree-message-rounds[OF htv, of as] dg-res roots-rounds by simp

```

```

have fri-lookups-dg:
   $\forall i < \text{length roots.}$ 
    fmlookup (HashMap s-dg)
      (CompositionFriChallenge
        (PCompositionFriCounter s-dg + i)
        (foldl concat (PState s-dg) (take (Suc i) roots))) =
        Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{length roots}$ 
  have lookup-replay:
    fmlookup (HashMap replay-state)
      (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat
        (concat (foldl concat (PState replay-state) as) ?dg)
        (take (Suc i) roots))) =
        Some (challenges ! i)
    using fri-lookups i-bound by simp
  have replay-s-dg:  $\text{replay-state} \leq s\text{-dg}$ 
  proof –
    have s-alpha-s-dg:  $s\text{-alpha} \leq s\text{-dg}$ 
    using dg-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
    show ?thesis
    using alpha-res s-alpha-s-dg by (meson p.hash-ext-trans)
  qed
  show fmlookup (HashMap s-dg)
    (CompositionFriChallenge
      (PCompositionFriCounter s-dg + i)
      (foldl concat (PState s-dg) (take (Suc i) roots))) =
      Some (challenges ! i)
    using p.hash-extension-lookup[OF lookup-replay replay-s-dg] alpha-res dg-res
    by simp
  qed
have fri-nf-start:
   $\text{None} \notin \text{dom (dist (execute$ 
    (ntimes v.receive-composition-fri-commits (length roots))
    (s-dg(PTranscript := roots @ ([final-msg] @ rest)))))
  using honest-receive-composition-fri-commits-list-no-failure[
    OF fri-lookups-dg len-challenges, of [final-msg] @ rest].
have fri-nf:
   $\text{None} \notin \text{dom (dist (execute$ 
    (ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1))) s-dg))
  using fri-nf-start dg-res rounds-eq by simp
have cont-nf:
   $\text{None} \notin \text{dom (dist (execute$ 
    (ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1)))  $\gg=$ 
    ( $\lambda fl. p.\text{read} \gg= (\lambda final. \text{return (as', dg, fl, final)})$ )) s-dg))
proof (rule no-failure-bindI[OF fri-nf])
  fix fl t
  assume fri-out:

```

```

    Some (fl, t) ∈
      set-dist (execute
        (ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1))) s-dg)
  have fri-out-start:
    Some (fl, t) ∈
      set-dist (execute
        (ntimes v.receive-composition-fri-commits (length roots))
        (s-dg(PTranscript := roots @ ([final-msg] @ rest))))
    using fri-out dg-res rounds-eq by simp
  have tr-t: PTranscript t = final-msg # rest
    using honest-receive-composition-fri-commits-list-outcome[
      OF fri-lookups-dg len-challenges fri-out-start]
    by simp
  show None ∉ dom
    (dist (execute (p.read >>=
      (λfinal. return (as', dg, fl, final))) t))
    by (rule no-failure-bind-returnI)
      (rule p.read-no-failure, simp add: tr-t)
  qed
  show None ∉ dom (dist (execute
    (assert (to-nat dg ≤ p.maxDegree) >>=
      (λ-. ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1)) >>=
        (λfl. p.read >>= (λfinal. return (as', dg, fl, final)))))) s-dg))
    using degree-ok cont-nf by (simp add: assert-def)
  qed
  qed
  qed

```

lemma *honest-trace-root-alpha-degree-fri-final-dynamic-prefix-outcome:*

```

  assumes htv: honest-trace-valid
  and len-as: length as = length spec
  and len-f-challenges: length f-challenges = length f-roots
  and f-roots-rounds: length f-roots = ceil-log clength
  and len-challenges: length challenges = length roots
  and roots-rounds:
    length roots = ceil-log (degree (p.cp as p.f-powers) + 1)
  and prefix:
    PTranscript replay-state =
      value f-merkle # f-roots @ [f-final-msg] @
        as @ [of-nat (degree (p.cp as p.f-powers))] @ roots @ [final-msg] @ rest
  and f-fri-lookups:
    ∀ i < length f-roots.
      fmlookup (HashMap replay-state)
        (TraceFriChallenge (PTraceFriCounter replay-state + i) (foldl concat
          (concat (PState replay-state) (value f-merkle))
          (take (Suc i) f-roots))) =
          Some (f-challenges ! i)
  and alpha-lookups:
    ∀ i < length as.

```

```

    fmllookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
          f-final-msg)
        (take i as))) =
      Some (as ! i)
  and fri-lookups:
    ∀ i < length roots.
      fmllookup (HashMap replay-state)
        (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat
          (concat
            (foldl concat
              (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
                f-final-msg)
              as)
            (of-nat (degree (p.cp as p.f-powers))))
          (take (Suc i) roots))) =
        Some (challenges ! i)
  and outcome:
    Some ((fr, f-fl, f-final, as', dg, fl, final), t) ∈ set-dist (execute
      (do {
        fr ← p.read;
        f-fl ← ntimes v.receive-trace-fri-commits (ceil-log clength);
        f-final ← p.read;
        as' ← mmap (replicate (length spec)
          (do {
            a0 ← p.receive-alpha-challenge;
            let a0' = a0;
            a1 ← p.read;
            let a1' = a1;
            assert (a0' = a1');
            return a1'
          })));
        dg ← p.read;
        assert (to-nat dg ≤ p.maxDegree);
        fl ← ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1));
        final ← p.read;
        return (fr, f-fl, f-final, as', dg, fl, final)
      }) replay-state)
  shows
    fr = value f-merkle ∧
    f-fl = zip f-challenges f-roots ∧
    f-final = f-final-msg ∧
    as' = as ∧
    dg = of-nat (degree (p.cp as p.f-powers)) ∧
    fl = zip challenges roots ∧
    final = final-msg ∧
    PState t =
      concat

```

```

    (foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
            f-final-msg)
          as)
        (of-nat (degree (p.cp as p.f-powers))))
      roots)
    final-msg ∧
    PTranscript t = rest ∧
    replay-state ≤ t
proof –
  let ?dg = of-nat (degree (p.cp as p.f-powers))
  let ?round =
    do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }
  let ?tail =
    do {
      as' ← mmap (replicate (length spec) ?round);
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      fl ← ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1));
      final ← p.read;
      return (as', dg, fl, final)
    }
  from outcome obtain s-root s-fri s-trace s-alpha s-dg s-assert s-comp where
    read-fr: Some (fr, s-root) ∈ set-dist (execute p.read replay-state)
    and trace-fri-out:
      Some (f-fl, s-fri) ∈
        set-dist (execute
          (ntimes v.receive-trace-fri-commits (ceil-log clength)) s-root)
    and read-f-final: Some (f-final, s-trace) ∈ set-dist (execute p.read s-fri)
    and alpha-out:
      Some (as', s-alpha) ∈
        set-dist (execute (mmap (replicate (length spec) ?round)) s-trace)
    and read-dg: Some (dg, s-dg) ∈ set-dist (execute p.read s-alpha)
    and assert-dg:
      Some ((), s-assert) ∈ set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s-dg)
    and fri-out:
      Some (fl, s-comp) ∈
        set-dist (execute
          (ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1)))
          s-assert)

```

```

and read-final: Some (final, t) ∈ set-dist (execute p.read s-comp)
by (auto elim!: p.set-dist-bindE)
have tail-out:
  Some ((as', dg, fl, final), t) ∈ set-dist (execute ?tail s-trace)
  apply (rule set-dist-bindI[OF alpha-out])
  apply (rule set-dist-bindI[OF read-dg])
  apply (rule set-dist-bindI[OF assert-dg])
  apply (rule set-dist-bindI[OF fri-out])
  apply (rule set-dist-bindI[OF read-final])
  by simp
have read-fr':
  Some (fr, s-root) ∈
    set-dist (execute p.read
      (replay-state(PTranscript :=
        value f-merkle # (f-roots @ [f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @ rest))))
proof –
  have replay-update:
    replay-state =
      replay-state(PTranscript :=
        value f-merkle # (f-roots @ [f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @ rest))
    using prefix by simp
  show ?thesis
    by (subst replay-update[symmetric]) (rule read-fr)
qed
have read-fr-res:
  fr = value f-merkle ∧
  s-root =
    replay-state(PState := concat (PState replay-state) (value f-merkle),
      PTranscript := f-roots @ [f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @ rest)
  using p.read-cons-outcome[OF read-fr'] by simp
have f-fri-lookups-root:
  ∀ i < length f-roots.
    fmlookup (HashMap s-root)
      (TraceFriChallenge
        (PTraceFriCounter s-root + i)
        (foldl concat (PState s-root) (take (Suc i) f-roots))) =
      Some (f-challenges ! i)
  using f-fri-lookups read-fr-res by simp
have trace-fri-out':
  Some (f-fl, s-fri) ∈
    set-dist (execute (ntimes v.receive-trace-fri-commits (length f-roots))
      (s-root(PTranscript := f-roots @ ([f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @
rest))))))
  using trace-fri-out f-roots-rounds read-fr-res by simp
have trace-fri-res:
  f-fl = zip f-challenges f-roots ∧
  PState s-fri = foldl concat (PState s-root) f-roots ∧
  PTranscript s-fri = [f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @ rest ∧
  s-root ≤ s-fri

```



```

using honest-receive-trace-fri-commits-list-outcome[
  OF f-fri-lookups-root len-f-challenges trace-fri-out'] .
have trace-fri-counters:
  PCompositionFriCounter s-fri = PCompositionFriCounter s-root ∧
  PAlphaCounter s-fri = PAlphaCounter s-root ∧
  PQueryCounter s-fri = PQueryCounter s-root
using receive-trace-fri-commits-preserves-other-counters[OF trace-fri-out']
by simp
have read-f-final':
  Some (f-final, s-trace) ∈
    set-dist (execute p.read (s-fri(PTranscript :=
      f-final-msg # (as @ [?dg] @ roots @ [final-msg] @ rest))))
proof -
  have s-fri-update:
    s-fri =
      s-fri(PTranscript := f-final-msg # (as @ [?dg] @ roots @ [final-msg] @ rest))
  using trace-fri-res by simp
show ?thesis
  by (subst s-fri-update[symmetric]) (rule read-f-final)
qed
have read-f-final-res:
  f-final = f-final-msg ∧
  s-trace =
    s-fri(PState := concat (PState s-fri) f-final-msg,
      PTranscript := as @ [?dg] @ roots @ [final-msg] @ rest)
using p.read-cons-outcome[OF read-f-final'] by simp
have s-trace-counters:
  PCompositionFriCounter s-trace = PCompositionFriCounter replay-state ∧
  PAlphaCounter s-trace = PAlphaCounter replay-state
using read-fr-res trace-fri-counters read-f-final-res by simp
have replay-s-trace: replay-state ≤ s-trace
proof -
  have replay-root: replay-state ≤ s-root
  using read-fr-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  have s-fri-trace: s-fri ≤ s-trace
  using read-f-final-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
show ?thesis
  using replay-root trace-fri-res s-fri-trace by (meson p.hash-ext-trans)
qed
have alpha-lookups-trace:
  ∀ i < length as.
    fmllookup (HashMap s-trace)
      (AlphaChallenge (PAlphaCounter s-trace + i)
        (foldl concat (PState s-trace) (take i as))) =
      Some (as ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length as
  have lookup-replay:

```

```

    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
          f-final-msg)
        (take i as))) =
      Some (as ! i)
  using alpha-lookups i-bound by simp
show fmlookup (HashMap s-trace)
  (AlphaChallenge (PAlphaCounter s-trace + i)
    (foldl concat (PState s-trace) (take i as))) =
  Some (as ! i)
  using p.hash-extension-lookup[OF lookup-replay replay-s-trace]
    read-fr-res trace-fri-res read-f-final-res s-trace-counters
  by simp
qed
have fri-lookups-trace:
  ∀ i < length roots.
    fmlookup (HashMap s-trace)
      (CompositionFriChallenge (PCompositionFriCounter s-trace + i) (foldl concat
        (concat (foldl concat (PState s-trace) as) ?dg)
        (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-replay:
    fmlookup (HashMap replay-state)
      (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat
        (concat
          (foldl concat
            (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
              f-final-msg)
            as)
          ?dg)
        (take (Suc i) roots))) =
      Some (challenges ! i)
    using fri-lookups i-bound by simp
  show fmlookup (HashMap s-trace)
    (CompositionFriChallenge (PCompositionFriCounter s-trace + i) (foldl concat
      (concat (foldl concat (PState s-trace) as) ?dg)
      (take (Suc i) roots))) =
    Some (challenges ! i)
    using p.hash-extension-lookup[OF lookup-replay replay-s-trace]
      read-fr-res trace-fri-res read-f-final-res s-trace-counters
    by simp
qed
have tail-prefix:
  PTranscript s-trace = as @ [?dg] @ roots @ [final-msg] @ rest
  using read-f-final-res by simp

```

```

have tail-res:
  as' = as ∧
  dg = ?dg ∧
  fl = zip challenges roots ∧
  final = final-msg ∧
  PState t =
    concat
      (foldl concat
        (concat (foldl concat (PState s-trace) as) ?dg)
        roots)
      final-msg ∧
  PTranscript t = rest ∧
  s-trace ≤ t
by (rule honest-alpha-degree-fri-final-dynamic-prefix-outcome[
  OF htv len-as len-challenges roots-rounds tail-prefix
  alpha-lookups-trace fri-lookups-trace tail-out])
have replay-t: replay-state ≤ t
proof –
  have replay-root: replay-state ≤ s-root
    using read-fr-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  have s-fri-trace: s-fri ≤ s-trace
    using read-f-final-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  show ?thesis
    using replay-root trace-fri-res s-fri-trace tail-res by (meson p.hash-ext-trans)
qed
show ?thesis
  using read-fr-res trace-fri-res read-f-final-res tail-res replay-t by simp
qed

```

lemma *honest-trace-root-alpha-degree-fri-final-dynamic-prefix-preserves-query-counter*:

```

assumes outcome:
  Some ((fr, f-fl, f-final, as', dg, fl, final), t) ∈ set-dist (execute
    (do {
      fr ← p.read;
      f-fl ← ntimes v.receive-trace-fri-commits (ceil-log clength);
      f-final ← p.read;
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        }));
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      fl ← ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1));
      final ← p.read;

```

```

    return (fr, f-fl, f-final, as', dg, fl, final)
  }) s)
shows PQueryCounter t = PQueryCounter s
proof –
  let ?round =
    do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }
from outcome obtain s-root s-fri s-trace s-alpha s-dg s-assert s-comp where
  read-fr: Some (fr, s-root) ∈ set-dist (execute p.read s)
  and trace-fri-out:
    Some (f-fl, s-fri) ∈
      set-dist (execute (ntimes v.receive-trace-fri-commits (ceil-log clength)) s-root)
  and read-f-final: Some (f-final, s-trace) ∈ set-dist (execute p.read s-fri)
  and alpha-out:
    Some (as', s-alpha) ∈
      set-dist (execute (mmap (replicate (length spec) ?round)) s-trace)
  and read-dg: Some (dg, s-dg) ∈ set-dist (execute p.read s-alpha)
  and assert-dg:
    Some ((), s-assert) ∈ set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) s-dg)
  and comp-out:
    Some (fl, s-comp) ∈
      set-dist (execute
        (ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1)))) s-assert)
  and read-final: Some (final, t) ∈ set-dist (execute p.read s-comp)
  by (auto elim!: p.set-dist-bindE)
have q-root: PQueryCounter s-root = PQueryCounter s
  using read-preserves-counters[OF read-fr] by simp
have q-fri: PQueryCounter s-fri = PQueryCounter s-root
  using receive-trace-fri-commits-preserves-other-counters[OF trace-fri-out] by simp
have q-trace: PQueryCounter s-trace = PQueryCounter s-fri
  using read-preserves-counters[OF read-f-final] by simp
have q-alpha: PQueryCounter s-alpha = PQueryCounter s-trace
  by (rule honest-alpha-mmap-replay-preserves-query-counter[OF alpha-out])
have q-dg: PQueryCounter s-dg = PQueryCounter s-alpha
  using read-preserves-counters[OF read-dg] by simp
have s-assert-eq: s-assert = s-dg
  using assert-outcomeD(2)[OF assert-dg] .
have q-comp: PQueryCounter s-comp = PQueryCounter s-assert
  using receive-composition-fri-commits-preserves-other-counters[OF comp-out] by simp
have q-final: PQueryCounter t = PQueryCounter s-comp
  using read-preserves-counters[OF read-final] by simp
show ?thesis
  using q-root q-fri q-trace q-alpha q-dg q-comp q-final

```

unfolding *s-assert-eq* by *simp*
 qed

lemma *honest-trace-root-alpha-degree-fri-final-dynamic-prefix-no-failure*:

assumes *htv*: *honest-trace-valid*

and *len-as*: *length as = length spec*

and *len-f-challenges*: *length f-challenges = length f-roots*

and *f-roots-rounds*: *length f-roots = ceil-log clength*

and *len-challenges*: *length challenges = length roots*

and *roots-rounds*:

length roots = ceil-log (degree (p.cp as p.f-powers) + 1)

and *prefix*:

PTranscript replay-state =

value f-merkle # f-roots @ [f-final-msg] @

as @ [of-nat (degree (p.cp as p.f-powers))] @ roots @ [final-msg] @ rest

and *f-fri-lookups*:

$\forall i < \text{length } f\text{-roots}.$

fmlookup (HashMap replay-state)

(TraceFriChallenge (PTraceFriCounter replay-state + i) (foldl concat

(concat (PState replay-state) (value f-merkle))

(take (Suc i) f-roots))) =

Some (f-challenges ! i)

and *alpha-lookups*:

$\forall i < \text{length } as.$

fmlookup (HashMap replay-state)

(AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat

(concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)

f-final-msg)

(take i as))) =

Some (as ! i)

and *fri-lookups*:

$\forall i < \text{length } roots.$

fmlookup (HashMap replay-state)

(CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat

(concat

(foldl concat

(concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)

f-final-msg)

as)

(of-nat (degree (p.cp as p.f-powers))))

(take (Suc i) roots))) =

Some (challenges ! i)

shows

None $\notin$ *dom (dist (execute*

(do {

fr $\leftarrow$ *p.read*;

f-fl $\leftarrow$ *ntimes v.receive-trace-fri-commits (ceil-log clength)*;

f-final $\leftarrow$ *p.read*;

as' $\leftarrow$ *mmap (replicate (length spec)*

```

      (do {
        a0 ← p.receive-alpha-challenge;
        let a0' = a0;
        a1 ← p.read;
        let a1' = a1;
        assert (a0' = a1');
        return a1'
      }));
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    fl ← ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1));
    final ← p.read;
    return (fr, f-fl, f-final, as', dg, fl, final)
  }) replay-state))
proof –
  let ?dg = of-nat (degree (p.cp as p.f-powers))
  let ?round =
    do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }
  show ?thesis
proof (rule no-failure-bindI)
  show None ∉ dom (dist (execute p.read replay-state))
  by (rule p.read-no-failure) (use prefix in simp)
next
  fix fr s-root
  assume read-fr: Some (fr, s-root) ∈ set-dist (execute p.read replay-state)
  have read-fr':
    Some (fr, s-root) ∈
      set-dist (execute p.read
        (replay-state(PTranscript :=
          value f-merkle # (f-roots @ [f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @ rest))))
proof –
  have replay-update:
    replay-state =
      replay-state(PTranscript :=
        value f-merkle # (f-roots @ [f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @ rest))
  using prefix by simp
  show ?thesis
  by (subst replay-update[symmetric]) (rule read-fr)
qed
have read-fr-res:
  s-root =
    replay-state(PState := concat (PState replay-state) (value f-merkle),

```

```

    PTranscript := f-roots @ [f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @ rest)
using p.read-cons-outcome[OF read-fr'] by simp
have f-fri-lookups-root:
  ∀ i < length f-roots.
    fmllookup (HashMap s-root)
      (TraceFriChallenge
        (PTraceFriCounter s-root + i)
        (foldl concat (PState s-root) (take (Suc i) f-roots))) =
      Some (f-challenges ! i)
using f-fri-lookups read-fr-res by simp
have trace-fri-nf-start:
  None ∉ dom (dist (execute
    (ntimes v.receive-trace-fri-commits (length f-roots))
    (s-root(PTranscript := f-roots @ ([f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @
rest))))))
using honest-receive-trace-fri-commits-list-no-failure[
  OF f-fri-lookups-root len-f-challenges, of [f-final-msg] @ as @ [?dg] @ roots @ [final-msg]
@ rest] .
have trace-fri-nf:
  None ∉ dom (dist (execute
    (ntimes v.receive-trace-fri-commits (ceil-log clength)) s-root))
using trace-fri-nf-start f-roots-rounds read-fr-res by simp
show None ∉ dom (dist (execute
  (ntimes v.receive-trace-fri-commits (ceil-log clength)) >>=
  (λf-fl. p.read >>=
    (λf-final. mmap (replicate (length spec) ?round) >>=
      (λas'. p.read >>=
        (λdg. assert (to-nat dg ≤ p.maxDegree) >>=
          (λ-. ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1)) >>=
            (λfl. p.read >>=
              (λfinal. return (fr, f-fl, f-final, as', dg, fl, final)))))))))) s-root))
proof (rule no-failure-bindI[OF trace-fri-nf])
fix f-fl s-fri
assume trace-fri-out:
  Some (f-fl, s-fri) ∈
    set-dist (execute
      (ntimes v.receive-trace-fri-commits (ceil-log clength)) s-root)
have trace-fri-out':
  Some (f-fl, s-fri) ∈
    set-dist (execute (ntimes v.receive-trace-fri-commits (length f-roots))
      (s-root(PTranscript := f-roots @ ([f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @
rest))))
using trace-fri-out f-roots-rounds read-fr-res by simp
have trace-fri-res:
  PState s-fri = foldl concat (PState s-root) f-roots ∧
  PTranscript s-fri = [f-final-msg] @ as @ [?dg] @ roots @ [final-msg] @ rest ∧
  s-root ≤ s-fri
using honest-receive-trace-fri-commits-list-outcome[
  OF f-fri-lookups-root len-f-challenges trace-fri-out'] by simp

```

```

have trace-fri-counters:
  PCompositionFriCounter s-fri = PCompositionFriCounter s-root ∧
  PAlphaCounter s-fri = PAlphaCounter s-root ∧
  PQueryCounter s-fri = PQueryCounter s-root
using receive-trace-fri-commits-preserves-other-counters[OF trace-fri-out']
by simp
show None ∉ dom (dist (execute
  (p.read >>=
    (λf-final. mmap (replicate (length spec) ?round) >>=
      (λas'. p.read >>=
        (λdg. assert (to-nat dg ≤ p.maxDegree) >>=
          (λ-. ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1)) >>=
            (λfl. p.read >>=
              (λfinal. return (fr, f-fl, f-final, as', dg, fl, final))))))) s-fri))
proof (rule no-failure-bindI)
  show None ∉ dom (dist (execute p.read s-fri))
  by (rule p.read-no-failure) (use trace-fri-res in simp)
next
  fix f-final s-trace
  assume read-f-final: Some (f-final, s-trace) ∈ set-dist (execute p.read s-fri)
  have read-f-final':
    Some (f-final, s-trace) ∈
      set-dist (execute p.read (s-fri(PTranscript :=
        f-final-msg # (as @ [?dg] @ roots @ [final-msg] @ rest))))
  proof –
    have s-fri-update:
      s-fri =
        s-fri(PTranscript := f-final-msg # (as @ [?dg] @ roots @ [final-msg] @ rest))
    using trace-fri-res by simp
    show ?thesis
    by (subst s-fri-update[symmetric]) (rule read-f-final)
  qed
  have read-f-final-res:
    s-trace =
      s-fri(PState := concat (PState s-fri) f-final-msg,
        PTranscript := as @ [?dg] @ roots @ [final-msg] @ rest)
    using p.read-cons-outcome[OF read-f-final'] by simp
  have s-trace-counters:
    PCompositionFriCounter s-trace = PCompositionFriCounter replay-state ∧
    PAlphaCounter s-trace = PAlphaCounter replay-state
    using read-fr-res trace-fri-counters read-f-final-res by simp
  have replay-s-trace: replay-state ≤ s-trace
  proof –
    have replay-root: replay-state ≤ s-root
    using read-fr-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
    have s-fri-trace: s-fri ≤ s-trace
    using read-f-final-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
    show ?thesis
    using replay-root trace-fri-res s-fri-trace by (meson p.hash-ext-trans)

```



```

qed
have alpha-lookups-trace:
   $\forall i < \text{length } as.$ 
    fmlookup (HashMap s-trace)
      (AlphaChallenge (PAlphaCounter s-trace + i)
        (foldl concat (PState s-trace) (take i as))) =
      Some (as ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length as
  have lookup-replay:
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
        f-final-msg)
        (take i as))) =
      Some (as ! i)
    using alpha-lookups i-bound by simp
  show fmlookup (HashMap s-trace)
    (AlphaChallenge (PAlphaCounter s-trace + i)
      (foldl concat (PState s-trace) (take i as))) =
    Some (as ! i)
    using p.hash-extension-lookup[OF lookup-replay replay-s-trace]
      read-fr-res trace-fri-res read-f-final-res s-trace-counters
    by simp
qed
have fri-lookups-trace:
   $\forall i < \text{length roots}.$ 
    fmlookup (HashMap s-trace)
      (CompositionFriChallenge (PCompositionFriCounter s-trace + i) (foldl concat
        (concat (foldl concat (PState s-trace) as) ?dg)
        (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-replay:
    fmlookup (HashMap replay-state)
      (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat
        (concat
          (foldl concat
            (concat (PState replay-state) (value f-merkle)) f-roots)
          f-final-msg)
          as)
          ?dg)
        (take (Suc i) roots))) =
      Some (challenges ! i)
    using fri-lookups i-bound by simp
  show fmlookup (HashMap s-trace)

```

```

    (CompositionFriChallenge (PCompositionFriCounter s-trace + i) (foldl concat
      (concat (foldl concat (PState s-trace) as) ?dg)
      (take (Suc i) roots))) =
    Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-replay replay-s-trace]
    read-fr-res trace-fri-res read-f-final-res s-trace-counters
  by simp
qed
have tail-prefix:
  PTranscript s-trace = as @ [?dg] @ roots @ [final-msg] @ rest
  using read-f-final-res by simp
have tail-nf:
  None  $\notin$  dom (dist (execute
    (do {
      as'  $\leftarrow$  mmap (replicate (length spec) ?round);
      dg  $\leftarrow$  p.read;
      assert (to-nat dg  $\leq$  p.maxDegree);
      fl  $\leftarrow$  ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1));
      final  $\leftarrow$  p.read;
      return (as', dg, fl, final)
    }) s-trace))
  by (rule honest-alpha-degree-fri-final-dynamic-prefix-no-failure[
    OF htv len-as len-challenges roots-rounds tail-prefix
    alpha-lookups-trace fri-lookups-trace])
show None  $\notin$  dom (dist (execute
  (mmap (replicate (length spec) ?round)  $\gg$ 
    ( $\lambda$ as'. p.read  $\gg$ 
      ( $\lambda$ dg. assert (to-nat dg  $\leq$  p.maxDegree)  $\gg$ 
        ( $\lambda$ -. ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1))  $\gg$ 
          ( $\lambda$ fl. p.read  $\gg$ 
            ( $\lambda$ final. return (fr, f-fl, f-final, as', dg, fl, final))))))) s-trace))
proof -
  let ?tail =
    do {
      as'  $\leftarrow$  mmap (replicate (length spec) ?round);
      dg  $\leftarrow$  p.read;
      assert (to-nat dg  $\leq$  p.maxDegree);
      fl  $\leftarrow$  ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1));
      final  $\leftarrow$  p.read;
      return (as', dg, fl, final)
    }
  have tail-full-eq:
    (mmap (replicate (length spec) ?round)  $\gg$ 
      ( $\lambda$ as'. p.read  $\gg$ 
        ( $\lambda$ dg. assert (to-nat dg  $\leq$  p.maxDegree)  $\gg$ 
          ( $\lambda$ -. ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1))  $\gg$ 
            ( $\lambda$ fl. p.read  $\gg$ 
              ( $\lambda$ final. return (fr, f-fl, f-final, as', dg, fl, final))))))) =
    (?tail  $\gg$ 

```

```

      (λout. return (case out of (as', dg, fl, final) ⇒
        (fr, f-fl, f-final, as', dg, fl, final))))
    by (simp add: sm-bind-assoc split: prod.splits)
  show ?thesis
    unfolding tail-full-eq
    by (rule no-failure-bind-returnI[OF tail-nf])
qed
qed
qed
qed
qed

```

lemma *honest-trace-root-alpha-degree-fri-final-dynamic-from-prover-tail-no-failure:*

```

assumes htv: honest-trace-valid
and create-f:
  Some (f-merkle, s1) ∈
    set-dist (execute (p.create p.f-eval) init-state)
and send-f:
  Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and f-nrounds-def: f-nrounds = ceil-log clength
and trace-fri:
  Some ((f-ps, f-ds, f-ls, f-ms), s-trace-fri) ∈
    set-dist (execute
      (p.trace-fri-commit f-nrounds [p.f] [p.eval-domain] [p.f-eval] [f-merkle]) s2)
and trace-final-send:
  Some ((), s-trace-final) ∈
    set-dist (execute (p.send (hd (last f-ls))) s-trace-fri)
and alphas:
  Some (as, s3) ∈
    set-dist (execute
      (mmap (replicate (length spec)
        (do {
          a ← p.receive-alpha-challenge;
          p.send a;
          return a
        }))) s-trace-final)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
and create-cp:
  Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and nrounds-def: nrounds = ceil-log (degree cp' + 1)
and fri:
  Some ((ps, ds, ls, ms), s6) ∈
    set-dist (execute
      (p.composition-fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and final-send:
  Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and tail:

```

Some (prover-result, prover-state) ∈
 set-dist (execute (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds) s-final)
and replay: replay-state = verifier-replay-state prover-state
shows
 None ∉ dom (dist (execute
 (do {
 fr ← p.read;
 f-fl ← ntimes v.receive-trace-fri-commits (ceil-log clength);
 f-final ← p.read;
 as' ← mmap (replicate (length spec)
 (do {
 a0 ← p.receive-alpha-challenge;
 let a0' = a0;
 a1 ← p.read;
 let a1' = a1;
 assert (a0' = a1');
 return a1'
 }));
 dg ← p.read;
 assert (to-nat dg ≤ p.maxDegree);
 fl ← ntimes v.receive-composition-fri-commits (ceil-log (to-nat dg + 1));
 final ← p.read;
 return (fr, f-fl, f-final, as', dg, fl, final)
 }) replay-state))
proof –
from trace-fri-commit-initial-replay-successor-data[OF trace-fri] **obtain** f-roots f-challenges
where
 f-len-roots: length f-roots = f-nrounds
and f-len-challenges: length f-challenges = f-nrounds
and tr-trace-fri: PTranscript s-trace-fri = rev f-roots @ PTranscript s2
and st-trace-fri: PState s-trace-fri = foldl concat (PState s2) f-roots
and trace-fri-lookups-state:
 ∀ i < length f-roots.
 fmlookup (HashMap s-trace-fri)
 (TraceFriChallenge (PTraceFriCounter s2 + i)
 (foldl concat (PState s2) (take (Suc i) f-roots))) =
 Some (f-challenges ! i)
by blast
from composition-fri-commit-initial-replay-successor-data[OF fri] **obtain** roots challenges **where**
 len-roots: length roots = nrounds
and len-challenges: length challenges = nrounds
and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
and st-s6: PState s6 = foldl concat (PState s5) roots
and fri-lookups-s6:
 ∀ i < length roots.
 fmlookup (HashMap s6)
 (CompositionFriChallenge
 (PCompositionFriCounter s5 + i)
 (foldl concat (PState s5) (take (Suc i) roots))) =

```

    Some (challenges ! i)
  by blast
have alpha-data:
  length as = length spec ∧
  s-trace-final ≤ s3 ∧
  PState s3 = foldl concat (PState s-trace-final) as ∧
  PTranscript s3 = rev as @ PTranscript s-trace-final ∧
  PCompositionFriCounter s3 = PCompositionFriCounter s-trace-final ∧
  (∀ i < length as.
    fmlookup (HashMap s3)
      (AlphaChallenge (PAlphaCounter s-trace-final + i)
        (foldl concat (PState s-trace-final) (take i as)))) =
    Some (as ! i))
  using alpha-mmap-outcome[OF alphas] by simp
have tail-hash: s-final ≤ prover-state
  by (rule ntimes-hash-extends[OF - tail])
  (rule prover-query-round-hash-extends)
have s2-trace-fri: s2 ≤ s-trace-fri
  using p.trace-fri-commit-extends[OF trace-fri] .
have trace-fri-final-hash: s-trace-fri ≤ s-trace-final
  using send-hash-extends[OF trace-final-send] .
have s3-s4: s3 ≤ s4
  using send-hash-extends[OF send-degree] .
have s4-s5: s4 ≤ s5
  using create-hash-extends[OF create-cp] .
have s5-s6: s5 ≤ s6
  using p.composition-fri-commit-extends[OF fri] .
have s6-final-hash: s6 ≤ s-final
  using send-hash-extends[OF final-send] .
have trace-fri-prover: s-trace-fri ≤ prover-state
  using trace-fri-final-hash alpha-data s3-s4 s4-s5 s5-s6 s6-final-hash tail-hash
  by (meson p.hash-ext-trans)
have s3-prover: s3 ≤ prover-state
  using s3-s4 s4-s5 s5-s6 s6-final-hash tail-hash by (meson p.hash-ext-trans)
have s6-prover: s6 ≤ prover-state
  using s6-final-hash tail-hash by (rule p.hash-ext-trans)
have final-tail-tr: transcript-extends prover-state s-final
  by (rule ntimes-transcript-extends[OF - tail])
  (rule prover-query-round-transcript-extends)
from final-tail-tr obtain ys where
  tr-prover: PTranscript prover-state = ys @ PTranscript s-final
  unfolding transcript-extends-def by auto
have tr-s1: PTranscript s1 = []
  using create-preserves-transcript[OF create-f]
  unfolding init-state-def by simp
have send-f-res:
  s2 = s1 (|
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1 |)

```

```

using p.send-outcome[OF send-f] .
have trace-final-res:
  s-trace-final = s-trace-fri(
    PState := concat (PState s-trace-fri) (hd (last f-ls)),
    PTranscript := hd (last f-ls) # PTranscript s-trace-fri)
  using p.send-outcome[OF trace-final-send] .
have send-degree-res:
  s4 = s3(
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3)
  using p.send-outcome[OF send-degree] .
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have final-send-res:
  s-final = s6(
    PState := concat (PState s6) (hd (last ls)),
    PTranscript := hd (last ls) # PTranscript s6)
  using p.send-outcome[OF final-send] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # f-roots @ [hd (last f-ls)] @
    as @ [of-nat (degree (p.cp as p.f-powers))) @ roots @ [hd (last ls)] @ rev ys
  using replay tr-prover final-send-res tr-s6 tr-s5 send-degree-res alpha-data
    trace-final-res tr-trace-fri send-f-res tr-s1 cp'-def
  unfolding verifier-replay-state-def by simp
have st-s1: PState s1 = 0
  using create-preserves-state[OF create-f]
  unfolding init-state-def by simp
have st-trace-final:
  PState s-trace-final =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
    (hd (last f-ls))
  using st-s1 send-f-res st-trace-fri trace-final-res replay
  unfolding verifier-replay-state-def by simp
have st-s5:
  PState s5 =
    concat
    (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
        (hd (last f-ls)))
      as)
    (of-nat (degree (p.cp as p.f-powers)))
  using st-trace-final alpha-data send-degree-res create-preserves-state[OF create-cp] cp'-def
  by simp
have trace-counter-s1-zero: PTraceFriCounter s1 = 0
  using p.create-preserves-channel(3)[OF create-f]
  unfolding init-state-def by simp
have trace-counter-s2-zero: PTraceFriCounter s2 = 0
  using send-f-res trace-counter-s1-zero by simp

```

```

have alpha-counter-s1-zero: PAlphaCounter s1 = 0
  using p.create-preserves-channel(5)[OF create-f]
  unfolding init-state-def by simp
have alpha-counter-s2-zero: PAlphaCounter s2 = 0
  using send-f-res alpha-counter-s1-zero by simp
have alpha-counter-trace-fri-zero: PAlphaCounter s-trace-fri = 0
  using trace-fri-commit-preserves-alpha-counter[OF trace-fri]
    alpha-counter-s2-zero by simp
have alpha-counter-trace-final-zero: PAlphaCounter s-trace-final = 0
  using trace-final-res alpha-counter-trace-fri-zero by simp
have comp-counter-s1-zero: PCompositionFriCounter s1 = 0
  using p.create-preserves-channel(4)[OF create-f]
  unfolding init-state-def by simp
have comp-counter-s2-zero: PCompositionFriCounter s2 = 0
  using send-f-res comp-counter-s1-zero by simp
have comp-counter-trace-fri-zero: PCompositionFriCounter s-trace-fri = 0
  using trace-fri-commit-preserves-composition-counter[OF trace-fri]
    comp-counter-s2-zero by simp
have comp-counter-trace-final-zero:
  PCompositionFriCounter s-trace-final = 0
  using trace-final-res comp-counter-trace-fri-zero by simp
have comp-counter-s3-zero: PCompositionFriCounter s3 = 0
  using alpha-data comp-counter-trace-final-zero by simp
have comp-counter-s4-zero: PCompositionFriCounter s4 = 0
  using send-degree-res comp-counter-s3-zero by simp
have comp-counter-s5-zero: PCompositionFriCounter s5 = 0
  using p.create-preserves-channel(4)[OF create-cp] comp-counter-s4-zero
  by simp
have f-fri-lookups:
  ∀ i < length f-roots.
    fmlookup (HashMap replay-state)
      (TraceFriChallenge (PTraceFriCounter replay-state + i) (foldl concat
        (concat (PState replay-state) (value f-merkle))
        (take (Suc i) f-roots)))) =
      Some (f-challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length f-roots
  have lookup-trace:
    fmlookup (HashMap s-trace-fri)
      (TraceFriChallenge (PTraceFriCounter s2 + i)
        (foldl concat (PState s2) (take (Suc i) f-roots)))) =
      Some (f-challenges ! i)
  using trace-fri-lookups-state i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state)
      (TraceFriChallenge (PTraceFriCounter s2 + i)
        (foldl concat (PState s2) (take (Suc i) f-roots)))) =
      Some (f-challenges ! i)

```

```

    using p.hash-extension-lookup[OF lookup-trace trace-fri-prover] .
show fmlookup (HashMap replay-state)
  (TraceFriChallenge (PTraceFriCounter replay-state + i) (foldl concat
    (concat (PState replay-state) (value f-merkle))
    (take (Suc i) f-roots))) =
  Some (f-challenges ! i)
using lookup-prover send-f-res st-s1 replay trace-counter-s2-zero
unfolding verifier-replay-state-def by simp
qed
have alpha-lookups:
   $\forall i < \text{length } as.$ 
  fmlookup (HashMap replay-state)
    (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
        (hd (last f-ls)))
      (take i as))) =
    Some (as ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{length } as$ 
  have alpha-lookups-s3:
     $\forall i < \text{length } as.$ 
    fmlookup (HashMap s3)
      (AlphaChallenge (PAlphaCounter s-trace-final + i)
        (foldl concat (PState s-trace-final) (take i as))) =
      Some (as ! i)
  using alpha-data by simp
  have lookup-s3:
    fmlookup (HashMap s3)
      (AlphaChallenge (PAlphaCounter s-trace-final + i)
        (foldl concat (PState s-trace-final) (take i as))) =
      Some (as ! i)
  using alpha-lookups-s3 i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state)
      (AlphaChallenge (PAlphaCounter s-trace-final + i)
        (foldl concat (PState s-trace-final) (take i as))) =
      Some (as ! i)
  using p.hash-extension-lookup[OF lookup-s3 s3-prover] .
show fmlookup (HashMap replay-state)
  (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat
    (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
      (hd (last f-ls)))
    (take i as))) =
  Some (as ! i)
using lookup-prover st-trace-final replay alpha-counter-trace-final-zero
unfolding verifier-replay-state-def by simp
qed
have fri-lookups:

```



```

   $\forall i < \text{length roots}.$ 
  fmlookup (HashMap replay-state)
    (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
            (hd (last f-ls)))
          as)
        (of-nat (degree (p.cp as p.f-powers))))
      (take (Suc i) roots))) =
    Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{length roots}$ 
  have lookup-s6:
    fmlookup (HashMap s6)
      (CompositionFriChallenge
        (PCompositionFriCounter s5 + i)
        (foldl concat (PState s5) (take (Suc i) roots))) =
      Some (challenges ! i)
  using fri-lookups-s6 i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state)
      (CompositionFriChallenge
        (PCompositionFriCounter s5 + i)
        (foldl concat (PState s5) (take (Suc i) roots))) =
      Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-s6 s6-prover] .
  show fmlookup (HashMap replay-state)
    (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
            (hd (last f-ls)))
          as)
        (of-nat (degree (p.cp as p.f-powers))))
      (take (Suc i) roots))) =
    Some (challenges ! i)
  using lookup-prover st-s5 replay comp-counter-s5-zero
  unfolding verifier-replay-state-def by simp
qed
have f-roots-rounds:  $\text{length f-roots} = \text{ceil-log clength}$ 
  using f-len-roots f-nrounds-def by simp
have f-len-challenges-roots:  $\text{length f-challenges} = \text{length f-roots}$ 
  using f-len-challenges f-len-roots by simp
have roots-rounds:
   $\text{length roots} = \text{ceil-log (degree (p.cp as p.f-powers) + 1)}$ 
  using len-roots nrounds-def cp'-def by simp
have len-challenges-roots:  $\text{length challenges} = \text{length roots}$ 

```

using *len-challenges len-roots* **by** *simp*
show *?thesis*
by (*rule honest-trace-root-alpha-degree-fri-final-dynamic-prefix-no-failure*[
OF htv conjunct1[OF alpha-data] f-len-challenges-roots f-roots-rounds
len-challenges-roots roots-rounds replay-prefix f-fri-lookups
alpha-lookups fri-lookups])
qed

lemma *honest-trace-root-alpha-degree-fri-final-dynamic-from-prover-tail-outcome:*
assumes *htv: honest-trace-valid*
and *create-f:*
Some (f-merkle, s1) ∈
set-dist (execute (p.create p.f-eval) init-state)
and *send-f:*
Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and *f-nrounds-def:* *f-nrounds = ceil-log clength*
and *trace-fri:*
Some ((f-ps, f-ds, f-ls, f-ms), s-trace-fri) ∈
set-dist (execute
(p.trace-fri-commit f-nrounds [p.f] [p.eval-domain] [p.f-eval] [f-merkle]) s2)
and *trace-final-send:*
Some ((), s-trace-final) ∈
set-dist (execute (p.send (hd (last f-ls))) s-trace-fri)
and *alphas:*
Some (as, s3) ∈
set-dist (execute
(mmap (replicate (length spec)
(do {
a ← p.receive-alpha-challenge;
p.send a;
return a
}})) s-trace-final)
and *cp'-def:* *cp' = p.cp as p.f-powers*
and *send-degree:*
Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
and *create-cp:*
Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and *nrounds-def:* *nrounds = ceil-log (degree cp' + 1)*
and *fri:*
Some ((ps, ds, ls, ms), s6) ∈
set-dist (execute
(p.composition-fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and *final-send:*
Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and *tail:*
Some (prover-result, prover-state) ∈
set-dist (execute (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds) s-final)
and *replay:* *replay-state = verifier-replay-state prover-state*
and *outcome:*

Some $((fr, f-fl, f-final, as', dg, fl, final), t) \in \text{set-dist}$ (execute
 (do {
 $fr \leftarrow p.\text{read}$;
 $f-fl \leftarrow \text{ntimes } v.\text{receive-trace-fri-commits } (\text{ceil-log } clength)$;
 $f-final \leftarrow p.\text{read}$;
 $as' \leftarrow \text{mmap } (\text{replicate } (length \text{ spec}))$
 (do {
 $a0 \leftarrow p.\text{receive-alpha-challenge}$;
 let $a0' = a0$;
 $a1 \leftarrow p.\text{read}$;
 let $a1' = a1$;
 assert $(a0' = a1')$;
 return $a1'$
 }));
 $dg \leftarrow p.\text{read}$;
 assert $(\text{to-nat } dg \leq p.\text{maxDegree})$;
 $fl \leftarrow \text{ntimes } v.\text{receive-composition-fri-commits } (\text{ceil-log } (\text{to-nat } dg + 1))$;
 $final \leftarrow p.\text{read}$;
 return $(fr, f-fl, f-final, as', dg, fl, final)$
 }) replay-state)
obtains $f\text{-roots } f\text{-challenges roots challenges ys}$ **where**
 $length \text{ } f\text{-roots} = f\text{-nrounds}$
 $length \text{ } f\text{-challenges} = f\text{-nrounds}$
 $f-fl = \text{zip } f\text{-challenges } f\text{-roots}$
 $\bigwedge j. j < f\text{-nrounds} \implies f\text{-roots } ! j = \text{value } (f\text{-ms } ! j)$
 $\bigwedge j. j < f\text{-nrounds} \implies$
 $p.\text{next-fri-layer } (f\text{-ps } ! j) (f\text{-ds } ! j) (f\text{-challenges } ! j) =$
 $(f\text{-ps } ! \text{Suc } j, f\text{-ds } ! \text{Suc } j, f\text{-ls } ! \text{Suc } j)$
 $f-final = \text{hd } (\text{last } f\text{-ls})$
 $length \text{ } roots = nrounds$
 $length \text{ } challenges = nrounds$
 $fl = \text{zip } challenges \text{ } roots$
 $\bigwedge j. j < nrounds \implies roots \text{ } ! j = \text{value } (ms \text{ } ! j)$
 $\bigwedge j. j < nrounds \implies$
 $p.\text{next-fri-layer } (ps \text{ } ! j) (ds \text{ } ! j) (challenges \text{ } ! j) =$
 $(ps \text{ } ! \text{Suc } j, ds \text{ } ! \text{Suc } j, ls \text{ } ! \text{Suc } j)$
 $fr = \text{value } f\text{-merkle}$
 $as' = as$
 $dg = \text{of-nat } (\text{degree } (p.\text{cp } as \text{ } p.f\text{-powers}))$
 $final = \text{hd } (\text{last } ls)$
 $PState \text{ } t = PState \text{ } s\text{-final}$
 $PTranscript \text{ } t = \text{rev } ys$
 $PTranscript \text{ } prover\text{-state} = ys @ PTranscript \text{ } s\text{-final}$
 $\text{replay-state} \leq t$
proof –
from $\text{trace-fri-commit-initial-replay-successor-data}[OF \text{ } trace\text{-fri}]$ **obtain** $f\text{-roots } f\text{-challenges}$
where
 $f\text{-len-roots}: length \text{ } f\text{-roots} = f\text{-nrounds}$
 and $f\text{-len-challenges}: length \text{ } f\text{-challenges} = f\text{-nrounds}$

```

and tr-trace-fri: PTranscript s-trace-fri = rev f-roots @ PTranscript s2
and st-trace-fri: PState s-trace-fri = foldl concat (PState s2) f-roots
and trace-fri-lookups-state:
   $\forall i < \text{length } f\text{-roots}.$ 
  fmlookup (HashMap s-trace-fri)
    (TraceFriChallenge (PTraceFriCounter s2 + i)
      (foldl concat (PState s2) (take (Suc i) f-roots))) =
      Some (f-challenges ! i)
and f-roots-aligned-all:  $\forall j < f\text{-nrounds}.$  f-roots ! j = value (f-ms ! j)
and f-successors-all:
   $\forall j < f\text{-nrounds}.$ 
  p.next-fri-layer (f-ps ! j) (f-ds ! j) (f-challenges ! j) =
    (f-ps ! Suc j, f-ds ! Suc j, f-ls ! Suc j)
by blast
from composition-fri-commit-initial-replay-successor-data[OF fri] obtain roots challenges where
  len-roots: length roots = nrounds
and len-challenges: length challenges = nrounds
and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
and st-s6: PState s6 = foldl concat (PState s5) roots
and fri-lookups-s6:
   $\forall i < \text{length roots}.$ 
  fmlookup (HashMap s6)
    (CompositionFriChallenge
      (PCompositionFriCounter s5 + i)
      (foldl concat (PState s5) (take (Suc i) roots))) =
      Some (challenges ! i)
and roots-aligned-all:  $\forall j < nrounds.$  roots ! j = value (ms ! j)
and successors-all:
   $\forall j < nrounds.$ 
  p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
    (ps ! Suc j, ds ! Suc j, ls ! Suc j)
by blast
have alpha-data:
  length as = length spec  $\wedge$ 
  s-trace-final  $\leq$  s3  $\wedge$ 
  PState s3 = foldl concat (PState s-trace-final) as  $\wedge$ 
  PTranscript s3 = rev as @ PTranscript s-trace-final  $\wedge$ 
  PCompositionFriCounter s3 = PCompositionFriCounter s-trace-final  $\wedge$ 
  ( $\forall i < \text{length as}.$ 
    fmlookup (HashMap s3)
      (AlphaChallenge (PAlphaCounter s-trace-final + i)
        (foldl concat (PState s-trace-final) (take i as))) =
        Some (as ! i))
using alpha-mmap-outcome[OF alphas] by simp
have tail-hash: s-final  $\leq$  prover-state
by (rule ntimes-hash-extends[OF - tail])
  (rule prover-query-round-hash-extends)
have trace-fri-final-hash: s-trace-fri  $\leq$  s-trace-final
using send-hash-extends[OF trace-final-send] .

```

```

have s3-s4:  $s3 \leq s4$ 
  using send-hash-extends[OF send-degree] .
have s4-s5:  $s4 \leq s5$ 
  using create-hash-extends[OF create-cp] .
have s5-s6:  $s5 \leq s6$ 
  using p.composition-fri-commit-extends[OF fri] .
have s6-final-hash:  $s6 \leq s\text{-final}$ 
  using send-hash-extends[OF final-send] .
have trace-fri-prover:  $s\text{-trace-fri} \leq \text{prover-state}$ 
  using trace-fri-final-hash alpha-data s3-s4 s4-s5 s5-s6 s6-final-hash tail-hash
  by (meson p.hash-ext-trans)
have s3-prover:  $s3 \leq \text{prover-state}$ 
  using s3-s4 s4-s5 s5-s6 s6-final-hash tail-hash by (meson p.hash-ext-trans)
have s6-prover:  $s6 \leq \text{prover-state}$ 
  using s6-final-hash tail-hash by (rule p.hash-ext-trans)
have final-tail-tr: transcript-extends prover-state s-final
  by (rule ntimes-transcript-extends[OF - tail])
  (rule prover-query-round-transcript-extends)
from final-tail-tr obtain ys where
  tr-prover: PTranscript prover-state = ys @ PTranscript s-final
  unfolding transcript-extends-def by auto
have tr-s1: PTranscript s1 = []
  using create-preserves-transcript[OF create-f]
  unfolding init-state-def by simp
have send-f-res:
  s2 = s1(|
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1|)
  using p.send-outcome[OF send-f] .
have trace-final-res:
  s-trace-final = s-trace-fri(|
    PState := concat (PState s-trace-fri) (hd (last f-ls)),
    PTranscript := hd (last f-ls) # PTranscript s-trace-fri|)
  using p.send-outcome[OF trace-final-send] .
have send-degree-res:
  s4 = s3(|
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3|)
  using p.send-outcome[OF send-degree] .
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have final-send-res:
  s-final = s6(|
    PState := concat (PState s6) (hd (last ls)),
    PTranscript := hd (last ls) # PTranscript s6|)
  using p.send-outcome[OF final-send] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # f-roots @ [hd (last f-ls)] @

```

```

    as @ [of-nat (degree (p.cp as p.f-powers))] @ roots @ [hd (last ls)] @ rev ys
using replay tr-prover final-send-res tr-s6 tr-s5 send-degree-res alpha-data
    trace-final-res tr-trace-fri send-f-res tr-s1 cp'-def
unfolding verifier-replay-state-def by simp
have st-s1: PState s1 = 0
    using create-preserves-state[OF create-f]
    unfolding init-state-def by simp
have st-trace-final:
    PState s-trace-final =
        concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
            (hd (last f-ls))
    using st-s1 send-f-res st-trace-fri trace-final-res replay
    unfolding verifier-replay-state-def by simp
have st-s5:
    PState s5 =
        concat
            (foldl concat
                (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
                    (hd (last f-ls)))
                as)
            (of-nat (degree (p.cp as p.f-powers)))
    using st-trace-final alpha-data send-degree-res create-preserves-state[OF create-cp] cp'-def
    by simp
have trace-counter-s1-zero: PTraceFriCounter s1 = 0
    using p.create-preserves-channel(3)[OF create-f]
    unfolding init-state-def by simp
have trace-counter-s2-zero: PTraceFriCounter s2 = 0
    using send-f-res trace-counter-s1-zero by simp
have alpha-counter-s1-zero: PAlphaCounter s1 = 0
    using p.create-preserves-channel(5)[OF create-f]
    unfolding init-state-def by simp
have alpha-counter-s2-zero: PAlphaCounter s2 = 0
    using send-f-res alpha-counter-s1-zero by simp
have alpha-counter-trace-fri-zero: PAlphaCounter s-trace-fri = 0
    using trace-fri-commit-preserves-alpha-counter[OF trace-fri]
        alpha-counter-s2-zero by simp
have alpha-counter-trace-final-zero: PAlphaCounter s-trace-final = 0
    using trace-final-res alpha-counter-trace-fri-zero by simp
have comp-counter-s1-zero: PCompositionFriCounter s1 = 0
    using p.create-preserves-channel(4)[OF create-f]
    unfolding init-state-def by simp
have comp-counter-s2-zero: PCompositionFriCounter s2 = 0
    using send-f-res comp-counter-s1-zero by simp
have comp-counter-trace-fri-zero: PCompositionFriCounter s-trace-fri = 0
    using trace-fri-commit-preserves-composition-counter[OF trace-fri]
        comp-counter-s2-zero by simp
have comp-counter-trace-final-zero:
    PCompositionFriCounter s-trace-final = 0
    using trace-final-res comp-counter-trace-fri-zero by simp

```

```

have comp-counter-s3-zero: PCompositionFriCounter s3 = 0
  using alpha-data comp-counter-trace-final-zero by simp
have comp-counter-s4-zero: PCompositionFriCounter s4 = 0
  using send-degree-res comp-counter-s3-zero by simp
have comp-counter-s5-zero: PCompositionFriCounter s5 = 0
  using p.create-preserves-channel(4)[OF create-cp] comp-counter-s4-zero
  by simp
have f-fri-lookups:
   $\forall i < \text{length } f\text{-roots}.$ 
    fmlookup (HashMap replay-state)
      (TraceFriChallenge (PTraceFriCounter replay-state + i) (foldl concat
        (concat (PState replay-state) (value f-merkle))
        (take (Suc i) f-roots))) =
      Some (f-challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length f-roots
  have lookup-trace:
    fmlookup (HashMap s-trace-fri)
      (TraceFriChallenge (PTraceFriCounter s2 + i)
        (foldl concat (PState s2) (take (Suc i) f-roots))) =
      Some (f-challenges ! i)
    using trace-fri-lookups-state i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state)
      (TraceFriChallenge (PTraceFriCounter s2 + i)
        (foldl concat (PState s2) (take (Suc i) f-roots))) =
      Some (f-challenges ! i)
    using p.hash-extension-lookup[OF lookup-trace trace-fri-prover] .
  show fmlookup (HashMap replay-state)
    (TraceFriChallenge (PTraceFriCounter replay-state + i) (foldl concat
      (concat (PState replay-state) (value f-merkle))
      (take (Suc i) f-roots))) =
    Some (f-challenges ! i)
    using lookup-prover send-f-res st-s1 replay trace-counter-s2-zero
    unfolding verifier-replay-state-def by simp
qed
have alpha-lookups:
   $\forall i < \text{length } as.$ 
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
        (hd (last f-ls)))
        (take i as))) =
      Some (as ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length as
  have alpha-lookups-s3:

```

```

  ∀ i < length as.
    fmlookup (HashMap s3)
      (AlphaChallenge (PAlphaCounter s-trace-final + i)
        (foldl concat (PState s-trace-final) (take i as))) =
      Some (as ! i)
  using alpha-data by simp
have lookup-s3:
  fmlookup (HashMap s3)
    (AlphaChallenge (PAlphaCounter s-trace-final + i)
      (foldl concat (PState s-trace-final) (take i as))) =
    Some (as ! i)
  using alpha-lookups-s3 i-bound by simp
have lookup-prover:
  fmlookup (HashMap prover-state)
    (AlphaChallenge (PAlphaCounter s-trace-final + i)
      (foldl concat (PState s-trace-final) (take i as))) =
    Some (as ! i)
  using p.hash-extension-lookup[OF lookup-s3 s3-prover] .
show fmlookup (HashMap replay-state)
  (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat
    (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
      (hd (last f-ls)))
    (take i as))) =
  Some (as ! i)
  using lookup-prover st-trace-final replay alpha-counter-trace-final-zero
  unfolding verifier-replay-state-def by simp
qed
have fri-lookups:
  ∀ i < length roots.
    fmlookup (HashMap replay-state)
      (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat
        (concat
          (foldl concat
            (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
              (hd (last f-ls)))
            as)
          (of-nat (degree (p.cp as p.f-powers))))
        (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-s6:
    fmlookup (HashMap s6)
      (CompositionFriChallenge
        (PCompositionFriCounter s5 + i)
        (foldl concat (PState s5) (take (Suc i) roots))) =
      Some (challenges ! i)
  using fri-lookups-s6 i-bound by simp

```



```

have lookup-prover:
  fmlookup (HashMap prover-state)
    (CompositionFriChallenge
      (PCompositionFriCounter s5 + i)
      (foldl concat (PState s5) (take (Suc i) roots))) =
    Some (challenges ! i)
using p.hash-extension-lookup[OF lookup-s6 s6-prover] .
show fmlookup (HashMap replay-state)
  (CompositionFriChallenge (PCompositionFriCounter replay-state + i) (foldl concat
    (concat
      (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
          (hd (last f-ls)))
        as)
        (of-nat (degree (p.cp as p.f-powers))))))
    (take (Suc i) roots))) =
  Some (challenges ! i)
using lookup-prover st-s5 replay comp-counter-s5-zero
unfolding verifier-replay-state-def by simp
qed
have f-roots-rounds: length f-roots = ceil-log clength
  using f-len-roots f-nrounds-def by simp
have f-len-challenges-roots: length f-challenges = length f-roots
  using f-len-challenges f-len-roots by simp
have roots-rounds:
  length roots = ceil-log (degree (p.cp as p.f-powers) + 1)
  using len-roots nrounds-def cp'-def by simp
have len-challenges-roots: length challenges = length roots
  using len-challenges len-roots by simp
have prefix-res:
  fr = value f-merkle ∧
  f-fl = zip f-challenges f-roots ∧
  f-final = hd (last f-ls) ∧
  as' = as ∧
  dg = of-nat (degree (p.cp as p.f-powers)) ∧
  fl = zip challenges roots ∧
  final = hd (last ls) ∧
  PState t =
    concat
      (foldl concat
        (concat
          (foldl concat
            (concat (foldl concat (concat (PState replay-state) (value f-merkle)) f-roots)
              (hd (last f-ls)))
            as)
            (of-nat (degree (p.cp as p.f-powers))))))
        roots)
      (hd (last ls)) ∧
  PTranscript t = rev ys ∧

```

```

    replay-state ≤ t
  by (rule honest-trace-root-alpha-degree-fri-final-dynamic-prefix-outcome[
    OF htv conjunct1[OF alpha-data] f-len-challenges-roots f-roots-rounds
    len-challenges-roots roots-rounds replay-prefix f-fri-lookups
    alpha-lookups fri-lookups outcome])
  have st-t-final: PState t = PState s-final
    using prefix-res st-s5 st-s6 final-send-res by simp
  show ?thesis
  by (rule that[
    OF f-len-roots f-len-challenges - - - len-roots len-challenges - - -
    - - - st-t-final - tr-prover -])
    (use prefix-res f-roots-aligned-all f-successors-all roots-aligned-all successors-all in simp-all)
qed

```

lemma *honest-root-alpha-degree-from-prover-no-failure:*

```

  assumes htv: honest-trace-valid
  and create-f:
    Some (f-merkle, s1) ∈
      set-dist (execute (p.create p.f-eval) init-state)
  and created-f: p.created-tree p.f-eval f-merkle s1
  and send-f:
    Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
  and alphas:
    Some (as, s3) ∈
      set-dist (execute
        (mmap (replicate (length spec)
          (do {
            a ← p.receive-alpha-challenge;
            p.send a;
            return a
          }))) s2)
  and cp'-def: cp' = p.cp as p.f-powers
  and send-degree:
    Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
  and create-cp:
    Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
  and fri:
    Some ((ps, ds, ls, ms), s6) ∈
      set-dist (execute
        (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
  and final-send:
    Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
  and random-idx:
    Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
  and idx-def: idx' = p.index (to-nat idx)
  and query-decommit:
    Some (qouts, s8) ∈
      set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
  and fri-decommit:

```

```

    Some (fri-idx, s9) ∈
      set-dist (execute
        (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
  and prover-state-eq: prover-state = s9
  and replay: replay-state = verifier-replay-state prover-state
shows
  None ∉ dom (dist (execute
    (do {
      fr ← p.read;
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        })));
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      return (fr, as', dg)
    }) replay-state))
proof -
  have root-alpha-tr:
    length as = length spec ∧
    PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
  by (rule prover-root-alpha-degree-transcript[
    OF create-f send-f alphas cp'-def send-degree])
  have later-tr: transcript-extends prover-state s4
  by (rule prover-later-phases-transcript-extends[
    OF create-cp fri final-send random-idx query-decommit fri-decommit prover-state-eq])
  from verifier-replay-root-alpha-degree-prefix[
    OF conjunct2[OF root-alpha-tr] later-tr replay]
  obtain rest where prefix:
    PTranscript replay-state =
      value f-merkle # as @ [of-nat (degree cp')] @ rest
  by blast
  have prefix-cp:
    PTranscript replay-state =
      value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @ rest
  using prefix cp'-def by simp
  have later-hash: s3 ≤ prover-state
  by (rule prover-later-phases-hash-extends[
    OF send-degree create-cp fri final-send random-idx query-decommit fri-decommit prover-state-eq])
  have lookups:
    ∀ i < length as.
      fmllookup (HashMap replay-state)
        (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
          play-state) (value f-merkle)) (take i as))) =

```

```

    Some (as ! i)
  by (rule prover-alpha-lookups-in-replay-state[
    OF create-f send-f alphas later-hash replay])
show ?thesis
  by (rule honest-root-alpha-degree-prefix-no-failure[
    OF htv conjunct1[OF root-alpha-tr] prefix-cp lookups])
qed

lemma honest-root-alpha-degree-from-prover-outcome:
  assumes htv: honest-trace-valid
  and create-f:
    Some (f-merkle, s1) ∈
      set-dist (execute (p.create p.f-eval) init-state)
  and send-f:
    Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
  and alphas:
    Some (as, s3) ∈
      set-dist (execute
        (mmap (replicate (length spec)
          (do {
            a ← p.receive-alpha-challenge;
            p.send a;
            return a
          }))) s2)
  and cp'-def: cp' = p.cp as p.f-powers
  and send-degree:
    Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
  and create-cp:
    Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
  and fri:
    Some ((ps, ds, ls, ms), s6) ∈
      set-dist (execute
        (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
  and final-send:
    Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
  and random-idx:
    Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
  and idx-def: idx' = p.index (to-nat idx)
  and query-decommit:
    Some (qouts, s8) ∈
      set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
  and fri-decommit:
    Some (fri-idx, s9) ∈
      set-dist (execute
        (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
  and prover-state-eq: prover-state = s9
  and replay: replay-state = verifier-replay-state prover-state
  and outcome:
    Some ((fr, as', dg), t) ∈ set-dist (execute

```

```

(do {
  fr ← p.read;
  as' ← mmap (replicate (length spec)
    (do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }));
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  return (fr, as', dg)
}) replay-state)

shows
fr = value f-merkle ∧
as' = as ∧
dg = of-nat (degree (p.cp as p.f-powers)) ∧
replay-state ≤ t

proof -
have root-alpha-tr:
length as = length spec ∧
PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
by (rule prover-root-alpha-degree-transcript[
  OF create-f send-f alphas cp'-def send-degree])
have later-tr: transcript-extends prover-state s4
by (rule prover-later-phases-transcript-extends[
  OF create-cp fri final-send random-idx query-decommit fri-decommit prover-state-eq])
from verifier-replay-root-alpha-degree-prefix[
  OF conjunct2[OF root-alpha-tr] later-tr replay]
obtain rest where prefix:
PTranscript replay-state =
value f-merkle # as @ [of-nat (degree cp')] @ rest
by blast
have prefix-cp:
PTranscript replay-state =
value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @ rest
using prefix cp'-def by simp
have later-hash: s3 ≤ prover-state
by (rule prover-later-phases-hash-extends[
  OF send-degree create-cp fri final-send random-idx query-decommit fri-decommit prover-state-eq])
have lookups:
∀ i < length as.
fmlookup (HashMap replay-state)
(AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
play-state) (value f-merkle)) (take i as))) =
Some (as ! i)
by (rule prover-alpha-lookups-in-replay-state[

```

$OF\ create\text{-}f\ send\text{-}f\ alphas\ later\text{-}hash\ replay])$
have *out*:
 $fr = value\ f\text{-}merkle \wedge$
 $as' = as \wedge$
 $dg = of\text{-}nat\ (degree\ (p.cp\ as\ p.f\text{-}powers)) \wedge$
 $PState\ t =$
 $\quad concat\ (foldl\ concat\ (concat\ (PState\ replay\text{-}state)\ (value\ f\text{-}merkle))\ as)$
 $\quad (of\text{-}nat\ (degree\ (p.cp\ as\ p.f\text{-}powers))) \wedge$
 $PTranscript\ t = rest \wedge$
 $replay\text{-}state \leq t$
by (*rule honest-root-alpha-degree-prefix-outcome*[
 $OF\ htv\ conjunct1[OF\ root\text{-}alpha\text{-}tr]\ prefix\text{-}cp\ lookups\ outcome]$)
then show *?thesis by simp*
qed

lemma *honest-root-alpha-degree-fri-from-prover-no-failure*:
assumes *htv: honest-trace-valid*
and *create-f*:
 $Some\ (f\text{-}merkle,\ s1) \in$
 $\quad set\text{-}dist\ (execute\ (p.create\ p.f\text{-}eval)\ init\text{-}state)$
and *send-f*:
 $Some\ ((),\ s2) \in set\text{-}dist\ (execute\ (p.send\ (value\ f\text{-}merkle))\ s1)$
and *alphas*:
 $Some\ (as,\ s3) \in$
 $\quad set\text{-}dist\ (execute$
 $\quad\quad (mmap\ (replicate\ (length\ spec)$
 $\quad\quad\quad (do\ \{$
 $\quad\quad\quad\quad a \leftarrow p.receive\text{-}alpha\text{-}challenge;$
 $\quad\quad\quad\quad p.send\ a;$
 $\quad\quad\quad\quad return\ a$
 $\quad\quad\quad\})\ s2)$
and *cp'-def*: $cp' = p.cp\ as\ p.f\text{-}powers$
and *send-degree*:
 $Some\ ((),\ s4) \in set\text{-}dist\ (execute\ (p.send\ (of\text{-}nat\ (degree\ cp')))\ s3)$
and *create-cp*:
 $Some\ (cp\text{-}merkle,\ s5) \in set\text{-}dist\ (execute\ (p.create\ (p.cp\text{-}eval\ as))\ s4)$
and *fri*:
 $Some\ ((ps,\ ds,\ ls,\ ms),\ s6) \in$
 $\quad set\text{-}dist\ (execute$
 $\quad\quad (p.fri\text{-}commit\ nrounds\ [cp']\ [p.eval\text{-}domain]\ [p.cp\text{-}eval\ as]\ [cp\text{-}merkle])\ s5)$
and *final-send*:
 $Some\ ((),\ s\text{-}final) \in set\text{-}dist\ (execute\ (p.send\ (hd\ (last\ ls)))\ s6)$
and *random-idx*:
 $Some\ (idx,\ s7) \in set\text{-}dist\ (execute\ p.receive\text{-}query\text{-}index\text{-}challenge\ s\text{-}final)$
and *idx-def*: $idx' = p.index\ (to\text{-}nat\ idx)$
and *query-decommit*:
 $Some\ (qouts,\ s8) \in$
 $\quad set\text{-}dist\ (execute\ (mmap\ (p.decommit\text{-}on\text{-}query\ idx'\ f\text{-}merkle))\ s7)$
and *fri-decommit*:

```

    Some (fri-idx, s9) ∈
      set-dist (execute
        (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
  and prover-state-eq: prover-state = s9
  and replay: replay-state = verifier-replay-state prover-state
shows
  None ∉ dom (dist (execute
    (do {
      fr ← p.read;
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        })));
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      fl ← ntimes v.receive-fri-commits nrounds;
      return (fr, as', dg, fl)
    }) replay-state))
proof -
  have root-alpha-tr:
    length as = length spec ∧
    PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
  by (rule prover-root-alpha-degree-transcript[
    OF create-f send-f alphas cp'-def send-degree])
  have later-hash: s3 ≤ prover-state
  by (rule prover-later-phases-hash-extends[
    OF send-degree create-cp fri final-send random-idx query-decommit fri-decommit prover-state-eq])
  have alpha-lookups:
    ∀ i < length as.
      fmlookup (HashMap replay-state)
        (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
          play-state) (value f-merkle)) (take i as))) =
        Some (as ! i)
  by (rule prover-alpha-lookups-in-replay-state[
    OF create-f send-f alphas later-hash replay])
  from fri-commit-replay-data[OF fri] obtain roots challenges where
    len-roots: length roots = nrounds
  and len-challenges: length challenges = nrounds
  and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
  and st-s6: PState s6 = foldl concat (PState s5) roots
  and fri-hash: s5 ≤ s6
  and fri-lookups-s6:
    ∀ i < length roots.
      fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)

```

```

roots))) =
  Some (challenges ! i)
  by blast
have after-tr: transcript-extends prover-state s6
  by (rule prover-after-fri-transcript-extends[
    OF final-send random-idx query-decommit fri-decommit prover-state-eq])
from after-tr obtain xs where
  tr-prover: PTranscript prover-state = xs @ PTranscript s6
  unfolding transcript-extends-def by auto
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @ roots @ rev xs
  using replay tr-prover tr-s6 tr-s5 root-alpha-tr cp'-def
  unfolding verifier-replay-state-def by simp
have st-s1: PState s1 = 0
  using create-preserves-state[OF create-f]
  unfolding init-state-def by simp
have send-f-res:
  s2 = s1(
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1)
  using p.send-outcome[OF send-f] .
have alpha-data:
  PState s3 = foldl concat (PState s2) as
  using alpha-mmap-outcome[OF alphas] by simp
have send-degree-res:
  s4 = s3(
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3)
  using p.send-outcome[OF send-degree] .
have st-s5:
  PState s5 =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers)))
  using st-s1 send-f-res alpha-data send-degree-res
  create-preserves-state[OF create-cp] replay cp'-def
  unfolding verifier-replay-state-def by simp
have after-hash: s6 ≤ prover-state
  by (rule prover-after-fri-hash-extends[
    OF final-send random-idx query-decommit fri-decommit prover-state-eq])
have fri-lookups:
  ∀ i < length roots.
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          (of-nat (degree (p.cp as p.f-powers))))
        (take (Suc i) roots))) =

```



```

    Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-s6:
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
    Some (challenges ! i)
  using fri-lookups-s6 i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state) (FiatShamirChallenge (foldl concat (PState s5) (take
(Suc i) roots))) =
    Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-s6 after-hash] .
show fmlookup (HashMap replay-state)
(FiatShamirChallenge (foldl concat
(concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
(of-nat (degree (p.cp as p.f-powers))))
(take (Suc i) roots))) =
  Some (challenges ! i)
  using lookup-prover st-s5 replay
  unfolding verifier-replay-state-def by simp
qed
have nf-roots:
  None  $\notin$  dom (dist (execute
(do {
  fr  $\leftarrow$  p.read;
  as'  $\leftarrow$  mmap (replicate (length spec)
(do {
  a0  $\leftarrow$  p.receive-alpha-challenge;
  let a0' = a0;
  a1  $\leftarrow$  p.read;
  let a1' = a1;
  assert (a0' = a1');
  return a1'
})));
  dg  $\leftarrow$  p.read;
  assert (to-nat dg  $\leq$  p.maxDegree);
  fl  $\leftarrow$  ntimes v.receive-fri-commits (length roots);
  return (fr, as', dg, fl)
}) replay-state))
by (rule honest-root-alpha-degree-fri-prefix-no-failure[
  OF htv conjunct1[OF root-alpha-tr] - replay-prefix alpha-lookups fri-lookups])
  (use len-roots len-challenges in simp-all)
show ?thesis
  using nf-roots len-roots by simp
qed

```

lemma honest-root-alpha-degree-fri-from-prover-tail-no-failure:

```

assumes htv: honest-trace-valid
and create-f:
  Some (f-merkle, s1) ∈
    set-dist (execute (p.create p.f-eval) init-state)
and send-f:
  Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and alphas:
  Some (as, s3) ∈
    set-dist (execute
      (mmap (replicate (length spec)
        (do {
          a ← p.receive-alpha-challenge;
          p.send a;
          return a
        }))) s2)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
and create-cp:
  Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and fri:
  Some ((ps, ds, ls, ms), s6) ∈
    set-dist (execute
      (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and final-send:
  Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and tail:
  Some (prover-result, prover-state) ∈
    set-dist (execute (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) nrounds) s-final)
and replay: replay-state = verifier-replay-state prover-state
shows
  None ∉ dom (dist (execute
    (do {
      fr ← p.read;
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        })))
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      fl ← ntimes v.receive-fri-commits nrounds;
      return (fr, as', dg, fl)
    }) replay-state))
proof –

```

```

have root-alpha-tr:
  length as = length spec ∧
  PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
  by (rule prover-root-alpha-degree-transcript[
    OF create-f send-f alphas cp'-def send-degree])
have tail-hash: s-final ≤ prover-state
  by (rule ntimes-hash-extends[OF - tail])
  (rule prover-query-round-hash-extends)
have s3-s4: s3 ≤ s4
  using send-hash-extends[OF send-degree] .
have s4-s5: s4 ≤ s5
  using create-hash-extends[OF create-cp] .
have s5-s6: s5 ≤ s6
  using honest-fri-commit-extends[OF fri] .
have s6-final-hash: s6 ≤ s-final
  using send-hash-extends[OF final-send] .
have later-hash: s3 ≤ prover-state
  using s3-s4 s4-s5 s5-s6 s6-final-hash tail-hash by (meson p.hash-ext-trans)
have alpha-lookups:
  ∀ i < length as.
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
        play-state) (value f-merkle)) (take i as))) =
      Some (as ! i)
  by (rule prover-alpha-lookups-in-replay-state[
    OF create-f send-f alphas later-hash replay])
from fri-commit-replay-data[OF fri] obtain roots challenges where
  len-roots: length roots = nrounds
  and len-challenges: length challenges = nrounds
  and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
  and st-s6: PState s6 = foldl concat (PState s5) roots
  and fri-hash: s5 ≤ s6
  and fri-lookups-s6:
    ∀ i < length roots.
      fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
        roots))) =
        Some (challenges ! i)
  by blast
have s6-final-tr: transcript-extends s-final s6
  using send-transcript-extends[OF final-send] .
have final-tail-tr: transcript-extends prover-state s-final
  by (rule ntimes-transcript-extends[OF - tail])
  (rule prover-query-round-transcript-extends)
have after-tr: transcript-extends prover-state s6
  using s6-final-tr final-tail-tr by (meson transcript-extends-trans)
from after-tr obtain xs where
  tr-prover: PTranscript prover-state = xs @ PTranscript s6
  unfolding transcript-extends-def by auto
have tr-s5: PTranscript s5 = PTranscript s4

```

```

using create-preserves-transcript[OF create-cp] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @ roots @ rev xs
using replay tr-prover tr-s6 tr-s5 root-alpha-tr cp'-def
unfolding verifier-replay-state-def by simp
have st-s1: PState s1 = 0
using create-preserves-state[OF create-f]
unfolding init-state-def by simp
have send-f-res:
  s2 = s1 (
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1 )
using p.send-outcome[OF send-f] .
have alpha-data:
  PState s3 = foldl concat (PState s2) as
using alpha-mmap-outcome[OF alphas] by simp
have send-degree-res:
  s4 = s3 (
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3 )
using p.send-outcome[OF send-degree] .
have st-s5:
  PState s5 =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers)))
using st-s1 send-f-res alpha-data send-degree-res
  create-preserves-state[OF create-cp] replay cp'-def
unfolding verifier-replay-state-def by simp
have after-hash: s6 ≤ prover-state
using s6-final-hash tail-hash by (rule p.hash-ext-trans)
have fri-lookups:
  ∀ i < length roots.
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          (of-nat (degree (p.cp as p.f-powers))))
        (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-s6:
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
      Some (challenges ! i)
using fri-lookups-s6 i-bound by simp
have lookup-prover:
  fmlookup (HashMap prover-state) (FiatShamirChallenge (foldl concat (PState s5) (take

```

```

(Suc i) roots))) =
  Some (challenges ! i)
using p.hash-extension-lookup[OF lookup-s6 after-hash] .
show fmlookup (HashMap replay-state)
  (FiatShamirChallenge (foldl concat
    (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers)))))
    (take (Suc i) roots))) =
  Some (challenges ! i)
using lookup-prover st-s5 replay
unfolding verifier-replay-state-def by simp
qed
have nf-roots:
  None  $\notin$  dom (dist (execute
    (do {
      fr  $\leftarrow$  p.read;
      as'  $\leftarrow$  mmap (replicate (length spec)
        (do {
          a0  $\leftarrow$  p.receive-alpha-challenge;
          let a0' = a0;
          a1  $\leftarrow$  p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        })))
      dg  $\leftarrow$  p.read;
      assert (to-nat dg  $\leq$  p.maxDegree);
      fl  $\leftarrow$  ntimes v.receive-fri-commits (length roots);
      return (fr, as', dg, fl)
    }) replay-state))
by (rule honest-root-alpha-degree-fri-prefix-no-failure[
  OF htv conjunct1[OF root-alpha-tr] - replay-prefix alpha-lookups fri-lookups])
  (use len-roots len-challenges in simp-all)
show ?thesis
using nf-roots len-roots by simp
qed

```

lemma honest-root-alpha-degree-fri-final-dynamic-from-prover-tail-no-failure:

```

assumes htv: honest-trace-valid
and create-f:
  Some (f-merkle, s1)  $\in$ 
    set-dist (execute (p.create p.f-eval) init-state)
and send-f:
  Some ((), s2)  $\in$  set-dist (execute (p.send (value f-merkle)) s1)
and alphas:
  Some (as, s3)  $\in$ 
    set-dist (execute
      (mmap (replicate (length spec)
        (do {

```

```

    a ← p.receive-alpha-challenge;
    p.send a;
    return a
  }))) s2)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some (((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp')))) s3)
and create-cp:
  Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and nrounds-def: nrounds = ceil-log (degree cp' + 1)
and fri:
  Some ((ps, ds, ls, ms), s6) ∈
    set-dist (execute
      (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and final-send:
  Some (((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and tail:
  Some (prover-result, prover-state) ∈
    set-dist (execute (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds) s-final)
and replay: replay-state = verifier-replay-state prover-state
shows
  None ∉ dom (dist (execute
    (do {
      fr ← p.read;
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        })))
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
      final ← p.read;
      return (fr, as', dg, fl, final)
    }) replay-state))
proof –
have root-alpha-tr:
  length as = length spec ∧
  PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
by (rule prover-root-alpha-degree-transcript[
  OF create-f send-f alphas cp'-def send-degree])
have tail-hash: s-final ≤ prover-state
by (rule ntimes-hash-extends[OF - tail])
  (rule prover-query-round-hash-extends)
have s3-s4: s3 ≤ s4

```

```

    using send-hash-extends[OF send-degree] .
have s4-s5:  $s_4 \leq s_5$ 
    using create-hash-extends[OF create-cp] .
have s5-s6:  $s_5 \leq s_6$ 
    using honest-fri-commit-extends[OF fri] .
have s6-final-hash:  $s_6 \leq s\text{-final}$ 
    using send-hash-extends[OF final-send] .
have later-hash:  $s_3 \leq \text{prover-state}$ 
    using s3-s4 s4-s5 s5-s6 s6-final-hash tail-hash by (meson p.hash-ext-trans)
have alpha-lookups:
   $\forall i < \text{length } as.$ 
    fmllookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
play-state) (value f-merkle)) (take i as))) =
        Some (as ! i)
    by (rule prover-alpha-lookups-in-replay-state[
      OF create-f send-f alphas later-hash replay])
from fri-commit-replay-data[OF fri] obtain roots challenges where
  len-roots: length roots = nrounds
  and len-challenges: length challenges = nrounds
  and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
  and st-s6: PState s6 = foldl concat (PState s5) roots
  and fri-lookups-s6:
     $\forall i < \text{length } roots.$ 
      fmllookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
        Some (challenges ! i)
    by blast
have final-tail-tr: transcript-extends prover-state s-final
  by (rule ntimes-transcript-extends[OF - tail])
    (rule prover-query-round-transcript-extends)
from final-tail-tr obtain ys where
  tr-prover: PTranscript prover-state = ys @ PTranscript s-final
  unfolding transcript-extends-def by auto
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have final-send-res:
  s-final = s6(
    PState := concat (PState s6) (hd (last ls)),
    PTranscript := hd (last ls) # PTranscript s6)
  using p.send-outcome[OF final-send] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
    roots @ [hd (last ls)] @ rev ys
  using replay tr-prover final-send-res tr-s6 tr-s5 root-alpha-tr cp'-def
  unfolding verifier-replay-state-def by simp
have st-s1: PState s1 = 0
  using create-preserves-state[OF create-f]

```

```

unfolding init-state-def by simp
have send-f-res:
  s2 = s1⟦
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1⟧
  using p.send-outcome[OF send-f] .
have alpha-data:
  PState s3 = foldl concat (PState s2) as
  using alpha-mmap-outcome[OF alphas] by simp
have send-degree-res:
  s4 = s3⟦
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3⟧
  using p.send-outcome[OF send-degree] .
have st-s5:
  PState s5 =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers))))
  using st-s1 send-f-res alpha-data send-degree-res
    create-preserves-state[OF create-cp] replay cp'-def
  unfolding verifier-replay-state-def by simp
have after-hash: s6 ≤ prover-state
  using s6-final-hash tail-hash by (rule p.hash-ext-trans)
have fri-lookups:
  ∀ i < length roots.
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          (of-nat (degree (p.cp as p.f-powers))))))
        (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-s6:
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
    roots)))) =
      Some (challenges ! i)
  using fri-lookups-s6 i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state) (FiatShamirChallenge (foldl concat (PState s5) (take
    (Suc i) roots)))) =
      Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-s6 after-hash] .
show fmlookup (HashMap replay-state)
  (FiatShamirChallenge (foldl concat
    (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers))))))
    (take (Suc i) roots))) =

```



```

    Some (challenges ! i)
    using lookup-prover st-s5 replay
    unfolding verifier-replay-state-def by simp
qed
have roots-rounds:
  length roots = ceil-log (degree (p.cp as p.f-powers) + 1)
  using len-roots nrounds-def cp'-def by simp
have len-challenges-roots: length challenges = length roots
  using len-challenges len-roots by simp
show ?thesis
  by (rule honest-root-alpha-degree-fri-final-dynamic-prefix-no-failure[
    OF htv conjunct1 [OF root-alpha-tr] len-challenges-roots roots-rounds
    replay-prefix alpha-lookups fri-lookups])
qed

lemma honest-root-alpha-degree-fri-final-dynamic-from-prover-tail-outcome:
  assumes htv: honest-trace-valid
  and create-f:
    Some (f-merkle, s1) ∈
      set-dist (execute (p.create p.f-eval) init-state)
  and send-f:
    Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
  and alphas:
    Some (as, s3) ∈
      set-dist (execute
        (mmap (replicate (length spec)
          (do {
            a ← p.receive-alpha-challenge;
            p.send a;
            return a
          }))) s2)
  and cp'-def: cp' = p.cp as p.f-powers
  and send-degree:
    Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp')) s3)
  and create-cp:
    Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
  and nrounds-def: nrounds = ceil-log (degree cp' + 1)
  and fri:
    Some ((ps, ds, ls, ms), s6) ∈
      set-dist (execute
        (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
  and final-send:
    Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
  and tail:
    Some (prover-result, prover-state) ∈
      set-dist (execute (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds) s-final)
  and replay: replay-state = verifier-replay-state prover-state
  and outcome:
    Some ((fr, as', dg, fl, final), t) ∈ set-dist (execute

```

```

(do {
  fr ← p.read;
  as' ← mmap (replicate (length spec)
    (do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }));
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
  final ← p.read;
  return (fr, as', dg, fl, final)
}) replay-state)
obtains roots challenges ys where
  length roots = nrounds
  length challenges = nrounds
  fl = zip challenges roots
   $\bigwedge j. j < nrounds \implies roots ! j = value (ms ! j)$ 
   $\bigwedge j. j < nrounds \implies$ 
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j)
  fr = value f-merkle
  as' = as
  dg = of-nat (degree (p.cp as p.f-powers))
  final = hd (last ls)
  PState t = PState s-final
  PTranscript t = rev ys
  PTranscript prover-state = ys @ PTranscript s-final
  replay-state ≤ t
proof –
  have root-alpha-tr:
    length as = length spec ∧
    PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
  by (rule prover-root-alpha-degree-transcript[
    OF create-f send-f alphas cp'-def send-degree])
  have tail-hash: s-final ≤ prover-state
  by (rule ntimes-hash-extends[OF - tail])
    (rule prover-query-round-hash-extends)
  have s3-s4: s3 ≤ s4
  using send-hash-extends[OF send-degree] .
  have s4-s5: s4 ≤ s5
  using create-hash-extends[OF create-cp] .
  have s5-s6: s5 ≤ s6
  using honest-fri-commit-extends[OF fri] .
  have s6-final-hash: s6 ≤ s-final

```

```

using send-hash-extends[OF final-send] .
have later-hash:  $s3 \leq \text{prover-state}$ 
using s3-s4 s4-s5 s5-s6 s6-final-hash tail-hash by (meson p.hash-ext-trans)
have alpha-lookups:
   $\forall i < \text{length as}.$ 
    fmllookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
play-state) (value f-merkle)) (take i as))) =
      Some (as ! i)
by (rule prover-alpha-lookups-in-replay-state[
  OF create-f send-f alphas later-hash replay])
from fri-commit-initial-replay-successor-data[OF fri] obtain roots challenges where
  len-roots: length roots = nrounds
  and len-challenges: length challenges = nrounds
  and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
  and st-s6: PState s6 = foldl concat (PState s5) roots
  and s5-s6':  $s5 \leq s6$ 
  and fri-lookups-s6:
     $\forall i < \text{length roots}.$ 
      fmllookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
      Some (challenges ! i)
  and roots-aligned-all:  $\forall j < \text{nrounds}. \text{roots} ! j = \text{value} (\text{ms} ! j)$ 
  and successors-all:
     $\forall j < \text{nrounds}.$ 
      p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j)
  by blast
have roots-aligned:
   $\bigwedge j. j < \text{nrounds} \implies \text{roots} ! j = \text{value} (\text{ms} ! j)$ 
using roots-aligned-all by simp
have successors:
   $\bigwedge j. j < \text{nrounds} \implies$ 
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
    (ps ! Suc j, ds ! Suc j, ls ! Suc j)
using successors-all by simp
have final-tail-tr: transcript-extends prover-state s-final
by (rule ntimes-transcript-extends[OF - tail])
  (rule prover-query-round-transcript-extends)
from final-tail-tr obtain ys where
  tr-prover: PTranscript prover-state = ys @ PTranscript s-final
  unfolding transcript-extends-def by auto
have tr-s5: PTranscript s5 = PTranscript s4
using create-preserves-transcript[OF create-cp] .
have final-send-res:
  s-final = s6(|
    PState := concat (PState s6) (hd (last ls)),
    PTranscript := hd (last ls) # PTranscript s6|)
using p.send-outcome[OF final-send] .

```

```

have replay-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
      roots @ [hd (last ls)] @ rev ys
  using replay tr-prover final-send-res tr-s6 tr-s5 root-alpha-tr cp'-def
  unfolding verifier-replay-state-def by simp
have st-s1: PState s1 = 0
  using create-preserves-state[OF create-f]
  unfolding init-state-def by simp
have send-f-res:
  s2 = s1 |
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1 |
  using p.send-outcome[OF send-f] .
have alpha-data:
  PState s3 = foldl concat (PState s2) as
  using alpha-mmap-outcome[OF alphas] by simp
have send-degree-res:
  s4 = s3 |
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3 |
  using p.send-outcome[OF send-degree] .
have st-s5:
  PState s5 =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers)))
  using st-s1 send-f-res alpha-data send-degree-res
    create-preserves-state[OF create-cp] replay cp'-def
  unfolding verifier-replay-state-def by simp
have after-hash: s6 ≤ prover-state
  using s6-final-hash tail-hash by (rule p.hash-ext-trans)
have fri-lookups:
  ∀ i < length roots.
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          (of-nat (degree (p.cp as p.f-powers)))))
        (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-s6:
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
      Some (challenges ! i)
  using fri-lookups-s6 i-bound by simp
have lookup-prover:
  fmlookup (HashMap prover-state) (FiatShamirChallenge (foldl concat (PState s5) (take

```

```

(Suc i) roots))) =
  Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-s6 after-hash] .
show fmlookup (HashMap replay-state)
  (FiatShamirChallenge (foldl concat
    (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers)))))
    (take (Suc i) roots))) =
  Some (challenges ! i)
  using lookup-prover st-s5 replay
  unfolding verifier-replay-state-def by simp
qed
have roots-rounds:
  length roots = ceil-log (degree (p.cp as p.f-powers) + 1)
  using len-roots nrounds-def cp'-def by simp
have len-challenges-roots: length challenges = length roots
  using len-challenges len-roots by simp
have prefix-res:
  fr = value f-merkle ∧
  as' = as ∧
  dg = of-nat (degree (p.cp as p.f-powers)) ∧
  fl = zip challenges roots ∧
  final = hd (last ls) ∧
  PState t =
    concat
      (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          (of-nat (degree (p.cp as p.f-powers)))))
        roots)
      (hd (last ls)) ∧
  PTranscript t = rev ys ∧
  replay-state ≤ t
by (rule honest-root-alpha-degree-fri-final-dynamic-prefix-outcome[
  OF htv conjunct1[OF root-alpha-tr] len-challenges-roots roots-rounds
  replay-prefix alpha-lookups fri-lookups outcome])
have st-t-final: PState t = PState s-final
  using prefix-res st-s5 st-s6 final-send-res by simp
show ?thesis
  by (rule that[
    OF len-roots len-challenges - roots-aligned successors - - -
    st-t-final - tr-prover -])
  (use prefix-res in simp-all)
qed

```

lemma honest-root-alpha-degree-fri-final-from-prover-no-failure:

assumes htv: honest-trace-valid

and create-f:

Some (f-merkle, s1) ∈

set-dist (execute (p.create p.f-eval) init-state)

and *send-f*:
Some $(((), s2) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{value } f\text{-merkle}))) s1)$
and *alphas*:
Some $(as, s3) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{mmap } (\text{replicate } (\text{length } \text{spec}))$
 $(\text{do } \{$
 $\quad a \leftarrow p.\text{receive-alpha-challenge};$
 $\quad p.\text{send } a;$
 $\quad \text{return } a$
 $\quad \}))) s2)$
and *cp'-def*: $cp' = p.cp \text{ as } p.f\text{-powers}$
and *send-degree*:
Some $(((), s4) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{of-nat } (\text{degree } cp')))) s3)$
and *create-cp*:
Some $(cp\text{-merkle}, s5) \in \text{set-dist } (\text{execute } (p.\text{create } (p.cp\text{-eval } as)) s4)$
and *fri*:
Some $((ps, ds, ls, ms), s6) \in$
 $\text{set-dist } (\text{execute}$
 $(p.fri\text{-commit } nrounds [cp'] [p.eval\text{-domain}] [p.cp\text{-eval } as] [cp\text{-merkle}]) s5)$
and *final-send*:
Some $(((), s\text{-final}) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{hd } (\text{last } ls)))) s6)$
and *random-idx*:
Some $(idx, s7) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s\text{-final})$
and *idx-def*: $idx' = p.index \text{ (to-nat } idx)$
and *query-decommit*:
Some $(qouts, s8) \in$
 $\text{set-dist } (\text{execute } (\text{mmap } (p.\text{decommit-on-query } idx' f\text{-merkle})) s7)$
and *fri-decommit*:
Some $(fri\text{-idx}, s9) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{mfold } idx' (p.\text{decommit-on-fri-layers } (\text{butlast } (\text{zip } ls ms)))) s8)$
and *prover-state-eq*: $\text{prover-state} = s9$
and *replay*: $\text{replay-state} = \text{verifier-replay-state } \text{prover-state}$
shows
 $\text{None} \notin \text{dom } (\text{dist } (\text{execute}$
 $(\text{do } \{$
 $\quad fr \leftarrow p.\text{read};$
 $\quad as' \leftarrow \text{mmap } (\text{replicate } (\text{length } \text{spec}))$
 $\quad (\text{do } \{$
 $\quad \quad a0 \leftarrow p.\text{receive-alpha-challenge};$
 $\quad \quad \text{let } a0' = a0;$
 $\quad \quad a1 \leftarrow p.\text{read};$
 $\quad \quad \text{let } a1' = a1;$
 $\quad \quad \text{assert } (a0' = a1');$
 $\quad \quad \text{return } a1'$
 $\quad \quad \}));$
 $\quad dg \leftarrow p.\text{read};$
 $\quad \text{assert } (\text{to-nat } dg \leq p.\text{maxDegree});$

```

    fl ← ntimes v.receive-fri-commits nrounds;
    final ← p.read;
    return (fr, as', dg, fl, final)
  }) replay-state))
proof -
  have root-alpha-tr:
    length as = length spec ∧
    PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
  by (rule prover-root-alpha-degree-transcript[
    OF create-f send-f alphas cp'-def send-degree])
  have later-hash: s3 ≤ prover-state
  by (rule prover-later-phases-hash-extends[
    OF send-degree create-cp fri final-send random-idx
    query-decommit fri-decommit prover-state-eq])
  have alpha-lookups:
    ∀ i < length as.
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
        play-state) (value f-merkle)) (take i as))) =
      Some (as ! i)
  by (rule prover-alpha-lookups-in-replay-state[
    OF create-f send-f alphas later-hash replay])
  from fri-commit-replay-data[OF fri] obtain roots challenges where
    len-roots: length roots = nrounds
  and len-challenges: length challenges = nrounds
  and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
  and st-s6: PState s6 = foldl concat (PState s5) roots
  and fri-hash: s5 ≤ s6
  and fri-lookups-s6:
    ∀ i < length roots.
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
      roots))) =
      Some (challenges ! i)
  by blast
  have final-tail-tr: transcript-extends prover-state s-final
proof -
  have final-s7: transcript-extends s7 s-final
  using receive-query-index-challenge-transcript-extends[OF random-idx] .
  have s7-s8: transcript-extends s8 s7
  by (rule mmap-transcript-extends[OF query-decommit])
  (rule decommit-on-query-step-transcript-extends)
  have s8-s9: transcript-extends s9 s8
  by (rule mfold-transcript-extends[OF fri-decommit])
  (rule decommit-on-fri-layers-step-transcript-extends)
  have s9-prover: transcript-extends prover-state s9
  using prover-state-eq by simp
  show ?thesis
  using final-s7 s7-s8 s8-s9 s9-prover by (meson transcript-extends-trans)
qed

```

```

from final-tail-tr obtain ys where
  tr-prover: PTranscript prover-state = ys @ PTranscript s-final
  unfolding transcript-extends-def by auto
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have final-send-res:
  s-final = s6()
  PState := concat (PState s6) (hd (last ls)),
  PTranscript := hd (last ls) # PTranscript s6()
  using p.send-outcome[OF final-send] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers)))] @
    roots @ [hd (last ls)] @ rev ys
  using replay tr-prover final-send-res tr-s6 tr-s5 root-alpha-tr cp'-def
  unfolding verifier-replay-state-def by simp
have st-s1: PState s1 = 0
  using create-preserves-state[OF create-f]
  unfolding init-state-def by simp
have send-f-res:
  s2 = s1()
  PState := concat (PState s1) (value f-merkle),
  PTranscript := value f-merkle # PTranscript s1()
  using p.send-outcome[OF send-f] .
have alpha-data:
  PState s3 = foldl concat (PState s2) as
  using alpha-mmap-outcome[OF alphas] by simp
have send-degree-res:
  s4 = s3()
  PState := concat (PState s3) (of-nat (degree cp')),
  PTranscript := of-nat (degree cp') # PTranscript s3()
  using p.send-outcome[OF send-degree] .
have st-s5:
  PState s5 =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers))))
  using st-s1 send-f-res alpha-data send-degree-res
  create-preserves-state[OF create-cp] replay cp'-def
  unfolding verifier-replay-state-def by simp
have after-hash: s6 ≤ prover-state
  by (rule prover-after-fri-hash-extends[
    OF final-send random-idx query-decommit fri-decommit prover-state-eq])
have fri-lookups:
  ∀ i < length roots.
  fmlookup (HashMap replay-state)
    (FiatShamirChallenge (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        (of-nat (degree (p.cp as p.f-powers))))))
      (take (Suc i) roots))) =

```



```

    Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-s6:
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
    Some (challenges ! i)
  using fri-lookups-s6 i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state) (FiatShamirChallenge (foldl concat (PState s5) (take
(Suc i) roots))) =
    Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-s6 after-hash] .
  show fmlookup (HashMap replay-state)
(FiatShamirChallenge (foldl concat
(concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
(of-nat (degree (p.cp as p.f-powers))))))
(take (Suc i) roots)) =
  Some (challenges ! i)
  using lookup-prover st-s5 replay
  unfolding verifier-replay-state-def by simp
qed
have len-challenges-roots: length challenges = length roots
  using len-challenges len-roots by simp
have nf-roots:
  None  $\notin$  dom (dist (execute
(do {
  fr  $\leftarrow$  p.read;
  as'  $\leftarrow$  mmap (replicate (length spec)
(do {
  a0  $\leftarrow$  p.receive-alpha-challenge;
  let a0' = a0;
  a1  $\leftarrow$  p.read;
  let a1' = a1;
  assert (a0' = a1');
  return a1'
})));
  dg  $\leftarrow$  p.read;
  assert (to-nat dg  $\leq$  p.maxDegree);
  fl  $\leftarrow$  ntimes v.receive-fri-commits (length roots);
  final  $\leftarrow$  p.read;
  return (fr, as', dg, fl, final)
}) replay-state))
by (rule honest-root-alpha-degree-fri-final-prefix-no-failure[
  OF htv conjunct1[OF root-alpha-tr] len-challenges-roots
  replay-prefix alpha-lookups fri-lookups])
show ?thesis
  using nf-roots len-roots by simp

```

qed

lemma *honest-root-alpha-degree-fri-final-dynamic-from-prover-no-failure:*

assumes *htv: honest-trace-valid*
and *create-f:*
 Some (f-merkle, s1) ∈
 set-dist (execute (p.create p.f-eval) init-state)
and *send-f:*
 Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and *alphas:*
 Some (as, s3) ∈
 set-dist (execute
 (mmap (replicate (length spec)
 (do {
 a ← p.receive-alpha-challenge;
 p.send a;
 return a
 }))) s2)
and *cp'-def: cp' = p.cp as p.f-powers*
and *send-degree:*
 Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
and *create-cp:*
 Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and *nrounds-def: nrounds = ceil-log (degree cp' + 1)*
and *fri:*
 Some ((ps, ds, ls, ms), s6) ∈
 set-dist (execute
 (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and *final-send:*
 Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and *random-idx:*
 Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
and *idx-def: idx' = p.index (to-nat idx)*
and *query-decommit:*
 Some (qouts, s8) ∈
 set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
and *fri-decommit:*
 Some (fri-idx, s9) ∈
 set-dist (execute
 (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
and *prover-state-eq: prover-state = s9*
and *replay: replay-state = verifier-replay-state prover-state*
shows
 None ∉ dom (dist (execute
 (do {
 fr ← p.read;
 as' ← mmap (replicate (length spec)
 (do {
 a0 ← p.receive-alpha-challenge;

```

    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }));
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
  final ← p.read;
  return (fr, as', dg, fl, final)
}) replay-state))
proof –
  have root-alpha-tr:
    length as = length spec ∧
    PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
  by (rule prover-root-alpha-degree-transcript[
    OF create-f send-f alphas cp'-def send-degree])
  have later-hash: s3 ≤ prover-state
  by (rule prover-later-phases-hash-extends[
    OF send-degree create-cp fri final-send random-idx
    query-decommit fri-decommit prover-state-eq])
  have alpha-lookups:
    ∀ i < length as.
      fmlookup (HashMap replay-state)
        (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
        play-state) (value f-merkle)) (take i as))) =
        Some (as ! i)
  by (rule prover-alpha-lookups-in-replay-state[
    OF create-f send-f alphas later-hash replay])
  from fri-commit-replay-data[OF fri] obtain roots challenges where
    len-roots: length roots = nrounds
    and len-challenges: length challenges = nrounds
    and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
    and st-s6: PState s6 = foldl concat (PState s5) roots
    and fri-hash: s5 ≤ s6
    and fri-lookups-s6:
      ∀ i < length roots.
        fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
        roots))) =
        Some (challenges ! i)
  by blast
  have final-tail-tr: transcript-extends prover-state s-final
proof –
  have final-s7: transcript-extends s7 s-final
  using receive-query-index-challenge-transcript-extends[OF random-idx] .
  have s7-s8: transcript-extends s8 s7
  by (rule mmap-transcript-extends[OF query-decommit])
    (rule decommit-on-query-step-transcript-extends)

```

```

have s8-s9: transcript-extends s9 s8
  by (rule mfold-transcript-extends[OF fri-decommit])
    (rule decommit-on-fri-layers-step-transcript-extends)
have s9-prover: transcript-extends prover-state s9
  using prover-state-eq by simp
show ?thesis
  using final-s7 s7-s8 s8-s9 s9-prover by (meson transcript-extends-trans)
qed
from final-tail-tr obtain ys where
  tr-prover: PTranscript prover-state = ys @ PTranscript s-final
  unfolding transcript-extends-def by auto
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have final-send-res:
  s-final = s6(
    PState := concat (PState s6) (hd (last ls)),
    PTranscript := hd (last ls) # PTranscript s6)
  using p.send-outcome[OF final-send] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
    roots @ [hd (last ls)] @ rev ys
  using replay tr-prover final-send-res tr-s6 tr-s5 root-alpha-tr cp'-def
  unfolding verifier-replay-state-def by simp
have st-s1: PState s1 = 0
  using create-preserves-state[OF create-f]
  unfolding init-state-def by simp
have send-f-res:
  s2 = s1(
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1)
  using p.send-outcome[OF send-f] .
have alpha-data:
  PState s3 = foldl concat (PState s2) as
  using alpha-mmap-outcome[OF alphas] by simp
have send-degree-res:
  s4 = s3(
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3)
  using p.send-outcome[OF send-degree] .
have st-s5:
  PState s5 =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
    (of-nat (degree (p.cp as p.f-powers)))
  using st-s1 send-f-res alpha-data send-degree-res
  create-preserves-state[OF create-cp] replay cp'-def
  unfolding verifier-replay-state-def by simp
have after-hash: s6 ≤ prover-state
  by (rule prover-after-fri-hash-extends[

```

```

    OF final-send random-idx query-decommit fri-decommit prover-state-eq])
have fri-lookups:
   $\forall i < \text{length roots}.$ 
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          (of-nat (degree (p.cp as p.f-powers))))))
        (take (Suc i) roots))) =
      Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{length roots}$ 
  have lookup-s6:
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
      Some (challenges ! i)
  using fri-lookups-s6 i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state) (FiatShamirChallenge (foldl concat (PState s5) (take
(Suc i) roots))) =
      Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-s6 after-hash] .
  show fmlookup (HashMap replay-state)
    (FiatShamirChallenge (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        (of-nat (degree (p.cp as p.f-powers))))))
      (take (Suc i) roots))) =
      Some (challenges ! i)
  using lookup-prover st-s5 replay
  unfolding verifier-replay-state-def by simp
qed
have roots-rounds:
   $\text{length roots} = \text{ceil-log } (\text{degree } (p.cp \text{ as } p.f\text{-powers}) + 1)$ 
  using len-roots nrounds-def cp'-def by simp
have len-challenges-roots:  $\text{length challenges} = \text{length roots}$ 
  using len-challenges len-roots by simp
show ?thesis
  by (rule honest-root-alpha-degree-fri-final-dynamic-prefix-no-failure[
    OF htv conjunct1[OF root-alpha-tr] len-challenges-roots roots-rounds
      replay-prefix alpha-lookups fri-lookups])
qed
end
end

```

```

theory Completeness-Query
imports Completeness-Replay

```

begin

2.5 Honest Query Checks

Honest query decommitment values and authenticated query replay.

context *verification*
begin

lemma *transcript-read-cons-wp-error*:
 $wp_error\ p.read\ (s \setminus PTranscript := x \# xs) = 0$
unfolding *p.read-def protocol-read-def assert-def* **by** (*simp add: wpsimps*)

lemma *merkle-authentication-path-outcome*:
assumes *p.created-tree xs r s0*
and $length\ xs = 2^n$
and $i < length\ xs$
and $v = xs[i]$
and $s0 \leq s$
and $Some\ (h, t) \in set_dist\ (execute\ (p.check_authentication_path\ (length\ xs)\ i\ v\ (get_authentication_path\ (length\ xs)\ i\ r))\ s)$
shows $h = value\ r$
using *p.check-created-tree-outcome[OF assms]* **by** *simp*

lemma *query-f-eval-nth*:
assumes $i < length\ p.eval_domain$
shows $p.f_eval\ !\ i = poly\ p.f\ (p.eval_domain\ !\ i)$
using *assms unfolding p.f-eval-def* **by** *simp*

lemma *f-powers-nth*:
assumes $k < powers$
shows $p.f_powers\ !\ k = p.f \circ_p XP\ (p.g \wedge k)$
using *assms unfolding p.f-powers-def* **by** *simp*

lemma *f-powers-length*:
 $length\ p.f_powers = powers$
unfolding *p.f-powers-def* **by** *simp*

lemma *f-powers-eval*:
assumes $k < powers$
shows $poly\ (p.f_powers\ !\ k)\ x = poly\ p.f\ (x * p.g \wedge k)$
using *f-powers-nth[OF assms]*
unfolding *XP-def*
by (*simp add: poly-pcompose poly-monom algebra-simps*)

lemma *query-cp-eval-nth*:
assumes $i < length\ p.eval_domain$
shows $p.cp_eval\ as\ !\ i = poly\ (p.cp\ as\ p.f_powers)\ (p.eval_domain\ !\ i)$
using *assms unfolding p.cp-eval-def* **by** *simp*

```

lemma powers-scaled-setD:
  assumes  $i \in \text{set } (p.\text{powers-scaled } \text{idx})$ 
  obtains  $k$  where  $k < \text{powers}$  and  $i = \text{idx} + k * \text{scale}$ 
  using assms unfolding p.powers-scaled-def by auto

lemma clength-pos:
   $0 < \text{clength}$ 
proof –
  obtain  $n$  where  $\text{len}: \text{clength} * \text{scale} = 2^n$ 
    using p.eval-domain-length-power by blast
  show ?thesis
    using  $\text{len}$  p.scale-pos by (cases clength) simp-all
qed

lemma max-powers-less-clength:
   $\text{Max } (\text{set } [0..<\text{powers}]) < \text{clength}$ 
proof –
  have  $\text{Max } (\text{set } [0..<\text{powers}]) = \text{powers} - 1$ 
    using p.powers-pos by (intro Max-eqI) auto
  also have  $\dots < \text{clength}$ 
    using p.powers-pos p.powers-le-clength by simp
  finally show ?thesis .
qed

lemma index-less-domain:
   $p.\text{index } x < \text{clength} * \text{scale}$ 
proof –
  have max-le:  $\text{Max } (\text{set } [0..<\text{powers}]) < \text{clength}$ 
    by (rule max-powers-less-clength)
  have mod-pos:  $0 < \text{clength} * \text{scale} - \text{Max } (\text{set } [0..<\text{powers}]) * \text{scale}$ 
    using max-le p.scale-pos
    by (simp add: diff-mult-distrib2)
  have idx-lt:
     $p.\text{index } x < \text{clength} * \text{scale} - \text{Max } (\text{set } [0..<\text{powers}]) * \text{scale}$ 
    unfolding p.index-def
    using mod-pos by simp
  have  $\text{clength} * \text{scale} - \text{Max } (\text{set } [0..<\text{powers}]) * \text{scale} \leq \text{clength} * \text{scale}$ 
    by simp
  then show ?thesis
    using idx-lt by linarith
qed

lemma index-less-domain-tail:
  assumes  $k < \text{powers}$ 
  shows  $p.\text{index } x + k * \text{scale} < \text{clength} * \text{scale}$ 
proof –
  have max-le:  $\text{Max } (\text{set } [0..<\text{powers}]) < \text{clength}$ 
    by (rule max-powers-less-clength)

```

```

have mod-pos:  $0 < \text{clength} * \text{scale} - \text{Max} (\text{set } [0..<\text{powers}]) * \text{scale}$ 
  using max-le p.scale-pos
  by (simp add: diff-mult-distrib2)
have idx-lt:
   $p.\text{index } x < \text{clength} * \text{scale} - \text{Max} (\text{set } [0..<\text{powers}]) * \text{scale}$ 
  unfolding p.index-def
  using mod-pos by simp
have k-le-max:  $k * \text{scale} \leq \text{Max} (\text{set } [0..<\text{powers}]) * \text{scale}$ 
proof (cases powers = 0)
  case True
  then show ?thesis using assms by simp
next
  case False
  then have  $k \leq \text{Max} (\text{set } [0..<\text{powers}])$ 
    using assms by simp
  then show ?thesis by simp
qed
have p.index  $x + k * \text{scale} <$ 
   $(\text{clength} * \text{scale} - \text{Max} (\text{set } [0..<\text{powers}]) * \text{scale}) +$ 
   $\text{Max} (\text{set } [0..<\text{powers}]) * \text{scale}$ 
  using idx-lt k-le-max by linarith
also have  $\dots = \text{clength} * \text{scale}$ 
  using max-le p.scale-pos
  by (simp add: diff-mult-distrib2)
finally show ?thesis .
qed

```

```

lemma powers-scaled-index-bound:
  assumes  $i \in \text{set } (p.\text{powers-scaled } (p.\text{index } x))$ 
  shows  $i < \text{clength} * \text{scale}$ 
  using assms
  by (auto elim!: powers-scaled-setD intro!: index-less-domain-tail)

```

```

lemma f-eval-length:
   $\text{length } p.f\text{-eval} = \text{clength} * \text{scale}$ 
  unfolding p.f-eval-def p.eval-domain-def p.H-def by simp

```

```

lemma trace-polynomial-degree-lt-clength:
   $\text{degree } p.f < \text{clength}$ 
proof -
  have vals-len:  $\text{length } \text{vals} = \text{clength}$ 
    using p.trace-length by simp
  have G-len:  $\text{length } p.G = \text{clength}$ 
    unfolding p.G-def by simp
  have deg-le:
     $\text{degree } p.f \leq \text{length } (\text{zip } p.G \text{ vals}) - 1$ 
    unfolding p.f-def by (rule degree-lagrange-interpolation-poly)
  have zip-len:  $\text{length } (\text{zip } p.G \text{ vals}) = \text{clength}$ 
    using vals-len G-len by simp

```



```

show ?thesis
  using deg-le zip-len p.clength-pos by simp
qed

lemma honest-initial-trace-fri-accumulator-value:
  assumes fv-eq:  $fv = \text{map } (\lambda i. p.f\text{-eval } ! i) (p.powers\text{-scaled } (p.index\ x))$ 
    and l0-eq:  $l0 = p.f\text{-eval}$ 
  shows  $hd\ fv = l0\ !\ p.index\ x$ 
proof -
  have range:  $[0..<powers] = 0 \# [1..<powers]$ 
    using p.powers-pos by (simp add: upt-conv-Cons)
  have ps:
     $p.powers\text{-scaled } (p.index\ x) =$ 
     $p.index\ x \# \text{map } (\lambda xa. p.index\ x + xa * scale) [1..<powers]$ 
  unfolding p.powers-scaled-def using range by simp
  show ?thesis
    using fv-eq l0-eq ps by simp
qed

lemma eval-domain-length:
   $length\ p.eval\text{-domain} = clength * scale$ 
  unfolding p.eval-domain-def p.H-def by simp

lemma eval-domain-nth:
  assumes  $i < clength * scale$ 
  shows  $p.eval\text{-domain } !\ i = p.h \wedge^i * shift$ 
  using assms
  unfolding p.eval-domain-def p.H-def
  by (simp add: mult.commute)

lemma query-f-power-eval-nth:
  assumes k-bound:  $k < powers$ 
    and idx-bound:  $idx + k * scale < clength * scale$ 
  shows
     $p.f\text{-eval } !\ (idx + k * scale) =$ 
     $poly\ (p.f\text{-powers } !\ k)\ (p.h \wedge^{idx} * shift)$ 
proof -
  have eval:
     $p.f\text{-eval } !\ (idx + k * scale) =$ 
     $poly\ p.f\ (p.h \wedge^{(idx + k * scale)} * shift)$ 
  using query-f-eval-nth[of idx + k * scale] eval-domain-nth[OF idx-bound]
    idx-bound eval-domain-length
  by simp
  have arg:
     $(p.h \wedge^{idx} * shift) * p.g \wedge^k = p.h \wedge^{(idx + k * scale)} * shift$ 
proof -
  have gpow:  $p.g \wedge^k = p.h \wedge^{(scale * k)}$ 
    using p.domain-alignment by (simp add: power-mult)
  have  $(p.h \wedge^{idx} * shift) * p.g \wedge^k =$ 

```

```

      p.h ^ idx * p.h ^ (scale * k) * shift
    using gpow by (simp add: algebra-simps)
  also have ... = p.h ^ (idx + k * scale) * shift
    by (simp add: power-add algebra-simps mult.commute)
  finally show ?thesis .
qed
show ?thesis
  using eval f-powers-eval[OF k-bound, of p.h ^ idx * shift] arg
  by simp
qed

lemma honest-query-values-f-powers:
  map (λi. p.f-eval ! i) (p.powers-scaled (p.index x)) =
    map (λq. poly q (p.h ^ p.index x * shift)) p.f-powers
proof (rule nth-equalityI)
  show length (map (λi. p.f-eval ! i) (p.powers-scaled (p.index x))) =
    length (map (λq. poly q (p.h ^ p.index x * shift)) p.f-powers)
  unfolding p.powers-scaled-def using f-powers-length by simp
next
  fix k
  assume k-bound:
    k < length (map (λi. p.f-eval ! i) (p.powers-scaled (p.index x)))
  then have k-lt: k < powers
    unfolding p.powers-scaled-def by simp
  have idx-bound: p.index x + k * scale < clength * scale
    by (rule index-less-domain-tail[OF k-lt])
  have left:
    map (λi. p.f-eval ! i) (p.powers-scaled (p.index x)) ! k =
      p.f-eval ! (p.index x + k * scale)
    using k-lt unfolding p.powers-scaled-def by simp
  have right:
    map (λq. poly q (p.h ^ p.index x * shift)) p.f-powers ! k =
      poly (p.f-powers ! k) (p.h ^ p.index x * shift)
    using k-lt f-powers-length by simp
  show
    map (λi. p.f-eval ! i) (p.powers-scaled (p.index x)) ! k =
      map (λq. poly q (p.h ^ p.index x * shift)) p.f-powers ! k
    using query-f-power-eval-nth[OF k-lt idx-bound] left right by simp
qed

lemma honest-cp-eval-agrees-lists:
  fixes specs :: (('f poly list ⇒ 'f poly) × nat list × nat) list
  and spec2s :: (('f list ⇒ 'f) × nat list) list
  and pacc :: 'f poly
  and vacc :: 'f
  assumes hta: honest-trace-algebra
  and idx-bound: idx < clength * scale
  and len: length spec2s = length specs
  and zip-subset: set (zip specs spec2s) ⊆ set (zip spec spec2)

```

```

and specs-subset: set specs  $\subseteq$  set spec
and acc-eq: vacc = poly pacc (p.h  $\wedge$  idx * shift)
shows
  fold
    ( $\lambda(a, (c2, roots2)).$ 
      (+) (a * (c2 (map ( $\lambda q. poly\ q\ (p.h\ \wedge\ idx\ * shift))\ p.f-powers)\ div$ 
        poly (prod (p.g-map roots2)) (p.h  $\wedge$  idx * shift))))
      (zip as spec2s) vacc =
    poly
      (fold
        ( $\lambda(a, (c, roots, -)).$ 
          (+) (CP a * ((c p.f-powers) div prod (p.g-map roots))))
        (zip as specs) pacc)
        (p.h  $\wedge$  idx * shift)
    using len zip-subset specs-subset acc-eq
proof (induction specs arbitrary: as spec2s pacc vacc)
  case Nil
  then have spec2s = []
  by simp
  then show ?case
    using Nil.prem(4) by simp
next
  case (Cons s specs')
  obtain c roots d where s-eq: s = (c, roots, d)
  by (cases s) auto
  show ?case
  proof (cases spec2s)
  case Nil
  then show ?thesis
    using Cons.prem(1) by simp
  next
  case spec2-Cons: (Cons s2 spec2s')
  then obtain c2 roots2 where s2-eq: s2 = (c2, roots2)
  by (cases s2) auto
  have len-tail: length spec2s' = length specs'
    using Cons.prem(1) spec2-Cons by simp
  show ?thesis
  proof (cases as)
  case Nil
  then show ?thesis
    using Cons.prem(4) by simp
  next
  case as-Cons: (Cons a as')
  have pair-in:
    ((c, roots, d), (c2, roots2))  $\in$  set (zip spec spec2)
  proof -
  have (s, s2)  $\in$  set (zip (s # specs') (s2 # spec2s'))
    by simp
  moreover have set (zip (s # specs') (s2 # spec2s'))  $\subseteq$  set (zip spec spec2)

```

```

    using Cons.premis(2) spec2-Cons by simp
ultimately have (s, s2) ∈ set (zip spec spec2)
  by blast
then show ?thesis
  using s-eq s2-eq by simp
qed
have spec-entry: (c, roots, d) ∈ set spec
proof -
  have s ∈ set (s # specs')
  by simp
  then have s ∈ set spec
  using Cons.premis(3) by blast
  then show ?thesis
  using s-eq by simp
qed
have roots-eq: roots2 = roots
and raw-eq:
  c2 (map (λq. poly q (p.h ^ idx * shift)) p.f-powers) =
    poly (c p.f-powers) (p.h ^ idx * shift)
  using v.specs-agree[OF pair-in, of p.f-powers p.h ^ idx * shift]
  unfolding p.f-powers-def by simp-all
have quotient-eq:
  poly ((c p.f-powers) div prod (p.g-map roots)) (p.h ^ idx * shift) =
    poly (c p.f-powers) (p.h ^ idx * shift) div
      poly (prod (p.g-map roots)) (p.h ^ idx * shift)
  by (rule honest-trace-algebra-quotient-eval[OF hta spec-entry idx-bound])
have cp-eval: poly (CP a) (p.h ^ idx * shift) = a
  by (simp add: CP-def poly-monom)
have term-eq:
  a * (c2 (map (λq. poly q (p.h ^ idx * shift)) p.f-powers) div
    poly (prod (p.g-map roots2)) (p.h ^ idx * shift)) =
    poly (CP a * ((c p.f-powers) div prod (p.g-map roots)))
      (p.h ^ idx * shift)
  using roots-eq raw-eq quotient-eq cp-eval by simp
have acc-step:
  a * (c2 (map (λq. poly q (p.h ^ idx * shift)) p.f-powers) div
    poly (prod (p.g-map roots2)) (p.h ^ idx * shift)) + vacc =
    poly (CP a * ((c p.f-powers) div prod (p.g-map roots)) + pacc)
      (p.h ^ idx * shift)
  using term-eq Cons.premis(4) by simp
have tail:
  fold
    (λ(a, (c2, roots2)).
      (+) (a * (c2 (map (λq. poly q (p.h ^ idx * shift)) p.f-powers) div
        poly (prod (p.g-map roots2)) (p.h ^ idx * shift))))
    (zip as' spec2s')
  (a * (c2 (map (λq. poly q (p.h ^ idx * shift)) p.f-powers) div
    poly (prod (p.g-map roots2)) (p.h ^ idx * shift)) + vacc) =
  poly

```

```

(fold
  (λ(a, (c, roots, -)).
    (+) (CP a * ((c p.f-powers) div prod (p.g-map roots))))
  (zip as' specs')
  (CP a * ((c p.f-powers) div prod (p.g-map roots)) + pacc))
(p.h ^ idx * shift)
proof (rule Cons.IH)
show length spec2s' = length specs'
using len-tail .
show set (zip specs' spec2s') ⊆ set (zip spec spec2)
using Cons.premis(2) spec2-Cons by auto
show set specs' ⊆ set spec
using Cons.premis(3) by auto
show a * (c2 (map (λq. poly q (p.h ^ idx * shift)) p.f-powers) /
  poly (prod (p.g-map roots2)) (p.h ^ idx * shift)) + vacc =
  poly (CP a * ((c p.f-powers) div prod (p.g-map roots)) + pacc)
  (p.h ^ idx * shift)
using acc-step by simp
qed
show ?thesis
using as-Cons spec2-Cons s-eq s2-eq tail by simp
qed
qed
qed

```

lemma *honest-cp-eval-agrees*:

assumes *hta*: *honest-trace-algebra*

and *idx-bound*: *idx* < *clength* * *scale*

shows

v.cp-eval as (map (λq. poly q (p.h ^ idx * shift)) p.f-powers)

(p.h ^ idx * shift) =

poly (p.cp as p.f-powers) (p.h ^ idx * shift)

unfolding *v.cp-eval-def* *p.cp-def*

by (rule *honest-cp-eval-agrees-lists*[OF *hta idx-bound*])

(use *v.spec2-length* **in** auto)

lemma *honest-query-cp-value*:

assumes *honest-trace-valid*

shows

v.cp-eval as (map (λi. p.f-eval ! i) (p.powers-scaled (p.index x)))

(p.h ^ p.index x * shift) =

poly (p.cp as p.f-powers) (p.h ^ p.index x * shift)

proof –

have *hta*: *honest-trace-algebra*

using *assms* **unfolding** *honest-trace-valid-def* **by** simp

have *idx-bound*: *p.index x* < *clength* * *scale*

by (rule *index-less-domain*)

show ?thesis

using *honest-cp-eval-agrees*[OF *hta idx-bound*, of *as*]

$honest_query_values_f_powers[of\ x]$
 by *simp*
 qed

lemma *cp-eval-index-value*:
 $p.cp_eval\ as\ !\ p.index\ x =$
 $poly\ (p.cp\ as\ p.f_powers)\ (p.h\ ^\wedge\ p.index\ x\ * shift)$
proof –
 have *idx-bound*: $p.index\ x < length\ p.eval_domain$
 using *index-less-domain eval-domain-length* by *simp*
 have $p.cp_eval\ as\ !\ p.index\ x =$
 $poly\ (p.cp\ as\ p.f_powers)\ (p.eval_domain\ !\ p.index\ x)$
 by (*rule query-cp-eval-nth[OF idx-bound]*)
 also have $\dots = poly\ (p.cp\ as\ p.f_powers)\ (p.h\ ^\wedge\ p.index\ x\ * shift)$
 using *eval-domain-nth[OF index-less-domain[of x]]* by *simp*
 finally show *?thesis* .
 qed

lemma *honest-initial-fri-accumulator-value*:
 assumes *htv*: *honest-trace-valid*
 and *as-eq*: $as' = as$
 and *fv-eq*: $fv = map\ (\lambda i. p.f_eval\ !\ i)\ (p.powers_scaled\ (p.index\ x))$
 and *l0-eq*: $l0 = p.cp_eval\ as$
 shows
 $v.cp_eval\ as'\ fv\ (p.h\ ^\wedge\ p.index\ x\ * shift) = l0\ !\ p.index\ x$
 using *honest-query-cp-value[OF htv, of as x]* *cp-eval-index-value[of as x]*
 $as-eq\ fv-eq\ l0-eq$
 by *simp*

lemma *h-nonzero*:
 $p.h \neq 0$
 unfolding *p.h-def* using *p.omega-nonzero-derived* by *simp*

lemma *verifier-domain-point-nonzero*:
 $((p.h\ ^\wedge\ i) * shift) ^\wedge\ pw \neq 0$
 using *h-nonzero p.shift-nonzero* by *simp*

lemma *eval-domain-size-pos*:
 $0 < clength * scale$
 using *maxDegree-less-eval-domain* by *linarith*

lemma *h-power-mod-eval-domain*:
 $p.h\ ^\wedge\ (k\ mod\ (clength * scale)) = p.h\ ^\wedge\ k$
proof –
 let *?N* = $clength * scale$
 have *order*: $p.h\ ^\wedge\ ?N = 1$
 using *p.h-order* .
 have *k-eq*: $k = (k\ div\ ?N) * ?N + k\ mod\ ?N$
 using *div-mult-mod-eq[of k ?N]* by (*simp add: mult.commute*)

```

have period:  $p.h \wedge ((k \text{ div } ?N) * ?N) = 1$ 
proof -
  have  $p.h \wedge ((k \text{ div } ?N) * ?N) = p.h \wedge (?N * (k \text{ div } ?N))$ 
    by (simp add: mult.commute)
  also have  $\dots = (p.h \wedge ?N) \wedge (k \text{ div } ?N)$ 
    by (simp only: power-mult)
  also have  $\dots = 1$ 
    using order by simp
  finally show ?thesis .
qed
have  $p.h \wedge k = p.h \wedge ((k \text{ div } ?N) * ?N + k \text{ mod } ?N)$ 
  using k-eq by simp
also have  $\dots = p.h \wedge ((k \text{ div } ?N) * ?N) * p.h \wedge (k \text{ mod } ?N)$ 
  by (simp add: power-add)
also have  $\dots = p.h \wedge (k \text{ mod } ?N)$ 
  using period by simp
finally show ?thesis
  by simp
qed

lemma h-half-shift:
   $p.h \wedge (k + (\text{clength} * \text{scale}) \text{ div } 2) = -(p.h \wedge k)$ 
  using p.h-half-order
  by (simp add: power-add algebra-simps)

lemma mod-mult-right-factor-nat:
  fixes a b c :: nat
  assumes b-pos:  $0 < b$ 
    and c-pos:  $0 < c$ 
  shows  $((a \text{ mod } b) * c) \text{ mod } (b * c) = (a * c) \text{ mod } (b * c)$ 
proof -
  have decomp:  $a = a \text{ div } b * b + a \text{ mod } b$ 
    using div-mult-mod-eq[of a b] by simp
  have r-less:  $(a \text{ mod } b) * c < b * c$ 
    using b-pos c-pos mod-less-divisor[OF b-pos]
    by (simp add: mult-less-mono2)
  have r-le:  $(a \text{ mod } b) * c \leq a * c$ 
proof -
  have  $a \text{ mod } b \leq a$ 
    using decomp by linarith
  then show ?thesis
    by simp
qed
have dvd-diff:  $b * c \text{ dvd } a * c - (a \text{ mod } b) * c$ 
proof -
  have ac-decomp:  $a * c = (a \text{ div } b) * (b * c) + (a \text{ mod } b) * c$ 
proof -
  have  $a * c = (a \text{ div } b * b + a \text{ mod } b) * c$ 
    using decomp by simp

```

```

    also have ... = (a div b * b) * c + (a mod b) * c
      by (simp only: distrib-right)
    also have ... = (a div b) * (b * c) + (a mod b) * c
      by (simp add: algebra-simps)
    finally show ?thesis .
  qed
  have a * c - (a mod b) * c = (a div b) * (b * c)
    using ac-decomp by simp
  then show ?thesis by simp
  qed
  have (a * c) mod (b * c) = (a mod b) * c
    by (rule mod-nat-eqI[OF r-less r-le dvd-diff])
  then show ?thesis by simp
  qed

lemma fri-sibling-domain-round:
  assumes len-pos: 0 < len
    and even-len: 2 dvd len
    and round: len * pw = clength * scale
  shows
    ((p.h ^ ((i + len div 2) mod len)) * shift) ^ pw =
      - (((p.h ^ i) * shift) ^ pw)
  proof -
    let ?N = clength * scale
    have pw-pos: 0 < pw
      using len-pos round eval-domain-size-pos by (cases pw) simp-all
    have half-round: (len div 2) * pw = ?N div 2
    proof -
      from even-len obtain k where len-eq: len = 2 * k
        by blast
      then have ?N = 2 * (k * pw)
        using round by simp
      then show ?thesis
        using len-eq by simp
    qed
    have exp-mod:
      (((i + len div 2) mod len) * pw) mod ?N =
        (i * pw + ?N div 2) mod ?N
    proof -
      have (((i + len div 2) mod len) * pw) mod ?N =
        ((i + len div 2) * pw) mod ?N
        using mod-mult-right-factor-nat[OF len-pos pw-pos, of i + len div 2]
        round by simp
      also have ... = (i * pw + ?N div 2) mod ?N
        using half-round by (simp add: algebra-simps)
      finally show ?thesis .
    qed
    have h-exp:
      p.h ^ (((i + len div 2) mod len) * pw) = -(p.h ^ (i * pw))

```



```

proof -
  have  $p.h \wedge (((i + \text{len div } 2) \bmod \text{len}) * pw) =$ 
     $p.h \wedge (((i + \text{len div } 2) \bmod \text{len}) * pw \bmod ?N)$ 
    using h-power-mod-eval-domain[of  $((i + \text{len div } 2) \bmod \text{len}) * pw$ ]
    by simp
  also have  $\dots = p.h \wedge ((i * pw + ?N \text{ div } 2) \bmod ?N)$ 
    using exp-mod by simp
  also have  $\dots = p.h \wedge (i * pw + ?N \text{ div } 2)$ 
    using h-power-mod-eval-domain[of  $i * pw + ?N \text{ div } 2$ ] by simp
  also have  $\dots = -(p.h \wedge (i * pw))$ 
    by (rule h-half-shift)
  finally show ?thesis .
qed
have lhs:
   $((p.h \wedge ((i + \text{len div } 2) \bmod \text{len})) * \text{shift}) \wedge pw =$ 
   $p.h \wedge (((i + \text{len div } 2) \bmod \text{len}) * pw) * \text{shift} \wedge pw$ 
  by (simp add: power-mult power-mult-distrib)
have rhs:
   $((p.h \wedge i) * \text{shift}) \wedge pw = p.h \wedge (i * pw) * \text{shift} \wedge pw$ 
  by (simp add: power-mult power-mult-distrib)
show ?thesis
  using h-exp by (simp add: power-mult power-mult-distrib)
qed

lemma fri-sibling-domain-at:
  assumes len-pos:  $0 < \text{len}$ 
    and even-len:  $2 \text{ dvd } \text{len}$ 
    and round:  $\text{len} * pw = \text{clength} * \text{scale}$ 
    and z-eq:  $z = ((p.h \wedge i) * \text{shift}) \wedge pw$ 
    and d-sib:
       $d \wedge ((i + \text{len div } 2) \bmod \text{len}) =$ 
       $((p.h \wedge ((i + \text{len div } 2) \bmod \text{len})) * \text{shift}) \wedge pw$ 
  shows  $d \wedge ((i + \text{len div } 2) \bmod \text{len}) = -z$ 
  using fri-sibling-domain-round[OF len-pos even-len round, of i] z-eq d-sib
  by simp

lemma fri-next-domain-round:
  assumes half-pos:  $0 < \text{len div } 2$ 
    and even-len:  $2 \text{ dvd } \text{len}$ 
    and round:  $\text{len} * pw = \text{clength} * \text{scale}$ 
  shows
     $((p.h \wedge i) * \text{shift}) \wedge pw * ((p.h \wedge i) * \text{shift}) \wedge pw =$ 
     $((p.h \wedge (i \bmod (\text{len div } 2))) * \text{shift}) \wedge (pw + pw)$ 
proof -
  let  $?N = \text{clength} * \text{scale}$ 
  have len-pos:  $0 < \text{len}$ 
    using half-pos by simp
  have pw-pos:  $0 < pw$ 
    using len-pos round eval-domain-size-pos by (cases pw) simp-all

```

```

have double-pw-pos:  $0 < pw + pw$ 
  using pw-pos by simp
have double-round:  $(len \text{ div } 2) * (pw + pw) = ?N$ 
proof -
  from even-len obtain k where len-eq:  $len = 2 * k$ 
    by blast
  have len-half:  $len \text{ div } 2 = k$ 
    using len-eq by simp
  have  $k * (pw + pw) = 2 * k * pw$ 
proof -
  have  $k * (pw + pw) = k * pw + k * pw$ 
    by (simp add: distrib-left)
  also have  $\dots = (k + k) * pw$ 
    by (simp add: distrib-right)
  also have  $\dots = 2 * k * pw$ 
    by simp
  finally show ?thesis .
qed
moreover have  $2 * k * pw = ?N$ 
  using len-eq round by simp
ultimately show ?thesis
  using len-half by simp
qed
have exp-mod:
   $((i \text{ mod } (len \text{ div } 2)) * (pw + pw)) \text{ mod } ?N =$ 
   $(i * (pw + pw)) \text{ mod } ?N$ 
proof -
  have  $((i \text{ mod } (len \text{ div } 2)) * (pw + pw)) \text{ mod } ?N =$ 
     $((i \text{ mod } (len \text{ div } 2)) * (pw + pw)) \text{ mod } ((len \text{ div } 2) * (pw + pw))$ 
    using double-round by simp
  also have  $\dots = (i * (pw + pw)) \text{ mod } ((len \text{ div } 2) * (pw + pw))$ 
    using mod-mult-right-factor-nat[OF half-pos double-pw-pos, of i]
    by simp
  also have  $\dots = (i * (pw + pw)) \text{ mod } ?N$ 
    using double-round by simp
  finally show ?thesis .
qed
have h-exp:
   $p.h \wedge ((i \text{ mod } (len \text{ div } 2)) * (pw + pw)) =$ 
   $p.h \wedge (i * (pw + pw))$ 
proof -
  have  $p.h \wedge ((i \text{ mod } (len \text{ div } 2)) * (pw + pw)) =$ 
     $p.h \wedge (((i \text{ mod } (len \text{ div } 2)) * (pw + pw)) \text{ mod } ?N)$ 
    using h-power-mod-eval-domain[of (i mod (len div 2)) * (pw + pw)]
    by simp
  also have  $\dots = p.h \wedge (i * (pw + pw)) \text{ mod } ?N$ 
    using exp-mod by simp
  also have  $\dots = p.h \wedge (i * (pw + pw))$ 
    using h-power-mod-eval-domain[of i * (pw + pw)] by simp

```

```

    finally show ?thesis .
qed
show ?thesis
proof -
  have lhs:
    
$$((p.h \wedge i) * shift) \wedge pw * ((p.h \wedge i) * shift) \wedge pw =$$


$$p.h \wedge (i * (pw + pw)) * shift \wedge (pw + pw)$$

  proof -
    have  $((p.h \wedge i) * shift) \wedge pw * ((p.h \wedge i) * shift) \wedge pw =$ 

$$((p.h \wedge i) * shift) \wedge (pw + pw)$$

    by (simp add: power-add)
    also have ... =  $(p.h \wedge i) \wedge (pw + pw) * shift \wedge (pw + pw)$ 
    by (simp add: power-mult-distrib)
    also have ... =  $p.h \wedge (i * (pw + pw)) * shift \wedge (pw + pw)$ 
    by (simp add: power-mult)
    finally show ?thesis .
  qed
  have rhs:
    
$$((p.h \wedge (i \bmod (len \div 2))) * shift) \wedge (pw + pw) =$$


$$p.h \wedge ((i \bmod (len \div 2)) * (pw + pw)) * shift \wedge (pw + pw)$$

    by (simp add: power-mult power-mult-distrib)
  show ?thesis
    using lhs rhs h-exp by simp
qed
qed

lemma fri-next-domain-at:
  assumes half-pos:  $0 < len \div 2$ 
    and even-len:  $2 \text{ dvd } len$ 
    and round:  $len * pw = clength * scale$ 
    and d-i:  $d ! i = ((p.h \wedge i) * shift) \wedge pw$ 
    and d-next:  $d' ! (i \bmod (len \div 2)) = d ! i * d ! i$ 
  shows
    
$$d' ! (i \bmod (len \div 2)) =$$


$$((p.h \wedge (i \bmod (len \div 2))) * shift) \wedge (pw + pw)$$

  using fri-next-domain-round[OF half-pos even-len round, of i] d-i d-next
  by simp

lemma cp-eval-length:
  length (p.cp-eval as) = clength * scale
  unfolding p.cp-eval-def using eval-domain-length by simp

lemma f-eval-length-power:
   $\exists n. \text{length } p.f\text{-eval} = 2 \wedge n$ 
  using p.eval-domain-length-power f-eval-length by simp

lemma cp-eval-length-power:
   $\exists n. \text{length } (p.cp\text{-eval } as) = 2 \wedge n$ 
  using p.eval-domain-length-power cp-eval-length by simp

```

```

lemma ceil-log-Suc-le-power:
  assumes  $d < 2^N$ 
  shows  $\text{ceil-log } (\text{Suc } d) \leq N$ 
proof (cases  $d = 0$ )
  case True
  then show ?thesis
    unfolding ceil-log-def by simp
next
  case False
  then have d-pos:  $0 < d$ 
    by simp
  have  $2^{\text{floor-log } d} \leq d$ 
    by (rule floor-log-exp2-le[OF d-pos])
  also have  $\dots < 2^N$ 
    using assms .
  finally have  $\text{floor-log } d < N$ 
    by (subst (asm) power-strict-increasing-iff) simp-all
  then show ?thesis
    using False unfolding ceil-log-def by simp
qed

```

```

lemma ceil-log-mono:
  assumes  $m \leq n$ 
  shows  $\text{ceil-log } m \leq \text{ceil-log } n$ 
proof (cases  $m \leq 1$ )
  case True
  then show ?thesis
    unfolding ceil-log-def by simp
next
  case False
  then have m-gt:  $1 < m$ 
    by simp
  then have n-gt:  $1 < n$ 
    using assms by simp
  have  $m - 1 \leq n - 1$ 
    using assms by simp
  then have  $\text{floor-log } (m - 1) \leq \text{floor-log } (n - 1)$ 
    using floor-log-le-iff by blast
  then show ?thesis
    using m-gt n-gt unfolding ceil-log-def by simp
qed

```

```

lemma fri-rounds-le-eval-domain-log:
  assumes htv: honest-trace-valid
  and cp'-def:  $\text{cp}' = p.\text{cp}$  as  $p.f.\text{powers}$ 
  and nrounds-def:  $\text{nrounds} = \text{ceil-log } (\text{degree } \text{cp}' + 1)$ 
  and init-len:  $\text{length } (p.\text{cp-eval } \text{as}) = 2^N$ 
  shows  $\text{nrounds} \leq N$ 

```

proof –
 have $\text{degree } cp' \leq p.\text{maxDegree}$
 using *composition-degree-from-honest-trace-valid*[*OF htv, of as*] *cp'-def* **by** *simp*
 also have $\dots < \text{clength} * \text{scale}$
 by (*rule maxDegree-less-eval-domain*)
 also have $\dots = 2^N$
 using *init-len cp-eval-length* **by** *simp*
 finally have $\text{degree } cp' < 2^N$.
 then show *?thesis*
 using *ceil-log-Suc-le-power*[*of degree cp' N*]
 unfolding *nrounds-def* **by** *simp*
qed

lemma *query-authentication-path-length*:
assumes *create-f*:
 Some (f-merkle, s1) ∈ set-dist (execute (p.create p.f-eval) init-state)
 and *i-in*: $i \in \text{set } (p.\text{powers-scaled } (p.\text{index } x))$
shows
 $\text{length } (\text{get-authentication-path } (\text{length } p.\text{f-eval}) \ i \ f\text{-merkle}) =$
 $\text{floor-log } (\text{length } p.\text{f-eval})$

proof –
 have *i-bound*: $i < \text{length } p.\text{f-eval}$
 using *i-in* **unfolding** *f-eval-length*
 by (*rule powers-scaled-index-bound*)
 then have *len-pos*: $0 < \text{length } p.\text{f-eval}$
 by (*cases p.f-eval*) *simp-all*
 show *?thesis*
 by (*rule p.create-get-authentication-path-len*[*OF create-f len-pos*])
 simp

qed

definition *query-decommitment-transcript*
where
query-decommitment-transcript idx f-merkle $\equiv$
List.concat (map
 $(\lambda i. p.\text{f-eval} \ ! \ i \ \#$
 $\text{get-authentication-path } (\text{length } p.\text{f-eval}) \ i \ f\text{-merkle})$
 $(p.\text{powers-scaled } \text{idx}))$

lemma *decommit-on-query-step-transcript*:
assumes *step*:
 Some (x, t) ∈
 set-dist (execute
 (do {
 $p.\text{send } (p.\text{f-eval} \ ! \ i);$
 $\text{mfold2 } p.\text{send } (\text{get-authentication-path } (\text{length } p.\text{f-eval}) \ i \ f\text{-merkle})$
 }) s)
shows
 $x = () \wedge$

```

    PTranscript t =
      rev (p.f-eval ! i #
        get-authentication-path (length p.f-eval) i f-merkle) @ PTranscript s
  proof -
  from step obtain u where
    send-leaf:
      Some ((), u) ∈ set-dist (execute (p.send (p.f-eval ! i)) s)
  and send-path:
    Some (x, t) ∈
      set-dist (execute
        (mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)) u)
  by (auto elim!: p.set-dist-bindE)
  have x-unit: x = ()
  by (cases x) simp
  have path-unit:
    Some ((), t) ∈
      set-dist (execute
        (mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)) u)
  using send-path x-unit by simp
  have tr-u: PTranscript u = p.f-eval ! i # PTranscript s
  using p.send-outcome[OF send-leaf] by simp
  have tr-t:
    PTranscript t =
      rev (get-authentication-path (length p.f-eval) i f-merkle) @ PTranscript u
  using p.mfold2-send-outcome[OF path-unit] by simp
  show ?thesis
  using x-unit tr-u tr-t by simp
qed

```

lemma *decommit-on-query-mmap-transcript*:

assumes *outcome*:

Some (qouts, t) ∈
 set-dist (execute (mmap (p.decommit-on-query idx f-merkle)) s)

shows

PTranscript t =
 rev (query-decommitment-transcript idx f-merkle) @ PTranscript s

proof -

have *aux*:

∧_{qs} qouts t s. Some (qouts, t) ∈
 set-dist (execute
 (mmap
 (map
 (λi. do {
 p.send (p.f-eval ! i);
 mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)
 })
 qs)) s) ⇒
 PTranscript t =
 rev (List.concat (map

```

      ( $\lambda i.$   $p.f\text{-eval} \ ! \ i \ \#$ 
         $get\text{-authentication-path} \ (length \ p.f\text{-eval}) \ i \ f\text{-merkle}$ 
         $qs$ )) @  $PTranscript \ s$ 
proof –
  fix  $qs \ qouts \ t \ s$ 
  assume  $outcome\text{-}qs$ :
     $Some \ (qouts, t) \in$ 
       $set\text{-dist} \ (execute$ 
         $(mmap$ 
           $(map$ 
             $(\lambda i.$   $do \ \{$ 
               $p.send \ (p.f\text{-eval} \ ! \ i);$ 
               $mfold2 \ p.send \ (get\text{-authentication-path} \ (length \ p.f\text{-eval}) \ i \ f\text{-merkle})$ 
             $\}$ 
           $qs$ ))  $s$ )
  then show
     $PTranscript \ t =$ 
       $rev \ (List.concat \ (map$ 
         $(\lambda i.$   $p.f\text{-eval} \ ! \ i \ \#$ 
           $get\text{-authentication-path} \ (length \ p.f\text{-eval}) \ i \ f\text{-merkle}$ 
           $qs$ )) @  $PTranscript \ s$ 
       $)$ 
  proof ( $induction \ qs \ arbitrary$ :  $qouts \ t \ s$ )
  case  $Nil$ 
  then show  $?case \ by \ simp$ 
next
  case  $(Cons \ i \ qs)$ 
  from  $Cons.premis$  obtain  $u0 \ u \ qouts'$  where
     $send\text{-leaf}$ :
       $Some \ ((), u0) \in set\text{-dist} \ (execute \ (p.send \ (p.f\text{-eval} \ ! \ i)) \ s)$ 
    and  $send\text{-path}$ :
       $Some \ ((), u) \in$ 
         $set\text{-dist} \ (execute$ 
           $(mfold2 \ p.send \ (get\text{-authentication-path} \ (length \ p.f\text{-eval}) \ i \ f\text{-merkle}))$ 
           $u0$ )
    and  $tail$ :
       $Some \ (qouts', t) \in$ 
         $set\text{-dist} \ (execute$ 
           $(mmap$ 
             $(map$ 
               $(\lambda i.$   $do \ \{$ 
                 $p.send \ (p.f\text{-eval} \ ! \ i);$ 
                 $mfold2 \ p.send \ (get\text{-authentication-path} \ (length \ p.f\text{-eval}) \ i \ f\text{-merkle})$ 
               $\}$ 
             $qs$ ))  $u$ )
           $)$ 
      by ( $auto \ elim!$ :  $p.set\text{-dist}\text{-}bindE$ )
  have  $head\text{-}tr$ :
     $PTranscript \ u =$ 
       $rev \ (p.f\text{-eval} \ ! \ i \ \#$ 
         $get\text{-authentication-path} \ (length \ p.f\text{-eval}) \ i \ f\text{-merkle})$  @  $PTranscript \ s$ 

```

```

proof –
  have tr-u0: PTranscript u0 = p.f-eval ! i # PTranscript s
    using p.send-outcome[OF send-leaf] by simp
  have tr-u:
    PTranscript u =
      rev (get-authentication-path (length p.f-eval) i f-merkle) @ PTranscript u0
    using p.mfold2-send-outcome[OF send-path] by simp
  show ?thesis
    using tr-u0 tr-u by simp
qed
have tail-tr:
  PTranscript t =
    rev (List.concat (map
      ( $\lambda i. p.f\text{-eval} \ ! \ i \ \#$ 
        get-authentication-path (length p.f-eval) i f-merkle)
      qs)) @ PTranscript u
    using Cons.IH[OF tail] .
  show ?case
    using head-tr tail-tr by simp
qed
qed
show ?thesis
  using aux[OF outcome[unfolded p.decommit-on-query-def]]
  unfolding query-decommitment-transcript-def .
qed

lemma decommit-on-query-mmap-state:
  assumes outcome:
    Some (qouts, t) ∈
      set-dist (execute (mmap (p.decommit-on-query idx f-merkle)) s)
  shows PState t = foldl concat (PState s) (query-decommitment-transcript idx f-merkle)
proof –
  have aux:
     $\bigwedge qs \ qouts \ t \ s. \ Some \ (qouts, \ t) \ \in$ 
      set-dist (execute
        (mmap
          (map
            ( $\lambda i. \ do \ \{$ 
              p.send (p.f-eval ! i);
              mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)
            })
          qs)) s)  $\implies$ 
    PState t =
      foldl concat (PState s)
        (List.concat (map
          ( $\lambda i. \ p.f\text{-eval} \ ! \ i \ \#$ 
            get-authentication-path (length p.f-eval) i f-merkle)
          qs))
proof –

```



```

fix qs qouts t s
assume outcome-qs:
  Some (qouts, t) ∈
    set-dist (execute
      (mmap
        (map
          (λi. do {
            p.send (p.f-eval ! i);
            mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)
          })
        qs)) s)
then show
  PState t =
    foldl concat (PState s)
      (List.concat (map
        (λi. p.f-eval ! i #
          get-authentication-path (length p.f-eval) i f-merkle)
        qs))
proof (induction qs arbitrary: qouts t s)
  case Nil
  then show ?case by simp
next
  case (Cons i qs)
  from Cons.prem obtain u0 u qouts' where
    send-leaf:
      Some ((), u0) ∈ set-dist (execute (p.send (p.f-eval ! i)) s)
  and send-path:
    Some ((), u) ∈
      set-dist (execute
        (mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle))
        u0)
  and tail:
    Some (qouts', t) ∈
      set-dist (execute
        (mmap
          (map
            (λi. do {
              p.send (p.f-eval ! i);
              mfold2 p.send (get-authentication-path (length p.f-eval) i f-merkle)
            })
          qs)) u)
  by (auto elim!: p.set-dist-bindE)
have head-state:
  PState u =
    foldl concat (PState s)
      (p.f-eval ! i # get-authentication-path (length p.f-eval) i f-merkle)
proof –
  have st-u0: PState u0 = concat (PState s) (p.f-eval ! i)
    using p.send-outcome[OF send-leaf] by simp

```

```

show ?thesis
  using p.mfold2-send-outcome[OF send-path] st-u0 by simp
qed
have tail-state:
  PState t =
    foldl concat (PState u)
      (List.concat (map
        (λi. p.f-eval ! i #
          get-authentication-path (length p.f-eval) i f-merkle)
        qs))
  using Cons.IH[OF tail] .
show ?case
  using head-state tail-state by simp
qed
qed
show ?thesis
  using aux[OF outcome[unfolded p.decommit-on-query-def]]
  unfolding query-decommitment-transcript-def .
qed

lemma prover-query-decommitment-transcript-in-replay-state:
assumes final-send:
  Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and random-idx:
  Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
and query-decommit:
  Some (qouts, s8) ∈
    set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
and fri-decommit:
  Some (fri-idx, s9) ∈
    set-dist (execute
      (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
and prover-state-eq: prover-state = s9
and replay: replay-state = verifier-replay-state prover-state
obtains rest where
  PTranscript replay-state =
    rev (PTranscript s6) @ [hd (last ls)] @
      query-decommitment-transcript idx' f-merkle @ rest
proof –
have final-tr:
  PTranscript s-final = hd (last ls) # PTranscript s6
  using p.send-outcome[OF final-send] by simp
have s7-tr: PTranscript s7 = PTranscript s-final
  using receive-query-index-challenge-preserves-transcript[OF random-idx] .
have s8-tr:
  PTranscript s8 =
    rev (query-decommitment-transcript idx' f-merkle) @ PTranscript s7
  using decommit-on-query-mmap-transcript[OF query-decommit] .
have s8-s9: transcript-extends s9 s8

```

```

    by (rule mfold-transcript-extends[OF fri-decommit])
      (rule decommit-on-fri-layers-step-transcript-extends)
  then obtain more where more:
    PTranscript s9 = more @ PTranscript s8
  unfolding transcript-extends-def by blast
have prover-tr:
  PTranscript prover-state =
    more @ rev (query-decommitment-transcript idx' f-merkle) @
      hd (last ls) # PTranscript s6
  using more s8-tr s7-tr final-tr prover-state-eq by simp
have replay-tr:
  PTranscript replay-state =
    rev (PTranscript s6) @ [hd (last ls)] @
      query-decommitment-transcript idx' f-merkle @ rev more
  using prover-tr replay
  unfolding verifier-replay-state-def by simp
show ?thesis
  by (rule that[OF replay-tr])
qed

```

lemma *honest-query-decommitment-list-no-failure:*

```

assumes created: p.created-tree p.f-eval f-merkle s0
and len-pow: length p.f-eval = 2 ^ n
and ext: s0 ≤ s
and paths:
  ∧ i. i ∈ set (p.powers-scaled (p.index x)) ⇒
    length (get-authentication-path (length p.f-eval) i f-merkle) =
      floor-log (length p.f-eval)
and root: fr = value f-merkle
shows
  None ∉ dom (dist (execute
    (mmap (v.check-decommit-on-query fr (p.index x)))
    (s⟦
      PTranscript :=
        query-decommitment-transcript (p.index x) f-merkle @ rest
    ⟧)))
proof –
have check-eq:
  v.check-decommit-on-query fr (p.index x) =
    map
      (λi. do {
        qh ← p.read;
        let len = scale * clength;
        qh-path ← ntimes p.read (floor-log len);
        ap ← p.check-authentication-path len i qh qh-path;
        assert (ap = fr);
        return qh
      })
    (p.powers-scaled (p.index x))

```

```

unfolding v.check-decommit-on-query-def p.powers-scaled-def by simp
define qs where qs = p.powers-scaled (p.index x)
have list-nf:
   $\bigwedge qs \text{ rest } s.$ 
     $set \text{ } qs \subseteq set \text{ } (p.powers-scaled \text{ } (p.index \text{ } x)) \implies$ 
     $s0 \leq s \implies$ 
     $None \notin dom \text{ } (dist \text{ } (execute$ 
      (mmap
        (map
          ( $\lambda i.$  do {
            qh  $\leftarrow$  p.read;
            let len = scale * clength;
            qh-path  $\leftarrow$  ntimes p.read (floor-log len);
            ap  $\leftarrow$  p.check-authentication-path len i qh qh-path;
            assert (ap = fr);
            return qh
          })
        qs))
      (s0
        PTranscript :=
        List.concat (map
          ( $\lambda i.$  p.f-eval ! i #
            get-authentication-path (length p.f-eval) i f-merkle)
          qs) @ rest
        ))))
proof –
  fix qs rest s
  assume qs-subset:  $set \text{ } qs \subseteq set \text{ } (p.powers-scaled \text{ } (p.index \text{ } x))$ 
  and s0-s:  $s0 \leq s$ 
  then show
     $None \notin dom \text{ } (dist \text{ } (execute$ 
      (mmap
        (map
          ( $\lambda i.$  do {
            qh  $\leftarrow$  p.read;
            let len = scale * clength;
            qh-path  $\leftarrow$  ntimes p.read (floor-log len);
            ap  $\leftarrow$  p.check-authentication-path len i qh qh-path;
            assert (ap = fr);
            return qh
          })
        qs))
      (s0
        PTranscript :=
        List.concat (map
          ( $\lambda i.$  p.f-eval ! i #
            get-authentication-path (length p.f-eval) i f-merkle)
          qs) @ rest
        ))))

```

```

proof (induction qs arbitrary: rest s)
  case Nil
  then show ?case by simp
next
  case (Cons i qs)
have i-in:  $i \in \text{set } (p.\text{powers-scaled } (p.\text{index } x))$ 
  using Cons.premis(1) by simp
let ?path = get-authentication-path (length p.f-eval) i f-merkle
let ?msg = p.f-eval ! i # ?path
let ?tail =
  List.concat (map
    ( $\lambda i.$  p.f-eval ! i #
      get-authentication-path (length p.f-eval) i f-merkle)
    qs) @ rest
have i-bound:  $i < \text{length } p.\text{f-eval}$ 
  using i-in unfolding f-eval-length
  by (rule powers-scaled-index-bound)
have path-len:
  length ?path = floor-log (length p.f-eval)
  using paths[OF i-in] .
have step-nf:
  None  $\notin \text{dom } (\text{dist } (\text{execute }
    (\text{do } \{
      qh \leftarrow p.\text{read};
      \text{let } \text{len} = \text{scale} * \text{clength};
      qh\text{-path} \leftarrow \text{ntimes } p.\text{read } (\text{floor-log } \text{len});
      ap \leftarrow p.\text{check-authentication-path } \text{len } i \text{ } qh \text{ } qh\text{-path};
      \text{assert } (ap = \text{fr});
      \text{return } qh
    \})))
    (s(PTranscript := ?msg @ ?tail))))
  using p.honest-query-decommitment-no-failure[
    OF created len-pow i-bound refl Cons.premis(2) path-len root,
    of ?tail]
  unfolding f-eval-length by (simp add: mult.commute Let-def)
let ?step =
  do {
    qh  $\leftarrow$  p.read;
    let len = scale * clength;
    qh-path  $\leftarrow$  ntimes p.read (floor-log len);
    ap  $\leftarrow$  p.check-authentication-path len i qh qh-path;
    assert (ap = fr);
    return qh
  }
let ?tail-m =
  mmap
    (map
      ( $\lambda i.$  do {
        qh  $\leftarrow$  p.read;$ 
```

```

    let len = scale * clength;
    qh-path ← ntimes p.read (floor-log len);
    ap ← p.check-authentication-path len i qh qh-path;
    assert (ap = fr);
    return qh
  })
  qs)
have case-eq:
  ?case =
    (None ∉ dom (dist (execute
      (?step ≫ (λqh. ?tail-m ≫ (λqhs. return (qh # qhs))))
      (s(PTranscript := ?msg @ ?tail)))))
  by (simp add: Let-def)
show ?case
  unfolding case-eq
proof (rule no-failure-bindI[OF step-nf])
fix qh t
assume step:
  Some (qh, t) ∈ set-dist (execute
    (do {
      qh ← p.read;
      let len = scale * clength;
      qh-path ← ntimes p.read (floor-log len);
      ap ← p.check-authentication-path len i qh qh-path;
      assert (ap = fr);
      return qh
    })
    (s(PTranscript := ?msg @ ?tail)))
have head-state: s0 ≤ t ∧ PTranscript t = ?tail
proof –
  from step obtain r u path u' ap u'' u''' where
  read-leaf:
    Some (r, u) ∈ set-dist (execute p.read (s(PTranscript := ?msg @ ?tail)))
  and read-path:
    Some (path, u') ∈
      set-dist (execute (ntimes p.read (floor-log (scale * clength))) u)
  and check:
    Some (ap, u'') ∈
      set-dist (execute (p.check-authentication-path (scale * clength) i r path) u')
  and assert-ok:
    Some ((), u''') ∈ set-dist (execute (assert (ap = fr)) u'')
  and ret:
    Some (qh, t) ∈ set-dist (execute (return r) u''')
  by (auto simp: Let-def elim!: p.set-dist-bindE)
have read-start-eq:
  s(PTranscript := ?msg @ ?tail) =
  s(PTranscript := p.f-eval ! i # (?path @ ?tail))
  by simp
have read-leaf-replay:

```

```

  r = p.f-eval ! i ∧
  u = s(
    PState := concat (PState s) (p.f-eval ! i),
    PTranscript := ?path @ ?tail)
  using p.read-cons-outcome[OF read-leaf[unfolded read-start-eq]] by simp
have r-eq: r = p.f-eval ! i
  using read-leaf-replay by simp
have read-leaf-res:
  u = s(
    PState := concat (PState s) (p.f-eval ! i),
    PTranscript := ?path @ ?tail)
  using read-leaf-replay by simp
have u-base:
  u =
    (s(PState := concat (PState s) (p.f-eval ! i)))
    (PTranscript := ?path @ ?tail)
  using read-leaf-res by simp
have scale-len: scale * clength = length p.f-eval
  using f-eval-length by (simp add: mult.commute)
have path-replay:
  path = ?path ∧
  u' =
    (s(PState := concat (PState s) (p.f-eval ! i)))
    (
      PState :=
        foldl concat
          (concat (PState s) (p.f-eval ! i))
          ?path,
      PTranscript := ?tail
    )
  using p.ntimes-read-prefix-outcome[
    OF read-path[unfolded scale-len path-len[symmetric] u-base]]
  by simp
have path-eq: path = ?path
  using path-replay by simp
have path-res:
  u' =
    (s(PState := concat (PState s) (p.f-eval ! i)))
    (
      PState :=
        foldl concat
          (concat (PState s) (p.f-eval ! i))
          ?path,
      PTranscript := ?tail
    )
  using path-replay by simp
have s0-u': s0 ≤ u'
  using Cons.premis path-res
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp

```

```

have u'-u'':  $u' \leq u''$ 
  using p.check-created-tree-outcome[
    OF created len-pow i-bound refl s0-u'
    check[unfolded scale-len r-eq path-eq]]
  unfolding root by simp
have check-transcript: PTranscript u'' = PTranscript u'
  using p.check-authentication-path-preserves-channel(2)[OF check] .
have assert-state: u''' = u''
  using assert-ok unfolding assert-def
  by (cases ap = fr)
    (auto simp: return.rep-eq dist-return-def throw.rep-eq dist-throw-def
      dist-delta-dist delta-map-def set-dist-def)
have ret-state: t = u'''
  using ret by simp
have u''-t:  $u'' \leq t$ 
  unfolding assert-state[symmetric] ret-state
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have t-transcript: PTranscript t = ?tail
  using path-res check-transcript assert-state ret-state by simp
show ?thesis
proof
  show s0 ≤ t
    using s0-u' u'-u'' u''-t
    unfolding less-eq-hash-ext-def less-eq-fmap-def
    by metis
  show PTranscript t = ?tail
    using t-transcript .
qed
qed
have s0-t:  $s0 \leq t$ 
  using head-state by simp
have t-transcript: PTranscript t = ?tail
  using head-state by simp
have tail-nf-start:
  None  $\notin$  dom (dist (execute ?tail-m (t(PTranscript := ?tail))))
  using Cons.IH[of t rest] Cons.prem(1) s0-t by simp
have t-update: t(PTranscript := ?tail) = t
  using t-transcript by simp
have tail-nf: None  $\notin$  dom (dist (execute ?tail-m t))
  using tail-nf-start unfolding t-update .
show None  $\notin$  dom (dist (execute (?tail-m  $\gg$  (λqhs. return (qh # qhs))) t))
  by (rule no-failure-bindI[OF tail-nf]) simp
qed
qed
qed
show ?thesis
  using list-nf[of p.powers-scaled (p.index x) s rest] ext
  unfolding query-decommitment-transcript-def check-eq qs-def by simp
qed

```


lemma *honest-query-decommitment-replay-no-failure*:

assumes *create-f*:

Some (*f-merkle*, *s0*) ∈
set-dist (*execute* (*p.create p.f-eval*) *init-state*)

and *ext*: *s0* ≤ *s*

and *root*: *fr* = *value f-merkle*

and *transcript*:

PTranscript s =
query-decommitment-transcript (*p.index x*) *f-merkle @ rest*

shows

None ∉ *dom* (*dist* (*execute*
(*mmap* (*v.check-decommit-on-query fr* (*p.index x*))) *s*))

proof –

have *created*: *p.created-tree p.f-eval f-merkle s0*

using *p.create-outcome[OF create-f]* **by** *simp*

from *f-eval-length-power* **obtain** *n* **where** *len-pow*: *length p.f-eval* = 2^n

by *blast*

have *paths*:

$\bigwedge i. i \in \text{set } (p.\text{powers-scaled } (p.\text{index } x)) \implies$
length (*get-authentication-path* (*length p.f-eval*) *i f-merkle*) =
floor-log (*length p.f-eval*)

using *query-authentication-path-length[OF create-f]* **by** *blast*

have *nf*:

None ∉ *dom* (*dist* (*execute*
(*mmap* (*v.check-decommit-on-query fr* (*p.index x*)))
(*s @*
PTranscript :=
query-decommitment-transcript (*p.index x*) *f-merkle @ rest*
))))

by (*rule honest-query-decommitment-list-no-failure*[
OF created len-pow ext paths root])

have *update*:

s @ PTranscript :=
query-decommitment-transcript (*p.index x*) *f-merkle @ rest @ s*

using *transcript* **by** *simp*

show *?thesis*

using *nf unfolding update* .

qed

lemma *honest-query-decommitment-replay-outcome*:

assumes *paths*:

$\bigwedge i. i \in \text{set } (p.\text{powers-scaled } idx) \implies$
length (*get-authentication-path* (*length p.f-eval*) *i f-merkle*) =
floor-log (*length p.f-eval*)

and *outcome*:

Some (*fv*, *t*) ∈
set-dist (*execute*
(*mmap* (*v.check-decommit-on-query fr idx*)))

```

      (s(PTranscript := query-decommitment-transcript idx f-merkle @ rest)))
shows
  fv = map (λi. p.f-eval ! i) (p.powers-scaled idx) ∧
  PTranscript t = rest ∧
  PState t = foldl concat (PState s) (query-decommitment-transcript idx f-merkle) ∧
  s(PTranscript := query-decommitment-transcript idx f-merkle @ rest) ≤ t
proof –
have check-eq:
  v.check-decommit-on-query fr idx =
    map
      (λi. do {
        gh ← p.read;
        let len = scale * clength;
        gh-path ← ntimes p.read (floor-log len);
        ap ← p.check-authentication-path len i gh gh-path;
        assert (ap = fr);
        return gh
      })
    (p.powers-scaled idx)
unfolding v.check-decommit-on-query-def p.powers-scaled-def by simp
have list-out:
  ∧qs fv t s rest.
  set qs ⊆ set (p.powers-scaled idx) ⇒
  Some (fv, t) ∈
    set-dist (execute
      (mmap
        (map
          (λi. do {
            gh ← p.read;
            let len = scale * clength;
            gh-path ← ntimes p.read (floor-log len);
            ap ← p.check-authentication-path len i gh gh-path;
            assert (ap = fr);
            return gh
          })
        qs))
      (s(
        PTranscript :=
        List.concat (map
          (λi. p.f-eval ! i #
            get-authentication-path (length p.f-eval) i f-merkle)
          qs) @ rest
      ))) ⇒
  fv = map (λi. p.f-eval ! i) qs ∧
  PTranscript t = rest ∧
  PState t =
    foldl concat (PState s)
      (List.concat (map
        (λi. p.f-eval ! i #

```

```

      get-authentication-path (length p.f-eval) i f-merkle)
    qs)) ∧
  s(|
    PTranscript :=
      List.concat (map
        (λi. p.f-eval ! i #
          get-authentication-path (length p.f-eval) i f-merkle)
        qs) @ rest
  ) ≤ t
proof –
  fix qs fv t s rest
  assume qs-subset: set qs ⊆ set (p.powers-scaled idx)
  and outcome-qs:
    Some (fv, t) ∈
      set-dist (execute
        (mmap
          (map
            (λi. do {
              qh ← p.read;
              let len = scale * clength;
              qh-path ← ntimes p.read (floor-log len);
              ap ← p.check-authentication-path len i qh qh-path;
              assert (ap = fr);
              return qh
            })
          qs))
        (s(|
          PTranscript :=
            List.concat (map
              (λi. p.f-eval ! i #
                get-authentication-path (length p.f-eval) i f-merkle)
              qs) @ rest
        )))
  then show
    fv = map (λi. p.f-eval ! i) qs ∧
    PTranscript t = rest ∧
    PState t =
      foldl concat (PState s)
        (List.concat (map
          (λi. p.f-eval ! i #
            get-authentication-path (length p.f-eval) i f-merkle)
          qs)) ∧
    s(|
      PTranscript :=
        List.concat (map
          (λi. p.f-eval ! i #
            get-authentication-path (length p.f-eval) i f-merkle)
          qs) @ rest
    ) ≤ t

```

```

proof (induction qs arbitrary: fv t s rest)
  case Nil
  then show ?case
    unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
next
  case (Cons i qs)
  let ?path = get-authentication-path (length p.f-eval) i f-merkle
  let ?msg = p.f-eval ! i # ?path
  let ?tail =
    List.concat (map
      (λi. p.f-eval ! i #
        get-authentication-path (length p.f-eval) i f-merkle)
      qs) @ rest
  let ?start = s(PTranscript := ?msg @ ?tail)
  let ?step =
    do {
      qh ← p.read;
      let len = scale * clength;
      qh-path ← ntimes p.read (floor-log len);
      ap ← p.check-authentication-path len i qh qh-path;
      assert (ap = fr);
      return qh
    }
  let ?tail-m =
    mmap
      (map
        (λi. do {
          qh ← p.read;
          let len = scale * clength;
          qh-path ← ntimes p.read (floor-log len);
          ap ← p.check-authentication-path len i qh qh-path;
          assert (ap = fr);
          return qh
        })
        qs)
  have cons-out:
    Some (fv, t) ∈
      set-dist (execute
        (?step ≫ (λqh. ?tail-m ≫ (λfv-tail. return (qh # fv-tail))))
        ?start)
    using Cons.premis(2) by simp
  from cons-out obtain qh u where
    head:
      Some (qh, u) ∈ set-dist (execute ?step ?start)
    and tail-bind:
      Some (fv, t) ∈
        set-dist (execute (?tail-m ≫ (λfv-tail. return (qh # fv-tail))) u)
  proof –
    show ?thesis

```

```

    using cons-out
  proof (rule p.set-dist-bindE)
    fix qh u
    assume head:
      Some (qh, u) ∈ set-dist (execute ?step ?start)
    and tail-bind:
      Some (fv, t) ∈
        set-dist (execute
          (?tail-m ≫ (λfv-tail. return (qh # fv-tail))) u)
    show ?thesis
      by (rule that[OF head tail-bind])
  qed
qed
from tail-bind obtain fv-tail u-tail where
  tail0:
    Some (fv-tail, u-tail) ∈ set-dist (execute ?tail-m u)
  and ret-tail:
    Some (fv, t) ∈ set-dist (execute (return (qh # fv-tail)) u-tail)
proof -
  show ?thesis
    using tail-bind
  proof (rule p.set-dist-bindE)
    fix fv-tail u-tail
    assume tail0:
      Some (fv-tail, u-tail) ∈ set-dist (execute ?tail-m u)
    and ret-tail:
      Some (fv, t) ∈
        set-dist (execute (return (qh # fv-tail)) u-tail)
    show ?thesis
      by (rule that[OF tail0 ret-tail])
  qed
qed
have ret-tail-res: fv = qh # fv-tail ∧ t = u-tail
  using ret-tail by simp
have tail:
  Some (fv-tail, t) ∈ set-dist (execute ?tail-m u)
  using tail0 ret-tail-res by simp
have fv-eq: fv = qh # fv-tail
  using ret-tail-res by simp
have i-in: i ∈ set (p.powers-scaled idx)
  using Cons.prem1 by simp
have path-len:
  length ?path = floor-log (scale * clength)
  using paths[OF i-in] f-eval-length by (simp add: mult.commute)
from head obtain qh-path ap s1 s2 s3 s4 where
  read-leaf: Some (qh, s1) ∈ set-dist (execute p.read ?start)
  and read-path:
    Some (qh-path, s2) ∈
      set-dist (execute (ntimes p.read (floor-log (scale * clength))) s1)

```

```

and check:
  Some (ap, s3) ∈
    set-dist (execute
      (p.check-authentication-path (scale * clength) i qh qh-path) s2)
and assert-ok:
  Some ((), s4) ∈ set-dist (execute (assert (ap = fr)) s3)
and ret: Some (qh, u) ∈ set-dist (execute (return qh) s4)
by (auto simp: Let-def elim!: p.set-dist-bindE)
have leaf-res:
  qh = p.f-eval ! i ∧
  s1 = s(
    PState := concat (PState s) (p.f-eval ! i),
    PTranscript := ?path @ ?tail)
proof –
  have start-eq:
    ?start =
      s(PTranscript := p.f-eval ! i # (?path @ ?tail))
  by simp
show ?thesis
  using p.read-cons-outcome[OF read-leaf[unfolded start-eq]] by simp
qed
let ?s-leaf = s(PState := concat (PState s) (p.f-eval ! i))
have s1-eq: s1 = ?s-leaf(PTranscript := ?path @ ?tail)
  using leaf-res by simp
have path-res:
  qh-path = ?path ∧
  s2 = ?s-leaf(
    PState := foldl concat (PState ?s-leaf) ?path,
    PTranscript := ?tail)
  using p.ntimes-read-prefix-outcome[
    OF read-path[unfolded path-len[symmetric] s1-eq]]
  by simp
have s3-tr: PTranscript s3 = ?tail
  using p.check-authentication-path-preserves-channel(2)[OF check] path-res
  by simp
have s4-tr: PTranscript s4 = ?tail
  using assert-outcomeD(2)[OF assert-ok] s3-tr by simp
have u-eq: u = s4
  using ret by simp
have u-tr: PTranscript u = ?tail
  using s4-tr u-eq by simp
have u-state:
  PState u = foldl concat (PState s) ?msg
proof –
  have s3-state: PState s3 = PState s2
    using p.check-authentication-path-preserves-channel(1)[OF check] .
  have s4-state: PState s4 = PState s3
    using assert-outcomeD(2)[OF assert-ok] by simp
  show ?thesis

```

```

    using leaf-res path-res s3-state s4-state u-eq by simp
  qed
have start-u: ?start ≤ u
proof -
  have ?start ≤ s1
    using read-leaf by (rule read-hash-extends)
  moreover have s1 ≤ s2
    using read-path by (rule ntimes-read-hash-extends)
  moreover have s2 ≤ s3
    using check by (rule check-authentication-path-hash-extends)
  moreover have s3 ≤ s4
    using assert-ok by (rule assert-hash-extends)
  moreover have s4 ≤ u
    using u-eq p.hash-ext-refl by simp
  ultimately show ?thesis
    by (meson p.hash-ext-trans)
qed
have u-update: u(PTranscript := ?tail) = u
  using u-tr by simp
have tail-start:
  Some (fv-tail, t) ∈
    set-dist (execute ?tail-m (u(PTranscript := ?tail)))
  using tail unfolding u-update .
have tail-out:
  fv-tail = map (λi. p.f-eval ! i) qs ∧
  PTranscript t = rest ∧
  PState t =
    foldl concat (PState u)
      (List.concat (map
        (λi. p.f-eval ! i #
          get-authentication-path (length p.f-eval) i f-merkle)
        qs)) ∧
  u(PTranscript := ?tail) ≤ t
proof (rule Cons.IH)
  show set qs ⊆ set (p.powers-scaled idx)
    using Cons.prem1 by simp
  show Some (fv-tail, t) ∈ set-dist (execute ?tail-m
    (u(PTranscript := ?tail)))
    using tail-start .
qed
have start-t: ?start ≤ t
  using start-u tail-out unfolding u-update by (meson p.hash-ext-trans)
show ?case
  using fv-eq leaf-res tail-out start-t u-state by simp
qed
qed
have outcome':
  Some (fv, t) ∈
    set-dist (execute

```

```

(mmap
  (map
    (λi. do {
      qh ← p.read;
      let len = scale * clength;
      qh-path ← ntimes p.read (floor-log len);
      ap ← p.check-authentication-path len i qh qh-path;
      assert (ap = fr);
      return qh
    })
    (p.powers-scaled idx)))
(s()
  PTranscript :=
    List.concat (map
      (λi. p.f-eval ! i #
        get-authentication-path (length p.f-eval) i f-merkle)
      (p.powers-scaled idx)) @ rest
  ))
using outcome
unfolding check-eq query-decommitment-transcript-def by simp
have result:
  fv = map (λi. p.f-eval ! i) (p.powers-scaled idx) ∧
  PTranscript t = rest ∧
  PState t =
    foldl concat (PState s)
      (List.concat (map
        (λi. p.f-eval ! i #
          get-authentication-path (length p.f-eval) i f-merkle)
        (p.powers-scaled idx))) ∧
  s()
  PTranscript :=
    List.concat (map
      (λi. p.f-eval ! i #
        get-authentication-path (length p.f-eval) i f-merkle)
      (p.powers-scaled idx)) @ rest
  ) ≤ t
by (rule list-out[OF - outcome']) simp
show ?thesis
  using result unfolding query-decommitment-transcript-def by simp
qed

```

lemma check-decommit-on-query-step-preserves-query-counter:

assumes step-in: $m \in \text{set } (v.\text{check-decommit-on-query } fr \text{ } idx)$

and outcome: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \text{ } s)$

shows $PQueryCounter \ t = PQueryCounter \ s$

proof –

from step-in **obtain** i **where**

$m\text{-eq}$:

$m = \text{do } \{$


```

     $qh \leftarrow p.read;$ 
     $let\ len = scale * clength;$ 
     $qh-path \leftarrow ntimes\ p.read\ (floor-log\ len);$ 
     $ap \leftarrow p.check-authentication-path\ len\ i\ qh\ qh-path;$ 
     $assert\ (ap = fr);$ 
     $return\ qh$ 
  }
  unfolding  $v.check-decommit-on-query-def\ p.powers-scaled-def$  by  $auto$ 
from  $outcome[unfolded\ m-eq]$  obtain  $qh\ s1\ qh-path\ s2\ ap\ s3\ s4$  where
   $read-qh: Some\ (qh, s1) \in set-dist\ (execute\ p.read\ s)$ 
and  $read-path:$ 
   $Some\ (qh-path, s2) \in$ 
   $set-dist\ (execute\ (ntimes\ p.read\ (floor-log\ (scale * clength))))\ s1)$ 
and  $check:$ 
   $Some\ (ap, s3) \in$ 
   $set-dist\ (execute\ (p.check-authentication-path\ (scale * clength)\ i\ qh\ qh-path)\ s2)$ 
and  $assert-ok: Some\ ((), s4) \in set-dist\ (execute\ (assert\ (ap = fr))\ s3)$ 
and  $ret: Some\ (x, t) \in set-dist\ (execute\ (return\ qh)\ s4)$ 
by  $(auto\ simp: Let-def\ elim!: p.set-dist-bindE)$ 
have  $q-s1: PQueryCounter\ s1 = PQueryCounter\ s$ 
  using  $read-preserves-counters[OF\ read-qh]$  by  $simp$ 
have  $q-s2: PQueryCounter\ s2 = PQueryCounter\ s1$ 
  by  $(rule\ ntimes-read-preserves-query-counter[OF\ read-path])$ 
have  $q-s3: PQueryCounter\ s3 = PQueryCounter\ s2$ 
  using  $p.check-authentication-path-preserves-channel(6)[OF\ check]$  .
have  $s4-eq: s4 = s3$ 
  using  $assert-outcomeD(2)[OF\ assert-ok]$  .
have  $t-eq: t = s4$ 
  using  $ret$  by  $simp$ 
show  $?thesis$ 
  using  $q-s1\ q-s2\ q-s3$  unfolding  $s4-eq\ t-eq$  by  $simp$ 
qed

```

lemma $honest-root-alpha-degree-fri-final-query-from-prover-no-failure:$

```

assumes  $htv: honest-trace-valid$ 
and  $create-f:$ 
   $Some\ (f-merkle, s1) \in$ 
   $set-dist\ (execute\ (p.create\ p.f-eval)\ init-state)$ 
and  $send-f:$ 
   $Some\ ((), s2) \in set-dist\ (execute\ (p.send\ (value\ f-merkle))\ s1)$ 
and  $alphas:$ 
   $Some\ (as, s3) \in$ 
   $set-dist\ (execute$ 
     $(mmap\ (replicate\ (length\ spec)$ 
       $(do\ \{$ 
         $a \leftarrow p.receive-alpha-challenge;$ 
         $p.send\ a;$ 
         $return\ a$ 
       $\})))\ s2)$ 

```

```

and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
and create-cp:
  Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and fri:
  Some ((ps, ds, ls, ms), s6) ∈
    set-dist (execute
      (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and final-send:
  Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and random-idx:
  Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
and idx-def: idx' = p.index (to-nat idx)
and query-decommit:
  Some (qouts, s8) ∈
    set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
and fri-decommit:
  Some (fri-idx, s9) ∈
    set-dist (execute
      (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms))))) s8)
and prover-state-eq: prover-state = s9
and replay: replay-state = verifier-replay-state prover-state
shows
  None ∉ dom (dist (execute
    (do {
      fr ← p.read;
      as' ← mmap (replicate (length spec)
        (do {
          a0 ← p.receive-alpha-challenge;
          let a0' = a0;
          a1 ← p.read;
          let a1' = a1;
          assert (a0' = a1');
          return a1'
        })));
      dg ← p.read;
      assert (to-nat dg ≤ p.maxDegree);
      fl ← ntimes v.receive-fri-commits nrounds;
      final ← p.read;
      idxv ← p.receive-query-index-challenge;
      let idxv' = p.index (to-nat idxv);
      fv ← mmap (v.check-decommit-on-query fr idxv');
      return (fr, as', dg, fl, final, idxv, fv)
    }) replay-state))
proof –
  let ?round =
    do {
      a0 ← p.receive-alpha-challenge;

```

```

    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }
let ?prefix-final =
do {
  fr ← p.read;
  as' ← mmap (replicate (length spec) ?round);
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits nrounds;
  final ← p.read;
  return (fr, as', dg, fl, final)
}
let ?full =
do {
  fr ← p.read;
  as' ← mmap (replicate (length spec) ?round);
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits nrounds;
  final ← p.read;
  idxv ← p.receive-query-index-challenge;
  let idxv' = p.index (to-nat idxv);
  fv ← mmap (v.check-decommit-on-query fr idxv');
  return (fr, as', dg, fl, final, idxv, fv)
}
have prefix-nf:
  None ∉ dom (dist (execute ?prefix-final replay-state))
  by (rule honest-root-alpha-degree-fri-final-from-prover-no-failure[
    OF htv create-f send-f alphas cp'-def send-degree create-cp fri
    final-send random-idx idx-def query-decommit fri-decommit prover-state-eq replay])
have root-alpha-tr:
  length as = length spec ∧
  PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
  by (rule prover-root-alpha-degree-transcript[
    OF create-f send-f alphas cp'-def send-degree])
have later-hash: s3 ≤ prover-state
  by (rule prover-later-phases-hash-extends[
    OF send-degree create-cp fri final-send random-idx query-decommit fri-decommit prover-state-eq])
have alpha-lookups:
  ∀ i < length as.
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
        play-state) (value f-merkle)) (take i as))) =
      Some (as ! i)
  by (rule prover-alpha-lookups-in-replay-state[

```

```

    OF create-f send-f alphas later-hash replay])
from fri-commit-replay-data[OF fri] obtain roots challenges where
  len-roots: length roots = nrounds
  and len-challenges: length challenges = nrounds
  and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
  and st-s6: PState s6 = foldl concat (PState s5) roots
  and fri-hash: s5 ≤ s6
  and fri-lookups-s6:
    ∀ i < length roots.
      fmllookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
        Some (challenges ! i)
  by blast
from prover-query-decommitment-transcript-in-replay-state[
  OF final-send random-idx query-decommit fri-decommit prover-state-eq replay]
obtain rest where replay-query-tr:
  PTranscript replay-state =
    rev (PTranscript s6) @ [hd (last ls)] @
      query-decommitment-transcript idx' f-merkle @ rest
  by blast
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
      roots @ [hd (last ls)] @ query-decommitment-transcript idx' f-merkle @ rest
  using replay-query-tr tr-s6 tr-s5 root-alpha-tr cp'-def by simp
have st-s1: PState s1 = 0
  using create-preserves-state[OF create-f]
  unfolding init-state-def by simp
have send-f-res:
  s2 = s1(
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1)
  using p.send-outcome[OF send-f] .
have alpha-data:
  s2 ≤ s3 ∧ PState s3 = foldl concat (PState s2) as
  using alpha-mmap-outcome[OF alphas] by simp
have send-degree-res:
  s4 = s3(
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3)
  using p.send-outcome[OF send-degree] .
have st-s5:
  PState s5 =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers)))
  using st-s1 send-f-res alpha-data send-degree-res
  create-preserves-state[OF create-cp] replay cp'-def

```

```

unfolding verifier-replay-state-def by simp
have after-hash:  $s6 \leq \text{prover-state}$ 
by (rule prover-after-fri-hash-extends[
  OF final-send random-idx query-decommit fri-decommit prover-state-eq])
have fri-lookups:
   $\forall i < \text{length roots}.$ 
  fmlookup (HashMap replay-state)
    (FiatShamirChallenge (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        (of-nat (degree (p.cp as p.f-powers))))))
      (take (Suc i) roots))) =
    Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{length roots}$ 
  have lookup-s6:
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
    Some (challenges ! i)
  using fri-lookups-s6 i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state) (FiatShamirChallenge (foldl concat (PState s5) (take
(Suc i) roots))) =
    Some (challenges ! i)
  using p.hash-extension-lookup[OF lookup-s6 after-hash] .
  show fmlookup (HashMap replay-state)
    (FiatShamirChallenge (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        (of-nat (degree (p.cp as p.f-powers))))))
      (take (Suc i) roots))) =
    Some (challenges ! i)
  using lookup-prover st-s5 replay
  unfolding verifier-replay-state-def by simp
qed
have final-send-res:
  s-final = s6(
    PState := concat (PState s6) (hd (last ls)),
    PTranscript := hd (last ls) # PTranscript s6)
  using p.send-outcome[OF final-send] .
have query-lookup-replay:
  fmlookup (HashMap replay-state) (QueryIndexChallenge (PQueryCounter s-final) (PState
s-final)) = Some idx
  by (rule prover-query-random-lookup-in-replay-state[
    OF random-idx query-decommit fri-decommit prover-state-eq replay])
have s1-s2:  $s1 \leq s2$ 
  using p.send-outcome[OF send-f]
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s1-s3:  $s1 \leq s3$ 
  using s1-s2 alpha-data by (meson p.hash-ext-trans)

```

```

have s3-prover: s3 ≤ prover-state
  using later-hash .
have s1-prover: s1 ≤ prover-state
  using s1-s3 s3-prover by (meson p.hash-ext-trans)
have s1-replay: s1 ≤ replay-state
  using s1-prover replay
  unfolding verifier-replay-state-def less-eq-hash-ext-def less-eq-fmap-def
  by simp
have full-eq:
  ?full =
    (?prefix-final >>=
      (λout. p.receive-query-index-challenge >>=
        (λidxv.
          let idxv' = p.index (to-nat idxv) in
          mmap (v.check-decommit-on-query (case out of (fr, as', dg, fl, final) ⇒ fr) idxv') >>=
            (λfv. return (case out of (fr, as', dg, fl, final) ⇒
              (fr, as', dg, fl, final, idxv, fv)))))))
    by (simp add: sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full-eq
proof (rule no-failure-bindI[OF prefix-nf])
  fix out s-prefix
  assume prefix-out: Some (out, s-prefix) ∈ set-dist (execute ?prefix-final replay-state)
  obtain fr as' dg fl final where out-eq: out = (fr, as', dg, fl, final)
    by (cases out) auto
  have prefix-out':
    Some ((fr, as', dg, fl, final), s-prefix) ∈
      set-dist (execute ?prefix-final replay-state)
    using prefix-out unfolding out-eq .
  have prefix-out-len:
    Some ((fr, as', dg, fl, final), s-prefix) ∈ set-dist (execute
      (do {
        fr ← p.read;
        as' ← mmap (replicate (length spec) ?round);
        dg ← p.read;
        assert (to-nat dg ≤ p.maxDegree);
        fl ← ntimes v.receive-fri-commits (length roots);
        final ← p.read;
        return (fr, as', dg, fl, final)
      }) replay-state)
    using prefix-out' len-roots by simp
  have prefix-res:
    fr = value f-merkle ∧
    as' = as ∧
    dg = of-nat (degree (p.cp as p.f-powers)) ∧
    fl = zip challenges roots ∧
    final = hd (last ls) ∧
    PState s-prefix =
      concat

```

```

    (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        (of-nat (degree (p.cp as p.f-powers))))
      roots)
    (hd (last ls)) ∧
    PTranscript s-prefix = query-decommitment-transcript idx' f-merkle @ rest ∧
    replay-state ≤ s-prefix
  by (rule honest-root-alpha-degree-fri-final-prefix-outcome[
    OF htv conjunct1[OF root-alpha-tr] - replay-prefix alpha-lookups fri-lookups
    prefix-out-len])
    (use len-roots len-challenges in simp-all)
  have st-prefix-final: PState s-prefix = PState s-final
    using prefix-res st-s5 st-s6 final-send-res by simp
  have q-prefix-replay: PQueryCounter s-prefix = PQueryCounter replay-state
    by (rule honest-root-alpha-degree-fri-final-prefix-preserves-query-counter[
      OF prefix-out-len])
  have q-final-replay: PQueryCounter s-final = PQueryCounter replay-state
    by (rule prover-final-query-counter-in-replay-state[
      OF create-f send-f alphas send-degree create-cp fri final-send replay])
  have lookup-prefix:
    fmllookup (HashMap s-prefix) (QueryIndexChallenge (PQueryCounter s-prefix) (PState
    s-prefix)) = Some idx
  proof -
    have lookup:
      fmllookup (HashMap s-prefix) (QueryIndexChallenge (PQueryCounter s-final) (PState
    s-final)) = Some idx
    using p.hash-extension-lookup[OF query-lookup-replay, of s-prefix] prefix-res
    by simp
    then show ?thesis
      using st-prefix-final q-prefix-replay q-final-replay by simp
  qed
  show None ∉ dom (dist (execute
    (p.receive-query-index-challenge >>=
      (λidxv.
        let idxv' = p.index (to-nat idxv)
        in mmap
          (v.check-decommit-on-query
            (case out of (fr, as', dg, fl, final) ⇒ fr) idxv') >>=
            (λfv.
              return
                (case out of (fr, as', dg, fl, final) ⇒
                  (fr, as', dg, fl, final, idxv, fv)))))) s-prefix))
  proof (rule no-failure-bindI)
    show None ∉ dom (dist (execute p.receive-query-index-challenge s-prefix))
      by (rule p.receive-query-index-challenge-no-failure)
  next
    fix idxv s-idx
    assume rand-v:
      Some (idxv, s-idx) ∈ set-dist (execute p.receive-query-index-challenge s-prefix)

```

```

have rand-res:
  idxv = idx ∧
  PState s-idx = PState s-prefix ∧
  PTranscript s-idx = PTranscript s-prefix ∧
  s-prefix ≤ s-idx
using p.receive-query-index-challenge-known-outcome[OF lookup-prefix rand-v]
by simp
have s1-s-idx: s1 ≤ s-idx
using s1-replay prefix-res rand-res by (meson p.hash-ext-trans)
have tr-idx:
  PTranscript s-idx =
    query-decommitment-transcript (p.index (to-nat idxv)) f-merkle @ rest
using rand-res prefix-res idx-def by simp
have query-nf:
  None ∉ dom (dist (execute
    (mmap (v.check-decommit-on-query fr (p.index (to-nat idxv)))) s-idx))
by (rule honest-query-decommitment-replay-no-failure[
    OF create-f s1-s-idx - tr-idx])
    (use prefix-res in simp)
show None ∉ dom (dist (execute
  ((let idxv' = p.index (to-nat idxv)
    in mmap
      (v.check-decommit-on-query
        (case out of (fr, as', dg, fl, final) ⇒ fr) idxv') ≫=
        (λfv.
          return
            (case out of (fr, as', dg, fl, final) ⇒
              (fr, as', dg, fl, final, idxv, fv)))))) s-idx))
unfolding out-eq
by (simp add: no-failure-bind-returnI[OF query-nf])
qed
qed
qed

```

```

lemma honest-root-alpha-degree-fri-final-query-dynamic-from-prover-no-failure:
assumes htv: honest-trace-valid
and create-f:
  Some (f-merkle, s1) ∈
    set-dist (execute (p.create p.f-eval) init-state)
and send-f:
  Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and alphas:
  Some (as, s3) ∈
    set-dist (execute
      (mmap (replicate (length spec)
        (do {
          a ← p.receive-alpha-challenge;
          p.send a;
          return a

```



```

    }))) s2)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
    Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
and create-cp:
    Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and nrounds-def: nrounds = ceil-log (degree cp' + 1)
and fri:
    Some ((ps, ds, ls, ms), s6) ∈
        set-dist (execute
            (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and final-send:
    Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and random-idx:
    Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
and idx-def: idx' = p.index (to-nat idx)
and query-decommit:
    Some (qouts, s8) ∈
        set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
and fri-decommit:
    Some (fri-idx, s9) ∈
        set-dist (execute
            (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
and prover-state-eq: prover-state = s9
and replay: replay-state = verifier-replay-state prover-state
shows
    None ∉ dom (dist (execute
        (do {
            fr ← p.read;
            as' ← mmap (replicate (length spec))
            (do {
                a0 ← p.receive-alpha-challenge;
                let a0' = a0;
                a1 ← p.read;
                let a1' = a1;
                assert (a0' = a1');
                return a1'
            }));
            dg ← p.read;
            assert (to-nat dg ≤ p.maxDegree);
            fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
            final ← p.read;
            idxv ← p.receive-query-index-challenge;
            let idxv' = p.index (to-nat idxv);
            fv ← mmap (v.check-decommit-on-query fr idxv');
            return (fr, as', dg, fl, final, idxv, fv)
        }) replay-state))
proof –
    let ?round =

```

```

do {
  a0 ← p.receive-alpha-challenge;
  let a0' = a0;
  a1 ← p.read;
  let a1' = a1;
  assert (a0' = a1');
  return a1'
}
let ?prefix-final =
do {
  fr ← p.read;
  as' ← mmap (replicate (length spec) ?round);
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
  final ← p.read;
  return (fr, as', dg, fl, final)
}
let ?full =
do {
  fr ← p.read;
  as' ← mmap (replicate (length spec) ?round);
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
  final ← p.read;
  idxv ← p.receive-query-index-challenge;
  let idxv' = p.index (to-nat idxv);
  fv ← mmap (v.check-decommit-on-query fr idxv');
  return (fr, as', dg, fl, final, idxv, fv)
}
have prefix-nf:
  None ∉ dom (dist (execute ?prefix-final replay-state))
by (rule honest-root-alpha-degree-fri-final-dynamic-from-prover-no-failure[
  OF htv create-f send-f alphas cp'-def send-degree create-cp nrounds-def fri
  final-send random-idx idx-def query-decommit fri-decommit prover-state-eq replay])
have root-alpha-tr:
  length as = length spec ∧
  PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
by (rule prover-root-alpha-degree-transcript[
  OF create-f send-f alphas cp'-def send-degree])
have later-hash: s3 ≤ prover-state
by (rule prover-later-phases-hash-extends[
  OF send-degree create-cp fri final-send random-idx query-decommit fri-decommit prover-state-eq])
have alpha-lookups:
  ∀ i < length as.
    fmlookup (HashMap replay-state)
      (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
play-state) (value f-merkle)) (take i as))) =

```

```

    Some (as ! i)
  by (rule prover-alpha-lookups-in-replay-state[
    OF create-f send-f alphas later-hash replay])
from fri-commit-replay-data[OF fri] obtain roots challenges where
  len-roots: length roots = nrounds
  and len-challenges: length challenges = nrounds
  and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
  and st-s6: PState s6 = foldl concat (PState s5) roots
  and fri-hash: s5 ≤ s6
  and fri-lookups-s6:
    ∀ i < length roots.
      fmllookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
        Some (challenges ! i)
  by blast
from prover-query-decommitment-transcript-in-replay-state[
  OF final-send random-idx query-decommit fri-decommit prover-state-eq replay]
obtain rest where replay-query-tr:
  PTranscript replay-state =
    rev (PTranscript s6) @ [hd (last ls)] @
      query-decommitment-transcript idx' f-merkle @ rest
  by blast
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
      roots @ [hd (last ls)] @ query-decommitment-transcript idx' f-merkle @ rest
  using replay-query-tr tr-s6 tr-s5 root-alpha-tr cp'-def by simp
have st-s1: PState s1 = 0
  using create-preserves-state[OF create-f]
  unfolding init-state-def by simp
have send-f-res:
  s2 = s1 (|
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1 |)
  using p.send-outcome[OF send-f] .
have alpha-data:
  s2 ≤ s3 ∧ PState s3 = foldl concat (PState s2) as
  using alpha-mmap-outcome[OF alphas] by simp
have send-degree-res:
  s4 = s3 (|
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3 |)
  using p.send-outcome[OF send-degree] .
have st-s5:
  PState s5 =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
      (of-nat (degree (p.cp as p.f-powers)))

```

```

using st-s1 send-f-res alpha-data send-degree-res
      create-preserves-state[OF create-cp] replay cp'-def
unfolding verifier-replay-state-def by simp
have after-hash: s6 ≤ prover-state
by (rule prover-after-fri-hash-extends[
      OF final-send random-idx query-decommit fri-decommit prover-state-eq])
have fri-lookups:
   $\forall i < \text{length roots.}$ 
  fmlookup (HashMap replay-state)
    (FiatShamirChallenge (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as
        (of-nat (degree (p.cp as p.f-powers))))
      (take (Suc i) roots))) =
      Some (challenges ! i))
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-s6:
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
      Some (challenges ! i)
    using fri-lookups-s6 i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state) (FiatShamirChallenge (foldl concat (PState s5) (take
(Suc i) roots))) =
      Some (challenges ! i)
    using p.hash-extension-lookup[OF lookup-s6 after-hash] .
  show fmlookup (HashMap replay-state)
    (FiatShamirChallenge (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as
        (of-nat (degree (p.cp as p.f-powers))))
      (take (Suc i) roots))) =
      Some (challenges ! i)
    using lookup-prover st-s5 replay
    unfolding verifier-replay-state-def by simp
qed
have final-send-res:
  s-final = s6(
    PState := concat (PState s6) (hd (last ls)),
    PTranscript := hd (last ls) # PTranscript s6)
  using p.send-outcome[OF final-send] .
have query-lookup-replay:
  fmlookup (HashMap replay-state) (QueryIndexChallenge (PQueryCounter s-final) (PState
s-final)) = Some idx
  by (rule prover-query-random-lookup-in-replay-state[
    OF random-idx query-decommit fri-decommit prover-state-eq replay])
have s1-s2: s1 ≤ s2
  using p.send-outcome[OF send-f]
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp

```

```

have s1-s3:  $s1 \leq s3$ 
  using s1-s2 alpha-data by (meson p.hash-ext-trans)
have s3-prover:  $s3 \leq \text{prover-state}$ 
  using later-hash .
have s1-prover:  $s1 \leq \text{prover-state}$ 
  using s1-s3 s3-prover by (meson p.hash-ext-trans)
have s1-replay:  $s1 \leq \text{replay-state}$ 
  using s1-prover replay
  unfolding verifier-replay-state-def less-eq-hash-ext-def less-eq-fmap-def
  by simp
have roots-rounds:
  length roots =  $\text{ceil-log } (\text{degree } (p.cp \text{ as } p.f\text{-powers}) + 1)$ 
  using len-roots nrounds-def cp'-def by simp
have full-eq:
  ?full =
    (?prefix-final  $\gg$ 
      ( $\lambda out. p.receive\text{-query-index-challenge} \gg$ 
        ( $\lambda idrv.$ 
          let  $idrv' = p.index \text{ (to-nat } idrv) \text{ in}$ 
          mmap ( $v.check\text{-decommit-on-query } (\text{case out of } (fr, as', dg, fl, final) \Rightarrow fr) idrv') \gg$ 
            ( $\lambda fv. \text{return } (\text{case out of } (fr, as', dg, fl, final) \Rightarrow$ 
              ( $fr, as', dg, fl, final, idrv, fv))))))$ 
        by (simp add: sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full-eq
proof (rule no-failure-bindI[OF prefix-nf])
  fix out s-prefix
  assume prefix-out:  $\text{Some } (out, s\text{-prefix}) \in \text{set-dist } (\text{execute } ?prefix\text{-final replay-state})$ 
  obtain fr as' dg fl final where out-eq:  $out = (fr, as', dg, fl, final)$ 
  by (cases out) auto
  have prefix-out':
     $\text{Some } ((fr, as', dg, fl, final), s\text{-prefix}) \in$ 
     $\text{set-dist } (\text{execute } ?prefix\text{-final replay-state})$ 
  using prefix-out unfolding out-eq .
  have prefix-res:
    fr = value f-merkle  $\wedge$ 
    as' = as  $\wedge$ 
    dg = of-nat (degree (p.cp as p.f-powers))  $\wedge$ 
    fl = zip challenges roots  $\wedge$ 
    final = hd (last ls)  $\wedge$ 
    PState s-prefix =
      concat
      (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
          (of-nat (degree (p.cp as p.f-powers))))
        roots)
      (hd (last ls))  $\wedge$ 
    PTranscript s-prefix = query-decommitment-transcript  $idx'$  f-merkle @ rest  $\wedge$ 
    replay-state  $\leq s\text{-prefix}$ 

```

```

by (rule honest-root-alpha-degree-fri-final-dynamic-prefix-outcome[
  OF htv conjunct1[OF root-alpha-tr] - roots-rounds replay-prefix
  alpha-lookups fri-lookups prefix-out'])
  (use len-roots len-challenges in simp-all)
have st-prefix-final: PState s-prefix = PState s-final
  using prefix-res st-s5 st-s6 final-send-res by simp
have q-prefix-replay: PQueryCounter s-prefix = PQueryCounter replay-state
  by (rule honest-root-alpha-degree-fri-final-dynamic-prefix-preserves-query-counter[
    OF prefix-out'])
have q-final-replay: PQueryCounter s-final = PQueryCounter replay-state
  by (rule prover-final-query-counter-in-replay-state[
    OF create-f send-f alphas send-degree create-cp fri final-send replay])
have lookup-prefix:
  fmllookup (HashMap s-prefix) (QueryIndexChallenge (PQueryCounter s-prefix) (PState
s-prefix)) = Some idx
  proof -
    have lookup:
      fmllookup (HashMap s-prefix) (QueryIndexChallenge (PQueryCounter s-final) (PState
s-final)) = Some idx
      using p.hash-extension-lookup[OF query-lookup-replay, of s-prefix] prefix-res
      by simp
    then show ?thesis
      using st-prefix-final q-prefix-replay q-final-replay by simp
  qed
show None  $\notin$  dom (dist (execute
  (p.receive-query-index-challenge  $\gg$ 
  ( $\lambda$ idxv.
    let idxv' = p.index (to-nat idxv)
    in mmap
      (v.check-decommit-on-query
        (case out of (fr, as', dg, fl, final)  $\Rightarrow$  fr) idxv')  $\gg$ 
      ( $\lambda$ fv.
        return
          (case out of (fr, as', dg, fl, final)  $\Rightarrow$ 
            (fr, as', dg, fl, final, idxv, fv)))))) s-prefix))
proof (rule no-failure-bindI)
  show None  $\notin$  dom (dist (execute p.receive-query-index-challenge s-prefix))
  by (rule p.receive-query-index-challenge-no-failure)
next
  fix idxv s-idx
  assume rand-v:
    Some (idxv, s-idx)  $\in$  set-dist (execute p.receive-query-index-challenge s-prefix)
  have rand-res:
    idxv = idx  $\wedge$ 
    PState s-idx = PState s-prefix  $\wedge$ 
    PTranscript s-idx = PTranscript s-prefix  $\wedge$ 
    s-prefix  $\leq$  s-idx
  using p.receive-query-index-challenge-known-outcome[OF lookup-prefix rand-v]
  by simp

```

```

have s1-s-idx:  $s1 \leq s\text{-}idx$ 
  using s1-replay prefix-res rand-res by (meson p.hash-ext-trans)
have tr-idx:
  PTranscript s-idx =
    query-decommitment-transcript (p.index (to-nat idxv)) f-merkle @ rest
  using rand-res prefix-res idx-def by simp
have query-nf:
  None  $\notin$  dom (dist (execute
    (mmap (v.check-decommit-on-query fr (p.index (to-nat idxv)))) s-idx))
  by (rule honest-query-decommitment-replay-no-failure[
    OF create-f s1-s-idx - tr-idx])
    (use prefix-res in simp)
show None  $\notin$  dom (dist (execute
  ((let idxv' = p.index (to-nat idxv)
    in mmap
      (v.check-decommit-on-query
        (case out of (fr, as', dg, fl, final)  $\Rightarrow$  fr) idxv')  $\gg$ 
        ( $\lambda$ fv.
          return
            (case out of (fr, as', dg, fl, final)  $\Rightarrow$ 
              (fr, as', dg, fl, final, idxv, fv)))))) s-idx))
  unfolding out-eq
  by (simp add: no-failure-bind-returnI[OF query-nf])
qed
qed
qed

```

lemma *honest-root-alpha-degree-fri-final-query-dynamic-outcome-to-fixed*:

```

assumes htv: honest-trace-valid
and create-f:
  Some (f-merkle, s1)  $\in$ 
    set-dist (execute (p.create p.f-eval) init-state)
and send-f:
  Some ((), s2)  $\in$  set-dist (execute (p.send (value f-merkle)) s1)
and alphas:
  Some (as, s3)  $\in$ 
    set-dist (execute
      (mmap (replicate (length spec)
        (do {
          a  $\leftarrow$  p.receive-alpha-challenge;
          p.send a;
          return a
        }))) s2)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some ((), s4)  $\in$  set-dist (execute (p.send (of-nat (degree cp'))) s3)
and create-cp:
  Some (cp-merkle, s5)  $\in$  set-dist (execute (p.create (p.cp-eval as)) s4)
and nrounds-def: nrounds = ceil-log (degree cp' + 1)

```

and *fri*:
Some ((ps, ds, ls, ms), s6) ∈
set-dist (execute
(p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and *final-send*:
Some (((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and *random-idx*:
Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
and *idx-def*: *idx' = p.index (to-nat idx)*
and *query-decommit*:
Some (qouts, s8) ∈
set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
and *fri-decommit*:
Some (fri-idx, s9) ∈
set-dist (execute
(mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
and *prover-state-eq*: *prover-state = s9*
and *replay*: *replay-state = verifier-replay-state prover-state*
and *outcome*:
Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
set-dist (execute
(do {
fr ← p.read;
as' ← mmap (replicate (length spec)
(do {
a0 ← p.receive-alpha-challenge;
let a0' = a0;
a1 ← p.read;
let a1' = a1;
assert (a0' = a1');
return a1'
}});
dg ← p.read;
assert (to-nat dg ≤ p.maxDegree);
fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
final ← p.read;
idxv ← p.receive-query-index-challenge;
let idxv' = p.index (to-nat idxv);
fv ← mmap (v.check-decommit-on-query fr idxv');
return (fr, as', dg, fl, final, idxv, fv)
}) replay-state)

shows

Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
set-dist (execute
(do {
fr ← p.read;
as' ← mmap (replicate (length spec)
(do {
a0 ← p.receive-alpha-challenge;


```

    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }));
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits nrounds;
  final ← p.read;
  idxv ← p.receive-query-index-challenge;
  let idxv' = p.index (to-nat idxv);
  fv ← mmap (v.check-decommit-on-query fr idxv');
  return (fr, as', dg, fl, final, idxv, fv)
}) replay-state)
proof –
let ?round =
  do {
    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }
from outcome obtain r1 r2 r3 r4 r5 r6 r7 where
  read-fr: Some (fr, r1) ∈ set-dist (execute p.read replay-state)
  and alpha-out:
    Some (as', r2) ∈
      set-dist (execute (mmap (replicate (length spec) ?round)) r1)
  and read-dg: Some (dg, r3) ∈ set-dist (execute p.read r2)
  and degree-assert:
    Some ((), r4) ∈ set-dist (execute (assert (to-nat dg ≤ p.maxDegree)) r3)
  and fri-reads:
    Some (fl, r5) ∈
      set-dist (execute (ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1))) r4)
  and read-final: Some (final, r6) ∈ set-dist (execute p.read r5)
  and random-v:
    Some (idxv, r7) ∈ set-dist (execute p.receive-query-index-challenge r6)
  and query-reads:
    Some (fv, t) ∈
      set-dist (execute (mmap (v.check-decommit-on-query fr (p.index (to-nat idxv)))) r7)
  by (auto simp: Let-def elim!: p.set-dist-bindE)
let ?rad =
  do {
    fr ← p.read;
    as' ← mmap (replicate (length spec) ?round);
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);

```

```

    return (fr, as', dg)
  }
have rad-out:
  Some ((fr, as', dg), r4) ∈ set-dist (execute ?rad replay-state)
  apply (rule set-dist-bindI[OF read-fr])
  apply (rule set-dist-bindI[OF alpha-out])
  apply (rule set-dist-bindI[OF read-dg])
  apply (rule set-dist-bindI[OF degree-assert])
  by simp
have dg-eq: dg = of-nat (degree (p.cp as p.f-powers))
  using honest-root-alpha-degree-from-prover-outcome[
    OF htv create-f send-f alphas cp'-def send-degree create-cp fri
    final-send random-idx idx-def query-decommit fri-decommit prover-state-eq
    replay rad-out]
  by blast
have rounds-eq: ceil-log (to-nat dg + 1) = nrounds
  using honest-degree-message-rounds[OF htv, of as] dg-eq cp'-def nrounds-def
  by simp
have fri-reads-fixed:
  Some (f1, r5) ∈ set-dist (execute (ntimes v.receive-fri-commits nrounds) r4)
  using fri-reads rounds-eq by simp
show ?thesis
  apply (rule set-dist-bindI[OF read-fr])
  apply (rule set-dist-bindI[OF alpha-out])
  apply (rule set-dist-bindI[OF read-dg])
  apply (rule set-dist-bindI[OF degree-assert])
  apply (rule set-dist-bindI[OF fri-reads-fixed])
  apply (rule set-dist-bindI[OF read-final])
  apply (rule set-dist-bindI[OF random-v])
  apply (simp add: Let-def)
  apply (rule set-dist-bindI[OF query-reads])
  by simp
qed

```

lemma honest-root-alpha-degree-fri-final-query-from-prover-outcome:

assumes htv: honest-trace-valid

and create-f:

Some (f-merkle, s1) ∈
 set-dist (execute (p.create p.f-eval) init-state)

and send-f:

Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)

and alphas:

Some (as, s3) ∈
 set-dist (execute
 (mmap (replicate (length spec)
 (do {
 a ← p.receive-alpha-challenge;
 p.send a;
 return a

```

    }))) s2)
and cp'-def:  $cp' = p.cp$  as  $p.f\text{-powers}$ 
and send-degree:
    Some ( $()$ ,  $s4$ )  $\in$  set-dist (execute ( $p.send$  (of-nat (degree  $cp'$ )))  $s3$ )
and create-cp:
    Some ( $cp\text{-merkle}$ ,  $s5$ )  $\in$  set-dist (execute ( $p.create$  ( $p.cp\text{-eval}$   $as$ ))  $s4$ )
and fri:
    Some ( $(ps, ds, ls, ms)$ ,  $s6$ )  $\in$ 
        set-dist (execute
            ( $p.fri\text{-commit}$   $nrounds$  [ $cp'$ ] [ $p.eval\text{-domain}$ ] [ $p.cp\text{-eval}$   $as$ ] [ $cp\text{-merkle}$ ]  $s5$ )
and final-send:
    Some ( $()$ ,  $s\text{-final}$ )  $\in$  set-dist (execute ( $p.send$  (hd (last  $ls$ )))  $s6$ )
and random-idx:
    Some ( $idx$ ,  $s7$ )  $\in$  set-dist (execute  $p.receive\text{-query-index-challenge}$   $s\text{-final}$ )
and idx-def:  $idx' = p.index$  (to-nat  $idx$ )
and query-decommit:
    Some ( $qouts$ ,  $s8$ )  $\in$ 
        set-dist (execute ( $mmap$  ( $p.decommit\text{-on-query}$   $idx'$   $f\text{-merkle}$ ))  $s7$ )
and fri-decommit:
    Some ( $fri\text{-idx}$ ,  $s9$ )  $\in$ 
        set-dist (execute
            ( $mfold$   $idx'$  ( $p.decommit\text{-on-fri-layers}$  (butlast (zip  $ls$   $ms$ ))))  $s8$ )
and prover-state-eq:  $prover\text{-state} = s9$ 
and replay:  $replay\text{-state} = verifier\text{-replay-state}$   $prover\text{-state}$ 
and outcome:
    Some ( $(fr, as', dg, fl, final, idxv, fv)$ ,  $t$ )  $\in$ 
        set-dist (execute
            (do {
                 $fr \leftarrow p.read$ ;
                 $as' \leftarrow mmap$  (replicate (length  $spec$ ))
                (do {
                     $a0 \leftarrow p.receive\text{-alpha-challenge}$ ;
                     $let\ a0' = a0$ ;
                     $a1 \leftarrow p.read$ ;
                     $let\ a1' = a1$ ;
                    assert ( $a0' = a1'$ );
                    return  $a1'$ 
                }));
                 $dg \leftarrow p.read$ ;
                assert (to-nat  $dg \leq p.maxDegree$ );
                 $fl \leftarrow ntimes\ v.receive\text{-fri-commits}$   $nrounds$ ;
                 $final \leftarrow p.read$ ;
                 $idxv \leftarrow p.receive\text{-query-index-challenge}$ ;
                 $let\ idxv' = p.index$  (to-nat  $idxv$ );
                 $fv \leftarrow mmap$  ( $v.check\text{-decommit-on-query}$   $fr$   $idxv'$ );
                return ( $fr, as', dg, fl, final, idxv, fv$ )
            })  $replay\text{-state}$ )
shows
     $fr = value\ f\text{-merkle} \wedge$ 

```

```

as' = as ∧
dg = of-nat (degree (p.cp as p.f-powers)) ∧
final = hd (last ls) ∧
idxv = idx ∧
fv = map (λi. p.f-eval ! i) (p.powers-scaled idx') ∧
replay-state ≤ t ∧
(∃ roots challenges rest.
  length roots = nrounds ∧
  length challenges = nrounds ∧
  fl = zip challenges roots ∧
  (∀ j < nrounds. roots ! j = value (ms ! j)) ∧
  (∀ j < nrounds.
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j)) ∧
  PTranscript t = rest ∧
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
    roots @ [hd (last ls)] @
    query-decommitment-transcript idx' f-merkle @ rest)
proof -
let ?round =
  do {
    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }
let ?prefix-final =
  do {
    fr ← p.read;
    as' ← mmap (replicate (length spec) ?round);
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    fl ← ntimes v.receive-fri-commits nrounds;
    final ← p.read;
    return (fr, as', dg, fl, final)
  }
let ?full =
  do {
    fr ← p.read;
    as' ← mmap (replicate (length spec) ?round);
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    fl ← ntimes v.receive-fri-commits nrounds;
    final ← p.read;
    idxv ← p.receive-query-index-challenge;
    let idxv' = p.index (to-nat idxv);

```

```

    fv ← mmap (v.check-decommit-on-query fr idxv');
    return (fr, as', dg, fl, final, idxv, fv)
  }
have root-alpha-tr:
  length as = length spec ∧
  PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
  by (rule prover-root-alpha-degree-transcript[
    OF create-f send-f alphas cp'-def send-degree])
have later-hash: s3 ≤ prover-state
  by (rule prover-later-phases-hash-extends[
    OF send-degree create-cp fri final-send random-idx query-decommit fri-decommit prover-state-eq])
have alpha-lookups:
  ∀ i < length as.
  fmlookup (HashMap replay-state)
    (AlphaChallenge (PAlphaCounter replay-state + i) (foldl concat (concat (PState re-
play-state) (value f-merkle)) (take i as))) =
    Some (as ! i)
  by (rule prover-alpha-lookups-in-replay-state[
    OF create-f send-f alphas later-hash replay])
from fri-commit-initial-replay-successor-data[OF fri] obtain roots challenges where
  len-roots: length roots = nrounds
  and len-challenges: length challenges = nrounds
  and tr-s6: PTranscript s6 = rev roots @ PTranscript s5
  and st-s6: PState s6 = foldl concat (PState s5) roots
  and fri-hash: s5 ≤ s6
  and fri-lookups-s6:
  ∀ i < length roots.
  fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
    Some (challenges ! i)
  and roots-aligned: ∀ j < nrounds. roots ! j = value (ms ! j)
  and successors:
  ∀ j < nrounds.
  p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
    (ps ! Suc j, ds ! Suc j, ls ! Suc j)
  by blast
from prover-query-decommitment-transcript-in-replay-state[
  OF final-send random-idx query-decommit fri-decommit prover-state-eq replay]
obtain rest where replay-query-tr:
  PTranscript replay-state =
    rev (PTranscript s6) @ [hd (last ls)] @
    query-decommitment-transcript idx' f-merkle @ rest
  by blast
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have replay-prefix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
    roots @ [hd (last ls)] @ query-decommitment-transcript idx' f-merkle @ rest

```

```

using replay-query-tr tr-s6 tr-s5 root-alpha-tr cp'-def by simp
have st-s1: PState s1 = 0
using create-preserves-state[OF create-f]
unfolding init-state-def by simp
have send-f-res:
  s2 = s1⟦
    PState := concat (PState s1) (value f-merkle),
    PTranscript := value f-merkle # PTranscript s1⟧
using p.send-outcome[OF send-f] .
have alpha-data:
  s2 ≤ s3 ∧ PState s3 = foldl concat (PState s2) as
using alpha-mmap-outcome[OF alphas] by simp
have send-degree-res:
  s4 = s3⟦
    PState := concat (PState s3) (of-nat (degree cp')),
    PTranscript := of-nat (degree cp') # PTranscript s3⟧
using p.send-outcome[OF send-degree] .
have st-s5:
  PState s5 =
    concat (foldl concat (concat (PState replay-state) (value f-merkle)) as
      (of-nat (degree (p.cp as p.f-powers))))
using st-s1 send-f-res alpha-data send-degree-res
  create-preserves-state[OF create-cp] replay cp'-def
unfolding verifier-replay-state-def by simp
have after-hash: s6 ≤ prover-state
by (rule prover-after-fri-hash-extends[
  OF final-send random-idx query-decommit fri-decommit prover-state-eq])
have fri-lookups:
  ∀ i < length roots.
    fmlookup (HashMap replay-state)
      (FiatShamirChallenge (foldl concat
        (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as
          (of-nat (degree (p.cp as p.f-powers))))
        (take (Suc i) roots))) =
        Some (challenges ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < length roots
  have lookup-s6:
    fmlookup (HashMap s6) (FiatShamirChallenge (foldl concat (PState s5) (take (Suc i)
roots))) =
      Some (challenges ! i)
    using fri-lookups-s6 i-bound by simp
  have lookup-prover:
    fmlookup (HashMap prover-state) (FiatShamirChallenge (foldl concat (PState s5) (take
(Suc i) roots))) =
      Some (challenges ! i)
    using p.hash-extension-lookup[OF lookup-s6 after-hash] .
  show fmlookup (HashMap replay-state)

```

```

    (FiatShamirChallenge (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        (of-nat (degree (p.cp as p.f-powers))))))
      (take (Suc i) roots))) =
    Some (challenges ! i)
  using lookup-prover st-s5 replay
  unfolding verifier-replay-state-def by simp
qed
have final-send-res:
  s-final = s6()
  PState := concat (PState s6) (hd (last ls)),
  PTranscript := hd (last ls) # PTranscript s6()
  using p.send-outcome[OF final-send] .
have query-lookup-replay:
  fmlookup (HashMap replay-state) (QueryIndexChallenge (PQueryCounter s-final) (PState
s-final)) = Some idx
  by (rule prover-query-random-lookup-in-replay-state[
    OF random-idx query-decommit fri-decommit prover-state-eq replay])
have s1-s2: s1 ≤ s2
  using p.send-outcome[OF send-f]
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s1-s3: s1 ≤ s3
  using s1-s2 alpha-data by (meson p.hash-ext-trans)
have s3-prover: s3 ≤ prover-state
  using later-hash .
have s1-prover: s1 ≤ prover-state
  using s1-s3 s3-prover by (meson p.hash-ext-trans)
have s1-replay: s1 ≤ replay-state
  using s1-prover replay
  unfolding verifier-replay-state-def less-eq-hash-ext-def less-eq-fmap-def
  by simp
have full-eq:
  ?full =
    (?prefix-final >>=
      (λout. p.receive-query-index-challenge >>=
        (λidxv.
          let idxv' = p.index (to-nat idxv) in
          mmap (v.check-decommit-on-query (case out of (fr, as', dg, fl, final) ⇒ fr) idxv') >>=
            (λfv. return (case out of (fr, as', dg, fl, final) ⇒
              (fr, as', dg, fl, final, idxv, fv)))))))
    by (simp add: sm-bind-assoc split: prod.splits)
have outcome-full:
  Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
    set-dist (execute ?full replay-state)
  using outcome by simp
from outcome-full[unfolded full-eq] obtain out s-prefix where
  prefix-out: Some (out, s-prefix) ∈ set-dist (execute ?prefix-final replay-state)
  and after-prefix:
    Some ((fr, as', dg, fl, final, idxv, fv), t) ∈

```

```

    set-dist (execute
      (p.receive-query-index-challenge  $\gg$ 
        ( $\lambda idv$ .
          let  $idv' = p.index (to-nat idv)$  in
          mmap (v.check-decommit-on-query
            (case out of (fr, as', dg, fl, final)  $\Rightarrow$  fr)  $idv'$ )  $\gg$ 
            ( $\lambda fv$ . return (case out of (fr, as', dg, fl, final)  $\Rightarrow$ 
              (fr, as', dg, fl, final,  $idv$ , fv)))))) s-prefix)
  proof -
    show ?thesis
      using outcome-full[unfolded full-eq]
    proof (rule p.set-dist-bindE)
      fix out s-prefix
      assume prefix-out:
        Some (out, s-prefix)  $\in$  set-dist (execute ?prefix-final replay-state)
      and after-prefix:
        Some ((fr, as', dg, fl, final,  $idv$ , fv), t)  $\in$ 
          set-dist (execute
            (p.receive-query-index-challenge  $\gg$ 
              ( $\lambda idv$ .
                let  $idv' = p.index (to-nat idv)$  in
                mmap (v.check-decommit-on-query
                  (case out of (fr, as', dg, fl, final)  $\Rightarrow$  fr)  $idv'$ )  $\gg$ 
                  ( $\lambda fv$ . return (case out of (fr, as', dg, fl, final)  $\Rightarrow$ 
                    (fr, as', dg, fl, final,  $idv$ , fv)))))) s-prefix)
              show ?thesis
                by (rule that[OF prefix-out after-prefix])
            qed
          qed
        obtain fr0 as0 dg0 fl0 final0 where out-eq:
          out = (fr0, as0, dg0, fl0, final0)
        by (cases out) auto
      have prefix-out':
        Some ((fr0, as0, dg0, fl0, final0), s-prefix)  $\in$ 
          set-dist (execute ?prefix-final replay-state)
      using prefix-out unfolding out-eq .
      have prefix-out-len:
        Some ((fr0, as0, dg0, fl0, final0), s-prefix)  $\in$  set-dist (execute
          (do {
            fr  $\leftarrow$  p.read;
            as'  $\leftarrow$  mmap (replicate (length spec) ?round);
            dg  $\leftarrow$  p.read;
            assert (to-nat dg  $\leq$  p.maxDegree);
            fl  $\leftarrow$  ntimes v.receive-fri-commits (length roots);
            final  $\leftarrow$  p.read;
            return (fr, as', dg, fl, final)
          }) replay-state)
      using prefix-out' len-roots by simp
    have prefix-res:

```



```

fr0 = value f-merkle ∧
as0 = as ∧
dg0 = of-nat (degree (p.cp as p.f-powers)) ∧
fl0 = zip challenges roots ∧
final0 = hd (last ls) ∧
PState s-prefix =
  concat
    (foldl concat
      (concat (foldl concat (concat (PState replay-state) (value f-merkle)) as)
        (of-nat (degree (p.cp as p.f-powers)))))
      roots)
    (hd (last ls)) ∧
PTranscript s-prefix = query-decommitment-transcript idx' f-merkle @ rest ∧
replay-state ≤ s-prefix
by (rule honest-root-alpha-degree-fri-final-prefix-outcome[
  OF htv conjunct1[OF root-alpha-tr] - replay-prefix alpha-lookups fri-lookups
  prefix-out-len])
  (use len-roots len-challenges in simp-all)
have st-prefix-final: PState s-prefix = PState s-final
  using prefix-res st-s5 st-s6 final-send-res by simp
have q-prefix-replay: PQueryCounter s-prefix = PQueryCounter replay-state
  by (rule honest-root-alpha-degree-fri-final-prefix-preserves-query-counter[
    OF prefix-out-len])
have q-final-replay: PQueryCounter s-final = PQueryCounter replay-state
  by (rule prover-final-query-counter-in-replay-state[
    OF create-f send-f alphas send-degree create-cp fri final-send replay])
have lookup-prefix:
  fmlookup (HashMap s-prefix) (QueryIndexChallenge (PQueryCounter s-prefix) (PState
s-prefix)) = Some idx
proof -
  have lookup:
    fmlookup (HashMap s-prefix) (QueryIndexChallenge (PQueryCounter s-final) (PState
s-final)) = Some idx
  using p.hash-extension-lookup[OF query-lookup-replay, of s-prefix] prefix-res
  by simp
  then show ?thesis
  using st-prefix-final q-prefix-replay q-final-replay by simp
qed
from after-prefix obtain idxv-state s-idx where
  rand-v:
    Some (idxv-state, s-idx) ∈
      set-dist (execute p.receive-query-index-challenge s-prefix)
and after-rand:
  Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
    set-dist (execute
      ((let idxv' = p.index (to-nat idxv-state) in
        mmap (v.check-decommit-on-query
          (case out of (fr, as', dg, fl, final) ⇒ fr) idxv')
          (λfv. return (case out of (fr, as', dg, fl, final) ⇒

```

```

      (fr, as', dg, fl, final, idxv-state, fv)))))) s-idx)
proof -
  show ?thesis
    using after-prefix
  proof (rule p.set-dist-bindE)
    fix idxv-state s-idx
    assume rand-v:
      Some (idxv-state, s-idx) ∈
        set-dist (execute p.receive-query-index-challenge s-prefix)
    and after-rand:
      Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
        set-dist (execute
          ((let idxv' = p.index (to-nat idxv-state) in
            mmap (v.check-decommit-on-query
              (case out of (fr, as', dg, fl, final) ⇒ fr) idxv') >>=
              (λfv. return (case out of (fr, as', dg, fl, final) ⇒
                (fr, as', dg, fl, final, idxv-state, fv)))))) s-idx)
    show ?thesis
      by (rule that[OF rand-v after-rand])
  qed
qed
have rand-res:
  idxv-state = idx ∧
  PState s-idx = PState s-prefix ∧
  PTranscript s-idx = PTranscript s-prefix ∧
  s-prefix ≤ s-idx
  using p.receive-query-index-challenge-known-outcome[OF lookup-prefix rand-v]
  by simp
have s1-s-idx: s1 ≤ s-idx
  using s1-replay prefix-res rand-res by (meson p.hash-ext-trans)
have tr-idx:
  PTranscript s-idx =
    query-decommitment-transcript (p.index (to-nat idxv-state)) f-merkle @ rest
  using rand-res prefix-res idx-def by simp
have paths:
  ∧i. i ∈ set (p.powers-scaled (p.index (to-nat idxv-state))) ⇒
    length (get-authentication-path (length p.f-eval) i f-merkle) =
      floor-log (length p.f-eval)
  using query-authentication-path-length[OF create-f] by blast
from after-rand obtain fv0 t0 where
  query-out:
    Some (fv0, t0) ∈
      set-dist (execute
        (mmap (v.check-decommit-on-query
          (case out of (fr, as', dg, fl, final) ⇒ fr)
          (p.index (to-nat idxv-state)))) s-idx)
  and ret-out:
    Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
      set-dist (execute

```

```

      (return (case out of (fr, as', dg, fl, final) ⇒
        (fr, as', dg, fl, final, idxv-state, fv0))) t0)
    using rand-res by (auto simp: Let-def elim!: p.set-dist-bindE)
  have query-out-replay:
    Some (fv0, t0) ∈
      set-dist (execute
        (mmap (v.check-decommit-on-query
          (case out of (fr, as', dg, fl, final) ⇒ fr)
          (p.index (to-nat idxv-state))))
        (s-idx(
          PTranscript :=
            query-decommitment-transcript (p.index (to-nat idxv-state)) f-merkle @ rest
          )))
  proof -
    have s-idx-update:
      s-idx(
        PTranscript :=
          query-decommitment-transcript (p.index (to-nat idxv-state)) f-merkle @ rest
        ) = s-idx
      using tr-idx by simp
    show ?thesis
      using query-out unfolding s-idx-update .
  qed
  have query-res:
    fv0 = map (λi. p.f-eval ! i) (p.powers-scaled (p.index (to-nat idxv-state))) ∧
    PTranscript t0 = rest ∧
    PState t0 =
      foldl concat (PState s-idx)
        (query-decommitment-transcript (p.index (to-nat idxv-state)) f-merkle) ∧
    s-idx(
      PTranscript :=
        query-decommitment-transcript (p.index (to-nat idxv-state)) f-merkle @ rest
      ) ≤ t0
  by (rule honest-query-decommitment-replay-outcome[
    OF paths query-out-replay])
  have t-res:
    (fr, as', dg, fl, final, idxv, fv) =
      (fr0, as0, dg0, fl0, final0, idxv-state, fv0) ∧
    t = t0
    using ret-out unfolding out-eq by simp
  have idxv-eq: idxv = idx
    using t-res rand-res by simp
  have fv-eq: fv = map (λi. p.f-eval ! i) (p.powers-scaled idx')
    using t-res query-res rand-res idx-def by simp
  have replay-t: replay-state ≤ t
  proof -
    have s-idx ≤ t0
    proof -
      have s-idx-update:

```

```

    s-idx(
      PTranscript :=
        query-decommitment-transcript (p.index (to-nat idxv-state)) f-merkle @ rest
    ) = s-idx
    using tr-idx by simp
  show ?thesis
    using query-res unfolding s-idx-update by simp
qed
then show ?thesis
  using prefix-res rand-res t-res by (meson p.hash-ext-trans)
qed
show ?thesis
  by (intro conjI exI[where x=roots] exI[where x=challenges] exI[where x=rest])
    (use prefix-res len-roots len-challenges query-res t-res idxv-eq fv-eq replay-t
      replay-prefix roots-aligned successors in simp-all)
qed

end

end

```

```

theory Completeness-FRI
  imports Completeness-Query
begin

```

2.6 Honest FRI Checks

Honest FRI decommitment, fold, degree-halving, and final-value facts.

```

context verification
begin

```

```

definition fri-layer-decommitment-transcript
  where
    fri-layer-decommitment-transcript l m i len ≡
      (let idx = i mod len;
       side = (idx + len div 2) mod len
       in l ! idx # get-authentication-path len idx m @
        l ! side # get-authentication-path len side m)

```

```

fun fri-decommitment-transcript
  where
    fri-decommitment-transcript idx [] = []
  | fri-decommitment-transcript idx ((l, m) # fs) =
    fri-layer-decommitment-transcript l m idx (length l) @
    fri-decommitment-transcript (idx mod length l) fs

```

```

lemma fri-layer-decommitment-transcript-cong:
  assumes idx mod len = idx' mod len

```

```

shows
  fri-layer-decommitment-transcript l m idx len =
    fri-layer-decommitment-transcript l m idx' len
using assms
unfolding fri-layer-decommitment-transcript-def by simp

lemma fri-decommitment-transcript-cong-mod-head:
  assumes idx mod length l = idx' mod length l
  shows
    fri-decommitment-transcript idx ((l, m) # fs) =
      fri-decommitment-transcript idx' ((l, m) # fs)
proof –
  have layer-eq:
    fri-layer-decommitment-transcript l m idx (length l) =
      fri-layer-decommitment-transcript l m idx' (length l)
  by (rule fri-layer-decommitment-transcript-cong[OF assms])
  have tail-eq:
    fri-decommitment-transcript (idx mod length l) fs =
      fri-decommitment-transcript (idx' mod length l) fs
  using assms by simp
  show ?thesis
  by (simp only: fri-decommitment-transcript.simps layer-eq tail-eq)
qed

lemma mod-mod-half-power:
  fixes len idx :: nat
  assumes len = 2 ^ Suc n
  shows (idx mod len) mod (len div 2) = idx mod (len div 2)
proof –
  have half: len div 2 = 2 ^ n
  using assms by (simp add: power-Suc)
  have dvd-half: len div 2 dvd len
  using assms half by (simp add: power-Suc)
  show ?thesis
  using mod-mod-cancel[of len div 2 len idx] dvd-half by simp
qed

lemma fri-decommitment-transcript-half-index-tail:
  assumes len-pow: len = 2 ^ Suc n
  and next-len: length l = len div 2
  shows
    fri-decommitment-transcript (idx mod len) ((l, m) # fs) =
      fri-decommitment-transcript (idx mod (len div 2)) ((l, m) # fs)
proof –
  have idx mod len mod length l = idx mod (len div 2) mod length l
  using mod-mod-half-power[OF len-pow, of idx] next-len by simp
  then show ?thesis
  by (rule fri-decommitment-transcript-cong-mod-head)
qed

```

definition *fri-fold-value*

where

$$\begin{aligned} \text{fri-fold-value } b \ l \ i \ len \ pw \equiv & \\ & ((l ! i) + (l ! ((i + len \text{ div } 2) \text{ mod } len))) \text{ div } 2 + \\ & b * (((l ! i) - (l ! ((i + len \text{ div } 2) \text{ mod } len))) \text{ div } \\ & (2 * ((p.h \hat{i}) * shift) \hat{pw})) \end{aligned}$$

definition *fri-verifier-layer*

where

```
fri-verifier-layer b fr i x len pw ≡
do {
  xp ← p.read;
  xp' ← ntimes p.read (floor-log len);
  xn ← p.read;
  xn' ← ntimes p.read (floor-log len);
  assert (xp = x);
  ap ← p.check-authentication-path len i xp xp';
  assert (ap = fr);
  let sidx = (i + len div 2) mod len;
  ap ← p.check-authentication-path len sidx xn xn';
  assert (ap = fr);
  let gp = (xp + xn) div 2;
  let hp = (xp - xn) div (2 * ((p.h ^ i) * shift) ^ pw);
  let x = gp + b * hp;
  return (i mod (len div 2), x, len div 2, pw + pw)
}
```

lemma *receive-query-commits-step-eq:*

$m \in \text{set } (v.\text{receive-query-commits } fl) \implies$
 $\exists b \text{ fr. } (b, fr) \in \text{set } fl \wedge$
 $m = (\lambda(i, x, len, pw). \text{fri-verifier-layer } b \text{ fr } i \ x \ len \ pw)$

unfolding *v.receive-query-commits-def fri-verifier-layer-def*
by *auto*

lemma *receive-query-commits-Cons:*

$v.\text{receive-query-commits } ((b, fr) \# fl) =$
 $(\lambda(i, x, len, pw). \text{fri-verifier-layer } b \text{ fr } i \ x \ len \ pw) \#$
 $v.\text{receive-query-commits } fl$

unfolding *v.receive-query-commits-def fri-verifier-layer-def* **by** *simp*

lemma *receive-query-commits-nth-apply:*

assumes $j < \text{length } fl$

shows

$((v.\text{receive-query-commits } fl) ! j) (i, x, len, pw) =$
 $\text{fri-verifier-layer } (\text{fst } (fl ! j)) (\text{snd } (fl ! j)) i \ x \ len \ pw$

proof –

obtain $bb \ rr$ **where** $\text{pair: } fl ! j = (bb, rr)$

by $(\text{cases } fl ! j)$

have *nth-eq*:
 ((*v.receive-query-commits fl*) ! *j*) =
 (λ(*i, x, len, pw*). *fri-verifier-layer bb rr i x len pw*)
using *assms pair*
unfolding *v.receive-query-commits-def fri-verifier-layer-def*
by *simp*
show *?thesis*
by (*subst nth-eq, simp add: pair*)
qed

lemma *receive-query-commits-zip-nth-apply*:
assumes *j < length challenges*
and *j < length roots*
shows
 ((*v.receive-query-commits (zip challenges roots)*) ! *j*) (*i, x, len, pw*) =
fri-verifier-layer (challenges ! j) (roots ! j) i x len pw
using *assms receive-query-commits-nth-apply*[
of j zip challenges roots i x len pw]
by *simp*

lemma *butlast-zip-suc-lengths*:
assumes *length xs = Suc n*
and *length ys = Suc n*
shows *butlast (zip xs ys) = take n (zip xs ys)*
using *assms by (simp add: butlast-conv-take)*

lemma *butlast-zip-suc-nth*:
assumes *length xs = Suc n*
and *length ys = Suc n*
and *j < n*
shows *butlast (zip xs ys) ! j = (xs ! j, ys ! j)*
proof –
have *butlast (zip xs ys) ! j = zip xs ys ! j*
using *butlast-zip-suc-lengths[OF assms(1,2)] assms(3) by simp*
also have ... = (*xs ! j, ys ! j*)
using *assms by (simp add: nth-zip)*
finally show *?thesis* .
qed

lemma *butlast-zip-suc-drop-nth*:
assumes *length xs = Suc n*
and *length ys = Suc n*
and *j < n*
shows
drop j (butlast (zip xs ys)) =
(xs ! j, ys ! j) # drop (Suc j) (butlast (zip xs ys))
proof –
have *bl-len: length (butlast (zip xs ys)) = n*
using *butlast-zip-suc-lengths[OF assms(1,2)] assms(1,2) by simp*

```

have drop-eq:
  drop j (butlast (zip xs ys)) =
    butlast (zip xs ys) ! j # drop (Suc j) (butlast (zip xs ys))
  using Cons-nth-drop-Suc[OF assms(3)[folded bl-len], symmetric] .
show ?thesis
  using drop-eq butlast-zip-suc-nth[OF assms] by simp
qed

lemma fri-decommitment-transcript-drop-half-index:
  assumes len-ls: length ls = Suc n
  and len-ms: length ms = Suc n
  and j-bound: j < n
  and len-pow: length (ls ! j) = 2 ^ Suc N
  and next-len: length (ls ! Suc j) = length (ls ! j) div 2
  shows
    fri-decommitment-transcript (idx mod length (ls ! j))
      (drop (Suc j) (butlast (zip ls ms))) =
    fri-decommitment-transcript (idx mod (length (ls ! j) div 2))
      (drop (Suc j) (butlast (zip ls ms)))
proof (cases Suc j < n)
  case True
  have bl-len: length (butlast (zip ls ms)) = n
    using butlast-zip-suc-lengths[OF len-ls len-ms] len-ls len-ms by simp
  have drop-eq:
    drop (Suc j) (butlast (zip ls ms)) =
      butlast (zip ls ms) ! Suc j #
        drop (Suc (Suc j)) (butlast (zip ls ms))
    using Cons-nth-drop-Suc[OF True[folded bl-len], symmetric] .
  have nth-eq:
    butlast (zip ls ms) ! Suc j = (ls ! Suc j, ms ! Suc j)
    by (rule butlast-zip-suc-nth[OF len-ls len-ms True])
  show ?thesis
    unfolding drop-eq nth-eq
    by (rule fri-decommitment-transcript-half-index-tail[
      OF len-pow next-len])
next
  case False
  have suc-eq: Suc j = n
    using False j-bound by simp
  have bl-len: length (butlast (zip ls ms)) = n
    using butlast-zip-suc-lengths[OF len-ls len-ms] len-ls len-ms by simp
  then show ?thesis
    using suc-eq by simp
qed

lemma decommit-on-fri-layer-transcript:
  assumes outcome:
    Some (x, t) ∈
      set-dist (execute

```



```

    (do {
      let len = length l;
      let idx' = idx mod len;
      let sidx = (idx' + (len div 2)) mod len;
      p.send (l ! idx');
      mfold2 p.send (get-authentication-path len idx' m);
      p.send (l ! sidx);
      mfold2 p.send (get-authentication-path len sidx m);
      return idx'
    }) s)
shows
  x = idx mod length l ∧
  PTranscript t =
    rev (fri-layer-decommitment-transcript l m idx (length l)) @ PTranscript s
proof -
  let ?len = length l
  let ?idx = idx mod ?len
  let ?sidx = (?idx + (?len div 2)) mod ?len
  from outcome[unfolded Let-def] obtain u1 u2 u3 u4 where
    send-xp: Some ((), u1) ∈ set-dist (execute (p.send (l ! ?idx)) s)
  and path-xp:
    Some ((), u2) ∈
      set-dist (execute (mfold2 p.send (get-authentication-path ?len ?idx m)) u1)
  and send-xn: Some ((), u3) ∈ set-dist (execute (p.send (l ! ?sidx)) u2)
  and path-xn:
    Some ((), u4) ∈
      set-dist (execute (mfold2 p.send (get-authentication-path ?len ?sidx m)) u3)
  and ret: Some (x, t) ∈ set-dist (execute (return ?idx) u4)
  by (auto elim!: p.set-dist-bindE)
  have x-eq: x = ?idx
    using ret by simp
  have t-eq: t = u4
    using ret by simp
  have tr-u1: PTranscript u1 = l ! ?idx # PTranscript s
    using p.send-outcome[OF send-xp] by simp
  have tr-u2:
    PTranscript u2 =
      rev (get-authentication-path ?len ?idx m) @ PTranscript u1
    using p.mfold2-send-outcome[OF path-xp] by simp
  have tr-u3: PTranscript u3 = l ! ?sidx # PTranscript u2
    using p.send-outcome[OF send-xn] by simp
  have tr-u4:
    PTranscript u4 =
      rev (get-authentication-path ?len ?sidx m) @ PTranscript u3
    using p.mfold2-send-outcome[OF path-xn] by simp
  have tr:
    PTranscript t =
      rev (fri-layer-decommitment-transcript l m idx ?len) @ PTranscript s
    using tr-u1 tr-u2 tr-u3 tr-u4 t-eq

```

```

unfolding fri-layer-decommitment-transcript-def
by (simp add: Let-def)
show ?thesis
using x-eq tr by simp
qed

```

lemma *decommit-on-fri-layer-state*:

assumes *outcome*:

```

Some (x, t) ∈
  set-dist (execute
    (do {
      let len = length l;
      let idx' = idx mod len;
      let sidx = (idx' + (len div 2)) mod len;
      p.send (l ! idx');
      mfold2 p.send (get-authentication-path len idx' m);
      p.send (l ! sidx);
      mfold2 p.send (get-authentication-path len sidx m);
      return idx'
    }) s)

```

shows

```

PState t =
  foldl concat (PState s)
    (fri-layer-decommitment-transcript l m idx (length l))

```

proof –

```

let ?len = length l
let ?idx = idx mod ?len
let ?sidx = (?idx + (?len div 2)) mod ?len
from outcome[unfolded Let-def] obtain u1 u2 u3 u4 where
  send-xp: Some ((), u1) ∈ set-dist (execute (p.send (l ! ?idx)) s)
and path-xp:
  Some ((), u2) ∈
    set-dist (execute (mfold2 p.send (get-authentication-path ?len ?idx m)) u1)
and send-xn: Some ((), u3) ∈ set-dist (execute (p.send (l ! ?sidx)) u2)
and path-xn:
  Some ((), u4) ∈
    set-dist (execute (mfold2 p.send (get-authentication-path ?len ?sidx m)) u3)
and ret: Some (x, t) ∈ set-dist (execute (return ?idx) u4)
by (auto elim!: p.set-dist-bindE)
have t-eq: t = u4
using ret by simp
have st-u1: PState u1 = concat (PState s) (l ! ?idx)
using p.send-outcome[OF send-xp] by simp
have st-u2:
  PState u2 =
    foldl concat (concat (PState s) (l ! ?idx))
      (get-authentication-path ?len ?idx m)
using p.mfold2-send-outcome[OF path-xp] st-u1 by simp
have st-u3: PState u3 = concat (PState u2) (l ! ?sidx)

```

```

using p.send-outcome[OF send-xn] by simp
have st-u4:
  PState u4 =
    foldl concat (concat (PState u2) (l ! ?sidx))
      (get-authentication-path ?len ?sidx m)
  using p.mfold2-send-outcome[OF path-xn] st-u3 by simp
show ?thesis
  using st-u2 st-u4 t-eq
  unfolding fri-layer-decommitment-transcript-def
  by (simp add: Let-def)
qed

```

```

lemma decommit-on-fri-layers-mfold-transcript:
assumes outcome:
  Some (idx', t) ∈
    set-dist (execute
      (mfold idx
        (p.decommit-on-fri-layers fs)) s)
shows
  PTranscript t =
    rev (fri-decommitment-transcript idx fs) @ PTranscript s
using outcome
proof (induction fs arbitrary: idx idx' s t)
case Nil
  then have t = s
    unfolding p.decommit-on-fri-layers-def by simp
  then show ?case
    by simp
next
case (Cons pair fs)
obtain l m where pair-eq: pair = (l, m)
  by (cases pair)
from Cons.prems obtain idx1 u where
  head:
    Some (idx1, u) ∈
      set-dist (execute
        (( $\lambda(l, m) \text{ idx. do } \{$ 
          let len = length l;
          let idx' = idx mod len;
          let sidx = (idx' + (len div 2)) mod len;
          p.send (l ! idx');
          mfold2 p.send (get-authentication-path len idx' m);
          p.send (l ! sidx);
          mfold2 p.send (get-authentication-path len sidx m);
          return idx'
        }) pair idx) s)
  and tail:
    Some (idx', t) ∈
      set-dist (execute

```

```

      (mfold idx1 (p.decommit-on-fri-layers fs)) u)
unfolding p.decommit-on-fri-layers-def
by (auto elim!: p.set-dist-bindE)
have head':
  Some (idx1, u) ∈
    set-dist (execute
      (do {
        let len = length l;
        let idx' = idx mod len;
        let sidx = (idx' + (len div 2)) mod len;
        p.send (l ! idx');
        mfold2 p.send (get-authentication-path len idx' m);
        p.send (l ! sidx);
        mfold2 p.send (get-authentication-path len sidx m);
        return idx'
      }) s)
using head unfolding pair-eq by simp
have head-res:
  idx1 = idx mod length l ∧
  PTranscript u =
    rev (fri-layer-decommitment-transcript l m idx (length l)) @ PTranscript s
by (rule decommit-on-fri-layer-transcript[OF head'])
have tail-tr:
  PTranscript t =
    rev (fri-decommitment-transcript idx1 fs) @ PTranscript u
using Cons.IH[OF tail] .
show ?case
using head-res tail-tr unfolding pair-eq by simp
qed

```

lemma decommit-on-fri-layers-mfold-state:

```

assumes outcome:
  Some (idx', t) ∈
    set-dist (execute
      (mfold idx
        (p.decommit-on-fri-layers fs)) s)
shows
  PState t =
    foldl concat (PState s) (fri-decommitment-transcript idx fs)
using outcome
proof (induction fs arbitrary: idx idx' s t)
case Nil
then show ?case
unfolding p.decommit-on-fri-layers-def by simp
next
case (Cons pair fs)
obtain l m where pair-eq: pair = (l, m)
by (cases pair)
from Cons.premis obtain idx1 u where

```

```

head:
  Some (idx1, u) ∈
    set-dist (execute
      ((λ(l, m) idx. do {
        let len = length l;
        let idx' = idx mod len;
        let sidx = (idx' + (len div 2)) mod len;
        p.send (l ! idx');
        mfold2 p.send (get-authentication-path len idx' m);
        p.send (l ! sidx);
        mfold2 p.send (get-authentication-path len sidx m);
        return idx'
      }) pair idx) s)
and tail:
  Some (idx', t) ∈
    set-dist (execute
      (mfold idx1 (p.decommit-on-fri-layers fs)) u)
unfolding p.decommit-on-fri-layers-def
by (auto elim!: p.set-dist-bindE)
have head':
  Some (idx1, u) ∈
    set-dist (execute
      (do {
        let len = length l;
        let idx' = idx mod len;
        let sidx = (idx' + (len div 2)) mod len;
        p.send (l ! idx');
        mfold2 p.send (get-authentication-path len idx' m);
        p.send (l ! sidx);
        mfold2 p.send (get-authentication-path len sidx m);
        return idx'
      }) s)
using head unfolding pair-eq by simp
have idx1-eq: idx1 = idx mod length l
using decommit-on-fri-layer-transcript[OF head'] by simp
have head-state:
  PState u =
    foldl concat (PState s)
      (fri-layer-decommitment-transcript l m idx (length l))
by (rule decommit-on-fri-layer-state[OF head'])
have tail-state:
  PState t =
    foldl concat (PState u) (fri-decommitment-transcript idx1 fs)
using Cons.IH[OF tail] .
show ?case
using idx1-eq head-state tail-state unfolding pair-eq by simp
qed

```

lemma prover-query-round-transcript:

```

assumes outcome:
  Some (x, t) ∈ set-dist (execute (prover-query-round f-merkle f-ls f-ms ls ms) s)
obtains idx where
  PTranscript t =
    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms))) @
    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms))) @
    rev (query-decommitment-transcript (p.index (to-nat idx)) f-merkle) @
    PTranscript s
proof (rule prover-query-round-outcomeE[OF outcome])
fix idx s1 qouts s2 f-fri-idx s3 fri-idx
assume
  rand: Some (idx, s1) ∈ set-dist (execute p.receive-query-index-challenge s)
and query:
  Some (qouts, s2) ∈
    set-dist (execute
      (mmap (p.decommit-on-query (p.index (to-nat idx)) f-merkle)) s1)
and trace-fri:
  Some (f-fri-idx, s3) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip f-ls f-ms)))) s2)
and fri:
  Some (fri-idx, t) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip ls ms)))) s3)
and unit: x = ()
have tr-s1: PTranscript s1 = PTranscript s
using receive-query-index-challenge-preserves-transcript[OF rand] .
have tr-s2:
  PTranscript s2 =
    rev (query-decommitment-transcript (p.index (to-nat idx)) f-merkle) @
    PTranscript s1
using decommit-on-query-mmap-transcript[OF query] .
have tr-s3:
  PTranscript s3 =
    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms))) @
    PTranscript s2
using decommit-on-fri-layers-mfold-transcript[OF trace-fri] .
have tr-t:
  PTranscript t =
    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms))) @
    PTranscript s3
using decommit-on-fri-layers-mfold-transcript[OF fri] .
show ?thesis
by (rule that[of idx]) (use tr-s1 tr-s2 tr-s3 tr-t in simp)
qed

```

lemma prover-query-round-lookup-transcript:

```

assumes outcome:
  Some (x, t) ∈ set-dist (execute (prover-query-round f-merkle f-ls f-ms ls ms) s)
obtains idx where
  x = ()
  fmlookup (HashMap t) (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some idx
  PTranscript t =
    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms))) @
    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms))) @
    rev (query-decommitment-transcript (p.index (to-nat idx)) f-merkle) @
    PTranscript s
  s ≤ t
proof (rule prover-query-round-outcomeE[OF outcome])
fix idx s1 qouts s2 f-fri-idx s3 fri-idx
assume rand: Some (idx, s1) ∈ set-dist (execute p.receive-query-index-challenge s)
and query:
  Some (qouts, s2) ∈
    set-dist (execute
      (mmap (p.decommit-on-query (p.index (to-nat idx)) f-merkle)) s1)
and trace-fri:
  Some (f-fri-idx, s3) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip f-ls f-ms)))) s2)
and fri:
  Some (fri-idx, t) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip ls ms)))) s3)
and x-unit: x = ()
have rand-res:
  s ≤ s1 ∧
  PState s1 = PState s ∧
  PTranscript s1 = PTranscript s ∧
  fmlookup (HashMap s1) (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some idx
using p.receive-query-index-challenge-outcome[OF rand] by simp
have query-ext: s1 ≤ s2
by (rule mmap-hash-extends[OF query])
  (rule decommit-on-query-step-hash-extends)
have trace-fri-ext: s2 ≤ s3
by (rule mfold-hash-extends[OF trace-fri])
  (rule decommit-on-fri-layers-step-hash-extends)
have fri-ext: s3 ≤ t
by (rule mfold-hash-extends[OF fri])
  (rule decommit-on-fri-layers-step-hash-extends)
have s-t: s ≤ t
using rand-res query-ext trace-fri-ext fri-ext by (meson p.hash-ext-trans)
have s1-t: s1 ≤ t
using query-ext trace-fri-ext fri-ext by (meson p.hash-ext-trans)
have lookup-s1: fmlookup (HashMap s1) (QueryIndexChallenge (PQueryCounter s) (PState

```

```

s)) = Some idx
  using rand-res by simp
  have lookup-t: fmllookup (HashMap t) (QueryIndexChallenge (PQueryCounter s) (PState s))
= Some idx
  by (rule p.hash-extension-lookup[OF lookup-s1 s1-t])
  have tr-s2:
    PTranscript s2 =
      rev (query-decommitment-transcript (p.index (to-nat idx)) f-merkle) @
        PTranscript s1
  by (rule decommit-on-query-mmap-transcript[OF query])
  have tr-s3:
    PTranscript s3 =
      rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms))) @
        PTranscript s2
  by (rule decommit-on-fri-layers-mfold-transcript[OF trace-fri])
  have tr-t:
    PTranscript t =
      rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms))) @
        PTranscript s3
  by (rule decommit-on-fri-layers-mfold-transcript[OF fri])
  show ?thesis
  by (rule that[OF x-unit lookup-t])
    (use rand-res tr-s2 tr-s3 tr-t s-t in simp-all)
qed

```

lemma prover-query-round-lookup-state-transcript:

assumes outcome:

Some (x, t) ∈ set-dist (execute (prover-query-round f-merkle f-ls f-ms ls ms) s)

obtains idx **where**

x = ()

fmllookup (HashMap t) (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some idx

PState t =

foldl concat (PState s)

(query-decommitment-transcript (p.index (to-nat idx)) f-merkle @

fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms))) @

fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms)))

PTranscript t =

rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms))) @

rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms))) @

rev (query-decommitment-transcript (p.index (to-nat idx)) f-merkle) @

PTranscript s

s ≤ t

proof (rule prover-query-round-outcomeE[OF outcome])

fix idx s1 qouts s2 f-fri-idx s3 fri-idx

assume rand: *Some (idx, s1) ∈ set-dist (execute p.receive-query-index-challenge s)*

and query:

Some (qouts, s2) ∈

set-dist (execute

(mmap (p.decommit-on-query (p.index (to-nat idx)) f-merkle)) s1)


```

and trace-fri:
  Some (f-fri-idx, s3) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip f-ls f-ms)))) s2)
and fri:
  Some (fri-idx, t) ∈
    set-dist (execute
      (mfold (p.index (to-nat idx))
        (p.decommit-on-fri-layers (butlast (zip ls ms)))) s3)
and x-unit: x = ()
have rand-res:
  s ≤ s1 ∧
    PState s1 = PState s ∧
    PTranscript s1 = PTranscript s ∧
    fmlookup (HashMap s1) (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some idx
using p.receive-query-index-challenge-outcome[OF rand] by simp
have query-ext: s1 ≤ s2
  by (rule mmap-hash-extends[OF query])
    (rule decommit-on-query-step-hash-extends)
have trace-fri-ext: s2 ≤ s3
  by (rule mfold-hash-extends[OF trace-fri])
    (rule decommit-on-fri-layers-step-hash-extends)
have fri-ext: s3 ≤ t
  by (rule mfold-hash-extends[OF fri])
    (rule decommit-on-fri-layers-step-hash-extends)
have s-t: s ≤ t
  using rand-res query-ext trace-fri-ext fri-ext by (meson p.hash-ext-trans)
have s1-t: s1 ≤ t
  using query-ext trace-fri-ext fri-ext by (meson p.hash-ext-trans)
have lookup-s1: fmlookup (HashMap s1) (QueryIndexChallenge (PQueryCounter s) (PState
s)) = Some idx
  using rand-res by simp
have lookup-t: fmlookup (HashMap t) (QueryIndexChallenge (PQueryCounter s) (PState s))
= Some idx
  by (rule p.hash-extension-lookup[OF lookup-s1 s1-t])
have query-state:
  PState s2 =
    foldl concat (PState s)
      (query-decommitment-transcript (p.index (to-nat idx)) f-merkle)
  using decommit-on-query-mmap-state[OF query] rand-res by simp
have trace-fri-state:
  PState s3 =
    foldl concat (PState s2)
      (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms)))
  using decommit-on-fri-layers-mfold-state[OF trace-fri] .
have fri-state:
  PState t =
    foldl concat (PState s3)

```

```

      (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms)))
using decommit-on-fri-layers-mfold-state[OF fri] .
have state-t:
  PState t =
    foldl concat (PState s)
      (query-decommitment-transcript (p.index (to-nat idx)) f-merkle @
        fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms)) @
        fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms)))
    using query-state trace-fri-state fri-state by simp
have tr-s2:
  PTranscript s2 =
    rev (query-decommitment-transcript (p.index (to-nat idx)) f-merkle) @
    PTranscript s1
    by (rule decommit-on-query-mmap-transcript[OF query])
have tr-s3:
  PTranscript s3 =
    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms))) @
    PTranscript s2
    by (rule decommit-on-fri-layers-mfold-transcript[OF trace-fri])
have tr-t:
  PTranscript t =
    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms))) @
    PTranscript s3
    by (rule decommit-on-fri-layers-mfold-transcript[OF fri])
show ?thesis
    by (rule that[OF x-unit lookup-t state-t])
      (use rand-res tr-s2 tr-s3 tr-t s-t in simp-all)
qed

lemma prover-fri-decommitment-transcript-in-replay-state:
assumes fri-decommit:
  Some (fri-idx, s9) ∈
    set-dist (execute
      (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
and prover-state-eq: prover-state = s9
and replay: replay-state = verifier-replay-state prover-state
shows
  PTranscript replay-state =
    rev (PTranscript s8) @
    fri-decommitment-transcript idx' (butlast (zip ls ms))
proof –
have s9-tr:
  PTranscript s9 =
    rev (fri-decommitment-transcript idx' (butlast (zip ls ms))) @
    PTranscript s8
    by (rule decommit-on-fri-layers-mfold-transcript[OF fri-decommit])
show ?thesis
    using s9-tr prover-state-eq replay
    unfolding verifier-replay-state-def by simp

```

qed

lemma *prover-query-fri-decommitment-transcript-in-replay-state:*

assumes *final-send:*

Some ($()$, s -final) $\in$ set-dist (execute (p .send (hd (last ls))) $s6$)

and *random-idx:*

Some (idx , $s7$) $\in$ set-dist (execute p .receive-query-index-challenge s -final)

and *query-decommit:*

Some ($qouts$, $s8$) $\in$

set-dist (execute (mmap (p .decommit-on-query idx' f -merkle))) $s7$)

and *fri-decommit:*

Some (fri -idx, $s9$) $\in$

set-dist (execute (mfold idx' (p .decommit-on-fri-layers (butlast (zip ls ms)))) $s8$)

and *prover-state-eq:* $prover$ -state = $s9$

and *replay:* $replay$ -state = $verifier$ -replay-state $prover$ -state

shows

$PTranscript$ $replay$ -state =

rev ($PTranscript$ $s6$) @ [hd (last ls)] @

$query$ -decommitment-transcript idx' f -merkle @

fri -decommitment-transcript idx' (butlast (zip ls ms))

proof –

have *final-tr:*

$PTranscript$ s -final = hd (last ls) # $PTranscript$ $s6$

using p .send-outcome[OF *final-send*] **by** *simp*

have *s7-tr:* $PTranscript$ $s7$ = $PTranscript$ s -final

using *receive-query-index-challenge-preserves-transcript*[OF *random-idx*] .

have *s8-tr:*

$PTranscript$ $s8$ =

rev ($query$ -decommitment-transcript idx' f -merkle) @ $PTranscript$ $s7$

using *decommit-on-query-mmap-transcript*[OF *query-decommit*] .

have *replay-tr:*

$PTranscript$ $replay$ -state =

rev ($PTranscript$ $s8$) @

fri -decommitment-transcript idx' (butlast (zip ls ms))

by (rule *prover-fri-decommitment-transcript-in-replay-state*[
OF *fri-decommit prover-state-eq replay*])

show *?thesis*

using *replay-tr s8-tr s7-tr final-tr* **by** *simp*

qed

lemma *honest-query-prefix-initial-fri-value:*

assumes *htv:* *honest-trace-valid*

and *create-f:*

Some (f -merkle, $s1$) $\in$

set-dist (execute (p .create p .f-eval) $init$ -state)

and *send-f:*

Some ($()$, $s2$) $\in$ set-dist (execute (p .send (value f -merkle)) $s1$)

and *alphas:*

```

Some (as, s3) ∈
  set-dist (execute
    (mmap (replicate (length spec)
      (do {
        a ← p.receive-alpha-challenge;
        p.send a;
        return a
      }))) s2)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
and create-cp:
  Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and fri:
  Some ((ps, ds, ls, ms), s6) ∈
    set-dist (execute
      (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and final-send:
  Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and random-idx:
  Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
and idx-def: idx' = p.index (to-nat idx)
and query-decommit:
  Some (qouts, s8) ∈
    set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
and fri-decommit:
  Some (fri-idx, s9) ∈
    set-dist (execute
      (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
and prover-state-eq: prover-state = s9
and replay: replay-state = verifier-replay-state prover-state
and outcome:
  Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
    set-dist (execute
      (do {
        fr ← p.read;
        as' ← mmap (replicate (length spec)
          (do {
            a0 ← p.receive-alpha-challenge;
            let a0' = a0;
            a1 ← p.read;
            let a1' = a1;
            assert (a0' = a1');
            return a1'
          }));
        dg ← p.read;
        assert (to-nat dg ≤ p.maxDegree);
        fl ← ntimes v.receive-fri-commits nrounds;
        final ← p.read;

```

```

    idxv ← p.receive-query-index-challenge;
    let idxv' = p.index (to-nat idxv);
    fv ← mmap (v.check-decommit-on-query fr idxv');
    return (fr, as', dg, fl, final, idxv, fv)
  }) replay-state)
shows
  v.cp-eval as' fv (p.h ^ idx' * shift) =
    poly (p.cp as p.f-powers) (p.h ^ idx' * shift)
proof –
  have replay-values:
    as' = as ∧
    idxv = idx ∧
    fv = map (λi. p.f-eval ! i) (p.powers-scaled idx')
  using honest-root-alpha-degree-fri-final-query-from-prover-outcome[
    OF htv create-f send-f alphas cp'-def send-degree create-cp fri
    final-send random-idx idx-def query-decommit fri-decommit
    prover-state-eq replay outcome]
  by blast
  have cp:
    v.cp-eval as (map (λi. p.f-eval ! i) (p.powers-scaled (p.index (to-nat idx))))
      (p.h ^ p.index (to-nat idx) * shift) =
      poly (p.cp as p.f-powers) (p.h ^ p.index (to-nat idx) * shift)
  by (rule honest-query-cp-value[OF htv])
  show ?thesis
  using replay-values idx-def cp by simp
qed

lemma honest-query-prefix-leaves-fri-decommitment-transcript:
assumes htv: honest-trace-valid
and create-f:
  Some (f-merkle, s1) ∈
    set-dist (execute (p.create p.f-eval) init-state)
and send-f:
  Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and alphas:
  Some (as, s3) ∈
    set-dist (execute
      (mmap (replicate (length spec)
        (do {
          a ← p.receive-alpha-challenge;
          p.send a;
          return a
        }))) s2)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
and create-cp:
  Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and fri:

```

```

    Some ((ps, ds, ls, ms), s6) ∈
      set-dist (execute
        (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
  and final-send:
    Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
  and random-idx:
    Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
  and idx-def: idx' = p.index (to-nat idx)
  and query-decommit:
    Some (qouts, s8) ∈
      set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
  and fri-decommit:
    Some (fri-idx, s9) ∈
      set-dist (execute
        (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
  and prover-state-eq: prover-state = s9
  and replay: replay-state = verifier-replay-state prover-state
  and outcome:
    Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
      set-dist (execute
        (do {
          fr ← p.read;
          as' ← mmap (replicate (length spec)
            (do {
              a0 ← p.receive-alpha-challenge;
              let a0' = a0;
              a1 ← p.read;
              let a1' = a1;
              assert (a0' = a1');
              return a1'
            })));
          dg ← p.read;
          assert (to-nat dg ≤ p.maxDegree);
          fl ← ntimes v.receive-fri-commits nrounds;
          final ← p.read;
          idxv ← p.receive-query-index-challenge;
          let idxv' = p.index (to-nat idxv);
          fv ← mmap (v.check-decommit-on-query fr idxv');
          return (fr, as', dg, fl, final, idxv, fv)
        }) replay-state)
  shows PTranscript t = fri-decommitment-transcript idx' (butlast (zip ls ms))
proof -
  from honest-root-alpha-degree-fri-final-query-from-prover-outcome[
    OF htv create-f send-f alphas cp'-def send-degree create-cp fri
      final-send random-idx idx-def query-decommit fri-decommit
      prover-state-eq replay outcome]
  obtain roots challenges rest where
    len-roots: length roots = nrounds
  and replay-prefix:

```

```

    PTranscript replay-state =
      value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
      roots @ [hd (last ls)] @
      query-decommitment-transcript idx' f-merkle @ rest
  and t-tr: PTranscript t = rest
  by blast
have root-alpha-tr:
  length as = length spec ∧
  PTranscript s4 = of-nat (degree cp') # rev as @ [value f-merkle]
  by (rule prover-root-alpha-degree-transcript[
    OF create-f send-f alphas cp'-def send-degree])
from fri-commit-replay-data[OF fri] obtain roots0 challenges0 where
  len-roots0: length roots0 = nrounds
  and tr-s6: PTranscript s6 = rev roots0 @ PTranscript s5
  by blast
have tr-s5: PTranscript s5 = PTranscript s4
  using create-preserves-transcript[OF create-cp] .
have replay-fri-suffix:
  PTranscript replay-state =
    value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))] @
    roots0 @ [hd (last ls)] @
    query-decommitment-transcript idx' f-merkle @
    fri-decommitment-transcript idx' (butlast (zip ls ms))
  using prover-query-fri-decommitment-transcript-in-replay-state[
    OF final-send random-idx query-decommit fri-decommit prover-state-eq replay]
    tr-s6 tr-s5 root-alpha-tr cp'-def
  by simp
let ?A = value f-merkle # as @ [of-nat (degree (p.cp as p.f-powers))]
let ?B = [hd (last ls)] @ query-decommitment-transcript idx' f-merkle
have eq:
  ?A @ roots @ ?B @ rest =
  ?A @ roots0 @ ?B @
  fri-decommitment-transcript idx' (butlast (zip ls ms))
  using replay-prefix replay-fri-suffix by simp
have after-A:
  roots @ ?B @ rest =
  roots0 @ ?B @ fri-decommitment-transcript idx' (butlast (zip ls ms))
  using arg-cong[OF eq, of λxs. drop (length ?A) xs] by simp
have after-roots:
  ?B @ rest =
  ?B @ fri-decommitment-transcript idx' (butlast (zip ls ms))
  using arg-cong[OF after-A, of λxs. drop nrounds xs]
  len-roots len-roots0 by simp
have rest = fri-decommitment-transcript idx' (butlast (zip ls ms))
  using arg-cong[OF after-roots, of λxs. drop (length ?B) xs] by simp
then show ?thesis
  using t-tr by simp
qed

```

```

lemma created-tree-get-authentication-path-len:
  assumes created:  $p.created-tree\ xs\ r\ s0$ 
    and len:  $length\ xs = 2^n$ 
    and i-bound:  $i < length\ xs$ 
  shows  $length\ (get-authentication-path\ (length\ xs)\ i\ r) = floor-log\ (length\ xs)$ 
  using created len i-bound
proof (induction n arbitrary: xs r i)
  case 0
  then obtain  $x$  where  $xs-eq: xs = [x]$ 
    by (cases xs) auto
  then obtain  $h$  where  $r-eq: r = \langle \rangle, h, \langle \rangle$ 
    using 0.premis(1)
    by (auto simp: p.created-tree-def protocol-created-tree-def
      protocol-merkle.created-tree.simps)
  show ?case
    unfolding  $xs-eq\ r-eq$  by (simp add: floor-log-Suc-zero)
next
  case (Suc n)
  let ? $m = 2^n$ 
  let ? $lxs = take\ ?m\ xs$ 
  let ? $rxs = drop\ ?m\ xs$ 
  have  $len-xs: length\ xs = 2 * ?m$ 
    using Suc.premis(2) by simp
  have  $mid: length\ xs\ div\ 2 = ?m$ 
    using  $len-xs$  by simp
  obtain  $a\ b\ cs$  where  $xs-cons: xs = a \# b \# cs$ 
    using  $len-xs$  by (cases xs; cases tl xs) auto
  have  $m-eq: ?m = Suc\ (length\ cs\ div\ 2)$ 
    using  $mid$  unfolding  $xs-cons$  by simp
  have  $take-eq: ?lxs = a \# take\ (length\ cs\ div\ 2)\ (b \# cs)$ 
    unfolding  $xs-cons\ m-eq$  by simp
  have  $drop-eq: ?rxs = drop\ (length\ cs\ div\ 2)\ (b \# cs)$ 
    unfolding  $xs-cons\ m-eq$  by simp
  have created-raw:
     $protocol-merkle.created-tree\ (map\ MerkleLeaf\ xs)\ r\ s0$ 
    using Suc.premis(1)
    unfolding  $p.created-tree-def\ protocol-created-tree-def$  .
  have raw-left-eq:
     $MerkleLeaf\ a \#$ 
       $take\ (length\ cs\ div\ 2)\ (MerkleLeaf\ b \# map\ MerkleLeaf\ cs) =$ 
       $map\ MerkleLeaf\ (a \# take\ (length\ cs\ div\ 2)\ (b \# cs))$ 
    by (metis list.simps(9) take-map)
  have raw-right-eq:
     $drop\ (length\ cs\ div\ 2)\ (MerkleLeaf\ b \# map\ MerkleLeaf\ cs) =$ 
     $map\ MerkleLeaf\ (drop\ (length\ cs\ div\ 2)\ (b \# cs))$ 
    by (metis list.simps(9) drop-map)
  from created-raw obtain  $l\ h\ rr$  where
     $r-eq: r = \langle l, h, rr \rangle$ 
    and l-created-raw0:

```



```

  protocol-merkle.created-tree
    (MerkleLeaf a #
      take (length cs div 2) (MerkleLeaf b # map MerkleLeaf cs)) l s0
and r-created-raw0:
  protocol-merkle.created-tree
    (drop (length cs div 2) (MerkleLeaf b # map MerkleLeaf cs)) rr s0
unfolding xs-cons
by (auto simp: protocol-merkle.created-tree.simps Let-def)
have l-created-raw:
  protocol-merkle.created-tree
    (map MerkleLeaf (a # take (length cs div 2) (b # cs))) l s0
using l-created-raw0 unfolding raw-left-eq .
have r-created-raw:
  protocol-merkle.created-tree
    (map MerkleLeaf (drop (length cs div 2) (b # cs))) rr s0
using r-created-raw0 unfolding raw-right-eq .
have l-created': p.created-tree (a # take (length cs div 2) (b # cs)) l s0
using l-created-raw
unfolding p.created-tree-def protocol-created-tree-def .
have r-created': p.created-tree (drop (length cs div 2) (b # cs)) rr s0
using r-created-raw
unfolding p.created-tree-def protocol-created-tree-def .
have l-created: p.created-tree ?lxs l s0
using l-created' unfolding take-eq .
have r-created: p.created-tree ?rxs rr s0
using r-created' unfolding drop-eq .
have len-l: length ?lxs = 2 ^ n
using len-xs by simp
have len-r: length ?rxs = 2 ^ n
using len-xs by simp
have lgth-ge: 2 ≤ length xs
using len-xs by simp
have floor-eq: floor-log (length xs) = Suc (floor-log (length xs div 2))
using floor-log-rec[OF lgth-ge] .
show ?case
proof (cases i < ?m)
  case True
    have i-l: i < length ?lxs
      using True len-l by simp
    have left-len:
      length (get-authentication-path (length ?lxs) i l) =
        floor-log (length ?lxs)
      by (rule Suc.IH[OF l-created len-l i-l])
    have left-len':
      length (get-authentication-path ?m i l) = floor-log ?m
      using left-len len-l by simp
    have length (get-authentication-path (length xs) i r) =
      Suc (floor-log ?m)
      using True left-len' unfolding r-eq

```

```

    by (simp add: mid len-xs)
  then show ?thesis
    using floor-eq mid len-l by simp
next
case False
have i-ge: ?m ≤ i
  using False by simp
have i-r: i - ?m < length ?rxs
  using Suc.premis(3) len-xs i-ge by simp
have right-len:
  length (get-authentication-path (length ?rxs) (i - ?m) rr) =
    floor-log (length ?rxs)
  by (rule Suc.IH[OF r-created len-r i-r])
have right-len':
  length (get-authentication-path ?m (i - ?m) rr) = floor-log ?m
  using right-len len-r by simp
have length (get-authentication-path (length xs) i r) =
  Suc (floor-log ?m)
  using False right-len' unfolding r-eq
  by (simp add: mid len-xs)
then show ?thesis
  using floor-eq mid len-r by simp
qed
qed

lemma honest-fri-layer-decommitment-no-failure:
  assumes created: p.created-tree l m s0
  and len-eq: len = length l
  and len-pow: length l = 2 ^ n
  and i-bound: i < len
  and x-eq: x = l ! i
  and ext: s0 ≤ s
  and path-i:
    length (get-authentication-path len i m) = floor-log len
  and path-s:
    length (get-authentication-path len ((i + len div 2) mod len) m) =
      floor-log len
  and root: fr = value m
shows
  None ∉ dom (dist (execute
    (do {
      xp ← p.read;
      xp' ← ntimes p.read (floor-log len);
      xn ← p.read;
      xn' ← ntimes p.read (floor-log len);
      assert (xp = x);
      ap ← p.check-authentication-path len i xp xp';
      assert (ap = fr);
      let sidx = (i + len div 2) mod len;

```

```

    ap ← p.check-authentication-path len sidx xn xn';
    assert (ap = fr);
    let gp = (xp + xn) div 2;
    let hp = (xp - xn) div (2 * ((p.h^i) * shift)^pw);
    let x = gp + b * hp;
    return (i mod (len div 2), x, len div 2, pw + pw)
  })
  (s(PTranscript := fri-layer-decommitment-transcript l m i len @ rest))))
proof –
  let ?sidx = (i + len div 2) mod len
  let ?path-i = get-authentication-path len i m
  let ?path-s = get-authentication-path len ?sidx m
  let ?prefix =
    do {
      xp ← p.read;
      xp' ← ntimes p.read (floor-log len);
      xn ← p.read;
      xn' ← ntimes p.read (floor-log len);
      return (xp, xp', xn, xn')
    }
  let ?start =
    s(PTranscript := l ! i # ?path-i @ l ! ?sidx # ?path-s @ rest)
  have len-pos: 0 < len
    using i-bound by simp
  have sidx-bound: ?sidx < length l
    using len-pos unfolding len-eq by simp
  have start-eq:
    s(PTranscript := fri-layer-decommitment-transcript l m i len @ rest) =
      ?start
    using i-bound
    unfolding fri-layer-decommitment-transcript-def
    by (simp add: Let-def)
  have prefix-nf: None ∉ dom (dist (execute ?prefix ?start))
  proof –
    have nf-read: None ∉ dom (dist (execute p.read ?start))
      by (rule p.read-no-failure) simp
    have nf-after-read:
      ∧ xp s-xp. Some (xp, s-xp) ∈ set-dist (execute p.read ?start) ⇒
        None ∉ dom
          (dist (execute
            (ntimes p.read (floor-log len) ≫=
              (λxp'. p.read ≫=
                (λxn. ntimes p.read (floor-log len) ≫=
                  (λxn'. return (xp, xp', xn, xn')))))) s-xp))
  proof –
    fix xp s-xp
    assume read-xp: Some (xp, s-xp) ∈ set-dist (execute p.read ?start)
    have read-xp-res:
      s-xp = s(PState := concat (PState s) (l ! i),

```

```

    PTranscript := ?path-i @ l ! ?sidx # ?path-s @ rest)
  using p.read-cons-outcome[OF read-xp] by simp
have enough-path-i: floor-log len ≤ length (PTranscript s-xp)
  using path-i unfolding read-xp-res by simp
have nf-path-i:
  None ∉ dom (dist (execute (ntimes p.read (floor-log len)) s-xp))
  by (rule p.ntimes-read-no-failure[OF enough-path-i])
have nf-after-path-i:
  ∧xp' s-path. Some (xp', s-path) ∈
    set-dist (execute (ntimes p.read (floor-log len)) s-xp) ⇒
    None ∉ dom
      (dist (execute
        (p.read ≫=
          (λxn. ntimes p.read (floor-log len) ≫=
            (λxn'. return (xp, xp', xn, xn')))) s-path))
proof -
  fix xp' s-path
  assume read-path:
    Some (xp', s-path) ∈
      set-dist (execute (ntimes p.read (floor-log len)) s-xp)
  let ?s-xp0 = s(PState := concat (PState s) (l ! i))
  have s-xp-eq:
    s-xp = ?s-xp0(PTranscript := ?path-i @ (l ! ?sidx # ?path-s @ rest))
  using read-xp-res by simp
  have path-res:
    s-path = ?s-xp0(
      PState := foldl concat (PState ?s-xp0) ?path-i,
      PTranscript := l ! ?sidx # ?path-s @ rest)
  using p.ntimes-read-prefix-outcome[
    OF read-path[unfolded path-i[symmetric] s-xp-eq]]
  by simp
  have nf-read-xn: None ∉ dom (dist (execute p.read s-path))
  by (rule p.read-no-failure) (simp add: path-res)
  have nf-after-read-xn:
    ∧xn s-xn. Some (xn, s-xn) ∈ set-dist (execute p.read s-path) ⇒
    None ∉ dom
      (dist (execute
        (ntimes p.read (floor-log len) ≫=
          (λxn'. return (xp, xp', xn, xn')))) s-xn))
proof -
  fix xn s-xn
  assume read-xn: Some (xn, s-xn) ∈ set-dist (execute p.read s-path)
  have read-xn-res:
    s-xn = s-path(
      PState := concat (PState s-path) (l ! ?sidx),
      PTranscript := ?path-s @ rest)
  using p.read-nonempty-outcome[OF read-xn] path-res
  by simp
  have enough-path-s: floor-log len ≤ length (PTranscript s-xn)

```

```

    using path-s unfolding read-xn-res by simp
  have nf-path-s:
    None  $\notin$  dom (dist (execute (ntimes p.read (floor-log len)) s-xn))
    by (rule p.ntimes-read-no-failure[OF enough-path-s])
  show None  $\notin$  dom
    (dist (execute
      (ntimes p.read (floor-log len)  $\gg$ 
        ( $\lambda$ xn'. return (xp, xp', xn, xn')))) s-xn))
    by (rule no-failure-bind-returnI[OF nf-path-s])
  qed
  show None  $\notin$  dom
    (dist (execute
      (p.read  $\gg$ 
        ( $\lambda$ xn. ntimes p.read (floor-log len)  $\gg$ 
          ( $\lambda$ xn'. return (xp, xp', xn, xn')))) s-path))
    using no-failure-bindI[OF nf-read-xn nf-after-read-xn] .
  qed
  show None  $\notin$  dom
    (dist (execute
      (ntimes p.read (floor-log len)  $\gg$ 
        ( $\lambda$ xp'. p.read  $\gg$ 
          ( $\lambda$ xn. ntimes p.read (floor-log len)  $\gg$ 
            ( $\lambda$ xn'. return (xp, xp', xn, xn'))))) s-xp))
    using no-failure-bindI[OF nf-path-i nf-after-path-i] .
  qed
  show ?thesis
    using no-failure-bindI[OF nf-read nf-after-read] by simp
  qed
  have full-eq:
    (do {
      xp  $\leftarrow$  p.read;
      xp'  $\leftarrow$  ntimes p.read (floor-log len);
      xn  $\leftarrow$  p.read;
      xn'  $\leftarrow$  ntimes p.read (floor-log len);
      assert (xp = x);
      ap  $\leftarrow$  p.check-authentication-path len i xp xp';
      assert (ap = fr);
      let idx = (i + len div 2) mod len;
      ap  $\leftarrow$  p.check-authentication-path len idx xn xn';
      assert (ap = fr);
      let gp = (xp + xn) div 2;
      let hp = (xp - xn) div (2 * ((p.h $\hat{=}$ i) * shift) $\hat{=}$ pw);
      let x = gp + b * hp;
      return (i mod (len div 2), x, len div 2, pw + pw)
    }) =
    (?prefix  $\gg$  ( $\lambda$ (xp, xp', xn, xn').
      do {
        assert (xp = x);
        ap  $\leftarrow$  p.check-authentication-path len i xp xp';

```

```

    assert (ap = fr);
    let sidx = (i + len div 2) mod len;
    ap ← p.check-authentication-path len sidx xn xn';
    assert (ap = fr);
    let gp = (xp + xn) div 2;
    let hp = (xp - xn) div (2 * ((p.h^i) * shift)^pw);
    let x = gp + b * hp;
    return (i mod (len div 2), x, len div 2, pw + pw)
  }))
by (simp add: sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding start-eq full-eq
proof (rule no-failure-bindI[OF prefix-nf])
  fix out t
  assume out: Some (out, t) ∈ set-dist (execute ?prefix ?start)
  obtain xp xp' xn xn' where out-eq: out = (xp, xp', xn, xn')
    by (cases out) auto
  from out[unfolded out-eq] obtain s1 s2 s3 where
    read-xp: Some (xp, s1) ∈ set-dist (execute p.read ?start)
    and read-xp-path:
      Some (xp', s2) ∈ set-dist (execute (ntimes p.read (floor-log len)) s1)
    and read-xn: Some (xn, s3) ∈ set-dist (execute p.read s2)
    and read-xn-path:
      Some (xn', t) ∈ set-dist (execute (ntimes p.read (floor-log len)) s3)
    by (auto elim!: p.set-dist-bindE)
  have read-xp-start:
    ?start = s(PTranscript := l ! i # (?path-i @ l ! ?sidx # ?path-s @ rest))
    by simp
  have read-xp':
    Some (xp, s1) ∈
      set-dist (execute p.read
        (s(PTranscript := l ! i # (?path-i @ l ! ?sidx # ?path-s @ rest))))
    by (subst read-xp-start[symmetric]) (rule read-xp)
  have xp-replay:
    xp = l ! i ∧
    s1 = s(
      PState := concat (PState s) (l ! i),
      PTranscript := ?path-i @ l ! ?sidx # ?path-s @ rest)
    by (rule p.read-cons-outcome[OF read-xp'])
  let ?s-xp = s(PState := concat (PState s) (l ! i))
  have s1-eq: s1 = ?s-xp(PTranscript := ?path-i @ l ! ?sidx # ?path-s @ rest)
    using xp-replay by simp
  have xp-path-replay:
    xp' = ?path-i ∧
    s2 = ?s-xp(
      PState := foldl concat (PState ?s-xp) ?path-i,
      PTranscript := l ! ?sidx # ?path-s @ rest)
    using p.ntimes-read-prefix-outcome[
      OF read-xp-path[unfolded path-i[symmetric] s1-eq]]

```

```

by simp
have read-xn-start:
  s2 = (s2(PTranscript := l ! ?sidx # (?path-s @ rest)))
  using xp-path-replay by simp
have read-xn':
  Some (xn, s3) ∈
    set-dist (execute p.read
      (s2(PTranscript := l ! ?sidx # (?path-s @ rest))))
  by (subst read-xn-start[symmetric]) (rule read-xn)
have xn-replay:
  xn = l ! ?sidx ∧
  s3 = s2(
    PState := concat (PState s2) (l ! ?sidx),
    PTranscript := ?path-s @ rest)
  by (rule p.read-cons-outcome[OF read-xn])
let ?s-xn = s2(PState := concat (PState s2) (l ! ?sidx))
have s3-eq: s3 = ?s-xn(PTranscript := ?path-s @ rest)
  using xn-replay by simp
have xn-path-replay:
  xn' = ?path-s ∧
  t = ?s-xn(
    PState := foldl concat (PState ?s-xn) ?path-s,
    PTranscript := rest)
  using p.ntimes-read-prefix-outcome[
    OF read-xn-path[unfolded path-s[symmetric] s3-eq]]
  by simp
have s0-t: s0 ≤ t
  using ext xp-replay xp-path-replay xn-replay xn-path-replay
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have xp-x: xp = x
  using xp-replay x-eq by simp
let ?after =
  do {
    assert (xp = x);
    ap ← p.check-authentication-path len i xp xp';
    assert (ap = fr);
    let sidx = (i + len div 2) mod len;
    ap ← p.check-authentication-path len sidx xn xn';
    assert (ap = fr);
    let gp = (xp + xn) div 2;
    let hp = (xp - xn) div (2 * ((p.h^i) * shift)^pw);
    let x = gp + b * hp;
    return (i mod (len div 2), x, len div 2, pw + pw)
  }
have after-eq:
  ?after =
  (assert (xp = x) >>= (λ-.
    p.check-authentication-path len i xp xp' >>= (λap.
      assert (ap = fr) >>= (λ-.

```

```

    p.check-authentication-path len ?sidx xn xn' >>= (λap.
    assert (ap = fr) >>= (λ-.
    let gp = (xp + xn) div 2;
      hp = (xp - xn) div (2 * ((p.h ^ i) * shift) ^ pw);
      x = gp + b * hp
    in return (i mod (len div 2), x, len div 2, pw + pw))))))
  by (simp add: Let-def sm-bind-assoc)
have after-nf: None ∉ dom (dist (execute ?after t))
  unfolding after-eq
proof (rule no-failure-assert-bindI[OF xp-x])
  show None ∉ dom
    (dist (execute
      (p.check-authentication-path len i xp xp' >>=
        (λapi.
          assert (api = fr) >>=
            (λ-.
              p.check-authentication-path len ?sidx xn xn' >>=
                (λap.
                  assert (ap = fr) >>=
                    (λ-.
                      let gp = (xp + xn) div 2;
                        hp = (xp - xn) div (2 * ((p.h ^ i) * shift) ^ pw);
                        x = gp + b * hp
                      in return (i mod (len div 2), x, len div 2, pw + pw)))))) t))
  proof (rule no-failure-bindI[OF p.check-authentication-path-no-failure])
  fix api u
  assume check-i:
    Some (api, u) ∈
      set-dist (execute (p.check-authentication-path len i xp xp') t)
  have check-i-honest:
    Some (api, u) ∈
      set-dist (execute
        (p.check-authentication-path (length l) i (l ! i) ?path-i) t)
    using check-i xp-replay xp-path-replay len-eq by simp
  have i-bound-l: i < length l
    using i-bound len-eq by simp
  have check-i-honest-l:
    Some (api, u) ∈
      set-dist (execute
        (p.check-authentication-path (length l) i (l ! i)
          (get-authentication-path (length l) i m)) t)
    using check-i-honest len-eq by simp
  have ap-i: api = value m ∧ t ≤ u
    using p.check-created-tree-outcome[
      OF created len-pow i-bound-l refl s0-t check-i-honest-l]
    by simp
  then have ap-fr: api = fr
    using root by simp
  have tail-nf: None ∉ dom

```



```

(dist (execute
  (p.check-authentication-path len ?sidx xn xn') >>=
    (λap.
      assert (ap = fr) >>=
        (λ-.
          let gp = (xp + xn) div 2;
          hp = (xp - xn) div (2 * ((p.h ^ i) * shift) ^ pw);
          x = gp + b * hp
          in return (i mod (len div 2), x, len div 2, pw + pw)))) u))
proof (rule no-failure-bindI)
show None ∉ dom
  (dist (execute (p.check-authentication-path len ?sidx xn xn') u))
  by (rule p.check-authentication-path-no-failure)
fix ap2 u2
assume check-s:
  Some (ap2, u2) ∈
    set-dist (execute (p.check-authentication-path len ?sidx xn xn') u)
have s0-u: s0 ≤ u
  using s0-t ap-i
  unfolding less-eq-hash-ext-def less-eq-fmap-def by metis
have check-s-honest:
  Some (ap2, u2) ∈
    set-dist (execute
      (p.check-authentication-path (length l) ?sidx (l ! ?sidx) ?path-s) u)
  using check-s xn-replay xn-path-replay len-eq by simp
have check-s-honest-l:
  Some (ap2, u2) ∈
    set-dist (execute
      (p.check-authentication-path (length l) ?sidx (l ! ?sidx)
        (get-authentication-path (length l) ?sidx m)) u)
  using check-s-honest len-eq by simp
have ap-s: ap2 = value m
  using p.check-created-tree-outcome[
    OF created len-pow sidx-bound refl s0-u check-s-honest-l]
  by simp
have ap-s-fr: ap2 = fr
  using ap-s root by simp
have ap-s-fr-true: (ap2 = fr) = True
  using ap-s-fr by simp
show None ∉ dom
  (dist (execute
    (assert (ap2 = fr) >>=
      (λ-.
        let gp = (xp + xn) div 2;
        hp = (xp - xn) div (2 * ((p.h ^ i) * shift) ^ pw);
        x = gp + b * hp
        in return (i mod (len div 2), x, len div 2, pw + pw)))) u2))
  using ap-s-fr by (simp add: assert-def)
qed

```

```

show  $\text{None} \notin \text{dom}$ 
  (dist (execute
    (assert ( $\text{api} = \text{fr}$ )  $\gg$ 
      ( $\lambda$ -.
        p.check-authentication-path len ?sid xn xn'  $\gg$ 
          ( $\lambda$ ap.
            assert ( $\text{ap} = \text{fr}$ )  $\gg$ 
              ( $\lambda$ -.
                let gp = (xp + xn) div 2;
                hp = (xp - xn) div (2 * ((p.h ^ i) * shift) ^ pw);
                x = gp + b * hp
                in return
                  (i mod (len div 2),
                    x,
                    len div 2, pw + pw)))))) u))
    by (rule no-failure-assert-bindI[OF ap-fr tail-nf])
  qed
qed
show  $\text{None} \notin \text{dom}$ 
  (dist (execute
    ((case out of (xp, xp', xn, xn')  $\Rightarrow$ 
      do {
        assert ( $\text{xp} = \text{x}$ );
        ap  $\leftarrow$  p.check-authentication-path len i xp xp';
        assert ( $\text{ap} = \text{fr}$ );
        let sid = (i + len div 2) mod len;
        ap  $\leftarrow$  p.check-authentication-path len sid xn xn';
        assert ( $\text{ap} = \text{fr}$ );
        let gp = (xp + xn) div 2;
        let hp = (xp - xn) div (2 * ((p.h ^ i) * shift) ^ pw);
        let x = gp + b * hp;
        return (i mod (len div 2), x, len div 2, pw + pw)
      }))) t))
    unfolding out-eq
    using after-nf by simp
  qed
qed

```

lemma *honest-fri-layer-decommitment-outcome*:

```

assumes len-eq: len = length l
and i-bound: i < len
and path-i:
  length (get-authentication-path len i m) = floor-log len
and path-s:
  length (get-authentication-path len ((i + len div 2) mod len) m) =
    floor-log len
and outcome:
  Some (res, t')  $\in$  set-dist (execute
    (do {

```

```

     $xp \leftarrow p.read;$ 
     $xp' \leftarrow \text{ntimes } p.read \text{ (floor-log len)};$ 
     $xn \leftarrow p.read;$ 
     $xn' \leftarrow \text{ntimes } p.read \text{ (floor-log len)};$ 
     $\text{assert } (xp = x);$ 
     $ap \leftarrow p.check\text{-authentication-path } len \ i \ xp \ xp';$ 
     $\text{assert } (ap = fr);$ 
     $\text{let } sid_x = (i + len \text{ div } 2) \text{ mod } len;$ 
     $ap \leftarrow p.check\text{-authentication-path } len \ sid_x \ xn \ xn';$ 
     $\text{assert } (ap = fr);$ 
     $\text{let } gp = (xp + xn) \text{ div } 2;$ 
     $\text{let } hp = (xp - xn) \text{ div } (2 * ((p.h^i) * shift)^{pw});$ 
     $\text{let } x = gp + b * hp;$ 
     $\text{return } (i \text{ mod } (len \text{ div } 2), x, len \text{ div } 2, pw + pw)$ 
  })
  (s(PTranscript := fri-layer-decommitment-transcript l m i len @ rest)))
shows
  res =
    (i mod (len div 2),
     ((l ! i) + (l ! ((i + len div 2) mod len))) div 2 +
     b * (((l ! i) - (l ! ((i + len div 2) mod len))) div
      (2 * ((p.h^i) * shift)^{pw})),
     len div 2,
     pw + pw)
  PTranscript t' = rest
  PState t' =
    foldl concat (PState s) (fri-layer-decommitment-transcript l m i len)
proof -
  let ?sid_x = (i + len div 2) mod len
  let ?path-i = get-authentication-path len i m
  let ?path-s = get-authentication-path len ?sid_x m
  let ?start =
    s(PTranscript := l ! i # ?path-i @ l ! ?sid_x # ?path-s @ rest)
  have len-pos: 0 < len
    using i-bound by simp
  have sid_x-bound: ?sid_x < length l
    using len-pos unfolding len-eq by simp
  have start-eq:
    s(PTranscript := fri-layer-decommitment-transcript l m i len @ rest) =
      ?start
  using i-bound
  unfolding fri-layer-decommitment-transcript-def
  by (simp add: Let-def)
  from outcome[unfolded start-eq] obtain xp xp' xn xn' api aps
    s1 s2 s3 s4 s5 s6 s7 s8 where
    read-xp: Some (xp, s1) ∈ set-dist (execute p.read ?start)
  and read-xp-path:
    Some (xp', s2) ∈
      set-dist (execute (ntimes p.read (floor-log len)) s1)

```

```

and read-xn: Some ( $xn, s3$ )  $\in$  set-dist (execute p.read  $s2$ )
and read-xn-path:
  Some ( $xn', s4$ )  $\in$ 
    set-dist (execute (ntimes p.read (floor-log len))  $s3$ )
and assert-x: Some ( $()$ ,  $s5$ )  $\in$  set-dist (execute (assert ( $xp = x$ ))  $s4$ )
and check-i:
  Some ( $api, s6$ )  $\in$ 
    set-dist (execute (p.check-authentication-path len  $i$   $xp$   $xp'$ )  $s5$ )
and assert-i: Some ( $()$ ,  $s7$ )  $\in$  set-dist (execute (assert ( $api = fr$ ))  $s6$ )
and check-s:
  Some ( $aps, s8$ )  $\in$ 
    set-dist (execute (p.check-authentication-path len  $?sidx$   $xn$   $xn'$ )  $s7$ )
and assert-s: Some ( $()$ ,  $t'$ )  $\in$  set-dist (execute (assert ( $aps = fr$ ))  $s8$ )
and res-eq:
   $res =$ 
    ( $i \bmod (len \div 2)$ ,
     ( $xp + xn$ )  $\div 2 + b * ((xp - xn) \div (2 * ((p.h \wedge i) * shift) \wedge pw))$ ,
      $len \div 2, pw + pw$ )
  by (auto simp: Let-def elim!: p.set-dist-bindE)
have xp-replay:
   $xp = l ! i \wedge$ 
   $s1 = s \langle$ 
     $PState := concat (PState\ s) (l ! i)$ ,
     $PTranscript := ?path-i @ l ! ?sidx \# ?path-s @ rest \rangle$ 
  by (rule p.read-cons-outcome[OF read-xp])
let  $?s-xp = s \langle PState := concat (PState\ s) (l ! i) \rangle$ 
have s1-eq:
   $s1 = ?s-xp \langle PTranscript := ?path-i @ l ! ?sidx \# ?path-s @ rest \rangle$ 
  using xp-replay by simp
have xp-path-replay:
   $xp' = ?path-i \wedge$ 
   $s2 = ?s-xp \langle$ 
     $PState := foldl\ concat\ (PState\ ?s-xp)\ ?path-i$ ,
     $PTranscript := l ! ?sidx \# ?path-s @ rest \rangle$ 
  using p.ntimes-read-prefix-outcome[
    OF read-xp-path[unfolded path-i[symmetric] s1-eq]
  by simp
have read-xn':
  Some ( $xn, s3$ )  $\in$ 
    set-dist (execute p.read
      ( $s2 \langle PTranscript := l ! ?sidx \# (?path-s @ rest) \rangle$ ))
  using read-xn xp-path-replay by simp
have xn-replay:
   $xn = l ! ?sidx \wedge$ 
   $s3 = s2 \langle$ 
     $PState := concat (PState\ s2) (l ! ?sidx)$ ,
     $PTranscript := ?path-s @ rest \rangle$ 
  by (rule p.read-cons-outcome[OF read-xn'])
let  $?s-xn = s2 \langle PState := concat (PState\ s2) (l ! ?sidx) \rangle$ 

```

```

have s3-eq:
  s3 = ?s-xn(|PTranscript := ?path-s @ rest)
  using xn-replay by simp
have xn-path-replay:
  xn' = ?path-s ∧
  s4 = ?s-xn(|
    PState := foldl concat (PState ?s-xn) ?path-s,
    PTranscript := rest)
  using p.ntimes-read-prefix-outcome[
    OF read-xn-path[unfolded path-s[symmetric] s3-eq]]
  by simp
have s5-tr: PTranscript s5 = rest
  using assert-outcomeD(2)[OF assert-x] xn-path-replay by simp
have s6-tr: PTranscript s6 = rest
  using p.check-authentication-path-preserves-channel(2)[OF check-i] s5-tr
  by simp
have s7-tr: PTranscript s7 = rest
  using assert-outcomeD(2)[OF assert-i] s6-tr by simp
have s8-tr: PTranscript s8 = rest
  using p.check-authentication-path-preserves-channel(2)[OF check-s] s7-tr
  by simp
have state-t:
  PState t' =
    foldl concat (PState s) (fri-layer-decommitment-transcript l m i len)
proof -
  have i-mod: i mod len = i
  using i-bound by simp
  have s5-state: PState s5 = PState s4
  using assert-outcomeD(2)[OF assert-x] by simp
  have s6-state: PState s6 = PState s5
  using p.check-authentication-path-preserves-channel(1)[OF check-i] .
  have s7-state: PState s7 = PState s6
  using assert-outcomeD(2)[OF assert-i] by simp
  have s8-state: PState s8 = PState s7
  using p.check-authentication-path-preserves-channel(1)[OF check-s] .
  have t-state: PState t' = PState s8
  using assert-outcomeD(2)[OF assert-s] by simp
  show ?thesis
  using xp-replay xp-path-replay xn-replay xn-path-replay
    s5-state s6-state s7-state s8-state t-state i-mod
  unfolding fri-layer-decommitment-transcript-def
  by (simp add: Let-def)
qed
show
  res =
    (i mod (len div 2),
    ((l ! i) + (l ! ?sidx)) div 2 +
    b * (((l ! i) - (l ! ?sidx)) div
    (2 * ((p.h∧i) * shift∧pw))),

```

```

    len div 2,
    pw + pw)
  using res-eq xp-replay xn-replay by simp
show PTranscript t' = rest
  using assert-outcomeD(2)[OF assert-s] s8-tr by simp
show PState t' =
  foldl concat (PState s) (fri-layer-decommitment-transcript l m i len)
  using state-t .
qed

```

lemma *fri-layer-decommitment-hash-extends:*

assumes *outcome:*

```

Some (res, t') ∈ set-dist (execute
  (do {
    xp ← p.read;
    xp' ← ntimes p.read (floor-log len);
    xn ← p.read;
    xn' ← ntimes p.read (floor-log len);
    assert (xp = x);
    ap ← p.check-authentication-path len i xp xp';
    assert (ap = fr);
    let sidx = (i + len div 2) mod len;
    ap ← p.check-authentication-path len sidx xn xn';
    assert (ap = fr);
    let gp = (xp + xn) div 2;
    let hp = (xp - xn) div (2 * ((p.h^i) * shift)^pw);
    let x = gp + b * hp;
    return (i mod (len div 2), x, len div 2, pw + pw)
  }) s)

```

shows $s \leq t'$

proof –

let $?sidx = (i + len \text{ div } 2) \text{ mod } len$

from *outcome* **obtain** $xp \ xp' \ xn \ xn' \ api \ aps$

$s1 \ s2 \ s3 \ s4 \ s5 \ s6 \ s7 \ s8$ **where**

read-xp: $Some \ (xp, s1) \in \text{set-dist} \ (\text{execute } p.\text{read } s)$

and *read-xp-path:*

$Some \ (xp', s2) \in$
 $\text{set-dist} \ (\text{execute} \ (ntimes \ p.\text{read} \ (\text{floor-log } len)) \ s1)$

and *read-xn:* $Some \ (xn, s3) \in \text{set-dist} \ (\text{execute } p.\text{read } s2)$

and *read-xn-path:*

$Some \ (xn', s4) \in$
 $\text{set-dist} \ (\text{execute} \ (ntimes \ p.\text{read} \ (\text{floor-log } len)) \ s3)$

and *assert-x:* $Some \ ((), s5) \in \text{set-dist} \ (\text{execute} \ (\text{assert} \ (xp = x)) \ s4)$

and *check-i:*

$Some \ (api, s6) \in$
 $\text{set-dist} \ (\text{execute} \ (p.\text{check-authentication-path} \ len \ i \ xp \ xp') \ s5)$

and *assert-i:* $Some \ ((), s7) \in \text{set-dist} \ (\text{execute} \ (\text{assert} \ (api = fr)) \ s6)$

and *check-s:*

$Some \ (aps, s8) \in$

```

    set-dist (execute (p.check-authentication-path len ?sidx xn xn') s7)
  and assert-s: Some ((), t') ∈ set-dist (execute (assert (aps = fr)) s8)
  by (auto simp: Let-def elim!: p.set-dist-bindE)
have s ≤ s1
  using read-xp by (rule read-hash-extends)
moreover have s1 ≤ s2
  using read-xp-path by (rule ntimes-read-hash-extends)
moreover have s2 ≤ s3
  using read-xn by (rule read-hash-extends)
moreover have s3 ≤ s4
  using read-xn-path by (rule ntimes-read-hash-extends)
moreover have s4 ≤ s5
  using assert-x by (rule assert-hash-extends)
moreover have s5 ≤ s6
  using check-i by (rule check-authentication-path-hash-extends)
moreover have s6 ≤ s7
  using assert-i by (rule assert-hash-extends)
moreover have s7 ≤ s8
  using check-s by (rule check-authentication-path-hash-extends)
moreover have s8 ≤ t'
  using assert-s by (rule assert-hash-extends)
ultimately show ?thesis
  by (meson p.hash-ext-trans)
qed

```

lemma *honest-fri-layer-decommitment-step*:

```

assumes created: p.created-tree l m s0
  and len-eq: len = length l
  and len-pow: length l = 2 ^ n
  and i-bound: i < len
  and x-eq: x = l ! i
  and ext: s0 ≤ s
  and path-i:
    length (get-authentication-path len i m) = floor-log len
  and path-s:
    length (get-authentication-path len ((i + len div 2) mod len) m) =
      floor-log len
  and root: fr = value m
  and tr:
    start =
      s(PTranscript := fri-layer-decommitment-transcript l m i len @ rest)
shows
  None ∉ dom (dist (execute (fri-verifier-layer b fr i x len pw) start))
  ∧ res t'. Some (res, t') ∈
    set-dist (execute (fri-verifier-layer b fr i x len pw) start) ⇒
  res =
    (i mod (len div 2), fri-fold-value b l i len pw, len div 2, pw + pw) ∧
  PTranscript t' = rest ∧
  start ≤ t'

```

proof –
show *nf*:
 None $\notin$ dom (*dist* (*execute* (*fri-verifier-layer* *b fr i x len pw*) *start*))
unfolding *fri-verifier-layer-def tr*
by (*rule honest-fri-layer-decommitment-no-failure* [
OF created len-eq len-pow i-bound x-eq ext path-i path-s root])
fix *res t'*
assume *outcome*:
Some (*res, t'*) $\in$
set-dist (*execute* (*fri-verifier-layer* *b fr i x len pw*) *start*)
have *res-tr*:
res =
 (*i mod* (*len div 2*), *fri-fold-value* *b l i len pw, len div 2, pw + pw*)
PTranscript t' = rest
using *honest-fri-layer-decommitment-outcome* [
OF len-eq i-bound path-i path-s outcome[*unfolded fri-verifier-layer-def tr*]]
unfolding *fri-fold-value-def* **by** *simp-all*
have *ext-start*: *start* $\leq$ *t'*
using *fri-layer-decommitment-hash-extends* [
OF outcome[*unfolded fri-verifier-layer-def*]] .
show
res =
 (*i mod* (*len div 2*), *fri-fold-value* *b l i len pw, len div 2, pw + pw*) $\wedge$
PTranscript t' = rest $\wedge$
start $\leq$ *t'*
using *res-tr ext-start* **by** *simp*
qed

lemma *honest-fri-layer-decommitment-step-created*:
assumes *created*: *p.created-tree l m s0*
and *len-eq*: *len = length l*
and *len-pow*: *length l = 2 ^ n*
and *i-bound*: *i < len*
and *x-eq*: *x = l ! i*
and *ext*: *s0* $\leq$ *s*
and *root*: *fr = value m*
and *tr*:
start =
s(*PTranscript := fri-layer-decommitment-transcript l m i len @ rest*)
shows
 None $\notin$ dom (*dist* (*execute* (*fri-verifier-layer* *b fr i x len pw*) *start*))
 $\wedge$ *res t'. Some* (*res, t'*) $\in$
set-dist (*execute* (*fri-verifier-layer* *b fr i x len pw*) *start*) $\implies$
res =
 (*i mod* (*len div 2*), *fri-fold-value* *b l i len pw, len div 2, pw + pw*) $\wedge$
PTranscript t' = rest $\wedge$
start $\leq$ *t'*

proof –
have *path-i*:


```

length (get-authentication-path len i m) = floor-log len
using created-tree-get-authentication-path-len[
  OF created len-pow, of i] i-bound len-eq by simp
have len-pos: 0 < len
  using i-bound by simp
have idx-bound:
  (i + len div 2) mod len < length l
  using len-pos len-eq by simp
have path-s:
  length (get-authentication-path len ((i + len div 2) mod len) m) =
    floor-log len
  using created-tree-get-authentication-path-len[
    OF created len-pow, of (i + len div 2) mod len] idx-bound len-eq
  by simp
show None ∉ dom (dist (execute (fri-verifier-layer b fr i x len pw) start))
  by (rule honest-fri-layer-decommitment-step(1)[
    OF created len-eq len-pow i-bound x-eq ext path-i path-s root tr])
fix res t'
assume outcome:
  Some (res, t') ∈
    set-dist (execute (fri-verifier-layer b fr i x len pw) start)
show
  res =
    (i mod (len div 2), fri-fold-value b l i len pw, len div 2, pw + pw) ∧
    PTranscript t' = rest ∧
    start ≤ t'
  by (rule honest-fri-layer-decommitment-step(2)[
    OF created len-eq len-pow i-bound x-eq ext path-i path-s root tr outcome])
qed

lemma receive-query-commits-step-hash-extends:
assumes step-in: m ∈ set (v.receive-query-commits fl)
and outcome: Some (y, t) ∈ set-dist (execute (m a) s)
shows s ≤ t
proof –
from receive-query-commits-step-eq[OF step-in] obtain b fr where
  m-eq: m = (λ(i, x, len, pw). fri-verifier-layer b fr i x len pw)
by blast
obtain i x len pw where a-eq: a = (i, x, len, pw)
by (cases a) auto
have layer-out:
  Some (y, t) ∈
    set-dist (execute (fri-verifier-layer b fr i x len pw) s)
using outcome unfolding m-eq a-eq by simp
show ?thesis
  using fri-layer-decommitment-hash-extends[
    OF layer-out[unfolded fri-verifier-layer-def]] .
qed

```

lemma *fri-verifier-layer-preserves-query-counter*:

assumes *outcome*:

Some (res, t) ∈ set-dist (execute (fri-verifier-layer b fr i x len pw) s)

shows *PQueryCounter t = PQueryCounter s*

proof –

from *outcome* **obtain** *xp s1 xp' s2 xn s3 xn' s4 s5 ap-i s6 s7 ap-s s8 s9* **where**

read-xp: Some (xp, s1) ∈ set-dist (execute p.read s)

and *read-xp-path*:

Some (xp', s2) ∈ set-dist (execute (ntimes p.read (floor-log len)) s1)

and *read-xn: Some (xn, s3) ∈ set-dist (execute p.read s2)*

and *read-xn-path*:

Some (xn', s4) ∈ set-dist (execute (ntimes p.read (floor-log len)) s3)

and *assert-x: Some ((), s5) ∈ set-dist (execute (assert (xp = x)) s4)*

and *check-i*:

Some (ap-i, s6) ∈

set-dist (execute (p.check-authentication-path len i xp xp') s5)

and *assert-i: Some ((), s7) ∈ set-dist (execute (assert (ap-i = fr)) s6)*

and *check-s*:

Some (ap-s, s8) ∈

set-dist (execute (p.check-authentication-path len ((i + len div 2) mod len) xn xn') s7)

and *assert-s: Some ((), s9) ∈ set-dist (execute (assert (ap-s = fr)) s8)*

and *ret*:

Some (res, t) ∈

set-dist (execute

(return

(i mod (len div 2),

*(xp + xn) div 2 + b * ((xp - xn) div (2 * ((p.h ^ i) * shift) ^ pw)),*

len div 2, pw + pw)) s9)

unfolding *fri-verifier-layer-def*

by (*auto simp: Let-def elim!: p.set-dist-bindE*)

have *q-s1: PQueryCounter s1 = PQueryCounter s*

using *read-preserves-counters[OF read-xp]* **by** *simp*

have *q-s2: PQueryCounter s2 = PQueryCounter s1*

by (*rule ntimes-read-preserves-query-counter[OF read-xp-path]*)

have *q-s3: PQueryCounter s3 = PQueryCounter s2*

using *read-preserves-counters[OF read-xn]* **by** *simp*

have *q-s4: PQueryCounter s4 = PQueryCounter s3*

by (*rule ntimes-read-preserves-query-counter[OF read-xn-path]*)

have *s5-eq: s5 = s4*

using *assert-outcomeD(2)[OF assert-x]* .

have *q-s6: PQueryCounter s6 = PQueryCounter s5*

using *p.check-authentication-path-preserves-channel(6)[OF check-i]* .

have *s7-eq: s7 = s6*

using *assert-outcomeD(2)[OF assert-i]* .

have *q-s8: PQueryCounter s8 = PQueryCounter s7*

using *p.check-authentication-path-preserves-channel(6)[OF check-s]* .

have *s9-eq: s9 = s8*

using *assert-outcomeD(2)[OF assert-s]* .

have *t-eq: t = s9*

```

    using ret by simp
  show ?thesis
    using q-s1 q-s2 q-s3 q-s4 q-s6 q-s8
    unfolding s5-eq s7-eq s9-eq t-eq by simp
qed

lemma receive-query-commits-step-preserves-query-counter:
  assumes step-in:  $m \in \text{set } (v.\text{receive-query-commits fl})$ 
    and outcome:  $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (m \ a) \ s)$ 
  shows  $P\text{QueryCounter } t = P\text{QueryCounter } s$ 
proof -
  from receive-query-commits-step-eq[OF step-in] obtain b fr where
    m-eq:  $m = (\lambda(i, x, \text{len}, pw). \text{fri-verifier-layer } b \text{ fr } i \ x \ \text{len } pw)$ 
    by blast
  obtain i x len pw where a-eq:  $a = (i, x, \text{len}, pw)$ 
    by (cases a) auto
  have layer-out:
     $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (\text{fri-verifier-layer } b \text{ fr } i \ x \ \text{len } pw) \ s)$ 
    using outcome unfolding m-eq a-eq by simp
  show ?thesis
    by (rule fri-verifier-layer-preserves-query-counter[OF layer-out])
qed

lemma receive-query-commits-mfold-preserves-query-counter:
  assumes outcome:
     $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (\text{mfold } a \ (v.\text{receive-query-commits fl})) \ s)$ 
  shows  $P\text{QueryCounter } t = P\text{QueryCounter } s$ 
  by (rule mfold-preserves-query-counter[OF outcome])
    (rule receive-query-commits-step-preserves-query-counter)

lemma receive-query-commits-mfold-hash-extends:
  assumes outcome:
     $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (\text{mfold } a \ (v.\text{receive-query-commits fl})) \ s)$ 
  shows  $s \leq t$ 
  by (rule mfold-hash-extends[OF outcome])
    (rule receive-query-commits-step-hash-extends)

lemma next-fri-domain-length:
  length (p.next-fri-domain d) = length d div 2
  unfolding p.next-fri-domain-def by simp

lemma next-fri-domain-nth:
  assumes  $i < \text{length } d \text{ div } 2$ 
  shows  $p.\text{next-fri-domain } d ! i = d ! i * d ! i$ 
  using assms unfolding p.next-fri-domain-def by simp

```

lemma *next-fri-layer-components*:
assumes $p.\text{next-fri-layer } q \ d \ b = (q', d', l')$
shows
 $q' = p.\text{next-fri-polynomial } q \ b$
 $d' = p.\text{next-fri-domain } d$
 $l' = \text{map } (\text{poly } q') \ d'$
using *assms*
unfolding $p.\text{next-fri-layer-def}$
by (*auto simp: Let-def split: prod.splits*)

lemma *next-fri-layer-length*:
assumes $p.\text{next-fri-layer } q \ d \ b = (q', d', l')$
shows $\text{length } l' = \text{length } d \ \text{div } 2$
using *next-fri-layer-components*[*OF assms*] *next-fri-domain-length* **by** *simp*

lemma *next-fri-layer-domain-nth*:
assumes *layer*: $p.\text{next-fri-layer } q \ d \ b = (q', d', l')$
and *i-bound*: $i < \text{length } d \ \text{div } 2$
shows $d' ! i = d ! i * d ! i$
using *next-fri-layer-components*(2)[*OF layer*] *next-fri-domain-nth*[*OF i-bound*]
by *simp*

lemma *fri-successor-domain-points*:
assumes *ds0*: $ds ! 0 = p.\text{eval-domain}$
and *successors*:
 $\bigwedge j. j < n \implies$
 $p.\text{next-fri-layer } (ps ! j) \ (ds ! j) \ (\text{challenges} ! j) =$
 $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$
and *j-le*: $j \leq n$
and *k-bound*: $k < \text{length } (ds ! j)$
shows $ds ! j ! k = ((p.h \wedge k) * \text{shift}) \wedge (2 \wedge j)$
using *j-le k-bound*
proof (*induction j arbitrary: k*)
case 0
have *k-lt*: $k < \text{clength} * \text{scale}$
using *0.premis*(2) *ds0 eval-domain-length* **by** *simp*
show ?*case*
using *eval-domain-nth*[*OF k-lt*] *ds0* **by** *simp*

next
case (*Suc j*)
have *j-lt-n*: $j < n$
using *Suc.premis*(1) **by** *simp*
have *ds-suc*:
 $ds ! \text{Suc } j = p.\text{next-fri-domain } (ds ! j)$
using *next-fri-layer-components*(2)[*OF successors*][*OF j-lt-n*]
by *simp*
have *k-half*: $k < \text{length } (ds ! j) \ \text{div } 2$
using *Suc.premis*(2) *ds-suc next-fri-domain-length* **by** *simp*
have *k-prev*: $k < \text{length } (ds ! j)$

```

    using k-half by linarith
have len-pos: 0 < length (ds ! j)
    using k-prev by (cases ds ! j) auto
have curr:
  ds ! Suc j ! k = ds ! j ! k * ds ! j ! k
    using ds-suc next-fri-domain-nth[OF k-half] by simp
have prev:
  ds ! j ! k = ((p.h ^ k) * shift) ^ (2 ^ j)
    by (rule Suc.IH) (use Suc.prem1 k-prev in simp-all)
have pow-suc: (2::nat) ^ j + (2::nat) ^ j = (2::nat) ^ Suc j
proof -
  have (2::nat) ^ Suc j = 2 * 2 ^ j
    by (simp add: power-Suc)
  also have ... = 2 ^ j + 2 ^ j
    by presburger
  finally show ?thesis
    by simp
qed
have next':
  ((p.h ^ k) * shift) ^ (2 ^ j) *
  ((p.h ^ k) * shift) ^ (2 ^ j) =
  ((p.h ^ k) * shift) ^ (2 ^ Suc j)
proof -
  have ((p.h ^ k) * shift) ^ (2 ^ j) *
  ((p.h ^ k) * shift) ^ (2 ^ j) =
  ((p.h ^ k) * shift) ^ (2 ^ j + 2 ^ j)
    by (simp add: power-add[symmetric])
  also have ... = ((p.h ^ k) * shift) ^ (2 ^ Suc j)
    by (simp only: pow-suc)
  finally show ?thesis .
qed
show ?case
  using curr prev next' by simp
qed

lemma fri-successor-layer-lengths-power:
  assumes init-len: length (ls ! 0) = 2 ^ N
  and n-le-N: n ≤ N
  and layer-lens:  $\bigwedge j. j < n \implies \text{length } (ds ! j) = \text{length } (ls ! j)$ 
  and successors:
     $\bigwedge j. j < n \implies$ 
     $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$ 
     $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 
  shows  $j \leq n \implies \text{length } (ls ! j) = 2 ^ (N - j)$ 
proof (induction j)
  case 0
  then show ?case
    using init-len by simp
next

```

```

case (Suc j)
have j-lt-n: j < n
  using Suc.premis by simp
have j-le-n: j ≤ n
  using Suc.premis by simp
have len-j: length (ls ! j) = 2 ^ (N - j)
  using Suc.IH[OF j-le-n] .
have len-next:
  length (ls ! Suc j) = length (ds ! j) div 2
  using next-fri-layer-length[OF successors[OF j-lt-n]] by simp
have j-lt-N: j < N
  using j-lt-n n-le-N by linarith
have diff-suc: N - j = Suc (N - Suc j)
  using j-lt-N by simp
show ?case
  using len-next layer-lens[OF j-lt-n] len-j diff-suc by simp
qed

```

lemma fri-successor-layer-lengths-are-powers:

```

assumes init-len: length (ls ! 0) = 2 ^ N
and n-le-N: n ≤ N
and layer-lens:  $\bigwedge j. j < n \implies \text{length } (ds ! j) = \text{length } (ls ! j)$ 
and successors:
   $\bigwedge j. j < n \implies$ 
  p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
  (ps ! Suc j, ds ! Suc j, ls ! Suc j)
and j-bound: j < n
shows  $\exists k. \text{length } (ls ! j) = 2 ^ k$ 
using fri-successor-layer-lengths-power[
  OF init-len n-le-N layer-lens successors, of j] j-bound
by auto

```

lemma fri-commit-initial-decommit-layer-lengths-are-powers:

```

assumes outcome:
  Some ((ps, ds, ls, ms), t) ∈
  set-dist (execute
    (p.fri-commit n [cp] [d] [l] [m]) s)
and layers:
   $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
  ls ! i = map (poly (ps ! i)) (ds ! i)
and init-len: length l = 2 ^ N
and n-le-N: n ≤ N
and j-bound: j < n
shows  $\exists k. \text{length } (ls ! j) = 2 ^ k$ 
proof -
have lens:
  length ps = Suc n
  length ds = Suc n
  length ls = Suc n

```

```

length ms = Suc n
using p.fri-commit-lengths[OF outcome] by simp-all
have ls0: ls ! 0 = l
proof -
  have take (length [l]) ls = [l]
    using p.fri-commit-preserves-prefix[OF outcome] by simp
  then show ?thesis
    using lens by (cases ls) auto
qed
from fri-commit-initial-replay-successor-data[OF outcome]
obtain roots challenges where
  successors:
     $\forall j < n.$ 
     $p.\text{next-fri-layer } (ps ! j) (ds ! j) (challenges ! j) =$ 
     $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 
  by blast
have layer-lens:
   $\bigwedge j. j < n \implies \text{length } (ds ! j) = \text{length } (ls ! j)$ 
proof -
  fix j
  assume j-lt: j < n
  have ls ! j = map (poly (ps ! j)) (ds ! j)
    by (rule layers) (use j-lt lens in simp-all)
  then show length (ds ! j) = length (ls ! j)
    by simp
qed
have init-len': length (ls ! 0) =  $2 \wedge N$ 
  using ls0 init-len by simp
have successors':
   $\bigwedge j. j < n \implies$ 
   $p.\text{next-fri-layer } (ps ! j) (ds ! j) (challenges ! j) =$ 
   $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 
  using successors by simp
show ?thesis
  by (rule fri-successor-layer-lengths-are-powers[
    where ps=ps and ds=ds and ls=ls and challenges=challenges,
    OF init-len' n-le-N layer-lens successors' j-bound])
qed

```

lemma *fri-commit-initial-layer-length-power:*

```

assumes outcome:
  Some ((ps, ds, ls, ms), t) ∈
  set-dist (execute
    (p.fri-commit n [cp] [d] [l] [m]) s)
and layers:
   $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
   $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
and init-len: length l =  $2 \wedge N$ 
and n-le-N: n ≤ N

```

and $j\text{-le}: j \leq n$
 shows $\text{length } (ls ! j) = 2^{\wedge} (N - j)$
proof –
 have $lens$:
 $\text{length } ps = \text{Suc } n$
 $\text{length } ds = \text{Suc } n$
 $\text{length } ls = \text{Suc } n$
 $\text{length } ms = \text{Suc } n$
 using $p.\text{fri-commit-lengths}[OF \text{ outcome}]$ by simp-all
 have $ls0: ls ! 0 = l$
 using $\text{fri-commit-initial-heads}(\mathcal{J})[OF \text{ outcome}]$.
 from $\text{fri-commit-initial-replay-successor-data}[OF \text{ outcome}]$
 obtain $roots \ challenges$ where
 $successors$:
 $\forall j < n.$
 $p.\text{next-fri-layer } (ps ! j) (ds ! j) (challenges ! j) =$
 $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$
 by blast
 have $layer\text{-}lens$:
 $\bigwedge j. j < n \implies \text{length } (ds ! j) = \text{length } (ls ! j)$
proof –
 fix j
 assume $j\text{-lt}: j < n$
 have $ls ! j = \text{map } (\text{poly } (ps ! j)) (ds ! j)$
 by (rule $layers$) (use $j\text{-lt lens}$ in simp-all)
 then show $\text{length } (ds ! j) = \text{length } (ls ! j)$
 by simp
 qed
 have $\text{init-len}' : \text{length } (ls ! 0) = 2^{\wedge} N$
 using $ls0 \text{ init-len}$ by simp
 have $successors'$:
 $\bigwedge j. j < n \implies$
 $p.\text{next-fri-layer } (ps ! j) (ds ! j) (challenges ! j) =$
 $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$
 using $successors$ by simp
 show $?thesis$
 by (rule $\text{fri-successor-layer-lengths-power}[$
 $OF \text{ init-len}' n\text{-le-}N \text{ layer-lens successors' } j\text{-le}]$)
 qed

lemma $\text{fri-commit-initial-domain-points}$:
 assumes $outcome$:
 $\text{Some } ((ps, ds, ls, ms), t) \in$
 $\text{set-dist } (\text{execute}$
 $(p.\text{fri-commit } n [cp] [p.\text{eval-domain}] [l] [m]) s)$
 and $j\text{-le}: j \leq n$
 and $k\text{-bound}: k < \text{length } (ds ! j)$
 shows $ds ! j ! k = ((p.h \wedge k) * \text{shift})^{\wedge} (2^{\wedge} j)$
proof –


```

have  $ds0$ :  $ds ! 0 = p.eval-domain$ 
  using  $fri-commit-initial-heads(2)[OF\ outcome]$  .
from  $fri-commit-initial-replay-successor-data[OF\ outcome]$ 
obtain  $roots\ challenges$  where
   $successors$ :
     $\bigwedge j. j < n \implies$ 
       $p.next-fri-layer\ (ps ! j)\ (ds ! j)\ (challenges ! j) =$ 
         $(ps ! Suc\ j, ds ! Suc\ j, ls ! Suc\ j)$ 
  by  $blast$ 
show  $?thesis$ 
  by ( $rule\ fri-successor-domain-points[$ 
     $OF\ ds0\ successors\ j-le\ k-bound]$ )
qed

lemma  $trace-fri-commit-initial-layer-length-power$ :
assumes  $outcome$ :
   $Some\ ((ps, ds, ls, ms), t) \in$ 
     $set-dist\ (execute$ 
       $(p.trace-fri-commit\ n\ [cp]\ [d]\ [l]\ [m])\ s)$ 
and  $layers$ :
   $\bigwedge i. i < length\ ps \implies i < length\ ds \implies i < length\ ls \implies$ 
     $ls ! i = map\ (poly\ (ps ! i))\ (ds ! i)$ 
and  $init-len$ :  $length\ l = 2^N$ 
and  $n-le-N$ :  $n \leq N$ 
and  $j-le$ :  $j \leq n$ 
shows  $length\ (ls ! j) = 2^{(N - j)}$ 
proof –
have  $lens$ :
   $length\ ps = Suc\ n$ 
   $length\ ds = Suc\ n$ 
   $length\ ls = Suc\ n$ 
   $length\ ms = Suc\ n$ 
  using  $p.trace-fri-commit-lengths[OF\ outcome]$  by  $simp-all$ 
have  $ls0$ :  $ls ! 0 = l$ 
  using  $trace-fri-commit-initial-heads(3)[OF\ outcome]$  .
from  $trace-fri-commit-initial-replay-successor-data[OF\ outcome]$ 
obtain  $roots\ challenges$  where
   $successors$ :
     $\forall j < n.$ 
       $p.next-fri-layer\ (ps ! j)\ (ds ! j)\ (challenges ! j) =$ 
         $(ps ! Suc\ j, ds ! Suc\ j, ls ! Suc\ j)$ 
  by  $blast$ 
have  $layer-lens$ :
   $\bigwedge j. j < n \implies length\ (ds ! j) = length\ (ls ! j)$ 
proof –
  fix  $j$ 
  assume  $j-lt$ :  $j < n$ 
  have  $ls ! j = map\ (poly\ (ps ! j))\ (ds ! j)$ 
  by ( $rule\ layers$ ) ( $use\ j-lt\ lens\ in\ simp-all$ )

```

```

    then show  $\text{length } (ds ! j) = \text{length } (ls ! j)$ 
      by simp
  qed
  have  $\text{init-len}'$ :  $\text{length } (ls ! 0) = 2 \wedge N$ 
    using  $ls0$   $\text{init-len}$  by simp
  have  $\text{successors}'$ :
     $\bigwedge j. j < n \implies$ 
       $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$ 
       $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 
    using  $\text{successors}$  by simp
  show ?thesis
    by (rule  $\text{fri-successor-layer-lengths-power}$ [
       $OF$   $\text{init-len}'$   $n\text{-le-}N$   $\text{layer-lens}$   $\text{successors}'$   $j\text{-le}$ ])
  qed

lemma  $\text{composition-fri-commit-initial-layer-length-power}$ :
  assumes  $\text{outcome}$ :
     $\text{Some } ((ps, ds, ls, ms), t) \in$ 
     $\text{set-dist } (\text{execute}$ 
       $(p.\text{composition-fri-commit } n [cp] [d] [l] [m]) s)$ 
  and  $\text{layers}$ :
     $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
     $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
  and  $\text{init-len}$ :  $\text{length } l = 2 \wedge N$ 
  and  $n\text{-le-}N$ :  $n \leq N$ 
  and  $j\text{-le}$ :  $j \leq n$ 
  shows  $\text{length } (ls ! j) = 2 \wedge (N - j)$ 
proof –
  have  $\text{lens}$ :
     $\text{length } ps = \text{Suc } n$ 
     $\text{length } ds = \text{Suc } n$ 
     $\text{length } ls = \text{Suc } n$ 
     $\text{length } ms = \text{Suc } n$ 
  using  $p.\text{composition-fri-commit-lengths}[OF \text{outcome}]$  by  $\text{simp-all}$ 
  have  $ls0$ :  $ls ! 0 = l$ 
  using  $\text{composition-fri-commit-initial-heads}(\mathcal{J})[OF \text{outcome}]$  .
  from  $\text{composition-fri-commit-initial-replay-successor-data}[OF \text{outcome}]$ 
  obtain  $\text{roots}$   $\text{challenges}$  where
     $\text{successors}$ :
       $\forall j < n.$ 
       $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$ 
       $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 
  by  $\text{blast}$ 
  have  $\text{layer-lens}$ :
     $\bigwedge j. j < n \implies \text{length } (ds ! j) = \text{length } (ls ! j)$ 
proof –
  fix  $j$ 
  assume  $j\text{-lt}$ :  $j < n$ 
  have  $ls ! j = \text{map } (\text{poly } (ps ! j)) (ds ! j)$ 

```

```

    by (rule layers) (use j-lt lens in simp-all)
  then show length (ds ! j) = length (ls ! j)
    by simp
qed
have init-len': length (ls ! 0) = 2 ^ N
  using ls0 init-len by simp
have successors':
   $\bigwedge j. j < n \implies$ 
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j)
  using successors by simp
show ?thesis
  by (rule fri-successor-layer-lengths-power[
    OF init-len' n-le-N layer-lens successors' j-le])
qed

lemma fri-commit-outcome-initial-layer-length-power:
  assumes outcome:
    fri-commit-outcome n cp d l m ps ds ls ms s t
  and layers:
     $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
      ls ! i = map (poly (ps ! i)) (ds ! i)
  and init-len: length l = 2 ^ N
  and n-le-N: n ≤ N
  and j-le: j ≤ n
  shows length (ls ! j) = 2 ^ (N - j)
proof -
  from outcome consider
    (old) Some ((ps, ds, ls, ms), t) ∈
      set-dist (execute (p.fri-commit n [cp] [d] [l] [m]) s)
  | (trace) Some ((ps, ds, ls, ms), t) ∈
      set-dist (execute (p.trace-fri-commit n [cp] [d] [l] [m]) s)
  | (composition) Some ((ps, ds, ls, ms), t) ∈
      set-dist (execute (p.composition-fri-commit n [cp] [d] [l] [m]) s)
  unfolding fri-commit-outcome-def by blast
then show ?thesis
proof cases
  case old
  show ?thesis
    by (rule fri-commit-initial-layer-length-power[
      OF old layers init-len n-le-N j-le])
next
  case trace
  show ?thesis
    by (rule trace-fri-commit-initial-layer-length-power[
      OF trace layers init-len n-le-N j-le])
next
  case composition
  show ?thesis

```

by (rule composition-fri-commit-initial-layer-length-power[
 OF composition layers init-len n-le-N j-le])
 qed
 qed
lemma trace-fri-commit-initial-domain-points:
 assumes outcome:
 Some ((ps, ds, ls, ms), t) ∈
 set-dist (execute
 (p.trace-fri-commit n [cp] [p.eval-domain] [l] [m]) s)
 and j-le: $j \leq n$
 and k-bound: $k < \text{length } (ds ! j)$
 shows $ds ! j ! k = ((p.h \wedge k) * \text{shift}) \wedge (2 \wedge j)$
proof –
 have ds0: $ds ! 0 = p.\text{eval-domain}$
 using trace-fri-commit-initial-heads(2)[OF outcome] .
from trace-fri-commit-initial-replay-successor-data[OF outcome]
obtain roots challenges **where**
 successors:
 $\bigwedge j. j < n \implies$
 $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$
 $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$
 by blast
show ?thesis
 by (rule fri-successor-domain-points[
 OF ds0 successors j-le k-bound])
 qed

lemma composition-fri-commit-initial-domain-points:
 assumes outcome:
 Some ((ps, ds, ls, ms), t) ∈
 set-dist (execute
 (p.composition-fri-commit n [cp] [p.eval-domain] [l] [m]) s)
 and j-le: $j \leq n$
 and k-bound: $k < \text{length } (ds ! j)$
 shows $ds ! j ! k = ((p.h \wedge k) * \text{shift}) \wedge (2 \wedge j)$
proof –
 have ds0: $ds ! 0 = p.\text{eval-domain}$
 using composition-fri-commit-initial-heads(2)[OF outcome] .
from composition-fri-commit-initial-replay-successor-data[OF outcome]
obtain roots challenges **where**
 successors:
 $\bigwedge j. j < n \implies$
 $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$
 $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$
 by blast
show ?thesis
 by (rule fri-successor-domain-points[
 OF ds0 successors j-le k-bound])

qed

lemma *fri-commit-outcome-initial-domain-points*:

assumes *outcome*:

fri-commit-outcome n cp $p.eval-domain$ l m ps ds ls ms s t

and *j-le*: $j \leq n$

and *k-bound*: $k < length\ (ds \ ! \ j)$

shows $ds \ ! \ j \ ! \ k = ((p.h \wedge k) * shift) \wedge (2 \wedge j)$

proof –

from *outcome* **consider**

(*old*) *Some* $((ps, ds, ls, ms), t) \in$

set-dist (*execute* ($p.fri-commit\ n\ [cp]\ [p.eval-domain]\ [l]\ [m]\ s$))

| (*trace*) *Some* $((ps, ds, ls, ms), t) \in$

set-dist (*execute* ($p.trace-fri-commit\ n\ [cp]\ [p.eval-domain]\ [l]\ [m]\ s$))

| (*composition*) *Some* $((ps, ds, ls, ms), t) \in$

set-dist (*execute*

$(p.composition-fri-commit\ n\ [cp]\ [p.eval-domain]\ [l]\ [m]\ s)$)

unfolding *fri-commit-outcome-def* **by** *blast*

then show *?thesis*

proof *cases*

case *old*

show *?thesis*

by (*rule* *fri-commit-initial-domain-points*[*OF old j-le k-bound*])

next

case *trace*

show *?thesis*

by (*rule* *trace-fri-commit-initial-domain-points*[*OF trace j-le k-bound*])

next

case *composition*

show *?thesis*

by (*rule* *composition-fri-commit-initial-domain-points*[
OF composition j-le k-bound])

qed

qed

lemma *fri-degree-decreases*:

assumes *degree* $q > 0$

and $p.next-fri-layer\ q\ d\ b = (q', d', l')$

shows *degree* $q' < degree\ q$

using *assms* $p.degree-less-degree\ next-fri-layer-components(1)$ **by** *blast*

lemma *degree-poly-of-list-less-length*:

assumes $xs \neq []$

shows *degree* (*poly-of-list* xs) $< length\ xs$

proof –

have *degree* (*poly-of-list* xs) $= length\ (coeffs\ (poly-of-list\ xs)) - 1$

by (*simp add: degree-eq-length-coeffs*)

also have $\dots \leq length\ xs - 1$

proof –

```

    have length (coeffs (poly-of-list xs)) ≤ length xs
      by (simp add: coeffs-Poly length-strip-while-le)
    with assms show ?thesis by linarith
  qed
  also have ... < length xs
    using assms by simp
  finally show ?thesis .
qed

lemma degree-nths-even-coeffs-le-half:
  degree (poly-of-list (nths-pred (coeffs q) even)) ≤ degree q div 2
proof (cases coeffs q = [])
  case True
  then show ?thesis by simp
next
  case False
  let ?xs = nths-pred (coeffs q) even
  have len: length ?xs = Suc (length (coeffs q)) div 2
    by (rule nths-pred-even-length-eq)
  show ?thesis
  proof (cases ?xs = [])
    case True
    then show ?thesis by simp
  next
    case xs-ne: False
    have dlt: degree (poly-of-list ?xs) < length ?xs
      using xs-ne by (rule degree-poly-of-list-less-length)
    have degree (poly-of-list ?xs) < Suc (length (coeffs q)) div 2
      using dlt len by simp
    then have degree (poly-of-list ?xs) ≤ (length (coeffs q) - 1) div 2
      by linarith
    then show ?thesis
      by (simp add: degree-eq-length-coeffs)
  qed
qed

lemma degree-nths-odd-coeffs-le-half:
  degree (poly-of-list (nths-pred (coeffs q) odd)) ≤ degree q div 2
proof (cases coeffs q = [])
  case True
  then show ?thesis by simp
next
  case False
  let ?xs = nths-pred (coeffs q) odd
  have len: length ?xs = length (coeffs q) div 2
    by (rule nths-pred-odd-length-eq)
  show ?thesis
  proof (cases ?xs = [])
    case True

```

```

    then show ?thesis by simp
next
case xs-ne: False
have dlt: degree (poly-of-list ?xs) < length ?xs
  using xs-ne by (rule degree-poly-of-list-less-length)
then have degree (poly-of-list ?xs) < Suc (length (coeffs q)) div 2
  using len by linarith
then have degree (poly-of-list ?xs) ≤ (length (coeffs q) - 1) div 2
  by linarith
then show ?thesis
  by (simp add: degree-eq-length-coeffs)
qed
qed

lemma fri-degree-halves-polynomial:
  degree (p.next-fri-polynomial q b) ≤ degree q div 2
proof -
  define odd-coefficients where odd-coefficients = nth-pred (coeffs q) odd
  define even-coefficients where even-coefficients = nth-pred (coeffs q) even
  define oddp where oddp = CP b * poly-of-list odd-coefficients
  define evenp where evenp = poly-of-list even-coefficients
  have odd-le: degree oddp ≤ degree q div 2
  proof -
    have degree oddp ≤ degree (CP b) + degree (poly-of-list odd-coefficients)
      unfolding oddp-def by (rule degree-mult-le)
    also have ... ≤ degree q div 2
      unfolding CP-def odd-coefficients-def
      using degree-nth-odd-coeffs-le-half[of q]
      by (simp add: monom-0)
    finally show ?thesis .
  qed
  have even-le: degree evenp ≤ degree q div 2
    unfolding evenp-def even-coefficients-def
    by (rule degree-nth-even-coeffs-le-half)
  have degree (p.next-fri-polynomial q b) = degree (oddp + evenp)
    unfolding p.next-fri-polynomial-def oddp-def evenp-def
    odd-coefficients-def even-coefficients-def
    by (simp add: Let-def)
  also have ... ≤ degree q div 2
    using odd-le even-le by (rule degree-add-le)
  finally show ?thesis .
qed

lemma fri-degree-halves-layer:
  assumes p.next-fri-layer q d b = (q', d', l')
  shows degree q' ≤ degree q div 2
  using next-fri-layer-components(1)[OF assms] fri-degree-halves-polynomial by simp

lemma ceil-log-Suc-power-gt:

```

```

     $d < 2^{\text{ceil-log } (\text{Suc } d)}$ 
proof (cases d)
  case 0
  then show ?thesis
    unfolding ceil-log-def by simp
next
  case (Suc d')
  then show ?thesis
    using floor-log-exp2-gt[of d]
    unfolding ceil-log-def
    by simp
qed

lemma trace-fri-rounds-enough:
   $\text{ceil-log } (\text{degree } p.f + 1) \leq \text{ceil-log } \text{clength}$ 
proof -
  have  $\text{degree } p.f + 1 \leq \text{clength}$ 
    using trace-polynomial-degree-lt-clength by simp
  then show ?thesis
    using ceil-log-mono by blast
qed

lemma div-power-two-ceil-log-Suc:
   $d \text{ div } 2^{\text{ceil-log } (\text{Suc } d)} = 0$ 
  using ceil-log-Suc-power-gt[of d] by simp

lemma div-div-two-power:
  fixes  $m \ n :: \text{nat}$ 
  shows  $(m \text{ div } 2^n) \text{ div } 2 = m \text{ div } 2^{\text{Suc } n}$ 
proof -
  have div-mult:
     $m \text{ div } (2^n * 2) = (m \text{ div } 2^n) \text{ div } 2$ 
    using div-mult2-eq'[of m  $2^n$  2] by simp
  have  $(m \text{ div } 2^n) \text{ div } 2 = m \text{ div } (2^n * 2)$ 
    using div-mult by simp
  also have ... =  $m \text{ div } 2^{\text{Suc } n}$ 
    by (simp add: power-Suc mult.commute)
  finally show ?thesis .
qed

lemma fri-degree-chain-halves:
  assumes successors:
     $\bigwedge j. j < n \implies$ 
     $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$ 
     $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 
  and j-le:  $j \leq n$ 
  shows  $\text{degree } (ps ! j) \leq \text{degree } (ps ! 0) \text{ div } 2^j$ 
  using j-le
proof (induction j)

```



```

    case 0
    then show ?case by simp
next
case (Suc j)
have j-lt:  $j < n$ 
  using Suc.premis by simp
have step:
  degree (ps ! Suc j)  $\leq$  degree (ps ! j) div 2
  by (rule fri-degree-halves-layer[OF successors[OF j-lt]])
also have ...  $\leq$  (degree (ps ! 0) div 2  $^j$ ) div 2
  by (rule div-le-mono) (rule Suc.IH, use Suc.premis in simp)
also have ... = degree (ps ! 0) div 2  $^{Suc j}$ 
  by (rule div-div-two-power)
finally show ?case .
qed

lemma fri-degree-final-zero:
  assumes successors:
     $\bigwedge j. j < n \implies$ 
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
    (ps ! Suc j, ds ! Suc j, ls ! Suc j)
    and n-def:  $n = \text{ceil-log } (\text{degree } (ps ! 0) + 1)$ 
  shows degree (ps ! n) = 0
proof -
  have degree (ps ! n)  $\leq$  degree (ps ! 0) div 2  $^n$ 
    by (rule fri-degree-chain-halves[OF successors le-refl])
  also have ... = 0
    using n-def div-power-two-ceil-log-Suc[of degree (ps ! 0)] by simp
  finally show ?thesis by simp
qed

lemma fri-degree-final-zero-ge:
  assumes successors:
     $\bigwedge j. j < n \implies$ 
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
    (ps ! Suc j, ds ! Suc j, ls ! Suc j)
    and enough:  $\text{ceil-log } (\text{degree } (ps ! 0) + 1) \leq n$ 
  shows degree (ps ! n) = 0
proof -
  have degree (ps ! n)  $\leq$  degree (ps ! 0) div 2  $^n$ 
    by (rule fri-degree-chain-halves[OF successors le-refl])
  also have ... = 0
    proof -
      have lt-ceil: degree (ps ! 0)  $< 2^{ceil-log } (\text{degree } (ps ! 0) + 1)$ 
        using ceil-log-Suc-power-gt[of degree (ps ! 0)] by simp
      also have ...  $\leq 2^n$ 
        using enough by simp
      finally show ?thesis
        by simp
    end
qed

```

```

qed
finally show ?thesis by simp
qed

```

```

lemma poly-degree-zero-const:
  assumes degree q = 0
  shows poly q x = poly q y
proof -
  have q-eq: q = [: coeff q 0 :]
    using degree-0-id[OF assms] by simp
  have poly q x = coeff q 0
    by (subst q-eq) simp
  also have ... = poly q y
    by (subst q-eq) simp
  finally show ?thesis .
qed

```

```

lemma fri-final-layer-constant-value:
  assumes degree0: degree (ps ! n) = 0
    and layer: ls ! n = map (poly (ps ! n)) (ds ! n)
    and i-bound: i < length (ls ! n)
    and len-pos: 0 < length (ls ! n)
  shows ls ! n ! i = hd (ls ! n)
proof -
  have ds-len: length (ds ! n) = length (ls ! n)
    using layer by simp
  then have ds-ne: ds ! n ≠ []
    using len-pos by auto
  have ls ! n ! i = poly (ps ! n) (ds ! n ! i)
    using layer i-bound by simp
  also have ... = poly (ps ! n) (hd (ds ! n))
    by (rule poly-degree-zero-const[OF degree0])
  also have ... = hd (map (poly (ps ! n)) (ds ! n))
  proof -
    have hd (map (poly (ps ! n)) (ds ! n)) =
      poly (ps ! n) (hd (ds ! n))
      by (rule hd-map[OF ds-ne])
    then show ?thesis by simp
  qed
  also have ... = hd (ls ! n)
    using layer by simp
  finally show ?thesis .
qed

```

```

lemma honest-fri-final-layer-equals-sent:
  assumes fri:
    Some ((ps, ds, ls, ms), s6) ∈
      set-dist (execute
        (p.fri-commit nrounds [cp] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)

```

```

and len-ps: length ps = Suc nrounds
and len-ds: length ds = Suc nrounds
and len-ls: length ls = Suc nrounds
and layers:
   $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
   $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
and init-len: length (p.cp-eval as) =  $2^N$ 
and n-le-N: nrounds  $\leq N$ 
and nrounds-def: nrounds = ceil-log (degree cp' + 1)
and successors:
   $\bigwedge j. j < \text{nrounds} \implies$ 
   $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$ 
   $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 
and i-bound:  $i < \text{length } (ls ! \text{nrounds})$ 
shows ls ! nrounds ! i = hd (last ls)
proof –
  have ps0: ps ! 0 = cp'
    using fri-commit-initial-heads(1)[OF fri] .
  have n-def: nrounds = ceil-log (degree (ps ! 0) + 1)
    using nrounds-def ps0 by simp
  have degree0: degree (ps ! nrounds) = 0
    by (rule fri-degree-final-zero[OF successors n-def])
  have layer-n: ls ! nrounds = map (poly (ps ! nrounds)) (ds ! nrounds)
    by (rule layers) (use len-ps len-ds len-ls in simp-all)
  have len-pos: 0 < length (ls ! nrounds)
    using fri-commit-initial-layer-length-power[OF fri layers init-len n-le-N, of nrounds]
    by simp
  have ls ! nrounds ! i = hd (ls ! nrounds)
    by (rule fri-final-layer-constant-value[OF degree0 layer-n i-bound len-pos])
  also have ... = hd (last ls)
proof –
  have ls-ne: ls  $\neq []$ 
    using len-ls by auto
  have last ls = ls ! nrounds
    using last-conv-nth[OF ls-ne] len-ls by simp
  then show ?thesis by simp
qed
finally show ?thesis .
qed

```

lemma honest-fri-final-layer-equals-sent-initial:

assumes fri:

Some ((ps, ds, ls, ms), s6) $\in$

set-dist (execute

(p.fri-commit nrounds [cp] [p.eval-domain] [l0] [m0]) s5)

and len-ps: length ps = Suc nrounds

and len-ds: length ds = Suc nrounds

and len-ls: length ls = Suc nrounds

and layers:

```

     $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
     $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
  and init-len:  $\text{length } l0 = 2 \wedge N$ 
  and n-le-N:  $nrounds \leq N$ 
  and nrounds-enough:  $\text{ceil-log } (\text{degree } cp + 1) \leq nrounds$ 
  and successors:
     $\bigwedge j. j < nrounds \implies$ 
     $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$ 
     $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 
  and i-bound:  $i < \text{length } (ls ! nrounds)$ 
  shows  $ls ! nrounds ! i = \text{hd } (\text{last } ls)$ 
proof -
  have ps0:  $ps ! 0 = cp$ 
    using fri-commit-initial-heads(1)[OF fri] .
  have n-enough:  $\text{ceil-log } (\text{degree } (ps ! 0) + 1) \leq nrounds$ 
    using nrounds-enough ps0 by simp
  have degree0:  $\text{degree } (ps ! nrounds) = 0$ 
    by (rule fri-degree-final-zero-ge[OF successors n-enough])
  have layer-n:  $ls ! nrounds = \text{map } (\text{poly } (ps ! nrounds)) (ds ! nrounds)$ 
    by (rule layers) (use len-ps len-ds len-ls in simp-all)
  have len-pos:  $0 < \text{length } (ls ! nrounds)$ 
    using fri-commit-initial-layer-length-power[OF fri layers init-len n-le-N, of nrounds]
    by simp
  have  $ls ! nrounds ! i = \text{hd } (ls ! nrounds)$ 
    by (rule fri-final-layer-constant-value[OF degree0 layer-n i-bound len-pos])
  also have  $\dots = \text{hd } (\text{last } ls)$ 
proof -
  have ls-ne:  $ls \neq []$ 
    using len-ls by auto
  have  $\text{last } ls = ls ! nrounds$ 
    using last-conv-nth[OF ls-ne] len-ls by simp
  then show ?thesis by simp
qed
finally show ?thesis .
qed

```

lemma *honest-fri-final-layer-equals-sent-shape*:

```

  assumes fri:
    fri-commit-outcome nrounds cp p.eval-domain l0 m0 ps ds ls ms s5 s6
  and len-ps:  $\text{length } ps = \text{Suc } nrounds$ 
  and len-ds:  $\text{length } ds = \text{Suc } nrounds$ 
  and len-ls:  $\text{length } ls = \text{Suc } nrounds$ 
  and layers:
     $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
     $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
  and init-len:  $\text{length } l0 = 2 \wedge N$ 
  and n-le-N:  $nrounds \leq N$ 
  and nrounds-def:  $nrounds = \text{ceil-log } (\text{degree } cp + 1)$ 
  and successors:

```

```

 $\bigwedge j. j < nrounds \implies$ 
   $p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =$ 
     $(ps ! Suc\ j, ds ! Suc\ j, ls ! Suc\ j)$ 
  and  $i-bound: i < length\ (ls ! nrounds)$ 
  shows  $ls ! nrounds ! i = hd\ (last\ ls)$ 
proof -
  have  $ps0: ps ! 0 = cp$ 
    using  $fri-commit-outcome-initial-heads(1)[OF\ fri]$  .
  have  $n-def: nrounds = ceil-log\ (degree\ (ps ! 0) + 1)$ 
    using  $nrounds-def\ ps0$  by  $simp$ 
  have  $degree0: degree\ (ps ! nrounds) = 0$ 
    by  $(rule\ fri-degree-final-zero[OF\ successors\ n-def])$ 
  have  $layer-n: ls ! nrounds = map\ (poly\ (ps ! nrounds))\ (ds ! nrounds)$ 
    by  $(rule\ layers)\ (use\ len-ps\ len-ds\ len-ls\ in\ simp-all)$ 
  have  $len-pos: 0 < length\ (ls ! nrounds)$ 
    using  $fri-commit-outcome-initial-layer-length-power[$ 
       $OF\ fri\ layers\ init-len\ n-le-N,\ of\ nrounds]$ 
    by  $simp$ 
  have  $ls ! nrounds ! i = hd\ (ls ! nrounds)$ 
    by  $(rule\ fri-final-layer-constant-value[OF\ degree0\ layer-n\ i-bound\ len-pos])$ 
  also have  $\dots = hd\ (last\ ls)$ 
proof -
  have  $ls-ne: ls \neq []$ 
    using  $len-ls$  by  $auto$ 
  have  $last\ ls = ls ! nrounds$ 
    using  $last-conv-nth[OF\ ls-ne]\ len-ls$  by  $simp$ 
  then show  $?thesis$  by  $simp$ 
qed
finally show  $?thesis$  .
qed

```

lemma *honest-fri-final-layer-equals-sent-initial-shape:*

assumes *fri:*

fri-commit-outcome nrounds cp p.eval-domain l0 m0 ps ds ls ms s5 s6

and *len-ps: length ps = Suc nrounds*

and *len-ds: length ds = Suc nrounds*

and *len-ls: length ls = Suc nrounds*

and *layers:*

$\bigwedge i. i < length\ ps \implies i < length\ ds \implies i < length\ ls \implies$

$ls ! i = map\ (poly\ (ps ! i))\ (ds ! i)$

and *init-len: length l0 = 2 ^ N*

and *n-le-N: nrounds ≤ N*

and *nrounds-enough: ceil-log (degree cp + 1) ≤ nrounds*

and *successors:*

$\bigwedge j. j < nrounds \implies$

$p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =$

$(ps ! Suc\ j, ds ! Suc\ j, ls ! Suc\ j)$

and *i-bound: i < length (ls ! nrounds)*

shows $ls ! nrounds ! i = hd\ (last\ ls)$

```

proof –
  have ps0: ps ! 0 = cp
    using fri-commit-outcome-initial-heads(1)[OF fri] .
  have n-enough: ceil-log (degree (ps ! 0) + 1) ≤ nrounds
    using nrounds-enough ps0 by simp
  have degree0: degree (ps ! nrounds) = 0
    by (rule fri-degree-final-zero-ge[OF successors n-enough])
  have layer-n: ls ! nrounds = map (poly (ps ! nrounds)) (ds ! nrounds)
    by (rule layers) (use len-ps len-ds len-ls in simp-all)
  have len-pos: 0 < length (ls ! nrounds)
    using fri-commit-outcome-initial-layer-length-power[
      OF fri layers init-len n-le-N, of nrounds]
    by simp
  have ls ! nrounds ! i = hd (ls ! nrounds)
    by (rule fri-final-layer-constant-value[OF degree0 layer-n i-bound len-pos])
  also have ... = hd (last ls)
  proof –
    have ls-ne: ls ≠ []
      using len-ls by auto
    have last ls = ls ! nrounds
      using last-conv-nth[OF ls-ne] len-ls by simp
    then show ?thesis by simp
  qed
  finally show ?thesis .
qed

```

```

lemma poly-of-list-even-odd-decompose:
  fixes xs :: 'f list
    and x :: 'f
  shows
    poly (poly-of-list xs) x =
      poly (poly-of-list (nths-pred xs even)) (x * x) +
      x * poly (poly-of-list (nths-pred xs odd)) (x * x)
proof (induction xs rule: measure-induct-rule[of length])
  case (less xs)
  show ?case
  proof (cases xs)
    case Nil
    then show ?thesis by simp
  next
    case (Cons a ys)
    note xs-eq = Cons
    show ?thesis
    proof (cases ys)
      case Nil
      then show ?thesis
        unfolding xs-eq by (simp add: nths-Cons)
    next
      case (Cons b zs)

```

```

note ys-eq = Cons
have even-tail:
  nths-pred (a # b # zs) even = a # nths-pred zs even
  unfolding nths-pred-def by (simp add: nths-Cons)
have odd-tail:
  nths-pred (a # b # zs) odd = b # nths-pred zs odd
  unfolding nths-pred-def by (simp add: nths-Cons)
have zs-less: length zs < length xs
  unfolding xs-eq ys-eq by simp
have poly-tail:
  poly (poly-of-list zs) x =
    poly (poly-of-list (nths-pred zs even)) (x * x) +
      x * poly (poly-of-list (nths-pred zs odd)) (x * x)
  by (rule less.IH[OF zs-less])
show ?thesis
  unfolding xs-eq ys-eq even-tail odd-tail
  using poly-tail
  by (simp add: algebra-simps power2-eq-square)
qed
qed
qed

lemma poly-even-part-reconstruct:
  fixes q :: 'f poly
  and x :: 'f
  assumes two-nonzero: (2::'f) ≠ 0
  shows
    poly (poly-of-list (nths-pred (coeffs q) even)) (x * x) =
      (poly q x + poly q (-x)) div 2
proof -
  have decomp-x:
    poly q x =
      poly (poly-of-list (nths-pred (coeffs q) even)) (x * x) +
        x * poly (poly-of-list (nths-pred (coeffs q) odd)) (x * x)
    using poly-of-list-even-odd-decompose[of coeffs q x]
    by simp
  have decomp-neg:
    poly q (-x) =
      poly (poly-of-list (nths-pred (coeffs q) even)) (x * x) -
        x * poly (poly-of-list (nths-pred (coeffs q) odd)) (x * x)
    using poly-of-list-even-odd-decompose[of coeffs q -x]
    by (simp add: algebra-simps power2-eq-square)
  show ?thesis
    using decomp-x decomp-neg two-nonzero
    by (simp add: field-simps)
qed

lemma poly-odd-part-reconstruct:
  fixes q :: 'f poly

```

```

and  $x :: 'f$ 
assumes  $two\text{-}nonzero: (2::'f) \neq 0$ 
and  $x\text{-}nonzero: x \neq 0$ 
shows  $poly (poly\text{-}of\text{-}list (nth\text{-}pred (coeffs\ q) odd)) (x * x) =$ 
 $(poly\ q\ x - poly\ q\ (-x))\ div\ (2 * x)$ 
proof -
have  $decomp\text{-}x:$ 
 $poly\ q\ x =$ 
 $poly (poly\text{-}of\text{-}list (nth\text{-}pred (coeffs\ q) even)) (x * x) +$ 
 $x * poly (poly\text{-}of\text{-}list (nth\text{-}pred (coeffs\ q) odd)) (x * x)$ 
using  $poly\text{-}of\text{-}list\text{-}even\text{-}odd\text{-}decompose[of\ coeffs\ q\ x]$ 
by  $simp$ 
have  $decomp\text{-}neg:$ 
 $poly\ q\ (-x) =$ 
 $poly (poly\text{-}of\text{-}list (nth\text{-}pred (coeffs\ q) even)) (x * x) -$ 
 $x * poly (poly\text{-}of\text{-}list (nth\text{-}pred (coeffs\ q) odd)) (x * x)$ 
using  $poly\text{-}of\text{-}list\text{-}even\text{-}odd\text{-}decompose[of\ coeffs\ q\ -x]$ 
by  $(simp\ add: algebra\text{-}simps\ power2\text{-}eq\text{-}square)$ 
show  $?thesis$ 
using  $decomp\text{-}x\ decomp\text{-}neg\ two\text{-}nonzero\ x\text{-}nonzero$ 
by  $(simp\ add: field\text{-}simps\ mult.\text{commute})$ 
qed

```

```

lemma  $next\text{-}fri\text{-}polynomial\text{-}fold:$ 
fixes  $q :: 'f\ poly$ 
and  $b\ z :: 'f$ 
assumes  $two\text{-}nonzero: (2::'f) \neq 0$ 
and  $z\text{-}nonzero: z \neq 0$ 
shows
 $((poly\ q\ z + poly\ q\ (-z))\ div\ 2 +$ 
 $b * ((poly\ q\ z - poly\ q\ (-z))\ div\ (2 * z))) =$ 
 $poly\ (p.\text{next}\text{-}fri\text{-}polynomial\ q\ b)\ (z * z)$ 
proof -
let  $?E = poly (poly\text{-}of\text{-}list (nth\text{-}pred (coeffs\ q) even)) (z * z)$ 
let  $?O = poly (poly\text{-}of\text{-}list (nth\text{-}pred (coeffs\ q) odd)) (z * z)$ 
have  $even:$ 
 $?E = (poly\ q\ z + poly\ q\ (-z))\ div\ 2$ 
by  $(rule\ poly\text{-}even\text{-}part\text{-}reconstruct[OF\ two\text{-}nonzero])$ 
have  $odd:$ 
 $?O = (poly\ q\ z - poly\ q\ (-z))\ div\ (2 * z)$ 
by  $(rule\ poly\text{-}odd\text{-}part\text{-}reconstruct[OF\ two\text{-}nonzero\ z\text{-}nonzero])$ 
have  $next\text{-}poly\text{-}eval:$ 
 $poly\ (p.\text{next}\text{-}fri\text{-}polynomial\ q\ b)\ (z * z) = b * ?O + ?E$ 
unfolding  $p.\text{next}\text{-}fri\text{-}polynomial\text{-}def\ CP\text{-}def$ 
by  $(simp\ add: Let\text{-}def\ monom\text{-}0\ algebra\text{-}simps)$ 
have  $lhs:$ 
 $((poly\ q\ z + poly\ q\ (-z))\ div\ 2 +$ 
 $b * ((poly\ q\ z - poly\ q\ (-z))\ div\ (2 * z))) =$ 
 $?E + b * ?O$ 

```



```

    using even odd by simp
  also have ... = poly (p.next-fri-polynomial q b) (z * z)
    using next-poly-eval by (simp add: algebra-simps)
  finally show ?thesis .
qed

```

```

lemma fri-fold-value-poly-next:
  fixes q :: 'f poly
  and b z :: 'f
  assumes two-nonzero: (2::'f) ≠ 0
  and z-nonzero: z ≠ 0
  and len-eq: len = length l
  and l-eval: l = map (poly q) d
  and d-len: length d = length l
  and i-bound: i < len
  and z-eq: z = ((p.h ^ i) * shift) ^ pw
  and d-i: d ! i = z
  and d-sib: d ! ((i + len div 2) mod len) = - z
  shows fri-fold-value b l i len pw =
    poly (p.next-fri-polynomial q b) (z * z)
proof -
  let ?sidx = (i + len div 2) mod len
  have len-pos: 0 < len
  using i-bound by simp
  have i-d-bound: i < length d
  using i-bound len-eq d-len by simp
  have sib-l-bound: ?sidx < length l
  using len-pos len-eq by simp
  have sib-d-bound: ?sidx < length d
  using sib-l-bound d-len by simp
  have l-i: l ! i = poly q z
  using l-eval d-i i-d-bound by simp
  have l-sib: l ! ?sidx = poly q (- z)
  using l-eval d-sib sib-d-bound by simp
  show ?thesis
  unfolding fri-fold-value-def
  using l-i l-sib z-eq next-fri-polynomial-fold[OF two-nonzero z-nonzero, of q b]
  by simp
qed

```

```

lemma fri-fold-value-next-layer-value:
  fixes q :: 'f poly
  and b z :: 'f
  assumes two-nonzero: (2::'f) ≠ 0
  and z-nonzero: z ≠ 0
  and layer: p.next-fri-layer q d b = (q', d', l')
  and len-eq: len = length l
  and l-eval: l = map (poly q) d
  and d-len: length d = length l

```

```

and i-bound:  $i < \text{len}$ 
and z-eq:  $z = ((p.h \wedge i) * \text{shift}) \wedge pw$ 
and d-i:  $d ! i = z$ 
and d-sib:  $d ! ((i + \text{len div } 2) \bmod \text{len}) = -z$ 
and next-idx-bound:  $i \bmod (\text{len div } 2) < \text{length } d'$ 
and d-next:  $d' ! (i \bmod (\text{len div } 2)) = z * z$ 
shows fri-fold-value  $b \ l \ i \ \text{len} \ pw = l' ! (i \bmod (\text{len div } 2))$ 
proof –
  have fold:
    fri-fold-value  $b \ l \ i \ \text{len} \ pw =$ 
       $\text{poly } (p.\text{next-fri-polynomial } q \ b) (z * z)$ 
    by (rule fri-fold-value-poly-next[
      OF two-nonzero z-nonzero len-eq l-eval d-len i-bound z-eq d-i d-sib])
  have q'-eq:  $q' = p.\text{next-fri-polynomial } q \ b$ 
    using next-fri-layer-components(1)[OF layer] .
  have l'-eq:  $l' = \text{map } (\text{poly } q') \ d'$ 
    using next-fri-layer-components(3)[OF layer] .
  have  $l' ! (i \bmod (\text{len div } 2)) = \text{poly } q' (z * z)$ 
    using l'-eq next-idx-bound d-next by simp
  then show ?thesis
    using fold q'-eq by simp
qed

lemma honest-fri-layer-decommitment-step-created-next-value:
  fixes  $q \ q' :: 'f \ \text{poly}$ 
  and  $d \ d' \ l \ l' :: 'f \ \text{list}$ 
assumes created:  $p.\text{created-tree } l \ m \ s0$ 
  and len-eq:  $\text{len} = \text{length } l$ 
  and len-pow:  $\text{length } l = 2 \wedge n$ 
  and i-bound:  $i < \text{len}$ 
  and x-eq:  $x = l ! i$ 
  and ext:  $s0 \leq s$ 
  and root:  $fr = \text{value } m$ 
  and tr:
    start =
       $s(P\text{Transcript} := \text{fri-layer-decommitment-transcript } l \ m \ i \ \text{len} \ @ \ \text{rest})$ 
  and layer:  $p.\text{next-fri-layer } q \ d \ b = (q', d', l')$ 
  and l-eval:  $l = \text{map } (\text{poly } q) \ d$ 
  and d-len:  $\text{length } d = \text{length } l$ 
  and z-eq:  $z = ((p.h \wedge i) * \text{shift}) \wedge pw$ 
  and d-i:  $d ! i = z$ 
  and d-sib:  $d ! ((i + \text{len div } 2) \bmod \text{len}) = -z$ 
  and next-idx-bound:  $i \bmod (\text{len div } 2) < \text{length } d'$ 
  and d-next:  $d' ! (i \bmod (\text{len div } 2)) = z * z$ 
shows
   $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{fri-verifier-layer } b \ fr \ i \ x \ \text{len} \ pw) \ \text{start}))$ 
   $\wedge \text{res } t'. \text{Some } (\text{res}, t') \in$ 
     $\text{set-dist } (\text{execute } (\text{fri-verifier-layer } b \ fr \ i \ x \ \text{len} \ pw) \ \text{start}) \implies$ 
     $\text{res} =$ 

```

```

      (i mod (len div 2), l' ! (i mod (len div 2)), len div 2, pw + pw) ∧
      PTranscript t' = rest ∧
      start ≤ t'
proof –
  show None ∉ dom (dist (execute (fri-verifier-layer b fr i x len pw) start))
    by (rule honest-fri-layer-decommitment-step-created(1)[
      OF created len-eq len-pow i-bound x-eq ext root tr])
  fix res t'
  assume outcome:
    Some (res, t') ∈
      set-dist (execute (fri-verifier-layer b fr i x len pw) start)
  have step:
    res =
      (i mod (len div 2), fri-fold-value b l i len pw, len div 2, pw + pw) ∧
      PTranscript t' = rest ∧
      start ≤ t'
    by (rule honest-fri-layer-decommitment-step-created(2)[
      OF created len-eq len-pow i-bound x-eq ext root tr outcome])
  have z-nonzero: z ≠ 0
    using z-eq verifier-domain-point-nonzero by simp
  have fold-eq:
    fri-fold-value b l i len pw = l' ! (i mod (len div 2))
    by (rule fri-fold-value-next-layer-value[
      OF p.two-nonzero z-nonzero layer len-eq l-eval d-len i-bound z-eq
      d-i d-sib next-idx-bound d-next])
  show
    res =
      (i mod (len div 2), l' ! (i mod (len div 2)), len div 2, pw + pw) ∧
      PTranscript t' = rest ∧
      start ≤ t'
    using step fold-eq by simp
qed

```

```

lemma honest-fri-layer-decommitment-step-created-next-value-domain:
  fixes q q' :: 'f poly
    and d d' l l' :: 'f list
  assumes created: p.created-tree l m s0
    and len-eq: len = length l
    and len-pow: length l = 2 ^ n
    and i-bound: i < len
    and x-eq: x = l ! i
    and ext: s0 ≤ s
    and root: fr = value m
    and tr:
      start =
        s(PTranscript := fri-layer-decommitment-transcript l m i len @ rest)
  and layer: p.next-fri-layer q d b = (q', d', l')
  and l-eval: l = map (poly q) d
  and d-len: length d = length l

```

```

and even-len: 2 dvd len
and round: len * pw = clength * scale
and d-i-point: d ! i = ((p.h ^ i) * shift) ^ pw
and d-sib-point:
  d ! ((i + len div 2) mod len) =
    ((p.h ^ ((i + len div 2) mod len)) * shift) ^ pw
and next-idx-bound: i mod (len div 2) < length d'
and d-next:
  d' ! (i mod (len div 2)) =
    ((p.h ^ i) * shift) ^ pw * ((p.h ^ i) * shift) ^ pw
shows
  None ∉ dom (dist (execute (fri-verifier-layer b fr i x len pw) start))
  ∧ res t'. Some (res, t') ∈
    set-dist (execute (fri-verifier-layer b fr i x len pw) start) ⇒
    res =
      (i mod (len div 2), l' ! (i mod (len div 2)), len div 2, pw + pw) ∧
      PTranscript t' = rest ∧
      start ≤ t'
proof –
  let ?z = ((p.h ^ i) * shift) ^ pw
  have len-pos: 0 < len
  using i-bound by simp
  have d-sib: d ! ((i + len div 2) mod len) = – ?z
  by (rule fri-sibling-domain-at[OF len-pos even-len round refl d-sib-point])
  show None ∉ dom (dist (execute (fri-verifier-layer b fr i x len pw) start))
  by (rule honest-fri-layer-decommitment-step-created-next-value(1)[
    OF created len-eq len-pow i-bound x-eq ext root tr layer l-eval d-len
    refl d-i-point d-sib next-idx-bound d-next])
  fix res t'
  assume outcome:
    Some (res, t') ∈
      set-dist (execute (fri-verifier-layer b fr i x len pw) start)
  show
    res =
      (i mod (len div 2), l' ! (i mod (len div 2)), len div 2, pw + pw) ∧
      PTranscript t' = rest ∧
      start ≤ t'
    by (rule honest-fri-layer-decommitment-step-created-next-value(2)[
      OF created len-eq len-pow i-bound x-eq ext root tr layer l-eval d-len
      refl d-i-point d-sib next-idx-bound d-next outcome])
qed

lemma honest-fri-layer-decommitment-step-created-next-value-domain-state:
  fixes q q' :: 'f poly
  and d d' l l' :: 'f list
  assumes created: p.created-tree l m s0
  and len-eq: len = length l
  and len-pow: length l = 2 ^ n
  and i-bound: i < len

```

```

and  $x\text{-eq}$ :  $x = l ! i$ 
and  $ext$ :  $s0 \leq s$ 
and  $root$ :  $fr = \text{value } m$ 
and  $tr$ :
   $start =$ 
     $s(PTranscript := \text{fri-layer-decommitment-transcript } l \ m \ i \ len \ @ \ rest)$ 
and  $layer$ :  $p.\text{next-fri-layer } q \ d \ b = (q', d', l')$ 
and  $l\text{-eval}$ :  $l = \text{map } (poly \ q) \ d$ 
and  $d\text{-len}$ :  $\text{length } d = \text{length } l$ 
and  $even\text{-len}$ :  $2 \ \text{dvd} \ len$ 
and  $round$ :  $len * pw = clength * scale$ 
and  $d\text{-i-point}$ :  $d ! i = ((p.h \wedge i) * shift) \wedge pw$ 
and  $d\text{-sib-point}$ :
   $d ! ((i + len \ \text{div} \ 2) \ \text{mod} \ len) =$ 
     $((p.h \wedge ((i + len \ \text{div} \ 2) \ \text{mod} \ len)) * shift) \wedge pw$ 
and  $next\text{-idx-bound}$ :  $i \ \text{mod} \ (len \ \text{div} \ 2) < \text{length } d'$ 
and  $d\text{-next}$ :
   $d' ! (i \ \text{mod} \ (len \ \text{div} \ 2)) =$ 
     $((p.h \wedge i) * shift) \wedge pw * ((p.h \wedge i) * shift) \wedge pw$ 
and  $outcome$ :
   $\text{Some } (res, t') \in$ 
     $\text{set-dist } (\text{execute } (\text{fri-verifier-layer } b \ fr \ i \ x \ len \ pw) \ start)$ 
shows
   $res =$ 
     $(i \ \text{mod} \ (len \ \text{div} \ 2), l' ! (i \ \text{mod} \ (len \ \text{div} \ 2)), len \ \text{div} \ 2, pw + pw) \wedge$ 
     $PTranscript \ t' = rest \wedge$ 
     $start \leq t' \wedge$ 
     $PState \ t' = \text{foldl concat } (PState \ s) \ (\text{fri-layer-decommitment-transcript } l \ m \ i \ len)$ 
proof –
  have  $old$ :
     $res =$ 
       $(i \ \text{mod} \ (len \ \text{div} \ 2), l' ! (i \ \text{mod} \ (len \ \text{div} \ 2)), len \ \text{div} \ 2, pw + pw) \wedge$ 
       $PTranscript \ t' = rest \wedge$ 
       $start \leq t'$ 
    by (rule honest-fri-layer-decommitment-step-created-next-value-domain(2)[
       $OF \ \text{created } len\text{-eq} \ len\text{-pow} \ i\text{-bound} \ x\text{-eq} \ ext \ root \ tr \ layer \ l\text{-eval} \ d\text{-len}$ 
       $even\text{-len} \ round \ d\text{-i-point} \ d\text{-sib-point} \ next\text{-idx-bound} \ d\text{-next} \ outcome]$ )
  have  $path\text{-}i$ :
     $\text{length } (\text{get-authentication-path } len \ i \ m) = \text{floor-log } len$ 
    using  $\text{created-tree-get-authentication-path-len}$  [
       $OF \ \text{created } len\text{-pow}, of \ i]$   $i\text{-bound} \ len\text{-eq}$  by  $\text{simp}$ 
  have  $len\text{-pos}$ :  $0 < len$ 
    using  $i\text{-bound}$  by  $\text{simp}$ 
  have  $sid\text{-bound}$ :
     $(i + len \ \text{div} \ 2) \ \text{mod} \ len < \text{length } l$ 
    using  $len\text{-pos} \ len\text{-eq}$  by  $\text{simp}$ 
  have  $path\text{-}s$ :
     $\text{length } (\text{get-authentication-path } len \ ((i + len \ \text{div} \ 2) \ \text{mod} \ len) \ m) =$ 
       $\text{floor-log } len$ 

```

```

    using created-tree-get-authentication-path-len[
      OF created len-pow, of (i + len div 2) mod len] sidx-bound len-eq
    by simp
  have state:
    PState t' = foldl concat (PState s) (fri-layer-decommitment-transcript l m i len)
    using honest-fri-layer-decommitment-outcome(3)[
      OF len-eq i-bound path-i path-s outcome[unfolded fri-verifier-layer-def tr]] .
  show ?thesis
    using old state by simp
qed

end

end

```

```

theory Completeness-Verifier
  imports Completeness-FRI
begin

```

2.7 Final Completeness Proof

Final honest verifier query-round replay and completeness theorem.

```

context verification
begin

```

```

lemma honest-fri-decommitments-mfold-no-failure:
  assumes fri:
    fri-commit-outcome n cp p.eval-domain l0 m0 ps ds ls ms s5 s6
    and len-ps: length ps = Suc n
    and len-ds: length ds = Suc n
    and len-ls: length ls = Suc n
    and len-ms: length ms = Suc n
    and layers:
       $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
       $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
    and created:
       $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$ 
       $p.\text{created-tree } (ls ! i) (ms ! i) s6$ 
    and init-len: length l0 =  $2^N$ 
    and init-domain:  $2^N = \text{clength} * \text{scale}$ 
    and n-le-N:  $n \leq N$ 
    and len-roots: length roots = n
    and len-challenges: length challenges = n
    and roots-aligned:  $\bigwedge j. j < n \implies \text{roots} ! j = \text{value } (ms ! j)$ 
    and successors:
       $\bigwedge j. j < n \implies$ 
       $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$ 
       $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 

```

```

and idx-bound: idx < length (ls ! 0)
and x0-eq: x0 = ls ! 0 ! idx
and len0-eq: len0 = length (ls ! 0)
and start-tr:
  PTranscript start = fri-decommitment-transcript idx (butlast (zip ls ms)) @ rest
and start-ext: s6 ≤ start
shows
  None ∉ dom (dist (execute
    (mfold (idx, x0, len0, 1) (v.receive-query-commits (zip challenges roots)))
    start))
proof –
  let ?fl =
    v.receive-query-commits (zip challenges roots) ::
      ((nat × 'f × nat × nat) ⇒
        (nat × 'f × nat × nat, 'f protocol-channel) state-monad) list
  let ?bl = butlast (zip ls ms)
  define I where
    I = (λ(j::nat) (a::nat × 'f × nat × nat)
      (s::'f protocol-channel).
      (case a of (i, x, len, pw) ⇒
        i = idx mod length (ls ! j) ∧
        x = ls ! j ! i ∧
        len = length (ls ! j) ∧
        pw = 2j ∧
        PTranscript s = fri-decommitment-transcript i (drop j ?bl) @ rest ∧
        s6 ≤ s))
  have fl-len: length ?fl = n
  using len-roots len-challenges
  unfolding v.receive-query-commits-def by simp
  have init-I: I 0 (idx, x0, len0, 1) start
  unfolding I-def
  using idx-bound x0-eq len0-eq start-tr start-ext by simp
  have step-nf:
    ∧j m a st. j < length ?fl ⇒
      m = ?fl ! j ⇒ I j a st ⇒
      None ∉ dom (dist (execute (m a) st))
  proof –
    fix j m a st
    assume j-lt-fl: j < length ?fl
    and m-eq: m = ?fl ! j
    and inv: I j a st
    have j-lt: j < n
    using j-lt-fl fl-len by presburger
    obtain i x len pw where a-eq: a = (i, x, len, pw)
    by (cases a) auto
    have invD:
      i = idx mod length (ls ! j)
      x = ls ! j ! i
      len = length (ls ! j)

```

```

  pw = 2 ^ j
  PTranscript st = fri-decommitment-transcript i (drop j ?bl) @ rest
  s6 ≤ st
  using inv unfolding I-def a-eq by simp-all
have j-le: j ≤ n
  using j-lt by simp
have len-exact:
  length (ls ! j) = 2 ^ (N - j)
  by (rule fri-commit-outcome-initial-layer-length-power[
    OF fri layers init-len n-le-N j-le])
have j-lt-N: j < N
  using j-lt n-le-N by linarith
have diff-suc: N - j = Suc (N - Suc j)
  using j-lt-N by simp
have len-pow-suc:
  length (ls ! j) = 2 ^ Suc (N - Suc j)
  using len-exact diff-suc by simp
have len-pos: 0 < length (ls ! j)
  using len-pow-suc by simp
have i-bound: i < len
  using invD len-pos by simp
have drop-eq:
  drop j ?bl = (ls ! j, ms ! j) # drop (Suc j) ?bl
  by (rule butlast-zip-suc-drop-nth[OF len-ls len-ms j-lt])
have tr-layer:
  st =
    st(PTranscript :=
      fri-layer-decommitment-transcript (ls ! j) (ms ! j) i len @
      fri-decommitment-transcript i (drop (Suc j) ?bl) @ rest)
  using invD drop-eq unfolding fri-decommitment-transcript.simps
  by simp
have len-pow: ∃ K. length (ls ! j) = 2 ^ K
  using len-pow-suc by blast
then obtain K where len-pow-K: length (ls ! j) = 2 ^ K
  by blast
have l-eval:
  ls ! j = map (poly (ps ! j)) (ds ! j)
  by (rule layers) (use j-lt len-ps len-ds len-ls in simp-all)
have d-len: length (ds ! j) = length (ls ! j)
  using l-eval by simp
have i-d-bound: i < length (ds ! j)
  using i-bound invD d-len by simp
have d-i:
  ds ! j ! i = ((p.h ^ i) * shift) ^ pw
  using fri-commit-outcome-initial-domain-points[OF fri j-le i-d-bound] invD by simp
have idx-bound:
  (i + len div 2) mod len < length (ds ! j)
  using len-pos invD d-len by simp
have d-sib:

```



```

  ds ! j ! ((i + len div 2) mod len) =
    ((p.h ^ ((i + len div 2) mod len)) * shift) ^ pw
  using fri-commit-outcome-initial-domain-points[OF fri j-le sidx-bound] invD
  by simp
have next-len:
  length (ls ! Suc j) = length (ls ! j) div 2
  using next-fri-layer-length[OF successors[OF j-lt]] d-len by simp
have half-pos: 0 < len div 2
  using invD len-pow-suc by simp
have even-len: 2 dvd len
  using invD len-pow-suc by simp
have round: len * pw = clength * scale
proof -
  have len * pw = 2 ^ (N - j) * 2 ^ j
    using invD len-exact by simp
  also have ... = 2 ^ ((N - j) + j)
    by (simp add: power-add)
  also have ... = 2 ^ N
    using j-lt-N by simp
  also have ... = clength * scale
    using init-domain .
  finally show ?thesis .
qed
have next-idx-bound:
  i mod (len div 2) < length (ds ! Suc j)
proof -
  have length (ds ! Suc j) = length (ls ! Suc j)
  proof -
    have ls ! Suc j = map (poly (ps ! Suc j)) (ds ! Suc j)
      by (rule layers) (use j-lt len-ps len-ds len-ls in simp-all)
    then show ?thesis by simp
  qed
  then show ?thesis
    using next-len half-pos invD by simp
qed
have next-domain:
  ds ! Suc j ! (i mod (len div 2)) =
    ((p.h ^ i) * shift) ^ pw * ((p.h ^ i) * shift) ^ pw
proof -
  have suc-le: Suc j ≤ n
    using j-lt by simp
  have pw-next: pw + pw = 2 ^ Suc j
  proof -
    have (2::nat) ^ j + 2 ^ j = 2 ^ Suc j
    proof -
      have (2::nat) ^ Suc j = 2 * 2 ^ j
        by (simp add: power-Suc)
      also have ... = 2 ^ j + 2 ^ j
        by presburger
    qed
  qed

```

```

    finally show ?thesis by simp
qed
then show ?thesis
  using invD by simp
qed
have point-next:
  ds ! Suc j ! (i mod (len div 2)) =
    ((p.h ^ (i mod (len div 2))) * shift) ^ (2 ^ Suc j)
  using fri-commit-outcome-initial-domain-points[OF fri suc-le next-idx-bound]
    invD next-len
  by simp
have fold-domain:
  ((p.h ^ i) * shift) ^ pw * ((p.h ^ i) * shift) ^ pw =
    ((p.h ^ (i mod (len div 2))) * shift) ^ (pw + pw)
  by (rule fri-next-domain-round[OF half-pos even-len round])
show ?thesis
  using point-next fold-domain pw-next by simp
qed
have step-eq:
  m a = fri-verifier-layer (challenges ! j) (roots ! j) i x len pw
  using receive-query-commits-zip-nth-apply[
    OF j-lt[folded len-challenges] j-lt[folded len-roots],
    of i x len pw]
  m-eq a-eq
  by simp
have created-j:
  p.created-tree (ls ! j) (ms ! j) s6
  by (rule created) (use j-lt len-ls len-ms in simp-all)
show None ∉ dom (dist (execute (m a) st))
  unfolding step-eq
proof (rule honest-fri-layer-decommitment-step-created-next-value-domain(1)[
  where rest=fri-decommitment-transcript i (drop (Suc j) ?bl) @ rest])
  show p.created-tree (ls ! j) (ms ! j) s6
    using created-j .
  show len = length (ls ! j)
    using invD by simp
  show length (ls ! j) = 2 ^ K
    using len-pow-K .
  show i < len
    using i-bound .
  show x = ls ! j ! i
    using invD by simp
  show s6 ≤ st
    using invD by simp
  show roots ! j = value (ms ! j)
    by (rule roots-aligned[OF j-lt])
  show st =
    st(PTranscript :=
      fri-layer-decommitment-transcript (ls ! j) (ms ! j) i len @

```

```

    fri-decommitment-transcript i (drop (Suc j) ?bl) @ rest)
  using tr-layer .
show p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
  (ps ! Suc j, ds ! Suc j, ls ! Suc j)
  by (rule successors[OF j-lt])
show ls ! j = map (poly (ps ! j)) (ds ! j)
  using l-eval .
show length (ds ! j) = length (ls ! j)
  using d-len .
show 2 dvd len
  using even-len .
show len * pw = clength * scale
  using round .
show ds ! j ! i = (p.h ^ i * shift) ^ pw
  using d-i .
show ds ! j ! ((i + len div 2) mod len) =
  (p.h ^ ((i + len div 2) mod len) * shift) ^ pw
  using d-sib .
show i mod (len div 2) < length (ds ! Suc j)
  using next-idx-bound .
show ds ! Suc j ! (i mod (len div 2)) =
  (p.h ^ i * shift) ^ pw * (p.h ^ i * shift) ^ pw
  using next-domain .
qed
qed
have nf-map:
  None ∉ dom (dist (execute
    (mfold (idx, x0, len0, 1) (map (λm. m) ?fl)) start))
proof (rule no-failure-mfold-map-indexI[
  where I=I and xs=?fl and F=λm. m, OF init-I])
show ∧j m a st. j < length ?fl ⇒
  m = ?fl ! j ⇒ I j a st ⇒
  None ∉ dom (dist (execute (m a) st))
  by (rule step-nf)
next
fix j m a st y t
assume j-lt-fl: j < length ?fl
and m-eq: m = ?fl ! j
and inv: I j a st
and out: Some (y, t) ∈ set-dist (execute (m a) st)
have j-lt: j < n
  using j-lt-fl fl-len by presburger
obtain i x len pw where a-eq: a = (i, x, len, pw)
  by (cases a) auto
have invD:
  i = idx mod length (ls ! j)
  x = ls ! j ! i
  len = length (ls ! j)
  pw = 2 ^ j

```

```

PTranscript st = fri-decommitment-transcript i (drop j ?bl) @ rest
s6 ≤ st
using inv unfolding I-def a-eq by simp-all
have j-le: j ≤ n
using j-lt by simp
have len-exact:
  length (ls ! j) =  $2^{(N - j)}$ 
by (rule fri-commit-outcome-initial-layer-length-power[
  OF fri layers init-len n-le-N j-le])
have j-lt-N: j < N
using j-lt n-le-N by linarith
have diff-suc: N - j = Suc (N - Suc j)
using j-lt-N by simp
have len-pow-suc:
  length (ls ! j) =  $2^{Suc\ (N - Suc\ j)}$ 
using len-exact diff-suc by simp
have len-pos:  $0 < \text{length}\ (ls\ !\ j)$ 
using len-pow-suc by simp
have i-bound: i < len
using invD len-pos by simp
have drop-eq:
  drop j ?bl = (ls ! j, ms ! j) # drop (Suc j) ?bl
by (rule butlast-zip-suc-drop-nth[OF len-ls len-ms j-lt])
have tr-layer:
  st =
    st(PTranscript :=
      fri-layer-decommitment-transcript (ls ! j) (ms ! j) i len @
      fri-decommitment-transcript i (drop (Suc j) ?bl) @ rest)
using invD drop-eq unfolding fri-decommitment-transcript.simps
by simp
obtain K where len-pow-K: length (ls ! j) =  $2^K$ 
using len-pow-suc by blast
have l-eval:
  ls ! j = map (poly (ps ! j)) (ds ! j)
by (rule layers) (use j-lt len-ps len-ds len-ls in simp-all)
have d-len: length (ds ! j) = length (ls ! j)
using l-eval by simp
have i-d-bound: i < length (ds ! j)
using i-bound invD d-len by simp
have d-i:
  ds ! j ! i = ((p.h ^ i) * shift) ^ pw
using fri-commit-outcome-initial-domain-points[OF fri j-le i-d-bound] invD by simp
have sidx-bound:
  (i + len div 2) mod len < length (ds ! j)
using len-pos invD d-len by simp
have d-sib:
  ds ! j ! ((i + len div 2) mod len) =
    ((p.h ^ ((i + len div 2) mod len)) * shift) ^ pw
using fri-commit-outcome-initial-domain-points[OF fri j-le sidx-bound] invD

```

```

    by simp
have next-len:
  length (ls ! Suc j) = length (ls ! j) div 2
  using next-fri-layer-length[OF successors[OF j-lt]] d-len by simp
have half-pos: 0 < len div 2
  using invD len-pow-suc by simp
have even-len: 2 dvd len
  using invD len-pow-suc by simp
have round: len * pw = clength * scale
proof -
  have len * pw = 2 ^ (N - j) * 2 ^ j
    using invD len-exact by simp
  also have ... = 2 ^ ((N - j) + j)
    by (simp add: power-add)
  also have ... = 2 ^ N
    using j-lt-N by simp
  also have ... = clength * scale
    using init-domain .
  finally show ?thesis .
qed
have next-idx-bound:
  i mod (len div 2) < length (ds ! Suc j)
proof -
  have length (ds ! Suc j) = length (ls ! Suc j)
  proof -
    have ls ! Suc j = map (poly (ps ! Suc j)) (ds ! Suc j)
      by (rule layers) (use j-lt len-ps len-ds len-ls in simp-all)
    then show ?thesis by simp
  qed
  then show ?thesis
    using next-len half-pos invD by simp
qed
have next-domain:
  ds ! Suc j ! (i mod (len div 2)) =
    ((p.h ^ i) * shift) ^ pw * ((p.h ^ i) * shift) ^ pw
proof -
  have suc-le: Suc j ≤ n
    using j-lt by simp
  have pw-next: pw + pw = 2 ^ Suc j
  proof -
    have (2::nat) ^ j + 2 ^ j = 2 ^ Suc j
    proof -
      have (2::nat) ^ Suc j = 2 * 2 ^ j
        by (simp add: power-Suc)
      also have ... = 2 ^ j + 2 ^ j
        by presburger
      finally show ?thesis by simp
    qed
  qed
  then show ?thesis

```

```

    using invD by simp
qed
have point-next:
  ds ! Suc j ! (i mod (len div 2)) =
    ((p.h ^ (i mod (len div 2))) * shift) ^ (2 ^ Suc j)
  using fri-commit-outcome-initial-domain-points[OF fri suc-le next-idx-bound]
    invD next-len
  by simp
have fold-domain:
  ((p.h ^ i) * shift) ^ pw * ((p.h ^ i) * shift) ^ pw =
    ((p.h ^ (i mod (len div 2))) * shift) ^ (pw + pw)
  by (rule fri-next-domain-round[OF half-pos even-len round])
show ?thesis
  using point-next fold-domain pw-next by simp
qed
have step-eq:
  m a = fri-verifier-layer (challenges ! j) (roots ! j) i x len pw
  using receive-query-commits-zip-nth-apply[
    OF j-lt[folded len-challenges] j-lt[folded len-roots],
    of i x len pw]
  m-eq a-eq
  by simp
have created-j:
  p.created-tree (ls ! j) (ms ! j) s6
  by (rule created) (use j-lt len-ls len-ms in simp-all)
have out':
  Some (y, t) ∈
    set-dist (execute
      (fri-verifier-layer (challenges ! j) (roots ! j) i x len pw) st)
  using out unfolding step-eq .
have step-res:
  y =
    (i mod (len div 2), ls ! Suc j ! (i mod (len div 2)),
      len div 2, pw + pw) ∧
    PTranscript t = fri-decommitment-transcript i (drop (Suc j) ?bl) @ rest ∧
    st ≤ t
proof (rule honest-fri-layer-decommitment-step-created-next-value-domain(2)[
  where rest=fri-decommitment-transcript i (drop (Suc j) ?bl) @ rest])
show p.created-tree (ls ! j) (ms ! j) s6
  using created-j .
show len = length (ls ! j)
  using invD by simp
show length (ls ! j) = 2 ^ K
  using len-pow-K .
show i < len
  using i-bound .
show x = ls ! j ! i
  using invD by simp
show s6 ≤ st

```

```

    using invD by simp
show roots ! j = value (ms ! j)
  by (rule roots-aligned[OF j-lt])
show st =
  st(PTranscript :=
    fri-layer-decommitment-transcript (ls ! j) (ms ! j) i len @
    fri-decommitment-transcript i (drop (Suc j) ?bl) @ rest)
  using tr-layer .
show p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
  (ps ! Suc j, ds ! Suc j, ls ! Suc j)
  by (rule successors[OF j-lt])
show ls ! j = map (poly (ps ! j)) (ds ! j)
  using l-eval .
show length (ds ! j) = length (ls ! j)
  using d-len .
show 2 dvd len
  using even-len .
show len * pw = clength * scale
  using round .
show ds ! j ! i = (p.h ^ i * shift) ^ pw
  using d-i .
show ds ! j ! ((i + len div 2) mod len) =
  (p.h ^ ((i + len div 2) mod len) * shift) ^ pw
  using d-sib .
show i mod (len div 2) < length (ds ! Suc j)
  using next-idx-bound .
show ds ! Suc j ! (i mod (len div 2)) =
  (p.h ^ i * shift) ^ pw * (p.h ^ i * shift) ^ pw
  using next-domain .
show Some (y, t) ∈
  set-dist (execute
    (fri-verifier-layer (challenges ! j) (roots ! j) i x len pw) st)
  using out' .
qed
have i-next:
  i mod (len div 2) = idx mod length (ls ! Suc j)
proof -
  have mod-half:
    (idx mod length (ls ! j)) mod (length (ls ! j) div 2) =
      idx mod (length (ls ! j) div 2)
    by (rule mod-mod-half-power[OF len-pow-suc])
  show ?thesis
    using invD next-len mod-half by simp
qed
have pw-next: pw + pw = 2 ^ Suc j
proof -
  have (2::nat) ^ j + 2 ^ j = 2 ^ Suc j
  proof -
    have (2::nat) ^ Suc j = 2 * 2 ^ j

```

```

    by (simp add: power-Suc)
  also have ... = 2 ^ j + 2 ^ j
    by presburger
  finally show ?thesis by simp
qed
then show ?thesis
  using invD by simp
qed
have tr-next:
  PTranscript t =
    fri-decommitment-transcript (i mod (len div 2))
    (drop (Suc j) ?bl) @ rest
proof -
  have bridge:
    fri-decommitment-transcript (idx mod length (ls ! j))
      (drop (Suc j) ?bl) =
    fri-decommitment-transcript
      (idx mod (length (ls ! j) div 2))
      (drop (Suc j) ?bl)
  by (rule fri-decommitment-transcript-drop-half-index[
    OF len-ls len-ms j-lt len-pow-suc next-len])
  have mod-half:
    (idx mod length (ls ! j)) mod (length (ls ! j) div 2) =
    idx mod (length (ls ! j) div 2)
  by (rule mod-mod-half-power[OF len-pow-suc])
  have bridge':
    fri-decommitment-transcript (idx mod (length (ls ! j) div 2))
      (drop (Suc j) ?bl) =
    fri-decommitment-transcript
      (idx mod length (ls ! j) mod (length (ls ! j) div 2))
      (drop (Suc j) ?bl)
  using mod-half by simp
  show ?thesis
    using step-res bridge bridge' invD by simp
qed
have ext-next: s6 ≤ t
  using invD step-res by (meson p.hash-ext-trans)
have len-half: len div 2 = length (ls ! Suc j)
  using invD next-len by simp
show I (Suc j) y t
  using step-res i-next pw-next tr-next ext-next next-len invD(3) len-half
  unfolding I-def
  by (simp add: split: prod.splits)
qed
show ?thesis
  using nf-map by simp
qed

```

lemma *honest-fri-decommitments-mfold-outcome:*

assumes *fri*:
fri-commit-outcome *n cp p.eval-domain l0 m0 ps ds ls ms s5 s6*
and *len-ps*: *length ps = Suc n*
and *len-ds*: *length ds = Suc n*
and *len-ls*: *length ls = Suc n*
and *len-ms*: *length ms = Suc n*
and *layers*:
 $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$
and *created*:
 $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$
 $p.\text{created-tree } (ls ! i) (ms ! i) s6$
and *init-len*: *length l0 = 2 ^ N*
and *init-domain*: $2 ^ N = \text{clength} * \text{scale}$
and *n-le-N*: $n \leq N$
and *len-roots*: *length roots = n*
and *len-challenges*: *length challenges = n*
and *roots-aligned*: $\bigwedge j. j < n \implies \text{roots} ! j = \text{value } (ms ! j)$
and *successors*:
 $\bigwedge j. j < n \implies$
 $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$
 $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$
and *idx-bound*: *idx < length (ls ! 0)*
and *x0-eq*: $x0 = ls ! 0 ! \text{idx}$
and *len0-eq*: $\text{len0} = \text{length } (ls ! 0)$
and *start-tr*:
 $P\text{Transcript start} = \text{fri-decommitment-transcript idx } (\text{butlast } (\text{zip } ls \ ms)) @ \text{rest}$
and *start-ext*: $s6 \leq \text{start}$
and *outcome*:
 $\text{Some } ((i, x, \text{len}, pw), t) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{mfold } (\text{idx}, x0, \text{len0}, 1)$
 $(v.\text{receive-query-commits } (\text{zip } \text{challenges } \text{roots}))) \text{ start})$
shows
 $i = \text{idx mod length } (ls ! n) \wedge$
 $x = ls ! n ! i \wedge$
 $\text{len} = \text{length } (ls ! n) \wedge$
 $pw = 2 ^ n \wedge$
 $P\text{Transcript } t =$
 $\text{fri-decommitment-transcript } i (\text{drop } n (\text{butlast } (\text{zip } ls \ ms))) @ \text{rest} \wedge$
 $P\text{State } t =$
 $\text{foldl concat } (P\text{State start})$
 $(\text{fri-decommitment-transcript idx } (\text{butlast } (\text{zip } ls \ ms))) \wedge$
 $s6 \leq t$
proof –
let *?fl* =
 $v.\text{receive-query-commits } (\text{zip } \text{challenges } \text{roots}) ::$
 $((\text{nat} \times 'f \times \text{nat} \times \text{nat}) \Rightarrow$
 $(\text{nat} \times 'f \times \text{nat} \times \text{nat}, 'f \text{ protocol-channel}) \text{ state-monad}) \text{ list}$

```

let ?bl = butlast (zip ls ms)
define I where
  I = (λ(j::nat) (a::nat × 'f × nat × nat)
    (s::'f protocol-channel).
    (case a of (i, x, len, pw) ⇒
      i = idx mod length (ls ! j) ∧
      x = ls ! j ! i ∧
      len = length (ls ! j) ∧
      pw = 2 ^ j ∧
      PTranscript s = fri-decommitment-transcript i (drop j ?bl) @ rest ∧
      (∃ consumed.
        fri-decommitment-transcript idx ?bl =
          consumed @ fri-decommitment-transcript i (drop j ?bl) ∧
          PState s = foldl concat (PState start) consumed) ∧
      s6 ≤ s))
have ft-len: length ?fl = n
  using len-roots len-challenges
  unfolding v.receive-query-commits-def by simp
have init-I: I 0 (idx, x0, len0, 1) start
  unfolding I-def
  using idx-bound x0-eq len0-eq start-tr start-ext by simp
have outcome-map:
  Some ((i, x, len, pw), t) ∈
    set-dist (execute
      (mfold (idx, x0, len0, 1) (map (λm. m) ?fl)) start)
  using outcome by simp
have final-I: I (length ?fl) (i, x, len, pw) t
proof (rule mfold-map-index-outcome-invariantI[
  where I=I and xs=?fl and F=λm. m, OF init-I - outcome-map])
  fix j m a st y t'
  assume j-lt-ft: j < length ?fl
    and m-eq: m = ?fl ! j
    and inv: I j a st
    and out: Some (y, t') ∈ set-dist (execute (m a) st)
  have j-lt: j < n
    using j-lt-ft ft-len by presburger
  obtain ii xx lenn pww where a-eq: a = (ii, xx, lenn, pww)
    by (cases a) auto
  have invD:
    ii = idx mod length (ls ! j)
    xx = ls ! j ! ii
    lenn = length (ls ! j)
    pww = 2 ^ j
    PTranscript st = fri-decommitment-transcript ii (drop j ?bl) @ rest
    s6 ≤ st
  using inv unfolding I-def a-eq by simp-all
  have consumed-ex:
    ∃ consumed.
      fri-decommitment-transcript idx ?bl =

```

```

    consumed @ fri-decommitment-transcript ii (drop j ?bl) ∧
    PState st = foldl concat (PState start) consumed
using inv invD(1) unfolding I-def a-eq by simp
from consumed-ex obtain consumed where
  consumed-total:
    fri-decommitment-transcript idx ?bl =
      consumed @ fri-decommitment-transcript ii (drop j ?bl)
and consumed-state:
    PState st = foldl concat (PState start) consumed
by blast
have j-le:  $j \leq n$ 
  using j-lt by simp
have len-exact:
  length (ls ! j) =  $2^{(N - j)}$ 
  by (rule fri-commit-outcome-initial-layer-length-power[
    OF fri layers init-len n-le-N j-le])
have j-lt-N:  $j < N$ 
  using j-lt n-le-N by linarith
have diff-suc:  $N - j = \text{Suc } (N - \text{Suc } j)$ 
  using j-lt-N by simp
have len-pow-suc:
  length (ls ! j) =  $2^{\text{Suc } (N - \text{Suc } j)}$ 
  using len-exact diff-suc by simp
have len-pos:  $0 < \text{length } (ls ! j)$ 
  using len-pow-suc by simp
have ii-bound:  $ii < \text{lenn}$ 
  using invD len-pos by simp
have drop-eq:
  drop j ?bl = (ls ! j, ms ! j) # drop (Suc j) ?bl
  by (rule butlast-zip-suc-drop-nth[OF len-ls len-ms j-lt])
have tr-layer:
  st =
    st(PTranscript :=
      fri-layer-decommitment-transcript (ls ! j) (ms ! j) ii lenn @
      fri-decommitment-transcript ii (drop (Suc j) ?bl) @ rest)
  using invD drop-eq unfolding fri-decommitment-transcript.simps
  by simp
obtain K where len-pow-K:  $\text{length } (ls ! j) = 2^K$ 
  using len-pow-suc by blast
have l-eval:
  ls ! j = map (poly (ps ! j)) (ds ! j)
  by (rule layers) (use j-lt len-ps len-ds len-ls in simp-all)
have d-len:  $\text{length } (ds ! j) = \text{length } (ls ! j)$ 
  using l-eval by simp
have ii-d-bound:  $ii < \text{length } (ds ! j)$ 
  using ii-bound invD d-len by simp
have d-i:
  ds ! j ! ii = ((p.h ^ ii) * shift) ^ pww
  using fri-commit-outcome-initial-domain-points[OF fri j-le ii-d-bound] invD by simp

```

```

have sidx-bound:
  (ii + lenn div 2) mod lenn < length (ds ! j)
  using len-pos invD d-len by simp
have d-sib:
  ds ! j ! ((ii + lenn div 2) mod lenn) =
    ((p.h ^ ((ii + lenn div 2) mod lenn)) * shift) ^ pww
  using fri-commit-outcome-initial-domain-points[OF fri j-le sidx-bound] invD
  by simp
have next-len:
  length (ls ! Suc j) = length (ls ! j) div 2
  using next-fri-layer-length[OF successors[OF j-lt]] d-len by simp
have half-pos: 0 < lenn div 2
  using invD len-pow-suc by simp
have even-len: 2 dvd lenn
  using invD len-pow-suc by simp
have round: lenn * pww = clength * scale
proof -
  have lenn * pww = 2 ^ (N - j) * 2 ^ j
    using invD len-exact by simp
  also have ... = 2 ^ ((N - j) + j)
    by (simp add: power-add)
  also have ... = 2 ^ N
    using j-lt-N by simp
  also have ... = clength * scale
    using init-domain .
  finally show ?thesis .
qed
have next-idx-bound:
  ii mod (lenn div 2) < length (ds ! Suc j)
proof -
  have length (ds ! Suc j) = length (ls ! Suc j)
  proof -
    have ls ! Suc j = map (poly (ps ! Suc j)) (ds ! Suc j)
      by (rule layers) (use j-lt len-ps len-ds len-ls in simp-all)
    then show ?thesis by simp
  qed
  then show ?thesis
    using next-len half-pos invD by simp
qed
have next-domain:
  ds ! Suc j ! (ii mod (lenn div 2)) =
    ((p.h ^ ii) * shift) ^ pww * ((p.h ^ ii) * shift) ^ pww
proof -
  have suc-le: Suc j ≤ n
    using j-lt by simp
  have pw-next: pww + pww = 2 ^ Suc j
  proof -
    have (2::nat) ^ j + 2 ^ j = 2 ^ Suc j
    proof -

```

```

    have (2::nat) ^ Suc j = 2 * 2 ^ j
      by (simp add: power-Suc)
    also have ... = 2 ^ j + 2 ^ j
      by presburger
    finally show ?thesis by simp
qed
then show ?thesis
  using invD by simp
qed
have point-next:
  ds ! Suc j ! (ii mod (lenn div 2)) =
    ((p.h ^ (ii mod (lenn div 2))) * shift) ^ (2 ^ Suc j)
  using fri-commit-outcome-initial-domain-points[OF fri suc-le next-idx-bound]
    invD next-len
  by simp
have fold-domain:
  ((p.h ^ ii) * shift) ^ pww * ((p.h ^ ii) * shift) ^ pww =
    ((p.h ^ (ii mod (lenn div 2))) * shift) ^ (pww + pww)
  by (rule fri-next-domain-round[OF half-pos even-len round])
show ?thesis
  using point-next fold-domain pw-next by simp
qed
have step-eq:
  m a = fri-verifier-layer (challenges ! j) (roots ! j) ii xx lenn pww
  using receive-query-commits-zip-nth-apply[
    OF j-lt[folded len-challenges] j-lt[folded len-roots],
    of ii xx lenn pww]
  m-eq a-eq
  by simp
have created-j:
  p.created-tree (ls ! j) (ms ! j) s6
  by (rule created) (use j-lt len-ls len-ms in simp-all)
have out':
  Some (y, t') ∈
    set-dist (execute
      (fri-verifier-layer (challenges ! j) (roots ! j) ii xx lenn pww) st)
  using out unfolding step-eq .
have step-res:
  y =
    (ii mod (lenn div 2), ls ! Suc j ! (ii mod (lenn div 2)),
      lenn div 2, pww + pww) ∧
  PTranscript t' = fri-decommitment-transcript ii (drop (Suc j) ?bl) @ rest ∧
  st ≤ t' ∧
  PState t' =
    foldl concat (PState st)
      (fri-layer-decommitment-transcript (ls ! j) (ms ! j) ii lenn)
proof (rule honest-fri-layer-decommitment-step-created-next-value-domain-state[
  where rest=fri-decommitment-transcript ii (drop (Suc j) ?bl) @ rest])
show p.created-tree (ls ! j) (ms ! j) s6

```

```

    using created-j .
show lenn = length (ls ! j)
    using invD by simp
show length (ls ! j) = 2 ^ K
    using len-pow-K .
show ii < lenn
    using ii-bound .
show xx = ls ! j ! ii
    using invD by simp
show s6 ≤ st
    using invD by simp
show roots ! j = value (ms ! j)
    by (rule roots-aligned[OF j-lt])
show st =
  st(PTranscript :=
    fri-layer-decommitment-transcript (ls ! j) (ms ! j) ii lenn @
    fri-decommitment-transcript ii (drop (Suc j) ?bl) @ rest)
    using tr-layer .
show p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
  (ps ! Suc j, ds ! Suc j, ls ! Suc j)
    by (rule successors[OF j-lt])
show ls ! j = map (poly (ps ! j)) (ds ! j)
    using l-eval .
show length (ds ! j) = length (ls ! j)
    using d-len .
show 2 dvd lenn
    using even-len .
show lenn * pww = clength * scale
    using round .
show ds ! j ! ii = (p.h ^ ii * shift) ^ pww
    using d-i .
show ds ! j ! ((ii + lenn div 2) mod lenn) =
  (p.h ^ ((ii + lenn div 2) mod lenn) * shift) ^ pww
    using d-sib .
show ii mod (lenn div 2) < length (ds ! Suc j)
    using next-idx-bound .
show ds ! Suc j ! (ii mod (lenn div 2)) =
  (p.h ^ ii * shift) ^ pww * (p.h ^ ii * shift) ^ pww
    using next-domain .
show Some (y, t') ∈
  set-dist (execute
    (fri-verifier-layer (challenges ! j) (roots ! j) ii xx lenn pww) st)
    using out' .
qed
have i-next:
  ii mod (lenn div 2) = idx mod length (ls ! Suc j)
proof -
  have mod-half:
    (idx mod length (ls ! j)) mod (length (ls ! j) div 2) =

```

```

      idx mod (length (ls ! j) div 2)
    by (rule mod-mod-half-power[OF len-pow-suc])
  show ?thesis
    using invD next-len mod-half by simp
qed
have pw-next: pww + pww = 2 ^ Suc j
proof -
  have (2::nat) ^ j + 2 ^ j = 2 ^ Suc j
  proof -
    have (2::nat) ^ Suc j = 2 * 2 ^ j
    by (simp add: power-Suc)
    also have ... = 2 ^ j + 2 ^ j
    by presburger
    finally show ?thesis by simp
  qed
then show ?thesis
  using invD by simp
qed
have tail-bridge:
  fri-decommitment-transcript ii (drop (Suc j) ?bl) =
  fri-decommitment-transcript (ii mod (lenn div 2))
  (drop (Suc j) ?bl)
proof -
  have bridge:
    fri-decommitment-transcript (idx mod length (ls ! j))
    (drop (Suc j) ?bl) =
    fri-decommitment-transcript
    (idx mod (length (ls ! j) div 2))
    (drop (Suc j) ?bl)
  by (rule fri-decommitment-transcript-drop-half-index[
    OF len-ls len-ms j-lt len-pow-suc next-len])
  have mod-half:
    (idx mod length (ls ! j)) mod (length (ls ! j) div 2) =
    idx mod (length (ls ! j) div 2)
  by (rule mod-mod-half-power[OF len-pow-suc])
  have bridge':
    fri-decommitment-transcript (idx mod (length (ls ! j) div 2))
    (drop (Suc j) ?bl) =
    fri-decommitment-transcript
    (idx mod length (ls ! j) mod (length (ls ! j) div 2))
    (drop (Suc j) ?bl)
  using mod-half by simp
  show ?thesis
    using bridge bridge' invD by simp
qed
have tr-next:
  PTranscript t' =
  fri-decommitment-transcript (ii mod (lenn div 2))
  (drop (Suc j) ?bl) @ rest

```

```

    using step-res tail-bridge by simp
have ext-next:  $s6 \leq t'$ 
    using invD step-res by (meson p.hash-ext-trans)
have len-half:  $\text{lenn div } 2 = \text{length } (ls ! \text{Suc } j)$ 
    using invD next-len by simp
have consumed-next:
  fri-decommitment-transcript idx ?bl =
    (consumed @
      fri-layer-decommitment-transcript (ls ! j) (ms ! j) ii lenn) @
    fri-decommitment-transcript (ii mod (lenn div 2))
    (drop (Suc j) ?bl)
proof -
  have tail-split:
    fri-decommitment-transcript ii (drop j ?bl) =
      fri-layer-decommitment-transcript (ls ! j) (ms ! j) ii lenn @
      fri-decommitment-transcript ii (drop (Suc j) ?bl)
    using drop-eq invD unfolding fri-decommitment-transcript.simps by simp
  show ?thesis
    using consumed-total tail-split tail-bridge by simp
qed
have state-next:
  PState t' =
    foldl concat (PState start)
      (consumed @
        fri-layer-decommitment-transcript (ls ! j) (ms ! j) ii lenn)
    using step-res consumed-state by simp
show I (Suc j) y t'
  using step-res i-next pw-next tr-next ext-next next-len invD(3) len-half
    consumed-next state-next
  unfolding I-def
  by (simp add: split: prod.splits)
qed
show ?thesis
proof -
  have bl-len:  $\text{length } ?bl = n$ 
    using butlast-zip-suc-lengths[OF len-ls len-ms] len-ls len-ms by simp
  obtain consumed where
    consumed-total:
      fri-decommitment-transcript idx ?bl =
        consumed @ fri-decommitment-transcript i (drop (length ?fl) ?bl)
    and consumed-state:
      PState t = foldl concat (PState start) consumed
    using final-I unfolding I-def by auto
  have drop-empty:  $\text{drop } (\text{length } ?fl) ?bl = []$ 
    using fl-len bl-len by simp
  have consumed-eq:  $\text{consumed} = \text{fri-decommitment-transcript idx } ?bl$ 
    using consumed-total drop-empty by simp
  show ?thesis
    using final-I fl-len consumed-state consumed-eq unfolding I-def by simp

```


qed
qed

lemma *honest-query-prefix-fri-mfold-no-failure:*
assumes *htv: honest-trace-valid*
and *create-f:*
 Some (f-merkle, s1) ∈
 set-dist (execute (p.create p.f-eval) init-state)
and *send-f:*
 Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and *alphas:*
 Some (as, s3) ∈
 set-dist (execute
 (mmap (replicate (length spec)
 (do {
 a ← p.receive-alpha-challenge;
 p.send a;
 return a
 }})) s2)
and *cp'-def: cp' = p.cp as p.f-powers*
and *send-degree:*
 Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)
and *create-cp:*
 Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and *fri:*
 Some ((ps, ds, ls, ms), s6) ∈
 set-dist (execute
 (p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)
and *len-ps: length ps = Suc nrounds*
and *len-ds: length ds = Suc nrounds*
and *len-ls: length ls = Suc nrounds*
and *len-ms: length ms = Suc nrounds*
and *layers:*
 $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$
and *created:*
 $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$
 $p.\text{created-tree } (ls ! i) (ms ! i) s6$
and *init-len: length (p.cp-eval as) = 2 ^ N*
and *n-le-N: nrounds ≤ N*
and *final-send:*
 Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)
and *random-idx:*
 Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)
and *idx-def: idx' = p.index (to-nat idx)*
and *query-decommit:*
 Some (qouts, s8) ∈
 set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
and *fri-decommit:*

```

Some (fri-idx, s9) ∈
  set-dist (execute
    (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
and prover-state-eq: prover-state = s9
and replay: replay-state = verifier-replay-state prover-state
and prefix-out:
  Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
    set-dist (execute
      (do {
        fr ← p.read;
        as' ← mmap (replicate (length spec)
          (do {
            a0 ← p.receive-alpha-challenge;
            let a0' = a0;
            a1 ← p.read;
            let a1' = a1;
            assert (a0' = a1');
            return a1'
          })));
        dg ← p.read;
        assert (to-nat dg ≤ p.maxDegree);
        fl ← ntimes v.receive-fri-commits nrounds;
        final ← p.read;
        idxv ← p.receive-query-index-challenge;
        let idxv' = p.index (to-nat idxv);
        fv ← mmap (v.check-decommit-on-query fr idxv');
        return (fr, as', dg, fl, final, idxv, fv)
      }) replay-state)
and fl-eq: fl = zip challenges roots
and len-roots: length roots = nrounds
and len-challenges: length challenges = nrounds
and roots-aligned:  $\bigwedge j. j < nrounds \implies roots ! j = value (ms ! j)$ 
and successors:
   $\bigwedge j. j < nrounds \implies$ 
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j)
shows
  None  $\notin$  dom (dist (execute
    (mfold
      (idx', v.cp-eval as' fv (p.h  $\wedge$  idx' * shift), clength * scale, 1)
      (v.receive-query-commits fl)) t))
proof –
  have ls0: ls ! 0 = p.cp-eval as
    using fri-commit-initial-heads(3)[OF fri] .
have idx-bound: idx' < length (ls ! 0)
    using idx-def index-less-domain cp-eval-length ls0 by simp
have x0-eq:
  v.cp-eval as' fv (p.h  $\wedge$  idx' * shift) = ls ! 0 ! idx'
proof –

```

```

have value-eq:
  v.cp-eval as' fv (p.h ^ idx' * shift) =
    poly (p.cp as p.f-powers) (p.h ^ idx' * shift)
by (rule honest-query-prefix-initial-fri-value[
  OF htv create-f send-f alphas cp'-def send-degree create-cp fri
  final-send random-idx idx-def query-decommit fri-decommit
  prover-state-eq replay prefix-out])
have ls0-value:
  ls ! 0 ! idx' =
    poly (p.cp as p.f-powers) (p.h ^ idx' * shift)
  using cp-eval-index-value[of as to-nat idx] idx-def ls0 by simp
show ?thesis
  using value-eq ls0-value by simp
qed
have len0-eq: clength * scale = length (ls ! 0)
  using ls0 cp-eval-length by simp
have init-domain:  $2^N = \text{clength} * \text{scale}$ 
  using init-len cp-eval-length by simp
have fri-shape:
  fri-commit-outcome nrounds cp' p.eval-domain (p.cp-eval as) cp-merkle
  ps ds ls ms s5 s6
by (rule fri-commit-outcomeI[OF fri])
have tr-t:
  PTranscript t = fri-decommitment-transcript idx' (butlast (zip ls ms))
by (rule honest-query-prefix-leaves-fri-decommitment-transcript[
  OF htv create-f send-f alphas cp'-def send-degree create-cp fri
  final-send random-idx idx-def query-decommit fri-decommit
  prover-state-eq replay prefix-out])
have replay-t: replay-state  $\leq$  t
  using honest-root-alpha-degree-fri-final-query-from-prover-outcome[
  OF htv create-f send-f alphas cp'-def send-degree create-cp fri
  final-send random-idx idx-def query-decommit fri-decommit
  prover-state-eq replay prefix-out]
  by blast
have s6-prover: s6  $\leq$  prover-state
  by (rule prover-after-fri-hash-extends[
  OF final-send random-idx query-decommit fri-decommit prover-state-eq])
have s6-replay: s6  $\leq$  replay-state
  using s6-prover replay
  unfolding verifier-replay-state-def less-eq-hash-ext-def less-eq-fmap-def
  by simp
have s6-t: s6  $\leq$  t
  using s6-replay replay-t by (meson p.hash-ext-trans)
have tr-t-empty:
  PTranscript t = fri-decommitment-transcript idx' (butlast (zip ls ms)) @ []
  using tr-t by simp
show ?thesis
  unfolding fl-eq
  by (rule honest-fri-decommitments-mfold-no-failure[

```

OF fri-shape len-ps len-ds len-ls len-ms layers created init-len init-domain n-le-N
len-roots len-challenges roots-aligned successors idx-bound x0-eq
len0-eq tr-t-empty s6-t])

qed

lemma *honest-query-prefix-fri-mfold-outcome:*

assumes *htv: honest-trace-valid*

and *create-f:*

Some (f-merkle, s1) ∈
set-dist (execute (p.create p.f-eval) init-state)

and *send-f:*

Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)

and *alphas:*

Some (as, s3) ∈
set-dist (execute
(mmap (replicate (length spec)
(do {
a ← p.receive-alpha-challenge;
p.send a;
return a
}))) s2)

and *cp'-def: cp' = p.cp as p.f-powers*

and *send-degree:*

Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)

and *create-cp:*

Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)

and *fri:*

Some ((ps, ds, ls, ms), s6) ∈
set-dist (execute
(p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)

and *len-ps: length ps = Suc nrounds*

and *len-ds: length ds = Suc nrounds*

and *len-ls: length ls = Suc nrounds*

and *len-ms: length ms = Suc nrounds*

and *layers:*

$\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$

and *created:*

$\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$
 $p.\text{created-tree } (ls ! i) (ms ! i) s6$

and *init-len: length (p.cp-eval as) = 2 ^ N*

and *n-le-N: nrounds ≤ N*

and *final-send:*

Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)

and *random-idx:*

Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)

and *idx-def: idx' = p.index (to-nat idx)*

and *query-decommit:*

Some (qouts, s8) ∈

```

    set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)
and fri-decommit:
    Some (fri-idx, s9) ∈
    set-dist (execute
      (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
and prover-state-eq: prover-state = s9
and replay: replay-state = verifier-replay-state prover-state
and prefix-out:
    Some ((fr, as', dg, fl, final, idrv, fv), t) ∈
    set-dist (execute
      (do {
        fr ← p.read;
        as' ← mmap (replicate (length spec)
          (do {
            a0 ← p.receive-alpha-challenge;
            let a0' = a0;
            a1 ← p.read;
            let a1' = a1;
            assert (a0' = a1');
            return a1'
          })));
        dg ← p.read;
        assert (to-nat dg ≤ p.maxDegree);
        fl ← ntimes v.receive-fri-commits nrounds;
        final ← p.read;
        idrv ← p.receive-query-index-challenge;
        let idrv' = p.index (to-nat idrv);
        fv ← mmap (v.check-decommit-on-query fr idrv');
        return (fr, as', dg, fl, final, idrv, fv)
      }) replay-state)
and fl-eq: fl = zip challenges roots
and len-roots: length roots = nrounds
and len-challenges: length challenges = nrounds
and roots-aligned:  $\bigwedge j. j < nrounds \implies roots ! j = value (ms ! j)$ 
and successors:
     $\bigwedge j. j < nrounds \implies$ 
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
    (ps ! Suc j, ds ! Suc j, ls ! Suc j)
and outcome:
    Some ((i, x, len, pw), u) ∈
    set-dist (execute
      (mfold
        (idx', v.cp-eval as' fv (p.h  $\wedge$  idx' * shift), clength * scale, 1)
        (v.receive-query-commits fl)) t)

```

shows

```

i = idx' mod length (ls ! nrounds) ∧
x = ls ! nrounds ! i ∧
len = length (ls ! nrounds) ∧
pw = 2  $\wedge$  nrounds ∧

```

```

PTranscript u =
  fri-decommitment-transcript i (drop nrounds (butlast (zip ls ms))) ∧
  s6 ≤ u
proof -
  have ls0: ls ! 0 = p.cp-eval as
    using fri-commit-initial-heads(3)[OF fri] .
  have idx-bound: idx' < length (ls ! 0)
    using idx-def index-less-domain cp-eval-length ls0 by simp
  have x0-eq:
    v.cp-eval as' fv (p.h ^ idx' * shift) = ls ! 0 ! idx'
  proof -
    have value-eq:
      v.cp-eval as' fv (p.h ^ idx' * shift) =
        poly (p.cp as p.f-powers) (p.h ^ idx' * shift)
    by (rule honest-query-prefix-initial-fri-value[
      OF htv create-f send-f alphas cp'-def send-degree create-cp fri
      final-send random-idx idx-def query-decommit fri-decommit
      prover-state-eq replay prefix-out])
    have ls0-value:
      ls ! 0 ! idx' =
        poly (p.cp as p.f-powers) (p.h ^ idx' * shift)
    using cp-eval-index-value[of as to-nat idx] idx-def ls0 by simp
  show ?thesis
    using value-eq ls0-value by simp
qed
have len0-eq: clength * scale = length (ls ! 0)
  using ls0 cp-eval-length by simp
have init-domain: 2 ^ N = clength * scale
  using init-len cp-eval-length by simp
have fri-shape:
  fri-commit-outcome nrounds cp' p.eval-domain (p.cp-eval as) cp-merkle
  ps ds ls ms s5 s6
  by (rule fri-commit-outcomeI[OF fri])
have tr-t:
  PTranscript t = fri-decommitment-transcript idx' (butlast (zip ls ms))
  by (rule honest-query-prefix-leaves-fri-decommitment-transcript[
    OF htv create-f send-f alphas cp'-def send-degree create-cp fri
    final-send random-idx idx-def query-decommit fri-decommit
    prover-state-eq replay prefix-out])
have replay-t: replay-state ≤ t
  using honest-root-alpha-degree-fri-final-query-from-prover-outcome[
    OF htv create-f send-f alphas cp'-def send-degree create-cp fri
    final-send random-idx idx-def query-decommit fri-decommit
    prover-state-eq replay prefix-out]
  by blast
have s6-prover: s6 ≤ prover-state
  by (rule prover-after-fri-hash-extends[
    OF final-send random-idx query-decommit fri-decommit prover-state-eq])
have s6-replay: s6 ≤ replay-state

```

```

using s6-prover replay
unfolding verifier-replay-state-def less-eq-hash-ext-def less-eq-fmap-def
by simp
have s6-t: s6 ≤ t
  using s6-replay replay-t by (meson p.hash-ext-trans)
have outcome-zip:
  Some ((i, x, len, pw), u) ∈
    set-dist (execute
      (mfold
        (idx', v.cp-eval as' fv (p.h ^ idx' * shift), clength * scale, 1)
        (v.receive-query-commits (zip challenges roots))) t)
  using outcome fl-eq by simp
have tr-t-empty:
  PTranscript t = fri-decommitment-transcript idx' (butlast (zip ls ms)) @ []
  using tr-t by simp
have res:
  i = idx' mod length (ls ! nrounds) ∧
  x = ls ! nrounds ! i ∧
  len = length (ls ! nrounds) ∧
  pw = 2 ^ nrounds ∧
  PTranscript u =
    fri-decommitment-transcript i (drop nrounds (butlast (zip ls ms))) @ [] ∧
  PState u =
    foldl concat (PState t)
    (fri-decommitment-transcript idx' (butlast (zip ls ms))) ∧
  s6 ≤ u
by (rule honest-fri-decommitments-mfold-outcome[
  OF fri-shape len-ps len-ds len-ls len-ms layers created init-len init-domain n-le-N
  len-roots len-challenges roots-aligned successors idx-bound x0-eq
  len0-eq tr-t-empty s6-t outcome-zip])
show ?thesis
  using res by simp
qed

```

```

lemma honest-query-prefix-final-assert-no-failure:
assumes htv: honest-trace-valid
and create-f:
  Some (f-merkle, s1) ∈
    set-dist (execute (p.create p.f-eval) init-state)
and send-f:
  Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)
and alphas:
  Some (as, s3) ∈
    set-dist (execute
      (mmap (replicate (length spec)
        (do {
          a ← p.receive-alpha-challenge;
          p.send a;
          return a

```

```

    }))) s2)
and cp'-def:  $cp' = p.cp$  as  $p.f-powers$ 
and send-degree:
    Some  $(((), s4) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{of-nat } (\text{degree } cp')))) s3)$ 
and create-cp:
    Some  $(cp\text{-merkle}, s5) \in \text{set-dist } (\text{execute } (p.\text{create } (p.cp\text{-eval } as)) s4)$ 
and nrounds-def:  $nrounds = \text{ceil-log } (\text{degree } cp' + 1)$ 
and fri:
    Some  $((ps, ds, ls, ms), s6) \in$ 
        set-dist  $(\text{execute } (p.fri\text{-commit } nrounds [cp'] [p.\text{eval-domain}] [p.cp\text{-eval } as] [cp\text{-merkle}]) s5)$ 
and len-ps:  $\text{length } ps = \text{Suc } nrounds$ 
and len-ds:  $\text{length } ds = \text{Suc } nrounds$ 
and len-ls:  $\text{length } ls = \text{Suc } nrounds$ 
and len-ms:  $\text{length } ms = \text{Suc } nrounds$ 
and layers:
     $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$ 
         $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$ 
and created:
     $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$ 
         $p.\text{created-tree } (ls ! i) (ms ! i) s6$ 
and init-len:  $\text{length } (p.cp\text{-eval } as) = 2^N$ 
and n-le-N:  $nrounds \leq N$ 
and final-send:
    Some  $(((), s\text{-final}) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{hd } (\text{last } ls)))) s6)$ 
and random-idx:
    Some  $(idx, s7) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s\text{-final})$ 
and idx-def:  $idx' = p.\text{index } (\text{to-nat } idx)$ 
and query-decommit:
    Some  $(qouts, s8) \in$ 
        set-dist  $(\text{execute } (\text{mmap } (p.\text{decommit-on-query } idx' f\text{-merkle})) s7)$ 
and fri-decommit:
    Some  $(fri\text{-idx}, s9) \in$ 
        set-dist  $(\text{execute } (mfold idx' (p.\text{decommit-on-fri-layers } (\text{butlast } (\text{zip } ls ms)))) s8)$ 
and prover-state-eq:  $\text{prover-state} = s9$ 
and replay:  $\text{replay-state} = \text{verifier-replay-state } \text{prover-state}$ 
and prefix-out:
    Some  $((fr, as', dg, fl, final, idxv, fv), t) \in$ 
        set-dist  $(\text{execute } (do \{$ 
             $fr \leftarrow p.\text{read};$ 
             $as' \leftarrow \text{mmap } (\text{replicate } (\text{length } spec))$ 
             $(do \{$ 
                 $a0 \leftarrow p.\text{receive-alpha-challenge};$ 
                 $\text{let } a0' = a0;$ 
                 $a1 \leftarrow p.\text{read};$ 
                 $\text{let } a1' = a1;$ 
                 $\text{assert } (a0' = a1');$ 
             $\})$ 
         $\})$ 

```



```

    return a1'
  }));
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits nrounds;
  final ← p.read;
  idxv ← p.receive-query-index-challenge;
  let idxv' = p.index (to-nat idxv);
  fv ← mmap (v.check-decommit-on-query fr idxv');
  return (fr, as', dg, fl, final, idxv, fv)
}) replay-state)
and fl-eq: fl = zip challenges roots
and len-roots: length roots = nrounds
and len-challenges: length challenges = nrounds
and roots-aligned:  $\bigwedge j. j < nrounds \implies roots ! j = value (ms ! j)$ 
and successors:
   $\bigwedge j. j < nrounds \implies$ 
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j)
and fold-out:
  Some ((i, x, len, pw), u) ∈
    set-dist (execute
      (mfold
        (idx', v.cp-eval as' fv (p.h ^ idx' * shift), clength * scale, 1)
        (v.receive-query-commits fl)) t)
shows None ∉ dom (dist (execute (assert (x = final)) u))
proof –
  have fold-res:
    i = idx' mod length (ls ! nrounds) ∧
    x = ls ! nrounds ! i ∧
    len = length (ls ! nrounds) ∧
    pw = 2 ^ nrounds ∧
    PTranscript u =
      fri-decommitment-transcript i (drop nrounds (butlast (zip ls ms))) ∧
    s6 ≤ u
  by (rule honest-query-prefix-fri-mfold-outcome[
    OF htv create-f send-f alphas cp'-def send-degree create-cp fri
    len-ps len-ds len-ls len-ms layers created init-len n-le-N
    final-send random-idx idx-def query-decommit fri-decommit prover-state-eq
    replay prefix-out fl-eq len-roots len-challenges roots-aligned successors
    fold-out])
  have prefix-res:
    final = hd (last ls)
  using honest-root-alpha-degree-fri-final-query-from-prover-outcome[
    OF htv create-f send-f alphas cp'-def send-degree create-cp fri
    final-send random-idx idx-def query-decommit fri-decommit prover-state-eq
    replay prefix-out]
  by blast
  have i-bound: i < length (ls ! nrounds)

```

```

proof –
  have len-pos:  $0 < \text{length } (ls \ ! \ nrounds)$ 
    using fri-commit-initial-layer-length-power[OF fri layers init-len n-le-N, of nrounds]
    by simp
  then show ?thesis
    using fold-res by simp
qed
have final-layer:
   $ls \ ! \ nrounds \ ! \ i = \text{hd } (\text{last } ls)$ 
  by (rule honest-fri-final-layer-equals-sent[
    OF fri len-ps len-ds len-ls layers init-len n-le-N nrounds-def
    successors i-bound])
have  $x = \text{final}$ 
  using fold-res prefix-res final-layer by simp
then show ?thesis
  by (simp add: assert-def)
qed

lemma honest-query-prefix-fri-mfold-final-no-failure:
assumes htv: honest-trace-valid
and create-f:
   $\text{Some } (f\text{-merkle}, s1) \in$ 
   $\text{set-dist } (\text{execute } (p.\text{create } p.f\text{-eval}) \text{ init-state})$ 
and send-f:
   $\text{Some } ((), s2) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{value } f\text{-merkle})) s1)$ 
and alphas:
   $\text{Some } (as, s3) \in$ 
   $\text{set-dist } (\text{execute}$ 
     $(\text{mmap } (\text{replicate } (\text{length } \text{spec}))$ 
     $(\text{do } \{$ 
       $a \leftarrow p.\text{receive-alpha-challenge};$ 
       $p.\text{send } a;$ 
       $\text{return } a$ 
     $\}))) s2)$ 
and cp'-def:  $cp' = p.cp \text{ as } p.f\text{-powers}$ 
and send-degree:
   $\text{Some } ((), s4) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{of-nat } (\text{degree } cp')))) s3)$ 
and create-cp:
   $\text{Some } (cp\text{-merkle}, s5) \in \text{set-dist } (\text{execute } (p.\text{create } (p.cp\text{-eval } as)) s4)$ 
and nrounds-def:  $nrounds = \text{ceil-log } (\text{degree } cp' + 1)$ 
and fri:
   $\text{Some } ((ps, ds, ls, ms), s6) \in$ 
   $\text{set-dist } (\text{execute}$ 
     $(p.fri\text{-commit } nrounds [cp'] [p.eval\text{-domain}] [p.cp\text{-eval } as] [cp\text{-merkle}]) s5)$ 
and len-ps:  $\text{length } ps = \text{Suc } nrounds$ 
and len-ds:  $\text{length } ds = \text{Suc } nrounds$ 
and len-ls:  $\text{length } ls = \text{Suc } nrounds$ 
and len-ms:  $\text{length } ms = \text{Suc } nrounds$ 
and layers:

```

$\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$
and *created*:
 $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$
 $p.\text{created-tree } (ls ! i) (ms ! i) s6$
and *init-len*: $\text{length } (p.\text{cp-eval } as) = 2^N$
and *n-le-N*: $n\text{rounds} \leq N$
and *final-send*:
 $\text{Some } ((), s\text{-final}) \in \text{set-dist } (\text{execute } (p.\text{send } (\text{hd } (\text{last } ls)))) s6)$
and *random-idx*:
 $\text{Some } (idx, s7) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s\text{-final})$
and *idx-def*: $idx' = p.\text{index } (\text{to-nat } idx)$
and *query-decommit*:
 $\text{Some } (qouts, s8) \in$
 $\text{set-dist } (\text{execute } (\text{mmap } (p.\text{decommit-on-query } idx' f\text{-merkle})) s7)$
and *fri-decommit*:
 $\text{Some } (fri\text{-idx}, s9) \in$
 $\text{set-dist } (\text{execute } (mfold\ idx' (p.\text{decommit-on-fri-layers } (\text{butlast } (\text{zip } ls\ ms)))) s8)$
and *prover-state-eq*: $\text{prover-state} = s9$
and *replay*: $\text{replay-state} = \text{verifier-replay-state } \text{prover-state}$
and *prefix-out*:
 $\text{Some } ((fr, as', dg, fl, final, idxv, fv), t) \in$
 $\text{set-dist } (\text{execute } (do \{$
 $\quad fr \leftarrow p.\text{read};$
 $\quad as' \leftarrow \text{mmap } (\text{replicate } (\text{length } spec)$
 $\quad (do \{$
 $\quad \quad a0 \leftarrow p.\text{receive-alpha-challenge};$
 $\quad \quad \text{let } a0' = a0;$
 $\quad \quad a1 \leftarrow p.\text{read};$
 $\quad \quad \text{let } a1' = a1;$
 $\quad \quad \text{assert } (a0' = a1');$
 $\quad \quad \text{return } a1'$
 $\quad \quad \}));$
 $\quad dg \leftarrow p.\text{read};$
 $\quad \text{assert } (\text{to-nat } dg \leq p.\text{maxDegree});$
 $\quad fl \leftarrow n\text{times } v.\text{receive-fri-commits } n\text{rounds};$
 $\quad final \leftarrow p.\text{read};$
 $\quad idxv \leftarrow p.\text{receive-query-index-challenge};$
 $\quad \text{let } idxv' = p.\text{index } (\text{to-nat } idxv);$
 $\quad fv \leftarrow \text{mmap } (v.\text{check-decommit-on-query } fr\ idxv');$
 $\quad \text{return } (fr, as', dg, fl, final, idxv, fv)$
 $\quad \}) \text{ replay-state})$
and *fl-eq*: $fl = \text{zip } challenges\ roots$
and *len-roots*: $\text{length } roots = n\text{rounds}$
and *len-challenges*: $\text{length } challenges = n\text{rounds}$
and *roots-aligned*: $\bigwedge j. j < n\text{rounds} \implies roots ! j = \text{value } (ms ! j)$
and *successors*:

$\bigwedge j. j < \text{nrounds} \implies$
 $p.\text{next-fri-layer } (ps ! j) (ds ! j) (\text{challenges} ! j) =$
 $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$

shows

$\text{None} \notin \text{dom } (\text{dist } (\text{execute}$
 $(\text{do } \{$
 $(i, x, \text{len}, pw) \leftarrow$
 mfold
 $(\text{idx}', v.\text{cp-eval as' fv } (p.h \hat{\ } \text{idx}' * \text{shift}), \text{clength} * \text{scale}, 1)$
 $(v.\text{receive-query-commits fl});$
 $\text{assert } (x = \text{final})$
 $\}) t))$

proof (rule no-failure-bindI)

show $\text{None} \notin \text{dom } (\text{dist } (\text{execute}$

$(\text{mfold}$
 $(\text{idx}', v.\text{cp-eval as' fv } (p.h \hat{\ } \text{idx}' * \text{shift}), \text{clength} * \text{scale}, 1)$
 $(v.\text{receive-query-commits fl}) t))$

by (rule honest-query-prefix-fri-mfold-no-failure[

$OF \text{ htv create-f send-f alphas cp'-def send-degree create-cp fri}$
 $\text{len-ps len-ds len-ls len-ms layers created init-len n-le-N}$
 $\text{final-send random-idx idx-def query-decommit fri-decommit prover-state-eq}$
 $\text{replay prefix-out fl-eq len-roots len-challenges roots-aligned successors}]$

next

fix fold-result u

assume fold-out:

$\text{Some } (\text{fold-result}, u) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{mfold}$
 $(\text{idx}', v.\text{cp-eval as' fv } (p.h \hat{\ } \text{idx}' * \text{shift}), \text{clength} * \text{scale}, 1)$
 $(v.\text{receive-query-commits fl}) t)$

obtain i x len pw **where** fold-result-eq: fold-result = (i, x, len, pw)

by (cases fold-result)

have fold-out':

$\text{Some } ((i, x, \text{len}, pw), u) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{mfold}$
 $(\text{idx}', v.\text{cp-eval as' fv } (p.h \hat{\ } \text{idx}' * \text{shift}), \text{clength} * \text{scale}, 1)$
 $(v.\text{receive-query-commits fl}) t)$

using fold-out **unfolding** fold-result-eq .

have assert-nf: $\text{None} \notin \text{dom } (\text{dist } (\text{execute } (\text{assert } (x = \text{final})) u))$

by (rule honest-query-prefix-final-assert-no-failure[

$OF \text{ htv create-f send-f alphas cp'-def send-degree create-cp nrounds-def}$
 $\text{fri len-ps len-ds len-ls len-ms layers created init-len n-le-N}$
 $\text{final-send random-idx idx-def query-decommit fri-decommit prover-state-eq}$
 $\text{replay prefix-out fl-eq len-roots len-challenges roots-aligned successors}$
 $\text{fold-out}]$

show $\text{None} \notin \text{dom } (\text{dist } (\text{execute}$

$((\text{case fold-result of } (i, x, \text{len}, pw) \Rightarrow \text{assert } (x = \text{final}))) u))$
using assert-nf **unfolding** fold-result-eq case-prod-unfold **by** simp

qed

lemma *honest-query-prefix-fri-mfold-final-dynamic-no-failure:*

assumes *htv: honest-trace-valid*

and *create-f:*

Some (f-merkle, s1) ∈
set-dist (execute (p.create p.f-eval) init-state)

and *send-f:*

Some ((), s2) ∈ set-dist (execute (p.send (value f-merkle)) s1)

and *alphas:*

Some (as, s3) ∈
set-dist (execute
(mmap (replicate (length spec)
(do {
a ← p.receive-alpha-challenge;
p.send a;
return a
}))) s2)

and *cp'-def: cp' = p.cp as p.f-powers*

and *send-degree:*

Some ((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp'))) s3)

and *create-cp:*

Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)

and *nrounds-def: nrounds = ceil-log (degree cp' + 1)*

and *fri:*

Some ((ps, ds, ls, ms), s6) ∈
set-dist (execute
(p.fri-commit nrounds [cp'] [p.eval-domain] [p.cp-eval as] [cp-merkle]) s5)

and *len-ps: length ps = Suc nrounds*

and *len-ds: length ds = Suc nrounds*

and *len-ls: length ls = Suc nrounds*

and *len-ms: length ms = Suc nrounds*

and *layers:*

$\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$

and *created:*

$\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$
 $p.\text{created-tree } (ls ! i) (ms ! i) s6$

and *init-len: length (p.cp-eval as) = 2 ^ N*

and *n-le-N: nrounds ≤ N*

and *final-send:*

Some ((), s-final) ∈ set-dist (execute (p.send (hd (last ls))) s6)

and *random-idx:*

Some (idx, s7) ∈ set-dist (execute p.receive-query-index-challenge s-final)

and *idx-def: idx' = p.index (to-nat idx)*

and *query-decommit:*

Some (qouts, s8) ∈
set-dist (execute (mmap (p.decommit-on-query idx' f-merkle)) s7)

and *fri-decommit:*

```

Some (fri-idx, s9) ∈
  set-dist (execute
    (mfold idx' (p.decommit-on-fri-layers (butlast (zip ls ms)))) s8)
and prover-state-eq: prover-state = s9
and replay: replay-state = verifier-replay-state prover-state
and prefix-out:
  Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
    set-dist (execute
      (do {
        fr ← p.read;
        as' ← mmap (replicate (length spec)
          (do {
            a0 ← p.receive-alpha-challenge;
            let a0' = a0;
            a1 ← p.read;
            let a1' = a1;
            assert (a0' = a1');
            return a1'
          })));
        dg ← p.read;
        assert (to-nat dg ≤ p.maxDegree);
        fl ← ntimes v.receive-fri-commits (ceil-log (to-nat dg + 1));
        final ← p.read;
        idxv ← p.receive-query-index-challenge;
        let idxv' = p.index (to-nat idxv);
        fv ← mmap (v.check-decommit-on-query fr idxv');
        return (fr, as', dg, fl, final, idxv, fv)
      }) replay-state)
and fl-eq: fl = zip challenges roots
and len-roots: length roots = nrounds
and len-challenges: length challenges = nrounds
and roots-aligned:  $\bigwedge j. j < nrounds \implies roots ! j = value (ms ! j)$ 
and successors:
   $\bigwedge j. j < nrounds \implies$ 
    p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j)
shows
  None  $\notin$  dom (dist (execute
    (do {
      (i, x, len, pw) ←
        mfold
          (idx', v.cp-eval as' fv (p.h  $\wedge$  idx' * shift), clength * scale, 1)
          (v.receive-query-commits fl);
      assert (x = final)
    }) t))
proof –
have prefix-fixed:
  Some ((fr, as', dg, fl, final, idxv, fv), t) ∈
    set-dist (execute

```

```

(do {
  fr ← p.read;
  as' ← mmap (replicate (length spec)
    (do {
      a0 ← p.receive-alpha-challenge;
      let a0' = a0;
      a1 ← p.read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }));
  dg ← p.read;
  assert (to-nat dg ≤ p.maxDegree);
  fl ← ntimes v.receive-fri-commits nrounds;
  final ← p.read;
  idxv ← p.receive-query-index-challenge;
  let idxv' = p.index (to-nat idxv);
  fv ← mmap (v.check-decommit-on-query fr idxv');
  return (fr, as', dg, fl, final, idxv, fv)
}) replay-state)
by (rule honest-root-alpha-degree-fri-final-query-dynamic-outcome-to-fixed[
  OF htv create-f send-f alphas cp'-def send-degree create-cp nrounds-def
  fri final-send random-idx idx-def query-decommit fri-decommit prover-state-eq
  replay prefix-out])
show ?thesis
by (rule honest-query-prefix-fri-mfold-final-no-failure[
  OF htv create-f send-f alphas cp'-def send-degree create-cp nrounds-def
  fri len-ps len-ds len-ls len-ms layers created init-len n-le-N
  final-send random-idx idx-def query-decommit fri-decommit prover-state-eq
  replay prefix-fixed fl-eq len-roots len-challenges roots-aligned successors])
qed

lemma honest-verifier-query-round-no-failure-from-state:
  assumes htv: honest-trace-valid
  and create-f:
    Some (f-merkle, s1) ∈
      set-dist (execute (p.create p.f-eval) init-state)
  and trace-fri:
    Some ((f-ps, f-ds, f-ls, f-ms), s-trace-fri) ∈
      set-dist (execute
        (p.trace-fri-commit f-nrounds [p.f] [p.eval-domain] [p.f-eval]
          [f-merkle]) s2)
  and f-len-ps: length f-ps = Suc f-nrounds
  and f-len-ds: length f-ds = Suc f-nrounds
  and f-len-ls: length f-ls = Suc f-nrounds
  and f-len-ms: length f-ms = Suc f-nrounds
  and f-layers:
    ∧ i. i < length f-ps ⇒ i < length f-ds ⇒ i < length f-ls ⇒
      f-ls ! i = map (poly (f-ps ! i)) (f-ds ! i)

```

and *f-created*:
 $\bigwedge i. i < \text{length } f\text{-ls} \implies i < \text{length } f\text{-ms} \implies$
 $p.\text{created-tree } (f\text{-ls} ! i) (f\text{-ms} ! i) s\text{-trace-fri}$
and *f-init-len*: $\text{length } p.f\text{-eval} = 2 \wedge FN$
and *f-n-le-N*: $f\text{-nrounds} \leq FN$
and *f-nrounds-def*: $f\text{-nrounds} = \text{ceil-log } clength$
and *f-fl-eq*: $f\text{-fl} = \text{zip } f\text{-challenges } f\text{-roots}$
and *f-final-eq*: $f\text{-final} = \text{hd } (\text{last } f\text{-ls})$
and *f-len-roots*: $\text{length } f\text{-roots} = f\text{-nrounds}$
and *f-len-challenges*: $\text{length } f\text{-challenges} = f\text{-nrounds}$
and *f-roots-aligned*: $\bigwedge j. j < f\text{-nrounds} \implies f\text{-roots} ! j = \text{value } (f\text{-ms} ! j)$
and *f-successors*:
 $\bigwedge j. j < f\text{-nrounds} \implies$
 $p.\text{next-fri-layer } (f\text{-ps} ! j) (f\text{-ds} ! j) (f\text{-challenges} ! j) =$
 $(f\text{-ps} ! \text{Suc } j, f\text{-ds} ! \text{Suc } j, f\text{-ls} ! \text{Suc } j)$
and *fri*:
 $\text{Some } ((ps, ds, ls, ms), s6) \in$
 $\text{set-dist } (\text{execute}$
 $(p.\text{composition-fri-commit } nrounds [cp'] [p.\text{eval-domain}]$
 $[p.\text{cp-eval as}] [cp\text{-merkle}]) s5)$
and *len-ps*: $\text{length } ps = \text{Suc } nrounds$
and *len-ds*: $\text{length } ds = \text{Suc } nrounds$
and *len-ls*: $\text{length } ls = \text{Suc } nrounds$
and *len-ms*: $\text{length } ms = \text{Suc } nrounds$
and *layers*:
 $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$
and *created*:
 $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$
 $p.\text{created-tree } (ls ! i) (ms ! i) s6$
and *init-len*: $\text{length } (p.\text{cp-eval as}) = 2 \wedge N$
and *n-le-N*: $nrounds \leq N$
and *nrounds-def*: $nrounds = \text{ceil-log } (\text{degree } cp' + 1)$
and *fr-eq*: $fr = \text{value } f\text{-merkle}$
and *as-eq*: $as' = as$
and *fl-eq*: $fl = \text{zip } challenges roots$
and *final-eq*: $final = \text{hd } (\text{last } ls)$
and *len-roots*: $\text{length } roots = nrounds$
and *len-challenges*: $\text{length } challenges = nrounds$
and *roots-aligned*: $\bigwedge j. j < nrounds \implies roots ! j = \text{value } (ms ! j)$
and *successors*:
 $\bigwedge j. j < nrounds \implies$
 $p.\text{next-fri-layer } (ps ! j) (ds ! j) (challenges ! j) =$
 $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$
and *lookup*: $\text{fmlookup } (\text{HashMap } s) (\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) =$
 $\text{Some } idx$
and *idx-def*: $idx' = p.\text{index } (\text{to-nat } idx)$
and *query-ext*: $s1 \leq s$
and *trace-fri-ext*: $s\text{-trace-fri} \leq s$


```

and fri-ext:  $s6 \leq s$ 
and transcript:
  PTranscript s =
    query-decommitment-transcript idx' f-merkle @
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest
shows
   $\text{None} \notin \text{dom} (\text{dist} (\text{execute} (\text{verifier-query-round } \text{fr } \text{f-fl } \text{f-final } \text{as'} \text{fl } \text{final}) \text{ s}))$ 
proof –
  have ls0:  $ls \neq 0 = p.\text{cp-eval } as$ 
    using composition-fri-commit-initial-heads(3)[OF fri] .
  have idx-bound:  $idx' < \text{length } (ls \neq 0)$ 
    using idx-def index-less-domain cp-eval-length ls0 by simp
  have len0-eq:  $\text{length} * \text{scale} = \text{length } (ls \neq 0)$ 
    using ls0 cp-eval-length by simp
  have init-domain:  $2^N = \text{length} * \text{scale}$ 
    using init-len cp-eval-length by simp
  have trace-fri-shape:
    fri-commit-outcome f-nrounds p.f p.eval-domain p.f-eval f-merkle
    f-ps f-ds f-ls f-ms s2 s-trace-fri
    by (rule trace-fri-commit-outcomeI[OF trace-fri])
  have fri-shape:
    fri-commit-outcome nrounds cp' p.eval-domain (p.cp-eval as) cp-merkle
    ps ds ls ms s5 s6
    by (rule composition-fri-commit-outcomeI[OF fri])
  have full-eq:
    verifier-query-round fr f-fl f-final as' fl final =
      (p.receive-query-index-challenge  $\gg (\lambda id xv.$ 
        let idxv' = p.index (to-nat idxv) in
        mmap (v.check-decommit-on-query fr idxv')  $\gg (\lambda fv.$ 
          mfold
            (idxv', hd fv,  $\text{clength} * \text{scale}$ , 1)
            (v.receive-query-commits f-fl)  $\gg$ 
            ( $\lambda(f-i, f-x, f-len, f-pw).$ 
              assert ( $f-x = f-final$ )  $\gg (\lambda-.$ 
                mfold
                  (idxv', v.cp-eval as' fv ( $p.h \wedge idxv' * \text{shift}$ ),  $\text{clength} * \text{scale}$ , 1)
                  (v.receive-query-commits fl)  $\gg$ 
                  ( $\lambda(i, x, len, pw). \text{assert } (x = \text{final})$ ))))))
    unfolding verifier-query-round-def by (simp add: Let-def sm-bind-assoc split: prod.splits)
  show ?thesis
    unfolding full-eq
  proof (rule no-failure-bindI)
    show  $\text{None} \notin \text{dom} (\text{dist} (\text{execute } p.\text{receive-query-index-challenge } s))$ 
      by (rule p.receive-query-index-challenge-no-failure)
  next
    fix idxv s-idx
    assume rand:
       $\text{Some } (idxv, s-idx) \in \text{set-dist } (\text{execute } p.\text{receive-query-index-challenge } s)$ 

```

```

have rand-res:
  idxv = idx ∧
  PState s-idx = PState s ∧
  PTranscript s-idx = PTranscript s ∧
  s ≤ s-idx
using p.receive-query-index-challenge-known-outcome[OF lookup rand] by simp
have idxv-def: p.index (to-nat idxv) = idx'
using rand-res idx-def by simp
have s1-idx: s1 ≤ s-idx
using query-ext rand-res by (meson p.hash-ext-trans)
have tr-query:
  PTranscript s-idx =
    query-decommitment-transcript (p.index (to-nat idxv)) f-merkle @
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest
using transcript rand-res idxv-def by simp
have query-nf:
  None ∉ dom (dist (execute
    (mmap (v.check-decommit-on-query fr (p.index (to-nat idxv)))) s-idx))
by (rule honest-query-decommitment-replay-no-failure[
  OF create-f s1-idx fr-eq tr-query])
show None ∉ dom (dist (execute
  ((let idxv' = p.index (to-nat idxv)
    in mmap (v.check-decommit-on-query fr idxv') >>=
    (λfv.
      mfold
        (idxv', hd fv, clength * scale, 1)
        (v.receive-query-commits f-fl) >>=
        (λ(f-i, f-x, f-len, f-pw).
          assert (f-x = f-final) >>= (λ-.
            mfold
              (idxv', v.cp-eval as' fv (p.h ^ idxv' * shift), clength * scale, 1)
              (v.receive-query-commits fl) >>=
              (λ(i, x, len, pw). assert (x = final)))))))) s-idx))
unfolding Let-def
proof (rule no-failure-bindI[OF query-nf])
fix fv t
assume query-out:
  Some (fv, t) ∈
    set-dist (execute
      (mmap (v.check-decommit-on-query fr (p.index (to-nat idxv)))) s-idx)
have paths:
  ∧ i. i ∈ set (p.powers-scaled (p.index (to-nat idxv))) ⇒
    length (get-authentication-path (length p.f-eval) i f-merkle) =
      floor-log (length p.f-eval)
using query-authentication-path-length[OF create-f] by blast
have query-out-replay:
  Some (fv, t) ∈
    set-dist (execute

```

```

      (mmap (v.check-decommit-on-query fr (p.index (to-nat idrv))))
      (s-idx(PTranscript :=
        query-decommitment-transcript (p.index (to-nat idrv)) f-merkle @
        fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
        fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest)))
proof -
  have upd:
    s-idx(PTranscript :=
      query-decommitment-transcript (p.index (to-nat idrv)) f-merkle @
      fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
      fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest) =
    s-idx
  using tr-query by simp
  show ?thesis
  using query-out unfolding upd .
qed
have query-res:
  fv = map (λi. p.f-eval ! i) (p.powers-scaled (p.index (to-nat idrv))) ∧
  PTranscript t =
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest ∧
  PState t =
    foldl concat (PState s-idx)
    (query-decommitment-transcript (p.index (to-nat idrv)) f-merkle) ∧
  s-idx(PTranscript :=
    query-decommitment-transcript (p.index (to-nat idrv)) f-merkle @
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest) ≤ t
  by (rule honest-query-decommitment-replay-outcome[
    OF paths query-out-replay])
have s-idx-t: s-idx ≤ t
proof -
  have upd:
    s-idx(PTranscript :=
      query-decommitment-transcript (p.index (to-nat idrv)) f-merkle @
      fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
      fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest) =
    s-idx
  using tr-query by simp
  show ?thesis
  using query-res unfolding upd by simp
qed
have f-ls0: f-ls ! 0 = p.f-eval
  using trace-fri-commit-initial-heads(3)[OF trace-fri] .
have f-idx-bound: idx' < length (f-ls ! 0)
  using idx-def index-less-domain f-eval-length f-ls0 by simp
have f-len0-eq: clength * scale = length (f-ls ! 0)
  using f-ls0 f-eval-length by simp
have f-init-domain: 2 ^ FN = clength * scale

```

```

    using f-init-len f-eval-length by simp
have s-trace-fri-t: s-trace-fri ≤ t
    using trace-fri-ext rand-res s-idx-t by (meson p.hash-ext-trans)
have s6-t: s6 ≤ t
    using fri-ext rand-res s-idx-t by (meson p.hash-ext-trans)
have f-tr-t:
  PTranscript t =
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest
    using query-res by simp
have f-x0-eq: hd fv = f-ls ! 0 ! idx'
proof -
  have raw:
    hd fv = p.f-eval ! p.index (to-nat idxv)
    by (rule honest-initial-trace-fri-accumulator-value)
    (use query-res in simp-all)
  show ?thesis
    using raw idxv-def f-ls0 by simp
qed
have f-fold-nf:
  None ∉ dom (dist (execute
    (mfold (idx', hd fv, clength * scale, 1)
      (v.receive-query-commits f-fl)) t))
  unfolding f-fl-eq
  by (rule honest-fri-decommitments-mfold-no-failure[
    OF trace-fri-shape f-len-ps f-len-ds f-len-ls f-len-ms f-layers f-created
    f-init-len f-init-domain f-n-le-N f-len-roots f-len-challenges
    f-roots-aligned f-successors f-idx-bound f-x0-eq f-len0-eq
    f-tr-t s-trace-fri-t])
show None ∉ dom (dist (execute
  (mfold
    (p.index (to-nat idxv), hd fv, clength * scale, 1)
    (v.receive-query-commits f-fl)) >>=
  (λ(f-i, f-x, f-len, f-pw).
    assert (f-x = f-final) >>=
    (λ-. mfold
      (p.index (to-nat idxv),
        v.cp-eval as' fv (p.h ^ p.index (to-nat idxv) * shift), clength * scale, 1)
      (v.receive-query-commits fl)) >>=
      (λ(i, x, len, pw). assert (x = final)))))) t))
  unfolding idxv-def
proof (rule no-failure-bindI[OF f-fold-nf])
  fix f-fold-result t-trace
  assume f-fold-out:
    Some (f-fold-result, t-trace) ∈
    set-dist (execute
      (mfold (idx', hd fv, clength * scale, 1)
        (v.receive-query-commits f-fl)) t)
  obtain f-i f-x f-len f-pw where f-fold-eq:

```

```

f-fold-result = (f-i, f-x, f-len, f-pw)
by (cases f-fold-result)
have f-fold-out':
  Some ((f-i, f-x, f-len, f-pw), t-trace) ∈
    set-dist (execute
      (mfold (idx', hd fv, clength * scale, 1)
        (v.receive-query-commits f-fl)) t)
    using f-fold-out unfolding f-fold-eq .
have f-fold-out-zip:
  Some ((f-i, f-x, f-len, f-pw), t-trace) ∈
    set-dist (execute
      (mfold (idx', hd fv, clength * scale, 1)
        (v.receive-query-commits (zip f-challenges f-roots))) t)
    using f-fold-out' f-fl-eq by simp
have f-fold-res:
  f-i = idx' mod length (f-ls ! f-nrounds) ∧
  f-x = f-ls ! f-nrounds ! f-i ∧
  f-len = length (f-ls ! f-nrounds) ∧
  f-pw = 2 ^ f-nrounds ∧
  PTranscript t-trace =
    fri-decommitment-transcript f-i (drop f-nrounds (butlast (zip f-ls f-ms))) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest ∧
  PState t-trace =
    foldl concat (PState t)
      (fri-decommitment-transcript idx' (butlast (zip f-ls f-ms))) ∧
  s-trace-fri ≤ t-trace
by (rule honest-fri-decommitments-mfold-outcome[
  OF trace-fri-shape f-len-ps f-len-ds f-len-ls f-len-ms f-layers f-created
  f-init-len f-init-domain f-n-le-N f-len-roots f-len-challenges
  f-roots-aligned f-successors f-idx-bound f-x0-eq f-len0-eq
  f-tr-t s-trace-fri-t f-fold-out-zip])
have f-i-bound: f-i < length (f-ls ! f-nrounds)
proof –
  have len-pos: 0 < length (f-ls ! f-nrounds)
  using fri-commit-outcome-initial-layer-length-power[
    OF trace-fri-shape f-layers f-init-len f-n-le-N, of f-nrounds]
  by simp
  then show ?thesis
  using f-fold-res by simp
qed
have f-rounds-enough:
  ceil-log (degree p.f + 1) ≤ f-nrounds
  using trace-fri-rounds-enough f-nrounds-def by simp
have f-final-layer:
  f-ls ! f-nrounds ! f-i = hd (last f-ls)
  by (rule honest-fri-final-layer-equals-sent-initial-shape[
    OF trace-fri-shape f-len-ps f-len-ds f-len-ls f-layers f-init-len
    f-n-le-N f-rounds-enough f-successors f-i-bound])
have f-x-final: f-x = f-final

```

```

using f-fold-res f-final-layer f-final-eq by simp
have after-trace-nf:
  None  $\notin$  dom (dist (execute
    (assert (f-x = f-final)  $\gg$ 
      ( $\lambda$ -. mfold
        (idx',
          v.cp-eval as' fv (p.h  $\wedge$  idx' * shift), clength * scale, 1)
          (v.receive-query-commits fl)  $\gg$ 
            ( $\lambda$ (i, x, len, pw). assert (x = final)))) t-trace))
proof (rule no-failure-bindI)
  show None  $\notin$  dom (dist (execute (assert (f-x = f-final)) t-trace))
    using f-x-final by (simp add: assert-def)
next
  fix unit-after-assert t-after-assert
  assume assert-out:
    Some (unit-after-assert, t-after-assert)  $\in$ 
      set-dist (execute (assert (f-x = f-final)) t-trace)
  have t-after-assert-eq: t-after-assert = t-trace
    using assert-out f-x-final by (simp add: assert-def)
  have f-bl-len: length (butlast (zip f-ls f-ms)) = f-nrounds
    using butlast-zip-suc-lengths[OF f-len-ls f-len-ms] f-len-ls f-len-ms
    by simp
  have f-drop-empty:
    drop f-nrounds (butlast (zip f-ls f-ms)) = []
    using f-bl-len by simp
  have tr-t-trace:
    PTranscript t-trace =
      fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest
    using f-fold-res f-drop-empty by simp
  have t-t-trace: t  $\leq$  t-trace
    by (rule receive-query-commits-mfold-hash-extends[OF f-fold-out])
  have s6-t-trace: s6  $\leq$  t-trace
    using s6-t t-t-trace by (meson p.hash-ext-trans)
  have tr-t-after-assert:
    PTranscript t-after-assert =
      fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest
    using t-after-assert-eq tr-t-trace by simp
  have s6-t-after-assert: s6  $\leq$  t-after-assert
    using t-after-assert-eq s6-t-trace by simp
  have x0-eq:
    v.cp-eval as' fv (p.h  $\wedge$  idx' * shift) = ls ! 0 ! idx'
  proof -
    have raw:
      v.cp-eval as' fv (p.h  $\wedge$  p.index (to-nat idxv) * shift) =
        ls ! 0 ! p.index (to-nat idxv)
      by (rule honest-initial-fri-accumulator-value[
        OF htv as-eq - ls0])
      (use query-res in simp)
    show ?thesis

```

```

    using raw idxv-def by simp
qed
have fold-nf:
  None  $\notin$  dom (dist (execute
    (mfold
      (idx', v.cp-eval as' fv (p.h  $\wedge$  idx' * shift), clength * scale, 1)
      (v.receive-query-commits fl)) t-after-assert))
  unfolding t-after-assert-eq fl-eq
  by (rule honest-fri-decommitments-mfold-no-failure[
    OF fri-shape len-ps len-ds len-ls len-ms layers created init-len init-domain
    n-le-N len-roots len-challenges roots-aligned successors idx-bound
    x0-eq len0-eq tr-t-trace s6-t-trace])
show None  $\notin$  dom (dist (execute
  (mfold
    (idx', v.cp-eval as' fv (p.h  $\wedge$  idx' * shift), clength * scale, 1)
    (v.receive-query-commits fl)  $\gg$ 
    ( $\lambda(i, x, len, pw).$  assert ( $x = final$ ))) t-after-assert))
proof (rule no-failure-bindI[OF fold-nf])
  fix fold-result u
  assume fold-out:
    Some (fold-result, u)  $\in$ 
    set-dist (execute
      (mfold
        (idx', v.cp-eval as' fv (p.h  $\wedge$  idx' * shift), clength * scale, 1)
        (v.receive-query-commits fl)) t-after-assert)
  obtain i x len pw where fold-eq: fold-result = (i, x, len, pw)
  by (cases fold-result)
  have fold-out':
    Some ((i, x, len, pw), u)  $\in$ 
    set-dist (execute
      (mfold
        (idx', v.cp-eval as' fv (p.h  $\wedge$  idx' * shift), clength * scale, 1)
        (v.receive-query-commits fl)) t-after-assert)
  using fold-out unfolding fold-eq .
  have fold-out-zip:
    Some ((i, x, len, pw), u)  $\in$ 
    set-dist (execute
      (mfold
        (idx', v.cp-eval as' fv (p.h  $\wedge$  idx' * shift), clength * scale, 1)
        (v.receive-query-commits (zip challenges roots))) t-after-assert)
  using fold-out' fl-eq by simp
  have fold-res:
    i = idx' mod length (ls ! nrounds)  $\wedge$ 
    x = ls ! nrounds ! i  $\wedge$ 
    len = length (ls ! nrounds)  $\wedge$ 
    pw = 2  $\wedge$  nrounds  $\wedge$ 
    PTranscript u =
      fri-decommitment-transcript i (drop nrounds (butlast (zip ls ms))) @ rest  $\wedge$ 
    PState u =

```

```

    foldl concat (PState t-after-assert)
      (fri-decommitment-transcript idx' (butlast (zip ls ms))) ∧
    s6 ≤ u
  by (rule honest-fri-decommitments-mfold-outcome[
    OF fri-shape len-ps len-ds len-ls len-ms layers created init-len init-domain
    n-le-N len-roots len-challenges roots-aligned successors idx-bound
    x0-eq len0-eq tr-t-after-assert s6-t-after-assert fold-out-zip])
  have i-bound: i < length (ls ! nrounds)
  proof -
    have len-pos: 0 < length (ls ! nrounds)
    using fri-commit-outcome-initial-layer-length-power[
      OF fri-shape layers init-len n-le-N, of nrounds]
    by simp
    then show ?thesis
    using fold-res by simp
  qed
  have final-layer:
    ls ! nrounds ! i = hd (last ls)
  by (rule honest-fri-final-layer-equals-sent-shape[
    OF fri-shape len-ps len-ds len-ls layers init-len n-le-N nrounds-def
    successors i-bound])
  have x-final: x = final
  using fold-res final-layer final-eq by simp
  show None ∉ dom (dist (execute
    ((case fold-result of (i, x, len, pw) ⇒ assert (x = final))) u))
  unfolding fold-eq using x-final by (simp add: assert-def)
  qed
  qed
  show None ∉ dom (dist (execute
    ((case f-fold-result of (f-i, f-x, f-len, f-pw) ⇒
      assert (f-x = f-final) ≫=
      (λ-. mfold
        (idx',
          v.cp-eval as' fv (p.h ^ idx' * shift), clength * scale, 1)
        (v.receive-query-commits fl) ≫=
        (λ(i, x, len, pw). assert (x = final)))))) t-trace))
  unfolding f-fold-eq using after-trace-nf by simp
  qed
  qed
  qed
  qed

```

lemma honest-verifier-query-round-outcome-from-state:

```

  assumes htv: honest-trace-valid
  and create-f:
    Some (f-merkle, s1) ∈
      set-dist (execute (p.create p.f-eval) init-state)
  and trace-fri:
    Some ((f-ps, f-ds, f-ls, f-ms), s-trace-fri) ∈

```


set-dist (*execute*
 (*p.trace-fri-commit* *f-nrounds* [*p.f*] [*p.eval-domain*] [*p.f-eval*]
 [*f-merkle*]) *s2*)
and *f-len-ps*: *length f-ps* = *Suc f-nrounds*
and *f-len-ds*: *length f-ds* = *Suc f-nrounds*
and *f-len-ls*: *length f-ls* = *Suc f-nrounds*
and *f-len-ms*: *length f-ms* = *Suc f-nrounds*
and *f-layers*:
 $\bigwedge i. i < \text{length } f\text{-ps} \implies i < \text{length } f\text{-ds} \implies i < \text{length } f\text{-ls} \implies$
 $f\text{-ls} ! i = \text{map } (\text{poly } (f\text{-ps} ! i)) (f\text{-ds} ! i)$
and *f-created*:
 $\bigwedge i. i < \text{length } f\text{-ls} \implies i < \text{length } f\text{-ms} \implies$
 $p.\text{created-tree } (f\text{-ls} ! i) (f\text{-ms} ! i) s\text{-trace-fri}$
and *f-init-len*: *length p.f-eval* = $2 \wedge FN$
and *f-n-le-N*: *f-nrounds* $\leq FN$
and *f-nrounds-def*: *f-nrounds* = *ceil-log clength*
and *f-fl-eq*: *f-fl* = *zip f-challenges f-roots*
and *f-final-eq*: *f-final* = *hd (last f-ls)*
and *f-len-roots*: *length f-roots* = *f-nrounds*
and *f-len-challenges*: *length f-challenges* = *f-nrounds*
and *f-roots-aligned*: $\bigwedge j. j < f\text{-nrounds} \implies f\text{-roots} ! j = \text{value } (f\text{-ms} ! j)$
and *f-successors*:
 $\bigwedge j. j < f\text{-nrounds} \implies$
 $p.\text{next-fri-layer } (f\text{-ps} ! j) (f\text{-ds} ! j) (f\text{-challenges} ! j) =$
 $(f\text{-ps} ! \text{Suc } j, f\text{-ds} ! \text{Suc } j, f\text{-ls} ! \text{Suc } j)$
and *fri*:
Some ((ps, ds, ls, ms), s6) ∈
set-dist (*execute*
 (*p.composition-fri-commit* *nrounds* [*cp'*] [*p.eval-domain*]
 [*p.cp-eval as*] [*cp-merkle*]) *s5*)
and *len-ps*: *length ps* = *Suc nrounds*
and *len-ds*: *length ds* = *Suc nrounds*
and *len-ls*: *length ls* = *Suc nrounds*
and *len-ms*: *length ms* = *Suc nrounds*
and *layers*:
 $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$
and *created*:
 $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$
 $p.\text{created-tree } (ls ! i) (ms ! i) s6$
and *init-len*: *length (p.cp-eval as)* = $2 \wedge N$
and *n-le-N*: *nrounds* $\leq N$
and *nrounds-def*: *nrounds* = *ceil-log (degree cp' + 1)*
and *fr-eq*: *fr* = *value f-merkle*
and *as-eq*: *as'* = *as*
and *fl-eq*: *fl* = *zip challenges roots*
and *final-eq*: *final* = *hd (last ls)*
and *len-roots*: *length roots* = *nrounds*
and *len-challenges*: *length challenges* = *nrounds*

```

and roots-aligned:  $\bigwedge j. j < \text{nrounds} \implies \text{roots} ! j = \text{value} (\text{ms} ! j)$ 
and successors:
   $\bigwedge j. j < \text{nrounds} \implies$ 
     $p.\text{next-fri-layer} (ps ! j) (ds ! j) (\text{challenges} ! j) =$ 
       $(ps ! \text{Suc } j, ds ! \text{Suc } j, ls ! \text{Suc } j)$ 
and lookup:  $\text{fmlookup} (\text{HashMap } s) (\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) =$ 
Some idx
  and idx-def:  $\text{idx}' = p.\text{index} (\text{to-nat } \text{idx})$ 
and query-ext:  $s1 \leq s$ 
and trace-fri-ext:  $s.\text{trace-fri} \leq s$ 
and fri-ext:  $s6 \leq s$ 
and transcript:
   $P\text{Transcript } s =$ 
     $\text{query-decommitment-transcript } \text{idx}' \text{ f-merkle } @$ 
     $\text{fri-decommitment-transcript } \text{idx}' (\text{butlast } (\text{zip } f\text{-ls } f\text{-ms})) @$ 
     $\text{fri-decommitment-transcript } \text{idx}' (\text{butlast } (\text{zip } ls \text{ ms})) @ \text{rest}$ 
and outcome:
   $\text{Some } (\text{res}, t') \in \text{set-dist } (\text{execute } (\text{verifier-query-round } fr \text{ f-fl } f\text{-final } as' \text{ fl } \text{final}) s)$ 
shows
   $P\text{State } t' =$ 
     $\text{foldl concat } (P\text{State } s)$ 
     $(\text{query-decommitment-transcript } \text{idx}' \text{ f-merkle } @$ 
       $\text{fri-decommitment-transcript } \text{idx}' (\text{butlast } (\text{zip } f\text{-ls } f\text{-ms})) @$ 
       $\text{fri-decommitment-transcript } \text{idx}' (\text{butlast } (\text{zip } ls \text{ ms})))$ 
   $P\text{Transcript } t' = \text{rest}$ 
   $s \leq t'$ 
proof –
  have f-ls0:  $f\text{-ls} ! 0 = p.\text{f-eval}$ 
    using trace-fri-commit-initial-heads(3)[OF trace-fri] .
  have f-idx-bound:  $\text{idx}' < \text{length } (f\text{-ls} ! 0)$ 
    using idx-def index-less-domain f-eval-length f-ls0 by simp
  have f-len0-eq:  $\text{clength} * \text{scale} = \text{length } (f\text{-ls} ! 0)$ 
    using f-ls0 f-eval-length by simp
  have f-init-domain:  $2^{\wedge} \text{FN} = \text{clength} * \text{scale}$ 
    using f-init-len f-eval-length by simp
  have ls0:  $ls ! 0 = p.\text{cp-eval } as$ 
    using composition-fri-commit-initial-heads(3)[OF fri] .
  have idx-bound:  $\text{idx}' < \text{length } (ls ! 0)$ 
    using idx-def index-less-domain cp-eval-length ls0 by simp
  have len0-eq:  $\text{clength} * \text{scale} = \text{length } (ls ! 0)$ 
    using ls0 cp-eval-length by simp
  have init-domain:  $2^{\wedge} N = \text{clength} * \text{scale}$ 
    using init-len cp-eval-length by simp
  have trace-fri-shape:
     $\text{fri-commit-outcome } f.\text{nrounds } p.\text{f } p.\text{eval-domain } p.\text{f-eval } f.\text{merkle}$ 
     $f.\text{ps } f.\text{ds } f.\text{ls } f.\text{ms } s2 \text{ s-trace-fri}$ 
    by (rule trace-fri-commit-outcomeI[OF trace-fri])
  have fri-shape:
     $\text{fri-commit-outcome } \text{nrounds } cp' \text{ p.eval-domain } (p.\text{cp-eval } as) \text{ cp-merkle}$ 

```

```

  ps ds ls ms s5 s6
  by (rule composition-fri-commit-outcomeI[OF fri])
  have full-eq:
    verifier-query-round fr f-fl f-final as' fl final =
      (p.receive-query-index-challenge  $\gg$  ( $\lambda$ idxv.
        let idxv' = p.index (to-nat idxv) in
        mmap (v.check-decommit-on-query fr idxv')  $\gg$  ( $\lambda$ fv.
          mfold
            (idxv', hd fv, clength * scale, 1)
            (v.receive-query-commits f-fl)  $\gg$ 
            ( $\lambda$ (f-i, f-x, f-len, f-pw).
              assert (f-x = f-final)  $\gg$  ( $\lambda$ -.
                mfold
                  (idxv', v.cp-eval as' fv (p.h  $\wedge$  idxv' * shift), clength * scale, 1)
                  (v.receive-query-commits fl)  $\gg$ 
                  ( $\lambda$ (i, x, len, pw). assert (x = final)))))))
  unfolding verifier-query-round-def by (simp add: Let-def sm-bind-assoc split: prod.splits)
  from outcome obtain idxv s-idx fv t f-fold-result t-trace t-after-trace fold-result u where
  rand:
    Some (idxv, s-idx)  $\in$  set-dist (execute p.receive-query-index-challenge s)
  and query-out:
    Some (fv, t)  $\in$ 
      set-dist (execute
        (mmap (v.check-decommit-on-query fr (p.index (to-nat idxv)))) s-idx)
  and f-fold-out:
    Some (f-fold-result, t-trace)  $\in$ 
      set-dist (execute
        (mfold
          (p.index (to-nat idxv), hd fv, clength * scale, 1)
          (v.receive-query-commits f-fl) t)
        (p.index (to-nat idxv),
          v.cp-eval as' fv (p.h  $\wedge$  p.index (to-nat idxv) * shift),
          clength * scale, 1)
          (v.receive-query-commits fl) t-after-trace)
  and f-assert-out:
    Some (( ), t-after-trace)  $\in$ 
      set-dist (execute
        ((case f-fold-result of (f-i, f-x, f-len, f-pw)  $\Rightarrow$ 
          assert (f-x = f-final))) t-trace)
  and fold-out:
    Some (fold-result, u)  $\in$ 
      set-dist (execute
        (mfold
          (p.index (to-nat idxv),
            v.cp-eval as' fv (p.h  $\wedge$  p.index (to-nat idxv) * shift),
            clength * scale, 1)
          (v.receive-query-commits fl) t-after-trace)
        (p.index (to-nat idxv),
          v.cp-eval as' fv (p.h  $\wedge$  p.index (to-nat idxv) * shift),
          clength * scale, 1)
          (v.receive-query-commits fl) t-after-trace)
  and assert-out:
    Some (res, t')  $\in$ 
      set-dist (execute
        ((case fold-result of (i, x, len, pw)  $\Rightarrow$  assert (x = final))) u)
  unfolding full-eq
  by (auto simp: Let-def elim!: p.set-dist-bindE split: prod.splits)

```

```

have rand-res:
  idxv = idx ∧
  PState s-idx = PState s ∧
  PTranscript s-idx = PTranscript s ∧
  s ≤ s-idx
using p.receive-query-index-challenge-known-outcome[OF lookup rand] by simp
have idxv-def: p.index (to-nat idxv) = idx'
using rand-res idx-def by simp
have s1-idx: s1 ≤ s-idx
using query-ext rand-res by (meson p.hash-ext-trans)
have tr-query:
  PTranscript s-idx =
    query-decommitment-transcript (p.index (to-nat idxv)) f-merkle @
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest
using transcript rand-res idxv-def by simp
have paths:
  ∧i. i ∈ set (p.powers-scaled (p.index (to-nat idxv))) ⇒
    length (get-authentication-path (length p.f-eval) i f-merkle) =
      floor-log (length p.f-eval)
using query-authentication-path-length[OF create-f] by blast
have query-out-replay:
  Some (fv, t) ∈
    set-dist (execute
      (mmap (v.check-decommit-on-query fr (p.index (to-nat idxv))))
      (s-idx(PTranscript :=
        query-decommitment-transcript (p.index (to-nat idxv)) f-merkle @
        fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
        fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest)))
proof -
have upd:
  s-idx(PTranscript :=
    query-decommitment-transcript (p.index (to-nat idxv)) f-merkle @
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest) =
    s-idx
using tr-query by simp
show ?thesis
using query-out unfolding upd .
qed
have query-res:
  fv = map (λi. p.f-eval ! i) (p.powers-scaled (p.index (to-nat idxv))) ∧
  PTranscript t =
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest ∧
  PState t =
    foldl concat (PState s-idx)
      (query-decommitment-transcript (p.index (to-nat idxv)) f-merkle) ∧
  s-idx(PTranscript :=

```

```

    query-decommitment-transcript (p.index (to-nat idxv)) f-merkle @
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest) ≤ t
  by (rule honest-query-decommitment-replay-outcome[
    OF paths query-out-replay])
have s-idx-t: s-idx ≤ t
proof -
  have upd:
    s-idx(PTranscript :=
      query-decommitment-transcript (p.index (to-nat idxv)) f-merkle @
      fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
      fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest) =
      s-idx
    using tr-query by simp
  show ?thesis
    using query-res unfolding upd by simp
qed
have s-trace-fri-t: s-trace-fri ≤ t
  using trace-fri-ext rand-res s-idx-t by (meson p.hash-ext-trans)
have tr-t:
  PTranscript t =
    fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest
  using query-res by simp
have f-x0-eq: hd fv = f-ls ! 0 ! idx'
proof -
  have raw: hd fv = p.f-eval ! p.index (to-nat idxv)
    by (rule honest-initial-trace-fri-accumulator-value)
    (use query-res in simp-all)
  show ?thesis
    using raw idxv-def f-ls0 by simp
qed
obtain f-i f-x f-len f-pw where f-fold-eq: f-fold-result = (f-i, f-x, f-len, f-pw)
  by (cases f-fold-result)
have f-fold-out':
  Some ((f-i, f-x, f-len, f-pw), t-trace) ∈
    set-dist (execute
      (mfold (idx', hd fv, clength * scale, 1)
        (v.receive-query-commits f-fl)) t)
  using f-fold-out unfolding f-fold-eq idxv-def by simp
have f-fold-out-zip:
  Some ((f-i, f-x, f-len, f-pw), t-trace) ∈
    set-dist (execute
      (mfold (idx', hd fv, clength * scale, 1)
        (v.receive-query-commits (zip f-challenges f-roots))) t)
  using f-fold-out' f-fl-eq by simp
have f-fold-res:
  f-i = idx' mod length (f-ls ! f-nrounds) ∧
  f-x = f-ls ! f-nrounds ! f-i ∧

```

```

f-len = length (f-ls ! f-nrounds) ∧
f-pw = 2 ^ f-nrounds ∧
PTranscript t-trace =
  fri-decommitment-transcript f-i
    (drop f-nrounds (butlast (zip f-ls f-ms))) @
  fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest ∧
PState t-trace =
  foldl concat (PState t)
    (fri-decommitment-transcript idx' (butlast (zip f-ls f-ms))) ∧
s-trace-fri ≤ t-trace
by (rule honest-fri-decommitments-mfold-outcome[
  OF trace-fri-shape f-len-ps f-len-ds f-len-ls f-len-ms f-layers f-created
    f-init-len f-init-domain f-n-le-N f-len-roots f-len-challenges
    f-roots-aligned f-successors f-idx-bound f-x0-eq f-len0-eq
    tr-t s-trace-fri-t f-fold-out-zip])
have f-i-bound: f-i < length (f-ls ! f-nrounds)
proof –
  have len-pos: 0 < length (f-ls ! f-nrounds)
    using fri-commit-outcome-initial-layer-length-power[
      OF trace-fri-shape f-layers f-init-len f-n-le-N, of f-nrounds]
    by simp
  then show ?thesis
    using f-fold-res by simp
qed
have f-rounds-enough:
  ceil-log (degree p.f + 1) ≤ f-nrounds
  using trace-fri-rounds-enough f-nrounds-def by simp
have f-final-layer:
  f-ls ! f-nrounds ! f-i = hd (last f-ls)
  by (rule honest-fri-final-layer-equals-sent-initial-shape[
    OF trace-fri-shape f-len-ps f-len-ds f-len-ls f-layers f-init-len
      f-n-le-N f-rounds-enough f-successors f-i-bound])
have f-x-final: f-x = f-final
  using f-fold-res f-final-layer f-final-eq by simp
have t-after-trace-eq: t-after-trace = t-trace
  using f-assert-out unfolding f-fold-eq f-x-final by (simp add: assert-def)
have f-bl-len: length (butlast (zip f-ls f-ms)) = f-nrounds
  using butlast-zip-suc-lengths[OF f-len-ls f-len-ms] f-len-ls f-len-ms by simp
have f-drop-empty:
  drop f-nrounds (butlast (zip f-ls f-ms)) = []
  using f-bl-len by simp
have tr-t-after-trace:
  PTranscript t-after-trace =
    fri-decommitment-transcript idx' (butlast (zip ls ms)) @ rest
  using f-fold-res f-drop-empty t-after-trace-eq by simp
have state-t-after-trace:
  PState t-after-trace =
    foldl concat (PState t)
      (fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)))

```

```

using f-fold-res t-after-trace-eq by simp
have s6-t-after-trace:  $s6 \leq t\text{-after-trace}$ 
proof –
  have t-t-trace:  $t \leq t\text{-trace}$ 
    by (rule receive-query-commits-mfold-hash-extends [OF f-fold-out])
  have t-t-after:  $t \leq t\text{-after-trace}$ 
    using t-after-trace-eq t-t-trace by simp
  show ?thesis
    using fri-ext rand-res s-idx-t t-t-after by (meson p.hash-ext-trans)
qed
have t-t-trace-ext:  $t \leq t\text{-trace}$ 
  by (rule receive-query-commits-mfold-hash-extends [OF f-fold-out])
have x0-eq:
   $v.cp\text{-eval } as' \text{ fv } (p.h \wedge idx' * shift) = ls ! 0 ! idx'$ 
proof –
  have raw:
     $v.cp\text{-eval } as' \text{ fv } (p.h \wedge p.index \text{ (to-nat } idxv) * shift) =$ 
     $ls ! 0 ! p.index \text{ (to-nat } idxv)$ 
    by (rule honest-initial-fri-accumulator-value [
      OF htv as-eq - ls0])
    (use query-res in simp)
  show ?thesis
    using raw idxv-def by simp
qed
obtain i x len pw where fold-eq:  $\text{fold-result} = (i, x, len, pw)$ 
  by (cases fold-result)
have fold-out':
   $\text{Some } ((i, x, len, pw), u) \in$ 
   $\text{set-dist (execute$ 
   $\text{(mfold$ 
   $\text{(idx', v.cp-eval } as' \text{ fv } (p.h \wedge idx' * shift), clength * scale, 1)$ 
   $\text{(v.receive-query-commits fl)) } t\text{-after-trace})}$ 
  using fold-out unfolding fold-eq idxv-def by simp
have fold-out-zip:
   $\text{Some } ((i, x, len, pw), u) \in$ 
   $\text{set-dist (execute$ 
   $\text{(mfold$ 
   $\text{(idx', v.cp-eval } as' \text{ fv } (p.h \wedge idx' * shift), clength * scale, 1)$ 
   $\text{(v.receive-query-commits (zip challenges roots))) } t\text{-after-trace})}$ 
  using fold-out' fl-eq by simp
have fold-res:
   $i = idx' \bmod length \text{ (ls ! nrounds)} \wedge$ 
   $x = ls ! nrounds ! i \wedge$ 
   $len = length \text{ (ls ! nrounds)} \wedge$ 
   $pw = 2 \wedge nrounds \wedge$ 
   $PTranscript \text{ } u =$ 
   $\text{fri-decommitment-transcript } i \text{ (drop nrounds (butlast (zip ls ms))) } @ \text{ rest} \wedge$ 
   $PState \text{ } u =$ 
   $\text{foldl concat (PState } t\text{-after-trace)}$ 

```

```

      (fri-decommitment-transcript idx' (butlast (zip ls ms))) ∧
s6 ≤ u
by (rule honest-fri-decommitments-mfold-outcome[
  OF fri-shape len-ps len-ds len-ls len-ms layers created init-len init-domain
  n-le-N len-roots len-challenges roots-aligned successors idx-bound
  x0-eq len0-eq tr-t-after-trace s6-t-after-trace fold-out-zip])
have i-bound: i < length (ls ! nrounds)
proof -
  have len-pos: 0 < length (ls ! nrounds)
  using fri-commit-outcome-initial-layer-length-power[
    OF fri-shape layers init-len n-le-N, of nrounds]
  by simp
  then show ?thesis
  using fold-res by simp
qed
have final-layer:
  ls ! nrounds ! i = hd (last ls)
by (rule honest-fri-final-layer-equals-sent-shape[
  OF fri-shape len-ps len-ds len-ls layers init-len n-le-N nrounds-def
  successors i-bound])
have x-final: x = final
  using fold-res final-layer final-eq by simp
have t'-eq: t' = u
  using assert-out unfolding fold-eq x-final by (simp add: assert-def)
have state-query:
  PState t =
    foldl concat (PState s)
      (query-decommitment-transcript idx' f-merkle)
  using query-res rand-res idxv-def by simp
show PState t' =
  foldl concat (PState s)
    (query-decommitment-transcript idx' f-merkle @
      fri-decommitment-transcript idx' (butlast (zip f-ls f-ms)) @
      fri-decommitment-transcript idx' (butlast (zip ls ms)))
  using fold-res state-query state-t-after-trace t'-eq by simp
show PTranscript t' = rest
proof -
  have bl-len: length (butlast (zip ls ms)) = nrounds
  using butlast-zip-suc-lengths[OF len-ls len-ms] len-ls len-ms by simp
  have drop-empty: drop nrounds (butlast (zip ls ms)) = []
  using bl-len by simp
  show ?thesis
  using fold-res t'-eq drop-empty by simp
qed
show s ≤ t'
proof -
  have s-t: s ≤ t
  using rand-res s-idx-t by (meson p.hash-ext-trans)
  have t-after-u: t-after-trace ≤ u

```



```

    by (rule receive-query-commits-mfold-hash-extends[OF fold-out])
  show ?thesis
    using s-t t-t-trace-ext t-after-trace-eq t-after-u t'-eq by (meson p.hash-ext-trans)
qed
qed

lemma verifier-query-round-query-counter:
  assumes outcome:
    Some (res, t') ∈ set-dist (execute (verifier-query-round fr f-fl f-final as' fl final) s)
  shows PQueryCounter t' = Suc (PQueryCounter s)
proof -
  have full-eq:
    verifier-query-round fr f-fl f-final as' fl final =
      (p.receive-query-index-challenge >>= (λidxv.
        let idxv' = p.index (to-nat idxv) in
        mmap (v.check-decommit-on-query fr idxv') >>= (λfv.
          mfold
            (idxv', hd fv, clength * scale, 1)
            (v.receive-query-commits f-fl) >>=
              (λ(f-i, f-x, f-len, f-pw).
                assert (f-x = f-final) >>= (λ-.
                  mfold
                    (idxv', v.cp-eval as' fv (p.h ^ idxv' * shift), clength * scale, 1)
                    (v.receive-query-commits fl) >>=
                      (λ(i, x, len, pw). assert (x = final))))))))))
  unfolding verifier-query-round-def by (simp add: Let-def sm-bind-assoc split: prod.splits)
  from outcome obtain idxv s-idx fv t f-fold-result t-trace t-after-trace fold-result u where
    rand:
      Some (idxv, s-idx) ∈ set-dist (execute p.receive-query-index-challenge s)
  and query-out:
      Some (fv, t) ∈
        set-dist (execute
          (mmap (v.check-decommit-on-query fr (p.index (to-nat idxv)))) s-idx)
  and f-fold-out:
      Some (f-fold-result, t-trace) ∈
        set-dist (execute
          (mfold
            (p.index (to-nat idxv), hd fv, clength * scale, 1)
            (v.receive-query-commits f-fl) t)
  and f-assert-out:
      Some ((), t-after-trace) ∈
        set-dist (execute
          ((case f-fold-result of (f-i, f-x, f-len, f-pw) ⇒
            assert (f-x = f-final))) t-trace)
  and fold-out:
      Some (fold-result, u) ∈
        set-dist (execute
          (mfold
            (p.index (to-nat idxv),

```

```

      v.cp-eval as' fv (p.h ^ p.index (to-nat idxv) * shift),
      clength * scale, 1)
    (v.receive-query-commits fl)) t-after-trace)
and assert-out:
  Some (res, t') ∈
    set-dist (execute
      ((case fold-result of (i, x, len, pw) ⇒ assert (x = final))) u)
unfolding full-eq
by (auto simp: Let-def elim!: p.set-dist-bindE split: prod.splits)
have q-idx: PQueryCounter s-idx = Suc (PQueryCounter s)
  using p.receive-query-index-challenge-counter-outcome[OF rand] by simp
have q-query: PQueryCounter t = PQueryCounter s-idx
  by (rule mmap-preserves-query-counter[OF query-out])
  (rule check-decommit-on-query-step-preserves-query-counter)
have q-f-fold: PQueryCounter t-trace = PQueryCounter t
  by (rule receive-query-commits-mfold-preserves-query-counter[OF f-fold-out])
have q-f-assert: PQueryCounter t-after-trace = PQueryCounter t-trace
proof –
  obtain f-i f-x f-len f-pw where f-fold-eq:
    f-fold-result = (f-i, f-x, f-len, f-pw)
  by (cases f-fold-result)
have assert':
  Some ((), t-after-trace) ∈
    set-dist (execute (assert (f-x = f-final)) t-trace)
  using f-assert-out unfolding f-fold-eq by simp
show ?thesis
  using assert-outcomeD(2)[OF assert'] by simp
qed
have q-fold: PQueryCounter u = PQueryCounter t-after-trace
  by (rule receive-query-commits-mfold-preserves-query-counter[OF fold-out])
have q-assert: PQueryCounter t' = PQueryCounter u
proof –
  obtain i x len pw where fold-eq: fold-result = (i, x, len, pw)
  by (cases fold-result)
have assert':
  Some ((), t') ∈ set-dist (execute (assert (x = final)) u)
  using assert-out unfolding fold-eq by simp
show ?thesis
  using assert-outcomeD(2)[OF assert'] by simp
qed
show ?thesis
  using q-idx q-query q-f-fold q-f-assert q-fold q-assert by simp
qed

```

lemma honest-verifier-query-tail-from-prover-tail-no-failure:

assumes htv: honest-trace-valid

and create-f:

Some (f-merkle, s1) ∈
 set-dist (execute (p.create p.f-eval) init-state)

and *trace-fri*:
Some $((f-ps, f-ds, f-ls, f-ms), s\text{-trace-fri}) \in$
set-dist (*execute*
 $(p.\text{trace-fri-commit } f\text{-nrounds } [p.f] [p.\text{eval-domain}] [p.f\text{-eval}]$
 $[f\text{-merkle}]) \ s2)$
and *f-len-ps*: $\text{length } f-ps = \text{Suc } f\text{-nrounds}$
and *f-len-ds*: $\text{length } f-ds = \text{Suc } f\text{-nrounds}$
and *f-len-ls*: $\text{length } f-ls = \text{Suc } f\text{-nrounds}$
and *f-len-ms*: $\text{length } f-ms = \text{Suc } f\text{-nrounds}$
and *f-layers*:
 $\bigwedge i. i < \text{length } f-ps \implies i < \text{length } f-ds \implies i < \text{length } f-ls \implies$
 $f-ls ! i = \text{map } (\text{poly } (f-ps ! i)) (f-ds ! i)$
and *f-created*:
 $\bigwedge i. i < \text{length } f-ls \implies i < \text{length } f-ms \implies$
 $p.\text{created-tree } (f-ls ! i) (f-ms ! i) \ s\text{-trace-fri}$
and *f-init-len*: $\text{length } p.f\text{-eval} = 2 \wedge FN$
and *f-n-le-N*: $f\text{-nrounds} \leq FN$
and *f-nrounds-def*: $f\text{-nrounds} = \text{ceil-log } clength$
and *f-fl-eq*: $f-fl = \text{zip } f\text{-challenges } f\text{-roots}$
and *f-final-eq*: $f\text{-final} = \text{hd } (\text{last } f-ls)$
and *f-len-roots*: $\text{length } f\text{-roots} = f\text{-nrounds}$
and *f-len-challenges*: $\text{length } f\text{-challenges} = f\text{-nrounds}$
and *f-roots-aligned*: $\bigwedge j. j < f\text{-nrounds} \implies f\text{-roots} ! j = \text{value } (f-ms ! j)$
and *f-successors*:
 $\bigwedge j. j < f\text{-nrounds} \implies$
 $p.\text{next-fri-layer } (f-ps ! j) (f-ds ! j) (f\text{-challenges} ! j) =$
 $(f-ps ! \text{Suc } j, f-ds ! \text{Suc } j, f-ls ! \text{Suc } j)$
and *fri*:
Some $((ps, ds, ls, ms), s6) \in$
set-dist (*execute*
 $(p.\text{composition-fri-commit } nrounds [cp'] [p.\text{eval-domain}]$
 $[p.cp\text{-eval } as] [cp\text{-merkle}]) \ s5)$
and *len-ps*: $\text{length } ps = \text{Suc } nrounds$
and *len-ds*: $\text{length } ds = \text{Suc } nrounds$
and *len-ls*: $\text{length } ls = \text{Suc } nrounds$
and *len-ms*: $\text{length } ms = \text{Suc } nrounds$
and *layers*:
 $\bigwedge i. i < \text{length } ps \implies i < \text{length } ds \implies i < \text{length } ls \implies$
 $ls ! i = \text{map } (\text{poly } (ps ! i)) (ds ! i)$
and *created*:
 $\bigwedge i. i < \text{length } ls \implies i < \text{length } ms \implies$
 $p.\text{created-tree } (ls ! i) (ms ! i) \ s6$
and *init-len*: $\text{length } (p.cp\text{-eval } as) = 2 \wedge N$
and *n-le-N*: $nrounds \leq N$
and *nrounds-def*: $nrounds = \text{ceil-log } (\text{degree } cp' + 1)$
and *fr-eq*: $fr = \text{value } f\text{-merkle}$
and *as-eq*: $as' = as$
and *fl-eq*: $fl = \text{zip } \text{challenges } \text{roots}$
and *final-eq*: $\text{final} = \text{hd } (\text{last } ls)$

```

and len-roots: length roots = nrounds
and len-challenges: length challenges = nrounds
and roots-aligned:  $\bigwedge j. j < nrounds \implies roots ! j = value (ms ! j)$ 
and successors:
   $\bigwedge j. j < nrounds \implies$ 
     $p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =$ 
       $(ps ! Suc\ j, ds ! Suc\ j, ls ! Suc\ j)$ 
and tail:
  Some (prover-result, prover-state)  $\in$ 
    set-dist (execute (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) n) p-s)
and state-eq: PState v-s = PState p-s
and transcript-v: PTranscript v-s = rev ys
and transcript-p: PTranscript prover-state = ys @ PTranscript p-s
and prover-ext: prover-state  $\leq$  v-s
and query-ext: s1  $\leq$  v-s
and trace-fri-ext: s-trace-fri  $\leq$  v-s
and fri-ext: s6  $\leq$  v-s
and query-counter-eq: PQueryCounter v-s = PQueryCounter p-s
shows
  None  $\notin$  dom (dist (execute (ntimes (verifier-query-round fr f-fl f-final as' fl final) n) v-s))
using tail state-eq transcript-v transcript-p prover-ext query-ext trace-fri-ext fri-ext
  query-counter-eq
proof (induction n arbitrary: prover-result prover-state p-s v-s ys)
case 0
then show ?case
by simp
next
case (Suc n)
from ntimes-Suc-outcomeE[OF Suc.premis(1)] obtain x xs p-next where
  step:
    Some (x, p-next)  $\in$ 
      set-dist (execute (prover-query-round f-merkle f-ls f-ms ls ms) p-s)
and tail-rest:
  Some (xs, prover-state)  $\in$ 
    set-dist (execute (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) n) p-next)
and prover-result-eq: prover-result = x # xs
by blast
from prover-query-round-lookup-state-transcript[OF step] obtain idx where
  x-eq: x = ()
and lookup-next: fmlookup (HashMap p-next) (QueryIndexChallenge (PQueryCounter p-s)
(PState p-s)) = Some idx
and state-next:
  PState p-next =
    foldl concat (PState p-s)
      (query-decommitment-transcript (p.index (to-nat idx)) f-merkle @
        fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms)) @
        fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms)))
and tr-next:
  PTranscript p-next =

```

```

    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip ls ms))) @
    rev (fri-decommitment-transcript (p.index (to-nat idx)) (butlast (zip f-ls f-ms))) @
    rev (query-decommitment-transcript (p.index (to-nat idx)) f-merkle) @
    PTranscript p-s
  and p-s-next: p-s ≤ p-next
  by blast
  let ?idx' = p.index (to-nat idx)
  let ?query = query-decommitment-transcript ?idx' f-merkle
  let ?f-fri = fri-decommitment-transcript ?idx' (butlast (zip f-ls f-ms))
  let ?fri = fri-decommitment-transcript ?idx' (butlast (zip ls ms))
  have p-next-prover: p-next ≤ prover-state
  by (rule ntimes-hash-extends[OF - tail-rest])
    (rule prover-query-round-hash-extends)
  have p-next-v: p-next ≤ v-s
  using p-next-prover Suc.premis(5) by (meson p.hash-ext-trans)
  have lookup-v: fmlookup (HashMap v-s) (QueryIndexChallenge (PQueryCounter v-s) (PState
v-s)) = Some idx
  using p.hash-extension-lookup[OF lookup-next p-next-v] Suc.premis(2) Suc.premis(9)
  by simp
  have tail-tr-ext: transcript-extends prover-state p-next
  by (rule ntimes-transcript-extends[OF - tail-rest])
    (rule prover-query-round-transcript-extends)
  from tail-tr-ext obtain ys-tail where
    tr-tail: PTranscript prover-state = ys-tail @ PTranscript p-next
  unfolding transcript-extends-def by auto
  have ys-eq: ys = ys-tail @ rev ?fri @ rev ?f-fri @ rev ?query
  using Suc.premis(4) tr-tail tr-next by simp
  have transcript-round:
    PTranscript v-s = ?query @ ?f-fri @ ?fri @ rev ys-tail
  using Suc.premis(3) ys-eq by simp
  have step-nf:
    None ∉ dom (dist (execute (verifier-query-round fr f-fl f-final as' fl final) v-s))
  by (rule honest-verifier-query-round-no-failure-from-state[
    OF htv create-f trace-fri f-len-ps f-len-ds f-len-ls f-len-ms f-layers
    f-created f-init-len f-n-le-N f-nrounds-def f-fl-eq f-final-eq
    f-len-roots f-len-challenges f-roots-aligned f-successors
    fri len-ps len-ds len-ls len-ms layers created init-len n-le-N
    nrounds-def fr-eq as-eq fl-eq final-eq len-roots len-challenges
    roots-aligned successors lookup-v refl Suc.premis(6) Suc.premis(7) Suc.premis(8)
    transcript-round])
  show ?case
  unfolding ntimes.simps
  proof (rule no-failure-bindI[OF step-nf])
    fix v-res v-next
    assume v-step:
      Some (v-res, v-next) ∈
      set-dist (execute (verifier-query-round fr f-fl f-final as' fl final) v-s)
    have v-state:
      PState v-next =

```

```

    foldl concat (PState v-s) (?query @ ?f-fri @ ?fri)
  by (rule honest-verifier-query-round-outcome-from-state(1)[
    OF htv create-f trace-fri f-len-ps f-len-ds f-len-ls f-len-ms f-layers
    f-created f-init-len f-n-le-N f-nrounds-def f-fl-eq f-final-eq
    f-len-roots f-len-challenges f-roots-aligned f-successors
    fri len-ps len-ds len-ls len-ms layers created init-len n-le-N
    nrounds-def fr-eq as-eq fl-eq final-eq len-roots len-challenges
    roots-aligned successors lookup-v refl Suc.prem(6) Suc.prem(7) Suc.prem(8)
    transcript-round v-step])
  have v-transcript: PTranscript v-next = rev ys-tail
    by (rule honest-verifier-query-round-outcome-from-state(2)[
      OF htv create-f trace-fri f-len-ps f-len-ds f-len-ls f-len-ms f-layers
      f-created f-init-len f-n-le-N f-nrounds-def f-fl-eq f-final-eq
      f-len-roots f-len-challenges f-roots-aligned f-successors
      fri len-ps len-ds len-ls len-ms layers created init-len n-le-N
      nrounds-def fr-eq as-eq fl-eq final-eq len-roots len-challenges
      roots-aligned successors lookup-v refl Suc.prem(6) Suc.prem(7) Suc.prem(8)
      transcript-round v-step])
  have v-ext: v-s ≤ v-next
    by (rule honest-verifier-query-round-outcome-from-state(3)[
      OF htv create-f trace-fri f-len-ps f-len-ds f-len-ls f-len-ms f-layers
      f-created f-init-len f-n-le-N f-nrounds-def f-fl-eq f-final-eq
      f-len-roots f-len-challenges f-roots-aligned f-successors
      fri len-ps len-ds len-ls len-ms layers created init-len n-le-N
      nrounds-def fr-eq as-eq fl-eq final-eq len-roots len-challenges
      roots-aligned successors lookup-v refl Suc.prem(6) Suc.prem(7) Suc.prem(8)
      transcript-round v-step])
  have state-v-next: PState v-next = PState p-next
    using v-state state-next Suc.prem(2) by simp
  have prover-ext-next: prover-state ≤ v-next
    using Suc.prem(5) v-ext by (meson p.hash-ext-trans)
  have query-ext-next: s1 ≤ v-next
    using Suc.prem(6) v-ext by (meson p.hash-ext-trans)
  have trace-fri-ext-next: s-trace-fri ≤ v-next
    using Suc.prem(7) v-ext by (meson p.hash-ext-trans)
  have fri-ext-next: s6 ≤ v-next
    using Suc.prem(8) v-ext by (meson p.hash-ext-trans)
  have q-p-next: PQueryCounter p-next = Suc (PQueryCounter p-s)
    by (rule prover-query-round-query-counter[OF step])
  have q-v-next: PQueryCounter v-next = Suc (PQueryCounter v-s)
    by (rule verifier-query-round-query-counter[OF v-step])
  have query-counter-next: PQueryCounter v-next = PQueryCounter p-next
    using q-p-next q-v-next Suc.prem(9) by simp
  have tail-nf:
    None ∉ dom
    (dist (execute (ntimes (verifier-query-round fr f-fl f-final as' fl final) n) v-next))
  by (rule Suc.IH[
    OF tail-rest state-v-next v-transcript tr-tail prover-ext-next
    query-ext-next trace-fri-ext-next fri-ext-next query-counter-next])

```

```

show None  $\notin$  dom (dist (execute
  (ntimes (verifier-query-round fr f-fl f-final as' fl final) n  $\gg$ 
    ( $\lambda xs$ . return (v-res # xs))) v-next))
by (rule no-failure-bind-returnI[OF tail-nf])
qed
qed

lemma honest-verify-from-prover-no-failure:
assumes htv: honest-trace-valid
and prover:
  Some (prover-result, prover-state)  $\in$ 
    set-dist (execute p.prover-monad init-state)
and replay: replay-state = verifier-replay-state prover-state
shows None  $\notin$  dom (dist (execute v.verify-monad replay-state))
proof (rule honest-prover-monad-outcome[OF prover])
fix f-merkle s1 s2 f-nrounds f-ps f-ds f-ls f-ms s-trace-fri
  s-trace-final as s3 cp' s4 cp-merkle s5 nrounds ps ds ls ms s6 s-final
assume create-f:
  Some (f-merkle, s1)  $\in$ 
    set-dist (execute (p.create p.f-eval) init-state)
and created-f: p.created-tree p.f-eval f-merkle s1
and send-f:
  Some ((), s2)  $\in$  set-dist (execute (p.send (value f-merkle)) s1)
and f-nrounds-def: f-nrounds = ceil-log clength
and trace-fri:
  Some ((f-ps, f-ds, f-ls, f-ms), s-trace-fri)  $\in$ 
    set-dist (execute
      (p.trace-fri-commit f-nrounds [p.f] [p.eval-domain] [p.f-eval]
        [f-merkle]) s2)
and f-len-ps: length f-ps = Suc f-nrounds
and f-len-ds: length f-ds = Suc f-nrounds
and f-len-ls: length f-ls = Suc f-nrounds
and f-len-ms: length f-ms = Suc f-nrounds
and f-layers:
   $\bigwedge i. i < \text{length } f\text{-ps} \implies i < \text{length } f\text{-ds} \implies i < \text{length } f\text{-ls} \implies$ 
     $f\text{-ls} ! i = \text{map } (\text{poly } (f\text{-ps} ! i)) (f\text{-ds} ! i)$ 
and f-created:
   $\bigwedge i. i < \text{length } f\text{-ls} \implies i < \text{length } f\text{-ms} \implies$ 
     $p.\text{created-tree } (f\text{-ls} ! i) (f\text{-ms} ! i) s\text{-trace-fri}$ 
and trace-final-send:
  Some ((), s-trace-final)  $\in$ 
    set-dist (execute (p.send (hd (last f-ls))) s-trace-fri)
and alphas:
  Some (as, s3)  $\in$ 
    set-dist (execute
      (mmap (replicate (length spec)
        (do {
          a  $\leftarrow$  p.receive-alpha-challenge;
          p.send a;

```

```

      return a
    }))) s-trace-final)
and cp'-def: cp' = p.cp as p.f-powers
and send-degree:
  Some (((), s4) ∈ set-dist (execute (p.send (of-nat (degree cp')))) s3)
and create-cp:
  Some (cp-merkle, s5) ∈ set-dist (execute (p.create (p.cp-eval as)) s4)
and created-cp: p.created-tree (p.cp-eval as) cp-merkle s5
and nrounds-def: nrounds = ceil-log (degree cp' + 1)
and fri:
  Some ((ps, ds, ls, ms), s6) ∈
    set-dist (execute
      (p.composition-fri-commit nrounds [cp'] [p.eval-domain]
        [p.cp-eval as] [cp-merkle]) s5)
and len-ps: length ps = Suc nrounds
and len-ds: length ds = Suc nrounds
and len-ls: length ls = Suc nrounds
and len-ms: length ms = Suc nrounds
and layers:
  ∧ i. i < length ps ⇒ i < length ds ⇒ i < length ls ⇒
    ls ! i = map (poly (ps ! i)) (ds ! i)
and created:
  ∧ i. i < length ls ⇒ i < length ms ⇒
    p.created-tree (ls ! i) (ms ! i) s6
and final-send:
  Some (((), s-final) ∈ set-dist (execute (p.send (hd (last ls)))) s6)
and tail:
  Some (prover-result, prover-state) ∈
    set-dist (execute
      (ntimes (prover-query-round f-merkle f-ls f-ms ls ms) rounds) s-final)
from cp-eval-length-power obtain N where init-len:
  length (p.cp-eval as) = 2 ^ N
  by blast
have n-le-N: nrounds ≤ N
  by (rule fri-rounds-le-eval-domain-log[OF htv cp'-def nrounds-def init-len])
from f-eval-length-power obtain FN where f-init-len:
  length p.f-eval = 2 ^ FN
  by blast
have f-n-le-N: f-nrounds ≤ FN
proof -
  have cl-le-len: clength ≤ length p.f-eval
    using f-eval-length p.scale-pos by simp
  have f-rounds-le-len: f-nrounds ≤ ceil-log (length p.f-eval)
    using ceil-log-mono[OF cl-le-len] f-nrounds-def by simp
  have len-pos: 0 < length p.f-eval
    using f-init-len by simp
  have ceil-len-le: ceil-log (length p.f-eval) ≤ FN
proof -
  have length p.f-eval - 1 < 2 ^ FN

```



```

    using f-init-len len-pos by simp
  then have ceil-log (Suc (length p.f-eval - 1)) ≤ FN
    by (rule ceil-log-Suc-le-power)
  then show ?thesis
    using len-pos by simp
qed
show ?thesis
  using f-rounds-le-len ceil-len-le by simp
qed
from trace-fri-commit-initial-replay-successor-data[OF trace-fri]
obtain f-roots f-challenges where
  f-len-roots: length f-roots = f-nrounds
  and f-len-challenges: length f-challenges = f-nrounds
  and f-roots-aligned: ∀ j < f-nrounds. f-roots ! j = value (f-ms ! j)
  and f-successors:
    ∀ j < f-nrounds.
      p.next-fri-layer (f-ps ! j) (f-ds ! j) (f-challenges ! j) =
        (f-ps ! Suc j, f-ds ! Suc j, f-ls ! Suc j)
  by blast
from composition-fri-commit-initial-replay-successor-data[OF fri]
obtain roots challenges where
  len-roots: length roots = nrounds
  and len-challenges: length challenges = nrounds
  and roots-aligned: ∀ j < nrounds. roots ! j = value (ms ! j)
  and successors:
    ∀ j < nrounds.
      p.next-fri-layer (ps ! j) (ds ! j) (challenges ! j) =
        (ps ! Suc j, ds ! Suc j, ls ! Suc j)
  by blast
let ?round =
  do {
    a0 ← p.receive-alpha-challenge;
    let a0' = a0;
    a1 ← p.read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
  }
let ?prefix =
  do {
    fr ← p.read;
    f-fl ← ntimes v.receive-trace-fri-commits (ceil-log clength);
    f-final ← p.read;
    as' ← mmap (replicate (length spec) ?round);
    dg ← p.read;
    assert (to-nat dg ≤ p.maxDegree);
    fl ← ntimes v.receive-composition-fri-commits
      (ceil-log (to-nat dg + 1));
    final ← p.read;

```

```

    return (fr, f-fl, f-final, as', dg, fl, final)
  }
have prefix-nf:
  None  $\notin$  dom (dist (execute ?prefix replay-state))
by (rule honest-trace-root-alpha-degree-fri-final-dynamic-from-prover-tail-no-failure[
    OF htv create-f send-f f-nrounds-def trace-fri trace-final-send alphas
    cp'-def send-degree create-cp nrounds-def fri final-send tail replay])
have full-eq:
  v.verify-monad =
    (?prefix  $\gg$ 
      ( $\lambda$ out. case out of (fr, f-fl, f-final, as', dg, fl, final)  $\Rightarrow$ 
        ntimes (verifier-query-round fr f-fl f-final as' fl final) rounds))
  unfolding v.verify-monad-def
by (simp add: Let-def verifier-query-round-def sm-bind-assoc split: prod.splits)
show ?thesis
  unfolding full-eq
proof (rule no-failure-bindI[OF prefix-nf])
  fix out t
  assume prefix-out-raw: Some (out, t)  $\in$  set-dist (execute ?prefix replay-state)
  obtain fr f-fl f-final as' dg fl final where out-eq:
    out = (fr, f-fl, f-final, as', dg, fl, final)
  by (cases out) auto
  show None  $\notin$  dom (dist (execute
    ((case out of (fr, f-fl, f-final, as', dg, fl, final)  $\Rightarrow$ 
      ntimes (verifier-query-round fr f-fl f-final as' fl final) rounds)) t))
  proof –
  have prefix-out:
    Some ((fr, f-fl, f-final, as', dg, fl, final), t)  $\in$ 
      set-dist (execute ?prefix replay-state)
  using prefix-out-raw unfolding out-eq .
  obtain f-roots0 f-challenges0 roots0 challenges0 ys where
    f-len-roots0: length f-roots0 = f-nrounds
    and f-len-challenges0: length f-challenges0 = f-nrounds
    and f-fl-eq0: f-fl = zip f-challenges0 f-roots0
    and f-roots-aligned0:
       $\bigwedge j. j < f\text{-nrounds} \Rightarrow f\text{-roots0} ! j = \text{value } (f\text{-ms} ! j)$ 
    and f-successors0:
       $\bigwedge j. j < f\text{-nrounds} \Rightarrow$ 
        p.next-fri-layer (f-ps ! j) (f-ds ! j) (f-challenges0 ! j) =
          (f-ps ! Suc j, f-ds ! Suc j, f-ls ! Suc j)
    and f-final-eq: f-final = hd (last f-ls)
    and
    len-roots0: length roots0 = nrounds
    and len-challenges0: length challenges0 = nrounds
    and fl-eq0: fl = zip challenges0 roots0
    and roots-aligned0:
       $\bigwedge j. j < nrounds \Rightarrow \text{roots0} ! j = \text{value } (ms ! j)$ 
    and successors0:
       $\bigwedge j. j < nrounds \Rightarrow$ 

```

```

    p.next-fri-layer (ps ! j) (ds ! j) (challenges0 ! j) =
      (ps ! Suc j, ds ! Suc j, ls ! Suc j)
  and fr-eq: fr = value f-merkle
  and as-eq: as' = as
  and dg-eq: dg = of-nat (degree (p.cp as p.f-powers))
  and final-eq: final = hd (last ls)
  and state-t: PState t = PState s-final
  and transcript-t: PTranscript t = rev ys
  and transcript-prover:
    PTranscript prover-state = ys @ PTranscript s-final
  and replay-t: replay-state ≤ t
  using honest-trace-root-alpha-degree-fri-final-dynamic-from-prover-tail-outcome[
    OF htv create-f send-f f-nrounds-def trace-fri trace-final-send alphas
    cp'-def send-degree create-cp nrounds-def fri final-send tail replay prefix-out]
  by blast
have tail-hash: s-final ≤ prover-state
  by (rule ntimes-hash-extends[OF - tail])
  (rule prover-query-round-hash-extends)
have prover-replay: prover-state ≤ replay-state
  using replay
  unfolding verifier-replay-state-def less-eq-hash-ext-def less-eq-fmap-def
  by simp
have prover-t: prover-state ≤ t
  using prover-replay replay-t by (meson p.hash-ext-trans)
have alpha-data:
  length as = length spec ∧
  s-trace-final ≤ s3 ∧
  PState s3 = foldl concat (PState s-trace-final) as ∧
  PTranscript s3 = rev as @ PTranscript s-trace-final ∧
  PTraceFriCounter s3 = PTraceFriCounter s-trace-final ∧
  PCompositionFriCounter s3 = PCompositionFriCounter s-trace-final ∧
  PAlphaCounter s3 = PAlphaCounter s-trace-final + length as ∧
  PQueryCounter s3 = PQueryCounter s-trace-final ∧
  (∀ i < length as.
    fmlookup (HashMap s3)
      (AlphaChallenge (PAlphaCounter s-trace-final + i)
        (foldl concat (PState s-trace-final) (take i as))) =
      Some (as ! i))
  using alpha-mmap-outcome[OF alphas] by simp
have s1-s2: s1 ≤ s2
  using send-hash-extends[OF send-f] .
have s2-trace-fri: s2 ≤ s-trace-fri
  using p.trace-fri-commit-extends[OF trace-fri] .
have trace-fri-final: s-trace-fri ≤ s-trace-final
  using send-hash-extends[OF trace-final-send] .
have trace-final-s3: s-trace-final ≤ s3
  using alpha-data by simp
have s3-s4: s3 ≤ s4
  using send-hash-extends[OF send-degree] .

```

```

have s4-s5:  $s_4 \leq s_5$ 
  using create-hash-extends[OF create-cp] .
have s5-s6:  $s_5 \leq s_6$ 
  using p.composition-fri-commit-extends[OF fri] .
have s6-final-hash:  $s_6 \leq s\text{-final}$ 
  using send-hash-extends[OF final-send] .
have s1-prover:  $s_1 \leq \text{prover-state}$ 
  using s1-s2 s2-trace-fri trace-fri-final trace-final-s3
    s3-s4 s4-s5 s5-s6 s6-final-hash tail-hash
  by (meson p.hash-ext-trans)
have trace-fri-prover:  $s\text{-trace-fri} \leq \text{prover-state}$ 
  using trace-fri-final trace-final-s3 s3-s4 s4-s5 s5-s6
    s6-final-hash tail-hash
  by (meson p.hash-ext-trans)
have s6-prover:  $s_6 \leq \text{prover-state}$ 
  using s6-final-hash tail-hash by (rule p.hash-ext-trans)
have s1-t:  $s_1 \leq t$ 
  using s1-prover prover-t by (meson p.hash-ext-trans)
have s-trace-fri-t:  $s\text{-trace-fri} \leq t$ 
  using trace-fri-prover prover-t by (meson p.hash-ext-trans)
have s6-t:  $s_6 \leq t$ 
  using s6-prover prover-t by (meson p.hash-ext-trans)
have q-t-replay:  $P\text{QueryCounter } t = P\text{QueryCounter replay-state}$ 
  by (rule honest-trace-root-alpha-degree-fri-final-dynamic-prefix-preserves-query-counter[
    OF prefix-out])
have q-s1-zero:  $P\text{QueryCounter } s_1 = 0$ 
  using p.create-preserves-channel(6)[OF create-f]
  unfolding init-state-def by simp
have q-s2-zero:  $P\text{QueryCounter } s_2 = 0$ 
  using p.send-outcome[OF send-f] q-s1-zero by simp
have q-trace-fri-zero:  $P\text{QueryCounter } s\text{-trace-fri} = 0$ 
  using trace-fri-commit-preserves-query-counter[OF trace-fri] q-s2-zero
  by simp
have q-trace-final-zero:  $P\text{QueryCounter } s\text{-trace-final} = 0$ 
  using p.send-outcome[OF trace-final-send] q-trace-fri-zero by simp
have q-s3-zero:  $P\text{QueryCounter } s_3 = 0$ 
  using alpha-data q-trace-final-zero by simp
have q-s4-zero:  $P\text{QueryCounter } s_4 = 0$ 
  using p.send-outcome[OF send-degree] q-s3-zero by simp
have q-s5-zero:  $P\text{QueryCounter } s_5 = 0$ 
  using p.create-preserves-channel(6)[OF create-cp] q-s4-zero by simp
have q-s6-zero:  $P\text{QueryCounter } s_6 = 0$ 
  using composition-fri-commit-preserves-query-counter[OF fri] q-s5-zero
  by simp
have q-s-final-zero:  $P\text{QueryCounter } s\text{-final} = 0$ 
  using p.send-outcome[OF final-send] q-s6-zero by simp
have query-counter-t-final:  $P\text{QueryCounter } t = P\text{QueryCounter } s\text{-final}$ 
  using q-t-replay q-s-final-zero replay
  unfolding verifier-replay-state-def by simp

```

```

have tail-nf:
  None  $\notin$  dom
  (dist (execute (ntimes (verifier-query-round fr f-fl f-final as' fl final) rounds) t))
by (rule honest-verifier-query-tail-from-prover-tail-no-failure[
  OF htv create-f trace-fri f-len-ps f-len-ds f-len-ls f-len-ms f-layers
  f-created f-init-len f-n-le-N f-nrounds-def f-fl-eq0 f-final-eq
  f-len-roots0 f-len-challenges0 f-roots-aligned0 f-successors0
  fri len-ps len-ds len-ls len-ms layers created init-len n-le-N
  nrounds-def fr-eq as-eq fl-eq0 final-eq len-roots0 len-challenges0
  roots-aligned0 successors0 tail state-t transcript-t transcript-prover
  prover-t s1-t s-trace-fri-t s6-t query-counter-t-final])
show ?thesis
  unfolding out-eq using tail-nf by simp
qed
qed
qed

lemma honest-trace-valid-implies-no-failure:
  assumes htv: honest-trace-valid
  shows None  $\notin$  dom (dist (execute exec init-state))
  unfolding exec-def
proof (rule no-failure-bindI[OF prover-monad-no-failure])
  fix prover-result prover-state
  assume prover:
    Some (prover-result, prover-state)  $\in$ 
    set-dist (execute p.prover-monad init-state)
  show None  $\notin$  dom (dist (execute
    (modify ( $\lambda$ c. c[PState := 0, PTranscript := rev (PTranscript c),
      PTraceFriCounter := 0, PCompositionFriCounter := 0,
      PAlphaCounter := 0, PQueryCounter := 0]))  $\gg$ 
    ( $\lambda$ -. v.verify-monad))
    prover-state))
  proof (rule no-failure-bindI)
  show None  $\notin$  dom (dist (execute
    (modify ( $\lambda$ c. c[PState := 0, PTranscript := rev (PTranscript c),
      PTraceFriCounter := 0, PCompositionFriCounter := 0,
      PAlphaCounter := 0, PQueryCounter := 0]))
    prover-state))
  by simp
next
  fix u replay-state
  assume reset:
    Some (u, replay-state)  $\in$ 
    set-dist (execute
      (modify ( $\lambda$ c. c[PState := 0, PTranscript := rev (PTranscript c),
        PTraceFriCounter := 0, PCompositionFriCounter := 0,
        PAlphaCounter := 0, PQueryCounter := 0]))
      prover-state)
  have replay: replay-state = verifier-replay-state prover-state

```

```

    using modify-verifier-replay-state-outcome[OF reset] by simp
  show None  $\notin$  dom (dist (execute v.verify-monad replay-state))
    by (rule honest-verify-from-prover-no-failure[OF htv prover replay])
qed
qed

lemma honest-query-decommitments-no-fail:
  assumes honest-trace-valid
  shows no-failure exec init-state
  using honest-trace-valid-implies-no-failure[OF assms]
  unfolding no-failure-def .

lemma honest-fri-decommitments-no-fail:
  assumes honest-trace-valid
  shows no-failure exec init-state
  using honest-trace-valid-implies-no-failure[OF assms]
  unfolding no-failure-def .

lemma honest-final-constant-no-fail:
  assumes honest-trace-valid
  shows no-failure exec init-state
  using honest-trace-valid-implies-no-failure[OF assms]
  unfolding no-failure-def .

lemma honest-verify-no-fail:
  assumes honest-trace-valid
  shows no-failure exec init-state
  using honest-trace-valid-implies-no-failure[OF assms]
  unfolding no-failure-def .

lemma wp-event-is-none-eq-wp-error:
  wp-event m ( $\lambda a.$  Option.is-none a) s = wp-error m s
  unfolding wp-event-def wp-error-def
  by (rule arg-cong[where f= $\lambda Q.$  wp m Q s]) (auto intro!: ext split: option.splits)

lemma wp-error-zero-if-no-failure:
  assumes None  $\notin$  dom (dist (execute m s))
  shows wp-error m s = 0
proof -
  have failure-prob m s = 0
    using assms
    unfolding failure-prob-def
    by (cases dist (execute m s) None) (auto simp: domIff)
  then show ?thesis
    by (simp add: wp-error-eq-failure-prob)
qed

lemma completeness-wp-reduction:
  assumes honest-trace-valid

```

```

shows wp-event exec ( $\lambda a. \text{Option.is-none } a$ ) init-state = 0
using honest-trace-valid-implies-no-failure[OF assms]
by (simp add: wp-event-is-none-eq-wp-error wp-error-zero-if-no-failure)

```

```

lemma completeness:
  assumes honest-trace-valid
  shows wp-event exec ( $\lambda a. \text{Option.is-none } a$ ) init-state  $\leq (0::prob)$ 
  using completeness-wp-reduction[OF assms]
  by simp

```

```

end

```

```

end

```

```

theory Completeness
  imports Completeness-Verifier
begin

```

2.8 Completeness Entry Point

Public entry point for the honest-completeness development. The imported theories reduce the probabilistic verifier failure event to deterministic algebraic, transcript, Merkle, query, and FRI facts for the honest prover.

```

end

```

3 Executable Example

```

theory Square-Sequence
  imports
    Stark-Core.Stark
    HOL-Number-Theory.Number-Theory
    Stark-Core.Galois-Field-5
begin

```

3.1 Square Sequence Example

In the following example we generate a proof for a program which computes the square sequence: $x, x^2, x^4, \dots$. Since the computation is probabilistic we keep the executable example small: a sequence of size 2 with scaling factor 1.

We use $\text{GF}(5)$ - the finite field of order 5 - for our example.

- It is a Proth field.
- It is large enough for a trace of length 2 with a scaling factor of 1 and a shifted evaluation domain disjoint from H .

- It is still feasible to execute probabilistically. Larger fields make the executable example substantially more expensive.

3.1.1 Polynomial Constraints

The first number in the sequence (at position 0) is 2

definition $p0:: gf5\ poly\ list \Rightarrow gf5\ poly$
where
 $p0\ f \equiv f!0 - 2$

definition $r0:: nat\ list$
where
 $r0 \equiv [0]$

The last number in the sequence (at position 1) is 4

definition $p1:: gf5\ poly\ list \Rightarrow gf5\ poly$
where
 $p1\ f \equiv f!0 - 4$

definition $r1:: nat\ list$
where
 $r1 \equiv [1]$

For all intermediate positions (which is only position 0) the next position is the square of the previous position.

definition $p2:: gf5\ poly\ list \Rightarrow gf5\ poly$
where
 $p2\ ps \equiv ps\ !\ 1 - (ps\ !\ 0 * ps\ !\ 0)$

definition $r2:: nat\ list$
where
 $r2 \equiv [0..<1]$

The spec consists of a list containing the above constraints. Note that the third parameter in each entry is the degree of the constraint polynomial.

definition $spec::(gf5\ poly\ list \Rightarrow gf5\ poly) \times nat\ list \times nat\ list$
where
 $spec = [(p0, r0, 1), (p1, r1, 1), (p2, r2, 2)]$

We also need the spec in an other format. These need to be equivalent and for the future it should be unified with the other spec.

definition $p0':: gf5\ list \Rightarrow gf5$
where
 $p0'\ f \equiv f!0 - 2$

definition $p1':: gf5\ list \Rightarrow gf5$

where

$$p1' f \equiv f!0 - 4$$

definition $p2':: gf5\ list \Rightarrow gf5$

where

$$p2' ps \equiv ps ! 1 - (ps ! 0 * ps ! 0)$$

definition $spec2::(gf5\ list \Rightarrow gf5) \times nat\ list \Rightarrow list$

where

$$spec2 = [(p0', r0), (p1', r1), (p2', r2)]$$

3.1.2 Interpreting the stark locale

abbreviation $tlength::nat$ **where** $tlength \equiv 2$

lemma *of-nat-to-nat*:

fixes n

assumes $n \leq Max\ (degrees\ tlength\ spec)$

shows $Galois-Field-5.to-nat\ (Galois-Field-5.of-nat\ n) = n$

proof –

have $1 * (tlength - 1) - length\ r0 = 0$ **unfolding** $r0-def$ **by** *simp*

moreover have $1 * (tlength - 1) - length\ r1 = 0$ **unfolding** $r1-def$ **by** *simp*

moreover have $tlength * (tlength - 1) - length\ r2 = 1$ **unfolding** $r2-def$ **by** *simp*

ultimately have $degrees\ tlength\ spec = \{0,1\}$ **unfolding** $degrees-def\ spec-def$

apply $(auto\ simp\ add:r0-def\ r1-def\ r2-def)$

by $(metis\ diff-Suc-1'\ length-Suc-conv\ list.size(3)\ numeral-2-eq-2)$

then have $Max\ (degrees\ tlength\ spec) = 1$ **by** *simp*

then consider $n = 0 \mid n = 1$ **using** *assms* **by** *linarith*

then show *?thesis*

proof *cases*

case *1*

then show *?thesis*

by $(metis\ nat-zero-as-int\ of-nat-to-nat\ to-nat.rep-eq\ zero-gf5.rep-eq)$

next

case *2*

then show *?thesis*

by $(metis\ nat-one-as-int\ of-nat-to-nat\ one-gf5.rep-eq\ to-nat.rep-eq)$

qed

qed

lemma *exact-order-gf5-2-4*:

exact-order gf5-2 4

unfolding *exact-order-def*

apply $(intro\ conjI)$

apply *simp*

apply *eval*

apply $(intro\ allI\ impI)$

apply $(subgoal-tac\ m = 1 \vee m = 2 \vee m = 3)$

apply $(elim\ disjE; hypsubst; eval)$

by *linarith*

lemma *gf5-2-nonzero*:

$gf5-2 \neq 0$

by *eval*

lemma *numeral-2-gf5-nonzero*:

$(2::gf5) \neq 0$

by *eval*

lemma *square-sequence-spec-degree-fits*:

assumes $(c, roots, d) \in set\ spec$

shows $d * (tlength - 1) - length\ roots < tlength * 1$

using *assms*

unfolding *spec-def r0-def r1-def r2-def*

by *auto*

lemma *square-sequence-constraint-degree-wellformed*:

assumes *deg-f*: $degree\ f < tlength$

and *spec-entry*: $(c, roots, d) \in set\ spec$

shows

degree

$(c\ (map\ (\lambda p. f\ \circ_p\ XP\ ((gf5-2 \wedge ((5 - 1)\ div\ tlength)) \wedge p))\ [0..<tlength])) \leq d * (tlength - 1)$

proof –

let *?fs* =

$map\ (\lambda p. f\ \circ_p\ XP\ ((gf5-2 \wedge ((5 - 1)\ div\ tlength)) \wedge p))\ [0..<tlength]$

have *deg-XP*: $degree\ (XP\ a :: gf5\ poly) \leq 1$ **for** *a*

unfolding *XP-def* **by** *(rule degree-monom-le)*

have *deg-f-le*: $degree\ f \leq 1$

using *deg-f* **by** *simp*

have *compose-le*: $degree\ (f\ \circ_p\ XP\ a) \leq 1$ **for** *a*

proof –

have $degree\ (f\ \circ_p\ XP\ a) \leq degree\ f * degree\ (XP\ a)$

by *(rule degree-pcompose-le)*

also have $\dots \leq 1$

using *deg-f-le deg-XP*[*of a*] **by** *(cases degree f; cases degree (XP a :: gf5 poly)) auto*

finally show *?thesis* .

qed

have *deg0*: $degree\ (?fs\ !\ 0) \leq 1$

using *compose-le*[*of ((gf5-2 $\wedge$ ((5 - 1) div tlength)) $\wedge$ 0)]* **by** *simp*

have *deg1*: $degree\ (?fs\ !\ 1) \leq 1$

using *compose-le*[*of ((gf5-2 $\wedge$ ((5 - 1) div tlength)) $\wedge$ 1)]* **by** *simp*

have *deg-square*: $degree\ (?fs\ !\ 0 * ?fs\ !\ 0) \leq 2$

proof –

have $degree\ (?fs\ !\ 0 * ?fs\ !\ 0) \leq degree\ (?fs\ !\ 0) + degree\ (?fs\ !\ 0)$

by *(rule degree-mult-le)*

also have $\dots \leq 2$

```

    using deg0 by linarith
    finally show ?thesis .
qed
show ?thesis
proof -
  have p0-bound: degree (?fs ! 0 - 2) ≤ 1
    using degree-diff-le[OF deg0, of 2] by simp
  have p1-bound: degree (?fs ! 0 - 4) ≤ 1
    using degree-diff-le[OF deg0, of 4] by simp
  have p2-bound: degree (?fs ! 1 - ?fs ! 0 * ?fs ! 0) ≤ 2
  proof -
    have degree (?fs ! 1 - ?fs ! 0 * ?fs ! 0) ≤ max (degree (?fs ! 1)) (degree (?fs ! 0 * ?fs !
0))
      by (rule degree-diff-le-max)
    also have ... ≤ 2
      using deg1 deg-square by simp
    finally show ?thesis .
  qed
show ?thesis
  using spec-entry p0-bound p1-bound p2-bound
  unfolding spec-def p0-def p1-def p2-def
  by auto
qed
qed

lemma square-sequence-spec-roots-in-range:
  assumes (c, roots, d) ∈ set spec
  and r ∈ set roots
  shows r < tlength
  using assms
  unfolding spec-def r0-def r1-def r2-def
  by auto

lemma square-sequence-spec-roots-distinct:
  assumes (c, roots, d) ∈ set spec
  shows distinct roots
  using assms
  unfolding spec-def r0-def r1-def r2-def
  by auto

lemma gf5-2-notin-scale1-H-raw:
  gf5-2 ∉
    set (map (( $\wedge$ ) (gf5-2  $\wedge$  ((5 - 1) div (tlength * 1)))) [0..<1 * tlength])
  by eval

lemma gf5-to-nat-range:
  range Galois-Field-5.to-nat = {0..<5}
proof -
  have vals:

```

```

    Galois-Field-5.to-nat gf5-0 = 0
    Galois-Field-5.to-nat gf5-1 = 1
    Galois-Field-5.to-nat gf5-2 = 2
    Galois-Field-5.to-nat gf5-3 = 3
    Galois-Field-5.to-nat gf5-4 = 4
  by eval+
have mems:
  0 ∈ range Galois-Field-5.to-nat
  1 ∈ range Galois-Field-5.to-nat
  2 ∈ range Galois-Field-5.to-nat
  3 ∈ range Galois-Field-5.to-nat
  4 ∈ range Galois-Field-5.to-nat
  using vals by (metis rangeI)+
have range Galois-Field-5.to-nat ⊆ {0..<5}
proof
  fix n :: nat
  assume n ∈ range Galois-Field-5.to-nat
  then obtain x where n-eq: n = Galois-Field-5.to-nat x
    by blast
  show n ∈ {0..<5}
    unfolding n-eq
    by (cases x; simp add: vals)
qed
moreover have {0..<5} ⊆ range Galois-Field-5.to-nat
proof
  fix n :: nat
  assume n ∈ {0..<5}
  then have n = 0 ∨ n = 1 ∨ n = 2 ∨ n = 3 ∨ n = 4
    by auto
  then show n ∈ range Galois-Field-5.to-nat
    using mems by auto
qed
ultimately show ?thesis
  by blast
qed

lemma square-sequence-query-sample-space-size-dvd:
  tlength * 1 - Max (set [0..

```

```

and stark-h = stark.h
and stark-H = stark.H
and stark-eval-domain = stark.eval-domain
and stark-powers-scaled = stark.powers-scaled
and stark-index = stark.index
and stark-g-map = stark.g-map
and stark-maxDegree = stark.maxDegree
and stark-cp = stark.cp
apply standard
  apply simp
  apply (rule exI[where x=1], simp)
  apply (rule exI[where x=0], simp)
  apply (simp add: exact-order-gf5-2-4)
  apply simp
  apply simp
  apply simp
  apply (rule gf5-to-nat-range)
  apply simp
  apply simp
  apply (rule square-sequence-query-sample-space-size-dvd)
  apply (rule gf5-2-nonzero)
  apply (rule numeral-2-gf5-nonzero)
  apply (erule square-sequence-spec-degree-fits)
  apply (erule (1) square-sequence-constraint-degree-wellformed)
  apply (erule (1) square-sequence-spec-roots-in-range)
  apply (erule square-sequence-spec-roots-distinct)
  apply (rule gf5-2-notin-scale1-H-raw)
using of-nat-to-nat by presburger

```

3.1.3 Interpretation of prover locale

definition *trace* where $trace = [2,4]$

lemma *length-trace*: $length\ trace = tlength\ unfolding\ trace-def\ by\ simp$

global-interpretation

prover:

prover

concat-gf5 gf5-2 gf5-2 1 tlength 2 Galois-Field-5.to-nat Galois-Field-5.of-nat 5 spec 1 trace

defines *prover-f* = *prover.f*

and *prover-f-eval* = *prover.f-eval*

and *prover-cp-eval* = *prover.cp-eval*

and *prover-next-fri-domain* = *prover.next-fri-domain*

and *prover-next-fri-polynomial* = *prover.next-fri-polynomial*

and *prover-next-fri-layer* = *prover.next-fri-layer*

and *prover-fri-commit* = *prover.fri-commit*

and *prover-decommit-on-fri-layers* = *prover.decommit-on-fri-layers*

and *prover-decommit-on-query* = *prover.decommit-on-query*

and *prover-sproof* = *prover.sproof*

```

    and prover-prover-monad = prover.prover-monad
    and prover-f-powers = prover.f-powers
  apply (rule prover.intro)
  apply standard
  using length-trace by metis

```

3.1.4 Interpretation of verifier locale

```

lemma nth-map-tlength-0[simp]:
  assumes tlength ≤ length xs
  shows map f xs ! 0 = f (xs ! 0)
  by (rule nth-map) (use assms in linarith)

```

```

lemma nth-map-tlength-1[simp]:
  assumes tlength ≤ length xs
  shows map f xs ! 1 = f (xs ! 1)
  by (rule nth-map) (use assms in linarith)

```

global-interpretation

```

verifier:
verifier
  concat-gf5 gf5-2 gf5-2 1 tlength 2 Galois-Field-5.to-nat Galois-Field-5.of-nat 5 spec 1 spec2
defines verifier-receive-fri-commits = verifier.receive-fri-commits
  and verifier-receive-query-commits = verifier.receive-query-commits
  and verifier-check-decommit-on-query = verifier.check-decommit-on-query
  and verifier-verify = verifier.verify
  and verifier-cp-eval = verifier.cp-eval
  and verifier-verify-monad = verifier.verify-monad
apply (rule verifier.intro)
apply standard
by (auto simp: spec-def spec2-def p0-def p1-def p2-def p0'-def p1'-def p2'-def)

```

3.1.5 Creating and Verifying Proof

The executable prover state can be inspected with *value prover-proof*. Concrete verifier transcripts are intentionally not hard-coded here: the transcript shape changes when protocol checks such as trace FRI are added.

end

4 Soundness

```

theory Soundness-Core-Base
  imports Stark-Core.Stark
begin

```

4.1 Core Soundness Infrastructure

Core definitions and deterministic relations used by the STARK soundness development. Later layers separate Merkle binding, verifier execution, bad events, FRI, random-oracle accounting, and the public reduction theorems.

```

locale soundness =
  verifier concat omega shift scale clength powers to-nat of-nat size spec rounds spec2
for concat :: 'f::proth-field  $\Rightarrow$  'f  $\Rightarrow$  'f
  and omega :: 'f
  and shift :: 'f
  and scale :: nat
  and clength :: nat
  and powers :: nat
  and to-nat :: 'f  $\Rightarrow$  nat
  and of-nat :: nat  $\Rightarrow$  'f
  and size :: nat
  and spec :: (('f poly list  $\Rightarrow$  'f poly)  $\times$  nat list  $\times$  nat) list
  and rounds :: nat
  and spec2 :: (('f list  $\Rightarrow$  'f)  $\times$  nat list) list
+
fixes trace-fri-error :: prob
  and composition-fri-error :: prob
  and merkle-binding-error :: prob
assumes rounds-positive: 0 < rounds
  and spec-query-margin:
    Max (degrees clength spec) +
    sum-list (map ( $\lambda$ (-, roots, -). length roots) spec)
    < clength * scale - Max (set [0.. $\text{powers}$ ]) * scale
begin

definition trace-satisfies-constraint
  :: 'f poly  $\Rightarrow$  ('f poly list  $\Rightarrow$  'f poly)  $\Rightarrow$  nat list  $\Rightarrow$  bool
  where
    trace-satisfies-constraint f c roots  $\longleftrightarrow$ 
      ( $\forall r \in \text{set roots.}$ 
        poly (c (trace-powers-of f)) (g  $\wedge$  r) = 0)

definition exists-valid-trace :: bool
  where
    exists-valid-trace  $\longleftrightarrow$ 
      ( $\exists f.$ 
        degree f < clength  $\wedge$ 
        ( $\forall (c, \text{roots}, d) \in \text{set spec.}$ 
          trace-satisfies-constraint f c roots))

definition composition-error-bound :: prob
  where composition-error-bound = 1 / nnreal size

definition query-sample-space-size :: nat

```

where
query-sample-space-size =
 length * scale - Max (set [0..*powers*]) * scale

definition *query-sample-space* :: nat set
where *query-sample-space* = {*idx*. *idx* < *query-sample-space-size*}

definition *all-constraint-roots* :: 'f list
where
all-constraint-roots =
 remdups (List.concat (map ($\lambda(c, \text{roots}, d).$ g-map roots) *spec*))

definition *query-agreement-bound* :: nat
where *query-agreement-bound* = maxDegree + length *all-constraint-roots*

definition *query-error-bound* :: prob
where
query-error-bound =
 nnreal *query-agreement-bound* / nnreal *query-sample-space-size*

definition *soundness-bound* :: prob
where
soundness-bound =
 composition-error-bound +
 trace-fri-error +
 composition-fri-error +
 nnreal rounds * *query-error-bound*

definition *soundness-bound-with-merkle* :: prob
where
soundness-bound-with-merkle =
soundness-bound + merkle-binding-error

Soundness should expose a transcript-only verifier interface: the adversary chooses the proof transcript, while the verifier starts from an empty random-oracle/hash map and the initial Fiat-Shamir state.

definition *verifier-initial-state* :: 'f list $\Rightarrow$ 'f protocol-channel
where
verifier-initial-state tr =
 ($\langle$ HashMap = fmempty, PState = 0, PTranscript = tr,
 PTraceFriCounter = 0, PCompositionFriCounter = 0,
 PAlphaCounter = 0, PQueryCounter = 0 $\rangle$)

definition *verify-transcript*
where
verify-transcript tr =
 execute verify-monad (*verifier-initial-state* tr)

lemma *verifier-initial-state-simps*[*simp*]:

$\text{HashMap } (\text{verifier-initial-state } tr) = \text{fmempty}$
 $\text{PState } (\text{verifier-initial-state } tr) = 0$
 $\text{PTranscript } (\text{verifier-initial-state } tr) = tr$
 $\text{PTraceFriCounter } (\text{verifier-initial-state } tr) = 0$
 $\text{PCompositionFriCounter } (\text{verifier-initial-state } tr) = 0$
 $\text{PAlphaCounter } (\text{verifier-initial-state } tr) = 0$
 $\text{PQueryCounter } (\text{verifier-initial-state } tr) = 0$
unfolding *verifier-initial-state-def* **by** *simp-all*

definition *alpha-future-fresh*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{alpha-future-fresh } s \longleftrightarrow$
 $(\forall i x.$
 $\quad \text{PAlphaCounter } s \leq i \longrightarrow$
 $\quad \text{fmlookup } (\text{HashMap } s) (\text{AlphaChallenge } i x) = \text{None})$

definition *trace-fri-future-fresh*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{trace-fri-future-fresh } s \longleftrightarrow$
 $(\forall i x.$
 $\quad \text{PTraceFriCounter } s \leq i \longrightarrow$
 $\quad \text{fmlookup } (\text{HashMap } s) (\text{TraceFriChallenge } i x) = \text{None})$

definition *composition-fri-future-fresh*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{composition-fri-future-fresh } s \longleftrightarrow$
 $(\forall i x.$
 $\quad \text{PCompositionFriCounter } s \leq i \longrightarrow$
 $\quad \text{fmlookup } (\text{HashMap } s) (\text{CompositionFriChallenge } i x) = \text{None})$

definition *query-future-fresh*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{query-future-fresh } s \longleftrightarrow$
 $(\forall i x.$
 $\quad \text{PQueryCounter } s \leq i \longrightarrow$
 $\quad \text{fmlookup } (\text{HashMap } s) (\text{QueryIndexChallenge } i x) = \text{None})$

lemma *alpha-future-fresh-current:*

assumes *alpha-future-fresh* *s*

shows

$\text{fmlookup } (\text{HashMap } s)$
 $(\text{AlphaChallenge } (\text{PAlphaCounter } s) (\text{PState } s)) = \text{None}$

using *assms* **unfolding** *alpha-future-fresh-def* **by** *simp*

lemma *query-future-fresh-current:*

```

assumes query-future-fresh s
shows
  fmlookup (HashMap s)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = None
using assms unfolding query-future-fresh-def by simp

```

```

lemma verifier-initial-alpha-future-fresh[simp]:
  alpha-future-fresh (verifier-initial-state tr)
unfolding alpha-future-fresh-def by simp

```

```

lemma verifier-initial-trace-fri-future-fresh[simp]:
  trace-fri-future-fresh (verifier-initial-state tr)
unfolding trace-fri-future-fresh-def by simp

```

```

lemma verifier-initial-composition-fri-future-fresh[simp]:
  composition-fri-future-fresh (verifier-initial-state tr)
unfolding composition-fri-future-fresh-def by simp

```

```

lemma verifier-initial-query-future-fresh[simp]:
  query-future-fresh (verifier-initial-state tr)
unfolding query-future-fresh-def by simp

```

The current bound has no separate Merkle-binding error term. Merkle binding is represented by the conditional premise *initial-merkle-binding-no-bad* below, because the present hash-map model does not assign a collision-resistance probability to Merkle roots.

4.1.1 Proof structure for soundness

The proof should decompose an accepting verifier execution for a false statement into four bad events:

- the trace FRI accepts a table that is not the evaluation of a trace polynomial satisfying *degree f < clength*;
- the composition FRI accepts a table that is not low-degree;
- the random linear combination hides a violated constraint;
- the verifier's sampled query indices miss all disagreement points.

The following predicates and lemmas are the intended interface between the probabilistic proof over *verify-monad* and the deterministic algebraic arguments.

```

definition accepted :: ('x × 's) option ⇒ bool
where accepted out ⇔ ¬ Option.is-none out

```

```

definition trace-table-low-degree :: 'f list ⇒ bool
where
  trace-table-low-degree ys ⇔
    (∃ f. degree f < clength ∧ ys = map (poly f) eval-domain)

```

definition *composition-table-low-degree* :: nat $\Rightarrow$ 'f list $\Rightarrow$ bool
where
composition-table-low-degree d ys $\longleftrightarrow$
 $(\exists q. \text{degree } q \leq d \wedge \text{ys} = \text{map } (\text{poly } q) \text{ eval-domain})$

definition *trace-table-valid* :: 'f list $\Rightarrow$ bool
where
trace-table-valid ys $\longleftrightarrow$
 $(\exists f. \text{degree } f < \text{clength} \wedge$
 $\text{ys} = \text{map } (\text{poly } f) \text{ eval-domain} \wedge$
 $(\forall (c, \text{roots}, d) \in \text{set spec.}$
 $\text{trace-satisfies-constraint } f \text{ c roots}))$

definition *query-consistent-at* :: 'f list $\Rightarrow$ 'f list $\Rightarrow$ 'f list $\Rightarrow$ nat $\Rightarrow$ bool
where
query-consistent-at trace-table composition-table as idx $\longleftrightarrow$
 $\text{idx} < \text{clength} * \text{scale} \wedge$
 $\text{length trace-table} = \text{clength} * \text{scale} \wedge$
 $\text{length composition-table} = \text{clength} * \text{scale} \wedge$
 $(\forall i \in \text{set } (\text{powers-scaled idx}). i < \text{length trace-table}) \wedge$
 $\text{composition-table} ! \text{idx} =$
 $\text{cp-eval as } (\text{map } (!) \text{ trace-table}) (\text{powers-scaled idx}) (h \wedge \text{idx} * \text{shift})$

lemma *query-consistent-at-cong-opened-values*:

assumes *trace-len*:
 $\text{length trace-table} = \text{length trace-table}'$
and *composition-len*:
 $\text{length composition-table} = \text{length composition-table}'$
and *trace-values*:
 $\text{map } (!) \text{ trace-table} (\text{powers-scaled idx}) =$
 $\text{map } (!) \text{ trace-table}' (\text{powers-scaled idx})$
and *composition-value*:
 $\text{composition-table} ! \text{idx} = \text{composition-table}' ! \text{idx}$

shows

$\text{query-consistent-at trace-table composition-table as idx} \longleftrightarrow$
 $\text{query-consistent-at trace-table}' \text{ composition-table}' \text{ as idx}$

proof –

have *cp-eq*:
 $\text{cp-eval as } (\text{map } (!) \text{ trace-table}) (\text{powers-scaled idx})$
 $(h \wedge \text{idx} * \text{shift}) =$
 $\text{cp-eval as } (\text{map } (!) \text{ trace-table}') (\text{powers-scaled idx})$
 $(h \wedge \text{idx} * \text{shift})$
by (rule arg-cong[**where**
 $f = \lambda xs. \text{cp-eval as } xs (h \wedge \text{idx} * \text{shift}), \text{ OF trace-values}]$)
show ?thesis
unfolding *query-consistent-at-def*
using *trace-len composition-len composition-value cp-eq* **by** simp
qed

definition *all-queries-consistent* :: 'f list $\Rightarrow$ 'f list $\Rightarrow$ 'f list $\Rightarrow$ bool

where

all-queries-consistent trace-table composition-table as $\longleftrightarrow$
 $(\forall \text{idx} \in \text{query-sample-space}.$
 $\text{query-consistent-at trace-table composition-table as idx})$

definition *bad-query-indices* :: 'f list $\Rightarrow$ 'f list $\Rightarrow$ 'f list $\Rightarrow$ nat set

where

bad-query-indices trace-table composition-table as =
 $\{\text{idx} \in \text{query-sample-space}.$
 $\neg \text{query-consistent-at trace-table composition-table as idx}\}$

definition *query-agreement-indices* :: 'f list $\Rightarrow$ 'f list $\Rightarrow$ 'f list $\Rightarrow$ nat set

where

query-agreement-indices trace-table composition-table as =
 $\{\text{idx} \in \text{query-sample-space}.$
 $\text{query-consistent-at trace-table composition-table as idx}\}$

definition *query-disagreement-indices* :: 'f list $\Rightarrow$ 'f list $\Rightarrow$ 'f list $\Rightarrow$ nat set

where

query-disagreement-indices trace-table composition-table as =
bad-query-indices trace-table composition-table as

definition *query-samples-miss-disagreements*

:: 'f list $\Rightarrow$ 'f list $\Rightarrow$ 'f list $\Rightarrow$ nat list $\Rightarrow$ bool

where

query-samples-miss-disagreements trace-table composition-table as query-idxs $\longleftrightarrow$
 $\text{query-disagreement-indices trace-table composition-table as} \neq \{\}$ $\wedge$
 $\text{set query-idxs} \subseteq \text{query-agreement-indices trace-table composition-table as}$

definition *violated-constraints*

:: 'f poly $\Rightarrow$ (('f poly list $\Rightarrow$ 'f poly) $\times$ nat list $\times$ nat) set

where

violated-constraints f =
 $\{\text{entry} \in \text{set spec}.$
 $\text{case entry of } (c, \text{roots}, d) \Rightarrow$
 $\neg \text{trace-satisfies-constraint f c roots}\}$

definition *constraint-remainder-bad*

:: 'f poly $\Rightarrow$ (('f poly list $\Rightarrow$ 'f poly) $\times$ nat list $\times$ nat) $\Rightarrow$ bool

where

constraint-remainder-bad f entry $\longleftrightarrow$
 $(\text{case entry of } (c, \text{roots}, d) \Rightarrow$
 $(\exists r \in \text{set roots}.$
 $\text{poly } (c \text{ (trace-powers-of f)}) (g \wedge r) \neq 0))$

definition *composition-vanishes-on-constraint-domain* :: 'f poly $\Rightarrow$ 'f list $\Rightarrow$ bool

where

composition-vanishes-on-constraint-domain f as $\longleftrightarrow$
 $(\forall x \in \text{set } G. \text{poly } (cp \text{ as } (\text{trace-powers-of } f)) \ x = 0)$

definition *constraint-root-residual*

$:: 'f \text{ poly} \Rightarrow \text{nat} \Rightarrow$
 $('f \times (('f \text{ poly list} \Rightarrow 'f \text{ poly}) \times \text{nat list} \times \text{nat})) \Rightarrow 'f$

where

constraint-root-residual f r *weighted-entry* =
 $(\text{case weighted-entry of } (a, (c, \text{roots}, d)) \Rightarrow$
 $\text{if } r \in \text{set roots}$
 $\text{then } a * \text{poly } (c \ (\text{trace-powers-of } f)) \ (g \wedge r)$
 $\text{else } 0)$

definition *composition-root-residual* $:: 'f \text{ poly} \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f$

where

composition-root-residual f as r =
 $\text{sum-list } (\text{map } (\text{constraint-root-residual } f \ r) \ (\text{zip as spec}))$

definition *random-combination-hides-violations* $:: 'f \text{ poly} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$

where

random-combination-hides-violations f as $\longleftrightarrow$
 $\text{violated-constraints } f \neq \{\}$ $\wedge$
 $(\forall \text{entry} \in \text{violated-constraints } f.$
 $\text{case entry of } (c, \text{roots}, d) \Rightarrow$
 $(\forall r \in \text{set roots. composition-root-residual } f \text{ as } r = 0))$

definition *constraint-root-cofactor* $:: \text{nat list} \Rightarrow 'f \text{ poly}$

where

constraint-root-cofactor roots =
 $\text{prod } (\text{filter } (\lambda x. x \notin \text{set } (g\text{-map roots})) \ \text{all-constraint-roots})$

definition *common-denominator-poly* $:: 'f \text{ poly}$

where *common-denominator-poly* = $\text{prod all-constraint-roots}$

definition *common-denominator-numerator-term*

$:: 'f \text{ poly} \Rightarrow$
 $('f \times (('f \text{ poly list} \Rightarrow 'f \text{ poly}) \times \text{nat list} \times \text{nat})) \Rightarrow$
 $'f \text{ poly}$

where

common-denominator-numerator-term f *weighted-entry* =
 $(\text{case weighted-entry of } (a, (c, \text{roots}, d)) \Rightarrow$
 $CP \ a * c \ (\text{trace-powers-of } f) * \text{constraint-root-cofactor roots})$

definition *common-denominator-numerator* $:: 'f \text{ poly} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ poly}$

where

common-denominator-numerator f as =
 sum-list
 $(\text{map } (\text{common-denominator-numerator-term } f) \ (\text{zip as spec}))$

definition *composition-spec-eval-term*

$:: 'f \text{ poly} \Rightarrow 'f \Rightarrow$
 $(('f \times ((('f \text{ poly list} \Rightarrow 'f \text{ poly}) \times \text{nat list} \times \text{nat})) \Rightarrow$
 $'f$

where

composition-spec-eval-term $f \ x \ \text{weighted-entry} =$
 $(\text{case weighted-entry of } (a, (c, \text{roots}, d)) \Rightarrow$
 $a * (\text{poly } (c \ (\text{trace-powers-of } f)) \ x \ /$
 $\text{poly } (\text{prod } (g\text{-map } \text{roots})) \ x))$

definition *composition-spec-eval* $:: 'f \text{ poly} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f$

where

composition-spec-eval $f \ \text{as} \ x =$
 $\text{sum-list } (\text{map } (\text{composition-spec-eval-term } f \ x) \ (\text{zip } \text{as } \text{spec}))$

definition *common-denominator-constraint-root-residual*

$:: 'f \text{ poly} \Rightarrow \text{nat} \Rightarrow$
 $(('f \times ((('f \text{ poly list} \Rightarrow 'f \text{ poly}) \times \text{nat list} \times \text{nat})) \Rightarrow 'f$

where

common-denominator-constraint-root-residual $f \ r \ \text{weighted-entry} =$
 $(\text{case weighted-entry of } (a, (c, \text{roots}, d)) \Rightarrow$
 $\text{if } r \in \text{set roots}$
 then
 $a * \text{poly } (c \ (\text{trace-powers-of } f)) \ (g \wedge r) *$
 $\text{poly } (\text{constraint-root-cofactor } \text{roots}) \ (g \wedge r)$
 $\text{else } 0)$

definition *common-denominator-composition-root-residual*

$:: 'f \text{ poly} \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f$

where

common-denominator-composition-root-residual $f \ \text{as} \ r =$
 sum-list
 $(\text{map } (\text{common-denominator-constraint-root-residual } f \ r) \ (\text{zip } \text{as } \text{spec}))$

definition *random-combination-common-denominator-hides-violations*

$:: 'f \text{ poly} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$

where

random-combination-common-denominator-hides-violations $f \ \text{as} \longleftrightarrow$
 $\text{violated-constraints } f \neq \{\} \wedge$
 $(\forall \text{entry} \in \text{violated-constraints } f.$
 $\text{case entry of } (c, \text{roots}, d) \Rightarrow$
 $(\forall r \in \text{set roots.}$
 $\text{common-denominator-composition-root-residual } f \ \text{as} \ r = 0))$

definition *common-denominator-degree-bounds* $:: 'f \text{ poly} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$

where

common-denominator-degree-bounds $f \ \text{as} \longleftrightarrow$
 $\text{degree } (\text{common-denominator-numerator } f \ \text{as}) < \text{query-sample-space-size} \wedge$
 $(\forall q. \text{degree } q \leq \text{maxDegree} \longrightarrow$

$\text{degree } (\text{common-denominator-poly} * q) < \text{query-sample-space-size}$

definition *query-agreement-margin* :: bool

where

query-agreement-margin $\longleftrightarrow$
query-agreement-bound < *query-sample-space-size*

definition *spec-degree-wellformed* :: bool

where

spec-degree-wellformed $\longleftrightarrow$ *query-agreement-margin*

lemma *spec-degree-wellformedI*:

assumes *query-agreement-margin*

shows *spec-degree-wellformed*

using *assms* **unfolding** *spec-degree-wellformed-def* **by** *simp*

lemma *spec-degree-wellformed-query-agreement-margin*:

assumes *spec-degree-wellformed*

shows *query-agreement-margin*

using *assms* **unfolding** *spec-degree-wellformed-def* **by** *simp*

lemma *spec-degree-wellformed-constraint-degree*:

assumes *wf*: *spec-degree-wellformed*

and *deg-f*: *degree f* < *clength*

and *spec-entry*: $(c, \text{roots}, d) \in \text{set spec}$

shows *degree (c (trace-powers-of f))* $\leq d * (\text{clength} - 1)$

by (rule *constraint-degree-wellformed[OF deg-f spec-entry]*)

lemma *query-agreement-marginD*:

assumes *query-agreement-margin*

shows *query-agreement-bound* < *query-sample-space-size*

using *assms* **unfolding** *query-agreement-margin-def* **by** *simp*

lemma *spec-degree-wellformed-query-margin*:

assumes *spec-degree-wellformed*

shows *query-agreement-bound* < *query-sample-space-size*

by (rule *query-agreement-marginD*

[*OF spec-degree-wellformed-query-agreement-margin[OF assms]*])

lemma *all-constraint-roots-length-le-sum-roots*:

length all-constraint-roots $\leq$

sum-list (map ($\lambda(-, \text{roots}, -). \text{length roots}$) spec)

proof –

have *length all-constraint-roots* $\leq$

length (List.concat (map ($\lambda(c, \text{roots}, d). g\text{-map roots}$) spec))

unfolding *all-constraint-roots-def*

by (rule *length-remdups-leq*)

also have ... =

sum-list (map ($\lambda(-, \text{roots}, -). \text{length roots}$) spec)

```

    unfolding g-map-def
    by (induction spec) auto
    finally show ?thesis .
qed

```

```

lemma spec-degree-wellformed-from-spec-query-margin:
  spec-degree-wellformed
proof -
  have margin: query-agreement-bound < query-sample-space-size
    using spec-query-margin all-constraint-roots-length-le-sum-roots
    unfolding query-agreement-bound-def query-sample-space-size-def
    maxDegree-def
    by linarith
  then show ?thesis
    unfolding spec-degree-wellformed-def query-agreement-margin-def by simp
qed

```

```

lemma exists-valid-traceI:
  assumes degree f < clength
    and  $\bigwedge c \text{ roots } d. (c, \text{roots}, d) \in \text{set spec} \implies$ 
      trace-satisfies-constraint f c roots
  shows exists-valid-trace
  using assms unfolding exists-valid-trace-def by auto

```

```

lemma trace-table-valid-imp-exists-valid-trace:
  assumes trace-table-valid ys
  shows exists-valid-trace
  using assms unfolding trace-table-valid-def
  by (auto intro!: exists-valid-traceI)

```

```

lemma false-statement-violated-constraints:
  assumes false-statement:  $\neg$  exists-valid-trace
    and low-degree: degree f < clength
  shows violated-constraints f  $\neq \{\}$ 
proof
  assume empty: violated-constraints f =  $\{\}$ 
  have  $\bigwedge c \text{ roots } d. (c, \text{roots}, d) \in \text{set spec} \implies$ 
    trace-satisfies-constraint f c roots
    using empty unfolding violated-constraints-def by force
  then have exists-valid-trace
    using low-degree unfolding exists-valid-trace-def by auto
  then show False
    using false-statement by contradiction
qed

```

```

lemma constraint-remainder-bad-iff-not-trace-satisfies:
  constraint-remainder-bad f (c, roots, d)  $\longleftrightarrow$ 
     $\neg$  trace-satisfies-constraint f c roots
  unfolding constraint-remainder-bad-def trace-satisfies-constraint-def

```


by auto

lemma *violated-constraints-has-remainder-bad*:
 assumes *violated-constraints* $f \neq \{\}$
 shows $\exists \text{entry} \in \text{violated-constraints } f. \text{constraint-remainder-bad } f \text{ entry}$
 using *assms*
 unfolding *violated-constraints-def*
 by (auto simp: *constraint-remainder-bad-iff-not-trace-satisfies*)

lemma *random-combination-hides-violations-has-remainder-bad*:
 assumes *random-combination-hides-violations* $f \text{ as}$
 shows $\exists \text{entry} \in \text{violated-constraints } f. \text{constraint-remainder-bad } f \text{ entry}$
 using *assms violated-constraints-has-remainder-bad*
 unfolding *random-combination-hides-violations-def*
 by *blast*

lemma *linear-equation-solution-unique*:
 fixes $c \ b \ x \ y :: 'f$
 assumes *c-nonzero*: $c \neq 0$
 and *x-sol*: $c * x + b = 0$
 and *y-sol*: $c * y + b = 0$
 shows $x = y$
proof –
 have *x-eq*: $c * x = - b$
 proof –
 have $c * x + b + (- b) = 0 + (- b)$
 using *x-sol* **by** *simp*
 then show *?thesis*
 by *simp*
 qed
 have *y-eq*: $c * y = - b$
 proof –
 have $c * y + b + (- b) = 0 + (- b)$
 using *y-sol* **by** *simp*
 then show *?thesis*
 by *simp*
 qed
 have $c * (x - y) = 0$
 using *x-eq y-eq* **by** (*simp add: algebra-simps*)
 then have $x - y = 0$
 using *c-nonzero* **by** *simp*
 then show *?thesis*
 by *simp*
qed

lemma *composition-root-residual-sum-nth*:
 assumes *len*: $\text{length } as = \text{length } spec$
 shows
 $\text{composition-root-residual } f \text{ as } r =$

```

    ( $\sum i \in \{0..<\text{length spec}\}$ .
       $\text{constraint-root-residual } f \ r \ (as ! i, \text{spec} ! i)$ )
  proof -
    have  $\text{composition-root-residual } f \ as \ r =$ 
       $\text{sum-list } (\text{map } (\text{constraint-root-residual } f \ r) \ (\text{zip } as \ \text{spec}))$ 
    by (simp add:  $\text{composition-root-residual-def}$ )
    also have ... =
      ( $\sum i \in \{0..<\text{length } (\text{zip } as \ \text{spec})\}$ .
         $\text{map } (\text{constraint-root-residual } f \ r) \ (\text{zip } as \ \text{spec}) ! i$ )
    by (simp add:  $\text{sum-list-sum-nth}$ )
    also have ... =
      ( $\sum i \in \{0..<\text{length spec}\}$ .
         $\text{constraint-root-residual } f \ r \ (as ! i, \text{spec} ! i)$ )
    using len by (intro  $\text{sum.cong}$ ) auto
    finally show ?thesis .
  qed

lemma  $\text{composition-root-residual-split-index}$ :
  assumes len:  $\text{length } as = \text{length spec}$ 
    and i-bound:  $i < \text{length spec}$ 
  shows
     $\text{composition-root-residual } f \ as \ r =$ 
       $\text{constraint-root-residual } f \ r \ (as ! i, \text{spec} ! i) +$ 
      ( $\sum j \in \{0..<\text{length spec}\} - \{i\}$ .
         $\text{constraint-root-residual } f \ r \ (as ! j, \text{spec} ! j)$ )
  proof -
    let ?F =  $\lambda j. \text{constraint-root-residual } f \ r \ (as ! j, \text{spec} ! j)$ 
    have  $\text{composition-root-residual } f \ as \ r = (\sum j \in \{0..<\text{length spec}\}. ?F \ j)$ 
    by (rule  $\text{composition-root-residual-sum-nth[OF len]}$ )
    also have ... =  $?F \ i + (\sum j \in \{0..<\text{length spec}\} - \{i\}. ?F \ j)$ 
    using i-bound by (simp add:  $\text{sum.remove}$ )
    finally show ?thesis .
  qed

lemma  $\text{composition-root-residual-zero-unique-nth}$ :
  fixes as bs :: 'f list
  assumes len-as:  $\text{length } as = \text{length spec}$ 
    and len-bs:  $\text{length } bs = \text{length spec}$ 
    and i-bound:  $i < \text{length spec}$ 
    and spec-i:  $\text{spec} ! i = (c, \text{roots}, d)$ 
    and r-in:  $r \in \text{set roots}$ 
    and nonzero:
       $\text{poly } (c \ (\text{trace-powers-of } f)) \ (g \wedge r) \neq 0$ 
    and outside:
       $\bigwedge j. j < \text{length spec} \implies j \neq i \implies as ! j = bs ! j$ 
    and zero-as:  $\text{composition-root-residual } f \ as \ r = 0$ 
    and zero-bs:  $\text{composition-root-residual } f \ bs \ r = 0$ 
  shows  $as ! i = bs ! i$ 
  proof -

```

```

let ?C = poly (c (trace-powers-of f)) (g ^ r)
let ?B =
  (∑ j ∈ {0..<length spec} - {i}.
    constraint-root-residual f r (as ! j, spec ! j))
have B-eq:
  (∑ j ∈ {0..<length spec} - {i}.
    constraint-root-residual f r (bs ! j, spec ! j)) = ?B
  by (intro sum.cong) (use outside in auto)
have as-eq:
  composition-root-residual f as r = ?C * (as ! i) + ?B
  using composition-root-residual-split-index[OF len-as i-bound]
    spec-i r-in
  unfolding constraint-root-residual-def
  by (simp add: algebra-simps)
have bs-eq:
  composition-root-residual f bs r = ?C * (bs ! i) + ?B
  using composition-root-residual-split-index[OF len-bs i-bound]
    spec-i r-in B-eq
  unfolding constraint-root-residual-def
  by (simp add: algebra-simps)
show ?thesis
proof (rule linear-equation-solution-unique[OF nonzero])
  show ?C * (as ! i) + ?B = 0
    using as-eq zero-as by simp
  show ?C * (bs ! i) + ?B = 0
    using bs-eq zero-bs by simp
qed
qed

```

lemma *random-combination-hides-violations-unique-nth:*

```

fixes as bs :: 'f list
assumes len-as: length as = length spec
  and len-bs: length bs = length spec
  and i-bound: i < length spec
  and spec-i: spec ! i = (c, roots, d)
  and violated: (c, roots, d) ∈ violated-constraints f
  and r-in: r ∈ set roots
  and nonzero:
    poly (c (trace-powers-of f)) (g ^ r) ≠ 0
  and outside:
    ∧ j. j < length spec ⇒ j ≠ i ⇒ as ! j = bs ! j
  and hides-as: random-combination-hides-violations f as
  and hides-bs: random-combination-hides-violations f bs
shows as ! i = bs ! i

```

proof –

```

have zero-as: composition-root-residual f as r = 0
  using hides-as violated r-in
  unfolding random-combination-hides-violations-def by auto
have zero-bs: composition-root-residual f bs r = 0

```

```

    using hides-bs violated r-in
    unfolding random-combination-hides-violations-def by auto
show ?thesis
  by (rule composition-root-residual-zero-unique-nth
      [OF len-as len-bs i-bound spec-i r-in nonzero outside zero-as zero-bs])
qed

lemma bad-query-indices-subset:
  bad-query-indices trace-table composition-table as  $\subseteq \{0..<\text{length} * \text{scale}\}$ 
  unfolding bad-query-indices-def query-sample-space-def query-sample-space-size-def
  by auto

lemma index-less-domain:
  index  $x < \text{length} * \text{scale}$ 
proof -
  have max-lt:  $\text{Max} (\text{set } [0..<\text{powers}]) < \text{length}$ 
  proof -
    have  $\text{Max} (\text{set } [0..<\text{powers}]) = \text{powers} - 1$ 
      using powers-pos by (intro Max-eqI) auto
    also have  $\dots < \text{length}$ 
      using powers-pos powers-le-length by simp
    finally show ?thesis .
  qed
  have mod-pos:  $0 < \text{length} * \text{scale} - \text{Max} (\text{set } [0..<\text{powers}]) * \text{scale}$ 
    using max-lt scale-pos
    by (simp add: diff-mult-distrib2)
  have index  $x < \text{length} * \text{scale} - \text{Max} (\text{set } [0..<\text{powers}]) * \text{scale}$ 
    unfolding index-def using mod-pos by simp
  moreover have  $\text{length} * \text{scale} - \text{Max} (\text{set } [0..<\text{powers}]) * \text{scale} \leq \text{length} * \text{scale}$ 
    by simp
  ultimately show ?thesis
    by linarith
qed

lemma query-sample-space-size-pos:
   $0 < \text{query-sample-space-size}$ 
proof -
  have max-lt:  $\text{Max} (\text{set } [0..<\text{powers}]) < \text{length}$ 
  proof -
    have  $\text{Max} (\text{set } [0..<\text{powers}]) = \text{powers} - 1$ 
      using powers-pos by (intro Max-eqI) auto
    also have  $\dots < \text{length}$ 
      using powers-pos powers-le-length by simp
    finally show ?thesis .
  qed
  show ?thesis
    unfolding query-sample-space-size-def
    using max-lt scale-pos by (simp add: diff-mult-distrib2)
qed

```

lemma *index-less-query-sample-space*:
index $x < \text{query-sample-space-size}$
unfolding *index-def query-sample-space-size-def*
using *query-sample-space-size-pos*[*unfolded query-sample-space-size-def*] **by** *simp*

lemma *query-sample-space-size-le-domain*:
query-sample-space-size $\leq \text{clength} * \text{scale}$
unfolding *query-sample-space-size-def* **by** *simp*

lemma *query-sample-spaceD*:
assumes *idx* $\in \text{query-sample-space}$
shows *idx* $< \text{query-sample-space-size}$
using *assms* **unfolding** *query-sample-space-def* **by** *simp*

lemma *finite-query-sample-space*[*simp*]:
finite query-sample-space
unfolding *query-sample-space-def* **by** *simp*

lemma *card-query-sample-space*[*simp*]:
card query-sample-space $= \text{query-sample-space-size}$
unfolding *query-sample-space-def* **by** *simp*

lemma *query-error-bound-query-sample-space*:
query-error-bound $=$
 $\text{nnreal } \text{query-agreement-bound} / \text{nnreal } (\text{card } \text{query-sample-space})$
unfolding *query-error-bound-def* **by** *simp*

lemma *query-sample-space-less-domain*:
assumes *idx* $\in \text{query-sample-space}$
shows *idx* $< \text{clength} * \text{scale}$
using *query-sample-spaceD*[*OF assms*] *query-sample-space-size-le-domain*
by *linarith*

lemma *query-sample-space-powers-scaled-bound*:
assumes *idx-in*: *idx* $\in \text{query-sample-space}$
and *i-in*: *i* $\in \text{set } (\text{powers-scaled } \text{idx})$
shows *i* $< \text{clength} * \text{scale}$
proof –
obtain *k* **where** *k-bound*: *k* $< \text{powers}$ **and** *i-eq*: *i* $= \text{idx} + k * \text{scale}$
using *i-in* **unfolding** *powers-scaled-def* **by** *auto*
have *max-eq*: $\text{Max } (\text{set } [0..<\text{powers}]) = \text{powers} - 1$
using *powers-pos* **by** (*intro Max-eqI*) *auto*
have *k-le-max*: *k* $\leq \text{Max } (\text{set } [0..<\text{powers}])$
using *k-bound* **unfolding** *max-eq* **by** *simp*
have *idx-bound*:
 $\text{idx} < \text{clength} * \text{scale} - \text{Max } (\text{set } [0..<\text{powers}]) * \text{scale}$
using *query-sample-spaceD*[*OF idx-in*]
unfolding *query-sample-space-size-def* .

```

have  $idx + k * scale \leq idx + \text{Max}(\text{set } [0..<powers]) * scale$ 
  using k-le-max by simp
also have  $\dots < \text{clength} * scale$ 
  using idx-bound by simp
finally show ?thesis
  unfolding i-eq .
qed

definition query-sample-domain :: 'f list
  where query-sample-domain = map ( $\lambda idx. h \wedge idx * \text{shift}$ )  $[0..<\text{query-sample-space-size}]$ 

lemma eval-domain-length:
   $\text{length } \text{eval-domain} = \text{clength} * \text{scale}$ 
  unfolding eval-domain-def H-def by simp

lemma eval-domain-nth:
  assumes  $i < \text{clength} * \text{scale}$ 
  shows  $\text{eval-domain} ! i = h \wedge i * \text{shift}$ 
  using assms
  unfolding eval-domain-def H-def
  by (simp add: mult.commute)

lemma h-nonzero:
   $h \neq 0$ 
  unfolding h-def using omega-nonzero-derived by simp

lemma h-power-inj-on-eval-domain:
  assumes i-bound:  $i < \text{clength} * \text{scale}$ 
    and j-bound:  $j < \text{clength} * \text{scale}$ 
    and eq:  $h \wedge i = h \wedge j$ 
  shows  $i = j$ 
proof (rule ccontr)
  let ?N =  $\text{clength} * \text{scale}$ 
  assume neg:  $i \neq j$ 
  consider  $i < j \mid j < i$ 
    using neg by linarith
  then show False
proof cases
  case 1
  have diff-pos:  $0 < j - i$ 
    using 1 by simp
  have diff-lt:  $j - i < ?N$ 
    using j-bound by simp
  have hi-nonzero:  $h \wedge i \neq 0$ 
    using h-nonzero by simp
  have j-decomp:  $j = i + (j - i)$ 
    using 1 by simp
  have mult-eq:  $h \wedge j = h \wedge i * h \wedge (j - i)$ 
    by (subst j-decomp) (simp add: power-add)

```

```

have cancel-eq:  $h^i * 1 = h^i * h^{(j-i)}$ 
  using eq mult-eq by simp
have  $h^i = 0 \vee 1 = h^{(j-i)}$ 
  using cancel-eq by (simp only: mult-cancel-left)
then have  $h^{(j-i)} = 1$ 
  using hi-nonzero by auto
then show False
  using h-exact-order diff-pos diff-lt unfolding exact-order-def by blast
next
case 2
have diff-pos:  $0 < i - j$ 
  using 2 by simp
have diff-lt:  $i - j < ?N$ 
  using i-bound by simp
have hj-nonzero:  $h^j \neq 0$ 
  using h-nonzero by simp
have i-decomp:  $i = j + (i - j)$ 
  using 2 by simp
have mult-eq:  $h^i = h^j * h^{(i-j)}$ 
  by (subst i-decomp) (simp add: power-add)
have cancel-eq:  $h^j * 1 = h^j * h^{(i-j)}$ 
  using eq mult-eq by simp
have  $h^j = 0 \vee 1 = h^{(i-j)}$ 
  using cancel-eq by (simp only: mult-cancel-left)
then have  $h^{(i-j)} = 1$ 
  using hj-nonzero by auto
then show False
  using h-exact-order diff-pos diff-lt unfolding exact-order-def by blast
qed
qed

lemma H-distinct:
  distinct H
proof -
  have inj: inj-on (( $\wedge$ ) h) (set [0.. $\text{scale} * \text{clength}$ ])
  proof (intro inj-onI)
    fix x y
    assume x-in:  $x \in \text{set } [0.. $\text{scale} * \text{clength}$ ]$ 
      and y-in:  $y \in \text{set } [0.. $\text{scale} * \text{clength}$ ]$ 
      and eq:  $h^x = h^y$ 
    have x-bound:  $x < \text{clength} * \text{scale}$ 
      using x-in by (simp add: mult.commute)
    have y-bound:  $y < \text{clength} * \text{scale}$ 
      using y-in by (simp add: mult.commute)
    show  $x = y$ 
      by (rule h-power-inj-on-eval-domain[OF x-bound y-bound eq])
  qed
  show ?thesis
    unfolding H-def using inj by (simp add: distinct-map)

```

qed

lemma *eval-domain-distinct*:

distinct eval-domain

proof –

have *inj*: *inj-on* $((*) \text{ shift})$ (*set H*)

using *shift-nonzero* **by** (*auto intro!*: *inj-onI*)

show *?thesis*

unfolding *eval-domain-def*

using *H-distinct inj* **by** (*simp add: distinct-map*)

qed

lemma *card-set-eval-domain*:

*card (set eval-domain) = clength * scale*

using *eval-domain-distinct eval-domain-length*

by (*simp add: distinct-card*)

lemma *poly-eq-on-eval-domainI*:

assumes *deg-p*: *degree p < clength * scale*

and *deg-q*: *degree q < clength * scale*

and *agree*:

$\bigwedge \text{idx. idx} < \text{clength} * \text{scale} \implies$

$\text{poly } p (h \wedge \text{idx} * \text{shift}) = \text{poly } q (h \wedge \text{idx} * \text{shift})$

shows $p = q$

proof (*rule poly-eqI-degree*[**where** $A = \text{set eval-domain}$])

fix *x*

assume *x-in*: $x \in \text{set eval-domain}$

then obtain *idx* **where** *idx-bound*: $\text{idx} < \text{clength} * \text{scale}$

and *x-eq*: $x = h \wedge \text{idx} * \text{shift}$

unfolding *eval-domain-def H-def*

by (*auto simp: mult.commute*)

show $\text{poly } p x = \text{poly } q x$

using *agree*[*OF idx-bound*] *x-eq* **by** *simp*

next

show $\text{degree } p < \text{card} (\text{set eval-domain})$

using *deg-p card-set-eval-domain* **by** *simp*

next

show $\text{degree } q < \text{card} (\text{set eval-domain})$

using *deg-q card-set-eval-domain* **by** *simp*

qed

lemma *query-sample-domain-distinct*:

distinct query-sample-domain

proof –

have *inj*: *inj-on* $(\lambda \text{idx. } h \wedge \text{idx} * \text{shift})$ (*set* $[0..<\text{query-sample-space-size}]$)

proof (*intro inj-onI*)

fix *i j*

assume *i-in*: $i \in \text{set } [0..<\text{query-sample-space-size}]$

and *j-in*: $j \in \text{set } [0..<\text{query-sample-space-size}]$


```

    and eq:  $h \wedge i * \text{shift} = h \wedge j * \text{shift}$ 
  have i-bound:  $i < \text{clength} * \text{scale}$ 
    using i-in query-sample-space-size-le-domain by simp
  have j-bound:  $j < \text{clength} * \text{scale}$ 
    using j-in query-sample-space-size-le-domain by simp
  have  $h \wedge i = h \wedge j$ 
    using eq shift-nonzero by simp
  then show  $i = j$ 
    by (rule h-power-inj-on-eval-domain[OF i-bound j-bound])
qed
show ?thesis
  unfolding query-sample-domain-def using inj by (simp add: distinct-map)
qed

lemma card-set-query-sample-domain:
  card (set query-sample-domain) = query-sample-space-size
proof -
  have card (set query-sample-domain) = length query-sample-domain
    using query-sample-domain-distinct by (simp add: distinct-card)
  also have ... = query-sample-space-size
    unfolding query-sample-domain-def by simp
  finally show ?thesis .
qed

lemma poly-eq-on-query-sample-domainI:
  assumes deg-p: degree  $p < \text{query-sample-space-size}$ 
  and deg-q: degree  $q < \text{query-sample-space-size}$ 
  and agree:
     $\bigwedge \text{idx}. \text{idx} \in \text{query-sample-space} \implies$ 
     $\text{poly } p (h \wedge \text{idx} * \text{shift}) = \text{poly } q (h \wedge \text{idx} * \text{shift})$ 
  shows  $p = q$ 
proof (rule poly-eqI-degree[where  $A = \text{set query-sample-domain}$ ])
  fix x
  assume x-in:  $x \in \text{set query-sample-domain}$ 
  then obtain idx where idx-bound:  $\text{idx} < \text{query-sample-space-size}$ 
    and x-eq:  $x = h \wedge \text{idx} * \text{shift}$ 
    unfolding query-sample-domain-def by auto
  have idx-in:  $\text{idx} \in \text{query-sample-space}$ 
    using idx-bound unfolding query-sample-space-def by simp
  show  $\text{poly } p x = \text{poly } q x$ 
    using agree[OF idx-in] x-eq by simp
next
  show degree  $p < \text{card (set query-sample-domain)}$ 
    using deg-p card-set-query-sample-domain by simp
next
  show degree  $q < \text{card (set query-sample-domain)}$ 
    using deg-q card-set-query-sample-domain by simp
qed

```

```

lemma prod-fold-acc:
  fixes roots :: 'f list
  shows  $\text{fold } (\lambda s \text{ acc. } (XP \ 1 - CP \ s) * acc) \ roots \ acc =$ 
     $\text{fold } (\lambda s \text{ acc. } (XP \ 1 - CP \ s) * acc) \ roots \ (1 :: 'f \text{ poly}) * acc$ 
proof (induction roots arbitrary: acc)
  case Nil
  then show ?case by simp
next
  case (Cons r roots)
  let ?F =  $\lambda s \text{ acc. } (XP \ 1 - CP \ s) * acc$ 
  let ?fac =  $XP \ 1 - CP \ r$ 
  have lhs:
     $\text{fold } ?F \ (r \ \# \ roots) \ acc =$ 
       $\text{fold } ?F \ roots \ 1 * (?fac * acc)$ 
    using Cons.IH[of ?fac * acc] by simp
  have  $\text{fold } ?F \ (r \ \# \ roots) \ acc =$ 
     $(\text{fold } ?F \ roots \ 1 * ?fac) * acc$ 
    using lhs by (simp only: mult.assoc)
  also have ... =  $\text{fold } ?F \ roots \ ?fac * acc$ 
    using Cons.IH[of ?fac, symmetric] by simp
  also have ... =  $\text{fold } ?F \ (r \ \# \ roots) \ 1 * acc$ 
    by simp
  finally show ?case .
qed

```

```

lemma prod-Cons:
  fixes r :: 'f and roots :: 'f list
  shows  $\text{prod } (r \ \# \ roots) = (XP \ 1 - CP \ r) * (\text{prod } roots :: 'f \text{ poly})$ 
proof -
  let ?F =  $\lambda s \text{ acc. } (XP \ 1 - CP \ s) * acc$ 
  have  $\text{prod } (r \ \# \ roots) = \text{fold } ?F \ roots \ ((XP \ 1 - CP \ r) * 1)$ 
    unfolding prod-def by simp
  also have ... =  $\text{fold } ?F \ roots \ 1 * ((XP \ 1 - CP \ r) * 1)$ 
    by (rule prod-fold-acc)
  also have ... =  $(\text{prod } roots :: 'f \text{ poly}) * (XP \ 1 - CP \ r)$ 
    unfolding prod-def by simp
  also have ... =  $(XP \ 1 - CP \ r) * (\text{prod } roots :: 'f \text{ poly})$ 
    by (rule mult.commute)
  finally show ?thesis .
qed

```

```

lemma prod-append:
  fixes xs ys :: 'f list
  shows  $\text{prod } (xs \ @ \ ys) = (\text{prod } xs :: 'f \text{ poly}) * \text{prod } ys$ 
proof (induction xs)
  case Nil
  then show ?case
    unfolding prod-def by simp
next

```

```

    case (Cons x xs)
  then show ?case
    by (simp add: prod-Cons algebra-simps)
qed

lemma prod-alt:
  fixes roots :: 'f list
  shows prod roots =
    prod-list (map ( $\lambda r. XP\ 1 - CP\ r$ ) roots :: 'f poly list)
proof (induction roots)
  case Nil
  then show ?case
    unfolding prod-def by simp
next
  case (Cons r roots)
  then show ?case
    by (simp add: prod-Cons)
qed

lemma degree-prod-le-length:
  fixes roots :: 'f list
  shows degree (prod roots :: 'f poly)  $\leq$  length roots
proof (induction roots)
  case Nil
  then show ?case
    unfolding prod-def by simp
next
  case (Cons r roots)
  have factor: degree ( $XP\ 1 - CP\ r :: 'f poly$ )  $\leq 1$ 
  proof -
    have deg-x: degree (Polynomial.monom (1::'f) 1)  $\leq 1$ 
      by (rule degree-monom-le)
    have deg-c: degree (Polynomial.monom r 0)  $\leq 1$ 
    proof -
      have degree (Polynomial.monom r 0)  $\leq 0$ 
        by (rule degree-monom-le)
      then show ?thesis by simp
    qed
    show ?thesis
      unfolding XP-def CP-def
      by (rule degree-diff-le[OF deg-x deg-c])
  qed
  have degree (prod (r # roots) :: 'f poly) =
    degree (( $XP\ 1 - CP\ r$ ) * (prod roots :: 'f poly))
    by (simp add: prod-Cons)
  also have ...  $\leq$ 
    degree ( $XP\ 1 - CP\ r :: 'f poly$ ) + degree (prod roots :: 'f poly)
    by (rule degree-mult-le)
  also have ...  $\leq$  Suc (length roots)

```

```

    using factor Cons.IH by simp
  finally show ?case by simp
qed

lemma degree-sum-list-less:
  fixes ps :: 'f poly list
  assumes  $\bigwedge p. p \in \text{set } ps \implies \text{degree } p < n$ 
    and  $0 < n$ 
  shows  $\text{degree } (\text{sum-list } ps) < n$ 
  using assms
proof (induction ps)
  case Nil
  then show ?case by simp
next
  case (Cons p ps)
  have p-lt:  $\text{degree } p < n$ 
    using Cons.prem by simp
  have ps-lt:  $\text{degree } (\text{sum-list } ps) < n$ 
    by (rule Cons.IH) (use Cons.prem in auto)
  show ?case
    using degree-add-less[OF p-lt ps-lt] by simp
qed

```

```

lemma degree-sum-list-le:
  fixes ps :: 'f poly list
  assumes  $\bigwedge p. p \in \text{set } ps \implies \text{degree } p \leq n$ 
  shows  $\text{degree } (\text{sum-list } ps) \leq n$ 
  using assms
proof (induction ps)
  case Nil
  then show ?case by simp
next
  case (Cons p ps)
  have p-le:  $\text{degree } p \leq n$ 
    using Cons.prem by simp
  have ps-le:  $\text{degree } (\text{sum-list } ps) \leq n$ 
    by (rule Cons.IH) (use Cons.prem in auto)
  show ?case
  proof -
    have  $\text{degree } (p + \text{sum-list } ps) \leq \max (\text{degree } p) (\text{degree } (\text{sum-list } ps))$ 
      by (rule degree-add-le-max)
    also have  $\dots \leq n$ 
      using p-le ps-le by simp
    finally show ?thesis by simp
  qed
qed

```

```

lemma prod-mset-eq:
  fixes xs ys :: 'f list

```

```

assumes mset xs = mset ys
shows prod xs = (prod ys :: 'f poly)
using assms
by (simp add: prod-alt flip: prod-mset-prod-list)

lemma prod-eval-nonzero:
  fixes roots :: 'f list
  assumes x ∉ set roots
  shows poly (prod roots :: 'f poly) x ≠ 0
  using assms
proof (induction roots)
  case Nil
  then show ?case
    unfolding prod-def by simp
next
  case (Cons r roots)
  have x-ne: x ≠ r
    using Cons.prems by simp
  have x-notin: x ∉ set roots
    using Cons.prems by simp
  have poly (prod (r # roots) :: 'f poly) x =
    (x - r) * poly (prod roots :: 'f poly) x
    unfolding prod-Cons XP-def CP-def
    by (simp add: poly-monom)
  then show ?case
    using x-ne Cons.IH[OF x-notin] by simp
qed

lemma prod-eval-zero:
  fixes roots :: 'f list
  assumes x ∈ set roots
  shows poly (prod roots :: 'f poly) x = 0
  using assms
proof (induction roots)
  case Nil
  then show ?case by simp
next
  case (Cons r roots)
  show ?case
  proof (cases x = r)
    case True
    then show ?thesis
      unfolding prod-Cons XP-def CP-def
      by (simp add: poly-monom)
  next
  case False
  then have x ∈ set roots
    using Cons.prems by simp
  then show ?thesis

```

```

    using Cons.IH
    unfolding prod-Cons XP-def CP-def
    by (simp add: poly-monom)
  qed
qed

lemma query-denominator-nonzero:
  assumes spec-entry:  $(c, \text{roots}, d) \in \text{set spec}$ 
    and idx-bound:  $\text{idx} < \text{clength} * \text{scale}$ 
  shows poly (prod (g-map roots))  $(h \wedge \text{idx} * \text{shift}) \neq 0$ 
proof -
  have  $h \wedge \text{idx} * \text{shift} \notin \text{set (g-map roots)}$ 
    by (rule query-domain-disjoint[OF spec-entry idx-bound])
  then show ?thesis
    by (rule prod-eval-nonzero)
qed

lemma common-denominator-nonzero-on-eval-domain:
  assumes idx-bound:  $\text{idx} < \text{clength} * \text{scale}$ 
  shows poly common-denominator-poly  $(h \wedge \text{idx} * \text{shift}) \neq 0$ 
proof -
  have notin:  $h \wedge \text{idx} * \text{shift} \notin \text{set all-constraint-roots}$ 
  proof
    assume in-roots:  $h \wedge \text{idx} * \text{shift} \in \text{set all-constraint-roots}$ 
    then obtain c roots d where spec-entry:  $(c, \text{roots}, d) \in \text{set spec}$ 
      and in-g-map:  $h \wedge \text{idx} * \text{shift} \in \text{set (g-map roots)}$ 
    unfolding all-constraint-roots-def by (auto split: prod.splits)
    have  $h \wedge \text{idx} * \text{shift} \notin \text{set (g-map roots)}$ 
      by (rule query-domain-disjoint[OF spec-entry idx-bound])
    then show False
      using in-g-map by simp
  qed
  then show ?thesis
    unfolding common-denominator-poly-def
    by (rule prod-eval-nonzero)
qed

lemma g-power-in-g-map:
  assumes  $r \in \text{set roots}$ 
  shows  $g \wedge r \in \text{set (g-map roots)}$ 
  using assms unfolding g-map-def by simp

lemma all-constraint-roots-distinct[simp]:
  distinct all-constraint-roots
  unfolding all-constraint-roots-def by simp

lemma g-map-subset-all-constraint-roots:
  assumes spec-entry:  $(c, \text{roots}, d) \in \text{set spec}$ 
  shows  $\text{set (g-map roots)} \subseteq \text{set all-constraint-roots}$ 

```

```

using spec-entry
unfolding all-constraint-roots-def
by (auto split: prod.splits)

lemma common-denominator-factor:
  assumes spec-entry: (c, roots, d) ∈ set spec
  shows
    common-denominator-poly =
      prod (g-map roots) * constraint-root-cofactor roots
proof –
  let ?rest = filter (λx. x ∉ set (g-map roots)) all-constraint-roots
  have subset: set (g-map roots) ⊆ set all-constraint-roots
    by (rule g-map-subset-all-constraint-roots[OF spec-entry])
  have distinct-roots: distinct (g-map roots)
    by (rule g-map-distinct[OF spec-entry])
  have distinct-all: distinct all-constraint-roots
    by simp
  have distinct-split: distinct (g-map roots @ ?rest)
    using distinct-roots distinct-all subset by auto
  have set-split: set (g-map roots @ ?rest) = set all-constraint-roots
    using subset by auto
  have mset-eq:
    mset all-constraint-roots = mset (g-map roots @ ?rest)
    using distinct-all distinct-split set-split
    by (metis set-eq-iff-mset-eq-distinct)
  have common-denominator-poly = prod all-constraint-roots
    unfolding common-denominator-poly-def by simp
  also have ... = prod (g-map roots @ ?rest)
    by (rule prod-mset-eq[OF mset-eq])
  also have ... = prod (g-map roots) * prod ?rest
    by (rule prod-append)
  finally show ?thesis
    unfolding constraint-root-cofactor-def .
qed

lemma declared-quotient-degree-le-maxDegree:
  assumes spec-entry: (c, roots, d) ∈ set spec
  shows d * (clength - 1) - length roots ≤ maxDegree
proof –
  have deg-set:
     $\{d * (clength - 1) - length\ rs \mid a\ rs\ d. (a, rs, d) \in set\ spec\} =$ 
     $(\lambda(a, rs, d). d * (clength - 1) - length\ rs)\ `set\ spec$ 
    by force
  have finite-degrees: finite (degrees clength spec)
    unfolding degrees-def deg-set by simp
  have entry-degree:
    d * (clength - 1) - length roots ∈ degrees clength spec
    using spec-entry unfolding degrees-def by blast
  show ?thesis

```

unfolding *maxDegree-def*
by (rule *Max-ge*[*OF finite-degrees entry-degree*])
qed

lemma *constraint-roots-length-le-all-constraint-roots*:

assumes *spec-entry*: $(c, \text{roots}, d) \in \text{set spec}$

shows $\text{length roots} \leq \text{length all-constraint-roots}$

proof –

have *subset*: $\text{set } (g\text{-map roots}) \subseteq \text{set all-constraint-roots}$

by (rule *g-map-subset-all-constraint-roots*[*OF spec-entry*])

have *distinct-roots*: $\text{distinct } (g\text{-map roots})$

by (rule *g-map-distinct*[*OF spec-entry*])

have *card-le*:

$\text{card } (\text{set } (g\text{-map roots})) \leq \text{card } (\text{set all-constraint-roots})$

by (rule *card-mono*) (use *subset* **in** *auto*)

have $\text{length roots} = \text{length } (g\text{-map roots})$

unfolding *g-map-def* **by** *simp*

also have $\dots = \text{card } (\text{set } (g\text{-map roots}))$

using *distinct-roots* **by** (*simp add: distinct-card*)

also have $\dots \leq \text{card } (\text{set all-constraint-roots})$

by (rule *card-le*)

also have $\dots = \text{length all-constraint-roots}$

by (*simp add: distinct-card*)

finally show *?thesis* .

qed

lemma *constraint-root-cofactor-degree-le*:

assumes *spec-entry*: $(c, \text{roots}, d) \in \text{set spec}$

shows

$\text{degree } (\text{constraint-root-cofactor roots}) \leq$
 $\text{length all-constraint-roots} - \text{length roots}$

proof –

let *?rest* = $\text{filter } (\lambda x. x \notin \text{set } (g\text{-map roots})) \text{ all-constraint-roots}$

have *subset*: $\text{set } (g\text{-map roots}) \subseteq \text{set all-constraint-roots}$

by (rule *g-map-subset-all-constraint-roots*[*OF spec-entry*])

have *distinct-roots*: $\text{distinct } (g\text{-map roots})$

by (rule *g-map-distinct*[*OF spec-entry*])

have *distinct-rest*: $\text{distinct } ?rest$

by *simp*

have *rest-set*: $\text{set } ?rest = \text{set all-constraint-roots} - \text{set } (g\text{-map roots})$

by *auto*

have *rest-len*:

$\text{length } ?rest = \text{length all-constraint-roots} - \text{length roots}$

proof –

have $\text{length } ?rest = \text{card } (\text{set } ?rest)$

using *distinct-card*[*OF distinct-rest*] **by** *simp*

also have $\dots = \text{card } (\text{set all-constraint-roots} - \text{set } (g\text{-map roots}))$

using *rest-set* **by** *simp*

also have $\dots =$


```

      card (set all-constraint-roots) - card (set (g-map roots))
    by (rule card-Diff-subset) (use subset in auto)
  also have ... = length all-constraint-roots - length roots
proof -
  have card (set (g-map roots)) = length (g-map roots)
    using distinct-roots by (simp add: distinct-card)
  also have ... = length roots
    unfolding g-map-def by simp
  finally show ?thesis
    by (simp add: distinct-card)
qed
finally show ?thesis .
qed
have degree (constraint-root-cofactor roots) ≤ length ?rest
  unfolding constraint-root-cofactor-def by (rule degree-prod-le-length)
then show ?thesis
  unfolding rest-len .
qed

lemma common-denominator-poly-degree-le:
  degree common-denominator-poly ≤ length all-constraint-roots
  unfolding common-denominator-poly-def
  by (rule degree-prod-le-length)

lemma common-denominator-numerator-term-degree-lt:
  assumes wf: spec-degree-wellformed
    and deg-f: degree f < clength
    and spec-entry: (c, roots, d) ∈ set spec
  shows
    degree (common-denominator-numerator-term f (a, (c, roots, d))) <
      query-sample-space-size
proof -
  let ?dc = d * (clength - 1)
  let ?all = length all-constraint-roots
  have deg-c:
    degree (c (trace-powers-of f)) ≤ ?dc
    by (rule spec-degree-wellformed-constraint-degree
      [OF wf deg-f spec-entry])
  have deg-cofactor:
    degree (constraint-root-cofactor roots) ≤ ?all - length roots
    by (rule constraint-root-cofactor-degree-le[OF spec-entry])
  have deg-cp: degree (CP a :: 'f poly) ≤ 0
    unfolding CP-def by (rule degree-monom-le)
  have term-le:
    degree (CP a * c (trace-powers-of f) *
      constraint-root-cofactor roots) ≤
      ?dc + (?all - length roots)
proof -
  have degree (CP a * c (trace-powers-of f) *

```

```

      constraint-root-cofactor roots) ≤
degree (CP a * c (trace-powers-of f)) +
degree (constraint-root-cofactor roots)
by (rule degree-mult-le)
also have ... ≤
degree (CP a :: 'f poly) + degree (c (trace-powers-of f)) +
degree (constraint-root-cofactor roots)
using degree-mult-le[of CP a :: 'f poly c (trace-powers-of f)]
by linarith
also have ... ≤ ?dc + (?all - length roots)
using deg-cp deg-c deg-cofactor by linarith
finally show ?thesis .
qed
have roots-le-all: length roots ≤ ?all
by (rule constraint-roots-length-le-all-constraint-roots[OF spec-entry])
have quotient-le:
?dc - length roots ≤ maxDegree
by (rule declared-quotient-degree-le-maxDegree[OF spec-entry])
have arith:
?dc + (?all - length roots) ≤ maxDegree + ?all
using roots-le-all quotient-le by linarith
have margin: maxDegree + ?all < query-sample-space-size
using spec-degree-wellformed-query-margin[OF wf]
unfolding query-agreement-bound-def by simp
show ?thesis
using term-le arith margin
unfolding common-denominator-numerator-term-def
by simp
qed

lemma common-denominator-numerator-degree-lt:
assumes wf: spec-degree-wellformed
and deg-f: degree f < clength
shows degree (common-denominator-numerator f as) < query-sample-space-size
proof -
have term-lt:
 $\bigwedge p. p \in$ 
set (map (common-denominator-numerator-term f) (zip as spec))  $\implies$ 
degree p < query-sample-space-size
proof -
fix p
assume p-in:
p ∈ set (map (common-denominator-numerator-term f) (zip as spec))
then obtain z where
z-in: z ∈ set (zip as spec)
and p-eq: p = common-denominator-numerator-term f z
by auto
obtain a entry where z-eq: z = (a, entry)
by (cases z) auto

```

```

obtain  $c$  roots  $d$  where entry-eq: entry = ( $c$ , roots,  $d$ )
  by (cases entry) auto
have spec-entry: ( $c$ , roots,  $d$ )  $\in$  set spec
  using z-in unfolding z-eq entry-eq by (auto simp: set-zip)
show degree  $p <$  query-sample-space-size
  unfolding p-eq z-eq entry-eq
  by (rule common-denominator-numerator-term-degree-lt
    [OF wf deg-f spec-entry])
qed
show ?thesis
  unfolding common-denominator-numerator-def
  by (rule degree-sum-list-less[OF term-lt query-sample-space-size-pos])
qed

lemma common-denominator-product-degree-lt:
  assumes wf: spec-degree-wellformed
  and deg-q: degree  $q \leq$  maxDegree
  shows degree (common-denominator-poly *  $q$ )  $<$  query-sample-space-size
proof -
  have degree (common-denominator-poly *  $q$ )  $\leq$ 
    degree common-denominator-poly + degree  $q$ 
  by (rule degree-mult-le)
  also have ...  $\leq$  length all-constraint-roots + maxDegree
  using common-denominator-poly-degree-le deg-q by linarith
  also have ... = maxDegree + length all-constraint-roots
  by simp
  also have ...  $<$  query-sample-space-size
  using spec-degree-wellformed-query-margin[OF wf]
  unfolding query-agreement-bound-def by simp
  finally show ?thesis .
qed

lemma common-denominator-numerator-term-degree-le-bound:
  assumes deg-f: degree  $f <$  clength
  and spec-entry: ( $c$ , roots,  $d$ )  $\in$  set spec
  shows
    degree (common-denominator-numerator-term  $f$  ( $a$ , ( $c$ , roots,  $d$ )))  $\leq$ 
      query-agreement-bound
proof -
  let ?dc =  $d * (clength - 1)$ 
  let ?all = length all-constraint-roots
  have deg-c:
    degree ( $c$  (trace-powers-of  $f$ ))  $\leq$  ?dc
  by (rule constraint-degree-wellformed[OF deg-f spec-entry])
  have deg-cofactor:
    degree (constraint-root-cofactor roots)  $\leq$  ?all - length roots
  by (rule constraint-root-cofactor-degree-le[OF spec-entry])
  have deg-cp: degree (CP  $a :: 'f$  poly)  $\leq 0$ 
  unfolding CP-def by (rule degree-monom-le)

```

```

have term-le:
  degree (CP a * c (trace-powers-of f) *
    constraint-root-cofactor roots) ≤
    ?dc + (?all - length roots)
proof -
  have degree (CP a * c (trace-powers-of f) *
    constraint-root-cofactor roots) ≤
    degree (CP a * c (trace-powers-of f)) +
    degree (constraint-root-cofactor roots)
  by (rule degree-mult-le)
  also have ... ≤
    degree (CP a :: 'f poly) + degree (c (trace-powers-of f)) +
    degree (constraint-root-cofactor roots)
  using degree-mult-le[of CP a :: 'f poly c (trace-powers-of f)]
  by linarith
  also have ... ≤ ?dc + (?all - length roots)
  using deg-cp deg-c deg-cofactor by linarith
  finally show ?thesis .
qed
have roots-le-all: length roots ≤ ?all
  by (rule constraint-roots-length-le-all-constraint-roots[OF spec-entry])
have quotient-le:
  ?dc - length roots ≤ maxDegree
  by (rule declared-quotient-degree-le-maxDegree[OF spec-entry])
have ?dc + (?all - length roots) ≤ maxDegree + ?all
  using roots-le-all quotient-le by linarith
then show ?thesis
  using term-le unfolding common-denominator-numerator-term-def
  query-agreement-bound-def
  by simp
qed

lemma common-denominator-numerator-degree-le-bound:
  assumes deg-f: degree f < clength
  shows degree (common-denominator-numerator f as) ≤ query-agreement-bound
proof -
  have term-le:
     $\bigwedge p. p \in$ 
    set (map (common-denominator-numerator-term f) (zip as spec))  $\implies$ 
    degree p ≤ query-agreement-bound
  proof -
    fix p
    assume p-in:
      p ∈ set (map (common-denominator-numerator-term f) (zip as spec))
    then obtain z where
      z-in: z ∈ set (zip as spec)
      and p-eq: p = common-denominator-numerator-term f z
      by auto
    obtain a entry where z-eq: z = (a, entry)

```

```

    by (cases z) auto
  obtain c roots d where entry-eq: entry = (c, roots, d)
    by (cases entry) auto
  have spec-entry: (c, roots, d) ∈ set spec
    using z-in unfolding z-eq entry-eq by (auto simp: set-zip)
  show degree p ≤ query-agreement-bound
    unfolding p-eq z-eq entry-eq
    by (rule common-denominator-numerator-term-degree-le-bound
        [OF deg-f spec-entry])
qed
show ?thesis
  unfolding common-denominator-numerator-def
  by (rule degree-sum-list-le[OF term-le])
qed

lemma common-denominator-product-degree-le-bound:
  assumes deg-q: degree q ≤ maxDegree
  shows degree (common-denominator-poly * q) ≤ query-agreement-bound
proof -
  have degree (common-denominator-poly * q) ≤
    degree common-denominator-poly + degree q
    by (rule degree-mult-le)
  also have ... ≤ length all-constraint-roots + maxDegree
    using common-denominator-poly-degree-le deg-q by linarith
  also have ... = query-agreement-bound
    unfolding query-agreement-bound-def by simp
  finally show ?thesis .
qed

lemma common-denominator-residual-degree-le-bound:
  assumes deg-f: degree f < clength
    and deg-q: degree q ≤ maxDegree
  shows
    degree (common-denominator-numerator f as - common-denominator-poly * q)
      ≤ query-agreement-bound
proof -
  have num-le:
    degree (common-denominator-numerator f as) ≤ query-agreement-bound
    by (rule common-denominator-numerator-degree-le-bound[OF deg-f])
  have prod-le:
    degree (common-denominator-poly * q) ≤ query-agreement-bound
    by (rule common-denominator-product-degree-le-bound[OF deg-q])
  show ?thesis
    using degree-diff-le[OF num-le prod-le] .
qed

lemma common-denominator-degree-bounds-from-spec-degree-wellformed:
  assumes wf: spec-degree-wellformed
    and deg-f: degree f < clength

```

shows *common-denominator-degree-bounds f as*
unfolding *common-denominator-degree-bounds-def*
using *common-denominator-numerator-degree-lt[OF wf deg-f]*
common-denominator-product-degree-lt[OF wf]
by *simp*

lemma *common-denominator-vanishes-on-constraint-root:*

assumes *spec-entry: (c, roots, d) ∈ set spec*
and *r-in: r ∈ set roots*
shows *poly common-denominator-poly (g ^ r) = 0*

proof –

have *factor:*
common-denominator-poly =
*prod (g-map roots) * constraint-root-cofactor roots*
by (*rule common-denominator-factor[OF spec-entry]*)
have *root-in: g ^ r ∈ set (g-map roots)*
by (*rule g-power-in-g-map[OF r-in]*)
have *poly (prod (g-map roots)) (g ^ r) = 0*
by (*rule prod-eval-zero[OF root-in]*)
then show *?thesis*
using *factor* **by** *simp*

qed

lemma *g-power-notin-g-map-if-root-notin:*

assumes *spec-entry: (c, roots, d) ∈ set spec*
and *r-bound: r < clength*
and *r-notin: r ∉ set roots*
shows *g ^ r ∉ set (g-map roots)*

proof

assume *g ^ r ∈ set (g-map roots)*
then obtain *r' where r'-in: r' ∈ set roots*
and *eq: g ^ r = g ^ r'*
unfolding *g-map-def* **by** *auto*
have *r'-bound: r' < clength*
by (*rule spec-roots-in-range[OF spec-entry r'-in]*)
have *r = r'*
by (*rule g-power-inj-on-range[OF r-bound r'-bound eq]*)
then show *False*
using *r-notin r'-in* **by** *simp*

qed

lemma *constraint-root-cofactor-eval-nonzero:*

assumes *spec-entry: (c, roots, d) ∈ set spec*
and *r-in: r ∈ set roots*
shows *poly (constraint-root-cofactor roots) (g ^ r) ≠ 0*

proof –

have *root-in: g ^ r ∈ set (g-map roots)*
by (*rule g-power-in-g-map[OF r-in]*)
have *g ^ r ∉*

```

      set (filter (λx. x ∉ set (g-map roots)) all-constraint-roots)
    using root-in by auto
  then show ?thesis
    unfolding constraint-root-cofactor-def
    by (rule prod-eval-nonzero)
qed

```

```

lemma constraint-root-cofactor-kills-other-root:
  assumes root-spec: (c0, roots0, d0) ∈ set spec
    and r-in0: r ∈ set roots0
    and spec-entry: (c, roots, d) ∈ set spec
    and r-notin: r ∉ set roots
  shows poly (constraint-root-cofactor roots) (g ^ r) = 0
proof -
  have r-bound: r < clength
    by (rule spec-roots-in-range[OF root-spec r-in0])
  have root-in-all: g ^ r ∈ set all-constraint-roots
    using g-map-subset-all-constraint-roots[OF root-spec]
      g-power-in-g-map[OF r-in0]
    by auto
  have root-notin-entry: g ^ r ∉ set (g-map roots)
    by (rule g-power-notin-g-map-if-root-notin
      [OF spec-entry r-bound r-notin])
  have g ^ r ∈
    set (filter (λx. x ∉ set (g-map roots)) all-constraint-roots)
    using root-in-all root-notin-entry by simp
  then show ?thesis
    unfolding constraint-root-cofactor-def
    by (rule prod-eval-zero)
qed

```

```

lemma common-denominator-residual-coefficient-nonzero:
  assumes spec-entry: (c, roots, d) ∈ set spec
    and r-in: r ∈ set roots
    and nonzero:
      poly (c (trace-powers-of f)) (g ^ r) ≠ 0
  shows
    poly (c (trace-powers-of f)) (g ^ r) *
      poly (constraint-root-cofactor roots) (g ^ r) ≠ 0
  using nonzero constraint-root-cofactor-eval-nonzero[OF spec-entry r-in]
  by simp

```

```

lemma common-denominator-numerator-root-eval:
  assumes len: length as = length spec
    and root-spec: (c0, roots0, d0) ∈ set spec
    and r-in0: r ∈ set roots0
  shows
    poly (common-denominator-numerator f as) (g ^ r) =
      common-denominator-composition-root-residual f as r

```

```

proof –
  have lhs:
    poly (common-denominator-numerator f as) (g ^ r) =
      (∑ i ∈ {0..length spec}.
        poly
          (case spec ! i of (c, roots, d) ⇒
            CP (as ! i) * c (trace-powers-of f) *
              constraint-root-cofactor roots)
            (g ^ r))
    using len
    unfolding common-denominator-numerator-def
    by (simp add: sum-list-sum-nth poly-sum
      common-denominator-numerator-term-def split-beta' case-prod-unfold
      CP-def poly-monom algebra-simps)
  have rhs:
    common-denominator-composition-root-residual f as r =
      (∑ i ∈ {0..length spec}.
        common-denominator-constraint-root-residual f r
          (as ! i, spec ! i))
    using len
    unfolding common-denominator-composition-root-residual-def
    by (simp add: sum-list-sum-nth split-beta')
  have pointwise:
    ∧i. i ∈ {0..length spec} ⇒
      poly
        (case spec ! i of (c, roots, d) ⇒
          CP (as ! i) * c (trace-powers-of f) *
            constraint-root-cofactor roots)
          (g ^ r) =
            common-denominator-constraint-root-residual f r
              (as ! i, spec ! i)
  proof –
    fix i
    assume i-in: i ∈ {0..length spec}
    then have i-bound: i < length spec
      by simp
    obtain c roots d where spec-i: spec ! i = (c, roots, d)
      by (cases spec ! i) auto
    have spec-entry: (c, roots, d) ∈ set spec
      using nth-mem[OF i-bound] spec-i by simp
    show
      poly
        (case spec ! i of (c, roots, d) ⇒
          CP (as ! i) * c (trace-powers-of f) *
            constraint-root-cofactor roots)
          (g ^ r) =
            common-denominator-constraint-root-residual f r
              (as ! i, spec ! i)
    proof (cases r ∈ set roots)

```



```

    case True
    then show ?thesis
      using spec-i
      unfolding common-denominator-constraint-root-residual-def
      by (simp add: CP-def poly-monom)
  next
  case False
  have kill:
    poly (constraint-root-cofactor roots) (g ^ r) = 0
    by (rule constraint-root-cofactor-kills-other-root
      [OF root-spec r-in0 spec-entry False])
  show ?thesis
    using spec-i False kill
    unfolding common-denominator-constraint-root-residual-def
    by (simp add: CP-def poly-monom)
qed
qed
show ?thesis
  unfolding lhs rhs
  by (intro sum.cong refl pointwise) simp
qed

lemma common-denominator-mult-composition-spec-eval:
  assumes denom-nonzero:
     $\bigwedge c \text{ roots } d. (c, \text{roots}, d) \in \text{set spec} \implies$ 
    poly (prod (g-map roots)) x  $\neq 0$ 
  shows
    poly common-denominator-poly x * composition-spec-eval f as x =
    poly (common-denominator-numerator f as) x
proof -
  let ?D = poly common-denominator-poly x
  have term-eval:
     $\bigwedge z. z \in \text{set (zip as spec)} \implies$ 
    ?D * composition-spec-eval-term f x z =
    poly (common-denominator-numerator-term f z) x
  proof -
    fix z
    assume z-in: z  $\in$  set (zip as spec)
    obtain a entry where z-eq: z = (a, entry)
      by (cases z) auto
    obtain c roots d where entry-eq: entry = (c, roots, d)
      by (cases entry) auto
    have entry-in: entry  $\in$  set spec
  proof -
    obtain i where i-bound: i < length as i < length spec
      and z-i: z = (as ! i, spec ! i)
      using z-in by (auto simp: set-zip)
    have entry = spec ! i
      using z-eq z-i by simp

```

```

then show ?thesis
  using nth-mem[OF i-bound(2)] by simp
qed
have spec-entry:  $(c, \text{roots}, d) \in \text{set spec}$ 
  using entry-in entry-eq by simp
have factor:
  ?D =
    poly (prod (g-map roots)) x *
    poly (constraint-root-cofactor roots) x
  using common-denominator-factor[OF spec-entry]
  by simp
have denom: poly (prod (g-map roots)) x  $\neq 0$ 
  by (rule denom-nonzero[OF spec-entry])
show
  ?D * composition-spec-eval-term f x z =
    poly (common-denominator-numerator-term f z) x
  using z-eq entry-eq factor denom
  unfolding composition-spec-eval-term-def
    common-denominator-numerator-term-def
  by (simp add: field-simps CP-def poly-monom)
qed
have mult-sum:
  ?D * sum-list (map (composition-spec-eval-term f x) (zip as spec)) =
    sum-list
      (map ( $\lambda z$ . poly (common-denominator-numerator-term f z) x)
        (zip as spec))
proof -
  have ?D * sum-list (map (composition-spec-eval-term f x) (zip as spec)) =
    sum-list
      (map ( $\lambda z$ . ?D * composition-spec-eval-term f x z)
        (zip as spec))
  by (simp add: sum-list-const-mult algebra-simps)
  also have ... =
    sum-list
      (map ( $\lambda z$ . poly (common-denominator-numerator-term f z) x)
        (zip as spec))
  using term-eval by (intro arg-cong[where f=sum-list] map-cong) auto
  finally show ?thesis .
qed
have rhs:
  poly (common-denominator-numerator f as) x =
    sum-list
      (map ( $\lambda z$ . poly (common-denominator-numerator-term f z) x)
        (zip as spec))
  unfolding common-denominator-numerator-def
  apply (simp add: poly-sum-list)
  apply (intro arg-cong[where f=sum-list] map-cong refl)
  apply (auto simp: common-denominator-numerator-term-def
    split-beta' case-prod-unfold algebra-simps CP-def poly-monom)

```

```

      split: prod.splits)
    done
  show ?thesis
    unfolding composition-spec-eval-def
    using mult-sum rhs by simp
qed

lemma common-denominator-composition-root-residual-sum-nth:
  assumes len: length as = length spec
  shows
    common-denominator-composition-root-residual f as r =
      ( $\sum i \in \{0..<\text{length spec}\}.$ 
        common-denominator-constraint-root-residual f r (as ! i, spec ! i))
  proof -
    have common-denominator-composition-root-residual f as r =
      sum-list
        (map (common-denominator-constraint-root-residual f r) (zip as spec))
    by (simp add: common-denominator-composition-root-residual-def)
    also have ... =
      ( $\sum i \in \{0..<\text{length (zip as spec)}\}.$ 
        map (common-denominator-constraint-root-residual f r) (zip as spec) ! i)
    by (simp add: sum-list-sum-nth)
    also have ... =
      ( $\sum i \in \{0..<\text{length spec}\}.$ 
        common-denominator-constraint-root-residual f r (as ! i, spec ! i))
    using len by (intro sum.cong) auto
    finally show ?thesis .
  qed

lemma common-denominator-composition-root-residual-split-index:
  assumes len: length as = length spec
  and i-bound: i < length spec
  shows
    common-denominator-composition-root-residual f as r =
      common-denominator-constraint-root-residual f r (as ! i, spec ! i) +
      ( $\sum j \in \{0..<\text{length spec}\} - \{i\}.$ 
        common-denominator-constraint-root-residual f r (as ! j, spec ! j))
  proof -
    let ?F =
       $\lambda j.$  common-denominator-constraint-root-residual f r (as ! j, spec ! j)
    have common-denominator-composition-root-residual f as r =
      ( $\sum j \in \{0..<\text{length spec}\}.$  ?F j)
    by (rule common-denominator-composition-root-residual-sum-nth[OF len])
    also have ... = ?F i + ( $\sum j \in \{0..<\text{length spec}\} - \{i\}.$  ?F j)
    using i-bound by (simp add: sum.remove)
    finally show ?thesis .
  qed

lemma common-denominator-composition-root-residual-zero-unique-nth:

```

```

fixes as bs :: 'f list
assumes len-as:  $\text{length } as = \text{length } spec$ 
  and len-bs:  $\text{length } bs = \text{length } spec$ 
  and i-bound:  $i < \text{length } spec$ 
  and spec-i:  $spec ! i = (c, roots, d)$ 
  and r-in:  $r \in \text{set } roots$ 
  and nonzero:
     $\text{poly } (c \text{ (trace-powers-of } f)) (g \wedge r) \neq 0$ 
  and outside:
     $\bigwedge j. j < \text{length } spec \implies j \neq i \implies as ! j = bs ! j$ 
  and zero-as:  $\text{common-denominator-composition-root-residual } f \text{ as } r = 0$ 
  and zero-bs:  $\text{common-denominator-composition-root-residual } f \text{ bs } r = 0$ 
shows  $as ! i = bs ! i$ 
proof -
  let ?C =
     $\text{poly } (c \text{ (trace-powers-of } f)) (g \wedge r) * \text{poly } (\text{constraint-root-cofactor } roots) (g \wedge r)$ 
  let ?B =
     $(\sum j \in \{0..<\text{length } spec\} - \{i\}. \text{common-denominator-constraint-root-residual } f \text{ r } (as ! j, spec ! j))$ 
  have spec-entry:  $(c, roots, d) \in \text{set } spec$ 
    using nth-mem[OF i-bound] spec-i by simp
  have C-nonzero:  $?C \neq 0$ 
    by (rule common-denominator-residual-coefficient-nonzero [OF spec-entry r-in nonzero])
  have B-eq:
     $(\sum j \in \{0..<\text{length } spec\} - \{i\}. \text{common-denominator-constraint-root-residual } f \text{ r } (bs ! j, spec ! j)) = ?B$ 
    by (intro sum.cong) (use outside in auto)
  have as-eq:
     $\text{common-denominator-composition-root-residual } f \text{ as } r = ?C * (as ! i) + ?B$ 
    using common-denominator-composition-root-residual-split-index [OF len-as i-bound] spec-i r-in
    unfolding common-denominator-constraint-root-residual-def
    by (simp add: algebra-simps)
  have bs-eq:
     $\text{common-denominator-composition-root-residual } f \text{ bs } r = ?C * (bs ! i) + ?B$ 
    using common-denominator-composition-root-residual-split-index [OF len-bs i-bound] spec-i r-in B-eq
    unfolding common-denominator-constraint-root-residual-def
    by (simp add: algebra-simps)
  show ?thesis
proof (rule linear-equation-solution-unique[OF C-nonzero])
  show  $?C * (as ! i) + ?B = 0$ 
    using as-eq zero-as by simp

```

```

    show ?C * (bs ! i) + ?B = 0
      using bs-eq zero-bs by simp
  qed
qed

lemma random-combination-common-denominator-hides-violations-unique-nth:
  fixes as bs :: 'f list
  assumes len-as: length as = length spec
    and len-bs: length bs = length spec
    and i-bound: i < length spec
    and spec-i: spec ! i = (c, roots, d)
    and violated: (c, roots, d) ∈ violated-constraints f
    and r-in: r ∈ set roots
    and nonzero:
      poly (c (trace-powers-of f)) (g ^ r) ≠ 0
    and outside:
       $\bigwedge j. j < \text{length spec} \implies j \neq i \implies \text{as} ! j = \text{bs} ! j$ 
    and hides-as: random-combination-common-denominator-hides-violations f as
    and hides-bs: random-combination-common-denominator-hides-violations f bs
  shows as ! i = bs ! i
proof -
  have zero-as:
    common-denominator-composition-root-residual f as r = 0
  using hides-as violated r-in
  unfolding random-combination-common-denominator-hides-violations-def
  by auto
  have zero-bs:
    common-denominator-composition-root-residual f bs r = 0
  using hides-bs violated r-in
  unfolding random-combination-common-denominator-hides-violations-def
  by auto
  show ?thesis
  by (rule common-denominator-composition-root-residual-zero-unique-nth
    [OF len-as len-bs i-bound spec-i r-in nonzero outside zero-as zero-bs])
qed

lemma violated-constraints-has-common-denominator-coefficient:
  assumes violated: violated-constraints f ≠ {}
  obtains i c roots d r
  where i < length spec
    and spec ! i = (c, roots, d)
    and (c, roots, d) ∈ violated-constraints f
    and r ∈ set roots
    and
      poly (c (trace-powers-of f)) (g ^ r) *
      poly (constraint-root-cofactor roots) (g ^ r) ≠ 0
proof -
  from violated-constraints-has-remainder-bad[OF violated]
  obtain entry where entry0:

```

```

    entry ∈ violated-constraints f
    constraint-remainder-bad f entry
  by blast
obtain c roots d where entry-eq: entry = (c, roots, d)
  by (cases entry) auto
have entry:
  (c, roots, d) ∈ violated-constraints f
  constraint-remainder-bad f (c, roots, d)
  using entry0 unfolding entry-eq by simp-all
then obtain r where r:
  r ∈ set roots
  poly (c (trace-powers-of f)) (g ^ r) ≠ 0
  unfolding constraint-remainder-bad-def by auto
have spec-entry: (c, roots, d) ∈ set spec
  using entry(1) unfolding violated-constraints-def by auto
then obtain i where i:
  i < length spec
  spec ! i = (c, roots, d)
  by (auto simp: in-set-conv-nth)
have coeff:
  poly (c (trace-powers-of f)) (g ^ r) *
  poly (constraint-root-cofactor roots) (g ^ r) ≠ 0
  by (rule common-denominator-residual-coefficient-nonzero
    [OF spec-entry r(1) r(2)])
show ?thesis
  by (rule that[OF i entry(1) r(1) coeff])
qed

lemma random-combination-common-denominator-hides-has-coefficient:
  assumes hides: random-combination-common-denominator-hides-violations f as
  obtains i c roots d r
  where i < length spec
  and spec ! i = (c, roots, d)
  and (c, roots, d) ∈ violated-constraints f
  and r ∈ set roots
  and
    poly (c (trace-powers-of f)) (g ^ r) *
    poly (constraint-root-cofactor roots) (g ^ r) ≠ 0
proof –
  have violated: violated-constraints f ≠ {}
  using hides
  unfolding random-combination-common-denominator-hides-violations-def
  by simp
show ?thesis
  by (rule violated-constraints-has-common-denominator-coefficient[OF violated])
    (rule that)
qed

definition alpha-space :: 'f list set

```

where $\text{alpha-space} = \{as. \text{length } as = \text{length spec}\}$

definition $\text{common-denominator-hiding-alpha-space} :: 'f \text{ poly} \Rightarrow 'f \text{ list set}$

where

$\text{common-denominator-hiding-alpha-space } f =$

$\{as \in \text{alpha-space}.$

$\text{random-combination-common-denominator-hides-violations } f \text{ as}\}$

lemma $\text{finite-length-lists-UNIV}:$

$\text{finite } \{xs :: 'f \text{ list}. \text{length } xs = n\}$

proof $-$

have $\{xs :: 'f \text{ list}. \text{length } xs = n\} =$

$\{xs. \text{set } xs \subseteq (\text{UNIV} :: 'f \text{ set}) \wedge \text{length } xs = n\}$

by auto

then show $?thesis$

using $\text{finite-lists-length-eq}$ [of $\text{UNIV} :: 'f \text{ set } n$] **by** simp

qed

lemma $\text{card-length-lists-UNIV}:$

$\text{card } \{xs :: 'f \text{ list}. \text{length } xs = n\} = \text{CARD}('f) \wedge n$

proof $-$

have $\{xs :: 'f \text{ list}. \text{length } xs = n\} =$

$\{xs. \text{set } xs \subseteq (\text{UNIV} :: 'f \text{ set}) \wedge \text{length } xs = n\}$

by auto

then show $?thesis$

using $\text{card-lists-length-eq}$ [of $\text{UNIV} :: 'f \text{ set } n$] **by** simp

qed

definition $\text{fri-challenge-space} :: \text{nat} \Rightarrow 'f \text{ list set}$

where $\text{fri-challenge-space } n = \{bs. \text{length } bs = n\}$

lemma $\text{finite-fri-challenge-space}[\text{simp}]:$

$\text{finite } (\text{fri-challenge-space } n)$

unfolding $\text{fri-challenge-space-def}$

by $(\text{rule } \text{finite-length-lists-UNIV})$

lemma $\text{card-fri-challenge-space}:$

$\text{card } (\text{fri-challenge-space } n) = \text{CARD}('f) \wedge n$

unfolding $\text{fri-challenge-space-def}$

by $(\text{rule } \text{card-length-lists-UNIV})$

lemma $\text{card-alpha-space}[\text{simp}]:$

$\text{card } \text{alpha-space} = \text{CARD}('f) \wedge \text{length spec}$

unfolding alpha-space-def **by** $(\text{rule } \text{card-length-lists-UNIV})$

lemma $\text{common-denominator-hiding-alpha-space-card-bound}:$

$\text{card } (\text{common-denominator-hiding-alpha-space } f) \leq$

$(\text{if } \text{length spec} = 0 \text{ then } 0 \text{ else } \text{CARD}('f) \wedge (\text{length spec} - 1))$

proof $(\text{cases } \text{common-denominator-hiding-alpha-space } f = \{\})$

```

case True
then show ?thesis by simp
next
case False
let ?S = common-denominator-hiding-alpha-space f
from False obtain as0 where as0-in: as0 ∈ ?S
  by auto
have hides0: random-combination-common-denominator-hides-violations f as0
  using as0-in unfolding common-denominator-hiding-alpha-space-def by simp
from random-combination-common-denominator-hides-has-coefficient[OF hides0]
obtain i c roots d r where
  i-bound: i < length spec
  and spec-i: spec ! i = (c, roots, d)
  and violated: (c, roots, d) ∈ violated-constraints f
  and r-in: r ∈ set roots
  and coeff-nonzero:
    poly (c (trace-powers-of f)) (g ^ r) *
    poly (constraint-root-cofactor roots) (g ^ r) ≠ 0
  by blast
have nonzero: poly (c (trace-powers-of f)) (g ^ r) ≠ 0
  using coeff-nonzero by auto
let ?key =  $\lambda xs :: 'f\ list. (take\ i\ xs, drop\ (Suc\ i)\ xs)$ 
have inj: inj-on ?key ?S
proof (rule inj-onI)
  fix xs ys
  assume xs-in: xs ∈ ?S
  and ys-in: ys ∈ ?S
  and key-eq: ?key xs = ?key ys
  have len-xs: length xs = length spec
    using xs-in
    unfolding common-denominator-hiding-alpha-space-def alpha-space-def
    by simp
  have len-ys: length ys = length spec
    using ys-in
    unfolding common-denominator-hiding-alpha-space-def alpha-space-def
    by simp
  have hides-xs:
    random-combination-common-denominator-hides-violations f xs
    using xs-in unfolding common-denominator-hiding-alpha-space-def by simp
  have hides-ys:
    random-combination-common-denominator-hides-violations f ys
    using ys-in unfolding common-denominator-hiding-alpha-space-def by simp
  have take-eq: take i xs = take i ys
    using key-eq by simp
  have drop-eq: drop (Suc i) xs = drop (Suc i) ys
    using key-eq by simp
  have outside:
     $\bigwedge j. j < \text{length spec} \implies j \neq i \implies xs ! j = ys ! j$ 
  proof –

```



```

fix j
assume j-bound: j < length spec
  and j-ne: j ≠ i
show xs ! j = ys ! j
proof (cases j < i)
  case True
    have xs ! j = take i xs ! j
      using True by simp
    also have ... = take i ys ! j
      using take-eq by simp
    also have ... = ys ! j
      using True by simp
    finally show ?thesis .
  next
    case False
    then have i-lt-j: i < j
      using j-ne by linarith
    have j-decomp: j = Suc i + (j - Suc i)
      using i-lt-j by simp
    have drop-idx-bound-xs: j - Suc i < length (drop (Suc i) xs)
      using j-bound i-lt-j len-xs by simp
    have drop-idx-bound-ys: j - Suc i < length (drop (Suc i) ys)
      using j-bound i-lt-j len-ys by simp
    have xs ! j = drop (Suc i) xs ! (j - Suc i)
      using drop-idx-bound-xs j-decomp by simp
    also have ... = drop (Suc i) ys ! (j - Suc i)
      using drop-eq by simp
    also have ... = ys ! j
      using drop-idx-bound-ys j-decomp by simp
    finally show ?thesis .
qed
qed
have nth-i: xs ! i = ys ! i
  by (rule random-combination-common-denominator-hides-violations-unique-nth
    [OF len-xs len-ys i-bound spec-i violated r-in nonzero outside
      hides-xs hides-ys])
show xs = ys
proof (rule nth-equalityI)
  show length xs = length ys
    using len-xs len-ys by simp
next
  fix j
  assume j-bound: j < length xs
  then have j-spec: j < length spec
    using len-xs by simp
  show xs ! j = ys ! j
  proof (cases j = i)
    case True
      then show ?thesis

```

```

    using nth-i by simp
  next
    case False
    then show ?thesis
      by (rule outside[OF j-spec])
    qed
  qed
qed
let ?A = {xs :: 'f list. length xs = i}
let ?B = {xs :: 'f list. length xs = length spec - Suc i}
have image-subset: ?key ' ?S  $\subseteq$  ?A  $\times$  ?B
proof
  fix x
  assume  $x \in ?key ' ?S$ 
  then obtain xs where xs-in:  $xs \in ?S$  and x-eq:  $x = ?key\ xs$ 
    by auto
  have len-xs:  $length\ xs = length\ spec$ 
    using xs-in
    unfolding common-denominator-hiding-alpha-space-def alpha-space-def
    by simp
  show  $x \in ?A \times ?B$ 
    using len-xs i-bound unfolding x-eq by simp
qed
have finite-A: finite ?A
  by (rule finite-length-lists-UNIV)
have finite-B: finite ?B
  by (rule finite-length-lists-UNIV)
have finite-prod: finite (?A  $\times$  ?B)
  using finite-A finite-B by simp
have finite-alpha: finite alpha-space
  unfolding alpha-space-def by (rule finite-length-lists-UNIV)
have finite-S: finite ?S
  unfolding common-denominator-hiding-alpha-space-def
  using finite-alpha by simp
have card-image-le:  $card\ (?key\ ' ?S) \leq card\ (?A \times ?B)$ 
  by (rule card-mono[OF finite-prod image-subset])
have card-S:  $card\ ?S = card\ (?key\ ' ?S)$ 
  using card-image[OF inj] by simp
have card-prod:
 $card\ (?A \times ?B) =$ 
 $CARD('f) \wedge i * CARD('f) \wedge (length\ spec - Suc\ i)$ 
  using finite-A finite-B by (simp add: card-length-lists-UNIV)
have exp-eq:
 $i + (length\ spec - Suc\ i) = length\ spec - 1$ 
  using i-bound by simp
have  $card\ ?S \leq CARD('f) \wedge i * CARD('f) \wedge (length\ spec - Suc\ i)$ 
  using card-image-le card-S card-prod by simp
also have  $\dots = CARD('f) \wedge (length\ spec - 1)$ 
  using exp-eq by (simp add: power-add[symmetric])

```

finally have *bound*:
 $\text{card } ?S \leq \text{CARD}('f) \wedge (\text{length spec} - 1) .$
have $\text{length spec} \neq 0$
using *i-bound* **by** *auto*
then show *?thesis*
using *bound* **by** *simp*
qed

lemma *nnreal-nat-divide-right-mono*:
assumes $a \leq b$
shows $\text{nnreal } a / \text{nnreal } c \leq \text{nnreal } b / \text{nnreal } c$
using *assms*
by *transfer (simp add: divide-right-mono)*

lemma *nnreal-nat-divide-self*:
assumes $k \neq 0$
shows $\text{nnreal } k / \text{nnreal } k = 1$
using *assms* **by** *transfer simp*

lemma *nnreal-power-fraction-cancel*:
assumes $k \neq 0$
shows $\text{nnreal } (k \wedge n) / \text{nnreal } (k \wedge \text{Suc } n) = 1 / \text{nnreal } k$
using *assms*
by *transfer (simp add: power-Suc field-simps)*

lemma *common-denominator-hiding-alpha-space-fraction-bound*:
 $\text{nnreal } (\text{card } (\text{common-denominator-hiding-alpha-space } f)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{length spec}) \leq \text{composition-error-bound}$
proof *(cases length spec)*
case 0
then have $\text{card } (\text{common-denominator-hiding-alpha-space } f) = 0$
using *common-denominator-hiding-alpha-space-card-bound[of f]* **by** *simp*
then show *?thesis*
unfolding *composition-error-bound-def* **by** *simp*
next
case *(Suc n)*
have *card-le*:
 $\text{card } (\text{common-denominator-hiding-alpha-space } f) \leq \text{CARD}('f) \wedge n$
using *common-denominator-hiding-alpha-space-card-bound[of f]*
unfolding *Suc* **by** *simp*
have *field-nonzero*: $\text{CARD}('f) \neq 0$
by *simp*
have $\text{nnreal } (\text{card } (\text{common-denominator-hiding-alpha-space } f)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{length spec}) \leq$
 $\text{nnreal } (\text{CARD}('f) \wedge n) / \text{nnreal } (\text{CARD}('f) \wedge \text{Suc } n)$
unfolding *Suc*
using *card-le*
by *(rule nnreal-nat-divide-right-mono)*
also have $\dots = 1 / \text{nnreal } (\text{CARD}('f))$

```

    by (rule nnreal-power-fraction-cancel[OF field-nonzero])
  also have ... = composition-error-bound
    using size-card unfolding composition-error-bound-def by simp
  finally show ?thesis .
qed

lemma low-degree-trace-table-unique:
  assumes deg-f: degree f < clength
    and deg-q: degree q < clength
    and table-eq: map (poly f) eval-domain = map (poly q) eval-domain
  shows f = q
proof -
  have clength-le-domain: clength ≤ clength * scale
    using scale-pos by simp
  have deg-f-domain: degree f < clength * scale
    using deg-f clength-le-domain by linarith
  have deg-q-domain: degree q < clength * scale
    using deg-q clength-le-domain by linarith
  show ?thesis
proof (rule poly-eq-on-eval-domainI[OF deg-f-domain deg-q-domain])
  fix idx
  assume idx-bound: idx < clength * scale
  have idx-len: idx < length eval-domain
    using idx-bound eval-domain-length by simp
  have map (poly f) eval-domain ! idx = map (poly q) eval-domain ! idx
    using table-eq idx-len by simp
  then show poly f (h ^ idx * shift) = poly q (h ^ idx * shift)
    using idx-len idx-bound eval-domain-nth by simp
qed
qed

definition low-degree-trace-witness :: 'f list ⇒ 'f poly
where
  low-degree-trace-witness trace-table =
    (SOME f. degree f < clength ∧
      trace-table = map (poly f) eval-domain)

lemma low-degree-trace-witness-eq:
  assumes deg-f: degree f < clength
    and trace-table: trace-table = map (poly f) eval-domain
  shows low-degree-trace-witness trace-table = f
  unfolding low-degree-trace-witness-def
proof (rule some-equality)
  show degree f < clength ∧ trace-table = map (poly f) eval-domain
    using deg-f trace-table by simp
next
  fix q
  assume q:
    degree q < clength ∧ trace-table = map (poly q) eval-domain

```

```

then have table-eq: map (poly f) eval-domain = map (poly q) eval-domain
using trace-table by simp
show q = f
using low-degree-trace-table-unique[OF deg-f conjunct1[OF q] table-eq]
by simp
qed

```

```

definition composition-trace-bad-alpha-space :: 'f list  $\Rightarrow$  'f list set
where
  composition-trace-bad-alpha-space trace-table =
    (let f = low-degree-trace-witness trace-table in
     if degree f < clength  $\wedge$ 
       trace-table = map (poly f) eval-domain  $\wedge$ 
       violated-constraints f  $\neq$  {}
     then
       {as  $\in$  common-denominator-hiding-alpha-space f.
        common-denominator-degree-bounds f as}
     else {})

```

```

lemma finite-alpha-space:
  finite alpha-space
unfolding alpha-space-def by (rule finite-length-lists-UNIV)

```

```

lemma common-denominator-hiding-alpha-space-subset-alpha-space:
  common-denominator-hiding-alpha-space f  $\subseteq$  alpha-space
unfolding common-denominator-hiding-alpha-space-def by auto

```

```

lemma finite-common-denominator-hiding-alpha-space:
  finite (common-denominator-hiding-alpha-space f)
using finite-alpha-space
unfolding common-denominator-hiding-alpha-space-def by simp

```

```

lemma composition-trace-bad-alpha-space-subset-alpha-space:
  composition-trace-bad-alpha-space trace-table  $\subseteq$  alpha-space
proof -
  let ?f = low-degree-trace-witness trace-table
  have composition-trace-bad-alpha-space trace-table  $\subseteq$ 
    common-denominator-hiding-alpha-space ?f
  unfolding composition-trace-bad-alpha-space-def Let-def by auto
  also have ...  $\subseteq$  alpha-space
  by (rule common-denominator-hiding-alpha-space-subset-alpha-space)
  finally show ?thesis .
qed

```

```

lemma composition-trace-bad-alpha-space-fraction-bound:
  nnreal (card (composition-trace-bad-alpha-space trace-table)) /
  nnreal (CARD('f) ^ length spec)  $\leq$  composition-error-bound
proof -
  let ?f = low-degree-trace-witness trace-table

```

```

have subset:
  composition-trace-bad-alpha-space trace-table  $\subseteq$ 
    common-denominator-hiding-alpha-space ?f
  unfolding composition-trace-bad-alpha-space-def Let-def by auto
have card-le:
  card (composition-trace-bad-alpha-space trace-table)  $\leq$ 
    card (common-denominator-hiding-alpha-space ?f)
  by (rule card-mono[OF finite-common-denominator-hiding-alpha-space subset])
have nnreal (card (composition-trace-bad-alpha-space trace-table)) /
  nnreal (CARD('f) ^ length spec)  $\leq$ 
  nnreal (card (common-denominator-hiding-alpha-space ?f)) /
  nnreal (CARD('f) ^ length spec)
  by (rule nnreal-nat-divide-right-mono[OF card-le])
also have ...  $\leq$  composition-error-bound
  by (rule common-denominator-hiding-alpha-space-fraction-bound)
finally show ?thesis .
qed

```

```

lemma composition-trace-bad-alpha-space-fraction-bound-alpha-space:
  nnreal (card (composition-trace-bad-alpha-space trace-table)) /
  nnreal (card alpha-space)  $\leq$  composition-error-bound
using composition-trace-bad-alpha-space-fraction-bound[of trace-table]
by simp

```

```

lemma composition-trace-bad-alpha-space-card-bound:
  card (composition-trace-bad-alpha-space trace-table)  $\leq$ 
    (if length spec = 0 then 0 else CARD('f) ^ (length spec - 1))
proof -
  let ?f = low-degree-trace-witness trace-table
  have subset:
    composition-trace-bad-alpha-space trace-table  $\subseteq$ 
      common-denominator-hiding-alpha-space ?f
    unfolding composition-trace-bad-alpha-space-def Let-def by auto
  have card-le:
    card (composition-trace-bad-alpha-space trace-table)  $\leq$ 
      card (common-denominator-hiding-alpha-space ?f)
    by (rule card-mono[OF finite-common-denominator-hiding-alpha-space subset])
  also have ...  $\leq$ 
    (if length spec = 0 then 0 else CARD('f) ^ (length spec - 1))
    by (rule common-denominator-hiding-alpha-space-card-bound)
  finally show ?thesis .
qed

```

```

lemma trace-powers-of-eval-nth:
  assumes k < powers
  shows poly (trace-powers-of f ! k) x = poly f (g ^ k * x)
  using assms
  unfolding trace-powers-of-def XP-def
  by (simp add: poly-pcompose poly-monom)

```

```

lemma query-trace-power-eval-nth:
  assumes k-bound:  $k < \text{powers}$ 
    and idx-bound:  $\text{idx} + k * \text{scale} < \text{clength} * \text{scale}$ 
  shows
     $\text{map } (\text{poly } f) \text{ eval-domain ! } (\text{idx} + k * \text{scale}) =$ 
       $\text{poly } (\text{trace-powers-of } f ! k) (h \wedge \text{idx} * \text{shift})$ 
proof -
  have eval:
     $\text{map } (\text{poly } f) \text{ eval-domain ! } (\text{idx} + k * \text{scale}) =$ 
       $\text{poly } f (h \wedge (\text{idx} + k * \text{scale}) * \text{shift})$ 
    using eval-domain-nth[OF idx-bound] idx-bound eval-domain-length by simp
  have arg:
     $g \wedge k * (h \wedge \text{idx} * \text{shift}) = h \wedge (\text{idx} + k * \text{scale}) * \text{shift}$ 
proof -
  have gpow:  $g \wedge k = h \wedge (\text{scale} * k)$ 
    using domain-alignment by (simp add: power-mult)
  have  $g \wedge k * (h \wedge \text{idx} * \text{shift}) =$ 
     $h \wedge (\text{scale} * k) * h \wedge \text{idx} * \text{shift}$ 
    using gpow by (simp add: algebra-simps)
  also have  $\dots = h \wedge (\text{idx} + k * \text{scale}) * \text{shift}$ 
    by (simp add: power-add algebra-simps mult.commute)
  finally show ?thesis .
qed
show ?thesis
  using eval trace-powers-of-eval-nth[OF k-bound, of f h \wedge idx * shift] arg
  by simp
qed

lemma query-values-trace-powers:
  assumes trace-table:  $\text{trace-table} = \text{map } (\text{poly } f) \text{ eval-domain}$ 
    and scaled-bound:
       $\bigwedge i. i \in \text{set } (\text{powers-scaled idx}) \implies i < \text{clength} * \text{scale}$ 
  shows
     $\text{map } (! \text{ trace-table}) (\text{powers-scaled idx}) =$ 
       $\text{map } (\lambda q. \text{poly } q (h \wedge \text{idx} * \text{shift})) (\text{trace-powers-of } f)$ 
proof (rule nth-equalityI)
  show  $\text{length } (\text{map } (! \text{ trace-table}) (\text{powers-scaled idx})) =$ 
     $\text{length } (\text{map } (\lambda q. \text{poly } q (h \wedge \text{idx} * \text{shift})) (\text{trace-powers-of } f))$ 
    unfolding powers-scaled-def by simp
next
  fix k
  assume k-bound:
     $k < \text{length } (\text{map } (! \text{ trace-table}) (\text{powers-scaled idx}))$ 
  then have k-lt:  $k < \text{powers}$ 
    unfolding powers-scaled-def by simp
  have scaled-idx-in:
     $\text{idx} + k * \text{scale} \in \text{set } (\text{powers-scaled idx})$ 
    using k-lt unfolding powers-scaled-def by auto

```

```

have idx-bound:  $idx + k * scale < clength * scale$ 
  by (rule scaled-bound[OF scaled-idx-in])
have left:
  map (!) trace-table (powers-scaled idx) !  $k =$ 
    trace-table ! ( $idx + k * scale$ )
  using k-lt unfolding powers-scaled-def by simp
have right:
  map ( $\lambda q. poly\ q\ (h \wedge idx * shift)$ ) (trace-powers-of f) !  $k =$ 
    poly (trace-powers-of f !  $k$ ) ( $h \wedge idx * shift$ )
  using k-lt by simp
show
  map (!) trace-table (powers-scaled idx) !  $k =$ 
    map ( $\lambda q. poly\ q\ (h \wedge idx * shift)$ ) (trace-powers-of f) !  $k$ 
  using query-trace-power-eval-nth[OF k-lt idx-bound] left right trace-table
  by simp
qed

lemma query-consistent-at-trace-powers:
  assumes qc: query-consistent-at trace-table composition-table as idx
  and trace-table: trace-table = map (poly f) eval-domain
  shows
    composition-table ! idx =
      cp-eval as (map ( $\lambda q. poly\ q\ (h \wedge idx * shift)$ ) (trace-powers-of f))
        ( $h \wedge idx * shift$ )
proof –
  have qc-unfolded:
     $idx < clength * scale \wedge$ 
     $length\ trace-table = clength * scale \wedge$ 
     $length\ composition-table = clength * scale \wedge$ 
     $(\forall i \in set\ (powers-scaled\ idx). i < length\ trace-table) \wedge$ 
    composition-table ! idx =
      cp-eval as (map (!) trace-table (powers-scaled idx)) ( $h \wedge idx * shift$ )
  using qc unfolding query-consistent-at-def .
have trace-len:  $length\ trace-table = clength * scale$ 
  using qc-unfolded by blast
have scaled-bound:
   $\bigwedge i. i \in set\ (powers-scaled\ idx) \implies i < clength * scale$ 
proof –
  fix i
  assume i-in:  $i \in set\ (powers-scaled\ idx)$ 
  have  $i < length\ trace-table$ 
  using qc-unfolded i-in by blast
  then show  $i < clength * scale$ 
  using trace-len by simp
qed
have vals-eq:
  map (!) trace-table (powers-scaled idx) =
    map ( $\lambda q. poly\ q\ (h \wedge idx * shift)$ ) (trace-powers-of f)
  by (rule query-values-trace-powers[OF trace-table scaled-bound])

```



```

have comp:
  composition-table ! idx =
    cp-eval as (map (! trace-table) (powers-scaled idx)) (h ^ idx * shift)
  using qc-unfolded by blast
have cp-eq:
  cp-eval as (map (! trace-table) (powers-scaled idx)) (h ^ idx * shift) =
    cp-eval as (map (λq. poly q (h ^ idx * shift)) (trace-powers-of f))
      (h ^ idx * shift)
  using vals-eq by (rule arg-cong)
show ?thesis
  using comp cp-eq by simp
qed

```

```

lemma fold-add-sum-list:
  fixes f :: 'x ⇒ 'b::comm-monoid-add
  shows fold (λx acc. f x + acc) xs acc = sum-list (map f xs) + acc
proof (induction xs arbitrary: acc)
  case Nil
  then show ?case by simp
next
  case (Cons x xs)
  then show ?case
    by (simp add: ac-simps)
qed

```

```

lemma cp-eval-agrees-with-spec-lists:
  fixes specs :: (('f poly list ⇒ 'f poly) × nat list × nat) list
  and spec2s :: (('f list ⇒ 'f) × nat list) list
  and qs :: 'f poly list
  assumes len: length spec2s = length specs
  and zip-subset: set (zip specs spec2s) ⊆ set (zip spec spec2)
  and qs-len: powers ≤ length qs
  shows
    fold
      (λ(a, (c2, roots2)).
        (+) (a * (c2 (map (λq. poly q x) qs) /
          poly (prod (g-map roots2)) x)))
      (zip as spec2s) vacc =
    sum-list
      (map
        (λ(a, (c, roots, d)).
          a * (poly (c qs) x / poly (prod (g-map roots)) x))
        (zip as specs)) + vacc
proof –
  let ?vterm =
    λz. case z of (a, entry) ⇒
      case entry of (c2, roots2) ⇒
        a * (c2 (map (λq. poly q x) qs) /
          poly (prod (g-map roots2)) x)

```

```

let ?sterm =
  λz. case z of (a, entry) ⇒
    case entry of (c, roots, d) ⇒
      a * (poly (c qs) x / poly (prod (g-map roots)) x)
have terms-eq:
  map ?vterm (zip as spec2s) = map ?sterm (zip as specs)
proof (rule nth-equalityI)
  show length (map ?vterm (zip as spec2s)) =
    length (map ?sterm (zip as specs))
  using len by simp
next
fix i
assume i-bound: i < length (map ?vterm (zip as spec2s))
then have i-as: i < length as
  by simp
have i-spec2: i < length spec2s
  using i-bound by simp
have i-specs: i < length specs
  using i-spec2 len by simp
obtain c roots d where specs-i: specs ! i = (c, roots, d)
  by (cases specs ! i) auto
obtain c2 roots2 where spec2s-i: spec2s ! i = (c2, roots2)
  by (cases spec2s ! i) auto
have pair-in:
  ((c, roots, d), (c2, roots2)) ∈ set (zip spec spec2)
proof -
  have (specs ! i, spec2s ! i) ∈ set (zip specs spec2s)
  proof -
    have i < length (zip specs spec2s)
    using i-specs i-spec2 by simp
    then have zip specs spec2s ! i ∈ set (zip specs spec2s)
    by (rule nth-mem)
    then show ?thesis
    using i-specs i-spec2 by simp
  qed
  then show ?thesis
  using zip-subset specs-i spec2s-i by auto
qed
have roots-eq: roots2 = roots
and raw-eq:
  c2 (map (λq. poly q x) qs) = poly (c qs) x
using specs-agree[OF pair-in qs-len, of x] by simp-all
show map ?vterm (zip as spec2s) ! i =
  map ?sterm (zip as specs) ! i
using i-as i-spec2 i-specs specs-i spec2s-i roots-eq raw-eq
by simp
qed
have fold-eq:
  fold

```

```

    (λ(a, (c2, roots2))).
    (+) (a * (c2 (map (λq. poly q x) qs) /
      poly (prod (g-map roots2)) x)))
    (zip as spec2s) vacc =
  fold (λz acc. ?vterm z + acc) (zip as spec2s) vacc
  by (rule fold-cong) (auto simp: fun-eq-iff split: prod.splits)
have
  fold
    (λ(a, (c2, roots2))).
    (+) (a * (c2 (map (λq. poly q x) qs) /
      poly (prod (g-map roots2)) x)))
    (zip as spec2s) vacc =
  sum-list (map ?vterm (zip as spec2s)) + vacc
  unfolding fold-eq
  by (rule fold-add-sum-list)
also have ... =
  sum-list
    (map
      (λ(a, (c, roots, d)).
        a * (poly (c qs) x / poly (prod (g-map roots)) x))
      (zip as specs)) + vacc
  using terms-eq by simp
  finally show ?thesis .
qed

lemma cp-eval-trace-powers-agrees-with-spec:
  cp-eval as (map (λq. poly q x) (trace-powers-of f)) x =
  composition-spec-eval f as x
proof -
  have aligned:
    fold
      (λ(a, (c2, roots2))).
      (+) (a * (c2
        (map (λq. poly q x) (trace-powers-of f)) /
        poly (prod (g-map roots2)) x)))
      (zip as spec2) 0 =
    sum-list
      (map
        (λ(a, (c, roots, d)).
          a * (poly (c (trace-powers-of f)) x /
            poly (prod (g-map roots)) x))
        (zip as spec)) + 0
      by (rule cp-eval-agrees-with-spec-lists[OF spec2-length subset-refl])
      simp
  have spec-sum:
    sum-list
      (map
        (λ(a, (c, roots, d)).
          a * (poly (c (trace-powers-of f)) x /

```

```

      poly (prod (g-map roots)) x))
    (zip as spec)) =
    sum-list (map (composition-spec-eval-term f x) (zip as spec))
  apply (intro arg-cong[where f=sum-list] map-cong refl)
  apply (auto simp: composition-spec-eval-term-def split-beta'
    case-prod-unfold split: prod.splits)
done
show ?thesis
  unfolding cp-eval-def composition-spec-eval-def
  using aligned spec-sum by simp
qed

lemma common-denominator-mult-cp-eval-trace-powers:
  assumes idx-bound: idx < clength * scale
  shows
    poly common-denominator-poly (h ^ idx * shift) *
      cp-eval as
      (map (λq. poly q (h ^ idx * shift)) (trace-powers-of f))
      (h ^ idx * shift) =
      poly (common-denominator-numerator f as) (h ^ idx * shift)
  proof -
    have denom:
      ∧ c roots d. (c, roots, d) ∈ set spec ⇒
      poly (prod (g-map roots)) (h ^ idx * shift) ≠ 0
    proof -
      fix c roots d
      assume spec-entry: (c, roots, d) ∈ set spec
      show poly (prod (g-map roots)) (h ^ idx * shift) ≠ 0
        by (rule query-denominator-nonzero[OF spec-entry idx-bound])
    qed
    have mult:
      poly common-denominator-poly (h ^ idx * shift) *
        composition-spec-eval f as (h ^ idx * shift) =
        poly (common-denominator-numerator f as) (h ^ idx * shift)
      by (rule common-denominator-mult-composition-spec-eval[OF denom])
    show ?thesis
      using mult cp-eval-trace-powers-agrees-with-spec[of as h ^ idx * shift f]
      by simp
  qed

lemma common-denominator-numerator-identity-residual-zero:
  assumes len: length as = length spec
  and identity:
    common-denominator-numerator f as = common-denominator-poly * q
  and spec-entry: (c, roots, d) ∈ set spec
  and r-in: r ∈ set roots
  shows common-denominator-composition-root-residual f as r = 0
  proof -
    have numerator-eval:

```

```

poly (common-denominator-numerator f as) (g ^ r) =
  common-denominator-composition-root-residual f as r
by (rule common-denominator-numerator-root-eval[OF len spec-entry r-in])
have root-zero: poly common-denominator-poly (g ^ r) = 0
by (rule common-denominator-vanishes-on-constraint-root
  [OF spec-entry r-in])
have poly (common-denominator-numerator f as) (g ^ r) = 0
  using identity root-zero by simp
then show ?thesis
  using numerator-eval by simp
qed

```

```

lemma common-denominator-numerator-identity-hides-violations:
  assumes len: length as = length spec
  and violated: violated-constraints f ≠ {}
  and identity:
    common-denominator-numerator f as = common-denominator-poly * q
  shows random-combination-common-denominator-hides-violations f as
proof -
  have residual-zero:
     $\bigwedge c \text{ roots } d \text{ r.}$ 
     $(c, \text{roots}, d) \in \text{violated-constraints } f \implies$ 
     $r \in \text{set roots} \implies$ 
     $\text{common-denominator-composition-root-residual } f \text{ as } r = 0$ 
  proof -
    fix c roots d r
    assume entry:  $(c, \text{roots}, d) \in \text{violated-constraints } f$ 
    and r-in:  $r \in \text{set roots}$ 
    have spec-entry:  $(c, \text{roots}, d) \in \text{set spec}$ 
    using entry unfolding violated-constraints-def by simp
    show common-denominator-composition-root-residual f as r = 0
    by (rule common-denominator-numerator-identity-residual-zero
      [OF len identity spec-entry r-in])
  qed
  show ?thesis
    using violated residual-zero
    unfolding random-combination-common-denominator-hides-violations-def
    by auto
qed

```

```

lemma composition-table-agrees-with-rational-expression:
  assumes composition-low:
    composition-table-low-degree maxDegree composition-table
  and trace-table: trace-table = map (poly f) eval-domain
  and all-queries: all-queries-consistent trace-table composition-table as
  obtains q where
    degree q ≤ maxDegree
  and  $\bigwedge \text{idx. idx} \in \text{query-sample-space} \implies$ 
    poly q (h ^ idx * shift) =

```

```

      cp-eval as (map (λp. poly p (h ^ idx * shift)) (trace-powers-of f))
      (h ^ idx * shift)
proof -
  from composition-low obtain q where deg-q: degree q ≤ maxDegree
  and table-q: composition-table = map (poly q) eval-domain
  unfolding composition-table-low-degree-def by blast
  have agree:
    ∧idx. idx ∈ query-sample-space ⇒
      poly q (h ^ idx * shift) =
        cp-eval as (map (λp. poly p (h ^ idx * shift)) (trace-powers-of f))
        (h ^ idx * shift)
  proof -
    fix idx
    assume idx-in: idx ∈ query-sample-space
    have idx-bound: idx < clength * scale
    by (rule query-sample-space-less-domain[OF idx-in])
    have qc: query-consistent-at trace-table composition-table as idx
    using all-queries idx-in unfolding all-queries-consistent-def by simp
    have comp-value:
      composition-table ! idx = poly q (h ^ idx * shift)
    using table-q eval-domain-nth[OF idx-bound] idx-bound eval-domain-length
    by simp
    show poly q (h ^ idx * shift) =
      cp-eval as (map (λp. poly p (h ^ idx * shift)) (trace-powers-of f))
      (h ^ idx * shift)
    using query-consistent-at-trace-powers[OF qc trace-table] comp-value
    by simp
  qed
  show ?thesis
  by (rule that[OF deg-q agree])
qed

lemma common-denominator-numerator-identity-from-queries:
  assumes composition-low:
    composition-table-low-degree maxDegree composition-table
  and trace-table: trace-table = map (poly f) eval-domain
  and all-queries: all-queries-consistent trace-table composition-table as
  and num-deg:
    degree (common-denominator-numerator f as) < query-sample-space-size
  and prod-deg:
    ∧q. degree q ≤ maxDegree ⇒
      degree (common-denominator-poly * q) < query-sample-space-size
  obtains q where
    degree q ≤ maxDegree
  and common-denominator-numerator f as = common-denominator-poly * q
proof -
  from composition-table-agrees-with-rational-expression
  [OF composition-low trace-table all-queries]
  obtain q where deg-q: degree q ≤ maxDegree

```

```

and agree:
   $\bigwedge idx. idx \in \text{query-sample-space} \implies$ 
     $\text{poly } q (h \wedge idx * \text{shift}) =$ 
      cp-eval as
         $(\text{map } (\lambda p. \text{poly } p (h \wedge idx * \text{shift})) (\text{trace-powers-of } f))$ 
         $(h \wedge idx * \text{shift})$ 
    by blast
have eval-agree:
   $\bigwedge idx. idx \in \text{query-sample-space} \implies$ 
     $\text{poly } (\text{common-denominator-numerator } f \text{ as}) (h \wedge idx * \text{shift}) =$ 
     $\text{poly } (\text{common-denominator-poly} * q) (h \wedge idx * \text{shift})$ 
proof –
  fix idx
  assume idx-in:  $idx \in \text{query-sample-space}$ 
  have idx-bound:  $idx < \text{clength} * \text{scale}$ 
    by (rule query-sample-space-less-domain[OF idx-in])
  have mult:
     $\text{poly } \text{common-denominator-poly} (h \wedge idx * \text{shift}) *$ 
    cp-eval as
       $(\text{map } (\lambda p. \text{poly } p (h \wedge idx * \text{shift})) (\text{trace-powers-of } f))$ 
       $(h \wedge idx * \text{shift}) =$ 
       $\text{poly } (\text{common-denominator-numerator } f \text{ as}) (h \wedge idx * \text{shift})$ 
    by (rule common-denominator-mult-cp-eval-trace-powers[OF idx-bound])
  have poly  $(\text{common-denominator-poly} * q) (h \wedge idx * \text{shift}) =$ 
     $\text{poly } \text{common-denominator-poly} (h \wedge idx * \text{shift}) *$ 
     $\text{poly } q (h \wedge idx * \text{shift})$ 
    by simp
  also have ... =
     $\text{poly } \text{common-denominator-poly} (h \wedge idx * \text{shift}) *$ 
    cp-eval as
       $(\text{map } (\lambda p. \text{poly } p (h \wedge idx * \text{shift})) (\text{trace-powers-of } f))$ 
       $(h \wedge idx * \text{shift})$ 
    using agree[OF idx-in] by simp
  also have ... =  $\text{poly } (\text{common-denominator-numerator } f \text{ as}) (h \wedge idx * \text{shift})$ 
    using mult by simp
  finally show
     $\text{poly } (\text{common-denominator-numerator } f \text{ as}) (h \wedge idx * \text{shift}) =$ 
     $\text{poly } (\text{common-denominator-poly} * q) (h \wedge idx * \text{shift})$ 
    by simp
qed
have identity:
   $\text{common-denominator-numerator } f \text{ as} = \text{common-denominator-poly} * q$ 
  by (rule poly-eq-on-query-sample-domainI[OF num-deg prod-deg[OF deg-q]])
  (rule eval-agree)
show ?thesis
  by (rule that[OF deg-q identity])
qed

```

lemma common-denominator-numerator-identity-from-queries-bounds:

```

assumes composition-low:
  composition-table-low-degree maxDegree composition-table
and trace-table: trace-table = map (poly f) eval-domain
and all-queries: all-queries-consistent trace-table composition-table as
and bounds: common-denominator-degree-bounds f as
obtains q where
  degree q ≤ maxDegree
and common-denominator-numerator f as = common-denominator-poly * q
proof –
  have num-deg:
    degree (common-denominator-numerator f as) < query-sample-space-size
    using bounds unfolding common-denominator-degree-bounds-def by simp
  have prod-deg:
     $\wedge q. \text{degree } q \leq \text{maxDegree} \implies$ 
    degree (common-denominator-poly * q) < query-sample-space-size
    using bounds unfolding common-denominator-degree-bounds-def by simp
  from common-denominator-numerator-identity-from-queries
    [OF composition-low trace-table all-queries num-deg prod-deg]
  obtain q where deg-q: degree q ≤ maxDegree
  and identity:
    common-denominator-numerator f as = common-denominator-poly * q
  by blast
  show ?thesis
  by (rule that[OF deg-q identity])
qed

```

lemma *random-combination-common-denominator-hides-from-queries*:

```

assumes len-as: length as = length spec
and false-statement:  $\neg$  exists-valid-trace
and deg-f: degree f < clength
and trace-table: trace-table = map (poly f) eval-domain
and composition-low:
  composition-table-low-degree maxDegree composition-table
and all-queries: all-queries-consistent trace-table composition-table as
and bounds: common-denominator-degree-bounds f as
shows random-combination-common-denominator-hides-violations f as

```

```

proof –
  have violated: violated-constraints f ≠ {}
  by (rule false-statement-violated-constraints[OF false-statement deg-f])
  from common-denominator-numerator-identity-from-queries-bounds
    [OF composition-low trace-table all-queries bounds]
  obtain q where
    degree q ≤ maxDegree
  and identity:
    common-denominator-numerator f as = common-denominator-poly * q
  by blast
  show ?thesis
  by (rule common-denominator-numerator-identity-hides-violations
    [OF len-as violated identity])

```


qed

lemma *all-queries-consistent-iff-no-bad-queries:*

all-queries-consistent trace-table composition-table as $\longleftrightarrow$

bad-query-indices trace-table composition-table as = $\{\}$

unfolding *all-queries-consistent-def bad-query-indices-def* **by** *auto*

lemma *query-disagreement-indices-nonempty-iff:*

query-disagreement-indices trace-table composition-table as $\neq \{\}$ $\longleftrightarrow$

$\neg$ all-queries-consistent trace-table composition-table as

unfolding *query-disagreement-indices-def*

using *all-queries-consistent-iff-no-bad-queries* **by** *simp*

lemma *query-consistent-at-from-common-denominator-residual-zero:*

assumes *trace-table: trace-table = map (poly f) eval-domain*

and *composition-table: composition-table = map (poly q) eval-domain*

and *residual-zero:*

*common-denominator-numerator f as - common-denominator-poly * q = 0*

and *idx-in: idx $\in$ query-sample-space*

shows *query-consistent-at trace-table composition-table as idx*

proof -

have *idx-bound: idx < clength * scale*

by (*rule query-sample-space-less-domain[OF idx-in]*)

have *trace-len: length trace-table = clength * scale*

using *trace-table eval-domain-length* **by** *simp*

have *comp-len: length composition-table = clength * scale*

using *composition-table eval-domain-length* **by** *simp*

have *powers-bound:*

$\forall i \in \text{set (powers-scaled idx)}. i < \text{length trace-table}$

proof

fix *i*

assume *i-in: i $\in$ set (powers-scaled idx)*

show *i < length trace-table*

using *query-sample-space-powers-scaled-bound[OF idx-in i-in]* *trace-len*

by *simp*

qed

have *comp-value:*

*composition-table ! idx = poly q (h $\wedge$ idx * shift)*

using *composition-table eval-domain-nth[OF idx-bound]* *idx-bound eval-domain-length*

by *simp*

have *trace-powers:*

map ((!) trace-table) (powers-scaled idx) =

*map ($\lambda p. \text{poly } p (h \wedge \text{idx} * \text{shift})$) (trace-powers-of f)*

by (*rule query-values-trace-powers[OF trace-table]*)

(*rule query-sample-space-powers-scaled-bound[OF idx-in]*)

have *mult:*

*poly common-denominator-poly (h $\wedge$ idx * shift) **

cp-eval as

(*map ($\lambda p. \text{poly } p (h \wedge \text{idx} * \text{shift})$) (trace-powers-of f)*)

```

    (h ^ idx * shift) =
    poly (common-denominator-numerator f as) (h ^ idx * shift)
  by (rule common-denominator-mult-cp-eval-trace-powers[OF idx-bound])
have denom-nonzero:
  poly common-denominator-poly (h ^ idx * shift) ≠ 0
  by (rule common-denominator-nonzero-on-eval-domain[OF idx-bound])
have residual-eval:
  poly (common-denominator-numerator f as) (h ^ idx * shift) =
  poly common-denominator-poly (h ^ idx * shift) *
  poly q (h ^ idx * shift)
  using residual-zero by simp
have cp-eq:
  cp-eval as
  (map (λp. poly p (h ^ idx * shift)) (trace-powers-of f))
  (h ^ idx * shift) =
  poly q (h ^ idx * shift)
  using mult residual-eval denom-nonzero by simp
show ?thesis
  unfolding query-consistent-at-def
  using idx-bound trace-len comp-len powers-bound comp-value trace-powers cp-eq
  by simp
qed

```

lemma *all-queries-consistent-from-common-denominator-residual-zero:*
assumes *trace-table:* trace-table = map (poly f) eval-domain
and *composition-table:* composition-table = map (poly q) eval-domain
and *residual-zero:*
 common-denominator-numerator f as – *common-denominator-poly * q = 0*
shows *all-queries-consistent* trace-table composition-table as
unfolding *all-queries-consistent-def*
using *query-consistent-at-from-common-denominator-residual-zero*
 [*OF trace-table composition-table residual-zero*]
by *blast*

lemma *common-denominator-residual-nonzero-from-query-disagreement:*
assumes *trace-table:* trace-table = map (poly f) eval-domain
and *composition-table:* composition-table = map (poly q) eval-domain
and *not-all:* ¬ *all-queries-consistent* trace-table composition-table as
shows *common-denominator-numerator f as* – *common-denominator-poly * q ≠ 0*
proof
assume *zero:*
 common-denominator-numerator f as – *common-denominator-poly * q = 0*
have *all-queries-consistent* trace-table composition-table as
by (rule *all-queries-consistent-from-common-denominator-residual-zero*
 [*OF trace-table composition-table zero*])
then show *False*
using *not-all* **by** *simp*
qed

lemma *query-agreement-indices-card-bound*:

assumes *trace-low*: *trace-table-low-degree trace-table*
and *composition-low*:
composition-table-low-degree maxDegree composition-table
and *not-all*:
 $\neg$ *all-queries-consistent trace-table composition-table as*

shows *card (query-agreement-indices trace-table composition-table as) $\leq$ query-agreement-bound*

proof –

from *trace-low* **obtain** *f* **where** *deg-f*: *degree f < clength*
and *trace-table*: *trace-table = map (poly f) eval-domain*
unfolding *trace-table-low-degree-def* **by** *blast*

from *composition-low* **obtain** *q* **where** *deg-q*: *degree q $\leq$ maxDegree*
and *composition-table*: *composition-table = map (poly q) eval-domain*
unfolding *composition-table-low-degree-def* **by** *blast*

let *?A* = *query-agreement-indices trace-table composition-table as*
let *?R* = *common-denominator-numerator f as – common-denominator-poly * q*
have *A-subset*: *?A $\subseteq$ query-sample-space*
unfolding *query-agreement-indices-def* **by** *auto*

have *finite-A*: *finite ?A*
unfolding *query-agreement-indices-def query-sample-space-def* **by** *simp*

have *inj*: *inj-on (λ idx. $h \wedge \text{idx} * \text{shift}$) ?A*

proof (*intro inj-onI*)
fix *i j*
assume *i-in*: *i $\in$?A*
and *j-in*: *j $\in$?A*
and *eq*: *$h \wedge i * \text{shift} = h \wedge j * \text{shift}$*
have *i-bound*: *i < clength * scale*
by (*rule query-sample-space-less-domain* [*OF set-mp* [*OF A-subset i-in*]])
have *j-bound*: *j < clength * scale*
by (*rule query-sample-space-less-domain* [*OF set-mp* [*OF A-subset j-in*]])
have *$h \wedge i = h \wedge j$*
using *eq shift-nonzero* **by** *simp*
then show *i = j*
by (*rule h-power-inj-on-eval-domain* [*OF i-bound j-bound*])

qed

have *residual-nonzero*:
?R $\neq$ 0
by (*rule common-denominator-residual-nonzero-from-query-disagreement* [*OF trace-table composition-table not-all*])

have *image-subset*:
*(λ idx. $h \wedge \text{idx} * \text{shift}$) ‘ ?A $\subseteq$ {x. poly ?R x = 0}*

proof
fix *x*
assume *x-in*: *x $\in$ (λ idx. $h \wedge \text{idx} * \text{shift}$) ‘ ?A*
then obtain *idx* **where** *idx-in*: *idx $\in$?A*
and *x-eq*: *x = $h \wedge \text{idx} * \text{shift}$*
by *blast*
have *idx-sample*: *idx $\in$ query-sample-space*

```

    by (rule set-mp[OF A-subset idx-in])
  have idx-bound:  $idx < clength * scale$ 
    by (rule query-sample-space-less-domain[OF idx-sample])
  have qc: query-consistent-at trace-table composition-table as idx
    using idx-in unfolding query-agreement-indices-def by simp
  have comp-value:
    composition-table ! idx = poly q ( $h \wedge idx * shift$ )
    using composition-table eval-domain-nth[OF idx-bound] idx-bound
      eval-domain-length by simp
  have trace-powers:
    map (! trace-table) (powers-scaled idx) =
      map ( $\lambda p. \text{poly } p (h \wedge idx * shift)$ ) (trace-powers-of f)
    by (rule query-values-trace-powers[OF trace-table])
      (rule query-sample-space-powers-scaled-bound[OF idx-sample])
  have mult:
    poly common-denominator-poly ( $h \wedge idx * shift$ ) *
      cp-eval as
      (map ( $\lambda p. \text{poly } p (h \wedge idx * shift)$ ) (trace-powers-of f))
      ( $h \wedge idx * shift$ ) =
      poly (common-denominator-numerator f as) ( $h \wedge idx * shift$ )
    by (rule common-denominator-mult-cp-eval-trace-powers[OF idx-bound])
  have cp-eq:
    cp-eval as
      (map ( $\lambda p. \text{poly } p (h \wedge idx * shift)$ ) (trace-powers-of f))
      ( $h \wedge idx * shift$ ) =
      poly q ( $h \wedge idx * shift$ )
    using qc comp-value trace-powers
    unfolding query-consistent-at-def by simp
  have poly (common-denominator-numerator f as) ( $h \wedge idx * shift$ ) =
    poly common-denominator-poly ( $h \wedge idx * shift$ ) *
    poly q ( $h \wedge idx * shift$ )
    using mult cp-eq by simp
  then have poly ?R ( $h \wedge idx * shift$ ) = 0
    by simp
  then show  $x \in \{x. \text{poly } ?R x = 0\}$ 
    unfolding x-eq by simp
qed
have card ?A = card (( $\lambda idx. h \wedge idx * shift$ ) ‘ ?A)
  using card-image[OF inj] by simp
also have  $\dots \leq \text{card } \{x. \text{poly } ?R x = 0\}$ 
  by (rule card-mono[OF poly-roots-finite[OF residual-nonzero] image-subset])
also have  $\dots \leq \text{degree } ?R$ 
  by (rule card-poly-roots-bound[OF residual-nonzero])
also have  $\dots \leq \text{query-agreement-bound}$ 
  by (rule common-denominator-residual-degree-le-bound[OF deg-f deg-q])
finally show ?thesis .
qed
end

```

end

theory *Soundness-Core-WP*
imports *Soundness-Core-Base*
begin

Generic probability and WP combinators used by soundness reductions.

context *soundness*
begin

lemma *wp-event-union-bound*:

fixes $P\ Q :: ('x \times 's)\ \text{option} \Rightarrow \text{bool}$

shows

$$\text{wp-event } m\ (\lambda \text{out. } P\ \text{out} \vee Q\ \text{out})\ s \leq \\ \text{wp-event } m\ P\ s + \text{wp-event } m\ Q\ s$$

proof –

have *pointwise*:

$$\bigwedge \text{out } p. p * (\text{if } P\ \text{out} \vee Q\ \text{out} \text{ then } 1 \text{ else } 0) \leq \\ p * ((\text{if } P\ \text{out} \text{ then } 1 \text{ else } 0) + (\text{if } Q\ \text{out} \text{ then } 1 \text{ else } 0)) :: \text{prob}$$

by (*simp add: mult-left-mono*)

have $\text{wp-event } m\ (\lambda \text{out. } P\ \text{out} \vee Q\ \text{out})\ s \leq$

$$\text{wp } m\ (\lambda \text{out. } (\text{if } P\ \text{out} \text{ then } 1 \text{ else } 0) + (\text{if } Q\ \text{out} \text{ then } 1 \text{ else } 0))\ s$$

unfolding *wp-event-def wp-def dist-expect-def*

by (*intro sum-mono pointwise*)

also have $\dots = \text{wp-event } m\ P\ s + \text{wp-event } m\ Q\ s$

unfolding *wp-event-def wp-def dist-expect-def*

by (*simp add: algebra-simps sum.distrib*)

finally show *?thesis* .

qed

lemma *wp-event-union-bound4*:

fixes $A\ B\ C\ D :: ('x \times 's)\ \text{option} \Rightarrow \text{bool}$

shows

$$\text{wp-event } m\ (\lambda \text{out. } A\ \text{out} \vee B\ \text{out} \vee C\ \text{out} \vee D\ \text{out})\ s \leq \\ \text{wp-event } m\ A\ s + \text{wp-event } m\ B\ s + \text{wp-event } m\ C\ s + \text{wp-event } m\ D\ s$$

proof –

have *pointwise*:

$$\bigwedge \text{out } p. p * (\text{if } A\ \text{out} \vee B\ \text{out} \vee C\ \text{out} \vee D\ \text{out} \text{ then } 1 \text{ else } 0) \leq \\ p * ((\text{if } A\ \text{out} \text{ then } 1 \text{ else } 0) + (\text{if } B\ \text{out} \text{ then } 1 \text{ else } 0) + \\ (\text{if } C\ \text{out} \text{ then } 1 \text{ else } 0) + (\text{if } D\ \text{out} \text{ then } 1 \text{ else } 0)) :: \text{prob}$$

by (*simp add: mult-left-mono*)

have $\text{wp-event } m\ (\lambda \text{out. } A\ \text{out} \vee B\ \text{out} \vee C\ \text{out} \vee D\ \text{out})\ s \leq$

$$\text{wp } m\ (\lambda \text{out. } (\text{if } A\ \text{out} \text{ then } 1 \text{ else } 0) + (\text{if } B\ \text{out} \text{ then } 1 \text{ else } 0) + \\ (\text{if } C\ \text{out} \text{ then } 1 \text{ else } 0) + (\text{if } D\ \text{out} \text{ then } 1 \text{ else } 0))\ s$$

unfolding *wp-event-def wp-def dist-expect-def*

by (*intro sum-mono pointwise*)

```

also have ... =
  wp-event m A s + wp-event m B s + wp-event m C s + wp-event m D s
  unfolding wp-event-def wp-def dist-expect-def
  by (simp add: algebra-simps sum.distrib)
finally show ?thesis
.
qed

lemma wp-event-finite-UN-bound:
  fixes P :: 'a ⇒ ('x × 's) option ⇒ bool
  and C :: 'a ⇒ prob
  assumes finite-A: finite A
  and bounds:  $\bigwedge a. a \in A \implies \text{wp-event } m (P a) s \leq C a$ 
  shows
    wp-event m ( $\lambda \text{out}. \exists a \in A. P a \text{ out}$ ) s  $\leq (\sum a \in A. C a)$ 
proof -
  have pointwise:
     $\bigwedge \text{out } p. p * (\text{if } \exists a \in A. P a \text{ out then } 1 \text{ else } 0) \leq$ 
     $p * (\sum a \in A. \text{if } P a \text{ out then } 1 \text{ else } 0 :: \text{prob})$ 
  proof -
    fix out p
    have indicator-le:
       $(\text{if } \exists a \in A. P a \text{ out then } 1 \text{ else } 0 :: \text{prob}) \leq$ 
       $(\sum a \in A. \text{if } P a \text{ out then } 1 \text{ else } 0)$ 
    proof (cases  $\exists a \in A. P a \text{ out}$ )
    case True
    then obtain a where a:  $a \in A \ P a \text{ out}$ 
    by blast
    have 1 =  $(\text{if } P a \text{ out then } 1 \text{ else } 0 :: \text{prob})$ 
    using a by simp
    also have ...  $\leq (\sum a \in A. \text{if } P a \text{ out then } 1 \text{ else } 0)$ 
    by (rule member-le-sum) (use a finite-A in simp-all)
    finally show ?thesis
    using True by simp
  next
  case False
  then show ?thesis by simp
qed
show p *  $(\text{if } \exists a \in A. P a \text{ out then } 1 \text{ else } 0) \leq$ 
  p *  $(\sum a \in A. \text{if } P a \text{ out then } 1 \text{ else } 0 :: \text{prob})$ 
  by (simp add: mult-left-mono[OF indicator-le])
qed
have wp-event m ( $\lambda \text{out}. \exists a \in A. P a \text{ out}$ ) s  $\leq$ 
  wp m ( $\lambda \text{out}. (\sum a \in A. \text{if } P a \text{ out then } 1 \text{ else } 0) s$ )
  unfolding wp-event-def wp-def dist-expect-def
  by (intro sum-mono pointwise)
also have ... =  $(\sum a \in A. \text{wp-event } m (P a) s)$ 
  using finite-A
proof (induction A)

```

```

case empty
then show ?case
  unfolding wp-event-def wp-def dist-expect-def by simp
next
case (insert a A)
have wp m (λout. ∑ x∈insert a A. if P x out then 1 else 0) s =
  wp m (λout.
    (if P a out then 1 else 0) +
    (∑ x∈A. if P x out then 1 else 0)) s
  using insert.hyps by simp
also have ... =
  wp-event m (P a) s +
  wp m (λout. ∑ x∈A. if P x out then 1 else 0) s
  unfolding wp-event-def wp-def dist-expect-def
  by (simp add: sum.distrib algebra-simps)
also have ... = wp-event m (P a) s + (∑ x∈A. wp-event m (P x) s)
  using insert.IH by simp
also have ... = (∑ x∈insert a A. wp-event m (P x) s)
  using insert.hyps by simp
finally show ?case .
qed
also have ... ≤ (∑ a∈A. C a)
  by (rule sum-mono) (rule bounds, assumption)
finally show ?thesis .
qed

lemma wp-event-mono:
  fixes P Q :: ('x × 's) option ⇒ bool
  assumes ∧out. P out ⇒ Q out
  shows wp-event m P s ≤ wp-event m Q s
proof –
  have pointwise:
    ∧out p. p * (if P out then 1 else 0) ≤
    p * (if Q out then 1 else 0 :: prob)
    using assms by (simp add: mult-left-mono)
  show ?thesis
    unfolding wp-event-def wp-def dist-expect-def
    by (intro sum-mono pointwise)
qed

lemma wp-event-le-1:
  fixes P :: ('x × 's) option ⇒ bool
  shows wp-event m P s ≤ 1
proof –
  have pointwise:
    ∧out p. p * (if P out then 1 else 0 :: prob) ≤ p
    by (simp add: mult-left-le)
  have wp-event m P s ≤
    (∑ out∈dom (dist (execute m s)).

```

```

    the (dist (execute m s) out))
  unfolding wp-event-def wp-def dist-expect-def
  by (intro sum-mono pointwise)
  also have ... = 1
    unfolding sum-map-def[symmetric] by simp
  finally show ?thesis .
qed

lemma wp-le-const-on-support:
  assumes  $\bigwedge out. out \in \text{set-dist } (execute\ m\ s) \implies Q\ out \leq C$ 
  shows  $wp\ m\ Q\ s \leq C$ 
proof -
  have pointwise:
     $\bigwedge out\ p. out \in \text{dom } (dist\ (execute\ m\ s)) \implies$ 
     $p * Q\ out \leq p * C$ 
  using assms unfolding set-dist-def by (simp add: mult-left-mono)
  have  $wp\ m\ Q\ s \leq$ 
     $(\sum out \in \text{dom } (dist\ (execute\ m\ s)).$ 
     $the\ (dist\ (execute\ m\ s)\ out) * C)$ 
  unfolding wp-def dist-expect-def
  by (intro sum-mono pointwise)
  also have ... =
     $(\sum out \in \text{dom } (dist\ (execute\ m\ s)).$ 
     $the\ (dist\ (execute\ m\ s)\ out)) * C$ 
  by (simp add: sum-distrib-right)
  also have ... = C
    unfolding sum-map-def[symmetric] by simp
  finally show ?thesis .
qed

```

```

lemma wp-mono-on-support:
  assumes  $\bigwedge out. out \in \text{set-dist } (execute\ m\ s) \implies Q\ out \leq R\ out$ 
  shows  $wp\ m\ Q\ s \leq wp\ m\ R\ s$ 
proof -
  have pointwise:
     $\bigwedge out\ p. out \in \text{dom } (dist\ (execute\ m\ s)) \implies$ 
     $p * Q\ out \leq p * R\ out$ 
  using assms unfolding set-dist-def by (simp add: mult-left-mono)
  show ?thesis
    unfolding wp-def dist-expect-def
    by (intro sum-mono pointwise) simp
qed

```

```

lemma wp-event-mono-on-support:
  fixes  $P\ Q :: ('x \times 's)\ \text{option} \Rightarrow \text{bool}$ 
  assumes  $\bigwedge out. out \in \text{set-dist } (execute\ m\ s) \implies P\ out \implies Q\ out$ 
  shows  $wp\text{-event}\ m\ P\ s \leq wp\text{-event}\ m\ Q\ s$ 
proof -
  have pointwise:

```


$\bigwedge out\ p. out \in \text{dom } (\text{dist } (\text{execute } m\ s)) \implies$
 $p * (\text{if } P\ out\ \text{then } 1\ \text{else } 0) \leq$
 $p * (\text{if } Q\ out\ \text{then } 1\ \text{else } 0 :: \text{prob})$
using *assms* **unfolding** *set-dist-def* **by** (*simp add: mult-left-mono*)
show *?thesis*
unfolding *wp-event-def wp-def dist-expect-def*
by (*intro sum-mono pointwise*) *simp*
qed

lemma *wp-event-pos-of-support*:
assumes *out*: $out \in \text{set-dist } (\text{execute } m\ s)$
and *event*: $P\ out$
shows $0 < \text{wp-event } m\ P\ s$
proof –
let $?D = \text{dom } (\text{dist } (\text{execute } m\ s))$
have *out-dom*: $out \in ?D$
using *out* **unfolding** *set-dist-def* .
have *out-weight-pos*: $0 < \text{the } (\text{dist } (\text{execute } m\ s)\ out)$
using *dist-nonzero-on-dom[OF out-dom]* **by** *simp*
have $0 < (\sum_{x \in ?D}.$
 $\text{the } (\text{dist } (\text{execute } m\ s)\ x) * (\text{if } P\ x\ \text{then } 1\ \text{else } 0 :: \text{prob}))$
proof (*rule sum-pos2*)
show *finite ?D*
by *simp*
show $out \in ?D$
by (*rule out-dom*)
show $0 < \text{the } (\text{dist } (\text{execute } m\ s)\ out) *$
 $(\text{if } P\ out\ \text{then } 1\ \text{else } 0 :: \text{prob})$
using *event out-weight-pos* **by** *simp*
qed *simp*
then show *?thesis*
unfolding *wp-event-def wp-def dist-expect-def* .
qed

lemma *wp-event-pos-imp-exists-support*:
assumes $0 < \text{wp-event } m\ P\ s$
obtains *out* **where** $out \in \text{set-dist } (\text{execute } m\ s)$ **and** $P\ out$
proof –
have $\exists out \in \text{set-dist } (\text{execute } m\ s). P\ out$
proof (*rule ccontr*)
assume $\neg (\exists out \in \text{set-dist } (\text{execute } m\ s). P\ out)$
then have *no-event*:
 $\bigwedge out. out \in \text{dom } (\text{dist } (\text{execute } m\ s)) \implies \neg P\ out$
unfolding *set-dist-def* **by** *blast*
have $\text{wp-event } m\ P\ s = 0$
unfolding *wp-event-def wp-def dist-expect-def*
by (*intro sum.neutral*) (*simp add: no-event*)
then show *False*
using *assms* **by** *simp*

```

qed
then show ?thesis
  using that by blast
qed

lemma wp-event-bind-bound-by-head-event:
  fixes m :: ('x, 's) state-monad
  and k :: 'x  $\Rightarrow$  ('y, 's) state-monad
  assumes m-bound: wp-event m P s  $\leq$  C
  and none-imp: Q None  $\Rightarrow$  P None
  and cont-imp:
     $\bigwedge x\ t\ out. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m\ s) \Rightarrow$ 
       $out \in \text{set-dist } (\text{execute } (k\ x)\ t) \Rightarrow$ 
       $Q\ out \Rightarrow P\ (\text{Some } (x, t))$ 
  shows wp-event (m  $\gg$  k) Q s  $\leq$  C
proof -
  have bind-eq:
    wp-event (m  $\gg$  k) Q s =
      wp m ( $\lambda out. \text{case out of}$ 
        None  $\Rightarrow$  if Q None then 1 else 0
        | Some (x, t)  $\Rightarrow$  wp-event (k x) Q t) s
    unfolding wp-event-def by (simp add: wpsimps)
  have bind-head:
    wp-event (m  $\gg$  k) Q s  $\leq$  wp-event m P s
proof -
  have wp-event (m  $\gg$  k) Q s =
    wp m ( $\lambda out. \text{case out of}$ 
      None  $\Rightarrow$  if Q None then 1 else 0
      | Some (x, t)  $\Rightarrow$  wp-event (k x) Q t) s
    by (rule bind-eq)
  also have ...  $\leq$  wp-event m P s
    unfolding wp-event-def
  proof (rule wp-mono-on-support)
    fix out
    assume out: out  $\in$  set-dist (execute m s)
    show
      (case out of None  $\Rightarrow$  if Q None then 1 else 0
        | Some (x, t)  $\Rightarrow$ 
          wp (k x) ( $\lambda out. \text{if } Q\ out \text{ then } 1 \text{ else } 0$ ) t)
         $\leq$  (if P out then 1 else 0)
    proof (cases out)
      case None
      then show ?thesis
        using none-imp by (cases Q None) simp-all
    next
      case (Some xt)
      then obtain x t where xt: xt = (x, t)
        by (cases xt) auto
      have out-eq: out = Some (x, t)

```

```

    using Some xt by simp
  have head-out: Some (x, t) ∈ set-dist (execute m s)
    using out out-eq by simp
  have wp-bound:
    wp (k x) (λout. if Q out then 1 else 0) t ≤
      (if P (Some (x, t)) then 1 else 0 :: prob)
  proof (rule wp-le-const-on-support)
    fix out'
    assume out': out' ∈ set-dist (execute (k x) t)
    show (if Q out' then 1 else 0) ≤
      (if P (Some (x, t)) then 1 else 0 :: prob)
    proof (cases Q out')
      case True
      then have P (Some (x, t))
        by (rule cont-imp[OF head-out out'])
      then show ?thesis
        using True by simp
    next
      case False
      then show ?thesis by simp
    qed
  qed
  show ?thesis
    using wp-bound out-eq by simp
  qed
  qed
  finally show ?thesis .
  qed
show ?thesis
  by (rule order-trans[OF bind-head m-bound])
qed

```

```

lemma wp-event-bind-cong-cont:
  fixes m :: ('x, 's) state-monad
  and k :: 'x ⇒ ('y, 's) state-monad
  and l :: 'x ⇒ ('z, 's) state-monad
  assumes none: Q None = R None
  and cont:
    ∧x t. wp-event (k x) Q t = wp-event (l x) R t
  shows wp-event (m ≫ k) Q s = wp-event (m ≫ l) R s
proof -
  have left:
    wp-event (m ≫ k) Q s =
    wp m (λout. case out of
      None ⇒ if Q None then 1 else 0
    | Some (x, t) ⇒ wp-event (k x) Q t) s
  unfolding wp-event-def by (simp add: wpsimps)
  have right:
    wp-event (m ≫ l) R s =

```

```

    wp m (λout. case out of
      None ⇒ if R None then 1 else 0
      | Some (x, t) ⇒ wp-event (l x) R t) s
  unfolding wp-event-def by (simp add: wpsimps)
have fun-eq:
  (λout. case out of
    None ⇒ if Q None then 1 else 0
    | Some (x, t) ⇒ wp-event (k x) Q t) =
  (λout. case out of
    None ⇒ if R None then 1 else 0
    | Some (x, t) ⇒ wp-event (l x) R t)
  by (rule ext) (simp add: none cont split: option.splits prod.splits)
show ?thesis
  unfolding left right
  by (simp add: fun-eq)
qed

```

lemma *wp-event-bind-bound-by-head-and-cont:*

```

fixes m :: ('x, 's) state-monad
and k :: 'x ⇒ ('y, 's) state-monad
assumes none: ¬ Q None
and cont-zero:
  λx t. Some (x, t) ∈ set-dist (execute m s) ⇒
    ¬ P (Some (x, t)) ⇒ wp-event (k x) Q t = 0
and cont-bound:
  λx t. Some (x, t) ∈ set-dist (execute m s) ⇒
    P (Some (x, t)) ⇒ wp-event (k x) Q t ≤ D
shows wp-event (m ≫ k) Q s ≤ wp-event m P s * D
proof -
  have bind-eq:
    wp-event (m ≫ k) Q s =
    wp m (λout. case out of
      None ⇒ if Q None then 1 else 0
      | Some (x, t) ⇒ wp-event (k x) Q t) s
  unfolding wp-event-def by (simp add: wpsimps)
have wp-event (m ≫ k) Q s ≤
  wp m (λout. (if P out then 1 else 0) * D) s
  unfolding bind-eq
proof (rule wp-mono-on-support)
  fix out
  assume out: out ∈ set-dist (execute m s)
  show (case out of None ⇒ if Q None then 1 else 0
    | Some (x, t) ⇒ wp-event (k x) Q t)
    ≤ (if P out then 1 else 0) * D
  proof (cases out)
    case None
    then show ?thesis
      using none by simp
  next

```

```

case (Some xt)
then obtain x t where xt: xt = (x, t)
  by (cases xt) auto
have head: Some (x, t) ∈ set-dist (execute m s)
  using out Some xt by simp
show ?thesis
proof (cases P (Some (x, t)))
  case True
  then show ?thesis
    using cont-bound[OF head True] Some xt by simp
next
  case False
  then show ?thesis
    using cont-zero[OF head False] Some xt by simp
qed
qed
qed
also have ... = D * wp-event m P s
  unfolding wp-event-def wp-def dist-expect-def
  by (simp add: sum-distrib-left algebra-simps)
also have ... = wp-event m P s * D
  by (simp add: mult.commute)
finally show ?thesis .
qed

```

```

lemma wp-event-bind-bound-by-cont:
  fixes m :: ('x, 's) state-monad
  and k :: 'x ⇒ ('y, 's) state-monad
  assumes none: ¬ Q None
  and cont-bound:
     $\bigwedge x t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$ 
     $\text{wp-event } (k \ x) \ Q \ t \leq C$ 
  shows wp-event (m  $\gg$  k) Q s ≤ C
proof -
  have wp-event (m  $\gg$  k) Q s =
    wp m (λout. case out of
      None ⇒ if Q None then 1 else 0
      | Some (x, t) ⇒ wp-event (k x) Q t) s
  unfolding wp-event-def by (simp add: wpsimps)
  also have ... ≤ C
proof (rule wp-le-const-on-support)
  fix out
  assume out: out ∈ set-dist (execute m s)
  show (case out of None ⇒ if Q None then 1 else 0
    | Some (x, t) ⇒ wp-event (k x) Q t) ≤ C
  proof (cases out)
    case None
    then show ?thesis
      using none by simp

```

```

next
  case (Some xt)
  then obtain x t where xt: xt = (x, t)
    by (cases xt) auto
  show ?thesis
    using cont-bound[of x t] out Some xt by simp
qed
qed
finally show ?thesis .
qed

lemma sm-bind-return-map:
  ((m >>= (λx. return (f x))) >>= k) = (m >>= (λx. k (f x)))
  by (simp add: sm-bind-assoc)

lemma wp-bind-return-map:
  fixes m :: ('x, 's) state-monad
  shows
    wp (m >>= (λx. return (f x))) Q s =
      wp m (λout. case out of None ⇒ Q None
        | Some (x, t) ⇒ Q (Some (f x, t))) s
proof -
  have cont:
    (λr. case r of None ⇒ Q None
      | Some (x, t) ⇒ wp (return (f x)) Q t) =
      (λout. case out of None ⇒ Q None
        | Some (x, t) ⇒ Q (Some (f x, t)))
  by (rule ext) (simp add: wp-return split: option.splits prod.splits)
  show ?thesis
    by (subst wp-bind) (simp add: cont)
qed

lemma wp-event-bind-return-map:
  fixes m :: ('x, 's) state-monad
  shows
    wp-event (m >>= (λx. return (f x))) P s =
      wp-event m (λout. case out of None ⇒ P None
        | Some (x, t) ⇒ P (Some (f x, t))) s
proof -
  have indicator-eq:
    (λout. case out of None ⇒ if P None then 1 else 0
      | Some (x, t) ⇒ if P (Some (f x, t)) then 1 else 0) =
    (λout. if (case out of None ⇒ P None
      | Some (x, t) ⇒ P (Some (f x, t))) then 1 else 0)
  by (rule ext) (simp split: option.splits prod.splits)
  show ?thesis
    unfolding wp-event-def
    by (simp add: wp-bind-return-map indicator-eq)
qed

```

```

lemma set-dist-bindI:
  fixes  $m :: ('x, 's) \text{state-monad}$ 
    and  $k :: 'x \Rightarrow ('y, 's) \text{state-monad}$ 
  assumes head:  $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s)$ 
    and tail:  $\text{out} \in \text{set-dist } (\text{execute } (k \ x) \ t)$ 
  shows  $\text{out} \in \text{set-dist } (\text{execute } (m \gg k) \ s)$ 
proof -
  let  $?M = \text{dist } (\text{execute } m \ s)$ 
  let  $?K = \lambda r. \text{bind-cont-map } (\lambda x \ t. \text{dist } (\text{execute } (k \ x) \ t)) \ r$ 
  let  $?h = \text{Some } (x, t)$ 
  have head-dom:  $?h \in \text{dom } ?M$ 
    using head unfolding set-dist-def .
  have tail-dom:  $\text{out} \in \text{dom } (?K \ ?h)$ 
    using tail unfolding set-dist-def bind-cont-map-def by simp
  have tail-dist-dom:  $\text{out} \in \text{dom } (\text{dist } (\text{execute } (k \ x) \ t))$ 
    using tail unfolding set-dist-def .
  have head-pos:  $0 < \text{the } (?M \ ?h)$ 
    using dist-nonzero-on-dom[OF head-dom] by simp
  have tail-dist-pos:  $0 < \text{the } (\text{dist } (\text{execute } (k \ x) \ t) \ \text{out})$ 
    using dist-nonzero-on-dom[OF tail-dist-dom] by simp
  obtain  $p$  where p-eq:  $\text{dist } (\text{execute } (k \ x) \ t) \ \text{out} = \text{Some } p$ 
    using tail-dist-dom by auto
  have p-pos:  $0 < p$ 
    using tail-dist-pos p-eq by simp
  have tail-pos:  $0 < \text{option-default } (?K \ ?h \ \text{out})$ 
    using p-eq p-pos by (simp add: bind-cont-map-def)
  have term-pos:
     $0 < \text{the } (?M \ ?h) * \text{option-default } (?K \ ?h \ \text{out})$ 
    using head-pos tail-pos by simp
  have sum-pos:
     $0 < (\sum r \in \text{dom } ?M. \text{the } (?M \ r) * \text{option-default } (?K \ r \ \text{out}))$ 
  proof (rule sum-pos2)
    show finite ( $\text{dom } ?M$ )
      by simp
    show  $?h \in \text{dom } ?M$ 
      by (rule head-dom)
    show  $0 < \text{the } (?M \ ?h) * \text{option-default } (?K \ ?h \ \text{out})$ 
      by (rule term-pos)
  qed simp
  let  $?p = (\sum r \in \text{dom } ?M. \text{the } (?M \ r) * \text{option-default } (?K \ r \ \text{out}))$ 
  have p-nonzero:  $?p \neq 0$ 
    using sum-pos by simp
  have dist-eq:  $\text{dist } (\text{execute } (m \gg k) \ s) \ \text{out} = \text{Some } ?p$ 
    unfolding sm-bind.rep-eq dist-bind.rep-eq map-bind-def
    by (simp add: Let-def o-def map-fun-def p-nonzero)
  then show ?thesis
    unfolding set-dist-def by auto
qed

```

end

end

theory *Soundness-Core-Events*
 imports *Soundness-Core-WP*
begin

Transcript-shape, Merkle binding, and alpha/header event definitions.

context *soundness*
begin

definition *verifier-header-messages*

$:: 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list}$

where

verifier-header-messages fr f-fri-roots f-final as dg composition-fri-roots final =
 [fr] @ f-fri-roots @ [f-final] @ as @ [dg] @
 composition-fri-roots @ [final]

definition *verifier-header-state*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f$

where

verifier-header-state s fr f-fri-roots f-final as dg composition-fri-roots final =
 foldl concat (PState s)
 (*verifier-header-messages* fr f-fri-roots f-final as dg
 composition-fri-roots final)

definition *verifier-header-transcript*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$

where

verifier-header-transcript s fr f-fri-roots f-final as dg
 composition-fri-roots final rest $\longleftrightarrow$
 PTranscript s =
 verifier-header-messages fr f-fri-roots f-final as dg
 composition-fri-roots final @ rest $\wedge$
 length f-fri-roots = ceil-log clength $\wedge$
 length as = length spec $\wedge$
 length composition-fri-roots = ceil-log (to-nat dg + 1)

definition *merkle-root-binds-table*

$:: 'f \Rightarrow 'f \text{ list} \Rightarrow ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

merkle-root-binds-table rt table s $\longleftrightarrow$
 ($\exists \text{ tree} :: 'f \text{ tree. created-tree table tree s} \wedge \text{rt} = \text{value tree}$)

definition *merkle-root-binding-collision*
 $:: 'f \Rightarrow ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
 $\text{merkle-root-binding-collision } rt \ s \longleftrightarrow$
 $(\exists \text{ table table'}. \text{merkle-root-binds-table } rt \ \text{table } s \wedge$
 $\text{merkle-root-binds-table } rt \ \text{table'} \ s \wedge$
 $\text{table} \neq \text{table'})$

definition *hash-map-output-collision*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
 $\text{hash-map-output-collision } s \longleftrightarrow$
 $(\exists (x :: 'f \text{ protocol-hash-input}) (y :: 'f \text{ protocol-hash-input}) \ z. \ x \neq y \wedge$
 $\text{fmlookup } (\text{HashMap } s) \ x = \text{Some } z \wedge$
 $\text{fmlookup } (\text{HashMap } s) \ y = \text{Some } z)$

definition *hash-map-output-values*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ set}$
where
 $\text{hash-map-output-values } s =$
 $\{z. \exists x :: 'f \text{ protocol-hash-input}. \text{fmlookup } (\text{HashMap } s) \ x = \text{Some } z\}$

lemma *hash-map-output-values-verifier-initial-state[simp]:*
 $\text{hash-map-output-values } (\text{verifier-initial-state } tr) = \{\}$
unfolding *hash-map-output-values-def* **by** *simp*

lemma *merkle-root-binding-collisionI:*
assumes *merkle-root-binds-table* $rt \ \text{table } s$
and *merkle-root-binds-table* $rt \ \text{table'} \ s$
and $\text{table} \neq \text{table'}$
shows *merkle-root-binding-collision* $rt \ s$
using *assms* **unfolding** *merkle-root-binding-collision-def* **by** *blast*

lemma *merkle-root-binds-table-unique-if-no-collision:*
assumes $\neg \text{merkle-root-binding-collision } rt \ s$
and *merkle-root-binds-table* $rt \ \text{table } s$
and *merkle-root-binds-table* $rt \ \text{table'} \ s$
shows $\text{table} = \text{table'}$
using *assms* **unfolding** *merkle-root-binding-collision-def* **by** *blast*

lemma *merkle-root-binds-table-mono:*
assumes *bind:* *merkle-root-binds-table* $rt \ \text{table } s$
and *ext:* $s \leq t$
shows *merkle-root-binds-table* $rt \ \text{table } t$
proof –
from *bind* **obtain** *tree* **where**

```

    created: created-tree table tree s
  and root: rt = value tree
  unfolding merkle-root-binds-table-def by blast
have created-tree table tree t
  by (rule created-tree-mono[OF created ext])
then show ?thesis
  unfolding merkle-root-binds-table-def
  using root by blast
qed

definition hash-maps-compatible
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  ('f, 'a) protocol-channel-scheme  $\Rightarrow$  bool
where
  hash-maps-compatible s t  $\longleftrightarrow$ 
    ( $\forall x y z.$ 
      fmlookup (HashMap s) x = Some y  $\longrightarrow$ 
      fmlookup (HashMap t) x = Some z  $\longrightarrow$ 
      y = z)

definition hash-map-value-conflict
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  ('f, 'a) protocol-channel-scheme  $\Rightarrow$  bool
where
  hash-map-value-conflict s t  $\longleftrightarrow$ 
    ( $\exists x y z.$ 
      fmlookup (HashMap s) x = Some y  $\wedge$ 
      fmlookup (HashMap t) x = Some z  $\wedge$ 
      y  $\neq$  z)

lemma hash-maps-compatible-iff-no-value-conflict:
  hash-maps-compatible s t  $\longleftrightarrow$   $\neg$  hash-map-value-conflict s t
  unfolding hash-maps-compatible-def hash-map-value-conflict-def by blast

lemma hash-map-value-conflict-sym:
  assumes hash-map-value-conflict s t
  shows hash-map-value-conflict t s
  using assms unfolding hash-map-value-conflict-def by blast

lemma hash-maps-compatible-sym:
  assumes hash-maps-compatible s t
  shows hash-maps-compatible t s
  using assms
  unfolding hash-maps-compatible-iff-no-value-conflict
  using hash-map-value-conflict-sym by blast

lemma hash-map-value-conflict-no-common-extension:
  assumes conflict: hash-map-value-conflict s t
  and s-ext: s  $\leq$  u

```

```

    and t-ext:  $t \leq u$ 
  shows False
proof -
  from conflict obtain x y z where
    s-lookup: fmlookup (HashMap s) x = Some y
    and t-lookup: fmlookup (HashMap t) x = Some z
    and neg:  $y \neq z$ 
    unfolding hash-map-value-conflict-def by blast
  have u-y: fmlookup (HashMap u) x = Some y
    by (rule hash-extension-lookup[OF s-lookup s-ext])
  have u-z: fmlookup (HashMap u) x = Some z
    by (rule hash-extension-lookup[OF t-lookup t-ext])
  show False
    using u-y u-z neg by simp
qed

lemma common-extension-imp-no-hash-map-value-conflict:
  assumes s-ext:  $s \leq u$ 
    and t-ext:  $t \leq u$ 
  shows  $\neg$  hash-map-value-conflict s t
  using hash-map-value-conflict-no-common-extension[OF s-ext t-ext] by blast

lemma hash-map-value-conflict-blocks-clean-merge-obligation:
  assumes hash-map-value-conflict s t
  shows  $\neg$  (hash-maps-compatible s t  $\wedge$ 
     $\neg$  hash-map-output-collision (hash-state-merge s t))
  using assms unfolding hash-maps-compatible-iff-no-value-conflict by simp

fun merkle-hash-key :: 'f protocol-hash-input  $\Rightarrow$  bool
  where
    merkle-hash-key (MerkleLeaf -)  $\longleftrightarrow$  True
  | merkle-hash-key (MerkleNode -)  $\longleftrightarrow$  True
  | merkle-hash-key (FiatShamirChallenge -)  $\longleftrightarrow$  False
  | merkle-hash-key (TraceFriChallenge -)  $\longleftrightarrow$  False
  | merkle-hash-key (CompositionFriChallenge -)  $\longleftrightarrow$  False
  | merkle-hash-key (AlphaChallenge -)  $\longleftrightarrow$  False
  | merkle-hash-key (QueryIndexChallenge -)  $\longleftrightarrow$  False

definition merkle-hash-maps-compatible
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    ('f, 'a) protocol-channel-scheme  $\Rightarrow$  bool
  where
    merkle-hash-maps-compatible s t  $\longleftrightarrow$ 
      ( $\forall$  x y z.
        merkle-hash-key x  $\longrightarrow$ 
        fmlookup (HashMap s) x = Some y  $\longrightarrow$ 
        fmlookup (HashMap t) x = Some z  $\longrightarrow$ 
        y = z)

```

definition *merkle-hash-value-conflict*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{merkle-hash-value-conflict } s \ t \longleftrightarrow$
 $(\exists x \ y \ z.$
 $\text{merkle-hash-key } x \wedge$
 $\text{fmlookup } (\text{HashMap } s) \ x = \text{Some } y \wedge$
 $\text{fmlookup } (\text{HashMap } t) \ x = \text{Some } z \wedge$
 $y \neq z)$

lemma *merkle-hash-maps-compatible-iff-no-value-conflict*:

$\text{merkle-hash-maps-compatible } s \ t \longleftrightarrow$
 $\neg \text{merkle-hash-value-conflict } s \ t$

unfolding *merkle-hash-maps-compatible-def merkle-hash-value-conflict-def*
by *blast*

lemma *merkle-hash-value-conflict-sym*:

assumes *merkle-hash-value-conflict* $s \ t$

shows *merkle-hash-value-conflict* $t \ s$

using *assms* **unfolding** *merkle-hash-value-conflict-def* **by** *blast*

lemma *merkle-hash-maps-compatible-sym*:

assumes *merkle-hash-maps-compatible* $s \ t$

shows *merkle-hash-maps-compatible* $t \ s$

using *assms* **unfolding** *merkle-hash-maps-compatible-def* **by** *blast*

definition *merkle-hash-extends*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{merkle-hash-extends } s \ t \longleftrightarrow$
 $(\forall x \ y.$
 $\text{merkle-hash-key } x \longrightarrow$
 $\text{fmlookup } (\text{HashMap } s) \ x = \text{Some } y \longrightarrow$
 $\text{fmlookup } (\text{HashMap } t) \ x = \text{Some } y)$

lemma *merkle-hash-value-conflict-no-common-extension*:

assumes *conflict*: *merkle-hash-value-conflict* $s \ t$

and *ext-s*: *merkle-hash-extends* $s \ u$

and *ext-t*: *merkle-hash-extends* $t \ u$

shows *False*

proof –

from *conflict* **obtain** $x \ y \ z$ **where**

merkle: *merkle-hash-key* x

and *s-lookup*: *fmlookup* $(\text{HashMap } s) \ x = \text{Some } y$

and *t-lookup*: *fmlookup* $(\text{HashMap } t) \ x = \text{Some } z$

and *neq*: $y \neq z$

unfolding *merkle-hash-value-conflict-def* **by** *blast*

```

have u-y: fmlookup (HashMap u) x = Some y
  using ext-s merkle s-lookup unfolding merkle-hash-extends-def by blast
have u-z: fmlookup (HashMap u) x = Some z
  using ext-t merkle t-lookup unfolding merkle-hash-extends-def by blast
show False
  using u-y u-z neq by simp
qed

```

```

lemma common-merkle-extension-imp-no-merkle-hash-value-conflict:
  assumes ext-s: merkle-hash-extends s u
    and ext-t: merkle-hash-extends t u
  shows  $\neg$  merkle-hash-value-conflict s t
  using merkle-hash-value-conflict-no-common-extension[OF ext-s ext-t]
  by blast

```

```

lemma merkle-hash-maps-compatible-if-common-merkle-extension:
  assumes ext-s: merkle-hash-extends s u
    and ext-t: merkle-hash-extends t u
  shows merkle-hash-maps-compatible s t
  using common-merkle-extension-imp-no-merkle-hash-value-conflict
    [OF ext-s ext-t]
  unfolding merkle-hash-maps-compatible-iff-no-value-conflict .

```

```

definition merkle-hash-state-merge
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  ('f, 'a) protocol-channel-scheme
where
  merkle-hash-state-merge s t =
    s(HashMap :=
      fmlfilter merkle-hash-key (HashMap s) ++f
      fmlfilter merkle-hash-key (HashMap t))

```

```

lemma merkle-hash-state-merge-extends-right:
  merkle-hash-extends t (merkle-hash-state-merge s t)
  unfolding merkle-hash-extends-def merkle-hash-state-merge-def
  by (simp add: fmllookup-dom-iff)

```

```

lemma merkle-hash-state-merge-extends-left-if-compatible:
  assumes compatible: merkle-hash-maps-compatible s t
  shows merkle-hash-extends s (merkle-hash-state-merge s t)
  unfolding merkle-hash-extends-def
proof (intro allI impI)
  fix x y
  assume merkle: merkle-hash-key x
    and s-lookup: fmlookup (HashMap s) x = Some y
  show fmlookup (HashMap (merkle-hash-state-merge s t)) x = Some y
  proof (cases fmlookup (HashMap t) x)
  case None

```

```

then show ?thesis
  using merkle s-lookup
  unfolding merkle-hash-state-merge-def
  by (simp add: fmlookup-dom-iff)
next
case (Some z)
have z = y
  using compatible merkle s-lookup Some
  unfolding merkle-hash-maps-compatible-def by blast
then show ?thesis
  using merkle Some
  unfolding merkle-hash-state-merge-def
  by (simp add: fmlookup-dom-iff)
qed
qed

lemma merkle-hash-state-merge-lookupD:
  assumes lookup:
    fmlookup (HashMap (merkle-hash-state-merge s t)) x = Some z
  obtains
    merkle-hash-key x fmlookup (HashMap s) x = Some z
  | merkle-hash-key x fmlookup (HashMap t) x = Some z
proof -
  have merkle: merkle-hash-key x
    using lookup
    unfolding merkle-hash-state-merge-def
    by (cases merkle-hash-key x) auto
  show ?thesis
proof (cases fmlookup (HashMap t) x)
  case (Some z')
  then have z-eq: z' = z
    using lookup merkle
    unfolding merkle-hash-state-merge-def
    by (simp add: fmlookup-dom-iff)
  show ?thesis
    by (rule that(2)[OF merkle Some[unfolded z-eq]])
next
case None
have s-lookup: fmlookup (HashMap s) x = Some z
  using lookup merkle None
  unfolding merkle-hash-state-merge-def
  by (simp add: fmlookup-dom-iff)
show ?thesis
  by (rule that(1)[OF merkle s-lookup])
qed
qed

lemma merkle-hash-state-merge-clean-if-common-clean-merkle-extension:
  assumes ext-s: merkle-hash-extends s u

```

```

    and ext-t: merkle-hash-extends t u
    and clean:  $\neg$  hash-map-output-collision u
  shows  $\neg$  hash-map-output-collision (merkle-hash-state-merge s t)
proof
  assume collision: hash-map-output-collision (merkle-hash-state-merge s t)
  then obtain x y z where neg:  $x \neq y$ 
  and x-lookup:
    fmllookup (HashMap (merkle-hash-state-merge s t)) x = Some z
  and y-lookup:
    fmllookup (HashMap (merkle-hash-state-merge s t)) y = Some z
  unfolding hash-map-output-collision-def by blast
  have x-lookup-u: fmllookup (HashMap u) x = Some z
  proof -
    from x-lookup show ?thesis
  proof (rule merkle-hash-state-merge-lookupD)
    assume merkle: merkle-hash-key x
    and lookup-s: fmllookup (HashMap s) x = Some z
    show ?thesis
    using ext-s merkle lookup-s
    unfolding merkle-hash-extends-def by blast
  next
    assume merkle: merkle-hash-key x
    and lookup-t: fmllookup (HashMap t) x = Some z
    show ?thesis
    using ext-t merkle lookup-t
    unfolding merkle-hash-extends-def by blast
  qed
  qed
  have y-lookup-u: fmllookup (HashMap u) y = Some z
  proof -
    from y-lookup show ?thesis
  proof (rule merkle-hash-state-merge-lookupD)
    assume merkle: merkle-hash-key y
    and lookup-s: fmllookup (HashMap s) y = Some z
    show ?thesis
    using ext-s merkle lookup-s
    unfolding merkle-hash-extends-def by blast
  next
    assume merkle: merkle-hash-key y
    and lookup-t: fmllookup (HashMap t) y = Some z
    show ?thesis
    using ext-t merkle lookup-t
    unfolding merkle-hash-extends-def by blast
  qed
  qed
  have hash-map-output-collision u
  unfolding hash-map-output-collision-def
  using neg x-lookup-u y-lookup-u by blast
  then show False

```

using *clean* by *contradiction*
qed

lemma *common-clean-merkle-extension-imp-pairwise-merkle-clean*:
assumes *ext-s*: *merkle-hash-extends s u*
and *ext-t*: *merkle-hash-extends t u*
and *clean*: $\neg$ *hash-map-output-collision u*
shows
 $\neg$ *merkle-hash-value-conflict s t* $\wedge$
 $\neg$ *hash-map-output-collision (merkle-hash-state-merge s t)*
using *common-merkle-extension-imp-no-merkle-hash-value-conflict*
[*OF ext-s ext-t*]
merkle-hash-state-merge-clean-if-common-clean-merkle-extension
[*OF ext-s ext-t clean*]
by *simp*

definition *merkle-hash-cross-output-collision*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
merkle-hash-cross-output-collision s t $\longleftrightarrow$
 $(\exists x y z.$
merkle-hash-key x $\wedge$ *merkle-hash-key y* $\wedge$ $x \neq y \wedge$
 $((\text{fmlookup (HashMap s) } x = \text{Some } z \wedge$
 $\text{fmlookup (HashMap t) } y = \text{Some } z) \vee$
 $(\text{fmlookup (HashMap t) } x = \text{Some } z \wedge$
 $\text{fmlookup (HashMap s) } y = \text{Some } z)))$

definition *merkle-hash-pairwise-coupling-bad*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
merkle-hash-pairwise-coupling-bad s t $\longleftrightarrow$
 $\text{merkle-hash-value-conflict s t} \vee$
merkle-hash-cross-output-collision s t

lemma *merkle-hash-cross-output-collision-sym*:
assumes *merkle-hash-cross-output-collision s t*
shows *merkle-hash-cross-output-collision t s*
using *assms* **unfolding** *merkle-hash-cross-output-collision-def* **by** *blast*

lemma *merkle-hash-pairwise-coupling-bad-sym*:
assumes *merkle-hash-pairwise-coupling-bad s t*
shows *merkle-hash-pairwise-coupling-bad t s*
using *assms*
unfolding *merkle-hash-pairwise-coupling-bad-def*
using *merkle-hash-value-conflict-sym*
merkle-hash-cross-output-collision-sym **by** *blast*


```

lemma merkle-hash-state-merge-collision-imp-local-or-cross:
  assumes collision:
    hash-map-output-collision (merkle-hash-state-merge s t)
  shows
    hash-map-output-collision s  $\vee$ 
    hash-map-output-collision t  $\vee$ 
    merkle-hash-cross-output-collision s t
proof -
  from collision obtain x y z where neq: x  $\neq$  y
  and x-lookup:
    fmlookup (HashMap (merkle-hash-state-merge s t)) x = Some z
  and y-lookup:
    fmlookup (HashMap (merkle-hash-state-merge s t)) y = Some z
  unfolding hash-map-output-collision-def by blast
from x-lookup consider
  (xs) merkle-hash-key x fmlookup (HashMap s) x = Some z
  | (xt) merkle-hash-key x fmlookup (HashMap t) x = Some z
  by (rule merkle-hash-state-merge-lookupD)
then show ?thesis
proof cases
  case xs
  from y-lookup consider
    (ys) merkle-hash-key y fmlookup (HashMap s) y = Some z
    | (yt) merkle-hash-key y fmlookup (HashMap t) y = Some z
    by (rule merkle-hash-state-merge-lookupD)
  then show ?thesis
proof cases
  case ys
  have hash-map-output-collision s
    unfolding hash-map-output-collision-def
    using neq xs ys by blast
  then show ?thesis by blast
next
  case yt
  have merkle-hash-cross-output-collision s t
    unfolding merkle-hash-cross-output-collision-def
    using neq xs yt by blast
  then show ?thesis by blast
qed
next
  case xt
  from y-lookup consider
    (ys) merkle-hash-key y fmlookup (HashMap s) y = Some z
    | (yt) merkle-hash-key y fmlookup (HashMap t) y = Some z
    by (rule merkle-hash-state-merge-lookupD)
  then show ?thesis
proof cases
  case ys
  have merkle-hash-cross-output-collision s t

```

```

    unfolding merkle-hash-cross-output-collision-def
    using neq xt ys by blast
  then show ?thesis by blast
next
case yt
have hash-map-output-collision t
  unfolding hash-map-output-collision-def
  using neq xt yt by blast
then show ?thesis by blast
qed
qed
qed

lemma merkle-hash-pairwise-bad-imp-local-collision-or-coupling-bad:
  assumes pair-bad:
    merkle-hash-value-conflict s t  $\vee$ 
    hash-map-output-collision (merkle-hash-state-merge s t)
  shows
    hash-map-output-collision s  $\vee$ 
    hash-map-output-collision t  $\vee$ 
    merkle-hash-pairwise-coupling-bad s t
proof (cases merkle-hash-value-conflict s t)
case True
  then show ?thesis
    unfolding merkle-hash-pairwise-coupling-bad-def by blast
next
case False
  then have merge-collision:
    hash-map-output-collision (merkle-hash-state-merge s t)
  using pair-bad by blast
from merkle-hash-state-merge-collision-imp-local-or-cross
  [OF merge-collision]
show ?thesis
  unfolding merkle-hash-pairwise-coupling-bad-def by blast
qed

definition hash-state-merge
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  ('f, 'a) protocol-channel-scheme
where
  hash-state-merge s t =
    s( $\llbracket$ HashMap := HashMap s ++f HashMap t $\rrbracket$ )

lemma hash-state-merge-extends-right:
  t  $\leq$  hash-state-merge s t
  unfolding hash-state-merge-def less-eq-hash-ext-def less-eq-fmap-def
  by auto

```

```

lemma hash-state-merge-extends-left-if-compatible:
  assumes compatible: hash-maps-compatible s t
  shows  $s \leq \text{hash-state-merge } s \ t$ 
proof –
  have lookup:
     $\bigwedge x \ y. \text{fmlookup } (\text{HashMap } s) \ x = \text{Some } y \implies$ 
     $\text{fmlookup } (\text{HashMap } (\text{hash-state-merge } s \ t)) \ x = \text{Some } y$ 
  proof –
    fix  $x \ y$ 
    assume s-lookup:  $\text{fmlookup } (\text{HashMap } s) \ x = \text{Some } y$ 
    show  $\text{fmlookup } (\text{HashMap } (\text{hash-state-merge } s \ t)) \ x = \text{Some } y$ 
    proof (cases  $x \in \text{fmdom } (\text{HashMap } t)$ )
      case True
      then obtain  $z$  where t-lookup:  $\text{fmlookup } (\text{HashMap } t) \ x = \text{Some } z$ 
      unfolding fmlookup-dom-iff by blast
      have  $z = y$ 
      using compatible s-lookup t-lookup
      unfolding hash-maps-compatible-def by blast
      then show ?thesis
      using True t-lookup
      unfolding hash-state-merge-def by simp
    next
      case False
      then show ?thesis
      using s-lookup unfolding hash-state-merge-def by simp
    qed
  qed
show ?thesis
  unfolding less-eq-hash-ext-def less-eq-fmap-def
proof
  fix  $x$ 
  show  $\text{fmlookup } (\text{HashMap } s) \ x = \text{None} \vee$ 
     $\text{fmlookup } (\text{HashMap } s) \ x =$ 
     $\text{fmlookup } (\text{HashMap } (\text{hash-state-merge } s \ t)) \ x$ 
  proof (cases  $\text{fmlookup } (\text{HashMap } s) \ x$ )
    case None
    then show ?thesis by simp
  next
    case (Some  $y$ )
    then have  $\text{fmlookup } (\text{HashMap } (\text{hash-state-merge } s \ t)) \ x = \text{Some } y$ 
    by (rule lookup)
    then show ?thesis
    using Some by simp
  qed
qed
qed

```

```

lemma hash-maps-compatible-if-common-extension:
  assumes s-ext:  $s \leq u$ 

```

```

    and t-ext:  $t \leq u$ 
  shows hash-maps-compatible s t
  unfolding hash-maps-compatible-def
proof (intro allI impI)
  fix x y z
  assume s-lookup: fmlookup (HashMap s) x = Some y
    and t-lookup: fmlookup (HashMap t) x = Some z
  have u-y: fmlookup (HashMap u) x = Some y
    by (rule hash-extension-lookup[OF s-lookup s-ext])
  have u-z: fmlookup (HashMap u) x = Some z
    by (rule hash-extension-lookup[OF t-lookup t-ext])
  show y = z
    using u-y u-z by simp
qed

lemma hash-state-merge-le-common-extension:
  assumes s-ext:  $s \leq u$ 
    and t-ext:  $t \leq u$ 
  shows hash-state-merge s t  $\leq u$ 
  unfolding less-eq-hash-ext-def less-eq-fmap-def
proof
  fix x
  show fmlookup (HashMap (hash-state-merge s t)) x = None  $\vee$ 
    fmlookup (HashMap (hash-state-merge s t)) x = fmlookup (HashMap u) x
proof (cases x | $\in$ | fndom (HashMap t))
  case True
  then obtain z where t-lookup: fmlookup (HashMap t) x = Some z
    unfolding fmlookup-dom-iff by blast
  have merge-lookup:
    fmlookup (HashMap (hash-state-merge s t)) x = Some z
    using True t-lookup unfolding hash-state-merge-def by simp
  have u-lookup: fmlookup (HashMap u) x = Some z
    by (rule hash-extension-lookup[OF t-lookup t-ext])
  show ?thesis
    using merge-lookup u-lookup by simp
next
  case False
  show ?thesis
proof (cases fmlookup (HashMap s) x)
  case None
  then have fmlookup (HashMap (hash-state-merge s t)) x = None
    using False unfolding hash-state-merge-def by simp
  then show ?thesis
    by simp
next
  case (Some y)
  then have merge-lookup:
    fmlookup (HashMap (hash-state-merge s t)) x = Some y
    using False unfolding hash-state-merge-def by simp

```

```

    have u-lookup: fmlookup (HashMap u) x = Some y
      by (rule hash-extension-lookup[OF Some s-ext])
    show ?thesis
      using merge-lookup u-lookup by simp
  qed
qed
qed

lemma hash-state-merge-clean-if-common-clean-extension:
  assumes s-ext:  $s \leq u$ 
    and t-ext:  $t \leq u$ 
    and clean:  $\neg \text{hash-map-output-collision } u$ 
  shows  $\neg \text{hash-map-output-collision } (\text{hash-state-merge } s \ t)$ 
proof
  assume collision: hash-map-output-collision (hash-state-merge s t)
  have merge-ext: hash-state-merge s t  $\leq u$ 
    by (rule hash-state-merge-le-common-extension[OF s-ext t-ext])
  from collision obtain x y z where neg:  $x \neq y$ 
    and x-lookup:
      fmlookup (HashMap (hash-state-merge s t)) x = Some z
    and y-lookup:
      fmlookup (HashMap (hash-state-merge s t)) y = Some z
  unfolding hash-map-output-collision-def by blast
  have x-lookup-u: fmlookup (HashMap u) x = Some z
    by (rule hash-extension-lookup[OF x-lookup merge-ext])
  have y-lookup-u: fmlookup (HashMap u) y = Some z
    by (rule hash-extension-lookup[OF y-lookup merge-ext])
  have hash-map-output-collision u
    unfolding hash-map-output-collision-def
    using neg x-lookup-u y-lookup-u by blast
  then show False
    using clean by contradiction
qed

lemma hash-map-output-collisionI:
  fixes x y :: 'f protocol-hash-input
  assumes  $x \neq y$ 
    and fmlookup (HashMap s) x = Some z
    and fmlookup (HashMap s) y = Some z
  shows hash-map-output-collision s
  using assms unfolding hash-map-output-collision-def by blast

lemma hash-map-output-collision-merkle-leafI:
  assumes  $x \neq y$ 
    and fmlookup (HashMap s) (MerkleLeaf x) = Some z
    and fmlookup (HashMap s) (MerkleLeaf y) = Some z
  shows hash-map-output-collision s
  by (rule hash-map-output-collisionI[OF - assms(2,3)])
    (use assms(1) in simp)

```

```

lemma hash-map-output-collision-merkle-nodeI:
  assumes  $(x1, x2) \neq (y1, y2)$ 
    and  $\text{fmlookup } (\text{HashMap } s) (\text{MerkleNode } x1 \ x2) = \text{Some } z$ 
    and  $\text{fmlookup } (\text{HashMap } s) (\text{MerkleNode } y1 \ y2) = \text{Some } z$ 
  shows hash-map-output-collision s
by (rule hash-map-output-collisionI [OF - assms(2,3)])
    (use assms(1) in auto)

lemma hash-map-output-collision-merkle-leaf-nodeI:
  assumes  $\text{fmlookup } (\text{HashMap } s) (\text{MerkleLeaf } x) = \text{Some } z$ 
    and  $\text{fmlookup } (\text{HashMap } s) (\text{MerkleNode } y1 \ y2) = \text{Some } z$ 
  shows hash-map-output-collision s
by (rule hash-map-output-collisionI [OF - assms])
    simp

lemma hash-map-output-valuesI:
  fixes  $x :: 'f \text{ protocol-hash-input}$ 
  assumes  $\text{fmlookup } (\text{HashMap } s) \ x = \text{Some } z$ 
  shows  $z \in \text{hash-map-output-values } s$ 
  using assms unfolding hash-map-output-values-def by blast

lemma hash-map-output-values-alt-def:
  hash-map-output-values s =
     $(\lambda x. \text{the } (\text{fmlookup } (\text{HashMap } s) \ x)) \text{ 'fmdom' } (\text{HashMap } s)$ 
  unfolding hash-map-output-values-def
proof
  show  $\{z. \exists x. \text{fmlookup } (\text{HashMap } s) \ x = \text{Some } z\}$ 
     $\subseteq (\lambda x. \text{the } (\text{fmlookup } (\text{HashMap } s) \ x)) \text{ 'fmdom' } (\text{HashMap } s)$ 
  proof
    fix  $z$ 
    assume  $z \in \{z. \exists x. \text{fmlookup } (\text{HashMap } s) \ x = \text{Some } z\}$ 
    then obtain  $x$  where lookup:  $\text{fmlookup } (\text{HashMap } s) \ x = \text{Some } z$ 
      by blast
    have  $x \in \text{fmdom' } (\text{HashMap } s)$ 
      using lookup by (simp add: fmlookup-dom'-iff)
    moreover have the  $(\text{fmlookup } (\text{HashMap } s) \ x) = z$ 
      using lookup by simp
    ultimately show  $z \in$ 
       $(\lambda x. \text{the } (\text{fmlookup } (\text{HashMap } s) \ x)) \text{ 'fmdom' } (\text{HashMap } s)$ 
      by blast
    qed
  show  $(\lambda x. \text{the } (\text{fmlookup } (\text{HashMap } s) \ x)) \text{ 'fmdom' } (\text{HashMap } s)$ 
     $\subseteq \{z. \exists x. \text{fmlookup } (\text{HashMap } s) \ x = \text{Some } z\}$ 
    by (auto simp: fmlookup-dom'-iff)
  qed

lemma finite-hash-map-output-values[simp]:
  finite (hash-map-output-values s)

```

unfolding *hash-map-output-values-alt-def* **by** *simp*

lemma *card-hash-map-output-values-le-fmdom*:
 $\text{card } (\text{hash-map-output-values } s) \leq \text{card } (\text{fmdom}' (\text{HashMap } s))$
unfolding *hash-map-output-values-alt-def*
by (*rule card-image-le*) *simp*

lemma *hash-map-output-values-after-update-subset*:
 fixes $x :: 'f \text{ protocol-hash-input}$
 shows
 hash-map-output-values
 $(s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s))) \subseteq$
 insert y (hash-map-output-values s)
unfolding *hash-map-output-values-def*
by *auto*

lemma *card-hash-map-output-values-after-update-le*:
 fixes $x :: 'f \text{ protocol-hash-input}$
 shows
 card (hash-map-output-values
 $(s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s)))) \leq$
 Suc (card (hash-map-output-values s))
proof –
 have *card (hash-map-output-values*
 $(s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s)))) \leq$
 card (insert y (hash-map-output-values s))
 by (*rule card-mono*)
 (*simp-all add: hash-map-output-values-after-update-subset*)
 also have $\dots \leq \text{Suc } (\text{card } (\text{hash-map-output-values } s))$
 by (*simp add: card-insert-if*)
 finally show *?thesis* .
qed

lemma *hash-map-output-collision-after-fresh-updateD*:
 fixes $x :: 'f \text{ protocol-hash-input}$
 assumes *clean*: $\neg \text{hash-map-output-collision } s$
 and *fresh*: $\text{fmlookup } (\text{HashMap } s) \ x = \text{None}$
 and *collision*:
 hash-map-output-collision
 $(s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s)))$
 shows $y \in \text{hash-map-output-values } s$
proof –
 obtain $a \ b \ z$ **where** *neg*: $a \neq b$
 and *a-lookup*:
 $\text{fmlookup } (\text{fmupd } x \ y \ (\text{HashMap } s)) \ a = \text{Some } z$
 and *b-lookup*:
 $\text{fmlookup } (\text{fmupd } x \ y \ (\text{HashMap } s)) \ b = \text{Some } z$
 using *collision* **unfolding** *hash-map-output-collision-def* **by** *auto*
show *?thesis*

```

proof (cases a = x)
  case True
  then have z-y: z = y
    using a-lookup by simp
  have b-ne-x: b ≠ x
    using neq True by simp
  have fmlookup (HashMap s) b = Some y
    using b-lookup b-ne-x z-y by simp
  then show ?thesis
    by (rule hash-map-output-valuesI)
next
  case a-ne-x: False
  show ?thesis
  proof (cases b = x)
    case True
    then have z-y: z = y
      using b-lookup by simp
    have fmlookup (HashMap s) a = Some y
      using a-lookup a-ne-x z-y by simp
    then show ?thesis
      by (rule hash-map-output-valuesI)
    next
    case b-ne-x: False
    have a-old: fmlookup (HashMap s) a = Some z
      using a-lookup a-ne-x by simp
    have b-old: fmlookup (HashMap s) b = Some z
      using b-lookup b-ne-x by simp
    have hash-map-output-collision s
      by (rule hash-map-output-collisionI[OF neq a-old b-old])
    then show ?thesis
      using clean by contradiction
  qed
qed
qed

```

lemma no-hash-map-output-collision-after-known-update:

```

fixes x :: 'f protocol-hash-input
assumes clean: ¬ hash-map-output-collision s
  and known: fmlookup (HashMap s) x = Some y
shows ¬ hash-map-output-collision
  (s(HashMap := fmupd x y (HashMap s)))

```

```

proof
assume collision:
  hash-map-output-collision
  (s(HashMap := fmupd x y (HashMap s)))
obtain a b z where neq: a ≠ b
and a-lookup:
  fmlookup (fmupd x y (HashMap s)) a = Some z
and b-lookup:

```



```

    fmlookup (fmupd x y (HashMap s)) b = Some z
  using collision unfolding hash-map-output-collision-def by auto
have a-old: fmlookup (HashMap s) a = Some z
  using a-lookup known by (cases a = x) simp-all
have b-old: fmlookup (HashMap s) b = Some z
  using b-lookup known by (cases b = x) simp-all
have hash-map-output-collision s
  by (rule hash-map-output-collisionI[OF neq a-old b-old])
then show False
  using clean by contradiction
qed

lemma dist-expect-uniform-fUNIV-indicator:
  fixes S :: 'f set
  shows
    dist-expect (dist-uniform (fUNIV :: 'f fset))
      (λx. if x ∈ S then 1 else 0) =
      nnreal (card S) / nnreal size
proof -
  let ?U = (fUNIV :: 'f fset)
  let ?w = 1 / nnreal (CARD('f))
  let ?B = ((λx :: 'f. (x, ?w)) |` ?U)

  have U-nonempty: ?U ≠ {}
  proof -
    have (undefined :: 'f) |∈| ?U
      by simp
    then show ?thesis
      by auto
  qed
  have fcard-U: fcard ?U = CARD('f)
    by (simp add: fcard.rep-eq)
  have total-nonzero:
    fsum snd ((λx :: 'f. (x, 1::prob)) |` ?U) ≠ 0
    using fsum-snd-uniform-ones[of ?U] fcard-U by simp
  have dist-eq:
    dist (dist-uniform ?U) = map-of ?B
    unfolding dist-uniform-def dist-of-fset.rep-eq map-of-fset-def
    using norm-fset-uniform-ones[OF U-nonempty] total-nonzero fcard-U
    by (simp add: Let-def)

  have lookup:
    dist (dist-uniform ?U) x = Some ?w for x :: 'f
  proof -
    have mem: (x, ?w) |∈| ?B
      by simp
    have uniq:
      ∧y z. (x, y) |∈| ?B ⇒ (x, z) |∈| ?B ⇒ y = z
      by auto
  qed

```

```

have val: map-of-val ?B x = ?w
  by (rule map-of-val-unique-key[OF uniq mem])
have x |∈| fimage fst ?B
  using mem by force
then have map-of ?B x = Some (map-of-val ?B x)
  unfolding map-of-def by simp
also have ... = Some ?w
  using val by simp
finally show ?thesis
  using dist-eq by simp
qed

have dom-eq: dom (dist (dist-uniform ?U)) = (UNIV :: 'f set)
  using lookup by auto

have dist-expect (dist-uniform ?U) (λx. if x ∈ S then 1 else 0) =
  (∑ x∈UNIV. ?w * (if x ∈ S then 1 else 0))
  unfolding dist-expect-def dom-eq
  by (intro sum.cong refl) (simp add: lookup)
also have ... = (∑ x∈UNIV. if x ∈ S then ?w else 0)
  by (intro sum.cong refl) (simp split: if-splits)
also have ... = (∑ x∈S. ?w)
proof -
  have (∑ x∈UNIV. if x ∈ S then ?w else 0) =
    (∑ x∈UNIV ∩ S. ?w)
  by (subst sum.inter-restrict[symmetric]) simp-all
  also have ... = (∑ x∈S. ?w)
  by simp
  finally show ?thesis .
qed
also have ... = (∑ x∈S. (1::prob)) / nnreal (CARD('f))
  by (rule sum-divide-nnreal) simp
also have ... = nnreal (card S) / nnreal (CARD('f))
  by (simp add: sum-1-nnreal)
also have ... = nnreal (card S) / nnreal size
  using size-card by simp
finally show ?thesis .
qed

lemma dist-expect-hash-dist-fresh-output-values:
  fixes x :: 'f protocol-hash-input
  assumes fresh: fmllookup (HashMap s) x = None
  shows
    dist-expect (hash-dist x s)
      (λy. if y ∈ hash-map-output-values s then 1 else 0) =
      nnreal (card (hash-map-output-values s)) / nnreal size
  using fresh
  unfolding hash-dist-def option-default-dist-def hash-default-def
  by (simp add: dist-expect-uniform-fUNIV-indicator)

```

lemma *dist-expect-hash-dist-fresh-indicator*:
fixes $x :: 'f$ *protocol-hash-input*
and $s :: ('f, 'a)$ *protocol-channel-scheme*
assumes *fresh*: $\text{fmlookup } (\text{HashMap } s) \ x = \text{None}$
shows
 $\text{dist-expect } (\text{hash-dist } x \ s) \ (\lambda y. \text{if } y \in B \text{ then } 1 \text{ else } 0) =$
 $\text{nnreal } (\text{card } B) / \text{nnreal size}$
using *fresh*
unfolding *hash-dist-def option-default-dist-def hash-default-def*
by (*simp add: dist-expect-uniform-fUNIV-indicator*)

lemma *wp-hash-fresh-set*:
fixes $x :: 'f$ *protocol-hash-input*
and $s :: ('f, 'a)$ *protocol-channel-scheme*
assumes *fresh*: $\text{fmlookup } (\text{HashMap } s) \ x = \text{None}$
shows
 $\text{wp-event } (\text{hash } x) \ (\lambda \text{out. case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (y, -) \Rightarrow y \in B) \ s =$
 $\text{nnreal } (\text{card } B) / \text{nnreal size}$
unfolding *wp-event-def*
by (*simp add: wp-hash dist-expect-hash-dist-fresh-indicator[OF fresh]*)

lemma *wp-hash-fresh-singleton-outcome*:
fixes $x :: 'f$ *protocol-hash-input*
and $s :: ('f, 'a)$ *protocol-channel-scheme*
assumes *fresh*: $\text{fmlookup } (\text{HashMap } s) \ x = \text{None}$
shows
 $\text{wp-event } (\text{hash } x)$
 $(\lambda \text{out. out} =$
 $\text{Some } (y, s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s)))) \ s =$
 $1 / \text{nnreal size}$

proof –
have *indicator*:
 $(\lambda xa. \text{if } xa = y \wedge$
 $s(\text{HashMap} := \text{fmupd } x \ xa \ (\text{HashMap } s)) =$
 $s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s))$
 $\text{then } 1 \text{ else } 0 :: \text{prob}) =$
 $(\lambda xa. \text{if } xa \in \{y\} \text{ then } 1 \text{ else } 0)$
by *auto*
have $\text{wp-event } (\text{hash } x)$
 $(\lambda \text{out. out} =$
 $\text{Some } (y, s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s)))) \ s =$
 $\text{dist-expect } (\text{hash-dist } x \ s) \ (\lambda xa. \text{if } xa \in \{y\} \text{ then } 1 \text{ else } 0)$
unfolding *wp-event-def* **by** (*simp add: wp-hash indicator*)
also have $\dots = \text{nnreal } (\text{card } (\{y\} :: 'f \text{ set})) / \text{nnreal size}$
by (*rule dist-expect-hash-dist-fresh-indicator[OF fresh]*)
also have $\dots = 1 / \text{nnreal size}$

by *simp*
 finally show *?thesis* .
 qed

lemma *hash-fresh-outcome*:

fixes $x :: 'f$ protocol-hash-input
 and $s :: ('f, 'a)$ protocol-channel-scheme
 assumes *fresh*: $\text{fmlookup } (\text{HashMap } s) \ x = \text{None}$
 shows
 $\text{Some } (y, s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s))) \in$
 $\text{set-dist } (\text{execute } (\text{hash } x) \ s)$

proof –

have *positive*:
 $0 < \text{wp-event } (\text{hash } x)$
 $(\lambda \text{out. out} =$
 $\text{Some } (y, s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s)))) \ s$
 using *wp-hash-fresh-singleton-outcome*[*OF fresh, of y*]
 size-card by *simp*
 then obtain *out* where *support*:
 $\text{out} \in \text{set-dist } (\text{execute } (\text{hash } x) \ s)$
 and *out-eq*:
 $\text{out} = \text{Some } (y, s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s)))$
 by (rule *wp-event-pos-imp-exists-support*)
 show *?thesis*
 using *support out-eq* by *simp*

qed

lemma *fresh-merkle-hash-has-pairwise-value-conflict*:

fixes $x :: 'f$ protocol-hash-input
 and $s :: ('f, 'a)$ protocol-channel-scheme
 assumes *merkle*: *merkle-hash-key* x
 and *fresh*: $\text{fmlookup } (\text{HashMap } s) \ x = \text{None}$
 and *neq*: $y \neq z$

shows

$\exists t \ u.$
 $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (\text{hash } x) \ s) \wedge$
 $\text{Some } (z, u) \in \text{set-dist } (\text{execute } (\text{hash } x) \ s) \wedge$
 $\text{merkle-hash-value-conflict } t \ u \wedge$
 $\text{merkle-hash-pairwise-coupling-bad } t \ u$

proof –

let $?t = s(\text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s))$
 let $?u = s(\text{HashMap} := \text{fmupd } x \ z \ (\text{HashMap } s))$
 have *out-y*: $\text{Some } (y, ?t) \in \text{set-dist } (\text{execute } (\text{hash } x) \ s)$
 by (rule *hash-fresh-outcome*[*OF fresh*])
 have *out-z*: $\text{Some } (z, ?u) \in \text{set-dist } (\text{execute } (\text{hash } x) \ s)$
 by (rule *hash-fresh-outcome*[*OF fresh*])
 have *conflict*: *merkle-hash-value-conflict* $?t \ ?u$
 unfolding *merkle-hash-value-conflict-def*
 using *merkle neq* by *auto*

then have *coupling: merkle-hash-pairwise-coupling-bad ?t ?u*
unfolding *merkle-hash-pairwise-coupling-bad-def* **by** *simp*
show *?thesis*
using *out-y out-z conflict coupling* **by** *blast*
qed

lemma *wp-receive-tagged-random-field-element-fresh-set:*
fixes *s :: ('f, 'a) protocol-channel-scheme*
assumes *fresh: fmlookup (HashMap s) (tag (PState s)) = None*
shows
wp-event (receive-tagged-random-field-element tag)
(λout. case out of None ⇒ False | Some (y, -) ⇒ y ∈ B) s =
nnreal (card B) / nnreal size
unfolding *wp-event-def receive-tagged-random-field-element-def*
protocol-receive-tagged-random-field-element-def
by (*simp add: wpsimps dist-expect-hash-dist-fresh-indicator[OF fresh]*)

lemma *wp-receive-counted-tagged-random-field-element-fresh-set:*
fixes *s :: ('f, 'a) protocol-channel-scheme*
assumes *fresh:*
fmlookup (HashMap s) (tag (counter s) (PState s)) = None
shows
wp-event
(protocol-receive-counted-tagged-random-field-element counter bump tag)
(λout. case out of None ⇒ False | Some (y, -) ⇒ y ∈ B) s =
nnreal (card B) / nnreal size
unfolding *wp-event-def protocol-receive-counted-tagged-random-field-element-def*
by (*simp add: wpsimps dist-expect-hash-dist-fresh-indicator[OF fresh]*)

lemma *wp-receive-alpha-challenge-fresh-set:*
fixes *s :: ('f, 'a) protocol-channel-scheme*
assumes *fresh: fmlookup (HashMap s) (AlphaChallenge (PAlphaCounter s) (PState s)) = None*
shows
wp-event receive-alpha-challenge
(λout. case out of None ⇒ False | Some (y, -) ⇒ y ∈ B) s =
nnreal (card B) / nnreal size
unfolding *receive-alpha-challenge-def*
by (*rule wp-receive-counted-tagged-random-field-element-fresh-set*
[where counter=PAlphaCounter
and bump=λs. s(PAlphaCounter := Suc (PAlphaCounter s))
and tag=AlphaChallenge and B=B and s=s, OF fresh])

lemma *wp-receive-query-index-challenge-fresh-set:*
fixes *s :: ('f, 'a) protocol-channel-scheme*
assumes *fresh:*
fmlookup (HashMap s) (QueryIndexChallenge (PQueryCounter s) (PState s)) = None
shows
wp-event receive-query-index-challenge
(λout. case out of None ⇒ False | Some (y, -) ⇒ y ∈ B) s =

```

    nnreal (card B) / nnreal size
unfolding receive-query-index-challenge-def
by (rule wp-receive-counted-tagged-random-field-element-fresh-set
    [where counter=PQueryCounter
     and bump= $\lambda s. s(PQueryCounter := Suc (PQueryCounter s))$ 
     and tag=QueryIndexChallenge and B=B and s=s, OF fresh])

definition query-index-raw-preimage :: nat set  $\Rightarrow$  'f set
  where
    query-index-raw-preimage B =
      {raw. index (to-nat raw)  $\in$  B}

definition query-index-nat-preimage :: nat set  $\Rightarrow$  nat set
  where
    query-index-nat-preimage B =
      {n  $\in$  range to-nat. n mod query-sample-space-size  $\in$  B}

lemma finite-query-index-raw-preimage[simp]:
  finite (query-index-raw-preimage B)
unfolding query-index-raw-preimage-def by simp

lemma finite-query-index-nat-preimage[simp]:
  finite (query-index-nat-preimage B)
unfolding query-index-nat-preimage-def by simp

lemma query-index-raw-preimage-image-to-nat:
  to-nat ' query-index-raw-preimage B = query-index-nat-preimage B
unfolding query-index-raw-preimage-def query-index-nat-preimage-def
  index-def query-sample-space-size-def
by auto

lemma card-query-index-raw-preimage-eq-nat-preimage:
  card (query-index-raw-preimage B) = card (query-index-nat-preimage B)
proof -
  have inj: inj-on to-nat (query-index-raw-preimage B)
    using to-nat-inj by (rule inj-on-subset) simp
  have card (query-index-raw-preimage B) =
    card (to-nat ' query-index-raw-preimage B)
    by (rule card-image[symmetric, OF inj])
  also have ... = card (query-index-nat-preimage B)
    unfolding query-index-raw-preimage-image-to-nat ..
  finally show ?thesis .
qed

lemma wp-receive-query-index-challenge-fresh-index-set:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes fresh:
    fmllookup (HashMap s) (QueryIndexChallenge (PQueryCounter s) (PState s)) = None
  shows

```

```

wp-event receive-query-index-challenge
  (λout. case out of
    None ⇒ False
    | Some (raw, -) ⇒ index (to-nat raw) ∈ B) s =
  nnreal (card (query-index-raw-preimage B)) / nnreal size
proof -
  have event-eq:
    (λout. case out of
      None ⇒ False
      | Some (raw, -) ⇒ index (to-nat raw) ∈ B) =
    (λout. case out of
      None ⇒ False
      | Some (raw, -) ⇒ raw ∈ query-index-raw-preimage B)
  unfolding query-index-raw-preimage-def
  by (rule ext) (auto split: option.splits prod.splits)
show ?thesis
  unfolding event-eq
  by (rule wp-receive-query-index-challenge-fresh-set[OF fresh])
qed

```

```

lemma card-mod-lessThan-dvd:
  assumes q-pos: 0 < q
  and dvd: q dvd N
  and b-lt: b < q
  shows card {n. n < N ∧ n mod q = b} = N div q
proof -
  let ?A = {0..<N div q}
  let ?B = {n. n < N ∧ n mod q = b}
  have N-eq: N = (N div q) * q
    using dvd q-pos by (simp add: dvd-eq-mod-eq-0)
  have image-eq: (λk. k * q + b) ' ?A = ?B
proof
  show (λk. k * q + b) ' ?A ⊆ ?B
proof
  fix n
  assume n ∈ (λk. k * q + b) ' ?A
  then obtain k where k-lt: k < N div q and n-eq: n = k * q + b
  by auto
  have k * q + b < (N div q) * q
proof -
  have sk-le: Suc k ≤ N div q
    using k-lt by simp
  have skq-le: Suc k * q ≤ (N div q) * q
    by (rule mult-le-mono1[OF sk-le])
  have k * q + b < k * q + q
    using b-lt by simp
  also have ... = Suc k * q
    by simp
  also have ... ≤ (N div q) * q

```

```

    by (rule skq-le)
    finally show ?thesis .
qed
then have  $n < N$ 
    using  $N\text{-eq } n\text{-eq}$  by simp
moreover have  $n \bmod q = b$ 
    using  $n\text{-eq } b\text{-lt}$  by simp
ultimately show  $n \in ?B$ 
    by simp
qed
show  $?B \subseteq (\lambda k. k * q + b) \text{ ` } ?A$ 
proof
  fix  $n$ 
  assume  $n\text{-in}: n \in ?B$ 
  then have  $n\text{-lt}: n < N$  and  $n\text{-mod}: n \bmod q = b$ 
    by simp-all
  have  $n\text{-eq}: n = (n \text{ div } q) * q + b$ 
    using  $\text{div-mult-mod-eq}[of\ n\ q]\ n\text{-mod}$  by simp
  have  $\text{div-lt}: n \text{ div } q < N \text{ div } q$ 
  proof (rule ccontr)
    assume  $\neg n \text{ div } q < N \text{ div } q$ 
    then have  $ge: N \text{ div } q \leq n \text{ div } q$ 
      by simp
    have  $N = (N \text{ div } q) * q$ 
      by (rule  $N\text{-eq}$ )
    also have  $\dots \leq (n \text{ div } q) * q$ 
      using  $ge$  by simp
    also have  $\dots \leq (n \text{ div } q) * q + b$ 
      by simp
    also have  $\dots = n$ 
      using  $n\text{-eq}$  by simp
    finally show  $False$ 
      using  $n\text{-lt}$  by simp
  qed
  then show  $n \in (\lambda k. k * q + b) \text{ ` } ?A$ 
    using  $n\text{-eq}$  by auto
qed
qed
have  $\text{inj}: \text{inj-on } (\lambda k. k * q + b) \text{ ` } ?A$ 
  using  $q\text{-pos}$  unfolding  $\text{inj-on-def}$  by auto
have  $\text{card } ?B = \text{card } ((\lambda k. k * q + b) \text{ ` } ?A)$ 
  using  $\text{image-eq}$  by simp
also have  $\dots = \text{card } ?A$ 
  by (rule  $\text{card-image}[OF\ \text{inj}]$ )
also have  $\dots = N \text{ div } q$ 
  by simp
finally show ?thesis .
qed

```


lemma *card-query-index-nat-preimage-uniform-range*:

assumes *range-eq*: $\text{range to-nat} = \{0..<\text{size}\}$
and *dvd*: $\text{query-sample-space-size} \text{ dvd } \text{size}$
and *subset*: $B \subseteq \text{query-sample-space}$
shows
 $\text{card } (\text{query-index-nat-preimage } B) =$
 $(\text{size div query-sample-space-size}) * \text{card } B$

proof –

let $?q = \text{query-sample-space-size}$
let $?A = \lambda b. \{n. n < \text{size} \wedge n \bmod ?q = b\}$
have *finite-space*: *finite query-sample-space*
unfolding *query-sample-space-def* **by** *simp*
have *finite-B*: *finite B*
by (*rule finite-subset[OF subset finite-space]*)
have *B-lt*: $\bigwedge b. b \in B \implies b < ?q$
using *subset* **unfolding** *query-sample-space-def* **by** *auto*
have *preimage-eq*: $\text{query-index-nat-preimage } B = \bigcup (?A \text{ ‘ } B)$
unfolding *query-index-nat-preimage-def range-eq* **by** *auto*
have *finite-A*: $\forall b \in B. \text{finite } (?A \text{ } b)$
by *simp*
have *disjoint*:
 $\forall b \in B. \forall c \in B. b \neq c \longrightarrow ?A \text{ } b \cap ?A \text{ } c = \{\}$
by *auto*
have $\text{card } (\text{query-index-nat-preimage } B) = \text{card } (\bigcup (?A \text{ ‘ } B))$
unfolding *preimage-eq ..*
also have $\dots = (\sum b \in B. \text{card } (?A \text{ } b))$
by (*rule card-UN-disjoint[OF finite-B finite-A disjoint]*)
also have $\dots = (\sum b \in B. \text{size div } ?q)$
by (*rule sum.cong*)
 $(\text{use } B\text{-lt in}$
 $\text{‹simp-all add: card-mod-lessThan-dvd[OF query-sample-space-size-pos dvd]›})$
also have $\dots = (\text{size div } ?q) * \text{card } B$
by *simp*
finally show *?thesis* .

qed

lemma *wp-receive-trace-fri-challenge-fresh-set*:

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$
assumes *fresh*:
 $\text{fmlookup } (\text{HashMap } s) (\text{TraceFriChallenge } (P\text{TraceFriCounter } s) (P\text{State } s)) = \text{None}$
shows
 $\text{wp-event receive-trace-fri-challenge}$
 $(\lambda \text{out. case out of None} \implies \text{False} \mid \text{Some } (y, -) \implies y \in B) \text{ } s =$
 $\text{nnreal } (\text{card } B) / \text{nnreal size}$
unfolding *receive-trace-fri-challenge-def*
by (*rule wp-receive-counted-tagged-random-field-element-fresh-set*
 $[\text{where counter} = P\text{TraceFriCounter}$
and $\text{bump} = \lambda s. s(P\text{TraceFriCounter} := \text{Suc } (P\text{TraceFriCounter } s))$
and $\text{tag} = \text{TraceFriChallenge}$ **and** $B = B$ **and** $s = s, \text{ OF fresh}]$)

```

lemma wp-receive-composition-fri-challenge-fresh-set:
  fixes  $s :: ('f, 'a) \text{ protocol-channel-scheme}$ 
  assumes fresh:
     $\text{fmlookup } (\text{HashMap } s) (\text{CompositionFriChallenge } (\text{PCompositionFriCounter } s) (\text{PState } s))$ 
   $= \text{None}$ 
  shows
     $\text{wp-event receive-composition-fri-challenge}$ 
     $(\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } (y, -) \Rightarrow y \in B) \ s =$ 
     $\text{nnreal } (\text{card } B) / \text{nnreal size}$ 
  unfolding receive-composition-fri-challenge-def
  by (rule wp-receive-counted-tagged-random-field-element-fresh-set
    [where  $\text{counter} = \text{PCompositionFriCounter}$ 
      and  $\text{bump} = \lambda s. s \llbracket \text{PCompositionFriCounter} := \text{Suc } (\text{PCompositionFriCounter } s) \rrbracket$ 
      and  $\text{tag} = \text{CompositionFriChallenge}$  and  $B = B$  and  $s = s$ , OF fresh])

fun merkle-path-root-in-map
   $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$ 
   $\text{nat} \Rightarrow \text{nat} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ option}$ 
  where
     $\text{merkle-path-root-in-map } s \text{ len idx leaf } [] =$ 
     $\text{fmlookup } (\text{HashMap } s) (\text{MerkleLeaf leaf})$ 
  |  $\text{merkle-path-root-in-map } s \text{ len idx leaf } (a \# \text{path}) =$ 
     $(\text{if } \text{idx} < \text{len div } 2$ 
       $\text{then case merkle-path-root-in-map } s (\text{len div } 2) \text{ idx leaf path of}$ 
         $\text{None} \Rightarrow \text{None}$ 
        |  $\text{Some } x \Rightarrow \text{fmlookup } (\text{HashMap } s) (\text{MerkleNode } x \ a)$ 
       $\text{else case merkle-path-root-in-map } s (\text{len div } 2) (\text{idx} - \text{len div } 2)$ 
         $\text{leaf path of}$ 
         $\text{None} \Rightarrow \text{None}$ 
        |  $\text{Some } x \Rightarrow \text{fmlookup } (\text{HashMap } s) (\text{MerkleNode } a \ x))$ 

lemma check-authentication-path-outcome-merkle-path-root:
  fixes  $\text{root leaf} :: 'f$ 
  and  $s \ t \ u :: ('f, 'a) \text{ protocol-channel-scheme}$ 
  assumes outcome:
     $\text{Some } (\text{root}, t) \in$ 
     $\text{set-dist } (\text{execute } (\text{check-authentication-path len idx leaf path}) \ s)$ 
  and ext:  $t \leq u$ 
  shows  $\text{merkle-path-root-in-map } u \text{ len idx leaf path} = \text{Some root}$ 
  using outcome ext
  unfolding check-authentication-path-def protocol-check-authentication-path-def
proof (induction path arbitrary: len idx leaf s root t)
  case Nil
  then have hash-step:
     $\text{Some } (\text{root}, t) \in \text{set-dist } (\text{execute } (\text{hash } (\text{MerkleLeaf leaf})) \ s)$ 
  by simp
  have lookup-t:
     $\text{fmlookup } (\text{HashMap } t) (\text{MerkleLeaf leaf}) = \text{Some root}$ 

```

```

    by (rule hash-outcome(2)[OF hash-step])
  show ?case
    using hash-extension-lookup[OF lookup-t Nil.prem(2)] by simp
next
case (Cons a path)
show ?case
proof (cases idx < len div 2)
  case True
  from Cons.prem(1) obtain child s1 where
    child-check:
      Some (child, s1) ∈
        set-dist (execute
          (protocol-merkle.check-authentication-path
            (len div 2) idx (MerkleLeaf leaf) path) s)
    and root-hash:
      Some (root, t) ∈
        set-dist (execute (hash (MerkleNode child a)) s1)
    using True by (auto elim!: set-dist-bindE)
  have s1-t: s1 ≤ t
    by (rule hash-outcome(1)[OF root-hash])
  have s1-u: s1 ≤ u
    using s1-t Cons.prem(2) by (meson hash-ext-trans)
  have child-root:
    merkle-path-root-in-map u (len div 2) idx leaf path = Some child
    by (rule Cons.IH[OF child-check s1-u])
  have lookup-t:
    fmllookup (HashMap t) (MerkleNode child a) = Some root
    by (rule hash-outcome(2)[OF root-hash])
  have lookup-u:
    fmllookup (HashMap u) (MerkleNode child a) = Some root
    by (rule hash-extension-lookup[OF lookup-t Cons.prem(2)])
  show ?thesis
    using True child-root lookup-u by simp
next
case False
  from Cons.prem(1) obtain child s1 where
    child-check:
      Some (child, s1) ∈
        set-dist (execute
          (protocol-merkle.check-authentication-path
            (len div 2) (idx - len div 2) (MerkleLeaf leaf) path) s)
    and root-hash:
      Some (root, t) ∈
        set-dist (execute (hash (MerkleNode a child)) s1)
    using False by (auto elim!: set-dist-bindE)
  have s1-t: s1 ≤ t
    by (rule hash-outcome(1)[OF root-hash])
  have s1-u: s1 ≤ u
    using s1-t Cons.prem(2) by (meson hash-ext-trans)

```

```

have child-root:
  merkle-path-root-in-map u (len div 2) (idx - len div 2) leaf path =
    Some child
  by (rule Cons.IH[OF child-check s1-u])
have lookup-t:
  fmlookup (HashMap t) (MerkleNode a child) = Some root
  by (rule hash-outcome(2)[OF root-hash])
have lookup-u:
  fmlookup (HashMap u) (MerkleNode a child) = Some root
  by (rule hash-extension-lookup[OF lookup-t Cons.prem(2)])
show ?thesis
  using False child-root lookup-u by simp
qed
qed

lemma merkle-path-root-same-index-eq-or-hash-collision:
  fixes root leaf leaf' :: 'f
  and s :: ('f, 'a) protocol-channel-scheme
  assumes root1:
    merkle-path-root-in-map s len idx leaf path = Some root
  and root2:
    merkle-path-root-in-map s len idx leaf' path' = Some root
  shows (leaf = leaf'  $\wedge$  path = path')  $\vee$  hash-map-output-collision s
  using root1 root2
proof (induction path arbitrary: len idx leaf path' leaf' root)
  case Nil
  show ?case
  proof (cases path')
  case Nil
  show ?thesis
  proof (cases leaf = leaf')
  case True
  then show ?thesis
    using Nil by simp
  next
  case False
  have lookup-leaf: fmlookup (HashMap s) (MerkleLeaf leaf) = Some root
    using Nil.prem(1) by simp
  have lookup-leaf':
    fmlookup (HashMap s) (MerkleLeaf leaf') = Some root
    using Nil.prem(2) unfolding Nil by simp
  have hash-map-output-collision s
    by (rule hash-map-output-collision-merkle-leafI
      [OF False lookup-leaf lookup-leaf'])
  then show ?thesis by simp
  qed
next
  case (Cons a rest)
  have lookup-leaf: fmlookup (HashMap s) (MerkleLeaf leaf) = Some root

```

```

    using Nil.premis(1) by simp
have hash-map-output-collision s
proof (cases idx < len div 2)
  case True
  from Nil.premis(2)[unfolded Cons] True obtain child where
    lookup-node: fmlookup (HashMap s) (MerkleNode child a) = Some root
    by (auto split: option.splits)
  show ?thesis
    by (rule hash-map-output-collision-merkle-leaf-nodeI
        [OF lookup-leaf lookup-node])
  next
  case False
  from Nil.premis(2)[unfolded Cons] False obtain child where
    lookup-node: fmlookup (HashMap s) (MerkleNode a child) = Some root
    by (auto split: option.splits)
  show ?thesis
    by (rule hash-map-output-collision-merkle-leaf-nodeI
        [OF lookup-leaf lookup-node])
qed
then show ?thesis by simp
qed
next
case (Cons a path)
show ?case
proof (cases path')
  case Nil
  have lookup-leaf': fmlookup (HashMap s) (MerkleLeaf leaf') = Some root
    using Cons.premis(2) unfolding Nil by simp
  have hash-map-output-collision s
  proof (cases idx < len div 2)
    case True
    from Cons.premis(1) True obtain child where
      lookup-node: fmlookup (HashMap s) (MerkleNode child a) = Some root
      by (auto split: option.splits)
    show ?thesis
      by (rule hash-map-output-collisionI
          [of MerkleNode child a MerkleLeaf leaf' s root])
        (use lookup-node lookup-leaf' in simp-all)
    next
    case False
    from Cons.premis(1) False obtain child where
      lookup-node: fmlookup (HashMap s) (MerkleNode a child) = Some root
      by (auto split: option.splits)
    show ?thesis
      by (rule hash-map-output-collisionI
          [of MerkleNode a child MerkleLeaf leaf' s root])
        (use lookup-node lookup-leaf' in simp-all)
  qed
qed
then show ?thesis by simp

```

```

next
case (Cons b path'')
show ?thesis
proof (cases idx < len div 2)
case True
from Cons.premis(1) obtain child where
child1:
merkle-path-root-in-map s (len div 2) idx leaf path =
Some child
and lookup1:
fmlookup (HashMap s) (MerkleNode child a) = Some root
using True by (auto split: option.splits)
from Cons.premis(2)[unfolded Cons] obtain child' where
child2:
merkle-path-root-in-map s (len div 2) idx leaf' path'' =
Some child'
and lookup2:
fmlookup (HashMap s) (MerkleNode child' b) = Some root
using True by (auto split: option.splits)
show ?thesis
proof (cases (child, a) = (child', b))
case False
have hash-map-output-collision s
by (rule hash-map-output-collision-merkle-nodeI
[OF False lookup1 lookup2])
then show ?thesis by simp
next
case pair-eq: True
have child-eq: child = child' and a-eq: a = b
using pair-eq by simp-all
have rec:
(leaf = leaf'  $\wedge$  path = path'')  $\vee$  hash-map-output-collision s
by (rule Cons.IH[OF child1 child2[unfolded child-eq[symmetric]]])
then show ?thesis
using a-eq Cons by auto
qed
next
case False
from Cons.premis(1) obtain child where
child1:
merkle-path-root-in-map s (len div 2) (idx - len div 2)
leaf path = Some child
and lookup1:
fmlookup (HashMap s) (MerkleNode a child) = Some root
using False by (auto split: option.splits)
from Cons.premis(2)[unfolded Cons] obtain child' where
child2:
merkle-path-root-in-map s (len div 2) (idx - len div 2)
leaf' path'' = Some child'

```

```

    and lookup2:
      fmlookup (HashMap s) (MerkleNode b child') = Some root
    using False by (auto split: option.splits)
  show ?thesis
  proof (cases (a, child) = (b, child'))
    case False
    have hash-map-output-collision s
    by (rule hash-map-output-collision-merkle-nodeI
      [OF False lookup1 lookup2])
    then show ?thesis by simp
  next
    case pair-eq: True
    have child-eq: child = child' and a-eq: a = b
    using pair-eq by simp-all
    have rec:
      (leaf = leaf'  $\wedge$  path = path'')  $\vee$  hash-map-output-collision s
    by (rule Cons.IH[OF child1 child2[unfolded child-eq[symmetric]]])
    then show ?thesis
  using a-eq Cons by auto
qed
qed
qed
qed

```

lemma *protocol-created-tree-singletonD*:

```

  assumes protocol-created-tree [x] t s
  obtains h where
    t = ⟨⟨⟩, h, ⟨⟩⟩
    fmlookup (HashMap s) (MerkleLeaf x) = Some h
  using assms
  unfolding protocol-created-tree-def
  by (auto simp: protocol-merkle.created-tree.simps)

```

lemma *protocol-created-tree-internalD*:

```

  assumes created: protocol-created-tree xs t s
  and len: 2 ≤ length xs
  obtains l h r i where
    i = length xs div 2
    t = ⟨l, h, r⟩
    protocol-created-tree (take i xs) l s
    protocol-created-tree (drop i xs) r s
    fmlookup (HashMap s) (MerkleNode (value l) (value r)) = Some h
  proof -
    obtain a b cs where xs-eq: xs = a # b # cs
    using len by (cases xs; cases tl xs) auto
    obtain l h r where
      t: t = ⟨l, h, r⟩
    and left:
      protocol-merkle.created-tree

```

```

    (take (length xs div 2) (map MerkleLeaf xs)) l s
  and right:
    protocol-merkle.created-tree
    (drop (length xs div 2) (map MerkleLeaf xs)) r s
  and root:
    fmllookup (HashMap s) (MerkleNode (value l) (value r)) = Some h
  using created len
  unfolding protocol-created-tree-def
  by (auto simp: xs-eq protocol-merkle.created-tree.simps Let-def)
have left':
  protocol-created-tree (take (length xs div 2) xs) l s
  using left
  unfolding protocol-created-tree-def by (simp add: take-map)
have right':
  protocol-created-tree (drop (length xs div 2) xs) r s
  using right
  unfolding protocol-created-tree-def by (simp add: drop-map)
show ?thesis
  by (rule that[OF refl t left' right' root])
qed

lemma protocol-created-tree-path-root-in-map:
  assumes created: protocol-created-tree xs tree s
  and len-pow: length xs = 2 ^ n
  and i-bound: i < length xs
  shows
    merkle-path-root-in-map s (length xs) i (xs ! i)
    (get-authentication-path (length xs) i tree) = Some (value tree)
  using created len-pow i-bound
proof (induction n arbitrary: xs tree i)
  case 0
  then have len-xs: length xs = 1
    by simp
  then obtain x where xs-eq: xs = [x]
    by (cases xs) auto
  from protocol-created-tree-singletonD[OF 0.prem1][unfolded xs-eq]
  obtain h where tree-eq:
    tree = (⟨⟩, h, ⟨⟩)
    and lookup: fmllookup (HashMap s) (MerkleLeaf x) = Some h
  by blast
  have i-eq: i = 0
    using 0.prem3 unfolding xs-eq by simp
  show ?case
    using lookup unfolding xs-eq tree-eq i-eq by simp
next
  case (Suc n)
  let ?mid = length xs div 2
  have len-xs: length xs = 2 ^ Suc n
    by (rule Suc.prem2)

```



```

have len-ge:  $2 \leq \text{length } xs$ 
  using len-xs by simp
have len-not-one:  $\text{length } xs \neq 1$ 
  using len-ge by simp
obtain l h r mid where mid-def:  $mid = ?mid$ 
  and tree-eq:  $tree = \langle l, h, r \rangle$ 
  and left:  $\text{protocol-created-tree } (take\ mid\ xs)\ l\ s$ 
  and right:  $\text{protocol-created-tree } (drop\ mid\ xs)\ r\ s$ 
  and root-lookup:
    fmlookup (HashMap s) (MerkleNode (value l) (value r)) = Some h
  by (rule protocol-created-tree-internalD[OF Suc.prem1 len-ge])
have mid-pow:  $mid = 2^n$ 
  using len-xs unfolding mid-def by simp
have mid-pos:  $0 < mid$ 
  using mid-pow by simp
have mid-lt:  $mid < \text{length } xs$ 
  using len-xs mid-pow by simp
have take-len:  $\text{length } (take\ mid\ xs) = 2^n$ 
  using mid-lt mid-pow by simp
have drop-len:  $\text{length } (drop\ mid\ xs) = 2^n$ 
  using len-xs mid-pow by simp
show ?case
proof (cases  $i < mid$ )
case True
  have idx-take:  $i < \text{length } (take\ mid\ xs)$ 
    using True take-len mid-pow by simp
  have child:
    merkle-path-root-in-map s (length (take mid xs)) i
      (take mid xs ! i)
      (get-authentication-path (length (take mid xs)) i l) =
        Some (value l)
    by (rule Suc.IH[OF left take-len idx-take])
  have child-mid:
    merkle-path-root-in-map s mid i (xs ! i)
      (get-authentication-path mid i l) = Some (value l)
    using child True take-len mid-pow by simp
  show ?thesis
    using True child-mid root-lookup len-not-one
    unfolding tree-eq mid-def by simp
next
case False
  have idx-drop:  $i - mid < \text{length } (drop\ mid\ xs)$ 
    using False Suc.prem3 drop-len mid-pow mid-def by simp
  have child:
    merkle-path-root-in-map s (length (drop mid xs)) (i - mid)
      (drop mid xs ! (i - mid))
      (get-authentication-path (length (drop mid xs)) (i - mid) r) =
        Some (value r)
    by (rule Suc.IH[OF right drop-len idx-drop])

```

```

have child-mid:
  merkle-path-root-in-map s mid (i - mid) (xs ! i)
    (get-authentication-path mid (i - mid) r) = Some (value r)
  using child False Suc.premis(3) drop-len mid-pow by simp
show ?thesis
  using False child-mid root-lookup len-not-one
  unfolding tree-eq mid-def by simp
qed
qed

lemma protocol-created-tree-merkle-mono:
  assumes created: protocol-created-tree xs t s
  and ext: merkle-hash-extends s u
  shows protocol-created-tree xs t u
  using created
proof (induction xs arbitrary: t rule: length-induct)
  case (1 xs)
  show ?case
  proof (cases xs = [])
    case True
    then show ?thesis
      using 1.premis
      unfolding protocol-created-tree-def
      by (simp add: protocol-merkle.created-tree.simps)
  next
  case xs-nonempty: False
  show ?thesis
  proof (cases length xs = 1)
    case True
    then obtain x where xs-eq: xs = [x]
      using xs-nonempty by (cases xs) auto
    from protocol-created-tree-singletonD[OF 1.premis[unfolded xs-eq]]
    obtain h where
      t-eq: t = ⟨⟨⟩, h, ⟨⟩⟩
      and lookup-s: fmllookup (HashMap s) (MerkleLeaf x) = Some h
      by blast
    have lookup-u: fmllookup (HashMap u) (MerkleLeaf x) = Some h
      using ext lookup-s unfolding merkle-hash-extends-def by simp
    show ?thesis
      unfolding protocol-created-tree-def xs-eq t-eq
      using lookup-u by (simp add: protocol-merkle.created-tree.simps)
  next
  case not-singleton: False
  have len-ge: 2 ≤ length xs
    using xs-nonempty not-singleton by (cases xs; cases tl xs) auto
  obtain l h r i where
    i-def: i = length xs div 2
    and t-eq: t = ⟨l, h, r⟩
    and left-created: protocol-created-tree (take i xs) l s

```

```

and right-created: protocol-created-tree (drop i xs) r s
and root-lookup:
  fmlookup (HashMap s) (MerkleNode (value l) (value r)) = Some h
  by (rule protocol-created-tree-internalD[OF 1.prems len-ge])
have i-pos:  $0 < i$ 
  using i-def len-ge by simp
have i-lt:  $i < \text{length } xs$ 
  using i-def len-ge by simp
have take-less:  $\text{length } (\text{take } i \text{ } xs) < \text{length } xs$ 
  using i-lt by simp
have drop-less:  $\text{length } (\text{drop } i \text{ } xs) < \text{length } xs$ 
  using i-pos i-lt by simp
have left-u: protocol-created-tree (take i xs) l u
  by (rule 1.IH[rule-format, OF take-less left-created])
have right-u: protocol-created-tree (drop i xs) r u
  by (rule 1.IH[rule-format, OF drop-less right-created])
have root-lookup-u:
  fmlookup (HashMap u) (MerkleNode (value l) (value r)) = Some h
  using ext root-lookup unfolding merkle-hash-extends-def by simp
obtain a b cs where xs-eq:  $xs = a \# b \# cs$ 
  using len-ge by (cases xs; cases tl xs) auto
have take-leaf-eq:
  MerkleLeaf a #
    take ( $\text{length } cs \text{ div } 2$ ) (MerkleLeaf b # map MerkleLeaf cs) =
  MerkleLeaf a #
    map MerkleLeaf (take ( $\text{length } cs \text{ div } 2$ ) (b # cs))
  by (metis list.map(2) take-map)
have drop-leaf-eq:
  drop ( $\text{length } cs \text{ div } 2$ ) (MerkleLeaf b # map MerkleLeaf cs) =
  map MerkleLeaf (drop ( $\text{length } cs \text{ div } 2$ ) (b # cs))
  by (metis list.map(2) drop-map)
have left-u-raw:
  protocol-merkle.created-tree
    (MerkleLeaf a #
      take ( $\text{length } cs \text{ div } 2$ ) (MerkleLeaf b # map MerkleLeaf cs))
    l u
  using left-u
  unfolding protocol-created-tree-def i-def xs-eq
  by (simp add: take-leaf-eq)
have right-u-raw:
  protocol-merkle.created-tree
    (drop ( $\text{length } cs \text{ div } 2$ ) (MerkleLeaf b # map MerkleLeaf cs))
    r u
  using right-u
  unfolding protocol-created-tree-def i-def xs-eq
  by (simp add: drop-leaf-eq)
show ?thesis
  using left-u-raw right-u-raw root-lookup-u
  unfolding protocol-created-tree-def t-eq xs-eq

```

```

      by (auto simp: protocol-merkle.created-tree.simps Let-def)
    qed
  qed
qed

lemma created-tree-merkle-mono:
  assumes created: created-tree xs t s
    and ext: merkle-hash-extends s u
  shows created-tree xs t u
  using protocol-created-tree-merkle-mono
    [OF created[unfolded created-tree-def] ext]
  unfolding created-tree-def .

lemma merkle-root-binds-table-merkle-mono:
  assumes bind: merkle-root-binds-table rt table s
    and ext: merkle-hash-extends s u
  shows merkle-root-binds-table rt table u
proof -
  from bind obtain tree where
    created: created-tree table tree s
    and root: rt = value tree
    unfolding merkle-root-binds-table-def by blast
  have created-tree table tree u
    by (rule created-tree-merkle-mono[OF created ext])
  then show ?thesis
    unfolding merkle-root-binds-table-def
    using root by blast
qed

lemma protocol-created-tree-same-length-root-eq-or-hash-collision:
  assumes left-created: protocol-created-tree xs tx s
    and right-created: protocol-created-tree ys ty s
    and same-len: length xs = length ys
    and nonempty-xs: xs ≠ []
    and nonempty-ys: ys ≠ []
    and same-root: value tx = value ty
  shows xs = ys ∨ hash-map-output-collision s
  using assms
proof (induction xs arbitrary: ys tx ty rule: length-induct)
  case (1 xs)
  show ?case
  proof (cases length xs = 1)
    case True
    then obtain x where xs-eq: xs = [x]
      using 1.prem(4) by (cases xs) auto
    from True 1.prem(3,5) obtain y where ys-eq: ys = [y]
      by (cases ys) auto
    from protocol-created-tree-singletonD[OF 1.prem(1)[unfolded xs-eq]]
    obtain hx where tx: tx = ⟨⟨⟩, hx, ⟨⟩⟩

```

```

    and hx-lookup: fmlookup (HashMap s) (MerkleLeaf x) = Some hx
  by blast
from protocol-created-tree-singletonD[OF 1.premis(2)[unfolded ys-eq]]
obtain hy where ty: ty = ⟨⟨⟩, hy, ⟨⟩⟩
  and hy-lookup: fmlookup (HashMap s) (MerkleLeaf y) = Some hy
  by blast
have h-eq: hx = hy
  using 1.premis(6) unfolding tx ty by simp
show ?thesis
proof (cases x = y)
  case True
  then show ?thesis
    using xs-eq ys-eq by simp
next
  case False
  have hy-lookup': fmlookup (HashMap s) (MerkleLeaf y) = Some hx
    using hy-lookup h-eq by simp
  then have hash-map-output-collision s
    by (rule hash-map-output-collision-merkle-leafI[OF False hx-lookup])
  then show ?thesis by simp
qed
next
case False
have 0 < length xs
  using 1.premis(4) by simp
have len-ge: 2 ≤ length xs
  using False ⟨0 < length xs⟩ by linarith
have len-ge-y: 2 ≤ length ys
  using len-ge 1.premis(3) by simp
obtain lx hx rx i where i-def: i = length xs div 2
  and tx: tx = ⟨lx, hx, rx⟩
  and lx-created: protocol-created-tree (take i xs) lx s
  and rx-created: protocol-created-tree (drop i xs) rx s
  and hx-lookup:
    fmlookup (HashMap s) (MerkleNode (value lx) (value rx)) = Some hx
  by (rule protocol-created-tree-internalD[OF 1.premis(1) len-ge])
obtain ly hy ry j where j-def: j = length ys div 2
  and ty: ty = ⟨ly, hy, ry⟩
  and ly-created: protocol-created-tree (take j ys) ly s
  and ry-created: protocol-created-tree (drop j ys) ry s
  and hy-lookup:
    fmlookup (HashMap s) (MerkleNode (value ly) (value ry)) = Some hy
  by (rule protocol-created-tree-internalD[OF 1.premis(2) len-ge-y])
have j-i: j = i
  using i-def j-def 1.premis(3) by simp
have ly-created-i: protocol-created-tree (take i ys) ly s
  using ly-created j-i by simp
have ry-created-i: protocol-created-tree (drop i ys) ry s
  using ry-created j-i by simp

```

```

have h-eq: hx = hy
  using 1.premis(6) unfolding tx ty by simp
show ?thesis
proof (cases (value lx, value rx) = (value ly, value ry))
  case False
  have hy-lookup':
    fmlookup (HashMap s) (MerkleNode (value ly) (value ry)) = Some hx
    using hy-lookup h-eq by simp
  then have hash-map-output-collision s
    by (rule hash-map-output-collision-merkle-nodeI[OF False hx-lookup])
  then show ?thesis by simp
next
case True
then have left-root: value lx = value ly
  and right-root: value rx = value ry
  by simp-all
have i-pos: 0 < i
  using i-def len-ge by simp
have i-lt: i < length xs
  using i-def len-ge by simp
have i-lt-y: i < length ys
  using i-lt 1.premis(3) by simp
have length-take-xs: length (take i xs) = i
  using i-lt by simp
have length-take-ys: length (take i ys) = i
  using i-lt-y by simp
have left-len: length (take i xs) = length (take i ys)
  using i-def j-i 1.premis(3) by simp
have right-len: length (drop i xs) = length (drop i ys)
  using i-def j-i 1.premis(3) by simp
have left-nonempty-xs: take i xs ≠ []
  using i-pos length-take-xs by auto
have left-nonempty-ys: take i ys ≠ []
  using i-pos length-take-ys by auto
have right-nonempty-xs: drop i xs ≠ []
  using i-lt by simp
have right-nonempty-ys: drop i ys ≠ []
  using i-lt-y by simp
have take-less: length (take i xs) < length xs
  using i-lt by simp
have drop-less: length (drop i xs) < length xs
  using i-pos i-lt by simp
show ?thesis
proof (cases take i xs = take i ys)
  case False
  have take i xs = take i ys ∨ hash-map-output-collision s
    by (rule 1.IH[rule-format, OF take-less lx-created ly-created-i
      left-len left-nonempty-xs left-nonempty-ys left-root])
  then show ?thesis

```

```

    using False by simp
next
case left-eq: True
show ?thesis
proof (cases drop i xs = drop i ys)
case True
then have xs = ys
  using left-eq i-def j-i 1.prems(3)
  by (metis append-take-drop-id)
then show ?thesis by simp
next
case False
have drop i xs = drop i ys  $\vee$  hash-map-output-collision s
  by (rule 1.IH[rule-format, OF drop-less rx-created ry-created-i
    right-len right-nonempty-xs right-nonempty-ys right-root])
then show ?thesis
  using False by simp
qed
qed
qed
qed
qed
qed

lemma merkle-root-binds-same-length-tables-collision-imp-hash-collision:
  assumes bind1: merkle-root-binds-table rt table s
    and bind2: merkle-root-binds-table rt table' s
    and same-len: length table = length table'
    and nonempty: table  $\neq$  []
    and neq: table  $\neq$  table'
  shows hash-map-output-collision s
proof -
  from bind1 obtain tree where tree:
    created-tree table tree s rt = value tree
  unfolding merkle-root-binds-table-def by blast
  from bind2 obtain tree' where tree':
    created-tree table' tree' s rt = value tree'
  unfolding merkle-root-binds-table-def by blast
  have nonempty': table'  $\neq$  []
    using same-len nonempty by auto
  have same-root: value tree = value tree'
    using tree tree' by simp
  have table = table'  $\vee$  hash-map-output-collision s
    by (rule protocol-created-tree-same-length-root-eq-or-hash-collision
      [OF tree(1)[unfolded created-tree-def]
        tree'(1)[unfolded created-tree-def]
          same-len nonempty nonempty' same-root])
  then show ?thesis
    using neq by simp
qed

```

lemma *merkle-root-binds-same-length-tables-unique-if-no-hash-collision:*
assumes *bind1*: merkle-root-binds-table *rt* *table* *s*
and *bind2*: merkle-root-binds-table *rt* *table'* *s*
and *same-len*: *length table* = *length table'*
and *clean*: $\neg$ *hash-map-output-collision s*
shows *table* = *table'*
proof (*cases table* = [])
case *True*
then show ?thesis
using *same-len* **by** *simp*
next
case *False*
show ?thesis
proof (*rule ccontr*)
assume *neq*: *table* $\neq$ *table'*
have *hash-map-output-collision s*
by (*rule merkle-root-binds-same-length-tables-collision-imp-hash-collision*
[*OF bind1 bind2 same-len False neq*])
then show *False*
using *clean* **by** *contradiction*
qed
qed

lemma *merkle-root-binds-same-length-tables-common-extension-collision:*
assumes *bind1*: merkle-root-binds-table *rt* *table* *s1*
and *bind2*: merkle-root-binds-table *rt* *table'* *s2*
and *ext1*: *s1* $\leq$ *u*
and *ext2*: *s2* $\leq$ *u*
and *same-len*: *length table* = *length table'*
and *nonempty*: *table* $\neq$ []
and *neq*: *table* $\neq$ *table'*
shows *hash-map-output-collision u*
proof –
have *bind1-u*: merkle-root-binds-table *rt* *table* *u*
by (*rule merkle-root-binds-table-mono*[*OF bind1 ext1*])
have *bind2-u*: merkle-root-binds-table *rt* *table'* *u*
by (*rule merkle-root-binds-table-mono*[*OF bind2 ext2*])
show ?thesis
by (*rule merkle-root-binds-same-length-tables-collision-imp-hash-collision*
[*OF bind1-u bind2-u same-len nonempty neq*])
qed

lemma *merkle-root-binds-same-length-tables-common-extension-unique-if-clean:*
assumes *clean*: $\neg$ *hash-map-output-collision u*
and *bind1*: merkle-root-binds-table *rt* *table* *s1*
and *bind2*: merkle-root-binds-table *rt* *table'* *s2*
and *ext1*: *s1* $\leq$ *u*
and *ext2*: *s2* $\leq$ *u*


```

    and same-len: length table = length table'
    and nonempty: table ≠ []
  shows table = table'
proof (rule ccontr)
  assume table ≠ table'
  then have hash-map-output-collision u
    by (rule merkle-root-binds-same-length-tables-common-extension-collision
        [OF bind1 bind2 ext1 ext2 same-len nonempty])
  then show False
    using clean by contradiction
qed

lemma merkle-root-binds-same-length-tables-compatible-states-unique-if-clean-merge:
  assumes compatible: hash-maps-compatible s1 s2
    and clean: ¬ hash-map-output-collision (hash-state-merge s1 s2)
    and bind1: merkle-root-binds-table rt table s1
    and bind2: merkle-root-binds-table rt table' s2
    and same-len: length table = length table'
    and nonempty: table ≠ []
  shows table = table'
proof -
  have ext1: s1 ≤ hash-state-merge s1 s2
    by (rule hash-state-merge-extends-left-if-compatible[OF compatible])
  have ext2: s2 ≤ hash-state-merge s1 s2
    by (rule hash-state-merge-extends-right)
  show ?thesis
    by (rule merkle-root-binds-same-length-tables-common-extension-unique-if-clean
        [OF clean bind1 bind2 ext1 ext2 same-len nonempty])
qed

lemma merkle-root-binds-same-length-tables-merkle-compatible-unique-if-clean-merge:
  assumes compatible: merkle-hash-maps-compatible s1 s2
    and clean:
      ¬ hash-map-output-collision (merkle-hash-state-merge s1 s2)
    and bind1: merkle-root-binds-table rt table s1
    and bind2: merkle-root-binds-table rt table' s2
    and same-len: length table = length table'
    and nonempty: table ≠ []
  shows table = table'
proof (rule ccontr)
  let ?u = merkle-hash-state-merge s1 s2
  assume neg: table ≠ table'
  have ext1: merkle-hash-extends s1 ?u
    by (rule merkle-hash-state-merge-extends-left-if-compatible[OF compatible])
  have ext2: merkle-hash-extends s2 ?u
    by (rule merkle-hash-state-merge-extends-right)
  have bind1-u: merkle-root-binds-table rt table ?u
    by (rule merkle-root-binds-table-merkle-mono[OF bind1 ext1])
  have bind2-u: merkle-root-binds-table rt table' ?u

```

```

    by (rule merkle-root-binds-table-merkle-mono[OF bind2 ext2])
  have hash-map-output-collision ?u
    by (rule merkle-root-binds-same-length-tables-collision-imp-hash-collision
        [OF bind1-u bind2-u same-len nonempty neq])
  then show False
    using clean by contradiction
qed

```

```

definition state-after-query-chunks :: 'f  $\Rightarrow$  'f list list  $\Rightarrow$  nat  $\Rightarrow$  'f
  where
    state-after-query-chunks st chunks i =
      foldl ( $\lambda$ acc chunk. foldl concat acc chunk) st (take i chunks)

```

lemma concat-append-eq-concat-same-chunk-lengths:

```

assumes len: length xs = length ys
and lens:
   $\bigwedge i. i < \text{length } xs \implies \text{length } (xs ! i) = \text{length } (ys ! i)$ 
and concat-eq: List.concat xs @ trailing = List.concat ys
shows xs = ys  $\wedge$  trailing = []
using len lens concat-eq
proof (induction xs arbitrary: ys trailing)
  case Nil
  then show ?case
    by (cases ys) simp-all
next
  case (Cons x xs)
  then obtain y ys' where ys-eq: ys = y # ys'
  proof (cases ys)
    case Nil
    then show ?thesis
      using Cons.prem1 by simp
  next
    case (Cons y ys')
    then show ?thesis
      by (rule that)
  qed

```

```

have head-len: length x = length y
  using Cons.prem2[of 0] unfolding ys-eq by simp
have append-eq:
  x @ (List.concat xs @ trailing) = y @ List.concat ys'
  using Cons.prem3 unfolding ys-eq by simp
have head-eq: x = y
  using append-eq head-len
  by (metis append-eq-conv-conj)
have tail-concat-eq:
  List.concat xs @ trailing = List.concat ys'
  using append-eq head-eq by simp
have tail-len: length xs = length ys'
  using Cons.prem1 unfolding ys-eq by simp

```

```

have tail-lens:
   $\bigwedge i. i < \text{length } xs \implies \text{length } (xs ! i) = \text{length } (ys' ! i)$ 
proof –
  fix i
  assume i-bound:  $i < \text{length } xs$ 
  then have  $\text{Suc } i < \text{length } (x \# xs)$ 
    by simp
  from Cons.prems(2)[OF this] show
     $\text{length } (xs ! i) = \text{length } (ys' ! i)$ 
    unfolding ys-eq by simp
qed
have  $xs = ys' \wedge \text{trailing} = []$ 
  by (rule Cons.IH[OF tail-len tail-lens tail-concat-eq])
then show ?case
  using head-eq unfolding ys-eq by simp
qed

fun fri-query-transcript-length ::  $\text{nat} \Rightarrow \text{nat} \Rightarrow \text{nat}$ 
  where
    fri-query-transcript-length 0 len = 0
    | fri-query-transcript-length (Suc n) len =
      2 + 2 * floor-log len + fri-query-transcript-length n (len div 2)

definition query-decommitment-transcript-length ::  $\text{nat} \Rightarrow \text{nat}$ 
  where
    query-decommitment-transcript-length idx =
       $\text{length } (\text{powers-scaled } idx) * (1 + \text{floor-log } (\text{clength} * \text{scale}))$ 

definition query-leaf-path-chunk ::  $\text{nat} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$ 
  where
    query-leaf-path-chunk len leaf path chunk  $\longleftrightarrow$ 
       $\text{length } path = \text{floor-log } len \wedge$ 
       $chunk = [leaf] @ path$ 

definition query-decommitment-transcript
  ::  $\text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$ 
  where
    query-decommitment-transcript idx leaves paths chunk  $\longleftrightarrow$ 
       $\text{length } leaves = \text{length } (\text{powers-scaled } idx) \wedge$ 
       $\text{length } paths = \text{length } (\text{powers-scaled } idx) \wedge$ 
       $chunk = \text{List.concat}$ 
        ( $\text{map } (\lambda(leaf, path). [leaf] @ path) (\text{zip } leaves paths)$ )  $\wedge$ 
       $(\forall path \in \text{set } paths. \text{length } path = \text{floor-log } (\text{clength} * \text{scale}))$ 

lemma query-decommitment-transcript-length:
  assumes query-decommitment-transcript idx leaves paths chunk
  shows  $\text{length } chunk = \text{query-decommitment-transcript-length } idx$ 
proof –

```

```

have leaves-len: length leaves = length (powers-scaled idx)
  using assms unfolding query-decommitment-transcript-def by simp
have paths-len: length paths = length (powers-scaled idx)
  using assms unfolding query-decommitment-transcript-def by simp
have leaves-paths-len: length leaves = length paths
  using leaves-len paths-len by simp
have map2-eq:
  map2 ( $\lambda$ leaf path. [leaf] @ path) leaves paths =
    map2 ( $\lambda$ leaf path. leaf # path) leaves paths
proof (induction leaves arbitrary: paths)
  case Nil
  then show ?case
    by (cases paths) simp-all
next
  case (Cons leaf leaves)
  then show ?case
    by (cases paths) simp-all
qed
have chunk-eq:
  chunk =
    List.concat (map2 ( $\lambda$ leaf path. leaf # path) leaves paths)
  using assms map2-eq unfolding query-decommitment-transcript-def
  by simp
have path-lens:
   $\bigwedge$ path. path  $\in$  set paths  $\implies$ 
    length path = floor-log (clength * scale)
  using assms unfolding query-decommitment-transcript-def by blast
have concat-len:
  length (List.concat (map2 ( $\lambda$ leaf path. leaf # path)
    leaves paths)) =
    length leaves * (1 + floor-log (clength * scale))
  using leaves-paths-len path-lens
proof (induction leaves arbitrary: paths)
  case Nil
  then show ?case
    by (cases paths) simp-all
next
  case (Cons leaf leaves)
  then obtain path paths' where paths-eq: paths = path # paths'
    by (cases paths) auto
  have len-rest: length leaves = length paths'
    using Cons.prem1 unfolding paths-eq by simp
  have path-len: length path = floor-log (clength * scale)
    using Cons.prem2 unfolding paths-eq by simp
  have paths'-len:
     $\bigwedge$ p. p  $\in$  set paths'  $\implies$ 
      length p = floor-log (clength * scale)
    using Cons.prem2 unfolding paths-eq by simp
  have rest:

```

```

    length (List.concat (map2 (λleaf path. leaf # path)
      leaves paths')) =
    length leaves * (1 + floor-log (clength * scale))
  by (rule Cons.IH[OF len-rest paths'-lens])
show ?case
  unfolding paths-eq using path-len rest by simp
qed
have length chunk =
  length leaves * (1 + floor-log (clength * scale))
  unfolding chunk-eq by (rule concat-len)
also have ... =
  length (powers-scaled idx) * (1 + floor-log (clength * scale))
  using leaves-len by simp
finally show ?thesis
  unfolding query-decommitment-transcript-length-def .
qed

definition fri-layer-opening-chunk
  :: nat ⇒ 'f ⇒ 'f list ⇒ 'f ⇒ 'f list ⇒ 'f list ⇒ bool
  where
    fri-layer-opening-chunk len xp xp-path xn xn-path chunk ⟷
      length xp-path = floor-log len ∧
      length xn-path = floor-log len ∧
      chunk = [xp] @ xp-path @ [xn] @ xn-path

lemma fri-layer-opening-chunk-length:
  assumes fri-layer-opening-chunk len xp xp-path xn xn-path chunk
  shows length chunk = 2 + 2 * floor-log len
  using assms unfolding fri-layer-opening-chunk-def by simp

fun fri-layers-transcript-length :: nat ⇒ nat ⇒ nat
  where
    fri-layers-transcript-length 0 len = 0
  | fri-layers-transcript-length (Suc n) len =
    2 + 2 * floor-log len + fri-layers-transcript-length n (len div 2)

fun fri-layer-lengths :: nat ⇒ nat ⇒ nat list
  where
    fri-layer-lengths 0 len = []
  | fri-layer-lengths (Suc n) len = len # fri-layer-lengths n (len div 2)

fun fri-layer-indices :: nat ⇒ nat ⇒ nat ⇒ nat list
  where
    fri-layer-indices 0 idx len = []
  | fri-layer-indices (Suc n) idx len =
    idx # fri-layer-indices n (idx mod (len div 2)) (len div 2)

definition fri-sibling-index :: nat ⇒ nat ⇒ nat
  where fri-sibling-index len idx = (idx + len div 2) mod len

```

definition *fri-fold-denominator* :: 'f $\Rightarrow$ nat $\Rightarrow$ 'f
where *fri-fold-denominator* x pw = 2 * x $\wedge$ pw

definition *fri-fold-value* :: 'f $\Rightarrow$ 'f $\Rightarrow$ 'f $\Rightarrow$ 'f $\Rightarrow$ 'f
where
fri-fold-value b xp xn denom =
 (xp + xn) div 2 + b * ((xp - xn) div denom)

lemma *length-fri-layer-lengths[simp]*:
length (*fri-layer-lengths* n len) = n
by (*induction* n *arbitrary*: len) *simp-all*

lemma *length-fri-layer-indices[simp]*:
length (*fri-layer-indices* n idx len) = n
by (*induction* n *arbitrary*: idx len) *simp-all*

definition *fri-layers-transcript*
 :: nat $\Rightarrow$ nat $\Rightarrow$ 'f list list $\Rightarrow$ 'f list $\Rightarrow$ bool
where
fri-layers-transcript n len layer-chunks chunk $\longleftrightarrow$
length layer-chunks = n $\wedge$
 chunk = List.concat layer-chunks $\wedge$
 ($\forall$ (layer-len, layer-chunk) $\in$ set (zip (*fri-layer-lengths* n len) layer-chunks).
 $\exists$ xp xp-path xn xn-path.
fri-layer-opening-chunk layer-len xp xp-path xn xn-path layer-chunk)

lemma *fri-layers-transcript-length*:
assumes *fri-layers-transcript* n len layer-chunks chunk
shows *length* chunk = *fri-layers-transcript-length* n len
using *assms*

proof (*induction* n *arbitrary*: len layer-chunks chunk)
case 0
then show ?case
unfolding *fri-layers-transcript-def* **by** *simp*

next
case (Suc n)
from Suc.premis **have** len-chunks:
length layer-chunks = Suc n
and chunk-eq: chunk = List.concat layer-chunks
and openings:
 $\forall$ (layer-len, layer-chunk) $\in$
 set (zip (*fri-layer-lengths* (Suc n) len) layer-chunks).
 $\exists$ xp xp-path xn xn-path.
fri-layer-opening-chunk layer-len xp xp-path xn xn-path
 layer-chunk
unfolding *fri-layers-transcript-def* **by** *simp-all*

obtain first rest **where** chunk-eq: layer-chunks = first # rest
using len-chunks **by** (cases layer-chunks) *auto*

```

from openings obtain  $xp$   $xp\text{-}path$   $xn$   $xn\text{-}path$  where first-open:
  fri-layer-opening-chunk  $len$   $xp$   $xp\text{-}path$   $xn$   $xn\text{-}path$  first
  unfolding chunks-eq by auto
have rest-layers:
  fri-layers-transcript  $n$  ( $len \text{ div } 2$ ) rest (List.concat rest)
  unfolding fri-layers-transcript-def
proof (intro conjI)
  show  $length\ rest = n$ 
    using len-chunks unfolding chunks-eq by simp
next
  show List.concat rest = List.concat rest
    by simp
next
  show  $\forall (layer\text{-}len, layer\text{-}chunk) \in$ 
     $set\ (zip\ (fri\text{-}layer\text{-}lengths\ n\ (len \text{ div } 2))\ rest).$ 
     $\exists xp\ xp\text{-}path\ xn\ xn\text{-}path.$ 
    fri-layer-opening-chunk  $layer\text{-}len$   $xp$   $xp\text{-}path$   $xn$   $xn\text{-}path$ 
     $layer\text{-}chunk$ 
    using openings unfolding chunks-eq by auto
qed
have  $length\ chunk = length\ first + length\ (List.concat\ rest)$ 
  unfolding chunk-eq chunks-eq by simp
also have ... =
  ( $2 + 2 * floor\text{-}log\ len$ ) +
  fri-layers-transcript-length  $n\ (len \text{ div } 2)$ 
  using fri-layer-opening-chunk-length[OF first-open]
  Suc.IH[OF rest-layers] by simp
finally show ?case
  by simp
qed

lemma fri-layers-transcript-nth-opening:
assumes layers: fri-layers-transcript  $n$   $len$  layer-chunks chunk
and j-bound:  $j < n$ 
obtains  $xp$   $xp\text{-}path$   $xn$   $xn\text{-}path$ 
where fri-layer-opening-chunk (fri-layer-lengths  $n$   $len$  !  $j$ )
   $xp$   $xp\text{-}path$   $xn$   $xn\text{-}path$  (layer-chunks !  $j$ )
proof –
have len-chunks:  $length\ layer\text{-}chunks = n$ 
  using layers unfolding fri-layers-transcript-def by simp
have pair-in:
  (fri-layer-lengths  $n$   $len$  !  $j$ , layer-chunks !  $j$ )  $\in$ 
   $set\ (zip\ (fri\text{-}layer\text{-}lengths\ n\ len)\ layer\text{-}chunks)$ 
proof –
have j-zip:
   $j < length\ (zip\ (fri\text{-}layer\text{-}lengths\ n\ len)\ layer\text{-}chunks)$ 
  using j-bound len-chunks by simp
have  $zip\ (fri\text{-}layer\text{-}lengths\ n\ len)\ layer\text{-}chunks\ !\ j =$ 
  (fri-layer-lengths  $n$   $len$  !  $j$ , layer-chunks !  $j$ )

```

```

    using j-bound len-chunks by simp
  then show ?thesis
    using nth-mem[OF j-zip] by metis
qed
then obtain xp xp-path xn xn-path where
  fri-layer-opening-chunk (fri-layer-lengths n len ! j)
  xp xp-path xn xn-path (layer-chunks ! j)
  using layers unfolding fri-layers-transcript-def by blast
then show ?thesis
  by (rule that)
qed

definition verifier-query-round-transcript-length
:: nat  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  nat
where
  verifier-query-round-transcript-length idx f-fri-roots composition-fri-roots =
    query-decommitment-transcript-length idx +
    fri-layers-transcript-length (length f-fri-roots) (clength * scale) +
    fri-layers-transcript-length (length composition-fri-roots) (clength * scale)

definition verifier-query-round-chunk
:: nat  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  bool
where
  verifier-query-round-chunk idx f-fri-roots composition-fri-roots chunk  $\longleftrightarrow$ 
    ( $\exists$  query-chunk trace-fri-chunk composition-fri-chunk
      leaves paths trace-layer-chunks composition-layer-chunks.
      chunk = query-chunk @ trace-fri-chunk @ composition-fri-chunk  $\wedge$ 
      query-decommitment-transcript idx leaves paths query-chunk  $\wedge$ 
      fri-layers-transcript (length f-fri-roots) (clength * scale)
      trace-layer-chunks trace-fri-chunk  $\wedge$ 
      fri-layers-transcript (length composition-fri-roots) (clength * scale)
      composition-layer-chunks composition-fri-chunk)

lemma verifier-query-round-chunk-length:
assumes verifier-query-round-chunk idx f-fri-roots composition-fri-roots chunk
shows
  length chunk =
    verifier-query-round-transcript-length idx f-fri-roots
    composition-fri-roots
proof -
from assms obtain query-chunk trace-fri-chunk composition-fri-chunk
  leaves paths trace-layer-chunks composition-layer-chunks where
  chunk-eq: chunk = query-chunk @ trace-fri-chunk @ composition-fri-chunk
and query:
    query-decommitment-transcript idx leaves paths query-chunk
and trace:
    fri-layers-transcript (length f-fri-roots) (clength * scale)
    trace-layer-chunks trace-fri-chunk
and composition:

```



```

    fri-layers-transcript (length composition-fri-roots) (clength * scale)
    composition-layer-chunks composition-fri-chunk
unfolding verifier-query-round-chunk-def by blast
show ?thesis
unfolding chunk-eq verifier-query-round-transcript-length-def
using query-decommitment-transcript-length[OF query]
    fri-layers-transcript-length[OF trace]
    fri-layers-transcript-length[OF composition]
by simp
qed

definition query-round-fri-layer-transcripts
  :: nat  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$ 
    'f list list  $\Rightarrow$  'f list list  $\Rightarrow$  bool
where
  query-round-fri-layer-transcripts idx trace-roots composition-roots
    chunk trace-layer-chunks composition-layer-chunks  $\longleftrightarrow$ 
    ( $\exists$  query-chunk trace-fri-chunk composition-fri-chunk leaves paths.
      chunk = query-chunk @ trace-fri-chunk @ composition-fri-chunk  $\wedge$ 
      query-decommitment-transcript idx leaves paths query-chunk  $\wedge$ 
      fri-layers-transcript (length trace-roots) (clength * scale)
        trace-layer-chunks trace-fri-chunk  $\wedge$ 
      fri-layers-transcript (length composition-roots) (clength * scale)
        composition-layer-chunks composition-fri-chunk)

lemma verifier-query-round-chunk-fri-layer-transcriptsE:
assumes verifier-query-round-chunk idx trace-roots composition-roots chunk
obtains trace-layer-chunks composition-layer-chunks
where query-round-fri-layer-transcripts idx trace-roots composition-roots
    chunk trace-layer-chunks composition-layer-chunks
using assms
unfolding verifier-query-round-chunk-def query-round-fri-layer-transcripts-def
by blast

definition verifier-query-indices-derived
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$ 
    'f  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  nat list  $\Rightarrow$  bool
where
  verifier-query-indices-derived s out query-start-state rest
    f-fri-roots composition-fri-roots query-idxs  $\longleftrightarrow$ 
    ( $\exists$  result final-state raw-idxs query-chunks trailing.
      out = Some (result, final-state)  $\wedge$ 
      length raw-idxs = rounds  $\wedge$ 
      query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs  $\wedge$ 
      length query-chunks = rounds  $\wedge$ 
      List.concat query-chunks @ trailing = rest  $\wedge$ 
      ( $\forall i < rounds$ .
        verifier-query-round-chunk (query-idxs ! i)

```

$f\text{-fri-roots } \text{composition-fri-roots } (\text{query-chunks } ! i) \wedge$
 $(\forall i < \text{rounds.}$
 $\text{fmlookup } (\text{HashMap } \text{final-state})$
 $(\text{QueryIndexChallenge } (\text{PQueryCounter } s + i)$
 $(\text{state-after-query-chunks } \text{query-start-state } \text{query-chunks } i)) =$
 $\text{Some } (\text{raw-idxs } ! i) \wedge$
 $(\forall \text{idx} \in \text{set } \text{query-idxs. } \text{idx} < \text{clength} * \text{scale}))$

definition *accepted-transcript-shape*

$:: (\text{'f}, \text{'a}) \text{protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{protocol-channel-scheme}) \text{option} \Rightarrow$
 $\text{'f list} \Rightarrow \text{nat list} \Rightarrow \text{bool}$

where

$\text{accepted-transcript-shape } s \text{ out alphas } \text{query-idxs} \longleftrightarrow$
 $(\exists \text{result } \text{final-state } \text{fr } f\text{-fri-roots } f\text{-final } \text{dg } \text{composition-fri-roots } \text{final } \text{rest.}$
 $\text{out} = \text{Some } (\text{result}, \text{final-state}) \wedge$
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final } \text{alphas } \text{dg}$
 $\text{composition-fri-roots } \text{final } \text{rest} \wedge$
 $\text{verifier-query-indices-derived } s \text{ out}$
 $(\text{verifier-header-state } s \text{ fr } f\text{-fri-roots } f\text{-final } \text{alphas } \text{dg}$
 $\text{composition-fri-roots } \text{final})$
 $\text{rest } f\text{-fri-roots } \text{composition-fri-roots } \text{query-idxs})$

lemma *accepted-transcript-shape-query-chunksE*:

assumes *shape*: $\text{accepted-transcript-shape } s \text{ out alphas } \text{query-idxs}$

obtains $\text{result } \text{final-state } \text{fr } f\text{-fri-roots } f\text{-final } \text{dg}$
 $\text{composition-fri-roots } \text{final } \text{rest } \text{raw-idxs } \text{query-chunks } \text{trailing}$

where

$\text{out} = \text{Some } (\text{result}, \text{final-state})$
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final } \text{alphas } \text{dg}$
 $\text{composition-fri-roots } \text{final } \text{rest}$
 $\text{length } \text{raw-idxs} = \text{rounds}$
 $\text{query-idxs} = \text{map } (\lambda \text{raw. } \text{index } (\text{to-nat } \text{raw})) \text{ raw-idxs}$
 $\text{length } \text{query-chunks} = \text{rounds}$
 $\text{List.concat } \text{query-chunks } @ \text{trailing} = \text{rest}$
 $\bigwedge i. i < \text{rounds} \implies$
 $\text{verifier-query-round-chunk } (\text{query-idxs } ! i)$
 $f\text{-fri-roots } \text{composition-fri-roots } (\text{query-chunks } ! i)$
 $\bigwedge i. i < \text{rounds} \implies$
 $\text{fmlookup } (\text{HashMap } \text{final-state})$
 $(\text{QueryIndexChallenge } (\text{PQueryCounter } s + i)$
 $(\text{state-after-query-chunks}$
 $(\text{verifier-header-state } s \text{ fr } f\text{-fri-roots } f\text{-final } \text{alphas } \text{dg}$
 $\text{composition-fri-roots } \text{final})$
 $\text{query-chunks } i)) =$
 $\text{Some } (\text{raw-idxs } ! i)$
 $\forall \text{idx} \in \text{set } \text{query-idxs. } \text{idx} < \text{clength} * \text{scale}$

proof —

from *shape* **obtain** $\text{result } \text{final-state } \text{fr } f\text{-fri-roots } f\text{-final } \text{dg}$

```

    composition-fri-roots final rest where
out-eq: out = Some (result, final-state)
and header:
    verifier-header-transcript s fr f-fri-roots f-final alphas dg
    composition-fri-roots final rest
and query:
    verifier-query-indices-derived s out
    (verifier-header-state s fr f-fri-roots f-final alphas dg
     composition-fri-roots final)
    rest f-fri-roots composition-fri-roots query-idxs
unfolding accepted-transcript-shape-def by blast
from query obtain result' final-state' raw-idxs query-chunks trailing where
out-eq': out = Some (result', final-state')
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-chunks: length query-chunks = rounds
and concat-chunks: List.concat query-chunks @ trailing = rest
and chunk-shape:
    ∧ i. i < rounds ⇒
        verifier-query-round-chunk (query-idxs ! i)
        f-fri-roots composition-fri-roots (query-chunks ! i)
and lookup:
    ∧ i. i < rounds ⇒
        fmlookup (HashMap final-state')
        (QueryIndexChallenge (PQueryCounter s + i)
         (state-after-query-chunks
          (verifier-header-state s fr f-fri-roots f-final alphas dg
           composition-fri-roots final)
           query-chunks i)) =
        Some (raw-idxs ! i)
and idx-bound:
    ∀ idx ∈ set query-idxs. idx < clength * scale
unfolding verifier-query-indices-derived-def by blast
have final-state-eq: final-state' = final-state
using out-eq out-eq' by simp
show ?thesis
by (rule that[OF out-eq header len-raw query-idxs-eq len-chunks
    concat-chunks chunk-shape])
    (use lookup idx-bound final-state-eq in simp-all)
qed

```

```

lemma accepted-transcript-shape-query-chunk-lengthsE:
assumes shape: accepted-transcript-shape s out alphas query-idxs
obtains result final-state fr f-fri-roots f-final dg
    composition-fri-roots final rest raw-idxs query-chunks trailing
where
    out = Some (result, final-state)
    verifier-header-transcript s fr f-fri-roots f-final alphas dg

```

```

    composition-fri-roots final rest
length raw-idxs = rounds
query-idxs = map (λraw. index (to-nat raw)) raw-idxs
length query-chunks = rounds
List.concat query-chunks @ trailing = rest
Λi. i < rounds ⇒
  length (query-chunks ! i) =
    verifier-query-round-transcript-length (query-idxs ! i)
    f-fri-roots composition-fri-roots
Λi. i < rounds ⇒
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks
        (verifier-header-state s fr f-fri-roots f-final alphas dg
          composition-fri-roots final)
        query-chunks i)) =
    Some (raw-idxs ! i)
  ∀ idx ∈ set query-idxs. idx < clength * scale
proof -
from accepted-transcript-shape-query-chunksE[OF shape]
obtain result final-state fr f-fri-roots f-final dg
  composition-fri-roots final rest raw-idxs query-chunks trailing where
  out-eq: out = Some (result, final-state)
and header:
  verifier-header-transcript s fr f-fri-roots f-final alphas dg
  composition-fri-roots final rest
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-chunks: length query-chunks = rounds
and concat-chunks: List.concat query-chunks @ trailing = rest
and chunk-shape:
  Λi. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
    f-fri-roots composition-fri-roots (query-chunks ! i)
and lookup:
  Λi. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks
          (verifier-header-state s fr f-fri-roots f-final alphas dg
            composition-fri-roots final)
          query-chunks i)) =
    Some (raw-idxs ! i)
and idx-bound:
  ∀ idx ∈ set query-idxs. idx < clength * scale
by blast
have chunk-len:
  Λi. i < rounds ⇒

```

$\text{length } (\text{query-chunks } ! i) =$
 $\text{verifier-query-round-transcript-length } (\text{query-idxs } ! i)$
 $\text{f-fri-roots composition-fri-roots}$
using $\text{verifier-query-round-chunk-length}[OF \text{ chunk-shape}]$ **by** blast
show $?thesis$
by $(\text{rule that}[OF \text{ out-eq header len-raw query-idxs-eq len-chunks}$
 $\text{concat-chunks chunk-len lookup idx-bound}])$
qed

definition *accepted-with-tables*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow$
 $\text{'f list} \Rightarrow \text{'f list} \Rightarrow \text{'f list} \Rightarrow \text{nat list} \Rightarrow \text{bool}$

where

$\text{accepted-with-tables } s \text{ out trace-table composition-table alphas query-idxs} \longleftrightarrow$
 $\text{accepted-transcript-shape } s \text{ out alphas query-idxs} \wedge$
 $(\forall \text{idx} \in \text{set query-idxs.}$
 $\text{query-consistent-at trace-table composition-table alphas idx}) \wedge$
 $\text{length trace-table} = \text{clength} * \text{scale} \wedge$
 $\text{length composition-table} = \text{clength} * \text{scale} \wedge$
 $\text{length alphas} = \text{length spec}$

definition *accepted-with-bound-tables*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow$
 $\text{'f list} \Rightarrow \text{'f list} \Rightarrow \text{'f list} \Rightarrow \text{nat list} \Rightarrow \text{bool}$

where

$\text{accepted-with-bound-tables } s \text{ out trace-table composition-table alphas query-idxs} \longleftrightarrow$
 $\text{accepted-with-tables } s \text{ out trace-table composition-table alphas query-idxs} \wedge$
 $(\exists \text{result final-state fr f-fri-roots f-final dg composition-fri-roots final rest.}$
 $\text{out} = \text{Some } (\text{result}, \text{final-state}) \wedge$
 $\text{verifier-header-transcript } s \text{ fr f-fri-roots f-final alphas dg}$
 $\text{composition-fri-roots final rest} \wedge$
 $\text{merkle-root-binds-table fr trace-table final-state} \wedge$
 $\text{composition-fri-roots} \neq [] \wedge$
 $\text{merkle-root-binds-table } (\text{hd composition-fri-roots}) \text{ composition-table final-state})$

definition *alpha-list-set-hit*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow \text{'f list set} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{alpha-list-set-hit } s \text{ B out} \longleftrightarrow$
 $(\exists \text{as query-idxs.}$
 $\text{accepted-transcript-shape } s \text{ out as query-idxs} \wedge \text{as} \in \text{B})$

definition *alpha-header-list-set-hit*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{'f} \Rightarrow \text{'f list} \Rightarrow \text{'f} \Rightarrow \text{'f list set}) \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where
 $\text{alpha-header-list-set-hit } s \ B \ \text{out} \longleftrightarrow$
 $(\exists \text{ as query-idxs result final-state fr f-fri-roots f-final dg}$
 $\text{composition-fri-roots final rest.}$
 $\text{accepted-transcript-shape } s \ \text{out as query-idxs} \wedge$
 $\text{out} = \text{Some } (\text{result}, \text{final-state}) \wedge$
 $\text{verifier-header-transcript } s \ \text{fr f-fri-roots f-final as dg}$
 $\text{composition-fri-roots final rest} \wedge$
 $\text{as} \in B \ \text{fr f-fri-roots f-final})$

end

end

theory *Soundness-Core*
imports *Soundness-Core-Events*
begin

Compatibility umbrella for core soundness definitions and lemmas.

end

theory *Soundness-Merkle*
imports
Soundness-Core
begin

context *soundness*
begin

A successful verifier execution only checks the sampled authentication paths. Turning those local checks into full trace and composition tables bound to the initial committed roots is a Merkle binding/collision-resistance property. The current random-oracle hash-map model records enough information to replay checked paths, but it does not by itself imply existence or uniqueness of a complete maliciously committed table. We therefore expose the missing part as a bad event over accepted verifier executions. Binding for intermediate FRI layers is not hidden here; it is part of the FRI bad-challenge-set reductions introduced below.

definition *initial-merkle-binding-bad*
 $:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where
 $\text{initial-merkle-binding-bad } s \ \text{out} \longleftrightarrow$
 $(\exists \text{ alphas query-idxs.}$
 $\text{accepted-transcript-shape } s \ \text{out alphas query-idxs} \wedge$
 $\neg (\exists \text{ trace-table composition-table.}$
 $\text{accepted-with-bound-tables } s \ \text{out trace-table composition-table})$

alphas query-idxs))

definition *initial-merkle-binding-no-bad* :: (*'f*, *'a*) *protocol-channel-scheme* $\Rightarrow$ *bool*

where

initial-merkle-binding-no-bad s $\longleftrightarrow$
 $(\forall out \in \text{set-dist } (\text{execute verify-monad } s).$
 $\neg \text{initial-merkle-binding-bad } s \text{ out})$

definition *initial-merkle-binding-collision-bad*

:: (*'f*, *'a*) *protocol-channel-scheme* $\Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

initial-merkle-binding-collision-bad s out $\longleftrightarrow$
 $\text{accepted out} \wedge$
 $(\exists \text{result final-state rt.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{merkle-root-binding-collision rt final-state})$

definition *merkle-root-binding-collision-bad*

:: (*'f*, *'a*) *protocol-channel-scheme* $\Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

merkle-root-binding-collision-bad s out $\longleftrightarrow$
 $\text{accepted out} \wedge$
 $(\exists \text{result final-state rt.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{merkle-root-binding-collision rt final-state})$

definition *hash-map-output-collision-bad*

:: (*'f*, *'a*) *protocol-channel-scheme* $\Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

hash-map-output-collision-bad s out $\longleftrightarrow$
 $\text{accepted out} \wedge$
 $(\exists \text{result final-state.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{hash-map-output-collision final-state})$

lemma *initial-merkle-binding-collision-bad-eq-root-collision-bad*:

initial-merkle-binding-collision-bad s out =
merkle-root-binding-collision-bad s out

unfolding *initial-merkle-binding-collision-bad-def*

merkle-root-binding-collision-bad-def

by *simp*

lemma *merkle-root-binding-collision-bad-imp-accepted*:

assumes *merkle-root-binding-collision-bad s out*

shows *accepted out*

using *assms* **unfolding** *merkle-root-binding-collision-bad-def* **by** *simp*

lemma *hash-map-output-collision-bad-imp-accepted*:
assumes *hash-map-output-collision-bad s out*
shows *accepted out*
using *assms* **unfolding** *hash-map-output-collision-bad-def* **by** *simp*

lemma *merkle-root-binding-collision-bad-mono-accepted*:
 $wp\text{-}event\ verify\text{-}monad\ (merkle\text{-}root\text{-}binding\text{-}collision\text{-}bad\ s)\ s \leq$
 $wp\text{-}event\ verify\text{-}monad\ accepted\ s$
by (rule *wp-event-mono*)
(simp add: merkle-root-binding-collision-bad-imp-accepted)

lemma *hash-map-output-collision-bad-mono-accepted*:
 $wp\text{-}event\ verify\text{-}monad\ (hash\text{-}map\text{-}output\text{-}collision\text{-}bad\ s)\ s \leq$
 $wp\text{-}event\ verify\text{-}monad\ accepted\ s$
by (rule *wp-event-mono*)
(simp add: hash-map-output-collision-bad-imp-accepted)

lemma *accepted-inconsistent-merkle-bindings-imp-collision-bad*:
assumes *out: out = Some (result, final-state)*
and *bind1: merkle-root-binds-table rt table final-state*
and *bind2: merkle-root-binds-table rt table' final-state*
and *neg: table $\neq$ table'*
shows *initial-merkle-binding-collision-bad s out*
proof –
have *collision: merkle-root-binding-collision rt final-state*
by (rule *merkle-root-binding-collisionI[OF bind1 bind2 neg]*)
show *?thesis*
proof (*unfold initial-merkle-binding-collision-bad-def accepted-def, intro conjI*)
show $\neg\ Option.is\ none\ out$
using *out* **by** *simp*
show $\exists\ result\ final\text{-}state\ rt.$
 $out = Some\ (result,\ final\text{-}state) \wedge$
 $merkle\text{-}root\text{-}binding\text{-}collision\ rt\ final\text{-}state$
using *out collision* **by** *blast*
qed
qed

lemma *accepted-inconsistent-merkle-bindings-imp-root-collision-bad*:
assumes *out: out = Some (result, final-state)*
and *bind1: merkle-root-binds-table rt table final-state*
and *bind2: merkle-root-binds-table rt table' final-state*
and *neg: table $\neq$ table'*
shows *merkle-root-binding-collision-bad s out*
using *accepted-inconsistent-merkle-bindings-imp-collision-bad*
 $[OF\ out\ bind1\ bind2\ neg]$
unfolding *initial-merkle-binding-collision-bad-eq-root-collision-bad* .

lemma *hash-map-output-collision-badI*:


```

assumes out: out = Some (result, final-state)
and neg: x ≠ y
and x-lookup: fmlookup (HashMap final-state) x = Some z
and y-lookup: fmlookup (HashMap final-state) y = Some z
shows hash-map-output-collision-bad s out
proof –
  have collision: hash-map-output-collision final-state
    by (rule hash-map-output-collisionI[OF neg x-lookup y-lookup])
  show ?thesis
    unfolding hash-map-output-collision-bad-def accepted-def
    using out collision by auto
qed

```

```

lemma accepted-inconsistent-same-length-merkle-bindings-imp-hash-collision-bad:
  assumes out: out = Some (result, final-state)
  and bind1: merkle-root-binds-table rt table final-state
  and bind2: merkle-root-binds-table rt table' final-state
  and same-len: length table = length table'
  and nonempty: table ≠ []
  and neg: table ≠ table'
  shows hash-map-output-collision-bad s out
proof –
  have collision: hash-map-output-collision final-state
    by (rule merkle-root-binds-same-length-tables-collision-imp-hash-collision
      [OF bind1 bind2 same-len nonempty neg])
  show ?thesis
    unfolding hash-map-output-collision-bad-def accepted-def
    using out collision by auto
qed

```

```

lemma initial-merkle-binding-collision-bad-no-collisionD:
  assumes ¬ initial-merkle-binding-collision-bad s (Some (result, final-state))
  shows ¬ merkle-root-binding-collision rt final-state
  using assms unfolding initial-merkle-binding-collision-bad-def accepted-def
  by auto

```

```

lemma accepted-with-tables-imp-accepted:
  assumes accepted-with-tables s out trace-table composition-table alphas query-idxs
  shows accepted out
  using assms unfolding accepted-with-tables-def accepted-transcript-shape-def
    verifier-query-indices-derived-def
    accepted-def
  by auto

```

```

lemma accepted-transcript-shape-imp-accepted:
  assumes accepted-transcript-shape s out alphas query-idxs
  shows accepted out
  using assms unfolding accepted-transcript-shape-def verifier-query-indices-derived-def
    accepted-def

```

by *auto*

lemma *accepted-with-tables-imp-accepted-transcript-shape:*
 assumes *accepted-with-tables s out trace-table composition-table alphas query-idxs*
 shows *accepted-transcript-shape s out alphas query-idxs*
 using *assms* **unfolding** *accepted-with-tables-def* **by** *simp*

lemma *accepted-with-tables-shapes:*
 assumes *accepted-with-tables s out trace-table composition-table alphas query-idxs*
 shows *length trace-table = clength * scale*
 and *length composition-table = clength * scale*
 and *length alphas = length spec*
 and *length query-idxs = rounds*
 and $\bigwedge idx. idx \in \text{set query-idxs} \implies idx < \text{clength} * \text{scale}$
 using *assms*
 unfolding *accepted-with-tables-def accepted-transcript-shape-def*
 verifier-query-indices-derived-def
 by *auto*

lemma *accepted-transcript-shape-query-shapes:*
 assumes *accepted-transcript-shape s out alphas query-idxs*
 shows *length query-idxs = rounds*
 and $\bigwedge idx. idx \in \text{set query-idxs} \implies idx < \text{clength} * \text{scale}$
 using *assms*
 unfolding *accepted-transcript-shape-def verifier-query-indices-derived-def*
 by *auto*

lemma *verifier-query-indices-derived-query-sample-space:*
 assumes *verifier-query-indices-derived s out query-start-state rest*
 f-fri-roots composition-fri-roots query-idxs
 shows $\bigwedge idx. idx \in \text{set query-idxs} \implies idx \in \text{query-sample-space}$
 using *assms index-less-query-sample-space*
 unfolding *verifier-query-indices-derived-def query-sample-space-def*
 by *auto*

lemma *accepted-transcript-shape-query-sample-space:*
 assumes *accepted-transcript-shape s out alphas query-idxs*
 shows $\bigwedge idx. idx \in \text{set query-idxs} \implies idx \in \text{query-sample-space}$
proof –
 fix *idx*
 assume *idx-in: idx ∈ set query-idxs*
 obtain *result final-state fr f-fri-roots f-final dg composition-fri-roots final rest*
 where *derived:*
 verifier-query-indices-derived s out
 (verifier-header-state s fr f-fri-roots f-final alphas dg
 composition-fri-roots final)
 rest f-fri-roots composition-fri-roots query-idxs
 using *assms* **unfolding** *accepted-transcript-shape-def* **by** *blast*
 show *idx ∈ query-sample-space*

by (rule verifier-query-indices-derived-query-sample-space[OF derived idx-in])
qed

lemma *accepted-with-tables-query-sample-space*:
assumes *accepted-with-tables s out trace-table composition-table alphas query-idxs*
shows $\bigwedge idx. idx \in \text{set query-idxs} \implies idx \in \text{query-sample-space}$
using *accepted-transcript-shape-query-sample-space*
 [OF *accepted-with-tables-imp-accepted-transcript-shape*[OF *assms*]]
by *blast*

lemma *accepted-with-bound-tables-query-sample-space*:
assumes *accepted-with-bound-tables s out trace-table composition-table alphas query-idxs*
shows $\bigwedge idx. idx \in \text{set query-idxs} \implies idx \in \text{query-sample-space}$
proof –
have *tables: accepted-with-tables s out trace-table composition-table alphas query-idxs*
using *assms unfolding accepted-with-bound-tables-def* **by** *simp*
show $\bigwedge idx. idx \in \text{set query-idxs} \implies idx \in \text{query-sample-space}$
by (rule *accepted-with-tables-query-sample-space*[OF *tables*])
 qed

lemma *accepted-with-tables-authenticated-roots*:
assumes *accepted-with-bound-tables s out trace-table composition-table alphas query-idxs*
obtains *result final-state fr composition-root*
where *out = Some (result, final-state)*
and *merkle-root-binds-table fr trace-table final-state*
and *merkle-root-binds-table composition-root composition-table final-state*
using *assms unfolding accepted-with-bound-tables-def* **by** *auto*

lemma *accepted-with-bound-tables-imp-accepted-with-tables*:
assumes *accepted-with-bound-tables s out trace-table composition-table alphas query-idxs*
shows *accepted-with-tables s out trace-table composition-table alphas query-idxs*
using *assms unfolding accepted-with-bound-tables-def* **by** *simp*

lemma *accepted-with-bound-tables-shapes*:
assumes *accepted-with-bound-tables s out trace-table composition-table alphas query-idxs*
shows *length trace-table = clength * scale*
and *length composition-table = clength * scale*
and *length alphas = length spec*
and *length query-idxs = rounds*
and $\bigwedge idx. idx \in \text{set query-idxs} \implies idx < \text{clength} * \text{scale}$
using *accepted-with-tables-shapes*
 [OF *accepted-with-bound-tables-imp-accepted-with-tables*[OF *assms*]]
by *blast+*

lemma *accepted-with-bound-tables-imp-accepted*:
assumes *accepted-with-bound-tables s out trace-table composition-table alphas query-idxs*
shows *accepted out*
using *accepted-with-bound-tables-imp-accepted-with-tables*[OF *assms*]
by (rule *accepted-with-tables-imp-accepted*)

definition *supported-hash-output-collision-possible*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{supported-hash-output-collision-possible } s \longleftrightarrow$

$(\exists \text{ out} \in \text{set-dist } (\text{execute verify-monad } s).$

$\text{hash-map-output-collision-bad } s \text{ out})$

lemma *accepted-with-bound-tables-hash-collision-imp-supported-hash-output-collision-possible:*

assumes *outcome:*

$\text{Some } (\text{result}, \text{final-state}) \in \text{set-dist } (\text{execute verify-monad } s)$

and *bound:*

$\text{accepted-with-bound-tables } s \text{ (Some } (\text{result}, \text{final-state}))$

$\text{trace-table composition-table as query-idxs}$

and *collision:* $\text{hash-map-output-collision final-state}$

shows *supported-hash-output-collision-possible* s

proof –

have *accepted:* $\text{accepted (Some (result, final-state))}$

by $(\text{rule accepted-with-bound-tables-imp-accepted}[OF \text{ bound}])$

have *bad:*

$\text{hash-map-output-collision-bad } s \text{ (Some (result, final-state))}$

unfolding *hash-map-output-collision-bad-def*

using *accepted collision* **by** *blast*

show *?thesis*

unfolding *supported-hash-output-collision-possible-def*

using *outcome bad* **by** *blast*

qed

definition *alpha-bad-set-hit*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(f \text{ list} \Rightarrow f \text{ list set}) \Rightarrow$

$(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{alpha-bad-set-hit } s \text{ bad-sets out} \longleftrightarrow$

$(\exists \text{ trace-table composition-table as query-idxs.}$

$\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs} \wedge$

$\text{as} \in \text{bad-sets trace-table})$

lemma *alpha-bad-set-hit-imp-alpha-list-set-hit:*

assumes *hit:* $\text{alpha-bad-set-hit } s \text{ bad-sets out}$

and *envelope:* $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq B$

shows *alpha-list-set-hit* $s B \text{ out}$

proof –

from *hit* **obtain** $\text{trace-table composition-table as query-idxs}$ **where**

bound:

$\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs}$

and *as-bad:* $\text{as} \in \text{bad-sets trace-table}$

unfolding *alpha-bad-set-hit-def* **by** *blast*

have *shape:*

```

    accepted-transcript-shape s out as query-idxs
using bound unfolding accepted-with-bound-tables-def
    accepted-with-tables-def by simp
have as  $\in B$ 
    using as-bad envelope[of trace-table] by auto
then show ?thesis
    unfolding alpha-list-set-hit-def using shape by blast
qed

definition alpha-header-trace-table-candidates
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
   'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list set
where
alpha-header-trace-table-candidates s fr f-fri-roots f-final =
  {trace-table.
    $\exists$  out composition-table as query-idxs dg composition-fri-roots final rest.
    accepted-with-bound-tables s out trace-table composition-table
    as query-idxs  $\wedge$ 
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest}

definition alpha-header-union-bad-sets
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
   ('f list  $\Rightarrow$  'f list set)  $\Rightarrow$ 
   'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list set
where
alpha-header-union-bad-sets s bad-sets fr f-fri-roots f-final =
  {as  $\in$  alpha-space.
    $\exists$  trace-table  $\in$ 
    alpha-header-trace-table-candidates s fr f-fri-roots f-final.
    as  $\in$  bad-sets trace-table}

definition alpha-header-supported-trace-table-candidates
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
   'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list set
where
alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final =
  {trace-table.
    $\exists$  out composition-table as query-idxs dg composition-fri-roots final rest.
    out  $\in$  set-dist (execute verify-monad s)  $\wedge$ 
    accepted-with-bound-tables s out trace-table composition-table
    as query-idxs  $\wedge$ 
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest}

definition alpha-header-supported-union-bad-sets
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
   ('f list  $\Rightarrow$  'f list set)  $\Rightarrow$ 
   'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list set

```

where

$\alpha\text{-header-supported-union-bad-sets } s \text{ bad-sets fr } f\text{-fri-roots } f\text{-final} =$
 $\{as \in \alpha\text{-space}.$
 $\exists \text{ trace-table} \in$
 $\alpha\text{-header-supported-trace-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final}.$
 $as \in \text{bad-sets trace-table}\}$

lemma $\alpha\text{-header-union-bad-sets-subset-}\alpha\text{-space}:$

$\alpha\text{-header-union-bad-sets } s \text{ bad-sets fr } f\text{-fri-roots } f\text{-final} \subseteq$
 $\alpha\text{-space}$

unfolding $\alpha\text{-header-union-bad-sets-def}$ **by** *auto*

lemma $\alpha\text{-header-supported-union-bad-sets-subset-}\alpha\text{-space}:$

$\alpha\text{-header-supported-union-bad-sets } s \text{ bad-sets fr } f\text{-fri-roots } f\text{-final}$
 $\subseteq \alpha\text{-space}$

unfolding $\alpha\text{-header-supported-union-bad-sets-def}$ **by** *auto*

lemma $\alpha\text{-header-union-bad-sets-fraction-bound-if-unique-candidate}:$

fixes $C :: \text{prob}$

assumes *unique*:

$\exists \text{ trace-table}.$

$\alpha\text{-header-trace-table-candidates } s \text{ fr } f\text{-fri-roots } f\text{-final} \subseteq$
 $\{\text{trace-table}\}$

and *subset*: $\bigwedge \text{trace-table}.$ $\text{bad-sets trace-table} \subseteq \alpha\text{-space}$

and *bound*:

$\bigwedge \text{trace-table}.$

$\text{nnreal } (\text{card } (\text{bad-sets trace-table})) / \text{nnreal } (\text{card } \alpha\text{-space}) \leq C$

shows

nnreal

$(\text{card } (\alpha\text{-header-union-bad-sets } s \text{ bad-sets fr } f\text{-fri-roots } f\text{-final})) /$
 $\text{nnreal } (\text{card } \alpha\text{-space}) \leq C$

proof –

from *unique* **obtain** *trace-table* **where** *candidates*:

$\alpha\text{-header-trace-table-candidates } s \text{ fr } f\text{-fri-roots } f\text{-final} \subseteq$
 $\{\text{trace-table}\}$

by *blast*

have *union-subset*:

$\alpha\text{-header-union-bad-sets } s \text{ bad-sets fr } f\text{-fri-roots } f\text{-final} \subseteq$
 $\text{bad-sets trace-table}$

unfolding $\alpha\text{-header-union-bad-sets-def}$

using *candidates* **by** *auto*

have *finite-bad*: *finite* (*bad-sets trace-table*)

by (*rule finite-subset*[*OF subset finite-alpha-space*])

have *card-le*:

$\text{card } (\alpha\text{-header-union-bad-sets } s \text{ bad-sets fr } f\text{-fri-roots } f\text{-final}) \leq$
 $\text{card } (\text{bad-sets trace-table})$

by (*rule card-mono*[*OF finite-bad union-subset*])

have *nnreal*

```

      (card (alpha-header-union-bad-sets s bad-sets fr f-fri-roots f-final)) /
      nnreal (card alpha-space) ≤
      nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)
    by (rule nnreal-nat-divide-right-mono[OF card-le])
  also have ... ≤ C
    by (rule bound)
  finally show ?thesis .
qed

lemma alpha-header-supported-union-bad-sets-fraction-bound-if-unique-candidate:
  fixes C :: prob
  assumes unique:
    ∃ trace-table.
      alpha-header-supported-trace-table-candidates s fr f-fri-roots
      f-final ⊆ {trace-table}
  and subset:  $\bigwedge$  trace-table. bad-sets trace-table ⊆ alpha-space
  and bound:
     $\bigwedge$  trace-table.
      nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space) ≤ C
  shows
    nnreal
      (card
        (alpha-header-supported-union-bad-sets s bad-sets fr f-fri-roots
          f-final)) /
      nnreal (card alpha-space) ≤ C
proof -
  from unique obtain trace-table where candidates:
    alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
    ⊆ {trace-table}
  by blast
  have union-subset:
    alpha-header-supported-union-bad-sets s bad-sets fr f-fri-roots f-final
    ⊆ bad-sets trace-table
  unfolding alpha-header-supported-union-bad-sets-def
  using candidates by auto
  have finite-bad: finite (bad-sets trace-table)
  by (rule finite-subset[OF subset finite-alpha-space])
  have card-le:
    card
      (alpha-header-supported-union-bad-sets s bad-sets fr f-fri-roots
        f-final) ≤ card (bad-sets trace-table)
  by (rule card-mono[OF finite-bad union-subset])
  have nnreal
    (card
      (alpha-header-supported-union-bad-sets s bad-sets fr f-fri-roots
        f-final)) /
    nnreal (card alpha-space) ≤
    nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)
  by (rule nnreal-nat-divide-right-mono[OF card-le])

```

also have $\dots \leq C$
 by (rule bound)
 finally show ?thesis .
 qed

lemma *composition-trace-alpha-header-union-fraction-bound-if-unique-candidate:*

assumes *unique:*
 $\exists$ *trace-table.*
 $\alpha\text{-header-trace-table-candidates } s \text{ fr } f\text{-fri-roots } f\text{-final} \subseteq \{\text{trace-table}\}$
shows
 $\text{nnreal } (\text{card } (\alpha\text{-header-union-bad-sets } s \text{ composition-trace-bad-alpha-space fr } f\text{-fri-roots } f\text{-final})) / \text{nnreal } (\text{card } \alpha\text{-space}) \leq \text{composition-error-bound}$
by (rule *alpha-header-union-bad-sets-fraction-bound-if-unique-candidate* [OF *unique composition-trace-bad-alpha-space-subset-alpha-space composition-trace-bad-alpha-space-fraction-bound-alpha-space*])

lemma *composition-trace-alpha-header-supported-union-fraction-bound-if-unique-candidate:*

assumes *unique:*
 $\exists$ *trace-table.*
 $\alpha\text{-header-supported-trace-table-candidates } s \text{ fr } f\text{-fri-roots } f\text{-final} \subseteq \{\text{trace-table}\}$
shows
 $\text{nnreal } (\text{card } (\alpha\text{-header-supported-union-bad-sets } s \text{ composition-trace-bad-alpha-space fr } f\text{-fri-roots } f\text{-final})) / \text{nnreal } (\text{card } \alpha\text{-space}) \leq \text{composition-error-bound}$
by (rule *alpha-header-supported-union-bad-sets-fraction-bound-if-unique-candidate* [OF *unique composition-trace-bad-alpha-space-subset-alpha-space composition-trace-bad-alpha-space-fraction-bound-alpha-space*])

lemma *alpha-bad-set-hit-imp-alpha-header-union-bad-set-hit:*

assumes *hit:* $\alpha\text{-bad-set-hit } s \text{ bad-sets out}$
and *subset:* $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \alpha\text{-space}$
shows
 $\alpha\text{-header-list-set-hit } s$
 $(\alpha\text{-header-union-bad-sets } s \text{ bad-sets}) \text{ out}$

proof –

from *hit* **obtain** *trace-table composition-table as query-idxs* **where**
bound:
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs}$
and *as-bad:* $as \in \text{bad-sets trace-table}$
unfolding *alpha-bad-set-hit-def* **by** *blast*
from *bound* **obtain** *result final-state fr f-fri-roots f-final dg*


```

    composition-fri-roots final rest where
    out-eq: out = Some (result, final-state)
and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
unfolding accepted-with-bound-tables-def by blast
have shape:
    accepted-transcript-shape s out as query-idxs
using accepted-with-bound-tables-imp-accepted-with-tables[OF bound]
    accepted-with-tables-imp-accepted-transcript-shape
by blast
have candidate:
    trace-table ∈
    alpha-header-trace-table-candidates s fr f-fri-roots f-final
unfolding alpha-header-trace-table-candidates-def
using bound header by blast
have as-union:
    as ∈ alpha-header-union-bad-sets s bad-sets fr f-fri-roots f-final
unfolding alpha-header-union-bad-sets-def
using as-bad candidate subset[of trace-table] by auto
show ?thesis
unfolding alpha-header-list-set-hit-def
using shape out-eq header as-union by blast
qed

```

lemma alpha-bad-set-hit-imp-alpha-header-supported-union-bad-set-hit:

```

assumes outcome: out ∈ set-dist (execute verify-monad s)
and hit: alpha-bad-set-hit s bad-sets out
and subset:  $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$ 
shows
    alpha-header-list-set-hit s
    (alpha-header-supported-union-bad-sets s bad-sets) out

```

proof –

```

from hit obtain trace-table composition-table as query-idxs where
    bound:
    accepted-with-bound-tables s out trace-table composition-table
    as query-idxs
and as-bad: as ∈ bad-sets trace-table
unfolding alpha-bad-set-hit-def by blast
from bound obtain result final-state fr f-fri-roots f-final dg
    composition-fri-roots final rest where
    out-eq: out = Some (result, final-state)
and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
unfolding accepted-with-bound-tables-def by blast
have shape:
    accepted-transcript-shape s out as query-idxs
using accepted-with-bound-tables-imp-accepted-with-tables[OF bound]

```

```

    accepted-with-tables-imp-accepted-transcript-shape
  by blast
have candidate:
  trace-table ∈
    alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
  unfolding alpha-header-supported-trace-table-candidates-def
  using outcome bound header by blast
have as-union:
  as ∈
    alpha-header-supported-union-bad-sets s bad-sets fr f-fri-roots f-final
  unfolding alpha-header-supported-union-bad-sets-def
  using as-bad candidate subset[of trace-table] by auto
show ?thesis
  unfolding alpha-header-list-set-hit-def
  using shape out-eq header as-union by blast
qed

```

definition *alpha-challenge-freshness-assumption*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
alpha-challenge-freshness-assumption $s \longleftrightarrow$
 $(\forall \text{ bad-sets.}$
 $(\forall \text{ trace-table. bad-sets trace-table} \subseteq \text{alpha-space}) \longrightarrow$
 $(\forall \text{ trace-table.}$
 $\text{nnreal (card (bad-sets trace-table)) /}$
 $\text{nnreal (card alpha-space)} \leq \text{composition-error-bound}) \longrightarrow$
 $\text{wp-event verify-monad (alpha-bad-set-hit s bad-sets) s} \leq$
 $\text{composition-error-bound})$

The protocol currently represents all Fiat-Shamir challenges through the same random-oracle hash map, without protocol-level domain-separated labels for alpha, query-index, and FRI challenges. The following predicates below state the precise freshness obligations needed for each challenge family. Phase 4E bundles them into one random-oracle model interface rather than adding labels to the protocol in this refactor. Merkle hash uses are not represented as freshness obligations here; their binding role remains the explicit *initial-merkle-binding-no-bad* premise.

lemma *initial-merkle-binding-no-badE*:
assumes *bind*: *initial-merkle-binding-no-bad* s
and *outcome*:
 $\text{Some (result, final-state)} \in \text{set-dist (execute verify-monad s)}$
and *shape*: *accepted-transcript-shape* s $(\text{Some (result, final-state)})$ *alphas query-idxs*
obtains *trace-table composition-table*
where *accepted-with-bound-tables* s $(\text{Some (result, final-state)})$
 $\text{trace-table composition-table alphas query-idxs}$
using *assms*
unfolding *initial-merkle-binding-no-bad-def initial-merkle-binding-bad-def*
by *blast*

lemma *initial-merkle-binding-bad-bound*:
assumes *bind: initial-merkle-binding-no-bad s*
shows *wp-event verify-monad (initial-merkle-binding-bad s) s ≤ 0*
proof –
have *wp-event verify-monad (initial-merkle-binding-bad s) s ≤*
wp-event verify-monad
(λ- :: (unit list × ('f, 'a) protocol-channel-scheme) option. False) s
by (*rule wp-event-mono-on-support*)
(use bind in ‹auto simp: initial-merkle-binding-no-bad-def›)
also have *... = 0*
unfolding *wp-event-def wp-def dist-expect-def* **by** *simp*
finally show *?thesis* .
qed

lemma *accepted-with-bound-tables-query-consistency*:
assumes *bound:*
accepted-with-bound-tables s out trace-table composition-table alphas query-idxs
shows $\bigwedge idx. idx \in \text{set query-idxs} \implies$
query-consistent-at trace-table composition-table alphas idx
using *bound* **unfolding** *accepted-with-bound-tables-def accepted-with-tables-def* **by** *simp*

lemma *accepted-with-bound-tables-bound-roots*:
assumes *bound:*
accepted-with-bound-tables s out trace-table composition-table alphas query-idxs
obtains *result final-state fr composition-root*
where *out = Some (result, final-state)*
and *merkle-root-binds-table fr trace-table final-state*
and *merkle-root-binds-table composition-root composition-table final-state*
using *accepted-with-tables-authenticated-roots[OF bound]* **by** *blast*

lemma *verifier-header-transcript-shapes*:
assumes *verifier-header-transcript s fr f-fri-roots f-final as dg*
composition-fri-roots final rest
shows *PTranscript s =*
verifier-header-messages fr f-fri-roots f-final as dg
composition-fri-roots final @ rest
and *length f-fri-roots = ceil-log clength*
and *length as = length spec*
and *length composition-fri-roots = ceil-log (to-nat dg + 1)*
using *assms* **unfolding** *verifier-header-transcript-def* **by** *simp-all*

lemma *verifier-header-transcript-unique*:
assumes *h1:*
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
and *h2:*
verifier-header-transcript s fr' f-fri-roots' f-final' as' dg'
composition-fri-roots' final' rest'
shows

```

fr' = fr ∧
f-fri-roots' = f-fri-roots ∧
f-final' = f-final ∧
as' = as ∧
dg' = dg ∧
composition-fri-roots' = composition-fri-roots ∧
final' = final ∧
rest' = rest
proof –
  have lengths:
    length f-fri-roots = ceil-log clength
    length as = length spec
    length composition-fri-roots = ceil-log (to-nat dg + 1)
    length f-fri-roots' = ceil-log clength
    length as' = length spec
    length composition-fri-roots' = ceil-log (to-nat dg' + 1)
  using h1 h2 unfolding verifier-header-transcript-def by simp-all
  have msg-eq:
    verifier-header-messages fr f-fri-roots f-final as dg
      composition-fri-roots final @ rest =
    verifier-header-messages fr' f-fri-roots' f-final' as' dg'
      composition-fri-roots' final' @ rest'
  using h1 h2 unfolding verifier-header-transcript-def by simp
  then show ?thesis
    using lengths
    unfolding verifier-header-messages-def
    by (auto simp: append-eq-append-conv-if split: if-splits)
qed

```

definition *alpha-header-trace-table-candidate-state*

```

:: ('f, 'a) protocol-channel-scheme ⇒
  'f ⇒ 'f list ⇒ 'f ⇒ 'f list ⇒
  ('f, 'a) protocol-channel-scheme ⇒ bool

```

where

```

alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
  trace-table final-state ⟷
  (∃ result composition-table as query-idxs dg composition-fri-roots final
    rest.
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs ∧
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest ∧
    merkle-root-binds-table fr trace-table final-state)

```

definition *alpha-header-supported-trace-table-candidate-state*

```

:: ('f, 'a) protocol-channel-scheme ⇒
  'f ⇒ 'f list ⇒ 'f ⇒ 'f list ⇒
  ('f, 'a) protocol-channel-scheme ⇒ bool

```

where

alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
f-final trace-table final-state $\longleftrightarrow$
($\exists$ result composition-table as query-idxs dg composition-fri-roots final
rest.
Some (result, final-state) $\in$ set-dist (execute verify-monad s) $\wedge$
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs $\wedge$
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest $\wedge$
merkle-root-binds-table fr trace-table final-state)

lemma *alpha-header-trace-table-candidates-iff-state:*

trace-table $\in$
alpha-header-trace-table-candidates s fr f-fri-roots f-final $\longleftrightarrow$
($\exists$ final-state.
alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
trace-table final-state))

proof

assume *candidate:*

trace-table $\in$
alpha-header-trace-table-candidates s fr f-fri-roots f-final

then obtain *out composition-table as query-idxs dg composition-fri-roots*
final rest **where**

bound:

accepted-with-bound-tables s out trace-table composition-table as
query-idxs

and *header:*

verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest

unfolding *alpha-header-trace-table-candidates-def* **by** *blast*
from *bound* **obtain** *result final-state fr' f-fri-roots' f-final' dg'*
composition-fri-roots' final' rest' where

out-eq: out = Some (result, final-state)

and *header':*

verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
composition-fri-roots' final' rest'

and *bind:*

merkle-root-binds-table fr' trace-table final-state

unfolding *accepted-with-bound-tables-def* **by** *blast*

have *fr-eq: fr' = fr*

using *verifier-header-transcript-unique[OF header' header]* **by** *simp*

have *bound-some:*

accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs

using *bound out-eq* **by** *simp*

have *state:*

alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
trace-table final-state

unfolding *alpha-header-trace-table-candidate-state-def*

```

    by (intro exI[of - result] exI[of - composition-table] exI[of - as]
        exI[of - query-idxs] exI[of - dg] exI[of - composition-fri-roots]
        exI[of - final] exI[of - rest])
    (use bound-some header bind fr-eq in simp)
  then show
     $\exists$  final-state.
      alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
      trace-table final-state
    by blast
next
assume
   $\exists$  final-state.
    alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table final-state
then obtain final-state result composition-table as query-idxs dg
  composition-fri-roots final rest where
  bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
unfolding alpha-header-trace-table-candidate-state-def by blast
show
  trace-table  $\in$ 
    alpha-header-trace-table-candidates s fr f-fri-roots f-final
unfolding alpha-header-trace-table-candidates-def
using bound header by blast
qed

```

lemma *alpha-header-supported-trace-table-candidates-iff-state:*

```

  trace-table  $\in$ 
    alpha-header-supported-trace-table-candidates s fr f-fri-roots
    f-final  $\longleftrightarrow$ 
  ( $\exists$  final-state.
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table final-state)

```

proof

```

  assume candidate:
    trace-table  $\in$ 
      alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
  then obtain out composition-table as query-idxs dg composition-fri-roots
    final rest where
    outcome: out  $\in$  set-dist (execute verify-monad s)
  and bound:
    accepted-with-bound-tables s out trace-table composition-table as
    query-idxs
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg

```

```

    composition-fri-roots final rest
  unfolding alpha-header-supported-trace-table-candidates-def by blast
from bound obtain result final-state fr' f-fri-roots' f-final' dg'
  composition-fri-roots' final' rest' where
out-eq: out = Some (result, final-state)
and header':
  verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
  composition-fri-roots' final' rest'
and bind:
  merkle-root-binds-table fr' trace-table final-state
  unfolding accepted-with-bound-tables-def by blast
have fr-eq: fr' = fr
  using verifier-header-transcript-unique[OF header' header] by simp
have state:
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table final-state
  unfolding alpha-header-supported-trace-table-candidate-state-def
by (intro exI[of - result] exI[of - composition-table] exI[of - as]
  exI[of - query-idxs] exI[of - dg] exI[of - composition-fri-roots]
  exI[of - final] exI[of - rest])
  (use outcome bound header bind fr-eq out-eq in simp)
then show
  ∃ final-state.
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table final-state
  by blast
next
assume
  ∃ final-state.
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table final-state
then obtain final-state result composition-table as query-idxs dg
  composition-fri-roots final rest where
outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
  unfolding alpha-header-supported-trace-table-candidate-state-def by blast
show
  trace-table ∈
    alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
  unfolding alpha-header-supported-trace-table-candidates-def
  using outcome bound header by blast
qed

```

lemma *alpha-header-supported-trace-table-candidate-state-imp-candidate-state:*
assumes
alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
f-final trace-table final-state
shows
alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
trace-table final-state
using *assms*
unfolding *alpha-header-supported-trace-table-candidate-state-def*
alpha-header-trace-table-candidate-state-def
by *blast*

lemma *alpha-header-trace-table-candidate-states-common-extension-unique-if-clean:*
assumes *clean: $\neg$ hash-map-output-collision u*
and *candidate1:*
alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
trace-table final-state
and *candidate2:*
alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
trace-table' final-state'
and *ext1: final-state $\leq$ u*
and *ext2: final-state' $\leq$ u*
shows *trace-table = trace-table'*
proof –
from *candidate1* **obtain** *result composition-table as query-idxs dg*
composition-fri-roots final rest **where**
bound1:
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs
and *bind1: merkle-root-binds-table fr trace-table final-state*
unfolding *alpha-header-trace-table-candidate-state-def* **by** *blast*
from *candidate2* **obtain** *result' composition-table' as' query-idxs' dg'*
composition-fri-roots' final' rest' **where**
bound2:
accepted-with-bound-tables s (Some (result', final-state'))
trace-table' composition-table' as' query-idxs'
and *bind2: merkle-root-binds-table fr trace-table' final-state'*
unfolding *alpha-header-trace-table-candidate-state-def* **by** *blast*
have *same-len: length trace-table = length trace-table'*
using *accepted-with-bound-tables-shapes(1)[OF bound1]*
accepted-with-bound-tables-shapes(1)[OF bound2]
by *simp*
have *nonempty: trace-table $\neq$ []*
using *accepted-with-bound-tables-shapes(1)[OF bound1]*
eval-domain-nontrivial **by** *auto*
show *?thesis*
by (*rule merkle-root-binds-same-length-tables-common-extension-unique-if-clean*
[OF clean bind1 bind2 ext1 ext2 same-len nonempty])
qed

lemma *alpha-header-trace-table-candidate-states-compatible-unique-if-clean-merge:*

assumes *compatible: hash-maps-compatible final-state final-state'*

and *clean:*

$\neg$ *hash-map-output-collision*
 (*hash-state-merge final-state final-state'*)

and *candidate1:*

alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
 trace-table final-state

and *candidate2:*

alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
 trace-table' final-state'

shows *trace-table = trace-table'*

proof –

from *candidate1* **obtain** *result composition-table as query-idxs dg*

composition-fri-roots final rest **where**

bound1:

accepted-with-bound-tables s (Some (result, final-state))
 trace-table composition-table as query-idxs

and *bind1: merkle-root-binds-table fr trace-table final-state*

unfolding *alpha-header-trace-table-candidate-state-def* **by** *blast*

from *candidate2* **obtain** *result' composition-table' as' query-idxs' dg'*

composition-fri-roots' final' rest' **where**

bound2:

accepted-with-bound-tables s (Some (result', final-state'))
 trace-table' composition-table' as' query-idxs'

and *bind2: merkle-root-binds-table fr trace-table' final-state'*

unfolding *alpha-header-trace-table-candidate-state-def* **by** *blast*

have *same-len: length trace-table = length trace-table'*

using *accepted-with-bound-tables-shapes(1)[OF bound1]*

accepted-with-bound-tables-shapes(1)[OF bound2]

by *simp*

have *nonempty: trace-table $\neq$ []*

using *accepted-with-bound-tables-shapes(1)[OF bound1]*

eval-domain-nontrivial **by** *auto*

show *?thesis*

by (*rule*

merkle-root-binds-same-length-tables-compatible-states-unique-if-clean-merge
 [*OF compatible clean bind1 bind2 same-len nonempty*])

qed

lemma *alpha-header-trace-table-candidate-states-merkle-compatible-unique-if-clean-merge:*

assumes *compatible:*

merkle-hash-maps-compatible final-state final-state'

and *clean:*

$\neg$ *hash-map-output-collision*
 (*merkle-hash-state-merge final-state final-state'*)

and *candidate1:*

alpha-header-trace-table-candidate-state s fr f-fri-roots f-final

```

    trace-table final-state
  and candidate2:
    alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table' final-state'
  shows trace-table = trace-table'
proof -
  from candidate1 obtain result composition-table as query-idxs dg
    composition-fri-roots final rest where
  bound1:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and bind1: merkle-root-binds-table fr trace-table final-state
  unfolding alpha-header-trace-table-candidate-state-def by blast
  from candidate2 obtain result' composition-table' as' query-idxs' dg'
    composition-fri-roots' final' rest' where
  bound2:
    accepted-with-bound-tables s (Some (result', final-state'))
    trace-table' composition-table' as' query-idxs'
  and bind2: merkle-root-binds-table fr trace-table' final-state'
  unfolding alpha-header-trace-table-candidate-state-def by blast
  have same-len: length trace-table = length trace-table'
  using accepted-with-bound-tables-shapes(1)[OF bound1]
    accepted-with-bound-tables-shapes(1)[OF bound2]
  by simp
  have nonempty: trace-table ≠ []
  using accepted-with-bound-tables-shapes(1)[OF bound1]
    eval-domain-nontrivial by auto
  show ?thesis
  by (rule
    merkle-root-binds-same-length-tables-merkle-compatible-unique-if-clean-merge
    [OF compatible clean bind1 bind2 same-len nonempty])
qed

```

lemma *alpha-header-trace-table-candidates-subset-singleton-if-common-clean-extension:*

```

  assumes clean: ¬ hash-map-output-collision u
  and common-ext:
     $\bigwedge$  trace-table final-state.
    alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table final-state  $\implies$ 
    final-state  $\leq$  u
  shows
     $\exists$  trace-table.
    alpha-header-trace-table-candidates s fr f-fri-roots f-final  $\subseteq$ 
    {trace-table}
proof (cases
  alpha-header-trace-table-candidates s fr f-fri-roots f-final = {})
case True
then show ?thesis
  by (intro exI[of - []]) simp

```

```

next
  case False
  then obtain trace-table0 where table0:
    trace-table0 ∈
      alpha-header-trace-table-candidates s fr f-fri-roots f-final
    by blast
  then obtain final-state0 where state0:
    alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table0 final-state0
    using alpha-header-trace-table-candidates-iff-state by blast
  have ext0: final-state0 ≤ u
    by (rule common-ext[OF state0])
  have subset:
    alpha-header-trace-table-candidates s fr f-fri-roots f-final ⊆
      {trace-table0}
  proof
    fix trace-table
    assume table:
      trace-table ∈
        alpha-header-trace-table-candidates s fr f-fri-roots f-final
    then obtain final-state where state:
      alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
      trace-table final-state
      using alpha-header-trace-table-candidates-iff-state by blast
    have ext: final-state ≤ u
      by (rule common-ext[OF state])
    have trace-table = trace-table0
      by (rule
        alpha-header-trace-table-candidate-states-common-extension-unique-if-clean
        [OF clean state state0 ext ext0])
    then show trace-table ∈ {trace-table0}
      by simp
  qed
  then show ?thesis
    by blast
qed

```

lemma *alpha-header-trace-table-candidates-subset-singleton-if-pairwise-clean-merge:*

```

assumes pairwise:
  ∧ trace-table final-state trace-table' final-state'.
    alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table final-state ⇒
    alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table' final-state' ⇒
    hash-maps-compatible final-state final-state' ∧
    ¬ hash-map-output-collision
    (hash-state-merge final-state final-state')

```

shows

∃ trace-table.

```

      alpha-header-trace-table-candidates s fr f-fri-roots f-final  $\subseteq$ 
        {trace-table}
proof (cases
  alpha-header-trace-table-candidates s fr f-fri-roots f-final = {})
case True
then show ?thesis
  by (intro exI[of - []]) simp
next
case False
then obtain trace-table0 where table0:
  trace-table0  $\in$ 
    alpha-header-trace-table-candidates s fr f-fri-roots f-final
by blast
then obtain final-state0 where state0:
  alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table0 final-state0
using alpha-header-trace-table-candidates-iff-state by blast
have subset:
  alpha-header-trace-table-candidates s fr f-fri-roots f-final  $\subseteq$ 
    {trace-table0}
proof
  fix trace-table
  assume table:
    trace-table  $\in$ 
      alpha-header-trace-table-candidates s fr f-fri-roots f-final
  then obtain final-state where state:
    alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
      trace-table final-state
  using alpha-header-trace-table-candidates-iff-state by blast
  have compatible:
    hash-maps-compatible final-state final-state0
  using pairwise[OF state state0] by simp
  have clean:
     $\neg$  hash-map-output-collision
      (hash-state-merge final-state final-state0)
  using pairwise[OF state state0] by simp
  have trace-table = trace-table0
    by (rule
      alpha-header-trace-table-candidate-states-compatible-unique-if-clean-merge
      [OF compatible clean state state0])
  then show trace-table  $\in$  {trace-table0}
    by simp
qed
then show ?thesis
  by blast
qed

lemma alpha-header-supported-trace-table-candidates-subset-singleton-if-pairwise-clean-merge:
  assumes pairwise:

```

```

 $\wedge$  trace-table final-state trace-table' final-state'.
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table final-state  $\impl$ 
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table' final-state'  $\impl$ 
  hash-maps-compatible final-state final-state'  $\wedge$ 
 $\neg$  hash-map-output-collision
    (hash-state-merge final-state final-state')
shows
 $\exists$  trace-table.
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
 $\subseteq \{ \text{trace-table} \}$ 
proof (cases
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final =
 $\{ \}$ )
case True
then show ?thesis
by (intro exI[of -  $\square$ ]) simp
next
case False
then obtain trace-table0 where table0:
  trace-table0  $\in$ 
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
by blast
then obtain final-state0 where state0:
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table0 final-state0
using alpha-header-supported-trace-table-candidates-iff-state by blast
have state0-plain:
  alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
  trace-table0 final-state0
by (rule alpha-header-supported-trace-table-candidate-state-imp-candidate-state
  [OF state0])
have subset:
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
 $\subseteq \{ \text{trace-table0} \}$ 
proof
fix trace-table
assume table:
  trace-table  $\in$ 
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
then obtain final-state where state:
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table final-state
using alpha-header-supported-trace-table-candidates-iff-state by blast
have state-plain:
  alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
  trace-table final-state
by (rule

```

```

      alpha-header-supported-trace-table-candidate-state-imp-candidate-state
      [OF state])
  have compatible:
    hash-maps-compatible final-state final-state0
    using pairwise[OF state state0] by simp
  have clean:
    ¬ hash-map-output-collision
    (hash-state-merge final-state final-state0)
    using pairwise[OF state state0] by simp
  have trace-table = trace-table0
    by (rule
      alpha-header-trace-table-candidate-states-compatible-unique-if-clean-merge
      [OF compatible clean state-plain state0-plain])
  then show trace-table ∈ {trace-table0}
    by simp
qed
then show ?thesis
  by blast
qed

```

lemma *alpha-header-supported-trace-table-candidates-subset-singleton-if-pairwise-no-value-conflict-clean-merge:*

assumes *pairwise:*

```

  ∧ trace-table final-state trace-table' final-state'.
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table final-state ⇒
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table' final-state' ⇒
  ¬ hash-map-value-conflict final-state final-state' ∧
  ¬ hash-map-output-collision
  (hash-state-merge final-state final-state')

```

shows

```

  ∃ trace-table.
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
  ⊆ {trace-table}

```

proof (rule

alpha-header-supported-trace-table-candidates-subset-singleton-if-pairwise-clean-merge)

fix *trace-table final-state trace-table' final-state'*

assume *candidate:*

```

  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table final-state

```

and *candidate':*

```

  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table' final-state'

```

have *no-conflict:*

```

  ¬ hash-map-value-conflict final-state final-state'
  using pairwise[OF candidate candidate'] by simp

```

have *compatible:*

```

  hash-maps-compatible final-state final-state'
  using no-conflict

```

```

unfolding hash-maps-compatible-iff-no-value-conflict .
have clean:
   $\neg$  hash-map-output-collision
    (hash-state-merge final-state final-state')
using pairwise[OF candidate candidate'] by simp
show hash-maps-compatible final-state final-state'  $\wedge$ 
   $\neg$  hash-map-output-collision
    (hash-state-merge final-state final-state')
using compatible clean by simp
qed

lemma alpha-header-supported-trace-table-candidates-subset-singleton-if-pairwise-merkle-clean-merge:
assumes pairwise:
   $\wedge$  trace-table final-state trace-table' final-state'.
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table final-state  $\implies$ 
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table' final-state'  $\implies$ 
  merkle-hash-maps-compatible final-state final-state'  $\wedge$ 
   $\neg$  hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
shows
   $\exists$  trace-table.
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
     $\subseteq$  {trace-table}
proof (cases
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final =
    {})
case True
then show ?thesis
  by (intro exI[of - []]) simp
next
case False
then obtain trace-table0 where table0:
  trace-table0  $\in$ 
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
by blast
then obtain final-state0 where state0:
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table0 final-state0
using alpha-header-supported-trace-table-candidates-iff-state by blast
have state0-plain:
  alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table0 final-state0
by (rule alpha-header-supported-trace-table-candidate-state-imp-candidate-state
    [OF state0])
have subset:
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
     $\subseteq$  {trace-table0}

```

proof
fix *trace-table*
assume *table*:
 trace-table $\in$
 alpha-header-supported-trace-table-candidates *s* *fr* *f-fri-roots* *f-final*
then obtain *final-state* **where** *state*:
 alpha-header-supported-trace-table-candidate-state *s* *fr* *f-fri-roots*
 f-final *trace-table* *final-state*
using *alpha-header-supported-trace-table-candidates-iff-state* **by** *blast*
have *state-plain*:
 alpha-header-trace-table-candidate-state *s* *fr* *f-fri-roots* *f-final*
 trace-table *final-state*
by (*rule*
 alpha-header-supported-trace-table-candidate-state-imp-candidate-state
 [*OF state*])
have *compatible*:
 merkle-hash-maps-compatible *final-state* *final-state0*
using *pairwise*[*OF state state0*] **by** *simp*
have *clean*:
 $\neg$ *hash-map-output-collision*
 (*merkle-hash-state-merge* *final-state* *final-state0*)
using *pairwise*[*OF state state0*] **by** *simp*
have *trace-table* = *trace-table0*
by (*rule*
 alpha-header-trace-table-candidate-states-merkle-compatible-unique-if-clean-merge
 [*OF compatible clean state-plain state0-plain*])
then show *trace-table* $\in$ {*trace-table0*}
by *simp*
qed
then show *?thesis*
by *blast*
qed

lemma *alpha-header-supported-trace-table-candidates-subset-singleton-if-pairwise-no-merkle-value-conflict-clean-merge*

assumes *pairwise*:

$\bigwedge$ *trace-table* *final-state* *trace-table'* *final-state'*.
 alpha-header-supported-trace-table-candidate-state *s* *fr* *f-fri-roots*
 f-final *trace-table* *final-state* $\implies$
 alpha-header-supported-trace-table-candidate-state *s* *fr* *f-fri-roots*
 f-final *trace-table'* *final-state'* $\implies$
 $\neg$ *merkle-hash-value-conflict* *final-state* *final-state'* $\wedge$
 $\neg$ *hash-map-output-collision*
 (*merkle-hash-state-merge* *final-state* *final-state'*)

shows

$\exists$ *trace-table*.
 alpha-header-supported-trace-table-candidates *s* *fr* *f-fri-roots* *f-final*
 $\subseteq$ {*trace-table*}

proof (*rule*

alpha-header-supported-trace-table-candidates-subset-singleton-if-pairwise-merkle-clean-merge)


```

fix trace-table final-state trace-table' final-state'
assume candidate:
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table final-state
and candidate':
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table' final-state'
have no-conflict:
  ¬ merkle-hash-value-conflict final-state final-state'
  using pairwise[OF candidate candidate'] by simp
have compatible:
  merkle-hash-maps-compatible final-state final-state'
  using no-conflict
  unfolding merkle-hash-maps-compatible-iff-no-value-conflict .
have clean:
  ¬ hash-map-output-collision
  (merkle-hash-state-merge final-state final-state')
  using pairwise[OF candidate candidate'] by simp
show merkle-hash-maps-compatible final-state final-state' ∧
  ¬ hash-map-output-collision
  (merkle-hash-state-merge final-state final-state')
  using compatible clean by simp
qed

lemma alpha-header-supported-trace-table-candidates-subset-singleton-if-common-clean-merkle-extension:
assumes clean: ¬ hash-map-output-collision u
and common-ext:
  ∧ trace-table final-state.
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table final-state ⇒
  merkle-hash-extends final-state u
shows
  ∃ trace-table.
  alpha-header-supported-trace-table-candidates s fr f-fri-roots f-final
  ⊆ {trace-table}
proof (rule
  alpha-header-supported-trace-table-candidates-subset-singleton-if-pairwise-no-merkle-value-conflict-clean-merge)
fix trace-table final-state trace-table' final-state'
assume candidate:
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table final-state
and candidate':
  alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
  f-final trace-table' final-state'
have ext: merkle-hash-extends final-state u
  by (rule common-ext[OF candidate])
have ext': merkle-hash-extends final-state' u
  by (rule common-ext[OF candidate'])
show ¬ merkle-hash-value-conflict final-state final-state' ∧

```

```

    ¬ hash-map-output-collision
      (merkle-hash-state-merge final-state final-state')
  by (rule common-clean-merkle-extension-imp-pairwise-merkle-clean
      [OF ext ext' clean])
qed

lemma alpha-header-supported-trace-table-candidate-states-distinct-imp-merkle-conflict-or-merge-collision:
  assumes candidate:
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table final-state
  and candidate':
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table' final-state'
  and distinct-tables: trace-table ≠ trace-table'
  shows
    merkle-hash-value-conflict final-state final-state' ∨
    hash-map-output-collision
      (merkle-hash-state-merge final-state final-state')
proof (rule ccontr)
  assume no-bad:
    ¬ (merkle-hash-value-conflict final-state final-state' ∨
        hash-map-output-collision
          (merkle-hash-state-merge final-state final-state'))
  have compatible:
    merkle-hash-maps-compatible final-state final-state'
  using no-bad unfolding merkle-hash-maps-compatible-iff-no-value-conflict
  by simp
  have clean:
    ¬ hash-map-output-collision
      (merkle-hash-state-merge final-state final-state')
  using no-bad by simp
  have candidate-plain:
    alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table final-state
  by (rule alpha-header-supported-trace-table-candidate-state-imp-candidate-state
      [OF candidate])
  have candidate-plain':
    alpha-header-trace-table-candidate-state s fr f-fri-roots f-final
    trace-table' final-state'
  by (rule alpha-header-supported-trace-table-candidate-state-imp-candidate-state
      [OF candidate'])
  have trace-table = trace-table'
  by (rule alpha-header-trace-table-candidate-states-merkle-compatible-unique-if-clean-merge
      [OF compatible clean candidate-plain candidate-plain'])
  then show False
  using distinct-tables by contradiction
qed

```

lemma alpha-header-supported-trace-table-candidates-not-singleton-imp-pairwise-merkle-bad:

assumes *not-unique*:
 $\neg (\exists \text{ trace-table.}$
 $\text{alpha-header-supported-trace-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final} \subseteq \{\text{trace-table}\})$

shows
 $\exists \text{ trace-table final-state trace-table' final-state'.$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final trace-table final-state} \wedge$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final trace-table' final-state'} \wedge$
 $\text{trace-table} \neq \text{trace-table'} \wedge$
 $(\text{merkle-hash-value-conflict final-state final-state'} \vee$
 $\text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge final-state final-state'}))$

proof –
let $?C =$
 $\text{alpha-header-supported-trace-table-candidates } s \text{ fr } f\text{-fri-roots } f\text{-final}$
have *nonempty*: $?C \neq \{\}$
using *not-unique* **by** *auto*
then obtain *trace-table* **where** $\text{table: trace-table} \in ?C$
by *blast*
have $\neg ?C \subseteq \{\text{trace-table}\}$
using *not-unique* **by** *blast*
then obtain *trace-table'* **where** table':
 $\text{trace-table'} \in ?C$
and *distinct-tables*: $\text{trace-table} \neq \text{trace-table'}$
by *auto*
from *table* **obtain** *final-state* **where** *candidate*:
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final trace-table final-state}$
using *alpha-header-supported-trace-table-candidates-iff-state* **by** *blast*
from *table'* **obtain** *final-state'* **where** *candidate'*:
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final trace-table' final-state'}$
using *alpha-header-supported-trace-table-candidates-iff-state* **by** *blast*
have *bad*:
 $\text{merkle-hash-value-conflict final-state final-state'} \vee$
 $\text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge final-state final-state'})$
by (*rule*
 $\text{alpha-header-supported-trace-table-candidate-states-distinct-imp-merkle-conflict-or-merge-collision}$
 $[\text{OF candidate candidate' distinct-tables}])$
show *?thesis*
using *candidate candidate' distinct-tables bad* **by** *blast*
qed

definition *alpha-header-supported-pairwise-merkle-bad*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $f \Rightarrow f \text{ list} \Rightarrow f \Rightarrow \text{bool}$

where

alpha-header-supported-pairwise-merkle-bad *s* *fr* *f-fri-roots* *f-final*
 $\longleftrightarrow$
 $(\exists \text{ trace-table } \text{final-state } \text{trace-table}' \text{ final-state}'.$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final } \text{trace-table } \text{final-state} \wedge$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final } \text{trace-table}' \text{ final-state}' \wedge$
 $\text{trace-table} \neq \text{trace-table}' \wedge$
 $(\text{merkle-hash-value-conflict } \text{final-state } \text{final-state}' \vee$
 $\text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge } \text{final-state } \text{final-state}'))))$

definition *alpha-supported-pairwise-merkle-bad*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

alpha-supported-pairwise-merkle-bad *s* $\longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final}.$
 $\text{alpha-header-supported-pairwise-merkle-bad } s \text{ fr } f\text{-fri-roots } f\text{-final})$

lemma *alpha-header-supported-candidate-unique-if-no-pairwise-merkle-bad*:

assumes *no-bad*:

$\neg \text{alpha-header-supported-pairwise-merkle-bad } s \text{ fr } f\text{-fri-roots } f\text{-final}$

shows

$\exists \text{ trace-table}.$
 $\text{alpha-header-supported-trace-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final} \subseteq \{\text{trace-table}\}$

proof (rule *ccontr*)

assume *not-unique*:

$\neg (\exists \text{ trace-table}.$
 $\text{alpha-header-supported-trace-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final} \subseteq \{\text{trace-table}\})$

have *alpha-header-supported-pairwise-merkle-bad* *s* *fr* *f-fri-roots* *f-final*

using

alpha-header-supported-trace-table-candidates-not-singleton-imp-pairwise-merkle-bad
 $[OF \text{ not-unique}]$

unfolding *alpha-header-supported-pairwise-merkle-bad-def*

by *blast*

then show *False*

using *no-bad* **by** *contradiction*

qed

lemma *alpha-header-supported-candidate-unique-if-no-global-pairwise-merkle-bad*:

assumes *no-bad*: $\neg \text{alpha-supported-pairwise-merkle-bad } s$

shows

$\exists \text{ trace-table}.$
 $\text{alpha-header-supported-trace-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final} \subseteq \{\text{trace-table}\}$

proof (rule *alpha-header-supported-candidate-unique-if-no-pairwise-merkle-bad*)

show $\neg$ *alpha-header-supported-pairwise-merkle-bad* *s* *fr* *f-fri-roots* *f-final*
using *no-bad* **unfolding** *alpha-supported-pairwise-merkle-bad-def* **by** *blast*
qed

lemma *accepted-with-bound-tables-common-roots*:

assumes *bound1*:

accepted-with-bound-tables *s* *out* *trace-table* *composition-table*

alphas *query-idxs*

and *bound2*:

accepted-with-bound-tables *s* *out* *trace-table'* *composition-table'*

alphas' *query-idxs'*

obtains *result* *final-state* *fr* *composition-root*

where *out* = *Some* (*result*, *final-state*)

and *merkle-root-binds-table* *fr* *trace-table* *final-state*

and *merkle-root-binds-table* *fr* *trace-table'* *final-state*

and *merkle-root-binds-table* *composition-root* *composition-table* *final-state*

and *merkle-root-binds-table* *composition-root* *composition-table'* *final-state*

proof –

from *bound1* **obtain** *result1* *final-state1* *fr* *f-fri-roots* *f-final* *dg*

composition-fri-roots *final* *rest* **where**

out1: *out* = *Some* (*result1*, *final-state1*)

and *header1*:

verifier-header-transcript *s* *fr* *f-fri-roots* *f-final* *alphas* *dg*

composition-fri-roots *final* *rest*

and *trace-bind1*:

merkle-root-binds-table *fr* *trace-table* *final-state1*

and *comp-bind1*:

merkle-root-binds-table (*hd* *composition-fri-roots*)

composition-table *final-state1*

unfolding *accepted-with-bound-tables-def* **by** *blast*

from *bound2* **obtain** *result2* *final-state2* *fr'* *f-fri-roots'* *f-final'* *dg'*

composition-fri-roots' *final'* *rest'* **where**

out2: *out* = *Some* (*result2*, *final-state2*)

and *header2*:

verifier-header-transcript *s* *fr'* *f-fri-roots'* *f-final'* *alphas'* *dg'*

composition-fri-roots' *final'* *rest'*

and *trace-bind2*:

merkle-root-binds-table *fr'* *trace-table'* *final-state2*

and *comp-bind2*:

merkle-root-binds-table (*hd* *composition-fri-roots'*)

composition-table' *final-state2*

unfolding *accepted-with-bound-tables-def* **by** *blast*

have *final-state-eq*: *final-state2* = *final-state1*

using *out1* *out2* **by** *simp*

have *roots-eq*:

fr' = *fr* $\wedge$ *composition-fri-roots'* = *composition-fri-roots*

using *verifier-header-transcript-unique*[*OF* *header1* *header2*] **by** *simp*

have *trace-bind2'*:

merkle-root-binds-table *fr* *trace-table'* *final-state1*

```

    using trace-bind2 roots-eq final-state-eq by simp
  have comp-bind2':
    merkle-root-binds-table (hd composition-fri-roots)
      composition-table' final-state1
    using comp-bind2 roots-eq final-state-eq by simp
  show ?thesis
    by (rule that[OF out1 trace-bind1 trace-bind2' comp-bind1 comp-bind2'])
qed

lemma accepted-with-bound-tables-trace-table-unique-or-hash-collision-bad:
  assumes bound1:
    accepted-with-bound-tables s out trace-table composition-table
    alphas query-idxs
  and bound2:
    accepted-with-bound-tables s out trace-table' composition-table'
    alphas' query-idxs'
  shows trace-table = trace-table'  $\vee$  hash-map-output-collision-bad s out
proof (cases trace-table = trace-table')
  case True
  then show ?thesis by simp
next
  case neg: False
  from accepted-with-bound-tables-common-roots[OF bound1 bound2]
  obtain result final-state fr composition-root where
    out: out = Some (result, final-state)
    and bind1: merkle-root-binds-table fr trace-table final-state
    and bind2: merkle-root-binds-table fr trace-table' final-state
    by blast
  have same-len: length trace-table = length trace-table'
  using accepted-with-bound-tables-shapes(1)[OF bound1]
    accepted-with-bound-tables-shapes(1)[OF bound2]
  by simp
  have nonempty: trace-table  $\neq$  []
  using accepted-with-bound-tables-shapes(1)[OF bound1]
    eval-domain-nontrivial
  by auto
  have hash-map-output-collision-bad s out
  by (rule accepted-inconsistent-same-length-merkle-bindings-imp-hash-collision-bad
    [OF out bind1 bind2 same-len nonempty neg])
  then show ?thesis by simp
qed

```

```

lemma accepted-with-bound-tables-composition-table-unique-or-hash-collision-bad:
  assumes bound1:
    accepted-with-bound-tables s out trace-table composition-table
    alphas query-idxs
  and bound2:
    accepted-with-bound-tables s out trace-table' composition-table'
    alphas' query-idxs'

```

```

shows composition-table = composition-table'  $\vee$ 
  hash-map-output-collision-bad s out
proof (cases composition-table = composition-table')
  case True
  then show ?thesis by simp
next
  case neg: False
  from accepted-with-bound-tables-common-roots[OF bound1 bound2]
  obtain result final-state fr composition-root where
    out: out = Some (result, final-state)
  and bind1:
    merkle-root-binds-table composition-root composition-table final-state
  and bind2:
    merkle-root-binds-table composition-root composition-table' final-state
  by blast
  have same-len: length composition-table = length composition-table'
  using accepted-with-bound-tables-shapes(2)[OF bound1]
    accepted-with-bound-tables-shapes(2)[OF bound2]
  by simp
  have nonempty: composition-table  $\neq$  []
  using accepted-with-bound-tables-shapes(2)[OF bound1]
    eval-domain-nontrivial
  by auto
  have hash-map-output-collision-bad s out
  by (rule accepted-inconsistent-same-length-merkle-bindings-imp-hash-collision-bad
    [OF out bind1 bind2 same-len nonempty neg])
  then show ?thesis by simp
qed

lemma accepted-with-bound-tables-tables-unique-or-hash-collision-bad:
  assumes bound1:
    accepted-with-bound-tables s out trace-table composition-table
    alphas query-idxs
  and bound2:
    accepted-with-bound-tables s out trace-table' composition-table'
    alphas' query-idxs'
  shows
    (trace-table = trace-table'  $\wedge$  composition-table = composition-table')  $\vee$ 
    hash-map-output-collision-bad s out
proof –
  have trace-unique:
    trace-table = trace-table'  $\vee$  hash-map-output-collision-bad s out
  by (rule accepted-with-bound-tables-trace-table-unique-or-hash-collision-bad
    [OF bound1 bound2])
  have comp-unique:
    composition-table = composition-table'  $\vee$ 
    hash-map-output-collision-bad s out
  by (rule accepted-with-bound-tables-composition-table-unique-or-hash-collision-bad
    [OF bound1 bound2])

```

```

show ?thesis
  using trace-unique comp-unique by blast
qed

lemma accepted-with-bound-tables-trace-table-unique-if-no-hash-collision-bad:
assumes no-collision:  $\neg$  hash-map-output-collision-bad  $s$  out
and bound1:
  accepted-with-bound-tables  $s$  out trace-table composition-table
  alphas query-idxs
and bound2:
  accepted-with-bound-tables  $s$  out trace-table' composition-table'
  alphas' query-idxs'
shows trace-table = trace-table'
using accepted-with-bound-tables-trace-table-unique-or-hash-collision-bad
  [OF bound1 bound2] no-collision
by blast

lemma accepted-with-bound-tables-composition-table-unique-if-no-hash-collision-bad:
assumes no-collision:  $\neg$  hash-map-output-collision-bad  $s$  out
and bound1:
  accepted-with-bound-tables  $s$  out trace-table composition-table
  alphas query-idxs
and bound2:
  accepted-with-bound-tables  $s$  out trace-table' composition-table'
  alphas' query-idxs'
shows composition-table = composition-table'
using accepted-with-bound-tables-composition-table-unique-or-hash-collision-bad
  [OF bound1 bound2] no-collision
by blast

definition accepted-bound-table-ambiguity-bad
:: ( $f$ ,  $a$ ) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ( $f$ ,  $a$ ) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  accepted-bound-table-ambiguity-bad  $s$  out  $\longleftrightarrow$ 
  ( $\exists$  trace-table composition-table alphas query-idxs
    trace-table' composition-table' alphas' query-idxs'.
    accepted-with-bound-tables  $s$  out trace-table composition-table
    alphas query-idxs  $\wedge$ 
    accepted-with-bound-tables  $s$  out trace-table' composition-table'
    alphas' query-idxs'  $\wedge$ 
    (trace-table  $\neq$  trace-table'  $\vee$ 
    composition-table  $\neq$  composition-table'))

lemma accepted-bound-table-ambiguity-bad-imp-hash-map-output-collision-bad:
assumes ambiguity: accepted-bound-table-ambiguity-bad  $s$  out
shows hash-map-output-collision-bad  $s$  out
proof —
from ambiguity obtain trace-table composition-table alphas query-idxs

```



```

    trace-table' composition-table' alphas' query-idxs' where
bound:
    accepted-with-bound-tables s out trace-table composition-table
    alphas query-idxs
and bound':
    accepted-with-bound-tables s out trace-table' composition-table'
    alphas' query-idxs'
and distinct:
    trace-table  $\neq$  trace-table'  $\vee$ 
    composition-table  $\neq$  composition-table'
unfolding accepted-bound-table-ambiguity-bad-def by blast
have
    (trace-table = trace-table'  $\wedge$ 
     composition-table = composition-table')  $\vee$ 
    hash-map-output-collision-bad s out
by (rule accepted-with-bound-tables-tables-unique-or-hash-collision-bad
     [OF bound bound'])
then show ?thesis
    using distinct by blast
qed

lemma accepted-bound-table-ambiguity-bad-mono-hash-map-output-collision-bad:
    wp-event verify-monad (accepted-bound-table-ambiguity-bad s) s  $\leq$ 
    wp-event verify-monad (hash-map-output-collision-bad s) s
by (rule wp-event-mono)
    (rule
     accepted-bound-table-ambiguity-bad-imp-hash-map-output-collision-bad)

end

end

theory Soundness-Execution-Query
    imports Soundness-Merkle
begin

Verifier query-round and authenticated-opening execution lemmas.

context soundness
begin

lemma receive-random-field-element-outcome:
    assumes outcome: Some (a, t)  $\in$  set-dist (execute receive-random-field-element s)
shows
    s  $\leq$  t  $\wedge$ 
    PState t = PState s  $\wedge$ 
    PTranscript t = PTranscript s  $\wedge$ 
    fmlookup (HashMap t) (FiatShamirChallenge (PState s)) = Some a
using protocol-receive-random-field-element-outcome

```

```

    [OF outcome[unfolded receive-random-field-element-def]]
  by simp

lemma throw-no-outcome:
  Some (x, t)  $\notin$  set-dist (execute throw s)
  unfolding set-dist-def throw.rep-eq dist-throw-def dist-delta-dist delta-map-def
  by simp

lemma read-outcome:
  assumes outcome: Some (x, t)  $\in$  set-dist (execute read s)
  shows
     $\exists$  rest.
      PTranscript s = x # rest  $\wedge$ 
      PState t = concat (PState s) x  $\wedge$ 
      PTranscript t = rest  $\wedge$ 
      s  $\leq$  t  $\wedge$ 
      PTraceFriCounter t = PTraceFriCounter s  $\wedge$ 
      PCompositionFriCounter t = PCompositionFriCounter s  $\wedge$ 
      PAlphaCounter t = PAlphaCounter s  $\wedge$ 
      PQueryCounter t = PQueryCounter s
  proof -
    obtain rest where tr-s: PTranscript s = x # rest
    proof -
      have PTranscript s  $\neq$  []
        using outcome unfolding read-def protocol-read-def assert-def
        by (cases PTranscript s) (auto simp: throw-no-outcome elim!: set-dist-bindE)
      then obtain y ys where ys: PTranscript s = y # ys
        by (cases PTranscript s) auto
      have x-y: x = y
        using read-nonempty-outcome[OF outcome ys] by simp
      show ?thesis
        using that ys x-y by auto
    qed
    have t-res:
      t = s(PState := concat (PState s) x, PTranscript := rest)
      using read-nonempty-outcome[OF outcome tr-s] by simp
    have s  $\leq$  t
      unfolding t-res less-eq-hash-ext-def less-eq-fmap-def by simp
    then show ?thesis
      using tr-s t-res by auto
    qed

lemma read-preserves-hash-map:
  assumes outcome: Some (x, t)  $\in$  set-dist (execute read s)
  shows HashMap t = HashMap s
  proof -
    obtain rest where tr-s: PTranscript s = x # rest
    proof -
      have PTranscript s  $\neq$  []

```

```

    using outcome unfolding read-def protocol-read-def assert-def
    by (cases PTranscript s) (auto simp: throw-no-outcome elim!: set-dist-bindE)
then obtain y ys where ys: PTranscript s = y # ys
    by (cases PTranscript s) auto
have x-y: x = y
    using read-nonempty-outcome[OF outcome ys] by simp
show ?thesis
    using that ys x-y by auto
qed
show ?thesis
    using read-nonempty-outcome[OF outcome tr-s] by simp
qed

```

```

lemma read-preserves-alpha-future-fresh:
  assumes future: alpha-future-fresh s
    and outcome: Some (x, t) ∈ set-dist (execute read s)
  shows alpha-future-fresh t
proof -
  have hash-eq: HashMap t = HashMap s
    by (rule read-preserves-hash-map[OF outcome])
  from read-outcome[OF outcome] have counter-eq:
    PAlphaCounter t = PAlphaCounter s
    by blast
  show ?thesis
    using future hash-eq counter-eq
    unfolding alpha-future-fresh-def by simp
qed

```

```

lemma read-preserves-trace-fri-future-fresh:
  assumes future: trace-fri-future-fresh s
    and outcome: Some (x, t) ∈ set-dist (execute read s)
  shows trace-fri-future-fresh t
proof -
  have hash-eq: HashMap t = HashMap s
    by (rule read-preserves-hash-map[OF outcome])
  from read-outcome[OF outcome] have counter-eq:
    PTraceFriCounter t = PTraceFriCounter s
    by blast
  show ?thesis
    using future hash-eq counter-eq
    unfolding trace-fri-future-fresh-def by simp
qed

```

```

lemma read-preserves-composition-fri-future-fresh:
  assumes future: composition-fri-future-fresh s
    and outcome: Some (x, t) ∈ set-dist (execute read s)
  shows composition-fri-future-fresh t
proof -
  have hash-eq: HashMap t = HashMap s

```

```

    by (rule read-preserves-hash-map[OF outcome])
  from read-outcome[OF outcome] have counter-eq:
    PCompositionFriCounter t = PCompositionFriCounter s
  by blast
show ?thesis
  using future hash-eq counter-eq
  unfolding composition-fri-future-fresh-def by simp
qed

lemma read-preserves-query-future-fresh:
  assumes future: query-future-fresh s
    and outcome: Some (x, t) ∈ set-dist (execute read s)
  shows query-future-fresh t
proof -
  have hash-eq: HashMap t = HashMap s
  by (rule read-preserves-hash-map[OF outcome])
  from read-outcome[OF outcome] have counter-eq:
    PQueryCounter t = PQueryCounter s
  by blast
show ?thesis
  using future hash-eq counter-eq
  unfolding query-future-fresh-def by simp
qed

lemma ntimes-read-any-outcome:
  assumes outcome: Some (xs, t) ∈ set-dist (execute (ntimes read n) s)
  shows
    length xs = n ∧
    PTranscript s = xs @ PTranscript t ∧
    PState t = foldl concat (PState s) xs ∧
    s ≤ t ∧
    PTraceFriCounter t = PTraceFriCounter s ∧
    PCompositionFriCounter t = PCompositionFriCounter s ∧
    PAlphaCounter t = PAlphaCounter s ∧
    PQueryCounter t = PQueryCounter s
  using outcome
proof (induction n arbitrary: s xs t)
  case 0
  then show ?case
    by (simp add: hash-ext-refl)
next
  case (Suc n)
  from Suc.premis obtain x xs' s1 where
    read-x: Some (x, s1) ∈ set-dist (execute read s)
  and reads:
    Some (xs', t) ∈ set-dist (execute (ntimes read n) s1)
  and xs-eq: xs = x # xs'
  by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-x] obtain rest1 where

```

```



```

```

lemma ntimes-read-preserves-lookup:
assumes outcome: Some (xs, t) ∈ set-dist (execute (ntimes read n) s)
shows fmlookup (HashMap t) key = fmlookup (HashMap s) key
proof (rule ntimes-preserves-lookup[OF - outcome])
fix x s t
assume read-x: Some (x, t) ∈ set-dist (execute read s)
show fmlookup (HashMap t) key = fmlookup (HashMap s) key
using read-preserves-hash-map[OF read-x] by simp
qed

```

```

lemma check-authentication-path-hash-extends:
assumes outcome:
Some (hh, t) ∈
set-dist (execute (check-authentication-path len i v path) s)
shows s ≤ t
using outcome
unfolding check-authentication-path-def protocol-check-authentication-path-def
proof (induction path arbitrary: len i v s hh t)
case Nil
then have hash-step:
Some (hh, t) ∈ set-dist (execute (hash (MerkleLeaf v)) s)
by simp

```

```

show ?case
  by (rule hash-outcome(1)[OF hash-step])
next
case (Cons a path)
show ?case
proof (cases i < len div 2)
  case True
  from Cons.premis obtain x u where
    rec:
      Some (x, u) ∈
        set-dist (execute
          (protocol-merkle.check-authentication-path
            (len div 2) i (MerkleLeaf v) path) s)
    and hash:
      Some (hh, t) ∈ set-dist (execute (hash (MerkleNode x a)) u)
    using True by (auto elim!: set-dist-bindE)
  have s ≤ u
    by (rule Cons.IH[OF rec])
  moreover have u ≤ t
    using hash-outcome(1)[OF hash] .
  ultimately show ?thesis
    by (rule hash-ext-trans)
next
case False
  from Cons.premis obtain x u where
    rec:
      Some (x, u) ∈
        set-dist (execute
          (protocol-merkle.check-authentication-path
            (len div 2) (i - len div 2) (MerkleLeaf v) path) s)
    and hash:
      Some (hh, t) ∈ set-dist (execute (hash (MerkleNode a x)) u)
    using False by (auto elim!: set-dist-bindE)
  have s ≤ u
    by (rule Cons.IH[OF rec])
  moreover have u ≤ t
    using hash-outcome(1)[OF hash] .
  ultimately show ?thesis
    by (rule hash-ext-trans)
qed
qed

```

lemma check-decommit-on-query-step-outcome:

assumes outcome:

```

Some (qh, t) ∈ set-dist (execute
  (do {
    qh ← read;
    let len = scale * clength;
    qh-path ← ntimes read (floor-log len);

```

```

    ap ← check-authentication-path len i qh qh-path;
    assert (ap = fr);
    return qh
  }) s)
shows
  ∃ path.
    length path = floor-log (scale * clength) ∧
    PTranscript s = ([qh] @ path) @ PTranscript t ∧
    PState t = foldl concat (PState s) ([qh] @ path) ∧
    s ≤ t ∧
    PQueryCounter t = PQueryCounter s
proof -
  let ?len = scale * clength
  from outcome obtain s1 path s2 ap s3 s4 where
    read-gh: Some (qh, s1) ∈ set-dist (execute read s)
  and read-path:
    Some (path, s2) ∈
      set-dist (execute (ntimes read (floor-log ?len)) s1)
  and check:
    Some (ap, s3) ∈
      set-dist (execute (check-authentication-path ?len i qh path) s2)
  and assert-ap: Some ((), s4) ∈ set-dist (execute (assert (ap = fr)) s3)
  and ret-gh: Some (qh, t) ∈ set-dist (execute (return qh) s4)
  by (auto simp: Let-def elim!: set-dist-bindE)
  from read-outcome[OF read-gh] obtain rest1 where
    tr-s: PTranscript s = qh # rest1
  and st-s1: PState s1 = concat (PState s) qh
  and tr-s1: PTranscript s1 = rest1
  and ext-s-s1: s ≤ s1
  and query-count-s1: PQueryCounter s1 = PQueryCounter s
  by blast
  have path-res:
    length path = floor-log ?len ∧
    PTranscript s1 = path @ PTranscript s2 ∧
    PState s2 = foldl concat (PState s1) path ∧
    s1 ≤ s2 ∧
    PQueryCounter s2 = PQueryCounter s1
  using ntimes-read-any-outcome[OF read-path] by simp
  have s2-s3: s2 ≤ s3
  by (rule check-authentication-path-hash-extends[OF check])
  have st-s3: PState s3 = PState s2
  using check-authentication-path-preserves-channel(1)[OF check] .
  have tr-s3: PTranscript s3 = PTranscript s2
  using check-authentication-path-preserves-channel(2)[OF check] .
  have query-count-s3: PQueryCounter s3 = PQueryCounter s2
  using check-authentication-path-preserves-channel(6)[OF check] .
  have ap-eq: ap = fr
  using assert-ap unfolding assert-def
  by (cases ap = fr) (auto simp: throw-no-outcome)

```

```

have s4-eq: s4 = s3
  using assert-ap ap-eq unfolding assert-def by simp
have t-eq: t = s4
  using ret-gh by simp
have s ≤ t
  using ext-s-s1 path-res s2-s3 unfolding t-eq s4-eq
  by (meson hash-ext-trans)
then show ?thesis
  using tr-s st-s1 tr-s1 path-res st-s3 tr-s3 query-count-s1
  query-count-s3 s4-eq t-eq
  by auto
qed

```

definition *query-decommitment-step*

where

```

query-decommitment-step fr i =
  (do {
    gh ← read;
    let len = scale * clength;
    gh-path ← ntimes read (floor-log len);
    ap ← check-authentication-path len i gh gh-path;
    assert (ap = fr);
    return gh
  })

```

lemma *query-decommitment-step-outcome:*

assumes *outcome:*

Some (gh, t) ∈ set-dist (execute (query-decommitment-step fr i) s)

shows

```

∃ path.
  length path = floor-log (scale * clength) ∧
  PTranscript s = ([gh] @ path) @ PTranscript t ∧
  PState t = foldl concat (PState s) ([gh] @ path) ∧
  s ≤ t ∧
  PQueryCounter t = PQueryCounter s

```

using *check-decommit-on-query-step-outcome[OF outcome[unfolded query-decommitment-step-def]]*

.

lemma *mmap-query-decommitment-steps-outcome:*

assumes *outcome:*

Some (leaves, t) ∈ set-dist (execute (mmap (map (query-decommitment-step fr) idxs)) s)

shows

```

∃ paths chunk.
  length leaves = length idxs ∧
  length paths = length idxs ∧
  chunk = List.concat (map (λ(leaf, path). [leaf] @ path) (zip leaves paths)) ∧
  (∀ path ∈ set paths. length path = floor-log (scale * clength)) ∧
  PTranscript s = chunk @ PTranscript t ∧

```



```

    PState t = foldl concat (PState s) chunk ∧
    s ≤ t ∧
    PQueryCounter t = PQueryCounter s
  using outcome
proof (induction idxs arbitrary: s leaves t)
  case Nil
  then show ?case
    by (intro exI[of - []]) (simp add: hash-ext-refl)
next
  case (Cons i idxs)
  from Cons.premis obtain leaf leaves' s1 where
    head:
      Some (leaf, s1) ∈
        set-dist (execute (query-decommitment-step fr i) s)
    and tail:
      Some (leaves', t) ∈
        set-dist (execute (mmap (map (query-decommitment-step fr) idxs)) s1)
    and leaves-eq: leaves = leaf # leaves'
    by (auto elim!: set-dist-bindE)
  from query-decommitment-step-outcome[OF head] obtain path where
    path-len: length path = floor-log (scale * clength)
    and tr-s: PTranscript s = ([leaf] @ path) @ PTranscript s1
    and st-s1: PState s1 = foldl concat (PState s) ([leaf] @ path)
    and ext-s-s1: s ≤ s1
    and query-count-s1: PQueryCounter s1 = PQueryCounter s
    by blast
  from Cons.IH[OF tail] obtain paths chunk-tail where
    len-leaves': length leaves' = length idxs
    and len-paths: length paths = length idxs
    and chunk-tail:
      chunk-tail =
        List.concat (map (λ(leaf, path). [leaf] @ path) (zip leaves' paths))
    and paths-len:
      ∀ path ∈ set paths. length path = floor-log (scale * clength)
    and tr-s1: PTranscript s1 = chunk-tail @ PTranscript t
    and st-t: PState t = foldl concat (PState s1) chunk-tail
    and ext-s1-t: s1 ≤ t
    and query-count-t: PQueryCounter t = PQueryCounter s1
    by blast
  let ?paths = path # paths
  let ?chunk = ([leaf] @ path) @ chunk-tail
  have s ≤ t
    using ext-s-s1 ext-s1-t by (rule hash-ext-trans)
  then show ?case
    apply (intro exI[of - ?paths] exI[of - ?chunk])
    using leaves-eq path-len tr-s tr-s1 st-s1 st-t len-leaves' len-paths
      chunk-tail paths-len query-count-s1 query-count-t
    by simp
qed

```

lemma *check-decommit-on-query-outcome*:

assumes *outcome*:

Some (*leaves*, *t*) ∈ *set-dist* (*execute* (*mmap* (*check-decommit-on-query* *fr idx*)) *s*)

shows

∃ *paths chunk*.
query-decommitment-transcript idx leaves paths chunk ∧
PTranscript s = chunk @ PTranscript t ∧
PState t = foldl concat (PState s) chunk ∧
s ≤ t ∧
PQueryCounter t = PQueryCounter s

proof –

have *map-eq*:

check-decommit-on-query fr idx =
map (query-decommitment-step fr) (powers-scaled idx)

unfolding *check-decommit-on-query-def query-decommitment-step-def* **by** *simp*

from *mmap-query-decommitment-steps-outcome[OF outcome[unfolded map-eq]]*

obtain *paths chunk* **where**

len-leaves: length leaves = length (powers-scaled idx)
and len-paths: length paths = length (powers-scaled idx)
and chunk:
chunk = List.concat
(map (λ(leaf, path). [leaf] @ path) (zip leaves paths))
and paths-len:
*∀ path ∈ set paths. length path = floor-log (scale * clength)*
and tr-s: PTranscript s = chunk @ PTranscript t
and st-t: PState t = foldl concat (PState s) chunk
and ext: s ≤ t
and query-count-t: PQueryCounter t = PQueryCounter s
by *blast*

have *qtr: query-decommitment-transcript idx leaves paths chunk*

using *len-leaves len-paths chunk paths-len*
unfolding *query-decommitment-transcript-def*
by (*simp add: mult.commute case-prod-unfold*)

show *?thesis*

using *qtr tr-s st-t ext query-count-t* **by** *blast*

qed

lemma *query-decommitment-step-preserves-query-lookup*:

assumes *outcome*:

Some (*qh*, *t*) ∈ *set-dist* (*execute* (*query-decommitment-step fr i*) *s*)

shows

fmlookup (HashMap t) (QueryIndexChallenge j x) =
fmlookup (HashMap s) (QueryIndexChallenge j x)

proof –

let *?len = scale * clength*

from *outcome* **obtain** *s1 path s2 ap s3 s4* **where**

read-qh: Some (*qh*, *s1*) ∈ *set-dist* (*execute read s*)

```

and read-path:
  Some (path, s2) ∈
    set-dist (execute (ntimes read (floor-log ?len)) s1)
and check:
  Some (ap, s3) ∈
    set-dist (execute (check-authentication-path ?len i qh path) s2)
and assert-ap: Some ((), s4) ∈ set-dist (execute (assert (ap = fr)) s3)
and ret-qh: Some (qh, t) ∈ set-dist (execute (return qh) s4)
unfolding query-decommitment-step-def
by (auto simp: Let-def elim!: set-dist-bindE)
have lookup-s1:
  fmlookup (HashMap s1) (QueryIndexChallenge j x) =
    fmlookup (HashMap s) (QueryIndexChallenge j x)
  using read-preserves-hash-map[OF read-qh] by simp
have lookup-s2:
  fmlookup (HashMap s2) (QueryIndexChallenge j x) =
    fmlookup (HashMap s1) (QueryIndexChallenge j x)
  by (rule ntimes-read-preserves-lookup[OF read-path])
have lookup-s3:
  fmlookup (HashMap s3) (QueryIndexChallenge j x) =
    fmlookup (HashMap s2) (QueryIndexChallenge j x)
  by (rule check-authentication-path-preserves-challenge-lookups(2)
    [OF check])
have s4-eq: s4 = s3
  using assert-ap unfolding assert-def
  by (cases ap = fr) (auto simp: throw-no-outcome)
have t-eq: t = s4
  using ret-qh by simp
show ?thesis
  using lookup-s1 lookup-s2 lookup-s3 s4-eq t-eq by simp
qed

```

```

lemma mmap-query-decommitment-steps-preserves-query-lookup:
assumes outcome:
  Some (leaves, t) ∈
    set-dist (execute (mmap (map (query-decommitment-step fr) idxs)) s)
shows
  fmlookup (HashMap t) (QueryIndexChallenge j x) =
    fmlookup (HashMap s) (QueryIndexChallenge j x)
using outcome
proof (induction idxs arbitrary: s leaves t)
case Nil
then show ?case
  by simp
next
case (Cons i idxs)
from Cons.prems obtain leaf leaves' s1 where
  head:
    Some (leaf, s1) ∈

```

```

      set-dist (execute (query-decommitment-step fr i) s)
and tail:
  Some (leaves', t) ∈
    set-dist (execute (mmap (map (query-decommitment-step fr) idxs)) s1)
by (auto elim!: set-dist-bindE)
have head-pres:
  fmlookup (HashMap s1) (QueryIndexChallenge j x) =
  fmlookup (HashMap s) (QueryIndexChallenge j x)
by (rule query-decommitment-step-preserves-query-lookup[OF head])
have tail-pres:
  fmlookup (HashMap t) (QueryIndexChallenge j x) =
  fmlookup (HashMap s1) (QueryIndexChallenge j x)
by (rule Cons.IH[OF tail])
show ?case
  using head-pres tail-pres by simp
qed

lemma check-decommit-on-query-preserves-query-lookup:
  assumes outcome:
    Some (leaves, t) ∈
      set-dist (execute (mmap (check-decommit-on-query fr idx)) s)
  shows
    fmlookup (HashMap t) (QueryIndexChallenge j x) =
    fmlookup (HashMap s) (QueryIndexChallenge j x)
proof -
  have map-eq:
    check-decommit-on-query fr idx =
    map (query-decommitment-step fr) (powers-scaled idx)
  unfolding check-decommit-on-query-def query-decommitment-step-def by simp
  show ?thesis
  by (rule mmap-query-decommitment-steps-preserves-query-lookup
    [OF outcome[unfolded map-eq]])
qed

```

definition *fri-layer-opening-step*

where

```

fri-layer-opening-step bf st =
  (case bf of (b, f) ⇒
    case st of (i, x, len, pw) ⇒
      do {
        xp ← read;
        xp' ← ntimes read (floor-log len);
        xn ← read;
        xn' ← ntimes read (floor-log len);
        assert (xp = x);
        ap ← check-authentication-path len i xp xp';
        assert (ap = f);
        let sidx = (i + len div 2) mod len;
        ap ← check-authentication-path len sidx xn xn';
      }
  )

```

```

    assert (ap = f);
    let gp = (xp + xn) div 2;
    let hp = (xp - xn) div (2*((h ^ i) * shift) ^ pw);
    let x = gp + b * hp;
    return (i mod (len div 2), x, len div 2, pw+pw)
  })

```

definition *fri-layer-opening-finish*

where

```

fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path =
  do {
    assert (xp = x);
    ap ← check-authentication-path len i xp xp-path;
    assert (ap = f);
    let idx = (i + len div 2) mod len;
    ap ← check-authentication-path len idx xn xn-path;
    assert (ap = f);
    let gp = (xp + xn) div 2;
    let hp = (xp - xn) div (2*((h ^ i) * shift) ^ pw);
    let x = gp + b * hp;
    return (i mod (len div 2), x, len div 2, pw+pw)
  }

```

lemma *fri-layer-opening-step-unfold-finish*:

```

fri-layer-opening-step (b, f) (i, x, len, pw) =
  do {
    xp ← read;
    xp-path ← ntimes read (floor-log len);
    xn ← read;
    xn-path ← ntimes read (floor-log len);
    fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path
  }

```

unfolding *fri-layer-opening-step-def* *fri-layer-opening-finish-def*
by *simp*

lemma *fri-layer-opening-step-outcome*:

assumes *outcome*:

```

Some (out, t) ∈
  set-dist (execute (fri-layer-opening-step (b, f) (i, x, len, pw)) s)

```

shows

```

∃ xp xp-path xn xn-path x'.
  out = (i mod (len div 2), x', len div 2, pw + pw) ∧
  xp = x ∧
  x' = fri-fold-value b xp xn
    (fri-fold-denominator ((h ^ i) * shift) pw) ∧
  fri-layer-opening-chunk len xp xp-path xn xn-path
    ([xp] @ xp-path @ [xn] @ xn-path) ∧
  PTranscript s = ([xp] @ xp-path @ [xn] @ xn-path) @ PTranscript t ∧
  PState t = foldl concat (PState s) ([xp] @ xp-path @ [xn] @ xn-path) ∧

```

$s \leq t \wedge$
 $PQueryCounter\ t = PQueryCounter\ s$
proof –
from *outcome* **obtain** $xp\ s1\ xp\text{-}path\ s2\ xn\ s3\ xn\text{-}path\ s4\ s5\ ap1\ s6\ s7$
 $ap2\ s8\ s9$ **where**
read-xp: $Some\ (xp,\ s1) \in set\text{-}dist\ (execute\ read\ s)$
and *read-xp-path*:
 $Some\ (xp\text{-}path,\ s2) \in$
 $set\text{-}dist\ (execute\ (ntimes\ read\ (floor\text{-}log\ len))\ s1)$
and *read-xn*: $Some\ (xn,\ s3) \in set\text{-}dist\ (execute\ read\ s2)$
and *read-xn-path*:
 $Some\ (xn\text{-}path,\ s4) \in$
 $set\text{-}dist\ (execute\ (ntimes\ read\ (floor\text{-}log\ len))\ s3)$
and *assert-xp*: $Some\ ((),\ s5) \in set\text{-}dist\ (execute\ (assert\ (xp = x))\ s4)$
and *check1*:
 $Some\ (ap1,\ s6) \in$
 $set\text{-}dist\ (execute\ (check\text{-}authentication\text{-}path\ len\ i\ xp\ xp\text{-}path)\ s5)$
and *assert-ap1*: $Some\ ((),\ s7) \in set\text{-}dist\ (execute\ (assert\ (ap1 = f))\ s6)$
and *check2*:
 $Some\ (ap2,\ s8) \in$
 $set\text{-}dist\ (execute$
 $(check\text{-}authentication\text{-}path\ len\ ((i + len\ div\ 2)\ mod\ len)\ xn\ xn\text{-}path)\ s7)$
and *assert-ap2*: $Some\ ((),\ s9) \in set\text{-}dist\ (execute\ (assert\ (ap2 = f))\ s8)$
and *ret*:
 $Some\ (out,\ t) \in$
 $set\text{-}dist\ (execute$
 $(return$
 $(i\ mod\ (len\ div\ 2),$
 $(xp + xn)\ div\ 2 +$
 $b * ((xp - xn)\ div\ (2 * ((h \wedge i) * shift) \wedge pw)),$
 $len\ div\ 2,\ pw + pw))\ s9)$
unfolding *fri-layer-opening-step-def*
by (*auto simp: Let-def elim!: set-dist-bindE*)
from *read-outcome*[*OF read-xp*] **obtain** *rest1* **where**
tr-s: $PTranscript\ s = xp \# rest1$
and *st-s1*: $PState\ s1 = concat\ (PState\ s)\ xp$
and *tr-s1*: $PTranscript\ s1 = rest1$
and *ext-s-s1*: $s \leq s1$
and *query-count-s1*: $PQueryCounter\ s1 = PQueryCounter\ s$
by *blast*
have *xp-path-res*:
 $length\ xp\text{-}path = floor\text{-}log\ len \wedge$
 $PTranscript\ s1 = xp\text{-}path\ @\ PTranscript\ s2 \wedge$
 $PState\ s2 = foldl\ concat\ (PState\ s1)\ xp\text{-}path \wedge$
 $s1 \leq s2 \wedge$
 $PQueryCounter\ s2 = PQueryCounter\ s1$
using *ntimes-read-any-outcome*[*OF read-xp-path*] **by** *simp*
from *read-outcome*[*OF read-xn*] **obtain** *rest3* **where**
tr-s2: $PTranscript\ s2 = xn \# rest3$

```

and st-s3: PState s3 = concat (PState s2) xn
and tr-s3: PTranscript s3 = rest3
and ext-s2-s3: s2 ≤ s3
and query-count-s3: PQueryCounter s3 = PQueryCounter s2
by blast
have xn-path-res:
  length xn-path = floor-log len ∧
  PTranscript s3 = xn-path @ PTranscript s4 ∧
  PState s4 = foldl concat (PState s3) xn-path ∧
  s3 ≤ s4 ∧
  PQueryCounter s4 = PQueryCounter s3
  using ntimes-read-any-outcome[OF read-xn-path] by simp
have xp-eq: xp = x
  using assert-xp unfolding assert-def
  by (cases xp = x) (auto simp: throw-no-outcome)
have s5-eq: s5 = s4
  using assert-xp xp-eq unfolding assert-def by simp
have s5-s6: s5 ≤ s6
  by (rule check-authentication-path-hash-extends[OF check1])
have st-s6: PState s6 = PState s5
  using check-authentication-path-preserves-channel(1)[OF check1] .
have tr-s6: PTranscript s6 = PTranscript s5
  using check-authentication-path-preserves-channel(2)[OF check1] .
have query-count-s6: PQueryCounter s6 = PQueryCounter s5
  using check-authentication-path-preserves-channel(6)[OF check1] .
have ap1-eq: ap1 = f
  using assert-ap1 unfolding assert-def
  by (cases ap1 = f) (auto simp: throw-no-outcome)
have s7-eq: s7 = s6
  using assert-ap1 ap1-eq unfolding assert-def by simp
have s7-s8: s7 ≤ s8
  by (rule check-authentication-path-hash-extends[OF check2])
have st-s8: PState s8 = PState s7
  using check-authentication-path-preserves-channel(1)[OF check2] .
have tr-s8: PTranscript s8 = PTranscript s7
  using check-authentication-path-preserves-channel(2)[OF check2] .
have query-count-s8: PQueryCounter s8 = PQueryCounter s7
  using check-authentication-path-preserves-channel(6)[OF check2] .
have ap2-eq: ap2 = f
  using assert-ap2 unfolding assert-def
  by (cases ap2 = f) (auto simp: throw-no-outcome)
have s9-eq: s9 = s8
  using assert-ap2 ap2-eq unfolding assert-def by simp
let ?x' =
  (xp + xn) div 2 +
  b * ((xp - xn) div (2 * ((h ^ i) * shift) ^ pw))
have out-t: out = (i mod (len div 2), ?x', len div 2, pw + pw) ∧ t = s9
  using ret by simp
have x'-fold:

```

```

?x' = fri-fold-value b xp xn
(fri-fold-denominator ((h ^ i) * shift) pw)
unfolding fri-fold-value-def fri-fold-denominator-def by simp
have chunk-shape:
  fri-layer-opening-chunk len xp xp-path xn xn-path
  ([xp] @ xp-path @ [xn] @ xn-path)
  using xp-path-res xn-path-res unfolding fri-layer-opening-chunk-def by simp
have tr-t:
  PTranscript s = ([xp] @ xp-path @ [xn] @ xn-path) @ PTranscript t
  using tr-s tr-s1 xp-path-res tr-s2 tr-s3 xn-path-res s5-eq tr-s6
  s7-eq tr-s8 s9-eq out-t
  by simp
have st-t:
  PState t = foldl concat (PState s) ([xp] @ xp-path @ [xn] @ xn-path)
  using st-s1 xp-path-res st-s3 xn-path-res s5-eq st-s6 s7-eq st-s8
  s9-eq out-t
  by simp
have s-t: s ≤ t
  using ext-s-s1 xp-path-res ext-s2-s3 xn-path-res s5-s6 s7-s8 out-t
  unfolding s5-eq s7-eq s9-eq
  by (meson hash-ext-trans)
have query-count-t: PQueryCounter t = PQueryCounter s
  using out-t query-count-s1
  xp-path-res query-count-s3 xn-path-res query-count-s6 query-count-s8
  s5-eq s7-eq s9-eq
  by simp
show ?thesis
  apply (rule exI[where x=xp])
  apply (rule exI[where x=xp-path])
  apply (rule exI[where x=xn])
  apply (rule exI[where x=xn-path])
  apply (rule exI[where x=?x'])
  using out-t xp-eq x'-fold chunk-shape tr-t st-t s-t query-count-t
  by simp
qed

```

lemma *mfold-fri-layer-openings-outcome*:

assumes *outcome*:

Some (*out*, *t*) ∈

set-dist (*execute* (*mfold* (*i*, *x*, *len*, *pw*) (*map* *fri-layer-opening-step* *bfs*)) *s*)

shows

∃ *layer-chunks chunk*.

fri-layers-transcript (*length* *bfs*) *len* *layer-chunks chunk* ∧

PTranscript *s* = *chunk* @ *PTranscript* *t* ∧

PState *t* = *foldl* *concat* (*PState* *s*) *chunk* ∧

s ≤ *t* ∧

PQueryCounter *t* = *PQueryCounter* *s*

using *outcome*

proof (*induction* *bfs* *arbitrary*: *i x len pw s out t*)


```

case Nil
then show ?case
  by (intro exI[of - []]) (simp add: fri-layers-transcript-def hash-ext-refl)
next
case (Cons bf bfs)
obtain b f where bf-eq: bf = (b, f)
  by (cases bf)
from Cons.prems obtain out1 s1 where
  head:
    Some (out1, s1) ∈
      set-dist (execute (fri-layer-opening-step bf (i, x, len, pw)) s)
  and tail:
    Some (out, t) ∈
      set-dist (execute (mfold out1 (map fri-layer-opening-step bfs)) s1)
  by (auto elim!: set-dist-bindE)
from fri-layer-opening-step-outcome[OF head[unfolded bf-eq]]
obtain xp xp-path xn xn-path x' where
  out1-eq: out1 = (i mod (len div 2), x', len div 2, pw + pw)
  and head-chunk:
    fri-layer-opening-chunk len xp xp-path xn xn-path
      ([xp] @ xp-path @ [xn] @ xn-path)
  and tr-s: PTranscript s = ([xp] @ xp-path @ [xn] @ xn-path) @ PTranscript s1
  and st-s1:
    PState s1 = foldl concat (PState s) ([xp] @ xp-path @ [xn] @ xn-path)
  and ext-s-s1: s ≤ s1
  and query-count-s1: PQueryCounter s1 = PQueryCounter s
  by blast
from Cons.IH[OF tail[unfolded out1-eq]]
obtain layer-chunks chunk-tail where
  tail-layers:
    fri-layers-transcript (length bfs) (len div 2) layer-chunks chunk-tail
  and tr-s1: PTranscript s1 = chunk-tail @ PTranscript t
  and st-t: PState t = foldl concat (PState s1) chunk-tail
  and ext-s1-t: s1 ≤ t
  and query-count-t: PQueryCounter t = PQueryCounter s1
  by blast
let ?head-chunk = [xp] @ xp-path @ [xn] @ xn-path
let ?chunks = ?head-chunk # layer-chunks
let ?chunk = ?head-chunk @ chunk-tail
have layers:
  fri-layers-transcript (length (bf # bfs)) len ?chunks ?chunk
  using head-chunk tail-layers
  unfolding fri-layers-transcript-def
  by auto
have s ≤ t
  using ext-s-s1 ext-s1-t by (rule hash-ext-trans)
then show ?case
  apply (intro exI[of - ?chunks] exI[of - ?chunk])
  using layers tr-s tr-s1 st-s1 st-t query-count-s1 query-count-t

```

by simp
qed

lemma *receive-query-commits-layers-outcome:*

assumes *outcome:*

Some (out, t) ∈

set-dist (execute (mfold (i, x, len, pw) (receive-query-commits bfs)) s)

shows

$\exists$ *layer-chunks chunk.*

fri-layers-transcript (length bfs) len layer-chunks chunk ∧

PTranscript s = chunk @ PTranscript t ∧

PState t = foldl concat (PState s) chunk ∧

s ≤ t ∧

PQueryCounter t = PQueryCounter s

proof –

have *map-eq: receive-query-commits bfs = map fri-layer-opening-step bfs*

unfolding *receive-query-commits-def fri-layer-opening-step-def*

by (*simp add: fun-eq-iff split: prod.splits*)

show *?thesis*

using *mfold-fri-layer-openings-outcome[OF outcome[unfolded map-eq]] .*

qed

lemma *fri-layer-opening-step-preserves-query-lookup:*

assumes *outcome:*

Some (out, t) ∈ set-dist (execute (fri-layer-opening-step bf st) s)

shows

fmllookup (HashMap t) (QueryIndexChallenge j x) =

fmllookup (HashMap s) (QueryIndexChallenge j x)

proof –

obtain *b f where bf-eq: bf = (b, f)*

by (*cases bf*)

obtain *i0 x0 len pw where st-eq: st = (i0, x0, len, pw)*

by (*cases st*)

from *outcome[unfolded bf-eq st-eq fri-layer-opening-step-def]*

obtain *xp s1 xp-path s2 xn s3 xn-path s4 s5 ap1 s6 s7 ap2 s8 s9 where*

read-xp: Some (xp, s1) ∈ set-dist (execute read s)

and *read-xp-path:*

Some (xp-path, s2) ∈

set-dist (execute (ntimes read (floor-log len)) s1)

and *read-xn: Some (xn, s3) ∈ set-dist (execute read s2)*

and *read-xn-path:*

Some (xn-path, s4) ∈

set-dist (execute (ntimes read (floor-log len)) s3)

and *assert-xp:*

Some ((), s5) ∈ set-dist (execute (assert (xp = x0)) s4)

and *check1:*

Some (ap1, s6) ∈

set-dist (execute (check-authentication-path len i0 xp xp-path) s5)

and *assert-ap1:*

```

    Some ((), s7) ∈ set-dist (execute (assert (ap1 = f)) s6)
and check2:
    Some (ap2, s8) ∈
      set-dist (execute
        (check-authentication-path len ((i0 + len div 2) mod len) xn
          xn-path) s7)
and assert-ap2:
    Some ((), s9) ∈ set-dist (execute (assert (ap2 = f)) s8)
and ret:
    Some (out, t) ∈
      set-dist
        (execute
          (return (i0 mod (len div 2),
            (xp + xn) div 2 +
              b * ((xp - xn) div (2 * ((h ^ i0) * shift) ^ pw)),
            len div 2, pw + pw)) s9)
by (auto elim!: set-dist-bindE)
have lookup-s1:
  fmlookup (HashMap s1) (QueryIndexChallenge j x) =
    fmlookup (HashMap s) (QueryIndexChallenge j x)
using read-preserves-hash-map[OF read-xp] by simp
have lookup-s2:
  fmlookup (HashMap s2) (QueryIndexChallenge j x) =
    fmlookup (HashMap s1) (QueryIndexChallenge j x)
by (rule ntimes-read-preserves-lookup[OF read-xp-path])
have lookup-s3:
  fmlookup (HashMap s3) (QueryIndexChallenge j x) =
    fmlookup (HashMap s2) (QueryIndexChallenge j x)
using read-preserves-hash-map[OF read-xn] by simp
have lookup-s4:
  fmlookup (HashMap s4) (QueryIndexChallenge j x) =
    fmlookup (HashMap s3) (QueryIndexChallenge j x)
by (rule ntimes-read-preserves-lookup[OF read-xn-path])
have s5-eq: s5 = s4
using assert-xp unfolding assert-def
by (cases xp = x0) (auto simp: throw-no-outcome)
have lookup-s6:
  fmlookup (HashMap s6) (QueryIndexChallenge j x) =
    fmlookup (HashMap s5) (QueryIndexChallenge j x)
by (rule check-authentication-path-preserves-challenge-lookups(2)
  [OF check1])
have s7-eq: s7 = s6
using assert-ap1 unfolding assert-def
by (cases ap1 = f) (auto simp: throw-no-outcome)
have lookup-s8:
  fmlookup (HashMap s8) (QueryIndexChallenge j x) =
    fmlookup (HashMap s7) (QueryIndexChallenge j x)
by (rule check-authentication-path-preserves-challenge-lookups(2)
  [OF check2])

```

```

have s9-eq: s9 = s8
  using assert-ap2 unfolding assert-def
  by (cases ap2 = f) (auto simp: throw-no-outcome)
have t-eq: t = s9
  using ret by simp
show ?thesis
  using lookup-s1 lookup-s2 lookup-s3 lookup-s4 lookup-s6 lookup-s8
    s5-eq s7-eq s9-eq t-eq
  by simp
qed

lemma mfold-fri-layer-openings-preserves-query-lookup:
  assumes outcome:
    Some (out, t) ∈
      set-dist (execute (mfold st (map fri-layer-opening-step bfs)) s)
  shows
    fmlookup (HashMap t) (QueryIndexChallenge j x) =
      fmlookup (HashMap s) (QueryIndexChallenge j x)
  using outcome
proof (induction bfs arbitrary: st s out t)
  case Nil
  then show ?case
    by simp
next
  case (Cons bf bfs)
  from Cons.prem obtain out1 s1 where
    head:
      Some (out1, s1) ∈
        set-dist (execute (fri-layer-opening-step bf st) s)
  and tail:
    Some (out, t) ∈
      set-dist (execute (mfold out1 (map fri-layer-opening-step bfs)) s1)
  by (auto elim!: set-dist-bindE)
  have head-pres:
    fmlookup (HashMap s1) (QueryIndexChallenge j x) =
      fmlookup (HashMap s) (QueryIndexChallenge j x)
  by (rule fri-layer-opening-step-preserves-query-lookup[OF head])
  have tail-pres:
    fmlookup (HashMap t) (QueryIndexChallenge j x) =
      fmlookup (HashMap s1) (QueryIndexChallenge j x)
  by (rule Cons.IH[OF tail])
  show ?case
    using head-pres tail-pres by simp
qed

lemma receive-query-commits-layers-preserves-query-lookup:
  assumes outcome:
    Some (out, t) ∈
      set-dist (execute (mfold st (receive-query-commits bfs)) s)

```

shows
 $fmlookup \ (HashMap \ t) \ (QueryIndexChallenge \ j \ x) =$
 $fmlookup \ (HashMap \ s) \ (QueryIndexChallenge \ j \ x)$
proof –
have *map-eq*: *receive-query-commits bfs = map fri-layer-opening-step bfs*
unfolding *receive-query-commits-def fri-layer-opening-step-def*
by (*simp add: fun-eq-iff split: prod.splits*)
show *?thesis*
by (*rule mfold-fri-layer-openings-preserves-query-lookup*
 $[OF \ outcome[unfolded \ map-eq]])$
qed

definition *verifier-query-round-after-index-program*
where
verifier-query-round-after-index-program fr f-fl f-final as fl final raw =
 $(let \ idx' = index \ (to-nat \ raw) \ in$
 $do \ \{$
 $fv \leftarrow mmap \ (check-decommit-on-query \ fr \ idx');$

 $(f-i, f-x, f-len, f-pow) \leftarrow$
 $mfold \ (idx', hd \ fv, clength * scale, 1)$
 $(receive-query-commits \ f-fl);$
 $assert \ (f-x = f-final);$

 $(i, x, len, pow) \leftarrow$
 $mfold \ (idx', cp-eval \ as \ fv \ (h^{idx'} * shift), clength * scale, 1)$
 $(receive-query-commits \ fl);$

 $assert \ (x = final)$
 $\})$

definition *verifier-query-round-program*
where
verifier-query-round-program fr f-fl f-final as fl final =
 $(do \ \{$
 $idx \leftarrow receive-query-index-challenge;$
 $let \ idx' = index \ (to-nat \ idx);$
 $fv \leftarrow mmap \ (check-decommit-on-query \ fr \ idx');$

 $(f-i, f-x, f-len, f-pow) \leftarrow$
 $mfold \ (idx', hd \ fv, clength * scale, 1) \ (receive-query-commits \ f-fl);$
 $assert \ (f-x = f-final);$

 $(i, x, len, pow) \leftarrow$
 $mfold \ (idx', cp-eval \ as \ fv \ (h^{idx'} * shift), clength * scale, 1)$
 $(receive-query-commits \ fl);$

 $assert \ (x = final)$
 $\})$

lemma *verifier-query-round-program-alt-def*:
verifier-query-round-program fr f-fl f-final as fl final =
 (receive-query-index-challenge $\gg$
 verifier-query-round-after-index-program fr f-fl f-final as fl final)
unfolding *verifier-query-round-program-def*
 verifier-query-round-after-index-program-def
by (*simp* add: *Let-def*)

lemma *verifier-query-round-after-index-program-outcome*:
assumes *outcome*:
 Some ($()$, t) $\in$
 set-dist (*execute*
 (*verifier-query-round-after-index-program* fr f-fl f-final as fl final raw) s)
shows $s \leq t \wedge PQueryCounter\ t = PQueryCounter\ s$
proof –
let $?idx = index\ (to-nat\ raw)$
from *outcome* **obtain** $fv\ s1\ f-out\ s2\ s3\ c-out\ s4$ **where**
 query-decommit:
 Some (fv , $s1$) $\in$
 set-dist (*execute* (*mmap* (*check-decommit-on-query* fr $?idx$)) s)
and *trace-fri*:
 Some ($f-out$, $s2$) $\in$
 set-dist (*execute*
 (*mfold* ($?idx$, $hd\ fv$, $clength * scale$, 1)
 (*receive-query-commits* f-fl)) $s1$)
and *assert-trace*:
 Some ($()$, $s3$) $\in$
 set-dist
 (*execute*
 (*assert* (*case* f-out of $(-, f-x, -, -) \Rightarrow f-x = f-final$)) $s2$)
and *comp-fri*:
 Some ($c-out$, $s4$) $\in$
 set-dist (*execute*
 (*mfold* ($?idx$, *cp-eval* as $fv\ (h \wedge ?idx * shift)$,
 $clength * scale$, 1)
 (*receive-query-commits* fl)) $s3$)
and *assert-comp*:
 Some ($()$, t) $\in$
 set-dist
 (*execute*
 (*assert* (*case* c-out of $(-, x, -, -) \Rightarrow x = final$)) $s4$)
unfolding *verifier-query-round-after-index-program-def*
by (*auto simp: Let-def split: prod.splits elim!: set-dist-bindE*)
from *check-decommit-on-query-outcome*[*OF* *query-decommit*]
obtain *query-paths* *query-chunk* **where**
 ext-s-s1: $s \leq s1$
and *query-count-s1*: $PQueryCounter\ s1 = PQueryCounter\ s$
by *blast*

```

from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where
  ext-s1-s2:  $s1 \leq s2$ 
  and query-count-s2: PQueryCounter  $s2 = PQueryCounter\ s1$ 
  by blast
have s3-eq:  $s3 = s2$ 
  using assert-trace unfolding assert-def
  by (cases case f-out of  $(-, f-x, -, -) \Rightarrow f-x = f-final$ )
    (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  ext-s3-s4:  $s3 \leq s4$ 
  and query-count-s4: PQueryCounter  $s4 = PQueryCounter\ s3$ 
  by blast
have t-eq:  $t = s4$ 
  using assert-comp unfolding assert-def
  by (cases case c-out of  $(-, x, -, -) \Rightarrow x = final$ )
    (auto simp: throw-no-outcome)
have  $s \leq t$ 
  using ext-s-s1 ext-s1-s2 ext-s3-s4 t-eq unfolding s3-eq
  by (meson hash-ext-trans)
moreover have PQueryCounter  $t = PQueryCounter\ s$ 
  using query-count-s1 query-count-s2 query-count-s4 s3-eq t-eq by simp
ultimately show ?thesis
  by simp
qed

```

```

lemma verifier-query-round-after-index-program-preserves-query-lookup:
  assumes outcome:
    Some  $(((), t) \in$ 
      set-dist (execute
        (verifier-query-round-after-index-program fr f-fl f-final as fl final raw) s)
  shows
    fmlookup (HashMap t) (QueryIndexChallenge j x) =
      fmlookup (HashMap s) (QueryIndexChallenge j x)
proof -
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
    query-decommit:
      Some  $(fv, s1) \in$ 
        set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
  and trace-fri:
    Some  $(f-out, s2) \in$ 
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
  and assert-trace:
    Some  $(((), s3) \in$ 
      set-dist

```

```

      (execute
        (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
    and comp-fri:
      Some (c-out, s4) ∈
        set-dist (execute
          (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
            clength * scale, 1)
            (receive-query-commits fl)) s3)
    and assert-comp:
      Some (( ), t) ∈
        set-dist
          (execute
            (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
    unfolding verifier-query-round-after-index-program-def
    by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
  have lookup-s1:
    fmlookup (HashMap s1) (QueryIndexChallenge j x) =
      fmlookup (HashMap s) (QueryIndexChallenge j x)
  by (rule check-decommit-on-query-preserves-query-lookup
    [OF query-decommit])
  have lookup-s2:
    fmlookup (HashMap s2) (QueryIndexChallenge j x) =
      fmlookup (HashMap s1) (QueryIndexChallenge j x)
  by (rule receive-query-commits-layers-preserves-query-lookup
    [OF trace-fri])
  have s3-eq: s3 = s2
    using assert-trace unfolding assert-def
    by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
      (auto simp: throw-no-outcome)
  have lookup-s4:
    fmlookup (HashMap s4) (QueryIndexChallenge j x) =
      fmlookup (HashMap s3) (QueryIndexChallenge j x)
  by (rule receive-query-commits-layers-preserves-query-lookup
    [OF comp-fri])
  have t-eq: t = s4
    using assert-comp unfolding assert-def
    by (cases case c-out of (-, x, -, -) ⇒ x = final)
      (auto simp: throw-no-outcome)
  show ?thesis
    using lookup-s1 lookup-s2 lookup-s4 s3-eq t-eq by simp
qed

lemma receive-query-index-challenge-preserves-query-future-fresh:
  assumes future: query-future-fresh s
  and outcome:
    Some (raw, t) ∈ set-dist (execute receive-query-index-challenge s)
  shows query-future-fresh t
proof -
  have counter-t: PQueryCounter t = Suc (PQueryCounter s)

```



```

    using receive-query-index-challenge-counter-outcome[OF outcome] by simp
show ?thesis
  unfolding query-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-ge-t: PQueryCounter t ≤ i
  have i-ge-s: PQueryCounter s ≤ i
    using i-ge-t counter-t by simp
  have neg:
    QueryIndexChallenge i x ≠
      QueryIndexChallenge (PQueryCounter s) (PState s)
    using i-ge-t counter-t by auto
  have fmllookup (HashMap t) (QueryIndexChallenge i x) =
    fmllookup (HashMap s) (QueryIndexChallenge i x)
    by (rule receive-query-index-challenge-preserves-other-lookup
      [OF outcome neg])
  also have ... = None
    using future i-ge-s unfolding query-future-fresh-def by blast
  finally show fmllookup (HashMap t) (QueryIndexChallenge i x) = None .
qed
qed

lemma verifier-query-round-program-preserves-query-future-fresh:
  assumes future: query-future-fresh s
  and outcome:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  shows query-future-fresh t
proof -
  from outcome[unfolded verifier-query-round-program-alt-def]
  obtain raw s0 where
    rand:
      Some (raw, s0) ∈
        set-dist (execute receive-query-index-challenge s)
  and tail:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program fr f-fl f-final as fl
            final raw) s0)
  by (auto elim!: set-dist-bindE)
  have future-s0: query-future-fresh s0
    by (rule receive-query-index-challenge-preserves-query-future-fresh
      [OF future rand])
  have counter-t: PQueryCounter t = PQueryCounter s0
    using verifier-query-round-after-index-program-outcome[OF tail] by simp
  show ?thesis

```

```

unfolding query-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-ge-t: PQueryCounter t ≤ i
  have i-ge-s0: PQueryCounter s0 ≤ i
    using i-ge-t counter-t by simp
  have fmlookup (HashMap t) (QueryIndexChallenge i x) =
    fmlookup (HashMap s0) (QueryIndexChallenge i x)
    by (rule verifier-query-round-after-index-program-preserves-query-lookup
      [OF tail])
  also have ... = None
    using future-s0 i-ge-s0 unfolding query-future-fresh-def by blast
  finally show fmlookup (HashMap t) (QueryIndexChallenge i x) = None .
qed
qed

lemma ntimes-verifier-query-round-program-preserves-query-future-fresh:
assumes outcome:
  Some (results, t) ∈
    set-dist
      (execute
        (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
          n) s)
  and future: query-future-fresh s
shows query-future-fresh t
using outcome future
proof (induction n arbitrary: s results t)
case 0
then show ?case
  by simp
next
case (Suc n)
from Suc.premis obtain u results' s1 where
  head:
    Some (u, s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results', t) ∈
      set-dist
        (execute
          (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
            n) s1)
  by (auto elim!: set-dist-bindE)
have u-eq: u = ()
  by (cases u) simp
have future-s1: query-future-fresh s1
  by (rule verifier-query-round-program-preserves-query-future-fresh

```

```

    [OF Suc.premis(2) head[unfolded u-eq]])
  show ?case
  by (rule Suc.IH[OF tail future-s1])
qed

lemma verifier-query-round-program-outcome:
  assumes outcome:
    Some ((), t) ∈
      set-dist (execute (verifier-query-round-program fr f-fl f-final as fl final) s)
  obtains raw-idx idx chunk
  where
    idx = index (to-nat raw-idx)
    and verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
    and PTranscript s = chunk @ PTranscript t
    and PState t = foldl concat (PState s) chunk
    and s ≤ t
    and PQueryCounter t = Suc (PQueryCounter s)
    and fmlookup (HashMap t) (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some
raw-idx
  proof -
    from outcome obtain raw-idx s0 idx fv s1 f-out s2 s3 c-out s4 where
      rand:
        Some (raw-idx, s0) ∈
          set-dist (execute receive-query-index-challenge s)
    and idx-def: idx = index (to-nat raw-idx)
    and query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr idx)) s0)
    and trace-fri:
      Some (f-out, s2) ∈
        set-dist (execute
          (mfold (idx, hd fv, clength * scale, 1) (receive-query-commits f-fl)) s1)
    and assert-trace:
      Some ((), s3) ∈ set-dist (execute (assert (case f-out of (_, f-x, -, -) ⇒ f-x = f-final)) s2)
    and comp-fri:
      Some (c-out, s4) ∈
        set-dist (execute
          (mfold
            (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
            (receive-query-commits fl)) s3)
    and assert-comp:
      Some ((), t) ∈ set-dist (execute (assert (case c-out of (_, x, -, -) ⇒ x = final)) s4)
  unfolding verifier-query-round-program-def
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
  have rand-res:
    s ≤ s0 ∧
    PState s0 = PState s ∧
    PTranscript s0 = PTranscript s ∧
    fmlookup (HashMap s0) (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some

```

raw-idx

```

using receive-query-index-challenge-outcome[OF rand] .
have rand-counter:
  PQueryCounter s0 = Suc (PQueryCounter s)
  using receive-query-index-challenge-counter-outcome[OF rand] by simp
from check-decommit-on-query-outcome[OF query-decommit]
obtain query-paths query-chunk where
  query-chunk-shape:
    query-decommitment-transcript idx fv query-paths query-chunk
  and tr-s0: PTranscript s0 = query-chunk @ PTranscript s1
  and st-s1: PState s1 = foldl concat (PState s0) query-chunk
  and ext-s0-s1: s0 ≤ s1
  and query-count-s1: PQueryCounter s1 = PQueryCounter s0
  by blast
from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where
  trace-layers:
    fri-layers-transcript (length f-fl) (clength * scale)
    trace-layer-chunks trace-fri-chunk
  and tr-s1: PTranscript s1 = trace-fri-chunk @ PTranscript s2
  and st-s2: PState s2 = foldl concat (PState s1) trace-fri-chunk
  and ext-s1-s2: s1 ≤ s2
  and query-count-s2: PQueryCounter s2 = PQueryCounter s1
  by blast
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def
  by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
    (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  comp-layers:
    fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
  and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
  and st-s4: PState s4 = foldl concat (PState s3) comp-fri-chunk
  and ext-s3-s4: s3 ≤ s4
  and query-count-s4: PQueryCounter s4 = PQueryCounter s3
  by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -) ⇒ x = final)
    (auto simp: throw-no-outcome)
let ?chunk = query-chunk @ trace-fri-chunk @ comp-fri-chunk
have round-chunk:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl) ?chunk
  unfolding verifier-query-round-chunk-def
  apply (rule exI[where x=query-chunk])
  apply (rule exI[where x=trace-fri-chunk])
  apply (rule exI[where x=comp-fri-chunk])

```

```

apply (rule exI[where  $x=fv$ ])
apply (rule exI[where  $x=query\text{-}paths$ ])
apply (rule exI[where  $x=trace\text{-}layer\text{-}chunks$ ])
apply (rule exI[where  $x=comp\text{-}layer\text{-}chunks$ ])
using query-chunk-shape trace-layers comp-layers
by simp
have tr-t: PTranscript  $s = ?chunk @ PTranscript\ t$ 
  using rand-res tr-s0 tr-s1 tr-s3 s3-eq t-eq by simp
have st-t: PState  $t = foldl\ concat\ (PState\ s)\ ?chunk$ 
  using rand-res st-s1 st-s2 st-s4 s3-eq t-eq by simp
have ext-s0-t:  $s0 \leq t$ 
  using ext-s0-s1 ext-s1-s2 ext-s3-s4 t-eq unfolding s3-eq
  by (meson hash-ext-trans)
have ext-s-t:  $s \leq t$ 
  using rand-res ext-s0-t by (meson hash-ext-trans)
have lookup-t: fmlookup (HashMap  $t$ ) (QueryIndexChallenge (PQueryCounter  $s$ ) (PState  $s$ ))
= Some raw-idx
  using rand-res hash-extension-lookup[OF conjunct2[OF conjunct2[OF conjunct2[OF rand-res]]]
ext-s0-t]
  by simp
have query-count-t: PQueryCounter  $t = Suc\ (PQueryCounter\ s)$ 
  using rand-counter query-count-s1 query-count-s2 query-count-s4 s3-eq t-eq
  by simp
show ?thesis
  by (rule that[OF idx-def round-chunk tr-t st-t ext-s-t query-count-t lookup-t])
qed

```

lemma *ntimes-verifier-query-rounds-outcome:*

assumes *outcome:*

Some (results, t) ∈

set-dist (execute

(ntimes (verifier-query-round-program fr f-fl f-final as fl final) n) s)

shows

$\exists$ raw-idxs query-idxs query-chunks.

$length\ raw\text{-}idxs = n \wedge$

$query\text{-}idxs = map\ (\lambda raw.\ index\ (to\text{-}nat\ raw))\ raw\text{-}idxs \wedge$

$length\ query\text{-}chunks = n \wedge$

$PTranscript\ s = List.concat\ query\text{-}chunks @ PTranscript\ t \wedge$

$PState\ t = state\text{-}after\text{-}query\text{-}chunks\ (PState\ s)\ query\text{-}chunks\ n \wedge$

$s \leq t \wedge$

$PQueryCounter\ t = PQueryCounter\ s + n \wedge$

$(\forall i < n.$

$verifier\text{-}query\text{-}round\text{-}chunk\ (query\text{-}idxs\ !\ i)$

$(map\ snd\ f\text{-}fl)\ (map\ snd\ fl)\ (query\text{-}chunks\ !\ i)) \wedge$

$(\forall i < n.$

$fmlookup\ (HashMap\ t)$

$(QueryIndexChallenge\ (PQueryCounter\ s + i)$

$(state\text{-}after\text{-}query\text{-}chunks\ (PState\ s)\ query\text{-}chunks\ i)) =$

$Some\ (raw\text{-}idxs\ !\ i)) \wedge$

```

    (∀ idx ∈ set query-idxs. idx < clength * scale)
  using outcome
proof (induction n arbitrary: s results t)
  case 0
  then show ?case
    by (intro exI[of - []]) (simp add: state-after-query-chunks-def hash-ext-refl)
next
  case (Suc n)
  from Suc.premis obtain u results' s1 where
    head:
      Some (u, s1) ∈
        set-dist (execute (verifier-query-round-program fr f-fl f-final as fl final) s)
    and tail:
      Some (results', t) ∈
        set-dist (execute
          (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n) s1)
    and results-eq: results = u # results'
  by (auto elim!: set-dist-bindE)
  have u-unit: u = ()
  by (cases u) simp
  from verifier-query-round-program-outcome[OF head[unfolded u-unit]]
  obtain raw idx chunk where
    idx-def: idx = index (to-nat raw)
    and round-chunk:
      verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
    and tr-s: PTranscript s = chunk @ PTranscript s1
    and st-s1: PState s1 = foldl concat (PState s) chunk
    and ext-s-s1: s ≤ s1
    and query-count-s1: PQueryCounter s1 = Suc (PQueryCounter s)
    and lookup-s1: fmlookup (HashMap s1) (QueryIndexChallenge (PQueryCounter s) (PState
s)) = Some raw
  by this
  from Suc.IH[OF tail] obtain raw-tail idx-tail chunk-tail where
    len-raw-tail: length raw-tail = n
    and idx-tail-def: idx-tail = map (λraw. index (to-nat raw)) raw-tail
    and len-chunk-tail: length chunk-tail = n
    and tr-s1: PTranscript s1 = List.concat chunk-tail @ PTranscript t
    and st-t-tail:
      PState t = state-after-query-chunks (PState s1) chunk-tail n
    and ext-s1-t: s1 ≤ t
    and query-count-t: PQueryCounter t = PQueryCounter s1 + n
    and round-tail:
      ∧ i. i < n ⇒
        verifier-query-round-chunk (idx-tail ! i)
          (map snd f-fl) (map snd fl) (chunk-tail ! i)
    and lookup-tail:
      ∧ i. i < n ⇒
        fmlookup (HashMap t)
          (QueryIndexChallenge (PQueryCounter s1 + i)

```

```

      (state-after-query-chunks (PState s1) chunk-tail i)) =
    Some (raw-tail ! i)
  and idx-tail-bound:
     $\forall idx \in \text{set } \text{idx-tail}. \text{idx} < \text{clength} * \text{scale}$ 
  by blast
  let ?raws = raw # raw-tail
  let ?idxs = idx # idx-tail
  let ?chunks = chunk # chunk-tail
  have idxs-def: ?idxs = map ( $\lambda \text{raw}. \text{index } (\text{to-nat } \text{raw})$ ) ?raws
    using idx-def idx-tail-def by simp
  have tr-all: PTranscript s = List.concat ?chunks @ PTranscript t
    using tr-s tr-s1 by simp
  have state-shift:
     $\bigwedge i. \text{state-after-query-chunks } (PState s) \text{ ?chunks } (Suc i) =$ 
      state-after-query-chunks (PState s1) chunk-tail i
    using st-s1 unfolding state-after-query-chunks-def by simp
  have st-all:
    PState t = state-after-query-chunks (PState s) ?chunks (Suc n)
    using st-t-tail state-shift[of n] by simp
  have ext-s-t:  $s \leq t$ 
    using ext-s-s1 ext-s1-t by (rule hash-ext-trans)
  have query-count-all: PQueryCounter t = PQueryCounter s + Suc n
    using query-count-s1 query-count-t by simp
  have round-all:
     $\bigwedge i. i < Suc n \implies$ 
      verifier-query-round-chunk (?idxs ! i)
      (map snd f-fl) (map snd fl) (?chunks ! i)
  proof -
    fix i
    assume i-bound:  $i < Suc n$ 
    show verifier-query-round-chunk (?idxs ! i)
      (map snd f-fl) (map snd fl) (?chunks ! i)
    proof (cases i)
      case 0
      then show ?thesis
        using round-chunk by simp
    next
      case (Suc j)
      then show ?thesis
        using round-tail[of j] i-bound by simp
    qed
  qed
  have lookup-head-t:
    fmlookup (HashMap t) (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
    using hash-extension-lookup[OF lookup-s1 ext-s1-t] .
  have lookup-all:
     $\bigwedge i. i < Suc n \implies$ 
      fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)

```

```

      (state-after-query-chunks (PState s) ?chunks i)) =
      Some (?rows ! i)
proof -
  fix i
  assume i-bound: i < Suc n
  show fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks (PState s) ?chunks i)) =
      Some (?rows ! i)
  proof (cases i)
    case 0
    then show ?thesis
      using lookup-head-t unfolding state-after-query-chunks-def by simp
  next
    case (Suc j)
    then show ?thesis
      using lookup-tail[of j] i-bound state-shift[of j] query-count-s1 by simp
  qed
qed
have idx-bound-all:
   $\forall idx \in \text{set } ?idxs. \text{idx} < \text{clength} * \text{scale}$ 
  using idx-tail-bound idx-def index-less-domain by auto
show ?case
  apply (intro exI[of - ?rows] exI[of - ?idxs] exI[of - ?chunks] conjI)
    apply (simp add: len-raw-tail)
    apply (rule idxs-def)
    apply (simp add: len-chunk-tail)
    apply (rule tr-all)
    apply (rule st-all)
    apply (rule ext-s-t)
    apply (rule query-count-all)
    apply (intro allI impI, rule round-all, assumption)
    apply (intro allI impI, rule lookup-all, assumption)
  using idx-bound-all by simp
qed

lemma verifier-query-round-program-preserves-past-query-lookup:
  assumes past: j < PQueryCounter s
  and outcome:
    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  shows
    fmlookup (HashMap t) (QueryIndexChallenge j x) =
    fmlookup (HashMap s) (QueryIndexChallenge j x)
proof -
  from outcome[unfolded verifier-query-round-program-alt-def]
  obtain raw s0 where

```



```

  rand:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s)
  and tail:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program fr f-fl f-final as fl
            final raw) s0)
  by (auto elim!: set-dist-bindE)
  have neg:
    QueryIndexChallenge j x ≠
      QueryIndexChallenge (PQueryCounter s) (PState s)
  using past by auto
  have lookup-s0:
    fmlookup (HashMap s0) (QueryIndexChallenge j x) =
      fmlookup (HashMap s) (QueryIndexChallenge j x)
  by (rule receive-query-index-challenge-preserves-other-lookup
    [OF rand neg])
  have lookup-t:
    fmlookup (HashMap t) (QueryIndexChallenge j x) =
      fmlookup (HashMap s0) (QueryIndexChallenge j x)
  by (rule verifier-query-round-after-index-program-preserves-query-lookup
    [OF tail])
  show ?thesis
    using lookup-s0 lookup-t by simp
qed

lemma ntimes-verifier-query-round-program-preserves-past-query-lookup:
  assumes past: j < PQueryCounter s
  and outcome:
    Some (results, t) ∈
      set-dist
        (execute
          (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
            n) s)
  shows
    fmlookup (HashMap t) (QueryIndexChallenge j x) =
      fmlookup (HashMap s) (QueryIndexChallenge j x)
  using outcome past
proof (induction n arbitrary: s results t)
  case 0
  then show ?case
    by simp
next
  case (Suc n)
  from Suc.premis obtain u results' s1 where
    head:
      Some (u, s1) ∈

```

```

    set-dist
    (execute
      (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results', t) ∈
      set-dist
      (execute
        (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
          n) s1)
  by (auto elim!: set-dist-bindE)
have u-eq: u = ()
  by (cases u) simp
have head-pres:
  fmlookup (HashMap s1) (QueryIndexChallenge j x) =
  fmlookup (HashMap s) (QueryIndexChallenge j x)
  by (rule verifier-query-round-program-preserves-past-query-lookup
    [OF Suc.premis(2) head[unfolded u-eq]])
have counter-s1:
  PQueryCounter s1 = Suc (PQueryCounter s)
  using verifier-query-round-program-outcome[OF head[unfolded u-eq]]
  by blast
have past-s1: j < PQueryCounter s1
  using Suc.premis(2) counter-s1 by simp
have tail-pres:
  fmlookup (HashMap t) (QueryIndexChallenge j x) =
  fmlookup (HashMap s1) (QueryIndexChallenge j x)
  by (rule Suc.IH[OF tail past-s1])
show ?case
  using head-pres tail-pres by simp
qed

lemma receive-fri-commits-outcome:
  assumes outcome: Some ((b, r), t) ∈ set-dist (execute receive-fri-commits s)
  shows
    ∃ rest.
      PTranscript s = r # rest ∧
      PState t = concat (PState s) r ∧
      PTranscript t = rest ∧
      s ≤ t ∧
      fmlookup (HashMap t) (FiatShamirChallenge (concat (PState s) r)) = Some b
proof -
  from outcome obtain s-read where
    read-r: Some (r, s-read) ∈ set-dist (execute read s)
  and rand-b:
    Some (b, t) ∈ set-dist (execute receive-random-field-element s-read)
  unfolding receive-fri-commits-def
  by (auto elim!: set-dist-bindE)
obtain rest where tr-s: PTranscript s = r # rest
proof -

```

```

have PTranscript s ≠ []
  using read-r unfolding read-def protocol-read-def assert-def
  by (cases PTranscript s) (auto simp: throw-no-outcome elim!: set-dist-bindE)
then obtain x xs where xs: PTranscript s = x # xs
  by (cases PTranscript s) auto
have r-x: r = x
  using read-nonempty-outcome[OF read-r xs] by simp
show ?thesis
  using that xs r-x by auto
qed
have read-res:
  r = r ∧
  s-read = s(PState := concat (PState s) r, PTranscript := rest)
  using read-nonempty-outcome[OF read-r tr-s] by simp
have rand-res:
  s-read ≤ t ∧
  PState t = PState s-read ∧
  PTranscript t = PTranscript s-read ∧
  fmlookup (HashMap t) (FiatShamirChallenge (PState s-read)) = Some b
  using receive-random-field-element-outcome[OF rand-b] .
have s-read-ext: s ≤ s-read
  using read-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s-t: s ≤ t
  using s-read-ext rand-res by (meson hash-ext-trans)
show ?thesis
  using tr-s read-res rand-res s-t by auto
qed

lemma ntimes-receive-fri-commits-outcome:
  assumes outcome:
    Some (brs, t) ∈
      set-dist (execute (ntimes receive-fri-commits n) s)
  shows
    length brs = n ∧
    PTranscript s = map snd brs @ PTranscript t ∧
    PState t = foldl concat (PState s) (map snd brs) ∧
    s ≤ t ∧
    (∀ i < n.
      fmlookup (HashMap t)
        (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) (map snd brs)))) =
        Some (fst (brs ! i)))
  using outcome
proof (induction n arbitrary: s brs t)
  case 0
  then show ?case
    by (simp add: hash-ext-refl)
next
  case (Suc n)
  from Suc.premis obtain br brs' s1 where

```

```

  head: Some (br, s1) ∈ set-dist (execute receive-fri-commits s)
and tail:
  Some (brs', t) ∈
    set-dist (execute (ntimes receive-fri-commits n) s1)
and brs-eq: brs = br # brs'
by (auto elim!: set-dist-bindE)
obtain b r where br-eq: br = (b, r)
by (cases br)
from receive-fri-commits-outcome[OF head[unfolded br-eq]]
obtain rest1 where
  tr-s: PTranscript s = r # rest1
and st-s1: PState s1 = concat (PState s) r
and tr-s1: PTranscript s1 = rest1
and ext-head: s ≤ s1
and lookup-head:
  fmlookup (HashMap s1) (FiatShamirChallenge (concat (PState s) r)) = Some b
by blast
have tail-res:
  length brs' = n ∧
  PTranscript s1 = map snd brs' @ PTranscript t ∧
  PState t = foldl concat (PState s1) (map snd brs') ∧
  s1 ≤ t ∧
  (∀ i < n.
    fmlookup (HashMap t)
      (FiatShamirChallenge (foldl concat (PState s1) (take (Suc i) (map snd brs')))) =
      Some (fst (brs' ! i)))
  using Suc.IH[OF tail] .
have ext-s1-t: s1 ≤ t
  using tail-res by simp
have lookup-head-t:
  fmlookup (HashMap t) (FiatShamirChallenge (concat (PState s) r)) = Some b
  using hash-extension-lookup[OF lookup-head ext-s1-t] .
have ext-combined: s ≤ t
  using ext-head ext-s1-t by (meson hash-ext-trans)
have lookups:
  ∀ i < Suc n.
    fmlookup (HashMap t)
      (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) (map snd brs')))) =
      Some (fst (brs ! i))
proof (intro allI impI)
  fix i
  assume i-bound: i < Suc n
  show fmlookup (HashMap t)
    (FiatShamirChallenge (foldl concat (PState s) (take (Suc i) (map snd brs')))) =
    Some (fst (brs ! i))
  proof (cases i)
    case 0
    then show ?thesis
      using lookup-head-t brs-eq br-eq by simp

```

```

next
  case (Suc j)
  then have j-bound:  $j < n$ 
    using i-bound by simp
  have key-eq:
    foldl concat (PState s) (take (Suc i) (map snd brs)) =
      foldl concat (PState s1) (take (Suc j) (map snd brs'))
    using Suc brs-eq br-eq st-s1 by simp
  show ?thesis
    using tail-res j-bound Suc brs-eq key-eq by simp
qed
qed
show ?case
  using tail-res tr-s tr-s1 st-s1 ext-combined lookups brs-eq br-eq
  by simp
qed

end

end

```

```

theory Soundness-Execution-Prefix
imports Soundness-Execution-Query
begin

```

Verifier alpha/FRI prefix execution and challenge-space bounds.

```

context soundness
begin

```

```

definition alpha-round

```

```

where
  alpha-round =
    (do {
      a0 ← receive-alpha-challenge;
      let a0' = a0;
      a1 ← read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    })

```

```

definition verifier-alpha-prefix

```

```

:: ('f × ('f × 'f) list × 'f × 'f list,
   ('f, 'a) protocol-channel-scheme) state-monad

```

```

where

```

```

  verifier-alpha-prefix =
    (do {
      fr ← read;

```

```

    f-fl ← ntimes receive-trace-fri-commits (ceil-log clength);
    f-final ← read;
    as ← mmap (replicate (length spec) alpha-round);
    return (fr, f-fl, f-final, as)
  })

```

definition *verifier-trace-fri-prefix*

```

:: ('f × ('f × 'f) list,
   ('f, 'a) protocol-channel-scheme) state-monad

```

where

```

verifier-trace-fri-prefix =
  (do {
    fr ← read;
    f-fl ← ntimes receive-trace-fri-commits (ceil-log clength);
    return (fr, f-fl)
  })

```

definition *verifier-after-trace-fri*

```

:: 'f × ('f × 'f) list ⇒
   (unit list, ('f, 'a) protocol-channel-scheme) state-monad

```

where

```

verifier-after-trace-fri header =
  (case header of (fr, f-fl) ⇒
    do {
      f-final ← read;
      as ← mmap (replicate (length spec) alpha-round);
      dg ← read;
      assert (to-nat dg ≤ maxDegree);
      fl ← ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1));
      final ← read;
      ntimes (verifier-query-round-program fr f-fl f-final as fl final)
        rounds
    })

```

definition *verifier-composition-fri-prefix*

```

:: ('f × ('f × 'f) list × 'f × 'f list ×
   'f × ('f × 'f) list,
   ('f, 'a) protocol-channel-scheme) state-monad

```

where

```

verifier-composition-fri-prefix =
  (do {
    fr ← read;
    f-fl ← ntimes receive-trace-fri-commits (ceil-log clength);
    f-final ← read;
    as ← mmap (replicate (length spec) alpha-round);
    dg ← read;
    assert (to-nat dg ≤ maxDegree);
    fl ← ntimes receive-composition-fri-commits

```

```

    (ceil-log (to-nat dg + 1));
  return (fr, f-fl, f-final, as, dg, fl)
})

```

definition *verifier-after-composition-fri*

```

:: 'f × ('f × 'f) list × 'f × 'f list ×
   'f × ('f × 'f) list ⇒
   (unit list, ('f, 'a) protocol-channel-scheme) state-monad

```

where

```

verifier-after-composition-fri header =
  (case header of (fr, f-fl, f-final, as, dg, fl) ⇒
    do {
      final ← read;
      ntimes (verifier-query-round-program fr f-fl f-final as fl final)
        rounds
    })

```

definition *verifier-after-alpha*

```

:: 'f × ('f × 'f) list × 'f × 'f list ⇒
   (unit list, ('f, 'a) protocol-channel-scheme) state-monad

```

where

```

verifier-after-alpha header =
  (case header of (fr, f-fl, f-final, as) ⇒
    do {
      dg ← read;
      assert (to-nat dg ≤ maxDegree);
      fl ← ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1));
      final ← read;
      ntimes (verifier-query-round-program fr f-fl f-final as fl final)
        rounds
    })

```

lemma *verify-monad-alpha-decomposition:*

```

verify-monad = (verifier-alpha-prefix ≫= verifier-after-alpha)
unfolding verify-monad-def verifier-alpha-prefix-def verifier-after-alpha-def
  alpha-round-def verifier-query-round-program-def
by (simp add: Let-def sm-bind-assoc sm-bind-return-map split: prod.splits)

```

lemma *verify-monad-trace-fri-decomposition:*

```

verify-monad = (verifier-trace-fri-prefix ≫= verifier-after-trace-fri)
unfolding verify-monad-def verifier-trace-fri-prefix-def
  verifier-after-trace-fri-def alpha-round-def verifier-query-round-program-def
by (simp add: Let-def sm-bind-assoc sm-bind-return-map split: prod.splits)

```

lemma *verify-monad-composition-fri-decomposition:*

```

verify-monad =
  (verifier-composition-fri-prefix ≫= verifier-after-composition-fri)
unfolding verify-monad-def verifier-composition-fri-prefix-def

```

```

verifier-after-composition-fri-def alpha-round-def
verifier-query-round-program-def
by (simp add: Let-def sm-bind-assoc sm-bind-return-map split: prod.splits)

lemma alpha-round-outcome:
  assumes outcome: Some (a, t) ∈ set-dist (execute alpha-round s)
  shows
    ∃ rest.
      PTranscript s = a # rest ∧
      PState t = concat (PState s) a ∧
      PTranscript t = rest ∧
      s ≤ t ∧
      PTraceFriCounter t = PTraceFriCounter s ∧
      PCompositionFriCounter t = PCompositionFriCounter s ∧
      PAlphaCounter t = Suc (PAlphaCounter s) ∧
      PQueryCounter t = PQueryCounter s ∧
      fmlookup (HashMap t) (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a
  proof -
    from outcome obtain a0 s0 a1 s1 s2 where
      rand: Some (a0, s0) ∈ set-dist (execute receive-alpha-challenge s)
      and read-a: Some (a1, s1) ∈ set-dist (execute read s0)
      and assert-a: Some ((), s2) ∈ set-dist (execute (assert (a0 = a1)) s1)
      and ret-a: Some (a, t) ∈ set-dist (execute (return a1) s2)
      unfolding alpha-round-def
      by (auto simp: Let-def elim!: set-dist-bindE)
    have rand-res:
      s ≤ s0 ∧
      PState s0 = PState s ∧
      PTranscript s0 = PTranscript s ∧
      fmlookup (HashMap s0) (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a0
      using receive-alpha-challenge-outcome[OF rand] .
    have rand-counter:
      PTraceFriCounter s0 = PTraceFriCounter s ∧
      PCompositionFriCounter s0 = PCompositionFriCounter s ∧
      PAlphaCounter s0 = Suc (PAlphaCounter s) ∧
      PQueryCounter s0 = PQueryCounter s
      using receive-alpha-challenge-counter-outcome[OF rand] by simp
    obtain rest where tr-s0: PTranscript s0 = a1 # rest
    proof -
      have PTranscript s0 ≠ []
        using read-a unfolding read-def protocol-read-def assert-def
        by (cases PTranscript s0) (auto simp: throw-no-outcome elim!: set-dist-bindE)
      then obtain x xs where xs: PTranscript s0 = x # xs
        by (cases PTranscript s0) auto
      have a1-x: a1 = x
        using read-nonempty-outcome[OF read-a xs] by simp
      show ?thesis
        using that xs a1-x by auto
    qed
  qed

```



```

have read-res:
  s1 = s0(PState := concat (PState s0) a1, PTranscript := rest)
  using read-nonempty-outcome[OF read-a tr-s0] by simp
have a-eq: a0 = a1
  using assert-a unfolding assert-def
  by (cases a0 = a1) (auto simp: throw-no-outcome)
have s2-eq: s2 = s1
  using assert-a a-eq unfolding assert-def by simp
have t-eq: t = s1 and a-ret: a = a1
  using ret-a s2-eq by simp-all
have lookup-t: fmllookup (HashMap t) (AlphaChallenge (PAlphaCounter s) (PState s)) = Some
a
  using rand-res read-res t-eq a-ret a-eq
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have ext: s ≤ t
  using rand-res read-res t-eq a-ret a-eq
  unfolding less-eq-hash-ext-def less-eq-fmap-def by auto
show ?thesis
  using rand-res rand-counter tr-s0 read-res t-eq a-ret a-eq lookup-t ext
  by auto
qed

lemma alpha-round-preserves-alpha-future-fresh:
  assumes future: alpha-future-fresh s
  and outcome: Some (a, t) ∈ set-dist (execute alpha-round s)
  shows alpha-future-fresh t
proof -
  from outcome obtain a0 s0 a1 s1 s2 where
    rand: Some (a0, s0) ∈ set-dist (execute receive-alpha-challenge s)
    and read-a: Some (a1, s1) ∈ set-dist (execute read s0)
    and assert-a: Some ((), s2) ∈ set-dist (execute (assert (a0 = a1)) s1)
    and ret-a: Some (a, t) ∈ set-dist (execute (return a1) s2)
    unfolding alpha-round-def
    by (auto simp: Let-def elim!: set-dist-bindE)
  have counter-t: PAlphaCounter t = Suc (PAlphaCounter s)
    using alpha-round-outcome[OF outcome] by blast
  have future-s0:
    ∧ i x. Suc (PAlphaCounter s) ≤ i ⇒
      fmllookup (HashMap s0) (AlphaChallenge i x) = None
  proof -
    fix i x
    assume i-ge: Suc (PAlphaCounter s) ≤ i
    have neg:
      AlphaChallenge i x ≠
        AlphaChallenge (PAlphaCounter s) (PState s)
      using i-ge by auto
    have fmllookup (HashMap s0) (AlphaChallenge i x) =
      fmllookup (HashMap s) (AlphaChallenge i x)
      by (rule receive-alpha-challenge-preserves-other-lookup[OF rand neg])
  end

```

```

also have ... = None
  using future i-ge unfolding alpha-future-fresh-def by auto
finally show
  fmlookup (HashMap s0) (AlphaChallenge i x) = None .
qed
have hash-s1: HashMap s1 = HashMap s0
  by (rule read-preserves-hash-map[OF read-a])
have s2-eq: s2 = s1
  using assert-a unfolding assert-def
  by (cases a0 = a1) (auto simp: throw-no-outcome)
have t-eq: t = s2
  using ret-a by simp
show ?thesis
  unfolding alpha-future-fresh-def
proof (intro allI impI)
  fix i x
  assume PAlphaCounter t ≤ i
  then have Suc (PAlphaCounter s) ≤ i
    using counter-t by simp
  then show fmlookup (HashMap t) (AlphaChallenge i x) = None
    using future-s0[of i x] hash-s1 unfolding t-eq s2-eq by simp
qed
qed

lemma alpha-round-preserves-composition-fri-future-fresh:
  assumes future: composition-fri-future-fresh s
  and outcome: Some (a, t) ∈ set-dist (execute alpha-round s)
  shows composition-fri-future-fresh t
proof -
  from outcome obtain a0 s0 a1 s1 s2 where
    rand: Some (a0, s0) ∈ set-dist (execute receive-alpha-challenge s)
  and read-a: Some (a1, s1) ∈ set-dist (execute read s0)
  and assert-a: Some ((), s2) ∈ set-dist (execute (assert (a0 = a1)) s1)
  and ret-a: Some (a, t) ∈ set-dist (execute (return a1) s2)
  unfolding alpha-round-def
  by (auto simp: Let-def elim!: set-dist-bindE)
have counter-s0:
  PCompositionFriCounter s0 = PCompositionFriCounter s
  using receive-alpha-challenge-counter-outcome[OF rand] by simp
have future-s0:
  composition-fri-future-fresh s0
  unfolding composition-fri-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-ge-s0: PCompositionFriCounter s0 ≤ i
  have i-ge-s: PCompositionFriCounter s ≤ i
    using i-ge-s0 counter-s0 by simp
  have neg:
    CompositionFriChallenge i x ≠

```

```

    AlphaChallenge (PAlphaCounter s) (PState s)
  by simp
  have fmllookup (HashMap s0) (CompositionFriChallenge i x) =
    fmllookup (HashMap s) (CompositionFriChallenge i x)
  by (rule receive-alpha-challenge-preserves-other-lookup[OF rand neg])
  also have ... = None
  using future i-ge-s
  unfolding composition-fri-future-fresh-def by blast
  finally show
    fmllookup (HashMap s0) (CompositionFriChallenge i x) = None .
qed
have future-s1:
  composition-fri-future-fresh s1
  by (rule read-preserves-composition-fri-future-fresh
    [OF future-s0 read-a])
have s2-eq: s2 = s1
  using assert-a unfolding assert-def
  by (cases a0 = a1) (auto simp: throw-no-outcome)
have t-eq: t = s2
  using ret-a by simp
show ?thesis
  using future-s1 unfolding t-eq s2-eq .
qed

lemma alpha-round-preserves-query-future-fresh:
  assumes future: query-future-fresh s
  and outcome: Some (a, t) ∈ set-dist (execute alpha-round s)
  shows query-future-fresh t
proof -
  from outcome obtain a0 s0 a1 s1 s2 where
    rand: Some (a0, s0) ∈ set-dist (execute receive-alpha-challenge s)
    and read-a: Some (a1, s1) ∈ set-dist (execute read s0)
    and assert-a: Some ((), s2) ∈ set-dist (execute (assert (a0 = a1)) s1)
    and ret-a: Some (a, t) ∈ set-dist (execute (return a1) s2)
  unfolding alpha-round-def
  by (auto simp: Let-def elim!: set-dist-bindE)
  have counter-s0:
    PQueryCounter s0 = PQueryCounter s
  using receive-alpha-challenge-counter-outcome[OF rand] by simp
  have future-s0:
    query-future-fresh s0
  unfolding query-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-ge-s0: PQueryCounter s0 ≤ i
  have i-ge-s: PQueryCounter s ≤ i
    using i-ge-s0 counter-s0 by simp
  have neg:
    QueryIndexChallenge i x ≠

```

```

    AlphaChallenge (PAlphaCounter s) (PState s)
  by simp
  have fmllookup (HashMap s0) (QueryIndexChallenge i x) =
    fmllookup (HashMap s) (QueryIndexChallenge i x)
  by (rule receive-alpha-challenge-preserves-other-lookup[OF rand neq])
  also have ... = None
  using future i-ge-s
  unfolding query-future-fresh-def by blast
  finally show
    fmllookup (HashMap s0) (QueryIndexChallenge i x) = None .
qed
have future-s1:
  query-future-fresh s1
  by (rule read-preserves-query-future-fresh[OF future-s0 read-a])
have s2-eq: s2 = s1
  using assert-a unfolding assert-def
  by (cases a0 = a1) (auto simp: throw-no-outcome)
have t-eq: t = s2
  using ret-a by simp
show ?thesis
  using future-s1 unfolding t-eq s2-eq .
qed

lemma wp-alpha-round-fresh-set:
  assumes future: alpha-future-fresh s
  shows
    wp-event alpha-round
      ( $\lambda out. \text{case out of None} \Rightarrow \text{False} \mid \text{Some } (a, -) \Rightarrow a \in B$ ) s
       $\leq$  nnreal (card B) / nnreal size
proof -
  let ?P =  $\lambda out. \text{case out of None} \Rightarrow \text{False} \mid \text{Some } (a, -) \Rightarrow a \in B$ 
  let ?k =
     $\lambda a0. \text{read} \gg (\lambda a1. \text{assert } (a0 = a1) \gg (\lambda -. \text{return } a1))$ 
  have head-bound:
    wp-event receive-alpha-challenge ?P s  $\leq$ 
      nnreal (card B) / nnreal size
  using wp-receive-alpha-challenge-fresh-set
    [OF alpha-future-fresh-current[OF future], of B]
  by simp
  have round-eq: alpha-round = (receive-alpha-challenge  $\gg$  ?k)
  unfolding alpha-round-def by (simp add: Let-def)
  show ?thesis
    unfolding round-eq
  proof (rule wp-event-bind-bound-by-head-event[OF head-bound])
    show ?P None  $\implies$  ?P None
    by simp
  next
    fix a0 s0 out

```

```

assume cont: out ∈ set-dist (execute (?k a0) s0)
  and hit: ?P out
show ?P (Some (a0, s0))
proof (cases out)
  case None
  then show ?thesis
    using hit by simp
next
  case (Some pair)
  obtain a t where pair-eq: pair = (a, t)
    by (cases pair)
  from cont[unfolded Some pair-eq]
  obtain a1 s1 s2 where
    read-a: Some (a1, s1) ∈ set-dist (execute read s0)
    and assert-a:
      Some ((), s2) ∈ set-dist (execute (assert (a0 = a1)) s1)
    and ret-a: Some (a, t) ∈ set-dist (execute (return a1) s2)
    by (auto elim!: set-dist-bindE)
  have a0-eq: a0 = a1
    using assert-a unfolding assert-def
    by (cases a0 = a1) (auto simp: throw-no-outcome)
  have a-eq: a = a1
    using ret-a by simp
  have a ∈ B
    using hit Some pair-eq by simp
  then show ?thesis
    using a0-eq a-eq by simp
qed
qed
qed

lemma wp-alpha-round-success-bound:
  assumes future: alpha-future-fresh s
  shows
    wp-event alpha-round
    ( $\lambda out. \text{case out of } None \Rightarrow False \mid \text{Some } - \Rightarrow True$ ) s
    ≤ 1 / nnreal size
proof (cases PTranscript s)
  case Nil
  have no-success:
     $\bigwedge out. out \in \text{set-dist } (\text{execute } \alpha\text{-round } s) \implies$ 
    ( $\text{case out of } None \Rightarrow False \mid \text{Some } - \Rightarrow True$ )  $\implies False$ 
  proof –
    fix out
    assume out: out ∈ set-dist (execute alpha-round s)
    and success: ( $\text{case out of } None \Rightarrow False \mid \text{Some } - \Rightarrow True$ )
    from success obtain a t where out-eq: out = Some (a, t)
      by (cases out) auto
    from alpha-round-outcome[OF out[unfolded out-eq]]

```

```

obtain rest where  $P\text{Transcript } s = a \# \text{rest}$ 
  by blast
then show False
  using Nil by simp
qed
have wp-event alpha-round
  ( $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } - \Rightarrow \text{True}$ ) s
   $\leq$  wp-event alpha-round
  ( $\lambda - :: ('f \times ('f, 'a) \text{protocol-channel-scheme}) \text{option. False}$ ) s
  by (rule wp-event-mono-on-support) (rule no-success)
also have  $\dots = 0$ 
  unfolding wp-event-def wp-def dist-expect-def by simp
also have  $\dots \leq 1 / \text{nnreal size}$ 
  by simp
finally show ?thesis .
next
case (Cons x rest)
let ?Q =  $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } - \Rightarrow \text{True}$ 
let ?P =  $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } (a, -) \Rightarrow a \in \{x\}$ 
let ?k =
   $\lambda a0. \text{read} \gg (\lambda a1. \text{assert } (a0 = a1) \gg (\lambda -. \text{return } a1))$ 
have head-bound:
  wp-event receive-alpha-challenge ?P s  $\leq 1 / \text{nnreal size}$ 
proof –
  have wp-event receive-alpha-challenge ?P s =
    nnreal (card ({x} :: 'f set)) / nnreal size
  using wp-receive-alpha-challenge-fresh-set
    [OF alpha-future-fresh-current[OF future], of {x}]
  by simp
  then show ?thesis
  by simp
qed
have round-eq: alpha-round = (receive-alpha-challenge  $\gg$  ?k)
  unfolding alpha-round-def by (simp add: Let-def)
show ?thesis
  unfolding round-eq
proof (rule wp-event-bind-bound-by-head-event[OF head-bound])
  show ?Q None  $\implies$  ?P None
  by simp
next
fix a0 s0 out
assume rand:
  Some (a0, s0)  $\in$  set-dist (execute receive-alpha-challenge s)
  and cont: out  $\in$  set-dist (execute (?k a0) s0)
  and success: ?Q out
from success obtain a t where out-eq: out = Some (a, t)
  by (cases out) auto
from cont[unfolded out-eq]

```

```

obtain  $a1\ s1\ s2$  where
   $read\text{-}a$ :  $Some\ (a1,\ s1) \in set\text{-}dist\ (execute\ read\ s0)$ 
  and  $assert\text{-}a$ :
     $Some\ ((),\ s2) \in set\text{-}dist\ (execute\ (assert\ (a0 = a1))\ s1)$ 
  and  $ret\text{-}a$ :  $Some\ (a,\ t) \in set\text{-}dist\ (execute\ (return\ a1)\ s2)$ 
  by ( $auto\ elim!$ :  $set\text{-}dist\ bindE$ )
have  $tr\text{-}s0$ :  $PTranscript\ s0 = x \# rest$ 
  using  $receive\text{-}\alpha\text{-}challenge\text{-}outcome$ [ $OF\ rand$ ] Cons by  $simp$ 
have  $a1\text{-}eq$ :  $a1 = x$ 
  using  $read\text{-}nonempty\text{-}outcome$ [ $OF\ read\text{-}a\ tr\text{-}s0$ ] by  $simp$ 
have  $a0\text{-}eq$ :  $a0 = a1$ 
  using  $assert\text{-}a$  unfolding  $assert\text{-}def$ 
  by ( $cases\ a0 = a1$ ) ( $auto\ simp$ :  $throw\text{-}no\text{-}outcome$ )
show  $?P\ (Some\ (a0,\ s0))$ 
  using  $a0\text{-}eq\ a1\text{-}eq$  by  $simp$ 
qed
qed

lemma  $mmap\text{-}\alpha\text{-}round\text{-}outcome$ :
assumes  $outcome$ :
   $Some\ (as,\ t) \in set\text{-}dist\ (execute\ (mmap\ (replicate\ n\ \alpha\text{-}round))\ s)$ 
shows
   $length\ as = n \wedge$ 
   $PTranscript\ s = as\ @\ PTranscript\ t \wedge$ 
   $PState\ t = foldl\ concat\ (PState\ s)\ as \wedge$ 
   $s \leq t \wedge$ 
   $PTraceFriCounter\ t = PTraceFriCounter\ s \wedge$ 
   $PCompositionFriCounter\ t = PCompositionFriCounter\ s \wedge$ 
   $PAlphaCounter\ t = PAlphaCounter\ s + n \wedge$ 
   $PQueryCounter\ t = PQueryCounter\ s \wedge$ 
   $(\forall i < n.$ 
     $fmlookup\ (HashMap\ t)$ 
     $(AlphaChallenge\ (PAlphaCounter\ s + i)$ 
       $(foldl\ concat\ (PState\ s)\ (take\ i\ as))) =$ 
       $Some\ (as\ !\ i))$ 
  using  $outcome$ 
proof ( $induction\ n\ arbitrary$ :  $s\ as\ t$ )
case 0
then show  $?case$ 
  by ( $simp\ add$ :  $hash\text{-}ext\text{-}refl$ )
next
case ( $Suc\ n$ )
from  $Suc.prem$ s obtain  $a\ as'\ s1$  where
   $head$ :  $Some\ (a,\ s1) \in set\text{-}dist\ (execute\ \alpha\text{-}round\ s)$ 
  and  $tail$ :
     $Some\ (as',\ t) \in set\text{-}dist\ (execute\ (mmap\ (replicate\ n\ \alpha\text{-}round))\ s1)$ 
  and  $as\text{-}eq$ :  $as = a \# as'$ 
  by ( $auto\ elim!$ :  $set\text{-}dist\ bindE$ )
from  $\alpha\text{-}round\text{-}outcome$ [ $OF\ head$ ] obtain  $rest1$  where

```

```
|s: PTranscript s = a # rest1
and st-s1: PState s1 = concat (PState s) a
and tr-s1: PTranscript s1 = rest1
and ext-head: s ≤ s1
and trace-count-s1: PTraceFriCounter s1 = PTraceFriCounter s
and comp-count-s1: PCompositionFriCounter s1 = PCompositionFriCounter s
and alpha-count-s1: PAlphaCounter s1 = Suc (PAlphaCounter s)
and query-count-s1: PQueryCounter s1 = PQueryCounter s
and lookup-head: fmlookup (HashMap s1) (AlphaChallenge (PAlphaCounter s) (PState s))
= Some a
  by blast
have tail-res:
  length as' = n ∧
  PTranscript s1 = as' @ PTranscript t ∧
  PState t = foldl concat (PState s1) as' ∧
  s1 ≤ t ∧
  PTraceFriCounter t = PTraceFriCounter s1 ∧
  PCompositionFriCounter t = PCompositionFriCounter s1 ∧
  PAlphaCounter t = PAlphaCounter s1 + n ∧
  PQueryCounter t = PQueryCounter s1 ∧
  (∀ i < n.
    fmlookup (HashMap t)
      (AlphaChallenge (PAlphaCounter s1 + i)
        (foldl concat (PState s1) (take i as')))) =
    Some (as' ! i))
  using Suc.IH[OF tail] .
have ext-s1-t: s1 ≤ t
  using tail-res by simp
have lookup-head-t:
  fmlookup (HashMap t) (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a
  using hash-extension-lookup[OF lookup-head ext-s1-t] .
have ext-combined: s ≤ t
  using ext-head ext-s1-t by (meson hash-ext-trans)
have counters-t:
  PTraceFriCounter t = PTraceFriCounter s ∧
  PCompositionFriCounter t = PCompositionFriCounter s ∧
  PAlphaCounter t = PAlphaCounter s + Suc n ∧
  PQueryCounter t = PQueryCounter s
  using tail-res trace-count-s1 comp-count-s1 alpha-count-s1 query-count-s1
  by simp
have lookups:
  ∀ i < Suc n.
    fmlookup (HashMap t)
      (AlphaChallenge (PAlphaCounter s + i)
        (foldl concat (PState s) (take i as')))) =
    Some (as ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < Suc n

|  |

```



```

show fmlookup (HashMap t)
  (AlphaChallenge (PAlphaCounter s + i)
    (foldl concat (PState s) (take i as))) =
  Some (as ! i)
proof (cases i)
  case 0
  then show ?thesis
    using lookup-head-t as-eq by simp
next
  case (Suc j)
  then have j-bound: j < n
    using i-bound by simp
  have key-eq:
    foldl concat (PState s) (take i as) =
      foldl concat (PState s1) (take j as')
    using Suc as-eq st-s1 by simp
  show ?thesis
    using tail-res j-bound Suc as-eq key-eq alpha-count-s1 by simp
qed
qed
show ?case
  using tail-res tr-s tr-s1 st-s1 ext-combined lookups as-eq
    counters-t
  by simp
qed

lemma mmap-alpha-round-preserves-alpha-future-fresh:
  assumes future: alpha-future-fresh s
  and outcome:
    Some (as, t) ∈
      set-dist (execute (mmap (replicate n alpha-round))) s)
  shows alpha-future-fresh t
  using outcome future
proof (induction n arbitrary: s as t)
  case 0
  then show ?case
    by simp
next
  case (Suc n)
  from Suc.prems obtain a as' s1 where
    head: Some (a, s1) ∈ set-dist (execute alpha-round s)
    and tail:
      Some (as', t) ∈
        set-dist (execute (mmap (replicate n alpha-round))) s1)
    by (auto elim!: set-dist-bindE)
  have future-s1: alpha-future-fresh s1
    by (rule alpha-round-preserves-alpha-future-fresh[OF Suc.prems(2) head])
  show ?case
    by (rule Suc.IH[OF tail future-s1])

```

qed

lemma *mmap-alpha-round-preserves-composition-fri-future-fresh:*

assumes *future: composition-fri-future-fresh s*

and *outcome:*

Some (as, t) ∈
set-dist (execute (mmap (replicate n alpha-round)) s)

shows *composition-fri-future-fresh t*

using *outcome future*

proof (*induction n arbitrary: s as t*)

case 0

then show ?*case*

by *simp*

next

case (*Suc n*)

from *Suc.prem*s **obtain** *a as' s1* **where**

head: Some (a, s1) ∈ set-dist (execute alpha-round s)

and *tail:*

Some (as', t) ∈
set-dist (execute (mmap (replicate n alpha-round)) s1)

by (*auto elim!: set-dist-bindE*)

have *future-s1: composition-fri-future-fresh s1*

by (*rule alpha-round-preserves-composition-fri-future-fresh*
*[OF Suc.prem*s(2) *head]*)

show ?*case*

by (*rule Suc.IH [OF tail future-s1]*)

qed

lemma *mmap-alpha-round-preserves-query-future-fresh:*

assumes *future: query-future-fresh s*

and *outcome:*

Some (as, t) ∈
set-dist (execute (mmap (replicate n alpha-round)) s)

shows *query-future-fresh t*

using *outcome future*

proof (*induction n arbitrary: s as t*)

case 0

then show ?*case*

by *simp*

next

case (*Suc n*)

from *Suc.prem*s **obtain** *a as' s1* **where**

head: Some (a, s1) ∈ set-dist (execute alpha-round s)

and *tail:*

Some (as', t) ∈
set-dist (execute (mmap (replicate n alpha-round)) s1)

by (*auto elim!: set-dist-bindE*)

have *future-s1: query-future-fresh s1*

by (*rule alpha-round-preserves-query-future-fresh*)

```

    [OF Suc.premis(2) head])
  show ?case
  by (rule Suc.IH[OF tail future-s1])
qed

lemma wp-mmap-alpha-round-success-bound:
  assumes future: alpha-future-fresh s
  shows
    wp-event (mmap (replicate n alpha-round))
      (λout. case out of None ⇒ False | Some - ⇒ True) s
      ≤ (1 / nnreal size) ^ n
  using future
proof (induction n arbitrary: s)
  case 0
  then show ?case
  by (simp add: wp-event-def wpsimps)
next
  case (Suc n)
  let ?ListSuccess =
    λout :: ('f list × ('f, 'a) protocol-channel-scheme) option.
      case out of None ⇒ False | Some - ⇒ True
  let ?HeadSuccess =
    λout :: ('f × ('f, 'a) protocol-channel-scheme) option.
      case out of None ⇒ False | Some - ⇒ True
  let ?tail =
    (mmap (replicate n alpha-round) ::
      ('f list, ('f, 'a) protocol-channel-scheme) state-monad)
  let ?D = (1 / nnreal size) ^ n
  have bind-bound:
    wp-event
      (alpha-round >>=
        (λa :: 'f. ?tail >>= (λas :: 'f list. return (a # as))))
      ?ListSuccess s
      ≤ wp-event alpha-round ?HeadSuccess s * ?D
  proof (rule wp-event-bind-bound-by-head-and-cont
    [where P = ?HeadSuccess])
    show ¬ ?ListSuccess None
    by simp
  next
    fix a t
    assume ¬ ?HeadSuccess (Some (a, t))
    then show
      wp-event (?tail >>= (λas :: 'f list. return (a # as)))
        ?ListSuccess t = 0
    by simp
  next
    fix a t
    assume head: Some (a, t) ∈ set-dist (execute alpha-round s)
    have future-t: alpha-future-fresh t

```

```

  by (rule alpha-round-preserves-alpha-future-fresh[OF Suc.prem head])
have tail-bound: wp-event ?tail ?ListSuccess t ≤ ?D
  by (rule Suc.IH[OF future-t])
have cont-eq:
  (λout. case out of
    None ⇒ (0 :: prob)
  | Some (as, u) ⇒
    wp (return (a # as))
    (λout. if ?ListSuccess out then 1 else 0) u) =
  (λout. if ?ListSuccess out then 1 else 0)
  by (rule ext) (auto simp: wpsimps split: option.splits prod.splits)
have wp-event (?tail ≫ (λas :: 'f list. return (a # as)))
  ?ListSuccess t =
  wp-event ?tail ?ListSuccess t
  unfolding wp-event-def
  by (simp add: wpsimps cont-eq)
also have ... ≤ ?D
  by (rule tail-bound)
finally show
  wp-event (?tail ≫ (λas :: 'f list. return (a # as)))
  ?ListSuccess t ≤ ?D .
qed
have wp-event (mmap (replicate (Suc n) alpha-round)) ?ListSuccess s
  ≤ wp-event alpha-round ?HeadSuccess s * ?D
  using bind-bound by simp
also have ... ≤ (1 / nnreal size) * ?D
  by (rule mult-right-mono[OF wp-alpha-round-success-bound[OF Suc.prem]])
  simp
also have ... = (1 / nnreal size) ^ Suc n
  by simp
finally show ?case .
qed

lemma wp-mmap-alpha-round-finite-set-bound:
  assumes future: alpha-future-fresh s
  and finite-B: finite B
  shows
    wp-event (mmap (replicate n alpha-round))
      (λout. case out of None ⇒ False | Some (as, -) ⇒ as ∈ B) s
      ≤ nnreal (card B) / (nnreal size) ^ n
proof (cases B = {})
case True
have wp-event (mmap (replicate n alpha-round))
  (λout. case out of None ⇒ False | Some (as, -) ⇒ as ∈ B) s
  ≤ wp-event (mmap (replicate n alpha-round))
    (λ- :: ('f list × ('f, 'a) protocol-channel-scheme) option. False) s
  unfolding True
  by (rule wp-event-mono) (auto split: option.splits prod.splits)
also have ... = 0

```

```

    unfolding wp-event-def wp-def dist-expect-def by simp
  finally show ?thesis
    unfolding True by simp
next
case False
have card-pos:  $0 < \text{card } B$ 
  using finite-B False by (simp add: card-gt-0-iff)
have event-success:
  wp-event (mmap (replicate n alpha-round))
    ( $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } (as, -) \Rightarrow as \in B$ ) s
     $\leq$  wp-event (mmap (replicate n alpha-round))
    ( $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } - \Rightarrow \text{True}$ ) s
  by (rule wp-event-mono) (auto split: option.splits prod.splits)
also have ...  $\leq (1 / \text{nnreal size})^n$ 
  by (rule wp-mmap-alpha-round-success-bound[OF future])
also have ...  $= 1 / (\text{nnreal size})^n$ 
  using size-card by transfer (simp add: power-divide)
also have ...  $= \text{nnreal } (1::\text{nat}) / \text{nnreal } (\text{size}^n)$ 
  by simp
also have ...  $\leq \text{nnreal } (\text{card } B) / \text{nnreal } (\text{size}^n)$ 
  by (rule nnreal-nat-divide-right-mono) (use card-pos in simp)
also have ...  $\leq \text{nnreal } (\text{card } B) / (\text{nnreal size})^n$ 
  by simp
finally show ?thesis .
qed

```

```

lemma wp-mmap-alpha-round-alpha-space-set-bound:
  assumes future: alpha-future-fresh s
  and subset:  $B \subseteq \text{alpha-space}$ 
  shows
    wp-event (mmap (replicate (length spec) alpha-round))
      ( $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } (as, -) \Rightarrow as \in B$ ) s
       $\leq \text{nnreal } (\text{card } B) / \text{nnreal } (\text{card alpha-space})$ 
proof -
  have finite-B: finite B
  by (rule finite-subset[OF subset finite-alpha-space])
  have wp-event (mmap (replicate (length spec) alpha-round))
    ( $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } (as, -) \Rightarrow as \in B$ ) s
     $\leq \text{nnreal } (\text{card } B) / (\text{nnreal size})^{\text{length spec}}$ 
  by (rule wp-mmap-alpha-round-finite-set-bound[OF future finite-B])
  also have ...  $= \text{nnreal } (\text{card } B) / \text{nnreal } (\text{card alpha-space})$ 
  using size-card by simp
  finally show ?thesis .
qed

```

```

lemma receive-fri-commits-with-outcome:
  assumes receiver-outcome:
     $\bigwedge b \ s\text{-read } t.$ 
    Some  $(b, t) \in \text{set-dist } (\text{execute receive-challenge } s\text{-read}) \implies$ 

```

```

    s-read ≤ t ∧
    PState t = PState s-read ∧
    PTranscript t = PTranscript s-read ∧
    fmlookup (HashMap t) (tag (PState s-read)) = Some b
  and outcome:
    Some ((b, r), t) ∈
      set-dist (execute (receive-fri-commits-with receive-challenge) s)
  shows
    ∃ rest.
      PTranscript s = r # rest ∧
      PState t = concat (PState s) r ∧
      PTranscript t = rest ∧
      s ≤ t ∧
      fmlookup (HashMap t) (tag (concat (PState s) r)) = Some b
  proof -
    from outcome obtain s-read where
      read-r: Some (r, s-read) ∈ set-dist (execute read s)
    and rand-b:
      Some (b, t) ∈ set-dist (execute receive-challenge s-read)
    unfolding receive-fri-commits-with-def
    by (auto elim!: set-dist-bindE)
  obtain rest where tr-s: PTranscript s = r # rest
  proof -
    have PTranscript s ≠ []
      using read-r unfolding read-def protocol-read-def assert-def
      by (cases PTranscript s) (auto simp: throw-no-outcome elim!: set-dist-bindE)
    then obtain x xs where xs: PTranscript s = x # xs
      by (cases PTranscript s) auto
    have r-x: r = x
      using read-nonempty-outcome[OF read-r xs] by simp
    show ?thesis
      using that xs r-x by auto
  qed
  have read-res:
    r = r ∧
    s-read = s(PState := concat (PState s) r, PTranscript := rest)
    using read-nonempty-outcome[OF read-r tr-s] by simp
  have rand-res:
    s-read ≤ t ∧
    PState t = PState s-read ∧
    PTranscript t = PTranscript s-read ∧
    fmlookup (HashMap t) (tag (PState s-read)) = Some b
    using receiver-outcome[OF rand-b] .
  have s-read-ext: s ≤ s-read
    using read-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  have s-t: s ≤ t
    using s-read-ext rand-res by (meson hash-ext-trans)
  show ?thesis
    using tr-s read-res rand-res s-t by auto

```

qed

lemma *receive-trace-fri-commits-outcome*:

assumes *outcome*:

Some ((b, r), t) ∈ set-dist (execute receive-trace-fri-commits s)

shows

$\exists \text{rest.}$

$P\text{Transcript } s = r \# \text{rest} \wedge$

$P\text{State } t = \text{concat } (P\text{State } s) \ r \wedge$

$P\text{Transcript } t = \text{rest} \wedge$

$s \leq t \wedge$

$P\text{TraceFriCounter } t = \text{Suc } (P\text{TraceFriCounter } s) \wedge$

$P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s \wedge$

$P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$

$P\text{QueryCounter } t = P\text{QueryCounter } s \wedge$

$\text{fmlookup } (\text{HashMap } t)$

$(\text{TraceFriChallenge } (P\text{TraceFriCounter } s) (\text{concat } (P\text{State } s) \ r)) =$

$\text{Some } b$

proof –

from *outcome* **obtain** *s-read* **where**

read-r: $\text{Some } (r, s\text{-read}) \in \text{set-dist } (\text{execute read } s)$

and *rand-b*:

$\text{Some } (b, t) \in \text{set-dist } (\text{execute receive-trace-fri-challenge } s\text{-read})$

unfolding *receive-trace-fri-commits-def* *receive-fri-commits-with-def*

by (*auto elim!*: *set-dist-bindE*)

obtain *rest* **where** *tr-s*: $P\text{Transcript } s = r \# \text{rest}$

proof –

have $P\text{Transcript } s \neq []$

using *read-r* **unfolding** *read-def* *protocol-read-def* *assert-def*

by (*cases* $P\text{Transcript } s$) (*auto simp*: *throw-no-outcome elim!*: *set-dist-bindE*)

then obtain *x xs* **where** *xs*: $P\text{Transcript } s = x \# xs$

by (*cases* $P\text{Transcript } s$) *auto*

have *r-x*: $r = x$

using *read-nonempty-outcome*[*OF* *read-r xs*] **by** *simp*

show *?thesis*

using *that xs r-x* **by** *auto*

qed

have *read-res*:

$s\text{-read} = s(P\text{State} := \text{concat } (P\text{State } s) \ r, P\text{Transcript} := \text{rest})$

using *read-nonempty-outcome*[*OF* *read-r tr-s*] **by** *simp*

have *rand-res*:

$s\text{-read} \leq t \wedge$

$P\text{State } t = P\text{State } s\text{-read} \wedge$

$P\text{Transcript } t = P\text{Transcript } s\text{-read} \wedge$

$\text{fmlookup } (\text{HashMap } t)$

$(\text{TraceFriChallenge } (P\text{TraceFriCounter } s\text{-read}) (P\text{State } s\text{-read})) =$

$\text{Some } b$

using *receive-trace-fri-challenge-outcome*[*OF* *rand-b*] .

```

have rand-counter:
  PTraceFriCounter t = Suc (PTraceFriCounter s-read) ∧
  PCompositionFriCounter t = PCompositionFriCounter s-read ∧
  PAlphaCounter t = PAlphaCounter s-read ∧
  PQueryCounter t = PQueryCounter s-read
  using receive-trace-fri-challenge-counter-outcome[OF rand-b] by simp
have s-read-ext: s ≤ s-read
  using read-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s-t: s ≤ t
  using s-read-ext rand-res by (meson hash-ext-trans)
show ?thesis
  using tr-s read-res rand-res rand-counter s-t by auto
qed

lemma receive-trace-fri-commits-preserves-trace-fri-future-fresh:
  assumes future: trace-fri-future-fresh s
  and outcome:
    Some ((b, r), t) ∈
      set-dist (execute receive-trace-fri-commits s)
  shows trace-fri-future-fresh t
proof –
  from outcome obtain s-read where
    read-r: Some (r, s-read) ∈ set-dist (execute read s)
  and rand-b:
    Some (b, t) ∈ set-dist (execute receive-trace-fri-challenge s-read)
  unfolding receive-trace-fri-commits-def receive-fri-commits-with-def
  by (auto elim!: set-dist-bindE)
from read-outcome[OF read-r] have read-counts:
  PTraceFriCounter s-read = PTraceFriCounter s
  by blast
have read-hash: HashMap s-read = HashMap s
  by (rule read-preserves-hash-map[OF read-r])
have counter-t:
  PTraceFriCounter t = Suc (PTraceFriCounter s)
  using receive-trace-fri-challenge-counter-outcome[OF rand-b] read-counts
  by simp
show ?thesis
  unfolding trace-fri-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-ge-t: PTraceFriCounter t ≤ i
  have i-ge-s: PTraceFriCounter s ≤ i
  using i-ge-t counter-t by simp
  have neg:
    TraceFriChallenge i x ≠
      TraceFriChallenge (PTraceFriCounter s-read) (PState s-read)
  using i-ge-t counter-t read-counts by auto
  have fmlookup (HashMap t) (TraceFriChallenge i x) =
    fmlookup (HashMap s-read) (TraceFriChallenge i x)

```



```

    by (rule receive-trace-fri-challenge-preserves-other-lookup
        [OF rand-b neq])
  also have ... = fmlookup (HashMap s) (TraceFriChallenge i x)
    using read-hash by simp
  also have ... = None
    using future i-ge-s unfolding trace-fri-future-fresh-def by blast
  finally show fmlookup (HashMap t) (TraceFriChallenge i x) = None .
qed
qed

```

```

lemma receive-trace-fri-commits-preserves-alpha-future-fresh:
  assumes future: alpha-future-fresh s
  and outcome:
    Some ((b, r), t) ∈
      set-dist (execute receive-trace-fri-commits s)
  shows alpha-future-fresh t
proof -
  from outcome obtain s-read where
    read-r: Some (r, s-read) ∈ set-dist (execute read s)
  and rand-b:
    Some (b, t) ∈ set-dist (execute receive-trace-fri-challenge s-read)
  unfolding receive-trace-fri-commits-def receive-fri-commits-with-def
  by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-r] have read-counts:
    PAlphaCounter s-read = PAlphaCounter s
  by blast
  have read-hash: HashMap s-read = HashMap s
  by (rule read-preserves-hash-map[OF read-r])
  have counter-t:
    PAlphaCounter t = PAlphaCounter s
  using receive-trace-fri-challenge-counter-outcome[OF rand-b]
  read-counts
  by simp
  show ?thesis
  unfolding alpha-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-ge-t: PAlphaCounter t ≤ i
  have i-ge-s: PAlphaCounter s ≤ i
    using i-ge-t counter-t by simp
  have neg:
    AlphaChallenge i x ≠
      TraceFriChallenge (PTraceFriCounter s-read) (PState s-read)
  by simp
  have fmlookup (HashMap t) (AlphaChallenge i x) =
    fmlookup (HashMap s-read) (AlphaChallenge i x)
  by (rule receive-trace-fri-challenge-preserves-other-lookup
      [OF rand-b neq])
  also have ... = fmlookup (HashMap s) (AlphaChallenge i x)

```

```

    using read-hash by simp
  also have ... = None
    using future i-ge-s unfolding alpha-future-fresh-def by blast
  finally show fmlookup (HashMap t) (AlphaChallenge i x) = None .
qed
qed

lemma receive-trace-fri-commits-preserves-composition-fri-future-fresh:
  assumes future: composition-fri-future-fresh s
  and outcome:
    Some ((b, r), t) ∈
      set-dist (execute receive-trace-fri-commits s)
  shows composition-fri-future-fresh t
proof -
  from outcome obtain s-read where
    read-r: Some (r, s-read) ∈ set-dist (execute read s)
  and rand-b:
    Some (b, t) ∈ set-dist (execute receive-trace-fri-challenge s-read)
  unfolding receive-trace-fri-commits-def receive-fri-commits-with-def
  by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-r] have read-counts:
    PCompositionFriCounter s-read = PCompositionFriCounter s
  by blast
  have read-hash: HashMap s-read = HashMap s
  by (rule read-preserves-hash-map[OF read-r])
  have counter-t:
    PCompositionFriCounter t = PCompositionFriCounter s
  using receive-trace-fri-challenge-counter-outcome[OF rand-b]
  read-counts
  by simp
  show ?thesis
  unfolding composition-fri-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-ge-t: PCompositionFriCounter t ≤ i
  have i-ge-s: PCompositionFriCounter s ≤ i
    using i-ge-t counter-t by simp
  have neg:
    CompositionFriChallenge i x ≠
      TraceFriChallenge (PTraceFriCounter s-read) (PState s-read)
  by simp
  have fmlookup (HashMap t) (CompositionFriChallenge i x) =
    fmlookup (HashMap s-read) (CompositionFriChallenge i x)
  by (rule receive-trace-fri-challenge-preserves-other-lookup
    [OF rand-b neg])
  also have ... = fmlookup (HashMap s) (CompositionFriChallenge i x)
  using read-hash by simp
  also have ... = None
  using future i-ge-s

```

```

    unfolding composition-fri-future-fresh-def by blast
  finally show
    fmllookup (HashMap t) (CompositionFriChallenge i x) = None .
qed
qed

lemma receive-trace-fri-commits-preserves-query-future-fresh:
  assumes future: query-future-fresh s
  and outcome:
    Some ((b, r), t) ∈
      set-dist (execute receive-trace-fri-commits s)
  shows query-future-fresh t
proof -
  from outcome obtain s-read where
    read-r: Some (r, s-read) ∈ set-dist (execute read s)
  and rand-b:
    Some (b, t) ∈ set-dist (execute receive-trace-fri-challenge s-read)
  unfolding receive-trace-fri-commits-def receive-fri-commits-with-def
  by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-r] have read-counts:
    PQueryCounter s-read = PQueryCounter s
  by blast
  have read-hash: HashMap s-read = HashMap s
  by (rule read-preserves-hash-map[OF read-r])
  have counter-t:
    PQueryCounter t = PQueryCounter s
  using receive-trace-fri-challenge-counter-outcome[OF rand-b]
  read-counts
  by simp
  show ?thesis
  unfolding query-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-ge-t: PQueryCounter t ≤ i
  have i-ge-s: PQueryCounter s ≤ i
  using i-ge-t counter-t by simp
  have neg:
    QueryIndexChallenge i x ≠
      TraceFriChallenge (PTraceFriCounter s-read) (PState s-read)
  by simp
  have fmllookup (HashMap t) (QueryIndexChallenge i x) =
    fmllookup (HashMap s-read) (QueryIndexChallenge i x)
  by (rule receive-trace-fri-challenge-preserves-other-lookup
    [OF rand-b neg])
  also have ... = fmllookup (HashMap s) (QueryIndexChallenge i x)
  using read-hash by simp
  also have ... = None
  using future i-ge-s unfolding query-future-fresh-def by blast
  finally show fmllookup (HashMap t) (QueryIndexChallenge i x) = None .

```

```

qed
qed

lemma wp-receive-trace-fri-commits-fresh-set:
  assumes future: trace-fri-future-fresh s
  shows
    wp-event receive-trace-fri-commits
      (λout. case out of
        None ⇒ False
        | Some ((b, r), -) ⇒ b ∈ B) s
      ≤ nnreal (card B) / nnreal size
  unfolding receive-trace-fri-commits-def receive-fri-commits-with-def
proof (rule wp-event-bind-bound-by-cont)
  show ¬ (case None of
    None ⇒ False | Some ((b, r), -) ⇒ b ∈ B)
  by simp
next
  fix r s-read
  assume read-r: Some (r, s-read) ∈ set-dist (execute read s)
  have read-counts:
    PTraceFriCounter s-read = PTraceFriCounter s
  using read-outcome[OF read-r] by blast
  have read-hash: HashMap s-read = HashMap s
  by (rule read-preserves-hash-map[OF read-r])
  have fresh:
    fmlookup (HashMap s-read)
      (TraceFriChallenge (PTraceFriCounter s-read) (PState s-read)) = None
  using future read-counts read-hash
  unfolding trace-fri-future-fresh-def by simp
  let ?Q =
    λout :: (('f × 'f) × ('f, 'a) protocol-channel-scheme) option.
      case out of None ⇒ False | Some ((b, r), -) ⇒ b ∈ B
  have cont-eq:
    (λout. case out of
      None ⇒ (0 :: prob)
      | Some (b, u) ⇒
        wp (return (b, r)) (λout. if ?Q out then 1 else 0) u) =
    (λout :: ('f × ('f, 'a) protocol-channel-scheme) option.
      if (case out of None ⇒ False | Some (b, -) ⇒ b ∈ B)
      then 1 else 0)
  by (rule ext) (auto simp: wpsimps split: option.splits prod.splits)
  have wp-event
    (receive-trace-fri-challenge ≫ (λb :: 'f. return (b, r)))
    ?Q s-read =
    wp-event receive-trace-fri-challenge
      (λout. case out of None ⇒ False | Some (b, -) ⇒ b ∈ B)
      s-read
  unfolding wp-event-def by (simp add: wpsimps cont-eq)
  also have ... = nnreal (card B) / nnreal size

```

```

    by (rule wp-receive-trace-fri-challenge-fresh-set[OF fresh])
  finally show
    wp-event
      (receive-trace-fri-challenge  $\gg$  ( $\lambda b :: 'f. \text{return } (b, r)$ ))
      ( $\lambda \text{out. case out of None} \Rightarrow \text{False} \mid \text{Some } ((b, r), -) \Rightarrow b \in B$ )
      s-read
       $\leq \text{nnreal } (\text{card } B) / \text{nnreal size}$ 
    by simp
qed

lemma ntimes-receive-trace-fri-commits-preserves-trace-fri-future-fresh:
  assumes future: trace-fri-future-fresh s
  and outcome:
    Some (brs, t)  $\in$ 
      set-dist (execute (ntimes receive-trace-fri-commits n) s)
  shows trace-fri-future-fresh t
  using outcome future
proof (induction n arbitrary: s brs t)
  case 0
  then show ?case
    by simp
next
  case (Suc n)
  from Suc.prem1 obtain br brs' s1 where
    head:
      Some (br, s1)  $\in$  set-dist (execute receive-trace-fri-commits s)
  and tail:
    Some (brs', t)  $\in$ 
      set-dist (execute (ntimes receive-trace-fri-commits n) s1)
  by (auto elim!: set-dist-bindE)
  obtain b r where br-eq: br = (b, r)
  by (cases br)
  have future-s1:
    trace-fri-future-fresh s1
  by (rule receive-trace-fri-commits-preserves-trace-fri-future-fresh
      [OF Suc.prem2(2) head[unfolded br-eq]])
  show ?case
    by (rule Suc.IH[OF tail future-s1])
qed

lemma ntimes-receive-trace-fri-commits-preserves-alpha-future-fresh:
  assumes outcome:
    Some (brs, t)  $\in$ 
      set-dist (execute (ntimes receive-trace-fri-commits n) s)
  and future: alpha-future-fresh s
  shows alpha-future-fresh t
  using outcome future
proof (induction n arbitrary: s brs t)
  case 0

```

```

then show ?case
  by simp
next
case (Suc n)
from Suc.premis(1) obtain br brs' s1 where
  head:
    Some (br, s1) ∈ set-dist (execute receive-trace-fri-commits s)
  and tail:
    Some (brs', t) ∈
      set-dist (execute (ntimes receive-trace-fri-commits n) s1)
  by (auto elim!: set-dist-bindE)
obtain b r where br-eq: br = (b, r)
  by (cases br)
have future-s1:
  alpha-future-fresh s1
  by (rule receive-trace-fri-commits-preserves-alpha-future-fresh
    [OF Suc.premis(2) head[unfolded br-eq]])
show ?case
  by (rule Suc.IH[OF tail future-s1])
qed

lemma ntimes-receive-trace-fri-commits-preserves-composition-fri-future-fresh:
assumes outcome:
  Some (brs, t) ∈
    set-dist (execute (ntimes receive-trace-fri-commits n) s)
and future: composition-fri-future-fresh s
shows composition-fri-future-fresh t
using outcome future
proof (induction n arbitrary: s brs t)
case 0
then show ?case
  by simp
next
case (Suc n)
from Suc.premis(1) obtain br brs' s1 where
  head:
    Some (br, s1) ∈ set-dist (execute receive-trace-fri-commits s)
  and tail:
    Some (brs', t) ∈
      set-dist (execute (ntimes receive-trace-fri-commits n) s1)
  by (auto elim!: set-dist-bindE)
obtain b r where br-eq: br = (b, r)
  by (cases br)
have future-s1:
  composition-fri-future-fresh s1
  by (rule receive-trace-fri-commits-preserves-composition-fri-future-fresh
    [OF Suc.premis(2) head[unfolded br-eq]])
show ?case
  by (rule Suc.IH[OF tail future-s1])

```

qed

lemma *ntimes-receive-trace-fri-commits-preserves-query-future-fresh:*

assumes *outcome:*

Some (brs, t) ∈

set-dist (execute (ntimes receive-trace-fri-commits n) s)

and *future: query-future-fresh s*

shows *query-future-fresh t*

using *outcome future*

proof (*induction n arbitrary: s brs t*)

case *0*

then show *?case*

by *simp*

next

case (*Suc n*)

from *Suc.prem1* **obtain** *br brs' s1* **where**

head:

Some (br, s1) ∈ set-dist (execute receive-trace-fri-commits s)

and *tail:*

Some (brs', t) ∈

set-dist (execute (ntimes receive-trace-fri-commits n) s1)

by (*auto elim!: set-dist-bindE*)

obtain *b r* **where** *br-eq: br = (b, r)*

by (*cases br*)

have *future-s1:*

query-future-fresh s1

by (*rule receive-trace-fri-commits-preserves-query-future-fresh*
[*OF Suc.prem2*] *head[unfolded br-eq]*)

show *?case*

by (*rule Suc.IH[OF tail future-s1]*)

qed

lemma *verifier-alpha-prefix-preserves-query-future-fresh:*

assumes *future: query-future-fresh s*

and *outcome:*

Some (header, t) ∈ set-dist (execute verifier-alpha-prefix s)

shows *query-future-fresh t*

proof –

from *outcome* **obtain** *fr s1 f-fl s2 f-final s3 as* **where**

read-fr: Some (fr, s1) ∈ set-dist (execute read s)

and *trace-fri:*

Some (f-fl, s2) ∈

set-dist (execute (ntimes receive-trace-fri-commits (ceil-log clength)) s1)

and *read-final: Some (f-final, s3) ∈ set-dist (execute read s2)*

and *alpha:*

Some (as, t) ∈

set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)

unfolding *verifier-alpha-prefix-def*

by (*auto elim!: set-dist-bindE*)

```

have future-s1: query-future-fresh s1
  by (rule read-preserves-query-future-fresh[OF future read-fr])
have future-s2: query-future-fresh s2
  by (rule ntimes-receive-trace-fri-commits-preserves-query-future-fresh
      [OF trace-fri future-s1])
have future-s3: query-future-fresh s3
  by (rule read-preserves-query-future-fresh[OF future-s2 read-final])
show ?thesis
  by (rule mmap-alpha-round-preserves-query-future-fresh
      [OF future-s3 alpha])
qed

```

lemma *wp-verifier-alpha-prefix-alpha-space-set-bound:*

assumes *future: alpha-future-fresh s*

and *subset: B ⊆ alpha-space*

shows

wp-event verifier-alpha-prefix
 $(\lambda out. \text{case out of}$
 $\quad \text{None} \Rightarrow \text{False}$
 $\quad | \text{Some } ((-, -, -, as'), -) \Rightarrow as' \in B) \ s$
 $\leq \text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{alpha-space})$

proof –

let *?Q =*

$\lambda out. \text{case out of}$
 $\quad \text{None} \Rightarrow \text{False}$
 $\quad | \text{Some } ((-, -, -, as'), -) \Rightarrow as' \in B$

let *?C = nnreal (card B) / nnreal (card alpha-space)*

show *?thesis*

unfolding *verifier-alpha-prefix-def*

proof (rule *wp-event-bind-bound-by-cont*)

show $\neg ?Q \text{ None}$

by *simp*

next

fix *fr s1*

assume *read-fr: Some (fr, s1) ∈ set-dist (execute read s)*

have *future-s1: alpha-future-fresh s1*

by (rule *read-preserves-alpha-future-fresh*[OF *future read-fr*])

show *wp-event*

$(\text{ntimes receive-trace-fri-commits } (\text{ceil-log } \text{clength}) \gg=$

$(\lambda f.fl. \text{read } \gg=$

$(\lambda f.final.$

$\text{mmap } (\text{replicate } (\text{length spec}) \text{ alpha-round}) \gg=$

$(\lambda as. \text{return } (fr, f.fl, f.final, as))))$

$?Q \ s1 \leq ?C$

proof (rule *wp-event-bind-bound-by-cont*)

show $\neg ?Q \text{ None}$

by *simp*

next

fix *f-fl s2*


```

assume trace-fri:
  Some (f-fl, s2) ∈
    set-dist
    (execute (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
have future-s2: alpha-future-fresh s2
  by (rule ntimes-receive-trace-fri-commits-preserves-alpha-future-fresh
    [OF trace-fri future-s1])
show wp-event
  (read  $\gg$ 
    (λf-final.
      mmap (replicate (length spec) alpha-round)  $\gg$ 
        (λas. return (fr, f-fl, f-final, as))))
  ?Q s2 ≤ ?C
proof (rule wp-event-bind-bound-by-cont)
  show ¬ ?Q None
  by simp
next
  fix f-final s3
  assume read-final:
    Some (f-final, s3) ∈ set-dist (execute read s2)
  have future-s3: alpha-future-fresh s3
  by (rule read-preserves-alpha-future-fresh [OF future-s2 read-final])
  let ?A =
    λout. case out of
      None ⇒ False
    | Some (as, -) ⇒ as ∈ B
  have alpha-bound:
    wp-event (mmap (replicate (length spec) alpha-round)) ?A s3 ≤ ?C
  by (rule wp-mmap-alpha-round-alpha-space-set-bound
    [OF future-s3 subset])
  show wp-event
    (mmap (replicate (length spec) alpha-round)  $\gg$ 
      (λas. return (fr, f-fl, f-final, as)))
    ?Q s3 ≤ ?C
  proof (rule wp-event-bind-bound-by-head-event [OF alpha-bound])
  show ?Q None  $\implies$  ?A None
  by simp
next
  fix as t out
  assume cont:
    out ∈
      set-dist
      (execute (return (fr, f-fl, f-final, as)) t)
  and hit: ?Q out
  have out-eq: out = Some ((fr, f-fl, f-final, as), t)
  using cont
  unfolding set-dist-def return.rep-eq dist-return-def
    dist-delta-dist delta-map-def
  by simp

```

```

      show ?A (Some (as, t))
      using hit unfolding out-eq by simp
    qed
  qed
  qed
  qed
  qed

lemma wp-verifier-alpha-prefix-dependent-alpha-space-set-bound:
  fixes C :: prob
  assumes future: alpha-future-fresh s
  and subset:
     $\bigwedge fr\ f-fl\ f-final. B\ fr\ f-fl\ f-final \subseteq \text{alpha-space}$ 
  and bound:
     $\bigwedge fr\ f-fl\ f-final. \text{nnreal} (\text{card} (B\ fr\ f-fl\ f-final)) / \text{nnreal} (\text{card} \text{alpha-space}) \leq C$ 
  shows
    wp-event verifier-alpha-prefix
      ( $\lambda out. \text{case out of}$ 
        None  $\Rightarrow$  False
        | Some ((fr, f-fl, f-final, as'), -)  $\Rightarrow$ 
           $as' \in B\ fr\ f-fl\ f-final$ ) s
       $\leq C$ 
  proof -
    let ?Q =
       $\lambda out. \text{case out of}$ 
        None  $\Rightarrow$  False
        | Some ((fr, f-fl, f-final, as'), -)  $\Rightarrow$ 
           $as' \in B\ fr\ f-fl\ f-final$ 
    show ?thesis
      unfolding verifier-alpha-prefix-def
    proof (rule wp-event-bind-bound-by-cont)
      show  $\neg ?Q\ None$ 
        by simp
    next
      fix fr s1
      assume read-fr: Some (fr, s1)  $\in$  set-dist (execute read s)
      have future-s1: alpha-future-fresh s1
        by (rule read-preserves-alpha-future-fresh[OF future read-fr])
      show wp-event
        ( $\text{ntimes receive-trace-fri-commits} (\text{ceil-log clength}) \gg=$ 
          ( $\lambda f-fl. \text{read} \gg=$ 
            ( $\lambda f-final. \text{mmap} (\text{replicate} (\text{length spec}) \text{alpha-round}) \gg=$ 
              ( $\lambda as. \text{return} (fr, f-fl, f-final, as))))$ 
           $?Q\ s1 \leq C$ 
        proof (rule wp-event-bind-bound-by-cont)
          show  $\neg ?Q\ None$ 
            by simp
        end
      end
    end
  end

```

```

next
  fix f-fl s2
  assume trace-fri:
    Some (f-fl, s2) ∈
      set-dist
      (execute (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
  have future-s2: alpha-future-fresh s2
  by (rule ntimes-receive-trace-fri-commits-preserves-alpha-future-fresh
      [OF trace-fri future-s1])
  show wp-event
    (read ≫=
      (λf-final.
        mmap (replicate (length spec) alpha-round) ≫=
          (λas. return (fr, f-fl, f-final, as))))
    ?Q s2 ≤ C
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ ?Q None
    by simp
  next
    fix f-final s3
    assume read-final:
      Some (f-final, s3) ∈ set-dist (execute read s2)
    have future-s3: alpha-future-fresh s3
    by (rule read-preserves-alpha-future-fresh[OF future-s2 read-final])
    let ?B = B fr f-fl f-final
    let ?A =
      λout. case out of
        None ⇒ False
        | Some (as, -) ⇒ as ∈ ?B
    have alpha-bound:
      wp-event (mmap (replicate (length spec) alpha-round)) ?A s3 ≤
        nnreal (card ?B) / nnreal (card alpha-space)
    by (rule wp-mmap-alpha-round-alpha-space-set-bound
        [OF future-s3 subset[of fr f-fl f-final]])
    have alpha-bound-C:
      wp-event (mmap (replicate (length spec) alpha-round)) ?A s3 ≤ C
    by (rule order-trans[OF alpha-bound bound[of fr f-fl f-final]])
    show wp-event
      (mmap (replicate (length spec) alpha-round) ≫=
        (λas. return (fr, f-fl, f-final, as)))
      ?Q s3 ≤ C
    proof (rule wp-event-bind-bound-by-head-event[OF alpha-bound-C])
      show ?Q None ⇒ ?A None
      by simp
    next
      fix as t out
      assume cont:
        out ∈
          set-dist

```

```

      (execute (return (fr, f-fl, f-final, as)) t)
    and hit: ?Q out
  have out-eq: out = Some ((fr, f-fl, f-final, as), t)
    using cont
    unfolding set-dist-def return.rep-eq dist-return-def
      dist-delta-dist delta-map-def
    by simp
  show ?A (Some (as, t))
    using hit unfolding out-eq by simp
qed
qed
qed
qed
qed

lemma wp-ntimes-receive-trace-fri-commits-exact-challenges-bound:
  assumes future: trace-fri-future-fresh s
  and len: length bs = n
  shows
    wp-event (ntimes receive-trace-fri-commits n)
      (λout. case out of
        None ⇒ False
        | Some (brs, -) ⇒ map fst brs = bs) s
      ≤ (1 / nnreal size) ^ n
  using len future
proof (induction n arbitrary: s bs)
  case 0
  then have bs-empty: bs = []
    by simp
  show ?case
    unfolding bs-empty by (simp add: wp-event-def wpsimps)
next
  case (Suc n)
  obtain b bs' where bs-eq: bs = b # bs'
    using Suc.prem1 by (cases bs) auto
  have len-bs': length bs' = n
    using Suc.prem1 bs-eq by simp
  let ?tail =
    (ntimes receive-trace-fri-commits n ::
      (('f × 'f) list, ('f, 'a) protocol-channel-scheme) state-monad)
  let ?Q =
    λout :: (('f × 'f) list) ×
      ('f, 'a) protocol-channel-scheme) option.
      case out of None ⇒ False | Some (brs, -) ⇒ map fst brs = bs
  let ?Head =
    λout :: (('f × 'f) × ('f, 'a) protocol-channel-scheme) option.
      case out of None ⇒ False | Some ((b0, r0), -) ⇒ b0 ∈ {b}
  have head-bound:
    wp-event receive-trace-fri-commits ?Head s ≤ 1 / nnreal size

```

```

proof –
  have wp-event receive-trace-fri-commits ?Head s ≤
    nnreal (card ({b} :: 'f set)) / nnreal size
    by (rule wp-receive-trace-fri-commits-fresh-set[OF Suc.prem(2)])
  then show ?thesis by simp
qed
have bind-bound:
  wp-event
    (receive-trace-fri-commits  $\gg$ 
      ( $\lambda br :: 'f \times 'f.$ 
         $?tail \gg (\lambda brs :: ('f \times 'f) \text{ list. return } (br \# brs))$ )))
   $?Q \ s$ 
   $\leq wp\text{-event receive-trace-fri-commits } ?Head \ s \ *$ 
   $(1 / nnreal \ size) \wedge^n$ 
proof (rule wp-event-bind-bound-by-head-and-cont[where P = ?Head])
  show  $\neg ?Q \ None$ 
  by simp
next
  fix br t
  assume not-head:  $\neg ?Head \ (Some \ (br, \ t))$ 
  obtain b0 r0 where br-eq:  $br = (b0, \ r0)$ 
  by (cases br)
  have b0-ne:  $b0 \neq b$ 
  using not-head unfolding br-eq by simp
  have cont-false:
    wp-event ( $?tail \gg (\lambda brs :: ('f \times 'f) \text{ list. return } (br \# brs))$ )
     $?Q \ t$ 
     $\leq wp\text{-event } (?tail \gg (\lambda brs :: ('f \times 'f) \text{ list. return } (br \# brs)))$ 
     $(\lambda - :: ((('f \times 'f) \text{ list}) \times$ 
       $(('f, 'a) \text{ protocol-channel-scheme}) \text{ option. False}) \ t$ 
  by (rule wp-event-mono-on-support)
    (use b0-ne bs-eq br-eq in
       $\langle auto \text{ simp: wpsimps elim!: set-dist-bindE split: option.splits prod.splits} \rangle$ )
  also have  $\dots = 0$ 
  unfolding wp-event-def wp-def dist-expect-def by simp
  finally have le-zero:
    wp-event ( $?tail \gg (\lambda brs :: ('f \times 'f) \text{ list. return } (br \# brs))$ )
     $?Q \ t \leq 0$  .
  show
    wp-event ( $?tail \gg (\lambda brs :: ('f \times 'f) \text{ list. return } (br \# brs))$ )
     $?Q \ t = 0$ 
    by (rule antisym[OF le-zero]) simp
next
  fix br t
  assume head:
     $Some \ (br, \ t) \in set\text{-dist } (execute \ receive\text{-trace-fri-commits } s)$ 
    and head-hit:  $?Head \ (Some \ (br, \ t))$ 
  obtain b0 r0 where br-eq:  $br = (b0, \ r0)$ 
  by (cases br)

```

```

have b0-eq: b0 = b
  using head-hit unfolding br-eq by simp
have future-t: trace-fri-future-fresh t
  by (rule receive-trace-fri-commits-preserves-trace-fri-future-fresh
    [OF Suc.prems(2) head[unfolded br-eq]])
have tail-bound:
  wp-event ?tail
    ( $\lambda out. \text{case out of } None \Rightarrow False$ 
      |  $\text{Some } (brs, -) \Rightarrow \text{map fst } brs = bs'$ ) t
     $\leq (1 / \text{nnreal size}) ^ n$ 
  by (rule Suc.IH[OF len-bs' future-t])
have cont-eq:
  ( $\lambda out. \text{case out of}$ 
    None  $\Rightarrow (0 :: \text{prob})$ 
    |  $\text{Some } (brs, u) \Rightarrow$ 
       $\text{wp } (\text{return } (br \# brs)) (\lambda out. \text{if } ?Q \text{ out then } 1 \text{ else } 0) u =$ 
      ( $\lambda out :: ((f \times 'f) \text{ list}) \times$ 
        ( $'f, 'a$ ) protocol-channel-scheme) option.
        if (case out of None  $\Rightarrow False$ 
          |  $\text{Some } (brs, -) \Rightarrow \text{map fst } brs = bs'$ )
          then 1 else 0)
  by (rule ext)
  (auto simp: wpsimps b0-eq bs-eq br-eq
    split: option.splits prod.splits)
have wp-event (?tail  $\gg= (\lambda brs :: (f \times 'f) \text{ list. return } (br \# brs))$ )
  ?Q t =
  wp-event ?tail
    ( $\lambda out. \text{case out of } None \Rightarrow False$ 
      |  $\text{Some } (brs, -) \Rightarrow \text{map fst } brs = bs'$ ) t
  unfolding wp-event-def by (simp add: wpsimps cont-eq)
also have ...  $\leq (1 / \text{nnreal size}) ^ n$ 
  by (rule tail-bound)
finally show
  wp-event (?tail  $\gg= (\lambda brs :: (f \times 'f) \text{ list. return } (br \# brs))$ )
  ?Q t  $\leq (1 / \text{nnreal size}) ^ n$  .
qed
have wp-event (ntimes receive-trace-fri-commits (Suc n)) ?Q s
   $\leq \text{wp-event receive-trace-fri-commits } ?\text{Head } s *$ 
   $(1 / \text{nnreal size}) ^ n$ 
  using bind-bound by simp
also have ...  $\leq (1 / \text{nnreal size}) * (1 / \text{nnreal size}) ^ n$ 
  by (rule mult-right-mono[OF head-bound]) simp
also have ... =  $(1 / \text{nnreal size}) ^ {Suc n}$ 
  by simp
finally show ?case .
qed

lemma wp-ntimes-receive-trace-fri-commits-finite-set-bound:
  assumes future: trace-fri-future-fresh s

```

and *finite-B*: *finite B*
and *lengths*: $\bigwedge bs. bs \in B \implies \text{length } bs = n$
shows
 $wp\text{-event } (ntimes \text{ receive-trace-fri-commits } n)$
 $(\lambda out. \text{ case out of}$
 $\quad None \Rightarrow False$
 $\quad | \text{ Some } (brs, -) \Rightarrow \text{map fst } brs \in B) \ s$
 $\leq nnreal \ (card \ B) / (nnreal \ size) ^ n$
proof –
let *?P* =
 $\lambda bs \text{ out. case out of}$
 $\quad None \Rightarrow False$
 $\quad | \text{ Some } (brs, -) \Rightarrow \text{map fst } brs = bs$
have *event-mono*:
 $wp\text{-event } (ntimes \text{ receive-trace-fri-commits } n)$
 $(\lambda out. \text{ case out of}$
 $\quad None \Rightarrow False$
 $\quad | \text{ Some } (brs, -) \Rightarrow \text{map fst } brs \in B) \ s$
 $\leq wp\text{-event } (ntimes \text{ receive-trace-fri-commits } n)$
 $(\lambda out. \exists bs \in B. ?P \ bs \ out) \ s$
by (*rule wp-event-mono*) (*auto split: option.splits prod.splits*)
also have $\dots \leq (\sum bs \in B. (1 / nnreal \ size) ^ n)$
by (*rule wp-event-finite-UN-bound[OF finite-B]*)
 $(\text{rule } wp\text{-ntimes-receive-trace-fri-commits-exact-challenges-bound}$
 $\quad [OF \ \text{future lengths}])$
also have $\dots = nnreal \ (card \ B) * ((1 / nnreal \ size) ^ n)$
using *finite-B* **by** *simp*
also have $\dots = nnreal \ (card \ B) / (nnreal \ size) ^ n$
using *size-card* **by** *transfer (simp add: power-divide)*
finally show *?thesis* .
qed

lemma *wp-ntimes-receive-trace-fri-commits-challenge-space-set-bound*:

assumes *future: trace-fri-future-fresh s*

and *subset*: $B \subseteq \text{fri-challenge-space } n$

shows

$wp\text{-event } (ntimes \text{ receive-trace-fri-commits } n)$
 $(\lambda out. \text{ case out of}$
 $\quad None \Rightarrow False$
 $\quad | \text{ Some } (brs, -) \Rightarrow \text{map fst } brs \in B) \ s$
 $\leq nnreal \ (card \ B) / nnreal \ (CARD(f) ^ n)$

proof –

have *finite-B*: *finite B*

by (*rule finite-subset[OF subset finite-fri-challenge-space]*)

have *lengths*: $\bigwedge bs. bs \in B \implies \text{length } bs = n$

using *subset* **unfolding** *fri-challenge-space-def* **by** *auto*

have *wp-event* (*ntimes receive-trace-fri-commits n*)

$(\lambda out. \text{ case out of}$
 $\quad None \Rightarrow False$

```

    | Some (brs, -)  $\Rightarrow$  map fst brs  $\in B$  s
   $\leq$  nnreal (card B) / (nnreal size)  $^n$ 
  by (rule wp-ntimes-receive-trace-fri-commits-finite-set-bound
    [OF future finite-B lengths])
  also have ... = nnreal (card B) / nnreal (CARD('f)  $^n$ )
    using size-card by simp
  finally show ?thesis .
qed

lemma wp-verifier-trace-fri-prefix-challenge-space-set-bound:
  assumes future: trace-fri-future-fresh s
    and subset:  $B \subseteq$  fri-challenge-space (ceil-log clength)
  shows
    wp-event verifier-trace-fri-prefix
      ( $\lambda$ out. case out of
        None  $\Rightarrow$  False
        | Some ((-, f-fl), -)  $\Rightarrow$  map fst f-fl  $\in B$  s
       $\leq$  nnreal (card B) / nnreal (CARD('f)  $^{\text{ceil-log clength}}$ )
  proof -
    let ?Q =
       $\lambda$ out. case out of
        None  $\Rightarrow$  False
        | Some ((-, f-fl), -)  $\Rightarrow$  map fst f-fl  $\in B$ 
    let ?C = nnreal (card B) / nnreal (CARD('f)  $^{\text{ceil-log clength}}$ )
    show ?thesis
      unfolding verifier-trace-fri-prefix-def
    proof (rule wp-event-bind-bound-by-cont)
      show  $\neg$  ?Q None
        by simp
    next
      fix fr s1
      assume read-fr: Some (fr, s1)  $\in$  set-dist (execute read s)
      have future-s1: trace-fri-future-fresh s1
        by (rule read-preserves-trace-fri-future-fresh[OF future read-fr])
      let ?Head =
         $\lambda$ out. case out of
          None  $\Rightarrow$  False
          | Some (f-fl, -)  $\Rightarrow$  map fst f-fl  $\in B$ 
      have trace-bound:
        wp-event (ntimes receive-trace-fri-commits (ceil-log clength))
          ?Head s1  $\leq$  ?C
      by (rule wp-ntimes-receive-trace-fri-commits-challenge-space-set-bound
        [OF future-s1 subset])
      show wp-event
        (ntimes receive-trace-fri-commits (ceil-log clength)  $\gg$ 
          ( $\lambda$ f-fl. return (fr, f-fl)))
        ?Q s1  $\leq$  ?C
    proof (rule wp-event-bind-bound-by-head-event[OF trace-bound])
      show ?Q None  $\Rightarrow$  ?Head None

```



```

    by simp
next
  fix f-fl t out
  assume cont:
    out ∈ set-dist (execute (return (fr, f-fl)) t)
  and hit: ?Q out
  have out-eq: out = Some ((fr, f-fl), t)
  using cont
  unfolding set-dist-def return.rep-eq dist-return-def
    dist-delta-dist delta-map-def
  by simp
  show ?Head (Some (f-fl, t))
    using hit unfolding out-eq by simp
qed
qed
qed

lemma wp-verifier-trace-fri-prefix-dependent-challenge-space-set-bound:
  fixes C :: prob
  assumes future: trace-fri-future-fresh s
  and subset:  $\bigwedge fr. B \text{ fr} \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$ 
  and bound:
     $\bigwedge fr. \text{nnreal } (\text{card } (B \text{ fr})) / \text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log } \text{clength}) \leq C$ 
  shows
    wp-event verifier-trace-fri-prefix
      ( $\lambda out. \text{case out of}$ 
        None  $\Rightarrow$  False
        | Some ((fr, f-fl), -)  $\Rightarrow \text{map fst f-fl} \in B \text{ fr}$ ) s
       $\leq C$ 
  proof -
    let ?Q =
       $\lambda out. \text{case out of}$ 
        None  $\Rightarrow$  False
        | Some ((fr, f-fl), -)  $\Rightarrow \text{map fst f-fl} \in B \text{ fr}$ 
    show ?thesis
      unfolding verifier-trace-fri-prefix-def
    proof (rule wp-event-bind-bound-by-cont)
      show  $\neg ?Q \text{ None}$ 
        by simp
    next
      fix fr s1
      assume read-fr: Some (fr, s1) ∈ set-dist (execute read s)
      have future-s1: trace-fri-future-fresh s1
        by (rule read-preserves-trace-fri-future-fresh[OF future read-fr])
      let ?Head =
         $\lambda out. \text{case out of}$ 
          None  $\Rightarrow$  False
          | Some (f-fl, -)  $\Rightarrow \text{map fst f-fl} \in B \text{ fr}$ 

```

```

have trace-bound:
  wp-event (ntimes receive-trace-fri-commits (ceil-log clength))
    ?Head s1 ≤
      nnreal (card (B fr)) / nnreal (CARD('f) ^ ceil-log clength)
  by (rule wp-ntimes-receive-trace-fri-commits-challenge-space-set-bound
    [OF future-s1 subset[of fr]])
have trace-bound-C:
  wp-event (ntimes receive-trace-fri-commits (ceil-log clength))
    ?Head s1 ≤ C
  by (rule order-trans[OF trace-bound bound[of fr]])
show wp-event
  (ntimes receive-trace-fri-commits (ceil-log clength) ≫=
    (λf-fl. return (fr, f-fl)))
  ?Q s1 ≤ C
proof (rule wp-event-bind-bound-by-head-event[OF trace-bound-C])
  show ?Q None ⇒ ?Head None
  by simp
next
  fix f-fl t out
  assume cont:
    out ∈ set-dist (execute (return (fr, f-fl)) t)
  and hit: ?Q out
  have out-eq: out = Some ((fr, f-fl), t)
  using cont
  unfolding set-dist-def return.rep-eq dist-return-def
    dist-delta-dist delta-map-def
  by simp
  show ?Head (Some (f-fl, t))
  using hit unfolding out-eq by simp
qed
qed
qed

lemma receive-composition-fri-commits-outcome:
assumes outcome:
  Some ((b, r), t) ∈
    set-dist (execute receive-composition-fri-commits s)
shows
  ∃ rest.
    PTranscript s = r # rest ∧
    PState t = concat (PState s) r ∧
    PTranscript t = rest ∧
    s ≤ t ∧
    PTraceFriCounter t = PTraceFriCounter s ∧
    PCompositionFriCounter t = Suc (PCompositionFriCounter s) ∧
    PAlphaCounter t = PAlphaCounter s ∧
    PQueryCounter t = PQueryCounter s ∧
    fmlookup (HashMap t)
      (CompositionFriChallenge (PCompositionFriCounter s))

```

```

      (concat (PState s) r)) =
    Some b
proof -
  from outcome obtain s-read where
    read-r: Some (r, s-read) ∈ set-dist (execute read s)
  and rand-b:
    Some (b, t) ∈ set-dist (execute receive-composition-fri-challenge s-read)
  unfolding receive-composition-fri-commits-def receive-fri-commits-with-def
  by (auto elim!: set-dist-bindE)
  obtain rest where tr-s: PTranscript s = r # rest
proof -
  have PTranscript s ≠ []
    using read-r unfolding read-def protocol-read-def assert-def
    by (cases PTranscript s) (auto simp: throw-no-outcome elim!: set-dist-bindE)
  then obtain x xs where xs: PTranscript s = x # xs
    by (cases PTranscript s) auto
  have r-x: r = x
    using read-nonempty-outcome[OF read-r xs] by simp
  show ?thesis
    using that xs r-x by auto
qed
have read-res:
  s(PState := concat (PState s) r, PTranscript := rest)
  using read-nonempty-outcome[OF read-r tr-s] by simp
have rand-res:
  s-read ≤ t ∧
  PState t = PState s-read ∧
  PTranscript t = PTranscript s-read ∧
  fmlookup (HashMap t)
  (CompositionFriChallenge (PCompositionFriCounter s-read)
    (PState s-read)) =
  Some b
  using receive-composition-fri-challenge-outcome[OF rand-b] .
have rand-counter:
  PTraceFriCounter t = PTraceFriCounter s-read ∧
  PCompositionFriCounter t = Suc (PCompositionFriCounter s-read) ∧
  PAlphaCounter t = PAlphaCounter s-read ∧
  PQueryCounter t = PQueryCounter s-read
  using receive-composition-fri-challenge-counter-outcome[OF rand-b] by simp
have s-read-ext: s ≤ s-read
  using read-res unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
have s-t: s ≤ t
  using s-read-ext rand-res by (meson hash-ext-trans)
show ?thesis
  using tr-s read-res rand-res rand-counter s-t by auto
qed

lemma receive-composition-fri-commits-preserves-composition-fri-future-fresh:
  assumes future: composition-fri-future-fresh s

```

```

and outcome:
  Some ((b, r), t) ∈
    set-dist (execute receive-composition-fri-commits s)
shows composition-fri-future-fresh t
proof –
from outcome obtain s-read where
  read-r: Some (r, s-read) ∈ set-dist (execute read s)
and rand-b:
  Some (b, t) ∈
    set-dist (execute receive-composition-fri-challenge s-read)
unfolding receive-composition-fri-commits-def receive-fri-commits-with-def
by (auto elim!: set-dist-bindE)
from read-outcome[OF read-r] have read-counts:
  PCompositionFriCounter s-read = PCompositionFriCounter s
by blast
have read-hash: HashMap s-read = HashMap s
by (rule read-preserves-hash-map[OF read-r])
have counter-t:
  PCompositionFriCounter t = Suc (PCompositionFriCounter s)
using receive-composition-fri-challenge-counter-outcome[OF rand-b]
  read-counts
by simp
show ?thesis
unfolding composition-fri-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-ge-t: PCompositionFriCounter t ≤ i
  have i-ge-s: PCompositionFriCounter s ≤ i
    using i-ge-t counter-t by simp
  have neg:
    CompositionFriChallenge i x ≠
      CompositionFriChallenge
        (PCompositionFriCounter s-read) (PState s-read)
    using i-ge-t counter-t read-counts by auto
  have fmlookup (HashMap t) (CompositionFriChallenge i x) =
    fmlookup (HashMap s-read) (CompositionFriChallenge i x)
    by (rule receive-composition-fri-challenge-preserves-other-lookup
      [OF rand-b neg])
  also have ... = fmlookup (HashMap s) (CompositionFriChallenge i x)
    using read-hash by simp
  also have ... = None
    using future i-ge-s unfolding composition-fri-future-fresh-def by blast
  finally show fmlookup (HashMap t) (CompositionFriChallenge i x) = None .
qed
qed

```

```

lemma receive-composition-fri-commits-preserves-query-future-fresh:
  assumes future: query-future-fresh s
  and outcome:

```

```

    Some ((b, r), t) ∈
      set-dist (execute receive-composition-fri-commits s)
  shows query-future-fresh t
proof -
  from outcome obtain s-read where
    read-r: Some (r, s-read) ∈ set-dist (execute read s)
  and rand-b:
    Some (b, t) ∈
      set-dist (execute receive-composition-fri-challenge s-read)
  unfolding receive-composition-fri-commits-def receive-fri-commits-with-def
  by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-r] have read-counts:
    PQueryCounter s-read = PQueryCounter s
  by blast
  have read-hash: HashMap s-read = HashMap s
  by (rule read-preserves-hash-map[OF read-r])
  have counter-t:
    PQueryCounter t = PQueryCounter s
  using receive-composition-fri-challenge-counter-outcome[OF rand-b]
    read-counts
  by simp
  show ?thesis
  unfolding query-future-fresh-def
  proof (intro allI impI)
    fix i x
    assume i-ge-t: PQueryCounter t ≤ i
    have i-ge-s: PQueryCounter s ≤ i
      using i-ge-t counter-t by simp
    have neg:
      QueryIndexChallenge i x ≠
        CompositionFriChallenge
          (PCompositionFriCounter s-read) (PState s-read)
    by simp
    have fmllookup (HashMap t) (QueryIndexChallenge i x) =
      fmllookup (HashMap s-read) (QueryIndexChallenge i x)
    by (rule receive-composition-fri-challenge-preserves-other-lookup
      [OF rand-b neg])
    also have ... = fmllookup (HashMap s) (QueryIndexChallenge i x)
      using read-hash by simp
    also have ... = None
      using future i-ge-s unfolding query-future-fresh-def by blast
    finally show fmllookup (HashMap t) (QueryIndexChallenge i x) = None .
  qed
qed

```

```

lemma wp-receive-composition-fri-commits-fresh-set:
  assumes future: composition-fri-future-fresh s
  shows
    wp-event receive-composition-fri-commits

```

```

    (λout. case out of
      None ⇒ False
      | Some ((b, r), -) ⇒ b ∈ B) s
    ≤ nnreal (card B) / nnreal size
  unfolding receive-composition-fri-commits-def receive-fri-commits-with-def
proof (rule wp-event-bind-bound-by-cont)
  show ¬ (case None of
    None ⇒ False | Some ((b, r), -) ⇒ b ∈ B)
  by simp
next
fix r s-read
assume read-r: Some (r, s-read) ∈ set-dist (execute read s)
have read-counts:
  PCompositionFriCounter s-read = PCompositionFriCounter s
  using read-outcome[OF read-r] by blast
have read-hash: HashMap s-read = HashMap s
  by (rule read-preserves-hash-map[OF read-r])
have fresh:
  fmlookup (HashMap s-read)
    (CompositionFriChallenge
      (PCompositionFriCounter s-read) (PState s-read)) = None
  using future read-counts read-hash
  unfolding composition-fri-future-fresh-def by simp
let ?Q =
  λout :: (('f × 'f) × ('f, 'a) protocol-channel-scheme) option.
    case out of None ⇒ False | Some ((b, r), -) ⇒ b ∈ B
have cont-eq:
  (λout. case out of
    None ⇒ (0 :: prob)
    | Some (b, u) ⇒
      wp (return (b, r)) (λout. if ?Q out then 1 else 0) u) =
  (λout :: (('f × 'f) × ('f, 'a) protocol-channel-scheme) option.
    if (case out of None ⇒ False | Some (b, -) ⇒ b ∈ B)
    then 1 else 0)
  by (rule ext) (auto simp: wpsimps split: option.splits prod.splits)
have wp-event
  (receive-composition-fri-challenge ≫= (λb :: 'f. return (b, r)))
  ?Q s-read =
  wp-event receive-composition-fri-challenge
    (λout. case out of None ⇒ False | Some (b, -) ⇒ b ∈ B)
    s-read
  unfolding wp-event-def by (simp add: wpsimps cont-eq)
also have ... = nnreal (card B) / nnreal size
  by (rule wp-receive-composition-fri-challenge-fresh-set[OF fresh])
finally show
  wp-event
    (receive-composition-fri-challenge ≫= (λb :: 'f. return (b, r)))
    (λout. case out of None ⇒ False | Some ((b, r), -) ⇒ b ∈ B)
    s-read

```

$\leq \text{nnreal } (\text{card } B) / \text{nnreal size}$
by *simp*
qed

lemma *ntimes-receive-composition-fri-commits-preserves-composition-fri-future-fresh*:
assumes *outcome*:
 $\text{Some } (brs, t) \in$
 $\text{set-dist } (\text{execute } (ntimes \text{ receive-composition-fri-commits } n) \ s)$
and *future*: *composition-fri-future-fresh s*
shows *composition-fri-future-fresh t*
using *outcome future*
proof (*induction n arbitrary: s brs t*)
case 0
then show ?*case* **by** *simp*
next
case (*Suc n*)
from *Suc.prem1* **obtain** *br brs' s1* **where**
head:
 $\text{Some } (br, s1) \in$
 $\text{set-dist } (\text{execute } \text{receive-composition-fri-commits } s)$
and *tail*:
 $\text{Some } (brs', t) \in$
 $\text{set-dist } (\text{execute } (ntimes \text{ receive-composition-fri-commits } n) \ s1)$
and *brs-eq*: $brs = br \ \# \ brs'$
by (*auto elim!: set-dist-bindE*)
obtain *b r* **where** *br-eq*: $br = (b, r)$
by (*cases br*)
have *future-s1*:
composition-fri-future-fresh s1
by (*rule receive-composition-fri-commits-preserves-composition-fri-future-fresh*
 $[OF \ Suc.prem2) \ \text{head}[\text{unfolded } br\text{-eq}]]$)
show ?*case*
by (*rule Suc.IH* [*OF tail future-s1*])
qed

lemma *ntimes-receive-composition-fri-commits-preserves-query-future-fresh*:
assumes *outcome*:
 $\text{Some } (brs, t) \in$
 $\text{set-dist } (\text{execute } (ntimes \text{ receive-composition-fri-commits } n) \ s)$
and *future*: *query-future-fresh s*
shows *query-future-fresh t*
using *outcome future*
proof (*induction n arbitrary: s brs t*)
case 0
then show ?*case* **by** *simp*
next
case (*Suc n*)
from *Suc.prem1* **obtain** *br brs' s1* **where**
head:

```

    Some (br, s1) ∈
      set-dist (execute receive-composition-fri-commits s)
  and tail:
    Some (brs', t) ∈
      set-dist (execute (ntimes receive-composition-fri-commits n) s1)
  by (auto elim!: set-dist-bindE)
  obtain b r where br-eq: br = (b, r)
  by (cases br)
  have future-s1:
    query-future-fresh s1
  by (rule receive-composition-fri-commits-preserves-query-future-fresh
    [OF Suc.premis(2) head[unfolded br-eq]])
  show ?case
  by (rule Suc.IH[OF tail future-s1])
qed

lemma wp-ntimes-receive-composition-fri-commits-exact-challenges-bound:
  assumes future: composition-fri-future-fresh s
  and len: length bs = n
  shows
    wp-event (ntimes receive-composition-fri-commits n)
      (λout. case out of
        None ⇒ False
        | Some (brs, -) ⇒ map fst brs = bs) s
      ≤ (1 / nnreal size) ^ n
  using len future
proof (induction n arbitrary: s bs)
  case 0
  then have bs-empty: bs = []
  by simp
  show ?case
  unfolding bs-empty by (simp add: wp-event-def wpsimps)
next
  case (Suc n)
  obtain b bs' where bs-eq: bs = b # bs'
  using Suc.premis(1) by (cases bs) auto
  have len-bs': length bs' = n
  using Suc.premis(1) bs-eq by simp
  let ?tail =
    (ntimes receive-composition-fri-commits n ::
      (('f × 'f) list, ('f, 'a) protocol-channel-scheme) state-monad)
  let ?Q =
    λout :: (((('f × 'f) list) ×
      ('f, 'a) protocol-channel-scheme) option.
      case out of None ⇒ False | Some (brs, -) ⇒ map fst brs = bs
  let ?Head =
    λout :: (((('f × 'f) × ('f, 'a) protocol-channel-scheme) option.
      case out of None ⇒ False | Some ((b0, r0), -) ⇒ b0 ∈ {b}
  have head-bound:

```



```

wp-event receive-composition-fri-commits ?Head s ≤ 1 / nnreal size
proof -
  have wp-event receive-composition-fri-commits ?Head s ≤
    nnreal (card ({b} :: 'f set)) / nnreal size
    by (rule wp-receive-composition-fri-commits-fresh-set[OF Suc.premis(2)])
  then show ?thesis by simp
qed
have bind-bound:
  wp-event
    (receive-composition-fri-commits ≫
      (λbr :: 'f × 'f.
        ?tail ≫ (λbrs :: ('f × 'f) list. return (br # brs))))
    ?Q s
  ≤ wp-event receive-composition-fri-commits ?Head s *
    (1 / nnreal size) ^ n
proof (rule wp-event-bind-bound-by-head-and-cont[where P = ?Head])
  show ¬ ?Q None
    by simp
next
  fix br t
  assume not-head: ¬ ?Head (Some (br, t))
  obtain b0 r0 where br-eq: br = (b0, r0)
    by (cases br)
  have b0-ne: b0 ≠ b
    using not-head unfolding br-eq by simp
  have cont-false:
    wp-event (?tail ≫ (λbrs :: ('f × 'f) list. return (br # brs)))
      ?Q t
    ≤ wp-event (?tail ≫ (λbrs :: ('f × 'f) list. return (br # brs)))
      (λ- :: ((('f × 'f) list) ×
        ('f, 'a) protocol-channel-scheme) option. False) t
    by (rule wp-event-mono-on-support)
    (use b0-ne bs-eq br-eq in
      ‹auto simp: wpsimps elim!: set-dist-bindE split: option.splits prod.splits›)
  also have ... = 0
    unfolding wp-event-def wp-def dist-expect-def by simp
  finally have le-zero:
    wp-event (?tail ≫ (λbrs :: ('f × 'f) list. return (br # brs)))
      ?Q t ≤ 0 .
  show
    wp-event (?tail ≫ (λbrs :: ('f × 'f) list. return (br # brs)))
      ?Q t = 0
    by (rule antisym[OF le-zero]) simp
next
  fix br t
  assume head:
    Some (br, t) ∈
      set-dist (execute receive-composition-fri-commits s)
    and head-hit: ?Head (Some (br, t))

```

```

obtain  $b0\ r0$  where  $br\text{-}eq$ :  $br = (b0, r0)$ 
  by (cases  $br$ )
have  $b0\text{-}eq$ :  $b0 = b$ 
  using head-hit unfolding  $br\text{-}eq$  by simp
have  $future\text{-}t$ : composition-fri-future-fresh  $t$ 
  by (rule receive-composition-fri-commits-preserves-composition-fri-future-fresh
    [OF Suc.prems(2) head[unfolded  $br\text{-}eq$ ]])
have tail-bound:
  wp-event  $?tail$ 
    ( $\lambda out.$  case out of None  $\Rightarrow$  False
      | Some ( $brs, -$ )  $\Rightarrow$  map fst  $brs = bs'$ )  $t$ 
       $\leq (1 / \text{nnreal } size) ^ n$ 
  by (rule Suc.IH[OF len-bs'  $future\text{-}t$ ])
have cont-eq:
  ( $\lambda out.$  case out of
    None  $\Rightarrow (0 :: \text{prob})$ 
    | Some ( $brs, u$ )  $\Rightarrow$ 
      wp (return ( $br \# brs$ )) ( $\lambda out.$  if  $?Q$  out then 1 else 0)  $u$ ) =
  ( $\lambda out :: ((f \times 'f) \text{ list}) \times$ 
    ( $f, 'a$ ) protocol-channel-scheme) option.
    if (case out of None  $\Rightarrow$  False
      | Some ( $brs, -$ )  $\Rightarrow$  map fst  $brs = bs'$ )
      then 1 else 0)
  by (rule ext)
  (auto simp: wpsimps  $b0\text{-}eq$   $bs\text{-}eq$   $br\text{-}eq$ 
    split: option.splits prod.splits)
have wp-event ( $?tail \gg= (\lambda brs :: (f \times 'f) \text{ list. return } (br \# brs))$ )
   $?Q\ t =$ 
  wp-event  $?tail$ 
    ( $\lambda out.$  case out of None  $\Rightarrow$  False
      | Some ( $brs, -$ )  $\Rightarrow$  map fst  $brs = bs'$ )  $t$ 
  unfolding wp-event-def by (simp add: wpsimps cont-eq)
also have  $\dots \leq (1 / \text{nnreal } size) ^ n$ 
  by (rule tail-bound)
finally show
  wp-event ( $?tail \gg= (\lambda brs :: (f \times 'f) \text{ list. return } (br \# brs))$ )
   $?Q\ t \leq (1 / \text{nnreal } size) ^ n$  .
qed
have wp-event (ntimes receive-composition-fri-commits (Suc  $n$ ))  $?Q\ s$ 
   $\leq$  wp-event receive-composition-fri-commits  $?Head\ s *$ 
  ( $1 / \text{nnreal } size$ )  $^ n$ 
  using bind-bound by simp
also have  $\dots \leq (1 / \text{nnreal } size) * (1 / \text{nnreal } size) ^ n$ 
  by (rule mult-right-mono[OF head-bound]) simp
also have  $\dots = (1 / \text{nnreal } size) ^ {Suc\ n}$ 
  by simp
finally show  $?case$  .
qed

```

lemma *wp-ntimes-receive-composition-fri-commits-finite-set-bound:*
assumes *future: composition-fri-future-fresh s*
and *finite-B: finite B*
and *lengths: $\bigwedge bs. bs \in B \implies \text{length } bs = n$*
shows

$$\text{wp-event } (ntimes \text{ receive-composition-fri-commits } n)$$

$$(\lambda out. \text{ case out of } \begin{array}{l} \text{None} \Rightarrow \text{False} \\ | \text{Some } (brs, -) \Rightarrow \text{map fst } brs \in B \end{array}) s$$

$$\leq \text{nnreal } (\text{card } B) / (\text{nnreal size}) ^ n$$
proof –
let *?P* =

$$\lambda bs \text{ out. case out of } \begin{array}{l} \text{None} \Rightarrow \text{False} \\ | \text{Some } (brs, -) \Rightarrow \text{map fst } brs = bs \end{array}$$
have *event-mono:*

$$\text{wp-event } (ntimes \text{ receive-composition-fri-commits } n)$$

$$(\lambda out. \text{ case out of } \begin{array}{l} \text{None} \Rightarrow \text{False} \\ | \text{Some } (brs, -) \Rightarrow \text{map fst } brs \in B \end{array}) s$$

$$\leq \text{wp-event } (ntimes \text{ receive-composition-fri-commits } n)$$

$$(\lambda out. \exists bs \in B. ?P \text{ } bs \text{ out}) s$$
by (rule *wp-event-mono*) (auto *split: option.splits prod.splits*)
also have $\dots \leq (\sum bs \in B. (1 / \text{nnreal size}) ^ n)$
by (rule *wp-event-finite-UN-bound[OF finite-B]*)
(rule *wp-ntimes-receive-composition-fri-commits-exact-challenges-bound*
[OF future lengths])
also have $\dots = \text{nnreal } (\text{card } B) * ((1 / \text{nnreal size}) ^ n)$
using *finite-B* **by** *simp*
also have $\dots = \text{nnreal } (\text{card } B) / (\text{nnreal size}) ^ n$
using *size-card* **by** *transfer (simp add: power-divide)*
finally show *?thesis* .
qed

lemma *wp-ntimes-receive-composition-fri-commits-challenge-space-set-bound:*
assumes *future: composition-fri-future-fresh s*
and *subset: $B \subseteq \text{fri-challenge-space } n$*
shows

$$\text{wp-event } (ntimes \text{ receive-composition-fri-commits } n)$$

$$(\lambda out. \text{ case out of } \begin{array}{l} \text{None} \Rightarrow \text{False} \\ | \text{Some } (brs, -) \Rightarrow \text{map fst } brs \in B \end{array}) s$$

$$\leq \text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}(f) ^ n)$$
proof –
have *finite-B: finite B*
by (rule *finite-subset[OF subset finite-fri-challenge-space]*)
have *lengths: $\bigwedge bs. bs \in B \implies \text{length } bs = n$*
using *subset* **unfolding** *fri-challenge-space-def* **by** *auto*
have $\text{wp-event } (ntimes \text{ receive-composition-fri-commits } n)$

```

    (λout. case out of
      None ⇒ False
      | Some (brs, -) ⇒ map fst brs ∈ B) s
    ≤ nnreal (card B) / (nnreal size) ^ n
  by (rule wp-ntimes-receive-composition-fri-commits-finite-set-bound
      [OF future finite-B lengths])
  also have ... = nnreal (card B) / nnreal (CARD('f) ^ n)
    using size-card by simp
  finally show ?thesis .
qed

```

lemma *wp-verifier-composition-fri-prefix-challenge-space-set-bound:*

```

assumes future: composition-fri-future-fresh s
and subset:
  ∧ dg. B dg ⊆ fri-challenge-space (ceil-log (to-nat dg + 1))
and bound:
  ∧ dg. nnreal (card (B dg)) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C

```

shows

```

wp-event verifier-composition-fri-prefix
(λout. case out of
  None ⇒ False
  | Some ((-, -, -, dg, fl), -) ⇒ map fst fl ∈ B dg) s
≤ C

```

proof –

```

let ?Q =
  λout. case out of
    None ⇒ False
    | Some ((-, -, -, dg, fl), -) ⇒ map fst fl ∈ B dg

```

show ?thesis

unfolding verifier-composition-fri-prefix-def

proof (rule wp-event-bind-bound-by-cont)

show ¬ ?Q None

by simp

next

fix fr s1

assume read-fr: Some (fr, s1) ∈ set-dist (execute read s)

have future-s1: composition-fri-future-fresh s1

by (rule read-preserves-composition-fri-future-fresh
[OF future read-fr])

show wp-event

(ntimes receive-trace-fri-commits (ceil-log clength) ≫=

(λf.fl. read ≫=

(λf.final.

mmap (replicate (length spec) alpha-round) ≫=

(λas. read ≫=

(λdg.

assert (to-nat dg ≤ maxDegree) ≫=

(λ-. ntimes receive-composition-fri-commits

```

      (ceil-log (to-nat dg + 1)) >=>
      (λfl.
        return (fr, f-fl, f-final, as, dg, fl))))))
    ?Q s1 ≤ C
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ ?Q None
      by simp
  next
    fix f-fl s2
    assume trace-fri:
      Some (f-fl, s2) ∈
        set-dist
        (execute (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
    have future-s2: composition-fri-future-fresh s2
      by (rule ntimes-receive-trace-fri-commits-preserves-composition-fri-future-fresh
        [OF trace-fri future-s1])
    show wp-event
      (read >=>
        (λf-final.
          mmap (replicate (length spec) alpha-round) >=>
            (λas. read >=>
              (λdg.
                assert (to-nat dg ≤ maxDegree) >=>
                (λ-. ntimes receive-composition-fri-commits
                  (ceil-log (to-nat dg + 1)) >=>
                  (λfl.
                    return (fr, f-fl, f-final, as, dg, fl)))))))
        ?Q s2 ≤ C
    proof (rule wp-event-bind-bound-by-cont)
      show ¬ ?Q None
        by simp
    next
      fix f-final s3
      assume read-final: Some (f-final, s3) ∈ set-dist (execute read s2)
      have future-s3: composition-fri-future-fresh s3
        by (rule read-preserves-composition-fri-future-fresh
          [OF future-s2 read-final])
      show wp-event
        (mmap (replicate (length spec) alpha-round) >=>
          (λas. read >=>
            (λdg.
              assert (to-nat dg ≤ maxDegree) >=>
              (λ-. ntimes receive-composition-fri-commits
                (ceil-log (to-nat dg + 1)) >=>
                (λfl.
                  return (fr, f-fl, f-final, as, dg, fl))))))
        ?Q s3 ≤ C
    proof (rule wp-event-bind-bound-by-cont)
      show ¬ ?Q None

```

```

    by simp
next
  fix as s4
  assume alpha:
    Some (as, s4) ∈
      set-dist
        (execute (mmap (replicate (length spec) alpha-round)) s3)
  have future-s4: composition-fri-future-fresh s4
    by (rule mmap-alpha-round-preserves-composition-fri-future-fresh
        [OF future-s3 alpha])
  show wp-event
    (read ≫=
      (λdg.
        assert (to-nat dg ≤ maxDegree) ≫=
          (λ-. ntimes receive-composition-fri-commits
            (ceil-log (to-nat dg + 1)) ≫=
              (λfl.
                return (fr, f-fl, f-final, as, dg, fl))))))
    ?Q s4 ≤ C
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ ?Q None
      by simp
  next
    fix dg s5
    assume read-dg: Some (dg, s5) ∈ set-dist (execute read s4)
    have future-s5: composition-fri-future-fresh s5
      by (rule read-preserves-composition-fri-future-fresh
          [OF future-s4 read-dg])
    show wp-event
      (assert (to-nat dg ≤ maxDegree) ≫=
        (λ-. ntimes receive-composition-fri-commits
          (ceil-log (to-nat dg + 1)) ≫=
            (λfl.
              return (fr, f-fl, f-final, as, dg, fl))))
      ?Q s5 ≤ C
    proof (rule wp-event-bind-bound-by-cont)
      show ¬ ?Q None
        by simp
    next
      fix u s6
      assume assert-out:
        Some (u, s6) ∈
          set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
      have s6-eq: s6 = s5
        using assert-out unfolding assert-def
        by (cases to-nat dg ≤ maxDegree)
          (auto simp: throw-no-outcome)
      have future-s6: composition-fri-future-fresh s6
        using future-s5 unfolding s6-eq .

```

```

let ?Head =
  λout. case out of
    None ⇒ False
  | Some (fl, -) ⇒ map fst fl ∈ B dg
have comp-bound:
  wp-event
    (ntimes receive-composition-fri-commits
      (ceil-log (to-nat dg + 1)))
  ?Head s6 ≤
    nnreal (card (B dg)) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
by (rule
  wp-ntimes-receive-composition-fri-commits-challenge-space-set-bound
  [OF future-s6 subset])
have comp-bound-C:
  wp-event
    (ntimes receive-composition-fri-commits
      (ceil-log (to-nat dg + 1)))
  ?Head s6 ≤ C
by (rule order-trans[OF comp-bound bound])
show wp-event
  (ntimes receive-composition-fri-commits
    (ceil-log (to-nat dg + 1)) >>=
    (λfl.
      return (fr, f-fl, f-final, as, dg, fl)))
  ?Q s6 ≤ C
proof (rule wp-event-bind-bound-by-head-event[OF comp-bound-C])
  show ?Q None ⇒ ?Head None
  by simp
next
  fix fl t out
  assume cont:
    out ∈
      set-dist
        (execute
          (return (fr, f-fl, f-final, as, dg, fl)) t)
  and hit: ?Q out
  have out-eq:
    out = Some ((fr, f-fl, f-final, as, dg, fl), t)
  using cont
  unfolding set-dist-def return.rep-eq dist-return-def
    dist-delta-dist delta-map-def
  by simp
  show ?Head (Some (fl, t))
  using hit unfolding out-eq by simp
qed
qed
qed
qed

```

qed
 qed
 qed
 qed

lemma *ntimes-receive-fri-commits-with-outcome:*

fixes *receive-commits* :: ('f, 'f × 'f, 'a) protocol-c-monad
and *counter* :: ('f, 'a) protocol-channel-scheme ⇒ nat
and *preserve1* :: ('f, 'a) protocol-channel-scheme ⇒ nat
and *preserve2* :: ('f, 'a) protocol-channel-scheme ⇒ nat
and *preserve3* :: ('f, 'a) protocol-channel-scheme ⇒ nat
and *tag* :: nat ⇒ 'f ⇒ 'f protocol-hash-input

assumes *step-outcome:*

$\bigwedge s\ b\ r\ t. \text{Some } ((b, r), t) \in \text{set-dist } (\text{execute } \text{receive-commits } s) \implies$
 $\exists \text{rest.}$
 $P\text{Transcript } s = r \# \text{rest} \wedge$
 $P\text{State } t = \text{concat } (P\text{State } s) \ r \wedge$
 $P\text{Transcript } t = \text{rest} \wedge$
 $s \leq t \wedge$
 $\text{counter } t = \text{Suc } (\text{counter } s) \wedge$
 $\text{preserve1 } t = \text{preserve1 } s \wedge$
 $\text{preserve2 } t = \text{preserve2 } s \wedge$
 $\text{preserve3 } t = \text{preserve3 } s \wedge$
 $\text{fmlookup } (\text{HashMap } t) (\text{tag } (\text{counter } s) (\text{concat } (P\text{State } s) \ r)) = \text{Some } b$

and *outcome:*

$\text{Some } (\text{brs}, t) \in \text{set-dist } (\text{execute } (\text{ntimes } \text{receive-commits } n) \ s)$

shows

$\text{length } \text{brs} = n \wedge$
 $P\text{Transcript } s = \text{map } \text{snd } \text{brs} \ @ \ P\text{Transcript } t \wedge$
 $P\text{State } t = \text{foldl } \text{concat } (P\text{State } s) (\text{map } \text{snd } \text{brs}) \wedge$
 $s \leq t \wedge$
 $\text{counter } t = \text{counter } s + n \wedge$
 $\text{preserve1 } t = \text{preserve1 } s \wedge$
 $\text{preserve2 } t = \text{preserve2 } s \wedge$
 $\text{preserve3 } t = \text{preserve3 } s \wedge$
 $(\forall i < n.$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{tag } (\text{counter } s + i)$
 $(\text{foldl } \text{concat } (P\text{State } s) (\text{take } (\text{Suc } i) (\text{map } \text{snd } \text{brs})))) =$
 $\text{Some } (\text{fst } (\text{brs } ! \ i)))$

using *outcome*

proof (*induction n arbitrary: s brs t*)

case 0

then show ?case

by (*simp add: hash-ext-refl*)

next

case (*Suc n*)

from *Suc.prem*s **obtain** *br brs' s1* **where**

head: Some (br, s1) ∈ set-dist (execute receive-commits s)


```

and tail:
  Some (brs', t) ∈
    set-dist (execute (ntimes receive-commits n) s1)
and brs-eq: brs = br # brs'
by (auto elim!: set-dist-bindE)
obtain b r where br-eq: br = (b, r)
by (cases br)
from step-outcome[OF head[unfolded br-eq]]
obtain rest1 where
  tr-s: PTranscript s = r # rest1
and st-s1: PState s1 = concat (PState s) r
and tr-s1: PTranscript s1 = rest1
and ext-head: s ≤ s1
and counter-s1: counter s1 = Suc (counter s)
and preserve1-s1: preserve1 s1 = preserve1 s
and preserve2-s1: preserve2 s1 = preserve2 s
and preserve3-s1: preserve3 s1 = preserve3 s
and lookup-head:
  fmlookup (HashMap s1) (tag (counter s) (concat (PState s) r)) = Some b
by blast
have tail-res:
  length brs' = n ∧
    PTranscript s1 = map snd brs' @ PTranscript t ∧
    PState t = foldl concat (PState s1) (map snd brs') ∧
    s1 ≤ t ∧
    counter t = counter s1 + n ∧
    preserve1 t = preserve1 s1 ∧
    preserve2 t = preserve2 s1 ∧
    preserve3 t = preserve3 s1 ∧
    (∀ i < n.
      fmlookup (HashMap t)
        (tag (counter s1 + i)
          (foldl concat (PState s1) (take (Suc i) (map snd brs')))) =
          Some (fst (brs' ! i)))
    using Suc.IH[OF tail] .
have ext-s1-t: s1 ≤ t
using tail-res by simp
have lookup-head-t:
  fmlookup (HashMap t) (tag (counter s) (concat (PState s) r)) = Some b
using hash-extension-lookup[OF lookup-head ext-s1-t] .
have ext-combined: s ≤ t
using ext-head ext-s1-t by (meson hash-ext-trans)
have counter-t: counter t = counter s + Suc n
using tail-res counter-s1 by simp
have preserves-t:
  preserve1 t = preserve1 s ∧
  preserve2 t = preserve2 s ∧
  preserve3 t = preserve3 s
using tail-res preserve1-s1 preserve2-s1 preserve3-s1 by simp

```

```

have lookups:
   $\forall i < \text{Suc } n.$ 
    fmlookup (HashMap t)
      (tag (counter s + i)
        (foldl concat (PState s) (take (Suc i) (map snd brs)))) =
      Some (fst (brs ! i))
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{Suc } n$ 
  show fmlookup (HashMap t)
    (tag (counter s + i)
      (foldl concat (PState s) (take (Suc i) (map snd brs)))) =
    Some (fst (brs ! i))
proof (cases i)
  case 0
  then show ?thesis
    using brs-eq br-eq lookup-head-t by simp
next
  case (Suc j)
  have j-bound:  $j < n$ 
    using i-bound Suc by simp
  have state-shift:
    foldl concat (PState s) (take (Suc (Suc j)) (map snd brs)) =
    foldl concat (PState s1) (take (Suc j) (map snd brs'))
    using brs-eq br-eq st-s1 by simp
  have brs-nth:  $\text{brs} ! \text{Suc } j = \text{brs}' ! j$ 
    using brs-eq j-bound by simp
  show ?thesis
    using tail-res j-bound Suc state-shift brs-nth counter-s1 by simp
qed
qed
have as-eq:
  PTranscript s = map snd brs @ PTranscript t  $\wedge$ 
  PState t = foldl concat (PState s) (map snd brs)
  using tail-res tr-s tr-s1 st-s1 brs-eq br-eq by simp
show ?case
  using tail-res tr-s tr-s1 st-s1 ext-combined lookups as-eq brs-eq br-eq
    counter-t preserves-t
  by simp
qed

```

```

lemma ntimes-receive-trace-fri-commits-outcome:
assumes outcome:
  Some (brs, t)  $\in$ 
    set-dist (execute (ntimes receive-trace-fri-commits n) s)
shows
  length brs = n  $\wedge$ 
  PTranscript s = map snd brs @ PTranscript t  $\wedge$ 
  PState t = foldl concat (PState s) (map snd brs)  $\wedge$ 

```

$s \leq t \wedge$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s + n \wedge$
 $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = P\text{QueryCounter } s \wedge$
 $(\forall i < n.$
 $\quad \text{fmlookup } (\text{HashMap } t)$
 $\quad (\text{TraceFriChallenge } (P\text{TraceFriCounter } s + i)$
 $\quad \quad (\text{foldl concat } (P\text{State } s) (\text{take } (\text{Suc } i) (\text{map snd } \text{brs})))) =$
 $\quad \quad \text{Some } (\text{fst } (\text{brs } ! i)))$

proof –

have *res*:

$\text{length } \text{brs} = n \wedge$
 $P\text{Transcript } s = \text{map snd } \text{brs} @ P\text{Transcript } t \wedge$
 $P\text{State } t = \text{foldl concat } (P\text{State } s) (\text{map snd } \text{brs}) \wedge$
 $s \leq t \wedge$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s + n \wedge$
 $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = P\text{QueryCounter } s \wedge$
 $(\forall i < n.$
 $\quad \text{fmlookup } (\text{HashMap } t)$
 $\quad (\text{TraceFriChallenge } (P\text{TraceFriCounter } s + i)$
 $\quad \quad (\text{foldl concat } (P\text{State } s) (\text{take } (\text{Suc } i) (\text{map snd } \text{brs})))) =$
 $\quad \quad \text{Some } (\text{fst } (\text{brs } ! i)))$

by (rule *ntimes-receive-fri-commits-with-outcome*
 [of *receive-trace-fri-commits* $P\text{TraceFriCounter}$ $P\text{CompositionFriCounter}$
 $P\text{AlphaCounter}$ $P\text{QueryCounter}$ *TraceFriChallenge*,
OF receive-trace-fri-commits-outcome outcome])

then show *?thesis* .

qed

lemma *ntimes-receive-composition-fri-commits-outcome*:

assumes *outcome*:

$\text{Some } (\text{brs}, t) \in$
 $\text{set-dist } (\text{execute } (\text{ntimes receive-composition-fri-commits } n) s)$

shows

$\text{length } \text{brs} = n \wedge$
 $P\text{Transcript } s = \text{map snd } \text{brs} @ P\text{Transcript } t \wedge$
 $P\text{State } t = \text{foldl concat } (P\text{State } s) (\text{map snd } \text{brs}) \wedge$
 $s \leq t \wedge$
 $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s + n \wedge$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = P\text{QueryCounter } s \wedge$
 $(\forall i < n.$
 $\quad \text{fmlookup } (\text{HashMap } t)$
 $\quad (\text{CompositionFriChallenge } (P\text{CompositionFriCounter } s + i)$
 $\quad \quad (\text{foldl concat } (P\text{State } s) (\text{take } (\text{Suc } i) (\text{map snd } \text{brs})))) =$

Some (fst (brs ! i)))

proof –

have *composition-step-outcome*:

$\bigwedge s \ b \ r \ t. \text{Some } ((b, r), t) \in$
 $\text{set-dist } (\text{execute receive-composition-fri-commits } s) \implies$
 $\exists \text{rest.}$
 $P\text{Transcript } s = r \# \text{rest} \wedge$
 $P\text{State } t = \text{concat } (P\text{State } s) \ r \wedge$
 $P\text{Transcript } t = \text{rest} \wedge$
 $s \leq t \wedge$
 $P\text{CompositionFriCounter } t = \text{Suc } (P\text{CompositionFriCounter } s) \wedge$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = P\text{QueryCounter } s \wedge$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{CompositionFriChallenge } (P\text{CompositionFriCounter } s)$
 $(\text{concat } (P\text{State } s) \ r)) =$
 $\text{Some } b$

using *receive-composition-fri-commits-outcome* **by** *blast*

have *res*:

$\text{length } \text{brs} = n \wedge$
 $P\text{Transcript } s = \text{map snd } \text{brs} \ @ \ P\text{Transcript } t \wedge$
 $P\text{State } t = \text{foldl concat } (P\text{State } s) \ (\text{map snd } \text{brs}) \wedge$
 $s \leq t \wedge$
 $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s + n \wedge$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = P\text{QueryCounter } s \wedge$
 $(\forall i < n.$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{CompositionFriChallenge } (P\text{CompositionFriCounter } s + i)$
 $(\text{foldl concat } (P\text{State } s) \ (\text{take } (\text{Suc } i) \ (\text{map snd } \text{brs})))) =$
 $\text{Some } (\text{fst } (\text{brs } ! \ i)))$

using *composition-step-outcome* *outcome*

by (rule *ntimes-receive-fri-commits-with-outcome*
[*of receive-composition-fri-commits PCompositionFriCounter*
PTraceFriCounter PAlphaCounter PQueryCounter CompositionFriChallenge])

then show *?thesis* .

qed

lemma *ntimes-receive-composition-fri-commits-roots-outcome*:

assumes *outcome*:

$\text{Some } (\text{brs}, t) \in$

$\text{set-dist } (\text{execute } (\text{ntimes receive-composition-fri-commits } n) \ s)$

shows $\text{map snd } \text{brs} = \text{take } n \ (P\text{Transcript } s)$

proof –

have *res*:

$\text{length } \text{brs} = n \wedge P\text{Transcript } s = \text{map snd } \text{brs} \ @ \ P\text{Transcript } t$

using *ntimes-receive-composition-fri-commits-outcome*[*OF outcome*] **by** *simp*

```

have len: length (map snd brs) = n
  using res by simp
have take n (PTranscript s) = take n (map snd brs @ PTranscript t)
  using res by simp
also have ... = map snd brs
  using len by simp
finally show ?thesis
  by simp
qed

lemma wp-ntimes-receive-composition-fri-commits-root-dependent-challenge-space-set-bound:
  fixes B :: 'f list  $\Rightarrow$  'f list set
  assumes future: composition-fri-future-fresh s
  and subset:  $\bigwedge$  roots. B roots  $\subseteq$  fri-challenge-space n
  shows
    wp-event (ntimes receive-composition-fri-commits n)
      ( $\lambda$ out. case out of
        None  $\Rightarrow$  False
        | Some (brs, -)  $\Rightarrow$  map fst brs  $\in$  B (map snd brs)) s
       $\leq$  nnreal (card (B (take n (PTranscript s)))) /
        nnreal (CARD('f) ^ n)
  proof -
    let ?roots = take n (PTranscript s)
    let ?Q =
       $\lambda$ out. case out of
        None  $\Rightarrow$  False
        | Some (brs, -)  $\Rightarrow$  map fst brs  $\in$  B (map snd brs)
    let ?Q' =
       $\lambda$ out. case out of
        None  $\Rightarrow$  False
        | Some (brs, -)  $\Rightarrow$  map fst brs  $\in$  B ?roots
    have fixed-bound:
      wp-event (ntimes receive-composition-fri-commits n) ?Q' s
       $\leq$  nnreal (card (B ?roots)) / nnreal (CARD('f) ^ n)
    by (rule wp-ntimes-receive-composition-fri-commits-challenge-space-set-bound
      [OF future subset])
    have event-le:
      wp-event (ntimes receive-composition-fri-commits n) ?Q s
       $\leq$  wp-event (ntimes receive-composition-fri-commits n) ?Q' s
    proof (rule wp-event-mono-on-support)
      fix out
      assume out-supp:
        out  $\in$  set-dist (execute (ntimes receive-composition-fri-commits n) s)
      and hit: ?Q out
      show ?Q' out
    proof (cases out)
      case None
      then show ?thesis
        using hit by simp
    end
  end

```

```

next
  case (Some pair)
  then obtain brs t where out-eq: out = Some (brs, t)
    by (cases pair) auto
  have roots-eq: map snd brs = ?roots
    by (rule ntimes-receive-composition-fri-commits-roots-outcome)
      (use out-supp out-eq in simp)
  show ?thesis
    using hit roots-eq out-eq by simp
qed
qed
show ?thesis
  by (rule order-trans[OF event-le fixed-bound])
qed

lemma wp-ntimes-receive-composition-fri-commits-root-dependent-bound:
  fixes C :: prob
  assumes future: composition-fri-future-fresh s
    and subset:  $\bigwedge \text{roots}. B \text{ roots} \subseteq \text{fri-challenge-space } n$ 
    and bound:
      
$$\bigwedge \text{roots}. \text{nnreal} (\text{card } (B \text{ roots})) / \text{nnreal} (\text{CARD}(f) ^ n) \leq C$$

  shows
    wp-event (ntimes receive-composition-fri-commits n)
      ( $\lambda \text{out}. \text{case out of}$ 
        None  $\Rightarrow$  False
        | Some (brs, -)  $\Rightarrow$  map fst brs  $\in B$  (map snd brs)) s
       $\leq C$ 
  proof -
    have root-bound:
      wp-event (ntimes receive-composition-fri-commits n)
        ( $\lambda \text{out}. \text{case out of}$ 
          None  $\Rightarrow$  False
          | Some (brs, -)  $\Rightarrow$  map fst brs  $\in B$  (map snd brs)) s
         $\leq \text{nnreal} (\text{card } (B (\text{take } n (\text{PTranscript } s)))) / \text{nnreal} (\text{CARD}(f) ^ n)$ 
      by (rule
        wp-ntimes-receive-composition-fri-commits-root-dependent-challenge-space-set-bound
        [OF future subset])
    also have ...  $\leq C$ 
      by (rule bound)
    finally show ?thesis .
  qed

```

```

lemma wp-verifier-composition-fri-prefix-root-dependent-challenge-space-set-bound:
  fixes C :: prob
  assumes future: composition-fri-future-fresh s
    and subset:
       $\bigwedge \text{fr } f\text{-fl } f\text{-final as dg roots}.$ 

```

```

    B fr f-fl f-final as dg roots  $\subseteq$ 
      fri-challenge-space (ceil-log (to-nat dg + 1))
  and bound:
     $\bigwedge$  fr f-fl f-final as dg roots.
      nnreal (card (B fr f-fl f-final as dg roots)) /
        nnreal (CARD('f)  $\wedge$  ceil-log (to-nat dg + 1))  $\leq C$ 
  shows
    wp-event verifier-composition-fri-prefix
      ( $\lambda$ out. case out of
        None  $\Rightarrow$  False
        | Some ((fr, f-fl, f-final, as, dg, fl), -)  $\Rightarrow$ 
          map fst fl  $\in$  B fr f-fl f-final as dg (map snd fl)) s
       $\leq C$ 
  proof -
    let ?Q =
       $\lambda$ out. case out of
        None  $\Rightarrow$  False
        | Some ((fr, f-fl, f-final, as, dg, fl), -)  $\Rightarrow$ 
          map fst fl  $\in$  B fr f-fl f-final as dg (map snd fl)
    show ?thesis
      unfolding verifier-composition-fri-prefix-def
    proof (rule wp-event-bind-bound-by-cont)
      show  $\neg$  ?Q None
        by simp
    next
      fix s1
      assume read-fr: Some (fr, s1)  $\in$  set-dist (execute read s)
      have future-s1: composition-fri-future-fresh s1
        by (rule read-preserves-composition-fri-future-fresh
          [OF future read-fr])
      show wp-event
        (ntimes receive-trace-fri-commits (ceil-log clength)  $\gg$ 
          ( $\lambda$ f-fl. read  $\gg$ 
            ( $\lambda$ f-final.
              mmap (replicate (length spec) alpha-round)  $\gg$ 
                ( $\lambda$ as. read  $\gg$ 
                  ( $\lambda$ dg.
                    assert (to-nat dg  $\leq$  maxDegree)  $\gg$ 
                      ( $\lambda$ -. ntimes receive-composition-fri-commits
                        (ceil-log (to-nat dg + 1))  $\gg$ 
                          ( $\lambda$ fl.
                            return (fr, f-fl, f-final, as, dg, fl))))))))))
        ?Q s1  $\leq C$ 
    proof (rule wp-event-bind-bound-by-cont)
      show  $\neg$  ?Q None
        by simp
    next
      fix f-fl s2
      assume trace-fri:

```

```

    Some (f-fl, s2) ∈
      set-dist
        (execute (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
  have future-s2: composition-fri-future-fresh s2
    by (rule ntimes-receive-trace-fri-commits-preserves-composition-fri-future-fresh
        [OF trace-fri future-s1])
  show wp-event
    (read ≫=
      (λf-final.
        mmap (replicate (length spec) alpha-round) ≫=
          (λas. read ≫=
            (λdg.
              assert (to-nat dg ≤ maxDegree) ≫=
                (λ-. ntimes receive-composition-fri-commits
                  (ceil-log (to-nat dg + 1)) ≫=
                    (λfl.
                      return (fr, f-fl, f-final, as, dg, fl)))))))
    ?Q s2 ≤ C
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ ?Q None
      by simp
  next
  fix f-final s3
  assume read-final: Some (f-final, s3) ∈ set-dist (execute read s2)
  have future-s3: composition-fri-future-fresh s3
    by (rule read-preserves-composition-fri-future-fresh
        [OF future-s2 read-final])
  show wp-event
    (mmap (replicate (length spec) alpha-round) ≫=
      (λas. read ≫=
        (λdg.
          assert (to-nat dg ≤ maxDegree) ≫=
            (λ-. ntimes receive-composition-fri-commits
              (ceil-log (to-nat dg + 1)) ≫=
                (λfl.
                  return (fr, f-fl, f-final, as, dg, fl)))))))
    ?Q s3 ≤ C
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ ?Q None
      by simp
  next
  fix as s4
  assume alpha:
    Some (as, s4) ∈
      set-dist
        (execute (mmap (replicate (length spec) alpha-round)) s3)
  have future-s4: composition-fri-future-fresh s4
    by (rule mmap-alpha-round-preserves-composition-fri-future-fresh
        [OF future-s3 alpha])

```



```

show wp-event
  (read  $\gg$ 
    ( $\lambda dg.$ 
      assert (to-nat dg  $\leq$  maxDegree)  $\gg$ 
        ( $\lambda.$  ntimes receive-composition-fri-commits
          (ceil-log (to-nat dg + 1))  $\gg$ 
            ( $\lambda fl.$ 
              return (fr, f-fl, f-final, as, dg, fl))))))
  ?Q s4  $\leq$  C
proof (rule wp-event-bind-bound-by-cont)
  show  $\neg$  ?Q None
    by simp
next
  fix dg s5
  assume read-dg: Some (dg, s5)  $\in$  set-dist (execute read s4)
  have future-s5: composition-fri-future-fresh s5
    by (rule read-preserves-composition-fri-future-fresh
      [OF future-s4 read-dg])
  show wp-event
    (assert (to-nat dg  $\leq$  maxDegree)  $\gg$ 
      ( $\lambda.$  ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1))  $\gg$ 
          ( $\lambda fl.$ 
            return (fr, f-fl, f-final, as, dg, fl))))))
    ?Q s5  $\leq$  C
  proof (rule wp-event-bind-bound-by-cont)
    show  $\neg$  ?Q None
      by simp
  next
  fix u s6
  assume assert-out:
    Some (u, s6)  $\in$ 
      set-dist (execute (assert (to-nat dg  $\leq$  maxDegree)) s5)
  have s6-eq: s6 = s5
    using assert-out unfolding assert-def
    by (cases to-nat dg  $\leq$  maxDegree)
      (auto simp: throw-no-outcome)
  have future-s6: composition-fri-future-fresh s6
    using future-s5 unfolding s6-eq .
  let ?Head =
     $\lambda out.$  case out of
      None  $\Rightarrow$  False
      | Some (fl,  $-$ )  $\Rightarrow$ 
        map fst fl  $\in$  B fr f-fl f-final as dg (map snd fl)
  have comp-bound-C:
    wp-event
      (ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1)))
    ?Head s6  $\leq$  C

```

```

    by (rule
      wp-ntimes-receive-composition-fri-commits-root-dependent-bound
      [where B = B fr f-fl f-final as dg,
        OF future-s6])
      (use subset bound in simp-all)
  show wp-event
    (ntimes receive-composition-fri-commits
      (ceil-log (to-nat dg + 1)) ≥
      (λfl.
        return (fr, f-fl, f-final, as, dg, fl)))
    ?Q s6 ≤ C
  proof (rule wp-event-bind-bound-by-head-event[OF comp-bound-C])
    show ?Q None ⇒ ?Head None
      by simp
  next
    fix fl t out
    assume cont:
      out ∈
        set-dist
          (execute
            (return (fr, f-fl, f-final, as, dg, fl)) t)
    and hit: ?Q out
    have out-eq:
      out = Some ((fr, f-fl, f-final, as, dg, fl), t)
    using cont
    unfolding set-dist-def return.rep-eq dist-return-def
      dist-delta-dist delta-map-def
    by simp
    show ?Head (Some (fl, t))
      using hit unfolding out-eq by simp
  qed
qed
qed
qed
qed
qed
qed
qed
qed

```

lemma *verifier-alpha-prefix-outcome:*

assumes *outcome:*

Some ((fr, f-fl, f-final, as), t) ∈
set-dist (execute verifier-alpha-prefix s)

shows

length f-fl = ceil-log clength ∧
length as = length spec ∧
PTranscript s =
[fr] @ map snd f-fl @ [f-final] @ as @ PTranscript t ∧
PState t =

```

    foldl concat
      (concat (foldl concat (concat (PState s) fr) (map snd f-fl))
        f-final)
      as ∧
    PQueryCounter t = PQueryCounter s
  proof -
  from outcome obtain s1 s2 s3 s4 where
    read-fr: Some (fr, s1) ∈ set-dist (execute read s)
  and trace-fri:
    Some (f-fl, s2) ∈
      set-dist (execute
        (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
  and read-final: Some (f-final, s3) ∈ set-dist (execute read s2)
  and alpha-out:
    Some (as, s4) ∈
      set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)
  and ret:
    Some ((fr, f-fl, f-final, as), t) ∈
      set-dist (execute (return (fr, f-fl, f-final, as)) s4)
  unfolding verifier-alpha-prefix-def
  by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-fr] obtain rest1 where
    tr-s: PTranscript s = fr # rest1
  and st-s1: PState s1 = concat (PState s) fr
  and tr-s1: PTranscript s1 = rest1
  and query-count-s1: PQueryCounter s1 = PQueryCounter s
  by blast
  have trace-res:
    length f-fl = ceil-log clength ∧
    PTranscript s1 = map snd f-fl @ PTranscript s2 ∧
    PState s2 = foldl concat (PState s1) (map snd f-fl) ∧
    PQueryCounter s2 = PQueryCounter s1
  using ntimes-receive-trace-fri-commits-outcome[OF trace-fri] by simp
  from read-outcome[OF read-final] obtain rest3 where
    tr-s2: PTranscript s2 = f-final # rest3
  and st-s3: PState s3 = concat (PState s2) f-final
  and tr-s3: PTranscript s3 = rest3
  and query-count-s3: PQueryCounter s3 = PQueryCounter s2
  by blast
  have alpha-res:
    length as = length spec ∧
    PTranscript s3 = as @ PTranscript s4 ∧
    PState s4 = foldl concat (PState s3) as ∧
    PQueryCounter s4 = PQueryCounter s3
  using mmap-alpha-round-outcome[OF alpha-out] by simp
  have t-eq: t = s4
  using ret by simp
  show ?thesis
  using tr-s tr-s1 trace-res tr-s2 tr-s3 alpha-res st-s1 st-s3

```

query-count-s1 query-count-s3 t-eq
 by simp
 qed

lemma *verifier-trace-fri-prefix-outcome*:

assumes *outcome*:

Some ((fr, f-fl), t) ∈
 set-dist (execute *verifier-trace-fri-prefix* s)

shows

length f-fl = ceil-log clength ∧
 PTranscript s = [fr] @ map snd f-fl @ PTranscript t ∧
 PState t = foldl concat (concat (PState s) fr) (map snd f-fl) ∧
 s ≤ t ∧
 PTraceFriCounter t = PTraceFriCounter s + ceil-log clength ∧
 PCompositionFriCounter t = PCompositionFriCounter s ∧
 PAlphaCounter t = PAlphaCounter s ∧
 PQueryCounter t = PQueryCounter s ∧
 (∀ i < length f-fl.
 fmlookup (HashMap t)
 (TraceFriChallenge (PTraceFriCounter s + i)
 (foldl concat (concat (PState s) fr)
 (take (Suc i) (map snd f-fl)))) =
 Some (fst (f-fl ! i)))

proof –

from *outcome* **obtain** s1 s2 **where**

read-fr: Some (fr, s1) ∈ set-dist (execute read s)

and trace-fri:

Some (f-fl, s2) ∈
 set-dist (execute
 (ntimes receive-trace-fri-commits (ceil-log clength)) s1)

and ret:

Some ((fr, f-fl), t) ∈ set-dist (execute (return (fr, f-fl)) s2)

unfolding *verifier-trace-fri-prefix-def*

by (auto elim!: set-dist-bindE)

from read-outcome[OF read-fr] **obtain** rest1 **where**

tr-s: PTranscript s = fr # rest1

and st-s1: PState s1 = concat (PState s) fr

and tr-s1: PTranscript s1 = rest1

and ext-s-s1: s ≤ s1

and trace-count-s1: PTraceFriCounter s1 = PTraceFriCounter s

and comp-count-s1: PCompositionFriCounter s1 = PCompositionFriCounter s

and alpha-count-s1: PAlphaCounter s1 = PAlphaCounter s

and query-count-s1: PQueryCounter s1 = PQueryCounter s

by blast

have trace-res:

length f-fl = ceil-log clength ∧
 PTranscript s1 = map snd f-fl @ PTranscript s2 ∧
 PState s2 = foldl concat (PState s1) (map snd f-fl) ∧
 s1 ≤ s2 ∧

```

PTraceFriCounter s2 = PTraceFriCounter s1 + ceil-log clength ∧
PCompositionFriCounter s2 = PCompositionFriCounter s1 ∧
PAlphaCounter s2 = PAlphaCounter s1 ∧
PQueryCounter s2 = PQueryCounter s1 ∧
(∀ i < ceil-log clength.
  fmllookup (HashMap s2)
    (TraceFriChallenge (PTraceFriCounter s1 + i)
      (foldl concat (PState s1) (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i)))
using ntimes-receive-trace-fri-commits-outcome[OF trace-fri] by simp
have t-eq: t = s2
using ret by simp
have ext-s-t: s ≤ t
using ext-s-s1 trace-res t-eq by (meson hash-ext-trans)
have lookups:
  ∀ i < length f-fl.
    fmllookup (HashMap t)
      (TraceFriChallenge (PTraceFriCounter s + i)
        (foldl concat (concat (PState s) fr)
          (take (Suc i) (map snd f-fl)))) =
        Some (fst (f-fl ! i))
using trace-res st-s1 trace-count-s1 t-eq by simp
show ?thesis
using tr-s tr-s1 st-s1 trace-res t-eq ext-s-t trace-count-s1
  comp-count-s1 alpha-count-s1 query-count-s1 lookups
by simp
qed

```

lemma verifier-composition-fri-prefix-outcome:

assumes outcome:

Some ((fr, f-fl, f-final, as, dg, fl), t) ∈
 set-dist (execute verifier-composition-fri-prefix s)

shows

length f-fl = ceil-log clength ∧
 length as = length spec ∧
 length fl = ceil-log (to-nat dg + 1) ∧
 PTranscript s =
 [fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
 map snd fl @ PTranscript t ∧
 PState t =
 foldl concat
 (concat
 (foldl concat
 (concat
 (foldl concat (concat (PState s) fr) (map snd f-fl))
 f-final)
 as)
 dg)
 (map snd fl) ∧

```

s ≤ t ∧
PQueryCounter t = PQueryCounter s ∧
(∀ i < length f-fl.
  fmllookup (HashMap t)
    (TraceFriChallenge (PTraceFriCounter s + i)
      (foldl concat (concat (PState s) fr)
        (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))) ∧
(∀ i < length fl.
  fmllookup (HashMap t)
    (CompositionFriChallenge (PCompositionFriCounter s + i)
      (foldl concat
        (concat
          (foldl concat
            (concat
              (foldl concat (concat (PState s) fr) (map snd f-fl))
                f-final)
            as)
          dg)
        (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i)))
proof –
from outcome obtain s1 s2 s3 s4 s5 s6 s7 where
  read-fr: Some (fr, s1) ∈ set-dist (execute read s)
  and trace-fri:
    Some (f-fl, s2) ∈
      set-dist (execute
        (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
  and read-f-final: Some (f-final, s3) ∈ set-dist (execute read s2)
  and alpha-out:
    Some (as, s4) ∈
      set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)
  and read-dg: Some (dg, s5) ∈ set-dist (execute read s4)
  and degree-assert:
    Some ((), s6) ∈
      set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
  and comp-fri:
    Some (fl, s7) ∈
      set-dist (execute
        (ntimes receive-composition-fri-commits
          (ceil-log (to-nat dg + 1)))) s6)
  and ret:
    Some ((fr, f-fl, f-final, as, dg, fl), t) ∈
      set-dist (execute (return (fr, f-fl, f-final, as, dg, fl)) s7)
  unfolding verifier-composition-fri-prefix-def
  by (auto elim!: set-dist-bindE)
from read-outcome[OF read-fr] obtain rest1 where
  tr-s: PTranscript s = fr # rest1
  and st-s1: PState s1 = concat (PState s) fr

```

```

and tr-s1: PTranscript s1 = rest1
and ext-s-s1: s ≤ s1
and trace-count-s1: PTraceFriCounter s1 = PTraceFriCounter s
and comp-count-s1: PCompositionFriCounter s1 = PCompositionFriCounter s
and alpha-count-s1: PAlphaCounter s1 = PAlphaCounter s
and query-count-s1: PQueryCounter s1 = PQueryCounter s
by blast
have trace-res:
  length f-fl = ceil-log clength ∧
    PTranscript s1 = map snd f-fl @ PTranscript s2 ∧
    PState s2 = foldl concat (PState s1) (map snd f-fl) ∧
    s1 ≤ s2 ∧
    PTraceFriCounter s2 = PTraceFriCounter s1 + ceil-log clength ∧
    PCompositionFriCounter s2 = PCompositionFriCounter s1 ∧
    PAlphaCounter s2 = PAlphaCounter s1 ∧
    PQueryCounter s2 = PQueryCounter s1 ∧
    (∀ i < ceil-log clength.
      fmlookup (HashMap s2)
        (TraceFriChallenge (PTraceFriCounter s1 + i)
          (foldl concat (PState s1) (take (Suc i) (map snd f-fl)))) =
          Some (fst (f-fl ! i)))
    using ntimes-receive-trace-fri-commits-outcome[OF trace-fri] by simp
from read-outcome[OF read-f-final] obtain rest3 where
  tr-s2: PTranscript s2 = f-final # rest3
and st-s3: PState s3 = concat (PState s2) f-final
and tr-s3: PTranscript s3 = rest3
and ext-s2-s3: s2 ≤ s3
and comp-count-s3: PCompositionFriCounter s3 = PCompositionFriCounter s2
and query-count-s3: PQueryCounter s3 = PQueryCounter s2
by blast
have alpha-res:
  length as = length spec ∧
    PTranscript s3 = as @ PTranscript s4 ∧
    PState s4 = foldl concat (PState s3) as ∧
    s3 ≤ s4 ∧
    PCompositionFriCounter s4 = PCompositionFriCounter s3 ∧
    PQueryCounter s4 = PQueryCounter s3
    using mmap-alpha-round-outcome[OF alpha-out] by simp
from read-outcome[OF read-dg] obtain rest5 where
  tr-s4: PTranscript s4 = dg # rest5
and st-s5: PState s5 = concat (PState s4) dg
and tr-s5: PTranscript s5 = rest5
and ext-s4-s5: s4 ≤ s5
and comp-count-s5: PCompositionFriCounter s5 = PCompositionFriCounter s4
and query-count-s5: PQueryCounter s5 = PQueryCounter s4
by blast
have s6-eq: s6 = s5
using degree-assert unfolding assert-def
by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)

```

```

have ext-s5-s6:  $s5 \leq s6$ 
  unfolding s6-eq by (rule hash-ext-refl)
have comp-res:
  length fl = ceil-log (to-nat dg + 1)  $\wedge$ 
    PTranscript s6 = map snd fl @ PTranscript s7  $\wedge$ 
    PState s7 = foldl concat (PState s6) (map snd fl)  $\wedge$ 
    s6  $\leq$  s7  $\wedge$ 
    PCompositionFriCounter s7 =
      PCompositionFriCounter s6 + ceil-log (to-nat dg + 1)  $\wedge$ 
    PQueryCounter s7 = PQueryCounter s6  $\wedge$ 
    ( $\forall i < \text{ceil-log (to-nat dg + 1)}$ .
      fmlookup (HashMap s7)
        (CompositionFriChallenge (PCompositionFriCounter s6 + i)
          (foldl concat (PState s6) (take (Suc i) (map snd fl)))) =
        Some (fst (fl ! i)))
  using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
  by simp
have t-eq:  $t = s7$ 
  using ret by simp
have ext-s-t:  $s \leq t$ 
  using ext-s-s1 trace-res ext-s2-s3 alpha-res ext-s4-s5 ext-s5-s6
    comp-res t-eq
  by (meson hash-ext-trans)
have ext-s2-t:  $s2 \leq t$ 
  using ext-s2-s3 alpha-res ext-s4-s5 ext-s5-s6 comp-res t-eq
  by (meson hash-ext-trans)
have trace-lookup:
   $\forall i < \text{length f-fl}$ .
    fmlookup (HashMap t)
      (TraceFriChallenge (PTraceFriCounter s + i)
        (foldl concat (concat (PState s) fr)
          (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{length f-fl}$ 
  have lookup-s2:
    fmlookup (HashMap s2)
      (TraceFriChallenge (PTraceFriCounter s1 + i)
        (foldl concat (PState s1) (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
  using trace-res i-bound by simp
show fmlookup (HashMap t)
  (TraceFriChallenge (PTraceFriCounter s + i)
    (foldl concat (concat (PState s) fr)
      (take (Suc i) (map snd f-fl)))) =
  Some (fst (f-fl ! i))
  using hash-extension-lookup[OF lookup-s2 ext-s2-t] st-s1
    trace-count-s1

```



```

    by simp
qed
have comp-lookup:
  ∀ i < length fl.
    fmlookup (HashMap t)
      (CompositionFriChallenge (PCompositionFriCounter s + i)
        (foldl concat
          (concat
            (foldl concat
              (concat
                (foldl concat (concat (PState s) fr) (map snd f-fl))
                f-final)
              as)
            dg)
          (take (Suc i) (map snd fl)))))) =
    Some (fst (fl ! i))
proof (intro allI impI)
  fix i
  assume i-bound: i < length fl
  have lookup-s7:
    fmlookup (HashMap s7)
      (CompositionFriChallenge (PCompositionFriCounter s6 + i)
        (foldl concat (PState s6) (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i))
  using comp-res i-bound by simp
show fmlookup (HashMap t)
  (CompositionFriChallenge (PCompositionFriCounter s + i)
    (foldl concat
      (concat
        (foldl concat
          (concat
            (foldl concat (concat (PState s) fr) (map snd f-fl))
            f-final)
          as)
        dg)
      (take (Suc i) (map snd fl)))) =
  Some (fst (fl ! i))
  using lookup-s7 t-eq st-s1 trace-res st-s3 alpha-res st-s5 s6-eq
    comp-count-s1 comp-count-s3 comp-count-s5
  by simp
qed
have query-count-t: PQueryCounter t = PQueryCounter s
  using query-count-s1 trace-res query-count-s3 alpha-res query-count-s5
    s6-eq comp-res t-eq
  by simp
show ?thesis
  using tr-s tr-s1 trace-res tr-s2 tr-s3 alpha-res tr-s4 tr-s5 s6-eq
    comp-res st-s1 st-s3 st-s5 t-eq ext-s-t query-count-t trace-lookup
    comp-lookup

```

by simp
qed

end

end

theory *Soundness-Execution-Extraction*
 imports *Soundness-Execution-Prefix*
begin

Accepted verifier execution transcript extraction and witness lemmas.

context *soundness*
begin

lemma *verify-monad-hash-extends*:

assumes *outcome*:

Some (result, final-state) ∈ set-dist (execute verify-monad s)

shows $s \leq \text{final-state}$

proof –

from *outcome[unfolded verify-monad-composition-fri-decomposition]*

obtain *header prefix-state* **where**

prefix:

Some (header, prefix-state) ∈
 set-dist (execute verifier-composition-fri-prefix s)

and *suffix*:

Some (result, final-state) ∈
 set-dist
 (execute (verifier-after-composition-fri header) prefix-state)

by *(auto elim!: set-dist-bindE)*

obtain *fr f-fl f-final as dg fl* **where** *header-eq*:

header = (fr, f-fl, f-final, as, dg, fl)

by *(cases header) auto*

have *ext-prefix: s ≤ prefix-state*

using *verifier-composition-fri-prefix-outcome*
 [OF prefix[unfolded header-eq]]

by *simp*

from *suffix[unfolded header-eq verifier-after-composition-fri-def]*

obtain *final query-state* **where**

read-final:

Some (final, query-state) ∈ set-dist (execute read prefix-state)

and *query-out*:

Some (result, final-state) ∈
 set-dist
 (execute
 (ntimes
 (verifier-query-round-program fr f-fl f-final as fl final)
 rounds)

```

      query-state)
    by (auto elim!: set-dist-bindE)
  have ext-query: prefix-state  $\leq$  query-state
    using read-outcome[OF read-final] by blast
  have ext-final: query-state  $\leq$  final-state
    using ntimes-verifier-query-rounds-outcome[OF query-out] by blast
  show ?thesis
    using ext-prefix ext-query ext-final by (meson hash-ext-trans)
qed

```

lemma *alpha-header-supported-trace-table-candidate-state-hash-extends:*

```

  assumes
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table final-state
  shows  $s \leq$  final-state
proof -
  from assms obtain result composition-table as query-idxs dg
    composition-fri-roots final rest where
    outcome:
      Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
  unfolding alpha-header-supported-trace-table-candidate-state-def
  by blast
  show ?thesis
    by (rule verify-monad-hash-extends[OF outcome])
qed

```

lemma *alpha-header-supported-pairwise-merkle-bad-extends-initial:*

```

  assumes bad:
    alpha-header-supported-pairwise-merkle-bad s fr f-fri-roots f-final
  obtains trace-table final-state trace-table' final-state'
  where
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table final-state
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table' final-state'
    trace-table  $\neq$  trace-table'
    merkle-hash-value-conflict final-state final-state'  $\vee$ 
    hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
     $s \leq$  final-state
     $s \leq$  final-state'

```

```

proof -
  from bad obtain trace-table final-state trace-table' final-state' where
    candidate:
      alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
      f-final trace-table final-state
  and candidate':
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table' final-state'

```

and *distinct*: $\text{trace-table} \neq \text{trace-table}'$
and *pair-bad*:
 $\text{merkle-hash-value-conflict } \text{final-state } \text{final-state}' \vee$
 $\text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge } \text{final-state } \text{final-state}')$
unfolding *alpha-header-supported-pairwise-merkle-bad-def* **by** *blast*
have *ext*: $s \leq \text{final-state}$
by (rule *alpha-header-supported-trace-table-candidate-state-hash-extends*
 $[\text{OF candidate}]$)
have *ext'*: $s \leq \text{final-state}'$
by (rule *alpha-header-supported-trace-table-candidate-state-hash-extends*
 $[\text{OF candidate}]$)
show *?thesis*
by (rule *that*[*OF candidate candidate' distinct pair-bad ext ext'*])
qed

definition *alpha-header-supported-pairwise-coupling-bad*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $\text{'f} \Rightarrow \text{'f list} \Rightarrow \text{'f} \Rightarrow \text{bool}$

where

$\text{alpha-header-supported-pairwise-coupling-bad } s \text{ fr } \text{f-fri-roots } \text{f-final}$
 $\longleftrightarrow$
 $(\exists \text{ trace-table } \text{final-state } \text{trace-table}' \text{ final-state}'.$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } \text{f-fri-roots}$
 $\text{f-final } \text{trace-table } \text{final-state} \wedge$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } \text{f-fri-roots}$
 $\text{f-final } \text{trace-table}' \text{ final-state}' \wedge$
 $\text{trace-table} \neq \text{trace-table}' \wedge$
 $\text{merkle-hash-pairwise-coupling-bad } \text{final-state } \text{final-state}')$

definition *alpha-supported-pairwise-coupling-bad*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{alpha-supported-pairwise-coupling-bad } s \longleftrightarrow$
 $(\exists \text{ fr } \text{f-fri-roots } \text{f-final}.$
 $\text{alpha-header-supported-pairwise-coupling-bad } s \text{ fr } \text{f-fri-roots}$
 $\text{f-final})$

lemma *alpha-header-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad*:

assumes *bad*:

$\text{alpha-header-supported-pairwise-merkle-bad } s \text{ fr } \text{f-fri-roots } \text{f-final}$

shows

$(\exists \text{ trace-table } \text{final-state } \text{trace-table}' \text{ final-state}'.$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } \text{f-fri-roots}$
 $\text{f-final } \text{trace-table } \text{final-state} \wedge$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } \text{f-fri-roots}$
 $\text{f-final } \text{trace-table}' \text{ final-state}' \wedge$
 $\text{trace-table} \neq \text{trace-table}' \wedge$
 $(\text{hash-map-output-collision } \text{final-state} \vee$

$\text{hash-map-output-collision final-state'}) \vee$
 $\text{alpha-header-supported-pairwise-coupling-bad } s \text{ fr } f\text{-fri-roots } f\text{-final}$
proof –
from *bad* **obtain** *trace-table final-state trace-table' final-state'*
where *candidate*:
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final trace-table final-state}$
and *candidate'*:
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final trace-table' final-state'}$
and *distinct*: $\text{trace-table} \neq \text{trace-table'}$
and *pair-bad*:
 $\text{merkle-hash-value-conflict final-state final-state'} \vee$
 $\text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge final-state final-state'})$
unfolding *alpha-header-supported-pairwise-merkle-bad-def* **by** *blast*
have *decomp*:
 $\text{hash-map-output-collision final-state} \vee$
 $\text{hash-map-output-collision final-state'} \vee$
 $\text{merkle-hash-pairwise-coupling-bad final-state final-state'}$
by (*rule merkle-hash-pairwise-bad-imp-local-collision-or-coupling-bad*
 $[\text{OF pair-bad}]$)
show *?thesis*
using *candidate candidate' distinct decomp*
unfolding *alpha-header-supported-pairwise-coupling-bad-def* **by** *blast*
qed

lemma *alpha-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad*:

assumes *bad*: *alpha-supported-pairwise-merkle-bad s*

shows

$(\exists \text{ fr } f\text{-fri-roots } f\text{-final trace-table final-state trace-table'}$
 $\text{final-state'.$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final trace-table final-state} \wedge$
 $\text{alpha-header-supported-trace-table-candidate-state } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final trace-table' final-state'} \wedge$
 $\text{trace-table} \neq \text{trace-table'} \wedge$
 $(\text{hash-map-output-collision final-state} \vee$
 $\text{hash-map-output-collision final-state'}) \vee$
 $\text{alpha-supported-pairwise-coupling-bad } s$

proof –

from *bad* **obtain** *fr f-fri-roots f-final* **where** *header-bad*:

$\text{alpha-header-supported-pairwise-merkle-bad } s \text{ fr } f\text{-fri-roots } f\text{-final}$

unfolding *alpha-supported-pairwise-merkle-bad-def* **by** *blast*

from

$\text{alpha-header-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad}$
 $[\text{OF header-bad}]$

show *?thesis*

unfolding *alpha-supported-pairwise-coupling-bad-def* **by** *blast*

qed

lemma *alpha-supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad:*

assumes *bad: alpha-supported-pairwise-merkle-bad s*

shows

supported-hash-output-collision-possible s $\vee$

alpha-supported-pairwise-coupling-bad s

proof –

have *decomp:*

$(\exists \text{fr } f\text{-fri-roots } f\text{-final trace-table final-state trace-table'}$
 $\text{final-state'.$

alpha-header-supported-trace-table-candidate-state s fr f-fri-roots

f-final trace-table final-state $\wedge$

alpha-header-supported-trace-table-candidate-state s fr f-fri-roots

f-final trace-table' final-state' $\wedge$

trace-table $\neq$ *trace-table'* $\wedge$

$(\text{hash-map-output-collision final-state} \vee$

$\text{hash-map-output-collision final-state'}) \vee$

alpha-supported-pairwise-coupling-bad s

by (*rule*

alpha-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad

[*OF bad*])

then show *?thesis*

proof

assume *local:*

$\exists \text{fr } f\text{-fri-roots } f\text{-final trace-table final-state trace-table'}$
 $\text{final-state'.$

alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
f-final trace-table final-state $\wedge$

alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
f-final trace-table' final-state' $\wedge$

trace-table $\neq$ *trace-table'* $\wedge$

$(\text{hash-map-output-collision final-state} \vee$

$\text{hash-map-output-collision final-state'})$

then obtain *fr f-fri-roots f-final trace-table final-state trace-table'*
final-state' **where**

candidate:

alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
f-final trace-table final-state

and *candidate':*

alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
f-final trace-table' final-state'

and *collision:*

hash-map-output-collision final-state $\vee$

hash-map-output-collision final-state'

by *blast*

from *collision* **show** *?thesis*

proof

assume *collision: hash-map-output-collision final-state*

```

from candidate obtain result composition-table as query-idxs dg
  composition-fri-roots final rest where
  outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
and bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
unfolding alpha-header-supported-trace-table-candidate-state-def
by blast
have supported-hash-output-collision-possible s
by (rule
  accepted-with-bound-tables-hash-collision-imp-supported-hash-output-collision-possible
  [OF outcome bound collision])
then show ?thesis by simp
next
assume collision': hash-map-output-collision final-state'
from candidate' obtain result' composition-table' as' query-idxs' dg'
  composition-fri-roots' final' rest' where
  outcome':
    Some (result', final-state') ∈ set-dist (execute verify-monad s)
and bound':
  accepted-with-bound-tables s (Some (result', final-state'))
  trace-table' composition-table' as' query-idxs'
unfolding alpha-header-supported-trace-table-candidate-state-def
by blast
have supported-hash-output-collision-possible s
by (rule
  accepted-with-bound-tables-hash-collision-imp-supported-hash-output-collision-possible
  [OF outcome' bound' collision'])
then show ?thesis by simp
qed
next
assume alpha-supported-pairwise-coupling-bad s
then show ?thesis by simp
qed
qed

lemma verifier-composition-fri-prefix-preserves-query-future-fresh:
assumes future: query-future-fresh s
and outcome:
  Some (header, t) ∈
  set-dist (execute verifier-composition-fri-prefix s)
shows query-future-fresh t
proof –
obtain fr f-fl f-final as dg fl where header-eq:
  header = (fr, f-fl, f-final, as, dg, fl)
by (cases header) auto
from outcome[unfolded header-eq] obtain s1 s2 s3 s4 s5 s6 where
  read-fr: Some (fr, s1) ∈ set-dist (execute read s)

```

```

and trace-fri:
  Some (f-fl, s2) ∈
    set-dist (execute
      (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
and read-f-final:
  Some (f-final, s3) ∈ set-dist (execute read s2)
and alpha-out:
  Some (as, s4) ∈
    set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)
and read-dg: Some (dg, s5) ∈ set-dist (execute read s4)
and degree-assert:
  Some ((), s6) ∈
    set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
and comp-fri:
  Some (fl, t) ∈
    set-dist (execute
      (ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1))) s6)
unfolding verifier-composition-fri-prefix-def
by (auto elim!: set-dist-bindE)
have future-s1: query-future-fresh s1
by (rule read-preserves-query-future-fresh[OF future read-fr])
have future-s2: query-future-fresh s2
by (rule ntimes-receive-trace-fri-commits-preserves-query-future-fresh
  [OF trace-fri future-s1])
have future-s3: query-future-fresh s3
by (rule read-preserves-query-future-fresh[OF future-s2 read-f-final])
have future-s4: query-future-fresh s4
by (rule mmap-alpha-round-preserves-query-future-fresh
  [OF future-s3 alpha-out])
have future-s5: query-future-fresh s5
by (rule read-preserves-query-future-fresh[OF future-s4 read-dg])
have s6-eq: s6 = s5
using degree-assert unfolding assert-def
by (cases to-nat dg ≤ maxDegree)
  (auto simp: throw-no-outcome)
have future-s6: query-future-fresh s6
using future-s5 unfolding s6-eq .
show ?thesis
by (rule ntimes-receive-composition-fri-commits-preserves-query-future-fresh
  [OF comp-fri future-s6])
qed

lemma verify-monad-header-extraction:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
obtains fr f-fl f-final as dg fl final query-state
where

```



```

verifier-header-transcript s fr (map snd f-fl) f-final as dg
  (map snd fl) final (PTranscript query-state)
and PState query-state =
  verifier-header-state s fr (map snd f-fl) f-final as dg (map snd fl) final
and Some (result, final-state) ∈
  set-dist (execute
    (ntimes (verifier-query-round-program fr f-fl f-final as fl final) rounds)
    query-state)
and PQueryCounter query-state = PQueryCounter s
and length f-fl = ceil-log clength
and length as = length spec
and length fl = ceil-log (to-nat dg + 1)
proof –
  let ?alpha =
    do {
      a0 ← receive-alpha-challenge;
      let a0' = a0;
      a1 ← read;
      let a1' = a1;
      assert (a0' = a1');
      return a1'
    }
from outcome obtain fr s1 f-fl s2 f-final s3 as s4 dg s5 s6 fl s7 final query-state
where read-fr: Some (fr, s1) ∈ set-dist (execute read s)
and trace-fri:
  Some (f-fl, s2) ∈
    set-dist (execute
      (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
and read-f-final: Some (f-final, s3) ∈ set-dist (execute read s2)
and alpha-out:
  Some (as, s4) ∈
    set-dist (execute (mmap (replicate (length spec) ?alpha)) s3)
and read-dg: Some (dg, s5) ∈ set-dist (execute read s4)
and degree-assert:
  Some ((), s6) ∈ set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
and comp-fri:
  Some (fl, s7) ∈
    set-dist (execute
      (ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1))) s6)
and read-final: Some (final, query-state) ∈ set-dist (execute read s7)
and query-out:
  Some (result, final-state) ∈
    set-dist (execute
      (ntimes (verifier-query-round-program fr f-fl f-final as fl final) rounds)
      query-state)
unfolding verify-monad-def verifier-query-round-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
from read-outcome[OF read-fr] obtain rest1 where

```

```
|s: PTranscript s = fr # rest1
and st-s1: PState s1 = concat (PState s) fr
and tr-s1: PTranscript s1 = rest1
and query-count-s1: PQueryCounter s1 = PQueryCounter s
by blast
have trace-fri-res:
length f-fl = ceil-log clength ∧
PTranscript s1 = map snd f-fl @ PTranscript s2 ∧
PState s2 = foldl concat (PState s1) (map snd f-fl)
using ntimes-receive-trace-fri-commits-outcome[OF trace-fri] by simp
have query-count-s2: PQueryCounter s2 = PQueryCounter s1
using ntimes-receive-trace-fri-commits-outcome[OF trace-fri] by simp
from read-outcome[OF read-f-final] obtain rest3 where
tr-s2: PTranscript s2 = f-final # rest3
and st-s3: PState s3 = concat (PState s2) f-final
and tr-s3: PTranscript s3 = rest3
and query-count-s3: PQueryCounter s3 = PQueryCounter s2
by blast
have alpha-out':
Some (as, s4) ∈
set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)
using alpha-out by (simp add: alpha-round-def)
have alpha-res:
length as = length spec ∧
PTranscript s3 = as @ PTranscript s4 ∧
PState s4 = foldl concat (PState s3) as
using mmap-alpha-round-outcome[OF alpha-out'] by simp
have query-count-s4: PQueryCounter s4 = PQueryCounter s3
using mmap-alpha-round-outcome[OF alpha-out'] by simp
from read-outcome[OF read-dg] obtain rest5 where
tr-s4: PTranscript s4 = dg # rest5
and st-s5: PState s5 = concat (PState s4) dg
and tr-s5: PTranscript s5 = rest5
and query-count-s5: PQueryCounter s5 = PQueryCounter s4
by blast
have s6-eq: s6 = s5
using degree-assert unfolding assert-def
by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
have comp-fri-res:
length fl = ceil-log (to-nat dg + 1) ∧
PTranscript s6 = map snd fl @ PTranscript s7 ∧
PState s7 = foldl concat (PState s6) (map snd fl)
using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
by simp
have query-count-s7: PQueryCounter s7 = PQueryCounter s6
using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
by simp
from read-outcome[OF read-final] obtain rest-query where
tr-s7: PTranscript s7 = final # rest-query

|  |

```

```

and st-query: PState query-state = concat (PState s7) final
and tr-query: PTranscript query-state = rest-query
and query-count-query: PQueryCounter query-state = PQueryCounter s7
by blast
have header-tr:
  verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
using tr-s tr-s1 trace-fri-res tr-s2 tr-s3 alpha-res tr-s4 tr-s5
  s6-eq comp-fri-res tr-s7 tr-query
unfolding verifier-header-transcript-def verifier-header-messages-def
by simp
have header-state:
  PState query-state =
    verifier-header-state s fr (map snd f-fl) f-final as dg (map snd fl) final
using st-s1 trace-fri-res st-s3 alpha-res st-s5 s6-eq comp-fri-res st-query
unfolding verifier-header-state-def verifier-header-messages-def
by simp
have query-count-header:
  PQueryCounter query-state = PQueryCounter s
using query-count-s1 query-count-s2 query-count-s3 query-count-s4
  query-count-s5 s6-eq query-count-s7 query-count-query
by simp
show ?thesis
by (rule that[OF header-tr header-state query-out query-count-header])
  (use trace-fri-res alpha-res comp-fri-res in simp-all)
qed

```

```

lemma verify-monad-header-query-extraction:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  obtains fr f-fl f-final as dg fl final query-state
  where
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
      (map snd fl) final (PTranscript query-state)
    and Some (result, final-state) ∈
      set-dist (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final) rounds)
        query-state)
proof (rule verify-monad-header-extraction[OF outcome])
fix fr f-fl f-final as dg fl final query-state
assume header:
  verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
and query-out:
  Some (result, final-state) ∈
    set-dist
    (execute

```

```

      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
      query-state)
show ?thesis
  by (rule that[OF header query-out])
qed

lemma verify-monad-header-random-oracle-replay:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  obtains fr f-fl f-final as dg fl final query-state
  where
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
      (map snd fl) final (PTranscript query-state)
  and PState query-state =
    verifier-header-state s fr (map snd f-fl) f-final as dg (map snd fl) final
  and Some (result, final-state) ∈
    set-dist (execute
      (ntimes (verifier-query-round-program fr f-fl f-final as fl final) rounds)
      query-state)
  and PQueryCounter query-state = PQueryCounter s
  and s ≤ query-state
  and  $\bigwedge i. i < \text{length } f\text{-fl} \implies$ 
    fmlookup (HashMap query-state)
      (TraceFriChallenge (PTraceFriCounter s + i)
        (foldl concat (concat (PState s) fr) (take (Suc i) (map snd f-fl)))) =
      Some (fst (f-fl ! i))
  and  $\bigwedge i. i < \text{length } as \implies$ 
    fmlookup (HashMap query-state)
      (AlphaChallenge (PAlphaCounter s + i) (foldl concat
        (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
        (take i as))) =
      Some (as ! i)
  and  $\bigwedge i. i < \text{length } fl \implies$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
        (concat
          (foldl concat
            (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
            as)
          dg)
        (take (Suc i) (map snd fl)))) =
      Some (fst (fl ! i))
  and to-nat dg ≤ maxDegree
proof -
  let ?alpha =
  do {

```

```

    a0 ← receive-alpha-challenge;
    let a0' = a0;
    a1 ← read;
    let a1' = a1;
    assert (a0' = a1');
    return a1'
}
from outcome obtain fr s1 f-fl s2 f-final s3 as s4 dg s5 s6 fl s7 final query-state
where read-fr: Some (fr, s1) ∈ set-dist (execute read s)
and trace-fri:
    Some (f-fl, s2) ∈
    set-dist (execute
        (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
and read-f-final: Some (f-final, s3) ∈ set-dist (execute read s2)
and alpha-out:
    Some (as, s4) ∈
    set-dist (execute (mmap (replicate (length spec) ?alpha)) s3)
and read-dg: Some (dg, s5) ∈ set-dist (execute read s4)
and degree-assert:
    Some ((), s6) ∈ set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
and comp-fri:
    Some (fl, s7) ∈
    set-dist (execute
        (ntimes receive-composition-fri-commits
            (ceil-log (to-nat dg + 1))) s6)
and read-final: Some (final, query-state) ∈ set-dist (execute read s7)
and query-out:
    Some (result, final-state) ∈
    set-dist (execute
        (ntimes (verifier-query-round-program fr f-fl f-final as fl final) rounds)
        query-state)
unfolding verify-monad-def verifier-query-round-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
from read-outcome[OF read-fr] obtain rest1 where
    tr-s: PTranscript s = fr # rest1
and st-s1: PState s1 = concat (PState s) fr
and tr-s1: PTranscript s1 = rest1
and ext-s-s1: s ≤ s1
and trace-count-s1: PTraceFriCounter s1 = PTraceFriCounter s
and comp-count-s1: PCompositionFriCounter s1 = PCompositionFriCounter s
and alpha-count-s1: PAlphaCounter s1 = PAlphaCounter s
and query-count-s1: PQueryCounter s1 = PQueryCounter s
by blast
have trace-fri-res:
    length f-fl = ceil-log clength ∧
    PTranscript s1 = map snd f-fl @ PTranscript s2 ∧
    PState s2 = foldl concat (PState s1) (map snd f-fl) ∧
    s1 ≤ s2 ∧
    PTraceFriCounter s2 = PTraceFriCounter s1 + ceil-log clength ∧

```

```

    PCompositionFriCounter s2 = PCompositionFriCounter s1 ∧
    PAlphaCounter s2 = PAlphaCounter s1 ∧
    PQueryCounter s2 = PQueryCounter s1 ∧
    (∀ i < ceil-log clength.
      fmllookup (HashMap s2)
        (TraceFriChallenge (PTraceFriCounter s1 + i)
          (foldl concat (PState s1) (take (Suc i) (map snd f-fl)))) =
        Some (fst (f-fl ! i)))
  using ntimes-receive-trace-fri-commits-outcome[OF trace-fri] by simp
from read-outcome[OF read-f-final] obtain rest3 where
  tr-s2: PTranscript s2 = f-final # rest3
  and st-s3: PState s3 = concat (PState s2) f-final
  and tr-s3: PTranscript s3 = rest3
  and ext-s2-s3: s2 ≤ s3
  and trace-count-s3: PTraceFriCounter s3 = PTraceFriCounter s2
  and comp-count-s3: PCompositionFriCounter s3 = PCompositionFriCounter s2
  and alpha-count-s3: PAlphaCounter s3 = PAlphaCounter s2
  and query-count-s3: PQueryCounter s3 = PQueryCounter s2
  by blast
have alpha-out':
  Some (as, s4) ∈
    set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)
  using alpha-out by (simp add: alpha-round-def)
have alpha-res:
  length as = length spec ∧
  PTranscript s3 = as @ PTranscript s4 ∧
  PState s4 = foldl concat (PState s3) as ∧
  s3 ≤ s4 ∧
  PTraceFriCounter s4 = PTraceFriCounter s3 ∧
  PCompositionFriCounter s4 = PCompositionFriCounter s3 ∧
  PAlphaCounter s4 = PAlphaCounter s3 + length spec ∧
  PQueryCounter s4 = PQueryCounter s3 ∧
  (∀ i < length spec.
    fmllookup (HashMap s4)
      (AlphaChallenge (PAlphaCounter s3 + i)
        (foldl concat (PState s3) (take i as))) =
      Some (as ! i))
  using mmap-alpha-round-outcome[OF alpha-out'] by simp
from read-outcome[OF read-dg] obtain rest5 where
  tr-s4: PTranscript s4 = dg # rest5
  and st-s5: PState s5 = concat (PState s4) dg
  and tr-s5: PTranscript s5 = rest5
  and ext-s4-s5: s4 ≤ s5
  and trace-count-s5: PTraceFriCounter s5 = PTraceFriCounter s4
  and comp-count-s5: PCompositionFriCounter s5 = PCompositionFriCounter s4
  and alpha-count-s5: PAlphaCounter s5 = PAlphaCounter s4
  and query-count-s5: PQueryCounter s5 = PQueryCounter s4
  by blast
have s6-eq: s6 = s5

```

```

using degree-assert unfolding assert-def
by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
have degree-bound: to-nat dg ≤ maxDegree
using degree-assert unfolding assert-def
by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
have ext-s5-s6: s5 ≤ s6
unfolding s6-eq by (rule hash-ext-refl)
have comp-fri-res:
  length fl = ceil-log (to-nat dg + 1) ∧
  PTranscript s6 = map snd fl @ PTranscript s7 ∧
  PState s7 = foldl concat (PState s6) (map snd fl) ∧
  s6 ≤ s7 ∧
  PTraceFriCounter s7 = PTraceFriCounter s6 ∧
  PCompositionFriCounter s7 = PCompositionFriCounter s6 + ceil-log (to-nat dg + 1) ∧
  PAlphaCounter s7 = PAlphaCounter s6 ∧
  PQueryCounter s7 = PQueryCounter s6 ∧
  (∀ i < ceil-log (to-nat dg + 1).
    fmllookup (HashMap s7)
      (CompositionFriChallenge (PCompositionFriCounter s6 + i)
        (foldl concat (PState s6) (take (Suc i) (map snd fl)))) =
      Some (fst (fl ! i)))
using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-s7: PTranscript s7 = final # rest-query
and st-query: PState query-state = concat (PState s7) final
and tr-query: PTranscript query-state = rest-query
and ext-s7-query: s7 ≤ query-state
and query-count-query: PQueryCounter query-state = PQueryCounter s7
by blast
have ext-s2-query: s2 ≤ query-state
using ext-s2-s3 alpha-res ext-s4-s5 ext-s5-s6 comp-fri-res ext-s7-query
by (meson hash-ext-trans)
have ext-s4-query: s4 ≤ query-state
using ext-s4-s5 ext-s5-s6 comp-fri-res ext-s7-query
by (meson hash-ext-trans)
have ext-s7-query': s7 ≤ query-state
using ext-s7-query .
have ext-s1-s2: s1 ≤ s2
using trace-fri-res by simp
have ext-s3-s4: s3 ≤ s4
using alpha-res by simp
have ext-s6-s7: s6 ≤ s7
using comp-fri-res by simp
have ext-s-s2: s ≤ s2
by (rule hash-ext-trans[OF ext-s-s1 ext-s1-s2])
have ext-s-s3: s ≤ s3
by (rule hash-ext-trans[OF ext-s-s2 ext-s2-s3])
have ext-s-s4: s ≤ s4

```

```

  by (rule hash-ext-trans[OF ext-s-s3 ext-s3-s4])
have ext-s-s5:  $s \leq s5$ 
  by (rule hash-ext-trans[OF ext-s-s4 ext-s4-s5])
have ext-s-s6:  $s \leq s6$ 
  by (rule hash-ext-trans[OF ext-s-s5 ext-s5-s6])
have ext-s-s7:  $s \leq s7$ 
  by (rule hash-ext-trans[OF ext-s-s6 ext-s6-s7])
have ext-s-query:  $s \leq \text{query-state}$ 
  by (rule hash-ext-trans[OF ext-s-s7 ext-s7-query])
have header-tr:
  verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
  using tr-s tr-s1 trace-fri-res tr-s2 tr-s3 alpha-res tr-s4 tr-s5
    s6-eq comp-fri-res tr-s7 tr-query
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
have header-state:
  PState query-state =
    verifier-header-state s fr (map snd f-fl) f-final as dg (map snd fl) final
  using st-s1 trace-fri-res st-s3 alpha-res st-s5 s6-eq comp-fri-res st-query
  unfolding verifier-header-state-def verifier-header-messages-def
  by simp
have trace-lookup:
 $\bigwedge i. i < \text{length } f\text{-fl} \implies$ 
  fmlookup (HashMap query-state)
    (TraceFriChallenge (PTraceFriCounter s + i)
      (foldl concat (concat (PState s) fr) (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
proof -
  fix i
  assume i-bound:  $i < \text{length } f\text{-fl}$ 
  have lookup-s2:
    fmlookup (HashMap s2)
      (TraceFriChallenge (PTraceFriCounter s1 + i)
        (foldl concat (PState s1) (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
  using trace-fri-res i-bound by simp
  show fmlookup (HashMap query-state)
    (TraceFriChallenge (PTraceFriCounter s + i)
      (foldl concat (concat (PState s) fr) (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
  using hash-extension-lookup[OF lookup-s2 ext-s2-query] st-s1
    trace-count-s1
  by simp
qed
have alpha-lookup:
 $\bigwedge i. i < \text{length } as \implies$ 
  fmlookup (HashMap query-state)
    (AlphaChallenge (PAlphaCounter s + i) (foldl concat

```



```

      (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
      (take i as))) =
    Some (as ! i)
proof -
  fix i
  assume i-bound: i < length as
  have lookup-s4:
    fmlookup (HashMap s4)
      (AlphaChallenge (PAlphaCounter s3 + i)
        (foldl concat (PState s3) (take i as))) =
      Some (as ! i)
  using alpha-res i-bound by simp
  show fmlookup (HashMap query-state)
    (AlphaChallenge (PAlphaCounter s + i) (foldl concat
      (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
      (take i as))) =
    Some (as ! i)
  using hash-extension-lookup[OF lookup-s4 ext-s4-query]
    st-s1 trace-fri-res st-s3 alpha-count-s1 alpha-count-s3
  by simp
qed
have comp-lookup:
   $\bigwedge i. i < \text{length } fl \implies$ 
  fmlookup (HashMap query-state)
    (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
          as)
        dg)
      (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i))
proof -
  fix i
  assume i-bound: i < length fl
  have lookup-s7:
    fmlookup (HashMap s7)
      (CompositionFriChallenge (PCompositionFriCounter s6 + i)
        (foldl concat (PState s6) (take (Suc i) (map snd fl)))) =
      Some (fst (fl ! i))
  using comp-fri-res i-bound by simp
  show fmlookup (HashMap query-state)
    (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
          as)
        dg)
      (take (Suc i) (map snd fl)))) =

```

```

    Some (fst (fl ! i))
using hash-extension-lookup[OF lookup-s7 ext-s7-query]
    st-s1 trace-fri-res st-s3 alpha-res st-s5 s6-eq
    comp-count-s1 comp-count-s3 comp-count-s5
by simp
qed
have query-count-header:
  PQueryCounter query-state = PQueryCounter s
using query-count-s1 trace-fri-res query-count-s3 alpha-res query-count-s5
  s6-eq comp-fri-res query-count-query
by simp
show ?thesis
by (rule that[OF header-tr header-state query-out query-count-header
  ext-s-query trace-lookup alpha-lookup comp-lookup degree-bound])
qed

lemma verify-monad-random-oracle-key-extraction:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
obtains fr :: 'f
  and f-fl :: ('f × 'f) list
  and f-final :: 'f
  and as :: 'f list
  and dg :: 'f
  and fl :: ('f × 'f) list
  and final :: 'f
  and query-state :: ('f, 'a) protocol-channel-scheme
  and raw-idxs :: 'f list
  and query-idxs :: nat list
  and query-chunks :: 'f list list
where
  verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
and PState query-state =
  verifier-header-state s fr (map snd f-fl) f-final as dg (map snd fl) final
and length raw-idxs = rounds
and query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and length query-chunks = rounds
and PTranscript query-state =
  List.concat query-chunks @ PTranscript final-state
and ∧i. i < length f-fl ⇒
  fmlookup (HashMap query-state)
    (TraceFriChallenge
      (PTraceFriCounter s + i)
      (foldl concat (concat (PState s) fr)
        (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
and ∧i. i < length as ⇒

```

```

    fmlookup (HashMap query-state)
      (AlphaChallenge (PAlphaCounter s + i) (foldl concat
        (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
        (take i as))) =
      Some (as ! i)
  and  $\bigwedge i. i < \text{length } fl \implies$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
        (concat
          (foldl concat
            (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
            as)
          dg)
        (take (Suc i) (map snd fl)))) =
      Some (fst (fl ! i))
  and  $\bigwedge i. i < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
  and  $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge
        (PQueryCounter query-state + i) (state-after-query-chunks (PState query-state) query-chunks
          i)) =
      Some (raw-idxs ! i)
  and  $\bigwedge idx. idx \in \text{set query-idxs} \implies idx < \text{clength} * \text{scale}$ 
proof –
from verify-monad-header-random-oracle-replay[OF outcome]
obtain fr f-fl f-final as dg fl final query-state where
  header:
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
      (map snd fl) final (PTranscript query-state)
  and header-state:
    PState query-state =
      verifier-header-state s fr (map snd f-fl) f-final as dg
        (map snd fl) final
  and query-out:
    Some (result, final-state)  $\in$ 
      set-dist (execute
        (ntimes (verifier-query-round-program fr f-fl f-final as fl final) rounds)
        query-state)
  and query-count-header:
    PQueryCounter query-state = PQueryCounter s
  and ext-s-query-header:  $s \leq \text{query-state}$ 
  and trace-lookup:
     $\bigwedge i. i < \text{length } f-fl \implies$ 
      fmlookup (HashMap query-state)
        (TraceFriChallenge
          (PTraceFriCounter s + i) (foldl concat (concat (PState s) fr)
            (take (Suc i) (map snd f-fl)))) =

```

```

    Some (fst (f-fl ! i))
and alpha-lookup:
   $\bigwedge i. i < \text{length } as \implies$ 
    fmlookup (HashMap query-state)
      (AlphaChallenge (PAlphaCounter s + i) (foldl concat
        (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
        (take i as))) =
    Some (as ! i)
and comp-lookup:
   $\bigwedge i. i < \text{length } fl \implies$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
        (concat
          (foldl concat
            (concat
              (foldl concat (concat (PState s) fr) (map snd f-fl))
              f-final)
            as)
          dg)
        (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i))
and degree-bound: to-nat dg  $\leq$  maxDegree
by blast
from ntimes-verifier-query-rounds-outcome[OF query-out]
obtain raw-idxs query-idxs query-chunks where
  len-raw: length raw-idxs = rounds
and query-idxs-def: query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and len-chunks: length query-chunks = rounds
and tr-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and round-chunks:
   $\forall i < \text{rounds}.$ 
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and query-lookup:
   $\forall i < \text{rounds}.$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge
        (PQueryCounter query-state + i) (state-after-query-chunks (PState query-state)
          query-chunks i)) =
    Some (raw-idxs ! i)
and idx-bounds:  $\forall idx \in \text{set query-idxs}. idx < \text{clength} * \text{scale}$ 
by blast
show ?thesis
proof (rule that[of fr f-fl f-final as dg fl final query-state
  raw-idxs query-idxs query-chunks])
show verifier-header-transcript s fr (map snd f-fl) f-final as dg
  (map snd fl) final (PTranscript query-state)

```

```

    by (rule header)
next
show PState query-state =
  verifier-header-state s fr (map snd f-fl) f-final as dg
  (map snd fl) final
  by (rule header-state)
next
show length raw-idxs = rounds
  by (rule len-raw)
next
show query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  by (rule query-idxs-def)
next
show length query-chunks = rounds
  by (rule len-chunks)
next
show PTranscript query-state =
  List.concat query-chunks @ PTranscript final-state
  by (rule tr-query)
next
fix i
assume i < length f-fl
then show fmlookup (HashMap query-state)
  (TraceFriChallenge
    (PTraceFriCounter s + i)
    (foldl concat (concat (PState s) fr)
      (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
  by (rule trace-lookup)
next
fix i
assume i < length as
then show fmlookup (HashMap query-state)
  (AlphaChallenge (PAlphaCounter s + i) (foldl concat
    (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
    (take i as))) =
    Some (as ! i)
  by (rule alpha-lookup)
next
fix i
assume i < length fl
then show fmlookup (HashMap query-state)
  (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
    (concat
      (foldl concat
        (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
        as)
      dg)
    (take (Suc i) (map snd fl)))) =

```

```

    Some (fst (fl ! i))
  by (rule comp-lookup)
next
fix i
assume i < rounds
then show verifier-query-round-chunk (query-idxs ! i)
  (map snd f-fl) (map snd fl) (query-chunks ! i)
  using round-chunks by blast
next
fix i
assume i < rounds
then show fmlookup (HashMap final-state)
  (QueryIndexChallenge
    (PQueryCounter query-state + i) (state-after-query-chunks (PState query-state) query-chunks
i)) =
  Some (raw-idxs ! i)
  using query-lookup by blast
next
fix idx
assume idx ∈ set query-idxs
then show idx < clength * scale
  using idx-bounds by blast
qed
qed

```

lemma *verify-monad-accepted-transcript-shape*:

fixes $s :: ('f, 'a) \text{protocol-channel-scheme}$

assumes *outcome*:

$\text{Some (result, final-state)} \in \text{set-dist (execute verify-monad s)}$

shows $\exists \text{alphas query-idxs.}$

$\text{accepted-transcript-shape s (Some (result, final-state)) alphas query-idxs}$

proof –

from *verify-monad-header-extraction*[OF *outcome*]

obtain $\text{fr f-fl f-final as dg fl final query-state}$ **where**

header:

$\text{verifier-header-transcript s fr (map snd f-fl) f-final as dg}$

$(\text{map snd fl}) \text{ final (PTranscript query-state)}$

and *header-state*:

$\text{PState query-state} =$

$\text{verifier-header-state s fr (map snd f-fl) f-final as dg (map snd fl) final}$

and *query-out*:

$\text{Some (result, final-state)} \in$

set-dist (execute

$(\text{ntimes (verifier-query-round-program fr f-fl f-final as fl final) rounds})$

query-state)

and *query-count-header*:

$\text{PQueryCounter query-state} = \text{PQueryCounter s}$

and *len-f-fl*: $\text{length f-fl} = \text{ceil-log clength}$

and *len-as*: $\text{length as} = \text{length spec}$

```

and len-fl: length fl = ceil-log (to-nat dg + 1)
by blast
from ntimes-verifier-query-rounds-outcome[OF query-out]
obtain raw-idxs query-idxs query-chunks where
  len-raw: length raw-idxs = rounds
  and query-idxs-def: query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
  and len-chunks: length query-chunks = rounds
  and tr-query:
    PTranscript query-state = List.concat query-chunks @ PTranscript final-state
  and round-chunks:
     $\forall i < \text{rounds}.$ 
      verifier-query-round-chunk (query-idxs ! i)
        (map snd f-fl) (map snd fl) (query-chunks ! i)
  and lookups:
     $\forall i < \text{rounds}.$ 
      fmllookup (HashMap final-state)
        (QueryIndexChallenge (PQueryCounter query-state + i) (state-after-query-chunks (PState
query-state) query-chunks i)) =
        Some (raw-idxs ! i)
  and idx-bounds:  $\forall \text{idx} \in \text{set query-idxs}. \text{idx} < \text{clength} * \text{scale}$ 
by auto
have derived:
  verifier-query-indices-derived s (Some (result, final-state))
    (verifier-header-state s fr (map snd f-fl) f-final as dg (map snd fl) final)
    (PTranscript query-state) (map snd f-fl) (map snd fl) query-idxs
proof –
  have tr-query':
    List.concat query-chunks @ PTranscript final-state = PTranscript query-state
  using tr-query by simp
  have lookups':
     $\forall i < \text{rounds}.$ 
      fmllookup (HashMap final-state)
        (QueryIndexChallenge (PQueryCounter s + i) (state-after-query-chunks
(verifier-header-state s fr (map snd f-fl) f-final as dg (map snd fl) final)
query-chunks i)) =
        Some (raw-idxs ! i)
  using lookups header-state query-count-header by simp
show ?thesis
unfolding verifier-query-indices-derived-def
apply (rule exI[where x=result])
apply (rule exI[where x=final-state])
apply (rule exI[where x=raw-idxs])
apply (rule exI[where x=query-chunks])
apply (rule exI[where x=PTranscript final-state])
apply (intro conjI)
  apply simp
  apply (rule len-raw)
  apply (rule query-idxs-def)
  apply (rule len-chunks)

```

```

    apply (rule tr-query')
    apply (rule round-chunks)
    apply (rule lookups')
    apply (rule idx-bounds)
  done
qed
have shape:
  accepted-transcript-shape s (Some (result, final-state)) as query-idxs
  unfolding accepted-transcript-shape-def
  apply (rule exI[where x=result])
  apply (rule exI[where x=final-state])
  apply (rule exI[where x=fr])
  apply (rule exI[where x=map snd f-fl])
  apply (rule exI[where x=f-final])
  apply (rule exI[where x=dg])
  apply (rule exI[where x=map snd fl])
  apply (rule exI[where x=final])
  apply (rule exI[where x=PTranscript query-state])
  apply (intro conjI)
    apply simp
    apply (rule header)
    apply (rule derived)
  done
show ?thesis
  using shape by blast
qed

```

```

lemma verify-monad-supplied-header-transcript-shape:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and header0:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  obtains query-idxs raw-idxs query-chunks where
    accepted-transcript-shape s (Some (result, final-state)) as query-idxs
    length raw-idxs = rounds
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
    length query-chunks = rounds
    List.concat query-chunks @ PTranscript final-state = rest
    ∧ i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
      f-fri-roots composition-fri-roots (query-chunks ! i)
    ∧ i. i < rounds ⇒
      fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks
      (verifier-header-state s fr f-fri-roots f-final as dg
      composition-fri-roots final)

```



```

      query-chunks i)) =
      Some (raw-idxs ! i)
proof –
from verify-monad-header-extraction[OF outcome]
obtain fr' f-fl f-final' as' dg' fl final' query-state where
  header:
    verifier-header-transcript s fr' (map snd f-fl) f-final' as' dg'
      (map snd fl) final' (PTranscript query-state)
and header-state:
    PState query-state =
      verifier-header-state s fr' (map snd f-fl) f-final' as' dg'
        (map snd fl) final'
and query-out:
    Some (result, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr' f-fl f-final' as' fl
              final') rounds)
            query-state)
and query-count-header:
    PQueryCounter query-state = PQueryCounter s
by metis
have eqs:
  fr' = fr ∧
    map snd f-fl = f-fri-roots ∧
    f-final' = f-final ∧
    as' = as ∧
    dg' = dg ∧
    map snd fl = composition-fri-roots ∧
    final' = final ∧
    PTranscript query-state = rest
using verifier-header-transcript-unique[OF header header0] by simp
from ntimes-verifier-query-rounds-outcome[OF query-out]
obtain raw-idxs query-idxs query-chunks where
  len-raw: length raw-idxs = rounds
and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-chunks: length query-chunks = rounds
and tr-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
and chunks:
  ∀ i < rounds.
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and lookups:
  ∀ i < rounds.
    fmlookup (HashMap final-state)

```

```

      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
    and idx-bounds:  $\forall \text{idx} \in \text{set query-idxs. idx} < \text{clength} * \text{scale}$ 
    by blast
  have tr-query':
    List.concat query-chunks @ PTranscript final-state = rest
  using tr-query eqs by simp
  have chunks':
     $\wedge i. i < \text{rounds} \implies$ 
      verifier-query-round-chunk (query-idxs ! i)
      f-fri-roots composition-fri-roots (query-chunks ! i)
  using chunks eqs by simp
  have lookups':
     $\wedge i. i < \text{rounds} \implies$ 
      fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks
          (verifier-header-state s fr f-fri-roots f-final as dg
            composition-fri-roots final)
          query-chunks i)) =
      Some (raw-idxs ! i)
  using lookups header-state query-count-header eqs by simp
  have derived:
    verifier-query-indices-derived s (Some (result, final-state))
    (verifier-header-state s fr f-fri-roots f-final as dg
      composition-fri-roots final)
    rest f-fri-roots composition-fri-roots query-idxs
  unfolding verifier-query-indices-derived-def
  by (intro exI[of - result] exI[of - final-state]
    exI[of - raw-idxs] exI[of - query-chunks]
    exI[of - PTranscript final-state] conjI)
    (use len-raw query-idxs-eq len-chunks tr-query' chunks' lookups'
      idx-bounds in auto)
  have shape:
    accepted-transcript-shape s (Some (result, final-state)) as query-idxs
  unfolding accepted-transcript-shape-def
  by (intro exI[of - result] exI[of - final-state] exI[of - fr]
    exI[of - f-fri-roots] exI[of - f-final] exI[of - dg]
    exI[of - composition-fri-roots] exI[of - final]
    exI[of - rest] conjI)
    (use header0 derived in simp-all)
  show ?thesis
  by (rule that[OF shape len-raw query-idxs-eq len-chunks tr-query'
    chunks' lookups'])
qed

```

lemma *verify-monad-supplied-header-query-rounds:*
fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *outcome*:
Some (result, final-state) ∈ set-dist (execute verify-monad s)
and *header0*:
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
obtains *f-fl fl query-state* **where**
map snd f-fl = f-fri-roots
map snd fl = composition-fri-roots
PTranscript query-state = rest
PState query-state =
verifier-header-state s fr f-fri-roots f-final as dg
composition-fri-roots final
PQueryCounter query-state = PQueryCounter s
Some (result, final-state) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final)
rounds)
query-state)
proof –
from *verify-monad-header-extraction[OF outcome]*
obtain *fr' f-fl f-final' as' dg' fl final' query-state* **where**
header:
verifier-header-transcript s fr' (map snd f-fl) f-final' as' dg'
(map snd fl) final' (PTranscript query-state)
and *header-state*:
PState query-state =
verifier-header-state s fr' (map snd f-fl) f-final' as' dg'
(map snd fl) final'
and *query-out*:
Some (result, final-state) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr' f-fl f-final' as' fl
final') rounds)
query-state)
and *query-count-header*:
PQueryCounter query-state = PQueryCounter s
by *metis*
have *eqs*:
fr' = fr ∧
map snd f-fl = f-fri-roots ∧
f-final' = f-final ∧
as' = as ∧
dg' = dg ∧
map snd fl = composition-fri-roots ∧
final' = final ∧

```

    PTranscript query-state = rest
  using verifier-header-transcript-unique[OF header header0] by simp
show ?thesis
  by (rule that[of f-fl fl query-state])
    (use eqs header-state query-count-header query-out in simp-all)
qed

```

```

lemma verify-monad-supplied-empty-header-query-rounds:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and header0:
    verifier-header-transcript s fr f-fri-roots f-final as dg []
    final rest
  obtains f-fl query-state where
    map snd f-fl = f-fri-roots
    PTranscript query-state = rest
    PState query-state =
      verifier-header-state s fr f-fri-roots f-final as dg [] final
    PQueryCounter query-state = PQueryCounter s
    Some (result, final-state) ∈
      set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as [] final)
          rounds)
        query-state)
proof –
  from verify-monad-supplied-header-query-rounds[OF outcome header0]
  obtain f-fl fl query-state where f-roots:
    map snd f-fl = f-fri-roots
  and fl-roots: map snd fl = []
  and tr: PTranscript query-state = rest
  and st:
    PState query-state =
      verifier-header-state s fr f-fri-roots f-final as dg [] final
  and qc: PQueryCounter query-state = PQueryCounter s
  and out:
    Some (result, final-state) ∈
      set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
  by blast
  have fl-empty: fl = []
  using fl-roots by (cases fl) simp-all
  show ?thesis

```

```

    by (rule that[of f-fl query-state])
      (use f-roots tr st qc out fl-empty in simp-all)
qed

lemma verifier-after-alpha-accepted-transcript-shape:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes prefix:
    Some ((fr, f-fl, f-final, as), prefix-state) ∈
      set-dist (execute verifier-alpha-prefix s)
  and suffix:
    Some (result, final-state) ∈
      set-dist
        (execute (verifier-after-alpha (fr, f-fl, f-final, as))
          prefix-state)
  shows ∃ query-idxs.
    accepted-transcript-shape s (Some (result, final-state)) as query-idxs
proof -
  have prefix-res:
    length f-fl = ceil-log clength ∧
    length as = length spec ∧
    PTranscript s =
      [fr] @ map snd f-fl @ [f-final] @ as @ PTranscript prefix-state ∧
    PState prefix-state =
      foldl concat
        (concat (foldl concat (concat (PState s) fr) (map snd f-fl))
          f-final)
        as ∧
    PQueryCounter prefix-state = PQueryCounter s
  by (rule verifier-alpha-prefix-outcome[OF prefix])
  from suffix obtain dg s5 s6 fl s7 final query-state where
    read-dg: Some (dg, s5) ∈ set-dist (execute read prefix-state)
  and degree-assert:
    Some ((), s6) ∈
      set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
  and comp-fri:
    Some (fl, s7) ∈
      set-dist (execute
        (ntimes receive-composition-fri-commits
          (ceil-log (to-nat dg + 1))) s6)
  and read-final: Some (final, query-state) ∈ set-dist (execute read s7)
  and query-out:
    Some (result, final-state) ∈
      set-dist (execute
        (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
  unfolding verifier-after-alpha-def
  by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-dg] obtain rest5 where

```

```

tr-prefix: PTranscript prefix-state = dg # rest5
and st-s5: PState s5 = concat (PState prefix-state) dg
and tr-s5: PTranscript s5 = rest5
and query-count-s5: PQueryCounter s5 = PQueryCounter prefix-state
by blast
have s6-eq: s6 = s5
  using degree-assert unfolding assert-def
  by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
have comp-res:
  length fl = ceil-log (to-nat dg + 1) ∧
  PTranscript s6 = map snd fl @ PTranscript s7 ∧
  PState s7 = foldl concat (PState s6) (map snd fl) ∧
  PQueryCounter s7 = PQueryCounter s6
  using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
  by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-s7: PTranscript s7 = final # rest-query
  and st-query: PState query-state = concat (PState s7) final
  and tr-query: PTranscript query-state = rest-query
  and query-count-query: PQueryCounter query-state = PQueryCounter s7
  by blast
have header:
  verifier-header-transcript s fr (map snd f-fl) f-final as dg
  (map snd fl) final (PTranscript query-state)
  using prefix-res tr-prefix tr-s5 s6-eq comp-res tr-s7 tr-query
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
have header-state:
  PState query-state =
    verifier-header-state s fr (map snd f-fl) f-final as dg
    (map snd fl) final
  using prefix-res st-s5 s6-eq comp-res st-query
  unfolding verifier-header-state-def verifier-header-messages-def
  by simp
have query-count-header:
  PQueryCounter query-state = PQueryCounter s
  using prefix-res query-count-s5 s6-eq comp-res query-count-query
  by simp
from ntimes-verifier-query-rounds-outcome[OF query-out]
obtain raw-idxs query-idxs query-chunks where
  len-raw: length raw-idxs = rounds
  and query-idxs-def: query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-chunks: length query-chunks = rounds
  and tr-query-rounds:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
and round-chunks:
  ∀ i < rounds.
    verifier-query-round-chunk (query-idxs ! i)

```

```

      (map snd f-fl) (map snd fl) (query-chunks ! i)
and lookups:
  ∀ i < rounds.
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
and idx-bounds: ∀ idx ∈ set query-idxs. idx < clength * scale
by blast
have derived:
  verifier-query-indices-derived s (Some (result, final-state))
    (verifier-header-state s fr (map snd f-fl) f-final as dg
      (map snd fl) final)
    (PTranscript query-state) (map snd f-fl) (map snd fl) query-idxs
proof -
have tr-query-rounds':
  List.concat query-chunks @ PTranscript final-state =
    PTranscript query-state
using tr-query-rounds by simp
have lookups':
  ∀ i < rounds.
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks
          (verifier-header-state s fr (map snd f-fl) f-final as dg
            (map snd fl) final)
          query-chunks i)) =
      Some (raw-idxs ! i)
using lookups header-state query-count-header by simp
show ?thesis
unfolding verifier-query-indices-derived-def
apply (rule exI[where x=result])
apply (rule exI[where x=final-state])
apply (rule exI[where x=raw-idxs])
apply (rule exI[where x=query-chunks])
apply (rule exI[where x=PTranscript final-state])
apply (intro conjI)
  apply simp
  apply (rule len-raw)
  apply (rule query-idxs-def)
  apply (rule len-chunks)
  apply (rule tr-query-rounds')
  apply (rule round-chunks)
  apply (rule lookups')
  apply (rule idx-bounds)
done
qed
have shape:
  accepted-transcript-shape s (Some (result, final-state)) as query-idxs

```

```

unfolding accepted-transcript-shape-def
apply (rule exI[where  $x=result$ ])
apply (rule exI[where  $x=final-state$ ])
apply (rule exI[where  $x=fr$ ])
apply (rule exI[where  $x=map\ snd\ f-fl$ ])
apply (rule exI[where  $x=f-final$ ])
apply (rule exI[where  $x=dg$ ])
apply (rule exI[where  $x=map\ snd\ fl$ ])
apply (rule exI[where  $x=final$ ])
apply (rule exI[where  $x=PTranscript\ query-state$ ])
apply (intro conjI)
  apply simp
  apply (rule header)
apply (rule derived)
done
show ?thesis
  using shape by blast
qed

lemma verifier-after-alpha-accepted-transcript-header:
fixes  $s :: ('f, 'a)\ protocol-channel-scheme$ 
assumes prefix:
   $Some\ ((fr, f-fl, f-final, as), prefix-state) \in$ 
     $set-dist\ (execute\ verifier-alpha-prefix\ s)$ 
and suffix:
   $Some\ (result, final-state) \in$ 
     $set-dist$ 
     $(execute\ (verifier-after-alpha\ (fr, f-fl, f-final, as))$ 
       $prefix-state)$ 
obtains dg composition-fri-roots final rest query-idxs
where
  verifier-header-transcript  $s\ fr\ (map\ snd\ f-fl)\ f-final\ as\ dg$ 
    composition-fri-roots final rest
and accepted-transcript-shape  $s\ (Some\ (result, final-state))$ 
    as query-idxs
proof –
have prefix-res:
   $length\ f-fl = ceil-log\ clength \wedge$ 
   $length\ as = length\ spec \wedge$ 
   $PTranscript\ s =$ 
     $[fr] @ map\ snd\ f-fl @ [f-final] @ as @ PTranscript\ prefix-state$ 
using verifier-alpha-prefix-outcome[OF prefix] by simp
from suffix obtain  $dg\ s5\ s6\ fl\ s7\ final\ query-state$  where
  read-dg:  $Some\ (dg, s5) \in set-dist\ (execute\ read\ prefix-state)$ 
and degree-assert:
   $Some\ ((), s6) \in$ 
     $set-dist\ (execute\ (assert\ (to-nat\ dg \leq maxDegree))\ s5)$ 
and comp-fri:
   $Some\ (fl, s7) \in$ 

```



```

    set-dist (execute
      (ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1))) s6)
  and read-final: Some (final, query-state) ∈ set-dist (execute read s7)
  unfolding verifier-after-alpha-def
  by (auto elim!: set-dist-bindE)
from read-outcome[OF read-dg] obtain rest5 where
  tr-prefix: PTranscript prefix-state = dg # rest5
  and tr-s5: PTranscript s5 = rest5
  by blast
have s6-eq: s6 = s5
  using degree-assert unfolding assert-def
  by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
have comp-res:
  length fl = ceil-log (to-nat dg + 1) ∧
  PTranscript s6 = map snd fl @ PTranscript s7
  using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
  by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-s7: PTranscript s7 = final # rest-query
  and tr-query: PTranscript query-state = rest-query
  by blast
have header:
  verifier-header-transcript s fr (map snd f-fl) f-final as dg
  (map snd fl) final (PTranscript query-state)
  using prefix-res tr-prefix tr-s5 s6-eq comp-res tr-s7 tr-query
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
from verifier-after-alpha-accepted-transcript-shape[OF prefix suffix]
obtain query-idxs where shape:
  accepted-transcript-shape s (Some (result, final-state)) as query-idxs
  by blast
show ?thesis
  by (rule that[of dg map snd fl final PTranscript query-state query-idxs,
    OF header shape])
qed

```

```

lemma verify-monad-merkle-bound-witnesses:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes bind: initial-merkle-binding-no-bad s
  and outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  shows ∃ trace-table composition-table alphas query-idxs.
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table alphas query-idxs
proof -
  from verify-monad-accepted-transcript-shape[OF outcome]
  obtain alphas query-idxs where
    shape:

```

```

    accepted-transcript-shape s (Some (result, final-state)) alphas query-idxs
  by blast
from initial-merkle-binding-no-badE[OF bind outcome shape]
obtain trace-table composition-table where
  bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table alphas query-idxs
  by blast
show ?thesis
  using bound by blast
qed

```

```

lemma verify-monad-merkle-bound-tables:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes bind: initial-merkle-binding-no-bad s
  and outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  obtains trace-table composition-table alphas query-idxs
  where
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table alphas query-idxs
  and accepted-with-tables s (Some (result, final-state))
    trace-table composition-table alphas query-idxs
  and  $\bigwedge idx. idx \in \text{set query-idxs} \implies$ 
    query-consistent-at trace-table composition-table alphas idx
  and  $\exists fr \text{ composition-root.}$ 
    merkle-root-binds-table fr trace-table final-state  $\wedge$ 
    merkle-root-binds-table composition-root composition-table final-state
proof -
  from verify-monad-merkle-bound-witnesses[OF bind outcome]
  obtain trace-table composition-table alphas query-idxs where
    bound:
      accepted-with-bound-tables s (Some (result, final-state))
      trace-table composition-table alphas query-idxs
    by blast
  have tables:
    accepted-with-tables s (Some (result, final-state))
    trace-table composition-table alphas query-idxs
  by (rule accepted-with-bound-tables-imp-accepted-with-tables[OF bound])
  have query-cons:
     $\bigwedge idx. idx \in \text{set query-idxs} \implies$ 
    query-consistent-at trace-table composition-table alphas idx
  by (rule accepted-with-bound-tables-query-consistency[OF bound])
  have roots:
     $\exists fr \text{ composition-root.}$ 
    merkle-root-binds-table fr trace-table final-state  $\wedge$ 
    merkle-root-binds-table composition-root composition-table final-state
  using accepted-with-bound-tables-bound-roots[OF bound] by blast
  show ?thesis

```

by (rule that[OF bound tables query-cons roots])
qed

lemma *verifier-query-indices-derivedE*:
assumes *verifier-query-indices-derived s out query-start-state rest*
f-fri-roots composition-fri-roots query-idxs
obtains *result final-state raw-idxs query-chunks trailing*
where *out = Some (result, final-state)*
and *length raw-idxs = rounds*
and *query-idxs = map (λraw. index (to-nat raw)) raw-idxs*
and *length query-chunks = rounds*
and *List.concat query-chunks @ trailing = rest*
and $\bigwedge i. i < \text{rounds} \implies$
verifier-query-round-chunk (query-idxs ! i)
f-fri-roots composition-fri-roots (query-chunks ! i)
and $\bigwedge i. i < \text{rounds} \implies$
fmlookup (HashMap final-state)
(QueryIndexChallenge (PQueryCounter s + i)
(state-after-query-chunks query-start-state query-chunks i)) =
Some (raw-idxs ! i)
and $\bigwedge \text{idx}. \text{idx} \in \text{set query-idxs} \implies \text{idx} < \text{clength} * \text{scale}$
using *assms unfolding verifier-query-indices-derived-def* **by** *blast*

lemma *accepted-transcript-shape-header-query-extraction*:
assumes *accepted-transcript-shape s out alphas query-idxs*
obtains *result final-state fr f-fri-roots f-final dg composition-fri-roots final rest*
where *out = Some (result, final-state)*
and *verifier-header-transcript s fr f-fri-roots f-final alphas dg*
composition-fri-roots final rest
and *verifier-query-indices-derived s out*
(verifier-header-state s fr f-fri-roots f-final alphas dg composition-fri-roots final)
rest f-fri-roots composition-fri-roots query-idxs
using *assms unfolding accepted-transcript-shape-def* **by** *blast*

lemma *accepted-transcript-shape-alphas-unique*:
assumes *shape1: accepted-transcript-shape s out alphas query-idxs*
and *shape2: accepted-transcript-shape s out alphas' query-idxs'*
shows *alphas' = alphas*
proof –
from *shape1* **obtain** *result final-state fr f-fri-roots f-final dg*
composition-fri-roots final rest **where**
out1: out = Some (result, final-state)
and *header1:*
verifier-header-transcript s fr f-fri-roots f-final alphas dg
composition-fri-roots final rest
by (*elim accepted-transcript-shape-header-query-extraction*)
from *shape2* **obtain** *result' final-state' fr' f-fri-roots' f-final' dg'*
composition-fri-roots' final' rest' **where**
out2: out = Some (result', final-state')

```

and header2:
  verifier-header-transcript s fr' f-fri-roots' f-final' alphas' dg'
  composition-fri-roots' final' rest'
by (elim accepted-transcript-shape-header-query-extraction)
show ?thesis
  using verifier-header-transcript-unique[OF header1 header2] by simp
qed

lemma accepted-with-bound-tables-alphas-unique:
  assumes bound1:
    accepted-with-bound-tables s out trace-table composition-table
    alphas query-idxs
  and bound2:
    accepted-with-bound-tables s out trace-table' composition-table'
    alphas' query-idxs'
  shows alphas' = alphas
proof -
  have shape1: accepted-transcript-shape s out alphas query-idxs
  using bound1 unfolding accepted-with-bound-tables-def
    accepted-with-tables-def by simp
  have shape2: accepted-transcript-shape s out alphas' query-idxs'
  using bound2 unfolding accepted-with-bound-tables-def
    accepted-with-tables-def by simp
  show ?thesis
  by (rule accepted-transcript-shape-alphas-unique[OF shape1 shape2])
qed

lemma wp-verify-monad-alpha-list-set-bound:
  assumes future: alpha-future-fresh s
  and subset:  $B \subseteq \text{alpha-space}$ 
  shows
    wp-event verify-monad (alpha-list-set-hit s B) s ≤
      nnreal (card B) / nnreal (card alpha-space)
proof -
  let ?Head =
    λout. case out of
      None ⇒ False
      | Some ((-, -, -, as), -) ⇒ as ∈ B
  have prefix-bound:
    wp-event verifier-alpha-prefix ?Head s ≤
      nnreal (card B) / nnreal (card alpha-space)
  by (rule wp-verifier-alpha-prefix-alpha-space-set-bound
    [OF future subset])
  show ?thesis
  unfolding verify-monad-alpha-decomposition
proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
  show alpha-list-set-hit s B None ⇒ ?Head None
  unfolding alpha-list-set-hit-def accepted-transcript-shape-def by blast
next

```

```

fix header prefix-state out
assume prefix:
  Some (header, prefix-state) ∈
    set-dist (execute verifier-alpha-prefix s)
and suffix:
  out ∈
    set-dist (execute (verifier-after-alpha header) prefix-state)
and hit: alpha-list-set-hit s B out
obtain fr f-fl f-final as where header-eq:
  header = (fr, f-fl, f-final, as)
  by (cases header) auto
from hit obtain as' query-idxs where
  shape-hit: accepted-transcript-shape s out as' query-idxs
  and as'-B: as' ∈ B
  unfolding alpha-list-set-hit-def by blast
from shape-hit obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-transcript-shape-def by blast
have prefix':
  Some ((fr, f-fl, f-final, as), prefix-state) ∈
    set-dist (execute verifier-alpha-prefix s)
  using prefix header-eq by simp
have suffix':
  Some (result, final-state) ∈
    set-dist
      (execute (verifier-after-alpha (fr, f-fl, f-final, as))
        prefix-state)
  using suffix header-eq out-eq by simp
from verifier-after-alpha-accepted-transcript-shape[OF prefix' suffix']
obtain query-idxs' where
  shape-prefix:
    accepted-transcript-shape s (Some (result, final-state))
    as query-idxs'
  by blast
have as' = as
  by (rule accepted-transcript-shape-alphas-unique
    [OF shape-prefix shape-hit[unfolded out-eq]])
then show ?Head (Some (header, prefix-state))
  using as'-B header-eq by simp
qed
qed

```

```

lemma wp-verify-monad-alpha-header-list-set-bound:
  fixes C :: prob
  assumes future: alpha-future-fresh s
  and subset:
     $\bigwedge$  fr f-fri-roots f-final.
      B fr f-fri-roots f-final  $\subseteq$  alpha-space
  and bound:

```

```

 $\bigwedge fr \text{ } f\text{-}fri\text{-}roots \text{ } f\text{-}final.$ 
  nnreal (card (B fr f-fri-roots f-final)) /
  nnreal (card alpha-space)  $\leq C$ 
shows
  wp-event verify-monad (alpha-header-list-set-hit s B) s  $\leq C$ 
proof -
  let ?Head =
     $\lambda out.$  case out of
      None  $\Rightarrow$  False
    | Some ((fr, f-fl, f-final, as), -)  $\Rightarrow$ 
      as  $\in B$  fr (map snd f-fl) f-final
  have prefix-bound:
    wp-event verifier-alpha-prefix ?Head s  $\leq C$ 
  by (rule wp-verifier-alpha-prefix-dependent-alpha-space-set-bound
      [OF future])
    (use subset bound in simp-all)
  show ?thesis
    unfolding verify-monad-alpha-decomposition
  proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
    show alpha-header-list-set-hit s B None  $\implies$  ?Head None
      unfolding alpha-header-list-set-hit-def
        accepted-transcript-shape-def by blast
  next
    fix header prefix-state out
    assume prefix:
      Some (header, prefix-state)  $\in$ 
        set-dist (execute verifier-alpha-prefix s)
    and suffix:
      out  $\in$ 
        set-dist (execute (verifier-after-alpha header) prefix-state)
    and hit: alpha-header-list-set-hit s B out
    obtain fr f-fl f-final as where header-eq:
      header = (fr, f-fl, f-final, as)
    by (cases header) auto
    from hit obtain as' query-idxs result final-state fr' f-fri-roots'
      f-final' dg' composition-fri-roots' final' rest' where
      shape-hit:
        accepted-transcript-shape s out as' query-idxs
    and out-eq: out = Some (result, final-state)
    and header-hit:
      verifier-header-transcript s fr' f-fri-roots' f-final' as' dg'
        composition-fri-roots' final' rest'
    and as'-B: as'  $\in B$  fr' f-fri-roots' f-final'
    unfolding alpha-header-list-set-hit-def by blast
    have prefix':
      Some ((fr, f-fl, f-final, as), prefix-state)  $\in$ 
        set-dist (execute verifier-alpha-prefix s)
    using prefix header-eq by simp
    have suffix':

```

Some (result, final-state) ∈
 set-dist
 (execute (verifier-after-alpha (fr, f-fl, f-final, as))
 prefix-state)
 using suffix header-eq out-eq by simp
 from verifier-after-alpha-accepted-transcript-header[OF prefix' suffix']
 obtain dg composition-fri-roots final rest query-idxs' where
 header-prefix:
 verifier-header-transcript s fr (map snd f-fl) f-final as dg
 composition-fri-roots final rest
 and shape-prefix:
 accepted-transcript-shape s (Some (result, final-state))
 as query-idxs'
 by metis
 have unique-header:
 fr' = fr ∧
 f-fri-roots' = map snd f-fl ∧
 f-final' = f-final ∧
 as' = as
 using verifier-header-transcript-unique[OF header-prefix header-hit]
 by simp
 then show ?Head (Some (header, prefix-state))
 using as'-B header-eq by simp
 qed
 qed

lemma wp-alpha-bad-set-hit-bound-via-header-union:
 fixes C :: prob
 assumes future: alpha-future-fresh s
 and subset: $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$
 and union-bound:
 $\bigwedge \text{fr f-fri-roots f-final.}$
 nnreal
 $(\text{card}$
 $(\text{alpha-header-union-bad-sets } s \text{ bad-sets fr f-fri-roots f-final})) /$
 $\text{nnreal } (\text{card } \text{alpha-space}) \leq C$
 shows wp-event verify-monad (alpha-bad-set-hit s bad-sets) s ≤ C
proof –
 have header-bound:
 wp-event verify-monad
 (alpha-header-list-set-hit s
 (alpha-header-union-bad-sets s bad-sets)) s ≤ C
 by (rule wp-verify-monad-alpha-header-list-set-bound[OF future])
 (use alpha-header-union-bad-sets-subset-alpha-space union-bound in
 simp-all)
 have wp-event verify-monad (alpha-bad-set-hit s bad-sets) s ≤
 wp-event verify-monad
 (alpha-header-list-set-hit s
 (alpha-header-union-bad-sets s bad-sets)) s

by (rule wp-event-mono)
 (rule alpha-bad-set-hit-imp-alpha-header-union-bad-set-hit[OF - subset])
 also have ... $\leq C$
 by (rule header-bound)
 finally show ?thesis .
 qed

lemma wp-alpha-bad-set-hit-bound-via-supported-header-union:

fixes $C :: \text{prob}$
 assumes future: alpha-future-fresh s
 and subset: $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$
 and union-bound:
 $\bigwedge \text{fr f-fri-roots f-final.}$
 nnreal
 $(\text{card}$
 $(\text{alpha-header-supported-union-bad-sets } s \text{ bad-sets fr f-fri-roots}$
 $\text{f-final})) /$
 $\text{nnreal } (\text{card } \text{alpha-space}) \leq C$

shows wp-event verify-monad (alpha-bad-set-hit s bad-sets) $s \leq C$

proof –

have header-bound:
 wp-event verify-monad
 (alpha-header-list-set-hit s
 (alpha-header-supported-union-bad-sets s bad-sets)) $s \leq C$
 by (rule wp-verify-monad-alpha-header-list-set-bound[OF future])
 (use alpha-header-supported-union-bad-sets-subset-alpha-space
 union-bound in simp-all)
 have wp-event verify-monad (alpha-bad-set-hit s bad-sets) $s \leq$
 wp-event verify-monad
 (alpha-header-list-set-hit s
 (alpha-header-supported-union-bad-sets s bad-sets)) s
 by (rule wp-event-mono-on-support)
 (rule alpha-bad-set-hit-imp-alpha-header-supported-union-bad-set-hit
 [OF - - subset])
 also have ... $\leq C$
 by (rule header-bound)
 finally show ?thesis .

qed

lemma accepted-with-tables-header-query-extraction:

assumes accepted-with-tables s out trace-table composition-table alphas query-idxs
 obtains result final-state fr f-fri-roots f-final dg composition-fri-roots final rest
 where out = Some (result, final-state)
 and verifier-header-transcript s fr f-fri-roots f-final alphas dg
 composition-fri-roots final rest
 and verifier-query-indices-derived s out
 (verifier-header-state s fr f-fri-roots f-final alphas dg composition-fri-roots final)
 rest f-fri-roots composition-fri-roots query-idxs
 and $\bigwedge \text{idx. idx} \in \text{set query-idxs} \implies$


```

    query-consistent-at trace-table composition-table alphas idx
  and length trace-table = clength * scale
  and length composition-table = clength * scale
  and length alphas = length spec
using assms
unfolding accepted-with-tables-def accepted-transcript-shape-def
by blast

```

lemma *accepted-with-tables-existential-witnesses:*
assumes *accepted-with-tables s out trace-table composition-table alphas query-idxs*
shows $\exists$ *trace-table composition-table as query-idxs.*
accepted-with-tables s out trace-table composition-table as query-idxs
using *assms* **by** *blast*

lemma *accepted-transcript-shape-header-extraction:*
assumes *shape: accepted-transcript-shape s (Some (result, final-state)) alphas query-idxs*
shows $\exists$ *fr f-fri-roots f-final as dg composition-fri-roots final rest.*
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
using *shape* **unfolding** *accepted-transcript-shape-def* **by** *blast*

lemma *accepted-transcript-shape-query-indices-extraction:*
assumes *shape: accepted-transcript-shape s (Some (result, final-state)) alphas query-idxs*
shows $\exists$ *fr f-fri-roots f-final dg composition-fri-roots final rest.*
verifier-query-indices-derived s (Some (result, final-state))
(verifier-header-state s fr f-fri-roots f-final alphas dg
composition-fri-roots final)
rest f-fri-roots composition-fri-roots query-idxs
using *shape* **unfolding** *accepted-transcript-shape-def* **by** *blast*

lemma *accepted-with-tables-has-witnesses:*
assumes *tables: accepted-with-tables s (Some (result, final-state))*
trace-table composition-table alphas query-idxs
shows $\exists$ *trace-table composition-table as query-idxs.*
accepted-with-tables s (Some (result, final-state))
trace-table composition-table as query-idxs
using *tables* **by** *blast*

end

end

theory *Soundness-Execution*
imports *Soundness-Execution-Extraction*
begin

Compatibility umbrella for verifier execution soundness lemmas.

end

```

theory Soundness-Bad-Events
  imports
    Soundness-Execution
begin

```

```

context soundness
begin

```

At this stage *accepted-with-tables* is the boundary between the verifier execution and the algebraic proof. It records the trace table, composition table, and verifier challenges associated with an accepted execution. It now also records that the initial verifier transcript has the expected header shape and that the sampled query positions are locally consistent with the associated tables. Merkle binding is separated into *accepted-with-bound-tables*, because malicious binding requires additional Merkle/collision-resistance reasoning.

definition *composition-bad-context*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$ 
  'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  nat list  $\Rightarrow$ 
  'f poly  $\Rightarrow$  bool

```

where

```

composition-bad-context s out trace-table composition-table as query-idxs f  $\longleftrightarrow$ 
  accepted-with-bound-tables s out trace-table composition-table as query-idxs f  $\wedge$ 
  degree f < clength  $\wedge$ 
  trace-table = map (poly f) eval-domain  $\wedge$ 
  violated-constraints f  $\neq$  {}  $\wedge$ 
  composition-table-low-degree maxDegree composition-table  $\wedge$ 
  all-queries-consistent trace-table composition-table as

```

definition *composition-degree-bad*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool

```

where

```

composition-degree-bad s out  $\longleftrightarrow$ 
  ( $\exists$  trace-table composition-table as query-idxs f.
    composition-bad-context s out trace-table composition-table as query-idxs f  $\wedge$ 
     $\neg$  common-denominator-degree-bounds f as)

```

definition *composition-randomization-bad*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool

```

where

```

composition-randomization-bad s out  $\longleftrightarrow$ 
  ( $\exists$  trace-table composition-table as query-idxs f.
    composition-bad-context s out trace-table composition-table as query-idxs f  $\wedge$ 
    common-denominator-degree-bounds f as  $\wedge$ 
    random-combination-common-denominator-hides-violations f as)

```

definition *composition-bad*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{composition-bad } s \text{ out} \longleftrightarrow$
 $\text{composition-degree-bad } s \text{ out} \vee \text{composition-randomization-bad } s \text{ out}$

lemma *composition-bad-has-common-denominator-witness:*

assumes *composition-bad* $s \text{ out}$
shows $\exists \text{ trace-table composition-table as query-idxs } f.$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs} \wedge$
 $\text{degree } f < \text{clength} \wedge$
 $\text{trace-table} = \text{map } (\text{poly } f) \text{ eval-domain} \wedge$
 $\text{violated-constraints } f \neq \{\}$ $\wedge$
 $\text{composition-table-low-degree } \text{maxDegree } \text{composition-table} \wedge$
 $\text{all-queries-consistent trace-table composition-table as} \wedge$
 $(\neg \text{common-denominator-degree-bounds } f \text{ as} \vee$
 $\text{random-combination-common-denominator-hides-violations } f \text{ as})$
using *assms*
unfolding *composition-bad-def composition-degree-bad-def*
 $\text{composition-randomization-bad-def composition-bad-context-def}$
by *blast*

lemma *composition-bad-imp-spec-nonempty:*

assumes *composition-bad* $s \text{ out}$
shows $\text{spec} \neq []$
proof –
obtain *trace-table composition-table as query-idxs* f
where $\text{violated-constraints } f \neq \{\}$
using *composition-bad-has-common-denominator-witness* $[OF \text{ assms}]$ **by** *blast*
then obtain *entry* **where** $\text{entry} \in \text{violated-constraints } f$
by *auto*
then have $\text{entry} \in \text{set spec}$
unfolding *violated-constraints-def* **by** *simp*
then show *?thesis*
by *auto*
qed

definition *trace-fri-bad*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-bad } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-table composition-table as query-idxs.}$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs} \wedge$
 $\neg \text{trace-table-low-degree trace-table})$

definition *composition-fri-bad*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{composition-fri-bad } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-table composition-table as query-idxs.}$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\neg \text{composition-table-low-degree maxDegree composition-table})$

definition *query-bad*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{query-bad } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-table composition-table as query-idxs.}$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\text{composition-table-low-degree maxDegree composition-table} \wedge$
 $\neg \text{all-queries-consistent trace-table composition-table as})$

definition *soundness-bad-event*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{soundness-bad-event } s \text{ out} \longleftrightarrow$
 $\text{composition-bad } s \text{ out} \vee$
 $\text{trace-fri-bad } s \text{ out} \vee$
 $\text{composition-fri-bad } s \text{ out} \vee$
 $\text{query-bad } s \text{ out}$

definition *query-sampling-success-space*
 $:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat set}$
where
 $\text{query-sampling-success-space trace-table composition-table as} =$
 $(\text{if trace-table-low-degree trace-table} \wedge$
 $\text{composition-table-low-degree maxDegree composition-table} \wedge$
 $\neg \text{all-queries-consistent trace-table composition-table as}$
 $\text{then query-agreement-indices trace-table composition-table as}$
 $\text{else } \{\})$

definition *query-index-set-hit*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{query-index-set-hit } s \text{ good-sets out} \longleftrightarrow$
 $(\exists \text{ trace-table composition-table as query-idxs.}$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs} \wedge$
 $\text{set query-idxs} \subseteq \text{good-sets trace-table composition-table as})$

definition *query-index-round-set-hit-at*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat set}) \Rightarrow$
 $\text{nat} \Rightarrow (\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{query-index-round-set-hit-at } s \text{ good-sets } i \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-table composition-table as query-idxs.}$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as}$
 $\text{query-idxs} \wedge$
 $i < \text{rounds} \wedge$
 $\text{query-idxs} ! i \in \text{good-sets trace-table composition-table as})$

definition *query-index-round-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{query-index-round-set-hit } s \text{ good-sets out} \longleftrightarrow$
 $(\exists i < \text{rounds. query-index-round-set-hit-at } s \text{ good-sets } i \text{ out})$

definition *query-round-index-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{nat set} \Rightarrow$
 $(\text{unit} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{query-round-index-set-hit } s \text{ B out} \longleftrightarrow$
 $(\exists \text{ raw } t.$
 $\text{out} = \text{Some } ((), t) \wedge$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) = \text{Some raw} \wedge$
 $\text{index } (\text{to-nat raw}) \in B)$

definition *query-round-any-index-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{nat set} \Rightarrow$
 $(\text{unit} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{query-round-any-index-set-hit } s \text{ B out} \longleftrightarrow$
 $(\exists \text{ raw } t \text{ x.}$
 $\text{out} = \text{Some } ((), t) \wedge$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } s) x) = \text{Some raw} \wedge$
 $\text{index } (\text{to-nat raw}) \in B)$

definition *query-rounds-any-index-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{nat set} \Rightarrow \text{nat} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{query-rounds-any-index-set-hit } s \text{ B n out} \longleftrightarrow$
 $(\exists \text{ results } t \text{ i x raw.}$
 $\text{out} = \text{Some } (\text{results}, t) \wedge$

$i < n \wedge$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } (\text{PQueryCounter } s + i) \ x) = \text{Some } \text{raw} \wedge$
 $\text{index } (\text{to-nat } \text{raw}) \in B$

lemma *accepted-with-bound-tables-query-rounds-any-index-set-hit:*

assumes *bound:*

accepted-with-bound-tables s *out* *trace-table composition-table* *as*
query-idxs

and *i-bound:* $i < \text{rounds}$

and *hit:* $\text{query-idxs} \ ! \ i \in B$

shows *query-rounds-any-index-set-hit* $s \ B$ *rounds out*

proof –

have *shape:* *accepted-transcript-shape* s *out as* *query-idxs*

using *accepted-with-bound-tables-imp-accepted-with-tables*[*OF bound*]

accepted-with-tables-imp-accepted-transcript-shape **by** *blast*

then obtain *result final-state* *fr* *f-fri-roots f-final dg*

composition-fri-roots final rest raw-idxs query-chunks trailing **where**

out-eq: $\text{out} = \text{Some } (\text{result}, \text{final-state})$

and *len-raw:* $\text{length } \text{raw-idxs} = \text{rounds}$

and *query-idxs-eq:* $\text{query-idxs} = \text{map } (\lambda \text{raw. index } (\text{to-nat } \text{raw})) \ \text{raw-idxs}$

and *lookup:*

$\forall i < \text{rounds.}$

$\text{fmlookup } (\text{HashMap } \text{final-state})$

$(\text{QueryIndexChallenge } (\text{PQueryCounter } s + i)$

$(\text{state-after-query-chunks}$

$(\text{verifier-header-state } s \ \text{fr } \text{f-fri-roots } \text{f-final } \text{as } \text{dg}$

$\text{composition-fri-roots } \text{final})$

$\text{query-chunks } i)) =$

$\text{Some } (\text{raw-idxs} \ ! \ i)$

unfolding *accepted-transcript-shape-def verifier-query-indices-derived-def*

by *blast*

have *raw-hit:* $\text{index } (\text{to-nat } (\text{raw-idxs} \ ! \ i)) \in B$

using *hit query-idxs-eq len-raw i-bound* **by** *simp*

have *lookup-i:*

$\text{fmlookup } (\text{HashMap } \text{final-state})$

$(\text{QueryIndexChallenge } (\text{PQueryCounter } s + i)$

$(\text{state-after-query-chunks}$

$(\text{verifier-header-state } s \ \text{fr } \text{f-fri-roots } \text{f-final } \text{as } \text{dg}$

$\text{composition-fri-roots } \text{final})$

$\text{query-chunks } i)) =$

$\text{Some } (\text{raw-idxs} \ ! \ i)$

using *lookup i-bound* **by** *simp*

show *?thesis*

unfolding *query-rounds-any-index-set-hit-def*

apply $(\text{rule } \text{exI}[\text{where } x=\text{result}])$

apply $(\text{rule } \text{exI}[\text{where } x=\text{final-state}])$

apply $(\text{rule } \text{exI}[\text{where } x=i])$

apply $(\text{rule } \text{exI}[$

where $x = \text{state-after-query-chunks}$
 (verifier-header-state s fr f -fri-roots f -final as dg
 composition-fri-roots final)
 query-chunks i)
apply (rule exI [**where** $x = \text{raw-idxs} ! i$])
using out-eq i -bound lookup- i raw-hit **by** simp
qed

lemma query-index-round-set-hit-at-imp-query-rounds-any-index-set-hit:
assumes hit: query-index-round-set-hit-at s good-sets i out
shows
 $\exists$ trace-table composition-table as.
 query-rounds-any-index-set-hit s
 (good-sets trace-table composition-table as) rounds out
proof –
from hit **obtain** trace-table composition-table as query-idxs **where**
 bound:
 accepted-with-bound-tables s out trace-table composition-table as
 query-idxs
and i -bound: $i < \text{rounds}$
and query-hit:
 query-idxs $! i \in \text{good-sets trace-table composition-table as}$
unfolding query-index-round-set-hit-at-def **by** blast
have round-hit:
 query-rounds-any-index-set-hit s
 (good-sets trace-table composition-table as) rounds out
by (rule accepted-with-bound-tables-query-rounds-any-index-set-hit
 [OF bound i -bound query-hit])
show ?thesis
using round-hit **by** blast
qed

definition query-header-supported-table-candidates
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow ('f \text{ list} \times 'f \text{ list}) \text{ set}$
where
 query-header-supported-table-candidates s fr f -fri-roots f -final as dg
 composition-fri-roots final =
 $\{(\text{trace-table}, \text{composition-table}).$
 $\exists \text{ out query-idxs rest.}$
 $\text{out} \in \text{set-dist (execute verify-monad } s) \wedge$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as}$
 $\text{query-idxs} \wedge$
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final as } dg$
 $\text{composition-fri-roots final rest} \}$

definition query-header-supported-table-candidate-state
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

query-header-supported-table-candidate-state *s* *fr* *f-fri-roots* *f-final*
as *dg* *composition-fri-roots* *final* *trace-table* *composition-table*
final-state $\longleftrightarrow$
 $(\exists \text{ result } \text{query-idxs } \text{rest}.$
 $\text{Some } (\text{result}, \text{final-state}) \in \text{set-dist } (\text{execute } \text{verify-monad } s) \wedge$
 $\text{accepted-with-bound-tables } s (\text{Some } (\text{result}, \text{final-state}))$
 $\text{trace-table } \text{composition-table } \text{as } \text{query-idxs} \wedge$
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final } \text{as } dg$
 $\text{composition-fri-roots } \text{final } \text{rest} \wedge$
 $\text{merkle-root-binds-table } \text{fr } \text{trace-table } \text{final-state} \wedge$
 $\text{composition-fri-roots} \neq [] \wedge$
 $\text{merkle-root-binds-table } (\text{hd } \text{composition-fri-roots}) \text{ composition-table}$
 $\text{final-state})$

lemma *query-header-supported-table-candidates-iff-state:*

$(\text{trace-table}, \text{composition-table}) \in$
 $\text{query-header-supported-table-candidates } s \text{ fr } f\text{-fri-roots } f\text{-final } \text{as } dg$
 $\text{composition-fri-roots } \text{final} \longleftrightarrow$
 $(\exists \text{ final-state}.$
 $\text{query-header-supported-table-candidate-state } s \text{ fr } f\text{-fri-roots } f\text{-final}$
 $\text{as } dg \text{ composition-fri-roots } \text{final } \text{trace-table } \text{composition-table}$
 $\text{final-state})$

proof

assume *candidate:*

$(\text{trace-table}, \text{composition-table}) \in$
 $\text{query-header-supported-table-candidates } s \text{ fr } f\text{-fri-roots } f\text{-final } \text{as } dg$
 $\text{composition-fri-roots } \text{final}$

then obtain *out* *query-idxs* *rest* **where**

outcome: *out* $\in \text{set-dist } (\text{execute } \text{verify-monad } s)$

and *bound:*

$\text{accepted-with-bound-tables } s \text{ out } \text{trace-table } \text{composition-table } \text{as}$
 query-idxs

and *header:*

$\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final } \text{as } dg$
 $\text{composition-fri-roots } \text{final } \text{rest}$

unfolding *query-header-supported-table-candidates-def* **by** *blast*

from *bound* **obtain** *result* *final-state* *fr'* *f-fri-roots'* *f-final'* *dg'*

composition-fri-roots' *final'* *rest'* **where**

out-eq: *out* = $\text{Some } (\text{result}, \text{final-state})$

and *header':*

$\text{verifier-header-transcript } s \text{ fr' } f\text{-fri-roots' } f\text{-final' } \text{as } dg'$
 $\text{composition-fri-roots' } \text{final' } \text{rest'}$

and *trace-bind:*

$\text{merkle-root-binds-table } \text{fr' } \text{trace-table } \text{final-state}$

and *comp-nonempty:* $\text{composition-fri-roots'} \neq []$


```

and comp-bind:
  merkle-root-binds-table (hd composition-fri-roots')
  composition-table final-state
unfolding accepted-with-bound-tables-def by blast
have eqs:
  fr' = fr ∧
  f-fri-roots' = f-fri-roots ∧
  f-final' = f-final ∧
  dg' = dg ∧
  composition-fri-roots' = composition-fri-roots ∧
  final' = final ∧
  rest' = rest
using verifier-header-transcript-unique[OF header' header] by simp
have state:
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table composition-table
  final-state
unfolding query-header-supported-table-candidate-state-def
by (intro exI[of - result] exI[of - query-idxs] exI[of - rest])
  (use outcome out-eq bound header trace-bind comp-nonempty comp-bind eqs
   in simp)
then show
  ∃ final-state.
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table composition-table
  final-state
by blast
next
assume
  ∃ final-state.
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table composition-table
  final-state
then obtain final-state result query-idxs rest where
  outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
and bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
unfolding query-header-supported-table-candidate-state-def by blast
show (trace-table, composition-table) ∈
  query-header-supported-table-candidates s fr f-fri-roots f-final as dg
  composition-fri-roots final
unfolding query-header-supported-table-candidates-def
using outcome bound header by blast
qed

```

lemma *query-header-supported-table-candidate-state-hash-extends:*
assumes
query-header-supported-table-candidate-state s fr f-fri-roots f-final
as dg composition-fri-roots final trace-table composition-table
final-state
shows $s \leq \text{final-state}$
proof –
from *assms* **obtain** *result query-idxs rest* **where**
outcome:
Some (result, final-state) ∈ set-dist (execute verify-monad s)
unfolding *query-header-supported-table-candidate-state-def* **by** *blast*
show *?thesis*
by *(rule verify-monad-hash-extends[OF outcome])*
qed

lemma *query-header-supported-table-candidate-states-merkle-compatible-unique-if-clean-merge:*
assumes *compatible:*
merkle-hash-maps-compatible final-state final-state'
and *clean:*
 $\neg \text{hash-map-output-collision}$
(merkle-hash-state-merge final-state final-state')
and *candidate:*
query-header-supported-table-candidate-state s fr f-fri-roots f-final
as dg composition-fri-roots final trace-table composition-table
final-state
and *candidate':*
query-header-supported-table-candidate-state s fr f-fri-roots f-final
as dg composition-fri-roots final trace-table' composition-table'
final-state'
shows
 $\text{trace-table} = \text{trace-table}' \wedge$
 $\text{composition-table} = \text{composition-table}'$
proof –
from *candidate* **obtain** *result query-idxs rest* **where**
bound:
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs
and *trace-bind:*
merkle-root-binds-table fr trace-table final-state
and *comp-nonempty:* $\text{composition-fri-roots} \neq []$
and *comp-bind:*
merkle-root-binds-table (hd composition-fri-roots) composition-table
final-state
unfolding *query-header-supported-table-candidate-state-def* **by** *blast*
from *candidate'* **obtain** *result' query-idxs' rest'* **where**
bound':
accepted-with-bound-tables s (Some (result', final-state'))
trace-table' composition-table' as query-idxs'

```

and trace-bind':
  merkle-root-binds-table fr trace-table' final-state'
and comp-bind':
  merkle-root-binds-table (hd composition-fri-roots) composition-table'
    final-state'
unfolding query-header-supported-table-candidate-state-def by blast
have same-trace-len: length trace-table = length trace-table'
using accepted-with-bound-tables-shapes(1)[OF bound]
  accepted-with-bound-tables-shapes(1)[OF bound']
by simp
have trace-nonempty: trace-table ≠ []
using accepted-with-bound-tables-shapes(1)[OF bound]
  eval-domain-nontrivial by auto
have trace-eq: trace-table = trace-table'
by (rule
  merkle-root-binds-same-length-tables-merkle-compatible-unique-if-clean-merge
  [OF compatible clean trace-bind trace-bind' same-trace-len
    trace-nonempty'])
have same-comp-len: length composition-table = length composition-table'
using accepted-with-bound-tables-shapes(2)[OF bound]
  accepted-with-bound-tables-shapes(2)[OF bound']
by simp
have comp-nonempty': composition-table ≠ []
using accepted-with-bound-tables-shapes(2)[OF bound]
  eval-domain-nontrivial by auto
have comp-eq: composition-table = composition-table'
by (rule
  merkle-root-binds-same-length-tables-merkle-compatible-unique-if-clean-merge
  [OF compatible clean comp-bind comp-bind' same-comp-len
    comp-nonempty'])
show ?thesis
using trace-eq comp-eq by simp
qed

```

lemma *query-header-supported-table-candidates-subset-singleton-if-pairwise-merkle-clean-merge*:
assumes *pairwise*:

```

  ∧ trace-table composition-table final-state trace-table'
    composition-table' final-state'.
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table composition-table
    final-state ⇒
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table' composition-table'
    final-state' ⇒
  merkle-hash-maps-compatible final-state final-state' ∧
  ¬ hash-map-output-collision
  (merkle-hash-state-merge final-state final-state')

```

shows

∃ *trace-table composition-table*.

```

    query-header-supported-table-candidates s fr f-fri-roots f-final as dg
    composition-fri-roots final  $\subseteq$ 
    {(trace-table, composition-table)}
proof (cases
    query-header-supported-table-candidates s fr f-fri-roots f-final as dg
    composition-fri-roots final = {})
case True
then show ?thesis
    by (intro exI[of -] exI[of -]) simp
next
case False
then obtain table-pair0 where table-pair0:
    table-pair0  $\in$ 
    query-header-supported-table-candidates s fr f-fri-roots f-final as dg
    composition-fri-roots final
by blast
obtain trace-table0 composition-table0 where table-pair0-eq:
    table-pair0 = (trace-table0, composition-table0)
by (cases table-pair0)
have table0:
    (trace-table0, composition-table0)  $\in$ 
    query-header-supported-table-candidates s fr f-fri-roots f-final as dg
    composition-fri-roots final
using table-pair0 unfolding table-pair0-eq .
then obtain final-state0 where state0:
    query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table0 composition-table0
    final-state0
using query-header-supported-table-candidates-iff-state by blast
have subset:
    query-header-supported-table-candidates s fr f-fri-roots f-final as dg
    composition-fri-roots final  $\subseteq$ 
    {(trace-table0, composition-table0)}
proof
fix table-pair
assume pair-in:
    table-pair  $\in$ 
    query-header-supported-table-candidates s fr f-fri-roots f-final as
    dg composition-fri-roots final
obtain trace-table composition-table where pair-eq:
    table-pair = (trace-table, composition-table)
by (cases table-pair)
from pair-in obtain final-state where state:
    query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table composition-table
    final-state
unfolding pair-eq
using query-header-supported-table-candidates-iff-state by blast
have compatible:

```

```

merkle-hash-maps-compatible final-state final-state0
using pairwise[OF state state0] by simp
have clean:
  ¬ hash-map-output-collision
    (merkle-hash-state-merge final-state final-state0)
using pairwise[OF state state0] by simp
have eqs:
  trace-table = trace-table0 ∧
  composition-table = composition-table0
by (rule
  query-header-supported-table-candidate-states-merkle-compatible-unique-if-clean-merge
  [OF compatible clean state state0])
show table-pair ∈ {(trace-table0, composition-table0)}
using pair-eq eqs by simp
qed
then show ?thesis
by blast
qed

lemma query-header-supported-table-candidates-subset-singleton-if-pairwise-no-merkle-value-conflict-clean-merge:
assumes pairwise:
  ∧ trace-table composition-table final-state trace-table'
    composition-table' final-state'.
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table composition-table
  final-state ⇒
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table' composition-table'
  final-state' ⇒
  ¬ merkle-hash-value-conflict final-state final-state' ∧
  ¬ hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
shows
  ∃ trace-table composition-table.
  query-header-supported-table-candidates s fr f-fri-roots f-final as dg
  composition-fri-roots final ⊆
  {(trace-table, composition-table)}
proof (rule
  query-header-supported-table-candidates-subset-singleton-if-pairwise-merkle-clean-merge)
fix trace-table composition-table final-state trace-table'
  composition-table' final-state'
assume candidate:
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table composition-table
  final-state
and candidate':
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table' composition-table'
  final-state'

```

```

have no-conflict:
   $\neg$  merkle-hash-value-conflict final-state final-state'
  using pairwise[OF candidate candidate'] by simp
have compatible:
  merkle-hash-maps-compatible final-state final-state'
  using no-conflict
  unfolding merkle-hash-maps-compatible-iff-no-value-conflict .
have clean:
   $\neg$  hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
  using pairwise[OF candidate candidate'] by simp
show merkle-hash-maps-compatible final-state final-state'  $\wedge$ 
   $\neg$  hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
  using compatible clean by simp
qed

lemma query-header-supported-table-candidates-subset-singleton-if-common-clean-merkle-extension:
assumes clean:  $\neg$  hash-map-output-collision u
and common-ext:
   $\bigwedge$  trace-table composition-table final-state.
    query-header-supported-table-candidate-state s fr f-fri-roots f-final
      as dg composition-fri-roots final trace-table composition-table
        final-state  $\implies$ 
          merkle-hash-extends final-state u
shows
   $\exists$  trace-table composition-table.
    query-header-supported-table-candidates s fr f-fri-roots f-final as dg
      composition-fri-roots final  $\subseteq$ 
         $\{(trace-table, composition-table)\}$ 
proof (rule
  query-header-supported-table-candidates-subset-singleton-if-pairwise-no-merkle-value-conflict-clean-merge)
fix trace-table composition-table final-state trace-table'
  composition-table' final-state'
assume candidate:
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table composition-table
      final-state
and candidate':
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table' composition-table'
      final-state'
have ext: merkle-hash-extends final-state u
  by (rule common-ext[OF candidate])
have ext': merkle-hash-extends final-state' u
  by (rule common-ext[OF candidate'])
show  $\neg$  merkle-hash-value-conflict final-state final-state'  $\wedge$ 
   $\neg$  hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')

```

by (rule common-clean-merkle-extension-imp-pairwise-merkle-clean
 [OF ext ext' clean])

qed

lemma *query-header-supported-table-candidate-states-distinct-imp-merkle-conflict-or-merge-collision*:

assumes *candidate*:

query-header-supported-table-candidate-state s fr f-fri-roots f-final
 as dg composition-fri-roots final trace-table composition-table
 final-state

and *candidate'*:

query-header-supported-table-candidate-state s fr f-fri-roots f-final
 as dg composition-fri-roots final trace-table' composition-table'
 final-state'

and *distinct-tables*:

trace-table $\neq$ *trace-table'* $\vee$
 composition-table $\neq$ *composition-table'*

shows

merkle-hash-value-conflict final-state final-state' $\vee$
 hash-map-output-collision
 (*merkle-hash-state-merge final-state final-state'*)

proof (rule ccontr)

assume *no-bad*:

$\neg$ (*merkle-hash-value-conflict final-state final-state'* $\vee$
 hash-map-output-collision
 (*merkle-hash-state-merge final-state final-state'*))

have *compatible*:

merkle-hash-maps-compatible final-state final-state'

using *no-bad* **unfolding** *merkle-hash-maps-compatible-iff-no-value-conflict*

by *simp*

have *clean*:

$\neg$ *hash-map-output-collision*
 (*merkle-hash-state-merge final-state final-state'*)

using *no-bad* **by** *simp*

have *eqs*:

trace-table = *trace-table'* $\wedge$
 composition-table = *composition-table'*

by (rule
 query-header-supported-table-candidate-states-merkle-compatible-unique-if-clean-merge
 [OF *compatible clean candidate candidate'*])

then show *False*

using *distinct-tables* **by** *simp*

qed

definition *query-header-supported-pairwise-merkle-bad*

$::$ (*f*, *a*) *protocol-channel-scheme* $\Rightarrow$
 f $\Rightarrow$ *f list* $\Rightarrow$ *f* $\Rightarrow$ *f list* $\Rightarrow$ *f* $\Rightarrow$
 f list $\Rightarrow$ *f* $\Rightarrow$ *bool*

where

query-header-supported-pairwise-merkle-bad s fr f-fri-roots f-final as

$dg \text{ composition-fri-roots final} \longleftrightarrow$
 $(\exists \text{ trace-table composition-table final-state trace-table'}$
 $\quad \text{composition-table' final-state'.$
 $\text{query-header-supported-table-candidate-state } s \text{ fr f-fri-roots f-final}$
 $\quad \text{as } dg \text{ composition-fri-roots final trace-table composition-table}$
 $\quad \text{final-state} \wedge$
 $\text{query-header-supported-table-candidate-state } s \text{ fr f-fri-roots f-final}$
 $\quad \text{as } dg \text{ composition-fri-roots final trace-table' composition-table'}$
 $\quad \text{final-state' } \wedge$
 $(\text{trace-table} \neq \text{trace-table'} \vee$
 $\quad \text{composition-table} \neq \text{composition-table'}) \wedge$
 $(\text{merkle-hash-value-conflict final-state final-state'} \vee$
 $\quad \text{hash-map-output-collision}$
 $\quad (\text{merkle-hash-state-merge final-state final-state'})))$

definition *query-supported-pairwise-merkle-bad*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{query-supported-pairwise-merkle-bad } s \longleftrightarrow$
 $(\exists \text{ fr f-fri-roots f-final as } dg \text{ composition-fri-roots final.}$
 $\quad \text{query-header-supported-pairwise-merkle-bad } s \text{ fr f-fri-roots f-final}$
 $\quad \text{as } dg \text{ composition-fri-roots final})$

lemma *query-header-supported-table-candidates-not-singleton-imp-pairwise-merkle-bad:*

assumes *not-unique:*

$\neg (\exists \text{ trace-table composition-table.}$
 $\quad \text{query-header-supported-table-candidates } s \text{ fr f-fri-roots f-final as } dg$
 $\quad \text{composition-fri-roots final} \subseteq$
 $\quad \{(\text{trace-table}, \text{composition-table} \})$

shows

$\exists \text{ trace-table composition-table final-state trace-table'}$
 $\quad \text{composition-table' final-state'.$
 $\text{query-header-supported-table-candidate-state } s \text{ fr f-fri-roots f-final}$
 $\quad \text{as } dg \text{ composition-fri-roots final trace-table composition-table}$
 $\quad \text{final-state} \wedge$
 $\text{query-header-supported-table-candidate-state } s \text{ fr f-fri-roots f-final}$
 $\quad \text{as } dg \text{ composition-fri-roots final trace-table' composition-table'}$
 $\quad \text{final-state' } \wedge$
 $(\text{trace-table} \neq \text{trace-table'} \vee$
 $\quad \text{composition-table} \neq \text{composition-table'}) \wedge$
 $(\text{merkle-hash-value-conflict final-state final-state'} \vee$
 $\quad \text{hash-map-output-collision}$
 $\quad (\text{merkle-hash-state-merge final-state final-state'}))$

proof –

let $?C =$

$\text{query-header-supported-table-candidates } s \text{ fr f-fri-roots f-final as } dg$
 $\quad \text{composition-fri-roots final}$

have *nonempty:* $?C \neq \{\}$

using *not-unique* **by** *auto*


```

then obtain table-pair where table-pair:
  table-pair  $\in ?C$ 
  by blast
obtain trace-table composition-table where table-pair-eq:
  table-pair = (trace-table, composition-table)
  by (cases table-pair)
have table:
  (trace-table, composition-table)  $\in ?C$ 
  using table-pair unfolding table-pair-eq .
have  $\neg ?C \subseteq \{(trace-table, composition-table)\}$ 
  using not-unique by blast
then obtain table-pair' where table-pair':
  table-pair'  $\in ?C$ 
  and table-pair'-not:
  table-pair'  $\notin \{(trace-table, composition-table)\}$ 
  by auto
obtain trace-table' composition-table' where table-pair'-eq:
  table-pair' = (trace-table', composition-table')
  by (cases table-pair')
have table':
  (trace-table', composition-table')  $\in ?C$ 
  using table-pair' unfolding table-pair'-eq .
have distinct-tables:
  trace-table  $\neq trace-table'$   $\vee$ 
  composition-table  $\neq composition-table'$ 
  using table-pair'-not unfolding table-pair'-eq by auto
from table obtain final-state where candidate:
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table composition-table
  final-state
  using query-header-supported-table-candidates-iff-state by blast
from table' obtain final-state' where candidate':
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table' composition-table'
  final-state'
  using query-header-supported-table-candidates-iff-state by blast
have bad:
  merkle-hash-value-conflict final-state final-state'  $\vee$ 
  hash-map-output-collision
  (merkle-hash-state-merge final-state final-state')
  by (rule
  query-header-supported-table-candidate-states-distinct-imp-merkle-conflict-or-merge-collision
  [OF candidate candidate' distinct-tables])
show ?thesis
  using candidate candidate' distinct-tables bad by blast
qed

lemma query-header-supported-pairwise-merkle-bad-extends-initial:
  assumes bad:

```

query-header-supported-pairwise-merkle-bad s *fr* f -*fri-roots* f -*final* *as*
dg *composition-fri-roots* *final*
obtains *trace-table* *composition-table* *final-state* *trace-table'*
composition-table' *final-state'*
where
query-header-supported-table-candidate-state s *fr* f -*fri-roots* f -*final*
as *dg* *composition-fri-roots* *final* *trace-table* *composition-table*
final-state
query-header-supported-table-candidate-state s *fr* f -*fri-roots* f -*final*
as *dg* *composition-fri-roots* *final* *trace-table'* *composition-table'*
final-state'
trace-table $\neq$ *trace-table'* $\vee$
composition-table $\neq$ *composition-table'*
merkle-hash-value-conflict *final-state* *final-state'* $\vee$
hash-map-output-collision
(*merkle-hash-state-merge* *final-state* *final-state'*)
 $s \leq$ *final-state*
 $s \leq$ *final-state'*
proof –
from *bad* **obtain** *trace-table* *composition-table* *final-state* *trace-table'*
composition-table' *final-state'* **where**
candidate:
query-header-supported-table-candidate-state s *fr* f -*fri-roots* f -*final*
as *dg* *composition-fri-roots* *final* *trace-table* *composition-table*
final-state
and *candidate'*:
query-header-supported-table-candidate-state s *fr* f -*fri-roots* f -*final*
as *dg* *composition-fri-roots* *final* *trace-table'* *composition-table'*
final-state'
and *distinct*:
trace-table $\neq$ *trace-table'* $\vee$
composition-table $\neq$ *composition-table'*
and *pair-bad*:
merkle-hash-value-conflict *final-state* *final-state'* $\vee$
hash-map-output-collision
(*merkle-hash-state-merge* *final-state* *final-state'*)
unfolding *query-header-supported-pairwise-merkle-bad-def* **by** *blast*
have *ext*: $s \leq$ *final-state*
by (*rule* *query-header-supported-table-candidate-state-hash-extends*
[*OF candidate*])
have *ext'*: $s \leq$ *final-state'*
by (*rule* *query-header-supported-table-candidate-state-hash-extends*
[*OF candidate'*])
show *?thesis*
by (*rule* *that*[*OF candidate candidate'* *distinct pair-bad ext ext'*])
qed

definition *query-header-supported-pairwise-coupling-bad*
 $:: (f, 'a)$ *protocol-channel-scheme* $\Rightarrow$

'f $\Rightarrow$ 'f list $\Rightarrow$ 'f $\Rightarrow$ 'f list $\Rightarrow$ 'f $\Rightarrow$
 'f list $\Rightarrow$ 'f $\Rightarrow$ bool

where

query-header-supported-pairwise-coupling-bad s fr f-fri-roots f-final as
 dg composition-fri-roots final $\longleftrightarrow$
 ($\exists$ trace-table composition-table final-state trace-table'
 composition-table' final-state'.
 query-header-supported-table-candidate-state s fr f-fri-roots f-final
 as dg composition-fri-roots final trace-table composition-table
 final-state $\wedge$
 query-header-supported-table-candidate-state s fr f-fri-roots f-final
 as dg composition-fri-roots final trace-table' composition-table'
 final-state' $\wedge$
 (trace-table $\neq$ trace-table' $\vee$
 composition-table $\neq$ composition-table') $\wedge$
 merkle-hash-pairwise-coupling-bad final-state final-state')

definition query-supported-pairwise-coupling-bad

:: ('f, 'a) protocol-channel-scheme $\Rightarrow$ bool

where

query-supported-pairwise-coupling-bad s $\longleftrightarrow$
 ($\exists$ fr f-fri-roots f-final as dg composition-fri-roots final.
 query-header-supported-pairwise-coupling-bad s fr f-fri-roots f-final
 as dg composition-fri-roots final)

lemma query-header-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad:

assumes bad:

query-header-supported-pairwise-merkle-bad s fr f-fri-roots f-final as
 dg composition-fri-roots final

shows

($\exists$ trace-table composition-table final-state trace-table'
 composition-table' final-state'.
 query-header-supported-table-candidate-state s fr f-fri-roots f-final
 as dg composition-fri-roots final trace-table composition-table
 final-state $\wedge$
 query-header-supported-table-candidate-state s fr f-fri-roots f-final
 as dg composition-fri-roots final trace-table' composition-table'
 final-state' $\wedge$
 (trace-table $\neq$ trace-table' $\vee$
 composition-table $\neq$ composition-table') $\wedge$
 (hash-map-output-collision final-state $\vee$
 hash-map-output-collision final-state')) $\vee$
 query-header-supported-pairwise-coupling-bad s fr f-fri-roots f-final as
 dg composition-fri-roots final

proof —

from bad **obtain** trace-table composition-table final-state trace-table'
 composition-table' final-state'

where candidate:

query-header-supported-table-candidate-state s fr f-fri-roots f-final

```

    as dg composition-fri-roots final trace-table composition-table
    final-state
  and candidate':
    query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table' composition-table'
    final-state'
  and distinct:
    trace-table  $\neq$  trace-table'  $\vee$ 
    composition-table  $\neq$  composition-table'
  and pair-bad:
    merkle-hash-value-conflict final-state final-state'  $\vee$ 
    hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
  unfolding query-header-supported-pairwise-merkle-bad-def by blast
have decomp:
  hash-map-output-collision final-state  $\vee$ 
  hash-map-output-collision final-state'  $\vee$ 
  merkle-hash-pairwise-coupling-bad final-state final-state'
by (rule merkle-hash-pairwise-bad-imp-local-collision-or-coupling-bad
  [OF pair-bad])
show ?thesis
  using candidate candidate' distinct decomp
  unfolding query-header-supported-pairwise-coupling-bad-def by blast
qed

```

lemma query-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad:

assumes bad: query-supported-pairwise-merkle-bad s

shows

```

  ( $\exists$  fr f-fri-roots f-final as dg composition-fri-roots final
    trace-table composition-table final-state trace-table'
    composition-table' final-state'.
    query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table composition-table
    final-state  $\wedge$ 
    query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table' composition-table'
    final-state'  $\wedge$ 
    (trace-table  $\neq$  trace-table'  $\vee$ 
     composition-table  $\neq$  composition-table')  $\wedge$ 
    (hash-map-output-collision final-state  $\vee$ 
     hash-map-output-collision final-state'))  $\vee$ 
    query-supported-pairwise-coupling-bad s

```

proof —

from bad **obtain** fr f-fri-roots f-final as dg composition-fri-roots final

where header-bad:

```

  query-header-supported-pairwise-merkle-bad s fr f-fri-roots f-final
  as dg composition-fri-roots final

```

unfolding query-supported-pairwise-merkle-bad-def **by** blast

from

```

    query-header-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad
      [OF header-bad]
  show ?thesis
    unfolding query-supported-pairwise-coupling-bad-def by blast
qed

lemma query-supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad:
  assumes bad: query-supported-pairwise-merkle-bad s
  shows
    supported-hash-output-collision-possible s ∨
    query-supported-pairwise-coupling-bad s
proof -
  have decomp:
    (∃ fr f-fri-roots f-final as dg composition-fri-roots final
      trace-table composition-table final-state trace-table'
      composition-table' final-state'.
      query-header-supported-table-candidate-state s fr f-fri-roots f-final
        as dg composition-fri-roots final trace-table composition-table
        final-state ∧
      query-header-supported-table-candidate-state s fr f-fri-roots f-final
        as dg composition-fri-roots final trace-table' composition-table'
        final-state' ∧
      (trace-table ≠ trace-table' ∨
        composition-table ≠ composition-table') ∧
      (hash-map-output-collision final-state ∨
        hash-map-output-collision final-state')) ∨
      query-supported-pairwise-coupling-bad s
  by (rule query-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad
      [OF bad])
  then show ?thesis
proof
  assume local:
    ∃ fr f-fri-roots f-final as dg composition-fri-roots final
      trace-table composition-table final-state trace-table'
      composition-table' final-state'.
      query-header-supported-table-candidate-state s fr f-fri-roots f-final
        as dg composition-fri-roots final trace-table composition-table
        final-state ∧
      query-header-supported-table-candidate-state s fr f-fri-roots f-final
        as dg composition-fri-roots final trace-table' composition-table'
        final-state' ∧
      (trace-table ≠ trace-table' ∨
        composition-table ≠ composition-table') ∧
      (hash-map-output-collision final-state ∨
        hash-map-output-collision final-state')
  then obtain fr f-fri-roots f-final as dg composition-fri-roots final
    trace-table composition-table final-state trace-table'
    composition-table' final-state' where
    candidate:

```

```

    query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table composition-table
    final-state
  and candidate':
    query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table' composition-table'
    final-state'
  and collision:
    hash-map-output-collision final-state  $\vee$ 
    hash-map-output-collision final-state'
  by blast
from collision show ?thesis
proof
  assume collision-final: hash-map-output-collision final-state
  from candidate obtain result query-idxs rest where
    outcome:
      Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
  and bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
  unfolding query-header-supported-table-candidate-state-def by blast
have supported-hash-output-collision-possible s
  by (rule
    accepted-with-bound-tables-hash-collision-imp-supported-hash-output-collision-possible
    [OF outcome bound collision-final])
  then show ?thesis by simp
next
  assume collision-final': hash-map-output-collision final-state'
  from candidate' obtain result' query-idxs' rest' where
    outcome':
      Some (result', final-state')  $\in$  set-dist (execute verify-monad s)
  and bound':
    accepted-with-bound-tables s (Some (result', final-state'))
    trace-table' composition-table' as query-idxs'
  unfolding query-header-supported-table-candidate-state-def by blast
have supported-hash-output-collision-possible s
  by (rule
    accepted-with-bound-tables-hash-collision-imp-supported-hash-output-collision-possible
    [OF outcome' bound' collision-final'])
  then show ?thesis by simp
qed
next
  assume query-supported-pairwise-coupling-bad s
  then show ?thesis by simp
qed
qed

lemma query-header-supported-candidate-unique-if-no-pairwise-merkle-bad:
  assumes no-bad:

```

$\neg$ *query-header-supported-pairwise-merkle-bad* *s* fr *f-fri-roots* *f-final*
as *dg* *composition-fri-roots* *final*
shows
 $\exists$ *trace-table* *composition-table*.
query-header-supported-table-candidates *s* fr *f-fri-roots* *f-final* as *dg*
composition-fri-roots *final* $\subseteq$
 $\{(trace-table, composition-table)\}$
proof (rule *ccontr*)
assume *not-unique*:
 $\neg$ ($\exists$ *trace-table* *composition-table*.
query-header-supported-table-candidates *s* fr *f-fri-roots* *f-final* as *dg*
composition-fri-roots *final* $\subseteq$
 $\{(trace-table, composition-table)\}$)
have *query-header-supported-pairwise-merkle-bad* *s* fr *f-fri-roots* *f-final*
as *dg* *composition-fri-roots* *final*
using
query-header-supported-table-candidates-not-singleton-imp-pairwise-merkle-bad
[*OF not-unique*]
unfolding *query-header-supported-pairwise-merkle-bad-def*
by *blast*
then show *False*
using *no-bad* **by** *contradiction*
qed

lemma *query-header-supported-candidate-unique-if-no-global-pairwise-merkle-bad*:
assumes *no-bad*: $\neg$ *query-supported-pairwise-merkle-bad* *s*
shows
 $\exists$ *trace-table* *composition-table*.
query-header-supported-table-candidates *s* fr *f-fri-roots* *f-final* as *dg*
composition-fri-roots *final* $\subseteq$
 $\{(trace-table, composition-table)\}$
proof (rule *query-header-supported-candidate-unique-if-no-pairwise-merkle-bad*)
show $\neg$ *query-header-supported-pairwise-merkle-bad* *s* fr *f-fri-roots*
f-final as *dg* *composition-fri-roots* *final*
using *no-bad* **unfolding** *query-supported-pairwise-merkle-bad-def* **by** *blast*
qed

definition *query-header-supported-union-good-sets*
 $:: (f, 'a)$ *protocol-channel-scheme* $\Rightarrow$
 $(f\ list \Rightarrow 'f\ list \Rightarrow 'f\ list \Rightarrow nat\ set) \Rightarrow$
 $'f \Rightarrow 'f\ list \Rightarrow 'f \Rightarrow 'f\ list \Rightarrow 'f \Rightarrow$
 $'f\ list \Rightarrow 'f \Rightarrow nat\ set$
where
query-header-supported-union-good-sets *s* *good-sets* fr *f-fri-roots*
f-final as *dg* *composition-fri-roots* *final* =
 $\{idx \in query-sample-space.$
 $\exists$ *trace-table* *composition-table*.
 $(trace-table, composition-table) \in$
query-header-supported-table-candidates *s* fr *f-fri-roots* *f-final*

$as \text{ dg composition-fri-roots final} \wedge$
 $idx \in \text{good-sets trace-table composition-table as}\}$

lemma *query-header-supported-union-good-sets-subset:*
query-header-supported-union-good-sets s good-sets fr f-fri-roots
f-final as dg composition-fri-roots final $\subseteq$ query-sample-space
unfolding *query-header-supported-union-good-sets-def* **by** *auto*

definition *query-header-rounds-any-index-set-hit*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat set}) \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

query-header-rounds-any-index-set-hit s B out $\longleftrightarrow$
 $(\exists \text{ fr f-fri-roots f-final as dg composition-fri-roots final rest.}$
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest $\wedge$
query-rounds-any-index-set-hit s
 $(B \text{ fr f-fri-roots f-final as dg composition-fri-roots final})$
rounds out)

lemma *query-header-supported-union-good-sets-fraction-bound-if-unique-candidate:*

assumes *unique:*

$\exists \text{ trace-table composition-table.}$
query-header-supported-table-candidates s fr f-fri-roots f-final as
dg composition-fri-roots final $\subseteq$
 $\{(\text{trace-table, composition-table})\}$

and *subset:*

$\bigwedge \text{ trace-table composition-table as.}$
good-sets trace-table composition-table as $\subseteq$ query-sample-space

and *bound:*

$\bigwedge \text{ trace-table composition-table as.}$
 $\text{nnreal} (\text{card} (\text{good-sets trace-table composition-table as})) /$
 $\text{nnreal} (\text{card query-sample-space}) \leq \text{query-error-bound}$

shows

nnreal
 $(\text{card}$
 $(\text{query-header-supported-union-good-sets s good-sets fr f-fri-roots}$
 $\text{f-final as dg composition-fri-roots final})) /$
 $\text{nnreal} (\text{card query-sample-space}) \leq \text{query-error-bound}$

proof –

from *unique* **obtain** *trace-table composition-table* **where** *candidates:*

query-header-supported-table-candidates s fr f-fri-roots f-final as dg
composition-fri-roots final $\subseteq$
 $\{(\text{trace-table, composition-table})\}$

by *blast*

have *union-subset:*

query-header-supported-union-good-sets s good-sets fr f-fri-roots


```

    f-final as dg composition-fri-roots final  $\subseteq$ 
    good-sets trace-table composition-table as
unfolding query-header-supported-union-good-sets-def
using candidates by auto
have finite-good: finite (good-sets trace-table composition-table as)
by (rule finite-subset[OF subset finite-query-sample-space])
have card-le:
  card
    (query-header-supported-union-good-sets s good-sets fr f-fri-roots
     f-final as dg composition-fri-roots final)  $\leq$ 
    card (good-sets trace-table composition-table as)
by (rule card-mono[OF finite-good union-subset])
have nnreal
  (card
    (query-header-supported-union-good-sets s good-sets fr f-fri-roots
     f-final as dg composition-fri-roots final)) /
  nnreal (card query-sample-space)  $\leq$ 
  nnreal (card (good-sets trace-table composition-table as)) /
  nnreal (card query-sample-space)
by (rule nnreal-nat-divide-right-mono[OF card-le])
also have ...  $\leq$  query-error-bound
by (rule bound)
finally show ?thesis .
qed

lemma query-header-supported-union-good-sets-fraction-bound-if-no-pairwise-merkle-bad:
assumes no-bad:
   $\neg$  query-header-supported-pairwise-merkle-bad s fr f-fri-roots
  f-final as dg composition-fri-roots final
and subset:
   $\bigwedge$  trace-table composition-table as.
  good-sets trace-table composition-table as  $\subseteq$  query-sample-space
and bound:
   $\bigwedge$  trace-table composition-table as.
  nnreal (card (good-sets trace-table composition-table as)) /
  nnreal (card query-sample-space)  $\leq$  query-error-bound
shows
  nnreal
    (card
      (query-header-supported-union-good-sets s good-sets fr f-fri-roots
       f-final as dg composition-fri-roots final)) /
    nnreal (card query-sample-space)  $\leq$  query-error-bound
proof –
have unique:
   $\exists$  trace-table composition-table.
  query-header-supported-table-candidates s fr f-fri-roots f-final as dg
  composition-fri-roots final  $\subseteq$ 
  {(trace-table, composition-table)}
by (rule query-header-supported-candidate-unique-if-no-pairwise-merkle-bad

```

```

    [OF no-bad])
  show ?thesis
  by (rule query-header-supported-union-good-sets-fraction-bound-if-unique-candidate
      [OF unique subset bound])
qed

lemma query-header-supported-union-good-sets-fraction-bound-if-no-global-pairwise-merkle-bad:
  assumes no-bad:  $\neg$  query-supported-pairwise-merkle-bad s
  and subset:
     $\bigwedge$  trace-table composition-table as.
      good-sets trace-table composition-table as  $\subseteq$  query-sample-space
  and bound:
     $\bigwedge$  trace-table composition-table as.
      nnreal (card (good-sets trace-table composition-table as)) /
        nnreal (card query-sample-space)  $\leq$  query-error-bound
  shows
    nnreal
      (card
        (query-header-supported-union-good-sets s good-sets fr f-fri-roots
          f-final as dg composition-fri-roots final)) /
        nnreal (card query-sample-space)  $\leq$  query-error-bound
  proof -
    have no-header-bad:
       $\neg$  query-header-supported-pairwise-merkle-bad s fr f-fri-roots
        f-final as dg composition-fri-roots final
    using no-bad unfolding query-supported-pairwise-merkle-bad-def by blast
    show ?thesis
    by (rule
        query-header-supported-union-good-sets-fraction-bound-if-no-pairwise-merkle-bad
        [OF no-header-bad subset bound])
  qed

definition query-header-committed-table-candidates
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    'f  $\Rightarrow$  'f list  $\Rightarrow$  ('f list  $\times$  'f list) set
  where
    query-header-committed-table-candidates s fr composition-fri-roots =
      {(trace-table, composition-table).
        merkle-root-binds-table fr trace-table s  $\wedge$ 
        composition-fri-roots  $\neq$  []  $\wedge$ 
        merkle-root-binds-table (hd composition-fri-roots) composition-table
        s}

definition query-header-committed-union-good-sets
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    ('f list  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  nat set)  $\Rightarrow$ 
    'f  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  nat set
  where
    query-header-committed-union-good-sets s good-sets fr as

```

$composition-fri-roots =$
 $\{idx \in query-sample-space.$
 $\exists trace-table\ composition-table.$
 $(trace-table, composition-table) \in$
 $query-header-committed-table-candidates\ s\ fr$
 $composition-fri-roots \wedge$
 $idx \in good-sets\ trace-table\ composition-table\ as\}$

lemma *query-header-committed-union-good-sets-subset:*
query-header-committed-union-good-sets s good-sets fr as
composition-fri-roots $\subseteq$ query-sample-space
unfolding *query-header-committed-union-good-sets-def* **by** *auto*

lemma *query-header-committed-union-good-sets-fraction-bound-if-unique-candidate:*
assumes *unique:*
 $\exists trace-table\ composition-table.$
query-header-committed-table-candidates s fr composition-fri-roots
 $\subseteq \{(trace-table, composition-table)\}$
and *subset:*
 $\bigwedge trace-table\ composition-table\ as.$
good-sets trace-table composition-table as $\subseteq$ query-sample-space
and *bound:*
 $\bigwedge trace-table\ composition-table\ as.$
 $nnreal\ (card\ (good-sets\ trace-table\ composition-table\ as))\ /$
 $nnreal\ (card\ query-sample-space) \leq query-error-bound$
shows
 $nnreal$
 $(card$
 $(query-header-committed-union-good-sets\ s\ good-sets\ fr\ as$
 $composition-fri-roots))\ /$
 $nnreal\ (card\ query-sample-space) \leq query-error-bound$

proof –
from *unique* **obtain** *trace-table composition-table* **where** *candidates:*
query-header-committed-table-candidates s fr composition-fri-roots
 $\subseteq \{(trace-table, composition-table)\}$
by *blast*
have *union-subset:*
query-header-committed-union-good-sets s good-sets fr as
composition-fri-roots $\subseteq$ good-sets trace-table composition-table as
unfolding *query-header-committed-union-good-sets-def*
using *candidates* **by** *auto*
have *finite-good:* *finite (good-sets trace-table composition-table as)*
by *(rule finite-subset[OF subset finite-query-sample-space])*
have *card-le:*
 $card$
 $(query-header-committed-union-good-sets\ s\ good-sets\ fr\ as$
 $composition-fri-roots) \leq$
 $card\ (good-sets\ trace-table\ composition-table\ as)$
by *(rule card-mono[OF finite-good union-subset])*

```

have nnreal
  (card
    (query-header-committed-union-good-sets s good-sets fr as
      composition-fri-roots)) /
  nnreal (card query-sample-space) ≤
  nnreal (card (good-sets trace-table composition-table as)) /
  nnreal (card query-sample-space)
  by (rule nnreal-nat-divide-right-mono[OF card-le])
also have ...  $\leq$  query-error-bound
  by (rule bound)
finally show ?thesis .
qed

lemma query-header-committed-table-candidates-subset-singleton-if-no-root-collisions:
assumes trace-clean:  $\neg$  merkle-root-binding-collision fr s
and comp-clean:
   $\bigwedge rt. \text{composition-fri-roots} \neq [] \implies$ 
   $rt = \text{hd composition-fri-roots} \implies$ 
   $\neg \text{merkle-root-binding-collision } rt \ s$ 
shows
   $\exists \text{trace-table composition-table.}$ 
   $\text{query-header-committed-table-candidates } s \text{ fr composition-fri-roots}$ 
   $\subseteq \{(\text{trace-table}, \text{composition-table})\}$ 
proof (cases
   $\text{query-header-committed-table-candidates } s \text{ fr composition-fri-roots} = \{\}$ )
case True
then show ?thesis
  by (intro exI[of -] exI[of -] simp)
next
case False
then obtain pair where pair-in:
   $\text{pair} \in$ 
   $\text{query-header-committed-table-candidates } s \text{ fr composition-fri-roots}$ 
by blast
then obtain trace-table composition-table where pair-eq:
   $\text{pair} = (\text{trace-table}, \text{composition-table})$ 
by (cases pair)
have candidate:
   $(\text{trace-table}, \text{composition-table}) \in$ 
   $\text{query-header-committed-table-candidates } s \text{ fr composition-fri-roots}$ 
using pair-in unfolding pair-eq .
have trace-bind:
   $\text{merkle-root-binds-table fr trace-table } s$ 
using candidate unfolding query-header-committed-table-candidates-def
by simp
have comp-nonempty:  $\text{composition-fri-roots} \neq []$ 
using candidate unfolding query-header-committed-table-candidates-def
by simp
have comp-bind:

```

```

merkle-root-binds-table (hd composition-fri-roots) composition-table s
using candidate unfolding query-header-committed-table-candidates-def
by simp
have candidates:
  query-header-committed-table-candidates s fr composition-fri-roots
     $\subseteq \{(trace-table, composition-table)\}$ 
proof
  fix pair
  assume pair-in:
    pair  $\in$ 
      query-header-committed-table-candidates s fr composition-fri-roots
  obtain trace-table' composition-table' where pair-eq:
    pair = (trace-table', composition-table')
    by (cases pair)
  have trace-bind':
    merkle-root-binds-table fr trace-table' s
    using pair-in unfolding pair-eq
      query-header-committed-table-candidates-def
    by simp
  have comp-bind':
    merkle-root-binds-table (hd composition-fri-roots) composition-table'
      s
    using pair-in unfolding pair-eq
      query-header-committed-table-candidates-def
    by simp
  have trace-eq: trace-table' = trace-table
    using merkle-root-binds-table-unique-if-no-collision
      [OF trace-clean trace-bind' trace-bind] by simp
  have comp-eq: composition-table' = composition-table
    using merkle-root-binds-table-unique-if-no-collision
      [OF comp-clean [OF comp-nonempty refl] comp-bind' comp-bind]
    by simp
  show pair  $\in \{(trace-table, composition-table)\}$ 
    unfolding pair-eq trace-eq comp-eq by simp
qed
show ?thesis
  using candidates by blast
qed

lemma query-index-round-set-hit-at-imp-query-header-supported-union-hit:
  assumes outcome: out  $\in$  set-dist (execute verify-monad s)
  and hit: query-index-round-set-hit-at s good-sets i out
  shows
     $\exists$  fr f-fri-roots f-final as dg composition-fri-roots final.
      query-rounds-any-index-set-hit s
        (query-header-supported-union-good-sets s good-sets fr f-fri-roots
          f-final as dg composition-fri-roots final)
        rounds out
proof —

```

```

from hit obtain trace-table composition-table as query-idxs where
  bound:
    accepted-with-bound-tables s out trace-table composition-table as
      query-idxs
  and i-bound: i < rounds
  and query-hit:
    query-idxs ! i ∈ good-sets trace-table composition-table as
  unfolding query-index-round-set-hit-at-def by blast
from bound obtain result final-state fr f-fri-roots f-final dg
  composition-fri-roots final rest where
  out-eq: out = Some (result, final-state)
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  unfolding accepted-with-bound-tables-def by blast
have candidate:
  (trace-table, composition-table) ∈
    query-header-supported-table-candidates s fr f-fri-roots f-final as dg
    composition-fri-roots final
  unfolding query-header-supported-table-candidates-def
  using outcome bound header by blast
have len: length query-idxs = rounds
  by (rule accepted-with-bound-tables-shapes(4)[OF bound])
have idx-mem: query-idxs ! i ∈ set query-idxs
  by (rule nth-mem) (use i-bound len in simp)
have idx-sample: query-idxs ! i ∈ query-sample-space
  by (rule accepted-with-bound-tables-query-sample-space
    [OF bound idx-mem])
have union-hit:
  query-idxs ! i ∈
    query-header-supported-union-good-sets s good-sets fr f-fri-roots
    f-final as dg composition-fri-roots final
  unfolding query-header-supported-union-good-sets-def
  using idx-sample candidate query-hit by blast
have round-hit:
  query-rounds-any-index-set-hit s
  (query-header-supported-union-good-sets s good-sets fr f-fri-roots
    f-final as dg composition-fri-roots final)
  rounds out
  by (rule accepted-with-bound-tables-query-rounds-any-index-set-hit
    [OF bound i-bound union-hit])
show ?thesis
  using round-hit by blast
qed

lemma query-index-round-set-hit-at-imp-query-header-supported-rounds-hit:
assumes outcome: out ∈ set-dist (execute verify-monad s)
and hit: query-index-round-set-hit-at s good-sets i out
shows

```

```

    query-header-rounds-any-index-set-hit s
    (query-header-supported-union-good-sets s good-sets) out
proof –
from hit obtain trace-table composition-table as query-idxs where
    bound:
        accepted-with-bound-tables s out trace-table composition-table as
            query-idxs
    and i-bound:  $i < \text{rounds}$ 
    and query-hit:
        query-idxs !  $i \in \text{good-sets}$  trace-table composition-table as
    unfolding query-index-round-set-hit-at-def by blast
from bound obtain result final-state fr f-fri-roots f-final dg
    composition-fri-roots final rest where
    header:
        verifier-header-transcript s fr f-fri-roots f-final as dg
        composition-fri-roots final rest
    unfolding accepted-with-bound-tables-def by blast
from query-index-round-set-hit-at-imp-query-header-supported-union-hit
    [OF outcome hit]
obtain fr' f-fri-roots' f-final' as' dg' composition-fri-roots' final'
where round-hit':
    query-rounds-any-index-set-hit s
    (query-header-supported-union-good-sets s good-sets fr' f-fri-roots'
        f-final' as' dg' composition-fri-roots' final')
    rounds out
by blast
have candidate:
    (trace-table, composition-table)  $\in$ 
    query-header-supported-table-candidates s fr f-fri-roots f-final as dg
    composition-fri-roots final
    unfolding query-header-supported-table-candidates-def
    using outcome bound header by blast
have len: length query-idxs = rounds
    by (rule accepted-with-bound-tables-shapes(4)) [OF bound])
have idx-mem: query-idxs !  $i \in \text{set query-idxs}$ 
    by (rule nth-mem) (use i-bound len in simp)
have idx-sample: query-idxs !  $i \in \text{query-sample-space}$ 
    by (rule accepted-with-bound-tables-query-sample-space
        [OF bound idx-mem])
have union-hit:
    query-idxs !  $i \in$ 
    query-header-supported-union-good-sets s good-sets fr f-fri-roots
    f-final as dg composition-fri-roots final
    unfolding query-header-supported-union-good-sets-def
    using idx-sample candidate query-hit by blast
have round-hit:
    query-rounds-any-index-set-hit s
    (query-header-supported-union-good-sets s good-sets fr f-fri-roots
        f-final as dg composition-fri-roots final)

```

```

    rounds out
  by (rule accepted-with-bound-tables-query-rounds-any-index-set-hit
      [OF bound i-bound union-hit])
  show ?thesis
    unfolding query-header-rounds-any-index-set-hit-def
    using header round-hit by blast
qed

lemma query-index-round-set-hit-imp-query-header-supported-rounds-hit:
  assumes outcome: out ∈ set-dist (execute verify-monad s)
  and hit: query-index-round-set-hit s good-sets out
  shows
    query-header-rounds-any-index-set-hit s
      (query-header-supported-union-good-sets s good-sets) out
  proof -
    from hit obtain i where
      hit-at: query-index-round-set-hit-at s good-sets i out
    unfolding query-index-round-set-hit-def by blast
    show ?thesis
      by (rule query-index-round-set-hit-at-imp-query-header-supported-rounds-hit
          [OF outcome hit-at])
  qed

lemma query-index-set-hit-empty-queries:
  assumes bound:
    accepted-with-bound-tables s out trace-table composition-table as []
  shows query-index-set-hit s good-sets out
  unfolding query-index-set-hit-def
  apply (rule exI[where x=trace-table])
  apply (rule exI[where x=composition-table])
  apply (rule exI[where x=as])
  apply (rule exI[where x=[]])
  using bound by simp

lemma query-index-set-hit-vacuous-if-rounds-zero:
  assumes rounds-zero: rounds = 0
  and bound:
    accepted-with-bound-tables s out trace-table composition-table as query-idxs
  shows query-index-set-hit s good-sets out
  proof -
    have length query-idxs = 0
      using accepted-with-bound-tables-shapes(4)[OF bound] rounds-zero by simp
    then have query-idxs-empty: query-idxs = []
      by simp
    show ?thesis
      using bound unfolding query-idxs-empty
      by (rule query-index-set-hit-empty-queries)
  qed

```


lemma *query-index-set-hit-imp-round-set-hit*:
assumes *hit*: *query-index-set-hit s good-sets out*
shows *query-index-round-set-hit s good-sets out*
proof –
from *hit* **obtain** *trace-table composition-table as query-idxs* **where**
bound:
accepted-with-bound-tables s out trace-table composition-table as
query-idxs
and *subset*:
set query-idxs $\subseteq$ good-sets trace-table composition-table as
unfolding *query-index-set-hit-def* **by** *blast*
have *len*: *length query-idxs = rounds*
by (rule *accepted-with-bound-tables-shapes*(4)[OF *bound*])
have *zero-bound*: *0 < length query-idxs*
using *len rounds-positive* **by** *simp*
have *nth-in*: *query-idxs ! 0 $\in$ set query-idxs*
by (rule *nth-mem*[OF *zero-bound*])
have *round-hit*:
query-index-round-set-hit-at s good-sets 0 out
unfolding *query-index-round-set-hit-at-def*
apply (rule *exI*[**where** *x=trace-table*])
apply (rule *exI*[**where** *x=composition-table*])
apply (rule *exI*[**where** *x=as*])
apply (rule *exI*[**where** *x=query-idxs*])
using *bound subset nth-in rounds-positive* **by** *auto*
show *?thesis*
unfolding *query-index-round-set-hit-def*
using *round-hit rounds-positive* **by** *blast*
qed

lemma *accepted-with-bound-tables-query-samples-miss-if-not-all-consistent*:
assumes *bound*:
accepted-with-bound-tables s out trace-table composition-table as query-idxs
and *not-all*:
 $\neg$ *all-queries-consistent trace-table composition-table as*
shows *query-samples-miss-disagreements trace-table composition-table as query-idxs*
proof –
have *disagreements*:
query-disagreement-indices trace-table composition-table as $\neq$ {}
using *not-all query-disagreement-indices-nonempty-iff* **by** *blast*
have *samples-agree*:
set query-idxs $\subseteq$ query-agreement-indices trace-table composition-table as
proof
fix *idx*
assume *idx-in*: *idx $\in$ set query-idxs*
have *sample*: *idx $\in$ query-sample-space*
by (rule *accepted-with-bound-tables-query-sample-space*[OF *bound idx-in*])
have *consistent*:
query-consistent-at trace-table composition-table as idx

```

    by (rule accepted-with-bound-tables-query-consistency[OF bound idx-in])
  show  $idx \in \text{query-agreement-indices trace-table composition-table as}$ 
    using sample consistent unfolding query-agreement-indices-def by simp
qed
show ?thesis
  using disagreements samples-agree
  unfolding query-samples-miss-disagreements-def by simp
qed

lemma query-sampling-success-space-subset:
  query-sampling-success-space trace-table composition-table as  $\subseteq$  query-sample-space
  unfolding query-sampling-success-space-def query-agreement-indices-def by auto

lemma query-sampling-success-space-fraction-bound:
  shows
    nnreal (card (query-sampling-success-space trace-table composition-table as)) /
      nnreal query-sample-space-size  $\leq$  query-error-bound
  proof (cases
    trace-table-low-degree trace-table  $\wedge$ 
    composition-table-low-degree maxDegree composition-table  $\wedge$ 
     $\neg$  all-queries-consistent trace-table composition-table as)
  case True
  then have space-eq:
    query-sampling-success-space trace-table composition-table as =
      query-agreement-indices trace-table composition-table as
    unfolding query-sampling-success-space-def by simp
  have card-le:
    card (query-sampling-success-space trace-table composition-table as)  $\leq$ 
      query-agreement-bound
    unfolding space-eq
    by (rule query-agreement-indices-card-bound)
      (use True in auto)
  have nnreal (card (query-sampling-success-space trace-table composition-table as)) /
    nnreal query-sample-space-size  $\leq$ 
    nnreal query-agreement-bound / nnreal query-sample-space-size
    by (rule nnreal-nat-divide-right-mono[OF card-le])
  then show ?thesis
    unfolding query-error-bound-def .
next
  case False
  then have space-eq:
    query-sampling-success-space trace-table composition-table as = {}
    unfolding query-sampling-success-space-def by auto
  show ?thesis
    unfolding space-eq by simp
qed

lemma query-sampling-success-space-fraction-bound-query-sample-space:
  shows

```

$\text{nnreal } (\text{card } (\text{query-sampling-success-space trace-table composition-table as})) /$
 $\text{nnreal } (\text{card query-sample-space}) \leq \text{query-error-bound}$
using *query-sampling-success-space-fraction-bound*[*of trace-table composition-table as*]
by *simp*

lemma *query-sampling-success-header-supported-union-fraction-bound-if-no-global-pairwise-merkle-bad*:

assumes *no-bad*: $\neg \text{query-supported-pairwise-merkle-bad } s$

shows

nnreal
 $(\text{card}$
 $(\text{query-header-supported-union-good-sets } s \text{ query-sampling-success-space}$
 $\text{fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final})) /$
 $\text{nnreal } (\text{card query-sample-space}) \leq \text{query-error-bound}$
by (*rule*
query-header-supported-union-good-sets-fraction-bound-if-no-global-pairwise-merkle-bad
 $[OF \text{ no-bad query-sampling-success-space-subset}$
 $\text{query-sampling-success-space-fraction-bound-query-sample-space}]$)

definition *query-index-raw-preimage-bound* :: *bool*

where

$\text{query-index-raw-preimage-bound} \longleftrightarrow$
 $(\forall B.$
 $B \subseteq \text{query-sample-space} \longrightarrow$
 $\text{nnreal } (\text{card } (\text{query-index-raw-preimage } B)) / \text{nnreal size} \leq$
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card query-sample-space}))$

lemma *query-index-raw-preimage-bound-if-uniform-range*:

assumes *range-eq*: $\text{range to-nat} = \{0..<\text{size}\}$

and *dvd*: $\text{query-sample-space-size} \text{ dvd } \text{size}$

shows *query-index-raw-preimage-bound*

unfolding *query-index-raw-preimage-bound-def*

proof (*intro allI impI*)

fix *B* :: *nat set*

assume *subset*: $B \subseteq \text{query-sample-space}$

let *?q* = *query-sample-space-size*

have *card-raw*:

$\text{card } (\text{query-index-raw-preimage } B) =$
 $(\text{size div } ?q) * \text{card } B$

unfolding *card-query-index-raw-preimage-eq-nat-preimage*

by (*rule card-query-index-nat-preimage-uniform-range*
 $[OF \text{ range-eq } \text{dvd } \text{subset}]$)

have *size-eq*: $\text{size} = (\text{size div } ?q) * ?q$

using *dvd query-sample-space-size-pos* **by** (*simp add: dvd-eq-mod-eq-0*)

have *size-pos*: $0 < \text{size}$

using *size-card* **by** *simp*

have *div-pos*: $0 < \text{size div } ?q$

proof (*cases size div ?q*)

case *0*

then show *?thesis*

```

    using size-eq size-pos by simp
next
case (Suc n)
then show ?thesis
  by simp
qed
have frac-eq:
  nnreal ((size div ?q) * card B) / nnreal ((size div ?q) * ?q) =
    nnreal (card B) / nnreal ?q
  using div-pos query-sample-space-size-pos
  by transfer (simp add: field-simps)
have lhs-eq:
  nnreal (card (query-index-raw-preimage B)) / nnreal size =
    nnreal ((size div ?q) * card B) / nnreal ((size div ?q) * ?q)
  using card-raw size-eq by simp
show
  nnreal (card (query-index-raw-preimage B)) / nnreal size ≤
    nnreal (card B) / nnreal (card query-sample-space)
  using lhs-eq frac-eq by simp
qed

lemma query-index-raw-preimage-bound-from-sampler-wellformed:
  query-index-raw-preimage-bound
proof (rule query-index-raw-preimage-bound-if-uniform-range)
  show range to-nat = {0..

```

```

assumes raw-bound: query-index-raw-preimage-bound
and subset:  $B \subseteq \text{query-sample-space}$ 
and frac:
   $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{query-sample-space}) \leq$ 
   $\text{query-error-bound}$ 
shows
   $\text{nnreal } (\text{card } (\text{query-index-raw-preimage } B)) / \text{nnreal size} \leq$ 
   $\text{query-error-bound}$ 
proof –
  have  $\text{nnreal } (\text{card } (\text{query-index-raw-preimage } B)) / \text{nnreal size} \leq$ 
   $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{query-sample-space})$ 
  using raw-bound subset unfolding query-index-raw-preimage-bound-def by blast
  also have  $\dots \leq \text{query-error-bound}$ 
  by (rule frac)
  finally show ?thesis .
qed

```

lemma wp-receive-query-index-challenge-fresh-index-set-raw-bound:

```

assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and subset:  $B \subseteq \text{query-sample-space}$ 
shows
  wp-event receive-query-index-challenge
  ( $\lambda \text{out. case out of None} \Rightarrow \text{False}$ 
   |  $\text{Some } (\text{raw}, -) \Rightarrow \text{index } (\text{to-nat raw}) \in B$ ) s  $\leq$ 
   $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{query-sample-space})$ 
proof –
  have fresh:
    fmlookup (HashMap s)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = None
  by (rule query-future-fresh-current[OF future])
  have exact:
    wp-event receive-query-index-challenge
    ( $\lambda \text{out. case out of None} \Rightarrow \text{False}$ 
     |  $\text{Some } (\text{raw}, -) \Rightarrow \text{index } (\text{to-nat raw}) \in B$ ) s =
     $\text{nnreal } (\text{card } (\text{query-index-raw-preimage } B)) / \text{nnreal size}$ 
  by (rule wp-receive-query-index-challenge-fresh-index-set[OF fresh])
  have raw:
     $\text{nnreal } (\text{card } (\text{query-index-raw-preimage } B)) / \text{nnreal size} \leq$ 
     $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{query-sample-space})$ 
  using raw-bound subset
  unfolding query-index-raw-preimage-bound-def by blast
  show ?thesis
  unfolding exact by (rule raw)
qed

```

lemma wp-receive-query-index-challenge-fresh-index-set-query-error-bound:

```

assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound

```

```

and subset:  $B \subseteq \text{query-sample-space}$ 
and frac:
   $\text{nnreal} (\text{card } B) / \text{nnreal} (\text{card } \text{query-sample-space}) \leq$ 
   $\text{query-error-bound}$ 
shows
   $\text{wp-event receive-query-index-challenge}$ 
   $(\lambda \text{out. case out of None} \Rightarrow \text{False}$ 
     $| \text{Some (raw, -)} \Rightarrow \text{index (to-nat raw)} \in B) \ s \leq$ 
     $\text{query-error-bound}$ 
by (rule order-trans
  [OF wp-receive-query-index-challenge-fresh-index-set-row-bound
  [OF future raw-bound subset] frac])

lemma wp-verifier-query-round-program-index-set-bound:
assumes future: query-future-fresh  $s$ 
and raw-bound: query-index-raw-preimage-bound
and subset:  $B \subseteq \text{query-sample-space}$ 
shows
   $\text{wp-event}$ 
   $(\text{verifier-query-round-program fr f-fl f-final as fl final})$ 
   $(\text{query-round-index-set-hit } s \ B) \ s \leq$ 
   $\text{nnreal} (\text{card } B) / \text{nnreal} (\text{card } \text{query-sample-space})$ 
proof -
let ?P =
   $\lambda \text{out. case out of None} \Rightarrow \text{False}$ 
   $| \text{Some (raw, -)} \Rightarrow \text{index (to-nat raw)} \in B$ 
have head-bound:
   $\text{wp-event receive-query-index-challenge } ?P \ s \leq$ 
   $\text{nnreal} (\text{card } B) / \text{nnreal} (\text{card } \text{query-sample-space})$ 
by (rule wp-receive-query-index-challenge-fresh-index-set-row-bound
  [OF future raw-bound subset])
show ?thesis
unfolding verifier-query-round-program-alt-def
proof (rule wp-event-bind-bound-by-head-event[OF head-bound])
show query-round-index-set-hit  $s \ B \ \text{None} \Longrightarrow ?P \ \text{None}$ 
unfolding query-round-index-set-hit-def by simp
next
fix raw s0 out
assume head:
   $\text{Some (raw, s0)} \in$ 
   $\text{set-dist (execute receive-query-index-challenge } s)$ 
and tail:
   $\text{out} \in$ 
   $\text{set-dist}$ 
   $(\text{execute}$ 
     $(\text{verifier-query-round-after-index-program fr f-fl f-final as fl}$ 
       $\text{final raw) s0})$ 
and hit: query-round-index-set-hit  $s \ B \ \text{out}$ 
from hit obtain raw' t where out-eq:  $\text{out} = \text{Some } ((), t)$ 

```

```

and lookup-t:
  fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw'
and raw'-in: index (to-nat raw') ∈ B
unfolding query-round-index-set-hit-def by blast
have tail-out:
  Some ((), t) ∈
    set-dist
    (execute
      (verifier-query-round-after-index-program fr f-fl f-final as fl
        final raw) s0)
using tail out-eq by simp
have ext-s0-t: s0 ≤ t
using verifier-query-round-after-index-program-outcome[OF tail-out]
by simp
have lookup-s0:
  fmlookup (HashMap s0)
  (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
using receive-query-index-challenge-outcome[OF head] by blast
have lookup-t-row:
  fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
by (rule hash-extension-lookup[OF lookup-s0 ext-s0-t])
have raw = raw'
using lookup-t-row lookup-t by simp
then show ?P (Some (raw, s0))
using raw'-in by simp
qed
qed

lemma wp-verifier-query-round-program-index-set-query-error-bound:
assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and subset: B ⊆ query-sample-space
and frac:
  nnreal (card B) / nnreal (card query-sample-space) ≤
    query-error-bound
shows
  wp-event
    (verifier-query-round-program fr f-fl f-final as fl final)
    (query-round-index-set-hit s B) s ≤ query-error-bound
by (rule order-trans
  [OF wp-verifier-query-round-program-index-set-bound
    [OF future raw-bound subset] frac])

lemma wp-verifier-query-round-program-any-index-set-bound:
assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and subset: B ⊆ query-sample-space

```

```

shows
  wp-event
    (verifier-query-round-program fr f-fl f-final as fl final)
    (query-round-any-index-set-hit s B) s ≤
    nnreal (card B) / nnreal (card query-sample-space)
proof -
  let ?P =
    λout. case out of None ⇒ False
      | Some (raw, -) ⇒ index (to-nat raw) ∈ B
  have head-bound:
    wp-event receive-query-index-challenge ?P s ≤
      nnreal (card B) / nnreal (card query-sample-space)
  by (rule wp-receive-query-index-challenge-fresh-index-set-raw-bound
      [OF future raw-bound subset])
  show ?thesis
    unfolding verifier-query-round-program-alt-def
  proof (rule wp-event-bind-bound-by-head-event[OF head-bound])
    show query-round-any-index-set-hit s B None ⇒ ?P None
      unfolding query-round-any-index-set-hit-def by simp
  next
    fix raw s0 out
    assume head:
      Some (raw, s0) ∈
        set-dist (execute receive-query-index-challenge s)
    and tail:
      out ∈
        set-dist
          (execute
            (verifier-query-round-after-index-program fr f-fl f-final as fl
              final raw) s0)
    and hit: query-round-any-index-set-hit s B out
  from hit obtain raw' t x where out-eq: out = Some ((), t)
  and lookup-t:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) x) = Some raw'
  and raw'-in: index (to-nat raw') ∈ B
  unfolding query-round-any-index-set-hit-def by blast
  have tail-out:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program fr f-fl f-final as fl
            final raw) s0)
  using tail out-eq by simp
  have lookup-s0-x:
    fmlookup (HashMap s0)
      (QueryIndexChallenge (PQueryCounter s) x) = Some raw'
  using verifier-query-round-after-index-program-preserves-query-lookup
    [OF tail-out, of PQueryCounter s x]

```



```

    lookup-t
  by simp
have lookup-current:
  fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  using receive-query-index-challenge-outcome[OF head] by blast
have key-eq:
  QueryIndexChallenge (PQueryCounter s) x =
    QueryIndexChallenge (PQueryCounter s) (PState s)
proof (rule ccontr)
  assume neg:
    QueryIndexChallenge (PQueryCounter s) x ≠
      QueryIndexChallenge (PQueryCounter s) (PState s)
  have lookup-s0-x':
    fmlookup (HashMap s0)
      (QueryIndexChallenge (PQueryCounter s) x) =
        fmlookup (HashMap s)
          (QueryIndexChallenge (PQueryCounter s) x)
    by (rule receive-query-index-challenge-preserves-other-lookup
        [OF head neg])
  have fmlookup (HashMap s)
    (QueryIndexChallenge (PQueryCounter s) x) = None
  using future unfolding query-future-fresh-def by simp
  then show False
    using lookup-s0-x lookup-s0-x' by simp
qed
have raw' = raw
  using lookup-s0-x lookup-current key-eq by simp
then show ?P (Some (raw, s0))
  using raw'-in by simp
qed
qed

lemma wp-ntimes-verifier-query-round-program-any-index-set-bound:
  assumes future: query-future-fresh s
  and raw-bound: query-index-raw-preimage-bound
  and subset:  $B \subseteq \text{query-sample-space}$ 
  shows
    wp-event
      (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n)
      (query-rounds-any-index-set-hit s B n)  $s \leq$ 
      nnreal n *
        (nnreal (card B) / nnreal (card query-sample-space))
  using future
proof (induction n arbitrary: s)
  case 0
  show ?case
    unfolding query-rounds-any-index-set-hit-def
    by (simp add: wp-event-def wpsimps)

```

```

next
  case (Suc n)
  let ?prog = verifier-query-round-program fr f-fl f-final as fl final
  let ?C = nnreal (card B) / nnreal (card query-sample-space)
  let ?Q = query-rounds-any-index-set-hit s B (Suc n)
  let ?Head =
    λout :: (unit list × ('f, 'a) protocol-channel-scheme) option.
      case out of None ⇒ False
      | Some (-, t) ⇒
        (∃ raw x.
          fmlookup (HashMap t)
            (QueryIndexChallenge (PQueryCounter s) x) = Some raw ∧
            index (to-nat raw) ∈ B)
  let ?Tail =
    λout :: (unit list × ('f, 'a) protocol-channel-scheme) option.
      case out of None ⇒ False
      | Some (-, t) ⇒
        (∃ i < n. ∃ x raw.
          fmlookup (HashMap t)
            (QueryIndexChallenge (PQueryCounter s + Suc i) x) =
              Some raw ∧
              index (to-nat raw) ∈ B)
  have split-hit:
    ∧out. ?Q out ⇒ ?Head out ∨ ?Tail out
  proof -
    fix out
    assume ?Q out
    then obtain results t i x raw where out-eq: out = Some (results, t)
      and i-bound: i < Suc n
      and lookup:
        fmlookup (HashMap t)
          (QueryIndexChallenge (PQueryCounter s + i) x) = Some raw
      and raw-in: index (to-nat raw) ∈ B
    unfolding query-rounds-any-index-set-hit-def by blast
    show ?Head out ∨ ?Tail out
  proof (cases i)
    case 0
    then have ?Head out
      using out-eq lookup raw-in by simp blast
    then show ?thesis by simp
  next
    case (Suc j)
    then have j-bound: j < n
      using i-bound by simp
    have ?Tail out
      using out-eq lookup raw-in j-bound Suc by simp blast
    then show ?thesis by simp
  qed
qed

```

```

have head-bound:
  wp-event (ntimes ?prog (Suc n)) ?Head s ≤ ?C
proof -
  have one-bound:
    wp-event ?prog (query-round-any-index-set-hit s B) s ≤ ?C
    by (rule wp-verifier-query-round-program-any-index-set-bound
        [OF Suc.premis raw-bound subset])
  show ?thesis
    unfolding ntimes.simps
  proof (rule wp-event-bind-bound-by-head-event[OF one-bound])
    show ?Head None ⇒ query-round-any-index-set-hit s B None
      by simp
  next
    fix u s1 out
    assume head:
      Some (u, s1) ∈ set-dist (execute ?prog s)
    and tail:
      out ∈
        set-dist
          (execute
            (ntimes ?prog n ≫ (λxs. return (u # xs)))) s1)
    and hit: ?Head out
  have u-eq: u = ()
    by (cases u) simp
  from hit obtain ys0 t0 where out-some: out = Some (ys0, t0)
    by (cases out) auto
  from tail[unfolded out-some] obtain results t where
    tail-rounds:
      Some (results, t) ∈ set-dist (execute (ntimes ?prog n) s1)
    and out-ret:
      Some (ys0, t0) ∈ set-dist (execute (return (u # results)) t)
    by (rule set-dist-bindE)
  have out-eq: out = Some (u # results, t)
    using out-ret out-some
    by simp
  from hit obtain raw x where lookup-t:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) x) = Some raw
    and raw-in: index (to-nat raw) ∈ B
    using out-eq by auto
  have counter-s1: PQueryCounter s1 = Suc (PQueryCounter s)
    using verifier-query-round-program-outcome[OF head[unfolded u-eq]]
    by blast
  have past: PQueryCounter s < PQueryCounter s1
    using counter-s1 by simp
  have lookup-s1:
    fmlookup (HashMap s1)
      (QueryIndexChallenge (PQueryCounter s) x) = Some raw
    using ntimes-verifier-query-round-program-preserves-past-query-lookup

```

```

      [OF past tail-rounds, of x]
      lookup-t
    by simp
  show query-round-any-index-set-hit s B (Some (u, s1))
    unfolding query-round-any-index-set-hit-def
    using lookup-s1 raw-in u-eq by blast
qed
qed
have tail-bound:
  wp-event (ntimes ?prog (Suc n)) ?Tail s ≤ nnreal n * ?C
proof -
  show ?thesis
    unfolding ntimes.simps
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ ?Tail None
      by simp
  next
    fix u s1
    assume head:
      Some (u, s1) ∈ set-dist (execute ?prog s)
    have u-eq: u = ()
      by (cases u) simp
    have future-s1: query-future-fresh s1
      by (rule verifier-query-round-program-preserves-query-future-fresh
        [OF Suc.prem head[unfolded u-eq]])
    have counter-s1: PQueryCounter s1 = Suc (PQueryCounter s)
      using verifier-query-round-program-outcome[OF head[unfolded u-eq]]
      by blast
    have tail-induct:
      wp-event (ntimes ?prog n)
        (query-rounds-any-index-set-hit s1 B n) s1 ≤
        nnreal n * ?C
      by (rule Suc.IH[OF future-s1])
    show wp-event
      (ntimes ?prog n ≫ (λxs. return (u # xs))) ?Tail s1
      ≤ nnreal n * ?C
  proof (rule wp-event-bind-bound-by-head-event[OF tail-induct])
    show ?Tail None ⇒ query-rounds-any-index-set-hit s1 B n None
      by simp
  next
    fix results t out
    assume tail-rounds:
      Some (results, t) ∈
        set-dist (execute (ntimes ?prog n) s1)
    and ret:
      out ∈ set-dist (execute (return (u # results)) t)
    and hit: ?Tail out
    have out-eq: out = Some (u # results, t)
      using ret

```

```

unfolding set-dist-def return.rep-eq dist-return-def
  dist-delta-dist delta-map-def
by simp
from hit obtain i x raw where i-bound: i < n
and lookup:
  fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s + Suc i) x) =
    Some raw
and raw-in: index (to-nat raw) ∈ B
using out-eq by auto
have key-eq:
  Suc (PQueryCounter s + i) = PQueryCounter s1 + i
using counter-s1 by presburger
have lookup':
  fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s1 + i) x) = Some raw
using lookup key-eq by simp
show query-rounds-any-index-set-hit s1 B n (Some (results, t))
  unfolding query-rounds-any-index-set-hit-def
  apply (rule exI[where x=results])
  apply (rule exI[where x=t])
  apply (rule exI[where x=i])
  apply (rule exI[where x=x])
  apply (rule exI[where x=raw])
  using i-bound lookup' raw-in by simp
qed
qed
qed
have wp-event (ntimes ?prog (Suc n)) ?Q s ≤
  wp-event (ntimes ?prog (Suc n))
  (λout. ?Head out ∨ ?Tail out) s
by (rule wp-event-mono) (use split-hit in blast)
also have ... ≤ wp-event (ntimes ?prog (Suc n)) ?Head s +
  wp-event (ntimes ?prog (Suc n)) ?Tail s
by (rule wp-event-union-bound)
also have ... ≤ ?C + nnreal n * ?C
by (intro add-mono head-bound tail-bound)
also have ... = nnreal (Suc n) * ?C
by (simp add: algebra-simps)
finally show ?case .
qed

lemma wp-ntimes-verifier-query-round-program-any-index-set-query-error-bound:
assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and subset: B ⊆ query-sample-space
and frac:
  nnreal (card B) / nnreal (card query-sample-space) ≤
  query-error-bound

```

```

shows
  wp-event
    (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n)
    (query-rounds-any-index-set-hit s B n) s ≤
    nnreal n * query-error-bound
proof –
  have base:
    wp-event
      (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n)
      (query-rounds-any-index-set-hit s B n) s ≤
      nnreal n *
        (nnreal (card B) / nnreal (card query-sample-space))
  by (rule wp-ntimes-verifier-query-round-program-any-index-set-bound
    [OF future raw-bound subset])
  have mono:
    nnreal n *
      (nnreal (card B) / nnreal (card query-sample-space)) ≤
      nnreal n * query-error-bound
  by (rule mult-left-mono) (use frac in simp-all)
  show ?thesis
  by (rule order-trans[OF base mono])
qed

lemma wp-verifier-after-composition-fri-query-rounds-any-index-set-bound:
assumes future: query-future-fresh t
and counter: PQueryCounter t = PQueryCounter s
and raw-bound: query-index-raw-preimage-bound
and subset: B ⊆ query-sample-space
shows
  wp-event (verifier-after-composition-fri header)
    (query-rounds-any-index-set-hit s B rounds) t ≤
    nnreal rounds *
      (nnreal (card B) / nnreal (card query-sample-space))
proof –
  obtain fr f-fl f-final as dg fl where header-eq:
    header = (fr, f-fl, f-final, as, dg, fl)
  by (cases header) auto
  have after-eq:
    verifier-after-composition-fri header =
      (read >>=
        (λfinal.
          ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            rounds))
  unfolding header-eq verifier-after-composition-fri-def by simp
  show ?thesis
  unfolding after-eq
proof (rule wp-event-bind-bound-by-cont)
  show ¬ query-rounds-any-index-set-hit s B rounds None

```

unfolding *query-rounds-any-index-set-hit-def* **by** *simp*
next
fix *final query-state*
assume *read-final*:
Some (final, query-state) ∈ set-dist (execute read t)
have *future-query: query-future-fresh query-state*
by (*rule read-preserves-query-future-fresh[OF future read-final]*)
from *read-outcome[OF read-final]* **obtain** *rest* **where**
counter-query-t: PQueryCounter query-state = PQueryCounter t
by *blast*
have *counter-query: PQueryCounter query-state = PQueryCounter s*
using *counter-query-t counter* **by** *simp*
have *event-imp*:
 $\bigwedge \text{out. query-rounds-any-index-set-hit } s \text{ } B \text{ rounds out} \implies$
 $\text{query-rounds-any-index-set-hit query-state } B \text{ rounds out}$
using *counter-query*
unfolding *query-rounds-any-index-set-hit-def* **by** *auto*
have *wp-event*
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final)
rounds)
(query-rounds-any-index-set-hit s B rounds) query-state ≤
wp-event
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final)
rounds)
(query-rounds-any-index-set-hit query-state B rounds) query-state
by (*rule wp-event-mono*) (*rule event-imp*)
also have $\dots \leq$
 $\text{nnreal rounds} * (\text{nnreal (card } B) / \text{nnreal (card query-sample-space)})$
by (*rule wp-ntimes-verifier-query-round-program-any-index-set-bound*
 $[OF \text{ future-query raw-bound subset}]$)
finally show
wp-event
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final)
rounds)
(query-rounds-any-index-set-hit s B rounds) query-state ≤
 $\text{nnreal rounds} * (\text{nnreal (card } B) / \text{nnreal (card query-sample-space)}) .$
qed
qed

lemma *wp-verify-monad-query-rounds-any-index-set-bound*:
assumes *future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *subset: B ⊆ query-sample-space*
shows

```

wp-event verify-monad (query-rounds-any-index-set-hit s B rounds) s ≤
  nnreal rounds *
    (nnreal (card B) / nnreal (card query-sample-space))
unfolding verify-monad-composition-fri-decomposition
proof (rule wp-event-bind-bound-by-cont)
  show ¬ query-rounds-any-index-set-hit s B rounds None
    unfolding query-rounds-any-index-set-hit-def by simp
next
fix header t
assume prefix:
  Some (header, t) ∈
    set-dist (execute verifier-composition-fri-prefix s)
obtain fr f-fl f-final as dg fl where header-eq:
  header = (fr, f-fl, f-final, as, dg, fl)
by (cases header) auto
have future-t: query-future-fresh t
  by (rule verifier-composition-fri-prefix-preserves-query-future-fresh
    [OF future prefix])
have counter-t: PQueryCounter t = PQueryCounter s
  using verifier-composition-fri-prefix-outcome[OF prefix[unfolded header-eq]]
  by simp
show
  wp-event (verifier-after-composition-fri header)
    (query-rounds-any-index-set-hit s B rounds) t ≤
    nnreal rounds *
      (nnreal (card B) / nnreal (card query-sample-space))
  by (rule
    wp-verifier-after-composition-fri-query-rounds-any-index-set-bound
    [OF future-t counter-t raw-bound subset])
qed

lemma wp-verify-monad-query-rounds-any-index-set-query-error-bound:
assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and subset: B ⊆ query-sample-space
and frac:
  nnreal (card B) / nnreal (card query-sample-space) ≤
    query-error-bound
shows
  wp-event verify-monad (query-rounds-any-index-set-hit s B rounds) s ≤
    nnreal rounds * query-error-bound
proof –
have base:
  wp-event verify-monad (query-rounds-any-index-set-hit s B rounds) s ≤
    nnreal rounds *
      (nnreal (card B) / nnreal (card query-sample-space))
  by (rule wp-verify-monad-query-rounds-any-index-set-bound
    [OF future raw-bound subset])
have mono:

```



```

nnreal rounds *
  (nnreal (card B) / nnreal (card query-sample-space)) ≤
nnreal rounds * query-error-bound
by (rule mult-left-mono) (use frac in simp-all)
show ?thesis
by (rule order-trans[OF base mono])
qed

lemma wp-verify-monad-query-header-rounds-any-index-set-bound:
fixes C :: prob
assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and subset:
  ∧fr f-fri-roots f-final as dg composition-fri-roots final.
  B fr f-fri-roots f-final as dg composition-fri-roots final
  ⊆ query-sample-space
and bound:
  ∧fr f-fri-roots f-final as dg composition-fri-roots final.
  nnreal
  (card
    (B fr f-fri-roots f-final as dg composition-fri-roots final)) /
  nnreal (card query-sample-space) ≤ C
shows
  wp-event verify-monad (query-header-rounds-any-index-set-hit s B) s ≤
  nnreal rounds * C
unfolding verify-monad-composition-fri-decomposition
proof (rule wp-event-bind-bound-by-cont)
show ¬ query-header-rounds-any-index-set-hit s B None
unfolding query-header-rounds-any-index-set-hit-def
  query-rounds-any-index-set-hit-def
by auto
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state) ∈
  set-dist (execute verifier-composition-fri-prefix s)
obtain fr f-fl f-final as dg fl where header-eq:
  header = (fr, f-fl, f-final, as, dg, fl)
by (cases header) auto
have prefix-res:
  length f-fl = ceil-log clength ∧
  length as = length spec ∧
  length fl = ceil-log (to-nat dg + 1) ∧
  PTranscript s =
    [fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
    map snd fl @ PTranscript prefix-state ∧
  PQueryCounter prefix-state = PQueryCounter s
using verifier-composition-fri-prefix-outcome
  [OF prefix[unfolded header-eq]]

```

```

  by simp
have future-prefix: query-future-fresh prefix-state
  by (rule verifier-composition-fri-prefix-preserves-query-future-fresh
      [OF future prefix])
have after-eq:
  verifier-after-composition-fri header =
    (read  $\gg$ 
      ( $\lambda$ final.
        ntimes (verifier-query-round-program fr f-fl f-final as fl final)
          rounds))
  unfolding header-eq verifier-after-composition-fri-def by simp
show
  wp-event (verifier-after-composition-fri header)
    (query-header-rounds-any-index-set-hit s B) prefix-state  $\leq$ 
    nnreal rounds * C
  unfolding after-eq
proof (rule wp-event-bind-bound-by-cont)
  show  $\neg$  query-header-rounds-any-index-set-hit s B None
    unfolding query-header-rounds-any-index-set-hit-def
      query-rounds-any-index-set-hit-def
    by auto
next
  fix final query-state
  assume read-final:
    Some (final, query-state)  $\in$  set-dist (execute read prefix-state)
  from read-outcome[OF read-final] obtain rest where
    tr-prefix: PTranscript prefix-state = final # rest
    and tr-query: PTranscript query-state = rest
    and query-counter:
      PQueryCounter query-state = PQueryCounter prefix-state
    by blast
  have future-query: query-future-fresh query-state
    by (rule read-preserves-query-future-fresh[OF future-prefix read-final])
  have counter-query: PQueryCounter query-state = PQueryCounter s
    using prefix-res query-counter by simp
  have header-prefix:
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
      (map snd fl) final (PTranscript query-state)
    using prefix-res tr-prefix tr-query
    unfolding verifier-header-transcript-def verifier-header-messages-def
    by simp
  let ?B = B fr (map snd f-fl) f-final as dg (map snd fl) final
  let ?prog = verifier-query-round-program fr f-fl f-final as fl final
  have base:
    wp-event (ntimes ?prog rounds)
      (query-rounds-any-index-set-hit query-state ?B rounds)
      query-state  $\leq$ 
      nnreal rounds *
        (nnreal (card ?B) / nnreal (card query-sample-space))

```

```

  by (rule wp-ntimes-verifier-query-round-program-any-index-set-bound
      [OF future-query raw-bound subset])
have base-C:
  wp-event (ntimes ?prog rounds)
    (query-rounds-any-index-set-hit query-state ?B rounds)
    query-state ≤ nreal rounds * C
proof -
  have nreal rounds *
    (nreal (card ?B) / nreal (card query-sample-space)) ≤
    nreal rounds * C
  by (rule mult-left-mono) (use bound in simp-all)
  then show ?thesis
    by (rule order-trans[OF base])
qed
show
  wp-event
    (ntimes ?prog rounds)
    (query-header-rounds-any-index-set-hit s B) query-state ≤
    nreal rounds * C
proof (rule order-trans[OF - base-C])
  show
    wp-event
      (ntimes ?prog rounds)
      (query-header-rounds-any-index-set-hit s B) query-state ≤
    wp-event
      (ntimes ?prog rounds)
      (query-rounds-any-index-set-hit query-state ?B rounds) query-state
proof (rule wp-event-mono-on-support)
  fix out
  assume out-support:
    out ∈ set-dist (execute (ntimes ?prog rounds) query-state)
  and hit: query-header-rounds-any-index-set-hit s B out
  from hit obtain fr' f-fri-roots' f-final' as' dg'
    composition-fri-roots' final' rest' where
  header-hit:
    verifier-header-transcript s fr' f-fri-roots' f-final' as' dg'
    composition-fri-roots' final' rest'
  and round-hit:
    query-rounds-any-index-set-hit s
      (B fr' f-fri-roots' f-final' as' dg'
        composition-fri-roots' final')
    rounds out
  unfolding query-header-rounds-any-index-set-hit-def by blast
  have eqs:
    fr' = fr ∧
    f-fri-roots' = map snd f-fl ∧
    f-final' = f-final ∧
    as' = as ∧
    dg' = dg ∧

```

```

      composition-fri-roots' = map snd fl ∧
      final' = final
    using verifier-header-transcript-unique[OF header-prefix header-hit]
    by simp
  have round-hit-s:
    query-rounds-any-index-set-hit s ?B rounds out
    using round-hit eqs by simp
  show
    query-rounds-any-index-set-hit query-state ?B rounds out
    using round-hit-s counter-query
    unfolding query-rounds-any-index-set-hit-def by auto
qed
qed
qed
qed

```

lemma *wp-query-index-round-set-hit-bound-via-supported-header-union:*

```

  assumes future: query-future-fresh s
  and raw-bound: query-index-raw-preimage-bound
  and union-bound:
    ∧fr f-fri-roots f-final as dg composition-fri-roots final.
    nnreal
    (card
      (query-header-supported-union-good-sets s good-sets fr
        f-fri-roots f-final as dg composition-fri-roots final)) /
    nnreal (card query-sample-space) ≤ query-error-bound

```

shows

```

  wp-event verify-monad (query-index-round-set-hit s good-sets) s ≤
  nnreal rounds * query-error-bound

```

proof –

```

  have header-bound:
    wp-event verify-monad
      (query-header-rounds-any-index-set-hit s
        (query-header-supported-union-good-sets s good-sets)) s ≤
    nnreal rounds * query-error-bound
  by (rule wp-verify-monad-query-header-rounds-any-index-set-bound
    [OF future raw-bound])
    (use query-header-supported-union-good-sets-subset union-bound in
      simp-all)
  have wp-event verify-monad (query-index-round-set-hit s good-sets) s ≤
    wp-event verify-monad
      (query-header-rounds-any-index-set-hit s
        (query-header-supported-union-good-sets s good-sets)) s
  by (rule wp-event-mono-on-support)
    (rule query-index-round-set-hit-imp-query-header-supported-rounds-hit)
  also have ... ≤ nnreal rounds * query-error-bound
  by (rule header-bound)
  finally show ?thesis .

```

qed

lemma *verifier-initial-query-rounds-any-index-set-error-bound:*

assumes *raw-bound: query-index-raw-preimage-bound*

and *subset: $B \subseteq \text{query-sample-space}$*

and *frac:*

*nnreal (card B) / nnreal (card $\text{query-sample-space}$) $\leq$
*query-error-bound**

shows

wp-event verify-monad

(query-rounds-any-index-set-hit (verifier-initial-state tr) B rounds)

(verifier-initial-state tr) $\leq$

*nnreal rounds * query-error-bound*

by (*rule wp-verify-monad-query-rounds-any-index-set-query-error-bound*
[OF verifier-initial-query-future-fresh raw-bound subset frac])

definition *query-index-freshness-assumption*

:: ('f, 'a) protocol-channel-scheme $\Rightarrow$ bool

where

query-index-freshness-assumption $s \longleftrightarrow$

($\forall$ good-sets.

($\forall$ trace-table composition-table as .

good-sets trace-table composition-table $as \subseteq \text{query-sample-space}$) $\longrightarrow$

($\forall$ trace-table composition-table as .

nnreal (card (good-sets trace-table composition-table as)) /

nnreal (card $\text{query-sample-space}$) $\leq$ query-error-bound) $\longrightarrow$

wp-event verify-monad (query-index-set-hit s good-sets) $s \leq$

*nnreal rounds * query-error-bound)*

definition *query-index-round-freshness-assumption*

:: ('f, 'a) protocol-channel-scheme $\Rightarrow$ bool

where

query-index-round-freshness-assumption $s \longleftrightarrow$

($\forall$ good-sets.

($\forall$ trace-table composition-table as .

good-sets trace-table composition-table $as \subseteq \text{query-sample-space}$) $\longrightarrow$

($\forall$ trace-table composition-table as .

nnreal (card (good-sets trace-table composition-table as)) /

nnreal (card $\text{query-sample-space}$) $\leq$ query-error-bound) $\longrightarrow$

wp-event verify-monad (query-index-round-set-hit s good-sets) $s \leq$

*nnreal rounds * query-error-bound)*

lemma *query-index-freshness-from-round-freshness:*

assumes *fresh-round: query-index-round-freshness-assumption s*

shows *query-index-freshness-assumption s*

unfolding *query-index-freshness-assumption-def*

proof (*intro allI impI*)

fix *good-sets :: 'f list $\Rightarrow$ 'f list $\Rightarrow$ 'f list $\Rightarrow$ nat set*

assume *subset:*

$\forall$ trace-table composition-table as .

good-sets trace-table composition-table as $\subseteq$ *query-sample-space*
and *frac*:
 $\forall$ *trace-table composition-table as*.
 $\text{nnreal} (\text{card} (\text{good-sets trace-table composition-table as})) /$
 $\text{nnreal} (\text{card query-sample-space}) \leq \text{query-error-bound}$
have *wp-event verify-monad (query-index-set-hit s good-sets) s* $\leq$
wp-event verify-monad (query-index-round-set-hit s good-sets) s
by (*rule wp-event-mono*) (*rule query-index-set-hit-imp-round-set-hit*)
also have $\dots \leq \text{nnreal rounds} * \text{query-error-bound}$
using *fresh-round subset frac*
unfolding *query-index-round-freshness-assumption-def* **by** *blast*
finally show *wp-event verify-monad (query-index-set-hit s good-sets) s* $\leq$
 $\text{nnreal rounds} * \text{query-error-bound}$.
qed

lemma *query-index-round-freshness-if-no-query-supported-pairwise-merkle-bad*:
assumes *future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *no-bad: $\neg$ query-supported-pairwise-merkle-bad s*
shows *query-index-round-freshness-assumption s*
unfolding *query-index-round-freshness-assumption-def*
proof (*intro allI impI*)
fix *good-sets :: 'f list $\Rightarrow$ 'f list $\Rightarrow$ 'f list $\Rightarrow$ nat set*
assume *subset*:
 $\forall$ *trace-table composition-table as*.
good-sets trace-table composition-table as $\subseteq$ *query-sample-space*
and *bounded*:
 $\forall$ *trace-table composition-table as*.
 $\text{nnreal} (\text{card} (\text{good-sets trace-table composition-table as})) /$
 $\text{nnreal} (\text{card query-sample-space}) \leq \text{query-error-bound}$
have *union-bound*:
 $\bigwedge \text{fr f-fri-roots f-final as dg composition-fri-roots final.}$
 nnreal
 $(\text{card}$
 $(\text{query-header-supported-union-good-sets s good-sets fr f-fri-roots}$
 $\text{f-final as dg composition-fri-roots final})) /$
 $\text{nnreal} (\text{card query-sample-space}) \leq \text{query-error-bound}$
by (*rule*
query-header-supported-union-good-sets-fraction-bound-if-no-global-pairwise-merkle-bad
 $[OF \text{ no-bad}]$)
(use subset bounded in blast)+
show *wp-event verify-monad (query-index-round-set-hit s good-sets) s* $\leq$
 $\text{nnreal rounds} * \text{query-error-bound}$
by (*rule wp-query-index-round-set-hit-bound-via-supported-header-union*
 $[OF \text{ future raw-bound union-bound}]$)
qed

lemma *query-index-freshness-if-no-query-supported-pairwise-merkle-bad*:
assumes *future: query-future-fresh s*

and *raw-bound*: *query-index-raw-preimage-bound*
and *no-bad*: $\neg$ *query-supported-pairwise-merkle-bad* *s*
shows *query-index-freshness-assumption* *s*
by (rule *query-index-freshness-from-round-freshness*)
(rule *query-index-round-freshness-if-no-query-supported-pairwise-merkle-bad*
[OF *future raw-bound no-bad*])

lemma *verifier-initial-query-index-freshness-if-no-query-supported-pairwise-merkle-bad*:
assumes *raw-bound*: *query-index-raw-preimage-bound*
and *no-bad*:
 $\neg$ *query-supported-pairwise-merkle-bad* (*verifier-initial-state tr*)
shows *query-index-freshness-assumption* (*verifier-initial-state tr*)
by (rule *query-index-freshness-if-no-query-supported-pairwise-merkle-bad*)
(use *raw-bound no-bad* **in** *simp-all*)

lemma *verifier-initial-query-index-freshness-if-uniform-sampler-and-no-query-supported-pairwise-merkle-bad*:
assumes *range-eq*: *range to-nat* = $\{0..<size\}$
and *dvd*: *query-sample-space-size dvd size*
and *no-bad*:
 $\neg$ *query-supported-pairwise-merkle-bad* (*verifier-initial-state tr*)
shows *query-index-freshness-assumption* (*verifier-initial-state tr*)
proof –
have *raw-bound*: *query-index-raw-preimage-bound*
by (rule *query-index-raw-preimage-bound-if-uniform-range*
[OF *range-eq dvd*])
show *?thesis*
by (rule
verifier-initial-query-index-freshness-if-no-query-supported-pairwise-merkle-bad
[OF *raw-bound no-bad*])

qed

lemma *query-index-freshness-rounds-zero-obstruction*:
assumes *rounds-zero*: *rounds* = 0
and *outcome*: *out* $\in$ *set-dist* (*execute verify-monad s*)
and *bound*:
accepted-with-bound-tables s out trace-table composition-table as query-idxs
shows $\neg$ *query-index-freshness-assumption s*
proof
assume *fresh*: *query-index-freshness-assumption s*
let *?good-sets* = λ - - . {}
have *hit*: *query-index-set-hit s ?good-sets out*
by (rule *query-index-set-hit-vacuous-if-rounds-zero*
[OF *rounds-zero bound*])
have *positive*:
0 < *wp-event verify-monad* (*query-index-set-hit s ?good-sets*) *s*
by (rule *wp-event-pos-of-support*
[**where** *out*=*out* **and** *m*=*verify-monad* **and** *s*=*s*
and *P*=*query-index-set-hit s ?good-sets*, OF *outcome hit*])
have *subset*:

```

 $\wedge$  trace-table composition-table as.
  ?good-sets trace-table composition-table as  $\subseteq$  query-sample-space
by simp
have frac:
 $\wedge$  trace-table composition-table as.
  nnreal (card (?good-sets trace-table composition-table as)) /
    nnreal (card query-sample-space)  $\leq$  query-error-bound
by simp
have zero-bound:
  wp-event verify-monad (query-index-set-hit s ?good-sets) s  $\leq$  0
  using fresh subset frac rounds-zero
  unfolding query-index-freshness-assumption-def by simp
have zero:
  wp-event verify-monad (query-index-set-hit s ?good-sets) s = 0
  by (rule antisym[OF zero-bound]) simp
show False
  using positive zero by simp
qed

lemma query-bad-imp-query-index-set-hit:
  assumes query-bad s out
  shows query-index-set-hit s query-sampling-success-space out
proof -
  from assms obtain trace-table composition-table as query-idxs
  where bound:
    accepted-with-bound-tables s out trace-table composition-table as query-idxs
  and trace-low: trace-table-low-degree trace-table
  and composition-low:
    composition-table-low-degree maxDegree composition-table
  and not-all:
     $\neg$  all-queries-consistent trace-table composition-table as
  unfolding query-bad-def by blast
have miss:
  query-samples-miss-disagreements trace-table composition-table as query-idxs
  by (rule accepted-with-bound-tables-query-samples-miss-if-not-all-consistent
    [OF bound not-all])
have success-space:
  query-sampling-success-space trace-table composition-table as =
    query-agreement-indices trace-table composition-table as
  using trace-low composition-low not-all
  unfolding query-sampling-success-space-def by simp
have samples:
  set query-idxs  $\subseteq$ 
    query-sampling-success-space trace-table composition-table as
  using miss unfolding query-samples-miss-disagreements-def success-space by simp
show ?thesis
  unfolding query-index-set-hit-def
  using bound samples by blast
qed

```


lemma *query-bad-imp-query-index-round-set-hit*:
assumes *query-bad s out*
shows *query-index-round-set-hit s query-sampling-success-space out*
by (*rule query-index-set-hit-imp-round-set-hit*)
(rule query-bad-imp-query-index-set-hit[OF assms])

lemma *query-bad-bound-from-phase4B*:
assumes *fresh: query-index-freshness-assumption s*
shows *wp-event verify-monad (query-bad s) s ≤*
*nnreal rounds * query-error-bound*

proof –
have *hit-bound*:
wp-event verify-monad
(query-index-set-hit s query-sampling-success-space) s ≤
*nnreal rounds * query-error-bound*
using *fresh query-sampling-success-space-subset*
query-sampling-success-space-fraction-bound-query-sample-space
unfolding *query-index-freshness-assumption-def*
by *blast*
have *wp-event verify-monad (query-bad s) s ≤*
wp-event verify-monad
(query-index-set-hit s query-sampling-success-space) s
by (*rule wp-event-mono*) (*rule query-bad-imp-query-index-set-hit*)
also have *... ≤ nnreal rounds * query-error-bound*
by (*rule hit-bound*)
finally show *?thesis* .
qed

lemma *accepted-bound-trace-low-degree-or-trace-fri-bad*:
assumes *bound*:
accepted-with-bound-tables s out trace-table composition-table as query-idxs
shows *trace-table-low-degree trace-table ∨ trace-fri-bad s out*
proof (*cases trace-table-low-degree trace-table*)
case *True*
then show *?thesis* **by** *simp*
next
case *False*
then have *trace-fri-bad s out*
unfolding *trace-fri-bad-def* **using** *bound* **by** *blast*
then show *?thesis* **by** *simp*
qed

lemma *accepted-bound-composition-low-degree-or-composition-fri-bad*:
assumes *bound*:
accepted-with-bound-tables s out trace-table composition-table as query-idxs
and *trace-low: trace-table-low-degree trace-table*
shows *composition-table-low-degree maxDegree composition-table ∨*
composition-fri-bad s out

```

proof (cases composition-table-low-degree maxDegree composition-table)
  case True
  then show ?thesis by simp
next
  case False
  then have composition-fri-bad s out
    unfolding composition-fri-bad-def using bound trace-low by blast
  then show ?thesis by simp
qed

lemma accepted-bound-all-queries-consistent-or-query-bad:
  assumes bound:
    accepted-with-bound-tables s out trace-table composition-table as query-idxs
    and trace-low: trace-table-low-degree trace-table
    and composition-low: composition-table-low-degree maxDegree composition-table
  shows all-queries-consistent trace-table composition-table as  $\vee$ 
    query-bad s out
proof (cases all-queries-consistent trace-table composition-table as)
  case True
  then show ?thesis by simp
next
  case False
  then have query-bad s out
    unfolding query-bad-def using bound trace-low composition-low by blast
  then show ?thesis by simp
qed

lemma accepted-bound-composition-bad-if-globally-consistent:
  assumes false-statement:  $\neg$  exists-valid-trace
  and bound:
    accepted-with-bound-tables s out trace-table composition-table as query-idxs
    and trace-low: trace-table-low-degree trace-table
    and composition-low: composition-table-low-degree maxDegree composition-table
    and all-queries: all-queries-consistent trace-table composition-table as
  shows composition-bad s out
proof –
  from trace-low obtain f where
    deg-f: degree f < clength
    and trace-table: trace-table = map (poly f) eval-domain
    unfolding trace-table-low-degree-def by blast
  have violated: violated-constraints f  $\neq$  {}
    by (rule false-statement-violated-constraints[OF false-statement deg-f])
  have len-as: length as = length spec
    using accepted-with-tables-shapes(3)
    [OF accepted-with-bound-tables-imp-accepted-with-tables[OF bound]]
    by simp
  have hide-if-bounds:
    common-denominator-degree-bounds f as  $\implies$ 
    random-combination-common-denominator-hides-violations f as

```

```

    by (rule random-combination-common-denominator-hides-from-queries
        [OF len-as false-statement deg-f trace-table composition-low all-queries])
  show ?thesis
    unfolding composition-bad-def composition-degree-bad-def
      composition-randomization-bad-def composition-bad-context-def
    using bound deg-f trace-table violated composition-low all-queries hide-if-bounds
    by blast
qed

```

```

lemma accepted-bound-table-partition:
  assumes false-statement:  $\neg$  exists-valid-trace
  and bound:
    accepted-with-bound-tables s out trace-table composition-table as query-idxs
  shows
    composition-bad s out  $\vee$  trace-fri-bad s out  $\vee$ 
    composition-fri-bad s out  $\vee$  query-bad s out
proof -
  from accepted-bound-trace-low-degree-or-trace-fri-bad[OF bound]
  consider
    (trace-low) trace-table-low-degree trace-table
  | (trace-bad) trace-fri-bad s out
  by blast
  then show ?thesis
proof cases
  case trace-bad
  then show ?thesis by blast
next
  case trace-low
  from accepted-bound-composition-low-degree-or-composition-fri-bad
    [OF bound trace-low]
  consider
    (composition-low) composition-table-low-degree maxDegree composition-table
  | (composition-bad) composition-fri-bad s out
  by blast
  then show ?thesis
proof cases
  case composition-bad
  then show ?thesis by blast
next
  case composition-low
  from accepted-bound-all-queries-consistent-or-query-bad
    [OF bound trace-low composition-low]
  consider
    (all-queries) all-queries-consistent trace-table composition-table as
  | (query-bad) query-bad s out
  by blast
  then show ?thesis
proof cases
  case query-bad

```

```

    then show ?thesis by blast
  next
    case all-queries
    have composition-bad s out
      by (rule accepted-bound-composition-bad-if-globally-consistent
        [OF false-statement bound trace-low composition-low all-queries])
    then show ?thesis by blast
  qed
qed
qed
qed
qed

lemma composition-randomization-bad-imp-alpha-bad-set-hit:
  assumes bad: composition-randomization-bad s out
  shows alpha-bad-set-hit s composition-trace-bad-alpha-space out
proof -
  from bad obtain trace-table composition-table as query-idxs f where
    ctx:
      composition-bad-context s out trace-table composition-table as query-idxs f
    and bounds: common-denominator-degree-bounds f as
    and hides: random-combination-common-denominator-hides-violations f as
    unfolding composition-randomization-bad-def by blast
  have bound:
    accepted-with-bound-tables s out trace-table composition-table as query-idxs
    using ctx unfolding composition-bad-context-def by blast
  have deg-f: degree f < clength
    using ctx unfolding composition-bad-context-def by blast
  have trace-table: trace-table = map (poly f) eval-domain
    using ctx unfolding composition-bad-context-def by blast
  have violated: violated-constraints f ≠ {}
    using ctx unfolding composition-bad-context-def by blast
  have len-as: length as = length spec
    using accepted-with-tables-shapes(3)
    [OF accepted-with-bound-tables-imp-accepted-with-tables[OF bound]] .
  have as-in:
    as ∈ common-denominator-hiding-alpha-space f
    using len-as hides
    unfolding common-denominator-hiding-alpha-space-def alpha-space-def
    by simp
  have witness-eq: low-degree-trace-witness trace-table = f
    by (rule low-degree-trace-witness-eq[OF deg-f trace-table])
  have as ∈ composition-trace-bad-alpha-space trace-table
    using deg-f trace-table violated bounds as-in
    unfolding composition-trace-bad-alpha-space-def witness-eq
    by simp
  then show ?thesis
    using bound unfolding alpha-bad-set-hit-def by blast
qed

```

lemma *composition-randomization-bad-bound-from-alpha-freshness:*
assumes *fresh:* *alpha-challenge-freshness-assumption s*
shows *wp-event verify-monad (composition-randomization-bad s) s ≤ composition-error-bound*
proof –
have *hit-bound:*
wp-event verify-monad
(alpha-bad-set-hit s composition-trace-bad-alpha-space) s ≤ composition-error-bound
using *fresh composition-trace-bad-alpha-space-subset-alpha-space composition-trace-bad-alpha-space-fraction-bound-alpha-space*
unfolding *alpha-challenge-freshness-assumption-def*
by *blast*
have *wp-event verify-monad (composition-randomization-bad s) s ≤ wp-event verify-monad (alpha-bad-set-hit s composition-trace-bad-alpha-space) s*
by *(rule wp-event-mono)*
(rule composition-randomization-bad-imp-alpha-bad-set-hit)
also have *... ≤ composition-error-bound*
by *(rule hit-bound)*
finally show *?thesis .*
qed

lemma *composition-randomization-bad-bound-via-header-union:*
assumes *future:* *alpha-future-fresh s*
and *union-bound:*
 $\bigwedge_{fr} f\text{-fri-roots } f\text{-final.}$
 $nnreal$
 $(card$
 $(alpha\text{-header-union-bad-sets } s$
 $composition\text{-trace-bad-alpha-space } fr\ f\text{-fri-roots } f\text{-final})) /$
 $nnreal (card\ alpha\text{-space}) \leq composition\text{-error-bound}$
shows *wp-event verify-monad (composition-randomization-bad s) s ≤ composition-error-bound*
proof –
have *hit-bound:*
wp-event verify-monad
(alpha-bad-set-hit s composition-trace-bad-alpha-space) s ≤ composition-error-bound
by *(rule wp-alpha-bad-set-hit-bound-via-header-union[OF future])*
(use composition-trace-bad-alpha-space-subset-alpha-space union-bound in simp-all)
have *wp-event verify-monad (composition-randomization-bad s) s ≤ wp-event verify-monad (alpha-bad-set-hit s composition-trace-bad-alpha-space) s*
by *(rule wp-event-mono)*
(rule composition-randomization-bad-imp-alpha-bad-set-hit)
also have *... ≤ composition-error-bound*
by *(rule hit-bound)*

finally show ?thesis .
qed

lemma *composition-randomization-bad-bound-via-supported-header-union:*

assumes *future: alpha-future-fresh s*

and *union-bound:*

$\bigwedge fr\ f\text{-fri-roots}\ f\text{-final}.$

nnreal

(*card*

(*alpha-header-supported-union-bad-sets s*

composition-trace-bad-alpha-space fr f-fri-roots f-final)) /

nnreal (card alpha-space) ≤ composition-error-bound

shows *wp-event verify-monad (composition-randomization-bad s) s ≤ composition-error-bound*

proof –

have *hit-bound:*

wp-event verify-monad

(*alpha-bad-set-hit s composition-trace-bad-alpha-space*) *s ≤*

composition-error-bound

by (rule *wp-alpha-bad-set-hit-bound-via-supported-header-union[OF future]*)

(use *composition-trace-bad-alpha-space-subset-alpha-space union-bound* in *simp-all*)

have *wp-event verify-monad (composition-randomization-bad s) s ≤*

wp-event verify-monad

(*alpha-bad-set-hit s composition-trace-bad-alpha-space*) *s*

by (rule *wp-event-mono*)

(rule *composition-randomization-bad-imp-alpha-bad-set-hit*)

also have ... *≤ composition-error-bound*

by (rule *hit-bound*)

finally show ?thesis .

qed

lemma *composition-randomization-bad-bound-if-alpha-header-candidate-unique:*

assumes *future: alpha-future-fresh s*

and *unique:*

$\bigwedge fr\ f\text{-fri-roots}\ f\text{-final}.$

$\exists trace\text{-table}.$

alpha-header-trace-table-candidates s fr f-fri-roots f-final $\subseteq$

{*trace-table*}

shows *wp-event verify-monad (composition-randomization-bad s) s ≤*

composition-error-bound

by (rule *composition-randomization-bad-bound-via-header-union[OF future]*)

(rule *composition-trace-alpha-header-union-fraction-bound-if-unique-candidate* [OF *unique*])

lemma *composition-randomization-bad-bound-if-alpha-header-supported-candidate-unique:*

assumes *future: alpha-future-fresh s*

and *unique:*

$\bigwedge fr\ f\text{-fri-roots}\ f\text{-final}.$

$\exists$ *trace-table*.
 α -header-supported-trace-table-candidates *s* fr *f*-fri-roots
 f -final $\subseteq \{\text{trace-table}\}$
shows *wp-event verify-monad* (*composition-randomization-bad s*) $s \leq$
composition-error-bound
by (*rule composition-randomization-bad-bound-via-supported-header-union*
 $[OF \text{ future}]$)
rule
composition-trace-alpha-header-supported-union-fraction-bound-if-unique-candidate
 $[OF \text{ unique}]$)

lemma *composition-randomization-bad-bound-if-no-alpha-supported-pairwise-merkle-bad*:
assumes *future*: α -future-fresh *s*
and *no-bad*: $\neg \alpha$ -supported-pairwise-merkle-bad *s*
shows *wp-event verify-monad* (*composition-randomization-bad s*) $s \leq$
composition-error-bound
proof (*rule composition-randomization-bad-bound-if-alpha-header-supported-candidate-unique*
 $[OF \text{ future}]$)
fix *fr f*-fri-roots *f*-final
show
 $\exists$ *trace-table*.
 α -header-supported-trace-table-candidates *s* fr *f*-fri-roots *f*-final
 $\subseteq \{\text{trace-table}\}$
by (*rule alpha-header-supported-candidate-unique-if-no-global-pairwise-merkle-bad*
 $[OF \text{ no-bad}]$)

qed

lemma *composition-degree-bad-false*:
assumes *wf*: *spec-degree-wellformed*
shows $\neg$ *composition-degree-bad s out*
using *common-denominator-degree-bounds-from-spec-degree-wellformed* $[OF \text{ wf}]$
unfolding *composition-degree-bad-def composition-bad-context-def*
by *blast*

lemma *composition-degree-bad-bound*:
assumes *wf*: *spec-degree-wellformed*
shows *wp-event verify-monad* (*composition-degree-bad s*) $s \leq 0$
proof –
have *composition-degree-bad s* =
 $(\lambda \text{out} :: (\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option}. \text{ False})$
by (*rule ext*) (*simp add: composition-degree-bad-false* $[OF \text{ wf}]$)
then show *?thesis*
unfolding *wp-event-def wp-def dist-expect-def* **by** *simp*

qed

lemma *composition-bad-bound-from-phase4A*:
assumes *wf*: *spec-degree-wellformed*
and α -fresh: α -challenge-freshness-assumption *s*
shows *wp-event verify-monad* (*composition-bad s*) $s \leq$ *composition-error-bound*

```

proof –
  have degree-bound:
    wp-event verify-monad (composition-degree-bad s) s ≤ 0
    by (rule composition-degree-bad-bound [OF wf])
  have randomization-bound:
    wp-event verify-monad (composition-randomization-bad s) s ≤
      composition-error-bound
    by (rule composition-randomization-bad-bound-from-alpha-freshness [OF alpha-fresh])
  have wp-event verify-monad (composition-bad s) s =
    wp-event verify-monad
      ( $\lambda out. composition-degree-bad\ s\ out \vee$ 
       composition-randomization-bad s out) s
    unfolding composition-bad-def by simp
  also have ...  $\leq$ 
    wp-event verify-monad (composition-degree-bad s) s +
    wp-event verify-monad (composition-randomization-bad s) s
    by (rule wp-event-union-bound)
  also have ...  $\leq 0 + composition-error-bound$ 
    by (intro add-mono degree-bound randomization-bound)
  finally show ?thesis by simp
qed

end

end

```

```

theory Soundness-FRI
  imports
    Soundness-Bad-Events
begin

```

4.2 FRI Interfaces

```

context soundness
begin

```

4.2.1 Algebraic FRI Soundness Interface

The next definitions isolate the algebraic content of the verifier’s FRI fold equation from transcript extraction and Merkle authentication. They are intentionally protocol-independent except for using the same affine fold as *receive-query-commits*.

```

definition fri-table-fold-value
  :: 'f  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  'f
  where
    fri-table-fold-value b layer fri-dom len pw i =
      fri-fold-value b (layer ! i) (layer ! fri-sibling-index len i)
      (fri-fold-denominator (fri-dom ! i) 1)

```


definition *fri-algebraic-layer-relation*
 $:: \text{nat} \Rightarrow \text{nat} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$
where
fri-algebraic-layer-relation *len pw b fri-dom layer next-dom*
next-layer $\longleftrightarrow$
 $0 < \text{len} \wedge$
 $\text{len} \leq \text{length } \text{fri-dom} \wedge$
 $\text{len} \leq \text{length } \text{layer} \wedge$
 $\text{len} \text{ div } 2 \leq \text{length } \text{next-dom} \wedge$
 $\text{len} \text{ div } 2 \leq \text{length } \text{next-layer} \wedge$
 $(\forall i < \text{len} \text{ div } 2.$
 $\text{next-dom} ! i = (\text{fri-dom} ! i)^2 \wedge$
 $\text{next-layer} ! i = \text{fri-table-fold-value } b \text{ layer } \text{fri-dom } \text{len } pw \ i)$

lemma *fri-algebraic-layer-relationD*:
assumes *fri-algebraic-layer-relation* *len pw b fri-dom layer next-dom*
next-layer
and $i < \text{len} \text{ div } 2$
shows $\text{next-dom} ! i = (\text{fri-dom} ! i)^2$
and $\text{next-layer} ! i = \text{fri-table-fold-value } b \text{ layer } \text{fri-dom } \text{len } pw \ i$
using *assms* **unfolding** *fri-algebraic-layer-relation-def* **by** *auto*

definition *fri-opening-matches-table*
 $:: \text{nat} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \Rightarrow \text{bool}$
where
fri-opening-matches-table *len idx layer xp xn* $\longleftrightarrow$
 $\text{idx} < \text{len} \wedge$
 $\text{len} \leq \text{length } \text{layer} \wedge$
 $\text{layer} ! \text{idx} = xp \wedge$
 $\text{layer} ! \text{fri-sibling-index } \text{len } \text{idx} = xn$

lemma *fri-opening-matches-tableD*:
assumes *fri-opening-matches-table* *len idx layer xp xn*
shows $\text{idx} < \text{len}$
and $\text{len} \leq \text{length } \text{layer}$
and $\text{layer} ! \text{idx} = xp$
and $\text{layer} ! \text{fri-sibling-index } \text{len } \text{idx} = xn$
using *assms* **unfolding** *fri-opening-matches-table-def* **by** *auto*

lemma *fri-fold-value-eq-table-fold-value*:
assumes *dom-idx*: $\text{fri-dom} ! \text{idx} = (h \wedge \text{idx} * \text{shift}) \wedge pw$
and *match*: *fri-opening-matches-table* *len idx layer xp xn*
shows
 $\text{fri-fold-value } b \text{ xp } xn \ (\text{fri-fold-denominator } (h \wedge \text{idx} * \text{shift}) \ pw) =$
 $\text{fri-table-fold-value } b \text{ layer } \text{fri-dom } \text{len } pw \ \text{idx}$
using *dom-idx* *fri-opening-matches-tableD*(3,4)[*OF match*]
unfolding *fri-table-fold-value-def* *fri-fold-denominator-def* **by** *simp*

lemma *fri-layer-opening-step-outcome-fold-table-if-opening-matches*:
assumes *outcome*:
Some (out, t) ∈
set-dist (execute (fri-layer-opening-step (b, f) (i, x, len, pw)) s)
and *dom-i*: *fri-dom ! i = (h ^ i * shift) ^ pw*
obtains *xp xp-path xn xn-path x'*
where
out = (i mod (len div 2), x', len div 2, pw + pw)
and *xp = x*
and *fri-layer-opening-chunk len xp xp-path xn xn-path*
([xp] @ xp-path @ [xn] @ xn-path)
and *PTranscript s = ([xp] @ xp-path @ [xn] @ xn-path) @ PTranscript t*
and *PState t = foldl concat (PState s) ([xp] @ xp-path @ [xn] @ xn-path)*
and *s ≤ t*
and *fri-opening-matches-table len i layer xp xn ⇒*
x' = fri-table-fold-value b layer fri-dom len pw i
proof –
from *fri-layer-opening-step-outcome[OF outcome]*
obtain *xp xp-path xn xn-path x'* **where**
out-eq: out = (i mod (len div 2), x', len div 2, pw + pw)
and *xp-eq: xp = x*
and *x'-eq:*
x' = fri-fold-value b xp xn
*(fri-fold-denominator ((h ^ i) * shift) pw)*
and *chunk:*
fri-layer-opening-chunk len xp xp-path xn xn-path
([xp] @ xp-path @ [xn] @ xn-path)
and *tr:*
PTranscript s = ([xp] @ xp-path @ [xn] @ xn-path) @ PTranscript t
and *st:*
PState t = foldl concat (PState s) ([xp] @ xp-path @ [xn] @ xn-path)
and *ext: s ≤ t*
by *blast*
have *table-fold:*
fri-opening-matches-table len i layer xp xn ⇒
x' = fri-table-fold-value b layer fri-dom len pw i
proof –
assume *match: fri-opening-matches-table len i layer xp xn*
have *fri-fold-value b xp xn*
*(fri-fold-denominator ((h ^ i) * shift) pw) =*
fri-table-fold-value b layer fri-dom len pw i
by *(rule fri-fold-value-eq-table-fold-value[OF dom-i match])*
then show *x' = fri-table-fold-value b layer fri-dom len pw i*
using *x'-eq* **by** *simp*
qed
show *?thesis*
by *(rule that[OF out-eq xp-eq chunk tr st ext table-fold])*
qed

definition *fri-fold-coeff-pair* :: 'f $\Rightarrow$ 'f $\Rightarrow$ 'f $\Rightarrow$ 'f $\times$ 'f
where
fri-fold-coeff-pair *xp xn denom* =
 $((xp + xn) \text{ div } 2, (xp - xn) \text{ div } denom)$

definition *fri-local-hiding-challenges*
:: 'f $\Rightarrow$ 'f $\Rightarrow$ 'f $\Rightarrow$ 'f $\Rightarrow$ 'f $\Rightarrow$ 'f *set*
where
fri-local-hiding-challenges *xp xn yp yn denom* =
 $\{b. \text{fri-fold-value } b \text{ } xp \text{ } xn \text{ } denom = \text{fri-fold-value } b \text{ } yp \text{ } yn \text{ } denom\}$

lemma *fri-affine-collision-challenges-card-le-one*:
fixes *a c a' c'* :: 'f
assumes *pair-diff*: $(a, c) \neq (a', c')$
shows $\text{card } \{b. a + b * c = a' + b * c'\} \leq (1::nat)$
proof –
show ?thesis
proof (*cases* $c = c'$)
case *True*
then have *a-diff*: $a \neq a'$
using *pair-diff* **by** *auto*
have $\{b. a + b * c = a' + b * c'\} = \{\}$
proof (*rule ccontr*)
assume $\{b. a + b * c = a' + b * c'\} \neq \{\}$
then obtain *b* **where**
b-in: $b \in \{b. a + b * c = a' + b * c'\}$
by *blast*
then have *eq*: $a + b * c = a' + b * c'$
by *simp*
then have $a + b * c' = a' + b * c'$
using *True* **by** *simp*
then have $(a + b * c') - b * c' = (a' + b * c') - b * c'$
by *simp*
then have $a = a'$
by *simp*
then show *False*
using *a-diff* **by** *contradiction*
qed
then show ?thesis **by** *simp*
next
case *False*
have *subset*:
 $\{b. a + b * c = a' + b * c'\} \subseteq$
 $\{(a' - a) \text{ div } (c - c')\}$
proof
fix *b*
assume *b-in*: $b \in \{b. a + b * c = a' + b * c'\}$
then have *eq*: $a + b * c = a' + b * c'$
by *simp*

```

    have  $b * (c - c') = a' - a$ 
      using eq by (simp add: algebra-simps)
    then have  $b = (a' - a) \text{ div } (c - c')$ 
      using False by (simp add: field-simps)
    then show  $b \in \{(a' - a) \text{ div } (c - c')\}$ 
      by simp
  qed
  have card  $\{b. a + b * c = a' + b * c'\} \leq$ 
    card  $\{(a' - a) \text{ div } (c - c')\}$ 
    by (rule card-mono) (use subset in auto)
  then show ?thesis by simp
qed
qed

lemma fri-local-hiding-challenges-card-le-one:
  assumes pair-diff:
    fri-fold-coeff-pair  $xp \ xn \ denom \neq$  fri-fold-coeff-pair  $yp \ yn \ denom$ 
  shows card (fri-local-hiding-challenges  $xp \ xn \ yp \ yn \ denom$ )  $\leq 1$ 
proof -
  let ?a =  $(xp + xn) \text{ div } 2$ 
  let ?c =  $(xp - xn) \text{ div } denom$ 
  let ?a' =  $(yp + yn) \text{ div } 2$ 
  let ?c' =  $(yp - yn) \text{ div } denom$ 
  have pair-diff':  $(?a, ?c) \neq (?a', ?c')$ 
    using pair-diff unfolding fri-fold-coeff-pair-def by simp
  have fri-local-hiding-challenges  $xp \ xn \ yp \ yn \ denom =$ 
     $\{b. ?a + b * ?c = ?a' + b * ?c'\}$ 
  unfolding fri-local-hiding-challenges-def fri-fold-value-def by simp
  then show ?thesis
    using fri-affine-collision-challenges-card-le-one[OF pair-diff'] by simp
qed

lemma fri-one-step-disagreement-hiding-bound:
  assumes
    fri-fold-coeff-pair  $xp \ xn \ denom \neq$  fri-fold-coeff-pair  $yp \ yn \ denom$ 
  shows card (fri-local-hiding-challenges  $xp \ xn \ yp \ yn \ denom$ )  $\leq 1$ 
  by (rule fri-local-hiding-challenges-card-le-one[OF assms])

lemma fri-one-step-disagreement-outside-hiding:
  assumes  $b \notin$  fri-local-hiding-challenges  $xp \ xn \ yp \ yn \ denom$ 
  shows fri-fold-value  $b \ xp \ xn \ denom \neq$  fri-fold-value  $b \ yp \ yn \ denom$ 
  using assms unfolding fri-local-hiding-challenges-def by simp

definition fri-table-local-hiding-challenges
  :: 'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f list  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$ 
    'f set
where
  fri-table-local-hiding-challenges layer claimed fri-dom len pw  $i =$ 
    fri-local-hiding-challenges

```

$(\text{layer} \ ! \ i) \ (\text{layer} \ ! \ \text{fri-sibling-index} \ \text{len} \ i)$
 $(\text{claimed} \ ! \ i) \ (\text{claimed} \ ! \ \text{fri-sibling-index} \ \text{len} \ i)$
 $(\text{fri-fold-denominator} \ (\text{fri-dom} \ ! \ i) \ pw)$

lemma *fri-table-local-hiding-challenges-card-le-one*:

assumes

$\text{fri-fold-coeff-pair}$
 $(\text{layer} \ ! \ i) \ (\text{layer} \ ! \ \text{fri-sibling-index} \ \text{len} \ i)$
 $(\text{fri-fold-denominator} \ (\text{fri-dom} \ ! \ i) \ pw) \neq$
 $\text{fri-fold-coeff-pair}$
 $(\text{claimed} \ ! \ i) \ (\text{claimed} \ ! \ \text{fri-sibling-index} \ \text{len} \ i)$
 $(\text{fri-fold-denominator} \ (\text{fri-dom} \ ! \ i) \ pw)$

shows $\text{card} \ (\text{fri-table-local-hiding-challenges} \ \text{layer} \ \text{claimed} \ \text{fri-dom} \ \text{len} \ pw \ i)$
 ≤ 1

unfolding *fri-table-local-hiding-challenges-def*

by $(\text{rule} \ \text{fri-local-hiding-challenges-card-le-one}[\text{OF} \ \text{assms}])$

definition *fri-one-step-disagreement-challenges*

$:: \text{nat} \Rightarrow \text{nat} \Rightarrow 'f \ \text{list} \Rightarrow 'f \ \text{list} \Rightarrow 'f \ \text{list} \Rightarrow 'f \ \text{set}$

where

$\text{fri-one-step-disagreement-challenges} \ \text{len} \ pw \ \text{layer} \ \text{claimed} \ \text{fri-dom} =$
 $\{b. \exists i < \text{len} \ \text{div} \ 2.$
 $\text{fri-fold-coeff-pair}$
 $(\text{layer} \ ! \ i) \ (\text{layer} \ ! \ \text{fri-sibling-index} \ \text{len} \ i)$
 $(\text{fri-fold-denominator} \ (\text{fri-dom} \ ! \ i) \ pw) \neq$
 $\text{fri-fold-coeff-pair}$
 $(\text{claimed} \ ! \ i) \ (\text{claimed} \ ! \ \text{fri-sibling-index} \ \text{len} \ i)$
 $(\text{fri-fold-denominator} \ (\text{fri-dom} \ ! \ i) \ pw) \wedge$
 $b \in \text{fri-table-local-hiding-challenges} \ \text{layer} \ \text{claimed} \ \text{fri-dom} \ \text{len} \ pw \ i\}$

lemma *fri-one-step-disagreement-challenges-card-bound*:

$\text{card} \ (\text{fri-one-step-disagreement-challenges} \ \text{len} \ pw \ \text{layer} \ \text{claimed} \ \text{fri-dom})$
 $\leq \text{CARD}('f)$

proof –

have $\text{finite-univ} : \text{finite} \ (UNIV :: 'f \ \text{set})$

by *simp*

have $\text{card} \ (\text{fri-one-step-disagreement-challenges} \ \text{len} \ pw \ \text{layer} \ \text{claimed} \ \text{fri-dom})$
 $\leq \text{card} \ (UNIV :: 'f \ \text{set})$

by $(\text{rule} \ \text{card-mono}[\text{OF} \ \text{finite-univ}]) \ \text{auto}$

then show *?thesis* **by** *simp*

qed

lemma *fri-one-step-disagreement-challenges-card-bound-local*:

$\text{card} \ (\text{fri-one-step-disagreement-challenges} \ \text{len} \ pw \ \text{layer} \ \text{claimed} \ \text{fri-dom})$
 $\leq \text{len} \ \text{div} \ 2$

proof –

let $?S =$

$\lambda i. \text{if}$
 $\text{fri-fold-coeff-pair}$

```

      (layer ! i) (layer ! fri-sibling-index len i)
      (fri-fold-denominator (fri-dom ! i) pw) ≠
    fri-fold-coeff-pair
      (claimed ! i) (claimed ! fri-sibling-index len i)
      (fri-fold-denominator (fri-dom ! i) pw)
    then fri-table-local-hiding-challenges layer claimed fri-dom len pw i
  else {}
have set-eq:
  fri-one-step-disagreement-challenges len pw layer claimed fri-dom =
    (⋃ i ∈ {.. $\text{len div } 2$ }. ?S i)
  unfolding fri-one-step-disagreement-challenges-def by auto
have local-bound:
   $\bigwedge i. i < \text{len div } 2 \implies \text{card } (?S i) \leq 1$ 
proof -
  fix i
  assume i < len div 2
  show card (?S i) ≤ 1
  proof (cases
    fri-fold-coeff-pair
      (layer ! i) (layer ! fri-sibling-index len i)
      (fri-fold-denominator (fri-dom ! i) pw) ≠
    fri-fold-coeff-pair
      (claimed ! i) (claimed ! fri-sibling-index len i)
      (fri-fold-denominator (fri-dom ! i) pw))
  case True
  then have card
    (fri-table-local-hiding-challenges layer claimed fri-dom len pw i)
    ≤ 1
    by (rule fri-table-local-hiding-challenges-card-le-one)
  then show ?thesis
    using True by simp
  next
  case False
  then show ?thesis by simp
qed
qed
have card (fri-one-step-disagreement-challenges len pw layer claimed fri-dom) =
  card (⋃ i ∈ {.. $\text{len div } 2$ }. ?S i)
  using set-eq by simp
also have ... ≤ (∑ i ∈ {.. $\text{len div } 2$ }. card (?S i))
  by (rule card-UN-le) simp
also have ... ≤ (∑ i ∈ {.. $\text{len div } 2$ }. 1)
  by (rule sum-mono) (rule local-bound, simp)
also have ... = len div 2
  by simp
finally show ?thesis .
qed

```

definition *fri-multiround-bad-challenge-lists*

```

:: nat ⇒ (nat ⇒ 'f list ⇒ 'f set) ⇒ 'f list set
where
  fri-mulround-bad-challenge-lists n bad =
    {bs ∈ fri-challenge-space n.
     ∃ i < n. bs ! i ∈ bad i (take i bs)}

definition fri-bad-challenge-lists-bounded-by
:: nat ⇒ 'f list set ⇒ nat ⇒ bool
where
  fri-bad-challenge-lists-bounded-by n bad bd ⇔
    bad ⊆ fri-challenge-space n ∧ card bad ≤ bd

lemma fri-mulround-bad-challenge-lists-subset:
  fri-mulround-bad-challenge-lists n bad ⊆ fri-challenge-space n
unfolding fri-mulround-bad-challenge-lists-def by auto

lemma fri-mulround-bad-challenge-lists-card-bound:
  card (fri-mulround-bad-challenge-lists n bad) ≤ CARD('f) ^ n
proof -
  have card (fri-mulround-bad-challenge-lists n bad) ≤
    card (fri-challenge-space n)
  by (rule card-mono[OF finite-fri-challenge-space])
    (auto simp: fri-mulround-bad-challenge-lists-subset)
  also have ... = CARD('f) ^ n
  by (rule card-fri-challenge-space)
  finally show ?thesis .
qed

lemma fri-mulround-bad-challenge-lists-bounded-by:
  fri-bad-challenge-lists-bounded-by n
  (fri-mulround-bad-challenge-lists n bad) (CARD('f) ^ n)
unfolding fri-bad-challenge-lists-bounded-by-def
using fri-mulround-bad-challenge-lists-subset
  fri-mulround-bad-challenge-lists-card-bound
by simp

definition fri-final-constant-consistent :: 'f list ⇒ 'f ⇒ bool
where
  fri-final-constant-consistent layer final ⇔
    (∀ x ∈ set layer. x = final)

lemma fri-final-constant-consistent-nth:
assumes fri-final-constant-consistent layer final
and i < length layer
shows layer ! i = final
using assms unfolding fri-final-constant-consistent-def
by (simp add: nth-mem)

lemma fri-final-constant-consistent-hd:

```

assumes *fri-final-constant-consistent layer final*
and *layer* $\neq []$
shows *hd layer = final*
using *assms fri-final-constant-consistent-nth[of layer final 0]*
by (*cases layer*) *auto*

definition *fri-table-low-degree-on* :: *nat* $\Rightarrow$ '*f list* $\Rightarrow$ '*f list* $\Rightarrow$ *bool*
where
fri-table-low-degree-on d fri-dom table $\longleftrightarrow$
 $(\exists p. \text{degree } p \leq d \wedge \text{table} = \text{map } (\text{poly } p) \text{ fri-dom})$

fun *fri-degree-after* :: *nat* $\Rightarrow$ *nat* $\Rightarrow$ *nat*
where
fri-degree-after 0 d = *d*
| *fri-degree-after (Suc n) d* = *fri-degree-after n d div 2*

lemma *fri-degree-after-Suc[simp]*:
fri-degree-after (Suc n) d = *fri-degree-after n d div 2*
by *simp*

definition *fri-one-round-proximity-step*
:: *nat* $\Rightarrow$ *nat* $\Rightarrow$ *nat* $\Rightarrow$ '*f list* $\Rightarrow$ '*f list* $\Rightarrow$
'*f list* $\Rightarrow$ '*f list* $\Rightarrow$ '*f set* $\Rightarrow$ *bool*
where
fri-one-round-proximity-step d len pw fri-dom layer next-dom next-layer bad $\longleftrightarrow$
 $(\forall b. b \notin \text{bad} \longrightarrow$
fri-algebraic-layer-relation len pw b fri-dom layer next-dom next-layer $\longrightarrow$
fri-table-low-degree-on (d div 2) next-dom next-layer $\longrightarrow$
fri-table-low-degree-on d fri-dom layer)

lemma *fri-one-round-proximity-step-bad-or-next-not-low*:
assumes *step*:
fri-one-round-proximity-step d len pw fri-dom layer next-dom next-layer bad
and *relation*:
fri-algebraic-layer-relation len pw b fri-dom layer next-dom next-layer
and *current-not-low*: $\neg \text{fri-table-low-degree-on } d \text{ fri-dom layer}$
and *challenge-not-bad*: $b \notin \text{bad}$
shows $\neg \text{fri-table-low-degree-on } (d \text{ div } 2) \text{ next-dom next-layer}$
using *assms unfolding fri-one-round-proximity-step-def* **by** *blast*

definition *fri-multiround-proximity-steps*
:: *nat* $\Rightarrow$ *nat* $\Rightarrow$ *nat list* $\Rightarrow$ *nat list* $\Rightarrow$
'*f list list* $\Rightarrow$ '*f list list* $\Rightarrow$
 $(\text{nat} \Rightarrow \text{'f list} \Rightarrow \text{'f set}) \Rightarrow \text{bool}$
where
fri-multiround-proximity-steps n d lens pws doms layers bad $\longleftrightarrow$
 $\text{length lens} = n \wedge$
 $\text{length pws} = n \wedge$
 $\text{length doms} = \text{Suc } n \wedge$

$length\ layers = Suc\ n \wedge$
 $(\forall bs \in fri_challenge_space\ n.$
 $\quad \forall i < n.$
 $\quad\quad fri_one_round_proximity_step\ (fri_degree_after\ i\ d)$
 $\quad\quad\quad (lens\ !\ i)\ (pws\ !\ i)\ (doms\ !\ i)\ (layers\ !\ i)$
 $\quad\quad\quad (doms\ !\ Suc\ i)\ (layers\ !\ Suc\ i)\ (bad\ i\ (take\ i\ bs)))$

lemma *fri-multiround-bad-challenge-lists-notinD*:
assumes *challenges*: $bs \in fri_challenge_space\ n$
and *not-bad*: $bs \notin fri_multiround_bad_challenge_lists\ n\ bad$
and *i-bound*: $i < n$
shows $bs\ !\ i \notin bad\ i\ (take\ i\ bs)$
using *assms* **unfolding** *fri-multiround-bad-challenge-lists-def* **by** *blast*

lemma *fri-multiround-proximity-skeleton*:
assumes *steps*: *fri-multiround-proximity-steps* $n\ d\ lens\ pws\ doms\ layers\ bad$
and *challenges*: $bs \in fri_challenge_space\ n$
and *relations*:
 $\bigwedge i. i < n \implies$
 $\quad fri_algebraic_layer_relation\ (lens\ !\ i)\ (pws\ !\ i)\ (bs\ !\ i)$
 $\quad\quad (doms\ !\ i)\ (layers\ !\ i)\ (doms\ !\ Suc\ i)\ (layers\ !\ Suc\ i)$
and *start-not-low*:
 $\neg fri_table_low_degree_on\ d\ (doms\ !\ 0)\ (layers\ !\ 0)$
and *final-low*:
 $fri_table_low_degree_on\ (fri_degree_after\ n\ d)\ (doms\ !\ n)\ (layers\ !\ n)$
shows $bs \in fri_multiround_bad_challenge_lists\ n\ bad$
proof (*rule ccontr*)
assume *not-bad*: $bs \notin fri_multiround_bad_challenge_lists\ n\ bad$
have *step-at*:
 $\bigwedge i. i < n \implies$
 $\quad fri_one_round_proximity_step\ (fri_degree_after\ i\ d)$
 $\quad\quad (lens\ !\ i)\ (pws\ !\ i)\ (doms\ !\ i)\ (layers\ !\ i)$
 $\quad\quad (doms\ !\ Suc\ i)\ (layers\ !\ Suc\ i)\ (bad\ i\ (take\ i\ bs))$
using *steps* *challenges* **unfolding** *fri-multiround-proximity-steps-def* **by** *blast*
have *not-low*:
 $\bigwedge i. i \leq n \implies$
 $\quad \neg fri_table_low_degree_on\ (fri_degree_after\ i\ d)\ (doms\ !\ i)\ (layers\ !\ i)$
proof –
fix *i*
assume $i \leq n$
then show
 $\neg fri_table_low_degree_on\ (fri_degree_after\ i\ d)\ (doms\ !\ i)\ (layers\ !\ i)$
proof (*induction i*)
case 0
then show ?*case*
using *start-not-low* **by** *simp*
next
case (*Suc i*)
have *i-bound*: $i < n$

```

    using Suc.premis by simp
  have prev-not-low:
    ¬ fri-table-low-degree-on (fri-degree-after i d)
      (doms ! i) (layers ! i)
    using Suc.IH Suc.premis by simp
  have challenge-not-bad: bs ! i ∉ bad i (take i bs)
    by (rule fri-multiround-bad-challenge-lists-notinD
      [OF challenges not-bad i-bound])
  have next-not-low:
    ¬ fri-table-low-degree-on (fri-degree-after i d div 2)
      (doms ! Suc i) (layers ! Suc i)
    by (rule fri-one-round-proximity-step-bad-or-next-not-low
      [OF step-at [OF i-bound] relations [OF i-bound] prev-not-low
        challenge-not-bad])
  then show ?case
    by simp
qed
qed
have ¬ fri-table-low-degree-on (fri-degree-after n d) (doms ! n) (layers ! n)
  by (rule not-low) simp
then show False
  using final-low by contradiction
qed

```

definition *fri-round-count-for-degree-bound* :: *nat* ⇒ *nat*
 where *fri-round-count-for-degree-bound* d = *ceil-log* (d + 1)

definition *trace-fri-algebraic-round-count* :: *nat*
 where
trace-fri-algebraic-round-count =
fri-round-count-for-degree-bound (clength - 1)

definition *composition-fri-algebraic-round-count* :: 'f ⇒ *nat*
 where
composition-fri-algebraic-round-count dg =
fri-round-count-for-degree-bound (to-nat dg)

lemma *trace-fri-algebraic-round-count-eq*:
trace-fri-algebraic-round-count = *ceil-log* clength
unfolding *trace-fri-algebraic-round-count-def*
fri-round-count-for-degree-bound-def
using *clength-pos* **by** *simp*

lemma *composition-fri-algebraic-round-count-eq*:
composition-fri-algebraic-round-count dg =
ceil-log (to-nat dg + 1)
unfolding *composition-fri-algebraic-round-count-def*
fri-round-count-for-degree-bound-def
by *simp*

lemma *trace-fri-algebraic-challenge-space*:
 $\text{fri-challenge-space } \text{trace-fri-algebraic-round-count} =$
 $\text{fri-challenge-space } (\text{ceil-log } \text{clength})$
by (*simp add: trace-fri-algebraic-round-count-eq*)

lemma *composition-fri-algebraic-challenge-space*:
 $\text{fri-challenge-space } (\text{composition-fri-algebraic-round-count } dg) =$
 $\text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
by (*simp add: composition-fri-algebraic-round-count-eq*)

definition *trace-fri-multiround-bad-sets*
 $:: ('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list set}$
where
 $\text{trace-fri-multiround-bad-sets } \text{bad } \text{trace-table} =$
 $\text{fri-multiround-bad-challenge-lists } \text{trace-fri-algebraic-round-count}$
 $(\text{bad } \text{trace-table})$

definition *composition-fri-multiround-bad-sets*
 $:: ('f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list set}$
where
 $\text{composition-fri-multiround-bad-sets } \text{bad } dg \text{ composition-table} =$
 $\text{fri-multiround-bad-challenge-lists}$
 $(\text{composition-fri-algebraic-round-count } dg)$
 $(\text{bad } dg \text{ composition-table})$

lemma *trace-fri-multiround-bad-sets-subset*:
 $\text{trace-fri-multiround-bad-sets } \text{bad } \text{trace-table} \subseteq$
 $\text{fri-challenge-space } (\text{ceil-log } \text{clength})$
proof –
have $\text{trace-fri-multiround-bad-sets } \text{bad } \text{trace-table} \subseteq$
 $\text{fri-challenge-space } \text{trace-fri-algebraic-round-count}$
unfolding *trace-fri-multiround-bad-sets-def*
by (*rule fri-multiround-bad-challenge-lists-subset*)
also have $\dots = \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
by (*rule trace-fri-algebraic-challenge-space*)
finally show *?thesis* .
qed

lemma *composition-fri-multiround-bad-sets-subset*:
 $\text{composition-fri-multiround-bad-sets } \text{bad } dg \text{ composition-table} \subseteq$
 $\text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
proof –
have $\text{composition-fri-multiround-bad-sets } \text{bad } dg \text{ composition-table} \subseteq$
 $\text{fri-challenge-space } (\text{composition-fri-algebraic-round-count } dg)$
unfolding *composition-fri-multiround-bad-sets-def*
by (*rule fri-multiround-bad-challenge-lists-subset*)

also have ... = fri-challenge-space (ceil-log (to-nat dg + 1))
 by (rule composition-fri-algebraic-challenge-space)
 finally show ?thesis .
 qed

lemma trace-fri-multiround-bad-sets-card-bound:
 card (trace-fri-multiround-bad-sets bad trace-table) ≤
 CARD('f) ^ ceil-log clength
unfolding trace-fri-multiround-bad-sets-def
using fri-multiround-bad-challenge-lists-card-bound
by (simp add: trace-fri-algebraic-round-count-eq)

lemma composition-fri-multiround-bad-sets-card-bound:
 card (composition-fri-multiround-bad-sets bad dg composition-table) ≤
 CARD('f) ^ ceil-log (to-nat dg + 1)
unfolding composition-fri-multiround-bad-sets-def
using fri-multiround-bad-challenge-lists-card-bound
by (simp add: composition-fri-algebraic-round-count-eq)

definition accepted-fri-challenges
 :: ('f, 'a) protocol-channel-scheme ⇒
 (unit list × ('f, 'a) protocol-channel-scheme) option ⇒
 'f list ⇒ 'f ⇒ 'f list ⇒ bool
where

accepted-fri-challenges s out trace-bs dg comp-bs ⟷
 (∃ result final-state fr f-fl f-final as fl final query-state.
 out = Some (result, final-state) ∧
 verifier-header-transcript s fr (map snd f-fl) f-final as dg
 (map snd fl) final (PTranscript query-state) ∧
 PState query-state =
 verifier-header-state s fr (map snd f-fl) f-final as dg
 (map snd fl) final ∧
 Some (result, final-state) ∈
 set-dist (execute
 (ntimes
 (verifier-query-round-program fr f-fl f-final as fl final)
 rounds)
 query-state) ∧
 PQueryCounter query-state = PQueryCounter s ∧
 trace-bs = map fst f-fl ∧
 comp-bs = map fst fl ∧
 (∀ i < length f-fl.
 fmlookup (HashMap query-state)
 (TraceFriChallenge (PTraceFriCounter s + i) (foldl concat (concat (PState s) fr)
 (take (Suc i) (map snd f-fl)))) =
 Some (fst (f-fl ! i))) ∧
 (∀ i < length fl.
 fmlookup (HashMap query-state)
 (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat

```

      (concat
        (foldl concat
          (concat
            (foldl concat (concat (PState s) fr) (map snd f-fl))
            f-final)
          as)
        dg)
      (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i)))

```

lemma *verify-monad-accepted-fri-challenges*:

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *outcome*:

$\text{Some (result, final-state)} \in \text{set-dist (execute verify-monad s)}$

obtains *trace-bs dg comp-bs*

where *accepted-fri-challenges s (Some (result, final-state))*

trace-bs dg comp-bs

proof –

from *verify-monad-header-random-oracle-replay[OF outcome]*

obtain *fr f-fl f-final as dg fl final query-state* **where**

header:

verifier-header-transcript s fr (map snd f-fl) f-final as dg
(map snd fl) final (PTranscript query-state)

and *query-state-eq*:

PState query-state =
verifier-header-state s fr (map snd f-fl) f-final as dg
(map snd fl) final

and *query-out*:

Some (result, final-state) ∈
set-dist (execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final)
rounds)
query-state)

and *query-count-header*:

PQueryCounter query-state = PQueryCounter s

and *ext-s-query-header: s ≤ query-state*

and *trace-lookup*:

$\bigwedge i. i < \text{length } f\text{-fl} \implies$
fmlookup (HashMap query-state)
(TraceFriChallenge (PTraceFriCounter s + i) (foldl concat (concat (PState s) fr)
(take (Suc i) (map snd f-fl)))) =
Some (fst (f-fl ! i))

and *alpha-lookup*:

$\bigwedge i. i < \text{length } as \implies$
fmlookup (HashMap query-state)
(AlphaChallenge (PAlphaCounter s + i) (foldl concat
(concat (foldl concat (concat (PState s) fr) (map snd f-fl))
f-final)

```

      (take i as))) =
    Some (as ! i)
and comp-lookup:
   $\bigwedge i. i < \text{length } fl \implies$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
        (concat
          (foldl concat
            (concat
              (foldl concat (concat (PState s) fr) (map snd f-fl))
              f-final)
            as)
          dg)
        (take (Suc i) (map snd fl)))))) =
    Some (fst (fl ! i))
and degree-bound: to-nat dg  $\leq$  maxDegree
by blast
have accepted-fri-challenges s (Some (result, final-state))
  (map fst f-fl) dg (map fst fl)
proof -
have trace-lookup-all:
   $\forall i < \text{length } f\text{-}fl.$ 
    fmlookup (HashMap query-state)
      (TraceFriChallenge (PTraceFriCounter s + i) (foldl concat (concat (PState s) fr)
        (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
using trace-lookup by blast
have comp-lookup-all:
   $\forall i < \text{length } fl.$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
        (concat
          (foldl concat
            (concat
              (foldl concat (concat (PState s) fr) (map snd f-fl))
              f-final)
            as)
          dg)
        (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i))
using comp-lookup by blast
show ?thesis
unfolding accepted-fri-challenges-def
by (intro exI[of - result] exI[of - final-state] exI[of - fr]
  exI[of - f-fl] exI[of - f-final] exI[of - as] exI[of - fl]
  exI[of - final] exI[of - query-state])
  (use header query-state-eq query-out query-count-header trace-lookup-all
    comp-lookup-all in simp)
qed

```

```

then show ?thesis
  by (rule that)
qed

lemma accepted-fri-challenges-trace-space:
  assumes accepted-fri-challenges s out trace-bs dg comp-bs
  shows trace-bs  $\in$  fri-challenge-space (ceil-log clength)
  using assms
  unfolding accepted-fri-challenges-def verifier-header-transcript-def
    fri-challenge-space-def
  by auto

lemma accepted-fri-challenges-composition-space:
  assumes accepted-fri-challenges s out trace-bs dg comp-bs
  shows comp-bs  $\in$  fri-challenge-space (ceil-log (to-nat dg + 1))
  using assms
  unfolding accepted-fri-challenges-def verifier-header-transcript-def
    fri-challenge-space-def
  by auto

lemma accepted-fri-challenges-unique:
  assumes c1:
    accepted-fri-challenges s (Some (result, final-state))
    trace-bs dg comp-bs
  and c2:
    accepted-fri-challenges s (Some (result, final-state))
    trace-bs' dg' comp-bs'
  shows trace-bs' = trace-bs  $\wedge$  dg' = dg  $\wedge$  comp-bs' = comp-bs
proof –
from c1 obtain fr f-fl f-final as fl final query-state where
  header1:
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
  and query-state1:
    PState query-state =
      verifier-header-state s fr (map snd f-fl) f-final as dg
      (map snd fl) final
  and query-out1:
    Some (result, final-state)  $\in$ 
      set-dist (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
  and trace-bs-eq: trace-bs = map fst f-fl
  and comp-bs-eq: comp-bs = map fst fl
  and trace-lookup1:
     $\bigwedge i. i < \text{length } f\text{-fl} \implies$ 
      fmlookup (HashMap query-state)

```

```

    (TraceFriChallenge (PTraceFriCounter s + i)
      (foldl concat (concat (PState s) fr)
        (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
and comp-lookup1:
   $\bigwedge i. i < \text{length } fl \implies$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter s + i)
        (foldl concat
          (concat
            (foldl concat
              (concat
                (foldl concat (concat (PState s) fr) (map snd f-fl))
                f-final)
              as)
            dg)
          (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i))
unfolding accepted-fri-challenges-def by blast
from c2 obtain fr' f-fl' f-final' as' fl' final' query-state' where
  header2:
    verifier-header-transcript s fr' (map snd f-fl') f-final' as' dg'
    (map snd fl') final' (PTranscript query-state')
and query-state2:
  PState query-state' =
    verifier-header-state s fr' (map snd f-fl') f-final' as' dg'
    (map snd fl') final'
and query-out2:
  Some (result, final-state)  $\in$ 
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr' f-fl' f-final' as' fl' final')
        rounds)
      query-state')
and trace-bs'-eq: trace-bs' = map fst f-fl'
and comp-bs'-eq: comp-bs' = map fst fl'
and trace-lookup2:
   $\bigwedge i. i < \text{length } f-fl' \implies$ 
    fmlookup (HashMap query-state')
      (TraceFriChallenge (PTraceFriCounter s + i)
        (foldl concat (concat (PState s) fr')
          (take (Suc i) (map snd f-fl')))) =
    Some (fst (f-fl' ! i))
and comp-lookup2:
   $\bigwedge i. i < \text{length } fl' \implies$ 
    fmlookup (HashMap query-state')
      (CompositionFriChallenge (PCompositionFriCounter s + i)
        (foldl concat
          (concat

```



```

      (foldl concat
        (concat
          (foldl concat (concat (PState s) fr') (map snd f-fl'))
          f-final')
        as')
      dg')
    (take (Suc i) (map snd fl')))) =
  Some (fst (fl' ! i))
unfolding accepted-fri-challenges-def by blast
have header-unique:
  fr' = fr ∧
  map snd f-fl' = map snd f-fl ∧
  f-final' = f-final ∧
  as' = as ∧
  dg' = dg ∧
  map snd fl' = map snd fl ∧
  final' = final ∧
  PTranscript query-state' = PTranscript query-state
by (rule verifier-header-transcript-unique[OF header1 header2])
have ext1: query-state ≤ final-state
  using ntimes-verifier-query-rounds-outcome[OF query-out1] by blast
have ext2: query-state' ≤ final-state
  using ntimes-verifier-query-rounds-outcome[OF query-out2] by blast
have len-f-fl': length f-fl' = length f-fl
  using header-unique by (metis length-map)
have f-fl-fst-eq: map fst f-fl' = map fst f-fl
proof (rule nth-equalityI)
  show length (map fst f-fl') = length (map fst f-fl)
  using len-f-fl' by simp
next
fix i
assume i-bound: i < length (map fst f-fl')
then have i-bound': i < length f-fl'
  by simp
have i-bound-old: i < length f-fl
  using i-bound' len-f-fl' by simp
let ?key =
  TraceFriChallenge (PTraceFriCounter s + i)
  (foldl concat (concat (PState s) fr)
    (take (Suc i) (map snd f-fl)))
have lookup1-final:
  fmlookup (HashMap final-state) ?key =
  Some (fst (f-fl ! i))
  by (rule hash-extension-lookup[OF trace-lookup1[OF i-bound-old] ext1])
have key-eq:
  TraceFriChallenge (PTraceFriCounter s + i)
  (foldl concat (concat (PState s) fr')
    (take (Suc i) (map snd f-fl')))) = ?key
  using header-unique by simp

```

```

have lookup2-final:
  fmlookup (HashMap final-state) ?key =
    Some (fst (f-fl' ! i))
  using hash-extension-lookup[OF trace-lookup2[OF i-bound'] ext2]
  unfolding key-eq .
show map fst f-fl' ! i = map fst f-fl ! i
  using lookup1-final lookup2-final i-bound' i-bound-old by simp
qed
have len-fl': length fl' = length fl
  using header-unique by (metis length-map)
have fl-fst-eq: map fst fl' = map fst fl
proof (rule nth-equalityI)
  show length (map fst fl') = length (map fst fl)
  using len-fl' by simp
next
fix i
assume i-bound: i < length (map fst fl')
then have i-bound': i < length fl'
  by simp
have i-bound-old: i < length fl
  using i-bound' len-fl' by simp
let ?key =
  CompositionFriChallenge (PCompositionFriCounter s + i)
  (foldl concat
    (concat
      (foldl concat
        (concat
          (foldl concat (concat (PState s) fr) (map snd f-fl))
          f-final)
        as)
      dg)
    (take (Suc i) (map snd fl)))
have lookup1-final:
  fmlookup (HashMap final-state) ?key =
    Some (fst (fl ! i))
  by (rule hash-extension-lookup[OF comp-lookup1[OF i-bound-old] ext1])
have key-eq:
  CompositionFriChallenge (PCompositionFriCounter s + i)
  (foldl concat
    (concat
      (foldl concat
        (concat
          (foldl concat (concat (PState s) fr') (map snd f-fl'))
          f-final')
        as')
      dg')
    (take (Suc i) (map snd fl'))) = ?key
  using header-unique by simp
have lookup2-final:

```

```

    fmlookup (HashMap final-state) ?key =
      Some (fst (fl' ! i))
    using hash-extension-lookup[OF comp-lookup2[OF i-bound'] ext2]
    unfolding key-eq .
  show map fst fl' ! i = map fst fl ! i
    using lookup1-final lookup2-final i-bound' i-bound-old by simp
qed
show ?thesis
  using header-unique trace-bs-eq trace-bs'-eq comp-bs-eq comp-bs'-eq
    f-fl-fst-eq fl-fst-eq
  by simp
qed

lemma verify-monad-trace-fri-prequery-lookup:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and challenges:
    accepted-fri-challenges s (Some (result, final-state))
    trace-bs dg comp-bs
  and header:
    verifier-header-transcript s fr roots trace-final as dg
    composition-roots final rest
  and i-len: i < length roots
  and lookup-s:
    fmlookup (HashMap s)
      (TraceFriChallenge (PTraceFriCounter s + i)
        (foldl concat (concat (PState s) fr) (take (Suc i) roots))) =
      Some y
  shows i < length trace-bs ∧ y = trace-bs ! i
proof (rule verify-monad-header-random-oracle-replay[OF outcome])
  fix fr' f-fl trace-final' as' dg' fl' final' query-state
  assume header-replay:
    verifier-header-transcript s fr' (map snd f-fl) trace-final' as' dg'
    (map snd fl') final' (PTranscript query-state)
  and query-state-eq:
    PState query-state =
      verifier-header-state s fr' (map snd f-fl) trace-final' as' dg'
      (map snd fl') final'
  and query-out:
    Some (result, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr' f-fl trace-final' as' fl'
              final')
            rounds)
          query-state)
  and query-count-header: PQueryCounter query-state = PQueryCounter s

```

```

and ext-s-query:  $s \leq \text{query-state}$ 
and trace-lookup:
   $\bigwedge j. j < \text{length } f\text{-}fl \implies$ 
    fmlookup (HashMap query-state)
      (TraceFriChallenge (PTraceFriCounter  $s + j$ )
        (foldl concat (concat (PState  $s$ )  $fr'$ )
          (take (Suc  $j$ ) (map snd  $f\text{-}fl$ )))) =
        Some (fst ( $f\text{-}fl ! j$ ))
and comp-lookup:
   $\bigwedge j. j < \text{length } fl' \implies$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter  $s + j$ )
        (foldl concat
          (concat
            (foldl concat
              (concat
                (foldl concat (concat (PState  $s$ )  $fr'$ ) (map snd  $f\text{-}fl$ ))
                trace-final')
              as')
            dg')
          (take (Suc  $j$ ) (map snd  $fl'$ )))) =
        Some (fst ( $fl' ! j$ ))
and degree-bound: to-nat  $dg' \leq \text{maxDegree}$ 
have replay-challenges:
  accepted-fri-challenges  $s$  (Some (result, final-state))
    (map fst  $f\text{-}fl$ )  $dg'$  (map fst  $fl'$ )
unfolding accepted-fri-challenges-def
by (intro exI[of - result] exI[of - final-state] exI[of -  $fr'$ ]
  exI[of -  $f\text{-}fl$ ] exI[of - trace-final'] exI[of -  $as'$ ]
  exI[of -  $fl'$ ] exI[of - final'] exI[of - query-state])
  (use header-replay query-state-eq query-out query-count-header
    trace-lookup comp-lookup in auto)
have trace-bs-eq: trace-bs = map fst  $f\text{-}fl$ 
using accepted-fri-challenges-unique[OF replay-challenges challenges]
by simp
have header-unique:
   $fr = fr' \wedge$ 
   $roots = \text{map } \text{snd } f\text{-}fl \wedge$ 
   $\text{trace-final} = \text{trace-final}' \wedge$ 
   $as = as' \wedge$ 
   $dg = dg' \wedge$ 
   $\text{composition-roots} = \text{map } \text{snd } fl' \wedge$ 
   $\text{final} = \text{final}' \wedge$ 
   $\text{rest} = \text{PTranscript } \text{query-state}$ 
using verifier-header-transcript-unique[OF header header-replay]
by simp
have i-f-fl:  $i < \text{length } f\text{-}fl$ 
using i-len header-unique by simp
have lookup-query-from-s:

```

```

fmlookup (HashMap query-state)
  (TraceFriChallenge (PTraceFriCounter s + i)
    (foldl concat (concat (PState s) fr) (take (Suc i) roots))) =
  Some y
by (rule hash-extension-lookup[OF lookup-s ext-s-query])
have lookup-query-from-trace:
fmlookup (HashMap query-state)
  (TraceFriChallenge (PTraceFriCounter s + i)
    (foldl concat (concat (PState s) fr) (take (Suc i) roots))) =
  Some (trace-bs ! i)
using trace-lookup[OF i-f-fl] header-unique trace-bs-eq i-f-fl
by simp
then have y-eq: y = trace-bs ! i
  using lookup-query-from-s by simp
have i < length trace-bs
  using i-f-fl trace-bs-eq by simp
then show ?thesis
  using y-eq by simp
qed

```

```

lemma verify-monad-composition-fri-prequery-lookup:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and challenges:
  accepted-fri-challenges s (Some (result, final-state))
  trace-bs dg comp-bs
and header:
  verifier-header-transcript s fr trace-roots trace-final as dg
  composition-roots final rest
and i-len: i < length composition-roots
and lookup-s:
fmlookup (HashMap s)
  (CompositionFriChallenge (PCompositionFriCounter s + i)
    (foldl concat
      (concat
        (foldl concat
          (concat
            (foldl concat (concat (PState s) fr) trace-roots)
            trace-final)
          as)
        dg)
      (take (Suc i) composition-roots))) =
  Some y
shows i < length comp-bs ∧ y = comp-bs ! i
proof (rule verify-monad-header-random-oracle-replay[OF outcome])
fix fr' f-fl trace-final' as' dg' fl' final' query-state
assume header-replay:
  verifier-header-transcript s fr' (map snd f-fl) trace-final' as' dg'

```

```

    (map snd fl') final' (PTranscript query-state)
and query-state-eq:
  PState query-state =
    verifier-header-state s fr' (map snd f-fl) trace-final' as' dg'
    (map snd fl') final'
and query-out:
  Some (result, final-state) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr' f-fl trace-final' as' fl'
          final')
        rounds)
      query-state)
and query-count-header: PQueryCounter query-state = PQueryCounter s
and ext-s-query: s ≤ query-state
and trace-lookup:
   $\bigwedge j. j < \text{length } f\text{-}fl \implies$ 
    fmlookup (HashMap query-state)
      (TraceFriChallenge (PTraceFriCounter s + j)
        (foldl concat (concat (PState s) fr')
          (take (Suc j) (map snd f-fl)))) =
      Some (fst (f-fl ! j))
and comp-lookup:
   $\bigwedge j. j < \text{length } fl' \implies$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter s + j)
        (foldl concat
          (concat
            (foldl concat
              (concat
                (foldl concat (concat (PState s) fr') (map snd f-fl))
                trace-final')
              as')
            dg')
          (take (Suc j) (map snd fl')))) =
      Some (fst (fl' ! j))
and degree-bound: to-nat dg' ≤ maxDegree
have replay-challenges:
  accepted-fri-challenges s (Some (result, final-state))
    (map fst f-fl) dg' (map fst fl')
unfolding accepted-fri-challenges-def
by (intro exI[of - result] exI[of - final-state] exI[of - fr']
    exI[of - f-fl] exI[of - trace-final'] exI[of - as']
    exI[of - fl'] exI[of - final'] exI[of - query-state])
  (use header-replay query-state-eq query-out query-count-header
    trace-lookup comp-lookup in auto)
have comp-bs-eq: comp-bs = map fst fl'
using accepted-fri-challenges-unique[OF replay-challenges challenges]

```

```

by simp
have header-unique:
  fr = fr' ∧
  trace-roots = map snd f-fl ∧
  trace-final = trace-final' ∧
  as = as' ∧
  dg = dg' ∧
  composition-roots = map snd fl' ∧
  final = final' ∧
  rest = PTranscript query-state
using verifier-header-transcript-unique[OF header header-replay]
by simp
have i-fl: i < length fl'
  using i-len header-unique by simp
have lookup-query-from-s:
  fmlookup (HashMap query-state)
    (CompositionFriChallenge (PCompositionFriCounter s + i)
      (foldl concat
        (concat
          (foldl concat
            (concat
              (foldl concat (concat (PState s) fr) trace-roots)
              trace-final)
            as)
          dg)
        (take (Suc i) composition-roots)))) =
    Some y
  by (rule hash-extension-lookup[OF lookup-s ext-s-query])
have lookup-query-from-comp:
  fmlookup (HashMap query-state)
    (CompositionFriChallenge (PCompositionFriCounter s + i)
      (foldl concat
        (concat
          (foldl concat
            (concat
              (foldl concat (concat (PState s) fr) trace-roots)
              trace-final)
            as)
          dg)
        (take (Suc i) composition-roots)))) =
    Some (comp-bs ! i)
  using comp-lookup[OF i-fl] header-unique comp-bs-eq i-fl
by simp
then have y-eq: y = comp-bs ! i
  using lookup-query-from-s by simp
have i < length comp-bs
  using i-fl comp-bs-eq by simp
then show ?thesis
  using y-eq by simp

```

qed

lemma *verify-monad-accepted-fri-challenges-degree-bound*:
fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$
assumes *outcome*:
 $\text{Some } (\text{result}, \text{final-state}) \in \text{set-dist } (\text{execute } \text{verify-monad } s)$
and *challenges*:
 $\text{accepted-fri-challenges } s (\text{Some } (\text{result}, \text{final-state}))$
 $\text{trace-bs } dg \text{ comp-bs}$
shows $\text{to-nat } dg \leq \text{maxDegree}$
proof –
from *verify-monad-header-random-oracle-replay* [*OF outcome*]
obtain $\text{fr } f\text{-fl } f\text{-final as } dg' \text{ fl final query-state}$ **where**
header:
 $\text{verifier-header-transcript } s \text{ fr } (\text{map } \text{snd } f\text{-fl}) \text{ } f\text{-final as } dg'$
 $(\text{map } \text{snd } \text{fl}) \text{ final } (P\text{Transcript } \text{query-state})$
and *query-state-eq*:
 $P\text{State } \text{query-state} =$
 $\text{verifier-header-state } s \text{ fr } (\text{map } \text{snd } f\text{-fl}) \text{ } f\text{-final as } dg'$
 $(\text{map } \text{snd } \text{fl}) \text{ final}$
and *query-out*:
 $\text{Some } (\text{result}, \text{final-state}) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{ntimes}$
 $(\text{verifier-query-round-program } \text{fr } f\text{-fl } f\text{-final as } \text{fl final})$
 $\text{rounds})$
 $\text{query-state})$
and *query-count-header*:
 $P\text{QueryCounter } \text{query-state} = P\text{QueryCounter } s$
and *trace-lookup*:
 $\bigwedge i. i < \text{length } f\text{-fl} \implies$
 $\text{fmlookup } (\text{HashMap } \text{query-state})$
 $(\text{TraceFriChallenge } (P\text{TraceFriCounter } s + i) (\text{foldl } \text{concat}$
 $(\text{concat } (P\text{State } s) \text{ fr } (\text{take } (\text{Suc } i) (\text{map } \text{snd } f\text{-fl})))) =$
 $\text{Some } (\text{fst } (f\text{-fl } ! i))$
and *comp-lookup*:
 $\bigwedge i. i < \text{length } \text{fl} \implies$
 $\text{fmlookup } (\text{HashMap } \text{query-state})$
 $(\text{CompositionFriChallenge } (P\text{CompositionFriCounter } s + i) (\text{foldl } \text{concat}$
 $(\text{concat}$
 $(\text{foldl } \text{concat}$
 $(\text{concat}$
 $(\text{foldl } \text{concat } (\text{concat } (P\text{State } s) \text{ fr } (\text{map } \text{snd } f\text{-fl}))$
 $f\text{-final})$
 $\text{as})$
 $dg')$
 $(\text{take } (\text{Suc } i) (\text{map } \text{snd } \text{fl})))) =$
 $\text{Some } (\text{fst } (\text{fl } ! i))$
and *degree-bound*: $\text{to-nat } dg' \leq \text{maxDegree}$


```

by metis
have trace-lookup-all:
   $\forall i < \text{length } f\text{-}fl.$ 
    fmlookup (HashMap query-state)
      (TraceFriChallenge (PTraceFriCounter s + i) (foldl concat
        (concat (PState s) fr) (take (Suc i) (map snd f-fl)))) =
      Some (fst (f-fl ! i))
    using trace-lookup by blast
have comp-lookup-all:
   $\forall i < \text{length } fl.$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
        (concat
          (foldl concat
            (concat
              (foldl concat (concat (PState s) fr) (map snd f-fl))
              f-final)
            as)
          dg')
        (take (Suc i) (map snd fl)))) =
      Some (fst (fl ! i))
    using comp-lookup by blast
have challenges':
  accepted-fri-challenges s (Some (result, final-state))
    (map fst f-fl) dg' (map fst fl)
unfolding accepted-fri-challenges-def
by (intro exI [of - result] exI [of - final-state] exI [of - fr]
  exI [of - f-fl] exI [of - f-final] exI [of - as] exI [of - fl]
  exI [of - final] exI [of - query-state])
  (use header query-state-eq query-out query-count-header
    trace-lookup-all comp-lookup-all in simp)
have dg' = dg
  using accepted-fri-challenges-unique [OF challenges challenges'] by simp
then show ?thesis
  using degree-bound by simp
qed

```

```

lemma verifier-after-trace-fri-accepted-fri-challenges:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes prefix:
  Some ((fr, f-fl), prefix-state)  $\in$ 
    set-dist (execute verifier-trace-fri-prefix s)
and suffix:
  Some (result, final-state)  $\in$ 
    set-dist
      (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
shows  $\exists dg \text{ comp-bs.}$ 
  accepted-fri-challenges s (Some (result, final-state))
    (map fst f-fl) dg comp-bs

```

proof –

have *prefix-res*:

length f-fl = *ceil-log clength* $\wedge$

PTranscript s = [*fr*] @ *map snd f-fl* @ *PTranscript prefix-state* $\wedge$

PState prefix-state =

foldl concat (concat (PState s) fr) (map snd f-fl) $\wedge$

s $\leq$ *prefix-state* $\wedge$

PTraceFriCounter prefix-state = *PTraceFriCounter s* + *ceil-log clength* $\wedge$

PCompositionFriCounter prefix-state = *PCompositionFriCounter s* $\wedge$

PAlphaCounter prefix-state = *PAlphaCounter s* $\wedge$

PQueryCounter prefix-state = *PQueryCounter s* $\wedge$

($\forall i < \text{length } f\text{-fl}.$

fmlookup (HashMap prefix-state)

(*TraceFriChallenge (PTraceFriCounter s* + *i)*

(*foldl concat (concat (PState s) fr)*

(*take (Suc i) (map snd f-fl)*))) =

Some (fst (f-fl ! i)))

by (*rule verifier-trace-fri-prefix-outcome[OF prefix]*)

from *suffix* **obtain** *f-final s3* **as** *s4 dg s5 s6 fl s7 final query-state* **where**

read-trace-final:

Some (f-final, s3) $\in$ *set-dist (execute read prefix-state)*

and *alpha-out*:

Some (as, s4) $\in$

set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)

and *read-dg*: *Some (dg, s5)* $\in$ *set-dist (execute read s4)*

and *degree-assert*:

Some ((), s6) $\in$

set-dist (execute (assert (to-nat dg $\leq$ maxDegree)) s5)

and *comp-fri*:

Some (fl, s7) $\in$

set-dist (execute

(*ntimes receive-composition-fri-commits*

(*ceil-log (to-nat dg* + 1))) *s6)*

and *read-final*: *Some (final, query-state)* $\in$ *set-dist (execute read s7)*

and *query-out*:

Some (result, final-state) $\in$

set-dist (execute

(*ntimes (verifier-query-round-program fr f-fl f-final as fl final)*

rounds)

query-state)

unfolding *verifier-after-trace-fri-def*

by (*auto elim!: set-dist-bindE*)

from *read-outcome[OF read-trace-final]* **obtain** *rest3* **where**

tr-prefix: *PTranscript prefix-state* = *f-final* # *rest3*

and *st-s3*: *PState s3* = *concat (PState prefix-state) f-final*

and *tr-s3*: *PTranscript s3* = *rest3*

and *ext-prefix-s3*: *prefix-state* $\leq$ *s3*

and *trace-count-s3*: *PTraceFriCounter s3* = *PTraceFriCounter prefix-state*

and *comp-count-s3*: *PCompositionFriCounter s3* = *PCompositionFriCounter prefix-state*

```

and alpha-count-s3: PAlphaCounter s3 = PAlphaCounter prefix-state
and query-count-s3: PQueryCounter s3 = PQueryCounter prefix-state
by blast
have alpha-res:
  length as = length spec  $\wedge$ 
  PTranscript s3 = as @ PTranscript s4  $\wedge$ 
  PState s4 = foldl concat (PState s3) as  $\wedge$ 
  s3  $\leq$  s4  $\wedge$ 
  PTraceFriCounter s4 = PTraceFriCounter s3  $\wedge$ 
  PCompositionFriCounter s4 = PCompositionFriCounter s3  $\wedge$ 
  PAlphaCounter s4 = PAlphaCounter s3 + length spec  $\wedge$ 
  PQueryCounter s4 = PQueryCounter s3
using mmap-alpha-round-outcome[OF alpha-out] by simp
from read-outcome[OF read-dg] obtain rest5 where
  tr-s4: PTranscript s4 = dg # rest5
and st-s5: PState s5 = concat (PState s4) dg
and tr-s5: PTranscript s5 = rest5
and ext-s4-s5: s4  $\leq$  s5
and trace-count-s5: PTraceFriCounter s5 = PTraceFriCounter s4
and comp-count-s5: PCompositionFriCounter s5 = PCompositionFriCounter s4
and alpha-count-s5: PAlphaCounter s5 = PAlphaCounter s4
and query-count-s5: PQueryCounter s5 = PQueryCounter s4
by blast
have s6-eq: s6 = s5
using degree-assert unfolding assert-def
by (cases to-nat dg  $\leq$  maxDegree) (auto simp: throw-no-outcome)
have ext-s5-s6: s5  $\leq$  s6
unfolding s6-eq by (rule hash-ext-refl)
have comp-res:
  length fl = ceil-log (to-nat dg + 1)  $\wedge$ 
  PTranscript s6 = map snd fl @ PTranscript s7  $\wedge$ 
  PState s7 = foldl concat (PState s6) (map snd fl)  $\wedge$ 
  s6  $\leq$  s7  $\wedge$ 
  PTraceFriCounter s7 = PTraceFriCounter s6  $\wedge$ 
  PCompositionFriCounter s7 =
    PCompositionFriCounter s6 + ceil-log (to-nat dg + 1)  $\wedge$ 
  PAlphaCounter s7 = PAlphaCounter s6  $\wedge$ 
  PQueryCounter s7 = PQueryCounter s6  $\wedge$ 
  ( $\forall i < \text{ceil-log (to-nat dg + 1)}$ ).
    fmllookup (HashMap s7)
      (CompositionFriChallenge (PCompositionFriCounter s6 + i)
        (foldl concat (PState s6) (take (Suc i) (map snd fl)))) =
      Some (fst (fl ! i))
using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-s7: PTranscript s7 = final # rest-query
and st-query: PState query-state = concat (PState s7) final
and tr-query: PTranscript query-state = rest-query

```

```

and ext-s7-query:  $s7 \leq \text{query-state}$ 
and query-count-query:  $P\text{QueryCounter query-state} = P\text{QueryCounter } s7$ 
by blast
have ext-prefix-query:  $\text{prefix-state} \leq \text{query-state}$ 
using ext-prefix-s3 alpha-res ext-s4-s5 ext-s5-s6 comp-res ext-s7-query
by (meson hash-ext-trans)
have header:
  verifier-header-transcript  $s$  fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
using prefix-res tr-prefix tr-s3 alpha-res tr-s4 tr-s5 s6-eq comp-res
  tr-s7 tr-query
unfolding verifier-header-transcript-def verifier-header-messages-def
by simp
have header-state:
  PState query-state =
    verifier-header-state  $s$  fr (map snd f-fl) f-final as dg
      (map snd fl) final
using prefix-res st-s3 alpha-res st-s5 s6-eq comp-res st-query
unfolding verifier-header-state-def verifier-header-messages-def
by simp
have query-count-header:
  PQueryCounter query-state = PQueryCounter  $s$ 
using prefix-res query-count-s3 alpha-res query-count-s5 s6-eq comp-res
  query-count-query
by simp
have trace-lookup:
   $\bigwedge i. i < \text{length } f\text{-fl} \implies$ 
    fmlookup (HashMap query-state)
      (TraceFriChallenge (PTraceFriCounter  $s + i$ )
        (foldl concat (concat (PState  $s$ ) fr)
          (take (Suc  $i$ ) (map snd f-fl)))) =
      Some (fst (f-fl !  $i$ ))
proof -
  fix  $i$ 
assume i-bound:  $i < \text{length } f\text{-fl}$ 
have lookup-prefix:
  fmlookup (HashMap prefix-state)
    (TraceFriChallenge (PTraceFriCounter  $s + i$ )
      (foldl concat (concat (PState  $s$ ) fr)
        (take (Suc  $i$ ) (map snd f-fl)))) =
    Some (fst (f-fl !  $i$ ))
using prefix-res i-bound by simp
show fmlookup (HashMap query-state)
  (TraceFriChallenge (PTraceFriCounter  $s + i$ )
    (foldl concat (concat (PState  $s$ ) fr)
      (take (Suc  $i$ ) (map snd f-fl)))) =
  Some (fst (f-fl !  $i$ ))
by (rule hash-extension-lookup[OF lookup-prefix ext-prefix-query])
qed

```

```

have comp-lookup:
   $\wedge i. i < \text{length } fl \implies$ 
    fmlookup (HashMap query-state)
      (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
        (concat
          (foldl concat
            (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
            as)
          dg)
        (take (Suc i) (map snd fl)))) =
      Some (fst (fl ! i))
proof -
  fix i
  assume i-bound:  $i < \text{length } fl$ 
  have lookup-s7:
    fmlookup (HashMap s7)
      (CompositionFriChallenge (PCompositionFriCounter s6 + i)
        (foldl concat (PState s6) (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i))
  using comp-res i-bound by simp
  show fmlookup (HashMap query-state)
    (CompositionFriChallenge (PCompositionFriCounter s + i) (foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat (PState s) fr) (map snd f-fl)) f-final)
          as)
        dg)
        (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i))
  using hash-extension-lookup[OF lookup-s7 ext-s7-query]
    prefix-res st-s3 alpha-res st-s5 s6-eq comp-count-s3 comp-count-s5
  by simp
qed
have challenges:
  accepted-fri-challenges s (Some (result, final-state))
    (map fst f-fl) dg (map fst fl)
  unfolding accepted-fri-challenges-def
  apply (rule exI[where  $x = \text{result}$ ])
  apply (rule exI[where  $x = \text{final-state}$ ])
  apply (rule exI[where  $x = \text{fr}$ ])
  apply (rule exI[where  $x = f\text{-}fl$ ])
  apply (rule exI[where  $x = f\text{-}final$ ])
  apply (rule exI[where  $x = \text{as}$ ])
  apply (rule exI[where  $x = fl$ ])
  apply (rule exI[where  $x = \text{final}$ ])
  apply (rule exI[where  $x = \text{query-state}$ ])
  apply (intro conjI)
    apply simp
    apply (rule header)

```

```

      apply (rule header-state)
      apply (rule query-out)
      apply (rule query-count-header)
      apply simp
      apply simp
      apply (intro allI impI, rule trace-lookup, assumption)
      apply (intro allI impI, rule comp-lookup, assumption)
    done
  show ?thesis
    using challenges by blast
qed

```

lemma *verifier-after-composition-fri-accepted-fri-challenges:*

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *prefix:*

Some ((fr, f-fl, f-final, as, dg, fl), prefix-state) ∈
set-dist (execute verifier-composition-fri-prefix s)

and *suffix:*

Some (result, final-state) ∈
set-dist
(execute
(verifier-after-composition-fri
(fr, f-fl, f-final, as, dg, fl))
prefix-state)

shows *accepted-fri-challenges s (Some (result, final-state))*
(map fst f-fl) dg (map fst fl)

proof –

have *prefix-res:*

length f-fl = ceil-log clength ∧
length as = length spec ∧
length fl = ceil-log (to-nat dg + 1) ∧
PTranscript s =
[fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
map snd fl @ PTranscript prefix-state ∧
PState prefix-state =
foldl concat
(concat
(foldl concat
(concat
(foldl concat (concat (PState s) fr) (map snd f-fl))
f-final)
as)
dg)
(map snd fl) ∧
s ≤ prefix-state ∧
PQueryCounter prefix-state = PQueryCounter s ∧
(∀ i < length f-fl.
fmlookup (HashMap prefix-state)
(TraceFriChallenge (PTraceFriCounter s + i)

```

    (foldl concat (concat (PState s) fr)
      (take (Suc i) (map snd f-fl)))) =
  Some (fst (f-fl ! i)) ∧
(∀ i < length fl.
  fmllookup (HashMap prefix-state)
    (CompositionFriChallenge (PCompositionFriCounter s + i)
      (foldl concat
        (concat
          (foldl concat
            (concat
              (foldl concat (concat (PState s) fr) (map snd f-fl))
                f-final)
            as)
          dg)
        (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i)))
by (rule verifier-composition-fri-prefix-outcome[OF prefix])
from suffix obtain final query-state where
  read-final:
    Some (final, query-state) ∈
      set-dist (execute read prefix-state)
and query-out:
    Some (result, final-state) ∈
      set-dist (execute
        (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
unfolding verifier-after-composition-fri-def
by (auto elim!: set-dist-bindE)
from read-outcome[OF read-final] obtain rest-query where
  tr-prefix: PTranscript prefix-state = final # rest-query
and st-query: PState query-state = concat (PState prefix-state) final
and tr-query: PTranscript query-state = rest-query
and ext-prefix-query: prefix-state ≤ query-state
and query-count-query:
  PQueryCounter query-state = PQueryCounter prefix-state
by blast
have header:
  verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
using prefix-res tr-prefix tr-query
unfolding verifier-header-transcript-def verifier-header-messages-def
by simp
have header-state:
  PState query-state =
    verifier-header-state s fr (map snd f-fl) f-final as dg
      (map snd fl) final
using prefix-res st-query
unfolding verifier-header-state-def verifier-header-messages-def

```

```

    by simp
  have query-count-header:
    PQueryCounter query-state = PQueryCounter s
    using prefix-res query-count-query by simp
  have trace-lookup:
     $\bigwedge i. i < \text{length } f\text{-fl} \implies$ 
      fmllookup (HashMap query-state)
        (TraceFriChallenge (PTraceFriCounter s + i)
          (foldl concat (concat (PState s) fr)
            (take (Suc i) (map snd f-fl)))) =
        Some (fst (f-fl ! i))
  proof -
    fix i
    assume i-bound:  $i < \text{length } f\text{-fl}$ 
    have lookup-prefix:
      fmllookup (HashMap prefix-state)
        (TraceFriChallenge (PTraceFriCounter s + i)
          (foldl concat (concat (PState s) fr)
            (take (Suc i) (map snd f-fl)))) =
        Some (fst (f-fl ! i))
      using prefix-res i-bound by simp
    show fmllookup (HashMap query-state)
      (TraceFriChallenge (PTraceFriCounter s + i)
        (foldl concat (concat (PState s) fr)
          (take (Suc i) (map snd f-fl)))) =
        Some (fst (f-fl ! i))
      by (rule hash-extension-lookup[OF lookup-prefix ext-prefix-query])
  qed
  have comp-lookup:
     $\bigwedge i. i < \text{length } fl \implies$ 
      fmllookup (HashMap query-state)
        (CompositionFriChallenge (PCompositionFriCounter s + i)
          (foldl concat
            (concat
              (foldl concat
                (concat
                  (foldl concat (concat (PState s) fr) (map snd f-fl))
                  f-final)
                as)
              dg)
            (take (Suc i) (map snd fl)))) =
        Some (fst (fl ! i))
  proof -
    fix i
    assume i-bound:  $i < \text{length } fl$ 
    have lookup-prefix:
      fmllookup (HashMap prefix-state)
        (CompositionFriChallenge (PCompositionFriCounter s + i)
          (foldl concat

```



```

      (concat
        (foldl concat
          (concat
            (foldl concat (concat (PState s) fr) (map snd f-fl))
            f-final)
          as)
        dg)
      (take (Suc i) (map snd fl)))) =
  Some (fst (fl ! i))
using prefix-res i-bound by simp
show fmlookup (HashMap query-state)
  (CompositionFriChallenge (PCompositionFriCounter s + i)
    (foldl concat
      (concat
        (foldl concat
          (concat
            (foldl concat (concat (PState s) fr) (map snd f-fl))
            f-final)
          as)
        dg)
      (take (Suc i) (map snd fl)))) =
  Some (fst (fl ! i))
by (rule hash-extension-lookup[OF lookup-prefix ext-prefix-query])
qed
show ?thesis
  unfolding accepted-fri-challenges-def
  apply (rule exI[where x=result])
  apply (rule exI[where x=final-state])
  apply (rule exI[where x=fr])
  apply (rule exI[where x=f-fl])
  apply (rule exI[where x=f-final])
  apply (rule exI[where x=as])
  apply (rule exI[where x=fl])
  apply (rule exI[where x=final])
  apply (rule exI[where x=query-state])
  apply (intro conjI)
  apply simp
  apply (rule header)
  apply (rule header-state)
  apply (rule query-out)
  apply (rule query-count-header)
  apply simp
  apply simp
  apply (intro allI impI, rule trace-lookup, assumption)
  apply (intro allI impI, rule comp-lookup, assumption)
done
qed

```

definition trace-fri-challenge-list-set-hit

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list set} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-challenge-list-set-hit } s \ B \ \text{out} \longleftrightarrow$
 $(\exists \text{ trace-bs } dg \ \text{comp-bs.}$
 $\text{accepted-fri-challenges } s \ \text{out trace-bs } dg \ \text{comp-bs} \wedge$
 $\text{trace-bs} \in B)$

definition *trace-fri-root-list-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-root-list-set-hit } s \ B \ \text{out} \longleftrightarrow$
 $(\exists \text{ trace-bs } dg \ \text{comp-bs } fr \ f\text{-fri-roots } f\text{-final } as \ \text{composition-fri-roots}$
 final rest.
 $\text{accepted-fri-challenges } s \ \text{out trace-bs } dg \ \text{comp-bs} \wedge$
 $\text{verifier-header-transcript } s \ fr \ f\text{-fri-roots } f\text{-final } as \ dg$
 $\text{composition-fri-roots final rest} \wedge$
 $\text{trace-bs} \in B \ fr)$

definition *composition-fri-challenge-list-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow ('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-challenge-list-set-hit } s \ B \ \text{out} \longleftrightarrow$
 $(\exists \text{ trace-bs } dg \ \text{comp-bs.}$
 $\text{accepted-fri-challenges } s \ \text{out trace-bs } dg \ \text{comp-bs} \wedge$
 $\text{comp-bs} \in B \ dg)$

definition *composition-fri-root-list-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-root-list-set-hit } s \ B \ \text{out} \longleftrightarrow$
 $(\exists \text{ trace-bs } dg \ \text{comp-bs } fr \ f\text{-fri-roots } f\text{-final } as$
 $\text{composition-fri-roots final rest.}$
 $\text{accepted-fri-challenges } s \ \text{out trace-bs } dg \ \text{comp-bs} \wedge$
 $\text{verifier-header-transcript } s \ fr \ f\text{-fri-roots } f\text{-final } as \ dg$
 $\text{composition-fri-roots final rest} \wedge$
 $\text{comp-bs} \in B \ fr \ f\text{-fri-roots } f\text{-final } as \ dg \ \text{composition-fri-roots})$

lemma *wp-verify-monad-trace-fri-challenge-list-set-bound:*

assumes *future:* $\text{trace-fri-future-fresh } s$

and *subset:* $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$

shows

$\text{wp-event verify-monad } (\text{trace-fri-challenge-list-set-hit } s \ B) \ s \leq$

```

      nnreal (card B) / nnreal (CARD('f) ^ ceil-log clength)
proof -
  let ?Head =
    λout. case out of
      None ⇒ False
    | Some ((-, f-fl), -) ⇒ map fst f-fl ∈ B
  have prefix-bound:
    wp-event verifier-trace-fri-prefix ?Head s ≤
      nnreal (card B) / nnreal (CARD('f) ^ ceil-log clength)
  by (rule wp-verifier-trace-fri-prefix-challenge-space-set-bound
    [OF future subset])
  show ?thesis
    unfolding verify-monad-trace-fri-decomposition
  proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
  show trace-fri-challenge-list-set-hit s B None ⇒ ?Head None
    unfolding trace-fri-challenge-list-set-hit-def
    accepted-fri-challenges-def by blast
  next
    fix header prefix-state out
    assume prefix:
      Some (header, prefix-state) ∈
        set-dist (execute verifier-trace-fri-prefix s)
    and suffix:
      out ∈
        set-dist (execute (verifier-after-trace-fri header) prefix-state)
    and hit: trace-fri-challenge-list-set-hit s B out
  obtain fr f-fl where header-eq: header = (fr, f-fl)
    by (cases header) auto
  from hit obtain trace-bs dg comp-bs where
    challenges-hit: accepted-fri-challenges s out trace-bs dg comp-bs
    and trace-bs-B: trace-bs ∈ B
    unfolding trace-fri-challenge-list-set-hit-def by blast
  from challenges-hit obtain result final-state where out-eq:
    out = Some (result, final-state)
    unfolding accepted-fri-challenges-def by blast
  have prefix':
    Some ((fr, f-fl), prefix-state) ∈
      set-dist (execute verifier-trace-fri-prefix s)
    using prefix header-eq by simp
  have suffix':
    Some (result, final-state) ∈
      set-dist
        (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
    using suffix header-eq out-eq by simp
  from verifier-after-trace-fri-accepted-fri-challenges[OF prefix' suffix']
  obtain dg' comp-bs' where
    challenges-prefix:
      accepted-fri-challenges s (Some (result, final-state))
      (map fst f-fl) dg' comp-bs'

```

```

    by blast
  have trace-bs = map fst f-fl
    using accepted-fri-challenges-unique
      [OF challenges-prefix challenges-hit[unfolded out-eq]]
    by simp
  then show ?Head (Some (header, prefix-state))
    using trace-bs-B header-eq by simp
qed
qed

lemma wp-verify-monad-trace-fri-root-list-set-bound:
  fixes C :: prob
  assumes future: trace-fri-future-fresh s
    and subset:  $\bigwedge fr. B \text{ fr} \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$ 
    and bound:
       $\bigwedge fr. \text{nnreal } (\text{card } (B \text{ fr})) / \text{nnreal } (\text{CARD}'f) ^ \text{ceil-log } \text{clength} \leq C$ 
  shows
    wp-event verify-monad (trace-fri-root-list-set-hit s B) s  $\leq C$ 
proof -
  let ?Head =
     $\lambda out. \text{case out of}$ 
      None  $\Rightarrow$  False
    | Some ((fr, f-fl), -)  $\Rightarrow$  map fst f-fl  $\in B \text{ fr}$ 
  have prefix-bound:
    wp-event verifier-trace-fri-prefix ?Head s  $\leq C$ 
  by (rule wp-verifier-trace-fri-prefix-dependent-challenge-space-set-bound
      [OF future subset bound])
  show ?thesis
    unfolding verify-monad-trace-fri-decomposition
  proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
    show trace-fri-root-list-set-hit s B None  $\implies$  ?Head None
      unfolding trace-fri-root-list-set-hit-def
        accepted-fri-challenges-def by blast
  next
    fix header prefix-state out
    assume prefix:
      Some (header, prefix-state)  $\in$ 
        set-dist (execute verifier-trace-fri-prefix s)
    and suffix:
      out  $\in$ 
        set-dist (execute (verifier-after-trace-fri header) prefix-state)
    and hit: trace-fri-root-list-set-hit s B out
    obtain fr f-fl where header-eq: header = (fr, f-fl)
      by (cases header) auto
    from hit obtain trace-bs dg comp-bs fr' f-fri-roots' f-final' as'
      composition-fri-roots' final' rest' where
      challenges-hit: accepted-fri-challenges s out trace-bs dg comp-bs
    and header-hit:

```

```

    verifier-header-transcript s fr' f-fri-roots' f-final' as' dg
    composition-fri-roots' final' rest'
  and trace-bs-B: trace-bs ∈ B fr'
  unfolding trace-fri-root-list-set-hit-def by blast
from challenges-hit obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-fri-challenges-def by blast
have prefix':
  Some ((fr, f-fl), prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
  using prefix header-eq by simp
have suffix':
  Some (result, final-state) ∈
    set-dist
    (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
  using suffix header-eq out-eq by simp
have prefix-res:
  PTranscript s = [fr] @ map snd f-fl @ PTranscript prefix-state
  using verifier-trace-fri-prefix-outcome[OF prefix'] by simp
have fr-eq: fr' = fr
  using prefix-res header-hit
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
from verifier-after-trace-fri-accepted-fri-challenges[OF prefix' suffix']
obtain dg' comp-bs' where
  challenges-prefix:
    accepted-fri-challenges s (Some (result, final-state))
    (map fst f-fl) dg' comp-bs'
  by blast
have trace-bs-eq: trace-bs = map fst f-fl
  using accepted-fri-challenges-unique
  [OF challenges-prefix challenges-hit[unfolded out-eq]]
  by simp
then show ?Head (Some (header, prefix-state))
  using trace-bs-B fr-eq header-eq by simp
qed
qed

```

lemma *wp-verify-monad-composition-fri-root-list-set-bound*:

fixes $C :: \text{prob}$

assumes *future*: *composition-fri-future-fresh* s

and *subset*:

$$\bigwedge fr \text{ f-fri-roots } f\text{-final } as \text{ dg } \text{composition-fri-roots.}$$

$$B \text{ fr f-fri-roots } f\text{-final } as \text{ dg } \text{composition-fri-roots} \subseteq$$

$$\text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$$

and *bound*:

$$\bigwedge fr \text{ f-fri-roots } f\text{-final } as \text{ dg } \text{composition-fri-roots.}$$

$$\text{nnreal}$$

$$(\text{card}$$

```

      (B fr f-fri-roots f-final as dg composition-fri-roots)) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C
shows
  wp-event verify-monad (composition-fri-root-list-set-hit s B) s ≤ C
proof -
  let ?Head =
    λout. case out of
      None ⇒ False
    | Some ((fr, f-fl, f-final, as, dg, fl), -) ⇒
      map fst fl ∈ B fr (map snd f-fl) f-final as dg (map snd fl)
  have prefix-bound:
    wp-event verifier-composition-fri-prefix ?Head s ≤ C
  by (rule wp-verifier-composition-fri-prefix-root-dependent-challenge-space-set-bound
      [OF future subset bound])
  show ?thesis
    unfolding verify-monad-composition-fri-decomposition
  proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
    show composition-fri-root-list-set-hit s B None ⇒ ?Head None
      unfolding composition-fri-root-list-set-hit-def
        accepted-fri-challenges-def by blast
  next
    fix header prefix-state out
    assume prefix:
      Some (header, prefix-state) ∈
        set-dist (execute verifier-composition-fri-prefix s)
    and suffix:
      out ∈
        set-dist (execute
          (verifier-after-composition-fri header) prefix-state)
    and hit: composition-fri-root-list-set-hit s B out
    obtain fr f-fl f-final as dg fl where header-eq:
      header = (fr, f-fl, f-final, as, dg, fl)
    by (cases header) auto
    from hit obtain trace-bs dg' comp-bs fr' f-fri-roots' f-final' as'
      composition-fri-roots' final' rest' where
      challenges-hit:
        accepted-fri-challenges s out trace-bs dg' comp-bs
    and header-hit:
      verifier-header-transcript s fr' f-fri-roots' f-final' as' dg'
        composition-fri-roots' final' rest'
    and comp-bs-B:
      comp-bs ∈
        B fr' f-fri-roots' f-final' as' dg' composition-fri-roots'
    unfolding composition-fri-root-list-set-hit-def by blast
    from challenges-hit obtain result final-state where out-eq:
      out = Some (result, final-state)
    unfolding accepted-fri-challenges-def by blast
  have prefix':
    Some ((fr, f-fl, f-final, as, dg, fl), prefix-state) ∈

```

```

    set-dist (execute verifier-composition-fri-prefix s)
using prefix header-eq by simp
have suffix':
  Some (result, final-state) ∈
    set-dist
      (execute
        (verifier-after-composition-fri
          (fr, f-fl, f-final, as, dg, fl))
        prefix-state)
    using suffix header-eq out-eq by simp
from suffix'[unfolding verifier-after-composition-fri-def]
obtain final query-state where
  read-final:
    Some (final, query-state) ∈ set-dist (execute read prefix-state)
  and query-out:
    Some (result, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            rounds)
          query-state)
    by (auto elim!: set-dist-bindE)
have prefix-res:
  length f-fl = ceil-log clength ∧
  length as = length spec ∧
  length fl = ceil-log (to-nat dg + 1) ∧
  PTranscript s =
    [fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
    map snd fl @ PTranscript prefix-state
    using verifier-composition-fri-prefix-outcome[OF prefix'] by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-prefix: PTranscript prefix-state = final # rest-query
  and tr-query: PTranscript query-state = rest-query
  by blast
have header-prefix:
  verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
    using prefix-res tr-prefix tr-query
    unfolding verifier-header-transcript-def verifier-header-messages-def
    by simp
have header-egs:
  fr' = fr ∧
  f-fri-roots' = map snd f-fl ∧
  f-final' = f-final ∧
  as' = as ∧
  dg' = dg ∧
  composition-fri-roots' = map snd fl
  using verifier-header-transcript-unique[OF header-prefix header-hit]

```

```

    by simp
  have challenges-prefix:
    accepted-fri-challenges s (Some (result, final-state))
      (map fst f-fl) dg (map fst fl)
  by (rule verifier-after-composition-fri-accepted-fri-challenges
      [OF prefix' suffix'])
  have comp-bs-eq: comp-bs = map fst fl
  using accepted-fri-challenges-unique
    [OF challenges-prefix challenges-hit[unfolded out-eq]]
  by simp
  show ?Head (Some (header, prefix-state))
  using comp-bs-B comp-bs-eq header-eqs header-eq by simp
qed
qed

lemma wp-verify-monad-composition-fri-challenge-list-set-bound:
  fixes C :: prob
  assumes future: composition-fri-future-fresh s
  and subset:
     $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$ 
  and bound:
     $\bigwedge dg. \text{nnreal } (\text{card } (B \ dg)) / \text{nnreal } (\text{CARD('f)} \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$ 
  shows
    wp-event verify-monad (composition-fri-challenge-list-set-hit s B) s  $\leq C$ 
proof -
  let ?Head =
     $\lambda out. \text{case out of}$ 
      None  $\Rightarrow$  False
    | Some ((-, -, -, dg, fl), -)  $\Rightarrow$  map fst fl  $\in B \ dg$ 
  have prefix-bound:
    wp-event verifier-composition-fri-prefix ?Head s  $\leq C$ 
  by (rule wp-verifier-composition-fri-prefix-challenge-space-set-bound
      [OF future subset bound])
  show ?thesis
  unfolding verify-monad-composition-fri-decomposition
proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
  show composition-fri-challenge-list-set-hit s B None  $\implies$  ?Head None
  unfolding composition-fri-challenge-list-set-hit-def
    accepted-fri-challenges-def by blast
next
  fix header prefix-state out
  assume prefix:
    Some (header, prefix-state)  $\in$ 
      set-dist (execute verifier-composition-fri-prefix s)
  and suffix:
    out  $\in$ 
      set-dist (execute
        (verifier-after-composition-fri header) prefix-state)

```



```

    and hit: composition-fri-challenge-list-set-hit s B out
  obtain fr f-fl f-final as dg fl where header-eq:
    header = (fr, f-fl, f-final, as, dg, fl)
    by (cases header) auto
  from hit obtain trace-bs dg' comp-bs where
    challenges-hit: accepted-fri-challenges s out trace-bs dg' comp-bs
    and comp-bs-B: comp-bs ∈ B dg'
    unfolding composition-fri-challenge-list-set-hit-def by blast
  from challenges-hit obtain result final-state where out-eq:
    out = Some (result, final-state)
    unfolding accepted-fri-challenges-def by blast
  have prefix':
    Some ((fr, f-fl, f-final, as, dg, fl), prefix-state) ∈
      set-dist (execute verifier-composition-fri-prefix s)
    using prefix header-eq by simp
  have suffix':
    Some (result, final-state) ∈
      set-dist
        (execute
          (verifier-after-composition-fri
            (fr, f-fl, f-final, as, dg, fl))
          prefix-state)
    using suffix header-eq out-eq by simp
  have challenges-prefix:
    accepted-fri-challenges s (Some (result, final-state))
      (map fst f-fl) dg (map fst fl)
    by (rule verifier-after-composition-fri-accepted-fri-challenges
      [OF prefix' suffix'])
  have eqs:
    trace-bs = map fst f-fl ∧ dg' = dg ∧ comp-bs = map fst fl
    using accepted-fri-challenges-unique
      [OF challenges-prefix challenges-hit[unfolded out-eq]]
    by simp
  then show ?Head (Some (header, prefix-state))
    using comp-bs-B header-eq by simp
qed
qed

lemma verifier-initial-alpha-list-set-error-bound:
  assumes subset: B ⊆ alpha-space
  and bounded:
    nnreal (card B) / nnreal (card alpha-space) ≤
      composition-error-bound
  shows
    wp-event verify-monad
      (alpha-list-set-hit (verifier-initial-state tr) B)
      (verifier-initial-state tr) ≤ composition-error-bound
proof -
  have wp-event verify-monad

```

$(\text{alpha-list-set-hit } (\text{verifier-initial-state } tr) B)$
 $(\text{verifier-initial-state } tr) \leq$
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{alpha-space})$
by (rule *wp-verify-monad-alpha-list-set-bound*) (use subset **in** *simp-all*)
also have $\dots \leq \text{composition-error-bound}$
by (rule *bounded*)
finally show ?thesis .
qed

lemma *verifier-initial-trace-fri-challenge-list-set-error-bound*:
assumes subset: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and *bounded*:
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength}) \leq$
 trace-fri-error
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-challenge-list-set-hit } (\text{verifier-initial-state } tr) B)$
 $(\text{verifier-initial-state } tr) \leq \text{trace-fri-error}$

proof –
have $\text{wp-event verify-monad}$
 $(\text{trace-fri-challenge-list-set-hit } (\text{verifier-initial-state } tr) B)$
 $(\text{verifier-initial-state } tr) \leq$
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
by (rule *wp-verify-monad-trace-fri-challenge-list-set-bound*)
 (use subset **in** *simp-all*)
also have $\dots \leq \text{trace-fri-error}$
by (rule *bounded*)
finally show ?thesis .
qed

lemma *verifier-initial-composition-fri-challenge-list-set-error-bound*:
assumes subset:
 $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and *bounded*:
 $\bigwedge dg. \text{nnreal } (\text{card } (B \ dg)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq$
 $\text{composition-fri-error}$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-challenge-list-set-hit } (\text{verifier-initial-state } tr) B)$
 $(\text{verifier-initial-state } tr) \leq \text{composition-fri-error}$
by (rule *wp-verify-monad-composition-fri-challenge-list-set-bound*)
 (use subset *bounded* **in** *simp-all*)

definition *accepted-fri-opening-transcript*
 $:: ('f, 'a) \text{protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{protocol-channel-scheme}) \text{option} \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$

$\text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow 'f \text{ list list list} \Rightarrow \text{bool}$
where
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs trace-final}$
 $\text{dg composition-roots composition-bs composition-final query-idxs}$
 $\text{trace-round-layers composition-round-layers} \longleftrightarrow$
 $(\exists \text{result final-state fr f-fl as fl query-state raw-idxs query-chunks.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{trace-roots} = \text{map snd f-fl} \wedge$
 $\text{trace-bs} = \text{map fst f-fl} \wedge$
 $\text{composition-roots} = \text{map snd fl} \wedge$
 $\text{composition-bs} = \text{map fst fl} \wedge$
 $\text{accepted-fri-challenges } s \text{ out trace-bs dg composition-bs} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final (PTranscript query-state)} \wedge$
 $\text{PState query-state} =$
 $\text{verifier-header-state } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final} \wedge$
 $\text{PQueryCounter query-state} = \text{PQueryCounter } s \wedge$
 $\text{Some (result, final-state)} \in$
 set-dist (execute
 (ntimes
 $\text{(verifier-query-round-program fr f-fl trace-final as fl}$
 $\text{composition-final)}$
 rounds)
 $\text{query-state)} \wedge$
 $\text{verifier-query-indices-derived } s \text{ out}$
 $\text{(verifier-header-state } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final)}$
 $\text{(PTranscript query-state) trace-roots composition-roots query-idxs} \wedge$
 $\text{length trace-round-layers} = \text{rounds} \wedge$
 $\text{length composition-round-layers} = \text{rounds} \wedge$
 $\text{length raw-idxs} = \text{rounds} \wedge$
 $\text{query-idxs} = \text{map } (\lambda \text{raw. index (to-nat raw)}) \text{ raw-idxs} \wedge$
 $\text{length query-chunks} = \text{rounds} \wedge$
 $\text{PTranscript query-state} =$
 $\text{List.concat query-chunks @ PTranscript final-state} \wedge$
 $(\forall i < \text{rounds.}$
 $\text{verifier-query-round-chunk (query-idxs ! } i)$
 $\text{trace-roots composition-roots (query-chunks ! } i)) \wedge$
 $(\forall i < \text{rounds.}$
 $\text{query-round-fri-layer-transcripts (query-idxs ! } i)$
 $\text{trace-roots composition-roots (query-chunks ! } i)$
 $\text{(trace-round-layers ! } i) \text{ (composition-round-layers ! } i)) \wedge$
 $(\forall i < \text{rounds.}$
 $\text{fmlookup (HashMap final-state)}$
 $\text{(QueryIndexChallenge (PQueryCounter query-state + } i) \text{ (state-after-query-chunks}$
 $\text{(PState query-state) query-chunks } i)) =$
 $\text{Some (raw-idxs ! } i)) \wedge$
 $(\forall \text{idx} \in \text{set query-idxs. idx} < \text{clength * scale}))$

lemma *verify-monad-accepted-fri-opening-transcript*:

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *outcome*:

$\text{Some } (\text{result}, \text{final-state}) \in \text{set-dist } (\text{execute } \text{verify-monad } s)$

obtains *trace-roots trace-bs trace-final dg composition-roots composition-bs composition-final query-idxs trace-round-layers composition-round-layers*

where *accepted-fri-opening-transcript s* ($\text{Some } (\text{result}, \text{final-state})$)

$\text{trace-roots trace-bs trace-final dg composition-roots composition-bs composition-final query-idxs trace-round-layers composition-round-layers}$

proof –

from *verify-monad-accepted-fri-challenges*[*OF outcome*]

obtain *trace-bs dg composition-bs* **where** *challenges*:

$\text{accepted-fri-challenges } s (\text{Some } (\text{result}, \text{final-state}))$

$\text{trace-bs dg composition-bs}$

by *blast*

then obtain *result' final-state'* *fr f-fl trace-final as fl composition-final query-state* **where**

$\text{out-eq: } \text{Some } (\text{result}, \text{final-state}) = \text{Some } (\text{result}', \text{final-state}')$

and *trace-bs-eq*: $\text{trace-bs} = \text{map fst } f\text{-fl}$

and *composition-bs-eq*: $\text{composition-bs} = \text{map fst } fl$

and *header*:

$\text{verifier-header-transcript } s \text{ fr } (\text{map snd } f\text{-fl}) \text{ trace-final as dg}$

$(\text{map snd } fl) \text{ composition-final } (P\text{Transcript } \text{query-state})$

and *query-state-eq*:

$P\text{State } \text{query-state} =$

$\text{verifier-header-state } s \text{ fr } (\text{map snd } f\text{-fl}) \text{ trace-final as dg}$

$(\text{map snd } fl) \text{ composition-final}$

and *query-out*:

$\text{Some } (\text{result}', \text{final-state}') \in$

$\text{set-dist } (\text{execute}$

$(\text{ntimes}$

$(\text{verifier-query-round-program } \text{fr } f\text{-fl } \text{trace-final as fl}$

composition-final

$\text{rounds})$

$\text{query-state})$

and *query-count-header*:

$P\text{QueryCounter } \text{query-state} = P\text{QueryCounter } s$

unfolding *accepted-fri-challenges-def* **by** *blast*

have *result'-eq*: $\text{result}' = \text{result}$

using *out-eq* **by** *simp*

have *final-state'-eq*: $\text{final-state}' = \text{final-state}$

using *out-eq* **by** *simp*

have *query-out'*:

$\text{Some } (\text{result}, \text{final-state}) \in$

$\text{set-dist } (\text{execute}$

$(\text{ntimes}$

$(\text{verifier-query-round-program } \text{fr } f\text{-fl } \text{trace-final as fl}$

$\text{composition-final})$

```

    rounds)
    query-state)
  using query-out result'-eq final-state'-eq by simp
from ntimes-verifier-query-rounds-outcome[OF query-out']
obtain raw-idxs query-idxs query-chunks where
  len-raw: length raw-idxs = rounds
  and query-idxs-def: query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-chunks: length query-chunks = rounds
  and tr-query:
    PTranscript query-state = List.concat query-chunks @ PTranscript final-state
  and round-chunks:
    ∧ i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
  and lookups:
    ∧ i. i < rounds ⇒
      fmllookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i) (state-after-query-chunks (PState
query-state) query-chunks i)) =
      Some (raw-idxs ! i)
  and idx-bounds:
    ∀ idx ∈ set query-idxs. idx < clength * scale
  by auto
have layer-exists:
  ∀ i < rounds.
    ∃ trace-layers composition-layers.
      query-round-fri-layer-transcripts (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
      trace-layers composition-layers
proof (intro allI impI)
  fix i
  assume i-bound: i < rounds
  from verifier-query-round-chunk-fri-layer-transcriptsE
  [OF round-chunks[OF i-bound]]
  show ∃ trace-layers composition-layers.
    query-round-fri-layer-transcripts (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
    trace-layers composition-layers
  by blast
qed
then obtain trace-layers-of composition-layers-of where layer-of:
  ∧ i. i < rounds ⇒
    query-round-fri-layer-transcripts (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
    (trace-layers-of i) (composition-layers-of i)
  by metis
let ?trace-round-layers = map trace-layers-of [0..<rounds]
let ?composition-round-layers = map composition-layers-of [0..<rounds]
have layers:

```

```

   $\forall i < \text{rounds}.$ 
  query-round-fri-layer-transcripts (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
    (?trace-round-layers ! i) (?composition-round-layers ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  then show query-round-fri-layer-transcripts (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
    (?trace-round-layers ! i) (?composition-round-layers ! i)
    using layer-of[OF i-bound] by simp
qed
have derived:
  verifier-query-indices-derived s (Some (result, final-state))
    (verifier-header-state s fr (map snd f-fl) trace-final as dg
      (map snd fl) composition-final)
    (PTranscript query-state) (map snd f-fl) (map snd fl) query-idxs
proof -
  have tr-query':
    List.concat query-chunks @ PTranscript final-state = PTranscript query-state
    using tr-query by simp
  have lookups':
     $\forall i < \text{rounds}.$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter s + i) (state-after-query-chunks
        (verifier-header-state s fr (map snd f-fl) trace-final as dg
          (map snd fl) composition-final)
          query-chunks i)) =
      Some (raw-idxs ! i)
    using lookups query-state-eq query-count-header by simp
  show ?thesis
  unfolding verifier-query-indices-derived-def
  apply (rule exI[where x=result])
  apply (rule exI[where x=final-state])
  apply (rule exI[where x=raw-idxs])
  apply (rule exI[where x=query-chunks])
  apply (rule exI[where x=PTranscript final-state])
  apply (intro conjI)
    apply simp
    apply (rule len-raw)
    apply (rule query-idxs-def)
    apply (rule len-chunks)
    apply (rule tr-query')
  apply (intro allI impI, rule round-chunks, assumption)
  apply (rule lookups')
  using idx-bounds by simp
qed
have tr-query-tail:
  List.concat query-chunks @ PTranscript final-state = PTranscript query-state

```

```

using tr-query by simp
have lookups-all:
   $\forall i < \text{rounds}.$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i) (state-after-query-chunks (PState
query-state) query-chunks i)) =
      Some (raw-idxs ! i)
    using lookups by blast
have lookups-all-header:
   $\forall i < \text{rounds}.$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (verifier-header-state s fr (map snd f-fl) trace-final as dg
            (map snd fl) composition-final)
          query-chunks i)) =
      Some (raw-idxs ! i)
    using lookups-all query-state-eq by simp
have challenges-maps:
  accepted-fri-challenges s (Some (result, final-state))
    (map fst f-fl) dg (map fst fl)
  using challenges trace-bs-eq composition-bs-eq by simp
have derived-maps:
  verifier-query-indices-derived s (Some (result, final-state))
    (verifier-header-state s fr (map snd f-fl) trace-final as dg
      (map snd fl) composition-final)
    (PTranscript query-state) (map snd f-fl) (map snd fl)
    (map ( $\lambda \text{raw}.$  index (to-nat raw)) raw-idxs)
  using derived query-idxs-def by simp
have layers-maps:
   $\forall i < \text{rounds}.$ 
    query-round-fri-layer-transcripts (index (to-nat (raw-idxs ! i)))
      (map snd f-fl) (map snd fl) (query-chunks ! i)
      (trace-layers-of i) (composition-layers-of i)
    using layers query-idxs-def len-raw by auto
have round-chunks-raw:
   $\forall i < \text{rounds}.$ 
    verifier-query-round-chunk (index (to-nat (raw-idxs ! i)))
      (map snd f-fl) (map snd fl) (query-chunks ! i)
    using round-chunks query-idxs-def len-raw by auto
have lookups-header:
   $\forall i < \text{rounds}.$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter s + i) (state-after-query-chunks
        (verifier-header-state s fr (map snd f-fl) trace-final as dg
          (map snd fl) composition-final)
        query-chunks i)) =
      Some (raw-idxs ! i)
    using lookups-all query-state-eq query-count-header by simp

```

```

have accepted:
  accepted-fri-opening-transcript s (Some (result, final-state))
    (map snd f-fl) trace-bs trace-final dg (map snd fl) composition-bs
      composition-final query-idxs ?trace-round-layers ?composition-round-layers
unfolding accepted-fri-opening-transcript-def
apply (rule exI[where x=result])
apply (rule exI[where x=final-state])
apply (rule exI[where x=fr])
apply (rule exI[where x=f-fl])
apply (rule exI[where x=as])
apply (rule exI[where x=fl])
apply (rule exI[where x=query-state])
apply (rule exI[where x=raw-idxs])
apply (rule exI[where x=query-chunks])
apply (simp add: challenges header query-state-eq query-out' derived len-raw
  query-count-header query-idxs-def len-chunks tr-query-tail layers
  lookups-all idx-bounds trace-bs-eq composition-bs-eq challenges-maps
  derived-maps layers-maps lookups-header lookups-all-header round-chunks)
apply (rule round-chunks-raw)
done
then show ?thesis
  by (rule that)
qed

```

```

lemma accepted-fri-opening-transcript-challenges:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers
shows accepted-fri-challenges s out trace-bs dg composition-bs
using fri-openings
unfolding accepted-fri-opening-transcript-def
apply (elim exE conjE)
apply assumption
done

```

```

lemma accepted-fri-opening-transcript-trace-challenge-list-set-hit:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers
and member: trace-bs ∈ B
shows trace-fri-challenge-list-set-hit s B out
using fri-openings
unfolding accepted-fri-opening-transcript-def trace-fri-challenge-list-set-hit-def
apply (elim exE conjE)
apply (intro exI conjI)
apply assumption
apply (rule member)

```


done

lemma *accepted-fri-opening-transcript-composition-challenge-list-set-hit:*

assumes *fri-openings:*

*accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers*

and *member: composition-bs ∈ B dg*

shows *composition-fri-challenge-list-set-hit s B out*

using *fri-openings*

unfolding *accepted-fri-opening-transcript-def*

composition-fri-challenge-list-set-hit-def

apply (*elim exE conjE*)

apply (*intro exI conjI*)

apply *assumption*

apply (*rule member*)

done

lemma *accepted-fri-opening-transcript-shapes:*

assumes *fri-openings:*

*accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers*

shows *length trace-roots = ceil-log clength*

and *length trace-bs = length trace-roots*

and *length composition-roots = ceil-log (to-nat dg + 1)*

and *length composition-bs = length composition-roots*

and *length trace-round-layers = rounds*

and *length composition-round-layers = rounds*

and *length query-idxs = rounds*

using *fri-openings*

unfolding *accepted-fri-opening-transcript-def verifier-header-transcript-def*

by *auto*

lemma *accepted-fri-opening-transcript-algebraic-round-counts:*

assumes *fri-openings:*

*accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers*

shows *length trace-roots = trace-fri-algebraic-round-count*

and *length trace-bs = trace-fri-algebraic-round-count*

and *length composition-roots = composition-fri-algebraic-round-count dg*

and *length composition-bs = composition-fri-algebraic-round-count dg*

and *length trace-round-layers = rounds*

and *length composition-round-layers = rounds*

and *length query-idxs = rounds*

proof —

have *shapes:*

length trace-roots = ceil-log clength

```

length trace-bs = length trace-roots
length composition-roots = ceil-log (to-nat dg + 1)
length composition-bs = length composition-roots
length trace-round-layers = rounds
length composition-round-layers = rounds
length query-idxs = rounds
using accepted-fri-opening-transcript-shapes[OF fri-openings] by blast+
show length trace-roots = trace-fri-algebraic-round-count
  using shapes(1) by (simp add: trace-fri-algebraic-round-count-eq)
show length trace-bs = trace-fri-algebraic-round-count
  using shapes(1,2) by (simp add: trace-fri-algebraic-round-count-eq)
show length composition-roots = composition-fri-algebraic-round-count dg
  using shapes(3) by (simp add: composition-fri-algebraic-round-count-eq)
show length composition-bs = composition-fri-algebraic-round-count dg
  using shapes(3,4) by (simp add: composition-fri-algebraic-round-count-eq)
show length trace-round-layers = rounds
  using shapes(5) .
show length composition-round-layers = rounds
  using shapes(6) .
show length query-idxs = rounds
  using shapes(7) .
qed

```

lemma *accepted-fri-opening-transcript-trace-round-layer-length:*

assumes *fri-openings:*

*accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers*

and *round-bound: round-idx < rounds*

shows *length (trace-round-layers ! round-idx) = length trace-roots*

proof –

obtain *result final-state fr f-fl as fl query-state raw-idxs query-chunks*

where *layers:*

$\bigwedge i. i < rounds \implies$

*query-round-fri-layer-transcripts (query-idxs ! i)
trace-roots composition-roots (query-chunks ! i)
(trace-round-layers ! i) (composition-round-layers ! i)*

using *fri-openings* **unfolding** *accepted-fri-opening-transcript-def* **by** *blast*
from *layers[OF round-bound]*

obtain *query-chunk trace-fri-chunk composition-fri-chunk leaves paths* **where**
trace-layers:

*fri-layers-transcript (length trace-roots) (clength * scale)
(trace-round-layers ! round-idx) trace-fri-chunk*

unfolding *query-round-fri-layer-transcripts-def* **by** *blast*

then show *?thesis*

unfolding *fri-layers-transcript-def* **by** *simp*

qed

lemma *accepted-fri-opening-transcript-composition-round-layer-length:*

assumes *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers
and *round-bound*: *round-idx < rounds*
shows *length (composition-round-layers ! round-idx) = length composition-roots*
proof –
obtain *result final-state fr f-fl as fl query-state raw-idxs query-chunks*
where *layers*:
 $\bigwedge i. i < rounds \implies$
query-round-fri-layer-transcripts (query-idxs ! i)
trace-roots composition-roots (query-chunks ! i)
(trace-round-layers ! i) (composition-round-layers ! i)
using *fri-openings* **unfolding** *accepted-fri-opening-transcript-def* **by** *blast*
from *layers[OF round-bound]*
obtain *query-chunk trace-fri-chunk composition-fri-chunk leaves paths* **where**
composition-layers:
*fri-layers-transcript (length composition-roots) (clength * scale)*
(composition-round-layers ! round-idx) composition-fri-chunk
unfolding *query-round-fri-layer-transcripts-def* **by** *blast*
then show *?thesis*
unfolding *fri-layers-transcript-def* **by** *simp*
qed

lemma *accepted-fri-opening-transcript-trace-layer-opening*:
assumes *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers
and *round-bound*: *round-idx < rounds*
and *layer-bound*: *layer-idx < length trace-roots*
obtains *root challenge len idx sibling-idx xp xp-path xn xn-path*
where *root = trace-roots ! layer-idx*
and *challenge = trace-bs ! layer-idx*
and *len = fri-layer-lengths (length trace-roots) (clength * scale) ! layer-idx*
and *idx =*
fri-layer-indices (length trace-roots) (query-idxs ! round-idx)
*(clength * scale) ! layer-idx*
and *sibling-idx = fri-sibling-index len idx*
and *fri-layer-opening-chunk len xp xp-path xn xn-path*
(trace-round-layers ! round-idx ! layer-idx)
proof –
obtain *result final-state fr f-fl as fl query-state raw-idxs query-chunks*
where *layers*:
 $\bigwedge i. i < rounds \implies$
query-round-fri-layer-transcripts (query-idxs ! i)
trace-roots composition-roots (query-chunks ! i)
(trace-round-layers ! i) (composition-round-layers ! i)
using *fri-openings* **unfolding** *accepted-fri-opening-transcript-def* **by** *blast*

```

from layers[OF round-bound]
obtain query-chunk trace-fri-chunk composition-fri-chunk leaves paths where
  trace-layers:
    fri-layers-transcript (length trace-roots) (clength * scale)
      (trace-round-layers ! round-idx) trace-fri-chunk
    unfolding query-round-fri-layer-transcripts-def by blast
from fri-layers-transcript-nth-opening[OF trace-layers layer-bound]
obtain xp xp-path xn xn-path where layer-opening:
  fri-layer-opening-chunk
    (fri-layer-lengths (length trace-roots) (clength * scale) ! layer-idx)
    xp xp-path xn xn-path (trace-round-layers ! round-idx ! layer-idx)
  by blast
let ?len = fri-layer-lengths (length trace-roots) (clength * scale) ! layer-idx
let ?idx =
  fri-layer-indices (length trace-roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx
let ?sidx = fri-sibling-index ?len ?idx
show ?thesis
  by (rule that[of trace-roots ! layer-idx trace-bs ! layer-idx
    ?len ?idx ?sidx xp xp-path xn xn-path])
    (use layer-opening in simp-all)
qed

lemma accepted-fri-opening-transcript-composition-layer-opening:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
and round-bound: round-idx < rounds
and layer-bound: layer-idx < length composition-roots
obtains root challenge len idx sibling-idx xp xp-path xn xn-path
where root = composition-roots ! layer-idx
and challenge = composition-bs ! layer-idx
and len =
  fri-layer-lengths (length composition-roots) (clength * scale) ! layer-idx
and idx =
  fri-layer-indices (length composition-roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx
and sibling-idx = fri-sibling-index len idx
and fri-layer-opening-chunk len xp xp-path xn xn-path
  (composition-round-layers ! round-idx ! layer-idx)
proof –
obtain result final-state fr f-fl as fl query-state raw-idxs query-chunks
where layers:
   $\bigwedge i. i < rounds \implies$ 
    query-round-fri-layer-transcripts (query-idxs ! i)
      trace-roots composition-roots (query-chunks ! i)
      (trace-round-layers ! i) (composition-round-layers ! i)
using fri-openings unfolding accepted-fri-opening-transcript-def by blast

```

```

from layers[OF round-bound]
obtain query-chunk trace-fri-chunk composition-fri-chunk leaves paths where
  composition-layers:
    fri-layers-transcript (length composition-roots) (clength * scale)
      (composition-round-layers ! round-idx) composition-fri-chunk
    unfolding query-round-fri-layer-transcripts-def by blast
from fri-layers-transcript-nth-opening[OF composition-layers layer-bound]
obtain xp xp-path xn xn-path where layer-opening:
  fri-layer-opening-chunk
    (fri-layer-lengths (length composition-roots) (clength * scale) ! layer-idx)
    xp xp-path xn xn-path
    (composition-round-layers ! round-idx ! layer-idx)
  by blast
let ?len =
  fri-layer-lengths (length composition-roots) (clength * scale) ! layer-idx
let ?idx =
  fri-layer-indices (length composition-roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx
let ?sidx = fri-sibling-index ?len ?idx
show ?thesis
  by (rule that[of composition-roots ! layer-idx composition-bs ! layer-idx
    ?len ?idx ?sidx xp xp-path xn xn-path])
    (use layer-opening in simp-all)
qed

```

4.2.2 Global FRI table bindings

The verifier sees authenticated local openings. The following predicates describe the stronger deterministic object needed by the algebraic FRI argument: complete layer tables bound to the Merkle roots, with all local openings matching the corresponding table entries.

```

definition fri-layer-tables-shape
  :: nat  $\Rightarrow$  'f list list  $\Rightarrow$  bool
where
  fri-layer-tables-shape n tables  $\longleftrightarrow$ 
    length tables = n  $\wedge$ 
    ( $\forall j < n.$ 
      length (tables ! j) =
        fri-layer-lengths n (clength * scale) ! j)

definition fri-roots-bind-tables
  :: 'f list  $\Rightarrow$  'f list list  $\Rightarrow$ 
    ('f, 'a) protocol-channel-scheme  $\Rightarrow$  bool
where
  fri-roots-bind-tables roots tables final-state  $\longleftrightarrow$ 
    length tables = length roots  $\wedge$ 
    ( $\forall j < \text{length roots}.$ 
      merkle-root-binds-table (roots ! j) (tables ! j) final-state)

```

definition *fri-bound-layer-tables*

$:: 'f \text{ list} \Rightarrow 'f \text{ list list} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

fri-bound-layer-tables roots tables final-state $\longleftrightarrow$
fri-layer-tables-shape (length roots) tables $\wedge$
fri-roots-bind-tables roots tables final-state

definition *fri-layer-opening-matches-bound-table*

$:: \text{nat} \Rightarrow \text{nat} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

fri-layer-opening-matches-bound-table query-idx layer-idx tables
layer-chunks $\longleftrightarrow$
layer-idx < length tables $\wedge$
length layer-chunks = length tables $\wedge$
(let len =
 fri-layer-lengths (length tables) (clength * scale) ! *layer-idx*;
 idx =
 fri-layer-indices (length tables) *query-idx* (clength * scale) !
 layer-idx
in $\exists xp \text{ } xp\text{-path } xn \text{ } xn\text{-path}.$
 fri-layer-opening-chunk len *xp xp-path xn xn-path*
 (*layer-chunks* ! *layer-idx*) $\wedge$
 fri-opening-matches-table len *idx* (*tables* ! *layer-idx*) *xp xn*)

definition *fri-round-openings-match-tables*

$:: 'f \text{ list list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

fri-round-openings-match-tables tables query-idx layer-chunks $\longleftrightarrow$
length layer-chunks = length tables $\wedge$
 $(\forall j < \text{length tables}.$
 fri-layer-opening-matches-bound-table query-idx j tables
 layer-chunks)

definition *fri-all-round-openings-match-tables*

$:: 'f \text{ list list} \Rightarrow \text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow \text{bool}$

where

fri-all-round-openings-match-tables tables query-idxs round-layers $\longleftrightarrow$
length round-layers = length query-idxs $\wedge$
 $(\forall i < \text{length query-idxs}.$
 fri-round-openings-match-tables tables (*query-idxs* ! *i*)
 (*round-layers* ! *i*))

definition *fri-final-table-matches*

$:: 'f \text{ list} \Rightarrow 'f \Rightarrow \text{bool}$

where

fri-final-table-matches final-table final $\longleftrightarrow$
final-table $\neq [] \wedge$

fri-final-constant-consistent final-table final

definition *trace-fri-tables-bound*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$

where

trace-fri-tables-bound s out trace-roots trace-bs trace-final
query-idxs trace-round-layers trace-tables trace-final-table $\longleftrightarrow$
 $(\exists \text{result final-state dg composition-roots composition-bs}$
 $\text{composition-final composition-round-layers.}$
 $\text{out} = \text{Some}(\text{result}, \text{final-state}) \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{fri-bound-layer-tables trace-roots trace-tables final-state} \wedge$
 $\text{fri-all-round-openings-match-tables trace-tables query-idxs}$
 $\text{trace-round-layers} \wedge$
 $\text{fri-final-table-matches trace-final-table trace-final})$

definition *composition-fri-tables-bound*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$

where

composition-fri-tables-bound s out dg composition-roots
composition-bs composition-final query-idxs composition-round-layers
composition-tables composition-final-table $\longleftrightarrow$
 $(\exists \text{result final-state trace-roots trace-bs trace-final}$
 $\text{trace-round-layers.}$
 $\text{out} = \text{Some}(\text{result}, \text{final-state}) \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{fri-bound-layer-tables composition-roots composition-tables}$
 $\text{final-state} \wedge$
 $\text{fri-all-round-openings-match-tables composition-tables query-idxs}$
 $\text{composition-round-layers} \wedge$
 $\text{fri-final-table-matches composition-final-table composition-final})$

definition *fri-initial-table-agrees*

$:: 'f \text{ list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

fri-initial-table-agrees initial-table tables $\longleftrightarrow$
 $\text{tables} = [] \vee \text{hd tables} = \text{initial-table}$

definition *fri-merkle-binding-good*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  fri-merkle-binding-good s out  $\longleftrightarrow$ 
    accepted out  $\longrightarrow$ 
      ( $\exists$  trace-table composition-table as query-idxs
        trace-roots trace-bs trace-final dg composition-roots
        composition-bs composition-final trace-round-layers
        composition-round-layers trace-tables composition-tables
        trace-final-table composition-final-table.
        accepted-with-bound-tables s out trace-table composition-table as
        query-idxs  $\wedge$ 
        accepted-fri-opening-transcript s out trace-roots trace-bs
        trace-final dg composition-roots composition-bs
        composition-final query-idxs trace-round-layers
        composition-round-layers  $\wedge$ 
        trace-fri-tables-bound s out trace-roots trace-bs trace-final
        query-idxs trace-round-layers trace-tables trace-final-table  $\wedge$ 
        composition-fri-tables-bound s out dg composition-roots
        composition-bs composition-final query-idxs
        composition-round-layers composition-tables
        composition-final-table  $\wedge$ 
        fri-initial-table-agrees trace-table trace-tables  $\wedge$ 
        fri-initial-table-agrees composition-table composition-tables)

definition fri-merkle-binding-bad
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  fri-merkle-binding-bad s out  $\longleftrightarrow$ 
    accepted out  $\wedge \neg$  fri-merkle-binding-good s out

definition fri-merkle-binding-bound
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$  bool
where
  fri-merkle-binding-bound s  $\longleftrightarrow$ 
    wp-event verify-monad (fri-merkle-binding-bad s) s  $\leq$ 
    merkle-binding-error

end

end

theory Soundness-Oracle-Target
  imports Soundness-FRI
begin

```


4.3 Oracle Budgets and Security Experiment

Random-oracle target-hit events and target-budget calculus.

context *soundness*

begin

definition *hash-map-output-collision-bound*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{hash-map-output-collision-bound } s \longleftrightarrow$

$\text{wp-event verify-monad } (\text{hash-map-output-collision-bad } s) \ s \leq$

$\text{merkle-binding-error}$

definition *hash-map-new-output-collision*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{hash-map-new-output-collision initial final} \longleftrightarrow$

$\neg \text{hash-map-output-collision initial} \wedge$

$\text{hash-map-output-collision final}$

definition *hash-new-collision-event*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(r \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{hash-new-collision-event } s \text{ out} \longleftrightarrow$

$(\text{case out of}$

$\text{None} \Rightarrow \text{False}$

$| \text{Some } (-, t) \Rightarrow \text{hash-map-new-output-collision } s \ t)$

definition *hash-map-new-output-hit*

$:: f \text{ set} \Rightarrow$

$(f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{hash-map-new-output-hit } B \text{ initial final} \longleftrightarrow$

$(\exists x \ y.$

$\text{fmlookup } (\text{HashMap initial}) \ x = \text{None} \wedge$

$\text{fmlookup } (\text{HashMap final}) \ x = \text{Some } y \wedge$

$y \in B)$

definition *hash-new-output-hit-event*

$:: f \text{ set} \Rightarrow$

$(f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(r \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{hash-new-output-hit-event } B \ s \text{ out} \longleftrightarrow$

$(\text{case out of}$

$\text{None} \Rightarrow \text{False}$

| *Some* $(-, t) \Rightarrow \text{hash-map-new-output-hit } B \ s \ t)$

lemma *hash-map-new-output-hit-subset*:

assumes *subset*: $A \subseteq B$

and *hit*: *hash-map-new-output-hit* $A \ s \ t$

shows *hash-map-new-output-hit* $B \ s \ t$

using *subset hit unfolding hash-map-new-output-hit-def* **by** *blast*

lemma *protocol-merkle-created-tree-pullback-or-new-output-hit*:

assumes *created*: *protocol-merkle.created-tree* $xs \ tree \ t$

and *ext*: $s \leq t$

shows

protocol-merkle.created-tree $xs \ tree \ s \ \vee$

hash-map-new-output-hit $(\text{set-tree } tree) \ s \ t$

using *created ext*

proof (*induction length xs arbitrary: xs tree s t rule: less-induct*)

case *less*

show *?case*

proof (*cases xs*)

case *Nil*

then show *?thesis*

using *less.prem*s

by (*simp add: protocol-merkle.created-tree.simps*)

next

case (*Cons x rest*)

have *xs-cons*: $xs = x \# \text{rest}$

using *Cons* **by** *simp*

show *?thesis*

proof (*cases rest*)

case *Nil*

then obtain *h* **where**

tree-eq: $tree = \langle \langle \rangle, h, \langle \rangle \rangle$

and *lookup-t*: *fmlookup* (*HashMap* *t*) *x* = *Some* *h*

using *less.prem*s *Cons*

by (*auto simp: protocol-merkle.created-tree.simps*)

show *?thesis*

proof (*cases fmlookup (HashMap s) x*)

case *None*

then have *hash-map-new-output-hit* $(\text{set-tree } tree) \ s \ t$

unfolding *hash-map-new-output-hit-def tree-eq*

using *lookup-t* **by** *auto*

then show *?thesis* **by** *simp*

next

case (*Some h'*)

have *lookup-t'*: *fmlookup* (*HashMap* *t*) *x* = *Some* *h'*

by (*rule hash-extension-lookup[OF Some less.prem*s(2)]

then have *h'-eq*: $h' = h$

using *lookup-t* **by** *simp*

have *protocol-merkle.created-tree* $[x] \ tree \ s$

```

    unfolding tree-eq
    using Some h'-eq
    by (simp add: protocol-merkle.created-tree.simps)
  then show ?thesis
    using Cons Nil by simp
qed
next
case (Cons y ys)
let ?xs = x # y # ys
let ?i = length ?xs div 2
have xs-eq: xs = ?xs
  using xs-cons Cons by simp
from less.prem1(1) obtain l h r where
  tree-eq: tree = ⟨l, h, r⟩
  and left-t:
    protocol-merkle.created-tree (take ?i ?xs) l t
  and right-t:
    protocol-merkle.created-tree (drop ?i ?xs) r t
  and lookup-t:
    fmllookup (HashMap t) (MerkleNode (value l) (value r)) =
      Some h
  unfolding xs-eq
  by (auto simp: protocol-merkle.created-tree.simps Let-def)
have i-pos: 0 < ?i
  by simp
have i-lt: ?i < length ?xs
  by simp
have len-left: length (take ?i ?xs) < length xs
  using i-lt unfolding xs-eq by simp
have len-right: length (drop ?i ?xs) < length xs
  using i-pos unfolding xs-eq by simp
have left-split:
  protocol-merkle.created-tree (take ?i ?xs) l s ∨
  hash-map-new-output-hit (set-tree l) s t
  by (rule less.hyps[OF len-left])
  (use left-t less.prem1(2) in simp-all)
from left-split show ?thesis
proof
  assume left-s:
    protocol-merkle.created-tree (take ?i ?xs) l s
  have right-split:
    protocol-merkle.created-tree (drop ?i ?xs) r s ∨
    hash-map-new-output-hit (set-tree r) s t
    by (rule less.hyps[OF len-right])
    (use right-t less.prem1(2) in simp-all)
  from right-split show ?thesis
proof
    assume right-s:
      protocol-merkle.created-tree (drop ?i ?xs) r s

```

```

show ?thesis
proof (cases
  fmlookup (HashMap s)
    (MerkleNode (value l) (value r)))
  case None
  then have hash-map-new-output-hit (set-tree tree) s t
    unfolding hash-map-new-output-hit-def tree-eq
    using lookup-t by auto
  then show ?thesis by simp
next
  case (Some h')
  have lookup-t':
    fmlookup (HashMap t) (MerkleNode (value l) (value r)) =
      Some h'
    by (rule hash-extension-lookup[OF Some less.premis(2)])
  then have h'-eq: h' = h
    using lookup-t by simp
  have protocol-merkle.created-tree ?xs tree s
    unfolding tree-eq
    using left-s right-s Some h'-eq
    by (auto simp add: protocol-merkle.created-tree.simps Let-def)
  then have protocol-merkle.created-tree xs tree s
    unfolding xs-eq .
  then show ?thesis by blast
qed
next
  assume right-hit: hash-map-new-output-hit (set-tree r) s t
  have hash-map-new-output-hit (set-tree tree) s t
    by (rule hash-map-new-output-hit-subset[OF - right-hit])
    (auto simp add: tree-eq)
  then show ?thesis by simp
qed
next
  assume left-hit: hash-map-new-output-hit (set-tree l) s t
  have hash-map-new-output-hit (set-tree tree) s t
    by (rule hash-map-new-output-hit-subset[OF - left-hit])
    (auto simp add: tree-eq)
  then show ?thesis by simp
qed
qed
qed
qed

lemma protocol-created-tree-pullback-or-new-output-hit:
  assumes created: protocol-created-tree xs tree t
  and ext: s ≤ t
  shows
    protocol-created-tree xs tree s ∨
    hash-map-new-output-hit (set-tree tree) s t

```

```

using protocol-merkle-created-tree-pullback-or-new-output-hit
  [OF created[unfolded protocol-created-tree-def] ext]
unfolding protocol-created-tree-def .

lemma protocol-created-tree-nonempty-root-lookup:
  assumes created: protocol-created-tree xs tree s
  and nonempty: xs ≠ []
  obtains input where
    fmlookup (HashMap s) input = Some (value tree)
proof (cases length xs = 1)
case True
  then obtain x where xs-eq: xs = [x]
  using nonempty by (cases xs) auto
  from protocol-created-tree-singletonD[OF created[unfolded xs-eq]]
  obtain h where
    tree-eq: tree = ⟨⟨⟩, h, ⟨⟩⟩
  and lookup: fmlookup (HashMap s) (MerkleLeaf x) = Some h
  by blast
  show ?thesis
  by (rule that[of MerkleLeaf x]) (use lookup in ⟨simp add: tree-eq⟩)
next
case False
  have len: 2 ≤ length xs
  using nonempty False by (cases xs; cases tl xs) auto
  from protocol-created-tree-internalD[OF created len]
  obtain l h r i where
    tree-eq: tree = ⟨l, h, r⟩
  and lookup:
    fmlookup (HashMap s) (MerkleNode (value l) (value r)) = Some h
  by blast
  show ?thesis
  by (rule that[of MerkleNode (value l) (value r)])
    (use lookup in ⟨simp add: tree-eq⟩)
qed

lemma merkle-root-binds-nonempty-table-output-lookup:
  fixes rt :: 'f
  assumes bind: merkle-root-binds-table rt table s
  and nonempty: table ≠ []
  obtains input where
    fmlookup (HashMap s) input = Some rt
proof –
  from bind obtain tree where
    created: created-tree table tree s
  and root-eq: rt = value tree
  unfolding merkle-root-binds-table-def by blast
  from protocol-created-tree-nonempty-root-lookup
    [OF created[unfolded created-tree-def] nonempty]
  obtain input where

```

```

    fmlookup (HashMap s) input = Some (value tree)
  by blast
then have lookup: fmlookup (HashMap s) input = Some rt
  using root-eq by simp
show ?thesis
  by (rule that[OF lookup])
qed

lemma merkle-root-binds-table-pullback-or-new-tree-output-hit:
  assumes ext: prefix-state  $\leq$  final-state
  and bind: merkle-root-binds-table rt table final-state
  obtains tree where
    created-tree table tree final-state
    rt = value tree
    merkle-root-binds-table rt table prefix-state  $\vee$ 
    hash-map-new-output-hit (set-tree tree) prefix-state final-state
proof -
  from bind obtain tree where
    created: created-tree table tree final-state
    and rt-eq: rt = value tree
    unfolding merkle-root-binds-table-def by blast
  have pullback:
    created-tree table tree prefix-state  $\vee$ 
    hash-map-new-output-hit (set-tree tree) prefix-state final-state
  using protocol-created-tree-pullback-or-new-output-hit
    [OF created[unfolded created-tree-def] ext]
  unfolding created-tree-def .
  have
    merkle-root-binds-table rt table prefix-state  $\vee$ 
    hash-map-new-output-hit (set-tree tree) prefix-state final-state
  proof -
    from pullback show ?thesis
  proof
    assume created-tree table tree prefix-state
    then have merkle-root-binds-table rt table prefix-state
      unfolding merkle-root-binds-table-def
      using rt-eq by blast
    then show ?thesis by simp
  next
    assume hash-map-new-output-hit (set-tree tree) prefix-state final-state
    then show ?thesis by simp
  qed
qed
then show ?thesis
  by (rule that[OF created rt-eq])
qed

lemma merkle-root-binds-nonempty-table-late-root-output-preexisting-or-new-hit:
  fixes rt :: 'f

```

```

assumes ext:  $s \leq t$ 
and bind: merkle-root-binds-table rt table t
and nonempty: table  $\neq []$ 
shows
   $(\exists \text{input. fmlookup (HashMap } s) \text{ input} = \text{Some } rt) \vee$ 
  hash-map-new-output-hit {rt} s t
proof –
from merkle-root-binds-nonempty-table-output-lookup[OF bind nonempty]
obtain input where lookup-t:
  fmlookup (HashMap t) input = Some rt
by blast
show ?thesis
proof (cases fmlookup (HashMap s) input)
case None
then have hash-map-new-output-hit {rt} s t
  unfolding hash-map-new-output-hit-def
  using lookup-t by blast
then show ?thesis by simp
next
case (Some old)
have lookup-t-old: fmlookup (HashMap t) input = Some old
  by (rule hash-extension-lookup[OF Some ext])
then have old-eq: old = rt
  using lookup-t by simp
then show ?thesis
  using Some by blast
qed
qed

lemma accepted-with-bound-tables-trace-root-late-output-preexisting-or-new-hit:
assumes ext: prefix-state  $\leq$  final-state
and bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
obtains fr f-fri-roots f-final dg composition-fri-roots final rest
where
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
  merkle-root-binds-table fr trace-table final-state
   $(\exists \text{input. fmlookup (HashMap } \text{prefix-state}) \text{ input} = \text{Some } \text{fr}) \vee$ 
  hash-map-new-output-hit {fr} prefix-state final-state
proof –
from bound obtain fr f-fri-roots f-final dg composition-fri-roots final
  rest where
  header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
and bind: merkle-root-binds-table fr trace-table final-state
unfolding accepted-with-bound-tables-def by blast

```

```

have nonempty: trace-table ≠ []
using accepted-with-bound-tables-shapes(1)[OF bound]
      eval-domain-nontrivial by auto
have late:
  (∃ input. fmlookup (HashMap prefix-state) input = Some fr) ∨
    hash-map-new-output-hit {fr} prefix-state final-state
by (rule
      merkle-root-binds-nonempty-table-late-root-output-preexisting-or-new-hit
      [OF ext bind nonempty])
show ?thesis
by (rule that[OF header bind late])
qed

lemma accepted-with-bound-tables-composition-root-late-output-preexisting-or-new-hit:
assumes ext: prefix-state ≤ final-state
and bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
obtains fr f-fri-roots f-final dg composition-fri-roots final rest
where
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
  composition-fri-roots ≠ []
  merkle-root-binds-table (hd composition-fri-roots) composition-table
    final-state
  (∃ input.
    fmlookup (HashMap prefix-state) input =
      Some (hd composition-fri-roots)) ∨
    hash-map-new-output-hit {hd composition-fri-roots} prefix-state
      final-state
proof —
from bound obtain fr f-fri-roots f-final dg composition-fri-roots final
  rest where
    header:
      verifier-header-transcript s fr f-fri-roots f-final as dg
      composition-fri-roots final rest
    and comp-nonempty: composition-fri-roots ≠ []
    and bind:
      merkle-root-binds-table (hd composition-fri-roots)
      composition-table final-state
    unfolding accepted-with-bound-tables-def by blast
have nonempty: composition-table ≠ []
using accepted-with-bound-tables-shapes(2)[OF bound]
      eval-domain-nontrivial by auto
have late:
  (∃ input.
    fmlookup (HashMap prefix-state) input =
      Some (hd composition-fri-roots)) ∨
    hash-map-new-output-hit {hd composition-fri-roots} prefix-state

```



```

    final-state
  by (rule
    merkle-root-binds-nonempty-table-late-root-output-preexisting-or-new-hit
    [OF ext bind nonempty])
  show ?thesis
  by (rule that[OF header comp-nonempty bind late])
qed

lemma accepted-with-bound-tables-initial-roots-late-output-preexisting-or-new-hit:
  assumes ext: prefix-state ≤ final-state
  and bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
  obtains fr f-fri-roots f-final dg composition-fri-roots final rest
  where
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
    composition-fri-roots ≠ []
    merkle-root-binds-table fr trace-table final-state
    merkle-root-binds-table (hd composition-fri-roots) composition-table
    final-state
    (∃ input. fmlookup (HashMap prefix-state) input = Some fr) ∨
    hash-map-new-output-hit {fr} prefix-state final-state
    (∃ input.
      fmlookup (HashMap prefix-state) input =
        Some (hd composition-fri-roots)) ∨
    hash-map-new-output-hit {hd composition-fri-roots} prefix-state
    final-state
  proof -
  from accepted-with-bound-tables-trace-root-late-output-preexisting-or-new-hit
    [OF ext bound]
  obtain fr f-fri-roots f-final dg composition-fri-roots final rest where
    header:
      verifier-header-transcript s fr f-fri-roots f-final as dg
      composition-fri-roots final rest
  and trace-bind:
    merkle-root-binds-table fr trace-table final-state
  and trace-late:
    (∃ input. fmlookup (HashMap prefix-state) input = Some fr) ∨
    hash-map-new-output-hit {fr} prefix-state final-state
  by blast
  from accepted-with-bound-tables-composition-root-late-output-preexisting-or-new-hit
    [OF ext bound]
  obtain fr' f-fri-roots' f-final' dg' composition-fri-roots' final' rest'
  where header':
    verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
    composition-fri-roots' final' rest'
  and comp-nonempty': composition-fri-roots' ≠ []
  and comp-bind':

```

```

merkle-root-binds-table (hd composition-fri-roots')
  composition-table final-state
and comp-late':
  (∃ input.
    fmlookup (HashMap prefix-state) input =
      Some (hd composition-fri-roots')) ∨
    hash-map-new-output-hit {hd composition-fri-roots'} prefix-state
      final-state
  by blast
have eqs:
  fr' = fr ∧
  f-fri-roots' = f-fri-roots ∧
  f-final' = f-final ∧
  dg' = dg ∧
  composition-fri-roots' = composition-fri-roots ∧
  final' = final ∧
  rest' = rest
  using verifier-header-transcript-unique[OF header header'] by simp
show ?thesis
  by (rule that[OF header])
    (use comp-nonempty' comp-bind' comp-late' trace-bind trace-late eqs
      in simp-all)
qed

lemma accepted-with-bound-tables-initial-roots-late-output-preexisting-or-union-hit:
assumes ext: prefix-state ≤ final-state
and bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
obtains fr f-fri-roots f-final dg composition-fri-roots final rest
where
  verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  composition-fri-roots ≠ []
  merkle-root-binds-table fr trace-table final-state
  merkle-root-binds-table (hd composition-fri-roots) composition-table
    final-state
  ((∃ input. fmlookup (HashMap prefix-state) input = Some fr) ∧
    (∃ input.
      fmlookup (HashMap prefix-state) input =
        Some (hd composition-fri-roots))) ∨
    hash-map-new-output-hit {fr, hd composition-fri-roots} prefix-state
      final-state
proof –
from accepted-with-bound-tables-initial-roots-late-output-preexisting-or-new-hit
  [OF ext bound]
obtain fr f-fri-roots f-final dg composition-fri-roots final rest where
  header:
    verifier-header-transcript s fr f-fri-roots f-final as dg

```

```

    composition-fri-roots final rest
and comp-nonempty: composition-fri-roots  $\neq []$ 
and trace-bind:
    merkle-root-binds-table fr trace-table final-state
and comp-bind:
    merkle-root-binds-table (hd composition-fri-roots)
    composition-table final-state
and trace-late:
    ( $\exists$  input. fmlookup (HashMap prefix-state) input = Some fr)  $\vee$ 
    hash-map-new-output-hit {fr} prefix-state final-state
and comp-late:
    ( $\exists$  input.
    fmlookup (HashMap prefix-state) input =
    Some (hd composition-fri-roots))  $\vee$ 
    hash-map-new-output-hit {hd composition-fri-roots} prefix-state
    final-state
by blast
have late-union:
    (( $\exists$  input. fmlookup (HashMap prefix-state) input = Some fr)  $\wedge$ 
    ( $\exists$  input.
    fmlookup (HashMap prefix-state) input =
    Some (hd composition-fri-roots)))  $\vee$ 
    hash-map-new-output-hit {fr, hd composition-fri-roots} prefix-state
    final-state
proof (cases
    ( $\exists$  input. fmlookup (HashMap prefix-state) input = Some fr)  $\wedge$ 
    ( $\exists$  input.
    fmlookup (HashMap prefix-state) input =
    Some (hd composition-fri-roots)))
case True
then show ?thesis by simp
next
case False
then have
    hash-map-new-output-hit {fr} prefix-state final-state  $\vee$ 
    hash-map-new-output-hit {hd composition-fri-roots} prefix-state
    final-state
using trace-late comp-late by blast
then show ?thesis
proof
assume hash-map-new-output-hit {fr} prefix-state final-state
then have
    hash-map-new-output-hit {fr, hd composition-fri-roots}
    prefix-state final-state
using hash-map-new-output-hit-subset[of {fr}
    {fr, hd composition-fri-roots} prefix-state final-state]
by auto
then show ?thesis by simp
next

```

```

assume
  hash-map-new-output-hit {hd composition-fri-roots} prefix-state
  final-state
then have
  hash-map-new-output-hit {fr, hd composition-fri-roots}
  prefix-state final-state
using hash-map-new-output-hit-subset
  [of {hd composition-fri-roots}
   {fr, hd composition-fri-roots} prefix-state final-state]
by auto
then show ?thesis by simp
qed
qed
show ?thesis
by (rule that[OF header comp-nonempty trace-bind comp-bind late-union])
qed

lemma accepted-with-bound-tables-initial-roots-pullback-or-new-tree-output-hit:
assumes ext: prefix-state  $\leq$  final-state
and bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
obtains fr f-fri-roots f-final dg composition-fri-roots final rest
  trace-tree composition-tree
where
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
  composition-fri-roots  $\neq$  []
  created-tree trace-table trace-tree final-state
  fr = value trace-tree
  created-tree composition-table composition-tree final-state
  hd composition-fri-roots = value composition-tree
  (merkle-root-binds-table fr trace-table prefix-state  $\wedge$ 
   merkle-root-binds-table (hd composition-fri-roots) composition-table
   prefix-state)  $\vee$ 
  hash-map-new-output-hit
  (set-tree trace-tree  $\cup$  set-tree composition-tree)
  prefix-state final-state
proof –
from bound obtain fr f-fri-roots f-final dg composition-fri-roots final
  rest where
  header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
and comp-nonempty: composition-fri-roots  $\neq$  []
and trace-bind:
  merkle-root-binds-table fr trace-table final-state
and comp-bind:
  merkle-root-binds-table (hd composition-fri-roots)

```

```

    composition-table final-state
  unfolding accepted-with-bound-tables-def by blast
from merkle-root-binds-table-pullback-or-new-tree-output-hit
  [OF ext trace-bind]
obtain trace-tree where
  trace-created: created-tree trace-table trace-tree final-state
  and trace-root: fr = value trace-tree
  and trace-split:
    merkle-root-binds-table fr trace-table prefix-state  $\vee$ 
    hash-map-new-output-hit (set-tree trace-tree) prefix-state
    final-state
  by blast
from merkle-root-binds-table-pullback-or-new-tree-output-hit
  [OF ext comp-bind]
obtain composition-tree where
  comp-created:
    created-tree composition-table composition-tree final-state
  and comp-root:
    hd composition-fri-roots = value composition-tree
  and comp-split:
    merkle-root-binds-table (hd composition-fri-roots)
    composition-table prefix-state  $\vee$ 
    hash-map-new-output-hit (set-tree composition-tree) prefix-state
    final-state
  by blast
have split:
  (merkle-root-binds-table fr trace-table prefix-state  $\wedge$ 
   merkle-root-binds-table (hd composition-fri-roots) composition-table
   prefix-state)  $\vee$ 
  hash-map-new-output-hit
  (set-tree trace-tree  $\cup$  set-tree composition-tree)
  prefix-state final-state
proof -
  from trace-split show ?thesis
proof
  assume trace-prefix:
    merkle-root-binds-table fr trace-table prefix-state
  from comp-split show ?thesis
proof
  assume comp-prefix:
    merkle-root-binds-table (hd composition-fri-roots)
    composition-table prefix-state
  then show ?thesis
  using trace-prefix by simp
next
  assume comp-hit:
    hash-map-new-output-hit (set-tree composition-tree)
    prefix-state final-state
  have hash-map-new-output-hit

```

```

      (set-tree trace-tree  $\cup$  set-tree composition-tree)
      prefix-state final-state
    by (rule hash-map-new-output-hit-subset[OF - comp-hit])
      simp
    then show ?thesis by simp
  qed
next
  assume trace-hit:
    hash-map-new-output-hit (set-tree trace-tree) prefix-state
    final-state
  have hash-map-new-output-hit
    (set-tree trace-tree  $\cup$  set-tree composition-tree)
    prefix-state final-state
  by (rule hash-map-new-output-hit-subset[OF - trace-hit])
    simp
  then show ?thesis by simp
  qed
qed
show ?thesis
  by (rule that[OF header comp-nonempty trace-created trace-root
    comp-created comp-root split])
qed

lemma hash-map-new-output-hit-after-update:
  hash-map-new-output-hit B s
  (s( $\downarrow$ HashMap := fmupd x y (HashMap s)))  $\longleftrightarrow$ 
  fmlookup (HashMap s) x = None  $\wedge$  y  $\in$  B
proof
  assume hit:
    hash-map-new-output-hit B s
    (s( $\downarrow$ HashMap := fmupd x y (HashMap s)))
  then obtain k z where fresh:
    fmlookup (HashMap s) k = None
  and lookup:
    fmlookup (fmupd x y (HashMap s)) k = Some z
  and z-in: z  $\in$  B
  unfolding hash-map-new-output-hit-def by auto
  have k = x
  proof (rule ccontr)
    assume k  $\neq$  x
    then have fmlookup (HashMap s) k = Some z
      using lookup by simp
    then show False
      using fresh by simp
  qed
  then show fmlookup (HashMap s) x = None  $\wedge$  y  $\in$  B
    using fresh lookup z-in by simp
next
  assume fmlookup (HashMap s) x = None  $\wedge$  y  $\in$  B

```

then show
 $\text{hash-map-new-output-hit } B \ s$
 $(s \ll \text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s))$
unfolding $\text{hash-map-new-output-hit-def}$ **by** *auto*
qed

lemma *hash-extension-none*:
assumes $\text{ext}: s \leq t$
and $\text{none}: \text{fmlookup } (\text{HashMap } t) \ x = \text{None}$
shows $\text{fmlookup } (\text{HashMap } s) \ x = \text{None}$
proof (*cases fmlookup (HashMap s) x*)
case *None*
then show *?thesis* .
next
case (*Some y*)
have $\text{fmlookup } (\text{HashMap } t) \ x = \text{Some } y$
by (*rule hash-extension-lookup[OF Some ext]*)
then show *?thesis*
using *none* **by** *simp*
qed

lemma *hash-map-new-output-hit-extend-initial*:
assumes $\text{ext}: s \leq t$
and $\text{hit}: \text{hash-map-new-output-hit } B \ t \ u$
shows $\text{hash-map-new-output-hit } B \ s \ u$
proof –
from *hit* **obtain** $x \ y$ **where** *fresh-t*:
 $\text{fmlookup } (\text{HashMap } t) \ x = \text{None}$
and $\text{lookup-u}: \text{fmlookup } (\text{HashMap } u) \ x = \text{Some } y$
and $y \in B$
unfolding $\text{hash-map-new-output-hit-def}$ **by** *blast*
have $\text{fresh-s}: \text{fmlookup } (\text{HashMap } s) \ x = \text{None}$
by (*rule hash-extension-none[OF ext fresh-t]*)
show *?thesis*
unfolding $\text{hash-map-new-output-hit-def}$
using *fresh-s lookup-u y-in* **by** *blast*
qed

lemma *hash-map-new-output-hit-trans-decomp*:
assumes $\text{ext-st}: s \leq t$
and $\text{ext-tu}: t \leq u$
and $\text{hit}: \text{hash-map-new-output-hit } B \ s \ u$
shows
 $\text{hash-map-new-output-hit } B \ s \ t \ \vee$
 $\text{hash-map-new-output-hit } B \ t \ u$
proof –
from *hit* **obtain** $x \ y$ **where** *fresh-s*:
 $\text{fmlookup } (\text{HashMap } s) \ x = \text{None}$
and $\text{lookup-u}: \text{fmlookup } (\text{HashMap } u) \ x = \text{Some } y$

```

    and  $y\text{-in}: y \in B$ 
    unfolding hash-map-new-output-hit-def by blast
show ?thesis
proof (cases fmlookup (HashMap t) x)
  case None
  then have hash-map-new-output-hit B t u
    unfolding hash-map-new-output-hit-def
    using lookup-u  $y\text{-in}$  by blast
  then show ?thesis by simp
next
  case (Some z)
  have lookup-u-z: fmlookup (HashMap u) x = Some z
    by (rule hash-extension-lookup[OF Some ext-tu])
  have z = y
    using lookup-u lookup-u-z by simp
  then have hash-map-new-output-hit B s t
    unfolding hash-map-new-output-hit-def
    using fresh-s Some  $y\text{-in}$  by blast
  then show ?thesis by simp
qed
qed

definition hash-extension-preserving
:: ('r, ('f, 'a) protocol-channel-scheme) state-monad  $\Rightarrow$  bool
where
  hash-extension-preserving m  $\longleftrightarrow$ 
  ( $\forall s\ x\ t.$ 
    Some (x, t)  $\in$  set-dist (execute m s)  $\longrightarrow s \leq t$ )

definition hash-map-preserving
:: ('r, ('f, 'a) protocol-channel-scheme) state-monad  $\Rightarrow$  bool
where
  hash-map-preserving m  $\longleftrightarrow$ 
  ( $\forall s\ x\ t.$ 
    Some (x, t)  $\in$  set-dist (execute m s)  $\longrightarrow$ 
    HashMap t = HashMap s)

lemma hash-map-preserving-imp-hash-extension-preserving:
assumes hash-map-preserving m
shows hash-extension-preserving m
unfolding hash-extension-preserving-def
proof (intro allI impI)
  fix s x t
  assume outcome: Some (x, t)  $\in$  set-dist (execute m s)
  have HashMap t = HashMap s
    using assms outcome unfolding hash-map-preserving-def by blast
  then show s  $\leq$  t
    unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
qed

```


lemma *hash-extension-preserving-return*:
hash-extension-preserving (return x)
unfolding *hash-extension-preserving-def*
by (intro allI impI) (simp add: hash-ext-refl)

lemma *hash-extension-preserving-hash*:
fixes $x :: 'f$ protocol-hash-input
shows
hash-extension-preserving
 (hash $x :: ('f, ('f, 'a)$ protocol-channel-scheme) state-monad)
unfolding *hash-extension-preserving-def*
by (intro allI impI) (rule hash-outcome(1))

lemma *hash-extension-preserving-bind*:
fixes $m :: ('x, ('f, 'a)$ protocol-channel-scheme) state-monad
and $k :: 'x \Rightarrow ('y, ('f, 'a)$ protocol-channel-scheme) state-monad
assumes *m-ext*: *hash-extension-preserving* m
and *k-ext*: $\bigwedge x. \text{hash-extension-preserving } (k\ x)$
shows *hash-extension-preserving* ($m \gg k$)
unfolding *hash-extension-preserving-def*
proof (intro allI impI)
fix $s\ y\ u$
assume *outcome*: Some $(y, u) \in \text{set-dist } (\text{execute } (m \gg k)\ s)$
from *outcome* **obtain** $x\ t$ **where** *m-out*:
 Some $(x, t) \in \text{set-dist } (\text{execute } m\ s)$
and *k-out*: Some $(y, u) \in \text{set-dist } (\text{execute } (k\ x)\ t)$
by (auto elim!: set-dist-bindE)
have *ext-st*: $s \leq t$
using *m-ext* *m-out* **unfolding** *hash-extension-preserving-def* **by** blast
have *ext-tu*: $t \leq u$
using *k-ext*[of x] *k-out* **unfolding** *hash-extension-preserving-def* **by** blast
show $s \leq u$
by (rule hash-ext-trans[OF *ext-st* *ext-tu*])
qed

lemma *wp-hash-new-output-hit*:
fixes $x :: 'f$ protocol-hash-input
and $s :: ('f, 'a)$ protocol-channel-scheme
shows
wp-event (hash x) (hash-new-output-hit-event $B\ s$) $s =$
 (if fmlookup (HashMap s) $x = \text{None}$
 then nnreal (card B) / nnreal size
 else 0)
proof (cases fmlookup (HashMap s) x)
case None
have *indicator*:
 ($\lambda y. \text{if hash-map-new-output-hit } B\ s$
 ($s \downarrow \text{HashMap} := \text{fmupd } x\ y\ (\text{HashMap } s) \downarrow$))

```

    then 1 else 0 :: prob) =
    (λy. if y ∈ B then 1 else 0)
  by (rule ext)
    (simp add: hash-map-new-output-hit-after-update None)
  have wp-event (hash x) (hash-new-output-hit-event B s) s =
    dist-expect (hash-dist x s) (λy. if y ∈ B then 1 else 0)
  unfolding wp-event-def hash-new-output-hit-event-def
  by (simp add: wp-hash indicator)
  also have ... = nnreal (card B) / nnreal size
  by (rule dist-expect-hash-dist-fresh-indicator[OF None])
  finally show ?thesis
  using None by simp
next
case (Some y)
have no-hit:
  ¬ hash-map-new-output-hit B s
  (s(HashMap := fmupd x y (HashMap s)))
  by (simp add: hash-map-new-output-hit-after-update Some)
show ?thesis
  unfolding wp-event-def hash-new-output-hit-event-def
  by (simp add: wp-hash Some hash-dist-def option-default-dist-def no-hit)
qed

```

lemma *wp-hash-new-output-hit-bound*:

fixes $x :: 'f \text{ protocol-hash-input}$
and $s :: ('f, 'a) \text{ protocol-channel-scheme}$
shows
 $\text{wp-event (hash } x \text{) (hash-new-output-hit-event } B \text{ } s) \text{ } s \leq$
 $\text{nnreal (card } B \text{) / nnreal size}$
unfolding *wp-hash-new-output-hit*
by *simp*

definition *hash-target-budget-value*
 $:: 'f \text{ set} \Rightarrow \text{nat} \Rightarrow \text{prob}$
where
 $\text{hash-target-budget-value } B \text{ } n =$
 $\text{nnreal (n * card } B \text{) / nnreal size}$

definition *hash-target-budget*
 $:: 'f \text{ set} \Rightarrow \text{nat} \Rightarrow$
 $(r, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad} \Rightarrow \text{bool}$
where
 $\text{hash-target-budget } B \text{ } n \text{ } m \longleftrightarrow$
 $(\forall s.$
 $\text{wp-event } m \text{ (hash-new-output-hit-event } B \text{ } s) \text{ } s \leq$
 $\text{hash-target-budget-value } B \text{ } n)$

lemma *hash-target-budget-value-add*:
 $\text{hash-target-budget-value } B \text{ } n + \text{hash-target-budget-value } B \text{ } m =$

```

    hash-target-budget-value  $B (n + m)$ 
unfolding hash-target-budget-value-def
by (simp add: add-divide-nnreal algebra-simps)

lemma hash-new-output-hit-event-step-bound:
  fixes  $m :: ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$ 
  assumes  $\text{ext-st}: s \leq t$ 
  and  $\text{ext-m}: \text{hash-extension-preserving } m$ 
  shows
     $\text{wp-event } m (\text{hash-new-output-hit-event } B s) t \leq$ 
     $(\text{if } \text{hash-map-new-output-hit } B s t \text{ then } 1$ 
     $\text{ else } \text{wp-event } m (\text{hash-new-output-hit-event } B t) t)$ 
proof (cases hash-map-new-output-hit  $B s t$ )
  case True
  then show ?thesis
    using wp-event-le-1 [of  $m$  hash-new-output-hit-event  $B s t$ ]
    by simp
next
  case False
  have mono:
     $\text{wp-event } m (\text{hash-new-output-hit-event } B s) t \leq$ 
     $\text{wp-event } m (\text{hash-new-output-hit-event } B t) t$ 
proof (rule wp-event-mono-on-support)
  fix out
  assume support:  $\text{out} \in \text{set-dist } (\text{execute } m t)$ 
  and hit-s:  $\text{hash-new-output-hit-event } B s \text{ out}$ 
  show  $\text{hash-new-output-hit-event } B t \text{ out}$ 
proof (cases out)
  case None
  then show ?thesis
    using hit-s unfolding hash-new-output-hit-event-def by simp
next
  case (Some  $xu$ )
  then obtain  $x u$  where  $xu: xu = (x, u)$ 
  by (cases  $xu$ ) simp
  have out:  $\text{Some } (x, u) \in \text{set-dist } (\text{execute } m t)$ 
  using support Some  $xu$  by simp
  have ext-tu:  $t \leq u$ 
  using ext-m out unfolding hash-extension-preserving-def by blast
  have hit-su:  $\text{hash-map-new-output-hit } B s u$ 
  using hit-s Some  $xu$  unfolding hash-new-output-hit-event-def by simp
  have  $\text{hash-map-new-output-hit } B s t \vee$ 
   $\text{hash-map-new-output-hit } B t u$ 
  by (rule hash-map-new-output-hit-trans-decomp
    [OF ext-st ext-tu hit-su])
  then have  $\text{hash-map-new-output-hit } B t u$ 
  using False by blast
  then show ?thesis
    using Some  $xu$  unfolding hash-new-output-hit-event-def by simp

```

```

    qed
  qed
  show ?thesis
    using False mono by simp
  qed

lemma hash-map-new-output-hit-refl[simp]:
  ¬ hash-map-new-output-hit B s s
  unfolding hash-map-new-output-hit-def by auto

lemma hash-target-budget-return:
  hash-target-budget B 0 (return x)
  unfolding hash-target-budget-def hash-target-budget-value-def
    hash-new-output-hit-event-def
  by (simp add: wp-event-def wpsimps)

lemma hash-map-preserving-no-new-output-hit-on-support:
  assumes preserving: hash-map-preserving m
    and support: out ∈ set-dist (execute m s)
  shows ¬ hash-new-output-hit-event B s out
proof (cases out)
  case None
  then show ?thesis
    unfolding hash-new-output-hit-event-def by simp
next
  case (Some xt)
  then obtain x t where xt: xt = (x, t)
    by (cases xt) simp
  have out: Some (x, t) ∈ set-dist (execute m s)
    using support Some xt by simp
  have map-eq: HashMap t = HashMap s
    using preserving out unfolding hash-map-preserving-def by blast
  have ¬ hash-map-new-output-hit B s t
    using map-eq unfolding hash-map-new-output-hit-def by auto
  then show ?thesis
    using Some xt unfolding hash-new-output-hit-event-def by simp
qed

lemma hash-map-preserving-imp-hash-target-budget-zero:
  assumes preserving: hash-map-preserving m
  shows hash-target-budget B 0 m
  unfolding hash-target-budget-def hash-target-budget-value-def
  apply (intro allI)
  apply (rule order-trans[
    where y=wp-event m (λ-. False) -])
  apply (rule wp-event-mono-on-support)
  apply (meson hash-map-preserving-no-new-output-hit-on-support preserving)
  unfolding wp-event-def wp-def dist-expect-def
  by simp

```

lemma *hash-target-budget-hash*:
fixes $x :: 'f \text{ protocol-hash-input}$
shows
 $\text{hash-target-budget } B \ 1$
 $(\text{hash } x :: ('f, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
unfolding *hash-target-budget-def*
proof (*intro allI*)
fix $s :: ('f, 'a) \text{ protocol-channel-scheme}$
show
 $\text{wp-event } (\text{hash } x) (\text{hash-new-output-hit-event } B \ s) \ s \leq$
 $\text{hash-target-budget-value } B \ 1$
using *wp-hash-new-output-hit-bound* [**where** $x=x$ **and** $s=s$ **and** $B=B$]
unfolding *hash-target-budget-value-def* **by** *simp*
qed

lemma *hash-target-budget-bind*:
fixes $m :: ('x, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
and $k :: 'x \Rightarrow ('y, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
assumes *m-ext*: *hash-extension-preserving* m
and *m-budget*: *hash-target-budget* $B \ n \ m$
and *k-ext*: $\bigwedge x. \text{hash-extension-preserving } (k \ x)$
and *k-budget*: $\bigwedge x. \text{hash-target-budget } B \ n' \ (k \ x)$
shows *hash-target-budget* $B \ (n + n') \ (m \ggg k)$
unfolding *hash-target-budget-def*
proof (*intro allI*)
fix $s :: ('f, 'a) \text{ protocol-channel-scheme}$
let $?E = \text{hash-new-output-hit-event } B \ s$
let $?Tail =$
 $\lambda out. \text{case out of}$
 $\text{None} \Rightarrow 0$
 $| \text{Some } (x, t) \Rightarrow$
 $\text{if } \text{hash-map-new-output-hit } B \ s \ t \text{ then } 0$
 $\text{else } \text{wp-event } (k \ x) (\text{hash-new-output-hit-event } B \ t) \ t$
let $?R = \lambda out. (\text{if } ?E \ out \text{ then } 1 \text{ else } 0) + ?Tail \ out$
have *split*:
 $\text{wp-event } (m \ggg k) \ ?E \ s \leq \text{wp } m \ ?R \ s$
proof –
have *exact*:
 $\text{wp-event } (m \ggg k) \ ?E \ s =$
 $\text{wp } m$
 $(\lambda out. \text{case out of}$
 $\text{None} \Rightarrow 0$
 $| \text{Some } (x, t) \Rightarrow \text{wp-event } (k \ x) \ ?E \ t) \ s$
unfolding *wp-event-def* *hash-new-output-hit-event-def*
by (*simp add: wpsimps*)
have *wp m*
 $(\lambda out. \text{case out of}$
 $\text{None} \Rightarrow 0$

```

      | Some (x, t) ⇒ wp-event (k x) ?E t) s
    ≤ wp m ?R s
proof (rule wp-mono-on-support)
  fix out
  assume support: out ∈ set-dist (execute m s)
  show (case out of
    None ⇒ 0
    | Some (x, t) ⇒ wp-event (k x) ?E t) ≤ ?R out
  proof (cases out)
    case None
    then show ?thesis
      unfolding hash-new-output-hit-event-def by simp
  next
    case (Some xt)
    then obtain x t where xt: xt = (x, t)
      by (cases xt) simp
    have m-out: Some (x, t) ∈ set-dist (execute m s)
      using support Some xt by simp
    have ext-st: s ≤ t
      using m-ext m-out unfolding hash-extension-preserving-def by blast
    have step:
      wp-event (k x) ?E t ≤
        (if hash-map-new-output-hit B s t then 1
         else wp-event (k x) (hash-new-output-hit-event B t) t)
      by (rule hash-new-output-hit-event-step-bound
        [OF ext-st k-ext[of x]])
    show ?thesis
  proof (cases hash-map-new-output-hit B s t)
    case True
    then show ?thesis
      using Some xt step
      unfolding hash-new-output-hit-event-def by simp
  next
    case False
    then show ?thesis
      using Some xt step
      unfolding hash-new-output-hit-event-def by simp
  qed
qed
qed
then show ?thesis
  unfolding exact .
qed
have head-bound:
  wp-event m ?E s ≤ hash-target-budget-value B n
  using m-budget unfolding hash-target-budget-def by blast
have tail-bound:
  wp m ?Tail s ≤ hash-target-budget-value B n'
proof (rule wp-le-const-on-support)

```

```

fix out
assume support: out  $\in$  set-dist (execute m s)
show ?Tail out  $\leq$  hash-target-budget-value B n'
proof (cases out)
  case None
  then show ?thesis
    unfolding hash-target-budget-value-def by simp
next
  case (Some xt)
  then obtain x t where xt: xt = (x, t)
    by (cases xt) simp
  have tail:
    wp-event (k x) (hash-new-output-hit-event B t) t  $\leq$ 
      hash-target-budget-value B n'
    using k-budget[of x] unfolding hash-target-budget-def by blast
  show ?thesis
    using Some xt tail by simp
qed
qed
have wp-event (m  $\ggg$  k) ?E s  $\leq$  wp m ?R s
  by (rule split)
also have ... = wp-event m ?E s + wp m ?Tail s
  unfolding wp-event-def wp-def dist-expect-def
  by (simp add: sum.distrib algebra-simps
    hash-new-output-hit-event-def)
also have ...  $\leq$ 
  hash-target-budget-value B n + hash-target-budget-value B n'
  by (intro add-mono head-bound tail-bound)
also have ... = hash-target-budget-value B (n + n')
  by (rule hash-target-budget-value-add)
finally show
  wp-event (m  $\ggg$  k) (hash-new-output-hit-event B s) s  $\leq$ 
    hash-target-budget-value B (n + n') .
qed

lemma hash-target-budget-value-mono:
  assumes m  $\leq$  n
  shows hash-target-budget-value B m  $\leq$  hash-target-budget-value B n
  unfolding hash-target-budget-value-def
  apply (subst nn2real-le-iff[symmetric])
  using assms by (simp add: divide-right-mono mult-right-mono)

lemma hash-target-budget-mono:
  assumes m  $\leq$  n
  and budget: hash-target-budget B m p
  shows hash-target-budget B n p
  unfolding hash-target-budget-def
proof (intro allI)
  fix s

```

have $wp\text{-event } p \text{ (hash-new-output-hit-event } B \text{ } s) \text{ } s \leq$
 $\text{hash-target-budget-value } B \text{ } m$
using *budget unfolding hash-target-budget-def* **by** *blast*
also have $\dots \leq \text{hash-target-budget-value } B \text{ } n$
by $(\text{rule hash-target-budget-value-mono}[OF \text{ } assms(1)])$
finally show
 $wp\text{-event } p \text{ (hash-new-output-hit-event } B \text{ } s) \text{ } s \leq$
 $\text{hash-target-budget-value } B \text{ } n \text{ .}$
qed

definition *hash-target-program*
 $:: 'f \text{ set} \Rightarrow \text{nat} \Rightarrow$
 $(r, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad} \Rightarrow \text{bool}$
where
 $\text{hash-target-program } B \text{ } n \text{ } m \longleftrightarrow$
 $\text{hash-extension-preserving } m \wedge \text{hash-target-budget } B \text{ } n \text{ } m$

lemma *hash-target-program-extension:*
assumes $\text{hash-target-program } B \text{ } n \text{ } m$
shows $\text{hash-extension-preserving } m$
using *assms unfolding hash-target-program-def* **by** *simp*

lemma *hash-target-program-budget:*
assumes $\text{hash-target-program } B \text{ } n \text{ } m$
shows $\text{hash-target-budget } B \text{ } n \text{ } m$
using *assms unfolding hash-target-program-def* **by** *simp*

lemma *hash-target-program-mono:*
assumes $m \leq n$
and *program:* $\text{hash-target-program } B \text{ } m \text{ } p$
shows $\text{hash-target-program } B \text{ } n \text{ } p$
unfolding *hash-target-program-def*
proof
show $\text{hash-extension-preserving } p$
using *program unfolding hash-target-program-def* **by** *simp*
have $\text{hash-target-budget } B \text{ } m \text{ } p$
using *program unfolding hash-target-program-def* **by** *simp*
then show $\text{hash-target-budget } B \text{ } n \text{ } p$
by $(\text{rule hash-target-budget-mono}[OF \text{ } assms(1)])$
qed

lemma *hash-target-program-return:*
 $\text{hash-target-program } B \text{ } 0 \text{ (return } x)$
unfolding *hash-target-program-def*
by $(\text{intro conjI hash-extension-preserving-return hash-target-budget-return})$

lemma *hash-target-program-hash:*
fixes $x :: 'f \text{ protocol-hash-input}$
shows

$\text{hash-target-program } B \ 1$
 $(\text{hash } x :: ('f, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
unfolding $\text{hash-target-program-def}$
by $(\text{intro conjI hash-extension-preserving-hash hash-target-budget-hash})$

lemma $\text{hash-map-preserving-imp-hash-target-program-zero}$:
assumes $\text{hash-map-preserving } m$
shows $\text{hash-target-program } B \ 0 \ m$
unfolding $\text{hash-target-program-def}$
using $\text{assms hash-map-preserving-imp-hash-extension-preserving}$
 $\text{hash-map-preserving-imp-hash-target-budget-zero}$
by blast

lemma $\text{hash-target-program-bind}$:
fixes $m :: ('x, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
and $k :: 'x \Rightarrow ('y, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
assumes $m: \text{hash-target-program } B \ n \ m$
and $k: \bigwedge x. \text{hash-target-program } B \ n' \ (k \ x)$
shows $\text{hash-target-program } B \ (n + n') \ (m \gg k)$
unfolding $\text{hash-target-program-def}$
proof (intro conjI)
show $\text{hash-extension-preserving } (m \gg k)$
by $(\text{rule hash-extension-preserving-bind})$
 $(\text{use } m \ k \ \text{in } \langle \text{simp-all add: hash-target-program-def} \rangle)$
show $\text{hash-target-budget } B \ (n + n') \ (m \gg k)$
by $(\text{rule hash-target-budget-bind})$
 $(\text{use } m \ k \ \text{in } \langle \text{simp-all add: hash-target-program-def} \rangle)$
qed

lemma $\text{hash-extension-preserving-bind-on-outcomes}$:
fixes $m :: ('x, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
and $k :: 'x \Rightarrow ('y, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
assumes $m\text{-ext}: \text{hash-extension-preserving } m$
and $k\text{-ext}$:
 $\bigwedge s \ x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{hash-extension-preserving } (k \ x)$
shows $\text{hash-extension-preserving } (m \gg k)$
unfolding $\text{hash-extension-preserving-def}$
proof (intro allI impI)
fix $s \ y \ u$
assume $\text{outcome}: \text{Some } (y, u) \in \text{set-dist } (\text{execute } (m \gg k) \ s)$
from outcome **obtain** $x \ t$ **where** $m\text{-out}$:
 $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s)$
and $k\text{-out}$: $\text{Some } (y, u) \in \text{set-dist } (\text{execute } (k \ x) \ t)$
by $(\text{auto elim!: set-dist-bindE})$
have $\text{ext-st}: s \leq t$
using $m\text{-ext } m\text{-out}$ **unfolding** $\text{hash-extension-preserving-def}$ **by** blast
have $\text{ext-tu}: t \leq u$
using $k\text{-ext}[OF \ m\text{-out}] \ k\text{-out}$

```

unfolding hash-extension-preserving-def by blast
show  $s \leq u$ 
by (rule hash-ext-trans[OF ext-st ext-tu])
qed

lemma hash-target-budget-bind-on-outcomes:
fixes  $m :: ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$ 
and  $k :: 'x \Rightarrow ('y, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$ 
assumes  $m\text{-ext}$ : hash-extension-preserving  $m$ 
and  $m\text{-budget}$ : hash-target-budget  $B \ n \ m$ 
and  $k\text{-ext}$ :
 $\bigwedge s \ x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$ 
 $\text{hash-extension-preserving } (k \ x)$ 
and  $k\text{-budget}$ :
 $\bigwedge s \ x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$ 
 $\text{hash-target-budget } B \ n' \ (k \ x)$ 
shows hash-target-budget  $B \ (n + n') \ (m \ggg k)$ 
unfolding hash-target-budget-def
proof (intro allI)
fix  $s :: ('f, 'a) \text{protocol-channel-scheme}$ 
let  $?E = \text{hash-new-output-hit-event } B \ s$ 
let  $?Tail =$ 
 $\lambda out. \text{case out of}$ 
 $\text{None} \Rightarrow 0$ 
 $| \text{Some } (x, t) \Rightarrow$ 
 $\text{if hash-map-new-output-hit } B \ s \ t \text{ then } 0$ 
 $\text{else wp-event } (k \ x) \ (\text{hash-new-output-hit-event } B \ t) \ t$ 
let  $?R = \lambda out. (\text{if } ?E \text{ out then } 1 \text{ else } 0) + ?Tail \text{ out}$ 
have split:
 $\text{wp-event } (m \ggg k) \ ?E \ s \leq \text{wp } m \ ?R \ s$ 
proof –
have exact:
 $\text{wp-event } (m \ggg k) \ ?E \ s =$ 
 $\text{wp } m$ 
 $(\lambda out. \text{case out of}$ 
 $\text{None} \Rightarrow 0$ 
 $| \text{Some } (x, t) \Rightarrow \text{wp-event } (k \ x) \ ?E \ t) \ s$ 
unfolding wp-event-def hash-new-output-hit-event-def
by (simp add: wpsimps)
have wp  $m$ 
 $(\lambda out. \text{case out of}$ 
 $\text{None} \Rightarrow 0$ 
 $| \text{Some } (x, t) \Rightarrow \text{wp-event } (k \ x) \ ?E \ t) \ s$ 
 $\leq \text{wp } m \ ?R \ s$ 
proof (rule wp-mono-on-support)
fix  $out$ 
assume support:  $out \in \text{set-dist } (\text{execute } m \ s)$ 
show (case out of
 $\text{None} \Rightarrow 0$ 

```

```

    | Some (x, t)  $\Rightarrow$  wp-event (k x) ?E t  $\leq$  ?R out
proof (cases out)
  case None
  then show ?thesis
    unfolding hash-new-output-hit-event-def by simp
next
  case (Some xt)
  then obtain x t where xt: xt = (x, t)
    by (cases xt) simp
  have m-out: Some (x, t)  $\in$  set-dist (execute m s)
    using support Some xt by simp
  have ext-st: s  $\leq$  t
    using m-ext m-out unfolding hash-extension-preserving-def by blast
  have step:
    wp-event (k x) ?E t  $\leq$ 
      (if hash-map-new-output-hit B s t then 1
       else wp-event (k x) (hash-new-output-hit-event B t) t)
    by (rule hash-new-output-hit-event-step-bound
      [OF ext-st k-ext[OF m-out]])
  show ?thesis
    using Some xt step
    unfolding hash-new-output-hit-event-def
    by (cases hash-map-new-output-hit B s t) simp-all
qed
qed
then show ?thesis
  unfolding exact .
qed
have head-bound:
  wp-event m ?E s  $\leq$  hash-target-budget-value B n
    using m-budget unfolding hash-target-budget-def by blast
have tail-bound:
  wp m ?Tail s  $\leq$  hash-target-budget-value B n'
proof (rule wp-le-const-on-support)
  fix out
  assume support: out  $\in$  set-dist (execute m s)
  show ?Tail out  $\leq$  hash-target-budget-value B n'
  proof (cases out)
    case None
    then show ?thesis
      unfolding hash-target-budget-value-def by simp
  next
    case (Some xt)
    then obtain x t where xt: xt = (x, t)
      by (cases xt) simp
    have m-out: Some (x, t)  $\in$  set-dist (execute m s)
      using support Some xt by simp
    have tail:
      wp-event (k x) (hash-new-output-hit-event B t) t  $\leq$ 

```

```

      hash-target-budget-value B n'
    using k-budget[OF m-out] unfolding hash-target-budget-def by blast
  show ?thesis
    using Some xt tail by simp
qed
qed
have wp-event (m  $\ggg$  k) ?E s  $\leq$  wp m ?R s
  by (rule split)
also have ... = wp-event m ?E s + wp m ?Tail s
  unfolding wp-event-def wp-def dist-expect-def
  by (simp add: sum.distrib algebra-simps
      hash-new-output-hit-event-def)
also have ...  $\leq$ 
  hash-target-budget-value B n + hash-target-budget-value B n'
  by (intro add-mono head-bound tail-bound)
also have ... = hash-target-budget-value B (n + n')
  by (rule hash-target-budget-value-add)
finally show
  wp-event (m  $\ggg$  k) (hash-new-output-hit-event B s) s  $\leq$ 
  hash-target-budget-value B (n + n') .
qed

lemma hash-target-program-bind-on-outcomes:
  fixes m :: ('x, ('f, 'a) protocol-channel-scheme) state-monad
  and k :: 'x  $\Rightarrow$  ('y, ('f, 'a) protocol-channel-scheme) state-monad
  assumes m: hash-target-program B n m
  and k:
     $\bigwedge s \ x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$ 
    hash-target-program B n' (k x)
  shows hash-target-program B (n + n') (m  $\ggg$  k)
  unfolding hash-target-program-def
proof (intro conjI)
  show hash-extension-preserving (m  $\ggg$  k)
    by (rule hash-extension-preserving-bind-on-outcomes)
    (use m k in <simp-all add: hash-target-program-def>)
  show hash-target-budget B (n + n') (m  $\ggg$  k)
    by (rule hash-target-budget-bind-on-outcomes)
    (use m k in <simp-all add: hash-target-program-def>)
qed

lemma hash-target-program-ntimes:
  fixes m :: ('x, ('f, 'a) protocol-channel-scheme) state-monad
  assumes m: hash-target-program B n m
  shows hash-target-program B (k * n) (ntimes m k)
  using m
proof (induction k)
  case 0
  then show ?case
    by (simp add: hash-target-program-return)

```

```

next
  case (Suc k)
  have tail: hash-target-program B (k * n) (ntimes m k)
    using Suc.IH Suc.premis by simp
  have cont:
    hash-target-program B (k * n + 0)
      (ntimes m k  $\gg$  ( $\lambda xs.$  return (x # xs))) for x
    by (rule hash-target-program-bind[OF tail hash-target-program-return])
  have hash-target-program B (n + (k * n + 0))
    (m  $\gg$  ( $\lambda x.$  ntimes m k  $\gg$  ( $\lambda xs.$  return (x # xs))))
    by (rule hash-target-program-bind[OF Suc.premis cont])
  then show ?case by simp
qed

```

```

lemma hash-target-program-mmap:
  fixes ms :: ('x, ('f, 'a) protocol-channel-scheme) state-monad list
  assumes ms:  $\bigwedge m. m \in \text{set } ms \implies \text{hash-target-program } B \ n \ m$ 
  shows hash-target-program B (length ms * n) (mmap ms)
  using ms
proof (induction ms)
  case Nil
  then show ?case
    by (simp add: hash-target-program-return)
next
  case (Cons m ms)
  have head: hash-target-program B n m
    using Cons.premis by simp
  have tail: hash-target-program B (length ms * n) (mmap ms)
    using Cons.IH Cons.premis by simp
  have cont:
    hash-target-program B (length ms * n + 0)
      (mmap ms  $\gg$  ( $\lambda xs.$  return (x # xs))) for x
    by (rule hash-target-program-bind[OF tail hash-target-program-return])
  have hash-target-program B (n + (length ms * n + 0))
    (m  $\gg$  ( $\lambda x.$  mmap ms  $\gg$  ( $\lambda xs.$  return (x # xs))))
    by (rule hash-target-program-bind[OF head cont])
  then show ?case by simp
qed

```

```

lemma hash-target-program-mfold:
  fixes steps ::
    ('x  $\Rightarrow$  ('x, ('f, 'a) protocol-channel-scheme) state-monad) list
  assumes steps:
     $\bigwedge \text{step } x. \text{step} \in \text{set steps} \implies$ 
      hash-target-program B n (step x)
  shows hash-target-program B (length steps * n) (mfold x steps)
  using steps
proof (induction steps arbitrary: x)
  case Nil

```

```

    then show ?case
      by (simp add: hash-target-program-return)
next
case (Cons step steps)
have head: hash-target-program B n (step x)
  using Cons.prem by simp
have tail:
   $\bigwedge y. \text{hash-target-program } B \text{ (length steps * n) (mfold y steps)}$ 
  using Cons.IH Cons.prem by simp
have hash-target-program B (n + length steps * n)
  (step x  $\gg$  ( $\lambda y. \text{mfold y steps}$ ))
  by (rule hash-target-program-bind[OF head tail])
then show ?case by simp
qed

end

end

```

```

theory Soundness-Oracle-Budgets
  imports Soundness-Oracle-Target
begin

```

Random-oracle range and collision budget calculus for protocol programs.

```

context soundness
begin

```

```

definition hash-collision-budget-value :: nat  $\Rightarrow$  nat  $\Rightarrow$  prob
where
  hash-collision-budget-value c n =
    nnreal (n * (c + n)) / nnreal size

```

```

definition hash-range-budget
:: nat  $\Rightarrow$  ('r, ('f, 'a) protocol-channel-scheme) state-monad  $\Rightarrow$  bool
where
  hash-range-budget n m  $\longleftrightarrow$ 
    ( $\forall s x t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m s) \longrightarrow$ 
      card (hash-map-output-values t)  $\leq$ 
      card (hash-map-output-values s) + n)

```

```

definition hash-collision-budget
:: nat  $\Rightarrow$  ('r, ('f, 'a) protocol-channel-scheme) state-monad  $\Rightarrow$  bool
where
  hash-collision-budget n m  $\longleftrightarrow$ 
    ( $\forall s. \neg \text{hash-map-output-collision } s \longrightarrow$ 
      wp-event m (hash-new-collision-event s) s  $\leq$ 
      hash-collision-budget-value (card (hash-map-output-values s)) n)

```

lemma *hash-collision-budget-value-mono-left*:
assumes $c \leq d$
shows *hash-collision-budget-value* $c\ n \leq$
hash-collision-budget-value $d\ n$
unfolding *hash-collision-budget-value-def*
apply (*subst nn2real-le-iff*[*symmetric*])
using *assms* **by** (*simp add: divide-right-mono add-mono mult-left-mono*)

lemma *hash-collision-budget-value-mono-right*:
assumes $m \leq n$
shows *hash-collision-budget-value* $c\ m \leq$
hash-collision-budget-value $c\ n$
unfolding *hash-collision-budget-value-def*
apply (*subst nn2real-le-iff*[*symmetric*])
using *assms* **by** (*simp add: divide-right-mono add-mono mult-mono*)

lemma *hash-range-budget-mono*:
fixes $p :: ('r, ('f, 'a)\ \text{protocol-channel-scheme})\ \text{state-monad}$
assumes $m \leq n$
and *budget*: *hash-range-budget* $m\ p$
shows *hash-range-budget* $n\ p$
using *assms* **unfolding** *hash-range-budget-def* **by** *force*

lemma *hash-collision-budget-mono*:
fixes $p :: ('r, ('f, 'a)\ \text{protocol-channel-scheme})\ \text{state-monad}$
assumes $m \leq n$
and *budget*: *hash-collision-budget* $m\ p$
shows *hash-collision-budget* $n\ p$
unfolding *hash-collision-budget-def*
proof (*intro allI impI*)
fix $s :: ('f, 'a)\ \text{protocol-channel-scheme}$
assume *clean*: $\neg\ \text{hash-map-output-collision}\ s$
have *wp-event* $p\ (\text{hash-new-collision-event}\ s)\ s \leq$
hash-collision-budget-value (*card* (*hash-map-output-values* s)) m
using *budget clean* **unfolding** *hash-collision-budget-def* **by** *blast*
also have $\dots \leq$
hash-collision-budget-value (*card* (*hash-map-output-values* s)) n
by (*rule hash-collision-budget-value-mono-right*[*OF assms*(1)])
finally show *wp-event* $p\ (\text{hash-new-collision-event}\ s)\ s \leq$
hash-collision-budget-value (*card* (*hash-map-output-values* s)) n .
qed

lemma *hash-collision-budget-value-bind-le*:
hash-collision-budget-value $c\ n +$
hash-collision-budget-value $(c + n)\ n' \leq$
hash-collision-budget-value $c\ (n + n')$
unfolding *hash-collision-budget-value-def*
by (*simp add: add-divide-nnreal algebra-simps*)

lemma *hash-range-budget-bind*:
fixes $m :: ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
and $k :: 'x \Rightarrow ('y, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
assumes $m: \text{hash-range-budget } n \ m$
and $k: \bigwedge x. \text{hash-range-budget } n' (k \ x)$
shows $\text{hash-range-budget } (n + n') (m \ggg k)$
unfolding *hash-range-budget-def*
proof (*intro allI impI*)
fix $s \ y \ t$
assume *outcome*: $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (m \ggg k) \ s)$
from *outcome* **obtain** $x \ s1$ **where**
 $m\text{-out}: \text{Some } (x, s1) \in \text{set-dist } (\text{execute } m \ s)$
and $k\text{-out}: \text{Some } (y, t) \in \text{set-dist } (\text{execute } (k \ x) \ s1)$
by (*auto elim!: set-dist-bindE*)
have *m-range*:
 $\text{card } (\text{hash-map-output-values } s1) \leq$
 $\text{card } (\text{hash-map-output-values } s) + n$
using $m \ m\text{-out}$ **unfolding** *hash-range-budget-def* **by** *blast*
have *k-range*:
 $\text{card } (\text{hash-map-output-values } t) \leq$
 $\text{card } (\text{hash-map-output-values } s1) + n'$
using $k \ k\text{-out}$ **unfolding** *hash-range-budget-def* **by** *blast*
show $\text{card } (\text{hash-map-output-values } t)$
 $\leq \text{card } (\text{hash-map-output-values } s) + (n + n')$
using *m-range k-range* **by** *linarith*
qed

lemma *hash-range-budget-bind-on-outcomes*:
fixes $m :: ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
and $k :: 'x \Rightarrow ('y, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
assumes $m: \text{hash-range-budget } n \ m$
and $k:$
 $\bigwedge s \ x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{hash-range-budget } n' (k \ x)$
shows $\text{hash-range-budget } (n + n') (m \ggg k)$
unfolding *hash-range-budget-def*
proof (*intro allI impI*)
fix $s \ y \ t$
assume *outcome*: $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (m \ggg k) \ s)$
from *outcome* **obtain** $x \ s1$ **where**
 $m\text{-out}: \text{Some } (x, s1) \in \text{set-dist } (\text{execute } m \ s)$
and $k\text{-out}: \text{Some } (y, t) \in \text{set-dist } (\text{execute } (k \ x) \ s1)$
by (*auto elim!: set-dist-bindE*)
have *m-range*:
 $\text{card } (\text{hash-map-output-values } s1) \leq$
 $\text{card } (\text{hash-map-output-values } s) + n$
using $m \ m\text{-out}$ **unfolding** *hash-range-budget-def* **by** *blast*
have *k-range*:


```

    card (hash-map-output-values t) ≤
      card (hash-map-output-values s1) + n'
    using k[OF m-out] k-out unfolding hash-range-budget-def by blast
  show card (hash-map-output-values t)
    ≤ card (hash-map-output-values s) + (n + n')
    using m-range k-range by linarith
qed

lemma hash-new-collision-event-step-bound:
  fixes m :: ('x, ('f, 'a) protocol-channel-scheme) state-monad
    and s t :: ('f, 'a) protocol-channel-scheme
  assumes clean: ¬ hash-map-output-collision s
  shows
    wp-event m (hash-new-collision-event s) t ≤
      (if hash-map-new-output-collision s t then 1
       else wp-event m (hash-new-collision-event t) t)
proof (cases hash-map-new-output-collision s t)
  case True
  then show ?thesis
    using wp-event-le-1[of m hash-new-collision-event s t] by simp
next
  case False
  then have clean-t: ¬ hash-map-output-collision t
    using clean unfolding hash-map-new-output-collision-def by simp
  have same:
    hash-new-collision-event s out = hash-new-collision-event t out
    for out :: ('x × ('f, 'a) protocol-channel-scheme) option
    using clean clean-t
    unfolding hash-new-collision-event-def hash-map-new-output-collision-def
    by (cases out) auto
  have wp-event m (hash-new-collision-event s) t =
    wp-event m (hash-new-collision-event t) t
    unfolding wp-event-def by (simp add: same)
  show ?thesis
    using False ⟨wp-event m (hash-new-collision-event s) t =
      wp-event m (hash-new-collision-event t) t⟩ by simp
qed

```

```

lemma hash-collision-budget-bind:
  fixes m :: ('x, ('f, 'a) protocol-channel-scheme) state-monad
    and k :: 'x ⇒ ('y, ('f, 'a) protocol-channel-scheme) state-monad
  assumes range-m: hash-range-budget n m
    and coll-m: hash-collision-budget n m
    and range-k: ∧x. hash-range-budget n' (k x)
    and coll-k: ∧x. hash-collision-budget n' (k x)
  shows hash-collision-budget (n + n') (m ≫ k)
    unfolding hash-collision-budget-def
proof (intro allI impI)
  fix s :: ('f, 'a) protocol-channel-scheme

```

```

assume clean:  $\neg$  hash-map-output-collision s
let ?c = card (hash-map-output-values s)
let ?B1 = hash-collision-budget-value ?c n
let ?B2 = hash-collision-budget-value (?c + n) n'
let ?Em = hash-new-collision-event s
let ?S =
   $\lambda out.$  case out of
    None  $\Rightarrow 0$ 
  | Some (x, s1)  $\Rightarrow$ 
    if hash-map-new-output-collision s s1 then 0
    else wp-event (k x) (hash-new-collision-event s1) s1
let ?R =
   $\lambda out.$  (if ?Em out then 1 else 0) + ?S out
have split:
  wp-event (m  $\gg$  k) (hash-new-collision-event s) s  $\leq$  wp m ?R s
proof –
  have exact:
    wp-event (m  $\gg$  k) (hash-new-collision-event s) s =
      wp m
      ( $\lambda out.$  case out of
        None  $\Rightarrow 0$ 
      | Some (x, s1)  $\Rightarrow$ 
        wp-event (k x) (hash-new-collision-event s) s1) s
    unfolding wp-event-def hash-new-collision-event-def
    by (simp add: wpsimps)
  have wp m
    ( $\lambda out.$  case out of
      None  $\Rightarrow 0$ 
    | Some (x, s1)  $\Rightarrow$ 
      wp-event (k x) (hash-new-collision-event s) s1) s
     $\leq$  wp m ?R s
  proof (rule wp-mono-on-support)
  fix out
  assume out  $\in$  set-dist (execute m s)
  show (case out of
    None  $\Rightarrow 0$ 
  | Some (x, s1)  $\Rightarrow$ 
    wp-event (k x) (hash-new-collision-event s) s1)  $\leq$  ?R out
  proof (cases out)
  case None
  then show ?thesis
    unfolding hash-new-collision-event-def by simp
  next
  case (Some xs)
  then obtain x s1 where xs: xs = (x, s1)
  by (cases xs) simp
  have step:
    wp-event (k x) (hash-new-collision-event s) s1  $\leq$ 
      (if hash-map-new-output-collision s s1 then 1

```

```

      else wp-event (k x) (hash-new-collision-event s1) s1)
    by (rule hash-new-collision-event-step-bound[OF clean])
  have R-eq:
    ?R out =
      (if hash-map-new-output-collision s s1 then 1
       else wp-event (k x) (hash-new-collision-event s1) s1)
    using Some xs unfolding hash-new-collision-event-def by simp
  have L-eq:
    (case out of
     None  $\Rightarrow$  0
    | Some (x, s1)  $\Rightarrow$ 
      wp-event (k x) (hash-new-collision-event s) s1) =
      wp-event (k x) (hash-new-collision-event s) s1
    using Some xs by simp
  show ?thesis
    using step unfolding L-eq R-eq .
qed
qed
then show ?thesis
  unfolding exact .
qed
have m-bound:
  wp-event m ?Em s  $\leq$  ?B1
  using coll-m clean unfolding hash-collision-budget-def by blast
have S-bound: wp m ?S s  $\leq$  ?B2
proof (rule wp-le-const-on-support)
  fix out
  assume out-support: out  $\in$  set-dist (execute m s)
  show ?S out  $\leq$  ?B2
  proof (cases out)
    case None
    then show ?thesis by simp
  next
    case (Some xs)
    then obtain x s1 where xs: xs = (x, s1)
      by (cases xs) simp
    show ?thesis
    proof (cases hash-map-new-output-collision s s1)
      case True
      then show ?thesis
        using Some xs by simp
    next
      case False
      then have clean-s1:  $\neg$  hash-map-output-collision s1
        using clean unfolding hash-map-new-output-collision-def by simp
      have k-bound:
        wp-event (k x) (hash-new-collision-event s1) s1  $\leq$ 
        hash-collision-budget-value
        (card (hash-map-output-values s1)) n'

```

```

    using coll-k[of x] clean-s1 unfolding hash-collision-budget-def
    by blast
  have m-out:
    Some (x, s1) ∈ set-dist (execute m s)
    using out-support Some xs by simp
  have range:
    card (hash-map-output-values s1) ≤ ?c + n
    using range-m m-out unfolding hash-range-budget-def by blast
  have hash-collision-budget-value
    (card (hash-map-output-values s1)) n' ≤ ?B2
    by (rule hash-collision-budget-value-mono-left[OF range])
  then show ?thesis
    using Some xs False k-bound by simp
qed
qed
qed
have wp-event (m ≫ k) (hash-new-collision-event s) s
  ≤ wp m ?R s
  by (rule split)
also have ... =
  wp-event m ?Em s + wp m ?S s
  unfolding wp-event-def wp-def dist-expect-def
  by (simp add: sum.distrib algebra-simps hash-new-collision-event-def)
also have ... ≤ ?B1 + ?B2
  using m-bound S-bound by (intro add-mono)
also have ... ≤ hash-collision-budget-value ?c (n + n')
  by (rule hash-collision-budget-value-bind-le)
finally show
  wp-event (m ≫ k) (hash-new-collision-event s) s
  ≤ hash-collision-budget-value ?c (n + n') .
qed

lemma hash-collision-budget-bind-on-outcomes:
  fixes m :: ('x, ('f, 'a) protocol-channel-scheme) state-monad
  and k :: 'x ⇒ ('y, ('f, 'a) protocol-channel-scheme) state-monad
  assumes range-m: hash-range-budget n m
  and coll-m: hash-collision-budget n m
  and range-k:
    ∧ s x t. Some (x, t) ∈ set-dist (execute m s) ⇒
      hash-range-budget n' (k x)
  and coll-k:
    ∧ s x t. Some (x, t) ∈ set-dist (execute m s) ⇒
      hash-collision-budget n' (k x)
  shows hash-collision-budget (n + n') (m ≫ k)
  unfolding hash-collision-budget-def
proof (intro allI impI)
  fix s :: ('f, 'a) protocol-channel-scheme
  assume clean: ¬ hash-map-output-collision s
  let ?c = card (hash-map-output-values s)

```

```

let ?B1 = hash-collision-budget-value ?c n
let ?B2 = hash-collision-budget-value (?c + n) n'
let ?Em = hash-new-collision-event s
let ?S =
  λout. case out of
    None ⇒ 0
  | Some (x, s1) ⇒
    if hash-map-new-output-collision s s1 then 0
    else wp-event (k x) (hash-new-collision-event s1) s1
let ?R =
  λout. (if ?Em out then 1 else 0) + ?S out
have split:
  wp-event (m ≫ k) (hash-new-collision-event s) s ≤ wp m ?R s
proof -
  have exact:
    wp-event (m ≫ k) (hash-new-collision-event s) s =
      wp m
      (λout. case out of
        None ⇒ 0
        | Some (x, s1) ⇒
          wp-event (k x) (hash-new-collision-event s) s1) s
    unfolding wp-event-def hash-new-collision-event-def
    by (simp add: wpsimps)
  have wp m
    (λout. case out of
      None ⇒ 0
      | Some (x, s1) ⇒
        wp-event (k x) (hash-new-collision-event s) s1) s
    ≤ wp m ?R s
  proof (rule wp-mono-on-support)
    fix out
    assume out ∈ set-dist (execute m s)
    show (case out of
      None ⇒ 0
      | Some (x, s1) ⇒
        wp-event (k x) (hash-new-collision-event s) s1) ≤ ?R out
  proof (cases out)
    case None
    then show ?thesis
      unfolding hash-new-collision-event-def by simp
  next
    case (Some xs)
    then obtain x s1 where xs: xs = (x, s1)
      by (cases xs) simp
    have step:
      wp-event (k x) (hash-new-collision-event s) s1 ≤
        (if hash-map-new-output-collision s s1 then 1
         else wp-event (k x) (hash-new-collision-event s1) s1)
      by (rule hash-new-collision-event-step-bound[OF clean])
  
```

```

have R-eq:
  ?R out =
    (if hash-map-new-output-collision s s1 then 1
     else wp-event (k x) (hash-new-collision-event s1) s1)
  using Some xs unfolding hash-new-collision-event-def by simp
have L-eq:
  (case out of
   None  $\Rightarrow$  0
  | Some (x, s1)  $\Rightarrow$ 
    wp-event (k x) (hash-new-collision-event s) s1) =
    wp-event (k x) (hash-new-collision-event s) s1
  using Some xs by simp
show ?thesis
  using step unfolding L-eq R-eq .
qed
qed
then show ?thesis
  unfolding exact .
qed
have m-bound:
  wp-event m ?Em s  $\leq$  ?B1
  using coll-m clean unfolding hash-collision-budget-def by blast
have S-bound: wp m ?S s  $\leq$  ?B2
proof (rule wp-le-const-on-support)
  fix out
  assume out-support: out  $\in$  set-dist (execute m s)
  show ?S out  $\leq$  ?B2
  proof (cases out)
    case None
    then show ?thesis by simp
  next
    case (Some xs)
    then obtain x s1 where xs: xs = (x, s1)
    by (cases xs) simp
    have m-out:
      Some (x, s1)  $\in$  set-dist (execute m s)
      using out-support Some xs by simp
    show ?thesis
    proof (cases hash-map-new-output-collision s s1)
      case True
      then show ?thesis
        using Some xs by simp
    next
      case False
      then have clean-s1:  $\neg$  hash-map-output-collision s1
      using clean unfolding hash-map-new-output-collision-def by simp
      have k-bound:
        wp-event (k x) (hash-new-collision-event s1) s1  $\leq$ 
        hash-collision-budget-value

```

```

      (card (hash-map-output-values s1)) n'
    using coll-k[OF m-out] clean-s1
    unfolding hash-collision-budget-def by blast
  have range:
    card (hash-map-output-values s1) ≤ ?c + n
    using range-m m-out unfolding hash-range-budget-def by blast
  have hash-collision-budget-value
    (card (hash-map-output-values s1)) n' ≤ ?B2
    by (rule hash-collision-budget-value-mono-left[OF range])
  then show ?thesis
    using Some xs False k-bound by simp
qed
qed
qed
have wp-event (m ≫ k) (hash-new-collision-event s) s
  ≤ wp m ?R s
  by (rule split)
also have ... =
  wp-event m ?Em s + wp m ?S s
  unfolding wp-event-def wp-def dist-expect-def
  by (simp add: sum.distrib algebra-simps hash-new-collision-event-def)
also have ... ≤ ?B1 + ?B2
  using m-bound S-bound by (intro add-mono)
also have ... ≤ hash-collision-budget-value ?c (n + n')
  by (rule hash-collision-budget-value-bind-le)
finally show
  wp-event (m ≫ k) (hash-new-collision-event s) s
  ≤ hash-collision-budget-value ?c (n + n') .
qed

lemma wp-hash-new-output-collision-bound:
  fixes x :: 'f protocol-hash-input
  and s :: ('f, 'a) protocol-channel-scheme
  assumes clean: ¬ hash-map-output-collision s
  shows
    wp-event (hash x)
      (hash-new-collision-event s) s
    ≤ nnreal (card (hash-map-output-values s)) / nnreal size
proof (cases fmllookup (HashMap s) x)
case None
let ?P =
  λy. hash-map-new-output-collision s
    (s(HashMap := fmuupd x y (HashMap s)))
let ?Q = λy. y ∈ hash-map-output-values s
have mono:
  dist-expect (hash-dist x s) (λy. if ?P y then 1 else 0) ≤
  dist-expect (hash-dist x s) (λy. if ?Q y then 1 else 0)
  unfolding dist-expect-def
proof (intro sum-mono)

```

```

fix y
assume y ∈ dom (dist (hash-dist x s))
show the (dist (hash-dist x s) y) * (if ?P y then 1 else 0)
  ≤ the (dist (hash-dist x s) y) * (if ?Q y then 1 else 0)
proof (cases ?P y)
  case True
  then have ?Q y
    using hash-map-output-collision-after-fresh-updateD
      [OF clean None]
    unfolding hash-map-new-output-collision-def by simp
  then show ?thesis
    using True by simp
  next
  case False
  then show ?thesis by simp
qed
qed
show ?thesis
proof -
  have wp-event (hash x)
    (hash-new-collision-event s) s =
    dist-expect (hash-dist x s) (λy. if ?P y then 1 else 0)
  unfolding wp-event-def hash-new-collision-event-def
  by (simp add: wp-hash)
  also have ... ≤
    dist-expect (hash-dist x s) (λy. if ?Q y then 1 else 0)
  by (rule mono)
  also have ... =
    nnreal (card (hash-map-output-values s)) / nnreal size
  by (rule dist-expect-hash-dist-fresh-output-values[OF None])
  finally show ?thesis .
qed
next
case (Some y)
have no-new:
  ¬ hash-map-new-output-collision s
  (s(HashMap := fmupd x y (HashMap s)))
  using no-hash-map-output-collision-after-known-update[OF clean Some]
  unfolding hash-map-new-output-collision-def by simp
show ?thesis
  unfolding wp-event-def hash-new-collision-event-def
  by (simp add: wp-hash Some hash-dist-def option-default-dist-def no-new)
qed

lemma hash-range-budget-return:
  hash-range-budget 0 (return x)
  unfolding hash-range-budget-def by simp

lemma hash-collision-budget-return:

```



```

hash-collision-budget 0 (return x)
unfolding hash-collision-budget-def hash-new-collision-event-def
  hash-collision-budget-value-def hash-map-new-output-collision-def
by (simp add: wp-event-def wpsimps)

```

```

lemma hash-range-budget-hash:
  fixes x :: 'f protocol-hash-input
  shows
    hash-range-budget 1
    (hash x :: ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding hash-range-budget-def
proof (intro allI impI)
  fix s y t
  assume outcome:
    Some (y, t) ∈
      set-dist (execute
        (hash x :: ('f, ('f, 'a) protocol-channel-scheme) state-monad) s)
  have t-eq: t = s(HashMap := fmupd x y (HashMap s))
  proof -
    from outcome obtain a s1 where
      apply-step:
        Some (a, s1) ∈ set-dist (execute (apply-hash x) s)
    and rest:
      Some (y, t) ∈
        set-dist (execute
          (modify-HashMap x a >>= (λ-. return a)) s1)
  proof -
    have Some (y, t) ∈
      set-dist (execute
        (apply-hash x >>=
          (λa. modify-HashMap x a >>= (λ-. return a)))) s)
    using outcome unfolding hash-def .
    then show ?thesis
      by (rule set-dist-bindE) (rule that)
  qed
  have apply-img:
    Some (a, s1) ∈ (λa. Some (a, s)) ‘ set-dist (hash-dist x s)
    using apply-step
    unfolding apply-hash-def lift.rep-eq lift-dist-def
    by (simp add: set-dist-dist-map)
  then have s1-eq: s1 = s
    by auto
  from rest obtain u s2 where
    mod:
      Some (u, s2) ∈ set-dist (execute (modify-HashMap x a) s1)
    and ret:
      Some (y, t) ∈ set-dist (execute (return a) s2)
    by (elim set-dist-bindE)
  have s2-eq: s2 = s1(HashMap := fmupd x a (HashMap s1))

```

```

    using mod unfolding modify-HashMap-def modify-def
    by (auto elim!: set-dist-bindE)
  have y-eq:  $y = a$  and t-eq-s2:  $t = s^2$ 
    using ret by auto
  show ?thesis
    using s1-eq s2-eq y-eq t-eq-s2 by simp
qed
show card (hash-map-output-values t)
  ≤ card (hash-map-output-values s) + 1
  unfolding t-eq
  using card-hash-map-output-values-after-update-le[of x y s]
  by simp
qed

lemma hash-collision-budget-hash:
  fixes x :: 'f protocol-hash-input
  shows
    hash-collision-budget 1
      (hash x :: ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding hash-collision-budget-def
proof (intro allI impI)
  fix s :: ('f, 'a) protocol-channel-scheme
  assume clean:  $\neg$  hash-map-output-collision s
  have one-step:
    wp-event (hash x) (hash-new-collision-event s) s ≤
      nnreal (card (hash-map-output-values s)) / nnreal size
    by (rule wp-hash-new-output-collision-bound[OF clean])
  also have ... ≤
    hash-collision-budget-value (card (hash-map-output-values s)) 1
  unfolding hash-collision-budget-value-def
  apply (subst nn2real-le-iff[symmetric])
  by (simp add: divide-right-mono)
  finally show
    wp-event (hash x) (hash-new-collision-event s) s ≤
      hash-collision-budget-value (card (hash-map-output-values s)) 1 .
qed

lemma hash-range-budget-get:
  hash-range-budget 0
    (get :: (('f, 'a) protocol-channel-scheme,
      ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding hash-range-budget-def by simp

lemma hash-collision-budget-get:
  hash-collision-budget 0
    (get :: (('f, 'a) protocol-channel-scheme,
      ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding hash-collision-budget-def hash-new-collision-event-def
    hash-map-new-output-collision-def hash-collision-budget-value-def

```

```

by (simp add: wp-event-def wpsimps)

lemma hash-range-budget-assert:
  hash-range-budget 0
  (assert b :: (unit, ('f, 'a) protocol-channel-scheme) state-monad)
unfolding hash-range-budget-def assert-def
by (auto simp: throw-no-outcome)

lemma hash-collision-budget-assert:
  hash-collision-budget 0
  (assert b :: (unit, ('f, 'a) protocol-channel-scheme) state-monad)
unfolding hash-collision-budget-def hash-new-collision-event-def
  hash-map-new-output-collision-def hash-collision-budget-value-def
  assert-def
by (simp add: wp-event-def wpsimps)

lemma hash-range-budget-send:
  hash-range-budget 0 (send x)
unfolding hash-range-budget-def send-def protocol-send-def modify-def
by (auto simp: hash-map-output-values-def elim!: set-dist-bindE)

lemma hash-collision-budget-send:
  hash-collision-budget 0 (send x)
unfolding hash-collision-budget-def hash-new-collision-event-def
  hash-map-new-output-collision-def hash-collision-budget-value-def
  hash-map-output-collision-def
  send-def protocol-send-def
by (simp add: wp-event-def wpsimps)

lemma hash-range-budget-read:
  hash-range-budget 0 (read :: ('f, 'f, 'a) protocol-c-monad)
unfolding hash-range-budget-def
proof (intro allI impI)
  fix s x t
  assume out: Some (x, t) ∈
    set-dist (execute (read :: ('f, 'f, 'a) protocol-c-monad) s)
  obtain y ys where tr: PTranscript s = y # ys
  proof –
    have PTranscript s ≠ []
    using out unfolding read-def protocol-read-def assert-def
    by (cases PTranscript s)
    (auto simp: throw-no-outcome elim!: set-dist-bindE)
    then show ?thesis
    by (cases PTranscript s) (auto intro: that)
  qed
have t-eq:
  t = s(PState := concat (PState s) y, PTranscript := ys)
  using read-nonempty-outcome[OF out tr] by simp
show card (hash-map-output-values t)

```

$\leq \text{card } (\text{hash-map-output-values } s) + 0$
unfolding *t-eq hash-map-output-values-def* **by** *simp*
qed

lemma *hash-collision-budget-read*:
hash-collision-budget 0
(read :: ('f, 'f, 'a) protocol-c-monad)
unfolding *hash-collision-budget-def hash-new-collision-event-def*
hash-map-new-output-collision-def hash-collision-budget-value-def
hash-map-output-collision-def
read-def protocol-read-def assert-def
by (*simp add: wp-event-def wpsimps*)

lemma *hash-range-budget-modify-preserves-hash-map*:
assumes $\bigwedge s. \text{HashMap } (f \ s) = \text{HashMap } s$
shows *hash-range-budget 0 (modify f)*
using *assms*
unfolding *hash-range-budget-def modify-def hash-map-output-values-def*
by (*auto elim!: set-dist-bindE*)

lemma *hash-collision-budget-modify-preserves-hash-map*:
assumes $\bigwedge s. \text{HashMap } (f \ s) = \text{HashMap } s$
shows *hash-collision-budget 0 (modify f)*
using *assms*
unfolding *hash-collision-budget-def hash-new-collision-event-def*
hash-map-new-output-collision-def hash-collision-budget-value-def
hash-map-output-collision-def modify-def
by (*simp add: wp-event-def wpsimps*)

lemma *hash-map-preserving-get*:
hash-map-preserving
(get :: (('f, 'a) protocol-channel-scheme,
('f, 'a) protocol-channel-scheme) state-monad)
unfolding *hash-map-preserving-def* **by** *simp*

lemma *hash-map-preserving-assert*:
hash-map-preserving
(assert b :: (unit, ('f, 'a) protocol-channel-scheme) state-monad)
unfolding *hash-map-preserving-def assert-def*
by (*auto simp: throw-no-outcome elim!: set-dist-bindE*)

lemma *hash-map-preserving-send*:
hash-map-preserving (send x)
unfolding *hash-map-preserving-def send-def protocol-send-def modify-def*
by (*auto elim!: set-dist-bindE*)

lemma *hash-map-preserving-read*:
hash-map-preserving (read :: ('f, 'f, 'a) protocol-c-monad)
unfolding *hash-map-preserving-def*

```

proof (intro allI impI)
  fix s x t
  assume out:
    Some (x, t) ∈
      set-dist (execute (read :: ('f, 'f, 'a) protocol-c-monad) s)
  obtain y ys where tr: PTranscript s = y # ys
  proof –
    have PTranscript s ≠ []
    using out unfolding read-def protocol-read-def assert-def
    by (cases PTranscript s)
      (auto simp: throw-no-outcome elim!: set-dist-bindE)
    then show ?thesis
    by (cases PTranscript s) (auto intro: that)
  qed
  have t-eq:
    t = s(PState := concat (PState s) y, PTranscript := ys)
    using read-nonempty-outcome[OF out tr] by simp
  show HashMap t = HashMap s
    unfolding t-eq by simp
  qed

```

```

lemma hash-map-preserving-modify:
  assumes  $\bigwedge s. \text{HashMap } (f\ s) = \text{HashMap } s$ 
  shows hash-map-preserving (modify f)
  using assms unfolding hash-map-preserving-def modify-def
  by (auto elim!: set-dist-bindE)

```

```

lemma hash-target-program-get:
  hash-target-program B 0
  (get :: (('f, 'a) protocol-channel-scheme,
    ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-map-preserving-imp-hash-target-program-zero)
  (rule hash-map-preserving-get)

```

```

lemma hash-target-program-assert:
  hash-target-program B 0
  (assert b :: (unit, ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-map-preserving-imp-hash-target-program-zero)
  (rule hash-map-preserving-assert)

```

```

lemma hash-target-program-send:
  hash-target-program B 0 (send x)
  by (rule hash-map-preserving-imp-hash-target-program-zero)
  (rule hash-map-preserving-send)

```

```

lemma hash-target-program-read:
  hash-target-program B 0 (read :: ('f, 'f, 'a) protocol-c-monad)
  by (rule hash-map-preserving-imp-hash-target-program-zero)
  (rule hash-map-preserving-read)

```

lemma *hash-target-program-ntimes-read-bind*:
fixes $k :: 'f \text{ list} \Rightarrow$
 $('y, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
assumes k :
 $\bigwedge xs. \text{length } xs = n \implies \text{hash-target-program } B \ b \ (k \ xs)$
shows
 $\text{hash-target-program } B \ b$
 $((\text{ntimes read } n ::$
 $('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}) \ggg k)$
proof –
have *reads*:
 $\text{hash-target-program } B \ (n * 0)$
 $(\text{ntimes read } n ::$
 $('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
by (rule *hash-target-program-ntimes*[*OF* *hash-target-program-read*])
have $\text{hash-target-program } B \ (n * 0 + b)$
 $((\text{ntimes read } n ::$
 $('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}) \ggg k)$
proof (rule *hash-target-program-bind-on-outcomes*[*OF* *reads*])
fix $s \ xs \ t$
assume *out*:
 $\text{Some } (xs, t) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{ntimes read } n ::$
 $('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}) \ s)$
show $\text{hash-target-program } B \ b \ (k \ xs)$
by (rule k) (use *ntimes-read-any-outcome*[*OF* *out*] **in** *simp*)
qed
then show *?thesis* **by** *simp*
qed

lemma *hash-target-program-modify*:
assumes $\bigwedge s. \text{HashMap } (f \ s) = \text{HashMap } s$
shows $\text{hash-target-program } B \ 0 \ (\text{modify } f)$
by (rule *hash-map-preserving-imp-hash-target-program-zero*)
(rule *hash-map-preserving-modify*[*OF* *assms*])

lemma *hash-target-program-receive-random-field-element*:
 $\text{hash-target-program } B \ 1$
 $(\text{receive-random-field-element} :: ('f, 'f, 'a) \text{ protocol-c-monad})$
proof –
have *tail*:
 $\text{hash-target-program } B \ (1 + 0)$
 $(\text{hash } (\text{FiatShamirChallenge } (PState \ s)) \ggg \text{return} ::$
 $('f, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
for $s :: ('f, 'a) \text{ protocol-channel-scheme}$
by (rule *hash-target-program-bind*)
(rule *hash-target-program-hash*, rule *hash-target-program-return*)

```

have hash-target-program B (0 + (1 + 0))
  (get  $\gg$ 
    ( $\lambda s :: ('f, 'a)$  protocol-channel-scheme.
      hash (FiatShamirChallenge (PState s))  $\gg$  return))
  by (rule hash-target-program-bind[OF hash-target-program-get tail])
then show ?thesis
  unfolding receive-random-field-element-def
    protocol-receive-random-field-element-def
    protocol-receive-tagged-random-field-element-def
  by simp
qed

```

lemma hash-target-program-receive-tagged-random-field-element:

```

  hash-target-program B 1
  (receive-tagged-random-field-element tag ::
    ('f, 'f, 'a) protocol-c-monad)
proof –
  have tail:
    hash-target-program B (1 + 0)
    (hash (tag (PState s))  $\gg$  return ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for s :: ('f, 'a) protocol-channel-scheme
  by (rule hash-target-program-bind)
    (rule hash-target-program-hash, rule hash-target-program-return)
have hash-target-program B (0 + (1 + 0))
  (get  $\gg$ 
    ( $\lambda s :: ('f, 'a)$  protocol-channel-scheme.
      hash (tag (PState s))  $\gg$  return))
  by (rule hash-target-program-bind[OF hash-target-program-get tail])
then show ?thesis
  unfolding receive-tagged-random-field-element-def
    protocol-receive-tagged-random-field-element-def
  by simp
qed

```

lemma hash-target-program-receive-counted-tagged-random-field-element:

```

  assumes bump:  $\bigwedge s.$  HashMap (bump s) = HashMap s
  shows hash-target-program B 1
  (protocol-receive-counted-tagged-random-field-element counter bump tag ::
    ('f, 'f, 'a) protocol-c-monad)
proof –
  have modify-return:
    hash-target-program B (0 + 0)
    (modify bump  $\gg$  ( $\lambda -.$  return r) ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for r :: 'f
  by (rule hash-target-program-bind)
    (rule hash-target-program-modify[OF bump],
      rule hash-target-program-return)

```

```

have hash-tail:
  hash-target-program B (1 + (0 + 0))
  (hash (tag (counter s) (PState s))  $\gg$ =
    ( $\lambda r$ . modify bump  $\gg$ = ( $\lambda$ -. return r)) ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for s :: ('f, 'a) protocol-channel-scheme
  by (rule hash-target-program-bind)
  (rule hash-target-program-hash, rule modify-return)
have hash-target-program B (0 + (1 + (0 + 0)))
  (get  $\gg$ =
    ( $\lambda s$  :: ('f, 'a) protocol-channel-scheme.
      hash (tag (counter s) (PState s))  $\gg$ =
        ( $\lambda r$ . modify bump  $\gg$ = ( $\lambda$ -. return r))))
  by (rule hash-target-program-bind[OF hash-target-program-get hash-tail])
then show ?thesis
  unfolding protocol-receive-counted-tagged-random-field-element-def
  by simp
qed

```

```

lemma hash-target-program-receive-alpha-challenge:
  hash-target-program B 1
  (receive-alpha-challenge :: ('f, 'f, 'a) protocol-c-monad)
  unfolding receive-alpha-challenge-def
  by (rule hash-target-program-receive-counted-tagged-random-field-element)
  simp

```

```

lemma hash-target-program-receive-query-index-challenge:
  hash-target-program B 1
  (receive-query-index-challenge :: ('f, 'f, 'a) protocol-c-monad)
  unfolding receive-query-index-challenge-def
  by (rule hash-target-program-receive-counted-tagged-random-field-element)
  simp

```

```

lemma hash-target-program-receive-trace-fri-challenge:
  hash-target-program B 1
  (receive-trace-fri-challenge :: ('f, 'f, 'a) protocol-c-monad)
  unfolding receive-trace-fri-challenge-def
  by (rule hash-target-program-receive-counted-tagged-random-field-element)
  simp

```

```

lemma hash-target-program-receive-composition-fri-challenge:
  hash-target-program B 1
  (receive-composition-fri-challenge :: ('f, 'f, 'a) protocol-c-monad)
  unfolding receive-composition-fri-challenge-def
  by (rule hash-target-program-receive-counted-tagged-random-field-element)
  simp

```

```

lemma hash-range-budget-receive-random-field-element:
  hash-range-budget 1

```



```

    (receive-random-field-element :: ('f, 'f, 'a) protocol-c-monad)
proof –
  have get-step:
    hash-range-budget 0
    (get :: (('f, 'a) protocol-channel-scheme,
      ('f, 'a) protocol-channel-scheme) state-monad)
    by (rule hash-range-budget-get)
  have hash-then-return:
    hash-range-budget (1 + 0)
    (hash (FiatShamirChallenge (PState s))  $\gg=$  return ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
    for s :: ('f, 'a) protocol-channel-scheme
    by (rule hash-range-budget-bind)
      (rule hash-range-budget-hash, rule hash-range-budget-return)
  have hash-range-budget (0 + (1 + 0))
    (get  $\gg=$ 
      ( $\lambda$ s :: ('f, 'a) protocol-channel-scheme.
        hash (FiatShamirChallenge (PState s))  $\gg=$  return))
    by (rule hash-range-budget-bind[OF get-step hash-then-return])
  then show ?thesis
    unfolding receive-random-field-element-def
      protocol-receive-random-field-element-def
      protocol-receive-tagged-random-field-element-def
    by simp
qed

lemma hash-range-budget-receive-tagged-random-field-element:
  hash-range-budget 1
  (receive-tagged-random-field-element tag :: ('f, 'f, 'a) protocol-c-monad)
proof –
  have get-step:
    hash-range-budget 0
    (get :: (('f, 'a) protocol-channel-scheme,
      ('f, 'a) protocol-channel-scheme) state-monad)
    by (rule hash-range-budget-get)
  have hash-then-return:
    hash-range-budget (1 + 0)
    (hash (tag (PState s))  $\gg=$  return ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
    for s :: ('f, 'a) protocol-channel-scheme
    by (rule hash-range-budget-bind)
      (rule hash-range-budget-hash, rule hash-range-budget-return)
  have hash-range-budget (0 + (1 + 0))
    (get  $\gg=$ 
      ( $\lambda$ s :: ('f, 'a) protocol-channel-scheme.
        hash (tag (PState s))  $\gg=$  return))
    by (rule hash-range-budget-bind[OF get-step hash-then-return])
  then show ?thesis
    unfolding receive-tagged-random-field-element-def

```

```

    protocol-receive-tagged-random-field-element-def
  by simp
qed

lemma hash-range-budget-receive-counted-tagged-random-field-element:
  assumes bump:  $\bigwedge s. \text{HashMap } (\text{bump } s) = \text{HashMap } s$ 
  shows hash-range-budget 1
    (protocol-receive-counted-tagged-random-field-element counter bump tag ::
      ('f, 'f, 'a) protocol-c-monad)
proof -
  have get-step:
    hash-range-budget 0
    (get :: (('f, 'a) protocol-channel-scheme,
      ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-get)
  have modify-return:
    hash-range-budget (0 + 0)
    (modify bump  $\gg=$  ( $\lambda\cdot$ . return r) ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for r :: 'f
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-modify-preserves-hash-map[OF bump],
      rule hash-range-budget-return)
  have hash-then-tail:
    hash-range-budget (1 + (0 + 0))
    (hash (tag (counter s) (PState s))  $\gg=$ 
      ( $\lambda r$ . modify bump  $\gg=$  ( $\lambda\cdot$ . return r)) ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for s :: ('f, 'a) protocol-channel-scheme
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-hash, rule modify-return)
  have hash-range-budget (0 + (1 + (0 + 0)))
    (get  $\gg=$ 
      ( $\lambda s :: ('f, 'a) \text{ protocol-channel-scheme.}$ 
        hash (tag (counter s) (PState s))  $\gg=$ 
          ( $\lambda r$ . modify bump  $\gg=$  ( $\lambda\cdot$ . return r))))
  by (rule hash-range-budget-bind[OF get-step hash-then-tail])
  then show ?thesis
  unfolding protocol-receive-counted-tagged-random-field-element-def
  by simp
qed

```

```

lemma hash-range-budget-receive-alpha-challenge:
  hash-range-budget 1
  (receive-alpha-challenge :: ('f, 'f, 'a) protocol-c-monad)
  unfolding receive-alpha-challenge-def
  by (rule hash-range-budget-receive-counted-tagged-random-field-element) simp

```

```

lemma hash-range-budget-receive-query-index-challenge:

```

hash-range-budget 1
 (receive-query-index-challenge :: ('f, 'f, 'a) protocol-c-monad)
unfolding receive-query-index-challenge-def
by (rule hash-range-budget-receive-counted-tagged-random-field-element) simp

lemma hash-range-budget-receive-trace-fri-challenge:
hash-range-budget 1
 (receive-trace-fri-challenge :: ('f, 'f, 'a) protocol-c-monad)
unfolding receive-trace-fri-challenge-def
by (rule hash-range-budget-receive-counted-tagged-random-field-element) simp

lemma hash-range-budget-receive-composition-fri-challenge:
hash-range-budget 1
 (receive-composition-fri-challenge :: ('f, 'f, 'a) protocol-c-monad)
unfolding receive-composition-fri-challenge-def
by (rule hash-range-budget-receive-counted-tagged-random-field-element) simp

lemma hash-collision-budget-receive-random-field-element:
hash-collision-budget 1
 (receive-random-field-element :: ('f, 'f, 'a) protocol-c-monad)

proof –

have get-range:
hash-range-budget 0
 (get :: (('f, 'a) protocol-channel-scheme,
 ('f, 'a) protocol-channel-scheme) state-monad)
by (rule hash-range-budget-get)
have get-coll:
hash-collision-budget 0
 (get :: (('f, 'a) protocol-channel-scheme,
 ('f, 'a) protocol-channel-scheme) state-monad)
by (rule hash-collision-budget-get)
have hash-then-return-range:
hash-range-budget (1 + 0)
 (hash (FiatShamirChallenge (PState s)) $\gg=$ return ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad)
for s :: ('f, 'a) protocol-channel-scheme
by (rule hash-range-budget-bind)
 (rule hash-range-budget-hash, rule hash-range-budget-return)
have hash-then-return-coll:
hash-collision-budget (1 + 0)
 (hash (FiatShamirChallenge (PState s)) $\gg=$ return ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad)
for s :: ('f, 'a) protocol-channel-scheme
by (rule hash-collision-budget-bind)
 (rule hash-range-budget-hash, rule hash-collision-budget-hash,
 rule hash-range-budget-return, rule hash-collision-budget-return)
have hash-collision-budget (0 + (1 + 0))
 (get $\gg=$
 (λ s :: ('f, 'a) protocol-channel-scheme.

```

      hash (FiatShamirChallenge (PState s))  $\gg=$  return))
by (rule hash-collision-budget-bind
  [OF get-range get-coll hash-then-return-range
    hash-then-return-coll])
then show ?thesis
  unfolding receive-random-field-element-def
    protocol-receive-random-field-element-def
    protocol-receive-tagged-random-field-element-def
  by simp
qed

lemma hash-collision-budget-receive-tagged-random-field-element:
  hash-collision-budget 1
  (receive-tagged-random-field-element tag :: ('f, 'a) protocol-c-monad)
proof -
  have get-range:
    hash-range-budget 0
    (get :: (('f, 'a) protocol-channel-scheme,
      ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-get)
  have get-coll:
    hash-collision-budget 0
    (get :: (('f, 'a) protocol-channel-scheme,
      ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-collision-budget-get)
  have hash-then-return-range:
    hash-range-budget (1 + 0)
    (hash (tag (PState s))  $\gg=$  return ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for s :: ('f, 'a) protocol-channel-scheme
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-hash, rule hash-range-budget-return)
  have hash-then-return-coll:
    hash-collision-budget (1 + 0)
    (hash (tag (PState s))  $\gg=$  return ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for s :: ('f, 'a) protocol-channel-scheme
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-hash, rule hash-collision-budget-hash,
      rule hash-range-budget-return, rule hash-collision-budget-return)
  have hash-collision-budget (0 + (1 + 0))
    (get  $\gg=$ 
      ( $\lambda$ s :: ('f, 'a) protocol-channel-scheme.
        hash (tag (PState s))  $\gg=$  return))
  by (rule hash-collision-budget-bind
    [OF get-range get-coll hash-then-return-range
      hash-then-return-coll])
then show ?thesis
  unfolding receive-tagged-random-field-element-def

```

```

    protocol-receive-tagged-random-field-element-def
  by simp
qed

lemma hash-collision-budget-receive-counted-tagged-random-field-element:
  assumes bump:  $\bigwedge s. \text{HashMap } (\text{bump } s) = \text{HashMap } s$ 
  shows hash-collision-budget 1
    (protocol-receive-counted-tagged-random-field-element counter bump tag ::
      ('f, 'f, 'a) protocol-c-monad)
proof -
  have get-range:
    hash-range-budget 0
    (get :: (('f, 'a) protocol-channel-scheme,
      ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-get)
  have get-coll:
    hash-collision-budget 0
    (get :: (('f, 'a) protocol-channel-scheme,
      ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-collision-budget-get)
  have modify-return-range:
    hash-range-budget (0 + 0)
    (modify bump  $\gg=$  ( $\lambda$ -. return r) ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for r :: 'f
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-modify-preserves-hash-map[OF bump],
      rule hash-range-budget-return)
  have modify-return-coll:
    hash-collision-budget (0 + 0)
    (modify bump  $\gg=$  ( $\lambda$ -. return r) ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for r :: 'f
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-modify-preserves-hash-map[OF bump],
      rule hash-collision-budget-modify-preserves-hash-map[OF bump],
      rule hash-range-budget-return, rule hash-collision-budget-return)
  have hash-then-tail-range:
    hash-range-budget (1 + (0 + 0))
    (hash (tag (counter s) (PState s))  $\gg=$ 
      ( $\lambda$ r. modify bump  $\gg=$  ( $\lambda$ -. return r)) ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for s :: ('f, 'a) protocol-channel-scheme
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-hash, rule modify-return-range)
  have hash-then-tail-coll:
    hash-collision-budget (1 + (0 + 0))
    (hash (tag (counter s) (PState s))  $\gg=$ 
      ( $\lambda$ r. modify bump  $\gg=$  ( $\lambda$ -. return r)) ::

```

```

    ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  for s :: ('f, 'a) protocol-channel-scheme
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-hash, rule hash-collision-budget-hash,
      rule modify-return-range, rule modify-return-coll)
  have hash-collision-budget (0 + (1 + (0 + 0)))
    (get  $\gg$ 
      ( $\lambda s :: ('f, 'a) \text{ protocol-channel-scheme.}$ 
        hash (tag (counter s) (PState s))  $\gg$ 
          ( $\lambda r. \text{ modify bump } \gg (\lambda-. \text{ return } r)$ )))
  by (rule hash-collision-budget-bind
    [OF get-range get-coll hash-then-tail-range hash-then-tail-coll])
  then show ?thesis
    unfolding protocol-receive-counted-tagged-random-field-element-def
  by simp
qed

```

lemma hash-collision-budget-receive-alpha-challenge:

```

  hash-collision-budget 1
  (receive-alpha-challenge :: ('f, 'f, 'a) protocol-c-monad)
  unfolding receive-alpha-challenge-def
  by (rule hash-collision-budget-receive-counted-tagged-random-field-element) simp

```

lemma hash-collision-budget-receive-query-index-challenge:

```

  hash-collision-budget 1
  (receive-query-index-challenge :: ('f, 'f, 'a) protocol-c-monad)
  unfolding receive-query-index-challenge-def
  by (rule hash-collision-budget-receive-counted-tagged-random-field-element) simp

```

lemma hash-collision-budget-receive-trace-fri-challenge:

```

  hash-collision-budget 1
  (receive-trace-fri-challenge :: ('f, 'f, 'a) protocol-c-monad)
  unfolding receive-trace-fri-challenge-def
  by (rule hash-collision-budget-receive-counted-tagged-random-field-element) simp

```

lemma hash-collision-budget-receive-composition-fri-challenge:

```

  hash-collision-budget 1
  (receive-composition-fri-challenge :: ('f, 'f, 'a) protocol-c-monad)
  unfolding receive-composition-fri-challenge-def
  by (rule hash-collision-budget-receive-counted-tagged-random-field-element) simp

```

lemma hash-range-budget-ntimes:

```

  fixes m :: ('x, ('f, 'a) protocol-channel-scheme) state-monad
  assumes m: hash-range-budget n m
  shows hash-range-budget (k * n) (ntimes m k)
  using m
proof (induction k)
  case 0
  then show ?case

```

```

    by (simp add: hash-range-budget-return)
next
case (Suc k)
have tail: hash-range-budget (k * n) (ntimes m k)
  using Suc.IH Suc.prem by simp
have cont:
  hash-range-budget (k * n + 0)
    (ntimes m k  $\gg$  ( $\lambda x s$ . return (x # xs))) for x
  by (rule hash-range-budget-bind[OF tail hash-range-budget-return])
have hash-range-budget (n + (k * n + 0))
  (m  $\gg$  ( $\lambda x$ . ntimes m k  $\gg$  ( $\lambda x s$ . return (x # xs))))
  by (rule hash-range-budget-bind[OF Suc.prem cont])
then show ?case
  by simp
qed

lemma hash-collision-budget-ntimes:
  fixes m :: ('x, ('f, 'a) protocol-channel-scheme) state-monad
  assumes range-m: hash-range-budget n m
  and coll-m: hash-collision-budget n m
  shows hash-collision-budget (k * n) (ntimes m k)
  using range-m coll-m
proof (induction k)
case 0
then show ?case
  by (simp add: hash-collision-budget-return)
next
case (Suc k)
have tail-range: hash-range-budget (k * n) (ntimes m k)
  by (rule hash-range-budget-ntimes[OF Suc.prem(1)])
have tail-coll: hash-collision-budget (k * n) (ntimes m k)
  using Suc.IH Suc.prem by simp
have cont-range:
  hash-range-budget (k * n + 0)
    (ntimes m k  $\gg$  ( $\lambda x s$ . return (x # xs))) for x
  by (rule hash-range-budget-bind[OF tail-range hash-range-budget-return])
have cont-coll:
  hash-collision-budget (k * n + 0)
    (ntimes m k  $\gg$  ( $\lambda x s$ . return (x # xs))) for x
  by (rule hash-collision-budget-bind
    [OF tail-range tail-coll hash-range-budget-return
      hash-collision-budget-return])
have hash-collision-budget (n + (k * n + 0))
  (m  $\gg$  ( $\lambda x$ . ntimes m k  $\gg$  ( $\lambda x s$ . return (x # xs))))
  by (rule hash-collision-budget-bind
    [OF Suc.prem(1) Suc.prem(2) cont-range cont-coll])
then show ?case
  by simp
qed

```

lemma *hash-range-budget-ntimes-read-bind*:
fixes $k :: 'f \text{ list} \Rightarrow$
 $('y, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
assumes $k: \bigwedge xs. \text{length } xs = n \implies \text{hash-range-budget } b (k \text{ } xs)$
shows
 $\text{hash-range-budget } b$
 $((\text{ntimes read } n :: ('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
 $\gg k)$
proof –
have *read-range*:
 $\text{hash-range-budget } (n * 0)$
 $(\text{ntimes read } n :: ('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
by (rule *hash-range-budget-ntimes*[OF *hash-range-budget-read*])
have *hash-range-budget* $(n * 0 + b)$
 $((\text{ntimes read } n :: ('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
 $\gg k)$
proof (rule *hash-range-budget-bind-on-outcomes*[OF *read-range*])
fix $s \text{ } xs \text{ } t$
assume *out*:
 $\text{Some } (xs, t) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{ntimes read } n ::$
 $('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}) \text{ } s)$
show *hash-range-budget* $b (k \text{ } xs)$
by (rule k) (use *ntimes-read-any-outcome*[OF *out*] **in** *simp*)
qed
then show *?thesis* **by** *simp*
qed

lemma *hash-collision-budget-ntimes-read-bind*:
fixes $k :: 'f \text{ list} \Rightarrow$
 $('y, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
assumes *range-k*: $\bigwedge xs. \text{length } xs = n \implies \text{hash-range-budget } b (k \text{ } xs)$
and *coll-k*: $\bigwedge xs. \text{length } xs = n \implies \text{hash-collision-budget } b (k \text{ } xs)$
shows
 $\text{hash-collision-budget } b$
 $((\text{ntimes read } n :: ('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
 $\gg k)$
proof –
have *read-range*:
 $\text{hash-range-budget } (n * 0)$
 $(\text{ntimes read } n :: ('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
by (rule *hash-range-budget-ntimes*[OF *hash-range-budget-read*])
have *read-coll*:
 $\text{hash-collision-budget } (n * 0)$
 $(\text{ntimes read } n :: ('f \text{ list}, ('f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
by (rule *hash-collision-budget-ntimes*
 $[\text{OF } \text{hash-range-budget-read } \text{hash-collision-budget-read}]$)


```

have hash-collision-budget ( $n * 0 + b$ )
  (( $ntimes$  read  $n :: ('f$  list,  $('f, 'a)$  protocol-channel-scheme) state-monad)
     $\gg=$   $k$ )
proof (rule hash-collision-budget-bind-on-outcomes
  [OF read-range read-coll])
fix  $s$   $xs$   $t$ 
assume out:
  Some ( $xs, t$ )  $\in$ 
    set-dist (execute
      ( $ntimes$  read  $n ::$ 
        ( $'f$  list,  $('f, 'a)$  protocol-channel-scheme) state-monad)  $s$ )
have len: length  $xs = n$ 
  using  $ntimes$ -read-any-outcome[OF out] by simp
show hash-range-budget  $b$  ( $k$   $xs$ )
  by (rule range-k[OF len])
show hash-collision-budget  $b$  ( $k$   $xs$ )
  by (rule coll-k[OF len])
qed
then show ?thesis by simp
qed

```

```

lemma hash-range-budget-mmap:
  fixes  $ms :: ('x, ('f, 'a)$  protocol-channel-scheme) state-monad list
  assumes  $ms: \bigwedge m. m \in \text{set } ms \implies \text{hash-range-budget } n \ m$ 
  shows hash-range-budget (length  $ms * n$ ) (mmap  $ms$ )
  using  $ms$ 
proof (induction  $ms$ )
  case Nil
  then show ?case
    by (simp add: hash-range-budget-return)
next
  case (Cons  $m$   $ms$ )
  have head: hash-range-budget  $n$   $m$ 
    using Cons.prem by simp
  have tail: hash-range-budget (length  $ms * n$ ) (mmap  $ms$ )
    using Cons.IH Cons.prem by simp
  have cont:
    hash-range-budget (length  $ms * n + 0$ )
      (mmap  $ms \gg= (\lambda xs. \text{return } (x \# xs))$ ) for  $x$ 
    by (rule hash-range-budget-bind[OF tail hash-range-budget-return])
  have hash-range-budget ( $n + (\text{length } ms * n + 0)$ )
    ( $m \gg= (\lambda x. \text{mmap } ms \gg= (\lambda xs. \text{return } (x \# xs)))$ )
    by (rule hash-range-budget-bind[OF head cont])
  then show ?case
    by simp
qed

```

```

lemma hash-collision-budget-mmap:
  fixes  $ms :: ('x, ('f, 'a)$  protocol-channel-scheme) state-monad list

```

```

assumes range-ms:  $\bigwedge m. m \in \text{set } ms \implies \text{hash-range-budget } n \ m$ 
and coll-ms:  $\bigwedge m. m \in \text{set } ms \implies \text{hash-collision-budget } n \ m$ 
shows hash-collision-budget (length ms * n) (mmap ms)
using range-ms coll-ms
proof (induction ms)
  case Nil
  then show ?case
    by (simp add: hash-collision-budget-return)
next
  case (Cons m ms)
  have head-range: hash-range-budget n m
    using Cons.prem1 by simp
  have head-coll: hash-collision-budget n m
    using Cons.prem1 by simp
  have tail-range: hash-range-budget (length ms * n) (mmap ms)
    by (rule hash-range-budget-mmap) (use Cons.prem1 in auto)
  have tail-coll: hash-collision-budget (length ms * n) (mmap ms)
    using Cons.IH Cons.prem1 by simp
  have cont-range:
    hash-range-budget (length ms * n + 0)
    (mmap ms  $\gg (\lambda xs. \text{return } (x \# xs))$ ) for x
    by (rule hash-range-budget-bind[OF tail-range hash-range-budget-return])
  have cont-coll:
    hash-collision-budget (length ms * n + 0)
    (mmap ms  $\gg (\lambda xs. \text{return } (x \# xs))$ ) for x
    by (rule hash-collision-budget-bind
      [OF tail-range tail-coll hash-range-budget-return
        hash-collision-budget-return])
  have hash-collision-budget (n + (length ms * n + 0))
    (m  $\gg (\lambda x. \text{mmap } ms \gg (\lambda xs. \text{return } (x \# xs)))$ )
    by (rule hash-collision-budget-bind
      [OF head-range head-coll cont-range cont-coll])
  then show ?case
    by simp
qed

```

```

lemma hash-range-budget-mfold:
  fixes steps :: ('x  $\Rightarrow$  ('f, 'a) protocol-channel-scheme) state-monad) list
  assumes steps:  $\bigwedge \text{step } x. \text{step} \in \text{set steps} \implies$ 
    hash-range-budget n (step x)
  shows hash-range-budget (length steps * n) (mfold x steps)
  using steps
proof (induction steps arbitrary: x)
  case Nil
  then show ?case
    by (simp add: hash-range-budget-return)
next
  case (Cons step steps)
  have head: hash-range-budget n (step x)

```

```

    using Cons.premis by simp
  have tail:  $\bigwedge y. \text{hash-range-budget } (\text{length steps} * n) (\text{mfold } y \text{ steps})$ 
    using Cons.IH Cons.premis by simp
  have hash-range-budget  $(n + \text{length steps} * n)$ 
     $(\text{step } x \gg (\lambda y. \text{mfold } y \text{ steps}))$ 
    by (rule hash-range-budget-bind[OF head tail])
  then show ?case
    by simp
qed

lemma hash-collision-budget-mfold:
  fixes steps ::  $('x \Rightarrow ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}) \text{list}$ 
  assumes range-steps:  $\bigwedge \text{step } x. \text{step} \in \text{set steps} \Rightarrow$ 
     $\text{hash-range-budget } n (\text{step } x)$ 
  and coll-steps:  $\bigwedge \text{step } x. \text{step} \in \text{set steps} \Rightarrow$ 
     $\text{hash-collision-budget } n (\text{step } x)$ 
  shows hash-collision-budget  $(\text{length steps} * n) (\text{mfold } x \text{ steps})$ 
  using range-steps coll-steps
proof (induction steps arbitrary: x)
  case Nil
  then show ?case
    by (simp add: hash-collision-budget-return)
next
  case (Cons step steps)
  have head-range:  $\text{hash-range-budget } n (\text{step } x)$ 
    using Cons.premis by simp
  have head-coll:  $\text{hash-collision-budget } n (\text{step } x)$ 
    using Cons.premis by simp
  have tail-range:
     $\bigwedge y. \text{hash-range-budget } (\text{length steps} * n) (\text{mfold } y \text{ steps})$ 
    by (rule hash-range-budget-mfold) (use Cons.premis in auto)
  have tail-coll:
     $\bigwedge y. \text{hash-collision-budget } (\text{length steps} * n) (\text{mfold } y \text{ steps})$ 
    using Cons.IH Cons.premis by simp
  have hash-collision-budget  $(n + \text{length steps} * n)$ 
     $(\text{step } x \gg (\lambda y. \text{mfold } y \text{ steps}))$ 
    by (rule hash-collision-budget-bind
      [OF head-range head-coll tail-range tail-coll])
  then show ?case
    by simp
qed

lemma hash-range-budget-mfold-invariant:
  fixes steps ::  $('x \Rightarrow ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}) \text{list}$ 
  assumes init:  $I x$ 
  and step-budget:
     $\bigwedge \text{step } y. \text{step} \in \text{set steps} \Rightarrow I y \Rightarrow$ 
     $\text{hash-range-budget } n (\text{step } y)$ 
  and step-inv:

```

```

 $\bigwedge \text{step } y \ z \ s \ t. \text{ step} \in \text{set steps} \implies I \ y \implies$ 
 $\text{Some } (z, t) \in \text{set-dist } (\text{execute } (\text{step } y) \ s) \implies I \ z$ 
shows  $\text{hash-range-budget } (\text{length steps} * n) \ (\text{mfold } x \text{ steps})$ 
using  $\text{init step-budget step-inv}$ 
proof ( $\text{induction steps arbitrary: } x$ )
  case  $\text{Nil}$ 
  then show  $?case$ 
    by ( $\text{simp add: hash-range-budget-return}$ )
next
  case ( $\text{Cons step steps}$ )
  have  $Ix: I \ x$ 
    using  $\text{Cons.premis}$  by  $\text{simp}$ 
  have  $\text{head: hash-range-budget } n \ (\text{step } x)$ 
    by ( $\text{rule Cons.premis}(2)$ ) ( $\text{use } Ix \text{ in auto}$ )
  have  $\text{tail:}$ 
     $\text{hash-range-budget } (\text{length steps} * n) \ (\text{mfold } y \text{ steps})$ 
    if out:  $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (\text{step } x) \ s)$ 
    for  $s \ y \ t$ 
  proof ( $\text{rule Cons.IH}$ )
    show  $I \ y$ 
    proof –
      have  $\text{mem: step} \in \text{set } (\text{step} \# \text{steps})$ 
        by  $\text{simp}$ 
      have  $\text{inv: } \bigwedge z \ s \ t.$ 
         $\text{Some } (z, t) \in \text{set-dist } (\text{execute } (\text{step } x) \ s) \implies I \ z$ 
        by ( $\text{rule Cons.premis}(3)[\text{OF mem } Ix]$ )
      show  $?thesis$ 
        by ( $\text{rule inv}[\text{OF out}]$ )
    qed
  show  $\bigwedge \text{step } y. \text{ step} \in \text{set steps} \implies I \ y \implies$ 
     $\text{hash-range-budget } n \ (\text{step } y)$ 
    by ( $\text{rule Cons.premis}(2)$ )  $\text{simp-all}$ 
  show  $\bigwedge \text{step } y \ z \ s \ t.$ 
     $\text{step} \in \text{set steps} \implies$ 
     $I \ y \implies \text{Some } (z, t) \in \text{set-dist } (\text{execute } (\text{step } y) \ s) \implies I \ z$ 
    by ( $\text{rule Cons.premis}(3)$ )  $\text{simp-all}$ 
  qed
  have  $\text{hash-range-budget } (n + \text{length steps} * n)$ 
    ( $\text{step } x \gg (\lambda y. \text{mfold } y \text{ steps})$ )
    by ( $\text{rule hash-range-budget-bind-on-outcomes}[\text{OF head}]$ ) ( $\text{rule tail}$ )
  then show  $?case$ 
    by  $\text{simp}$ 
qed

```

lemma $\text{hash-collision-budget-mfold-invariant:}$
fixes $\text{steps} :: ('x \Rightarrow ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}) \text{list}$
assumes $\text{init: } I \ x$
and range-step:
 $\bigwedge \text{step } y. \text{ step} \in \text{set steps} \implies I \ y \implies$

```

    hash-range-budget n (step y)
  and coll-step:
     $\bigwedge \text{step } y. \text{ step} \in \text{set steps} \implies I y \implies$ 
      hash-collision-budget n (step y)
  and step-inv:
     $\bigwedge \text{step } y \ z \ s \ t. \text{ step} \in \text{set steps} \implies I y \implies$ 
      Some (z, t)  $\in$  set-dist (execute (step y) s)  $\implies I z$ 
  shows hash-collision-budget (length steps * n) (mfold x steps)
  using init range-step coll-step step-inv
proof (induction steps arbitrary: x)
  case Nil
  then show ?case
    by (simp add: hash-collision-budget-return)
next
  case (Cons step steps)
  have Ix: I x
    using Cons.prem1 by simp
  have head-range: hash-range-budget n (step x)
    by (rule Cons.prem2) (use Ix in auto)
  have head-coll: hash-collision-budget n (step x)
    by (rule Cons.prem3) (use Ix in auto)
  have tail-range:
    hash-range-budget (length steps * n) (mfold y steps)
    if out: Some (y, t)  $\in$  set-dist (execute (step x) s)
    for s y t
  proof (rule hash-range-budget-mfold-invariant)
    show I y
  proof -
    have mem: step  $\in$  set (step # steps)
      by simp
    have inv:  $\bigwedge z \ s \ t. \text{ Some } (z, t) \in \text{set-dist } (\text{execute } (\text{step } x) \ s) \implies I z$ 
      by (rule Cons.prem4)[OF mem Ix]
    show ?thesis
      by (rule inv[OF out])
  qed
  show  $\bigwedge \text{step } y. \text{ step} \in \text{set steps} \implies I y \implies$ 
    hash-range-budget n (step y)
    by (rule Cons.prem2) simp-all
  show  $\bigwedge \text{step } y \ z \ s \ t. \text{ step} \in \text{set steps} \implies$ 
    I y  $\implies \text{Some } (z, t) \in \text{set-dist } (\text{execute } (\text{step } y) \ s) \implies I z$ 
    by (rule Cons.prem4) simp-all
  qed
  have tail-coll:
    hash-collision-budget (length steps * n) (mfold y steps)
    if out: Some (y, t)  $\in$  set-dist (execute (step x) s)
    for s y t
  proof (rule Cons.IH)

```

```

show  $I\ y$ 
proof –
  have  $mem: step \in set\ steps$ 
  by simp
  have  $inv: \bigwedge z\ s\ t.$ 
     $Some\ (z, t) \in set-dist\ (execute\ (step\ x)\ s) \implies I\ z$ 
  by (rule Cons.prem4) [OF mem Ix]
  show ?thesis
  by (rule inv [OF out])
qed
show  $\bigwedge step\ y. step \in set\ steps \implies I\ y \implies$ 
   $hash-range-budget\ n\ (step\ y)$ 
  by (rule Cons.prem2) simp-all
show  $\bigwedge step\ y. step \in set\ steps \implies I\ y \implies$ 
   $hash-collision-budget\ n\ (step\ y)$ 
  by (rule Cons.prem3) simp-all
show  $\bigwedge step\ y\ z\ s\ t.$ 
   $step \in set\ steps \implies$ 
   $I\ y \implies Some\ (z, t) \in set-dist\ (execute\ (step\ y)\ s) \implies I\ z$ 
  by (rule Cons.prem4) simp-all
qed
have  $hash-collision-budget\ (n + length\ steps * n)$ 
   $(step\ x \gg (\lambda y. mfold\ y\ steps))$ 
  by (rule hash-collision-budget-bind-on-outcomes
    [OF head-range head-coll tail-range tail-coll])
then show ?case
  by simp
qed

lemma hash-range-budget-fri-state-mfold-invariant:
fixes  $st :: nat \times 'f \times nat \times nat$ 
and  $steps ::$ 
   $(nat \times 'f \times nat \times nat \Rightarrow$ 
     $(nat \times 'f \times nat \times nat,$ 
       $(f, 'a)\ protocol-channel-scheme)\ state-monad)\ list$ 
and  $I :: nat \times 'f \times nat \times nat \Rightarrow bool$ 
assumes init:  $I\ st$ 
and step-budget:
   $\bigwedge step\ y. step \in set\ steps \implies I\ y \implies$ 
   $hash-range-budget\ n\ (step\ y)$ 
and step-inv:
   $\bigwedge step\ y\ z\ s\ t. step \in set\ steps \implies I\ y \implies$ 
   $Some\ (z, t) \in set-dist\ (execute\ (step\ y)\ s) \implies I\ z$ 
shows  $hash-range-budget\ (length\ steps * n)\ (mfold\ st\ steps)$ 
using assms by (rule hash-range-budget-mfold-invariant)

```

```

lemma hash-collision-budget-fri-state-mfold-invariant:
fixes  $st :: nat \times 'f \times nat \times nat$ 
and  $steps ::$ 

```

```

    (nat × 'f × nat × nat ⇒
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) list
  and I :: nat × 'f × nat × nat ⇒ bool
assumes init: I st
and range-step:
  ∧ step y. step ∈ set steps ⇒ I y ⇒
    hash-range-budget n (step y)
and coll-step:
  ∧ step y. step ∈ set steps ⇒ I y ⇒
    hash-collision-budget n (step y)
and step-inv:
  ∧ step y z s t. step ∈ set steps ⇒ I y ⇒
    Some (z, t) ∈ set-dist (execute (step y) s) ⇒ I z
shows hash-collision-budget (length steps * n) (mfold st steps)
using assms by (rule hash-collision-budget-mfold-invariant)

```

lemma hash-target-program-receive-fri-commits:

hash-target-program B 1 receive-fri-commits

proof –

have tail:

hash-target-program B (1 + 0)

(receive-random-field-element ≫ (λb. return (b, r)))

for r

by (rule hash-target-program-bind)

(rule hash-target-program-receive-random-field-element,

rule hash-target-program-return)

have hash-target-program B (0 + (1 + 0))

(read ≫

(λr. receive-random-field-element ≫

(λb. return (b, r))))

by (rule hash-target-program-bind[OF hash-target-program-read tail])

then show ?thesis

unfolding receive-fri-commits-def **by** simp

qed

lemma hash-target-program-receive-fri-commits-with:

assumes challenge: hash-target-program B 1 receive-challenge

shows hash-target-program B 1

(receive-fri-commits-with receive-challenge)

proof –

have tail:

hash-target-program B (1 + 0)

(receive-challenge ≫ (λb. return (b, r)))

for r

by (rule hash-target-program-bind[OF challenge hash-target-program-return])

have hash-target-program B (0 + (1 + 0))

```

      (read  $\gg$ 
        ( $\lambda r$ . receive-challenge  $\gg$ 
          ( $\lambda b$ . return (b, r))))
    by (rule hash-target-program-bind[OF hash-target-program-read tail])
  then show ?thesis
    unfolding receive-fri-commits-with-def by simp
qed

```

lemma hash-target-program-receive-trace-fri-commits:
 hash-target-program B 1 receive-trace-fri-commits
unfolding receive-trace-fri-commits-def
by (rule hash-target-program-receive-fri-commits-with)
 (rule hash-target-program-receive-trace-fri-challenge)

lemma hash-target-program-receive-composition-fri-commits:
 hash-target-program B 1 receive-composition-fri-commits
unfolding receive-composition-fri-commits-def
by (rule hash-target-program-receive-fri-commits-with)
 (rule hash-target-program-receive-composition-fri-challenge)

lemma hash-range-budget-receive-fri-commits:
 hash-range-budget 1 receive-fri-commits

proof –
have read-then-random:
 hash-range-budget (0 + (1 + 0))
 (read $\gg$
 (λr . receive-random-field-element $\gg$
 (λb . return (b, r))))
by (rule hash-range-budget-bind)
 (rule hash-range-budget-read,
 rule hash-range-budget-bind
 [OF hash-range-budget-receive-random-field-element
 hash-range-budget-return])
then show ?thesis
 unfolding receive-fri-commits-def by simp
qed

lemma hash-collision-budget-receive-fri-commits:
 hash-collision-budget 1 receive-fri-commits

proof –
have random-then-return-range:
 hash-range-budget (1 + 0)
 (receive-random-field-element $\gg$ (λb . return (b, r))) **for** r
by (rule hash-range-budget-bind
 [OF hash-range-budget-receive-random-field-element
 hash-range-budget-return])
have random-then-return-coll:
 hash-collision-budget (1 + 0)
 (receive-random-field-element $\gg$ (λb . return (b, r))) **for** r


```

    by (rule hash-collision-budget-bind
        [OF hash-range-budget-receive-random-field-element
            hash-collision-budget-receive-random-field-element
            hash-range-budget-return hash-collision-budget-return])
  have hash-collision-budget (0 + (1 + 0))
    (read  $\gg$ 
      ( $\lambda r$ . receive-random-field-element  $\gg$ 
        ( $\lambda b$ . return (b, r))))
  by (rule hash-collision-budget-bind
      [OF hash-range-budget-read hash-collision-budget-read
          random-then-return-range random-then-return-coll])
  then show ?thesis
    unfolding receive-fri-commits-def by simp
qed

```

lemma *hash-range-budget-receive-fri-commits-with:*
assumes challenge: hash-range-budget 1 receive-challenge
shows hash-range-budget 1 (receive-fri-commits-with receive-challenge)
proof –
have read-then-challenge:
 hash-range-budget (0 + (1 + 0))
 (read $\gg$
 (λr . receive-challenge $\gg$
 (λb . return (b, r))))
by (rule hash-range-budget-bind)
 (rule hash-range-budget-read,
 rule hash-range-budget-bind [OF challenge hash-range-budget-return])
then show ?thesis
 unfolding receive-fri-commits-with-def by simp
qed

lemma *hash-collision-budget-receive-fri-commits-with:*
assumes challenge-range: hash-range-budget 1 receive-challenge
and challenge-collision: hash-collision-budget 1 receive-challenge
shows hash-collision-budget 1 (receive-fri-commits-with receive-challenge)
proof –
have challenge-then-return-range:
 hash-range-budget (1 + 0)
 (receive-challenge $\gg$ (λb . return (b, r))) **for** r
by (rule hash-range-budget-bind
 [OF challenge-range hash-range-budget-return])
have challenge-then-return-coll:
 hash-collision-budget (1 + 0)
 (receive-challenge $\gg$ (λb . return (b, r))) **for** r
by (rule hash-collision-budget-bind
 [OF challenge-range challenge-collision
 hash-range-budget-return hash-collision-budget-return])
have hash-collision-budget (0 + (1 + 0))
 (read $\gg$

```

    (λr. receive-challenge ≫
      (λb. return (b, r)))
  by (rule hash-collision-budget-bind
    [OF hash-range-budget-read hash-collision-budget-read
      challenge-then-return-range challenge-then-return-coll])
  then show ?thesis
    unfolding receive-fri-commits-with-def by simp
qed

lemma hash-range-budget-receive-trace-fri-commits:
  hash-range-budget 1 receive-trace-fri-commits
  unfolding receive-trace-fri-commits-def
  by (rule hash-range-budget-receive-fri-commits-with)
    (rule hash-range-budget-receive-trace-fri-challenge)

lemma hash-collision-budget-receive-trace-fri-commits:
  hash-collision-budget 1 receive-trace-fri-commits
  unfolding receive-trace-fri-commits-def
  by (rule hash-collision-budget-receive-fri-commits-with)
    (rule hash-range-budget-receive-trace-fri-challenge,
      rule hash-collision-budget-receive-trace-fri-challenge)

lemma hash-range-budget-receive-composition-fri-commits:
  hash-range-budget 1 receive-composition-fri-commits
  unfolding receive-composition-fri-commits-def
  by (rule hash-range-budget-receive-fri-commits-with)
    (rule hash-range-budget-receive-composition-fri-challenge)

lemma hash-collision-budget-receive-composition-fri-commits:
  hash-collision-budget 1 receive-composition-fri-commits
  unfolding receive-composition-fri-commits-def
  by (rule hash-collision-budget-receive-fri-commits-with)
    (rule hash-range-budget-receive-composition-fri-challenge,
      rule hash-collision-budget-receive-composition-fri-challenge)

lemma hash-target-program-protocol-check-authentication-path-raw:
  fixes leaf :: 'f protocol-hash-input
  shows hash-target-program B (Suc (length path))
    (protocol-merkle.check-authentication-path len i leaf path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
proof (induction path arbitrary: len i)
  case Nil
  have hash-target-program B 1
    (hash leaf :: ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-target-program-hash)
  then show ?case
    by (simp add: protocol-merkle.check-authentication-path.simps)
next
  case (Cons a path)

```

```

show ?case
proof (cases i < len div 2)
  case True
  have rec:
    hash-target-program B (Suc (length path))
    (protocol-merkle.check-authentication-path (len div 2) i leaf path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  using Cons.IH[of len div 2 i] by simp
  have bound:
    hash-target-program B (Suc (length path) + 1)
    ((protocol-merkle.check-authentication-path (len div 2) i leaf path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
      ( $\lambda x$ . hash (MerkleNode x a)))
  by (rule hash-target-program-bind[OF rec hash-target-program-hash])
show ?thesis
  using True bound
  by (simp add: protocol-merkle.check-authentication-path.simps)
next
  case False
  have rec:
    hash-target-program B (Suc (length path))
    (protocol-merkle.check-authentication-path
      (len div 2) (i - len div 2) leaf path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  using Cons.IH[of len div 2 i - len div 2] by simp
  have bound:
    hash-target-program B (Suc (length path) + 1)
    ((protocol-merkle.check-authentication-path
      (len div 2) (i - len div 2) leaf path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
      ( $\lambda x$ . hash (MerkleNode a x)))
  by (rule hash-target-program-bind[OF rec hash-target-program-hash])
show ?thesis
  using False bound
  by (simp add: protocol-merkle.check-authentication-path.simps)
qed
qed

lemma hash-target-program-check-authentication-path:
  hash-target-program B (Suc (length path))
  (check-authentication-path len i v path)
unfolding check-authentication-path-def protocol-check-authentication-path-def
by (rule hash-target-program-protocol-check-authentication-path-raw)

lemma hash-target-program-check-authentication-path-assert-return:
  fixes path :: 'f list
  and v root :: 'f
  and r :: 'r
shows

```

```

hash-target-program B (Suc (length path))
  ((check-authentication-path len i v path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = root) >>= (λ-. return r)))
proof -
  have tail:
    hash-target-program B (0 + 0)
    (assert (ap = root) >>= (λ-. return r) ::
      ('r, ('f, 'a) protocol-channel-scheme) state-monad)
  for ap
  by (rule hash-target-program-bind)
  (rule hash-target-program-assert, rule hash-target-program-return)
have hash-target-program B (Suc (length path) + (0 + 0))
  ((check-authentication-path len i v path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = root) >>= (λ-. return r)))
  by (rule hash-target-program-bind)
  (rule hash-target-program-check-authentication-path, rule tail)
then show ?thesis by simp
qed

```

lemma *hash-target-program-check-authentication-path-assert-bind:*

```

fixes path :: 'f list
and v root :: 'f
and k :: unit ⇒
  ('r, ('f, 'a) protocol-channel-scheme) state-monad
assumes k: ∧u. hash-target-program B n (k u)
shows
  hash-target-program B (Suc (length path) + n)
  ((check-authentication-path len i v path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = root) >>= k))
proof -
  have tail:
    hash-target-program B (0 + n)
    (assert (ap = root) >>= k ::
      ('r, ('f, 'a) protocol-channel-scheme) state-monad)
  for ap
  by (rule hash-target-program-bind[OF hash-target-program-assert k])
have hash-target-program B (Suc (length path) + (0 + n))
  ((check-authentication-path len i v path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = root) >>= k))
  by (rule hash-target-program-bind)
  (rule hash-target-program-check-authentication-path, rule tail)
then show ?thesis by simp
qed

```

lemma *hash-target-program-fri-layer-opening-finish:*

```

assumes xp-path-len: length xp-path = floor-log len
and xn-path-len: length xn-path = floor-log len
shows
  hash-target-program B
    (Suc (floor-log len) + Suc (floor-log len))
    (fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)
proof –
  let ?H = Suc (floor-log len)
  let ?sidx = (i + len div 2) mod len
  let ?out =
    (i mod (len div 2),
     (xp + xn) div 2 +
      b * ((xp - xn) div (2 * ((h ^ i) * shift) ^ pw)),
     len div 2, pw + pw)
  have second:
    hash-target-program B ?H
      ((check-authentication-path len ?sidx xn xn-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
        ( $\lambda$ ap. assert (ap = f)  $\gg$  ( $\lambda$ -. return ?out)))
    by (subst xn-path-len[symmetric])
      (rule hash-target-program-check-authentication-path-assert-return)
  have first0:
    hash-target-program B (Suc (length xp-path) + ?H)
      ((check-authentication-path len i xp xp-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
        ( $\lambda$ ap. assert (ap = f)  $\gg$ 
          ( $\lambda$ -.
            (check-authentication-path len ?sidx xn xn-path ::
              ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
              ( $\lambda$ ap. assert (ap = f)  $\gg$  ( $\lambda$ -. return ?out))))))
    by (rule hash-target-program-check-authentication-path-assert-bind)
      (use second in simp)
  have first:
    hash-target-program B (?H + ?H)
      ((check-authentication-path len i xp xp-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
        ( $\lambda$ ap. assert (ap = f)  $\gg$ 
          ( $\lambda$ -.
            (check-authentication-path len ?sidx xn xn-path ::
              ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
              ( $\lambda$ ap. assert (ap = f)  $\gg$  ( $\lambda$ -. return ?out))))))
    using first0 xp-path-len by simp
  have raw:
    hash-target-program B (0 + (?H + ?H))
      ((assert (xp = x) ::
        (unit, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
        ( $\lambda$ -.
          (check-authentication-path len i xp xp-path ::
            ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
            ( $\lambda$ -. return ?out))))

```

```

    (λap. assert (ap = f) >>=
      (λ-.
        (check-authentication-path len ?sidx xn xn-path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = f) >>= (λ-. return ?out))))))
  by (rule hash-target-program-bind[OF hash-target-program-assert first])
show ?thesis
  using raw unfolding fri-layer-opening-finish-def
  by (simp add: Let-def)
qed

```

lemma hash-target-program-fri-layer-opening-step:

```

hash-target-program B
  (Suc (floor-log len) + Suc (floor-log len))
  (fri-layer-opening-step (b, f) (i, x, len, pw))

```

proof –

```

let ?N = Suc (floor-log len) + Suc (floor-log len)

```

have after-xn-path:

```

hash-target-program B ?N
  ((ntimes read (floor-log len) ::
    ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λxn-path.
      fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))

```

```

if xp-path-len: length xp-path = floor-log len

```

```

for xp xp-path xn

```

proof (rule hash-target-program-ntimes-read-bind)

```

fix xn-path :: 'f list

```

```

assume xn-path-len: length xn-path = floor-log len

```

```

show hash-target-program B ?N

```

```

  (fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)

```

```

  by (rule hash-target-program-fri-layer-opening-finish
    [OF xp-path-len xn-path-len])

```

qed

have after-xn:

```

hash-target-program B ?N
  (read >>=
    (λxn.
      (ntimes read (floor-log len) ::
        ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λxn-path.
          fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)))

```

```

if xp-path-len: length xp-path = floor-log len

```

```

for xp xp-path

```

proof –

```

have hash-target-program B (0 + ?N)

```

```

  (read >>=
    (λxn.
      (ntimes read (floor-log len) ::
        ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=

```

```

      (λxn-path.
        fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)))
  by (rule hash-target-program-bind
    [OF hash-target-program-read after-xn-path[OF xp-path-len]])
  then show ?thesis by simp
qed
have after-xp-path:
  hash-target-program B ?N
  ((ntimes read (floor-log len) ::
    ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λxp-path.
      read >>=
      (λxn.
        (ntimes read (floor-log len) ::
          ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λxn-path.
            fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))))
  for xp
proof (rule hash-target-program-ntimes-read-bind)
  fix xp-path :: 'f list
  assume xp-path-len: length xp-path = floor-log len
  show hash-target-program B ?N
    (read >>=
      (λxn.
        (ntimes read (floor-log len) ::
          ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λxn-path.
            fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
  by (rule after-xn[OF xp-path-len])
qed
have hash-target-program B (0 + ?N)
  (read >>=
    (λxp.
      (ntimes read (floor-log len) ::
        ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λxp-path.
          read >>=
          (λxn.
            (ntimes read (floor-log len) ::
              ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
              (λxn-path.
                fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))))
    by (rule hash-target-program-bind
      [OF hash-target-program-read after-xp-path])
  then show ?thesis
    unfolding fri-layer-opening-step-unfold-finish by simp
qed

```

lemma hash-range-budget-protocol-check-authentication-path-raw:

```

fixes leaf :: 'f protocol-hash-input
shows hash-range-budget (Suc (length path))
  (protocol-merkle.check-authentication-path len i leaf path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)
proof (induction path arbitrary: len i)
case Nil
have hash-budget:
  hash-range-budget 1
  (hash leaf :: ('f, ('f, 'a) protocol-channel-scheme) state-monad)
by (rule hash-range-budget-hash)
then show ?case
  by (simp add: protocol-merkle.check-authentication-path.simps)
next
case (Cons a path)
show ?case
proof (cases i < len div 2)
case True
have rec:
  hash-range-budget (Suc (length path))
  (protocol-merkle.check-authentication-path (len div 2) i leaf path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)
using Cons.IH[of len div 2 i] by simp
have bound:
  hash-range-budget (Suc (length path) + 1)
  ((protocol-merkle.check-authentication-path (len div 2) i leaf path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda x$ . hash (MerkleNode x a)))
by (rule hash-range-budget-bind[OF rec hash-range-budget-hash])
show ?thesis
using True bound
by (simp add: protocol-merkle.check-authentication-path.simps)
next
case False
have rec:
  hash-range-budget (Suc (length path))
  (protocol-merkle.check-authentication-path
    (len div 2) (i - len div 2) leaf path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)
using Cons.IH[of len div 2 i - len div 2] by simp
have bound:
  hash-range-budget (Suc (length path) + 1)
  ((protocol-merkle.check-authentication-path
    (len div 2) (i - len div 2) leaf path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda x$ . hash (MerkleNode a x)))
by (rule hash-range-budget-bind[OF rec hash-range-budget-hash])
show ?thesis
using False bound
by (simp add: protocol-merkle.check-authentication-path.simps)

```


qed
qed

lemma *hash-range-budget-check-authentication-path*:
hash-range-budget (Suc (length path))
 (check-authentication-path len i v path)
unfolding *check-authentication-path-def* *protocol-check-authentication-path-def*
by (rule *hash-range-budget-protocol-check-authentication-path-raw*)

lemma *hash-collision-budget-protocol-check-authentication-path-raw*:

fixes *leaf* :: 'f protocol-hash-input
shows *hash-collision-budget* (Suc (length path))
 (protocol-merkle.check-authentication-path len i leaf path ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad)

proof (induction path arbitrary: len i)
case Nil
have *hash-budget*:
hash-collision-budget 1
 (hash leaf :: ('f, ('f, 'a) protocol-channel-scheme) state-monad)
by (rule *hash-collision-budget-hash*)
then show ?case
by (simp add: protocol-merkle.check-authentication-path.simps)

next

case (Cons a path)

show ?case

proof (cases i < len div 2)

case True

have *rec-range*:

hash-range-budget (Suc (length path))
 (protocol-merkle.check-authentication-path (len div 2) i leaf path ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad)
by (rule *hash-range-budget-protocol-check-authentication-path-raw*)

have *rec-coll*:

hash-collision-budget (Suc (length path))
 (protocol-merkle.check-authentication-path (len div 2) i leaf path ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad)

using Cons.IH[of len div 2 i] **by** simp

have *bound*:

hash-collision-budget (Suc (length path) + 1)
 ((protocol-merkle.check-authentication-path (len div 2) i leaf path ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad) $\gg$
 (λx . hash (MerkleNode x a)))

by (rule *hash-collision-budget-bind*
 [OF *rec-range* *rec-coll* *hash-range-budget-hash*
hash-collision-budget-hash])

show ?thesis

using True *bound*

by (simp add: protocol-merkle.check-authentication-path.simps)

next

```

case False
have rec-range:
  hash-range-budget (Suc (length path))
    (protocol-merkle.check-authentication-path
      (len div 2) (i - len div 2) leaf path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-protocol-check-authentication-path-raw)
have rec-coll:
  hash-collision-budget (Suc (length path))
    (protocol-merkle.check-authentication-path
      (len div 2) (i - len div 2) leaf path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  using Cons.IH[of len div 2 i - len div 2] by simp
have bound:
  hash-collision-budget (Suc (length path) + 1)
    ((protocol-merkle.check-authentication-path
      (len div 2) (i - len div 2) leaf path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
      ( $\lambda x.$  hash (MerkleNode a x)))
  by (rule hash-collision-budget-bind
    [OF rec-range rec-coll hash-range-budget-hash
      hash-collision-budget-hash])
show ?thesis
  using False bound
  by (simp add: protocol-merkle.check-authentication-path.simps)
qed
qed

lemma hash-collision-budget-check-authentication-path:
  hash-collision-budget (Suc (length path))
    (check-authentication-path len i v path)
unfolding check-authentication-path-def protocol-check-authentication-path-def
by (rule hash-collision-budget-protocol-check-authentication-path-raw)

lemma hash-range-budget-check-authentication-path-assert-return:
fixes path :: 'f list
and v root :: 'f
and r :: 'r
shows
  hash-range-budget (Suc (length path))
    (((check-authentication-path len i v path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
      ( $\lambda ap.$  assert (ap = root)  $\gg$  ( $\lambda -.$  return r))))
proof -
have check-budget:
  hash-range-budget (Suc (length path))
    (check-authentication-path len i v path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-check-authentication-path)

```

```

have assert-return:
  hash-range-budget ( $0 + 0$ )
    (assert ( $ap = root$ )  $\gg$  ( $\lambda\text{-}$ . return  $r$ ) ::
      ( $'r$ , ( $'f$ ,  $'a$ ) protocol-channel-scheme) state-monad) for  $ap$ 
  by (rule hash-range-budget-bind
    [OF hash-range-budget-assert hash-range-budget-return])
have hash-range-budget (Suc (length path) + ( $0 + 0$ ))
  (((check-authentication-path len i v path ::
    ( $'f$ , ( $'f$ ,  $'a$ ) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda ap$ . assert ( $ap = root$ )  $\gg$  ( $\lambda\text{-}$ . return  $r$ ))))
  by (rule hash-range-budget-bind [OF check-budget assert-return])
then show ?thesis by simp
qed

```

lemma *hash-collision-budget-check-authentication-path-assert-return*:

```

fixes  $path :: 'f$  list
and  $v$  root ::  $'f$ 
and  $r :: 'r$ 
shows
  hash-collision-budget (Suc (length path))
    (((check-authentication-path len i v path ::
      ( $'f$ , ( $'f$ ,  $'a$ ) protocol-channel-scheme) state-monad)  $\gg$ 
      ( $\lambda ap$ . assert ( $ap = root$ )  $\gg$  ( $\lambda\text{-}$ . return  $r$ ))))

```

proof –

```

have check-range:
  hash-range-budget (Suc (length path))
    (check-authentication-path len i v path ::
      ( $'f$ , ( $'f$ ,  $'a$ ) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-check-authentication-path)
have check-coll:
  hash-collision-budget (Suc (length path))
    (check-authentication-path len i v path ::
      ( $'f$ , ( $'f$ ,  $'a$ ) protocol-channel-scheme) state-monad)
  by (rule hash-collision-budget-check-authentication-path)
have assert-return-range:
  hash-range-budget ( $0 + 0$ )
    (assert ( $ap = root$ )  $\gg$  ( $\lambda\text{-}$ . return  $r$ ) ::
      ( $'r$ , ( $'f$ ,  $'a$ ) protocol-channel-scheme) state-monad) for  $ap$ 
  by (rule hash-range-budget-bind
    [OF hash-range-budget-assert hash-range-budget-return])
have assert-return-coll:
  hash-collision-budget ( $0 + 0$ )
    (assert ( $ap = root$ )  $\gg$  ( $\lambda\text{-}$ . return  $r$ ) ::
      ( $'r$ , ( $'f$ ,  $'a$ ) protocol-channel-scheme) state-monad) for  $ap$ 
  by (rule hash-collision-budget-bind
    [OF hash-range-budget-assert hash-collision-budget-assert
      hash-range-budget-return hash-collision-budget-return])
have hash-collision-budget (Suc (length path) + ( $0 + 0$ ))
  (((check-authentication-path len i v path ::

```

```

      ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λap. assert (ap = root) >>= (λ-. return r))))
  by (rule hash-collision-budget-bind
      [OF check-range check-coll assert-return-range assert-return-coll])
  then show ?thesis by simp
qed

```

lemma *hash-range-budget-check-authentication-path-assert-bind*:

```

  fixes path :: 'f list
  and v root :: 'f
  and k :: unit ⇒ ('r, ('f, 'a) protocol-channel-scheme) state-monad
  assumes k: ∧u. hash-range-budget n (k u)
  shows
    hash-range-budget (Suc (length path) + n)
      (((check-authentication-path len i v path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λap. assert (ap = root) >>= k))))
proof -
  have check-budget:
    hash-range-budget (Suc (length path))
      (check-authentication-path len i v path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-check-authentication-path)
  have assert-cont:
    hash-range-budget (0 + n)
      (assert (ap = root) >>= k ::
        ('r, ('f, 'a) protocol-channel-scheme) state-monad) for ap
  by (rule hash-range-budget-bind[OF hash-range-budget-assert k])
  have hash-range-budget (Suc (length path) + (0 + n))
    (((check-authentication-path len i v path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λap. assert (ap = root) >>= k))))
  by (rule hash-range-budget-bind[OF check-budget assert-cont])
  then show ?thesis by simp
qed

```

lemma *hash-collision-budget-check-authentication-path-assert-bind*:

```

  fixes path :: 'f list
  and v root :: 'f
  and k :: unit ⇒ ('r, ('f, 'a) protocol-channel-scheme) state-monad
  assumes range-k: ∧u. hash-range-budget n (k u)
  and coll-k: ∧u. hash-collision-budget n (k u)
  shows
    hash-collision-budget (Suc (length path) + n)
      (((check-authentication-path len i v path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λap. assert (ap = root) >>= k))))
proof -
  have check-range:

```

```

hash-range-budget (Suc (length path))
  (check-authentication-path len i v path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)
by (rule hash-range-budget-check-authentication-path)
have check-coll:
hash-collision-budget (Suc (length path))
  (check-authentication-path len i v path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)
by (rule hash-collision-budget-check-authentication-path)
have assert-range:
hash-range-budget (0 + n)
  (assert (ap = root)  $\gg$  k ::
    ('r, ('f, 'a) protocol-channel-scheme) state-monad) for ap
by (rule hash-range-budget-bind[OF hash-range-budget-assert range-k])
have assert-coll:
hash-collision-budget (0 + n)
  (assert (ap = root)  $\gg$  k ::
    ('r, ('f, 'a) protocol-channel-scheme) state-monad) for ap
by (rule hash-collision-budget-bind
  [OF hash-range-budget-assert hash-collision-budget-assert
    range-k coll-k])
have hash-collision-budget (Suc (length path) + (0 + n))
  (((check-authentication-path len i v path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda$ ap. assert (ap = root)  $\gg$  k)))
  by (rule hash-collision-budget-bind
    [OF check-range check-coll assert-range assert-coll])
then show ?thesis by simp
qed

```

```

lemma hash-range-budget-fri-layer-opening-finish:
  assumes xp-path-len: length xp-path = floor-log len
  and xn-path-len: length xn-path = floor-log len
  shows
    hash-range-budget (Suc (floor-log len) + Suc (floor-log len))
      (fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)
proof -
  let ?b = Suc (floor-log len)
  let ?sidx = (i + len div 2) mod len
  let ?out =
    (i mod (len div 2),
     (xp + xn) div 2 +
     b * ((xp - xn) div (2 * ((h ^ i) * shift) ^ pw)),
     len div 2, pw + pw)
  have second:
    hash-range-budget ?b
      (((check-authentication-path len ?sidx xn xn-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
        ( $\lambda$ ap. assert (ap = f)  $\gg$  ( $\lambda$ -. return ?out))))

```

```

by (subst xn-path-len[symmetric])
  (rule hash-range-budget-check-authentication-path-assert-return)
have second-cont:
   $\wedge u.$  hash-range-budget ?b
  (((check-authentication-path len ?sid xn xn-path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda ap.$  assert (ap = f)  $\gg$  ( $\lambda -.$  return ?out))))
  using second by simp
have first0:
  hash-range-budget (Suc (length xp-path) + ?b)
  (((check-authentication-path len i xp xp-path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda ap.$  assert (ap = f)  $\gg$ 
      ( $\lambda -.$ 
        (check-authentication-path len ?sid xn xn-path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
          ( $\lambda ap.$  assert (ap = f)  $\gg$  ( $\lambda -.$  return ?out)))))))
  by (rule hash-range-budget-check-authentication-path-assert-bind
    [OF second-cont])
have first:
  hash-range-budget (?b + ?b)
  (((check-authentication-path len i xp xp-path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda ap.$  assert (ap = f)  $\gg$ 
      ( $\lambda -.$ 
        (check-authentication-path len ?sid xn xn-path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
          ( $\lambda ap.$  assert (ap = f)  $\gg$  ( $\lambda -.$  return ?out)))))))
  using first0 xp-path-len by simp
have raw:
  hash-range-budget (0 + (?b + ?b))
  ((assert (xp = x) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda -.$ 
      (check-authentication-path len i xp xp-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
        ( $\lambda ap.$  assert (ap = f)  $\gg$ 
          ( $\lambda -.$ 
            (check-authentication-path len ?sid xn xn-path ::
              ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
              ( $\lambda ap.$  assert (ap = f)  $\gg$  ( $\lambda -.$  return ?out)))))))
  by (rule hash-range-budget-bind[OF hash-range-budget-assert first])
have all:
  hash-range-budget (?b + ?b)
  ((assert (xp = x) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda -.$ 
      (check-authentication-path len i xp xp-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
        ( $\lambda ap.$  assert (ap = f)  $\gg$  ( $\lambda -.$  return ?out))))

```

```

    (λap. assert (ap = f) >>=
      (λ-.
        (check-authentication-path len ?sidx xn xn-path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = f) >>= (λ-. return ?out))))))
  using raw by simp
from all show ?thesis
  unfolding fri-layer-opening-finish-def
  by (simp add: Let-def)
qed

lemma hash-collision-budget-fri-layer-opening-finish:
  assumes xp-path-len: length xp-path = floor-log len
  and xn-path-len: length xn-path = floor-log len
  shows
    hash-collision-budget (Suc (floor-log len) + Suc (floor-log len))
      (fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)
proof -
  let ?b = Suc (floor-log len)
  let ?sidx = (i + len div 2) mod len
  let ?out =
    (i mod (len div 2),
     (xp + xn) div 2 +
     b * ((xp - xn) div (2 * ((h ^ i) * shift) ^ pw)),
     len div 2, pw + pw)
  have second-range:
    hash-range-budget ?b
      (((check-authentication-path len ?sidx xn xn-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λap. assert (ap = f) >>= (λ-. return ?out))))
  by (subst xn-path-len[symmetric])
    (rule hash-range-budget-check-authentication-path-assert-return)
  have second-coll:
    hash-collision-budget ?b
      (((check-authentication-path len ?sidx xn xn-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λap. assert (ap = f) >>= (λ-. return ?out))))
  by (subst xn-path-len[symmetric])
    (rule hash-collision-budget-check-authentication-path-assert-return)
  have second-cont-range:
    ∧u. hash-range-budget ?b
      (((check-authentication-path len ?sidx xn xn-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λap. assert (ap = f) >>= (λ-. return ?out))))
  using second-range by simp
  have second-cont-coll:
    ∧u. hash-collision-budget ?b
      (((check-authentication-path len ?sidx xn xn-path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=

```

```

    (λap. assert (ap = f) >>= (λ-. return ?out))))
using second-coll by simp
have first-range0:
  hash-range-budget (Suc (length xp-path) + ?b)
  (((check-authentication-path len i xp xp-path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = f) >>=
      (λ-.
        (check-authentication-path len ?sidx xn xn-path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = f) >>= (λ-. return ?out)))))))
  by (rule hash-range-budget-check-authentication-path-assert-bind
    [OF second-cont-range])
have first-coll0:
  hash-collision-budget (Suc (length xp-path) + ?b)
  (((check-authentication-path len i xp xp-path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = f) >>=
      (λ-.
        (check-authentication-path len ?sidx xn xn-path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = f) >>= (λ-. return ?out)))))))
  by (rule hash-collision-budget-check-authentication-path-assert-bind
    [OF second-cont-range second-cont-coll])
have first-range:
  hash-range-budget (?b + ?b)
  (((check-authentication-path len i xp xp-path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = f) >>=
      (λ-.
        (check-authentication-path len ?sidx xn xn-path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = f) >>= (λ-. return ?out)))))))
  using first-range0 xp-path-len by simp
have first-coll:
  hash-collision-budget (?b + ?b)
  (((check-authentication-path len i xp xp-path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = f) >>=
      (λ-.
        (check-authentication-path len ?sidx xn xn-path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = f) >>= (λ-. return ?out)))))))
  using first-coll0 xp-path-len by simp
have raw:
  hash-collision-budget (0 + (?b + ?b))
  ((assert (xp = x) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λ-.

```



```

    (check-authentication-path len i xp xp-path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = f) >>=
      (λ-.
        (check-authentication-path len ?sidx xn xn-path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = f) >>= (λ-. return ?out))))))
  by (rule hash-collision-budget-bind
    [OF hash-range-budget-assert hash-collision-budget-assert
      first-range first-coll])
  then show ?thesis
    unfolding fri-layer-opening-finish-def
    by (simp add: Let-def)
qed

lemma hash-range-budget-fri-layer-opening-step:
  hash-range-budget (Suc (floor-log len) + Suc (floor-log len))
  (fri-layer-opening-step (b, f) (i, x, len, pw))
proof -
  let ?B = Suc (floor-log len) + Suc (floor-log len)
  have after-xn-path:
    hash-range-budget ?B
    ((ntimes read (floor-log len) ::
      ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λxn-path.
        fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)))
  if xp-path-len: length xp-path = floor-log len
  for xp xp-path xn
proof (rule hash-range-budget-ntimes-read-bind)
  fix xn-path :: 'f list
  assume xn-path-len: length xn-path = floor-log len
  show hash-range-budget ?B
    (fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)
  by (rule hash-range-budget-fri-layer-opening-finish
    [OF xp-path-len xn-path-len])
qed
have after-xn:
  hash-range-budget ?B
  (read >>=
    (λxn.
      (ntimes read (floor-log len) ::
        ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λxn-path.
          fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
  if xp-path-len: length xp-path = floor-log len
  for xp xp-path
proof -
  have hash-range-budget (0 + ?B)
    (read >>=

```

```

(λxn.
  (ntimes read (floor-log len) ::
    ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
  (λxn-path.
    fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)))
by (rule hash-range-budget-bind
  [OF hash-range-budget-read after-xn-path[OF xp-path-len]])
then show ?thesis by simp
qed
have after-xp-path:
  hash-range-budget ?B
  ((ntimes read (floor-log len) ::
    ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
  (λxp-path.
    read >>=
    (λxn.
      (ntimes read (floor-log len) ::
        ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λxn-path.
          fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))))
for xp
proof (rule hash-range-budget-ntimes-read-bind)
fix xp-path :: 'f list
assume xp-path-len: length xp-path = floor-log len
show hash-range-budget ?B
  (read >>=
  (λxn.
    (ntimes read (floor-log len) ::
      ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λxn-path.
        fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
by (rule after-xn[OF xp-path-len])
qed
have hash-range-budget (0 + ?B)
  (read >>=
  (λxp.
    (ntimes read (floor-log len) ::
      ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λxp-path.
        read >>=
        (λxn.
          (ntimes read (floor-log len) ::
            ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
            (λxn-path.
              fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))))
by (rule hash-range-budget-bind
  [OF hash-range-budget-read after-xp-path])
then show ?thesis
unfolding fri-layer-opening-step-unfold-finish by simp

```

qed

lemma *hash-collision-budget-fri-layer-opening-step:*

hash-collision-budget (Suc (floor-log len) + Suc (floor-log len))
(fri-layer-opening-step (b, f) (i, x, len, pw))

proof –

let *?B = Suc (floor-log len) + Suc (floor-log len)*

have *after-xn-path-range:*

hash-range-budget ?B

((ntimes read (floor-log len) ::
(f list, (f, 'a) protocol-channel-scheme) state-monad) >>=
(λxn-path.

fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))

if *xp-path-len: length xp-path = floor-log len*

for *xp xp-path xn*

proof (*rule hash-range-budget-ntimes-read-bind*)

fix *xn-path :: f list*

assume *xn-path-len: length xn-path = floor-log len*

show *hash-range-budget ?B*

(fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)

by (*rule hash-range-budget-fri-layer-opening-finish*
[OF xp-path-len xn-path-len])

qed

have *after-xn-path-coll:*

hash-collision-budget ?B

((ntimes read (floor-log len) ::
(f list, (f, 'a) protocol-channel-scheme) state-monad) >>=
(λxn-path.

fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))

if *xp-path-len: length xp-path = floor-log len*

for *xp xp-path xn*

proof (*rule hash-collision-budget-ntimes-read-bind*)

fix *xn-path :: f list*

assume *xn-path-len: length xn-path = floor-log len*

show *hash-range-budget ?B*

(fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)

by (*rule hash-range-budget-fri-layer-opening-finish*
[OF xp-path-len xn-path-len])

show *hash-collision-budget ?B*

(fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)

by (*rule hash-collision-budget-fri-layer-opening-finish*
[OF xp-path-len xn-path-len])

qed

have *after-xn-range:*

hash-range-budget ?B

(read >>=

(λxn.

(ntimes read (floor-log len) ::

(f list, (f, 'a) protocol-channel-scheme) state-monad) >>=

```

      (λxn-path.
        fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path)))
  if xp-path-len: length xp-path = floor-log len
  for xp xp-path
  proof -
    have hash-range-budget (0 + ?B)
      (read >>=
        (λxn.
          (ntimes read (floor-log len) ::
            ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
            (λxn-path.
              fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
    by (rule hash-range-budget-bind
      [OF hash-range-budget-read after-xn-path-range[OF xp-path-len]])
    then show ?thesis by simp
  qed
  have after-xn-coll:
    hash-collision-budget ?B
      (read >>=
        (λxn.
          (ntimes read (floor-log len) ::
            ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
            (λxn-path.
              fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
    if xp-path-len: length xp-path = floor-log len
    for xp xp-path
  proof -
    have hash-collision-budget (0 + ?B)
      (read >>=
        (λxn.
          (ntimes read (floor-log len) ::
            ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
            (λxn-path.
              fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
    by (rule hash-collision-budget-bind
      [OF hash-range-budget-read hash-collision-budget-read
        after-xn-path-range[OF xp-path-len]
        after-xn-path-coll[OF xp-path-len]])
    then show ?thesis by simp
  qed
  have after-xp-path-range:
    hash-range-budget ?B
      ((ntimes read (floor-log len) ::
        ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λxp-path.
          read >>=
            (λxn.
              (ntimes read (floor-log len) ::
                ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=

```

```

      (λxn-path.
        fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
  for xp
  proof (rule hash-range-budget-ntimes-read-bind)
    fix xp-path :: 'f list
    assume xp-path-len: length xp-path = floor-log len
    show hash-range-budget ?B
      (read >=>
        (λxn.
          (ntimes read (floor-log len) ::
            ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >=>
            (λxn-path.
              fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
    by (rule after-xn-range[OF xp-path-len])
  qed
  have after-xp-path-coll:
    hash-collision-budget ?B
      ((ntimes read (floor-log len) ::
        ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >=>
        (λxp-path.
          read >=>
            (λxn.
              (ntimes read (floor-log len) ::
                ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >=>
                (λxn-path.
                  fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))))
  for xp
  proof (rule hash-collision-budget-ntimes-read-bind)
    fix xp-path :: 'f list
    assume xp-path-len: length xp-path = floor-log len
    show hash-range-budget ?B
      (read >=>
        (λxn.
          (ntimes read (floor-log len) ::
            ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >=>
            (λxn-path.
              fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
    by (rule after-xn-range[OF xp-path-len])
  show hash-collision-budget ?B
    (read >=>
      (λxn.
        (ntimes read (floor-log len) ::
          ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >=>
          (λxn-path.
            fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))
    by (rule after-xn-coll[OF xp-path-len])
  qed
  have hash-collision-budget (0 + ?B)
    (read >=>

```

```

( $\lambda xp.$ 
  ( $ntimes$  read ( $floor\text{-}log$  len) ::
    ( $'f$  list, ( $'f$ ,  $'a$ ) protocol-channel-scheme) state-monad)  $\gg$ 
  ( $\lambda xp\text{-}path.$ 
    read  $\gg$ 
    ( $\lambda xn.$ 
      ( $ntimes$  read ( $floor\text{-}log$  len) ::
        ( $'f$  list, ( $'f$ ,  $'a$ ) protocol-channel-scheme) state-monad)  $\gg$ 
        ( $\lambda xn\text{-}path.$ 
          fri-layer-opening-finish b f i x len pw xp xp-path xn xn-path))))))
by (rule hash-collision-budget-bind
  [OF hash-range-budget-read hash-collision-budget-read
    after-xp-path-range after-xp-path-coll])
then show ?thesis
  unfolding fri-layer-opening-step-unfold-finish by simp
qed

```

lemma floor-log-div2-le-self:
 floor-log ($n \text{ div } 2$) $\leq$ floor-log n
 by (rule floor-log-le-iff) simp

lemma receive-query-commits-eq-fri-layer-opening-steps:
 receive-query-commits fl = map fri-layer-opening-step fl
 unfolding receive-query-commits-def fri-layer-opening-step-def
 by (simp add: fun-eq-iff split-def)

lemma hash-target-program-fri-layer-opening-steps-mfold-state:
 fixes st :: $\text{nat} \times 'f \times \text{nat} \times \text{nat}$
 assumes init:
 case st of (i, x, len, pw) $\Rightarrow$ floor-log len $\leq L$
 shows
 hash-target-program B
 (length fl * (Suc L + Suc L))
 (mfold st (map fri-layer-opening-step fl) ::
 ($\text{nat} \times 'f \times \text{nat} \times \text{nat}$,
 ($'f$, $'a$) protocol-channel-scheme) state-monad)
 using init
proof (induction fl arbitrary: st)
 case Nil
 then show ?case
 by (simp add: hash-target-program-return)
next
 case (Cons bf fl)
 obtain i x len pw where st-eq: st = (i, x, len, pw)
 by (cases st)
 obtain b f where bf-eq: bf = (b, f)
 by (cases bf)
 let ?N = Suc L + Suc L
 have head-exact:

```

hash-target-program B
  (Suc (floor-log len) + Suc (floor-log len))
  (fri-layer-opening-step (b, f) (i, x, len, pw))
by (rule hash-target-program-fri-layer-opening-step)
have head-le:
  Suc (floor-log len) + Suc (floor-log len) ≤ ?N
  using Cons.premis unfolding st-eq by simp
have head:
  hash-target-program B ?N
  (fri-layer-opening-step bf st ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding bf-eq st-eq
  by (rule hash-target-program-mono[OF head-le head-exact])
have tail:
  hash-target-program B (length fl * ?N)
  (mfold z (map fri-layer-opening-step fl) ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad)
  if out:
    Some (z, t) ∈
      set-dist (execute (fri-layer-opening-step bf st) s)
  for z and s :: ('f, 'a) protocol-channel-scheme
  and t :: ('f, 'a) protocol-channel-scheme
proof -
  from fri-layer-opening-step-outcome[OF out[unfolded bf-eq st-eq]]
  obtain xp xp-path xn xn-path x' where
    z-eq: z = (i mod (len div 2), x', len div 2, pw + pw)
  by blast
  have len-le: floor-log len ≤ L
    using Cons.premis unfolding st-eq by simp
  have next-len: floor-log (len div 2) ≤ L
    using floor-log-div2-le-self[of len] len-le by linarith
  have z-inv:
    case z of (i, x, len, pw) ⇒ floor-log len ≤ L
    unfolding z-eq using next-len by simp
  show ?thesis
    by (rule Cons.IH[OF z-inv])
qed
have hash-target-program B (?N + length fl * ?N)
  ((fri-layer-opening-step bf st ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad) ≫=
    (λz. mfold z (map fri-layer-opening-step fl)))
  by (rule hash-target-program-bind-on-outcomes[OF head]) (rule tail)
then show ?case
  by (simp add: add.assoc add.commute add.left-commute)
qed

```

lemma *hash-target-program-receive-query-commits-mfold:*
assumes *init:* $\text{floor-log } \text{len} \leq L$
shows
 $\text{hash-target-program } B$
 $(\text{length } \text{fl} * (\text{Suc } L + \text{Suc } L))$
 $(\text{mfold } (i, x, \text{len}, \text{pw}) (\text{receive-query-commits } \text{fl}))$
unfolding *receive-query-commits-eq-fri-layer-opening-steps*
proof –
have *st-inv:*
 $\text{case } (i, x, \text{len}, \text{pw}) \text{ of}$
 $(i, x, \text{len}, \text{pw}) \Rightarrow \text{floor-log } \text{len} \leq L$
using *init* **by** *simp*
show $\text{hash-target-program } B (\text{length } \text{fl} * (\text{Suc } L + \text{Suc } L))$
 $(\text{mfold } (i, x, \text{len}, \text{pw}) (\text{map } \text{fri-layer-opening-step } \text{fl})) ::$
 $(\text{nat} \times 'f \times \text{nat} \times \text{nat},$
 $(f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
by (*rule hash-target-program-fri-layer-opening-steps-mfold-state*
 $[\text{OF } \text{st-inv}]$)
qed

lemma *hash-range-budget-fri-layer-opening-steps-mfold-state:*
fixes *st* :: $\text{nat} \times 'f \times \text{nat} \times \text{nat}$
assumes *init:*
 $\text{case } \text{st} \text{ of } (i, x, \text{len}, \text{pw}) \Rightarrow \text{floor-log } \text{len} \leq L$
shows
 hash-range-budget
 $(\text{length } \text{fl} * (\text{Suc } L + \text{Suc } L))$
 $(\text{mfold } \text{st} (\text{map } \text{fri-layer-opening-step } \text{fl})) ::$
 $(\text{nat} \times 'f \times \text{nat} \times \text{nat},$
 $(f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
using *init*
proof (*induction fl arbitrary: st*)
case *Nil*
then show *?case*
by (*simp add: hash-range-budget-return*)
next
case (*Cons bf fl*)
obtain *i x len pw* **where** *st-eq:* $\text{st} = (i, x, \text{len}, \text{pw})$
by (*cases st*)
obtain *b f* **where** *bf-eq:* $\text{bf} = (b, f)$
by (*cases bf*)
let $?B = \text{Suc } L + \text{Suc } L$
have *head-exact:*
 $\text{hash-range-budget } (\text{Suc } (\text{floor-log } \text{len}) + \text{Suc } (\text{floor-log } \text{len}))$
 $(\text{fri-layer-opening-step } (b, f) (i, x, \text{len}, \text{pw}))$
by (*rule hash-range-budget-fri-layer-opening-step*)
have *head-le:*
 $\text{Suc } (\text{floor-log } \text{len}) + \text{Suc } (\text{floor-log } \text{len}) \leq ?B$
using *Cons.prem*s **unfolding** *st-eq* **by** *simp*


```

have head:
  hash-range-budget ?B
  (fri-layer-opening-step bf st ::
    (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding bf-eq st-eq
  by (rule hash-range-budget-mono[OF head-le head-exact])
have tail:
  hash-range-budget (length fl * ?B)
  (mfold z (map fri-layer-opening-step fl) ::
    (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad)
  if out:
    Some (z, t) ∈
      set-dist (execute (fri-layer-opening-step bf st) s)
  for z and s :: ('f, 'a) protocol-channel-scheme
  and t :: ('f, 'a) protocol-channel-scheme
proof –
  from fri-layer-opening-step-outcome[OF out[unfolded bf-eq st-eq]]
  obtain xp xp-path xn xn-path x' where
    z-eq: z = (i mod (len div 2), x', len div 2, pw + pw)
    by blast
  have len-le: floor-log len ≤ L
    using Cons.prems unfolding st-eq by simp
  have next-len: floor-log (len div 2) ≤ L
    using floor-log-div2-le-self[of len] len-le by linarith
  have z-inv:
    case z of (i, x, len, pw) ⇒ floor-log len ≤ L
    unfolding z-eq using next-len by simp
  have ih-z:
    hash-range-budget (length fl * ?B)
    (mfold z (map fri-layer-opening-step fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad)
    using Cons.IH[OF z-inv] by assumption
  show ?thesis
    by (rule ih-z)
qed
have hash-range-budget (?B + length fl * ?B)
  ((fri-layer-opening-step bf st ::
    (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad) ≫=
    (λz. mfold z (map fri-layer-opening-step fl)))
  by (rule hash-range-budget-bind-on-outcomes[OF head]) (rule tail)
then show ?case
  by (simp add: add.assoc add.commute add.left-commute)
qed

```

lemma *hash-range-budget-receive-query-commits-mfold*:

assumes *init*: $\text{floor-log len} \leq L$
shows
hash-range-budget
 $(\text{length fl} * (\text{Suc } L + \text{Suc } L))$
 $(\text{mfold } (i, x, \text{len}, \text{pw}) (\text{receive-query-commits fl}))$
unfolding *receive-query-commits-eq-fri-layer-opening-steps*
proof –
have *st-inv*:
case $(i, x, \text{len}, \text{pw})$ *of*
 $(i, x, \text{len}, \text{pw}) \Rightarrow \text{floor-log len} \leq L$
using *init* **by** *simp*
show *hash-range-budget* $(\text{length fl} * (\text{Suc } L + \text{Suc } L))$
 $(\text{mfold } (i, x, \text{len}, \text{pw}) (\text{map fri-layer-opening-step fl})) ::$
 $(\text{nat} \times 'f \times \text{nat} \times \text{nat},$
 $(f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
by $(\text{rule hash-range-budget-fri-layer-opening-steps-mfold-state}$
 $[OF \text{ st-inv}])$
qed

lemma *hash-collision-budget-fri-layer-opening-steps-mfold-state*:
fixes $st :: \text{nat} \times 'f \times \text{nat} \times \text{nat}$
assumes *init*:
case st *of* $(i, x, \text{len}, \text{pw}) \Rightarrow \text{floor-log len} \leq L$
shows
hash-collision-budget
 $(\text{length fl} * (\text{Suc } L + \text{Suc } L))$
 $(\text{mfold } st (\text{map fri-layer-opening-step fl})) ::$
 $(\text{nat} \times 'f \times \text{nat} \times \text{nat},$
 $(f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}$
using *init*
proof $(\text{induction fl arbitrary: st})$
case *Nil*
then show *?case*
by $(\text{simp add: hash-collision-budget-return})$
next
case $(\text{Cons } bf \text{ fl})$
obtain $i \ x \ \text{len} \ \text{pw}$ **where** $st\text{-eq}: st = (i, x, \text{len}, \text{pw})$
by $(\text{cases } st)$
obtain $b \ f$ **where** $bf\text{-eq}: bf = (b, f)$
by $(\text{cases } bf)$
let $?B = \text{Suc } L + \text{Suc } L$
have *head-range-exact*:
 $\text{hash-range-budget } (\text{Suc } (\text{floor-log len}) + \text{Suc } (\text{floor-log len}))$
 $(\text{fri-layer-opening-step } (b, f) (i, x, \text{len}, \text{pw}))$
by $(\text{rule hash-range-budget-fri-layer-opening-step})$
have *head-coll-exact*:
 $\text{hash-collision-budget } (\text{Suc } (\text{floor-log len}) + \text{Suc } (\text{floor-log len}))$
 $(\text{fri-layer-opening-step } (b, f) (i, x, \text{len}, \text{pw}))$
by $(\text{rule hash-collision-budget-fri-layer-opening-step})$

```

have head-le:
  Suc (floor-log len) + Suc (floor-log len) ≤ ?B
  using Cons.premis unfolding st-eq by simp
have head-range:
  hash-range-budget ?B
  (fri-layer-opening-step bf st ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding bf-eq st-eq
  by (rule hash-range-budget-mono[OF head-le head-range-exact])
have head-coll:
  hash-collision-budget ?B
  (fri-layer-opening-step bf st ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding bf-eq st-eq
  by (rule hash-collision-budget-mono[OF head-le head-coll-exact])
have tail-range:
  hash-range-budget (length fl * ?B)
  (mfold z (map fri-layer-opening-step fl) ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad)
  if out:
    Some (z, t) ∈
      set-dist (execute (fri-layer-opening-step bf st) s)
  for z and s :: ('f, 'a) protocol-channel-scheme
  and t :: ('f, 'a) protocol-channel-scheme
proof –
  from fri-layer-opening-step-outcome[OF out[unfolded bf-eq st-eq]]
  obtain xp xp-path xn xn-path x' where
    z-eq: z = (i mod (len div 2), x', len div 2, pw + pw)
    by blast
  have len-le: floor-log len ≤ L
    using Cons.premis unfolding st-eq by simp
  have next-len: floor-log (len div 2) ≤ L
    using floor-log-div2-le-self[of len] len-le by linarith
  have z-inv:
    case z of (i, x, len, pw) ⇒ floor-log len ≤ L
    unfolding z-eq using next-len by simp
  show ?thesis
    by (rule hash-range-budget-fri-layer-opening-steps-mfold-state
      [OF z-inv])
qed
have tail-coll:
  hash-collision-budget (length fl * ?B)
  (mfold z (map fri-layer-opening-step fl) ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad)
  if out:

```

```

    Some (z, t) ∈
      set-dist (execute (fri-layer-opening-step bf st) s)
  for z and s :: ('f, 'a) protocol-channel-scheme
    and t :: ('f, 'a) protocol-channel-scheme
proof -
  from fri-layer-opening-step-outcome[OF out[unfolded bf-eq st-eq]]
  obtain xp xp-path xn xn-path x' where
    z-eq: z = (i mod (len div 2), x', len div 2, pw + pw)
    by blast
  have len-le: floor-log len ≤ L
    using Cons.premis unfolding st-eq by simp
  have next-len: floor-log (len div 2) ≤ L
    using floor-log-div2-le-self[of len] len-le by linarith
  have z-inv:
    case z of (i, x, len, pw) ⇒ floor-log len ≤ L
    unfolding z-eq using next-len by simp
  have ih-z:
    hash-collision-budget (length fl * ?B)
    (mfold z (map fri-layer-opening-step fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad)
    using Cons.IH[OF z-inv] by assumption
  show ?thesis
    by (rule ih-z)
qed
have hash-collision-budget (?B + length fl * ?B)
  ((fri-layer-opening-step bf st ::
    (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad) ≫
    (λz. mfold z (map fri-layer-opening-step fl)))
  by (rule hash-collision-budget-bind-on-outcomes
    [OF head-range head-coll tail-range tail-coll])
then show ?case
  by (simp add: add.assoc add.commute add.left-commute)
qed

lemma hash-collision-budget-receive-query-commits-mfold:
  assumes init: floor-log len ≤ L
  shows
    hash-collision-budget
      (length fl * (Suc L + Suc L))
      (mfold (i, x, len, pw) (receive-query-commits fl))
  unfolding receive-query-commits-eq-fri-layer-opening-steps
proof -
  have st-inv:
    case (i, x, len, pw) of
      (i, x, len, pw) ⇒ floor-log len ≤ L
    using init by simp
  show hash-collision-budget (length fl * (Suc L + Suc L))

```

```

(mfold (i, x, len, pw) (map fri-layer-opening-step fl) ::
  (nat × 'f × nat × nat,
   ('f, 'a) protocol-channel-scheme) state-monad)
by (rule hash-collision-budget-fri-layer-opening-steps-mfold-state
    [OF st-inv])
qed

lemma hash-target-program-query-decommitment-step:
  hash-target-program B (Suc (floor-log (scale * clength)))
  (query-decommitment-step fr i)
proof -
  let ?len = scale * clength
  have after-path:
    hash-target-program B (Suc (floor-log ?len))
    ((ntimes read (floor-log ?len) ::
      ('f list, ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λpath.
        (check-authentication-path ?len i qh path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) ≫=
          (λap. assert (ap = fr) ≫= (λ-. return qh))))))
  for qh
proof (rule hash-target-program-ntimes-read-bind)
  fix path :: 'f list
  assume path-len: length path = floor-log ?len
  have hash-target-program B (Suc (length path))
    ((check-authentication-path ?len i qh path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λap. assert (ap = fr) ≫= (λ-. return qh)))
  by (rule hash-target-program-check-authentication-path-assert-return)
  then show hash-target-program B (Suc (floor-log ?len))
    ((check-authentication-path ?len i qh path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λap. assert (ap = fr) ≫= (λ-. return qh)))
  by (subst path-len[symmetric])
qed
have hash-target-program B (0 + Suc (floor-log ?len))
  (read ≫=
   (λqh. (ntimes read (floor-log ?len) ::
     ('f list, ('f, 'a) protocol-channel-scheme) state-monad) ≫=
     (λpath.
       (check-authentication-path ?len i qh path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad) ≫=
        (λap. assert (ap = fr) ≫= (λ-. return qh))))))
  by (rule hash-target-program-bind
      [OF hash-target-program-read after-path])
then show ?thesis
  unfolding query-decommitment-step-def by (simp add: Let-def)
qed

```

```

lemma hash-target-program-check-decommit-on-query:
  hash-target-program B
    (powers * Suc (floor-log (scale * clength)))
    (mmap (check-decommit-on-query fr idx))
proof –
  let ?b = Suc (floor-log (scale * clength))
  let ?steps =
    (map (query-decommitment-step fr) (powers-scaled idx) ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad list)
  have step-budget:
     $\wedge (m :: ('f, ('f, 'a) protocol-channel-scheme) state-monad).$ 
     $m \in \text{set } ?steps \implies$ 
    hash-target-program B ?b m
  proof –
    fix m :: ('f, ('f, 'a) protocol-channel-scheme) state-monad
    assume m  $\in$  set ?steps
    then obtain i where i  $\in$  set (powers-scaled idx)
      and m-eq: m = query-decommitment-step fr i
      by auto
    show hash-target-program B ?b m
      unfolding m-eq by (rule hash-target-program-query-decommitment-step)
  qed
  have map-budget:
    hash-target-program B (length ?steps * ?b) (mmap ?steps)
    by (rule hash-target-program-mmap) (rule step-budget)
  then have hash-target-program B
    (length (powers-scaled idx) * ?b) (mmap ?steps)
    by simp
  then show ?thesis
    unfolding check-decommit-on-query-def query-decommitment-step-def
      powers-scaled-def
    by simp
qed

lemma hash-range-budget-query-decommitment-step:
  hash-range-budget (Suc (floor-log (scale * clength)))
    (query-decommitment-step fr i)
proof –
  let ?len = scale * clength
  have after-path:
    hash-range-budget (Suc (floor-log ?len))
      ((ntimes read (floor-log ?len) ::
        ('f list, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg=$ 
        ( $\lambda$ path.
          (check-authentication-path ?len i qh path ::
            ('f, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg=$ 
            ( $\lambda$ ap. assert (ap = fr)  $\gg=$  ( $\lambda$ -. return qh)))) for qh
  proof (rule hash-range-budget-ntimes-read-bind)
    fix path :: 'f list

```

```

assume path-len: length path = floor-log ?len
have hash-range-budget (Suc (length path))
  (((check-authentication-path ?len i qh path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = fr) >>= (λ-. return qh))))
  by (rule hash-range-budget-check-authentication-path-assert-return)
note path-budget = this
show hash-range-budget (Suc (floor-log ?len))
  ((check-authentication-path ?len i qh path ::
    ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λap. assert (ap = fr) >>= (λ-. return qh)))
  by (subst path-len[symmetric]) (rule path-budget)
qed
have hash-range-budget (0 + Suc (floor-log ?len))
  (read >>=
    (λqh. (ntimes read (floor-log ?len) ::
      ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λpath.
        (check-authentication-path ?len i qh path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = fr) >>= (λ-. return qh)))))
  by (rule hash-range-budget-bind[OF hash-range-budget-read after-path])
then show ?thesis
  unfolding query-decommitment-step-def by (simp add: Let-def)
qed

```

```

lemma hash-collision-budget-query-decommitment-step:
  hash-collision-budget (Suc (floor-log (scale * clength)))
  (query-decommitment-step fr i)
proof –
  let ?len = scale * clength
  have after-path-range:
    hash-range-budget (Suc (floor-log ?len))
    ((ntimes read (floor-log ?len) ::
      ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λpath.
        (check-authentication-path ?len i qh path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = fr) >>= (λ-. return qh))))) for qh
  proof (rule hash-range-budget-ntimes-read-bind)
  fix path :: 'f list
  assume path-len: length path = floor-log ?len
  have hash-range-budget (Suc (length path))
    (((check-authentication-path ?len i qh path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λap. assert (ap = fr) >>= (λ-. return qh))))
    by (rule hash-range-budget-check-authentication-path-assert-return)
  note path-budget = this
  show hash-range-budget (Suc (floor-log ?len))

```

```

    ((check-authentication-path ?len i qh path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λap. assert (ap = fr) >>= (λ-. return qh)))
  by (subst path-len[symmetric]) (rule path-budget)
qed
have after-path-coll:
  hash-collision-budget (Suc (floor-log ?len))
  ((ntimes read (floor-log ?len) ::
    ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λpath.
      (check-authentication-path ?len i qh path ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λap. assert (ap = fr) >>= (λ-. return qh)))) for qh
proof (rule hash-collision-budget-ntimes-read-bind)
  fix path :: 'f list
  assume path-len: length path = floor-log ?len
  have range-bind:
    hash-range-budget (Suc (length path))
    (((check-authentication-path ?len i qh path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λap. assert (ap = fr) >>= (λ-. return qh))))
    by (rule hash-range-budget-check-authentication-path-assert-return)
  show hash-range-budget (Suc (floor-log ?len))
    ((check-authentication-path ?len i qh path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λap. assert (ap = fr) >>= (λ-. return qh)))
    by (subst path-len[symmetric]) (rule range-bind)
  have hash-collision-budget (Suc (length path))
    (((check-authentication-path ?len i qh path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λap. assert (ap = fr) >>= (λ-. return qh))))
    by (rule hash-collision-budget-check-authentication-path-assert-return)
  note path-budget = this
  show hash-collision-budget (Suc (floor-log ?len))
    ((check-authentication-path ?len i qh path ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λap. assert (ap = fr) >>= (λ-. return qh)))
    by (subst path-len[symmetric]) (rule path-budget)
qed
have hash-collision-budget (0 + Suc (floor-log ?len))
  (read >>=
    (λqh. (ntimes read (floor-log ?len) ::
      ('f list, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λpath.
        (check-authentication-path ?len i qh path ::
          ('f, ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λap. assert (ap = fr) >>= (λ-. return qh))))))
  by (rule hash-collision-budget-bind
    [OF hash-range-budget-read hash-collision-budget-read

```



```

      after-path-range after-path-coll])
  then show ?thesis
    unfolding query-decommitment-step-def by (simp add: Let-def)
qed

lemma hash-range-budget-check-decommit-on-query:
  hash-range-budget
    (powers * Suc (floor-log (scale * clength)))
    (mmap (check-decommit-on-query fr idx))
proof -
  let ?b = Suc (floor-log (scale * clength))
  let ?steps =
    (map (query-decommitment-step fr) (powers-scaled idx) ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad list)
  have step-budget:
     $\bigwedge (m :: ('f, ('f, 'a) protocol-channel-scheme) state-monad).$ 
     $m \in \text{set } ?steps \implies$ 
    hash-range-budget ?b m
  proof -
    fix m :: ('f, ('f, 'a) protocol-channel-scheme) state-monad
    assume m  $\in$  set ?steps
    then obtain i where i  $\in$  set (powers-scaled idx)
      and m-eq: m = query-decommitment-step fr i
      by auto
    show hash-range-budget ?b m
      unfolding m-eq by (rule hash-range-budget-query-decommitment-step)
  qed
  have map-budget:
    hash-range-budget
      (length ?steps * ?b) (mmap ?steps)
    by (rule hash-range-budget-mmap) (rule step-budget)
  then have hash-range-budget (length (powers-scaled idx) * ?b)
    (mmap ?steps)
    by simp
  then show ?thesis
    unfolding check-decommit-on-query-def query-decommitment-step-def
      powers-scaled-def
    by simp
qed

lemma hash-collision-budget-check-decommit-on-query:
  hash-collision-budget
    (powers * Suc (floor-log (scale * clength)))
    (mmap (check-decommit-on-query fr idx))
proof -
  let ?b = Suc (floor-log (scale * clength))
  let ?steps =
    (map (query-decommitment-step fr) (powers-scaled idx) ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad list)

```

```

have step-range:
   $\wedge(m :: ('f, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}).$ 
   $m \in \text{set } ?\text{steps} \implies$ 
  hash-range-budget ?b m
proof -
  fix m :: ('f, ('f, 'a) protocol-channel-scheme) state-monad
  assume  $m \in \text{set } ?\text{steps}$ 
  then obtain i where  $i \in \text{set } (\text{powers-scaled } \text{id}x)$ 
  and m-eq:  $m = \text{query-decommitment-step } \text{fr } i$ 
  by auto
  show hash-range-budget ?b m
  unfolding m-eq by (rule hash-range-budget-query-decommitment-step)
qed
have step-coll:
   $\wedge(m :: ('f, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}).$ 
   $m \in \text{set } ?\text{steps} \implies$ 
  hash-collision-budget ?b m
proof -
  fix m :: ('f, ('f, 'a) protocol-channel-scheme) state-monad
  assume  $m \in \text{set } ?\text{steps}$ 
  then obtain i where  $i \in \text{set } (\text{powers-scaled } \text{id}x)$ 
  and m-eq:  $m = \text{query-decommitment-step } \text{fr } i$ 
  by auto
  show hash-collision-budget ?b m
  unfolding m-eq by (rule hash-collision-budget-query-decommitment-step)
qed
have map-budget:
  hash-collision-budget
  (length ?steps * ?b) (mmap ?steps)
  apply (rule hash-collision-budget-mmap)
  apply (rule step-range, assumption)
  apply (rule step-coll, assumption)
  done
then have hash-collision-budget (length (powers-scaled idx) * ?b)
  (mmap ?steps)
  by simp
then show ?thesis
  unfolding check-decommit-on-query-def query-decommitment-step-def
  powers-scaled-def
  by simp
qed

end

end

```

```

theory Soundness-Oracle-Verifier
imports Soundness-Oracle-Budgets

```

begin

Concrete verifier hash-query budget bounds.

context *soundness*

begin

lemma *ceil-log-mono*:

assumes $m \leq n$

shows $\text{ceil-log } m \leq \text{ceil-log } n$

proof (*cases* $m \leq 1$)

case *True*

then show *?thesis*

unfolding *ceil-log-def* **by** *simp*

next

case *False*

then have *m-pos*: $0 < m$ **and** *m-minus*: $m - 1 \leq n - 1$

using *assms* **by** *simp-all*

have *n-gt*: $1 < n$

using *False assms* **by** *linarith*

have $\text{floor-log } (m - 1) \leq \text{floor-log } (n - 1)$

using *floor-log-le-iff m-minus* **by** *blast*

then show *?thesis*

using *False n-gt* **unfolding** *ceil-log-def* **by** *simp*

qed

definition *verifier-fri-layer-hash-budget* :: *nat*

where *verifier-fri-layer-hash-budget* =

Suc ($\text{floor-log } (\text{clength} * \text{scale})$) + *Suc* ($\text{floor-log } (\text{clength} * \text{scale})$)

definition *verifier-query-decommit-hash-budget* :: *nat*

where *verifier-query-decommit-hash-budget* =

powers * *Suc* ($\text{floor-log } (\text{clength} * \text{scale})$)

definition *verifier-query-round-hash-budget* :: *nat*

where *verifier-query-round-hash-budget* =

$1 + \text{verifier-query-decommit-hash-budget} +$
 $(\text{ceil-log } \text{clength} + \text{ceil-log } (\text{maxDegree} + 1)) * \text{verifier-fri-layer-hash-budget}$

definition *verifier-header-hash-budget* :: *nat*

where *verifier-header-hash-budget* =

$\text{ceil-log } \text{clength} + \text{length spec} + \text{ceil-log } (\text{maxDegree} + 1)$

definition *verifier-hash-query-budget* :: *nat*

where *verifier-hash-query-budget* =

verifier-header-hash-budget + *rounds* * *verifier-query-round-hash-budget*

definition *concrete-merkle-binding-error* :: *prob*

where

concrete-merkle-binding-error =
hash-collision-budget-value 0 *verifier-hash-query-budget*

definition *concrete-transcript-target-error* :: 'f list $\Rightarrow$ prob
where
concrete-transcript-target-error tr =
 nnreal (card (set tr) * *verifier-hash-query-budget*) / nnreal size

lemma *hash-target-program-alpha-challenge-step*:

hash-target-program B 1
 (do {
 a0 $\leftarrow$ *receive-alpha-challenge*;
 let a0' = a0;
 a1 $\leftarrow$ *read*;
 let a1' = a1;
 assert (a0' = a1');
 return a1'
 })

proof –

have *assert-return*:
hash-target-program B (0 + 0)
 (assert (a0 = a1) $\gg$ (λ -. return a1)) ::
 ('f, ('f, 'a) *protocol-channel-scheme*) *state-monad*
for a0 a1
by (rule *hash-target-program-bind*
 [OF *hash-target-program-assert hash-target-program-return*])

have *after-read*:
hash-target-program B (0 + 0 + 0)
 (read $\gg$ (λ a1. assert (a0 = a1) $\gg$ (λ -. return a1))) ::
 ('f, ('f, 'a) *protocol-channel-scheme*) *state-monad*
for a0

proof –

have *hash-target-program* B (0 + (0 + 0))
 (read $\gg$ (λ a1. assert (a0 = a1) $\gg$ (λ -. return a1))) ::
 ('f, ('f, 'a) *protocol-channel-scheme*) *state-monad*
by (rule *hash-target-program-bind*
 [OF *hash-target-program-read assert-return*])

then show ?thesis **by** *simp*

qed

have *hash-target-program* B (1 + (0 + 0 + 0))
 (*receive-alpha-challenge* $\gg$
 (λ a0. read $\gg$ (λ a1.
 assert (a0 = a1) $\gg$ (λ -. return a1)))) ::
 ('f, ('f, 'a) *protocol-channel-scheme*) *state-monad*
by (rule *hash-target-program-bind*
 [OF *hash-target-program-receive-alpha-challenge after-read*])

then show ?thesis

by *simp*

qed

lemma *hash-range-budget-alpha-challenge-step:*
hash-range-budget 1
 (do {
a0 ← *receive-alpha-challenge*;
 let *a0'* = *a0*;
a1 ← *read*;
 let *a1'* = *a1*;
 assert (*a0'* = *a1'*);
 return *a1'*
 })
proof –
have *assert-return:*
hash-range-budget (0 + 0)
 (assert (*a0* = *a1*) $\gg$ (λ -. return *a1*)) ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad **for** *a0 a1*
by (rule *hash-range-budget-bind*
 [OF *hash-range-budget-assert hash-range-budget-return*])
have *after-read:*
hash-range-budget (0 + 0 + 0)
 (read $\gg$ ($\lambda a1$. assert (*a0* = *a1*) $\gg$ (λ -. return *a1*))) ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad **for** *a0*
proof –
have *hash-range-budget (0 + (0 + 0))*
 (read $\gg$ ($\lambda a1$. assert (*a0* = *a1*) $\gg$ (λ -. return *a1*))) ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad
by (rule *hash-range-budget-bind*
 [OF *hash-range-budget-read assert-return*])
then show ?thesis **by** *simp*
qed
have *hash-range-budget (1 + (0 + 0 + 0))*
 (receive-alpha-challenge $\gg$
 ($\lambda a0$. read $\gg$ ($\lambda a1$.
 assert (*a0* = *a1*) $\gg$ (λ -. return *a1*)))) ::
 ('f, ('f, 'a) protocol-channel-scheme) state-monad
by (rule *hash-range-budget-bind*
 [OF *hash-range-budget-receive-alpha-challenge after-read*])
then show ?thesis
by *simp*
qed

lemma *hash-collision-budget-alpha-challenge-step:*
hash-collision-budget 1
 (do {
a0 ← *receive-alpha-challenge*;
 let *a0'* = *a0*;
a1 ← *read*;
 let *a1'* = *a1*;
 assert (*a0'* = *a1'*);
 })

```

    return a1'
  })
proof –
  have assert-return-range:
    hash-range-budget (0 + 0)
    (assert (a0 = a1)  $\gg$  (λ-. return a1)) ::
      (f, (f, 'a) protocol-channel-scheme) state-monad) for a0 a1
  by (rule hash-range-budget-bind
    [OF hash-range-budget-assert hash-range-budget-return])
  have assert-return-coll:
    hash-collision-budget (0 + 0)
    (assert (a0 = a1)  $\gg$  (λ-. return a1)) ::
      (f, (f, 'a) protocol-channel-scheme) state-monad) for a0 a1
  by (rule hash-collision-budget-bind
    [OF hash-range-budget-assert hash-collision-budget-assert
      hash-range-budget-return hash-collision-budget-return])
  have after-read-range:
    hash-range-budget (0 + 0 + 0)
    (read  $\gg$  (λa1. assert (a0 = a1)  $\gg$  (λ-. return a1))) ::
      (f, (f, 'a) protocol-channel-scheme) state-monad) for a0
  proof –
    have hash-range-budget (0 + (0 + 0))
      (read  $\gg$  (λa1. assert (a0 = a1)  $\gg$  (λ-. return a1))) ::
        (f, (f, 'a) protocol-channel-scheme) state-monad)
    by (rule hash-range-budget-bind
      [OF hash-range-budget-read assert-return-range])
    then show ?thesis by simp
  qed
  have after-read-coll:
    hash-collision-budget (0 + 0 + 0)
    (read  $\gg$  (λa1. assert (a0 = a1)  $\gg$  (λ-. return a1))) ::
      (f, (f, 'a) protocol-channel-scheme) state-monad) for a0
  proof –
    have hash-collision-budget (0 + (0 + 0))
      (read  $\gg$  (λa1. assert (a0 = a1)  $\gg$  (λ-. return a1))) ::
        (f, (f, 'a) protocol-channel-scheme) state-monad)
    by (rule hash-collision-budget-bind
      [OF hash-range-budget-read hash-collision-budget-read
        assert-return-range assert-return-coll])
    then show ?thesis by simp
  qed
  have hash-collision-budget (1 + (0 + 0 + 0))
    (receive-alpha-challenge  $\gg$ 
      (λa0. read  $\gg$  (λa1.
        assert (a0 = a1)  $\gg$  (λ-. return a1)))) ::
      (f, (f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-collision-budget-bind
    [OF hash-range-budget-receive-alpha-challenge
      hash-collision-budget-receive-alpha-challenge])

```

```

      after-read-range after-read-coll])
  then show ?thesis
  by simp
qed

lemma hash-target-program-verifier-query-round-program-exact:
  hash-target-program B
  (1 + verifier-query-decommit-hash-budget +
   (length f-fl + length fl) * verifier-fri-layer-hash-budget)
  (verifier-query-round-program fr f-fl f-final as fl final)
proof -
  let ?L = floor-log (clength * scale)
  let ?B = verifier-fri-layer-hash-budget
  have B-eq: ?B = Suc ?L + Suc ?L
    unfolding verifier-fri-layer-hash-budget-def by simp
  have after-comp:
    hash-target-program B (length fl * ?B + 0)
    ((mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
       ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(i, x, len, pw). assert (x = final))))
    for idx fv
  proof (rule hash-target-program-bind)
    show hash-target-program B (length fl * ?B)
      (mfold
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat,
         ('f, 'a) protocol-channel-scheme) state-monad)
      unfolding B-eq
      by (rule hash-target-program-receive-query-commits-mfold) simp
    show ∧x. hash-target-program B 0
      ((case x of (i, x, len, pw) ⇒ assert (x = final)) ::
       (unit, ('f, 'a) protocol-channel-scheme) state-monad)
      by (simp add: hash-target-program-assert split: prod.splits)
  qed
  have after-trace-assert:
    hash-target-program B (0 + (length fl * ?B + 0))
    ((assert (f-x = f-final) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ-. (mfold
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat,
         ('f, 'a) protocol-channel-scheme) state-monad) ≫=
        (λ(i, x, len, pw). assert (x = final)))))
    for idx fv f-i f-x f-len f-pow

```

```

by (rule hash-target-program-bind
  [OF hash-target-program-assert after-comp])
have after-trace:
  hash-target-program B
  (length f-fl * ?B + (0 + (length fl * ?B + 0)))
  ((mfold (idx, hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat,
    ('f, 'a) protocol-channel-scheme) state-monad) ≫=
  (λ(f-i, f-x, f-len, f-pow).
    assert (f-x = f-final) ≫=
    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(i, x, len, pw). assert (x = final))))))
for idx fv
proof (rule hash-target-program-bind)
show hash-target-program B (length f-fl * ?B)
  (mfold (idx, hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat,
    ('f, 'a) protocol-channel-scheme) state-monad)
unfolding B-eq
by (rule hash-target-program-receive-query-commits-mfold) simp
fix x :: nat × 'f × nat × nat
obtain f-i f-x f-len f-pow where x-eq:
  x = (f-i, f-x, f-len, f-pow)
by (cases x) auto
have case-budget:
  hash-target-program B (0 + (length fl * ?B + 0))
  ((assert (f-x = f-final) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad) ≫=
  (λ-. (mfold
    (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
    (receive-query-commits fl) ::
    (nat × 'f × nat × nat,
    ('f, 'a) protocol-channel-scheme) state-monad) ≫=
    (λ(i, x, len, pw). assert (x = final))))))
by (rule hash-target-program-bind
  [OF hash-target-program-assert after-comp])
show hash-target-program B (0 + (length fl * ?B + 0))
  ((case x of (f-i, f-x, f-len, f-pow) ⇒
    assert (f-x = f-final) ≫=
    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,

```



```

      ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ(i, x, len, pw). assert (x = final)))) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad)
using case-budget
unfolding x-eq
by (simp split: prod.splits)
qed
have after-query:
  hash-target-program B
    (verifier-query-decommit-hash-budget +
      (length f-fl * ?B + (0 + (length fl * ?B + 0))))
    (mmap (check-decommit-on-query fr idx) >>=
      (λfv. (mfold (idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl) ::
        (nat × 'f × nat × nat,
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λ(f-i, f-x, f-len, f-pow).
          assert (f-x = f-final) >>=
            (λ-. (mfold
              (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
              (receive-query-commits fl) ::
              (nat × 'f × nat × nat,
                ('f, 'a) protocol-channel-scheme) state-monad) >>=
              (λ(i, x, len, pw). assert (x = final)))))))
    for idx
proof -
  have decommit:
    hash-target-program B verifier-query-decommit-hash-budget
      (mmap (check-decommit-on-query fr idx))
  proof -
    have exact:
      hash-target-program B
        (powers * Suc (floor-log (scale * clength)))
        (mmap (check-decommit-on-query fr idx))
      by (rule hash-target-program-check-decommit-on-query)
    show ?thesis
      using exact
      unfolding verifier-query-decommit-hash-budget-def
      by (simp add: mult.commute)
  qed
  show ?thesis
    by (rule hash-target-program-bind[OF decommit after-trace])
qed
have after-query-simple:
  hash-target-program B
    (verifier-query-decommit-hash-budget +
      (length f-fl * ?B + length fl * ?B))
    (mmap (check-decommit-on-query fr idx) >>=
      (λfv. (mfold (idx, hd fv, clength * scale, 1)

```

```

    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λ(f-i, f-x, f-len, f-pow).
     assert (f-x = f-final) >>=
     (λ-. (mfold
           (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
           (receive-query-commits fl) ::
           (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) >>=
           (λ(i, x, len, pw). assert (x = final))))))
  for idx
  using after-query[of idx] by simp
have after-random:
  hash-target-program B
  (verifier-query-decommit-hash-budget +
   (length f-fl * ?B + length fl * ?B))
  ((let idx' = index (to-nat idx) in
   mmap (check-decommit-on-query fr idx') >>=
   (λfv. (mfold (idx', hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λ(f-i, f-x, f-len, f-pow).
     assert (f-x = f-final) >>=
     (λ-. (mfold
           (idx', cp-eval as fv (h ^ idx' * shift), clength * scale, 1)
           (receive-query-commits fl) ::
           (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) >>=
           (λ(i, x, len, pw). assert (x = final))))))))
  for idx
  using after-query-simple[of index (to-nat idx)]
  by (simp add: Let-def algebra-simps mult.commute split: prod.splits)
have hash-target-program B
  (1 + (verifier-query-decommit-hash-budget +
   (length f-fl * ?B + length fl * ?B)))
  (receive-query-index-challenge >>=
   (λidx.
    let idx' = index (to-nat idx) in
    mmap (check-decommit-on-query fr idx') >>=
    (λfv. (mfold (idx', hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λ(f-i, f-x, f-len, f-pow).
     assert (f-x = f-final) >>=
     (λ-. (mfold
           (idx', cp-eval as fv (h ^ idx' * shift), clength * scale, 1)

```

```

      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
       ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(i, x, len, pw). assert (x = final))))))
  by (rule hash-target-program-bind
      [OF hash-target-program-receive-query-index-challenge])
      (rule after-random)
  then show ?thesis
    unfolding verifier-query-round-program-def
    by (simp add: Let-def algebra-simps mult.commute split: prod.splits)
qed

```

lemma hash-range-budget-verifier-query-round-program-exact:

```

hash-range-budget
  (1 + verifier-query-decommit-hash-budget +
   (length f-fl + length fl) * verifier-fri-layer-hash-budget)
  (verifier-query-round-program fr f-fl f-final as fl final)

```

proof –

```

let ?L = floor-log (clength * scale)
let ?B = verifier-fri-layer-hash-budget
have B-eq: ?B = Suc ?L + Suc ?L
  unfolding verifier-fri-layer-hash-budget-def by simp
have after-comp:
  hash-range-budget (length fl * ?B + 0)
    ((mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
       ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(i, x, len, pw). assert (x = final))))

```

for idx fv

```

proof (rule hash-range-budget-bind)
  show hash-range-budget (length fl * ?B)
    (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
       ('f, 'a) protocol-channel-scheme) state-monad)
    unfolding B-eq
    by (rule hash-range-budget-receive-query-commits-mfold) simp
  show ∧x. hash-range-budget 0
    ((case x of (i, x, len, pw) ⇒ assert (x = final)) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad)
    by (simp add: hash-range-budget-assert split: prod.splits)

```

qed

have after-trace-assert:

```

hash-range-budget (0 + (length fl * ?B + 0))
  ((assert (f-x = f-final) ::

```

```

    (unit, ('f, 'a) protocol-channel-scheme) state-monad) >>=
  (λ-. (mfold
    (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
    (receive-query-commits fl) ::
    (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λ(i, x, len, pw). assert (x = final))))
for idx fv f-i f-x f-len f-pow
by (rule hash-range-budget-bind
  [OF hash-range-budget-assert after-comp])
have after-trace:
  hash-range-budget
    (length f-fl * ?B + (0 + (length fl * ?B + 0)))
    ((mfold (idx, hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ(f-i, f-x, f-len, f-pow).
        assert (f-x = f-final) >>=
        (λ-. (mfold
          (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
          (receive-query-commits fl) ::
          (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) >>=
            (λ(i, x, len, pw). assert (x = final)))))))
for idx fv
proof (rule hash-range-budget-bind)
show hash-range-budget (length f-fl * ?B)
  (mfold (idx, hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad)
unfolding B-eq
by (rule hash-range-budget-receive-query-commits-mfold) simp
fix x :: nat × 'f × nat × nat
obtain f-i f-x f-len f-pow where x-eq:
  x = (f-i, f-x, f-len, f-pow)
by (cases x) auto
have case-budget:
  hash-range-budget (0 + (length fl * ?B + 0))
    ((assert (f-x = f-final) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ-. (mfold
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat,
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λ(i, x, len, pw). assert (x = final))))))
by (rule hash-range-budget-bind)

```

```

    [OF hash-range-budget-assert after-comp])
show hash-range-budget (0 + (length fl * ?B + 0))
  ((case x of (f-i, f-x, f-len, f-pow) =>
    assert (f-x = f-final) >>=
    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ(i, x, len, pw). assert (x = final)))) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad)
using case-budget
unfolding x-eq
by (simp split: prod.splits)
qed
have after-query:
  hash-range-budget
    (verifier-query-decommit-hash-budget +
      (length f-fl * ?B + (0 + (length fl * ?B + 0))))
    (mmap (check-decommit-on-query fr idx) >>=
      (λfv. (mfold (idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl) ::
        (nat × 'f × nat × nat,
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λ(f-i, f-x, f-len, f-pow).
          assert (f-x = f-final) >>=
          (λ-. (mfold
            (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
            (receive-query-commits fl) ::
            (nat × 'f × nat × nat,
              ('f, 'a) protocol-channel-scheme) state-monad) >>=
            (λ(i, x, len, pw). assert (x = final)))))))
for idx
proof -
have decommit:
  hash-range-budget verifier-query-decommit-hash-budget
    (mmap (check-decommit-on-query fr idx))
proof -
have exact:
  hash-range-budget
    (powers * Suc (floor-log (scale * clength)))
    (mmap (check-decommit-on-query fr idx))
by (rule hash-range-budget-check-decommit-on-query)
show ?thesis
using exact
unfolding verifier-query-decommit-hash-budget-def
by (simp add: mult.commute)
qed
show ?thesis

```

```

    by (rule hash-range-budget-bind[OF decommit after-trace])
qed
have after-query-simple:
  hash-range-budget
    (verifier-query-decommit-hash-budget +
     (length f-fl * ?B + length fl * ?B))
    (mmap (check-decommit-on-query fr idx) >=>
     (λfv. (mfold (idx, hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat,
       ('f, 'a) protocol-channel-scheme) state-monad) >=>
      (λ(f-i, f-x, f-len, f-pow).
       assert (f-x = f-final) >=>
       (λ-. (mfold
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat,
         ('f, 'a) protocol-channel-scheme) state-monad) >=>
        (λ(i, x, len, pw). assert (x = final)))))))
  for idx
  using after-query[of idx] by simp
have after-random:
  hash-range-budget
    (verifier-query-decommit-hash-budget +
     (length f-fl * ?B + length fl * ?B))
    ((let idx' = index (to-nat idx) in
     mmap (check-decommit-on-query fr idx') >=>
     (λfv. (mfold (idx', hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat,
       ('f, 'a) protocol-channel-scheme) state-monad) >=>
      (λ(f-i, f-x, f-len, f-pow).
       assert (f-x = f-final) >=>
       (λ-. (mfold
        (idx', cp-eval as fv (h ^ idx' * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat,
         ('f, 'a) protocol-channel-scheme) state-monad) >=>
        (λ(i, x, len, pw). assert (x = final)))))))
  for idx
  using after-query-simple[of index (to-nat idx)]
  by (simp add: Let-def algebra-simps mult.commute split: prod.splits)
have hash-range-budget
  (1 + (verifier-query-decommit-hash-budget +
   (length f-fl * ?B + length fl * ?B)))
  (receive-query-index-challenge >=>
   (λidx.
    let idx' = index (to-nat idx) in
    mmap (check-decommit-on-query fr idx') >=>

```

```

    (λfv. (mfold (idx', hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ(f-i, f-x, f-len, f-pow).
        assert (f-x = f-final) >>=
        (λ-. (mfold
          (idx', cp-eval as fv (h ^ idx' * shift), clength * scale, 1)
          (receive-query-commits fl) ::
          (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λ(i, x, len, pw). assert (x = final))))))
  by (rule hash-range-budget-bind
    [OF hash-range-budget-receive-query-index-challenge])
  (rule after-random)
then show ?thesis
  unfolding verifier-query-round-program-def
  by (simp add: Let-def algebra-simps mult.commute split: prod.splits)
qed

```

lemma hash-target-program-verifier-query-round-program:

assumes len-f: length f-fl ≤ ceil-log clength
and len-c: length fl ≤ ceil-log (maxDegree + 1)

shows

hash-target-program B verifier-query-round-hash-budget
 (verifier-query-round-program fr f-fl f-final as fl final)

proof –

let ?N = verifier-fri-layer-hash-budget

have exact:

hash-target-program B
 (1 + verifier-query-decommit-hash-budget +
 (length f-fl + length fl) * ?N)
 (verifier-query-round-program fr f-fl f-final as fl final)
by (rule hash-target-program-verifier-query-round-program-exact)

have le:

1 + verifier-query-decommit-hash-budget +
 (length f-fl + length fl) * ?N
 ≤ verifier-query-round-hash-budget

using len-f len-c

unfolding verifier-query-round-hash-budget-def

by simp

show ?thesis

by (rule hash-target-program-mono[OF le exact])

qed

lemma hash-range-budget-verifier-query-round-program:

assumes len-f: length f-fl ≤ ceil-log clength
and len-c: length fl ≤ ceil-log (maxDegree + 1)

shows

```

    hash-range-budget verifier-query-round-hash-budget
      (verifier-query-round-program fr f-fl f-final as fl final)
proof –
  let ?B = verifier-fri-layer-hash-budget
  have exact:
    hash-range-budget
      (1 + verifier-query-decommit-hash-budget +
        (length f-fl + length fl) * ?B)
      (verifier-query-round-program fr f-fl f-final as fl final)
  by (rule hash-range-budget-verifier-query-round-program-exact)
  have le:
    1 + verifier-query-decommit-hash-budget +
      (length f-fl + length fl) * ?B
    ≤ verifier-query-round-hash-budget
  using len-f len-c
  unfolding verifier-query-round-hash-budget-def
  by simp
  show ?thesis
    by (rule hash-range-budget-mono[OF le exact])
qed

```

```

lemma hash-collision-budget-verifier-query-round-program-exact:
  hash-collision-budget
    (1 + verifier-query-decommit-hash-budget +
      (length f-fl + length fl) * verifier-fri-layer-hash-budget)
    (verifier-query-round-program fr f-fl f-final as fl final)
proof –
  let ?L = floor-log (clength * scale)
  let ?B = verifier-fri-layer-hash-budget
  have B-eq: ?B = Suc ?L + Suc ?L
    unfolding verifier-fri-layer-hash-budget-def by simp
  have after-comp-range:
    hash-range-budget (length fl * ?B + 0)
      ((mfold
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat,
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λ(i, x, len, pw). assert (x = final))))
    for idx fv
  proof (rule hash-range-budget-bind)
  show hash-range-budget (length fl * ?B)
    (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad)
    unfolding B-eq
    by (rule hash-range-budget-receive-query-commits-mfold) simp

```



```

show  $\bigwedge x. \text{hash-range-budget } 0$ 
  ((case x of (i, x, len, pw)  $\Rightarrow$  assert (x = final)) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad)
  by (simp add: hash-range-budget-assert split: prod.splits)
qed
have after-comp-coll:
  hash-collision-budget (length fl * ?B + 0)
    ((mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat  $\times$  'f  $\times$  nat  $\times$  nat,
       ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
      ( $\lambda(i, x, len, pw). \text{assert } (x = \text{final})$ )))
  for idx fv
proof (rule hash-collision-budget-bind)
show hash-range-budget (length fl * ?B)
  (mfold
    (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
    (receive-query-commits fl) ::
    (nat  $\times$  'f  $\times$  nat  $\times$  nat,
     ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding B-eq
  by (rule hash-range-budget-receive-query-commits-mfold) simp
show hash-collision-budget (length fl * ?B)
  (mfold
    (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
    (receive-query-commits fl) ::
    (nat  $\times$  'f  $\times$  nat  $\times$  nat,
     ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding B-eq
  by (rule hash-collision-budget-receive-query-commits-mfold) simp
show  $\bigwedge x. \text{hash-range-budget } 0$ 
  ((case x of (i, x, len, pw)  $\Rightarrow$  assert (x = final)) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad)
  by (simp add: hash-range-budget-assert split: prod.splits)
show  $\bigwedge x. \text{hash-collision-budget } 0$ 
  ((case x of (i, x, len, pw)  $\Rightarrow$  assert (x = final)) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad)
  by (simp add: hash-collision-budget-assert split: prod.splits)
qed
have trace-cont-range:
  hash-range-budget (0 + (length fl * ?B + 0))
    ((assert (f-x = f-final) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
      ( $\lambda-. \text{mfold}$ 
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat  $\times$  'f  $\times$  nat  $\times$  nat,
         ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 

```

```

    (λ(i, x, len, pw). assert (x = final))))
for idx fv f-x
by (rule hash-range-budget-bind
    [OF hash-range-budget-assert after-comp-range])
have trace-cont-coll:
  hash-collision-budget (0 + (length fl * ?B + 0))
  ((assert (f-x = f-final) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ(i, x, len, pw). assert (x = final)))))
for idx fv f-x
by (rule hash-collision-budget-bind
    [OF hash-range-budget-assert hash-collision-budget-assert
      after-comp-range after-comp-coll])
have after-trace-range:
  hash-range-budget
  (length f-fl * ?B + (0 + (length fl * ?B + 0)))
  ((mfold (idx, hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λ(f-i, f-x, f-len, fpw).
      assert (f-x = f-final) >>=
      (λ-. (mfold
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat,
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λ(i, x, len, pw). assert (x = final))))))
for idx fv
proof (rule hash-range-budget-bind)
show hash-range-budget (length f-fl * ?B)
  (mfold (idx, hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad)
unfolding B-eq
by (rule hash-range-budget-receive-query-commits-mfold) simp
fix x :: nat × 'f × nat × nat
obtain f-i f-x f-len fpw where x-eq: x = (f-i, f-x, f-len, fpw)
by (cases x) auto
have case-budget:
  hash-range-budget (0 + (length fl * ?B + 0))
  ((assert (f-x = f-final) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad) >>=

```

```

    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(i, x, len, pw). assert (x = final))))
  by (rule trace-cont-range)
show hash-range-budget (0 + (length fl * ?B + 0))
((case x of (f-i, f-x, f-len, fpw) ⇒
  assert (f-x = f-final) ≫=
  (λ-. (mfold
    (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
    (receive-query-commits fl) ::
    (nat × 'f × nat × nat,
      ('f, 'a) protocol-channel-scheme) state-monad) ≫=
    (λ(i, x, len, pw). assert (x = final)))) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad)
  using case-budget unfolding x-eq by (simp split: prod.splits)
qed
have after-trace-coll:
  hash-collision-budget
    (length f-fl * ?B + (0 + (length fl * ?B + 0)))
    ((mfold (idx, hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(f-i, f-x, f-len, fpw).
        assert (f-x = f-final) ≫=
        (λ-. (mfold
          (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
          (receive-query-commits fl) ::
          (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) ≫=
          (λ(i, x, len, pw). assert (x = final)))))))
  for idx fv
proof (rule hash-collision-budget-bind)
show hash-range-budget (length f-fl * ?B)
(mfold (idx, hd fv, clength * scale, 1)
  (receive-query-commits f-fl) ::
  (nat × 'f × nat × nat,
    ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding B-eq
  by (rule hash-range-budget-receive-query-commits-mfold) simp
show hash-collision-budget (length f-fl * ?B)
(mfold (idx, hd fv, clength * scale, 1)
  (receive-query-commits f-fl) ::
  (nat × 'f × nat × nat,
    ('f, 'a) protocol-channel-scheme) state-monad)
  unfolding B-eq

```

```

  by (rule hash-collision-budget-receive-query-commits-mfold) simp
fix x :: nat × 'f × nat × nat
obtain f-i f-x f-len fpw where x-eq: x = (f-i, f-x, f-len, fpw)
  by (cases x) auto
have range-case:
  hash-range-budget (0 + (length fl * ?B + 0))
    ((assert (f-x = f-final) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ-. (mfold
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat,
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λ(i, x, len, pw). assert (x = final))))))
  by (rule trace-cont-range)
show hash-range-budget (0 + (length fl * ?B + 0))
  ((case x of (f-i, f-x, f-len, fpw) ⇒
    assert (f-x = f-final) >>=
    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ(i, x, len, pw). assert (x = final)))))) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad)
  using range-case unfolding x-eq by (simp split: prod.splits)
have coll-case:
  hash-collision-budget (0 + (length fl * ?B + 0))
    ((assert (f-x = f-final) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ-. (mfold
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat,
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λ(i, x, len, pw). assert (x = final))))))
  by (rule trace-cont-coll)
show hash-collision-budget (0 + (length fl * ?B + 0))
  ((case x of (f-i, f-x, f-len, fpw) ⇒
    assert (f-x = f-final) >>=
    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ(i, x, len, pw). assert (x = final)))))) ::
    (unit, ('f, 'a) protocol-channel-scheme) state-monad)
  using coll-case unfolding x-eq by (simp split: prod.splits)
qed

```

```

have after-query-range:
  hash-range-budget
    (verifier-query-decommit-hash-budget +
      (length f-fl * ?B + (0 + (length fl * ?B + 0))))
    (mmap (check-decommit-on-query fr idx) >=>
      (λfv. (mfold (idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl) ::
          (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) >=>
        (λ(f-i, f-x, f-len, fpw).
          assert (f-x = f-final) >=>
            (λ-. (mfold
              (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
              (receive-query-commits fl) ::
                (nat × 'f × nat × nat,
                  ('f, 'a) protocol-channel-scheme) state-monad) >=>
                (λ(i, x, len, pw). assert (x = final)))))))
    for idx
  proof -
    have decommit:
      hash-range-budget verifier-query-decommit-hash-budget
        (mmap (check-decommit-on-query fr idx))
    proof -
      have exact:
        hash-range-budget
          (powers * Suc (floor-log (scale * clength)))
          (mmap (check-decommit-on-query fr idx))
      by (rule hash-range-budget-check-decommit-on-query)
    show ?thesis
      using exact
      unfolding verifier-query-decommit-hash-budget-def
      by (simp add: mult.commute)
    qed
  show ?thesis
    by (rule hash-range-budget-bind[OF decommit after-trace-range])
  qed
have after-query-coll:
  hash-collision-budget
    (verifier-query-decommit-hash-budget +
      (length f-fl * ?B + (0 + (length fl * ?B + 0))))
    (mmap (check-decommit-on-query fr idx) >=>
      (λfv. (mfold (idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl) ::
          (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) >=>
        (λ(f-i, f-x, f-len, fpw).
          assert (f-x = f-final) >=>
            (λ-. (mfold
              (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)

```

```

      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
       ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(i, x, len, pw). assert (x = final))))))
for idx
proof -
  have decommit-range:
    hash-range-budget verifier-query-decommit-hash-budget
      (mmap (check-decommit-on-query fr idx))
  proof -
    have exact:
      hash-range-budget
        (powers * Suc (floor-log (scale * clength)))
        (mmap (check-decommit-on-query fr idx))
    by (rule hash-range-budget-check-decommit-on-query)
  show ?thesis
    using exact
    unfolding verifier-query-decommit-hash-budget-def
    by (simp add: mult.commute)
qed
have decommit-coll:
  hash-collision-budget verifier-query-decommit-hash-budget
    (mmap (check-decommit-on-query fr idx))
proof -
  have exact:
    hash-collision-budget
      (powers * Suc (floor-log (scale * clength)))
      (mmap (check-decommit-on-query fr idx))
  by (rule hash-collision-budget-check-decommit-on-query)
show ?thesis
  using exact
  unfolding verifier-query-decommit-hash-budget-def
  by (simp add: mult.commute)
qed
show ?thesis
  by (rule hash-collision-budget-bind
      [OF decommit-range decommit-coll after-trace-range after-trace-coll])
qed
have after-query-range-simple:
  hash-range-budget
    (verifier-query-decommit-hash-budget +
     (length f-fl * ?B + length fl * ?B))
  (mmap (check-decommit-on-query fr idx)) ≫=
  (λfv. (mfold (idx, hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat,
     ('f, 'a) protocol-channel-scheme) state-monad) ≫=
    (λ(f-i, f-x, f-len, fpw).
     assert (f-x = f-final) ≫=

```

```

    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(i, x, len, pw). assert (x = final))))))
  for idx
  using after-query-range[of idx] by simp
have after-query-coll-simple:
  hash-collision-budget
  (verifier-query-decommit-hash-budget +
    (length f-fl * ?B + length fl * ?B))
  (mmap (check-decommit-on-query fr idx) ≫=
    (λfv. (mfold (idx, hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(f-i, f-x, f-len, fpw).
        assert (f-x = f-final) ≫=
        (λ-. (mfold
          (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
          (receive-query-commits fl) ::
          (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) ≫=
          (λ(i, x, len, pw). assert (x = final)))))))
    for idx
    using after-query-coll[of idx] by simp
have after-random-range:
  hash-range-budget
  (verifier-query-decommit-hash-budget +
    (length f-fl * ?B + length fl * ?B))
  ((let idx' = index (to-nat idx) in
    mmap (check-decommit-on-query fr idx') ≫=
    (λfv. (mfold (idx', hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λ(f-i, f-x, f-len, fpw).
        assert (f-x = f-final) ≫=
        (λ-. (mfold
          (idx', cp-eval as fv (h ^ idx' * shift), clength * scale, 1)
          (receive-query-commits fl) ::
          (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) ≫=
          (λ(i, x, len, pw). assert (x = final)))))))
    for idx
    using after-query-range-simple[of index (to-nat idx)]
  by (simp add: Let-def algebra-simps mult.commute split: prod.splits)
have after-random-coll:

```

```

hash-collision-budget
  (verifier-query-decommit-hash-budget +
    (length f-fl * ?B + length fl * ?B))
  ((let idx' = index (to-nat idx) in
    mmap (check-decommit-on-query fr idx') >>=
    (λfv. (mfold (idx', hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat,
        ('f, 'a) protocol-channel-scheme) state-monad) >>=
      (λ(f-i, f-x, f-len, fpw).
        assert (f-x = f-final) >>=
        (λ-. (mfold
          (idx', cp-eval as fv (h ^ idx' * shift), clength * scale, 1)
          (receive-query-commits fl) ::
          (nat × 'f × nat × nat,
            ('f, 'a) protocol-channel-scheme) state-monad) >>=
          (λ(i, x, len, pw). assert (x = final))))))))))
  for idx
  using after-query-coll-simple[of index (to-nat idx)]
  by (simp add: Let-def algebra-simps mult.commute split: prod.splits)
have hash-collision-budget
  (1 + (verifier-query-decommit-hash-budget +
    (length f-fl * ?B + length fl * ?B)))
  (receive-query-index-challenge >>=
    (λidx.
      let idx' = index (to-nat idx) in
      mmap (check-decommit-on-query fr idx') >>=
      (λfv. (mfold (idx', hd fv, clength * scale, 1)
        (receive-query-commits f-fl) ::
        (nat × 'f × nat × nat,
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λ(f-i, f-x, f-len, fpw).
          assert (f-x = f-final) >>=
          (λ-. (mfold
            (idx', cp-eval as fv (h ^ idx' * shift), clength * scale, 1)
            (receive-query-commits fl) ::
            (nat × 'f × nat × nat,
              ('f, 'a) protocol-channel-scheme) state-monad) >>=
            (λ(i, x, len, pw). assert (x = final))))))))))
  by (rule hash-collision-budget-bind
    [OF hash-range-budget-receive-query-index-challenge
      hash-collision-budget-receive-query-index-challenge])
  (rule after-random-range, rule after-random-coll)
then show ?thesis
  unfolding verifier-query-round-program-def
  by (simp add: Let-def algebra-simps mult.commute split: prod.splits)
qed

```

lemma hash-collision-budget-verifier-query-round-program:


```

assumes len-f:  $\text{length } f\text{-fl} \leq \text{ceil-log } \text{clength}$ 
and len-c:  $\text{length } fl \leq \text{ceil-log } (\text{maxDegree} + 1)$ 
shows
  hash-collision-budget verifier-query-round-hash-budget
  (verifier-query-round-program fr f-fl f-final as fl final)
proof –
  let ?B = verifier-fri-layer-hash-budget
  have exact:
    hash-collision-budget
    ( $1 + \text{verifier-query-decommit-hash-budget} +$ 
     ( $\text{length } f\text{-fl} + \text{length } fl$ ) * ?B)
    (verifier-query-round-program fr f-fl f-final as fl final)
  by (rule hash-collision-budget-verifier-query-round-program-exact)
  have le:
     $1 + \text{verifier-query-decommit-hash-budget} +$ 
    ( $\text{length } f\text{-fl} + \text{length } fl$ ) * ?B
     $\leq \text{verifier-query-round-hash-budget}$ 
  using len-f len-c
  unfolding verifier-query-round-hash-budget-def
  by simp
  show ?thesis
  by (rule hash-collision-budget-mono[OF le exact])
qed

lemma ceil-log-to-nat-degree-bound:
  assumes to-nat dg  $\leq \text{maxDegree}$ 
  shows  $\text{ceil-log } (\text{to-nat } dg + 1) \leq \text{ceil-log } (\text{maxDegree} + 1)$ 
  by (rule ceil-log-mono) (use assms in simp)

lemma hash-range-budget-ntimes-receive-fri-commits-bound:
  assumes  $n \leq N$ 
  shows hash-range-budget N (ntimes receive-fri-commits n)
proof –
  have exact: hash-range-budget (n * 1) (ntimes receive-fri-commits n)
  by (rule hash-range-budget-ntimes[OF hash-range-budget-receive-fri-commits])
  show ?thesis
  by (rule hash-range-budget-mono[OF - exact]) (use assms in simp)
qed

lemma hash-target-program-ntimes-receive-trace-fri-commits-bound:
  assumes  $n \leq N$ 
  shows hash-target-program B N (ntimes receive-trace-fri-commits n)
proof –
  have exact:
    hash-target-program B (n * 1) (ntimes receive-trace-fri-commits n)
  by (rule hash-target-program-ntimes
    [OF hash-target-program-receive-trace-fri-commits])
  show ?thesis
  by (rule hash-target-program-mono[OF - exact]) (use assms in simp)

```

qed

lemma *hash-target-program-ntimes-receive-composition-fri-commits-bound*:
assumes $n \leq N$
shows *hash-target-program B N (ntimes receive-composition-fri-commits n)*
proof –
have *exact*:
*hash-target-program B (n * 1) (ntimes receive-composition-fri-commits n)*
by (rule *hash-target-program-ntimes*
 $[OF \text{ hash-target-program-receive-composition-fri-commits}]$)
show *?thesis*
by (rule *hash-target-program-mono* $[OF - exact]$) (use *assms* **in** *simp*)
qed

lemma *hash-target-program-alpha-round*:
hash-target-program B 1 alpha-round
unfolding *alpha-round-def*
by (rule *hash-target-program-alpha-challenge-step*)

lemma *hash-target-program-verify-monad*:
hash-target-program B verifier-hash-query-budget verify-monad
proof –
let $?Q = \text{rounds} * \text{verifier-query-round-hash-budget}$
have *alpha-map-range*:
hash-target-program B (length spec)
 $(\text{mmap } (\text{replicate } (\text{length spec})$
 $(\text{alpha-round} ::$
 $(f, (f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad}))$)
proof –
let $?steps =$
 $\text{replicate } (\text{length spec})$
 $(\text{alpha-round} ::$
 $(f, (f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
have *step-range*:
 $\bigwedge m. m \in \text{set } ?steps \implies \text{hash-target-program B 1 } m$
proof –
fix m
assume $m \in \text{set } ?steps$
then have *m-eq*:
 $m =$
 $(\text{alpha-round} ::$
 $(f, (f, 'a) \text{ protocol-channel-scheme}) \text{ state-monad})$
by *simp*
show *hash-target-program B 1 m*
unfolding *m-eq*
by (rule *hash-target-program-alpha-round*)
qed
have *hash-target-program B (length ?steps * 1) (mmap ?steps)*
by (rule *hash-target-program-mmap*)

```

    (rule step-range)
  then show ?thesis by simp
qed
have after-final:
  hash-target-program B (0 + ?Q)
  (read  $\gg$ 
    ( $\lambda$ final. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds))
  if len-f: length f-fl  $\leq$  ceil-log clength
    and len-fl: length fl  $\leq$  ceil-log (maxDegree + 1)
  for fr f-fl f-final as fl
  by (rule hash-target-program-bind
    [OF hash-target-program-read
      hash-target-program-ntimes
      [OF hash-target-program-verifier-query-round-program
        [OF len-f len-fl]]])
have after-comp-fri:
  hash-target-program B (ceil-log (maxDegree + 1) + (0 + ?Q))
  ((ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f  $\times$  'f) list),
      (f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda$ fl. read  $\gg$ 
      ( $\lambda$ final. ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds))))
  if len-f: length f-fl  $\leq$  ceil-log clength
    and dg-bound: to-nat dg  $\leq$  maxDegree
  for fr f-fl f-final as dg
proof (rule hash-target-program-bind-on-outcomes)
  show hash-target-program B (ceil-log (maxDegree + 1))
    (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
      (((f  $\times$  'f) list),
        (f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-target-program-ntimes-receive-composition-fri-commits-bound)
    (rule ceil-log-to-nat-degree-bound[OF dg-bound])
fix s fl t
assume out:
  Some (fl, t)  $\in$ 
  set-dist (execute
    (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
      (((f  $\times$  'f) list),
        (f, 'a) protocol-channel-scheme) state-monad) s)
have len-fl: length fl  $\leq$  ceil-log (maxDegree + 1)
  using ntimes-receive-composition-fri-commits-outcome[OF out]
  ceil-log-to-nat-degree-bound[OF dg-bound]
  by simp
show hash-target-program B (0 + ?Q)
  (read  $\gg$ 

```

```

    (λfinal. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds))
  by (rule after-final[OF len-f len-fl])
qed
have after-degree-assert:
  hash-target-program B (0 + (ceil-log (maxDegree + 1) + (0 + ?Q)))
  (assert (to-nat dg ≤ maxDegree) ≫=
    (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
      (((f × 'f) list),
        ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λfl. read ≫=
        (λfinal. ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds))))))
  if len-f: length f-fl ≤ ceil-log clength
  for fr f-fl f-final as dg
proof (rule hash-target-program-bind-on-outcomes)
  show hash-target-program B 0
    (assert (to-nat dg ≤ maxDegree) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad)
    by (rule hash-target-program-assert)
  fix s u t
  assume out:
    Some (u, t) ∈
      set-dist (execute
        (assert (to-nat dg ≤ maxDegree) ::
          (unit, ('f, 'a) protocol-channel-scheme) state-monad) s)
  have dg-bound: to-nat dg ≤ maxDegree
    using out unfolding assert-def
    by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
  show hash-target-program B (ceil-log (maxDegree + 1) + (0 + ?Q))
    ((ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
      (((f × 'f) list),
        ('f, 'a) protocol-channel-scheme) state-monad) ≫=
      (λfl. read ≫=
        (λfinal. ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds))))
    by (rule after-comp-fri[OF len-f dg-bound])
qed
have after-read-dg:
  hash-target-program B
  (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))
  (read ≫=
    (λdg. assert (to-nat dg ≤ maxDegree) ≫=
      (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
        (((f × 'f) list),
          ('f, 'a) protocol-channel-scheme) state-monad) ≫=

```

```

    (λfl. read >>=
      (λfinal. ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds))))))
  if len-f: length f-fl ≤ ceil-log clength
  for fr f-fl f-final as
  by (rule hash-target-program-bind
    [OF hash-target-program-read after-degree-assert[OF len-f]])
  have after-alpha:
    hash-target-program B
      (length spec +
        (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q)))))
      (mmap (replicate (length spec) alpha-round) >>=
        (λas. read >>=
          (λdg. assert (to-nat dg ≤ maxDegree) >>=
            (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
              (((f × 'f) list),
                ('f, 'a) protocol-channel-scheme) state-monad) >>=
              (λfl. read >>=
                (λfinal. ntimes
                  (verifier-query-round-program fr f-fl f-final as fl final)
                  rounds)))))))
          if len-f: length f-fl ≤ ceil-log clength
          for fr f-fl f-final
          by (rule hash-target-program-bind
            [OF alpha-map-range after-read-dg[OF len-f]])
          have after-read-f-final:
            hash-target-program B
              (0 + (length spec +
                (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q)))))
              (read >>=
                (λf-final. mmap (replicate (length spec) alpha-round) >>=
                  (λas. read >>=
                    (λdg. assert (to-nat dg ≤ maxDegree) >>=
                      (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
                        (((f × 'f) list),
                          ('f, 'a) protocol-channel-scheme) state-monad) >>=
                        (λfl. read >>=
                          (λfinal. ntimes
                            (verifier-query-round-program fr f-fl f-final as fl final)
                            rounds)))))))
                    if len-f: length f-fl ≤ ceil-log clength
                    for fr f-fl
                    by (rule hash-target-program-bind
                      [OF hash-target-program-read after-alpha[OF len-f]])
                    have after-trace-fri:
                      hash-target-program B
                        (ceil-log clength +
                          (0 + (length spec +

```

```

    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  ((ntimes receive-trace-fri-commits (ceil-log clength) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad) ≫=
   (λf-fl. read ≫=
    (λf-final. mmap (replicate (length spec) alpha-round) ≫=
     (λas. read ≫=
      (λdg. assert (to-nat dg ≤ maxDegree) ≫=
       (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
        (((f × 'f) list),
         (f, 'a) protocol-channel-scheme) state-monad) ≫=
        (λfl. read ≫=
         (λfinal. ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds))))))))))
  for fr
proof (rule hash-target-program-bind-on-outcomes)
show hash-target-program B (ceil-log clength)
  (ntimes receive-trace-fri-commits (ceil-log clength) ::
   (((f × 'f) list),
    (f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-target-program-ntimes-receive-trace-fri-commits-bound) simp
fix s f-fl t
assume out:
  Some (f-fl, t) ∈
  set-dist (execute
    (ntimes receive-trace-fri-commits (ceil-log clength) ::
     (((f × 'f) list),
      (f, 'a) protocol-channel-scheme) state-monad) s)
have len-f: length f-fl ≤ ceil-log clength
  using ntimes-receive-trace-fri-commits-outcome[OF out] by simp
show hash-target-program B
  (0 + (length spec +
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  (read ≫=
   (λf-final. mmap (replicate (length spec) alpha-round) ≫=
    (λas. read ≫=
     (λdg. assert (to-nat dg ≤ maxDegree) ≫=
      (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
       (((f × 'f) list),
        (f, 'a) protocol-channel-scheme) state-monad) ≫=
       (λfl. read ≫=
        (λfinal. ntimes
         (verifier-query-round-program fr f-fl f-final as fl final)
         rounds))))))))))
  by (rule after-read-f-final[OF len-f])
qed
have top:
  hash-target-program B

```

```

(0 + (ceil-log clength +
  (0 + (length spec +
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q)))))))
(read >>=
  (λfr. (ntimes receive-trace-fri-commits (ceil-log clength) ::
    (((f × 'f) list),
      (f, 'a) protocol-channel-scheme) state-monad) >>=
  (λf-fl. read >>=
    (λf-final. mmap (replicate (length spec) alpha-round) >>=
      (λas. read >>=
        (λdg. assert (to-nat dg ≤ maxDegree) >>=
          (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
            (((f × 'f) list),
              (f, 'a) protocol-channel-scheme) state-monad) >>=
            (λfl. read >>=
              (λfinal. ntimes
                (verifier-query-round-program fr f-fl f-final as fl final)
                rounds))))))))))
by (rule hash-target-program-bind
  [OF hash-target-program-read after-trace-fri])
show ?thesis
using top
unfolding verify-monad-def verifier-query-round-program-def alpha-round-def
  verifier-hash-query-budget-def verifier-header-hash-budget-def
by (simp add: algebra-simps)
qed

```

lemma hash-target-budget-verify-monad:
 hash-target-budget B verifier-hash-query-budget verify-monad
 by (rule hash-target-program-budget[OF hash-target-program-verify-monad])

lemma wp-verify-monad-hash-new-output-hit-bound:
 wp-event verify-monad (hash-new-output-hit-event B s) $s \leq$
 hash-target-budget-value B verifier-hash-query-budget
 using hash-target-budget-verify-monad
 unfolding hash-target-budget-def by blast

lemma wp-verify-monad-transcript-hash-target-hit-bound:
 wp-event verify-monad
 (hash-new-output-hit-event (set tr) (verifier-initial-state tr))
 (verifier-initial-state tr)
 $\leq$ nnreal (card (set tr) * verifier-hash-query-budget) / nnreal size
 using wp-verify-monad-hash-new-output-hit-bound
 [of set tr verifier-initial-state tr]
 unfolding hash-target-budget-value-def
 by (simp add: mult.commute)

lemma hash-range-budget-ntimes-receive-trace-fri-commits-bound:

```

assumes  $n \leq N$ 
shows hash-range-budget  $N$  (ntimes receive-trace-fri-commits  $n$ )
proof –
  have exact:
    hash-range-budget ( $n * 1$ ) (ntimes receive-trace-fri-commits  $n$ )
  by (rule hash-range-budget-ntimes
    [OF hash-range-budget-receive-trace-fri-commits])
  show ?thesis
  by (rule hash-range-budget-mono[OF - exact]) (use assms in simp)
qed

lemma hash-range-budget-ntimes-receive-composition-fri-commits-bound:
  assumes  $n \leq N$ 
  shows hash-range-budget  $N$  (ntimes receive-composition-fri-commits  $n$ )
proof –
  have exact:
    hash-range-budget ( $n * 1$ ) (ntimes receive-composition-fri-commits  $n$ )
  by (rule hash-range-budget-ntimes
    [OF hash-range-budget-receive-composition-fri-commits])
  show ?thesis
  by (rule hash-range-budget-mono[OF - exact]) (use assms in simp)
qed

lemma hash-range-budget-alpha-round:
  hash-range-budget 1 alpha-round
  unfolding alpha-round-def
  by (rule hash-range-budget-alpha-challenge-step)

lemma hash-range-budget-verify-monad:
  hash-range-budget verifier-hash-query-budget verify-monad
proof –
  let  $?Q = \text{rounds} * \text{verifier-query-round-hash-budget}$ 
  have alpha-map-range:
    hash-range-budget (length spec)
    (mmap (replicate (length spec)
      (alpha-round ::
        ( $'f, ('f, 'a)$  protocol-channel-scheme) state-monad)))
  proof –
    let  $?steps =$ 
      replicate (length spec)
      (alpha-round ::
        ( $'f, ('f, 'a)$  protocol-channel-scheme) state-monad)
    have step-range:
       $\bigwedge m. m \in \text{set } ?steps \implies \text{hash-range-budget } 1 \ m$ 
  proof –
    fix  $m$ 
    assume  $m \in \text{set } ?steps$ 
    then have m-eq:
       $m =$ 

```



```

    (alpha-round ::
      ('f, ('f, 'a) protocol-channel-scheme) state-monad)
  by simp
show hash-range-budget 1 m
  unfolding m-eq
  by (rule hash-range-budget-alpha-round)
qed
have hash-range-budget (length ?steps * 1) (mmap ?steps)
  by (rule hash-range-budget-mmap)
  (rule step-range)
then show ?thesis by simp
qed
have after-final:
  hash-range-budget (0 + ?Q)
  (read  $\gg$ 
    ( $\lambda$ final. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds))
  if len-f: length f-fl  $\leq$  ceil-log clength
  and len-fl: length fl  $\leq$  ceil-log (maxDegree + 1)
  for fr f-fl f-final as fl
  by (rule hash-range-budget-bind
    [OF hash-range-budget-read
      hash-range-budget-ntimes
      [OF hash-range-budget-verifier-query-round-program
        [OF len-f len-fl]]])
have after-comp-fri:
  hash-range-budget (ceil-log (maxDegree + 1) + (0 + ?Q))
  ((ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f  $\times$  'f) list),
      ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda$ fl. read  $\gg$ 
      ( $\lambda$ final. ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds))))
  if len-f: length f-fl  $\leq$  ceil-log clength
  and dg-bound: to-nat dg  $\leq$  maxDegree
  for fr f-fl f-final as dg
proof (rule hash-range-budget-bind-on-outcomes)
show hash-range-budget (ceil-log (maxDegree + 1))
  (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f  $\times$  'f) list),
      ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-ntimes-receive-composition-fri-commits-bound)
  (rule ceil-log-to-nat-degree-bound[OF dg-bound])
fix s fl t
assume out:
  Some (fl, t)  $\in$ 
  set-dist (execute

```

```

      (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
        (((f × 'f) list),
         ('f, 'a) protocol-channel-scheme) state-monad) s)
have len-fl: length fl ≤ ceil-log (maxDegree + 1)
  using ntimes-receive-composition-fri-commits-outcome[OF out]
    ceil-log-to-nat-degree-bound[OF dg-bound]
  by simp
show hash-range-budget (0 + ?Q)
  (read ≥=
    (λfinal. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds))
  by (rule after-final[OF len-f len-fl])
qed
have after-degree-assert:
  hash-range-budget (0 + (ceil-log (maxDegree + 1) + (0 + ?Q)))
  (assert (to-nat dg ≤ maxDegree) ≥=
    (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
      (((f × 'f) list),
       ('f, 'a) protocol-channel-scheme) state-monad) ≥=
      (λfl. read ≥=
        (λfinal. ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds))))))
  if len-f: length f-fl ≤ ceil-log clength
  for fr f-fl f-final as dg
proof (rule hash-range-budget-bind-on-outcomes)
  show hash-range-budget 0
    (assert (to-nat dg ≤ maxDegree) ::
      (unit, ('f, 'a) protocol-channel-scheme) state-monad)
    by (rule hash-range-budget-assert)
  fix s u t
  assume out:
    Some (u, t) ∈
      set-dist (execute
        (assert (to-nat dg ≤ maxDegree) ::
          (unit, ('f, 'a) protocol-channel-scheme) state-monad) s)
  have dg-bound: to-nat dg ≤ maxDegree
    using out unfolding assert-def
    by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
  show hash-range-budget (ceil-log (maxDegree + 1) + (0 + ?Q))
    ((ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
      (((f × 'f) list),
       ('f, 'a) protocol-channel-scheme) state-monad) ≥=
      (λfl. read ≥=
        (λfinal. ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds))))
    by (rule after-comp-fri[OF len-f dg-bound])

```

```

qed
have after-read-dg:
  hash-range-budget
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))
    (read >=
      (λdg. assert (to-nat dg ≤ maxDegree) >=
        (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
          (((f × 'f) list),
            ('f, 'a) protocol-channel-scheme) state-monad) >=
          (λfl. read >=
            (λfinal. ntimes
              (verifier-query-round-program fr f-fl f-final as fl final)
              rounds))))))
    if len-f: length f-fl ≤ ceil-log clength
    for fr f-fl f-final as
    by (rule hash-range-budget-bind
      [OF hash-range-budget-read after-degree-assert[OF len-f]])
have after-alpha:
  hash-range-budget
    (length spec +
      (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))
      (mmap (replicate (length spec) alpha-round) >=
        (λas. read >=
          (λdg. assert (to-nat dg ≤ maxDegree) >=
            (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
              (((f × 'f) list),
                ('f, 'a) protocol-channel-scheme) state-monad) >=
                (λfl. read >=
                  (λfinal. ntimes
                    (verifier-query-round-program fr f-fl f-final as fl final)
                    rounds))))))
          if len-f: length f-fl ≤ ceil-log clength
          for fr f-fl f-final
          by (rule hash-range-budget-bind
            [OF alpha-map-range after-read-dg[OF len-f]])
have after-read-f-final:
  hash-range-budget
    (0 + (length spec +
      (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))
      (read >=
        (λf-final. mmap (replicate (length spec) alpha-round) >=
          (λas. read >=
            (λdg. assert (to-nat dg ≤ maxDegree) >=
              (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
                (((f × 'f) list),
                  ('f, 'a) protocol-channel-scheme) state-monad) >=
                  (λfl. read >=
                    (λfinal. ntimes
                      (verifier-query-round-program fr f-fl f-final as fl final)
                      rounds))))))
            if len-f: length f-fl ≤ ceil-log clength
            for fr f-fl f-final
            by (rule hash-range-budget-bind
              [OF alpha-map-range after-read-dg[OF len-f]])
          if len-f: length f-fl ≤ ceil-log clength
          for fr f-fl f-final
          by (rule hash-range-budget-bind
            [OF alpha-map-range after-read-dg[OF len-f]])
        if len-f: length f-fl ≤ ceil-log clength
        for fr f-fl f-final
        by (rule hash-range-budget-bind
          [OF alpha-map-range after-read-dg[OF len-f]])
      if len-f: length f-fl ≤ ceil-log clength
      for fr f-fl f-final
      by (rule hash-range-budget-bind
        [OF alpha-map-range after-read-dg[OF len-f]])
    if len-f: length f-fl ≤ ceil-log clength
    for fr f-fl f-final
    by (rule hash-range-budget-bind
      [OF alpha-map-range after-read-dg[OF len-f]])

```

```

rounds)))))))))
if len-f: length f-fl ≤ ceil-log clength
for fr f-fl
by (rule hash-range-budget-bind
  [OF hash-range-budget-read after-alpha[OF len-f]])
have after-trace-fri:
  hash-range-budget
    (ceil-log clength +
      (0 + (length spec +
        (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q)))))))
    ((ntimes receive-trace-fri-commits (ceil-log clength) ::
      (((f × 'f) list),
        (f, 'a) protocol-channel-scheme) state-monad) >>=
      (λf-fl. read >>=
        (λf-final. mmap (replicate (length spec) alpha-round) >>=
          (λas. read >>=
            (λdg. assert (to-nat dg ≤ maxDegree) >>=
              (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
                (((f × 'f) list),
                  (f, 'a) protocol-channel-scheme) state-monad) >>=
                  (λfl. read >>=
                    (λfinal. ntimes
                      (verifier-query-round-program fr f-fl f-final as fl final)
                      rounds))))))))))
for fr
proof (rule hash-range-budget-bind-on-outcomes)
show hash-range-budget (ceil-log clength)
  (ntimes receive-trace-fri-commits (ceil-log clength) ::
    (((f × 'f) list),
      (f, 'a) protocol-channel-scheme) state-monad)
by (rule hash-range-budget-ntimes-receive-trace-fri-commits-bound) simp
fix s f-fl t
assume out:
  Some (f-fl, t) ∈
    set-dist (execute
      (ntimes receive-trace-fri-commits (ceil-log clength) ::
        (((f × 'f) list),
          (f, 'a) protocol-channel-scheme) state-monad) s)
have len-f: length f-fl ≤ ceil-log clength
using ntimes-receive-trace-fri-commits-outcome[OF out] by simp
show hash-range-budget
  (0 + (length spec +
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  (read >>=
    (λf-final. mmap (replicate (length spec) alpha-round) >>=
      (λas. read >>=
        (λdg. assert (to-nat dg ≤ maxDegree) >>=
          (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
            (((f × 'f) list),
              (f, 'a) protocol-channel-scheme) state-monad) >>=
              (λfl. read >>=
                (λfinal. ntimes
                  (verifier-query-round-program fr f-fl f-final as fl final)
                  rounds))))))))

```

```

      ('f, 'a) protocol-channel-scheme) state-monad) >>=
    (λfl. read >>=
      (λfinal. ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds))))))
  by (rule after-read-f-final[OF len-f])
qed
have top:
  hash-range-budget
    (0 + (ceil-log clength +
      (0 + (length spec +
        (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))))
    (read >>=
      (λfr. (ntimes receive-trace-fri-commits (ceil-log clength) ::
        (((f × 'f) list),
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λf-fl. read >>=
          (λf-final. mmap (replicate (length spec) alpha-round) >>=
            (λas. read >>=
              (λdg. assert (to-nat dg ≤ maxDegree) >>=
                (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
                  (((f × 'f) list),
                    ('f, 'a) protocol-channel-scheme) state-monad) >>=
                    (λfl. read >>=
                      (λfinal. ntimes
                        (verifier-query-round-program fr f-fl f-final as fl final)
                        rounds))))))))))
        by (rule hash-range-budget-bind
          [OF hash-range-budget-read after-trace-fri])
    show ?thesis
    using top
    unfolding verify-monad-def verifier-query-round-program-def alpha-round-def
      verifier-hash-query-budget-def verifier-header-hash-budget-def
    by (simp add: algebra-simps)
qed

lemma hash-collision-budget-ntimes-receive-fri-commits-bound:
  assumes n ≤ N
  shows hash-collision-budget N (ntimes receive-fri-commits n)
proof -
  have exact: hash-collision-budget (n * 1) (ntimes receive-fri-commits n)
  by (rule hash-collision-budget-ntimes
    [OF hash-range-budget-receive-fri-commits
      hash-collision-budget-receive-fri-commits])
  show ?thesis
  by (rule hash-collision-budget-mono[OF - exact]) (use assms in simp)
qed

lemma hash-collision-budget-ntimes-receive-trace-fri-commits-bound:

```

```

assumes  $n \leq N$ 
shows hash-collision-budget  $N$  (ntimes receive-trace-fri-commits  $n$ )
proof –
  have exact:
    hash-collision-budget ( $n * 1$ ) (ntimes receive-trace-fri-commits  $n$ )
  by (rule hash-collision-budget-ntimes
    [OF hash-range-budget-receive-trace-fri-commits
      hash-collision-budget-receive-trace-fri-commits])
  show ?thesis
    by (rule hash-collision-budget-mono[OF - exact]) (use assms in simp)
qed

lemma hash-collision-budget-ntimes-receive-composition-fri-commits-bound:
  assumes  $n \leq N$ 
  shows hash-collision-budget  $N$  (ntimes receive-composition-fri-commits  $n$ )
proof –
  have exact:
    hash-collision-budget ( $n * 1$ )
      (ntimes receive-composition-fri-commits  $n$ )
  by (rule hash-collision-budget-ntimes
    [OF hash-range-budget-receive-composition-fri-commits
      hash-collision-budget-receive-composition-fri-commits])
  show ?thesis
    by (rule hash-collision-budget-mono[OF - exact]) (use assms in simp)
qed

lemma hash-collision-budget-alpha-round:
  hash-collision-budget 1 alpha-round
  unfolding alpha-round-def
  by (rule hash-collision-budget-alpha-challenge-step)

lemma hash-collision-budget-verify-monad:
  hash-collision-budget verifier-hash-query-budget verify-monad
proof –
  let  $?Q = \text{rounds} * \text{verifier-query-round-hash-budget}$ 
  let  $?alpha\text{-steps} =$ 
    replicate (length spec)
      (alpha-round ::
        ( $'f, ('f, 'a) \text{protocol-channel-scheme}$ ) state-monad)
  have alpha-step-range:
     $\bigwedge m. m \in \text{set } ?alpha\text{-steps} \implies \text{hash-range-budget } 1\ m$ 
  proof –
    fix  $m$ 
    assume  $m \in \text{set } ?alpha\text{-steps}$ 
    then have m-eq:
       $m =$ 
        (alpha-round ::
          ( $'f, ('f, 'a) \text{protocol-channel-scheme}$ ) state-monad)
    by simp

```

```

show hash-range-budget 1 m
  unfolding m-eq by (rule hash-range-budget-alpha-round)
qed
have alpha-step-coll:
   $\bigwedge m. m \in \text{set } ?\text{alpha-steps} \implies \text{hash-collision-budget } 1 \ m$ 
proof –
  fix m
  assume m  $\in \text{set } ?\text{alpha-steps}$ 
  then have m-eq:
    m =
      (alpha-round ::
        ('f, ('f, 'a) protocol-channel-scheme) state-monad)
    by simp
  show hash-collision-budget 1 m
    unfolding m-eq by (rule hash-collision-budget-alpha-round)
qed
have alpha-map-range:
  hash-range-budget (length spec) (mmap ?alpha-steps)
proof –
  have hash-range-budget (length ?alpha-steps * 1) (mmap ?alpha-steps)
    by (rule hash-range-budget-mmap) (rule alpha-step-range)
  then show ?thesis by simp
qed
have alpha-map-coll:
  hash-collision-budget (length spec) (mmap ?alpha-steps)
proof –
  have hash-collision-budget (length ?alpha-steps * 1) (mmap ?alpha-steps)
  proof (rule hash-collision-budget-mmap)
    fix m
    assume m-in: m  $\in \text{set } ?\text{alpha-steps}$ 
    show hash-range-budget 1 m
      by (rule alpha-step-range[OF m-in])
    show hash-collision-budget 1 m
      by (rule alpha-step-coll[OF m-in])
  qed
  then show ?thesis by simp
qed
have after-final-range:
  hash-range-budget (0 + ?Q)
    (read  $\gg$ 
      ( $\lambda$ final. ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds))
  if len-f: length f-fl  $\leq \text{ceil-log } \text{clength}$ 
    and len-fl: length fl  $\leq \text{ceil-log } (\text{maxDegree} + 1)$ 
  for fr f-fl f-final as fl
  by (rule hash-range-budget-bind
    [OF hash-range-budget-read
      hash-range-budget-ntimes

```

```

      [OF hash-range-budget-verifier-query-round-program
        [OF len-f len-fl]]])
have after-final-coll:
  hash-collision-budget (0 + ?Q)
  (read  $\gg$ 
    ( $\lambda$ final. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds))
  if len-f: length f-fl  $\leq$  ceil-log clength
    and len-fl: length fl  $\leq$  ceil-log (maxDegree + 1)
  for fr f-fl f-final as fl
proof -
  have query-range:
     $\bigwedge$ final. hash-range-budget verifier-query-round-hash-budget
      (verifier-query-round-program fr f-fl f-final as fl final)
  by (rule hash-range-budget-verifier-query-round-program
    [OF len-f len-fl])
  have query-coll:
     $\bigwedge$ final. hash-collision-budget verifier-query-round-hash-budget
      (verifier-query-round-program fr f-fl f-final as fl final)
  by (rule hash-collision-budget-verifier-query-round-program
    [OF len-f len-fl])
  have rounds-range:
     $\bigwedge$ final. hash-range-budget ?Q
      (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
  by (rule hash-range-budget-ntimes) (rule query-range)
  have rounds-coll:
     $\bigwedge$ final. hash-collision-budget ?Q
      (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
  by (rule hash-collision-budget-ntimes)
    (rule query-range, rule query-coll)
  show ?thesis
  by (rule hash-collision-budget-bind
    [OF hash-range-budget-read hash-collision-budget-read
      rounds-range rounds-coll])
qed
have after-comp-fri-range:
  hash-range-budget (ceil-log (maxDegree + 1) + (0 + ?Q))
  ((ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f  $\times$  'f) list),
      ('f, 'a) protocol-channel-scheme) state-monad)  $\gg$ 
    ( $\lambda$ fl. read  $\gg$ 
      ( $\lambda$ final. ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds))))
  if len-f: length f-fl  $\leq$  ceil-log clength
    and dg-bound: to-nat dg  $\leq$  maxDegree

```



```

for fr f-fl f-final as dg
proof (rule hash-range-budget-bind-on-outcomes)
show hash-range-budget (ceil-log (maxDegree + 1))
  (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad)
by (rule hash-range-budget-ntimes-receive-composition-fri-commits-bound)
  (rule ceil-log-to-nat-degree-bound[OF dg-bound])
fix s fl t
assume out:
  Some (fl, t) ∈
    set-dist (execute
      (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
        (((f × 'f) list),
         (f, 'a) protocol-channel-scheme) state-monad) s)
have len-fl: length fl ≤ ceil-log (maxDegree + 1)
using ntimes-receive-composition-fri-commits-outcome[OF out]
  ceil-log-to-nat-degree-bound[OF dg-bound]
by simp
show hash-range-budget (0 + ?Q)
  (read ≥=
    (λfinal. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final
        rounds)))
by (rule after-final-range[OF len-f len-fl])
qed
have after-comp-fri-coll:
  hash-collision-budget (ceil-log (maxDegree + 1) + (0 + ?Q))
  ((ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad) ≥=
    (λfl. read ≥=
      (λfinal. ntimes
        (verifier-query-round-program fr f-fl f-final as fl final
          rounds))))
if len-f: length f-fl ≤ ceil-log clength
  and dg-bound: to-nat dg ≤ maxDegree
for fr f-fl f-final as dg
proof (rule hash-collision-budget-bind-on-outcomes)
show hash-range-budget (ceil-log (maxDegree + 1))
  (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad)
by (rule hash-range-budget-ntimes-receive-composition-fri-commits-bound)
  (rule ceil-log-to-nat-degree-bound[OF dg-bound])
show hash-collision-budget (ceil-log (maxDegree + 1))
  (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad)

```

```

  by (rule hash-collision-budget-ntimes-receive-composition-fri-commits-bound)
    (rule ceil-log-to-nat-degree-bound[OF dg-bound])
fix s fl t
assume out:
  Some (fl, t) ∈
    set-dist (execute
      (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
        (((f × f) list),
         (f, 'a) protocol-channel-scheme) state-monad) s)
have len-fl: length fl ≤ ceil-log (maxDegree + 1)
  using ntimes-receive-composition-fri-commits-outcome[OF out]
    ceil-log-to-nat-degree-bound[OF dg-bound]
  by simp
show hash-range-budget (0 + ?Q)
  (read ≥=
    (λfinal. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds))
  by (rule after-final-range[OF len-f len-fl])
show hash-collision-budget (0 + ?Q)
  (read ≥=
    (λfinal. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds))
  by (rule after-final-coll[OF len-f len-fl])
qed
have after-degree-assert-range:
  hash-range-budget (0 + (ceil-log (maxDegree + 1) + (0 + ?Q)))
  (assert (to-nat dg ≤ maxDegree) ≥=
    (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
      (((f × f) list),
       (f, 'a) protocol-channel-scheme) state-monad) ≥=
    (λfl. read ≥=
      (λfinal. ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds))))))
if len-f: length f-fl ≤ ceil-log clength
for fr f-fl f-final as dg
proof (rule hash-range-budget-bind-on-outcomes)
show hash-range-budget 0
  (assert (to-nat dg ≤ maxDegree) ::
    (unit, (f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-assert)
fix s u t
assume out:
  Some (u, t) ∈
    set-dist (execute
      (assert (to-nat dg ≤ maxDegree) ::
        (unit, (f, 'a) protocol-channel-scheme) state-monad) s)

```

```

have dg-bound: to-nat dg ≤ maxDegree
  using out unfolding assert-def
  by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
show hash-range-budget (ceil-log (maxDegree + 1) + (0 + ?Q))
  ((ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad) ≫=
    (λfl. read ≫=
     (λfinal. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds))))
  by (rule after-comp-fri-range[OF len-f dg-bound])
qed
have after-degree-assert-coll:
  hash-collision-budget (0 + (ceil-log (maxDegree + 1) + (0 + ?Q)))
  (assert (to-nat dg ≤ maxDegree) ≫=
   (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
     (((f × 'f) list),
      (f, 'a) protocol-channel-scheme) state-monad) ≫=
     (λfl. read ≫=
      (λfinal. ntimes
       (verifier-query-round-program fr f-fl f-final as fl final)
       rounds))))))
  if len-f: length f-fl ≤ ceil-log clength
  for fr f-fl f-final as dg
proof (rule hash-collision-budget-bind-on-outcomes)
  show hash-range-budget 0
  (assert (to-nat dg ≤ maxDegree) ::
   (unit, (f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-assert)
  show hash-collision-budget 0
  (assert (to-nat dg ≤ maxDegree) ::
   (unit, (f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-collision-budget-assert)
fix s u t
assume out:
  Some (u, t) ∈
  set-dist (execute
    (assert (to-nat dg ≤ maxDegree) ::
     (unit, (f, 'a) protocol-channel-scheme) state-monad) s)
have dg-bound: to-nat dg ≤ maxDegree
  using out unfolding assert-def
  by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
show hash-range-budget (ceil-log (maxDegree + 1) + (0 + ?Q))
  ((ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad) ≫=
    (λfl. read ≫=
     (λfinal. ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds))))

```

```

      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds)))
  by (rule after-comp-fri-range[OF len-f dg-bound])
show hash-collision-budget (ceil-log (maxDegree + 1) + (0 + ?Q))
  ((ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad) ≫=
   (λfl. read ≫=
    (λfinal. ntimes
     (verifier-query-round-program fr f-fl f-final as fl final)
     rounds))))
  by (rule after-comp-fri-coll[OF len-f dg-bound])
qed
have after-read-dg-range:
  hash-range-budget
  (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))
  (read ≫=
   (λdg. assert (to-nat dg ≤ maxDegree) ≫=
    (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
      (((f × 'f) list),
       (f, 'a) protocol-channel-scheme) state-monad) ≫=
     (λfl. read ≫=
      (λfinal. ntimes
       (verifier-query-round-program fr f-fl f-final as fl final)
       rounds))))))
  if len-f: length f-fl ≤ ceil-log clength
  for fr f-fl f-final as
  by (rule hash-range-budget-bind
    [OF hash-range-budget-read after-degree-assert-range[OF len-f]])
have after-read-dg-coll:
  hash-collision-budget
  (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))
  (read ≫=
   (λdg. assert (to-nat dg ≤ maxDegree) ≫=
    (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
      (((f × 'f) list),
       (f, 'a) protocol-channel-scheme) state-monad) ≫=
     (λfl. read ≫=
      (λfinal. ntimes
       (verifier-query-round-program fr f-fl f-final as fl final)
       rounds))))))
  if len-f: length f-fl ≤ ceil-log clength
  for fr f-fl f-final as
  by (rule hash-collision-budget-bind
    [OF hash-range-budget-read hash-collision-budget-read
     after-degree-assert-range[OF len-f]
     after-degree-assert-coll[OF len-f]])
have after-alpha-range:
  hash-range-budget

```

```

(length spec +
  (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
(mmap ?alpha-steps >>=
  (λas. read >>=
    (λdg. assert (to-nat dg ≤ maxDegree) >>=
      (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
        (((f × 'f) list),
          ('f, 'a) protocol-channel-scheme) state-monad) >>=
        (λfl. read >>=
          (λfinal. ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            rounds)))))))
if len-f: length f-fl ≤ ceil-log clength
for fr f-fl f-final
by (rule hash-range-budget-bind
  [OF alpha-map-range after-read-dg-range[OF len-f]])
have after-alpha-coll:
  hash-collision-budget
  (length spec +
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  (mmap ?alpha-steps >>=
    (λas. read >>=
      (λdg. assert (to-nat dg ≤ maxDegree) >>=
        (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
          (((f × 'f) list),
            ('f, 'a) protocol-channel-scheme) state-monad) >>=
            (λfl. read >>=
              (λfinal. ntimes
                (verifier-query-round-program fr f-fl f-final as fl final)
                rounds)))))))
if len-f: length f-fl ≤ ceil-log clength
for fr f-fl f-final
by (rule hash-collision-budget-bind
  [OF alpha-map-range alpha-map-coll
    after-read-dg-range[OF len-f] after-read-dg-coll[OF len-f]])
have after-read-f-final-range:
  hash-range-budget
  (0 + (length spec +
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  (read >>=
    (λf-final. mmap ?alpha-steps >>=
      (λas. read >>=
        (λdg. assert (to-nat dg ≤ maxDegree) >>=
          (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
            (((f × 'f) list),
              ('f, 'a) protocol-channel-scheme) state-monad) >>=
              (λfl. read >>=
                (λfinal. ntimes
                  (verifier-query-round-program fr f-fl f-final as fl final)
                  rounds)))))))

```

```

rounds))))))
if len-f: length f-fl ≤ ceil-log clength
for fr f-fl
by (rule hash-range-budget-bind
  [OF hash-range-budget-read after-alpha-range[OF len-f]])
have after-read-f-final-coll:
  hash-collision-budget
  (0 + (length spec +
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  (read ≥=
    (λf-final. mmap ?alpha-steps ≥=
      (λas. read ≥=
        (λdg. assert (to-nat dg ≤ maxDegree) ≥=
          (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
            (((f × 'f) list),
              (f, 'a) protocol-channel-scheme) state-monad) ≥=
            (λfl. read ≥=
              (λfinal. ntimes
                (verifier-query-round-program fr f-fl f-final as fl final)
                rounds))))))))
if len-f: length f-fl ≤ ceil-log clength
for fr f-fl
by (rule hash-collision-budget-bind
  [OF hash-range-budget-read hash-collision-budget-read
    after-alpha-range[OF len-f] after-alpha-coll[OF len-f]])
have after-trace-fri-range:
  hash-range-budget
  (ceil-log clength +
    (0 + (length spec +
      (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  ((ntimes receive-trace-fri-commits (ceil-log clength) ::
    (((f × 'f) list),
      (f, 'a) protocol-channel-scheme) state-monad) ≥=
    (λf-fl. read ≥=
      (λf-final. mmap ?alpha-steps ≥=
        (λas. read ≥=
          (λdg. assert (to-nat dg ≤ maxDegree) ≥=
            (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
              (((f × 'f) list),
                (f, 'a) protocol-channel-scheme) state-monad) ≥=
                (λfl. read ≥=
                  (λfinal. ntimes
                    (verifier-query-round-program fr f-fl f-final as fl final)
                    rounds))))))))
for fr
proof (rule hash-range-budget-bind-on-outcomes)
show hash-range-budget (ceil-log clength)
  (ntimes receive-trace-fri-commits (ceil-log clength) ::
    (((f × 'f) list),

```

```

    ('f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-ntimes-receive-trace-fri-commits-bound) simp
fix s f-fl t
assume out:
  Some (f-fl, t) ∈
    set-dist (execute
      (ntimes receive-trace-fri-commits (ceil-log clength) ::
        (((f × 'f) list),
          ('f, 'a) protocol-channel-scheme) state-monad) s)
have len-f: length f-fl ≤ ceil-log clength
  using ntimes-receive-trace-fri-commits-outcome[OF out] by simp
show hash-range-budget
  (0 + (length spec +
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  (read >=
    (λf-final. mmap ?alpha-steps >=
      (λas. read >=
        (λdg. assert (to-nat dg ≤ maxDegree) >=
          (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
            (((f × 'f) list),
              ('f, 'a) protocol-channel-scheme) state-monad) >=
            (λfl. read >=
              (λfinal. ntimes
                (verifier-query-round-program fr f-fl f-final as fl final)
                rounds))))))))
  by (rule after-read-f-final-range[OF len-f])
qed
have after-trace-fri-coll:
  hash-collision-budget
    (ceil-log clength +
      (0 + (length spec +
        (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
    ((ntimes receive-trace-fri-commits (ceil-log clength) ::
      (((f × 'f) list),
        ('f, 'a) protocol-channel-scheme) state-monad) >=
      (λf-fl. read >=
        (λf-final. mmap ?alpha-steps >=
          (λas. read >=
            (λdg. assert (to-nat dg ≤ maxDegree) >=
              (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
                (((f × 'f) list),
                  ('f, 'a) protocol-channel-scheme) state-monad) >=
                (λfl. read >=
                  (λfinal. ntimes
                    (verifier-query-round-program fr f-fl f-final as fl final)
                    rounds))))))))
    for fr
proof (rule hash-collision-budget-bind-on-outcomes)
  show hash-range-budget (ceil-log clength)

```

```

  (ntimes receive-trace-fri-commits (ceil-log clength) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-range-budget-ntimes-receive-trace-fri-commits-bound) simp
show hash-collision-budget (ceil-log clength)
  (ntimes receive-trace-fri-commits (ceil-log clength) ::
    (((f × 'f) list),
     (f, 'a) protocol-channel-scheme) state-monad)
  by (rule hash-collision-budget-ntimes-receive-trace-fri-commits-bound) simp
fix s f-fl t
assume out:
  Some (f-fl, t) ∈
    set-dist (execute
      (ntimes receive-trace-fri-commits (ceil-log clength) ::
        (((f × 'f) list),
         (f, 'a) protocol-channel-scheme) state-monad) s)
have len-f: length f-fl ≤ ceil-log clength
  using ntimes-receive-trace-fri-commits-outcome[OF out] by simp
show hash-range-budget
  (0 + (length spec +
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  (read ≥=
    (λf-final. mmap ?alpha-steps ≥=
      (λas. read ≥=
        (λdg. assert (to-nat dg ≤ maxDegree) ≥=
          (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
            (((f × 'f) list),
             (f, 'a) protocol-channel-scheme) state-monad) ≥=
            (λfl. read ≥=
              (λfinal. ntimes
                (verifier-query-round-program fr f-fl f-final as fl final)
                rounds))))))))
  by (rule after-read-f-final-range[OF len-f])
show hash-collision-budget
  (0 + (length spec +
    (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))
  (read ≥=
    (λf-final. mmap ?alpha-steps ≥=
      (λas. read ≥=
        (λdg. assert (to-nat dg ≤ maxDegree) ≥=
          (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
            (((f × 'f) list),
             (f, 'a) protocol-channel-scheme) state-monad) ≥=
            (λfl. read ≥=
              (λfinal. ntimes
                (verifier-query-round-program fr f-fl f-final as fl final)
                rounds))))))))
  by (rule after-read-f-final-coll[OF len-f])
qed

```



```

have top-range:
  hash-range-budget
    (0 + (ceil-log clength +
      (0 + (length spec +
        (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))))
    (read >=
      (λfr. (ntimes receive-trace-fri-commits (ceil-log clength) ::
        (((f × 'f) list),
          (f, 'a) protocol-channel-scheme) state-monad) >=
      (λf-fl. read >=
        (λf-final. mmap ?alpha-steps >=
          (λas. read >=
            (λdg. assert (to-nat dg ≤ maxDegree) >=
              (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
                (((f × 'f) list),
                  (f, 'a) protocol-channel-scheme) state-monad) >=
                (λfl. read >=
                  (λfinal. ntimes
                    (verifier-query-round-program fr f-fl f-final as fl final)
                    rounds))))))))))
    by (rule hash-range-budget-bind
      [OF hash-range-budget-read after-trace-fri-range])
have top-coll:
  hash-collision-budget
    (0 + (ceil-log clength +
      (0 + (length spec +
        (0 + (0 + (ceil-log (maxDegree + 1) + (0 + ?Q))))))))
    (read >=
      (λfr. (ntimes receive-trace-fri-commits (ceil-log clength) ::
        (((f × 'f) list),
          (f, 'a) protocol-channel-scheme) state-monad) >=
      (λf-fl. read >=
        (λf-final. mmap ?alpha-steps >=
          (λas. read >=
            (λdg. assert (to-nat dg ≤ maxDegree) >=
              (λ-. (ntimes receive-composition-fri-commits (ceil-log (to-nat dg + 1)) ::
                (((f × 'f) list),
                  (f, 'a) protocol-channel-scheme) state-monad) >=
                (λfl. read >=
                  (λfinal. ntimes
                    (verifier-query-round-program fr f-fl f-final as fl final)
                    rounds))))))))))
    by (rule hash-collision-budget-bind
      [OF hash-range-budget-read hash-collision-budget-read
        after-trace-fri-range after-trace-fri-coll])
show ?thesis
using top-coll
unfolding verify-monad-def verifier-query-round-program-def alpha-round-def
  verifier-hash-query-budget-def verifier-header-hash-budget-def

```

by (simp add: algebra-simps)
qed

end

end

theory Soundness-Oracle-Merkle
 imports Soundness-Oracle-Verifier
begin

Merkle-binding collision covers and concrete binding interfaces.

context soundness
begin

definition hash-map-new-output-collision-bad
:: ('f, 'a) protocol-channel-scheme $\Rightarrow$
 (unit list $\times$ ('f, 'a) protocol-channel-scheme) option $\Rightarrow$ bool
where
 hash-map-new-output-collision-bad s out $\longleftrightarrow$
 accepted out $\wedge$
 ($\exists$ result final-state.
 out = Some (result, final-state) $\wedge$
 hash-map-new-output-collision s final-state)

definition hash-map-new-output-collision-bound
:: ('f, 'a) protocol-channel-scheme $\Rightarrow$ bool
where
 hash-map-new-output-collision-bound s $\longleftrightarrow$
 wp-event verify-monad (hash-map-new-output-collision-bad s) s $\leq$
 merkle-binding-error

lemma hash-map-new-output-collision-bound-if-error-ge-one:
 assumes 1 $\leq$ merkle-binding-error
 shows hash-map-new-output-collision-bound s
 unfolding hash-map-new-output-collision-bound-def
 using order-trans[OF wp-event-le-1 assms] .

lemma hash-map-new-output-collision-bad-imp-hash-new-collision-event:
 assumes hash-map-new-output-collision-bad s out
 shows hash-new-collision-event s out
 using assms
 unfolding hash-map-new-output-collision-bad-def hash-new-collision-event-def
 by auto

lemma hash-map-new-output-collision-event-bound-from-verify-hash-budget:
 assumes clean: $\neg$ hash-map-output-collision s
 shows

$wp\text{-}event\ verify\text{-}monad\ (hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bad\ s)\ s \leq$
 $hash\text{-}collision\text{-}budget\text{-}value\ (card\ (hash\text{-}map\text{-}output\text{-}values\ s))$
 $verifier\text{-}hash\text{-}query\text{-}budget$

proof –

have $wp\text{-}event\ verify\text{-}monad\ (hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bad\ s)\ s \leq$
 $wp\text{-}event\ verify\text{-}monad\ (hash\text{-}new\text{-}collision\text{-}event\ s)\ s$
by (rule $wp\text{-}event\text{-}mono$)
(rule $hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bad\text{-}imp\text{-}hash\text{-}new\text{-}collision\text{-}event$)

also have $\dots \leq$
 $hash\text{-}collision\text{-}budget\text{-}value\ (card\ (hash\text{-}map\text{-}output\text{-}values\ s))$
 $verifier\text{-}hash\text{-}query\text{-}budget$
using $hash\text{-}collision\text{-}budget\text{-}verify\text{-}monad\ clean$
unfolding $hash\text{-}collision\text{-}budget\text{-}def$ **by** $blast$
finally show $?thesis$.

qed

lemma $hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bound\text{-}from\text{-}verify\text{-}hash\text{-}budget$:

assumes $clean: \neg\ hash\text{-}map\text{-}output\text{-}collision\ s$
and $error$:
 $hash\text{-}collision\text{-}budget\text{-}value\ (card\ (hash\text{-}map\text{-}output\text{-}values\ s))$
 $verifier\text{-}hash\text{-}query\text{-}budget \leq merkle\text{-}binding\text{-}error$

shows $hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bound\ s$

proof –

have $wp\text{-}event\ verify\text{-}monad\ (hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bad\ s)\ s \leq$
 $hash\text{-}collision\text{-}budget\text{-}value\ (card\ (hash\text{-}map\text{-}output\text{-}values\ s))$
 $verifier\text{-}hash\text{-}query\text{-}budget$
by (rule $hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}event\text{-}bound\text{-}from\text{-}verify\text{-}hash\text{-}budget$
 $[OF\ clean]$)

also have $\dots \leq merkle\text{-}binding\text{-}error$
by (rule $error$)

finally show $?thesis$
unfolding $hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bound\text{-}def$.

qed

lemma $hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bound\text{-}from\text{-}verify\text{-}hash\text{-}budget\text{-}empty$:

assumes $empty\text{-}hashmap: HashMap\ s = fmempty$
and $error$:
 $hash\text{-}collision\text{-}budget\text{-}value\ (card\ (hash\text{-}map\text{-}output\text{-}values\ s))$
 $verifier\text{-}hash\text{-}query\text{-}budget \leq merkle\text{-}binding\text{-}error$

shows $hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bound\ s$

proof –

have $clean: \neg\ hash\text{-}map\text{-}output\text{-}collision\ s$
using $empty\text{-}hashmap$ **unfolding** $hash\text{-}map\text{-}output\text{-}collision\text{-}def$ **by** $simp$
show $?thesis$
by (rule $hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bound\text{-}from\text{-}verify\text{-}hash\text{-}budget$
 $[OF\ clean\ error]$)

qed

lemma $verifier\text{-}initial\text{-}hash\text{-}map\text{-}new\text{-}output\text{-}collision\text{-}bound$:

assumes *error*:

hash-collision-budget-value 0 verifier-hash-query-budget ≤
merkle-binding-error

shows

hash-map-new-output-collision-bound (verifier-initial-state tr)

by (*rule hash-map-new-output-collision-bound-from-verify-hash-budget-empty*)
(*use error in simp-all*)

lemma *verifier-initial-hash-map-new-output-collision-bad-bound-concrete*:

wp-event verify-monad

(*hash-map-new-output-collision-bad (verifier-initial-state tr)*)

(*verifier-initial-state tr*) ≤ *concrete-merkle-binding-error*

proof –

have *clean*:

¬ *hash-map-output-collision (verifier-initial-state tr)*

unfolding *verifier-initial-state-def hash-map-output-collision-def*

by *simp*

have *bound*:

wp-event verify-monad

(*hash-map-new-output-collision-bad (verifier-initial-state tr)*)

(*verifier-initial-state tr*) ≤

hash-collision-budget-value

(*card (hash-map-output-values (verifier-initial-state tr))*)

verifier-hash-query-budget

by (*rule hash-map-new-output-collision-event-bound-from-verify-hash-budget*
[*OF clean*])

show *?thesis*

using *bound unfolding concrete-merkle-binding-error-def* **by** *simp*

qed

lemma *hash-map-output-collision-bad-imp-new-collision-bad-if-initial-clean*:

assumes *clean*: ¬ *hash-map-output-collision s*

and *bad*: *hash-map-output-collision-bad s out*

shows *hash-map-new-output-collision-bad s out*

using *assms*

unfolding *hash-map-output-collision-bad-def*

hash-map-new-output-collision-bad-def

hash-map-new-output-collision-def

by *auto*

lemma *hash-map-output-collision-bound-from-new-collision-bound*:

assumes *clean*: ¬ *hash-map-output-collision s*

and *new-bound*: *hash-map-new-output-collision-bound s*

shows *hash-map-output-collision-bound s*

proof –

have *wp-event verify-monad (hash-map-output-collision-bad s) s ≤*

wp-event verify-monad (hash-map-new-output-collision-bad s) s

by (*rule wp-event-mono*)

(*rule hash-map-output-collision-bad-imp-new-collision-bad-if-initial-clean*)

[OF clean])
 also have ... $\leq$ merkle-binding-error
 using new-bound unfolding hash-map-new-output-collision-bound-def .
 finally show ?thesis
 unfolding hash-map-output-collision-bound-def .
 qed

definition fri-merkle-binding-bad-covered-by-hash-collision
 :: ('f, 'a) protocol-channel-scheme $\Rightarrow$ bool
 where
 fri-merkle-binding-bad-covered-by-hash-collision s $\longleftrightarrow$
 ($\forall$ out $\in$ set-dist (execute verify-monad s).
 fri-merkle-binding-bad s out $\longrightarrow$
 hash-map-output-collision-bad s out)

lemma fri-merkle-binding-bound-from-hash-collision-cover:
 assumes cover: fri-merkle-binding-bad-covered-by-hash-collision s
 and collision-bound: hash-map-output-collision-bound s
 shows fri-merkle-binding-bound s
proof –
 have wp-event verify-monad (fri-merkle-binding-bad s) s $\leq$
 wp-event verify-monad (hash-map-output-collision-bad s) s
 by (rule wp-event-mono-on-support)
 (use cover in
 $\langle$ auto simp: fri-merkle-binding-bad-covered-by-hash-collision-def $\rangle$)
 also have ... $\leq$ merkle-binding-error
 using collision-bound unfolding hash-map-output-collision-bound-def .
 finally show ?thesis
 unfolding fri-merkle-binding-bound-def .
 qed

lemma fri-merkle-binding-bound-from-new-hash-collision-cover:
 assumes clean: $\neg$ hash-map-output-collision s
 and cover: fri-merkle-binding-bad-covered-by-hash-collision s
 and new-bound: hash-map-new-output-collision-bound s
 shows fri-merkle-binding-bound s
proof –
 have collision-bound: hash-map-output-collision-bound s
 by (rule hash-map-output-collision-bound-from-new-collision-bound
 [OF clean new-bound])
 show ?thesis
 by (rule fri-merkle-binding-bound-from-hash-collision-cover
 [OF cover collision-bound])
 qed

lemma hash-map-output-collision-empty-hashmap[simp]:
 assumes HashMap s = fmempty
 shows $\neg$ hash-map-output-collision s
 using assms unfolding hash-map-output-collision-def by simp

lemma *verifier-initial-no-hash-map-output-collision*[simp]:
 $\neg$ *hash-map-output-collision* (*verifier-initial-state tr*)
by *simp*

lemma *verifier-initial-fri-merkle-binding-bound-concrete*:
assumes *cover*:
fri-merkle-binding-bad-covered-by-hash-collision
(*verifier-initial-state tr*)
shows
wp-event verify-monad
(*fri-merkle-binding-bad* (*verifier-initial-state tr*))
(*verifier-initial-state tr*) $\leq$ *concrete-merkle-binding-error*

proof –
let *?s* = *verifier-initial-state tr*
have *wp-event verify-monad* (*fri-merkle-binding-bad ?s*) *?s* $\leq$
wp-event verify-monad (*hash-map-output-collision-bad ?s*) *?s*
by (*rule wp-event-mono-on-support*)
(*use cover in*
 $\langle$ *auto simp: fri-merkle-binding-bad-covered-by-hash-collision-def* $\rangle$)
also have ... $\leq$
wp-event verify-monad (*hash-map-new-output-collision-bad ?s*) *?s*
by (*rule wp-event-mono*)
(*rule hash-map-output-collision-bad-imp-new-collision-bad-if-initial-clean,*
simp, assumption)
also have ... $\leq$ *concrete-merkle-binding-error*
by (*rule verifier-initial-hash-map-new-output-collision-bad-bound-concrete*)
finally show *?thesis* .
qed

lemma *fri-merkle-binding-bound-from-new-hash-collision-cover-empty*:
assumes *HashMap s = fmempty*
and *cover*: *fri-merkle-binding-bad-covered-by-hash-collision s*
and *new-bound*: *hash-map-new-output-collision-bound s*
shows *fri-merkle-binding-bound s*
by (*rule fri-merkle-binding-bound-from-new-hash-collision-cover*
[*OF - cover new-bound*])
(*use assms(1) in simp*)

lemma *fri-bound-layer-tables-length*:
assumes *fri-bound-layer-tables roots tables final-state*
shows *length tables = length roots*
using *assms*
unfolding *fri-bound-layer-tables-def fri-layer-tables-shape-def* **by** *simp*

lemma *fri-bound-layer-tables-root*:
assumes *fri-bound-layer-tables roots tables final-state*
and *j < length roots*
shows *merkle-root-binds-table* (*roots ! j*) (*tables ! j*) *final-state*

```

using assms
unfolding fri-bound-layer-tables-def fri-roots-bind-tables-def by simp

lemma fri-bound-layer-tables-table-length:
  assumes fri-bound-layer-tables roots tables final-state
    and j < length roots
  shows
    length (tables ! j) =
      fri-layer-lengths (length roots) (clength * scale) ! j
  using assms
  unfolding fri-bound-layer-tables-def fri-layer-tables-shape-def
  by simp

lemma fri-bound-layer-tables-same-root-collision-imp-hash-collision-bad:
  assumes out: out = Some (result, final-state)
    and bound1: fri-bound-layer-tables roots tables final-state
    and bound2: fri-bound-layer-tables roots tables' final-state
    and j-bound: j < length roots
    and nonempty: tables ! j ≠ []
    and neq: tables ! j ≠ tables' ! j
  shows hash-map-output-collision-bad s out
proof –
  have bind1:
    merkle-root-binds-table (roots ! j) (tables ! j) final-state
    by (rule fri-bound-layer-tables-root[OF bound1 j-bound])
  have bind2:
    merkle-root-binds-table (roots ! j) (tables' ! j) final-state
    by (rule fri-bound-layer-tables-root[OF bound2 j-bound])
  have same-len: length (tables ! j) = length (tables' ! j)
    using fri-bound-layer-tables-table-length[OF bound1 j-bound]
      fri-bound-layer-tables-table-length[OF bound2 j-bound]
    by simp
  show ?thesis
    by (rule accepted-inconsistent-same-length-merkle-bindings-imp-hash-collision-bad
      [OF out bind1 bind2 same-len nonempty neq])
qed

lemma fri-layer-opening-matches-bound-tableD:
  assumes match:
    fri-layer-opening-matches-bound-table query-idx layer-idx tables
      layer-chunks
  obtains len idx xp xp-path xn xn-path
  where layer-idx < length tables
    and length layer-chunks = length tables
    and len =
      fri-layer-lengths (length tables) (clength * scale) ! layer-idx
    and idx =
      fri-layer-indices (length tables) query-idx (clength * scale) !
        layer-idx

```

```

and fri-layer-opening-chunk len xp xp-path xn xn-path
  (layer-chunks ! layer-idx)
and fri-opening-matches-table len idx (tables ! layer-idx) xp xn
using match
unfolding fri-layer-opening-matches-bound-table-def Let-def
by blast

```

lemma *fri-round-openings-match-tablesD*:

```

assumes fri-round-openings-match-tables tables query-idx layer-chunks
and j < length tables
shows
  fri-layer-opening-matches-bound-table query-idx j tables layer-chunks
using assms unfolding fri-round-openings-match-tables-def by simp

```

lemma *fri-all-round-openings-match-tablesD*:

```

assumes
  fri-all-round-openings-match-tables tables query-idxs round-layers
and i < length query-idxs
shows
  fri-round-openings-match-tables tables (query-idxs ! i)
  (round-layers ! i)
using assms unfolding fri-all-round-openings-match-tables-def by simp

```

lemma *trace-fri-tables-bound-opening-matches*:

```

assumes bound:
  trace-fri-tables-bound s out trace-roots trace-bs trace-final
  query-idxs trace-round-layers trace-tables trace-final-table
and i-bound: i < length query-idxs
and j-bound: j < length trace-tables
shows
  fri-layer-opening-matches-bound-table (query-idxs ! i) j
  trace-tables (trace-round-layers ! i)
proof –
  have rounds:
    fri-all-round-openings-match-tables trace-tables query-idxs
    trace-round-layers
    using bound unfolding trace-fri-tables-bound-def by blast
  from fri-all-round-openings-match-tablesD[OF rounds i-bound]
  have fri-round-openings-match-tables trace-tables (query-idxs ! i)
    (trace-round-layers ! i) .
  then show ?thesis
    using j-bound by (rule fri-round-openings-match-tablesD)
qed

```

lemma *composition-fri-tables-bound-opening-matches*:

```

assumes bound:
  composition-fri-tables-bound s out dg composition-roots composition-bs
  composition-final query-idxs composition-round-layers
  composition-tables composition-final-table

```


and $i\text{-bound}$: $i < \text{length } \text{query-idxs}$
and $j\text{-bound}$: $j < \text{length } \text{composition-tables}$
shows
 $\text{fri-layer-opening-matches-bound-table } (\text{query-idxs } ! i) j$
 $\text{composition-tables } (\text{composition-round-layers } ! i)$
proof –
have rounds :
 $\text{fri-all-round-openings-match-tables } \text{composition-tables } \text{query-idxs}$
 $\text{composition-round-layers}$
using $\text{bound unfolding composition-fri-tables-bound-def}$ **by** blast
from $\text{fri-all-round-openings-match-tablesD}[OF \text{ rounds } i\text{-bound}]$
have $\text{fri-round-openings-match-tables } \text{composition-tables } (\text{query-idxs } ! i)$
 $(\text{composition-round-layers } ! i)$.
then show $?thesis$
using $j\text{-bound}$ **by** $(\text{rule fri-round-openings-match-tablesD})$
qed

lemma $\text{fri-merkle-binding-bad-imp-accepted}$:
assumes $\text{fri-merkle-binding-bad } s \text{ out}$
shows accepted out
using $\text{assms unfolding fri-merkle-binding-bad-def}$ **by** simp

lemma $\text{fri-merkle-binding-bad-mono-accepted}$:
 $\text{wp-event verify-monad } (\text{fri-merkle-binding-bad } s) s \leq$
 $\text{wp-event verify-monad accepted } s$
by $(\text{rule wp-event-mono})$
 $(\text{simp add: fri-merkle-binding-bad-imp-accepted})$

lemma $\text{accepted-good-or-fri-merkle-binding-bad}$:
assumes accepted out
shows $\text{fri-merkle-binding-good } s \text{ out} \vee \text{fri-merkle-binding-bad } s \text{ out}$
using $\text{assms unfolding fri-merkle-binding-bad-def}$ **by** blast

lemma $\text{fri-merkle-binding-goodE}$:
assumes $\text{good: fri-merkle-binding-good } s \text{ out}$
and acc: accepted out
obtains $\text{trace-table } \text{composition-table as } \text{query-idxs}$
 $\text{trace-roots } \text{trace-bs } \text{trace-final dg } \text{composition-roots } \text{composition-bs}$
 $\text{composition-final } \text{trace-round-layers } \text{composition-round-layers}$
 $\text{trace-tables } \text{composition-tables } \text{trace-final-table } \text{composition-final-table}$
where
 $\text{accepted-with-bound-tables } s \text{ out } \text{trace-table } \text{composition-table as}$
 query-idxs
and $\text{accepted-fri-opening-transcript } s \text{ out } \text{trace-roots } \text{trace-bs}$
 $\text{trace-final dg } \text{composition-roots } \text{composition-bs } \text{composition-final}$
 $\text{query-idxs } \text{trace-round-layers } \text{composition-round-layers}$
and $\text{trace-fri-tables-bound } s \text{ out } \text{trace-roots } \text{trace-bs } \text{trace-final}$
 $\text{query-idxs } \text{trace-round-layers } \text{trace-tables } \text{trace-final-table}$
and $\text{composition-fri-tables-bound } s \text{ out dg } \text{composition-roots}$

```

    composition-bs composition-final query-idxs composition-round-layers
    composition-tables composition-final-table
    and fri-initial-table-agrees trace-table trace-tables
    and fri-initial-table-agrees composition-table composition-tables
    using good acc unfolding fri-merkle-binding-good-def by blast

```

end

end

```

theory Soundness-Oracle
  imports Soundness-Oracle-Merkle
begin

```

Compatibility umbrella for random-oracle soundness infrastructure.

end

```

theory Controlled-RO
  imports
    Soundness-Oracle
begin

```

```

context soundness
begin

```

Controlled random-oracle programs provide a capability-safe interface for attacker stages. The constructors expose ordinary probabilistic computation and oracle queries, but no operation that can directly modify the oracle map or protocol-local channel fields.

```

inductive controlled-ro-program
  :: nat  $\Rightarrow$ 
    ('r, 'f protocol-channel) state-monad  $\Rightarrow$  bool
where
  Return[intro]:
    controlled-ro-program 0 (return x)
| Fail[intro]:
    controlled-ro-program 0 throw
| Sample[intro]:
    controlled-ro-program 0 (lift ( $\lambda$ -. d))
| Query[intro]:
    ( $\forall y.$  controlled-ro-program q (k y))  $\Rightarrow$ 
    controlled-ro-program (Suc q)
    ((hash x :: ('f, 'f protocol-channel) state-monad)
      $\ggg$  k)
| Bind[intro]:
    controlled-ro-program q m  $\Rightarrow$ 
    ( $\forall x.$  controlled-ro-program r (k x))  $\Rightarrow$ 
    controlled-ro-program (q + r) (m  $\ggg$  k)

```

| *Weaken:*
 $\text{controlled-ro-program } q \ m \implies q \leq r \implies \text{controlled-ro-program } r \ m$

lemma *hash-map-preserving-throw:*
hash-map-preserving
 $(\text{throw} :: ('r, 'f \text{ protocol-channel}) \text{ state-monad})$
unfolding *hash-map-preserving-def*
by (*simp add: throw-no-outcome*)

lemma *hash-map-preserving-return:*
hash-map-preserving
 $(\text{return } x :: ('r, 'f \text{ protocol-channel}) \text{ state-monad})$
unfolding *hash-map-preserving-def* **by** *simp*

lemma *hash-map-preserving-state-independent-sample:*
hash-map-preserving
 $(\text{lift } (\lambda-. d) :: ('r, 'f \text{ protocol-channel}) \text{ state-monad})$
unfolding *hash-map-preserving-def lift.rep-eq lift-dist-def*
by (*auto simp: set-dist-dist-map*)

definition *protocol-fields-preserving*
 $:: ('r, 'f \text{ protocol-channel}) \text{ state-monad} \Rightarrow \text{bool}$
where
 $\text{protocol-fields-preserving } m \longleftrightarrow$
 $(\forall s \ x \ t.$
 $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \longrightarrow$
 $P\text{State } t = P\text{State } s \wedge$
 $P\text{Transcript } t = P\text{Transcript } s \wedge$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s \wedge$
 $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = P\text{QueryCounter } s)$

lemma *controlled-ro-program-preserves-protocol-fields:*
assumes *controlled-ro-program* $q \ m$
shows *protocol-fields-preserving* m
using *assms*
proof (*induction rule: controlled-ro-program.induct*)
case (*Return* x)
then show *?case*
unfolding *protocol-fields-preserving-def* **by** *simp*
next
case *Fail*
then show *?case*
unfolding *protocol-fields-preserving-def*
by (*simp add: throw-no-outcome*)
next

```

case (Sample d)
then show ?case
  unfolding protocol-fields-preserving-def lift.rep-eq lift-dist-def
  by (auto simp: set-dist-dist-map)
next
case (Query q k x)
show ?case
  unfolding protocol-fields-preserving-def
proof (intro allI impI)
  fix s result t
  assume outcome:
    Some (result, t) ∈
      set-dist
      (execute
        ((hash x :: ('f, 'f protocol-channel) state-monad) >>= k) s)
  then obtain h u where
    hash-out:
      Some (h, u) ∈
        set-dist
        (execute (hash x :: ('f, 'f protocol-channel) state-monad) s)
  and rest: Some (result, t) ∈ set-dist (execute (k h) u)
  by (auto elim!: set-dist-bindE)
have fields-u:
  PState u = PState s ∧
  PTranscript u = PTranscript s ∧
  PTraceFriCounter u = PTraceFriCounter s ∧
  PCompositionFriCounter u = PCompositionFriCounter s ∧
  PAlphaCounter u = PAlphaCounter s ∧
  PQueryCounter u = PQueryCounter s
  using hash-channel-preserves[OF hash-out] by simp
have preserving-k: protocol-fields-preserving (k h)
  using Query.IH by blast
have fields-t:
  PState t = PState u ∧
  PTranscript t = PTranscript u ∧
  PTraceFriCounter t = PTraceFriCounter u ∧
  PCompositionFriCounter t = PCompositionFriCounter u ∧
  PAlphaCounter t = PAlphaCounter u ∧
  PQueryCounter t = PQueryCounter u
  using preserving-k rest
  unfolding protocol-fields-preserving-def by blast
show
  PState t = PState s ∧
  PTranscript t = PTranscript s ∧
  PTraceFriCounter t = PTraceFriCounter s ∧
  PCompositionFriCounter t = PCompositionFriCounter s ∧
  PAlphaCounter t = PAlphaCounter s ∧
  PQueryCounter t = PQueryCounter s
  using fields-u fields-t by simp

```

```

qed
next
  case (Bind q m r k)
  show ?case
    unfolding protocol-fields-preserving-def
  proof (intro allI impI)
    fix s result t
    assume outcome:
      Some (result, t) ∈ set-dist (execute (m ≫ k) s)
    then obtain x u where
      first: Some (x, u) ∈ set-dist (execute m s)
      and rest: Some (result, t) ∈ set-dist (execute (k x) u)
      by (auto elim!: set-dist-bindE)
    have fields-u:
      PState u = PState s ∧
      PTranscript u = PTranscript s ∧
      PTraceFriCounter u = PTraceFriCounter s ∧
      PCompositionFriCounter u = PCompositionFriCounter s ∧
      PAlphaCounter u = PAlphaCounter s ∧
      PQueryCounter u = PQueryCounter s
    using Bind.IH(1) first
    unfolding protocol-fields-preserving-def by blast
    have preserving-k: protocol-fields-preserving (k x)
    using Bind.IH(2) by blast
    have fields-t:
      PState t = PState u ∧
      PTranscript t = PTranscript u ∧
      PTraceFriCounter t = PTraceFriCounter u ∧
      PCompositionFriCounter t = PCompositionFriCounter u ∧
      PAlphaCounter t = PAlphaCounter u ∧
      PQueryCounter t = PQueryCounter u
    using preserving-k rest
    unfolding protocol-fields-preserving-def by blast
  show
    PState t = PState s ∧
    PTranscript t = PTranscript s ∧
    PTraceFriCounter t = PTraceFriCounter s ∧
    PCompositionFriCounter t = PCompositionFriCounter s ∧
    PAlphaCounter t = PAlphaCounter s ∧
    PQueryCounter t = PQueryCounter s
    using fields-u fields-t by simp
  qed
next
  case (Weaken q m r)
  then show ?case by simp
qed

```

lemma hash-range-budget-throw:
 hash-range-budget 0

```

    (throw :: ('r, 'f protocol-channel) state-monad)
unfolding hash-range-budget-def
by (simp add: throw-no-outcome)

lemma hash-collision-budget-throw:
  hash-collision-budget 0
  (throw :: ('r, 'f protocol-channel) state-monad)
unfolding hash-collision-budget-def wp-event-def wp-def
  throw.rep-eq dist-throw-def dist-delta-dist delta-map-def
  hash-collision-budget-value-def hash-new-collision-event-def
by simp

lemma hash-target-program-throw:
  hash-target-program B 0
  (throw :: ('r, 'f protocol-channel) state-monad)
by (rule hash-map-preserving-imp-hash-target-program-zero)
    (rule hash-map-preserving-throw)

lemma hash-range-budget-state-independent-sample:
  hash-range-budget 0
  (lift (λ-. d) ::
    ('r, 'f protocol-channel) state-monad)
unfolding hash-range-budget-def lift.rep-eq lift-dist-def
by (auto simp: set-dist-dist-map hash-map-output-values-def)

lemma hash-map-preserving-imp-hash-collision-budget-zero:
  fixes m :: ('r, 'f protocol-channel) state-monad
  assumes preserving: hash-map-preserving m
  shows hash-collision-budget 0 m
  unfolding hash-collision-budget-def
proof (intro allI impI)
  fix s :: 'f protocol-channel
  assume clean: ¬ hash-map-output-collision s
  have no-event:
    ∧ out. out ∈ set-dist (execute m s) ⇒
      ¬ hash-new-collision-event s out
proof -
  fix out
  assume support: out ∈ set-dist (execute m s)
  show ¬ hash-new-collision-event s out
  proof (cases out)
  case None
  then show ?thesis
    unfolding hash-new-collision-event-def by simp
  next
  case (Some xt)
  then obtain x t where xt: xt = (x, t)
    by (cases xt) simp
  have same: HashMap t = HashMap s

```

```

    using preserving support Some xt
    unfolding hash-map-preserving-def by blast
show ?thesis
    using clean same Some xt
    unfolding hash-new-collision-event-def
      hash-map-new-output-collision-def hash-map-output-collision-def
    by simp
qed
qed
have no-event-dom:
   $\bigwedge out. out \in dom (dist (execute\ m\ s)) \implies$ 
     $\neg hash\_new\_collision\_event\ s\ out$ 
  using no-event unfolding set-dist-def by blast
have wp-event m (hash-new-collision-event s) s = 0
  unfolding wp-event-def wp-def dist-expect-def
  by (intro sum.neutral) (simp add: no-event-dom)
then show
  wp-event m (hash-new-collision-event s) s  $\leq$ 
    hash-collision-budget-value
      (card (hash-map-output-values s)) 0
  unfolding hash-collision-budget-value-def by simp
qed

lemma hash-collision-budget-state-independent-sample:
  hash-collision-budget 0
  (lift ( $\lambda\cdot. d$ ) ::
    ('r, 'f protocol-channel) state-monad)
by (rule hash-map-preserving-imp-hash-collision-budget-zero)
  (rule hash-map-preserving-state-independent-sample)

lemma hash-target-program-state-independent-sample:
  hash-target-program B 0
  (lift ( $\lambda\cdot. d$ ) ::
    ('r, 'f protocol-channel) state-monad)
by (rule hash-map-preserving-imp-hash-target-program-zero)
  (rule hash-map-preserving-state-independent-sample)

lemma controlled-ro-program-extension:
  assumes controlled-ro-program q m
  shows hash-extension-preserving m
  using assms
proof (induction rule: controlled-ro-program.induct)
  case (Return x)
  then show ?case by (rule hash-extension-preserving-return)
next
  case Fail
  then show ?case
  by (rule hash-map-preserving-imp-hash-extension-preserving)
    (rule hash-map-preserving-throw)

```

```

next
  case (Sample d)
  then show ?case
    by (rule hash-map-preserving-imp-hash-extension-preserving)
      (rule hash-map-preserving-state-independent-sample)
next
  case (Query q k x)
  have ext-k:  $\bigwedge y. \text{hash-extension-preserving } (k y)$ 
    using Query.IH by blast
  show ?case
    by (rule hash-extension-preserving-bind)
      (rule hash-extension-preserving-hash, rule ext-k)
next
  case (Bind q m r k)
  have ext-k:  $\bigwedge x. \text{hash-extension-preserving } (k x)$ 
    using Bind.IH(2) by blast
  show ?case
    by (rule hash-extension-preserving-bind[OF Bind.IH(1) ext-k])
next
  case (Weaken q m r)
  then show ?case by simp
qed

lemma controlled-ro-program-range:
  assumes controlled-ro-program q m
  shows hash-range-budget q m
  using assms
proof (induction rule: controlled-ro-program.induct)
  case (Return x)
  then show ?case by (rule hash-range-budget-return)
next
  case Fail
  then show ?case by (rule hash-range-budget-throw)
next
  case (Sample d)
  then show ?case by (rule hash-range-budget-state-independent-sample)
next
  case (Query q k x)
  have range-k:  $\bigwedge y. \text{hash-range-budget } q (k y)$ 
    using Query.IH by blast
  have hash-range-budget (1 + q)
    ((hash x :: ('f, 'f protocol-channel) state-monad)
      $\gg=$  k)
    by (rule hash-range-budget-bind)
      (rule hash-range-budget-hash, rule range-k)
  then show ?case by simp
next
  case (Bind q m r k)
  have range-k:  $\bigwedge x. \text{hash-range-budget } r (k x)$ 

```



```

    using Bind.IH(2) by blast
  show ?case
    by (rule hash-range-budget-bind[OF Bind.IH(1) range-k])
next
  case (Weaken q m r)
  show ?case
    by (rule hash-range-budget-mono[OF Weaken.hyps(2) Weaken.IH])
qed

lemma controlled-ro-program-collision:
  assumes controlled-ro-program q m
  shows hash-collision-budget q m
  using assms
proof (induction rule: controlled-ro-program.induct)
  case (Return x)
  then show ?case by (rule hash-collision-budget-return)
next
  case Fail
  then show ?case by (rule hash-collision-budget-throw)
next
  case (Sample d)
  then show ?case by (rule hash-collision-budget-state-independent-sample)
next
  case (Query q k x)
  have range-k:  $\bigwedge y. \text{hash-range-budget } q \ (k \ y)$ 
    using Query(1) by (blast intro: controlled-ro-program-range)
  have collision-k:  $\bigwedge y. \text{hash-collision-budget } q \ (k \ y)$ 
    using Query.IH by blast
  have hash-collision-budget (1 + q)
    ((hash x :: ('f, 'f protocol-channel) state-monad)
      $\ggg$  k)
    by (rule hash-collision-budget-bind)
      (rule hash-range-budget-hash, rule hash-collision-budget-hash,
       rule range-k, rule collision-k)
  then show ?case by simp
next
  case (Bind q m r k)
  have range-m: hash-range-budget q m
    by (rule controlled-ro-program-range[OF Bind(1)])
  have range-k:  $\bigwedge x. \text{hash-range-budget } r \ (k \ x)$ 
    using Bind(3) by (blast intro: controlled-ro-program-range)
  have collision-k:  $\bigwedge x. \text{hash-collision-budget } r \ (k \ x)$ 
    using Bind.IH(2) by blast
  show ?case
    by (rule hash-collision-budget-bind)
      (rule range-m, rule Bind.IH(1), rule range-k, rule collision-k)
next
  case (Weaken q m r)
  show ?case

```

```

    by (rule hash-collision-budget-mono[OF Weaken.hyps(2) Weaken.IH])
qed

lemma controlled-ro-program-target:
  assumes controlled-ro-program q m
  shows hash-target-program B q m
  using assms
proof (induction rule: controlled-ro-program.induct)
  case (Return x)
  then show ?case by (rule hash-target-program-return)
next
  case Fail
  then show ?case by (rule hash-target-program-throw)
next
  case (Sample d)
  then show ?case by (rule hash-target-program-state-independent-sample)
next
  case (Query q k x)
  have target-k:  $\bigwedge y. \text{hash-target-program } B \ q \ (k \ y)$ 
    using Query.IH by blast
  have hash-target-program B (1 + q)
    ((hash x :: ('f, 'f protocol-channel) state-monad)
      $\ggg$  k)
    by (rule hash-target-program-bind)
    (rule hash-target-program-hash, rule target-k)
  then show ?case by simp
next
  case (Bind q m r k)
  have target-k:  $\bigwedge x. \text{hash-target-program } B \ r \ (k \ x)$ 
    using Bind.IH(2) by blast
  show ?case
    by (rule hash-target-program-bind[OF Bind.IH(1) target-k])
next
  case (Weaken q m r)
  show ?case
    by (rule hash-target-program-mono[OF Weaken.hyps(2) Weaken.IH])
qed

```

```

definition controlled-ro-admissible
  :: nat  $\Rightarrow$ 
    ('r, 'f protocol-channel) state-monad  $\Rightarrow$  bool
where
  controlled-ro-admissible q m  $\longleftrightarrow$ 
    hash-extension-preserving m  $\wedge$ 
    hash-range-budget q m  $\wedge$ 
    hash-collision-budget q m  $\wedge$ 
    ( $\forall B. \text{hash-target-program } B \ q \ m$ )

```

lemma *controlled-ro-program-admissible:*

assumes *controlled-ro-program* $q\ m$
shows *controlled-ro-admissible* $q\ m$
unfolding *controlled-ro-admissible-def*
using *controlled-ro-program-extension* [*OF* *assms*]
controlled-ro-program-range [*OF* *assms*]
controlled-ro-program-collision [*OF* *assms*]
controlled-ro-program-target [*OF* *assms*]
by *blast*

definition *hash-relation-hit*
 $:: ('f\ \text{protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}) \Rightarrow$
 $'f\ \text{protocol-channel} \Rightarrow 'f\ \text{protocol-channel} \Rightarrow \text{bool}$
where
hash-relation-hit $R\ \text{initial}\ \text{final} \longleftrightarrow$
 $(\exists x\ y.$
 $\text{fmlookup}\ (\text{HashMap}\ \text{initial})\ x = \text{None} \wedge$
 $\text{fmlookup}\ (\text{HashMap}\ \text{final})\ x = \text{Some}\ y \wedge$
 $R\ x\ y)$

definition *hash-relation-hit-event*
 $:: ('f\ \text{protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}) \Rightarrow$
 $'f\ \text{protocol-channel} \Rightarrow$
 $('r \times 'f\ \text{protocol-channel})\ \text{option} \Rightarrow \text{bool}$
where
hash-relation-hit-event $R\ \text{initial}\ \text{out} \longleftrightarrow$
 $(\text{case}\ \text{out}\ \text{of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some}\ (-, \text{final}) \Rightarrow \text{hash-relation-hit}\ R\ \text{initial}\ \text{final})$

lemma *hash-relation-hit-after-update*:
hash-relation-hit $R\ s$
 $(s(\text{HashMap} := \text{fmupd}\ x\ y\ (\text{HashMap}\ s))) \longleftrightarrow$
 $\text{fmlookup}\ (\text{HashMap}\ s)\ x = \text{None} \wedge R\ x\ y$

proof
assume *hit*:
hash-relation-hit $R\ s$
 $(s(\text{HashMap} := \text{fmupd}\ x\ y\ (\text{HashMap}\ s)))$
then obtain $k\ z$ **where** *fresh*:
 $\text{fmlookup}\ (\text{HashMap}\ s)\ k = \text{None}$
and *lookup*:
 $\text{fmlookup}\ (\text{fmupd}\ x\ y\ (\text{HashMap}\ s))\ k = \text{Some}\ z$
and *rel*: $R\ k\ z$
by (*auto simp: hash-relation-hit-def*)
have $k = x$
proof (*rule ccontr*)
assume $k \neq x$
then have $\text{fmlookup}\ (\text{HashMap}\ s)\ k = \text{Some}\ z$
using *lookup* **by** *simp*
then show *False* **using** *fresh* **by** *simp*

```

qed
then show  $\text{fmlookup } (\text{HashMap } s) \ x = \text{None} \wedge R \ x \ y$ 
  using fresh lookup rel by simp
next
assume  $\text{fmlookup } (\text{HashMap } s) \ x = \text{None} \wedge R \ x \ y$ 
then show
   $\text{hash-relation-hit } R \ s$ 
  ( $s \lfloor \text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s) \rfloor$ )
  unfolding hash-relation-hit-def by auto
qed

lemma wp-hash-relation-hit:
  fixes  $x :: 'f \text{ protocol-hash-input}$ 
  and  $s :: 'f \text{ protocol-channel}$ 
  shows
     $\text{wp-event } (\text{hash } x) \ (\text{hash-relation-hit-event } R \ s) \ s =$ 
    ( $\text{if } \text{fmlookup } (\text{HashMap } s) \ x = \text{None}$ 
     then  $\text{nnreal } (\text{card } \{y. R \ x \ y\}) / \text{nnreal size}$ 
     else  $0$ )
  proof (cases  $\text{fmlookup } (\text{HashMap } s) \ x$ )
  case None
  have indicator:
    ( $\lambda y. \text{if } \text{hash-relation-hit } R \ s$ 
     ( $s \lfloor \text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s) \rfloor$ )
     then  $1$  else  $0 :: \text{prob}$ ) =
    ( $\lambda y. \text{if } y \in \{y. R \ x \ y\}$  then  $1$  else  $0$ )
  by (rule ext)
  (simp add: hash-relation-hit-after-update None)
  have  $\text{wp-event } (\text{hash } x) \ (\text{hash-relation-hit-event } R \ s) \ s =$ 
     $\text{dist-expect } (\text{hash-dist } x \ s)$ 
    ( $\lambda y. \text{if } y \in \{y. R \ x \ y\}$  then  $1$  else  $0$ )
  unfolding wp-event-def hash-relation-hit-event-def
  by (simp add: wp-hash indicator)
  also have  $\dots = \text{nnreal } (\text{card } \{y. R \ x \ y\}) / \text{nnreal size}$ 
  by (rule dist-expect-hash-dist-fresh-indicator[OF None])
  finally show ?thesis using None by simp
next
case (Some y)
have no-hit:
   $\neg \text{hash-relation-hit } R \ s$ 
  ( $s \lfloor \text{HashMap} := \text{fmupd } x \ y \ (\text{HashMap } s) \rfloor$ )
  by (simp add: hash-relation-hit-after-update Some)
show ?thesis
  unfolding wp-event-def hash-relation-hit-event-def
  by (simp add: wp-hash Some hash-dist-def option-default-dist-def no-hit)
qed

definition hash-relation-budget-value ::  $\text{nat} \Rightarrow \text{nat} \Rightarrow \text{prob}$ 
  where

```

hash-relation-budget-value $b\ q =$
 $\text{nnreal } (q * b) / \text{nnreal size}$

definition *hash-relation-budget*
 $:: ('f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}) \Rightarrow$
 $\text{nat} \Rightarrow \text{nat} \Rightarrow$
 $('r, 'f \text{ protocol-channel}) \text{ state-monad} \Rightarrow \text{bool}$
where
 $\text{hash-relation-budget } R\ b\ q\ m \longleftrightarrow$
 $(\forall s.$
 $\text{wp-event } m\ (\text{hash-relation-hit-event } R\ s)\ s \leq$
 $\text{hash-relation-budget-value } b\ q)$

definition *hash-relation-program*
 $:: ('f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}) \Rightarrow$
 $\text{nat} \Rightarrow \text{nat} \Rightarrow$
 $('r, 'f \text{ protocol-channel}) \text{ state-monad} \Rightarrow \text{bool}$
where
 $\text{hash-relation-program } R\ b\ q\ m \longleftrightarrow$
 $\text{hash-extension-preserving } m \wedge \text{hash-relation-budget } R\ b\ q\ m$

lemma *hash-relation-budget-value-add:*
 $\text{hash-relation-budget-value } b\ q + \text{hash-relation-budget-value } b\ r =$
 $\text{hash-relation-budget-value } b\ (q + r)$
unfolding *hash-relation-budget-value-def*
by (*simp add: add-divide-nnreal algebra-simps*)

lemma *hash-relation-budget-value-mono:*
assumes $q \leq r$
shows $\text{hash-relation-budget-value } b\ q \leq \text{hash-relation-budget-value } b\ r$
unfolding *hash-relation-budget-value-def*
apply (*subst nn2real-le-iff[symmetric]*)
using *assms* **by** (*simp add: divide-right-mono mult-right-mono*)

lemma *hash-relation-budget-value-mono-left:*
assumes $b \leq c$
shows $\text{hash-relation-budget-value } b\ q \leq \text{hash-relation-budget-value } c\ q$
unfolding *hash-relation-budget-value-def*
apply (*subst nn2real-le-iff[symmetric]*)
using *assms* **by** (*simp add: divide-right-mono mult-left-mono*)

lemma *hash-relation-hit-event-step-bound:*
fixes $m :: ('r, 'f \text{ protocol-channel}) \text{ state-monad}$
assumes *ext-st:* $s \leq t$
and *ext-m:* $\text{hash-extension-preserving } m$
shows
 $\text{wp-event } m\ (\text{hash-relation-hit-event } R\ s)\ t \leq$
 $(\text{if } \text{hash-relation-hit } R\ s\ t \text{ then } 1$
 $\text{else } \text{wp-event } m\ (\text{hash-relation-hit-event } R\ t)\ t)$

```

proof (cases hash-relation-hit R s t)
  case True
  then show ?thesis
    using wp-event-le-1[of m hash-relation-hit-event R s t] by simp
next
  case False
  have mono:
    wp-event m (hash-relation-hit-event R s) t ≤
      wp-event m (hash-relation-hit-event R t) t
  proof (rule wp-event-mono-on-support)
    fix out
    assume support: out ∈ set-dist (execute m t)
    and hit-s: hash-relation-hit-event R s out
    show hash-relation-hit-event R t out
  proof (cases out)
    case None
    then show ?thesis
      using hit-s unfolding hash-relation-hit-event-def by simp
  next
    case (Some xu)
    then obtain x u where xu: xu = (x, u)
    by (cases xu) simp
    have out: Some (x, u) ∈ set-dist (execute m t)
    using support Some xu by simp
    have ext-tu: t ≤ u
    using ext-m out unfolding hash-extension-preserving-def by blast
    from hit-s obtain k y where fresh-s:
      fmlookup (HashMap s) k = None
    and lookup-u: fmlookup (HashMap u) k = Some y
    and rel: R k y
    unfolding hash-relation-hit-event-def hash-relation-hit-def
    using Some xu by auto
  show ?thesis
  proof (cases fmlookup (HashMap t) k)
    case None
    then show ?thesis
      using Some xu lookup-u rel
      unfolding hash-relation-hit-event-def hash-relation-hit-def
      by auto
  next
    case (Some z)
    have lookup-u-z: fmlookup (HashMap u) k = Some z
    by (rule hash-extension-lookup[OF Some ext-tu])
    have z = y using lookup-u lookup-u-z by simp
    then have hash-relation-hit R s t
    unfolding hash-relation-hit-def
    using fresh-s Some rel by blast
    then show ?thesis using False by contradiction
  qed

```

```

qed
qed
then show ?thesis using False by simp
qed

lemma hash-relation-budget-bind:
  fixes m :: ('x, 'f protocol-channel) state-monad
    and k :: 'x  $\Rightarrow$  ('y, 'f protocol-channel) state-monad
  assumes m-ext: hash-extension-preserving m
    and m-budget: hash-relation-budget R b q m
    and k-ext:  $\bigwedge x. \text{hash-extension-preserving } (k\ x)$ 
    and k-budget:  $\bigwedge x. \text{hash-relation-budget } R\ b\ r\ (k\ x)$ 
  shows hash-relation-budget R b (q + r) (m  $\ggg$  k)
  unfolding hash-relation-budget-def
proof (intro allI)
  fix s :: 'f protocol-channel
  let ?E = hash-relation-hit-event R s
  let ?Tail =
     $\lambda out. \text{case out of}$ 
     $\text{None} \Rightarrow 0$ 
     $| \text{Some } (x, t) \Rightarrow$ 
     $\text{if hash-relation-hit } R\ s\ t \text{ then } 0$ 
     $\text{else wp-event } (k\ x)\ (\text{hash-relation-hit-event } R\ t)\ t$ 
  let ?R =  $\lambda out. (\text{if } ?E\ out \text{ then } 1 \text{ else } 0) + ?Tail\ out$ 
  have split:
     $\text{wp-event } (m\ \ggg\ k)\ ?E\ s \leq \text{wp } m\ ?R\ s$ 
  proof -
    have exact:
       $\text{wp-event } (m\ \ggg\ k)\ ?E\ s =$ 
       $\text{wp } m$ 
       $(\lambda out. \text{case out of}$ 
       $\text{None} \Rightarrow 0$ 
       $| \text{Some } (x, t) \Rightarrow \text{wp-event } (k\ x)\ ?E\ t)\ s$ 
    unfolding wp-event-def hash-relation-hit-event-def
    by (simp add: wpsimps)
  have wp m
     $(\lambda out. \text{case out of}$ 
     $\text{None} \Rightarrow 0$ 
     $| \text{Some } (x, t) \Rightarrow \text{wp-event } (k\ x)\ ?E\ t)\ s$ 
     $\leq \text{wp } m\ ?R\ s$ 
  proof (rule wp-mono-on-support)
    fix out
    assume support:  $out \in \text{set-dist } (\text{execute } m\ s)$ 
    show  $(\text{case out of}$ 
     $\text{None} \Rightarrow 0$ 
     $| \text{Some } (x, t) \Rightarrow \text{wp-event } (k\ x)\ ?E\ t) \leq ?R\ out$ 
  proof (cases out)
    case None
    then show ?thesis

```

```

    unfolding hash-relation-hit-event-def by simp
next
  case (Some xt)
  then obtain x t where xt: xt = (x, t)
    by (cases xt) simp
  have m-out: Some (x, t) ∈ set-dist (execute m s)
    using support Some xt by simp
  have ext-st: s ≤ t
    using m-ext m-out unfolding hash-extension-preserving-def by blast
  have step:
    wp-event (k x) ?E t ≤
      (if hash-relation-hit R s t then 1
       else wp-event (k x) (hash-relation-hit-event R t) t)
    by (rule hash-relation-hit-event-step-bound
        [OF ext-st k-ext[of x]])
  show ?thesis
    using Some xt step
    unfolding hash-relation-hit-event-def
    by (cases hash-relation-hit R s t) simp-all
qed
qed
then show ?thesis unfolding exact .
qed
have head-bound:
  wp-event m ?E s ≤ hash-relation-budget-value b q
  using m-budget unfolding hash-relation-budget-def by blast
have tail-bound:
  wp m ?Tail s ≤ hash-relation-budget-value b r
proof (rule wp-le-const-on-support)
  fix out
  assume support: out ∈ set-dist (execute m s)
  show ?Tail out ≤ hash-relation-budget-value b r
  proof (cases out)
    case None
    then show ?thesis
      unfolding hash-relation-budget-value-def by simp
  next
    case (Some xt)
    then obtain x t where xt: xt = (x, t)
      by (cases xt) simp
    have tail:
      wp-event (k x) (hash-relation-hit-event R t) t ≤
        hash-relation-budget-value b r
      using k-budget[of x] unfolding hash-relation-budget-def by blast
    show ?thesis using Some xt tail by simp
  qed
qed
have wp-event (m ≫= k) ?E s ≤ wp m ?R s
  by (rule split)

```


also have ... = *wp-event* *m* ?*E* *s* + *wp* *m* ?*Tail* *s*
unfolding *wp-event-def* *wp-def* *dist-expect-def*
by (*simp* *add*: *sum.distrib* *algebra-simps*
hash-relation-hit-event-def)
also have ... ≤
hash-relation-budget-value *b* *q* + *hash-relation-budget-value* *b* *r*
by (*intro* *add-mono* *head-bound* *tail-bound*)
also have ... = *hash-relation-budget-value* *b* (*q* + *r*)
by (*rule* *hash-relation-budget-value-add*)
finally show
wp-event (*m* $\ggg$ *k*) (*hash-relation-hit-event* *R* *s*) *s* ≤
hash-relation-budget-value *b* (*q* + *r*) .
qed

lemma *hash-relation-budget-bind-on-outcomes*:
fixes *m* :: ('*x*, '*f* *protocol-channel*) *state-monad*
and *k* :: '*x* ⇒ ('*y*, '*f* *protocol-channel*) *state-monad*
assumes *m-ext*: *hash-extension-preserving* *m*
and *m-budget*: *hash-relation-budget* *R* *b* *q* *m*
and *k-ext*:
 $\bigwedge s\ x\ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m\ s) \implies$
hash-extension-preserving (*k* *x*)
and *k-budget*:
 $\bigwedge s\ x\ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m\ s) \implies$
hash-relation-budget *R* *b* *r* (*k* *x*)
shows *hash-relation-budget* *R* *b* (*q* + *r*) (*m* $\ggg$ *k*)
unfolding *hash-relation-budget-def*
proof (*intro* *allI*)
fix *s* :: '*f* *protocol-channel*
let ?*E* = *hash-relation-hit-event* *R* *s*
let ?*Tail* =
 $\lambda out. \text{case out of}$
 $\text{None} \Rightarrow 0$
 $| \text{Some } (x, t) \Rightarrow$
 $\text{if } \text{hash-relation-hit } R\ s\ t \text{ then } 0$
 $\text{else } \text{wp-event } (k\ x) (\text{hash-relation-hit-event } R\ t)\ t$
let ?*R* = $\lambda out. (\text{if } ?E\ out \text{ then } 1 \text{ else } 0) + ?Tail\ out$
have *split*:
wp-event (*m* $\ggg$ *k*) ?*E* *s* ≤ *wp* *m* ?*R* *s*
proof –
have *exact*:
wp-event (*m* $\ggg$ *k*) ?*E* *s* =
wp *m*
 $(\lambda out. \text{case out of}$
 $\text{None} \Rightarrow 0$
 $| \text{Some } (x, t) \Rightarrow \text{wp-event } (k\ x) ?E\ t)\ s$
unfolding *wp-event-def* *hash-relation-hit-event-def*
by (*simp* *add*: *wpsimps*)
have *wp* *m*

```

    ( $\lambda out.$  case out of
      None  $\Rightarrow 0$ 
      | Some (x, t)  $\Rightarrow wp\text{-event } (k\ x) \ ?E\ t) \ s$ 
     $\leq wp\ m \ ?R\ s$ 
  proof (rule wp-mono-on-support)
    fix out
    assume support: out  $\in set\text{-dist } (execute\ m\ s)$ 
    show (case out of
      None  $\Rightarrow 0$ 
      | Some (x, t)  $\Rightarrow wp\text{-event } (k\ x) \ ?E\ t) \leq ?R\ out$ 
    proof (cases out)
      case None
      then show ?thesis
        unfolding hash-relation-hit-event-def by simp
    next
      case (Some xt)
      then obtain x t where xt: xt = (x, t)
        by (cases xt) simp
      have m-out: Some (x, t)  $\in set\text{-dist } (execute\ m\ s)$ 
        using support Some xt by simp
      have ext-st: s  $\leq t$ 
        using m-ext m-out unfolding hash-extension-preserving-def by blast
      have step:
        wp-event (k x) ?E t  $\leq$ 
        (if hash-relation-hit R s t then 1
         else wp-event (k x) (hash-relation-hit-event R t) t)
        by (rule hash-relation-hit-event-step-bound
          [OF ext-st k-ext[OF m-out]])
      show ?thesis
        using Some xt step
        unfolding hash-relation-hit-event-def
        by (cases hash-relation-hit R s t) simp-all
    qed
  qed
  then show ?thesis unfolding exact .
qed
have head-bound:
  wp-event m ?E s  $\leq hash\text{-relation-budget-value } b\ q$ 
  using m-budget unfolding hash-relation-budget-def by blast
have tail-bound:
  wp m ?Tail s  $\leq hash\text{-relation-budget-value } b\ r$ 
proof (rule wp-le-const-on-support)
  fix out
  assume support: out  $\in set\text{-dist } (execute\ m\ s)$ 
  show ?Tail out  $\leq hash\text{-relation-budget-value } b\ r$ 
  proof (cases out)
    case None
    then show ?thesis
      unfolding hash-relation-budget-value-def by simp
  end
end

```

```

next
  case (Some xt)
  then obtain x t where xt: xt = (x, t)
    by (cases xt) simp
  have m-out: Some (x, t) ∈ set-dist (execute m s)
    using support Some xt by simp
  have tail:
    wp-event (k x) (hash-relation-hit-event R t) t ≤
      hash-relation-budget-value b r
    using k-budget[OF m-out] unfolding hash-relation-budget-def
    by blast
  show ?thesis using Some xt tail by simp
qed
qed
have wp-event (m ≫ k) ?E s ≤ wp m ?R s
  by (rule split)
also have ... = wp-event m ?E s + wp m ?Tail s
  unfolding wp-event-def wp-def dist-expect-def
  by (simp add: sum.distrib algebra-simps
    hash-relation-hit-event-def)
also have ... ≤
  hash-relation-budget-value b q + hash-relation-budget-value b r
  by (intro add-mono head-bound tail-bound)
also have ... = hash-relation-budget-value b (q + r)
  by (rule hash-relation-budget-value-add)
finally show
  wp-event (m ≫ k) (hash-relation-hit-event R s) s ≤
    hash-relation-budget-value b (q + r) .
qed

lemma hash-relation-program-bind:
  assumes m: hash-relation-program R b q m
    and k:  $\bigwedge x. \text{hash-relation-program } R \text{ b r } (k \ x)$ 
  shows hash-relation-program R b (q + r) (m ≫ k)
  unfolding hash-relation-program-def
proof
  show hash-extension-preserving (m ≫ k)
    by (rule hash-extension-preserving-bind)
      (use m k in ⟨simp-all add: hash-relation-program-def⟩)
  show hash-relation-budget R b (q + r) (m ≫ k)
    by (rule hash-relation-budget-bind)
      (use m k in ⟨simp-all add: hash-relation-program-def⟩)
qed

lemma hash-relation-program-bind-on-outcomes:
  fixes m :: ('x, 'f protocol-channel) state-monad
    and k :: 'x ⇒ ('y, 'f protocol-channel) state-monad
  assumes m: hash-relation-program R b q m
    and k:

```

```

 $\bigwedge s \ x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$ 
  hash-relation-program  $R \ b \ r \ (k \ x)$ 
shows hash-relation-program  $R \ b \ (q + r) \ (m \ggg k)$ 
unfolding hash-relation-program-def
proof
  show hash-extension-preserving  $(m \ggg k)$ 
    by (rule hash-extension-preserving-bind-on-outcomes)
      (use  $m \ k$  in  $\langle \text{simp-all add: hash-relation-program-def} \rangle$ )
  show hash-relation-budget  $R \ b \ (q + r) \ (m \ggg k)$ 
    by (rule hash-relation-budget-bind-on-outcomes)
      (use  $m \ k$  in  $\langle \text{simp-all add: hash-relation-program-def} \rangle$ )
qed

lemma hash-relation-program-zero:
  assumes preserving: hash-map-preserving  $m$ 
  shows hash-relation-program  $R \ b \ 0 \ m$ 
  unfolding hash-relation-program-def hash-relation-budget-def
proof
  show hash-extension-preserving  $m$ 
    by (rule hash-map-preserving-imp-hash-extension-preserving[OF preserving])
  show  $\forall s.$ 
    wp-event  $m \ (\text{hash-relation-hit-event } R \ s) \ s \leq$ 
    hash-relation-budget-value  $b \ 0$ 
proof
  fix  $s$ 
  have no-event-support:
     $\bigwedge \text{out}. \text{out} \in \text{set-dist } (\text{execute } m \ s) \implies$ 
     $\neg \text{hash-relation-hit-event } R \ s \ \text{out}$ 
proof –
  fix  $\text{out}$ 
  assume support:  $\text{out} \in \text{set-dist } (\text{execute } m \ s)$ 
  show  $\neg \text{hash-relation-hit-event } R \ s \ \text{out}$ 
  proof (cases  $\text{out}$ )
    case None
    then show ?thesis
      unfolding hash-relation-hit-event-def by simp
  next
  case (Some  $xt$ )
  then obtain  $x \ t$  where  $xt: xt = (x, t)$ 
    by (cases  $xt$ ) simp
  have same:  $\text{HashMap } t = \text{HashMap } s$ 
    using preserving support Some  $xt$ 
    unfolding hash-map-preserving-def by blast
  show ?thesis
    using same Some  $xt$ 
    unfolding hash-relation-hit-event-def hash-relation-hit-def
    by auto
qed
qed

```

```

have no-event:
   $\bigwedge out. out \in dom (dist (execute\ m\ s)) \implies$ 
     $\neg hash\_relation\_hit\_event\ R\ s\ out$ 
  using no-event-support unfolding set-dist-def by blast
have wp-event  $m\ (hash\_relation\_hit\_event\ R\ s)\ s = 0$ 
  unfolding wp-event-def wp-def dist-expect-def
  by (intro sum.neutral) (simp add: no-event)
then show
  wp-event  $m\ (hash\_relation\_hit\_event\ R\ s)\ s \leq$ 
    hash-relation-budget-value  $b\ 0$ 
  unfolding hash-relation-budget-value-def by simp
qed
qed

lemma hash-relation-program-hash:
  assumes fibers:  $\bigwedge x. card\ \{y. R\ x\ y\} \leq b$ 
  shows
    hash-relation-program  $R\ b\ 1$ 
    ( $hash\ x :: ('f, 'f\ protocol\_channel)\ state\_monad$ )
  unfolding hash-relation-program-def hash-relation-budget-def
proof
  show hash-extension-preserving
    ( $hash\ x :: ('f, 'f\ protocol\_channel)\ state\_monad$ )
    by (rule hash-extension-preserving-hash)
  show  $\forall s.$ 
    wp-event ( $hash\ x$ ) ( $hash\_relation\_hit\_event\ R\ s$ )  $s \leq$ 
      hash-relation-budget-value  $b\ 1$ 
  proof
    fix  $s :: 'f\ protocol\_channel$ 
    have
      wp-event ( $hash\ x$ ) ( $hash\_relation\_hit\_event\ R\ s$ )  $s \leq$ 
        nnreal (card  $\{y. R\ x\ y\}$ ) / nnreal size
      unfolding wp-hash-relation-hit by simp
    also have  $\dots \leq nnreal\ b / nnreal\ size$ 
      by (rule nnreal-nat-divide-right-mono)
      (rule fibers)
    also have  $\dots = hash\_relation\_budget\_value\ b\ 1$ 
      unfolding hash-relation-budget-value-def by simp
    finally show
      wp-event ( $hash\ x$ ) ( $hash\_relation\_hit\_event\ R\ s$ )  $s \leq$ 
        hash-relation-budget-value  $b\ 1$  .
  qed
qed

lemma hash-relation-program-mono:
  assumes qr:  $q \leq r$ 
  and program: hash-relation-program  $R\ b\ q\ m$ 
  shows hash-relation-program  $R\ b\ r\ m$ 
  unfolding hash-relation-program-def hash-relation-budget-def

```

```

proof
  show hash-extension-preserving m
    using program unfolding hash-relation-program-def by simp
  show  $\forall s.$ 
    wp-event m (hash-relation-hit-event R s) s  $\leq$ 
      hash-relation-budget-value b r
proof
  fix s
  have
    wp-event m (hash-relation-hit-event R s) s  $\leq$ 
      hash-relation-budget-value b q
    using program
    unfolding hash-relation-program-def hash-relation-budget-def by blast
  also have  $\dots \leq$  hash-relation-budget-value b r
    by (rule hash-relation-budget-value-mono[OF qr])
  finally show
    wp-event m (hash-relation-hit-event R s) s  $\leq$ 
      hash-relation-budget-value b r .
qed
qed

lemma controlled-ro-program-relation:
  assumes controlled: controlled-ro-program q m
    and fibers:  $\bigwedge x. \text{card } \{y. R\ x\ y\} \leq b$ 
  shows hash-relation-program R b q m
  using controlled
proof (induction rule: controlled-ro-program.induct)
  case (Return x)
  show ?case
    by (rule hash-relation-program-zero)
      (rule hash-map-preserving-return)
next
  case Fail
  show ?case
    by (rule hash-relation-program-zero)
      (rule hash-map-preserving-throw)
next
  case (Sample d)
  show ?case
    by (rule hash-relation-program-zero)
      (rule hash-map-preserving-state-independent-sample)
next
  case (Query q k x)
  have tail:  $\bigwedge y. \text{hash-relation-program } R\ b\ q\ (k\ y)$ 
    using Query.IH fibers by blast
  have hash-relation-program R b ( $1 + q$ )
    (hash x :: ('f, 'f protocol-channel) state-monad)  $\gg= k$ )
  by (rule hash-relation-program-bind)
    (rule hash-relation-program-hash[OF fibers], rule tail)

```

```

    then show ?case by simp
next
  case (Bind q m r k)
  have tail:  $\bigwedge x. \text{hash-relation-program } R \ b \ r \ (k \ x)$ 
    using Bind.IH(2) by blast
  show ?case
    by (rule hash-relation-program-bind[OF Bind.IH(1) tail])
next
  case (Weaken q m r)
  show ?case
    by (rule hash-relation-program-mono[OF Weaken.hyps(2)
      Weaken.IH])
qed

theorem controlled-ro-program-relation-bound:
  assumes controlled: controlled-ro-program q m
  and fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
  shows
    wp-event m (hash-relation-hit-event R s)  $s \leq$ 
      nnreal (q * b) / nnreal size
proof -
  have program: hash-relation-program R b q m
    by (rule controlled-ro-program-relation[OF controlled fibers])
  have budget: hash-relation-budget R b q m
    using program unfolding hash-relation-program-def by simp
  show ?thesis
    using budget unfolding hash-relation-budget-def
      hash-relation-budget-value-def by blast
qed

end

end

```

```

theory Security-Experiment
imports
  Soundness-Oracle
begin

```

```

type-synonym 'f semantic-adversary =
  ('f list, 'f protocol-channel) state-monad

```

```

context soundness
begin

```

A semantic adversary is an arbitrary probabilistic state-monad program that returns a complete proof transcript. Admissibility states the random-oracle laws needed by the security proof: oracle consistency, bounded range growth, bounded collisions, and

bounded hits on every fixed target set.

This interface deliberately does not prescribe an attacker programming language. A future capability-safe command language may expose only probabilistic computation and random-oracle queries, with a refinement theorem showing that every query-bounded command program satisfies *admissible-adversary*.

definition *adversary-initial-state* :: 'f protocol-channel
where

adversary-initial-state =
 ($\text{HashMap} = \text{fempty}$, $\text{PState} = 0$, $\text{PTranscript} = []$,
 $\text{PTraceFriCounter} = 0$, $\text{PCompositionFriCounter} = 0$,
 $\text{PAlphaCounter} = 0$, $\text{PQueryCounter} = 0$)

definition *admissible-adversary*
 :: nat $\Rightarrow$ 'f semantic-adversary $\Rightarrow$ bool
where

admissible-adversary $q\ A \longleftrightarrow$
 $\text{hash-extension-preserving}\ A \wedge$
 $\text{hash-range-budget}\ q\ A \wedge$
 $\text{hash-collision-budget}\ q\ A \wedge$
 $(\forall B. \text{hash-target-program}\ B\ q\ A)$

lemma *adversary-initial-state-simps[simp]*:
 $\text{HashMap}\ \text{adversary-initial-state} = \text{fempty}$
 $\text{PState}\ \text{adversary-initial-state} = 0$
 $\text{PTranscript}\ \text{adversary-initial-state} = []$
 $\text{PTraceFriCounter}\ \text{adversary-initial-state} = 0$
 $\text{PCompositionFriCounter}\ \text{adversary-initial-state} = 0$
 $\text{PAlphaCounter}\ \text{adversary-initial-state} = 0$
 $\text{PQueryCounter}\ \text{adversary-initial-state} = 0$
unfolding *adversary-initial-state-def* **by** *simp-all*

lemma *adversary-initial-state-eq-verifier-initial-state*:
 $\text{adversary-initial-state} = \text{verifier-initial-state}\ []$
unfolding *adversary-initial-state-def* *verifier-initial-state-def* **by** *simp*

lemma *admissible-adversary-extension*:
assumes *admissible-adversary* $q\ A$
shows *hash-extension-preserving* A
using *assms* **unfolding** *admissible-adversary-def* **by** *simp*

lemma *admissible-adversary-range*:
assumes *admissible-adversary* $q\ A$
shows *hash-range-budget* $q\ A$
using *assms* **unfolding** *admissible-adversary-def* **by** *simp*

lemma *admissible-adversary-collision*:
assumes *admissible-adversary* $q\ A$
shows *hash-collision-budget* $q\ A$

using *assms* **unfolding** *admissible-adversary-def* **by** *simp*

lemma *admissible-adversary-target*:

assumes *admissible-adversary* *q A*

shows *hash-target-program* *B q A*

using *assms* **unfolding** *admissible-adversary-def* **by** *simp*

definition *verifier-state-from-adversary*

$:: 'f \text{ protocol-channel} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ protocol-channel}$

where

verifier-state-from-adversary *s tr* =
 $(\text{HashMap} = \text{HashMap } s, \text{PState} = 0, \text{PTranscript} = tr,$
 $\text{PTraceFriCounter} = 0, \text{PCompositionFriCounter} = 0,$
 $\text{PAlphaCounter} = 0, \text{PQueryCounter} = 0)$

lemma *verifier-state-from-adversary-simps*[*simp*]:

HashMap (*verifier-state-from-adversary* *s tr*) = *HashMap* *s*

PState (*verifier-state-from-adversary* *s tr*) = 0

PTranscript (*verifier-state-from-adversary* *s tr*) = *tr*

PTraceFriCounter (*verifier-state-from-adversary* *s tr*) = 0

PCompositionFriCounter (*verifier-state-from-adversary* *s tr*) = 0

PAlphaCounter (*verifier-state-from-adversary* *s tr*) = 0

PQueryCounter (*verifier-state-from-adversary* *s tr*) = 0

unfolding *verifier-state-from-adversary-def* **by** *simp-all*

lemma *verifier-state-from-adversary-empty*:

verifier-state-from-adversary *adversary-initial-state* *tr* =

verifier-initial-state *tr*

unfolding *verifier-state-from-adversary-def* *adversary-initial-state-def*

verifier-initial-state-def

by *simp*

lemma *verifier-state-from-adversary-hash-extends*:

assumes $s \leq t$

shows *verifier-state-from-adversary* *s tr* $\leq$

verifier-state-from-adversary *t tr*

using *assms*

unfolding *verifier-state-from-adversary-def* *less-eq-hash-ext-def*

less-eq-fmap-def

by *simp*

definition *verifier-state-transfer*

$:: 'f \text{ list} \Rightarrow (\text{unit}, 'f \text{ protocol-channel}) \text{ state-monad}$

where

verifier-state-transfer *tr* =
 $\text{get} \gg (\lambda s. \text{put } (\text{verifier-state-from-adversary } s \text{ tr}))$

lemma *verifier-state-transfer-outcomeE*:

assumes

Some $(((), t) \in \text{set-dist } (\text{execute } (\text{verifier-state-transfer } tr) s))$
shows $t = \text{verifier-state-from-adversary } s \ tr$
using *assms* **unfolding** *verifier-state-transfer-def*
by *(auto elim!: set-dist-bindE)*

lemma *hash-map-preserving-verifier-state-transfer:*
hash-map-preserving *(verifier-state-transfer tr)*
unfolding *hash-map-preserving-def*
using *verifier-state-transfer-outcomeE* **by** *fastforce*

lemma *hash-map-preserving-imp-hash-range-budget-zero-semantic:*
fixes $m :: ('r, 'f \text{ protocol-channel}) \text{ state-monad}$
assumes *preserving: hash-map-preserving m*
shows *hash-range-budget 0 m*
using *preserving*
unfolding *hash-range-budget-def hash-map-preserving-def*
hash-map-output-values-def
by *auto*

lemma *hash-map-preserving-imp-hash-collision-budget-zero-semantic:*
fixes $m :: ('r, 'f \text{ protocol-channel}) \text{ state-monad}$
assumes *preserving: hash-map-preserving m*
shows *hash-collision-budget 0 m*
unfolding *hash-collision-budget-def*

proof *(intro allI impI)*
fix $s :: 'f \text{ protocol-channel}$
assume *clean: $\neg \text{hash-map-output-collision } s$*
have *no-event:*
 $\bigwedge \text{out. out} \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\neg \text{hash-new-collision-event } s \ \text{out}$
proof $-$
fix *out*
assume *support: out $\in$ set-dist (execute m s)*
show $\neg \text{hash-new-collision-event } s \ \text{out}$
proof *(cases out)*
case *None*
then show *?thesis*
unfolding *hash-new-collision-event-def* **by** *simp*
next
case *(Some xt)*
then obtain $x \ t$ **where** $xt: xt = (x, t)$
by *(cases xt) simp*
have *same: HashMap t = HashMap s*
using *preserving support Some xt*
unfolding *hash-map-preserving-def* **by** *blast*
show *?thesis*
using *clean same Some xt*
unfolding *hash-new-collision-event-def*
hash-map-new-output-collision-def hash-map-output-collision-def

```

    by simp
  qed
qed
have no-event-dom:
   $\bigwedge out. out \in \text{dom} (\text{dist} (\text{execute } m \ s)) \implies$ 
   $\neg \text{hash-new-collision-event } s \ out$ 
  using no-event unfolding set-dist-def by blast
have wp-event m (hash-new-collision-event s) s = 0
  unfolding wp-event-def wp-def dist-expect-def
  by (intro sum.neutral) (simp add: no-event-dom)
then show
  wp-event m (hash-new-collision-event s) s  $\leq$ 
  hash-collision-budget-value
  (card (hash-map-output-values s)) 0
  unfolding hash-collision-budget-value-def by simp
qed

lemma hash-range-budget-verifier-state-transfer:
  hash-range-budget 0 (verifier-state-transfer tr)
  by (rule hash-map-preserving-imp-hash-range-budget-zero-semantic)
  (rule hash-map-preserving-verifier-state-transfer)

lemma hash-collision-budget-verifier-state-transfer:
  hash-collision-budget 0 (verifier-state-transfer tr)
  by (rule hash-map-preserving-imp-hash-collision-budget-zero-semantic)
  (rule hash-map-preserving-verifier-state-transfer)

lemma hash-range-budget-verifier-after-adversary:
  hash-range-budget verifier-hash-query-budget
  (verifier-state-transfer tr  $\ggg$  ( $\lambda\cdot$ . verify-monad))
proof -
  have range:
    hash-range-budget (0 + verifier-hash-query-budget)
    (verifier-state-transfer tr  $\ggg$  ( $\lambda\cdot$ . verify-monad))
  by (rule hash-range-budget-bind)
  (rule hash-range-budget-verifier-state-transfer,
  rule hash-range-budget-verify-monad)
  then show ?thesis by simp
qed

lemma hash-collision-budget-verifier-after-adversary:
  hash-collision-budget verifier-hash-query-budget
  (verifier-state-transfer tr  $\ggg$  ( $\lambda\cdot$ . verify-monad))
proof -
  have collision:
    hash-collision-budget (0 + verifier-hash-query-budget)
    (verifier-state-transfer tr  $\ggg$  ( $\lambda\cdot$ . verify-monad))
  by (rule hash-collision-budget-bind)
  (rule hash-range-budget-verifier-state-transfer,

```

```

    rule hash-collision-budget-verifier-state-transfer,
    rule hash-range-budget-verify-monad,
    rule hash-collision-budget-verify-monad)
  then show ?thesis by simp
qed

definition security-experiment
  :: 'f semantic-adversary  $\Rightarrow$ 
    (unit list, 'f protocol-channel) state-monad
  where
    security-experiment A =
      A  $\gg=$  ( $\lambda$ tr.
        verifier-state-transfer tr  $\gg=$  ( $\lambda$ -. verify-monad))

definition adversary-acceptance-probability
  :: 'f semantic-adversary  $\Rightarrow$  prob
  where
    adversary-acceptance-probability A =
      wp-event (security-experiment A)
        ( $\lambda$ out.  $\neg$  Option.is-none out) adversary-initial-state

lemma security-experiment-outcomeE:
  assumes outcome:
    Some (result, final-state)  $\in$ 
      set-dist (execute (security-experiment A) initial-state)
  obtains tr attacker-state where
    Some (tr, attacker-state)  $\in$  set-dist (execute A initial-state)
    Some (result, final-state)  $\in$ 
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state tr))
  using outcome
  unfolding security-experiment-def verifier-state-transfer-def
  by (auto elim!: set-dist-bindE intro: that)

lemma accepted-security-experimentE:
  assumes outcome:
    out  $\in$  set-dist
      (execute (security-experiment A) adversary-initial-state)
  and accepted:  $\neg$  Option.is-none out
  obtains result final-state tr attacker-state where
    out = Some (result, final-state)
    Some (tr, attacker-state)  $\in$ 
      set-dist (execute A adversary-initial-state)
    Some (result, final-state)  $\in$ 
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state tr))
proof –

```

```

from accepted obtain result final-state where
  out: out = Some (result, final-state)
  by (cases out) auto
from security-experiment-outcomeE[OF outcome[unfolded out]]
obtain tr attacker-state where
  attacker:
    Some (tr, attacker-state) ∈
      set-dist (execute A adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state tr))
  by blast
show ?thesis
  by (rule that[OF out attacker verifier])
qed

```

```

lemma attacker-failure-not-accepted[simp]:
  ¬ (¬ Option.is-none
    (None :: (unit list × 'f protocol-channel) option))
by simp

```

```

lemma security-experiment-success-requires-attacker-success:
  assumes no-success:
    ∧ tr attacker-state.
      Some (tr, attacker-state) ∉ set-dist (execute A initial-state)
  shows
    Some (result, final-state) ∉
      set-dist (execute (security-experiment A) initial-state)
proof
  assume outcome:
    Some (result, final-state) ∈
      set-dist (execute (security-experiment A) initial-state)
  from security-experiment-outcomeE[OF outcome]
  obtain tr attacker-state where
    Some (tr, attacker-state) ∈ set-dist (execute A initial-state)
  by blast
  then show False
    using no-success[of tr attacker-state] by blast
qed

```

```

lemma admissible-adversary-outcome-extends:
  assumes admissible: admissible-adversary q A
  and outcome:
    Some (tr, attacker-state) ∈
      set-dist (execute A adversary-initial-state)
  shows adversary-initial-state ≤ attacker-state
  using admissible-adversary-extension[OF admissible] outcome

```

unfolding *hash-extension-preserving-def* **by** *blast*

lemma *hash-map-output-values-adversary-initial-state*[*simp*]:
hash-map-output-values adversary-initial-state = {}
unfolding *adversary-initial-state-eq-verifier-initial-state* **by** *simp*

lemma *adversary-initial-state-no-output-collision*[*simp*]:
 $\neg$ *hash-map-output-collision adversary-initial-state*
unfolding *hash-map-output-collision-def* **by** *simp*

lemma *admissible-adversary-output-range-bound*:
assumes *admissible*: *admissible-adversary* *q* *A*
and *outcome*:
Some (*tr*, *attacker-state*) $\in$
set-dist (*execute* *A* *adversary-initial-state*)
shows *card* (*hash-map-output-values attacker-state*) $\leq$ *q*
proof –
have
card (*hash-map-output-values attacker-state*) $\leq$
card (*hash-map-output-values adversary-initial-state*) + *q*
using *admissible-adversary-range*[*OF* *admissible*] *outcome*
unfolding *hash-range-budget-def* **by** *blast*
then show ?*thesis* **by** *simp*
qed

lemma *admissible-adversary-collision-bound*:
assumes *admissible*: *admissible-adversary* *q* *A*
shows
wp-event *A* (*hash-new-collision-event adversary-initial-state*)
adversary-initial-state $\leq$ *hash-collision-budget-value* 0 *q*
proof –
have *collision-budget*: *hash-collision-budget* *q* *A*
by (*rule* *admissible-adversary-collision*[*OF* *admissible*])
have
wp-event *A* (*hash-new-collision-event adversary-initial-state*)
adversary-initial-state $\leq$
hash-collision-budget-value
(*card* (*hash-map-output-values adversary-initial-state*)) *q*
using *collision-budget*
adversary-initial-state-no-output-collision
unfolding *hash-collision-budget-def* **by** *blast*
then show ?*thesis* **by** *simp*
qed

lemma *admissible-adversary-target-bound*:
assumes *admissible*: *admissible-adversary* *q* *A*
shows
wp-event *A*
(*hash-new-output-hit-event* *B* *adversary-initial-state*)

$\text{adversary-initial-state} \leq \text{hash-target-budget-value } B \ q$
proof –
have $\text{hash-target-budget } B \ q \ A$
by (rule $\text{hash-target-program-budget}$)
(rule $\text{admissible-adversary-target}[OF \text{ admissible}]$)
then show $?thesis$
unfolding $\text{hash-target-budget-def}$ **by** blast
qed

lemma $\text{hash-range-budget-security-experiment}$:
assumes $\text{admissible: admissible-adversary } q \ A$
shows
 $\text{hash-range-budget } (q + \text{verifier-hash-query-budget})$
 $(\text{security-experiment } A)$
unfolding $\text{security-experiment-def}$
by (rule $\text{hash-range-budget-bind}$)
(rule $\text{admissible-adversary-range}[OF \text{ admissible}]$,
rule $\text{hash-range-budget-verifier-after-adversary}$)

lemma $\text{hash-collision-budget-security-experiment}$:
assumes $\text{admissible: admissible-adversary } q \ A$
shows
 $\text{hash-collision-budget } (q + \text{verifier-hash-query-budget})$
 $(\text{security-experiment } A)$
unfolding $\text{security-experiment-def}$
by (rule $\text{hash-collision-budget-bind}$)
(rule $\text{admissible-adversary-range}[OF \text{ admissible}]$,
rule $\text{admissible-adversary-collision}[OF \text{ admissible}]$,
rule $\text{hash-range-budget-verifier-after-adversary}$,
rule $\text{hash-collision-budget-verifier-after-adversary}$)

lemma $\text{security-experiment-hash-new-collision-bound}$:
assumes $\text{admissible: admissible-adversary } q \ A$
shows
 $\text{wp-event } (\text{security-experiment } A)$
 $(\text{hash-new-collision-event adversary-initial-state})$
 $\text{adversary-initial-state} \leq$
 $\text{hash-collision-budget-value } 0 \ (q + \text{verifier-hash-query-budget})$
proof –
have collision-budget :
 $\text{hash-collision-budget } (q + \text{verifier-hash-query-budget})$
 $(\text{security-experiment } A)$
by (rule $\text{hash-collision-budget-security-experiment}[OF \text{ admissible}]$)
have
 $\text{wp-event } (\text{security-experiment } A)$
 $(\text{hash-new-collision-event adversary-initial-state})$
 $\text{adversary-initial-state} \leq$
 $\text{hash-collision-budget-value}$
 $(\text{card } (\text{hash-map-output-values adversary-initial-state}))$

```

      (q + verifier-hash-query-budget)
    using collision-budget adversary-initial-state-no-output-collision
    unfolding hash-collision-budget-def by blast
  then show ?thesis by simp
qed

lemma hash-map-output-collision-bad-imp-security-new-collision:
  assumes bad: hash-map-output-collision-bad adversary-initial-state out
  shows hash-new-collision-event adversary-initial-state out
proof -
  have new-bad: hash-map-new-output-collision-bad adversary-initial-state out
    by (rule hash-map-output-collision-bad-imp-new-collision-bad-if-initial-clean)
    (use bad in simp-all)
  show ?thesis
    by (rule hash-map-new-output-collision-bad-imp-hash-new-collision-event
        [OF new-bad])
qed

lemma security-experiment-hash-map-output-collision-bad-bound:
  assumes admissible: admissible-adversary q A
  shows
    wp-event (security-experiment A)
      (hash-map-output-collision-bad adversary-initial-state)
      adversary-initial-state ≤
      hash-collision-budget-value 0 (q + verifier-hash-query-budget)
proof -
  have wp-event (security-experiment A)
    (hash-map-output-collision-bad adversary-initial-state)
    adversary-initial-state ≤
    wp-event (security-experiment A)
      (hash-new-collision-event adversary-initial-state)
      adversary-initial-state
    by (rule wp-event-mono)
      (rule hash-map-output-collision-bad-imp-security-new-collision)
  also have ... ≤
    hash-collision-budget-value 0 (q + verifier-hash-query-budget)
    by (rule security-experiment-hash-new-collision-bound[OF admissible])
  finally show ?thesis .
qed

end

end

```

```

theory Staged-Security-Experiment-Core
  imports Controlled-RO Security-Experiment
begin

```

Core staged-adversary records, transcript builders, and experiment projections.


```

record staged-budgets =
  trace-root-budget :: nat
  trace-fri-budgets :: nat list
  trace-final-budget :: nat
  degree-budget :: nat
  composition-fri-budgets :: nat list
  composition-final-budget :: nat
  query-opening-budgets :: nat list

record 'f staged-adversary =
  trace-root-stage :: ('f, 'f protocol-channel) state-monad
  trace-fri-root-stage ::
    nat ⇒ 'f list ⇒ ('f, 'f protocol-channel) state-monad
  trace-final-stage ::
    'f list ⇒ ('f, 'f protocol-channel) state-monad
  degree-stage ::
    'f list ⇒ ('f, 'f protocol-channel) state-monad
  composition-fri-root-stage ::
    'f ⇒ nat ⇒ 'f list ⇒
      ('f, 'f protocol-channel) state-monad
  composition-final-stage ::
    'f ⇒ 'f list ⇒ ('f, 'f protocol-channel) state-monad
  query-opening-stage ::
    nat ⇒ 'f ⇒ ('f list, 'f protocol-channel) state-monad

record 'f staged-proof-data =
  staged-trace-root :: 'f
  staged-trace-fri-roots :: 'f list
  staged-trace-fri-challenges :: 'f list
  staged-trace-final :: 'f
  staged-alphas :: 'f list
  staged-degree :: 'f
  staged-composition-fri-roots :: 'f list
  staged-composition-fri-challenges :: 'f list
  staged-composition-final :: 'f
  staged-query-chunks :: 'f list list

record 'f staged-composition-prefix-data =
  scp-trace-root :: 'f
  scp-trace-fri-roots :: 'f list
  scp-trace-fri-challenges :: 'f list
  scp-trace-final :: 'f
  scp-alphas :: 'f list
  scp-degree :: 'f
  scp-composition-fri-roots :: 'f list
  scp-composition-fri-challenges :: 'f list

record 'f staged-query-prefix-data =
  sqp-trace-root :: 'f

```

```

sqp-trace-fri-roots :: 'f list
sqp-trace-fri-challenges :: 'f list
sqp-trace-final :: 'f
sqp-alphas :: 'f list
sqp-degree :: 'f
sqp-composition-fri-roots :: 'f list
sqp-composition-fri-challenges :: 'f list
sqp-composition-final :: 'f
sqp-query-chunks :: 'f list list

context soundness
begin

lemma wp-event-bind-return-map:
  fixes m :: ('x, 'f protocol-channel) state-monad
  shows
    wp-event (m  $\gg$  (λx. return (f x))) P s =
      wp-event m (λout. case out of None  $\Rightarrow$  P None
        | Some (x, t)  $\Rightarrow$  P (Some (f x, t))) s
  proof -
    have indicator-eq:
      (λout. case out of None  $\Rightarrow$  if P None then 1 else 0
        | Some (x, t)  $\Rightarrow$  if P (Some (f x, t)) then 1 else 0) =
      (λout. if (case out of None  $\Rightarrow$  P None
        | Some (x, t)  $\Rightarrow$  P (Some (f x, t))) then 1 else 0)
    by (rule ext) (simp split: option.splits prod.splits)
  show ?thesis
    unfolding wp-event-def
    apply (subst wp-bind)
    apply (rule arg-cong[where f=λQ. wp m Q s])
    apply (rule ext)
    apply (simp add: wp-return indicator-eq
      split: option.splits prod.splits)
  done
qed

lemma wp-event-bind-mono-cont:
  fixes m :: ('x, 'f protocol-channel) state-monad
  and k :: 'x  $\Rightarrow$  ('y, 'f protocol-channel) state-monad
  and l :: 'x  $\Rightarrow$  ('z, 'f protocol-channel) state-monad
  assumes none: P None  $\Longrightarrow$  Q None
  and cont:
     $\bigwedge x t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \Longrightarrow$ 
      wp-event (k x) P t  $\leq$  wp-event (l x) Q t
  shows wp-event (m  $\gg$  k) P s  $\leq$  wp-event (m  $\gg$  l) Q s
  proof -
    have left:
      wp-event (m  $\gg$  k) P s =
      wp m (λout. case out of

```

```

      None  $\Rightarrow$  if  $P$  None then 1 else 0
    | Some  $(x, t) \Rightarrow$  wp-event  $(k\ x)\ P\ t)\ s$ 
  unfolding wp-event-def by (simp add: wpsimps)
have right:
  wp-event  $(m \gg l)\ Q\ s =$ 
    wp  $m\ (\lambda out. \text{case out of}$ 
      None  $\Rightarrow$  if  $Q$  None then 1 else 0
    | Some  $(x, t) \Rightarrow$  wp-event  $(l\ x)\ Q\ t)\ s$ 
  unfolding wp-event-def by (simp add: wpsimps)
show ?thesis
  unfolding left right
proof (rule wp-mono-on-support)
  fix out
  assume out:  $out \in \text{set-dist}\ (\text{execute } m\ s)$ 
  show (case out of None  $\Rightarrow$  if  $P$  None then 1 else 0
    | Some  $(x, t) \Rightarrow$  wp-event  $(k\ x)\ P\ t)$ 
     $\leq$  (case out of None  $\Rightarrow$  if  $Q$  None then 1 else 0
    | Some  $(x, t) \Rightarrow$  wp-event  $(l\ x)\ Q\ t)$ 
  proof (cases out)
    case None
    then show ?thesis
      using none by (cases  $P$  None) simp-all
  next
    case (Some  $xt$ )
    then obtain  $x\ t$  where  $xt: xt = (x, t)$ 
      by (cases  $xt$ ) auto
    show ?thesis
      using cont[ $of\ x\ t$ ] out Some  $xt$  by simp
  qed
qed
qed
qed

lemma wp-event-assert-bind-le:
  fixes  $k :: \text{unit} \Rightarrow ('x, 'f\ \text{protocol-channel})\ \text{state-monad}$ 
  assumes none:  $\neg P\ \text{None}$ 
  shows wp-event  $(\text{assert } b \gg k)\ P\ s \leq \text{wp-event}\ (k\ ())\ P\ s$ 
proof (cases  $b$ )
  case True
  then show ?thesis
    unfolding assert-def by simp
next
  case False
  then show ?thesis
    using none unfolding assert-def
    by (simp add: wp-event-def wpsimps throw-no-outcome)
qed

definition staged-budget-wellformed ::  $\text{staged-budgets} \Rightarrow \text{bool}$ 
  where

```

staged-budget-wellformed budgets $\longleftrightarrow$
length (trace-fri-budgets budgets) = ceil-log clength $\wedge$
length (composition-fri-budgets budgets) =
ceil-log (maxDegree + 1) $\wedge$
length (query-opening-budgets budgets) = rounds

definition *staged-adversary-controlled*

$:: \text{staged-budgets} \Rightarrow 'f \text{ staged-adversary} \Rightarrow \text{bool}$

where

staged-adversary-controlled budgets A $\longleftrightarrow$
controlled-ro-program (trace-root-budget budgets)
(trace-root-stage A) $\wedge$
 $(\forall i \text{ bs.}$
i < length (trace-fri-budgets budgets) $\longrightarrow$
controlled-ro-program (trace-fri-budgets budgets ! i)
(trace-fri-root-stage A i bs)) $\wedge$
 $(\forall \text{ bs.}$
controlled-ro-program (trace-final-budget budgets)
(trace-final-stage A bs)) $\wedge$
 $(\forall \text{ as.}$
controlled-ro-program (degree-budget budgets)
(degree-stage A as)) $\wedge$
 $(\forall \text{ dg } i \text{ bs.}$
i < length (composition-fri-budgets budgets) $\longrightarrow$
controlled-ro-program (composition-fri-budgets budgets ! i)
(composition-fri-root-stage A dg i bs)) $\wedge$
 $(\forall \text{ dg } \text{bs.}$
controlled-ro-program (composition-final-budget budgets)
(composition-final-stage A dg bs)) $\wedge$
 $(\forall i \text{ raw.}$
i < length (query-opening-budgets budgets) $\longrightarrow$
controlled-ro-program (query-opening-budgets budgets ! i)
(query-opening-stage A i raw))

definition *staged-attacker-query-budget* $:: \text{staged-budgets} \Rightarrow \text{nat}$

where

staged-attacker-query-budget budgets =
trace-root-budget budgets +
sum-list (trace-fri-budgets budgets) +
trace-final-budget budgets +
degree-budget budgets +
sum-list (composition-fri-budgets budgets) +
composition-final-budget budgets +
sum-list (query-opening-budgets budgets)

definition *staged-challenge-query-budget* $:: \text{nat}$

where

staged-challenge-query-budget =
ceil-log clength + length spec +

$\text{ceil-log } (\text{maxDegree} + 1) + \text{rounds}$

definition *staged-trace-fri-search-queries*

$:: \text{staged-budgets} \Rightarrow \text{nat} \Rightarrow \text{nat}$

where

$\text{staged-trace-fri-search-queries budgets } i =$
 $\text{trace-root-budget budgets} +$
 $\text{sum-list (take (Suc } i) (\text{trace-fri-budgets budgets}))} + i$

definition *staged-alpha-search-queries*

$:: \text{staged-budgets} \Rightarrow \text{nat} \Rightarrow \text{nat}$

where

$\text{staged-alpha-search-queries budgets } i =$
 $\text{trace-root-budget budgets} +$
 $\text{sum-list (trace-fri-budgets budgets)} +$
 $\text{trace-final-budget budgets} +$
 $\text{ceil-log clength} + i$

definition *staged-composition-fri-search-queries*

$:: \text{staged-budgets} \Rightarrow \text{nat} \Rightarrow \text{nat}$

where

$\text{staged-composition-fri-search-queries budgets } i =$
 $\text{trace-root-budget budgets} +$
 $\text{sum-list (trace-fri-budgets budgets)} +$
 $\text{trace-final-budget budgets} +$
 $\text{degree-budget budgets} +$
 $\text{sum-list (take (Suc } i) (\text{composition-fri-budgets budgets}))} +$
 $\text{ceil-log clength} + \text{length spec} + i$

definition *staged-composition-fri-vector-search-queries*

$:: \text{staged-budgets} \Rightarrow \text{nat}$

where

$\text{staged-composition-fri-vector-search-queries budgets} =$
 $\text{staged-alpha-search-queries budgets } 0 +$
 $\text{length spec} +$
 $\text{degree-budget budgets} +$
 $\text{sum-list (composition-fri-budgets budgets)} +$
 $\text{ceil-log (maxDegree} + 1)$

definition *staged-query-search-queries*

$:: \text{staged-budgets} \Rightarrow \text{nat} \Rightarrow \text{nat}$

where

$\text{staged-query-search-queries budgets } i =$
 $\text{trace-root-budget budgets} +$
 $\text{sum-list (trace-fri-budgets budgets)} +$
 $\text{trace-final-budget budgets} +$
 $\text{degree-budget budgets} +$
 $\text{sum-list (composition-fri-budgets budgets)} +$
 $\text{composition-final-budget budgets} +$

$\text{sum-list } (\text{take } i \text{ (query-opening-budgets budgets)}) +$
 $\text{ceil-log clength} + \text{length spec} +$
 $\text{ceil-log (maxDegree} + 1) + i$

definition *staged-phase-relation-error*

$:: \text{nat} \Rightarrow \text{nat} \Rightarrow \text{prob}$

where

$\text{staged-phase-relation-error fiber-bound query-bound} =$
 $\text{hash-relation-budget-value fiber-bound query-bound}$

definition *staged-phase-target-error*

$:: 'f \text{ set} \Rightarrow \text{nat} \Rightarrow \text{prob}$

where

$\text{staged-phase-target-error } B \text{ query-bound} =$
 $\text{hash-target-budget-value } B \text{ query-bound}$

lemma *staged-phase-query-index-target-error-fraction-bound:*

assumes *raw-bound: query-index-raw-preimage-bound*

and *subset: $B \subseteq \text{query-sample-space}$*

shows

$\text{staged-phase-target-error (query-index-raw-preimage } B) q \leq$
 $\text{nnreal } q * (\text{nnreal (card } B) / \text{nnreal (card query-sample-space)})$

proof —

have *raw:*

$\text{nnreal (card (query-index-raw-preimage } B)) / \text{nnreal size} \leq$
 $\text{nnreal (card } B) / \text{nnreal (card query-sample-space)}$

using *raw-bound subset unfolding query-index-raw-preimage-bound-def*

by *blast*

have *eq:*

$\text{staged-phase-target-error (query-index-raw-preimage } B) q =$
 $\text{nnreal } q *$

$(\text{nnreal (card (query-index-raw-preimage } B)) / \text{nnreal size})$

unfolding *staged-phase-target-error-def hash-target-budget-value-def*

by *transfer (simp add: field-simps)*

show *?thesis*

unfolding *eq by (rule mult-left-mono) (use raw in simp-all)*

qed

lemma *staged-phase-query-index-target-error-query-bound:*

assumes *raw-bound: query-index-raw-preimage-bound*

and *subset: $B \subseteq \text{query-sample-space}$*

and *frac:*

$\text{nnreal (card } B) / \text{nnreal (card query-sample-space)} \leq C$

shows

$\text{staged-phase-target-error (query-index-raw-preimage } B) q \leq$
 $\text{nnreal } q * C$

proof —

have

$\text{staged-phase-target-error (query-index-raw-preimage } B) q \leq$

```

      nnreal q * (nnreal (card B) / nnreal (card query-sample-space))
    by (rule staged-phase-query-index-target-error-fraction-bound
        [OF raw-bound subset])
  also have ... ≤ nnreal q * C
    by (rule mult-left-mono) (use frac in simp-all)
  finally show ?thesis .
qed

```

definition *record-staged-message*

```

:: 'f ⇒ (unit, 'f protocol-channel) state-monad
where
  record-staged-message x =
    modify
      (λs. s⟦
        PState := concat (PState s) x,
        PTranscript := PTranscript s @ [x]⟧)

```

definition *record-staged-messages*

```

:: 'f list ⇒ (unit, 'f protocol-channel) state-monad
where
  record-staged-messages xs = mfold2 record-staged-message xs

```

primrec *staged-trace-fri-program*

```

:: 'f staged-adversary ⇒ nat ⇒ nat ⇒ 'f list ⇒
   ('f list × 'f list, 'f protocol-channel) state-monad
where
  staged-trace-fri-program A i 0 bs = return ([], bs)
| staged-trace-fri-program A i (Suc n) bs =
  do {
    root ← trace-fri-root-stage A i bs;
    record-staged-message root;
    b ← receive-trace-fri-challenge;
    (roots, bs') ← staged-trace-fri-program A (Suc i) n (bs @ [b]);
    return (root # roots, bs')
  }

```

primrec *staged-alpha-program*

```

:: nat ⇒ ('f list, 'f protocol-channel) state-monad
where
  staged-alpha-program 0 = return []
| staged-alpha-program (Suc n) =
  do {
    a ← receive-alpha-challenge;
    record-staged-message a;
    as ← staged-alpha-program n;
    return (a # as)
  }

```

primrec *staged-composition-fri-program*

```

:: 'f staged-adversary ⇒ 'f ⇒ nat ⇒ nat ⇒
   'f list ⇒
   ('f list × 'f list, 'f protocol-channel) state-monad
where
  staged-composition-fri-program A dg i 0 bs = return ([], bs)
| staged-composition-fri-program A dg i (Suc n) bs =
  do {
    root ← composition-fri-root-stage A dg i bs;
    record-staged-message root;
    b ← receive-composition-fri-challenge;
    (roots, bs') ←
      staged-composition-fri-program A dg (Suc i) n (bs @ [b]);
    return (root # roots, bs')
  }

primrec staged-query-program
:: 'f staged-adversary ⇒ nat ⇒ nat ⇒
   ('f list list, 'f protocol-channel) state-monad
where
  staged-query-program A i 0 = return []
| staged-query-program A i (Suc n) =
  do {
    raw ← receive-query-index-challenge;
    chunk ← query-opening-stage A i raw;
    record-staged-messages chunk;
    chunks ← staged-query-program A (Suc i) n;
    return (chunk # chunks)
  }

primrec checked-staged-query-program
:: 'f staged-adversary ⇒ 'f list ⇒ 'f list ⇒
   nat ⇒ nat ⇒ ('f list list, 'f protocol-channel) state-monad
where
  checked-staged-query-program A trace-roots composition-roots i 0 =
    return []
| checked-staged-query-program A trace-roots composition-roots i (Suc n) =
  do {
    raw ← receive-query-index-challenge;
    let idx = index (to-nat raw);
    chunk ← query-opening-stage A i raw;
    assert
      (verifier-query-round-chunk idx trace-roots composition-roots chunk);
    record-staged-messages chunk;
    chunks ←
      checked-staged-query-program A trace-roots composition-roots
        (Suc i) n;
    return (chunk # chunks)
  }

```


definition *staged-transcript-program*

```

:: 'f staged-adversary =>
  ('f staged-proof-data, 'f protocol-channel) state-monad
where
  staged-transcript-program A =
    do {
      fr ← trace-root-stage A;
      record-staged-message fr;
      (trace-roots, trace-bs) ←
        staged-trace-fri-program A 0 (ceil-log clength) [];
      trace-final ← trace-final-stage A trace-bs;
      record-staged-message trace-final;
      as ← staged-alpha-program (length spec);
      dg ← degree-stage A as;
      record-staged-message dg;
      let composition-rounds = ceil-log (to-nat dg + 1);
      assert (composition-rounds ≤ ceil-log (maxDegree + 1));
      (composition-roots, composition-bs) ←
        staged-composition-fri-program A dg 0 composition-rounds [];
      composition-final ←
        composition-final-stage A dg composition-bs;
      record-staged-message composition-final;
      query-chunks ← staged-query-program A 0 rounds;
      return
        (|staged-trace-root = fr,
         staged-trace-fri-roots = trace-roots,
         staged-trace-fri-challenges = trace-bs,
         staged-trace-final = trace-final,
         staged-alphas = as,
         staged-degree = dg,
         staged-composition-fri-roots = composition-roots,
         staged-composition-fri-challenges = composition-bs,
         staged-composition-final = composition-final,
         staged-query-chunks = query-chunks|)
    }

```

definition *checked-staged-transcript-program*

```

:: 'f staged-adversary =>
  ('f staged-proof-data, 'f protocol-channel) state-monad
where
  checked-staged-transcript-program A =
    do {
      fr ← trace-root-stage A;
      record-staged-message fr;
      (trace-roots, trace-bs) ←
        staged-trace-fri-program A 0 (ceil-log clength) [];
      trace-final ← trace-final-stage A trace-bs;
      record-staged-message trace-final;
      as ← staged-alpha-program (length spec);

```

```

    dg ← degree-stage A as;
    record-staged-message dg;
    let composition-rounds = ceil-log (to-nat dg + 1);
    assert (composition-rounds ≤ ceil-log (maxDegree + 1));
    (composition-roots, composition-bs) ←
      staged-composition-fri-program A dg 0 composition-rounds [];
    composition-final ←
      composition-final-stage A dg composition-bs;
    record-staged-message composition-final;
    query-chunks ←
      checked-staged-query-program A trace-roots composition-roots
        0 rounds;
    return
      (|staged-trace-root = fr,
        staged-trace-fri-roots = trace-roots,
        staged-trace-fri-challenges = trace-bs,
        staged-trace-final = trace-final,
        staged-alphas = as,
        staged-degree = dg,
        staged-composition-fri-roots = composition-roots,
        staged-composition-fri-challenges = composition-bs,
        staged-composition-final = composition-final,
        staged-query-chunks = query-chunks)
  }

```

definition *staged-alpha-prefix-program*

```

:: 'f staged-adversary ⇒
  ('f × 'f list × 'f list × 'f, 'f protocol-channel)
  state-monad

```

where

```

staged-alpha-prefix-program A =
  do {
    fr ← trace-root-stage A;
    record-staged-message fr;
    (trace-roots, trace-bs) ←
      staged-trace-fri-program A 0 (ceil-log clength) [];
    trace-final ← trace-final-stage A trace-bs;
    record-staged-message trace-final;
    return (fr, trace-roots, trace-bs, trace-final)
  }

```

definition *staged-trace-fri-challenge-prefix-program*

```

:: 'f staged-adversary ⇒ nat ⇒
  ('f × 'f list × 'f, 'f protocol-channel) state-monad

```

where

```

staged-trace-fri-challenge-prefix-program A i =
  do {
    fr ← trace-root-stage A;
    record-staged-message fr;

```

```

    (-, trace-bs) ← staged-trace-fri-program A 0 i [];
    root ← trace-fri-root-stage A i trace-bs;
    record-staged-message root;
    return (fr, trace-bs, root)
  }

```

definition *staged-alpha-challenge-prefix-program*

```

:: 'f staged-adversary ⇒ nat ⇒
  (('f × 'f list × 'f list × 'f) × 'f list,
   'f protocol-channel) state-monad

```

where

```

staged-alpha-challenge-prefix-program A i =
  do {
    prefix ← staged-alpha-prefix-program A;
    as-prefix ← staged-alpha-program i;
    return (prefix, as-prefix)
  }

```

definition *staged-composition-fri-challenge-prefix-program*

```

:: 'f staged-adversary ⇒ nat ⇒
  ('f × 'f list × 'f, 'f protocol-channel) state-monad

```

where

```

staged-composition-fri-challenge-prefix-program A i =
  do {
    (-, -, trace-bs, -) ← staged-alpha-prefix-program A;
    as ← staged-alpha-program (length spec);
    dg ← degree-stage A as;
    record-staged-message dg;
    let composition-rounds = ceil-log (to-nat dg + 1);
    assert (Suc i ≤ composition-rounds);
    assert (composition-rounds ≤ ceil-log (maxDegree + 1));
    (-, composition-bs) ← staged-composition-fri-program A dg 0 i [];
    root ← composition-fri-root-stage A dg i composition-bs;
    record-staged-message root;
    return (dg, composition-bs, root)
  }

```

definition *staged-composition-fri-vector-program*

```

:: 'f staged-adversary ⇒
  ('f × 'f list, 'f protocol-channel) state-monad

```

where

```

staged-composition-fri-vector-program A =
  do {
    (-, -, trace-bs, -) ← staged-alpha-prefix-program A;
    as ← staged-alpha-program (length spec);
    dg ← degree-stage A as;
    record-staged-message dg;
    let composition-rounds = ceil-log (to-nat dg + 1);
    assert (composition-rounds ≤ ceil-log (maxDegree + 1));
  }

```

```

    (-, composition-bs) ←
      staged-composition-fri-program A dg 0 composition-rounds [];
    return (dg, composition-bs)
  }

```

definition *staged-composition-fri-full-vector-program*

```

:: 'f staged-adversary ⇒
  ('f staged-composition-prefix-data, 'f protocol-channel) state-monad

```

where

```

staged-composition-fri-full-vector-program A =
  do {
    (fr, trace-roots, trace-bs, trace-final) ←
      staged-alpha-prefix-program A;
    as ← staged-alpha-program (length spec);
    dg ← degree-stage A as;
    record-staged-message dg;
    let composition-rounds = ceil-log (to-nat dg + 1);
    assert (composition-rounds ≤ ceil-log (maxDegree + 1));
    (composition-roots, composition-bs) ←
      staged-composition-fri-program A dg 0 composition-rounds [];
    return
      (|scp-trace-root = fr,
        scp-trace-fri-roots = trace-roots,
        scp-trace-fri-challenges = trace-bs,
        scp-trace-final = trace-final,
        scp-alphas = as,
        scp-degree = dg,
        scp-composition-fri-roots = composition-roots,
        scp-composition-fri-challenges = composition-bs|)
  }

```

definition *staged-query-challenge-prefix-program*

```

:: 'f staged-adversary ⇒ nat ⇒
  ('f × 'f × 'f list list, 'f protocol-channel) state-monad

```

where

```

staged-query-challenge-prefix-program A i =
  do {
    (-, -, trace-bs, -) ← staged-alpha-prefix-program A;
    as ← staged-alpha-program (length spec);
    dg ← degree-stage A as;
    record-staged-message dg;
    let composition-rounds = ceil-log (to-nat dg + 1);
    assert (composition-rounds ≤ ceil-log (maxDegree + 1));
    (-, composition-bs) ←
      staged-composition-fri-program A dg 0 composition-rounds [];
    composition-final ←
      composition-final-stage A dg composition-bs;
    record-staged-message composition-final;
    query-chunks ← staged-query-program A 0 i;
  }

```

```

    return (dg, composition-final, query-chunks)
}

```

definition *checked-staged-query-challenge-prefix-program*

```

:: 'f staged-adversary  $\Rightarrow$  nat  $\Rightarrow$ 
   ('f staged-query-prefix-data, 'f protocol-channel) state-monad

```

where

```

checked-staged-query-challenge-prefix-program A i =
do {
  (fr, trace-roots, trace-bs, trace-final)  $\leftarrow$ 
    staged-alpha-prefix-program A;
  as  $\leftarrow$  staged-alpha-program (length spec);
  dg  $\leftarrow$  degree-stage A as;
  record-staged-message dg;
  let composition-rounds = ceil-log (to-nat dg + 1);
  assert (composition-rounds  $\leq$  ceil-log (maxDegree + 1));
  (composition-roots, composition-bs)  $\leftarrow$ 
    staged-composition-fri-program A dg 0 composition-rounds [];
  composition-final  $\leftarrow$ 
    composition-final-stage A dg composition-bs;
  record-staged-message composition-final;
  query-chunks  $\leftarrow$ 
    checked-staged-query-program A trace-roots composition-roots 0 i;
  return
    (\sqq-trace-root = fr,
     sqq-trace-fri-roots = trace-roots,
     sqq-trace-fri-challenges = trace-bs,
     sqq-trace-final = trace-final,
     sqq-alphas = as,
     sqq-degree = dg,
     sqq-composition-fri-roots = composition-roots,
     sqq-composition-fri-challenges = composition-bs,
     sqq-composition-final = composition-final,
     sqq-query-chunks = query-chunks)
}

```

definition *staged-proof-transcript* :: 'f staged-proof-data $\Rightarrow$ 'f list

where

```

staged-proof-transcript data =
  verifier-header-messages
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data) @
  List.concat (staged-query-chunks data)

```

definition *staged-trace-fri-start-hash* :: 'f staged-proof-data $\Rightarrow$ 'f
where
staged-trace-fri-start-hash data =
concat 0 (staged-trace-root data)

definition *staged-composition-fri-start-hash*
:: 'f staged-proof-data $\Rightarrow$ 'f
where
staged-composition-fri-start-hash data =
concat
(foldl concat
(concat
(foldl concat
(staged-trace-fri-start-hash data)
(staged-trace-fri-roots data))
(staged-trace-final data))
(staged-alphas data))
(staged-degree data)

definition *staged-query-start-hash* :: 'f staged-proof-data $\Rightarrow$ 'f
where
staged-query-start-hash data =
concat
(foldl concat
(staged-composition-fri-start-hash data)
(staged-composition-fri-roots data))
(staged-composition-final data)

definition *staged-query-chunks-match-verifier-lengths*
:: 'f staged-proof-data $\Rightarrow$ nat list $\Rightarrow$ bool
where
staged-query-chunks-match-verifier-lengths data query-idxs $\longleftrightarrow$
length (staged-query-chunks data) = rounds $\wedge$
($\forall i < \text{rounds}.$
length (staged-query-chunks data ! i) =
verifier-query-round-transcript-length (query-idxs ! i)
(staged-trace-fri-roots data)
(staged-composition-fri-roots data))

lemma *query-chunks-eq-staged-if-matching-lengths:*
assumes parser-len: length query-chunks = rounds
and concat-eq:
List.concat query-chunks @ trailing =
List.concat (staged-query-chunks data)
and parser-chunk-len:
 $\bigwedge i. i < \text{rounds} \implies$
length (query-chunks ! i) =
verifier-query-round-transcript-length (query-idxs ! i)
(staged-trace-fri-roots data)

```

      (staged-composition-fri-roots data)
    and staged-len:
      staged-query-chunks-match-verifier-lengths data query-idxs
    shows query-chunks = staged-query-chunks data  $\wedge$  trailing = []
  proof -
    have staged-chunks-len:
      length (staged-query-chunks data) = rounds
    using staged-len
    unfolding staged-query-chunks-match-verifier-lengths-def by simp
    have same-len:
      length query-chunks = length (staged-query-chunks data)
    using parser-len staged-chunks-len by simp
    have same-chunk-lens:
       $\bigwedge i. i < \text{length query-chunks} \implies$ 
      length (query-chunks ! i) =
      length (staged-query-chunks data ! i)
    proof -
      fix i
      assume i-bound:  $i < \text{length query-chunks}$ 
      then have i-round:  $i < \text{rounds}$ 
      using parser-len by simp
      show
        length (query-chunks ! i) =
        length (staged-query-chunks data ! i)
      using parser-chunk-len[OF i-round] staged-len i-round
      unfolding staged-query-chunks-match-verifier-lengths-def by simp
    qed
  show ?thesis
    by (rule concat-append-eq-concat-same-chunk-lengths
      [OF same-len same-chunk-lens concat-eq])
  qed

lemma verifier-query-round-transcript-length-index-irrelevant:
  verifier-query-round-transcript-length idx trace-roots composition-roots =
  verifier-query-round-transcript-length idx' trace-roots composition-roots
  unfolding verifier-query-round-transcript-length-def
  query-decommitment-transcript-length-def powers-scaled-def
  by simp

lemma staged-query-chunks-match-verifier-lengths-transfer:
  assumes match: staged-query-chunks-match-verifier-lengths data query-idxs
  and len: length query-idxs' = rounds
  shows staged-query-chunks-match-verifier-lengths data query-idxs'
  proof -
    have chunks-len: length (staged-query-chunks data) = rounds
    using match unfolding staged-query-chunks-match-verifier-lengths-def
    by simp
    have chunk-lens:
       $\forall i < \text{rounds}.$ 

```

```

length (staged-query-chunks data ! i) =
  verifier-query-round-transcript-length (query-idxs' ! i)
    (staged-trace-fri-roots data)
    (staged-composition-fri-roots data)
proof (intro allI impI)
  fix i
  assume i-bound: i < rounds
  have old:
    length (staged-query-chunks data ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
  using match i-bound
  unfolding staged-query-chunks-match-verifier-lengths-def by simp
show
  length (staged-query-chunks data ! i) =
    verifier-query-round-transcript-length (query-idxs' ! i)
      (staged-trace-fri-roots data)
      (staged-composition-fri-roots data)
  using old
    verifier-query-round-transcript-length-index-irrelevant
    [of query-idxs ! i
      staged-trace-fri-roots data
      staged-composition-fri-roots data query-idxs' ! i]
  by simp
qed
show ?thesis
  unfolding staged-query-chunks-match-verifier-lengths-def
  using chunks-len chunk-lens by simp
qed

```

definition *staged-transcript-trace-fri-vector-hit*
 $:: 'f \text{ list set} \Rightarrow$
 $('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
staged-transcript-trace-fri-vector-hit B out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (data, -) $\Rightarrow$ staged-trace-fri-challenges data $\in$ B)

definition *staged-transcript-alpha-list-hit*
 $:: 'f \text{ list set} \Rightarrow$
 $('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
staged-transcript-alpha-list-hit B out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (data, -) $\Rightarrow$ staged-alphas data $\in$ B)

definition *staged-transcript-composition-fri-vector-hit*
 $:: ('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
staged-transcript-composition-fri-vector-hit $B \text{ out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (data, -) \Rightarrow$
 $\text{staged-composition-fri-challenges } data \in$
 $B (\text{staged-degree } data))$

definition *staged-transcript-composition-fri-vector-guarded-hit*
 $:: ('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
staged-transcript-composition-fri-vector-guarded-hit $B \text{ out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (data, -) \Rightarrow$
 $\text{staged-composition-fri-challenges } data \in$
 $B (\text{staged-degree } data) \wedge$
 $0 < \text{ceil-log } (\text{to-nat } (\text{staged-degree } data) + 1) \wedge$
 $\text{ceil-log } (\text{to-nat } (\text{staged-degree } data) + 1) \leq$
 $\text{ceil-log } (\text{maxDegree} + 1))$

definition *staged-transcript-query-index-set-hit*
 $:: \text{nat set} \Rightarrow$
 $('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
staged-transcript-query-index-set-hit $B \text{ out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (data, t) \Rightarrow$
 $(\exists i \text{ raw. } i < \text{rounds} \wedge$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } i$
 $(\text{state-after-query-chunks}$
 $(\text{staged-query-start-hash } data)$
 $(\text{staged-query-chunks } data) \ i)) =$
 $\text{Some raw} \wedge$
 $\text{index } (\text{to-nat } \text{raw}) \in B))$

definition *staged-security-with-data-trace-fri-vector-hit*
 $:: 'f \text{ list set} \Rightarrow$
 $(('f \text{ staged-proof-data} \times \text{unit list}) \times 'f \text{ protocol-channel})$
 $\text{option} \Rightarrow \text{bool}$
where
staged-security-with-data-trace-fri-vector-hit $B \text{ out} \longleftrightarrow$
 $(\text{case out of}$

$None \Rightarrow False$
 $| Some ((data, -), -) \Rightarrow staged\text{-}trace\text{-}fri\text{-}challenges\ data \in B$

definition *staged-security-with-data-alpha-list-hit*

$:: 'f\ list\ set \Rightarrow$
 $((('f\ staged\text{-}proof\text{-}data \times unit\ list) \times 'f\ protocol\text{-}channel)$
 $option \Rightarrow bool$

where

$staged\text{-}security\text{-}with\text{-}data\text{-}alpha\text{-}list\text{-}hit\ B\ out \longleftrightarrow$
 $(case\ out\ of$
 $None \Rightarrow False$
 $| Some ((data, -), -) \Rightarrow staged\text{-}alphas\ data \in B)$

definition *staged-security-with-data-alpha-bad-set-hit*

$:: ('f\ list \Rightarrow 'f\ list\ set) \Rightarrow$
 $((('f\ staged\text{-}proof\text{-}data \times unit\ list) \times 'f\ protocol\text{-}channel)$
 $option \Rightarrow bool$

where

$staged\text{-}security\text{-}with\text{-}data\text{-}alpha\text{-}bad\text{-}set\text{-}hit\ bad\text{-}sets\ out \longleftrightarrow$
 $(case\ out\ of$
 $None \Rightarrow False$
 $| Some ((data, -), -) \Rightarrow$
 $(\exists trace\text{-}table. staged\text{-}alphas\ data \in bad\text{-}sets\ trace\text{-}table))$

definition *staged-security-with-data-composition-fri-vector-guarded-hit*

$:: ('f \Rightarrow 'f\ list\ set) \Rightarrow$
 $((('f\ staged\text{-}proof\text{-}data \times unit\ list) \times 'f\ protocol\text{-}channel)$
 $option \Rightarrow bool$

where

$staged\text{-}security\text{-}with\text{-}data\text{-}composition\text{-}fri\text{-}vector\text{-}guarded\text{-}hit\ B\ out \longleftrightarrow$
 $(case\ out\ of$
 $None \Rightarrow False$
 $| Some ((data, -), -) \Rightarrow$
 $staged\text{-}composition\text{-}fri\text{-}challenges\ data \in$
 $B\ (staged\text{-}degree\ data) \wedge$
 $0 < ceil\text{-}log\ (to\text{-}nat\ (staged\text{-}degree\ data) + 1) \wedge$
 $ceil\text{-}log\ (to\text{-}nat\ (staged\text{-}degree\ data) + 1) \leq$
 $ceil\text{-}log\ (maxDegree + 1))$

definition *staged-security-with-data-query-index-set-hit*

$:: nat\ set \Rightarrow$
 $((('f\ staged\text{-}proof\text{-}data \times unit\ list) \times 'f\ protocol\text{-}channel)$
 $option \Rightarrow bool$

where

$staged\text{-}security\text{-}with\text{-}data\text{-}query\text{-}index\text{-}set\text{-}hit\ B\ out \longleftrightarrow$
 $(case\ out\ of$
 $None \Rightarrow False$
 $| Some ((data, -), t) \Rightarrow$
 $(\exists i\ raw. i < rounds \wedge$

```

fmlookup (HashMap t)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
  Some raw ∧
  index (to-nat raw) ∈ B))

```

definition *staged-security-with-data-query-bad-hit*
 $:: ((\text{'f staged-proof-data} \times \text{unit list}) \times \text{'f protocol-channel})$
 $\text{option} \Rightarrow \text{bool}$

where

```

staged-security-with-data-query-bad-hit out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some ((-, result), final-state)  $\Rightarrow$ 
    ( $\exists$  s.
      Some (result, final-state)  $\in$  set-dist (execute verify-monad s)  $\wedge$ 
      query-bad s (Some (result, final-state))))

```

definition *staged-security-with-data-query-sampling-hit*
 $:: ((\text{'f staged-proof-data} \times \text{unit list}) \times \text{'f protocol-channel})$
 $\text{option} \Rightarrow \text{bool}$

where

```

staged-security-with-data-query-sampling-hit out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some ((-, result), final-state)  $\Rightarrow$ 
    ( $\exists$  s trace-table composition-table as.
      query-rounds-any-index-set-hit s
        (query-sampling-success-space trace-table composition-table as)
        rounds
        (Some (result, final-state))))

```

definition *staged-security-with-data-query-header-sampling-hit*
 $:: ((\text{'f staged-proof-data} \times \text{unit list}) \times \text{'f protocol-channel})$
 $\text{option} \Rightarrow \text{bool}$

where

```

staged-security-with-data-query-header-sampling-hit out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some ((-, result), final-state)  $\Rightarrow$ 
    ( $\exists$  s.
      query-header-rounds-any-index-set-hit s
        (query-header-supported-union-good-sets s
          query-sampling-success-space)
        (Some (result, final-state))))

```

definition *staged-security-with-data-state-query-bad-hit*

$:: (((\text{'f staged-proof-data} \times \text{'f protocol-channel}) \times \text{unit list}) \times \text{'f protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{staged-security-with-data-state-query-bad-hit out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((\text{data}, \text{attacker-state}), \text{result}), \text{final-state}) \Rightarrow$
 $(\text{let } s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $\text{(staged-proof-transcript data)}$
 in
 $\text{Some (result, final-state)} \in \text{set-dist (execute verify-monad s)} \wedge$
 $\text{query-bad s (Some (result, final-state))}))$

definition *staged-security-with-data-state-query-header-sampling-hit*

$:: (((\text{'f staged-proof-data} \times \text{'f protocol-channel}) \times \text{unit list}) \times \text{'f protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{staged-security-with-data-state-query-header-sampling-hit out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((\text{data}, \text{attacker-state}), \text{result}), \text{final-state}) \Rightarrow$
 $(\text{let } s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $\text{(staged-proof-transcript data)}$
 in
 $\text{query-header-rounds-any-index-set-hit s}$
 $\text{(query-header-supported-union-good-sets s}$
 $\text{query-sampling-success-space)}$
 $\text{(Some (result, final-state))}))$

definition *staged-security-with-data-state-query-header-key-hit*

$:: (((\text{'f staged-proof-data} \times \text{'f protocol-channel}) \times \text{unit list}) \times \text{'f protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{staged-security-with-data-state-query-header-key-hit out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((\text{data}, \text{attacker-state}), -), \text{final-state}) \Rightarrow$
 $(\text{let } s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $\text{(staged-proof-transcript data);}$
 $B =$
 $\text{query-header-supported-union-good-sets s}$
 $\text{query-sampling-success-space}$
 $\text{(staged-trace-root data)}$
 $\text{(staged-trace-fri-roots data)}$
 $\text{(staged-trace-final data)}$
 $\text{(staged-alphas data)})$

(staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
 in
 (∃ i raw. i < rounds ∧
 fmlookup (HashMap final-state)
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data)
 (staged-query-chunks data) i)) =
 Some raw ∧
 index (to-nat raw) ∈ B)))

definition staged-security-with-data-state-verifier-event

:: ('f protocol-channel ⇒
 (unit list × 'f protocol-channel) option ⇒ bool) ⇒
 (((('f staged-proof-data × 'f protocol-channel) × unit list) ×
 'f protocol-channel) option ⇒ bool

where

staged-security-with-data-state-verifier-event P out ↔
 (case out of
 None ⇒ False
 | Some (((data, attacker-state), result), final-state) ⇒
 P
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (Some (result, final-state)))

definition staged-security-with-data-state-transcript-pre-hit

:: (((('f staged-proof-data × 'f protocol-channel) × unit list) ×
 'f protocol-channel) option ⇒ bool

where

staged-security-with-data-state-transcript-pre-hit out ↔
 (case out of
 None ⇒ False
 | Some (((data, attacker-state), -), -) ⇒
 hash-map-output-values
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ∩
 set (staged-proof-transcript data) ≠ {})

definition staged-security-with-data-state-transcript-new-hit

:: (((('f staged-proof-data × 'f protocol-channel) × unit list) ×
 'f protocol-channel) option ⇒ bool

where

staged-security-with-data-state-transcript-new-hit out ↔
 (case out of
 None ⇒ False
 | Some (((data, attacker-state), -), final-state) ⇒

$\text{hash-map-new-output-hit } (\text{set } (\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $\text{final-state})$

definition *staged-composition-prefix-fri-vector-hit*

$:: ('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $('f \text{ staged-composition-prefix-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow$
 bool

where

$\text{staged-composition-prefix-fri-vector-hit } B \text{ out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (\text{prefix}, -) \Rightarrow$
 $\text{scp-composition-fri-challenges prefix} \in$
 $B (\text{scp-degree prefix}) \wedge$
 $0 < \text{ceil-log } (\text{to-nat } (\text{scp-degree prefix}) + 1) \wedge$
 $\text{ceil-log } (\text{to-nat } (\text{scp-degree prefix}) + 1) \leq$
 $\text{ceil-log } (\text{maxDegree} + 1))$

definition *staged-after-alpha-prefix-program*

$:: 'f \text{ staged-adversary} \Rightarrow$
 $'f \times 'f \text{ list} \times 'f \text{ list} \times 'f \Rightarrow$
 $('f \text{ staged-proof-data}, 'f \text{ protocol-channel}) \text{ state-monad}$

where

$\text{staged-after-alpha-prefix-program } A \text{ prefix} =$
 $(\text{case prefix of } (\text{fr}, \text{trace-roots}, \text{trace-bs}, \text{trace-final}) \Rightarrow$
 $\text{do } \{$
 $\text{as} \leftarrow \text{staged-alpha-program } (\text{length spec});$
 $\text{dg} \leftarrow \text{degree-stage } A \text{ as};$
 $\text{record-staged-message dg};$
 $\text{let composition-rounds} = \text{ceil-log } (\text{to-nat dg} + 1);$
 $\text{assert } (\text{composition-rounds} \leq \text{ceil-log } (\text{maxDegree} + 1));$
 $(\text{composition-roots}, \text{composition-bs}) \leftarrow$
 $\text{staged-composition-fri-program } A \text{ dg } 0 \text{ composition-rounds } [];$
 $\text{composition-final} \leftarrow$
 $\text{composition-final-stage } A \text{ dg composition-bs};$
 $\text{record-staged-message composition-final};$
 $\text{query-chunks} \leftarrow \text{staged-query-program } A \text{ } 0 \text{ rounds};$
 return
 $(\text{staged-trace-root} = \text{fr},$
 $\text{staged-trace-fri-roots} = \text{trace-roots},$
 $\text{staged-trace-fri-challenges} = \text{trace-bs},$
 $\text{staged-trace-final} = \text{trace-final},$
 $\text{staged-alphas} = \text{as},$
 $\text{staged-degree} = \text{dg},$
 $\text{staged-composition-fri-roots} = \text{composition-roots},$
 $\text{staged-composition-fri-challenges} = \text{composition-bs},$
 $\text{staged-composition-final} = \text{composition-final},$

staged-query-chunks = query-chunks)
 })

definition *staged-after-composition-fri-full-vector-program*
 $:: 'f \text{ staged-adversary} \Rightarrow 'f \text{ staged-composition-prefix-data} \Rightarrow$
 $('f \text{ staged-proof-data}, 'f \text{ protocol-channel}) \text{ state-monad}$

where

staged-after-composition-fri-full-vector-program *A* *prefix* =
 do {
 composition-final $\leftarrow$
 composition-final-stage *A*
 (*scp-degree* *prefix*)
 (*scp-composition-fri-challenges* *prefix*);
 record-staged-message *composition-final*;
 query-chunks $\leftarrow$ *staged-query-program* *A* 0 rounds;
 return
 (*staged-trace-root* = *scp-trace-root* *prefix*,
 staged-trace-fri-roots = *scp-trace-fri-roots* *prefix*,
 staged-trace-fri-challenges = *scp-trace-fri-challenges* *prefix*,
 staged-trace-final = *scp-trace-final* *prefix*,
 staged-alphas = *scp-alphas* *prefix*,
 staged-degree = *scp-degree* *prefix*,
 staged-composition-fri-roots =
 scp-composition-fri-roots *prefix*,
 staged-composition-fri-challenges =
 scp-composition-fri-challenges *prefix*,
 staged-composition-final = *composition-final*,
 staged-query-chunks = *query-chunks*)
 }

lemma *staged-transcript-program-alpha-prefix-decomp*:

staged-transcript-program *A* =
staged-alpha-prefix-program *A* $\gg$
staged-after-alpha-prefix-program *A*

unfolding *staged-transcript-program-def* *staged-alpha-prefix-program-def*
staged-after-alpha-prefix-program-def *Let-def*

by (*simp* add: *sm-bind-assoc* *split-def* *split*: *prod.splits*)

lemma *staged-transcript-program-composition-fri-full-vector-decomp*:

staged-transcript-program *A* =
staged-composition-fri-full-vector-program *A* $\gg$
staged-after-composition-fri-full-vector-program *A*

unfolding *staged-transcript-program-def*
staged-composition-fri-full-vector-program-def
staged-after-composition-fri-full-vector-program-def
staged-alpha-prefix-program-def *Let-def*

by (*simp* add: *sm-bind-assoc* *split-def* *split*: *prod.splits*)

lemma *staged-composition-fri-vector-program-projection*:

```

staged-composition-fri-vector-program A =
  staged-composition-fri-full-vector-program A  $\gg$ 
    ( $\lambda$ prefix. return
      (scp-degree prefix, scp-composition-fri-challenges prefix))
unfolding staged-composition-fri-vector-program-def
  staged-composition-fri-full-vector-program-def
  staged-alpha-prefix-program-def Let-def
by (simp add: sm-bind-assoc split-def split: prod.splits)

lemma staged-after-alpha-prefix-program-trace-fri-vector-hitD:
assumes outcome:
  out  $\in$ 
    set-dist
    (execute
      (staged-after-alpha-prefix-program A
        (fr, trace-roots, trace-bs, trace-final)) s)
    and hit: staged-transcript-trace-fri-vector-hit B out
shows trace-bs  $\in$  B
proof (cases out)
case None
then show ?thesis
  using hit unfolding staged-transcript-trace-fri-vector-hit-def by simp
next
case (Some result)
then obtain data u where out-eq: out = Some (data, u)
  by (cases result) simp
from outcome[unfolded out-eq]
obtain as s1 dg s2 s3 composition-roots composition-bs s4
  composition-final s5 s6 query-chunks where
ret:
  Some (data, u)  $\in$ 
    set-dist
    (execute
      (return
        (staged-trace-root = fr,
          staged-trace-fri-roots = trace-roots,
          staged-trace-fri-challenges = trace-bs,
          staged-trace-final = trace-final,
          staged-alphas = as,
          staged-degree = dg,
          staged-composition-fri-roots = composition-roots,
          staged-composition-fri-challenges = composition-bs,
          staged-composition-final = composition-final,
          staged-query-chunks = query-chunks)))
      s6)
  unfolding staged-after-alpha-prefix-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have data-eq:
  data =

```



```

(|staged-trace-root = fr,
 staged-trace-fri-roots = trace-roots,
 staged-trace-fri-challenges = trace-bs,
 staged-trace-final = trace-final,
 staged-alphas = as,
 staged-degree = dg,
 staged-composition-fri-roots = composition-roots,
 staged-composition-fri-challenges = composition-bs,
 staged-composition-final = composition-final,
 staged-query-chunks = query-chunks)
using ret by simp
show ?thesis
using hit unfolding out-eq data-eq
  staged-transcript-trace-fri-vector-hit-def
by simp
qed

lemma staged-after-alpha-prefix-program-alpha-list-hitD:
assumes outcome:
  out ∈
    set-dist
      (execute
        (staged-after-alpha-prefix-program A
          (fr, trace-roots, trace-bs, trace-final)) s)
and hit: staged-transcript-alpha-list-hit B out
shows
  ∃ as s1.
    Some (as, s1) ∈
      set-dist (execute (staged-alpha-program (length spec)) s) ∧
      as ∈ B
proof (cases out)
case None
then show ?thesis
  using hit unfolding staged-transcript-alpha-list-hit-def by simp
next
case (Some result)
then obtain data u where out-eq: out = Some (data, u)
  by (cases result) simp
from outcome[unfolded out-eq]
obtain as s1 dg s2 s3 composition-roots composition-bs s4
  composition-final s5 s6 query-chunks where
  alpha-out:
    Some (as, s1) ∈
      set-dist (execute (staged-alpha-program (length spec)) s)
and ret:
  Some (data, u) ∈
    set-dist
      (execute
        (return

```

```

    (staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots = composition-roots,
     staged-composition-fri-challenges = composition-bs,
     staged-composition-final = composition-final,
     staged-query-chunks = query-chunks))
  s6)
unfolding staged-after-alpha-prefix-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have data-eq:
  data =
    (staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots = composition-roots,
     staged-composition-fri-challenges = composition-bs,
     staged-composition-final = composition-final,
     staged-query-chunks = query-chunks)
using ret by simp
have as-in: as ∈ B
using hit unfolding out-eq data-eq
  staged-transcript-alpha-list-hit-def
by simp
show ?thesis
using alpha-out as-in by blast
qed

lemma staged-after-alpha-prefix-program-alpha-list-bound-from-alpha-program:
assumes alpha-bound:
  wp-event (staged-alpha-program (length spec))
  (λout. case out of None ⇒ False
    | Some (as, -) ⇒ as ∈ B) s ≤ C
shows
  wp-event
    (staged-after-alpha-prefix-program A
     (fr, trace-roots, trace-bs, trace-final))
    (staged-transcript-alpha-list-hit B) s ≤ C
proof –
have decomp:
  staged-after-alpha-prefix-program A
    (fr, trace-roots, trace-bs, trace-final) =
    (staged-alpha-program (length spec) ≫=

```

```

( $\lambda$ as. degree-stage A as  $\gg$ 
( $\lambda$ dg. record-staged-message dg  $\gg$ 
( $\lambda$ -. let composition-rounds =
  ceil-log (to-nat dg + 1)
in assert
  (composition-rounds  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
( $\lambda$ -. staged-composition-fri-program A dg 0
  composition-rounds []  $\gg$ 
( $\lambda$ (composition-roots, composition-bs).
  composition-final-stage A dg composition-bs  $\gg$ 
( $\lambda$ composition-final.
  record-staged-message composition-final  $\gg$ 
( $\lambda$ -. staged-query-program A 0 rounds  $\gg$ 
( $\lambda$ query-chunks.
  return
    (staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots =
       composition-roots,
     staged-composition-fri-challenges =
       composition-bs,
     staged-composition-final =
       composition-final,
     staged-query-chunks =
       query-chunks)))))))))
unfolding staged-after-alpha-prefix-program-def Let-def
by simp
show ?thesis
unfolding decomp
proof (rule wp-event-bind-bound-by-head-event[OF alpha-bound])
show staged-transcript-alpha-list-hit B None  $\impl$ 
  (case None of None  $\impl$  False | Some (as, -)  $\impl$  as  $\in$  B)
unfolding staged-transcript-alpha-list-hit-def by simp
next
fix as s1 out
assume alpha-out:
  Some (as, s1)  $\in$ 
    set-dist (execute (staged-alpha-program (length spec)) s)
and cont:
  out  $\in$ 
    set-dist
      (execute
        (degree-stage A as  $\gg$ 
          ( $\lambda$ dg. record-staged-message dg  $\gg$ 
            ( $\lambda$ -. let composition-rounds =

```

```

      ceil-log (to-nat dg + 1)
in assert
  (composition-rounds ≤ ceil-log (maxDegree + 1)) ≫=
  (λ-. staged-composition-fri-program A dg 0
    composition-rounds [] ≫=
    (λ(composition-roots, composition-bs).
      composition-final-stage A dg composition-bs ≫=
      (λcomposition-final.
        record-staged-message composition-final ≫=
        (λ-. staged-query-program A 0 rounds ≫=
          (λquery-chunks.
            return
              (|staged-trace-root = fr,
                staged-trace-fri-roots = trace-roots,
                staged-trace-fri-challenges = trace-bs,
                staged-trace-final = trace-final,
                staged-alphas = as,
                staged-degree = dg,
                staged-composition-fri-roots =
                  composition-roots,
                staged-composition-fri-challenges =
                  composition-bs,
                staged-composition-final =
                  composition-final,
                staged-query-chunks =
                  query-chunks|)))))))))
    s1)
  and hit: staged-transcript-alpha-list-hit B out
show (case Some (as, s1) of None ⇒ False
  | Some (as, -) ⇒ as ∈ B)
proof (cases out)
case None
then show ?thesis
  using hit unfolding staged-transcript-alpha-list-hit-def by simp
next
case (Some result)
then obtain data u where out-eq: out = Some (data, u)
  by (cases result) simp
from cont[unfolded out-eq]
obtain dg s2 s3 composition-roots composition-bs s4
  composition-final s5 s6 query-chunks where
ret:
  Some (data, u) ∈
  set-dist
  (execute
    (return
      (|staged-trace-root = fr,
        staged-trace-fri-roots = trace-roots,
        staged-trace-fri-challenges = trace-bs,

```

```

    staged-trace-final = trace-final,
    staged-alphas = as,
    staged-degree = dg,
    staged-composition-fri-roots = composition-roots,
    staged-composition-fri-challenges = composition-bs,
    staged-composition-final = composition-final,
    staged-query-chunks = query-chunks))
  s6)
  by (auto simp: Let-def assert-def
    elim!: set-dist-bindE split: prod.splits)
  have data-eq:
    data =
      (|staged-trace-root = fr,
        staged-trace-fri-roots = trace-roots,
        staged-trace-fri-challenges = trace-bs,
        staged-trace-final = trace-final,
        staged-alphas = as,
        staged-degree = dg,
        staged-composition-fri-roots = composition-roots,
        staged-composition-fri-challenges = composition-bs,
        staged-composition-final = composition-final,
        staged-query-chunks = query-chunks|)
    using ret by simp
  show ?thesis
    using hit unfolding out-eq data-eq
      staged-transcript-alpha-list-hit-def by simp
qed
qed
qed

lemma staged-after-composition-fri-full-vector-program-hitD:
  assumes outcome:
    out ∈
      set-dist
        (execute
          (staged-after-composition-fri-full-vector-program A prefix) s)
  and hit:
    staged-transcript-composition-fri-vector-guarded-hit B out
  shows staged-composition-prefix-fri-vector-hit B (Some (prefix, s))
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding staged-transcript-composition-fri-vector-guarded-hit-def
    by simp
next
  case (Some result)
  then obtain data u where out-eq: out = Some (data, u)
    by (cases result) simp

```

```

from outcome[unfolded out-eq]
obtain composition-final s1 s2 query-chunks where
  ret:
    Some (data, u) ∈
      set-dist
      (execute
        (return
          (|staged-trace-root = scp-trace-root prefix,
            staged-trace-fri-roots = scp-trace-fri-roots prefix,
            staged-trace-fri-challenges =
              scp-trace-fri-challenges prefix,
            staged-trace-final = scp-trace-final prefix,
            staged-alphas = scp-alphas prefix,
            staged-degree = scp-degree prefix,
            staged-composition-fri-roots =
              scp-composition-fri-roots prefix,
            staged-composition-fri-challenges =
              scp-composition-fri-challenges prefix,
            staged-composition-final = composition-final,
            staged-query-chunks = query-chunks))
          s2)
    unfolding staged-after-composition-fri-full-vector-program-def
    by (auto elim!: set-dist-bindE split: prod.splits)
have data-eq:
  data =
    (|staged-trace-root = scp-trace-root prefix,
      staged-trace-fri-roots = scp-trace-fri-roots prefix,
      staged-trace-fri-challenges = scp-trace-fri-challenges prefix,
      staged-trace-final = scp-trace-final prefix,
      staged-alphas = scp-alphas prefix,
      staged-degree = scp-degree prefix,
      staged-composition-fri-roots =
        scp-composition-fri-roots prefix,
      staged-composition-fri-challenges =
        scp-composition-fri-challenges prefix,
      staged-composition-final = composition-final,
      staged-query-chunks = query-chunks)
    using ret by simp
show ?thesis
  using hit
  unfolding out-eq data-eq
    staged-transcript-composition-fri-vector-guarded-hit-def
    staged-composition-prefix-fri-vector-hit-def
  by simp
qed

lemma staged-proof-transcript-verifier-header-transcript:
  assumes trace-len:
    length (staged-trace-fri-roots data) = ceil-log clength

```

and *alpha-len*:
 $\text{length } (\text{staged-alphas } \text{data}) = \text{length } \text{spec}$
and *composition-len*:
 $\text{length } (\text{staged-composition-fri-roots } \text{data}) =$
 $\text{ceil-log } (\text{to-nat } (\text{staged-degree } \text{data}) + 1)$
shows
verifier-header-transcript
 $(\text{verifier-state-from-adversary } s \ (\text{staged-proof-transcript } \text{data}))$
 $(\text{staged-trace-root } \text{data})$
 $(\text{staged-trace-fri-roots } \text{data})$
 $(\text{staged-trace-final } \text{data})$
 $(\text{staged-alphas } \text{data})$
 $(\text{staged-degree } \text{data})$
 $(\text{staged-composition-fri-roots } \text{data})$
 $(\text{staged-composition-final } \text{data})$
 $(\text{List.concat } (\text{staged-query-chunks } \text{data}))$
using *trace-len alpha-len composition-len*
unfolding *verifier-header-transcript-def staged-proof-transcript-def*
by *simp*

definition *staged-security-experiment*
 $:: 'f \text{ staged-adversary} \Rightarrow$
 $(\text{unit list}, 'f \text{ protocol-channel}) \text{ state-monad}$
where
staged-security-experiment $A =$
staged-transcript-program $A \gg= (\lambda \text{data}.$
 $\text{get} \gg= (\lambda s.$
 put
 $(\text{verifier-state-from-adversary } s$
 $(\text{staged-proof-transcript } \text{data})) \gg=$
 $(\lambda -. \text{verify-monad}))$

definition *staged-security-experiment-with-data*
 $:: 'f \text{ staged-adversary} \Rightarrow$
 $('f \text{ staged-proof-data} \times \text{unit list}, 'f \text{ protocol-channel}) \text{ state-monad}$
where
staged-security-experiment-with-data $A =$
staged-transcript-program $A \gg= (\lambda \text{data}.$
 $\text{get} \gg= (\lambda s.$
 put
 $(\text{verifier-state-from-adversary } s$
 $(\text{staged-proof-transcript } \text{data})) \gg=$
 $(\lambda -. \text{verify-monad} \gg= (\lambda \text{result}. \text{return } (\text{data}, \text{result}))))$

definition *staged-security-experiment-with-data-state*
 $:: 'f \text{ staged-adversary} \Rightarrow$
 $(('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \times \text{unit list},$
 $'f \text{ protocol-channel}) \text{ state-monad}$
where

staged-security-experiment-with-data-state $A =$
staged-transcript-program $A \gg= (\lambda data.$
get $\gg= (\lambda s.$
put
(verifier-state-from-adversary s
(staged-proof-transcript $data)) \gg=$
($\lambda.$
verify-monad $\gg=$
($\lambda result.$ *return* *((data, s), result))))*

definition *checked-staged-security-experiment*
 $:: 'f \text{ staged-adversary} \Rightarrow$
 $(\text{unit list}, 'f \text{ protocol-channel}) \text{ state-monad}$
where
checked-staged-security-experiment $A =$
checked-staged-transcript-program $A \gg= (\lambda data.$
get $\gg= (\lambda s.$
put
(verifier-state-from-adversary s
(staged-proof-transcript $data)) \gg=$
($\lambda.$ *verify-monad**))*

definition *checked-staged-security-experiment-with-data*
 $:: 'f \text{ staged-adversary} \Rightarrow$
 $('f \text{ staged-proof-data} \times \text{unit list}, 'f \text{ protocol-channel}) \text{ state-monad}$
where
checked-staged-security-experiment-with-data $A =$
checked-staged-transcript-program $A \gg= (\lambda data.$
get $\gg= (\lambda s.$
put
(verifier-state-from-adversary s
(staged-proof-transcript $data)) \gg=$
($\lambda.$ *verify-monad* $\gg= (\lambda result.$ *return* *(data, result))))*

definition *checked-staged-security-experiment-with-data-state*
 $:: 'f \text{ staged-adversary} \Rightarrow$
 $(('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \times \text{unit list},$
 $'f \text{ protocol-channel}) \text{ state-monad}$
where
checked-staged-security-experiment-with-data-state $A =$
checked-staged-transcript-program $A \gg= (\lambda data.$
get $\gg= (\lambda s.$
put
(verifier-state-from-adversary s
(staged-proof-transcript $data)) \gg=$
($\lambda.$
verify-monad $\gg=$
($\lambda result.$ *return* *((data, s), result))))*

lemma *staged-security-experiment-projection:*

staged-security-experiment-with-data $A \gg=$
 $(\lambda x. \text{return } (\text{snd } x)) =$
 staged-security-experiment A
unfolding *staged-security-experiment-with-data-def*
 staged-security-experiment-def
by (*simp add: sm-bind-assoc split: prod.splits*)

lemma *staged-security-experiment-with-data-state-projection:*

staged-security-experiment-with-data-state $A \gg=$
 $(\lambda x. \text{return } (\text{snd } x)) =$
 staged-security-experiment A
unfolding *staged-security-experiment-with-data-state-def*
 staged-security-experiment-def
by (*simp add: sm-bind-assoc split: prod.splits*)

lemma *staged-security-experiment-with-data-state-projection-to-data:*

staged-security-experiment-with-data-state $A \gg=$
 $(\lambda x. \text{case } x \text{ of } ((\text{data}, -), \text{result}) \Rightarrow$
 $\text{return } (\text{data}, \text{result})) =$
 staged-security-experiment-with-data A
unfolding *staged-security-experiment-with-data-state-def*
 staged-security-experiment-with-data-def
by (*simp add: sm-bind-assoc split: prod.splits*)

lemma *checked-staged-security-experiment-projection:*

checked-staged-security-experiment-with-data $A \gg=$
 $(\lambda x. \text{return } (\text{snd } x)) =$
 checked-staged-security-experiment A
unfolding *checked-staged-security-experiment-with-data-def*
 checked-staged-security-experiment-def
by (*simp add: sm-bind-assoc split: prod.splits*)

lemma *checked-staged-security-experiment-with-data-state-projection:*

checked-staged-security-experiment-with-data-state $A \gg=$
 $(\lambda x. \text{return } (\text{snd } x)) =$
 checked-staged-security-experiment A
unfolding *checked-staged-security-experiment-with-data-state-def*
 checked-staged-security-experiment-def
by (*simp add: sm-bind-assoc split: prod.splits*)

lemma *checked-staged-security-experiment-with-data-state-projection-to-data:*

checked-staged-security-experiment-with-data-state $A \gg=$
 $(\lambda x. \text{case } x \text{ of } ((\text{data}, -), \text{result}) \Rightarrow$
 $\text{return } (\text{data}, \text{result})) =$
 checked-staged-security-experiment-with-data A
unfolding *checked-staged-security-experiment-with-data-state-def*
 checked-staged-security-experiment-with-data-def
by (*simp add: sm-bind-assoc split: prod.splits*)

lemma *staged-security-experiment-acceptance-with-data:*

$$\text{wp-event } (\text{staged-security-experiment } A) \text{ accepted adversary-initial-state} =$$

$$\text{wp-event } (\text{staged-security-experiment-with-data } A) \text{ accepted}$$

$$\text{adversary-initial-state}$$
proof –
have *proj*:

$$\text{staged-security-experiment } A =$$

$$\text{staged-security-experiment-with-data } A \gg=$$

$$(\lambda x. \text{return } (\text{snd } x))$$
using *staged-security-experiment-projection* **by** *simp*
show *?thesis*
unfolding *proj*
by (*subst wp-event-bind-return-map[where f=snd]*)
(unfold wp-event-def accepted-def,
rule arg-cong[where
f= $\lambda Q. \text{wp } (\text{staged-security-experiment-with-data } A) \text{ } Q$
adversary-initial-state],
rule ext, simp split: option.splits prod.splits)
qed

lemma *staged-security-experiment-acceptance-with-data-state:*

$$\text{wp-event } (\text{staged-security-experiment } A) \text{ accepted adversary-initial-state} =$$

$$\text{wp-event } (\text{staged-security-experiment-with-data-state } A) \text{ accepted}$$

$$\text{adversary-initial-state}$$
proof –
have *proj*:

$$\text{staged-security-experiment } A =$$

$$\text{staged-security-experiment-with-data-state } A \gg=$$

$$(\lambda x. \text{return } (\text{snd } x))$$
using *staged-security-experiment-with-data-state-projection* **by** *simp*
show *?thesis*
unfolding *proj*
by (*subst wp-event-bind-return-map[where f=snd]*)
(unfold wp-event-def accepted-def,
rule arg-cong[where
f= $\lambda Q. \text{wp } (\text{staged-security-experiment-with-data-state } A) \text{ } Q$
adversary-initial-state],
rule ext, simp split: option.splits prod.splits)
qed

lemma *checked-staged-security-experiment-acceptance-with-data:*

$$\text{wp-event } (\text{checked-staged-security-experiment } A) \text{ accepted}$$

$$\text{adversary-initial-state} =$$

$$\text{wp-event } (\text{checked-staged-security-experiment-with-data } A) \text{ accepted}$$

$$\text{adversary-initial-state}$$
proof –
have *proj*:

$$\text{checked-staged-security-experiment } A =$$

```

    checked-staged-security-experiment-with-data A  $\gg$ 
    ( $\lambda x$ . return (snd x))
  using checked-staged-security-experiment-projection by simp
show ?thesis
  unfolding proj
  by (subst wp-event-bind-return-map[where f=snd])
    (unfold wp-event-def accepted-def,
     rule arg-cong[where
      f= $\lambda Q$ . wp (checked-staged-security-experiment-with-data A) Q
      adversary-initial-state],
     rule ext, simp split: option.splits prod.splits)
qed

lemma checked-staged-security-experiment-acceptance-with-data-state:
  wp-event (checked-staged-security-experiment A) accepted
  adversary-initial-state =
  wp-event (checked-staged-security-experiment-with-data-state A) accepted
  adversary-initial-state
proof -
  have proj:
    checked-staged-security-experiment A =
    checked-staged-security-experiment-with-data-state A  $\gg$ 
    ( $\lambda x$ . return (snd x))
  using checked-staged-security-experiment-with-data-state-projection
  by simp
show ?thesis
  unfolding proj
  by (subst wp-event-bind-return-map[where f=snd])
    (unfold wp-event-def accepted-def,
     rule arg-cong[where
      f= $\lambda Q$ . wp (checked-staged-security-experiment-with-data-state A) Q
      adversary-initial-state],
     rule ext, simp split: option.splits prod.splits)
qed

lemma staged-security-experiment-with-data-event-from-data-state:
  wp-event (staged-security-experiment-with-data A) P adversary-initial-state =
  wp-event (staged-security-experiment-with-data-state A)
  ( $\lambda out$ . case out of None  $\Rightarrow$  P None
    | Some (((data, -), result), t)  $\Rightarrow$  P (Some ((data, result), t)))
  adversary-initial-state
proof -
  have proj:
    staged-security-experiment-with-data A =
    staged-security-experiment-with-data-state A  $\gg$ 
    ( $\lambda x$ . case x of ((data, -), result)  $\Rightarrow$ 
      return (data, result))
  using staged-security-experiment-with-data-state-projection-to-data
  by simp

```

```

show ?thesis
  unfolding proj wp-event-def
  apply (subst wp-bind)
  apply (rule arg-cong[where
     $f = \lambda Q. wp \ (staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A) \ Q$ 
    adversary-initial-state])
  apply (rule ext)
  apply (simp add: wp-return split: option.splits prod.splits)
  done
qed

lemma checked-staged-security-experiment-with-data-event-from-data-state:
  wp-event (checked-staged-security-experiment-with-data A) P
    adversary-initial-state =
  wp-event (checked-staged-security-experiment-with-data-state A)
    ( $\lambda out. case\ out\ of\ None \Rightarrow P\ None$ 
      |  $Some\ (((data, -), result), t) \Rightarrow P\ (Some\ ((data, result), t))$ )
    adversary-initial-state
proof -
  have proj:
    checked-staged-security-experiment-with-data A =
    checked-staged-security-experiment-with-data-state A  $\ggg$ 
    ( $\lambda x. case\ x\ of\ ((data, -), result) \Rightarrow$ 
      return (data, result))
  using checked-staged-security-experiment-with-data-state-projection-to-data
  by simp
show ?thesis
  unfolding proj wp-event-def
  apply (subst wp-bind)
  apply (rule arg-cong[where
     $f = \lambda Q. wp \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A) \ Q$ 
    adversary-initial-state])
  apply (rule ext)
  apply (simp add: wp-return split: option.splits prod.splits)
  done
qed

lemma staged-security-experiment-with-data-state-outcomeE:
  assumes
    Some (((data, attacker-state), result), final-state)  $\in$ 
      set-dist
      (execute (staged-security-experiment-with-data-state A) initial-state)
  obtains
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (staged-transcript-program A) initial-state)
    Some (result, final-state)  $\in$ 
      set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state

```

```

      (staged-proof-transcript data)))
using assms unfolding staged-security-experiment-with-data-state-def
by (auto elim!: set-dist-bindE intro: that)

lemma checked-staged-security-experiment-with-data-state-outcomeE:
assumes
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-data-state A)
      initial-state)
obtains
  Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A) initial-state)
  Some (result, final-state) ∈
    set-dist
    (execute verify-monad
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
using assms
unfolding checked-staged-security-experiment-with-data-state-def
by (auto elim!: set-dist-bindE intro: that)

lemma checked-staged-security-with-data-state-verifier-event-bound-from-cont:
assumes no-none:  $\bigwedge s. \neg P\ s\ \text{None}$ 
and cont-bound:
 $\bigwedge \text{data}\ \text{attacker-state}.$ 
  Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    wp-event verify-monad
      (P
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))  $\leq C$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event P)
    adversary-initial-state  $\leq C$ 
unfolding checked-staged-security-experiment-with-data-state-def
proof (rule wp-event-bind-bound-by-cont)
show  $\neg$  staged-security-with-data-state-verifier-event P None
unfolding staged-security-with-data-state-verifier-event-def by simp
next
fix data attacker-state
assume builder:

```

```

Some (data, attacker-state) ∈
  set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have cont-eq:
  wp-event
    (get >=>
      (λs. put
        (verifier-state-from-adversary s
          (staged-proof-transcript data)) >=>
        (λ-. verify-monad >=>
          (λresult. return ((data, s), result))))))
    (staged-security-with-data-state-verifier-event P) attacker-state =
  wp-event verify-monad (P ?s) ?s
unfolding wp-event-def staged-security-with-data-state-verifier-event-def
apply (simp add: wpsimps)
apply (rule arg-cong[where f=λQ. wp verify-monad Q ?s])
apply (rule ext)
apply (simp add: no-none wp-return split: option.splits prod.splits)
done
show
  wp-event
    (get >=>
      (λs. put
        (verifier-state-from-adversary s
          (staged-proof-transcript data)) >=>
        (λ-. verify-monad >=>
          (λresult. return ((data, s), result))))))
    (staged-security-with-data-state-verifier-event P) attacker-state ≤ C
unfolding cont-eq by (rule cont-bound[OF builder])
qed

```

lemma checked-staged-security-with-data-state-bound-from-data-cont:

assumes none: $\neg P \text{ None}$

and cont-bound:

$\bigwedge \text{data attacker-state.}$

Some (data, attacker-state) ∈

set-dist

(execute (checked-staged-transcript-program A)

adversary-initial-state) $\implies$

wp-event verify-monad

(λout.

P (case out of

None $\implies$ None

| Some (result, final-state) $\implies$

Some (((data, attacker-state), result), final-state)))

```

      (verifier-state-from-adversary attacker-state
       (staged-proof-transcript data)) ≤ C
shows
  wp-event (checked-staged-security-experiment-with-data-state A) P
    adversary-initial-state ≤ C
unfolding checked-staged-security-experiment-with-data-state-def
proof (rule wp-event-bind-bound-by-cont)
  show ¬ P None
    by (rule none)
next
  fix data attacker-state
  assume builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
         adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have cont-eq:
    wp-event
      (get ≥=
        (λs. put
          (verifier-state-from-adversary s
            (staged-proof-transcript data)) ≥=
            (λ-. verify-monad ≥=
              (λresult. return ((data, s), result))))))
    P attacker-state =
    wp-event verify-monad
      (λout.
        P (case out of
          None ⇒ None
          | Some (result, final-state) ⇒
            Some (((data, attacker-state), result), final-state)))
      ?s
  unfolding wp-event-def
  apply (simp add: none wpsimps)
  apply (rule arg-cong[where f=λQ. wp verify-monad Q ?s])
  apply (rule ext)
  apply (rename-tac out)
  apply (case-tac out)
  apply (simp add: none)
  apply (rename-tac result-state)
  apply (case-tac result-state)
  by (simp add: wp-return)
show
  wp-event
    (get ≥=
      (λs. put

```

```

      (verifier-state-from-adversary s
        (staged-proof-transcript data))  $\gg$ 
      ( $\lambda$ -. verify-monad  $\gg$ 
        ( $\lambda$ result. return ((data, s), result))))
    P attacker-state  $\leq$  C
  unfolding cont-eq by (rule cont-bound[OF builder])
qed

lemma checked-staged-security-with-data-state-query-bad-verifier-event-bound:
  assumes query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-bad-hit
    adversary-initial-state  $\leq$  C
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event query-bad)
    adversary-initial-state  $\leq$  C
proof -
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event query-bad)
    adversary-initial-state  $\leq$ 
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-bad-hit
    adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
  and event:
    staged-security-with-data-state-verifier-event query-bad out
  show staged-security-with-data-state-query-bad-hit out
proof (cases out)
  case None
  then show ?thesis
    using event unfolding staged-security-with-data-state-verifier-event-def
    by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have verifier:

```



```

    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  using checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
  by blast
have bad: query-bad ?s (Some (result, final-state))
  using event
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
show ?thesis
  unfolding out-eq staged-security-with-data-state-query-bad-hit-def
  Let-def
  using verifier bad by simp
qed
qed
also have ... ≤ C
  by (rule query-bound)
finally show ?thesis .
qed

```

definition *staged-semantic-adversary*
 :: 'f staged-adversary $\Rightarrow$ 'f semantic-adversary
 where
 staged-semantic-adversary A =
 staged-transcript-program A $\gg$
 (λ data. return (staged-proof-transcript data))

lemma *staged-security-experiment-eq-security-experiment:*
 staged-security-experiment A =
 security-experiment (staged-semantic-adversary A)
unfolding staged-security-experiment-def security-experiment-def
 staged-semantic-adversary-def verifier-state-transfer-def
 by (simp add: sm-bind-assoc)

definition *staged-adversary-acceptance-probability*
 :: 'f staged-adversary $\Rightarrow$ prob
 where
 staged-adversary-acceptance-probability A =
 wp-event (staged-security-experiment A)
 (λ out. $\neg$ Option.is-none out) adversary-initial-state

definition *checked-staged-adversary-acceptance-probability*
 :: 'f staged-adversary $\Rightarrow$ prob
 where
 checked-staged-adversary-acceptance-probability A =
 wp-event (checked-staged-security-experiment A)
 (λ out. $\neg$ Option.is-none out) adversary-initial-state

end

end

theory *Staged-Security-Experiment-Budgets*
imports *Staged-Security-Experiment-Core*
begin

Hash range, collision, relation, and target-budget calculus for staged programs.

context *soundness*
begin

lemma *record-staged-message-outcome:*

assumes

Some ((), t) ∈ set-dist (execute (record-staged-message x) s)

shows

t = s⟦

PState := concat (PState s) x,

PTranscript := PTranscript s @ [x]⟧

using *assms* **unfolding** *record-staged-message-def modify-def*
by *(auto elim!: set-dist-bindE)*

lemma *record-staged-message-simps[simp]:*

HashMap

(s⟦PState := concat (PState s) x,

PTranscript := PTranscript s @ [x]⟧) = HashMap s

PState

(s⟦PState := concat (PState s) x,

PTranscript := PTranscript s @ [x]⟧) =

concat (PState s) x

PTranscript

(s⟦PState := concat (PState s) x,

PTranscript := PTranscript s @ [x]⟧) =

PTranscript s @ [x]

PTraceFriCounter

(s⟦PState := concat (PState s) x,

PTranscript := PTranscript s @ [x]⟧) =

PTraceFriCounter s

PCompositionFriCounter

(s⟦PState := concat (PState s) x,

PTranscript := PTranscript s @ [x]⟧) =

PCompositionFriCounter s

PAlphaCounter

(s⟦PState := concat (PState s) x,

PTranscript := PTranscript s @ [x]⟧) =

PAlphaCounter s

PQueryCounter

(s⟦PState := concat (PState s) x,

PTranscript := PTranscript s @ [x]⟧) =

PQueryCounter s

by *simp-all*

lemma *record-staged-message-preserves-hash-map:*

hash-map-preserving (*record-staged-message* *x*)

unfolding *hash-map-preserving-def* *record-staged-message-def* *modify-def*

by (*auto elim!*: *set-dist-bindE*)

lemma *hash-range-budget-record-staged-message:*

hash-range-budget 0 (*record-staged-message* *x*)

unfolding *record-staged-message-def*

by (*rule* *hash-range-budget-modify-preserves-hash-map*) *simp*

lemma *hash-collision-budget-record-staged-message:*

hash-collision-budget 0 (*record-staged-message* *x*)

unfolding *record-staged-message-def*

by (*rule* *hash-collision-budget-modify-preserves-hash-map*) *simp*

lemma *hash-target-program-record-staged-message:*

hash-target-program *B* 0 (*record-staged-message* *x*)

by (*rule* *hash-map-preserving-imp-hash-target-program-zero*)

(*rule* *record-staged-message-preserves-hash-map*)

lemma *hash-relation-program-get:*

hash-relation-program *R* *b* 0

(*get* :: ('f *protocol-channel*, 'f *protocol-channel*) *state-monad*)

by (*rule* *hash-relation-program-zero*)

(*rule* *hash-map-preserving-get*)

lemma *hash-relation-program-assert:*

hash-relation-program *R* *b* 0

(*assert* *x* :: (*unit*, 'f *protocol-channel*) *state-monad*)

by (*rule* *hash-relation-program-zero*)

(*rule* *hash-map-preserving-assert*)

lemma *hash-relation-program-modify:*

assumes $\bigwedge s. \text{HashMap } (f\ s) = \text{HashMap } s$

shows *hash-relation-program* *R* *b* 0 (*modify* *f*)

by (*rule* *hash-relation-program-zero*)

(*rule* *hash-map-preserving-modify*[*OF* *assms*])

lemma *hash-relation-program-record-staged-message:*

hash-relation-program *R* *b* 0 (*record-staged-message* *x*)

by (*rule* *hash-relation-program-zero*)

(*rule* *record-staged-message-preserves-hash-map*)

lemma *record-staged-messages-outcome:*

assumes

Some (*()*, *t*) ∈

set-dist (*execute* (*record-staged-messages* *xs*) *s*)

```

shows
  t = s⟦
    PState := foldl concat (PState s) xs,
    PTranscript := PTranscript s @ xs⟧
using assms
proof (induction xs arbitrary: s)
case Nil
then show ?case
  unfolding record-staged-messages-def by simp
next
case (Cons x xs)
from Cons.prem obtain u where
  head:
    Some ((), u) ∈
      set-dist (execute (record-staged-message x) s)
and tail:
    Some ((), t) ∈
      set-dist (execute (record-staged-messages xs) u)
unfolding record-staged-messages-def
by (auto elim!: set-dist-bindE)
have u:
  u = s⟦
    PState := concat (PState s) x,
    PTranscript := PTranscript s @ [x]⟧
by (rule record-staged-message-outcome[OF head])
have t:
  t = u⟦
    PState := foldl concat (PState u) xs,
    PTranscript := PTranscript u @ xs⟧
by (rule Cons.IH[OF tail])
show ?case
  unfolding t u by simp
qed

lemma record-staged-messages-preserves-hash-map:
  hash-map-preserving (record-staged-messages xs)
unfolding hash-map-preserving-def
using record-staged-messages-outcome by fastforce

lemma hash-range-budget-record-staged-messages:
  hash-range-budget 0 (record-staged-messages xs)
using record-staged-messages-preserves-hash-map
unfolding hash-map-preserving-def hash-range-budget-def
  hash-map-output-values-def
by auto

lemma hash-collision-budget-record-staged-messages:
  hash-collision-budget 0 (record-staged-messages xs)
by (rule hash-map-preserving-imp-hash-collision-budget-zero-semantic)

```

(rule record-staged-messages-preserves-hash-map)

lemma *hash-target-program-record-staged-messages:*
hash-target-program B 0 (record-staged-messages xs)
by (rule hash-map-preserving-imp-hash-target-program-zero)
(rule record-staged-messages-preserves-hash-map)

lemma *hash-relation-program-record-staged-messages:*
hash-relation-program R b 0 (record-staged-messages xs)
by (rule hash-relation-program-zero)
(rule record-staged-messages-preserves-hash-map)

lemma *hash-relation-program-receive-counted-tagged-random-field-element:*
assumes *bump: $\bigwedge s. \text{HashMap } (\text{bump } s) = \text{HashMap } s$*
and fibers: *$\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$*
shows *hash-relation-program R b 1*
(protocol-receive-counted-tagged-random-field-element counter bump tag ::
('f, 'f, unit) protocol-c-monad)

proof –

have *modify-return:*
hash-relation-program R b (0 + 0)
(modify bump $\gg=$ (λ -. return r) ::
('f, 'f protocol-channel) state-monad)
for *r :: 'f*
by (rule hash-relation-program-bind)
(rule hash-relation-program-modify[OF bump],
rule hash-relation-program-zero[OF hash-map-preserving-return])
have *hash-tail:*
hash-relation-program R b (1 + (0 + 0))
(hash (tag (counter s) (PState s)) $\gg=$
(λ r. modify bump $\gg=$ (λ -. return r)) ::
('f, 'f protocol-channel) state-monad)
for *s :: 'f protocol-channel*
by (rule hash-relation-program-bind)
(rule hash-relation-program-hash[OF fibers], rule modify-return)
have *hash-relation-program R b (0 + (1 + (0 + 0)))*
(get $\gg=$
(λ s :: 'f protocol-channel.
hash (tag (counter s) (PState s)) $\gg=$
(λ r. modify bump $\gg=$ (λ -. return r))))
by (rule hash-relation-program-bind
[OF hash-relation-program-get hash-tail])
then show *?thesis*
unfolding *protocol-receive-counted-tagged-random-field-element-def*
by *simp*
qed

lemma *hash-relation-program-receive-query-index-challenge:*
assumes *fibers: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$*

shows *hash-relation-program* $R \ b \ 1$
 (*receive-query-index-challenge* :: ($'f, 'f, unit$) *protocol-c-monad*)
unfolding *receive-query-index-challenge-def*
by (*rule hash-relation-program-receive-counted-tagged-random-field-element*)
 (*simp-all add: fibers*)

lemma *hash-relation-program-receive-trace-fri-challenge*:
assumes *fibers*: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$
shows *hash-relation-program* $R \ b \ 1$
 (*receive-trace-fri-challenge* :: ($'f, 'f, unit$) *protocol-c-monad*)
unfolding *receive-trace-fri-challenge-def*
by (*rule hash-relation-program-receive-counted-tagged-random-field-element*)
 (*simp-all add: fibers*)

lemma *hash-relation-program-receive-composition-fri-challenge*:
assumes *fibers*: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$
shows *hash-relation-program* $R \ b \ 1$
 (*receive-composition-fri-challenge* :: ($'f, 'f, unit$) *protocol-c-monad*)
unfolding *receive-composition-fri-challenge-def*
by (*rule hash-relation-program-receive-counted-tagged-random-field-element*)
 (*simp-all add: fibers*)

lemma *hash-relation-program-receive-alpha-challenge*:
assumes *fibers*: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$
shows *hash-relation-program* $R \ b \ 1$
 (*receive-alpha-challenge* :: ($'f, 'f, unit$) *protocol-c-monad*)
unfolding *receive-alpha-challenge-def*
by (*rule hash-relation-program-receive-counted-tagged-random-field-element*)
 (*simp-all add: fibers*)

lemma *controlled-stage-outcome-fields*:
assumes *controlled*: *controlled-ro-program* $q \ m$
and *outcome*: *Some* $(x, t) \in \text{set-dist } (\text{execute } m \ s)$
shows
 $PState \ t = PState \ s$
 $PTranscript \ t = PTranscript \ s$
 $PTraceFriCounter \ t = PTraceFriCounter \ s$
 $PCompositionFriCounter \ t = PCompositionFriCounter \ s$
 $PAlphaCounter \ t = PAlphaCounter \ s$
 $PQueryCounter \ t = PQueryCounter \ s$
using *controlled-ro-program-preserves-protocol-fields*[*OF controlled*]
outcome
unfolding *protocol-fields-preserving-def*
by *blast+*

lemma *sum-list-take-Suc-drop*:
assumes $i < \text{length } xs$
shows $\text{sum-list } (\text{take } (Suc \ n) \ (\text{drop } i \ xs)) =$
 $xs ! i + \text{sum-list } (\text{take } n \ (\text{drop } (Suc \ i) \ xs))$

```

proof –
  have drop-i:
    drop i xs = xs ! i # drop (Suc i) xs
    using Cons-nth-drop-Suc[OF assms] by simp
  show ?thesis
    unfolding drop-i by simp
qed

lemma sum-list-take-le:
  fixes xs :: nat list
  shows sum-list (take n xs) ≤ sum-list xs
proof (induction n arbitrary: xs)
  case 0
  then show ?case by simp
next
  case (Suc n)
  then show ?case
    by (cases xs) simp-all
qed

lemma hash-range-budget-staged-trace-fri-program:
  assumes controlled: staged-adversary-controlled budgets A
  and bound: i + n ≤ length (trace-fri-budgets budgets)
  shows
    hash-range-budget
      (sum-list (take n (drop i (trace-fri-budgets budgets))) + n)
      (staged-trace-fri-program A i n bs)
  using bound
proof (induction n arbitrary: i bs)
  case 0
  then show ?case by (simp add: hash-range-budget-return)
next
  case (Suc n)
  have i-bound: i < length (trace-fri-budgets budgets)
    using Suc.prems by simp
  have stage:
    hash-range-budget (trace-fri-budgets budgets ! i)
      (trace-fri-root-stage A i bs)
    using controlled i-bound
    unfolding staged-adversary-controlled-def
    by (blast intro: controlled-ro-program-range)
  have record-budget:
     $\wedge$ root. hash-range-budget 0 (record-staged-message root)
    by (rule hash-range-budget-record-staged-message)
  have challenge: hash-range-budget 1 receive-trace-fri-challenge
    by (rule hash-range-budget-receive-trace-fri-challenge)
  have tail:
     $\wedge$ b. hash-range-budget
      (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) + n)

```

```

    (staged-trace-fri-program A (Suc i) n (bs @ [b]))
  by (rule Suc.IH) (use Suc.premis in simp)
have tail-return:
   $\wedge b$  root. hash-range-budget
    (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) + n + 0)
    (staged-trace-fri-program A (Suc i) n (bs @ [b])  $\gg$ 
      ( $\lambda$ (roots, bs'). return (root # roots, bs'))))
  by (rule hash-range-budget-bind)
    (rule tail, simp add: hash-range-budget-return split: prod.splits)
have after-challenge:
   $\wedge$ root. hash-range-budget
    (1 +
      (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) +
        n + 0))
    (receive-trace-fri-challenge  $\gg$ 
      ( $\lambda b$ . staged-trace-fri-program A (Suc i) n (bs @ [b])  $\gg$ 
        ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
  by (rule hash-range-budget-bind)
    (rule challenge, rule tail-return)
have after-record:
   $\wedge$ root. hash-range-budget
    (0 +
      (1 +
        (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) +
          n + 0))))
    (record-staged-message root  $\gg$ 
      ( $\lambda$ -. receive-trace-fri-challenge  $\gg$ 
        ( $\lambda b$ . staged-trace-fri-program A (Suc i) n (bs @ [b])  $\gg$ 
          ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
  by (rule hash-range-budget-bind)
    (rule record-budget, rule after-challenge)
have whole:
  hash-range-budget
    (trace-fri-budgets budgets ! i +
      (0 +
        (1 +
          (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) +
            n + 0))))
    (staged-trace-fri-program A i (Suc n) bs)
  unfolding staged-trace-fri-program.simps
  by (rule hash-range-budget-bind)
    (rule stage, rule after-record)
have budget-eq:
  trace-fri-budgets budgets ! i +
    (0 +
      (1 +
        (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) +
          n + 0))) =
    sum-list (take (Suc n) (drop i (trace-fri-budgets budgets))) + Suc n

```



```

    using sum-list-take-Suc-drop[OF i-bound, of n] by simp
  show ?case
    using whole unfolding budget-eq .
qed

lemma hash-collision-budget-staged-trace-fri-program:
  assumes controlled: staged-adversary-controlled budgets A
    and bound:  $i + n \leq \text{length } (\text{trace-fri-budgets budgets})$ 
  shows
    hash-collision-budget
      (sum-list (take n (drop i (trace-fri-budgets budgets))) + n)
      (staged-trace-fri-program A i n bs)
  using bound
proof (induction n arbitrary: i bs)
  case 0
  then show ?case by (simp add: hash-collision-budget-return)
next
  case (Suc n)
  have i-bound:  $i < \text{length } (\text{trace-fri-budgets budgets})$ 
  using Suc.prem1 by simp
  have stage-range:
    hash-range-budget (trace-fri-budgets budgets ! i)
      (trace-fri-root-stage A i bs)
  using controlled i-bound
  unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
  have stage-coll:
    hash-collision-budget (trace-fri-budgets budgets ! i)
      (trace-fri-root-stage A i bs)
  using controlled i-bound
  unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-collision)
  have record-range:
     $\wedge \text{root. hash-range-budget } 0 \text{ (record-staged-message root)}$ 
  by (rule hash-range-budget-record-staged-message)
  have record-coll:
     $\wedge \text{root. hash-collision-budget } 0 \text{ (record-staged-message root)}$ 
  by (rule hash-collision-budget-record-staged-message)
  have challenge-range: hash-range-budget 1 receive-trace-fri-challenge
  by (rule hash-range-budget-receive-trace-fri-challenge)
  have challenge-coll: hash-collision-budget 1 receive-trace-fri-challenge
  by (rule hash-collision-budget-receive-trace-fri-challenge)
  have tail-range:
     $\wedge b. \text{hash-range-budget}$ 
      (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) + n)
      (staged-trace-fri-program A (Suc i) n (bs @ [b]))
  by (rule hash-range-budget-staged-trace-fri-program[OF controlled])
    (use Suc.prem1 in simp)
  have tail-coll:

```

$\wedge b.$ *hash-collision-budget*
 $(\text{sum-list } (\text{take } n \ (\text{drop } (\text{Suc } i) \ (\text{trace-fri-budgets budgets}))) + n)$
 $(\text{staged-trace-fri-program } A \ (\text{Suc } i) \ n \ (bs \ @ \ [b]))$
by (rule *Suc.IH*) (use *Suc.premis in simp*)
have *tail-return-range*:
 $\wedge b$ *root.* *hash-range-budget*
 $(\text{sum-list } (\text{take } n \ (\text{drop } (\text{Suc } i) \ (\text{trace-fri-budgets budgets}))) + n + 0)$
 $(\text{staged-trace-fri-program } A \ (\text{Suc } i) \ n \ (bs \ @ \ [b])) \gg=$
 $(\lambda(\text{roots}, bs'). \text{return } (\text{root} \ \# \ \text{roots}, bs'))$
by (rule *hash-range-budget-bind*)
(rule *tail-range*, *simp add: hash-range-budget-return split: prod.splits*)
have *tail-return-coll*:
 $\wedge b$ *root.* *hash-collision-budget*
 $(\text{sum-list } (\text{take } n \ (\text{drop } (\text{Suc } i) \ (\text{trace-fri-budgets budgets}))) + n + 0)$
 $(\text{staged-trace-fri-program } A \ (\text{Suc } i) \ n \ (bs \ @ \ [b])) \gg=$
 $(\lambda(\text{roots}, bs'). \text{return } (\text{root} \ \# \ \text{roots}, bs'))$
by (rule *hash-collision-budget-bind*)
(rule *tail-range*, rule *tail-coll*,
simp add: hash-range-budget-return split: prod.splits,
simp add: hash-collision-budget-return split: prod.splits)
have *after-challenge-range*:
 $\wedge \text{root.}$ *hash-range-budget*
 $(1 +$
 $(\text{sum-list } (\text{take } n \ (\text{drop } (\text{Suc } i) \ (\text{trace-fri-budgets budgets}))) +$
 $n + 0))$
 $(\text{receive-trace-fri-challenge} \gg=$
 $(\lambda b. \text{staged-trace-fri-program } A \ (\text{Suc } i) \ n \ (bs \ @ \ [b])) \gg=$
 $(\lambda(\text{roots}, bs'). \text{return } (\text{root} \ \# \ \text{roots}, bs'))))$
by (rule *hash-range-budget-bind*)
(rule *challenge-range*, rule *tail-return-range*)
have *after-challenge-coll*:
 $\wedge \text{root.}$ *hash-collision-budget*
 $(1 +$
 $(\text{sum-list } (\text{take } n \ (\text{drop } (\text{Suc } i) \ (\text{trace-fri-budgets budgets}))) +$
 $n + 0))$
 $(\text{receive-trace-fri-challenge} \gg=$
 $(\lambda b. \text{staged-trace-fri-program } A \ (\text{Suc } i) \ n \ (bs \ @ \ [b])) \gg=$
 $(\lambda(\text{roots}, bs'). \text{return } (\text{root} \ \# \ \text{roots}, bs'))))$
by (rule *hash-collision-budget-bind*)
(rule *challenge-range*, rule *challenge-coll*, rule *tail-return-range*,
rule *tail-return-coll*)
have *after-record-range*:
 $\wedge \text{root.}$ *hash-range-budget*
 $(0 +$
 $(1 +$
 $(\text{sum-list } (\text{take } n \ (\text{drop } (\text{Suc } i) \ (\text{trace-fri-budgets budgets}))) +$
 $n + 0)))$
 $(\text{record-staged-message root} \gg=$
 $(\lambda-. \text{receive-trace-fri-challenge} \gg=$

```

      (λb. staged-trace-fri-program A (Suc i) n (bs @ [b]) ≫=
        (λ(roots, bs'). return (root # roots, bs')))))
  by (rule hash-range-budget-bind)
      (rule record-range, rule after-challenge-range)
  have after-record-coll:
    ∧ root. hash-collision-budget
      (0 +
        (1 +
          (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) +
            n + 0)))
        (record-staged-message root ≫=
          (λ-. receive-trace-fri-challenge ≫=
            (λb. staged-trace-fri-program A (Suc i) n (bs @ [b]) ≫=
              (λ(roots, bs'). return (root # roots, bs'))))))
  by (rule hash-collision-budget-bind)
      (rule record-range, rule record-coll, rule after-challenge-range,
        rule after-challenge-coll)
  have whole:
    hash-collision-budget
      (trace-fri-budgets budgets ! i +
        (0 +
          (1 +
            (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) +
              n + 0))))
        (staged-trace-fri-program A i (Suc n) bs)
  unfolding staged-trace-fri-program.simps
  by (rule hash-collision-budget-bind)
      (rule stage-range, rule stage-coll, rule after-record-range,
        rule after-record-coll)
  have budget-eq:
    trace-fri-budgets budgets ! i +
      (0 +
        (1 +
          (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) +
            n + 0))) =
      sum-list (take (Suc n) (drop i (trace-fri-budgets budgets))) + Suc n
  using sum-list-take-Suc-drop[OF i-bound, of n] by simp
  show ?case
    using whole unfolding budget-eq .
qed

lemma hash-range-budget-staged-alpha-program:
  hash-range-budget n (staged-alpha-program n)
proof (induction n)
  case 0
  then show ?case by (simp add: hash-range-budget-return)
next
  case (Suc n)
  have challenge: hash-range-budget 1 receive-alpha-challenge

```

```

  by (rule hash-range-budget-receive-alpha-challenge)
have record-budget:
   $\wedge a.$  hash-range-budget 0 (record-staged-message a)
  by (rule hash-range-budget-record-staged-message)
have tail-return:
   $\wedge a.$  hash-range-budget (n + 0)
    (staged-alpha-program n  $\gg$  ( $\lambda as.$  return (a # as)))
  by (rule hash-range-budget-bind)
    (rule Suc.IH, rule hash-range-budget-return)
have after-record:
   $\wedge a.$  hash-range-budget (0 + (n + 0))
    (record-staged-message a  $\gg$ 
      ( $\lambda -. staged-alpha-program n \gg (\lambda as. return (a \# as))$ )))
  by (rule hash-range-budget-bind)
    (rule record-budget, rule tail-return)
have hash-range-budget (1 + (0 + (n + 0)))
  (staged-alpha-program (Suc n))
  unfolding staged-alpha-program.simps
  by (rule hash-range-budget-bind)
    (rule challenge, rule after-record)
then show ?case by simp
qed

lemma hash-collision-budget-staged-alpha-program:
  hash-collision-budget n (staged-alpha-program n)
proof (induction n)
  case 0
  then show ?case by (simp add: hash-collision-budget-return)
next
  case (Suc n)
  have challenge-range: hash-range-budget 1 receive-alpha-challenge
    by (rule hash-range-budget-receive-alpha-challenge)
  have challenge-coll: hash-collision-budget 1 receive-alpha-challenge
    by (rule hash-collision-budget-receive-alpha-challenge)
  have record-range:
     $\wedge a.$  hash-range-budget 0 (record-staged-message a)
    by (rule hash-range-budget-record-staged-message)
  have record-coll:
     $\wedge a.$  hash-collision-budget 0 (record-staged-message a)
    by (rule hash-collision-budget-record-staged-message)
  have tail-return-range:
     $\wedge a.$  hash-range-budget (n + 0)
      (staged-alpha-program n  $\gg$  ( $\lambda as.$  return (a # as)))
    by (rule hash-range-budget-bind)
      (rule hash-range-budget-staged-alpha-program,
        rule hash-range-budget-return)
  have tail-return-coll:
     $\wedge a.$  hash-collision-budget (n + 0)
      (staged-alpha-program n  $\gg$  ( $\lambda as.$  return (a # as)))

```

```

by (rule hash-collision-budget-bind)
  (rule hash-range-budget-staged-alpha-program, rule Suc.IH,
   rule hash-range-budget-return, rule hash-collision-budget-return)
have after-record-range:
 $\bigwedge a.$  hash-range-budget  $(0 + (n + 0))$ 
  (record-staged-message  $a \gg=$ 
    $(\lambda-. \text{staged-alpha-program } n \gg= (\lambda as. \text{return } (a \# as))))$ 
by (rule hash-range-budget-bind)
  (rule record-range, rule tail-return-range)
have after-record-coll:
 $\bigwedge a.$  hash-collision-budget  $(0 + (n + 0))$ 
  (record-staged-message  $a \gg=$ 
    $(\lambda-. \text{staged-alpha-program } n \gg= (\lambda as. \text{return } (a \# as))))$ 
proof (rule hash-collision-budget-bind)
  fix a
  show hash-range-budget 0 (record-staged-message a)
    by (rule record-range)
  show hash-collision-budget 0 (record-staged-message a)
    by (rule record-coll)
  show  $\bigwedge x.$  hash-range-budget  $(n + 0)$ 
    (staged-alpha-program  $n \gg= (\lambda as. \text{return } (a \# as))))$ 
    by (rule tail-return-range)
  show  $\bigwedge x.$  hash-collision-budget  $(n + 0)$ 
    (staged-alpha-program  $n \gg= (\lambda as. \text{return } (a \# as))))$ 
    by (rule tail-return-coll)
qed
have hash-collision-budget  $(1 + (0 + (n + 0)))$ 
  (staged-alpha-program (Suc n))
  unfolding staged-alpha-program.simps
  by (rule hash-collision-budget-bind)
    (rule challenge-range, rule challenge-coll, rule after-record-range,
     rule after-record-coll)
then show ?case by simp
qed

lemma hash-range-budget-staged-composition-fri-program:
  assumes controlled: staged-adversary-controlled budgets A
    and bound:  $i + n \leq \text{length } (\text{composition-fri-budgets budgets})$ 
  shows
    hash-range-budget
      (sum-list (take n (drop i (composition-fri-budgets budgets))) + n)
      (staged-composition-fri-program A dg i n bs)
  using bound
proof (induction n arbitrary: i bs)
  case 0
  then show ?case by (simp add: hash-range-budget-return)
next
  case (Suc n)
  have i-bound:  $i < \text{length } (\text{composition-fri-budgets budgets})$ 

```

```

using Suc.prems by simp
have stage:
  hash-range-budget (composition-fri-budgets budgets ! i)
    (composition-fri-root-stage A dg i bs)
using controlled i-bound
unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-range)
have record-budget:
   $\wedge$ root. hash-range-budget 0 (record-staged-message root)
by (rule hash-range-budget-record-staged-message)
have challenge:
  hash-range-budget 1 receive-composition-fri-challenge
by (rule hash-range-budget-receive-composition-fri-challenge)
have tail:
   $\wedge$ b. hash-range-budget
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) + n)
    (staged-composition-fri-program A dg (Suc i) n (bs @ [b]))
by (rule Suc.IH) (use Suc.prems in simp)
have tail-return:
   $\wedge$ b root. hash-range-budget
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) + n + 0)
    (staged-composition-fri-program A dg (Suc i) n (bs @ [b])  $\gg$ 
      ( $\lambda$ (roots, bs'). return (root # roots, bs')))
by (rule hash-range-budget-bind)
    (rule tail, simp add: hash-range-budget-return split: prod.splits)
have after-challenge:
   $\wedge$ root. hash-range-budget
    (1 +
      (sum-list
        (take n (drop (Suc i) (composition-fri-budgets budgets))) +
        n + 0))
    (receive-composition-fri-challenge  $\gg$ 
      ( $\lambda$ b. staged-composition-fri-program A dg (Suc i) n
        (bs @ [b])  $\gg$ 
        ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
by (rule hash-range-budget-bind)
    (rule challenge, rule tail-return)
have after-record:
   $\wedge$ root. hash-range-budget
    (0 +
      (1 +
        (sum-list
          (take n (drop (Suc i) (composition-fri-budgets budgets))) +
          n + 0)))
    (record-staged-message root  $\gg$ 
      ( $\lambda$ -. receive-composition-fri-challenge  $\gg$ 
        ( $\lambda$ b. staged-composition-fri-program A dg (Suc i) n
          (bs @ [b])  $\gg$ 
          ( $\lambda$ (roots, bs'). return (root # roots, bs')))))

```

```

      (bs @ [b]) >>=
      (λ(roots, bs'). return (root # roots, bs^))))
  by (rule hash-range-budget-bind)
    (rule record-budget, rule after-challenge)
have whole:
  hash-range-budget
    (composition-fri-budgets budgets ! i +
      (0 +
        (1 +
          (sum-list
            (take n (drop (Suc i) (composition-fri-budgets budgets))) +
              n + 0))))
      (staged-composition-fri-program A dg i (Suc n) bs)
  unfolding staged-composition-fri-program.simps
  by (rule hash-range-budget-bind)
    (rule stage, rule after-record)
have budget-eq:
  composition-fri-budgets budgets ! i +
    (0 +
      (1 +
        (sum-list
          (take n (drop (Suc i) (composition-fri-budgets budgets))) +
            n + 0))) =
    sum-list (take (Suc n) (drop i (composition-fri-budgets budgets))) +
      Suc n
  using sum-list-take-Suc-drop[OF i-bound, of n] by simp
show ?case
  using whole unfolding budget-eq .
qed

lemma hash-collision-budget-staged-composition-fri-program:
  assumes controlled: staged-adversary-controlled budgets A
    and bound: i + n ≤ length (composition-fri-budgets budgets)
  shows
    hash-collision-budget
      (sum-list (take n (drop i (composition-fri-budgets budgets))) + n)
      (staged-composition-fri-program A dg i n bs)
  using bound
proof (induction n arbitrary: i bs)
  case 0
  then show ?case by (simp add: hash-collision-budget-return)
next
  case (Suc n)
  have i-bound: i < length (composition-fri-budgets budgets)
  using Suc.premis by simp
  have stage-range:
    hash-range-budget (composition-fri-budgets budgets ! i)
      (composition-fri-root-stage A dg i bs)
  using controlled i-bound

```

```

unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-range)
have stage-coll:
  hash-collision-budget (composition-fri-budgets budgets ! i)
    (composition-fri-root-stage A dg i bs)
using controlled i-bound
unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-collision)
have record-range:
   $\wedge$  root. hash-range-budget 0 (record-staged-message root)
by (rule hash-range-budget-record-staged-message)
have record-coll:
   $\wedge$  root. hash-collision-budget 0 (record-staged-message root)
by (rule hash-collision-budget-record-staged-message)
have challenge-range:
  hash-range-budget 1 receive-composition-fri-challenge
by (rule hash-range-budget-receive-composition-fri-challenge)
have challenge-coll:
  hash-collision-budget 1 receive-composition-fri-challenge
by (rule hash-collision-budget-receive-composition-fri-challenge)
have tail-range:
   $\wedge$  b. hash-range-budget
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) + n)
      (staged-composition-fri-program A dg (Suc i) n (bs @ [b]))
by (rule hash-range-budget-staged-composition-fri-program
      [OF controlled])
    (use Suc.premis in simp)
have tail-coll:
   $\wedge$  b. hash-collision-budget
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) + n)
      (staged-composition-fri-program A dg (Suc i) n (bs @ [b]))
by (rule Suc.IH) (use Suc.premis in simp)
have tail-return-range:
   $\wedge$  b root. hash-range-budget
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) + n + 0)
      (staged-composition-fri-program A dg (Suc i) n (bs @ [b])  $\gg$ 
        ( $\lambda$ (roots, bs'). return (root # roots, bs'))))
by (rule hash-range-budget-bind)
    (rule tail-range, simp add: hash-range-budget-return split: prod.splits)
have tail-return-coll:
   $\wedge$  b root. hash-collision-budget
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) + n + 0)
      (staged-composition-fri-program A dg (Suc i) n (bs @ [b])  $\gg$ 
        ( $\lambda$ (roots, bs'). return (root # roots, bs'))))
by (rule hash-collision-budget-bind)

```



```

(rule tail-range, rule tail-coll,
  simp add: hash-range-budget-return split: prod.splits,
  simp add: hash-collision-budget-return split: prod.splits)
have after-challenge-range:
 $\wedge$ root. hash-range-budget
  (1 +
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) +
      n + 0))
    (receive-composition-fri-challenge  $\gg$ 
      ( $\lambda$ b. staged-composition-fri-program A dg (Suc i) n
        (bs @ [b])  $\gg$ 
          ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
by (rule hash-range-budget-bind)
  (rule challenge-range, rule tail-return-range)
have after-challenge-coll:
 $\wedge$ root. hash-collision-budget
  (1 +
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) +
      n + 0))
    (receive-composition-fri-challenge  $\gg$ 
      ( $\lambda$ b. staged-composition-fri-program A dg (Suc i) n
        (bs @ [b])  $\gg$ 
          ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
by (rule hash-collision-budget-bind)
  (rule challenge-range, rule challenge-coll, rule tail-return-range,
    rule tail-return-coll)
have after-record-range:
 $\wedge$ root. hash-range-budget
  (0 +
    (1 +
      (sum-list
        (take n (drop (Suc i) (composition-fri-budgets budgets))) +
        n + 0)))
    (record-staged-message root  $\gg$ 
      ( $\lambda$ -. receive-composition-fri-challenge  $\gg$ 
        ( $\lambda$ b. staged-composition-fri-program A dg (Suc i) n
          (bs @ [b])  $\gg$ 
            ( $\lambda$ (roots, bs'). return (root # roots, bs'))))))
by (rule hash-range-budget-bind)
  (rule record-range, rule after-challenge-range)
have after-record-coll:
 $\wedge$ root. hash-collision-budget
  (0 +
    (1 +
      (sum-list
        (take n (drop (Suc i) (composition-fri-budgets budgets))) +
        n + 0)))

```

```

    (record-staged-message root  $\gg$ 
      ( $\lambda$ -. receive-composition-fri-challenge  $\gg$ 
        ( $\lambda$ b. staged-composition-fri-program A dg (Suc i) n
          (bs @ [b])  $\gg$ 
            ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
  by (rule hash-collision-budget-bind)
    (rule record-range, rule record-coll, rule after-challenge-range,
      rule after-challenge-coll)
have whole:
  hash-collision-budget
    (composition-fri-budgets budgets ! i +
      (0 +
        (1 +
          (sum-list
            (take n (drop (Suc i) (composition-fri-budgets budgets))) +
              n + 0))))
    (staged-composition-fri-program A dg i (Suc n) bs)
unfolding staged-composition-fri-program.simps
by (rule hash-collision-budget-bind)
    (rule stage-range, rule stage-coll, rule after-record-range,
      rule after-record-coll)
have budget-eq:
  composition-fri-budgets budgets ! i +
    (0 +
      (1 +
        (sum-list
          (take n (drop (Suc i) (composition-fri-budgets budgets))) +
            n + 0))) =
    sum-list (take (Suc n) (drop i (composition-fri-budgets budgets))) +
      Suc n
using sum-list-take-Suc-drop[OF i-bound, of n] by simp
show ?case
using whole unfolding budget-eq .
qed

lemma hash-range-budget-staged-query-program:
assumes controlled: staged-adversary-controlled budgets A
and bound:  $i + n \leq \text{length } (\text{query-opening-budgets budgets})$ 
shows
  hash-range-budget
    (sum-list (take n (drop i (query-opening-budgets budgets))) + n)
    (staged-query-program A i n)
using bound
proof (induction n arbitrary: i)
case 0
then show ?case by (simp add: hash-range-budget-return)
next
case (Suc n)
have i-bound:  $i < \text{length } (\text{query-opening-budgets budgets})$ 

```

```

using Suc.prems by simp
have challenge: hash-range-budget 1 receive-query-index-challenge
by (rule hash-range-budget-receive-query-index-challenge)
have stage:
 $\wedge$ raw. hash-range-budget (query-opening-budgets budgets ! i)
  (query-opening-stage A i raw)
using controlled i-bound
unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-range)
have record-budget:
 $\wedge$ chunk. hash-range-budget 0 (record-staged-messages chunk)
by (rule hash-range-budget-record-staged-messages)
have tail:
hash-range-budget
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n)
  (staged-query-program A (Suc i) n)
by (rule Suc.IH) (use Suc.prems in simp)
have tail-return:
 $\wedge$ chunk. hash-range-budget
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n + 0)
  (staged-query-program A (Suc i) n  $\gg$ 
    ( $\lambda$ chunks. return (chunk # chunks))))
by (rule hash-range-budget-bind)
  (rule tail, rule hash-range-budget-return)
have after-record:
 $\wedge$ chunk. hash-range-budget
  (0 +
    (sum-list
      (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0))
    (record-staged-messages chunk  $\gg$ 
      ( $\lambda$ -. staged-query-program A (Suc i) n  $\gg$ 
        ( $\lambda$ chunks. return (chunk # chunks)))))
by (rule hash-range-budget-bind)
  (rule record-budget, rule tail-return)
have after-stage:
 $\wedge$ raw. hash-range-budget
  (query-opening-budgets budgets ! i +
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
          n + 0)))
    (query-opening-stage A i raw  $\gg$ 
      ( $\lambda$ chunk. record-staged-messages chunk  $\gg$ 
        ( $\lambda$ -. staged-query-program A (Suc i) n  $\gg$ 
          ( $\lambda$ chunks. return (chunk # chunks)))))))
by (rule hash-range-budget-bind)

```

```

    (rule stage, rule after-record)
have whole:
  hash-range-budget
    (1 +
      (query-opening-budgets budgets ! i +
        (0 +
          (sum-list
            (take n (drop (Suc i) (query-opening-budgets budgets))) +
              n + 0))))
      (staged-query-program A i (Suc n))
    unfolding staged-query-program.simps
    by (rule hash-range-budget-bind)
      (rule challenge, rule after-stage)
have budget-eq:
  1 +
    (query-opening-budgets budgets ! i +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
            n + 0))) =
    sum-list (take (Suc n) (drop i (query-opening-budgets budgets))) +
      Suc n
    using sum-list-take-Suc-drop[OF i-bound, of n] by simp
show ?case
    using whole unfolding budget-eq .
qed

lemma hash-collision-budget-staged-query-program:
  assumes controlled: staged-adversary-controlled budgets A
    and bound:  $i + n \leq \text{length } (\text{query-opening-budgets budgets})$ 
  shows
    hash-collision-budget
      (sum-list (take n (drop i (query-opening-budgets budgets))) + n)
      (staged-query-program A i n)
    using bound
proof (induction n arbitrary: i)
  case 0
    then show ?case by (simp add: hash-collision-budget-return)
next
  case (Suc n)
    have i-bound:  $i < \text{length } (\text{query-opening-budgets budgets})$ 
      using Suc.prem1 by simp
    have challenge-range: hash-range-budget 1 receive-query-index-challenge
      by (rule hash-range-budget-receive-query-index-challenge)
    have challenge-coll: hash-collision-budget 1 receive-query-index-challenge
      by (rule hash-collision-budget-receive-query-index-challenge)
    have stage-range:
       $\bigwedge \text{raw. hash-range-budget } (\text{query-opening-budgets budgets ! } i) \text{ (query-opening-stage A i raw)}$ 

```

```

using controlled i-bound
unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-range)
have stage-coll:
 $\wedge$ raw. hash-collision-budget (query-opening-budgets budgets ! i)
  (query-opening-stage A i raw)
using controlled i-bound
unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-collision)
have record-range:
 $\wedge$ chunk. hash-range-budget 0 (record-staged-messages chunk)
by (rule hash-range-budget-record-staged-messages)
have record-coll:
 $\wedge$ chunk. hash-collision-budget 0 (record-staged-messages chunk)
by (rule hash-collision-budget-record-staged-messages)
have tail-range:
hash-range-budget
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n)
  (staged-query-program A (Suc i) n)
by (rule hash-range-budget-staged-query-program[OF controlled])
  (use Suc.premis in simp)
have tail-coll:
hash-collision-budget
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n)
  (staged-query-program A (Suc i) n)
by (rule Suc.IH) (use Suc.premis in simp)
have tail-return-range:
 $\wedge$ chunk. hash-range-budget
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n + 0)
  (staged-query-program A (Suc i) n  $\gg=$ 
    ( $\lambda$ chunks. return (chunk # chunks)))
by (rule hash-range-budget-bind)
  (rule tail-range, rule hash-range-budget-return)
have tail-return-coll:
 $\wedge$ chunk. hash-collision-budget
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n + 0)
  (staged-query-program A (Suc i) n  $\gg=$ 
    ( $\lambda$ chunks. return (chunk # chunks)))
by (rule hash-collision-budget-bind)
  (rule tail-range, rule tail-coll, rule hash-range-budget-return,
    rule hash-collision-budget-return)
have after-record-range:
 $\wedge$ chunk. hash-range-budget
  (0 +
    (sum-list

```

```

      (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n + 0))
    (record-staged-messages chunk >=>
      (λ-. staged-query-program A (Suc i) n >=>
        (λchunks. return (chunk # chunks))))
  by (rule hash-range-budget-bind)
    (rule record-range, rule tail-return-range)
have after-record-coll:
  ∧chunk. hash-collision-budget
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0))
    (record-staged-messages chunk >=>
      (λ-. staged-query-program A (Suc i) n >=>
        (λchunks. return (chunk # chunks))))
  by (rule hash-collision-budget-bind)
    (rule record-range, rule record-coll, rule tail-return-range,
      rule tail-return-coll)
have after-stage-range:
  ∧raw. hash-range-budget
    (query-opening-budgets budgets ! i +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
          n + 0))))
    (query-opening-stage A i raw >=>
      (λchunk. record-staged-messages chunk >=>
        (λ-. staged-query-program A (Suc i) n >=>
          (λchunks. return (chunk # chunks))))))
  by (rule hash-range-budget-bind)
    (rule stage-range, rule after-record-range)
have after-stage-coll:
  ∧raw. hash-collision-budget
    (query-opening-budgets budgets ! i +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
          n + 0))))
    (query-opening-stage A i raw >=>
      (λchunk. record-staged-messages chunk >=>
        (λ-. staged-query-program A (Suc i) n >=>
          (λchunks. return (chunk # chunks))))))
  by (rule hash-collision-budget-bind)
    (rule stage-range, rule stage-coll, rule after-record-range,
      rule after-record-coll)
have whole:
  hash-collision-budget
    (1 +

```

```

    (query-opening-budgets budgets ! i +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
            n + 0))))
    (staged-query-program A i (Suc n))
  unfolding staged-query-program.simps
  by (rule hash-collision-budget-bind)
    (rule challenge-range, rule challenge-coll, rule after-stage-range,
      rule after-stage-coll)
  have budget-eq:
    1 +
      (query-opening-budgets budgets ! i +
        (0 +
          (sum-list
            (take n (drop (Suc i) (query-opening-budgets budgets))) +
              n + 0))) =
      sum-list (take (Suc n) (drop i (query-opening-budgets budgets))) +
        Suc n
    using sum-list-take-Suc-drop[OF i-bound, of n] by simp
  show ?case
    using whole unfolding budget-eq .
qed

lemma hash-target-program-staged-trace-fri-program:
  assumes controlled: staged-adversary-controlled budgets A
    and bound:  $i + n \leq \text{length } (\text{trace-fri-budgets budgets})$ 
  shows
    hash-target-program B
      (sum-list (take n (drop i (trace-fri-budgets budgets))) + n)
      (staged-trace-fri-program A i n bs)
    using bound
  proof (induction n arbitrary: i bs)
    case 0
    then show ?case by (simp add: hash-target-program-return)
  next
    case (Suc n)
    have i-bound:  $i < \text{length } (\text{trace-fri-budgets budgets})$ 
      using Suc.prem1 by simp
    have stage:
      hash-target-program B (trace-fri-budgets budgets ! i)
        (trace-fri-root-stage A i bs)
      using controlled i-bound
    unfolding staged-adversary-controlled-def
    by (blast intro: controlled-ro-program-target)
    have record-budget:
       $\wedge \text{root. hash-target-program B 0 (record-staged-message root)}$ 
      by (rule hash-target-program-record-staged-message)
    have challenge: hash-target-program B 1 receive-trace-fri-challenge

```

```

by (rule hash-target-program-receive-trace-fri-challenge)
have tail:
   $\wedge b.$  hash-target-program  $B$ 
    (sum-list (take  $n$  (drop (Suc  $i$ ) (trace-fri-budgets budgets))) +  $n$ )
    (staged-trace-fri-program  $A$  (Suc  $i$ )  $n$  (bs @ [b]))
  by (rule Suc.IH) (use Suc.premis in simp)
have tail-return:
   $\wedge b$  root. hash-target-program  $B$ 
    (sum-list (take  $n$  (drop (Suc  $i$ ) (trace-fri-budgets budgets))) +  $n$  + 0)
    (staged-trace-fri-program  $A$  (Suc  $i$ )  $n$  (bs @ [b])  $\gg$ 
      ( $\lambda$ (roots, bs'). return (root # roots, bs')))
  by (rule hash-target-program-bind)
    (rule tail, simp add: hash-target-program-return split: prod.splits)
have after-challenge:
   $\wedge$ root. hash-target-program  $B$ 
    (1 +
      (sum-list (take  $n$  (drop (Suc  $i$ ) (trace-fri-budgets budgets))) +
         $n$  + 0))
    (receive-trace-fri-challenge  $\gg$ 
      ( $\lambda b.$  staged-trace-fri-program  $A$  (Suc  $i$ )  $n$  (bs @ [b])  $\gg$ 
        ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
  by (rule hash-target-program-bind)
    (rule challenge, rule tail-return)
have after-record:
   $\wedge$ root. hash-target-program  $B$ 
    (0 +
      (1 +
        (sum-list (take  $n$  (drop (Suc  $i$ ) (trace-fri-budgets budgets))) +
           $n$  + 0))))
    (record-staged-message root  $\gg$ 
      ( $\lambda.$  receive-trace-fri-challenge  $\gg$ 
        ( $\lambda b.$  staged-trace-fri-program  $A$  (Suc  $i$ )  $n$  (bs @ [b])  $\gg$ 
          ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
  by (rule hash-target-program-bind)
    (rule record-budget, rule after-challenge)
have whole:
  hash-target-program  $B$ 
    (trace-fri-budgets budgets !  $i$  +
      (0 +
        (1 +
          (sum-list (take  $n$  (drop (Suc  $i$ ) (trace-fri-budgets budgets))) +
             $n$  + 0))))
    (staged-trace-fri-program  $A$   $i$  (Suc  $n$ ) bs)
unfolding staged-trace-fri-program.simps
by (rule hash-target-program-bind)
    (rule stage, rule after-record)
have budget-eq:
  trace-fri-budgets budgets !  $i$  +
    (0 +

```



```

      (1 +
        (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) +
          n + 0))) =
      sum-list (take (Suc n) (drop i (trace-fri-budgets budgets))) + Suc n
    using sum-list-take-Suc-drop[OF i-bound, of n] by simp
  show ?case
    using whole unfolding budget-eq .
qed

```

```

lemma hash-target-program-staged-alpha-program:
  hash-target-program B n (staged-alpha-program n)
proof (induction n)
  case 0
  then show ?case by (simp add: hash-target-program-return)
next
  case (Suc n)
  have challenge: hash-target-program B 1 receive-alpha-challenge
    by (rule hash-target-program-receive-alpha-challenge)
  have record-budget:
     $\wedge a.$  hash-target-program B 0 (record-staged-message a)
    by (rule hash-target-program-record-staged-message)
  have tail-return:
     $\wedge a.$  hash-target-program B (n + 0)
      (staged-alpha-program n  $\gg=$  ( $\lambda as.$  return (a # as)))
    by (rule hash-target-program-bind)
      (rule Suc.IH, rule hash-target-program-return)
  have after-record:
     $\wedge a.$  hash-target-program B (0 + (n + 0))
      (record-staged-message a  $\gg=$ 
        ( $\lambda -.$  staged-alpha-program n  $\gg=$  ( $\lambda as.$  return (a # as))))
    by (rule hash-target-program-bind)
      (rule record-budget, rule tail-return)
  have hash-target-program B (1 + (0 + (n + 0)))
    (staged-alpha-program (Suc n))
    unfolding staged-alpha-program.simps
    by (rule hash-target-program-bind)
      (rule challenge, rule after-record)
  then show ?case by simp
qed

```

```

lemma hash-target-program-staged-composition-fri-program:
  assumes controlled: staged-adversary-controlled budgets A
  and bound:  $i + n \leq \text{length } (\text{composition-fri-budgets budgets})$ 
  shows
    hash-target-program B
      (sum-list (take n (drop i (composition-fri-budgets budgets))) + n)
      (staged-composition-fri-program A dg i n bs)
  using bound
proof (induction n arbitrary: i bs)

```

```

case 0
then show ?case by (simp add: hash-target-program-return)
next
case (Suc n)
have i-bound:  $i < \text{length } (\text{composition-fri-budgets } \text{budgets})$ 
  using Suc.prems by simp
have stage:
  hash-target-program B (composition-fri-budgets budgets ! i)
    (composition-fri-root-stage A dg i bs)
  using controlled i-bound
  unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-target)
have record-budget:
   $\bigwedge \text{root. } \text{hash-target-program } B \ 0 \ (\text{record-staged-message } \text{root})$ 
  by (rule hash-target-program-record-staged-message)
have challenge:
  hash-target-program B 1 receive-composition-fri-challenge
  by (rule hash-target-program-receive-composition-fri-challenge)
have tail:
   $\bigwedge b. \text{hash-target-program } B$ 
    (sum-list
      ( $\text{take } n \ (\text{drop } (\text{Suc } i) \ (\text{composition-fri-budgets } \text{budgets}))) + n$ )
      (staged-composition-fri-program A dg (Suc i) n (bs @ [b]))
    )
  by (rule Suc.IH) (use Suc.prems in simp)
have tail-return:
   $\bigwedge b \text{ root. } \text{hash-target-program } B$ 
    (sum-list
      ( $\text{take } n \ (\text{drop } (\text{Suc } i) \ (\text{composition-fri-budgets } \text{budgets}))) + n + 0$ )
      (staged-composition-fri-program A dg (Suc i) n (bs @ [b]))  $\gg=$ 
      ( $\lambda(\text{roots}, \text{bs}'). \text{return } (\text{root} \# \text{roots}, \text{bs}')$ ))
  by (rule hash-target-program-bind)
    (rule tail, simp add: hash-target-program-return split: prod.splits)
have after-challenge:
   $\bigwedge \text{root. } \text{hash-target-program } B$ 
    ( $1 +$ 
      (sum-list
        ( $\text{take } n \ (\text{drop } (\text{Suc } i) \ (\text{composition-fri-budgets } \text{budgets}))) +$ 
           $n + 0$ ))
      (receive-composition-fri-challenge)  $\gg=$ 
      ( $\lambda b. \text{staged-composition-fri-program } A \ dg \ (\text{Suc } i) \ n$ 
        (bs @ [b])  $\gg=$ 
        ( $\lambda(\text{roots}, \text{bs}'). \text{return } (\text{root} \# \text{roots}, \text{bs}')$ )))
  by (rule hash-target-program-bind)
    (rule challenge, rule tail-return)
have after-record:
   $\bigwedge \text{root. } \text{hash-target-program } B$ 
    ( $0 +$ 
      ( $1 +$ 
        (sum-list

```

```

      (take n (drop (Suc i) (composition-fri-budgets budgets))) +
      n + 0)))
  (record-staged-message root  $\gg$ 
    ( $\lambda$ -. receive-composition-fri-challenge  $\gg$ 
      ( $\lambda$ b. staged-composition-fri-program A dg (Suc i) n
        (bs @ [b])  $\gg$ 
          ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
  by (rule hash-target-program-bind)
  (rule record-budget, rule after-challenge)
have whole:
  hash-target-program B
  (composition-fri-budgets budgets ! i +
    (0 +
      (1 +
        (sum-list
          (take n (drop (Suc i) (composition-fri-budgets budgets))) +
          n + 0))))
    (staged-composition-fri-program A dg i (Suc n) bs)
unfolding staged-composition-fri-program.simps
by (rule hash-target-program-bind)
  (rule stage, rule after-record)
have budget-eq:
  composition-fri-budgets budgets ! i +
    (0 +
      (1 +
        (sum-list
          (take n (drop (Suc i) (composition-fri-budgets budgets))) +
          n + 0))) =
    sum-list (take (Suc n) (drop i (composition-fri-budgets budgets))) +
    Suc n
using sum-list-take-Suc-drop[OF i-bound, of n] by simp
show ?case
using whole unfolding budget-eq .
qed

```

```

lemma hash-target-program-staged-query-program:
assumes controlled: staged-adversary-controlled budgets A
and bound:  $i + n \leq \text{length } (\text{query-opening-budgets } \text{budgets})$ 
shows
  hash-target-program B
  (sum-list (take n (drop i (query-opening-budgets budgets))) + n)
  (staged-query-program A i n)
using bound
proof (induction n arbitrary: i)
case 0
then show ?case by (simp add: hash-target-program-return)
next
case (Suc n)
have i-bound:  $i < \text{length } (\text{query-opening-budgets } \text{budgets})$ 

```

```

using Suc.prems by simp
have challenge: hash-target-program B 1 receive-query-index-challenge
by (rule hash-target-program-receive-query-index-challenge)
have stage:
 $\wedge$ raw. hash-target-program B (query-opening-budgets budgets ! i)
  (query-opening-stage A i raw)
using controlled i-bound
unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-target)
have record-budget:
 $\wedge$ chunk. hash-target-program B 0 (record-staged-messages chunk)
by (rule hash-target-program-record-staged-messages)
have tail:
hash-target-program B
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n)
  (staged-query-program A (Suc i) n)
by (rule Suc.IH) (use Suc.prems in simp)
have tail-return:
 $\wedge$ chunk. hash-target-program B
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n + 0)
  (staged-query-program A (Suc i) n  $\gg=$ 
    ( $\lambda$ chunks. return (chunk # chunks))))
by (rule hash-target-program-bind)
  (rule tail, rule hash-target-program-return)
have after-record:
 $\wedge$ chunk. hash-target-program B
  (0 +
    (sum-list
      (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0))
    (record-staged-messages chunk  $\gg=$ 
      ( $\lambda$ -. staged-query-program A (Suc i) n  $\gg=$ 
        ( $\lambda$ chunks. return (chunk # chunks)))))
by (rule hash-target-program-bind)
  (rule record-budget, rule tail-return)
have after-stage:
 $\wedge$ raw. hash-target-program B
  (query-opening-budgets budgets ! i +
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
          n + 0)))
    (query-opening-stage A i raw  $\gg=$ 
      ( $\lambda$ chunk. record-staged-messages chunk  $\gg=$ 
        ( $\lambda$ -. staged-query-program A (Suc i) n  $\gg=$ 
          ( $\lambda$ chunks. return (chunk # chunks)))))))
by (rule hash-target-program-bind)

```

```

    (rule stage, rule after-record)
have whole:
  hash-target-program B
  (1 +
    (query-opening-budgets budgets ! i +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
            n + 0))))
    (staged-query-program A i (Suc n))
  unfolding staged-query-program.simps
  by (rule hash-target-program-bind)
    (rule challenge, rule after-stage)
have budget-eq:
  1 +
    (query-opening-budgets budgets ! i +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
            n + 0))) =
    sum-list (take (Suc n) (drop i (query-opening-budgets budgets))) +
      Suc n
  using sum-list-take-Suc-drop[OF i-bound, of n] by simp
show ?case
  using whole unfolding budget-eq .
qed

lemma hash-target-program-checked-staged-query-program:
  assumes controlled: staged-adversary-controlled budgets A
  and bound:  $i + n \leq \text{length } (\text{query-opening-budgets budgets})$ 
  shows
    hash-target-program B
    (sum-list (take n (drop i (query-opening-budgets budgets))) + n)
    (checked-staged-query-program A trace-roots composition-roots i n)
  using bound
proof (induction n arbitrary: i)
  case 0
  then show ?case by (simp add: hash-target-program-return)
next
  case (Suc n)
  have i-bound:  $i < \text{length } (\text{query-opening-budgets budgets})$ 
  using Suc.prem1 by simp
  have challenge: hash-target-program B 1 receive-query-index-challenge
  by (rule hash-target-program-receive-query-index-challenge)
  have stage:
     $\bigwedge \text{raw. hash-target-program B } (\text{query-opening-budgets budgets ! } i)$ 
    (query-opening-stage A i raw)
  using controlled i-bound
  unfolding staged-adversary-controlled-def

```

```

by (blast intro: controlled-ro-program-target)
have check:
   $\bigwedge$ raw chunk. hash-target-program B 0
    (assert
      (verifier-query-round-chunk (index (to-nat raw))
        trace-roots composition-roots chunk))
    by (rule hash-target-program-assert)
have record-budget:
   $\bigwedge$ chunk. hash-target-program B 0 (record-staged-messages chunk)
    by (rule hash-target-program-record-staged-messages)
have tail:
  hash-target-program B
    (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n)
    (checked-staged-query-program A trace-roots composition-roots
      (Suc i) n)
    by (rule Suc.IH) (use Suc.premis in simp)
have tail-return:
   $\bigwedge$ chunk. hash-target-program B
    (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n + 0)
    (checked-staged-query-program A trace-roots composition-roots
      (Suc i) n  $\gg=$ 
      ( $\lambda$ chunks. return (chunk # chunks)))
    by (rule hash-target-program-bind)
      (rule tail, rule hash-target-program-return)
have after-record:
   $\bigwedge$ chunk. hash-target-program B
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0))
    (record-staged-messages chunk  $\gg=$ 
      ( $\lambda$ -. checked-staged-query-program A trace-roots composition-roots
        (Suc i) n  $\gg=$ 
        ( $\lambda$ chunks. return (chunk # chunks)))))
    by (rule hash-target-program-bind)
      (rule record-budget, rule tail-return)
have after-check:
   $\bigwedge$ raw chunk. hash-target-program B
    (0 +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
          n + 0)))
    (assert
      (verifier-query-round-chunk (index (to-nat raw))
        trace-roots composition-roots chunk)  $\gg=$ 
      ( $\lambda$ -. record-staged-messages chunk  $\gg=$ 

```

```

    (λ-. checked-staged-query-program A trace-roots
      composition-roots (Suc i) n >>=
        (λchunks. return (chunk # chunks))))))
  by (rule hash-target-program-bind)
    (rule check, rule after-record)
have after-stage:
  ∧ raw. hash-target-program B
    (query-opening-budgets budgets ! i +
      (0 +
        (0 +
          (sum-list
            (take n (drop (Suc i) (query-opening-budgets budgets))) +
              n + 0))))))
    (query-opening-stage A i raw >>=
      (λchunk. assert
        (verifier-query-round-chunk (index (to-nat raw))
          trace-roots composition-roots chunk) >>=
          (λ-. record-staged-messages chunk >>=
            (λ-. checked-staged-query-program A trace-roots
              composition-roots (Suc i) n >>=
                (λchunks. return (chunk # chunks)))))))
  by (rule hash-target-program-bind)
    (rule stage, rule after-check)
have whole:
  hash-target-program B
    (1 +
      (query-opening-budgets budgets ! i +
        (0 +
          (0 +
            (sum-list
              (take n (drop (Suc i) (query-opening-budgets budgets))) +
                n + 0))))))
    (checked-staged-query-program A trace-roots composition-roots
      i (Suc n))
unfolding checked-staged-query-program.simps Let-def
by (rule hash-target-program-bind)
    (rule challenge, rule after-stage)
have budget-eq:
  1 +
    (query-opening-budgets budgets ! i +
      (0 +
        (0 +
          (sum-list
            (take n (drop (Suc i) (query-opening-budgets budgets))) +
              n + 0)))) =
    sum-list (take (Suc n) (drop i (query-opening-budgets budgets))) +
      Suc n
using sum-list-take-Suc-drop[OF i-bound, of n] by simp
show ?case

```

```

    using whole unfolding budget-eq .
qed

lemma hash-relation-program-checked-staged-query-program:
  assumes controlled: staged-adversary-controlled budgets A
    and bound:  $i + n \leq \text{length } (\text{query-opening-budgets budgets})$ 
    and fibers:  $\bigwedge x. \text{card } \{y. R x y\} \leq b$ 
  shows
    hash-relation-program R b
      (sum-list (take n (drop i (query-opening-budgets budgets))) + n)
      (checked-staged-query-program A trace-roots composition-roots i n)
  using bound
proof (induction n arbitrary: i)
  case 0
  then show ?case
  by (simp add:
      hash-relation-program-zero[OF hash-map-preserving-return])
next
  case (Suc n)
  have i-bound:  $i < \text{length } (\text{query-opening-budgets budgets})$ 
    using Suc.premis by simp
  have challenge:
    hash-relation-program R b 1 receive-query-index-challenge
  by (rule hash-relation-program-receive-query-index-challenge[OF fibers])
  have stage:
     $\bigwedge \text{raw}. \text{hash-relation-program R b}$ 
      (query-opening-budgets budgets ! i)
      (query-opening-stage A i raw)
  proof -
    fix raw
    have stage-controlled:
      controlled-ro-program (query-opening-budgets budgets ! i)
        (query-opening-stage A i raw)
    using controlled i-bound
    unfolding staged-adversary-controlled-def by blast
    show hash-relation-program R b
      (query-opening-budgets budgets ! i)
      (query-opening-stage A i raw)
    by (rule controlled-ro-program-relation
        [OF stage-controlled fibers])
  qed
  have check:
     $\bigwedge \text{raw chunk}. \text{hash-relation-program R b 0}$ 
      (assert
        (verifier-query-round-chunk (index (to-nat raw))
          trace-roots composition-roots chunk))
    by (rule hash-relation-program-assert)
  have record-budget:
     $\bigwedge \text{chunk}. \text{hash-relation-program R b 0}$ 

```



```

    (record-staged-messages chunk)
  by (rule hash-relation-program-record-staged-messages)
have tail:
  hash-relation-program R b
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n)
  (checked-staged-query-program A trace-roots composition-roots
    (Suc i) n)
  by (rule Suc.IH) (use Suc.premis in simp)
have tail-return:
   $\bigwedge$  chunk. hash-relation-program R b
  (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
    n + 0)
  (checked-staged-query-program A trace-roots composition-roots
    (Suc i) n  $\gg=$ 
    ( $\lambda$  chunks. return (chunk # chunks)))
  by (rule hash-relation-program-bind)
  (rule tail,
    rule hash-relation-program-zero[OF hash-map-preserving-return])
have after-record:
   $\bigwedge$  chunk. hash-relation-program R b
  (0 +
    (sum-list
      (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n + 0))
  (record-staged-messages chunk  $\gg=$ 
    ( $\lambda$ -. checked-staged-query-program A trace-roots composition-roots
      (Suc i) n  $\gg=$ 
      ( $\lambda$  chunks. return (chunk # chunks)))))
  by (rule hash-relation-program-bind)
  (rule record-budget, rule tail-return)
have after-check:
   $\bigwedge$  raw chunk. hash-relation-program R b
  (0 +
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0))))
  (assert
    (verifier-query-round-chunk (index (to-nat raw))
      trace-roots composition-roots chunk)  $\gg=$ 
    ( $\lambda$ -. record-staged-messages chunk  $\gg=$ 
      ( $\lambda$ -. checked-staged-query-program A trace-roots
        composition-roots (Suc i) n  $\gg=$ 
        ( $\lambda$  chunks. return (chunk # chunks)))))
  by (rule hash-relation-program-bind)
  (rule check, rule after-record)
have after-stage:
   $\bigwedge$  raw. hash-relation-program R b

```

```

(query-opening-budgets budgets ! i +
  (0 +
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0))))
(query-opening-stage A i raw >=>
  (λchunk. assert
    (verifier-query-round-chunk (index (to-nat raw))
      trace-roots composition-roots chunk) >=>
    (λ-. record-staged-messages chunk >=>
      (λ-. checked-staged-query-program A trace-roots
        composition-roots (Suc i) n >=>
        (λchunks. return (chunk # chunks))))))
by (rule hash-relation-program-bind)
  (rule stage, rule after-check)
have whole:
  hash-relation-program R b
  (1 +
    (query-opening-budgets budgets ! i +
      (0 +
        (0 +
          (sum-list
            (take n (drop (Suc i) (query-opening-budgets budgets))) +
            n + 0))))))
  (checked-staged-query-program A trace-roots composition-roots
    i (Suc n))
unfolding checked-staged-query-program.simps Let-def
by (rule hash-relation-program-bind)
  (rule challenge, rule after-stage)
have budget-eq:
  1 +
    (query-opening-budgets budgets ! i +
      (0 +
        (0 +
          (sum-list
            (take n (drop (Suc i) (query-opening-budgets budgets))) +
            n + 0)))) =
    sum-list (take (Suc n) (drop i (query-opening-budgets budgets))) +
    Suc n
  using sum-list-take-Suc-drop[OF i-bound, of n] by simp
show ?case
  using whole unfolding budget-eq .
qed

```

lemma *hash-relation-program-staged-trace-fri-program:*
assumes *controlled: staged-adversary-controlled budgets A*
and *bound: $i + n \leq \text{length } (\text{trace-fri-budgets budgets})$*
and *fibers: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$*

```

shows
  hash-relation-program R b
    (sum-list (take n (drop i (trace-fri-budgets budgets))) + n)
    (staged-trace-fri-program A i n bs)
using bound
proof (induction n arbitrary: i bs)
  case 0
  then show ?case
    by (simp add:
      hash-relation-program-zero[OF hash-map-preserving-return])
next
  case (Suc n)
  have i-bound: i < length (trace-fri-budgets budgets)
    using Suc.prem by simp
  have stage:
    hash-relation-program R b (trace-fri-budgets budgets ! i)
    (trace-fri-root-stage A i bs)
  proof -
    have stage-controlled:
      controlled-ro-program (trace-fri-budgets budgets ! i)
      (trace-fri-root-stage A i bs)
    using controlled i-bound
    unfolding staged-adversary-controlled-def by blast
  show ?thesis
    by (rule controlled-ro-program-relation
      [OF stage-controlled fibers])
qed
have record-budget:
   $\wedge$ root. hash-relation-program R b 0 (record-staged-message root)
  by (rule hash-relation-program-record-staged-message)
have challenge:
  hash-relation-program R b 1 receive-trace-fri-challenge
  by (rule hash-relation-program-receive-trace-fri-challenge[OF fibers])
have tail:
   $\wedge$ root b'. hash-relation-program R b
    (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) + n)
    (staged-trace-fri-program A (Suc i) n (bs @ [b'])))
  by (rule Suc.IH) (use Suc.prem in simp)
have tail-return:
   $\wedge$ root b'. hash-relation-program R b
    (sum-list (take n (drop (Suc i) (trace-fri-budgets budgets))) +
      n + 0)
    (staged-trace-fri-program A (Suc i) n (bs @ [b'])  $\gg$ 
      ( $\lambda$ (roots, bs'). return (root # roots, bs'))))
  by (rule hash-relation-program-bind)
  (rule tail,
    simp add: hash-relation-program-zero[OF hash-map-preserving-return]
    split: prod.splits)
have after-challenge:

```

```

 $\wedge$ root. hash-relation-program  $R$   $b$ 
  (1 +
    (sum-list (take  $n$  (drop (Suc  $i$ ) (trace-fri-budgets budgets))) +
       $n + 0$ ))
    (receive-trace-fri-challenge  $\gg=$ 
      ( $\lambda b'$ . staged-trace-fri-program  $A$  (Suc  $i$ )  $n$  ( $bs @ [b']$ )  $\gg=$ 
        ( $\lambda$ (roots,  $bs'$ ). return (root # roots,  $bs'$ ))))
  by (rule hash-relation-program-bind)
    (rule challenge, rule tail-return)
have after-record:
 $\wedge$ root. hash-relation-program  $R$   $b$ 
  (0 +
    (1 +
      (sum-list (take  $n$  (drop (Suc  $i$ ) (trace-fri-budgets budgets))) +
         $n + 0$ )))
    (record-staged-message root  $\gg=$ 
      ( $\lambda$ -. receive-trace-fri-challenge  $\gg=$ 
        ( $\lambda b'$ . staged-trace-fri-program  $A$  (Suc  $i$ )  $n$  ( $bs @ [b']$ )  $\gg=$ 
          ( $\lambda$ (roots,  $bs'$ ). return (root # roots,  $bs'$ ))))
  by (rule hash-relation-program-bind)
    (rule record-budget, rule after-challenge)
have whole:
  hash-relation-program  $R$   $b$ 
    (trace-fri-budgets budgets !  $i +$ 
      (0 +
        (1 +
          (sum-list (take  $n$  (drop (Suc  $i$ ) (trace-fri-budgets budgets))) +
             $n + 0$ ))))
      (staged-trace-fri-program  $A$   $i$  (Suc  $n$ )  $bs$ )
  unfolding staged-trace-fri-program.simps
  by (rule hash-relation-program-bind)
    (rule stage, rule after-record)
have budget-eq:
  trace-fri-budgets budgets !  $i +$ 
    (0 +
      (1 +
        (sum-list (take  $n$  (drop (Suc  $i$ ) (trace-fri-budgets budgets))) +
           $n + 0$ ))) =
    sum-list (take (Suc  $n$ ) (drop  $i$  (trace-fri-budgets budgets))) +
      Suc  $n$ 
  using sum-list-take-Suc-drop[OF  $i$ -bound, of  $n$ ] by simp
show ?case
  using whole unfolding budget-eq .
qed

```

lemma hash-relation-program-staged-alpha-program:
 assumes fibers: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$
 shows hash-relation-program R b n (staged-alpha-program n)
 proof (induction n)

```

case 0
then show ?case
  by (simp add:
    hash-relation-program-zero[OF hash-map-preserving-return])
next
case (Suc n)
have challenge: hash-relation-program R b 1 receive-alpha-challenge
  by (rule hash-relation-program-receive-alpha-challenge[OF fibers])
have record-budget:
   $\bigwedge a.$  hash-relation-program R b 0 (record-staged-message a)
  by (rule hash-relation-program-record-staged-message)
have tail-return:
   $\bigwedge a.$  hash-relation-program R b (n + 0)
    (staged-alpha-program n  $\gg$  ( $\lambda as.$  return (a # as)))
  by (rule hash-relation-program-bind)
    (rule Suc.IH,
      rule hash-relation-program-zero[OF hash-map-preserving-return])
have after-record:
   $\bigwedge a.$  hash-relation-program R b (0 + (n + 0))
    (record-staged-message a  $\gg$ 
      ( $\lambda -.$  staged-alpha-program n  $\gg$  ( $\lambda as.$  return (a # as))))
  by (rule hash-relation-program-bind)
    (rule record-budget, rule tail-return)
have hash-relation-program R b (1 + (0 + (n + 0)))
  (staged-alpha-program (Suc n))
  unfolding staged-alpha-program.simps
  by (rule hash-relation-program-bind)
    (rule challenge, rule after-record)
then show ?case by simp
qed

```

```

lemma hash-relation-program-staged-composition-fri-program:
assumes controlled: staged-adversary-controlled budgets A
and bound:  $i + n \leq \text{length } (\text{composition-fri-budgets budgets})$ 
and fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
shows
  hash-relation-program R b
    (sum-list (take n (drop i (composition-fri-budgets budgets))) + n)
    (staged-composition-fri-program A dg i n bs)
using bound
proof (induction n arbitrary: i bs)
case 0
then show ?case
  by (simp add:
    hash-relation-program-zero[OF hash-map-preserving-return])
next
case (Suc n)
have i-bound:  $i < \text{length } (\text{composition-fri-budgets budgets})$ 
  using Suc.premis by simp

```

```

have stage:
  hash-relation-program R b (composition-fri-budgets budgets ! i)
    (composition-fri-root-stage A dg i bs)
proof —
  have stage-controlled:
    controlled-ro-program (composition-fri-budgets budgets ! i)
      (composition-fri-root-stage A dg i bs)
    using controlled i-bound
    unfolding staged-adversary-controlled-def by blast
  show ?thesis
    by (rule controlled-ro-program-relation
      [OF stage-controlled fibers])
qed
have record-budget:
   $\wedge$ root. hash-relation-program R b 0 (record-staged-message root)
  by (rule hash-relation-program-record-staged-message)
have challenge:
  hash-relation-program R b 1 receive-composition-fri-challenge
  by (rule
    hash-relation-program-receive-composition-fri-challenge[OF fibers])
have tail:
   $\wedge$ root b'. hash-relation-program R b
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) + n)
    (staged-composition-fri-program A dg (Suc i) n (bs @ [b'])))
  by (rule Suc.IH) (use Suc.prem in simp)
have tail-return:
   $\wedge$ root b'. hash-relation-program R b
    (sum-list
      (take n (drop (Suc i) (composition-fri-budgets budgets))) + n + 0)
    (staged-composition-fri-program A dg (Suc i) n (bs @ [b'])  $\gg$ 
      ( $\lambda$ (roots, bs'). return (root # roots, bs'))))
  by (rule hash-relation-program-bind)
    (rule tail,
      simp add: hash-relation-program-zero[OF hash-map-preserving-return]
        split: prod.splits)
have after-challenge:
   $\wedge$ root. hash-relation-program R b
    (1 +
      (sum-list
        (take n (drop (Suc i) (composition-fri-budgets budgets))) +
        n + 0))
    (receive-composition-fri-challenge  $\gg$ 
      ( $\lambda$ b'. staged-composition-fri-program A dg (Suc i) n
        (bs @ [b'])  $\gg$ 
        ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
  by (rule hash-relation-program-bind)
    (rule challenge, rule tail-return)
have after-record:

```

```

 $\wedge$ root. hash-relation-program  $R$   $b$ 
  (0 +
    (1 +
      (sum-list
        (take  $n$  (drop (Suc  $i$ ) (composition-fri-budgets budgets))) +
         $n + 0$ )))
    (record-staged-message root  $\gg$ 
      ( $\lambda$ -. receive-composition-fri-challenge  $\gg$ 
        ( $\lambda$  $b'$ . staged-composition-fri-program  $A$  dg (Suc  $i$ )  $n$ 
          (bs @ [b'])  $\gg$ 
            ( $\lambda$ (roots, bs'). return (root # roots, bs')))))
  by (rule hash-relation-program-bind)
  (rule record-budget, rule after-challenge)
have whole:
  hash-relation-program  $R$   $b$ 
  (composition-fri-budgets budgets !  $i$  +
    (0 +
      (1 +
        (sum-list
          (take  $n$  (drop (Suc  $i$ ) (composition-fri-budgets budgets))) +
           $n + 0$ )))
      (staged-composition-fri-program  $A$  dg  $i$  (Suc  $n$ ) bs)
  unfolding staged-composition-fri-program.simps
  by (rule hash-relation-program-bind)
  (rule stage, rule after-record)
have budget-eq:
  composition-fri-budgets budgets !  $i$  +
    (0 +
      (1 +
        (sum-list
          (take  $n$  (drop (Suc  $i$ ) (composition-fri-budgets budgets))) +
           $n + 0$ ))) =
    sum-list (take (Suc  $n$ ) (drop  $i$  (composition-fri-budgets budgets))) +
    Suc  $n$ 
  using sum-list-take-Suc-drop[OF  $i$ -bound, of  $n$ ] by simp
show ?case
  using whole unfolding budget-eq .
qed

```

lemma hash-relation-program-guarded-staged-composition-fri-program:

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and fibers: $\wedge x. \text{card } \{y. R \ x \ y\} \leq b$

shows

hash-relation-program R b

(sum-list (composition-fri-budgets budgets) +

ceil-log (maxDegree + 1))

(assert (ceil-log (to-nat dg + 1) $\leq$ ceil-log (maxDegree + 1)) $\gg$

(λ -. staged-composition-fri-program A dg 0

```

      (ceil-log (to-nat dg + 1)) [])
proof -
  let ?B =
    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
  have hash-relation-program R b (0 + ?B)
    (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))
  proof (rule hash-relation-program-bind-on-outcomes
    [where q=0 and r=?B])
  show hash-relation-program R b 0
    (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)))
  by (rule hash-relation-program-assert)
next
  fix s x t
  assume out:
    Some (x, t) ∈
      set-dist
      (execute
        (assert
          (ceil-log (to-nat dg + 1) ≤
            ceil-log (maxDegree + 1))) s)
  have round-bound:
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using out unfolding assert-def
  by (cases ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
  have len:
    0 + ceil-log (to-nat dg + 1) ≤
      length (composition-fri-budgets budgets)
  using round-bound wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-relation-program R b
      (sum-list
        (take (ceil-log (to-nat dg + 1))
          (drop 0 (composition-fri-budgets budgets))) +
        ceil-log (to-nat dg + 1))
      (staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [])
  by (rule hash-relation-program-staged-composition-fri-program
    [OF controlled len fibers])
  have le:
    sum-list
      (take (ceil-log (to-nat dg + 1))
        (drop 0 (composition-fri-budgets budgets))) +
      ceil-log (to-nat dg + 1) ≤
    sum-list (composition-fri-budgets budgets) +
      ceil-log (maxDegree + 1)

```



```

    using round-bound sum-list-take-le[of ceil-log (to-nat dg + 1)
      composition-fri-budgets budgets]
  by simp
show hash-relation-program R b ?B
  (staged-composition-fri-program A dg 0
    (ceil-log (to-nat dg + 1)) [])
  by (rule hash-relation-program-mono[OF le exact])
qed
then show ?thesis by simp
qed

lemma hash-relation-program-staged-alpha-prefix-program:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
  shows
    hash-relation-program R b
      (staged-alpha-search-queries budgets 0)
      (staged-alpha-prefix-program A)
proof -
  let ?trace-root = trace-root-budget budgets
  let ?trace-fri =
    sum-list (trace-fri-budgets budgets) + ceil-log clength
  let ?trace-final = trace-final-budget budgets
  have trace-root:
    hash-relation-program R b ?trace-root (trace-root-stage A)
proof -
  have controlled-root:
    controlled-ro-program ?trace-root (trace-root-stage A)
  using controlled unfolding staged-adversary-controlled-def by blast
  show ?thesis
  by (rule controlled-ro-program-relation
    [OF controlled-root fibers])
qed
have record-root:
   $\bigwedge fr. \text{hash-relation-program } R \ b \ 0 \ (\text{record-staged-message } fr)$ 
  by (rule hash-relation-program-record-staged-message)
have trace-fri:
  hash-relation-program R b ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])
proof -
  have len:
     $0 + \text{ceil-log clength} \leq \text{length } (\text{trace-fri-budgets budgets})$ 
  using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-relation-program R b
      (sum-list
        (take (ceil-log clength) (drop 0 (trace-fri-budgets budgets)))) +
        ceil-log clength)

```

```

    (staged-trace-fri-program A 0 (ceil-log clength) [])
  by (rule hash-relation-program-staged-trace-fri-program
      [OF controlled len fibers])
have le:
  sum-list
    (take (ceil-log clength) (drop 0 (trace-fri-budgets budgets))) +
    ceil-log clength ≤
  sum-list (trace-fri-budgets budgets) + ceil-log clength
using sum-list-take-le[of ceil-log clength
    trace-fri-budgets budgets]
  by simp
show ?thesis
  by (rule hash-relation-program-mono[OF le exact])
qed
have trace-final:
  ∧ trace-roots trace-bs. hash-relation-program R b ?trace-final
  (trace-final-stage A trace-bs)
proof -
  fix trace-roots trace-bs
  have controlled-final:
    controlled-ro-program ?trace-final
    (trace-final-stage A trace-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
show hash-relation-program R b ?trace-final
  (trace-final-stage A trace-bs)
  by (rule controlled-ro-program-relation
      [OF controlled-final fibers])
qed
have record-final:
  ∧ trace-final. hash-relation-program R b 0
  (record-staged-message trace-final)
  by (rule hash-relation-program-record-staged-message)
have after-final-record:
  ∧ fr trace-roots trace-bs trace-final.
  hash-relation-program R b 0
  (record-staged-message trace-final >=>
    (λ-. return (fr, trace-roots, trace-bs, trace-final)))
proof -
  fix fr trace-roots trace-bs trace-final
  have hash-relation-program R b (0 + 0)
    (record-staged-message trace-final >=>
      (λ-. return (fr, trace-roots, trace-bs, trace-final)))
  by (rule hash-relation-program-bind)
  (rule record-final,
    rule hash-relation-program-zero[OF hash-map-preserving-return])
then show hash-relation-program R b 0
  (record-staged-message trace-final >=>
    (λ-. return (fr, trace-roots, trace-bs, trace-final)))
  by simp

```

```

qed
have after-final:
   $\wedge$ fr trace-roots trace-bs. hash-relation-program R b
    (?trace-final + 0)
    (trace-final-stage A trace-bs  $\gg$ 
      ( $\lambda$ trace-final. record-staged-message trace-final  $\gg$ 
        ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
  by (rule hash-relation-program-bind)
    (rule trace-final, rule after-final-record)
have after-trace-fri:
   $\wedge$ fr. hash-relation-program R b
    (?trace-fri + (?trace-final + 0))
    (staged-trace-fri-program A 0 (ceil-log clength) []  $\gg$ 
      ( $\lambda$ (trace-roots, trace-bs).
        trace-final-stage A trace-bs  $\gg$ 
          ( $\lambda$ trace-final. record-staged-message trace-final  $\gg$ 
            ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))))
proof -
  fix fr
  have cont:
     $\wedge$ x. hash-relation-program R b (?trace-final + 0)
      (case x of (trace-roots, trace-bs)  $\Rightarrow$ 
        trace-final-stage A trace-bs  $\gg$ 
          ( $\lambda$ trace-final. record-staged-message trace-final  $\gg$ 
            ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
  proof -
    fix x
    show hash-relation-program R b (?trace-final + 0)
      (case x of (trace-roots, trace-bs)  $\Rightarrow$ 
        trace-final-stage A trace-bs  $\gg$ 
          ( $\lambda$ trace-final. record-staged-message trace-final  $\gg$ 
            ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
  proof (cases x)
    case (Pair trace-roots trace-bs)
    have step:
      hash-relation-program R b (?trace-final + 0)
        (trace-final-stage A trace-bs  $\gg$ 
          ( $\lambda$ trace-final. record-staged-message trace-final  $\gg$ 
            ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
    by (rule after-final)
    show ?thesis
      unfolding Pair using step by simp
  qed
qed
have hash-relation-program R b (?trace-fri + (?trace-final + 0))
  (staged-trace-fri-program A 0 (ceil-log clength) []  $\gg$ 
    ( $\lambda$ (trace-roots, trace-bs).
      trace-final-stage A trace-bs  $\gg$ 
        ( $\lambda$ trace-final. record-staged-message trace-final  $\gg$ 
          ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))

```

```

    (λ-. return (fr, trace-roots, trace-bs, trace-final))))))
  by (rule hash-relation-program-bind)
    (rule trace-fri, rule cont)
then show hash-relation-program R b
  (?trace-fri + (?trace-final + 0))
  (staged-trace-fri-program A 0 (ceil-log clength) [] ≫=
    (λ(trace-roots, trace-bs).
      trace-final-stage A trace-bs ≫=
        (λtrace-final. record-staged-message trace-final ≫=
          (λ-. return (fr, trace-roots, trace-bs, trace-final)))))) .
qed
have after-root-record:
  ∧fr. hash-relation-program R b
  (0 + (?trace-fri + (?trace-final + 0)))
  (record-staged-message fr ≫=
    (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] ≫=
      (λ(trace-roots, trace-bs).
        trace-final-stage A trace-bs ≫=
          (λtrace-final. record-staged-message trace-final ≫=
            (λ-. return (fr, trace-roots, trace-bs, trace-final))))))
  by (rule hash-relation-program-bind)
    (rule record-root, rule after-trace-fri)
have whole:
  hash-relation-program R b
  (?trace-root + (0 + (?trace-fri + (?trace-final + 0))))
  (staged-alpha-prefix-program A)
unfolding staged-alpha-prefix-program-def
by (rule hash-relation-program-bind)
  (rule trace-root, rule after-root-record)
show ?thesis
using whole
unfolding staged-alpha-search-queries-def
by (simp add: add.assoc add.commute add.left-commute)
qed

lemma hash-range-budget-staged-alpha-prefix-program:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  hash-range-budget
  (staged-alpha-search-queries budgets 0)
  (staged-alpha-prefix-program A)
proof –
let ?trace-root = trace-root-budget budgets
let ?trace-fri =
  sum-list (trace-fri-budgets budgets) + ceil-log clength
let ?trace-final = trace-final-budget budgets
have trace-root:
  hash-range-budget ?trace-root (trace-root-stage A)

```

```

using controlled unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-range)
have trace-fri:
  hash-range-budget ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])
proof –
  have len:
    0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-range-budget
      (sum-list
        (take (ceil-log clength)
          (drop 0 (trace-fri-budgets budgets))) +
        ceil-log clength)
      (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule hash-range-budget-staged-trace-fri-program
      [OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have trace-final:
  ∧bs. hash-range-budget ?trace-final (trace-final-stage A bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
have after-trace-final-record:
  ∧fr trace-roots trace-bs trace-final.
    hash-range-budget 0
      (record-staged-message trace-final ≫=
        (λ-. return (fr, trace-roots, trace-bs, trace-final)))
proof –
  fix fr trace-roots trace-bs trace-final
  have hash-range-budget (0 + 0)
    (record-staged-message trace-final ≫=
      (λ-. return (fr, trace-roots, trace-bs, trace-final)))
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule hash-range-budget-return)
  then show
    hash-range-budget 0
      (record-staged-message trace-final ≫=
        (λ-. return (fr, trace-roots, trace-bs, trace-final)))
    by simp
qed
have after-trace-final:
  ∧fr trace-roots trace-bs.
    hash-range-budget (?trace-final + 0)
      (trace-final-stage A trace-bs ≫=
        (λtrace-final.

```

```

      record-staged-message trace-final  $\gg=$ 
        ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
  by (rule hash-range-budget-bind)
    (rule trace-final, rule after-trace-final-record)
have after-trace-fri:
 $\wedge$ fr. hash-range-budget (?trace-fri + (?trace-final + 0))
  (staged-trace-fri-program A 0 (ceil-log clength) []  $\gg=$ 
    ( $\lambda$ (trace-roots, trace-bs).
      trace-final-stage A trace-bs  $\gg=$ 
        ( $\lambda$ trace-final.
          record-staged-message trace-final  $\gg=$ 
            ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))))
proof (rule hash-range-budget-bind)
  fix fr
  show hash-range-budget ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])
  by (rule trace-fri)
next
  fix fr x
  show hash-range-budget (?trace-final + 0)
    (case x of (trace-roots, trace-bs)  $\Rightarrow$ 
      trace-final-stage A trace-bs  $\gg=$ 
        ( $\lambda$ trace-final.
          record-staged-message trace-final  $\gg=$ 
            ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
  proof (cases x)
    case (Pair trace-roots trace-bs)
    have step:
      hash-range-budget (?trace-final + 0)
        (trace-final-stage A trace-bs  $\gg=$ 
          ( $\lambda$ trace-final.
            record-staged-message trace-final  $\gg=$ 
              ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
    by (rule after-trace-final)
  show ?thesis
    unfolding Pair using step by simp
qed
qed
have after-trace-root-record:
 $\wedge$ fr. hash-range-budget (0 + (?trace-fri + (?trace-final + 0)))
  (record-staged-message fr  $\gg=$ 
    ( $\lambda$ -. staged-trace-fri-program A 0 (ceil-log clength) []  $\gg=$ 
      ( $\lambda$ (trace-roots, trace-bs).
        trace-final-stage A trace-bs  $\gg=$ 
          ( $\lambda$ trace-final.
            record-staged-message trace-final  $\gg=$ 
              ( $\lambda$ -. return
                (fr, trace-roots, trace-bs, trace-final))))))
  by (rule hash-range-budget-bind)

```

```

    (rule hash-range-budget-record-staged-message, rule after-trace-fri)
have whole:
  hash-range-budget
    (?trace-root + (0 + (?trace-fri + (?trace-final + 0))))
    (staged-alpha-prefix-program A)
  unfolding staged-alpha-prefix-program-def
  by (rule hash-range-budget-bind)
    (rule trace-root, rule after-trace-root-record)
show ?thesis
  using whole
  unfolding staged-alpha-search-queries-def
  by (simp add: add.assoc add.commute add.left-commute)
qed

```

```

lemma hash-collision-budget-staged-alpha-prefix-program:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    hash-collision-budget
      (staged-alpha-search-queries budgets 0)
      (staged-alpha-prefix-program A)
proof –
  let ?trace-root = trace-root-budget budgets
  let ?trace-fri =
    sum-list (trace-fri-budgets budgets) + ceil-log clength
  let ?trace-final = trace-final-budget budgets
  have trace-root-range:
    hash-range-budget ?trace-root (trace-root-stage A)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
  have trace-root-coll:
    hash-collision-budget ?trace-root (trace-root-stage A)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-collision)
  have trace-fri-range':
    hash-range-budget ?trace-fri
      (staged-trace-fri-program A 0 (ceil-log clength) [])
proof –
  have len:
    0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
  using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-range-budget
      (sum-list
        (take (ceil-log clength)
          (drop 0 (trace-fri-budgets budgets))) +
        ceil-log clength)
      (staged-trace-fri-program A 0 (ceil-log clength) [])
  by (rule hash-range-budget-staged-trace-fri-program

```

```

    [OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have trace-fri-coll:
  hash-collision-budget ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])
proof -
  have len:
    0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-collision-budget
      (sum-list
        (take (ceil-log clength)
          (drop 0 (trace-fri-budgets budgets))) +
        ceil-log clength)
      (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule hash-collision-budget-staged-trace-fri-program
      [OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have trace-final-range:
  ∧bs. hash-range-budget ?trace-final (trace-final-stage A bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
have trace-final-coll:
  ∧bs. hash-collision-budget ?trace-final (trace-final-stage A bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-collision)
have after-trace-final-record-range:
  ∧fr trace-roots trace-bs trace-final.
  hash-range-budget 0
    (record-staged-message trace-final ≫=
      (λ-. return (fr, trace-roots, trace-bs, trace-final)))
proof -
  fix fr trace-roots trace-bs trace-final
  have hash-range-budget (0 + 0)
    (record-staged-message trace-final ≫=
      (λ-. return (fr, trace-roots, trace-bs, trace-final)))
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule hash-range-budget-return)
  then show hash-range-budget 0
    (record-staged-message trace-final ≫=
      (λ-. return (fr, trace-roots, trace-bs, trace-final)))
  by simp
qed

```



```

have after-trace-final-record-coll:
   $\bigwedge$ fr trace-roots trace-bs trace-final.
    hash-collision-budget 0
    (record-staged-message trace-final  $\gg$ 
      ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final)))
proof -
  fix fr trace-roots trace-bs trace-final
  have hash-collision-budget (0 + 0)
    (record-staged-message trace-final  $\gg$ 
      ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final)))
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule hash-collision-budget-record-staged-message,
      rule hash-range-budget-return, rule hash-collision-budget-return)
  then show hash-collision-budget 0
    (record-staged-message trace-final  $\gg$ 
      ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final)))
  by simp
qed
have after-trace-final-range:
   $\bigwedge$ fr trace-roots trace-bs.
    hash-range-budget (?trace-final + 0)
    (trace-final-stage A trace-bs  $\gg$ 
      ( $\lambda$ trace-final.
        record-staged-message trace-final  $\gg$ 
          ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
  by (rule hash-range-budget-bind)
    (rule trace-final-range, rule after-trace-final-record-range)
have after-trace-final-coll:
   $\bigwedge$ fr trace-roots trace-bs.
    hash-collision-budget (?trace-final + 0)
    (trace-final-stage A trace-bs  $\gg$ 
      ( $\lambda$ trace-final.
        record-staged-message trace-final  $\gg$ 
          ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
  by (rule hash-collision-budget-bind)
    (rule trace-final-range, rule trace-final-coll,
      rule after-trace-final-record-range,
      rule after-trace-final-record-coll)
have after-trace-fri-range:
   $\bigwedge$ fr. hash-range-budget (?trace-fri + (?trace-final + 0))
    (staged-trace-fri-program A 0 (ceil-log clength)  $\square$   $\gg$ 
      ( $\lambda$ (trace-roots, trace-bs).
        trace-final-stage A trace-bs  $\gg$ 
          ( $\lambda$ trace-final.
            record-staged-message trace-final  $\gg$ 
              ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))))
proof (rule hash-range-budget-bind)
  fix fr

```

```

show hash-range-budget ?trace-fri
  (staged-trace-fri-program A 0 (ceil-log clength) [])
  by (rule trace-fri-range')
next
fix fr x
show hash-range-budget (?trace-final + 0)
  (case x of (trace-roots, trace-bs)  $\Rightarrow$ 
    trace-final-stage A trace-bs  $\gg$ 
    ( $\lambda$ trace-final.
      record-staged-message trace-final  $\gg$ 
      ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
proof (cases x)
  case (Pair trace-roots trace-bs)
  have step:
    hash-range-budget (?trace-final + 0)
    (trace-final-stage A trace-bs  $\gg$ 
      ( $\lambda$ trace-final.
        record-staged-message trace-final  $\gg$ 
        ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
    by (rule after-trace-final-range)
  show ?thesis
    unfolding Pair using step by simp
qed
qed
have after-trace-fri-coll:
   $\bigwedge$ fr. hash-collision-budget (?trace-fri + (?trace-final + 0))
  (staged-trace-fri-program A 0 (ceil-log clength) [])  $\gg$ 
  ( $\lambda$ (trace-roots, trace-bs).
    trace-final-stage A trace-bs  $\gg$ 
    ( $\lambda$ trace-final.
      record-staged-message trace-final  $\gg$ 
      ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
proof (rule hash-collision-budget-bind)
fix fr
show hash-range-budget ?trace-fri
  (staged-trace-fri-program A 0 (ceil-log clength) [])
  by (rule trace-fri-range')
show hash-collision-budget ?trace-fri
  (staged-trace-fri-program A 0 (ceil-log clength) [])
  by (rule trace-fri-coll)
next
fix fr x
show hash-range-budget (?trace-final + 0)
  (case x of (trace-roots, trace-bs)  $\Rightarrow$ 
    trace-final-stage A trace-bs  $\gg$ 
    ( $\lambda$ trace-final.
      record-staged-message trace-final  $\gg$ 
      ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
proof (cases x)

```

```

case (Pair trace-roots trace-bs)
have step:
  hash-range-budget (?trace-final + 0)
    (trace-final-stage A trace-bs  $\gg$ 
      ( $\lambda$ trace-final.
        record-staged-message trace-final  $\gg$ 
          ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
  by (rule after-trace-final-range)
show ?thesis
  unfolding Pair using step by simp
qed
show hash-collision-budget (?trace-final + 0)
  (case x of (trace-roots, trace-bs)  $\Rightarrow$ 
    trace-final-stage A trace-bs  $\gg$ 
      ( $\lambda$ trace-final.
        record-staged-message trace-final  $\gg$ 
          ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
proof (cases x)
case (Pair trace-roots trace-bs)
have step:
  hash-collision-budget (?trace-final + 0)
    (trace-final-stage A trace-bs  $\gg$ 
      ( $\lambda$ trace-final.
        record-staged-message trace-final  $\gg$ 
          ( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
  by (rule after-trace-final-coll)
show ?thesis
  unfolding Pair using step by simp
qed
qed
have after-trace-root-record-range:
   $\wedge$ fr. hash-range-budget (0 + (?trace-fri + (?trace-final + 0)))
    (record-staged-message fr  $\gg$ 
      ( $\lambda$ -. staged-trace-fri-program A 0 (ceil-log clength)  $\square$   $\gg$ 
        ( $\lambda$ (trace-roots, trace-bs).
          trace-final-stage A trace-bs  $\gg$ 
            ( $\lambda$ trace-final.
              record-staged-message trace-final  $\gg$ 
                ( $\lambda$ -. return
                  (fr, trace-roots, trace-bs, trace-final))))))
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule after-trace-fri-range)
have after-trace-root-record-coll:
   $\wedge$ fr. hash-collision-budget
    (0 + (?trace-fri + (?trace-final + 0)))
    (record-staged-message fr  $\gg$ 
      ( $\lambda$ -. staged-trace-fri-program A 0 (ceil-log clength)  $\square$   $\gg$ 
        ( $\lambda$ (trace-roots, trace-bs).
          trace-final-stage A trace-bs  $\gg$ 
            ( $\lambda$ trace-final.
              record-staged-message trace-final  $\gg$ 
                ( $\lambda$ -. return
                  (fr, trace-roots, trace-bs, trace-final))))))

```

```

    trace-final-stage A trace-bs  $\gg$ 
    ( $\lambda$ trace-final.
      record-staged-message trace-final  $\gg$ 
      ( $\lambda$ -. return
        (fr, trace-roots, trace-bs, trace-final))))))
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-record-staged-message,
     rule hash-collision-budget-record-staged-message,
     rule after-trace-fri-range, rule after-trace-fri-coll)
  have whole:
    hash-collision-budget
      (?trace-root + (0 + (?trace-fri + (?trace-final + 0))))
      (staged-alpha-prefix-program A)
  unfolding staged-alpha-prefix-program-def
  by (rule hash-collision-budget-bind)
    (rule trace-root-range, rule trace-root-coll,
     rule after-trace-root-record-range,
     rule after-trace-root-record-coll)
  show ?thesis
  using whole
  unfolding staged-alpha-search-queries-def
  by (simp add: add.assoc add.commute add.left-commute)
qed

lemma hash-range-budget-checked-staged-query-program:
  assumes controlled: staged-adversary-controlled budgets A
    and bound:  $i + n \leq \text{length } (\text{query-opening-budgets budgets})$ 
  shows
    hash-range-budget
      (sum-list (take n (drop i (query-opening-budgets budgets))) + n)
      (checked-staged-query-program A trace-roots composition-roots i n)
  using bound
proof (induction n arbitrary: i)
  case 0
  then show ?case by (simp add: hash-range-budget-return)
next
  case (Suc n)
  have i-bound:  $i < \text{length } (\text{query-opening-budgets budgets})$ 
  using Suc.prem1 by simp
  have challenge: hash-range-budget 1 receive-query-index-challenge
  by (rule hash-range-budget-receive-query-index-challenge)
  have stage:
     $\bigwedge \text{raw. hash-range-budget } (\text{query-opening-budgets budgets } ! i)$ 
     $(\text{query-opening-stage } A \ i \ \text{raw})$ 
  using controlled i-bound
  unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
  have check:
     $\bigwedge \text{raw chunk. hash-range-budget } 0$ 

```

```

    (assert
      (verifier-query-round-chunk (index (to-nat raw))
        trace-roots composition-roots chunk))
  by (rule hash-range-budget-assert)
have record-budget:
   $\wedge$  chunk. hash-range-budget 0 (record-staged-messages chunk)
  by (rule hash-range-budget-record-staged-messages)
have tail:
  hash-range-budget
    (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n)
    (checked-staged-query-program A trace-roots composition-roots
      (Suc i) n)
  by (rule Suc.IH) (use Suc.prem in simp)
have tail-return:
   $\wedge$  chunk. hash-range-budget
    (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n + 0)
    (checked-staged-query-program A trace-roots composition-roots
      (Suc i) n  $\gg=$ 
      ( $\lambda$  chunks. return (chunk # chunks)))
  by (rule hash-range-budget-bind)
    (rule tail, rule hash-range-budget-return)
have after-record:
   $\wedge$  chunk. hash-range-budget
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0))
    (record-staged-messages chunk  $\gg=$ 
      ( $\lambda$ -. checked-staged-query-program A trace-roots composition-roots
        (Suc i) n  $\gg=$ 
        ( $\lambda$  chunks. return (chunk # chunks)))))
  by (rule hash-range-budget-bind)
    (rule record-budget, rule tail-return)
have after-check:
   $\wedge$  raw chunk. hash-range-budget
    (0 +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
          n + 0)))
    (assert
      (verifier-query-round-chunk (index (to-nat raw))
        trace-roots composition-roots chunk)  $\gg=$ 
      ( $\lambda$ -. record-staged-messages chunk  $\gg=$ 
        ( $\lambda$ -. checked-staged-query-program A trace-roots
          composition-roots (Suc i) n  $\gg=$ 
          ( $\lambda$  chunks. return (chunk # chunks)))))

```

```

by (rule hash-range-budget-bind)
  (rule check, rule after-record)
have after-stage:
 $\wedge$ raw. hash-range-budget
  (query-opening-budgets budgets ! i +
    (0 +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
            n + 0))))
  (query-opening-stage A i raw  $\gg$ =
    ( $\lambda$ chunk. assert
      (verifier-query-round-chunk (index (to-nat raw))
        trace-roots composition-roots chunk)  $\gg$ =
      ( $\lambda$ -. record-staged-messages chunk  $\gg$ =
        ( $\lambda$ -. checked-staged-query-program A trace-roots
          composition-roots (Suc i) n  $\gg$ =
            ( $\lambda$ chunks. return (chunk # chunks))))))
by (rule hash-range-budget-bind)
  (rule stage, rule after-check)
have whole:
  hash-range-budget
    (1 +
      (query-opening-budgets budgets ! i +
        (0 +
          (0 +
            (sum-list
              (take n (drop (Suc i) (query-opening-budgets budgets))) +
                n + 0))))
      (checked-staged-query-program A trace-roots composition-roots
        i (Suc n))
    )
unfolding checked-staged-query-program.simps Let-def
by (rule hash-range-budget-bind)
  (rule challenge, rule after-stage)
have budget-eq:
  1 +
    (query-opening-budgets budgets ! i +
      (0 +
        (0 +
          (sum-list
            (take n (drop (Suc i) (query-opening-budgets budgets))) +
              n + 0))))
    =
    sum-list (take (Suc n) (drop i (query-opening-budgets budgets))) +
      Suc n
using sum-list-take-Suc-drop[OF i-bound, of n] by simp
show ?case
using whole unfolding budget-eq .
qed

```

lemma *hash-collision-budget-checked-staged-query-program:*
assumes *controlled: staged-adversary-controlled budgets A*
and *bound: $i + n \leq \text{length } (\text{query-opening-budgets budgets})$*
shows
hash-collision-budget
 $(\text{sum-list } (\text{take } n \ (\text{drop } i \ (\text{query-opening-budgets budgets}))) + n)$
 $(\text{checked-staged-query-program } A \ \text{trace-roots composition-roots } i \ n)$
using *bound*
proof (*induction n arbitrary: i*)
case *0*
then show ?*case* **by** (*simp add: hash-collision-budget-return*)
next
case (*Suc n*)
have *i-bound: $i < \text{length } (\text{query-opening-budgets budgets})$*
using *Suc.prem* **by** *simp*
have *challenge-range: hash-range-budget 1 receive-query-index-challenge*
by (*rule hash-range-budget-receive-query-index-challenge*)
have *challenge-coll: hash-collision-budget 1 receive-query-index-challenge*
by (*rule hash-collision-budget-receive-query-index-challenge*)
have *stage-range:*
 $\bigwedge \text{raw. hash-range-budget } (\text{query-opening-budgets budgets } ! \ i)$
 $(\text{query-opening-stage } A \ i \ \text{raw})$
using *controlled i-bound*
unfolding *staged-adversary-controlled-def*
by (*blast intro: controlled-ro-program-range*)
have *stage-coll:*
 $\bigwedge \text{raw. hash-collision-budget } (\text{query-opening-budgets budgets } ! \ i)$
 $(\text{query-opening-stage } A \ i \ \text{raw})$
using *controlled i-bound*
unfolding *staged-adversary-controlled-def*
by (*blast intro: controlled-ro-program-collision*)
have *check-range:*
 $\bigwedge \text{raw chunk. hash-range-budget } 0$
 $(\text{assert } (\text{verifier-query-round-chunk } (\text{index } (\text{to-nat raw})))$
 $\text{trace-roots composition-roots chunk}))$
by (*rule hash-range-budget-assert*)
have *check-coll:*
 $\bigwedge \text{raw chunk. hash-collision-budget } 0$
 $(\text{assert } (\text{verifier-query-round-chunk } (\text{index } (\text{to-nat raw})))$
 $\text{trace-roots composition-roots chunk}))$
by (*rule hash-collision-budget-assert*)
have *record-range:*
 $\bigwedge \text{chunk. hash-range-budget } 0 \ (\text{record-staged-messages chunk})$
by (*rule hash-range-budget-record-staged-messages*)
have *record-coll:*
 $\bigwedge \text{chunk. hash-collision-budget } 0 \ (\text{record-staged-messages chunk})$
by (*rule hash-collision-budget-record-staged-messages*)

```

have tail-range:
  hash-range-budget
    (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n)
    (checked-staged-query-program A trace-roots composition-roots
      (Suc i) n)
  by (rule hash-range-budget-checked-staged-query-program[OF controlled])
    (use Suc.premis in simp)
have tail-coll:
  hash-collision-budget
    (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n)
    (checked-staged-query-program A trace-roots composition-roots
      (Suc i) n)
  by (rule Suc.IH) (use Suc.premis in simp)
have tail-return-range:
   $\bigwedge$ chunk. hash-range-budget
    (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n + 0)
    (checked-staged-query-program A trace-roots composition-roots
      (Suc i) n  $\gg=$ 
      ( $\lambda$ chunks. return (chunk # chunks)))
  by (rule hash-range-budget-bind)
    (rule tail-range, rule hash-range-budget-return)
have tail-return-coll:
   $\bigwedge$ chunk. hash-collision-budget
    (sum-list (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n + 0)
    (checked-staged-query-program A trace-roots composition-roots
      (Suc i) n  $\gg=$ 
      ( $\lambda$ chunks. return (chunk # chunks)))
  by (rule hash-collision-budget-bind)
    (rule tail-range, rule tail-coll, rule hash-range-budget-return,
      rule hash-collision-budget-return)
have after-record-range:
   $\bigwedge$ chunk. hash-range-budget
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0))
    (record-staged-messages chunk  $\gg=$ 
      ( $\lambda$ -. checked-staged-query-program A trace-roots composition-roots
        (Suc i) n  $\gg=$ 
        ( $\lambda$ chunks. return (chunk # chunks)))))
  by (rule hash-range-budget-bind)
    (rule record-range, rule tail-return-range)
have after-record-coll:
   $\bigwedge$ chunk. hash-collision-budget
    (0 +

```



```

    (sum-list
      (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n + 0))
  (record-staged-messages chunk >=>
    (λ-. checked-staged-query-program A trace-roots composition-roots
      (Suc i) n >=>
      (λchunks. return (chunk # chunks))))
by (rule hash-collision-budget-bind)
  (rule record-range, rule record-coll, rule tail-return-range,
    rule tail-return-coll)
have after-check-range:
  ∧raw chunk. hash-range-budget
  (0 +
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0)))
  (assert
    (verifier-query-round-chunk (index (to-nat raw))
      trace-roots composition-roots chunk) >=>
    (λ-. record-staged-messages chunk >=>
      (λ-. checked-staged-query-program A trace-roots
        composition-roots (Suc i) n >=>
        (λchunks. return (chunk # chunks))))))
by (rule hash-range-budget-bind)
  (rule check-range, rule after-record-range)
have after-check-coll:
  ∧raw chunk. hash-collision-budget
  (0 +
    (0 +
      (sum-list
        (take n (drop (Suc i) (query-opening-budgets budgets))) +
        n + 0)))
  (assert
    (verifier-query-round-chunk (index (to-nat raw))
      trace-roots composition-roots chunk) >=>
    (λ-. record-staged-messages chunk >=>
      (λ-. checked-staged-query-program A trace-roots
        composition-roots (Suc i) n >=>
        (λchunks. return (chunk # chunks))))))
by (rule hash-collision-budget-bind)
  (rule check-range, rule check-coll, rule after-record-range,
    rule after-record-coll)
have after-stage-range:
  ∧raw. hash-range-budget
  (query-opening-budgets budgets ! i +
    (0 +
      (0 +
        (sum-list

```

```

      (take n (drop (Suc i) (query-opening-budgets budgets))) +
      n + 0))))
(query-opening-stage A i raw >>=
  (λchunk. assert
    (verifier-query-round-chunk (index (to-nat raw))
      trace-roots composition-roots chunk) >>=
    (λ-. record-staged-messages chunk >>=
      (λ-. checked-staged-query-program A trace-roots
        composition-roots (Suc i) n >>=
        (λchunks. return (chunk # chunks))))))
by (rule hash-range-budget-bind)
  (rule stage-range, rule after-check-range)
have after-stage-coll:
  ∧raw. hash-collision-budget
  (query-opening-budgets budgets ! i +
    (0 +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
          n + 0))))
  (query-opening-stage A i raw >>=
    (λchunk. assert
      (verifier-query-round-chunk (index (to-nat raw))
        trace-roots composition-roots chunk) >>=
      (λ-. record-staged-messages chunk >>=
        (λ-. checked-staged-query-program A trace-roots
          composition-roots (Suc i) n >>=
          (λchunks. return (chunk # chunks))))))
by (rule hash-collision-budget-bind)
  (rule stage-range, rule stage-coll, rule after-check-range,
    rule after-check-coll)
have whole:
  hash-collision-budget
  (1 +
    (query-opening-budgets budgets ! i +
      (0 +
        (0 +
          (sum-list
            (take n (drop (Suc i) (query-opening-budgets budgets))) +
            n + 0))))
    (checked-staged-query-program A trace-roots composition-roots
      i (Suc n))
  unfolding checked-staged-query-program.simps Let-def
by (rule hash-collision-budget-bind)
  (rule challenge-range, rule challenge-coll, rule after-stage-range,
    rule after-stage-coll)
have budget-eq:
  1 +
    (query-opening-budgets budgets ! i +

```

```

    (0 +
      (0 +
        (sum-list
          (take n (drop (Suc i) (query-opening-budgets budgets))) +
          n + 0)))) =
    sum-list (take (Suc n) (drop i (query-opening-budgets budgets))) +
    Suc n
  using sum-list-take-Suc-drop[OF i-bound, of n] by simp
  show ?case
  using whole unfolding budget-eq .
qed

```

lemma *hash-range-budget-guarded-staged-composition-fri-program:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows

hash-range-budget

```

  (sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1))
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))

```

proof –

let *?B* =

```

  sum-list (composition-fri-budgets budgets) +
  ceil-log (maxDegree + 1)

```

have *hash-range-budget (0 + ?B)*

```

  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))

```

proof (*rule hash-range-budget-bind-on-outcomes*

[where n=0 and n'=?B])

show *hash-range-budget 0*

```

  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)))

```

unfolding *hash-range-budget-def assert-def*

by (*auto simp: throw-no-outcome*)

next

fix *s x t*

assume *out:*

Some (x, t) ∈

set-dist

(execute

(assert

```

  (ceil-log (to-nat dg + 1) ≤
    ceil-log (maxDegree + 1))) s)

```

have *round-bound:*

```

  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)

```

using *out unfolding assert-def*

by (*cases ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)*)

```

    (auto simp: throw-no-outcome)
  have len:
    0 + ceil-log (to-nat dg + 1) ≤
      length (composition-fri-budgets budgets)
    using round-bound wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-range-budget
      (sum-list
        (take (ceil-log (to-nat dg + 1))
          (drop 0 (composition-fri-budgets budgets)))) +
        ceil-log (to-nat dg + 1))
      (staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [])
    by (rule hash-range-budget-staged-composition-fri-program
      [OF controlled len])
  have le:
    sum-list
      (take (ceil-log (to-nat dg + 1))
        (drop 0 (composition-fri-budgets budgets))) +
        ceil-log (to-nat dg + 1) ≤
      sum-list (composition-fri-budgets budgets) +
        ceil-log (maxDegree + 1)
    using round-bound sum-list-take-le[of ceil-log (to-nat dg + 1)
      composition-fri-budgets budgets]
    by simp
  show hash-range-budget
    (sum-list (composition-fri-budgets budgets) +
      ceil-log (maxDegree + 1))
    (staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) [])
    by (rule hash-range-budget-mono[OF le exact])
  qed
  then show ?thesis by simp
qed

```

lemma *hash-collision-budget-guarded-staged-composition-fri-program:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows

```

  hash-collision-budget
    (sum-list (composition-fri-budgets budgets) +
      ceil-log (maxDegree + 1))
    (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))

```

proof –

```

  let ?B =
    sum-list (composition-fri-budgets budgets) +
      ceil-log (maxDegree + 1)

```

```

have hash-collision-budget ( $0 + ?B$ )
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
proof (rule hash-collision-budget-bind-on-outcomes
  [where n=0 and n'=?B])
show hash-range-budget 0
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)))
  unfolding hash-range-budget-def assert-def
  by (auto simp: throw-no-outcome)
show hash-collision-budget 0
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)))
  unfolding hash-collision-budget-def hash-new-collision-event-def
    hash-map-new-output-collision-def hash-collision-budget-value-def
    assert-def
  by (simp add: wp-event-def wpsimps)
next
fix s x t
assume out:
  Some (x, t) ∈
    set-dist
    (execute
      (assert
        (ceil-log (to-nat dg + 1) ≤
          ceil-log (maxDegree + 1)))) s)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using out unfolding assert-def
  by (cases ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have len:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
  using round-bound wf unfolding staged-budget-wellformed-def by simp
have exact:
  hash-range-budget
    (sum-list
      (take (ceil-log (to-nat dg + 1))
        (drop 0 (composition-fri-budgets budgets))) +
      ceil-log (to-nat dg + 1))
    (staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) [])
  by (rule hash-range-budget-staged-composition-fri-program
    [OF controlled len])
have le:
  sum-list
    (take (ceil-log (to-nat dg + 1))
      (drop 0 (composition-fri-budgets budgets))) +
    ceil-log (to-nat dg + 1) ≤

```

```

    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
using round-bound sum-list-take-le[of ceil-log (to-nat dg + 1)
    composition-fri-budgets budgets]
by simp
show hash-range-budget
  (sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1))
  (staged-composition-fri-program A dg 0
    (ceil-log (to-nat dg + 1)) [])
by (rule hash-range-budget-mono[OF le exact])
next
fix s x t
assume out:
  Some (x, t) ∈
    set-dist
    (execute
      (assert
        (ceil-log (to-nat dg + 1) ≤
          ceil-log (maxDegree + 1))) s)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
using out unfolding assert-def
by (cases ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
  (auto simp: throw-no-outcome)
have len:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
using round-bound wf unfolding staged-budget-wellformed-def by simp
have exact:
  hash-collision-budget
  (sum-list
    (take (ceil-log (to-nat dg + 1))
      (drop 0 (composition-fri-budgets budgets))) +
    ceil-log (to-nat dg + 1))
  (staged-composition-fri-program A dg 0
    (ceil-log (to-nat dg + 1)) [])
by (rule hash-collision-budget-staged-composition-fri-program
  [OF controlled len])
have le:
  sum-list
    (take (ceil-log (to-nat dg + 1))
      (drop 0 (composition-fri-budgets budgets))) +
    ceil-log (to-nat dg + 1) ≤
    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
using round-bound sum-list-take-le[of ceil-log (to-nat dg + 1)
  composition-fri-budgets budgets]
by simp

```

```

show hash-collision-budget
  (sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1))
  (staged-composition-fri-program A dg 0
    (ceil-log (to-nat dg + 1)) [])
  by (rule hash-collision-budget-mono[OF le exact])
qed
then show ?thesis by simp
qed

lemma hash-target-program-guarded-staged-composition-fri-program:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  hash-target-program B
  (sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1))
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) ≫=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
proof -
let ?B =
  sum-list (composition-fri-budgets budgets) +
  ceil-log (maxDegree + 1)
have hash-target-program B (0 + ?B)
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) ≫=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
proof (rule hash-target-program-bind-on-outcomes
  [where n=0 and n'=?B])
show hash-target-program B 0
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)))
  by (rule hash-target-program-assert)
next
fix s x t
assume out:
  Some (x, t) ∈
  set-dist
  (execute
    (assert
      (ceil-log (to-nat dg + 1) ≤
        ceil-log (maxDegree + 1))) s)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
using out unfolding assert-def
by (cases ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
  (auto simp: throw-no-outcome)
have len:
  0 + ceil-log (to-nat dg + 1) ≤

```

```

    length (composition-fri-budgets budgets)
using round-bound wf unfolding staged-budget-wellformed-def by simp
have exact:
  hash-target-program B
    (sum-list
      (take (ceil-log (to-nat dg + 1))
        (drop 0 (composition-fri-budgets budgets))) +
      ceil-log (to-nat dg + 1))
    (staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) [])
by (rule hash-target-program-staged-composition-fri-program
  [OF controlled len])
have le:
  sum-list
    (take (ceil-log (to-nat dg + 1))
      (drop 0 (composition-fri-budgets budgets))) +
    ceil-log (to-nat dg + 1) ≤
    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
using round-bound sum-list-take-le[of ceil-log (to-nat dg + 1)
  composition-fri-budgets budgets]
by simp
show hash-target-program B
  (sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1))
  (staged-composition-fri-program A dg 0
    (ceil-log (to-nat dg + 1)) [])
by (rule hash-target-program-mono[OF le exact])
qed
then show ?thesis by simp
qed

lemma hash-target-program-staged-alpha-prefix-program:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  hash-target-program B
    (staged-alpha-search-queries budgets 0)
    (staged-alpha-prefix-program A)
proof -
let ?trace-root = trace-root-budget budgets
let ?trace-fri =
  sum-list (trace-fri-budgets budgets) + ceil-log clength
let ?trace-final = trace-final-budget budgets
have trace-root-target:
  hash-target-program B ?trace-root (trace-root-stage A)
using controlled unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-target)
have trace-fri-target:

```



```

hash-target-program B ?trace-fri
  (staged-trace-fri-program A 0 (ceil-log clength) [])
proof –
  have len:
    0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-target-program B
      (sum-list
        (take (ceil-log clength)
          (drop 0 (trace-fri-budgets budgets))) +
        ceil-log clength)
      (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule hash-target-program-staged-trace-fri-program
      [OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have trace-final-target:
  ∧bs. hash-target-program B ?trace-final (trace-final-stage A bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-target)
have after-trace-final-record:
  ∧fr trace-roots trace-bs trace-final.
  hash-target-program B 0
    (record-staged-message trace-final ≫=
      (λ-. return (fr, trace-roots, trace-bs, trace-final)))
proof –
  fix fr trace-roots trace-bs trace-final
  have hash-target-program B (0 + 0)
    (record-staged-message trace-final ≫=
      (λ-. return (fr, trace-roots, trace-bs, trace-final)))
  by (rule hash-target-program-bind)
    (rule hash-target-program-record-staged-message,
      rule hash-target-program-return)
  then show
    hash-target-program B 0
      (record-staged-message trace-final ≫=
        (λ-. return (fr, trace-roots, trace-bs, trace-final)))
    by simp
qed
have after-trace-final:
  ∧fr trace-roots trace-bs.
  hash-target-program B (?trace-final + 0)
    (trace-final-stage A trace-bs ≫=
      (λtrace-final.
        record-staged-message trace-final ≫=
          (λ-. return (fr, trace-roots, trace-bs, trace-final))))
  by (rule hash-target-program-bind)

```

```

(rule trace-final-target, rule after-trace-final-record)
have after-trace-fri:
 $\wedge$ fr. hash-target-program B (?trace-fri + (?trace-final + 0))
(staged-trace-fri-program A 0 (ceil-log clength) []  $\gg$ 
( $\lambda$ (trace-roots, trace-bs).
trace-final-stage A trace-bs  $\gg$ 
( $\lambda$ trace-final.
record-staged-message trace-final  $\gg$ 
( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))))
proof (rule hash-target-program-bind)
fix fr
show hash-target-program B ?trace-fri
(staged-trace-fri-program A 0 (ceil-log clength) [])
by (rule trace-fri-target)
next
fix fr x
show hash-target-program B (?trace-final + 0)
(case x of (trace-roots, trace-bs)  $\Rightarrow$ 
trace-final-stage A trace-bs  $\gg$ 
( $\lambda$ trace-final.
record-staged-message trace-final  $\gg$ 
( $\lambda$ -. return (fr, trace-roots, trace-bs, trace-final))))
proof (cases x)
case (Pair trace-roots trace-bs)
have target:
hash-target-program B (?trace-final + 0)
(trace-final-stage A trace-bs  $\gg$ 
( $\lambda$ trace-final.
record-staged-message trace-final  $\gg$ 
( $\lambda$ -. return
(fr, trace-roots, trace-bs, trace-final))))
by (rule hash-target-program-bind)
(rule trace-final-target, rule after-trace-final-record)
show ?thesis
unfolding Pair using target by simp
qed
qed
have after-trace-root-record:
 $\wedge$ fr. hash-target-program B (0 + (?trace-fri + (?trace-final + 0)))
(record-staged-message fr  $\gg$ 
( $\lambda$ -. staged-trace-fri-program A 0 (ceil-log clength) []  $\gg$ 
( $\lambda$ (trace-roots, trace-bs).
trace-final-stage A trace-bs  $\gg$ 
( $\lambda$ trace-final.
record-staged-message trace-final  $\gg$ 
( $\lambda$ -. return
(fr, trace-roots, trace-bs, trace-final))))))
by (rule hash-target-program-bind)
(rule hash-target-program-record-staged-message, rule after-trace-fri)

```

```

have whole:
  hash-target-program B
    (?trace-root + (0 + (?trace-fri + (?trace-final + 0))))
    (staged-alpha-prefix-program A)
  unfolding staged-alpha-prefix-program-def
  by (rule hash-target-program-bind)
    (rule trace-root-target, rule after-trace-root-record)
show ?thesis
  using whole
  unfolding staged-alpha-search-queries-def
  by (simp add: add.assoc add.commute add.left-commute)
qed

lemma staged-alpha-prefix-target-hit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (staged-alpha-prefix-program A)
      (hash-new-output-hit-event B s) s ≤
      staged-phase-target-error B
      (staged-alpha-search-queries budgets 0)
proof –
  have budget:
    hash-target-budget B (staged-alpha-search-queries budgets 0)
      (staged-alpha-prefix-program A)
  using hash-target-program-staged-alpha-prefix-program[OF wf controlled]
  unfolding hash-target-program-def by simp
  show ?thesis
  using budget unfolding hash-target-budget-def
    staged-phase-target-error-def by blast
qed

lemma hash-target-program-staged-alpha-challenge-prefix-program:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i ≤ length spec
  shows
    hash-target-program B (staged-alpha-search-queries budgets i)
      (staged-alpha-challenge-prefix-program A i)
proof –
  let ?alpha-prefix = staged-alpha-search-queries budgets 0
  have prefix-target:
    hash-target-program B ?alpha-prefix (staged-alpha-prefix-program A)
  by (rule hash-target-program-staged-alpha-prefix-program[OF wf controlled])
  have alpha-target:
     $\bigwedge$  prefix. hash-target-program B i
      (staged-alpha-program i  $\gg$ 
        ( $\lambda$  as-prefix. return (prefix, as-prefix)))
  proof –

```

```

fix prefix
have hash-target-program B (i + 0)
  (staged-alpha-program i  $\gg$ 
    ( $\lambda$ as-prefix. return (prefix, as-prefix)))
by (rule hash-target-program-bind)
  (rule hash-target-program-staged-alpha-program,
    rule hash-target-program-return)
then show hash-target-program B i
  (staged-alpha-program i  $\gg$ 
    ( $\lambda$ as-prefix. return (prefix, as-prefix)))
by simp
qed
have whole:
  hash-target-program B (?alpha-prefix + i)
  (staged-alpha-challenge-prefix-program A i)
unfolding staged-alpha-challenge-prefix-program-def
by (rule hash-target-program-bind[OF prefix-target alpha-target])
show ?thesis
using whole i-bound
unfolding staged-alpha-search-queries-def
by simp
qed

lemma staged-alpha-prefix-target-hit-bound-i:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i ≤ length spec
shows
  wp-event (staged-alpha-challenge-prefix-program A i)
  (hash-new-output-hit-event B s) s ≤
  staged-phase-target-error B
  (staged-alpha-search-queries budgets i)
proof –
have budget:
  hash-target-budget B (staged-alpha-search-queries budgets i)
  (staged-alpha-challenge-prefix-program A i)
using hash-target-program-staged-alpha-challenge-prefix-program
  [OF wf controlled i-bound]
unfolding hash-target-program-def by simp
show ?thesis
using budget unfolding hash-target-budget-def
  staged-phase-target-error-def by blast
qed

lemma hash-target-program-staged-composition-fri-vector-program:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  hash-target-program B

```

```

    (staged-composition-fri-vector-search-queries budgets)
    (staged-composition-fri-vector-program A)
proof -
  let ?alpha-prefix = staged-alpha-search-queries budgets 0
  let ?alpha = length spec
  let ?degree = degree-budget budgets
  let ?composition =
    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
  have alpha-prefix-target:
    hash-target-program B ?alpha-prefix (staged-alpha-prefix-program A)
    by (rule hash-target-program-staged-alpha-prefix-program[OF wf controlled])
  have alpha-target:
    hash-target-program B ?alpha (staged-alpha-program (length spec))
    by (rule hash-target-program-staged-alpha-program)
  have degree-target:
     $\wedge$ as. hash-target-program B ?degree (degree-stage A as)
    using controlled unfolding staged-adversary-controlled-def
    by (blast intro: controlled-ro-program-target)
  have guarded-return:
     $\wedge$ dg. hash-target-program B (?composition + 0)
    ((assert
      (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ =
      ( $\lambda$ -. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))  $\gg$ =
      ( $\lambda$ x. case x of (-, composition-bs)  $\Rightarrow$ 
        return (dg, composition-bs)))
    by (rule hash-target-program-bind)
    (rule hash-target-program-guarded-staged-composition-fri-program
      [OF wf controlled], simp add: hash-target-program-return
      split: prod.splits)
  have after-record:
     $\wedge$ dg. hash-target-program B (0 + (?composition + 0))
    (record-staged-message dg  $\gg$ =
      ( $\lambda$ -. (assert
        (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ =
        ( $\lambda$ -. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) []))  $\gg$ =
        ( $\lambda$ x. case x of (-, composition-bs)  $\Rightarrow$ 
          return (dg, composition-bs))))
    by (rule hash-target-program-bind)
    (rule hash-target-program-record-staged-message, rule guarded-return)
  have after-degree:
     $\wedge$ as. hash-target-program B
      (?degree + (0 + (?composition + 0)))
      (degree-stage A as  $\gg$ =
        ( $\lambda$ dg. record-staged-message dg  $\gg$ =
          ( $\lambda$ -. let composition-rounds = ceil-log (to-nat dg + 1) in
            assert (composition-rounds  $\leq$  ceil-log (maxDegree + 1))  $\gg$ =

```

```

      (λ-. staged-composition-fri-program A dg 0
        composition-rounds [] >=
        (λ(-, composition-bs). return (dg, composition-bs))))))
  unfolding Let-def
proof (rule hash-target-program-bind)
  fix as
  show hash-target-program B ?degree (degree-stage A as)
    by (rule degree-target)
next
  fix as dg
  show hash-target-program B (0 + (?composition + 0))
    (record-staged-message dg >=
      (λ-. assert
        (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >=
        (λ-. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [] >=
          (λ(-, composition-bs). return (dg, composition-bs))))))
    using after-record[of dg] by (simp add: sm-bind-assoc)
qed
have after-alpha:
  ∧x. hash-target-program B
    (?alpha + (?degree + (0 + (?composition + 0))))
    (staged-alpha-program (length spec) >=
      (λas. degree-stage A as >=
        (λdg. record-staged-message dg >=
          (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
            assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >=
            (λ-. staged-composition-fri-program A dg 0
              composition-rounds [] >=
              (λ(-, composition-bs). return (dg, composition-bs))))))
        proof (rule hash-target-program-bind)
          fix x
          show hash-target-program B ?alpha
            (staged-alpha-program (length spec))
            by (rule alpha-target)
        next
          fix x as
          show hash-target-program B (?degree + (0 + (?composition + 0)))
            (degree-stage A as >=
              (λdg. record-staged-message dg >=
                (λ-. let composition-rounds = ceil-log (to-nat dg + 1)
                  in assert
                    (composition-rounds ≤ ceil-log (maxDegree + 1)) >=
                    (λ-. staged-composition-fri-program A dg 0
                      composition-rounds [] >=
                      (λ(-, composition-bs). return (dg, composition-bs))))))
                by (rule after-degree)
        qed
      have whole:

```

```

hash-target-program B
  (?alpha-prefix +
    (?alpha + (?degree + (0 + (?composition + 0)))))
  (staged-composition-fri-vector-program A)
unfolding staged-composition-fri-vector-program-def
proof (rule hash-target-program-bind)
  show hash-target-program B ?alpha-prefix
    (staged-alpha-prefix-program A)
    by (rule alpha-prefix-target)
next
fix x
show hash-target-program B
  (?alpha + (?degree + (0 + (?composition + 0)))))
  (case x of
    (a, b, trace-bs, c) =>
      staged-alpha-program (length spec) >>=
      (λas. degree-stage A as >>=
        (λdg. record-staged-message dg >>=
          (λ-. let composition-rounds =
            ceil-log (to-nat dg + 1)
            in assert
              (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
              (λ-. staged-composition-fri-program A dg 0
                composition-rounds [] >>=
                (λ(r, composition-bs).
                  return (dg, composition-bs)))))))
    using after-alpha by (cases x) simp
qed
show ?thesis
  using whole
  unfolding staged-composition-fri-vector-search-queries-def
  by (simp add: add.assoc add.commute add.left-commute)
qed

lemma staged-composition-fri-vector-target-hit-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event (staged-composition-fri-vector-program A)
    (hash-new-output-hit-event B s) s ≤
    staged-phase-target-error B
    (staged-composition-fri-vector-search-queries budgets)
proof -
have budget:
  hash-target-budget B
    (staged-composition-fri-vector-search-queries budgets)
    (staged-composition-fri-vector-program A)
using hash-target-program-staged-composition-fri-vector-program
  [OF wf controlled]

```

```

    unfolding hash-target-program-def by simp
  show ?thesis
    using budget unfolding hash-target-budget-def
      staged-phase-target-error-def by blast
qed

lemma hash-target-program-staged-trace-fri-challenge-prefix-program:
  assumes controlled: staged-adversary-controlled budgets A
    and i-bound:  $i < \text{length } (\text{trace-fri-budgets } \text{budgets})$ 
  shows
    hash-target-program B
      (staged-trace-fri-search-queries budgets i)
      (staged-trace-fri-challenge-prefix-program A i)
  proof -
    let ?trace-root = trace-root-budget budgets
    let ?completed = sum-list (take i (trace-fri-budgets budgets)) + i
    let ?current = trace-fri-budgets budgets ! i
    have trace-root-target:
      hash-target-program B ?trace-root (trace-root-stage A)
      using controlled unfolding staged-adversary-controlled-def
      by (blast intro: controlled-ro-program-target)
    have completed-target:
      hash-target-program B ?completed
      (staged-trace-fri-program A 0 i [])
    proof -
      have len:  $0 + i \leq \text{length } (\text{trace-fri-budgets } \text{budgets})$ 
      using i-bound by simp
      have exact:
        hash-target-program B
          (sum-list (take i (drop 0 (trace-fri-budgets budgets))) + i)
          (staged-trace-fri-program A 0 i [])
      by (rule hash-target-program-staged-trace-fri-program
        [OF controlled len])
      then show ?thesis by simp
    qed
    have current-target:
       $\bigwedge \text{trace-bs. hash-target-program B ?current}$ 
      (trace-fri-root-stage A i trace-bs)
      using controlled i-bound unfolding staged-adversary-controlled-def
      by (blast intro: controlled-ro-program-target)
    have after-current-record:
       $\bigwedge \text{fr trace-bs root. hash-target-program B 0}$ 
      (record-staged-message root  $\gg=$ 
        ( $\lambda \text{-. return (fr, trace-bs, root)}$ )))
    proof -
      fix fr trace-bs root
      have hash-target-program B (0 + 0)
        (record-staged-message root  $\gg=$ 
          ( $\lambda \text{-. return (fr, trace-bs, root)}$ )))

```



```

    by (rule hash-target-program-bind)
      (rule hash-target-program-record-staged-message,
        rule hash-target-program-return)
  then show hash-target-program B 0
    (record-staged-message root  $\gg$ 
      ( $\lambda$ -. return (fr, trace-bs, root)))
  by simp
qed
have after-current:
 $\bigwedge$ fr trace-bs. hash-target-program B (?current + 0)
  (trace-fri-root-stage A i trace-bs  $\gg$ 
    ( $\lambda$ root. record-staged-message root  $\gg$ 
      ( $\lambda$ -. return (fr, trace-bs, root))))
  by (rule hash-target-program-bind)
    (rule current-target, rule after-current-record)
have after-completed:
 $\bigwedge$ fr. hash-target-program B (?completed + (?current + 0))
  (staged-trace-fri-program A 0 i []  $\gg$ 
    ( $\lambda$ (-, trace-bs).
      trace-fri-root-stage A i trace-bs  $\gg$ 
        ( $\lambda$ root. record-staged-message root  $\gg$ 
          ( $\lambda$ -. return (fr, trace-bs, root))))))
proof (rule hash-target-program-bind)
  fix fr
  show hash-target-program B ?completed
    (staged-trace-fri-program A 0 i [])
  by (rule completed-target)
next
  fix fr x
  show hash-target-program B (?current + 0)
    (case x of (-, trace-bs)  $\Rightarrow$ 
      trace-fri-root-stage A i trace-bs  $\gg$ 
        ( $\lambda$ root. record-staged-message root  $\gg$ 
          ( $\lambda$ -. return (fr, trace-bs, root))))
  proof (cases x)
    case (Pair roots trace-bs)
    have target:
      hash-target-program B (?current + 0)
        (trace-fri-root-stage A i trace-bs  $\gg$ 
          ( $\lambda$ root. record-staged-message root  $\gg$ 
            ( $\lambda$ -. return (fr, trace-bs, root))))
    by (rule after-current)
    show ?thesis
      unfolding Pair using target by simp
  qed
qed
have after-trace-root-record:
 $\bigwedge$ fr. hash-target-program B (0 + (?completed + (?current + 0)))
  (record-staged-message fr  $\gg$ 

```

```

    (λ-. staged-trace-fri-program A 0 i [] >>=
      (λ(-, trace-bs).
        trace-fri-root-stage A i trace-bs >>=
          (λroot. record-staged-message root >>=
            (λ-. return (fr, trace-bs, root))))))
  by (rule hash-target-program-bind)
    (rule hash-target-program-record-staged-message, rule after-completed)
have whole:
  hash-target-program B
    (?trace-root + (0 + (?completed + (?current + 0))))
    (staged-trace-fri-challenge-prefix-program A i)
  unfolding staged-trace-fri-challenge-prefix-program-def
  by (rule hash-target-program-bind)
    (rule trace-root-target, rule after-trace-root-record)
have sum-eq:
  ?current + sum-list (take i (trace-fri-budgets budgets)) =
    sum-list (take (Suc i) (trace-fri-budgets budgets))
proof -
  have take (Suc i) (trace-fri-budgets budgets) =
    take i (trace-fri-budgets budgets) @
    [trace-fri-budgets budgets ! i]
  by (rule take-Suc-conv-app-nth[OF i-bound])
  then show ?thesis by simp
qed
show ?thesis
  using whole
  unfolding staged-trace-fri-search-queries-def
  by (simp add: sum-eq add.assoc add.commute add.left-commute)
qed

lemma staged-trace-fri-prefix-target-hit-bound:
  assumes controlled: staged-adversary-controlled budgets A
  and i-bound: i < length (trace-fri-budgets budgets)
  shows
    wp-event (staged-trace-fri-challenge-prefix-program A i)
      (hash-new-output-hit-event B s) s ≤
      staged-phase-target-error B
      (staged-trace-fri-search-queries budgets i)
proof -
  have budget:
    hash-target-budget B (staged-trace-fri-search-queries budgets i)
      (staged-trace-fri-challenge-prefix-program A i)
  using hash-target-program-staged-trace-fri-challenge-prefix-program
    [OF controlled i-bound]
  unfolding hash-target-program-def by simp
  show ?thesis
    using budget unfolding hash-target-budget-def
      staged-phase-target-error-def by blast
qed

```

lemma *hash-target-program-staged-composition-fri-challenge-prefix-program:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < ceil-log (maxDegree + 1)*
shows
hash-target-program B
(staged-composition-fri-search-queries budgets i)
(staged-composition-fri-challenge-prefix-program A i)
proof –
let *?alpha-prefix = staged-alpha-search-queries budgets 0*
let *?alpha = length spec*
let *?degree = degree-budget budgets*
let *?completed =*
sum-list (take i (composition-fri-budgets budgets)) + i
let *?current = composition-fri-budgets budgets ! i*
have *comp-i-bound: i < length (composition-fri-budgets budgets)*
using *wf i-bound unfolding staged-budget-wellformed-def by simp*
have *alpha-prefix-target:*
hash-target-program B ?alpha-prefix (staged-alpha-prefix-program A)
by *(rule hash-target-program-staged-alpha-prefix-program[OF wf controlled])*
have *alpha-target:*
hash-target-program B ?alpha (staged-alpha-program (length spec))
by *(rule hash-target-program-staged-alpha-program)*
have *degree-target:*
 $\bigwedge as.$ *hash-target-program B ?degree (degree-stage A as)*
using *controlled unfolding staged-adversary-controlled-def*
by *(blast intro: controlled-ro-program-target)*
have *completed-target:*
 $\bigwedge dg.$ *hash-target-program B ?completed*
(staged-composition-fri-program A dg 0 i [])
proof –
fix *dg*
have *len: 0 + i ≤ length (composition-fri-budgets budgets)*
using *comp-i-bound by simp*
have *exact:*
hash-target-program B
(sum-list (take i (drop 0 (composition-fri-budgets budgets))) + i)
(staged-composition-fri-program A dg 0 i [])
by *(rule hash-target-program-staged-composition-fri-program*
[OF controlled len])
then show
hash-target-program B ?completed
(staged-composition-fri-program A dg 0 i [])
by *simp*
qed
have *current-target:*
 $\bigwedge dg$ *composition-bs. hash-target-program B ?current*
(composition-fri-root-stage A dg i composition-bs)

```

using controlled comp-i-bound
unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-target)
have after-current-record:
   $\bigwedge dg \text{ composition-bs } \text{root}. \text{hash-target-program } B \ 0$ 
  (record-staged-message root  $\gg=$ 
    ( $\lambda\text{-}. \text{return } (dg, \text{composition-bs}, \text{root}))$ )
proof –
  fix dg composition-bs root
  have hash-target-program B (0 + 0)
    (record-staged-message root  $\gg=$ 
      ( $\lambda\text{-}. \text{return } (dg, \text{composition-bs}, \text{root}))$ )
  by (rule hash-target-program-bind)
    (rule hash-target-program-record-staged-message,
      rule hash-target-program-return)
  then show
    hash-target-program B 0
    (record-staged-message root  $\gg=$ 
      ( $\lambda\text{-}. \text{return } (dg, \text{composition-bs}, \text{root}))$ )
  by simp
qed
have after-current:
   $\bigwedge dg \text{ composition-bs}. \text{hash-target-program } B \ (?current + 0)$ 
  (composition-fri-root-stage A dg i composition-bs  $\gg=$ 
    ( $\lambda\text{root}. \text{record-staged-message } \text{root} \gg=$ 
      ( $\lambda\text{-}. \text{return } (dg, \text{composition-bs}, \text{root}))$ )))
  by (rule hash-target-program-bind)
    (rule current-target, rule after-current-record)
have after-completed:
   $\bigwedge dg. \text{hash-target-program } B \ (?completed + (?current + 0))$ 
  (staged-composition-fri-program A dg 0 i []  $\gg=$ 
    ( $\lambda(-, \text{composition-bs}).$ 
      composition-fri-root-stage A dg i composition-bs  $\gg=$ 
      ( $\lambda\text{root}. \text{record-staged-message } \text{root} \gg=$ 
        ( $\lambda\text{-}. \text{return } (dg, \text{composition-bs}, \text{root}))$ )))
proof (rule hash-target-program-bind)
  fix dg
  show hash-target-program B ?completed
    (staged-composition-fri-program A dg 0 i [])
  by (rule completed-target)
next
  fix dg x
  show hash-target-program B (?current + 0)
    (case x of  $(-, \text{composition-bs}) \Rightarrow$ 
      composition-fri-root-stage A dg i composition-bs  $\gg=$ 
      ( $\lambda\text{root}. \text{record-staged-message } \text{root} \gg=$ 
        ( $\lambda\text{-}. \text{return } (dg, \text{composition-bs}, \text{root}))$ )))
  proof (cases x)
    case (Pair roots composition-bs)

```

```

have target:
  hash-target-program B (?current + 0)
  (composition-fri-root-stage A dg i composition-bs >>=
    (λroot. record-staged-message root >>=
      (λ-. return (dg, composition-bs, root))))
  by (rule after-current)
show ?thesis
  unfolding Pair using target by simp
qed
qed
have after-second-assert:
  ∧dg. hash-target-program B (0 + (?completed + (?current + 0)))
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0 i [] >>=
      (λ(-, composition-bs).
        composition-fri-root-stage A dg i composition-bs >>=
          (λroot. record-staged-message root >>=
            (λ-. return (dg, composition-bs, root)))))))
  by (rule hash-target-program-bind)
  (rule hash-target-program-assert, rule after-completed)
have after-first-assert:
  ∧dg. hash-target-program B
  (0 + (0 + (?completed + (?current + 0))))
  (assert (Suc i ≤ ceil-log (to-nat dg + 1)) >>=
    (λ-. assert
      (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
        (λ-. staged-composition-fri-program A dg 0 i [] >>=
          (λ(-, composition-bs).
            composition-fri-root-stage A dg i composition-bs >>=
              (λroot. record-staged-message root >>=
                (λ-. return (dg, composition-bs, root))))))))
  by (rule hash-target-program-bind)
  (rule hash-target-program-assert, rule after-second-assert)
have after-degree-record:
  ∧dg. hash-target-program B
  (0 + (0 + (0 + (?completed + (?current + 0)))))
  (record-staged-message dg >>=
    (λ-. assert (Suc i ≤ ceil-log (to-nat dg + 1)) >>=
      (λ-. assert
        (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
          (λ-. staged-composition-fri-program A dg 0 i [] >>=
            (λ(-, composition-bs).
              composition-fri-root-stage A dg i composition-bs >>=
                (λroot. record-staged-message root >>=
                  (λ-. return (dg, composition-bs, root))))))))
  by (rule hash-target-program-bind)
  (rule hash-target-program-record-staged-message, rule after-first-assert)
have after-degree:
  ∧as. hash-target-program B

```

```

(?degree + (0 + (0 + (0 + (?completed + (?current + 0))))))
(degree-stage A as >>=
  (λdg. record-staged-message dg >>=
    (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
      assert (Suc i ≤ composition-rounds) >>=
      (λ-. assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
        (λ-. staged-composition-fri-program A dg 0 i [] >>=
          (λ(-, composition-bs).
            composition-fri-root-stage A dg i composition-bs >>=
            (λroot. record-staged-message root >>=
              (λ-. return (dg, composition-bs, root))))))))))
unfolding Let-def
by (rule hash-target-program-bind)
  (rule degree-target, rule after-degree-record)
have after-alpha:
  ∧x. hash-target-program B
  (?alpha + (?degree + (0 + (0 + (0 + (?completed + (?current + 0)))))))
  (staged-alpha-program (length spec) >>=
    (λas. degree-stage A as >>=
      (λdg. record-staged-message dg >>=
        (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert (Suc i ≤ composition-rounds) >>=
          (λ-. assert
            (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0 i [] >>=
              (λ(-, composition-bs).
                composition-fri-root-stage A dg i composition-bs >>=
                (λroot. record-staged-message root >>=
                  (λ-. return
                    (dg, composition-bs, root))))))))))
    (dg, composition-bs, root))))))
by (rule hash-target-program-bind)
  (rule alpha-target, rule after-degree)
have after-alpha-prefix:
  hash-target-program B
  (?alpha-prefix +
    (?alpha + (?degree +
      (0 + (0 + (0 + (?completed + (?current + 0)))))))
    (staged-composition-fri-challenge-prefix-program A i)
unfolding staged-composition-fri-challenge-prefix-program-def
proof (rule hash-target-program-bind)
show hash-target-program B ?alpha-prefix
  (staged-alpha-prefix-program A)
by (rule alpha-prefix-target)
next
fix x
show hash-target-program B
  (?alpha + (?degree +
    (0 + (0 + (0 + (?completed + (?current + 0)))))))
  (case x of (-, -, trace-bs, -) ⇒

```

```

    staged-alpha-program (length spec) >>=
      (λas. degree-stage A as >>=
        (λdg. record-staged-message dg >>=
          (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
            assert (Suc i ≤ composition-rounds) >>=
              (λ-. assert
                (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
                  (λ-. staged-composition-fri-program A dg 0 i [] >>=
                    (λ(-, composition-bs).
                      composition-fri-root-stage A dg i composition-bs >>=
                        (λroot. record-staged-message root >>=
                          (λ-. return
                            (dg, composition-bs, root))))))))))
  proof (cases x)
  case (fields a b c d)
  have target:
    hash-target-program B
      (?alpha + (?degree +
        (0 + (0 + (0 + (?completed + (?current + 0)))))))
    (staged-alpha-program (length spec) >>=
      (λas. degree-stage A as >>=
        (λdg. record-staged-message dg >>=
          (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
            assert (Suc i ≤ composition-rounds) >>=
              (λ-. assert
                (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
                  (λ-. staged-composition-fri-program A dg 0 i [] >>=
                    (λ(-, composition-bs).
                      composition-fri-root-stage A dg i composition-bs >>=
                        (λroot. record-staged-message root >>=
                          (λ-. return
                            (dg, composition-bs, root))))))))))
      by (rule after-alpha)
    show ?thesis
      unfolding fields
      using target
      by (simp add: add.assoc add.commute add.left-commute)
  qed
qed
have sum-eq:
  ?current + sum-list (take i (composition-fri-budgets budgets)) =
    sum-list (take (Suc i) (composition-fri-budgets budgets))
proof -
  have take (Suc i) (composition-fri-budgets budgets) =
    take i (composition-fri-budgets budgets) @
    [composition-fri-budgets budgets ! i]
  by (rule take-Suc-conv-app-nth[OF comp-i-bound])
  then show ?thesis by simp
qed

```

```

show ?thesis
  using after-alpha-prefix
  unfolding staged-composition-fri-search-queries-def
    staged-alpha-search-queries-def
  by (simp add: sum-eq add.assoc add.commute add.left-commute)
qed

lemma staged-composition-fri-prefix-target-hit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound:  $i < \text{ceil-log } (\text{maxDegree} + 1)$ 
  shows
    wp-event (staged-composition-fri-challenge-prefix-program A i)
      (hash-new-output-hit-event B s)  $s \leq$ 
      staged-phase-target-error B
      (staged-composition-fri-search-queries budgets i)
proof –
  have budget:
    hash-target-budget B (staged-composition-fri-search-queries budgets i)
      (staged-composition-fri-challenge-prefix-program A i)
  using
    hash-target-program-staged-composition-fri-challenge-prefix-program
      [OF wf controlled i-bound]
  unfolding hash-target-program-def by simp
  show ?thesis
    using budget unfolding hash-target-budget-def
      staged-phase-target-error-def by blast
qed

lemma hash-target-program-staged-query-challenge-prefix-program:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound:  $i \leq \text{rounds}$ 
  shows
    hash-target-program B
      (staged-query-search-queries budgets i)
      (staged-query-challenge-prefix-program A i)
proof –
  let ?alpha-prefix = staged-alpha-search-queries budgets 0
  let ?alpha = length spec
  let ?degree = degree-budget budgets
  let ?composition-fri =
    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
  let ?composition-final = composition-final-budget budgets
  let ?query = sum-list (take i (query-opening-budgets budgets)) + i
  have alpha-prefix-target:
    hash-target-program B ?alpha-prefix (staged-alpha-prefix-program A)
  by (rule hash-target-program-staged-alpha-prefix-program[OF wf controlled])

```



```

have alpha-target:
  hash-target-program B ?alpha (staged-alpha-program (length spec))
  by (rule hash-target-program-staged-alpha-program)
have degree-target:
   $\wedge$  as. hash-target-program B ?degree (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-target)
have composition-fri-target:
   $\wedge$  dg. hash-target-program B ?composition-fri
    (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))  $\gg$ 
      ( $\lambda$ -. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))
  by (rule hash-target-program-guarded-staged-composition-fri-program
    [OF wf controlled])
have composition-final-target:
   $\wedge$  dg bs. hash-target-program B ?composition-final
    (composition-final-stage A dg bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-target)
have query-target:
  hash-target-program B ?query (staged-query-program A 0 i)
proof –
  have len: 0 + i ≤ length (query-opening-budgets budgets)
    using wf i-bound unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-target-program B
      (sum-list (take i (drop 0 (query-opening-budgets budgets))) + i
        (staged-query-program A 0 i))
    by (rule hash-target-program-staged-query-program
      [OF controlled len])
  then show ?thesis by simp
qed
have after-query:
   $\wedge$  dg composition-final. hash-target-program B (?query + 0)
    (staged-query-program A 0 i  $\gg$ 
      ( $\lambda$  query-chunks.
        return (dg, composition-final, query-chunks))))
  by (rule hash-target-program-bind
    (rule query-target, rule hash-target-program-return))
have after-composition-final-record:
   $\wedge$  dg composition-final. hash-target-program B (0 + (?query + 0))
    (record-staged-message composition-final  $\gg$ 
      ( $\lambda$ -. staged-query-program A 0 i  $\gg$ 
        ( $\lambda$  query-chunks.
          return (dg, composition-final, query-chunks))))
  by (rule hash-target-program-bind
    (rule hash-target-program-record-staged-message, rule after-query))
have after-composition-final:
   $\wedge$  dg composition-bs. hash-target-program B

```

```

    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 i >>=
            (λquery-chunks.
              return (dg, composition-final, query-chunks))))))
  by (rule hash-target-program-bind)
  (rule composition-final-target, rule after-composition-final-record)
have after-composition-fri:
  ∧ dg. hash-target-program B
  (?composition-fri + (?composition-final + (0 + (?query + 0))))
  ((assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) [])) >>=
    (λ(-, composition-bs).
      composition-final-stage A dg composition-bs >>=
        (λcomposition-final.
          record-staged-message composition-final >>=
            (λ-. staged-query-program A 0 i >>=
              (λquery-chunks.
                return (dg, composition-final, query-chunks)))))))
proof (rule hash-target-program-bind)
  fix dg
  show hash-target-program B ?composition-fri
    (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))
    by (rule composition-fri-target)
next
  fix dg x
  show hash-target-program B
    (?composition-final + (0 + (?query + 0)))
    (case x of (-, composition-bs) ⇒
      composition-final-stage A dg composition-bs >>=
        (λcomposition-final.
          record-staged-message composition-final >>=
            (λ-. staged-query-program A 0 i >>=
              (λquery-chunks.
                return (dg, composition-final, query-chunks))))))
proof (cases x)
  case (Pair composition-roots composition-bs)
  have target:
    hash-target-program B
    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 i >>=
            (λquery-chunks.
              return (dg, composition-final, query-chunks))))))

```

```

      (λquery-chunks.
        return (dg, composition-final, query-chunks))))))
  by (rule after-composition-final)
show ?thesis
  unfolding Pair using target by simp
qed
qed
have after-degree-record:
  ∧dg. hash-target-program B
  (0 + (?composition-fri + (?composition-final + (0 + (?query + 0)))))
  (record-staged-message dg ≫=
    (λ-. (assert
      (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) ≫=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [])) ≫=
      (λ(-, composition-bs).
        composition-final-stage A dg composition-bs ≫=
        (λcomposition-final.
          record-staged-message composition-final ≫=
          (λ-. staged-query-program A 0 i ≫=
            (λquery-chunks.
              return (dg, composition-final, query-chunks))))))))))
  by (rule hash-target-program-bind)
  (rule hash-target-program-record-staged-message,
    rule after-composition-fri)
have after-degree-record-expanded:
  ∧dg. hash-target-program B
  (0 + (?composition-fri + (?composition-final + (0 + (?query + 0)))))
  (record-staged-message dg ≫=
    (λ-. assert
      (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) ≫=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [] ≫=
        (λ(-, composition-bs).
          composition-final-stage A dg composition-bs ≫=
          (λcomposition-final.
            record-staged-message composition-final ≫=
            (λ-. staged-query-program A 0 i ≫=
              (λquery-chunks.
                return (dg, composition-final, query-chunks))))))))))
  using after-degree-record
  by (simp add: sm-bind-assoc split: prod.splits)
have after-degree:
  ∧as. hash-target-program B
  (?degree +
    (0 + (?composition-fri + (?composition-final + (0 + (?query + 0)))))
  (degree-stage A as ≫=
    (λdg. record-staged-message dg ≫=
      (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in

```

```

    assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      composition-rounds [] >>=
      (λ(-, composition-bs).
        composition-final-stage A dg composition-bs >>=
        (λcomposition-final.
          record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 i >>=
            (λquery-chunks.
              return
              (dg, composition-final, query-chunks))))))))))
unfolding Let-def
by (rule hash-target-program-bind)
    (rule degree-target, rule after-degree-record-expanded)
have after-alpha:
  ∧x. hash-target-program B
    (?alpha +
      (?degree +
        (0 + (?composition-fri +
          (?composition-final + (0 + (?query + 0)))))))
    (staged-alpha-program (length spec) >>=
      (λas. degree-stage A as >>=
        (λdg. record-staged-message dg >>=
          (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
            assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0
              composition-rounds [] >>=
              (λ(-, composition-bs).
                composition-final-stage A dg composition-bs >>=
                (λcomposition-final.
                  record-staged-message composition-final >>=
                  (λ-. staged-query-program A 0 i >>=
                    (λquery-chunks.
                      return
                      (dg, composition-final, query-chunks))))))))))
        by (rule hash-target-program-bind)
          (rule alpha-target, rule after-degree)
have after-alpha-prefix:
  hash-target-program B
    (?alpha-prefix +
      (?alpha +
        (?degree +
          (0 + (?composition-fri +
            (?composition-final + (0 + (?query + 0)))))))
    (staged-query-challenge-prefix-program A i)
unfolding staged-query-challenge-prefix-program-def
proof (rule hash-target-program-bind)
show hash-target-program B ?alpha-prefix
  (staged-alpha-prefix-program A)

```

```

    by (rule alpha-prefix-target)
next
fix x
show hash-target-program B
  (?alpha +
    (?degree +
      (0 + (?composition-fri +
        (?composition-final + (0 + (?query + 0)))))))
  (case x of (-, -, trace-bs, -) =>
    staged-alpha-program (length spec) >=>
    (λas. degree-stage A as >=>
      (λdg. record-staged-message dg >=>
        (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >=>
            (λ-. staged-composition-fri-program A dg 0
              composition-rounds [] >=>
                (λ(-, composition-bs).
                  composition-final-stage A dg composition-bs >=>
                    (λcomposition-final.
                      record-staged-message composition-final >=>
                        (λ-. staged-query-program A 0 i >=>
                          (λquery-chunks.
                            return
                              (dg, composition-final,
                                query-chunks))))))))))
proof (cases x)
case (fields a b c d)
have target:
  hash-target-program B
  (?alpha +
    (?degree +
      (0 + (?composition-fri +
        (?composition-final + (0 + (?query + 0)))))))
  (staged-alpha-program (length spec) >=>
    (λas. degree-stage A as >=>
      (λdg. record-staged-message dg >=>
        (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert
            (composition-rounds ≤ ceil-log (maxDegree + 1)) >=>
              (λ-. staged-composition-fri-program A dg 0
                composition-rounds [] >=>
                  (λ(-, composition-bs).
                    composition-final-stage A dg composition-bs >=>
                      (λcomposition-final.
                        record-staged-message composition-final >=>
                          (λ-. staged-query-program A 0 i >=>
                            (λquery-chunks.
                              return
                                (dg, composition-final,
                                  query-chunks))))))))))

```

```

      query-chunks))))))))))
    by (rule after-alpha)
  show ?thesis
    unfolding fields
    using target
    by (simp add: add.assoc add.commute add.left-commute)
qed
qed
show ?thesis
  using after-alpha-prefix
  unfolding staged-query-search-queries-def staged-alpha-search-queries-def
  by (simp add: add.assoc add.commute add.left-commute)
qed

lemma hash-range-budget-staged-query-challenge-prefix-program:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound:  $i \leq \text{rounds}$ 
  shows
    hash-range-budget
      (staged-query-search-queries budgets i)
      (staged-query-challenge-prefix-program A i)
proof -
  let ?alpha-prefix = staged-alpha-search-queries budgets 0
  let ?alpha = length spec
  let ?degree = degree-budget budgets
  let ?composition-fri =
    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
  let ?composition-final = composition-final-budget budgets
  let ?query = sum-list (take i (query-opening-budgets budgets)) + i
  have alpha-prefix:
    hash-range-budget ?alpha-prefix (staged-alpha-prefix-program A)
  by (rule hash-range-budget-staged-alpha-prefix-program
    [OF wf controlled])
  have alpha: hash-range-budget ?alpha (staged-alpha-program (length spec))
  by (rule hash-range-budget-staged-alpha-program)
  have degree:
     $\wedge as.$  hash-range-budget ?degree (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
  have composition-fri:
     $\wedge dg.$  hash-range-budget ?composition-fri
      (assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg=$ 
        ( $\lambda -.$  staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) []))
  by (rule hash-range-budget-guarded-staged-composition-fri-program
    [OF wf controlled])
  have composition-final:

```

```

 $\wedge dg$  bs. hash-range-budget ?composition-final
  (composition-final-stage A dg bs)
using controlled unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-range)
have query: hash-range-budget ?query (staged-query-program A 0 i)
proof –
  have len: 0 + i  $\leq$  length (query-opening-budgets budgets)
    using wf i-bound unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-range-budget
      (sum-list (take i (drop 0 (query-opening-budgets budgets))) + i)
      (staged-query-program A 0 i)
    by (rule hash-range-budget-staged-query-program[OF controlled len])
  then show ?thesis by simp
qed
have after-query:
 $\wedge dg$  composition-final. hash-range-budget (?query + 0)
  (staged-query-program A 0 i  $\gg$ 
    ( $\lambda$ query-chunks. return (dg, composition-final, query-chunks)))
by (rule hash-range-budget-bind)
  (rule query, rule hash-range-budget-return)
have after-composition-final-record:
 $\wedge dg$  composition-final. hash-range-budget (0 + (?query + 0))
  (record-staged-message composition-final  $\gg$ 
    ( $\lambda$ -. staged-query-program A 0 i  $\gg$ 
      ( $\lambda$ query-chunks. return
        (dg, composition-final, query-chunks))))
by (rule hash-range-budget-bind)
  (rule hash-range-budget-record-staged-message, rule after-query)
have after-composition-final:
 $\wedge dg$  bs. hash-range-budget
  (?composition-final + (0 + (?query + 0)))
  (composition-final-stage A dg bs  $\gg$ 
    ( $\lambda$ composition-final.
      record-staged-message composition-final  $\gg$ 
        ( $\lambda$ -. staged-query-program A 0 i  $\gg$ 
          ( $\lambda$ query-chunks. return
            (dg, composition-final, query-chunks))))))
by (rule hash-range-budget-bind)
  (rule composition-final, rule after-composition-final-record)
have after-composition-fri:
 $\wedge dg$ . hash-range-budget
  (?composition-fri + (?composition-final + (0 + (?query + 0))))
  ((assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
    ( $\lambda$ -. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))  $\gg$ 
    ( $\lambda$ (-, composition-bs).
      composition-final-stage A dg composition-bs  $\gg$ 
        ( $\lambda$ composition-final.

```

```

      record-staged-message composition-final  $\gg$ 
      ( $\lambda$ -. staged-query-program A 0 i  $\gg$ 
      ( $\lambda$ query-chunks. return
      (dg, composition-final, query-chunks))))))
proof (rule hash-range-budget-bind)
  fix dg
  show hash-range-budget ?composition-fri
    (assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
    ( $\lambda$ -. staged-composition-fri-program A dg 0
    (ceil-log (to-nat dg + 1)) []))
    by (rule composition-fri)
next
  fix dg x
  show hash-range-budget
    (?composition-final + (0 + (?query + 0)))
    (case x of (-, composition-bs)  $\Rightarrow$ 
    composition-final-stage A dg composition-bs  $\gg$ 
    ( $\lambda$ composition-final.
    record-staged-message composition-final  $\gg$ 
    ( $\lambda$ -. staged-query-program A 0 i  $\gg$ 
    ( $\lambda$ query-chunks. return
    (dg, composition-final, query-chunks))))))
  proof (cases x)
  case (Pair composition-roots composition-bs)
  have step:
    hash-range-budget
    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg composition-bs  $\gg$ 
    ( $\lambda$ composition-final.
    record-staged-message composition-final  $\gg$ 
    ( $\lambda$ -. staged-query-program A 0 i  $\gg$ 
    ( $\lambda$ query-chunks. return
    (dg, composition-final, query-chunks))))))
    by (rule after-composition-final)
  show ?thesis
    unfolding Pair using step by simp
  qed
qed
have after-degree-record:
   $\wedge$  dg. hash-range-budget
  (0 + (?composition-fri + (?composition-final + (0 + (?query + 0)))))
  (record-staged-message dg  $\gg$ 
  ( $\lambda$ -. (assert
  (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
  ( $\lambda$ -. staged-composition-fri-program A dg 0
  (ceil-log (to-nat dg + 1)) []))  $\gg$ 
  ( $\lambda$ (-, composition-bs).
  composition-final-stage A dg composition-bs  $\gg$ 
  ( $\lambda$ composition-final.

```



```

      record-staged-message composition-final >>=
      (λ-. staged-query-program A 0 i >>=
      (λquery-chunks. return
      (dg, composition-final, query-chunks))))))
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule after-composition-fri)
  have after-degree-record-expanded:
  ∧dg. hash-range-budget
    (0 + (?composition-fri + (?composition-final + (0 + (?query + 0)))))
    (record-staged-message dg >>=
    (λ-. assert
      (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) [] >>=
      (λ(-, composition-bs).
      composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
      record-staged-message composition-final >>=
      (λ-. staged-query-program A 0 i >>=
      (λquery-chunks. return
      (dg, composition-final, query-chunks))))))))))
  using after-degree-record
  by (simp add: sm-bind-assoc split: prod.splits)
  have after-degree:
  ∧as. hash-range-budget
    (?degree +
    (0 + (?composition-fri +
    (?composition-final + (0 + (?query + 0)))))
    (degree-stage A as >>=
    (λdg. record-staged-message dg >>=
    (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
    assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
    composition-rounds [] >>=
    (λ(-, composition-bs).
    composition-final-stage A dg composition-bs >>=
    (λcomposition-final.
    record-staged-message composition-final >>=
    (λ-. staged-query-program A 0 i >>=
    (λquery-chunks. return
    (dg, composition-final, query-chunks))))))))))
  unfolding Let-def
  by (rule hash-range-budget-bind)
    (rule degree, rule after-degree-record-expanded)
  have after-alpha:
  ∧x. hash-range-budget
    (?alpha +
    (?degree +

```

```

    (0 + (?composition-fri +
      (?composition-final + (0 + (?query + 0))))))
  (staged-alpha-program (length spec) >>=
    (λas. degree-stage A as >>=
      (λdg. record-staged-message dg >>=
        (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0
              composition-rounds [] >>=
                (λ(-, composition-bs).
                  composition-final-stage A dg composition-bs >>=
                    (λcomposition-final.
                      record-staged-message composition-final >>=
                        (λ-. staged-query-program A 0 i >>=
                          (λquery-chunks. return
                            (dg, composition-final, query-chunks))))))))))
  by (rule hash-range-budget-bind)
  (rule alpha, rule after-degree)
have after-alpha-prefix:
  hash-range-budget
  (?alpha-prefix +
    (?alpha +
      (?degree +
        (0 + (?composition-fri +
          (?composition-final + (0 + (?query + 0))))))
      (staged-query-challenge-prefix-program A i)
    )
  )
unfolding staged-query-challenge-prefix-program-def
proof (rule hash-range-budget-bind)
show hash-range-budget ?alpha-prefix
  (staged-alpha-prefix-program A)
  by (rule alpha-prefix)
next
fix x
show hash-range-budget
  (?alpha +
    (?degree +
      (0 + (?composition-fri +
        (?composition-final + (0 + (?query + 0))))))
    (case x of (-, -, trace-bs, -) ⇒
      staged-alpha-program (length spec) >>=
        (λas. degree-stage A as >>=
          (λdg. record-staged-message dg >>=
            (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
              assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
                (λ-. staged-composition-fri-program A dg 0
                  composition-rounds [] >>=
                    (λ(-, composition-bs).
                      composition-final-stage A dg composition-bs >>=
                        (λcomposition-final.

```

```

      record-staged-message composition-final >>=
      (λ-. staged-query-program A 0 i >>=
      (λquery-chunks. return
      (dg, composition-final,
      query-chunks)))))))))
proof (cases x)
  case (fields a b c d)
  have step:
    hash-range-budget
    (?alpha +
    (?degree +
    (0 + (?composition-fri +
    (?composition-final + (0 + (?query + 0)))))))
    (staged-alpha-program (length spec) >>=
    (λas. degree-stage A as >>=
    (λdg. record-staged-message dg >>=
    (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
    assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
    composition-rounds [] >>=
    (λ(-, composition-bs).
    composition-final-stage A dg composition-bs >>=
    (λcomposition-final.
    record-staged-message composition-final >>=
    (λ-. staged-query-program A 0 i >>=
    (λquery-chunks. return
    (dg, composition-final,
    query-chunks)))))))))
    by (rule after-alpha)
  show ?thesis
    unfolding fields using step
    by (simp add: add.assoc add.commute add.left-commute)
qed
qed
show ?thesis
  using after-alpha-prefix
  unfolding staged-query-search-queries-def staged-alpha-search-queries-def
  by (simp add: add.assoc add.commute add.left-commute)
qed

lemma hash-collision-budget-staged-query-challenge-prefix-program:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i ≤ rounds
  shows
    hash-collision-budget
    (staged-query-search-queries budgets i)
    (staged-query-challenge-prefix-program A i)
proof -

```

```

let ?alpha-prefix = staged-alpha-search-queries budgets 0
let ?alpha = length spec
let ?degree = degree-budget budgets
let ?composition-fri =
  sum-list (composition-fri-budgets budgets) +
  ceil-log (maxDegree + 1)
let ?composition-final = composition-final-budget budgets
let ?query = sum-list (take i (query-opening-budgets budgets)) + i
have alpha-prefix-range:
  hash-range-budget ?alpha-prefix (staged-alpha-prefix-program A)
  by (rule hash-range-budget-staged-alpha-prefix-program
      [OF wf controlled])
have alpha-prefix-coll:
  hash-collision-budget ?alpha-prefix (staged-alpha-prefix-program A)
  by (rule hash-collision-budget-staged-alpha-prefix-program
      [OF wf controlled])
have alpha-range:
  hash-range-budget ?alpha (staged-alpha-program (length spec))
  by (rule hash-range-budget-staged-alpha-program)
have alpha-coll:
  hash-collision-budget ?alpha (staged-alpha-program (length spec))
  by (rule hash-collision-budget-staged-alpha-program)
have degree-range:
   $\wedge$  as. hash-range-budget ?degree (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
have degree-coll:
   $\wedge$  as. hash-collision-budget ?degree (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-collision)
have composition-fri-range:
   $\wedge$  dg. hash-range-budget ?composition-fri
  (assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
    ( $\lambda$ -. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
  by (rule hash-range-budget-guarded-staged-composition-fri-program
      [OF wf controlled])
have composition-fri-coll:
   $\wedge$  dg. hash-collision-budget ?composition-fri
  (assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
    ( $\lambda$ -. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
  by (rule hash-collision-budget-guarded-staged-composition-fri-program
      [OF wf controlled])
have composition-final-range:
   $\wedge$  dg bs. hash-range-budget ?composition-final
  (composition-final-stage A dg bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)

```

```

have composition-final-coll:
   $\wedge dg$  bs. hash-collision-budget ?composition-final
    (composition-final-stage A dg bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-collision)
have query-range: hash-range-budget ?query (staged-query-program A 0 i)
proof –
  have len:  $0 + i \leq \text{length } (\text{query-opening-budgets budgets})$ 
    using wf i-bound unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-range-budget
      (sum-list (take i (drop 0 (query-opening-budgets budgets))) + i)
      (staged-query-program A 0 i)
    by (rule hash-range-budget-staged-query-program[OF controlled len])
  then show ?thesis by simp
qed
have query-coll: hash-collision-budget ?query (staged-query-program A 0 i)
proof –
  have len:  $0 + i \leq \text{length } (\text{query-opening-budgets budgets})$ 
    using wf i-bound unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-collision-budget
      (sum-list (take i (drop 0 (query-opening-budgets budgets))) + i)
      (staged-query-program A 0 i)
    by (rule hash-collision-budget-staged-query-program[OF controlled len])
  then show ?thesis by simp
qed
have after-query-range:
   $\wedge dg$  composition-final. hash-range-budget (?query + 0)
    (staged-query-program A 0 i  $\gg$ 
      ( $\lambda$ query-chunks. return (dg, composition-final, query-chunks)))
  by (rule hash-range-budget-bind)
    (rule query-range, rule hash-range-budget-return)
have after-query-coll:
   $\wedge dg$  composition-final. hash-collision-budget (?query + 0)
    (staged-query-program A 0 i  $\gg$ 
      ( $\lambda$ query-chunks. return (dg, composition-final, query-chunks)))
  by (rule hash-collision-budget-bind)
    (rule query-range, rule query-coll,
      rule hash-range-budget-return, rule hash-collision-budget-return)
have after-composition-final-record-range:
   $\wedge dg$  composition-final. hash-range-budget (0 + (?query + 0))
    (record-staged-message composition-final  $\gg$ 
      ( $\lambda$ -. staged-query-program A 0 i  $\gg$ 
        ( $\lambda$ query-chunks. return
          (dg, composition-final, query-chunks))))
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message, rule after-query-range)
have after-composition-final-record-coll:

```

```

 $\wedge dg$  composition-final. hash-collision-budget ( $0 + (?query + 0)$ )
  (record-staged-message composition-final  $\gg$ 
    ( $\lambda$ -. staged-query-program  $A$   $0$   $i$   $\gg$ 
      ( $\lambda$ query-chunks. return
        (dg, composition-final, query-chunks))))
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule hash-collision-budget-record-staged-message,
      rule after-query-range, rule after-query-coll)
  have after-composition-final-range:
     $\wedge dg$  bs. hash-range-budget
      (?composition-final + ( $0 + (?query + 0)$ ))
      (composition-final-stage  $A$  dg bs  $\gg$ 
        ( $\lambda$ composition-final.
          record-staged-message composition-final  $\gg$ 
            ( $\lambda$ -. staged-query-program  $A$   $0$   $i$   $\gg$ 
              ( $\lambda$ query-chunks. return
                (dg, composition-final, query-chunks))))))
      by (rule hash-range-budget-bind)
        (rule composition-final-range,
          rule after-composition-final-record-range)
  have after-composition-final-coll:
     $\wedge dg$  bs. hash-collision-budget
      (?composition-final + ( $0 + (?query + 0)$ ))
      (composition-final-stage  $A$  dg bs  $\gg$ 
        ( $\lambda$ composition-final.
          record-staged-message composition-final  $\gg$ 
            ( $\lambda$ -. staged-query-program  $A$   $0$   $i$   $\gg$ 
              ( $\lambda$ query-chunks. return
                (dg, composition-final, query-chunks))))))
      by (rule hash-collision-budget-bind)
        (rule composition-final-range, rule composition-final-coll,
          rule after-composition-final-record-range,
          rule after-composition-final-record-coll)
  have after-composition-fri-range:
     $\wedge dg$ . hash-range-budget
      (?composition-fri + (?composition-final + ( $0 + (?query + 0)$ )))
      ((assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
        ( $\lambda$ -. staged-composition-fri-program  $A$  dg  $0$ 
          (ceil-log (to-nat dg + 1)) []))  $\gg$ 
        ( $\lambda$ (-, composition-bs).
          composition-final-stage  $A$  dg composition-bs  $\gg$ 
            ( $\lambda$ composition-final.
              record-staged-message composition-final  $\gg$ 
                ( $\lambda$ -. staged-query-program  $A$   $0$   $i$   $\gg$ 
                  ( $\lambda$ query-chunks. return
                    (dg, composition-final, query-chunks))))))
      proof (rule hash-range-budget-bind)
        fix dg

```

```

show hash-range-budget ?composition-fri
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) ≫=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
  by (rule composition-fri-range)
next
fix dg x
show hash-range-budget
  (?composition-final + (0 + (?query + 0)))
  (case x of (-, composition-bs) ⇒
    composition-final-stage A dg composition-bs ≫=
      (λcomposition-final.
        record-staged-message composition-final ≫=
          (λ-. staged-query-program A 0 i ≫=
            (λquery-chunks. return
              (dg, composition-final, query-chunks)))))
proof (cases x)
case (Pair composition-roots composition-bs)
have step:
  hash-range-budget
  (?composition-final + (0 + (?query + 0)))
  (composition-final-stage A dg composition-bs ≫=
    (λcomposition-final.
      record-staged-message composition-final ≫=
        (λ-. staged-query-program A 0 i ≫=
          (λquery-chunks. return
            (dg, composition-final, query-chunks)))))
  by (rule after-composition-final-range)
show ?thesis
  unfolding Pair using step by simp
qed
qed
have after-composition-fri-coll:
  ∧ dg. hash-collision-budget
  (?composition-fri + (?composition-final + (0 + (?query + 0))))
  ((assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) ≫=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) [])) ≫=
    (λ(-, composition-bs).
      composition-final-stage A dg composition-bs ≫=
        (λcomposition-final.
          record-staged-message composition-final ≫=
            (λ-. staged-query-program A 0 i ≫=
              (λquery-chunks. return
                (dg, composition-final, query-chunks)))))
  proof (rule hash-collision-budget-bind)
  fix dg
  show hash-range-budget ?composition-fri
    (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) ≫=

```

```

    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
  by (rule composition-fri-range)
show hash-collision-budget ?composition-fri
  (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
  by (rule composition-fri-coll)
next
fix dg x
show hash-range-budget
  (?composition-final + (0 + (?query + 0)))
  (case x of (-, composition-bs) ⇒
    composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 i >>=
            (λquery-chunks. return
              (dg, composition-final, query-chunks)))))
proof (cases x)
case (Pair composition-roots composition-bs)
have step:
  hash-range-budget
    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 i >>=
            (λquery-chunks. return
              (dg, composition-final, query-chunks)))))
  by (rule after-composition-final-range)
show ?thesis
  unfolding Pair using step by simp
qed
show hash-collision-budget
  (?composition-final + (0 + (?query + 0)))
  (case x of (-, composition-bs) ⇒
    composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 i >>=
            (λquery-chunks. return
              (dg, composition-final, query-chunks)))))
proof (cases x)
case (Pair composition-roots composition-bs)
have step:
  hash-collision-budget
    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg composition-bs >>=

```



```

      (λcomposition-final.
        record-staged-message composition-final >>=
        (λ-. staged-query-program A 0 i >>=
          (λquery-chunks. return
            (dg, composition-final, query-chunks))))))
    by (rule after-composition-final-coll)
  show ?thesis
    unfolding Pair using step by simp
qed
qed
have after-composition-fri-range-expanded:
  ∧ dg. hash-range-budget
    (?composition-fri + (?composition-final + (0 + (?query + 0))))
    (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [] >>=
        (λ(-, composition-bs).
          composition-final-stage A dg composition-bs >>=
          (λcomposition-final.
            record-staged-message composition-final >>=
            (λ-. staged-query-program A 0 i >>=
              (λquery-chunks. return
                (dg, composition-final, query-chunks))))))))))
  using after-composition-fri-range
  by (simp add: sm-bind-assoc split: prod.splits)
have after-composition-fri-coll-expanded:
  ∧ dg. hash-collision-budget
    (?composition-fri + (?composition-final + (0 + (?query + 0))))
    (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [] >>=
        (λ(-, composition-bs).
          composition-final-stage A dg composition-bs >>=
          (λcomposition-final.
            record-staged-message composition-final >>=
            (λ-. staged-query-program A 0 i >>=
              (λquery-chunks. return
                (dg, composition-final, query-chunks))))))))))
  using after-composition-fri-coll
  by (simp add: sm-bind-assoc split: prod.splits)
have after-degree-record-range:
  ∧ dg. hash-range-budget
    (0 + (?composition-fri + (?composition-final + (0 + (?query + 0))))))
    (record-staged-message dg >>=
      (λ-. assert
        (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
        (λ-. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [] >>=
          (λ(-, composition-bs).
            composition-final-stage A dg composition-bs >>=
            (λcomposition-final.
              record-staged-message composition-final >>=
              (λ-. staged-query-program A 0 i >>=
                (λquery-chunks. return
                  (dg, composition-final, query-chunks))))))))))

```

```

    composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 i >>=
            (λquery-chunks. return
              (dg, composition-final, query-chunks))))))
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule after-composition-fri-range-expanded)
have after-degree-record-coll:
  ∧ dg. hash-collision-budget
    (0 + (?composition-fri + (?composition-final + (0 + (?query + 0)))))
    (record-staged-message dg >>=
      (λ-. assert
        (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
          (λ-. staged-composition-fri-program A dg 0
            (ceil-log (to-nat dg + 1)) [] >>=
              (λ(-, composition-bs).
                composition-final-stage A dg composition-bs >>=
                  (λcomposition-final.
                    record-staged-message composition-final >>=
                      (λ-. staged-query-program A 0 i >>=
                        (λquery-chunks. return
                          (dg, composition-final, query-chunks)))))))))
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule hash-collision-budget-record-staged-message,
      rule after-composition-fri-range-expanded,
      rule after-composition-fri-coll-expanded)
have after-degree-range:
  ∧ as. hash-range-budget
    (?degree +
      (0 + (?composition-fri +
        (?composition-final + (0 + (?query + 0)))))
    (degree-stage A as >>=
      (λ dg. record-staged-message dg >>=
        (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0
              composition-rounds [] >>=
                (λ(-, composition-bs).
                  composition-final-stage A dg composition-bs >>=
                    (λcomposition-final.
                      record-staged-message composition-final >>=
                        (λ-. staged-query-program A 0 i >>=
                          (λquery-chunks. return
                            (dg, composition-final, query-chunks)))))))))
  unfolding Let-def
  by (rule hash-range-budget-bind)

```

```

(rule degree-range, rule after-degree-record-range)
have after-degree-coll:
   $\wedge$ as. hash-collision-budget
    (?degree +
      (0 + (?composition-fri +
        (?composition-final + (0 + (?query + 0))))))
    (degree-stage A as  $\gg$ 
      ( $\lambda$ dg. record-staged-message dg  $\gg$ 
        ( $\lambda$ -. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert (composition-rounds  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
            ( $\lambda$ -. staged-composition-fri-program A dg 0
              composition-rounds []  $\gg$ 
                ( $\lambda$ (-, composition-bs).
                  composition-final-stage A dg composition-bs  $\gg$ 
                    ( $\lambda$ composition-final.
                      record-staged-message composition-final  $\gg$ 
                        ( $\lambda$ -. staged-query-program A 0 i  $\gg$ 
                          ( $\lambda$ query-chunks. return
                            (dg, composition-final, query-chunks))))))))))
    unfolding Let-def
    by (rule hash-collision-budget-bind)
      (rule degree-range, rule degree-coll,
        rule after-degree-record-range, rule after-degree-record-coll)
have after-alpha-range:
   $\wedge$ x. hash-range-budget
    (?alpha +
      (?degree +
        (0 + (?composition-fri +
          (?composition-final + (0 + (?query + 0))))))
    (staged-alpha-program (length spec)  $\gg$ 
      ( $\lambda$ as. degree-stage A as  $\gg$ 
        ( $\lambda$ dg. record-staged-message dg  $\gg$ 
          ( $\lambda$ -. let composition-rounds = ceil-log (to-nat dg + 1) in
            assert (composition-rounds  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
              ( $\lambda$ -. staged-composition-fri-program A dg 0
                composition-rounds []  $\gg$ 
                  ( $\lambda$ (-, composition-bs).
                    composition-final-stage A dg composition-bs  $\gg$ 
                      ( $\lambda$ composition-final.
                        record-staged-message composition-final  $\gg$ 
                          ( $\lambda$ -. staged-query-program A 0 i  $\gg$ 
                            ( $\lambda$ query-chunks. return
                              (dg, composition-final, query-chunks))))))))))
      by (rule hash-range-budget-bind)
        (rule alpha-range, rule after-degree-range)
have after-alpha-coll:
   $\wedge$ x. hash-collision-budget
    (?alpha +
      (?degree +

```

```

    (0 + (?composition-fri +
      (?composition-final + (0 + (?query + 0))))))
  (staged-alpha-program (length spec) >>=
    (λas. degree-stage A as >>=
      (λdg. record-staged-message dg >>=
        (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0
              composition-rounds [] >>=
                (λ(-, composition-bs).
                  composition-final-stage A dg composition-bs >>=
                    (λcomposition-final.
                      record-staged-message composition-final >>=
                        (λ-. staged-query-program A 0 i >>=
                          (λquery-chunks. return
                            (dg, composition-final, query-chunks))))))))))
  by (rule hash-collision-budget-bind)
    (rule alpha-range, rule alpha-coll,
      rule after-degree-range, rule after-degree-coll)
have after-alpha-prefix:
  hash-collision-budget
    (?alpha-prefix +
      (?alpha +
        (?degree +
          (0 + (?composition-fri +
            (?composition-final + (0 + (?query + 0))))))
        (staged-query-challenge-prefix-program A i)
      )
    )
  unfolding staged-query-challenge-prefix-program-def
proof (rule hash-collision-budget-bind)
  show hash-range-budget ?alpha-prefix
    (staged-alpha-prefix-program A)
  by (rule alpha-prefix-range)
  show hash-collision-budget ?alpha-prefix
    (staged-alpha-prefix-program A)
  by (rule alpha-prefix-coll)
next
fix x
show hash-range-budget
  (?alpha +
    (?degree +
      (0 + (?composition-fri +
        (?composition-final + (0 + (?query + 0))))))
    (case x of (-, -, trace-bs, -) ⇒
      staged-alpha-program (length spec) >>=
        (λas. degree-stage A as >>=
          (λdg. record-staged-message dg >>=
            (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
              assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
                (λ-. staged-composition-fri-program A dg 0

```

```

composition-rounds [] >>=
(λ(-, composition-bs).
  composition-final-stage A dg composition-bs >>=
  (λcomposition-final.
    record-staged-message composition-final >>=
    (λ-. staged-query-program A 0 i >>=
      (λquery-chunks. return
        (dg, composition-final,
          query-chunks)))))))))
proof (cases x)
  case (fields a b c d)
  show ?thesis
    unfolding fields using after-alpha-range
    by (simp add: add.assoc add.commute add.left-commute)
qed
show hash-collision-budget
  (?alpha +
    (?degree +
      (0 + (?composition-fri +
        (?composition-final + (0 + (?query + 0)))))))
  (case x of (-, -, trace-bs, -) =>
    staged-alpha-program (length spec) >>=
    (λas. degree-stage A as >>=
      (λdg. record-staged-message dg >>=
        (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
          (λ-. staged-composition-fri-program A dg 0
            composition-rounds [] >>=
            (λ(-, composition-bs).
              composition-final-stage A dg composition-bs >>=
              (λcomposition-final.
                record-staged-message composition-final >>=
                (λ-. staged-query-program A 0 i >>=
                  (λquery-chunks. return
                    (dg, composition-final,
                      query-chunks)))))))))
  proof (cases x)
    case (fields a b c d)
    show ?thesis
      unfolding fields using after-alpha-coll
      by (simp add: add.assoc add.commute add.left-commute)
    qed
  qed
show ?thesis
  using after-alpha-prefix
  unfolding staged-query-search-queries-def staged-alpha-search-queries-def
  by (simp add: add.assoc add.commute add.left-commute)
qed

```

lemma *staged-query-prefix-target-hit-bound*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: $i \leq \text{rounds}$
shows
 $\text{wp-event } (\text{staged-query-challenge-prefix-program } A \ i)$
 $(\text{hash-new-output-hit-event } B \ s) \ s \leq$
 $\text{staged-phase-target-error } B$
 $(\text{staged-query-search-queries budgets } i)$
proof –
have *budget*:
 $\text{hash-target-budget } B \ (\text{staged-query-search-queries budgets } i)$
 $(\text{staged-query-challenge-prefix-program } A \ i)$
using *hash-target-program-staged-query-challenge-prefix-program*
 $[OF \ wf \ controlled \ i\text{-bound}]$
unfolding *hash-target-program-def* **by** *simp*
show *?thesis*
using *budget unfolding hash-target-budget-def*
 $\text{staged-phase-target-error-def}$ **by** *blast*
qed
end
end

theory *Staged-Security-Experiment-Targets*
imports *Staged-Security-Experiment-Budgets*
begin

Bad-value target events and checked-prefix probability bounds.

context *soundness*
begin

definition *staged-query-prefix-bad-index-error*
 $:: \text{staged-budgets} \Rightarrow \text{nat set} \Rightarrow \text{nat} \Rightarrow \text{prob}$
where
 $\text{staged-query-prefix-bad-index-error budgets } B \ i =$
 $\text{staged-phase-target-error } (\text{query-index-raw-preimage } B)$
 $(\text{staged-query-search-queries budgets } i)$

definition *fri-vector-position-values* $:: 'f \text{ list set} \Rightarrow \text{nat} \Rightarrow 'f \text{ set}$
where
 $\text{fri-vector-position-values } B \ i =$
 $\{b. \exists bs \in B. i < \text{length } bs \wedge bs ! i = b\}$

lemma *fri-vector-position-valueI*:
assumes $bs \in B$
and $i < \text{length } bs$

shows $bs ! i \in \text{fri-vector-position-values } B i$
using *assms* **unfolding** *fri-vector-position-values-def* **by** *blast*

lemma *fri-vector-member-imp-position-value*:
assumes *subset*: $B \subseteq \text{fri-challenge-space } n$
and *member*: $bs \in B$
and *i-bound*: $i < n$
shows $bs ! i \in \text{fri-vector-position-values } B i$
proof –
have $\text{length } bs = n$
using *subset member* **unfolding** *fri-challenge-space-def* **by** *auto*
then have $i < \text{length } bs$
using *i-bound* **by** *simp*
then show *?thesis*
by (rule *fri-vector-position-valueI*[*OF member*])
qed

lemma *fri-vector-member-imp-first-position-value*:
assumes *subset*: $B \subseteq \text{fri-challenge-space } n$
and *member*: $bs \in B$
and *nonempty*: $0 < n$
shows $bs ! 0 \in \text{fri-vector-position-values } B 0$
by (rule *fri-vector-member-imp-position-value*[*OF subset member*])
(use nonempty in simp)

lemma *fri-vector-extend-member-imp-position-value*:
assumes *member*: $bs @ [b] \in B$
and *length*: $\text{length } bs = i$
shows $b \in \text{fri-vector-position-values } B i$
using *assms* **unfolding** *fri-vector-position-values-def* **by** *force*

lemma *fri-vector-position-values-card-le*:
assumes *subset*: $B \subseteq \text{fri-challenge-space } n$
and *i-bound*: $i < n$
shows $\text{card } (\text{fri-vector-position-values } B i) \leq \text{card } B$
proof –
have *finite-B*: *finite* B
by (rule *finite-subset*[*OF subset finite-fri-challenge-space*])
have *values-eq*:
 $\text{fri-vector-position-values } B i = (\lambda bs. bs ! i) ` B$
using *subset i-bound*
unfolding *fri-vector-position-values-def fri-challenge-space-def*
by *auto*
show *?thesis*
unfolding *values-eq* **by** (rule *card-image-le*[*OF finite-B*])
qed

lemma *fri-vector-member-coordinate-cover*:
assumes *subset*: $B \subseteq \text{fri-challenge-space } n$

and *member*: $bs \in B$
and *nonempty*: $0 < n$
shows $\exists i < n. bs ! i \in \text{fri-vector-position-values } B \ i$
proof –
have $bs ! 0 \in \text{fri-vector-position-values } B \ 0$
by (rule *fri-vector-member-imp-first-position-value*
 $[OF \ \text{subset} \ \text{member} \ \text{nonempty}]$)
then show *?thesis*
using *nonempty* **by** *blast*
qed

definition *trace-fri-challenge-values-absent*
 $:: 'f \ \text{set} \Rightarrow ('f, 'a) \ \text{protocol-channel-scheme} \Rightarrow \text{bool}$
where
 $\text{trace-fri-challenge-values-absent } B \ s \longleftrightarrow$
 $(\forall i \ x \ y.$
 $\text{fmlookup } (\text{HashMap } s) \ (\text{TraceFriChallenge } i \ x) = \text{Some } y \longrightarrow$
 $y \notin B)$

definition *composition-fri-challenge-values-absent*
 $:: 'f \ \text{set} \Rightarrow ('f, 'a) \ \text{protocol-channel-scheme} \Rightarrow \text{bool}$
where
 $\text{composition-fri-challenge-values-absent } B \ s \longleftrightarrow$
 $(\forall i \ x \ y.$
 $\text{fmlookup } (\text{HashMap } s) \ (\text{CompositionFriChallenge } i \ x) = \text{Some } y \longrightarrow$
 $y \notin B)$

definition *alpha-challenge-values-absent*
 $:: 'f \ \text{set} \Rightarrow ('f, 'a) \ \text{protocol-channel-scheme} \Rightarrow \text{bool}$
where
 $\text{alpha-challenge-values-absent } B \ s \longleftrightarrow$
 $(\forall i \ x \ y.$
 $\text{fmlookup } (\text{HashMap } s) \ (\text{AlphaChallenge } i \ x) = \text{Some } y \longrightarrow$
 $y \notin B)$

definition *query-index-raw-preimage-absent*
 $:: \text{nat} \ \text{set} \Rightarrow ('f, 'a) \ \text{protocol-channel-scheme} \Rightarrow \text{bool}$
where
 $\text{query-index-raw-preimage-absent } B \ s \longleftrightarrow$
 $(\forall i \ x \ \text{raw}.$
 $\text{fmlookup } (\text{HashMap } s) \ (\text{QueryIndexChallenge } i \ x) = \text{Some } \text{raw} \longrightarrow$
 $\text{raw} \notin \text{query-index-raw-preimage } B)$

lemma *staged-query-prefix-bad-index-error-eq*:
 $\text{staged-query-prefix-bad-index-error } \text{budgets } B \ i =$
 $\text{hash-target-budget-value } (\text{query-index-raw-preimage } B)$
 $(\text{staged-query-search-queries } \text{budgets } i)$
unfolding *staged-query-prefix-bad-index-error-def*
 $\text{staged-phase-target-error-def}$ **by** *simp*

lemma *trace-fri-challenge-values-absent-empty*[simp]:
trace-fri-challenge-values-absent B
 $(s \setminus \text{HashMap} := \text{fmempty})$
unfolding *trace-fri-challenge-values-absent-def* **by** *simp*

lemma *composition-fri-challenge-values-absent-empty*[simp]:
composition-fri-challenge-values-absent B
 $(s \setminus \text{HashMap} := \text{fmempty})$
unfolding *composition-fri-challenge-values-absent-def* **by** *simp*

lemma *alpha-challenge-values-absent-empty*[simp]:
alpha-challenge-values-absent B
 $(s \setminus \text{HashMap} := \text{fmempty})$
unfolding *alpha-challenge-values-absent-def* **by** *simp*

lemma *staged-query-prefix-bad-index-target-bound*:
assumes *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* A
and *i-bound*: $i \leq \text{rounds}$
shows
 $\text{wp-event } (\text{staged-query-challenge-prefix-program } A \ i)$
 $(\text{hash-new-output-hit-event } (\text{query-index-raw-preimage } B) \ s) \ s \leq$
 $\text{staged-query-prefix-bad-index-error } \text{budgets } B \ i$
unfolding *staged-query-prefix-bad-index-error-def*
by $(\text{rule } \text{staged-query-prefix-target-hit-bound}[\text{OF } wf \text{ controlled } i\text{-bound}])$

lemma *query-index-raw-preimage-absent-empty*[simp]:
query-index-raw-preimage-absent B
 $(s \setminus \text{HashMap} := \text{fmempty})$
unfolding *query-index-raw-preimage-absent-def* **by** *simp*

lemma *staged-trace-fri-prefix-recv-bad-value-imp-new-output-hit*:
assumes *controlled*: *staged-adversary-controlled* *budgets* A
and *i-bound*: $i < \text{length } (\text{trace-fri-budgets } \text{budgets})$
and *absent*: *trace-fri-challenge-values-absent* $B \ s$
and *out*:
 $\text{Some } (b, t) \in$
 set-dist
 $(\text{execute}$
 $(\text{staged-trace-fri-challenge-prefix-program } A \ i \gg=$
 $(\lambda \cdot. \text{receive-trace-fri-challenge})) \ s)$
and *bad*: $b \in B$
shows $\text{hash-new-output-hit-event } B \ s \ (\text{Some } (b, t))$
proof –
obtain *prefix-out* u **where** *prefix*:
 $\text{Some } (\text{prefix-out}, u) \in$
 set-dist
 $(\text{execute } (\text{staged-trace-fri-challenge-prefix-program } A \ i) \ s)$

```

and recv:
  Some (b, t) ∈ set-dist (execute receive-trace-fri-challenge u)
using out by (auto elim!: set-dist-bindE)
have prefix-ext: s ≤ u
using hash-target-program-staged-trace-fri-challenge-prefix-program
  [OF controlled i-bound, of B] prefix
unfolding hash-target-program-def hash-extension-preserving-def
by blast
have recv-out:
  u ≤ t ∧
  fmlookup (HashMap t)
    (TraceFriChallenge (PTraceFriCounter u) (PState u)) = Some b
using receive-trace-fri-challenge-outcome[OF recv] by blast
show ?thesis
proof (cases
  fmlookup (HashMap s)
    (TraceFriChallenge (PTraceFriCounter u) (PState u)))
case None
then have hash-map-new-output-hit B s t
  unfolding hash-map-new-output-hit-def
  using recv-out bad by blast
then show ?thesis
  unfolding hash-new-output-hit-event-def by simp
next
case (Some old-b)
have old-lookup-t:
  fmlookup (HashMap t)
    (TraceFriChallenge (PTraceFriCounter u) (PState u)) = Some old-b
by (rule hash-extension-lookup[OF Some])
  (rule hash-ext-trans[OF prefix-ext conjunct1[OF recv-out]])
then have old-eq: old-b = b
  using recv-out by simp
have old-b ∉ B
  using absent Some
  unfolding trace-fri-challenge-values-absent-def by blast
then show ?thesis
  using old-eq bad by simp
qed
qed

lemma staged-trace-fri-prefix-receive-bad-value-bound:
assumes controlled: staged-adversary-controlled budgets A
and i-bound: i < length (trace-fri-budgets budgets)
and absent: trace-fri-challenge-values-absent B s
shows
  wp-event
    (staged-trace-fri-challenge-prefix-program A i ≫=
      (λ-. receive-trace-fri-challenge))
  (λout. case out of None ⇒ False)

```

```

    | Some (b, -)  $\Rightarrow$   $b \in B$ )  $s \leq$ 
    staged-phase-target-error  $B$ 
    (Suc (staged-trace-fri-search-queries budgets  $i$ ))
proof -
  let ?m =
    staged-trace-fri-challenge-prefix-program  $A$   $i$   $\gg$ 
    ( $\lambda$ -. receive-trace-fri-challenge)
  let ?bad =
     $\lambda$ out. case out of None  $\Rightarrow$  False
    | Some (b, -)  $\Rightarrow$   $b \in B$ 
  let ?hit = hash-new-output-hit-event  $B$   $s$ 
  have bad-imp-hit:
    wp-event ?m ?bad  $s \leq$  wp-event ?m ?hit  $s$ 
proof (rule wp-event-mono-on-support)
  fix out
  assume support: out  $\in$  set-dist (execute ?m  $s$ )
  and bad: ?bad out
  show ?hit out
proof (cases out)
  case None
  then show ?thesis using bad by simp
next
  case (Some bt)
  then obtain b t where out-eq: out = Some (b, t)
  by (cases bt) simp
  have out':
    Some (b, t)  $\in$  set-dist (execute ?m  $s$ )
  using support out-eq by simp
  have b-in:  $b \in B$ 
  using bad out-eq by simp
  show ?thesis
  unfolding out-eq
  by (rule staged-trace-fri-prefix-receive-bad-value-imp-new-output-hit
    [OF controlled i-bound absent out' b-in])
qed
qed
have prefix-target:
  hash-target-program  $B$ 
  (staged-trace-fri-search-queries budgets  $i$ )
  (staged-trace-fri-challenge-prefix-program  $A$   $i$ )
  by (rule hash-target-program-staged-trace-fri-challenge-prefix-program
    [OF controlled i-bound])
have target:
  hash-target-program  $B$ 
  (staged-trace-fri-search-queries budgets  $i + 1$ ) ?m
  by (rule hash-target-program-bind[OF prefix-target])
  (rule hash-target-program-receive-trace-fri-challenge)
have hit-bound:
  wp-event ?m ?hit  $s \leq$ 

```

```

    staged-phase-target-error B
    (staged-trace-fri-search-queries budgets i + 1)
using target unfolding hash-target-program-def hash-target-budget-def
    staged-phase-target-error-def by blast
show ?thesis
    using order-trans[OF bad-imp-hit hit-bound] by simp
qed

lemma staged-composition-fri-prefix-receive-bad-value-imp-new-output-hit:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < ceil-log (maxDegree + 1)
and absent: composition-fri-challenge-values-absent B s
and out:
    Some (b, t) ∈
    set-dist
    (execute
    (staged-composition-fri-challenge-prefix-program A i >>=
    (λ-. receive-composition-fri-challenge)) s)
and bad: b ∈ B
shows hash-new-output-hit-event B s (Some (b, t))
proof –
obtain prefix-out u where prefix:
    Some (prefix-out, u) ∈
    set-dist
    (execute (staged-composition-fri-challenge-prefix-program A i) s)
and recv:
    Some (b, t) ∈
    set-dist (execute receive-composition-fri-challenge u)
using out by (auto elim!: set-dist-bindE)
have prefix-ext: s ≤ u
using hash-target-program-staged-composition-fri-challenge-prefix-program
    [OF wf controlled i-bound, of B] prefix
unfolding hash-target-program-def hash-extension-preserving-def
by blast
have recv-out:
    u ≤ t ∧
    fmlookup (HashMap t)
    (CompositionFriChallenge (PCompositionFriCounter u) (PState u)) =
    Some b
using receive-composition-fri-challenge-outcome[OF recv] by blast
show ?thesis
proof (cases
    fmlookup (HashMap s)
    (CompositionFriChallenge (PCompositionFriCounter u) (PState u)))
case None
then have hash-map-new-output-hit B s t
unfolding hash-map-new-output-hit-def
using recv-out bad by blast

```

```

then show ?thesis
  unfolding hash-new-output-hit-event-def by simp
next
case (Some old-b)
have old-lookup-t:
  fmlookup (HashMap t)
    (CompositionFriChallenge (PCompositionFriCounter u) (PState u)) =
    Some old-b
  by (rule hash-extension-lookup[OF Some])
    (rule hash-ext-trans[OF prefix-ext conjunct1[OF recv-out]])
then have old-eq: old-b = b
  using recv-out by simp
have old-b  $\notin$  B
  using absent Some
  unfolding composition-fri-challenge-values-absent-def by blast
then show ?thesis
  using old-eq bad by simp
qed
qed

```

lemma staged-composition-fri-prefix-receive-bad-value-bound:

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i < \text{ceil-log } (\text{maxDegree} + 1)$
and absent: composition-fri-challenge-values-absent B s
shows

wp-event
 (staged-composition-fri-challenge-prefix-program A i $\gg$
 (λ -. receive-composition-fri-challenge))
 (λ out. case out of None $\Rightarrow$ False
 | Some (b, -) $\Rightarrow$ $b \in B$) s $\leq$
 staged-phase-target-error B
 (Suc (staged-composition-fri-search-queries budgets i))

proof –

let ?m =
 staged-composition-fri-challenge-prefix-program A i $\gg$
 (λ -. receive-composition-fri-challenge)

let ?bad =
 λ out. case out of None $\Rightarrow$ False
 | Some (b, -) $\Rightarrow$ $b \in B$

let ?hit = hash-new-output-hit-event B s

have bad-imp-hit:
 wp-event ?m ?bad s $\leq$ wp-event ?m ?hit s

proof (rule wp-event-mono-on-support)

fix out
assume support: out $\in$ set-dist (execute ?m s)
and bad: ?bad out
show ?hit out
proof (cases out)

```

    case None
    then show ?thesis using bad by simp
next
case (Some bt)
then obtain b t where out-eq: out = Some (b, t)
  by (cases bt) simp
have out':
  Some (b, t) ∈ set-dist (execute ?m s)
  using support out-eq by simp
have b-in: b ∈ B
  using bad out-eq by simp
show ?thesis
  unfolding out-eq
  by (rule
    staged-composition-fri-prefix-receive-bad-value-imp-new-output-hit
    [OF wf controlled i-bound absent out' b-in])
qed
qed
have prefix-target:
  hash-target-program B
  (staged-composition-fri-search-queries budgets i)
  (staged-composition-fri-challenge-prefix-program A i)
  by (rule
    hash-target-program-staged-composition-fri-challenge-prefix-program
    [OF wf controlled i-bound])
have target:
  hash-target-program B
  (staged-composition-fri-search-queries budgets i + 1) ?m
  by (rule hash-target-program-bind[OF prefix-target])
  (rule hash-target-program-receive-composition-fri-challenge)
have hit-bound:
  wp-event ?m ?hit s ≤
  staged-phase-target-error B
  (staged-composition-fri-search-queries budgets i + 1)
  using target unfolding hash-target-program-def hash-target-budget-def
  staged-phase-target-error-def by blast
show ?thesis
  using order-trans[OF bad-imp-hit hit-bound] by simp
qed

lemma staged-query-prefix-receive-bad-index-imp-new-output-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i ≤ rounds
  and absent: query-index-raw-preimage-absent B s
  and out:
    Some (raw, t) ∈
    set-dist
    (execute

```

```

      (staged-query-challenge-prefix-program A i >>=
        (λ-. receive-query-index-challenge)) s)
    and bad: index (to-nat raw) ∈ B
  shows
    hash-new-output-hit-event (query-index-raw-preimage B) s
      (Some (raw, t))
  proof -
    obtain prefix-out u where prefix:
      Some (prefix-out, u) ∈
        set-dist (execute (staged-query-challenge-prefix-program A i) s)
    and recv:
      Some (raw, t) ∈
        set-dist (execute receive-query-index-challenge u)
    using out by (auto elim!: set-dist-bindE)
  have prefix-ext:
    s ≤ u
  using hash-target-program-staged-query-challenge-prefix-program
    [OF wf controlled i-bound, of query-index-raw-preimage B]
    prefix
  unfolding hash-target-program-def hash-extension-preserving-def
  by blast
  have recv-out:
    u ≤ t ∧
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter u) (PState u)) = Some raw
  using receive-query-index-challenge-outcome[OF recv] by blast
  have raw-in: raw ∈ query-index-raw-preimage B
  using bad unfolding query-index-raw-preimage-def by simp
  show ?thesis
  proof (cases
    fmlookup (HashMap s)
      (QueryIndexChallenge (PQueryCounter u) (PState u)))
    case None
    then have hash-map-new-output-hit (query-index-raw-preimage B) s t
      unfolding hash-map-new-output-hit-def
      using recv-out raw-in by blast
    then show ?thesis
      unfolding hash-new-output-hit-event-def by simp
  next
    case (Some old-raw)
    have old-lookup-t:
      fmlookup (HashMap t)
        (QueryIndexChallenge (PQueryCounter u) (PState u)) =
        Some old-raw
    by (rule hash-extension-lookup[OF Some])
      (rule hash-ext-trans[OF prefix-ext conjunct1[OF recv-out]])
    then have old-eq: old-raw = raw
    using recv-out by simp
    have old-raw ∉ query-index-raw-preimage B

```

```

    using absent Some
    unfolding query-index-raw-preimage-absent-def by blast
  then show ?thesis
    using old-eq raw-in by simp
qed
qed

lemma staged-query-prefix-receive-bad-index-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound:  $i \leq \text{rounds}$ 
    and absent: query-index-raw-preimage-absent B s
  shows
    wp-event
      (staged-query-challenge-prefix-program A i  $\gg$ 
        ( $\lambda$ -. receive-query-index-challenge))
      ( $\lambda$ out. case out of None  $\Rightarrow$  False
        | Some (raw, -)  $\Rightarrow$  index (to-nat raw)  $\in$  B) s  $\leq$ 
        staged-phase-target-error (query-index-raw-preimage B)
        (Suc (staged-query-search-queries budgets i))
  proof -
    let ?m =
      staged-query-challenge-prefix-program A i  $\gg$ 
        ( $\lambda$ -. receive-query-index-challenge)
    let ?B = query-index-raw-preimage B
    let ?bad =
       $\lambda$ out. case out of None  $\Rightarrow$  False
        | Some (raw, -)  $\Rightarrow$  index (to-nat raw)  $\in$  B
    let ?hit = hash-new-output-hit-event ?B s
    have bad-imp-hit:
      wp-event ?m ?bad s  $\leq$  wp-event ?m ?hit s
    proof (rule wp-event-mono-on-support)
      fix out
      assume support: out  $\in$  set-dist (execute ?m s)
        and bad: ?bad out
      show ?hit out
      proof (cases out)
        case None
        then show ?thesis using bad by simp
      next
        case (Some rt)
        then obtain raw t where out-eq: out = Some (raw, t)
          by (cases rt) simp
        have out':
          Some (raw, t)  $\in$  set-dist (execute ?m s)
          using support out-eq by simp
        have bad-raw: index (to-nat raw)  $\in$  B
          using bad out-eq by simp
        show ?thesis

```



```

    unfolding out-eq
  by (rule staged-query-prefix-receive-bad-index-imp-new-output-hit
      [OF wf controlled i-bound absent out' bad-raw])
qed
qed
have prefix-target:
  hash-target-program ?B
    (staged-query-search-queries budgets i)
    (staged-query-challenge-prefix-program A i)
  by (rule hash-target-program-staged-query-challenge-prefix-program
      [OF wf controlled i-bound])
have target:
  hash-target-program ?B
    (staged-query-search-queries budgets i + 1) ?m
  by (rule hash-target-program-bind[OF prefix-target])
    (rule hash-target-program-receive-query-index-challenge)
have hit-bound:
  wp-event ?m ?hit s ≤
    staged-phase-target-error ?B
    (staged-query-search-queries budgets i + 1)
  using target unfolding hash-target-program-def hash-target-budget-def
    staged-phase-target-error-def by blast
show ?thesis
  using order-trans[OF bad-imp-hit hit-bound] by simp
qed

lemma receive-alpha-challenge-bad-value-imp-new-output-hit:
  assumes absent: alpha-challenge-values-absent B s
  and out:
    Some (a, t) ∈ set-dist (execute receive-alpha-challenge s)
  and bad: a ∈ B
  shows hash-new-output-hit-event B s (Some (a, t))
proof -
  have recv-out:
    s ≤ t ∧
    fmlookup (HashMap t)
      (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a
  using receive-alpha-challenge-outcome[OF out] by blast
show ?thesis
proof (cases
  fmlookup (HashMap s)
    (AlphaChallenge (PAlphaCounter s) (PState s)))
case None
then have hash-map-new-output-hit B s t
  unfolding hash-map-new-output-hit-def
  using recv-out bad by blast
then show ?thesis
  unfolding hash-new-output-hit-event-def by simp
next

```

```

case (Some old-a)
have old-lookup-t:
  fmlookup (HashMap t)
  (AlphaChallenge (PAlphaCounter s) (PState s)) = Some old-a
  by (rule hash-extension-lookup[OF Some conjunct1[OF recv-out]])
then have old-eq: old-a = a
  using recv-out by simp
have old-a  $\notin$  B
  using absent Some
  unfolding alpha-challenge-values-absent-def by blast
then show ?thesis
  using old-eq bad by simp
qed
qed

lemma receive-alpha-challenge-bad-value-bound:
  assumes absent: alpha-challenge-values-absent B s
  shows
    wp-event receive-alpha-challenge
      ( $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
       |  $\text{Some } (a, -) \Rightarrow a \in B$ ) s  $\leq$ 
      staged-phase-target-error B 1
proof -
  let ?bad =
     $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
    |  $\text{Some } (a, -) \Rightarrow a \in B$ 
  let ?hit = hash-new-output-hit-event B s
  have bad-imp-hit:
    wp-event receive-alpha-challenge ?bad s  $\leq$ 
    wp-event receive-alpha-challenge ?hit s
  proof (rule wp-event-mono-on-support)
    fix out
    assume support: out  $\in$  set-dist (execute receive-alpha-challenge s)
    and bad: ?bad out
    show ?hit out
    proof (cases out)
      case None
      then show ?thesis using bad by simp
    next
      case (Some pair)
      obtain a t where pair-eq: pair = (a, t)
      by (cases pair)
      have out-eq: out = Some (a, t)
      using Some pair-eq by simp
      have out':
        Some (a, t)  $\in$  set-dist (execute receive-alpha-challenge s)
      using support out-eq by simp
      have a-in: a  $\in$  B
      using bad out-eq by simp

```

```

show ?thesis
  unfolding out-eq
  by (rule receive-alpha-challenge-bad-value-imp-new-output-hit
    [OF absent out' a-in])
qed
qed
have target:
  hash-target-program B 1 receive-alpha-challenge
  by (rule hash-target-program-receive-alpha-challenge)
have hit-bound:
  wp-event receive-alpha-challenge ?hit s ≤
    staged-phase-target-error B 1
  using target unfolding hash-target-program-def hash-target-budget-def
    staged-phase-target-error-def by blast
show ?thesis
  using order-trans[OF bad-imp-hit hit-bound] .
qed

lemma staged-alpha-prefix-receive-bad-value-imp-new-output-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i ≤ length spec
  and absent: alpha-challenge-values-absent B s
  and out:
    Some (a, t) ∈
      set-dist
        (execute
          (staged-alpha-challenge-prefix-program A i ≫
            (λ-. receive-alpha-challenge)) s)
  and bad: a ∈ B
  shows hash-new-output-hit-event B s (Some (a, t))
proof –
  obtain prefix-out u where prefix:
    Some (prefix-out, u) ∈
      set-dist (execute (staged-alpha-challenge-prefix-program A i) s)
  and recv:
    Some (a, t) ∈ set-dist (execute receive-alpha-challenge u)
  using out by (auto elim!: set-dist-bindE)
  have prefix-ext:
    s ≤ u
  using hash-target-program-staged-alpha-challenge-prefix-program
    [OF wf controlled i-bound, of B] prefix
  unfolding hash-target-program-def hash-extension-preserving-def
  by blast
  have recv-out:
    u ≤ t ∧
      fmlookup (HashMap t)
        (AlphaChallenge (PAlphaCounter u) (PState u)) = Some a
  using receive-alpha-challenge-outcome[OF recv] by blast

```

```

show ?thesis
proof (cases
  fmlookup (HashMap s)
    (AlphaChallenge (PAlphaCounter u) (PState u)))
case None
then have hash-map-new-output-hit B s t
  unfolding hash-map-new-output-hit-def
  using recv-out bad by blast
then show ?thesis
  unfolding hash-new-output-hit-event-def by simp
next
case (Some old-a)
have old-lookup-t:
  fmlookup (HashMap t)
    (AlphaChallenge (PAlphaCounter u) (PState u)) = Some old-a
by (rule hash-extension-lookup[OF Some])
  (rule hash-ext-trans[OF prefix-ext conjunct1[OF recv-out]])
then have old-eq: old-a = a
  using recv-out by simp
have old-a  $\notin$  B
  using absent Some
  unfolding alpha-challenge-values-absent-def by blast
then show ?thesis
  using old-eq bad by simp
qed
qed

lemma staged-alpha-prefix-receive-bad-value-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i \leq \text{length spec}$ 
and absent: alpha-challenge-values-absent B s
shows
  wp-event
    (staged-alpha-challenge-prefix-program A i  $\gg$ 
      ( $\lambda$ -. receive-alpha-challenge))
    ( $\lambda$ out. case out of None  $\Rightarrow$  False
      | Some (a, -)  $\Rightarrow$   $a \in B$ ) s  $\leq$ 
    staged-phase-target-error B
    (Suc (staged-alpha-search-queries budgets i))
proof -
let ?m =
  staged-alpha-challenge-prefix-program A i  $\gg$ 
    ( $\lambda$ -. receive-alpha-challenge)
let ?bad =
   $\lambda$ out. case out of None  $\Rightarrow$  False
    | Some (a, -)  $\Rightarrow$   $a \in B$ 
let ?hit = hash-new-output-hit-event B s
have bad-imp-hit:

```

```

wp-event ?m ?bad s ≤ wp-event ?m ?hit s
proof (rule wp-event-mono-on-support)
  fix out
  assume support: out ∈ set-dist (execute ?m s)
  and bad: ?bad out
  show ?hit out
  proof (cases out)
    case None
    then show ?thesis using bad by simp
  next
    case (Some pair)
    obtain a t where pair-eq: pair = (a, t)
    by (cases pair)
    have out-eq: out = Some (a, t)
    using Some pair-eq by simp
    have out':
      Some (a, t) ∈ set-dist (execute ?m s)
    using support out-eq by simp
    have a-in: a ∈ B
    using bad out-eq by simp
    show ?thesis
    unfolding out-eq
    by (rule staged-alpha-prefix-recv-bad-value-imp-new-output-hit
      [OF wf controlled i-bound absent out' a-in])
  qed
qed
have prefix-target:
  hash-target-program B (staged-alpha-search-queries budgets i)
    (staged-alpha-challenge-prefix-program A i)
  by (rule hash-target-program-staged-alpha-challenge-prefix-program
    [OF wf controlled i-bound])
have target:
  hash-target-program B
    (staged-alpha-search-queries budgets i + 1) ?m
  by (rule hash-target-program-bind [OF prefix-target])
    (rule hash-target-program-recv-alpha-challenge)
have hit-bound:
  wp-event ?m ?hit s ≤
    staged-phase-target-error B
    (staged-alpha-search-queries budgets i + 1)
  using target unfolding hash-target-program-def hash-target-budget-def
    staged-phase-target-error-def by blast
show ?thesis
  using order-trans [OF bad-imp-hit hit-bound] by simp
qed

lemma hash-range-budget-staged-transcript-program:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A

```

```

shows
  hash-range-budget
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    (staged-transcript-program A)
proof -
  let ?trace-root = trace-root-budget budgets
  let ?trace-fri =
    sum-list (trace-fri-budgets budgets) + ceil-log clength
  let ?trace-final = trace-final-budget budgets
  let ?alpha = length spec
  let ?degree = degree-budget budgets
  let ?composition-fri =
    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
  let ?composition-final = composition-final-budget budgets
  let ?query =
    sum-list (query-opening-budgets budgets) + rounds
  have trace-root-range:
    hash-range-budget ?trace-root (trace-root-stage A)
    using controlled unfolding staged-adversary-controlled-def
    by (blast intro: controlled-ro-program-range)
  have trace-fri-range:
    hash-range-budget ?trace-fri
      (staged-trace-fri-program A 0 (ceil-log clength) [])
  proof -
    have len:
      0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
      using wf unfolding staged-budget-wellformed-def by simp
    have exact:
      hash-range-budget
        (sum-list
          (take (ceil-log clength)
            (drop 0 (trace-fri-budgets budgets)))) +
        ceil-log clength
        (staged-trace-fri-program A 0 (ceil-log clength) [])
      by (rule hash-range-budget-staged-trace-fri-program
        [OF controlled len])
    then show ?thesis
      using wf unfolding staged-budget-wellformed-def by simp
  qed
  have trace-final-range:
    ∧ bs. hash-range-budget ?trace-final (trace-final-stage A bs)
    using controlled unfolding staged-adversary-controlled-def
    by (blast intro: controlled-ro-program-range)
  have alpha-range:
    hash-range-budget ?alpha (staged-alpha-program (length spec))
    by (rule hash-range-budget-staged-alpha-program)
  have degree-range:
    ∧ as. hash-range-budget ?degree (degree-stage A as)

```

```

using controlled unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-range)
have composition-fri-range:
 $\wedge dg.$  hash-range-budget ?composition-fri
  (assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
    ( $\lambda$ -. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
by (rule hash-range-budget-guarded-staged-composition-fri-program
  [OF wf controlled])
have composition-final-range:
 $\wedge dg$  bs. hash-range-budget ?composition-final
  (composition-final-stage A dg bs)
using controlled unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-range)
have query-range:
  hash-range-budget ?query (staged-query-program A 0 rounds)
proof –
  have len:
    0 + rounds  $\leq$  length (query-opening-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-range-budget
      (sum-list
        (take rounds (drop 0 (query-opening-budgets budgets))) +
        rounds)
      (staged-query-program A 0 rounds)
    by (rule hash-range-budget-staged-query-program[OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have query-tail:
 $\wedge F.$  hash-range-budget (?query + 0)
  (staged-query-program A 0 rounds  $\gg$ 
    ( $\lambda$ query-chunks. return (F query-chunks)))
by (rule hash-range-budget-bind)
  (rule query-range, rule hash-range-budget-return)
have after-composition-final-record:
 $\wedge$  composition-final F. hash-range-budget (0 + (?query + 0))
  (record-staged-message composition-final  $\gg$ 
    ( $\lambda$ -. staged-query-program A 0 rounds  $\gg$ 
      ( $\lambda$ query-chunks. return (F query-chunks))))
by (rule hash-range-budget-bind)
  (rule hash-range-budget-record-staged-message, rule query-tail)
have after-composition-final:
 $\wedge dg$  bs F. hash-range-budget
  (?composition-final + (0 + (?query + 0)))
  (composition-final-stage A dg bs  $\gg$ 
    ( $\lambda$ composition-final.
      record-staged-message composition-final  $\gg$ 

```

```

    (λ-. staged-query-program A 0 rounds >>=
      (λquery-chunks.
        return (F composition-final query-chunks))))))
  by (rule hash-range-budget-bind)
    (rule composition-final-range, rule after-composition-final-record)
have after-composition-fri:
  ∧dg F. hash-range-budget
    (?composition-fri +
      (?composition-final + (0 + (?query + 0))))
    ((assert (ceil-log (to-nat dg + 1) ≤
      ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [])) >>=
      (λ(composition-roots, composition-bs).
        composition-final-stage A dg composition-bs >>=
          (λcomposition-final.
            record-staged-message composition-final >>=
              (λ-. staged-query-program A 0 rounds >>=
                (λquery-chunks.
                  return
                    (F composition-roots composition-bs composition-final
                      query-chunks)))))))
proof (rule hash-range-budget-bind)
  fix dg F
  show hash-range-budget ?composition-fri
    (assert (ceil-log (to-nat dg + 1) ≤
      ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))
    by (rule composition-fri-range)
  fix dg F x
  show hash-range-budget
    (?composition-final + (0 + (?query + 0)))
    (case x of (composition-roots, composition-bs) ⇒
      composition-final-stage A dg composition-bs >>=
        (λcomposition-final.
          record-staged-message composition-final >>=
            (λ-. staged-query-program A 0 rounds >>=
              (λquery-chunks.
                return
                  (F composition-roots composition-bs composition-final
                    query-chunks))))))
proof (cases x)
  case (Pair composition-roots composition-bs)
  have hash-range-budget
    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=

```



```

    (λ-. staged-query-program A 0 rounds >>=
      (λquery-chunks.
        return
          (F composition-roots composition-bs composition-final
            query-chunks))))))
  by (rule after-composition-final)
then show ?thesis
  unfolding Pair by simp
qed
qed
have after-degree-record:
  ∧dg F. hash-range-budget
    (0 +
      (?composition-fri +
        (?composition-final + (0 + (?query + 0)))))
  (record-staged-message dg >>=
    (λ-. (assert (ceil-log (to-nat dg + 1) ≤
      ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [])) >>=
      (λ(composition-roots, composition-bs).
        composition-final-stage A dg composition-bs >>=
        (λcomposition-final.
          record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 rounds >>=
            (λquery-chunks.
              return
                (F composition-roots composition-bs composition-final
                  query-chunks)))))))
  by (rule hash-range-budget-bind)
  (rule hash-range-budget-record-staged-message, rule after-composition-fri)
have after-degree:
  ∧as F. hash-range-budget
    (?degree +
      (0 +
        (?composition-fri +
          (?composition-final + (0 + (?query + 0)))))
    (degree-stage A as >>=
      (λdg. record-staged-message dg >>=
        (λ-. (assert (ceil-log (to-nat dg + 1) ≤
          ceil-log (maxDegree + 1)) >>=
          (λ-. staged-composition-fri-program A dg 0
            (ceil-log (to-nat dg + 1)) [])) >>=
          (λ(composition-roots, composition-bs).
            composition-final-stage A dg composition-bs >>=
            (λcomposition-final.
              record-staged-message composition-final >>=
              (λ-. staged-query-program A 0 rounds >>=
                (λquery-chunks.

```

```

      return
      (F dg composition-roots composition-bs
       composition-final query-chunks)))))))))
  by (rule hash-range-budget-bind)
    (rule degree-range, rule after-degree-record)
  have after-alpha:
     $\wedge F.$  hash-range-budget
    (?alpha +
     (?degree +
      (0 +
       (?composition-fri +
        (?composition-final + (0 + (?query + 0)))))))
    (staged-alpha-program (length spec)  $\gg$ 
     ( $\lambda$ as. degree-stage A as  $\gg$ 
      ( $\lambda$ dg. record-staged-message dg  $\gg$ 
       ( $\lambda$ -. (assert (ceil-log (to-nat dg + 1)  $\leq$ 
        ceil-log (maxDegree + 1))  $\gg$ 
        ( $\lambda$ -. staged-composition-fri-program A dg 0
         (ceil-log (to-nat dg + 1)) []))  $\gg$ 
        ( $\lambda$ (composition-roots, composition-bs).
         composition-final-stage A dg composition-bs  $\gg$ 
         ( $\lambda$ composition-final.
          record-staged-message composition-final  $\gg$ 
          ( $\lambda$ -. staged-query-program A 0 rounds  $\gg$ 
           ( $\lambda$ query-chunks.
            return
            (F as dg composition-roots composition-bs
             composition-final query-chunks)))))))))
    by (rule hash-range-budget-bind)
      (rule alpha-range, rule after-degree)
  have after-trace-final-record:
     $\wedge$ trace-final F. hash-range-budget
    (0 +
     (?alpha +
      (?degree +
       (0 +
        (?composition-fri +
         (?composition-final + (0 + (?query + 0)))))))
    (record-staged-message trace-final  $\gg$ 
     ( $\lambda$ -. staged-alpha-program (length spec)  $\gg$ 
      ( $\lambda$ as. degree-stage A as  $\gg$ 
       ( $\lambda$ dg. record-staged-message dg  $\gg$ 
        ( $\lambda$ -. (assert (ceil-log (to-nat dg + 1)  $\leq$ 
         ceil-log (maxDegree + 1))  $\gg$ 
         ( $\lambda$ -. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) []))  $\gg$ 
          ( $\lambda$ (composition-roots, composition-bs).
           composition-final-stage A dg composition-bs  $\gg$ 
           ( $\lambda$ composition-final.
            return
            (F as dg composition-roots composition-bs
             composition-final query-chunks)))))))))

```

```

      record-staged-message composition-final >>=
      (λ-. staged-query-program A 0 rounds >>=
      (λquery-chunks.
      return
      (F as dg composition-roots composition-bs
      composition-final query-chunks)))))))))
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message, rule after-alpha)
  have after-trace-final:
    ∧ trace-bs F. hash-range-budget
    (?trace-final +
    (0 +
    (?alpha +
    (?degree +
    (0 +
    (?composition-fri +
    (?composition-final + (0 + (?query + 0))))))))))
    (trace-final-stage A trace-bs >>=
    (λtrace-final. record-staged-message trace-final >>=
    (λ-. staged-alpha-program (length spec) >>=
    (λas. degree-stage A as >>=
    (λdg. record-staged-message dg >>=
    (λ-. (assert (ceil-log (to-nat dg + 1)) ≤
    ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
    (ceil-log (to-nat dg + 1)) [])) >>=
    (λ(composition-roots, composition-bs).
    composition-final-stage A dg composition-bs >>=
    (λcomposition-final.
    record-staged-message composition-final >>=
    (λ-. staged-query-program A 0 rounds >>=
    (λquery-chunks.
    return
    (F trace-final as dg composition-roots
    composition-bs composition-final
    query-chunks)))))))))
    by (rule hash-range-budget-bind)
      (rule trace-final-range, rule after-trace-final-record)
  have after-trace-fri:
    ∧ F. hash-range-budget
    (?trace-fri +
    (?trace-final +
    (0 +
    (?alpha +
    (?degree +
    (0 +
    (?composition-fri +
    (?composition-final + (0 + (?query + 0))))))))))
    (staged-trace-fri-program A 0 (ceil-log clength) [] >>=

```

```

(λ(trace-roots, trace-bs).
  trace-final-stage A trace-bs >>=
    (λtrace-final. record-staged-message trace-final >>=
      (λ-. staged-alpha-program (length spec) >>=
        (λas. degree-stage A as >>=
          (λdg. record-staged-message dg >>=
            (λ-. (assert (ceil-log (to-nat dg + 1) ≤
              ceil-log (maxDegree + 1)) >>=
                (λ-. staged-composition-fri-program A dg 0
                  (ceil-log (to-nat dg + 1)) [])) >>=
                  (λ(composition-roots, composition-bs).
                    composition-final-stage A dg composition-bs >>=
                      (λcomposition-final.
                        record-staged-message composition-final >>=
                          (λ-. staged-query-program A 0 rounds >>=
                            (λquery-chunks.
                              return
                                (F trace-roots trace-bs trace-final as dg
                                  composition-roots composition-bs
                                  composition-final query-chunks))))))))))
proof (rule hash-range-budget-bind)
fix F
show hash-range-budget ?trace-fri
  (staged-trace-fri-program A 0 (ceil-log clength) [])
by (rule trace-fri-range)
fix F x
show hash-range-budget
  (?trace-final +
    (0 +
      (?alpha +
        (?degree +
          (0 +
            (?composition-fri +
              (?composition-final + (0 + (?query + 0))))))))))
  (case x of (trace-roots, trace-bs) ⇒
    trace-final-stage A trace-bs >>=
      (λtrace-final. record-staged-message trace-final >>=
        (λ-. staged-alpha-program (length spec) >>=
          (λas. degree-stage A as >>=
            (λdg. record-staged-message dg >>=
              (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                ceil-log (maxDegree + 1)) >>=
                  (λ-. staged-composition-fri-program A dg 0
                    (ceil-log (to-nat dg + 1)) [])) >>=
                    (λ(composition-roots, composition-bs).
                      composition-final-stage A dg composition-bs >>=
                        (λcomposition-final.
                          record-staged-message composition-final >>=
                            (λ-. staged-query-program A 0 rounds >>=
                              (λquery-chunks.
                                return
                                  (F trace-roots trace-bs trace-final as dg
                                    composition-roots composition-bs
                                    composition-final query-chunks))))))))))

```

```

      (λquery-chunks.
        return
          (F trace-roots trace-bs trace-final as dg
            composition-roots composition-bs
            composition-final query-chunks)))))))))
proof (cases x)
case (Pair trace-roots trace-bs)
have hash-range-budget
  (?trace-final +
    (0 +
      (?alpha +
        (?degree +
          (0 +
            (?composition-fri +
              (?composition-final + (0 + (?query + 0))))))))))
  (trace-final-stage A trace-bs >=>
    (λtrace-final. record-staged-message trace-final >=>
      (λ-. staged-alpha-program (length spec) >=>
        (λas. degree-stage A as >=>
          (λdg. record-staged-message dg >=>
            (λ-. (assert (ceil-log (to-nat dg + 1)) ≤
              ceil-log (maxDegree + 1)) >=>
              (λ-. staged-composition-fri-program A dg 0
                (ceil-log (to-nat dg + 1)) [])) >=>
              (λ(composition-roots, composition-bs).
                composition-final-stage A dg composition-bs >=>
                  (λcomposition-final.
                    record-staged-message composition-final >=>
                      (λ-. staged-query-program A 0 rounds >=>
                        (λquery-chunks.
                          return
                            (F trace-roots trace-bs trace-final as dg
                              composition-roots composition-bs
                              composition-final query-chunks)))))))))))))
by (rule after-trace-final)
then show ?thesis
unfolding Pair by simp
qed
qed
have after-trace-root-record:
  ∧fr F. hash-range-budget
  (0 +
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +

```

```

      (?composition-final + (0 + (?query + 0)))))))))
(record-staged-message fr >>=
  (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] >>=
    (λ(trace-roots, trace-bs).
      trace-final-stage A trace-bs >>=
        (λtrace-final. record-staged-message trace-final >>=
          (λ-. staged-alpha-program (length spec) >>=
            (λas. degree-stage A as >>=
              (λdg. record-staged-message dg >>=
                (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                  ceil-log (maxDegree + 1)) >>=
                    (λ-. staged-composition-fri-program A dg 0
                      (ceil-log (to-nat dg + 1)) [])) >>=
                      (λ(composition-roots, composition-bs).
                        composition-final-stage A dg composition-bs >>=
                          (λcomposition-final.
                            record-staged-message composition-final >>=
                              (λ-. staged-query-program A 0 rounds >>=
                                (λquery-chunks.
                                  return
                                    (F trace-roots trace-bs trace-final as dg
                                      composition-roots composition-bs
                                      composition-final query-chunks)))))))))))))
  by (rule hash-range-budget-bind)
  (rule hash-range-budget-record-staged-message, rule after-trace-fri)
have whole:
  hash-range-budget
  (?trace-root +
    (0 +
      (?trace-fri +
        (?trace-final +
          (0 +
            (?alpha +
              (?degree +
                (0 +
                  (?composition-fri +
                    (?composition-final + (0 + (?query + 0)))))))))
    (staged-transcript-program A)
  unfolding staged-transcript-program-def Let-def
proof (rule hash-range-budget-bind)
show hash-range-budget ?trace-root (trace-root-stage A)
  by (rule trace-root-range)
fix fr
show hash-range-budget
  (0 +
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +

```

```

    (?degree +
      (0 +
        (?composition-fri +
          (?composition-final + (0 + (?query + 0))))))))))
  (record-staged-message fr >>=
    (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] >>=
      (λ(trace-roots, trace-bs).
        trace-final-stage A trace-bs >>=
          (λtrace-final.
            record-staged-message trace-final >>=
              (λ-. staged-alpha-program (length spec) >>=
                (λas. degree-stage A as >>=
                  (λdg. record-staged-message dg >>=
                    (λ-. assert (ceil-log (to-nat dg + 1) ≤
                      ceil-log (maxDegree + 1)) >>=
                      (λ-. staged-composition-fri-program A dg 0
                        (ceil-log (to-nat dg + 1)) [] >>=
                          (λ(composition-roots, composition-bs).
                            composition-final-stage A dg composition-bs >>=
                              (λcomposition-final.
                                record-staged-message
                                  composition-final >>=
                                    (λ-. staged-query-program A 0
                                      rounds >>=
                                        (λquery-chunks.
                                          return
                                            (|staged-trace-root = fr,
                                              staged-trace-fri-roots =
                                                trace-roots,
                                              staged-trace-fri-challenges =
                                                trace-bs,
                                              staged-trace-final =
                                                trace-final,
                                              staged-alphas = as,
                                              staged-degree = dg,
                                              staged-composition-fri-roots =
                                                composition-roots,
                                              staged-composition-fri-challenges =
                                                composition-bs,
                                              staged-composition-final =
                                                composition-final,
                                              staged-query-chunks =
                                                query-chunks|)))))))))))))
  using after-trace-root-record
  [of fr
    λtrace-roots trace-bs trace-final as dg composition-roots
      composition-bs composition-final query-chunks.
    (|staged-trace-root = fr,
      staged-trace-fri-roots = trace-roots,

```

```

    staged-trace-fri-challenges = trace-bs,
    staged-trace-final = trace-final,
    staged-alphas = as,
    staged-degree = dg,
    staged-composition-fri-roots = composition-roots,
    staged-composition-fri-challenges = composition-bs,
    staged-composition-final = composition-final,
    staged-query-chunks = query-chunks)]
  by (simp add: sm-bind-assoc)
qed
have budget-eq:
  ?trace-root +
  (0 +
  (?trace-fri +
  (?trace-final +
  (0 +
  (?alpha +
  (?degree +
  (0 +
  (?composition-fri +
  (?composition-final + (0 + (?query + 0)))))))))) =
  staged-attacker-query-budget budgets + staged-challenge-query-budget
unfolding staged-attacker-query-budget-def
  staged-challenge-query-budget-def
  by (simp add: add.assoc add.commute add.left-commute)
show ?thesis
  using whole unfolding budget-eq .
qed

lemma hash-range-budget-staged-semantic-adversary:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    hash-range-budget
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (staged-semantic-adversary A)
proof -
  have hash-range-budget
    ((staged-attacker-query-budget budgets + staged-challenge-query-budget) +
    0)
    (staged-semantic-adversary A)
  unfolding staged-semantic-adversary-def
proof (rule hash-range-budget-bind
  [where n=staged-attacker-query-budget budgets +
    staged-challenge-query-budget and n'=0])
  show hash-range-budget
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    (staged-transcript-program A)
  by (rule hash-range-budget-staged-transcript-program[OF wf controlled])

```



```

    show  $\bigwedge x.$  hash-range-budget 0 (return (staged-proof-transcript x))
    by (rule hash-range-budget-return)
qed
then show ?thesis by simp
qed

lemma hash-collision-budget-staged-transcript-program:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    hash-collision-budget
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (staged-transcript-program A)
proof -
  let ?trace-root = trace-root-budget budgets
  let ?trace-fri =
    sum-list (trace-fri-budgets budgets) + ceil-log clength
  let ?trace-final = trace-final-budget budgets
  let ?alpha = length spec
  let ?degree = degree-budget budgets
  let ?composition-fri =
    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
  let ?composition-final = composition-final-budget budgets
  let ?query =
    sum-list (query-opening-budgets budgets) + rounds
  have trace-root-range:
    hash-range-budget ?trace-root (trace-root-stage A)
    using controlled unfolding staged-adversary-controlled-def
    by (blast intro: controlled-ro-program-range)
  have trace-root-coll:
    hash-collision-budget ?trace-root (trace-root-stage A)
    using controlled unfolding staged-adversary-controlled-def
    by (blast intro: controlled-ro-program-collision)
  have trace-fri-range:
    hash-range-budget ?trace-fri
      (staged-trace-fri-program A 0 (ceil-log clength) [])
proof -
  have len:
    0 + ceil-log clength  $\leq$  length (trace-fri-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-range-budget
      (sum-list
        (take (ceil-log clength)
          (drop 0 (trace-fri-budgets budgets)))) +
      ceil-log clength
      (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule hash-range-budget-staged-trace-fri-program

```

```

    [OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have trace-fri-coll:
  hash-collision-budget ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])
proof -
  have len:
    0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-collision-budget
      (sum-list
        (take (ceil-log clength)
          (drop 0 (trace-fri-budgets budgets))) +
        ceil-log clength)
      (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule hash-collision-budget-staged-trace-fri-program
      [OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have trace-final-range:
  ∧bs. hash-range-budget ?trace-final (trace-final-stage A bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
have trace-final-coll:
  ∧bs. hash-collision-budget ?trace-final (trace-final-stage A bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-collision)
have alpha-range:
  hash-range-budget ?alpha (staged-alpha-program (length spec))
  by (rule hash-range-budget-staged-alpha-program)
have alpha-coll:
  hash-collision-budget ?alpha (staged-alpha-program (length spec))
  by (rule hash-collision-budget-staged-alpha-program)
have degree-range:
  ∧as. hash-range-budget ?degree (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
have degree-coll:
  ∧as. hash-collision-budget ?degree (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-collision)
have composition-fri-range:
  ∧dg. hash-range-budget ?composition-fri
    (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0

```

```

    (ceil-log (to-nat dg + 1)) [])
  by (rule hash-range-budget-guarded-staged-composition-fri-program
      [OF wf controlled])
have composition-fri-coll:
   $\wedge dg.$  hash-collision-budget ?composition-fri
    (assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
      ( $\lambda -.$  staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))
  by (rule hash-collision-budget-guarded-staged-composition-fri-program
      [OF wf controlled])
have composition-final-range:
   $\wedge dg$  bs. hash-range-budget ?composition-final
    (composition-final-stage A dg bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-range)
have composition-final-coll:
   $\wedge dg$  bs. hash-collision-budget ?composition-final
    (composition-final-stage A dg bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-collision)
have query-range:
  hash-range-budget ?query (staged-query-program A 0 rounds)
proof -
  have len:
    0 + rounds  $\leq$  length (query-opening-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-range-budget
      (sum-list
        (take rounds (drop 0 (query-opening-budgets budgets))) +
        rounds)
      (staged-query-program A 0 rounds)
    by (rule hash-range-budget-staged-query-program[OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have query-coll:
  hash-collision-budget ?query (staged-query-program A 0 rounds)
proof -
  have len:
    0 + rounds  $\leq$  length (query-opening-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-collision-budget
      (sum-list
        (take rounds (drop 0 (query-opening-budgets budgets))) +
        rounds)
      (staged-query-program A 0 rounds)
    by (rule hash-collision-budget-staged-query-program[OF controlled len])

```

```

then show ?thesis
  using wf unfolding staged-budget-wellformed-def by simp
qed
have query-tail-range:
   $\wedge F.$  hash-range-budget (?query + 0)
    (staged-query-program A 0 rounds  $\gg=$ 
      ( $\lambda$ query-chunks. return (F query-chunks)))
  by (rule hash-range-budget-bind)
    (rule query-range, rule hash-range-budget-return)
have query-tail-coll:
   $\wedge F.$  hash-collision-budget (?query + 0)
    (staged-query-program A 0 rounds  $\gg=$ 
      ( $\lambda$ query-chunks. return (F query-chunks)))
  by (rule hash-collision-budget-bind)
    (rule query-range, rule query-coll, rule hash-range-budget-return,
      rule hash-collision-budget-return)
have after-composition-final-record-range:
   $\wedge$ composition-final F. hash-range-budget (0 + (?query + 0))
    (record-staged-message composition-final  $\gg=$ 
      ( $\lambda$ -. staged-query-program A 0 rounds  $\gg=$ 
        ( $\lambda$ query-chunks. return (F query-chunks)))))
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message, rule query-tail-range)
have after-composition-final-record-coll:
   $\wedge$ composition-final F. hash-collision-budget (0 + (?query + 0))
    (record-staged-message composition-final  $\gg=$ 
      ( $\lambda$ -. staged-query-program A 0 rounds  $\gg=$ 
        ( $\lambda$ query-chunks. return (F query-chunks)))))
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule hash-collision-budget-record-staged-message,
      rule query-tail-range, rule query-tail-coll)
have after-composition-final-range:
   $\wedge$ dg bs F. hash-range-budget
    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg bs  $\gg=$ 
      ( $\lambda$ composition-final.
        record-staged-message composition-final  $\gg=$ 
          ( $\lambda$ -. staged-query-program A 0 rounds  $\gg=$ 
            ( $\lambda$ query-chunks.
              return (F composition-final query-chunks))))))
  by (rule hash-range-budget-bind)
    (rule composition-final-range,
      rule after-composition-final-record-range)
have after-composition-final-coll:
   $\wedge$ dg bs F. hash-collision-budget
    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg bs  $\gg=$ 
      ( $\lambda$ composition-final.

```

```

    record-staged-message composition-final  $\gg=$ 
      ( $\lambda$ -. staged-query-program A 0 rounds  $\gg=$ 
        ( $\lambda$ query-chunks.
          return (F composition-final query-chunks))))))
  by (rule hash-collision-budget-bind)
    (rule composition-final-range, rule composition-final-coll,
      rule after-composition-final-record-range,
      rule after-composition-final-record-coll)
  have after-composition-fri-range:
     $\wedge$  dg F. hash-range-budget
      (?composition-fri +
        (?composition-final + (0 + (?query + 0))))
      ((assert (ceil-log (to-nat dg + 1)  $\leq$ 
        ceil-log (maxDegree + 1))  $\gg=$ 
        ( $\lambda$ -. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) []))  $\gg=$ 
        ( $\lambda$ (composition-roots, composition-bs).
          composition-final-stage A dg composition-bs  $\gg=$ 
            ( $\lambda$ composition-final.
              record-staged-message composition-final  $\gg=$ 
                ( $\lambda$ -. staged-query-program A 0 rounds  $\gg=$ 
                  ( $\lambda$ query-chunks.
                    return
                      (F composition-roots composition-bs composition-final
                        query-chunks)))))))
  proof (rule hash-range-budget-bind)
    fix dg F
    show hash-range-budget ?composition-fri
      (assert (ceil-log (to-nat dg + 1)  $\leq$ 
        ceil-log (maxDegree + 1))  $\gg=$ 
        ( $\lambda$ -. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) []))
      by (rule composition-fri-range)
    fix dg F x
    show hash-range-budget
      (?composition-final + (0 + (?query + 0)))
      (case x of (composition-roots, composition-bs)  $\Rightarrow$ 
        composition-final-stage A dg composition-bs  $\gg=$ 
          ( $\lambda$ composition-final.
            record-staged-message composition-final  $\gg=$ 
              ( $\lambda$ -. staged-query-program A 0 rounds  $\gg=$ 
                ( $\lambda$ query-chunks.
                  return
                    (F composition-roots composition-bs composition-final
                      query-chunks))))))
  proof (cases x)
    case (Pair composition-roots composition-bs)
    have hash-range-budget
      (?composition-final + (0 + (?query + 0)))

```

```

    (composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 rounds >>=
            (λquery-chunks.
              return
                (F composition-roots composition-bs composition-final
                  query-chunks))))))
  by (rule after-composition-final-range)
then show ?thesis
  unfolding Pair by simp
qed
qed
have after-composition-fri-coll:
  ∧dg F. hash-collision-budget
    (?composition-fri +
      (?composition-final + (0 + (?query + 0))))
  ((assert (ceil-log (to-nat dg + 1) ≤
    ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) [])) >>=
    (λ(composition-roots, composition-bs).
      composition-final-stage A dg composition-bs >>=
        (λcomposition-final.
          record-staged-message composition-final >>=
            (λ-. staged-query-program A 0 rounds >>=
              (λquery-chunks.
                return
                  (F composition-roots composition-bs composition-final
                    query-chunks)))))))
proof (rule hash-collision-budget-bind)
  fix dg F
  show hash-range-budget ?composition-fri
    (assert (ceil-log (to-nat dg + 1) ≤
      ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))
    by (rule composition-fri-range)
  fix dg F
  show hash-collision-budget ?composition-fri
    (assert (ceil-log (to-nat dg + 1) ≤
      ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))
    by (rule composition-fri-coll)
  fix dg F x
  show hash-range-budget
    (?composition-final + (0 + (?query + 0)))
    (case x of (composition-roots, composition-bs) ⇒

```

```

composition-final-stage A dg composition-bs >>=
  (λcomposition-final.
    record-staged-message composition-final >>=
      (λ-. staged-query-program A 0 rounds >>=
        (λquery-chunks.
          return
            (F composition-roots composition-bs composition-final
              query-chunks))))))
proof (cases x)
case (Pair composition-roots composition-bs)
have hash-range-budget
  (?composition-final + (0 + (?query + 0)))
  (composition-final-stage A dg composition-bs >>=
    (λcomposition-final.
      record-staged-message composition-final >>=
        (λ-. staged-query-program A 0 rounds >>=
          (λquery-chunks.
            return
              (F composition-roots composition-bs composition-final
                query-chunks))))))
by (rule after-composition-final-range)
then show ?thesis
unfolding Pair by simp
qed
fix dg F x
show hash-collision-budget
  (?composition-final + (0 + (?query + 0)))
  (case x of (composition-roots, composition-bs) =>
    composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 rounds >>=
            (λquery-chunks.
              return
                (F composition-roots composition-bs composition-final
                  query-chunks))))))
proof (cases x)
case (Pair composition-roots composition-bs)
have hash-collision-budget
  (?composition-final + (0 + (?query + 0)))
  (composition-final-stage A dg composition-bs >>=
    (λcomposition-final.
      record-staged-message composition-final >>=
        (λ-. staged-query-program A 0 rounds >>=
          (λquery-chunks.
            return
              (F composition-roots composition-bs composition-final
                query-chunks))))))
by (rule after-composition-final-coll)

```

```

then show ?thesis
  unfolding Pair by simp
qed
qed
have after-degree-record-range:
   $\wedge dg F.$  hash-range-budget
  (0 +
    (?composition-fri +
      (?composition-final + (0 + (?query + 0)))))
  (record-staged-message dg  $\gg$ 
    ( $\lambda$ -. (assert (ceil-log (to-nat dg + 1)  $\leq$ 
      ceil-log (maxDegree + 1))  $\gg$ 
      ( $\lambda$ -. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))  $\gg$ 
      ( $\lambda$ (composition-roots, composition-bs).
        composition-final-stage A dg composition-bs  $\gg$ 
          ( $\lambda$ composition-final.
            record-staged-message composition-final  $\gg$ 
              ( $\lambda$ -. staged-query-program A 0 rounds  $\gg$ 
                ( $\lambda$ query-chunks.
                  return
                    (F composition-roots composition-bs composition-final
                      query-chunks)))))))
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule after-composition-fri-range)
have after-degree-record-coll:
   $\wedge dg F.$  hash-collision-budget
  (0 +
    (?composition-fri +
      (?composition-final + (0 + (?query + 0)))))
  (record-staged-message dg  $\gg$ 
    ( $\lambda$ -. (assert (ceil-log (to-nat dg + 1)  $\leq$ 
      ceil-log (maxDegree + 1))  $\gg$ 
      ( $\lambda$ -. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))  $\gg$ 
      ( $\lambda$ (composition-roots, composition-bs).
        composition-final-stage A dg composition-bs  $\gg$ 
          ( $\lambda$ composition-final.
            record-staged-message composition-final  $\gg$ 
              ( $\lambda$ -. staged-query-program A 0 rounds  $\gg$ 
                ( $\lambda$ query-chunks.
                  return
                    (F composition-roots composition-bs composition-final
                      query-chunks)))))))
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule hash-collision-budget-record-staged-message,
      rule after-composition-fri-range, rule after-composition-fri-coll)

```



```

have after-degree-range:
   $\wedge$ as F. hash-range-budget
    (?degree +
      (0 +
        (?composition-fri +
          (?composition-final + (0 + (?query + 0))))))
    (degree-stage A as  $\gg$ 
      ( $\lambda$ dg. record-staged-message dg  $\gg$ 
        ( $\lambda$ -. (assert (ceil-log (to-nat dg + 1)  $\leq$ 
          ceil-log (maxDegree + 1))  $\gg$ 
          ( $\lambda$ -. staged-composition-fri-program A dg 0
            (ceil-log (to-nat dg + 1)) []))  $\gg$ 
          ( $\lambda$ (composition-roots, composition-bs).
            composition-final-stage A dg composition-bs  $\gg$ 
              ( $\lambda$ composition-final.
                record-staged-message composition-final  $\gg$ 
                  ( $\lambda$ -. staged-query-program A 0 rounds  $\gg$ 
                    ( $\lambda$ query-chunks.
                      return
                        (F dg composition-roots composition-bs
                          composition-final query-chunks)))))))
    by (rule hash-range-budget-bind)
      (rule degree-range, rule after-degree-record-range)
have after-degree-coll:
   $\wedge$ as F. hash-collision-budget
    (?degree +
      (0 +
        (?composition-fri +
          (?composition-final + (0 + (?query + 0))))))
    (degree-stage A as  $\gg$ 
      ( $\lambda$ dg. record-staged-message dg  $\gg$ 
        ( $\lambda$ -. (assert (ceil-log (to-nat dg + 1)  $\leq$ 
          ceil-log (maxDegree + 1))  $\gg$ 
          ( $\lambda$ -. staged-composition-fri-program A dg 0
            (ceil-log (to-nat dg + 1)) []))  $\gg$ 
          ( $\lambda$ (composition-roots, composition-bs).
            composition-final-stage A dg composition-bs  $\gg$ 
              ( $\lambda$ composition-final.
                record-staged-message composition-final  $\gg$ 
                  ( $\lambda$ -. staged-query-program A 0 rounds  $\gg$ 
                    ( $\lambda$ query-chunks.
                      return
                        (F dg composition-roots composition-bs
                          composition-final query-chunks)))))))
    by (rule hash-collision-budget-bind)
      (rule degree-range, rule degree-coll, rule after-degree-record-range,
        rule after-degree-record-coll)
have after-alpha-range:
   $\wedge$ F. hash-range-budget

```

```

(?alpha +
  (?degree +
    (0 +
      (?composition-fri +
        (?composition-final + (0 + (?query + 0)))))))
(staged-alpha-program (length spec) >>=
  (λas. degree-stage A as >>=
    (λdg. record-staged-message dg >>=
      (λ-. (assert (ceil-log (to-nat dg + 1) ≤
        ceil-log (maxDegree + 1)) >>=
        (λ-. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [])) >>=
        (λ(composition-roots, composition-bs).
          composition-final-stage A dg composition-bs >>=
            (λcomposition-final.
              record-staged-message composition-final >>=
                (λ-. staged-query-program A 0 rounds >>=
                  (λquery-chunks.
                    return
                      (F as dg composition-roots composition-bs
                        composition-final query-chunks))))))))
  by (rule hash-range-budget-bind)
  (rule alpha-range, rule after-degree-range)
have after-alpha-coll:
  ∧F. hash-collision-budget
  (?alpha +
    (?degree +
      (0 +
        (?composition-fri +
          (?composition-final + (0 + (?query + 0)))))))
  (staged-alpha-program (length spec) >>=
    (λas. degree-stage A as >>=
      (λdg. record-staged-message dg >>=
        (λ-. (assert (ceil-log (to-nat dg + 1) ≤
          ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0
              (ceil-log (to-nat dg + 1)) [])) >>=
              (λ(composition-roots, composition-bs).
                composition-final-stage A dg composition-bs >>=
                  (λcomposition-final.
                    record-staged-message composition-final >>=
                      (λ-. staged-query-program A 0 rounds >>=
                        (λquery-chunks.
                          return
                            (F as dg composition-roots composition-bs
                              composition-final query-chunks))))))))
    by (rule hash-collision-budget-bind)
    (rule alpha-range, rule alpha-coll, rule after-degree-range,
      rule after-degree-coll)

```

```

have after-trace-final-record-range:
   $\wedge$  trace-final  $F$ . hash-range-budget
    (0 +
      (?alpha +
        (?degree +
          (0 +
            (?composition-fri +
              (?composition-final + (0 + (?query + 0)))))))
      (record-staged-message trace-final  $\gg$ 
        ( $\lambda$ -. staged-alpha-program (length spec)  $\gg$ 
          ( $\lambda$ as. degree-stage  $A$  as  $\gg$ 
            ( $\lambda$ dg. record-staged-message dg  $\gg$ 
              ( $\lambda$ -. (assert (ceil-log (to-nat dg + 1)  $\leq$ 
                ceil-log (maxDegree + 1))  $\gg$ 
                ( $\lambda$ -. staged-composition-fri-program  $A$  dg 0
                  (ceil-log (to-nat dg + 1)) []))  $\gg$ 
                ( $\lambda$ (composition-roots, composition-bs).
                  composition-final-stage  $A$  dg composition-bs  $\gg$ 
                    ( $\lambda$ composition-final.
                      record-staged-message composition-final  $\gg$ 
                        ( $\lambda$ -. staged-query-program  $A$  0 rounds  $\gg$ 
                          ( $\lambda$ query-chunks.
                            return
                              (F as dg composition-roots composition-bs
                                composition-final query-chunks)))))))))
    )
  by (rule hash-range-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule after-alpha-range)
have after-trace-final-record-coll:
   $\wedge$  trace-final  $F$ . hash-collision-budget
    (0 +
      (?alpha +
        (?degree +
          (0 +
            (?composition-fri +
              (?composition-final + (0 + (?query + 0)))))))
      (record-staged-message trace-final  $\gg$ 
        ( $\lambda$ -. staged-alpha-program (length spec)  $\gg$ 
          ( $\lambda$ as. degree-stage  $A$  as  $\gg$ 
            ( $\lambda$ dg. record-staged-message dg  $\gg$ 
              ( $\lambda$ -. (assert (ceil-log (to-nat dg + 1)  $\leq$ 
                ceil-log (maxDegree + 1))  $\gg$ 
                ( $\lambda$ -. staged-composition-fri-program  $A$  dg 0
                  (ceil-log (to-nat dg + 1)) []))  $\gg$ 
                ( $\lambda$ (composition-roots, composition-bs).
                  composition-final-stage  $A$  dg composition-bs  $\gg$ 
                    ( $\lambda$ composition-final.
                      record-staged-message composition-final  $\gg$ 
                        ( $\lambda$ -. staged-query-program  $A$  0 rounds  $\gg$ 
                          ( $\lambda$ query-chunks.
                            return
                              (F as dg composition-roots composition-bs
                                composition-final query-chunks)))))))))
    )

```

```

      (λquery-chunks.
        return
          (F as dg composition-roots composition-bs
            composition-final query-chunks)))))))))
  by (rule hash-collision-budget-bind)
    (rule hash-range-budget-record-staged-message,
      rule hash-collision-budget-record-staged-message,
      rule after-alpha-range, rule after-alpha-coll)
  have after-trace-final-range:
    ∧ trace-bs F. hash-range-budget
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0))))))))))
      (trace-final-stage A trace-bs >=
        (λtrace-final. record-staged-message trace-final >=
          (λ-. staged-alpha-program (length spec) >=
            (λas. degree-stage A as >=
              (λdg. record-staged-message dg >=
                (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                  ceil-log (maxDegree + 1)) >=
                  (λ-. staged-composition-fri-program A dg 0
                    (ceil-log (to-nat dg + 1)) [])) >=
                  (λ(composition-roots, composition-bs).
                    composition-final-stage A dg composition-bs >=
                      (λcomposition-final.
                        record-staged-message composition-final >=
                          (λ-. staged-query-program A 0 rounds >=
                            (λquery-chunks.
                              return
                                (F trace-final as dg composition-roots
                                  composition-bs composition-final
                                    query-chunks)))))))))))))
  by (rule hash-range-budget-bind)
    (rule trace-final-range, rule after-trace-final-record-range)
  have after-trace-final-coll:
    ∧ trace-bs F. hash-collision-budget
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0))))))))))
      (trace-final-stage A trace-bs >=
        (λtrace-final. record-staged-message trace-final >=

```

```

(λ-. staged-alpha-program (length spec) >>=
  (λas. degree-stage A as >>=
    (λdg. record-staged-message dg >>=
      (λ-. (assert (ceil-log (to-nat dg + 1) ≤
        ceil-log (maxDegree + 1)) >>=
        (λ-. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [])) >>=
        (λ(composition-roots, composition-bs).
          composition-final-stage A dg composition-bs >>=
            (λcomposition-final.
              record-staged-message composition-final >>=
                (λ-. staged-query-program A 0 rounds >>=
                  (λquery-chunks.
                    return
                      (F trace-final as dg composition-roots
                        composition-bs composition-final
                          query-chunks))))))))))
by (rule hash-collision-budget-bind)
  (rule trace-final-range, rule trace-final-coll,
    rule after-trace-final-record-range,
    rule after-trace-final-record-coll)
have after-trace-fri-range:
  ∧F. hash-range-budget
  ( ?trace-fri +
    ( ?trace-final +
      (0 +
        ( ?alpha +
          ( ?degree +
            (0 +
              ( ?composition-fri +
                ( ?composition-final + (0 + ( ?query + 0))))))))))
  (staged-trace-fri-program A 0 (ceil-log clength) [] >>=
    (λ(trace-roots, trace-bs).
      trace-final-stage A trace-bs >>=
        (λtrace-final. record-staged-message trace-final >>=
          (λ-. staged-alpha-program (length spec) >>=
            (λas. degree-stage A as >>=
              (λdg. record-staged-message dg >>=
                (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                  ceil-log (maxDegree + 1)) >>=
                  (λ-. staged-composition-fri-program A dg 0
                    (ceil-log (to-nat dg + 1)) [])) >>=
                  (λ(composition-roots, composition-bs).
                    composition-final-stage A dg composition-bs >>=
                      (λcomposition-final.
                        record-staged-message composition-final >>=
                          (λ-. staged-query-program A 0 rounds >>=
                            (λquery-chunks.
                              return
                                (F trace-final as dg composition-roots
                                  composition-bs composition-final
                                    query-chunks))))))))))

```

```

(F trace-roots trace-bs trace-final as dg
 composition-roots composition-bs
 composition-final query-chunks))))))))))
proof (rule hash-range-budget-bind)
  fix F
  show hash-range-budget ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])
  by (rule trace-fri-range)
  fix F x
  show hash-range-budget
    (?trace-final +
      (0 +
        (?alpha +
          (?degree +
            (0 +
              (?composition-fri +
                (?composition-final + (0 + (?query + 0))))))))))
    (case x of (trace-roots, trace-bs) =>
      trace-final-stage A trace-bs >=>
      (λtrace-final. record-staged-message trace-final >=>
        (λ-. staged-alpha-program (length spec) >=>
          (λas. degree-stage A as >=>
            (λdg. record-staged-message dg >=>
              (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                ceil-log (maxDegree + 1)) >=>
                (λ-. staged-composition-fri-program A dg 0
                  (ceil-log (to-nat dg + 1) [])) >=>
                (λ(composition-roots, composition-bs).
                  composition-final-stage A dg composition-bs >=>
                    (λcomposition-final.
                      record-staged-message composition-final >=>
                        (λ-. staged-query-program A 0 rounds >=>
                          (λquery-chunks.
                            return
                              (F trace-roots trace-bs trace-final as dg
                                composition-roots composition-bs
                                composition-final query-chunks))))))))))
proof (cases x)
  case (Pair trace-roots trace-bs)
  have hash-range-budget
    (?trace-final +
      (0 +
        (?alpha +
          (?degree +
            (0 +
              (?composition-fri +
                (?composition-final + (0 + (?query + 0))))))))))
    (trace-final-stage A trace-bs >=>
      (λtrace-final. record-staged-message trace-final >=>

```

```

(λ-. staged-alpha-program (length spec) >>=
  (λas. degree-stage A as >>=
    (λdg. record-staged-message dg >>=
      (λ-. (assert (ceil-log (to-nat dg + 1)) ≤
        ceil-log (maxDegree + 1)) >>=
        (λ-. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [])) >>=
        (λ(composition-roots, composition-bs).
          composition-final-stage A dg composition-bs >>=
            (λcomposition-final.
              record-staged-message composition-final >>=
                (λ-. staged-query-program A 0 rounds >>=
                  (λquery-chunks.
                    return
                      (F trace-roots trace-bs trace-final as dg
                        composition-roots composition-bs
                        composition-final query-chunks))))))))))
  by (rule after-trace-final-range)
  then show ?thesis
  unfolding Pair by simp
qed
qed
have after-trace-fri-coll:
  ∧F. hash-collision-budget
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0))))))))))
    (staged-trace-fri-program A 0 (ceil-log clength) [] >>=
      (λ(trace-roots, trace-bs).
        trace-final-stage A trace-bs >>=
          (λtrace-final. record-staged-message trace-final >>=
            (λ-. staged-alpha-program (length spec) >>=
              (λas. degree-stage A as >>=
                (λdg. record-staged-message dg >>=
                  (λ-. (assert (ceil-log (to-nat dg + 1)) ≤
                    ceil-log (maxDegree + 1)) >>=
                    (λ-. staged-composition-fri-program A dg 0
                      (ceil-log (to-nat dg + 1)) [])) >>=
                    (λ(composition-roots, composition-bs).
                      composition-final-stage A dg composition-bs >>=
                        (λcomposition-final.
                          record-staged-message composition-final >>=
                            (λ-. staged-query-program A 0 rounds >>=
                              (λquery-chunks.

```

```

      return
      (F trace-roots trace-bs trace-final as dg
       composition-roots composition-bs
       composition-final query-chunks)))))))))
proof (rule hash-collision-budget-bind)
  fix F
  show hash-range-budget ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule trace-fri-range)
  fix F
  show hash-collision-budget ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule trace-fri-coll)
  fix F x
  show hash-range-budget
    (?trace-final +
     (0 +
      (?alpha +
       (?degree +
        (0 +
         (?composition-fri +
          (?composition-final + (0 + (?query + 0))))))))))
    (case x of (trace-roots, trace-bs) =>
     trace-final-stage A trace-bs >=>
     (λtrace-final. record-staged-message trace-final >=>
      (λ-. staged-alpha-program (length spec) >=>
       (λas. degree-stage A as >=>
        (λdg. record-staged-message dg >=>
         (λ-. (assert (ceil-log (to-nat dg + 1)) ≤
                    ceil-log (maxDegree + 1)) >=>
          (λ-. staged-composition-fri-program A dg 0
              (ceil-log (to-nat dg + 1)) [])) >=>
          (λ(composition-roots, composition-bs).
            composition-final-stage A dg composition-bs >=>
            (λcomposition-final.
              record-staged-message composition-final >=>
              (λ-. staged-query-program A 0 rounds >=>
               (λquery-chunks.
                 return
                 (F trace-roots trace-bs trace-final as dg
                  composition-roots composition-bs
                  composition-final query-chunks)))))))))
proof (cases x)
  case (Pair trace-roots trace-bs)
  have hash-range-budget
    (?trace-final +
     (0 +
      (?alpha +
       (?degree +

```



```

(0 +
  (?composition-fri +
    (?composition-final + (0 + (?query + 0)))))))))
(trace-final-stage A trace-bs >=>
  (λtrace-final. record-staged-message trace-final >=>
    (λ-. staged-alpha-program (length spec) >=>
      (λas. degree-stage A as >=>
        (λdg. record-staged-message dg >=>
          (λ-. (assert (ceil-log (to-nat dg + 1) ≤
            ceil-log (maxDegree + 1)) >=>
              (λ-. staged-composition-fri-program A dg 0
                (ceil-log (to-nat dg + 1)) [])) >=>
                (λ(composition-roots, composition-bs).
                  composition-final-stage A dg composition-bs >=>
                    (λcomposition-final.
                      record-staged-message composition-final >=>
                        (λ-. staged-query-program A 0 rounds >=>
                          (λquery-chunks.
                            return
                              (F trace-roots trace-bs trace-final as dg
                                composition-roots composition-bs
                                composition-final query-chunks)))))))))))))
  by (rule after-trace-final-range)
then show ?thesis
  unfolding Pair by simp
qed
fix F x
show hash-collision-budget
  (?trace-final +
    (0 +
      (?alpha +
        (?degree +
          (0 +
            (?composition-fri +
              (?composition-final + (0 + (?query + 0)))))))))
  (case x of (trace-roots, trace-bs) =>
    trace-final-stage A trace-bs >=>
      (λtrace-final. record-staged-message trace-final >=>
        (λ-. staged-alpha-program (length spec) >=>
          (λas. degree-stage A as >=>
            (λdg. record-staged-message dg >=>
              (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                ceil-log (maxDegree + 1)) >=>
                  (λ-. staged-composition-fri-program A dg 0
                    (ceil-log (to-nat dg + 1)) [])) >=>
                    (λ(composition-roots, composition-bs).
                      composition-final-stage A dg composition-bs >=>
                        (λcomposition-final.
                          record-staged-message composition-final >=>

```

```

      (λ-. staged-query-program A 0 rounds >>=
      (λquery-chunks.
      return
      (F trace-roots trace-bs trace-final as dg
      composition-roots composition-bs
      composition-final query-chunks)))))))))
proof (cases x)
case (Pair trace-roots trace-bs)
have hash-collision-budget
  (?trace-final +
  (0 +
  (?alpha +
  (?degree +
  (0 +
  (?composition-fri +
  (?composition-final + (0 + (?query + 0))))))))))
  (trace-final-stage A trace-bs >>=
  (λtrace-final. record-staged-message trace-final >>=
  (λ-. staged-alpha-program (length spec) >>=
  (λas. degree-stage A as >>=
  (λdg. record-staged-message dg >>=
  (λ-. (assert (ceil-log (to-nat dg + 1)) ≤
  ceil-log (maxDegree + 1)) >>=
  (λ-. staged-composition-fri-program A dg 0
  (ceil-log (to-nat dg + 1)) [])) >>=
  (λ(composition-roots, composition-bs).
  composition-final-stage A dg composition-bs >>=
  (λcomposition-final.
  record-staged-message composition-final >>=
  (λ-. staged-query-program A 0 rounds >>=
  (λquery-chunks.
  return
  (F trace-roots trace-bs trace-final as dg
  composition-roots composition-bs
  composition-final query-chunks)))))))))
  by (rule after-trace-final-coll)
then show ?thesis
unfolding Pair by simp
qed
qed
have after-trace-root-record-range:
  ∧fr F. hash-range-budget
  (0 +
  (?trace-fri +
  (?trace-final +
  (0 +
  (?alpha +
  (?degree +
  (0 +

```

```

      (?composition-fri +
        (?composition-final + (0 + (?query + 0)))))))))
(record-staged-message fr >=
  (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] >=
    (λ(trace-roots, trace-bs).
      trace-final-stage A trace-bs >=
        (λtrace-final. record-staged-message trace-final >=
          (λ-. staged-alpha-program (length spec) >=
            (λas. degree-stage A as >=
              (λdg. record-staged-message dg >=
                (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                  ceil-log (maxDegree + 1)) >=
                    (λ-. staged-composition-fri-program A dg 0
                      (ceil-log (to-nat dg + 1) [])) >=
                        (λ(composition-roots, composition-bs).
                          composition-final-stage A dg composition-bs >=
                            (λcomposition-final.
                              record-staged-message composition-final >=
                                (λ-. staged-query-program A 0 rounds >=
                                  (λquery-chunks.
                                    return
                                      (F trace-roots trace-bs trace-final as dg
                                        composition-roots composition-bs
                                        composition-final query-chunks)))))))))))))
by (rule hash-range-budget-bind)
  (rule hash-range-budget-record-staged-message,
    rule after-trace-fri-range)
have after-trace-root-record-coll:
  ∧fr F. hash-collision-budget
  (0 +
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0)))))))))
    (record-staged-message fr >=
      (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] >=
        (λ(trace-roots, trace-bs).
          trace-final-stage A trace-bs >=
            (λtrace-final. record-staged-message trace-final >=
              (λ-. staged-alpha-program (length spec) >=
                (λas. degree-stage A as >=
                  (λdg. record-staged-message dg >=
                    (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                      ceil-log (maxDegree + 1)) >=
                        (λ-. staged-composition-fri-program A dg 0
                          (ceil-log (to-nat dg + 1) [])) >=
                            (λ(composition-roots, composition-bs).
                              composition-final-stage A dg composition-bs >=
                                (λcomposition-final.
                                  record-staged-message composition-final >=
                                    (λ-. staged-query-program A 0 rounds >=
                                      (λquery-chunks.
                                        return
                                          (F trace-roots trace-bs trace-final as dg
                                            composition-roots composition-bs
                                            composition-final query-chunks)))))))))))))

```

```

      (ceil-log (to-nat dg + 1)) []) >=>
      (λ(composition-roots, composition-bs).
        composition-final-stage A dg composition-bs >=>
        (λcomposition-final.
          record-staged-message composition-final >=>
          (λ-. staged-query-program A 0 rounds >=>
            (λquery-chunks.
              return
              (F trace-roots trace-bs trace-final as dg
                composition-roots composition-bs
                composition-final query-chunks))))))))))
by (rule hash-collision-budget-bind)
      (rule hash-range-budget-record-staged-message,
        rule hash-collision-budget-record-staged-message,
        rule after-trace-fri-range, rule after-trace-fri-coll)
have whole:
  hash-collision-budget
    (?trace-root +
      (0 +
        (?trace-fri +
          (?trace-final +
            (0 +
              (?alpha +
                (?degree +
                  (0 +
                    (?composition-fri +
                      (?composition-final + (0 + (?query + 0))))))))))
            (staged-transcript-program A)
          unfolding staged-transcript-program-def Let-def
        proof (rule hash-collision-budget-bind)
        show hash-range-budget ?trace-root (trace-root-stage A)
          by (rule trace-root-range)
        show hash-collision-budget ?trace-root (trace-root-stage A)
          by (rule trace-root-coll)
        fix fr
        show hash-range-budget
          (0 +
            (?trace-fri +
              (?trace-final +
                (0 +
                  (?alpha +
                    (?degree +
                      (0 +
                        (?composition-fri +
                          (?composition-final + (0 + (?query + 0))))))))
                (record-staged-message fr >=>
                  (λ-. staged-trace-fri-program A 0 (ceil-log clength)) [] >=>
                    (λ(trace-roots, trace-bs).
                      trace-final-stage A trace-bs >=>

```

```

( $\lambda$ trace-final.
  record-staged-message trace-final  $\gg$ 
  ( $\lambda$ -. staged-alpha-program (length spec)  $\gg$ 
    ( $\lambda$ as. degree-stage A as  $\gg$ 
      ( $\lambda$ dg. record-staged-message dg  $\gg$ 
        ( $\lambda$ -. assert (ceil-log (to-nat dg + 1)  $\leq$ 
          ceil-log (maxDegree + 1))  $\gg$ 
          ( $\lambda$ -. staged-composition-fri-program A dg 0
            (ceil-log (to-nat dg + 1))  $\square$   $\gg$ 
            ( $\lambda$ (composition-roots, composition-bs).
              composition-final-stage A dg composition-bs  $\gg$ 
              ( $\lambda$ composition-final.
                record-staged-message
                composition-final  $\gg$ 
                ( $\lambda$ -. staged-query-program A 0
                  rounds  $\gg$ 
                  ( $\lambda$ query-chunks.
                    return
                    ( $\square$ staged-trace-root = fr,
                      staged-trace-fri-roots =
                      trace-roots,
                      staged-trace-fri-challenges =
                      trace-bs,
                      staged-trace-final =
                      trace-final,
                      staged-alphas = as,
                      staged-degree = dg,
                      staged-composition-fri-roots =
                      composition-roots,
                      staged-composition-fri-challenges =
                      composition-bs,
                      staged-composition-final =
                      composition-final,
                      staged-query-chunks =
                      query-chunks)))))))))))))
  )
)

using after-trace-root-record-range
[of fr
   $\lambda$ trace-roots trace-bs trace-final as dg composition-roots
    composition-bs composition-final query-chunks.
    ( $\square$ staged-trace-root = fr,
      staged-trace-fri-roots = trace-roots,
      staged-trace-fri-challenges = trace-bs,
      staged-trace-final = trace-final,
      staged-alphas = as,
      staged-degree = dg,
      staged-composition-fri-roots = composition-roots,
      staged-composition-fri-challenges = composition-bs,
      staged-composition-final = composition-final,
      staged-query-chunks = query-chunks)]

```

```

    by (simp add: sm-bind-assoc)
  fix fr
  show hash-collision-budget
    (0 +
      (?trace-fri +
        (?trace-final +
          (0 +
            (?alpha +
              (?degree +
                (0 +
                  (?composition-fri +
                    (?composition-final + (0 + (?query + 0))))))))))
    (record-staged-message fr >=
      (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] >=
        (λ(trace-roots, trace-bs).
          trace-final-stage A trace-bs >=
            (λtrace-final.
              record-staged-message trace-final >=
                (λ-. staged-alpha-program (length spec) >=
                  (λas. degree-stage A as >=
                    (λdg. record-staged-message dg >=
                      (λ-. assert (ceil-log (to-nat dg + 1) ≤
                        ceil-log (maxDegree + 1)) >=
                        (λ-. staged-composition-fri-program A dg 0
                          (ceil-log (to-nat dg + 1)) [] >=
                            (λ(composition-roots, composition-bs).
                              composition-final-stage A dg composition-bs >=
                                (λcomposition-final.
                                  record-staged-message
                                    composition-final >=
                                      (λ-. staged-query-program A 0
                                        rounds >=
                                          (λquery-chunks.
                                            return
                                              (|staged-trace-root = fr,
                                                staged-trace-fri-roots =
                                                  trace-roots,
                                                staged-trace-fri-challenges =
                                                  trace-bs,
                                                staged-trace-final =
                                                  trace-final,
                                                staged-alphas = as,
                                                staged-degree = dg,
                                                staged-composition-fri-roots =
                                                  composition-roots,
                                                staged-composition-fri-challenges =
                                                  composition-bs,
                                                staged-composition-final =
                                                  composition-final,

```

```

                                staged-query-chunks =
                                query-chunks))))))))))
using after-trace-root-record-coll
  [of fr
    λtrace-roots trace-bs trace-final as dg composition-roots
      composition-bs composition-final query-chunks.
    (|staged-trace-root = fr,
      staged-trace-fri-roots = trace-roots,
      staged-trace-fri-challenges = trace-bs,
      staged-trace-final = trace-final,
      staged-alphas = as,
      staged-degree = dg,
      staged-composition-fri-roots = composition-roots,
      staged-composition-fri-challenges = composition-bs,
      staged-composition-final = composition-final,
      staged-query-chunks = query-chunks|)]
  by (simp add: sm-bind-assoc)
qed
have budget-eq:
  ?trace-root +
  (0 +
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0)))))))))) =
    staged-attacker-query-budget budgets + staged-challenge-query-budget
unfolding staged-attacker-query-budget-def
  staged-challenge-query-budget-def
by (simp add: add.assoc add.commute add.left-commute)
show ?thesis
  using whole unfolding budget-eq .
qed

lemma hash-collision-budget-staged-semantic-adversary:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    hash-collision-budget
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    (staged-semantic-adversary A)
proof -
  have hash-collision-budget
    ((staged-attacker-query-budget budgets + staged-challenge-query-budget) +
     0)
    (staged-semantic-adversary A)

```

```

unfolding staged-semantic-adversary-def
proof (rule hash-collision-budget-bind
  [where  $n = \text{staged-attacker-query-budget budgets} +$ 
     $\text{staged-challenge-query-budget}$  and  $n' = 0$ ])
show hash-range-budget
  (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  (staged-transcript-program A)
  by (rule hash-range-budget-staged-transcript-program[OF wf controlled])
show hash-collision-budget
  (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  (staged-transcript-program A)
  by (rule hash-collision-budget-staged-transcript-program
    [OF wf controlled])
show  $\bigwedge x. \text{hash-range-budget } 0 \text{ (return (staged-proof-transcript } x))$ 
  by (rule hash-range-budget-return)
show  $\bigwedge x. \text{hash-collision-budget } 0 \text{ (return (staged-proof-transcript } x))$ 
  by (rule hash-collision-budget-return)
qed
then show ?thesis by simp
qed

lemma hash-target-program-staged-transcript-program:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  hash-target-program B
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    (staged-transcript-program A)
proof –
let ?trace-root = trace-root-budget budgets
let ?trace-fri =
  sum-list (trace-fri-budgets budgets) + ceil-log clength
let ?trace-final = trace-final-budget budgets
let ?alpha = length spec
let ?degree = degree-budget budgets
let ?composition-fri =
  sum-list (composition-fri-budgets budgets) +
  ceil-log (maxDegree + 1)
let ?composition-final = composition-final-budget budgets
let ?query =
  sum-list (query-opening-budgets budgets) + rounds
have trace-root-target:
  hash-target-program B ?trace-root (trace-root-stage A)
using controlled unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-target)
have trace-fri-target:
  hash-target-program B ?trace-fri
  (staged-trace-fri-program A 0 (ceil-log clength) [])

```



```

proof –
  have len:
     $0 + \text{ceil-log } \text{clength} \leq \text{length } (\text{trace-fri-budgets } \text{budgets})$ 
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-target-program B
      (sum-list
        (take (ceil-log clength)
          (drop 0 (trace-fri-budgets budgets))) +
        ceil-log clength)
      (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule hash-target-program-staged-trace-fri-program
      [OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have trace-final-target:
   $\wedge bs. \text{hash-target-program } B \text{ ?trace-final } (\text{trace-final-stage } A \text{ } bs)$ 
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-target)
have alpha-target:
  hash-target-program B ?alpha (staged-alpha-program (length spec))
  by (rule hash-target-program-staged-alpha-program)
have degree-target:
   $\wedge as. \text{hash-target-program } B \text{ ?degree } (\text{degree-stage } A \text{ } as)$ 
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-target)
have composition-fri-target:
   $\wedge dg. \text{hash-target-program } B \text{ ?composition-fri}$ 
  (assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
    ( $\lambda-. \text{staged-composition-fri-program } A \text{ } dg \text{ } 0$ 
      (ceil-log (to-nat dg + 1)) []))
  by (rule hash-target-program-guarded-staged-composition-fri-program
    [OF wf controlled])
have composition-final-target:
   $\wedge dg \text{ } bs. \text{hash-target-program } B \text{ ?composition-final}$ 
  (composition-final-stage A dg bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-target)
have query-target:
  hash-target-program B ?query (staged-query-program A 0 rounds)
proof –
  have len:
     $0 + \text{rounds} \leq \text{length } (\text{query-opening-budgets } \text{budgets})$ 
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-target-program B
      (sum-list
        (take rounds (drop 0 (query-opening-budgets budgets))) +

```

```

    rounds)
  (staged-query-program A 0 rounds)
  by (rule hash-target-program-staged-query-program[OF controlled len])
then show ?thesis
  using wf unfolding staged-budget-wellformed-def by simp
qed
have query-tail:
   $\wedge F.$  hash-target-program B ( $?query + 0$ )
  (staged-query-program A 0 rounds  $\gg=$ 
    ( $\lambda$ query-chunks. return (F query-chunks)))
  by (rule hash-target-program-bind)
  (rule query-target, rule hash-target-program-return)
have after-composition-final-record:
   $\wedge$ composition-final F. hash-target-program B ( $0 + (?query + 0)$ )
  (record-staged-message composition-final  $\gg=$ 
    ( $\lambda$ -. staged-query-program A 0 rounds  $\gg=$ 
      ( $\lambda$ query-chunks. return (F query-chunks))))
  by (rule hash-target-program-bind)
  (rule hash-target-program-record-staged-message, rule query-tail)
have after-composition-final:
   $\wedge$ dg bs F. hash-target-program B
  ( $?composition$ -final + ( $0 + (?query + 0)$ ))
  (composition-final-stage A dg bs  $\gg=$ 
    ( $\lambda$ composition-final.
      record-staged-message composition-final  $\gg=$ 
        ( $\lambda$ -. staged-query-program A 0 rounds  $\gg=$ 
          ( $\lambda$ query-chunks.
            return (F composition-final query-chunks))))))
  by (rule hash-target-program-bind)
  (rule composition-final-target, rule after-composition-final-record)
have after-composition-fri:
   $\wedge$ dg F. hash-target-program B
  ( $?composition$ -fri +
    ( $?composition$ -final + ( $0 + (?query + 0)$ )))
  ((assert (ceil-log (to-nat dg + 1)  $\leq$ 
    ceil-log (maxDegree + 1))  $\gg=$ 
    ( $\lambda$ -. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))  $\gg=$ 
    ( $\lambda$ (composition-roots, composition-bs).
      composition-final-stage A dg composition-bs  $\gg=$ 
        ( $\lambda$ composition-final.
          record-staged-message composition-final  $\gg=$ 
            ( $\lambda$ -. staged-query-program A 0 rounds  $\gg=$ 
              ( $\lambda$ query-chunks.
                return
                  (F composition-roots composition-bs composition-final
                    query-chunks)))))))
proof (rule hash-target-program-bind)
fix dg F

```

```

show hash-target-program B ?composition-fri
  (assert (ceil-log (to-nat dg + 1) ≤
    ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
  by (rule composition-fri-target)
fix dg F x
show hash-target-program B
  (?composition-final + (0 + (?query + 0)))
  (case x of (composition-roots, composition-bs) ⇒
    composition-final-stage A dg composition-bs >>=
    (λcomposition-final.
      record-staged-message composition-final >>=
      (λ-. staged-query-program A 0 rounds >>=
        (λquery-chunks.
          return
            (F composition-roots composition-bs composition-final
              query-chunks))))))
proof (cases x)
case (Pair composition-roots composition-bs)
have hash-target-program B
  (?composition-final + (0 + (?query + 0)))
  (composition-final-stage A dg composition-bs >>=
  (λcomposition-final.
    record-staged-message composition-final >>=
    (λ-. staged-query-program A 0 rounds >>=
      (λquery-chunks.
        return
          (F composition-roots composition-bs composition-final
            query-chunks))))))
  by (rule after-composition-final)
then show ?thesis
  unfolding Pair by simp
qed
qed
have after-degree-record:
  ∧ dg F. hash-target-program B
  (0 +
    (?composition-fri +
      (?composition-final + (0 + (?query + 0)))))
  (record-staged-message dg >>=
  (λ-. (assert (ceil-log (to-nat dg + 1) ≤
    ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) [])) >>=
    (λ(composition-roots, composition-bs).
      composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=

```

```

    (λ-. staged-query-program A 0 rounds >>=
      (λquery-chunks.
        return
          (F composition-roots composition-bs composition-final
            query-chunks))))))
  by (rule hash-target-program-bind)
    (rule hash-target-program-record-staged-message, rule after-composition-fri)
have after-degree:
  ∧as F. hash-target-program B
    (?degree +
      (0 +
        (?composition-fri +
          (?composition-final + (0 + (?query + 0)))))))
    (degree-stage A as >>=
      (λdg. record-staged-message dg >>=
        (λ-. (assert (ceil-log (to-nat dg + 1) ≤
          ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0
              (ceil-log (to-nat dg + 1)) [])) >>=
              (λ(composition-roots, composition-bs).
                composition-final-stage A dg composition-bs >>=
                  (λcomposition-final.
                    record-staged-message composition-final >>=
                      (λ-. staged-query-program A 0 rounds >>=
                        (λquery-chunks.
                          return
                            (F dg composition-roots composition-bs
                              composition-final query-chunks))))))))))
    by (rule hash-target-program-bind)
      (rule degree-target, rule after-degree-record)
have after-alpha:
  ∧F. hash-target-program B
    (?alpha +
      (?degree +
        (0 +
          (?composition-fri +
            (?composition-final + (0 + (?query + 0)))))))
    (staged-alpha-program (length spec) >>=
      (λas. degree-stage A as >>=
        (λdg. record-staged-message dg >>=
          (λ-. (assert (ceil-log (to-nat dg + 1) ≤
            ceil-log (maxDegree + 1)) >>=
              (λ-. staged-composition-fri-program A dg 0
                (ceil-log (to-nat dg + 1)) [])) >>=
                (λ(composition-roots, composition-bs).
                  composition-final-stage A dg composition-bs >>=
                    (λcomposition-final.
                      record-staged-message composition-final >>=
                        (λ-. staged-query-program A 0 rounds >>=
                          (λquery-chunks.
                            return
                              (F dg composition-roots composition-bs
                                composition-final query-chunks))))))))))

```

```

      (λquery-chunks.
        return
          (F as dg composition-roots composition-bs
            composition-final query-chunks)))))))))
  by (rule hash-target-program-bind)
    (rule alpha-target, rule after-degree)
have after-trace-final-record:
  ∧ trace-final F. hash-target-program B
  (0 +
    (?alpha +
      (?degree +
        (0 +
          (?composition-fri +
            (?composition-final + (0 + (?query + 0))))))))))
  (record-staged-message trace-final >=>
    (λ-. staged-alpha-program (length spec) >=>
      (λas. degree-stage A as >=>
        (λdg. record-staged-message dg >=>
          (λ-. (assert (ceil-log (to-nat dg + 1) ≤
            ceil-log (maxDegree + 1)) >=>
            (λ-. staged-composition-fri-program A dg 0
              (ceil-log (to-nat dg + 1)) [])) >=>
            (λ(composition-roots, composition-bs).
              composition-final-stage A dg composition-bs >=>
                (λcomposition-final.
                  record-staged-message composition-final >=>
                    (λ-. staged-query-program A 0 rounds >=>
                      (λquery-chunks.
                        return
                          (F as dg composition-roots composition-bs
                            composition-final query-chunks)))))))))))))
  by (rule hash-target-program-bind)
    (rule hash-target-program-record-staged-message, rule after-alpha)
have after-trace-final:
  ∧ trace-bs F. hash-target-program B
  (?trace-final +
    (0 +
      (?alpha +
        (?degree +
          (0 +
            (?composition-fri +
              (?composition-final + (0 + (?query + 0))))))))))
  (trace-final-stage A trace-bs >=>
    (λtrace-final. record-staged-message trace-final >=>
      (λ-. staged-alpha-program (length spec) >=>
        (λas. degree-stage A as >=>
          (λdg. record-staged-message dg >=>
            (λ-. (assert (ceil-log (to-nat dg + 1) ≤
              ceil-log (maxDegree + 1)) >=>

```

```

    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []) >=>
    (λ(composition-roots, composition-bs).
      composition-final-stage A dg composition-bs >=>
      (λcomposition-final.
        record-staged-message composition-final >=>
        (λ-. staged-query-program A 0 rounds >=>
          (λquery-chunks.
            return
              (F trace-final as dg composition-roots
                composition-bs composition-final
                query-chunks))))))))))
  by (rule hash-target-program-bind)
    (rule trace-final-target, rule after-trace-final-record)
have after-trace-fri:
  ∧F. hash-target-program B
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0))))))))))
    (staged-trace-fri-program A 0 (ceil-log clength) []) >=>
    (λ(trace-roots, trace-bs).
      trace-final-stage A trace-bs >=>
      (λtrace-final. record-staged-message trace-final >=>
        (λ-. staged-alpha-program (length spec) >=>
          (λas. degree-stage A as >=>
            (λdg. record-staged-message dg >=>
              (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                ceil-log (maxDegree + 1)) >=>
                (λ-. staged-composition-fri-program A dg 0
                  (ceil-log (to-nat dg + 1)) [])) >=>
                (λ(composition-roots, composition-bs).
                  composition-final-stage A dg composition-bs >=>
                  (λcomposition-final.
                    record-staged-message composition-final >=>
                    (λ-. staged-query-program A 0 rounds >=>
                      (λquery-chunks.
                        return
                          (F trace-roots trace-bs trace-final as dg
                            composition-roots composition-bs
                            composition-final query-chunks))))))))))
proof (rule hash-target-program-bind)
  fix F
  show hash-target-program B ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])

```

```

    by (rule trace-fri-target)
  fix F x
  show hash-target-program B
    (?trace-final +
      (0 +
        (?alpha +
          (?degree +
            (0 +
              (?composition-fri +
                (?composition-final + (0 + (?query + 0))))))))))
  (case x of (trace-roots, trace-bs) ⇒
    trace-final-stage A trace-bs ≫=
      (λtrace-final. record-staged-message trace-final ≫=
        (λ-. staged-alpha-program (length spec) ≫=
          (λas. degree-stage A as ≫=
            (λdg. record-staged-message dg ≫=
              (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                ceil-log (maxDegree + 1)) ≫=
                (λ-. staged-composition-fri-program A dg 0
                  (ceil-log (to-nat dg + 1)) [])) ≫=
                (λ(composition-roots, composition-bs).
                  composition-final-stage A dg composition-bs ≫=
                    (λcomposition-final.
                      record-staged-message composition-final ≫=
                        (λ-. staged-query-program A 0 rounds ≫=
                          (λquery-chunks.
                            return
                              (F trace-roots trace-bs trace-final as dg
                                composition-roots composition-bs
                                composition-final query-chunks))))))))))
  proof (cases x)
  case (Pair trace-roots trace-bs)
  have hash-target-program B
    (?trace-final +
      (0 +
        (?alpha +
          (?degree +
            (0 +
              (?composition-fri +
                (?composition-final + (0 + (?query + 0))))))))))
  (trace-final-stage A trace-bs ≫=
    (λtrace-final. record-staged-message trace-final ≫=
      (λ-. staged-alpha-program (length spec) ≫=
        (λas. degree-stage A as ≫=
          (λdg. record-staged-message dg ≫=
            (λ-. (assert (ceil-log (to-nat dg + 1) ≤
              ceil-log (maxDegree + 1)) ≫=
              (λ-. staged-composition-fri-program A dg 0
                (ceil-log (to-nat dg + 1)) [])) ≫=
              (λcomposition-roots, composition-bs).
                composition-final-stage A dg composition-bs ≫=
                  (λcomposition-final.
                    record-staged-message composition-final ≫=
                      (λ-. staged-query-program A 0 rounds ≫=
                        (λquery-chunks.
                          return
                            (F trace-roots trace-bs trace-final as dg
                              composition-roots composition-bs
                              composition-final query-chunks))))))))))

```

```

      (λ(composition-roots, composition-bs).
        composition-final-stage A dg composition-bs >>=
        (λcomposition-final.
          record-staged-message composition-final >>=
          (λ-. staged-query-program A 0 rounds >>=
            (λquery-chunks.
              return
              (F trace-roots trace-bs trace-final as dg
                composition-roots composition-bs
                composition-final query-chunks))))))))))
    by (rule after-trace-final)
  then show ?thesis
    unfolding Pair by simp
qed
qed
have after-trace-root-record:
  ∧fr F. hash-target-program B
  (0 +
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0))))))))))
  (record-staged-message fr >>=
    (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] >>=
      (λ(trace-roots, trace-bs).
        trace-final-stage A trace-bs >>=
        (λtrace-final. record-staged-message trace-final >>=
          (λ-. staged-alpha-program (length spec) >>=
            (λas. degree-stage A as >>=
              (λdg. record-staged-message dg >>=
                (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                  ceil-log (maxDegree + 1)) >>=
                  (λ-. staged-composition-fri-program A dg 0
                    (ceil-log (to-nat dg + 1) [])) >>=
                    (λ(composition-roots, composition-bs).
                      composition-final-stage A dg composition-bs >>=
                      (λcomposition-final.
                        record-staged-message composition-final >>=
                        (λ-. staged-query-program A 0 rounds >>=
                          (λquery-chunks.
                            return
                            (F trace-roots trace-bs trace-final as dg
                              composition-roots composition-bs
                              composition-final query-chunks))))))))))
      by (rule hash-target-program-bind)

```



```

(rule hash-target-program-record-staged-message, rule after-trace-fri)
have whole:
  hash-target-program B
  (?trace-root +
    (0 +
      (?trace-fri +
        (?trace-final +
          (0 +
            (?alpha +
              (?degree +
                (0 +
                  (?composition-fri +
                    (?composition-final + (0 + (?query + 0))))))))))
    (staged-transcript-program A)
  unfolding staged-transcript-program-def Let-def
proof (rule hash-target-program-bind)
show hash-target-program B ?trace-root (trace-root-stage A)
  by (rule trace-root-target)
fix fr
show hash-target-program B
  (0 +
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0))))))))))
  (record-staged-message fr >>=
    (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] >>=
      (λ(trace-roots, trace-bs).
        trace-final-stage A trace-bs >>=
          (λtrace-final.
            record-staged-message trace-final >>=
              (λ-. staged-alpha-program (length spec) >>=
                (λas. degree-stage A as >>=
                  (λdg. record-staged-message dg >>=
                    (λ-. assert (ceil-log (to-nat dg + 1)) ≤
                      ceil-log (maxDegree + 1)) >>=
                      (λ-. staged-composition-fri-program A dg 0
                        (ceil-log (to-nat dg + 1)) [] >>=
                          (λ(composition-roots, composition-bs).
                            composition-final-stage A dg composition-bs >>=
                              (λcomposition-final.
                                record-staged-message
                                  composition-final >>=
                                    (λ-. staged-query-program A 0
                                      rounds >>=

```

```

(λquery-chunks.
  return
    (λ(staged-trace-root = fr,
      staged-trace-fri-roots =
        trace-roots,
      staged-trace-fri-challenges =
        trace-bs,
      staged-trace-final =
        trace-final,
      staged-alphas = as,
      staged-degree = dg,
      staged-composition-fri-roots =
        composition-roots,
      staged-composition-fri-challenges =
        composition-bs,
      staged-composition-final =
        composition-final,
      staged-query-chunks =
        query-chunks)))))))))
using after-trace-root-record
  [of fr
    λtrace-roots trace-bs trace-final as dg composition-roots
      composition-bs composition-final query-chunks.
      (λ(staged-trace-root = fr,
        staged-trace-fri-roots = trace-roots,
        staged-trace-fri-challenges = trace-bs,
        staged-trace-final = trace-final,
        staged-alphas = as,
        staged-degree = dg,
        staged-composition-fri-roots = composition-roots,
        staged-composition-fri-challenges = composition-bs,
        staged-composition-final = composition-final,
        staged-query-chunks = query-chunks))]
    by (simp add: sm-bind-assoc)
qed
have budget-eq:
  ?trace-root +
    (0 +
      (?trace-fri +
        (?trace-final +
          (0 +
            (?alpha +
              (?degree +
                (0 +
                  (?composition-fri +
                    (?composition-final + (0 + (?query + 0)))))))))) =
    staged-attacker-query-budget budgets + staged-challenge-query-budget
unfolding staged-attacker-query-budget-def
  staged-challenge-query-budget-def

```

```

    by (simp add: add.assoc add.commute add.left-commute)
  show ?thesis
    using whole unfolding budget-eq .
qed

lemma hash-target-program-checked-staged-transcript-program:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    hash-target-program B
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (checked-staged-transcript-program A)
proof -
  let ?trace-root = trace-root-budget budgets
  let ?trace-fri =
    sum-list (trace-fri-budgets budgets) + ceil-log clength
  let ?trace-final = trace-final-budget budgets
  let ?alpha = length spec
  let ?degree = degree-budget budgets
  let ?composition-fri =
    sum-list (composition-fri-budgets budgets) +
    ceil-log (maxDegree + 1)
  let ?composition-final = composition-final-budget budgets
  let ?query =
    sum-list (query-opening-budgets budgets) + rounds
  have trace-root-target:
    hash-target-program B ?trace-root (trace-root-stage A)
    using controlled unfolding staged-adversary-controlled-def
    by (blast intro: controlled-ro-program-target)
  have trace-fri-target:
    hash-target-program B ?trace-fri
      (staged-trace-fri-program A 0 (ceil-log clength) [])
proof -
  have len:
    0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-target-program B
      (sum-list
        (take (ceil-log clength)
          (drop 0 (trace-fri-budgets budgets)))) +
      ceil-log clength
      (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule hash-target-program-staged-trace-fri-program
      [OF controlled len])
  then show ?thesis
    using wf unfolding staged-budget-wellformed-def by simp
qed
have trace-final-target:

```

```

 $\wedge$ bs. hash-target-program B ?trace-final (trace-final-stage A bs)
using controlled unfolding staged-adversary-controlled-def
by (blast intro: controlled-ro-program-target)
have alpha-target:
  hash-target-program B ?alpha (staged-alpha-program (length spec))
  by (rule hash-target-program-staged-alpha-program)
have degree-target:
   $\wedge$ as. hash-target-program B ?degree (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-target)
have composition-fri-target:
   $\wedge$ dg. hash-target-program B ?composition-fri
    (assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
      ( $\lambda$ -. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []))
  by (rule hash-target-program-guarded-staged-composition-fri-program
    [OF wf controlled])
have composition-final-target:
   $\wedge$ dg bs. hash-target-program B ?composition-final
    (composition-final-stage A dg bs)
  using controlled unfolding staged-adversary-controlled-def
  by (blast intro: controlled-ro-program-target)
have query-target:
   $\wedge$ trace-roots composition-roots.
    hash-target-program B ?query
    (checked-staged-query-program A trace-roots composition-roots
      0 rounds)
proof –
  fix trace-roots composition-roots
  have len:
    0 + rounds  $\leq$  length (query-opening-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-target-program B
      (sum-list
        (take rounds (drop 0 (query-opening-budgets budgets))) +
        rounds)
      (checked-staged-query-program A trace-roots composition-roots
        0 rounds)
    by (rule hash-target-program-checked-staged-query-program
      [OF controlled len])
  then show
    hash-target-program B ?query
      (checked-staged-query-program A trace-roots composition-roots
        0 rounds)
    using wf unfolding staged-budget-wellformed-def by simp
qed
have query-tail:
   $\wedge$ trace-roots composition-roots F. hash-target-program B (?query + 0)

```

```

    (checked-staged-query-program A trace-roots composition-roots
      0 rounds  $\gg$ 
      ( $\lambda$ query-chunks. return (F query-chunks)))
  by (rule hash-target-program-bind)
    (rule query-target, rule hash-target-program-return)
have after-composition-final-record:
 $\wedge$ trace-roots composition-roots composition-final F.
  hash-target-program B (0 + (?query + 0))
  (record-staged-message composition-final  $\gg$ 
    ( $\lambda$ -. checked-staged-query-program A trace-roots
      composition-roots 0 rounds  $\gg$ 
      ( $\lambda$ query-chunks. return (F query-chunks))))
  by (rule hash-target-program-bind)
    (rule hash-target-program-record-staged-message, rule query-tail)
have after-composition-final:
 $\wedge$ trace-roots dg composition-roots bs F. hash-target-program B
  (?composition-final + (0 + (?query + 0)))
  (composition-final-stage A dg bs  $\gg$ 
    ( $\lambda$ composition-final.
      record-staged-message composition-final  $\gg$ 
      ( $\lambda$ -. checked-staged-query-program A trace-roots
        composition-roots 0 rounds  $\gg$ 
        ( $\lambda$ query-chunks.
          return (F composition-final query-chunks))))))
  by (rule hash-target-program-bind)
    (rule composition-final-target, rule after-composition-final-record)
have after-composition-fri:
 $\wedge$ trace-roots dg F. hash-target-program B
  (?composition-fri +
    (?composition-final + (0 + (?query + 0))))
  ((assert (ceil-log (to-nat dg + 1)  $\leq$ 
    ceil-log (maxDegree + 1))  $\gg$ 
    ( $\lambda$ -. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))  $\gg$ 
    ( $\lambda$ (composition-roots, composition-bs).
      composition-final-stage A dg composition-bs  $\gg$ 
      ( $\lambda$ composition-final.
        record-staged-message composition-final  $\gg$ 
        ( $\lambda$ -. checked-staged-query-program A trace-roots
          composition-roots 0 rounds  $\gg$ 
          ( $\lambda$ query-chunks.
            return
              (F composition-roots composition-bs composition-final
                query-chunks)))))))
proof (rule hash-target-program-bind)
fix trace-roots dg F
show hash-target-program B ?composition-fri
  (assert (ceil-log (to-nat dg + 1)  $\leq$ 
    ceil-log (maxDegree + 1))  $\gg$ 
    )

```

```

    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))
  by (rule composition-fri-target)
next
fix trace-roots dg F x
show hash-target-program B
  (?composition-final + (0 + (?query + 0)))
  (case x of (composition-roots, composition-bs) ⇒
    composition-final-stage A dg composition-bs ≫=
      (λcomposition-final.
        record-staged-message composition-final ≫=
          (λ-. checked-staged-query-program A trace-roots
            composition-roots 0 rounds ≫=
              (λquery-chunks.
                return
                  (F composition-roots composition-bs composition-final
                    query-chunks))))))
proof (cases x)
case (Pair composition-roots composition-bs)
have hash-target-program B
  (?composition-final + (0 + (?query + 0)))
  (composition-final-stage A dg composition-bs ≫=
    (λcomposition-final.
      record-staged-message composition-final ≫=
        (λ-. checked-staged-query-program A trace-roots
          composition-roots 0 rounds ≫=
            (λquery-chunks.
              return
                (F composition-roots composition-bs composition-final
                  query-chunks))))))
  by (rule after-composition-final)
then show ?thesis
  unfolding Pair by simp
qed
qed
have after-degree-record:
  ∧ trace-roots dg F. hash-target-program B
  (0 +
    (?composition-fri +
      (?composition-final + (0 + (?query + 0)))))
  (record-staged-message dg ≫=
    (λ-. (assert (ceil-log (to-nat dg + 1) ≤
      ceil-log (maxDegree + 1)) ≫=
        (λ-. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [])) ≫=
      (λ(composition-roots, composition-bs).
        composition-final-stage A dg composition-bs ≫=
          (λcomposition-final.
            record-staged-message composition-final ≫=

```

```

      (λ-. checked-staged-query-program A trace-roots
        composition-roots 0 rounds >>=
        (λquery-chunks.
          return
            (F composition-roots composition-bs composition-final
              query-chunks))))))
  by (rule hash-target-program-bind)
    (rule hash-target-program-record-staged-message,
      rule after-composition-fri)
  have after-degree:
    ∧ trace-roots as F. hash-target-program B
      (?degree +
        (0 +
          (?composition-fri +
            (?composition-final + (0 + (?query + 0))))))
        (degree-stage A as >>=
          (λdg. record-staged-message dg >>=
            (λ-. (assert (ceil-log (to-nat dg + 1)) ≤
              ceil-log (maxDegree + 1)) >>=
              (λ-. staged-composition-fri-program A dg 0
                (ceil-log (to-nat dg + 1)) [])) >>=
              (λ(composition-roots, composition-bs).
                composition-final-stage A dg composition-bs >>=
                (λcomposition-final.
                  record-staged-message composition-final >>=
                  (λ-. checked-staged-query-program A trace-roots
                    composition-roots 0 rounds >>=
                    (λquery-chunks.
                      return
                        (F dg composition-roots composition-bs
                          composition-final query-chunks))))))))
            by (rule hash-target-program-bind)
              (rule degree-target, rule after-degree-record)
  have after-alpha:
    ∧ trace-roots F. hash-target-program B
      (?alpha +
        (?degree +
          (0 +
            (?composition-fri +
              (?composition-final + (0 + (?query + 0))))))
          (staged-alpha-program (length spec) >>=
            (λas. degree-stage A as >>=
              (λdg. record-staged-message dg >>=
                (λ-. (assert (ceil-log (to-nat dg + 1)) ≤
                  ceil-log (maxDegree + 1)) >>=
                  (λ-. staged-composition-fri-program A dg 0
                    (ceil-log (to-nat dg + 1)) [])) >>=
                  (λ(composition-roots, composition-bs).
                    composition-final-stage A dg composition-bs >>=

```

```

      (λcomposition-final.
        record-staged-message composition-final >>=
        (λ-. checked-staged-query-program A trace-roots
          composition-roots 0 rounds >>=
          (λquery-chunks.
            return
            (F as dg composition-roots composition-bs
              composition-final query-chunks)))))))))
  by (rule hash-target-program-bind)
    (rule alpha-target, rule after-degree)
have after-trace-final-record:
  ∧trace-roots trace-final F. hash-target-program B
  (0 +
    (?alpha +
      (?degree +
        (0 +
          (?composition-fri +
            (?composition-final + (0 + (?query + 0))))))))))
  (record-staged-message trace-final >>=
    (λ-. staged-alpha-program (length spec) >>=
      (λas. degree-stage A as >>=
        (λdg. record-staged-message dg >>=
          (λ-. (assert (ceil-log (to-nat dg + 1) ≤
            ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0
              (ceil-log (to-nat dg + 1)) [])) >>=
            (λ(composition-roots, composition-bs).
              composition-final-stage A dg composition-bs >>=
              (λcomposition-final.
                record-staged-message composition-final >>=
                (λ-. checked-staged-query-program A trace-roots
                  composition-roots 0 rounds >>=
                  (λquery-chunks.
                    return
                    (F as dg composition-roots composition-bs
                      composition-final query-chunks)))))))))))))
  by (rule hash-target-program-bind)
    (rule hash-target-program-record-staged-message, rule after-alpha)
have after-trace-final:
  ∧trace-roots trace-bs F. hash-target-program B
  (?trace-final +
    (0 +
      (?alpha +
        (?degree +
          (0 +
            (?composition-fri +
              (?composition-final + (0 + (?query + 0))))))))))
  (trace-final-stage A trace-bs >>=
    (λtrace-final. record-staged-message trace-final >>=

```



```

(F trace-roots trace-bs trace-final as dg
  composition-roots composition-bs
  composition-final query-chunks)))))))))
proof (rule hash-target-program-bind)
  fix F
  show hash-target-program B ?trace-fri
    (staged-trace-fri-program A 0 (ceil-log clength) [])
    by (rule trace-fri-target)
next
  fix F x
  show hash-target-program B
    (?trace-final +
      (0 +
        (?alpha +
          (?degree +
            (0 +
              (?composition-fri +
                (?composition-final + (0 + (?query + 0))))))))))
    (case x of (trace-roots, trace-bs) =>
      trace-final-stage A trace-bs >=>
      (λtrace-final. record-staged-message trace-final >=>
        (λ-. staged-alpha-program (length spec) >=>
          (λas. degree-stage A as >=>
            (λdg. record-staged-message dg >=>
              (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                ceil-log (maxDegree + 1)) >=>
                (λ-. staged-composition-fri-program A dg 0
                  (ceil-log (to-nat dg + 1) [])) >=>
                (λ(composition-roots, composition-bs).
                  composition-final-stage A dg composition-bs >=>
                    (λcomposition-final.
                      record-staged-message composition-final >=>
                        (λ-. checked-staged-query-program A
                          trace-roots composition-roots 0 rounds >=>
                            (λquery-chunks.
                              return
                                (F trace-roots trace-bs trace-final as dg
                                  composition-roots composition-bs
                                  composition-final query-chunks))))))))))
proof (cases x)
  case (Pair trace-roots trace-bs)
  have hash-target-program B
    (?trace-final +
      (0 +
        (?alpha +
          (?degree +
            (0 +
              (?composition-fri +
                (?composition-final + (0 + (?query + 0))))))))))

```

```

(trace-final-stage A trace-bs >>=
  (λtrace-final. record-staged-message trace-final >>=
    (λ-. staged-alpha-program (length spec) >>=
      (λas. degree-stage A as >>=
        (λdg. record-staged-message dg >>=
          (λ-. (assert (ceil-log (to-nat dg + 1) ≤
            ceil-log (maxDegree + 1)) >>=
              (λ-. staged-composition-fri-program A dg 0
                (ceil-log (to-nat dg + 1)) [])) >>=
                (λ(composition-roots, composition-bs).
                  composition-final-stage A dg composition-bs >>=
                    (λcomposition-final.
                      record-staged-message composition-final >>=
                        (λ-. checked-staged-query-program A
                          trace-roots composition-roots 0 rounds >>=
                            (λquery-chunks.
                              return
                                (F trace-roots trace-bs trace-final as dg
                                  composition-roots composition-bs
                                  composition-final query-chunks))))))))))
  by (rule after-trace-final)
then show ?thesis
  unfolding Pair by simp
qed
qed
have after-trace-root-record:
  ∧fr F. hash-target-program B
  (0 +
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0))))))))))
  (record-staged-message fr >>=
    (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] >>=
      (λ(trace-roots, trace-bs).
        trace-final-stage A trace-bs >>=
          (λtrace-final. record-staged-message trace-final >>=
            (λ-. staged-alpha-program (length spec) >>=
              (λas. degree-stage A as >>=
                (λdg. record-staged-message dg >>=
                  (λ-. (assert (ceil-log (to-nat dg + 1) ≤
                    ceil-log (maxDegree + 1)) >>=
                      (λ-. staged-composition-fri-program A dg 0
                        (ceil-log (to-nat dg + 1)) [])) >>=
                        (λ(composition-roots, composition-bs).

```

```

composition-final-stage A dg composition-bs ≫=
(λcomposition-final.
  record-staged-message composition-final ≫=
  (λ-. checked-staged-query-program A
    trace-roots composition-roots 0 rounds ≫=
    (λquery-chunks.
      return
      (F trace-roots trace-bs trace-final as dg
        composition-roots composition-bs
        composition-final query-chunks)))))))))
by (rule hash-target-program-bind)
  (rule hash-target-program-record-staged-message, rule after-trace-fri)
have whole:
  hash-target-program B
  (?trace-root +
    (0 +
      (?trace-fri +
        (?trace-final +
          (0 +
            (?alpha +
              (?degree +
                (0 +
                  (?composition-fri +
                    (?composition-final + (0 + (?query + 0)))))))))))
    (checked-staged-transcript-program A)
  unfolding checked-staged-transcript-program-def Let-def
proof (rule hash-target-program-bind)
  show hash-target-program B ?trace-root (trace-root-stage A)
  by (rule trace-root-target)
next
fix fr
show hash-target-program B
  (0 +
    (?trace-fri +
      (?trace-final +
        (0 +
          (?alpha +
            (?degree +
              (0 +
                (?composition-fri +
                  (?composition-final + (0 + (?query + 0))))))))))
  (record-staged-message fr ≫=
    (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] ≫=
      (λ(trace-roots, trace-bs).
        trace-final-stage A trace-bs ≫=
        (λtrace-final.
          record-staged-message trace-final ≫=
          (λ-. staged-alpha-program (length spec) ≫=
            (λas. degree-stage A as ≫=

```

```

(λdg. record-staged-message dg >>=
  (λ-. assert (ceil-log (to-nat dg + 1) ≤
    ceil-log (maxDegree + 1)) >>=
    (λ-. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) [] >>=
        (λ(composition-roots, composition-bs).
          composition-final-stage A dg composition-bs >>=
            (λcomposition-final.
              record-staged-message
                composition-final >>=
                  (λ-. checked-staged-query-program A
                    trace-roots composition-roots 0
                    rounds >>=
                      (λquery-chunks.
                        return
                          (λ(staged-trace-root = fr,
                            staged-trace-fri-roots =
                              trace-roots,
                            staged-trace-fri-challenges =
                              trace-bs,
                            staged-trace-final =
                              trace-final,
                            staged-alphas = as,
                            staged-degree = dg,
                            staged-composition-fri-roots =
                              composition-roots,
                            staged-composition-fri-challenges =
                              composition-bs,
                            staged-composition-final =
                              composition-final,
                            staged-query-chunks =
                              query-chunks)))))))))))))
using after-trace-root-record
[of fr
  λtrace-roots trace-bs trace-final as dg composition-roots
    composition-bs composition-final query-chunks.
    (λ(staged-trace-root = fr,
      staged-trace-fri-roots = trace-roots,
      staged-trace-fri-challenges = trace-bs,
      staged-trace-final = trace-final,
      staged-alphas = as,
      staged-degree = dg,
      staged-composition-fri-roots = composition-roots,
      staged-composition-fri-challenges = composition-bs,
      staged-composition-final = composition-final,
      staged-query-chunks = query-chunks))]
by (simp add: sm-bind-assoc)
qed
have budget-eq:

```

```

    ?trace-root +
      (0 +
        (?trace-fri +
          (?trace-final +
            (0 +
              (?alpha +
                (?degree +
                  (0 +
                    (?composition-fri +
                      (?composition-final + (0 + (?query + 0)))))))))) =
      staged-attacker-query-budget budgets + staged-challenge-query-budget
unfolding staged-attacker-query-budget-def
    staged-challenge-query-budget-def
by (simp add: add.assoc add.commute add.left-commute)
show ?thesis
    using whole unfolding budget-eq .
qed

lemma hash-target-program-staged-semantic-adversary:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    hash-target-program B
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (staged-semantic-adversary A)
proof -
  have hash-target-program B
    ((staged-attacker-query-budget budgets + staged-challenge-query-budget) +
     0)
    (staged-semantic-adversary A)
  unfolding staged-semantic-adversary-def
proof (rule hash-target-program-bind
  [where n=staged-attacker-query-budget budgets +
    staged-challenge-query-budget and n'=0])
  show hash-target-program B
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    (staged-transcript-program A)
    by (rule hash-target-program-staged-transcript-program[OF wf controlled])
  show  $\bigwedge x.$  hash-target-program B 0
    (return (staged-proof-transcript x))
    by (rule hash-target-program-return)
qed
then show ?thesis by simp
qed

lemma hash-extension-preserving-staged-semantic-adversary:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows hash-extension-preserving (staged-semantic-adversary A)

```

by (*rule hash-target-program-extension*
 [*OF hash-target-program-staged-semantic-adversary*
 [*OF wf controlled, of {}*]])

lemma *admissible-staged-semantic-adversary*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
shows
admissible-adversary
 (*staged-attacker-query-budget budgets + staged-challenge-query-budget*)
 (*staged-semantic-adversary A*)
unfolding *admissible-adversary-def*
proof (*intro conjI allI*)
show *hash-extension-preserving (staged-semantic-adversary A)*
by (*rule hash-extension-preserving-staged-semantic-adversary*
 [*OF wf controlled*])
show *hash-range-budget*
 (*staged-attacker-query-budget budgets + staged-challenge-query-budget*)
 (*staged-semantic-adversary A*)
by (*rule hash-range-budget-staged-semantic-adversary*[*OF wf controlled*])
show *hash-collision-budget*
 (*staged-attacker-query-budget budgets + staged-challenge-query-budget*)
 (*staged-semantic-adversary A*)
by (*rule hash-collision-budget-staged-semantic-adversary*
 [*OF wf controlled*])
fix *B*
show *hash-target-program B*
 (*staged-attacker-query-budget budgets + staged-challenge-query-budget*)
 (*staged-semantic-adversary A*)
by (*rule hash-target-program-staged-semantic-adversary*[*OF wf controlled*])
qed

lemma *staged-trace-fri-program-Suc-outcomeE*:
assumes *outcome*:
Some ((roots, bs'), t) ∈
set-dist (execute (staged-trace-fri-program A i (Suc n) bs) s)
obtains *root b roots' s1 s2 s3* **where**
Some (root, s1) ∈
set-dist (execute (trace-fri-root-stage A i bs) s)
Some ((), s2) ∈
set-dist (execute (record-staged-message root) s1)
Some (b, s3) ∈
set-dist (execute receive-trace-fri-challenge s2)
Some ((roots', bs'), t) ∈
set-dist
(execute
(staged-trace-fri-program A (Suc i) n (bs @ [b])) s3)
roots = root # roots'
using *outcome*

by (*auto elim!*: *set-dist-bindE* *intro*: *that*)

lemma *staged-composition-fri-program-Suc-outcomeE*:

assumes *outcome*:

Some ((*roots*, *bs'*), *t*) ∈

set-dist

(*execute*

(*staged-composition-fri-program* *A* *dg* *i* (*Suc* *n*) *bs*) *s*)

obtains *root* *b* *roots'* *s1* *s2* *s3* **where**

Some (*root*, *s1*) ∈

set-dist (*execute* (*composition-fri-root-stage* *A* *dg* *i* *bs*) *s*)

Some (*()*, *s2*) ∈

set-dist (*execute* (*record-staged-message* *root*) *s1*)

Some (*b*, *s3*) ∈

set-dist (*execute* *receive-composition-fri-challenge* *s2*)

Some ((*roots'*, *bs'*), *t*) ∈

set-dist

(*execute*

(*staged-composition-fri-program*

A *dg* (*Suc* *i*) *n* (*bs* @ [*b*])) *s3*)

roots = *root* # *roots'*

using *outcome*

by (*auto elim!*: *set-dist-bindE* *intro*: *that*)

lemma *staged-alpha-program-Suc-outcomeE*:

assumes *outcome*:

Some (*as*, *t*) ∈

set-dist (*execute* (*staged-alpha-program* (*Suc* *n*)) *s*)

obtains *a* *as'* *s1* *s2* **where**

Some (*a*, *s1*) ∈

set-dist (*execute* *receive-alpha-challenge* *s*)

Some (*()*, *s2*) ∈

set-dist (*execute* (*record-staged-message* *a*) *s1*)

Some (*as'*, *t*) ∈

set-dist (*execute* (*staged-alpha-program* *n*) *s2*)

as = *a* # *as'*

using *outcome*

by (*auto elim!*: *set-dist-bindE* *intro*: *that*)

lemma *staged-query-program-Suc-outcomeE*:

assumes *outcome*:

Some (*chunks*, *t*) ∈

set-dist (*execute* (*staged-query-program* *A* *i* (*Suc* *n*)) *s*)

obtains *raw* *chunk* *chunks'* *s1* *s2* *s3* **where**

Some (*raw*, *s1*) ∈

set-dist (*execute* *receive-query-index-challenge* *s*)

Some (*chunk*, *s2*) ∈

set-dist (*execute* (*query-opening-stage* *A* *i* *raw*) *s1*)

Some (*()*, *s3*) ∈


```

    set-dist (execute (record-staged-messages chunk) s2)
  Some (chunks', t) ∈
    set-dist (execute (staged-query-program A (Suc i) n) s3)
  chunks = chunk # chunks'
using outcome
by (auto elim!: set-dist-bindE intro: that)

```

lemma *checked-staged-query-program-Suc-outcomeE*:

```

assumes outcome:
  Some (chunks, t) ∈
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          i (Suc n)) s)
obtains raw chunk chunks' s1 s2 s3 where
  Some (raw, s1) ∈
    set-dist (execute receive-query-index-challenge s)
  Some (chunk, s2) ∈
    set-dist (execute (query-opening-stage A i raw) s1)
  verifier-query-round-chunk (index (to-nat raw))
    trace-roots composition-roots chunk
  Some ((), s3) ∈
    set-dist (execute (record-staged-messages chunk) s2)
  Some (chunks', t) ∈
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          (Suc i) n) s3)
  chunks = chunk # chunks'
using outcome
unfolding checked-staged-query-program.simps assert-def Let-def
by (auto simp: throw-no-outcome elim!: set-dist-bindE intro: that
    split: if-splits)

```

lemma *checked-staged-query-program-outcome-imp-staged-query-program*:

```

assumes outcome:
  Some (chunks, t) ∈
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          i n) s)
shows
  Some (chunks, t) ∈
    set-dist (execute (staged-query-program A i n) s)
using outcome
proof (induction n arbitrary: i chunks s t)
case 0
then show ?case
by simp

```

```

next
case (Suc n)
obtain raw chunk chunks' s1 s2 s3 where
  challenge-out:
    Some (raw, s1) ∈
      set-dist (execute receive-query-index-challenge s)
  and stage-out:
    Some (chunk, s2) ∈
      set-dist (execute (query-opening-stage A i raw) s1)
  and record-out:
    Some ((), s3) ∈
      set-dist (execute (record-staged-messages chunk) s2)
  and rest-out:
    Some (chunks', t) ∈
      set-dist
        (execute
          (checked-staged-query-program A trace-roots composition-roots
            (Suc i) n) s3)
  and chunks-eq: chunks = chunk # chunks'
proof (rule checked-staged-query-program-Suc-outcomeE[OF Suc.prem])
fix raw chunk chunks' s1 s2 s3
assume challenge:
  Some (raw, s1) ∈
    set-dist (execute receive-query-index-challenge s)
  and stage:
    Some (chunk, s2) ∈
      set-dist (execute (query-opening-stage A i raw) s1)
  and recorded:
    Some ((), s3) ∈
      set-dist (execute (record-staged-messages chunk) s2)
  and rest:
    Some (chunks', t) ∈
      set-dist
        (execute
          (checked-staged-query-program A trace-roots composition-roots
            (Suc i) n) s3)
  and chunks: chunks = chunk # chunks'
show ?thesis
by (rule that[OF challenge stage recorded rest chunks])
qed
have rest-unchecked:
  Some (chunks', t) ∈
    set-dist (execute (staged-query-program A (Suc i) n) s3)
by (rule Suc.IH[OF rest-out])
show ?case
unfolding chunks-eq staged-query-program.simps
by (intro set-dist-bindI[OF challenge-out]
    set-dist-bindI[OF stage-out]
    set-dist-bindI[OF record-out])

```

```

      set-dist-bindI[OF rest-unchecked])
    simp
qed

lemma checked-staged-transcript-program-outcome-imp-staged-transcript-program:
  assumes outcome:
    Some (data, t) ∈
      set-dist
        (execute (checked-staged-transcript-program A) s)
  shows
    Some (data, t) ∈
      set-dist (execute (staged-transcript-program A) s)
  using outcome
  unfolding checked-staged-transcript-program-def staged-transcript-program-def
    Let-def
  by (auto elim!: set-dist-bindE
      intro!: set-dist-bindI
      intro: checked-staged-query-program-outcome-imp-staged-query-program
      split: prod.splits)

lemma
  checked-staged-query-challenge-prefix-outcome-imp-staged-query-prefix:
  assumes outcome:
    Some (prefix, t) ∈
      set-dist
        (execute (checked-staged-query-challenge-prefix-program A i) s)
  shows
    Some ((sqp-degree prefix, sqp-composition-final prefix,
      sqp-query-chunks prefix), t) ∈
      set-dist (execute (staged-query-challenge-prefix-program A i) s)
  using outcome
  unfolding checked-staged-query-challenge-prefix-program-def
    staged-query-challenge-prefix-program-def Let-def
  by (auto elim!: set-dist-bindE
      intro!: set-dist-bindI
      intro: checked-staged-query-program-outcome-imp-staged-query-program
      split: prod.splits)

lemma checked-staged-query-program-wp-event-le-staged:
  assumes none:  $\neg P \text{ None}$ 
  shows
    wp-event
      (checked-staged-query-program A trace-roots composition-roots i n)
       $P \ s \leq$ 
      wp-event (staged-query-program A i n) P s
  using none
  proof (induction n arbitrary: i s P)
    case 0
    then show ?case by simp
  end

```

```

next
  case (Suc n)
  let ?checked =
    checked-staged-query-program A trace-roots composition-roots i (Suc n)
  let ?plain = staged-query-program A i (Suc n)
  show ?case
    unfolding checked-staged-query-program.simps
      staged-query-program.simps Let-def
  proof (rule wp-event-bind-mono-cont)
    show P None  $\implies$  P None .
  next
    fix raw s1
    assume raw-out:
      Some (raw, s1)  $\in$ 
      set-dist (execute receive-query-index-challenge s)
    show
      wp-event
      (query-opening-stage A i raw  $\gg=$ 
        ( $\lambda$ chunk.
          assert
            (verifier-query-round-chunk (index (to-nat raw))
              trace-roots composition-roots chunk)  $\gg=$ 
            ( $\lambda$ -.
              record-staged-messages chunk  $\gg=$ 
              ( $\lambda$ -.
                checked-staged-query-program A trace-roots
                  composition-roots (Suc i) n  $\gg=$ 
                  ( $\lambda$ chunks. return (chunk # chunks)))))))
        P s1
         $\leq$ 
        wp-event
        (query-opening-stage A i raw  $\gg=$ 
          ( $\lambda$ chunk.
            record-staged-messages chunk  $\gg=$ 
            ( $\lambda$ -.
              staged-query-program A (Suc i) n  $\gg=$ 
              ( $\lambda$ chunks. return (chunk # chunks))))))
        P s1
    proof (rule wp-event-bind-mono-cont)
      show P None  $\implies$  P None .
    next
      fix chunk s2
      assume chunk-out:
        Some (chunk, s2)  $\in$ 
        set-dist (execute (query-opening-stage A i raw) s1)
      let ?tailP =
         $\lambda$ out. case out of None  $\Rightarrow$  P None
          | Some (chunks, t)  $\Rightarrow$  P (Some (chunk # chunks, t))
      have tail-none:  $\neg$  ?tailP None

```

```

using Suc.prems by simp
have checked-tail-le:
   $\wedge s'$ .
  wp-event
    (checked-staged-query-program A trace-roots composition-roots
      (Suc i) n)
    ?tailP s'
   $\leq$ 
  wp-event (staged-query-program A (Suc i) n) ?tailP s'
by (rule Suc.IH) (rule tail-none)
have after-record:
  wp-event
    (record-staged-messages chunk  $\gg$ 
      ( $\lambda$ -.
        checked-staged-query-program A trace-roots
          composition-roots (Suc i) n  $\gg$ 
            ( $\lambda$ chunks. return (chunk # chunks))))
    P s2
   $\leq$ 
  wp-event
    (record-staged-messages chunk  $\gg$ 
      ( $\lambda$ -.
        staged-query-program A (Suc i) n  $\gg$ 
          ( $\lambda$ chunks. return (chunk # chunks))))
    P s2
proof (rule wp-event-bind-mono-cont)
  show P None  $\implies$  P None .
next
  fix u s3
  assume Some (u, s3)  $\in$ 
    set-dist (execute (record-staged-messages chunk) s2)
  show
    wp-event
      (checked-staged-query-program A trace-roots composition-roots
        (Suc i) n  $\gg$ 
          ( $\lambda$ chunks. return (chunk # chunks))))
      P s3
     $\leq$ 
    wp-event
      (staged-query-program A (Suc i) n  $\gg$ 
        ( $\lambda$ chunks. return (chunk # chunks))))
      P s3
    unfolding wp-event-bind-return-map
    using checked-tail-le[of s3]
    by simp
qed
have checked-assert:
  wp-event
    (assert

```

```

    (verifier-query-round-chunk (index (to-nat raw))
      trace-roots composition-roots chunk) >=>
    (λ-.
      record-staged-messages chunk >=>
      (λ-.
        checked-staged-query-program A trace-roots
          composition-roots (Suc i) n >=>
          (λchunks. return (chunk # chunks))))))
  P s2
≤
wp-event
(record-staged-messages chunk >=>
  (λ-.
    checked-staged-query-program A trace-roots
      composition-roots (Suc i) n >=>
      (λchunks. return (chunk # chunks))))
  P s2
by (rule wp-event-assert-bind-le) (rule Suc.prem)
show
wp-event
(assert
  (verifier-query-round-chunk (index (to-nat raw))
    trace-roots composition-roots chunk) >=>
  (λ-.
    record-staged-messages chunk >=>
    (λ-.
      checked-staged-query-program A trace-roots
        composition-roots (Suc i) n >=>
        (λchunks. return (chunk # chunks))))))
  P s2
≤
wp-event
(record-staged-messages chunk >=>
  (λ-.
    staged-query-program A (Suc i) n >=>
    (λchunks. return (chunk # chunks))))
  P s2
by (rule order-trans[OF checked-assert after-record])
qed
qed
qed

lemma checked-staged-transcript-program-wp-event-le-staged:
assumes none: ¬ P None
shows
  wp-event (checked-staged-transcript-program A) P s ≤
  wp-event (staged-transcript-program A) P s
unfolding checked-staged-transcript-program-def staged-transcript-program-def
  Let-def

```

```

apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac fr s1)
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac u1 s2)
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac trace-pair s3)
apply (case-tac trace-pair)
apply (rename-tac trace-roots trace-bs)
apply simp
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac trace-final s4)
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac u2 s5)
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac as s6)
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac dg s7)
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac u3 s8)
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac u4 s9)
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac composition-pair s10)
apply (case-tac composition-pair)
apply (rename-tac composition-roots composition-bs)
apply simp
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac composition-final s11)
apply (rule wp-event-bind-mono-cont)
  apply simp
apply (rename-tac u5 s12)
apply (subst wp-event-bind-return-map)
apply (subst wp-event-bind-return-map)
apply (rule checked-staged-query-program-wp-event-le-staged)
apply (simp add: none)
done

```

lemma *hash-range-budget-checked-staged-query-challenge-prefix-program:*

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *i-bound*: $i \leq \text{rounds}$
shows
hash-range-budget
(staged-query-search-queries budgets i)
(checked-staged-query-challenge-prefix-program A i)
unfolding *hash-range-budget-def*
proof (*intro allI impI*)
fix *s prefix t*
assume *outcome*:
Some (prefix, t) ∈
set-dist
(execute (checked-staged-query-challenge-prefix-program A i) s)
have *unchecked-out*:
Some ((spp-degree prefix, spp-composition-final prefix,
spp-query-chunks prefix), t) ∈
set-dist (execute (staged-query-challenge-prefix-program A i) s)
by (*rule checked-staged-query-challenge-prefix-outcome-imp-staged-query-prefix*
[OF outcome])
have *budget*:
hash-range-budget
(staged-query-search-queries budgets i)
(staged-query-challenge-prefix-program A i)
by (*rule hash-range-budget-staged-query-challenge-prefix-program*
[OF wf controlled i-bound])
show *card (hash-map-output-values t)*
 $\leq \text{card (hash-map-output-values s) +}$
 $\text{staged-query-search-queries budgets i}$
using *budget unchecked-out unfolding hash-range-budget-def by blast*
qed

lemma *hash-collision-budget-checked-staged-query-challenge-prefix-program*:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *i-bound*: $i \leq \text{rounds}$

shows

hash-collision-budget
(staged-query-search-queries budgets i)
(checked-staged-query-challenge-prefix-program A i)

unfolding *hash-collision-budget-def*

proof (*intro allI impI*)

fix *s :: 'f protocol-channel*

assume *clean*: $\neg \text{hash-map-output-collision s}$

have *checked-le*:

wp-event (checked-staged-query-challenge-prefix-program A i)
(hash-new-collision-event s) s ≤
wp-event (staged-query-challenge-prefix-program A i)
(hash-new-collision-event s) s


```

unfolding checked-staged-query-challenge-prefix-program-def
  staged-query-challenge-prefix-program-def Let-def
apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-collision-event-def)
apply (rename-tac alpha-prefix s1)
apply (case-tac alpha-prefix)
apply (rename-tac fr trace-roots trace-bs trace-final)
apply simp
apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-collision-event-def)
apply (rename-tac as s2)
apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-collision-event-def)
apply (rename-tac dg s3)
apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-collision-event-def)
apply (rename-tac u s4)
apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-collision-event-def)
apply (rename-tac u2 s5)
apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-collision-event-def)
apply (rename-tac composition-pair s6)
apply (case-tac composition-pair)
apply (rename-tac composition-roots composition-bs)
apply simp
apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-collision-event-def)
apply (rename-tac composition-final s7)
apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-collision-event-def)
apply (rename-tac u3 s8)
apply (subst wp-event-bind-return-map)
apply (subst wp-event-bind-return-map)
apply (simp add: hash-new-collision-event-def)
apply (rule checked-staged-query-program-wp-event-le-staged)
apply (simp add: hash-new-collision-event-def)
done
have unchecked-bound:
  wp-event (staged-query-challenge-prefix-program A i)
    (hash-new-collision-event s)  $s \leq$ 
    hash-collision-budget-value (card (hash-map-output-values s))
    (staged-query-search-queries budgets i)
using hash-collision-budget-staged-query-challenge-prefix-program
  [OF wf controlled i-bound] clean
unfolding hash-collision-budget-def by blast
show wp-event (checked-staged-query-challenge-prefix-program A i)
  (hash-new-collision-event s)  $s$ 
   $\leq$  hash-collision-budget-value (card (hash-map-output-values s))

```

```

      (staged-query-search-queries budgets i)
    by (rule order-trans[OF checked-le unchecked-bound])
qed

lemma hash-range-budget-checked-staged-transcript-program:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    hash-range-budget
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (checked-staged-transcript-program A)
  unfolding hash-range-budget-def
proof (intro allI impI)
  fix s data t
  assume outcome:
    Some (data, t) ∈
      set-dist (execute (checked-staged-transcript-program A) s)
  have unchecked-out:
    Some (data, t) ∈ set-dist (execute (staged-transcript-program A) s)
  by (rule checked-staged-transcript-program-outcome-imp-staged-transcript-program
      [OF outcome])
  have budget:
    hash-range-budget
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (staged-transcript-program A)
  by (rule hash-range-budget-staged-transcript-program[OF wf controlled])
  show card (hash-map-output-values t)
    ≤ card (hash-map-output-values s) +
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  using budget unchecked-out unfolding hash-range-budget-def by blast
qed

```

```

lemma hash-collision-budget-checked-staged-transcript-program:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    hash-collision-budget
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (checked-staged-transcript-program A)
  unfolding hash-collision-budget-def
proof (intro allI impI)
  fix s :: 'f protocol-channel
  assume clean: ¬ hash-map-output-collision s
  let ?budget =
    staged-attacker-query-budget budgets + staged-challenge-query-budget
  have checked-le:
    wp-event (checked-staged-transcript-program A)
      (hash-new-collision-event s) s ≤
    wp-event (staged-transcript-program A)

```

```

    (hash-new-collision-event s) s
  by (rule checked-staged-transcript-program-wp-event-le-staged)
    (simp add: hash-new-collision-event-def)
have unchecked-bound:
  wp-event (staged-transcript-program A)
    (hash-new-collision-event s) s ≤
    hash-collision-budget-value (card (hash-map-output-values s)) ?budget
using hash-collision-budget-staged-transcript-program[OF wf controlled]
  clean
  unfolding hash-collision-budget-def by blast
show wp-event (checked-staged-transcript-program A)
  (hash-new-collision-event s) s
  ≤ hash-collision-budget-value (card (hash-map-output-values s)) ?budget
  by (rule order-trans[OF checked-le unchecked-bound])
qed

lemma checked-staged-query-prefix-target-hit-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound: i ≤ rounds
  shows
    wp-event (checked-staged-query-challenge-prefix-program A i)
      (hash-new-output-hit-event B s) s ≤
      staged-phase-target-error B
      (staged-query-search-queries budgets i)
proof -
  have checked-le-unchecked:
    wp-event (checked-staged-query-challenge-prefix-program A i)
      (hash-new-output-hit-event B s) s ≤
      wp-event (staged-query-challenge-prefix-program A i)
        (hash-new-output-hit-event B s) s
  unfolding checked-staged-query-challenge-prefix-program-def
    staged-query-challenge-prefix-program-def Let-def
  apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-output-hit-event-def)
  apply (rename-tac alpha-prefix s1)
  apply (case-tac alpha-prefix)
  apply (rename-tac fr trace-roots trace-bs trace-final)
  apply simp
  apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-output-hit-event-def)
  apply (rename-tac as s2)
  apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-output-hit-event-def)
  apply (rename-tac dg s3)
  apply (rule wp-event-bind-mono-cont)
  apply (simp add: hash-new-output-hit-event-def)
  apply (rename-tac u s4)
  apply (rule wp-event-bind-mono-cont)

```

```

    apply (simp add: hash-new-output-hit-event-def)
  apply (rename-tac u2 s5)
  apply (rule wp-event-bind-mono-cont)
    apply (simp add: hash-new-output-hit-event-def)
  apply (rename-tac composition-pair s6)
  apply (case-tac composition-pair)
  apply (rename-tac composition-roots composition-bs)
  apply simp
  apply (rule wp-event-bind-mono-cont)
    apply (simp add: hash-new-output-hit-event-def)
  apply (rename-tac composition-final s7)
  apply (rule wp-event-bind-mono-cont)
    apply (simp add: hash-new-output-hit-event-def)
  apply (rename-tac u3 s8)
  apply (subst wp-event-bind-return-map)
  apply (subst wp-event-bind-return-map)
  apply (simp add: hash-new-output-hit-event-def)
  apply (rule checked-staged-query-program-wp-event-le-staged)
  apply simp
done
also have ... ≤
  staged-phase-target-error B
  (staged-query-search-queries budgets i)
  by (rule staged-query-prefix-target-hit-bound[OF wf controlled i-bound])
finally show ?thesis .
qed

```

```

lemma checked-staged-query-prefix-bad-index-target-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound:  $i \leq \text{rounds}$ 
  shows
    wp-event (checked-staged-query-challenge-prefix-program A i)
      (hash-new-output-hit-event (query-index-raw-preimage B) s) s ≤
      staged-query-prefix-bad-index-error budgets B i
  unfolding staged-query-prefix-bad-index-error-def
  by (rule checked-staged-query-prefix-target-hit-bound
      [OF wf controlled i-bound])

```

end

end

```

theory Staged-Security-Experiment-Checked-Prefix
  imports Staged-Security-Experiment-Targets
begin

```

Checked staged-prefix replay, alignment, and transcript extraction lemmas.

context *soundness*
begin

lemma *hash-relation-program-checked-staged-query-challenge-prefix-program:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i ≤ rounds*
and *fibers: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$*

shows

hash-relation-program R b
(staged-query-search-queries budgets i)
(checked-staged-query-challenge-prefix-program A i)

proof –

let *?alpha-prefix = staged-alpha-search-queries budgets 0*
let *?alpha = length spec*
let *?degree = degree-budget budgets*
let *?composition-fri =*
sum-list (composition-fri-budgets budgets) +
ceil-log (maxDegree + 1)
let *?composition-final = composition-final-budget budgets*
let *?query = sum-list (take i (query-opening-budgets budgets)) + i*
have *alpha-prefix:*
hash-relation-program R b ?alpha-prefix
(staged-alpha-prefix-program A)
by *(rule hash-relation-program-staged-alpha-prefix-program*
[OF wf controlled fibers])

have *alpha:*

hash-relation-program R b ?alpha
(staged-alpha-program (length spec))
by *(rule hash-relation-program-staged-alpha-program[OF fibers])*

have *degree:*

$\bigwedge as. \text{hash-relation-program } R \ b \ ?degree \ (\text{degree-stage } A \ as)$

proof –

fix *as*

have *controlled-degree:*

controlled-ro-program ?degree (degree-stage A as)
using *controlled unfolding staged-adversary-controlled-def* **by** *blast*
show *hash-relation-program R b ?degree (degree-stage A as)*
by *(rule controlled-ro-program-relation*
[OF controlled-degree fibers])

qed

have *composition-fri:*

$\bigwedge dg. \text{hash-relation-program } R \ b \ ?composition-fri$
(assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) $\gg$
($\lambda-. \text{staged-composition-fri-program } A \ dg \ 0$
(ceil-log (to-nat dg + 1)) []))
by *(rule hash-relation-program-guarded-staged-composition-fri-program*
[OF wf controlled fibers])

have *composition-final:*

```

 $\wedge dg \text{ composition-bs. hash-relation-program } R \ b \ ?\text{composition-final}$ 
  (composition-final-stage A dg composition-bs)
proof –
  fix dg composition-bs
  have controlled-final:
    controlled-ro-program ?composition-final
    (composition-final-stage A dg composition-bs)
    using controlled unfolding staged-adversary-controlled-def by blast
  show hash-relation-program R b ?composition-final
    (composition-final-stage A dg composition-bs)
    by (rule controlled-ro-program-relation
      [OF controlled-final fibers])
qed
have query:
   $\wedge \text{trace-roots composition-roots.}$ 
    hash-relation-program R b ?query
    (checked-staged-query-program A trace-roots composition-roots 0 i)
proof –
  fix trace-roots composition-roots
  have len:  $0 + i \leq \text{length (query-opening-budgets budgets)}$ 
    using wf i-bound unfolding staged-budget-wellformed-def by simp
  have exact:
    hash-relation-program R b
    (sum-list (take i (drop 0 (query-opening-budgets budgets))) + i)
    (checked-staged-query-program A trace-roots composition-roots 0 i)
    by (rule hash-relation-program-checked-staged-query-program
      [OF controlled len fibers])
  then show hash-relation-program R b ?query
    (checked-staged-query-program A trace-roots composition-roots 0 i)
    by simp
qed
have after-query:
   $\wedge fr \text{ trace-roots trace-bs trace-final as dg composition-roots}$ 
    composition-bs composition-final.
    hash-relation-program R b (?query + 0)
    (checked-staged-query-program A trace-roots composition-roots 0 i  $\gg$ 
      ( $\lambda \text{query-chunks. return}$ 
        ( $\downarrow \text{sqp-trace-root} = fr,$ 
           $\text{sqp-trace-fri-roots} = \text{trace-roots},$ 
           $\text{sqp-trace-fri-challenges} = \text{trace-bs},$ 
           $\text{sqp-trace-final} = \text{trace-final},$ 
           $\text{sqp-alphas} = as,$ 
           $\text{sqp-degree} = dg,$ 
           $\text{sqp-composition-fri-roots} = \text{composition-roots},$ 
           $\text{sqp-composition-fri-challenges} = \text{composition-bs},$ 
           $\text{sqp-composition-final} = \text{composition-final},$ 
           $\text{sqp-query-chunks} = \text{query-chunks}))$ ))
    by (rule hash-relation-program-bind)
    (rule query,

```

```

    rule hash-relation-program-zero[OF hash-map-preserving-return])
have after-composition-final-record:
 $\bigwedge$ fr trace-roots trace-bs trace-final as dg composition-roots
    composition-bs composition-final.
    hash-relation-program R b (0 + (?query + 0))
    (record-staged-message composition-final  $\gg$ 
      ( $\lambda$ -. checked-staged-query-program A trace-roots
        composition-roots 0 i  $\gg$ 
          ( $\lambda$ query-chunks. return
            (sqp-trace-root = fr,
              sqp-trace-fri-roots = trace-roots,
              sqp-trace-fri-challenges = trace-bs,
              sqp-trace-final = trace-final,
              sqp-alphas = as,
              sqp-degree = dg,
              sqp-composition-fri-roots = composition-roots,
              sqp-composition-fri-challenges = composition-bs,
              sqp-composition-final = composition-final,
              sqp-query-chunks = query-chunks))))))
by (rule hash-relation-program-bind)
    (rule hash-relation-program-record-staged-message, rule after-query)
have after-composition-final:
 $\bigwedge$ fr trace-roots trace-bs trace-final as dg composition-roots
    composition-bs.
    hash-relation-program R b
    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg composition-bs  $\gg$ 
      ( $\lambda$ composition-final.
        record-staged-message composition-final  $\gg$ 
          ( $\lambda$ -. checked-staged-query-program A trace-roots
            composition-roots 0 i  $\gg$ 
              ( $\lambda$ query-chunks. return
                (sqp-trace-root = fr,
                  sqp-trace-fri-roots = trace-roots,
                  sqp-trace-fri-challenges = trace-bs,
                  sqp-trace-final = trace-final,
                  sqp-alphas = as,
                  sqp-degree = dg,
                  sqp-composition-fri-roots = composition-roots,
                  sqp-composition-fri-challenges = composition-bs,
                  sqp-composition-final = composition-final,
                  sqp-query-chunks = query-chunks))))))
by (rule hash-relation-program-bind)
    (rule composition-final, rule after-composition-final-record)
have after-composition-fri:
 $\bigwedge$ fr trace-roots trace-bs trace-final as dg.
    hash-relation-program R b
    (?composition-fri +
      (?composition-final + (0 + (?query + 0))))

```

```

((assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
  (λ-. staged-composition-fri-program A dg 0
    (ceil-log (to-nat dg + 1)) [])) >>=
  (λ(composition-roots, composition-bs).
    composition-final-stage A dg composition-bs >>=
    (λcomposition-final.
      record-staged-message composition-final >>=
      (λ-. checked-staged-query-program A trace-roots
        composition-roots 0 i >>=
        (λquery-chunks. return
          (λsqp-trace-root = fr,
            sqp-trace-fri-roots = trace-roots,
            sqp-trace-fri-challenges = trace-bs,
            sqp-trace-final = trace-final,
            sqp-alphas = as,
            sqp-degree = dg,
            sqp-composition-fri-roots = composition-roots,
            sqp-composition-fri-challenges = composition-bs,
            sqp-composition-final = composition-final,
            sqp-query-chunks = query-chunks))))))

```

proof –

fix *fr trace-roots trace-bs trace-final as dg*

have *cont*:

```

  (λx. hash-relation-program R b
    (?composition-final + (0 + (?query + 0)))
    (case x of (composition-roots, composition-bs) ⇒
      composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
        (λ-. checked-staged-query-program A trace-roots
          composition-roots 0 i >>=
          (λquery-chunks. return
            (λsqp-trace-root = fr,
              sqp-trace-fri-roots = trace-roots,
              sqp-trace-fri-challenges = trace-bs,
              sqp-trace-final = trace-final,
              sqp-alphas = as,
              sqp-degree = dg,
              sqp-composition-fri-roots = composition-roots,
              sqp-composition-fri-challenges = composition-bs,
              sqp-composition-final = composition-final,
              sqp-query-chunks = query-chunks))))))

```

proof –

fix *x*

show *hash-relation-program R b*

```

  (?composition-final + (0 + (?query + 0)))
  (case x of (composition-roots, composition-bs) ⇒
    composition-final-stage A dg composition-bs >>=
    (λcomposition-final.

```



```

record-staged-message composition-final  $\gg$ 
  ( $\lambda$ -. checked-staged-query-program A trace-roots
    composition-roots 0 i  $\gg$ 
    ( $\lambda$ query-chunks. return
      ( $\lambda$ sqp-trace-root = fr,
        sqp-trace-fri-roots = trace-roots,
        sqp-trace-fri-challenges = trace-bs,
        sqp-trace-final = trace-final,
        sqp-alphas = as,
        sqp-degree = dg,
        sqp-composition-fri-roots = composition-roots,
        sqp-composition-fri-challenges = composition-bs,
        sqp-composition-final = composition-final,
        sqp-query-chunks = query-chunks))))))
proof (cases x)
case (Pair composition-roots composition-bs)
have step:
  hash-relation-program R b
    (?composition-final + (0 + (?query + 0)))
    (composition-final-stage A dg composition-bs  $\gg$ 
      ( $\lambda$ composition-final.
        record-staged-message composition-final  $\gg$ 
          ( $\lambda$ -. checked-staged-query-program A trace-roots
            composition-roots 0 i  $\gg$ 
              ( $\lambda$ query-chunks. return
                ( $\lambda$ sqp-trace-root = fr,
                  sqp-trace-fri-roots = trace-roots,
                  sqp-trace-fri-challenges = trace-bs,
                  sqp-trace-final = trace-final,
                  sqp-alphas = as,
                  sqp-degree = dg,
                  sqp-composition-fri-roots = composition-roots,
                  sqp-composition-fri-challenges = composition-bs,
                  sqp-composition-final = composition-final,
                  sqp-query-chunks = query-chunks))))))
by (rule after-composition-final)
show ?thesis
  unfolding Pair using step by simp
qed
qed
show hash-relation-program R b
  (?composition-fri +
    (?composition-final + (0 + (?query + 0))))
  ((assert (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
    ( $\lambda$ -. staged-composition-fri-program A dg 0
      (ceil-log (to-nat dg + 1)) []))  $\gg$ 
    ( $\lambda$ (composition-roots, composition-bs).
      composition-final-stage A dg composition-bs  $\gg$ 
        ( $\lambda$ composition-final.

```

```

    record-staged-message composition-final  $\gg$ 
    ( $\lambda$ -. checked-staged-query-program A trace-roots
      composition-roots 0 i  $\gg$ 
      ( $\lambda$ query-chunks. return
        ( $\downarrow$ sqp-trace-root = fr,
          sqp-trace-fri-roots = trace-roots,
          sqp-trace-fri-challenges = trace-bs,
          sqp-trace-final = trace-final,
          sqp-alphas = as,
          sqp-degree = dg,
          sqp-composition-fri-roots = composition-roots,
          sqp-composition-fri-challenges = composition-bs,
          sqp-composition-final = composition-final,
          sqp-query-chunks = query-chunks))))))
  by (rule hash-relation-program-bind)
    (rule composition-fri, rule cont)
qed
have after-degree-record:
 $\wedge$ fr trace-roots trace-bs trace-final as dg.
  hash-relation-program R b
    (0 + (?composition-fri +
      (?composition-final + (0 + (?query + 0)))))
  (record-staged-message dg  $\gg$ 
    ( $\lambda$ -. (assert
      (ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
      ( $\lambda$ -. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1))  $\square$ ))  $\gg$ 
      ( $\lambda$ (composition-roots, composition-bs).
        composition-final-stage A dg composition-bs  $\gg$ 
        ( $\lambda$ composition-final.
          record-staged-message composition-final  $\gg$ 
          ( $\lambda$ -. checked-staged-query-program A trace-roots
            composition-roots 0 i  $\gg$ 
            ( $\lambda$ query-chunks. return
              ( $\downarrow$ sqp-trace-root = fr,
                sqp-trace-fri-roots = trace-roots,
                sqp-trace-fri-challenges = trace-bs,
                sqp-trace-final = trace-final,
                sqp-alphas = as,
                sqp-degree = dg,
                sqp-composition-fri-roots = composition-roots,
                sqp-composition-fri-challenges = composition-bs,
                sqp-composition-final = composition-final,
                sqp-query-chunks = query-chunks)))))))))
  by (rule hash-relation-program-bind)
    (rule hash-relation-program-record-staged-message,
      rule after-composition-fri)
have after-degree:
 $\wedge$ fr trace-roots trace-bs trace-final as.

```

```

hash-relation-program R b
  (?degree + (0 + (?composition-fri +
    (?composition-final + (0 + (?query + 0))))))
  (degree-stage A as >>=
    (λdg. record-staged-message dg >>=
      (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
        assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
          (λ-. staged-composition-fri-program A dg 0
            composition-rounds [] >>=
              (λ(composition-roots, composition-bs).
                composition-final-stage A dg composition-bs >>=
                  (λcomposition-final.
                    record-staged-message composition-final >>=
                      (λ-. checked-staged-query-program A trace-roots
                        composition-roots 0 i >>=
                          (λquery-chunks. return
                            (|spp-trace-root = fr,
                               spp-trace-fri-roots = trace-roots,
                               spp-trace-fri-challenges = trace-bs,
                               spp-trace-final = trace-final,
                               spp-alphas = as,
                               spp-degree = dg,
                               spp-composition-fri-roots = composition-roots,
                               spp-composition-fri-challenges = composition-bs,
                               spp-composition-final = composition-final,
                               spp-query-chunks = query-chunks|))))))))))

```

proof –

fix fr trace-roots trace-bs trace-final as

have cont:

```

Λdg. hash-relation-program R b
  (0 + (?composition-fri +
    (?composition-final + (0 + (?query + 0))))))
  (record-staged-message dg >>=
    (λ-. assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >>=
      (λ-. staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [] >>=
          (λ(composition-roots, composition-bs).
            composition-final-stage A dg composition-bs >>=
              (λcomposition-final.
                record-staged-message composition-final >>=
                  (λ-. checked-staged-query-program A trace-roots
                    composition-roots 0 i >>=
                      (λquery-chunks. return
                        (|spp-trace-root = fr,
                           spp-trace-fri-roots = trace-roots,
                           spp-trace-fri-challenges = trace-bs,
                           spp-trace-final = trace-final,
                           spp-alphas = as,
                           spp-degree = dg,

```

```

      sqp-composition-fri-roots = composition-roots,
      sqp-composition-fri-challenges = composition-bs,
      sqp-composition-final = composition-final,
      sqp-query-chunks = query-chunks)))))))))
proof -
  fix dg
  have step:
    hash-relation-program R b
    (0 + (?composition-fri +
      (?composition-final + (0 + (?query + 0)))))
    (record-staged-message dg >=>
      (λ-. (assert
        (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >=>
        (λ-. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [])) >=>
        (λ(composition-roots, composition-bs).
          composition-final-stage A dg composition-bs >=>
          (λcomposition-final.
            record-staged-message composition-final >=>
            (λ-. checked-staged-query-program A trace-roots
              composition-roots 0 i >=>
              (λquery-chunks. return
                (sqp-trace-root = fr,
                 sqp-trace-fri-roots = trace-roots,
                 sqp-trace-fri-challenges = trace-bs,
                 sqp-trace-final = trace-final,
                 sqp-alphas = as,
                 sqp-degree = dg,
                 sqp-composition-fri-roots = composition-roots,
                 sqp-composition-fri-challenges = composition-bs,
                 sqp-composition-final = composition-final,
                 sqp-query-chunks = query-chunks)))))))))
    by (rule after-degree-record)
  show hash-relation-program R b
    (0 + (?composition-fri +
      (?composition-final + (0 + (?query + 0)))))
    (record-staged-message dg >=>
      (λ-. assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) >=>
        (λ-. staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [] >=>
          (λ(composition-roots, composition-bs).
            composition-final-stage A dg composition-bs >=>
            (λcomposition-final.
              record-staged-message composition-final >=>
              (λ-. checked-staged-query-program A trace-roots
                composition-roots 0 i >=>
                (λquery-chunks. return
                  (sqp-trace-root = fr,
                   sqp-trace-fri-roots = trace-roots,

```

```

      sqp-trace-fri-challenges = trace-bs,
      sqp-trace-final = trace-final,
      sqp-alphas = as,
      sqp-degree = dg,
      sqp-composition-fri-roots = composition-roots,
      sqp-composition-fri-challenges = composition-bs,
      sqp-composition-final = composition-final,
      sqp-query-chunks = query-chunks)))))))))
  using step by (simp add: sm-bind-assoc)
qed
show hash-relation-program R b
  (?degree + (0 + (?composition-fri +
    (?composition-final + (0 + (?query + 0))))))
  (degree-stage A as >=
    (λdg. record-staged-message dg >=
      (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
        assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >=
          (λ-. staged-composition-fri-program A dg 0
            composition-rounds [] >=
              (λ(composition-roots, composition-bs).
                composition-final-stage A dg composition-bs >=
                  (λcomposition-final.
                    record-staged-message composition-final >=
                      (λ-. checked-staged-query-program A trace-roots
                        composition-roots 0 i >=
                          (λquery-chunks. return
                            (sqp-trace-root = fr,
                              sqp-trace-fri-roots = trace-roots,
                              sqp-trace-fri-challenges = trace-bs,
                              sqp-trace-final = trace-final,
                              sqp-alphas = as,
                              sqp-degree = dg,
                              sqp-composition-fri-roots = composition-roots,
                              sqp-composition-fri-challenges = composition-bs,
                              sqp-composition-final = composition-final,
                              sqp-query-chunks = query-chunks))))))))))
  unfolding Let-def
  by (rule hash-relation-program-bind)
    (rule degree, rule cont)
qed
have after-alpha:
  ∧fr trace-roots trace-bs trace-final.
  hash-relation-program R b
    (?alpha + (?degree + (0 + (?composition-fri +
      (?composition-final + (0 + (?query + 0))))))
    (staged-alpha-program (length spec) >=
      (λas. degree-stage A as >=
        (λdg. record-staged-message dg >=
          (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in

```

```

    assert (composition-rounds ≤ ceil-log (maxDegree + 1)) ≫=
    (λ-. staged-composition-fri-program A dg 0
      composition-rounds [] ≫=
      (λ(composition-roots, composition-bs).
        composition-final-stage A dg composition-bs ≫=
        (λcomposition-final.
          record-staged-message composition-final ≫=
          (λ-. checked-staged-query-program A trace-roots
            composition-roots 0 i ≫=
            (λquery-chunks. return
              (sqp-trace-root = fr,
               sqp-trace-fri-roots = trace-roots,
               sqp-trace-fri-challenges = trace-bs,
               sqp-trace-final = trace-final,
               sqp-alphas = as,
               sqp-degree = dg,
               sqp-composition-fri-roots = composition-roots,
               sqp-composition-fri-challenges = composition-bs,
               sqp-composition-final = composition-final,
               sqp-query-chunks = query-chunks))))))))))
  by (rule hash-relation-program-bind)
  (rule alpha, rule after-degree)
have after-alpha-prefix:
  hash-relation-program R b
  (?alpha-prefix +
    (?alpha + (?degree + (0 + (?composition-fri +
      (?composition-final + (0 + (?query + 0))))))))))
  (checked-staged-query-challenge-prefix-program A i)
  unfolding checked-staged-query-challenge-prefix-program-def
proof (rule hash-relation-program-bind)
  show hash-relation-program R b ?alpha-prefix
    (staged-alpha-prefix-program A)
  by (rule alpha-prefix)
next
fix x
show hash-relation-program R b
  (?alpha + (?degree + (0 + (?composition-fri +
    (?composition-final + (0 + (?query + 0))))))))
  (case x of (fr, trace-roots, trace-bs, trace-final) ⇒
    staged-alpha-program (length spec) ≫=
    (λas. degree-stage A as ≫=
      (λdg. record-staged-message dg ≫=
        (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert (composition-rounds ≤ ceil-log (maxDegree + 1)) ≫=
          (λ-. staged-composition-fri-program A dg 0
            composition-rounds [] ≫=
            (λ(composition-roots, composition-bs).
              composition-final-stage A dg composition-bs ≫=
              (λcomposition-final.

```

```

    record-staged-message composition-final >>=
  (λ-. checked-staged-query-program A trace-roots
    composition-roots 0 i >>=
    (λquery-chunks. return
      (sqp-trace-root = fr,
        sqp-trace-fri-roots = trace-roots,
        sqp-trace-fri-challenges = trace-bs,
        sqp-trace-final = trace-final,
        sqp-alphas = as,
        sqp-degree = dg,
        sqp-composition-fri-roots = composition-roots,
        sqp-composition-fri-challenges = composition-bs,
        sqp-composition-final = composition-final,
        sqp-query-chunks = query-chunks)))))))))
proof (cases x)
case (fields fr trace-roots trace-bs trace-final)
have step:
  hash-relation-program R b
  (?alpha + (?degree + (0 + (?composition-fri +
    (?composition-final + (0 + (?query + 0)))))))
  (staged-alpha-program (length spec) >>=
    (λas. degree-stage A as >>=
      (λdg. record-staged-message dg >>=
        (λ-. let composition-rounds = ceil-log (to-nat dg + 1) in
          assert (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0
              composition-rounds [] >>=
                (λ(composition-roots, composition-bs).
                  composition-final-stage A dg composition-bs >>=
                    (λcomposition-final.
                      record-staged-message composition-final >>=
                        (λ-. checked-staged-query-program A trace-roots
                          composition-roots 0 i >>=
                            (λquery-chunks. return
                              (sqp-trace-root = fr,
                                sqp-trace-fri-roots = trace-roots,
                                sqp-trace-fri-challenges = trace-bs,
                                sqp-trace-final = trace-final,
                                sqp-alphas = as,
                                sqp-degree = dg,
                                sqp-composition-fri-roots = composition-roots,
                                sqp-composition-fri-challenges = composition-bs,
                                sqp-composition-final = composition-final,
                                sqp-query-chunks = query-chunks)))))))))
        by (rule after-alpha)
      show ?thesis
        unfolding fields using step by simp
    qed
  qed

```

```

show ?thesis
  using after-alpha-prefix
  unfolding staged-query-search-queries-def
    staged-alpha-search-queries-def
  by (simp add: add.assoc add.commute add.left-commute)
qed

lemma staged-trace-fri-program-challenge-prefix-pre-alignment:
  assumes outcome:
    Some ((roots, bs'), t) ∈
      set-dist (execute (staged-trace-fri-program A i n bs) s)
  shows take (length bs) bs' = bs
  using outcome
proof (induction n arbitrary: i bs roots bs' s t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.premis obtain root b roots' s1 s2 s3 where
    rest-out:
      Some ((roots', bs'), t) ∈
        set-dist
          (execute
            (staged-trace-fri-program A (Suc i) n (bs @ [b])) s3)
  unfolding staged-trace-fri-program.simps
  by (auto elim!: set-dist-bindE)
  have prefix-suc:
    take (length (bs @ [b])) bs' = bs @ [b]
  by (rule Suc.IH[OF rest-out])
  have take (length bs) bs' =
    take (length bs) (take (length (bs @ [b])) bs')
  by simp
  also have ... = bs
  using prefix-suc by simp
  finally show ?case .
qed

lemma staged-trace-fri-program-alignment:
  assumes controlled: staged-adversary-controlled budgets A
  and bound: i + n ≤ length (trace-fri-budgets budgets)
  and outcome:
    Some ((roots, bs'), t) ∈
      set-dist (execute (staged-trace-fri-program A i n bs) s)
  shows
    length roots = n ∧
    length bs' = length bs + n ∧
    PTranscript t = PTranscript s @ roots ∧
    PState t = foldl concat (PState s) roots ∧
    s ≤ t ∧

```



```

PTraceFriCounter t = PTraceFriCounter s + n ∧
PCompositionFriCounter t = PCompositionFriCounter s ∧
PAlphaCounter t = PAlphaCounter s ∧
PQueryCounter t = PQueryCounter s ∧
(∀ j < n.
  fmlookup (HashMap t)
    (TraceFriChallenge (PTraceFriCounter s + j)
      (foldl concat (PState s) (take (Suc j) roots))) =
      Some (bs' ! (length bs + j)))
using bound outcome
proof (induction n arbitrary: i bs roots bs' s t)
case 0
then show ?case by (simp add: hash-ext-refl)
next
case (Suc n)
from staged-trace-fri-program-Suc-outcomeE[OF Suc.prems(2)]
obtain root b roots' s1 s2 s3 where
  stage-out:
    Some (root, s1) ∈
      set-dist (execute (trace-fri-root-stage A i bs) s)
and record-out:
    Some ((), s2) ∈
      set-dist (execute (record-staged-message root) s1)
and challenge-out:
    Some (b, s3) ∈
      set-dist (execute receive-trace-fri-challenge s2)
and rest-out:
    Some ((roots', bs'), t) ∈
      set-dist
        (execute
          (staged-trace-fri-program A (Suc i) n (bs @ [b])) s3)
and roots-eq: roots = root # roots'
  .
have i-bound: i < length (trace-fri-budgets budgets)
using Suc.prems(1) by simp
have stage-controlled:
  controlled-ro-program (trace-fri-budgets budgets ! i)
    (trace-fri-root-stage A i bs)
using controlled i-bound
unfolding staged-adversary-controlled-def by blast
have stage-fields:
  PState s1 = PState s ∧
  PTranscript s1 = PTranscript s ∧
  PTraceFriCounter s1 = PTraceFriCounter s ∧
  PCompositionFriCounter s1 = PCompositionFriCounter s ∧
  PAlphaCounter s1 = PAlphaCounter s ∧
  PQueryCounter s1 = PQueryCounter s
using controlled-stage-outcome-fields[OF stage-controlled stage-out]
by simp

```

```

have ext-s-s1:  $s \leq s1$ 
  using controlled-ro-program-extension[OF stage-controlled] stage-out
  unfolding hash-extension-preserving-def by blast
have s2:
  s2 = s1 (|
    PState := concat (PState s1) root,
    PTranscript := PTranscript s1 @ [root])
  by (rule record-staged-message-outcome[OF record-out])
have ext-s1-s2:  $s1 \leq s2$ 
  unfolding s2 less-eq-hash-ext-def less-eq-fmap-def by simp
have s2-fields:
  PState s2 = concat (PState s) root ∧
  PTranscript s2 = PTranscript s @ [root] ∧
  PTraceFriCounter s2 = PTraceFriCounter s ∧
  PCompositionFriCounter s2 = PCompositionFriCounter s ∧
  PAlphaCounter s2 = PAlphaCounter s ∧
  PQueryCounter s2 = PQueryCounter s
  unfolding s2 using stage-fields by simp
have challenge-fields:
  PState s3 = PState s2 ∧
  PTranscript s3 = PTranscript s2
  using receive-trace-fri-challenge-outcome[OF challenge-out] by simp
have ext-s2-s3:  $s2 \leq s3$ 
  using receive-trace-fri-challenge-outcome[OF challenge-out] by simp
have challenge-counters:
  PTraceFriCounter s3 = Suc (PTraceFriCounter s2) ∧
  PCompositionFriCounter s3 = PCompositionFriCounter s2 ∧
  PAlphaCounter s3 = PAlphaCounter s2 ∧
  PQueryCounter s3 = PQueryCounter s2
  using receive-trace-fri-challenge-counter-outcome[OF challenge-out] by simp
have rest-bound:
  Suc i + n ≤ length (trace-fri-budgets budgets)
  using Suc.prem1 by simp
have rest-alignment:
  length roots' = n ∧
  length bs' = length (bs @ [b]) + n ∧
  PTranscript t = PTranscript s3 @ roots' ∧
  PState t = foldl concat (PState s3) roots' ∧
   $s3 \leq t$  ∧
  PTraceFriCounter t = PTraceFriCounter s3 + n ∧
  PCompositionFriCounter t = PCompositionFriCounter s3 ∧
  PAlphaCounter t = PAlphaCounter s3 ∧
  PQueryCounter t = PQueryCounter s3 ∧
  ( $\forall j < n.$ 
    fmlookup (HashMap t)
      (TraceFriChallenge (PTraceFriCounter s3 + j)
        (foldl concat (PState s3) (take (Suc j) roots'))) =
        Some (bs' ! (length (bs @ [b]) + j)))
  by (rule Suc.IH[OF rest-bound rest-out])

```

```

have ext-s3-t:  $s3 \leq t$ 
  using rest-alignment by simp
have ext-s-t:  $s \leq t$ 
  by (rule hash-ext-trans[OF ext-s-s1])
    (rule hash-ext-trans[OF ext-s1-s2],
      rule hash-ext-trans[OF ext-s2-s3 ext-s3-t])
have challenge-lookup-s3:
  fmlookup (HashMap s3)
    (TraceFriChallenge (PTraceFriCounter s)
      (concat (PState s) root)) = Some b
  using receive-trace-fri-challenge-outcome[OF challenge-out]
    s2-fields by simp
have challenge-lookup-t:
  fmlookup (HashMap t)
    (TraceFriChallenge (PTraceFriCounter s)
      (concat (PState s) root)) = Some b
  by (rule hash-extension-lookup[OF challenge-lookup-s3 ext-s3-t])
have prefix-suc:
  take (length (bs @ [b])) bs' = bs @ [b]
  by (rule staged-trace-fri-program-challenge-prefix-pre-alignment
    [OF rest-out])
have lookup-all:
   $\forall j < \text{Suc } n.$ 
  fmlookup (HashMap t)
    (TraceFriChallenge (PTraceFriCounter s + j)
      (foldl concat (PState s) (take (Suc j) (root # roots')))) =
    Some (bs' ! (length bs + j))
proof (intro allI impI)
  fix j
  assume j-bound:  $j < \text{Suc } n$ 
  show
    fmlookup (HashMap t)
      (TraceFriChallenge (PTraceFriCounter s + j)
        (foldl concat (PState s) (take (Suc j) (root # roots')))) =
        Some (bs' ! (length bs + j))
  proof (cases j)
  case 0
  have bs' ! length bs = b
  proof –
    have idx-lt: length bs < length (bs @ [b])
      by simp
    have bs' ! length bs =
      take (length (bs @ [b])) bs' ! length bs
      using idx-lt by simp
    also have ... = (bs @ [b]) ! length bs
      using prefix-suc by simp
    also have ... = b
      by simp
    finally show ?thesis .

```

```

    qed
  then show ?thesis
    using challenge-lookup-t unfolding 0 by simp
next
  case (Suc k)
  then have k-bound:  $k < n$ 
    using j-bound by simp
  show ?thesis
    using rest-alignment k-bound s2-fields challenge-fields
      challenge-counters
    unfolding Suc
    by simp
  qed
qed
show ?case
  unfolding roots-eq
  using s2-fields challenge-fields challenge-counters rest-alignment
    ext-s-t lookup-all
  by simp
qed

lemma staged-trace-fri-program-challenge-prefix:
  assumes outcome:
    Some ((roots, bs'), t) ∈
      set-dist (execute (staged-trace-fri-program A i n bs) s)
  shows take (length bs) bs' = bs
  using outcome
proof (induction n arbitrary: i bs roots bs' s t)
  case 0
  then show ?case by (simp add: hash-ext-refl)
next
  case (Suc n)
  from Suc.premis obtain root b roots' s1 s2 s3 where
    rest-out:
      Some ((roots', bs'), t) ∈
        set-dist
          (execute
            (staged-trace-fri-program A (Suc i) n (bs @ [b])) s3)
    unfolding staged-trace-fri-program.simps
    by (auto elim!: set-dist-bindE)
  have prefix-suc:
    take (length (bs @ [b])) bs' = bs @ [b]
    by (rule Suc.IH[OF rest-out])
  have take (length bs) bs' =
    take (length bs) (take (length (bs @ [b])) bs')
    by simp
  also have ... = bs
    using prefix-suc by simp
  finally show ?case .

```

qed

lemma *staged-trace-fri-program-challenge-length:*

assumes *outcome:*

Some ((roots, bs'), t) ∈

set-dist (execute (staged-trace-fri-program A i n bs) s)

shows *length bs' = length bs + n*

using *outcome*

proof (*induction n arbitrary: i bs roots bs' s t*)

case 0

then show ?*case* **by** (*simp add: hash-ext-refl*)

next

case (*Suc n*)

from *Suc.prem*s **obtain** *root b roots' s1 s2 s3* **where**

rest-out:

Some ((roots', bs'), t) ∈

set-dist

(execute

(staged-trace-fri-program A (Suc i) n (bs @ [b])) s3)

unfolding *staged-trace-fri-program.simp*s

by (*auto elim!: set-dist-bindE*)

have *length bs' = length (bs @ [b]) + n*

by (*rule Suc.IH[OF rest-out]*)

then show ?*case* **by** *simp*

qed

lemma *staged-alpha-prefix-program-trace-challenge-length:*

assumes *outcome:*

Some ((fr, trace-roots, trace-bs, trace-final), t) ∈

set-dist (execute (staged-alpha-prefix-program A) s)

shows *length trace-bs = ceil-log clength*

proof –

from *outcome* **obtain** *s1 s2 s3* **where**

fri-out:

Some ((trace-roots, trace-bs), s2) ∈

set-dist

(execute

(staged-trace-fri-program A 0 (ceil-log clength) []) s1)

unfolding *staged-alpha-prefix-program-def*

by (*auto elim!: set-dist-bindE*)

have *length trace-bs = length ([] :: 'f list) + ceil-log clength*

by (*rule staged-trace-fri-program-challenge-length[OF fri-out]*)

then show ?*thesis* **by** *simp*

qed

lemma *staged-alpha-prefix-trace-vector-member-imp-position-value:*

assumes *outcome:*

Some ((fr, trace-roots, trace-bs, trace-final), t) ∈

set-dist (execute (staged-alpha-prefix-program A) s)

and *subset*: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and *member*: $\text{trace-bs} \in B$
and *i-bound*: $i < \text{ceil-log } \text{clength}$
shows $\text{trace-bs} ! i \in \text{fri-vector-position-values } B \ i$
by (*rule fri-vector-member-imp-position-value*
 $[OF \ \text{subset} \ \text{member} \ i\text{-bound}]$)

lemma *staged-alpha-prefix-bad-trace-vector-coordinate-cover-bound*:

assumes *subset*: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and *nonempty*: $0 < \text{ceil-log } \text{clength}$
shows
 $\text{wp-event } (\text{staged-alpha-prefix-program } A)$
 $(\lambda \text{out. case out of None} \Rightarrow \text{False}$
 $\mid \text{Some } ((-, -, \text{trace-bs}, -), -) \Rightarrow \text{trace-bs} \in B) \ s \leq$
 $(\sum i < \text{ceil-log } \text{clength}.$
 $\text{wp-event } (\text{staged-alpha-prefix-program } A)$
 $(\lambda \text{out. case out of None} \Rightarrow \text{False}$
 $\mid \text{Some } ((-, -, \text{trace-bs}, -), -) \Rightarrow$
 $\text{trace-bs} ! i \in \text{fri-vector-position-values } B \ i) \ s)$

proof –

let *?bad* =
 $\lambda \text{out. case out of None} \Rightarrow \text{False}$
 $\mid \text{Some } ((-, -, \text{trace-bs}, -), -) \Rightarrow \text{trace-bs} \in B$
let *?coord* =
 $\lambda i \text{ out. case out of None} \Rightarrow \text{False}$
 $\mid \text{Some } ((-, -, \text{trace-bs}, -), -) \Rightarrow$
 $\text{trace-bs} ! i \in \text{fri-vector-position-values } B \ i$
have *bad-to-coord*:
 $\text{wp-event } (\text{staged-alpha-prefix-program } A) \ ?bad \ s \leq$
 $\text{wp-event } (\text{staged-alpha-prefix-program } A)$
 $(\lambda \text{out. } \exists i \in \{.. < \text{ceil-log } \text{clength}\}. \ ?coord \ i \ \text{out}) \ s$

proof (*rule wp-event-mono*)

fix *out*

assume *bad*: *?bad out*

show $\exists i \in \{.. < \text{ceil-log } \text{clength}\}. \ ?coord \ i \ \text{out}$

proof (*cases out*)

case *None*

then show *?thesis* **using** *bad* **by** *simp*

next

case (*Some result*)

then obtain *fr trace-roots trace-bs trace-final t* **where**

out-eq:

$\text{out} = \text{Some } ((\text{fr}, \text{trace-roots}, \text{trace-bs}, \text{trace-final}), t)$

by (*cases result, auto split: prod.splits*)

have *member*: $\text{trace-bs} \in B$

using *bad unfolding out-eq* **by** *simp*

from *fri-vector-member-coordinate-cover*[*OF subset member nonempty*]

obtain *i* **where** *i-bound*: $i < \text{ceil-log } \text{clength}$

and *coord*: $\text{trace-bs} ! i \in \text{fri-vector-position-values } B \ i$

```

    by blast
  show ?thesis
    by (rule beXI[of - i]) (use i-bound coord out-eq in simp-all)
qed
qed
also have ... ≤
  (∑ i < ceil-log clength.
    wp-event (staged-alpha-prefix-program A) (?coord i) s)
  by (rule wp-event-finite-UN-bound) simp-all
finally show ?thesis .
qed

```

lemma *staged-alpha-prefix-trace-coordinate-hit-imp-new-output-hit:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < ceil-log clength*

and *absent:*

trace-fri-challenge-values-absent

(fri-vector-position-values B i) s

and *outcome:*

Some ((fr, trace-roots, trace-bs, trace-final), t) ∈

set-dist (execute (staged-alpha-prefix-program A) s)

and *hit: trace-bs ! i ∈ fri-vector-position-values B i*

shows

hash-new-output-hit-event (fri-vector-position-values B i) s

(Some ((fr, trace-roots, trace-bs, trace-final), t))

proof –

from *outcome* **obtain** *s1 s2 s3 s4* **where**

root-out:

Some (fr, s1) ∈ set-dist (execute (trace-root-stage A) s)

and *record-root:*

Some ((), s2) ∈ set-dist (execute (record-staged-message fr) s1)

and *fri-out:*

Some ((trace-roots, trace-bs), s3) ∈

set-dist

(execute

(staged-trace-fri-program A 0 (ceil-log clength) []) s2)

and *final-out:*

Some (trace-final, s4) ∈

set-dist (execute (trace-final-stage A trace-bs) s3)

and *record-final:*

Some ((), t) ∈

set-dist (execute (record-staged-message trace-final) s4)

unfolding *staged-alpha-prefix-program-def*

by *(auto elim!: set-dist-bindE)*

have *root-controlled:*

controlled-ro-program (trace-root-budget budgets)

(trace-root-stage A)

using *controlled* **unfolding** *staged-adversary-controlled-def* **by** *simp*

```

have final-controlled:
  controlled-ro-program (trace-final-budget budgets)
    (trace-final-stage A trace-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have ext-s-s1:  $s \leq s1$ 
  using controlled-ro-program-extension[OF root-controlled] root-out
  unfolding hash-extension-preserving-def by blast
have s2-eq:
   $s2 = s1 \langle$ 
     $PState := \text{concat } (PState\ s1)\ fr,$ 
     $PTranscript := PTranscript\ s1\ @\ [fr]\rangle$ 
  by (rule record-staged-message-outcome[OF record-root])
have ext-s1-s2:  $s1 \leq s2$ 
  unfolding s2-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have trace-bound:
   $0 + \text{ceil-log length} \leq \text{length } (trace-fri-budgets\ budgets)$ 
  using wf unfolding staged-budget-wellformed-def by simp
have trace-alignment:
   $\text{length } trace-roots = \text{ceil-log length} \wedge$ 
   $\text{length } trace-bs = \text{length } ([\ ] :: 'f\ list) + \text{ceil-log length} \wedge$ 
   $PTranscript\ s3 = PTranscript\ s2\ @\ trace-roots \wedge$ 
   $PState\ s3 = \text{foldl concat } (PState\ s2)\ trace-roots \wedge$ 
   $s2 \leq s3 \wedge$ 
   $PTraceFriCounter\ s3 = PTraceFriCounter\ s2 + \text{ceil-log length} \wedge$ 
   $PCompositionFriCounter\ s3 = PCompositionFriCounter\ s2 \wedge$ 
   $PAlphaCounter\ s3 = PAlphaCounter\ s2 \wedge$ 
   $PQueryCounter\ s3 = PQueryCounter\ s2 \wedge$ 
   $(\forall j < \text{ceil-log length}.$ 
     $\text{fmlookup } (HashMap\ s3)$ 
     $(TraceFriChallenge\ (PTraceFriCounter\ s2 + j)$ 
       $(\text{foldl concat } (PState\ s2)\ (\text{take } (Suc\ j)\ trace-roots))) =$ 
       $Some\ (trace-bs\ !\ (\text{length } ([\ ] :: 'f\ list) + j)))$ 
  by (rule staged-trace-fri-program-alignment
    [OF controlled trace-bound fri-out])
let ?key =
   $TraceFriChallenge\ (PTraceFriCounter\ s2 + i)$ 
   $(\text{foldl concat } (PState\ s2)\ (\text{take } (Suc\ i)\ trace-roots))$ 
have lookup-s3:
   $\text{fmlookup } (HashMap\ s3)\ ?key = Some\ (trace-bs\ !\ i)$ 
  using trace-alignment i-bound by simp
have ext-s-s3:  $s \leq s3$ 
  by (rule hash-ext-trans[OF ext-s-s1])
  (rule hash-ext-trans[OF ext-s1-s2], use trace-alignment in simp)
have ext-s3-s4:  $s3 \leq s4$ 
  using controlled-ro-program-extension[OF final-controlled] final-out
  unfolding hash-extension-preserving-def by blast
have t-eq:
   $t = s4 \langle$ 
     $PState := \text{concat } (PState\ s4)\ trace-final,$ 

```



```

    PTranscript := PTranscript s4 @ [trace-final])
  by (rule record-staged-message-outcome[OF record-final])
have ext-s4-t: s4 ≤ t
  unfolding t-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have ext-s3-t: s3 ≤ t
  by (rule hash-ext-trans[OF ext-s3-s4 ext-s4-t])
have lookup-t:
  fmllookup (HashMap t) ?key = Some (trace-bs ! i)
  by (rule hash-extension-lookup[OF lookup-s3 ext-s3-t])
have none-s: fmllookup (HashMap s) ?key = None
proof (cases fmllookup (HashMap s) ?key)
  case None
  then show ?thesis .
next
case (Some old)
have lookup-s3-old: fmllookup (HashMap s3) ?key = Some old
  by (rule hash-extension-lookup[OF Some ext-s-s3])
then have old-eq: old = trace-bs ! i
  using lookup-s3 by simp
have old ∉ fri-vector-position-values B i
  using absent Some
  unfolding trace-fri-challenge-values-absent-def by blast
then show ?thesis
  using old-eq hit by simp
qed
have hash-map-new-output-hit (fri-vector-position-values B i) s t
  unfolding hash-map-new-output-hit-def
  using none-s lookup-t hit by blast
then show ?thesis
  unfolding hash-new-output-hit-event-def by simp
qed

lemma staged-alpha-prefix-trace-coordinate-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < ceil-log clength
  and absent:
    trace-fri-challenge-values-absent
    (fri-vector-position-values B i) s
  shows
    wp-event (staged-alpha-prefix-program A)
    (λout. case out of None ⇒ False
      | Some ((-, -, trace-bs, -), -) ⇒
        trace-bs ! i ∈ fri-vector-position-values B i) s ≤
    staged-phase-target-error (fri-vector-position-values B i)
    (staged-alpha-search-queries budgets 0)
proof -
  let ?P =
    λout. case out of None ⇒ False

```

```

| Some ((-, -, trace-bs, -), -) ⇒
  trace-bs ! i ∈ fri-vector-position-values B i
let ?H =
  hash-new-output-hit-event (fri-vector-position-values B i) s
have event-imp-hit:
  wp-event (staged-alpha-prefix-program A) ?P s ≤
    wp-event (staged-alpha-prefix-program A) ?H s
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈ set-dist (execute (staged-alpha-prefix-program A) s)
  and event: ?P out
  show ?H out
  proof (cases out)
    case None
    then show ?thesis using event by simp
  next
    case (Some result)
    then obtain fr trace-roots trace-bs trace-final t where
      out-eq:
        out = Some ((fr, trace-roots, trace-bs, trace-final), t)
    by (cases result, auto split: prod.splits)
  have outcome:
    Some ((fr, trace-roots, trace-bs, trace-final), t) ∈
      set-dist (execute (staged-alpha-prefix-program A) s)
    using support unfolding out-eq .
  have hit: trace-bs ! i ∈ fri-vector-position-values B i
    using event unfolding out-eq by simp
  show ?thesis
    unfolding out-eq
    by (rule
      staged-alpha-prefix-trace-coordinate-hit-imp-new-output-hit
      [OF wf controlled i-bound absent outcome hit])
  qed
qed
also have
  wp-event (staged-alpha-prefix-program A) ?H s ≤
    staged-phase-target-error (fri-vector-position-values B i)
    (staged-alpha-search-queries budgets 0)
  by (rule staged-alpha-prefix-target-hit-bound[OF wf controlled])
  finally show ?thesis .
qed

lemma staged-trace-fri-challenge-prefix-program-length:
  assumes outcome:
    Some ((fr, trace-bs, rt), t) ∈
      set-dist
        (execute (staged-trace-fri-challenge-prefix-program A i) s)
  shows length trace-bs = i

```

proof –
from *outcome* **obtain** *roots pre-state post-state* **where**
fri-out:
Some ((roots, trace-bs), post-state) ∈
set-dist
(execute (staged-trace-fri-program A 0 i []) pre-state)
unfolding *staged-trace-fri-challenge-prefix-program-def*
by *(auto elim!: set-dist-bindE)*
have *length trace-bs = length ([] :: 'f list) + i*
by *(rule staged-trace-fri-program-challenge-length[OF fri-out])*
then show *?thesis* **by** *simp*
qed

lemma *staged-trace-fri-prefix-vector-member-imp-position-value:*
assumes *outcome:*
Some ((fr, trace-bs, rt), t) ∈
set-dist
(execute (staged-trace-fri-challenge-prefix-program A i) s)
and *subset: B ⊆ fri-challenge-space i*
and *member: trace-bs ∈ B*
and *j-bound: j < i*
shows *trace-bs ! j ∈ fri-vector-position-values B j*
by *(rule fri-vector-member-imp-position-value*
[OF subset member j-bound])

lemma *staged-trace-fri-prefix-head-challenge-imp-position-value:*
assumes *outcome:*
Some (((fr, trace-bs, rt), b), t) ∈
set-dist
(execute
(staged-trace-fri-challenge-prefix-program A i >>=
(λprefix. receive-trace-fri-challenge >>=
(λb. return (prefix, b)))) s)
and *member: trace-bs @ [b] ∈ B*
shows *b ∈ fri-vector-position-values B i*

proof –
from *outcome* **obtain** *s1* **where**
prefix-out:
Some ((fr, trace-bs, rt), s1) ∈
set-dist
(execute (staged-trace-fri-challenge-prefix-program A i) s)
by *(auto elim!: set-dist-bindE)*
have *len: length trace-bs = i*
by *(rule staged-trace-fri-challenge-prefix-program-length*
[OF prefix-out])
show *?thesis*
by *(rule fri-vector-extend-member-imp-position-value*
[OF member len])
qed

lemma *staged-alpha-program-alignment*:

assumes *outcome*:

Some (as, t) ∈ set-dist (execute (staged-alpha-program n) s)

shows

length as = n ∧
PTranscript t = PTranscript s @ as ∧
PState t = foldl concat (PState s) as ∧
PTraceFriCounter t = PTraceFriCounter s ∧
PCompositionFriCounter t = PCompositionFriCounter s ∧
PAlphaCounter t = PAlphaCounter s + n ∧
PQueryCounter t = PQueryCounter s

using *outcome*

proof (*induction n arbitrary: as s t*)

case 0

then show ?*case* **by** *simp*

next

case (*Suc n*)

from *staged-alpha-program-Suc-outcomeE[OF Suc.premis]*

obtain *a as' s1 s2* **where**

challenge-out:
Some (a, s1) ∈
set-dist (execute receive-alpha-challenge s)

and *record-out:*
Some ((), s2) ∈
set-dist (execute (record-staged-message a) s1)

and *rest-out:*
Some (as', t) ∈
set-dist (execute (staged-alpha-program n) s2)

and *as-eq: as = a # as'*

.

have *challenge-fields:*
PState s1 = PState s ∧
PTranscript s1 = PTranscript s

using *receive-alpha-challenge-outcome[OF challenge-out]* **by** *simp*

have *challenge-counters:*
PTraceFriCounter s1 = PTraceFriCounter s ∧
PCompositionFriCounter s1 = PCompositionFriCounter s ∧
PAlphaCounter s1 = Suc (PAlphaCounter s) ∧
PQueryCounter s1 = PQueryCounter s

using *receive-alpha-challenge-counter-outcome[OF challenge-out]* **by** *simp*

have *s2:*
s2 = s1 |
PState := concat (PState s1) a,
PTranscript := PTranscript s1 @ [a]

by (*rule record-staged-message-outcome[OF record-out]*)

have *s2-fields:*
PState s2 = concat (PState s) a ∧
PTranscript s2 = PTranscript s @ [a] ∧

```

    PTraceFriCounter s2 = PTraceFriCounter s ∧
    PCompositionFriCounter s2 = PCompositionFriCounter s ∧
    PAlphaCounter s2 = Suc (PAlphaCounter s) ∧
    PQueryCounter s2 = PQueryCounter s
  unfolding s2 using challenge-fields challenge-counters by simp
have rest-alignment:
  length as' = n ∧
  PTranscript t = PTranscript s2 @ as' ∧
  PState t = foldl concat (PState s2) as' ∧
  PTraceFriCounter t = PTraceFriCounter s2 ∧
  PCompositionFriCounter t = PCompositionFriCounter s2 ∧
  PAlphaCounter t = PAlphaCounter s2 + n ∧
  PQueryCounter t = PQueryCounter s2
  by (rule Suc.IH[OF rest-out])
show ?case
  unfolding as-eq
  using s2-fields rest-alignment
  by simp
qed

lemma staged-receive-alpha-challenge-preserves-alpha-future-fresh:
  assumes future: alpha-future-fresh s
  and outcome: Some (a, t) ∈ set-dist (execute receive-alpha-challenge s)
  shows alpha-future-fresh t
proof -
  have counter-t: PAlphaCounter t = Suc (PAlphaCounter s)
  using receive-alpha-challenge-counter-outcome[OF outcome] by simp
  show ?thesis
  unfolding alpha-future-fresh-def
proof (intro allI impI)
  fix i x
  assume i-future: PAlphaCounter t ≤ i
  have i-ge-s: PAlphaCounter s ≤ i
  using counter-t i-future by simp
  have neg:
    AlphaChallenge i x ≠
    AlphaChallenge (PAlphaCounter s) (PState s)
  using counter-t i-future by auto
  have fmlookup (HashMap t) (AlphaChallenge i x) =
    fmlookup (HashMap s) (AlphaChallenge i x)
  by (rule receive-alpha-challenge-preserves-other-lookup
    [OF outcome neg])
  also have ... = None
  using future i-ge-s unfolding alpha-future-fresh-def by blast
  finally show fmlookup (HashMap t) (AlphaChallenge i x) = None .
qed
qed

lemma record-staged-message-preserves-alpha-future-fresh:

```

```

assumes future: alpha-future-fresh s
and outcome:
  Some ((), t)  $\in$  set-dist (execute (record-staged-message x) s)
shows alpha-future-fresh t
proof –
  have t-eq:
    t = s⟦
      PState := concat (PState s) x,
      PTranscript := PTranscript s @ [x]⟧
    by (rule record-staged-message-outcome[OF outcome])
  show ?thesis
    using future unfolding t-eq alpha-future-fresh-def by simp
qed

lemma staged-alpha-program-preserves-alpha-future-fresh:
  assumes future: alpha-future-fresh s
  and outcome:
    Some (as, t)  $\in$  set-dist (execute (staged-alpha-program n) s)
  shows alpha-future-fresh t
  using future outcome
proof (induction n arbitrary: as s t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  show ?case
  proof (cases rule:
    staged-alpha-program-Suc-outcomeE[OF Suc.prems(2)])
  case (1 a as' s1 s2)
  have future-s1:
    alpha-future-fresh s1
    by (rule staged-receive-alpha-challenge-preserves-alpha-future-fresh
      [OF Suc.prems(1) 1(1)])
  have future-s2:
    alpha-future-fresh s2
    by (rule record-staged-message-preserves-alpha-future-fresh
      [OF future-s1 1(2)])
  show ?thesis
    by (rule Suc.IH[OF future-s2 1(3)])
  qed
qed

```

```

lemma wp-staged-alpha-program-exact-challenges-bound:
  assumes len: length as = n
  and future: alpha-future-fresh s
  shows
    wp-event (staged-alpha-program n)
    ( $\lambda out. \text{case } out \text{ of } None \Rightarrow False$ 
      | Some (as', -)  $\Rightarrow as' = as$ ) s  $\leq$ 

```

```

      (1 / nnreal size) ^ n
    using len future
  proof (induction n arbitrary: as s)
    case 0
    then have as-empty: as = []
      by simp
    show ?case
      unfolding as-empty by (simp add: wp-event-def wpsimps)
  next
  case (Suc n)
  obtain a as-tail where as-eq: as = a # as-tail
    using Suc.prem1(1) by (cases as) auto
  have len-tail: length as-tail = n
    using Suc.prem1(1) as-eq by simp
  let ?tail =
    staged-alpha-program n ::
      ('f list, 'f protocol-channel) state-monad
  let ?Q =
    λout :: ('f list ×
      'f protocol-channel) option.
      case out of None ⇒ False | Some (as', -) ⇒ as' = as
  let ?Head =
    λout :: ('f ×
      'f protocol-channel) option.
      case out of None ⇒ False | Some (a0, -) ⇒ a0 ∈ {a}
  have head-bound:
    wp-event receive-alpha-challenge ?Head s ≤ 1 / nnreal size
  proof -
    have current-fresh:
      fmlookup (HashMap s)
        (AlphaChallenge (PAlphaCounter s) (PState s)) = None
      by (rule alpha-future-fresh-current[OF Suc.prem2])
    have wp-event receive-alpha-challenge ?Head s =
      nnreal (card ({a} :: 'f set)) / nnreal size
      by (rule wp-receive-alpha-challenge-fresh-set[OF current-fresh])
    then show ?thesis by simp
  qed
  have bind-bound:
    wp-event
      (receive-alpha-challenge >=>
        (λa0. record-staged-message a0 >=>
          (λ-. ?tail >=>
            (λas'. return (a0 # as')))))
      ?Q s ≤
      wp-event receive-alpha-challenge ?Head s *
      (1 / nnreal size) ^ n
  proof (rule wp-event-bind-bound-by-head-and-cont[where P = ?Head])
    show ¬ ?Q None
      by simp
  end

```

```

next
  fix a0 t
  assume recv:
    Some (a0, t) ∈ set-dist (execute receive-alpha-challenge s)
    and not-head: ¬ ?Head (Some (a0, t))
  have a0-ne: a0 ≠ a
    using not-head by simp
  have cont-false:
    wp-event
      (record-staged-message a0 ≫=
        (λ-. ?tail ≫= (λas'. return (a0 # as'))))
      ?Q t ≤
    wp-event
      (record-staged-message a0 ≫=
        (λ-. ?tail ≫= (λas'. return (a0 # as'))))
      (λ- :: ('f list ×
        'f protocol-channel) option. False) t
    by (rule wp-event-mono-on-support)
      (use a0-ne as-eq in
        ‹auto simp: wpsimps elim!: set-dist-bindE
          split: option.splits prod.splits›)
  also have ... = 0
    unfolding wp-event-def wp-def dist-expect-def by simp
  finally have le-zero:
    wp-event
      (record-staged-message a0 ≫=
        (λ-. ?tail ≫= (λas'. return (a0 # as'))))
      ?Q t ≤ 0 .
  show
    wp-event
      (record-staged-message a0 ≫=
        (λ-. ?tail ≫= (λas'. return (a0 # as'))))
      ?Q t = 0
    by (rule antisym[OF le-zero]) simp
next
  fix a0 t
  assume recv:
    Some (a0, t) ∈ set-dist (execute receive-alpha-challenge s)
    and head: ?Head (Some (a0, t))
  have a0-eq: a0 = a
    using head by simp
  have future-t: alpha-future-fresh t
    by (rule staged-receive-alpha-challenge-preserves-alpha-future-fresh
      [OF Suc.prem2(2) recv])
  have record-cont-bound:
    wp-event
      (record-staged-message a0 ≫=
        (λ-. ?tail ≫= (λas'. return (a0 # as'))))
      ?Q t ≤ (1 / nnreal size) ^ n

```



```

proof (rule wp-event-bind-bound-by-cont)
  show  $\neg ?Q \text{ None}$ 
    by simp
next
  fix uu u
  assume record-out:
    Some (uu, u)  $\in$ 
      set-dist (execute (record-staged-message a0) t)
  have record-unit:
    Some ((), u)  $\in$ 
      set-dist (execute (record-staged-message a0) t)
  using record-out by (cases uu) simp
  have future-u: alpha-future-fresh u
    by (rule record-staged-message-preserves-alpha-future-fresh
      [OF future-t record-unit])
  have tail-bound:
    wp-event ?tail
      ( $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
        |  $\text{Some } (as', -) \Rightarrow as' = as\text{-tail}$ ) u
       $\leq (1 / \text{nnreal size}) \wedge^n$ 
    by (rule Suc.IH[OF len-tail future-u])
  have cont-eq:
    wp-event
      ( $?tail \ggg (\lambda as'. \text{return } (a0 \# as'))$ ) ?Q u =
    wp-event ?tail
      ( $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
        |  $\text{Some } (as', -) \Rightarrow as' = as\text{-tail}$ ) u
    by (simp add: wp-event-bind-return-map a0-eq as-eq
      split: option.splits prod.splits)
  show
    wp-event ( $?tail \ggg (\lambda as'. \text{return } (a0 \# as'))$ ) ?Q u
       $\leq (1 / \text{nnreal size}) \wedge^n$ 
    unfolding cont-eq by (rule tail-bound)
  qed
show
  wp-event
    (record-staged-message a0  $\ggg$ 
      ( $\lambda -. ?tail \ggg (\lambda as'. \text{return } (a0 \# as'))$ ))
    ?Q t  $\leq (1 / \text{nnreal size}) \wedge^n$ 
  by (rule record-cont-bound)
qed
have wp-event (staged-alpha-program (Suc n)) ?Q s
   $\leq$  wp-event receive-alpha-challenge ?Head s *
     $(1 / \text{nnreal size}) \wedge^n$ 
  unfolding staged-alpha-program.simps by (rule bind-bound)
also have ...  $\leq (1 / \text{nnreal size}) * (1 / \text{nnreal size}) \wedge^n$ 
  by (rule mult-right-mono[OF head-bound]) simp
also have ... =  $(1 / \text{nnreal size}) \wedge^{Suc\ n}$ 
  by simp

```

finally show ?case .
qed

lemma *wp-staged-alpha-program-finite-set-bound:*

assumes *future: alpha-future-fresh s*

and *finite-B: finite B*

and *lengths: $\bigwedge as. as \in B \implies \text{length } as = n$*

shows

wp-event (staged-alpha-program n)
 $(\lambda out. \text{case out of None} \Rightarrow \text{False}$
 $\quad | \text{Some } (as, -) \Rightarrow as \in B) \ s \leq$
 $\text{nnreal } (\text{card } B) / (\text{nnreal size}) ^ n$

proof –

let ?P =

$\lambda as \text{ out. case out of None} \Rightarrow \text{False}$
 $\quad | \text{Some } (as', -) \Rightarrow as' = as$

have *event-mono:*

wp-event (staged-alpha-program n)
 $(\lambda out. \text{case out of None} \Rightarrow \text{False}$
 $\quad | \text{Some } (as, -) \Rightarrow as \in B) \ s \leq$
wp-event (staged-alpha-program n)
 $(\lambda out. \exists as \in B. ?P \ as \ out) \ s$

by (rule *wp-event-mono*) (auto split: option.splits prod.splits)

also have $\dots \leq (\sum as \in B. (1 / \text{nnreal size}) ^ n)$

by (rule *wp-event-finite-UN-bound[OF finite-B]*)
(rule *wp-staged-alpha-program-exact-challenges-bound*
[*OF lengths future*])

also have $\dots = \text{nnreal } (\text{card } B) * ((1 / \text{nnreal size}) ^ n)$

using *finite-B* **by** *simp*

also have $\dots = \text{nnreal } (\text{card } B) / (\text{nnreal size}) ^ n$

using *size-card* **by** *transfer (simp add: power-divide)*

finally show ?thesis .

qed

lemma *wp-staged-alpha-program-alpha-space-set-bound:*

assumes *future: alpha-future-fresh s*

and *subset: $B \subseteq \text{alpha-space}$*

shows

wp-event (staged-alpha-program (length spec))
 $(\lambda out. \text{case out of None} \Rightarrow \text{False}$
 $\quad | \text{Some } (as, -) \Rightarrow as \in B) \ s \leq$
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD('f)} ^ \text{length spec})$

proof –

have *finite-B: finite B*

by (rule *finite-subset[OF subset]*)
(simp add: *alpha-space-def finite-length-lists-UNIV*)

have *lengths: $\bigwedge as. as \in B \implies \text{length } as = \text{length spec}$*

using *subset unfolding alpha-space-def* **by** *auto*

have *wp-event (staged-alpha-program (length spec))*

```

    (λout. case out of None ⇒ False
      | Some (as, -) ⇒ as ∈ B) s ≤
    nnreal (card B) / (nnreal size) ^ length spec
  by (rule wp-staged-alpha-program-finite-set-bound
      [OF future finite-B lengths])
  also have ... =
    nnreal (card B) / nnreal (CARD('f) ^ length spec)
  using size-card by simp
  finally show ?thesis .
qed

lemma staged-after-alpha-prefix-program-alpha-space-list-bound-from-fresh:
  assumes future: alpha-future-fresh s
  and subset: B ⊆ alpha-space
  shows
    wp-event
      (staged-after-alpha-prefix-program A
        (fr, trace-roots, trace-bs, trace-final))
      (staged-transcript-alpha-list-hit B) s ≤
      nnreal (card B) / nnreal (CARD('f) ^ length spec)
  proof -
    have alpha-bound:
      wp-event (staged-alpha-program (length spec))
        (λout. case out of None ⇒ False
          | Some (as, -) ⇒ as ∈ B) s ≤
          nnreal (card B) / nnreal (CARD('f) ^ length spec)
    by (rule wp-staged-alpha-program-alpha-space-set-bound
        [OF future subset])
    show ?thesis
      by (rule staged-after-alpha-prefix-program-alpha-list-bound-from-alpha-program
          [OF alpha-bound])
  qed

lemma staged-transcript-program-alpha-space-list-bound-from-fresh-prefix:
  assumes prefix-future:
    ∧ prefix t.
      Some (prefix, t) ∈
        set-dist (execute (staged-alpha-prefix-program A)
          adversary-initial-state) ⇒
        alpha-future-fresh t
  and subset: B ⊆ alpha-space
  shows
    wp-event (staged-transcript-program A)
      (staged-transcript-alpha-list-hit B) adversary-initial-state ≤
      nnreal (card B) / nnreal (CARD('f) ^ length spec)
  unfolding staged-transcript-program-alpha-prefix-decomp
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ staged-transcript-alpha-list-hit B None
      unfolding staged-transcript-alpha-list-hit-def by simp
  end

```

```

next
  fix prefix t
  assume prefix-out:
    Some (prefix, t) ∈
      set-dist (execute (staged-alpha-prefix-program A)
        adversary-initial-state)
  obtain fr trace-roots trace-bs trace-final
    where prefix-eq: prefix = (fr, trace-roots, trace-bs, trace-final)
    by (cases prefix) auto
  have future-t: alpha-future-fresh t
    by (rule prefix-future[OF prefix-out])
  show
    wp-event (staged-after-alpha-prefix-program A prefix)
      (staged-transcript-alpha-list-hit B) t ≤
      nnreal (card B) / nnreal (CARD('f) ^ length spec)
  unfolding prefix-eq
  by (rule staged-after-alpha-prefix-program-alpha-space-list-bound-from-fresh
    [OF future-t subset])
qed

lemma staged-composition-fri-program-challenge-prefix-pre-alignment:
  assumes outcome:
    Some ((roots, bs'), t) ∈
      set-dist
        (execute (staged-composition-fri-program A dg i n bs) s)
  shows take (length bs) bs' = bs
  using outcome
proof (induction n arbitrary: i bs roots bs' s t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.premis obtain root b roots' s1 s2 s3 where
    rest-out:
      Some ((roots', bs'), t) ∈
        set-dist
          (execute
            (staged-composition-fri-program
              A dg (Suc i) n (bs @ [b])) s3)
  unfolding staged-composition-fri-program.simps
  by (auto elim!: set-dist-bindE)
  have prefix-suc:
    take (length (bs @ [b])) bs' = bs @ [b]
  by (rule Suc.IH[OF rest-out])
  have take (length bs) bs' =
    take (length bs) (take (length (bs @ [b])) bs')
  by simp
  also have ... = bs
  using prefix-suc by simp

```

finally show ?case .
qed

lemma *staged-composition-fri-program-alignment:*
assumes *controlled: staged-adversary-controlled budgets A*
and *bound:*
 $i + n \leq \text{length } (\text{composition-fri-budgets } \text{budgets})$
and *outcome:*
 $\text{Some } ((\text{roots}, \text{bs}'), t) \in$
 set-dist
 $(\text{execute}$
 $(\text{staged-composition-fri-program } A \text{ dg } i \ n \ \text{bs}) \ s)$
shows
 $\text{length } \text{roots} = n \wedge$
 $\text{length } \text{bs}' = \text{length } \text{bs} + n \wedge$
 $P\text{Transcript } t = P\text{Transcript } s @ \text{roots} \wedge$
 $P\text{State } t = \text{foldl } \text{concat } (P\text{State } s) \ \text{roots} \wedge$
 $s \leq t \wedge$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s \wedge$
 $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s + n \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = P\text{QueryCounter } s \wedge$
 $(\forall j < n.$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{CompositionFriChallenge } (P\text{CompositionFriCounter } s + j)$
 $(\text{foldl } \text{concat } (P\text{State } s) (\text{take } (\text{Suc } j) \ \text{roots}))) =$
 $\text{Some } (\text{bs}' ! (\text{length } \text{bs} + j)))$
using *bound outcome*
proof (*induction n arbitrary: i bs roots bs' s t*)
case 0
then show ?case **by** (*simp add: hash-ext-refl*)
next
case (*Suc n*)
from *staged-composition-fri-program-Suc-outcomeE[OF Suc.premis(2)]*
obtain *root b roots' s1 s2 s3* **where**
stage-out:
 $\text{Some } (\text{root}, s1) \in$
 set-dist
 $(\text{execute } (\text{composition-fri-root-stage } A \text{ dg } i \ \text{bs}) \ s)$
and *record-out:*
 $\text{Some } ((), s2) \in$
 $\text{set-dist } (\text{execute } (\text{record-staged-message } \text{root}) \ s1)$
and *challenge-out:*
 $\text{Some } (b, s3) \in$
 $\text{set-dist } (\text{execute } \text{receive-composition-fri-challenge } s2)$
and *rest-out:*
 $\text{Some } ((\text{roots}', \text{bs}'), t) \in$
 set-dist
 $(\text{execute}$

```

      (staged-composition-fri-program
       A dg (Suc i) n (bs @ [b])) s3)
    and roots-eq: roots = root # roots'
  .
have i-bound: i < length (composition-fri-budgets budgets)
  using Suc.premis(1) by simp
have stage-controlled:
  controlled-ro-program (composition-fri-budgets budgets ! i)
    (composition-fri-root-stage A dg i bs)
  using controlled i-bound
  unfolding staged-adversary-controlled-def by blast
have stage-fields:
  PState s1 = PState s ∧
  PTranscript s1 = PTranscript s ∧
  PTraceFriCounter s1 = PTraceFriCounter s ∧
  PCompositionFriCounter s1 = PCompositionFriCounter s ∧
  PAlphaCounter s1 = PAlphaCounter s ∧
  PQueryCounter s1 = PQueryCounter s
  using controlled-stage-outcome-fields[OF stage-controlled stage-out]
  by simp
have ext-s-s1: s ≤ s1
  using controlled-ro-program-extension[OF stage-controlled] stage-out
  unfolding hash-extension-preserving-def by blast
have s2:
  s2 = s1(|
    PState := concat (PState s1) root,
    PTranscript := PTranscript s1 @ [root])
  by (rule record-staged-message-outcome[OF record-out])
have ext-s1-s2: s1 ≤ s2
  unfolding s2 less-eq-hash-ext-def less-eq-fmap-def by simp
have s2-fields:
  PState s2 = concat (PState s) root ∧
  PTranscript s2 = PTranscript s @ [root] ∧
  PTraceFriCounter s2 = PTraceFriCounter s ∧
  PCompositionFriCounter s2 = PCompositionFriCounter s ∧
  PAlphaCounter s2 = PAlphaCounter s ∧
  PQueryCounter s2 = PQueryCounter s
  unfolding s2 using stage-fields by simp
have challenge-fields:
  PState s3 = PState s2 ∧
  PTranscript s3 = PTranscript s2
  using receive-composition-fri-challenge-outcome[OF challenge-out]
  by simp
have ext-s2-s3: s2 ≤ s3
  using receive-composition-fri-challenge-outcome[OF challenge-out]
  by simp
have challenge-counters:
  PTraceFriCounter s3 = PTraceFriCounter s2 ∧
  PCompositionFriCounter s3 =

```

```

    Suc (PCompositionFriCounter s2) ∧
    PAlphaCounter s3 = PAlphaCounter s2 ∧
    PQueryCounter s3 = PQueryCounter s2
using
    receive-composition-fri-challenge-counter-outcome[OF challenge-out]
by simp
have rest-bound:
    Suc i + n ≤ length (composition-fri-budgets budgets)
    using Suc.premis(1) by simp
have rest-alignment:
    length roots' = n ∧
    length bs' = length (bs @ [b]) + n ∧
    PTranscript t = PTranscript s3 @ roots' ∧
    PState t = foldl concat (PState s3) roots' ∧
    s3 ≤ t ∧
    PTraceFriCounter t = PTraceFriCounter s3 ∧
    PCompositionFriCounter t = PCompositionFriCounter s3 + n ∧
    PAlphaCounter t = PAlphaCounter s3 ∧
    PQueryCounter t = PQueryCounter s3 ∧
    (∀ j < n.
      fmllookup (HashMap t)
        (CompositionFriChallenge (PCompositionFriCounter s3 + j)
          (foldl concat (PState s3) (take (Suc j) roots'))) =
        Some (bs' ! (length (bs @ [b]) + j)))
    by (rule Suc.IH[OF rest-bound rest-out])
have ext-s3-t: s3 ≤ t
    using rest-alignment by simp
have ext-s-t: s ≤ t
    by (rule hash-ext-trans[OF ext-s-s1])
    (rule hash-ext-trans[OF ext-s1-s2],
     rule hash-ext-trans[OF ext-s2-s3 ext-s3-t])
have challenge-lookup-s3:
    fmllookup (HashMap s3)
      (CompositionFriChallenge (PCompositionFriCounter s)
        (concat (PState s) root)) = Some b
    using receive-composition-fri-challenge-counter-outcome[OF challenge-out]
    s2-fields by simp
have challenge-lookup-t:
    fmllookup (HashMap t)
      (CompositionFriChallenge (PCompositionFriCounter s)
        (concat (PState s) root)) = Some b
    by (rule hash-extension-lookup[OF challenge-lookup-s3 ext-s3-t])
have prefix-suc:
    take (length (bs @ [b])) bs' = bs @ [b]
    by (rule staged-composition-fri-program-challenge-prefix-pre-alignment
      [OF rest-out])
have lookup-all:
    ∀ j < Suc n.
      fmllookup (HashMap t)

```

```

      (CompositionFriChallenge (PCompositionFriCounter s + j)
        (foldl concat (PState s) (take (Suc j) (root # roots')))) =
      Some (bs' ! (length bs + j))
proof (intro allI impI)
  fix j
  assume j-bound: j < Suc n
  show
    fmlookup (HashMap t)
      (CompositionFriChallenge (PCompositionFriCounter s + j)
        (foldl concat (PState s) (take (Suc j) (root # roots')))) =
      Some (bs' ! (length bs + j))
proof (cases j)
  case 0
  have bs' ! length bs = b
  proof -
    have idx-lt: length bs < length (bs @ [b])
    by simp
    have bs' ! length bs =
      take (length (bs @ [b])) bs' ! length bs
    using idx-lt by simp
    also have ... = (bs @ [b]) ! length bs
    using prefix-suc by simp
    also have ... = b
    by simp
    finally show ?thesis .
  qed
  then show ?thesis
    using challenge-lookup-t unfolding 0 by simp
next
  case (Suc k)
  then have k-bound: k < n
  using j-bound by simp
  show ?thesis
    using rest-alignment k-bound s2-fields challenge-fields
      challenge-counters
    unfolding Suc
    by simp
  qed
qed
show ?case
  unfolding roots-eq
  using s2-fields challenge-fields challenge-counters rest-alignment
    ext-s-t lookup-all
  by simp
qed

lemma staged-composition-fri-program-challenge-prefix:
assumes outcome:
  Some ((roots, bs'), t) ∈

```



```

      set-dist
      (execute (staged-composition-fri-program A dg i n bs) s)
shows take (length bs) bs' = bs
using outcome
proof (induction n arbitrary: i bs roots bs' s t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.premis obtain root b roots' s1 s2 s3 where
    rest-out:
      Some ((roots', bs'), t) ∈
        set-dist
        (execute
          (staged-composition-fri-program
            A dg (Suc i) n (bs @ [b])) s3)
    unfolding staged-composition-fri-program.simps
    by (auto elim!: set-dist-bindE)
  have prefix-suc:
    take (length (bs @ [b])) bs' = bs @ [b]
    by (rule Suc.IH[OF rest-out])
  have take (length bs) bs' =
    take (length bs) (take (length (bs @ [b])) bs')
    by simp
  also have ... = bs
    using prefix-suc by simp
  finally show ?case .
qed

```

```

lemma staged-composition-fri-program-challenge-length:
  assumes outcome:
    Some ((roots, bs'), t) ∈
      set-dist
      (execute (staged-composition-fri-program A dg i n bs) s)
  shows length bs' = length bs + n
  using outcome
proof (induction n arbitrary: i bs roots bs' s t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.premis obtain root b roots' s1 s2 s3 where
    rest-out:
      Some ((roots', bs'), t) ∈
        set-dist
        (execute
          (staged-composition-fri-program
            A dg (Suc i) n (bs @ [b])) s3)
    unfolding staged-composition-fri-program.simps

```

```

    by (auto elim!: set-dist-bindE)
  have length bs' = length (bs @ [b]) + n
    by (rule Suc.IH[OF rest-out])
  then show ?case by simp
qed

lemma staged-composition-fri-challenge-prefix-program-length:
  assumes outcome:
    Some ((dg, composition-bs, rt), t) ∈
      set-dist
        (execute (staged-composition-fri-challenge-prefix-program A i) s)
  shows length composition-bs = i
proof -
  from outcome obtain as s1 dg0 s2 s3 where
    inner:
      Some ((dg, composition-bs, rt), t) ∈
        set-dist
          (execute
            (let composition-rounds = ceil-log (to-nat dg0 + 1)
              in assert (Suc i ≤ composition-rounds) >>=
                (λ-. assert
                  (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
                    (λ-.
                      staged-composition-fri-program A dg0 0 i [] >>=
                        (λ(composition-roots, composition-bs).
                          composition-fri-root-stage A dg0 i
                            composition-bs >>=
                              (λrt.
                                record-staged-message rt >>=
                                  (λ-. return
                                    (dg0, composition-bs, rt)))))))
              s3)
            unfolding staged-composition-fri-challenge-prefix-program-def
            by (auto elim!: set-dist-bindE)
          let ?rounds = ceil-log (to-nat dg0 + 1)
          have fri-out:
            ∃ composition-roots s4.
              Some ((composition-roots, composition-bs), s4) ∈
                set-dist
                  (execute (staged-composition-fri-program A dg0 0 i []) s3)
          proof (cases Suc i ≤ ?rounds)
            case False
            then show ?thesis
              using inner
              by (auto simp: assert-def throw-no-outcome elim!: set-dist-bindE)
          next
            case first-guard: True
            show ?thesis
              proof (cases ?rounds ≤ ceil-log (maxDegree + 1))

```

```

    case False
  then show ?thesis
    using inner first-guard
    by (auto simp: assert-def throw-no-outcome elim!: set-dist-bindE)
next
case second-guard: True
from inner obtain composition-roots s4 where
  Some ((composition-roots, composition-bs), s4) ∈
    set-dist
      (execute (staged-composition-fri-program A dg0 0 i []) s3)
using first-guard second-guard
by (auto simp: assert-def elim!: set-dist-bindE split: prod.splits)
then show ?thesis by blast
qed
qed
then obtain composition-roots s4 where fri-out:
  Some ((composition-roots, composition-bs), s4) ∈
    set-dist
      (execute (staged-composition-fri-program A dg0 0 i []) s3)
by blast
have length composition-bs = length ([] :: 'f list) + i
  by (rule staged-composition-fri-program-challenge-length[OF fri-out])
then show ?thesis by simp
qed

lemma staged-composition-fri-prefix-vector-member-imp-position-value:
  assumes outcome:
    Some ((dg, composition-bs, rt), t) ∈
      set-dist
        (execute (staged-composition-fri-challenge-prefix-program A i) s)
  and subset: B ⊆ fri-challenge-space i
  and member: composition-bs ∈ B
  and j-bound: j < i
  shows composition-bs ! j ∈ fri-vector-position-values B j
  by (rule fri-vector-member-imp-position-value
      [OF subset member j-bound])

lemma staged-composition-fri-prefix-head-challenge-imp-position-value:
  assumes outcome:
    Some (((dg, composition-bs, rt), b), t) ∈
      set-dist
        (execute
          (staged-composition-fri-challenge-prefix-program A i ≫=
            (λprefix. receive-composition-fri-challenge ≫=
              (λb. return (prefix, b)))) s)
  and member: composition-bs @ [b] ∈ B
  shows b ∈ fri-vector-position-values B i
proof -
  from outcome obtain s1 where

```

```

prefix-out:
  Some ((dg, composition-bs, rt), s1) ∈
    set-dist
    (execute (staged-composition-fri-challenge-prefix-program A i) s)
  by (auto elim!: set-dist-bindE)
have len: length composition-bs = i
  by (rule staged-composition-fri-challenge-prefix-program-length
      [OF prefix-out])
show ?thesis
  by (rule fri-vector-extend-member-imp-position-value
      [OF member len])
qed

```

lemma *staged-composition-fri-vector-bad-coordinate-cover-bound:*

assumes *subset:*

$\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$

shows

```

wp-event (staged-composition-fri-vector-program A)
  (λout. case out of None ⇒ False
    | Some ((dg, composition-bs), -) ⇒
      composition-bs ∈ B dg ∧
      0 < ceil-log (to-nat dg + 1) ∧
      ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)) s ≤
  (∑ i < ceil-log (maxDegree + 1).
    wp-event (staged-composition-fri-vector-program A)
      (λout. case out of None ⇒ False
        | Some ((dg, composition-bs), -) ⇒
          i < ceil-log (to-nat dg + 1) ∧
          composition-bs ! i ∈
            fri-vector-position-values (B dg) i) s)

```

proof –

let *?bad* =

```

λout. case out of None ⇒ False
  | Some ((dg, composition-bs), -) ⇒
    composition-bs ∈ B dg ∧
    0 < ceil-log (to-nat dg + 1) ∧
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)

```

let *?coord* =

```

λi out. case out of None ⇒ False
  | Some ((dg, composition-bs), -) ⇒
    i < ceil-log (to-nat dg + 1) ∧
    composition-bs ! i ∈ fri-vector-position-values (B dg) i

```

have *bad-to-coord:*

```

wp-event (staged-composition-fri-vector-program A) ?bad s ≤
  wp-event (staged-composition-fri-vector-program A)
    (λout. ∃ i ∈ {.. $\text{ceil-log } (\text{maxDegree} + 1)$ }. ?coord i out) s

```

proof (rule *wp-event-mono*)

fix *out*

assume *bad: ?bad out*

```

show  $\exists i \in \{..<\text{ceil-log } (\text{maxDegree} + 1)\}$ .  $?coord\ i\ out$ 
proof (cases out)
  case None
  then show  $?thesis$  using bad by simp
next
  case (Some result)
  then obtain dg composition-bs t where
    out-eq: out = Some ((dg, composition-bs), t)
    by (cases result, auto split: prod.splits)
  let  $?n = \text{ceil-log } (\text{to-nat } dg + 1)$ 
  have member: composition-bs  $\in B\ dg$ 
    and nonempty:  $0 < ?n$ 
    and rounds-bound:  $?n \leq \text{ceil-log } (\text{maxDegree} + 1)$ 
    using bad unfolding out-eq by simp-all
  from fri-vector-member-coordinate-cover
    [OF subset[of dg] member nonempty]
  obtain i where i-bound:  $i < ?n$ 
    and coord:
      composition-bs !  $i \in \text{fri-vector-position-values } (B\ dg)\ i$ 
    by blast
  have i-max:  $i < \text{ceil-log } (\text{maxDegree} + 1)$ 
    using i-bound rounds-bound by simp
  show  $?thesis$ 
    by (rule bexI[of - i])
    (use i-bound i-max coord out-eq in simp-all)
  qed
qed
also have  $\dots \leq$ 
  ( $\sum i < \text{ceil-log } (\text{maxDegree} + 1)$ .
    wp-event (staged-composition-fri-vector-program A) (?coord i) s)
  by (rule wp-event-finite-UN-bound) simp-all
finally show  $?thesis$  .
qed

```

```

lemma
  staged-composition-fri-vector-coordinate-hit-imp-new-output-hit:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i < \text{ceil-log } (\text{to-nat } dg + 1)$ 
and absent:
  composition-fri-challenge-values-absent
  (fri-vector-position-values (B dg) i) s
and outcome:
  Some ((dg, composition-bs), t)  $\in$ 
  set-dist (execute (staged-composition-fri-vector-program A) s)
and hit:
  composition-bs !  $i \in \text{fri-vector-position-values } (B\ dg)\ i$ 
shows
  hash-new-output-hit-event (fri-vector-position-values (B dg) i) s

```

```

    (Some ((dg, composition-bs), t))
proof –
from outcome obtain alpha-prefix s1 as s2 s3 s4 s5 roots where
  alpha-prefix-out:
    Some (alpha-prefix, s1) ∈
      set-dist (execute (staged-alpha-prefix-program A) s)
and alpha-out:
    Some (as, s2) ∈
      set-dist (execute (staged-alpha-program (length spec)) s1)
and degree-out:
    Some (dg, s3) ∈ set-dist (execute (degree-stage A as) s2)
and record-degree:
    Some ((), s4) ∈
      set-dist (execute (record-staged-message dg) s3)
and assert-out:
    Some ((), s5) ∈
      set-dist
        (execute
          (assert
            (ceil-log (to-nat dg + 1) ≤
              ceil-log (maxDegree + 1))) s4)
and fri-out:
    Some ((roots, composition-bs), t) ∈
      set-dist
        (execute
          (staged-composition-fri-program A dg 0
            (ceil-log (to-nat dg + 1)) [] s5)
unfolding staged-composition-fri-vector-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
using assert-out unfolding assert-def
by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
  (auto simp: throw-no-outcome)
have len:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
using round-bound wf unfolding staged-budget-wellformed-def by simp
have alignment:
  length roots = ceil-log (to-nat dg + 1) ∧
  length composition-bs =
    length ([] :: 'f list) + ceil-log (to-nat dg + 1) ∧
  PTranscript t = PTranscript s5 @ roots ∧
  PState t = foldl concat (PState s5) roots ∧
  s5 ≤ t ∧
  PTraceFriCounter t = PTraceFriCounter s5 ∧
  PCompositionFriCounter t =
    PCompositionFriCounter s5 + ceil-log (to-nat dg + 1) ∧

```

```

PAlphaCounter t = PAlphaCounter s5 ∧
PQueryCounter t = PQueryCounter s5 ∧
(∀ j < ceil-log (to-nat dg + 1).
  fmlookup (HashMap t)
    (CompositionFriChallenge (PCompositionFriCounter s5 + j)
      (foldl concat (PState s5) (take (Suc j) roots))) =
      Some (composition-bs ! (length ([]) :: 'f list) + j)))
by (rule staged-composition-fri-program-alignment
  [OF controlled len fri-out])
let ?key =
  CompositionFriChallenge (PCompositionFriCounter s5 + i)
    (foldl concat (PState s5) (take (Suc i) roots))
have lookup-t:
  fmlookup (HashMap t) ?key = Some (composition-bs ! i)
  using alignment i-bound by simp
have ext-s-t: s ≤ t
  using hash-target-program-staged-composition-fri-vector-program
    [OF wf controlled,
      of fri-vector-position-values (B dg) i] outcome
  unfolding hash-target-program-def hash-extension-preserving-def
  by blast
have none-s: fmlookup (HashMap s) ?key = None
proof (cases fmlookup (HashMap s) ?key)
  case None
    then show ?thesis .
next
  case (Some old)
    have old-lookup-t:
      fmlookup (HashMap t) ?key = Some old
      by (rule hash-extension-lookup[OF Some ext-s-t])
    then have old-eq: old = composition-bs ! i
    using lookup-t by simp
    have old ∉ fri-vector-position-values (B dg) i
    using absent Some
    unfolding composition-fri-challenge-values-absent-def by blast
    then show ?thesis
    using old-eq hit by simp
qed
have hash-map-new-output-hit
  (fri-vector-position-values (B dg) i) s t
  unfolding hash-map-new-output-hit-def
  using none-s lookup-t hit by blast
then show ?thesis
  unfolding hash-new-output-hit-event-def by simp
qed

lemma staged-composition-fri-vector-coordinate-bound-fixed-degree:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A

```

```

and i-bound:  $i < \text{ceil-log } (\text{to-nat } dg + 1)$ 
and absent:
  composition-fri-challenge-values-absent
  (fri-vector-position-values ( $B \ dg$ )  $i$ )  $s$ 
shows
  wp-event (staged-composition-fri-vector-program  $A$ )
    ( $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
      | Some ( $(dg', \text{composition-bs}), -$ )  $\Rightarrow$ 
         $dg' = dg \wedge$ 
         $\text{composition-bs} ! i \in \text{fri-vector-position-values } (B \ dg) \ i \ s \leq$ 
        staged-phase-target-error (fri-vector-position-values ( $B \ dg$ )  $i$ )
        (staged-composition-fri-vector-search-queries budgets)
proof -
  let  $?P =$ 
     $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
    | Some ( $(dg', \text{composition-bs}), -$ )  $\Rightarrow$ 
       $dg' = dg \wedge$ 
       $\text{composition-bs} ! i \in \text{fri-vector-position-values } (B \ dg) \ i$ 
  let  $?H =$ 
    hash-new-output-hit-event (fri-vector-position-values ( $B \ dg$ )  $i$ )  $s$ 
have event-imp-hit:
  wp-event (staged-composition-fri-vector-program  $A$ )  $?P \ s \leq$ 
  wp-event (staged-composition-fri-vector-program  $A$ )  $?H \ s$ 
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
     $out \in \text{set-dist}$ 
    (execute (staged-composition-fri-vector-program  $A$ )  $s$ )
  and event:  $?P \ out$ 
  show  $?H \ out$ 
  proof (cases out)
    case None
    then show ?thesis using event by simp
  next
    case (Some result)
    then obtain  $dg' \ \text{composition-bs} \ t$  where
      out-eq:  $out = \text{Some } ((dg', \text{composition-bs}), t)$ 
      by (cases result, auto split: prod.splits)
    have dg-eq:  $dg' = dg$ 
      using event unfolding out-eq by simp
    have outcome:
       $\text{Some } ((dg, \text{composition-bs}), t) \in$ 
      set-dist
      (execute (staged-composition-fri-vector-program  $A$ )  $s$ )
      using support unfolding out-eq dg-eq .
    have hit:
       $\text{composition-bs} ! i \in \text{fri-vector-position-values } (B \ dg) \ i$ 
      using event unfolding out-eq by simp
    show ?thesis

```



```

unfolding out-eq dg-eq
by (rule
  staged-composition-fri-vector-coordinate-hit-imp-new-output-hit
  [where  $B=B$ ,  $OF$  wf controlled  $i$ -bound absent outcome hit])
qed
qed
also have
  wp-event (staged-composition-fri-vector-program  $A$ ) ? $H$   $s \leq$ 
    staged-phase-target-error (fri-vector-position-values ( $B$  dg)  $i$ )
    (staged-composition-fri-vector-search-queries budgets)
by (rule staged-composition-fri-vector-target-hit-bound
  [OF wf controlled])
finally show ?thesis .
qed

lemma composition-fri-challenge-values-absent-subset:
  assumes subset:  $A \subseteq B$ 
  and absent: composition-fri-challenge-values-absent  $B$   $s$ 
shows composition-fri-challenge-values-absent  $A$   $s$ 
using subset absent
unfolding composition-fri-challenge-values-absent-def by blast

lemma hash-map-new-output-hit-mono-set:
  assumes subset:  $A \subseteq B$ 
  and hit: hash-map-new-output-hit  $A$   $s$   $t$ 
shows hash-map-new-output-hit  $B$   $s$   $t$ 
using subset hit unfolding hash-map-new-output-hit-def by blast

lemma hash-new-output-hit-event-mono-set:
  assumes subset:  $A \subseteq B$ 
  and hit: hash-new-output-hit-event  $A$   $s$  out
shows hash-new-output-hit-event  $B$   $s$  out
proof (cases out)
  case None
  then show ?thesis
    using hit unfolding hash-new-output-hit-event-def by simp
next
  case (Some result)
  then obtain  $x$   $t$  where out-eq: out = Some ( $x$ ,  $t$ )
    by (cases result) simp
  have map-hit: hash-map-new-output-hit  $A$   $s$   $t$ 
    using hit unfolding out-eq hash-new-output-hit-event-def by simp
  have hash-map-new-output-hit  $B$   $s$   $t$ 
    by (rule hash-map-new-output-hit-mono-set[OF subset map-hit])
  show ?thesis
    using ⟨hash-map-new-output-hit  $B$   $s$   $t$ ⟩
    unfolding out-eq hash-new-output-hit-event-def by simp
qed

```

lemma *staged-composition-fri-vector-coordinate-bound-union*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *absent*:
composition-fri-challenge-values-absent
 $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i) \ s$
shows
 $wp\text{-event } (staged\text{-composition-fri-vector-program } A)$
 $(\lambda out. \text{case out of None} \Rightarrow \text{False}$
 $\quad | \text{Some } ((dg, \text{composition-bs}), -) \Rightarrow$
 $\quad \quad i < \text{ceil-log } (to\text{-nat } dg + 1) \wedge$
 $\quad \quad \text{composition-bs} \ ! \ i \in$
 $\quad \quad \text{fri-vector-position-values } (B \ dg) \ i) \ s \leq$
 $staged\text{-phase-target-error}$
 $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$
 $(staged\text{-composition-fri-vector-search-queries budgets})$
proof –
let $?U = (\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$
let $?P =$
 $\lambda out. \text{case out of None} \Rightarrow \text{False}$
 $\quad | \text{Some } ((dg, \text{composition-bs}), -) \Rightarrow$
 $\quad \quad i < \text{ceil-log } (to\text{-nat } dg + 1) \wedge$
 $\quad \quad \text{composition-bs} \ ! \ i \in \text{fri-vector-position-values } (B \ dg) \ i$
let $?H = \text{hash-new-output-hit-event } ?U \ s$
have *event-imp-hit*:
 $wp\text{-event } (staged\text{-composition-fri-vector-program } A) \ ?P \ s \leq$
 $wp\text{-event } (staged\text{-composition-fri-vector-program } A) \ ?H \ s$
proof (rule *wp-event-mono-on-support*)
fix *out*
assume *support*:
 $out \in \text{set-dist}$
 $(\text{execute } (staged\text{-composition-fri-vector-program } A) \ s)$
and *event*: $?P \ out$
show $?H \ out$
proof (cases *out*)
case *None*
then show *?thesis* **using** *event* **by** *simp*
next
case (Some *result*)
then obtain *dg composition-bs t* **where**
 $out\text{-eq}: out = \text{Some } ((dg, \text{composition-bs}), t)$
by (cases *result*, *auto split: prod.splits*)
have *outcome*:
 $\text{Some } ((dg, \text{composition-bs}), t) \in$
 set-dist
 $(\text{execute } (staged\text{-composition-fri-vector-program } A) \ s)$
using *support* **unfolding** *out-eq* .
have *i-bound*: $i < \text{ceil-log } (to\text{-nat } dg + 1)$
using *event* **unfolding** *out-eq* **by** *simp*

```

have hit:
  composition-bs ! i ∈ fri-vector-position-values (B dg) i
  using event unfolding out-eq by simp
have small-absent:
  composition-fri-challenge-values-absent
    (fri-vector-position-values (B dg) i) s
  by (rule composition-fri-challenge-values-absent-subset
    [OF - absent]) blast
have small-hit:
  hash-new-output-hit-event
    (fri-vector-position-values (B dg) i) s
    (Some ((dg, composition-bs), t))
  by (rule
    staged-composition-fri-vector-coordinate-hit-imp-new-output-hit
    [where B=B,
      OF wf controlled i-bound small-absent outcome hit])
have subset:
  fri-vector-position-values (B dg) i ⊆ ?U
  by blast
show ?thesis
  unfolding out-eq
  by (rule hash-new-output-hit-event-mono-set
    [OF subset small-hit])
qed
qed
also have
  wp-event (staged-composition-fri-vector-program A) ?H s ≤
    staged-phase-target-error ?U
    (staged-composition-fri-vector-search-queries budgets)
  by (rule staged-composition-fri-vector-target-hit-bound
    [OF wf controlled])
finally show ?thesis .
qed

```

```

lemma staged-composition-fri-vector-bad-bound-union:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and subset:
   $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$ 
and absent:
   $\bigwedge i. i < \text{ceil-log } (\text{maxDegree} + 1) \implies$ 
    composition-fri-challenge-values-absent
      ( $\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i$ ) s
shows
  wp-event (staged-composition-fri-vector-program A)
    ( $\lambda out. \text{case out of None} \implies \text{False}$ 
      |  $\text{Some } ((dg, \text{composition-bs}), -) \implies$ 
        composition-bs ∈ B dg ∧
         $0 < \text{ceil-log } (\text{to-nat } dg + 1) \wedge$ 

```

$\text{ceil-log } (\text{to-nat } dg + 1) \leq \text{ceil-log } (\text{maxDegree} + 1)) \ s \leq$
 $(\sum i < \text{ceil-log } (\text{maxDegree} + 1)).$
 $\text{staged-phase-target-error}$
 $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$
 $(\text{staged-composition-fri-vector-search-queries budgets}))$
proof –
let $?coord =$
 $\lambda i \text{ out. case out of None} \Rightarrow \text{False}$
 $| \text{Some } ((dg, \text{composition-bs}), -) \Rightarrow$
 $i < \text{ceil-log } (\text{to-nat } dg + 1) \wedge$
 $\text{composition-bs} ! i \in \text{fri-vector-position-values } (B \ dg) \ i$
have cover:
 $\text{wp-event } (\text{staged-composition-fri-vector-program } A)$
 $(\lambda \text{out. case out of None} \Rightarrow \text{False}$
 $| \text{Some } ((dg, \text{composition-bs}), -) \Rightarrow$
 $\text{composition-bs} \in B \ dg \wedge$
 $0 < \text{ceil-log } (\text{to-nat } dg + 1) \wedge$
 $\text{ceil-log } (\text{to-nat } dg + 1) \leq$
 $\text{ceil-log } (\text{maxDegree} + 1)) \ s \leq$
 $(\sum i < \text{ceil-log } (\text{maxDegree} + 1)).$
 $\text{wp-event } (\text{staged-composition-fri-vector-program } A)$
 $(?coord \ i) \ s)$
by $(\text{rule staged-composition-fri-vector-bad-coordinate-cover-bound}$
 $[OF \ \text{subset}])$
also have $\dots \leq$
 $(\sum i < \text{ceil-log } (\text{maxDegree} + 1)).$
 $\text{staged-phase-target-error}$
 $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$
 $(\text{staged-composition-fri-vector-search-queries budgets}))$
proof (rule sum-mono)
fix i
assume $i\text{-in: } i \in \{.. < \text{ceil-log } (\text{maxDegree} + 1)\}$
then have $i\text{-bound: } i < \text{ceil-log } (\text{maxDegree} + 1)$
by simp
show
 $\text{wp-event } (\text{staged-composition-fri-vector-program } A)$
 $(?coord \ i) \ s \leq$
 $\text{staged-phase-target-error}$
 $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$
 $(\text{staged-composition-fri-vector-search-queries budgets}))$
by $(\text{rule staged-composition-fri-vector-coordinate-bound-union}$
 $[OF \ \text{wf controlled absent}[OF \ i\text{-bound}]])$
qed
finally show $?thesis$.
qed

lemma $\text{staged-composition-fri-full-vector-bad-bound-union:}$
assumes $\text{wf: staged-budget-wellformed budgets}$
and $\text{controlled: staged-adversary-controlled budgets } A$

```

and subset:
   $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$ 
and absent:
   $\bigwedge i. i < \text{ceil-log } (\text{maxDegree} + 1) \implies$ 
    composition-fri-challenge-values-absent
     $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$ 
    adversary-initial-state
shows
  wp-event (staged-composition-fri-full-vector-program A)
    (staged-composition-prefix-fri-vector-hit B)
    adversary-initial-state  $\leq$ 
     $(\sum i < \text{ceil-log } (\text{maxDegree} + 1).$ 
      staged-phase-target-error
       $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$ 
      (staged-composition-fri-vector-search-queries budgets))
proof -
  let ?Q =
     $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
    | Some ((dg, composition-bs), -)  $\Rightarrow$ 
      composition-bs  $\in B \ dg \wedge$ 
       $0 < \text{ceil-log } (\text{to-nat } dg + 1) \wedge$ 
       $\text{ceil-log } (\text{to-nat } dg + 1) \leq \text{ceil-log } (\text{maxDegree} + 1)$ 
  have projected:
    wp-event (staged-composition-fri-vector-program A) ?Q
    adversary-initial-state  $\leq$ 
     $(\sum i < \text{ceil-log } (\text{maxDegree} + 1).$ 
      staged-phase-target-error
       $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$ 
      (staged-composition-fri-vector-search-queries budgets))
  by (rule staged-composition-fri-vector-bad-bound-union
    [OF wf controlled subset absent])
  have eq:
    wp-event (staged-composition-fri-vector-program A) ?Q
    adversary-initial-state =
    wp-event (staged-composition-fri-full-vector-program A)
      (staged-composition-prefix-fri-vector-hit B)
      adversary-initial-state
  unfolding staged-composition-fri-vector-program-projection
    wp-event-bind-return-map
    staged-composition-prefix-fri-vector-hit-def
  by (simp split: option.splits prod.splits)
  show ?thesis
  using projected unfolding eq .
qed

```

```

lemma staged-transcript-program-composition-fri-vector-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and subset:

```

$\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and *absent*:
 $\bigwedge i. i < \text{ceil-log } (\text{maxDegree} + 1) \implies$
 $\text{composition-fri-challenge-values-absent}$
 $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$
 $\text{adversary-initial-state}$
shows
 $\text{wp-event } (\text{staged-transcript-program } A)$
 $(\text{staged-transcript-composition-fri-vector-guarded-hit } B)$
 $\text{adversary-initial-state} \leq$
 $(\sum i < \text{ceil-log } (\text{maxDegree} + 1).$
 $\text{staged-phase-target-error}$
 $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$
 $(\text{staged-composition-fri-vector-search-queries budgets}))$
proof –
have *prefix-bound*:
 $\text{wp-event } (\text{staged-composition-fri-full-vector-program } A)$
 $(\text{staged-composition-prefix-fri-vector-hit } B)$
 $\text{adversary-initial-state} \leq$
 $(\sum i < \text{ceil-log } (\text{maxDegree} + 1).$
 $\text{staged-phase-target-error}$
 $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$
 $(\text{staged-composition-fri-vector-search-queries budgets}))$
by (rule *staged-composition-fri-full-vector-bad-bound-union*
 $[OF \ \text{wf controlled subset absent}]$)
show *?thesis*
unfolding *staged-transcript-program-composition-fri-full-vector-decomp*
proof (rule *wp-event-bind-bound-by-head-event*[*OF prefix-bound*])
show
 $\text{staged-transcript-composition-fri-vector-guarded-hit } B \ \text{None} \implies$
 $\text{staged-composition-prefix-fri-vector-hit } B \ \text{None}$
unfolding *staged-transcript-composition-fri-vector-guarded-hit-def*
 $\text{staged-composition-prefix-fri-vector-hit-def}$
by *simp*
next
fix *prefix t out*
assume *cont*:
 $out \in$
 set-dist
 $(\text{execute}$
 $(\text{staged-after-composition-fri-full-vector-program } A \ \text{prefix}) \ t)$
and *hit*:
 $\text{staged-transcript-composition-fri-vector-guarded-hit } B \ out$
show $\text{staged-composition-prefix-fri-vector-hit } B \ (\text{Some } (\text{prefix}, t))$
by (rule *staged-after-composition-fri-full-vector-program-hitD*
 $[OF \ cont \ hit]$)
qed
qed

lemma *staged-query-program-alignment*:

assumes *controlled*: *staged-adversary-controlled budgets A*
and *bound*: $i + n \leq \text{length } (\text{query-opening-budgets budgets})$
and *outcome*:
Some (chunks, t) ∈
set-dist (execute (staged-query-program A i n) s)

shows
 $\text{length chunks} = n \wedge$
 $P\text{Transcript } t = P\text{Transcript } s @ \text{List.concat chunks} \wedge$
 $P\text{State } t = \text{foldl concat } (P\text{State } s) (\text{List.concat chunks}) \wedge$
 $P\text{TraceFriCounter } t = P\text{TraceFriCounter } s \wedge$
 $P\text{CompositionFriCounter } t = P\text{CompositionFriCounter } s \wedge$
 $P\text{AlphaCounter } t = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } t = P\text{QueryCounter } s + n$

using *bound outcome*

proof (*induction n arbitrary: i chunks s t*)
case 0
then show ?case **by** *simp*

next
case (*Suc n*)
from *staged-query-program-Suc-outcomeE[OF Suc.prem(2)]*
obtain *raw chunk chunks' s1 s2 s3* **where**
challenge-out:
Some (raw, s1) ∈
set-dist (execute receive-query-index-challenge s)
and *stage-out*:
Some (chunk, s2) ∈
set-dist (execute (query-opening-stage A i raw) s1)
and *record-out*:
Some ((), s3) ∈
set-dist (execute (record-staged-messages chunk) s2)
and *rest-out*:
Some (chunks', t) ∈
set-dist (execute (staged-query-program A (Suc i) n) s3)
and *chunks-eq*: $\text{chunks} = \text{chunk} \# \text{chunks}'$
.

have *challenge-fields*:
 $P\text{State } s1 = P\text{State } s \wedge$
 $P\text{Transcript } s1 = P\text{Transcript } s$
using *receive-query-index-challenge-outcome[OF challenge-out]* **by** *simp*

have *challenge-counters*:
 $P\text{TraceFriCounter } s1 = P\text{TraceFriCounter } s \wedge$
 $P\text{CompositionFriCounter } s1 = P\text{CompositionFriCounter } s \wedge$
 $P\text{AlphaCounter } s1 = P\text{AlphaCounter } s \wedge$
 $P\text{QueryCounter } s1 = \text{Suc } (P\text{QueryCounter } s)$
using *receive-query-index-challenge-counter-outcome[OF challenge-out]*
by *simp*

have *i-bound*: $i < \text{length } (\text{query-opening-budgets budgets})$
using *Suc.prem(1)* **by** *simp*

```

have stage-controlled:
  controlled-ro-program (query-opening-budgets budgets ! i)
    (query-opening-stage A i raw)
  using controlled i-bound
  unfolding staged-adversary-controlled-def by blast
have stage-fields:
  PState s2 = PState s1 ∧
  PTranscript s2 = PTranscript s1 ∧
  PTraceFriCounter s2 = PTraceFriCounter s1 ∧
  PCompositionFriCounter s2 = PCompositionFriCounter s1 ∧
  PAlphaCounter s2 = PAlphaCounter s1 ∧
  PQueryCounter s2 = PQueryCounter s1
  using controlled-stage-outcome-fields[OF stage-controlled stage-out]
  by simp
have s3:
  s3 = s2(|
    PState := foldl concat (PState s2) chunk,
    PTranscript := PTranscript s2 @ chunk|)
  by (rule record-staged-messages-outcome[OF record-out])
have s3-fields:
  PState s3 = foldl concat (PState s) chunk ∧
  PTranscript s3 = PTranscript s @ chunk ∧
  PTraceFriCounter s3 = PTraceFriCounter s ∧
  PCompositionFriCounter s3 = PCompositionFriCounter s ∧
  PAlphaCounter s3 = PAlphaCounter s ∧
  PQueryCounter s3 = Suc (PQueryCounter s)
  unfolding s3
  using challenge-fields challenge-counters stage-fields by simp
have rest-bound:
  Suc i + n ≤ length (query-opening-budgets budgets)
  using Suc.prem(1) by simp
have rest-alignment:
  length chunks' = n ∧
  PTranscript t = PTranscript s3 @ List.concat chunks' ∧
  PState t =
    foldl concat (PState s3) (List.concat chunks') ∧
  PTraceFriCounter t = PTraceFriCounter s3 ∧
  PCompositionFriCounter t = PCompositionFriCounter s3 ∧
  PAlphaCounter t = PAlphaCounter s3 ∧
  PQueryCounter t = PQueryCounter s3 + n
  by (rule Suc.IH[OF rest-bound rest-out])
show ?case
  unfolding chunks-eq
  using s3-fields rest-alignment
  by (simp add: foldl-append)
qed

```

```

lemma staged-query-program-outcome-with-raws:
  assumes controlled: staged-adversary-controlled budgets A

```



```

and bound:  $i + n \leq \text{length } (\text{query-opening-budgets } \text{budgets})$ 
and outcome:
   $\text{Some } (\text{chunks}, t) \in$ 
     $\text{set-dist } (\text{execute } (\text{staged-query-program } A \ i \ n) \ s)$ 
shows
 $\exists \text{raw-idxs query-idxs.}$ 
   $\text{length raw-idxs} = n \wedge$ 
   $\text{query-idxs} = \text{map } (\lambda \text{raw. index } (\text{to-nat raw})) \text{ raw-idxs} \wedge$ 
   $\text{length chunks} = n \wedge$ 
   $P\text{Transcript } t = P\text{Transcript } s @ \text{List.concat chunks} \wedge$ 
   $P\text{State } t = \text{state-after-query-chunks } (P\text{State } s) \text{ chunks } n \wedge$ 
   $s \leq t \wedge$ 
   $P\text{QueryCounter } t = P\text{QueryCounter } s + n \wedge$ 
   $(\forall j < n.$ 
     $\text{fmlookup } (\text{HashMap } t)$ 
     $(\text{QueryIndexChallenge } (P\text{QueryCounter } s + j)$ 
       $(\text{state-after-query-chunks } (P\text{State } s) \text{ chunks } j)) =$ 
       $\text{Some } (\text{raw-idxs } ! j)) \wedge$ 
     $(\forall \text{idx} \in \text{set query-idxs. idx} < \text{length} * \text{scale})$ 
  using bound outcome
proof (induction  $n$  arbitrary:  $i \text{ chunks } s \ t$ )
case 0
then show ?case
  by (intro exI[of - []])
    (simp add: state-after-query-chunks-def hash-ext-refl)
next
case (Suc  $n$ )
from staged-query-program-Suc-outcomeE[OF Suc.prem(2)]
obtain raw chunk chunks'  $s1 \ s2 \ s3$  where
  challenge-out:
     $\text{Some } (\text{raw}, s1) \in$ 
       $\text{set-dist } (\text{execute receive-query-index-challenge } s)$ 
and stage-out:
     $\text{Some } (\text{chunk}, s2) \in$ 
       $\text{set-dist } (\text{execute } (\text{query-opening-stage } A \ i \ \text{raw}) \ s1)$ 
and record-out:
     $\text{Some } ((), s3) \in$ 
       $\text{set-dist } (\text{execute } (\text{record-staged-messages } \text{chunk}) \ s2)$ 
and rest-out:
     $\text{Some } (\text{chunks}', t) \in$ 
       $\text{set-dist } (\text{execute } (\text{staged-query-program } A \ (\text{Suc } i) \ n) \ s3)$ 
and chunks-eq:  $\text{chunks} = \text{chunk} \# \text{chunks}'$ 
  .
have i-bound:  $i < \text{length } (\text{query-opening-budgets } \text{budgets})$ 
  using Suc.prem(1) by simp
have stage-controlled:
    controlled-ro-program ( $\text{query-opening-budgets } \text{budgets} \ ! \ i$ )
      ( $\text{query-opening-stage } A \ i \ \text{raw}$ )
  using controlled i-bound

```

unfolding *staged-adversary-controlled-def* **by** *blast*
have *challenge*:
 $s \leq s1 \wedge$
 $PState\ s1 = PState\ s \wedge$
 $PTranscript\ s1 = PTranscript\ s \wedge$
 $fmlookup\ (HashMap\ s1)$
 $(QueryIndexChallenge\ (PQueryCounter\ s)\ (PState\ s)) = Some\ raw$
by (rule *receive-query-index-challenge-outcome*[*OF challenge-out*])
have *challenge-counters*:
 $PQueryCounter\ s1 = Suc\ (PQueryCounter\ s)$
using *receive-query-index-challenge-counter-outcome*[*OF challenge-out*]
by *simp*
have *ext-s1-s2*: $s1 \leq s2$
using *controlled-ro-program-extension*[*OF stage-controlled*] *stage-out*
unfolding *hash-extension-preserving-def* **by** *blast*
have *stage-fields*:
 $PState\ s2 = PState\ s1 \wedge$
 $PTranscript\ s2 = PTranscript\ s1 \wedge$
 $PQueryCounter\ s2 = PQueryCounter\ s1$
using *controlled-stage-outcome-fields*[*OF stage-controlled stage-out*]
by *simp*
have *s3-eq*:
 $s3 = s2 \mid$
 $PState := foldl\ concat\ (PState\ s2)\ chunk,$
 $PTranscript := PTranscript\ s2\ @\ chunk \mid$
by (rule *record-staged-messages-outcome*[*OF record-out*])
have *ext-s2-s3*: $s2 \leq s3$
unfolding *s3-eq less-eq-hash-ext-def less-eq-fmap-def* **by** *simp*
have *s3-fields*:
 $PState\ s3 = foldl\ concat\ (PState\ s)\ chunk \wedge$
 $PTranscript\ s3 = PTranscript\ s\ @\ chunk \wedge$
 $PQueryCounter\ s3 = Suc\ (PQueryCounter\ s)$
unfolding *s3-eq* **using** *challenge stage-fields challenge-counters* **by** *simp*
have *rest-bound*:
 $Suc\ i + n \leq length\ (query-opening-budgets\ budgets)$
using *Suc.prem1* **by** *simp*
from *Suc.IH*[*OF rest-bound rest-out*]
obtain *raw-tail idx-tail* **where**
 $len_raw_tail: length\ raw_tail = n$
and *idx-tail-def*:
 $idx_tail = map\ (\lambda raw. index\ (to_nat\ raw))\ raw_tail$
and *len-chunks-tail*: $length\ chunks' = n$
and *tr-tail*:
 $PTranscript\ t = PTranscript\ s3\ @\ List.concat\ chunks'$
and *st-tail*:
 $PState\ t = state_after_query_chunks\ (PState\ s3)\ chunks'\ n$
and *ext-s3-t*: $s3 \leq t$
and *query-count-tail*:
 $PQueryCounter\ t = PQueryCounter\ s3 + n$

```

and lookup-tail:
   $\bigwedge j. j < n \implies$ 
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s3 + j)
        (state-after-query-chunks (PState s3) chunks' j)) =
        Some (raw-tail ! j)
and idx-tail-bound:
   $\forall idx \in \text{set } \text{idx-tail}. \text{idx} < \text{clength} * \text{scale}$ 
by blast
let ?raws = raw # raw-tail
let ?idxs = index (to-nat raw) # idx-tail
let ?chunks = chunk # chunks'
have idxs-def: ?idxs = map ( $\lambda \text{raw}. \text{index } (\text{to-nat } \text{raw})$ ) ?raws
  using idx-tail-def by simp
have tr-all:
  PTranscript t = PTranscript s @ List.concat ?chunks
  using tr-tail s3-fields by simp
have state-shift:
   $\bigwedge j. \text{state-after-query-chunks } (PState s) \text{ ?chunks } (Suc j) =$ 
    state-after-query-chunks (PState s3) chunks' j
  using s3-fields unfolding state-after-query-chunks-def by simp
have st-all:
  PState t = state-after-query-chunks (PState s) ?chunks (Suc n)
  using st-tail state-shift[of n] by simp
have ext-s-s2:  $s \leq s2$ 
  by (rule hash-ext-trans[OF conjunct1[OF challenge] ext-s1-s2])
have ext-s-s3:  $s \leq s3$ 
  by (rule hash-ext-trans[OF ext-s-s2 ext-s2-s3])
have ext-s-t:  $s \leq t$ 
  by (rule hash-ext-trans[OF ext-s-s3 ext-s3-t])
have query-count-all: PQueryCounter t = PQueryCounter s + Suc n
  using query-count-tail s3-fields by simp
have ext-s1-t:  $s1 \leq t$ 
  by (rule hash-ext-trans[OF ext-s1-s2])
  (rule hash-ext-trans[OF ext-s2-s3 ext-s3-t])
have lookup-head-t:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) =
    Some raw
  by (rule hash-extension-lookup[OF conjunct2[OF conjunct2[OF conjunct2[OF challenge]]]
    ext-s1-t])
have lookup-all:
   $\bigwedge j. j < Suc n \implies$ 
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + j)
        (state-after-query-chunks (PState s) ?chunks j)) =
        Some (?raws ! j)
proof –
  fix j

```

```

assume j-bound:  $j < \text{Suc } n$ 
show
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + j)
      (state-after-query-chunks (PState s) ?chunks j)) =
      Some (?rows ! j)
proof (cases j)
  case 0
  then show ?thesis
    using lookup-head-t unfolding state-after-query-chunks-def by simp
next
  case (Suc k)
  then show ?thesis
    using lookup-tail[of k] j-bound state-shift[of k] s3-fields by simp
qed
qed
have idx-bound-all:
   $\forall \text{idx} \in \text{set } ?\text{idxs}. \text{idx} < \text{clength} * \text{scale}$ 
  using idx-tail-bound index-less-domain by auto
show ?case
  unfolding chunks-eq
  apply (intro exI[of - ?rows] exI[of - ?idxs] conjI)
    apply (simp add: len-raw-tail)
    apply (rule idxs-def)
    apply (simp add: len-chunks-tail)
    apply (rule tr-all)
    apply (rule st-all)
    apply (rule ext-s-t)
    apply (rule query-count-all)
    apply (intro allI impI, rule lookup-all, assumption)
  using idx-bound-all by simp
qed

lemma checked-staged-query-program-chunks-match-verifier:
assumes outcome:
  Some (chunks, t)  $\in$ 
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          i n) s)
shows
   $\exists \text{raw-idxs query-idxs}.$ 
   $\text{length raw-idxs} = n \wedge$ 
   $\text{query-idxs} = \text{map } (\lambda \text{raw}. \text{index } (\text{to-nat raw})) \text{ raw-idxs} \wedge$ 
   $\text{length chunks} = n \wedge$ 
   $(\forall j < n.$ 
     $\text{verifier-query-round-chunk } (\text{query-idxs } ! j)$ 
     $\text{trace-roots composition-roots } (\text{chunks } ! j)) \wedge$ 
   $(\forall \text{idx} \in \text{set query-idxs}. \text{idx} < \text{clength} * \text{scale})$ 

```

```

using outcome
proof (induction n arbitrary: i chunks s t)
  case 0
  then show ?case
    by (intro exI[of - []] exI[of - []] simp)
next
  case (Suc n)
  obtain raw chunk chunks' s1 s2 s3 where
    chunk-shape:
      verifier-query-round-chunk (index (to-nat raw))
      trace-roots composition-roots chunk
  and rest-out:
    Some (chunks', t) ∈
      set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          (Suc i) n) s3)
  and chunks-eq: chunks = chunk # chunks'
proof (rule checked-staged-query-program-Suc-outcomeE[OF Suc.premis])
  fix raw chunk chunks' s1 s2 s3
  assume shape:
    verifier-query-round-chunk (index (to-nat raw))
    trace-roots composition-roots chunk
  and rest:
    Some (chunks', t) ∈
      set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          (Suc i) n) s3)
  and chunks: chunks = chunk # chunks'
  show ?thesis
    by (rule that[OF shape rest chunks])
qed
from Suc.IH[OF rest-out]
obtain raw-tail idx-tail where
  len-raw-tail: length raw-tail = n
  and idx-tail-def:
    idx-tail = map (λraw. index (to-nat raw)) raw-tail
  and len-chunks-tail: length chunks' = n
  and tail-shape:
     $\forall j < n.$ 
    verifier-query-round-chunk (idx-tail ! j)
    trace-roots composition-roots (chunks' ! j)
  and idx-tail-bound:
     $\forall idx \in \text{set } idx\text{-tail}. idx < \text{clength} * \text{scale}$ 
  by blast
let ?raws = raw # raw-tail
let ?idxs = index (to-nat raw) # idx-tail
have idxs-def: ?idxs = map (λraw. index (to-nat raw)) ?raws

```

```

using idx-tail-def by simp
have shape-all:
   $\forall j < \text{Suc } n.$ 
    verifier-query-round-chunk (?idxs ! j)
      trace-roots composition-roots ((chunk # chunks') ! j)
proof (intro allI impI)
  fix j
  assume j-bound: j < Suc n
  show verifier-query-round-chunk (?idxs ! j)
    trace-roots composition-roots ((chunk # chunks') ! j)
proof (cases j)
  case 0
  then show ?thesis
    using chunk-shape by simp
next
  case (Suc k)
  then have k-bound: k < n
    using j-bound by simp
  then show ?thesis
    using tail-shape Suc by simp
qed
qed
have idx-bound-all:
   $\forall idx \in \text{set } ?idxs. idx < \text{clength} * \text{scale}$ 
  using idx-tail-bound index-less-domain by auto
show ?case
  unfolding chunks-eq
  by (intro exI[of - ?raws] exI[of - ?idxs] conjI)
    (use len-raw-tail idxs-def len-chunks-tail shape-all idx-bound-all
      in simp-all)
qed

lemma checked-staged-query-program-outcome-with-raws:
assumes controlled: staged-adversary-controlled budgets A
and bound:  $i + n \leq \text{length } (\text{query-opening-budgets } \text{budgets})$ 
and outcome:
  Some (chunks, t)  $\in$ 
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          i n) s)
shows
   $\exists \text{raw-idxs query-idxs}.$ 
     $\text{length raw-idxs} = n \wedge$ 
     $\text{query-idxs} = \text{map } (\lambda \text{raw}. \text{index } (\text{to-nat raw})) \text{ raw-idxs} \wedge$ 
     $\text{length chunks} = n \wedge$ 
     $P\text{Transcript } t = P\text{Transcript } s @ \text{List.concat chunks} \wedge$ 
     $P\text{State } t = \text{state-after-query-chunks } (P\text{State } s) \text{ chunks } n \wedge$ 
     $s \leq t \wedge$ 

```

```

    PQueryCounter t = PQueryCounter s + n ∧
    (∀ j < n.
      verifier-query-round-chunk (query-idxs ! j)
      trace-roots composition-roots (chunks ! j)) ∧
    (∀ j < n.
      fmllookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + j)
        (state-after-query-chunks (PState s) chunks j)) =
      Some (raw-idxs ! j)) ∧
    (∀ idx ∈ set query-idxs. idx < clength * scale)
  using bound outcome
proof (induction n arbitrary: i chunks s t)
  case 0
  then show ?case
  by (intro exI[of -] exI[of -])
    (simp add: state-after-query-chunks-def hash-ext-refl)
next
  case (Suc n)
  from checked-staged-query-program-Suc-outcomeE[OF Suc.prem1(2)]
  obtain raw chunk chunks' s1 s2 s3 where
    challenge-out:
      Some (raw, s1) ∈
      set-dist (execute receive-query-index-challenge s)
  and stage-out:
      Some (chunk, s2) ∈
      set-dist (execute (query-opening-stage A i raw) s1)
  and chunk-shape:
      verifier-query-round-chunk (index (to-nat raw))
      trace-roots composition-roots chunk
  and record-out:
      Some ((), s3) ∈
      set-dist (execute (record-staged-messages chunk) s2)
  and rest-out:
      Some (chunks', t) ∈
      set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          (Suc i) n) s3)
  and chunks-eq: chunks = chunk # chunks'
  .
  have i-bound: i < length (query-opening-budgets budgets)
  using Suc.prem1(1) by simp
  have stage-controlled:
    controlled-ro-program (query-opening-budgets budgets ! i)
    (query-opening-stage A i raw)
  using controlled i-bound
  unfolding staged-adversary-controlled-def by blast
  have challenge:
    s ≤ s1 ∧

```

```

PState s1 = PState s ∧
PTranscript s1 = PTranscript s ∧
fmlookup (HashMap s1)
  (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
by (rule receive-query-index-challenge-outcome[OF challenge-out])
have challenge-counters:
  PQueryCounter s1 = Suc (PQueryCounter s)
  using receive-query-index-challenge-counter-outcome[OF challenge-out]
  by simp
have ext-s1-s2: s1 ≤ s2
  using controlled-ro-program-extension[OF stage-controlled] stage-out
  unfolding hash-extension-preserving-def by blast
have stage-fields:
  PState s2 = PState s1 ∧
  PTranscript s2 = PTranscript s1 ∧
  PQueryCounter s2 = PQueryCounter s1
  using controlled-stage-outcome-fields[OF stage-controlled stage-out]
  by simp
have s3-eq:
  s3 = s2(|
    PState := foldl concat (PState s2) chunk,
    PTranscript := PTranscript s2 @ chunk|)
  by (rule record-staged-messages-outcome[OF record-out])
have ext-s2-s3: s2 ≤ s3
  unfolding s3-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have s3-fields:
  PState s3 = foldl concat (PState s) chunk ∧
  PTranscript s3 = PTranscript s @ chunk ∧
  PQueryCounter s3 = Suc (PQueryCounter s)
  unfolding s3-eq using challenge stage-fields challenge-counters by simp
have rest-bound:
  Suc i + n ≤ length (query-opening-budgets budgets)
  using Suc.prem1 by simp
from Suc.IH[OF rest-bound rest-out]
obtain raw-tail idx-tail where
  len-raw-tail: length raw-tail = n
  and idx-tail-def:
    idx-tail = map (λraw. index (to-nat raw)) raw-tail
  and len-chunks-tail: length chunks' = n
  and tr-tail:
    PTranscript t = PTranscript s3 @ List.concat chunks'
  and st-tail:
    PState t = state-after-query-chunks (PState s3) chunks' n
  and ext-s3-t: s3 ≤ t
  and query-count-tail:
    PQueryCounter t = PQueryCounter s3 + n
  and chunk-shape-tail:
    ∧j. j < n ⇒
      verifier-query-round-chunk (idx-tail ! j)

```



```

      trace-roots composition-roots (chunks' ! j)
and lookup-tail:
   $\bigwedge j. j < n \implies$ 
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s3 + j)
        (state-after-query-chunks (PState s3) chunks' j)) =
      Some (raw-tail ! j)
and idx-tail-bound:
   $\forall idx \in \text{set } \text{idx-tail}. \text{idx} < \text{clength} * \text{scale}$ 
by blast
let ?raws = raw # raw-tail
let ?idxs = index (to-nat raw) # idx-tail
let ?chunks = chunk # chunks'
have idxs-def: ?idxs = map ( $\lambda \text{raw}. \text{index } (\text{to-nat } \text{raw})$ ) ?raws
  using idx-tail-def by simp
have tr-all:
  PTranscript t = PTranscript s @ List.concat ?chunks
  using tr-tail s3-fields by simp
have state-shift:
   $\bigwedge j. \text{state-after-query-chunks } (PState s) \text{ ?chunks } (Suc j) =$ 
    state-after-query-chunks (PState s3) chunks' j
  using s3-fields unfolding state-after-query-chunks-def by simp
have st-all:
  PState t = state-after-query-chunks (PState s) ?chunks (Suc n)
  using st-tail state-shift[of n] by simp
have ext-s-s2:  $s \leq s2$ 
  by (rule hash-ext-trans[OF conjunct1[OF challenge] ext-s1-s2])
have ext-s-s3:  $s \leq s3$ 
  by (rule hash-ext-trans[OF ext-s-s2 ext-s2-s3])
have ext-s-t:  $s \leq t$ 
  by (rule hash-ext-trans[OF ext-s-s3 ext-s3-t])
have query-count-all: PQueryCounter t = PQueryCounter s + Suc n
  using query-count-tail s3-fields by simp
have ext-s1-t:  $s1 \leq t$ 
  by (rule hash-ext-trans[OF ext-s1-s2])
  (rule hash-ext-trans[OF ext-s2-s3 ext-s3-t])
have lookup-head-t:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) =
    Some raw
  by (rule hash-extension-lookup
    [OF conjunct2[OF conjunct2[OF conjunct2[OF challenge]]] ext-s1-t])
have chunk-shape-all:
   $\bigwedge j. j < Suc n \implies$ 
    verifier-query-round-chunk (?idxs ! j)
      trace-roots composition-roots (?chunks ! j)
proof -
  fix j
  assume j-bound:  $j < Suc n$ 

```

```

show
  verifier-query-round-chunk (?idxs ! j)
  trace-roots composition-roots (?chunks ! j)
proof (cases j)
  case 0
  then show ?thesis
    using chunk-shape by simp
next
  case (Suc k)
  then show ?thesis
    using chunk-shape-tail[of k] j-bound by simp
qed
qed
have lookup-all:
   $\bigwedge j. j < \text{Suc } n \implies$ 
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + j)
      (state-after-query-chunks (PState s) ?chunks j)) =
    Some (?raws ! j)
proof -
  fix j
  assume j-bound:  $j < \text{Suc } n$ 
  show
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + j)
        (state-after-query-chunks (PState s) ?chunks j)) =
      Some (?raws ! j)
proof (cases j)
  case 0
  then show ?thesis
    using lookup-head-t unfolding state-after-query-chunks-def by simp
next
  case (Suc k)
  then show ?thesis
    using lookup-tail[of k] j-bound state-shift[of k] s3-fields by simp
qed
qed
have idx-bound-all:
   $\forall \text{idx} \in \text{set } ?\text{idxs}. \text{idx} < \text{clength} * \text{scale}$ 
  using idx-tail-bound index-less-domain by auto
show ?case
  unfolding chunks-eq
  by (intro exI[of - ?raws] exI[of - ?idxs] conjI)
    (use len-raw-tail idxs-def len-chunks-tail tr-all st-all ext-s-t
      query-count-all chunk-shape-all lookup-all idx-bound-all in simp-all)
qed

lemma checked-staged-query-program-chunks-match-verifier-lengths:
  assumes outcome:

```

Some (chunks, t) ∈
set-dist
(execute
(checked-staged-query-program A trace-roots composition-roots
i rounds) s)
shows
 $\exists$ *query-idxs.*
staged-query-chunks-match-verifier-lengths
 $(\downarrow$ *staged-trace-root = fr,*
staged-trace-fri-roots = trace-roots,
staged-trace-fri-challenges = trace-bs,
staged-trace-final = trace-final,
staged-alphas = as,
staged-degree = dg,
staged-composition-fri-roots = composition-roots,
staged-composition-fri-challenges = composition-bs,
staged-composition-final = composition-final,
*staged-query-chunks = chunks)
query-idxs
proof –
from *checked-staged-query-program-chunks-match-verifier[OF outcome]*
obtain *raw-idxs query-idxs where*
len-chunks: length chunks = rounds
and *chunk-shape:*
 $\forall j < \text{rounds}.$
verifier-query-round-chunk (query-idxs ! j)
trace-roots composition-roots (chunks ! j)
by *blast*
have *chunk-lens:*
 $\forall j < \text{rounds}.$
length (chunks ! j) =
verifier-query-round-transcript-length (query-idxs ! j)
trace-roots composition-roots
using *chunk-shape verifier-query-round-chunk-length by blast*
show *?thesis*
by *(intro exI[of - query-idxs])*
(use len-chunks chunk-lens in
⟨simp add: staged-query-chunks-match-verifier-lengths-def⟩)
qed

lemma *checked-staged-transcript-program-query-chunks-match-verifier-lengths:*
assumes *outcome:*
Some (data, t) ∈
set-dist
(execute (checked-staged-transcript-program A) s)
shows $\exists$ *query-idxs.*
staged-query-chunks-match-verifier-lengths data query-idxs
proof –
from *outcome obtain fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5**

```

as s6 dg s7 s8 s9 composition-roots composition-bs s10
composition-final s11 s12 query-chunks where
query-out:
  Some (query-chunks, s12) ∈
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          0 rounds)
          s11)
and data-eq:
  data =
    (|staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots = composition-roots,
     staged-composition-fri-challenges = composition-bs,
     staged-composition-final = composition-final,
     staged-query-chunks = query-chunks)
unfolding checked-staged-transcript-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
show ?thesis
unfolding data-eq
by (rule checked-staged-query-program-chunks-match-verifier-lengths
      [OF query-out])
qed

lemma checked-staged-security-experiment-with-data-state-query-lengthsE:
assumes outcome:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
          initial-state)
obtains query-idxs where
  staged-query-chunks-match-verifier-lengths data query-idxs
proof –
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF outcome]
have builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A) initial-state)
by blast
from checked-staged-transcript-program-query-chunks-match-verifier-lengths
  [OF builder]
obtain query-idxs where

```

```

    staged-query-chunks-match-verifier-lengths data query-idxs
  by blast
then show ?thesis
  by (rule that)
qed

lemma verifier-query-round-program-replays-shaped-chunk:
  assumes outcome:
    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and lookup:
    fmlookup (HashMap s)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  and chunk-shape:
    verifier-query-round-chunk (index (to-nat raw))
      (map snd f-fl) (map snd fl) chunk
  and transcript: PTranscript s = chunk @ rest
  shows
    PTranscript t = rest ∧
    PState t = foldl concat (PState s) chunk ∧
    PQueryCounter t = Suc (PQueryCounter s) ∧
    s ≤ t
proof -
  from verifier-query-round-program-outcome[OF outcome]
  obtain raw' idx' chunk' where
    idx'-def: idx' = index (to-nat raw')
  and chunk'-shape:
    verifier-query-round-chunk idx' (map snd f-fl) (map snd fl)
    chunk'
  and transcript': PTranscript s = chunk' @ PTranscript t
  and state-t: PState t = foldl concat (PState s) chunk'
  and ext: s ≤ t
  and counter-t: PQueryCounter t = Suc (PQueryCounter s)
  and lookup-t:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw'
  by blast
  have lookup-t-from-s:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  by (rule hash-extension-lookup[OF lookup ext])
  have raw'-eq: raw' = raw
  using lookup-t lookup-t-from-s by simp
  have idx'-eq: idx' = index (to-nat raw)
  unfolding idx'-def raw'-eq by simp
  have len-eq: length chunk' = length chunk
  using verifier-query-round-chunk-length[OF chunk'-shape]

```

```

    verifier-query-round-chunk-length[OF chunk-shape]
  unfolding idx'-eq by simp
have chunk'-eq: chunk' = chunk
proof -
  have chunk' = take (length chunk') (PTranscript s)
    using transcript' by simp
  also have ... = chunk
    using transcript len-eq by simp
  finally show ?thesis .
qed
have transcript-t: PTranscript t = rest
  using transcript' transcript unfolding chunk'-eq by simp
show ?thesis
  using transcript-t state-t counter-t ext unfolding chunk'-eq by simp
qed

lemma ntimes-verifier-query-round-program-replays-shaped-chunks:
  assumes outcome:
    Some (results, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            n) s)
  and len-row: length raw-idxs = n
  and len-chunks: length chunks = n
  and chunk-shape:
     $\bigwedge i. i < n \implies$ 
    verifier-query-round-chunk (index (to-nat (raw-idxs ! i)))
      (map snd f-fl) (map snd fl) (chunks ! i)
  and lookup:
     $\bigwedge i. i < n \implies$ 
    fmllookup (HashMap s)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) chunks i)) =
    Some (raw-idxs ! i)
  and transcript: PTranscript s = List.concat chunks @ rest
shows
  PTranscript t = rest  $\wedge$ 
  PState t = state-after-query-chunks (PState s) chunks n  $\wedge$ 
  PQueryCounter t = PQueryCounter s + n  $\wedge$ 
  s  $\leq$  t
  using outcome len-row len-chunks chunk-shape lookup transcript
proof (induction n arbitrary: s results t raw-idxs chunks rest)
  case 0
  then show ?case
    by (simp add: state-after-query-chunks-def hash-ext-refl)
next
  case (Suc n)

```

```

from Suc.prems(2) obtain raw raw-tail where raw-eq:
  raw-idxs = raw # raw-tail
  by (cases raw-idxs) simp-all
from Suc.prems(3) obtain chunk chunk-tail where chunks-eq:
  chunks = chunk # chunk-tail
  by (cases chunks) simp-all
from Suc.prems(1) obtain u s1 results' where
  head:
    Some (u, s1) ∈
      set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results', t) ∈
      set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          n) s1)
    and results-eq: results = u # results'
    by (auto elim!: set-dist-bindE)
have u-eq: u = ()
  by (cases u) simp
have len-raw-tail: length raw-tail = n
  using Suc.prems(2) unfolding raw-eq by simp
have len-chunk-tail: length chunk-tail = n
  using Suc.prems(3) unfolding chunks-eq by simp
have head-shape:
  verifier-query-round-chunk (index (to-nat raw))
    (map snd f-fl) (map snd fl) chunk
  using Suc.prems(4)[of 0] unfolding raw-eq chunks-eq by simp
have head-lookup:
  fmlookup (HashMap s)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) =
    Some raw
  using Suc.prems(5)[of 0]
  unfolding raw-eq chunks-eq state-after-query-chunks-def by simp
have head-transcript:
  PTranscript s = chunk @ (List.concat chunk-tail @ rest)
  using Suc.prems(6) unfolding chunks-eq by simp
from verifier-query-round-program-replays-shaped-chunk
  [OF head[unfolded u-eq] head-lookup head-shape head-transcript]
have replay-head:
  PTranscript s1 = List.concat chunk-tail @ rest ∧
  PState s1 = foldl concat (PState s) chunk ∧
  PQueryCounter s1 = Suc (PQueryCounter s) ∧
  s ≤ s1 .
have tail-shape:
  ∧i. i < n ⇒

```

```

    verifier-query-round-chunk (index (to-nat (raw-tail ! i)))
    (map snd f-fl) (map snd fl) (chunk-tail ! i)
proof -
  fix i
  assume i-bound:  $i < n$ 
  then show
    verifier-query-round-chunk (index (to-nat (raw-tail ! i)))
    (map snd f-fl) (map snd fl) (chunk-tail ! i)
  using Suc.premis(4)[of Suc i]
  unfolding raw-eq chunks-eq by simp
qed
have state-shift:
   $\wedge i.$  state-after-query-chunks (PState s) (chunk # chunk-tail)
    (Suc i) =
    state-after-query-chunks (PState s1) chunk-tail i
  using replay-head unfolding state-after-query-chunks-def by simp
have tail-lookup:
   $\wedge i.$   $i < n \implies$ 
    fmlookup (HashMap s1)
      (QueryIndexChallenge (PQueryCounter s1 + i)
        (state-after-query-chunks (PState s1) chunk-tail i)) =
    Some (raw-tail ! i)
proof -
  fix i
  assume i-bound:  $i < n$ 
  have lookup-s:
    fmlookup (HashMap s)
      (QueryIndexChallenge (PQueryCounter s + Suc i)
        (state-after-query-chunks (PState s) (chunk # chunk-tail)
          (Suc i))) =
    Some (raw-tail ! i)
  using Suc.premis(5)[of Suc i] i-bound
  unfolding raw-eq chunks-eq by simp
  have key-eq:
    QueryIndexChallenge (PQueryCounter s + Suc i)
      (state-after-query-chunks (PState s) (chunk # chunk-tail)
        (Suc i)) =
    QueryIndexChallenge (PQueryCounter s1 + i)
      (state-after-query-chunks (PState s1) chunk-tail i)
  using replay-head state-shift[of i] by simp
  show
    fmlookup (HashMap s1)
      (QueryIndexChallenge (PQueryCounter s1 + i)
        (state-after-query-chunks (PState s1) chunk-tail i)) =
    Some (raw-tail ! i)
  using hash-extension-lookup[OF lookup-s conjunct2[OF conjunct2[OF conjunct2[OF re-
    play-head]]]]
  unfolding key-eq .
qed

```



```

from Suc.IH[OF tail len-raw-tail len-chunk-tail tail-shape tail-lookup
  conjunct1[OF replay-head]]
have replay-tail:
  PTranscript t = rest  $\wedge$ 
  PState t = state-after-query-chunks (PState s1) chunk-tail n  $\wedge$ 
  PQueryCounter t = PQueryCounter s1 + n  $\wedge$ 
  s1  $\leq$  t .
have state-all:
  PState t =
    state-after-query-chunks (PState s) (chunk # chunk-tail) (Suc n)
  using replay-tail state-shift[of n] by simp
have counter-all: PQueryCounter t = PQueryCounter s + Suc n
  using replay-head replay-tail by simp
have ext-all: s  $\leq$  t
  using replay-head replay-tail by (blast intro: hash-ext-trans)
show ?case
  unfolding raw-eq chunks-eq
  using replay-tail state-all counter-all ext-all by simp
qed

lemma checked-staged-query-program-ntimes-replay:
assumes controlled: staged-adversary-controlled budgets A
and bound: i + n  $\leq$  length (query-opening-budgets budgets)
and checked:
  Some (chunks, attacker-t)  $\in$ 
    set-dist
    (execute
      (checked-staged-query-program A (map snd f-fl) (map snd fl)
        i n) attacker-s)
and verifier:
  Some (results, verifier-t)  $\in$ 
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
          n) verifier-s)
and hash-ext: attacker-t  $\leq$  verifier-s
and state-eq: PState verifier-s = PState attacker-s
and counter-eq: PQueryCounter verifier-s = PQueryCounter attacker-s
and transcript-eq:
  PTranscript verifier-s = List.concat chunks @ rest
shows
  PTranscript verifier-t = rest  $\wedge$ 
  PState verifier-t =
    state-after-query-chunks (PState verifier-s) chunks n  $\wedge$ 
  PQueryCounter verifier-t = PQueryCounter verifier-s + n  $\wedge$ 
  verifier-s  $\leq$  verifier-t
proof –
from checked-staged-query-program-outcome-with-raws

```

```

    [OF controlled bound checked]
obtain raw-idxs query-idxs where
  len-raw: length raw-idxs = n
and query-idxs-def:
  query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and len-chunks: length chunks = n
and chunk-shape:
   $\bigwedge j. j < n \implies$ 
    verifier-query-round-chunk (query-idxs ! j)
    (map snd f-fl) (map snd fl) (chunks ! j)
and lookup:
   $\bigwedge j. j < n \implies$ 
    fmlookup (HashMap attacker-t)
    (QueryIndexChallenge (PQueryCounter attacker-s + j)
      (state-after-query-chunks (PState attacker-s) chunks j)) =
    Some (raw-idxs ! j)
by blast
have chunk-shape-raw:
   $\bigwedge j. j < n \implies$ 
    verifier-query-round-chunk (index (to-nat (raw-idxs ! j)))
    (map snd f-fl) (map snd fl) (chunks ! j)
proof –
  fix j
  assume j-bound: j < n
  have query-idxs ! j = index (to-nat (raw-idxs ! j))
    using query-idxs-def len-raw j-bound by simp
  then show
    verifier-query-round-chunk (index (to-nat (raw-idxs ! j)))
    (map snd f-fl) (map snd fl) (chunks ! j)
    using chunk-shape[OF j-bound] by simp
qed
have lookup-verifier:
   $\bigwedge j. j < n \implies$ 
    fmlookup (HashMap verifier-s)
    (QueryIndexChallenge (PQueryCounter verifier-s + j)
      (state-after-query-chunks (PState verifier-s) chunks j)) =
    Some (raw-idxs ! j)
proof –
  fix j
  assume j-bound: j < n
  have lookup-attacker:
    fmlookup (HashMap attacker-t)
    (QueryIndexChallenge (PQueryCounter attacker-s + j)
      (state-after-query-chunks (PState attacker-s) chunks j)) =
    Some (raw-idxs ! j)
    by (rule lookup[OF j-bound])
  have lookup-verifier-s:
    fmlookup (HashMap verifier-s)
    (QueryIndexChallenge (PQueryCounter attacker-s + j)

```

```

    (state-after-query-chunks (PState attacker-s) chunks j)) =
    Some (raw-idxs ! j)
  by (rule hash-extension-lookup[OF lookup-attacker hash-ext])
show
  fmlookup (HashMap verifier-s)
    (QueryIndexChallenge (PQueryCounter verifier-s + j)
      (state-after-query-chunks (PState verifier-s) chunks j)) =
    Some (raw-idxs ! j)
  using lookup-verifier-s state-eq counter-eq by simp
qed
show ?thesis
  by (rule ntimes-verifier-query-round-program-replays-shaped-chunks
    [OF verifier len-row len-chunks chunk-shape-row lookup-verifier
      transcript-eq])
qed

```

lemma *checked-staged-transcript-program-outcome-shape:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *outcome:*
Some (data, t) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)

shows

length (staged-trace-fri-roots data) = ceil-log clength ∧
length (staged-trace-fri-challenges data) = ceil-log clength ∧
length (staged-alphas data) = length spec ∧
length (staged-composition-fri-roots data) =
ceil-log (to-nat (staged-degree data) + 1) ∧
length (staged-composition-fri-challenges data) =
ceil-log (to-nat (staged-degree data) + 1) ∧
ceil-log (to-nat (staged-degree data) + 1) ≤
ceil-log (maxDegree + 1) ∧
length (staged-query-chunks data) = rounds

proof —

from *outcome* **obtain** *fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5*
as s6 dg s7 s8 s9 composition-roots composition-bs s10
composition-final s11 s12 query-chunks **where**
trace-out:
Some ((trace-roots, trace-bs), s3) ∈
set-dist
(execute
(staged-trace-fri-program A 0 (ceil-log clength) []) s2)
and *alpha-out:*
Some (as, s6) ∈
set-dist (execute (staged-alpha-program (length spec)) s5)
and *degree-eq: staged-degree data = dg*
and *assert-out:*

```

Some (((), s9) ∈
  set-dist
    (execute
      (assert
        (ceil-log (to-nat dg + 1) ≤
          ceil-log (maxDegree + 1))) s8)
and composition-out:
Some ((composition-roots, composition-bs), s10) ∈
  set-dist
    (execute
      (staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) [] s9)
and query-out:
Some (query-chunks, s12) ∈
  set-dist
    (execute
      (checked-staged-query-program A trace-roots composition-roots
        0 rounds)
        s11)
and data-eq:
data =
  (|staged-trace-root = fr,
    staged-trace-fri-roots = trace-roots,
    staged-trace-fri-challenges = trace-bs,
    staged-trace-final = trace-final,
    staged-alphas = as,
    staged-degree = dg,
    staged-composition-fri-roots = composition-roots,
    staged-composition-fri-challenges = composition-bs,
    staged-composition-final = composition-final,
    staged-query-chunks = query-chunks|)
unfolding checked-staged-transcript-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have trace-len:
length trace-roots = ceil-log clength ∧
length trace-bs = ceil-log clength
proof –
have len:
0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
using wf unfolding staged-budget-wellformed-def by simp
have alignment:
length trace-roots = ceil-log clength ∧
length trace-bs = length (| :: 'f list) + ceil-log clength ∧
PTranscript s3 = PTranscript s2 @ trace-roots ∧
PState s3 = foldl concat (PState s2) trace-roots ∧
s2 ≤ s3 ∧
PTraceFriCounter s3 = PTraceFriCounter s2 + ceil-log clength ∧
PCompositionFriCounter s3 = PCompositionFriCounter s2 ∧
PAlphaCounter s3 = PAlphaCounter s2 ∧

```

```

    PQueryCounter s3 = PQueryCounter s2 ∧
    (∀ j < ceil-log clength.
      fmllookup (HashMap s3)
        (TraceFriChallenge (PTraceFriCounter s2 + j)
          (foldl concat (PState s2) (take (Suc j) trace-roots))) =
        Some (trace-bs ! (length ([] :: 'f list) + j)))
  by (rule staged-trace-fri-program-alignment
      [OF controlled len trace-out])
  then show ?thesis by simp
qed
have alpha-len: length as = length spec
  using staged-alpha-program-alignment[OF alpha-out] by simp
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using assert-out unfolding assert-def
  by (cases
      ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
      (auto simp: throw-no-outcome)
have composition-len:
  length composition-roots = ceil-log (to-nat dg + 1) ∧
  length composition-bs = ceil-log (to-nat dg + 1)
proof -
  have len:
    0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
    using wf round-bound unfolding staged-budget-wellformed-def by simp
  have alignment:
    length composition-roots = ceil-log (to-nat dg + 1) ∧
    length composition-bs =
      length ([] :: 'f list) + ceil-log (to-nat dg + 1) ∧
    PTranscript s10 = PTranscript s9 @ composition-roots ∧
    PState s10 = foldl concat (PState s9) composition-roots ∧
    s9 ≤ s10 ∧
    PTraceFriCounter s10 = PTraceFriCounter s9 ∧
    PCompositionFriCounter s10 =
      PCompositionFriCounter s9 + ceil-log (to-nat dg + 1) ∧
    PAlphaCounter s10 = PAlphaCounter s9 ∧
    PQueryCounter s10 = PQueryCounter s9 ∧
    (∀ j < ceil-log (to-nat dg + 1).
      fmllookup (HashMap s10)
        (CompositionFriChallenge (PCompositionFriCounter s9 + j)
          (foldl concat (PState s9) (take (Suc j) composition-roots))) =
        Some (composition-bs ! (length ([] :: 'f list) + j)))
    by (rule staged-composition-fri-program-alignment
        [OF controlled len composition-out])
  then show ?thesis by simp
qed
have query-len: length query-chunks = rounds
  using checked-staged-query-program-chunks-match-verifier[OF query-out]

```

```

    by blast
  show ?thesis
    unfolding data-eq degree-eq
    using trace-len alpha-len composition-len round-bound query-len by simp
qed

```

lemma *checked-staged-transcript-program-outcome-header-transcript:*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and outcome:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)

```

shows

```

  verifier-header-transcript
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))

```

proof –

have *shape:*

```

  length (staged-trace-fri-roots data) = ceil-log clength ∧
  length (staged-trace-fri-challenges data) = ceil-log clength ∧
  length (staged-alphas data) = length spec ∧
  length (staged-composition-fri-roots data) =
    ceil-log (to-nat (staged-degree data) + 1) ∧
  length (staged-composition-fri-challenges data) =
    ceil-log (to-nat (staged-degree data) + 1) ∧
  ceil-log (to-nat (staged-degree data) + 1) ≤
    ceil-log (maxDegree + 1) ∧
  length (staged-query-chunks data) = rounds

```

```

by (rule checked-staged-transcript-program-outcome-shape
  [OF wf controlled outcome])

```

show ?thesis

```

by (rule staged-proof-transcript-verifier-header-transcript)
  (use shape in simp-all)

```

qed

lemma *staged-transcript-program-outcome-shape:*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and outcome:

```

Some (*data*, *t*) ∈
set-dist
 (*execute* (*staged-transcript-program* *A*)
adversary-initial-state)
shows
length (*staged-trace-fri-roots data*) = *ceil-log clength* ∧
length (*staged-trace-fri-challenges data*) = *ceil-log clength* ∧
length (*staged-alphas data*) = *length spec* ∧
length (*staged-composition-fri-roots data*) =
ceil-log (*to-nat* (*staged-degree data*) + 1) ∧
length (*staged-composition-fri-challenges data*) =
ceil-log (*to-nat* (*staged-degree data*) + 1) ∧
ceil-log (*to-nat* (*staged-degree data*) + 1) ≤
ceil-log (*maxDegree* + 1) ∧
length (*staged-query-chunks data*) = *rounds*
proof –
from *outcome* **obtain** *fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5*
as s6 dg s7 s8 s9 composition-roots composition-bs s10
composition-final s11 s12 query-chunks **where**
trace-out:
Some ((*trace-roots*, *trace-bs*), *s3*) ∈
set-dist
 (*execute*
 (*staged-trace-fri-program* *A* 0 (*ceil-log clength*) []) *s2*)
and *alpha-out:*
Some (*as*, *s6*) ∈
set-dist (*execute* (*staged-alpha-program* (*length spec*)) *s5*)
and *degree-eq:* *staged-degree data* = *dg*
and *assert-out:*
Some (*()*, *s9*) ∈
set-dist
 (*execute*
 (*assert*
 (*ceil-log* (*to-nat dg* + 1) ≤
ceil-log (*maxDegree* + 1))) *s8*)
and *composition-out:*
Some ((*composition-roots*, *composition-bs*), *s10*) ∈
set-dist
 (*execute*
 (*staged-composition-fri-program* *A dg* 0
 (*ceil-log* (*to-nat dg* + 1)) []) *s9*)
and *query-out:*
Some (*query-chunks*, *s12*) ∈
set-dist (*execute* (*staged-query-program* *A* 0 *rounds*) *s11*)
and *data-eq:*
data =
 (|*staged-trace-root* = *fr*,
staged-trace-fri-roots = *trace-roots*,
staged-trace-fri-challenges = *trace-bs*,

```

    staged-trace-final = trace-final,
    staged-alphas = as,
    staged-degree = dg,
    staged-composition-fri-roots = composition-roots,
    staged-composition-fri-challenges = composition-bs,
    staged-composition-final = composition-final,
    staged-query-chunks = query-chunks)
unfolding staged-transcript-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have trace-len:
  length trace-roots = ceil-log clength ∧
  length trace-bs = ceil-log clength
proof –
  have len:
    0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
    using wf unfolding staged-budget-wellformed-def by simp
  have alignment:
    length trace-roots = ceil-log clength ∧
    length trace-bs = length ([] :: 'f list) + ceil-log clength ∧
    PTranscript s3 = PTranscript s2 @ trace-roots ∧
    PState s3 = foldl concat (PState s2) trace-roots ∧
    s2 ≤ s3 ∧
    PTraceFriCounter s3 = PTraceFriCounter s2 + ceil-log clength ∧
    PCompositionFriCounter s3 = PCompositionFriCounter s2 ∧
    PAlphaCounter s3 = PAlphaCounter s2 ∧
    PQueryCounter s3 = PQueryCounter s2 ∧
    (∀ j < ceil-log clength.
      fmlookup (HashMap s3)
        (TraceFriChallenge (PTraceFriCounter s2 + j)
          (foldl concat (PState s2) (take (Suc j) trace-roots))) =
        Some (trace-bs ! (length ([] :: 'f list) + j)))
    by (rule staged-trace-fri-program-alignment
      [OF controlled len trace-out])
  then show ?thesis by simp
qed
have alpha-len: length as = length spec
  using staged-alpha-program-alignment[OF alpha-out] by simp
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have composition-len:
  length composition-roots = ceil-log (to-nat dg + 1) ∧
  length composition-bs = ceil-log (to-nat dg + 1)
proof –
  have len:
    0 + ceil-log (to-nat dg + 1) ≤

```



```

    length (composition-fri-budgets budgets)
  using wf round-bound unfolding staged-budget-wellformed-def by simp
have alignment:
  length composition-roots = ceil-log (to-nat dg + 1) ∧
  length composition-bs =
    length ([] :: 'f list) + ceil-log (to-nat dg + 1) ∧
  PTranscript s10 = PTranscript s9 @ composition-roots ∧
  PState s10 = foldl concat (PState s9) composition-roots ∧
  s9 ≤ s10 ∧
  PTraceFriCounter s10 = PTraceFriCounter s9 ∧
  PCompositionFriCounter s10 =
    PCompositionFriCounter s9 + ceil-log (to-nat dg + 1) ∧
  PAlphaCounter s10 = PAlphaCounter s9 ∧
  PQueryCounter s10 = PQueryCounter s9 ∧
  (∀ j < ceil-log (to-nat dg + 1).
    fmlookup (HashMap s10)
      (CompositionFriChallenge (PCompositionFriCounter s9 + j)
        (foldl concat (PState s9) (take (Suc j) composition-roots))) =
      Some (composition-bs ! (length ([] :: 'f list) + j)))
  by (rule staged-composition-fri-program-alignment
    [OF controlled len composition-out])
  then show ?thesis by simp
qed
have query-len:
  length query-chunks = rounds
proof -
  have len: 0 + rounds ≤ length (query-opening-budgets budgets)
  using wf unfolding staged-budget-wellformed-def by simp
  have alignment:
    length query-chunks = rounds ∧
    PTranscript s12 = PTranscript s11 @ List.concat query-chunks ∧
    PState s12 =
      foldl concat (PState s11) (List.concat query-chunks) ∧
    PTraceFriCounter s12 = PTraceFriCounter s11 ∧
    PCompositionFriCounter s12 = PCompositionFriCounter s11 ∧
    PAlphaCounter s12 = PAlphaCounter s11 ∧
    PQueryCounter s12 = PQueryCounter s11 + rounds
  by (rule staged-query-program-alignment[OF controlled len query-out])
  then show ?thesis by simp
qed
show ?thesis
  unfolding data-eq degree-eq
  using trace-len alpha-len composition-len round-bound query-len by simp
qed

lemma hash-target-program-outcome-extension:
  assumes target: hash-target-program B q m
  and outcome: Some (x, t) ∈ set-dist (execute m s)
  shows s ≤ t

```

using *hash-target-program-extension*[*OF target*] *outcome*
unfolding *hash-extension-preserving-def* **by** *blast*

lemma *staged-transcript-program-outcome-query-lookups*:

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *outcome*:

Some (data, attacker-state) ∈
set-dist
(execute (staged-transcript-program A)
adversary-initial-state)

shows

$\exists$ *raw-idxs query-idxs*.
length raw-idxs = rounds $\wedge$
query-idxs = map (λ raw. index (to-nat raw)) raw-idxs $\wedge$
length (staged-query-chunks data) = rounds $\wedge$
PState attacker-state =
state-after-query-chunks
(staged-query-start-hash data)
(staged-query-chunks data) rounds $\wedge$
PQueryCounter attacker-state = rounds $\wedge$
 $(\forall j < \text{rounds}.$
fmlookup (HashMap attacker-state)
(QueryIndexChallenge j
(state-after-query-chunks
(staged-query-start-hash data)
(staged-query-chunks data) j)) =
Some (raw-idxs ! j)) $\wedge$
 $(\forall \text{idx} \in \text{set query-idxs. idx} < \text{clength} * \text{scale})$

proof –

from *outcome* **obtain** *fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5*
as s6 dg s7 s8 s9 composition-roots composition-bs s10
composition-final s11 s12 query-chunks **where**

root-out:

Some (fr, s1) ∈
set-dist (execute (trace-root-stage A) adversary-initial-state)

and *record-root*:

Some ((), s2) ∈ set-dist (execute (record-staged-message fr) s1)

and *trace-out*:

Some ((trace-roots, trace-bs), s3) ∈
set-dist
(execute
(staged-trace-fri-program A 0 (ceil-log clength) []) s2)

and *final-out*:

Some (trace-final, s4) ∈
set-dist (execute (trace-final-stage A trace-bs) s3)

and *record-final*:

Some ((), s5) ∈
set-dist (execute (record-staged-message trace-final) s4)

```

and alpha-out:
  Some (as, s6) ∈
    set-dist (execute (staged-alpha-program (length spec)) s5)
and degree-out:
  Some (dg, s7) ∈ set-dist (execute (degree-stage A as) s6)
and record-degree:
  Some ((), s8) ∈ set-dist (execute (record-staged-message dg) s7)
and assert-out:
  Some ((), s9) ∈
    set-dist
      (execute
        (assert
          (ceil-log (to-nat dg + 1) ≤
            ceil-log (maxDegree + 1))) s8)
and composition-out:
  Some ((composition-roots, composition-bs), s10) ∈
    set-dist
      (execute
        (staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) []) s9)
and composition-final-out:
  Some (composition-final, s11) ∈
    set-dist
      (execute (composition-final-stage A dg composition-bs) s10)
and record-composition-final:
  Some ((), s12) ∈
    set-dist (execute (record-staged-message composition-final) s11)
and query-out:
  Some (query-chunks, attacker-state) ∈
    set-dist (execute (staged-query-program A 0 rounds) s12)
and data-eq:
  data =
    (staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots = composition-roots,
     staged-composition-fri-challenges = composition-bs,
     staged-composition-final = composition-final,
     staged-query-chunks = query-chunks)
unfolding staged-transcript-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have root-controlled:
  controlled-ro-program (trace-root-budget budgets)
    (trace-root-stage A)
using controlled unfolding staged-adversary-controlled-def by simp
have trace-final-controlled:

```

```

controlled-ro-program (trace-final-budget budgets)
  (trace-final-stage A trace-bs)
using controlled unfolding staged-adversary-controlled-def by blast
have degree-controlled:
  controlled-ro-program (degree-budget budgets) (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def by blast
have composition-final-controlled:
  controlled-ro-program (composition-final-budget budgets)
    (composition-final-stage A dg composition-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have root-state: PState s1 = 0
  using controlled-stage-outcome-fields[OF root-controlled root-out]
  by simp
have s2-eq:
  s2 = s1(
    PState := concat (PState s1) fr,
    PTranscript := PTranscript s1 @ [fr])
  by (rule record-staged-message-outcome[OF record-root])
have trace-bound:
   $0 + \text{ceil-log } \text{clength} \leq \text{length } (\text{trace-fri-budgets } \text{budgets})$ 
  using wf unfolding staged-budget-wellformed-def by simp
have trace-alignment:
  PState s3 = foldl concat (PState s2) trace-roots
  using staged-trace-fri-program-alignment
    [OF controlled trace-bound trace-out]
  by simp
have state-s3:
  PState s3 = foldl concat (concat 0 fr) trace-roots
  using root-state trace-alignment unfolding s2-eq by simp
have state-s4: PState s4 = PState s3
  using controlled-stage-outcome-fields
    [OF trace-final-controlled final-out]
  by simp
have s5-eq:
  s5 = s4(
    PState := concat (PState s4) trace-final,
    PTranscript := PTranscript s4 @ [trace-final])
  by (rule record-staged-message-outcome[OF record-final])
have alpha-alignment:
  PState s6 = foldl concat (PState s5) as
  using staged-alpha-program-alignment[OF alpha-out] by simp
have state-s6:
  PState s6 =
    foldl concat
      (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
      as
  using state-s3 state-s4 alpha-alignment unfolding s5-eq by simp
have state-s7: PState s7 = PState s6
  using controlled-stage-outcome-fields[OF degree-controlled degree-out]

```

```

by simp
have s8-eq:
  s8 = s7(
    PState := concat (PState s7) dg,
    PTranscript := PTranscript s7 @ [dg])
  by (rule record-staged-message-outcome[OF record-degree])
have s9-eq: s9 = s8
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have composition-bound:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
  using wf round-bound unfolding staged-budget-wellformed-def by simp
have composition-alignment:
  PState s10 = foldl concat (PState s9) composition-roots
  using staged-composition-fri-program-alignment
    [OF controlled composition-bound composition-out]
  by simp
have state-s10:
  PState s10 =
    foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
          as)
        dg)
      composition-roots
  using state-s6 state-s7 composition-alignment
  unfolding s8-eq s9-eq by simp
have state-s11: PState s11 = PState s10
  using controlled-stage-outcome-fields
    [OF composition-final-controlled composition-final-out]
  by simp
have s12-eq:
  s12 = s11(
    PState := concat (PState s11) composition-final,
    PTranscript := PTranscript s11 @ [composition-final])
  by (rule record-staged-message-outcome[OF record-composition-final])
have state-s12:
  PState s12 =
    concat

```

```

    (foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
          as)
        dg)
      composition-roots)
    composition-final
  using state-s10 state-s11 unfolding s12-eq by simp
have query-counter-s12: PQueryCounter s12 = 0
proof -
  have root-fields:
    PQueryCounter s1 = PQueryCounter adversary-initial-state
    using controlled-stage-outcome-fields[OF root-controlled root-out]
    by simp
  have trace-counters:
    PQueryCounter s3 = PQueryCounter s2
    using staged-trace-fri-program-alignment
      [OF controlled trace-bound trace-out]
    by simp
  have final-counter: PQueryCounter s4 = PQueryCounter s3
    using controlled-stage-outcome-fields
      [OF trace-final-controlled final-out]
    by simp
  have alpha-counter: PQueryCounter s6 = PQueryCounter s5
    using staged-alpha-program-alignment[OF alpha-out] by simp
  have degree-counter: PQueryCounter s7 = PQueryCounter s6
    using controlled-stage-outcome-fields[OF degree-controlled degree-out]
    by simp
  have comp-counter: PQueryCounter s10 = PQueryCounter s9
    using staged-composition-fri-program-alignment
      [OF controlled composition-bound composition-out]
    by simp
  have comp-final-counter: PQueryCounter s11 = PQueryCounter s10
    using controlled-stage-outcome-fields
      [OF composition-final-controlled composition-final-out]
    by simp
  show ?thesis
    using root-fields trace-counters final-counter alpha-counter
      degree-counter comp-counter comp-final-counter
    unfolding s2-eq s5-eq s8-eq s9-eq s12-eq by simp
qed
have query-bound: 0 + rounds ≤ length (query-opening-budgets budgets)
  using wf unfolding staged-budget-wellformed-def by simp
from staged-query-program-outcome-with-raws
  [OF controlled query-bound query-out]
obtain raw-idxs query-idxs where
  len-raw: length raw-idxs = rounds
  and query-idxs-def:

```

```

    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-chunks: length query-chunks = rounds
and state-attacker:
    PState attacker-state =
        state-after-query-chunks (PState s12) query-chunks rounds
and query-counter-attacker:
    PQueryCounter attacker-state = PQueryCounter s12 + rounds
and lookup:
    ∧j. j < rounds ⇒
        fmlookup (HashMap attacker-state)
            (QueryIndexChallenge (PQueryCounter s12 + j)
                (state-after-query-chunks (PState s12) query-chunks j)) =
                Some (raw-idxs ! j)
and idx-bound:
    ∀ idx ∈ set query-idxs. idx < clength * scale
by blast
have start-eq:
    PState s12 = staged-query-start-hash data
    unfolding data-eq staged-query-start-hash-def
        staged-composition-fri-start-hash-def staged-trace-fri-start-hash-def
    using state-s12 by simp
have chunks-eq: staged-query-chunks data = query-chunks
    unfolding data-eq by simp
have state-attacker':
    PState attacker-state =
        state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) rounds
    using state-attacker start-eq chunks-eq by simp
have query-counter-attacker':
    PQueryCounter attacker-state = rounds
    using query-counter-attacker query-counter-s12 by simp
have lookup':
    ∧j. j < rounds ⇒
        fmlookup (HashMap attacker-state)
            (QueryIndexChallenge j
                (state-after-query-chunks
                    (staged-query-start-hash data)
                    (staged-query-chunks data) j)) =
                Some (raw-idxs ! j)
    using lookup start-eq query-counter-s12 chunks-eq by simp
show ?thesis
by (intro exI[of - raw-idxs] exI[of - query-idxs] conjI)
   (use len-raw query-idxs-def chunks-eq len-chunks state-attacker'
       query-counter-attacker' lookup' idx-bound
       in simp-all)
qed

```

lemma *checked-staged-transcript-program-outcome-query-lookups:*

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *outcome*:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
shows
 $\exists$ *raw-idxs query-idxs*.
length raw-idxs = rounds ∧
query-idxs = map (λraw. index (to-nat raw)) raw-idxs ∧
length (staged-query-chunks data) = rounds ∧
PState attacker-state =
state-after-query-chunks
(staged-query-start-hash data)
(staged-query-chunks data) rounds ∧
PQueryCounter attacker-state = rounds ∧
 $(\forall j < \text{rounds}.$
fmlookup (HashMap attacker-state)
(QueryIndexChallenge j
(state-after-query-chunks
(staged-query-start-hash data)
(staged-query-chunks data) j)) =
Some (raw-idxs ! j) ∧
 $(\forall \text{idx} \in \text{set query-idxs. idx} < \text{clength} * \text{scale})$
proof –
have *unchecked-out*:
Some (data, attacker-state) ∈
set-dist
(execute (staged-transcript-program A) adversary-initial-state)
by (rule *checked-staged-transcript-program-outcome-imp-staged-transcript-program*
 $[OF \text{ outcome}]$)
show *?thesis*
by (rule *staged-transcript-program-outcome-query-lookups*
 $[OF \text{ wf controlled unchecked-out}]$)
qed

lemma *checked-staged-transcript-program-query-phaseE*:
assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *outcome*:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
obtains *fr trace-roots trace-bs trace-final as dg composition-roots*
composition-bs composition-final query-start query-chunks
where
Some (query-chunks, attacker-state) ∈


```

    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          0 rounds)
        query-start)
  data =
    (|staged-trace-root = fr,
      staged-trace-fri-roots = trace-roots,
      staged-trace-fri-challenges = trace-bs,
      staged-trace-final = trace-final,
      staged-alphas = as,
      staged-degree = dg,
      staged-composition-fri-roots = composition-roots,
      staged-composition-fri-challenges = composition-bs,
      staged-composition-final = composition-final,
      staged-query-chunks = query-chunks|)
  PState query-start = staged-query-start-hash data
  PQueryCounter query-start = 0
proof –
from outcome obtain fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5
  as s6 dg s7 s8 s9 composition-roots composition-bs s10
  composition-final s11 query-start query-chunks where
  root-out:
    Some (fr, s1) ∈
      set-dist (execute (trace-root-stage A) adversary-initial-state)
  and record-root:
    Some ((), s2) ∈ set-dist (execute (record-staged-message fr) s1)
  and trace-out:
    Some ((trace-roots, trace-bs), s3) ∈
      set-dist
        (execute
          (staged-trace-fri-program A 0 (ceil-log clength) []) s2)
  and final-out:
    Some (trace-final, s4) ∈
      set-dist (execute (trace-final-stage A trace-bs) s3)
  and record-final:
    Some ((), s5) ∈
      set-dist (execute (record-staged-message trace-final) s4)
  and alpha-out:
    Some (as, s6) ∈
      set-dist (execute (staged-alpha-program (length spec)) s5)
  and degree-out:
    Some (dg, s7) ∈ set-dist (execute (degree-stage A as) s6)
  and record-degree:
    Some ((), s8) ∈ set-dist (execute (record-staged-message dg) s7)
  and assert-out:
    Some ((), s9) ∈
      set-dist
        (execute

```

```

      (assert
        (ceil-log (to-nat dg + 1) ≤
          ceil-log (maxDegree + 1))) s8)
and composition-out:
  Some ((composition-roots, composition-bs), s10) ∈
    set-dist
      (execute
        (staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [] s9))
and composition-final-out:
  Some (composition-final, s11) ∈
    set-dist
      (execute (composition-final-stage A dg composition-bs) s10)
and record-composition-final:
  Some ((), query-start) ∈
    set-dist (execute (record-staged-message composition-final) s11)
and query-out:
  Some (query-chunks, attacker-state) ∈
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          0 rounds)
        query-start)
and data-eq:
  data =
    (|staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots = composition-roots,
     staged-composition-fri-challenges = composition-bs,
     staged-composition-final = composition-final,
     staged-query-chunks = query-chunks|)
unfolding checked-staged-transcript-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have root-controlled:
  controlled-ro-program (trace-root-budget budgets)
    (trace-root-stage A)
using controlled unfolding staged-adversary-controlled-def by simp
have trace-final-controlled:
  controlled-ro-program (trace-final-budget budgets)
    (trace-final-stage A trace-bs)
using controlled unfolding staged-adversary-controlled-def by blast
have degree-controlled:
  controlled-ro-program (degree-budget budgets) (degree-stage A as)
using controlled unfolding staged-adversary-controlled-def by blast
have composition-final-controlled:

```

```

controlled-ro-program (composition-final-budget budgets)
  (composition-final-stage A dg composition-bs)
using controlled unfolding staged-adversary-controlled-def by blast
have root-state: PState s1 = 0
  using controlled-stage-outcome-fields[OF root-controlled root-out]
  by simp
have s2-eq:
  s2 = s1(
    PState := concat (PState s1) fr,
    PTranscript := PTranscript s1 @ [fr])
  by (rule record-staged-message-outcome[OF record-root])
have trace-bound:
   $0 + \text{ceil-log } \text{clength} \leq \text{length } (\text{trace-fri-budgets } \text{budgets})$ 
  using wf unfolding staged-budget-wellformed-def by simp
have trace-alignment:
  PState s3 = foldl concat (PState s2) trace-roots
  using staged-trace-fri-program-alignment
    [OF controlled trace-bound trace-out]
  by simp
have state-s3:
  PState s3 = foldl concat (concat 0 fr) trace-roots
  using root-state trace-alignment unfolding s2-eq by simp
have state-s4: PState s4 = PState s3
  using controlled-stage-outcome-fields
    [OF trace-final-controlled final-out]
  by simp
have s5-eq:
  s5 = s4(
    PState := concat (PState s4) trace-final,
    PTranscript := PTranscript s4 @ [trace-final])
  by (rule record-staged-message-outcome[OF record-final])
have alpha-alignment:
  PState s6 = foldl concat (PState s5) as
  using staged-alpha-program-alignment[OF alpha-out] by simp
have state-s6:
  PState s6 =
    foldl concat
      (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
    as
  using state-s3 state-s4 alpha-alignment unfolding s5-eq by simp
have state-s7: PState s7 = PState s6
  using controlled-stage-outcome-fields[OF degree-controlled degree-out]
  by simp
have s8-eq:
  s8 = s7(
    PState := concat (PState s7) dg,
    PTranscript := PTranscript s7 @ [dg])
  by (rule record-staged-message-outcome[OF record-degree])
have s9-eq: s9 = s8

```

```

using assert-out unfolding assert-def
by (cases
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
  (auto simp: throw-no-outcome)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
using assert-out unfolding assert-def
by (cases
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
  (auto simp: throw-no-outcome)
have composition-bound:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
using wf round-bound unfolding staged-budget-wellformed-def by simp
have composition-alignment:
  PState s10 = foldl concat (PState s9) composition-roots
using staged-composition-fri-program-alignment
  [OF controlled composition-bound composition-out]
by simp
have state-s10:
  PState s10 =
    foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
          as)
        dg)
      composition-roots
using state-s6 state-s7 composition-alignment
unfolding s8-eq s9-eq by simp
have state-s11: PState s11 = PState s10
using controlled-stage-outcome-fields
  [OF composition-final-controlled composition-final-out]
by simp
have query-start-eq:
  query-start = s11(
    PState := concat (PState s11) composition-final,
    PTranscript := PTranscript s11 @ [composition-final])
by (rule record-staged-message-outcome[OF record-composition-final])
have state-query-start:
  PState query-start =
    concat
      (foldl concat
        (concat
          (foldl concat
            (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
            as)
          dg)
        composition-roots)

```

```

    composition-final
  using state-s10 state-s11 unfolding query-start-eq by simp
have query-counter-start: PQueryCounter query-start = 0
proof -
  have root-fields:
    PQueryCounter s1 = PQueryCounter adversary-initial-state
    using controlled-stage-outcome-fields[OF root-controlled root-out]
    by simp
  have trace-counters:
    PQueryCounter s3 = PQueryCounter s2
    using staged-trace-fri-program-alignment
      [OF controlled trace-bound trace-out]
    by simp
  have final-counter: PQueryCounter s4 = PQueryCounter s3
    using controlled-stage-outcome-fields
      [OF trace-final-controlled final-out]
    by simp
  have alpha-counter: PQueryCounter s6 = PQueryCounter s5
    using staged-alpha-program-alignment[OF alpha-out] by simp
  have degree-counter: PQueryCounter s7 = PQueryCounter s6
    using controlled-stage-outcome-fields[OF degree-controlled degree-out]
    by simp
  have comp-counter: PQueryCounter s10 = PQueryCounter s9
    using staged-composition-fri-program-alignment
      [OF controlled composition-bound composition-out]
    by simp
  have comp-final-counter: PQueryCounter s11 = PQueryCounter s10
    using controlled-stage-outcome-fields
      [OF composition-final-controlled composition-final-out]
    by simp
  show ?thesis
    using root-fields trace-counters final-counter alpha-counter
      degree-counter comp-counter comp-final-counter
    unfolding s2-eq s5-eq s8-eq s9-eq query-start-eq by simp
qed
have start-eq:
  PState query-start = staged-query-start-hash data
  unfolding data-eq staged-query-start-hash-def
    staged-composition-fri-start-hash-def staged-trace-fri-start-hash-def
  using state-query-start by simp
show ?thesis
  by (rule that[OF query-out data-eq start-eq query-counter-start])
qed

lemma checked-staged-security-with-data-state-accepted-shape-query-keys:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (((data, attacker-state), result), final-state) ∈

```

```

    set-dist (execute
      (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
  and shape:
    accepted-transcript-shape
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result, final-state)) as query-idxs
  shows
    as = staged-alphas data ∧
    (∃ raw-idxs.
      length raw-idxs = rounds ∧
      query-idxs = map (λraw. index (to-nat raw)) raw-idxs ∧
      (∀ i < rounds.
        fmllookup (HashMap final-state)
          (QueryIndexChallenge i
            (state-after-query-chunks
              (staged-query-start-hash data)
              (staged-query-chunks data) i)) =
          Some (raw-idxs ! i)) ∧
      (∀ idx ∈ set query-idxs. idx < clength * scale))
  proof -
  let ?s = verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF outcome]
  have builder:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈
      set-dist (execute verify-monad ?s)
  by blast+
  have staged-header:
    verifier-header-transcript ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
      (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
  from accepted-transcript-shape-query-chunksE[OF shape]
  obtain result' final-state' fr f-fri-roots f-final dg
    composition-fri-roots final rest raw-idxs query-chunks trailing

```

```

where out-eq:
  Some (result, final-state) = Some (result', final-state')
and header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-chunks: length query-chunks = rounds
and concat-chunks: List.concat query-chunks @ trailing = rest
and chunk-shape:
  ∧ i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
    f-fri-roots composition-fri-roots (query-chunks ! i)
and lookup:
  ∧ i. i < rounds ⇒
    fmlookup (HashMap final-state')
    (QueryIndexChallenge (PQueryCounter ?s + i)
    (state-after-query-chunks
    (verifier-header-state ?s fr f-fri-roots f-final as dg
    composition-fri-roots final)
    query-chunks i)) =
    Some (raw-idxs ! i)
and idx-bound:
  ∀ idx ∈ set query-idxs. idx < clength * scale
by blast
have header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  composition-fri-roots = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF staged-header header]
by simp
have final-state-eq: final-state' = final-state
using out-eq by simp
have header-state-eq:
  verifier-header-state ?s fr f-fri-roots f-final as dg
  composition-fri-roots final =
  staged-query-start-hash data
using header-eq
unfolding verifier-header-state-def verifier-header-messages-def
  staged-query-start-hash-def staged-composition-fri-start-hash-def
  staged-trace-fri-start-hash-def
by simp
from checked-staged-transcript-program-query-chunks-match-verifier-lengths

```

```

    [OF builder]
obtain staged-query-idxs where
  staged-match:
    staged-query-chunks-match-verifier-lengths data staged-query-idxs
  by blast
have query-idxs-len: length query-idxs = rounds
  using query-idxs-eq len-raw by simp
have staged-match-query-idxs:
  staged-query-chunks-match-verifier-lengths data query-idxs
  by (rule staged-query-chunks-match-verifier-lengths-transfer
    [OF staged-match query-idxs-len])
have parser-chunk-len:
   $\wedge i. i < rounds \implies$ 
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
proof -
  fix i
  assume i-bound: i < rounds
  show
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
    using verifier-query-round-chunk-length[OF chunk-shape[OF i-bound]]
    header-eq by simp
qed
have concat-staged:
  List.concat query-chunks @ trailing =
    List.concat (staged-query-chunks data)
  using concat-chunks header-eq by simp
have query-chunks-eq:
  query-chunks = staged-query-chunks data  $\wedge$  trailing = []
  by (rule query-chunks-eq-staged-if-matching-lengths
    [OF len-chunks concat-staged parser-chunk-len
      staged-match-query-idxs])
have lookup-staged:
   $\wedge i. i < rounds \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
proof -
  fix i
  assume i-bound: i < rounds
  have lookup-i:

```



```

    fmlookup (HashMap final-state')
      (QueryIndexChallenge (PQueryCounter ?s + i)
        (state-after-query-chunks
          (verifier-header-state ?s fr f-fri-roots f-final as dg
            composition-fri-roots final)
          query-chunks i)) =
        Some (raw-idxs ! i)
    by (rule lookup[OF i-bound])
  show
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
    using lookup-i final-state-eq header-state-eq query-chunks-eq by simp
  qed
  show ?thesis
    using header-eq len-raw query-idxs-eq lookup-staged idx-bound by blast
  qed

lemma checked-staged-security-with-data-state-accepted-shape-query-set-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  and shape:
    accepted-transcript-shape
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result, final-state)) as query-idxs
  and hit: ∃ i < rounds. query-idxs ! i ∈ B
  shows staged-transcript-query-index-set-hit B (Some (data, attacker-state))
proof -
  let ?s = verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF outcome]
  have builder:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈
      set-dist (execute verify-monad ?s)
  by blast+

```

```

from checked-staged-transcript-program-outcome-query-lookups
  [OF wf controlled builder]
obtain attacker-raw-idxs where
  lookup-attacker:
     $\forall j < \text{rounds}.$ 
      fmlookup (HashMap attacker-state)
        (QueryIndexChallenge j
          (state-after-query-chunks (staged-query-start-hash data)
            (staged-query-chunks data j))) =
          Some (attacker-raw-idxs ! j)
  by blast
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled outcome shape]
obtain raw-idxs where
  len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda \text{raw}. \text{index } (\text{to-nat } \text{raw})$ ) raw-idxs
  and lookup:
     $\bigwedge i. i < \text{rounds} \implies$ 
      fmlookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks (staged-query-start-hash data)
            (staged-query-chunks data i))) =
          Some (raw-idxs ! i)
  by blast
from hit obtain i where i-bound: i < rounds
  and query-hit: query-idxs ! i ∈ B
  by blast
have raw-i: i < length raw-idxs
  using len-raw i-bound by simp
have raw-hit: index (to-nat (raw-idxs ! i)) ∈ B
  using query-hit query-idxs-eq raw-i by (simp add: nth-map)
have ext: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
have lookup-initial:
  fmlookup (HashMap ?s)
    (QueryIndexChallenge i
      (state-after-query-chunks (staged-query-start-hash data)
        (staged-query-chunks data i))) =
      Some (attacker-raw-idxs ! i)
  using lookup-attacker i-bound by simp
have lookup-final-from-initial:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks (staged-query-start-hash data)
        (staged-query-chunks data i))) =
      Some (attacker-raw-idxs ! i)
  by (rule hash-extension-lookup[OF lookup-initial ext])
have raw-eq: attacker-raw-idxs ! i = raw-idxs ! i

```

```

using lookup-final-from-initial lookup[OF i-bound] by simp
have attacker-raw-hit: index (to-nat (attacker-raw-idxs ! i)) ∈ B
using raw-hit raw-eq by simp
have lookup-attacker-i:
  fmlookup (HashMap attacker-state)
    (QueryIndexChallenge i
      (state-after-query-chunks (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some (attacker-raw-idxs ! i)
using lookup-attacker i-bound by simp
show ?thesis
unfolding staged-transcript-query-index-set-hit-def
using i-bound lookup-attacker-i attacker-raw-hit by auto
qed

lemma staged-security-with-data-query-index-set-bound-from-transcript:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and transcript-bound:
  wp-event (staged-transcript-program A)
    (staged-transcript-query-index-set-hit B)
    adversary-initial-state ≤ C
shows
  wp-event (staged-security-experiment-with-data A)
    (staged-security-with-data-query-index-set-hit B)
    adversary-initial-state ≤ C
unfolding staged-security-experiment-with-data-def
proof (rule wp-event-bind-bound-by-head-event[OF transcript-bound])
show staged-security-with-data-query-index-set-hit B None ⇒
  staged-transcript-query-index-set-hit B None
unfolding staged-security-with-data-query-index-set-hit-def
  staged-transcript-query-index-set-hit-def
by simp
next
fix data attacker-state out
assume builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (staged-transcript-program A)
        adversary-initial-state)
and cont:
  out ∈
    set-dist
      (execute
        (get ≫=
          (λs. put
            (verifier-state-from-adversary s
              (staged-proof-transcript data)) ≫=
            (λ-. verify-monad ≫=

```

```

      (λresult. return (data, result))))))
    attacker-state)
  and hit: staged-security-with-data-query-index-set-hit B out
from staged-transcript-program-outcome-query-lookups
  [OF wf controlled builder]
obtain raw-idxs query-idxs where
  len-raw: length raw-idxs = rounds
  and lookup-attacker:
    ∀ j < rounds.
      fmlookup (HashMap attacker-state)
        (QueryIndexChallenge j
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) j)) =
        Some (raw-idxs ! j)
  by blast
from hit obtain data' result final-state i raw where
  out-eq: out = Some ((data', result), final-state)
  and i-bound: i < rounds
  and lookup-final:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data')
          (staged-query-chunks data') i)) =
      Some raw
  and raw-hit: index (to-nat raw) ∈ B
  unfolding staged-security-with-data-query-index-set-hit-def
  by (auto split: option.splits prod.splits)
from cont[unfolded out-eq] obtain verifier-state verifier-result where
  verifier:
    Some (verifier-result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
  and data'-eq: data' = data
  by (auto elim!: set-dist-bindE)
have ext:
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data) ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
have lookup-initial:
  fmlookup
    (HashMap
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (QueryIndexChallenge i
      (state-after-query-chunks

```

```

      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
  using lookup-attacker i-bound by simp
have lookup-final-from-initial:
  fmllookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
  by (rule hash-extension-lookup[OF lookup-initial ext])
have raw-eq: raw = raw-idxs ! i
  using lookup-final lookup-final-from-initial data'-eq by simp
have lookup-attacker-i:
  fmllookup (HashMap attacker-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
  using lookup-attacker i-bound by simp
have raw-idxs-hit: index (to-nat (raw-idxs ! i)) ∈ B
  using raw-hit raw-eq by simp
show staged-transcript-query-index-set-hit B
  (Some (data, attacker-state))
  unfolding staged-transcript-query-index-set-hit-def
  using i-bound lookup-attacker-i raw-idxs-hit by auto
qed

lemma checked-staged-security-with-data-query-index-set-bound-from-transcript:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and transcript-bound:
    wp-event (checked-staged-transcript-program A)
      (staged-transcript-query-index-set-hit B)
      adversary-initial-state ≤ C
  shows
    wp-event (checked-staged-security-experiment-with-data A)
      (staged-security-with-data-query-index-set-hit B)
      adversary-initial-state ≤ C
  unfolding checked-staged-security-experiment-with-data-def
proof (rule wp-event-bind-bound-by-head-event[OF transcript-bound])
  show staged-security-with-data-query-index-set-hit B None ⇒
    staged-transcript-query-index-set-hit B None
  unfolding staged-security-with-data-query-index-set-hit-def
    staged-transcript-query-index-set-hit-def
  by simp
next

```

```

fix data attacker-state out
assume builder:
  Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
and cont:
  out ∈
    set-dist
    (execute
      (get >>=
        (λs. put
          (verifier-state-from-adversary s
            (staged-proof-transcript data)) >>=
            (λ-. verify-monad >>=
              (λresult. return (data, result))))))
      attacker-state)
  and hit: staged-security-with-data-query-index-set-hit B out
from checked-staged-transcript-program-outcome-query-lookups
  [OF wf controlled builder]
obtain raw-idxs query-idxs where
  len-raw: length raw-idxs = rounds
and lookup-attacker:
  ∀ j < rounds.
    fmlookup (HashMap attacker-state)
      (QueryIndexChallenge j
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) j)) =
      Some (raw-idxs ! j)
by blast
from hit obtain data' result final-state i raw where
  out-eq: out = Some ((data', result), final-state)
and i-bound: i < rounds
and lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data')
        (staged-query-chunks data') i)) =
    Some raw
and raw-hit: index (to-nat raw) ∈ B
unfolding staged-security-with-data-query-index-set-hit-def
by (auto split: option.splits prod.splits)
from cont[unfolded out-eq] obtain verifier-state verifier-result where
  verifier:
    Some (verifier-result, final-state) ∈
      set-dist
      (execute verify-monad

```

```

      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    and data'-eq: data' = data
    by (auto elim!: set-dist-bindE)
  have ext:
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data) ≤ final-state
    by (rule verify-monad-hash-extends[OF verifier])
  have lookup-initial:
    fmllookup
      (HashMap
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
    using lookup-attacker i-bound by simp
  have lookup-final-from-initial:
    fmllookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
    by (rule hash-extension-lookup[OF lookup-initial ext])
  have raw-eq: raw = raw-idxs ! i
    using lookup-final lookup-final-from-initial data'-eq by simp
  have lookup-attacker-i:
    fmllookup (HashMap attacker-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
    using lookup-attacker i-bound by simp
  have raw-idxs-hit: index (to-nat (raw-idxs ! i)) ∈ B
    using raw-hit raw-eq by simp
  show staged-transcript-query-index-set-hit B
    (Some (data, attacker-state))
    unfolding staged-transcript-query-index-set-hit-def
    using i-bound lookup-attacker-i raw-idxs-hit by auto
qed

lemma staged-transcript-query-index-set-hit-imp-target-hit:
  assumes hit: staged-transcript-query-index-set-hit B out
  shows
    hash-new-output-hit-event (query-index-raw-preimage B)

```

```

    adversary-initial-state out
proof (cases out)
  case None
  then show ?thesis
    using hit unfolding staged-transcript-query-index-set-hit-def by simp
next
  case (Some result)
  then obtain data t where out-eq: out = Some (data, t)
    by (cases result) simp
  from hit[unfolded out-eq staged-transcript-query-index-set-hit-def]
  obtain i raw where lookup:
    fmlookup (HashMap t)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
        Some raw
    and raw-in: raw  $\in$  query-index-raw-preimage B
  unfolding query-index-raw-preimage-def by auto
  have new-hit:
    hash-map-new-output-hit (query-index-raw-preimage B)
      adversary-initial-state t
  unfolding hash-map-new-output-hit-def
  using lookup raw-in by auto
  show ?thesis
    unfolding out-eq hash-new-output-hit-event-def
    using new-hit by simp
qed

lemma staged-transcript-query-index-set-hit-bound-from-target-program:
  assumes target:
    hash-target-program (query-index-raw-preimage B) q m
  shows
    wp-event m (staged-transcript-query-index-set-hit B)
      adversary-initial-state  $\leq$ 
      staged-phase-target-error (query-index-raw-preimage B) q
proof –
  have event-mono:
    wp-event m (staged-transcript-query-index-set-hit B)
      adversary-initial-state  $\leq$ 
      wp-event m
        (hash-new-output-hit-event (query-index-raw-preimage B)
          adversary-initial-state)
      adversary-initial-state
  by (rule wp-event-mono)
    (rule staged-transcript-query-index-set-hit-imp-target-hit)
  have target-bound:
    wp-event m
      (hash-new-output-hit-event (query-index-raw-preimage B)

```


$\text{adversary-initial-state})$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-target-error } (\text{query-index-raw-preimage } B) \ q$
using **target unfolding** $\text{hash-target-program-def}$ $\text{hash-target-budget-def}$
 $\text{staged-phase-target-error-def}$ **by** blast
show $?thesis$
by $(\text{rule order-trans}[\text{OF event-mono target-bound}])$
qed

lemma $\text{staged-transcript-query-index-set-hit-bound}$:
assumes $\text{wf: staged-budget-wellformed budgets}$
and $\text{controlled: staged-adversary-controlled budgets } A$
shows
 $\text{wp-event } (\text{staged-transcript-program } A)$
 $(\text{staged-transcript-query-index-set-hit } B)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-target-error } (\text{query-index-raw-preimage } B)$
 $(\text{staged-attacker-query-budget budgets } +$
 $\text{staged-challenge-query-budget})$
by $(\text{rule staged-transcript-query-index-set-hit-bound-from-target-program})$
 $(\text{rule hash-target-program-staged-transcript-program}[\text{OF wf controlled}])$

lemma $\text{checked-staged-transcript-query-index-set-hit-bound-from-target-program}$:
assumes target:
 $\text{hash-target-program } (\text{query-index-raw-preimage } B) \ q$
 $(\text{checked-staged-transcript-program } A)$
shows
 $\text{wp-event } (\text{checked-staged-transcript-program } A)$
 $(\text{staged-transcript-query-index-set-hit } B)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-target-error } (\text{query-index-raw-preimage } B) \ q$
by $(\text{rule staged-transcript-query-index-set-hit-bound-from-target-program}$
 $[\text{OF target}])$

lemma $\text{checked-staged-transcript-query-index-set-hit-bound}$:
assumes $\text{wf: staged-budget-wellformed budgets}$
and $\text{controlled: staged-adversary-controlled budgets } A$
shows
 $\text{wp-event } (\text{checked-staged-transcript-program } A)$
 $(\text{staged-transcript-query-index-set-hit } B)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-target-error } (\text{query-index-raw-preimage } B)$
 $(\text{staged-attacker-query-budget budgets } +$
 $\text{staged-challenge-query-budget})$
by $(\text{rule checked-staged-transcript-query-index-set-hit-bound-from-target-program})$
 $(\text{rule hash-target-program-checked-staged-transcript-program}$
 $[\text{OF wf controlled}])$

lemma $\text{staged-security-with-data-query-index-set-hit-bound}$:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
shows
wp-event (*staged-security-experiment-with-data A*)
(*staged-security-with-data-query-index-set-hit B*)
adversary-initial-state $\leq$
staged-phase-target-error (*query-index-raw-preimage B*)
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)
by (*rule staged-security-with-data-query-index-set-bound-from-transcript*
[*OF wf controlled*
staged-transcript-query-index-set-hit-bound[OF wf controlled]])

lemma *checked-staged-security-with-data-query-index-set-hit-bound-from-target-program*:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *target*:
hash-target-program (*query-index-raw-preimage B*) *q*
(*checked-staged-transcript-program A*)
shows
wp-event (*checked-staged-security-experiment-with-data A*)
(*staged-security-with-data-query-index-set-hit B*)
adversary-initial-state $\leq$
staged-phase-target-error (*query-index-raw-preimage B*) *q*
by (*rule checked-staged-security-with-data-query-index-set-bound-from-transcript*
[*OF wf controlled*
checked-staged-transcript-query-index-set-hit-bound-from-target-program
[*OF target*]])

lemma *checked-staged-security-with-data-query-index-set-hit-bound*:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
shows
wp-event (*checked-staged-security-experiment-with-data A*)
(*staged-security-with-data-query-index-set-hit B*)
adversary-initial-state $\leq$
staged-phase-target-error (*query-index-raw-preimage B*)
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)
by (*rule checked-staged-security-with-data-query-index-set-hit-bound-from-target-program*
[*OF wf controlled*
hash-target-program-checked-staged-transcript-program
[*OF wf controlled*]])

end

end

```

theory Staged-Security-Experiment-Checked
  imports Staged-Security-Experiment-Checked-Prefix
begin

```

Final checked staged query-bad and collision bounds.

```

context soundness
begin

```

```

lemma query-bad-imp-query-rounds-sampling-success-hit:
  assumes bad: query-bad s out
  shows
     $\exists$  trace-table composition-table as.
      query-rounds-any-index-set-hit s
      (query-sampling-success-space trace-table composition-table as)
      rounds out

```

```

proof –
  have round-hit:
    query-index-round-set-hit s query-sampling-success-space out
  by (rule query-bad-imp-query-index-round-set-hit[OF bad])
  then obtain i where hit-at:
    query-index-round-set-hit-at s query-sampling-success-space i out
  unfolding query-index-round-set-hit-def by blast
  show ?thesis
    by (rule query-index-round-set-hit-at-imp-query-rounds-any-index-set-hit
      [OF hit-at])

```

qed

```

lemma query-bad-imp-query-header-sampling-success-hit:
  assumes outcome: out  $\in$  set-dist (execute verify-monad s)
  and bad: query-bad s out
  shows
    query-header-rounds-any-index-set-hit s
    (query-header-supported-union-good-sets s query-sampling-success-space)
    out

```

```

proof –
  have round-hit:
    query-index-round-set-hit s query-sampling-success-space out
  by (rule query-bad-imp-query-index-round-set-hit[OF bad])
  show ?thesis
    by (rule query-index-round-set-hit-imp-query-header-supported-rounds-hit
      [OF outcome round-hit])

```

qed

```

lemma staged-security-with-data-query-bad-imp-sampling-hit:
  assumes bad: staged-security-with-data-query-bad-hit out
  shows staged-security-with-data-query-sampling-hit out
proof (cases out)
  case None
  then show ?thesis

```

```

    using bad unfolding staged-security-with-data-query-bad-hit-def by simp
next
case (Some result)
then obtain data verifier-result final-state where out-eq:
  out = Some ((data, verifier-result), final-state)
  by (cases result, auto split: prod.splits)
from bad[unfolded out-eq staged-security-with-data-query-bad-hit-def]
obtain s where query-bad:
  query-bad s (Some (verifier-result, final-state))
  by auto
from query-bad-imp-query-rounds-sampling-success-hit[OF query-bad]
obtain trace-table composition-table as where hit:
  query-rounds-any-index-set-hit s
  (query-sampling-success-space trace-table composition-table as)
  rounds (Some (verifier-result, final-state))
  by blast
have witness:
  ∃ s trace-table composition-table as.
    query-rounds-any-index-set-hit s
    (query-sampling-success-space trace-table composition-table as)
    rounds (Some (verifier-result, final-state))
  by (intro exI[of - s] exI[of - trace-table]
      exI[of - composition-table] exI[of - as])
      (rule hit)
show ?thesis
  unfolding out-eq staged-security-with-data-query-sampling-hit-def
  using witness by simp
qed

```

```

lemma staged-security-with-data-query-bad-imp-header-sampling-hit:
  assumes bad: staged-security-with-data-query-bad-hit out
  shows staged-security-with-data-query-header-sampling-hit out
proof (cases out)
case None
then show ?thesis
  using bad unfolding staged-security-with-data-query-bad-hit-def by simp
next
case (Some result)
then obtain data verifier-result final-state where out-eq:
  out = Some ((data, verifier-result), final-state)
  by (cases result, auto split: prod.splits)
from bad[unfolded out-eq staged-security-with-data-query-bad-hit-def]
obtain s where outcome:
  Some (verifier-result, final-state) ∈
  set-dist (execute verify-monad s)
  and query-bad:
    query-bad s (Some (verifier-result, final-state))
  by auto
have hit:

```

```

  query-header-rounds-any-index-set-hit s
    (query-header-supported-union-good-sets s query-sampling-success-space)
    (Some (verifier-result, final-state))
  by (rule query-bad-imp-query-header-sampling-success-hit
      [OF outcome query-bad])
  then show ?thesis
    unfolding out-eq staged-security-with-data-query-header-sampling-hit-def
    by auto
qed

```

```

lemma staged-security-with-data-state-query-bad-imp-header-sampling-hit:
  assumes bad: staged-security-with-data-state-query-bad-hit out
  shows staged-security-with-data-state-query-header-sampling-hit out
proof (cases out)
  case None
  then show ?thesis
    using bad
    unfolding staged-security-with-data-state-query-bad-hit-def by simp
next
  case (Some result)
  then obtain data attacker-state verifier-result final-state where out-eq:
    out = Some (((data, attacker-state), verifier-result), final-state)
    by (cases result, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from bad[unfolded out-eq staged-security-with-data-state-query-bad-hit-def]
  have outcome:
    Some (verifier-result, final-state) ∈
    set-dist (execute verify-monad ?s)
  and query-bad:
    query-bad ?s (Some (verifier-result, final-state))
  by (simp-all add: Let-def)
  have hit:
    query-header-rounds-any-index-set-hit ?s
    (query-header-supported-union-good-sets ?s query-sampling-success-space)
    (Some (verifier-result, final-state))
  by (rule query-bad-imp-query-header-sampling-success-hit
      [OF outcome query-bad])
  then show ?thesis
    unfolding out-eq
    staged-security-with-data-state-query-header-sampling-hit-def
    by simp
qed

```

```

lemma checked-staged-security-with-data-state-query-bad-imp-header-key-hit-on-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:

```

```

    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
    and bad: staged-security-with-data-state-query-bad-hit out
  shows staged-security-with-data-state-query-header-key-hit out
proof (cases out)
case None
then show ?thesis
  using bad unfolding staged-security-with-data-state-query-bad-hit-def
  by simp
next
case (Some result-pack)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases result-pack, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have support-some:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  using support unfolding out-eq .
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support-some]
have builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast+
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])

```

```

from bad[unfolded out-eq
  staged-security-with-data-state-query-bad-hit-def Let-def]
have query-bad: query-bad ?s (Some (result, final-state))
  by simp
from query-bad-imp-query-index-round-set-hit[OF query-bad]
obtain i trace-table composition-table as query-idxs where
  bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
and i-bound: i < rounds
and query-hit:
  query-idxs ! i ∈
    query-sampling-success-space trace-table composition-table as
unfolding query-index-round-set-hit-def
  query-index-round-set-hit-at-def
by blast
have shape:
  accepted-transcript-shape ?s (Some (result, final-state))
    as query-idxs
using bound
unfolding accepted-with-bound-tables-def accepted-with-tables-def
by simp
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support-some shape]
obtain raw-idxs where
  as-eq: as = staged-alphas data
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and lookup:
   $\bigwedge j. j < rounds \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge j
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) j)) =
      Some (raw-idxs ! j)
and idx-bound:
   $\forall idx \in set\ query-idxs. idx < clength * scale$ 
by blast
have idx-sample:
  query-idxs ! i ∈ query-sample-space
proof —
  have len-query: length query-idxs = rounds
    using query-idxs-eq len-raw by simp
  have query-idxs ! i ∈ set query-idxs
    by (rule nth-mem) (use i-bound len-query in simp)
  then show ?thesis
    by (rule accepted-with-bound-tables-query-sample-space[OF bound])

```

```

qed
have candidate:
  (trace-table, composition-table) ∈
    query-header-supported-table-candidates ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
proof -
  have bound-staged:
    accepted-with-bound-tables ?s (Some (result, final-state))
      trace-table composition-table (staged-alphas data) query-idxs
  using bound as-eq by simp
  have witness:
    ∃ out query-idxs rest.
      out ∈ set-dist (execute verify-monad ?s) ∧
      accepted-with-bound-tables ?s out trace-table composition-table
        (staged-alphas data) query-idxs ∧
      verifier-header-transcript ?s
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data) rest
  by (intro exI[where x=Some (result, final-state)]
      exI[where x=query-idxs]
      exI[where x=List.concat (staged-query-chunks data)] conjI)
    (use verifier bound-staged staged-header in simp-all)
  show ?thesis
    using witness
    unfolding query-header-supported-table-candidates-def by simp
qed
have raw-hit:
  index (to-nat (raw-idxs ! i)) ∈
    query-header-supported-union-good-sets ?s
      query-sampling-success-space
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
proof -

```



```

have raw-idx-eq: index (to-nat (raw-idxs ! i)) = query-idxs ! i
  using query-idxs-eq len-raw i-bound by simp
show ?thesis
  unfolding query-header-supported-union-good-sets-def raw-idx-eq
  using idx-sample candidate query-hit as-eq by blast
qed
have lookup-i:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
  by (rule lookup[OF i-bound])
show ?thesis
  unfolding out-eq staged-security-with-data-state-query-header-key-hit-def
    Let-def
  using i-bound lookup-i raw-hit by auto
qed

lemma staged-security-with-data-state-query-bad-bound-from-header-sampling-hit:
  assumes header-bound:
    wp-event (staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-header-sampling-hit
    adversary-initial-state  $\leq C$ 
  shows
    wp-event (staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-bad-hit
    adversary-initial-state  $\leq C$ 
proof –
  have wp-event (staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-bad-hit
    adversary-initial-state  $\leq$ 
    wp-event (staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-header-sampling-hit
    adversary-initial-state
  by (rule wp-event-mono)
    (rule staged-security-with-data-state-query-bad-imp-header-sampling-hit)
  also have ...  $\leq C$ 
  by (rule header-bound)
  finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-query-bad-bound-from-header-key-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and header-key-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-header-key-hit

```

$adversary-initial-state \leq C$
shows
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $staged-security-with-data-state-query-bad-hit$
 $adversary-initial-state \leq C$
proof –
have $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $staged-security-with-data-state-query-bad-hit$
 $adversary-initial-state \leq$
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $staged-security-with-data-state-query-header-key-hit$
 $adversary-initial-state$
proof (rule $wp-event-mono-on-support$)
fix out
assume $support$:
 $out \in$
 $set-dist$
 $(execute$
 $(checked-staged-security-experiment-with-data-state \ A)$
 $adversary-initial-state)$
and bad : $staged-security-with-data-state-query-bad-hit \ out$
show $staged-security-with-data-state-query-header-key-hit \ out$
by (rule
 $checked-staged-security-with-data-state-query-bad-imp-header-key-hit-on-support$
 $[OF \ wf \ controlled \ support \ bad])$
qed
also have $\dots \leq C$
by (rule $header-key-bound$)
finally show $?thesis$.
qed

lemma
 $checked-staged-security-with-data-state-header-key-hit-bound-from-index-set$:
assumes $cover$:
 $\bigwedge data \ attacker-state.$
 $query-header-supported-union-good-sets$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data))$
 $query-sampling-success-space$
 $(staged-trace-root \ data)$
 $(staged-trace-fri-roots \ data)$
 $(staged-trace-final \ data)$
 $(staged-alphas \ data)$
 $(staged-degree \ data)$
 $(staged-composition-fri-roots \ data)$
 $(staged-composition-final \ data) \subseteq B$
and $index-bound$:
 $wp-event \ (checked-staged-security-experiment-with-data \ A)$
 $(staged-security-with-data-query-index-set-hit \ B)$

```

    adversary-initial-state ≤ C
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-header-key-hit
    adversary-initial-state ≤ C
proof -
  let ?projected =
    λout. case out of None ⇒
      staged-security-with-data-query-index-set-hit B None
    | Some (((data, -), result), t) ⇒
      staged-security-with-data-query-index-set-hit B
        (Some ((data, result), t))
  have event-le:
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-header-key-hit
      adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      ?projected adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume hit:
    staged-security-with-data-state-query-header-key-hit out
  show ?projected out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding staged-security-with-data-state-query-header-key-hit-def
    by simp
next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  let ?B =
    query-header-supported-union-good-sets ?s
      query-sampling-success-space
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  from hit[unfolded out-eq
    staged-security-with-data-state-query-header-key-hit-def Let-def]

```

```

obtain  $i$  raw where  $i$ -bound:  $i < \text{rounds}$ 
and lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge  $i$ 
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data)  $i$ )) =
      Some raw
and raw-hit:  $\text{index } (\text{to-nat } \text{raw}) \in ?B$ 
by auto
have raw-in-B:  $\text{index } (\text{to-nat } \text{raw}) \in B$ 
proof –
  have  $?B \subseteq B$ 
  by (rule cover)
  then show ?thesis
  using raw-hit by blast
qed
show ?thesis
  unfolding out-eq staged-security-with-data-query-index-set-hit-def
  using  $i$ -bound lookup raw-in-B by auto
qed
qed
also have
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
    ?projected adversary-initial-state =
  wp-event (checked-staged-security-experiment-with-data  $A$ )
    (staged-security-with-data-query-index-set-hit  $B$ )
    adversary-initial-state
using checked-staged-security-experiment-with-data-event-from-data-state
  [of  $A$  staged-security-with-data-query-index-set-hit  $B$ ]
by simp
also have  $\dots \leq C$ 
by (rule index-bound)
finally show ?thesis .
qed

lemma
  checked-staged-security-with-data-state-header-key-hit-bound-from-index-set-controlled:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and cover:
   $\bigwedge$  data attacker-state.
    query-header-supported-union-good-sets
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
    query-sampling-success-space
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)

```

(staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) $\subseteq B$

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-query-header-key-hit
 adversary-initial-state $\leq$
 staged-phase-target-error (query-index-raw-preimage B)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)

by (rule
 checked-staged-security-with-data-state-header-key-hit-bound-from-index-set
 [OF cover])
 (rule checked-staged-security-with-data-query-index-set-hit-bound
 [OF wf controlled])

lemma

checked-staged-security-with-data-state-query-bad-bound-from-index-set-controlled:

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and cover:

$\bigwedge$ data attacker-state.
 query-header-supported-union-good-sets
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 query-sampling-success-space
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) $\subseteq B$

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-query-bad-hit
 adversary-initial-state $\leq$
 staged-phase-target-error (query-index-raw-preimage B)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)

by (rule checked-staged-security-with-data-state-query-bad-bound-from-header-key-hit
 [OF wf controlled])
 (rule
 checked-staged-security-with-data-state-header-key-hit-bound-from-index-set-controlled
 [OF wf controlled cover])

lemma

checked-staged-security-with-data-state-query-bad-bound-from-fixed-query-error-cover:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *cover*:
 $\bigwedge$ data attacker-state.
query-header-supported-union-good-sets
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
query-sampling-success-space
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data) $\subseteq B$
and *raw-bound*: query-index-raw-preimage-bound
and *subset*: $B \subseteq \text{query-sample-space}$
and *frac*:
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{query-sample-space}) \leq$
query-error-bound
shows
wp-event (checked-staged-security-experiment-with-data-state *A*)
staged-security-with-data-state-query-bad-hit
adversary-initial-state $\leq$
 nnreal
(staged-attacker-query-budget budgets +
staged-challenge-query-budget) * query-error-bound
proof –
have *target*:
wp-event (checked-staged-security-experiment-with-data-state *A*)
staged-security-with-data-state-query-bad-hit
adversary-initial-state $\leq$
staged-phase-target-error (query-index-raw-preimage *B*)
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)
by (rule
checked-staged-security-with-data-state-query-bad-bound-from-index-set-controlled
[OF *wf controlled cover*])
also have ... $\leq$
 nnreal
(staged-attacker-query-budget budgets +
staged-challenge-query-budget) * query-error-bound
by (rule staged-phase-query-index-target-error-query-bound
[OF *raw-bound subset frac*])
finally show ?thesis .
qed

lemma
checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-index-set-controlled:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *cover*:
 $\bigwedge$ *data* *attacker-state*.
 $\text{query-header-supported-union-good-sets}$
 $(\text{verifier-state-from-adversary } \text{attacker-state}$
 $(\text{staged-proof-transcript } \text{data}))$
 $\text{query-sampling-success-space}$
 $(\text{staged-trace-root } \text{data})$
 $(\text{staged-trace-fri-roots } \text{data})$
 $(\text{staged-trace-final } \text{data})$
 $(\text{staged-alphas } \text{data})$
 $(\text{staged-degree } \text{data})$
 $(\text{staged-composition-fri-roots } \text{data})$
 $(\text{staged-composition-final } \text{data}) \subseteq B$
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event } \text{query-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-target-error } (\text{query-index-raw-preimage } B)$
 $(\text{staged-attacker-query-budget } \text{budgets} +$
 $\text{staged-challenge-query-budget})$
by (rule
 $\text{checked-staged-security-with-data-state-query-bad-verifier-event-bound})$
(rule
 $\text{checked-staged-security-with-data-state-query-bad-bound-from-index-set-controlled}$
 $[\text{OF } wf \text{ controlled } cover])$

lemma

checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-fixed-query-error-cover:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *cover*:
 $\bigwedge$ *data* *attacker-state*.
 $\text{query-header-supported-union-good-sets}$
 $(\text{verifier-state-from-adversary } \text{attacker-state}$
 $(\text{staged-proof-transcript } \text{data}))$
 $\text{query-sampling-success-space}$
 $(\text{staged-trace-root } \text{data})$
 $(\text{staged-trace-fri-roots } \text{data})$
 $(\text{staged-trace-final } \text{data})$
 $(\text{staged-alphas } \text{data})$
 $(\text{staged-degree } \text{data})$
 $(\text{staged-composition-fri-roots } \text{data})$
 $(\text{staged-composition-final } \text{data}) \subseteq B$
and *raw-bound*: $\text{query-index-raw-preimage-bound}$
and *subset*: $B \subseteq \text{query-sample-space}$
and *frac*:
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{query-sample-space}) \leq$

query-error-bound

shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event query-bad)
adversary-initial-state ≤
nnreal
(staged-attacker-query-budget budgets +
*staged-challenge-query-budget) * query-error-bound*
by (rule
checked-staged-security-with-data-state-query-bad-verifier-event-bound)
(rule
checked-staged-security-with-data-state-query-bad-bound-from-fixed-query-error-cover
[*OF wf controlled cover raw-bound subset frac*])

lemma *staged-security-with-data-query-bad-bound-from-sampling-hit:*
assumes *sampling-bound:*
wp-event (staged-security-experiment-with-data A)
staged-security-with-data-query-sampling-hit
adversary-initial-state ≤ C
shows
wp-event (staged-security-experiment-with-data A)
staged-security-with-data-query-bad-hit
adversary-initial-state ≤ C
proof –
have *wp-event (staged-security-experiment-with-data A)*
staged-security-with-data-query-bad-hit
adversary-initial-state ≤
wp-event (staged-security-experiment-with-data A)
staged-security-with-data-query-sampling-hit
adversary-initial-state
by (rule *wp-event-mono*)
(rule *staged-security-with-data-query-bad-imp-sampling-hit*)
also have *... ≤ C*
by (rule *sampling-bound*)
finally show *?thesis .*
qed

lemma *staged-security-with-data-query-bad-bound-from-header-sampling-hit:*
assumes *header-bound:*
wp-event (staged-security-experiment-with-data A)
staged-security-with-data-query-header-sampling-hit
adversary-initial-state ≤ C
shows
wp-event (staged-security-experiment-with-data A)
staged-security-with-data-query-bad-hit
adversary-initial-state ≤ C
proof –
have *wp-event (staged-security-experiment-with-data A)*
staged-security-with-data-query-bad-hit


```

    adversary-initial-state ≤
wp-event (staged-security-experiment-with-data A)
    staged-security-with-data-query-header-sampling-hit
    adversary-initial-state
by (rule wp-event-mono)
    (rule staged-security-with-data-query-bad-imp-header-sampling-hit)
also have ... ≤ C
    by (rule header-bound)
finally show ?thesis .
qed

```

lemma *staged-transcript-program-outcome-trace-challenge-lookup:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *outcome:*

Some (data, attacker-state) ∈
set-dist
(execute (staged-transcript-program A)
adversary-initial-state)

and *i-bound: i < ceil-log clength*

shows

fmllookup (HashMap attacker-state)
(TraceFriChallenge i
(foldl concat (staged-trace-fri-start-hash data)
(take (Suc i) (staged-trace-fri-roots data)))) =
Some (staged-trace-fri-challenges data ! i)

proof –

from *outcome* **obtain** *fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5*
as s6 dg s7 s8 s9 composition-roots composition-bs s10
composition-final s11 s12 query-chunks **where**

root-out:

Some (fr, s1) ∈
set-dist (execute (trace-root-stage A) adversary-initial-state)

and *record-root:*

Some ((), s2) ∈ set-dist (execute (record-staged-message fr) s1)

and *trace-out:*

Some ((trace-roots, trace-bs), s3) ∈
set-dist
(execute
(staged-trace-fri-program A 0 (ceil-log clength) []) s2)

and *final-out:*

Some (trace-final, s4) ∈
set-dist (execute (trace-final-stage A trace-bs) s3)

and *record-final:*

Some ((), s5) ∈
set-dist (execute (record-staged-message trace-final) s4)

and *alpha-out:*

Some (as, s6) ∈
set-dist (execute (staged-alpha-program (length spec)) s5)

```

and degree-out:
  Some (dg, s7) ∈ set-dist (execute (degree-stage A as) s6)
and record-degree:
  Some ((), s8) ∈ set-dist (execute (record-staged-message dg) s7)
and assert-out:
  Some ((), s9) ∈
    set-dist
      (execute
        (assert
          (ceil-log (to-nat dg + 1) ≤
            ceil-log (maxDegree + 1))) s8)
and composition-out:
  Some ((composition-roots, composition-bs), s10) ∈
    set-dist
      (execute
        (staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) []) s9)
and composition-final-out:
  Some (composition-final, s11) ∈
    set-dist
      (execute (composition-final-stage A dg composition-bs) s10)
and record-composition-final:
  Some ((), s12) ∈
    set-dist (execute (record-staged-message composition-final) s11)
and query-out:
  Some (query-chunks, attacker-state) ∈
    set-dist (execute (staged-query-program A 0 rounds) s12)
and data-eq:
  data =
    (|staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots = composition-roots,
     staged-composition-fri-challenges = composition-bs,
     staged-composition-final = composition-final,
     staged-query-chunks = query-chunks|)
unfolding staged-transcript-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have root-controlled:
  controlled-ro-program (trace-root-budget budgets)
    (trace-root-stage A)
using controlled unfolding staged-adversary-controlled-def by simp
have trace-final-controlled:
  controlled-ro-program (trace-final-budget budgets)
    (trace-final-stage A trace-bs)
using controlled unfolding staged-adversary-controlled-def by blast

```

```

have degree-controlled:
  controlled-ro-program (degree-budget budgets) (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def by blast
have composition-final-controlled:
  controlled-ro-program (composition-final-budget budgets)
    (composition-final-stage A dg composition-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have root-fields:
  PState s1 = 0  $\wedge$ 
  PTraceFriCounter s1 = 0
  using controlled-stage-outcome-fields[OF root-controlled root-out]
  by simp
have s2-eq:
  s2 = s1 ( $\lambda$ 
    PState := concat (PState s1) fr,
    PTranscript := PTranscript s1 @ [fr])
  by (rule record-staged-message-outcome[OF record-root])
have trace-bound:
  0 + ceil-log clength  $\leq$  length (trace-fri-budgets budgets)
  using wf unfolding staged-budget-wellformed-def by simp
have trace-alignment:
  length trace-roots = ceil-log clength  $\wedge$ 
  length trace-bs = length ( $\lambda :: 'f \text{ list}$ ) + ceil-log clength  $\wedge$ 
  PTranscript s3 = PTranscript s2 @ trace-roots  $\wedge$ 
  PState s3 = foldl concat (PState s2) trace-roots  $\wedge$ 
  s2  $\leq$  s3  $\wedge$ 
  PTraceFriCounter s3 = PTraceFriCounter s2 + ceil-log clength  $\wedge$ 
  PCompositionFriCounter s3 = PCompositionFriCounter s2  $\wedge$ 
  PAlphaCounter s3 = PAlphaCounter s2  $\wedge$ 
  PQueryCounter s3 = PQueryCounter s2  $\wedge$ 
  ( $\forall j < \text{ceil-log clength.}$ 
    fmlookup (HashMap s3)
      (TraceFriChallenge (PTraceFriCounter s2 + j)
        (foldl concat (PState s2) (take (Suc j) trace-roots))) =
        Some (trace-bs ! (length ( $\lambda :: 'f \text{ list}$ ) + j)))
  by (rule staged-trace-fri-program-alignment
    [OF controlled trace-bound trace-out])
let ?key =
  TraceFriChallenge i
    (foldl concat (concat 0 fr) (take (Suc i) trace-roots))
have lookup-s3: fmlookup (HashMap s3) ?key = Some (trace-bs ! i)
  using trace-alignment i-bound root-fields
  unfolding s2-eq by simp
have ext-s3-s4: s3  $\leq$  s4
  using controlled-ro-program-extension[OF trace-final-controlled]
    final-out
  unfolding hash-extension-preserving-def by blast
have s5-eq:
  s5 = s4 ( $\lambda$ 

```

```

    PState := concat (PState s4) trace-final,
    PTranscript := PTranscript s4 @ [trace-final])
  by (rule record-staged-message-outcome[OF record-final])
have ext-s4-s5: s4 ≤ s5
  unfolding s5-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have ext-s5-s6: s5 ≤ s6
  by (rule hash-target-program-outcome-extension
    [OF hash-target-program-staged-alpha-program alpha-out])
have ext-s6-s7: s6 ≤ s7
  using controlled-ro-program-extension[OF degree-controlled] degree-out
  unfolding hash-extension-preserving-def by blast
have s8-eq:
  s8 = s7(
    PState := concat (PState s7) dg,
    PTranscript := PTranscript s7 @ [dg])
  by (rule record-staged-message-outcome[OF record-degree])
have ext-s7-s8: s7 ≤ s8
  unfolding s8-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have s9-eq: s9 = s8
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have ext-s8-s9: s8 ≤ s9
  unfolding s9-eq by (simp add: hash-ext-refl)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have composition-bound:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
  using wf round-bound unfolding staged-budget-wellformed-def by simp
have ext-s9-s10: s9 ≤ s10
  using staged-composition-fri-program-alignment
    [OF controlled composition-bound composition-out]
  by simp
have ext-s10-s11: s10 ≤ s11
  using controlled-ro-program-extension[OF composition-final-controlled]
    composition-final-out
  unfolding hash-extension-preserving-def by blast
have s12-eq:
  s12 = s11(
    PState := concat (PState s11) composition-final,
    PTranscript := PTranscript s11 @ [composition-final])
  by (rule record-staged-message-outcome[OF record-composition-final])
have ext-s11-s12: s11 ≤ s12

```

```

  unfolding s12-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have query-bound: 0 + rounds ≤ length (query-opening-budgets budgets)
  using wf unfolding staged-budget-wellformed-def by simp
have ext-s12-attacker: s12 ≤ attacker-state
  by (rule hash-target-program-outcome-extension
      [OF hash-target-program-staged-query-program[OF controlled query-bound]
        query-out])
have ext-s3-s5: s3 ≤ s5
  by (rule hash-ext-trans[OF ext-s3-s4 ext-s4-s5])
have ext-s3-s6: s3 ≤ s6
  by (rule hash-ext-trans[OF ext-s3-s5 ext-s5-s6])
have ext-s3-s7: s3 ≤ s7
  by (rule hash-ext-trans[OF ext-s3-s6 ext-s6-s7])
have ext-s3-s8: s3 ≤ s8
  by (rule hash-ext-trans[OF ext-s3-s7 ext-s7-s8])
have ext-s3-s9: s3 ≤ s9
  by (rule hash-ext-trans[OF ext-s3-s8 ext-s8-s9])
have ext-s3-s10: s3 ≤ s10
  by (rule hash-ext-trans[OF ext-s3-s9 ext-s9-s10])
have ext-s3-s11: s3 ≤ s11
  by (rule hash-ext-trans[OF ext-s3-s10 ext-s10-s11])
have ext-s3-s12: s3 ≤ s12
  by (rule hash-ext-trans[OF ext-s3-s11 ext-s11-s12])
have ext-s3-attacker: s3 ≤ attacker-state
  by (rule hash-ext-trans[OF ext-s3-s12 ext-s12-attacker])
have lookup-attacker:
  fmlookup (HashMap attacker-state) ?key = Some (trace-bs ! i)
  by (rule hash-extension-lookup[OF lookup-s3 ext-s3-attacker])
show ?thesis
  unfolding data-eq staged-trace-fri-start-hash-def
  using lookup-attacker by simp
qed

```

```

lemma staged-transcript-program-outcome-composition-challenge-lookup:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state) ∈
      set-dist
        (execute (staged-transcript-program A)
          adversary-initial-state)
  and i-bound:
    i < ceil-log (to-nat (staged-degree data) + 1)
  shows
    fmlookup (HashMap attacker-state)
      (CompositionFriChallenge i
        (foldl concat (staged-composition-fri-start-hash data)
          (take (Suc i) (staged-composition-fri-roots data)))) =
      Some (staged-composition-fri-challenges data ! i)

```

proof –

from *outcome* **obtain** *fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5*
as s6 dg s7 s8 s9 composition-roots composition-bs s10
composition-final s11 s12 query-chunks **where**
root-out:
Some (fr, s1) ∈
set-dist (execute (trace-root-stage A) adversary-initial-state)
and *record-root:*
Some ((), s2) ∈ set-dist (execute (record-staged-message fr) s1)
and *trace-out:*
Some ((trace-roots, trace-bs), s3) ∈
set-dist
(execute
(staged-trace-fri-program A 0 (ceil-log clength) []) s2)
and *final-out:*
Some (trace-final, s4) ∈
set-dist (execute (trace-final-stage A trace-bs) s3)
and *record-final:*
Some ((), s5) ∈
set-dist (execute (record-staged-message trace-final) s4)
and *alpha-out:*
Some (as, s6) ∈
set-dist (execute (staged-alpha-program (length spec)) s5)
and *degree-out:*
Some (dg, s7) ∈ set-dist (execute (degree-stage A as) s6)
and *record-degree:*
Some ((), s8) ∈ set-dist (execute (record-staged-message dg) s7)
and *assert-out:*
Some ((), s9) ∈
set-dist
(execute
(assert
(ceil-log (to-nat dg + 1) ≤
ceil-log (maxDegree + 1))) s8)
and *composition-out:*
Some ((composition-roots, composition-bs), s10) ∈
set-dist
(execute
(staged-composition-fri-program A dg 0
(ceil-log (to-nat dg + 1)) []) s9)
and *composition-final-out:*
Some (composition-final, s11) ∈
set-dist
(execute (composition-final-stage A dg composition-bs) s10)
and *record-composition-final:*
Some ((), s12) ∈
set-dist (execute (record-staged-message composition-final) s11)
and *query-out:*
Some (query-chunks, attacker-state) ∈

```

    set-dist (execute (staged-query-program A 0 rounds) s12)
and data-eq:
  data =
    (|staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots = composition-roots,
     staged-composition-fri-challenges = composition-bs,
     staged-composition-final = composition-final,
     staged-query-chunks = query-chunks|)
  unfolding staged-transcript-program-def Let-def
  by (auto elim!: set-dist-bindE split: prod.splits)
have root-controlled:
  controlled-ro-program (trace-root-budget budgets)
    (trace-root-stage A)
  using controlled unfolding staged-adversary-controlled-def by simp
have trace-final-controlled:
  controlled-ro-program (trace-final-budget budgets)
    (trace-final-stage A trace-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have degree-controlled:
  controlled-ro-program (degree-budget budgets) (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def by blast
have composition-final-controlled:
  controlled-ro-program (composition-final-budget budgets)
    (composition-final-stage A dg composition-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have root-fields:
  PState s1 = 0  $\wedge$ 
  PCompositionFriCounter s1 = 0
  using controlled-stage-outcome-fields[OF root-controlled root-out]
  by simp
have s2-eq:
  s2 = s1 (|
    PState := concat (PState s1) fr,
    PTranscript := PTranscript s1 @ [fr]|)
  by (rule record-staged-message-outcome[OF record-root])
have s2-fields:
  PState s2 = concat 0 fr  $\wedge$ 
  PCompositionFriCounter s2 = 0
  unfolding s2-eq using root-fields by simp
have trace-bound:
  0 + ceil-log clength  $\leq$  length (trace-fri-budgets budgets)
  using wf unfolding staged-budget-wellformed-def by simp
have trace-alignment:
  length trace-roots = ceil-log clength  $\wedge$ 

```

```

length trace-bs = length ([] :: 'f list) + ceil-log clength ∧
PTranscript s3 = PTranscript s2 @ trace-roots ∧
PState s3 = foldl concat (PState s2) trace-roots ∧
s2 ≤ s3 ∧
PTraceFriCounter s3 = PTraceFriCounter s2 + ceil-log clength ∧
PCompositionFriCounter s3 = PCompositionFriCounter s2 ∧
PAlphaCounter s3 = PAlphaCounter s2 ∧
PQueryCounter s3 = PQueryCounter s2 ∧
(∀ j < ceil-log clength.
  fmllookup (HashMap s3)
    (TraceFriChallenge (PTraceFriCounter s2 + j)
      (foldl concat (PState s2) (take (Suc j) trace-roots))) =
    Some (trace-bs ! (length ([] :: 'f list) + j)))
by (rule staged-trace-fri-program-alignment
  [OF controlled trace-bound trace-out])
have final-fields:
  PState s4 = PState s3 ∧
  PCompositionFriCounter s4 = PCompositionFriCounter s3
using controlled-stage-outcome-fields
  [OF trace-final-controlled final-out]
by simp
have s5-eq:
  s5 = s4 (
    PState := concat (PState s4) trace-final,
    PTranscript := PTranscript s4 @ [trace-final])
by (rule record-staged-message-outcome[OF record-final])
have s5-fields:
  PState s5 =
    concat (foldl concat (concat 0 fr) trace-roots) trace-final ∧
  PCompositionFriCounter s5 = 0
unfolding s5-eq
using s2-fields trace-alignment final-fields by simp
have alpha-alignment:
  length as = length spec ∧
  PTranscript s6 = PTranscript s5 @ as ∧
  PState s6 = foldl concat (PState s5) as ∧
  PTraceFriCounter s6 = PTraceFriCounter s5 ∧
  PCompositionFriCounter s6 = PCompositionFriCounter s5 ∧
  PAlphaCounter s6 = PAlphaCounter s5 + length spec ∧
  PQueryCounter s6 = PQueryCounter s5
by (rule staged-alpha-program-alignment[OF alpha-out])
have degree-fields:
  PState s7 = PState s6 ∧
  PCompositionFriCounter s7 = PCompositionFriCounter s6
using controlled-stage-outcome-fields[OF degree-controlled degree-out]
by simp
have s8-eq:
  s8 = s7 (
    PState := concat (PState s7) dg,

```



```

    PTranscript := PTranscript s7 @ [dg])
  by (rule record-staged-message-outcome[OF record-degree])
have s9-eq: s9 = s8
  using assert-out unfolding assert-def
  by (cases
      ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
      (auto simp: throw-no-outcome)
have start-fields:
  PState s9 =
    concat
      (foldl concat
        (concat
          (foldl concat (concat 0 fr) trace-roots)
          trace-final)
        as)
      dg ∧
    PCompositionFriCounter s9 = 0
  unfolding s9-eq s8-eq
  using s5-fields alpha-alignment degree-fields
  by simp
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using assert-out unfolding assert-def
  by (cases
      ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
      (auto simp: throw-no-outcome)
have composition-bound:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
  using wf round-bound unfolding staged-budget-wellformed-def by simp
have composition-alignment:
  length composition-roots = ceil-log (to-nat dg + 1) ∧
  length composition-bs =
    length ([] :: 'f list) + ceil-log (to-nat dg + 1) ∧
  PTranscript s10 = PTranscript s9 @ composition-roots ∧
  PState s10 = foldl concat (PState s9) composition-roots ∧
  s9 ≤ s10 ∧
  PTraceFriCounter s10 = PTraceFriCounter s9 ∧
  PCompositionFriCounter s10 =
    PCompositionFriCounter s9 + ceil-log (to-nat dg + 1) ∧
  PAlphaCounter s10 = PAlphaCounter s9 ∧
  PQueryCounter s10 = PQueryCounter s9 ∧
  (∀ j < ceil-log (to-nat dg + 1).
    fmlookup (HashMap s10)
      (CompositionFriChallenge (PCompositionFriCounter s9 + j)
        (foldl concat (PState s9) (take (Suc j) composition-roots))) =
      Some (composition-bs ! (length ([] :: 'f list) + j)))
  by (rule staged-composition-fri-program-alignment
      [OF controlled composition-bound composition-out])

```

```

let ?key =
  CompositionFriChallenge i
  (foldl concat
    (concat
      (foldl concat
        (concat
          (foldl concat (concat 0 fr) trace-roots)
          trace-final)
        as)
      dg)
    (take (Suc i) composition-roots))
have i-bound':  $i < \text{ceil-log } (\text{to-nat } dg + 1)$ 
  using i-bound unfolding data-eq by simp
have lookup-s10:
  fmlookup (HashMap s10) ?key = Some (composition-bs ! i)
  using composition-alignment i-bound' start-fields by simp
have ext-s10-s11:  $s10 \leq s11$ 
  using controlled-ro-program-extension[OF composition-final-controlled]
  composition-final-out
  unfolding hash-extension-preserving-def by blast
have s12-eq:
  s12 = s11()
  PState := concat (PState s11) composition-final,
  PTranscript := PTranscript s11 @ [composition-final])
  by (rule record-staged-message-outcome[OF record-composition-final])
have ext-s11-s12:  $s11 \leq s12$ 
  unfolding s12-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have query-bound:  $0 + \text{rounds} \leq \text{length } (\text{query-opening-budgets budgets})$ 
  using wf unfolding staged-budget-wellformed-def by simp
have ext-s12-attacker:  $s12 \leq \text{attacker-state}$ 
  by (rule hash-target-program-outcome-extension
    [OF hash-target-program-staged-query-program[OF controlled query-bound]
    query-out])
have ext-s10-s12:  $s10 \leq s12$ 
  by (rule hash-ext-trans[OF ext-s10-s11 ext-s11-s12])
have ext-s10-attacker:  $s10 \leq \text{attacker-state}$ 
  by (rule hash-ext-trans[OF ext-s10-s12 ext-s12-attacker])
have lookup-attacker:
  fmlookup (HashMap attacker-state) ?key = Some (composition-bs ! i)
  by (rule hash-extension-lookup[OF lookup-s10 ext-s10-attacker])
show ?thesis
  unfolding data-eq staged-composition-fri-start-hash-def
  staged-trace-fri-start-hash-def
  using lookup-attacker by simp
qed

lemma staged-transcript-program-outcome-header-transcript:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A

```

and *outcome*:
Some (data, attacker-state) ∈
set-dist
(execute (staged-transcript-program A)
adversary-initial-state)
shows
verifier-header-transcript
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)
(List.concat (staged-query-chunks data))
proof –
have *shape*:
length (staged-trace-fri-roots data) = ceil-log clength ∧
length (staged-trace-fri-challenges data) = ceil-log clength ∧
length (staged-alphas data) = length spec ∧
length (staged-composition-fri-roots data) =
ceil-log (to-nat (staged-degree data) + 1) ∧
length (staged-composition-fri-challenges data) =
ceil-log (to-nat (staged-degree data) + 1) ∧
ceil-log (to-nat (staged-degree data) + 1) ≤
ceil-log (maxDegree + 1) ∧
length (staged-query-chunks data) = rounds
by (rule *staged-transcript-program-outcome-shape*
[*OF wf controlled outcome*])
show *?thesis*
by (rule *staged-proof-transcript-verifier-header-transcript*)
(use shape in simp-all)
qed

lemma *staged-security-experiment-outcomeE*:
assumes
Some (result, final-state) ∈
set-dist (execute (staged-security-experiment A) initial-state)
obtains *data attacker-state* **where**
Some (data, attacker-state) ∈
set-dist (execute (staged-transcript-program A) initial-state)
Some (result, final-state) ∈
set-dist
(execute verify-monad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
using *assms* **unfolding** *staged-security-experiment-def*

by (auto elim!: set-dist-bindE intro: that)

lemma *accepted-staged-security-experimentE*:

assumes *outcome*:

out $\in$ *set-dist*

(*execute* (*staged-security-experiment* *A*) *adversary-initial-state*)

and *accepted*: $\neg$ *Option.is-none* *out*

obtains *result final-state data attacker-state* **where**

out = *Some* (*result*, *final-state*)

Some (*data*, *attacker-state*) $\in$

set-dist (*execute* (*staged-transcript-program* *A*) *adversary-initial-state*)

Some (*result*, *final-state*) $\in$

set-dist

(*execute* *verify-monad*

(*verifier-state-from-adversary* *attacker-state*

(*staged-proof-transcript* *data*)))

proof –

from *accepted* **obtain** *result final-state* **where**

out: *out* = *Some* (*result*, *final-state*)

by (*cases* *out*) *auto*

from *staged-security-experiment-outcomeE*[*OF* *outcome*[*unfolded out*]]

obtain *data attacker-state* **where**

builder:

Some (*data*, *attacker-state*) $\in$

set-dist (*execute* (*staged-transcript-program* *A*)

adversary-initial-state)

and *verifier*:

Some (*result*, *final-state*) $\in$

set-dist

(*execute* *verify-monad*

(*verifier-state-from-adversary* *attacker-state*

(*staged-proof-transcript* *data*)))

by *blast*

show *?thesis* **by** (*rule* *that*[*OF* *out* *builder* *verifier*])

qed

lemma *accepted-staged-security-experiment-headerE*:

assumes *wf*: *staged-budget-wellformed* *budgets*

and *controlled*: *staged-adversary-controlled* *budgets* *A*

and *outcome*:

out $\in$ *set-dist*

(*execute* (*staged-security-experiment* *A*) *adversary-initial-state*)

and *accepted*: $\neg$ *Option.is-none* *out*

obtains *result final-state data attacker-state* **where**

out = *Some* (*result*, *final-state*)

Some (*data*, *attacker-state*) $\in$

set-dist (*execute* (*staged-transcript-program* *A*)

adversary-initial-state)

Some (*result*, *final-state*) $\in$

```

    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
  verifier-header-transcript
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
proof –
from accepted-staged-security-experimentE[OF outcome accepted]
obtain result final-state data attacker-state where
  out-eq: out = Some (result, final-state)
and builder:
  Some (data, attacker-state) ∈
    set-dist (execute (staged-transcript-program A)
      adversary-initial-state)
and verifier:
  Some (result, final-state) ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
by blast
have header:
  verifier-header-transcript
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
by (rule staged-transcript-program-outcome-header-transcript
  [OF wf controlled builder])
show ?thesis
by (rule that[OF out-eq builder verifier header])
qed

```

lemma accepted-staged-security-experiment-transcript-shapeE:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *outcome*:
 out ∈ *set-dist*
 (*execute* (*staged-security-experiment* *A*) *adversary-initial-state*)
and *accepted*: $\neg$ *Option.is-none* *out*
obtains *result final-state data attacker-state query-idxs* **where**
 out = *Some* (*result, final-state*)
 Some (*data, attacker-state*) ∈
 set-dist (*execute* (*staged-transcript-program* *A*)
 adversary-initial-state)
 Some (*result, final-state*) ∈
 set-dist
 (*execute* *verify-monad*
 (*verifier-state-from-adversary* *attacker-state*
 (*staged-proof-transcript* *data*)))
 accepted-transcript-shape
 (*verifier-state-from-adversary* *attacker-state*
 (*staged-proof-transcript* *data*))
 out
 (*staged-alphas* *data*)
 query-idxs
proof –
from *accepted-staged-security-experiment-headerE*
 [*OF wf controlled outcome accepted*]
obtain *result final-state data attacker-state* **where**
 out-eq: *out* = *Some* (*result, final-state*)
and *builder*:
 Some (*data, attacker-state*) ∈
 set-dist (*execute* (*staged-transcript-program* *A*)
 adversary-initial-state)
and *verifier*:
 Some (*result, final-state*) ∈
 set-dist
 (*execute* *verify-monad*
 (*verifier-state-from-adversary* *attacker-state*
 (*staged-proof-transcript* *data*)))
and *staged-header*:
 verifier-header-transcript
 (*verifier-state-from-adversary* *attacker-state*
 (*staged-proof-transcript* *data*))
 (*staged-trace-root* *data*)
 (*staged-trace-fri-roots* *data*)
 (*staged-trace-final* *data*)
 (*staged-alphas* *data*)
 (*staged-degree* *data*)
 (*staged-composition-fri-roots* *data*)
 (*staged-composition-final* *data*)
 (*List.concat* (*staged-query-chunks* *data*))

```

    by blast
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  from verify-monad-accepted-transcript-shape[OF verifier]
  obtain as query-idxs where shape:
    accepted-transcript-shape ?s (Some (result, final-state))
      as query-idxs
  by blast
  from shape obtain result' final-state' fr f-fri-roots f-final dg
    composition-fri-roots final rest where
    shape-out: Some (result, final-state) = Some (result', final-state')
  and shape-header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
      composition-fri-roots final rest
  by (elim accepted-transcript-shape-header-query-extraction)
  have as-eq: as = staged-alphas data
    using verifier-header-transcript-unique[OF staged-header shape-header]
  by simp
  have shape-staged:
    accepted-transcript-shape ?s out (staged-alphas data) query-idxs
    using shape unfolding out-eq as-eq .
  show ?thesis
    by (rule that[OF out-eq builder verifier shape-staged])
qed

```

```

lemma accepted-staged-security-experiment-query-chunksE:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    out ∈ set-dist
      (execute (staged-security-experiment A) adversary-initial-state)
  and accepted: ¬ Option.is-none out
  obtains result final-state data attacker-state raw-idxs query-idxs
    query-chunks trailing
  where
    out = Some (result, final-state)
    Some (data, attacker-state) ∈
      set-dist (execute (staged-transcript-program A)
        adversary-initial-state)
    Some (result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
    length raw-idxs = rounds
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
    length query-chunks = rounds
    List.concat query-chunks @ trailing =

```

```

    List.concat (staged-query-chunks data)
  ∧ i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
      (staged-trace-fri-roots data)
      (staged-composition-fri-roots data)
      (query-chunks ! i)
  ∧ i. i < rounds ⇒
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
  ∧ i. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) query-chunks i)) =
      Some (raw-idxs ! i)
  ∀ idx ∈ set query-idxs. idx < clength * scale
proof –
from accepted-staged-security-experiment-headerE
  [OF wf controlled outcome accepted]
obtain result final-state data attacker-state where
  out-eq: out = Some (result, final-state)
and builder:
  Some (data, attacker-state) ∈
    set-dist (execute (staged-transcript-program A)
      adversary-initial-state)
and verifier:
  Some (result, final-state) ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
and staged-header:
  verifier-header-transcript
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
by blast
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)

```



```

from verify-monad-accepted-transcript-shape[OF verifier]
obtain as accepted-query-idxs where shape:
  accepted-transcript-shape ?s (Some (result, final-state))
  as accepted-query-idxs
by blast
from accepted-transcript-shape-query-chunksE[OF shape]
obtain result' final-state' fr f-fri-roots f-final dg
  composition-fri-roots final rest raw-idxs query-chunks trailing
where shape-out:
  Some (result, final-state) = Some (result', final-state')
and header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  accepted-query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and len-chunks: length query-chunks = rounds
and concat-chunks: List.concat query-chunks @ trailing = rest
and chunk-shape:
   $\bigwedge i. i < \text{rounds} \implies$ 
    verifier-query-round-chunk (accepted-query-idxs ! i)
    f-fri-roots composition-fri-roots (query-chunks ! i)
and lookup:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state')
    (QueryIndexChallenge (PQueryCounter ?s + i)
    (state-after-query-chunks
    (verifier-header-state ?s fr f-fri-roots f-final as dg
    composition-fri-roots final)
    query-chunks i)) =
    Some (raw-idxs ! i)
and idx-bound:
   $\forall \text{idx} \in \text{set } \text{accepted-query-idxs}. \text{idx} < \text{clength} * \text{scale}$ 
by blast
have header-eq:
  fr = staged-trace-root data  $\wedge$ 
  f-fri-roots = staged-trace-fri-roots data  $\wedge$ 
  f-final = staged-trace-final data  $\wedge$ 
  as = staged-alphas data  $\wedge$ 
  dg = staged-degree data  $\wedge$ 
  composition-fri-roots = staged-composition-fri-roots data  $\wedge$ 
  final = staged-composition-final data  $\wedge$ 
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF staged-header header]
by simp
have final-state-eq: final-state' = final-state
using shape-out by simp
have header-state-eq:
  verifier-header-state ?s fr f-fri-roots f-final as dg

```

```

    composition-fri-roots final =
    staged-query-start-hash data
using header-eq
unfolding verifier-header-state-def verifier-header-messages-def
    staged-query-start-hash-def staged-composition-fri-start-hash-def
    staged-trace-fri-start-hash-def
by simp
have lookup':
   $\wedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) query-chunks i)) =
      Some (raw-idxs ! i)
using lookup final-state-eq header-state-eq by simp
have chunk-shape':
   $\wedge i. i < \text{rounds} \implies$ 
    verifier-query-round-chunk (accepted-query-idxs ! i)
      (staged-trace-fri-roots data)
      (staged-composition-fri-roots data)
      (query-chunks ! i)
using chunk-shape header-eq by simp
have chunk-len:
   $\wedge i. i < \text{rounds} \implies$ 
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (accepted-query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
by (rule verifier-query-round-chunk-length[OF chunk-shape'])
show ?thesis
by (rule that[OF out-eq builder verifier len-raw query-idxs-eq
  len-chunks])
  (use concat-chunks header-eq chunk-shape' chunk-len lookup' idx-bound
    in simp-all)
qed

```

```

lemma staged-transcript-shape-alpha-badD:
  assumes shape:
    accepted-transcript-shape s out staged-as staged-query-idxs
  and bad: alpha-bad-set-hit s B out
  shows  $\exists \text{trace-table}. \text{staged-as} \in B \text{ trace-table}$ 
proof –
from bad obtain trace-table composition-table as query-idxs where
  bound:
    accepted-with-bound-tables s out trace-table composition-table
    as query-idxs
  and as-bad: as  $\in B \text{ trace-table}$ 
  unfolding alpha-bad-set-hit-def by blast
have shape-bad: accepted-transcript-shape s out as query-idxs

```

```

    using bound unfolding accepted-with-bound-tables-def
      accepted-with-tables-def by simp
  have as = staged-as
    by (rule accepted-transcript-shape-alphas-unique[OF shape shape-bad])
  then show ?thesis
    using as-bad by blast
qed

lemma staged-security-with-data-alpha-bad-set-hit-imp-alpha-list-hit:
  assumes envelope:  $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq B$ 
    and hit: staged-security-with-data-alpha-bad-set-hit bad-sets out
  shows staged-security-with-data-alpha-list-hit B out
  using assms
  unfolding staged-security-with-data-alpha-bad-set-hit-def
    staged-security-with-data-alpha-list-hit-def
  by (auto split: option.splits prod.splits)

lemma staged-security-with-data-alpha-list-bound-from-transcript:
  assumes transcript-bound:
    wp-event (staged-transcript-program A)
      (staged-transcript-alpha-list-hit B) adversary-initial-state  $\leq C$ 
  shows
    wp-event (staged-security-experiment-with-data A)
      (staged-security-with-data-alpha-list-hit B)
      adversary-initial-state  $\leq C$ 
  unfolding staged-security-experiment-with-data-def
proof (rule wp-event-bind-bound-by-head-event[OF transcript-bound])
  show staged-security-with-data-alpha-list-hit B None  $\implies$ 
    staged-transcript-alpha-list-hit B None
  unfolding staged-security-with-data-alpha-list-hit-def
    staged-transcript-alpha-list-hit-def
  by simp
next
  fix data attacker-state out
  assume cont:
    out  $\in$ 
      set-dist
        (execute
          (get  $\gg=$ 
            ( $\lambda s.$  put
              (verifier-state-from-adversary s
                (staged-proof-transcript data))  $\gg=$ 
              ( $\lambda.$  verify-monad  $\gg=$ 
                ( $\lambda \text{result. return (data, result)}))))$ 
            attacker-state)
        and hit: staged-security-with-data-alpha-list-hit B out
  have data-bad: staged-alphas data  $\in B$ 
proof (cases out)
  case None

```

```

then show ?thesis
  using hit unfolding staged-security-with-data-alpha-list-hit-def
  by simp
next
case (Some result)
then obtain data' result' final-state where out-eq:
  out = Some ((data', result'), final-state)
  by (cases result, auto split: prod.splits)
from cont[unfolded out-eq]
obtain s3 where
  ret:
    Some ((data', result'), final-state) ∈
      set-dist (execute (return (data, result'))) s3
  by (auto elim!: set-dist-bindE)
have data'-eq: data' = data
  using ret by simp
show ?thesis
  using hit unfolding out-eq data'-eq
    staged-security-with-data-alpha-list-hit-def
  by simp
qed
show staged-transcript-alpha-list-hit B
  (Some (data, attacker-state))
  using data-bad unfolding staged-transcript-alpha-list-hit-def by simp
qed

lemma staged-security-with-data-alpha-bad-set-bound-from-alpha-list:
assumes envelope:  $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq B$ 
and list-bound:
  wp-event (staged-security-experiment-with-data A)
    (staged-security-with-data-alpha-list-hit B)
    adversary-initial-state  $\leq C$ 
shows
  wp-event (staged-security-experiment-with-data A)
    (staged-security-with-data-alpha-bad-set-hit bad-sets)
    adversary-initial-state  $\leq C$ 
proof –
have wp-event (staged-security-experiment-with-data A)
  (staged-security-with-data-alpha-bad-set-hit bad-sets)
  adversary-initial-state  $\leq$ 
  wp-event (staged-security-experiment-with-data A)
  (staged-security-with-data-alpha-list-hit B)
  adversary-initial-state
by (rule wp-event-mono)
  (rule staged-security-with-data-alpha-bad-set-hit-imp-alpha-list-hit
    [OF envelope])
also have ...  $\leq C$ 
by (rule list-bound)
finally show ?thesis .

```

qed

definition *checked-staged-security-with-data-state-alpha-list-hit*

$:: 'f \text{ list set} \Rightarrow$
 $((('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-data-state-alpha-list-hit $B \text{ out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((\text{data}, -), -), -) \Rightarrow \text{staged-alphas data} \in B)$

definition *checked-staged-security-with-data-state-alpha-bad-set-hit*

$:: ('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $((('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-data-state-alpha-bad-set-hit bad-sets out
 $\longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((\text{data}, -), -), -) \Rightarrow$
 $(\exists \text{ trace-table. staged-alphas data} \in \text{bad-sets trace-table}))$

definition *checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit*

$:: ('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $((('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
 $\text{bad-sets out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((\text{data}, \text{attacker-state}), -), -) \Rightarrow$
 $(\text{let } s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $\text{(staged-proof-transcript data)}$
 $\text{in staged-alphas data} \in$
 $\text{alpha-header-supported-union-bad-sets } s \text{ bad-sets}$
 $\text{(staged-trace-root data)}$
 $\text{(staged-trace-fri-roots data)}$
 $\text{(staged-trace-final data)}))$

lemma *checked-staged-security-with-data-state-alpha-bad-set-hit-imp-alpha-list-hit:*

assumes $\text{envelope: } \bigwedge \text{ trace-table. bad-sets trace-table} \subseteq B$

and hit:

checked-staged-security-with-data-state-alpha-bad-set-hit bad-sets out

shows *checked-staged-security-with-data-state-alpha-list-hit* $B \text{ out}$

using *assms*

```

unfolding checked-staged-security-with-data-state-alpha-bad-set-hit-def
  checked-staged-security-with-data-state-alpha-list-hit-def
by (auto split: option.splits prod.splits)

lemma checked-staged-security-with-data-state-alpha-list-bound-from-transcript:
assumes transcript-bound:
  wp-event (checked-staged-transcript-program A)
  (staged-transcript-alpha-list-hit B) adversary-initial-state  $\leq C$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (checked-staged-security-with-data-state-alpha-list-hit B)
  adversary-initial-state  $\leq C$ 
unfolding checked-staged-security-experiment-with-data-state-def
proof (rule wp-event-bind-bound-by-head-event[OF transcript-bound])
show
  checked-staged-security-with-data-state-alpha-list-hit B None  $\implies$ 
    staged-transcript-alpha-list-hit B None
unfolding checked-staged-security-with-data-state-alpha-list-hit-def
  staged-transcript-alpha-list-hit-def
by simp
next
fix data attacker-state out
assume cont:
  out  $\in$ 
    set-dist
    (execute
      (get  $\gg$ 
        ( $\lambda s$ . put
          (verifier-state-from-adversary s
            (staged-proof-transcript data))  $\gg$ 
            ( $\lambda$ -. verify-monad  $\gg$ 
              ( $\lambda$ result. return ((data, s), result))))))
      attacker-state)
  and hit: checked-staged-security-with-data-state-alpha-list-hit B out
have data-bad: staged-alphas data  $\in B$ 
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding checked-staged-security-with-data-state-alpha-list-hit-def
  by simp
next
case (Some result-pack)
then obtain data' attacker-state' result final-state where out-eq:
  out = Some (((data', attacker-state'), result), final-state)
  by (cases result-pack, auto split: prod.splits)
from cont[unfolded out-eq]
obtain saved-state state-after-get where
  ret:

```

```

    Some (((data', attacker-state'), result), final-state) ∈
      set-dist (execute (return ((data, saved-state), result))
        final-state)
  by (auto elim!: set-dist-bindE)
have data'-eq: data' = data
  using ret by simp
show ?thesis
  using hit
  unfolding out-eq data'-eq
    checked-staged-security-with-data-state-alpha-list-hit-def
  by simp
qed
show staged-transcript-alpha-list-hit B
  (Some (data, attacker-state))
  using data-bad unfolding staged-transcript-alpha-list-hit-def by simp
qed

```

lemma *checked-staged-security-with-data-state-alpha-bad-set-bound-from-alpha-list:*

```

assumes envelope:  $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq B$ 
and list-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (checked-staged-security-with-data-state-alpha-list-hit B)
  adversary-initial-state  $\leq C$ 

```

shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (checked-staged-security-with-data-state-alpha-bad-set-hit bad-sets)
  adversary-initial-state  $\leq C$ 

```

proof –

```

have wp-event (checked-staged-security-experiment-with-data-state A)
  (checked-staged-security-with-data-state-alpha-bad-set-hit bad-sets)
  adversary-initial-state  $\leq$ 
wp-event (checked-staged-security-experiment-with-data-state A)
  (checked-staged-security-with-data-state-alpha-list-hit B)
  adversary-initial-state
by (rule wp-event-mono)
  (rule
    checked-staged-security-with-data-state-alpha-bad-set-hit-imp-alpha-list-hit
    [OF envelope])

```

also have ... $\leq C$

by (rule list-bound)

finally show ?thesis .

qed

lemma *checked-staged-security-with-data-state-composition-randomization-bad-imp-alpha-header-supported-bad-set-h*

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and support:

```

out ∈
  set-dist

```

```

      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
    and bad:
      staged-security-with-data-state-verifier-event
      composition-randomization-bad out
    shows
      checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
      composition-trace-bad-alpha-space out
  proof (cases out)
  case None
  then show ?thesis
  using bad
  unfolding staged-security-with-data-state-verifier-event-def
  by simp
next
case (Some result-pack)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases result-pack, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have support-some:
  Some (((data, attacker-state), result), final-state) ∈
  set-dist
  (execute (checked-staged-security-experiment-with-data-state A)
    adversary-initial-state)
  using support unfolding out-eq .
from checked-staged-security-experiment-with-data-state-outcomeE
[OF support-some]
have builder:
  Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)
    adversary-initial-state)
and verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast+
have staged-header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript

```



```

    [OF wf controlled builder])
from bad[unfolded out-eq
    staged-security-with-data-state-verifier-event-def]
have random-bad:
    composition-randomization-bad ?s (Some (result, final-state))
by simp
have alpha-hit:
    alpha-bad-set-hit ?s composition-trace-bad-alpha-space
    (Some (result, final-state))
by (rule composition-randomization-bad-imp-alpha-bad-set-hit
    [OF random-bad])
from alpha-hit obtain trace-table composition-table as query-idxs where
    bound:
        accepted-with-bound-tables ?s (Some (result, final-state))
        trace-table composition-table as query-idxs
and as-bad: as ∈ composition-trace-bad-alpha-space trace-table
unfolding alpha-bad-set-hit-def by blast
have shape:
    accepted-transcript-shape ?s (Some (result, final-state))
    as query-idxs
using bound
unfolding accepted-with-bound-tables-def accepted-with-tables-def
by simp
from shape obtain result' final-state' fr f-fri-roots f-final dg
    composition-fri-roots final rest where
    header:
        verifier-header-transcript ?s fr f-fri-roots f-final as dg
        composition-fri-roots final rest
by (elim accepted-transcript-shape-header-query-extraction)
have as-eq: as = staged-alphas data
using verifier-header-transcript-unique[OF staged-header header]
by simp
have bound-staged:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table (staged-alphas data) query-idxs
using bound as-eq by simp
have staged-bad:
    staged-alphas data ∈ composition-trace-bad-alpha-space trace-table
using as-bad as-eq by simp
have candidate:
    trace-table ∈
        alpha-header-supported-trace-table-candidates ?s
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
unfolding alpha-header-supported-trace-table-candidates-def
using verifier bound-staged staged-header by blast
have union-bad:
    staged-alphas data ∈

```

```

    alpha-header-supported-union-bad-sets ?s
    composition-trace-bad-alpha-space
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
  unfolding alpha-header-supported-union-bad-sets-def
  using candidate staged-bad
    composition-trace-bad-alpha-space-subset-alpha-space[of trace-table]
  by auto
show ?thesis
  using union-bad
  unfolding out-eq
    checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-def
  by simp
qed

lemma checked-staged-security-with-data-state-composition-randomization-bad-bound-from-alpha-header-supported-ba
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and alpha-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
      composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq C$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-randomization-bad)
    adversary-initial-state  $\leq C$ 
proof -
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-randomization-bad)
    adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    (checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
      composition-trace-bad-alpha-space)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (rule
      checked-staged-security-with-data-state-composition-randomization-bad-imp-alpha-header-supported-bad-set-hit-on-support
      [OF wf controlled])
  also have ...  $\leq C$ 
  by (rule alpha-bound)
  finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-composition-randomization-bad-imp-alpha-bad-set-hit-on-support:
  assumes wf: staged-budget-wellformed budgets

```

```

and controlled: staged-adversary-controlled budgets A
and support:
  out ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
and bad:
  staged-security-with-data-state-verifier-event
  composition-randomization-bad out
shows
  checked-staged-security-with-data-state-alpha-bad-set-hit
  composition-trace-bad-alpha-space out
proof (cases out)
  case None
  then show ?thesis
    using bad
    unfolding staged-security-with-data-state-verifier-event-def
    by simp
next
  case (Some result-pack)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases result-pack, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have support-some:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
    using support unfolding out-eq .
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF support-some]
  have builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
    by blast
  have staged-header:
    verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)

```

```

    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
from bad[unfolded out-eq
  staged-security-with-data-state-verifier-event-def]
have random-bad:
  composition-randomization-bad ?s (Some (result, final-state))
  by simp
have alpha-hit:
  alpha-bad-set-hit ?s composition-trace-bad-alpha-space
    (Some (result, final-state))
  by (rule composition-randomization-bad-imp-alpha-bad-set-hit
    [OF random-bad])
from alpha-hit obtain trace-table composition-table as query-idxs where
  bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and as-bad: as ∈ composition-trace-bad-alpha-space trace-table
  unfolding alpha-bad-set-hit-def by blast
have shape:
  accepted-transcript-shape ?s (Some (result, final-state))
  as query-idxs
  using bound
  unfolding accepted-with-bound-tables-def accepted-with-tables-def
  by simp
from shape obtain result' final-state' fr f-fri-roots f-final dg
  composition-fri-roots final rest where
  header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  by (elim accepted-transcript-shape-header-query-extraction)
have as-eq: as = staged-alphas data
  using verifier-header-transcript-unique[OF staged-header header]
  by simp
have staged-bad:
  staged-alphas data ∈ composition-trace-bad-alpha-space trace-table
  using as-bad as-eq by simp
show ?thesis
  using staged-bad
  unfolding out-eq
  checked-staged-security-with-data-state-alpha-bad-set-hit-def
  by auto
qed

lemma checked-staged-security-with-data-state-composition-randomization-bad-bound-from-alpha-bad-set:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and alpha-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (checked-staged-security-with-data-state-alpha-bad-set-hit
     composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq C$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-randomization-bad)
    adversary-initial-state  $\leq C$ 
  proof -
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-randomization-bad)
    adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    (checked-staged-security-with-data-state-alpha-bad-set-hit
     composition-trace-bad-alpha-space)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (rule
     checked-staged-security-with-data-state-composition-randomization-bad-imp-alpha-bad-set-hit-on-support
     [OF wf controlled])
  also have ...  $\leq C$ 
    by (rule alpha-bound)
  finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-composition-degree-bad-bound:
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event composition-degree-bad)
    adversary-initial-state  $\leq 0$ 
  proof -
  have event-eq:
    staged-security-with-data-state-verifier-event composition-degree-bad =
      ( $\lambda$  out :: (((f staged-proof-data  $\times$  f protocol-channel)  $\times$ 
        unit list)  $\times$  f protocol-channel) option. False)
  by (rule ext)
    (auto simp: staged-security-with-data-state-verifier-event-def
      composition-degree-bad-false[OF spec-degree-wellformed-from-spec-query-margin]
      split: option.splits prod.splits)
  show ?thesis
    unfolding event-eq wp-event-def wp-def dist-expect-def by simp
qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-alpha-bad-set:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and alpha-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (checked-staged-security-with-data-state-alpha-bad-set-hit
     composition-trace-bad-alpha-space)
    adversary-initial-state ≤ composition-error-bound
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event composition-bad)
      adversary-initial-state ≤ composition-error-bound
  proof -
    let ?M = checked-staged-security-experiment-with-data-state A
    let ?degree =
      staged-security-with-data-state-verifier-event composition-degree-bad
    let ?random =
      staged-security-with-data-state-verifier-event
        composition-randomization-bad
    have event-eq:
      staged-security-with-data-state-verifier-event composition-bad =
        (λout. ?degree out ∨ ?random out)
    by (rule ext)
      (auto simp: staged-security-with-data-state-verifier-event-def
        composition-bad-def split: option.splits prod.splits)
    have degree-bound:
      wp-event ?M ?degree adversary-initial-state ≤ 0
    by (rule
      checked-staged-security-with-data-state-composition-degree-bad-bound)
    have random-bound:
      wp-event ?M ?random adversary-initial-state ≤
        composition-error-bound
    by (rule
      checked-staged-security-with-data-state-composition-randomization-bad-bound-from-alpha-bad-set
        [OF wf controlled alpha-bound])
    have wp-event ?M
      (staged-security-with-data-state-verifier-event composition-bad)
      adversary-initial-state ≤
      wp-event ?M ?degree adversary-initial-state +
      wp-event ?M ?random adversary-initial-state
    unfolding event-eq by (rule wp-event-union-bound)
    also have ... ≤ 0 + composition-error-bound
    by (intro add-mono degree-bound random-bound)
    finally show ?thesis by simp
  qed

  lemma checked-staged-security-with-data-state-composition-bad-bound-from-alpha-header-supported-bad-set:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and alpha-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
        composition-trace-bad-alpha-space)
      adversary-initial-state ≤ composition-error-bound

```

shows
 $wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A)$
 $(staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event } composition\text{-bad})$
 $adversary\text{-initial}\text{-state} \leq composition\text{-error}\text{-bound}$

proof –
let $?M = checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A$
let $?degree =$
 $staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event } composition\text{-degree}\text{-bad}$
let $?random =$
 $staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event}$
 $composition\text{-randomization}\text{-bad}$
have $event\text{-eq}$:
 $staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event } composition\text{-bad} =$
 $(\lambda out. ?degree \text{ out} \vee ?random \text{ out})$
by (rule ext)
 $(auto \text{ simp: staged}\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event}\text{-def}$
 $composition\text{-bad}\text{-def } split: option.splits \text{ prod.splits})$
have $degree\text{-bound}$:
 $wp\text{-event } ?M ?degree \text{ adversary}\text{-initial}\text{-state} \leq 0$
by (rule
 $checked\text{-staged}\text{-security}\text{-with}\text{-data}\text{-state}\text{-composition}\text{-degree}\text{-bad}\text{-bound})$
have $random\text{-bound}$:
 $wp\text{-event } ?M ?random \text{ adversary}\text{-initial}\text{-state} \leq$
 $composition\text{-error}\text{-bound}$
by (rule
 $checked\text{-staged}\text{-security}\text{-with}\text{-data}\text{-state}\text{-composition}\text{-randomization}\text{-bad}\text{-bound}\text{-from}\text{-alpha}\text{-header}\text{-supported}\text{-bad}\text{-s}$
 $[OF \text{ wf controlled } alpha\text{-bound}])$
have $wp\text{-event } ?M$
 $(staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event } composition\text{-bad})$
 $adversary\text{-initial}\text{-state} \leq$
 $wp\text{-event } ?M ?degree \text{ adversary}\text{-initial}\text{-state} +$
 $wp\text{-event } ?M ?random \text{ adversary}\text{-initial}\text{-state}$
unfolding $event\text{-eq}$ **by** (rule $wp\text{-event}\text{-union}\text{-bound}$)
also have $\dots \leq 0 + composition\text{-error}\text{-bound}$
by (intro $add\text{-mono } degree\text{-bound } random\text{-bound}$)
finally show $?thesis$ **by** $simp$
qed

lemma $checked\text{-staged}\text{-security}\text{-with}\text{-data}\text{-state}\text{-composition}\text{-bad}\text{-bound}\text{-from}\text{-alpha}\text{-header}\text{-supported}\text{-bad}\text{-set}\text{-general}$:
assumes wf : $staged\text{-budget}\text{-wellformed } budgets$
and $controlled$: $staged\text{-adversary}\text{-controlled } budgets \text{ } A$
and $alpha\text{-bound}$:
 $wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A)$
 $(checked\text{-staged}\text{-security}\text{-with}\text{-data}\text{-state}\text{-alpha}\text{-header}\text{-supported}\text{-bad}\text{-set}\text{-hit}$
 $composition\text{-trace}\text{-bad}\text{-alpha}\text{-space})$
 $adversary\text{-initial}\text{-state} \leq C$

shows

$wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A)$
 $(staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event } composition\text{-bad})$

```

    adversary-initial-state  $\leq C$ 
proof –
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?degree =
    staged-security-with-data-state-verifier-event composition-degree-bad
  let ?random =
    staged-security-with-data-state-verifier-event
    composition-randomization-bad
  have event-eq:
    staged-security-with-data-state-verifier-event composition-bad =
      ( $\lambda$ out. ?degree out  $\vee$  ?random out)
  by (rule ext)
    (auto simp: staged-security-with-data-state-verifier-event-def
      composition-bad-def split: option.splits prod.splits)
  have degree-bound:
    wp-event ?M ?degree adversary-initial-state  $\leq 0$ 
  by (rule
    checked-staged-security-with-data-state-composition-degree-bad-bound)
  have random-bound:
    wp-event ?M ?random adversary-initial-state  $\leq C$ 
  by (rule
    checked-staged-security-with-data-state-composition-randomization-bad-bound-from-alpha-header-supported-bad-s
    [OF wf controlled alpha-bound])
  have wp-event ?M
    (staged-security-with-data-state-verifier-event composition-bad)
    adversary-initial-state  $\leq$ 
    wp-event ?M ?degree adversary-initial-state +
    wp-event ?M ?random adversary-initial-state
  unfolding event-eq by (rule wp-event-union-bound)
  also have ...  $\leq 0 + C$ 
  by (intro add-mono degree-bound random-bound)
  finally show ?thesis by simp
qed

```

lemma staged-composition-alpha-header-supported-union-fraction-bound-if-no-pairwise-merkle-bad:

```

assumes no-bad:
   $\neg$  alpha-supported-pairwise-merkle-bad
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))

```

shows

```

  nnreal
  (card
    (alpha-header-supported-union-bad-sets
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      composition-trace-bad-alpha-space
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)))) /

```


$nnreal\ (card\ \alpha\text{-}space) \leq composition\text{-}error\text{-}bound$
by (rule
 $composition\text{-}trace\text{-}\alpha\text{-}header\text{-}supported\text{-}union\text{-}fraction\text{-}bound\text{-}if\text{-}unique\text{-}candidate$)
(rule
 $\alpha\text{-}header\text{-}supported\text{-}candidate\text{-}unique\text{-}if\text{-}no\text{-}global\text{-}pairwise\text{-}merkle\text{-}bad$
 $[OF\ no\text{-}bad]$)

lemma *accepted-staged-security-experiment-fri-challengesE*:

assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *outcome*:
 $out \in set\text{-}dist$
 $(execute\ (staged\text{-}security\text{-}experiment\ A)\ adversary\text{-}initial\text{-}state)$
and *accepted*: $\neg Option.is\ none\ out$
obtains *result final-state data attacker-state* **where**
 $out = Some\ (result,\ final\text{-}state)$
 $Some\ (data,\ attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (staged\text{-}transcript\text{-}program\ A)$
 $\quad adversary\text{-}initial\text{-}state)$
 $Some\ (result,\ final\text{-}state) \in$
 $set\text{-}dist$
 $(execute\ verify\text{-}monad$
 $\quad (verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $\quad\quad (staged\text{-}proof\text{-}transcript\ data)))$
accepted-fri-challenges
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $\quad (staged\text{-}proof\text{-}transcript\ data))$
 out
 $(staged\text{-}trace\text{-}fri\text{-}challenges\ data)$
 $(staged\text{-}degree\ data)$
 $(staged\text{-}composition\text{-}fri\text{-}challenges\ data)$

proof –

from *accepted-staged-security-experiment-headerE*
 $[OF\ wf\ controlled\ outcome\ accepted]$
obtain *result final-state data attacker-state* **where**
 $out\text{-}eq: out = Some\ (result,\ final\text{-}state)$
and *builder*:
 $Some\ (data,\ attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (staged\text{-}transcript\text{-}program\ A)$
 $\quad adversary\text{-}initial\text{-}state)$
and *verifier*:
 $Some\ (result,\ final\text{-}state) \in$
 $set\text{-}dist$
 $(execute\ verify\text{-}monad$
 $\quad (verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $\quad\quad (staged\text{-}proof\text{-}transcript\ data)))$
and *staged-header*:
 $verifier\text{-}header\text{-}transcript$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state)$

```

    (staged-proof-transcript data))
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  (List.concat (staged-query-chunks data))
  by blast
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from verify-monad-header-random-oracle-replay[OF verifier]
obtain fr f-fl f-final as dg fl final query-state where
  header:
    verifier-header-transcript ?s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
and query-state-eq:
  PState query-state =
    verifier-header-state ?s fr (map snd f-fl) f-final as dg
    (map snd fl) final
and query-out:
  Some (result, final-state) ∈
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
      query-state)
and query-count-header: PQueryCounter query-state = PQueryCounter ?s
and ext-s-query: ?s ≤ query-state
and trace-lookup:
  ∧i. i < length f-fl ⇒
    fmllookup (HashMap query-state)
      (TraceFriChallenge (PTraceFriCounter ?s + i)
        (foldl concat (concat (PState ?s) fr)
          (take (Suc i) (map snd f-fl)))) =
      Some (fst (f-fl ! i))
and alpha-lookup:
  ∧i. i < length as ⇒
    fmllookup (HashMap query-state)
      (AlphaChallenge (PAlphaCounter ?s + i) (foldl concat
        (concat (foldl concat (concat (PState ?s) fr) (map snd f-fl))
          f-final)
        (take i as)))) =
      Some (as ! i)
and comp-lookup:
  ∧i. i < length fl ⇒
    fmllookup (HashMap query-state)

```

```

    (CompositionFriChallenge (PCompositionFriCounter ?s + i)
      (foldl concat
        (concat
          (foldl concat
            (concat
              (foldl concat (concat (PState ?s) fr) (map snd f-fl))
              f-final)
            as)
          dg)
        (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i))
  and degree-bound: to-nat dg ≤ maxDegree
  by blast
  have header-unique:
    fr = staged-trace-root data ∧
    map snd f-fl = staged-trace-fri-roots data ∧
    f-final = staged-trace-final data ∧
    as = staged-alphas data ∧
    dg = staged-degree data ∧
    map snd fl = staged-composition-fri-roots data ∧
    final = staged-composition-final data ∧
    PTranscript query-state = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF staged-header header] by simp
  have shape:
    length (staged-trace-fri-roots data) = ceil-log clength ∧
    length (staged-trace-fri-challenges data) = ceil-log clength ∧
    length (staged-alphas data) = length spec ∧
    length (staged-composition-fri-roots data) =
      ceil-log (to-nat (staged-degree data) + 1) ∧
    length (staged-composition-fri-challenges data) =
      ceil-log (to-nat (staged-degree data) + 1) ∧
    ceil-log (to-nat (staged-degree data) + 1) ≤
      ceil-log (maxDegree + 1) ∧
    length (staged-query-chunks data) = rounds
  by (rule staged-transcript-program-outcome-shape
    [OF wf controlled builder])
  have trace-bs-eq:
    map fst f-fl = staged-trace-fri-challenges data
  proof (rule nth-equalityI)
    show length (map fst f-fl) =
      length (staged-trace-fri-challenges data)
    using header-unique shape by (metis length-map)
  next
    fix i
    assume i-bound: i < length (map fst f-fl)
    then have i-f-fl: i < length f-fl
      by simp
    have i-round: i < ceil-log clength
      using i-f-fl header-unique shape by (metis length-map)

```

```

have staged-lookup-attacker:
  fmlookup (HashMap attacker-state)
    (TraceFriChallenge i
      (foldl concat (staged-trace-fri-start-hash data)
        (take (Suc i) (staged-trace-fri-roots data)))) =
    Some (staged-trace-fri-challenges data ! i)
  by (rule staged-transcript-program-outcome-trace-challenge-lookup
    [OF wf controlled builder i-round])
have staged-lookup-s:
  fmlookup (HashMap ?s)
    (TraceFriChallenge i
      (foldl concat (staged-trace-fri-start-hash data)
        (take (Suc i) (staged-trace-fri-roots data)))) =
    Some (staged-trace-fri-challenges data ! i)
  using staged-lookup-attacker by simp
have staged-lookup-query:
  fmlookup (HashMap query-state)
    (TraceFriChallenge i
      (foldl concat (staged-trace-fri-start-hash data)
        (take (Suc i) (staged-trace-fri-roots data)))) =
    Some (staged-trace-fri-challenges data ! i)
  by (rule hash-extension-lookup[OF staged-lookup-s ext-s-query])
have verifier-lookup-query:
  fmlookup (HashMap query-state)
    (TraceFriChallenge i
      (foldl concat (staged-trace-fri-start-hash data)
        (take (Suc i) (staged-trace-fri-roots data)))) =
    Some (fst (f-fl ! i))
  using trace-lookup[OF i-f-fl] header-unique
  unfolding staged-trace-fri-start-hash-def
  by simp
show map fst f-fl ! i = staged-trace-fri-challenges data ! i
  using verifier-lookup-query staged-lookup-query i-f-fl
  by simp
qed
have comp-bs-eq:
  map fst fl = staged-composition-fri-challenges data
proof (rule nth-equalityI)
  show length (map fst fl) =
    length (staged-composition-fri-challenges data)
  using header-unique shape by (metis length-map)
next
  fix i
  assume i-bound: i < length (map fst fl)
  then have i-fl: i < length fl
  by simp
  have i-round:
    i < ceil-log (to-nat (staged-degree data) + 1)
  using i-fl header-unique shape by (metis length-map)

```

```

have staged-lookup-attacker:
  fmlookup (HashMap attacker-state)
    (CompositionFriChallenge i
      (foldl concat (staged-composition-fri-start-hash data)
        (take (Suc i) (staged-composition-fri-roots data)))) =
    Some (staged-composition-fri-challenges data ! i)
  by (rule staged-transcript-program-outcome-composition-challenge-lookup
    [OF wf controlled builder i-round])
have staged-lookup-s:
  fmlookup (HashMap ?s)
    (CompositionFriChallenge i
      (foldl concat (staged-composition-fri-start-hash data)
        (take (Suc i) (staged-composition-fri-roots data)))) =
    Some (staged-composition-fri-challenges data ! i)
  using staged-lookup-attacker by simp
have staged-lookup-query:
  fmlookup (HashMap query-state)
    (CompositionFriChallenge i
      (foldl concat (staged-composition-fri-start-hash data)
        (take (Suc i) (staged-composition-fri-roots data)))) =
    Some (staged-composition-fri-challenges data ! i)
  by (rule hash-extension-lookup[OF staged-lookup-s ext-s-query])
have verifier-lookup-query:
  fmlookup (HashMap query-state)
    (CompositionFriChallenge i
      (foldl concat (staged-composition-fri-start-hash data)
        (take (Suc i) (staged-composition-fri-roots data)))) =
    Some (fst (fl ! i))
  using comp-lookup[OF i-fl] header-unique
  unfolding staged-composition-fri-start-hash-def
    staged-trace-fri-start-hash-def
  by simp
show map fst fl ! i =
  staged-composition-fri-challenges data ! i
  using verifier-lookup-query staged-lookup-query i-fl
  by simp
qed
have trace-lookup-all:
   $\forall i < \text{length } f\text{-fl}.$ 
  fmlookup (HashMap query-state)
    (TraceFriChallenge (PTraceFriCounter ?s + i)
      (foldl concat (concat (PState ?s) fr)
        (take (Suc i) (map snd f-fl)))) =
    Some (fst (f-fl ! i))
  using trace-lookup by blast
have comp-lookup-all:
   $\forall i < \text{length } fl.$ 
  fmlookup (HashMap query-state)
    (CompositionFriChallenge (PCompositionFriCounter ?s + i)

```

```

    (foldl concat
      (concat
        (foldl concat
          (concat
            (foldl concat (concat (PState ?s) fr) (map snd f-fl))
            f-final)
          as)
        dg)
      (take (Suc i) (map snd fl)))) =
  Some (fst (fl ! i))
using comp-lookup by blast
have challenges:
  accepted-fri-challenges ?s (Some (result, final-state))
  (staged-trace-fri-challenges data)
  (staged-degree data)
  (staged-composition-fri-challenges data)
unfolding accepted-fri-challenges-def
by (intro exI[of - result] exI[of - final-state] exI[of - fr]
  exI[of - f-fl] exI[of - f-final] exI[of - as] exI[of - fl]
  exI[of - final] exI[of - query-state])
  (use header query-state-eq query-out query-count-header
    header-unique trace-bs-eq comp-bs-eq trace-lookup-all comp-lookup-all
    in simp)
have challenges-out:
  accepted-fri-challenges ?s out
  (staged-trace-fri-challenges data)
  (staged-degree data)
  (staged-composition-fri-challenges data)
using challenges out-eq by simp
show ?thesis
  by (rule that[OF out-eq builder verifier challenges-out])
qed

lemma staged-budget-trace-length:
  assumes staged-budget-wellformed budgets
  shows length (trace-fri-budgets budgets) = ceil-log clength
  using assms unfolding staged-budget-wellformed-def by simp

lemma staged-budget-composition-length:
  assumes staged-budget-wellformed budgets
  shows
    length (composition-fri-budgets budgets) =
      ceil-log (maxDegree + 1)
  using assms unfolding staged-budget-wellformed-def by simp

lemma staged-budget-query-length:
  assumes staged-budget-wellformed budgets
  shows length (query-opening-budgets budgets) = rounds
  using assms unfolding staged-budget-wellformed-def by simp

```

lemma *staged-trace-fri-search-queries-le-total*:
assumes *wf*: *staged-budget-wellformed budgets*
and *i-bound*: $i < \text{ceil-log } \text{clength}$
shows
 $\text{staged-trace-fri-search-queries budgets } i \leq$
 $\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget}$
proof –
have *take-le*:
 $\text{sum-list (take (Suc } i) (\text{trace-fri-budgets budgets}))} \leq$
 $\text{sum-list (trace-fri-budgets budgets)}$
by (rule *sum-list-take-le*)
have *i-le*: $i \leq \text{ceil-log } \text{clength}$
using *i-bound* **by** *simp*
show ?thesis
unfolding *staged-trace-fri-search-queries-def*
 $\text{staged-attacker-query-budget-def staged-challenge-query-budget-def}$
using *take-le i-le* **by** *simp*
qed

lemma *staged-alpha-search-queries-le-total*:
assumes $i \leq \text{length spec}$
shows
 $\text{staged-alpha-search-queries budgets } i \leq$
 $\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget}$
unfolding *staged-alpha-search-queries-def*
 $\text{staged-attacker-query-budget-def staged-challenge-query-budget-def}$
using *assms* **by** *simp*

lemma *staged-composition-fri-search-queries-le-total*:
assumes *wf*: *staged-budget-wellformed budgets*
and *i-bound*: $i < \text{ceil-log } (\text{maxDegree} + 1)$
shows
 $\text{staged-composition-fri-search-queries budgets } i \leq$
 $\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget}$
proof –
have *take-le*:
 $\text{sum-list (take (Suc } i) (\text{composition-fri-budgets budgets}))} \leq$
 $\text{sum-list (composition-fri-budgets budgets)}$
by (rule *sum-list-take-le*)
have *i-le*: $i \leq \text{ceil-log } (\text{maxDegree} + 1)$
using *i-bound* **by** *simp*
show ?thesis
unfolding *staged-composition-fri-search-queries-def*
 $\text{staged-attacker-query-budget-def staged-challenge-query-budget-def}$
using *take-le i-le* **by** *simp*
qed

lemma *staged-query-search-queries-le-total*:

```

assumes wf: staged-budget-wellformed budgets
and i-bound:  $i < \text{rounds}$ 
shows
  staged-query-search-queries budgets  $i \leq$ 
    staged-attacker-query-budget budgets + staged-challenge-query-budget
proof –
  have take-le:
    sum-list (take i (query-opening-budgets budgets))  $\leq$ 
      sum-list (query-opening-budgets budgets)
  by (rule sum-list-take-le)
  have i-le:  $i \leq \text{rounds}$ 
  using i-bound by simp
  show ?thesis
  unfolding staged-query-search-queries-def
    staged-attacker-query-budget-def staged-challenge-query-budget-def
  using take-le i-le by simp
qed

```

```

lemma staged-phase-relation-error-mono:
assumes  $q \leq r$ 
shows
  staged-phase-relation-error fiber-bound  $q \leq$ 
    staged-phase-relation-error fiber-bound  $r$ 
unfolding staged-phase-relation-error-def
by (rule hash-relation-budget-value-mono[OF assms])

```

```

lemma staged-phase-target-error-mono:
assumes  $q \leq r$ 
shows staged-phase-target-error  $B \ q \leq$  staged-phase-target-error  $B \ r$ 
unfolding staged-phase-target-error-def
by (rule hash-target-budget-value-mono[OF assms])

```

```

lemma staged-phase-target-error-card-mono:
assumes card  $B \leq b$ 
shows staged-phase-target-error  $B \ q \leq$ 
  staged-phase-relation-error  $b \ q$ 
unfolding staged-phase-target-error-def staged-phase-relation-error-def
  hash-target-budget-value-def hash-relation-budget-value-def
apply (subst nn2real-le-iff[symmetric])
using assms by (simp add: divide-right-mono mult-left-mono)

```

```

lemma staged-dynamic-query-set-subset:
  query-header-supported-union-good-sets
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  query-sampling-success-space
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)

```


(staged-*alphas* data)
 (staged-*degree* data)
 (staged-*composition-fri-roots* data)
 (staged-*composition-final* data) $\subseteq$ *query-sample-space*
by (rule *query-header-supported-union-good-sets-subset*)

lemma *staged-dynamic-query-set-fraction-bound-if-no-pairwise-merkle-bad*:

assumes *no-bad*:

$\neg$ *query-supported-pairwise-merkle-bad*
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))

shows

nnreal
 (card
 (query-header-supported-union-good-sets
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
query-sampling-success-space
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-*alphas* data)
 (staged-*degree* data)
 (staged-*composition-fri-roots* data)
 (staged-*composition-final* data))) /
nnreal (card *query-sample-space*) $\leq$ *query-error-bound*

by (rule

query-header-supported-union-good-sets-fraction-bound-if-no-global-pairwise-merkle-bad
 [OF *no-bad query-sampling-success-space-subset*
query-sampling-success-space-fraction-bound-query-sample-space])

lemma *staged-dynamic-query-set-target-error-bound-if-no-pairwise-merkle-bad*:

assumes *raw-bound*: *query-index-raw-preimage-bound*

and *no-bad*:

$\neg$ *query-supported-pairwise-merkle-bad*
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))

shows

staged-phase-target-error
 (query-index-raw-preimage
 (query-header-supported-union-good-sets
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
query-sampling-success-space
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-*alphas* data)
 (staged-*degree* data))

$(\text{staged-composition-fri-roots } \text{data})$
 $(\text{staged-composition-final } \text{data})) \text{ } q \leq$
 $\text{nnreal } q * \text{query-error-bound}$
by (*rule staged-phase-query-index-target-error-query-bound*
 $[\text{OF raw-bound staged-dynamic-query-set-subset}$
 $\text{staged-dynamic-query-set-fraction-bound-if-no-pairwise-merkle-bad}$
 $[\text{OF no-bad}]]$)

lemma *staged-alpha-prefix-receive-bad-vector-position-bound:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i < \text{length spec}$*
and *absent:*
 $\text{alpha-challenge-values-absent}$
 $(\text{fri-vector-position-values } B \ i) \ s$

shows

wp-event
 $(\text{staged-alpha-challenge-prefix-program } A \ i \gg=$
 $(\lambda-. \text{receive-alpha-challenge}))$
 $(\lambda \text{out. case out of None} \Rightarrow \text{False}$
 $\mid \text{Some } (a, -) \Rightarrow a \in \text{fri-vector-position-values } B \ i) \ s \leq$
 $\text{staged-phase-target-error } (\text{fri-vector-position-values } B \ i)$
 $(\text{Suc } (\text{staged-alpha-search-queries budgets } i)))$
by (*rule staged-alpha-prefix-receive-bad-value-bound*
 $[\text{OF wf controlled - absent}]$)
 $(\text{use } i\text{-bound in simp})$

lemma *staged-alpha-prefix-receive-bad-vector-position-relation-bound:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *subset: $B \subseteq \text{fri-challenge-space } (\text{length spec})$*
and *i-bound: $i < \text{length spec}$*
and *absent:*
 $\text{alpha-challenge-values-absent}$
 $(\text{fri-vector-position-values } B \ i) \ s$

shows

wp-event
 $(\text{staged-alpha-challenge-prefix-program } A \ i \gg=$
 $(\lambda-. \text{receive-alpha-challenge}))$
 $(\lambda \text{out. case out of None} \Rightarrow \text{False}$
 $\mid \text{Some } (a, -) \Rightarrow a \in \text{fri-vector-position-values } B \ i) \ s \leq$
 $\text{staged-phase-relation-error } (\text{card } B)$
 $(\text{Suc } (\text{staged-alpha-search-queries budgets } i)))$

proof –

have *target:*
 wp-event
 $(\text{staged-alpha-challenge-prefix-program } A \ i \gg=$
 $(\lambda-. \text{receive-alpha-challenge}))$
 $(\lambda \text{out. case out of None} \Rightarrow \text{False})$

```

    | Some (a, -) ⇒ a ∈ fri-vector-position-values B i) s ≤
    staged-phase-target-error (fri-vector-position-values B i)
    (Suc (staged-alpha-search-queries budgets i))
  by (rule staged-alpha-prefix-receive-bad-vector-position-bound
      [OF wf controlled i-bound absent])
  also have ... ≤
    staged-phase-relation-error (card B)
    (Suc (staged-alpha-search-queries budgets i))
  by (rule staged-phase-target-error-card-mono)
    (rule fri-vector-position-values-card-le[OF subset i-bound])
  finally show ?thesis .
qed

lemma staged-alpha-prefix-receive-bad-vector-position-relation-bound-total:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and subset: B ⊆ fri-challenge-space (length spec)
and i-bound: i < length spec
and absent:
  alpha-challenge-values-absent
  (fri-vector-position-values B i) s
shows
  wp-event
  (staged-alpha-challenge-prefix-program A i ≫=
    (λ-. receive-alpha-challenge))
  (λout. case out of None ⇒ False
    | Some (a, -) ⇒ a ∈ fri-vector-position-values B i) s ≤
  staged-phase-relation-error (card B)
  (Suc
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget))

proof –
have local:
  wp-event
  (staged-alpha-challenge-prefix-program A i ≫=
    (λ-. receive-alpha-challenge))
  (λout. case out of None ⇒ False
    | Some (a, -) ⇒ a ∈ fri-vector-position-values B i) s ≤
  staged-phase-relation-error (card B)
  (Suc (staged-alpha-search-queries budgets i))
by (rule staged-alpha-prefix-receive-bad-vector-position-relation-bound
      [OF wf controlled subset i-bound absent])
also have
  staged-phase-relation-error (card B)
  (Suc (staged-alpha-search-queries budgets i)) ≤
  staged-phase-relation-error (card B)
  (Suc
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget))

```

```

    by (rule staged-phase-relation-error-mono)
      (simp add: staged-alpha-search-queries-le-total[OF less-imp-le[OF i-bound]])
  finally show ?thesis .
qed

lemma staged-alpha-prefix-recv-keep-prefix-bad-vector-position-relation-bound-total:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and subset:  $B \subseteq \text{fri-challenge-space}$  (length spec)
    and i-bound:  $i < \text{length spec}$ 
    and absent:
      alpha-challenge-values-absent
      (fri-vector-position-values B i) s
  shows
    wp-event
      (staged-alpha-challenge-prefix-program A i  $\gg=$ 
        ( $\lambda \text{prefix. receive-alpha-challenge } \gg=$ 
          ( $\lambda a. \text{return (prefix, a)}$ ))))
      ( $\lambda \text{out. case out of None} \Rightarrow \text{False}$ 
        |  $\text{Some } ((-, a), -) \Rightarrow a \in \text{fri-vector-position-values B i}$ ) s  $\leq$ 
        staged-phase-relation-error (card B)
        (Suc
          (staged-attacker-query-budget budgets +
            staged-challenge-query-budget)))
  proof -
    let ?P =
       $\lambda \text{out. case out of None} \Rightarrow \text{False}$ 
      |  $\text{Some } (a, -) \Rightarrow a \in \text{fri-vector-position-values B i}$ 
    let ?Q =
       $\lambda \text{out. case out of None} \Rightarrow \text{False}$ 
      |  $\text{Some } ((-, a), -) \Rightarrow a \in \text{fri-vector-position-values B i}$ 
    have cont-eq:
       $\bigwedge \text{prefix } t. \text{wp-event}$ 
      (receive-alpha-challenge  $\gg=$  ( $\lambda a. \text{return (prefix, a)}$ )) ?Q t =
      wp-event receive-alpha-challenge ?P t
    proof -
      fix prefix t
      show wp-event
        (receive-alpha-challenge  $\gg=$  ( $\lambda a. \text{return (prefix, a)}$ )) ?Q t =
        wp-event receive-alpha-challenge ?P t
      unfolding wp-event-def
      apply (subst wp-bind)
      apply (rule arg-cong[where
        f= $\lambda Q. \text{wp receive-alpha-challenge } Q \ t$ ])
      apply (rule ext)
      apply (simp add: wp-return split: option.splits prod.splits)
      done
    qed
  have eq:

```

```

wp-event
  (staged-alpha-challenge-prefix-program A i >>=
    (λprefix. receive-alpha-challenge >>=
      (λa. return (prefix, a)))) ?Q s =
wp-event
  (staged-alpha-challenge-prefix-program A i >>=
    (λ-. receive-alpha-challenge)) ?P s
by (rule wp-event-bind-cong-cont) (simp-all add: cont-eq)
show ?thesis
  unfolding eq
  by (rule staged-alpha-prefix-receive-bad-vector-position-relation-bound-total
    [OF wf controlled subset i-bound absent])
qed

```

lemma *staged-alpha-prefix-bad-trace-vector-bound:*

```

assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and subset: B ⊆ fri-challenge-space (ceil-log clength)
  and nonempty: 0 < ceil-log clength
  and absent:
    ∧ i. i < ceil-log clength ⇒
      trace-fri-challenge-values-absent
      (fri-vector-position-values B i) s

```

shows

```

wp-event (staged-alpha-prefix-program A)
  (λout. case out of None ⇒ False
    | Some ((-, -, trace-bs, -), -) ⇒ trace-bs ∈ B) s ≤
  nnreal (ceil-log clength) *
  staged-phase-relation-error (card B)
  (staged-alpha-search-queries budgets 0)

```

proof –

```

let ?coord =
  λi out. case out of None ⇒ False
    | Some ((-, -, trace-bs, -), -) ⇒
      trace-bs ! i ∈ fri-vector-position-values B i

```

have *cover:*

```

wp-event (staged-alpha-prefix-program A)
  (λout. case out of None ⇒ False
    | Some ((-, -, trace-bs, -), -) ⇒ trace-bs ∈ B) s ≤
  (∑ i < ceil-log clength.
    wp-event (staged-alpha-prefix-program A) (?coord i) s)
by (rule staged-alpha-prefix-bad-trace-vector-coordinate-cover-bound
  [OF subset nonempty])

```

also have ... ≤

```

  (∑ i < ceil-log clength.
    staged-phase-relation-error (card B)
    (staged-alpha-search-queries budgets 0))

```

proof (rule sum-mono)

fix *i*

```

assume i-in:  $i \in \{..<\text{ceil-log } \text{clength}\}$ 
then have i-bound:  $i < \text{ceil-log } \text{clength}$ 
  by simp
have target:
  wp-event (staged-alpha-prefix-program A) (?coord i)  $s \leq$ 
    staged-phase-target-error (fri-vector-position-values B i)
      (staged-alpha-search-queries budgets 0)
  by (rule staged-alpha-prefix-trace-coordinate-bound
    [OF wf controlled i-bound absent[OF i-bound]])
also have ...  $\leq$ 
  staged-phase-relation-error (card B)
    (staged-alpha-search-queries budgets 0)
  by (rule staged-phase-target-error-card-mono)
    (rule fri-vector-position-values-card-le[OF subset i-bound])
finally show
  wp-event (staged-alpha-prefix-program A) (?coord i)  $s \leq$ 
    staged-phase-relation-error (card B)
      (staged-alpha-search-queries budgets 0) .
qed
also have ... =
  nnreal (ceil-log clength) *
    staged-phase-relation-error (card B)
      (staged-alpha-search-queries budgets 0)
  by simp
finally show ?thesis .
qed

lemma staged-transcript-program-trace-fri-vector-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and subset:  $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$ 
and nonempty:  $0 < \text{ceil-log } \text{clength}$ 
and absent:
   $\bigwedge i. i < \text{ceil-log } \text{clength} \implies$ 
    trace-fri-challenge-values-absent
      (fri-vector-position-values B i) adversary-initial-state
shows
  wp-event (staged-transcript-program A)
    (staged-transcript-trace-fri-vector-hit B)
    adversary-initial-state  $\leq$ 
      nnreal (ceil-log clength) *
        staged-phase-relation-error (card B)
          (staged-alpha-search-queries budgets 0)
proof –
let ?Head =
   $\lambda \text{out}. \text{case out of}$ 
    None  $\Rightarrow$  False
    | Some ((-, -, trace-bs, -), -)  $\Rightarrow$  trace-bs  $\in B$ 
have prefix-bound:

```

```

wp-event (staged-alpha-prefix-program A) ?Head
  adversary-initial-state ≤
  nnreal (ceil-log clength) *
  staged-phase-relation-error (card B)
  (staged-alpha-search-queries budgets 0)
by (rule staged-alpha-prefix-bad-trace-vector-bound
  [OF wf controlled subset nonempty absent])
show ?thesis
  unfolding staged-transcript-program-alpha-prefix-decomp
proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
  show staged-transcript-trace-fri-vector-hit B None ⇒
    ?Head None
    unfolding staged-transcript-trace-fri-vector-hit-def by simp
next
fix x t out
assume cont:
  out ∈
  set-dist
  (execute (staged-after-alpha-prefix-program A x) t)
  and hit: staged-transcript-trace-fri-vector-hit B out
obtain fr trace-roots trace-bs trace-final where x-eq:
  x = (fr, trace-roots, trace-bs, trace-final)
by (cases x) auto
have trace-bs-in: trace-bs ∈ B
by (rule staged-after-alpha-prefix-program-trace-fri-vector-hitD
  [OF cont[unfolded x-eq] hit])
show ?Head (Some (x, t))
  unfolding x-eq
  using trace-bs-in by simp
qed
qed

lemma staged-security-with-data-trace-fri-vector-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and subset: B ⊆ fri-challenge-space (ceil-log clength)
  and nonempty: 0 < ceil-log clength
  and absent:
    ∧ i. i < ceil-log clength ⇒
      trace-fri-challenge-values-absent
      (fri-vector-position-values B i) adversary-initial-state
  shows
    wp-event (staged-security-experiment-with-data A)
      (staged-security-with-data-trace-fri-vector-hit B)
      adversary-initial-state ≤
      nnreal (ceil-log clength) *
      staged-phase-relation-error (card B)
      (staged-alpha-search-queries budgets 0)
proof -

```

```

have transcript-bound:
  wp-event (staged-transcript-program A)
    (staged-transcript-trace-fri-vector-hit B)
    adversary-initial-state ≤
    nnreal (ceil-log clength) *
    staged-phase-relation-error (card B)
    (staged-alpha-search-queries budgets 0)
by (rule staged-transcript-program-trace-fri-vector-bound
    [OF wf controlled subset nonempty absent])
show ?thesis
  unfolding staged-security-experiment-with-data-def
proof (rule wp-event-bind-bound-by-head-event[OF transcript-bound])
show staged-security-with-data-trace-fri-vector-hit B None  $\implies$ 
  staged-transcript-trace-fri-vector-hit B None
  unfolding staged-security-with-data-trace-fri-vector-hit-def
    staged-transcript-trace-fri-vector-hit-def
  by simp
next
fix data attacker-state out
assume cont:
  out ∈
    set-dist
      (execute
        (get  $\gg=$ 
          ( $\lambda s$ . put
            (verifier-state-from-adversary s
              (staged-proof-transcript data))  $\gg=$ 
            ( $\lambda \cdot$ . verify-monad  $\gg=$ 
              ( $\lambda result$ . return (data, result))))))
          attacker-state)
  and hit: staged-security-with-data-trace-fri-vector-hit B out
have data-bad: staged-trace-fri-challenges data ∈ B
proof (cases out)
  case None
  then show ?thesis
    using hit unfolding staged-security-with-data-trace-fri-vector-hit-def
    by simp
next
  case (Some result)
  then obtain data' result' final-state where out-eq:
    out = Some ((data', result'), final-state)
    by (cases result, auto split: prod.splits)
  from cont[unfolded out-eq]
  obtain s1 s2 s3 where
    ret:
      Some ((data', result'), final-state) ∈
        set-dist (execute (return (data, result')) s3)
    by (auto elim!: set-dist-bindE)
  have data'-eq: data' = data

```



```

    using ret by simp
show ?thesis
    using hit unfolding out-eq data'-eq
    staged-security-with-data-trace-fri-vector-hit-def
    by simp
qed
show staged-transcript-trace-fri-vector-hit B
    (Some (data, attacker-state))
    using data-bad unfolding staged-transcript-trace-fri-vector-hit-def
    by simp
qed
qed

lemma staged-security-with-data-composition-fri-vector-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and subset:
     $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (to\text{-nat } dg + 1))$ 
  and absent:
     $\bigwedge i. i < \text{ceil-log } (\text{maxDegree} + 1) \implies$ 
      composition-fri-challenge-values-absent
       $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$ 
      adversary-initial-state
  shows
    wp-event (staged-security-experiment-with-data A)
      (staged-security-with-data-composition-fri-vector-guarded-hit B)
      adversary-initial-state  $\leq$ 
       $(\sum i < \text{ceil-log } (\text{maxDegree} + 1).$ 
        staged-phase-target-error
         $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$ 
        (staged-composition-fri-vector-search-queries budgets))
proof -
  have transcript-bound:
    wp-event (staged-transcript-program A)
      (staged-transcript-composition-fri-vector-guarded-hit B)
      adversary-initial-state  $\leq$ 
       $(\sum i < \text{ceil-log } (\text{maxDegree} + 1).$ 
        staged-phase-target-error
         $(\bigcup dg. \text{fri-vector-position-values } (B \ dg) \ i)$ 
        (staged-composition-fri-vector-search-queries budgets))
    by (rule staged-transcript-program-composition-fri-vector-bound
      [OF wf controlled subset absent])
  show ?thesis
    unfolding staged-security-experiment-with-data-def
  proof (rule wp-event-bind-bound-by-head-event[OF transcript-bound])
  show
    staged-security-with-data-composition-fri-vector-guarded-hit B None
       $\implies$  staged-transcript-composition-fri-vector-guarded-hit B None
    unfolding

```

```

    staged-security-with-data-composition-fri-vector-guarded-hit-def
    staged-transcript-composition-fri-vector-guarded-hit-def
  by simp
next
fix data attacker-state out
assume cont:
  out ∈
    set-dist
      (execute
        (get >>=
          (λs. put
            (verifier-state-from-adversary s
              (staged-proof-transcript data)) >>=
            (λ-. verify-monad >>=
              (λresult. return (data, result))))))
          attacker-state)
and hit:
  staged-security-with-data-composition-fri-vector-guarded-hit B out
have data-bad:
  staged-composition-fri-challenges data ∈
    B (staged-degree data) ∧
    0 < ceil-log (to-nat (staged-degree data) + 1) ∧
    ceil-log (to-nat (staged-degree data) + 1) ≤
    ceil-log (maxDegree + 1)
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    staged-security-with-data-composition-fri-vector-guarded-hit-def
  by simp
next
case (Some result)
then obtain data' result' final-state where out-eq:
  out = Some ((data', result'), final-state)
  by (cases result, auto split: prod.splits)
from cont[unfolded out-eq]
obtain s1 s2 s3 where
  ret:
    Some ((data', result'), final-state) ∈
      set-dist (execute (return (data, result')) s3)
  by (auto elim!: set-dist-bindE)
have data'-eq: data' = data
  using ret by simp
show ?thesis
  using hit unfolding out-eq data'-eq
    staged-security-with-data-composition-fri-vector-guarded-hit-def
  by simp
qed

```

```

show staged-transcript-composition-fri-vector-guarded-hit  $B$ 
  (Some (data, attacker-state))
using data-bad
unfolding staged-transcript-composition-fri-vector-guarded-hit-def
by simp
qed
qed

lemma staged-trace-fri-prefix-receive-bad-value-bound-total:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and i-bound:  $i < \text{ceil-log } \text{clength}$ 
and absent: trace-fri-challenge-values-absent  $B$   $s$ 
shows
  wp-event
  (staged-trace-fri-challenge-prefix-program  $A$   $i$   $\gg$ 
    ( $\lambda$ -. receive-trace-fri-challenge))
  ( $\lambda$ out. case out of None  $\Rightarrow$  False
    | Some (b, -)  $\Rightarrow$   $b \in B$ )  $s \leq$ 
  staged-phase-target-error  $B$ 
  (Suc
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget))

proof -
have i-len:  $i < \text{length } (\text{trace-fri-budgets } \text{budgets})$ 
using wf i-bound unfolding staged-budget-wellformed-def by simp
have local:
  wp-event
  (staged-trace-fri-challenge-prefix-program  $A$   $i$   $\gg$ 
    ( $\lambda$ -. receive-trace-fri-challenge))
  ( $\lambda$ out. case out of None  $\Rightarrow$  False
    | Some (b, -)  $\Rightarrow$   $b \in B$ )  $s \leq$ 
  staged-phase-target-error  $B$ 
  (Suc (staged-trace-fri-search-queries budgets  $i$ ))
by (rule staged-trace-fri-prefix-receive-bad-value-bound
  [OF controlled i-len absent])
also have
  staged-phase-target-error  $B$ 
  (Suc (staged-trace-fri-search-queries budgets  $i$ ))  $\leq$ 
  staged-phase-target-error  $B$ 
  (Suc
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget))
by (rule staged-phase-target-error-mono)
  (simp add: staged-trace-fri-search-queries-le-total [OF wf i-bound])
finally show ?thesis .
qed

```

lemma staged-trace-fri-prefix-receive-bad-vector-position-bound-total:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *i-bound*: $i < \text{ceil-log length}$
and *absent*:
 trace-fri-challenge-values-absent
 (*fri-vector-position-values* *B i*) *s*
shows
 wp-event
 (*staged-trace-fri-challenge-prefix-program* *A i* $\gg$
 ($\lambda\cdot$. *receive-trace-fri-challenge*))
 ($\lambda\text{out. case out of None} \Rightarrow \text{False}$
 | *Some* (*b, -*) $\Rightarrow b \in \text{fri-vector-position-values } B i$) $s \leq$
 staged-phase-target-error (*fri-vector-position-values* *B i*)
 (*Suc*
 (*staged-attacker-query-budget* *budgets* +
 staged-challenge-query-budget))
by (*rule staged-trace-fri-prefix-receive-bad-value-bound-total*
 [*OF wf controlled i-bound absent*])

lemma *staged-trace-fri-prefix-receive-bad-vector-position-relation-bound*:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *subset*: $B \subseteq \text{fri-challenge-space } (\text{ceil-log length})$
and *i-bound*: $i < \text{ceil-log length}$
and *absent*:
 trace-fri-challenge-values-absent
 (*fri-vector-position-values* *B i*) *s*

shows
 wp-event
 (*staged-trace-fri-challenge-prefix-program* *A i* $\gg$
 ($\lambda\cdot$. *receive-trace-fri-challenge*))
 ($\lambda\text{out. case out of None} \Rightarrow \text{False}$
 | *Some* (*b, -*) $\Rightarrow b \in \text{fri-vector-position-values } B i$) $s \leq$
 staged-phase-relation-error (*card* *B*)
 (*Suc*
 (*staged-attacker-query-budget* *budgets* +
 staged-challenge-query-budget))

proof –

have *target*:
 wp-event
 (*staged-trace-fri-challenge-prefix-program* *A i* $\gg$
 ($\lambda\cdot$. *receive-trace-fri-challenge*))
 ($\lambda\text{out. case out of None} \Rightarrow \text{False}$
 | *Some* (*b, -*) $\Rightarrow b \in \text{fri-vector-position-values } B i$) $s \leq$
 staged-phase-target-error (*fri-vector-position-values* *B i*)
 (*Suc*
 (*staged-attacker-query-budget* *budgets* +
 staged-challenge-query-budget))
by (*rule staged-trace-fri-prefix-receive-bad-vector-position-bound-total*)

```

    [OF wf controlled i-bound absent])
  also have ... ≤
    staged-phase-relation-error (card B)
    (Suc
      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget))
  by (rule staged-phase-target-error-card-mono)
    (rule fri-vector-position-values-card-le[OF subset i-bound])
  finally show ?thesis .
qed

lemma staged-trace-fri-prefix-receive-keep-prefix-bad-vector-position-relation-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and subset:  $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$ 
    and i-bound:  $i < \text{ceil-log } \text{clength}$ 
    and absent:
      trace-fri-challenge-values-absent
      (fri-vector-position-values B i) s
  shows
    wp-event
      (staged-trace-fri-challenge-prefix-program A i  $\gg$ 
        ( $\lambda \text{prefix. receive-trace-fri-challenge } \gg$ 
          ( $\lambda b. \text{return } (\text{prefix}, b)$ ))))
      ( $\lambda \text{out. case out of None } \Rightarrow \text{False}$ 
        |  $\text{Some } ((-, b), -) \Rightarrow b \in \text{fri-vector-position-values } B \ i$ ) s ≤
      staged-phase-relation-error (card B)
      (Suc
        (staged-attacker-query-budget budgets +
         staged-challenge-query-budget))

proof -
  let ?P =
     $\lambda \text{out. case out of None } \Rightarrow \text{False}$ 
    |  $\text{Some } (b, -) \Rightarrow b \in \text{fri-vector-position-values } B \ i$ 
  let ?Q =
     $\lambda \text{out. case out of None } \Rightarrow \text{False}$ 
    |  $\text{Some } ((-, b), -) \Rightarrow b \in \text{fri-vector-position-values } B \ i$ 
  have cont-eq:
     $\bigwedge \text{prefix } t. \text{wp-event}$ 
    ( $\text{receive-trace-fri-challenge } \gg (\lambda b. \text{return } (\text{prefix}, b))$ ) ?Q t =
    wp-event receive-trace-fri-challenge ?P t
  proof -
    fix prefix t
    show wp-event
      ( $\text{receive-trace-fri-challenge } \gg (\lambda b. \text{return } (\text{prefix}, b))$ ) ?Q t =
      wp-event receive-trace-fri-challenge ?P t
    unfolding wp-event-def
    apply (subst wp-bind)
    apply (rule arg-cong[where

```

```

    f = λQ. wp receive-trace-fri-challenge Q t])
  apply (rule ext)
  apply (simp add: wp-return split: option.splits prod.splits)
done
qed
have eq:
  wp-event
    (staged-trace-fri-challenge-prefix-program A i ≫=
      (λprefix. receive-trace-fri-challenge ≫=
        (λb. return (prefix, b)))) ?Q s =
  wp-event
    (staged-trace-fri-challenge-prefix-program A i ≫=
      (λ-. receive-trace-fri-challenge)) ?P s
  by (rule wp-event-bind-cong-cont) (simp-all add: cont-eq)
show ?thesis
  unfolding eq
  by (rule staged-trace-fri-prefix-receive-bad-vector-position-relation-bound
    [OF wf controlled subset i-bound absent])
qed

```

lemma *staged-trace-fri-bad-vector-coordinate-sum-bound:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *subset: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$*
and *absent:*
 $\bigwedge i. i < \text{ceil-log } \text{clength} \implies$
 $\text{trace-fri-challenge-values-absent}$
 $(\text{fri-vector-position-values } B \ i) \ s$

shows

$(\sum i < \text{ceil-log } \text{clength}.$
 wp-event
 $(\text{staged-trace-fri-challenge-prefix-program } A \ i \gg=$
 $(\lambda-. \text{receive-trace-fri-challenge}))$
 $(\lambda \text{out. case out of None} \Rightarrow \text{False}$
 $\mid \text{Some } (b, -) \Rightarrow b \in \text{fri-vector-position-values } B \ i) \ s) \leq$
 $\text{nnreal } (\text{ceil-log } \text{clength}) *$
 $\text{staged-phase-relation-error } (\text{card } B)$
 $(\text{Suc}$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget}))$

proof –

have

$(\sum i < \text{ceil-log } \text{clength}.$
 wp-event
 $(\text{staged-trace-fri-challenge-prefix-program } A \ i \gg=$
 $(\lambda-. \text{receive-trace-fri-challenge}))$
 $(\lambda \text{out. case out of None} \Rightarrow \text{False}$
 $\mid \text{Some } (b, -) \Rightarrow b \in \text{fri-vector-position-values } B \ i) \ s) \leq$
 $(\sum i < \text{ceil-log } \text{clength}.$

```

    staged-phase-relation-error (card B)
    (Suc
      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget)))
  by (rule sum-mono)
    (rule staged-trace-fri-prefix-receive-bad-vector-position-relation-bound
      [OF wf controlled subset], simp-all add: absent)
  also have ... =
    nnreal (ceil-log clength) *
    staged-phase-relation-error (card B)
    (Suc
      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget))
  by simp
  finally show ?thesis .
qed

```

lemma *staged-trace-fri-bad-vector-coordinate-sum-keep-prefix-bound:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *subset: $B \subseteq \text{fri-challenge-space (ceil-log clength)}$*
and *absent:*
 $\bigwedge i. i < \text{ceil-log clength} \implies$
 $\text{trace-fri-challenge-values-absent}$
 $(\text{fri-vector-position-values } B \ i) \ s$

shows

$(\sum i < \text{ceil-log clength}.$
wp-event
 $(\text{staged-trace-fri-challenge-prefix-program } A \ i \gg=$
 $(\lambda \text{prefix. receive-trace-fri-challenge } \gg=$
 $(\lambda b. \text{return (prefix, b)})))$
 $(\lambda \text{out. case out of None } \Rightarrow \text{False}$
 $| \text{Some } ((-, b), -) \Rightarrow$
 $b \in \text{fri-vector-position-values } B \ i) \ s) \leq$
 $\text{nnreal (ceil-log clength) *}$
 $\text{staged-phase-relation-error (card B)}$
 $(\text{Suc}$
 $(\text{staged-attacker-query-budget budgets +}$
 $\text{staged-challenge-query-budget}))$

proof –

have

$(\sum i < \text{ceil-log clength}.$
wp-event
 $(\text{staged-trace-fri-challenge-prefix-program } A \ i \gg=$
 $(\lambda \text{prefix. receive-trace-fri-challenge } \gg=$
 $(\lambda b. \text{return (prefix, b)})))$
 $(\lambda \text{out. case out of None } \Rightarrow \text{False}$
 $| \text{Some } ((-, b), -) \Rightarrow$
 $b \in \text{fri-vector-position-values } B \ i) \ s) \leq$

```

    ( $\sum i < \text{ceil-log } \text{clength}.$ 
     staged-phase-relation-error (card  $B$ )
     (Suc
      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget)))
  by (rule sum-mono)
  (rule
   staged-trace-fri-prefix-receive-keep-prefix-bad-vector-position-relation-bound
   [OF wf controlled subset], simp-all add: absent)
also have ... =
  nnreal (ceil-log clength) *
  staged-phase-relation-error (card  $B$ )
  (Suc
   (staged-attacker-query-budget budgets +
    staged-challenge-query-budget))
  by simp
finally show ?thesis .
qed

```

lemma staged-query-prefix-bad-index-error-mono:
assumes $i\text{-bound}$: $i < \text{rounds}$
and wf: staged-budget-wellformed budgets
shows
 staged-query-prefix-bad-index-error budgets B $i \leq$
 staged-phase-target-error (query-index-raw-preimage B)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)
unfolding staged-query-prefix-bad-index-error-def
by (rule staged-phase-target-error-mono)
 (rule staged-query-search-queries-le-total[OF wf $i\text{-bound}$])

lemma staged-query-prefix-bad-index-target-bound-total:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and $i\text{-bound}$: $i < \text{rounds}$
shows
 wp-event (staged-query-challenge-prefix-program A i)
 (hash-new-output-hit-event (query-index-raw-preimage B) s) $s \leq$
 staged-phase-target-error (query-index-raw-preimage B)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)

proof –
have local:
 wp-event (staged-query-challenge-prefix-program A i)
 (hash-new-output-hit-event (query-index-raw-preimage B) s) $s \leq$
 staged-query-prefix-bad-index-error budgets B i
by (rule staged-query-prefix-bad-index-target-bound)
 (use wf controlled $i\text{-bound}$ **in** simp-all)
also have

staged-query-prefix-bad-index-error budgets B i ≤
staged-phase-target-error (query-index-raw-preimage B)
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)
by (rule *staged-query-prefix-bad-index-error-mono*[*OF i-bound wf*])
finally show ?thesis .
qed

lemma *checked-staged-query-prefix-bad-index-target-bound-total*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: *i < rounds*
shows
wp-event (checked-staged-query-challenge-prefix-program A i)
(hash-new-output-hit-event (query-index-raw-preimage B) s) s ≤
staged-phase-target-error (query-index-raw-preimage B)
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)
proof –
have *local*:
wp-event (checked-staged-query-challenge-prefix-program A i)
(hash-new-output-hit-event (query-index-raw-preimage B) s) s ≤
staged-query-prefix-bad-index-error budgets B i
by (rule *checked-staged-query-prefix-bad-index-target-bound*)
(use wf controlled i-bound in simp-all)
also have
staged-query-prefix-bad-index-error budgets B i ≤
staged-phase-target-error (query-index-raw-preimage B)
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)
by (rule *staged-query-prefix-bad-index-error-mono*[*OF i-bound wf*])
finally show ?thesis .
qed

lemma *staged-query-prefix-receive-bad-index-bound-total*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: *i < rounds*
and *absent*: *query-index-raw-preimage-absent B s*
shows
wp-event
(staged-query-challenge-prefix-program A i ≫=
(λ-. receive-query-index-challenge))
(λout. case out of None ⇒ False
| Some (raw, -) ⇒ index (to-nat raw) ∈ B) s ≤
staged-phase-target-error (query-index-raw-preimage B)
(Suc
(staged-attacker-query-budget budgets +
staged-challenge-query-budget))

proof –
have *local*:
wp-event
 (*staged-query-challenge-prefix-program* *A i* $\gg$
 (λ -. *receive-query-index-challenge*))
 (λ *out*. *case out of None* $\Rightarrow$ *False*
 | *Some* (*raw*, -) $\Rightarrow$ *index* (*to-nat raw*) $\in$ *B*) *s* $\leq$
staged-phase-target-error (*query-index-raw-preimage B*)
 (*Suc* (*staged-query-search-queries budgets i*))
by (*rule staged-query-prefix-receive-bad-index-bound*)
 (*use wf controlled i-bound absent in simp-all*)
also have
staged-phase-target-error (*query-index-raw-preimage B*)
 (*Suc* (*staged-query-search-queries budgets i*)) $\leq$
staged-phase-target-error (*query-index-raw-preimage B*)
 (*Suc*
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget))
by (*rule staged-phase-target-error-mono*)
 (*simp add: staged-query-search-queries-le-total[OF wf i-bound]*)
finally show *?thesis* .
qed

lemma *staged-query-prefix-receive-bad-index-query-bound-total*:

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *i-bound*: *i* < *rounds*

and *absent*: *query-index-raw-preimage-absent B s*

and *raw-bound*: *query-index-raw-preimage-bound*

and *subset*: *B* $\subseteq$ *query-sample-space*

and *frac*:

$$\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{query-sample-space}) \leq C$$

shows

wp-event
 (*staged-query-challenge-prefix-program* *A i* $\gg$
 (λ -. *receive-query-index-challenge*))
 (λ *out*. *case out of None* $\Rightarrow$ *False*
 | *Some* (*raw*, -) $\Rightarrow$ *index* (*to-nat raw*) $\in$ *B*) *s* $\leq$
nnreal
 (*Suc*
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)) * *C*

proof –

have *local*:

wp-event
 (*staged-query-challenge-prefix-program* *A i* $\gg$
 (λ -. *receive-query-index-challenge*))
 (λ *out*. *case out of None* $\Rightarrow$ *False*
 | *Some* (*raw*, -) $\Rightarrow$ *index* (*to-nat raw*) $\in$ *B*) *s* $\leq$

```

    staged-phase-target-error (query-index-raw-preimage B)
    (Suc
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget))
  by (rule staged-query-prefix-receive-bad-index-bound-total
    [OF wf controlled i-bound absent])
  also have ... ≤
    nnreal
    (Suc
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) * C
  by (rule staged-phase-query-index-target-error-query-bound
    [OF raw-bound subset frac])
  finally show ?thesis .
qed

lemma staged-query-prefix-receive-header-supported-sampling-bound-total:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and absent:
    query-index-raw-preimage-absent
    (query-header-supported-union-good-sets verifier-state
      query-sampling-success-space fr f-fri-roots f-final as dg
      composition-fri-roots final)
    prefix-state
  and raw-bound: query-index-raw-preimage-bound
  and no-bad: ¬ query-supported-pairwise-merkle-bad verifier-state
  shows
    wp-event
    (staged-query-challenge-prefix-program A i ≫
      (λ-. receive-query-index-challenge))
    (λout. case out of None ⇒ False
      | Some (raw, -) ⇒
        index (to-nat raw) ∈
          query-header-supported-union-good-sets verifier-state
          query-sampling-success-space fr f-fri-roots f-final as dg
          composition-fri-roots final)
    prefix-state ≤
    nnreal
    (Suc
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) * query-error-bound
  proof -
    let ?B =
      query-header-supported-union-good-sets verifier-state
      query-sampling-success-space fr f-fri-roots f-final as dg
      composition-fri-roots final
    have subset: ?B ⊆ query-sample-space

```

```

    by (rule query-header-supported-union-good-sets-subset)
  have frac:
    nnreal (card ?B) / nnreal (card query-sample-space) ≤
      query-error-bound
  by (rule
    query-header-supported-union-good-sets-fraction-bound-if-no-global-pairwise-merkle-bad
    [OF no-bad query-sampling-success-space-subset
      query-sampling-success-space-fraction-bound-query-sample-space])
  show ?thesis
  by (rule staged-query-prefix-receive-bad-index-query-bound-total
    [OF wf controlled i-bound absent raw-bound subset frac])
qed

```

lemma *staged-composition-fri-prefix-receive-bad-value-bound-total:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < ceil-log (maxDegree + 1)*
and *absent: composition-fri-challenge-values-absent B s*
shows

```

wp-event
  (staged-composition-fri-challenge-prefix-program A i ≫=
    (λ-. receive-composition-fri-challenge))
  (λout. case out of None ⇒ False
    | Some (b, -) ⇒ b ∈ B) s ≤
  staged-phase-target-error B
  (Suc
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget))

```

proof –

have *local:*

```

wp-event
  (staged-composition-fri-challenge-prefix-program A i ≫=
    (λ-. receive-composition-fri-challenge))
  (λout. case out of None ⇒ False
    | Some (b, -) ⇒ b ∈ B) s ≤
  staged-phase-target-error B
  (Suc (staged-composition-fri-search-queries budgets i))
  by (rule staged-composition-fri-prefix-receive-bad-value-bound
    [OF wf controlled i-bound absent])

```

also have

```

  staged-phase-target-error B
  (Suc (staged-composition-fri-search-queries budgets i)) ≤
  staged-phase-target-error B
  (Suc
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget))
  by (rule staged-phase-target-error-mono)
  (simp add: staged-composition-fri-search-queries-le-total
    [OF wf i-bound])

```

finally show ?thesis .
qed

lemma *staged-composition-fri-prefix-recv-bad-vector-position-bound-total:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: $i < \text{ceil-log } (\text{maxDegree} + 1)$*

and *absent:*

composition-fri-challenge-values-absent
(fri-vector-position-values B i) s

shows

wp-event

(staged-composition-fri-challenge-prefix-program A i $\gg$
(λ -. receive-composition-fri-challenge))

(λ out. case out of None $\Rightarrow$ False

| Some (b, -) $\Rightarrow$ $b \in \text{fri-vector-position-values B i}$) s $\leq$

staged-phase-target-error (fri-vector-position-values B i)

(Suc

(staged-attacker-query-budget budgets +
staged-challenge-query-budget))

by (*rule staged-composition-fri-prefix-recv-bad-value-bound-total*
[OF wf controlled i-bound absent])

lemma

staged-composition-fri-prefix-recv-bad-vector-position-relation-bound:

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *subset:*

B $\subseteq$ fri-challenge-space (ceil-log (to-nat dg + 1))

and *i-bound: $i < \text{ceil-log } (\text{to-nat dg} + 1)$*

and *dg-bound: $\text{ceil-log } (\text{to-nat dg} + 1) \leq \text{ceil-log } (\text{maxDegree} + 1)$*

and *absent:*

composition-fri-challenge-values-absent
(fri-vector-position-values B i) s

shows

wp-event

(staged-composition-fri-challenge-prefix-program A i $\gg$
(λ -. receive-composition-fri-challenge))

(λ out. case out of None $\Rightarrow$ False

| Some (b, -) $\Rightarrow$ $b \in \text{fri-vector-position-values B i}$) s $\leq$

staged-phase-relation-error (card B)

(Suc

(staged-attacker-query-budget budgets +
staged-challenge-query-budget))

proof –

have *i-max: $i < \text{ceil-log } (\text{maxDegree} + 1)$*

using *i-bound dg-bound* **by** *simp*

have *target:*

wp-event

```

    (staged-composition-fri-challenge-prefix-program A i  $\gg$ 
      ( $\lambda$ -. receive-composition-fri-challenge))
    ( $\lambda$ out. case out of None  $\Rightarrow$  False
      | Some (b, -)  $\Rightarrow$  b  $\in$  fri-vector-position-values B i) s  $\leq$ 
      staged-phase-target-error (fri-vector-position-values B i)
      (Suc
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget))
  by (rule
      staged-composition-fri-prefix-receive-bad-vector-position-bound-total
      [OF wf controlled i-max absent])
  also have ...  $\leq$ 
    staged-phase-relation-error (card B)
    (Suc
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget))
  by (rule staged-phase-target-error-card-mono)
    (rule fri-vector-position-values-card-le[OF subset i-bound])
  finally show ?thesis .
qed

lemma
  staged-composition-fri-prefix-receive-keep-prefix-bad-vector-position-relation-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and subset:
    B  $\subseteq$  fri-challenge-space (ceil-log (to-nat dg + 1))
  and i-bound: i < ceil-log (to-nat dg + 1)
  and dg-bound: ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1)
  and absent:
    composition-fri-challenge-values-absent
    (fri-vector-position-values B i) s
  shows
    wp-event
    (staged-composition-fri-challenge-prefix-program A i  $\gg$ 
      ( $\lambda$ prefix. receive-composition-fri-challenge  $\gg$ 
        ( $\lambda$ b. return (prefix, b))))
    ( $\lambda$ out. case out of None  $\Rightarrow$  False
      | Some ((-, b), -)  $\Rightarrow$  b  $\in$  fri-vector-position-values B i) s  $\leq$ 
      staged-phase-relation-error (card B)
      (Suc
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget))

proof -
  let ?P =
     $\lambda$ out. case out of None  $\Rightarrow$  False
    | Some (b, -)  $\Rightarrow$  b  $\in$  fri-vector-position-values B i
  let ?Q =
     $\lambda$ out. case out of None  $\Rightarrow$  False

```

$| \text{Some } ((-, b), -) \Rightarrow b \in \text{fri-vector-position-values } B \ i$
have *cont-eq*:
 $\wedge \text{prefix } t. \text{wp-event}$
 $(\text{receive-composition-fri-challenge} \gg=$
 $(\lambda b. \text{return } (\text{prefix}, b))) \ ?Q \ t =$
 $\text{wp-event receive-composition-fri-challenge} \ ?P \ t$
proof –
fix *prefix* *t*
show *wp-event*
 $(\text{receive-composition-fri-challenge} \gg=$
 $(\lambda b. \text{return } (\text{prefix}, b))) \ ?Q \ t =$
 $\text{wp-event receive-composition-fri-challenge} \ ?P \ t$
unfolding *wp-event-def*
apply (*subst wp-bind*)
apply (*rule arg-cong*[**where**
 $f = \lambda Q. \text{wp receive-composition-fri-challenge } Q \ t$])
apply (*rule ext*)
apply (*simp add: wp-return split: option.splits prod.splits*)
done
qed
have *eq*:
 wp-event
 $(\text{staged-composition-fri-challenge-prefix-program } A \ i \gg=$
 $(\lambda \text{prefix}. \text{receive-composition-fri-challenge} \gg=$
 $(\lambda b. \text{return } (\text{prefix}, b)))) \ ?Q \ s =$
 wp-event
 $(\text{staged-composition-fri-challenge-prefix-program } A \ i \gg=$
 $(\lambda -. \text{receive-composition-fri-challenge})) \ ?P \ s$
by (*rule wp-event-bind-cong-cont*) (*simp-all add: cont-eq*)
show *?thesis*
unfolding *eq*
by (*rule*
 $\text{staged-composition-fri-prefix-receive-bad-vector-position-relation-bound}$
 $[\text{OF wf controlled subset i-bound dg-bound absent}]$)
qed

lemma *staged-composition-fri-bad-vector-coordinate-sum-bound*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *subset*:
 $B \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and *dg-bound*:
 $\text{ceil-log } (\text{to-nat } dg + 1) \leq \text{ceil-log } (\text{maxDegree} + 1)$
and *absent*:
 $\wedge i. i < \text{ceil-log } (\text{to-nat } dg + 1) \Rightarrow$
 $\text{composition-fri-challenge-values-absent}$
 $(\text{fri-vector-position-values } B \ i) \ s$
shows
 $(\sum i < \text{ceil-log } (\text{to-nat } dg + 1)).$

```

wp-event
  (staged-composition-fri-challenge-prefix-program A i >>=
    (λ-. receive-composition-fri-challenge))
  (λout. case out of None ⇒ False
    | Some (b, -) ⇒ b ∈ fri-vector-position-values B i) s ≤
nnreal (ceil-log (to-nat dg + 1)) *
  staged-phase-relation-error (card B)
  (Suc
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget))
proof –
have
  (∑ i < ceil-log (to-nat dg + 1).
    wp-event
      (staged-composition-fri-challenge-prefix-program A i >>=
        (λ-. receive-composition-fri-challenge))
      (λout. case out of None ⇒ False
        | Some (b, -) ⇒ b ∈ fri-vector-position-values B i) s ≤
      (∑ i < ceil-log (to-nat dg + 1).
        staged-phase-relation-error (card B)
        (Suc
          (staged-attacker-query-budget budgets +
            staged-challenge-query-budget))))
proof (rule sum-mono)
fix i
assume i-bound: i ∈ {.. $\text{ceil-log (to-nat dg + 1)}$ }
then have i-lt: i <  $\text{ceil-log (to-nat dg + 1)}$ 
by simp
show
  wp-event
    (staged-composition-fri-challenge-prefix-program A i >>=
      (λ-. receive-composition-fri-challenge))
    (λout. case out of None ⇒ False
      | Some (b, -) ⇒ b ∈ fri-vector-position-values B i) s ≤
    staged-phase-relation-error (card B)
    (Suc
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget))
by (rule
  staged-composition-fri-prefix-receive-bad-vector-position-relation-bound
  [OF wf controlled subset i-lt dg-bound absent[OF i-lt]])
qed
also have ... =
  nnreal (ceil-log (to-nat dg + 1)) *
    staged-phase-relation-error (card B)
    (Suc
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget))
by simp

```


finally show *?thesis* .
 qed

lemma *staged-composition-fri-bad-vector-coordinate-sum-keep-prefix-bound*:

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *subset*:

$B \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$

and *dg-bound*:

$\text{ceil-log } (\text{to-nat } dg + 1) \leq \text{ceil-log } (\text{maxDegree} + 1)$

and *absent*:

$\bigwedge i. i < \text{ceil-log } (\text{to-nat } dg + 1) \implies$
 $\text{composition-fri-challenge-values-absent}$
 $(\text{fri-vector-position-values } B \ i) \ s$

shows

$(\sum i < \text{ceil-log } (\text{to-nat } dg + 1)).$

wp-event

$(\text{staged-composition-fri-challenge-prefix-program } A \ i \gg=$

$(\lambda \text{prefix. receive-composition-fri-challenge } \gg=$

$(\lambda b. \text{return } (\text{prefix}, b))))$

$(\lambda \text{out. case out of None } \Rightarrow \text{False}$

$| \text{Some } ((-, b), -) \Rightarrow$

$b \in \text{fri-vector-position-values } B \ i) \ s) \leq$

$\text{nnreal } (\text{ceil-log } (\text{to-nat } dg + 1)) *$

$\text{staged-phase-relation-error } (\text{card } B)$

$(\text{Suc}$

$(\text{staged-attacker-query-budget budgets} +$

$\text{staged-challenge-query-budget}))$

proof –

have

$(\sum i < \text{ceil-log } (\text{to-nat } dg + 1)).$

wp-event

$(\text{staged-composition-fri-challenge-prefix-program } A \ i \gg=$

$(\lambda \text{prefix. receive-composition-fri-challenge } \gg=$

$(\lambda b. \text{return } (\text{prefix}, b))))$

$(\lambda \text{out. case out of None } \Rightarrow \text{False}$

$| \text{Some } ((-, b), -) \Rightarrow$

$b \in \text{fri-vector-position-values } B \ i) \ s) \leq$

$(\sum i < \text{ceil-log } (\text{to-nat } dg + 1)).$

$\text{staged-phase-relation-error } (\text{card } B)$

$(\text{Suc}$

$(\text{staged-attacker-query-budget budgets} +$

$\text{staged-challenge-query-budget}))$

proof (*rule sum-mono*)

fix *i*

assume *i-bound*: $i \in \{..<\text{ceil-log } (\text{to-nat } dg + 1)\}$

then have *i-lt*: $i < \text{ceil-log } (\text{to-nat } dg + 1)$

by *simp*

show

```

wp-event
  (staged-composition-fri-challenge-prefix-program A i >>=
    (λprefix. receive-composition-fri-challenge >>=
      (λb. return (prefix, b))))
  (λout. case out of None ⇒ False
    | Some ((-, b), -) ⇒
      b ∈ fri-vector-position-values B i) s ≤
    staged-phase-relation-error (card B)
  (Suc
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget))
  by (rule
    staged-composition-fri-prefix-keep-prefix-bad-vector-position-relation-bound
    [OF wf controlled subset i-lt dg-bound absent[OF i-lt]])
qed
also have ... =
  nnreal (ceil-log (to-nat dg + 1)) *
    staged-phase-relation-error (card B)
  (Suc
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget))
  by simp
finally show ?thesis .
qed

lemma staged-controlled-trace-root:
  assumes staged-adversary-controlled budgets A
  shows
    controlled-ro-program (trace-root-budget budgets)
    (trace-root-stage A)
  using assms unfolding staged-adversary-controlled-def by simp

lemma staged-controlled-trace-fri:
  assumes controlled: staged-adversary-controlled budgets A
  and i < length (trace-fri-budgets budgets)
  shows
    controlled-ro-program (trace-fri-budgets budgets ! i)
    (trace-fri-root-stage A i bs)
  using assms unfolding staged-adversary-controlled-def by blast

lemma staged-controlled-trace-final:
  assumes staged-adversary-controlled budgets A
  shows
    controlled-ro-program (trace-final-budget budgets)
    (trace-final-stage A bs)
  using assms unfolding staged-adversary-controlled-def by blast

lemma staged-controlled-degree:
  assumes staged-adversary-controlled budgets A

```

shows
controlled-ro-program (degree-budget budgets)
(degree-stage A as)
using *assms unfolding staged-adversary-controlled-def* **by** *blast*

lemma *staged-controlled-composition-fri:*
assumes *controlled: staged-adversary-controlled budgets A*
and *i < length (composition-fri-budgets budgets)*
shows
controlled-ro-program (composition-fri-budgets budgets ! i)
(composition-fri-root-stage A dg i bs)
using *assms unfolding staged-adversary-controlled-def* **by** *blast*

lemma *staged-controlled-composition-final:*
assumes *staged-adversary-controlled budgets A*
shows
controlled-ro-program (composition-final-budget budgets)
(composition-final-stage A dg bs)
using *assms unfolding staged-adversary-controlled-def* **by** *blast*

lemma *staged-controlled-query:*
assumes *controlled: staged-adversary-controlled budgets A*
and *i < length (query-opening-budgets budgets)*
shows
controlled-ro-program (query-opening-budgets budgets ! i)
(query-opening-stage A i raw)
using *assms unfolding staged-adversary-controlled-def* **by** *blast*

lemma *staged-security-experiment-hash-map-output-collision-bad-bound:*
assumes *admissible:*
admissible-adversary q (staged-semantic-adversary A)
shows
wp-event (staged-security-experiment A)
(hash-map-output-collision-bad adversary-initial-state)
adversary-initial-state ≤
hash-collision-budget-value 0 (q + verifier-hash-query-budget)
unfolding *staged-security-experiment-eq-security-experiment*
by *(rule security-experiment-hash-map-output-collision-bad-bound*
[OF admissible])

lemma *staged-security-experiment-hash-map-output-collision-bad-bound-controlled:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows
wp-event (staged-security-experiment A)
(hash-map-output-collision-bad adversary-initial-state)
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +

staged-challenge-query-budget + verifier-hash-query-budget)
by (*rule staged-security-experiment-hash-map-output-collision-bad-bound*
 [*OF admissible-staged-semantic-adversary[OF wf controlled]*])

lemma *hash-range-budget-checked-staged-security-experiment:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows

hash-range-budget
 (*staged-attacker-query-budget budgets +*
staged-challenge-query-budget + verifier-hash-query-budget)
 (*checked-staged-security-experiment A*)

proof –

have *cont:*

$\wedge$ *data. hash-range-budget verifier-hash-query-budget*
 (*get* $\gg$
 ($\lambda s.$ *put*
 (*verifier-state-from-adversary s*
 (*staged-proof-transcript data*)) $\gg$
 ($\lambda.$ *verify-monad*)))

using *hash-range-budget-verifier-after-adversary*

[*of staged-proof-transcript data for data*]

unfolding *verifier-state-transfer-def sm-bind-assoc .*

have *budget:*

hash-range-budget
 ((*staged-attacker-query-budget budgets +*
staged-challenge-query-budget) +
verifier-hash-query-budget)
 (*checked-staged-security-experiment A*)

unfolding *checked-staged-security-experiment-def*

by (*rule hash-range-budget-bind*)

(*rule hash-range-budget-checked-staged-transcript-program*
 [*OF wf controlled*], *rule cont*)

then show *?thesis by (simp add: add.assoc)*

qed

lemma *hash-collision-budget-checked-staged-security-experiment:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows

hash-collision-budget
 (*staged-attacker-query-budget budgets +*
staged-challenge-query-budget + verifier-hash-query-budget)
 (*checked-staged-security-experiment A*)

proof –

have *cont-range:*

$\wedge$ *data. hash-range-budget verifier-hash-query-budget*
 (*get* $\gg$
 ($\lambda s.$ *put*

```

      (verifier-state-from-adversary s
        (staged-proof-transcript data)) >>=
      (λ-. verify-monad)))
using hash-range-budget-verifier-after-adversary
  [of staged-proof-transcript data for data]
unfolding verifier-state-transfer-def sm-bind-assoc .
have cont-collision:
  ∧ data. hash-collision-budget verifier-hash-query-budget
    (get >>=
      (λs. put
        (verifier-state-from-adversary s
          (staged-proof-transcript data)) >>=
        (λ-. verify-monad))))
using hash-collision-budget-verifier-after-adversary
  [of staged-proof-transcript data for data]
unfolding verifier-state-transfer-def sm-bind-assoc .
have budget:
  hash-collision-budget
    ((staged-attacker-query-budget budgets +
      staged-challenge-query-budget) +
      verifier-hash-query-budget)
    (checked-staged-security-experiment A)
unfolding checked-staged-security-experiment-def
by (rule hash-collision-budget-bind)
  (rule hash-range-budget-checked-staged-transcript-program
    [OF wf controlled],
    rule hash-collision-budget-checked-staged-transcript-program
      [OF wf controlled],
    rule cont-range,
    rule cont-collision)
then show ?thesis by (simp add: add.assoc)
qed

lemma checked-staged-security-experiment-hash-new-collision-bound-controlled:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event (checked-staged-security-experiment A)
    (hash-new-collision-event adversary-initial-state)
    adversary-initial-state ≤
    hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget + verifier-hash-query-budget)
proof -
let ?q =
  staged-attacker-query-budget budgets +
  staged-challenge-query-budget + verifier-hash-query-budget
have collision-budget:
  hash-collision-budget ?q (checked-staged-security-experiment A)

```

```

    by (rule hash-collision-budget-checked-staged-security-experiment
        [OF wf controlled])
  have wp-event (checked-staged-security-experiment A)
    (hash-new-collision-event adversary-initial-state)
    adversary-initial-state ≤
    hash-collision-budget-value
    (card (hash-map-output-values adversary-initial-state)) ?q
  using collision-budget adversary-initial-state-no-output-collision
  unfolding hash-collision-budget-def by blast
  then show ?thesis by simp
qed

lemma checked-staged-security-experiment-hash-map-output-collision-bad-bound-controlled:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment A)
    (hash-map-output-collision-bad adversary-initial-state)
    adversary-initial-state ≤
    hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget + verifier-hash-query-budget)
proof -
  have wp-event (checked-staged-security-experiment A)
    (hash-map-output-collision-bad adversary-initial-state)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment A)
    (hash-new-collision-event adversary-initial-state)
    adversary-initial-state
  by (rule wp-event-mono)
    (rule hash-map-output-collision-bad-imp-security-new-collision)
  also have ... ≤
    hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget + verifier-hash-query-budget)
  by (rule checked-staged-security-experiment-hash-new-collision-bound-controlled
      [OF wf controlled])
  finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     hash-map-output-collision-bad)
    adversary-initial-state ≤
    hash-collision-budget-value 0

```

```

      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget + verifier-hash-query-budget)
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?P = hash-map-output-collision-bad adversary-initial-state
  let ?projected =
    λout. case out of None ⇒ ?P None
      | Some (x, t) ⇒ ?P (Some (snd x, t))
  have event-le-projected:
    wp-event ?M
      (staged-security-with-data-state-verifier-event
       hash-map-output-collision-bad)
    adversary-initial-state ≤
    wp-event ?M ?projected adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume bad:
    staged-security-with-data-state-verifier-event
    hash-map-output-collision-bad out
  show ?projected out
proof (cases out)
  case None
  then show ?thesis
    using bad unfolding staged-security-with-data-state-verifier-event-def
    by simp
next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  have local-bad:
    hash-map-output-collision-bad
      (verifier-state-from-adversary attacker-state
       (staged-proof-transcript data))
      (Some (result, final-state))
    using bad unfolding out-eq
    staged-security-with-data-state-verifier-event-def by simp
  then have global-bad:
    hash-map-output-collision-bad adversary-initial-state
      (Some (result, final-state))
    unfolding hash-map-output-collision-bad-def by simp
  show ?thesis
    unfolding out-eq using global-bad by simp
qed
qed
have projected-eq:
  wp-event ?M ?projected adversary-initial-state =
  wp-event (checked-staged-security-experiment A) ?P
  adversary-initial-state

```

```

proof –
  have  $?M \gg (\lambda x. \text{return } (\text{snd } x)) =$ 
    checked-staged-security-experiment A
    by (rule checked-staged-security-experiment-with-data-state-projection)
  then have wp-event (checked-staged-security-experiment A) ?P
    adversary-initial-state =
    wp-event (?M  $\gg (\lambda x. \text{return } (\text{snd } x))$ ) ?P
    adversary-initial-state
    by simp
  also have  $\dots = \text{wp-event } ?M \text{ ?projected adversary-initial-state}$ 
    by (subst wp-event-bind-return-map[where f=snd]) simp
  finally show ?thesis by simp
qed
have checked-bound:
  wp-event (checked-staged-security-experiment A)
  (hash-map-output-collision-bad adversary-initial-state)
  adversary-initial-state  $\leq$ 
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget)
  by (rule
    checked-staged-security-experiment-hash-map-output-collision-bad-bound-controlled
    [OF wf controlled])
have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    hash-map-output-collision-bad)
  adversary-initial-state  $\leq$ 
  wp-event ?M ?projected adversary-initial-state
  by (rule event-le-projected)
also have  $\dots =$ 
  wp-event (checked-staged-security-experiment A) ?P
  adversary-initial-state
  by (rule projected-eq)
also have  $\dots \leq$ 
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget)
  by (rule checked-bound)
  finally show ?thesis .
qed

end

end

```

```

theory Soundness-Oracle-Dynamic-Target
  imports Soundness-Oracle-Budgets
begin

```


State-dependent target-hit combinators.

These lemmas are deliberately generic: the selected target set may depend on an intermediate oracle state, but all probability bounds are still discharged through the existing random-oracle target-budget interface.

context *soundness*

begin

definition *state-dependent-output-hit*

$:: ((f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

state-dependent-output-hit $B \ s \ out \longleftrightarrow$
 $(\text{case out of}$
 $\quad \text{None} \Rightarrow \text{False}$
 $\quad | \text{Some } (y, -) \Rightarrow y \in B \ s)$

definition *state-dependent-new-output-hit*

$:: ((f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

state-dependent-new-output-hit $B \ s \ out \longleftrightarrow$
 $\text{hash-new-output-hit-event } (B \ s) \ s \ out$

definition *state-dependent-prequeried-hash-hit*

$:: ((f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ set}) \Rightarrow$
 $'f \text{ protocol-hash-input} \Rightarrow$
 $(f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

state-dependent-prequeried-hash-hit $B \ x \ s \longleftrightarrow$
 $(\exists y. \text{fmlookup } (\text{HashMap } s) \ x = \text{Some } y \wedge y \in B \ s)$

lemma *state-dependent-output-hit-None[simp]:*

$\neg \text{state-dependent-output-hit } B \ s \ \text{None}$
unfolding *state-dependent-output-hit-def* **by** *simp*

lemma *state-dependent-new-output-hit-None[simp]:*

$\neg \text{state-dependent-new-output-hit } B \ s \ \text{None}$
unfolding *state-dependent-new-output-hit-def* *hash-new-output-hit-event-def*
by *simp*

lemma *state-dependent-prequeried-hash-hitI:*

assumes $\text{fmlookup } (\text{HashMap } s) \ x = \text{Some } y$
and $y \in B \ s$
shows *state-dependent-prequeried-hash-hit* $B \ x \ s$
using *assms* **unfolding** *state-dependent-prequeried-hash-hit-def* **by** *blast*

```

lemma state-dependent-output-hit-mono:
  assumes subset:  $\bigwedge s. B\ s \subseteq C\ s$ 
    and hit: state-dependent-output-hit  $B\ s\ out$ 
  shows state-dependent-output-hit  $C\ s\ out$ 
  using subset hit
  unfolding state-dependent-output-hit-def
  by (cases out) auto

lemma state-dependent-new-output-hit-mono:
  assumes subset:  $\bigwedge s. B\ s \subseteq C\ s$ 
    and hit: state-dependent-new-output-hit  $B\ s\ out$ 
  shows state-dependent-new-output-hit  $C\ s\ out$ 
  using subset hit
  unfolding state-dependent-new-output-hit-def hash-new-output-hit-event-def
  by (cases out) (auto intro: hash-map-new-output-hit-subset)

lemma state-dependent-output-hit-union:
  state-dependent-output-hit  $(\lambda s. B\ s \cup C\ s)\ s\ out \longleftrightarrow$ 
    state-dependent-output-hit  $B\ s\ out \vee$ 
    state-dependent-output-hit  $C\ s\ out$ 
  unfolding state-dependent-output-hit-def
  by (cases out) auto

lemma state-dependent-new-output-hit-union-imp:
  assumes hit: state-dependent-new-output-hit  $(\lambda s. B\ s \cup C\ s)\ s\ out$ 
  shows
    state-dependent-new-output-hit  $B\ s\ out \vee$ 
    state-dependent-new-output-hit  $C\ s\ out$ 
proof (cases out)
  case None
  then show ?thesis
    using hit by simp
next
  case (Some rt)
  then obtain  $r\ t$  where rt:  $rt = (r, t)$ 
    by (cases rt) simp
  obtain  $x\ y$  where
    none: fmlookup (HashMap s)  $x = None$ 
    and some: fmlookup (HashMap t)  $x = Some\ y$ 
    and  $y: y \in B\ s \cup C\ s$ 
    using hit Some rt
  unfolding state-dependent-new-output-hit-def hash-new-output-hit-event-def
    hash-map-new-output-hit-def
  by auto
show ?thesis
proof ( $y \in B\ s$ )
  case True
  then show ?thesis
    using Some rt none some

```

```

    unfolding state-dependent-new-output-hit-def hash-new-output-hit-event-def
      hash-map-new-output-hit-def
    by auto
  next
  case False
  then have  $y \in C\ s$ 
    using  $y$  by simp
  then show ?thesis
    using Some rt none some
    unfolding state-dependent-new-output-hit-def hash-new-output-hit-event-def
      hash-map-new-output-hit-def
    by auto
qed
qed

```

lemma *wp-event-state-dependent-output-union-bound*:

```

wp-event m
  (state-dependent-output-hit ( $\lambda s. B\ s \cup C\ s$ )  $s$ )  $s \leq$ 
  wp-event m (state-dependent-output-hit  $B\ s$ )  $s$  +
  wp-event m (state-dependent-output-hit  $C\ s$ )  $s$ 
unfolding state-dependent-output-hit-union
by (rule wp-event-union-bound)

```

lemma *wp-event-state-dependent-new-output-union-bound*:

```

wp-event m
  (state-dependent-new-output-hit ( $\lambda s. B\ s \cup C\ s$ )  $s$ )  $s \leq$ 
  wp-event m (state-dependent-new-output-hit  $B\ s$ )  $s$  +
  wp-event m (state-dependent-new-output-hit  $C\ s$ )  $s$ 

```

proof —

```

have wp-event m
  (state-dependent-new-output-hit ( $\lambda s. B\ s \cup C\ s$ )  $s$ )  $s \leq$ 
  wp-event m
    ( $\lambda out. state-dependent-new-output-hit\ B\ s\ out \vee$ 
      state-dependent-new-output-hit  $C\ s\ out$ )  $s$ 
  by (rule wp-event-mono) (rule state-dependent-new-output-hit-union-imp)
also have ...  $\leq$ 
  wp-event m (state-dependent-new-output-hit  $B\ s$ )  $s$  +
  wp-event m (state-dependent-new-output-hit  $C\ s$ )  $s$ 
  by (rule wp-event-union-bound)
finally show ?thesis .

```

qed

lemma *wp-hash-state-dependent-new-output-hit*:

```

fixes  $x :: 'f\ protocol-hash-input$ 
shows
  wp-event (hash  $x$ ) (state-dependent-new-output-hit  $B\ s$ )  $s =$ 
    (if fmllookup (HashMap  $s$ )  $x = None$ 
      then nnreal (card ( $B\ s$ )) / nnreal size
      else 0)

```

```

unfolding state-dependent-new-output-hit-def
by (rule wp-hash-new-output-hit)

lemma wp-hash-state-dependent-new-output-hit-bound:
  fixes  $x :: 'f$  protocol-hash-input
  shows
     $wp\_event\ (hash\ x)\ (state\_dependent\_new\_output\_hit\ B\ s)\ s \leq$ 
       $nnreal\ (card\ (B\ s))\ /\ nnreal\ size$ 
  unfolding state-dependent-new-output-hit-def
  by (rule wp-hash-new-output-hit-bound)

lemma wp-hash-state-dependent-new-output-hit-bound-card:
  fixes  $x :: 'f$  protocol-hash-input
  assumes card-bound:  $card\ (B\ s) \leq N$ 
  shows
     $wp\_event\ (hash\ x)\ (state\_dependent\_new\_output\_hit\ B\ s)\ s \leq$ 
       $nnreal\ N\ /\ nnreal\ size$ 
proof -
  have  $wp\_event\ (hash\ x)\ (state\_dependent\_new\_output\_hit\ B\ s)\ s \leq$ 
     $nnreal\ (card\ (B\ s))\ /\ nnreal\ size$ 
    by (rule wp-hash-state-dependent-new-output-hit-bound)
  also have  $\dots \leq nnreal\ N\ /\ nnreal\ size$ 
    by (rule nnreal-nat-divide-right-mono[OF card-bound])
  finally show ?thesis .
qed

lemma hash-state-dependent-output-hit-fresh-or-prequeried:
  fixes  $x :: 'f$  protocol-hash-input
  assumes out:  $out \in set\_dist\ (execute\ (hash\ x)\ s)$ 
  and hit: state-dependent-output-hit  $B\ s\ out$ 
  shows
     $state\_dependent\_new\_output\_hit\ B\ s\ out \vee$ 
       $state\_dependent\_prequeried\_hash\_hit\ B\ x\ s$ 
proof (cases out)
  case None
  then show ?thesis
    using hit by simp
next
  case (Some ht)
  then obtain  $h\ t$  where  $ht: ht = (h, t)$ 
    by (cases ht) simp
  have hash-out:  $Some\ (h, t) \in set\_dist\ (execute\ (hash\ x)\ s)$ 
    using out Some ht by simp
  have h-in:  $h \in B\ s$ 
    using hit Some ht unfolding state-dependent-output-hit-def by simp
  have lookup-t:  $fmlookup\ (HashMap\ t)\ x = Some\ h$ 
    using protocol-merkle.hash-outcome(2) [OF hash-out] .
  show ?thesis
proof (cases fmlookup (HashMap s) x)

```

```

case None
then have hash-map-new-output-hit (B s) s t
  unfolding hash-map-new-output-hit-def
  using lookup-t h-in by blast
then show ?thesis
  using Some ht
  unfolding state-dependent-new-output-hit-def hash-new-output-hit-event-def
  by simp
next
case (Some y)
then have y = h
using lookup-t hash-extension-lookup[OF Some protocol-merkle.hash-outcome(1)[OF hash-out]]
  by simp
then show ?thesis
  using Some h-in
  unfolding state-dependent-prequeried-hash-hit-def
  by blast
qed
qed

lemma receive-alpha-challenge-state-dependent-hit-fresh-or-prequeried:
assumes out: Some (a, t) ∈ set-dist (execute receive-alpha-challenge s)
and hit: a ∈ B s
shows
  state-dependent-new-output-hit B s (Some (a, t)) ∨
  state-dependent-prequeried-hash-hit B
  (AlphaChallenge (PAlphaCounter s) (PState s)) s
proof –
have recv:
  s ≤ t ∧
  fmlookup (HashMap t)
  (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a
using receive-alpha-challenge-outcome[OF out] by blast
show ?thesis
proof (cases
  fmlookup (HashMap s)
  (AlphaChallenge (PAlphaCounter s) (PState s)))
case None
then have hash-map-new-output-hit (B s) s t
  unfolding hash-map-new-output-hit-def
  using recv hit by blast
then show ?thesis
  unfolding state-dependent-new-output-hit-def
  hash-new-output-hit-event-def
  by simp
next
case (Some old-a)
have old-lookup-t:
  fmlookup (HashMap t)

```

```

      (AlphaChallenge (PAlphaCounter s) (PState s)) = Some old-a
    by (rule hash-extension-lookup[OF Some conjunct1[OF recv]])
  then have old-a = a
    using recv by simp
  then show ?thesis
    using Some hit
    unfolding state-dependent-prequeried-hash-hit-def
    by blast
qed
qed

```

lemma *wp-receive-alpha-challenge-state-dependent-hit-split:*

```

wp-event receive-alpha-challenge
  (λout. case out of None ⇒ False
    | Some (a, -) ⇒ a ∈ B s) s ≤
wp-event receive-alpha-challenge
  (state-dependent-new-output-hit B s) s +
(if state-dependent-prequeried-hash-hit B
  (AlphaChallenge (PAlphaCounter s) (PState s)) s
  then 1 else 0)

```

proof –

```

let ?bad =
  λout. case out of None ⇒ False
    | Some (a, -) ⇒ a ∈ B s
let ?fresh = state-dependent-new-output-hit B s
let ?pre =
  state-dependent-prequeried-hash-hit B
  (AlphaChallenge (PAlphaCounter s) (PState s)) s
have wp-event receive-alpha-challenge ?bad s ≤
  wp-event receive-alpha-challenge
  (λout. ?fresh out ∨ ?pre) s

```

proof (rule wp-event-mono-on-support)

fix out

assume support: out ∈ set-dist (execute receive-alpha-challenge s)

and bad: ?bad out

show ?fresh out ∨ ?pre

proof (cases out)

case None

then show ?thesis

using bad **by** simp

next

case (Some pair)

then obtain a t **where** out-eq: out = Some (a, t)

by (cases pair) auto

have a-in: a ∈ B s

using bad **unfolding** out-eq **by** simp

show ?thesis

unfolding out-eq

by (rule

```

    receive-alpha-challenge-state-dependent-hit-fresh-or-prequeried
    [where B=B and s=s and a=a and t=t, OF - a-in])
    (use support out-eq in simp)
qed
qed
also have ... ≤
  wp-event receive-alpha-challenge ?fresh s +
  wp-event receive-alpha-challenge (λ-. ?pre) s
by (rule wp-event-union-bound)
also have ... ≤
  wp-event receive-alpha-challenge ?fresh s +
  (if ?pre then 1 else 0)
proof (cases ?pre)
case True
have wp-event receive-alpha-challenge (λ-. ?pre) s ≤ 1
by (rule wp-event-le-1)
then show ?thesis
using True by simp
next
case False
then show ?thesis
by (simp add: wp-event-def wp-def)
qed
finally show ?thesis .
qed

```

lemma *wp-event-bind-state-dependent-output-bound-by-cont:*
fixes $m :: ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
and $k :: 'x \Rightarrow ('f, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
assumes *none:* $\neg E \text{ None}$
and *event-imp:*
 $\bigwedge x \ t \text{ out. Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{out} \in \text{set-dist } (\text{execute } (k \ x) \ t) \implies$
 $E \text{ out} \implies \text{state-dependent-output-hit } B \ t \ \text{out}$
and *cont-bound:*
 $\bigwedge x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{wp-event } (k \ x) \ (\text{state-dependent-output-hit } B \ t) \ t \leq C$
shows $\text{wp-event } (m \gg k) \ E \ s \leq C$
proof (rule *wp-event-bind-bound-by-cont*)
[where $Q=E$ and $m=m$ and $k=k$ and $s=s$ and $C=C$]
show $\neg E \text{ None}$
by (rule *none*)
next
fix $x \ t$
assume *head:* $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s)$
have $\text{wp-event } (k \ x) \ E \ t \leq$
 $\text{wp-event } (k \ x) \ (\text{state-dependent-output-hit } B \ t) \ t$
by (rule *wp-event-mono-on-support*)
(use *head event-imp in blast*)

also have $\dots \leq C$
 by (rule cont-bound[OF head])
 finally show $\text{wp-event } (k \ x) \ E \ t \leq C$.
 qed

lemma *wp-event-bind-state-dependent-new-output-bound-by-cont:*
 fixes $m :: ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
 and $k :: 'x \Rightarrow ('y, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
 assumes none: $\neg E \text{ None}$
 and event-imp:
 $\bigwedge x \ t \ \text{out}. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{out} \in \text{set-dist } (\text{execute } (k \ x) \ t) \implies$
 $E \ \text{out} \implies \text{state-dependent-new-output-hit } B \ t \ \text{out}$
 and cont-bound:
 $\bigwedge x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{wp-event } (k \ x) \ (\text{state-dependent-new-output-hit } B \ t) \ t \leq C$
 shows $\text{wp-event } (m \gg k) \ E \ s \leq C$
proof (rule wp-event-bind-bound-by-cont
 [where $Q=E$ and $m=m$ and $k=k$ and $s=s$ and $C=C$])
 show $\neg E \text{ None}$
 by (rule none)
 next
 fix $x \ t$
 assume head: $\text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s)$
 have $\text{wp-event } (k \ x) \ E \ t \leq$
 $\text{wp-event } (k \ x) \ (\text{state-dependent-new-output-hit } B \ t) \ t$
 by (rule wp-event-mono-on-support)
 (use head event-imp in blast)
 also have $\dots \leq C$
 by (rule cont-bound[OF head])
 finally show $\text{wp-event } (k \ x) \ E \ t \leq C$.
 qed

lemma *wp-event-bind-state-dependent-new-output-bound-by-target-budget:*
 fixes $m :: ('x, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
 and $k :: 'x \Rightarrow ('y, ('f, 'a) \text{protocol-channel-scheme}) \text{state-monad}$
 assumes none: $\neg E \text{ None}$
 and event-imp:
 $\bigwedge x \ t \ \text{out}. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{out} \in \text{set-dist } (\text{execute } (k \ x) \ t) \implies$
 $E \ \text{out} \implies \text{state-dependent-new-output-hit } B \ t \ \text{out}$
 and budget:
 $\bigwedge x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{hash-target-budget } (B \ t) \ n \ (k \ x)$
 and value-bound:
 $\bigwedge x \ t. \text{Some } (x, t) \in \text{set-dist } (\text{execute } m \ s) \implies$
 $\text{hash-target-budget-value } (B \ t) \ n \leq C$
 shows
 $\text{wp-event } (m \gg k) \ E \ s \leq C$


```

proof (rule wp-event-bind-state-dependent-new-output-bound-by-cont
  [where  $E=E$  and  $m=m$  and  $k=k$  and  $s=s$  and  $B=B$  and  $C=C$ ])
  show  $\neg E \text{ None}$ 
    by (rule none)
next
  show  $\bigwedge x \ t \ out.$ 
     $Some \ (x, t) \in set-dist \ (execute \ m \ s) \implies$ 
     $out \in set-dist \ (execute \ (k \ x) \ t) \implies$ 
     $E \ out \implies state-dependent-new-output-hit \ B \ t \ out$ 
    by (rule event-imp)
next
  fix  $x \ t$ 
  assume  $head: Some \ (x, t) \in set-dist \ (execute \ m \ s)$ 
  have  $wp-event \ (k \ x) \ (state-dependent-new-output-hit \ B \ t) \ t \leq$ 
     $hash-target-budget-value \ (B \ t) \ n$ 
    using  $budget[OF \ head]$ 
    unfolding  $hash-target-budget-def \ state-dependent-new-output-hit-def$ 
    by blast
  also have  $\dots \leq C$ 
    by (rule value-bound[OF  $head$ ])
  finally show
     $wp-event \ (k \ x) \ (state-dependent-new-output-hit \ B \ t) \ t \leq C .$ 
qed

end

end

```

```

theory Staged-Security-Experiment-Alpha-Prequery
imports
  Staged-Security-Experiment-Checked-Prefix
  Soundness-Oracle-Dynamic-Target
begin

```

4.4 Staged Composition Accounting

Alpha-key prequery support for staged soundness.

This layer records deterministic facts about the alpha challenge key path. It is proof infrastructure for accounting adversarial prequeries to future alpha keys; it does not add assumptions or change the oracle model.

```

context soundness
begin

```

```

definition alpha-challenge-key-at
  :: 'f protocol-channel  $\Rightarrow$  nat  $\Rightarrow$  'f list  $\Rightarrow$ 
    'f protocol-hash-input
  where

```

alpha-challenge-key-at $s\ i\ as =$
 $AlphaChallenge\ (PAlphaCounter\ s + i)$
 $(foldl\ concat\ (PState\ s)\ (take\ i\ as))$

definition *alpha-vector-key-relation*

$:: 'f\ protocol-channel \Rightarrow 'f\ list\ set \Rightarrow nat \Rightarrow$
 $'f\ protocol-hash-input \Rightarrow 'f \Rightarrow bool$

where

alpha-vector-key-relation $s\ B\ n\ x\ y \longleftrightarrow$
 $(\exists as \in B. \exists i < n.$
 $i < length\ as \wedge$
 $x = alpha-challenge-key-at\ s\ i\ as \wedge$
 $y = as\ !\ i)$

definition *alpha-vector-prequery-hit-event*

$:: 'f\ protocol-channel \Rightarrow 'f\ list\ set \Rightarrow nat \Rightarrow$
 $('r \times 'f\ protocol-channel)\ option \Rightarrow bool$

where

alpha-vector-prequery-hit-event $s\ B\ n =$
 $hash-relation-hit-event\ (alpha-vector-key-relation\ s\ B\ n)\ s$

definition *alpha-vector-prequeried-in-state*

$:: 'f\ protocol-channel \Rightarrow 'f\ list \Rightarrow nat \Rightarrow bool$

where

alpha-vector-prequeried-in-state $s\ as\ n \longleftrightarrow$
 $(\exists i < n.$
 $i < length\ as \wedge$
 $fmlookup\ (HashMap\ s)\ (alpha-challenge-key-at\ s\ i\ as) =$
 $Some\ (as\ !\ i))$

definition *alpha-vector-path-fresh*

$:: 'f\ protocol-channel \Rightarrow 'f\ list \Rightarrow nat \Rightarrow bool$

where

alpha-vector-path-fresh $s\ as\ n \longleftrightarrow$
 $(\forall i < n.$
 $i < length\ as \longrightarrow$
 $fmlookup\ (HashMap\ s)\ (alpha-challenge-key-at\ s\ i\ as) = None)$

lemma *alpha-vector-key-relation-fiber-card-bound:*

assumes *finite-B:* *finite* B

shows $card\ \{y. alpha-vector-key-relation\ s\ B\ n\ x\ y\} \leq card\ B * n$

proof –

let $?pairs = B \times \{..<n\}$

let $?values = (\lambda(as, i). as\ !\ i) \text{ ‘ } ?pairs$

have *subset:*

$\{y. alpha-vector-key-relation\ s\ B\ n\ x\ y\} \subseteq ?values$

unfolding *alpha-vector-key-relation-def*

by *force*

have *finite-values:* *finite* $?values$

using *finite-B* by *simp*
 have $\text{card } \{y. \text{alpha-vector-key-relation } s \ B \ n \ x \ y\} \leq \text{card } ?\text{values}$
 by (rule *card-mono*[*OF finite-values subset*])
 also have $\dots \leq \text{card } ?\text{pairs}$
 by (rule *card-image-le*) (use *finite-B* in *simp*)
 also have $\dots = \text{card } B * n$
 using *finite-B* by *simp*
 finally show *?thesis* .
 qed

lemma *hash-relation-program-staged-alpha-prefix-alpha-vector-prequery*:
 assumes *wf*: *staged-budget-wellformed budgets*
 and *controlled*: *staged-adversary-controlled budgets A*
 and *finite-B*: *finite B*
 shows
 hash-relation-program (*alpha-vector-key-relation s B n*) (*card B * n*)
 (*staged-alpha-search-queries budgets 0*)
 (*staged-alpha-prefix-program A*)
 by (rule *hash-relation-program-staged-alpha-prefix-program*
 [*OF wf controlled*])
 (rule *alpha-vector-key-relation-fiber-card-bound*[*OF finite-B*])

lemma *staged-alpha-prefix-alpha-vector-prequery-hit-bound*:
 assumes *wf*: *staged-budget-wellformed budgets*
 and *controlled*: *staged-adversary-controlled budgets A*
 and *finite-B*: *finite B*
 shows
 wp-event (*staged-alpha-prefix-program A*)
 (*alpha-vector-prequery-hit-event s B n*) $s \leq$
 hash-relation-budget-value (*card B * n*)
 (*staged-alpha-search-queries budgets 0*)

proof –
 have *program*:
 hash-relation-program (*alpha-vector-key-relation s B n*) (*card B * n*)
 (*staged-alpha-search-queries budgets 0*)
 (*staged-alpha-prefix-program A*)
 by (rule *hash-relation-program-staged-alpha-prefix-alpha-vector-prequery*
 [*OF wf controlled finite-B*])
 then have *budget*:
 hash-relation-budget (*alpha-vector-key-relation s B n*) (*card B * n*)
 (*staged-alpha-search-queries budgets 0*)
 (*staged-alpha-prefix-program A*)
 unfolding *hash-relation-program-def* by *simp*
 show *?thesis*
 using *budget*
 unfolding *alpha-vector-prequery-hit-event-def*
 hash-relation-budget-def by *blast*
 qed

lemma *staged-alpha-prefix-event-bound-from-hash-relation:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *fibers: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$*
and *event-imp:*
 $\bigwedge x \ t. \text{Some } (x, t) \in$
 $\text{set-dist } (\text{execute } (\text{staged-alpha-prefix-program } A) \ s) \implies$
 $E \ (\text{Some } (x, t)) \implies \text{hash-relation-hit } R \ s \ t$
and *none: $\neg E \ \text{None}$*
shows
 $\text{wp-event } (\text{staged-alpha-prefix-program } A) \ E \ s \leq$
 $\text{hash-relation-budget-value } b \ (\text{staged-alpha-search-queries budgets } 0)$
proof –
have *program:*
 $\text{hash-relation-program } R \ b \ (\text{staged-alpha-search-queries budgets } 0)$
 $(\text{staged-alpha-prefix-program } A)$
by *(rule hash-relation-program-staged-alpha-prefix-program*
 $[\text{OF } wf \ \text{controlled fibers}])$
have
 $\text{wp-event } (\text{staged-alpha-prefix-program } A) \ E \ s \leq$
 $\text{wp-event } (\text{staged-alpha-prefix-program } A)$
 $(\text{hash-relation-hit-event } R \ s) \ s$
proof *(rule wp-event-mono-on-support)*
fix *out*
assume *support:*
 $out \in \text{set-dist } (\text{execute } (\text{staged-alpha-prefix-program } A) \ s)$
and *event: $E \ out$*
show *hash-relation-hit-event $R \ s \ out$*
proof *(cases out)*
case *None*
then show *?thesis using none event by simp*
next
case *(Some xt)*
then obtain *x t where xt: xt = (x, t)*
by *(cases xt) simp*
have *hash-relation-hit $R \ s \ t$*
by *(rule event-imp[of x t]) (use support event Some xt in simp-all)*
then show *?thesis*
unfolding *Some xt hash-relation-hit-event-def by simp*
qed
qed
also have $\dots \leq$
 $\text{hash-relation-budget-value } b \ (\text{staged-alpha-search-queries budgets } 0)$
using *program unfolding hash-relation-program-def hash-relation-budget-def*
by *blast*
finally show *?thesis .*
qed

lemma *alpha-vector-path-fresh-Cons-tail:*

```

assumes fresh: alpha-vector-path-fresh s (a # xs) (Suc n)
and recv:
  Some (a, t) ∈ set-dist (execute receive-alpha-challenge s)
and record-out:
  Some ((), u) ∈ set-dist (execute (record-staged-message a) t)
shows alpha-vector-path-fresh u xs n
proof –
have recv-fields:
  PAlphaCounter t = Suc (PAlphaCounter s)
  PState t = PState s
  fmlookup (HashMap t)
    (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a
using receive-alpha-challenge-outcome[OF recv]
  receive-alpha-challenge-counter-outcome[OF recv]
by simp-all
have u-eq:
  u = t |
    PState := concat (PState t) a,
    PTranscript := PTranscript t @ [a]
by (rule record-staged-message-outcome[OF record-out])
show ?thesis
  unfolding alpha-vector-path-fresh-def
proof (intro allI impI)
  fix i
  assume i-bound: i < n
  and i-len: i < length xs
  let ?tail-key = alpha-challenge-key-at u i xs
  let ?full-key = alpha-challenge-key-at s (Suc i) (a # xs)
  have key-eq: ?tail-key = ?full-key
    unfolding alpha-challenge-key-at-def u-eq
    using recv-fields
    by simp
  have full-none:
    fmlookup (HashMap s) ?full-key = None
    using fresh i-bound i-len
    unfolding alpha-vector-path-fresh-def
    by simp
  have key-ne:
    ?full-key ≠ AlphaChallenge (PAlphaCounter s) (PState s)
    unfolding alpha-challenge-key-at-def by simp
  have lookup-t:
    fmlookup (HashMap t) ?full-key = None
    using receive-alpha-challenge-preserves-other-lookup[OF recv key-ne]
    full-none
    by simp
  show fmlookup (HashMap u) ?tail-key = None
    using lookup-t key-eq unfolding u-eq by simp
qed
qed

```

lemma *wp-staged-alpha-program-exact-challenges-bound-from-path-fresh:*

assumes *len: length as = n*

and *fresh: alpha-vector-path-fresh s as n*

shows

wp-event (staged-alpha-program n)
 $(\lambda out. \text{case out of None} \Rightarrow \text{False}$
 $\quad | \text{Some (as', -)} \Rightarrow \text{as' = as}) s \leq$
 $(1 / \text{nnreal size})^n$

using *len fresh*

proof (*induction n arbitrary: as s*)

case *0*

then have *as-empty: as = []*

by *simp*

show *?case*

unfolding *as-empty* **by** (*simp add: wp-event-def wpsimps*)

next

case (*Suc n*)

obtain *a as-tail* **where** *as-eq: as = a # as-tail*

using *Suc.premis(1)* **by** (*cases as*) *auto*

have *len-tail: length as-tail = n*

using *Suc.premis(1)* *as-eq* **by** *simp*

let *?tail =*

staged-alpha-program n ::
 $(\text{'f list, 'f protocol-channel}) \text{ state-monad}$

let *?Q =*

$\lambda out :: (\text{'f list} \times$
 $\quad \text{'f protocol-channel}) \text{ option.}$
 $\text{case out of None} \Rightarrow \text{False} \mid \text{Some (as', -)} \Rightarrow \text{as' = as}$

let *?Head =*

$\lambda out :: (\text{'f} \times$
 $\quad \text{'f protocol-channel}) \text{ option.}$
 $\text{case out of None} \Rightarrow \text{False} \mid \text{Some (a0, -)} \Rightarrow a0 \in \{a\}$

have *head-fresh:*

fmlookup (HashMap s)
 $(\text{AlphaChallenge (PAlphaCounter s) (PState s)}) = \text{None}$

proof –

have *fmlookup (HashMap s)*
 $(\text{alpha-challenge-key-at s 0 (a # as-tail)}) = \text{None}$

using *Suc.premis(2)*

unfolding *as-eq alpha-vector-path-fresh-def* **by** *simp*

then show *?thesis*

unfolding *alpha-challenge-key-at-def* **by** *simp*

qed

have *head-bound:*

wp-event receive-alpha-challenge ?Head s $\leq 1 / \text{nnreal size}$

proof –

have *wp-event receive-alpha-challenge ?Head s =*
 $\text{nnreal (card (\{a\} :: 'f set))} / \text{nnreal size}$

```

    by (rule wp-receive-alpha-challenge-fresh-set[OF head-fresh])
  then show ?thesis by simp
qed
have bind-bound:
  wp-event
    (receive-alpha-challenge  $\gg$ 
      ( $\lambda a0$ . record-staged-message  $a0$   $\gg$ 
        ( $\lambda$ -. ?tail  $\gg$ 
          ( $\lambda as'$ . return ( $a0 \# as'$ ))))))
  ?Q s  $\leq$ 
  wp-event receive-alpha-challenge ?Head s *
    (1 / nnreal size) ^ n
proof (rule wp-event-bind-bound-by-head-and-cont[where P = ?Head])
  show  $\neg$  ?Q None
  by simp
next
fix a0 t
assume recv:
  Some (a0, t)  $\in$  set-dist (execute receive-alpha-challenge s)
  and not-head:  $\neg$  ?Head (Some (a0, t))
have a0-ne: a0  $\neq$  a
  using not-head by simp
have cont-false:
  wp-event
    (record-staged-message a0  $\gg$ 
      ( $\lambda$ -. ?tail  $\gg$  ( $\lambda as'$ . return ( $a0 \# as'$ ))))
  ?Q t  $\leq$ 
  wp-event
    (record-staged-message a0  $\gg$ 
      ( $\lambda$ -. ?tail  $\gg$  ( $\lambda as'$ . return ( $a0 \# as'$ ))))
    ( $\lambda$  :: ('f list  $\times$  'f protocol-channel) option. False) t
  by (rule wp-event-mono-on-support)
    (use a0-ne as-eq in
       $\langle$ auto simp: wpsimps elim!: set-dist-bindE
        split: option.splits prod.splits $\rangle$ )
also have ... = 0
  unfolding wp-event-def wp-def dist-expect-def by simp
finally have le-zero:
  wp-event
    (record-staged-message a0  $\gg$ 
      ( $\lambda$ -. ?tail  $\gg$  ( $\lambda as'$ . return ( $a0 \# as'$ ))))
  ?Q t  $\leq$  0 .
show
  wp-event
    (record-staged-message a0  $\gg$ 
      ( $\lambda$ -. ?tail  $\gg$  ( $\lambda as'$ . return ( $a0 \# as'$ ))))
  ?Q t = 0
  by (rule antisym[OF le-zero]) simp
next

```

```

fix a0 t
assume recv:
  Some (a0, t) ∈ set-dist (execute receive-alpha-challenge s)
  and head: ?Head (Some (a0, t))
have a0-eq: a0 = a
  using head by simp
have record-cont-bound:
  wp-event
  (record-staged-message a0  $\gg$ 
    ( $\lambda$ -. ?tail  $\gg$  ( $\lambda$ as'. return (a0 # as'))))
  ?Q t ≤ (1 / nnreal size) ^ n
proof (rule wp-event-bind-bound-by-cont)
  show  $\neg$  ?Q None
  by simp
next
fix uu u
assume record-out:
  Some (uu, u) ∈
    set-dist (execute (record-staged-message a0) t)
have record-unit:
  Some ((), u) ∈
    set-dist (execute (record-staged-message a0) t)
  using record-out by (cases uu) simp
have tail-fresh: alpha-vector-path-fresh u as-tail n
proof -
  have fresh-cons:
    alpha-vector-path-fresh s (a # as-tail) (Suc n)
    using Suc.premis(2) unfolding as-eq .
  have recv-a:
    Some (a, t) ∈ set-dist (execute receive-alpha-challenge s)
    using recv a0-eq by simp
  have record-a:
    Some ((), u) ∈ set-dist (execute (record-staged-message a) t)
    using record-unit a0-eq by simp
  show ?thesis
  by (rule alpha-vector-path-fresh-Cons-tail
    [OF fresh-cons recv-a record-a])
qed
have tail-bound:
  wp-event ?tail
  ( $\lambda$ out. case out of None  $\Rightarrow$  False
    | Some (as', -)  $\Rightarrow$  as' = as-tail) u
  ≤ (1 / nnreal size) ^ n
  by (rule Suc.IH[OF len-tail tail-fresh])
have cont-eq:
  wp-event
  (?tail  $\gg$  ( $\lambda$ as'. return (a0 # as')) ?Q u =
  wp-event ?tail
  ( $\lambda$ out. case out of None  $\Rightarrow$  False

```



```

    | Some (as', -) ⇒ as' = as-tail) u
  by (simp add: wp-event-bind-return-map a0-eq as-eq
    split: option.splits prod.splits)
show
  wp-event (?tail ≫ (λas'. return (a0 # as'))) ?Q u
    ≤ (1 / nnreal size) ^ n
  unfolding cont-eq by (rule tail-bound)
qed
show
  wp-event
    (record-staged-message a0 ≫
      (λ-. ?tail ≫ (λas'. return (a0 # as'))))
    ?Q t ≤ (1 / nnreal size) ^ n
  by (rule record-cont-bound)
qed
have wp-event (staged-alpha-program (Suc n)) ?Q s
  ≤ wp-event receive-alpha-challenge ?Head s *
    (1 / nnreal size) ^ n
  unfolding staged-alpha-program.simps by (rule bind-bound)
also have ... ≤ (1 / nnreal size) * (1 / nnreal size) ^ n
  by (rule mult-right-mono[OF head-bound]) simp
also have ... = (1 / nnreal size) ^ Suc n
  by simp
finally show ?case .
qed

lemma wp-staged-alpha-program-finite-set-bound-from-path-fresh:
  assumes finite-B: finite B
  and lengths: ⋀as. as ∈ B ⇒ length as = n
  and fresh: ⋀as. as ∈ B ⇒ alpha-vector-path-fresh s as n
  shows
    wp-event (staged-alpha-program n)
      (λout. case out of None ⇒ False
        | Some (as, -) ⇒ as ∈ B) s ≤
      nnreal (card B) / (nnreal size) ^ n
proof -
  let ?P =
    λa out. case out of None ⇒ False
      | Some (as', -) ⇒ as' = a
  have event-mono:
    wp-event (staged-alpha-program n)
      (λout. case out of None ⇒ False
        | Some (as, -) ⇒ as ∈ B) s ≤
    wp-event (staged-alpha-program n)
      (λout. ∃ as∈B. ?P as out) s
  by (rule wp-event-mono) (auto split: option.splits prod.splits)
also have ... ≤
  (∑ a∈B. wp-event (staged-alpha-program n) (?P a) s)
  by (rule wp-event-finite-UN-bound[where A = B and P = ?P])

```

```

    (use finite-B in simp-all)
  also have ... ≤ (∑ as∈B. (1 / nnreal size) ^ n)
  proof (rule sum-mono)
    fix as
    assume as-in: as ∈ B
    show
      wp-event (staged-alpha-program n) (?P as) s ≤
        (1 / nnreal size) ^ n
    by (rule wp-staged-alpha-program-exact-challenges-bound-from-path-fresh
        [OF lengths[OF as-in] fresh[OF as-in]])
  qed
  also have ... = nnreal (card B) * ((1 / nnreal size) ^ n)
    using finite-B by simp
  also have ... = nnreal (card B) / (nnreal size) ^ n
    using size-card by transfer (simp add: power-divide)
  finally show ?thesis .
qed

lemma alpha-vector-prequeried-in-state-imp-hash-relation-hit:
  assumes prequeried: alpha-vector-prequeried-in-state s as n
  and member: as ∈ B
  and initial-absent:
    ∧i. i < n ⇒ i < length as ⇒
      fmllookup (HashMap initial) (alpha-challenge-key-at s i as) = None
  shows hash-relation-hit (alpha-vector-key-relation s B n) initial s
proof -
  from prequeried obtain i where
    i-n: i < n
  and i-len: i < length as
  and lookup:
    fmllookup (HashMap s) (alpha-challenge-key-at s i as) =
      Some (as ! i)
  unfolding alpha-vector-prequeried-in-state-def by blast
  have relation:
    alpha-vector-key-relation s B n (alpha-challenge-key-at s i as)
      (as ! i)
  unfolding alpha-vector-key-relation-def
  using member i-n i-len by blast
  show ?thesis
    unfolding hash-relation-hit-def
    using initial-absent[OF i-n i-len] lookup relation by blast
qed

lemma alpha-vector-prequeried-in-state-imp-hash-relation-hit-empty-initial:
  assumes prequeried: alpha-vector-prequeried-in-state s as n
  and member: as ∈ B
  and empty: HashMap initial = fmempty
  shows hash-relation-hit (alpha-vector-key-relation s B n) initial s
  by (rule alpha-vector-prequeried-in-state-imp-hash-relation-hit

```

[OF prequeried member])
 (simp add: empty)

lemma staged-alpha-program-outcome-lookup:

assumes outcome:

Some (as, t) ∈ set-dist (execute (staged-alpha-program n) s)

and i-bound: i < n

shows

fmlookup (HashMap t)

(AlphaChallenge (PAlphaCounter s + i)

(foldl concat (PState s) (take i as))) = Some (as ! i)

using outcome i-bound

proof (induction n arbitrary: as s t i)

case 0

then show ?case **by** simp

next

case (Suc n)

from staged-alpha-program-Suc-outcomeE[OF Suc.premis(1)]

obtain a as' s1 s2 **where**

challenge-out:

Some (a, s1) ∈ set-dist (execute receive-alpha-challenge s)

and record-out:

Some ((), s2) ∈ set-dist (execute (record-staged-message a) s1)

and rest-out:

Some (as', t) ∈

set-dist (execute (staged-alpha-program n) s2)

and as-eq: as = a # as'

.

have challenge-lookup:

fmlookup (HashMap s1)

(AlphaChallenge (PAlphaCounter s) (PState s)) = Some a

using receive-alpha-challenge-outcome[OF challenge-out] **by** blast

have challenge-counter:

PAlphaCounter s1 = Suc (PAlphaCounter s)

using receive-alpha-challenge-counter-outcome[OF challenge-out] **by** simp

have record-eq:

s2 = s1 |

PState := concat (PState s1) a,

PTranscript := PTranscript s1 @ [a]

by (rule record-staged-message-outcome[OF record-out])

have s2-fields:

HashMap s2 = HashMap s1

PState s2 = concat (PState s) a

PAlphaCounter s2 = Suc (PAlphaCounter s)

unfolding record-eq

using receive-alpha-challenge-outcome[OF challenge-out]

challenge-counter

by simp-all

have rest-ext: s2 ≤ t

```

proof –
  have hash-target-program {} n (staged-alpha-program n)
    by (rule hash-target-program-staged-alpha-program)
  then show ?thesis
    using rest-out
    unfolding hash-target-program-def hash-extension-preserving-def
    by blast
qed
show ?case
proof (cases i)
  case 0
  have lookup-s2:
    fmlookup (HashMap s2)
      (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a
    using challenge-lookup unfolding s2-fields by simp
  have lookup-t:
    fmlookup (HashMap t)
      (AlphaChallenge (PAlphaCounter s) (PState s)) = Some a
    by (rule hash-extension-lookup[OF lookup-s2 rest-ext])
  then show ?thesis
    using 0 as-eq by simp
next
  case (Suc j)
  then have j-bound: j < n
    using Suc.premis(2) by simp
  have rest-lookup:
    fmlookup (HashMap t)
      (AlphaChallenge (PAlphaCounter s2 + j)
        (foldl concat (PState s2) (take j as'))) = Some (as' ! j)
    by (rule Suc.IH[OF rest-out j-bound])
  show ?thesis
    using rest-lookup
    unfolding Suc as-eq s2-fields
    by simp
qed
qed

lemma staged-alpha-program-outcome-not-path-fresh-imp-prequeried:
  assumes outcome:
    Some (as, t) ∈ set-dist (execute (staged-alpha-program n) s)
    and not-fresh:  $\neg$  alpha-vector-path-fresh s as n
  shows alpha-vector-prequeried-in-state s as n
proof –
  from not-fresh obtain i old where
    i-bound: i < n
    and i-len: i < length as
    and lookup-s:
      fmlookup (HashMap s) (alpha-challenge-key-at s i as) = Some old
    unfolding alpha-vector-path-fresh-def

```

```

    by (metis option.exhaust)
  have lookup-t:
    fmlookup (HashMap t) (alpha-challenge-key-at s i as) =
      Some (as ! i)
    using staged-alpha-program-outcome-lookup[OF outcome i-bound]
    unfolding alpha-challenge-key-at-def .
  have ext:  $s \leq t$ 
  proof -
    have hash-target-program {} n (staged-alpha-program n)
      by (rule hash-target-program-staged-alpha-program)
    then show ?thesis
      using outcome
      unfolding hash-target-program-def hash-extension-preserving-def
      by blast
  qed
  have lookup-t-old:
    fmlookup (HashMap t) (alpha-challenge-key-at s i as) = Some old
    by (rule hash-extension-lookup[OF lookup-s ext])
  have old-eq: old = as ! i
    using lookup-t lookup-t-old by simp
  show ?thesis
    unfolding alpha-vector-prequeried-in-state-def
    using i-bound i-len lookup-s old-eq by blast
  qed

lemma wp-staged-alpha-program-set-bound-from-path-fresh-or-prequery:
  assumes subset:  $B \subseteq \text{alpha-space}$ 
  shows
    wp-event (staged-alpha-program (length spec))
      ( $\lambda \text{out. case out of None} \Rightarrow \text{False}$ 
        |  $\text{Some (as, -)} \Rightarrow \text{as} \in B$ )  $s \leq$ 
      nnreal (card B) / nnreal (CARD('f) ^ length spec) +
      (if  $\exists \text{as} \in B.$ 
        alpha-vector-prequeried-in-state s as (length spec)
        then 1 else 0)
  proof (cases  $\exists \text{as} \in B.$ 
    alpha-vector-prequeried-in-state s as (length spec))
  case True
  have wp-event (staged-alpha-program (length spec))
    ( $\lambda \text{out. case out of None} \Rightarrow \text{False}$ 
      |  $\text{Some (as, -)} \Rightarrow \text{as} \in B$ )  $s \leq 1$ 
    by (rule wp-event-le-1)
  also have ...  $\leq$ 
    nnreal (card B) / nnreal (CARD('f) ^ length spec) + 1
    by simp
  finally show ?thesis
    using True by simp
next
case False

```

```

let ?BF =
  {as ∈ B. alpha-vector-path-fresh s as (length spec)}
have finite-B: finite B
  by (rule finite-subset[OF subset])
  (simp add: alpha-space-def finite-length-lists-UNIV)
have finite-BF: finite ?BF
  using finite-B by simp
have lengths:
  ∧as. as ∈ ?BF ⇒ length as = length spec
  using subset unfolding alpha-space-def by auto
have fresh:
  ∧as. as ∈ ?BF ⇒
    alpha-vector-path-fresh s as (length spec)
  by simp
have event-mono:
  wp-event (staged-alpha-program (length spec))
    (λout. case out of None ⇒ False
      | Some (as, -) ⇒ as ∈ B) s ≤
  wp-event (staged-alpha-program (length spec))
    (λout. case out of None ⇒ False
      | Some (as, -) ⇒ as ∈ ?BF) s
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈ set-dist
    (execute (staged-alpha-program (length spec)) s)
  and hit:
    (case out of None ⇒ False
      | Some (as, -) ⇒ as ∈ B)
  show
    (case out of None ⇒ False
      | Some (as, -) ⇒ as ∈ ?BF)
proof (cases out)
  case None
  then show ?thesis using hit by simp
next
  case (Some result)
  then obtain as t where out-eq: out = Some (as, t)
    by (cases result) simp
  have as-in: as ∈ B
    using hit unfolding out-eq by simp
  have outcome:
    Some (as, t) ∈
    set-dist (execute (staged-alpha-program (length spec)) s)
    using support unfolding out-eq .
  have path-fresh:
    alpha-vector-path-fresh s as (length spec)
proof (rule ccontr)
  assume ¬ alpha-vector-path-fresh s as (length spec)

```

```

then have alpha-vector-prequeried-in-state s as (length spec)
  by (rule
    staged-alpha-program-outcome-not-path-fresh-imp-prequeried
    [OF outcome])
then show False
  using False as-in by blast
qed
show ?thesis
  unfolding out-eq using as-in path-fresh by simp
qed
qed
also have ...  $\leq$ 
  nnreal (card ?BF) / (nnreal size) ^ length spec
by (rule wp-staged-alpha-program-finite-set-bound-from-path-fresh
  [OF finite-BF lengths fresh])
also have ...  $\leq$  nnreal (card B) / (nnreal size) ^ length spec
proof -
  have card ?BF  $\leq$  card B
    by (rule card-mono[OF finite-B]) blast
  then show ?thesis
    apply (subst nn2real-le-iff[symmetric])
    by (simp add: divide-right-mono)
qed
also have ... =
  nnreal (card B) / nnreal (CARD('f) ^ length spec)
  using size-card by simp
finally show ?thesis
  using False by simp
qed

lemma wp-staged-alpha-program-set-bound-excluding-prequeried:
assumes subset: B  $\subseteq$  alpha-space
shows
  wp-event (staged-alpha-program (length spec))
  (λout. case out of None  $\Rightarrow$  False
    | Some (as, -)  $\Rightarrow$ 
      as  $\in$  B  $\wedge$ 
       $\neg$  alpha-vector-prequeried-in-state s as (length spec))
  s  $\leq$  nnreal (card B) / nnreal (CARD('f) ^ length spec)
proof -
let ?BF =
  {as  $\in$  B. alpha-vector-path-fresh s as (length spec)}
have finite-B: finite B
by (rule finite-subset[OF subset])
  (simp add: alpha-space-def finite-length-lists-UNIV)
have finite-BF: finite ?BF
using finite-B by simp
have lengths:
   $\bigwedge as. as \in ?BF \implies length\ as = length\ spec$ 

```

```

using subset unfolding alpha-space-def by auto
have fresh:
   $\bigwedge as. as \in ?BF \implies$ 
    alpha-vector-path-fresh  $s$   $as$  (length spec)
by simp
have event-mono:
  wp-event (staged-alpha-program (length spec))
    ( $\lambda out. case\ out\ of\ None \Rightarrow False$ 
      |  $Some\ (as, -) \Rightarrow$ 
         $as \in B \wedge$ 
         $\neg$  alpha-vector-prequeried-in-state  $s$   $as$  (length spec))
     $s \leq$ 
    wp-event (staged-alpha-program (length spec))
      ( $\lambda out. case\ out\ of\ None \Rightarrow False$ 
        |  $Some\ (as, -) \Rightarrow as \in ?BF$ )  $s$ 
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out  $\in$  set-dist
  (execute (staged-alpha-program (length spec))  $s$ )
and hit:
  ( $case\ out\ of\ None \Rightarrow False$ 
    |  $Some\ (as, -) \Rightarrow$ 
       $as \in B \wedge$ 
       $\neg$  alpha-vector-prequeried-in-state  $s$   $as$  (length spec))
show
  ( $case\ out\ of\ None \Rightarrow False$ 
    |  $Some\ (as, -) \Rightarrow as \in ?BF$ )
proof (cases out)
case None
  then show ?thesis using hit by simp
next
case (Some result)
then obtain  $as\ t$  where out-eq: out = Some ( $as, t$ )
  by (cases result) simp
have as-in:  $as \in B$ 
and not-prequeried:
   $\neg$  alpha-vector-prequeried-in-state  $s$   $as$  (length spec)
using hit unfolding out-eq by simp-all
have outcome:
  Some ( $as, t$ )  $\in$ 
    set-dist (execute (staged-alpha-program (length spec))  $s$ )
using support unfolding out-eq .
have path-fresh:
  alpha-vector-path-fresh  $s$   $as$  (length spec)
proof (rule ccontr)
assume  $\neg$  alpha-vector-path-fresh  $s$   $as$  (length spec)
then have alpha-vector-prequeried-in-state  $s$   $as$  (length spec)
  by (rule

```



```

      staged-alpha-program-outcome-not-path-fresh-imp-prequeried
      [OF outcome])
    then show False using not-prequeried by contradiction
  qed
  show ?thesis
    unfolding out-eq using as-in path-fresh by simp
  qed
  qed
  also have ... ≤
    nnreal (card ?BF) / (nnreal size) ^ length spec
  by (rule wp-staged-alpha-program-finite-set-bound-from-path-fresh
      [OF finite-BF lengths fresh])
  also have ... ≤ nnreal (card B) / (nnreal size) ^ length spec
  proof -
    have card ?BF ≤ card B
      by (rule card-mono[OF finite-B]) blast
    then show ?thesis
      apply (subst nn2real-le-iff[symmetric])
      by (simp add: divide-right-mono)
    qed
  also have ... =
    nnreal (card B) / nnreal (CARD('f) ^ length spec)
  using size-card by simp
  finally show ?thesis .
  qed

lemma staged-after-alpha-prefix-program-alpha-list-bound-from-path-fresh-or-prequery:
  assumes subset: B ⊆ alpha-space
  shows
    wp-event
      (staged-after-alpha-prefix-program A
        (fr, trace-roots, trace-bs, trace-final))
      (staged-transcript-alpha-list-hit B) s ≤
      nnreal (card B) / nnreal (CARD('f) ^ length spec) +
      (if ∃ as ∈ B.
        alpha-vector-prequeried-in-state s as (length spec)
        then 1 else 0)
  proof -
    have alpha-bound:
      wp-event (staged-alpha-program (length spec))
        (λout. case out of None ⇒ False
          | Some (as, -) ⇒ as ∈ B) s ≤
      nnreal (card B) / nnreal (CARD('f) ^ length spec) +
      (if ∃ as ∈ B.
        alpha-vector-prequeried-in-state s as (length spec)
        then 1 else 0)
    by (rule wp-staged-alpha-program-set-bound-from-path-fresh-or-prequery
        [OF subset])
  show ?thesis

```

```

    by (rule staged-after-alpha-prefix-program-alpha-list-bound-from-alpha-program
        [OF alpha-bound])
qed
end
end

```

```

theory Staged-Security-Experiment-Composition
imports
    Staged-Security-Experiment-Checked
    Staged-Security-Experiment-Alpha-Prequery
begin

```

Prefix-fixed composition alpha targets for staged soundness.

```

context soundness
begin

```

```

definition checked-staged-after-alpha-prefix-program
:: 'f staged-adversary  $\Rightarrow$ 
    ('f  $\times$  'f list  $\times$  'f list  $\times$  'f)  $\Rightarrow$ 
    ('f staged-proof-data, 'f protocol-channel) state-monad
where
checked-staged-after-alpha-prefix-program A prefix =
    (case prefix of (fr, trace-roots, trace-bs, trace-final)  $\Rightarrow$ 
        do {
            as  $\leftarrow$  staged-alpha-program (length spec);
            dg  $\leftarrow$  degree-stage A as;
            record-staged-message dg;
            let composition-rounds = ceil-log (to-nat dg + 1);
            assert (composition-rounds  $\leq$  ceil-log (maxDegree + 1));
            (composition-roots, composition-bs)  $\leftarrow$ 
                staged-composition-fri-program A dg 0 composition-rounds [];
            composition-final  $\leftarrow$ 
                composition-final-stage A dg composition-bs;
            record-staged-message composition-final;
            query-chunks  $\leftarrow$ 
                checked-staged-query-program A trace-roots composition-roots
                    0 rounds;
            return
                (| staged-trace-root = fr,
                  staged-trace-fri-roots = trace-roots,
                  staged-trace-fri-challenges = trace-bs,
                  staged-trace-final = trace-final,
                  staged-alphas = as,
                  staged-degree = dg,
                  staged-composition-fri-roots = composition-roots,
                  staged-composition-fri-challenges = composition-bs,

```

```

    staged-composition-final = composition-final,
    staged-query-chunks = query-chunks)
})

```

lemma *checked-staged-transcript-program-alpha-prefix-decomp:*

```

checked-staged-transcript-program A =
  staged-alpha-prefix-program A  $\gg$ 
  checked-staged-after-alpha-prefix-program A
unfolding checked-staged-transcript-program-def
  staged-alpha-prefix-program-def
  checked-staged-after-alpha-prefix-program-def Let-def
by (simp add: sm-bind-assoc split-def split: prod.splits)

```

lemma *sm-bind-get-ignore:*

```

(get  $\gg$  ( $\lambda\cdot$ . m)) = m
apply transfer
apply (rule ext)
apply (rule dist-inject [THEN iffD1])
unfolding dist-bind.rep-eq dist-get-def
apply (simp add: o-def dist-delta-dist)
apply (subst map-bind-delta-left-state [where f =  $\lambda s$ . Some (s, s)])
apply (auto simp: bind-cont-map-def split: option.splits prod.splits)
done

```

lemma *checked-staged-after-alpha-prefix-program-alpha-list-bound-from-alpha-program:*

```

assumes alpha-bound:
  wp-event (staged-alpha-program (length spec))
  ( $\lambda out$ . case out of None  $\Rightarrow$  False
    | Some (as, -)  $\Rightarrow as \in B$ ) s  $\leq C$ 
shows
  wp-event (checked-staged-after-alpha-prefix-program A prefix)
  (staged-transcript-alpha-list-hit B) s  $\leq C$ 

```

proof –

```

obtain fr trace-roots trace-bs trace-final where prefix-eq:
  prefix = (fr, trace-roots, trace-bs, trace-final)
by (cases prefix) auto
let ?P =
   $\lambda out$ . case out of None  $\Rightarrow$  False
    | Some (as, -)  $\Rightarrow as \in B$ 
have after-eq:
  checked-staged-after-alpha-prefix-program A prefix =
  staged-alpha-program (length spec)  $\gg$ 
  ( $\lambda as$ . degree-stage A as  $\gg$ 
    ( $\lambda dg$ . record-staged-message dg  $\gg$ 
      ( $\lambda\cdot$ . let composition-rounds = ceil-log (to-nat dg + 1)
        in assert
          (composition-rounds  $\leq$  ceil-log (maxDegree + 1))  $\gg$ 
          ( $\lambda\cdot$ . staged-composition-fri-program A dg 0
            composition-rounds []  $\gg$ 

```

```

    (λ(composition-roots, composition-bs).
      composition-final-stage A dg composition-bs ≫=
      (λcomposition-final.
        record-staged-message composition-final ≫=
        (λ-. checked-staged-query-program A trace-roots
          composition-roots 0 rounds ≫=
          (λquery-chunks.
            return
            (λ(staged-trace-root = fr,
              staged-trace-fri-roots = trace-roots,
              staged-trace-fri-challenges = trace-bs,
              staged-trace-final = trace-final,
              staged-alphas = as,
              staged-degree = dg,
              staged-composition-fri-roots =
                composition-roots,
              staged-composition-fri-challenges =
                composition-bs,
              staged-composition-final = composition-final,
              staged-query-chunks =
                query-chunks)))))))))
  unfolding checked-staged-after-alpha-prefix-program-def prefix-eq Let-def
  by simp
show ?thesis
  unfolding after-eq staged-transcript-alpha-list-hit-def
proof (rule wp-event-bind-bound-by-head-event[OF alpha-bound])
  show (case None of None ⇒ False
    | Some (data, -) ⇒ staged-alphas data ∈ B) ⇒
    ?P None
  by simp
next
fix as t out
assume cont:
  out ∈
  set-dist
  (execute
    (degree-stage A as ≫=
      (λdg. record-staged-message dg ≫=
        (λ-. let composition-rounds =
          ceil-log (to-nat dg + 1)
        in assert
          (composition-rounds ≤ ceil-log (maxDegree + 1)) ≫=
          (λ-. staged-composition-fri-program A dg 0
            composition-rounds [] ≫=
            (λ(composition-roots, composition-bs).
              composition-final-stage A dg composition-bs ≫=
              (λcomposition-final.
                record-staged-message composition-final ≫=
                (λ-. checked-staged-query-program A trace-roots

```

```

composition-roots 0 rounds >>=
(λquery-chunks.
  return
    (λ(staged-trace-root = fr,
      staged-trace-fri-roots = trace-roots,
      staged-trace-fri-challenges = trace-bs,
      staged-trace-final = trace-final,
      staged-alphas = as,
      staged-degree = dg,
      staged-composition-fri-roots =
        composition-roots,
      staged-composition-fri-challenges =
        composition-bs,
      staged-composition-final = composition-final,
      staged-query-chunks =
        query-chunks))))))
t)
and hit:
  (case out of None ⇒ False
   | Some (data, -) ⇒ staged-alphas data ∈ B)
show ?P (Some (as, t))
proof (cases out)
  case None
  then show ?thesis using hit by simp
next
  case (Some result)
  then obtain data u where out-eq: out = Some (data, u)
  by (cases result) simp
  have data-as: staged-alphas data = as
  using cont unfolding out-eq Let-def
  by (auto elim!: set-dist-bindE split: prod.splits)
  have staged-alphas data ∈ B
  using hit unfolding out-eq by simp
  then show ?thesis
  using data-as by simp
qed
qed
qed

lemma checked-staged-after-alpha-prefix-program-alpha-list-bound-from-path-fresh-or-prequery:
  assumes subset: B ⊆ alpha-space
  shows
    wp-event (checked-staged-after-alpha-prefix-program A prefix)
      (staged-transcript-alpha-list-hit B) s ≤
      nnreal (card B) / nnreal (CARD('f) ^ length spec) +
      (if ∃ as ∈ B.
        alpha-vector-prequeried-in-state s as (length spec)
        then 1 else 0)
proof -

```

```

have alpha-bound:
  wp-event (staged-alpha-program (length spec))
    ( $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
      |  $\text{Some } (as, -) \Rightarrow as \in B$ )  $s \leq$ 
       $nnreal \text{ (card } B) / nnreal \text{ (CARD('f) } \wedge \text{length spec) +}$ 
      ( $\text{if } \exists as \in B.$ 
         $\alpha\text{-vector-prequeried-in-state } s \text{ as (length spec)}$ 
         $\text{then } 1 \text{ else } 0$ )
    by (rule wp-staged-alpha-program-set-bound-from-path-fresh-or-prequery
      [OF subset])
show ?thesis
  by (rule
    checked-staged-after-alpha-prefix-program-alpha-list-bound-from-alpha-program
    [OF alpha-bound])
qed

lemma checked-staged-after-alpha-prefix-program-alpha-list-bound-excluding-prequeried:
assumes subset:  $B \subseteq \alpha\text{-space}$ 
shows
  wp-event (checked-staged-after-alpha-prefix-program A prefix)
    ( $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
      |  $\text{Some } (data, -) \Rightarrow$ 
         $\text{staged-alphas } data \in B \wedge$ 
         $\neg \alpha\text{-vector-prequeried-in-state } s$ 
         $\text{(staged-alphas } data) \text{ (length spec)}$ )
       $s \leq nnreal \text{ (card } B) / nnreal \text{ (CARD('f) } \wedge \text{length spec)}$ 
proof –
  obtain fr trace-roots trace-bs trace-final where prefix-eq:
    prefix = (fr, trace-roots, trace-bs, trace-final)
  by (cases prefix) auto
let ?P =
   $\lambda out. \text{case out of None} \Rightarrow \text{False}$ 
  |  $\text{Some } (as, -) \Rightarrow$ 
     $as \in B \wedge$ 
     $\neg \alpha\text{-vector-prequeried-in-state } s \text{ as (length spec)}$ 
have alpha-bound:
  wp-event (staged-alpha-program (length spec))
    ?P  $s \leq nnreal \text{ (card } B) / nnreal \text{ (CARD('f) } \wedge \text{length spec)}$ 
  by (rule wp-staged-alpha-program-set-bound-excluding-prequeried
    [OF subset])
have after-eq:
  checked-staged-after-alpha-prefix-program A prefix =
    staged-alpha-program (length spec)  $\gg=$ 
    ( $\lambda as. \text{degree-stage } A \text{ as } \gg=$ 
      ( $\lambda dg. \text{record-staged-message } dg \gg=$ 
        ( $\lambda -. \text{let composition-rounds} = \text{ceil-log (to-nat } dg + 1)$ 
          in assert
            ( $\text{composition-rounds} \leq \text{ceil-log (maxDegree} + 1)$ )  $\gg=$ 
            ( $\lambda -. \text{staged-composition-fri-program } A \text{ dg } 0$ ))))

```

```

composition-rounds [] >>=
(λ(composition-roots, composition-bs).
  composition-final-stage A dg composition-bs >>=
  (λcomposition-final.
    record-staged-message composition-final >>=
    (λ-. checked-staged-query-program A trace-roots
      composition-roots 0 rounds >>=
      (λquery-chunks.
        return
        (λ(staged-trace-root = fr,
          staged-trace-fri-roots = trace-roots,
          staged-trace-fri-challenges = trace-bs,
          staged-trace-final = trace-final,
          staged-alphas = as,
          staged-degree = dg,
          staged-composition-fri-roots =
            composition-roots,
          staged-composition-fri-challenges =
            composition-bs,
          staged-composition-final = composition-final,
          staged-query-chunks =
            query-chunks)))))))))
unfolding checked-staged-after-alpha-prefix-program-def prefix-eq Let-def
by simp
show ?thesis
unfolding after-eq
proof (rule wp-event-bind-bound-by-head-event[OF alpha-bound])
show (case None of None ⇒ False
  | Some (data, -) ⇒
    staged-alphas data ∈ B ∧
    ¬ alpha-vector-prequeried-in-state s
    (staged-alphas data) (length spec)) ⇒
  ?P None
by simp
next
fix as t out
assume cont:
  out ∈
  set-dist
  (execute
    (degree-stage A as >>=
      (λdg. record-staged-message dg >>=
        (λ-. let composition-rounds =
          ceil-log (to-nat dg + 1)
          in assert
            (composition-rounds ≤ ceil-log (maxDegree + 1)) >>=
            (λ-. staged-composition-fri-program A dg 0
              composition-rounds [] >>=
              (λ(composition-roots, composition-bs).

```

```

composition-final-stage A dg composition-bs >>=
(λcomposition-final.
  record-staged-message composition-final >>=
  (λ-. checked-staged-query-program A trace-roots
    composition-roots 0 rounds >>=
    (λquery-chunks.
      return
        (|staged-trace-root = fr,
          staged-trace-fri-roots = trace-roots,
          staged-trace-fri-challenges = trace-bs,
          staged-trace-final = trace-final,
          staged-alphas = as,
          staged-degree = dg,
          staged-composition-fri-roots =
            composition-roots,
          staged-composition-fri-challenges =
            composition-bs,
          staged-composition-final = composition-final,
          staged-query-chunks =
            query-chunks|))))))
t)
and hit:
  (case out of None ⇒ False
   | Some (data, -) ⇒
     staged-alphas data ∈ B ∧
     ¬ alpha-vector-prequeried-in-state s
     (staged-alphas data) (length spec))
show ?P (Some (as, t))
proof (cases out)
  case None
  then show ?thesis using hit by simp
next
  case (Some result)
  then obtain data u where out-eq: out = Some (data, u)
  by (cases result) simp
have data-as: staged-alphas data = as
  using cont unfolding out-eq Let-def
  by (auto elim!: set-dist-bindE split: prod.splits)
have staged-alphas data ∈ B
  ¬ alpha-vector-prequeried-in-state s
  (staged-alphas data) (length spec)
  using hit unfolding out-eq by simp-all
then show ?thesis
  using data-as by simp
qed
qed
qed

```

definition *checked-staged-transcript-with-alpha-prefix-program*


```

:: 'f staged-adversary ⇒
  (((('f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data,
    'f protocol-channel) state-monad
where
  checked-staged-transcript-with-alpha-prefix-program A =
    staged-alpha-prefix-program A ≫=
      (λprefix. get ≫=
        (λprefix-state.
          checked-staged-after-alpha-prefix-program A prefix ≫=
            (λdata. return ((prefix, prefix-state), data))))))

```

definition *checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

```

:: 'f staged-adversary ⇒
  ((((((('f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list,
    'f protocol-channel) state-monad
where

```

```

  checked-staged-security-experiment-with-actual-alpha-prefix-data-state A =
    checked-staged-transcript-with-alpha-prefix-program A ≫=
      (λpacked. get ≫=
        (λs.
          put
            (verifier-state-from-adversary s
              (staged-proof-transcript (snd packed))) ≫=
            (λ-.
              verify-monad ≫=
                (λresult. return ((packed, s), result))))))

```

lemma *checked-staged-transcript-with-alpha-prefix-program-projection:*

```

checked-staged-transcript-with-alpha-prefix-program A ≫=
  (λpacked. return (snd packed)) =
  checked-staged-transcript-program A

```

unfolding *checked-staged-transcript-with-alpha-prefix-program-def*

```

checked-staged-transcript-program-alpha-prefix-decomp
by (simp add: sm-bind-assoc sm-bind-get-ignore)

```

lemma *checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection:*

```

checked-staged-security-experiment-with-actual-alpha-prefix-data-state A ≫=
  (λx. return ((snd (fst (fst x)), snd (fst x)), snd x)) =
  checked-staged-security-experiment-with-data-state A

```

proof –

```

let ?K =
  λdata. get ≫=
    (λs. put
      (verifier-state-from-adversary s
        (staged-proof-transcript data)) ≫=
      (λ-. verify-monad ≫=

```

```

      (λresult. return ((data, s), result))))
have
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state A ≫=
    (λx. return ((snd (fst (fst x)), snd (fst x)), snd x)) =
    checked-staged-transcript-with-alpha-prefix-program A ≫=
    (λpacked. ?K (snd packed))
unfolding
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
by (simp add: sm-bind-assoc split-def)
also have ... =
  (checked-staged-transcript-with-alpha-prefix-program A ≫=
    (λpacked. return (snd packed))) ≫= ?K
by (simp add: sm-bind-assoc)
also have ... =
  checked-staged-transcript-program A ≫= ?K
unfolding checked-staged-transcript-with-alpha-prefix-program-projection
by simp
also have ... =
  checked-staged-security-experiment-with-data-state A
unfolding checked-staged-security-experiment-with-data-state-def by simp
finally show ?thesis .
qed

```

lemma *alpha-prefix-hash-extends-verifier-state-from-adversary-right:*

```

  assumes ext:  $s \leq t$ 
  shows  $s \leq \text{verifier-state-from-adversary } t \text{ tr}$ 
  using ext
  unfolding verifier-state-from-adversary-def less-eq-hash-ext-def
    less-eq-fmap-def
  by simp

```

definition *alpha-prefix-trace-table-candidates*

```

:: 'f protocol-channel  $\Rightarrow$  'f  $\Rightarrow$  'f list set
where
  alpha-prefix-trace-table-candidates prefix-state fr =
    {trace-table.
      length trace-table = clength * scale  $\wedge$ 
      merkle-root-binds-table fr trace-table prefix-state}

```

lemma *alpha-prefix-trace-table-candidateI:*

```

  assumes len: length trace-table = clength * scale
  and bind: merkle-root-binds-table fr trace-table prefix-state
  shows trace-table  $\in$  alpha-prefix-trace-table-candidates prefix-state fr
  using assms unfolding alpha-prefix-trace-table-candidates-def by simp

```

definition *alpha-prefix-union-bad-sets*

```

:: 'f protocol-channel  $\Rightarrow$  ('f list  $\Rightarrow$  'f list set)  $\Rightarrow$ 
  'f  $\Rightarrow$  'f list set
where

```

alpha-prefix-union-bad-sets *prefix-state* *bad-sets* *fr* =
 { *as* ∈ *alpha-space*.
 ∃ *trace-table* ∈
 alpha-prefix-trace-table-candidates *prefix-state* *fr*.
 as ∈ *bad-sets* *trace-table* }

definition *checked-staged-transcript-actual-alpha-prefix-bad-set-hit*

:: ('f list ⇒ 'f list set) ⇒
 (((('f × 'f list × 'f list × 'f) ×
 'f protocol-channel) × 'f staged-proof-data) ×
 'f protocol-channel) option ⇒ bool

where

checked-staged-transcript-actual-alpha-prefix-bad-set-hit *bad-sets* *out*
 ⇔
 (case out of
 None ⇒ False
 | Some (((*fr*, -, -, -), *prefix-state*), *data*), -) ⇒
 staged-alphas *data* ∈
 alpha-prefix-union-bad-sets *prefix-state* *bad-sets* *fr*)

definition *checked-staged-transcript-actual-alpha-prefix-prequery-hit*

:: ('f list ⇒ 'f list set) ⇒
 (((('f × 'f list × 'f list × 'f) ×
 'f protocol-channel) × 'f staged-proof-data) ×
 'f protocol-channel) option ⇒ bool

where

checked-staged-transcript-actual-alpha-prefix-prequery-hit *bad-sets* *out*
 ⇔
 (case out of
 None ⇒ False
 | Some (((*fr*, -, -, -), *prefix-state*), *data*), -) ⇒
 staged-alphas *data* ∈
 alpha-prefix-union-bad-sets *prefix-state* *bad-sets* *fr* ∧
 alpha-vector-prequeried-in-state *prefix-state*
 (*staged-alphas* *data*) (length spec))

lemma *checked-staged-transcript-actual-alpha-prefix-prequery-hit-imp-bad-set-hit*:

assumes

checked-staged-transcript-actual-alpha-prefix-prequery-hit *bad-sets* *out*

shows *checked-staged-transcript-actual-alpha-prefix-bad-set-hit* *bad-sets* *out*

using *assms*

unfolding *checked-staged-transcript-actual-alpha-prefix-prequery-hit-def*

checked-staged-transcript-actual-alpha-prefix-bad-set-hit-def

by (auto split: option.splits prod.splits)

definition *checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit*

:: ('f list ⇒ 'f list set) ⇒
 (((('f × 'f list × 'f list × 'f) ×
 'f protocol-channel) × 'f staged-proof-data) ×

$\text{'f protocol-channel) option} \Rightarrow \text{bool}$
where
 $\text{checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit}$
 $\text{bad-sets out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((\text{prefix}, \text{prefix-state}), \text{data}), -) \Rightarrow$
 $\neg \text{hash-map-output-collision prefix-state} \wedge$
 $\text{staged-alphas data} \in$
 $\text{alpha-prefix-union-bad-sets prefix-state bad-sets (fst prefix)} \wedge$
 $\neg \text{alpha-vector-prequeried-in-state prefix-state}$
 $(\text{staged-alphas data}) (\text{length spec}))$

lemma $\text{checked-staged-transcript-actual-alpha-prefix-bad-set-hit-split:}$
assumes
 $\text{checked-staged-transcript-actual-alpha-prefix-bad-set-hit bad-sets out}$
shows
 $\text{checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit}$
 $\text{bad-sets out} \vee$
 $\text{checked-staged-transcript-actual-alpha-prefix-prequery-hit bad-sets out} \vee$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((-, \text{prefix-state}), -), -) \Rightarrow$
 $\text{hash-map-output-collision prefix-state})$

using assms
unfolding $\text{checked-staged-transcript-actual-alpha-prefix-bad-set-hit-def}$
 $\text{checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit-def}$
 $\text{checked-staged-transcript-actual-alpha-prefix-prequery-hit-def}$
by $(\text{auto split: option.splits prod.splits})$

lemma $\text{alpha-prefix-union-bad-sets-subset-alpha-space:}$
 $\text{alpha-prefix-union-bad-sets prefix-state bad-sets fr} \subseteq \text{alpha-space}$
unfolding $\text{alpha-prefix-union-bad-sets-def}$ **by** auto

lemma $\text{alpha-prefix-union-bad-sets-finite:}$
 $\text{finite } (\text{alpha-prefix-union-bad-sets prefix-state bad-sets fr})$
by $(\text{rule finite-subset}[OF \text{alpha-prefix-union-bad-sets-subset-alpha-space}$
 $\text{finite-alpha-space}])$

lemma $\text{alpha-prefix-union-bad-sets-fraction-bound-if-unique-candidate:}$
fixes $C :: \text{prob}$
assumes unique:
 $\exists \text{trace-table.}$
 $\text{alpha-prefix-trace-table-candidates prefix-state fr} \subseteq$
 $\{\text{trace-table}\}$
and $\text{subset: } \bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$
and bound:
 $\bigwedge \text{trace-table.}$
 $\text{nnreal } (\text{card } (\text{bad-sets trace-table})) / \text{nnreal } (\text{card } \text{alpha-space}) \leq C$

```

shows
  nnreal (card (alpha-prefix-union-bad-sets prefix-state bad-sets fr)) /
    nnreal (card alpha-space) ≤ C
proof -
  from unique obtain trace-table where candidates:
    alpha-prefix-trace-table-candidates prefix-state fr ⊆
      {trace-table}
  by blast
  have union-subset:
    alpha-prefix-union-bad-sets prefix-state bad-sets fr ⊆
      bad-sets trace-table
  unfolding alpha-prefix-union-bad-sets-def
  using candidates by auto
  have finite-bad: finite (bad-sets trace-table)
  by (rule finite-subset[OF subset finite-alpha-space])
  have card-le:
    card (alpha-prefix-union-bad-sets prefix-state bad-sets fr) ≤
      card (bad-sets trace-table)
  by (rule card-mono[OF finite-bad union-subset])
  have nnreal (card (alpha-prefix-union-bad-sets prefix-state bad-sets fr)) /
    nnreal (card alpha-space) ≤
      nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)
  by (rule nnreal-nat-divide-right-mono[OF card-le])
  also have ... ≤ C
  by (rule bound)
  finally show ?thesis .
qed

lemma alpha-prefix-trace-table-candidates-unique-if-no-hash-collision:
  assumes clean: ¬ hash-map-output-collision prefix-state
  shows
    ∃ trace-table.
      alpha-prefix-trace-table-candidates prefix-state fr ⊆
        {trace-table}
proof (cases alpha-prefix-trace-table-candidates prefix-state fr = {})
  case True
  then show ?thesis
  by blast
next
  case False
  then obtain trace-table where table-in:
    trace-table ∈ alpha-prefix-trace-table-candidates prefix-state fr
  by blast
  have table-len: length trace-table = clength * scale
  using table-in unfolding alpha-prefix-trace-table-candidates-def by simp
  have table-bind: merkle-root-binds-table fr trace-table prefix-state
  using table-in unfolding alpha-prefix-trace-table-candidates-def by simp
  have subset-single:
    alpha-prefix-trace-table-candidates prefix-state fr ⊆

```

```

    {trace-table}
  proof
    fix trace-table'
    assume table'-in:
      trace-table' ∈ alpha-prefix-trace-table-candidates prefix-state fr
    have table'-len: length trace-table' = clength * scale
      using table'-in unfolding alpha-prefix-trace-table-candidates-def
      by simp
    have table'-bind:
      merkle-root-binds-table fr trace-table' prefix-state
      using table'-in unfolding alpha-prefix-trace-table-candidates-def
      by simp
    have trace-table' = trace-table
      by (rule merkle-root-binds-same-length-tables-unique-if-no-hash-collision
        [OF table'-bind table-bind])
        (use table'-len table-len clean in simp-all)
    then show trace-table' ∈ {trace-table}
      by simp
  qed
  then show ?thesis by blast
qed

```

lemma *composition-alpha-prefix-union-fraction-bound-if-no-hash-collision:*
assumes clean: $\neg$ hash-map-output-collision prefix-state
shows

```

  nnreal
    (card
      (alpha-prefix-union-bad-sets prefix-state
        composition-trace-bad-alpha-space fr)) /
  nnreal (card alpha-space) ≤ composition-error-bound
by (rule alpha-prefix-union-bad-sets-fraction-bound-if-unique-candidate)
  (rule alpha-prefix-trace-table-candidates-unique-if-no-hash-collision
    [OF clean],
    rule composition-trace-bad-alpha-space-subset-alpha-space,
    rule composition-trace-bad-alpha-space-fraction-bound-alpha-space)

```

lemma *alpha-prefix-union-bad-sets-card-bound-if-unique-candidate:*

```

assumes unique:
  ∃ trace-table.
    alpha-prefix-trace-table-candidates prefix-state fr ⊆
      {trace-table}
and subset:  $\bigwedge$  trace-table. bad-sets trace-table ⊆ alpha-space
and bound:
   $\bigwedge$  trace-table. card (bad-sets trace-table) ≤ C

```

shows

```

  card (alpha-prefix-union-bad-sets prefix-state bad-sets fr) ≤ C

```

proof –

```

from unique obtain trace-table where candidates:
  alpha-prefix-trace-table-candidates prefix-state fr ⊆

```

```

    {trace-table}
  by blast
have union-subset:
  alpha-prefix-union-bad-sets prefix-state bad-sets fr  $\subseteq$ 
    bad-sets trace-table
  unfolding alpha-prefix-union-bad-sets-def
  using candidates by auto
have finite-bad: finite (bad-sets trace-table)
  by (rule finite-subset[OF subset finite-alpha-space])
have card (alpha-prefix-union-bad-sets prefix-state bad-sets fr)  $\leq$ 
  card (bad-sets trace-table)
  by (rule card-mono[OF finite-bad union-subset])
also have ...  $\leq C$ 
  by (rule bound)
finally show ?thesis .
qed

lemma composition-alpha-prefix-union-card-bound-if-no-hash-collision:
  assumes clean:  $\neg$  hash-map-output-collision prefix-state
  shows
    card
      (alpha-prefix-union-bad-sets prefix-state
        composition-trace-bad-alpha-space fr)  $\leq$ 
      (if length spec = 0 then 0 else CARD('f)  $^{\wedge}$  (length spec - 1))
  by (rule alpha-prefix-union-bad-sets-card-bound-if-unique-candidate)
  (rule alpha-prefix-trace-table-candidates-unique-if-no-hash-collision
    [OF clean],
    rule composition-trace-bad-alpha-space-subset-alpha-space,
    rule composition-trace-bad-alpha-space-card-bound)

lemma checked-staged-transcript-with-alpha-prefix-clean-fresh-composition-bound:
  shows
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
      (checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit
        composition-trace-bad-alpha-space)
      adversary-initial-state  $\leq$  composition-error-bound
  unfolding checked-staged-transcript-with-alpha-prefix-program-def
proof (rule wp-event-bind-bound-by-cont)
  show
     $\neg$  checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit
      composition-trace-bad-alpha-space None
  unfolding
    checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit-def
  by simp
next
fix prefix prefix-state
assume prefix-out:
  Some (prefix, prefix-state)  $\in$ 
    set-dist

```

```

    (execute (staged-alpha-prefix-program A) adversary-initial-state)
obtain fr trace-roots trace-bs trace-final where prefix-eq:
  prefix = (fr, trace-roots, trace-bs, trace-final)
  by (cases prefix) auto
let ?B =
  alpha-prefix-union-bad-sets prefix-state
  composition-trace-bad-alpha-space fr
let ?Q =
  checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit
  composition-trace-bad-alpha-space
let ?E =
  λout. case out of
    None ⇒ False
  | Some (data, -) ⇒
    ¬ hash-map-output-collision prefix-state ∧
    staged-alphas data ∈ ?B ∧
    ¬ alpha-vector-prequeried-in-state prefix-state
    (staged-alphas data) (length spec)
have get-eq:
  wp-event
  (get >=>
    (λprefix-state'.
      checked-staged-after-alpha-prefix-program A prefix >=>
        (λdata. return ((prefix, prefix-state'), data))))
  ?Q prefix-state =
  wp-event
  (checked-staged-after-alpha-prefix-program A prefix >=>
    (λdata. return ((prefix, prefix-state), data)))
  ?Q
  prefix-state
unfolding wp-event-def by (simp add: wpsimps)
have return-eq:
  wp-event
  (checked-staged-after-alpha-prefix-program A prefix >=>
    (λdata. return ((prefix, prefix-state), data)))
  ?Q prefix-state =
  wp-event (checked-staged-after-alpha-prefix-program A prefix) ?E
  prefix-state
apply (subst wp-event-bind-return-map)
unfolding
  checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit-def
  prefix-eq
by (simp split: option.splits prod.splits)
have cont-eq:
  wp-event
  (get >=>
    (λprefix-state'.
      checked-staged-after-alpha-prefix-program A prefix >=>
        (λdata. return ((prefix, prefix-state'), data))))

```



```

    ?Q prefix-state =
    wp-event (checked-staged-after-alpha-prefix-program A prefix) ?E
    prefix-state
  using get-eq return-eq by simp
show
  wp-event
  (get  $\gg$ 
    ( $\lambda$ prefix-state.
      checked-staged-after-alpha-prefix-program A prefix  $\gg$ 
      ( $\lambda$ data. return ((prefix, prefix-state), data))))
  ?Q prefix-state  $\leq$  composition-error-bound
proof (cases hash-map-output-collision prefix-state)
case True
have zero:
  wp-event (checked-staged-after-alpha-prefix-program A prefix) ?E
  prefix-state = 0
proof -
  have indicator-zero:
    ( $\lambda x$ . if ?E x then 1 else 0 :: prob) = ( $\lambda$ -. 0)
  using True
  by (auto simp: fun-eq-iff split: option.splits prod.splits)
show ?thesis
  unfolding wp-event-def indicator-zero wp-def by simp
qed
show ?thesis
  unfolding cont-eq zero by simp
next
case False
have subset: ?B  $\subseteq$  alpha-space
  by (rule alpha-prefix-union-bad-sets-subset-alpha-space)
have suffix-bound:
  wp-event (checked-staged-after-alpha-prefix-program A prefix)
  ( $\lambda$ out. case out of None  $\Rightarrow$  False
    | Some (data, -)  $\Rightarrow$ 
      staged-alphas data  $\in$  ?B  $\wedge$ 
       $\neg$  alpha-vector-prequeried-in-state prefix-state
      (staged-alphas data) (length spec))
  prefix-state  $\leq$ 
  nnreal (card ?B) / nnreal (CARD('f)  $\wedge$  length spec)
  by (rule
    checked-staged-after-alpha-prefix-program-alpha-list-bound-excluding-prequeried
    [OF subset])
have clean-bound:
  nnreal (card ?B) / nnreal (CARD('f)  $\wedge$  length spec)  $\leq$ 
  composition-error-bound
  using composition-alpha-prefix-union-fraction-bound-if-no-hash-collision
  [OF False, of fr]
  by (simp add: card-alpha-space)
have event-eq:

```

```

?E =
  (λout. case out of None ⇒ False
   | Some (data, -) ⇒
     staged-alphas data ∈ ?B ∧
     ¬ alpha-vector-prequeried-in-state prefix-state
       (staged-alphas data) (length spec))
  using False
  by (auto simp: fun-eq-iff split: option.splits prod.splits)
show ?thesis
  unfolding cont-eq event-eq
  by (rule order-trans[OF suffix-bound clean-bound])
qed
qed

lemma alpha-prefix-composition-vector-key-relation-fiber-bound-if-no-hash-collision:
  assumes clean: ¬ hash-map-output-collision prefix-state
  shows
    card
      {y. alpha-vector-key-relation prefix-state
        (alpha-prefix-union-bad-sets prefix-state
          composition-trace-bad-alpha-space fr)
        (length spec) x y}
    ≤ (if length spec = 0 then 0 else CARD('f) ^ (length spec - 1)) *
      length spec
  proof -
    have finite-bad:
      finite
        (alpha-prefix-union-bad-sets prefix-state
          composition-trace-bad-alpha-space fr)
    by (rule alpha-prefix-union-bad-sets-finite)
    have relation-bound:
      card
        {y. alpha-vector-key-relation prefix-state
          (alpha-prefix-union-bad-sets prefix-state
            composition-trace-bad-alpha-space fr)
          (length spec) x y}
      ≤
      card
        (alpha-prefix-union-bad-sets prefix-state
          composition-trace-bad-alpha-space fr) *
      length spec
    by (rule alpha-vector-key-relation-fiber-card-bound[OF finite-bad])
    also have ... ≤
      (if length spec = 0 then 0 else CARD('f) ^ (length spec - 1)) *
      length spec
  proof (cases length spec = 0)
    case True
    then show ?thesis by simp
  next

```

```

case False
have card
  (alpha-prefix-union-bad-sets prefix-state
   composition-trace-bad-alpha-space fr)
  ≤ CARD('f) ^ (length spec - 1)
using composition-alpha-prefix-union-card-bound-if-no-hash-collision
  [OF clean] False
by simp
then show ?thesis
  using False
  by (intro mult-right-mono) simp-all
qed
finally show ?thesis .
qed

```

definition *checked-staged-security-with-data-state-alpha-prefix-bad-set-hit*

```

:: 'f staged-adversary ⇒ ('f list ⇒ 'f list set) ⇒
  (((('f staged-proof-data × 'f protocol-channel) × unit list) ×
   'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
bad-sets out ⇔
  (case out of
   None ⇒ False
  | Some (((data, -), -), -) ⇒
    (∃ prefix prefix-state.
     Some (prefix, prefix-state) ∈
      set-dist
        (execute (staged-alpha-prefix-program A)
         adversary-initial-state) ∧
        staged-trace-root data = fst prefix ∧
        staged-trace-fri-roots data = fst (snd prefix) ∧
        staged-trace-fri-challenges data = fst (snd (snd prefix)) ∧
        staged-trace-final data = snd (snd (snd prefix)) ∧
        staged-alphas data ∈
          alpha-prefix-union-bad-sets prefix-state bad-sets
            (staged-trace-root data)))

```

definition *checked-staged-transcript-alpha-prefix-prequery-hit*

```

:: 'f staged-adversary ⇒ ('f list ⇒ 'f list set) ⇒
  ('f staged-proof-data × 'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-transcript-alpha-prefix-prequery-hit A bad-sets out
⇔
  (case out of
   None ⇒ False
  | Some (data, -) ⇒
    (∃ prefix prefix-state.
     Some (prefix, prefix-state) ∈

```

```

set-dist
  (execute (staged-alpha-prefix-program A)
    adversary-initial-state) ∧
staged-trace-root data = fst prefix ∧
staged-trace-fri-roots data = fst (snd prefix) ∧
staged-trace-fri-challenges data = fst (snd (snd prefix)) ∧
staged-trace-final data = snd (snd (snd prefix)) ∧
staged-alphas data ∈
  alpha-prefix-union-bad-sets prefix-state bad-sets
  (staged-trace-root data) ∧
alpha-vector-prequeried-in-state prefix-state
  (staged-alphas data) (length spec)))

```

definition *checked-staged-transcript-alpha-prefix-bad-set-hit*
 $:: 'f \text{ staged-adversary} \Rightarrow ('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
checked-staged-transcript-alpha-prefix-bad-set-hit A bad-sets out
 $\longleftrightarrow$
(case out of
None $\Rightarrow$ False
| Some (data, -) $\Rightarrow$
 $(\exists \text{ prefix prefix-state.}$
 $\text{Some (prefix, prefix-state)} \in$
 set-dist
 $(\text{execute (staged-alpha-prefix-program A)}$
 $\text{adversary-initial-state}) \wedge$
 $\text{staged-trace-root data} = \text{fst prefix} \wedge$
 $\text{staged-trace-fri-roots data} = \text{fst (snd prefix)} \wedge$
 $\text{staged-trace-fri-challenges data} = \text{fst (snd (snd prefix))} \wedge$
 $\text{staged-trace-final data} = \text{snd (snd (snd prefix))} \wedge$
 $\text{staged-alphas data} \in$
 $\text{alpha-prefix-union-bad-sets prefix-state bad-sets}$
 $(\text{staged-trace-root data}))$
)

lemma *checked-staged-transcript-alpha-prefix-prequery-hit-imp-bad-set-hit:*
assumes *hit:*
checked-staged-transcript-alpha-prefix-prequery-hit A bad-sets out
shows *checked-staged-transcript-alpha-prefix-bad-set-hit A bad-sets out*
using *hit*
unfolding *checked-staged-transcript-alpha-prefix-prequery-hit-def*
checked-staged-transcript-alpha-prefix-bad-set-hit-def
by (*auto split: option.splits prod.splits*)

lemma *checked-staged-transcript-alpha-prefix-prequery-hitE:*
assumes *hit:*
checked-staged-transcript-alpha-prefix-prequery-hit A bad-sets
 $(\text{Some (data, attacker-state)})$
obtains *prefix prefix-state* **where**

Some (prefix, prefix-state) ∈
set-dist
(execute (staged-alpha-prefix-program A) adversary-initial-state)
staged-trace-root data = fst prefix
staged-trace-fri-roots data = fst (snd prefix)
staged-trace-fri-challenges data = fst (snd (snd prefix))
staged-trace-final data = snd (snd (snd prefix))
staged-alphas data ∈
alpha-prefix-union-bad-sets prefix-state bad-sets
(staged-trace-root data)
alpha-vector-prequeried-in-state prefix-state
(staged-alphas data) (length spec)
hash-relation-hit
(alpha-vector-key-relation prefix-state
(alpha-prefix-union-bad-sets prefix-state bad-sets
(staged-trace-root data))
(length spec))
adversary-initial-state prefix-state
proof –
from *hit* **obtain** *prefix prefix-state* **where**
prefix-out:
Some (prefix, prefix-state) ∈
set-dist
(execute (staged-alpha-prefix-program A)
adversary-initial-state)
and *root-eq:* *staged-trace-root data = fst prefix*
and *trace-roots-eq:*
staged-trace-fri-roots data = fst (snd prefix)
and *trace-bs-eq:*
staged-trace-fri-challenges data = fst (snd (snd prefix))
and *trace-final-eq:*
staged-trace-final data = snd (snd (snd prefix))
and *member:*
staged-alphas data ∈
alpha-prefix-union-bad-sets prefix-state bad-sets
(staged-trace-root data)
and *prequeried:*
alpha-vector-prequeried-in-state prefix-state
(staged-alphas data) (length spec)
unfolding *checked-staged-transcript-alpha-prefix-prequery-hit-def*
by *auto*
have *relation-hit:*
hash-relation-hit
(alpha-vector-key-relation prefix-state
(alpha-prefix-union-bad-sets prefix-state bad-sets
(staged-trace-root data))
(length spec))
adversary-initial-state prefix-state
by *(rule alpha-vector-prequeried-in-state-imp-hash-relation-hit-empty-initial*

```

    [OF prequeried member])
  simp
show ?thesis
by (rule that[OF prefix-out root-eq trace-roots-eq trace-bs-eq
    trace-final-eq member prequeried relation-hit])
qed

definition staged-alpha-prefix-dynamic-prequery-hit
:: ('f list  $\Rightarrow$  'f list set)  $\Rightarrow$ 
  (('f  $\times$  'f list  $\times$  'f list  $\times$  'f)  $\times$ 
   'f protocol-channel) option  $\Rightarrow$  bool
where
  staged-alpha-prefix-dynamic-prequery-hit bad-sets out  $\longleftrightarrow$ 
    (case out of
      None  $\Rightarrow$  False
    | Some ((fr, -), prefix-state)  $\Rightarrow$ 
        ( $\exists$  as  $\in$  alpha-prefix-union-bad-sets prefix-state bad-sets fr.
          alpha-vector-prequeried-in-state prefix-state as
            (length spec)))

definition staged-alpha-prefix-dynamic-prequery-relation
:: 'f staged-adversary  $\Rightarrow$  'f protocol-channel  $\Rightarrow$ 
  ('f list  $\Rightarrow$  'f list set)  $\Rightarrow$ 
  'f protocol-hash-input  $\Rightarrow$  'f  $\Rightarrow$  bool
where
  staged-alpha-prefix-dynamic-prequery-relation A s bad-sets x y
 $\longleftrightarrow$ 
    ( $\exists$  fr trace-roots trace-bs trace-final prefix-state as i.
      Some ((fr, trace-roots, trace-bs, trace-final), prefix-state)  $\in$ 
        set-dist (execute (staged-alpha-prefix-program A) s)  $\wedge$ 
      as  $\in$  alpha-prefix-union-bad-sets prefix-state bad-sets fr  $\wedge$ 
      i < length spec  $\wedge$ 
      i < length as  $\wedge$ 
      x = alpha-challenge-key-at prefix-state i as  $\wedge$ 
      y = as ! i)

lemma staged-alpha-prefix-dynamic-prequery-hit-imp-relation-hit:
assumes empty: HashMap s = fmempty
and support:
  Some ((fr, trace-roots, trace-bs, trace-final), prefix-state)  $\in$ 
    set-dist (execute (staged-alpha-prefix-program A) s)
and hit:
  staged-alpha-prefix-dynamic-prequery-hit bad-sets
    (Some ((fr, trace-roots, trace-bs, trace-final), prefix-state))
shows
  hash-relation-hit-event
    (staged-alpha-prefix-dynamic-prequery-relation A s bad-sets)
    s (Some ((fr, trace-roots, trace-bs, trace-final), prefix-state))
proof -

```

```

from hit obtain as i where member:
  as  $\in$  alpha-prefix-union-bad-sets prefix-state bad-sets fr
and i-bound: i < length spec
and i-len: i < length as
and lookup:
  fmlookup (HashMap prefix-state)
  (alpha-challenge-key-at prefix-state i as) = Some (as ! i)
unfolding staged-alpha-prefix-dynamic-prequery-hit-def
  alpha-vector-prequeried-in-state-def
by auto
have initial-none:
  fmlookup (HashMap s) (alpha-challenge-key-at prefix-state i as) =
    None
using empty by simp
have relation:
  staged-alpha-prefix-dynamic-prequery-relation A s bad-sets
  (alpha-challenge-key-at prefix-state i as) (as ! i)
unfolding staged-alpha-prefix-dynamic-prequery-relation-def
using support member i-bound i-len by blast
have hit-relation:
  hash-relation-hit
  (staged-alpha-prefix-dynamic-prequery-relation A s bad-sets)
  s prefix-state
unfolding hash-relation-hit-def
by (intro exI[of - alpha-challenge-key-at prefix-state i as]
  exI[of - as ! i] conjI)
  (use initial-none lookup relation in simp-all)
show ?thesis
unfolding hash-relation-hit-event-def using hit-relation by simp
qed

lemma staged-alpha-prefix-dynamic-prequery-hit-bound-by-relation-hit:
assumes empty: HashMap s = fmempty
shows
  wp-event (staged-alpha-prefix-program A)
  (staged-alpha-prefix-dynamic-prequery-hit bad-sets) s  $\leq$ 
  wp-event (staged-alpha-prefix-program A)
  (hash-relation-hit-event
  (staged-alpha-prefix-dynamic-prequery-relation A s bad-sets) s) s
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out  $\in$  set-dist (execute (staged-alpha-prefix-program A) s)
and prehit: staged-alpha-prefix-dynamic-prequery-hit bad-sets out
show
  hash-relation-hit-event
  (staged-alpha-prefix-dynamic-prequery-relation A s bad-sets)
  s out
proof (cases out)

```

```

case None
then show ?thesis
  using prehit unfolding staged-alpha-prefix-dynamic-prequery-hit-def
  by simp
next
case (Some packed)
then obtain fr trace-roots trace-bs trace-final prefix-state
  where out-eq:
    out =
      Some ((fr, trace-roots, trace-bs, trace-final), prefix-state)
  by (cases packed, auto split: prod.splits)
show ?thesis
  unfolding out-eq
  by (rule staged-alpha-prefix-dynamic-prequery-hit-imp-relation-hit
    [OF empty])
    (use support prehit out-eq in simp-all)
qed
qed

lemma staged-alpha-prefix-dynamic-prequery-relation-fiber-bound-size:
  card
    {y. staged-alpha-prefix-dynamic-prequery-relation A s bad-sets x y}
  ≤ size
proof –
  have subset:
    {y. staged-alpha-prefix-dynamic-prequery-relation A s bad-sets x y}
    ⊆ (UNIV :: 'f set)
  by simp
  have finite-field: finite (UNIV :: 'f set)
  by simp
  have card
    {y. staged-alpha-prefix-dynamic-prequery-relation A s bad-sets x y}
    ≤ card (UNIV :: 'f set)
  by (rule card-mono[OF finite-field subset])
  also have ... = size
  using size-card by simp
  finally show ?thesis .
qed

lemma staged-alpha-prefix-dynamic-prequery-hit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (staged-alpha-prefix-program A)
    (staged-alpha-prefix-dynamic-prequery-hit bad-sets)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)
proof –

```



```

have prehit-le-relation:
  wp-event (staged-alpha-prefix-program A)
    (staged-alpha-prefix-dynamic-prequery-hit bad-sets)
    adversary-initial-state  $\leq$ 
  wp-event (staged-alpha-prefix-program A)
    (hash-relation-hit-event
      (staged-alpha-prefix-dynamic-prequery-relation A
        adversary-initial-state bad-sets)
        adversary-initial-state)
    adversary-initial-state
by (rule staged-alpha-prefix-dynamic-prequery-hit-bound-by-relation-hit)
  (simp add: adversary-initial-state-def)
also have ...  $\leq$ 
  staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)
proof —
  have program:
    hash-relation-program
      (staged-alpha-prefix-dynamic-prequery-relation A
        adversary-initial-state bad-sets)
      size (staged-alpha-search-queries budgets 0)
      (staged-alpha-prefix-program A)
    by (rule hash-relation-program-staged-alpha-prefix-program
      [OF wf controlled])
      (rule staged-alpha-prefix-dynamic-prequery-relation-fiber-bound-size)
  show ?thesis
    using program
    unfolding hash-relation-program-def hash-relation-budget-def
      staged-phase-relation-error-def
    by blast
qed
finally show ?thesis .
qed

```

definition *checked-staged-transcript-actual-alpha-prefix-collision-hit*
 $:: (((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $\quad 'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $\quad 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

```

checked-staged-transcript-actual-alpha-prefix-collision-hit out
 $\longleftrightarrow$ 
(case out of
  None  $\Rightarrow$  False
  | Some (((-, prefix-state), -), -)  $\Rightarrow$ 
    hash-map-output-collision prefix-state)

```

lemma *checked-staged-transcript-actual-alpha-prefix-bad-set-hit-split-events*:

assumes

checked-staged-transcript-actual-alpha-prefix-bad-set-hit *bad-sets* *out*

shows
checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit
bad-sets out $\vee$
checked-staged-transcript-actual-alpha-prefix-prequery-hit bad-sets out $\vee$
checked-staged-transcript-actual-alpha-prefix-collision-hit out
using *checked-staged-transcript-actual-alpha-prefix-bad-set-hit-split*
 $[OF\ assms]$
unfolding *checked-staged-transcript-actual-alpha-prefix-collision-hit-def*
by (auto split: option.splits prod.splits)

lemma *checked-staged-transcript-with-alpha-prefix-prequery-bound:*
shows
wp-event (checked-staged-transcript-with-alpha-prefix-program A)
(checked-staged-transcript-actual-alpha-prefix-prequery-hit bad-sets)
adversary-initial-state $\leq$
wp-event (staged-alpha-prefix-program A)
(staged-alpha-prefix-dynamic-prequery-hit bad-sets)
adversary-initial-state
unfolding *checked-staged-transcript-with-alpha-prefix-program-def*
proof (rule wp-event-bind-bound-by-head-event)
show
wp-event (staged-alpha-prefix-program A)
(staged-alpha-prefix-dynamic-prequery-hit bad-sets)
adversary-initial-state $\leq$
wp-event (staged-alpha-prefix-program A)
(staged-alpha-prefix-dynamic-prequery-hit bad-sets)
adversary-initial-state
by simp

next
show
checked-staged-transcript-actual-alpha-prefix-prequery-hit
bad-sets None $\impl$
staged-alpha-prefix-dynamic-prequery-hit bad-sets None
unfolding *checked-staged-transcript-actual-alpha-prefix-prequery-hit-def*
staged-alpha-prefix-dynamic-prequery-hit-def
by simp

next
fix prefix prefix-state out
assume prefix-out:
Some (prefix, prefix-state) $\in$
set-dist
(execute (staged-alpha-prefix-program A) adversary-initial-state)
and cont:
out $\in$
set-dist
(execute
(get $\gg$
(λ prefix-state.
checked-staged-after-alpha-prefix-program A prefix $\gg$

```

      (λdata. return ((prefix, prefix-state), data))))
    prefix-state)
  and hit:
    checked-staged-transcript-actual-alpha-prefix-prequery-hit
    bad-sets out
  obtain fr trace-roots trace-bs trace-final where prefix-eq:
    prefix = (fr, trace-roots, trace-bs, trace-final)
  by (cases prefix) auto
  from hit cont show
    staged-alpha-prefix-dynamic-prequery-hit bad-sets
    (Some (prefix, prefix-state))
  unfolding checked-staged-transcript-actual-alpha-prefix-prequery-hit-def
    staged-alpha-prefix-dynamic-prequery-hit-def prefix-eq
  by (auto elim!: set-dist-bindE split: option.splits prod.splits)
qed

lemma checked-staged-transcript-with-alpha-prefix-collision-bound:
  shows
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
      checked-staged-transcript-actual-alpha-prefix-collision-hit
      adversary-initial-state ≤
    wp-event (staged-alpha-prefix-program A)
      (hash-new-collision-event adversary-initial-state)
      adversary-initial-state
  unfolding checked-staged-transcript-with-alpha-prefix-program-def
  proof (rule wp-event-bind-bound-by-head-event)
  show
    wp-event (staged-alpha-prefix-program A)
      (hash-new-collision-event adversary-initial-state)
      adversary-initial-state ≤
    wp-event (staged-alpha-prefix-program A)
      (hash-new-collision-event adversary-initial-state)
      adversary-initial-state
  by simp
next
  show
    checked-staged-transcript-actual-alpha-prefix-collision-hit None ⇒
    hash-new-collision-event adversary-initial-state None
  unfolding checked-staged-transcript-actual-alpha-prefix-collision-hit-def
    hash-new-collision-event-def
  by simp
next
  fix prefix prefix-state out
  assume prefix-out:
    Some (prefix, prefix-state) ∈
      set-dist
      (execute (staged-alpha-prefix-program A) adversary-initial-state)
  and cont:
    out ∈

```

```

    set-dist
    (execute
      (get >>=
        (λprefix-state.
          checked-staged-after-alpha-prefix-program A prefix >>=
            (λdata. return ((prefix, prefix-state), data))))
        prefix-state)
    and hit:
    checked-staged-transcript-actual-alpha-prefix-collision-hit out
  have collision: hash-map-output-collision prefix-state
  using hit cont
  unfolding checked-staged-transcript-actual-alpha-prefix-collision-hit-def
  by (auto elim!: set-dist-bindE split: option.splits prod.splits)
show
  hash-new-collision-event adversary-initial-state
    (Some (prefix, prefix-state))
  using collision adversary-initial-state-no-output-collision
  unfolding hash-new-collision-event-def hash-map-new-output-collision-def
  by simp
qed

lemma checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound:
  shows
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
      (checked-staged-transcript-actual-alpha-prefix-bad-set-hit
        composition-trace-bad-alpha-space)
    adversary-initial-state ≤
    composition-error-bound +
    wp-event (staged-alpha-prefix-program A)
      (staged-alpha-prefix-dynamic-prequery-hit
        composition-trace-bad-alpha-space)
    adversary-initial-state +
    wp-event (staged-alpha-prefix-program A)
      (hash-new-collision-event adversary-initial-state)
    adversary-initial-state
  proof -
  let ?m = checked-staged-transcript-with-alpha-prefix-program A
  let ?bad =
    checked-staged-transcript-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space
  let ?clean =
    checked-staged-transcript-actual-alpha-prefix-clean-fresh-bad-set-hit
    composition-trace-bad-alpha-space
  let ?pre =
    checked-staged-transcript-actual-alpha-prefix-prequery-hit
    composition-trace-bad-alpha-space
  let ?coll = checked-staged-transcript-actual-alpha-prefix-collision-hit
  have bad-le-split:
    wp-event ?m ?bad adversary-initial-state ≤

```

```

    wp-event ?m (λout. ?clean out ∨ ?pre out ∨ ?coll out)
    adversary-initial-state
  by (rule wp-event-mono)
    (rule
      checked-staged-transcript-actual-alpha-prefix-bad-set-hit-split-events)
also have ... ≤
  wp-event ?m ?clean adversary-initial-state +
  wp-event ?m ?pre adversary-initial-state +
  wp-event ?m ?coll adversary-initial-state
proof -
  have ... ≤
    wp-event ?m ?clean adversary-initial-state +
    wp-event ?m (λout. ?pre out ∨ ?coll out)
    adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤
  wp-event ?m ?clean adversary-initial-state +
  (wp-event ?m ?pre adversary-initial-state +
    wp-event ?m ?coll adversary-initial-state)
  by (intro add-mono order.refl wp-event-union-bound)
finally show ?thesis
  by (simp add: add.assoc)
qed
also have ... ≤
  composition-error-bound +
  wp-event (staged-alpha-prefix-program A)
  (staged-alpha-prefix-dynamic-prequery-hit
    composition-trace-bad-alpha-space)
  adversary-initial-state +
  wp-event (staged-alpha-prefix-program A)
  (hash-new-collision-event adversary-initial-state)
  adversary-initial-state
  by (intro add-mono
    checked-staged-transcript-with-alpha-prefix-clean-fresh-composition-bound
    checked-staged-transcript-with-alpha-prefix-prequery-bound
    checked-staged-transcript-with-alpha-prefix-collision-bound)
finally show ?thesis .
qed

lemma checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
    (checked-staged-transcript-actual-alpha-prefix-bad-set-hit
      composition-trace-bad-alpha-space)
    adversary-initial-state ≤
    composition-error-bound +
    staged-phase-relation-error size

```

```

      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0)
proof -
  have prequery-bound:
    wp-event (staged-alpha-prefix-program A)
      (staged-alpha-prefix-dynamic-prequery-hit
        composition-trace-bad-alpha-space)
    adversary-initial-state ≤
    staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0)
  by (rule staged-alpha-prefix-dynamic-prequery-hit-bound
    [OF wf controlled])
  have collision-budget:
    hash-collision-budget
      (staged-alpha-search-queries budgets 0)
      (staged-alpha-prefix-program A)
  by (rule hash-collision-budget-staged-alpha-prefix-program
    [OF wf controlled])
  have collision-bound:
    wp-event (staged-alpha-prefix-program A)
      (hash-new-collision-event adversary-initial-state)
    adversary-initial-state ≤
    hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0)
  proof -
    have
      wp-event (staged-alpha-prefix-program A)
        (hash-new-collision-event adversary-initial-state)
        adversary-initial-state ≤
        hash-collision-budget-value
          (card (hash-map-output-values adversary-initial-state))
          (staged-alpha-search-queries budgets 0)
      using collision-budget adversary-initial-state-no-output-collision
      unfolding hash-collision-budget-def by blast
    also have ... =
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0)
    by simp
    finally show ?thesis .
  qed
  have
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
      (checked-staged-transcript-actual-alpha-prefix-bad-set-hit
        composition-trace-bad-alpha-space)
    adversary-initial-state ≤
    composition-error-bound +
    wp-event (staged-alpha-prefix-program A)
      (staged-alpha-prefix-dynamic-prequery-hit

```

```

      composition-trace-bad-alpha-space)
    adversary-initial-state +
    wp-event (staged-alpha-prefix-program A)
      (hash-new-collision-event adversary-initial-state)
    adversary-initial-state
  by (rule checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound)
also have ... ≤
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0)
  by (intro add-mono order.refl prequery-bound collision-bound)
finally show ?thesis .
qed

```

definition *checked-staged-security-with-actual-alpha-prefix-bad-set-hit*

```

:: ('f list ⇒ 'f list set) ⇒
  ((((((('f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  bad-sets out ⟷
  (case out of
    None ⇒ False
  | Some (x, -) ⇒
    (let packed = fst (fst x);
      prefix = fst (fst packed);
      prefix-state = snd (fst packed);
      data = snd packed
    in staged-alphas data ∈
      alpha-prefix-union-bad-sets prefix-state bad-sets (fst prefix)))

```

definition *checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit*

```

:: ('f list ⇒ 'f list set) ⇒
  ((((((('f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit
  bad-sets out ⟷
  (case out of
    None ⇒ False
  | Some (x, -) ⇒
    (let packed = fst (fst x);
      data = snd packed;

```

$attacker-state = snd\ (fst\ x)$
in $staged\ alphas\ data \in$
 $alpha\ header\ supported\ union\ bad\ sets$
 $(verifier\ state\ from\ adversary\ attacker\ state$
 $(staged\ proof\ transcript\ data))$
 $bad\ sets$
 $(staged\ trace\ root\ data)$
 $(staged\ trace\ fri\ roots\ data)$
 $(staged\ trace\ final\ data)))$

definition *checked-staged-security-with-actual-alpha-prefix-tree-output-hit*

$:: ((((((f \times 'f\ list \times 'f\ list \times 'f) \times$
 $'f\ protocol\ channel) \times 'f\ staged\ proof\ data) \times$
 $'f\ protocol\ channel) \times unit\ list) \times$
 $'f\ protocol\ channel) option \Rightarrow bool$

where

$checked\ staged\ security\ with\ actual\ alpha\ prefix\ tree\ output\ hit\ out$
 $\longleftrightarrow$
 $(case\ out\ of$
 $None \Rightarrow False$
 $| Some\ (x, -) \Rightarrow$
 $(let\ packed = fst\ (fst\ x);$
 $prefix\ state = snd\ (fst\ packed);$
 $data = snd\ packed;$
 $attacker\ state = snd\ (fst\ x);$
 $s =$
 $verifier\ state\ from\ adversary\ attacker\ state$
 $(staged\ proof\ transcript\ data)$
 $in\ \exists\ result\ final\ state\ trace\ table\ composition\ table\ as$
 $query\ idxs\ dg\ composition\ fri\ roots\ final\ rest\ trace\ tree$
 $composition\ tree.$
 $Some\ (result, final\ state) \in$
 $set\ dist\ (execute\ verify\ monad\ s) \wedge$
 $accepted\ with\ bound\ tables\ s\ (Some\ (result, final\ state))$
 $trace\ table\ composition\ table\ as\ query\ idxs \wedge$
 $verifier\ header\ transcript\ s$
 $(staged\ trace\ root\ data)$
 $(staged\ trace\ fri\ roots\ data)$
 $(staged\ trace\ final\ data)$
 $as\ dg\ composition\ fri\ roots\ final\ rest \wedge$
 $created\ tree\ trace\ table\ trace\ tree\ final\ state \wedge$
 $created\ tree\ composition\ table\ composition\ tree\ final\ state \wedge$
 $hash\ map\ new\ output\ hit$
 $(set\ tree\ trace\ tree \cup set\ tree\ composition\ tree)$
 $prefix\ state\ final\ state))$

lemma *checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript:*

assumes *transcript-bound:*

wp-event (checked-staged-transcript-with-alpha-prefix-program A)


```

    (checked-staged-transcript-actual-alpha-prefix-bad-set-hit bad-sets)
    adversary-initial-state ≤ C
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-bad-set-hit bad-sets)
    adversary-initial-state ≤ C
unfolding
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
proof (rule wp-event-bind-bound-by-head-event[OF transcript-bound])
show
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    bad-sets None ⇒
  checked-staged-transcript-actual-alpha-prefix-bad-set-hit bad-sets None
unfolding checked-staged-security-with-actual-alpha-prefix-bad-set-hit-def
  checked-staged-transcript-actual-alpha-prefix-bad-set-hit-def
by simp
next
fix packed attacker-state out
assume cont:
  out ∈
    set-dist
      (execute
        (get >>=
          (λs. put
            (verifier-state-from-adversary s
              (staged-proof-transcript (snd packed)))) >>=
            (λ-. verify-monad >>=
              (λresult. return ((packed, s), result))))))
        attacker-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    bad-sets out
show
  checked-staged-transcript-actual-alpha-prefix-bad-set-hit bad-sets
    (Some (packed, attacker-state))
using cont hit
unfolding checked-staged-security-with-actual-alpha-prefix-bad-set-hit-def
  checked-staged-transcript-actual-alpha-prefix-bad-set-hit-def
by (auto elim!: set-dist-bindE split: option.splits prod.splits)
qed

lemma checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-actual-projection:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
      bad-sets)
  adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

```

```

    (checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit
      bad-sets)
  adversary-initial-state
proof –
  let ?project =
    λx. ((snd (fst (fst x)), snd (fst x)), snd x)
  have
    wp-event (checked-staged-security-experiment-with-data-state A)
      (checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
        bad-sets)
    adversary-initial-state =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
        ≫ (λx. return (?project x)))
      (checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
        bad-sets)
    adversary-initial-state
  using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
      [of A]
  by simp
  also have ... =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (λout. case out of
        None ⇒
          checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
            bad-sets None
        | Some (x, t) ⇒
          checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
            bad-sets (Some (?project x, t)))
      adversary-initial-state
  by (rule wp-event-bind-return-map)
  also have ... =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit
        bad-sets)
      adversary-initial-state
  unfolding
    checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-def
    checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit-def
  apply (rule arg-cong[where
    f = λP. wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      P adversary-initial-state])
  apply (rule ext)
  apply (simp add: Let-def split: option.splits prod.splits)
  done

```

finally show *?thesis* .
qed

lemma *checked-staged-security-with-actual-alpha-prefix-tree-output-hitI*:

assumes *verify-out*:

Some (result, final-state) ∈
set-dist
(execute verify-monad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

and *bound*:

accepted-with-bound-tables
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(Some (result, final-state)) trace-table composition-table as
query-idxs

and *header*:

verifier-header-transcript
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
as dg composition-fri-roots final rest

and *trace-created*: *created-tree trace-table trace-tree final-state*

and *composition-created*:

created-tree composition-table composition-tree final-state

and *tree-hit*:

hash-map-new-output-hit
(set-tree trace-tree ∪ set-tree composition-tree)
prefix-state final-state

shows

checked-staged-security-with-actual-alpha-prefix-tree-output-hit
(Some (((((prefix, prefix-state), data), attacker-state),
checked-result), checked-final-state)))

proof –

have *witness*:

$\exists$ *result final-state trace-table composition-table as query-idxs dg*
composition-fri-roots final rest trace-tree composition-tree.

Some (result, final-state) ∈
set-dist

(execute verify-monad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))) $\wedge$

accepted-with-bound-tables

(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))

(Some (result, final-state)) trace-table composition-table as
query-idxs $\wedge$

```

    verifier-header-transcript
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      as dg composition-fri-roots final rest  $\wedge$ 
    created-tree trace-table trace-tree final-state  $\wedge$ 
    created-tree composition-table composition-tree final-state  $\wedge$ 
    hash-map-new-output-hit
      (set-tree trace-tree  $\cup$  set-tree composition-tree)
      prefix-state final-state
  using verify-out bound header trace-created composition-created tree-hit
  by blast
show ?thesis
unfolding
  checked-staged-security-with-actual-alpha-prefix-tree-output-hit-def
  Let-def
  using witness by simp
qed

```

definition *checked-staged-security-with-data-state-alpha-prefix-tree-output-hit*

```

:: 'f staged-adversary  $\Rightarrow$ 
  (((('f staged-proof-data  $\times$  'f protocol-channel)  $\times$  unit list)  $\times$ 
    'f protocol-channel) option  $\Rightarrow$  bool

```

where

```

checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A
  out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some (((data, attacker-state), -), -)  $\Rightarrow$ 
    (let s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    in  $\exists$  prefix prefix-state result final-state trace-table
      composition-table as query-idxs dg composition-fri-roots final
      rest trace-tree composition-tree.
      Some (prefix, prefix-state)  $\in$ 
        set-dist
          (execute (staged-alpha-prefix-program A)
            adversary-initial-state)  $\wedge$ 
          staged-trace-root data = fst prefix  $\wedge$ 
          staged-trace-fri-roots data = fst (snd prefix)  $\wedge$ 
          staged-trace-fri-challenges data = fst (snd (snd prefix))  $\wedge$ 
          staged-trace-final data = snd (snd (snd prefix))  $\wedge$ 
          Some (result, final-state)  $\in$ 
            set-dist (execute verify-monad s)  $\wedge$ 
            accepted-with-bound-tables s (Some (result, final-state))
            trace-table composition-table as query-idxs  $\wedge$ 

```

verifier-header-transcript s
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
as dg composition-fri-roots final rest $\wedge$
created-tree trace-table trace-tree final-state $\wedge$
created-tree composition-table composition-tree final-state $\wedge$
hash-map-new-output-hit
(set-tree trace-tree $\cup$ set-tree composition-tree)
prefix-state final-state))

lemma *checked-staged-security-with-data-state-alpha-prefix-bad-set-hit-alphas-in-space:*

assumes *hit:*

checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A

bad-sets out

shows

case out of

None $\Rightarrow$ False

| Some (((data, -), -), -) $\Rightarrow$ staged-alphas data $\in$ alpha-space

using *hit*

unfolding *checked-staged-security-with-data-state-alpha-prefix-bad-set-hit-def*

alpha-prefix-union-bad-sets-def

by *(auto split: option.splits prod.splits)*

lemma *checked-staged-security-with-data-state-alpha-prefix-bad-set-hit-bound-from-transcript:*

assumes *transcript-bound:*

wp-event (checked-staged-transcript-program A)

(checked-staged-transcript-alpha-prefix-bad-set-hit A bad-sets)

adversary-initial-state $\leq C$

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
bad-sets)

adversary-initial-state $\leq C$

unfolding *checked-staged-security-experiment-with-data-state-def*

proof *(rule wp-event-bind-bound-by-head-event[OF transcript-bound])*

show

checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A

bad-sets None $\implies$

checked-staged-transcript-alpha-prefix-bad-set-hit A bad-sets None

unfolding

checked-staged-security-with-data-state-alpha-prefix-bad-set-hit-def

checked-staged-transcript-alpha-prefix-bad-set-hit-def

by *simp*

next

fix *data attacker-state out*

assume *cont:*

out $\in$

set-dist

```

    (execute
      (get >>=
        (λs. put
          (verifier-state-from-adversary s
            (staged-proof-transcript data)) >>=
            (λ-. verify-monad >>=
              (λresult. return ((data, s), result))))))
      attacker-state)
  and hit:
    checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
    bad-sets out
show
  checked-staged-transcript-alpha-prefix-bad-set-hit A bad-sets
  (Some (data, attacker-state))
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-data-state-alpha-prefix-bad-set-hit-def
  by simp
next
case (Some result-pack)
then obtain data' attacker-state' result final-state where out-eq:
  out = Some (((data', attacker-state'), result), final-state)
  by (cases result-pack, auto split: prod.splits)
have data-eq: data' = data
  and attacker-eq: attacker-state' = attacker-state
  using cont unfolding out-eq
  by (auto elim!: set-dist-bindE)
show ?thesis
  using hit unfolding out-eq data-eq attacker-eq
    checked-staged-security-with-data-state-alpha-prefix-bad-set-hit-def
    checked-staged-transcript-alpha-prefix-bad-set-hit-def
  by simp
qed
qed

lemma staged-after-alpha-prefix-program-outcome-extension:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, t) ∈
      set-dist
        (execute (staged-after-alpha-prefix-program A prefix) s)
  shows s ≤ t
proof -
  obtain fr trace-roots trace-bs trace-final where prefix-eq:
    prefix = (fr, trace-roots, trace-bs, trace-final)

```

```

  by (cases prefix) simp
from outcome[unfolded prefix-eq]
obtain as s1 dg s2 s3 s4 composition-roots composition-bs s5
  composition-final s6 s7 query-chunks where
alpha-out:
  Some (as, s1) ∈
    set-dist (execute (staged-alpha-program (length spec)) s)
and degree-out:
  Some (dg, s2) ∈ set-dist (execute (degree-stage A as) s1)
and record-degree-out:
  Some ((), s3) ∈ set-dist (execute (record-staged-message dg) s2)
and assert-out:
  Some ((), s4) ∈
    set-dist
      (execute
        (assert
          (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)))
        s3)
and composition-out:
  Some ((composition-roots, composition-bs), s5) ∈
    set-dist
      (execute
        (staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) []))
      s4)
and composition-final-out:
  Some (composition-final, s6) ∈
    set-dist (execute (composition-final-stage A dg composition-bs) s5)
and record-composition-final-out:
  Some ((), s7) ∈
    set-dist (execute (record-staged-message composition-final) s6)
and query-out:
  Some (query-chunks, t) ∈
    set-dist (execute (staged-query-program A 0 rounds) s7)
unfolding staged-after-alpha-prefix-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have degree-controlled:
  controlled-ro-program (degree-budget budgets) (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def by blast
have composition-final-controlled:
  controlled-ro-program (composition-final-budget budgets)
    (composition-final-stage A dg composition-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have ext-s-s1: s ≤ s1
  by (rule hash-target-program-outcome-extension
    [OF hash-target-program-staged-alpha-program alpha-out])
have ext-s1-s2: s1 ≤ s2
  using controlled-ro-program-extension[OF degree-controlled] degree-out
  unfolding hash-extension-preserving-def by blast

```

```

have s3-eq:
  s3 = s2(|
    PState := concat (PState s2) dg,
    PTranscript := PTranscript s2 @ [dg])
  by (rule record-staged-message-outcome[OF record-degree-out])
have ext-s2-s3: s2 ≤ s3
  unfolding s3-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have s4-eq: s4 = s3
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have ext-s3-s4: s3 ≤ s4
  unfolding s4-eq by (simp add: hash-ext-refl)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have composition-bound:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
  using wf round-bound unfolding staged-budget-wellformed-def by simp
have ext-s4-s5: s4 ≤ s5
  using staged-composition-fri-program-alignment
    [OF controlled composition-bound composition-out]
  by simp
have ext-s5-s6: s5 ≤ s6
  using controlled-ro-program-extension[OF composition-final-controlled]
    composition-final-out
  unfolding hash-extension-preserving-def by blast
have s7-eq:
  s7 = s6(|
    PState := concat (PState s6) composition-final,
    PTranscript := PTranscript s6 @ [composition-final])
  by (rule record-staged-message-outcome[OF record-composition-final-out])
have ext-s6-s7: s6 ≤ s7
  unfolding s7-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have query-bound: 0 + rounds ≤ length (query-opening-budgets budgets)
  using wf unfolding staged-budget-wellformed-def by simp
have ext-s7-t: s7 ≤ t
  by (rule hash-target-program-outcome-extension
    [OF hash-target-program-staged-query-program[OF controlled query-bound]
      query-out])
show ?thesis
  by (rule hash-ext-trans[OF ext-s-s1])
    (rule hash-ext-trans[OF ext-s1-s2],
    rule hash-ext-trans[OF ext-s2-s3],

```



```

    rule hash-ext-trans[OF ext-s3-s4],
    rule hash-ext-trans[OF ext-s4-s5],
    rule hash-ext-trans[OF ext-s5-s6],
    rule hash-ext-trans[OF ext-s6-s7 ext-s7-t])
qed

lemma checked-staged-after-alpha-prefix-program-outcome-extension:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, t) ∈
      set-dist
        (execute (checked-staged-after-alpha-prefix-program A prefix) s)
  shows  $s \leq t$ 
proof –
  obtain fr trace-roots trace-bs trace-final where prefix-eq:
    prefix = (fr, trace-roots, trace-bs, trace-final)
  by (cases prefix) simp
from outcome[unfolded prefix-eq]
obtain as s1 dg s2 s3 s4 composition-roots composition-bs s5
  composition-final s6 s7 query-chunks where
  alpha-out:
    Some (as, s1) ∈
      set-dist (execute (staged-alpha-program (length spec)) s)
  and degree-out:
    Some (dg, s2) ∈ set-dist (execute (degree-stage A as) s1)
  and record-degree-out:
    Some ((), s3) ∈ set-dist (execute (record-staged-message dg) s2)
  and assert-out:
    Some ((), s4) ∈
      set-dist
        (execute
          (assert
            (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))))
          s3)
  and composition-out:
    Some ((composition-roots, composition-bs), s5) ∈
      set-dist
        (execute
          (staged-composition-fri-program A dg 0
            (ceil-log (to-nat dg + 1)) []))
          s4)
  and composition-final-out:
    Some (composition-final, s6) ∈
      set-dist (execute (composition-final-stage A dg composition-bs) s5)
  and record-composition-final-out:
    Some ((), s7) ∈
      set-dist (execute (record-staged-message composition-final) s6)
  and query-out:

```

```

    Some (query-chunks, t) ∈
      set-dist
        (execute
          (checked-staged-query-program A trace-roots composition-roots
            0 rounds)
          s7)
  unfolding checked-staged-after-alpha-prefix-program-def Let-def
  by (auto elim!: set-dist-bindE split: prod.splits)
have degree-controlled:
  controlled-ro-program (degree-budget budgets) (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def by blast
have composition-final-controlled:
  controlled-ro-program (composition-final-budget budgets)
    (composition-final-stage A dg composition-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have ext-s-s1: s ≤ s1
  by (rule hash-target-program-outcome-extension
    [OF hash-target-program-staged-alpha-program alpha-out])
have ext-s1-s2: s1 ≤ s2
  using controlled-ro-program-extension[OF degree-controlled] degree-out
  unfolding hash-extension-preserving-def by blast
have s3-eq:
  s3 = s2(|
    PState := concat (PState s2) dg,
    PTranscript := PTranscript s2 @ [dg])
  by (rule record-staged-message-outcome[OF record-degree-out])
have ext-s2-s3: s2 ≤ s3
  unfolding s3-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have s4-eq: s4 = s3
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have ext-s3-s4: s3 ≤ s4
  unfolding s4-eq by (simp add: hash-ext-refl)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have composition-bound:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
  using wf round-bound unfolding staged-budget-wellformed-def by simp
have ext-s4-s5: s4 ≤ s5
  using staged-composition-fri-program-alignment
    [OF controlled composition-bound composition-out]
  by simp

```

```

have ext-s5-s6:  $s5 \leq s6$ 
  using controlled-ro-program-extension[OF composition-final-controlled]
    composition-final-out
  unfolding hash-extension-preserving-def by blast
have s7-eq:
  s7 = s6(
    PState := concat (PState s6) composition-final,
    PTranscript := PTranscript s6 @ [composition-final])
  by (rule record-staged-message-outcome[OF record-composition-final-out])
have ext-s6-s7:  $s6 \leq s7$ 
  unfolding s7-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have query-bound:  $0 + \text{rounds} \leq \text{length } (\text{query-opening-budgets } \text{budgets})$ 
  using wf unfolding staged-budget-wellformed-def by simp
have query-program:
  hash-relation-program ( $\lambda - . \text{False}$ ) 0
    (sum-list (take rounds (drop 0 (query-opening-budgets budgets))) +
      rounds)
    (checked-staged-query-program A trace-roots composition-roots 0
      rounds)
  by (rule hash-relation-program-checked-staged-query-program
    [OF controlled query-bound])
    simp
have ext-s7-t:  $s7 \leq t$ 
  using query-program query-out
  unfolding hash-relation-program-def hash-extension-preserving-def
  by blast
show ?thesis
  by (rule hash-ext-trans[OF ext-s-s1])
    (rule hash-ext-trans[OF ext-s1-s2],
      rule hash-ext-trans[OF ext-s2-s3],
      rule hash-ext-trans[OF ext-s3-s4],
      rule hash-ext-trans[OF ext-s4-s5],
      rule hash-ext-trans[OF ext-s5-s6],
      rule hash-ext-trans[OF ext-s6-s7 ext-s7-t])

```

qed

lemma *checked-staged-transcript-with-alpha-prefix-program-support*:

assumes *wf*: *staged-budget-wellformed* *budgets*

and *controlled*: *staged-adversary-controlled* *budgets* *A*

and *outcome*:

Some (((*prefix*, *prefix-state*), *data*), *attacker-state*) $\in$

set-dist

(*execute* (*checked-staged-transcript-with-alpha-prefix-program* *A*)

adversary-initial-state)

obtains

Some (*prefix*, *prefix-state*) $\in$

set-dist

(*execute* (*staged-alpha-prefix-program* *A*) *adversary-initial-state*)

staged-trace-root *data* = *fst prefix*

```

    staged-trace-fri-roots data = fst (snd prefix)
    staged-trace-fri-challenges data = fst (snd (snd prefix))
    staged-trace-final data = snd (snd (snd prefix))
    prefix-state ≤ attacker-state
proof -
from outcome obtain after-out where prefix-out:
    Some (prefix, prefix-state) ∈
      set-dist
        (execute (staged-alpha-prefix-program A) adversary-initial-state)
and after-out:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-after-alpha-prefix-program A prefix)
          prefix-state)
unfolding checked-staged-transcript-with-alpha-prefix-program-def
by (auto elim!: set-dist-bindE)
have prefix-ext: prefix-state ≤ attacker-state
by (rule checked-staged-after-alpha-prefix-program-outcome-extension
      [OF - - after-out])
      (fact wf, fact controlled)
have root-eq:
    staged-trace-root data = fst prefix
and trace-roots-eq:
    staged-trace-fri-roots data = fst (snd prefix)
and trace-bs-eq:
    staged-trace-fri-challenges data = fst (snd (snd prefix))
and trace-final-eq:
    staged-trace-final data = snd (snd (snd prefix))
proof -
obtain fr trace-roots trace-bs trace-final where prefix-eq:
    prefix = (fr, trace-roots, trace-bs, trace-final)
by (cases prefix) simp
from after-out[unfolded prefix-eq]
obtain as s1 dg s2 s3 composition-roots composition-bs s4
    composition-final s5 s6 query-chunks where ret:
    Some (data, attacker-state) ∈
      set-dist
        (execute
          (return
            (|staged-trace-root = fr,
              staged-trace-fri-roots = trace-roots,
              staged-trace-fri-challenges = trace-bs,
              staged-trace-final = trace-final,
              staged-alphas = as,
              staged-degree = dg,
              staged-composition-fri-roots = composition-roots,
              staged-composition-fri-challenges = composition-bs,
              staged-composition-final = composition-final,
              staged-query-chunks = query-chunks|)))

```

```

s6)
unfolding checked-staged-after-alpha-prefix-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have data-eq:
  data =
    (|staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots = composition-roots,
     staged-composition-fri-challenges = composition-bs,
     staged-composition-final = composition-final,
     staged-query-chunks = query-chunks|)
  using ret by simp
show staged-trace-root data = fst prefix
  unfolding prefix-eq data-eq by simp
show staged-trace-fri-roots data = fst (snd prefix)
  unfolding prefix-eq data-eq by simp
show staged-trace-fri-challenges data = fst (snd (snd prefix))
  unfolding prefix-eq data-eq by simp
show staged-trace-final data = snd (snd (snd prefix))
  unfolding prefix-eq data-eq by simp
qed
show ?thesis
  by (rule that[OF prefix-out root-eq trace-roots-eq trace-bs-eq
    trace-final-eq prefix-ext])
qed

lemma checked-staged-transcript-program-alpha-prefix-support:
assumes outcome:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
obtains prefix prefix-state
where
  Some (prefix, prefix-state) ∈
    set-dist
      (execute (staged-alpha-prefix-program A)
        adversary-initial-state)
  staged-trace-root data = fst prefix
  staged-trace-fri-roots data = fst (snd prefix)
  staged-trace-fri-challenges data = fst (snd (snd prefix))
  staged-trace-final data = snd (snd (snd prefix))
proof –
have staged-out:
  Some (data, attacker-state) ∈

```

```

    set-dist
    (execute (staged-transcript-program A) adversary-initial-state)
  by (rule checked-staged-transcript-program-outcome-imp-staged-transcript-program
    [OF outcome])
then obtain prefix prefix-state where prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
    (execute (staged-alpha-prefix-program A)
      adversary-initial-state)
and after-out:
  Some (data, attacker-state) ∈
    set-dist
    (execute (staged-after-alpha-prefix-program A prefix)
      prefix-state)
unfolding staged-transcript-program-alpha-prefix-decomp
by (auto elim!: set-dist-bindE)
have root-eq:
  staged-trace-root data = fst prefix
and trace-roots-eq:
  staged-trace-fri-roots data = fst (snd prefix)
and trace-bs-eq:
  staged-trace-fri-challenges data = fst (snd (snd prefix))
and trace-final-eq:
  staged-trace-final data = snd (snd (snd prefix))
proof –
obtain fr trace-roots trace-bs trace-final where prefix-eq:
  prefix = (fr, trace-roots, trace-bs, trace-final)
by (cases prefix) simp
from after-out[unfolded prefix-eq]
obtain as s1 dg s2 s3 composition-roots composition-bs s4
  composition-final s5 s6 query-chunks where ret:
  Some (data, attacker-state) ∈
    set-dist
    (execute
      (return
        (|staged-trace-root = fr,
          staged-trace-fri-roots = trace-roots,
          staged-trace-fri-challenges = trace-bs,
          staged-trace-final = trace-final,
          staged-alphas = as,
          staged-degree = dg,
          staged-composition-fri-roots = composition-roots,
          staged-composition-fri-challenges = composition-bs,
          staged-composition-final = composition-final,
          staged-query-chunks = query-chunks|))
      s6)
unfolding staged-after-alpha-prefix-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have data-eq:

```

```

data =
  (|staged-trace-root = fr,
   staged-trace-fri-roots = trace-roots,
   staged-trace-fri-challenges = trace-bs,
   staged-trace-final = trace-final,
   staged-alphas = as,
   staged-degree = dg,
   staged-composition-fri-roots = composition-roots,
   staged-composition-fri-challenges = composition-bs,
   staged-composition-final = composition-final,
   staged-query-chunks = query-chunks|)
using ret by simp
show staged-trace-root data = fst prefix
  unfolding prefix-eq data-eq by simp
show staged-trace-fri-roots data = fst (snd prefix)
  unfolding prefix-eq data-eq by simp
show staged-trace-fri-challenges data = fst (snd (snd prefix))
  unfolding prefix-eq data-eq by simp-all
show staged-trace-final data = snd (snd (snd prefix))
  unfolding prefix-eq data-eq by simp
qed
show ?thesis
  by (rule that[OF prefix-out root-eq trace-roots-eq trace-bs-eq
    trace-final-eq])
qed

lemma checked-staged-transcript-program-alpha-prefix-match-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  obtains prefix prefix-state
  where
    Some (prefix, prefix-state) ∈
      set-dist
        (execute (staged-alpha-prefix-program A)
          adversary-initial-state)
    staged-trace-root data = fst prefix
    staged-trace-fri-roots data = fst (snd prefix)
    staged-trace-fri-challenges data = fst (snd (snd prefix))
    staged-trace-final data = snd (snd (snd prefix))
    prefix-state ≤ attacker-state
proof -
  have staged-out:
    Some (data, attacker-state) ∈
      set-dist

```

```

    (execute (staged-transcript-program A) adversary-initial-state)
  by (rule checked-staged-transcript-program-outcome-imp-staged-transcript-program
    [OF outcome])
then obtain prefix prefix-state where prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
      (execute (staged-alpha-prefix-program A)
        adversary-initial-state)
and after-out:
  Some (data, attacker-state) ∈
    set-dist
      (execute (staged-after-alpha-prefix-program A prefix)
        prefix-state)
unfolding staged-transcript-program-alpha-prefix-decomp
by (auto elim!: set-dist-bindE)
have prefix-ext: prefix-state ≤ attacker-state
by (rule staged-after-alpha-prefix-program-outcome-extension
    [OF wf controlled after-out])
have root-eq:
  staged-trace-root data = fst prefix
and trace-roots-eq:
  staged-trace-fri-roots data = fst (snd prefix)
and trace-bs-eq:
  staged-trace-fri-challenges data = fst (snd (snd prefix))
and trace-final-eq:
  staged-trace-final data = snd (snd (snd prefix))
proof –
obtain fr trace-roots trace-bs trace-final where prefix-eq:
  prefix = (fr, trace-roots, trace-bs, trace-final)
by (cases prefix) simp
from after-out[unfolded prefix-eq]
obtain as s1 dg s2 s3 composition-roots composition-bs s4
  composition-final s5 s6 query-chunks where ret:
  Some (data, attacker-state) ∈
    set-dist
      (execute
        (return
          (|staged-trace-root = fr,
            staged-trace-fri-roots = trace-roots,
            staged-trace-fri-challenges = trace-bs,
            staged-trace-final = trace-final,
            staged-alphas = as,
            staged-degree = dg,
            staged-composition-fri-roots = composition-roots,
            staged-composition-fri-challenges = composition-bs,
            staged-composition-final = composition-final,
            staged-query-chunks = query-chunks|))
          s6)
unfolding staged-after-alpha-prefix-program-def Let-def

```



```

    by (auto elim!: set-dist-bindE split: prod.splits)
  have data-eq:
    data =
      (|staged-trace-root = fr,
        staged-trace-fri-roots = trace-roots,
        staged-trace-fri-challenges = trace-bs,
        staged-trace-final = trace-final,
        staged-alphas = as,
        staged-degree = dg,
        staged-composition-fri-roots = composition-roots,
        staged-composition-fri-challenges = composition-bs,
        staged-composition-final = composition-final,
        staged-query-chunks = query-chunks|)
    using ret by simp
  show staged-trace-root data = fst prefix
    unfolding prefix-eq data-eq by simp
  show staged-trace-fri-roots data = fst (snd prefix)
    unfolding prefix-eq data-eq by simp
  show staged-trace-fri-challenges data = fst (snd (snd prefix))
    unfolding prefix-eq data-eq by simp
  show staged-trace-final data = snd (snd (snd prefix))
    unfolding prefix-eq data-eq by simp
qed
show ?thesis
  by (rule that[OF prefix-out root-eq trace-roots-eq trace-bs-eq
    trace-final-eq prefix-ext])
qed

lemma checked-staged-security-with-data-state-alpha-prefix-tree-output-hitI:
  assumes prefix-out:
    Some (prefix, prefix-state) ∈
      set-dist
        (execute (staged-alpha-prefix-program A)
          adversary-initial-state)
  and root-eq: staged-trace-root data = fst prefix
  and trace-roots-eq:
    staged-trace-fri-roots data = fst (snd prefix)
  and trace-bs-eq:
    staged-trace-fri-challenges data = fst (snd (snd prefix))
  and trace-final-eq:
    staged-trace-final data = snd (snd (snd prefix))
  and verify-out:
    Some (result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))))
  and bound:
    accepted-with-bound-tables

```

```

    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state)) trace-table composition-table as
    query-idxs
  and header:
    verifier-header-transcript
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    as dg composition-fri-roots final rest
  and trace-created: created-tree trace-table trace-tree final-state
  and composition-created:
    created-tree composition-table composition-tree final-state
  and tree-hit:
    hash-map-new-output-hit
    (set-tree trace-tree  $\cup$  set-tree composition-tree)
    prefix-state final-state
  shows
    checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A
    (Some (((data, attacker-state), checked-result), checked-final-state))
proof -
  have prefix-tuple:
    prefix =
      (staged-trace-root data, staged-trace-fri-roots data,
        staged-trace-fri-challenges data, staged-trace-final data)
  by (cases prefix; cases snd prefix; cases snd (snd prefix))
    (use root-eq trace-roots-eq trace-bs-eq trace-final-eq in auto)
  show ?thesis
  unfolding checked-staged-security-with-data-state-alpha-prefix-tree-output-hit-def
    Let-def
  apply (simp add: prefix-tuple)
  apply (rule exI[where x=prefix-state])
  apply (rule conjI)
  using prefix-out prefix-tuple
  apply simp
  apply (rule exI[where x=result])
  apply (rule exI[where x=final-state])
  apply (rule conjI)
  using verify-out
  apply simp
  apply (rule exI[where x=trace-table])
  apply (rule exI[where x=composition-table])
  apply (rule exI[where x=as])
  apply (rule conjI)
  apply (rule exI[where x=query-idxs])
  using bound
  apply simp

```

```

apply (rule conjI)
apply (rule exI[where x=dg])
apply (rule exI[where x=composition-fri-roots])
apply (rule exI[where x=final])
apply (rule exI[where x=rest])
  using header
  apply simp
apply (rule exI[where x=trace-tree])
apply (rule conjI)
  using trace-created
  apply simp
apply (rule exI[where x=composition-tree])
using composition-created tree-hit
apply simp
done
qed

lemma checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-imp-prefix-or-tree-output-hit-on-su
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  out ∈
    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
and hit:
  checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
  bad-sets out
shows
  checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
  bad-sets out ∨
  checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A
  out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-def
  by simp
next
case (Some result-pack)
then obtain data attacker-state checked-result checked-final-state where
  out-eq:
    out =
      Some (((data, attacker-state), checked-result),
        checked-final-state)
  by (cases result-pack, auto split: prod.splits)
let ?s =

```

```

    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
have support-some:
  Some (((data, attacker-state), checked-result),
    checked-final-state) ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
using support unfolding out-eq .
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support-some]
have builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and checked-verify:
  Some (checked-result, checked-final-state) ∈
    set-dist (execute verify-monad ?s)
by blast+
from checked-staged-transcript-program-alpha-prefix-match-support
  [OF wf controlled builder]
obtain prefix prefix-state where prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
      (execute (staged-alpha-prefix-program A)
        adversary-initial-state)
and root-eq: staged-trace-root data = fst prefix
and trace-roots-eq:
  staged-trace-fri-roots data = fst (snd prefix)
and trace-bs-eq:
  staged-trace-fri-challenges data = fst (snd (snd prefix))
and trace-final-eq:
  staged-trace-final data = snd (snd (snd prefix))
and prefix-ext: prefix-state ≤ attacker-state
by blast
have prefix-tuple:
  prefix =
    (staged-trace-root data, staged-trace-fri-roots data,
      staged-trace-fri-challenges data, staged-trace-final data)
by (cases prefix; cases snd prefix; cases snd (snd prefix))
  (use root-eq trace-roots-eq trace-bs-eq trace-final-eq in auto)
have union-bad:
  staged-alphas data ∈
    alpha-header-supported-union-bad-sets ?s bad-sets
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
using hit unfolding out-eq

```

```

    checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-def
  by simp
then have alphas-space: staged-alphas data ∈ alpha-space
  unfolding alpha-header-supported-union-bad-sets-def by auto
from union-bad obtain trace-table where header-candidate:
  trace-table ∈
    alpha-header-supported-trace-table-candidates ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
and bad:
  staged-alphas data ∈ bad-sets trace-table
  unfolding alpha-header-supported-union-bad-sets-def by blast
from header-candidate obtain candidate-out composition-table as query-idxs
  dg composition-fri-roots final rest where candidate-out:
  candidate-out ∈ set-dist (execute verify-monad ?s)
and bound:
  accepted-with-bound-tables ?s candidate-out trace-table
  composition-table as query-idxs
and candidate-header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  as dg composition-fri-roots final rest
  unfolding alpha-header-supported-trace-table-candidates-def by blast
from bound obtain result final-state where candidate-out-eq:
  candidate-out = Some (result, final-state)
  unfolding accepted-with-bound-tables-def by blast
have verifier-out:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  using candidate-out candidate-out-eq by simp
have bound-some:
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table as query-idxs
  using bound candidate-out-eq by simp
have prefix-s: prefix-state ≤ ?s
  by (rule
    alpha-prefix-hash-extends-verifier-state-from-adversary-right
    [OF prefix-ext])
have s-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier-out])
have prefix-final: prefix-state ≤ final-state
  by (rule hash-ext-trans[OF prefix-s s-final])
show ?thesis
proof (rule
  accepted-with-bound-tables-initial-roots-pullback-or-new-tree-output-hit
  [OF prefix-final bound-some])
  fix fr f-fri-roots f-final dg' composition-fri-roots' final'

```

```

    rest' trace-tree composition-tree
assume pull-header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg'
    composition-fri-roots' final' rest'
and trace-created:
    created-tree trace-table trace-tree final-state
and composition-created:
    created-tree composition-table composition-tree final-state
and split:
    (merkle-root-binds-table fr trace-table prefix-state  $\wedge$ 
     merkle-root-binds-table (hd composition-fri-roots')
     composition-table prefix-state)  $\vee$ 
    hash-map-new-output-hit
    (set-tree trace-tree  $\cup$  set-tree composition-tree)
    prefix-state final-state
have pull-header-eq:
    fr = staged-trace-root data  $\wedge$ 
    f-fri-roots = staged-trace-fri-roots data  $\wedge$ 
    f-final = staged-trace-final data  $\wedge$ 
    dg' = dg  $\wedge$ 
    composition-fri-roots' = composition-fri-roots  $\wedge$ 
    final' = final  $\wedge$ 
    rest' = rest
using verifier-header-transcript-unique[OF candidate-header pull-header]
by simp
then have fr-eq: fr = staged-trace-root data
by simp
from split show ?thesis
proof
  assume binds:
    merkle-root-binds-table fr trace-table prefix-state  $\wedge$ 
    merkle-root-binds-table (hd composition-fri-roots')
    composition-table prefix-state
  have trace-len: length trace-table = clength * scale
    using bound-some unfolding accepted-with-bound-tables-def
    accepted-with-tables-def by simp
  have prefix-candidate:
    trace-table  $\in$ 
    alpha-prefix-trace-table-candidates prefix-state
    (staged-trace-root data)
    by (rule alpha-prefix-trace-table-candidateI[OF trace-len])
    (use binds fr-eq in simp)
  have prefix-bad:
    staged-alphas data  $\in$ 
    alpha-prefix-union-bad-sets prefix-state bad-sets
    (staged-trace-root data)
    unfolding alpha-prefix-union-bad-sets-def
    using alphas-space prefix-candidate bad by auto
have

```

```

checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
  bad-sets out
unfolding out-eq
  checked-staged-security-with-data-state-alpha-prefix-bad-set-hit-def
apply simp
apply (rule exI[where  $x = \text{prefix-state}$ ])
using prefix-out prefix-tuple prefix-bad
apply simp
done
then show ?thesis by simp
next
assume tree-hit:
  hash-map-new-output-hit
  ( $\text{set-tree trace-tree} \cup \text{set-tree composition-tree}$ )
  prefix-state final-state
have
  checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A
  out
unfolding out-eq
by (rule
  checked-staged-security-with-data-state-alpha-prefix-tree-output-hitI
  [OF prefix-out root-eq trace-roots-eq trace-bs-eq trace-final-eq
  verifier-out bound-some candidate-header trace-created
  composition-created tree-hit])
then show ?thesis by simp
qed
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit-imp-prefix-or-tree-output-hit-on-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  out  $\in$ 
  set-dist
  (execute
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
  A)
  adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit
  bad-sets out
shows
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  bad-sets out  $\vee$ 
  checked-staged-security-with-actual-alpha-prefix-tree-output-hit out
proof (cases out)
case None
then show ?thesis

```

```

using hit
unfolding
  checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit-def
by simp
next
case (Some result-pack)
then obtain prefix prefix-state data attacker-state checked-result
  checked-final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state),
        checked-result), checked-final-state)
    by (cases result-pack, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have support-some:
  Some (((((prefix, prefix-state), data), attacker-state),
    checked-result), checked-final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
    using support unfolding out-eq .
from support-some obtain trans-out where trans-out:
  Some (((prefix, prefix-state), data), attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-with-alpha-prefix-program A)
      adversary-initial-state)
and checked-verify:
  Some (checked-result, checked-final-state) ∈
    set-dist (execute verify-monad ?s)
unfolding
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
by (auto elim!: set-dist-bindE split: prod.splits)
from checked-staged-transcript-with-alpha-prefix-program-support
  [OF wf controlled trans-out]
have prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
    (execute (staged-alpha-prefix-program A)
      adversary-initial-state)
and root-eq: staged-trace-root data = fst prefix
and trace-roots-eq:
  staged-trace-fri-roots data = fst (snd prefix)
and trace-bs-eq:
  staged-trace-fri-challenges data = fst (snd (snd prefix))
and trace-final-eq:
  staged-trace-final data = snd (snd (snd prefix))

```



```

and prefix-ext:  $\text{prefix-state} \leq \text{attacker-state}$ 
by blast+
have union-bad:
  staged-alphas data  $\in$ 
    alpha-header-supported-union-bad-sets ?s bad-sets
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
using hit unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit-def
  Let-def
by simp
then have alphas-space: staged-alphas data  $\in$  alpha-space
  unfolding alpha-header-supported-union-bad-sets-def by auto
from union-bad obtain trace-table where header-candidate:
  trace-table  $\in$ 
    alpha-header-supported-trace-table-candidates ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
and bad:
  staged-alphas data  $\in$  bad-sets trace-table
unfolding alpha-header-supported-union-bad-sets-def by blast
from header-candidate obtain candidate-out composition-table as query-idxs
  dg composition-fri-roots final rest where candidate-out:
  candidate-out  $\in$  set-dist (execute verify-monad ?s)
and bound:
  accepted-with-bound-tables ?s candidate-out trace-table
  composition-table as query-idxs
and candidate-header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  as dg composition-fri-roots final rest
unfolding alpha-header-supported-trace-table-candidates-def by blast
from bound obtain result final-state where candidate-out-eq:
  candidate-out = Some (result, final-state)
unfolding accepted-with-bound-tables-def by blast
have verifier-out:
  Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)
using candidate-out candidate-out-eq by simp
have bound-some:
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table as query-idxs
using bound candidate-out-eq by simp
have prefix-s:  $\text{prefix-state} \leq ?s$ 
by (rule
  alpha-prefix-hash-extends-verifier-state-from-adversary-right

```

```

    [OF prefix-ext])
have s-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier-out])
have prefix-final: prefix-state ≤ final-state
  by (rule hash-ext-trans[OF prefix-s s-final])
show ?thesis
proof (rule
  accepted-with-bound-tables-initial-roots-pullback-or-new-tree-output-hit
  [OF prefix-final bound-some])
fix fr f-fri-roots f-final dg' composition-fri-roots' final'
  rest' trace-tree composition-tree
assume pull-header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg'
  composition-fri-roots' final' rest'
and trace-created:
  created-tree trace-table trace-tree final-state
and composition-created:
  created-tree composition-table composition-tree final-state
and split:
  (merkle-root-binds-table fr trace-table prefix-state ∧
   merkle-root-binds-table (hd composition-fri-roots')
   composition-table prefix-state) ∨
  hash-map-new-output-hit
  (set-tree trace-tree ∪ set-tree composition-tree)
  prefix-state final-state
have pull-header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  dg' = dg ∧
  composition-fri-roots' = composition-fri-roots ∧
  final' = final ∧
  rest' = rest
using verifier-header-transcript-unique[OF candidate-header pull-header]
by simp
then have fr-eq: fr = staged-trace-root data
  by simp
from split show ?thesis
proof
assume binds:
  merkle-root-binds-table fr trace-table prefix-state ∧
  merkle-root-binds-table (hd composition-fri-roots')
  composition-table prefix-state
have trace-len: length trace-table = clength * scale
  using bound-some unfolding accepted-with-bound-tables-def
  accepted-with-tables-def by simp
have prefix-candidate:
  trace-table ∈
  alpha-prefix-trace-table-candidates prefix-state (fst prefix)

```

```

    by (rule alpha-prefix-trace-table-candidateI[OF trace-len])
      (use binds fr-eq root-eq in simp)
  have prefix-bad:
    staged-alphas data ∈
      alpha-prefix-union-bad-sets prefix-state bad-sets (fst prefix)
  unfolding alpha-prefix-union-bad-sets-def
  using alphas-space prefix-candidate bad by auto
  have
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
      bad-sets out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit-def
    Let-def
  using prefix-bad
  by simp
  then show ?thesis by simp
next
  assume tree-hit:
    hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
    prefix-state final-state
  have
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit
      out
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-tree-output-hitI
    [OF verifier-out bound-some candidate-header trace-created
      composition-created tree-hit])
  then show ?thesis by simp
qed
qed
qed

lemma checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-bound-from-actual-prefix-and-tree:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and prefix-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
        bad-sets)
    adversary-initial-state ≤ P
  and tree-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit

```

$adversary-initial-state \leq T$
shows
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit$
 $bad-sets)$
 $adversary-initial-state \leq P + T$
proof –
let $?M =$
 $checked-staged-security-experiment-with-actual-alpha-prefix-data-state \ A$
let $?Prefix =$
 $checked-staged-security-with-actual-alpha-prefix-bad-set-hit \ bad-sets$
let $?Tree = checked-staged-security-with-actual-alpha-prefix-tree-output-hit$
let $?Header =$
 $checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit$
 $bad-sets$
have $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit$
 $bad-sets)$
 $adversary-initial-state =$
 $wp-event \ ?M \ ?Header \ adversary-initial-state$
by (rule
 $checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-actual-projection)$
also have $\dots \leq wp-event \ ?M \ (\lambda out. \ ?Prefix \ out \vee \ ?Tree \ out)$
 $adversary-initial-state$
proof (rule $wp-event-mono-on-support$)
fix out
assume $support: out \in set-dist \ (execute \ ?M \ adversary-initial-state)$
and $hit: ?Header \ out$
show $?Prefix \ out \vee \ ?Tree \ out$
by (rule
 $checked-staged-security-with-actual-alpha-prefix-header-supported-bad-set-hit-imp-prefix-or-tree-output-hit-on-su$
 $[OF \ wf \ controlled \ support \ hit])$
qed
also have $\dots \leq$
 $wp-event \ ?M \ ?Prefix \ adversary-initial-state +$
 $wp-event \ ?M \ ?Tree \ adversary-initial-state$
by (rule $wp-event-union-bound$)
also have $\dots \leq P + T$
by (rule $add-mono[OF \ prefix-bound \ tree-bound]$)
finally show $?thesis \ .$
qed

lemma $checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-bound-from-prefix-and-tree:$
assumes $wf: staged-budget-wellformed \ budgets$
and $controlled: staged-adversary-controlled \ budgets \ A$
and $prefix-bound:$
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(checked-staged-security-with-data-state-alpha-prefix-bad-set-hit \ A$
 $bad-sets)$

```

    adversary-initial-state  $\leq P$ 
and tree-bound:
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
      (checked-staged-security-with-data-state-alpha-prefix-tree-output-hit  $A$ )
      adversary-initial-state  $\leq T$ 
shows
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
      (checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
        bad-sets)
      adversary-initial-state  $\leq P + T$ 
proof –
  let ? $M$  = checked-staged-security-experiment-with-data-state  $A$ 
  let ?Prefix =
    checked-staged-security-with-data-state-alpha-prefix-bad-set-hit  $A$ 
    bad-sets
  let ?Tree =
    checked-staged-security-with-data-state-alpha-prefix-tree-output-hit  $A$ 
  let ?Header =
    checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit
    bad-sets
  have wp-event ? $M$  ?Header adversary-initial-state  $\leq$ 
    wp-event ? $M$  ( $\lambda out. ?Prefix\ out \vee ?Tree\ out$ )
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support:
      out  $\in$  set-dist (execute ? $M$  adversary-initial-state)
    and hit: ?Header out
    show ?Prefix out  $\vee$  ?Tree out
    by (rule
      checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-imp-prefix-or-tree-output-hit-on-suppp
      [OF wf controlled support hit])
  qed
  also have ...  $\leq$ 
    wp-event ? $M$  ?Prefix adversary-initial-state +
    wp-event ? $M$  ?Tree adversary-initial-state
    by (rule wp-event-union-bound)
  also have ...  $\leq P + T$ 
    by (rule add-mono[OF prefix-bound tree-bound])
  finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-alpha-prefix-and-tree:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and prefix-bound:
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
    (checked-staged-security-with-data-state-alpha-prefix-bad-set-hit  $A$ 
      composition-trace-bad-alpha-space)

```

$adversary-initial-state \leq P$
and *tree-bound*:
 $wp-event$ (*checked-staged-security-experiment-with-data-state A*)
 $(checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A)$
 $adversary-initial-state \leq T$
shows
 $wp-event$ (*checked-staged-security-experiment-with-data-state A*)
 $(staged-security-with-data-state-verifier-event composition-bad)$
 $adversary-initial-state \leq P + T$
proof –
have *alpha-bound*:
 $wp-event$ (*checked-staged-security-experiment-with-data-state A*)
 $(checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit$
 $composition-trace-bad-alpha-space)$
 $adversary-initial-state \leq P + T$
by (*rule*
 $checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-bound-from-prefix-and-tree$
 $[OF\ wf\ controlled\ prefix-bound\ tree-bound])$)
show *?thesis*
by (*rule*
 $checked-staged-security-with-data-state-composition-bad-bound-from-alpha-header-supported-bad-set-general$
 $[OF\ wf\ controlled\ alpha-bound])$)
qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-and-tree*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *prefix-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $(checked-staged-security-with-actual-alpha-prefix-bad-set-hit$
 $composition-trace-bad-alpha-space)$
 $adversary-initial-state \leq P$
and *tree-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-tree-output-hit$
 $adversary-initial-state \leq T$
shows
 $wp-event$ (*checked-staged-security-experiment-with-data-state A*)
 $(staged-security-with-data-state-verifier-event composition-bad)$
 $adversary-initial-state \leq P + T$
proof –
have *alpha-bound*:
 $wp-event$ (*checked-staged-security-experiment-with-data-state A*)
 $(checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit$
 $composition-trace-bad-alpha-space)$

```

    adversary-initial-state  $\leq P + T$ 
  by (rule
    checked-staged-security-with-data-state-alpha-header-supported-bad-set-hit-bound-from-actual-prefix-and-tree
    [OF wf controlled prefix-bound tree-bound])
  show ?thesis
  by (rule
    checked-staged-security-with-data-state-composition-bad-bound-from-alpha-header-supported-bad-set-general
    [OF wf controlled alpha-bound])
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Tree-Output
imports Staged-Security-Experiment-Composition
begin

```

Bridges for the composition tree-output side event.

The actual-alpha-prefix experiment carries the concrete alpha-prefix state through the verifier run. This layer relates that path-specific event back to the ordinary checked data-state experiment, so later bounds can be proved once for the data-state event.

```

context soundness
begin

```

```

lemma hash-map-new-output-hit-Un-iff:
  hash-map-new-output-hit  $(A \cup B) s t \longleftrightarrow$ 
  hash-map-new-output-hit  $A s t \vee$  hash-map-new-output-hit  $B s t$ 
unfolding hash-map-new-output-hit-def by blast

```

```

lemma hash-new-output-hit-event-Un-iff:
  hash-new-output-hit-event  $(A \cup B) s out \longleftrightarrow$ 
  hash-new-output-hit-event  $A s out \vee$ 
  hash-new-output-hit-event  $B s out$ 
unfolding hash-new-output-hit-event-def
by (cases out) (auto simp: hash-map-new-output-hit-Un-iff)

```

```

lemma hash-map-new-output-hit-tree-root-or-subtree:
assumes hit: hash-map-new-output-hit (set-tree tree)  $s t$ 
shows
  hash-map-new-output-hit {value tree}  $s t \vee$ 
   $(\exists l v r. tree = \langle l, v, r \rangle \wedge$ 
    hash-map-new-output-hit (set-tree  $l \cup$  set-tree  $r$ )  $s t)$ 
using hit
unfolding hash-map-new-output-hit-def
by (cases tree) auto

```

```

lemma hash-map-new-output-hit-two-trees-root-or-subtree:

```

```

assumes hit:
  hash-map-new-output-hit (set-tree trace-tree  $\cup$  set-tree composition-tree)
    s t
shows
  hash-map-new-output-hit {value trace-tree, value composition-tree} s t  $\vee$ 
    ( $\exists l\ v\ r.$  trace-tree =  $\langle l, v, r \rangle \wedge$ 
      hash-map-new-output-hit (set-tree l  $\cup$  set-tree r) s t)  $\vee$ 
    ( $\exists l\ v\ r.$  composition-tree =  $\langle l, v, r \rangle \wedge$ 
      hash-map-new-output-hit (set-tree l  $\cup$  set-tree r) s t)
proof –
from hit have split:
  hash-map-new-output-hit (set-tree trace-tree) s t  $\vee$ 
    hash-map-new-output-hit (set-tree composition-tree) s t
  unfolding hash-map-new-output-hit-Un-iff .
from split show ?thesis
proof
  assume trace-hit: hash-map-new-output-hit (set-tree trace-tree) s t
  from hash-map-new-output-hit-tree-root-or-subtree[OF trace-hit]
  show ?thesis
  proof
    assume root-hit: hash-map-new-output-hit {value trace-tree} s t
    have hash-map-new-output-hit
      {value trace-tree, value composition-tree} s t
    by (rule hash-map-new-output-hit-subset[OF - root-hit]) auto
    then show ?thesis by simp
  next
    assume  $\exists l\ v\ r.$  trace-tree =  $\langle l, v, r \rangle \wedge$ 
      hash-map-new-output-hit (set-tree l  $\cup$  set-tree r) s t
    then show ?thesis by blast
  qed
next
  assume composition-hit:
    hash-map-new-output-hit (set-tree composition-tree) s t
  from hash-map-new-output-hit-tree-root-or-subtree[OF composition-hit]
  show ?thesis
  proof
    assume root-hit:
      hash-map-new-output-hit {value composition-tree} s t
    have hash-map-new-output-hit
      {value trace-tree, value composition-tree} s t
    by (rule hash-map-new-output-hit-subset[OF - root-hit]) auto
    then show ?thesis by simp
  next
    assume  $\exists l\ v\ r.$  composition-tree =  $\langle l, v, r \rangle \wedge$ 
      hash-map-new-output-hit (set-tree l  $\cup$  set-tree r) s t
    then show ?thesis by blast
  qed
qed
qed

```



```

lemma card-set-tree-le-inorder-length:
  card (set-tree t) ≤ length (inorder t)
  using card-length[of inorder t]
  by simp

lemma protocol-created-tree-inorder-length-bound:
  assumes created: protocol-created-tree xs t s
  shows length (inorder t) ≤ 2 * length xs - 1
  using created
proof (induction xs arbitrary: t rule: length-induct)
  case (1 xs)
  show ?case
  proof (cases length xs = 0)
    case True
    then have xs-eq: xs = []
    by simp
    then show ?thesis
    using 1.prems
    unfolding protocol-created-tree-def
    by (simp add: protocol-merkle.created-tree.simps)
  next
  case False
  show ?thesis
  proof (cases length xs = 1)
    case True
    then obtain x where xs-eq: xs = [x]
    by (cases xs) auto
    from protocol-created-tree-singletonD[OF 1.prems[unfolded xs-eq]]
    obtain h where t-eq: t = ⟨⟨⟩, h, ⟨⟩⟩
    by blast
    then show ?thesis
    using xs-eq by simp
  next
  case False
  then have len-ge: 2 ≤ length xs
  using ⟨length xs ≠ 0⟩ by linarith
  let ?i = length xs div 2
  have i-pos: 0 < ?i
  using len-ge by simp
  have i-lt: ?i < length xs
  using len-ge by simp
  from 1.prems obtain l h r where
    i-def: ?i = length xs div 2
    and
    t-eq: t = ⟨l, h, r⟩
    and left: protocol-created-tree (take ?i xs) l s
    and right: protocol-created-tree (drop ?i xs) r s
    using protocol-created-tree-internalD[OF 1.prems len-ge]

```

```

    by blast
  have drop-lt: length (drop ?i xs) < length xs
    using i-pos by simp
  have right-bound: length (inorder r) ≤
    2 * length (drop ?i xs) - 1
    using 1.IH[rule-format, OF drop-lt, of r] right by simp
  have take-lt: length (take ?i xs) < length xs
    using i-lt by simp
  have left-bound: length (inorder l) ≤
    2 * length (take ?i xs) - 1
    using 1.IH[rule-format, OF take-lt, of l] left by simp
  have take-pos: 0 < length (take ?i xs)
    using i-pos i-lt by simp
  have drop-pos: 0 < length (drop ?i xs)
    using i-lt by simp
  have length (inorder t) =
    Suc (length (inorder l) + length (inorder r))
    using t-eq by simp
  also have ... ≤
    Suc ((2 * length (take ?i xs) - 1) +
      (2 * length (drop ?i xs) - 1))
    using left-bound right-bound by simp
  also have ... = 2 * length xs - 1
    using take-pos drop-pos i-lt by simp
  finally show ?thesis .
qed
qed
qed

lemma created-tree-inorder-length-bound:
  assumes created: created-tree xs t s
  shows length (inorder t) ≤ 2 * length xs - 1
  using protocol-created-tree-inorder-length-bound[
    OF created[unfolded created-tree-def]]
  .

lemma created-tree-card-set-tree-bound:
  assumes created: created-tree xs t s
  shows card (set-tree t) ≤ 2 * length xs - 1
  by (rule order.trans[OF card-set-tree-le-inorder-length
    created-tree-inorder-length-bound[OF created]])

lemma protocol-merkle-created-tree-same-inputs-value-eq:
  assumes left: protocol-merkle.created-tree xs left-tree s
    and right: protocol-merkle.created-tree xs right-tree s
  shows value left-tree = value right-tree
  using assms
proof (induction xs arbitrary: left-tree right-tree rule: length-induct)
  case (1 xs)

```

```

show ?case
proof (cases xs)
  case Nil
  then show ?thesis
    using 1.premis
    by (simp add: protocol-merkle.created-tree.simps)
next
case (Cons x rest)
have xs-cons: xs = x # rest
  using Cons by simp
show ?thesis
proof (cases rest)
  case Nil
  then obtain left-hash right-hash where
    left-tree:
      left-tree = ⟨⟨⟩, left-hash, ⟨⟩⟩
      fmlookup (HashMap s) x = Some left-hash
  and right-tree:
      right-tree = ⟨⟨⟩, right-hash, ⟨⟩⟩
      fmlookup (HashMap s) x = Some right-hash
  using 1.premis Cons Nil
  by (auto simp: protocol-merkle.created-tree.simps)
  then show ?thesis by simp
next
case (Cons y ys)
have xs-eq: xs = x # y # ys
  using xs-cons Cons by simp
let ?i = length xs div 2
have len: 2 ≤ length xs
  using xs-eq by simp
obtain ll lh lr i where i-def: i = ?i
  and left-tree-eq: left-tree = ⟨ll, lh, lr⟩
  and ll-created: protocol-merkle.created-tree (take i xs) ll s
  and lr-created: protocol-merkle.created-tree (drop i xs) lr s
  and lh-lookup:
    fmlookup (HashMap s) (MerkleNode (value ll) (value lr)) =
      Some lh
  using 1.premis(1) xs-eq
  by (auto simp: protocol-merkle.created-tree.simps Let-def)
obtain rl rh rr j where j-def: j = ?i
  and right-tree-eq: right-tree = ⟨rl, rh, rr⟩
  and rl-created: protocol-merkle.created-tree (take j xs) rl s
  and rr-created: protocol-merkle.created-tree (drop j xs) rr s
  and rh-lookup:
    fmlookup (HashMap s) (MerkleNode (value rl) (value rr)) =
      Some rh
  using 1.premis(2) xs-eq
  by (auto simp: protocol-merkle.created-tree.simps Let-def)
have j-i: j = i

```

```

    using i-def j-def by simp
  have i-pos: 0 < i
    using i-def len by simp
  have i-lt: i < length xs
    using i-def len by simp
  have take-less: length (take i xs) < length xs
    using i-lt by simp
  have drop-less: length (drop i xs) < length xs
    using i-pos i-lt by simp
  have left-value: value ll = value rl
    by (rule 1.IH[rule-format, OF take-less ll-created])
    (use rl-created j-i in simp)
  have right-value: value lr = value rr
    by (rule 1.IH[rule-format, OF drop-less lr-created])
    (use rr-created j-i in simp)
  have lh = rh
    using lh-lookup rh-lookup left-value right-value by simp
  then show ?thesis
    using left-tree-eq right-tree-eq by simp
qed
qed
qed

```

lemma *protocol-created-tree-same-table-value-eq:*
assumes *left: protocol-created-tree xs left-tree s*
and *right: protocol-created-tree xs right-tree s*
shows *value left-tree = value right-tree*
by (*rule protocol-merkle-created-tree-same-inputs-value-eq*
[OF left[unfolded protocol-created-tree-def]
right[unfolded protocol-created-tree-def]])

lemma *created-tree-same-table-value-eq:*
assumes *left: created-tree xs left-tree s*
and *right: created-tree xs right-tree s*
shows *value left-tree = value right-tree*
by (*rule protocol-created-tree-same-table-value-eq*
[OF left[unfolded created-tree-def] right[unfolded created-tree-def]])

definition *merkle-node-child-output-relation*
 $:: 'f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}$
where
merkle-node-child-output-relation $x \ y \longleftrightarrow$
 $(\exists z. x = \text{MerkleNode } y \ z \vee x = \text{MerkleNode } z \ y)$

lemma *merkle-node-child-output-relation-values-subset:*
 $\{y. \text{merkle-node-child-output-relation } x \ y\} \subseteq$
(case x *of*
 $\text{MerkleNode } l \ r \Rightarrow \{l, r\}$
 $| _ \Rightarrow \{\}$ *)*

unfolding merkle-node-child-output-relation-def
by (cases x) auto

lemma merkle-node-child-output-relation-fiber-card-bound:
 $\text{card } \{y. \text{merkle-node-child-output-relation } x \ y\} \leq 2$

proof –
have finite-values:
 finite
 (case x of
 MerkleNode l r $\Rightarrow \{l, r\}$
 | - $\Rightarrow \{\}$)
by (cases x) simp-all
have card {y. merkle-node-child-output-relation x y} $\leq$
 card
 (case x of
 MerkleNode l r $\Rightarrow \{l, r\}$
 | - $\Rightarrow \{\}$)
by (rule card-mono[OF finite-values])
 (rule merkle-node-child-output-relation-values-subset)
also have ... ≤ 2
proof (cases x)
 case (MerkleNode l r)
 then show ?thesis
 by (simp add: card-insert-if)
qed simp-all
finally show ?thesis .
qed

definition merkle-root-or-child-output-relation
 $:: 'f \text{ set} \Rightarrow 'f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}$
where
 merkle-root-or-child-output-relation roots x y $\longleftrightarrow$
 $y \in \text{roots} \vee \text{merkle-node-child-output-relation } x \ y$

lemma merkle-root-or-child-output-relation-values-subset:
 $\{y. \text{merkle-root-or-child-output-relation roots } x \ y\} \subseteq$
 $\text{roots} \cup$
 (case x of
 MerkleNode l r $\Rightarrow \{l, r\}$
 | - $\Rightarrow \{\}$)
unfolding merkle-root-or-child-output-relation-def
using merkle-node-child-output-relation-values-subset
by blast

lemma merkle-root-or-child-output-relation-fiber-card-bound:
assumes finite-roots: finite roots
shows card {y. merkle-root-or-child-output-relation roots x y} $\leq$
 card roots + 2
proof –

```

have finite-values:
  finite
    (roots  $\cup$ 
      (case x of
        MerkleNode l r  $\Rightarrow \{l, r\}$ 
        |  $- \Rightarrow \{\}$ )))
  using finite-roots by (cases x) simp-all
have card {y. merkle-root-or-child-output-relation roots x y}  $\leq$ 
  card
    (roots  $\cup$ 
      (case x of
        MerkleNode l r  $\Rightarrow \{l, r\}$ 
        |  $- \Rightarrow \{\}$ )))
  by (rule card-mono[OF finite-values])
    (rule merkle-root-or-child-output-relation-values-subset)
also have ...  $\leq$ 
  card roots +
  card
    (case x of
      MerkleNode l r  $\Rightarrow \{l, r\}$ 
      |  $- \Rightarrow \{\}$ )
proof -
  show ?thesis
    by (rule card-Un-le)
qed
also have ...  $\leq$  card roots + 2
proof (cases x)
  case (MerkleNode l r)
  then show ?thesis
    by (simp add: card-insert-if)
qed simp-all
finally show ?thesis .
qed

lemma hash-relation-program-protocol-check-authentication-path-raw:
  fixes leaf :: 'f protocol-hash-input
  assumes fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
  shows hash-relation-program R b (Suc (length path))
    (protocol-merkle.check-authentication-path len i leaf path ::
      ('f, 'f protocol-channel) state-monad)
proof (induction path arbitrary: len i)
  case Nil
  have hash-relation-program R b 1
    (hash leaf :: ('f, 'f protocol-channel) state-monad)
    by (rule hash-relation-program-hash[OF fibers])
  then show ?case
    by (simp add: protocol-merkle.check-authentication-path.simps)
next
  case (Cons a path)

```

```

show ?case
proof (cases i < len div 2)
  case True
  have rec:
    hash-relation-program R b (Suc (length path))
      (protocol-merkle.check-authentication-path (len div 2) i leaf path ::
        ('f, 'f protocol-channel) state-monad)
    using Cons.IH[of len div 2 i] by simp
  have bound:
    hash-relation-program R b (Suc (length path) + 1)
      ((protocol-merkle.check-authentication-path (len div 2) i leaf path ::
        ('f, 'f protocol-channel) state-monad)  $\gg$ 
        ( $\lambda x$ . hash (MerkleNode x a)))
    by (rule hash-relation-program-bind)
      (rule rec, rule hash-relation-program-hash[OF fibers])
  show ?thesis
    using True bound
    by (simp add: protocol-merkle.check-authentication-path.simps)
next
  case False
  have rec:
    hash-relation-program R b (Suc (length path))
      (protocol-merkle.check-authentication-path
        (len div 2) (i - len div 2) leaf path ::
        ('f, 'f protocol-channel) state-monad)
    using Cons.IH[of len div 2 i - len div 2] by simp
  have bound:
    hash-relation-program R b (Suc (length path) + 1)
      ((protocol-merkle.check-authentication-path
        (len div 2) (i - len div 2) leaf path ::
        ('f, 'f protocol-channel) state-monad)  $\gg$ 
        ( $\lambda x$ . hash (MerkleNode a x)))
    by (rule hash-relation-program-bind)
      (rule rec, rule hash-relation-program-hash[OF fibers])
  show ?thesis
    using False bound
    by (simp add: protocol-merkle.check-authentication-path.simps)
qed
qed

lemma hash-relation-program-check-authentication-path:
  assumes fibers:  $\bigwedge x$ . card {y. R x y}  $\leq$  b
  shows hash-relation-program R b (Suc (length path))
    (check-authentication-path len i v path)
  unfolding check-authentication-path-def protocol-check-authentication-path-def
  by (rule hash-relation-program-protocol-check-authentication-path-raw
    [OF fibers])

```

lemma hash-relation-program-check-authentication-path-assert-return:

```

fixes path :: 'f list
and v root :: 'f
and r :: 'r
assumes fibers:  $\bigwedge x. \text{card } \{y. R\ x\ y\} \leq b$ 
shows
  hash-relation-program R b (Suc (length path))
    ((check-authentication-path len i v path ::
      ('f, 'f protocol-channel) state-monad)  $\gg=$ 
      ( $\lambda ap. \text{assert } (ap = \text{root}) \gg= (\lambda -. \text{return } r)$ )))
proof -
  have tail:
    hash-relation-program R b (0 + 0)
      (assert (ap = root)  $\gg= (\lambda -. \text{return } r)$  ::
        ('r, 'f protocol-channel) state-monad)
  for ap
  by (rule hash-relation-program-bind)
    (rule hash-relation-program-assert, rule hash-relation-program-zero
      [OF hash-map-preserving-return])
  have hash-relation-program R b (Suc (length path) + (0 + 0))
    ((check-authentication-path len i v path ::
      ('f, 'f protocol-channel) state-monad)  $\gg=$ 
      ( $\lambda ap. \text{assert } (ap = \text{root}) \gg= (\lambda -. \text{return } r)$ )))
  by (rule hash-relation-program-bind)
    (rule hash-relation-program-check-authentication-path[OF fibers],
      rule tail)
  then show ?thesis by simp
qed

```

lemma hash-relation-program-check-authentication-path-assert-bind:

```

fixes path :: 'f list
and v root :: 'f
and k :: unit  $\Rightarrow$  ('r, 'f protocol-channel) state-monad
assumes fibers:  $\bigwedge x. \text{card } \{y. R\ x\ y\} \leq b$ 
and k:  $\bigwedge u. \text{hash-relation-program } R\ b\ n\ (k\ u)$ 
shows
  hash-relation-program R b (Suc (length path) + n)
    ((check-authentication-path len i v path ::
      ('f, 'f protocol-channel) state-monad)  $\gg=$ 
      ( $\lambda ap. \text{assert } (ap = \text{root}) \gg= k$ )))
proof -
  have tail:
    hash-relation-program R b (0 + n)
      (assert (ap = root)  $\gg= k$  ::
        ('r, 'f protocol-channel) state-monad)
  for ap
  by (rule hash-relation-program-bind)
    (rule hash-relation-program-assert, rule k)
  have hash-relation-program R b (Suc (length path) + (0 + n))
    ((check-authentication-path len i v path ::

```



```

      ('f, 'f protocol-channel) state-monad)  $\gg$ 
      ( $\lambda ap.$  assert ( $ap = \text{root}$ )  $\gg$   $k$ ))
  by (rule hash-relation-program-bind)
      (rule hash-relation-program-check-authentication-path[OF fibers],
       rule tail)
  then show ?thesis by simp
qed

lemma hash-relation-program-read:
  hash-relation-program  $R$   $b$  0
  (read :: ('f, 'f, unit) protocol-c-monad)
  by (rule hash-relation-program-zero)
      (rule hash-map-preserving-read)

lemma hash-relation-program-ntimes:
  fixes  $m :: ('x, 'f \text{ protocol-channel}) \text{ state-monad}$ 
  assumes  $m$ : hash-relation-program  $R$   $b$   $n$   $m$ 
  shows hash-relation-program  $R$   $b$  ( $k * n$ ) (ntimes  $m$   $k$ )
  using  $m$ 
proof (induction  $k$ )
  case 0
  then show ?case
    by (simp add: hash-relation-program-zero[OF hash-map-preserving-return])
next
  case (Suc  $k$ )
  have tail: hash-relation-program  $R$   $b$  ( $k * n$ ) (ntimes  $m$   $k$ )
    using Suc.IH Suc.prems by simp
  have cont:
    hash-relation-program  $R$   $b$  ( $k * n + 0$ )
    (ntimes  $m$   $k$   $\gg$  ( $\lambda xs.$  return ( $x \# xs$ ))) for  $x$ 
    by (rule hash-relation-program-bind)
        (rule tail, rule hash-relation-program-zero[OF hash-map-preserving-return])
  have hash-relation-program  $R$   $b$  ( $n + (k * n + 0)$ )
    ( $m$   $\gg$  ( $\lambda x.$  ntimes  $m$   $k$   $\gg$  ( $\lambda xs.$  return ( $x \# xs$ ))))
    by (rule hash-relation-program-bind[OF Suc.prems cont])
  then show ?case by simp
qed

lemma hash-relation-program-mmap:
  fixes  $ms :: ('x, 'f \text{ protocol-channel}) \text{ state-monad list}$ 
  assumes  $ms$ :  $\bigwedge m. m \in \text{set } ms \implies \text{hash-relation-program } R \text{ } b \text{ } n \text{ } m$ 
  shows hash-relation-program  $R$   $b$  (length  $ms * n$ ) (mmap  $ms$ )
  using  $ms$ 
proof (induction  $ms$ )
  case Nil
  then show ?case
    by (simp add: hash-relation-program-zero[OF hash-map-preserving-return])
next
  case (Cons  $m$   $ms$ )

```

```

have head: hash-relation-program R b n m
  using Cons.premis by simp
have tail: hash-relation-program R b (length ms * n) (mmap ms)
  using Cons.IH Cons.premis by simp
have cont:
  hash-relation-program R b (length ms * n + 0)
    (mmap ms  $\gg$  ( $\lambda xs.$  return ( $x \# xs$ ))) for x
  by (rule hash-relation-program-bind)
    (rule tail, rule hash-relation-program-zero[OF hash-map-preserving-return])
have hash-relation-program R b (n + (length ms * n + 0))
  (m  $\gg$  ( $\lambda x.$  mmap ms  $\gg$  ( $\lambda xs.$  return ( $x \# xs$ ))))
  by (rule hash-relation-program-bind[OF head cont])
then show ?case by simp
qed

```

```

lemma hash-relation-program-mfold:
  fixes steps :: ('x  $\Rightarrow$  ('x, 'f protocol-channel) state-monad) list
  assumes steps:
     $\bigwedge \text{step } x. \text{step} \in \text{set steps} \implies$ 
      hash-relation-program R b n (step x)
  shows hash-relation-program R b (length steps * n) (mfold x steps)
  using steps
proof (induction steps arbitrary: x)
  case Nil
  then show ?case
    by (simp add: hash-relation-program-zero[OF hash-map-preserving-return])
next
  case (Cons step steps)
  have head: hash-relation-program R b n (step x)
    using Cons.premis by simp
  have tail:
     $\bigwedge y. \text{hash-relation-program } R \text{ b } (length \text{ steps } * n) (mfold y \text{ steps})$ 
    using Cons.IH Cons.premis by simp
  have hash-relation-program R b (n + length steps * n)
    (step x  $\gg$  ( $\lambda y.$  mfold y steps))
    by (rule hash-relation-program-bind[OF head tail])
  then show ?case by simp
qed

```

```

lemma hash-relation-program-ntimes-read-bind:
  fixes k :: 'f list  $\Rightarrow$  ('y, 'f protocol-channel) state-monad
  assumes k:
     $\bigwedge xs. length \text{ xs} = n \implies \text{hash-relation-program } R \text{ b } q (k \text{ xs})$ 
  shows
    hash-relation-program R b q
      ((ntimes read n :: ('f list, 'f protocol-channel) state-monad)  $\gg$  k)
proof -
  have reads:
    hash-relation-program R b (n * 0)

```

```

    (ntimes read n :: ('f list, 'f protocol-channel) state-monad)
  by (rule hash-relation-program-ntimes[OF hash-relation-program-read])
have hash-relation-program R b (n * 0 + q)
  ((ntimes read n :: ('f list, 'f protocol-channel) state-monad)  $\gg=$  k)
proof (rule hash-relation-program-bind-on-outcomes[OF reads])
  fix s xs t
  assume out:
    Some (xs, t)  $\in$ 
      set-dist
        (execute
          (ntimes read n :: ('f list, 'f protocol-channel) state-monad)
            s)
  then have length xs = n
    using ntimes-read-any-outcome by blast
  then show hash-relation-program R b q (k xs)
    by (rule k)
qed
then show ?thesis by simp
qed

```

lemma *hash-relation-program-check-decommit-on-query-step:*
assumes *fibers:* $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$
shows
 hash-relation-program R b (Suc (floor-log (scale * clength)))
 (do {
 qh $\leftarrow$ read;
 let len = scale * clength;
 qh-path $\leftarrow$ ntimes read (floor-log len);
 ap $\leftarrow$ check-authentication-path len i qh qh-path;
 assert (ap = fr);
 return qh
 })
proof –
 let ?len = scale * clength
 have after-path:
 hash-relation-program R b (Suc (floor-log ?len))
 ((check-authentication-path ?len i qh qh-path ::
 ('f, 'f protocol-channel) state-monad) $\gg=$
 (λ ap. assert (ap = fr) $\gg=$ (λ -. return qh)))
 if path-len: length qh-path = floor-log ?len
 for qh qh-path
 using path-len
 by (subst path-len[symmetric])
 (rule hash-relation-program-check-authentication-path-assert-return
 [OF fibers])
 have after-reads:
 hash-relation-program R b (Suc (floor-log ?len))
 ((ntimes read (floor-log ?len) ::
 ('f list, 'f protocol-channel) state-monad) $\gg=$

```

    (λqh-path.
      (check-authentication-path ?len i qh qh-path ::
        ('f, 'f protocol-channel) state-monad) >>=
      (λap. assert (ap = fr) >>= (λ-. return qh))))
  for qh
  by (rule hash-relation-program-ntimes-read-bind)
    (rule after-path)
  have hash-relation-program R b (0 + Suc (floor-log ?len))
    (read >>=
      (λqh.
        (ntimes read (floor-log ?len) ::
          ('f list, 'f protocol-channel) state-monad) >>=
          (λqh-path.
            (check-authentication-path ?len i qh qh-path ::
              ('f, 'f protocol-channel) state-monad) >>=
              (λap. assert (ap = fr) >>= (λ-. return qh)))))
      by (rule hash-relation-program-bind)
        (rule hash-relation-program-read, rule after-reads)
    then show ?thesis
    by (simp add: Let-def)
qed

lemma hash-relation-program-check-decommit-on-query:
  assumes fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
  shows
    hash-relation-program R b
      (length (powers-scaled idx) * Suc (floor-log (scale * clength)))
      (mmap (check-decommit-on-query fr idx))
proof -
  let ?step =
    λi. do {
      qh ← read;
      let len = scale * clength;
      qh-path ← ntimes read (floor-log len);
      ap ← check-authentication-path len i qh qh-path;
      assert (ap = fr);
      return qh
    }
  have mapped-general:
     $\bigwedge xs. \text{hash-relation-program } R \ b$ 
      (length xs + length xs * floor-log (scale * clength))
      (mmap (map ?step xs))
proof -
  fix xs
  show hash-relation-program R b
    (length xs + length xs * floor-log (scale * clength))
    (mmap (map ?step xs))
proof (induction xs)
case Nil

```

```

then show ?case
  by (simp add: hash-relation-program-zero[OF hash-map-preserving-return])
next
case (Cons i rest)
have head:
  hash-relation-program R b (Suc (floor-log (scale * clength)))
    (?step i)
  by (rule hash-relation-program-check-decommit-on-query-step[OF fibers])
have tail:
  hash-relation-program R b
    (length rest + length rest * floor-log (scale * clength))
    (mmap (map ?step rest))
  by (rule Cons.IH)
have cont:
  hash-relation-program R b
    ((length rest + length rest * floor-log (scale * clength)) + 0)
    (mmap (map ?step rest)  $\gg$  ( $\lambda x$ . return (x # xs)))
  for x
  by (rule hash-relation-program-bind)
    (rule tail, rule hash-relation-program-zero[OF hash-map-preserving-return])
have hash-relation-program R b
  (Suc (floor-log (scale * clength)) +
    ((length rest + length rest * floor-log (scale * clength)) + 0))
  (?step i  $\gg$  ( $\lambda x$ . mmap (map ?step rest)  $\gg$ 
    ( $\lambda x$ . return (x # xs)))))
  by (rule hash-relation-program-bind[OF head cont])
then show ?case
  by (simp add: algebra-simps)
qed
qed
have mapped:
  hash-relation-program R b
    (length (powers-scaled idx) +
      length (powers-scaled idx) * floor-log (scale * clength))
    (mmap (map ?step (powers-scaled idx)))
  by (rule mapped-general)
show ?thesis
  using mapped
  unfolding check-decommit-on-query-def by simp
qed

```

```

lemma hash-relation-program-fri-layer-opening-finish:
assumes fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
  and xp-path-len: length xp-path = floor-log len
  and xn-path-len: length xn-path = floor-log len
shows
  hash-relation-program R b
    (Suc (floor-log len) + Suc (floor-log len))
    (fri-layer-opening-finish b' f i x len pw xp xp-path xn xn-path)

```

```

proof –
  let ?H = Suc (floor-log len)
  let ?sidx = (i + len div 2) mod len
  let ?out =
    (i mod (len div 2),
     (xp + xn) div 2 +
      b' * ((xp - xn) div (2 * ((h ^ i) * shift) ^ pw)),
     len div 2, pw + pw)
  have second:
    hash-relation-program R b ?H
    ((check-authentication-path len ?sidx xn xn-path ::
      ('f, 'f protocol-channel) state-monad) >>=
      (λap. assert (ap = f) >>= (λ-. return ?out)))
  by (subst xn-path-len[symmetric])
    (rule hash-relation-program-check-authentication-path-assert-return
      [OF fibers])
  have first0:
    hash-relation-program R b (Suc (length xp-path) + ?H)
    ((check-authentication-path len i xp xp-path ::
      ('f, 'f protocol-channel) state-monad) >>=
      (λap. assert (ap = f) >>=
        (λ-.
          (check-authentication-path len ?sidx xn xn-path ::
            ('f, 'f protocol-channel) state-monad) >>=
            (λap. assert (ap = f) >>= (λ-. return ?out))))))
  by (rule hash-relation-program-check-authentication-path-assert-bind
    [OF fibers])
    (use second in simp)
  have first:
    hash-relation-program R b (?H + ?H)
    ((check-authentication-path len i xp xp-path ::
      ('f, 'f protocol-channel) state-monad) >>=
      (λap. assert (ap = f) >>=
        (λ-.
          (check-authentication-path len ?sidx xn xn-path ::
            ('f, 'f protocol-channel) state-monad) >>=
            (λap. assert (ap = f) >>= (λ-. return ?out))))))
  using first0 xp-path-len by simp
  have raw:
    hash-relation-program R b (0 + (?H + ?H))
    ((assert (xp = x) ::
      (unit, 'f protocol-channel) state-monad) >>=
      (λ-.
        (check-authentication-path len i xp xp-path ::
          ('f, 'f protocol-channel) state-monad) >>=
          (λap. assert (ap = f) >>=
            (λ-.
              (check-authentication-path len ?sidx xn xn-path ::
                ('f, 'f protocol-channel) state-monad) >>=

```

```

      (λap. assert (ap = f) ≫ (λ-. return ?out))))))
  by (rule hash-relation-program-bind)
    (rule hash-relation-program-assert, rule first)
show ?thesis
  using raw unfolding fri-layer-opening-finish-def
  by (simp add: Let-def)
qed

lemma hash-relation-program-fri-layer-opening-step:
  assumes fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
  shows
    hash-relation-program  $R \ b$ 
      (Suc (floor-log len) + Suc (floor-log len))
      (fri-layer-opening-step (b', f) (i, x, len, pw))
proof -
  let ?N = Suc (floor-log len) + Suc (floor-log len)
  have after-xn-path:
    hash-relation-program  $R \ b \ ?N$ 
      ((ntimes read (floor-log len) ::
        ('f list, 'f protocol-channel) state-monad) ≫
        (λxn-path.
          fri-layer-opening-finish b' f i x len pw xp xp-path xn xn-path)))
  if xp-path-len: length xp-path = floor-log len
  for xp xp-path xn
proof (rule hash-relation-program-ntimes-read-bind)
  fix xn-path :: 'f list
  assume xn-path-len: length xn-path = floor-log len
  show hash-relation-program  $R \ b \ ?N$ 
    (fri-layer-opening-finish b' f i x len pw xp xp-path xn xn-path)
  by (rule hash-relation-program-fri-layer-opening-finish
      [OF fibers xp-path-len xn-path-len])
qed
have after-xn:
  hash-relation-program  $R \ b \ ?N$ 
    (read ≫
      (λxn.
        (ntimes read (floor-log len) ::
          ('f list, 'f protocol-channel) state-monad) ≫
          (λxn-path.
            fri-layer-opening-finish b' f i x len pw xp xp-path xn xn-path)))
  if xp-path-len: length xp-path = floor-log len
  for xp xp-path
proof -
  have hash-relation-program  $R \ b \ (0 + ?N)$ 
    (read ≫
      (λxn.
        (ntimes read (floor-log len) ::
          ('f list, 'f protocol-channel) state-monad) ≫
          (λxn-path.

```

```

      fri-layer-opening-finish b' f i x len pw xp xp-path xn xn-path))))
  by (rule hash-relation-program-bind)
    (rule hash-relation-program-read, rule after-xn-path[OF xp-path-len])
  then show ?thesis by simp
qed
have after-xp-path:
  hash-relation-program R b ?N
  ((ntimes read (floor-log len) ::
    ('f list, 'f protocol-channel) state-monad)  $\gg$ =
    (λxp-path.
      read  $\gg$ =
      (λxn.
        (ntimes read (floor-log len) ::
          ('f list, 'f protocol-channel) state-monad)  $\gg$ =
          (λxn-path.
            fri-layer-opening-finish b' f i x len pw xp xp-path xn xn-path))))))
  for xp
proof (rule hash-relation-program-ntimes-read-bind)
  fix xp-path :: 'f list
  assume xp-path-len: length xp-path = floor-log len
  show hash-relation-program R b ?N
    (read  $\gg$ =
      (λxn.
        (ntimes read (floor-log len) ::
          ('f list, 'f protocol-channel) state-monad)  $\gg$ =
          (λxn-path.
            fri-layer-opening-finish b' f i x len pw xp xp-path xn xn-path))))
  by (rule after-xn[OF xp-path-len])
qed
have hash-relation-program R b (0 + ?N)
  (read  $\gg$ =
    (λxp.
      (ntimes read (floor-log len) ::
        ('f list, 'f protocol-channel) state-monad)  $\gg$ =
        (λxp-path.
          read  $\gg$ =
          (λxn.
            (ntimes read (floor-log len) ::
              ('f list, 'f protocol-channel) state-monad)  $\gg$ =
              (λxn-path.
                fri-layer-opening-finish b' f i x len pw xp xp-path xn xn-path))))))
    by (rule hash-relation-program-bind)
      (rule hash-relation-program-read, rule after-xp-path)
  then show ?thesis
    unfolding fri-layer-opening-step-unfold-finish by simp
qed

```

lemma hash-relation-program-fri-layer-opening-steps-mfold-state:
 fixes st :: nat × 'f × nat × nat


```

assumes fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
and init:
   $\text{case } st \text{ of } (i, x, \text{len}, pw) \Rightarrow \text{floor-log } \text{len} \leq L$ 
shows
  hash-relation-program R b
    (length fl * (Suc L + Suc L))
    (mfold st (map fri-layer-opening-step fl) ::
      (nat × 'f × nat × nat, 'f protocol-channel) state-monad)
using init
proof (induction fl arbitrary: st)
case Nil
then show ?case
  by (simp add: hash-relation-program-zero[OF hash-map-preserving-return])
next
case (Cons bf fl)
obtain i x len pw where st-eq: st = (i, x, len, pw)
  by (cases st)
obtain b' f where bf-eq: bf = (b', f)
  by (cases bf)
let ?N = Suc L + Suc L
have head-exact:
  hash-relation-program R b
    (Suc (floor-log len) + Suc (floor-log len))
    (fri-layer-opening-step (b', f) (i, x, len, pw))
  by (rule hash-relation-program-fri-layer-opening-step[OF fibers])
have head-le:
  Suc (floor-log len) + Suc (floor-log len) ≤ ?N
  using Cons.prems unfolding st-eq by simp
have head:
  hash-relation-program R b ?N
    (fri-layer-opening-step bf st ::
      (nat × 'f × nat × nat, 'f protocol-channel) state-monad)
  unfolding bf-eq st-eq
  by (rule hash-relation-program-mono[OF head-le head-exact])
have tail:
  hash-relation-program R b (length fl * ?N)
    (mfold z (map fri-layer-opening-step fl) ::
      (nat × 'f × nat × nat, 'f protocol-channel) state-monad)
  if out:
    Some (z, t) ∈
      set-dist (execute (fri-layer-opening-step bf st) s)
  for z and s :: 'f protocol-channel and t :: 'f protocol-channel
proof –
  from fri-layer-opening-step-outcome[OF out[unfolded bf-eq st-eq]]
  obtain xp xp-path xn xn-path x' where
    z-eq: z = (i mod (len div 2), x', len div 2, pw + pw)
    by blast
  have len-le: floor-log len ≤ L
    using Cons.prems unfolding st-eq by simp

```

```

have next-len: floor-log (len div 2) ≤ L
  using floor-log-div2-le-self[of len] len-le by linarith
have z-inv:
  case z of (i, x, len, pw) ⇒ floor-log len ≤ L
  unfolding z-eq using next-len by simp
show ?thesis
  by (rule Cons.IH[OF z-inv])
qed
have hash-relation-program R b (?N + length fl * ?N)
  ((fri-layer-opening-step bf st ::
    (nat × 'f × nat × nat, 'f protocol-channel) state-monad) ≫=
    (λz. mfold z (map fri-layer-opening-step fl)))
  by (rule hash-relation-program-bind-on-outcomes[OF head]) (rule tail)
then show ?case
  by (simp add: add.assoc add.commute add.left-commute)
qed

lemma hash-relation-program-receive-query-commits-mfold:
assumes fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
and init: floor-log len ≤ L
shows
  hash-relation-program R b
    (length fl * (Suc L + Suc L))
    (mfold (i, x, len, pw) (receive-query-commits fl))
unfolding receive-query-commits-eq-fri-layer-opening-steps
proof –
have st-inv:
  case (i, x, len, pw) of
    (i, x, len, pw) ⇒ floor-log len ≤ L
  using init by simp
show hash-relation-program R b (length fl * (Suc L + Suc L))
  (mfold (i, x, len, pw) (map fri-layer-opening-step fl) ::
    (nat × 'f × nat × nat, 'f protocol-channel) state-monad)
  by (rule hash-relation-program-fri-layer-opening-steps-mfold-state
    [OF fibers st-inv])
qed

lemma hash-relation-program-check-decommit-on-query-exact:
assumes fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
shows
  hash-relation-program R b verifier-query-decommit-hash-budget
    (mmap (check-decommit-on-query fr idx))
proof –
have exact:
  hash-relation-program R b
    (length (powers-scaled idx) * Suc (floor-log (scale * clength)))
    (mmap (check-decommit-on-query fr idx))
  by (rule hash-relation-program-check-decommit-on-query[OF fibers])
show ?thesis

```

```

using exact
unfolding verifier-query-decommit-hash-budget-def powers-scaled-def
by (simp add: mult.commute)
qed

lemma hash-relation-program-verifier-query-round-program-exact:
  assumes fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
  shows
    hash-relation-program R b
      (1 + verifier-query-decommit-hash-budget +
        (length f-fl + length fl) * verifier-fri-layer-hash-budget)
      (verifier-query-round-program fr f-fl f-final as fl final)
proof -
  let ?L = floor-log (clength * scale)
  let ?B = verifier-fri-layer-hash-budget
  have B-eq: ?B = Suc ?L + Suc ?L
    unfolding verifier-fri-layer-hash-budget-def by simp
  have after-comp:
    hash-relation-program R b (length fl * ?B + 0)
      ((mfold
        (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
        (receive-query-commits fl) ::
        (nat × 'f × nat × nat, 'f protocol-channel) state-monad)  $\gg$ 
        ( $\lambda(i, x, \text{len}, \text{pw}). \text{assert } (x = \text{final})$ )))
    for idx fv
  proof (rule hash-relation-program-bind)
  show hash-relation-program R b (length fl * ?B)
    (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat, 'f protocol-channel) state-monad)
    unfolding B-eq
    by (rule hash-relation-program-receive-query-commits-mfold)
      (rule fibers, simp)
  show  $\bigwedge x. \text{hash-relation-program } R \ b \ 0$ 
    ((case x of (i, x, len, pw)  $\Rightarrow$  assert (x = final)) ::
      (unit, 'f protocol-channel) state-monad)
    by (simp add: hash-relation-program-assert split: prod.splits)
qed
have after-trace:
  hash-relation-program R b
    (length f-fl * ?B + (0 + (length fl * ?B + 0)))
    ((mfold (idx, hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat, 'f protocol-channel) state-monad)  $\gg$ 
      ( $\lambda(f-i, f-x, f-len, f-pow).$ 
        assert (f-x = f-final)  $\gg$ 
        ( $\lambda-. \text{mfold}$ 
          (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)

```

```

    (receive-query-commits fl) ::
    (nat × 'f × nat × nat, 'f protocol-channel) state-monad) ≫=
    (λ(i, x, len, pw). assert (x = final))))))
for idx fv
proof (rule hash-relation-program-bind)
show hash-relation-program R b (length f-fl * ?B)
  (mfold (idx, hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat, 'f protocol-channel) state-monad)
  unfolding B-eq
  by (rule hash-relation-program-receive-query-commits-mfold)
    (rule fibers, simp)
fix x :: nat × 'f × nat × nat
obtain f-i f-x f-len f-pow where x-eq:
  x = (f-i, f-x, f-len, f-pow)
  by (cases x) auto
have case-budget:
  hash-relation-program R b (0 + (length fl * ?B + 0))
    ((assert (f-x = f-final) ::
    (unit, 'f protocol-channel) state-monad) ≫=
    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat, 'f protocol-channel) state-monad) ≫=
      (λ(i, x, len, pw). assert (x = final))))))
  by (rule hash-relation-program-bind)
    (rule hash-relation-program-assert, rule after-comp)
show hash-relation-program R b (0 + (length fl * ?B + 0))
  ((case x of (f-i, f-x, f-len, f-pow) ⇒
  assert (f-x = f-final) ≫=
  (λ-. (mfold
    (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
    (receive-query-commits fl) ::
    (nat × 'f × nat × nat, 'f protocol-channel) state-monad) ≫=
    (λ(i, x, len, pw). assert (x = final)))))) ::
  (unit, 'f protocol-channel) state-monad)
  using case-budget
  unfolding x-eq
  by (simp split: prod.splits)
qed
have after-query:
  hash-relation-program R b
    (verifier-query-decommit-hash-budget +
    (length f-fl * ?B + (0 + (length fl * ?B + 0))))
  (mmap (check-decommit-on-query fr idx) ≫=
  (λfv. (mfold (idx, hd fv, clength * scale, 1)
    (receive-query-commits f-fl) ::
    (nat × 'f × nat × nat, 'f protocol-channel) state-monad) ≫=
    (λ(f-i, f-x, f-len, f-pow).

```

```

    assert (f-x = f-final) >>=
    (λ-. (mfold
      (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
      (receive-query-commits fl) ::
      (nat × 'f × nat × nat, 'f protocol-channel) state-monad) >>=
      (λ(i, x, len, pw). assert (x = final))))))
  for idx
  by (rule hash-relation-program-bind)
    (rule hash-relation-program-check-decommit-on-query-exact[OF fibers],
    rule after-trace)
  have after-query-simple:
    hash-relation-program R b
      (verifier-query-decommit-hash-budget +
      (length f-fl * ?B + length fl * ?B))
      (mmap (check-decommit-on-query fr idx) >>=
      (λfv. (mfold (idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl) ::
        (nat × 'f × nat × nat, 'f protocol-channel) state-monad) >>=
        (λ(f-i, f-x, f-len, f-pow).
          assert (f-x = f-final) >>=
          (λ-. (mfold
            (idx, cp-eval as fv (h ^ idx * shift), clength * scale, 1)
            (receive-query-commits fl) ::
            (nat × 'f × nat × nat, 'f protocol-channel) state-monad) >>=
            (λ(i, x, len, pw). assert (x = final)))))))
      for idx
      using after-query[of idx] by simp
  have after-random:
    hash-relation-program R b
      (verifier-query-decommit-hash-budget +
      (length f-fl * ?B + length fl * ?B))
      ((let idx' = index (to-nat idx) in
        mmap (check-decommit-on-query fr idx') >>=
        (λfv. (mfold (idx', hd fv, clength * scale, 1)
          (receive-query-commits f-fl) ::
          (nat × 'f × nat × nat, 'f protocol-channel) state-monad) >>=
          (λ(f-i, f-x, f-len, f-pow).
            assert (f-x = f-final) >>=
            (λ-. (mfold
              (idx', cp-eval as fv (h ^ idx' * shift), clength * scale, 1)
              (receive-query-commits fl) ::
              (nat × 'f × nat × nat, 'f protocol-channel) state-monad) >>=
              (λ(i, x, len, pw). assert (x = final)))))))
      for idx
      using after-query-simple[of index (to-nat idx)]
  by (simp add: Let-def algebra-simps mult.commute split: prod.splits)
  have hash-relation-program R b
    (1 + (verifier-query-decommit-hash-budget +
    (length f-fl * ?B + length fl * ?B)))

```

```

(receive-query-index-challenge >>=
  (λidx.
    let idx' = index (to-nat idx) in
    mmap (check-decommit-on-query fr idx') >>=
    (λfv. (mfold (idx', hd fv, clength * scale, 1)
      (receive-query-commits f-fl) ::
      (nat × 'f × nat × nat, 'f protocol-channel) state-monad) >>=
      (λ(f-i, f-x, f-len, f-pow).
        assert (f-x = f-final) >>=
        (λ-. (mfold
          (idx', cp-eval as fv (h ^ idx' * shift), clength * scale, 1)
          (receive-query-commits fl) ::
          (nat × 'f × nat × nat, 'f protocol-channel) state-monad) >>=
          (λ(i, x, len, pw). assert (x = final)))))))
  by (rule hash-relation-program-bind)
  (rule hash-relation-program-receive-query-index-challenge[OF fibers],
    rule after-random)
then show ?thesis
  unfolding verifier-query-round-program-def
  by (simp add: Let-def algebra-simps mult.commute split: prod.splits)
qed

```

lemma *hash-relation-program-verifier-query-round-program:*

```

assumes fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
and len-f:  $\text{length } f\text{-fl} \leq \text{ceil-log } \text{clength}$ 
and len-c:  $\text{length } fl \leq \text{ceil-log } (\text{maxDegree} + 1)$ 
shows
  hash-relation-program  $R \ b$  verifier-query-round-hash-budget
    (verifier-query-round-program fr f-fl f-final as fl final)
proof –
  let ?N = verifier-fri-layer-hash-budget
  have exact:
    hash-relation-program  $R \ b$ 
      (1 + verifier-query-decommit-hash-budget +
        (length f-fl + length fl) * ?N)
      (verifier-query-round-program fr f-fl f-final as fl final)
  by (rule hash-relation-program-verifier-query-round-program-exact
    [OF fibers])
  have le:
    1 + verifier-query-decommit-hash-budget +
      (length f-fl + length fl) * ?N
    ≤ verifier-query-round-hash-budget
  using len-f len-c
  unfolding verifier-query-round-hash-budget-def
  by simp
  show ?thesis
  by (rule hash-relation-program-mono[OF le exact])
qed

```

lemma *wp-verifier-query-round-program-hash-relation-hit-bound:*

assumes *fibers*: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$

and *len-f*: $\text{length } f\text{-fl} \leq \text{ceil-log } \text{clength}$

and *len-c*: $\text{length } fl \leq \text{ceil-log } (\text{maxDegree} + 1)$

shows

wp-event

(*verifier-query-round-program* *fr* *f-fl* *f-final* *as fl final*)

(*hash-relation-hit-event* *R s*) $s \leq$

hash-relation-budget-value *b* *verifier-query-round-hash-budget*

proof –

have *program*:

hash-relation-program *R b* *verifier-query-round-hash-budget*

(*verifier-query-round-program* *fr f-fl f-final as fl final*)

by (*rule hash-relation-program-verifier-query-round-program*
[*OF fibers len-f len-c*])

then show *?thesis*

unfolding *hash-relation-program-def hash-relation-budget-def* **by** *blast*

qed

lemma *wp-verifier-query-round-program-merkle-root-or-child-relation-bound:*

assumes *finite-roots*: *finite roots*

and *len-f*: $\text{length } f\text{-fl} \leq \text{ceil-log } \text{clength}$

and *len-c*: $\text{length } fl \leq \text{ceil-log } (\text{maxDegree} + 1)$

shows

wp-event

(*verifier-query-round-program* *fr f-fl f-final as fl final*)

(*hash-relation-hit-event*

(*merkle-root-or-child-output-relation roots*) *s*) $s \leq$

hash-relation-budget-value (*card roots* + 2)

verifier-query-round-hash-budget

by (*rule wp-verifier-query-round-program-hash-relation-hit-bound*)

(*rule merkle-root-or-child-output-relation-fiber-card-bound*

[*OF finite-roots*], *rule len-f*, *rule len-c*)

lemma *hash-relation-program-ntimes-verifier-query-round-program:*

assumes *fibers*: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$

and *len-f*: $\text{length } f\text{-fl} \leq \text{ceil-log } \text{clength}$

and *len-c*: $\text{length } fl \leq \text{ceil-log } (\text{maxDegree} + 1)$

shows

hash-relation-program *R b* (*n* * *verifier-query-round-hash-budget*)

(*ntimes*

(*verifier-query-round-program* *fr f-fl f-final as fl final*) *n*)

proof (*rule hash-relation-program-ntimes*)

show $\bigwedge x.$

hash-relation-program *R b* *verifier-query-round-hash-budget*

(*verifier-query-round-program* *fr f-fl f-final as fl final*)

by (*rule hash-relation-program-verifier-query-round-program*
[*OF fibers len-f len-c*])

qed

lemma *wp-ntimes-verifier-query-round-program-hash-relation-hit-bound:*

assumes *fibers*: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$
and *len-f*: $\text{length } f\text{-fl} \leq \text{ceil-log } \text{clength}$
and *len-c*: $\text{length } fl \leq \text{ceil-log } (\text{maxDegree} + 1)$

shows

wp-event
 (*ntimes*
 (*verifier-query-round-program* fr *f-fl* *f-final* as *fl* *final*) *n*)
 (*hash-relation-hit-event* *R* *s*) *s* $\leq$
hash-relation-budget-value *b*
 (*n* * *verifier-query-round-hash-budget*)

proof –

have *program*:

hash-relation-program *R* *b* (*n* * *verifier-query-round-hash-budget*)
 (*ntimes*
 (*verifier-query-round-program* fr *f-fl* *f-final* as *fl* *final*) *n*)
by (rule *hash-relation-program-ntimes-verifier-query-round-program*
 [*OF fibers len-f len-c*])

then show *?thesis*

unfolding *hash-relation-program-def hash-relation-budget-def* **by** *blast*

qed

lemma *wp-ntimes-verifier-query-round-program-merkle-root-or-child-relation-bound:*

assumes *finite-roots*: *finite roots*
and *len-f*: $\text{length } f\text{-fl} \leq \text{ceil-log } \text{clength}$
and *len-c*: $\text{length } fl \leq \text{ceil-log } (\text{maxDegree} + 1)$

shows

wp-event
 (*ntimes*
 (*verifier-query-round-program* fr *f-fl* *f-final* as *fl* *final*) *n*)
 (*hash-relation-hit-event*
 (*merkle-root-or-child-output-relation* *roots*) *s*) *s* $\leq$
hash-relation-budget-value (*card roots* + 2)
 (*n* * *verifier-query-round-hash-budget*)
by (rule *wp-ntimes-verifier-query-round-program-hash-relation-hit-bound*)
 (rule *merkle-root-or-child-output-relation-fiber-card-bound*
 [*OF finite-roots*], rule *len-f*, rule *len-c*)

definition *checked-staged-security-with-actual-alpha-prefix-tree-output-case*

where

checked-staged-security-with-actual-alpha-prefix-tree-output-case *P* *out*
 $\longleftrightarrow$
 (*case out of*
 None $\Rightarrow$ *False*
 | *Some* (*x*, *-*) $\Rightarrow$
 (*let* *packed* = *fst* (*fst x*);
 prefix-state = *snd* (*fst packed*);
 data = *snd* *packed*;

$attacker_state = snd\ (fst\ x);$
 $s =$
 $verifier_state_from_adversary\ attacker_state$
 $(staged_proof_transcript\ data)$
 $in\ \exists\ result\ final_state\ trace_table\ composition_table\ as$
 $query_idxs\ dg\ composition_fri_roots\ final\ rest\ trace_tree$
 $composition_tree.$
 $Some\ (result,\ final_state) \in$
 $set_dist\ (execute\ verify_monad\ s) \wedge$
 $accepted_with_bound_tables\ s\ (Some\ (result,\ final_state))$
 $trace_table\ composition_table\ as\ query_idxs \wedge$
 $verifier_header_transcript\ s$
 $(staged_trace_root\ data)$
 $(staged_trace_fri_roots\ data)$
 $(staged_trace_final\ data)$
 $as\ dg\ composition_fri_roots\ final\ rest \wedge$
 $created_tree\ trace_table\ trace_tree\ final_state \wedge$
 $created_tree\ composition_table\ composition_tree\ final_state \wedge$
 $P\ trace_tree\ composition_tree\ prefix_state\ final_state))$

definition *checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit*
where

$checked_staged_security_with_actual_alpha_prefix_tree_root_output_hit =$
 $checked_staged_security_with_actual_alpha_prefix_tree_output_case$
 $(\lambda trace_tree\ composition_tree\ prefix_state\ final_state.$
 $hash_map_new_output_hit$
 $\{value\ trace_tree,\ value\ composition_tree\}$
 $prefix_state\ final_state)$

definition

checked-staged-security-with-actual-alpha-prefix-tree-subtree-output-hit

where

$checked_staged_security_with_actual_alpha_prefix_tree_subtree_output_hit =$
 $checked_staged_security_with_actual_alpha_prefix_tree_output_case$
 $(\lambda trace_tree\ composition_tree\ prefix_state\ final_state.$
 $(\exists\ l\ v\ r.\ trace_tree = \langle l,\ v,\ r \rangle \wedge$
 $hash_map_new_output_hit\ (set_tree\ l \cup set_tree\ r)$
 $prefix_state\ final_state) \vee$
 $(\exists\ l\ v\ r.\ composition_tree = \langle l,\ v,\ r \rangle \wedge$
 $hash_map_new_output_hit\ (set_tree\ l \cup set_tree\ r)$
 $prefix_state\ final_state))$

lemma *checked-staged-security-with-actual-alpha-prefix-tree-output-hit-imp-root-or-subtree:*

assumes *hit:*

checked-staged-security-with-actual-alpha-prefix-tree-output-hit out

shows

$checked_staged_security_with_actual_alpha_prefix_tree_root_output_hit\ out \vee$
 $checked_staged_security_with_actual_alpha_prefix_tree_subtree_output_hit$
 out

```

proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-tree-output-hit-def
    by simp
next
  case (Some result-pack)
  then obtain x checked-final-state where out-eq:
    out = Some (x, checked-final-state)
    by (cases result-pack) simp
  let ?packed = fst (fst x)
  let ?prefix-state = snd (fst ?packed)
  let ?data = snd ?packed
  let ?attacker-state = snd (fst x)
  let ?s =
    verifier-state-from-adversary ?attacker-state
    (staged-proof-transcript ?data)
  from hit obtain result final-state trace-table composition-table as
    query-idxs dg composition-fri-roots final rest trace-tree
    composition-tree where
    verify-out:
      Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
    and bound:
      accepted-with-bound-tables ?s (Some (result, final-state))
      trace-table composition-table as query-idxs
    and header:
      verifier-header-transcript ?s
      (staged-trace-root ?data)
      (staged-trace-fri-roots ?data)
      (staged-trace-final ?data)
      as dg composition-fri-roots final rest
    and trace-created:
      created-tree trace-table trace-tree final-state
    and composition-created:
      created-tree composition-table composition-tree final-state
    and tree-hit:
      hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
      ?prefix-state final-state
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit-def
    Let-def
  apply simp
  by blast
have split:
  hash-map-new-output-hit
    {value trace-tree, value composition-tree}

```

```

    ?prefix-state final-state  $\vee$ 
    ( $\exists l\ v\ r.$  trace-tree =  $\langle l, v, r \rangle \wedge$ 
      hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
      ?prefix-state final-state)  $\vee$ 
    ( $\exists l\ v\ r.$  composition-tree =  $\langle l, v, r \rangle \wedge$ 
      hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
      ?prefix-state final-state)
  by (rule hash-map-new-output-hit-two-trees-root-or-subtree
    [OF tree-hit])
from split show ?thesis
proof
  assume root-hit:
    hash-map-new-output-hit
    {value trace-tree, value composition-tree}
    ?prefix-state final-state
  have checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit
    out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit-def
    checked-staged-security-with-actual-alpha-prefix-tree-output-case-def
    Let-def
  using verify-out bound header trace-created composition-created root-hit
  apply simp
  by blast
  then show ?thesis by simp
next
  assume subtree-hit:
    ( $\exists l\ v\ r.$  trace-tree =  $\langle l, v, r \rangle \wedge$ 
      hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
      ?prefix-state final-state)  $\vee$ 
    ( $\exists l\ v\ r.$  composition-tree =  $\langle l, v, r \rangle \wedge$ 
      hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
      ?prefix-state final-state)
  have
    checked-staged-security-with-actual-alpha-prefix-tree-subtree-output-hit
    out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-tree-subtree-output-hit-def
    checked-staged-security-with-actual-alpha-prefix-tree-output-case-def
    Let-def
  using verify-out bound header trace-created composition-created
    subtree-hit
  apply simp
  by blast
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-tree-output-hit-bound-from-root-and-subtree:

```

```

assumes root-bound:
  wp-event m
    checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit s
       $\leq R$ 
and subtree-bound:
  wp-event m
    checked-staged-security-with-actual-alpha-prefix-tree-subtree-output-hit s
       $\leq S$ 
shows
  wp-event m checked-staged-security-with-actual-alpha-prefix-tree-output-hit
     $s \leq R + S$ 
proof -
have
  wp-event m checked-staged-security-with-actual-alpha-prefix-tree-output-hit
     $s \leq$ 
  wp-event m
    ( $\lambda out.$ 
      checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit
         $out \vee$ 
        checked-staged-security-with-actual-alpha-prefix-tree-subtree-output-hit
           $out$ )
     $s$ 
by (rule wp-event-mono)
  (rule
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit-imp-root-or-subtree)
also have  $\dots \leq$ 
  wp-event m
    checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit s +
  wp-event m
    checked-staged-security-with-actual-alpha-prefix-tree-subtree-output-hit
       $s$ 
by (rule wp-event-union-bound)
also have  $\dots \leq R + S$ 
by (rule add-mono[OF root-bound subtree-bound])
finally show ?thesis .
qed

```

```

definition checked-staged-security-with-actual-alpha-prefix-transcript-root-output-hit
where
  checked-staged-security-with-actual-alpha-prefix-transcript-root-output-hit
     $out \longleftrightarrow$ 
    (case out of
      None  $\Rightarrow$  False
      | Some (x, -)  $\Rightarrow$ 
        (let packed = fst (fst x);
          prefix-state = snd (fst packed);
          data = snd packed;
          attacker-state = snd (fst x);
           $s =$ 

```

verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
in $\exists$ result final-state trace-table composition-table as
query-idxs dg composition-fri-roots final rest.
Some (result, final-state) $\in$
set-dist (execute verify-monad s) $\wedge$
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs $\wedge$
verifier-header-transcript s
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
as dg composition-fri-roots final rest $\wedge$
hash-map-new-output-hit
{staged-trace-root data, hd composition-fri-roots}
prefix-state final-state))

lemma *checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit-imp-transcript-root-output-hit:*

assumes *root-hit:*

checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit out

shows

checked-staged-security-with-actual-alpha-prefix-transcript-root-output-hit
out

proof (*cases out*)

case *None*

then show *?thesis*

using *root-hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit-def

checked-staged-security-with-actual-alpha-prefix-tree-output-case-def

by *simp*

next

case (*Some result-pack*)

then obtain *x checked-final-state* **where** *out-eq:*

out = Some (x, checked-final-state)

by (*cases result-pack*) *simp*

let *?packed = fst (fst x)*

let *?prefix-state = snd (fst ?packed)*

let *?data = snd ?packed*

let *?attacker-state = snd (fst x)*

let *?s =*

verifier-state-from-adversary ?attacker-state

(staged-proof-transcript ?data)

from *root-hit* **obtain** *result final-state trace-table composition-table as*

query-idxs dg composition-fri-roots final rest trace-tree

composition-tree **where**

verify-out:

Some (result, final-state) $\in$ set-dist (execute verify-monad ?s)

and *bound:*

```

    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
and header:
    verifier-header-transcript ?s
    (staged-trace-root ?data)
    (staged-trace-fri-roots ?data)
    (staged-trace-final ?data)
    as dg composition-fri-roots final rest
and trace-created:
    created-tree trace-table trace-tree final-state
and composition-created:
    created-tree composition-table composition-tree final-state
and tree-root-hit:
    hash-map-new-output-hit
    {value trace-tree, value composition-tree}
    ?prefix-state final-state
unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit-def
    checked-staged-security-with-actual-alpha-prefix-tree-output-case-def
    Let-def
apply simp
by blast
from bound obtain fr' f-fri-roots' f-final' dg'
    composition-fri-roots' final' rest' where
    header':
    verifier-header-transcript ?s fr' f-fri-roots' f-final' as dg'
    composition-fri-roots' final' rest'
and trace-bind:
    merkle-root-binds-table fr' trace-table final-state
and composition-bind:
    merkle-root-binds-table (hd composition-fri-roots')
    composition-table final-state
unfolding accepted-with-bound-tables-def by auto
have header-eq:
    fr' = staged-trace-root ?data ∧
    f-fri-roots' = staged-trace-fri-roots ?data ∧
    f-final' = staged-trace-final ?data ∧
    dg' = dg ∧
    composition-fri-roots' = composition-fri-roots ∧
    final' = final ∧
    rest' = rest
using verifier-header-transcript-unique[OF header' header]
by simp
then have fr-eq: fr' = staged-trace-root ?data
and comp-roots-eq:
    composition-fri-roots' = composition-fri-roots
by simp-all
from trace-bind obtain trace-root-tree where
    trace-root-created:

```

```

    created-tree trace-table trace-root-tree final-state
  and trace-root-eq: fr' = value trace-root-tree
  unfolding merkle-root-binds-table-def by blast
have trace-value-eq:
  value trace-tree = staged-trace-root ?data
  using created-tree-same-table-value-eq
    [OF trace-created trace-root-created]
    trace-root-eq fr-eq
  by simp
from composition-bind obtain composition-root-tree where
  composition-root-created:
    created-tree composition-table composition-root-tree final-state
  and composition-root-eq:
    hd composition-fri-roots' = value composition-root-tree
  unfolding merkle-root-binds-table-def by blast
have composition-value-eq:
  value composition-tree = hd composition-fri-roots
  using created-tree-same-table-value-eq
    [OF composition-created composition-root-created]
    composition-root-eq comp-roots-eq
  by simp
have transcript-root-hit:
  hash-map-new-output-hit
    {staged-trace-root ?data, hd composition-fri-roots}
    ?prefix-state final-state
  using tree-root-hit trace-value-eq composition-value-eq by simp
show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-transcript-root-output-hit-def
    Let-def
  using verify-out bound header transcript-root-hit
  apply simp
  by blast
qed

lemma checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit-bound-from-transcript-roots:
  assumes root-bound:
    wp-event m
    checked-staged-security-with-actual-alpha-prefix-transcript-root-output-hit
    s ≤ R
  shows
    wp-event m
    checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit s
    ≤ R
  by (rule order.trans[OF - root-bound])
  (rule wp-event-mono,
   rule
    checked-staged-security-with-actual-alpha-prefix-tree-root-output-hit-imp-transcript-root-output-hit)

```

definition*checked-staged-security-with-actual-alpha-prefix-transcript-root-preverifier-output-hit***where***checked-staged-security-with-actual-alpha-prefix-transcript-root-preverifier-output-hit**out* $\longleftrightarrow$

(case out of

None $\Rightarrow$ *False*| *Some* (*x*, -) $\Rightarrow$ (let *packed* = *fst* (*fst* *x*);*prefix-state* = *snd* (*fst* *packed*);*data* = *snd* *packed*;*attacker-state* = *snd* (*fst* *x*);*s* =*verifier-state-from-adversary* *attacker-state*(*staged-proof-transcript* *data*)in $\exists$ *result* *final-state* *trace-table* *composition-table* as*query-idxs* *dg* *composition-fri-roots* *final* *rest*.*Some* (*result*, *final-state*) $\in$ *set-dist* (*execute* *verify-monad* *s*) $\wedge$ *accepted-with-bound-tables* *s* (*Some* (*result*, *final-state*))*trace-table* *composition-table* as *query-idxs* $\wedge$ *verifier-header-transcript* *s*(*staged-trace-root* *data*)(*staged-trace-fri-roots* *data*)(*staged-trace-final* *data*)as *dg* *composition-fri-roots* *final* *rest* $\wedge$ *hash-map-new-output-hit*{*staged-trace-root* *data*, *hd* *composition-fri-roots*}*prefix-state* *s*))**definition***checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit***where***checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit**out* $\longleftrightarrow$

(case out of

None $\Rightarrow$ *False*| *Some* (*x*, -) $\Rightarrow$ (let *packed* = *fst* (*fst* *x*);*data* = *snd* *packed*;*attacker-state* = *snd* (*fst* *x*);*s* =*verifier-state-from-adversary* *attacker-state*(*staged-proof-transcript* *data*)in $\exists$ *result* *final-state* *trace-table* *composition-table* as*query-idxs* *dg* *composition-fri-roots* *final* *rest*.*Some* (*result*, *final-state*) $\in$ *set-dist* (*execute* *verify-monad* *s*) $\wedge$ *accepted-with-bound-tables* *s* (*Some* (*result*, *final-state*))

trace-table composition-table as query-idxs $\wedge$
verifier-header-transcript s
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
as dg composition-fri-roots final rest $\wedge$
hash-map-new-output-hit
{staged-trace-root data, hd composition-fri-roots}
s final-state))

lemma *checked-staged-security-with-actual-alpha-prefix-transcript-root-output-hit-imp-preverifier-or-verifier-on-suppo*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

out $\in$
set-dist
(execute
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
adversary-initial-state)

and *hit:*

checked-staged-security-with-actual-alpha-prefix-transcript-root-output-hit
out

shows

checked-staged-security-with-actual-alpha-prefix-transcript-root-preverifier-output-hit
out $\vee$
checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit
out

proof (*cases out*)

case *None*

then show *?thesis*

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-transcript-root-output-hit-def

by *simp*

next

case (*Some result-pack*)

then obtain *prefix prefix-state data attacker-state checked-result*

checked-final-state **where** *out-eq:*

out =
Some (((((prefix, prefix-state), data), attacker-state),
checked-result), checked-final-state)

by (*cases result-pack, auto split: prod.splits*)

let *?s* =

verifier-state-from-adversary attacker-state
(staged-proof-transcript data)

have *support-some:*

Some (((((prefix, prefix-state), data), attacker-state),
checked-result), checked-final-state) $\in$

```

    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  using support unfolding out-eq .
from support-some obtain trans-out where trans-out:
  Some (((prefix, prefix-state), data), attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-with-alpha-prefix-program A)
      adversary-initial-state)
  unfolding
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
  by (auto elim!: set-dist-bindE split: prod.splits)
from checked-staged-transcript-with-alpha-prefix-program-support
  [OF wf controlled trans-out]
have prefix-ext: prefix-state ≤ attacker-state
  by blast
have prefix-s: prefix-state ≤ ?s
  by (rule
    alpha-prefix-hash-extends-verifier-state-from-adversary-right
    [OF prefix-ext])
from hit obtain result final-state trace-table composition-table as
  query-idxs dg composition-fri-roots final rest where
  verify-out:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  and bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and header:
    verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    as dg composition-fri-roots final rest
  and root-hit:
    hash-map-new-output-hit
    {staged-trace-root data, hd composition-fri-roots}
    prefix-state final-state
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-transcript-root-output-hit-def
    Let-def
  by simp blast
have s-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verify-out])
from hash-map-new-output-hit-trans-decomp
  [OF prefix-s s-final root-hit]
show ?thesis
proof

```

```

assume pre:
  hash-map-new-output-hit
    {staged-trace-root data, hd composition-fri-roots}
    prefix-state ?s
have
  checked-staged-security-with-actual-alpha-prefix-transcript-root-preverifier-output-hit
    out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-transcript-root-preverifier-output-hit-def
  Let-def
using verify-out bound header pre
by simp blast
then show ?thesis by simp
next
assume verifier:
  hash-map-new-output-hit
    {staged-trace-root data, hd composition-fri-roots}
    ?s final-state
have
  checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit
    out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit-def
  Let-def
using verify-out bound header verifier
by simp blast
then show ?thesis by simp
qed
qed

lemma verifier-header-transcript-initial-roots-subset-transcript:
assumes header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and nonempty: composition-fri-roots  $\neq \square$ 
shows {fr, hd composition-fri-roots}  $\subseteq$  set (PTranscript s)
using header nonempty
unfolding verifier-header-transcript-def verifier-header-messages-def
by (cases composition-fri-roots) auto

definition
checked-staged-security-with-actual-alpha-prefix-verifier-transcript-output-hit
where
checked-staged-security-with-actual-alpha-prefix-verifier-transcript-output-hit
  out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some (x, -)  $\Rightarrow$ 
    (let packed = fst (fst x);
```

```

      data = snd packed;
      attacker-state = snd (fst x);
      s =
        verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)
    in  $\exists$  result final-state trace-table composition-table as
        query-idxs.
      Some (result, final-state)  $\in$ 
        set-dist (execute verify-monad s)  $\wedge$ 
        accepted-with-bound-tables s (Some (result, final-state))
        trace-table composition-table as query-idxs  $\wedge$ 
        hash-map-new-output-hit (set (PTranscript s)) s final-state))

lemma checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit-imp-verifier-transcript-outp
assumes hit:
  checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit
  out
shows
  checked-staged-security-with-actual-alpha-prefix-verifier-transcript-output-hit
  out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit-def
  by simp
next
case (Some result-pack)
then obtain x checked-final-state where out-eq:
  out = Some (x, checked-final-state)
  by (cases result-pack) simp
let ?packed = fst (fst x)
let ?data = snd ?packed
let ?attacker-state = snd (fst x)
let ?s =
  verifier-state-from-adversary ?attacker-state
    (staged-proof-transcript ?data)
from hit obtain result final-state trace-table composition-table as
  query-idxs dg composition-fri-roots final rest where
  verify-out:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)
  and bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and header:
    verifier-header-transcript ?s
    (staged-trace-root ?data)
    (staged-trace-fri-roots ?data)

```

```

    (staged-trace-final ?data)
    as dg composition-fri-roots final rest
and root-hit:
    hash-map-new-output-hit
    {staged-trace-root ?data, hd composition-fri-roots}
    ?s final-state
unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit-def
    Let-def
by simp blast
from bound obtain result' final-state' fr' f-fri-roots' f-final' dg'
    composition-fri-roots' final' rest' where
    out-eq':
    Some (result, final-state) = Some (result', final-state')
and header':
    verifier-header-transcript ?s fr' f-fri-roots' f-final' as dg'
    composition-fri-roots' final' rest'
and nonempty': composition-fri-roots' ≠ []
unfolding accepted-with-bound-tables-def by blast
have header-eq:
    fr' = staged-trace-root ?data ∧
    f-fri-roots' = staged-trace-fri-roots ?data ∧
    f-final' = staged-trace-final ?data ∧
    dg' = dg ∧
    composition-fri-roots' = composition-fri-roots ∧
    final' = final ∧
    rest' = rest
using verifier-header-transcript-unique[OF header' header]
by simp
have nonempty: composition-fri-roots ≠ []
using nonempty' header-eq by simp
have subset:
    {staged-trace-root ?data, hd composition-fri-roots}
    ⊆ set (PTranscript ?s)
by (rule verifier-header-transcript-initial-roots-subset-transcript
    [OF header nonempty])
have transcript-hit:
    hash-map-new-output-hit (set (PTranscript ?s)) ?s final-state
by (rule hash-map-new-output-hit-subset[OF subset root-hit])
show ?thesis
unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-verifier-transcript-output-hit-def
    Let-def
using verify-out bound transcript-hit
by simp blast
qed

```

lemma checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit-bound-from-verifier-transcript-root-verifier-transcript-bound:

```

    wp-event m
    checked-staged-security-with-actual-alpha-prefix-verifier-transcript-output-hit
    s ≤ C
shows
    wp-event m
    checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit
    s ≤ C
by (rule order.trans[OF - transcript-bound])
    (rule wp-event-mono,
     rule
     checked-staged-security-with-actual-alpha-prefix-transcript-root-verifier-output-hit-imp-verifier-transcript-output-hit)

lemma checked-staged-security-with-actual-alpha-prefix-tree-output-hit-imp-data-state-alpha-prefix-tree-output-hit-on-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
    out ∈
    set-dist
    (execute
     (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
     adversary-initial-state)
and hit: checked-staged-security-with-actual-alpha-prefix-tree-output-hit
    out
shows
    case out of
    None ⇒ False
    | Some (((((prefix, prefix-state), data), attacker-state),
              checked-result), checked-final-state) ⇒
              checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A
              (Some (((data, attacker-state), checked-result),
                      checked-final-state)))
proof (cases out)
case None
then show ?thesis
    using hit
    unfolding checked-staged-security-with-actual-alpha-prefix-tree-output-hit-def
    by simp
next
case (Some result-pack)
then obtain prefix prefix-state data attacker-state checked-result
    checked-final-state where out-eq:
    out =
    Some (((((prefix, prefix-state), data), attacker-state),
            checked-result), checked-final-state)
by (cases result-pack, auto split: prod.splits)
let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)

```

```

from support[unfolded out-eq] obtain trans-out where trans-out:
  Some (((prefix, prefix-state), data), attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-with-alpha-prefix-program A)
      adversary-initial-state)
unfolding
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
by (auto elim!: set-dist-bindE split: prod.splits)
from checked-staged-transcript-with-alpha-prefix-program-support
  [OF wf controlled trans-out]
have prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
    (execute (staged-alpha-prefix-program A)
      adversary-initial-state)
and root-eq: staged-trace-root data = fst prefix
and trace-roots-eq:
  staged-trace-fri-roots data = fst (snd prefix)
and trace-bs-eq:
  staged-trace-fri-challenges data = fst (snd (snd prefix))
and trace-final-eq:
  staged-trace-final data = snd (snd (snd prefix))
by blast+
have actual-body:
  ∃ result final-state trace-table composition-table as query-idxs dg
    composition-fri-roots final rest trace-tree composition-tree.
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s) ∧
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table as query-idxs ∧
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  as dg composition-fri-roots final rest ∧
  created-tree trace-table trace-tree final-state ∧
  created-tree composition-table composition-tree final-state ∧
  hash-map-new-output-hit
  (set-tree trace-tree ∪ set-tree composition-tree)
  prefix-state final-state
using hit
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-tree-output-hit-def
  Let-def
by simp
from actual-body obtain result final-state trace-table composition-table as
  query-idxs dg composition-fri-roots final rest trace-tree
  composition-tree where
  verify-out:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)

```

```

and bound:
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table as query-idxs
and header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  as dg composition-fri-roots final rest
and trace-created: created-tree trace-table trace-tree final-state
and composition-created:
  created-tree composition-table composition-tree final-state
and tree-hit:
  hash-map-new-output-hit
  (set-tree trace-tree  $\cup$  set-tree composition-tree)
  prefix-state final-state
by blast
have data-hit:
  checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A
  (Some (((data, attacker-state), checked-result),
    checked-final-state))
by (rule
  checked-staged-security-with-data-state-alpha-prefix-tree-output-hitI
  [OF prefix-out root-eq trace-roots-eq trace-bs-eq trace-final-eq
    verify-out bound header trace-created composition-created
    tree-hit])
show ?thesis
  unfolding out-eq
  using data-hit by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-tree-output-hit-bound-from-data-state:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and data-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A)
  adversary-initial-state  $\leq$  T
shows
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-tree-output-hit
  adversary-initial-state  $\leq$  T
proof -
let ?M =
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let ?project =
   $\lambda x. ((snd (fst (fst x)), snd (fst x)), snd x)$ 
let ?DataTree =

```



```

checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A
have wp-event ?M checked-staged-security-with-actual-alpha-prefix-tree-output-hit
  adversary-initial-state ≤
  wp-event ?M
    (λout. case out of
      None ⇒ False
      | Some (x, t) ⇒ ?DataTree (Some (?project x, t)))
    adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈ set-dist (execute ?M adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit out
  show
    (case out of None ⇒ False
     | Some (x, t) ⇒ ?DataTree (Some (?project x, t)))
  using
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit-imp-data-state-alpha-prefix-tree-output-hit-on-sup
    [OF wf controlled support hit]
  by (cases out) (auto split: prod.splits)
qed
also have ... =
  wp-event (?M ≫= (λx. return (?project x))) ?DataTree
    adversary-initial-state
unfolding
  checked-staged-security-with-data-state-alpha-prefix-tree-output-hit-def
by (subst wp-event-bind-return-map)
  (simp split: option.splits prod.splits)
also have ... =
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?DataTree adversary-initial-state
using
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
  [of A]
by simp
also have ... ≤ T
  by (rule data-bound)
finally show ?thesis .
qed

```

Branch-specific tree-output events.

The actual-prefix tree-output events are existential over verifier witnesses for a fixed prefix. They are useful for deterministic reductions, but too coarse for direct verifier-local WP bounds. The following auxiliary events tie the witness to the current verifier branch of the checked experiment, so target-budget lemmas can be applied without changing the public theorem interface.

definition

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

where

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

out $\longleftrightarrow$

(*case out of*

None $\Rightarrow$ *False*

| *Some* (*x*, *final-state*) $\Rightarrow$

(*let packed* = *fst* (*fst x*);

data = *snd packed*;

attacker-state = *snd* (*fst x*);

result = *snd x*;

s =

verifier-state-from-adversary attacker-state

(*staged-proof-transcript data*)

in $\exists$ *trace-table composition-table as query-idxs*.

Some (*result*, *final-state*) $\in$

set-dist (*execute verify-monad s*) $\wedge$

accepted-with-bound-tables s (*Some* (*result*, *final-state*))

trace-table composition-table as query-idxs $\wedge$

hash-map-new-output-hit (*set* (*PTranscript s*)) *s final-state*)

definition

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit

where

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit

out $\longleftrightarrow$

(*case out of*

None $\Rightarrow$ *False*

| *Some* (*x*, *final-state*) $\Rightarrow$

(*let packed* = *fst* (*fst x*);

data = *snd packed*;

attacker-state = *snd* (*fst x*);

result = *snd x*;

s =

verifier-state-from-adversary attacker-state

(*staged-proof-transcript data*)

in $\exists$ *trace-table composition-table as query-idxs dg*

composition-fri-roots final rest.

Some (*result*, *final-state*) $\in$

set-dist (*execute verify-monad s*) $\wedge$

accepted-with-bound-tables s (*Some* (*result*, *final-state*))

trace-table composition-table as query-idxs $\wedge$

verifier-header-transcript s

(*staged-trace-root data*)

(*staged-trace-fri-roots data*)

(*staged-trace-final data*)

as dg composition-fri-roots final rest $\wedge$

hash-map-new-output-hit

{*staged-trace-root data*, *hd composition-fri-roots*}

s final-state))

lemma *branch-verifier-transcript-output-hit-imp-verifier-transcript-output-hit:*

assumes *hit:*

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit
out

shows

checked-staged-security-with-actual-alpha-prefix-verifier-transcript-output-hit
out

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit-def
checked-staged-security-with-actual-alpha-prefix-verifier-transcript-output-hit-def
Let-def

by (*cases out*) (*fastforce split: prod.splits*)**+**

lemma *branch-transcript-root-verifier-output-hit-imp-branch-verifier-transcript-output-hit:*

assumes *hit:*

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit
out

shows

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit
out

proof (*cases out*)

case *None*

then show *?thesis*

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-def

by *simp*

next

case (*Some result-pack*)

then obtain *x final-state* **where** *out-eq: out = Some (x, final-state)*

by (*cases result-pack*) *simp*

let *?packed = fst (fst x)*

let *?data = snd ?packed*

let *?attacker-state = snd (fst x)*

let *?result = snd x*

let *?s =*

verifier-state-from-adversary ?attacker-state
(staged-proof-transcript ?data)

from *hit* **obtain** *trace-table composition-table as query-idxs dg*

composition-fri-roots final rest **where**

verify-out:

Some (?result, final-state) ∈ set-dist (execute verify-monad ?s)

and *bound:*

accepted-with-bound-tables ?s (Some (?result, final-state))

trace-table composition-table as query-idxs

and *header:*

```

    verifier-header-transcript ?s
      (staged-trace-root ?data)
      (staged-trace-fri-roots ?data)
      (staged-trace-final ?data)
      as dg composition-fri-roots final rest
  and root-hit:
    hash-map-new-output-hit
      {staged-trace-root ?data, hd composition-fri-roots}
      ?s final-state
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-def
    Let-def
  by simp blast
from bound obtain result' final-state' fr' f-fri-roots' f-final' dg'
  composition-fri-roots' final' rest' where
  out-eq':
    Some (?result, final-state) = Some (result', final-state')
  and header':
    verifier-header-transcript ?s fr' f-fri-roots' f-final' as dg'
    composition-fri-roots' final' rest'
  and nonempty': composition-fri-roots' ≠ []
  unfolding accepted-with-bound-tables-def by blast
have header-eq:
  fr' = staged-trace-root ?data ∧
  f-fri-roots' = staged-trace-fri-roots ?data ∧
  f-final' = staged-trace-final ?data ∧
  dg' = dg ∧
  composition-fri-roots' = composition-fri-roots ∧
  final' = final ∧
  rest' = rest
  using verifier-header-transcript-unique[OF header' header]
  by simp
have nonempty: composition-fri-roots ≠ []
  using nonempty' header-eq by simp
have subset:
  {staged-trace-root ?data, hd composition-fri-roots}
  ⊆ set (PTranscript ?s)
  by (rule verifier-header-transcript-initial-roots-subset-transcript
    [OF header nonempty])
have transcript-hit:
  hash-map-new-output-hit (set (PTranscript ?s)) ?s final-state
  by (rule hash-map-new-output-hit-subset[OF subset root-hit])
show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit-def
    Let-def
  using verify-out bound transcript-hit
  by simp blast
qed

```

lemma *branch-transcript-root-verifier-output-hit-bound-from-branch-verifier-transcript-output:*
assumes *transcript-bound:*

wp-event m

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

s ≤ C

shows

wp-event m

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit

s ≤ C

by (*rule order.trans[OF - transcript-bound]*)

(*rule wp-event-mono,*

rule

branch-transcript-root-verifier-output-hit-imp-branch-verifier-transcript-output-hit)

lemma *actual-alpha-prefix-branch-verifier-transcript-output-cont-imp-hash-new-output-hit:*

fixes *packed* **and** *attacker-state*

defines *s ≡*

verifier-state-from-adversary attacker-state

(*staged-proof-transcript (snd packed)*)

assumes *hit:*

(*case out of*

None ⇒

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

None

| *Some (result, t) ⇒*

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

(*Some (((packed, attacker-state), result), t)*)

shows *hash-new-output-hit-event (set (PTranscript s)) s out*

proof (*cases out*)

case *None*

then show *?thesis*

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit-def

by *simp*

next

case (*Some result-state*)

then obtain *result t* **where** *result-state: result-state = (result, t)*

by (*cases result-state*) *simp*

have *hash-map-new-output-hit (set (PTranscript s)) s t*

using *hit Some result-state*

unfolding

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit-def

s-def Let-def

by (*auto split: prod.splits*)

then show *?thesis*

using *Some result-state*

unfolding *hash-new-output-hit-event-def* **by** *simp*

qed

lemma *wp-verify-monad-bind-return-actual-alpha-prefix-branch-verifier-transcript-output-hit-bound:*

fixes *packed* **and** *attacker-state*

defines *s* $\equiv$

verifier-state-from-adversary *attacker-state*
(staged-proof-transcript (snd packed))

shows

wp-event

(verify-monad $\gg$

(λ result. return ((packed, attacker-state), result)))

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

s $\leq$

hash-target-budget-value (set (staged-proof-transcript (snd packed)))

verifier-hash-query-budget

proof —

have *mono:*

wp-event verify-monad

(λ out. case out of

None $\Rightarrow$

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

None

| Some (x, t) $\Rightarrow$

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

(Some (((packed, attacker-state), x), t))) s $\leq$

wp-event verify-monad

(hash-new-output-hit-event (set (PTranscript s)) s) s

proof (rule *wp-event-mono*)

fix *out*

assume *hit:*

(case out of

None $\Rightarrow$

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

None

| Some (x, t) $\Rightarrow$

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit

(Some (((packed, attacker-state), x), t)))

show *hash-new-output-hit-event (set (PTranscript s)) s out*

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit-def

hash-new-output-hit-event-def s-def Let-def

by (*cases out*) (*auto split: prod.splits*)

qed

have *target:*

wp-event verify-monad

(hash-new-output-hit-event (set (PTranscript s)) s) s $\leq$

hash-target-budget-value (set (PTranscript s))

verifier-hash-query-budget

```

    by (rule wp-verify-monad-hash-new-output-hit-bound)
  have cont-bound:
    wp-event verify-monad
      (λout. case out of
        None ⇒
          checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit
            None
        | Some (x, t) ⇒
          checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit
            (Some (((packed, attacker-state), x), t))) s ≤
          hash-target-budget-value (set (staged-proof-transcript (snd packed)))
            verifier-hash-query-budget
      using order.trans[OF mono target]
    unfolding s-def by simp
  show ?thesis
    apply (subst wp-event-bind-return-map)
    using cont-bound
    apply (simp split: option.splits prod.splits)
    done
qed

lemma wp-verify-monad-bind-return-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-bound:
  fixes packed and attacker-state
  defines s ≡
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript (snd packed))
  shows
    wp-event
      (verify-monad >>=
        (λresult. return ((packed, attacker-state), result)))
      checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit
      s ≤
      hash-target-budget-value (set (staged-proof-transcript (snd packed)))
        verifier-hash-query-budget
  proof -
    have transcript-bound:
      wp-event
        (verify-monad >>=
          (λresult. return ((packed, attacker-state), result)))
        checked-staged-security-with-actual-alpha-prefix-branch-verifier-transcript-output-hit
        s ≤
        hash-target-budget-value (set (staged-proof-transcript (snd packed)))
          verifier-hash-query-budget
      unfolding s-def
    by (rule
      wp-verify-monad-bind-return-actual-alpha-prefix-branch-verifier-transcript-output-hit-bound)
  show ?thesis
    by (rule
      branch-transcript-root-verifier-output-hit-bound-from-branch-verifier-transcript-output

```

[OF transcript-bound])

qed

lemma *checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-bound-from-transcript*

assumes *target-bound:*

$\bigwedge$ *packed attacker-state.*

Some (packed, attacker-state) $\in$

set-dist

(execute (checked-staged-transcript-with-alpha-prefix-program A)

adversary-initial-state) $\implies$

hash-target-budget-value (set (staged-proof-transcript (snd packed)))

verifier-hash-query-budget $\leq C$

shows

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit

adversary-initial-state $\leq C$

unfolding *checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def*

proof (rule *wp-event-bind-bound-by-cont*)

show

$\neg$

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit

None

unfolding

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-def

by *simp*

next

fix *packed attacker-state*

assume *support:*

Some (packed, attacker-state) $\in$

set-dist

(execute (checked-staged-transcript-with-alpha-prefix-program A)

adversary-initial-state)

let *?s =*

verifier-state-from-adversary attacker-state

(staged-proof-transcript (snd packed))

have *cont-eq:*

wp-event

(get $\gg=$

($\lambda s.$ put

(verifier-state-from-adversary s

(staged-proof-transcript (snd packed))) $\gg=$

($\lambda -. verify-monad \gg=$

($\lambda result.$ return ((packed, s), result))))

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit

attacker-state =

wp-event

(verify-monad $\gg=$

($\lambda result.$ return ((packed, attacker-state), result)))


```

    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit
    ?s
unfolding wp-event-def by (simp add: wpsimps)
have verifier-bound:
  wp-event
    (verify-monad  $\gg$ 
      ( $\lambda$ result. return ((packed, attacker-state), result)))
    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit
    ?s  $\leq$ 
    hash-target-budget-value (set (staged-proof-transcript (snd packed)))
    verifier-hash-query-budget
by (rule
  wp-verify-monad-bind-return-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-bound)
show
  wp-event
    (get  $\gg$ 
      ( $\lambda$ s. put
        (verifier-state-from-adversary s
          (staged-proof-transcript (snd packed)))  $\gg$ 
          ( $\lambda$ -. verify-monad  $\gg$ 
            ( $\lambda$ result. return ((packed, s), result)))))
    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit
    attacker-state  $\leq C$ 
unfolding cont-eq
by (rule order.trans[OF verifier-bound target-bound[OF support]])
qed

lemma checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-bound-from-concret
assumes target-bound:
   $\bigwedge$ packed attacker-state.
  Some (packed, attacker-state)  $\in$ 
    set-dist
      (execute (checked-staged-transcript-with-alpha-prefix-program A)
        adversary-initial-state)  $\implies$ 
    concrete-transcript-target-error (staged-proof-transcript (snd packed))
       $\leq C$ 
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit
    adversary-initial-state  $\leq C$ 
proof (rule
  checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-bound-from-transcript-tar
fix packed attacker-state
assume support:
  Some (packed, attacker-state)  $\in$ 
    set-dist
      (execute (checked-staged-transcript-with-alpha-prefix-program A)
        adversary-initial-state)

```

```

show
  hash-target-budget-value (set (staged-proof-transcript (snd packed)))
    verifier-hash-query-budget  $\leq C$ 
using target-bound[OF support]
unfolding concrete-transcript-target-error-def hash-target-budget-value-def
by (simp add: mult.commute)
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Bounds
imports Staged-Security-Experiment-Composition-Tree-Output
begin

```

Budget-level composition corollaries for the staged security experiment.

This layer consumes the actual-alpha-prefix prequery accounting theorem and exposes the remaining composition side condition as the tree-output event only. The tree-output event is still Merkle/candidate proof work; the alpha-key prequery accounting itself is derived from the current staged oracle-budget interface.

```

context soundness
begin

```

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-tree-and-budgets:*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and tree-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
  checked-staged-security-with-actual-alpha-prefix-tree-output-hit
  adversary-initial-state  $\leq T$ 

```

```

shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event composition-bad)
  adversary-initial-state  $\leq$ 
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    T

```

```

proof -
let ?P =
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +

```

hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0)
have *transcript-bound*:
wp-event (checked-staged-transcript-with-alpha-prefix-program A)
(checked-staged-transcript-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state ≤ ?P
by (rule
checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
[OF wf controlled])
have *prefix-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state ≤ ?P
by (rule
checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
[OF transcript-bound])
have
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event composition-bad)
adversary-initial-state ≤ ?P + T
by (rule
checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-and-tree
[OF wf controlled prefix-bound tree-bound])
then show *?thesis*
by (*simp add: add.assoc*)
qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-data-alpha-prefix-tree-and-budgets*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *data-tree-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(checked-staged-security-with-data-state-alpha-prefix-tree-output-hit A)
adversary-initial-state ≤ T
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event composition-bad)
adversary-initial-state ≤
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
T
proof –

```

have actual-tree-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit
    adversary-initial-state  $\leq T$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit-bound-from-data-state
    [OF wf controlled data-tree-bound])
show ?thesis
  by (rule
    checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-tree-and-budgets
    [OF wf controlled actual-tree-bound])
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Branch
  imports Staged-Security-Experiment-Composition-Bounds
begin

```

Branch-specific composition reductions.

This layer introduces branch-tied events for the verifier result actually sampled by the checked experiment. These events are suitable for verifier-local WP bounds and avoid broad existential side events.

```

context soundness
begin

```

```

definition staged-proof-data-of-query-prefix
  :: 'f staged-query-prefix-data  $\Rightarrow$  'f staged-proof-data
  where
    staged-proof-data-of-query-prefix prefix =
      (staged-trace-root = sqp-trace-root prefix,
       staged-trace-fri-roots = sqp-trace-fri-roots prefix,
       staged-trace-fri-challenges = sqp-trace-fri-challenges prefix,
       staged-trace-final = sqp-trace-final prefix,
       staged-alphas = sqp-alphas prefix,
       staged-degree = sqp-degree prefix,
       staged-composition-fri-roots = sqp-composition-fri-roots prefix,
       staged-composition-fri-challenges =
         sqp-composition-fri-challenges prefix,
       staged-composition-final = sqp-composition-final prefix,
       staged-query-chunks = sqp-query-chunks prefix)

```

```

lemma checked-staged-transcript-program-query-prefix-map:
  checked-staged-transcript-program A =
    checked-staged-query-challenge-prefix-program A rounds  $\gg$ 

```

```

    (λprefix. return (staged-proof-data-of-query-prefix prefix))
unfolding checked-staged-transcript-program-def
    checked-staged-query-challenge-prefix-program-def
    staged-alpha-prefix-program-def
    staged-proof-data-of-query-prefix-def Let-def
by (simp add: sm-bind-assoc split-def split: prod.splits)

```

lemma staged-query-search-queries-rounds-eq-total:

assumes wf: staged-budget-wellformed budgets

shows

staged-query-search-queries budgets rounds =

staged-attacker-query-budget budgets + staged-challenge-query-budget

using wf

unfolding staged-query-search-queries-def

staged-attacker-query-budget-def staged-challenge-query-budget-def

staged-budget-wellformed-def

by simp

lemma hash-relation-program-checked-staged-transcript-program:

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and fibers: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$

shows

hash-relation-program R b

(staged-attacker-query-budget budgets + staged-challenge-query-budget)

(checked-staged-transcript-program A)

proof –

have prefix:

hash-relation-program R b

(staged-query-search-queries budgets rounds)

(checked-staged-query-challenge-prefix-program A rounds)

by (rule hash-relation-program-checked-staged-query-challenge-prefix-program

[OF wf controlled - fibers])

simp

have mapped:

hash-relation-program R b

(staged-query-search-queries budgets rounds + 0)

(checked-staged-query-challenge-prefix-program A rounds $\gg=$

(λprefix. return (staged-proof-data-of-query-prefix prefix)))

by (rule hash-relation-program-bind)

(rule prefix,

simp add: hash-relation-program-zero[OF hash-map-preserving-return])

have budget-eq:

staged-query-search-queries budgets rounds =

staged-attacker-query-budget budgets + staged-challenge-query-budget

by (rule staged-query-search-queries-rounds-eq-total[OF wf])

show ?thesis

unfolding checked-staged-transcript-program-query-prefix-map

using mapped budget-eq **by** simp

qed

definition *checked-staged-transcript-initial-roots-relation*
 $:: 'f \text{ staged-adversary} \Rightarrow 'f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}$
where
checked-staged-transcript-initial-roots-relation $A \ x \ y \longleftrightarrow$
 $(\exists \text{ data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 set-dist
 $(\text{execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \wedge$
 $y \in \{\text{staged-trace-root data,}$
 $\text{hd (staged-composition-fri-roots data)}\})$

lemma *checked-staged-transcript-initial-roots-relation-fiber-bound-size:*
 $\text{card } \{y. \text{checked-staged-transcript-initial-roots-relation } A \ x \ y\}$
 $\leq \text{size}$

proof –

have *subset:*
 $\{y. \text{checked-staged-transcript-initial-roots-relation } A \ x \ y\}$
 $\subseteq (\text{UNIV} :: 'f \text{ set})$
by *simp*
have *finite-univ:* $\text{finite } (\text{UNIV} :: 'f \text{ set})$
by *simp*
have $\text{card } \{y. \text{checked-staged-transcript-initial-roots-relation } A \ x \ y\}$
 $\leq \text{card } (\text{UNIV} :: 'f \text{ set})$
by $(\text{rule card-mono}[\text{OF finite-univ subset}])$
also have $\dots = \text{size}$
using *size-card* **by** *simp*
finally show *?thesis* .

qed

lemma *checked-staged-transcript-initial-roots-relation-hit-bound:*
assumes *wf:* *staged-budget-wellformed budgets*
and *controlled:* *staged-adversary-controlled budgets A*
shows
 $\text{wp-event (checked-staged-transcript-program } A)$
 $(\text{hash-relation-hit-event}$
 $(\text{checked-staged-transcript-initial-roots-relation } A)$
 $\text{adversary-initial-state})$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$

proof –

have *program:*
 $\text{hash-relation-program}$
 $(\text{checked-staged-transcript-initial-roots-relation } A)$
 size
 $(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$

```

    (checked-staged-transcript-program A)
  by (rule hash-relation-program-checked-staged-transcript-program
      [OF wf controlled])
    (rule checked-staged-transcript-initial-roots-relation-fiber-bound-size)
show ?thesis
using program
unfolding hash-relation-program-def hash-relation-budget-def
  staged-phase-relation-error-def
by blast
qed

lemma checked-staged-transcript-with-alpha-prefix-initial-roots-relation-hit-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
      (λout. case out of
        None ⇒ False
      | Some (packed, attacker-state) ⇒
        hash-relation-hit
          (checked-staged-transcript-initial-roots-relation A)
          adversary-initial-state attacker-state)
        adversary-initial-state ≤
        staged-phase-relation-error size
        (staged-attacker-query-budget budgets + staged-challenge-query-budget)
proof -
  let ?R = checked-staged-transcript-initial-roots-relation A
  let ?H =
    λout. case out of
      None ⇒ False
    | Some (packed, attacker-state) ⇒
      hash-relation-hit ?R adversary-initial-state attacker-state
  have map-eq:
    wp-event (checked-staged-transcript-with-alpha-prefix-program A) ?H
      adversary-initial-state =
    wp-event
      (checked-staged-transcript-with-alpha-prefix-program A ≫=
        (λpacked. return (snd packed)))
      (hash-relation-hit-event ?R adversary-initial-state)
      adversary-initial-state
  by (subst wp-event-bind-return-map)
    (simp add: hash-relation-hit-event-def split: option.splits prod.splits)
  also have ... =
    wp-event (checked-staged-transcript-program A)
      (hash-relation-hit-event ?R adversary-initial-state)
      adversary-initial-state
  unfolding checked-staged-transcript-with-alpha-prefix-program-projection
  by simp
  also have ... ≤

```

```

    staged-phase-relation-error size
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule checked-staged-transcript-initial-roots-relation-hit-bound
    [OF wf controlled])
  finally show ?thesis .
qed

```

```

definition staged-proof-transcript-length-bound :: nat
where
  staged-proof-transcript-length-bound =
    4 + ceil-log clength + length spec + ceil-log (maxDegree + 1) +
    rounds *
    (query-decommitment-transcript-length 0 +
     fri-layers-transcript-length (ceil-log clength)
     (clength * scale) +
     fri-layers-transcript-length (ceil-log (maxDegree + 1))
     (clength * scale))

```

```

definition staged-concrete-transcript-target-error-bound :: prob
where
  staged-concrete-transcript-target-error-bound =
    nnreal
    (staged-proof-transcript-length-bound * verifier-hash-query-budget) /
    nnreal size

```

lemma *fri-layers-transcript-length-mono-layers:*

```

  assumes m ≤ n
  shows fri-layers-transcript-length m len ≤
    fri-layers-transcript-length n len
  using assms
proof (induction m arbitrary: n len)
  case 0
  then show ?case by simp
next
  case (Suc m)
  then obtain n' where n-eq: n = Suc n'
  by (cases n) auto
  have m-le: m ≤ n'
  using Suc.prem1 n-eq by simp
  have rec:
    fri-layers-transcript-length m (len div 2) ≤
    fri-layers-transcript-length n' (len div 2)
  by (rule Suc.IH[OF m-le])
  show ?case
  unfolding n-eq by simp (rule rec)
qed

```

lemma *length-concat-eq-sum-list-map-length:*

```

  length (List.concat xss) = sum-list (map length xss)

```


by (*induction xss*) *simp-all*

lemma *sum-list-map-const*:

$(\sum x \leftarrow xs. c) = \text{length } xs * c$

by (*induction xs*) *simp-all*

lemma *staged-query-chunks-total-length-le*:

assumes *match*: *staged-query-chunks-match-verifier-lengths data query-idxs*

and *trace-len*:

$\text{length } (\text{staged-trace-fri-roots } data) = \text{ceil-log } clength$

and *composition-len*:

$\text{length } (\text{staged-composition-fri-roots } data) \leq$
 $\text{ceil-log } (\text{maxDegree} + 1)$

shows

$\text{length } (\text{List.concat } (\text{staged-query-chunks } data)) \leq$
 $\text{rounds} * (\text{query-decommitment-transcript-length } 0 +$
 $\text{fri-layers-transcript-length } (\text{ceil-log } clength)$
 $(clength * scale) +$
 $\text{fri-layers-transcript-length } (\text{ceil-log } (\text{maxDegree} + 1))$
 $(clength * scale))$

proof –

let *?Q* =

$\text{query-decommitment-transcript-length } 0 +$
 $\text{fri-layers-transcript-length } (\text{ceil-log } clength) (clength * scale) +$
 $\text{fri-layers-transcript-length } (\text{ceil-log } (\text{maxDegree} + 1))$
 $(clength * scale)$

have *chunks-len*: $\text{length } (\text{staged-query-chunks } data) = \text{rounds}$

using *match* **unfolding** *staged-query-chunks-match-verifier-lengths-def*

by *simp*

have *chunk-bound*:

$\bigwedge \text{chunk}. \text{chunk} \in \text{set } (\text{staged-query-chunks } data) \implies$
 $\text{length } \text{chunk} \leq ?Q$

proof –

fix *chunk*

assume *chunk-in*: $\text{chunk} \in \text{set } (\text{staged-query-chunks } data)$

then obtain *i* **where** *i-bound*: $i < \text{rounds}$

and *chunk-eq*: $\text{chunk} = \text{staged-query-chunks } data ! i$

using *chunks-len*

by (*metis in-set-conv-nth*)

have *chunk-len*:

$\text{length } \text{chunk} =$
 $\text{verifier-query-round-transcript-length } (\text{query-idxs } ! i)$
 $(\text{staged-trace-fri-roots } data)$
 $(\text{staged-composition-fri-roots } data)$

using *match i-bound chunk-eq*

unfolding *staged-query-chunks-match-verifier-lengths-def* **by** *simp*

have *comp-mono*:

fri-layers-transcript-length

```

    (length (staged-composition-fri-roots data))
    (clength * scale)
  ≤
  fri-layers-transcript-length (ceil-log (maxDegree + 1))
  (clength * scale)
  by (rule fri-layers-transcript-length-mono-layers[OF composition-len])
show length chunk ≤ ?Q
  unfolding chunk-len verifier-query-round-transcript-length-def
    trace-len query-decommitment-transcript-length-def powers-scaled-def
  using comp-mono by simp
qed
have sum-bound:
  (∑ chunk←staged-query-chunks data. length chunk) ≤
  (∑ chunk←staged-query-chunks data. ?Q)
  by (rule sum-list-mono) (rule chunk-bound)
have concat-len:
  length (List.concat (staged-query-chunks data)) =
  (∑ chunk←staged-query-chunks data. length chunk)
  by (rule length-concat-eq-sum-list-map-length)
also have ... ≤ (∑ chunk←staged-query-chunks data. ?Q)
  by (rule sum-bound)
also have ... = length (staged-query-chunks data) * ?Q
  by (rule sum-list-map-const)
also have ... = rounds * ?Q
  using chunks-len by simp
finally show ?thesis .
qed

lemma checked-staged-transcript-program-staged-proof-transcript-length-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
    adversary-initial-state)
  shows length (staged-proof-transcript data) ≤
    staged-proof-transcript-length-bound
proof -
  have shape:
    length (staged-trace-fri-roots data) = ceil-log clength ∧
    length (staged-alphas data) = length spec ∧
    length (staged-composition-fri-roots data) =
      ceil-log (to-nat (staged-degree data) + 1) ∧
    ceil-log (to-nat (staged-degree data) + 1) ≤
      ceil-log (maxDegree + 1)
  using checked-staged-transcript-program-outcome-shape
    [OF wf controlled outcome]
  by blast

```

```

from checked-staged-transcript-program-query-chunks-match-verifier-lengths
  [OF outcome]
obtain query-idxs where match:
  staged-query-chunks-match-verifier-lengths data query-idxs
by blast
have query-len:
  length (List.concat (staged-query-chunks data)) ≤
    rounds * (query-decommitment-transcript-length 0 +
      fri-layers-transcript-length (ceil-log clength)
      (clength * scale) +
      fri-layers-transcript-length (ceil-log (maxDegree + 1))
      (clength * scale))
by (rule staged-query-chunks-total-length-le[OF match])
  (use shape in simp-all)
have header-len:
  length
    (verifier-header-messages
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data))
    ≤ 4 + ceil-log clength + length spec + ceil-log (maxDegree + 1)
using shape unfolding verifier-header-messages-def by simp
show ?thesis
  unfolding staged-proof-transcript-def
    staged-proof-transcript-length-bound-def
  using header-len query-len by simp
qed

lemma concrete-transcript-target-error-bound-from-length:
  assumes length tr ≤ n
  shows concrete-transcript-target-error tr ≤
    nnreal (n * verifier-hash-query-budget) / nnreal size
proof –
  have card (set tr) ≤ length tr
    by (rule card-length)
  then have card-le: card (set tr) ≤ n
    using assms by linarith
  have mult-le:
    card (set tr) * verifier-hash-query-budget ≤
      n * verifier-hash-query-budget
    by (rule mult-right-mono[OF card-le]) simp
  show ?thesis
    unfolding concrete-transcript-target-error-def
    by (rule nnreal-nat-divide-right-mono[OF mult-le])

```

qed

lemma *checked-staged-transcript-with-alpha-prefix-program-data-outcome:*

assumes *outcome:*

Some (packed, attacker-state) ∈

set-dist

(execute (checked-staged-transcript-with-alpha-prefix-program A)

adversary-initial-state)

shows

Some (snd packed, attacker-state) ∈

set-dist

(execute (checked-staged-transcript-program A)

adversary-initial-state)

using *outcome*

unfolding *checked-staged-transcript-with-alpha-prefix-program-projection*

[symmetric]

by (rule-tac *x=packed and t=attacker-state in set-dist-bindI*) *simp-all*

lemma *checked-staged-transcript-with-alpha-prefix-program-concrete-transcript-target-error-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *outcome:*

Some (packed, attacker-state) ∈

set-dist

(execute (checked-staged-transcript-with-alpha-prefix-program A)

adversary-initial-state)

shows

concrete-transcript-target-error (staged-proof-transcript (snd packed))

≤ staged-concrete-transcript-target-error-bound

proof –

have *data-out:*

Some (snd packed, attacker-state) ∈

set-dist

(execute (checked-staged-transcript-program A)

adversary-initial-state)

by (rule *checked-staged-transcript-with-alpha-prefix-program-data-outcome*

[OF outcome])

have *len:*

length (staged-proof-transcript (snd packed)) ≤

staged-proof-transcript-length-bound

by (rule *checked-staged-transcript-program-staged-proof-transcript-length-bound*

[OF wf controlled data-out])

show *?thesis*

unfolding *staged-concrete-transcript-target-error-bound-def*

by (rule *concrete-transcript-target-error-bound-from-length* *[OF len]*)

qed

definition *checked-staged-security-with-actual-alpha-prefix-verifier-event*

:: (f protocol-channel ⇒

$(\text{unit list} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool})$
 $\Rightarrow$
 $(((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $\quad 'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $\quad 'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $\quad 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
checked-staged-security-with-actual-alpha-prefix-verifier-event P *out*
 $\longleftrightarrow$
 $(\text{case out of}$
 $\quad \text{None} \Rightarrow \text{False}$
 $| \text{Some } (x, \text{final-state}) \Rightarrow$
 $\quad (\text{let packed} = \text{fst } (\text{fst } x);$
 $\quad \quad \text{data} = \text{snd packed};$
 $\quad \quad \text{attacker-state} = \text{snd } (\text{fst } x);$
 $\quad \quad \text{result} = \text{snd } x;$
 $\quad \quad s =$
 $\quad \quad \text{verifier-state-from-adversary attacker-state}$
 $\quad \quad (\text{staged-proof-transcript data})$
 $\quad \text{in } P \ s \ (\text{Some } (\text{result}, \text{final-state}))))$
definition *checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit*
 $:: ((((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $\quad 'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $\quad 'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $\quad 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
out $\longleftrightarrow$
 $(\text{case out of}$
 $\quad \text{None} \Rightarrow \text{False}$
 $| \text{Some } (x, \text{final-state}) \Rightarrow$
 $\quad (\text{let packed} = \text{fst } (\text{fst } x);$
 $\quad \quad \text{prefix-state} = \text{snd } (\text{fst packed});$
 $\quad \quad \text{data} = \text{snd packed};$
 $\quad \quad \text{attacker-state} = \text{snd } (\text{fst } x);$
 $\quad \quad \text{result} = \text{snd } x;$
 $\quad \quad s =$
 $\quad \quad \text{verifier-state-from-adversary attacker-state}$
 $\quad \quad (\text{staged-proof-transcript data})$
 $\quad \text{in } \exists \text{ trace-table composition-table as query-idxs dg}$
 $\quad \quad \text{composition-fri-roots final rest trace-tree composition-tree.}$
 $\quad \text{Some } (\text{result}, \text{final-state}) \in$
 $\quad \quad \text{set-dist } (\text{execute verify-monad } s) \wedge$
 $\quad \text{accepted-with-bound-tables } s \ (\text{Some } (\text{result}, \text{final-state}))$
 $\quad \quad \text{trace-table composition-table as query-idxs} \wedge$
 $\quad \text{verifier-header-transcript } s$
 $\quad \quad (\text{staged-trace-root data})$
 $\quad \quad (\text{staged-trace-fri-roots data})$

(staged-trace-final data)
 as dg composition-fri-roots final rest $\wedge$
 created-tree trace-table trace-tree final-state $\wedge$
 created-tree composition-table composition-tree final-state $\wedge$
 hash-map-new-output-hit
 (set-tree trace-tree $\cup$ set-tree composition-tree)
 prefix-state final-state))

definition *checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case*
where

checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case
 $P \text{ out} \longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (x, final-state) $\Rightarrow$
 (let packed = fst (fst x);
 prefix-state = snd (fst packed);
 data = snd packed;
 attacker-state = snd (fst x);
 result = snd x;
 s =
 verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)
 in $\exists$ trace-table composition-table as query-idxs dg
 composition-fri-roots final rest trace-tree composition-tree.
 Some (result, final-state) $\in$
 set-dist (execute verify-monad s) $\wedge$
 accepted-with-bound-tables s (Some (result, final-state))
 trace-table composition-table as query-idxs $\wedge$
 verifier-header-transcript s
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 as dg composition-fri-roots final rest $\wedge$
 created-tree trace-table trace-tree final-state $\wedge$
 created-tree composition-table composition-tree final-state $\wedge$
 P trace-tree composition-tree prefix-state final-state))

definition *checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit*
where

checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit =
checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case
 (λ trace-tree composition-tree prefix-state final-state.
 hash-map-new-output-hit
 {value trace-tree, value composition-tree}
 prefix-state final-state)

definition

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit

where

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit =
checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case
 $(\lambda \text{trace-tree composition-tree prefix-state final-state.}$
 $(\exists l \ v \ r. \text{trace-tree} = \langle l, v, r \rangle \wedge$
 $\text{hash-map-new-output-hit} (\text{set-tree } l \cup \text{set-tree } r)$
 $\text{prefix-state final-state}) \vee$
 $(\exists l \ v \ r. \text{composition-tree} = \langle l, v, r \rangle \wedge$
 $\text{hash-map-new-output-hit} (\text{set-tree } l \cup \text{set-tree } r)$
 $\text{prefix-state final-state}))$

definition

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit

where

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit
 $\text{out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (x, \text{final-state}) \Rightarrow$
 $(\text{let packed} = \text{fst } (\text{fst } x);$
 $\text{prefix-state} = \text{snd } (\text{fst } \text{packed});$
 $\text{data} = \text{snd } \text{packed};$
 $\text{attacker-state} = \text{snd } (\text{fst } x);$
 $\text{result} = \text{snd } x;$
 $s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
 $\text{in } \exists \text{trace-table composition-table as query-idxs dg}$
 $\text{composition-fri-roots final rest trace-tree composition-tree.}$
 $\text{Some } (\text{result}, \text{final-state}) \in$
 $\text{set-dist } (\text{execute verify-monad } s) \wedge$
 $\text{accepted-with-bound-tables } s (\text{Some } (\text{result}, \text{final-state}))$
 $\text{trace-table composition-table as query-idxs} \wedge$
 $\text{verifier-header-transcript } s$
 $(\text{staged-trace-root data})$
 $(\text{staged-trace-fri-roots data})$
 $(\text{staged-trace-final data})$
 $\text{as dg composition-fri-roots final rest} \wedge$
 $\text{created-tree trace-table trace-tree final-state} \wedge$
 $\text{created-tree composition-table composition-tree final-state} \wedge$
 $((\exists l \ v \ r. \text{trace-tree} = \langle l, v, r \rangle \wedge$
 $\text{hash-map-new-output-hit} (\text{set-tree } l \cup \text{set-tree } r)$
 $\text{prefix-state } s) \vee$
 $(\exists l \ v \ r. \text{composition-tree} = \langle l, v, r \rangle \wedge$
 $\text{hash-map-new-output-hit} (\text{set-tree } l \cup \text{set-tree } r)$
 $\text{prefix-state } s))))$

definition

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit

where

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
 $out \longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (x, final-state) $\Rightarrow$
 (let packed = fst (fst x);
 data = snd packed;
 attacker-state = snd (fst x);
 result = snd x;
 s =
 verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)
 in $\exists$ trace-table composition-table as query-idxs dg
 composition-fri-roots final rest trace-tree composition-tree.
 Some (result, final-state) $\in$
 set-dist (execute verify-monad s) $\wedge$
 accepted-with-bound-tables s (Some (result, final-state))
 trace-table composition-table as query-idxs $\wedge$
 verifier-header-transcript s
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 as dg composition-fri-roots final rest $\wedge$
 created-tree trace-table trace-tree final-state $\wedge$
 created-tree composition-table composition-tree final-state $\wedge$
 (($\exists l v r$. trace-tree = $\langle l, v, r \rangle$ $\wedge$
 hash-map-new-output-hit (set-tree l $\cup$ set-tree r)
 s final-state) $\vee$
 ($\exists l v r$. composition-tree = $\langle l, v, r \rangle$ $\wedge$
 hash-map-new-output-hit (set-tree l $\cup$ set-tree r)
 s final-state))))

definition

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit

where

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit
 $out \longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (x, final-state) $\Rightarrow$
 (let packed = fst (fst x);
 prefix-state = snd (fst packed);
 data = snd packed;
 attacker-state = snd (fst x);
 result = snd x;
 s =
 verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)

in $\exists$ *trace-table composition-table as query-idxs dg*
composition-fri-roots final rest.
Some (result, final-state) $\in$
set-dist (execute verify-monad s) $\wedge$
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs $\wedge$
verifier-header-transcript s
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
as dg composition-fri-roots final rest $\wedge$
hash-map-new-output-hit
{staged-trace-root data, hd composition-fri-roots}
prefix-state final-state))

definition

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit

where

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit

out $\longleftrightarrow$

(case out of

None $\Rightarrow$ False

| Some (x, final-state) $\Rightarrow$

(let packed = fst (fst x);

prefix-state = snd (fst packed);

data = snd packed;

attacker-state = snd (fst x);

result = snd x;

s =

verifier-state-from-adversary attacker-state

(staged-proof-transcript data)

in $\exists$ *trace-table composition-table as query-idxs dg*

composition-fri-roots final rest.

Some (result, final-state) $\in$

set-dist (execute verify-monad s) $\wedge$

accepted-with-bound-tables s (Some (result, final-state))

trace-table composition-table as query-idxs $\wedge$

verifier-header-transcript s

(staged-trace-root data)

(staged-trace-fri-roots data)

(staged-trace-final data)

as dg composition-fri-roots final rest $\wedge$

hash-map-new-output-hit

{staged-trace-root data, hd composition-fri-roots}

prefix-state s))

lemma *checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit-imp-initial-roots*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

```

and support:
  out ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
    out
shows
  case out of
    None ⇒ False
  | Some ((((prefix, prefix-state), data), attacker-state),
    result), final-state) ⇒
    hash-relation-hit
      (checked-staged-transcript-initial-roots-relation A)
      adversary-initial-state attacker-state
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit-def
    by simp
next
  case (Some result-pack)
  then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some ((((prefix, prefix-state), data), attacker-state),
        result), final-state)
  by (cases result-pack, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have support-some:
    Some ((((prefix, prefix-state), data), attacker-state),
      result), final-state) ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
            A)
          adversary-initial-state)
    using support unfolding out-eq .
  from support-some obtain trans-out where trans-out:
    Some (((prefix, prefix-state), data), attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-with-alpha-prefix-program A)

```

```

      adversary-initial-state)
unfolding
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
by (auto elim!: set-dist-bindE split: prod.splits)
have data-out:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
using checked-staged-transcript-with-alpha-prefix-program-data-outcome
  [OF trans-out]
by simp
from checked-staged-transcript-with-alpha-prefix-program-support
  [OF wf controlled trans-out]
have prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
      (execute (staged-alpha-prefix-program A)
        adversary-initial-state)
by blast
have initial-prefix: adversary-initial-state ≤ prefix-state
proof –
  have program:
    hash-target-program {} (staged-alpha-search-queries budgets 0)
    (staged-alpha-prefix-program A)
by (rule hash-target-program-staged-alpha-prefix-program
  [OF wf controlled])
  then show ?thesis
    using prefix-out
    unfolding hash-target-program-def hash-extension-preserving-def
    by blast
qed
from hit obtain trace-table composition-table as query-idxs dg
  composition-fri-roots final rest where
  verify-out:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
and bound:
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table as query-idxs
and header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
  as dg composition-fri-roots final rest
and root-hit:
  hash-map-new-output-hit
    {staged-trace-root data, hd composition-fri-roots}
  prefix-state ?s

```

```

unfolding out-eq
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit-def
  Let-def
by simp blast
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
  [OF wf controlled data-out])
have header-eq:
  as = staged-alphas data  $\wedge$ 
  dg = staged-degree data  $\wedge$ 
  composition-fri-roots = staged-composition-fri-roots data  $\wedge$ 
  final = staged-composition-final data  $\wedge$ 
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp
from root-hit obtain x y where
  y-in:  $y \in \{\textit{staged-trace-root data}, \textit{hd composition-fri-roots}\}$ 
  and lookup-prefix-none: fmlookup (HashMap prefix-state) x = None
  and lookup-s: fmlookup (HashMap ?s) x = Some y
  unfolding hash-map-new-output-hit-def by blast
have lookup-initial-none: fmlookup (HashMap adversary-initial-state) x = None
proof (cases fmlookup (HashMap adversary-initial-state) x)
  case None
    then show ?thesis .
next
  case (Some old)
    have fmlookup (HashMap prefix-state) x = Some old
      by (rule hash-extension-lookup[OF Some initial-prefix])
    then show ?thesis
      using lookup-prefix-none by simp
qed
have lookup-attacker: fmlookup (HashMap attacker-state) x = Some y
  using lookup-s by simp
have relation:
  checked-staged-transcript-initial-roots-relation A x y
proof –
  have  $y \in \{\textit{staged-trace-root data}, \textit{hd (staged-composition-fri-roots data)}\}$ 
    using y-in header-eq by auto
  then show ?thesis

```

```

    unfolding checked-staged-transcript-initial-roots-relation-def
    using data-out by blast
qed
have
  hash-relation-hit
    (checked-staged-transcript-initial-roots-relation A)
    adversary-initial-state attacker-state
  unfolding hash-relation-hit-def
  by (intro exI[of -] x exI[of -] y conjI)
    (use lookup-initial-none lookup-attacker relation in simp-all)
then show ?thesis
  unfolding out-eq by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
      adversary-initial-state ≤
      staged-phase-relation-error size
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  unfolding checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
proof (rule wp-event-bind-bound-by-head-event)
  show
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
      (λout. case out of
        None ⇒ False
      | Some (packed, attacker-state) ⇒
        hash-relation-hit
          (checked-staged-transcript-initial-roots-relation A)
          adversary-initial-state attacker-state)
      adversary-initial-state
      ≤ staged-phase-relation-error size
      (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  by (rule
    checked-staged-transcript-with-alpha-prefix-initial-roots-relation-hit-bound
    [OF wf controlled])
next
  show
    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
      None ⇒
      (case None of None ⇒ False
      | Some (packed, attacker-state) ⇒
        hash-relation-hit
          (checked-staged-transcript-initial-roots-relation A)

```

```

      adversary-initial-state attacker-state)
unfolding
  checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit-def
by simp
next
fix packed attacker-state out
assume trans-out:
  Some (packed, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-with-alpha-prefix-program A)
      adversary-initial-state)
and cont:
  out ∈
    set-dist
    (execute
      (get  $\gg$ 
        (λs. put
          (verifier-state-from-adversary s
            (staged-proof-transcript (snd packed))))  $\gg$ 
          (λ-. verify-monad  $\gg$ 
            (λresult. return ((packed, s), result))))))
      attacker-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
  out
have support:
  out ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
unfolding
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
by (rule-tac x=packed and t=attacker-state in set-dist-bindI)
  (use trans-out cont in simp-all)
have rel:
  case out of
    None  $\Rightarrow$  False
  | Some (((((prefix, prefix-state), data), attacker-state),
    result), final-state)  $\Rightarrow$ 
    hash-relation-hit
    (checked-staged-transcript-initial-roots-relation A)
    adversary-initial-state attacker-state
by (rule
  checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit-imp-initial-roots-rel
  [OF wf controlled support hit])
show
  (case Some (packed, attacker-state) of

```

```

    None  $\Rightarrow$  False
  | Some (packed, attacker-state)  $\Rightarrow$ 
    hash-relation-hit
      (checked-staged-transcript-initial-roots-relation A)
      adversary-initial-state attacker-state)
  using cont rel
  by (cases out)
    (auto elim!: set-dist-bindE split: option.splits prod.splits)
qed

lemma checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-imp-tree-output-hit:
  assumes hit:
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
      out
  shows
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit out
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-def
    checked-staged-security-with-actual-alpha-prefix-tree-output-hit-def
    Let-def
  by (cases out) (fastforce split: prod.splits)+

lemma checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-imp-root-or-subtree:
  assumes hit:
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
      out
  shows
    checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
      out  $\vee$ 
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit
      out
  proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-def
    by simp
  next
  case (Some result-pack)
  then obtain x final-state where out-eq:
    out = Some (x, final-state)
  by (cases result-pack) simp
  from hit obtain trace-table composition-table as query-idxs dg
    composition-fri-roots final rest trace-tree composition-tree where
    body:
      Some (snd x, final-state)  $\in$ 
        set-dist

```

```

      (execute verify-monad
        (verifier-state-from-adversary (snd (fst x))
          (staged-proof-transcript (snd (fst (fst x)))))))
accepted-with-bound-tables
  (verifier-state-from-adversary (snd (fst x))
    (staged-proof-transcript (snd (fst (fst x))))))
  (Some (snd x, final-state)) trace-table composition-table as
  query-idxs
verifier-header-transcript
  (verifier-state-from-adversary (snd (fst x))
    (staged-proof-transcript (snd (fst (fst x))))))
  (staged-trace-root (snd (fst (fst x))))
  (staged-trace-fri-roots (snd (fst (fst x))))
  (staged-trace-final (snd (fst (fst x))))
  as dg composition-fri-roots final rest
created-tree trace-table trace-tree final-state
created-tree composition-table composition-tree final-state
hash-map-new-output-hit
  (set-tree trace-tree  $\cup$  set-tree composition-tree)
  (snd (fst (fst (fst x)))) final-state
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-def
  Let-def
by (auto split: prod.splits)
from hash-map-new-output-hit-two-trees-root-or-subtree[OF body(6)]
show ?thesis
proof
  assume root-hit:
    hash-map-new-output-hit {value trace-tree, value composition-tree}
    (snd (fst (fst (fst x)))) final-state
  have
    checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
    out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-def
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case-def
    Let-def
  using body(1-5) root-hit
  by (cases x) (fastforce split: prod.splits)+
then show ?thesis by simp
next
assume subtree:
  ( $\exists l\ v\ r.$  trace-tree =  $\langle l, v, r \rangle \wedge$ 
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
    (snd (fst (fst (fst x)))) final-state)  $\vee$ 
  ( $\exists l\ v\ r.$  composition-tree =  $\langle l, v, r \rangle \wedge$ 
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
    (snd (fst (fst (fst x)))) final-state)
have

```



```

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit
  out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit-def
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case-def
  Let-def
using body(1-5) subtree
by (cases x) (fastforce split: prod.splits)+
then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-bound-from-root-and-subtree:
assumes root-bound:
  wp-event m
  checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
   $s \leq R$ 
and subtree-bound:
  wp-event m
  checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit
   $s \leq S$ 
shows
  wp-event m
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
   $s \leq R + S$ 
proof -
have wp-event m
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
   $s \leq$ 
  wp-event m
  ( $\lambda out.$ 
    checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
    out  $\vee$ 
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit
    out) s
by (rule wp-event-mono)
  (rule
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-imp-root-or-subtree)
also have ...  $\leq$ 
  wp-event m
  checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
   $s +$ 
  wp-event m
  checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit
  s
by (rule wp-event-union-bound)
also have ...  $\leq R + S$ 
by (intro add-mono root-bound subtree-bound)
finally show ?thesis .

```

qed

lemma *checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit-imp-preverifier-or-verifier-on-s*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

out $\in$

set-dist

(*execute*

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

adversary-initial-state)

and *hit:*

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit

out

shows

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit

out $\vee$

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit

out

proof (*cases out*)

case *None*

then show *?thesis*

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit-def

checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case-def

by *simp*

next

case (*Some result-pack*)

then obtain *prefix prefix-state data attacker-state result final-state* **where**

out-eq:

out =

Some (((((*prefix, prefix-state*), *data*), *attacker-state*),

result), *final-state*)

by (*cases result-pack, auto split: prod.splits*)

let *?s* =

verifier-state-from-adversary attacker-state

(*staged-proof-transcript data*)

have *support-some:*

Some (((((*prefix, prefix-state*), *data*), *attacker-state*),

result), *final-state*) $\in$

set-dist

(*execute*

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

adversary-initial-state)

using *support* **unfolding** *out-eq* .

from *support-some* **obtain** *trans-out* **where** *trans-out:*

```

    Some (((prefix, prefix-state), data), attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-with-alpha-prefix-program A)
          adversary-initial-state)
  unfolding
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
  by (auto elim!: set-dist-bindE split: prod.splits)
from checked-staged-transcript-with-alpha-prefix-program-support
  [OF wf controlled trans-out]
have prefix-ext: prefix-state ≤ attacker-state
  by blast
have prefix-s: prefix-state ≤ ?s
  by (rule
    alpha-prefix-hash-extends-verifier-state-from-adversary-right
    [OF prefix-ext])
from hit obtain trace-table composition-table as query-idxs dg
  composition-fri-roots final rest trace-tree composition-tree where
  verify-out:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
and bound:
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table as query-idxs
and header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  as dg composition-fri-roots final rest
and trace-created:
  created-tree trace-table trace-tree final-state
and composition-created:
  created-tree composition-table composition-tree final-state
and subtree:
  (∃ l v r. trace-tree = ⟨l, v, r⟩ ∧
    hash-map-new-output-hit (set-tree l ∪ set-tree r)
    prefix-state final-state) ∨
  (∃ l v r. composition-tree = ⟨l, v, r⟩ ∧
    hash-map-new-output-hit (set-tree l ∪ set-tree r)
    prefix-state final-state)
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit-def
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case-def
    Let-def
  by simp blast
have s-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verify-out])
from subtree show ?thesis
proof
  assume trace-subtree:

```

```

     $\exists l \ v \ r. \text{trace-tree} = \langle l, v, r \rangle \wedge$ 
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
    prefix-state final-state
then obtain l v r where trace-tree-eq: trace-tree =  $\langle l, v, r \rangle$ 
and tree-hit:
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
    prefix-state final-state
by blast
from hash-map-new-output-hit-trans-decomp
    [OF prefix-s s-final tree-hit]
show ?thesis
proof
  assume pre:
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
    prefix-state ?s
  have
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit
    out
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit-def
    Let-def
    using verify-out bound header trace-created composition-created
    trace-tree-eq pre
    by simp blast
  then show ?thesis by simp
next
  assume verifier:
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
    ?s final-state
  have
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
    out
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit-def
    Let-def
    using verify-out bound header trace-created composition-created
    trace-tree-eq verifier
    by simp blast
  then show ?thesis by simp
qed
next
  assume composition-subtree:
     $\exists l \ v \ r. \text{composition-tree} = \langle l, v, r \rangle \wedge$ 
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
    prefix-state final-state
  then obtain l v r where
    composition-tree-eq: composition-tree =  $\langle l, v, r \rangle$ 
  and tree-hit:
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)

```

```

    prefix-state final-state
  by blast
from hash-map-new-output-hit-trans-decomp
  [OF prefix-s s-final tree-hit]
show ?thesis
proof
  assume pre:
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
    prefix-state ?s
  have
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit
      out
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit-def
      Let-def
    using verify-out bound header trace-created composition-created
      composition-tree-eq pre
    by simp blast
  then show ?thesis by simp
next
  assume verifier:
    hash-map-new-output-hit (set-tree l  $\cup$  set-tree r)
    ?s final-state
  have
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
      out
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit-def
      Let-def
    using verify-out bound header trace-created composition-created
      composition-tree-eq verifier
    by simp blast
  then show ?thesis by simp
qed
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit-bound-from-preverifier-and-verifier
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and pre-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit
      adversary-initial-state  $\leq$  P
  and verifier-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state

```

A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
adversary-initial-state $\leq V$
shows
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit
adversary-initial-state $\leq P + V$
proof –
let ?M =
checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let ?Sub =
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit
let ?Pre =
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit
let ?Verifier =
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
have *wp-event* ?M ?Sub *adversary-initial-state* $\leq$
wp-event ?M ($\lambda out. ?Pre\ out \vee ?Verifier\ out$)
adversary-initial-state
by (rule *wp-event-mono-on-support*)
 (rule
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit-imp-preverifier-or-verifier-on-supp
 [OF *wf controlled*])
also have ... $\leq$
wp-event ?M ?Pre *adversary-initial-state* +
wp-event ?M ?Verifier *adversary-initial-state*
by (rule *wp-event-union-bound*)
also have ... $\leq P + V$
by (intro *add-mono pre-bound verifier-bound*)
finally show ?thesis .
qed

lemma *checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-imp-transcript-root-output-hit*:
assumes *root-hit*:
checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
out
shows
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit
out
proof (cases *out*)
case None
then show ?thesis
using *root-hit*
unfolding
checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-def
checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case-def
by *simp*
next

```

case (Some result-pack)
then obtain x final-state where out-eq:
  out = Some (x, final-state)
  by (cases result-pack) simp
let ?packed = fst (fst x)
let ?prefix-state = snd (fst ?packed)
let ?data = snd ?packed
let ?attacker-state = snd (fst x)
let ?result = snd x
let ?s =
  verifier-state-from-adversary ?attacker-state
  (staged-proof-transcript ?data)
from root-hit obtain trace-table composition-table as query-idxs dg
  composition-fri-roots final rest trace-tree composition-tree where
verify-out:
  Some (?result, final-state) ∈ set-dist (execute verify-monad ?s)
and bound:
  accepted-with-bound-tables ?s (Some (?result, final-state))
  trace-table composition-table as query-idxs
and header:
  verifier-header-transcript ?s
  (staged-trace-root ?data)
  (staged-trace-fri-roots ?data)
  (staged-trace-final ?data)
  as dg composition-fri-roots final rest
and trace-created:
  created-tree trace-table trace-tree final-state
and composition-created:
  created-tree composition-table composition-tree final-state
and tree-root-hit:
  hash-map-new-output-hit
  {value trace-tree, value composition-tree}
  ?prefix-state final-state
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-def
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-case-def
  Let-def
by simp blast
from bound obtain fr' f-fri-roots' f-final' dg'
  composition-fri-roots' final' rest' where
header':
  verifier-header-transcript ?s fr' f-fri-roots' f-final' as dg'
  composition-fri-roots' final' rest'
and trace-bind:
  merkle-root-binds-table fr' trace-table final-state
and composition-bind:
  merkle-root-binds-table (hd composition-fri-roots')
  composition-table final-state
unfolding accepted-with-bound-tables-def by auto

```

```

have header-eq:
  fr' = staged-trace-root ?data ∧
  f-fri-roots' = staged-trace-fri-roots ?data ∧
  f-final' = staged-trace-final ?data ∧
  dg' = dg ∧
  composition-fri-roots' = composition-fri-roots ∧
  final' = final ∧
  rest' = rest
using verifier-header-transcript-unique[OF header' header]
by simp
then have fr-eq: fr' = staged-trace-root ?data
and comp-roots-eq:
  composition-fri-roots' = composition-fri-roots
by simp-all
from trace-bind obtain trace-root-tree where
  trace-root-created:
    created-tree trace-table trace-root-tree final-state
and trace-root-eq: fr' = value trace-root-tree
unfolding merkle-root-binds-table-def by blast
have trace-value-eq:
  value trace-tree = staged-trace-root ?data
using created-tree-same-table-value-eq
  [OF trace-created trace-root-created]
  trace-root-eq fr-eq
by simp
from composition-bind obtain composition-root-tree where
  composition-root-created:
    created-tree composition-table composition-root-tree final-state
and composition-root-eq:
  hd composition-fri-roots' = value composition-root-tree
unfolding merkle-root-binds-table-def by blast
have composition-value-eq:
  value composition-tree = hd composition-fri-roots
using created-tree-same-table-value-eq
  [OF composition-created composition-root-created]
  composition-root-eq comp-roots-eq
by simp
have transcript-root-hit:
  hash-map-new-output-hit
    {staged-trace-root ?data, hd composition-fri-roots}
    ?prefix-state final-state
using tree-root-hit trace-value-eq composition-value-eq by simp
show ?thesis
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit-def
  Let-def
using verify-out bound header transcript-root-hit
by simp blast
qed

```


lemma *checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-bound-from-transcript-roots:*
assumes *root-bound:*

wp-event m
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit
 $s \leq R$

shows

wp-event m
checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
 $s \leq R$

by (*rule order.trans[OF - root-bound]*)

(*rule wp-event-mono,*
rule

checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-imp-transcript-root-output-hit)

lemma *checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit-imp-preverifier-or-verifier-o*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

out $\in$
set-dist
(*execute*
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
adversary-initial-state)

and *hit:*

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit
out

shows

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
out $\vee$
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit
out

proof (*cases out*)

case *None*

then show *?thesis*

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit-def

by *simp*

next

case (*Some result-pack*)

then obtain *prefix prefix-state data attacker-state result final-state* **where**

out-eq:

out =
Some (((((*prefix, prefix-state*), *data*), *attacker-state*),
result), *final-state*)

by (*cases result-pack, auto split: prod.splits*)

let *?s* =

```

    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
have support-some:
  Some (((((prefix, prefix-state), data), attacker-state),
    result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
using support unfolding out-eq .
from support-some obtain trans-out where trans-out:
  Some (((prefix, prefix-state), data), attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-with-alpha-prefix-program A)
        adversary-initial-state)
unfolding
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
by (auto elim!: set-dist-bindE split: prod.splits)
from checked-staged-transcript-with-alpha-prefix-program-support
  [OF wf controlled trans-out]
have prefix-ext: prefix-state ≤ attacker-state
by blast
have prefix-s: prefix-state ≤ ?s
by (rule
  alpha-prefix-hash-extends-verifier-state-from-adversary-right
  [OF prefix-ext])
from hit obtain trace-table composition-table as query-idxs dg
  composition-fri-roots final rest where
  verify-out:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
and bound:
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table as query-idxs
and header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
  as dg composition-fri-roots final rest
and root-hit:
  hash-map-new-output-hit
    {staged-trace-root data, hd composition-fri-roots}
  prefix-state final-state
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit-def
  Let-def
by simp blast
have s-final: ?s ≤ final-state

```

```

    by (rule verify-monad-hash-extends[OF verify-out])
  from hash-map-new-output-hit-trans-decomp
    [OF prefix-s s-final root-hit]
  show ?thesis
proof
  assume pre:
    hash-map-new-output-hit
    {staged-trace-root data, hd composition-fri-roots}
    prefix-state ?s
  have
    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
      out
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit-def
      Let-def
    using verify-out bound header pre
    by simp blast
  then show ?thesis by simp
next
  assume verifier:
    hash-map-new-output-hit
    {staged-trace-root data, hd composition-fri-roots}
    ?s final-state
  have
    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit
      out
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-def
      Let-def
    using verify-out bound header verifier
    by simp blast
  then show ?thesis by simp
qed
qed

```

```

lemma checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit-bound-from-preverifier-and-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and pre-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
  checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
    adversary-initial-state  $\leq P$ 
and verifier-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
  checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit

```

$adversary-initial-state \leq V$
shows
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state\ A)$
 $checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit$
 $adversary-initial-state \leq P + V$
proof –
let $?M =$
 $checked-staged-security-experiment-with-actual-alpha-prefix-data-state\ A$
let $?Root =$
 $checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit$
let $?Pre =$
 $checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit$
let $?Verifier =$
 $checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit$
have $wp-event\ ?M\ ?Root\ adversary-initial-state \leq$
 $wp-event\ ?M\ (\lambda out. ?Pre\ out \vee ?Verifier\ out)$
 $adversary-initial-state$
by (rule $wp-event-mono-on-support$)
 $(rule$
 $checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit-imp-preverifier-or-verifier-on-s$
 $[OF\ wf\ controlled])$
also have $\dots \leq$
 $wp-event\ ?M\ ?Pre\ adversary-initial-state +$
 $wp-event\ ?M\ ?Verifier\ adversary-initial-state$
by (rule $wp-event-union-bound$)
also have $\dots \leq P + V$
by (intro $add-mono\ pre-bound\ verifier-bound$)
finally show $?thesis$.
qed

lemma $checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-bound-from-preverifier-and-verifier$
assumes $wf: staged-budget-wellformed\ budgets$
and $controlled: staged-adversary-controlled\ budgets\ A$
and $pre-bound:$
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit$
 $adversary-initial-state \leq P$
and $verifier-bound:$
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit$
 $adversary-initial-state \leq V$
shows
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state\ A)$

checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
adversary-initial-state $\leq P + V$

proof –

have *root-bound*:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit
adversary-initial-state $\leq P + V$

by (rule
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-output-hit-bound-from-preverifier-and-verifier
[OF *wf controlled pre-bound verifier-bound*])

show *?thesis*

by (rule
checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-bound-from-transcript-roots
[OF *root-bound*])

qed

lemma *checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-bound-from-preverifier-and-concrete-transcript-target-error*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *pre-bound*:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
adversary-initial-state $\leq P$

and *target-bound*:

$\bigwedge$ *packed attacker-state.*
Some (packed, attacker-state) ∈ set-dist
(execute (checked-staged-transcript-with-alpha-prefix-program A) adversary-initial-state) ⇒
concrete-transcript-target-error (staged-proof-transcript (snd packed))
 $\leq V$

shows

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
adversary-initial-state $\leq P + V$

proof –

have *verifier-bound*:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit
adversary-initial-state $\leq V$

by (rule
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-verifier-output-hit-bound-from-concrete-transcript-target-error
[OF *target-bound*])

show *?thesis*

by (rule
 checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-bound-from-preverifier-and-verifier
 [*OF wf controlled pre-bound verifier-bound*])
 qed

lemma *checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-bound-from-preverifier-concrete-transcript*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *pre-bound:*
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
 adversary-initial-state $\leq P$
and *target-bound:*
 $\bigwedge$ packed attacker-state.
 Some (packed, attacker-state) $\in$
 set-dist
 (*execute (checked-staged-transcript-with-alpha-prefix-program A)*
 adversary-initial-state) $\implies$
 concrete-transcript-target-error (staged-proof-transcript (snd packed))
 $\leq V$
and *subtree-bound:*
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit
 adversary-initial-state $\leq S$
shows
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
 adversary-initial-state $\leq P + V + S$
proof –
have *root-bound:*
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit
 adversary-initial-state $\leq P + V$
by (rule
 checked-staged-security-with-actual-alpha-prefix-branch-tree-root-output-hit-bound-from-preverifier-and-concrete-transcript
 [*OF wf controlled pre-bound target-bound*])
have
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
 adversary-initial-state $\leq (P + V) + S$
by (rule
 checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-bound-from-root-and-subtree

$[OF \text{ root-bound subtree-bound}]])$
then show *?thesis*
by (*simp add: add.assoc*)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-bound-from-preverifier-concrete-transcript*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *root-pre-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
adversary-initial-state $\leq P$
and *target-bound:*
 $\bigwedge \text{packed attacker-state.}$
Some (packed, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-with-alpha-prefix-program A)
adversary-initial-state) $\implies$
concrete-transcript-target-error (staged-proof-transcript (snd packed))
 $\leq V$
and *subtree-pre-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit
adversary-initial-state $\leq SP$
and *subtree-verifier-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
adversary-initial-state $\leq SV$
shows
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
adversary-initial-state $\leq P + V + SP + SV$
proof –
have *subtree-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit
adversary-initial-state $\leq SP + SV$
by (*rule*
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-output-hit-bound-from-preverifier-and-verifier
 $[OF \text{ wf controlled subtree-pre-bound subtree-verifier-bound}]])$
have

```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
  adversary-initial-state  $\leq P + V + (SP + SV)$ 
by (rule
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-bound-from-preverifier-concrete-transcript
  [OF wf controlled root-pre-bound target-bound subtree-bound])
then show ?thesis
  by (simp add: add.assoc)
qed

```

lemma *checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection:*

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event P)
  adversary-initial-state =
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event P)
  adversary-initial-state
proof -
  let ?project =
     $\lambda x. ((\text{snd } (\text{fst } (\text{fst } x))), \text{snd } (\text{fst } x)), \text{snd } x)$ 
  have
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event P)
      adversary-initial-state =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
         $\gg (\lambda x. \text{return } (?project x))$ )
      (staged-security-with-data-state-verifier-event P)
      adversary-initial-state
  using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
  by simp
  also have ... =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      ( $\lambda \text{out. case out of}$ 
        None  $\Rightarrow$  staged-security-with-data-state-verifier-event P None
      | Some (x, t)  $\Rightarrow$ 
        staged-security-with-data-state-verifier-event P
        (Some (?project x, t)))
      adversary-initial-state
  by (rule wp-event-bind-return-map)
  also have ... =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-verifier-event P)

```



```

    adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
apply (rule arg-cong[where
    f= $\lambda Q.$  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    Q adversary-initial-state])
apply (rule ext)
apply (simp add: Let-def split: option.splits prod.splits)
done
finally show ?thesis .
qed

```

```

lemma checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-prefix-or-branch-tree-on-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
    out  $\in$ 
    set-dist
    (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
    adversary-initial-state)
and bad:
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-randomization-bad out
shows
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space out  $\vee$ 
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
    out
proof (cases out)
case None
then show ?thesis
    using bad
    unfolding
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    by simp
next
case (Some result-pack)
then obtain prefix prefix-state data attacker-state result final-state where
    out-eq:
    out =
    Some (((((prefix, prefix-state), data), attacker-state),
    result), final-state)
    by (cases result-pack, auto split: prod.splits)
let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)

```

```

have support-some:
  Some (((((prefix, prefix-state), data), attacker-state),
    result), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
using support unfolding out-eq .
from support-some obtain trans-out where trans-out:
  Some (((prefix, prefix-state), data), attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-with-alpha-prefix-program A)
      adversary-initial-state)
and verify-out:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
unfolding
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
by (auto elim!: set-dist-bindE split: prod.splits)
from checked-staged-transcript-with-alpha-prefix-program-support
  [OF wf controlled trans-out]
have prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
    (execute (staged-alpha-prefix-program A)
      adversary-initial-state)
and root-eq: staged-trace-root data = fst prefix
and trace-roots-eq:
  staged-trace-fri-roots data = fst (snd prefix)
and trace-bs-eq:
  staged-trace-fri-challenges data = fst (snd (snd prefix))
and trace-final-eq:
  staged-trace-final data = snd (snd (snd prefix))
and prefix-ext: prefix-state ≤ attacker-state
by blast+
have prefix-tuple:
  prefix =
    (staged-trace-root data, staged-trace-fri-roots data,
      staged-trace-fri-challenges data, staged-trace-final data)
by (cases prefix; cases snd prefix; cases snd (snd prefix))
    (use root-eq trace-roots-eq trace-bs-eq trace-final-eq in auto)
have random-bad:
  composition-randomization-bad ?s (Some (result, final-state))
using bad unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  Let-def
by simp
have alpha-hit:
  alpha-bad-set-hit ?s composition-trace-bad-alpha-space

```

```

    (Some (result, final-state))
  by (rule composition-randomization-bad-imp-alpha-bad-set-hit
    [OF random-bad])
from alpha-hit obtain trace-table composition-table as query-idxs where
  bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and as-bad: as ∈ composition-trace-bad-alpha-space trace-table
  unfolding alpha-bad-set-hit-def by blast
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
proof –
  have projected:
    Some (data, attacker-state) ∈
      set-dist
        (execute
          (checked-staged-transcript-with-alpha-prefix-program A ≫=
            (λpacked. return (snd packed)))
          adversary-initial-state)
  by (rule set-dist-bindI[OF trans-out]) simp
have checked-builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  using projected
  unfolding checked-staged-transcript-with-alpha-prefix-program-projection
  by simp
show ?thesis
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled checked-builder])
qed
from bound obtain fr f-fri-roots f-final dg composition-fri-roots final
  rest where
  header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  unfolding accepted-with-bound-tables-def by blast
have header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧

```

```

f-final = staged-trace-final data ∧
as = staged-alphas data ∧
dg = staged-degree data ∧
composition-fri-roots = staged-composition-fri-roots data ∧
final = staged-composition-final data ∧
rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp
have staged-bad:
  staged-alphas data ∈ composition-trace-bad-alpha-space trace-table
  using as-bad header-eq by simp
have alphas-space: staged-alphas data ∈ alpha-space
  using staged-bad composition-trace-bad-alpha-space-subset-alpha-space
  by blast
have prefix-s: prefix-state ≤ ?s
  by (rule
    alpha-prefix-hash-extends-verifier-state-from-adversary-right
    [OF prefix-ext])
have s-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verify-out])
have prefix-final: prefix-state ≤ final-state
  by (rule hash-ext-trans[OF prefix-s s-final])
show ?thesis
proof (rule
  accepted-with-bound-tables-initial-roots-pullback-or-new-tree-output-hit
  [OF prefix-final bound])
fix fr' f-fri-roots' f-final' dg' composition-fri-roots' final'
  rest' trace-tree composition-tree
assume pull-header:
  verifier-header-transcript ?s fr' f-fri-roots' f-final' as dg'
  composition-fri-roots' final' rest'
and trace-created:
  created-tree trace-table trace-tree final-state
and composition-created:
  created-tree composition-table composition-tree final-state
and split:
  (merkle-root-binds-table fr' trace-table prefix-state ∧
   merkle-root-binds-table (hd composition-fri-roots')
   composition-table prefix-state) ∨
  hash-map-new-output-hit
  (set-tree trace-tree ∪ set-tree composition-tree)
  prefix-state final-state
have pull-eq:
  fr' = staged-trace-root data ∧
  f-fri-roots' = staged-trace-fri-roots data ∧
  f-final' = staged-trace-final data ∧
  dg' = staged-degree data ∧
  composition-fri-roots' = staged-composition-fri-roots data ∧
  final' = staged-composition-final data ∧

```

```

    rest' = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF pull-header staged-header]
    header-eq
by simp
from split show ?thesis
proof
  assume binds:
    merkle-root-binds-table fr' trace-table prefix-state  $\wedge$ 
    merkle-root-binds-table (hd composition-fri-roots')
    composition-table prefix-state
  have trace-len: length trace-table = clength * scale
    using bound unfolding accepted-with-bound-tables-def
    accepted-with-tables-def by simp
  have prefix-candidate:
    trace-table  $\in$ 
    alpha-prefix-trace-table-candidates prefix-state (fst prefix)
    by (rule alpha-prefix-trace-table-candidateI[OF trace-len])
    (use binds pull-eq root-eq in simp)
  have prefix-bad:
    staged-alphas data  $\in$ 
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
    unfolding alpha-prefix-union-bad-sets-def
    using alphas-space prefix-candidate staged-bad by auto
  have
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space out
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit-def
    Let-def
    using prefix-bad by simp
  then show ?thesis by simp
next
  assume tree-hit:
    hash-map-new-output-hit
    (set-tree trace-tree  $\cup$  set-tree composition-tree)
    prefix-state final-state
  have
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
    out
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-def
    Let-def
    using verify-out bound pull-header trace-created composition-created
    tree-hit pull-eq
    by simp blast
  then show ?thesis by simp
qed
qed

```

qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-bound-from-prefix-and-branch*

assumes *wf*: *staged-budget-wellformed* budgets

and *controlled*: *staged-adversary-controlled* budgets *A*

and *prefix-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

(*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
composition-trace-bad-alpha-space)

adversary-initial-state $\leq P$

and *branch-tree-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
adversary-initial-state $\leq T$

shows

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)

(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
composition-randomization-bad)

adversary-initial-state $\leq P + T$

proof –

let *?M* =

checked-staged-security-experiment-with-actual-alpha-prefix-data-state *A*

let *?random* =

checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-randomization-bad

let *?prefix* =

checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space

let *?tree* =

checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit

have *wp-event* *?M* *?random* *adversary-initial-state* $\leq$

wp-event *?M* ($\lambda out. ?prefix\ out \vee ?tree\ out$)
adversary-initial-state

by (*rule wp-event-mono-on-support*)

(*rule*

checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-prefix-or-branch-tree-on-sup
[*OF wf controlled*])

also have ... $\leq$

wp-event *?M* *?prefix* *adversary-initial-state* +

wp-event *?M* *?tree* *adversary-initial-state*

by (*rule wp-event-union-bound*)

also have ... $\leq P + T$

by (*intro add-mono prefix-bound branch-tree-bound*)

finally show *?thesis* .

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-branch-tree:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *prefix-bound:*

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

(checked-staged-security-with-actual-alpha-prefix-bad-set-hit composition-trace-bad-alpha-space)

adversary-initial-state ≤ P

and *branch-tree-bound:*

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit

adversary-initial-state ≤ T

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event composition-bad)

adversary-initial-state ≤ P + T

proof —

let *?M =*

checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

let *?degree =*

checked-staged-security-with-actual-alpha-prefix-verifier-event composition-degree-bad

let *?random =*

checked-staged-security-with-actual-alpha-prefix-verifier-event composition-randomization-bad

let *?bad =*

checked-staged-security-with-actual-alpha-prefix-verifier-event composition-bad

have *projected:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event composition-bad)

adversary-initial-state =

wp-event ?M ?bad adversary-initial-state

by *(rule*

checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)

have *event-eq: ?bad = (λout. ?degree out ∨ ?random out)*

by *(rule ext)*

(auto simp:

checked-staged-security-with-actual-alpha-prefix-verifier-event-def composition-bad-def split: option.splits prod.splits)

have *degree-false: ?degree = (λout. False)*

by *(rule ext)*

(auto simp:

```

    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    composition-degree-bad-false[OF spec-degree-wellformed-from-spec-query-margin]
    split: option.splits prod.splits)
have degree-bound:
  wp-event ?M ?degree adversary-initial-state ≤ 0
  unfolding degree-false wp-event-def wp-def dist-expect-def by simp
have random-bound:
  wp-event ?M ?random adversary-initial-state ≤ P + T
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-bound-from-prefix-and-branch-tree-bound
    [OF wf controlled prefix-bound branch-tree-bound])
have wp-event ?M ?bad adversary-initial-state ≤
  wp-event ?M ?degree adversary-initial-state +
  wp-event ?M ?random adversary-initial-state
  unfolding event-eq by (rule wp-event-union-bound)
also have ... ≤ 0 + (P + T)
  by (intro add-mono degree-bound random-bound)
finally show ?thesis
  unfolding projected by simp
qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-branch-tree-and-budget
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and branch-tree-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
  adversary-initial-state ≤ T
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event composition-bad)
  adversary-initial-state ≤
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  T
proof –
let ?P =
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0)
have transcript-bound:
  wp-event (checked-staged-transcript-with-alpha-prefix-program A)

```



```

    (checked-staged-transcript-actual-alpha-prefix-bad-set-hit
     composition-trace-bad-alpha-space)
  adversary-initial-state  $\leq$  ?P
by (rule
    checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
    [OF wf controlled])
have prefix-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
     A)
    (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
     composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq$  ?P
by (rule
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
    [OF transcript-bound])
have
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event composition-bad)
    adversary-initial-state  $\leq$  ?P + T
by (rule
    checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-branch-tree
    [OF wf controlled prefix-bound branch-tree-bound])
then show ?thesis
  by (simp add: add.assoc)
qed

```

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-branch-split-side-events:*

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and root-pre-bound:

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
adversary-initial-state $\leq$ RP

and target-bound:

$\bigwedge$ packed attacker-state.

Some (packed, attacker-state) $\in$
set-dist

(execute (checked-staged-transcript-with-alpha-prefix-program A)
adversary-initial-state) $\implies$

concrete-transcript-target-error (staged-proof-transcript (snd packed))
 $\leq$ RV

and subtree-pre-bound:

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit

$adversary-initial-state \leq SP$
and subtree-verifier-bound:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit$
 $adversary-initial-state \leq SV$
shows
 $wp-event (checked-staged-security-experiment-with-data-state A)$
 $(staged-security-with-data-state-verifier-event composition-bad)$
 $adversary-initial-state \leq$
 $composition-error-bound +$
 $staged-phase-relation-error size$
 $(staged-alpha-search-queries budgets 0) +$
 $hash-collision-budget-value 0$
 $(staged-alpha-search-queries budgets 0) +$
 $(RP + RV + SP + SV)$
proof –
have branch-tree-bound:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)$
 $checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit$
 $adversary-initial-state \leq RP + RV + SP + SV$
by (rule
 $checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit-bound-from-preverifier-concrete-transcript$
 $[OF wf controlled root-pre-bound target-bound subtree-pre-bound$
 $subtree-verifier-bound])$
show ?thesis
by (rule
 $checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-branch-tree-and-budgets$
 $[OF wf controlled branch-tree-bound])$
qed
lemma checked-staged-security-with-data-state-composition-bad-bound-from-branch-split-side-events-and-transcript-b
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and root-pre-bound:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit$
 $adversary-initial-state \leq RP$
and subtree-pre-bound:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit$
 $adversary-initial-state \leq SP$
and subtree-verifier-bound:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
adversary-initial-state ≤ SV

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event composition-bad)
adversary-initial-state ≤
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(RP + staged-concrete-transcript-target-error-bound + SP + SV)

proof –

have *target-bound:*

$\bigwedge \text{packed attacker-state.}$
Some (packed, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-with-alpha-prefix-program A)
adversary-initial-state) ⇒
concrete-transcript-target-error (staged-proof-transcript (snd packed))
≤ staged-concrete-transcript-target-error-bound

by (rule
checked-staged-transcript-with-alpha-prefix-program-concrete-transcript-target-error-bound
[OF wf controlled])

show *?thesis*

by (rule
checked-staged-security-with-data-state-composition-bad-bound-from-branch-split-side-events
[OF wf controlled root-pre-bound target-bound subtree-pre-bound
subtree-verifier-bound])

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-branch-split-side-events-and-transcript-b*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *subtree-pre-bound:*

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit
adversary-initial-state ≤ SP

and *subtree-verifier-bound:*

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
adversary-initial-state ≤ SV

```

shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event composition-bad)
    adversary-initial-state  $\leq$ 
      composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
      (staged-phase-relation-error size
        (staged-attacker-query-budget budgets + staged-challenge-query-budget) +
        staged-concrete-transcript-target-error-bound + SP + SV)
proof –
  have root-pre-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
      adversary-initial-state  $\leq$ 
        staged-phase-relation-error size
        (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    by (rule
      checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit-bound
      [OF wf controlled])
  show ?thesis
    by (rule
      checked-staged-security-with-data-state-composition-bad-bound-from-branch-split-side-events-and-transcript-budg
      [OF wf controlled root-pre-bound subtree-pre-bound
        subtree-verifier-bound])
qed

end

end

theory Staged-Security-Experiment
  imports Staged-Security-Experiment-Composition-Branch
begin

  Compatibility umbrella for the staged-adversary security experiment.

end

theory Soundness-Partial-Merkle
  imports
    Staged-Security-Experiment
begin

```

4.5 Partial Merkle Openings and Algebraic Reductions

This theory records the authenticated openings that the verifier actually observes. It provides partial table candidates, consistency predicates, and reductions from inconsistent authenticated openings to hash-collision events.

```

record 'f authenticated-opening =
  opening-root :: 'f
  opening-length :: nat
  opening-index :: nat
  opening-value :: 'f
  opening-path :: 'f list

fun merkle-path-bound
  :: 'f  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  'f  $\Rightarrow$  'f list  $\Rightarrow$ 
    ('f, 'a) protocol-channel-scheme  $\Rightarrow$  bool
where
  merkle-path-bound rt len idx v [] s  $\longleftrightarrow$ 
    fmlookup (HashMap s) (MerkleLeaf v) = Some rt
  | merkle-path-bound rt len idx v (sibling # path) s  $\longleftrightarrow$ 
    (if idx < len div 2 then
      ( $\exists$  child.
        merkle-path-bound child (len div 2) idx v path s  $\wedge$ 
        fmlookup (HashMap s) (MerkleNode child sibling) = Some rt)
    else
      ( $\exists$  child.
        merkle-path-bound child (len div 2) (idx - len div 2)
          v path s  $\wedge$ 
        fmlookup (HashMap s) (MerkleNode sibling child) = Some rt))

context soundness
begin

definition authenticated-opening-in
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    'f authenticated-opening  $\Rightarrow$  bool
where
  authenticated-opening-in s opening  $\longleftrightarrow$ 
    opening-index opening < opening-length opening  $\wedge$ 
    length (opening-path opening) =
      floor-log (opening-length opening)  $\wedge$ 
    merkle-path-bound
      (opening-root opening)
      (opening-length opening)
      (opening-index opening)
      (opening-value opening)
      (opening-path opening)
    s

definition partial-authenticated-table

```

```

:: 'f ⇒ nat ⇒ 'f authenticated-opening list ⇒
  ('f, 'a) protocol-channel-scheme ⇒ bool
where
  partial-authenticated-table rt len openings s ⟷
    (∀ opening ∈ set openings.
      opening-root opening = rt ∧
      opening-length opening = len ∧
      authenticated-opening-in s opening)

lemma merkle-path-bound-mono:
  assumes bound: merkle-path-bound rt len idx v path s
    and ext: s ≤ t
  shows merkle-path-bound rt len idx v path t
  using bound
proof (induction path arbitrary: rt len idx)
  case Nil
  then show ?case
    using hash-extension-lookup[OF - ext] by simp
next
  case (Cons sibling path)
  show ?case
  proof (cases idx < len div 2)
  case True
  then obtain child where
    child:
      merkle-path-bound child (len div 2) idx v path s
    and lookup:
      fmlookup (HashMap s) (MerkleNode child sibling) = Some rt
    using Cons.premis by auto
  have child-t:
    merkle-path-bound child (len div 2) idx v path t
    by (rule Cons.IH[OF child])
  have lookup-t:
    fmlookup (HashMap t) (MerkleNode child sibling) = Some rt
    by (rule hash-extension-lookup[OF lookup ext])
  show ?thesis
    using True child-t lookup-t
    by (auto intro!: exI[of - child])
next
  case False
  then obtain child where
    child:
      merkle-path-bound child (len div 2) (idx - len div 2)
        v path s
    and lookup:
      fmlookup (HashMap s) (MerkleNode sibling child) = Some rt
    using Cons.premis by auto
  have child-t:
    merkle-path-bound child (len div 2) (idx - len div 2)

```

```

      v path t
    by (rule Cons.IH[OF child])
  have lookup-t:
    fmllookup (HashMap t) (MerkleNode sibling child) = Some rt
    by (rule hash-extension-lookup[OF lookup ext])
  show ?thesis
    using False child-t lookup-t
    by (auto intro!: exI[of - child])
qed
qed

lemma authenticated-opening-in-mono:
  assumes opn-auth: authenticated-opening-in s opn
    and ext:  $s \leq t$ 
  shows authenticated-opening-in t opn
  using opn-auth merkle-path-bound-mono[OF - ext]
  unfolding authenticated-opening-in-def by blast

lemma partial-authenticated-table-mono:
  assumes table: partial-authenticated-table rt len openings s
    and ext:  $s \leq t$ 
  shows partial-authenticated-table rt len openings t
  using table authenticated-opening-in-mono[OF - ext]
  unfolding partial-authenticated-table-def by blast

lemma merkle-path-bound-root-lookup:
  assumes merkle-path-bound rt len idx v path s
  shows  $\exists \text{input. fmllookup (HashMap s) input = Some rt}$ 
  using assms
proof (induction path arbitrary: rt len idx)
  case Nil
  then show ?case
    by (intro exI[of - MerkleLeaf v]) simp
next
  case (Cons sibling path)
  then show ?case
    by (cases idx < len div 2) auto
qed

lemma authenticated-opening-root-lookup:
  assumes authenticated-opening-in s opening
  shows  $\exists \text{input. fmllookup (HashMap s) input = Some (opening-root opening)}$ 
  using assms
  unfolding authenticated-opening-in-def
  by (blast intro: merkle-path-bound-root-lookup)

lemma authenticated-opening-root-late-output-preexisting-or-new-hit:
  assumes ext:  $s \leq t$ 
    and auth: authenticated-opening-in t opening

```

shows
 $(\exists \text{input}. \text{fmlookup} (\text{HashMap } s) \text{ input} = \text{Some} (\text{opening-root opening})) \vee$
 $\text{hash-map-new-output-hit} \{ \text{opening-root opening} \} s t$

proof –
from *authenticated-opening-root-lookup*[*OF auth*]
obtain *input* **where** *lookup-t*:
 $\text{fmlookup} (\text{HashMap } t) \text{ input} = \text{Some} (\text{opening-root opening})$
by *blast*
show *?thesis*
proof (*cases fmlookup (HashMap s) input*)
case *None*
then have *hash-map-new-output-hit {opening-root opening} s t*
unfolding *hash-map-new-output-hit-def*
using *lookup-t* **by** *blast*
then show *?thesis* **by** *simp*
next
case (*Some old*)
have *lookup-t-old: fmlookup (HashMap t) input = Some old*
by (*rule hash-extension-lookup[OF Some ext]*)
then have *old = opening-root opening*
using *lookup-t* **by** *simp*
then show *?thesis*
using *Some* **by** *blast*
qed
qed

lemma *partial-authenticated-table-root-late-output-preexisting-or-new-hit*:
assumes *ext: s ≤ t*
and *table: partial-authenticated-table rt len openings t*
and *opn-in: opening ∈ set openings*
shows
 $(\exists \text{input}. \text{fmlookup} (\text{HashMap } s) \text{ input} = \text{Some } rt) \vee$
 $\text{hash-map-new-output-hit} \{rt\} s t$

proof –
have *auth: authenticated-opening-in t opening*
using *table opn-in* **unfolding** *partial-authenticated-table-def* **by** *blast*
have *root-eq: opening-root opening = rt*
using *table opn-in* **unfolding** *partial-authenticated-table-def* **by** *blast*
show *?thesis*
using *authenticated-opening-root-late-output-preexisting-or-new-hit*
 $[OF \text{ ext } \text{auth}]$
unfolding *root-eq* .
qed

lemma *merkle-path-bound-root-in-map*:
assumes *merkle-path-bound rt len idx v path s*
shows *merkle-path-root-in-map s len idx v path = Some rt*
using *assms*
proof (*induction path arbitrary: rt len idx*)


```

    case Nil
    then show ?case by simp
next
    case (Cons sibling path)
    then show ?case
      by (cases idx < len div 2) auto
qed

lemma merkle-path-root-in-map-some-imp-bound:
  assumes merkle-path-root-in-map s len idx v path = Some rt
  shows merkle-path-bound rt len idx v path s
  using assms
proof (induction path arbitrary: rt len idx)
  case Nil
  then show ?case by simp
next
  case (Cons sibling path)
  show ?case
  proof (cases idx < len div 2)
    case True
    from Cons.prem1 obtain child where
      child-root:
        merkle-path-root-in-map s (len div 2) idx v path =
          Some child
    and lookup:
      fmlookup (HashMap s) (MerkleNode child sibling) = Some rt
    using True by (auto split: option.splits)
    have child-bound:
      merkle-path-bound child (len div 2) idx v path s
      by (rule Cons.IH[OF child-root])
    show ?thesis
      using True child-bound lookup by auto
  next
    case False
    from Cons.prem1 obtain child where
      child-root:
        merkle-path-root-in-map s (len div 2) (idx - len div 2)
          v path = Some child
    and lookup:
      fmlookup (HashMap s) (MerkleNode sibling child) = Some rt
    using False by (auto split: option.splits)
    have child-bound:
      merkle-path-bound child (len div 2) (idx - len div 2) v path s
      by (rule Cons.IH[OF child-root])
    show ?thesis
      using False child-bound lookup by auto
  qed
qed

```

```

fun merkle-path-target-roots
  :: ('f, 'a) protocol-channel-scheme ⇒ 'f ⇒ nat ⇒ nat ⇒
    'f ⇒ 'f list ⇒ 'f set
where
  merkle-path-target-roots t rt len idx v [] = {rt}
| merkle-path-target-roots t rt len idx v (sibling # path) =
  {rt} ∪
  (if idx < len div 2 then
    (case merkle-path-root-in-map t (len div 2) idx v path of
      None ⇒ {}
    | Some child ⇒
      merkle-path-target-roots t child (len div 2) idx v path)
  else
    (case merkle-path-root-in-map t (len div 2) (idx - len div 2)
      v path of
      None ⇒ {}
    | Some child ⇒
      merkle-path-target-roots t child (len div 2)
        (idx - len div 2) v path))

lemma finite-merkle-path-target-roots[simp]:
  finite (merkle-path-target-roots t rt len idx v path)
by (induction path arbitrary: rt len idx)
  (auto split: option.splits if-splits)

lemma card-merkle-path-target-roots-le:
  card (merkle-path-target-roots t rt len idx v path) ≤ Suc (length path)
proof (induction path arbitrary: rt len idx)
  case Nil
  then show ?case by simp
next
  case (Cons sibling path)
  show ?case
  proof (cases idx < len div 2)
  case True
  then show ?thesis
  proof (cases merkle-path-root-in-map t (len div 2) idx v path)
  case None
  then show ?thesis
  using True by simp
  next
  case (Some child)
  have finite-child:
    finite (merkle-path-target-roots t child (len div 2) idx v path)
  by simp
  have card-le:
    card ({rt} ∪
      merkle-path-target-roots t child (len div 2) idx v path) ≤
    Suc (card

```

```

      (merkle-path-target-roots t child (len div 2) idx v path))
    using finite-child by (simp add: card-insert-if)
  also have ... ≤ Suc (Suc (length path))
    using Cons.IH[of child len div 2 idx] by simp
  finally show ?thesis
    using True Some by simp
qed
next
case False
then show ?thesis
proof (cases
  merkle-path-root-in-map t (len div 2) (idx - len div 2) v path)
case None
then show ?thesis
  using False by simp
next
case (Some child)
have finite-child:
  finite (merkle-path-target-roots t child (len div 2)
    (idx - len div 2) v path)
  by simp
have card-le:
  card ({rt} ∪
    merkle-path-target-roots t child (len div 2)
    (idx - len div 2) v path) ≤
  Suc (card
    (merkle-path-target-roots t child (len div 2)
    (idx - len div 2) v path))
  using finite-child by (simp add: card-insert-if)
also have ... ≤ Suc (Suc (length path))
  using Cons.IH[of child len div 2 idx - len div 2] by simp
finally show ?thesis
  using False Some by simp
qed
qed
qed

lemma merkle-path-bound-pullback-or-new-output-hit:
  assumes ext: s ≤ t
  and bound: merkle-path-bound rt len idx v path t
  shows
    merkle-path-bound rt len idx v path s ∨
    hash-map-new-output-hit
      (merkle-path-target-roots t rt len idx v path) s t
  using bound
proof (induction path arbitrary: rt len idx)
case Nil
have lookup-t: fmllookup (HashMap t) (MerkleLeaf v) = Some rt
  using Nil.premis by simp

```

```

show ?case
proof (cases fmlookup (HashMap s) (MerkleLeaf v))
  case None
  then have
    hash-map-new-output-hit
      (merkle-path-target-roots t rt len idx v []) s t
    unfolding hash-map-new-output-hit-def
    using lookup-t by simp blast
  then show ?thesis by simp
next
case (Some old)
have lookup-t-old: fmlookup (HashMap t) (MerkleLeaf v) = Some old
  by (rule hash-extension-lookup[OF Some ext])
then have old = rt
  using lookup-t by simp
then have merkle-path-bound rt len idx v [] s
  using Some by simp
then show ?thesis by simp
qed
next
case (Cons sibling path)
show ?case
proof (cases idx < len div 2)
  case True
  from Cons.prem1 obtain child where child-bound:
    merkle-path-bound child (len div 2) idx v path t
  and lookup-t:
    fmlookup (HashMap t) (MerkleNode child sibling) = Some rt
  using True by auto
  have child-root:
    merkle-path-root-in-map t (len div 2) idx v path = Some child
  by (rule merkle-path-bound-root-in-map[OF child-bound])
  show ?thesis
  proof (cases fmlookup (HashMap s) (MerkleNode child sibling))
    case None
    then have root-hit:
      hash-map-new-output-hit {rt} s t
    unfolding hash-map-new-output-hit-def
    using lookup-t by blast
    have subset:
      {rt}  $\subseteq$ 
      merkle-path-target-roots t rt len idx v (sibling # path)
    using True child-root by simp
    have
      hash-map-new-output-hit
        (merkle-path-target-roots t rt len idx v (sibling # path)) s t
      by (rule hash-map-new-output-hit-subset[OF subset root-hit])
    then show ?thesis by simp
  next

```

```

case (Some old)
have lookup-t-old:
  fmlookup (HashMap t) (MerkleNode child sibling) = Some old
  by (rule hash-extension-lookup[OF Some ext])
then have old-eq: old = rt
  using lookup-t by simp
have lookup-s:
  fmlookup (HashMap s) (MerkleNode child sibling) = Some rt
  using Some old-eq by simp
from Cons.IH[OF child-bound]
have child-pullback:
  merkle-path-bound child (len div 2) idx v path s  $\vee$ 
  hash-map-new-output-hit
    (merkle-path-target-roots t child (len div 2) idx v path) s t
  .
then show ?thesis
proof
  assume child-s:
    merkle-path-bound child (len div 2) idx v path s
  then have merkle-path-bound rt len idx v (sibling # path) s
    using True lookup-s by auto
  then show ?thesis by simp
next
  assume child-hit:
    hash-map-new-output-hit
      (merkle-path-target-roots t child (len div 2) idx v path) s t
  have subset:
    merkle-path-target-roots t child (len div 2) idx v path
       $\subseteq$ 
    merkle-path-target-roots t rt len idx v (sibling # path)
    using True child-root by auto
  have
    hash-map-new-output-hit
      (merkle-path-target-roots t rt len idx v (sibling # path)) s t
    by (rule hash-map-new-output-hit-subset[OF subset child-hit])
  then show ?thesis by simp
qed
qed
next
case False
from Cons.prems obtain child where child-bound:
  merkle-path-bound child (len div 2) (idx - len div 2) v path t
and lookup-t:
  fmlookup (HashMap t) (MerkleNode sibling child) = Some rt
using False by auto
have child-root:
  merkle-path-root-in-map t (len div 2) (idx - len div 2) v path =
    Some child
by (rule merkle-path-bound-root-in-map[OF child-bound])

```

```

show ?thesis
proof (cases fmlookup (HashMap s) (MerkleNode sibling child))
  case None
  then have root-hit:
    hash-map-new-output-hit {rt} s t
    unfolding hash-map-new-output-hit-def
    using lookup-t by blast
  have subset:
    {rt}  $\subseteq$ 
    merkle-path-target-roots t rt len idx v (sibling # path)
    using False child-root by simp
  have
    hash-map-new-output-hit
    (merkle-path-target-roots t rt len idx v (sibling # path)) s t
    by (rule hash-map-new-output-hit-subset[OF subset root-hit])
  then show ?thesis by simp
next
  case (Some old)
  have lookup-t-old:
    fmlookup (HashMap t) (MerkleNode sibling child) = Some old
    by (rule hash-extension-lookup[OF Some ext])
  then have old-eq: old = rt
    using lookup-t by simp
  have lookup-s:
    fmlookup (HashMap s) (MerkleNode sibling child) = Some rt
    using Some old-eq by simp
  from Cons.IH[OF child-bound]
  have child-pullback:
    merkle-path-bound child (len div 2) (idx - len div 2) v path s  $\vee$ 
    hash-map-new-output-hit
    (merkle-path-target-roots t child (len div 2)
     (idx - len div 2) v path) s t
    .
  then show ?thesis
proof
  assume child-s:
    merkle-path-bound child (len div 2) (idx - len div 2) v path s
  then have merkle-path-bound rt len idx v (sibling # path) s
    using False lookup-s by auto
  then show ?thesis by simp
next
  assume child-hit:
    hash-map-new-output-hit
    (merkle-path-target-roots t child (len div 2)
     (idx - len div 2) v path) s t
  have subset:
    merkle-path-target-roots t child (len div 2)
    (idx - len div 2) v path
     $\subseteq$ 

```

```

      merkle-path-target-roots t rt len idx v (sibling # path)
    using False child-root by auto
  have
    hash-map-new-output-hit
      (merkle-path-target-roots t rt len idx v (sibling # path)) s t
    by (rule hash-map-new-output-hit-subset[OF subset child-hit])
  then show ?thesis by simp
qed
qed
qed
qed

```

lemma *authenticated-opening-pullback-or-new-output-hit*:

assumes *ext*: $s \leq t$

and *auth*: *authenticated-opening-in t opening*

shows

authenticated-opening-in s opening $\vee$
hash-map-new-output-hit
 (merkle-path-target-roots t
 (opening-root opening)
 (opening-length opening)
 (opening-index opening)
 (opening-value opening)
 (opening-path opening)) s t

proof —

have *idx-bound*: *opening-index opening* < *opening-length opening*

using *auth* **unfolding** *authenticated-opening-in-def* **by** *simp*

have *path-len*:

length (opening-path opening) =
floor-log (opening-length opening)

using *auth* **unfolding** *authenticated-opening-in-def* **by** *simp*

have *bound-t*:

merkle-path-bound
 (opening-root opening)
 (opening-length opening)
 (opening-index opening)
 (opening-value opening)
 (opening-path opening) t

using *auth* **unfolding** *authenticated-opening-in-def* **by** *simp*

from *merkle-path-bound-pullback-or-new-output-hit*[OF *ext bound-t*]

show ?thesis

proof

assume *bound-s*:

merkle-path-bound
 (opening-root opening)
 (opening-length opening)
 (opening-index opening)
 (opening-value opening)
 (opening-path opening) s

```

have authenticated-opening-in s opening
  unfolding authenticated-opening-in-def
  using idx-bound path-len bound-s by simp
then show ?thesis by simp
next
assume
  hash-map-new-output-hit
  (merkle-path-target-roots t
    (opening-root opening)
    (opening-length opening)
    (opening-index opening)
    (opening-value opening)
    (opening-path opening)) s t
  then show ?thesis by simp
qed
qed

lemma partial-authenticated-table-opening-pullback-or-new-output-hit:
assumes ext: s ≤ t
  and table: partial-authenticated-table rt len openings t
  and opn-in: opening ∈ set openings
shows
  authenticated-opening-in s opening ∨
  hash-map-new-output-hit
  (merkle-path-target-roots t rt len
    (opening-index opening)
    (opening-value opening)
    (opening-path opening)) s t
proof –
  have auth: authenticated-opening-in t opening
    using table opn-in unfolding partial-authenticated-table-def by blast
  have root-eq: opening-root opening = rt
    using table opn-in unfolding partial-authenticated-table-def by blast
  have len-eq: opening-length opening = len
    using table opn-in unfolding partial-authenticated-table-def by blast
  show ?thesis
    using authenticated-opening-pullback-or-new-output-hit[OF ext auth]
    unfolding root-eq len-eq .
qed

lemma partial-authenticated-table-pullback-or-new-output-hit:
assumes ext: s ≤ t
  and table: partial-authenticated-table rt len openings t
shows
  partial-authenticated-table rt len openings s ∨
  (∃ opening ∈ set openings.
    hash-map-new-output-hit
    (merkle-path-target-roots t rt len
      (opening-index opening)))

```



```

      (opening-value opening)
      (opening-path opening)) s t)
proof (cases
  ∃ opening ∈ set openings.
    hash-map-new-output-hit
      (merkle-path-target-roots t rt len
        (opening-index opening)
        (opening-value opening)
        (opening-path opening)) s t)
case True
then show ?thesis by simp
next
case False
have table-s:
  partial-authenticated-table rt len openings s
  unfolding partial-authenticated-table-def
proof
  fix opn
  assume opn-in: opn ∈ set openings
  have root-eq: opening-root opn = rt
    using table opn-in unfolding partial-authenticated-table-def by blast
  have len-eq: opening-length opn = len
    using table opn-in unfolding partial-authenticated-table-def by blast
  have auth-s: authenticated-opening-in s opn
  proof -
    have pullback:
      authenticated-opening-in s opn ∨
      hash-map-new-output-hit
        (merkle-path-target-roots t rt len
          (opening-index opn)
          (opening-value opn)
          (opening-path opn)) s t
    by (rule partial-authenticated-table-opening-pullback-or-new-output-hit
      [OF ext table opn-in])
  then show ?thesis
    using False opn-in by blast
qed
show
  opening-root opn = rt ∧
  opening-length opn = len ∧
  authenticated-opening-in s opn
  using root-eq len-eq auth-s by simp
qed
then show ?thesis by simp
qed

```

lemma *protocol-created-tree-get-authentication-path-len:*
assumes *created: protocol-created-tree xs tree s*
and *pos: 0 < lgth*

```

    and le: lgth ≤ length xs
  shows length (get-authentication-path lgth idx tree) = floor-log lgth
  using assms
proof (induction xs arbitrary: tree lgth idx rule: measure-induct-rule[of length])
  case (less xs)
  show ?case
  proof (cases xs)
    case Nil
    then show ?thesis
      using less.prem1 by simp
  next
  case (Cons x ys)
  show ?thesis
  proof (cases ys)
    case Nil
    then have lgth-eq: lgth = 1
      using less.prem1 Cons by simp
    obtain h where tree-eq: tree = ⟨⟨⟩, h, ⟨⟩⟩
      using protocol-created-tree-singletonD[OF less.prem1][unfolded Cons Nil]
      by blast
    show ?thesis
      unfolding tree-eq lgth-eq by (simp add: floor-log-Suc-zero)
  next
  case (Cons y zs)
  let ?mid = length xs div 2
  have xs-eq: xs = x # y # zs
    using ⟨xs = x # ys⟩ ⟨ys = y # zs⟩ by simp
  have len-ge: 2 ≤ length xs
    unfolding xs-eq by simp
  obtain l h r i where i-eq: i = ?mid
    and tree-eq: tree = ⟨l, h, r⟩
    and left-i: protocol-created-tree (take i xs) l s
    and right-i: protocol-created-tree (drop i xs) r s
    and root-lookup:
      fmlookup (HashMap s) (MerkleNode (value l) (value r)) =
        Some h
    by (rule protocol-created-tree-internalD[OF less.prem1 len-ge])
  have left: protocol-created-tree (take ?mid xs) l s
    using left-i i-eq by simp
  have right: protocol-created-tree (drop ?mid xs) r s
    using right-i i-eq by simp
  show ?thesis
  proof (cases lgth = 1)
    case True
    then show ?thesis
      unfolding tree-eq by (simp add: floor-log-Suc-zero)
  next
  case False
  note lgth-ne-one = False

```

```

have lgth-ge:  $2 \leq \text{lgth}$ 
  using lgth-ne-one less.prem by simp
have half-pos:  $0 < \text{lgth div } 2$ 
  using lgth-ge by simp
have take-less:  $\text{length (take ?mid xs)} < \text{length xs}$ 
  using len-ge by simp
have drop-less:  $\text{length (drop ?mid xs)} < \text{length xs}$ 
  using len-ge by simp
have half-le-take:  $\text{lgth div } 2 \leq \text{length (take ?mid xs)}$ 
  using less.prem(3) by simp
have half-le-drop:  $\text{lgth div } 2 \leq \text{length (drop ?mid xs)}$ 
  using less.prem(3) len-ge by simp
have left-len:
   $\text{length (get-authentication-path (lgth div } 2) \text{ idx l}) =$ 
   $\text{floor-log (lgth div } 2)$ 
  by (rule less.IH[OF take-less left half-pos half-le-take])
have right-len:
   $\text{length (get-authentication-path (lgth div } 2)$ 
   $(\text{idx} - \text{lgth div } 2) \text{ r}) =$ 
   $\text{floor-log (lgth div } 2)$ 
  by (rule less.IH[OF drop-less right half-pos half-le-drop])
have floor-eq:  $\text{floor-log lgth} = \text{Suc (floor-log (lgth div } 2))$ 
  using floor-log-rec[OF lgth-ge] .
show ?thesis
proof (cases  $\text{idx} < \text{lgth div } 2$ )
  case True
  then have  $\text{length (get-authentication-path lgth idx tree)} =$ 
     $\text{Suc (floor-log (lgth div } 2))$ 
    using lgth-ne-one left-len unfolding tree-eq by simp
  then show ?thesis
    by (simp only: floor-eq)
  next
  case False
  then have  $\text{length (get-authentication-path lgth idx tree)} =$ 
     $\text{Suc (floor-log (lgth div } 2))$ 
    using lgth-ne-one right-len unfolding tree-eq by simp
  then show ?thesis
    by (simp only: floor-eq)
qed
qed
qed
qed
qed

```

```

lemma created-tree-get-authentication-path-len:
  assumes created:  $\text{created-tree xs tree s}$ 
  and xs-nonempty:  $\text{xs} \neq []$ 
  shows  $\text{length (get-authentication-path (length xs) idx tree)} =$ 
   $\text{floor-log (length xs)}$ 

```

by (rule protocol-created-tree-get-authentication-path-len
 [OF created[unfolded created-tree-def]])
 (use xs-nonempty in simp-all)

lemma merkle-root-binds-table-authenticated-openings:
assumes bind: merkle-root-binds-table rt table final-state
and len-table: length table = scale * clength
and idxs-bound: $\bigwedge idx. idx \in \text{set } idxs \implies idx < \text{scale} * \text{clength}$
obtains openings **where**
 map opening-index openings = idxs
 map opening-value openings = map (! table) idxs
 partial-authenticated-table rt (scale * clength) openings final-state

proof –

obtain tree **where** created:
 created-tree table tree final-state
and rt-eq: rt = value tree
using bind **unfolding** merkle-root-binds-table-def **by** blast

let ?opening = $\lambda idx.$
 ($\mid$ opening-root = rt,
 opening-length = scale * clength,
 opening-index = idx,
 opening-value = table ! idx,
 opening-path = get-authentication-path (length table) idx tree) $\mid$

let ?openings = map ?opening idxs

have auth-all:
 $\forall opn \in \text{set } ?openings.$
 opening-root opn = rt $\wedge$
 opening-length opn = scale * clength $\wedge$
 authenticated-opening-in final-state opn

proof

fix opn
assume opn-in: opn $\in \text{set } ?openings$
then obtain idx **where** idx-in: idx $\in \text{set } idxs$
and opn-eq: opn = ?opening idx
by auto
have idx-bound-len: idx < length table
using idxs-bound[OF idx-in] len-table **by** simp
have idx-bound-domain: idx < scale * clength
by (rule idxs-bound[OF idx-in])
obtain n **where** len-pow: length table = 2^n
using len-table eval-domain-length-power
by (metis mult.commute)
have root-in-map:
 merkle-path-root-in-map final-state (length table) idx
 (table ! idx) (get-authentication-path (length table) idx tree) =
 Some rt
using protocol-created-tree-path-root-in-map
 [OF created[unfolded created-tree-def] len-pow idx-bound-len]
unfolding rt-eq .

```

have path-bound:
  merkle-path-bound rt (length table) idx (table ! idx)
    (get-authentication-path (length table) idx tree) final-state
  by (rule merkle-path-root-in-map-some-imp-bound[OF root-in-map])
have path-len:
  length (get-authentication-path (length table) idx tree) =
    floor-log (scale * clength)
proof –
  have table-nonempty: table ≠ []
    using idx-bound-len by auto
  show ?thesis
    using created-tree-get-authentication-path-len[OF created table-nonempty,
      of idx] len-table
    by simp
qed
have auth: authenticated-opening-in final-state opn
  unfolding authenticated-opening-in-def opn-eq
  using idx-bound-domain path-len path-bound len-table by simp
show opening-root opn = rt ∧
  opening-length opn = scale * clength ∧
  authenticated-opening-in final-state opn
  using auth unfolding opn-eq by simp
qed
have partial-table:
  partial-authenticated-table rt (scale * clength) ?openings
    final-state
  using auth-all unfolding partial-authenticated-table-def by blast
have indices: map opening-index ?openings = idxs
  by (induction idxs) simp-all
have value-map: map opening-value ?openings = map (! table) idxs
  by (induction idxs) simp-all
show ?thesis
  by (rule that[OF indices value-map partial-table])
qed

lemma authenticated-opening-root-in-map:
  assumes authenticated-opening-in s opening
  shows
    merkle-path-root-in-map s
      (opening-length opening)
      (opening-index opening)
      (opening-value opening)
      (opening-path opening) =
      Some (opening-root opening)
proof –
  have bound:
    merkle-path-bound
      (opening-root opening)
      (opening-length opening)

```

```

    (opening-index opening)
    (opening-value opening)
    (opening-path opening) s
  using assms unfolding authenticated-opening-in-def by simp
show ?thesis
  by (rule merkle-path-bound-root-in-map[OF bound])
qed

lemma check-authentication-path-outcome-bound:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (rt, t) ∈
      set-dist (execute (check-authentication-path len idx v path) s)
  shows merkle-path-bound rt len idx v path t
  using outcome
proof (induction path arbitrary: rt len idx s t)
  case Nil
  have hash-out:
    Some (rt, t) ∈
      set-dist
        (execute
          (hash (MerkleLeaf v) ::
            ('f, ('f, 'a) protocol-channel-scheme) state-monad) s)
  using Nil.premss
  unfolding check-authentication-path-def
    protocol-check-authentication-path-def
  by simp
  show ?case using hash-outcome(2)[OF hash-out] by simp
next
  case (Cons sibling path)
  show ?case
  proof (cases idx < len div 2)
    case True
    from Cons.premss obtain child u where
      rec-out:
        Some (child, u) ∈
          set-dist
            (execute
              (check-authentication-path (len div 2) idx v path) s)
    and hash-out:
      Some (rt, t) ∈
        set-dist
          (execute
            (hash (MerkleNode child sibling) ::
              ('f, ('f, 'a) protocol-channel-scheme) state-monad) u)
    unfolding check-authentication-path-def
      protocol-check-authentication-path-def
    using True
    by (auto simp: protocol-merkle.check-authentication-path.simpss)
  end
end

```

```

      elim!: set-dist-bindE)
have child-u:
  merkle-path-bound child (len div 2) idx v path u
  by (rule Cons.IH[OF rec-out])
have ext:  $u \leq t$ 
  by (rule hash-outcome(1)[OF hash-out])
have child-t:
  merkle-path-bound child (len div 2) idx v path t
  by (rule merkle-path-bound-mono[OF child-u ext])
have lookup:
  fmlookup (HashMap t) (MerkleNode child sibling) = Some rt
  by (rule hash-outcome(2)[OF hash-out])
show ?thesis
  using True child-t lookup
  by (auto intro!: exI[of - child])
next
case False
from Cons.premis obtain child u where
  rec-out:
    Some (child, u) ∈
      set-dist
        (execute
          (check-authentication-path
            (len div 2) (idx - len div 2) v path) s)
  and hash-out:
    Some (rt, t) ∈
      set-dist
        (execute
          (hash (MerkleNode sibling child) ::
            ('f, ('f, 'a) protocol-channel-scheme) state-monad) u)
  unfolding check-authentication-path-def
    protocol-check-authentication-path-def
  using False
  by (auto simp: protocol-merkle.check-authentication-path.simps
    elim!: set-dist-bindE)
have child-u:
  merkle-path-bound child (len div 2) (idx - len div 2)
    v path u
  by (rule Cons.IH[OF rec-out])
have ext:  $u \leq t$ 
  by (rule hash-outcome(1)[OF hash-out])
have child-t:
  merkle-path-bound child (len div 2) (idx - len div 2)
    v path t
  by (rule merkle-path-bound-mono[OF child-u ext])
have lookup:
  fmlookup (HashMap t) (MerkleNode sibling child) = Some rt
  by (rule hash-outcome(2)[OF hash-out])
show ?thesis

```

```

    using False child-t lookup
    by (auto intro!: exI[of - child])
qed
qed

lemma query-decommitment-step-authenticated-opening:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes idx-bound:  $idx < scale * clength$ 
  and outcome:
    Some (qh, t) ∈
      set-dist (execute (query-decommitment-step rt idx) s)
  obtains path opn where
    opn =
      (|opening-root = rt,
        opening-length = scale * clength,
        opening-index = idx,
        opening-value = qh,
        opening-path = path|)
    authenticated-opening-in t opn
proof -
  let ?len = scale * clength
  from outcome obtain s1 path s2 ap s3 s4 where
    read-gh:
      Some (qh, s1) ∈ set-dist (execute read s)
  and read-path:
    Some (path, s2) ∈
      set-dist (execute (ntimes read (floor-log ?len)) s1)
  and check:
    Some (ap, s3) ∈
      set-dist
        (execute (check-authentication-path ?len idx qh path) s2)
  and assert-ap:
    Some ((), s4) ∈ set-dist (execute (assert (ap = rt)) s3)
  and ret:
    Some (qh, t) ∈ set-dist (execute (return qh) s4)
  unfolding query-decommitment-step-def
  by (auto simp: Let-def elim!: set-dist-bindE)
  have path-length: length path = floor-log ?len
    using ntimes-read-any-outcome[OF read-path] by simp
  have ap-eq: ap = rt
    using assert-ap unfolding assert-def
    by (cases ap = rt) (auto simp: throw-no-outcome)
  have s4: s4 = s3
    using assert-ap ap-eq unfolding assert-def by simp
  have t: t = s3
    using ret unfolding s4 by simp
  have path-bound:
    merkle-path-bound rt ?len idx qh path t
proof -

```



```

have merkle-path-bound ap ?len idx qh path s3
  by (rule check-authentication-path-outcome-bound[OF check])
then show ?thesis unfolding ap-eq t .
qed
let ?opn =
  (opening-root = rt,
   opening-length = ?len,
   opening-index = idx,
   opening-value = qh,
   opening-path = path)
have authenticated: authenticated-opening-in t ?opn
unfolding authenticated-opening-in-def
using idx-bound path-length path-bound by simp
show ?thesis
  by (rule that[of ?opn path]) (simp-all add: authenticated)
qed

lemma mmap-query-decommitment-steps-authenticated-openings:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes idxs-bound:  $\bigwedge idx. idx \in \text{set } idxs \implies idx < \text{scale} * \text{clength}$ 
and outcome:
  Some (leaves, t)  $\in$ 
    set-dist (execute (mmap (map (query-decommitment-step rt) idxs)) s)
shows
   $\exists$  openings.
    length openings = length idxs  $\wedge$ 
    map opening-value openings = leaves  $\wedge$ 
    map opening-index openings = idxs  $\wedge$ 
    partial-authenticated-table rt (scale * clength) openings t
using outcome idxs-bound
proof (induction idxs arbitrary: s leaves t)
case Nil
then have leaves = [] and t = s
  by simp-all
moreover have
  partial-authenticated-table rt (scale * clength) [] t
unfolding partial-authenticated-table-def by simp
ultimately show ?case
  by (intro exI[of - []] simp)
next
case (Cons idx idxs)
from Cons.prems obtain leaf leaves' s1 where
  head:
    Some (leaf, s1)  $\in$ 
      set-dist (execute (query-decommitment-step rt idx) s)
and tail:
    Some (leaves', t)  $\in$ 
      set-dist (execute (mmap (map (query-decommitment-step rt) idxs)) s1)
and leaves-eq: leaves = leaf # leaves'

```

```

  by (auto elim!: set-dist-bindE)
have idx-bound:  $idx < scale * clength$ 
  by (rule Cons.premis(2)) simp
from query-decommitment-step-authenticated-opening[OF idx-bound head]
obtain path opn where
  opn-eq:
  opn =
    ( $\lambda$ opening-root = rt,
     opening-length = scale * clength,
     opening-index = idx,
     opening-value = leaf,
     opening-path = path)
  and opn-auth-s1: authenticated-opening-in s1 opn
  by blast
have tail-bound:
   $\bigwedge idx. idx \in \text{set } idxs \implies idx < scale * clength$ 
  by (rule Cons.premis(2)) simp
from Cons.IH[OF tail tail-bound] obtain openings where
  len-openings: length openings = length idxs
  and values-openings: map opening-value openings = leaves'
  and idx-openings: map opening-index openings = idxs
  and table-tail:
    partial-authenticated-table rt (scale * clength) openings t
  by blast
have s1-t:  $s1 \leq t$ 
  using mmap-query-decommitment-steps-outcome[OF tail] by blast
have opn-auth-t: authenticated-opening-in t opn
  by (rule authenticated-opening-in-mono[OF opn-auth-s1 s1-t])
let ?openings = opn # openings
have table-all:
  partial-authenticated-table rt (scale * clength) ?openings t
  using opn-auth-t table-tail opn-eq
  unfolding partial-authenticated-table-def by simp
show ?case
  using len-openings values-openings idx-openings leaves-eq opn-eq table-all
  by (intro exI[of - ?openings]) simp
qed

```

lemma *check-decommit-on-query-authenticated-openings:*
fixes $s\ t :: ('f, 'a) \text{protocol-channel-scheme}$
assumes *idx-in:* $idx \in \text{query-sample-space}$
and *outcome:*
 $\text{Some } (leaves, t) \in$
 $\text{set-dist } (\text{execute } (\text{mmap } (\text{check-decommit-on-query } rt\ idx)))\ s$
obtains *openings* **where**
 $\text{length } openings = \text{length } (\text{powers-scaled } idx)$
 $\text{map } \text{opening-value } openings = \text{leaves}$
 $\text{map } \text{opening-index } openings = \text{powers-scaled } idx$
 $\text{partial-authenticated-table } rt\ (scale * clength)\ openings\ t$

proof –
have *map-eq*:
 $check-decommit-on-query\ rt\ idx =$
 $map\ (query-decommitment-step\ rt)\ (powers-scaled\ idx)$
unfolding *check-decommit-on-query-def query-decommitment-step-def* **by** *simp*
have *idxs-bound*:
 $\bigwedge i. i \in set\ (powers-scaled\ idx) \implies i < scale * clength$
using *query-sample-space-powers-scaled-bound*[*OF idx-in*]
by (*simp add: mult.commute*)
from *mmap-query-decommitment-steps-authenticated-openings*
 $[OF\ idxs-bound\ outcome[unfolding\ map-eq]]$
obtain *openings* **where**
 $length\ openings = length\ (powers-scaled\ idx)$
 $map\ opening-value\ openings = leaves$
 $map\ opening-index\ openings = powers-scaled\ idx$
 $partial-authenticated-table\ rt\ (scale * clength)\ openings\ t$
by *blast*
then show *?thesis* **by** (*rule that*)
qed

lemma *fri-layer-opening-finish-authenticated-openings*:

fixes $s\ t :: ('f, 'a)\ protocol-channel-scheme$

assumes *idx-bound*: $idx < len$

and *sibling-bound*: $(idx + len\ div\ 2) \bmod len < len$

and *xp-path-length*: $length\ xp-path = floor-log\ len$

and *xn-path-length*: $length\ xn-path = floor-log\ len$

and *outcome*:

$Some\ (out, t) \in$

$set-dist$

$(execute$

$(fri-layer-opening-finish\ b\ rt\ idx\ x\ len\ pw$

$xp\ xp-path\ xn\ xn-path)\ s)$

obtains *xp-opening xn-opening* **where**

xp-opening =

$(\mid opening-root = rt,$

$opening-length = len,$

$opening-index = idx,$

$opening-value = xp,$

$opening-path = xp-path)\mid)$

xn-opening =

$(\mid opening-root = rt,$

$opening-length = len,$

$opening-index = (idx + len\ div\ 2) \bmod len,$

$opening-value = xn,$

$opening-path = xn-path)\mid)$

authenticated-opening-in t xp-opening

authenticated-opening-in t xn-opening

proof –

from *outcome* **obtain** $s1\ ap1\ s2\ s3\ ap2\ s4\ s5$ **where**

```

assert-xp:
  Some ((), s1)  $\in$  set-dist (execute (assert (xp = x)) s)
and check-xp:
  Some (ap1, s2)  $\in$ 
    set-dist
      (execute (check-authentication-path len idx xp xp-path) s1)
and assert-ap1:
  Some ((), s3)  $\in$  set-dist (execute (assert (ap1 = rt)) s2)
and check-xn:
  Some (ap2, s4)  $\in$ 
    set-dist
      (execute
        (check-authentication-path len
          ((idx + len div 2) mod len) xn xn-path) s3)
and assert-ap2:
  Some ((), s5)  $\in$  set-dist (execute (assert (ap2 = rt)) s4)
and ret:
  Some (out, t)  $\in$ 
    set-dist
      (execute
        (return
          (idx mod (len div 2),
            (xp + xn) div 2 +
              b * ((xp - xn) div
                (2 * ((h ^ idx) * shift) ^ pw)),
            len div 2, pw + pw)) s5)
    unfolding fri-layer-opening-finish-def
    by (auto simp: Let-def elim!: set-dist-bindE)
have ap1-eq: ap1 = rt
  using assert-ap1 unfolding assert-def
  by (cases ap1 = rt) (auto simp: throw-no-outcome)
have ap2-eq: ap2 = rt
  using assert-ap2 unfolding assert-def
  by (cases ap2 = rt) (auto simp: throw-no-outcome)
have xp-eq: xp = x
  using assert-xp unfolding assert-def
  by (cases xp = x) (auto simp: throw-no-outcome)
have s1: s1 = s
  using assert-xp xp-eq unfolding assert-def by simp
have s3: s3 = s2
  using assert-ap1 ap1-eq unfolding assert-def by simp
have s5: s5 = s4
  using assert-ap2 ap2-eq unfolding assert-def by simp
have t: t = s4
  using ret unfolding s5 by simp
have xp-bound-s2:
  merkle-path-bound rt len idx xp xp-path s2
proof –
  have merkle-path-bound ap1 len idx xp xp-path s2

```

```

    by (rule check-authentication-path-outcome-bound[OF check-xp])
  then show ?thesis unfolding ap1-eq .
qed
have s2-s4:  $s2 \leq s4$ 
  unfolding s3[symmetric]
  by (rule check-authentication-path-hash-extends[OF check-xn])
have xp-bound:
  merkle-path-bound rt len idx xp xp-path t
  unfolding t
  by (rule merkle-path-bound-mono[OF xp-bound-s2 s2-s4])
have xn-bound:
  merkle-path-bound rt len ((idx + len div 2) mod len)
  xn xn-path t
proof -
  have merkle-path-bound ap2 len ((idx + len div 2) mod len)
    xn xn-path s4
    by (rule check-authentication-path-outcome-bound[OF check-xn])
  then show ?thesis unfolding ap2-eq t .
qed
let ?xp-opening =
  (opening-root = rt,
   opening-length = len,
   opening-index = idx,
   opening-value = xp,
   opening-path = xp-path)
let ?xn-opening =
  (opening-root = rt,
   opening-length = len,
   opening-index = (idx + len div 2) mod len,
   opening-value = xn,
   opening-path = xn-path)
have xp-authenticated: authenticated-opening-in t ?xp-opening
  unfolding authenticated-opening-in-def
  using idx-bound xp-path-length xp-bound by simp
have xn-authenticated: authenticated-opening-in t ?xn-opening
  unfolding authenticated-opening-in-def
  using sibling-bound xn-path-length xn-bound by simp
show ?thesis
  by (rule that[of ?xp-opening ?xn-opening])
  (simp-all add: xp-authenticated xn-authenticated)
qed

lemma fri-layer-opening-step-authenticated-openings:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes len-pos:  $0 < len$ 
  and idx-bound:  $idx < len$ 
  and outcome:
    Some (out, t)  $\in$ 
    set-dist

```

```

      (execute (fri-layer-opening-step (b, rt) (idx, x, len, pw)) s)
obtains xp-opening xn-opening where
  opening-root xp-opening = rt
  opening-length xp-opening = len
  opening-index xp-opening = idx
  opening-root xn-opening = rt
  opening-length xn-opening = len
  opening-index xn-opening = (idx + len div 2) mod len
  authenticated-opening-in t xp-opening
  authenticated-opening-in t xn-opening
proof –
from outcome[unfolded fri-layer-opening-step-unfold-finish]
obtain xp s1 xp-path s2 xn s3 xn-path s4 where
  read-xp: Some (xp, s1) ∈ set-dist (execute read s)
and read-xp-path:
  Some (xp-path, s2) ∈
    set-dist (execute (ntimes read (floor-log len)) s1)
and read-xn: Some (xn, s3) ∈ set-dist (execute read s2)
and read-xn-path:
  Some (xn-path, s4) ∈
    set-dist (execute (ntimes read (floor-log len)) s3)
and finish:
  Some (out, t) ∈
    set-dist
      (execute
        (fri-layer-opening-finish b rt idx x len pw
          xp xp-path xn xn-path) s4)
by (auto elim!: set-dist-bindE)
have xp-path-length: length xp-path = floor-log len
  using ntimes-read-any-outcome[OF read-xp-path] by simp
have xn-path-length: length xn-path = floor-log len
  using ntimes-read-any-outcome[OF read-xn-path] by simp
have sibling-bound: (idx + len div 2) mod len < len
  using len-pos by simp
from fri-layer-opening-finish-authenticated-openings
  [OF idx-bound sibling-bound xp-path-length xn-path-length finish]
obtain xp-opening xn-opening where
  xp-eq:
    xp-opening =
      (|opening-root = rt,
        opening-length = len,
        opening-index = idx,
        opening-value = xp,
        opening-path = xp-path)
and xn-eq:
    xn-opening =
      (|opening-root = rt,
        opening-length = len,
        opening-index = (idx + len div 2) mod len,
```

```

    opening-value = xn,
    opening-path = xn-path)
  and xp-auth: authenticated-opening-in t xp-opening
  and xn-auth: authenticated-opening-in t xn-opening
  by blast
show ?thesis
  by (rule that[of xp-opening xn-opening])
    (use xp-eq xn-eq xp-auth xn-auth in simp-all)
qed

```

lemma *fri-layer-opening-step-partial-authenticated-table:*
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes $\text{len-pos}: 0 < \text{len}$
and $\text{idx-bound}: \text{idx} < \text{len}$
and $\text{outcome}: \text{Some } (\text{out}, t) \in \text{set-dist}$
 $(\text{execute } (\text{fri-layer-opening-step } (b, rt) (\text{idx}, x, \text{len}, pw))\ s)$
obtains openings **where**
 $\text{partial-authenticated-table } rt\ \text{len}\ \text{openings}\ t$
 $\text{map opening-index openings} = [\text{idx}, (\text{idx} + \text{len} \text{ div } 2) \bmod \text{len}]$
 $\text{length openings} = 2$
proof –
from *fri-layer-opening-step-authenticated-openings*
 $[\text{OF len-pos idx-bound outcome}]$
obtain $\text{xp-opening xn-opening}$ **where**
 $\text{xp-root}: \text{opening-root xp-opening} = rt$
and $\text{xp-len}: \text{opening-length xp-opening} = \text{len}$
and $\text{xp-idx}: \text{opening-index xp-opening} = \text{idx}$
and $\text{xn-root}: \text{opening-root xn-opening} = rt$
and $\text{xn-len}: \text{opening-length xn-opening} = \text{len}$
and $\text{xn-idx}: \text{opening-index xn-opening} = (\text{idx} + \text{len} \text{ div } 2) \bmod \text{len}$
and $\text{xp-auth}: \text{authenticated-opening-in } t\ \text{xp-opening}$
and $\text{xn-auth}: \text{authenticated-opening-in } t\ \text{xn-opening}$
by *blast*
let $?openings = [\text{xp-opening}, \text{xn-opening}]$
have $\text{table}: \text{partial-authenticated-table } rt\ \text{len}\ ?openings\ t$
unfolding *partial-authenticated-table-def*
using $\text{xp-root xp-len xp-auth xn-root xn-len xn-auth}$ **by** *simp*
have $\text{indices}: \text{map opening-index } ?openings = [\text{idx}, (\text{idx} + \text{len} \text{ div } 2) \bmod \text{len}]$
using xp-idx xn-idx **by** *simp*
show *?thesis*
by $(\text{rule that}[\text{OF table indices}])\ \text{simp}$
qed

lemma *verifier-query-round-after-index-authenticated-trace-openings:*
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$

```

assumes outcome:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s)
obtains idx leaves openings where
  idx = index (to-nat raw)
  length openings = length (powers-scaled idx)
  map opening-value openings = leaves
  map opening-index openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) openings t
proof –
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
    query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
    and trace-fri:
      Some (f-out, s2) ∈
        set-dist (execute
          (mfold (?idx, hd fv, clength * scale, 1)
            (receive-query-commits f-fl)) s1)
    and assert-trace:
      Some ((), s3) ∈
        set-dist
          (execute
            (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
    and comp-fri:
      Some (c-out, s4) ∈
        set-dist (execute
          (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
            clength * scale, 1)
            (receive-query-commits fl)) s3)
    and assert-comp:
      Some ((), t) ∈
        set-dist
          (execute
            (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
    unfolding verifier-query-round-after-index-program-def
    by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
  have idx-in: ?idx ∈ query-sample-space
    using index-less-query-sample-space
    unfolding query-sample-space-def by simp
  from check-decommit-on-query-authenticated-openings
    [OF idx-in query-decommit]
  obtain openings where
    len-openings: length openings = length (powers-scaled ?idx)
    and values-openings: map opening-value openings = fv

```



```

and idx-openings: map opening-index openings = powers-scaled ?idx
and table-s1:
  partial-authenticated-table fr (scale * clength) openings s1
by blast
from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where
  s1-s2: s1 ≤ s2
by blast
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def
  by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
    (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  s3-s4: s3 ≤ s4
by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -) ⇒ x = final)
    (auto simp: throw-no-outcome)
have s1-t: s1 ≤ t
  using s1-s2 s3-s4 unfolding s3-eq t-eq
  by (meson hash-ext-trans)
have table-t:
  partial-authenticated-table fr (scale * clength) openings t
  by (rule partial-authenticated-table-mono[OF table-s1 s1-t])
show ?thesis
  by (rule that[OF refl len-openings values-openings idx-openings table-t])
qed

```

lemma *verifier-query-round-after-index-authenticated-composition-openings*:

```

fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-eq: fl = (b, composition-root) # fl-tail
and outcome:
  Some (((), t) ∈
    set-dist
    (execute
      (verifier-query-round-after-index-program
        fr f-fl f-final as fl final raw) s))

```

```

obtains idx openings where
  idx = index (to-nat raw)
  partial-authenticated-table composition-root (scale * clength)
  openings t
  map opening-index openings =
    [idx, fri-sibling-index (scale * clength) idx]
  length openings = 2

```

proof –

```

let ?idx = index (to-nat raw)
from outcome obtain fv s1 f-out s2 s3 c-out s4 where

```

```

query-decommit:
  Some (fv, s1) ∈
    set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
and trace-fri:
  Some (f-out, s2) ∈
    set-dist (execute
      (mfold (?idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl)) s1)
and assert-trace:
  Some ((), s3) ∈
    set-dist
      (execute
        (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
  Some (c-out, s4) ∈
    set-dist (execute
      (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
        clength * scale, 1)
        (receive-query-commits fl)) s3)
and assert-comp:
  Some ((), t) ∈
    set-dist
      (execute
        (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
unfolding verifier-query-round-after-index-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have comp-steps:
  receive-query-commits fl =
    fri-layer-opening-step (b, composition-root) #
    receive-query-commits fl-tail
unfolding fl-eq receive-query-commits-def fri-layer-opening-step-def
by simp
from comp-fri[unfolded comp-steps]
obtain out1 s-head where
  head:
    Some (out1, s-head) ∈
      set-dist
        (execute
          (fri-layer-opening-step (b, composition-root)
            (?idx, cp-eval as fv (h ^ ?idx * shift),
              clength * scale, 1)) s3)
and tail:
    Some (c-out, s4) ∈
      set-dist (execute (mfold out1 (receive-query-commits fl-tail)
        s-head)
by (auto elim!: set-dist-bindE)
have len-pos: 0 < clength * scale
using eval-domain-nontrivial by linarith
have idx-bound: ?idx < clength * scale

```

```

    by (rule index-less-domain)
  from fri-layer-opening-step-partial-authenticated-table
    [OF len-pos idx-bound head]
  obtain openings where
    table-head:
      partial-authenticated-table composition-root (clength * scale)
      openings s-head
  and indices:
    map opening-index openings =
      [?idx, (?idx + (clength * scale) div 2) mod (clength * scale)]
  and len-openings: length openings = 2
  by (rule fri-layer-opening-step-partial-authenticated-table
    [OF len-pos idx-bound head])
  obtain i1 x1 len1 pw1 where out1-eq: out1 = (i1, x1, len1, pw1)
  by (cases out1)
  from receive-query-commits-layers-outcome[OF tail[unfolded out1-eq]]
  obtain tail-layer-chunks tail-chunk where s-head-s4: s-head ≤ s4
  by blast
  have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -) ⇒ x = final)
    (auto simp: throw-no-outcome)
  have table-t:
    partial-authenticated-table composition-root (scale * clength)
    openings t
  using partial-authenticated-table-mono[OF table-head s-head-s4]
  unfolding t-eq by (simp add: mult.commute)
  have indices':
    map opening-index openings =
      [?idx, fri-sibling-index (scale * clength) ?idx]
  using indices
  unfolding fri-sibling-index-def
  by (simp add: mult.commute)
  show ?thesis
  by (rule that[OF refl table-t indices' len-openings])
qed

```

```

lemma verifier-query-round-program-authenticated-trace-openings:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  obtains raw idx leaves openings where
    idx = index (to-nat raw)
    length openings = length (powers-scaled idx)
    map opening-value openings = leaves
    map opening-index openings = powers-scaled idx

```

```

    partial-authenticated-table fr (scale * clength) openings t
proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where
  rand:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s)
and tail:
    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s0)
  by (auto elim!: set-dist-bindE)
from verifier-query-round-after-index-authenticated-trace-openings[OF tail]
obtain idx leaves openings where
  idx-eq: idx = index (to-nat raw)
  and len-openings: length openings = length (powers-scaled idx)
  and values-openings: map opening-value openings = leaves
  and idx-openings: map opening-index openings = powers-scaled idx
  and table-t:
    partial-authenticated-table fr (scale * clength) openings t
  by blast
show ?thesis
  by (rule that[OF idx-eq len-openings values-openings idx-openings table-t])
qed

```

lemma verifier-query-round-program-authenticated-composition-openings:

```

fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-eq: fl = (b, composition-root) # fl-tail
and outcome:
  Some (( ), t) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
obtains raw idx openings where
  idx = index (to-nat raw)
  partial-authenticated-table composition-root (scale * clength)
    openings t
  map opening-index openings =
    [idx, fri-sibling-index (scale * clength) idx]
  length openings = 2

```

```

proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where
  rand:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s)
and tail:

```

```

    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s0)
  by (auto elim!: set-dist-bindE)
from verifier-query-round-after-index-authenticated-composition-openings
  [OF fl-eq tail]
obtain idx openings where
  idx-eq: idx = index (to-nat raw)
  and table-t:
    partial-authenticated-table composition-root (scale * clength)
      openings t
  and idx-openings:
    map opening-index openings =
      [idx, fri-sibling-index (scale * clength) idx]
  and len-openings: length openings = 2
  by blast
show ?thesis
  by (rule that[OF idx-eq table-t idx-openings len-openings])
qed

```

lemma *ntimes-verifier-query-rounds-authenticated-trace-openings:*

fixes *s t :: ('f, 'a) protocol-channel-scheme*

assumes *outcome:*

```

  Some (results, t) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final) n)
        s)

```

obtains *raw-idxs query-idxs trace-openings where*

```

  length raw-idxs = n
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  length trace-openings = n
  ∧ i. i < n ⇒
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
  ∧ i. i < n ⇒
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) t

```

proof –

have

```

  ∃ raw-idxs query-idxs trace-openings.
    length raw-idxs = n ∧
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs ∧
    length trace-openings = n ∧
    (∀ i < n.
      map opening-index (trace-openings ! i) =

```

```

    powers-scaled (query-idxs ! i)) ∧
  (∀ i < n.
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) t)
using outcome
proof (induction n arbitrary: s results t)
  case 0
  then show ?case
    by (intro exI[of - []]) simp
next
  case (Suc n)
  from Suc.premis obtain u results' s1 where
    head:
      Some (u, s1) ∈
        set-dist
          (execute
            (verifier-query-round-program fr f-fl f-final as fl final) s)
    and tail:
      Some (results', t) ∈
        set-dist
          (execute
            (ntimes
              (verifier-query-round-program fr f-fl f-final as fl final)
            n) s1)
    and results-eq: results = u # results'
    by (auto elim!: set-dist-bindE)
  have u-eq: u = ()
    by (cases u) simp
  from verifier-query-round-program-authenticated-trace-openings
    [OF head[unfolded u-eq]]
  obtain raw idx leaves openings where
    idx-eq: idx = index (to-nat raw)
    and idx-openings: map opening-index openings = powers-scaled idx
    and table-s1:
      partial-authenticated-table fr (scale * clength) openings s1
    by blast
  from Suc.IH[OF tail] obtain raw-tail idx-tail openings-tail where
    len-raw-tail: length raw-tail = n
    and idx-tail-eq:
      idx-tail = map (λraw. index (to-nat raw)) raw-tail
    and len-openings-tail: length openings-tail = n
    and tail-indices:
      ∀ i < n.
        map opening-index (openings-tail ! i) =
          powers-scaled (idx-tail ! i)
    and tail-tables:
      ∀ i < n.
        partial-authenticated-table fr (scale * clength)
          (openings-tail ! i) t

```

```

  by blast
have s1-t:  $s1 \leq t$ 
  using ntimes-verifier-query-rounds-outcome[OF tail] by blast
have table-t:
  partial-authenticated-table fr (scale * clength) openings t
  by (rule partial-authenticated-table-mono[OF table-s1 s1-t])
let ?rows = raw # raw-tail
let ?idxs = idx # idx-tail
let ?openings = openings # openings-tail
have idxs-eq:
  ?idxs = map ( $\lambda$ raw. index (to-nat raw)) ?rows
  using idx-eq idx-tail-eq by simp
have indices-all:
   $\forall i < \text{Suc } n.$ 
    map opening-index (?openings ! i) =
      powers-scaled (?idxs ! i)
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{Suc } n$ 
  show map opening-index (?openings ! i) =
    powers-scaled (?idxs ! i)
  proof (cases i)
    case 0
    then show ?thesis
      using idx-openings by simp
  next
    case (Suc j)
    then show ?thesis
      using i-bound tail-indices by simp
  qed
qed
have tables-all:
   $\forall i < \text{Suc } n.$ 
    partial-authenticated-table fr (scale * clength)
      (?openings ! i) t
proof (intro allI impI)
  fix i
  assume i-bound:  $i < \text{Suc } n$ 
  show partial-authenticated-table fr (scale * clength)
    (?openings ! i) t
  proof (cases i)
    case 0
    then show ?thesis
      using table-t by simp
  next
    case (Suc j)
    then show ?thesis
      using i-bound tail-tables by simp
  qed

```

```

qed
show ?case
  apply (intro exI[of - ?rows] exI[of - ?idxs] exI[of - ?openings]
    conjI)
    apply (simp add: len-row-tail)
    apply (rule idxs-eq)
    apply (simp add: len-openings-tail)
    apply (rule indices-all)
    apply (rule tables-all)
  done
qed
then obtain raw-idxs query-idxs trace-openings where
  length raw-idxs = n
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  length trace-openings = n
  ∧ i. i < n ⇒
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
  ∧ i. i < n ⇒
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) t
  by blast
then show ?thesis by (rule that)
qed

lemma ntimes-verifier-query-rounds-authenticated-composition-openings:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes fl-eq: fl = (b, composition-root) # fl-tail
  and outcome:
    Some (results, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s)
  obtains raw-idxs query-idxs composition-openings where
    length raw-idxs = n
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
    length composition-openings = n
    ∧ i. i < n ⇒
      map opening-index (composition-openings ! i) =
        [query-idxs ! i,
         fri-sibling-index (scale * clength) (query-idxs ! i)]
    ∧ i. i < n ⇒
      partial-authenticated-table composition-root (scale * clength)
        (composition-openings ! i) t
proof -
  have
    ∃ raw-idxs query-idxs composition-openings.

```



```

length raw-idxs = n ∧
query-idxs = map (λraw. index (to-nat raw)) raw-idxs ∧
length composition-openings = n ∧
(∀ i < n.
  map opening-index (composition-openings ! i) =
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)]) ∧
(∀ i < n.
  partial-authenticated-table composition-root (scale * clength)
    (composition-openings ! i) t)
using outcome
proof (induction n arbitrary: s results t)
  case 0
  then show ?case
    by (intro exI[of - []]) simp
next
case (Suc n)
from Suc.premis obtain u results' s1 where
  head:
    Some (u, s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results', t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            n) s1)
  by (auto elim!: set-dist-bindE)
have u-eq: u = ()
  by (cases u) simp
from verifier-query-round-program-authenticated-composition-openings
  [OF fl-eq head[unfolded u-eq]]
obtain raw idx openings where
  idx-eq: idx = index (to-nat raw)
  and table-s1:
    partial-authenticated-table composition-root (scale * clength)
      openings s1
  and idx-openings:
    map opening-index openings =
      [idx, fri-sibling-index (scale * clength) idx]
  and len-openings: length openings = 2
  by (rule verifier-query-round-program-authenticated-composition-openings
    [OF fl-eq head[unfolded u-eq]])
from Suc.IH[OF tail] obtain raw-tail idx-tail openings-tail where
  len-raw-tail: length raw-tail = n
  and idx-tail-eq:

```

```

    idx-tail = map ( $\lambda$ raw. index (to-nat raw)) raw-tail
and len-openings-tail: length openings-tail = n
and tail-indices:
   $\forall i < n.$ 
    map opening-index (openings-tail ! i) =
      [idx-tail ! i,
       fri-sibling-index (scale * clength) (idx-tail ! i)]
and tail-tables:
   $\forall i < n.$ 
    partial-authenticated-table composition-root (scale * clength)
      (openings-tail ! i) t
  by blast
have s1-t: s1  $\leq$  t
  using ntimes-verifier-query-rounds-outcome[OF tail] by blast
have table-t:
  partial-authenticated-table composition-root (scale * clength)
    openings t
  by (rule partial-authenticated-table-mono[OF table-s1 s1-t])
let ?rows = raw # raw-tail
let ?idxs = idx # idx-tail
let ?openings = openings # openings-tail
have idxs-eq:
  ?idxs = map ( $\lambda$ raw. index (to-nat raw)) ?rows
  using idx-eq idx-tail-eq by simp
have indices-all:
   $\forall i < \text{Suc } n.$ 
    map opening-index (?openings ! i) =
      [?idxs ! i, fri-sibling-index (scale * clength) (?idxs ! i)]
proof (intro allI impI)
  fix i
  assume i-bound: i < Suc n
  show map opening-index (?openings ! i) =
    [?idxs ! i, fri-sibling-index (scale * clength) (?idxs ! i)]
  proof (cases i)
  case 0
  then show ?thesis
    using idx-openings by simp
  next
  case (Suc j)
  then show ?thesis
    using i-bound tail-indices by simp
  qed
qed
have tables-all:
   $\forall i < \text{Suc } n.$ 
    partial-authenticated-table composition-root (scale * clength)
      (?openings ! i) t
  proof (intro allI impI)
  fix i

```

```

assume i-bound:  $i < \text{Suc } n$ 
show partial-authenticated-table composition-root ( $\text{scale} * \text{clength}$ )
  ( $?openings ! i$ ) t
proof (cases i)
  case 0
  then show ?thesis
    using table-t by simp
next
  case (Suc j)
  then show ?thesis
    using i-bound tail-tables by simp
qed
qed
show ?case
  apply (intro exI[of - ?rows] exI[of - ?idxs]
    exI[of - ?openings] conjI)
  apply (simp add: len-row-tail)
  apply (rule idxs-eq)
  apply (simp add: len-openings-tail)
  apply (rule indices-all)
  apply (rule tables-all)
  done
qed
then obtain raw-idxs query-idxs composition-openings where
  len-row:  $\text{length } \text{raw-idxs} = n$ 
and idxs-eq:  $\text{query-idxs} = \text{map } (\lambda \text{raw. index } (\text{to-nat } \text{raw})) \text{ raw-idxs}$ 
and len-openings:  $\text{length } \text{composition-openings} = n$ 
and indices:
   $\forall i < n.$ 
     $\text{map } \text{opening-index } (\text{composition-openings} ! i) =$ 
     $[\text{query-idxs} ! i,$ 
       $\text{fri-sibling-index } (\text{scale} * \text{clength}) (\text{query-idxs} ! i)]$ 
and tables:
   $\forall i < n.$ 
     $\text{partial-authenticated-table } \text{composition-root } (\text{scale} * \text{clength})$ 
     $(\text{composition-openings} ! i) \text{ } t$ 
by (elim exE conjE, blast)
show ?thesis
  by (rule that[OF len-row idxs-eq len-openings])
  (use indices tables in auto)
qed

lemma ntimes-verifier-query-rounds-authenticated-trace-openings-no-rows:
fixes  $s \text{ } t :: ('f, 'a) \text{ protocol-channel-scheme}$ 
assumes outcome:
   $\text{Some } (\text{results}, t) \in$ 
  set-dist
  (execute
    (ntimes

```

(verifier-query-round-program fr f-fl f-final as fl final) n)
 s)
obtains query-idxs trace-openings **where**
 length query-idxs = n
 length trace-openings = n
 $\bigwedge i. i < n \implies$
 map opening-index (trace-openings ! i) =
 powers-scaled (query-idxs ! i)
 $\bigwedge i. i < n \implies$
 partial-authenticated-table fr (scale * clength)
 (trace-openings ! i) t
proof (rule ntimes-verifier-query-rounds-authenticated-trace-openings
 [OF outcome])
fix raw-idxs query-idxs trace-openings
assume len-raw: length raw-idxs = n
and idxs-eq: query-idxs = map (λ raw. index (to-nat raw)) raw-idxs
and len-openings: length trace-openings = n
and indices:
 $\bigwedge i. i < n \implies$
 map opening-index (trace-openings ! i) =
 powers-scaled (query-idxs ! i)
and tables:
 $\bigwedge i. i < n \implies$
 partial-authenticated-table fr (scale * clength)
 (trace-openings ! i) t
have len-query-idxs: length query-idxs = n
using len-raw idxs-eq **by** simp
show ?thesis
by (rule that[OF len-query-idxs len-openings indices tables])
qed

lemma verify-monad-authenticated-trace-openings:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes outcome:
 Some (result, final-state) $\in$ set-dist (execute verify-monad s)
obtains fr query-idxs trace-openings **where**
 length query-idxs = rounds
 length trace-openings = rounds
 $\bigwedge i. i < rounds \implies$
 map opening-index (trace-openings ! i) =
 powers-scaled (query-idxs ! i)
 $\bigwedge i. i < rounds \implies$
 partial-authenticated-table fr (scale * clength)
 (trace-openings ! i) final-state
proof –
from verify-monad-header-extraction[OF outcome]
obtain fr f-fl f-final as dg fl final query-state **where**
 query-out:
 Some (result, final-state) $\in$

```

    set-dist (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
      query-state)
  by metis
from ntimes-verifier-query-rounds-authenticated-trace-openings
  [OF query-out]
obtain raw-idxs query-idxs trace-openings where
  len-raw: length raw-idxs = rounds
  and query-idxs-def:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-openings: length trace-openings = rounds
  and indices:
    ∧ i. i < rounds ⇒
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
  and tables:
    ∧ i. i < rounds ⇒
      partial-authenticated-table fr (scale * clength)
        (trace-openings ! i) final-state
  by blast
have len-query-idxs: length query-idxs = rounds
  using len-raw query-idxs-def by simp
show ?thesis
  by (rule that[OF len-query-idxs len-openings indices tables])
qed

```

definition *accepted-with-partial-trace-openings*

```

:: ('f, 'a) protocol-channel-scheme ⇒
  (unit list × ('f, 'a) protocol-channel-scheme) option ⇒
  'f ⇒ nat list ⇒ 'f authenticated-opening list list ⇒ bool

```

where

```

accepted-with-partial-trace-openings s out fr query-idxs trace-openings ⇔
  accepted out ∧
  (∃ result final-state.
    out = Some (result, final-state) ∧
    length query-idxs = rounds ∧
    length trace-openings = rounds ∧
    (∀ i < rounds.
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i) ∧
    (∀ i < rounds.
      partial-authenticated-table fr (scale * clength)
        (trace-openings ! i) final-state))

```

lemma *accepted-with-partial-trace-openings-imp-accepted:*

```

assumes accepted-with-partial-trace-openings s out fr query-idxs trace-openings
shows accepted out

```

using *assms* **unfolding** *accepted-with-partial-trace-openings-def* **by** *simp*

lemma *accepted-with-partial-trace-openings-shapes:*

assumes *accepted-with-partial-trace-openings s out fr query-idxs trace-openings*

shows *length query-idxs = rounds*

and *length trace-openings = rounds*

and $\bigwedge i. i < \text{rounds} \implies$

$\text{map opening-index } (\text{trace-openings } ! i) =$

$\text{powers-scaled } (\text{query-idxs } ! i)$

using *assms* **unfolding** *accepted-with-partial-trace-openings-def* **by** *blast+*

definition *table-agrees-with-authenticated-openings*

$:: 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ authenticated-opening list} \Rightarrow \text{bool}$

where

table-agrees-with-authenticated-openings table len openings $\longleftrightarrow$

length table = len $\wedge$

$(\forall \text{ opening} \in \text{set openings.}$

opening-index opening < len $\wedge$

table ! opening-index opening = opening-value opening)

lemma *table-agrees-with-authenticated-openings-subset:*

assumes *agrees:*

table-agrees-with-authenticated-openings table len openings

and *subset: set openings' $\subseteq$ set openings*

shows *table-agrees-with-authenticated-openings table len openings'*

using *assms*

unfolding *table-agrees-with-authenticated-openings-def* **by** *blast*

lemma *table-agrees-with-authenticated-openings-consistent-exists:*

assumes *idx-bound:*

$\bigwedge \text{opn. } \text{opn} \in \text{set openings} \implies \text{opening-index opn} < \text{len}$

and *consistent:*

$\bigwedge \text{opn } \text{opn}'. \text{opn} \in \text{set openings} \implies$

$\text{opn}' \in \text{set openings} \implies$

$\text{opening-index opn} = \text{opening-index opn}' \implies$

$\text{opening-value opn} = \text{opening-value opn}'$

shows $\exists \text{ table. table-agrees-with-authenticated-openings table len openings}$

proof –

let *?value* =

$\lambda j. \text{ if } \exists \text{ opn} \in \text{set openings. opening-index opn} = j$

$\text{ then opening-value}$

$(\text{SOME } \text{opn. } \text{opn} \in \text{set openings} \wedge \text{opening-index opn} = j)$

$\text{ else } 0$

let *?table* = $\text{map } ?\text{value } [0..<\text{len}]$

have *agrees:*

$\bigwedge \text{opn. } \text{opn} \in \text{set openings} \implies$

$?table ! \text{opening-index opn} = \text{opening-value opn}$

proof –

```

fix opn
assume opn-in: opn ∈ set openings
let ?chosen =
  SOME opn'. opn' ∈ set openings ∧
    opening-index opn' = opening-index opn
have ex-chosen:
  ∃ opn'. opn' ∈ set openings ∧
    opening-index opn' = opening-index opn
  using opn-in by blast
have chosen:
  ?chosen ∈ set openings ∧
    opening-index ?chosen = opening-index opn
  by (rule someI-ex[OF ex-chosen])
have value-eq:
  opening-value ?chosen = opening-value opn
  using consistent[OF conjunct1[OF chosen] opn-in conjunct2[OF chosen]]
  by simp
have ?table ! opening-index opn = ?value (opening-index opn)
  using idx-bound[OF opn-in] by simp
also have ... = opening-value ?chosen
proof –
  have ∃ opn' ∈ set openings. opening-index opn' =
    opening-index opn
  using opn-in by blast
  then show ?thesis by simp
qed
also have ... = opening-value opn
  by (rule value-eq)
finally show ?table ! opening-index opn = opening-value opn .
qed
have table-agrees-with-authenticated-openings ?table len openings
  unfolding table-agrees-with-authenticated-openings-def
  using idx-bound agrees by simp
then show ?thesis by blast
qed

definition partial-trace-table-candidate
  :: 'f list ⇒ 'f authenticated-opening list list ⇒ bool'
where
  partial-trace-table-candidate trace-table trace-openings  $\longleftrightarrow$ 
    length trace-table = scale * clength ∧
    length trace-openings = rounds ∧
    ( $\forall i < rounds.$ 
      table-agrees-with-authenticated-openings trace-table
        (scale * clength) (trace-openings ! i))

definition partial-trace-table-candidates
  :: 'f authenticated-opening list list ⇒ 'f list set'
where

```

partial-trace-table-candidates trace-openings =
 $\{\text{trace-table. } \text{partial-trace-table-candidate } \text{trace-table } \text{trace-openings}\}$

definition *accepted-with-partial-composition-openings*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow$
 $'f \Rightarrow \text{nat list} \Rightarrow 'f \text{ authenticated-opening list list} \Rightarrow \text{bool}$

where

accepted-with-partial-composition-openings s out composition-root
 $\text{query-idxs composition-openings} \longleftrightarrow$
 $\text{accepted out} \wedge$
 $(\exists \text{ result final-state.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{length query-idxs} = \text{rounds} \wedge$
 $\text{length composition-openings} = \text{rounds} \wedge$
 $(\forall i < \text{rounds.}$
 $\text{map opening-index (composition-openings ! i)} =$
 $[\text{query-idxs ! i,}$
 $\text{fri-sibling-index (scale * clength) (query-idxs ! i)}]) \wedge$
 $(\forall i < \text{rounds.}$
 $\text{partial-authenticated-table composition-root (scale * clength)}$
 $(\text{composition-openings ! i final-state}))$

definition *partial-composition-table-candidate*

$:: 'f \text{ list} \Rightarrow 'f \text{ authenticated-opening list list} \Rightarrow \text{bool}$

where

partial-composition-table-candidate composition-table composition-openings
 $\longleftrightarrow$
 $\text{length composition-table} = \text{scale * clength} \wedge$
 $\text{length composition-openings} = \text{rounds} \wedge$
 $(\forall i < \text{rounds.}$
 $\text{table-agrees-with-authenticated-openings composition-table}$
 $(\text{scale * clength) (composition-openings ! i)})$

definition *partial-composition-table-candidates*

$:: 'f \text{ authenticated-opening list list} \Rightarrow 'f \text{ list set}$

where

partial-composition-table-candidates composition-openings =
 $\{\text{composition-table.}$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings}\}$

lemma *partial-trace-table-candidateD:*

assumes *cand:* *partial-trace-table-candidate trace-table trace-openings*
and *i-bound:* $i < \text{rounds}$
and *opn-in:* $\text{opn} \in \text{set (trace-openings ! i)}$
shows $\text{length trace-table} = \text{scale * clength}$
and $\text{opening-index opn} < \text{scale * clength}$
and $\text{trace-table ! opening-index opn} = \text{opening-value opn}$


```

using assms
unfolding partial-trace-table-candidate-def
           table-agrees-with-authenticated-openings-def
by blast+

lemma table-agrees-with-authenticated-openings-map-values:
  assumes agrees:
    table-agrees-with-authenticated-openings table len openings
    and idxs: map opening-index openings = idxs
  shows map (! table) idxs = map opening-value openings
proof (rule nth-equalityI)
  show length (map (! table) idxs) =
    length (map opening-value openings)
proof –
  have length idxs = length openings
    using arg-cong[OF idxs, of length] by simp
  then show ?thesis by simp
qed
next
  fix k
  assume k-bound: k < length (map (! table) idxs)
  have len-idxs: length idxs = length openings
    using arg-cong[OF idxs, of length] by simp
  then have k-openings: k < length openings
    using k-bound len-idxs by simp
  have idx-eq: idxs ! k = opening-index (openings ! k)
    using idxs k-openings by (metis length-map nth-map)
  have opening-in: openings ! k ∈ set openings
    by (rule nth-mem[OF k-openings])
  have table ! opening-index (openings ! k) =
    opening-value (openings ! k)
    using agrees opening-in
    unfolding table-agrees-with-authenticated-openings-def by blast
  then show map (! table) idxs ! k =
    map opening-value openings ! k
    using k-bound k-openings idx-eq by simp
qed

lemma partial-trace-table-candidate-query-values:
  assumes cand: partial-trace-table-candidate trace-table trace-openings
    and i-bound: i < rounds
    and idxs:
      map opening-index (trace-openings ! i) = powers-scaled idx
  shows
    map (! trace-table) (powers-scaled idx) =
      map opening-value (trace-openings ! i)
proof –
  have agrees:
    table-agrees-with-authenticated-openings trace-table

```

(scale * clength) (trace-openings ! i)
using *cand i-bound* **unfolding** *partial-trace-table-candidate-def* **by** *blast*
show *?thesis*
by (rule *table-agrees-with-authenticated-openings-map-values*
 [OF *agrees idxs*])
qed

lemma *partial-trace-table-candidates-iff*:
trace-table ∈ *partial-trace-table-candidates* *trace-openings* $\longleftrightarrow$
partial-trace-table-candidate *trace-table* *trace-openings*
unfolding *partial-trace-table-candidates-def* **by** *simp*

lemma *accepted-with-partial-composition-openings-imp-accepted*:
assumes
accepted-with-partial-composition-openings *s* *out* *composition-root*
query-idxs *composition-openings*
shows *accepted* *out*
using *assms*
unfolding *accepted-with-partial-composition-openings-def* **by** *simp*

lemma *accepted-with-partial-composition-openings-shapes*:
assumes
accepted-with-partial-composition-openings *s* *out* *composition-root*
query-idxs *composition-openings*
shows *length query-idxs* = *rounds*
and *length composition-openings* = *rounds*
and $\bigwedge i. i < \text{rounds} \implies$
map opening-index (*composition-openings* ! *i*) =
 [*query-idxs* ! *i*,
fri-sibling-index (scale * clength) (*query-idxs* ! *i*)]
using *assms*
unfolding *accepted-with-partial-composition-openings-def* **by** *blast+*

definition *accepted-with-partial-initial-openings*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 (unit list $\times$ ('f, 'a) protocol-channel-scheme) option $\Rightarrow$
 'f $\Rightarrow$ 'f list $\Rightarrow$ 'f $\Rightarrow$ 'f list $\Rightarrow$ 'f $\Rightarrow$
 'f list $\Rightarrow$ 'f $\Rightarrow$ nat list $\Rightarrow$
 'f authenticated-opening list list $\Rightarrow$ nat list $\Rightarrow$
 'f authenticated-opening list list $\Rightarrow$ bool

where
accepted-with-partial-initial-openings *s* *out* *fr* *f-fri-roots* *f-final* *as* *dg*
composition-fri-roots *final* *trace-query-idxs* *trace-openings*
composition-query-idxs *composition-openings* $\longleftrightarrow$
accepted *out* $\wedge$
composition-fri-roots $\neq []$ $\wedge$
 ($\exists$ *rest*.
verifier-header-transcript *s* *fr* *f-fri-roots* *f-final* *as* *dg*
composition-fri-roots *final* *rest*) $\wedge$

accepted-with-partial-trace-openings s out fr trace-query-idxs
trace-openings $\wedge$
accepted-with-partial-composition-openings s out
(hd composition-fri-roots) composition-query-idxs composition-openings

lemma *accepted-with-partial-initial-openings-imp-accepted:*

assumes

accepted-with-partial-initial-openings s out fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings

shows *accepted out*

using *assms unfolding accepted-with-partial-initial-openings-def* **by** *simp*

lemma *accepted-with-partial-initial-openings-shapes:*

assumes

accepted-with-partial-initial-openings s out fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings

shows *composition-fri-roots $\neq []$*

and $\exists$ *rest. verifier-header-transcript s fr f-fri-roots f-final as dg*
composition-fri-roots final rest

and *accepted-with-partial-trace-openings s out fr trace-query-idxs*
trace-openings

and *accepted-with-partial-composition-openings s out*

(hd composition-fri-roots) composition-query-idxs composition-openings

using *assms unfolding accepted-with-partial-initial-openings-def* **by** *blast+*

lemma *query-rounds-accepted-with-partial-composition-openings:*

fixes *s query-state final-state :: ('f, 'a) protocol-channel-scheme*

assumes *fl-eq: fl = (b, composition-root) # fl-tail*

and *outcome:*

Some (results, final-state) $\in$

set-dist

(execute

(ntimes

(verifier-query-round-program fr f-fl f-final as fl final)

rounds)

query-state)

obtains *query-idxs composition-openings* **where**

accepted-with-partial-composition-openings s

(Some (results, final-state)) composition-root query-idxs

composition-openings

proof –

from *ntimes-verifier-query-rounds-authenticated-composition-openings*

[OF fl-eq outcome]

obtain *raw-idxs query-idxs composition-openings* **where**

len-raw: length raw-idxs = rounds

and *query-idxs-def:*

query-idxs = map (λ raw. index (to-nat raw)) raw-idxs

```

and len-openings: length composition-openings = rounds
and indices:
   $\bigwedge i. i < \text{rounds} \implies$ 
    map opening-index (composition-openings ! i) =
      [query-idxs ! i,
       fri-sibling-index (scale * clength) (query-idxs ! i)]
and tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table composition-root (scale * clength)
      (composition-openings ! i) final-state
by blast
have len-query-idxs: length query-idxs = rounds
  using len-raw query-idxs-def by simp
have partial:
  accepted-with-partial-composition-openings s
    (Some (results, final-state)) composition-root query-idxs
    composition-openings
unfolding accepted-with-partial-composition-openings-def accepted-def
apply (intro conjI exI)
  apply simp
  apply simp
  apply (rule len-query-idxs)
  apply (rule len-openings)
  using indices tables by simp-all
show ?thesis
  by (rule that[OF partial])
qed

lemma verify-monad-accepted-with-partial-composition-openings-if-bound-tables:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs-bound
obtains composition-root query-idxs composition-openings where
  accepted-with-partial-composition-openings s
    (Some (result, final-state)) composition-root query-idxs
    composition-openings
  merkle-root-binds-table composition-root composition-table final-state
proof –
from verify-monad-header-extraction[OF outcome]
obtain fr f-fl f-final as' dg fl final query-state where
  header:
    verifier-header-transcript s fr (map snd f-fl) f-final as' dg
    (map snd fl) final (PTranscript query-state)
and query-out:
  Some (result, final-state) ∈
    set-dist

```

```

    (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as' fl final)
        rounds)
      query-state)
  by metis
from bound obtain fr' f-fri-roots' f-final' dg'
  composition-fri-roots' final' rest' where
  header':
    verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
    composition-fri-roots' final' rest'
  and comp-nonempty: composition-fri-roots' ≠ []
  and comp-bind:
    merkle-root-binds-table (hd composition-fri-roots')
    composition-table final-state
  unfolding accepted-with-bound-tables-def by blast
have eqs:
  fr' = fr ∧
  f-fri-roots' = map snd f-fl ∧
  f-final' = f-final ∧
  as = as' ∧
  dg' = dg ∧
  composition-fri-roots' = map snd fl ∧
  final' = final ∧
  rest' = PTranscript query-state
  using verifier-header-transcript-unique[OF header' header] by simp
then have fl-nonempty: fl ≠ []
  using comp-nonempty by (cases fl) simp-all
then obtain b composition-root fl-tail where fl-eq:
  fl = (b, composition-root) # fl-tail
  by (cases fl) auto
have comp-root-eq: hd composition-fri-roots' = composition-root
  using eqs fl-eq comp-nonempty by simp
from query-rounds-accepted-with-partial-composition-openings
  [OF fl-eq query-out]
obtain query-idxs composition-openings where partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) composition-root query-idxs
  composition-openings
  by blast
have bind:
  merkle-root-binds-table composition-root composition-table final-state
  using comp-bind unfolding comp-root-eq .
show ?thesis
  by (rule that[OF partial bind])
qed

lemma verify-monad-accepted-with-partial-composition-openings-if-header-nonempty:
  fixes s :: ('f, 'a) protocol-channel-scheme

```

```

assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and header0:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and comp-nonempty: composition-fri-roots ≠ []
obtains query-idxs composition-openings where
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots) query-idxs
  composition-openings
proof –
from verify-monad-header-extraction[OF outcome]
obtain fr' f-fl f-final' as' dg' fl final' query-state where
  header:
    verifier-header-transcript s fr' (map snd f-fl) f-final' as' dg'
    (map snd fl) final' (PTranscript query-state)
and query-out:
  Some (result, final-state) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr' f-fl f-final' as' fl
          final') rounds)
      query-state)
by metis
have eqs:
  fr' = fr ∧
  map snd f-fl = f-fri-roots ∧
  f-final' = f-final ∧
  as' = as ∧
  dg' = dg ∧
  map snd fl = composition-fri-roots ∧
  final' = final ∧
  PTranscript query-state = rest
using verifier-header-transcript-unique[OF header header0] by simp
have fl-nonempty: fl ≠ []
using comp-nonempty eqs by (cases fl) simp-all
then obtain b composition-root fl-tail where fl-eq:
  fl = (b, composition-root) # fl-tail
by (cases fl) auto
have root-eq: composition-root = hd composition-fri-roots
proof –
  have map-eq: map snd fl = composition-fri-roots
    using eqs by simp
  then have composition-root # map snd fl-tail = composition-fri-roots
    unfolding fl-eq by simp
  then have hd composition-fri-roots = composition-root
    by (metis list.sel(1))
  then show ?thesis

```

```

    by simp
qed
from query-rounds-accepted-with-partial-composition-openings
  [OF fl-eq query-out]
obtain query-idxs composition-openings where partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) composition-root query-idxs
  composition-openings
by blast
show ?thesis
by (rule that[of query-idxs composition-openings])
  (use partial root-eq in simp)
qed

lemma partial-composition-table-candidateD:
  assumes
    cand:
      partial-composition-table-candidate composition-table
      composition-openings
    and i-bound:  $i < \text{rounds}$ 
    and opn-in:  $\text{opn} \in \text{set } (\text{composition-openings } ! i)$ 
  shows  $\text{length } \text{composition-table} = \text{scale} * \text{clength}$ 
    and  $\text{opening-index } \text{opn} < \text{scale} * \text{clength}$ 
    and  $\text{composition-table } ! \text{opening-index } \text{opn} = \text{opening-value } \text{opn}$ 
  using assms
  unfolding partial-composition-table-candidate-def
    table-agrees-with-authenticated-openings-def
  by blast+

lemma partial-composition-table-candidate-query-values:
  assumes cand:
    partial-composition-table-candidate composition-table
    composition-openings
  and i-bound:  $i < \text{rounds}$ 
  and idxs:
     $\text{map } \text{opening-index } (\text{composition-openings } ! i) =$ 
     $[\text{idx}, \text{fri-sibling-index } (\text{scale} * \text{clength}) \text{idx}]$ 
  shows
     $\text{map } (!) \text{ composition-table}$ 
     $[\text{idx}, \text{fri-sibling-index } (\text{scale} * \text{clength}) \text{idx}] =$ 
     $\text{map } \text{opening-value } (\text{composition-openings } ! i)$ 
proof -
  have agrees:
    table-agrees-with-authenticated-openings composition-table
    (scale * clength) (composition-openings ! i)
  using cand i-bound
  unfolding partial-composition-table-candidate-def by blast
  show ?thesis
  by (rule table-agrees-with-authenticated-openings-map-values)

```

[OF agrees idxs])

qed

lemma *partial-composition-table-candidate-query-value*:
assumes *cand*:
 partial-composition-table-candidate composition-table
 composition-openings
and *i-bound*: $i < \text{rounds}$
and *idxs*:
 $\text{map opening-index } (\text{composition-openings } ! i) =$
 $[\text{idx}, \text{fri-sibling-index } (\text{scale} * \text{clength}) \text{ idx}]$
shows
 $\text{composition-table } ! \text{idx} =$
 $\text{opening-value } ((\text{composition-openings } ! i) ! 0)$
proof –
 have *vals*:
 $\text{map } (!) \text{ composition-table}$
 $[\text{idx}, \text{fri-sibling-index } (\text{scale} * \text{clength}) \text{ idx}] =$
 $\text{map opening-value } (\text{composition-openings } ! i)$
 by (rule *partial-composition-table-candidate-query-values*
 [OF *cand i-bound idxs*])
 have *len-openings*: $\text{length } (\text{composition-openings } ! i) = 2$
 proof –
 have $\text{length } (\text{map opening-index } (\text{composition-openings } ! i)) =$
 $\text{length } [\text{idx}, \text{fri-sibling-index } (\text{scale} * \text{clength}) \text{ idx}]$
 using *arg-cong*[OF *idxs*, of *length*] **by** *simp*
 then show ?thesis **by** *simp*
 qed
 have *value0*:
 $\text{map } (!) \text{ composition-table}$
 $[\text{idx}, \text{fri-sibling-index } (\text{scale} * \text{clength}) \text{ idx}] ! 0 =$
 $\text{map opening-value } (\text{composition-openings } ! i) ! 0$
 using *vals* **by** *simp*
 show ?thesis
 using *value0 len-openings* **by** (cases *composition-openings* ! *i*) *auto*
qed

lemma *partial-composition-table-candidates-iff*:
 $\text{composition-table} \in$
 $\text{partial-composition-table-candidates } \text{composition-openings} \longleftrightarrow$
 $\text{partial-composition-table-candidate } \text{composition-table } \text{composition-openings}$
unfolding *partial-composition-table-candidates-def* **by** *simp*

lemma *verify-monad-accepted-with-partial-trace-openings*:
fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$
assumes *outcome*:
 $\text{Some } (\text{result}, \text{final-state}) \in \text{set-dist } (\text{execute verify-monad } s)$
obtains *fr query-idxs trace-openings* **where**
 accepted-with-partial-trace-openings s


```

    (Some (result, final-state)) fr query-idxs trace-openings
proof –
  show ?thesis
proof (rule verify-monad-authenticated-trace-openings[OF outcome])
  fix fr query-idxs trace-openings
  assume len-query-idxs: length query-idxs = rounds
  and len-openings: length trace-openings = rounds
  and indices:
     $\bigwedge i. i < \text{rounds} \implies$ 
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
  and tables:
     $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
  have partial:
    accepted-with-partial-trace-openings s
      (Some (result, final-state)) fr query-idxs trace-openings
  unfolding accepted-with-partial-trace-openings-def accepted-def
  apply (intro conjI exI)
  apply simp
  apply simp
  apply (rule len-query-idxs)
  apply (rule len-openings)
  using indices tables by simp-all
  show ?thesis
  by (rule that[OF partial])
qed
qed

lemma verify-monad-accepted-with-partial-trace-openings-for-header:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
and header0:
  verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
obtains query-idxs trace-openings where
  accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs trace-openings
proof –
from verify-monad-header-extraction[OF outcome]
obtain fr' f-fl f-final' as' dg' fl final' query-state where
  header:
    verifier-header-transcript s fr' (map snd f-fl) f-final' as' dg'
      (map snd fl) final' (PTranscript query-state)
and query-out:
  Some (result, final-state)  $\in$ 
    set-dist

```

```

      (execute
        (ntimes
          (verifier-query-round-program fr' f-fl f-final' as' fl
            final') rounds)
          query-state)
    by metis
  have fr-eq: fr' = fr
    using verifier-header-transcript-unique[OF header header0] by simp
  obtain query-idxs trace-openings where
    len-query-idxs: length query-idxs = rounds
    and len-openings: length trace-openings = rounds
    and indices:
       $\bigwedge i. i < \text{rounds} \implies$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
    and tables:
       $\bigwedge i. i < \text{rounds} \implies$ 
      partial-authenticated-table fr' (scale * clength)
        (trace-openings ! i) final-state
  proof (rule ntimes-verifier-query-rounds-authenticated-trace-openings-no-raws
    [OF query-out])
    fix query-idxs trace-openings
    assume len-query-idxs: length query-idxs = rounds
    and len-openings: length trace-openings = rounds
    and indices:
       $\bigwedge i. i < \text{rounds} \implies$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
    and tables:
       $\bigwedge i. i < \text{rounds} \implies$ 
      partial-authenticated-table fr' (scale * clength)
        (trace-openings ! i) final-state
    show ?thesis
      by (rule that[OF len-query-idxs len-openings indices tables])
  qed
  have partial:
    accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs trace-openings
  proof -
    have all-indices:
       $\forall i < \text{rounds}.$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
    using indices by blast
    have all-tables:
       $\forall i < \text{rounds}.$ 
      partial-authenticated-table fr (scale * clength)
        (trace-openings ! i) final-state
    using tables fr-eq by simp
  
```

```

show ?thesis
  unfolding accepted-with-partial-trace-openings-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state])
    (use len-query-idxs len-openings all-indices all-tables in simp-all)
qed
show ?thesis
  by (rule that[OF partial])
qed

lemma verify-monad-accepted-with-partial-initial-openings-if-header-nonempty:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and comp-nonempty: composition-fri-roots ≠ []
obtains trace-query-idxs trace-openings composition-query-idxs
  composition-openings where
  accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr trace-query-idxs trace-openings
  accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots)
  composition-query-idxs composition-openings
proof —
from verify-monad-accepted-with-partial-trace-openings-for-header
  [OF outcome header]
obtain trace-query-idxs trace-openings where trace-partial:
  accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr trace-query-idxs trace-openings
by blast
from verify-monad-accepted-with-partial-composition-openings-if-header-nonempty
  [OF outcome header comp-nonempty]
obtain composition-query-idxs composition-openings where composition-partial:
  accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots)
  composition-query-idxs composition-openings
by blast
show ?thesis
  by (rule that[OF trace-partial composition-partial])
qed

lemma verify-monad-accepted-with-partial-initial-openings:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest

```

```

and comp-nonempty: composition-fri-roots ≠ []
obtains trace-query-idxs trace-openings composition-query-idxs
  composition-openings where
    accepted-with-partial-initial-openings s
      (Some (result, final-state)) fr f-fri-roots f-final as dg
      composition-fri-roots final trace-query-idxs trace-openings
      composition-query-idxs composition-openings
proof –
from verify-monad-accepted-with-partial-initial-openings-if-header-nonempty
  [OF outcome header comp-nonempty]
obtain trace-query-idxs trace-openings composition-query-idxs
  composition-openings where
    trace-partial:
      accepted-with-partial-trace-openings s
        (Some (result, final-state)) fr trace-query-idxs trace-openings
    and composition-partial:
      accepted-with-partial-composition-openings s
        (Some (result, final-state)) (hd composition-fri-roots)
      composition-query-idxs composition-openings
    by blast
have accepted-out: accepted (Some (result, final-state))
    by (rule accepted-with-partial-trace-openings-imp-accepted
      [OF trace-partial])
have evidence:
    accepted-with-partial-initial-openings s
      (Some (result, final-state)) fr f-fri-roots f-final as dg
      composition-fri-roots final trace-query-idxs trace-openings
      composition-query-idxs composition-openings
    unfolding accepted-with-partial-initial-openings-def
    by (intro conjI exI[of - rest])
      (use accepted-out comp-nonempty header trace-partial
        composition-partial in simp-all)
    show ?thesis
    by (rule that[OF evidence])
qed

lemma merkle-path-bound-unique-value-if-clean:
  assumes clean: ¬ hash-map-output-collision s
    and first: merkle-path-bound rt len idx v path s
    and second: merkle-path-bound rt len idx v' path' s
    and same-length: length path = length path'
  shows v = v'
  using first second same-length
proof (induction path arbitrary: path' rt len idx)
case Nil
then have path': path' = [] by simp
have first-lookup:
  fmlookup (HashMap s) (MerkleLeaf v) = Some rt
  using Nil.premis(1) by simp

```

```

have second-lookup:
  fmlookup (HashMap s) (MerkleLeaf v') = Some rt
  using Nil.premis(2) path' by simp
show ?case
proof (rule ccontr)
  assume v ≠ v'
  then have keys: MerkleLeaf v ≠ MerkleLeaf v' by simp
  have hash-map-output-collision s
    by (rule hash-map-output-collisionI
      [OF keys first-lookup second-lookup])
  then show False using clean by contradiction
qed
next
case (Cons sibling path)
from Cons.premis(3) obtain sibling' path'' where
  path': path' = sibling' # path''
  and lengths: length path = length path''
  by (cases path') auto
show ?case
proof (cases idx < len div 2)
  case True
  obtain child where
    child:
      merkle-path-bound child (len div 2) idx v path s
  and first-lookup:
    fmlookup (HashMap s) (MerkleNode child sibling) = Some rt
  using Cons.premis(1) True by auto
  obtain child' where
    child':
      merkle-path-bound child' (len div 2) idx v' path'' s
  and second-lookup:
    fmlookup (HashMap s) (MerkleNode child' sibling') = Some rt
  using Cons.premis(2) True path' by auto
  have node-eq:
    MerkleNode child sibling = MerkleNode child' sibling'
  proof (rule ccontr)
    assume neq:
      MerkleNode child sibling ≠ MerkleNode child' sibling'
    have hash-map-output-collision s
      by (rule hash-map-output-collisionI
        [OF neq first-lookup second-lookup])
    then show False using clean by contradiction
  qed
  have child-eq: child = child' using node-eq by simp
  show ?thesis
    by (rule Cons.IH[OF child - lengths])
      (use child' child-eq in simp)
next
case False

```

```

obtain child where
  child:
    merkle-path-bound child (len div 2) (idx - len div 2)
    v path s
  and first-lookup:
    fmlookup (HashMap s) (MerkleNode sibling child) = Some rt
  using Cons.premis(1) False by auto
obtain child' where
  child':
    merkle-path-bound child' (len div 2) (idx - len div 2)
    v' path'' s
  and second-lookup:
    fmlookup (HashMap s) (MerkleNode sibling' child') = Some rt
  using Cons.premis(2) False path' by auto
have node-eq:
  MerkleNode sibling child = MerkleNode sibling' child'
proof (rule ccontr)
  assume neg:
    MerkleNode sibling child ≠ MerkleNode sibling' child'
  have hash-map-output-collision s
    by (rule hash-map-output-collisionI
      [OF neg first-lookup second-lookup])
  then show False using clean by contradiction
qed
have child-eq: child = child' using node-eq by simp
show ?thesis
  by (rule Cons.IH[OF child - lengths])
    (use child' child-eq in simp)
qed
qed

lemma inconsistent-authenticated-openings-imp-hash-collision:
  assumes first: authenticated-opening-in s opening
  and second: authenticated-opening-in s opening'
  and same-root: opening-root opening = opening-root opening'
  and same-length: opening-length opening = opening-length opening'
  and same-index: opening-index opening = opening-index opening'
  and different-value: opening-value opening ≠ opening-value opening'
  shows hash-map-output-collision s
proof (rule ccontr)
  assume clean: ¬ hash-map-output-collision s
  have path-lengths:
    length (opening-path opening) = length (opening-path opening')
  using first second same-length
  unfolding authenticated-opening-in-def by simp
have first-bound:
  merkle-path-bound
    (opening-root opening)
    (opening-length opening)

```

```

    (opening-index opening)
    (opening-value opening)
    (opening-path opening)
  s
  using first unfolding authenticated-opening-in-def by blast
have second-bound:
  merkle-path-bound
    (opening-root opening')
    (opening-length opening')
    (opening-index opening')
    (opening-value opening')
    (opening-path opening')
  s
  using second unfolding authenticated-opening-in-def by blast
have values-eq:
  opening-value opening = opening-value opening'
proof (rule merkle-path-bound-unique-value-if-clean[
  OF clean first-bound - path-lengths])
show
  merkle-path-bound
    (opening-root opening)
    (opening-length opening)
    (opening-index opening)
    (opening-value opening')
    (opening-path opening')
  s
  using second-bound same-root same-length same-index by simp
qed
show False using different-value values-eq by contradiction
qed

lemma partial-authenticated-table-values-unique-if-clean:
  assumes table: partial-authenticated-table rt len openings s
  and clean:  $\neg$  hash-map-output-collision s
  and first: opening  $\in$  set openings
  and second: opening'  $\in$  set openings
  and same-index: opening-index opening = opening-index opening'
  shows opening-value opening = opening-value opening'
proof (rule ccontr)
  assume different: opening-value opening  $\neq$  opening-value opening'
  have first-auth: authenticated-opening-in s opening
    using table first unfolding partial-authenticated-table-def by blast
  have second-auth: authenticated-opening-in s opening'
    using table second unfolding partial-authenticated-table-def by blast
  have roots: opening-root opening = opening-root opening'
  proof -
    have opening-root opening = rt
      using table first unfolding partial-authenticated-table-def by blast
    moreover have opening-root opening' = rt

```

```

    using table second unfolding partial-authenticated-table-def by blast
  ultimately show ?thesis by simp
qed
have lengths: opening-length opening = opening-length opening'
proof -
  have opening-length opening = len
    using table first unfolding partial-authenticated-table-def by blast
  moreover have opening-length opening' = len
    using table second unfolding partial-authenticated-table-def by blast
  ultimately show ?thesis by simp
qed
have hash-map-output-collision s
  by (rule inconsistent-authenticated-openings-imp-hash-collision[
    OF first-auth second-auth roots lengths same-index different])
then show False using clean by contradiction
qed

lemma accepted-with-partial-trace-openings-values-unique-if-clean:
  assumes partial:
    accepted-with-partial-trace-openings s (Some (result, final-state))
    fr query-idxs trace-openings
  and clean:  $\neg$  hash-map-output-collision final-state
  and i-bound:  $i < \text{rounds}$ 
  and opn-in:  $\text{opn} \in \text{set } (\text{trace-openings } ! i)$ 
  and opn'-in:  $\text{opn}' \in \text{set } (\text{trace-openings } ! i)$ 
  and same-index:  $\text{opening-index } \text{opn} = \text{opening-index } \text{opn}'$ 
  shows opening-value opn = opening-value opn'
proof -
  have table:
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) final-state
  using partial i-bound unfolding accepted-with-partial-trace-openings-def
  by blast
  show ?thesis
  by (rule partial-authenticated-table-values-unique-if-clean
    [OF table clean opn-in opn'-in same-index])
qed

definition partial-trace-openings-inconsistent-bad
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  partial-trace-openings-inconsistent-bad s out  $\longleftrightarrow$ 
    ( $\exists$  fr query-idxs trace-openings result final-state i opn opn'.
      out = Some (result, final-state)  $\wedge$ 
      accepted-with-partial-trace-openings s out fr query-idxs trace-openings  $\wedge$ 
       $i < \text{rounds}$   $\wedge$ 
       $\text{opn} \in \text{set } (\text{trace-openings } ! i)$   $\wedge$ 
       $\text{opn}' \in \text{set } (\text{trace-openings } ! i)$   $\wedge$ 

```


opening-index $opn = \text{opening-index } opn' \wedge$
opening-value $opn \neq \text{opening-value } opn'$)

lemma *accepted-with-partial-trace-openings-values-unique-if-no-bad:*

assumes *partial:*

accepted-with-partial-trace-openings s (*Some* (*result*, *final-state*))
fr *query-idxs* *trace-openings*

and *no-bad:*

$\neg$ *partial-trace-openings-inconsistent-bad* s
(*Some* (*result*, *final-state*))

and *i-bound:* $i < \text{rounds}$

and *opn-in:* $opn \in \text{set } (\text{trace-openings } ! i)$

and *opn'-in:* $opn' \in \text{set } (\text{trace-openings } ! i)$

and *same-index:* $\text{opening-index } opn = \text{opening-index } opn'$

shows $\text{opening-value } opn = \text{opening-value } opn'$

proof (*rule ccontr*)

assume *neg:* $\text{opening-value } opn \neq \text{opening-value } opn'$

have *partial-trace-openings-inconsistent-bad* s
(*Some* (*result*, *final-state*))

unfolding *partial-trace-openings-inconsistent-bad-def*

by (*intro* $exI[of - fr]$ $exI[of - query-idxs]$
 $exI[of - trace-openings]$ $exI[of - result]$
 $exI[of - final-state]$ $exI[of - i]$ $exI[of - opn]$
 $exI[of - opn']$)

(*use* *partial i-bound opn-in opn'-in same-index neg in simp*)

then show *False* **using** *no-bad* **by** *contradiction*

qed

lemma *merkle-bound-table-agrees-with-authenticated-opening-if-clean:*

assumes *bind:* *merkle-root-binds-table* rt *table* *final-state*

and *len-pow:* $\text{length } \text{table} = 2^n$

and *auth:* *authenticated-opening-in* *final-state* *opening*

and *root:* $\text{opening-root } \text{opening} = rt$

and *len:* $\text{opening-length } \text{opening} = \text{length } \text{table}$

and *clean:* $\neg \text{hash-map-output-collision } \text{final-state}$

shows

$\text{opening-index } \text{opening} < \text{length } \text{table} \wedge$
 $\text{table } ! \text{opening-index } \text{opening} = \text{opening-value } \text{opening}$

proof –

from *bind* **obtain** *tree* **where** *created:*

created-tree *table* *tree* *final-state*

and *rt-eq:* $rt = \text{value } \text{tree}$

unfolding *merkle-root-binds-table-def* **by** *blast*

have *idx-bound:* $\text{opening-index } \text{opening} < \text{length } \text{table}$

using *auth len* **unfolding** *authenticated-opening-in-def* **by** *simp*

have *table-root:*

merkle-path-root-in-map *final-state* ($\text{length } \text{table}$)
($\text{opening-index } \text{opening}$) ($\text{table } ! \text{opening-index } \text{opening}$)
(*get-authentication-path* ($\text{length } \text{table}$) ($\text{opening-index } \text{opening}$) *tree*) =

```

    Some rt
using protocol-created-tree-path-root-in-map[
    OF created[unfolded created-tree-def] len-pow idx-bound]
unfolding rt-eq .
have opening-root-in-map:
    merkle-path-root-in-map final-state (length table)
    (opening-index opening) (opening-value opening)
    (opening-path opening) = Some rt
using authenticated-opening-root-in-map[OF auth] root len by simp
have same:
    (table ! opening-index opening = opening-value opening  $\wedge$ 
    get-authentication-path (length table) (opening-index opening) tree =
    opening-path opening)  $\vee$ 
    hash-map-output-collision final-state
by (rule merkle-path-root-same-index-eq-or-hash-collision[
    OF table-root opening-root-in-map])
then have value-eq:
    table ! opening-index opening = opening-value opening
using clean by blast
show ?thesis
    using idx-bound value-eq by simp
qed

lemma accepted-bound-trace-table-partial-candidate-if-clean:
assumes bind: merkle-root-binds-table fr trace-table final-state
and len-table: length trace-table = scale * clength
and partial:
    accepted-with-partial-trace-openings s (Some (result, final-state))
    fr query-idxs trace-openings
and clean:  $\neg$  hash-map-output-collision final-state
shows partial-trace-table-candidate trace-table trace-openings
proof -
obtain n where len-pow: clength * scale =  $2^n$ 
using eval-domain-length-power by blast
have table-pow: length trace-table =  $2^n$ 
using len-table len-pow by (simp add: mult.commute)
have len-openings: length trace-openings = rounds
using partial unfolding accepted-with-partial-trace-openings-def by blast
have agrees:
     $\bigwedge i. i < \text{rounds} \implies$ 
    table-agrees-with-authenticated-openings trace-table
    (scale * clength) (trace-openings ! i)
proof -
fix i
assume i-bound:  $i < \text{rounds}$ 
have table-i:
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) final-state
using partial i-bound

```

```

unfolding accepted-with-partial-trace-openings-def by blast
show
  table-agrees-with-authenticated-openings trace-table
    (scale * clength) (trace-openings ! i)
proof –
  have opening-agrees:
     $\forall \text{opn} \in \text{set } (\text{trace-openings } ! i).$ 
      opening-index opn < scale * clength  $\wedge$ 
      trace-table ! opening-index opn = opening-value opn
  proof
  fix opn
  assume opn-in: opn  $\in$  set (trace-openings ! i)
  have root: opening-root opn = fr
    using table-i opn-in unfolding partial-authenticated-table-def
    by blast
  have len: opening-length opn = length trace-table
    using table-i opn-in len-table
    unfolding partial-authenticated-table-def by simp
  have auth: authenticated-opening-in final-state opn
    using table-i opn-in unfolding partial-authenticated-table-def
    by blast
  have agreement:
    opening-index opn < length trace-table  $\wedge$ 
    trace-table ! opening-index opn = opening-value opn
    by (rule merkle-bound-table-agrees-with-authenticated-opening-if-clean
      [OF bind table-pow auth root len clean])
  then show
    opening-index opn < scale * clength  $\wedge$ 
    trace-table ! opening-index opn = opening-value opn
    using len-table by simp
  qed
  show ?thesis
    unfolding table-agrees-with-authenticated-openings-def
    using len-table opening-agrees by simp
  qed
  show ?thesis
    unfolding partial-trace-table-candidate-def
    using len-table len-openings agrees by simp
qed

lemma accepted-bound-composition-table-partial-candidate-if-clean:
assumes bind:
  merkle-root-binds-table composition-root composition-table
  final-state
and len-table: length composition-table = scale * clength
and partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) composition-root query-idxs

```

```

    composition-openings
  and clean:  $\neg$  hash-map-output-collision final-state
shows
  partial-composition-table-candidate composition-table
  composition-openings
proof -
  obtain n where len-pow: clength * scale =  $2^n$ 
  using eval-domain-length-power by blast
  have table-pow: length composition-table =  $2^n$ 
  using len-table len-pow by (simp add: mult.commute)
  have len-openings: length composition-openings = rounds
  using partial
  unfolding accepted-with-partial-composition-openings-def by blast
  have agrees:
     $\bigwedge i. i < rounds \implies$ 
    table-agrees-with-authenticated-openings composition-table
    (scale * clength) (composition-openings ! i)
  proof -
    fix i
    assume i-bound: i < rounds
    have table-i:
      partial-authenticated-table composition-root (scale * clength)
      (composition-openings ! i) final-state
    using partial i-bound
    unfolding accepted-with-partial-composition-openings-def by blast
  show
    table-agrees-with-authenticated-openings composition-table
    (scale * clength) (composition-openings ! i)
  proof -
    have opening-agrees:
       $\forall \text{opn} \in \text{set } (composition-openings ! i).$ 
      opening-index opn < scale * clength  $\wedge$ 
      composition-table ! opening-index opn = opening-value opn
    proof
      fix opn
      assume opn-in: opn  $\in$  set (composition-openings ! i)
      have root: opening-root opn = composition-root
      using table-i opn-in unfolding partial-authenticated-table-def
      by blast
      have len: opening-length opn = length composition-table
      using table-i opn-in len-table
      unfolding partial-authenticated-table-def by simp
      have auth: authenticated-opening-in final-state opn
      using table-i opn-in unfolding partial-authenticated-table-def
      by blast
      have agreement:
        opening-index opn < length composition-table  $\wedge$ 
        composition-table ! opening-index opn = opening-value opn
      by (rule merkle-bound-table-agrees-with-authenticated-opening-if-clean)
    qed
  qed

```

```

    [OF bind table-pow auth root len clean])
  then show
    opening-index opn < scale * clength ∧
    composition-table ! opening-index opn = opening-value opn
    using len-table by simp
  qed
  show ?thesis
    unfolding table-agrees-with-authenticated-openings-def
    using len-table opening-agrees by simp
  qed
  qed
  show ?thesis
    unfolding partial-composition-table-candidate-def
    using len-table len-openings agrees by simp
  qed

definition partial-merkle-inconsistency-bad
  :: ('f, 'a) protocol-channel-scheme ⇒
    (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool
where
  partial-merkle-inconsistency-bad s out ⟷
    accepted out ∧
    (∃ result final-state opn opn'.
      out = Some (result, final-state) ∧
      authenticated-opening-in final-state opn ∧
      authenticated-opening-in final-state opn' ∧
      opening-root opn = opening-root opn' ∧
      opening-length opn = opening-length opn' ∧
      opening-index opn = opening-index opn' ∧
      opening-value opn ≠ opening-value opn')

lemma partial-trace-openings-inconsistent-bad-imp-partial-merkle-inconsistency-bad:
  assumes bad: partial-trace-openings-inconsistent-bad s out
  shows partial-merkle-inconsistency-bad s out
proof –
  obtain fr query-idxs trace-openings result final-state i opn opn' where
    out: out = Some (result, final-state)
  and partial:
    accepted-with-partial-trace-openings s out fr query-idxs trace-openings
  and i-bound: i < rounds
  and opn-in: opn ∈ set (trace-openings ! i)
  and opn'-in: opn' ∈ set (trace-openings ! i)
  and same-index: opening-index opn = opening-index opn'
  and diff-value: opening-value opn ≠ opening-value opn'
  using bad unfolding partial-trace-openings-inconsistent-bad-def by blast
  have table:
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) final-state
  proof –

```

```

obtain result' final-state' where
  partial-out: out = Some (result', final-state')
and tables:
   $\forall i < \text{rounds}.$ 
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) final-state'
using partial unfolding accepted-with-partial-trace-openings-def
by auto
have final-state-eq: final-state' = final-state
using out partial-out by simp
show ?thesis
using tables i-bound unfolding final-state-eq by simp
qed
have opn-auth: authenticated-opening-in final-state opn
using table opn-in unfolding partial-authenticated-table-def by blast
have opn'-auth: authenticated-opening-in final-state opn'
using table opn'-in unfolding partial-authenticated-table-def by blast
have same-root: opening-root opn = opening-root opn'
proof –
  have root-opn: opening-root opn = fr
using table opn-in unfolding partial-authenticated-table-def by simp
have root-opn': opening-root opn' = fr
using table opn'-in unfolding partial-authenticated-table-def by simp
show ?thesis
using root-opn root-opn' by simp
qed
have same-length: opening-length opn = opening-length opn'
proof –
  have length-opn: opening-length opn = scale * clength
using table opn-in unfolding partial-authenticated-table-def by simp
have length-opn': opening-length opn' = scale * clength
using table opn'-in unfolding partial-authenticated-table-def by simp
show ?thesis
using length-opn length-opn' by simp
qed
show ?thesis
unfolding partial-merkle-inconsistency-bad-def
using partial out opn-auth opn'-auth same-root same-length same-index
diff-value
unfolding accepted-with-partial-trace-openings-def
by blast
qed

lemma partial-merkle-inconsistency-bad-imp-hash-map-output-collision-bad:
assumes partial-merkle-inconsistency-bad s out
shows hash-map-output-collision-bad s out
proof –
obtain result final-state opn opn' where
  out: out = Some (result, final-state)

```

and *first*: *authenticated-opening-in final-state opn*
and *second*: *authenticated-opening-in final-state opn'*
and *same-root*: *opening-root opn = opening-root opn'*
and *same-length*: *opening-length opn = opening-length opn'*
and *same-index*: *opening-index opn = opening-index opn'*
and *different-value*:
 opening-value opn ≠ opening-value opn'
using *assms* **unfolding** *partial-merkle-inconsistency-bad-def* **by** *blast*
have *collision*: *hash-map-output-collision final-state*
by (*rule inconsistent-authenticated-openings-imp-hash-collision*[
 OF first second same-root same-length same-index different-value])
show *?thesis*
unfolding *hash-map-output-collision-bad-def* *accepted-def*
using *out collision* **by** *simp*
qed

lemma *partial-trace-openings-inconsistent-bad-imp-hash-map-output-collision-bad*:
assumes *partial-trace-openings-inconsistent-bad s out*
shows *hash-map-output-collision-bad s out*
by (*rule partial-merkle-inconsistency-bad-imp-hash-map-output-collision-bad*)
 (*rule partial-trace-openings-inconsistent-bad-imp-partial-merkle-inconsistency-bad*
 [*OF assms*])

lemma *partial-trace-openings-inconsistent-bad-mono-hash-map-output-collision-bad*:
wp-event verify-monad (partial-trace-openings-inconsistent-bad s) s ≤
wp-event verify-monad (hash-map-output-collision-bad s) s
by (*rule wp-event-mono*)
 (*rule partial-trace-openings-inconsistent-bad-imp-hash-map-output-collision-bad*)

lemma *partial-merkle-inconsistency-bad-mono-hash-map-output-collision-bad*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤
wp-event verify-monad (hash-map-output-collision-bad s) s
by (*rule wp-event-mono*)
 (*rule partial-merkle-inconsistency-bad-imp-hash-map-output-collision-bad*)

lemma *partial-trace-openings-inconsistent-bad-bound-from-hash-map-output-collision*:
fixes *H :: prob*
assumes *wp-event verify-monad (hash-map-output-collision-bad s) s ≤ H*
shows *wp-event verify-monad (partial-trace-openings-inconsistent-bad s) s ≤*
 H
by (*rule order-trans*
 [*OF partial-trace-openings-inconsistent-bad-mono-hash-map-output-collision-bad*
 assms])

lemma *partial-merkle-inconsistency-bad-bound-from-hash-map-output-collision*:
fixes *H :: prob*
assumes *wp-event verify-monad (hash-map-output-collision-bad s) s ≤ H*
shows *wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ H*
by (*rule order-trans*)

[*OF partial-merkle-inconsistency-bad-mono-hash-map-output-collision-bad*
assms])

lemma *partial-merkle-inconsistency-bad-imp-accepted:*

assumes *partial-merkle-inconsistency-bad s out*

shows *accepted out*

using *assms unfolding partial-merkle-inconsistency-bad-def by simp*

lemma *accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad:*

assumes *partial:*

accepted-with-partial-trace-openings s (Some (result, final-state))

fr query-idxs trace-openings

and *no-bad:*

$\neg$ *partial-merkle-inconsistency-bad s*

(Some (result, final-state))

shows $\exists$ *trace-table.*

partial-trace-table-candidate trace-table trace-openings

proof –

have *len-openings: length trace-openings = rounds*

using *partial unfolding accepted-with-partial-trace-openings-def by blast*

have *table:*

$\bigwedge i. i < \text{rounds} \implies$

*partial-authenticated-table fr (scale * clength)*

(trace-openings ! i) final-state

using *partial unfolding accepted-with-partial-trace-openings-def by blast*

have *accepted-out: accepted (Some (result, final-state))*

by *(rule accepted-with-partial-trace-openings-imp-accepted[OF partial])*

have *idx-bound:*

$\bigwedge \text{opn}. \text{opn} \in \text{set } (\text{List.concat trace-openings}) \implies$

*opening-index opn < scale * clength*

proof –

fix *opn*

assume *opn-in: opn ∈ set (List.concat trace-openings)*

then obtain *openings where openings-in: openings ∈ set trace-openings*

and *opn-openings: opn ∈ set openings*

by *auto*

then obtain *i where i-bound: i < length trace-openings*

and *openings-eq: trace-openings ! i = openings*

by *(meson in-set-conv-nth)*

have *table-i:*

*partial-authenticated-table fr (scale * clength)*

(trace-openings ! i) final-state

using *table i-bound len-openings by simp*

have *auth: authenticated-opening-in final-state opn*

using *table-i opn-openings unfolding openings-eq*

partial-authenticated-table-def by blast

have *len: opening-length opn = scale * clength*

using *table-i opn-openings unfolding openings-eq*

partial-authenticated-table-def by blast


```

show opening-index opn < scale * clength
  using auth len unfolding authenticated-opening-in-def by simp
qed
have consistent:
   $\bigwedge \text{opn opn}'. \text{opn} \in \text{set (List.concat trace-openings)} \implies$ 
     $\text{opn}' \in \text{set (List.concat trace-openings)} \implies$ 
    opening-index opn = opening-index opn'  $\implies$ 
    opening-value opn = opening-value opn'
proof (rule ccontr)
  fix opn opn'
  assume opn-in: opn  $\in$  set (List.concat trace-openings)
    and opn'-in: opn'  $\in$  set (List.concat trace-openings)
    and same-index: opening-index opn = opening-index opn'
    and neq: opening-value opn  $\neq$  opening-value opn'
  obtain openings where openings-in: openings  $\in$  set trace-openings
    and opn-openings: opn  $\in$  set openings
    using opn-in by auto
  then obtain i where i-bound: i < length trace-openings
    and openings-eq: trace-openings ! i = openings
    by (meson in-set-conv-nth)
  obtain openings' where openings'-in: openings'  $\in$  set trace-openings
    and opn'-openings: opn'  $\in$  set openings'
    using opn'-in by auto
  then obtain j where j-bound: j < length trace-openings
    and openings'-eq: trace-openings ! j = openings'
    by (meson in-set-conv-nth)
  have table-i:
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
    using table i-bound len-openings by simp
  have table-j:
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! j) final-state
    using table j-bound len-openings by simp
  have auth: authenticated-opening-in final-state opn
    using table-i opn-openings unfolding openings-eq
      partial-authenticated-table-def by blast
  have auth': authenticated-opening-in final-state opn'
    using table-j opn'-openings unfolding openings'-eq
      partial-authenticated-table-def by blast
  have root: opening-root opn = opening-root opn'
    using table-i table-j opn-openings opn'-openings
    unfolding openings-eq openings'-eq partial-authenticated-table-def
    by auto
  have len: opening-length opn = opening-length opn'
    using table-i table-j opn-openings opn'-openings
    unfolding openings-eq openings'-eq partial-authenticated-table-def
    by auto
  have bad:

```

```

    partial-merkle-inconsistency-bad s (Some (result, final-state))
  unfolding partial-merkle-inconsistency-bad-def
  by (intro conjI exI[of - result] exI[of - final-state]
      exI[of - opn] exI[of - opn'])
      (use accepted-out auth auth' root len same-index neg in simp-all)
  then show False using no-bad by contradiction
qed
obtain trace-table where agrees-all:
  table-agrees-with-authenticated-openings trace-table (scale * clength)
  (List.concat trace-openings)
using table-agrees-with-authenticated-openings-consistent-exists
  [OF idx-bound consistent]
by blast
have agrees:
   $\bigwedge i. i < \text{rounds} \implies$ 
  table-agrees-with-authenticated-openings trace-table
  (scale * clength) (trace-openings ! i)
proof -
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  have set (trace-openings ! i)  $\subseteq$ 
    set (List.concat trace-openings)
  proof
    fix opn
    assume opn-in:  $\text{opn} \in \text{set (trace-openings ! i)}$ 
    have trace-openings ! i  $\in \text{set trace-openings}$ 
      using i-bound len-openings by (intro nth-mem) simp
    then show  $\text{opn} \in \text{set (List.concat trace-openings)}$ 
      using opn-in by auto
  qed
  then show
    table-agrees-with-authenticated-openings trace-table
    (scale * clength) (trace-openings ! i)
  by (rule table-agrees-with-authenticated-openings-subset
      [OF agrees-all])
qed
have len-table:  $\text{length trace-table} = \text{scale} * \text{clength}$ 
  using agrees-all
  unfolding table-agrees-with-authenticated-openings-def by simp
have partial-trace-table-candidate trace-table trace-openings
  unfolding partial-trace-table-candidate-def
  using len-table len-openings agrees by simp
then show ?thesis by blast
qed

lemma accepted-with-partial-composition-openings-candidate-if-no-partial-merkle-bad:
  assumes partial:
    accepted-with-partial-composition-openings s
    (Some (result, final-state)) composition-root query-idxs

```

```

    composition-openings
  and no-bad:
    ¬ partial-merkle-inconsistency-bad s
    (Some (result, final-state))
  shows ∃ composition-table.
    partial-composition-table-candidate composition-table composition-openings
proof -
  have len-openings: length composition-openings = rounds
  using partial
  unfolding accepted-with-partial-composition-openings-def by blast
  have table:
    ∧ i. i < rounds ⇒
      partial-authenticated-table composition-root (scale * clength)
      (composition-openings ! i) final-state
  using partial
  unfolding accepted-with-partial-composition-openings-def by blast
  have accepted-out: accepted (Some (result, final-state))
  by (rule accepted-with-partial-composition-openings-imp-accepted
    [OF partial])
  have idx-bound:
    ∧ opn. opn ∈ set (List.concat composition-openings) ⇒
      opening-index opn < scale * clength
proof -
  fix opn
  assume opn-in: opn ∈ set (List.concat composition-openings)
  then obtain openings where openings-in:
    openings ∈ set composition-openings
  and opn-openings: opn ∈ set openings
  by auto
  then obtain i where i-bound: i < length composition-openings
  and openings-eq: composition-openings ! i = openings
  by (meson in-set-conv-nth)
  have table-i:
    partial-authenticated-table composition-root (scale * clength)
    (composition-openings ! i) final-state
  using table i-bound len-openings by simp
  have auth: authenticated-opening-in final-state opn
  using table-i opn-openings unfolding openings-eq
    partial-authenticated-table-def by blast
  have len: opening-length opn = scale * clength
  using table-i opn-openings unfolding openings-eq
    partial-authenticated-table-def by blast
  show opening-index opn < scale * clength
  using auth len unfolding authenticated-opening-in-def by simp
qed
have consistent:
  ∧ opn opn'. opn ∈ set (List.concat composition-openings) ⇒
  opn' ∈ set (List.concat composition-openings) ⇒
  opening-index opn = opening-index opn' ⇒

```

```

    opening-value opn = opening-value opn'
proof (rule ccontr)
  fix opn opn'
  assume opn-in: opn ∈ set (List.concat composition-openings)
    and opn'-in: opn' ∈ set (List.concat composition-openings)
    and same-index: opening-index opn = opening-index opn'
    and neg: opening-value opn ≠ opening-value opn'
  obtain openings where openings-in: openings ∈ set composition-openings
    and opn-openings: opn ∈ set openings
    using opn-in by auto
  then obtain i where i-bound: i < length composition-openings
    and openings-eq: composition-openings ! i = openings
    by (meson in-set-conv-nth)
  obtain openings' where openings'-in:
    openings' ∈ set composition-openings
    and opn'-openings: opn' ∈ set openings'
    using opn'-in by auto
  then obtain j where j-bound: j < length composition-openings
    and openings'-eq: composition-openings ! j = openings'
    by (meson in-set-conv-nth)
  have table-i:
    partial-authenticated-table composition-root (scale * clength)
      (composition-openings ! i) final-state
    using table i-bound len-openings by simp
  have table-j:
    partial-authenticated-table composition-root (scale * clength)
      (composition-openings ! j) final-state
    using table j-bound len-openings by simp
  have auth: authenticated-opening-in final-state opn
    using table-i opn-openings unfolding openings-eq
      partial-authenticated-table-def by blast
  have auth': authenticated-opening-in final-state opn'
    using table-j opn'-openings unfolding openings'-eq
      partial-authenticated-table-def by blast
  have root: opening-root opn = opening-root opn'
    using table-i table-j opn-openings opn'-openings
    unfolding openings-eq openings'-eq partial-authenticated-table-def
    by auto
  have len: opening-length opn = opening-length opn'
    using table-i table-j opn-openings opn'-openings
    unfolding openings-eq openings'-eq partial-authenticated-table-def
    by auto
  have bad:
    partial-merkle-inconsistency-bad s (Some (result, final-state))
    unfolding partial-merkle-inconsistency-bad-def
    by (intro conjI exI[of - result] exI[of - final-state]
      exI[of - opn] exI[of - opn'])
      (use accepted-out auth auth' root len same-index neg in simp-all)
  then show False using no-bad by contradiction

```

```

qed
obtain composition-table where agrees-all:
  table-agrees-with-authenticated-openings composition-table
    (scale * clength) (List.concat composition-openings)
using table-agrees-with-authenticated-openings-consistent-exists
  [OF idx-bound consistent]
by blast
have agrees:
   $\bigwedge i. i < \text{rounds} \implies$ 
    table-agrees-with-authenticated-openings composition-table
      (scale * clength) (composition-openings ! i)
proof -
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  have set (composition-openings ! i)  $\subseteq$ 
    set (List.concat composition-openings)
  proof
    fix opn
    assume opn-in:  $opn \in \text{set } (composition-openings ! i)$ 
    have  $composition-openings ! i \in \text{set } composition-openings$ 
      using i-bound len-openings by (intro nth-mem) simp
    then show  $opn \in \text{set } (List.concat composition-openings)$ 
      using opn-in by auto
  qed
then show
  table-agrees-with-authenticated-openings composition-table
    (scale * clength) (composition-openings ! i)
  by (rule table-agrees-with-authenticated-openings-subset
    [OF agrees-all])
qed
have len-table:  $\text{length } composition-table = scale * clength$ 
  using agrees-all
  unfolding table-agrees-with-authenticated-openings-def by simp
have partial-composition-table-candidate composition-table
  composition-openings
  unfolding partial-composition-table-candidate-def
  using len-table len-openings agrees by simp
then show ?thesis by blast
qed

lemma accepted-with-partial-initial-openings-candidates-if-no-partial-merkle-bad:
assumes partial:
  accepted-with-partial-initial-openings s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs trace-openings
  composition-query-idxs composition-openings
and no-bad:
   $\neg \text{partial-merkle-inconsistency-bad } s$ 
  (Some (result, final-state))

```

shows $\exists$ *trace-table composition-table*.
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings
proof –
have *trace-partial*:
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr trace-query-idxs trace-openings
using *accepted-with-partial-initial-openings-shapes(3)[OF partial]* .
have *comp-partial*:
accepted-with-partial-composition-openings s
(Some (result, final-state)) (hd composition-fri-roots)
composition-query-idxs composition-openings
using *accepted-with-partial-initial-openings-shapes(4)[OF partial]* .
obtain *trace-table* **where** *trace-candidate*:
partial-trace-table-candidate trace-table trace-openings
using *accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad*
 $[OF \text{ trace-partial no-bad}]$
by *blast*
obtain *composition-table* **where** *comp-candidate*:
partial-composition-table-candidate composition-table
composition-openings
using
accepted-with-partial-composition-openings-candidate-if-no-partial-merkle-bad
 $[OF \text{ comp-partial no-bad}]$
by *blast*
show *?thesis*
using *trace-candidate comp-candidate* **by** *blast*
qed

lemma *verify-monad-partial-initial-candidates-or-partial-merkle-bad*:
fixes *s :: ('f, 'a) protocol-channel-scheme*
assumes *outcome*:
Some (result, final-state) $\in$ set-dist (execute verify-monad s)
and *header*:
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
and *comp-nonempty*: *composition-fri-roots $\neq []$*
shows
partial-merkle-inconsistency-bad s (Some (result, final-state)) $\vee$
 $(\exists \text{ trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table.}$
accepted-with-partial-initial-openings s
(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings)

```

proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
  case True
  then show ?thesis by simp
next
  case False
  from verify-monad-accepted-with-partial-initial-openings
    [OF outcome header comp-nonempty]
  obtain trace-query-idxs trace-openings composition-query-idxs
    composition-openings where partial:
    accepted-with-partial-initial-openings s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings
  by blast
  obtain trace-table composition-table where candidates:
    partial-trace-table-candidate trace-table trace-openings
    partial-composition-table-candidate composition-table
    composition-openings
  using accepted-with-partial-initial-openings-candidates-if-no-partial-merkle-bad
    [OF partial False]
  by blast
  show ?thesis
  using partial candidates by blast
qed

```

```

lemma partial-merkle-inconsistency-bad-mono-accepted:
  wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤
  wp-event verify-monad accepted s
by (rule wp-event-mono)
  (rule partial-merkle-inconsistency-bad-imp-accepted)

```

end

end

```

theory Soundness-Reductions-Trace
  imports Soundness-Partial-Merkle
begin

```

FRI-table acceptance, transcript-target, and trace-table candidate reductions.

```

context soundness
begin

```

```

definition accepted-with-fri-tables
  :: ('f, 'a) protocol-channel-scheme ⇒
    (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool
  where

```

accepted-with-fri-tables s out $\longleftrightarrow$
accepted out $\wedge$ *fri-merkle-binding-good s out*

lemma *accepted-with-fri-tables-imp-accepted*:
assumes *accepted-with-fri-tables s out*
shows *accepted out*
using *assms* **unfolding** *accepted-with-fri-tables-def* **by** *simp*

lemma *accepted-with-fri-tablesE*:
assumes *tables: accepted-with-fri-tables s out*
obtains *trace-table composition-table as query-idxs*
trace-roots trace-bs trace-final dg composition-roots composition-bs
composition-final trace-round-layers composition-round-layers
trace-tables composition-tables trace-final-table composition-final-table
where
accepted-with-bound-tables s out trace-table composition-table as
query-idxs
and *accepted-fri-opening-transcript s out trace-roots trace-bs*
trace-final dg composition-roots composition-bs composition-final
query-idxs trace-round-layers composition-round-layers
and *trace-fri-tables-bound s out trace-roots trace-bs trace-final*
query-idxs trace-round-layers trace-tables trace-final-table
and *composition-fri-tables-bound s out dg composition-roots*
composition-bs composition-final query-idxs composition-round-layers
composition-tables composition-final-table
and *fri-initial-table-agrees trace-table trace-tables*
and *fri-initial-table-agrees composition-table composition-tables*
using *tables*
unfolding *accepted-with-fri-tables-def fri-merkle-binding-good-def*
by (*blast intro: that*)

lemma *accepted-with-fri-tables-obtain-bound-tables*:
assumes *accepted-with-fri-tables s out*
obtains *trace-table composition-table as query-idxs*
where *accepted-with-bound-tables s out trace-table composition-table as*
query-idxs
using *assms* **by** (*elim accepted-with-fri-tablesE*) *blast*

lemma *accepted-partition-fri-merkle-binding*:
assumes *accepted out*
shows *fri-merkle-binding-bad s out* $\vee$ *accepted-with-fri-tables s out*
using *assms*
unfolding *fri-merkle-binding-bad-def accepted-with-fri-tables-def*
by *blast*

lemma *accepted-with-bound-tables-trace-root-target-hit*:
assumes *bound*:
accepted-with-bound-tables (verifier-initial-state tr)
(Some (result, final-state)) trace-table composition-table as

query-idxs
shows
hash-map-new-output-hit (set tr) (verifier-initial-state tr) final-state
proof –
from *bound* **obtain** *fr f-fri-roots f-final dg composition-fri-roots final rest*
where *header:*
verifier-header-transcript (verifier-initial-state tr) fr f-fri-roots
f-final as dg composition-fri-roots final rest
and *bind:*
merkle-root-binds-table fr trace-table final-state
unfolding *accepted-with-bound-tables-def* **by** *blast*
have *table-nonempty: trace-table ≠ []*
using *accepted-with-bound-tables-shapes(1)[OF bound]*
eval-domain-nontrivial **by** *auto*
from *merkle-root-binds-nonempty-table-output-lookup[OF bind table-nonempty]*
obtain *input* **where** *lookup:*
fmlookup (HashMap final-state) input = Some fr
by *blast*
have *root-in: fr ∈ set tr*
using *header*
unfolding *verifier-header-transcript-def verifier-header-messages-def*
by *simp*
have *fresh:*
fmlookup (HashMap (verifier-initial-state tr)) input = None
by *simp*
show *?thesis*
unfolding *hash-map-new-output-hit-def*
using *fresh lookup root-in* **by** *blast*
qed

lemma *accepted-with-fri-tables-imp-transcript-target-hit:*
assumes *tables:*
accepted-with-fri-tables (verifier-initial-state tr) out
shows
hash-new-output-hit-event (set tr) (verifier-initial-state tr) out
proof –
from *accepted-with-fri-tables-obtain-bound-tables[OF tables]*
obtain *trace-table composition-table as query-idxs* **where** *bound:*
accepted-with-bound-tables (verifier-initial-state tr) out trace-table
composition-table as query-idxs
by *blast*
from *accepted-with-bound-tables-imp-accepted[OF bound]*
obtain *result final-state* **where** *out-eq: out = Some (result, final-state)*
unfolding *accepted-def* **by** *(cases out) auto*
have *hit:*
hash-map-new-output-hit (set tr) (verifier-initial-state tr) final-state
by *(rule accepted-with-bound-tables-trace-root-target-hit)*
(use bound out-eq in simp)
show *?thesis*

unfolding *hash-new-output-hit-event-def out-eq* **using** *hit* **by** *simp*
qed

lemma *accepted-imp-fri-merkle-binding-bad-or-transcript-target-hit:*
assumes *acc: accepted out*
shows
fri-merkle-binding-bad (verifier-initial-state tr) out $\vee$
hash-new-output-hit-event (set tr) (verifier-initial-state tr) out
proof –
from *accepted-partition-fri-merkle-binding*
[of out verifier-initial-state tr, OF acc]
show *?thesis*
proof
assume *fri-merkle-binding-bad (verifier-initial-state tr) out*
then show *?thesis* **by** *simp*
next
assume *tables:*
accepted-with-fri-tables (verifier-initial-state tr) out
have *hash-new-output-hit-event (set tr) (verifier-initial-state tr) out*
by *(rule accepted-with-fri-tables-imp-transcript-target-hit[OF tables])*
then show *?thesis* **by** *simp*
qed
qed

lemma *verify-monad-verifier-initial-trace-root-target-hit:*
assumes *outcome:*
Some (result, final-state) $\in$
set-dist (execute verify-monad (verifier-initial-state tr))
shows
hash-map-new-output-hit (set tr) (verifier-initial-state tr)
final-state
proof –
let *?s = verifier-initial-state tr*
from *verify-monad-header-extraction[OF outcome]*
obtain *fr f-fl f-final as dg fl final query-state* **where**
header:
verifier-header-transcript ?s fr (map snd f-fl) f-final as dg
(map snd fl) final (PTranscript query-state)
and *query-out:*
Some (result, final-state) $\in$
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final)
rounds)
query-state)
by *(rule verify-monad-header-query-extraction[OF outcome])*
obtain *query-idxs trace-openings* **where**
len-query-idxs: length query-idxs = rounds

```

and len-openings: length trace-openings = rounds
and indices:
   $\bigwedge i. i < \text{rounds} \implies$ 
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
and tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
proof (rule ntimes-verifier-query-rounds-authenticated-trace-openings-no-rows
  [OF query-out])
fix query-idxs trace-openings
assume len-query-idxs: length query-idxs = rounds
and len-openings: length trace-openings = rounds
and indices:
   $\bigwedge i. i < \text{rounds} \implies$ 
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
and tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
show ?thesis
  by (rule that[OF len-query-idxs len-openings indices tables])
qed
have round0: 0 < rounds
  by (rule rounds-positive)
have openings0-nonempty:
  trace-openings ! 0  $\neq []$ 
proof –
  have map opening-index (trace-openings ! 0) =
    powers-scaled (query-idxs ! 0)
    by (rule indices[OF round0])
  moreover have powers-scaled (query-idxs ! 0)  $\neq []$ 
    using powers-pos unfolding powers-scaled-def by simp
  ultimately show ?thesis
    by auto
qed
then obtain opn where opn-in:
  opn  $\in \text{set } (\text{trace-openings} ! 0)$ 
  by (cases trace-openings ! 0) auto
have table0:
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! 0) final-state
  using tables[OF round0] by simp
have opn-root: opening-root opn = fr
  using table0 opn-in unfolding partial-authenticated-table-def by blast
have opn-auth: authenticated-opening-in final-state opn
  using table0 opn-in unfolding partial-authenticated-table-def by blast

```

```

from authenticated-opening-root-lookup[OF opn-auth]
obtain input where lookup:
  fmlookup (HashMap final-state) input = Some fr
  using opn-root by auto
have root-in: fr ∈ set tr
  using header
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
have fresh: fmlookup (HashMap ?s) input = None
  by simp
show ?thesis
  unfolding hash-map-new-output-hit-def
  using fresh lookup root-in by blast
qed

lemma verify-monad-trace-root-target-preexisting-or-new:
assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and tr-eq: PTranscript s = tr
shows
  hash-map-output-values s ∩ set tr ≠ {} ∨
  hash-map-new-output-hit (set tr) s final-state
proof –
from verify-monad-header-extraction[OF outcome]
obtain fr f-fl f-final as dg fl final query-state where
  header:
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
and query-out:
  Some (result, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
  by (rule verify-monad-header-query-extraction[OF outcome])
obtain query-idxs trace-openings where
  indices:
     $\bigwedge i. i < \text{rounds} \implies$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
  and tables:
     $\bigwedge i. i < \text{rounds} \implies$ 
      partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
proof (rule ntimes-verifier-query-rounds-authenticated-trace-openings-no-raws
  [OF query-out])
fix query-idxs trace-openings

```

```

assume indices:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{map opening-index } (\text{trace-openings} ! i) =$ 
       $\text{powers-scaled } (\text{query-idxs} ! i)$ 
and tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{partial-authenticated-table fr } (\text{scale} * \text{clength})$ 
       $(\text{trace-openings} ! i) \text{ final-state}$ 
show ?thesis
  by (rule that[OF indices tables])
qed
have round0:  $0 < \text{rounds}$ 
  by (rule rounds-positive)
have openings0-nonempty:
   $\text{trace-openings} ! 0 \neq []$ 
proof –
  have  $\text{map opening-index } (\text{trace-openings} ! 0) =$ 
     $\text{powers-scaled } (\text{query-idxs} ! 0)$ 
    by (rule indices[OF round0])
  moreover have  $\text{powers-scaled } (\text{query-idxs} ! 0) \neq []$ 
    using powers-pos unfolding powers-scaled-def by simp
  ultimately show ?thesis
    by auto
qed
then obtain opn where opn-in:
   $\text{opn} \in \text{set } (\text{trace-openings} ! 0)$ 
  by (cases  $\text{trace-openings} ! 0$ ) auto
have table0:
   $\text{partial-authenticated-table fr } (\text{scale} * \text{clength})$ 
     $(\text{trace-openings} ! 0) \text{ final-state}$ 
  using tables[OF round0] by simp
have opn-root:  $\text{opening-root opn} = \text{fr}$ 
  using table0 opn-in unfolding partial-authenticated-table-def by blast
have opn-auth:  $\text{authenticated-opening-in final-state opn}$ 
  using table0 opn-in unfolding partial-authenticated-table-def by blast
from authenticated-opening-root-lookup[OF opn-auth]
obtain input where lookup-final:
   $\text{fmlookup } (\text{HashMap final-state}) \text{ input} = \text{Some fr}$ 
  using opn-root by auto
have root-in:  $\text{fr} \in \text{set tr}$ 
  using header tr-eq
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
have ext:  $s \leq \text{final-state}$ 
  by (rule verify-monad-hash-extends[OF outcome])
show ?thesis
proof (cases  $\text{fmlookup } (\text{HashMap } s) \text{ input}$ )
  case None
  then have hash-map-new-output-hit ( $\text{set tr}$ ) s final-state

```

```

    unfolding hash-map-new-output-hit-def
    using lookup-final root-in by blast
  then show ?thesis by simp
next
case (Some z)
have lookup-final-z: fmllookup (HashMap final-state) input = Some z
  by (rule hash-extension-lookup[OF Some ext])
then have z = fr
  using lookup-final by simp
then have fr ∈ hash-map-output-values s
  using Some by (auto intro: hash-map-output-valuesI)
then show ?thesis
  using root-in by blast
qed
qed

definition soundness-bad-event-with-merkle
:: ('f, 'a) protocol-channel-scheme ⇒
  (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool
where
  soundness-bad-event-with-merkle s out ⟷
    fri-merkle-binding-bad s out ∨ soundness-bad-event s out

lemma soundness-bad-event-imp-soundness-bad-event-with-merkle:
  assumes soundness-bad-event s out
  shows soundness-bad-event-with-merkle s out
  using assms unfolding soundness-bad-event-with-merkle-def by simp

lemma fri-merkle-binding-bad-imp-soundness-bad-event-with-merkle:
  assumes fri-merkle-binding-bad s out
  shows soundness-bad-event-with-merkle s out
  using assms unfolding soundness-bad-event-with-merkle-def by simp

lemma accepted-with-fri-tables-table-partition:
  assumes false-statement: ¬ exists-valid-trace
  and tables: accepted-with-fri-tables s out
  shows soundness-bad-event s out
proof –
  from accepted-with-fri-tables-obtain-bound-tables[OF tables]
  obtain trace-table composition-table as query-idxs where bound:
    accepted-with-bound-tables s out trace-table composition-table as
    query-idxs
  by blast
  have partition:
    composition-bad s out ∨ trace-fri-bad s out ∨
    composition-fri-bad s out ∨ query-bad s out
  by (rule accepted-bound-table-partition[OF false-statement bound])
  then show ?thesis
    unfolding soundness-bad-event-def by blast

```

qed

lemma *accepted-partition-soundness-bad-event-with-merkle:*

assumes *false-statement*: $\neg \text{exists-valid-trace}$

and *acc*: *accepted out*

shows *soundness-bad-event-with-merkle s out*

proof –

from *accepted-partition-fri-merkle-binding*[*OF acc*]

consider

(*merkle-bad*) *fri-merkle-binding-bad s out*

| (*with-tables*) *accepted-with-fri-tables s out*

by *blast*

then show *?thesis*

proof *cases*

case *merkle-bad*

then show *?thesis*

by (*rule fri-merkle-binding-bad-imp-soundness-bad-event-with-merkle*)

next

case *with-tables*

have *soundness-bad-event s out*

by (*rule accepted-with-fri-tables-table-partition*
[*OF false-statement with-tables*])

then show *?thesis*

by (*rule soundness-bad-event-imp-soundness-bad-event-with-merkle*)

qed

qed

definition *trace-fri-challenge-set-hit*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(f \text{ list} \Rightarrow f \text{ list set}) \Rightarrow$

$(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-challenge-set-hit s bad-sets out $\longleftrightarrow$

$(\exists \text{ trace-table composition-table as query-idxs trace-bs dg comp-bs.}$

accepted-with-bound-tables s out trace-table composition-table as query-idxs $\wedge$

accepted-fri-challenges s out trace-bs dg comp-bs $\wedge$

$\neg \text{trace-table-low-degree trace-table} \wedge$

trace-bs $\in \text{bad-sets trace-table}$)

definition *composition-fri-challenge-set-hit*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(f \Rightarrow f \text{ list} \Rightarrow f \text{ list set}) \Rightarrow$

$(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-challenge-set-hit s bad-sets out $\longleftrightarrow$

$(\exists \text{ trace-table composition-table as query-idxs trace-bs dg comp-bs.}$

accepted-with-bound-tables s out trace-table composition-table as query-idxs $\wedge$

accepted-fri-challenges s out trace-bs dg comp-bs $\wedge$

trace-table-low-degree trace-table $\wedge$

$\neg$ composition-table-low-degree maxDegree composition-table $\wedge$
 comp-bs $\in$ bad-sets dg composition-table)

lemma trace-fri-challenge-set-hit-imp-list-set-hit:

assumes hit: trace-fri-challenge-set-hit s bad-sets out

and envelope: $\bigwedge$ trace-table. bad-sets trace-table $\subseteq B$

shows trace-fri-challenge-list-set-hit s B out

proof –

from hit **obtain** trace-table composition-table as query-idxs trace-bs dg comp-bs

where challenges:

accepted-fri-challenges s out trace-bs dg comp-bs

and trace-bad: trace-bs $\in$ bad-sets trace-table

unfolding trace-fri-challenge-set-hit-def **by** blast

have trace-bs $\in B$

using trace-bad envelope[of trace-table] **by** auto

then show ?thesis

unfolding trace-fri-challenge-list-set-hit-def

using challenges **by** blast

qed

definition trace-fri-root-trace-table-candidates

$:: (f, 'a)$ protocol-channel-scheme $\Rightarrow f \Rightarrow 'f$ list set

where

trace-fri-root-trace-table-candidates s fr =

{trace-table.

$\exists$ out composition-table as query-idxs trace-bs dg comp-bs

f-fri-roots f-final header-as composition-fri-roots final rest.

accepted-with-bound-tables s out trace-table composition-table

as query-idxs $\wedge$

accepted-fri-challenges s out trace-bs dg comp-bs $\wedge$

verifier-header-transcript s fr f-fri-roots f-final header-as dg

composition-fri-roots final rest}

definition trace-fri-supported-root-trace-table-candidates

$:: (f, 'a)$ protocol-channel-scheme $\Rightarrow f \Rightarrow 'f$ list set

where

trace-fri-supported-root-trace-table-candidates s fr =

{trace-table.

$\exists$ out composition-table as query-idxs trace-bs dg comp-bs

f-fri-roots f-final header-as composition-fri-roots final rest.

out $\in$ set-dist (execute verify-monad s) $\wedge$

accepted-with-bound-tables s out trace-table composition-table

as query-idxs $\wedge$

accepted-fri-challenges s out trace-bs dg comp-bs $\wedge$

verifier-header-transcript s fr f-fri-roots f-final header-as dg

composition-fri-roots final rest}

definition trace-fri-supported-root-partial-trace-table-candidates

$:: (f, 'a)$ protocol-channel-scheme $\Rightarrow f \Rightarrow 'f$ list set

where

trace-fri-supported-root-partial-trace-table-candidates s fr =
 $\{ \text{trace-table.}$
 $\exists \text{ out query-idxs trace-openings trace-bs dg comp-bs}$
 $\text{f-fri-roots f-final header-as composition-fri-roots final rest.}$
 $\text{out} \in \text{set-dist (execute verify-monad s)} \wedge$
 $\text{accepted-with-partial-trace-openings s out fr query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{accepted-fri-challenges s out trace-bs dg comp-bs} \wedge$
 $\text{verifier-header-transcript s fr f-fri-roots f-final header-as dg}$
 $\text{composition-fri-roots final rest} \}$

definition *trace-fri-supported-root-partial-trace-table-candidate-state*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow \text{'f} \Rightarrow \text{'f list} \Rightarrow$
 $(\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

trace-fri-supported-root-partial-trace-table-candidate-state s fr
 $\text{trace-table final-state} \longleftrightarrow$
 $(\exists \text{ result query-idxs trace-openings trace-bs dg comp-bs}$
 $\text{f-fri-roots f-final header-as composition-fri-roots final rest.}$
 $\text{Some (result, final-state)} \in \text{set-dist (execute verify-monad s)} \wedge$
 $\text{accepted-with-partial-trace-openings s}$
 $(\text{Some (result, final-state)}) \text{ fr query-idxs trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{accepted-fri-challenges s (Some (result, final-state)) trace-bs dg}$
 $\text{comp-bs} \wedge$
 $\text{verifier-header-transcript s fr f-fri-roots f-final header-as dg}$
 $\text{composition-fri-roots final rest})$

definition *alpha-header-supported-partial-trace-table-candidates*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $\text{'f} \Rightarrow \text{'f list} \Rightarrow \text{'f} \Rightarrow \text{'f list set}$

where

alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
 $\text{f-final} =$
 $\{ \text{trace-table.}$
 $\exists \text{ out query-idxs trace-openings as dg composition-fri-roots final rest.}$
 $\text{out} \in \text{set-dist (execute verify-monad s)} \wedge$
 $\text{accepted-with-partial-trace-openings s out fr query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{verifier-header-transcript s fr f-fri-roots f-final as dg}$
 $\text{composition-fri-roots final rest} \}$

lemma *alpha-header-supported-partial-trace-table-candidatesE:*

assumes

$\text{trace-table} \in$
 $\text{alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots}$

f-final
obtains *out query-idxs trace-openings as dg composition-fri-roots final rest*
where
out $\in$ *set-dist* (*execute verify-monad s*)
accepted-with-partial-trace-openings s out fr query-idxs
trace-openings
partial-trace-table-candidate trace-table trace-openings
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
using *assms*
unfolding *alpha-header-supported-partial-trace-table-candidates-def*
by *blast*

definition *alpha-header-supported-partial-union-bad-sets*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list set}$

where

alpha-header-supported-partial-union-bad-sets s bad-sets fr
f-fri-roots f-final =
 $\{as \in \text{alpha-space}.$
 $\exists \text{ trace-table} \in$
 $\text{alpha-header-supported-partial-trace-table-candidates } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final}.$
 $as \in \text{bad-sets trace-table}\}$

definition *trace-fri-root-trace-table-candidate-state*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow$
 $(f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

trace-fri-root-trace-table-candidate-state s fr trace-table
final-state $\longleftrightarrow$
 $(\exists \text{ result composition-table as query-idxs trace-bs dg comp-bs}$
 $f\text{-fri-roots } f\text{-final header-as composition-fri-roots final rest}.$
 $\text{accepted-with-bound-tables } s \text{ (Some (result, final-state))}$
 $\text{trace-table composition-table as query-idxs} \wedge$
 $\text{accepted-fri-challenges } s \text{ (Some (result, final-state)) trace-bs dg}$
 $\text{comp-bs} \wedge$
 $\text{verifier-header-transcript } s \text{ fr f-fri-roots f-final header-as dg}$
 $\text{composition-fri-roots final rest} \wedge$
 $\text{merkle-root-binds-table fr trace-table final-state})$

definition *trace-fri-supported-root-trace-table-candidate-state*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow$
 $(f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state $\longleftrightarrow$
 $(\exists \text{ result composition-table as query-idxs trace-bs dg comp-bs}$

f-fri-roots f-final header-as composition-fri-roots final rest.
Some (result, final-state) ∈ set-dist (execute verify-monad s) ∧
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs ∧
accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
comp-bs ∧
verifier-header-transcript s fr f-fri-roots f-final header-as dg
composition-fri-roots final rest ∧
merkle-root-binds-table fr trace-table final-state)

lemma *trace-fri-root-trace-table-candidates-iff-state:*

trace-table ∈ trace-fri-root-trace-table-candidates s fr $\longleftrightarrow$
(∃ final-state.
trace-fri-root-trace-table-candidate-state s fr trace-table
final-state)

proof

assume *candidate:*

trace-table ∈ trace-fri-root-trace-table-candidates s fr

then obtain *out composition-table as query-idxs trace-bs dg comp-bs*
f-fri-roots f-final header-as composition-fri-roots final rest **where**
bound:

accepted-with-bound-tables s out trace-table composition-table as
query-idxs

and *challenges:*

accepted-fri-challenges s out trace-bs dg comp-bs

and *header:*

verifier-header-transcript s fr f-fri-roots f-final header-as dg
composition-fri-roots final rest

unfolding *trace-fri-root-trace-table-candidates-def* **by** *blast*
from *bound* **obtain** *result final-state fr' f-fri-roots' f-final' dg'*
composition-fri-roots' final' rest' **where**

out-eq: out = Some (result, final-state)

and *header':*

verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
composition-fri-roots' final' rest'

and *bind:*

merkle-root-binds-table fr' trace-table final-state

unfolding *accepted-with-bound-tables-def* **by** *blast*

have *fr-eq: fr' = fr*

using *verifier-header-transcript-unique[OF header' header]* **by** *simp*

have *bound-some:*

accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs

using *bound out-eq* **by** *simp*

have *challenges-some:*

accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
comp-bs

using *challenges out-eq* **by** *simp*

have *bind-fr: merkle-root-binds-table fr trace-table final-state*

```

using bind fr-eq by simp
have state:
  trace-fri-root-trace-table-candidate-state s fr trace-table
  final-state
unfolding trace-fri-root-trace-table-candidate-state-def
apply (rule exI[where x=result])
apply (rule exI[where x=composition-table])
apply (rule exI[where x=as])
apply (rule exI[where x=query-idxs])
apply (rule exI[where x=trace-bs])
apply (rule exI[where x=dg])
apply (rule exI[where x=comp-bs])
apply (rule exI[where x=f-fri-roots])
apply (rule exI[where x=f-final])
apply (rule exI[where x=header-as])
apply (rule exI[where x=composition-fri-roots])
apply (rule exI[where x=final])
apply (rule exI[where x=rest])
apply (intro conjI)
  apply (rule bound-some)
  apply (rule challenges-some)
  apply (rule header)
apply (rule bind-fr)
done
then show
   $\exists$  final-state.
    trace-fri-root-trace-table-candidate-state s fr trace-table
    final-state
by blast
next
assume
   $\exists$  final-state.
    trace-fri-root-trace-table-candidate-state s fr trace-table
    final-state
then obtain final-state result composition-table as query-idxs trace-bs dg
  comp-bs f-fri-roots f-final header-as composition-fri-roots final rest
where
  bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
and challenges:
    accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
    comp-bs
and header:
    verifier-header-transcript s fr f-fri-roots f-final header-as dg
    composition-fri-roots final rest
unfolding trace-fri-root-trace-table-candidate-state-def by blast
show trace-table  $\in$  trace-fri-root-trace-table-candidates s fr
unfolding trace-fri-root-trace-table-candidates-def

```

using bound challenges header by blast
 qed

lemma trace-fri-supported-root-trace-table-candidates-iff-state:
 $\text{trace-table} \in \text{trace-fri-supported-root-trace-table-candidates } s \text{ fr} \longleftrightarrow$
 $(\exists \text{ final-state.}$
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table}$
 $\text{final-state})$

proof
 assume candidate:
 $\text{trace-table} \in \text{trace-fri-supported-root-trace-table-candidates } s \text{ fr}$
then obtain out composition-table as query-idxs trace-bs dg comp-bs
 $f\text{-fri-roots } f\text{-final header-as composition-fri-roots final rest$ **where**
 outcome: out $\in$ set-dist (execute verify-monad s)
and bound:
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as}$
 query-idxs
and challenges:
 $\text{accepted-fri-challenges } s \text{ out trace-bs dg comp-bs}$
and header:
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final header-as dg}$
 $\text{composition-fri-roots final rest}$
unfolding trace-fri-supported-root-trace-table-candidates-def by blast
from bound **obtain** result final-state fr' f-fri-roots' f-final' dg'
 $\text{composition-fri-roots' final' rest'}$ **where**
 out-eq: out = Some (result, final-state)
and header':
 $\text{verifier-header-transcript } s \text{ fr' } f\text{-fri-roots' } f\text{-final' as dg'}$
 $\text{composition-fri-roots' final' rest'}$
and bind:
 $\text{merkle-root-binds-table } fr' \text{ trace-table final-state}$
unfolding accepted-with-bound-tables-def by blast
have fr-eq: fr' = fr
 using verifier-header-transcript-unique[OF header' header] by simp
have state:
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table}$
 final-state
unfolding trace-fri-supported-root-trace-table-candidate-state-def
apply (rule exI[**where** x=result])
apply (rule exI[**where** x=composition-table])
apply (rule exI[**where** x=as])
apply (rule exI[**where** x=query-idxs])
apply (rule exI[**where** x=trace-bs])
apply (rule exI[**where** x=dg])
apply (rule exI[**where** x=comp-bs])
apply (rule exI[**where** x=f-fri-roots])
apply (rule exI[**where** x=f-final])
apply (rule exI[**where** x=header-as])
apply (rule exI[**where** x=composition-fri-roots])

```

apply (rule exI[where x=final])
apply (rule exI[where x=rest])
apply (intro conjI)
  apply (use outcome out-eq in simp)
  apply (use bound out-eq in simp)
  apply (use challenges out-eq in simp)
  apply (rule header)
apply (use bind fr-eq in simp)
done
then show
   $\exists$  final-state.
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table
    final-state
  by blast
next
assume
   $\exists$  final-state.
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table
    final-state
then obtain final-state result composition-table as query-idxs trace-bs dg
  comp-bs f-fri-roots f-final header-as composition-fri-roots final rest
where
  outcome:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
and bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
and challenges:
    accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
    comp-bs
and header:
    verifier-header-transcript s fr f-fri-roots f-final header-as dg
    composition-fri-roots final rest
unfolding trace-fri-supported-root-trace-table-candidate-state-def by blast
show trace-table  $\in$  trace-fri-supported-root-trace-table-candidates s fr
unfolding trace-fri-supported-root-trace-table-candidates-def
using outcome bound challenges header by blast
qed

```

lemma trace-fri-supported-root-partial-trace-table-candidates-iff-state:

```

trace-table  $\in$ 
  trace-fri-supported-root-partial-trace-table-candidates s fr  $\longleftrightarrow$ 
  ( $\exists$  final-state.
    trace-fri-supported-root-partial-trace-table-candidate-state s fr
    trace-table final-state)

```

proof

```

assume candidate:
  trace-table  $\in$ 
    trace-fri-supported-root-partial-trace-table-candidates s fr

```

then obtain *out query-idxs trace-openings trace-bs dg comp-bs*
f-fri-roots f-final header-as composition-fri-roots final rest **where**
outcome: out ∈ set-dist (execute verify-monad s)
and *partial:*
accepted-with-partial-trace-openings s out fr query-idxs
trace-openings
and *table:*
partial-trace-table-candidate trace-table trace-openings
and *challenges:*
accepted-fri-challenges s out trace-bs dg comp-bs
and *header:*
verifier-header-transcript s fr f-fri-roots f-final header-as dg
composition-fri-roots final rest
unfolding *trace-fri-supported-root-partial-trace-table-candidates-def*
by *blast*
from *partial* **obtain** *result final-state* **where** *out-eq:*
out = Some (result, final-state)
unfolding *accepted-with-partial-trace-openings-def* **by** *blast*
have *state:*
trace-fri-supported-root-partial-trace-table-candidate-state s fr
trace-table final-state
proof –
have *outcome-some:*
Some (result, final-state) ∈ set-dist (execute verify-monad s)
using *outcome out-eq* **by** *simp*
have *partial-some:*
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr query-idxs trace-openings
using *partial out-eq* **by** *simp*
have *challenges-some:*
accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
comp-bs
using *challenges out-eq* **by** *simp*
show *?thesis*
unfolding
trace-fri-supported-root-partial-trace-table-candidate-state-def
using *outcome-some partial-some table challenges-some header* **by** *blast*
qed
then show
 $\exists$ *final-state.*
trace-fri-supported-root-partial-trace-table-candidate-state s fr
trace-table final-state
by *blast*
next
assume
 $\exists$ *final-state.*
trace-fri-supported-root-partial-trace-table-candidate-state s fr
trace-table final-state
then obtain *final-state result query-idxs trace-openings trace-bs dg*

comp-bs f-fri-roots f-final header-as composition-fri-roots final rest
where
outcome:
Some (result, final-state) ∈ set-dist (execute verify-monad s)
and *partial:*
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr query-idxs trace-openings
and *table:*
partial-trace-table-candidate trace-table trace-openings
and *challenges:*
accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
comp-bs
and *header:*
verifier-header-transcript s fr f-fri-roots f-final header-as dg
composition-fri-roots final rest
unfolding
trace-fri-supported-root-partial-trace-table-candidate-state-def
by *blast*
show
trace-table ∈
trace-fri-supported-root-partial-trace-table-candidates s fr
unfolding *trace-fri-supported-root-partial-trace-table-candidates-def*
using *outcome partial table challenges header* **by** *blast*
qed

lemma *trace-fri-supported-root-partial-trace-table-candidate-length:*
assumes
trace-fri-supported-root-partial-trace-table-candidate-state s fr
trace-table final-state
shows *length trace-table = scale * clength*
using *assms*
unfolding *trace-fri-supported-root-partial-trace-table-candidate-state-def*
partial-trace-table-candidate-def
by *blast*

lemma *alpha-header-supported-partial-union-bad-sets-subset-alpha-space:*
alpha-header-supported-partial-union-bad-sets s bad-sets fr f-fri-roots
f-final ⊆ alpha-space
unfolding *alpha-header-supported-partial-union-bad-sets-def* **by** *auto*

lemma *trace-fri-supported-root-trace-table-candidate-state-imp-candidate-state:*
assumes
trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state
shows
trace-fri-root-trace-table-candidate-state s fr trace-table final-state
using *assms*
unfolding *trace-fri-supported-root-trace-table-candidate-state-def*
trace-fri-root-trace-table-candidate-state-def

by blast

lemma *trace-fri-supported-root-trace-table-candidate-state-hash-extends:*

assumes

*trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state*

shows $s \leq \text{final-state}$

proof –

from *assms* **obtain** *result composition-table as query-idxs trace-bs dg comp-bs
f-fri-roots f-final header-as composition-fri-roots final rest* **where**
outcome:

Some (result, final-state) ∈ set-dist (execute verify-monad s)

unfolding *trace-fri-supported-root-trace-table-candidate-state-def*

by blast

show *?thesis*

by (rule *verify-monad-hash-extends*[OF *outcome*])

qed

lemma *trace-fri-supported-root-trace-table-candidate-state-imp-partial-candidate:*

assumes *state:*

*trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state*

and *clean:* $\neg \text{hash-map-output-collision final-state}$

shows

trace-table ∈

trace-fri-supported-root-partial-trace-table-candidates s fr

proof –

from *state* **obtain** *result composition-table as query-idxs trace-bs dg comp-bs
f-fri-roots f-final header-as composition-fri-roots final rest* **where**
outcome:

Some (result, final-state) ∈ set-dist (execute verify-monad s)

and *bound:*

accepted-with-bound-tables s (Some (result, final-state))

trace-table composition-table as query-idxs

and *challenges:*

*accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
comp-bs*

and *header:*

*verifier-header-transcript s fr f-fri-roots f-final header-as dg
composition-fri-roots final rest*

and *bind:*

merkle-root-binds-table fr trace-table final-state

unfolding *trace-fri-supported-root-trace-table-candidate-state-def*

by blast

from *verify-monad-header-extraction*[OF *outcome*]

obtain *fr' f-fl f-final' as' dg' fl final' query-state* **where**

header':

*verifier-header-transcript s fr' (map snd f-fl) f-final' as' dg'
(map snd fl) final' (PTranscript query-state)*

```

and query-out:
  Some (result, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr' f-fl f-final' as' fl final')
          rounds)
        query-state)
    by (rule verify-monad-header-query-extraction[OF outcome])
have fr-eq: fr' = fr
  using verifier-header-transcript-unique[OF header' header] by simp
obtain query-idxs' trace-openings where
  len-query-idxs: length query-idxs' = rounds
  and len-openings: length trace-openings = rounds
  and indices:
     $\bigwedge i. i < \text{rounds} \implies$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs' ! i)
  and tables:
     $\bigwedge i. i < \text{rounds} \implies$ 
      partial-authenticated-table fr'
        (scale * clength) (trace-openings ! i) final-state
proof (rule ntimes-verifier-query-rounds-authenticated-trace-openings-no-raws
  [OF query-out])
fix query-idxs' trace-openings
assume len-query-idxs: length query-idxs' = rounds
  and len-openings: length trace-openings = rounds
  and indices:
     $\bigwedge i. i < \text{rounds} \implies$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs' ! i)
  and tables:
     $\bigwedge i. i < \text{rounds} \implies$ 
      partial-authenticated-table fr'
        (scale * clength) (trace-openings ! i) final-state
show ?thesis
  by (rule that[OF len-query-idxs len-openings indices tables])
qed
have partial:
  accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs' trace-openings
proof –
  have all-indices:
     $\forall i < \text{rounds}.$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs' ! i)
  using indices by blast
have all-tables:
   $\forall i < \text{rounds}.$ 

```

```

    partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) final-state
  using tables fr-eq by simp
show ?thesis
  unfolding accepted-with-partial-trace-openings-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state])
    (use len-query-idxs len-openings all-indices all-tables in simp-all)
qed
have len-table: length trace-table = scale * clength
  using accepted-with-bound-tables-shapes(1)[OF bound] by simp
have partial-candidate:
  partial-trace-table-candidate trace-table trace-openings
  by (rule accepted-bound-trace-table-partial-candidate-if-clean
    [OF bind len-table partial clean])
show ?thesis
proof -
  have witness:
     $\exists$  out query-idxs trace-openings trace-bs dg comp-bs
      f-fri-roots f-final header-as composition-fri-roots final rest.
    out  $\in$  set-dist (execute verify-monad s)  $\wedge$ 
    accepted-with-partial-trace-openings s out fr query-idxs
      trace-openings  $\wedge$ 
    partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
    accepted-fri-challenges s out trace-bs dg comp-bs  $\wedge$ 
    verifier-header-transcript s fr f-fri-roots f-final header-as dg
      composition-fri-roots final rest
  using outcome partial partial-candidate challenges header by blast
show ?thesis
  using witness
  unfolding trace-fri-supported-root-partial-trace-table-candidates-def
  by simp
qed
qed

lemma alpha-header-supported-trace-table-candidate-state-imp-partial-candidate:
  assumes state:
    alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
    f-final trace-table final-state
  and clean:  $\neg$  hash-map-output-collision final-state
  shows
    trace-table  $\in$ 
    alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
    f-final
proof -
  from state obtain result composition-table as query-idxs dg
    composition-fri-roots final rest where
    outcome:
      Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
  and bound:

```

```

    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
and bind:
    merkle-root-binds-table fr trace-table final-state
unfolding alpha-header-supported-trace-table-candidate-state-def
by blast
from verify-monad-header-extraction[OF outcome]
obtain fr' f-fl f-final' as' dg' fl final' query-state where
  header':
    verifier-header-transcript s fr' (map snd f-fl) f-final' as' dg'
    (map snd fl) final' (PTranscript query-state)
and query-out:
    Some (result, final-state) ∈
      set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr' f-fl f-final' as' fl final')
          rounds)
        query-state)
by (rule verify-monad-header-query-extraction[OF outcome])
have eqs:
  fr' = fr ∧ map snd f-fl = f-fri-roots ∧
  f-final' = f-final ∧ as' = as ∧ dg' = dg ∧
  map snd fl = composition-fri-roots ∧ final' = final ∧
  PTranscript query-state = rest
using verifier-header-transcript-unique[OF header' header] by simp
obtain query-idxs' trace-openings where
  len-query-idxs: length query-idxs' = rounds
and len-openings: length trace-openings = rounds
and indices:
  ∧ i. i < rounds ⇒
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs' ! i)
and tables:
  ∧ i. i < rounds ⇒
    partial-authenticated-table fr'
      (scale * clength) (trace-openings ! i) final-state
proof (rule ntimes-verifier-query-rounds-authenticated-trace-openings-no-raws
  [OF query-out])
fix query-idxs' trace-openings
assume len-query-idxs: length query-idxs' = rounds
and len-openings: length trace-openings = rounds
and indices:
  ∧ i. i < rounds ⇒
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs' ! i)

```

```

and tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table fr'
      (scale * clength) (trace-openings ! i) final-state
show ?thesis
  by (rule that[OF len-query-idxs len-openings indices tables])
qed
have partial:
  accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs' trace-openings
proof –
  have all-indices:
     $\forall i < \text{rounds}.$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs' ! i)
    using indices by blast
  have all-tables:
     $\forall i < \text{rounds}.$ 
      partial-authenticated-table fr (scale * clength)
        (trace-openings ! i) final-state
    using tables eqs by simp
  show ?thesis
  unfolding accepted-with-partial-trace-openings-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state])
    (use len-query-idxs len-openings all-indices all-tables in simp-all)
qed
have len-table: length trace-table = scale * clength
  using accepted-with-bound-tables-shapes(1)[OF bound] by simp
have partial-candidate:
  partial-trace-table-candidate trace-table trace-openings
  by (rule accepted-bound-trace-table-partial-candidate-if-clean
    [OF bind len-table partial clean])
show ?thesis
  unfolding alpha-header-supported-partial-trace-table-candidates-def
  by (intro CollectI exI[of - Some (result, final-state)]
    exI[of - query-idxs] exI[of - trace-openings]
    exI[of - as] exI[of - dg]
    exI[of - composition-fri-roots] exI[of - final]
    exI[of - rest] conjI)
    (use outcome partial partial-candidate header in simp-all)
qed

lemma trace-fri-root-trace-table-candidate-states-common-extension-unique-if-clean:
assumes clean:  $\neg \text{hash-map-output-collision } u$ 
and candidate1:
  trace-fri-root-trace-table-candidate-state s fr trace-table
  final-state
and candidate2:
  trace-fri-root-trace-table-candidate-state s fr trace-table'

```

```

    final-state'
  and ext1: final-state ≤ u
  and ext2: final-state' ≤ u
  shows trace-table = trace-table'
proof -
  from candidate1 obtain result composition-table as query-idxs trace-bs dg
    comp-bs f-fri-roots f-final header-as composition-fri-roots final rest
  where
    bound1:
      accepted-with-bound-tables s (Some (result, final-state))
      trace-table composition-table as query-idxs
    and bind1: merkle-root-binds-table fr trace-table final-state
    unfolding trace-fri-root-trace-table-candidate-state-def by blast
  from candidate2 obtain result' composition-table' as' query-idxs' trace-bs'
    dg' comp-bs' f-fri-roots' f-final' header-as'
    composition-fri-roots' final' rest' where
    bound2:
      accepted-with-bound-tables s (Some (result', final-state'))
      trace-table' composition-table' as' query-idxs'
    and bind2: merkle-root-binds-table fr trace-table' final-state'
    unfolding trace-fri-root-trace-table-candidate-state-def by blast
  have same-len: length trace-table = length trace-table'
  using accepted-with-bound-tables-shapes(1)[OF bound1]
    accepted-with-bound-tables-shapes(1)[OF bound2]
  by simp
  have nonempty: trace-table ≠ []
  using accepted-with-bound-tables-shapes(1)[OF bound1]
    eval-domain-nontrivial by auto
  show ?thesis
  by (rule merkle-root-binds-same-length-tables-common-extension-unique-if-clean
    [OF clean bind1 bind2 ext1 ext2 same-len nonempty])
qed

```

```

lemma trace-fri-root-trace-table-candidate-states-compatible-unique-if-clean-merge:
  assumes compatible: hash-maps-compatible final-state final-state'
  and clean:
    ¬ hash-map-output-collision
    (hash-state-merge final-state final-state')
  and candidate1:
    trace-fri-root-trace-table-candidate-state s fr trace-table
    final-state
  and candidate2:
    trace-fri-root-trace-table-candidate-state s fr trace-table'
    final-state'
  shows trace-table = trace-table'
proof -
  from candidate1 obtain result composition-table as query-idxs trace-bs dg
    comp-bs f-fri-roots f-final header-as composition-fri-roots final rest
  where

```

```

bound1:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
and bind1: merkle-root-binds-table fr trace-table final-state
unfolding trace-fri-root-trace-table-candidate-state-def by blast
from candidate2 obtain result' composition-table' as' query-idxs' trace-bs'
  dg' comp-bs' f-fri-roots' f-final' header-as'
  composition-fri-roots' final' rest' where
bound2:
  accepted-with-bound-tables s (Some (result', final-state'))
  trace-table' composition-table' as' query-idxs'
and bind2: merkle-root-binds-table fr trace-table' final-state'
unfolding trace-fri-root-trace-table-candidate-state-def by blast
have same-len: length trace-table = length trace-table'
using accepted-with-bound-tables-shapes(1)[OF bound1]
  accepted-with-bound-tables-shapes(1)[OF bound2]
by simp
have nonempty: trace-table ≠ []
using accepted-with-bound-tables-shapes(1)[OF bound1]
  eval-domain-nontrivial by auto
show ?thesis
by (rule
  merkle-root-binds-same-length-tables-compatible-states-unique-if-clean-merge
  [OF compatible clean bind1 bind2 same-len nonempty])
qed

```

lemma trace-fri-root-trace-table-candidate-states-merkle-compatible-unique-if-clean-merge:

```

assumes compatible:
  merkle-hash-maps-compatible final-state final-state'
and clean:
  ¬ hash-map-output-collision
  (merkle-hash-state-merge final-state final-state')
and candidate1:
  trace-fri-root-trace-table-candidate-state s fr trace-table
  final-state
and candidate2:
  trace-fri-root-trace-table-candidate-state s fr trace-table'
  final-state'
shows trace-table = trace-table'
proof –
from candidate1 obtain result composition-table as query-idxs trace-bs dg
  comp-bs f-fri-roots f-final header-as composition-fri-roots final rest
where
bound1:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
and bind1: merkle-root-binds-table fr trace-table final-state
unfolding trace-fri-root-trace-table-candidate-state-def by blast
from candidate2 obtain result' composition-table' as' query-idxs' trace-bs'

```

```

    dg' comp-bs' f-fri-roots' f-final' header-as'
    composition-fri-roots' final' rest' where
bound2:
    accepted-with-bound-tables s (Some (result', final-state'))
    trace-table' composition-table' as' query-idxs'
and bind2: merkle-root-binds-table fr trace-table' final-state'
unfolding trace-fri-root-trace-table-candidate-state-def by blast
have same-len: length trace-table = length trace-table'
using accepted-with-bound-tables-shapes(1)[OF bound1]
    accepted-with-bound-tables-shapes(1)[OF bound2]
by simp
have nonempty: trace-table  $\neq$  []
using accepted-with-bound-tables-shapes(1)[OF bound1]
    eval-domain-nontrivial by auto
show ?thesis
by (rule
    merkle-root-binds-same-length-tables-merkle-compatible-unique-if-clean-merge
    [OF compatible clean bind1 bind2 same-len nonempty])
qed

lemma trace-fri-root-trace-table-candidates-subset-singleton-if-common-clean-extension:
assumes clean:  $\neg$  hash-map-output-collision u
and common-ext:
     $\bigwedge$  trace-table final-state.
    trace-fri-root-trace-table-candidate-state s fr trace-table
    final-state  $\implies$ 
    final-state  $\leq$  u
shows
     $\exists$  trace-table.
    trace-fri-root-trace-table-candidates s fr  $\subseteq$  {trace-table}
proof (cases trace-fri-root-trace-table-candidates s fr = {})
case True
then show ?thesis
by (intro exI[of - []]) simp
next
case False
then obtain trace-table0 where table0:
    trace-table0  $\in$  trace-fri-root-trace-table-candidates s fr
by blast
then obtain final-state0 where state0:
    trace-fri-root-trace-table-candidate-state s fr trace-table0
    final-state0
using trace-fri-root-trace-table-candidates-iff-state by blast
have ext0: final-state0  $\leq$  u
by (rule common-ext[OF state0])
have subset:
    trace-fri-root-trace-table-candidates s fr  $\subseteq$  {trace-table0}
proof
fix trace-table

```



```

assume table:
  trace-table ∈ trace-fri-root-trace-table-candidates s fr
then obtain final-state where state:
  trace-fri-root-trace-table-candidate-state s fr trace-table
    final-state
using trace-fri-root-trace-table-candidates-iff-state by blast
have ext: final-state ≤ u
  by (rule common-ext[OF state])
have trace-table = trace-table0
  by (rule
    trace-fri-root-trace-table-candidate-states-common-extension-unique-if-clean
    [OF clean state state0 ext ext0])
then show trace-table ∈ {trace-table0}
  by simp
qed
then show ?thesis
  by blast
qed

lemma trace-fri-root-trace-table-candidates-subset-singleton-if-pairwise-clean-merge:
assumes pairwise:
   $\bigwedge \text{trace-table final-state trace-table' final-state'.$ 
  trace-fri-root-trace-table-candidate-state s fr trace-table
    final-state  $\implies$ 
  trace-fri-root-trace-table-candidate-state s fr trace-table'
    final-state'  $\implies$ 
  hash-maps-compatible final-state final-state' ∧
   $\neg$  hash-map-output-collision
    (hash-state-merge final-state final-state')
shows
   $\exists \text{trace-table.}$ 
  trace-fri-root-trace-table-candidates s fr ⊆ {trace-table}
proof (cases trace-fri-root-trace-table-candidates s fr = {})
  case True
    then show ?thesis
      by (intro exI[of - []]) simp
  next
    case False
      then obtain trace-table0 where table0:
        trace-table0 ∈ trace-fri-root-trace-table-candidates s fr
        by blast
      then obtain final-state0 where state0:
        trace-fri-root-trace-table-candidate-state s fr trace-table0
          final-state0
        using trace-fri-root-trace-table-candidates-iff-state by blast
      have subset:
        trace-fri-root-trace-table-candidates s fr ⊆ {trace-table0}
      proof
        fix trace-table

```

```

assume table:
  trace-table  $\in$  trace-fri-root-trace-table-candidates s fr
then obtain final-state where state:
  trace-fri-root-trace-table-candidate-state s fr trace-table
  final-state
using trace-fri-root-trace-table-candidates-iff-state by blast
have compatible:
  hash-maps-compatible final-state final-state0
using pairwise[OF state state0] by simp
have clean:
   $\neg$  hash-map-output-collision
  (hash-state-merge final-state final-state0)
using pairwise[OF state state0] by simp
have trace-table = trace-table0
by (rule
  trace-fri-root-trace-table-candidate-states-compatible-unique-if-clean-merge
  [OF compatible clean state state0])
then show trace-table  $\in$  {trace-table0}
by simp
qed
then show ?thesis
by blast
qed

lemma trace-fri-supported-root-trace-table-candidates-subset-singleton-if-pairwise-clean-merge:
assumes pairwise:
   $\bigwedge$  trace-table final-state trace-table' final-state'.
  trace-fri-supported-root-trace-table-candidate-state s fr trace-table
  final-state  $\implies$ 
  trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
  final-state'  $\implies$ 
  hash-maps-compatible final-state final-state'  $\wedge$ 
   $\neg$  hash-map-output-collision
  (hash-state-merge final-state final-state')
shows
   $\exists$  trace-table.
  trace-fri-supported-root-trace-table-candidates s fr  $\subseteq$ 
  {trace-table}
proof (cases trace-fri-supported-root-trace-table-candidates s fr = {})
case True
then show ?thesis
by (intro exI[of - []]) simp
next
case False
then obtain trace-table0 where table0:
  trace-table0  $\in$  trace-fri-supported-root-trace-table-candidates s fr
by blast
then obtain final-state0 where state0:
  trace-fri-supported-root-trace-table-candidate-state s fr trace-table0

```

```

    final-state0
  using trace-fri-supported-root-trace-table-candidates-iff-state by blast
have state0-plain:
  trace-fri-root-trace-table-candidate-state s fr trace-table0
  final-state0
by (rule
  trace-fri-supported-root-trace-table-candidate-state-imp-candidate-state
  [OF state0])
have subset:
  trace-fri-supported-root-trace-table-candidates s fr  $\subseteq$ 
  {trace-table0}
proof
  fix trace-table
  assume table:
    trace-table  $\in$  trace-fri-supported-root-trace-table-candidates s fr
  then obtain final-state where state:
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table
    final-state
  using trace-fri-supported-root-trace-table-candidates-iff-state by blast
have state-plain:
  trace-fri-root-trace-table-candidate-state s fr trace-table
  final-state
by (rule
  trace-fri-supported-root-trace-table-candidate-state-imp-candidate-state
  [OF state])
have compatible:
  hash-maps-compatible final-state final-state0
  using pairwise[OF state state0] by simp
have clean:
   $\neg$  hash-map-output-collision
  (hash-state-merge final-state final-state0)
  using pairwise[OF state state0] by simp
have trace-table = trace-table0
  by (rule
    trace-fri-root-trace-table-candidate-states-compatible-unique-if-clean-merge
    [OF compatible clean state-plain state0-plain])
then show trace-table  $\in$  {trace-table0}
  by simp
qed
then show ?thesis
  by blast
qed

lemma trace-fri-supported-root-trace-table-candidates-subset-singleton-if-pairwise-no-value-conflict-clean-merge:
  assumes pairwise:
     $\bigwedge$  trace-table final-state trace-table' final-state'.
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table
    final-state  $\implies$ 
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table'

```

```

    final-state'  $\implies$ 
     $\neg$  hash-map-value-conflict final-state final-state'  $\wedge$ 
     $\neg$  hash-map-output-collision
    (hash-state-merge final-state final-state')
  shows
     $\exists$  trace-table.
    trace-fri-supported-root-trace-table-candidates s fr  $\subseteq$ 
    {trace-table}
  proof (rule
    trace-fri-supported-root-trace-table-candidates-subset-singleton-if-pairwise-clean-merge)
  fix trace-table final-state trace-table' final-state'
  assume candidate:
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table
    final-state
  and candidate':
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
    final-state'
  have no-conflict:
     $\neg$  hash-map-value-conflict final-state final-state'
  using pairwise[OF candidate candidate'] by simp
  have compatible:
    hash-maps-compatible final-state final-state'
  using no-conflict
  unfolding hash-maps-compatible-iff-no-value-conflict .
  have clean:
     $\neg$  hash-map-output-collision
    (hash-state-merge final-state final-state')
  using pairwise[OF candidate candidate'] by simp
  show hash-maps-compatible final-state final-state'  $\wedge$ 
     $\neg$  hash-map-output-collision
    (hash-state-merge final-state final-state')
  using compatible clean by simp
qed

lemma trace-fri-supported-root-trace-table-candidates-subset-singleton-if-pairwise-merkle-clean-merge:
  assumes pairwise:
     $\bigwedge$  trace-table final-state trace-table' final-state'.
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table
    final-state  $\implies$ 
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
    final-state'  $\implies$ 
    merkle-hash-maps-compatible final-state final-state'  $\wedge$ 
     $\neg$  hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
  shows
     $\exists$  trace-table.
    trace-fri-supported-root-trace-table-candidates s fr  $\subseteq$ 
    {trace-table}
  proof (cases trace-fri-supported-root-trace-table-candidates s fr = {})

```

```

case True
then show ?thesis
  by (intro exI[of - []]) simp
next
case False
then obtain trace-table0 where table0:
  trace-table0  $\in$  trace-fri-supported-root-trace-table-candidates s fr
  by blast
then obtain final-state0 where state0:
  trace-fri-supported-root-trace-table-candidate-state s fr trace-table0
  final-state0
  using trace-fri-supported-root-trace-table-candidates-iff-state by blast
have state0-plain:
  trace-fri-root-trace-table-candidate-state s fr trace-table0
  final-state0
  by (rule
    trace-fri-supported-root-trace-table-candidate-state-imp-candidate-state
    [OF state0])
have subset:
  trace-fri-supported-root-trace-table-candidates s fr  $\subseteq$ 
    {trace-table0}
proof
  fix trace-table
  assume table:
    trace-table  $\in$  trace-fri-supported-root-trace-table-candidates s fr
  then obtain final-state where state:
    trace-fri-supported-root-trace-table-candidate-state s fr trace-table
    final-state
    using trace-fri-supported-root-trace-table-candidates-iff-state by blast
  have state-plain:
    trace-fri-root-trace-table-candidate-state s fr trace-table
    final-state
    by (rule
      trace-fri-supported-root-trace-table-candidate-state-imp-candidate-state
      [OF state])
  have compatible:
    merkle-hash-maps-compatible final-state final-state0
    using pairwise[OF state state0] by simp
  have clean:
     $\neg$  hash-map-output-collision
      (merkle-hash-state-merge final-state final-state0)
    using pairwise[OF state state0] by simp
  have trace-table = trace-table0
    by (rule
      trace-fri-root-trace-table-candidate-states-merkle-compatible-unique-if-clean-merge
      [OF compatible clean state-plain state0-plain])
  then show trace-table  $\in$  {trace-table0}
    by simp
qed

```

then show *?thesis*
by *blast*
qed

lemma *trace-fri-supported-root-trace-table-candidates-subset-singleton-if-pairwise-no-merkle-value-conflict-clean-mer*

assumes *pairwise*:

$\bigwedge \text{trace-table } \text{final-state } \text{trace-table}' \text{ final-state'}$.
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr } \text{trace-table}$
 $\text{final-state} \implies$
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr } \text{trace-table}'$
 $\text{final-state}' \implies$
 $\neg \text{merkle-hash-value-conflict } \text{final-state } \text{final-state}' \wedge$
 $\neg \text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge } \text{final-state } \text{final-state}')$

shows

$\exists \text{trace-table}$.
 $\text{trace-fri-supported-root-trace-table-candidates } s \text{ fr } \subseteq$
 $\{\text{trace-table}\}$

proof (*rule*

trace-fri-supported-root-trace-table-candidates-subset-singleton-if-pairwise-merkle-clean-merge)

fix *trace-table final-state trace-table' final-state'*

assume *candidate*:

$\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr } \text{trace-table}$
 final-state

and *candidate'*:

$\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr } \text{trace-table}'$
 $\text{final-state}'$

have *no-conflict*:

$\neg \text{merkle-hash-value-conflict } \text{final-state } \text{final-state}'$
using *pairwise*[*OF candidate candidate'*] **by** *simp*

have *compatible*:

$\text{merkle-hash-maps-compatible } \text{final-state } \text{final-state}'$
using *no-conflict*
unfolding *merkle-hash-maps-compatible-iff-no-value-conflict* .

have *clean*:

$\neg \text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge } \text{final-state } \text{final-state}')$
using *pairwise*[*OF candidate candidate'*] **by** *simp*

show $\text{merkle-hash-maps-compatible } \text{final-state } \text{final-state}' \wedge$

$\neg \text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge } \text{final-state } \text{final-state}')$

using *compatible clean* **by** *simp*

qed

lemma *trace-fri-supported-root-trace-table-candidates-subset-singleton-if-common-clean-merkle-extension*:

assumes *clean*: $\neg \text{hash-map-output-collision } u$

and *common-ext*:

$\bigwedge \text{trace-table } \text{final-state}$.
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr } \text{trace-table}$

$final-state \implies$
 $merkle-hash-extends\ final-state\ u$
shows
 $\exists\ trace-table.$
 $trace-fri-supported-root-trace-table-candidates\ s\ fr \subseteq$
 $\{trace-table\}$
proof (rule
 $trace-fri-supported-root-trace-table-candidates-subset-singleton-if-pairwise-no-merkle-value-conflict-clean-merge)$
fix $trace-table\ final-state\ trace-table'\ final-state'$
assume $candidate:$
 $trace-fri-supported-root-trace-table-candidate-state\ s\ fr\ trace-table$
 $final-state$
and $candidate':$
 $trace-fri-supported-root-trace-table-candidate-state\ s\ fr\ trace-table'$
 $final-state'$
have $ext:$ $merkle-hash-extends\ final-state\ u$
by (rule $common-ext[OF\ candidate]$)
have $ext':$ $merkle-hash-extends\ final-state'\ u$
by (rule $common-ext[OF\ candidate']$)
show $\neg\ merkle-hash-value-conflict\ final-state\ final-state' \wedge$
 $\neg\ hash-map-output-collision$
 $(merkle-hash-state-merge\ final-state\ final-state')$
by (rule $common-clean-merkle-extension-imp-pairwise-merkle-clean$
 $[OF\ ext\ ext'\ clean]$)
qed

lemma $trace-fri-supported-root-trace-table-candidate-states-distinct-imp-merkle-conflict-or-merge-collision:$
assumes $candidate:$
 $trace-fri-supported-root-trace-table-candidate-state\ s\ fr\ trace-table$
 $final-state$
and $candidate':$
 $trace-fri-supported-root-trace-table-candidate-state\ s\ fr\ trace-table'$
 $final-state'$
and $distinct-tables:$ $trace-table \neq trace-table'$
shows
 $merkle-hash-value-conflict\ final-state\ final-state' \vee$
 $hash-map-output-collision$
 $(merkle-hash-state-merge\ final-state\ final-state')$
proof (rule $ccontr$)
assume $no-bad:$
 $\neg\ (merkle-hash-value-conflict\ final-state\ final-state' \vee$
 $hash-map-output-collision$
 $(merkle-hash-state-merge\ final-state\ final-state'))$
have $compatible:$
 $merkle-hash-maps-compatible\ final-state\ final-state'$
using $no-bad$ **unfolding** $merkle-hash-maps-compatible-iff-no-value-conflict$
by $simp$
have $clean:$
 $\neg\ hash-map-output-collision$

```

    (merkle-hash-state-merge final-state final-state')
  using no-bad by simp
have candidate-plain:
  trace-fri-root-trace-table-candidate-state s fr trace-table
  final-state
by (rule
  trace-fri-supported-root-trace-table-candidate-state-imp-candidate-state
  [OF candidate])
have candidate-plain':
  trace-fri-root-trace-table-candidate-state s fr trace-table'
  final-state'
by (rule
  trace-fri-supported-root-trace-table-candidate-state-imp-candidate-state
  [OF candidate'])
have trace-table = trace-table'
by (rule
  trace-fri-root-trace-table-candidate-states-merkle-compatible-unique-if-clean-merge
  [OF compatible clean candidate-plain candidate-plain'])
then show False
  using distinct-tables by contradiction
qed

lemma trace-fri-supported-root-trace-table-candidates-not-singleton-imp-pairwise-merkle-bad:
  assumes not-unique:
     $\neg (\exists \text{ trace-table. } \text{trace-fri-supported-root-trace-table-candidates } s \text{ fr } \subseteq \{\text{trace-table}\})$ 
  shows
     $\exists \text{ trace-table final-state trace-table' final-state'. } \text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table final-state} \wedge \text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table' final-state'} \wedge \text{trace-table} \neq \text{trace-table'} \wedge (\text{merkle-hash-value-conflict final-state final-state'} \vee \text{hash-map-output-collision (merkle-hash-state-merge final-state final-state')})$ 
proof -
  let ?C = trace-fri-supported-root-trace-table-candidates s fr
  have nonempty: ?C  $\neq \{\}$ 
  using not-unique by auto
  then obtain trace-table where table: trace-table  $\in$  ?C
  by blast
  have  $\neg ?C \subseteq \{\text{trace-table}\}$ 
  using not-unique by blast
  then obtain trace-table' where table':
    trace-table'  $\in$  ?C
  and distinct-tables: trace-table  $\neq$  trace-table'
  by auto

```


from *table* **obtain** *final-state* **where** *candidate*:
trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state
using *trace-fri-supported-root-trace-table-candidates-iff-state* **by** *blast*
from *table'* **obtain** *final-state'* **where** *candidate'*:
trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
final-state'
using *trace-fri-supported-root-trace-table-candidates-iff-state* **by** *blast*
have *bad*:
merkle-hash-value-conflict final-state final-state' ∨
hash-map-output-collision
(merkle-hash-state-merge final-state final-state')
by *(rule*
trace-fri-supported-root-trace-table-candidate-states-distinct-imp-merkle-conflict-or-merge-collision
[OF candidate candidate' distinct-tables])
show *?thesis*
using *candidate candidate' distinct-tables bad* **by** *blast*
qed

definition *trace-fri-supported-root-pairwise-merkle-bad*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
trace-fri-supported-root-pairwise-merkle-bad s fr $\longleftrightarrow$
 $(\exists \text{ trace-table final-state trace-table' final-state'.$
trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state $\wedge$
trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
final-state' $\wedge$
trace-table $\neq$ *trace-table'* $\wedge$
 $(\text{merkle-hash-value-conflict final-state final-state'} \vee$
 $\text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge final-state final-state'})))$

definition *trace-fri-supported-pairwise-merkle-bad*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
trace-fri-supported-pairwise-merkle-bad s $\longleftrightarrow$
 $(\exists \text{ fr. trace-fri-supported-root-pairwise-merkle-bad s fr})$

lemma *trace-fri-supported-root-pairwise-merkle-bad-extends-initial*:
assumes *bad: trace-fri-supported-root-pairwise-merkle-bad s fr*
obtains *trace-table final-state trace-table' final-state'*
where
trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state
trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
final-state'
trace-table $\neq$ *trace-table'*
merkle-hash-value-conflict final-state final-state' ∨

hash-map-output-collision
 (merkle-hash-state-merge final-state final-state')
 $s \leq \text{final-state}$
 $s \leq \text{final-state}'$
proof –
from *bad* **obtain** *trace-table final-state trace-table' final-state'* **where**
candidate:
trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state
and *candidate':*
trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
final-state'
and *distinct:* $\text{trace-table} \neq \text{trace-table}'$
and *pair-bad:*
merkle-hash-value-conflict final-state final-state' $\vee$
hash-map-output-collision
 (merkle-hash-state-merge final-state final-state')
unfolding *trace-fri-supported-root-pairwise-merkle-bad-def* **by** *blast*
have *ext:* $s \leq \text{final-state}$
by (rule *trace-fri-supported-root-trace-table-candidate-state-hash-extends*
 [OF *candidate*])
have *ext':* $s \leq \text{final-state}'$
by (rule *trace-fri-supported-root-trace-table-candidate-state-hash-extends*
 [OF *candidate'*])
show *?thesis*
by (rule *that*[OF *candidate candidate' distinct pair-bad ext ext'*])
qed

definition *trace-fri-supported-root-pairwise-coupling-bad*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow \text{bool}$
where
trace-fri-supported-root-pairwise-coupling-bad s fr $\longleftrightarrow$
 ($\exists \text{ trace-table final-state trace-table' final-state'.$
trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state $\wedge$
trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
final-state' $\wedge$
trace-table $\neq$ trace-table' $\wedge$
merkle-hash-pairwise-coupling-bad final-state final-state')

definition *trace-fri-supported-pairwise-coupling-bad*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
trace-fri-supported-pairwise-coupling-bad s $\longleftrightarrow$
 ($\exists \text{ fr. trace-fri-supported-root-pairwise-coupling-bad s fr}$)

lemma *trace-fri-supported-root-pairwise-merkle-bad-imp-local-collision-or-coupling-bad:*
assumes *bad:* *trace-fri-supported-root-pairwise-merkle-bad s fr*
shows

$(\exists \text{ trace-table final-state trace-table' final-state'.$
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table}$
 $\text{final-state} \wedge$
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table'}$
 $\text{final-state'} \wedge$
 $\text{trace-table} \neq \text{trace-table'} \wedge$
 $(\text{hash-map-output-collision final-state} \vee$
 $\text{hash-map-output-collision final-state'})) \vee$
 $\text{trace-fri-supported-root-pairwise-coupling-bad } s \text{ fr}$
proof –
from *bad* **obtain** $\text{trace-table final-state trace-table' final-state'}$
where *candidate*:
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table}$
 final-state
and *candidate'*:
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table'}$
 final-state'
and *distinct*: $\text{trace-table} \neq \text{trace-table'}$
and *pair-bad*:
 $\text{merkle-hash-value-conflict final-state final-state'} \vee$
 $\text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge final-state final-state'})$
unfolding *trace-fri-supported-root-pairwise-merkle-bad-def* **by** *blast*
have *decomp*:
 $\text{hash-map-output-collision final-state} \vee$
 $\text{hash-map-output-collision final-state'} \vee$
 $\text{merkle-hash-pairwise-coupling-bad final-state final-state'}$
by (*rule merkle-hash-pairwise-bad-imp-local-collision-or-coupling-bad*
 $[\text{OF pair-bad}]$)
show *?thesis*
using *candidate candidate' distinct decomp*
unfolding *trace-fri-supported-root-pairwise-coupling-bad-def* **by** *blast*
qed

lemma *trace-fri-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad*:
assumes *bad*: *trace-fri-supported-pairwise-merkle-bad s*
shows

$(\exists \text{ fr trace-table final-state trace-table' final-state'.$
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table}$
 $\text{final-state} \wedge$
 $\text{trace-fri-supported-root-trace-table-candidate-state } s \text{ fr trace-table'}$
 $\text{final-state'} \wedge$
 $\text{trace-table} \neq \text{trace-table'} \wedge$
 $(\text{hash-map-output-collision final-state} \vee$
 $\text{hash-map-output-collision final-state'})) \vee$
 $\text{trace-fri-supported-pairwise-coupling-bad } s$

proof –
from *bad* **obtain** *fr* **where** *root-bad*:
 $\text{trace-fri-supported-root-pairwise-merkle-bad } s \text{ fr}$

```

    unfolding trace-fri-supported-pairwise-merkle-bad-def by blast
  from
    trace-fri-supported-root-pairwise-merkle-bad-imp-local-collision-or-coupling-bad
    [OF root-bad]
  show ?thesis
    unfolding trace-fri-supported-pairwise-coupling-bad-def by blast
qed

lemma trace-fri-supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad:
  assumes bad: trace-fri-supported-pairwise-merkle-bad s
  shows
    supported-hash-output-collision-possible s ∨
    trace-fri-supported-pairwise-coupling-bad s
proof -
  have decomp:
    (∃ fr trace-table final-state trace-table' final-state'.
      trace-fri-supported-root-trace-table-candidate-state s fr trace-table
        final-state ∧
      trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
        final-state' ∧
      trace-table ≠ trace-table' ∧
      (hash-map-output-collision final-state ∨
       hash-map-output-collision final-state')) ∨
      trace-fri-supported-pairwise-coupling-bad s
  by (rule
      trace-fri-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad
      [OF bad])
  then show ?thesis
proof
  assume local:
    ∃ fr trace-table final-state trace-table' final-state'.
      trace-fri-supported-root-trace-table-candidate-state s fr trace-table
        final-state ∧
      trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
        final-state' ∧
      trace-table ≠ trace-table' ∧
      (hash-map-output-collision final-state ∨
       hash-map-output-collision final-state')
  then obtain fr trace-table final-state trace-table' final-state' where
    candidate:
      trace-fri-supported-root-trace-table-candidate-state s fr trace-table
        final-state
  and candidate':
      trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
        final-state'
  and collision:
      hash-map-output-collision final-state ∨
      hash-map-output-collision final-state'
  by blast

```

from *collision* **show** ?thesis
proof
 assume *collision-final*: *hash-map-output-collision final-state*
 from *candidate* **obtain** *result composition-table as query-idxs trace-bs dg*
 comp-bs f-fri-roots f-final header-as composition-fri-roots final
 rest **where**
 outcome:
 Some (result, final-state) ∈ set-dist (execute verify-monad s)
 and *bound*:
 accepted-with-bound-tables s (Some (result, final-state))
 trace-table composition-table as query-idxs
 unfolding *trace-fri-supported-root-trace-table-candidate-state-def*
 by *blast*
 have *supported-hash-output-collision-possible s*
 by (rule
 accepted-with-bound-tables-hash-collision-imp-supported-hash-output-collision-possible
 [*OF outcome bound collision-final*])
 then show ?thesis **by** *simp*
next
 assume *collision-final'*: *hash-map-output-collision final-state'*
 from *candidate'* **obtain** *result' composition-table' as' query-idxs'*
 trace-bs' dg' comp-bs' f-fri-roots' f-final' header-as'
 composition-fri-roots' final' rest' **where**
 outcome':
 Some (result', final-state') ∈ set-dist (execute verify-monad s)
 and *bound'*:
 accepted-with-bound-tables s (Some (result', final-state'))
 trace-table' composition-table' as' query-idxs'
 unfolding *trace-fri-supported-root-trace-table-candidate-state-def*
 by *blast*
 have *supported-hash-output-collision-possible s*
 by (rule
 accepted-with-bound-tables-hash-collision-imp-supported-hash-output-collision-possible
 [*OF outcome' bound' collision-final'*])
 then show ?thesis **by** *simp*
qed
next
 assume *trace-fri-supported-pairwise-coupling-bad s*
 then show ?thesis **by** *simp*
qed
qed

lemma *trace-fri-supported-root-candidate-unique-if-no-pairwise-merkle-bad*:
 assumes *no-bad*: $\neg$ *trace-fri-supported-root-pairwise-merkle-bad s fr*
 shows
 $\exists$ *trace-table*.
 trace-fri-supported-root-trace-table-candidates s fr $\subseteq$
 {*trace-table*}
proof (rule *ccontr*)

assume *not-unique*:
 $\neg (\exists \text{ trace-table.}$
 $\text{trace-fri-supported-root-trace-table-candidates } s \text{ fr} \subseteq$
 $\{\text{trace-table}\})$
have *trace-fri-supported-root-pairwise-merkle-bad* *s fr*
using
 $\text{trace-fri-supported-root-trace-table-candidates-not-singleton-imp-pairwise-merkle-bad}$
 $[OF \text{ not-unique}]$
unfolding *trace-fri-supported-root-pairwise-merkle-bad-def*
by *blast*
then show *False*
using *no-bad* **by** *contradiction*
qed

lemma *trace-fri-supported-root-candidate-unique-if-no-global-pairwise-merkle-bad*:
assumes *no-bad*: $\neg \text{trace-fri-supported-pairwise-merkle-bad } s$
shows
 $\exists \text{ trace-table.}$
 $\text{trace-fri-supported-root-trace-table-candidates } s \text{ fr} \subseteq$
 $\{\text{trace-table}\}$
proof (*rule trace-fri-supported-root-candidate-unique-if-no-pairwise-merkle-bad*)
show $\neg \text{trace-fri-supported-root-pairwise-merkle-bad } s \text{ fr}$
using *no-bad* **unfolding** *trace-fri-supported-pairwise-merkle-bad-def* **by** *blast*
qed

definition *trace-fri-root-union-bad-sets*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow 'f \Rightarrow 'f \text{ list set}$
where
 $\text{trace-fri-root-union-bad-sets } s \text{ bad-sets fr} =$
 $\{\text{trace-bs} \in \text{fri-challenge-space } (\text{ceil-log } \text{clength}).$
 $\exists \text{ trace-table} \in \text{trace-fri-root-trace-table-candidates } s \text{ fr.}$
 $\text{trace-bs} \in \text{bad-sets trace-table}\}$

definition *trace-fri-supported-root-union-bad-sets*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow 'f \Rightarrow 'f \text{ list set}$
where
 $\text{trace-fri-supported-root-union-bad-sets } s \text{ bad-sets fr} =$
 $\{\text{trace-bs} \in \text{fri-challenge-space } (\text{ceil-log } \text{clength}).$
 $\exists \text{ trace-table} \in$
 $\text{trace-fri-supported-root-trace-table-candidates } s \text{ fr.}$
 $\text{trace-bs} \in \text{bad-sets trace-table}\}$

definition *trace-fri-supported-root-partial-union-bad-sets*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow 'f \Rightarrow 'f \text{ list set}$
where
 $\text{trace-fri-supported-root-partial-union-bad-sets } s \text{ bad-sets fr} =$

$\{ \text{trace-bs} \in \text{fri-challenge-space } (\text{ceil-log } \text{clength}).$
 $\exists \text{ trace-table} \in$
 $\text{trace-fri-supported-root-partial-trace-table-candidates } s \text{ fr.}$
 $\text{trace-bs} \in \text{bad-sets } \text{trace-table} \}$

lemma *trace-fri-root-union-bad-sets-subset:*
 $\text{trace-fri-root-union-bad-sets } s \text{ bad-sets } \text{fr} \subseteq$
 $\text{fri-challenge-space } (\text{ceil-log } \text{clength})$
unfolding *trace-fri-root-union-bad-sets-def* **by** *auto*

lemma *trace-fri-supported-root-union-bad-sets-subset:*
 $\text{trace-fri-supported-root-union-bad-sets } s \text{ bad-sets } \text{fr} \subseteq$
 $\text{fri-challenge-space } (\text{ceil-log } \text{clength})$
unfolding *trace-fri-supported-root-union-bad-sets-def* **by** *auto*

lemma *trace-fri-supported-root-partial-union-bad-sets-subset:*
 $\text{trace-fri-supported-root-partial-union-bad-sets } s \text{ bad-sets } \text{fr} \subseteq$
 $\text{fri-challenge-space } (\text{ceil-log } \text{clength})$
unfolding *trace-fri-supported-root-partial-union-bad-sets-def* **by** *auto*

lemma *trace-fri-challenge-set-hit-imp-root-union-list-set-hit:*
assumes *hit:* $\text{trace-fri-challenge-set-hit } s \text{ bad-sets } \text{out}$
shows

$\text{trace-fri-root-list-set-hit } s$
 $(\text{trace-fri-root-union-bad-sets } s \text{ bad-sets}) \text{ out}$

proof –

from *hit* **obtain** *trace-table composition-table as query-idxs trace-bs dg comp-bs*

where *bound:*

$\text{accepted-with-bound-tables } s \text{ out } \text{trace-table } \text{composition-table as}$
 query-idxs

and *challenges:* $\text{accepted-fri-challenges } s \text{ out } \text{trace-bs } \text{dg } \text{comp-bs}$

and *trace-bs-bad:* $\text{trace-bs} \in \text{bad-sets } \text{trace-table}$

unfolding *trace-fri-challenge-set-hit-def* **by** *blast*

have *header-ex:*

$\exists \text{ fr } \text{f-fri-roots } \text{f-final } \text{header-as } \text{composition-fri-roots } \text{final rest.}$
 $\text{verifier-header-transcript } s \text{ fr } \text{f-fri-roots } \text{f-final } \text{header-as } \text{dg}$
 $\text{composition-fri-roots } \text{final rest}$

proof –

from *challenges[unfolded accepted-fri-challenges-def]* **show** *?thesis*

proof (*elim exE conjE*)

fix *result0 final-state0 fr0 f-fl0 f-final0 as0 fl0 final0*
 query-state0

assume *out0:* $\text{out} = \text{Some } (\text{result0}, \text{final-state0})$

and *header0:*

$\text{verifier-header-transcript } s \text{ fr0 } (\text{map } \text{snd } \text{f-fl0}) \text{ f-final0 } \text{as0 } \text{dg}$
 $(\text{map } \text{snd } \text{fl0}) \text{ final0 } (\text{PTranscript } \text{query-state0})$

show *?thesis*

apply (*rule exI[where x=fr0]*)

apply (*rule exI[where x=map snd f-fl0]*)

```

    apply (rule exI[where  $x=f\text{-final}0$ ])
    apply (rule exI[where  $x=as0$ ])
    apply (rule exI[where  $x=\text{map } \text{snd } fl0$ ])
    apply (rule exI[where  $x=\text{final}0$ ])
    apply (rule exI[where  $x=P\text{Transcript } \text{query-state}0$ ])
    apply (rule header0)
  done
qed
qed
then obtain  $fr\ f\text{-fri-roots}\ f\text{-final}\ \text{header-as}\ \text{composition-fri-roots}\ \text{final}$ 
   $rest$  where
  header:
    verifier-header-transcript  $s\ fr\ f\text{-fri-roots}\ f\text{-final}\ \text{header-as}\ dg$ 
    composition-fri-roots  $\text{final}\ rest$ 
  by blast
have candidate:
   $trace\text{-table} \in trace\text{-fri-root-trace-table-candidates}\ s\ fr$ 
  unfolding trace-fri-root-trace-table-candidates-def
  using bound challenges header by blast
have trace-bs-space:
   $trace\text{-bs} \in \text{fri-challenge-space}\ (\text{ceil-log } clength)$ 
  by (rule accepted-fri-challenges-trace-space[OF challenges])
have trace-bs-union:
   $trace\text{-bs} \in trace\text{-fri-root-union-bad-sets}\ s\ \text{bad-sets}\ fr$ 
  unfolding trace-fri-root-union-bad-sets-def
  using trace-bs-space candidate trace-bs-bad by blast
show ?thesis
  unfolding trace-fri-root-list-set-hit-def
  using challenges header trace-bs-union by blast
qed

lemma trace-fri-challenge-set-hit-imp-supported-root-union-list-set-hit:
assumes outcome:  $out \in \text{set-dist}\ (\text{execute } \text{verify-monad } s)$ 
and hit: trace-fri-challenge-set-hit  $s\ \text{bad-sets}\ out$ 
shows
  trace-fri-root-list-set-hit  $s$ 
   $(trace\text{-fri-supported-root-union-bad-sets}\ s\ \text{bad-sets})\ out$ 
proof –
from hit obtain trace-table composition-table as query-idxs trace-bs dg comp-bs
where bound:
  accepted-with-bound-tables  $s\ out\ trace\text{-table}\ \text{composition-table}\ as$ 
  query-idxs
and challenges: accepted-fri-challenges  $s\ out\ trace\text{-bs}\ dg\ \text{comp-bs}$ 
and not-low:  $\neg trace\text{-table-low-degree}\ trace\text{-table}$ 
and trace-bs-bad:  $trace\text{-bs} \in \text{bad-sets}\ trace\text{-table}$ 
unfolding trace-fri-challenge-set-hit-def by blast
have header-ex:
   $\exists fr\ f\text{-fri-roots}\ f\text{-final}\ \text{header-as}\ \text{composition-fri-roots}\ \text{final}\ rest.$ 
  verifier-header-transcript  $s\ fr\ f\text{-fri-roots}\ f\text{-final}\ \text{header-as}\ dg$ 

```



```

    composition-fri-roots final rest
proof –
  from challenges[unfolded accepted-fri-challenges-def] show ?thesis
proof (elim exE conjE)
  fix result0 final-state0 fr0 f-fl0 f-final0 as0 fl0 final0
    query-state0
  assume
    header0:
      verifier-header-transcript s fr0 (map snd f-fl0) f-final0 as0 dg
      (map snd fl0) final0 (PTranscript query-state0)
  show ?thesis
    apply (rule exI[where x=fr0])
    apply (rule exI[where x=map snd f-fl0])
    apply (rule exI[where x=f-final0])
    apply (rule exI[where x=as0])
    apply (rule exI[where x=map snd fl0])
    apply (rule exI[where x=final0])
    apply (rule exI[where x=PTranscript query-state0])
    apply (rule header0)
    done
  qed
qed
then obtain fr f-fri-roots f-final header-as composition-fri-roots final
  rest where
    header:
      verifier-header-transcript s fr f-fri-roots f-final header-as dg
      composition-fri-roots final rest
  by blast
have candidate:
  trace-table ∈ trace-fri-supported-root-trace-table-candidates s fr
  unfolding trace-fri-supported-root-trace-table-candidates-def
  using outcome bound challenges header by blast
have trace-bs-space:
  trace-bs ∈ fri-challenge-space (ceil-log clength)
  by (rule accepted-fri-challenges-trace-space[OF challenges])
have trace-bs-union:
  trace-bs ∈ trace-fri-supported-root-union-bad-sets s bad-sets fr
  unfolding trace-fri-supported-root-union-bad-sets-def
  using trace-bs-space candidate trace-bs-bad by blast
show ?thesis
  unfolding trace-fri-root-list-set-hit-def
  using challenges header trace-bs-union by blast
qed

lemma trace-fri-challenge-set-hit-imp-partial-union-or-collision:
assumes outcome: out ∈ set-dist (execute verify-monad s)
  and hit: trace-fri-challenge-set-hit s bad-sets out
shows
  trace-fri-root-list-set-hit s

```

```

    (trace-fri-supported-root-partial-union-bad-sets s bad-sets) out ∨
    hash-map-output-collision-bad s out
proof –
from hit obtain trace-table composition-table as query-idxs trace-bs dg comp-bs
where bound:
    accepted-with-bound-tables s out trace-table composition-table as
    query-idxs
and challenges: accepted-fri-challenges s out trace-bs dg comp-bs
and not-low:  $\neg$  trace-table-low-degree trace-table
and trace-bs-bad: trace-bs  $\in$  bad-sets trace-table
unfolding trace-fri-challenge-set-hit-def by blast
from accepted-with-bound-tables-imp-accepted[OF bound]
obtain result final-state where out-eq: out = Some (result, final-state)
unfolding accepted-def by (cases out) auto
show ?thesis
proof (cases hash-map-output-collision final-state)
case True
then have hash-map-output-collision-bad s out
unfolding hash-map-output-collision-bad-def accepted-def out-eq by simp
then show ?thesis by simp
next
case clean: False
have header-ex:
   $\exists$  fr f-fri-roots f-final header-as composition-fri-roots final rest.
  verifier-header-transcript s fr f-fri-roots f-final header-as dg
  composition-fri-roots final rest
proof –
from challenges[unfolded accepted-fri-challenges-def] show ?thesis
proof (elim exE conjE)
fix result0 final-state0 fr0 f-fl0 f-final0 as0 fl0 final0
    query-state0
assume
  header0:
    verifier-header-transcript s fr0 (map snd f-fl0) f-final0 as0 dg
    (map snd fl0) final0 (PTranscript query-state0)
show ?thesis
apply (rule exI[where x=fr0])
apply (rule exI[where x=map snd f-fl0])
apply (rule exI[where x=f-final0])
apply (rule exI[where x=as0])
apply (rule exI[where x=map snd fl0])
apply (rule exI[where x=final0])
apply (rule exI[where x=PTranscript query-state0])
apply (rule header0)
done
qed
qed
then obtain fr f-fri-roots f-final header-as composition-fri-roots final
  rest where

```

```

header:
  verifier-header-transcript s fr f-fri-roots f-final header-as dg
  composition-fri-roots final rest
by blast
from bound obtain fr' f-fri-roots' f-final' dg'
  composition-fri-roots' final' rest' where
header-bound:
  verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
  composition-fri-roots' final' rest'
and bind:
  merkle-root-binds-table fr' trace-table final-state
unfolding accepted-with-bound-tables-def out-eq by blast
have fr-eq: fr' = fr
  using verifier-header-transcript-unique[OF header-bound header] by simp
have outcome-some:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using outcome out-eq by simp
have bound-some:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
  using bound out-eq by simp
have challenges-some:
  accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
  comp-bs
  using challenges out-eq by simp
have bind-fr: merkle-root-binds-table fr trace-table final-state
  using bind fr-eq by simp
have state:
  trace-fri-supported-root-trace-table-candidate-state s fr trace-table
  final-state
  unfolding trace-fri-supported-root-trace-table-candidate-state-def
  using outcome-some bound-some challenges-some header bind-fr by blast
have candidate:
  trace-table ∈
  trace-fri-supported-root-partial-trace-table-candidates s fr
  by (rule
    trace-fri-supported-root-trace-table-candidate-state-imp-partial-candidate
    [OF state clean])
have trace-bs-space:
  trace-bs ∈ fri-challenge-space (ceil-log clength)
  by (rule accepted-fri-challenges-trace-space[OF challenges])
have trace-bs-union:
  trace-bs ∈
  trace-fri-supported-root-partial-union-bad-sets s bad-sets fr
  unfolding trace-fri-supported-root-partial-union-bad-sets-def
  using trace-bs-space candidate trace-bs-bad by blast
show ?thesis
  unfolding trace-fri-root-list-set-hit-def
  using challenges header trace-bs-union by blast

```

qed
qed

lemma *alpha-bad-set-hit-imp-partial-union-or-collision:*
assumes *outcome:* $out \in \text{set-dist}$ (*execute verify-monad s*)
and *hit:* *alpha-bad-set-hit s bad-sets out*
and *subset:* $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$
shows
alpha-header-list-set-hit s
(alpha-header-supported-partial-union-bad-sets s bad-sets) out $\vee$
hash-map-output-collision-bad s out

proof –
from *hit* **obtain** *trace-table composition-table as query-idxs* **where**
bound:
accepted-with-bound-tables s out trace-table composition-table
as query-idxs
and *as-bad:* *as* $\in$ *bad-sets trace-table*
unfolding *alpha-bad-set-hit-def* **by** *blast*
from *accepted-with-bound-tables-imp-accepted*[*OF bound*]
obtain *result final-state* **where** *out-eq:* *out = Some (result, final-state)*
unfolding *accepted-def* **by** (*cases out*) *auto*
show *?thesis*

proof (*cases hash-map-output-collision final-state*)
case *True*
then have *hash-map-output-collision-bad s out*
unfolding *hash-map-output-collision-bad-def* *accepted-def out-eq* **by** *simp*
then show *?thesis* **by** *simp*

next
case *clean: False*
from *bound* **obtain** *fr f-fri-roots f-final dg composition-fri-roots final*
rest **where**
header:
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
and *bind:*
merkle-root-binds-table fr trace-table final-state
unfolding *accepted-with-bound-tables-def out-eq* **by** *blast*
have *shape:*
accepted-transcript-shape s out as query-idxs
using *accepted-with-bound-tables-imp-accepted-with-tables*[*OF bound*]
accepted-with-tables-imp-accepted-transcript-shape
by *blast*
have *state:*
alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
f-final trace-table final-state
unfolding *alpha-header-supported-trace-table-candidate-state-def*
by (*intro exI*[*of - result*] *exI*[*of - composition-table*]
exI[*of - as*] *exI*[*of - query-idxs*] *exI*[*of - dg*]
exI[*of - composition-fri-roots*] *exI*[*of - final*])

```

      exI[of - rest])
    (use outcome out-eq bound header bind in simp)
  have candidate:
    trace-table ∈
      alpha-header-supported-partial-trace-table-candidates s fr
      f-fri-roots f-final
  by (rule
      alpha-header-supported-trace-table-candidate-state-imp-partial-candidate
      [OF state clean])
  have as-union:
    as ∈
      alpha-header-supported-partial-union-bad-sets s bad-sets fr
      f-fri-roots f-final
  unfolding alpha-header-supported-partial-union-bad-sets-def
  using as-bad candidate subset[of trace-table] by auto
  show ?thesis
  unfolding alpha-header-list-set-hit-def
  using shape out-eq header as-union by blast
qed
qed

lemma wp-trace-fri-challenge-set-hit-bound-via-partial-union-or-collision:
  fixes C H :: prob
  assumes partial-bound:
    wp-event verify-monad
      (trace-fri-root-list-set-hit s
        (trace-fri-supported-root-partial-union-bad-sets s bad-sets)) s ≤ C
  and collision-bound:
    wp-event verify-monad (hash-map-output-collision-bad s) s ≤ H
  shows
    wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s ≤
      C + H
  proof -
    have mono:
      wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s ≤
        wp-event verify-monad
          (λout.
            trace-fri-root-list-set-hit s
              (trace-fri-supported-root-partial-union-bad-sets s bad-sets) out ∨
              hash-map-output-collision-bad s out) s
    by (rule wp-event-mono-on-support)
      (use trace-fri-challenge-set-hit-imp-partial-union-or-collision in blast)
  also have ... ≤ C + H
  proof -
    have wp-event verify-monad
      (λout.
        trace-fri-root-list-set-hit s
          (trace-fri-supported-root-partial-union-bad-sets s bad-sets) out ∨
          hash-map-output-collision-bad s out) s ≤

```

```

    wp-event verify-monad
      (trace-fri-root-list-set-hit s
        (trace-fri-supported-root-partial-union-bad-sets s bad-sets)) s +
    wp-event verify-monad (hash-map-output-collision-bad s) s
  by (rule wp-event-union-bound)
also have ...  $\leq C + H$ 
  by (intro add-mono partial-bound collision-bound)
finally show ?thesis .
qed
finally show ?thesis .
qed

```

lemma *accepted-verifier-initial-imp-transcript-target-hit-on-support:*
assumes *support:*
 $out \in \text{set-dist } (\text{execute verify-monad } (\text{verifier-initial-state } tr))$
and *acc:* *accepted out*
shows *hash-new-output-hit-event (set tr) (verifier-initial-state tr) out*
proof –
from *acc* **obtain** *result final-state* **where** *out-eq:*
 $out = \text{Some } (result, final-state)$
unfolding *accepted-def* **by** (cases *out*) *auto*
have *outcome:*
 $\text{Some } (result, final-state) \in$
 $\text{set-dist } (\text{execute verify-monad } (\text{verifier-initial-state } tr))$
using *support out-eq* **by** *simp*
have *hit:*
 $\text{hash-map-new-output-hit } (set\ tr) (\text{verifier-initial-state } tr)$
 $final-state$
by (rule *verify-monad-verifier-initial-trace-root-target-hit*[*OF outcome*])
show ?thesis
unfolding *hash-new-output-hit-event-def out-eq*
using *hit* **by** *simp*
qed

lemma *wp-alpha-bad-set-hit-bound-via-partial-union-or-collision:*
fixes *C H :: prob*
assumes *subset:* $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$
and *partial-bound:*
 $\text{wp-event verify-monad}$
 $(\text{alpha-header-list-set-hit } s$
 $(\text{alpha-header-supported-partial-union-bad-sets } s \text{ bad-sets})) s \leq C$
and *collision-bound:*
 $\text{wp-event verify-monad } (\text{hash-map-output-collision-bad } s) s \leq H$
shows
 $\text{wp-event verify-monad } (\text{alpha-bad-set-hit } s \text{ bad-sets}) s \leq C + H$
proof –
have *mono:*
 $\text{wp-event verify-monad } (\text{alpha-bad-set-hit } s \text{ bad-sets}) s \leq$
 $\text{wp-event verify-monad}$

```

    (λout.
      alpha-header-list-set-hit s
      (alpha-header-supported-partial-union-bad-sets s bad-sets) out ∨
      hash-map-output-collision-bad s out) s
  proof (rule wp-event-mono-on-support)
  fix out
  assume support: out ∈ set-dist (execute verify-monad s)
  and hit: alpha-bad-set-hit s bad-sets out
  show
    alpha-header-list-set-hit s
    (alpha-header-supported-partial-union-bad-sets s bad-sets) out ∨
    hash-map-output-collision-bad s out
  by (rule alpha-bad-set-hit-imp-partial-union-or-collision
      [OF support hit subset])
qed
also have ... ≤ C + H
proof -
  have wp-event verify-monad
    (λout.
      alpha-header-list-set-hit s
      (alpha-header-supported-partial-union-bad-sets s bad-sets) out ∨
      hash-map-output-collision-bad s out) s ≤
    wp-event verify-monad
      (alpha-header-list-set-hit s
        (alpha-header-supported-partial-union-bad-sets s bad-sets)) s +
    wp-event verify-monad (hash-map-output-collision-bad s) s
  by (rule wp-event-union-bound)
  also have ... ≤ C + H
  by (intro add-mono partial-bound collision-bound)
  finally show ?thesis .
qed
finally show ?thesis .
qed

lemma wp-trace-fri-challenge-set-hit-bound-via-supported-partial-union-or-collision:
  fixes C H :: prob
  assumes future: trace-fri-future-fresh s
  and union-bound:
    ⋀fr. nnreal
      (card (trace-fri-supported-root-partial-union-bad-sets s bad-sets fr)) /
      nnreal (CARD('f) ^ ceil-log clength) ≤ C
  and collision-bound:
    wp-event verify-monad (hash-map-output-collision-bad s) s ≤ H
  shows
    wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s ≤
      C + H
proof -
  have partial-bound:
    wp-event verify-monad

```

(trace-fri-root-list-set-hit s
 (trace-fri-supported-root-partial-union-bad-sets s bad-sets)) $s \leq C$
by (rule wp-verify-monad-trace-fri-root-list-set-bound[OF future])
 (use trace-fri-supported-root-partial-union-bad-sets-subset union-bound
 in simp-all)
show ?thesis
by (rule wp-trace-fri-challenge-set-hit-bound-via-partial-union-or-collision
 [OF partial-bound collision-bound])
qed

lemma wp-alpha-bad-set-hit-bound-via-supported-partial-union-or-collision:
fixes $C H :: \text{prob}$
assumes future: alpha-future-fresh s
and subset: $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$
and union-bound:
 $\bigwedge \text{fr f-fri-roots f-final.}$
 nnreal
 $(\text{card}$
 $(\text{alpha-header-supported-partial-union-bad-sets } s \text{ bad-sets fr}$
 $\text{f-fri-roots f-final})) /$
 $\text{nnreal } (\text{card } \text{alpha-space}) \leq C$
and collision-bound:
 $\text{wp-event verify-monad } (\text{hash-map-output-collision-bad } s) s \leq H$
shows $\text{wp-event verify-monad } (\text{alpha-bad-set-hit } s \text{ bad-sets}) s \leq C + H$
proof –
have partial-bound:
 $\text{wp-event verify-monad}$
 $(\text{alpha-header-list-set-hit } s$
 $(\text{alpha-header-supported-partial-union-bad-sets } s \text{ bad-sets})) s \leq C$
by (rule wp-verify-monad-alpha-header-list-set-bound[OF future])
 (use alpha-header-supported-partial-union-bad-sets-subset-alpha-space
 union-bound in simp-all)
show ?thesis
by (rule wp-alpha-bad-set-hit-bound-via-partial-union-or-collision
 [OF subset partial-bound collision-bound])
qed

lemma trace-fri-supported-root-partial-union-bad-sets-fraction-bound-if-unique-candidate:
fixes $C :: \text{prob}$
assumes unique:
 $\exists \text{trace-table.}$
 $\text{trace-fri-supported-root-partial-trace-table-candidates } s \text{ fr} \subseteq$
 $\{\text{trace-table}\}$
and subset:
 $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq$
 $\text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and bound:
 $\bigwedge \text{trace-table.}$
 $\text{nnreal } (\text{card } (\text{bad-sets trace-table})) /$


```

      nnreal (CARD('f) ^ ceil-log clength) ≤ C
shows
  nnreal
    (card (trace-fri-supported-root-partial-union-bad-sets s bad-sets fr)) /
    nnreal (CARD('f) ^ ceil-log clength) ≤ C
proof -
  from unique obtain trace-table where candidates:
    trace-fri-supported-root-partial-trace-table-candidates s fr ⊆
      {trace-table}
  by blast
  have union-subset:
    trace-fri-supported-root-partial-union-bad-sets s bad-sets fr ⊆
      bad-sets trace-table
  unfolding trace-fri-supported-root-partial-union-bad-sets-def
  using candidates by auto
  have finite-bad: finite (bad-sets trace-table)
  by (rule finite-subset[OF subset finite-fri-challenge-space])
  have card-le:
    card (trace-fri-supported-root-partial-union-bad-sets s bad-sets fr) ≤
      card (bad-sets trace-table)
  by (rule card-mono[OF finite-bad union-subset])
  have nnreal
    (card
      (trace-fri-supported-root-partial-union-bad-sets s bad-sets fr)) /
    nnreal (CARD('f) ^ ceil-log clength) ≤
    nnreal (card (bad-sets trace-table)) /
    nnreal (CARD('f) ^ ceil-log clength)
  by (rule nnreal-nat-divide-right-mono[OF card-le])
  also have ... ≤ C
  by (rule bound)
  finally show ?thesis .
qed

end

end

```

```

theory Soundness-Reductions-Composition
  imports Soundness-Reductions-Trace
begin

```

Composition-table, query-header, and supported candidate reductions.

```

context soundness
begin

```

```

definition trace-fri-bad-challenge-sets-bounded
  :: ('f list ⇒ 'f list set) ⇒ bool
  where

```

$\text{trace-fri-bad-challenge-sets-bounded bad-sets} \longleftrightarrow$
 $(\forall \text{ trace-table.}$
 $\text{bad-sets trace-table} \subseteq \text{fri-challenge-space (ceil-log clength)} \wedge$
 $\text{nnreal (card (bad-sets trace-table)) /}$
 $\text{nnreal (CARD('f) } \wedge \text{ ceil-log clength)} \leq \text{trace-fri-error})$

definition *composition-fri-bad-challenge-sets-bounded*

$:: ('f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow \text{bool}$

where

$\text{composition-fri-bad-challenge-sets-bounded bad-sets} \longleftrightarrow$
 $(\forall \text{ dg composition-table.}$
 $\text{bad-sets dg composition-table} \subseteq$
 $\text{fri-challenge-space (ceil-log (to-nat dg + 1))} \wedge$
 $\text{nnreal (card (bad-sets dg composition-table)) /}$
 $\text{nnreal (CARD('f) } \wedge \text{ ceil-log (to-nat dg + 1))} \leq$
 $\text{composition-fri-error})$

lemma *wp-trace-fri-challenge-set-hit-bound-if-supported-partial-root-candidate-unique-or-collision:*

fixes $H :: \text{prob}$

assumes *future: trace-fri-future-fresh s*

and *unique:*

$\bigwedge_{fr.}$
 $\exists \text{ trace-table.}$
 $\text{trace-fri-supported-root-partial-trace-table-candidates s fr} \subseteq$
 $\{\text{trace-table}\}$

and *bounded: trace-fri-bad-challenge-sets-bounded bad-sets*

and *collision-bound:*

$\text{wp-event verify-monad (hash-map-output-collision-bad s) s} \leq H$

shows

$\text{wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s} \leq$
 $\text{trace-fri-error} + H$

proof –

have *union-bound:*

$\bigwedge_{fr.}$
 nnreal
 $(\text{card}$
 $(\text{trace-fri-supported-root-partial-union-bad-sets s bad-sets fr})) /$
 $\text{nnreal (CARD('f) } \wedge \text{ ceil-log clength)} \leq \text{trace-fri-error}$

proof –

fix fr

show

nnreal
 $(\text{card}$
 $(\text{trace-fri-supported-root-partial-union-bad-sets s bad-sets fr})) /$
 $\text{nnreal (CARD('f) } \wedge \text{ ceil-log clength)} \leq \text{trace-fri-error}$

by (rule

$\text{trace-fri-supported-root-partial-union-bad-sets-fraction-bound-if-unique-candidate}$
 $[\text{OF unique}]$

(use *bounded* **in**

```

      (auto simp: trace-fri-bad-challenge-sets-bounded-def))
qed
show ?thesis
  by (rule
      wp-trace-fri-challenge-set-hit-bound-via-supported-partial-union-or-collision
      [OF future union-bound collision-bound])
qed

lemma alpha-header-supported-partial-union-bad-sets-fraction-bound-if-unique-candidate:
  fixes C :: prob
  assumes unique:
     $\exists$  trace-table.
      alpha-header-supported-partial-trace-table-candidates s fr
      f-fri-roots f-final  $\subseteq$  {trace-table}
  and subset:  $\bigwedge$  trace-table. bad-sets trace-table  $\subseteq$  alpha-space
  and bound:
     $\bigwedge$  trace-table.
      nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)  $\leq$  C
  shows
    nnreal
      (card
        (alpha-header-supported-partial-union-bad-sets s bad-sets fr
          f-fri-roots f-final)) /
      nnreal (card alpha-space)  $\leq$  C
proof -
  from unique obtain trace-table where candidates:
    alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
    f-final  $\subseteq$  {trace-table}
  by blast
  have union-subset:
    alpha-header-supported-partial-union-bad-sets s bad-sets fr f-fri-roots
    f-final  $\subseteq$  bad-sets trace-table
  unfolding alpha-header-supported-partial-union-bad-sets-def
  using candidates by auto
  have finite-bad: finite (bad-sets trace-table)
  by (rule finite-subset[OF subset finite-alpha-space])
  have card-le:
    card
      (alpha-header-supported-partial-union-bad-sets s bad-sets fr
        f-fri-roots f-final)  $\leq$ 
    card (bad-sets trace-table)
  by (rule card-mono[OF finite-bad union-subset])
  have nnreal
    (card
      (alpha-header-supported-partial-union-bad-sets s bad-sets fr
        f-fri-roots f-final)) /
    nnreal (card alpha-space)  $\leq$ 
    nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)
  by (rule nnreal-nat-divide-right-mono[OF card-le])

```

also have $\dots \leq C$
 by (rule bound)
 finally show ?thesis .
 qed

lemma *wp-alpha-bad-set-hit-bound-if-supported-partial-header-candidate-unique-or-collision:*

fixes $C\ H :: \text{prob}$

assumes *future: alpha-future-fresh s*

and *unique:*

$\bigwedge \text{fr } f\text{-fri-roots } f\text{-final.}$

$\exists \text{ trace-table.}$

alpha-header-supported-partial-trace-table-candidates s fr

$f\text{-fri-roots } f\text{-final} \subseteq \{\text{trace-table}\}$

and *subset: $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$*

and *bounded:*

$\bigwedge \text{trace-table.}$

$\text{nnreal } (\text{card } (\text{bad-sets trace-table})) / \text{nnreal } (\text{card } \text{alpha-space}) \leq$

C

and *collision-bound:*

wp-event verify-monad (hash-map-output-collision-bad s) s $\leq H$

shows *wp-event verify-monad (alpha-bad-set-hit s bad-sets) s $\leq C + H$*

proof –

have *union-bound:*

$\bigwedge \text{fr } f\text{-fri-roots } f\text{-final.}$

nnreal

$(\text{card}$

(alpha-header-supported-partial-union-bad-sets s bad-sets fr

f-fri-roots f-final)) /

$\text{nnreal } (\text{card } \text{alpha-space}) \leq C$

proof –

fix *fr f-fri-roots f-final*

show

nnreal

$(\text{card}$

(alpha-header-supported-partial-union-bad-sets s bad-sets fr

f-fri-roots f-final)) /

$\text{nnreal } (\text{card } \text{alpha-space}) \leq C$

by (rule

alpha-header-supported-partial-union-bad-sets-fraction-bound-if-unique-candidate

[OF unique subset bounded])

qed

show ?thesis

by (rule *wp-alpha-bad-set-hit-bound-via-supported-partial-union-or-collision*

[OF future subset union-bound collision-bound])

qed

lemma *composition-randomization-bad-bound-if-alpha-header-supported-partial-candidate-unique-or-collision:*

fixes $H :: \text{prob}$

assumes *future: alpha-future-fresh s*

and *unique*:
 $\bigwedge fr \ f\text{-fri-roots} \ f\text{-final}.$
 $\exists \text{ trace-table}.$
 $\text{alpha-header-supported-partial-trace-table-candidates } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final} \subseteq \{\text{trace-table}\}$
and *collision-bound*:
 $\text{wp-event verify-monad } (\text{hash-map-output-collision-bad } s) \ s \leq H$
shows
 $\text{wp-event verify-monad } (\text{composition-randomization-bad } s) \ s \leq$
 $\text{composition-error-bound} + H$
proof –
have *alpha-bound*:
 $\text{wp-event verify-monad}$
 $(\text{alpha-bad-set-hit } s \text{ composition-trace-bad-alpha-space}) \ s \leq$
 $\text{composition-error-bound} + H$
by (rule
 $\text{wp-alpha-bad-set-hit-bound-if-supported-partial-header-candidate-unique-or-collision}$
 $[OF \text{ future unique composition-trace-bad-alpha-space-subset-alpha-space}$
 $\text{composition-trace-bad-alpha-space-fraction-bound-alpha-space}$
 $\text{collision-bound}]$)
have $\text{wp-event verify-monad } (\text{composition-randomization-bad } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{alpha-bad-set-hit } s \text{ composition-trace-bad-alpha-space}) \ s$
by (rule *wp-event-mono*)
 $(\text{rule composition-randomization-bad-imp-alpha-bad-set-hit})$
also have $\dots \leq \text{composition-error-bound} + H$
by (rule *alpha-bound*)
finally show *?thesis* .
qed

lemma *wp-trace-fri-challenge-set-hit-bound-via-root-union*:
fixes $C :: \text{prob}$
assumes *future*: $\text{trace-fri-future-fresh } s$
and *union-bound*:
 $\bigwedge fr. \text{nnreal}$
 $(\text{card } (\text{trace-fri-root-union-bad-sets } s \text{ bad-sets } fr)) /$
 $\text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log } \text{clength}) \leq C$
shows $\text{wp-event verify-monad } (\text{trace-fri-challenge-set-hit } s \text{ bad-sets}) \ s \leq C$
proof –
have *root-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-root-list-set-hit } s$
 $(\text{trace-fri-root-union-bad-sets } s \text{ bad-sets})) \ s \leq C$
by (rule *wp-verify-monad-trace-fri-root-list-set-bound*[*OF future*])
 $(\text{use trace-fri-root-union-bad-sets-subset union-bound in simp-all})$
have $\text{wp-event verify-monad } (\text{trace-fri-challenge-set-hit } s \text{ bad-sets}) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-root-list-set-hit } s$
 $(\text{trace-fri-root-union-bad-sets } s \text{ bad-sets})) \ s$

by (rule wp-event-mono)
 (rule trace-fri-challenge-set-hit-imp-root-union-list-set-hit)
 also have ... $\leq C$
 by (rule root-bound)
 finally show ?thesis .
 qed

lemma wp-trace-fri-challenge-set-hit-bound-via-supported-root-union:
 fixes $C :: prob$
 assumes future: trace-fri-future-fresh s
 and union-bound:
 $\bigwedge fr. nnreal$
 $(card (trace-fri-supported-root-union-bad-sets s bad-sets fr)) /$
 $nnreal (CARD('f) \wedge \text{ceil-log clength}) \leq C$
 shows wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) $s \leq C$
proof –
 have root-bound:
 wp-event verify-monad
 (trace-fri-root-list-set-hit s
 (trace-fri-supported-root-union-bad-sets s bad-sets)) $s \leq C$
 by (rule wp-verify-monad-trace-fri-root-list-set-bound[OF future])
 (use trace-fri-supported-root-union-bad-sets-subset union-bound in simp-all)
 have wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) $s \leq$
 wp-event verify-monad
 (trace-fri-root-list-set-hit s
 (trace-fri-supported-root-union-bad-sets s bad-sets)) s
 by (rule wp-event-mono-on-support)
 (rule trace-fri-challenge-set-hit-imp-supported-root-union-list-set-hit)
 also have ... $\leq C$
 by (rule root-bound)
 finally show ?thesis .
 qed

lemma trace-fri-root-union-bad-sets-fraction-bound-if-unique-candidate:
 fixes $C :: prob$
 assumes unique:
 $\exists trace\text{-table}.$
 $trace\text{-fri-root-trace-table-candidates } s \text{ fr} \subseteq \{trace\text{-table}\}$
 and subset:
 $\bigwedge trace\text{-table}. bad\text{-sets } trace\text{-table} \subseteq$
 $fri\text{-challenge-space } (ceil\text{-log clength})$
 and bound:
 $\bigwedge trace\text{-table}.$
 $nnreal (card (bad\text{-sets } trace\text{-table})) /$
 $nnreal (CARD('f) \wedge \text{ceil-log clength}) \leq C$
 shows
 $nnreal (card (trace\text{-fri-root-union-bad-sets } s \text{ bad-sets fr})) /$
 $nnreal (CARD('f) \wedge \text{ceil-log clength}) \leq C$

proof –

from *unique* **obtain** *trace-table* **where** *candidates:*
 trace-fri-root-trace-table-candidates s fr $\subseteq$ {*trace-table*}
by *blast*
have *union-subset:*
 trace-fri-root-union-bad-sets s bad-sets fr $\subseteq$ *bad-sets trace-table*
unfolding *trace-fri-root-union-bad-sets-def*
using *candidates* **by** *auto*
have *finite-bad:* *finite (bad-sets trace-table)*
by (*rule finite-subset*[*OF subset finite-fri-challenge-space*])
have *card-le:*
 card (trace-fri-root-union-bad-sets s bad-sets fr) $\leq$
 card (bad-sets trace-table)
by (*rule card-mono*[*OF finite-bad union-subset*])
have *nnreal (card (trace-fri-root-union-bad-sets s bad-sets fr)) /*
 nnreal (CARD('f) ^ ceil-log clength) ≤
 nnreal (card (bad-sets trace-table)) /
 nnreal (CARD('f) ^ ceil-log clength)
by (*rule nnreal-nat-divide-right-mono*[*OF card-le*])
also have ... $\leq C$
by (*rule bound*)
finally show ?*thesis* .
qed

lemma *trace-fri-supported-root-union-bad-sets-fraction-bound-if-unique-candidate:*

fixes *C :: prob*
assumes *unique:*
 $\exists$ *trace-table.*
 trace-fri-supported-root-trace-table-candidates s fr $\subseteq$
 {*trace-table*}
and *subset:*
 $\bigwedge$ *trace-table. bad-sets trace-table* $\subseteq$
 fri-challenge-space (ceil-log clength)
and *bound:*
 $\bigwedge$ *trace-table.*
 nnreal (card (bad-sets trace-table)) /
 nnreal (CARD('f) ^ ceil-log clength) ≤ C

shows

nnreal (card (trace-fri-supported-root-union-bad-sets s bad-sets fr)) /
nnreal (CARD('f) ^ ceil-log clength) ≤ C

proof –

from *unique* **obtain** *trace-table* **where** *candidates:*
 trace-fri-supported-root-trace-table-candidates s fr $\subseteq$
 {*trace-table*}
by *blast*
have *union-subset:*
 trace-fri-supported-root-union-bad-sets s bad-sets fr $\subseteq$
 bad-sets trace-table
unfolding *trace-fri-supported-root-union-bad-sets-def*

```

using candidates by auto
have finite-bad: finite (bad-sets trace-table)
  by (rule finite-subset[OF subset finite-fri-challenge-space])
have card-le:
  card (trace-fri-supported-root-union-bad-sets s bad-sets fr) ≤
    card (bad-sets trace-table)
  by (rule card-mono[OF finite-bad union-subset])
have nnreal
  (card (trace-fri-supported-root-union-bad-sets s bad-sets fr)) /
  nnreal (CARD('f) ^ ceil-log clength) ≤
  nnreal (card (bad-sets trace-table)) /
  nnreal (CARD('f) ^ ceil-log clength)
  by (rule nnreal-nat-divide-right-mono[OF card-le])
also have ... ≤ C
  by (rule bound)
finally show ?thesis .
qed

lemma composition-fri-challenge-set-hit-imp-list-set-hit:
  assumes hit: composition-fri-challenge-set-hit s bad-sets out
  and envelope:
     $\bigwedge dg$  composition-table. bad-sets dg composition-table  $\subseteq B$  dg
  shows composition-fri-challenge-list-set-hit s B out
proof –
  from hit obtain trace-table composition-table as query-idxs trace-bs dg comp-bs
  where challenges:
    accepted-fri-challenges s out trace-bs dg comp-bs
  and comp-bad: comp-bs ∈ bad-sets dg composition-table
  unfolding composition-fri-challenge-set-hit-def by blast
  have comp-bs ∈ B dg
  using comp-bad envelope[of dg composition-table] by auto
  then show ?thesis
    unfolding composition-fri-challenge-list-set-hit-def
    using challenges by blast
qed

definition composition-fri-degree-table-candidates
  :: ('f, 'a) protocol-channel-scheme ⇒ 'f ⇒ 'f list set
  where
    composition-fri-degree-table-candidates s dg =
      { composition-table.
        ∃ out trace-table as query-idxs trace-bs comp-bs.
          accepted-with-bound-tables s out trace-table composition-table
            as query-idxs ∧
            accepted-fri-challenges s out trace-bs dg comp-bs }

definition composition-fri-degree-union-bad-sets
  :: ('f, 'a) protocol-channel-scheme ⇒
    ('f ⇒ 'f list ⇒ 'f list set) ⇒ 'f ⇒ 'f list set

```


where

composition-fri-degree-union-bad-sets s bad-sets dg =
 $\{ \text{comp-bs} \in \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1)) .$
 $\exists \text{ composition-table} \in \text{composition-fri-degree-table-candidates } s \text{ dg} .$
 $\text{comp-bs} \in \text{bad-sets } dg \text{ composition-table} \}$

lemma *composition-fri-degree-union-bad-sets-subset:*

composition-fri-degree-union-bad-sets s bad-sets dg $\subseteq$
fri-challenge-space (ceil-log (to-nat dg + 1))

unfolding *composition-fri-degree-union-bad-sets-def* **by** *auto*

lemma *composition-fri-challenge-set-hit-imp-degree-union-list-set-hit:*

assumes *hit: composition-fri-challenge-set-hit s bad-sets out*

shows

composition-fri-challenge-list-set-hit s
(composition-fri-degree-union-bad-sets s bad-sets) out

proof –

from *hit* **obtain** *trace-table composition-table as query-idxs trace-bs dg comp-bs*

where *bound:*

accepted-with-bound-tables s out trace-table composition-table as
query-idxs

and *challenges: accepted-fri-challenges s out trace-bs dg comp-bs*

and *comp-bs-bad: comp-bs $\in$ bad-sets dg composition-table*

unfolding *composition-fri-challenge-set-hit-def* **by** *blast*

have *candidate:*

composition-table $\in$ composition-fri-degree-table-candidates s dg

unfolding *composition-fri-degree-table-candidates-def*

using *bound challenges* **by** *blast*

have *comp-bs-space:*

comp-bs $\in$ fri-challenge-space (ceil-log (to-nat dg + 1))

by *(rule accepted-fri-challenges-composition-space[OF challenges])*

have *comp-bs-union:*

comp-bs $\in$ composition-fri-degree-union-bad-sets s bad-sets dg

unfolding *composition-fri-degree-union-bad-sets-def*

using *comp-bs-space candidate comp-bs-bad* **by** *blast*

show *?thesis*

unfolding *composition-fri-challenge-list-set-hit-def*

using *challenges comp-bs-union* **by** *blast*

qed

lemma *wp-composition-fri-challenge-set-hit-bound-via-degree-union:*

fixes *C :: prob*

assumes *future: composition-fri-future-fresh s*

and *union-bound:*

$\bigwedge dg. \text{nnreal}$

$(\text{card } (\text{composition-fri-degree-union-bad-sets } s \text{ bad-sets } dg)) /$

$\text{nnreal } (\text{CARD('f)} \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$

shows

wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s $\leq$

C
proof –
have *list-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-challenge-list-set-hit } s$
 $(composition\text{-fri-degree-union-bad-sets } s \text{ bad-sets})) \ s \leq C$
by (rule *wp-verify-monad-composition-fri-challenge-list-set-bound*[*OF future*])
 $(use \ composition\text{-fri-degree-union-bad-sets-subset } union\text{-bound} \text{ in}$
 $simp\text{-all})$
have $wp\text{-event } verify\text{-monad } (composition\text{-fri-challenge-set-hit } s \text{ bad-sets}) \ s \leq$
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-challenge-list-set-hit } s$
 $(composition\text{-fri-degree-union-bad-sets } s \text{ bad-sets})) \ s$
by (rule *wp-event-mono*)
 $(rule \ composition\text{-fri-challenge-set-hit-imp-degree-union-list-set-hit})$
also have $\dots \leq C$
by (rule *list-bound*)
finally show *?thesis* .
qed

lemma *composition-fri-degree-union-bad-sets-fraction-bound-if-unique-candidate*:

fixes $C :: prob$

assumes *unique*:

$\exists \ composition\text{-table}.$

$composition\text{-fri-degree-table-candidates } s \ dg \subseteq \{composition\text{-table}\}$

and *subset*:

$\bigwedge \ composition\text{-table}. \ bad\text{-sets } dg \ composition\text{-table} \subseteq$
 $fri\text{-challenge-space } (ceil\text{-log } (to\text{-nat } dg + 1))$

and *bound*:

$\bigwedge \ composition\text{-table}.$
 $nnreal \ (card \ (bad\text{-sets } dg \ composition\text{-table})) \ /$
 $nnreal \ (CARD('f) \wedge ceil\text{-log } (to\text{-nat } dg + 1)) \leq C$

shows

$nnreal \ (card \ (composition\text{-fri-degree-union-bad-sets } s \text{ bad-sets } dg)) \ /$
 $nnreal \ (CARD('f) \wedge ceil\text{-log } (to\text{-nat } dg + 1)) \leq C$

proof –

from *unique* **obtain** *composition-table* **where** *candidates*:

$composition\text{-fri-degree-table-candidates } s \ dg \subseteq \{composition\text{-table}\}$

by *blast*

have *union-subset*:

$composition\text{-fri-degree-union-bad-sets } s \text{ bad-sets } dg \subseteq$
 $bad\text{-sets } dg \ composition\text{-table}$

unfolding *composition-fri-degree-union-bad-sets-def*

using *candidates* **by** *auto*

have *finite-bad*: *finite* (*bad-sets dg composition-table*)

by (rule *finite-subset*[*OF subset finite-fri-challenge-space*])

have *card-le*:

$card \ (composition\text{-fri-degree-union-bad-sets } s \text{ bad-sets } dg) \leq$
 $card \ (bad\text{-sets } dg \ composition\text{-table})$

by (rule card-mono[OF finite-bad union-subset])
 have nnreal (card (composition-fri-degree-union-bad-sets s bad-sets dg)) /
 nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤
 nnreal (card (bad-sets dg composition-table)) /
 nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
 by (rule nnreal-nat-divide-right-mono[OF card-le])
 also have ... ≤ C
 by (rule bound)
 finally show ?thesis .
 qed

definition *composition-fri-supported-root-composition-table-candidates*

:: ('f, 'a) protocol-channel-scheme ⇒
 'f ⇒ 'f list ⇒ 'f ⇒ 'f list ⇒ 'f ⇒
 'f list ⇒ 'f list set

where

composition-fri-supported-root-composition-table-candidates s fr
 f-fri-roots f-final as dg *composition-fri-roots* =
 { *composition-table*.
 ∃ out *trace-table query-idxs trace-bs comp-bs final rest*.
 out ∈ *set-dist (execute verify-monad s)* ∧
 accepted-with-bound-tables s out *trace-table composition-table* as
 query-idxs ∧
 accepted-fri-challenges s out *trace-bs dg comp-bs* ∧
 verifier-header-transcript s fr f-fri-roots f-final as dg
 composition-fri-roots final rest }

definition *composition-fri-supported-root-composition-table-candidate-state*

:: ('f, 'a) protocol-channel-scheme ⇒
 'f ⇒ 'f list ⇒ 'f ⇒ 'f list ⇒ 'f ⇒
 'f list ⇒ 'f list ⇒
 ('f, 'a) protocol-channel-scheme ⇒ bool

where

composition-fri-supported-root-composition-table-candidate-state s fr
 f-fri-roots f-final as dg *composition-fri-roots composition-table*
final-state $\longleftrightarrow$
 (∃ result *trace-table query-idxs trace-bs comp-bs final rest*.
 Some (result, final-state) ∈ *set-dist (execute verify-monad s)* ∧
 accepted-with-bound-tables s (*Some (result, final-state)*) *trace-table*
 composition-table as *query-idxs* ∧
 accepted-fri-challenges s (*Some (result, final-state)*) *trace-bs dg*
 comp-bs ∧
 verifier-header-transcript s fr f-fri-roots f-final as dg
 composition-fri-roots final rest ∧
 composition-fri-roots ≠ [] ∧
 merkle-root-binds-table (hd *composition-fri-roots*) *composition-table*
 final-state)

lemma *composition-fri-supported-root-composition-table-candidates-iff-state:*

composition-table $\in$
composition-fri-supported-root-composition-table-candidates *s* *fr*
f-fri-roots *f-final* *as dg composition-fri-roots* $\longleftrightarrow$
 $(\exists$ *final-state*.
composition-fri-supported-root-composition-table-candidate-state *s* *fr*
f-fri-roots *f-final* *as dg composition-fri-roots composition-table*
final-state)

proof

assume *candidate*:

composition-table $\in$
composition-fri-supported-root-composition-table-candidates *s* *fr*
f-fri-roots *f-final* *as dg composition-fri-roots*

then obtain *out trace-table query-idxs trace-bs comp-bs final rest* **where**
outcome: *out* $\in$ *set-dist* (*execute verify-monad s*)

and *bound*:

accepted-with-bound-tables *s out trace-table composition-table* *as*
query-idxs

and *challenges*: *accepted-fri-challenges* *s out trace-bs dg comp-bs*

and *header*:

verifier-header-transcript *s fr f-fri-roots f-final* *as dg*
composition-fri-roots final rest

unfolding *composition-fri-supported-root-composition-table-candidates-def*
by *blast*

from *bound* **obtain** *result final-state fr' f-fri-roots' f-final' dg'*
composition-fri-roots' final' rest' where
out-eq: *out* = *Some (result, final-state)*

and *header'*:

verifier-header-transcript *s fr' f-fri-roots' f-final' as dg'*
composition-fri-roots' final' rest'

and *comp-nonempty*: *composition-fri-roots'* $\neq []$

and *comp-bind*:

merkle-root-binds-table (*hd composition-fri-roots'*)
composition-table final-state

unfolding *accepted-with-bound-tables-def* **by** *blast*

have *eqs*:

fr' = *fr* $\wedge$
f-fri-roots' = *f-fri-roots* $\wedge$
f-final' = *f-final* $\wedge$
dg' = *dg* $\wedge$
composition-fri-roots' = *composition-fri-roots* $\wedge$
final' = *final* $\wedge$
rest' = *rest*

using *verifier-header-transcript-unique[OF header' header]* **by** *simp*

have *state*:

composition-fri-supported-root-composition-table-candidate-state *s* *fr*
f-fri-roots *f-final* *as dg composition-fri-roots composition-table*
final-state

unfolding *composition-fri-supported-root-composition-table-candidate-state-def*
by (*intro exI[of - result] exI[of - trace-table]*)

```

    exI[of - query-idxs] exI[of - trace-bs] exI[of - comp-bs]
    exI[of - final] exI[of - rest])
  (use outcome out-eq bound challenges header comp-nonempty comp-bind eqs
    in simp)
then show
   $\exists$  final-state.
    composition-fri-supported-root-composition-table-candidate-state s fr
      f-fri-roots f-final as dg composition-fri-roots composition-table
      final-state
  by blast
next
assume
   $\exists$  final-state.
    composition-fri-supported-root-composition-table-candidate-state s fr
      f-fri-roots f-final as dg composition-fri-roots composition-table
      final-state
then obtain final-state result trace-table query-idxs trace-bs comp-bs final
  rest where
    outcome:
      Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
    and bound:
      accepted-with-bound-tables s (Some (result, final-state))
      trace-table composition-table as query-idxs
    and challenges:
      accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
      comp-bs
    and header:
      verifier-header-transcript s fr f-fri-roots f-final as dg
      composition-fri-roots final rest
    unfolding composition-fri-supported-root-composition-table-candidate-state-def
    by blast
show
  composition-table  $\in$ 
    composition-fri-supported-root-composition-table-candidates s fr
      f-fri-roots f-final as dg composition-fri-roots
    unfolding composition-fri-supported-root-composition-table-candidates-def
    using outcome bound challenges header by blast
qed

```

definition *composition-fri-supported-root-partial-composition-table-candidates*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$ 
  'f list  $\Rightarrow$  'f list set

```

where

```

composition-fri-supported-root-partial-composition-table-candidates s fr
  f-fri-roots f-final as dg composition-fri-roots =
  { composition-table.
    composition-fri-roots  $\neq$  []  $\wedge$ 
    ( $\exists$  out query-idxs composition-openings trace-bs comp-bs final rest.

```

$out \in \text{set-dist } (\text{execute verify-monad } s) \wedge$
 $\text{accepted-with-partial-composition-openings } s \text{ out}$
 $(\text{hd composition-fri-roots}) \text{ query-idxs composition-openings } \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings } \wedge$
 $\text{accepted-fri-challenges } s \text{ out trace-bs dg comp-bs } \wedge$
 $\text{verifier-header-transcript } s \text{ fr f-fri-roots f-final as dg}$
 $\text{composition-fri-roots final rest})\}$

definition *composition-fri-supported-root-partial-composition-table-candidate-state*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{composition-fri-supported-root-partial-composition-table-candidate-state}$
 $s \text{ fr f-fri-roots f-final as dg composition-fri-roots composition-table}$
 $\text{final-state} \longleftrightarrow$
 $\text{composition-fri-roots} \neq [] \wedge$
 $(\exists \text{result query-idxs composition-openings trace-bs comp-bs final rest.}$
 $\text{Some (result, final-state)} \in \text{set-dist } (\text{execute verify-monad } s) \wedge$
 $\text{accepted-with-partial-composition-openings } s$
 $(\text{Some (result, final-state)}) (\text{hd composition-fri-roots}) \text{ query-idxs}$
 $\text{composition-openings } \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings } \wedge$
 $\text{accepted-fri-challenges } s (\text{Some (result, final-state)}) \text{ trace-bs dg}$
 $\text{comp-bs } \wedge$
 $\text{verifier-header-transcript } s \text{ fr f-fri-roots f-final as dg}$
 $\text{composition-fri-roots final rest})$

lemma *composition-fri-supported-root-partial-composition-table-candidates-iff-state:*

$\text{composition-table} \in$
 $\text{composition-fri-supported-root-partial-composition-table-candidates } s \text{ fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots} \longleftrightarrow$
 $(\exists \text{final-state.}$
 $\text{composition-fri-supported-root-partial-composition-table-candidate-state}$
 $s \text{ fr f-fri-roots f-final as dg composition-fri-roots}$
 $\text{composition-table final-state})$

proof

assume *candidate:*

$\text{composition-table} \in$
 $\text{composition-fri-supported-root-partial-composition-table-candidates } s \text{ fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots}$

then obtain *out query-idxs composition-openings trace-bs comp-bs final rest*

where *comp-nonempty:* $\text{composition-fri-roots} \neq []$

and *outcome:* $out \in \text{set-dist } (\text{execute verify-monad } s)$

and *partial:*

$\text{accepted-with-partial-composition-openings } s \text{ out}$

```

      (hd composition-fri-roots) query-idxs composition-openings
and table:
  partial-composition-table-candidate composition-table
    composition-openings
and challenges: accepted-fri-challenges s out trace-bs dg comp-bs
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
unfolding
  composition-fri-supported-root-partial-composition-table-candidates-def
by blast
from partial obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-with-partial-composition-openings-def by blast
have outcome-some:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using outcome out-eq by simp
have partial-some:
  accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots) query-idxs
    composition-openings
  using partial out-eq by simp
have challenges-some:
  accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
    comp-bs
  using challenges out-eq by simp
have state:
  composition-fri-supported-root-partial-composition-table-candidate-state
    s fr f-fri-roots f-final as dg composition-fri-roots
    composition-table final-state
unfolding
  composition-fri-supported-root-partial-composition-table-candidate-state-def
using comp-nonempty outcome-some partial-some table challenges-some header
by blast
then show
  ∃ final-state.
    composition-fri-supported-root-partial-composition-table-candidate-state
      s fr f-fri-roots f-final as dg composition-fri-roots
      composition-table final-state
  by blast
next
assume
  ∃ final-state.
    composition-fri-supported-root-partial-composition-table-candidate-state
      s fr f-fri-roots f-final as dg composition-fri-roots
      composition-table final-state
then obtain final-state result query-idxs composition-openings trace-bs
  comp-bs final rest where
  comp-nonempty: composition-fri-roots ≠ []

```

and *outcome*:
Some (result, final-state) ∈ set-dist (execute verify-monad s)
and *partial*:
accepted-with-partial-composition-openings s
(Some (result, final-state)) (hd composition-fri-roots) query-idxs
composition-openings
and *table*:
partial-composition-table-candidate composition-table
composition-openings
and *challenges*:
accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
comp-bs
and *header*:
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
unfolding
composition-fri-supported-root-partial-composition-table-candidate-state-def
by *blast*
show
composition-table ∈
composition-fri-supported-root-partial-composition-table-candidates s fr
f-fri-roots f-final as dg composition-fri-roots
unfolding
composition-fri-supported-root-partial-composition-table-candidates-def
using *comp-nonempty outcome partial table challenges header* **by** *blast*
qed

lemma *composition-fri-supported-root-composition-table-candidate-state-imp-partial-candidate-state*:
assumes *state*:
composition-fri-supported-root-composition-table-candidate-state s fr
f-fri-roots f-final as dg composition-fri-roots composition-table
final-state
and *clean*: $\neg$ *hash-map-output-collision final-state*
shows
composition-fri-supported-root-partial-composition-table-candidate-state
s fr f-fri-roots f-final as dg composition-fri-roots composition-table
final-state
proof –
from *state* **obtain** *result trace-table query-idxs trace-bs comp-bs final rest*
where *outcome*:
Some (result, final-state) ∈ set-dist (execute verify-monad s)
and *bound*:
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs
and *challenges*:
accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
comp-bs
and *header*:
verifier-header-transcript s fr f-fri-roots f-final as dg


```

    composition-fri-roots final rest
and comp-nonempty: composition-fri-roots  $\neq$  []
and bind:
    merkle-root-binds-table (hd composition-fri-roots) composition-table
    final-state
unfolding composition-fri-supported-root-composition-table-candidate-state-def
by blast
from verify-monad-header-extraction[OF outcome]
obtain fr' f-fl' f-final' as' dg' fl' final' query-state' where
    header':
        verifier-header-transcript s fr' (map snd f-fl') f-final' as' dg'
        (map snd fl') final' (PTranscript query-state')
and query-out:
    Some (result, final-state)  $\in$ 
        set-dist
        (execute
            (ntimes
                (verifier-query-round-program fr' f-fl' f-final' as' fl'
                    final') rounds)
            query-state')
by metis
have eqs:
    fr' = fr  $\wedge$ 
    map snd f-fl' = f-fri-roots  $\wedge$ 
    f-final' = f-final  $\wedge$ 
    as' = as  $\wedge$ 
    dg' = dg  $\wedge$ 
    map snd fl' = composition-fri-roots  $\wedge$ 
    final' = final  $\wedge$ 
    PTranscript query-state' = rest
using verifier-header-transcript-unique[OF header' header] by simp
then have fl-nonempty: fl'  $\neq$  []
using comp-nonempty by (cases fl') simp-all
then obtain b composition-root fl-tail where fl-eq:
    fl' = (b, composition-root) # fl-tail
by (cases fl') auto
have root-eq: composition-root = hd composition-fri-roots
proof -
    have map-eq: map snd fl' = composition-fri-roots
        using eqs by simp
    have cons-eq: composition-root # map snd fl-tail = composition-fri-roots
        using map-eq fl-eq by simp
    then have hd composition-fri-roots = composition-root
        by (metis list.sel(1))
    then show ?thesis by simp
qed
from query-rounds-accepted-with-partial-composition-openings
    [OF fl-eq query-out]
obtain partial-query-idxs composition-openings where partial:

```

```

    accepted-with-partial-composition-openings s
      (Some (result, final-state)) composition-root partial-query-idxs
      composition-openings
  by blast
have partial-root:
  accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots)
    partial-query-idxs composition-openings
  using partial root-eq by simp
have len-table: length composition-table = scale * clength
  using accepted-with-bound-tables-shapes(2)[OF bound]
  by (simp add: mult.commute)
have table:
  partial-composition-table-candidate composition-table
  composition-openings
  by (rule accepted-bound-composition-table-partial-candidate-if-clean
    [OF bind len-table partial-root clean])
show ?thesis
  unfolding
    composition-fri-supported-root-partial-composition-table-candidate-state-def
  using outcome partial-root table challenges header comp-nonempty by blast
qed

```

definition *query-header-supported-partial-table-candidates*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$ 
  'f list  $\Rightarrow$  'f  $\Rightarrow$  ('f list  $\times$  'f list) set

```

where

```

query-header-supported-partial-table-candidates s fr f-fri-roots
  f-final as dg composition-fri-roots final =
  {(trace-table, composition-table).
    composition-fri-roots  $\neq$  []  $\wedge$ 
    ( $\exists$  out trace-query-idxs composition-query-idxs trace-openings
      composition-openings rest.
      out  $\in$  set-dist (execute verify-monad s)  $\wedge$ 
      accepted-with-partial-trace-openings s out fr trace-query-idxs
        trace-openings  $\wedge$ 
      partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
      accepted-with-partial-composition-openings s out
        (hd composition-fri-roots) composition-query-idxs
          composition-openings  $\wedge$ 
      partial-composition-table-candidate composition-table
        composition-openings  $\wedge$ 
      verifier-header-transcript s fr f-fri-roots f-final as dg
        composition-fri-roots final rest)}

```

definition *query-header-supported-partial-table-candidate-state*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$ 

```

'f list $\Rightarrow$ 'f $\Rightarrow$ 'f list $\Rightarrow$ 'f list $\Rightarrow$
('f, 'a) protocol-channel-scheme $\Rightarrow$ bool

where

query-header-supported-partial-table-candidate-state s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state $\longleftrightarrow$
composition-fri-roots $\neq []$ $\wedge$
($\exists$ result trace-query-idxs composition-query-idxs trace-openings
composition-openings rest.
Some (result, final-state) $\in$ set-dist (execute verify-monad s) $\wedge$
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr trace-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
accepted-with-partial-composition-openings s
(Some (result, final-state)) (hd composition-fri-roots)
composition-query-idxs composition-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest)

lemma query-header-supported-partial-table-candidates-iff-state:

(trace-table, composition-table) $\in$
query-header-supported-partial-table-candidates s fr f-fri-roots
f-final as dg composition-fri-roots final $\longleftrightarrow$
($\exists$ final-state.
query-header-supported-partial-table-candidate-state s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state)

proof

assume candidate:

(trace-table, composition-table) $\in$
query-header-supported-partial-table-candidates s fr f-fri-roots
f-final as dg composition-fri-roots final

then obtain out trace-query-idxs composition-query-idxs trace-openings
composition-openings rest **where**

comp-nonempty: composition-fri-roots $\neq []$

and outcome: out $\in$ set-dist (execute verify-monad s)

and trace-partial:

accepted-with-partial-trace-openings s out fr trace-query-idxs
trace-openings

and trace-table:

partial-trace-table-candidate trace-table trace-openings

and comp-partial:

accepted-with-partial-composition-openings s out
(hd composition-fri-roots) composition-query-idxs
composition-openings

and comp-table:

```

    partial-composition-table-candidate composition-table
    composition-openings
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  unfolding query-header-supported-partial-table-candidates-def by blast
from trace-partial obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-with-partial-trace-openings-def by blast
have outcome-some:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using outcome out-eq by simp
have trace-partial-some:
  accepted-with-partial-trace-openings s (Some (result, final-state)) fr
  trace-query-idxs trace-openings
  using trace-partial out-eq by simp
have comp-partial-some:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots)
  composition-query-idxs composition-openings
  using comp-partial out-eq by simp
have state:
  query-header-supported-partial-table-candidate-state s fr f-fri-roots
  f-final as dg composition-fri-roots final trace-table
  composition-table final-state
  unfolding query-header-supported-partial-table-candidate-state-def
  using comp-nonempty outcome-some trace-partial-some trace-table
  comp-partial-some comp-table header
  by blast
then show
  ∃ final-state.
    query-header-supported-partial-table-candidate-state s fr f-fri-roots
    f-final as dg composition-fri-roots final trace-table
    composition-table final-state
  by blast
next
assume
  ∃ final-state.
    query-header-supported-partial-table-candidate-state s fr f-fri-roots
    f-final as dg composition-fri-roots final trace-table
    composition-table final-state
then obtain final-state result trace-query-idxs composition-query-idxs
  trace-openings composition-openings rest where
  comp-nonempty: composition-fri-roots ≠ []
and outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings

```

```

and trace-table:
  partial-trace-table-candidate trace-table trace-openings
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots)
  composition-query-idxs composition-openings
and comp-table:
  partial-composition-table-candidate composition-table
  composition-openings
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
unfolding query-header-supported-partial-table-candidate-state-def
by blast
show
  (trace-table, composition-table) ∈
  query-header-supported-partial-table-candidates s fr f-fri-roots
  f-final as dg composition-fri-roots final
unfolding query-header-supported-partial-table-candidates-def
using comp-nonempty outcome trace-partial trace-table comp-partial
  comp-table header
by blast
qed

```

lemma *query-header-supported-partial-table-candidate-state-hash-extends*:

```

assumes
  query-header-supported-partial-table-candidate-state s fr f-fri-roots
  f-final as dg composition-fri-roots final trace-table
  composition-table final-state
shows  $s \leq \text{final-state}$ 
proof –
from assms obtain result trace-query-idxs composition-query-idxs
  trace-openings composition-openings rest where
  outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
unfolding query-header-supported-partial-table-candidate-state-def
by blast
show ?thesis
by (rule verify-monad-hash-extends[OF outcome])
qed

```

lemma *query-header-supported-table-candidate-state-imp-partial-candidate-state*:

```

assumes state:
  query-header-supported-table-candidate-state s fr f-fri-roots f-final
  as dg composition-fri-roots final trace-table composition-table
  final-state
and clean:  $\neg \text{hash-map-output-collision final-state}$ 
shows
  query-header-supported-partial-table-candidate-state s fr f-fri-roots

```

f-final as dg composition-fri-roots final trace-table
composition-table final-state
proof –
from *state* **obtain** *result query-idxs rest* **where**
outcome:
Some (result, final-state) ∈ set-dist (execute verify-monad s)
and *bound:*
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs
and *header:*
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
and *trace-bind:*
merkle-root-binds-table fr trace-table final-state
and *comp-nonempty:* *composition-fri-roots ≠ []*
and *comp-bind:*
merkle-root-binds-table (hd composition-fri-roots) composition-table
final-state
unfolding *query-header-supported-table-candidate-state-def* **by** *blast*
from *verify-monad-header-extraction[OF outcome]*
obtain *fr' f-fl f-final' as' dg' fl final' query-state* **where**
header':
verifier-header-transcript s fr' (map snd f-fl) f-final' as' dg'
(map snd fl) final' (PTranscript query-state)
and *query-out:*
Some (result, final-state) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr' f-fl f-final' as' fl
final') rounds)
query-state)
by *metis*
have *eqs:*
fr' = fr ∧
map snd f-fl = f-fri-roots ∧
f-final' = f-final ∧
as' = as ∧
dg' = dg ∧
map snd fl = composition-fri-roots ∧
final' = final ∧
PTranscript query-state = rest
using *verifier-header-transcript-unique[OF header' header]* **by** *simp*
obtain *trace-query-idxs trace-openings* **where**
len-trace-query-idxs: length trace-query-idxs = rounds
and *len-trace-openings: length trace-openings = rounds*
and *trace-indices:*
∧ i. i < rounds ⇒
map opening-index (trace-openings ! i) =

```

      powers-scaled (trace-query-idxs ! i)
and trace-tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table fr'
    (scale * clength) (trace-openings ! i) final-state
proof (rule ntimes-verifier-query-rounds-authenticated-trace-openings-no-rows
  [OF query-out])
fix trace-query-idxs trace-openings
assume len-trace-query-idxs: length trace-query-idxs = rounds
and len-trace-openings: length trace-openings = rounds
and trace-indices:
   $\bigwedge i. i < \text{rounds} \implies$ 
    map opening-index (trace-openings ! i) =
    powers-scaled (trace-query-idxs ! i)
and trace-tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table fr'
    (scale * clength) (trace-openings ! i) final-state
show ?thesis
  by (rule that[OF len-trace-query-idxs len-trace-openings
    trace-indices trace-tables])
qed
have trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
proof -
  have all-indices:
     $\forall i < \text{rounds}.$ 
      map opening-index (trace-openings ! i) =
      powers-scaled (trace-query-idxs ! i)
  using trace-indices by blast
  have all-tables:
     $\forall i < \text{rounds}.$ 
      partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
  using trace-tables eqs by simp
  show ?thesis
    unfolding accepted-with-partial-trace-openings-def accepted-def
    by (intro conjI exI[of - result] exI[of - final-state])
      (use len-trace-query-idxs len-trace-openings all-indices all-tables
        in simp-all)
qed
have trace-len: length trace-table = scale * clength
  using accepted-with-bound-tables-shapes(1)[OF bound] by simp
have trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
  by (rule accepted-bound-trace-table-partial-candidate-if-clean
    [OF trace-bind trace-len trace-partial clean])
have fl-nonempty: fl  $\neq$  []

```

```

    using comp-nonempty eqs by (cases fl) simp-all
  then obtain b composition-root fl-tail where fl-eq:
    fl = (b, composition-root) # fl-tail
    by (cases fl) auto
  have root-eq: composition-root = hd composition-fri-roots
  proof -
    have cons-eq: composition-root # map snd fl-tail = composition-fri-roots
      using eqs fl-eq by simp
    then have hd composition-fri-roots = composition-root
      by (metis list.sel(1))
    then show ?thesis by simp
  qed
  from query-rounds-accepted-with-partial-composition-openings
    [OF fl-eq query-out]
  obtain composition-query-idxs composition-openings where comp-partial':
    accepted-with-partial-composition-openings s
    (Some (result, final-state)) composition-root composition-query-idxs
    composition-openings
  by blast
  have comp-partial:
    accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots)
    composition-query-idxs composition-openings
  using comp-partial' root-eq by simp
  have comp-len: length composition-table = scale * clength
  using accepted-with-bound-tables-shapes(2) [OF bound]
  by (simp add: mult.commute)
  have comp-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  by (rule accepted-bound-composition-table-partial-candidate-if-clean
    [OF comp-bind comp-len comp-partial clean])
  show ?thesis
    unfolding query-header-supported-partial-table-candidate-state-def
    using comp-nonempty outcome trace-partial trace-candidate comp-partial
    comp-candidate header
  by blast
  qed

  lemma accepted-with-bound-tables-header-composition-fri-roots-nonempty:
    assumes bound:
      accepted-with-bound-tables s out trace-table composition-table as
      query-idxs
    and header:
      verifier-header-transcript s fr f-fri-roots f-final as dg
      composition-fri-roots final rest
    shows composition-fri-roots ≠ []
  proof -
    from bound obtain result final-state fr' f-fri-roots' f-final' dg'

```



```

    composition-fri-roots' final' rest' where
out-eq: out = Some (result, final-state)
and header':
    verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
    composition-fri-roots' final' rest'
and comp-nonempty': composition-fri-roots' ≠ []
unfolding accepted-with-bound-tables-def by blast
have composition-fri-roots' = composition-fri-roots
    using verifier-header-transcript-unique[OF header' header] by simp
then show ?thesis
    using comp-nonempty' by simp
qed

```

definition query-header-supported-partial-union-good-sets

```

:: ('f, 'a) protocol-channel-scheme ⇒
  ('f list ⇒ 'f list ⇒ 'f list ⇒ nat set) ⇒
  'f ⇒ 'f list ⇒ 'f ⇒ 'f list ⇒ 'f ⇒
  'f list ⇒ 'f ⇒ nat set

```

where

```

query-header-supported-partial-union-good-sets s good-sets fr
f-fri-roots f-final as dg composition-fri-roots final =
{idx ∈ query-sample-space.
  ∃ trace-table composition-table.
    (trace-table, composition-table) ∈
    query-header-supported-partial-table-candidates s fr
    f-fri-roots f-final as dg composition-fri-roots final ∧
    idx ∈ good-sets trace-table composition-table as}

```

lemma query-header-supported-partial-union-good-sets-subset:

```

query-header-supported-partial-union-good-sets s good-sets fr f-fri-roots
f-final as dg composition-fri-roots final ⊆ query-sample-space
unfolding query-header-supported-partial-union-good-sets-def by auto

```

lemma query-header-supported-partial-union-good-sets-fraction-bound-if-unique-candidate:

```

assumes unique:
  ∃ trace-table composition-table.
    query-header-supported-partial-table-candidates s fr f-fri-roots
    f-final as dg composition-fri-roots final ⊆
    {(trace-table, composition-table)}
and subset:
  ∧ trace-table composition-table as.
    good-sets trace-table composition-table as ⊆ query-sample-space
and bound:
  ∧ trace-table composition-table as.
    nnreal (card (good-sets trace-table composition-table as)) /
    nnreal (card query-sample-space) ≤ query-error-bound

```

shows

```

nnreal
(card

```

$(\text{query-header-supported-partial-union-good-sets } s \text{ good-sets } fr$
 $f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots } final)) /$
 $nnreal (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
proof –
from *unique* **obtain** *trace-table composition-table* **where** *candidates:*
 $\text{query-header-supported-partial-table-candidates } s \text{ fr } f\text{-fri-roots } f\text{-final}$
 $\text{as } dg \text{ composition-fri-roots } final \subseteq$
 $\{(trace\text{-table}, \text{composition-table})\}$
by *blast*
have *union-subset:*
 $\text{query-header-supported-partial-union-good-sets } s \text{ good-sets } fr$
 $f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots } final \subseteq$
 $\text{good-sets } trace\text{-table } \text{composition-table as}$
unfolding *query-header-supported-partial-union-good-sets-def*
using *candidates* **by** *auto*
have *finite-good:* $\text{finite } (\text{good-sets } trace\text{-table } \text{composition-table as})$
by $(\text{rule } \text{finite-subset}[OF \text{ subset } \text{finite-query-sample-space}])$
have *card-le:*
 card
 $(\text{query-header-supported-partial-union-good-sets } s \text{ good-sets } fr$
 $f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots } final) \leq$
 $\text{card } (\text{good-sets } trace\text{-table } \text{composition-table as})$
by $(\text{rule } \text{card-mono}[OF \text{ finite-good } \text{union-subset}])$
have *nnreal*
 $(\text{card}$
 $(\text{query-header-supported-partial-union-good-sets } s \text{ good-sets } fr$
 $f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots } final)) /$
 $nnreal (\text{card } \text{query-sample-space}) \leq$
 $nnreal (\text{card } (\text{good-sets } trace\text{-table } \text{composition-table as})) /$
 $nnreal (\text{card } \text{query-sample-space})$
by $(\text{rule } \text{nnreal-nat-divide-right-mono}[OF \text{ card-le}])$
also have $\dots \leq \text{query-error-bound}$
by $(\text{rule } \text{bound})$
finally show *?thesis* .
qed

lemma *query-index-round-set-hit-at-imp-partial-union-or-collision:*
assumes *outcome:* $out \in \text{set-dist } (\text{execute } \text{verify-monad } s)$
and *hit:* $\text{query-index-round-set-hit-at } s \text{ good-sets } i \text{ out}$
shows
 $\text{query-header-rounds-any-index-set-hit } s$
 $(\text{query-header-supported-partial-union-good-sets } s \text{ good-sets}) \text{ out } \vee$
 $\text{hash-map-output-collision-bad } s \text{ out}$
proof –
from *hit* **obtain** *trace-table composition-table as query-idxs* **where**
 bound:
 $\text{accepted-with-bound-tables } s \text{ out } trace\text{-table } \text{composition-table as}$
 query-idxs
and *i-bound:* $i < \text{rounds}$

```

and query-hit:
  query-idxs ! i ∈ good-sets trace-table composition-table as
  unfolding query-index-round-set-hit-at-def by blast
from bound obtain result final-state fr f-fri-roots f-final dg
  composition-fri-roots final rest where
  out-eq: out = Some (result, final-state)
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and trace-bind:
  merkle-root-binds-table fr trace-table final-state
and comp-nonempty: composition-fri-roots ≠ []
and comp-bind:
  merkle-root-binds-table (hd composition-fri-roots) composition-table
  final-state
unfolding accepted-with-bound-tables-def by blast
show ?thesis
proof (cases hash-map-output-collision final-state)
  case True
  then have hash-map-output-collision-bad s out
    unfolding hash-map-output-collision-bad-def accepted-def out-eq by simp
  then show ?thesis by simp
next
  case clean: False
  have state:
    query-header-supported-table-candidate-state s fr f-fri-roots f-final
    as dg composition-fri-roots final trace-table composition-table
    final-state
    unfolding query-header-supported-table-candidate-state-def
    by (intro exI[of - result] exI[of - query-idxs] exI[of - rest])
    (use outcome out-eq bound header trace-bind comp-nonempty comp-bind
      in simp)
  have candidate:
    (trace-table, composition-table) ∈
    query-header-supported-partial-table-candidates s fr f-fri-roots
    f-final as dg composition-fri-roots final
    using
    query-header-supported-table-candidate-state-imp-partial-candidate-state
    [OF state clean]
    query-header-supported-partial-table-candidates-iff-state
    by blast
  have len: length query-idxs = rounds
    by (rule accepted-with-bound-tables-shapes(4)[OF bound])
  have idx-mem: query-idxs ! i ∈ set query-idxs
    by (rule nth-mem) (use i-bound len in simp)
  have idx-sample: query-idxs ! i ∈ query-sample-space
    by (rule accepted-with-bound-tables-query-sample-space
      [OF bound idx-mem])
  have union-hit:

```

```

    query-idxs ! i ∈
      query-header-supported-partial-union-good-sets s good-sets fr
      f-fri-roots f-final as dg composition-fri-roots final
  unfolding query-header-supported-partial-union-good-sets-def
  using idx-sample candidate query-hit by blast
have round-hit:
  query-rounds-any-index-set-hit s
    (query-header-supported-partial-union-good-sets s good-sets fr
      f-fri-roots f-final as dg composition-fri-roots final)
    rounds out
  by (rule accepted-with-bound-tables-query-rounds-any-index-set-hit
    [OF bound i-bound union-hit])
show ?thesis
  unfolding query-header-rounds-any-index-set-hit-def
  using header round-hit by blast
qed
qed

```

```

lemma query-index-round-set-hit-imp-partial-union-or-collision:
  assumes outcome: out ∈ set-dist (execute verify-monad s)
  and hit: query-index-round-set-hit s good-sets out
  shows
    query-header-rounds-any-index-set-hit s
      (query-header-supported-partial-union-good-sets s good-sets) out ∨
      hash-map-output-collision-bad s out
proof –
  from hit obtain i where
    query-index-round-set-hit-at s good-sets i out
  unfolding query-index-round-set-hit-def by blast
  then show ?thesis
  by (rule query-index-round-set-hit-at-imp-partial-union-or-collision
    [OF outcome])
qed

```

```

lemma wp-query-index-round-set-hit-bound-via-partial-union-or-collision:
  fixes C H :: prob
  assumes partial-bound:
    wp-event verify-monad
      (query-header-rounds-any-index-set-hit s
        (query-header-supported-partial-union-good-sets s good-sets)) s ≤
      C
  and collision-bound:
    wp-event verify-monad (hash-map-output-collision-bad s) s ≤ H
  shows
    wp-event verify-monad (query-index-round-set-hit s good-sets) s ≤
      C + H
proof –
  have mono:
    wp-event verify-monad (query-index-round-set-hit s good-sets) s ≤

```

$wp\text{-event verify-monad}$
 $(\lambda out.$
 $\quad query\text{-header-rounds-any-index-set-hit } s$
 $\quad (query\text{-header-supported-partial-union-good-sets } s \text{ good-sets})$
 $\quad out \vee$
 $\quad hash\text{-map-output-collision-bad } s \text{ out}) \ s$
by (rule $wp\text{-event-mono-on-support}$)
 $(use \text{ query-index-round-set-hit-imp-partial-union-or-collision in blast})$
also have $\dots \leq C + H$
proof –
have $wp\text{-event verify-monad}$
 $(\lambda out.$
 $\quad query\text{-header-rounds-any-index-set-hit } s$
 $\quad (query\text{-header-supported-partial-union-good-sets } s \text{ good-sets})$
 $\quad out \vee$
 $\quad hash\text{-map-output-collision-bad } s \text{ out}) \ s \leq$
 $wp\text{-event verify-monad}$
 $\quad (query\text{-header-rounds-any-index-set-hit } s$
 $\quad (query\text{-header-supported-partial-union-good-sets } s \text{ good-sets})) \ s +$
 $wp\text{-event verify-monad } (hash\text{-map-output-collision-bad } s) \ s$
by (rule $wp\text{-event-union-bound}$)
also have $\dots \leq C + H$
by (intro $add\text{-mono partial-bound collision-bound}$)
finally show $?thesis$.
qed
finally show $?thesis$.
qed

lemma $wp\text{-query-index-round-set-hit-bound-via-supported-partial-union-or-collision}$:
fixes $H :: prob$
assumes $future: query\text{-future-fresh } s$
and $raw\text{-bound}: query\text{-index-raw-preimage-bound}$
and $union\text{-bound}$:
 $\bigwedge fr \ f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots final.}$
 $nnreal$
 $(card$
 $\quad (query\text{-header-supported-partial-union-good-sets } s \text{ good-sets } fr$
 $\quad f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots final})) /$
 $nnreal \ (card \ query\text{-sample-space}) \leq query\text{-error-bound}$
and $collision\text{-bound}$:
 $wp\text{-event verify-monad } (hash\text{-map-output-collision-bad } s) \ s \leq H$
shows
 $wp\text{-event verify-monad } (query\text{-index-round-set-hit } s \text{ good-sets}) \ s \leq$
 $nnreal \ rounds * query\text{-error-bound} + H$
proof –
have $partial\text{-bound}$:
 $wp\text{-event verify-monad}$
 $\quad (query\text{-header-rounds-any-index-set-hit } s$
 $\quad (query\text{-header-supported-partial-union-good-sets } s \text{ good-sets})) \ s \leq$

$nnreal\ rounds * query\ error\ bound$
by (rule *wp-verify-monad-query-header-rounds-any-index-set-bound*
 $[OF\ future\ raw\ bound]$)
 (use *query-header-supported-partial-union-good-sets-subset union-bound*
 in *simp-all*)
show *?thesis*
by (rule *wp-query-index-round-set-hit-bound-via-partial-union-or-collision*
 $[OF\ partial\ bound\ collision\ bound]$)
qed

lemma *wp-query-index-set-hit-bound-via-partial-union-or-collision*:
fixes $C\ H :: prob$
assumes *partial-bound*:
 $wp\ event\ verify\ monad$
 $(query\ header\ rounds\ any\ index\ set\ hit\ s$
 $(query\ header\ supported\ partial\ union\ good\ sets\ s\ good\ sets))\ s \leq$
 C
and *collision-bound*:
 $wp\ event\ verify\ monad\ (hash\ map\ output\ collision\ bad\ s)\ s \leq H$
shows
 $wp\ event\ verify\ monad\ (query\ index\ set\ hit\ s\ good\ sets)\ s \leq C + H$
proof –
have $wp\ event\ verify\ monad\ (query\ index\ set\ hit\ s\ good\ sets)\ s \leq$
 $wp\ event\ verify\ monad\ (query\ index\ round\ set\ hit\ s\ good\ sets)\ s$
by (rule *wp-event-mono*) (rule *query-index-set-hit-imp-round-set-hit*)
also have $\dots \leq C + H$
by (rule *wp-query-index-round-set-hit-bound-via-partial-union-or-collision*
 $[OF\ partial\ bound\ collision\ bound]$)
finally show *?thesis* .
qed

lemma *wp-query-index-round-set-hit-bound-if-supported-partial-header-candidate-unique-or-collision*:
fixes $H :: prob$
assumes *future*: *query-future-fresh* s
and *raw-bound*: *query-index-raw-preimage-bound*
and *unique*:
 $\bigwedge fr\ f\ fri\ roots\ f\ final\ as\ dg\ composition\ fri\ roots\ final.$
 $\exists\ trace\ table\ composition\ table.$
 $query\ header\ supported\ partial\ table\ candidates\ s\ fr\ f\ fri\ roots$
 $f\ final\ as\ dg\ composition\ fri\ roots\ final \subseteq$
 $\{(trace\ table,\ composition\ table)\}$
and *subset*:
 $\bigwedge trace\ table\ composition\ table\ as.$
 $good\ sets\ trace\ table\ composition\ table\ as \subseteq query\ sample\ space$
and *bounded*:
 $\bigwedge trace\ table\ composition\ table\ as.$
 $nnreal\ (card\ (good\ sets\ trace\ table\ composition\ table\ as)) /$
 $nnreal\ (card\ query\ sample\ space) \leq query\ error\ bound$
and *collision-bound*:

$wp\text{-}event\ verify\text{-}monad\ (hash\text{-}map\text{-}output\text{-}collision\text{-}bad\ s)\ s \leq H$
shows
 $wp\text{-}event\ verify\text{-}monad\ (query\text{-}index\text{-}round\text{-}set\text{-}hit\ s\ good\text{-}sets)\ s \leq$
 $nnreal\ rounds * query\text{-}error\text{-}bound + H$
proof –
have *union-bound*:
 $\bigwedge fr\ f\text{-}fri\text{-}roots\ f\text{-}final\ as\ dg\ composition\text{-}fri\text{-}roots\ final.$
 $nnreal$
 $(card$
 $(query\text{-}header\text{-}supported\text{-}partial\text{-}union\text{-}good\text{-}sets\ s\ good\text{-}sets\ fr$
 $f\text{-}fri\text{-}roots\ f\text{-}final\ as\ dg\ composition\text{-}fri\text{-}roots\ final)) /$
 $nnreal\ (card\ query\text{-}sample\text{-}space) \leq query\text{-}error\text{-}bound$
by (rule
 $query\text{-}header\text{-}supported\text{-}partial\text{-}union\text{-}good\text{-}sets\text{-}fraction\text{-}bound\text{-}if\text{-}unique\text{-}candidate$
 $[OF\ unique\ subset\ bounded]$)
show *?thesis*
by (rule
 $wp\text{-}query\text{-}index\text{-}round\text{-}set\text{-}hit\text{-}bound\text{-}via\text{-}supported\text{-}partial\text{-}union\text{-}or\text{-}collision$
 $[OF\ future\ raw\text{-}bound\ union\text{-}bound\ collision\text{-}bound]$)
qed

lemma *wp-query-index-set-hit-bound-if-supported-partial-header-candidate-unique-or-collision*:
fixes $H :: prob$
assumes *future*: $query\text{-}future\text{-}fresh\ s$
and *raw-bound*: $query\text{-}index\text{-}raw\text{-}preimage\text{-}bound$
and *unique*:
 $\bigwedge fr\ f\text{-}fri\text{-}roots\ f\text{-}final\ as\ dg\ composition\text{-}fri\text{-}roots\ final.$
 $\exists\ trace\text{-}table\ composition\text{-}table.$
 $query\text{-}header\text{-}supported\text{-}partial\text{-}table\text{-}candidates\ s\ fr\ f\text{-}fri\text{-}roots$
 $f\text{-}final\ as\ dg\ composition\text{-}fri\text{-}roots\ final \subseteq$
 $\{(trace\text{-}table,\ composition\text{-}table)\}$
and *subset*:
 $\bigwedge trace\text{-}table\ composition\text{-}table\ as.$
 $good\text{-}sets\ trace\text{-}table\ composition\text{-}table\ as \subseteq query\text{-}sample\text{-}space$
and *bounded*:
 $\bigwedge trace\text{-}table\ composition\text{-}table\ as.$
 $nnreal\ (card\ (good\text{-}sets\ trace\text{-}table\ composition\text{-}table\ as)) /$
 $nnreal\ (card\ query\text{-}sample\text{-}space) \leq query\text{-}error\text{-}bound$
and *collision-bound*:
 $wp\text{-}event\ verify\text{-}monad\ (hash\text{-}map\text{-}output\text{-}collision\text{-}bad\ s)\ s \leq H$
shows
 $wp\text{-}event\ verify\text{-}monad\ (query\text{-}index\text{-}set\text{-}hit\ s\ good\text{-}sets)\ s \leq$
 $nnreal\ rounds * query\text{-}error\text{-}bound + H$
proof –
have $wp\text{-}event\ verify\text{-}monad\ (query\text{-}index\text{-}set\text{-}hit\ s\ good\text{-}sets)\ s \leq$
 $wp\text{-}event\ verify\text{-}monad\ (query\text{-}index\text{-}round\text{-}set\text{-}hit\ s\ good\text{-}sets)\ s$
by (rule *wp-event-mono*) (rule *query-index-set-hit-imp-round-set-hit*)
also have $\dots \leq nnreal\ rounds * query\text{-}error\text{-}bound + H$
by (rule

$wp\text{-}query\text{-}index\text{-}round\text{-}set\text{-}hit\text{-}bound\text{-}if\text{-}supported\text{-}partial\text{-}header\text{-}candidate\text{-}unique\text{-}or\text{-}collision$
 $[OF\ future\ raw\text{-}bound\ unique\ subset\ bounded\ collision\text{-}bound])$
finally show $?thesis$.
qed

lemma $wp\text{-}query\text{-}sampling\text{-}success\text{-}set\text{-}hit\text{-}bound\text{-}if\text{-}supported\text{-}partial\text{-}header\text{-}candidate\text{-}unique\text{-}or\text{-}collision$:
fixes $H :: prob$
assumes $future$: $query\text{-}future\text{-}fresh\ s$
and $raw\text{-}bound$: $query\text{-}index\text{-}raw\text{-}preimage\text{-}bound$
and $unique$:
 $\bigwedge fr\ f\text{-}fri\text{-}roots\ f\text{-}final\ as\ dg\ composition\text{-}fri\text{-}roots\ final.$
 $\exists\ trace\text{-}table\ composition\text{-}table.$
 $query\text{-}header\text{-}supported\text{-}partial\text{-}table\text{-}candidates\ s\ fr\ f\text{-}fri\text{-}roots$
 $f\text{-}final\ as\ dg\ composition\text{-}fri\text{-}roots\ final \subseteq$
 $\{(trace\text{-}table, composition\text{-}table)\}$
and $collision\text{-}bound$:
 $wp\text{-}event\ verify\text{-}monad\ (hash\text{-}map\text{-}output\text{-}collision\text{-}bad\ s)\ s \leq H$
shows
 $wp\text{-}event\ verify\text{-}monad$
 $(query\text{-}index\text{-}set\text{-}hit\ s\ query\text{-}sampling\text{-}success\text{-}space)\ s \leq$
 $nnreal\ rounds * query\text{-}error\text{-}bound + H$
by (rule
 $wp\text{-}query\text{-}index\text{-}set\text{-}hit\text{-}bound\text{-}if\text{-}supported\text{-}partial\text{-}header\text{-}candidate\text{-}unique\text{-}or\text{-}collision$
 $[OF\ future\ raw\text{-}bound\ unique\ query\text{-}sampling\text{-}success\text{-}space\text{-}subset$
 $query\text{-}sampling\text{-}success\text{-}space\text{-}fraction\text{-}bound\text{-}query\text{-}sample\text{-}space$
 $collision\text{-}bound])$

lemma $composition\text{-}fri\text{-}supported\text{-}root\text{-}composition\text{-}table\text{-}candidate\text{-}state\text{-}hash\text{-}extends$:
assumes
 $composition\text{-}fri\text{-}supported\text{-}root\text{-}composition\text{-}table\text{-}candidate\text{-}state\ s\ fr$
 $f\text{-}fri\text{-}roots\ f\text{-}final\ as\ dg\ composition\text{-}fri\text{-}roots\ composition\text{-}table$
 $final\text{-}state$
shows $s \leq final\text{-}state$
proof –
from $assms$ **obtain** $result\ trace\text{-}table\ query\text{-}idxs\ trace\text{-}bs\ comp\text{-}bs\ final\ rest$
where $outcome$:
 $Some\ (result, final\text{-}state) \in set\text{-}dist\ (execute\ verify\text{-}monad\ s)$
unfolding $composition\text{-}fri\text{-}supported\text{-}root\text{-}composition\text{-}table\text{-}candidate\text{-}state\text{-}def$
by $blast$
show $?thesis$
by (rule $verify\text{-}monad\text{-}hash\text{-}extends[OF\ outcome]$)
qed

lemma $composition\text{-}fri\text{-}supported\text{-}root\text{-}composition\text{-}table\text{-}candidate\text{-}states\text{-}merkle\text{-}compatible\text{-}unique\text{-}if\text{-}clean\text{-}merge$:
assumes $compatible$:
 $merkle\text{-}hash\text{-}maps\text{-}compatible\ final\text{-}state\ final\text{-}state'$
and $clean$:
 $\neg\ hash\text{-}map\text{-}output\text{-}collision$
 $(merkle\text{-}hash\text{-}state\text{-}merge\ final\text{-}state\ final\text{-}state')$


```

and candidate:
  composition-fri-supported-root-composition-table-candidate-state s fr
    f-fri-roots f-final as dg composition-fri-roots composition-table
    final-state
and candidate':
  composition-fri-supported-root-composition-table-candidate-state s fr
    f-fri-roots f-final as dg composition-fri-roots composition-table'
    final-state'
shows composition-table = composition-table'
proof –
from candidate obtain result trace-table query-idxs trace-bs comp-bs final
  rest where
  bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
and comp-nonempty: composition-fri-roots ≠ []
and bind:
  merkle-root-binds-table (hd composition-fri-roots) composition-table
    final-state
unfolding composition-fri-supported-root-composition-table-candidate-state-def
by blast
from candidate' obtain result' trace-table' query-idxs' trace-bs' comp-bs'
  final' rest' where
  bound':
    accepted-with-bound-tables s (Some (result', final-state'))
    trace-table' composition-table' as query-idxs'
and bind':
  merkle-root-binds-table (hd composition-fri-roots) composition-table'
    final-state'
unfolding composition-fri-supported-root-composition-table-candidate-state-def
by blast
have same-len: length composition-table = length composition-table'
using accepted-with-bound-tables-shapes(2)[OF bound]
  accepted-with-bound-tables-shapes(2)[OF bound']
by simp
have nonempty: composition-table ≠ []
using accepted-with-bound-tables-shapes(2)[OF bound]
  eval-domain-nontrivial by auto
show ?thesis
by (rule
  merkle-root-binds-same-length-tables-merkle-compatible-unique-if-clean-merge
  [OF compatible clean bind bind' same-len nonempty])
qed

```

```

lemma composition-fri-supported-root-composition-table-candidates-subset-singleton-if-common-clean-merkle-extension
assumes clean: ¬ hash-map-output-collision u
and common-ext:
  ∧ composition-table final-state.
  composition-fri-supported-root-composition-table-candidate-state s fr

```

```

    f-fri-roots f-final as dg composition-fri-roots composition-table
    final-state  $\implies$ 
    merkle-hash-extends final-state u
  shows
     $\exists$  composition-table.
    composition-fri-supported-root-composition-table-candidates s fr
    f-fri-roots f-final as dg composition-fri-roots  $\subseteq$ 
    {composition-table}
  proof (cases
    composition-fri-supported-root-composition-table-candidates s fr
    f-fri-roots f-final as dg composition-fri-roots = {})
  case True
  then show ?thesis
    by (intro exI[of - []]) simp
next
  case False
  then obtain composition-table0 where table0:
    composition-table0  $\in$ 
    composition-fri-supported-root-composition-table-candidates s fr
    f-fri-roots f-final as dg composition-fri-roots
  by blast
  then obtain final-state0 where state0:
    composition-fri-supported-root-composition-table-candidate-state s fr
    f-fri-roots f-final as dg composition-fri-roots composition-table0
    final-state0
  using composition-fri-supported-root-composition-table-candidates-iff-state
  by blast
  have subset:
    composition-fri-supported-root-composition-table-candidates s fr
    f-fri-roots f-final as dg composition-fri-roots  $\subseteq$ 
    {composition-table0}
  proof
    fix composition-table
    assume table:
      composition-table  $\in$ 
      composition-fri-supported-root-composition-table-candidates s fr
      f-fri-roots f-final as dg composition-fri-roots
    then obtain final-state where state:
      composition-fri-supported-root-composition-table-candidate-state s fr
      f-fri-roots f-final as dg composition-fri-roots composition-table
      final-state
    using composition-fri-supported-root-composition-table-candidates-iff-state
    by blast
    have ext: merkle-hash-extends final-state u
      by (rule common-ext[OF state])
    have ext0: merkle-hash-extends final-state0 u
      by (rule common-ext[OF state0])
    have clean-pair:
       $\neg$  merkle-hash-value-conflict final-state final-state0  $\wedge$ 

```

```

  ¬ hash-map-output-collision
    (merkle-hash-state-merge final-state final-state0)
  by (rule common-clean-merkle-extension-imp-pairwise-merkle-clean
    [OF ext ext0 clean])
  have compatible:
    merkle-hash-maps-compatible final-state final-state0
  using clean-pair
  unfolding merkle-hash-maps-compatible-iff-no-value-conflict by simp
  have clean-merge:
    ¬ hash-map-output-collision
      (merkle-hash-state-merge final-state final-state0)
  using clean-pair by simp
  have composition-table = composition-table0
  by (rule
    composition-fri-supported-root-composition-table-candidate-states-merkle-compatible-unique-if-clean-merge
    [OF compatible clean-merge state state0])
  then show composition-table ∈ {composition-table0}
  by simp
qed
then show ?thesis
  by blast
qed

lemma composition-fri-supported-root-composition-table-candidate-states-distinct-imp-merkle-conflict-or-merge-collis
  assumes candidate:
    composition-fri-supported-root-composition-table-candidate-state s fr
    f-fri-roots f-final as dg composition-fri-roots composition-table
    final-state
  and candidate':
    composition-fri-supported-root-composition-table-candidate-state s fr
    f-fri-roots f-final as dg composition-fri-roots composition-table'
    final-state'
  and distinct-tables: composition-table ≠ composition-table'
  shows
    merkle-hash-value-conflict final-state final-state' ∨
    hash-map-output-collision
      (merkle-hash-state-merge final-state final-state')
  proof (rule ccontr)
    assume no-bad:
      ¬ (merkle-hash-value-conflict final-state final-state' ∨
        hash-map-output-collision
          (merkle-hash-state-merge final-state final-state'))
    have compatible:
      merkle-hash-maps-compatible final-state final-state'
    using no-bad unfolding merkle-hash-maps-compatible-iff-no-value-conflict
    by simp
    have clean:
      ¬ hash-map-output-collision
        (merkle-hash-state-merge final-state final-state')

```

```

using no-bad by simp
have composition-table = composition-table'
by (rule
  composition-fri-supported-root-composition-table-candidate-states-merkle-compatible-unique-if-clean-merge
  [OF compatible clean candidate candidate'])
then show False
  using distinct-tables by contradiction
qed

```

lemma *composition-fri-supported-root-composition-table-candidates-not-singleton-imp-pairwise-merkle-bad*:
assumes *not-unique*:

```

   $\neg (\exists \textit{composition-table.}$ 
    composition-fri-supported-root-composition-table-candidates s fr
      f-fri-roots f-final as dg composition-fri-roots  $\subseteq$ 
        {composition-table})

```

shows

```

 $\exists \textit{composition-table}$  final-state composition-table' final-state'.
  composition-fri-supported-root-composition-table-candidate-state s fr
    f-fri-roots f-final as dg composition-fri-roots composition-table
      final-state  $\wedge$ 
    composition-fri-supported-root-composition-table-candidate-state s fr
      f-fri-roots f-final as dg composition-fri-roots composition-table'
        final-state'  $\wedge$ 
    composition-table  $\neq$  composition-table'  $\wedge$ 
    (merkle-hash-value-conflict final-state final-state'  $\vee$ 
      hash-map-output-collision
        (merkle-hash-state-merge final-state final-state'))

```

proof —

```

let ?C =
  composition-fri-supported-root-composition-table-candidates s fr
    f-fri-roots f-final as dg composition-fri-roots
have nonempty: ?C  $\neq$  {}
  using not-unique by auto
then obtain composition-table where table: composition-table  $\in$  ?C
  by blast
have  $\neg ?C \subseteq \{\textit{composition-table}\}$ 
  using not-unique by blast
then obtain composition-table' where table':
  composition-table'  $\in$  ?C
  and distinct-tables: composition-table  $\neq$  composition-table'
  by auto
from table obtain final-state where candidate:
  composition-fri-supported-root-composition-table-candidate-state s fr
    f-fri-roots f-final as dg composition-fri-roots composition-table
      final-state
  using composition-fri-supported-root-composition-table-candidates-iff-state
  by blast
from table' obtain final-state' where candidate':
  composition-fri-supported-root-composition-table-candidate-state s fr

```

f-fri-roots f-final as dg composition-fri-roots composition-table'
final-state'
using *composition-fri-supported-root-composition-table-candidates-iff-state*
by *blast*
have *bad:*
merkle-hash-value-conflict final-state final-state' ∨
hash-map-output-collision
(merkle-hash-state-merge final-state final-state')
by *(rule*
composition-fri-supported-root-composition-table-candidate-states-distinct-imp-merkle-conflict-or-merge-collision
[OF candidate candidate' distinct-tables])
show *?thesis*
using *candidate candidate' distinct-tables bad by blast*
qed

definition *composition-fri-supported-root-pairwise-merkle-bad*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $f \Rightarrow f \text{ list} \Rightarrow f \Rightarrow f \text{ list} \Rightarrow f \Rightarrow$
 $f \text{ list} \Rightarrow \text{bool}$

where

composition-fri-supported-root-pairwise-merkle-bad s fr f-fri-roots
f-final as dg composition-fri-roots $\longleftrightarrow$
($\exists$ composition-table final-state composition-table' final-state'.
composition-fri-supported-root-composition-table-candidate-state s fr
f-fri-roots f-final as dg composition-fri-roots composition-table
final-state $\wedge$
composition-fri-supported-root-composition-table-candidate-state s fr
f-fri-roots f-final as dg composition-fri-roots composition-table'
final-state' $\wedge$
composition-table $\neq$ composition-table' $\wedge$
(merkle-hash-value-conflict final-state final-state' $\vee$
hash-map-output-collision
(merkle-hash-state-merge final-state final-state'))))

definition *composition-fri-supported-pairwise-merkle-bad*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

composition-fri-supported-pairwise-merkle-bad s $\longleftrightarrow$
($\exists$ fr f-fri-roots f-final as dg composition-fri-roots.
composition-fri-supported-root-pairwise-merkle-bad s fr f-fri-roots
f-final as dg composition-fri-roots)

lemma *composition-fri-supported-root-pairwise-merkle-bad-extends-initial:*

assumes *bad:*

composition-fri-supported-root-pairwise-merkle-bad s fr f-fri-roots
f-final as dg composition-fri-roots

obtains *composition-table final-state composition-table' final-state'*

where

composition-fri-supported-root-composition-table-candidate-state s fr

*f-fri-roots f-final as dg composition-fri-roots composition-table
 final-state*
*composition-fri-supported-root-composition-table-candidate-state s fr
 f-fri-roots f-final as dg composition-fri-roots composition-table'
 final-state'*
composition-table $\neq$ composition-table'
*merkle-hash-value-conflict final-state final-state' $\vee$
 hash-map-output-collision
 (merkle-hash-state-merge final-state final-state')*
s $\leq$ final-state
s $\leq$ final-state'
proof –
from *bad* **obtain** *composition-table final-state composition-table' final-state'*
where *candidate:*
*composition-fri-supported-root-composition-table-candidate-state s fr
 f-fri-roots f-final as dg composition-fri-roots composition-table
 final-state*
and *candidate':*
*composition-fri-supported-root-composition-table-candidate-state s fr
 f-fri-roots f-final as dg composition-fri-roots composition-table'
 final-state'*
and *distinct: composition-table $\neq$ composition-table'*
and *pair-bad:*
*merkle-hash-value-conflict final-state final-state' $\vee$
 hash-map-output-collision
 (merkle-hash-state-merge final-state final-state')*
unfolding *composition-fri-supported-root-pairwise-merkle-bad-def*
by *blast*
have *ext: s $\leq$ final-state*
by (*rule*
composition-fri-supported-root-composition-table-candidate-state-hash-extends
[OF candidate])
have *ext': s $\leq$ final-state'*
by (*rule*
composition-fri-supported-root-composition-table-candidate-state-hash-extends
[OF candidate'])
show *?thesis*
by (*rule that[OF candidate candidate' distinct pair-bad ext ext']*)
qed

definition *composition-fri-supported-root-pairwise-coupling-bad*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow \text{bool}$
where
*composition-fri-supported-root-pairwise-coupling-bad s fr f-fri-roots
 f-final as dg composition-fri-roots $\longleftrightarrow$
 ($\exists$ *composition-table final-state composition-table' final-state'*
*composition-fri-supported-root-composition-table-candidate-state s fr**

$f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table}$
 $\text{final-state} \wedge$
 $\text{composition-fri-supported-root-composition-table-candidate-state } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table'}$
 $\text{final-state}' \wedge$
 $\text{composition-table} \neq \text{composition-table}' \wedge$
 $\text{merkle-hash-pairwise-coupling-bad final-state final-state}'$

definition $\text{composition-fri-supported-pairwise-coupling-bad}$

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{composition-fri-supported-pairwise-coupling-bad } s \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots.}$
 $\text{composition-fri-supported-root-pairwise-coupling-bad } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots})$

lemma $\text{composition-fri-supported-root-pairwise-merkle-bad-imp-local-collision-or-coupling-bad:}$

assumes bad:

$\text{composition-fri-supported-root-pairwise-merkle-bad } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots}$

shows

$(\exists \text{ composition-table final-state composition-table' final-state'.
 $\text{composition-fri-supported-root-composition-table-candidate-state } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table}$
 $\text{final-state} \wedge$
 $\text{composition-fri-supported-root-composition-table-candidate-state } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table'}$
 $\text{final-state}' \wedge$
 $\text{composition-table} \neq \text{composition-table}' \wedge$
 $(\text{hash-map-output-collision final-state} \vee$
 $\text{hash-map-output-collision final-state}')) \vee$
 $\text{composition-fri-supported-root-pairwise-coupling-bad } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots}$$

proof —

from bad **obtain** $\text{composition-table final-state composition-table' final-state'}$

where candidate:

$\text{composition-fri-supported-root-composition-table-candidate-state } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table}$
 final-state

and candidate':

$\text{composition-fri-supported-root-composition-table-candidate-state } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table'}$
 final-state'

and $\text{distinct: composition-table} \neq \text{composition-table'}$

and pair-bad:

$\text{merkle-hash-value-conflict final-state final-state}' \vee$
 $\text{hash-map-output-collision}$
 $(\text{merkle-hash-state-merge final-state final-state}')$

unfolding $\text{composition-fri-supported-root-pairwise-merkle-bad-def}$

```

    by blast
  have decomp:
    hash-map-output-collision final-state ∨
    hash-map-output-collision final-state' ∨
    merkle-hash-pairwise-coupling-bad final-state final-state'
  by (rule merkle-hash-pairwise-bad-imp-local-collision-or-coupling-bad
    [OF pair-bad])
  then show ?thesis
proof
  assume hash-map-output-collision final-state
  then show ?thesis
    using candidate candidate' distinct by blast
next
  assume rest:
    hash-map-output-collision final-state' ∨
    merkle-hash-pairwise-coupling-bad final-state final-state'
  then show ?thesis
proof
  assume hash-map-output-collision final-state'
  then show ?thesis
    using candidate candidate' distinct by blast
next
  assume merkle-hash-pairwise-coupling-bad final-state final-state'
  then have
    composition-fri-supported-root-pairwise-coupling-bad s fr f-fri-roots
    f-final as dg composition-fri-roots
  unfolding composition-fri-supported-root-pairwise-coupling-bad-def
  using candidate candidate' distinct by blast
  then show ?thesis by simp
qed
qed
qed

lemma composition-fri-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad:
  assumes bad: composition-fri-supported-pairwise-merkle-bad s
  shows
    (∃ fr f-fri-roots f-final as dg composition-fri-roots
      composition-table final-state composition-table' final-state'.
      composition-fri-supported-root-composition-table-candidate-state s fr
      f-fri-roots f-final as dg composition-fri-roots composition-table
      final-state ∧
      composition-fri-supported-root-composition-table-candidate-state s fr
      f-fri-roots f-final as dg composition-fri-roots composition-table'
      final-state' ∧
      composition-table ≠ composition-table' ∧
      (hash-map-output-collision final-state ∨
      hash-map-output-collision final-state')) ∨
      composition-fri-supported-pairwise-coupling-bad s
proof -

```


from *bad* **obtain** *fr f-fri-roots f-final as dg composition-fri-roots* **where**
root-bad:
composition-fri-supported-root-pairwise-merkle-bad s fr f-fri-roots
f-final as dg composition-fri-roots
unfolding *composition-fri-supported-pairwise-merkle-bad-def* **by** *blast*
from
composition-fri-supported-root-pairwise-merkle-bad-imp-local-collision-or-coupling-bad
[OF root-bad]
show *?thesis*
unfolding *composition-fri-supported-pairwise-coupling-bad-def* **by** *blast*
qed

lemma *composition-fri-supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad:*
assumes *bad: composition-fri-supported-pairwise-merkle-bad s*
shows
supported-hash-output-collision-possible s ∨
composition-fri-supported-pairwise-coupling-bad s
proof *—*
have *decomp:*
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots}$
 $\text{composition-table final-state composition-table' final-state'.$
 $\text{composition-fri-supported-root-composition-table-candidate-state } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table}$
 $\text{final-state} \wedge$
 $\text{composition-fri-supported-root-composition-table-candidate-state } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table'}$
 $\text{final-state'} \wedge$
 $\text{composition-table} \neq \text{composition-table'} \wedge$
 $(\text{hash-map-output-collision final-state} \vee$
 $\text{hash-map-output-collision final-state'})) \vee$
 $\text{composition-fri-supported-pairwise-coupling-bad } s$
by *(rule*
 $\text{composition-fri-supported-pairwise-merkle-bad-imp-local-collision-or-coupling-bad}$
 $[\text{OF bad}])$
then show *?thesis*
proof
assume *local:*
 $\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots}$
 $\text{composition-table final-state composition-table' final-state'.$
 $\text{composition-fri-supported-root-composition-table-candidate-state } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table}$
 $\text{final-state} \wedge$
 $\text{composition-fri-supported-root-composition-table-candidate-state } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots composition-table'}$
 $\text{final-state'} \wedge$
 $\text{composition-table} \neq \text{composition-table'} \wedge$
 $(\text{hash-map-output-collision final-state} \vee$
 $\text{hash-map-output-collision final-state'})$
then obtain *fr f-fri-roots f-final as dg composition-fri-roots*

```

    composition-table final-state composition-table' final-state' where
candidate:
    composition-fri-supported-root-composition-table-candidate-state s fr
    f-fri-roots f-final as dg composition-fri-roots composition-table
    final-state
and candidate':
    composition-fri-supported-root-composition-table-candidate-state s fr
    f-fri-roots f-final as dg composition-fri-roots composition-table'
    final-state'
and collision:
    hash-map-output-collision final-state  $\vee$ 
    hash-map-output-collision final-state'
by blast
from collision show ?thesis
proof
    assume collision-final: hash-map-output-collision final-state
    from candidate obtain result trace-table query-idxs trace-bs comp-bs final
    rest where
    outcome:
        Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
    and bound:
        accepted-with-bound-tables s (Some (result, final-state))
        trace-table composition-table as query-idxs
    unfolding
        composition-fri-supported-root-composition-table-candidate-state-def
    by blast
have supported-hash-output-collision-possible s
    by (rule
        accepted-with-bound-tables-hash-collision-imp-supported-hash-output-collision-possible
        [OF outcome bound collision-final])
    then show ?thesis by simp
next
    assume collision-final': hash-map-output-collision final-state'
    from candidate' obtain result' trace-table' query-idxs' trace-bs'
    comp-bs' final' rest' where
    outcome':
        Some (result', final-state')  $\in$  set-dist (execute verify-monad s)
    and bound':
        accepted-with-bound-tables s (Some (result', final-state'))
        trace-table' composition-table' as query-idxs'
    unfolding
        composition-fri-supported-root-composition-table-candidate-state-def
    by blast
have supported-hash-output-collision-possible s
    by (rule
        accepted-with-bound-tables-hash-collision-imp-supported-hash-output-collision-possible
        [OF outcome' bound' collision-final'])
    then show ?thesis by simp
qed

```

next
assume *composition-fri-supported-pairwise-coupling-bad s*
then show *?thesis* **by** *simp*
qed
qed

definition *alpha-output-local-candidate-ambiguity-bad*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{alpha-output-local-candidate-ambiguity-bad } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-table composition-table alphas query-idxs}$
 $\text{ trace-table}' \text{ composition-table}' \text{ alphas}' \text{ query-idxs}'.$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table}$
 $\text{ alphas query-idxs} \wedge$
 $\text{accepted-with-bound-tables } s \text{ out trace-table}' \text{ composition-table}'$
 $\text{ alphas}' \text{ query-idxs}' \wedge$
 $\text{trace-table} \neq \text{trace-table}')$

definition *query-output-local-candidate-ambiguity-bad*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{query-output-local-candidate-ambiguity-bad } s \text{ out} \longleftrightarrow$
 $\text{accepted-bound-table-ambiguity-bad } s \text{ out}$

definition *trace-fri-output-local-candidate-ambiguity-bad*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-output-local-candidate-ambiguity-bad } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-table composition-table alphas query-idxs}$
 $\text{ trace-table}' \text{ composition-table}' \text{ alphas}' \text{ query-idxs}'$
 $\text{ trace-bs dg composition-bs}.$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table}$
 $\text{ alphas query-idxs} \wedge$
 $\text{accepted-with-bound-tables } s \text{ out trace-table}' \text{ composition-table}'$
 $\text{ alphas}' \text{ query-idxs}' \wedge$
 $\text{accepted-fri-challenges } s \text{ out trace-bs dg composition-bs} \wedge$
 $\text{trace-table} \neq \text{trace-table}')$

definition *composition-fri-output-local-candidate-ambiguity-bad*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{composition-fri-output-local-candidate-ambiguity-bad } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-table composition-table alphas query-idxs}$
 $\text{ trace-table}' \text{ composition-table}' \text{ alphas}' \text{ query-idxs}'$
 $\text{ trace-bs dg composition-bs}.$

$accepted-with-bound-tables\ s\ out\ trace-table\ composition-table$
 $\alpha\ query-idxs\ \wedge$
 $accepted-with-bound-tables\ s\ out\ trace-table'\ composition-table'$
 $\alpha\ query-idxs'\ \wedge$
 $accepted-fri-challenges\ s\ out\ trace-bs\ dg\ composition-bs\ \wedge$
 $composition-table \neq composition-table')$

definition *supported-output-local-candidate-ambiguity-bad*

$:: (f, 'a)\ protocol-channel-scheme \Rightarrow$
 $(unit\ list \times (f, 'a)\ protocol-channel-scheme)\ option \Rightarrow bool$

where

$supported-output-local-candidate-ambiguity-bad\ s\ out \longleftrightarrow$
 $\alpha-output-local-candidate-ambiguity-bad\ s\ out\ \vee$
 $query-output-local-candidate-ambiguity-bad\ s\ out\ \vee$
 $trace-fri-output-local-candidate-ambiguity-bad\ s\ out\ \vee$
 $composition-fri-output-local-candidate-ambiguity-bad\ s\ out$

lemma *alpha-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad:*

assumes $\alpha-output-local-candidate-ambiguity-bad\ s\ out$
shows $accepted-bound-table-ambiguity-bad\ s\ out$
using *assms*
unfolding $\alpha-output-local-candidate-ambiguity-bad-def$
 $accepted-bound-table-ambiguity-bad-def$
by *blast*

lemma *query-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad:*

assumes $query-output-local-candidate-ambiguity-bad\ s\ out$
shows $accepted-bound-table-ambiguity-bad\ s\ out$
using *assms* **unfolding** $query-output-local-candidate-ambiguity-bad-def$.

lemma *trace-fri-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad:*

assumes $trace-fri-output-local-candidate-ambiguity-bad\ s\ out$
shows $accepted-bound-table-ambiguity-bad\ s\ out$
using *assms*
unfolding $trace-fri-output-local-candidate-ambiguity-bad-def$
 $accepted-bound-table-ambiguity-bad-def$
by *blast*

lemma *composition-fri-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad:*

assumes $composition-fri-output-local-candidate-ambiguity-bad\ s\ out$
shows $accepted-bound-table-ambiguity-bad\ s\ out$
using *assms*
unfolding $composition-fri-output-local-candidate-ambiguity-bad-def$
 $accepted-bound-table-ambiguity-bad-def$
by *blast*

lemma *supported-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad:*

assumes $supported-output-local-candidate-ambiguity-bad\ s\ out$
shows $accepted-bound-table-ambiguity-bad\ s\ out$

using *assms*
unfolding *supported-output-local-candidate-ambiguity-bad-def*
using
alpha-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad
query-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad
trace-fri-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad
composition-fri-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad
by *blast*

lemma *supported-output-local-candidate-ambiguity-bad-imp-hash-map-output-collision-bad:*
assumes *supported-output-local-candidate-ambiguity-bad s out*
shows *hash-map-output-collision-bad s out*
by (rule
accepted-bound-table-ambiguity-bad-imp-hash-map-output-collision-bad)
(rule
supported-output-local-candidate-ambiguity-bad-imp-accepted-bound-table-ambiguity-bad
[*OF assms*])

lemma *supported-output-local-candidate-ambiguity-bad-mono-hash-map-output-collision-bad:*
wp-event verify-monad
(supported-output-local-candidate-ambiguity-bad s) s ≤
wp-event verify-monad (hash-map-output-collision-bad s) s
by (rule *wp-event-mono*)
(rule
supported-output-local-candidate-ambiguity-bad-imp-hash-map-output-collision-bad)

definition *supported-output-local-side-bad*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
supported-output-local-side-bad s out $\longleftrightarrow$
hash-map-output-collision-bad s out $\vee$
supported-output-local-candidate-ambiguity-bad s out

lemma *supported-output-local-side-bad-iff-hash-map-output-collision-bad:*
supported-output-local-side-bad s out $\longleftrightarrow$
hash-map-output-collision-bad s out
unfolding *supported-output-local-side-bad-def*
using
supported-output-local-candidate-ambiguity-bad-imp-hash-map-output-collision-bad
by *blast*

lemma *supported-output-local-side-bad-imp-accepted:*
assumes *supported-output-local-side-bad s out*
shows *accepted out*
using *assms*
unfolding *supported-output-local-side-bad-iff-hash-map-output-collision-bad*
by (rule *hash-map-output-collision-bad-imp-accepted*)

lemma *supported-output-local-side-bad-mono-hash-map-output-collision-bad*:
 $wp\text{-}event\ verify\text{-}monad\ (supported\text{-}output\text{-}local\text{-}side\text{-}bad\ s)\ s \leq$
 $wp\text{-}event\ verify\text{-}monad\ (hash\text{-}map\text{-}output\text{-}collision\text{-}bad\ s)\ s$
by (rule *wp-event-mono*)
(simp add: *supported-output-local-side-bad-iff-hash-map-output-collision-bad*)

end

end

theory *Soundness-Reductions-Side*
imports *Soundness-Reductions-Composition*
begin

Supported pairwise/output-local side-event reductions.

context *soundness*
begin

The following cross-continuation predicates are retained temporarily as a record of the earlier proof route. The refined route below does not charge same-key/different-output lazy-oracle continuations as bad events; it uses *supported-output-local-side-bad* instead.

definition *supported-pairwise-merkle-bad*
 $:: (f, 'a)\ protocol\text{-}channel\text{-}scheme \Rightarrow bool$
where
 $supported\text{-}pairwise\text{-}merkle\text{-}bad\ s \longleftrightarrow$
 $\alpha\text{-}supported\text{-}pairwise\text{-}merkle\text{-}bad\ s \vee$
 $query\text{-}supported\text{-}pairwise\text{-}merkle\text{-}bad\ s \vee$
 $trace\text{-}fri\text{-}supported\text{-}pairwise\text{-}merkle\text{-}bad\ s \vee$
 $composition\text{-}fri\text{-}supported\text{-}pairwise\text{-}merkle\text{-}bad\ s$

definition *supported-pairwise-coupling-bad*
 $:: (f, 'a)\ protocol\text{-}channel\text{-}scheme \Rightarrow bool$
where
 $supported\text{-}pairwise\text{-}coupling\text{-}bad\ s \longleftrightarrow$
 $\alpha\text{-}supported\text{-}pairwise\text{-}coupling\text{-}bad\ s \vee$
 $query\text{-}supported\text{-}pairwise\text{-}coupling\text{-}bad\ s \vee$
 $trace\text{-}fri\text{-}supported\text{-}pairwise\text{-}coupling\text{-}bad\ s \vee$
 $composition\text{-}fri\text{-}supported\text{-}pairwise\text{-}coupling\text{-}bad\ s$

lemma *supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad*:
assumes *bad*: *supported-pairwise-merkle-bad s*
shows
 $supported\text{-}hash\text{-}output\text{-}collision\text{-}possible\ s \vee$
 $supported\text{-}pairwise\text{-}coupling\text{-}bad\ s$
using *assms*
unfolding *supported-pairwise-merkle-bad-def*
 $supported\text{-}pairwise\text{-}coupling\text{-}bad\text{-}def$
using

alpha-supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad
query-supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad
trace-fri-supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad
composition-fri-supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad
 by *blast*

lemma *no-supported-pairwise-merkle-bad-if-no-hash-collision-possible-no-coupling-bad:*

assumes *no-collision: $\neg$ supported-hash-output-collision-possible s*

and *no-coupling: $\neg$ supported-pairwise-coupling-bad s*

shows $\neg$ *supported-pairwise-merkle-bad s*

proof

assume *bad: supported-pairwise-merkle-bad s*

have *supported-hash-output-collision-possible $s \vee$*
supported-pairwise-coupling-bad s

by (rule

supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad
 [OF *bad*])

then show *False*

using *no-collision no-coupling* **by** *blast*

qed

lemma *no-alpha-supported-pairwise-merkle-bad-if-no-pairwise-side-bad:*

assumes *no-collision: $\neg$ supported-hash-output-collision-possible s*

and *no-coupling: $\neg$ supported-pairwise-coupling-bad s*

shows $\neg$ *alpha-supported-pairwise-merkle-bad s*

proof —

have $\neg$ *supported-pairwise-merkle-bad s*

by (rule

no-supported-pairwise-merkle-bad-if-no-hash-collision-possible-no-coupling-bad
 [OF *no-collision no-coupling*])

then show *?thesis*

unfolding *supported-pairwise-merkle-bad-def* **by** *blast*

qed

lemma *no-query-supported-pairwise-merkle-bad-if-no-pairwise-side-bad:*

assumes *no-collision: $\neg$ supported-hash-output-collision-possible s*

and *no-coupling: $\neg$ supported-pairwise-coupling-bad s*

shows $\neg$ *query-supported-pairwise-merkle-bad s*

proof —

have $\neg$ *supported-pairwise-merkle-bad s*

by (rule

no-supported-pairwise-merkle-bad-if-no-hash-collision-possible-no-coupling-bad
 [OF *no-collision no-coupling*])

then show *?thesis*

unfolding *supported-pairwise-merkle-bad-def* **by** *blast*

qed

lemma *no-trace-fri-supported-pairwise-merkle-bad-if-no-pairwise-side-bad:*

assumes *no-collision: $\neg$ supported-hash-output-collision-possible s*

and *no-coupling*: $\neg$ *supported-pairwise-coupling-bad* *s*
shows $\neg$ *trace-fri-supported-pairwise-merkle-bad* *s*
proof –
have $\neg$ *supported-pairwise-merkle-bad* *s*
by (*rule*
no-supported-pairwise-merkle-bad-if-no-hash-collision-possible-no-coupling-bad
 $[OF\ no-collision\ no-coupling]$)
then show ?*thesis*
unfolding *supported-pairwise-merkle-bad-def* **by** *blast*
qed

lemma *no-composition-fri-supported-pairwise-merkle-bad-if-no-pairwise-side-bad*:
assumes *no-collision*: $\neg$ *supported-hash-output-collision-possible* *s*
and *no-coupling*: $\neg$ *supported-pairwise-coupling-bad* *s*
shows $\neg$ *composition-fri-supported-pairwise-merkle-bad* *s*
proof –
have $\neg$ *supported-pairwise-merkle-bad* *s*
by (*rule*
no-supported-pairwise-merkle-bad-if-no-hash-collision-possible-no-coupling-bad
 $[OF\ no-collision\ no-coupling]$)
then show ?*thesis*
unfolding *supported-pairwise-merkle-bad-def* **by** *blast*
qed

definition *alpha-supported-pairwise-coupling-event*
 $:: (f, 'a)\ protocol-channel-scheme \Rightarrow$
 $(unit\ list \times (f, 'a)\ protocol-channel-scheme)\ option \Rightarrow bool$
where
alpha-supported-pairwise-coupling-event *s* *out* $\longleftrightarrow$
 $(\exists\ result\ final-state\ fr\ f-fri-roots\ f-final\ trace-table$
 $\quad trace-table'\ final-state'.$
 $\quad out = Some\ (result, final-state) \wedge$
 $\quad alpha-header-supported-trace-table-candidate-state\ s\ fr\ f-fri-roots$
 $\quad \quad f-final\ trace-table\ final-state \wedge$
 $\quad alpha-header-supported-trace-table-candidate-state\ s\ fr\ f-fri-roots$
 $\quad \quad f-final\ trace-table'\ final-state' \wedge$
 $\quad trace-table \neq trace-table' \wedge$
 $\quad merkle-hash-pairwise-coupling-bad\ final-state\ final-state')$

definition *query-supported-pairwise-coupling-event*
 $:: (f, 'a)\ protocol-channel-scheme \Rightarrow$
 $(unit\ list \times (f, 'a)\ protocol-channel-scheme)\ option \Rightarrow bool$
where
query-supported-pairwise-coupling-event *s* *out* $\longleftrightarrow$
 $(\exists\ result\ final-state\ fr\ f-fri-roots\ f-final\ as\ dg$
 $\quad composition-fri-roots\ final\ trace-table\ composition-table$
 $\quad trace-table'\ composition-table'\ final-state'.$
 $\quad out = Some\ (result, final-state) \wedge$
 $\quad query-header-supported-table-candidate-state\ s\ fr\ f-fri-roots\ f-final$

as dg composition-fri-roots final trace-table composition-table
 final-state $\wedge$
 query-header-supported-table-candidate-state s fr f-fri-roots f-final
 as dg composition-fri-roots final trace-table' composition-table'
 final-state' $\wedge$
 (trace-table $\neq$ trace-table' $\vee$
 composition-table $\neq$ composition-table') $\wedge$
 merkle-hash-pairwise-coupling-bad final-state final-state')

definition trace-fri-supported-pairwise-coupling-event

:: (f, 'a) protocol-channel-scheme $\Rightarrow$
 (unit list $\times$ (f, 'a) protocol-channel-scheme) option $\Rightarrow$ bool

where

trace-fri-supported-pairwise-coupling-event s out $\longleftrightarrow$
 ($\exists$ result final-state fr trace-table trace-table' final-state'.
 out = Some (result, final-state) $\wedge$
 trace-fri-supported-root-trace-table-candidate-state s fr trace-table
 final-state $\wedge$
 trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
 final-state' $\wedge$
 trace-table $\neq$ trace-table' $\wedge$
 merkle-hash-pairwise-coupling-bad final-state final-state')

definition composition-fri-supported-pairwise-coupling-event

:: (f, 'a) protocol-channel-scheme $\Rightarrow$
 (unit list $\times$ (f, 'a) protocol-channel-scheme) option $\Rightarrow$ bool

where

composition-fri-supported-pairwise-coupling-event s out $\longleftrightarrow$
 ($\exists$ result final-state fr f-fri-roots f-final as dg
 composition-fri-roots composition-table composition-table'
 final-state'.
 out = Some (result, final-state) $\wedge$
 composition-fri-supported-root-composition-table-candidate-state s fr
 f-fri-roots f-final as dg composition-fri-roots composition-table
 final-state $\wedge$
 composition-fri-supported-root-composition-table-candidate-state s fr
 f-fri-roots f-final as dg composition-fri-roots composition-table'
 final-state' $\wedge$
 composition-table $\neq$ composition-table' $\wedge$
 merkle-hash-pairwise-coupling-bad final-state final-state')

definition supported-pairwise-coupling-event

:: (f, 'a) protocol-channel-scheme $\Rightarrow$
 (unit list $\times$ (f, 'a) protocol-channel-scheme) option $\Rightarrow$ bool

where

supported-pairwise-coupling-event s out $\longleftrightarrow$
 alpha-supported-pairwise-coupling-event s out $\vee$
 query-supported-pairwise-coupling-event s out $\vee$
 trace-fri-supported-pairwise-coupling-event s out $\vee$

composition-fri-supported-pairwise-coupling-event s out

definition *supported-pairwise-side-bad*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{supported-pairwise-side-bad } s \text{ out} \longleftrightarrow$
 $\text{hash-map-output-collision-bad } s \text{ out} \vee$
 $\text{supported-pairwise-coupling-event } s \text{ out}$

lemma *alpha-supported-pairwise-coupling-event-imp-accepted:*

assumes *alpha-supported-pairwise-coupling-event s out*

shows *accepted out*

using *assms*

unfolding *alpha-supported-pairwise-coupling-event-def accepted-def*

by *auto*

lemma *query-supported-pairwise-coupling-event-imp-accepted:*

assumes *query-supported-pairwise-coupling-event s out*

shows *accepted out*

using *assms*

unfolding *query-supported-pairwise-coupling-event-def accepted-def*

by *auto*

lemma *trace-fri-supported-pairwise-coupling-event-imp-accepted:*

assumes *trace-fri-supported-pairwise-coupling-event s out*

shows *accepted out*

using *assms*

unfolding *trace-fri-supported-pairwise-coupling-event-def accepted-def*

by *auto*

lemma *composition-fri-supported-pairwise-coupling-event-imp-accepted:*

assumes *composition-fri-supported-pairwise-coupling-event s out*

shows *accepted out*

using *assms*

unfolding *composition-fri-supported-pairwise-coupling-event-def accepted-def*

by *auto*

lemma *supported-pairwise-coupling-event-imp-accepted:*

assumes *supported-pairwise-coupling-event s out*

shows *accepted out*

using *assms*

unfolding *supported-pairwise-coupling-event-def*

using *alpha-supported-pairwise-coupling-event-imp-accepted*

query-supported-pairwise-coupling-event-imp-accepted

trace-fri-supported-pairwise-coupling-event-imp-accepted

composition-fri-supported-pairwise-coupling-event-imp-accepted

by *blast*

lemma *supported-pairwise-side-bad-imp-accepted:*
assumes *supported-pairwise-side-bad s out*
shows *accepted out*
using *assms*
unfolding *supported-pairwise-side-bad-def*
using *hash-map-output-collision-bad-imp-accepted*
supported-pairwise-coupling-event-imp-accepted
by *blast*

lemma *supported-pairwise-side-bad-mono-accepted:*
wp-event verify-monad (supported-pairwise-side-bad s) s ≤
wp-event verify-monad accepted s
by (rule *wp-event-mono*)
(rule *supported-pairwise-side-bad-imp-accepted*)

lemma *supported-pairwise-side-bad-union-bound:*
fixes *hash-error coupling-error :: prob*
assumes *hash-bound:*
wp-event verify-monad (hash-map-output-collision-bad s) s ≤
hash-error
and *coupling-bound:*
wp-event verify-monad (supported-pairwise-coupling-event s) s ≤
coupling-error
shows
wp-event verify-monad (supported-pairwise-side-bad s) s ≤
hash-error + coupling-error
proof –
have *wp-event verify-monad (supported-pairwise-side-bad s) s ≤*
wp-event verify-monad (hash-map-output-collision-bad s) s +
wp-event verify-monad (supported-pairwise-coupling-event s) s
unfolding *supported-pairwise-side-bad-def*
supported-pairwise-coupling-event-def
by (rule *wp-event-union-bound*)
also have *... ≤ hash-error + coupling-error*
by (intro *add-mono hash-bound coupling-bound*)
finally show *?thesis* .
qed

lemma *verifier-initial-hash-map-output-collision-bad-bound-concrete:*
wp-event verify-monad
(hash-map-output-collision-bad (verifier-initial-state tr))
(verifier-initial-state tr) ≤ concrete-merkle-binding-error
proof –
let *?s = verifier-initial-state tr*
have *wp-event verify-monad (hash-map-output-collision-bad ?s) ?s ≤*
wp-event verify-monad (hash-map-new-output-collision-bad ?s) ?s
by (rule *wp-event-mono*)
(rule *hash-map-output-collision-bad-imp-new-collision-bad-if-initial-clean,*
simp, assumption)

also have ... $\leq$ *concrete-merkle-binding-error*
by (rule *verifier-initial-hash-map-new-output-collision-bad-bound-concrete*)
finally show ?thesis .
qed

lemma *verifier-initial-supported-output-local-candidate-ambiguity-bad-bound-concrete:*

wp-event verify-monad
 (supported-output-local-candidate-ambiguity-bad
 (verifier-initial-state tr))
 (verifier-initial-state tr) $\leq$
concrete-merkle-binding-error

proof –

have *wp-event verify-monad*
 (supported-output-local-candidate-ambiguity-bad
 (verifier-initial-state tr))
 (verifier-initial-state tr) $\leq$
wp-event verify-monad
 (hash-map-output-collision-bad (verifier-initial-state tr))
 (verifier-initial-state tr)
by (rule
 supported-output-local-candidate-ambiguity-bad-mono-hash-map-output-collision-bad)

also have ... $\leq$ *concrete-merkle-binding-error*
by (rule *verifier-initial-hash-map-output-collision-bad-bound-concrete*)
finally show ?thesis .

qed

lemma *verifier-initial-supported-output-local-side-bad-bound-concrete:*

wp-event verify-monad
 (supported-output-local-side-bad (verifier-initial-state tr))
 (verifier-initial-state tr) $\leq$
concrete-merkle-binding-error

unfolding *supported-output-local-side-bad-iff-hash-map-output-collision-bad*
by (rule *verifier-initial-hash-map-output-collision-bad-bound-concrete*)

lemma *verifier-initial-supported-pairwise-side-bad-bound-if-coupling-event-bound:*

assumes *coupling-bound:*
wp-event verify-monad
 (supported-pairwise-coupling-event (verifier-initial-state tr))
 (verifier-initial-state tr) $\leq$ *coupling-error*

shows

wp-event verify-monad
 (supported-pairwise-side-bad (verifier-initial-state tr))
 (verifier-initial-state tr) $\leq$
concrete-merkle-binding-error + *coupling-error*
by (rule *supported-pairwise-side-bad-union-bound*
 [OF *verifier-initial-hash-map-output-collision-bad-bound-concrete*
coupling-bound])

lemma *supported-hash-output-collision-possible-imp-exists-supported-pairwise-side-bad:*

assumes *supported-hash-output-collision-possible s*
shows $\exists \text{out} \in \text{set-dist } (\text{execute verify-monad } s).$
supported-pairwise-side-bad s out
using *assms*
unfolding *supported-hash-output-collision-possible-def*
supported-pairwise-side-bad-def
by *blast*

lemma *alpha-supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad:*

assumes *alpha-bad: alpha-supported-pairwise-coupling-bad s*
shows $\exists \text{out} \in \text{set-dist } (\text{execute verify-monad } s).$
supported-pairwise-side-bad s out

proof —

from *alpha-bad* **obtain** *fr f-fri-roots f-final trace-table final-state*
trace-table' final-state' **where**
candidate:
alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
f-final trace-table final-state
and *candidate':*
alpha-header-supported-trace-table-candidate-state s fr f-fri-roots
f-final trace-table' final-state'
and *distinct: trace-table $\neq$ trace-table'*
and *pair-bad:*
merkle-hash-pairwise-coupling-bad final-state final-state'
unfolding *alpha-supported-pairwise-coupling-bad-def*
alpha-header-supported-pairwise-coupling-bad-def
by *blast*
from *candidate* **obtain** *result composition-table as query-idxs dg*
composition-fri-roots final rest **where**
outcome:
Some (result, final-state) $\in$ set-dist (execute verify-monad s)
and *bound:*
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs
unfolding *alpha-header-supported-trace-table-candidate-state-def* **by** *blast*
have *coupling:*
supported-pairwise-coupling-event s (Some (result, final-state))
unfolding *supported-pairwise-coupling-event-def*
alpha-supported-pairwise-coupling-event-def
using *candidate candidate' distinct pair-bad* **by** *blast*
have *side:*
supported-pairwise-side-bad s (Some (result, final-state))
unfolding *supported-pairwise-side-bad-def*
using *coupling* **by** *simp*
show *?thesis*
using *outcome side* **by** *blast*
qed

lemma *query-supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad:*

assumes *query-bad: query-supported-pairwise-coupling-bad s*
shows $\exists \text{out} \in \text{set-dist (execute verify-monad s)}.$
supported-pairwise-side-bad s out
proof –
from *query-bad* **obtain** *fr f-fri-roots f-final as dg composition-fri-roots*
final trace-table composition-table final-state trace-table'
composition-table' final-state' where
candidate:
query-header-supported-table-candidate-state s fr f-fri-roots f-final
as dg composition-fri-roots final trace-table composition-table
final-state
and *candidate':*
query-header-supported-table-candidate-state s fr f-fri-roots f-final
as dg composition-fri-roots final trace-table' composition-table'
final-state'
and *distinct:*
trace-table $\neq$ trace-table' $\vee$
composition-table $\neq$ composition-table'
and *pair-bad:*
merkle-hash-pairwise-coupling-bad final-state final-state'
unfolding *query-supported-pairwise-coupling-bad-def*
query-header-supported-pairwise-coupling-bad-def
by *blast*
from *candidate* **obtain** *result query-idxs rest where*
outcome:
Some (result, final-state) $\in$ set-dist (execute verify-monad s)
and *bound:*
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs
unfolding *query-header-supported-table-candidate-state-def* **by** *blast*
have *coupling:*
supported-pairwise-coupling-event s (Some (result, final-state))
unfolding *supported-pairwise-coupling-event-def*
query-supported-pairwise-coupling-event-def
using *candidate candidate' distinct pair-bad* **by** *blast*
have *side:*
supported-pairwise-side-bad s (Some (result, final-state))
unfolding *supported-pairwise-side-bad-def*
using *coupling* **by** *simp*
show *?thesis*
using *outcome side* **by** *blast*
qed

lemma *trace-fri-supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad:*
assumes *trace-bad: trace-fri-supported-pairwise-coupling-bad s*
shows $\exists \text{out} \in \text{set-dist (execute verify-monad s)}.$
supported-pairwise-side-bad s out
proof –
from *trace-bad* **obtain** *fr trace-table final-state trace-table' final-state'*

where *candidate*:
trace-fri-supported-root-trace-table-candidate-state s fr trace-table
final-state
and *candidate'*:
trace-fri-supported-root-trace-table-candidate-state s fr trace-table'
final-state'
and *distinct*: *trace-table* $\neq$ *trace-table'*
and *pair-bad*:
merkle-hash-pairwise-coupling-bad final-state final-state'
unfolding *trace-fri-supported-pairwise-coupling-bad-def*
trace-fri-supported-root-pairwise-coupling-bad-def
by *blast*
from *candidate* **obtain** *result composition-table as query-idxs trace-bs dg*
comp-bs f-fri-roots f-final header-as composition-fri-roots final rest
where *outcome*:
Some (result, final-state) $\in$ set-dist (execute verify-monad s)
and *bound*:
accepted-with-bound-tables s (Some (result, final-state))
trace-table composition-table as query-idxs
unfolding *trace-fri-supported-root-trace-table-candidate-state-def*
by *blast*
have *coupling*:
supported-pairwise-coupling-event s (Some (result, final-state))
unfolding *supported-pairwise-coupling-event-def*
trace-fri-supported-pairwise-coupling-event-def
using *candidate candidate' distinct pair-bad* **by** *blast*
have *side*:
supported-pairwise-side-bad s (Some (result, final-state))
unfolding *supported-pairwise-side-bad-def*
using *coupling* **by** *simp*
show *?thesis*
using *outcome side* **by** *blast*
qed

lemma *composition-fri-supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad*:
assumes *composition-bad: composition-fri-supported-pairwise-coupling-bad s*
shows $\exists out \in \text{set-dist (execute verify-monad s)}$.
supported-pairwise-side-bad s out
proof –
from *composition-bad* **obtain** *fr f-fri-roots f-final as dg*
composition-fri-roots composition-table final-state composition-table'
final-state' where
candidate:
composition-fri-supported-root-composition-table-candidate-state s fr
f-fri-roots f-final as dg composition-fri-roots composition-table
final-state
and *candidate'*:
composition-fri-supported-root-composition-table-candidate-state s fr
f-fri-roots f-final as dg composition-fri-roots composition-table'

```

    final-state'
  and distinct: composition-table  $\neq$  composition-table'
  and pair-bad:
    merkle-hash-pairwise-coupling-bad final-state final-state'
  unfolding composition-fri-supported-pairwise-coupling-bad-def
    composition-fri-supported-root-pairwise-coupling-bad-def
  by blast
from candidate obtain result trace-table query-idxs trace-bs comp-bs final
  rest where
  outcome:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
  and bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
  unfolding
    composition-fri-supported-root-composition-table-candidate-state-def
  by blast
have coupling:
  supported-pairwise-coupling-event s (Some (result, final-state))
  unfolding supported-pairwise-coupling-event-def
    composition-fri-supported-pairwise-coupling-event-def
  using candidate candidate' distinct pair-bad by blast
have side:
  supported-pairwise-side-bad s (Some (result, final-state))
  unfolding supported-pairwise-side-bad-def
  using coupling by simp
show ?thesis
  using outcome side by blast
qed

lemma supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad:
  assumes coupling: supported-pairwise-coupling-bad s
  shows  $\exists$  out  $\in$  set-dist (execute verify-monad s).
    supported-pairwise-side-bad s out
  using coupling
  unfolding supported-pairwise-coupling-bad-def
  using
    alpha-supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad
    query-supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad
    trace-fri-supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad
    composition-fri-supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad
  by blast

lemma supported-pairwise-merkle-bad-imp-exists-supported-pairwise-side-bad:
  assumes bad: supported-pairwise-merkle-bad s
  shows  $\exists$  out  $\in$  set-dist (execute verify-monad s).
    supported-pairwise-side-bad s out
proof -
  have side:

```


supported-hash-output-collision-possible s $\vee$
supported-pairwise-coupling-bad s
by (rule
supported-pairwise-merkle-bad-imp-supported-hash-collision-possible-or-coupling-bad
 $[OF\ bad]$)
then show *?thesis*
proof
assume *supported-hash-output-collision-possible s*
then show *?thesis*
by (rule
supported-hash-output-collision-possible-imp-exists-supported-pairwise-side-bad)
next
assume *supported-pairwise-coupling-bad s*
then show *?thesis*
by (rule
supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad)
qed
qed

lemma *no-supported-pairwise-side-bad-on-support-imp-no-hash-collision-possible:*

assumes *no-side:*
 $\bigwedge out. out \in set-dist\ (execute\ verify-monad\ s) \implies$
 $\neg supported-pairwise-side-bad\ s\ out$
shows $\neg supported-hash-output-collision-possible\ s$
proof
assume *possible: supported-hash-output-collision-possible s*
from *supported-hash-output-collision-possible-imp-exists-supported-pairwise-side-bad*
 $[OF\ possible]$
obtain *out where out:*
 $out \in set-dist\ (execute\ verify-monad\ s)$
and *side: supported-pairwise-side-bad s out*
by *blast*
show *False*
using *no-side[OF out] side by contradiction*
qed

lemma *no-supported-pairwise-side-bad-on-support-imp-no-coupling-bad:*

assumes *no-side:*
 $\bigwedge out. out \in set-dist\ (execute\ verify-monad\ s) \implies$
 $\neg supported-pairwise-side-bad\ s\ out$
shows $\neg supported-pairwise-coupling-bad\ s$
proof
assume *coupling: supported-pairwise-coupling-bad s*
from *supported-pairwise-coupling-bad-imp-exists-supported-pairwise-side-bad*
 $[OF\ coupling]$
obtain *out where out:*
 $out \in set-dist\ (execute\ verify-monad\ s)$
and *side: supported-pairwise-side-bad s out*
by *blast*

show *False*
using *no-side*[*OF out*] *side* **by** *contradiction*
qed

lemma *no-supported-pairwise-side-bad-on-support-imp-no-pairwise-side-bad*:

assumes *no-side*:

$\bigwedge out. out \in \text{set-dist } (\text{execute verify-monad } s) \implies$
 $\neg \text{supported-pairwise-side-bad } s \text{ out}$

shows

$\neg \text{supported-hash-output-collision-possible } s$
 $\neg \text{supported-pairwise-coupling-bad } s$

proof $-$

show $\neg \text{supported-hash-output-collision-possible } s$

by (rule

$\text{no-supported-pairwise-side-bad-on-support-imp-no-hash-collision-possible}$
 $[\text{OF no-side}]$)

show $\neg \text{supported-pairwise-coupling-bad } s$

by (rule

$\text{no-supported-pairwise-side-bad-on-support-imp-no-coupling-bad}$
 $[\text{OF no-side}]$)

qed

lemma *composition-fri-supported-root-candidate-unique-if-no-pairwise-merkle-bad*:

assumes *no-bad*:

$\neg \text{composition-fri-supported-root-pairwise-merkle-bad } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots}$

shows

$\exists \text{composition-table.}$
 $\text{composition-fri-supported-root-composition-table-candidates } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots } \subseteq$
 $\{\text{composition-table}\}$

proof (rule *ccontr*)

assume *not-unique*:

$\neg (\exists \text{composition-table.}$
 $\text{composition-fri-supported-root-composition-table-candidates } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots } \subseteq$
 $\{\text{composition-table}\})$

have *composition-fri-supported-root-pairwise-merkle-bad* *s fr f-fri-roots*
f-final as dg composition-fri-roots

using

$\text{composition-fri-supported-root-composition-table-candidates-not-singleton-imp-pairwise-merkle-bad}$
 $[\text{OF not-unique}]$

unfolding *composition-fri-supported-root-pairwise-merkle-bad-def*

by *blast*

then show *False*

using *no-bad* **by** *contradiction*

qed

lemma *composition-fri-supported-root-candidate-unique-if-no-global-pairwise-merkle-bad*:

assumes *no-bad*: $\neg$ *composition-fri-supported-pairwise-merkle-bad* *s*
shows
 $\exists$ *composition-table*.
composition-fri-supported-root-composition-table-candidates *s* *fr*
f-fri-roots *f-final* *as dg* *composition-fri-roots* $\subseteq$
 $\{ \textit{composition-table} \}$
proof (rule
composition-fri-supported-root-candidate-unique-if-no-pairwise-merkle-bad)
show
 $\neg$ *composition-fri-supported-root-pairwise-merkle-bad* *s* *fr* *f-fri-roots*
f-final *as dg* *composition-fri-roots*
using *no-bad*
unfolding *composition-fri-supported-pairwise-merkle-bad-def* **by** *blast*
qed

definition *composition-fri-supported-root-union-bad-sets*

$:: (f, 'a) \textit{protocol-channel-scheme} \Rightarrow$
 $(f \Rightarrow 'f \textit{list} \Rightarrow 'f \textit{list set}) \Rightarrow$
 $'f \Rightarrow 'f \textit{list} \Rightarrow 'f \Rightarrow 'f \textit{list} \Rightarrow 'f \Rightarrow$
 $'f \textit{list} \Rightarrow 'f \textit{list set}$

where

composition-fri-supported-root-union-bad-sets *s* *bad-sets* *fr* *f-fri-roots*
f-final *as dg* *composition-fri-roots* =
 $\{ \textit{comp-bs} \in \textit{fri-challenge-space} (\textit{ceil-log} (\textit{to-nat dg} + 1)) .$
 $\exists \textit{composition-table} \in$
 $\textit{composition-fri-supported-root-composition-table-candidates } s \textit{ fr}$
 $\textit{f-fri-roots } \textit{f-final as dg composition-fri-roots} .$
 $\textit{comp-bs} \in \textit{bad-sets dg composition-table} \}$

lemma *composition-fri-supported-root-union-bad-sets-subset*:

composition-fri-supported-root-union-bad-sets *s* *bad-sets* *fr* *f-fri-roots*
f-final *as dg* *composition-fri-roots* $\subseteq$
 $\textit{fri-challenge-space} (\textit{ceil-log} (\textit{to-nat dg} + 1))$

unfolding *composition-fri-supported-root-union-bad-sets-def* **by** *auto*

lemma *composition-fri-challenge-set-hit-imp-supported-root-union-list-set-hit*:

assumes *outcome*: *out* $\in$ *set-dist* (*execute verify-monad* *s*)

and *hit*: *composition-fri-challenge-set-hit* *s* *bad-sets* *out*

shows

composition-fri-root-list-set-hit *s*
 $(\textit{composition-fri-supported-root-union-bad-sets } s \textit{ bad-sets}) \textit{ out}$

proof –

from *hit* **obtain** *trace-table* *composition-table* *as* *query-idxs* *trace-bs* *dg* *comp-bs*

where *bound*:

$\textit{accepted-with-bound-tables } s \textit{ out trace-table composition-table as}$
 $\textit{query-idxs}$

and *challenges*: *accepted-fri-challenges* *s* *out* *trace-bs* *dg* *comp-bs*

and *comp-bs-bad*: *comp-bs* $\in$ *bad-sets* *dg* *composition-table*

unfolding *composition-fri-challenge-set-hit-def* **by** *blast*

```

from bound obtain result-b final-state-b fr f-fri-roots f-final dg-b
  composition-fri-roots final rest where
  header-bound:
    verifier-header-transcript s fr f-fri-roots f-final as dg-b
    composition-fri-roots final rest
  unfolding accepted-with-bound-tables-def by blast
from challenges[unfolded accepted-fri-challenges-def]
obtain result-c final-state-c fr-c f-fl-c f-final-c as-c fl-c final-c
  query-state-c where challenge-body:
  out = Some (result-c, final-state-c) ∧
  verifier-header-transcript s fr-c (map snd f-fl-c) f-final-c as-c dg
    (map snd fl-c) final-c (PTranscript query-state-c) ∧
  PState query-state-c =
    verifier-header-state s fr-c (map snd f-fl-c) f-final-c as-c dg
      (map snd fl-c) final-c ∧
  Some (result-c, final-state-c) ∈
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr-c f-fl-c f-final-c as-c fl-c
          final-c)
        rounds)
      query-state-c) ∧
  PQueryCounter query-state-c = PQueryCounter s ∧
  trace-bs = map fst f-fl-c ∧
  comp-bs = map fst fl-c ∧
  (∀ i < length f-fl-c.
    fmlookup (HashMap query-state-c)
      (TraceFriChallenge (PTraceFriCounter s + i)
        (foldl concat (concat (PState s) fr-c)
          (take (Suc i) (map snd f-fl-c))))) =
      Some (fst (f-fl-c ! i))) ∧
  (∀ i < length fl-c.
    fmlookup (HashMap query-state-c)
      (CompositionFriChallenge (PCompositionFriCounter s + i)
        (foldl concat
          (concat
            (foldl concat
              (concat
                (foldl concat (concat (PState s) fr-c)
                  (map snd f-fl-c))
                f-final-c)
              as-c)
            dg)
          (take (Suc i) (map snd fl-c))))) =
      Some (fst (fl-c ! i)))
by (elim exE)
have out-challenges: out = Some (result-c, final-state-c)
using challenge-body by simp
have header-challenges:

```

```

    verifier-header-transcript s fr-c (map snd f-fl-c) f-final-c as-c dg
      (map snd fl-c) final-c (PTranscript query-state-c)
  using challenge-body by simp
have dg-eq: dg-b = dg
  using verifier-header-transcript-unique[OF header-challenges header-bound]
  by simp
have header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  using header-bound dg-eq by simp
have candidate:
  composition-table ∈
    composition-fri-supported-root-composition-table-candidates s fr
      f-fri-roots f-final as dg composition-fri-roots
  unfolding composition-fri-supported-root-composition-table-candidates-def
  using outcome bound challenges header by blast
have comp-bs-space:
  comp-bs ∈ fri-challenge-space (ceil-log (to-nat dg + 1))
  by (rule accepted-fri-challenges-composition-space[OF challenges])
have comp-bs-union:
  comp-bs ∈
    composition-fri-supported-root-union-bad-sets s bad-sets fr
      f-fri-roots f-final as dg composition-fri-roots
  unfolding composition-fri-supported-root-union-bad-sets-def
  using comp-bs-space candidate comp-bs-bad by blast
show ?thesis
  unfolding composition-fri-root-list-set-hit-def
  using challenges header comp-bs-union by blast
qed

```

lemma wp-composition-fri-challenge-set-hit-bound-via-supported-root-union:

```

  fixes C :: prob
  assumes future: composition-fri-future-fresh s
  and union-bound:
     $\bigwedge_{fr} f-fri-roots f-final as dg \text{ composition-fri-roots.}$ 
      nnreal
      (card
        (composition-fri-supported-root-union-bad-sets s bad-sets fr
          f-fri-roots f-final as dg composition-fri-roots)) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C

```

shows

```

  wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s ≤
    C

```

proof —

```

  have root-bound:
    wp-event verify-monad
      (composition-fri-root-list-set-hit s
        (composition-fri-supported-root-union-bad-sets s bad-sets)) s ≤ C
  by (rule wp-verify-monad-composition-fri-root-list-set-bound[OF future])

```

```

    (use composition-fri-supported-root-union-bad-sets-subset union-bound in
      simp-all)
  have wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s ≤
    wp-event verify-monad
      (composition-fri-root-list-set-hit s
        (composition-fri-supported-root-union-bad-sets s bad-sets)) s
  by (rule wp-event-mono-on-support)
    (rule composition-fri-challenge-set-hit-imp-supported-root-union-list-set-hit)
  also have ... ≤ C
    by (rule root-bound)
  finally show ?thesis .
qed

lemma composition-fri-supported-root-union-bad-sets-fraction-bound-if-unique-candidate:
  fixes C :: prob
  assumes unique:
    ∃ composition-table.
      composition-fri-supported-root-composition-table-candidates s fr
        f-fri-roots f-final as dg composition-fri-roots ⊆
          {composition-table}
  and subset:
    ∧ composition-table. bad-sets dg composition-table ⊆
      fri-challenge-space (ceil-log (to-nat dg + 1))
  and bound:
    ∧ composition-table.
      nnreal (card (bad-sets dg composition-table)) /
        nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C
  shows
    nnreal
      (card
        (composition-fri-supported-root-union-bad-sets s bad-sets fr
          f-fri-roots f-final as dg composition-fri-roots)) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C
proof -
  from unique obtain composition-table where candidates:
    composition-fri-supported-root-composition-table-candidates s fr
      f-fri-roots f-final as dg composition-fri-roots ⊆
        {composition-table}
  by blast
  have union-subset:
    composition-fri-supported-root-union-bad-sets s bad-sets fr f-fri-roots
      f-final as dg composition-fri-roots ⊆
        bad-sets dg composition-table
  unfolding composition-fri-supported-root-union-bad-sets-def
  using candidates by auto
  have finite-bad: finite (bad-sets dg composition-table)
    by (rule finite-subset[OF subset finite-fri-challenge-space])
  have card-le:
    card

```

$(\text{composition-fri-supported-root-union-bad-sets } s \text{ bad-sets } fr$
 $f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots}) \leq$
 $\text{card } (\text{bad-sets } dg \text{ composition-table})$
by (rule card-mono[OF finite-bad union-subset])
have $nnreal$
 $(\text{card}$
 $(\text{composition-fri-supported-root-union-bad-sets } s \text{ bad-sets } fr$
 $f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots})) /$
 $nnreal (CARD('f) \wedge \text{ceil-log } (to\text{-nat } dg + 1)) \leq$
 $nnreal (\text{card } (\text{bad-sets } dg \text{ composition-table})) /$
 $nnreal (CARD('f) \wedge \text{ceil-log } (to\text{-nat } dg + 1))$
by (rule nnreal-nat-divide-right-mono[OF card-le])
also have $\dots \leq C$
by (rule bound)
finally show ?thesis .
qed

definition *composition-fri-supported-root-partial-union-bad-sets*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list set}$

where

$\text{composition-fri-supported-root-partial-union-bad-sets } s \text{ bad-sets } fr$
 $f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots} =$
 $\{ \text{comp-bs} \in \text{fri-challenge-space } (\text{ceil-log } (to\text{-nat } dg + 1)) .$
 $\exists \text{ composition-table} \in$
 $\text{composition-fri-supported-root-partial-composition-table-candidates}$
 $s \text{ fr } f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots} .$
 $\text{comp-bs} \in \text{bad-sets } dg \text{ composition-table} \}$

lemma *composition-fri-supported-root-partial-union-bad-sets-subset:*

$\text{composition-fri-supported-root-partial-union-bad-sets } s \text{ bad-sets } fr$
 $f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots} \subseteq$
 $\text{fri-challenge-space } (\text{ceil-log } (to\text{-nat } dg + 1))$

unfolding *composition-fri-supported-root-partial-union-bad-sets-def* **by** auto

lemma *composition-fri-supported-root-partial-union-bad-sets-fraction-bound-if-unique-candidate:*

fixes $C :: \text{prob}$

assumes *unique:*

$\exists \text{ composition-table} .$
 $\text{composition-fri-supported-root-partial-composition-table-candidates } s$
 $fr \text{ f-fri-roots } f\text{-final as } dg \text{ composition-fri-roots} \subseteq$
 $\{ \text{composition-table} \}$

and *subset:*

$\bigwedge \text{composition-table} . \text{bad-sets } dg \text{ composition-table} \subseteq$
 $\text{fri-challenge-space } (\text{ceil-log } (to\text{-nat } dg + 1))$

and *bound:*

$\bigwedge \text{composition-table} .$

```

      nnreal (card (bad-sets dg composition-table)) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C
shows
  nnreal
    (card
      (composition-fri-supported-root-partial-union-bad-sets s bad-sets fr
        f-fri-roots f-final as dg composition-fri-roots)) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C
proof -
  from unique obtain composition-table where candidates:
    composition-fri-supported-root-partial-composition-table-candidates s fr
    f-fri-roots f-final as dg composition-fri-roots ⊆
    {composition-table}
  by blast
  have union-subset:
    composition-fri-supported-root-partial-union-bad-sets s bad-sets fr
    f-fri-roots f-final as dg composition-fri-roots ⊆
    bad-sets dg composition-table
  unfolding composition-fri-supported-root-partial-union-bad-sets-def
  using candidates by auto
  have finite-bad: finite (bad-sets dg composition-table)
  by (rule finite-subset[OF subset finite-fri-challenge-space])
  have card-le:
    card
      (composition-fri-supported-root-partial-union-bad-sets s bad-sets fr
        f-fri-roots f-final as dg composition-fri-roots) ≤
    card (bad-sets dg composition-table)
  by (rule card-mono[OF finite-bad union-subset])
  have nnreal
    (card
      (composition-fri-supported-root-partial-union-bad-sets s bad-sets fr
        f-fri-roots f-final as dg composition-fri-roots)) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤
    nnreal (card (bad-sets dg composition-table)) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
  by (rule nnreal-nat-divide-right-mono[OF card-le])
  also have ... ≤ C
  by (rule bound)
  finally show ?thesis .
qed

```

```

lemma composition-fri-challenge-set-hit-imp-partial-union-or-collision:
  assumes outcome: out ∈ set-dist (execute verify-monad s)
  and hit: composition-fri-challenge-set-hit s bad-sets out
  shows
    composition-fri-root-list-set-hit s
    (composition-fri-supported-root-partial-union-bad-sets s bad-sets) out ∨
    hash-map-output-collision-bad s out
proof -

```



```

from hit obtain trace-table composition-table as query-idxs trace-bs dg comp-bs
where bound:
  accepted-with-bound-tables s out trace-table composition-table as
    query-idxs
and challenges: accepted-fri-challenges s out trace-bs dg comp-bs
and trace-low: trace-table-low-degree trace-table
and not-low:
   $\neg$  composition-table-low-degree maxDegree composition-table
and comp-bs-bad: comp-bs  $\in$  bad-sets dg composition-table
unfolding composition-fri-challenge-set-hit-def by blast
from accepted-with-bound-tables-imp-accepted[OF bound]
obtain result final-state where out-eq: out = Some (result, final-state)
unfolding accepted-def by (cases out) auto
show ?thesis
proof (cases hash-map-output-collision final-state)
case True
then have hash-map-output-collision-bad s out
unfolding hash-map-output-collision-bad-def accepted-def out-eq by simp
then show ?thesis by simp
next
case clean: False
from challenges obtain result-c final-state-c fr-c f-fl-c f-final-c as-c
  fl-c final-c query-state-c where
    out-c: out = Some (result-c, final-state-c)
and
  header-challenges:
    verifier-header-transcript s fr-c (map snd f-fl-c) f-final-c as-c dg
    (map snd fl-c) final-c (PTranscript query-state-c)
and query-state-c:
  PState query-state-c =
    verifier-header-state s fr-c (map snd f-fl-c) f-final-c as-c dg
    (map snd fl-c) final-c
and query-out-c:
  Some (result-c, final-state-c)  $\in$ 
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr-c f-fl-c f-final-c as-c fl-c
          final-c) rounds)
      query-state-c)
and query-count-c:
  PQueryCounter query-state-c = PQueryCounter s
and trace-eq-c: trace-bs = map fst f-fl-c
and comp-eq-c: comp-bs = map fst fl-c
and trace-lookup-c:
   $\bigwedge i. i < \text{length } f\text{-}fl\text{-}c \implies$ 
    fmlookup (HashMap query-state-c)
    (TraceFriChallenge (PTraceFriCounter s + i)
      (foldl concat (concat (PState s) fr-c)

```

```

      (take (Suc i) (map snd f-fl-c)))) =
    Some (fst (f-fl-c ! i))
and comp-lookup-c:
   $\bigwedge i. i < \text{length } \text{fl-c} \implies$ 
    fmlookup (HashMap query-state-c)
      (CompositionFriChallenge (PCompositionFriCounter s + i)
        (foldl concat
          (concat
            (foldl concat
              (concat
                (foldl concat (concat (PState s) fr-c)
                  (map snd f-fl-c))
                f-final-c)
              as-c)
            dg)
          (take (Suc i) (map snd fl-c)))) =
    Some (fst (fl-c ! i))
unfolding accepted-fri-challenges-def by blast
from bound obtain fr-b f-fri-roots-b f-final-b dg-b
  composition-fri-roots-b final-b rest-b where
  header-bound:
    verifier-header-transcript s fr-b f-fri-roots-b f-final-b as dg-b
    composition-fri-roots-b final-b rest-b
and comp-nonempty:
  composition-fri-roots-b  $\neq []$ 
and comp-bind:
  merkle-root-binds-table (hd composition-fri-roots-b) composition-table
    final-state
unfolding accepted-with-bound-tables-def out-eq by blast
have dg-eq: dg-b = dg
using verifier-header-transcript-unique
  [OF header-bound header-challenges]
by simp
have header:
  verifier-header-transcript s fr-b f-fri-roots-b f-final-b as dg
  composition-fri-roots-b final-b rest-b
using header-bound dg-eq by simp
have outcome-some:
  Some (result, final-state)  $\in \text{set-dist } (\text{execute verify-monad } s)$ 
using outcome out-eq by simp
have bound-some:
  accepted-with-bound-tables s (Some (result, final-state)) trace-table
    composition-table as query-idxs
using bound out-eq by simp
have challenges-some:
  accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
  comp-bs
using challenges out-eq by simp
have state:

```

```

composition-fri-supported-root-composition-table-candidate-state s
  fr-b f-fri-roots-b f-final-b as dg composition-fri-roots-b
  composition-table final-state
unfolding composition-fri-supported-root-composition-table-candidate-state-def
using outcome-some bound-some challenges-some header comp-nonempty
  comp-bind
by blast
have candidate:
  composition-table ∈
    composition-fri-supported-root-partial-composition-table-candidates s
    fr-b f-fri-roots-b f-final-b as dg composition-fri-roots-b
using
  composition-fri-supported-root-composition-table-candidate-state-imp-partial-candidate-state
  [OF state clean]
  composition-fri-supported-root-partial-composition-table-candidates-iff-state
by blast
have comp-bs-space:
  comp-bs ∈ fri-challenge-space (ceil-log (to-nat dg + 1))
by (rule accepted-fri-challenges-composition-space[OF challenges])
have comp-bs-union:
  comp-bs ∈
    composition-fri-supported-root-partial-union-bad-sets s bad-sets
    fr-b f-fri-roots-b f-final-b as dg composition-fri-roots-b
unfolding composition-fri-supported-root-partial-union-bad-sets-def
using comp-bs-space candidate comp-bs-bad by blast
show ?thesis
unfolding composition-fri-root-list-set-hit-def
using challenges header comp-bs-union by blast
qed
qed

lemma wp-composition-fri-challenge-set-hit-bound-via-partial-union-or-collision:
fixes C H :: prob
assumes partial-bound:
  wp-event verify-monad
    (composition-fri-root-list-set-hit s
      (composition-fri-supported-root-partial-union-bad-sets s bad-sets))
  s ≤ C
and collision-bound:
  wp-event verify-monad (hash-map-output-collision-bad s) s ≤ H
shows
  wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s ≤
    C + H
proof –
have mono:
  wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s ≤
    wp-event verify-monad
      (λout.
        composition-fri-root-list-set-hit s

```

```

      (composition-fri-supported-root-partial-union-bad-sets s bad-sets)
      out ∨
      hash-map-output-collision-bad s out) s
  by (rule wp-event-mono-on-support)
    (use composition-fri-challenge-set-hit-imp-partial-union-or-collision
      in blast)
  also have ... ≤ C + H
  proof -
    have wp-event verify-monad
      (λout.
        composition-fri-root-list-set-hit s
        (composition-fri-supported-root-partial-union-bad-sets s bad-sets)
        out ∨
        hash-map-output-collision-bad s out) s ≤
      wp-event verify-monad
        (composition-fri-root-list-set-hit s
        (composition-fri-supported-root-partial-union-bad-sets s bad-sets))
      s +
      wp-event verify-monad (hash-map-output-collision-bad s) s
    by (rule wp-event-union-bound)
  also have ... ≤ C + H
  by (intro add-mono partial-bound collision-bound)
  finally show ?thesis .
qed
finally show ?thesis .
qed

lemma wp-composition-fri-challenge-set-hit-bound-via-supported-partial-union-or-collision:
  fixes C H :: prob
  assumes future: composition-fri-future-fresh s
  and union-bound:
    ∧fr f-fri-roots f-final as dg composition-fri-roots.
    nnreal
    (card
      (composition-fri-supported-root-partial-union-bad-sets s bad-sets
        fr f-fri-roots f-final as dg composition-fri-roots)) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C
  and collision-bound:
    wp-event verify-monad (hash-map-output-collision-bad s) s ≤ H
  shows
    wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s ≤
      C + H
  proof -
    have partial-bound:
      wp-event verify-monad
        (composition-fri-root-list-set-hit s
        (composition-fri-supported-root-partial-union-bad-sets s bad-sets))
      s ≤ C
    by (rule wp-verify-monad-composition-fri-root-list-set-bound[OF future])

```

```

    (use composition-fri-supported-root-partial-union-bad-sets-subset
      union-bound in simp-all)
  show ?thesis
  by (rule
      wp-composition-fri-challenge-set-hit-bound-via-partial-union-or-collision
      [OF partial-bound collision-bound])
qed

lemma wp-composition-fri-challenge-set-hit-bound-if-supported-partial-root-candidate-unique-or-collision:
  fixes H :: prob
  assumes future: composition-fri-future-fresh s
  and unique:
     $\bigwedge$  fr f-fri-roots f-final as dg composition-fri-roots.
     $\exists$  composition-table.
    composition-fri-supported-root-partial-composition-table-candidates
      s fr f-fri-roots f-final as dg composition-fri-roots  $\subseteq$ 
      {composition-table}
  and bounded: composition-fri-bad-challenge-sets-bounded bad-sets
  and collision-bound:
    wp-event verify-monad (hash-map-output-collision-bad s) s  $\leq$  H
  shows
    wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s  $\leq$ 
      composition-fri-error + H
proof -
  have union-bound:
     $\bigwedge$  fr f-fri-roots f-final as dg composition-fri-roots.
    nnreal
      (card
        (composition-fri-supported-root-partial-union-bad-sets s bad-sets
          fr f-fri-roots f-final as dg composition-fri-roots)) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))  $\leq$ 
      composition-fri-error
  proof -
    fix fr f-fri-roots f-final as dg composition-fri-roots
    show nnreal
      (card
        (composition-fri-supported-root-partial-union-bad-sets s bad-sets
          fr f-fri-roots f-final as dg composition-fri-roots)) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))  $\leq$ 
        composition-fri-error
    by (rule
        composition-fri-supported-root-partial-union-bad-sets-fraction-bound-if-unique-candidate
        [OF unique])
    (use bounded in
       $\langle$ auto simp: composition-fri-bad-challenge-sets-bounded-def $\rangle$ )
  qed
  show ?thesis
  by (rule
      wp-composition-fri-challenge-set-hit-bound-via-supported-partial-union-or-collision

```

[OF future union-bound collision-bound])

qed

definition *trace-fri-bad-with-tables*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-bad-with-tables } s \text{ out} \longleftrightarrow$
 $\text{accepted-with-fri-tables } s \text{ out} \wedge \text{trace-fri-bad } s \text{ out}$

definition *composition-fri-bad-with-tables*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{composition-fri-bad-with-tables } s \text{ out} \longleftrightarrow$
 $\text{accepted-with-fri-tables } s \text{ out} \wedge \text{composition-fri-bad } s \text{ out}$

definition *trace-fri-bad-with-partial-openings*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-bad-with-partial-openings } s \text{ out} \longleftrightarrow$
 $\text{trace-fri-bad } s \text{ out} \wedge$
 $(\exists \text{ fr query-idxs trace-openings.}$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr query-idxs}$
 $\text{trace-openings})$

definition *composition-fri-bad-with-partial-openings*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{composition-fri-bad-with-partial-openings } s \text{ out} \longleftrightarrow$
 $\text{composition-fri-bad } s \text{ out} \wedge$
 $(\exists \text{ fr query-idxs trace-openings.}$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr query-idxs}$
 $\text{trace-openings})$

end

end

theory *Soundness-Reductions-Partial*
imports *Soundness-Reductions-Side*
begin

Partial-opening bad-event definitions and accepted-execution partitions.

context *soundness*
begin

Reachable-state FRI interfaces for the partial-opening proof path. These predicates express the verifier-local FRI low-degree reductions over authenticated partial openings and deliberately avoid complete-table witnesses in their statement.

definition *trace-fri-partial-opening-reduction-assumption*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{trace-fri-partial-opening-reduction-assumption } s \longleftrightarrow$
 $\text{wp-event verify-monad } (\text{trace-fri-bad-with-partial-openings } s) \ s \leq$
 trace-fri-error

definition *composition-fri-partial-opening-reduction-assumption*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{composition-fri-partial-opening-reduction-assumption } s \longleftrightarrow$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-partial-openings } s) \ s \leq$
 $\text{composition-fri-error}$

lemma *trace-fri-bad-with-partial-openings-bound-from-partial-reduction:*

assumes *trace-fri-partial-opening-reduction-assumption* s

shows $\text{wp-event verify-monad } (\text{trace-fri-bad-with-partial-openings } s) \ s \leq$
 trace-fri-error

using *assms* **unfolding** *trace-fri-partial-opening-reduction-assumption-def*
by *simp*

lemma *composition-fri-bad-with-partial-openings-bound-from-partial-reduction:*

assumes *composition-fri-partial-opening-reduction-assumption* s

shows

$\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-partial-openings } s) \ s \leq$
 $\text{composition-fri-error}$

using *assms*

unfolding *composition-fri-partial-opening-reduction-assumption-def* **by** *simp*

lemma *trace-fri-bad-with-partial-openings-imp-trace-fri-bad:*

assumes *trace-fri-bad-with-partial-openings* s *out*

shows *trace-fri-bad* s *out*

using *assms* **unfolding** *trace-fri-bad-with-partial-openings-def* **by** *simp*

lemma *composition-fri-bad-with-partial-openings-imp-composition-fri-bad:*

assumes *composition-fri-bad-with-partial-openings* s *out*

shows *composition-fri-bad* s *out*

using *assms* **unfolding** *composition-fri-bad-with-partial-openings-def* **by** *simp*

lemma *wp-trace-fri-bad-with-partial-openings-le-trace-fri-bad:*

$\text{wp-event verify-monad } (\text{trace-fri-bad-with-partial-openings } s) \ s \leq$

$\text{wp-event verify-monad } (\text{trace-fri-bad } s) \ s$

by (rule *wp-event-mono*)

(rule *trace-fri-bad-with-partial-openings-imp-trace-fri-bad*)

lemma *wp-composition-fri-bad-with-partial-openings-le-composition-fri-bad*:
wp-event verify-monad (composition-fri-bad-with-partial-openings s) s ≤
wp-event verify-monad (composition-fri-bad s) s
by (rule *wp-event-mono*)
(rule *composition-fri-bad-with-partial-openings-imp-composition-fri-bad*)

lemma *trace-fri-bad-with-tables-imp-partial-openings-on-support*:
assumes *support*: *out ∈ set-dist (execute verify-monad s)*
and *bad*: *trace-fri-bad-with-tables s out*
shows *trace-fri-bad-with-partial-openings s out*

proof –

have *tables*: *accepted-with-fri-tables s out*
using *bad* **unfolding** *trace-fri-bad-with-tables-def* **by** *simp*
have *accepted*: *accepted out*
by (rule *accepted-with-fri-tables-imp-accepted[OF tables]*)
then obtain *result final-state* **where** *out-eq*:
out = Some (result, final-state)
unfolding *accepted-def* **by** (cases *out*) *auto*
from *verify-monad-accepted-with-partial-trace-openings*
[OF support[unfolded out-eq]]
obtain *fr query-idxs trace-openings* **where** *partial*:
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr query-idxs trace-openings
by *blast*
have *trace-bad*: *trace-fri-bad s out*
using *bad* **unfolding** *trace-fri-bad-with-tables-def* **by** *simp*
show *?thesis*
unfolding *trace-fri-bad-with-partial-openings-def out-eq*
using *trace-bad[unfolded out-eq]* *partial* **by** *blast*

qed

lemma *composition-fri-bad-with-tables-imp-partial-openings-on-support*:
assumes *support*: *out ∈ set-dist (execute verify-monad s)*
and *bad*: *composition-fri-bad-with-tables s out*
shows *composition-fri-bad-with-partial-openings s out*

proof –

have *tables*: *accepted-with-fri-tables s out*
using *bad* **unfolding** *composition-fri-bad-with-tables-def* **by** *simp*
have *accepted*: *accepted out*
by (rule *accepted-with-fri-tables-imp-accepted[OF tables]*)
then obtain *result final-state* **where** *out-eq*:
out = Some (result, final-state)
unfolding *accepted-def* **by** (cases *out*) *auto*
from *verify-monad-accepted-with-partial-trace-openings*
[OF support[unfolded out-eq]]
obtain *fr query-idxs trace-openings* **where** *partial*:
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr query-idxs trace-openings

by blast
 have *composition-bad: composition-fri-bad s out*
 using *bad unfolding composition-fri-bad-with-tables-def* by simp
 show ?thesis
 unfolding *composition-fri-bad-with-partial-openings-def out-eq*
 using *composition-bad[unfolded out-eq] partial* by blast
 qed

lemma *wp-trace-fri-bad-with-tables-le-partial-openings:*
wp-event verify-monad (trace-fri-bad-with-tables s) s ≤
wp-event verify-monad (trace-fri-bad-with-partial-openings s) s
 by (rule *wp-event-mono-on-support*)
 (rule *trace-fri-bad-with-tables-imp-partial-openings-on-support*)

lemma *wp-composition-fri-bad-with-tables-le-partial-openings:*
wp-event verify-monad (composition-fri-bad-with-tables s) s ≤
wp-event verify-monad (composition-fri-bad-with-partial-openings s) s
 by (rule *wp-event-mono-on-support*)
 (rule *composition-fri-bad-with-tables-imp-partial-openings-on-support*)

definition *soundness-bad-event-table-aware*
 :: (*f, 'a*) *protocol-channel-scheme* $\Rightarrow$
 (*unit list* $\times$ (*f, 'a*) *protocol-channel-scheme*) *option* $\Rightarrow$ *bool*
 where
soundness-bad-event-table-aware s out $\longleftrightarrow$
composition-bad s out $\vee$
trace-fri-bad-with-tables s out $\vee$
composition-fri-bad-with-tables s out $\vee$
query-bad s out

definition *soundness-bad-event-partial-opening*
 :: (*f, 'a*) *protocol-channel-scheme* $\Rightarrow$
 (*unit list* $\times$ (*f, 'a*) *protocol-channel-scheme*) *option* $\Rightarrow$ *bool*
 where
soundness-bad-event-partial-opening s out $\longleftrightarrow$
composition-bad s out $\vee$
trace-fri-bad-with-partial-openings s out $\vee$
composition-fri-bad-with-partial-openings s out $\vee$
query-bad s out

definition *soundness-bad-event-table-aware-with-merkle*
 :: (*f, 'a*) *protocol-channel-scheme* $\Rightarrow$
 (*unit list* $\times$ (*f, 'a*) *protocol-channel-scheme*) *option* $\Rightarrow$ *bool*
 where
soundness-bad-event-table-aware-with-merkle s out $\longleftrightarrow$
fri-merkle-binding-bad s out $\vee$
soundness-bad-event-table-aware s out

definition *soundness-bad-event-partial-opening-with-merkle*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{soundness-bad-event-partial-opening-with-merkle } s \text{ out} \longleftrightarrow$
 $\text{fri-merkle-binding-bad } s \text{ out} \vee$
 $\text{soundness-bad-event-partial-opening } s \text{ out}$

definition *soundness-bad-event-partial-opening-with-initial-merkle*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{soundness-bad-event-partial-opening-with-initial-merkle } s \text{ out} \longleftrightarrow$
 $\text{initial-merkle-binding-bad } s \text{ out} \vee$
 $\text{soundness-bad-event-partial-opening } s \text{ out}$

definition *soundness-bad-event-partial-opening-with-partial-merkle*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{soundness-bad-event-partial-opening-with-partial-merkle } s \text{ out} \longleftrightarrow$
 $\text{partial-merkle-inconsistency-bad } s \text{ out} \vee$
 $\text{soundness-bad-event-partial-opening } s \text{ out}$

definition *composition-bad-with-partial-candidates*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-bad-with-partial-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table } f.$
 $\text{accepted-with-partial-initial-openings } s \text{ out fr } f\text{-fri-roots } f\text{-final}$
 $\text{as dg composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{composition-query-idxs composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\text{degree } f < \text{clength} \wedge$
 $\text{trace-table} = \text{map } (\text{poly } f) \text{ eval-domain} \wedge$
 $\text{violated-constraints } f \neq \{\}$
 $\text{composition-table-low-degree maxDegree composition-table} \wedge$
 $\text{all-queries-consistent trace-table composition-table as})$

definition *composition-degree-bad-with-partial-candidates*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-degree-bad-with-partial-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$

trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table f.
accepted-with-partial-initial-openings s out fr f-fri-roots f-final
as dg composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
degree f < clength $\wedge$
trace-table = map (poly f) eval-domain $\wedge$
violated-constraints f $\neq \{\}$ $\wedge$
composition-table-low-degree maxDegree composition-table $\wedge$
all-queries-consistent trace-table composition-table as $\wedge$
 $\neg$ *common-denominator-degree-bounds f as)*

definition *composition-randomization-bad-with-partial-candidates*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-randomization-bad-with-partial-candidates s out $\longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg } \text{composition-fri-roots final}$
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table f.
accepted-with-partial-initial-openings s out fr f-fri-roots f-final
as dg composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
degree f < clength $\wedge$
trace-table = map (poly f) eval-domain $\wedge$
violated-constraints f $\neq \{\}$ $\wedge$
composition-table-low-degree maxDegree composition-table $\wedge$
all-queries-consistent trace-table composition-table as $\wedge$
common-denominator-degree-bounds f as $\wedge$
random-combination-common-denominator-hides-violations f as)

definition *trace-fri-bad-with-partial-candidates*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-bad-with-partial-candidates s out $\longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg } \text{composition-fri-roots final}$
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table.
accepted-with-partial-initial-openings s out fr f-fri-roots f-final
as dg composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$

$\neg \text{trace-table-low-degree trace-table})$

definition *composition-fri-bad-with-partial-candidates*

$:: ('f, 'a) \text{protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{protocol-channel-scheme}) \text{option} \Rightarrow \text{bool}$

where

$\text{composition-fri-bad-with-partial-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table.}$
 $\text{accepted-with-partial-initial-openings } s \text{ out fr } f\text{-fri-roots } f\text{-final}$
 $\text{as dg composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{composition-query-idxs composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\neg \text{composition-table-low-degree maxDegree composition-table})$

definition *query-bad-with-partial-candidates*

$:: ('f, 'a) \text{protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{protocol-channel-scheme}) \text{option} \Rightarrow \text{bool}$

where

$\text{query-bad-with-partial-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table.}$
 $\text{accepted-with-partial-initial-openings } s \text{ out fr } f\text{-fri-roots } f\text{-final}$
 $\text{as dg composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{composition-query-idxs composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\text{composition-table-low-degree maxDegree composition-table} \wedge$
 $\neg \text{all-queries-consistent trace-table composition-table as})$

definition *soundness-bad-event-partial-candidate*

$:: ('f, 'a) \text{protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{protocol-channel-scheme}) \text{option} \Rightarrow \text{bool}$

where

$\text{soundness-bad-event-partial-candidate } s \text{ out} \longleftrightarrow$
 $\text{composition-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{trace-fri-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{composition-fri-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{query-bad-with-partial-candidates } s \text{ out}$

definition *soundness-bad-event-partial-candidate-with-partial-merkle*

$:: ('f, 'a) \text{protocol-channel-scheme} \Rightarrow$

$(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{soundness-bad-event-partial-candidate-with-partial-merkle } s \text{ out} \longleftrightarrow$
 $\text{partial-merkle-inconsistency-bad } s \text{ out} \vee$
 $\text{soundness-bad-event-partial-candidate } s \text{ out}$

definition *accepted-with-empty-composition-header-candidates*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \Rightarrow \text{nat list} \Rightarrow 'f \text{ authenticated-opening list list} \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$

where
 $\text{accepted-with-empty-composition-header-candidates } s \text{ out fr } f\text{-fri-roots}$
 $f\text{-final as dg final trace-query-idxs trace-openings trace-table}$
 $\text{composition-table} \longleftrightarrow$
 $\text{accepted out} \wedge$
 $(\exists \text{ rest.}$
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final as dg } []$
 $\text{final rest}) \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr trace-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{composition-table} = \text{replicate } (\text{scale} * \text{clength}) \text{ final}$

definition *composition-bad-with-empty-composition-header-candidates*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where
 $\text{composition-bad-with-empty-composition-header-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{trace-openings trace-table composition-table } f.$
 $\text{accepted-with-empty-composition-header-candidates } s \text{ out fr}$
 $f\text{-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{trace-table composition-table} \wedge$
 $\text{degree } f < \text{clength} \wedge$
 $\text{trace-table} = \text{map } (\text{poly } f) \text{ eval-domain} \wedge$
 $\text{violated-constraints } f \neq \{\}$
 $\text{composition-table-low-degree maxDegree composition-table} \wedge$
 $\text{all-queries-consistent trace-table composition-table as})$

definition *composition-degree-bad-with-empty-composition-header-candidates*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where
 $\text{composition-degree-bad-with-empty-composition-header-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{trace-openings trace-table composition-table } f.$
 $\text{accepted-with-empty-composition-header-candidates } s \text{ out fr}$

$f\text{-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{trace-table composition-table } \wedge$
 $\text{degree } f < \text{clength } \wedge$
 $\text{trace-table} = \text{map } (\text{poly } f) \text{ eval-domain } \wedge$
 $\text{violated-constraints } f \neq \{\}$ $\wedge$
 $\text{composition-table-low-degree maxDegree composition-table } \wedge$
 $\text{all-queries-consistent trace-table composition-table as } \wedge$
 $\neg \text{common-denominator-degree-bounds } f \text{ as})$

definition *composition-randomization-bad-with-empty-composition-header-candidates*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-randomization-bad-with-empty-composition-header-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{trace-openings trace-table composition-table } f.$
 $\text{accepted-with-empty-composition-header-candidates } s \text{ out fr}$
 $f\text{-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{trace-table composition-table } \wedge$
 $\text{degree } f < \text{clength } \wedge$
 $\text{trace-table} = \text{map } (\text{poly } f) \text{ eval-domain } \wedge$
 $\text{violated-constraints } f \neq \{\}$ $\wedge$
 $\text{composition-table-low-degree maxDegree composition-table } \wedge$
 $\text{all-queries-consistent trace-table composition-table as } \wedge$
 $\text{common-denominator-degree-bounds } f \text{ as } \wedge$
 $\text{random-combination-common-denominator-hides-violations } f \text{ as})$

definition *composition-alpha-bad-set-hit-with-empty-composition-header-candidates*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-alpha-bad-set-hit-with-empty-composition-header-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{trace-openings trace-table composition-table.}$
 $\text{accepted-with-empty-composition-header-candidates } s \text{ out fr}$
 $f\text{-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{trace-table composition-table } \wedge$
 $\text{as} \in \text{composition-trace-bad-alpha-space trace-table})$

definition *composition-alpha-bad-set-hit-with-partial-candidates*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-alpha-bad-set-hit-with-partial-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table.}$
 $\text{accepted-with-partial-initial-openings } s \text{ out fr } f\text{-fri-roots } f\text{-final}$
 $\text{as dg composition-fri-roots final trace-query-idxs trace-openings}$

composition-query-idxs composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
as $\in$ *composition-trace-bad-alpha-space trace-table*)

definition *partial-trace-opening-alpha-union-bad-sets*

$:: 'f$ *authenticated-opening list list* $\Rightarrow 'f$ *list set*

where

partial-trace-opening-alpha-union-bad-sets trace-openings =
 $(\bigcup \text{trace-table} \in \text{partial-trace-table-candidates } \text{trace-openings}.$
composition-trace-bad-alpha-space trace-table)

definition *composition-alpha-partial-opening-union-hit-with-empty-header*

$:: ('f, 'a)$ *protocol-channel-scheme* $\Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-alpha-partial-opening-union-hit-with-empty-header s out $\longleftrightarrow$
 $(\exists \text{fr } f\text{-fri-roots } f\text{-final } \text{as } dg \text{ final } \text{trace-query-idxs}$
 $\text{trace-openings } \text{trace-table } \text{composition-table}.$
 $\text{accepted-with-empty-composition-header-candidates } s \text{ out fr}$
 $f\text{-fri-roots } f\text{-final } \text{as } dg \text{ final } \text{trace-query-idxs } \text{trace-openings}$
 $\text{trace-table } \text{composition-table} \wedge$
 $\text{as} \in \text{partial-trace-opening-alpha-union-bad-sets } \text{trace-openings})$

lemma *partial-trace-opening-alpha-union-bad-sets-subset-alpha-space:*

partial-trace-opening-alpha-union-bad-sets trace-openings $\subseteq$
alpha-space

unfolding *partial-trace-opening-alpha-union-bad-sets-def*

using *composition-trace-bad-alpha-space-subset-alpha-space* **by** *auto*

lemma *partial-trace-opening-alpha-union-bad-sets-finite:*

finite (*partial-trace-opening-alpha-union-bad-sets trace-openings*)

by (*rule finite-subset*

[*OF partial-trace-opening-alpha-union-bad-sets-subset-alpha-space*
finite-alpha-space])

lemma *partial-trace-opening-alpha-union-bad-sets-fraction-bound-if-unique-candidate:*

assumes *unique:*

$\exists \text{trace-table}.$

partial-trace-table-candidates trace-openings $\subseteq \{\text{trace-table}\}$

shows

nnreal

$(\text{card } (\text{partial-trace-opening-alpha-union-bad-sets } \text{trace-openings})) /$

nnreal $(\text{card } \text{alpha-space}) \leq \text{composition-error-bound}$

proof –

from *unique* **obtain** *trace-table* **where** *candidates:*

partial-trace-table-candidates trace-openings $\subseteq \{\text{trace-table}\}$

by *blast*

have *union-subset:*

partial-trace-opening-alpha-union-bad-sets trace-openings $\subseteq$

```

    composition-trace-bad-alpha-space trace-table
unfolding partial-trace-opening-alpha-union-bad-sets-def
using candidates by auto
have finite-bad: finite (composition-trace-bad-alpha-space trace-table)
by (rule finite-subset
    [OF composition-trace-bad-alpha-space-subset-alpha-space
    finite-alpha-space])
have card (partial-trace-opening-alpha-union-bad-sets trace-openings) ≤
    card (composition-trace-bad-alpha-space trace-table)
by (rule card-mono[OF finite-bad union-subset])
then have
    nnreal
    (card (partial-trace-opening-alpha-union-bad-sets trace-openings)) /
    nnreal (card alpha-space) ≤
    nnreal (card (composition-trace-bad-alpha-space trace-table)) /
    nnreal (card alpha-space)
by (rule nnreal-nat-divide-right-mono)
also have ... ≤ composition-error-bound
by (rule composition-trace-bad-alpha-space-fraction-bound-alpha-space)
finally show ?thesis .
qed

lemma composition-alpha-bad-set-hit-with-empty-header-imp-partial-opening-union-hit:
assumes
    composition-alpha-bad-set-hit-with-empty-composition-header-candidates
    s out
shows
    composition-alpha-partial-opening-union-hit-with-empty-header s out
proof —
obtain fr f-fri-roots f-final as dg final trace-query-idxs
    trace-openings trace-table composition-table where partial:
    accepted-with-empty-composition-header-candidates s out fr
    f-fri-roots f-final as dg final trace-query-idxs trace-openings
    trace-table composition-table
and as-bad: as ∈ composition-trace-bad-alpha-space trace-table
using assms
unfolding
    composition-alpha-bad-set-hit-with-empty-composition-header-candidates-def
by blast
have candidate:
    trace-table ∈ partial-trace-table-candidates trace-openings
using partial
unfolding accepted-with-empty-composition-header-candidates-def
    partial-trace-table-candidates-def
by simp
have
    as ∈ partial-trace-opening-alpha-union-bad-sets trace-openings
unfolding partial-trace-opening-alpha-union-bad-sets-def
using candidate as-bad by blast

```



```

then show ?thesis
  unfolding composition-alpha-partial-opening-union-hit-with-empty-header-def
  using partial by blast
qed

lemma wp-composition-alpha-bad-set-hit-with-empty-header-le-partial-opening-union-hit:
  wp-event verify-monad
    (composition-alpha-bad-set-hit-with-empty-composition-header-candidates s)
  s ≤
  wp-event verify-monad
    (composition-alpha-partial-opening-union-hit-with-empty-header s) s
by (rule wp-event-mono)
(rule
  composition-alpha-bad-set-hit-with-empty-header-imp-partial-opening-union-hit)

lemma composition-alpha-partial-opening-union-hit-with-empty-header-imp-alpha-header-partial-union-on-support:
  assumes support: out ∈ set-dist (execute verify-monad s)
  and hit:
    composition-alpha-partial-opening-union-hit-with-empty-header s out
  shows
    alpha-header-list-set-hit s
    (alpha-header-supported-partial-union-bad-sets s
      composition-trace-bad-alpha-space) out
proof -
  obtain fr f-fri-roots f-final as dg final trace-query-idxs
    trace-openings trace-table composition-table rest where partial:
    accepted-with-empty-composition-header-candidates s out fr
    f-fri-roots f-final as dg final trace-query-idxs trace-openings
    trace-table composition-table
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg []
    final rest
  and trace-partial:
    accepted-with-partial-trace-openings s out fr trace-query-idxs
    trace-openings
  and as-bad:
    as ∈ partial-trace-opening-alpha-union-bad-sets trace-openings
  using hit
  unfolding composition-alpha-partial-opening-union-hit-with-empty-header-def
    accepted-with-empty-composition-header-candidates-def
  by blast
  from trace-partial obtain result final-state where out-eq:
    out = Some (result, final-state)
  unfolding accepted-with-partial-trace-openings-def by blast
  from verify-monad-accepted-transcript-shape[OF support[unfolded out-eq]]
  obtain as' query-idxs where shape':
    accepted-transcript-shape s (Some (result, final-state)) as' query-idxs
  by blast
  from shape' obtain result' final-state' fr' f-fri-roots' f-final' dg'

```

```

    composition-fri-roots' final' rest' where out-shape:
    Some (result, final-state) = Some (result', final-state')
and header':
    verifier-header-transcript s fr' f-fri-roots' f-final' as' dg'
    composition-fri-roots' final' rest'
by (elim accepted-transcript-shape-header-query-extraction)
have as'-eq: as' = as
    using verifier-header-transcript-unique[OF header' header] by simp
have shape:
    accepted-transcript-shape s out as query-idxs
    using shape' as'-eq out-eq by simp
from as-bad obtain trace-table' where candidate':
    trace-table' ∈ partial-trace-table-candidates trace-openings
and as-bad':
    as ∈ composition-trace-bad-alpha-space trace-table'
    unfolding partial-trace-opening-alpha-union-bad-sets-def by blast
have candidate-header:
    trace-table' ∈
    alpha-header-supported-partial-trace-table-candidates s fr
    f-fri-roots f-final
proof –
    have partial-candidate':
    partial-trace-table-candidate trace-table' trace-openings
    using candidate'
    unfolding partial-trace-table-candidates-def by simp
    show ?thesis
    unfolding alpha-header-supported-partial-trace-table-candidates-def
    using support trace-partial partial-candidate' header by blast
qed
have as-union:
    as ∈
    alpha-header-supported-partial-union-bad-sets s
    composition-trace-bad-alpha-space fr f-fri-roots f-final
    unfolding alpha-header-supported-partial-union-bad-sets-def
    using candidate-header as-bad'
    composition-trace-bad-alpha-space-subset-alpha-space[of trace-table']
    by auto
    show ?thesis
    unfolding alpha-header-list-set-hit-def
proof (intro exI conjI)
    show accepted-transcript-shape s out as query-idxs
    by (rule shape)
next
    show out = Some (result, final-state)
    by (rule out-eq)
next
    show
    verifier-header-transcript s fr f-fri-roots f-final as dg []
    final rest

```

```

    by (rule header)
  next
  show
    as ∈
      alpha-header-supported-partial-union-bad-sets s
      composition-trace-bad-alpha-space fr f-fri-roots f-final
    by (rule as-union)
  qed
qed

```

lemma *wp-composition-alpha-partial-opening-union-hit-with-empty-header-le-alpha-header-partial-union*:

```

wp-event verify-monad
  (composition-alpha-partial-opening-union-hit-with-empty-header s) s ≤
wp-event verify-monad
  (alpha-header-list-set-hit s
    (alpha-header-supported-partial-union-bad-sets s
      composition-trace-bad-alpha-space)) s
by (rule wp-event-mono-on-support)
(rule
  composition-alpha-partial-opening-union-hit-with-empty-header-imp-alpha-header-partial-union-on-support)

```

definition *trace-fri-bad-with-empty-composition-header-candidates*

```

:: ('f, 'a) protocol-channel-scheme ⇒
  (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool

```

where

```

trace-fri-bad-with-empty-composition-header-candidates s out ⟷
  (∃ fr f-fri-roots f-final as dg final trace-query-idxs
    trace-openings trace-table composition-table.
    accepted-with-empty-composition-header-candidates s out fr
      f-fri-roots f-final as dg final trace-query-idxs trace-openings
        trace-table composition-table ∧
    ¬ trace-table-low-degree trace-table)

```

definition *query-bad-with-empty-composition-header-candidates*

```

:: ('f, 'a) protocol-channel-scheme ⇒
  (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool

```

where

```

query-bad-with-empty-composition-header-candidates s out ⟷
  (∃ fr f-fri-roots f-final as dg final trace-query-idxs
    trace-openings trace-table composition-table.
    accepted-with-empty-composition-header-candidates s out fr
      f-fri-roots f-final as dg final trace-query-idxs trace-openings
        trace-table composition-table ∧
    trace-table-low-degree trace-table ∧
    composition-table-low-degree maxDegree composition-table ∧
    ¬ all-queries-consistent trace-table composition-table as)

```

definition *soundness-bad-event-partial-candidate-empty-header*

```

:: ('f, 'a) protocol-channel-scheme ⇒

```

$(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{soundness-bad-event-partial-candidate-empty-header } s \text{ out} \longleftrightarrow$
 $\text{composition-bad-with-empty-composition-header-candidates } s \text{ out} \vee$
 $\text{trace-fri-bad-with-empty-composition-header-candidates } s \text{ out} \vee$
 $\text{query-bad-with-empty-composition-header-candidates } s \text{ out}$

definition *soundness-bad-event-table-aware-with-pairwise-side*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{soundness-bad-event-table-aware-with-pairwise-side } s \text{ out} \longleftrightarrow$
 $\text{supported-pairwise-side-bad } s \text{ out} \vee$
 $\text{soundness-bad-event-table-aware-with-merkle } s \text{ out}$

definition *soundness-bad-event-table-aware-with-output-local-side*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{soundness-bad-event-table-aware-with-output-local-side } s \text{ out} \longleftrightarrow$
 $\text{supported-output-local-side-bad } s \text{ out} \vee$
 $\text{soundness-bad-event-table-aware-with-merkle } s \text{ out}$

lemma *soundness-bad-event-table-aware-imp-with-merkle:*
assumes *soundness-bad-event-table-aware* $s \text{ out}$
shows *soundness-bad-event-table-aware-with-merkle* $s \text{ out}$
using *assms* **unfolding** *soundness-bad-event-table-aware-with-merkle-def* **by** *simp*

lemma *soundness-bad-event-partial-opening-imp-with-merkle:*
assumes *soundness-bad-event-partial-opening* $s \text{ out}$
shows *soundness-bad-event-partial-opening-with-merkle* $s \text{ out}$
using *assms*
unfolding *soundness-bad-event-partial-opening-with-merkle-def* **by** *simp*

lemma *initial-merkle-binding-bad-imp-partial-opening-with-initial-merkle:*
assumes *initial-merkle-binding-bad* $s \text{ out}$
shows *soundness-bad-event-partial-opening-with-initial-merkle* $s \text{ out}$
using *assms*
unfolding *soundness-bad-event-partial-opening-with-initial-merkle-def*
by *simp*

lemma *soundness-bad-event-partial-opening-imp-with-initial-merkle:*
assumes *soundness-bad-event-partial-opening* $s \text{ out}$
shows *soundness-bad-event-partial-opening-with-initial-merkle* $s \text{ out}$
using *assms*
unfolding *soundness-bad-event-partial-opening-with-initial-merkle-def*
by *simp*

lemma *partial-merkle-inconsistency-bad-imp-partial-opening-with-partial-merkle:*

assumes *partial-merkle-inconsistency-bad s out*
shows *soundness-bad-event-partial-opening-with-partial-merkle s out*
using *assms*
unfolding *soundness-bad-event-partial-opening-with-partial-merkle-def*
by *simp*

lemma *soundness-bad-event-partial-opening-imp-with-partial-merkle:*
assumes *soundness-bad-event-partial-opening s out*
shows *soundness-bad-event-partial-opening-with-partial-merkle s out*
using *assms*
unfolding *soundness-bad-event-partial-opening-with-partial-merkle-def*
by *simp*

lemma *soundness-bad-event-partial-candidate-imp-with-partial-merkle:*
assumes *soundness-bad-event-partial-candidate s out*
shows *soundness-bad-event-partial-candidate-with-partial-merkle s out*
using *assms*
unfolding *soundness-bad-event-partial-candidate-with-partial-merkle-def*
by *simp*

lemma *partial-merkle-inconsistency-bad-imp-partial-candidate-with-partial-merkle:*
assumes *partial-merkle-inconsistency-bad s out*
shows *soundness-bad-event-partial-candidate-with-partial-merkle s out*
using *assms*
unfolding *soundness-bad-event-partial-candidate-with-partial-merkle-def*
by *simp*

lemma *accepted-with-partial-initial-openings-alpha-length:*
assumes
 accepted-with-partial-initial-openings s out fr f-fri-roots f-final as dg
 composition-fri-roots final trace-query-idxs trace-openings
 composition-query-idxs composition-openings
shows *length as = length spec*
proof –
 obtain *rest* **where**
 verifier-header-transcript s fr f-fri-roots f-final as dg
 composition-fri-roots final rest
 using *assms*
 unfolding *accepted-with-partial-initial-openings-def* **by** *blast*
 then show *?thesis*
 using *verifier-header-transcript-shapes(3)* **by** *blast*
qed

lemma *composition-bad-with-partial-candidates-split:*
assumes *false-statement: $\neg$ exists-valid-trace*
and *bad: composition-bad-with-partial-candidates s out*
shows
 composition-degree-bad-with-partial-candidates s out $\vee$
 composition-randomization-bad-with-partial-candidates s out

```

proof –
obtain fr f-fri-roots f-final as dg composition-fri-roots final
  trace-query-idxs trace-openings composition-query-idxs
  composition-openings trace-table composition-table f
where partial:
  accepted-with-partial-initial-openings s out fr f-fri-roots f-final
    as dg composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and composition-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and deg-f: degree f < clength
and trace-table: trace-table = map (poly f) eval-domain
and violated: violated-constraints f ≠ {}
and composition-low:
  composition-table-low-degree maxDegree composition-table
and all-queries:
  all-queries-consistent trace-table composition-table as
using bad
unfolding composition-bad-with-partial-candidates-def by blast
show ?thesis
proof (cases common-denominator-degree-bounds f as)
  case False
  then have composition-degree-bad-with-partial-candidates s out
    unfolding composition-degree-bad-with-partial-candidates-def
    using partial trace-candidate composition-candidate deg-f trace-table
      violated composition-low all-queries
    by blast
  then show ?thesis by blast
next
  case True
  note bounds = True
  have len-as: length as = length spec
    by (rule accepted-with-partial-initial-openings-alpha-length
      [OF partial])
  have hides:
    random-combination-common-denominator-hides-violations f as
    by (rule random-combination-common-denominator-hides-from-queries
      [OF len-as false-statement deg-f trace-table composition-low
        all-queries bounds])
  have composition-randomization-bad-with-partial-candidates s out
    unfolding composition-randomization-bad-with-partial-candidates-def
    using partial trace-candidate composition-candidate deg-f trace-table
      violated composition-low all-queries bounds hides
    by blast
  then show ?thesis by blast
qed

```

qed

lemma *composition-degree-bad-with-partial-candidates-false:*

¬ *composition-degree-bad-with-partial-candidates s out*

using *common-denominator-degree-bounds-from-spec-degree-wellformed*
[*OF spec-degree-wellformed-from-spec-query-margin*]

unfolding *composition-degree-bad-with-partial-candidates-def* **by** *blast*

lemma *composition-randomization-bad-with-partial-candidates-imp-alpha-hit:*

assumes *bad: composition-randomization-bad-with-partial-candidates s out*

shows *composition-alpha-bad-set-hit-with-partial-candidates s out*

proof –

obtain *fr f-fri-roots f-final as dg composition-fri-roots final*

trace-query-idxs trace-openings composition-query-idxs

composition-openings trace-table composition-table f

where *partial:*

accepted-with-partial-initial-openings s out fr f-fri-roots f-final

as dg composition-fri-roots final trace-query-idxs trace-openings

composition-query-idxs composition-openings

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *deg-f: degree f < clength*

and *trace-table: trace-table = map (poly f) eval-domain*

and *violated: violated-constraints f ≠ {}*

and *bounds: common-denominator-degree-bounds f as*

and *hides:*

random-combination-common-denominator-hides-violations f as

using *bad*

unfolding *composition-randomization-bad-with-partial-candidates-def*

by *blast*

have *as-in:*

as ∈ common-denominator-hiding-alpha-space f

using *accepted-with-partial-initial-openings-alpha-length[OF partial]*

hides

unfolding *common-denominator-hiding-alpha-space-def alpha-space-def*

by *simp*

have *witness-eq: low-degree-trace-witness trace-table = f*

by (*rule low-degree-trace-witness-eq[OF deg-f trace-table]*)

have *as ∈ composition-trace-bad-alpha-space trace-table*

using *deg-f trace-table violated bounds as-in*

unfolding *composition-trace-bad-alpha-space-def witness-eq*

by *simp*

then show *?thesis*

unfolding *composition-alpha-bad-set-hit-with-partial-candidates-def*

using *partial trace-candidate* **by** *blast*

qed

lemma *composition-alpha-bad-set-hit-with-partial-candidates-imp-alpha-header-partial-union-on-support:*

assumes *support: out ∈ set-dist (execute verify-monad s)*

and *hit*: *composition-alpha-bad-set-hit-with-partial-candidates s out*
shows
alpha-header-list-set-hit s
(alpha-header-supported-partial-union-bad-sets s
composition-trace-bad-alpha-space) out
proof –
obtain *fr f-fri-roots f-final as dg composition-fri-roots final*
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table rest
where *partial*:
accepted-with-partial-initial-openings s out fr f-fri-roots f-final
as dg composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings
and *header*:
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
and *trace-partial*:
accepted-with-partial-trace-openings s out fr trace-query-idxs
trace-openings
and *trace-candidate*:
partial-trace-table-candidate trace-table trace-openings
and *as-bad*:
as ∈ composition-trace-bad-alpha-space trace-table
using *hit*
unfolding *composition-alpha-bad-set-hit-with-partial-candidates-def*
accepted-with-partial-initial-openings-def
by *blast*
from *trace-partial* **obtain** *result final-state* **where** *out-eq*:
out = Some (result, final-state)
unfolding *accepted-with-partial-trace-openings-def* **by** *blast*
from *verify-monad-accepted-transcript-shape[OF support[unfolded out-eq]]*
obtain *as' query-idxs* **where** *shape'*:
accepted-transcript-shape s (Some (result, final-state)) as' query-idxs
by *blast*
from *shape'* **obtain** *result' final-state' fr' f-fri-roots' f-final' dg'*
composition-fri-roots' final' rest' where out-shape:
Some (result, final-state) = Some (result', final-state')
and *header'*:
verifier-header-transcript s fr' f-fri-roots' f-final' as' dg'
composition-fri-roots' final' rest'
by *(elim accepted-transcript-shape-header-query-extraction)*
have *as'-eq*: *as' = as*
using *verifier-header-transcript-unique[OF header' header]* **by** *simp*
have *shape*:
accepted-transcript-shape s out as query-idxs
using *shape' as'-eq out-eq* **by** *simp*
have *candidate-header*:
trace-table ∈
alpha-header-supported-partial-trace-table-candidates s fr


```

    f-fri-roots f-final
  unfolding alpha-header-supported-partial-trace-table-candidates-def
  using support trace-partial trace-candidate header by blast
have as-union:
  as ∈
    alpha-header-supported-partial-union-bad-sets s
    composition-trace-bad-alpha-space fr f-fri-roots f-final
  unfolding alpha-header-supported-partial-union-bad-sets-def
  using candidate-header as-bad
    composition-trace-bad-alpha-space-subset-alpha-space[of trace-table]
  by auto
show ?thesis
  unfolding alpha-header-list-set-hit-def
  using shape out-eq header as-union by blast
qed

lemma wp-composition-randomization-bad-with-partial-candidates-le-alpha-header-partial-union:
  wp-event verify-monad
    (composition-randomization-bad-with-partial-candidates s) s ≤
  wp-event verify-monad
    (alpha-header-list-set-hit s
      (alpha-header-supported-partial-union-bad-sets s
        composition-trace-bad-alpha-space)) s
  by (rule wp-event-mono-on-support)
    (rule
      composition-alpha-bad-set-hit-with-partial-candidates-imp-alpha-header-partial-union-on-support
        [OF - composition-randomization-bad-with-partial-candidates-imp-alpha-hit])

lemma accepted-partial-candidate-partition:
  assumes false-statement: ¬ exists-valid-trace
  and partial:
    accepted-with-partial-initial-openings s out fr f-fri-roots f-final
    as dg composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings
  and trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and composition-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  shows soundness-bad-event-partial-candidate s out
proof (cases trace-table-low-degree trace-table)
case False
have trace-fri-bad-with-partial-candidates s out
  unfolding trace-fri-bad-with-partial-candidates-def
  using partial trace-candidate False by blast
then show ?thesis
  unfolding soundness-bad-event-partial-candidate-def by blast
next
case True

```

```

note trace-low = True
show ?thesis
proof (cases composition-table-low-degree maxDegree composition-table)
  case False
  have composition-fri-bad-with-partial-candidates s out
    unfolding composition-fri-bad-with-partial-candidates-def
    using partial trace-candidate composition-candidate trace-low False
    by blast
  then show ?thesis
    unfolding soundness-bad-event-partial-candidate-def by blast
next
  case True
  note composition-low = True
  show ?thesis
  proof (cases all-queries-consistent trace-table composition-table as)
    case False
    have query-bad-with-partial-candidates s out
      unfolding query-bad-with-partial-candidates-def
      using partial trace-candidate composition-candidate trace-low
        composition-low False
      by blast
    then show ?thesis
      unfolding soundness-bad-event-partial-candidate-def by blast
    next
    case True
    note all-queries = True
    from trace-low obtain f where
      deg-f: degree f < clength
      and trace-table-eq: trace-table = map (poly f) eval-domain
      unfolding trace-table-low-degree-def by blast
    have violated: violated-constraints f ≠ {}
      by (rule false-statement-violated-constraints
        [OF false-statement deg-f])
    have composition-bad-with-partial-candidates s out
      unfolding composition-bad-with-partial-candidates-def
      using partial trace-candidate composition-candidate deg-f
        trace-table-eq violated composition-low all-queries
      by blast
    then show ?thesis
      unfolding soundness-bad-event-partial-candidate-def by blast
    qed
  qed
qed

lemma constant-composition-table-low-degree:
  composition-table-low-degree maxDegree (replicate (scale * clength) final)
proof –
  have map (poly [:final:]) eval-domain =
    replicate (scale * clength) final

```

```

proof –
  have map (poly [:final:]) eval-domain =
    replicate (length eval-domain) final
  proof (rule nth-equalityI)
    show length (map (poly [:final:]) eval-domain) =
      length (replicate (length eval-domain) final)
    by simp
  next
    fix i
    assume i < length (map (poly [:final:]) eval-domain)
    then show map (poly [:final:]) eval-domain ! i =
      replicate (length eval-domain) final ! i
    by simp
  qed
  then show ?thesis
    using eval-domain-length by (simp add: mult.commute)
  qed
  then show ?thesis
    unfolding composition-table-low-degree-def
    by (intro exI[of - [:final:]]) simp
  qed

lemma accepted-with-empty-composition-header-candidates-alpha-length:
  assumes
    accepted-with-empty-composition-header-candidates s out fr
    f-fri-roots f-final as dg final trace-query-idxs trace-openings
    trace-table composition-table
  shows length as = length spec
proof –
  obtain rest where
    verifier-header-transcript s fr f-fri-roots f-final as dg []
    final rest
  using assms
  unfolding accepted-with-empty-composition-header-candidates-def by blast
  then show ?thesis
    using verifier-header-transcript-shapes(3) by blast
  qed

lemma composition-bad-with-empty-composition-header-candidates-split:
  assumes false-statement:  $\neg$  exists-valid-trace
  and bad: composition-bad-with-empty-composition-header-candidates s out
  shows
    composition-degree-bad-with-empty-composition-header-candidates s out  $\vee$ 
    composition-randomization-bad-with-empty-composition-header-candidates s out
proof –
  obtain fr f-fri-roots f-final as dg final trace-query-idxs
    trace-openings trace-table composition-table f
  where partial:
    accepted-with-empty-composition-header-candidates s out fr

```

```

    f-fri-roots f-final as dg final trace-query-idxs trace-openings
    trace-table composition-table
  and deg-f: degree f < clength
  and trace-table: trace-table = map (poly f) eval-domain
  and violated: violated-constraints f ≠ {}
  and composition-low:
    composition-table-low-degree maxDegree composition-table
  and all-queries:
    all-queries-consistent trace-table composition-table as
  using bad
  unfolding composition-bad-with-empty-composition-header-candidates-def
  by blast
show ?thesis
proof (cases common-denominator-degree-bounds f as)
  case False
  then have
    composition-degree-bad-with-empty-composition-header-candidates s out
  unfolding composition-degree-bad-with-empty-composition-header-candidates-def
  using partial deg-f trace-table violated composition-low all-queries
  by blast
  then show ?thesis by blast
next
case True
note bounds = True
have len-as: length as = length spec
  by (rule accepted-with-empty-composition-header-candidates-alpha-length
    [OF partial])
have hides:
  random-combination-common-denominator-hides-violations f as
  by (rule random-combination-common-denominator-hides-from-queries
    [OF len-as false-statement deg-f trace-table composition-low
      all-queries bounds])
have
  composition-randomization-bad-with-empty-composition-header-candidates
  s out
  unfolding
    composition-randomization-bad-with-empty-composition-header-candidates-def
  using partial deg-f trace-table violated composition-low all-queries
  bounds hides
  by blast
  then show ?thesis by blast
qed
qed

lemma composition-degree-bad-with-empty-composition-header-candidates-false:
  ¬ composition-degree-bad-with-empty-composition-header-candidates s out
  using common-denominator-degree-bounds-from-spec-degree-wellformed
    [OF spec-degree-wellformed-from-spec-query-margin]
  unfolding composition-degree-bad-with-empty-composition-header-candidates-def

```

by *blast*

lemma *composition-degree-bad-with-empty-composition-header-candidates-bound:*

wp-event verify-monad

(composition-degree-bad-with-empty-composition-header-candidates s) s ≤ 0

proof —

have *composition-degree-bad-with-empty-composition-header-candidates s =*
(λout :: (unit list × ('f, 'a) protocol-channel-scheme) option.
False)

by *(rule ext)*

(simp add:

composition-degree-bad-with-empty-composition-header-candidates-false)

then show *?thesis*

unfolding *wp-event-def wp-def dist-expect-def* **by** *simp*

qed

lemma *composition-randomization-bad-with-empty-composition-header-candidates-imp-alpha-hit:*

assumes *bad:*

composition-randomization-bad-with-empty-composition-header-candidates
s out

shows

composition-alpha-bad-set-hit-with-empty-composition-header-candidates
s out

proof —

obtain *fr f-fri-roots f-final as dg final trace-query-idxs*

trace-openings trace-table composition-table f

where *partial:*

accepted-with-empty-composition-header-candidates s out fr
f-fri-roots f-final as dg final trace-query-idxs trace-openings
trace-table composition-table

and *deg-f: degree f < clength*

and *trace-table: trace-table = map (poly f) eval-domain*

and *violated: violated-constraints f ≠ {}*

and *bounds: common-denominator-degree-bounds f as*

and *hides:*

random-combination-common-denominator-hides-violations f as

using *bad*

unfolding

composition-randomization-bad-with-empty-composition-header-candidates-def

by *blast*

have *as-in:*

as ∈ common-denominator-hiding-alpha-space f

using *accepted-with-empty-composition-header-candidates-alpha-length[OF partial]*
hides

unfolding *common-denominator-hiding-alpha-space-def alpha-space-def*

by *simp*

have *witness-eq: low-degree-trace-witness trace-table = f*

by *(rule low-degree-trace-witness-eq[OF deg-f trace-table])*

```

have as ∈ composition-trace-bad-alpha-space trace-table
  using deg-f trace-table violated bounds as-in
  unfolding composition-trace-bad-alpha-space-def witness-eq
  by simp
then show ?thesis
  unfolding
    composition-alpha-bad-set-hit-with-empty-composition-header-candidates-def
  using partial by blast
qed

lemma wp-composition-randomization-bad-with-empty-composition-header-candidates-le-alpha-hit:
  wp-event verify-monad
    (composition-randomization-bad-with-empty-composition-header-candidates s)
    s ≤
  wp-event verify-monad
    (composition-alpha-bad-set-hit-with-empty-composition-header-candidates s)
    s
by (rule wp-event-mono)
  (rule
    composition-randomization-bad-with-empty-composition-header-candidates-imp-alpha-hit)

lemma accepted-empty-composition-header-candidate-partition:
  assumes false-statement:  $\neg$  exists-valid-trace
  and partial:
    accepted-with-empty-composition-header-candidates s out fr
    f-fri-roots f-final as dg final trace-query-idxs trace-openings
    trace-table composition-table
  shows soundness-bad-event-partial-candidate-empty-header s out
proof (cases trace-table-low-degree trace-table)
  case False
  have trace-fri-bad-with-empty-composition-header-candidates s out
    unfolding trace-fri-bad-with-empty-composition-header-candidates-def
    using partial False by blast
  then show ?thesis
    unfolding soundness-bad-event-partial-candidate-empty-header-def by blast
next
  case True
  note trace-low = True
  have composition-low:
    composition-table-low-degree maxDegree composition-table
  proof –
    have composition-table = replicate (scale * clength) final
      using partial
      unfolding accepted-with-empty-composition-header-candidates-def by simp
    then show ?thesis
      using constant-composition-table-low-degree by simp
  qed
show ?thesis
proof (cases all-queries-consistent trace-table composition-table as)

```

```

case False
have query-bad-with-empty-composition-header-candidates s out
  unfolding query-bad-with-empty-composition-header-candidates-def
  using partial trace-low composition-low False by blast
then show ?thesis
  unfolding soundness-bad-event-partial-candidate-empty-header-def
  by blast
next
case True
note all-queries = True
from trace-low obtain f where
  deg-f: degree f < clength
  and trace-table-eq: trace-table = map (poly f) eval-domain
  unfolding trace-table-low-degree-def by blast
have violated: violated-constraints f ≠ {}
  by (rule false-statement-violated-constraints
    [OF false-statement deg-f])
have composition-bad-with-empty-composition-header-candidates s out
  unfolding composition-bad-with-empty-composition-header-candidates-def
  using partial deg-f trace-table-eq violated composition-low all-queries
  by blast
then show ?thesis
  unfolding soundness-bad-event-partial-candidate-empty-header-def
  by blast
qed
qed

lemma fri-merkle-binding-bad-imp-table-aware-with-merkle:
  assumes fri-merkle-binding-bad s out
  shows soundness-bad-event-table-aware-with-merkle s out
  using assms unfolding soundness-bad-event-table-aware-with-merkle-def by simp

lemma fri-merkle-binding-bad-imp-partial-opening-with-merkle:
  assumes fri-merkle-binding-bad s out
  shows soundness-bad-event-partial-opening-with-merkle s out
  using assms
  unfolding soundness-bad-event-partial-opening-with-merkle-def by simp

lemma soundness-bad-event-table-aware-imp-partial-opening-on-support:
  assumes support: out ∈ set-dist (execute verify-monad s)
  and bad: soundness-bad-event-table-aware s out
  shows soundness-bad-event-partial-opening s out
  using bad
proof (unfold soundness-bad-event-table-aware-def, elim disjE)
  assume composition-bad s out
  then show ?thesis
    unfolding soundness-bad-event-partial-opening-def by blast
next
  assume trace-bad: trace-fri-bad-with-tables s out

```

then have *trace-fri-bad-with-partial-openings s out*
by (rule *trace-fri-bad-with-tables-imp-partial-openings-on-support*
 $[OF\ support]$)
then show *?thesis*
unfolding *soundness-bad-event-partial-opening-def* **by** *blast*
next
assume *composition-bad: composition-fri-bad-with-tables s out*
then have *composition-fri-bad-with-partial-openings s out*
by (rule *composition-fri-bad-with-tables-imp-partial-openings-on-support*
 $[OF\ support]$)
then show *?thesis*
unfolding *soundness-bad-event-partial-opening-def* **by** *blast*
next
assume *query-bad s out*
then show *?thesis*
unfolding *soundness-bad-event-partial-opening-def* **by** *blast*
qed

lemma *soundness-bad-event-table-aware-with-merkle-imp-partial-opening-with-merkle-on-support:*
assumes *support: out $\in$ set-dist (execute verify-monad s)*
and *bad: soundness-bad-event-table-aware-with-merkle s out*
shows *soundness-bad-event-partial-opening-with-merkle s out*
using *bad*
proof (unfold *soundness-bad-event-table-aware-with-merkle-def*, elim *disjE*)
assume *fri-merkle-binding-bad s out*
then show *?thesis*
by (rule *fri-merkle-binding-bad-imp-partial-opening-with-merkle*)
next
assume *soundness-bad-event-table-aware s out*
then have *soundness-bad-event-partial-opening s out*
by (rule *soundness-bad-event-table-aware-imp-partial-opening-on-support*
 $[OF\ support]$)
then show *?thesis*
by (rule *soundness-bad-event-partial-opening-imp-with-merkle*)
qed

lemma *wp-soundness-bad-event-table-aware-le-partial-opening:*
 $wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad-event-table-aware}\ s)\ s \leq$
 $wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad-event-partial-opening}\ s)\ s$
by (rule *wp-event-mono-on-support*)
(rule *soundness-bad-event-table-aware-imp-partial-opening-on-support*)

lemma *wp-soundness-bad-event-table-aware-with-merkle-le-partial-opening-with-merkle:*
 $wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad-event-table-aware-with-merkle}\ s)\ s \leq$
 $wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad-event-partial-opening-with-merkle}\ s)\ s$
by (rule *wp-event-mono-on-support*)
(rule *soundness-bad-event-table-aware-with-merkle-imp-partial-opening-with-merkle-on-support*)

lemma *soundness-bad-event-table-aware-with-merkle-imp-with-pairwise-side:*
assumes *soundness-bad-event-table-aware-with-merkle s out*
shows *soundness-bad-event-table-aware-with-pairwise-side s out*
using *assms*
unfolding *soundness-bad-event-table-aware-with-pairwise-side-def* **by** *simp*

lemma *supported-pairwise-side-bad-imp-table-aware-with-pairwise-side:*
assumes *supported-pairwise-side-bad s out*
shows *soundness-bad-event-table-aware-with-pairwise-side s out*
using *assms*
unfolding *soundness-bad-event-table-aware-with-pairwise-side-def* **by** *simp*

lemma *soundness-bad-event-table-aware-with-merkle-imp-with-output-local-side:*
assumes *soundness-bad-event-table-aware-with-merkle s out*
shows *soundness-bad-event-table-aware-with-output-local-side s out*
using *assms*
unfolding *soundness-bad-event-table-aware-with-output-local-side-def*
by *simp*

lemma *supported-output-local-side-bad-imp-table-aware-with-output-local-side:*
assumes *supported-output-local-side-bad s out*
shows *soundness-bad-event-table-aware-with-output-local-side s out*
using *assms*
unfolding *soundness-bad-event-table-aware-with-output-local-side-def*
by *simp*

lemma *accepted-with-fri-tables-table-aware-partition:*
assumes *false-statement: $\neg$ exists-valid-trace*
and *tables: accepted-with-fri-tables s out*
shows *soundness-bad-event-table-aware s out*
proof –
from *accepted-with-fri-tables-obtain-bound-tables[OF tables]*
obtain *trace-table composition-table as query-idxs* **where** *bound:*
accepted-with-bound-tables s out trace-table composition-table as
query-idxs
by *blast*
have *partition:*
composition-bad s out $\vee$ trace-fri-bad s out $\vee$
composition-fri-bad s out $\vee$ query-bad s out
by *(rule accepted-bound-table-partition[OF false-statement bound])*
then show *?thesis*
proof
assume *composition-bad s out*
then show *?thesis*
unfolding *soundness-bad-event-table-aware-def* **by** *blast*
next
assume *rest: trace-fri-bad s out $\vee$*
composition-fri-bad s out $\vee$ query-bad s out

```

then show ?thesis
proof
  assume trace-bad: trace-fri-bad s out
  then have trace-fri-bad-with-tables s out
    using tables unfolding trace-fri-bad-with-tables-def by simp
  then show ?thesis
    unfolding soundness-bad-event-table-aware-def by blast
next
assume rest': composition-fri-bad s out  $\vee$  query-bad s out
then show ?thesis
proof
  assume composition-bad: composition-fri-bad s out
  then have composition-fri-bad-with-tables s out
    using tables unfolding composition-fri-bad-with-tables-def by simp
  then show ?thesis
    unfolding soundness-bad-event-table-aware-def by blast
next
assume query-bad s out
then show ?thesis
  unfolding soundness-bad-event-table-aware-def by blast
qed
qed
qed
qed

lemma accepted-partition-soundness-bad-event-table-aware-with-merkle:
  assumes false-statement:  $\neg$  exists-valid-trace
  and acc: accepted out
  shows soundness-bad-event-table-aware-with-merkle s out
proof -
  from accepted-partition-fri-merkle-binding[OF acc]
  consider
    (merkle-bad) fri-merkle-binding-bad s out
  | (with-tables) accepted-with-fri-tables s out
  by blast
then show ?thesis
proof cases
  case merkle-bad
  then show ?thesis
    by (rule fri-merkle-binding-bad-imp-table-aware-with-merkle)
next
  case with-tables
  have soundness-bad-event-table-aware s out
    by (rule accepted-with-fri-tables-table-aware-partition
      [OF false-statement with-tables])
  then show ?thesis
    by (rule soundness-bad-event-table-aware-imp-with-merkle)
qed
qed

```

lemma *accepted-partition-soundness-bad-event-table-aware-with-pairwise-side*:
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *acc*: *accepted out*
shows *soundness-bad-event-table-aware-with-pairwise-side s out*
by (rule *soundness-bad-event-table-aware-with-merkle-imp-with-pairwise-side*)
(rule *accepted-partition-soundness-bad-event-table-aware-with-merkle*
[OF *false-statement acc*])

lemma *accepted-partition-soundness-bad-event-table-aware-with-output-local-side*:
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *acc*: *accepted out*
shows *soundness-bad-event-table-aware-with-output-local-side s out*
by (rule
soundness-bad-event-table-aware-with-merkle-imp-with-output-local-side)
(rule *accepted-partition-soundness-bad-event-table-aware-with-merkle*
[OF *false-statement acc*])

lemma *accepted-partition-soundness-bad-event-partial-opening-with-merkle*:
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *supp*: *out* $\in$ *set-dist (execute verify-monad s)*
and *acc*: *accepted out*
shows *soundness-bad-event-partial-opening-with-merkle s out*
proof –
have *soundness-bad-event-table-aware-with-merkle s out*
by (rule *accepted-partition-soundness-bad-event-table-aware-with-merkle*
[OF *false-statement acc*])
then show *?thesis*
by (rule
soundness-bad-event-table-aware-with-merkle-imp-partial-opening-with-merkle-on-support
[OF *supp*])
qed

lemma *accepted-partition-soundness-bad-event-partial-opening-with-initial-merkle*:
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *supp*: *out* $\in$ *set-dist (execute verify-monad s)*
and *acc*: *accepted out*
shows *soundness-bad-event-partial-opening-with-initial-merkle s out*
proof –
from *acc* **obtain** *result final-state* **where** *out-eq*:
out = Some (result, final-state)
unfolding *accepted-def* **by** (cases *out*) *auto*
from *verify-monad-accepted-transcript-shape* [OF *supp* [unfolding *out-eq*]]
obtain *as query-idxs* **where** *shape*:
accepted-transcript-shape s (Some (result, final-state)) as query-idxs
by *blast*
show *?thesis*
proof (cases
 $\exists$ *trace-table composition-table*.)

```

    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
case False
have initial-merkle-binding-bad s out
  unfolding initial-merkle-binding-bad-def out-eq
  using shape False by blast
then show ?thesis
  by (rule initial-merkle-binding-bad-imp-partial-opening-with-initial-merkle)
next
case True
then obtain trace-table composition-table where bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table as query-idxs
  by blast
have partial-bad:
  soundness-bad-event-partial-opening s out
proof -
  have partition:
    composition-bad s (Some (result, final-state)) ∨
    trace-fri-bad s (Some (result, final-state)) ∨
    composition-fri-bad s (Some (result, final-state)) ∨
    query-bad s (Some (result, final-state))
  by (rule accepted-bound-table-partition[OF false-statement bound])
then show ?thesis
proof
  assume composition-bad s (Some (result, final-state))
  then show ?thesis
    unfolding soundness-bad-event-partial-opening-def out-eq by blast
next
  assume rest:
    trace-fri-bad s (Some (result, final-state)) ∨
    composition-fri-bad s (Some (result, final-state)) ∨
    query-bad s (Some (result, final-state))
  then show ?thesis
  proof
    assume trace-bad:
      trace-fri-bad s (Some (result, final-state))
    from verify-monad-accepted-with-partial-trace-openings
      [OF supp[unfolded out-eq]]
    obtain fr query-idxs' trace-openings where partial:
      accepted-with-partial-trace-openings s
      (Some (result, final-state)) fr query-idxs' trace-openings
    by blast
    have trace-fri-bad-with-partial-openings s out
      unfolding trace-fri-bad-with-partial-openings-def out-eq
      using trace-bad partial by blast
    then show ?thesis
      unfolding soundness-bad-event-partial-opening-def by blast
  next

```

```

assume rest':
  composition-fri-bad s (Some (result, final-state))  $\vee$ 
  query-bad s (Some (result, final-state))
then show ?thesis
proof
  assume composition-bad:
    composition-fri-bad s (Some (result, final-state))
  from verify-monad-accepted-with-partial-trace-openings
    [OF supp[unfolded out-eq]]
  obtain fr query-idxs' trace-openings where partial:
    accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs' trace-openings
  by blast
  have composition-fri-bad-with-partial-openings s out
    unfolding composition-fri-bad-with-partial-openings-def out-eq
    using composition-bad partial by blast
  then show ?thesis
    unfolding soundness-bad-event-partial-opening-def by blast
  next
    assume query-bad s (Some (result, final-state))
    then show ?thesis
      unfolding soundness-bad-event-partial-opening-def out-eq
      by blast
    qed
  qed
  qed
  qed
  then show ?thesis
    by (rule soundness-bad-event-partial-opening-imp-with-initial-merkle)
  qed
qed

lemma accepted-partition-soundness-bad-event-partial-candidate-with-partial-merkle-if-header-nonempty:
assumes false-statement:  $\neg$  exists-valid-trace
and supp: out  $\in$  set-dist (execute verify-monad s)
and acc: accepted out
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and comp-nonempty: composition-fri-roots  $\neq$  []
shows soundness-bad-event-partial-candidate-with-partial-merkle s out
proof –
from acc obtain result final-state where out-eq:
  out = Some (result, final-state)
unfolding accepted-def by (cases out) auto
from verify-monad-partial-initial-candidates-or-partial-merkle-bad
  [OF supp[unfolded out-eq] header comp-nonempty]
consider
  (partial-merkle)

```

```

    partial-merkle-inconsistency-bad s (Some (result, final-state))
  | (candidates) trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table where
    accepted-with-partial-initial-openings s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings
    partial-trace-table-candidate trace-table trace-openings
    partial-composition-table-candidate composition-table
    composition-openings
by blast
then show ?thesis
proof cases
  case partial-merkle
  then show ?thesis
    unfolding out-eq
    by (rule
      partial-merkle-inconsistency-bad-imp-partial-candidate-with-partial-merkle)
next
  case (candidates trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table)
  have bad:
    soundness-bad-event-partial-candidate s
    (Some (result, final-state))
    by (rule accepted-partial-candidate-partition
      [OF false-statement candidates(1) candidates(2) candidates(3)])
  then show ?thesis
    unfolding out-eq
    by (rule soundness-bad-event-partial-candidate-imp-with-partial-merkle)
qed
qed

lemma accepted-partition-soundness-bad-event-partial-candidate-with-partial-merkle-or-empty-header:
assumes false-statement:  $\neg$  exists-valid-trace
and supp: out  $\in$  set-dist (execute verify-monad s)
and acc: accepted out
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
shows
  soundness-bad-event-partial-candidate-with-partial-merkle s out  $\vee$ 
  soundness-bad-event-partial-candidate-empty-header s out
proof (cases composition-fri-roots = [])
  case False
  then have comp-nonempty: composition-fri-roots  $\neq$  []
    by simp
  have soundness-bad-event-partial-candidate-with-partial-merkle s out
    by (rule
      accepted-partition-soundness-bad-event-partial-candidate-with-partial-merkle-if-header-nonempty)

```

```

      [OF false-statement supp acc header comp-nonempty])
    then show ?thesis by simp
next
case True
note comp-empty = True
from acc obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then have soundness-bad-event-partial-candidate-with-partial-merkle s out
  unfolding out-eq
  by (rule
    partial-merkle-inconsistency-bad-imp-partial-candidate-with-partial-merkle)
then show ?thesis by simp
next
case False
from verify-monad-accepted-with-partial-trace-openings-for-header
  [OF supp[unfolded out-eq] header]
obtain trace-query-idxs trace-openings where trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
  by blast
obtain trace-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
  using accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
  [OF trace-partial False]
  by blast
have accepted-out: accepted (Some (result, final-state))
  by (rule accepted-with-partial-trace-openings-imp-accepted
    [OF trace-partial])
let ?composition-table = replicate (scale * clength) final
have empty-partial:
  accepted-with-empty-composition-header-candidates s
  (Some (result, final-state)) fr f-fri-roots f-final as dg final
  trace-query-idxs trace-openings trace-table ?composition-table
  unfolding accepted-with-empty-composition-header-candidates-def
  by (intro conjI exI[of - rest])
  (use accepted-out header comp-empty trace-partial trace-candidate
    in simp-all)
have soundness-bad-event-partial-candidate-empty-header s
  (Some (result, final-state))
  by (rule accepted-empty-composition-header-candidate-partition
    [OF false-statement empty-partial])
then show ?thesis
  unfolding out-eq by simp
qed

```

qed

definition *trace-fri-with-tables-opening-algebraic-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow \text{bool}$

where

trace-fri-with-tables-opening-algebraic-cover s *bad-sets* $\longleftrightarrow$
 $(\forall \text{ out trace-table composition-table as query-idxs opening-query-idxs}$
 $\text{ trace-roots trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final trace-round-layers composition-round-layers.}$
 $\text{accepted-with-fri-tables } s \text{ out} \longrightarrow$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as}$
 $\text{ query-idxs} \longrightarrow$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs trace-final}$
 $\text{ dg composition-roots composition-bs composition-final opening-query-idxs}$
 $\text{ trace-round-layers composition-round-layers} \longrightarrow$
 $\neg \text{ trace-table-low-degree trace-table} \longrightarrow$
 $\text{ trace-bs} \in \text{ bad-sets trace-table})$

definition *composition-fri-with-tables-opening-algebraic-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow \text{bool}$

where

composition-fri-with-tables-opening-algebraic-cover s *bad-sets* $\longleftrightarrow$
 $(\forall \text{ out trace-table composition-table as query-idxs opening-query-idxs}$
 $\text{ trace-roots trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final trace-round-layers composition-round-layers.}$
 $\text{accepted-with-fri-tables } s \text{ out} \longrightarrow$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as}$
 $\text{ query-idxs} \longrightarrow$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs trace-final}$
 $\text{ dg composition-roots composition-bs composition-final opening-query-idxs}$
 $\text{ trace-round-layers composition-round-layers} \longrightarrow$
 $\text{ trace-table-low-degree trace-table} \longrightarrow$
 $\neg \text{ composition-table-low-degree maxDegree composition-table} \longrightarrow$
 $\text{ composition-bs} \in \text{ bad-sets dg composition-table})$

definition *trace-fri-with-tables-bad-sets*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list set}$

where

trace-fri-with-tables-bad-sets s *trace-table* =
 $\{ \text{ trace-bs.}$
 $\exists \text{ out composition-table as query-idxs opening-query-idxs}$
 $\text{ trace-roots trace-final dg composition-roots composition-bs}$
 $\text{ composition-final trace-round-layers composition-round-layers.}$
 $\text{accepted-with-fri-tables } s \text{ out} \wedge$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as}$
 $\text{ query-idxs} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$

$\text{trace-final dg composition-roots composition-bs}$
 $\text{composition-final opening-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\neg \text{trace-table-low-degree trace-table}\}$

definition *composition-fri-with-tables-bad-sets*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list set}$

where

$\text{composition-fri-with-tables-bad-sets } s \text{ dg composition-table} =$
 $\{\text{composition-bs.}$

$\exists \text{ out trace-table as query-idxs opening-query-idxs trace-roots}$
 $\text{trace-bs trace-final composition-roots composition-final}$
 $\text{trace-round-layers composition-round-layers.}$

$\text{accepted-with-fri-tables } s \text{ out} \wedge$

$\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as}$
 $\text{query-idxs} \wedge$

$\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{opening-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$

$\neg \text{composition-table-low-degree maxDegree composition-table}\}$

lemma *trace-fri-with-tables-bad-sets-cover:*

$\text{trace-fri-with-tables-opening-algebraic-cover } s$

$(\text{trace-fri-with-tables-bad-sets } s)$

unfolding *trace-fri-with-tables-opening-algebraic-cover-def*

$\text{trace-fri-with-tables-bad-sets-def}$

by *blast*

lemma *composition-fri-with-tables-bad-sets-cover:*

$\text{composition-fri-with-tables-opening-algebraic-cover } s$

$(\text{composition-fri-with-tables-bad-sets } s)$

unfolding *composition-fri-with-tables-opening-algebraic-cover-def*

$\text{composition-fri-with-tables-bad-sets-def}$

by *blast*

lemma *trace-fri-with-tables-bad-sets-subset:*

$\text{trace-fri-with-tables-bad-sets } s \text{ trace-table} \subseteq$

$\text{fri-challenge-space } (\text{ceil-log clength})$

unfolding *trace-fri-with-tables-bad-sets-def*

using *accepted-fri-opening-transcript-challenges*

$\text{accepted-fri-challenges-trace-space}$

by *blast*

lemma *composition-fri-with-tables-bad-sets-subset:*

$\text{composition-fri-with-tables-bad-sets } s \text{ dg composition-table} \subseteq$

$\text{fri-challenge-space } (\text{ceil-log } (\text{to-nat dg} + 1))$

unfolding *composition-fri-with-tables-bad-sets-def*

using *accepted-fri-opening-transcript-challenges*

accepted-fri-challenges-composition-space
by *blast*

lemma *trace-fri-with-tables-bad-sets-card-bound:*
 $\text{card} (\text{trace-fri-with-tables-bad-sets } s \text{ trace-table}) \leq$
 $\text{CARD}('f) \wedge \text{ceil-log clength}$

proof –
have $\text{card} (\text{trace-fri-with-tables-bad-sets } s \text{ trace-table}) \leq$
 $\text{card} (\text{fri-challenge-space } (\text{ceil-log clength}))$
by (*rule card-mono[OF finite-fri-challenge-space]*)
(rule trace-fri-with-tables-bad-sets-subset)
also have $\dots = \text{CARD}('f) \wedge \text{ceil-log clength}$
by (*rule card-fri-challenge-space*)
finally show *?thesis* .
qed

lemma *composition-fri-with-tables-bad-sets-card-bound:*
 $\text{card} (\text{composition-fri-with-tables-bad-sets } s \text{ dg composition-table}) \leq$
 $\text{CARD}('f) \wedge \text{ceil-log (to-nat dg + 1)}$

proof –
have $\text{card} (\text{composition-fri-with-tables-bad-sets } s \text{ dg composition-table}) \leq$
 $\text{card} (\text{fri-challenge-space } (\text{ceil-log (to-nat dg + 1)}))$
by (*rule card-mono[OF finite-fri-challenge-space]*)
(rule composition-fri-with-tables-bad-sets-subset)
also have $\dots = \text{CARD}('f) \wedge \text{ceil-log (to-nat dg + 1)}$
by (*rule card-fri-challenge-space*)
finally show *?thesis* .
qed

lemma *trace-fri-with-tables-bad-sets-fraction-le-one:*
 $\text{nnreal} (\text{card} (\text{trace-fri-with-tables-bad-sets } s \text{ trace-table})) /$
 $\text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log clength}) \leq 1$

proof –
have $\text{nnreal} (\text{card} (\text{trace-fri-with-tables-bad-sets } s \text{ trace-table})) /$
 $\text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log clength}) \leq$
 $\text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log clength}) /$
 $\text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log clength})$
by (*rule nnreal-nat-divide-right-mono*)
(rule trace-fri-with-tables-bad-sets-card-bound)
also have $\dots = 1$
by (*rule nnreal-nat-divide-self*) *simp*
finally show *?thesis* .
qed

lemma *composition-fri-with-tables-bad-sets-fraction-le-one:*
 $\text{nnreal} (\text{card} (\text{composition-fri-with-tables-bad-sets } s \text{ dg composition-table})) /$
 $\text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log (to-nat dg + 1)}) \leq 1$

proof –
have *nnreal*

```

      (card (composition-fri-with-tables-bad-sets s dg composition-table)) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
    by (rule nnreal-nat-divide-right-mono)
      (rule composition-fri-with-tables-bad-sets-card-bound)
  also have ... = 1
    by (rule nnreal-nat-divide-self) simp
  finally show ?thesis .
qed

definition trace-fri-bad-with-tables-challenge-cover
:: ('f, 'a) protocol-channel-scheme ⇒
  ('f list ⇒ 'f list set) ⇒ bool
where
  trace-fri-bad-with-tables-challenge-cover s bad-sets ⟷
  (∀ out ∈ set-dist (execute verify-monad s).
    trace-fri-bad-with-tables s out ⟶
    trace-fri-challenge-set-hit s bad-sets out)

definition composition-fri-bad-with-tables-challenge-cover
:: ('f, 'a) protocol-channel-scheme ⇒
  ('f ⇒ 'f list ⇒ 'f list set) ⇒ bool
where
  composition-fri-bad-with-tables-challenge-cover s bad-sets ⟷
  (∀ out ∈ set-dist (execute verify-monad s).
    composition-fri-bad-with-tables s out ⟶
    composition-fri-challenge-set-hit s bad-sets out)

lemma trace-fri-with-tables-opening-algebraic-cover-imp-bad-challenge-cover:
assumes cover:
  trace-fri-with-tables-opening-algebraic-cover s bad-sets
shows trace-fri-bad-with-tables-challenge-cover s bad-sets
unfolding trace-fri-bad-with-tables-challenge-cover-def
proof (intro ballI impI)
  fix out :: (unit list × ('f, 'a) protocol-channel-scheme) option
  assume outcome: out ∈ set-dist (execute verify-monad s)
  and bad: trace-fri-bad-with-tables s out
  then have with-tables: accepted-with-fri-tables s out
  and trace-bad: trace-fri-bad s out
  unfolding trace-fri-bad-with-tables-def by simp-all
  from trace-bad obtain trace-table composition-table as query-idxs where bound:
    accepted-with-bound-tables s out trace-table composition-table as
    query-idxs
  and not-low: ¬ trace-table-low-degree trace-table
  unfolding trace-fri-bad-def by blast
  from accepted-with-bound-tables-imp-accepted[OF bound]
  obtain result final-state where out-eq: out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto

```

```

have verify-out:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using outcome out-eq by simp
from verify-monad-accepted-fri-opening-transcript[OF verify-out]
obtain trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs' trace-round-layers composition-round-layers
where openings:
  accepted-fri-opening-transcript s (Some (result, final-state))
  trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs' trace-round-layers
  composition-round-layers
by blast
have trace-bs-bad: trace-bs ∈ bad-sets trace-table
  using cover with-tables bound openings not-low out-eq
  unfolding trace-fri-with-tables-opening-algebraic-cover-def by blast
have challenges:
  accepted-fri-challenges s out trace-bs dg composition-bs
  using accepted-fri-opening-transcript-challenges[OF openings] out-eq
  by simp
show trace-fri-challenge-set-hit s bad-sets out
  unfolding trace-fri-challenge-set-hit-def
  using bound challenges not-low trace-bs-bad by blast
qed

lemma composition-fri-with-tables-opening-algebraic-cover-imp-bad-challenge-cover:
  assumes cover:
    composition-fri-with-tables-opening-algebraic-cover s bad-sets
  shows composition-fri-bad-with-tables-challenge-cover s bad-sets
  unfolding composition-fri-bad-with-tables-challenge-cover-def
proof (intro ballI impI)
  fix out :: (unit list × ('f, 'a) protocol-channel-scheme) option
  assume outcome: out ∈ set-dist (execute verify-monad s)
  and bad: composition-fri-bad-with-tables s out
  then have with-tables: accepted-with-fri-tables s out
  and composition-bad: composition-fri-bad s out
  unfolding composition-fri-bad-with-tables-def by simp-all
from composition-bad obtain trace-table composition-table as query-idxs
  where bound:
    accepted-with-bound-tables s out trace-table composition-table as
    query-idxs
  and trace-low: trace-table-low-degree trace-table
  and not-low: ¬ composition-table-low-degree maxDegree composition-table
  unfolding composition-fri-bad-def by blast
from accepted-with-bound-tables-imp-accepted[OF bound]
obtain result final-state where out-eq: out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto
have verify-out:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using outcome out-eq by simp

```

```

from verify-monad-accepted-fri-opening-transcript[OF verify-out]
obtain trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs' trace-round-layers composition-round-layers
where openings:
  accepted-fri-opening-transcript s (Some (result, final-state))
  trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs' trace-round-layers
  composition-round-layers
by blast
have composition-bs-bad:
  composition-bs ∈ bad-sets dg composition-table
using cover with-tables bound openings trace-low not-low out-eq
unfolding composition-fri-with-tables-opening-algebraic-cover-def by blast
have challenges:
  accepted-fri-challenges s out trace-bs dg composition-bs
using accepted-fri-opening-transcript-challenges[OF openings] out-eq
by simp
show composition-fri-challenge-set-hit s bad-sets out
unfolding composition-fri-challenge-set-hit-def
using bound challenges trace-low not-low composition-bs-bad by blast
qed

lemma wp-trace-fri-challenge-set-hit-bound-if-root-candidate-unique:
assumes future: trace-fri-future-fresh s
and unique:
   $\bigwedge_{fr}. \exists \text{ trace-table. } \text{trace-fri-root-trace-table-candidates } s \text{ } fr \subseteq \{\text{trace-table}\}$ 
and bounded: trace-fri-bad-challenge-sets-bounded bad-sets
shows wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s ≤
  trace-fri-error
proof –
have union-bound:
   $\bigwedge_{fr}. \text{nnreal } (\text{card } (\text{trace-fri-root-union-bad-sets } s \text{ bad-sets } fr)) /$ 
   $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength}) \leq \text{trace-fri-error}$ 
proof –
fix fr
show nnreal
   $(\text{card } (\text{trace-fri-root-union-bad-sets } s \text{ bad-sets } fr)) /$ 
   $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength}) \leq \text{trace-fri-error}$ 
by (rule trace-fri-root-union-bad-sets-fraction-bound-if-unique-candidate
  [OF unique])
  (use bounded in
    ⟨auto simp: trace-fri-bad-challenge-sets-bounded-def⟩)
qed
show ?thesis
by (rule wp-trace-fri-challenge-set-hit-bound-via-root-union
  [OF future union-bound])

```

qed

lemma *wp-trace-fri-challenge-set-hit-bound-if-supported-root-candidate-unique:*

assumes *future: trace-fri-future-fresh s*

and *unique:*

$\bigwedge_{fr.}$

$\exists$ *trace-table.*

trace-fri-supported-root-trace-table-candidates s fr $\subseteq$
 $\{\text{trace-table}\}$

and *bounded: trace-fri-bad-challenge-sets-bounded bad-sets*

shows *wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s* $\leq$
trace-fri-error

proof —

have *union-bound:*

$\bigwedge_{fr.}$ *nnreal*

$(\text{card } (\text{trace-fri-supported-root-union-bad-sets } s \text{ bad-sets } fr)) /$
 $\text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log clength}) \leq \text{trace-fri-error}$

proof —

fix *fr*

show *nnreal*

$(\text{card } (\text{trace-fri-supported-root-union-bad-sets } s \text{ bad-sets } fr)) /$
 $\text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log clength}) \leq \text{trace-fri-error}$

by (*rule*

trace-fri-supported-root-union-bad-sets-fraction-bound-if-unique-candidate
 $[OF \text{ unique}]$)

(*use bounded in*

$\langle \text{auto simp: trace-fri-bad-challenge-sets-bounded-def} \rangle$)

qed

show *?thesis*

by (*rule wp-trace-fri-challenge-set-hit-bound-via-supported-root-union*
 $[OF \text{ future union-bound}]$)

qed

lemma *wp-trace-fri-challenge-set-hit-bound-if-no-supported-pairwise-merkle-bad:*

assumes *future: trace-fri-future-fresh s*

and *no-bad: $\neg$ trace-fri-supported-pairwise-merkle-bad s*

and *bounded: trace-fri-bad-challenge-sets-bounded bad-sets*

shows *wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s* $\leq$
trace-fri-error

proof (*rule wp-trace-fri-challenge-set-hit-bound-if-supported-root-candidate-unique*
 $[OF \text{ future - bounded}]$)

fix *fr*

show

$\exists$ *trace-table.*

trace-fri-supported-root-trace-table-candidates s fr $\subseteq$
 $\{\text{trace-table}\}$

by (*rule*

trace-fri-supported-root-candidate-unique-if-no-global-pairwise-merkle-bad
 $[OF \text{ no-bad}]$)

qed

lemma *wp-composition-fri-challenge-set-hit-bound-if-degree-candidate-unique:*

assumes *future: composition-fri-future-fresh s*

and *unique:*

$\bigwedge dg.$

$\exists$ *composition-table.*

composition-fri-degree-table-candidates s dg $\subseteq$
 $\{ \text{composition-table} \}$

and *bounded: composition-fri-bad-challenge-sets-bounded bad-sets*

shows

wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s $\leq$
composition-fri-error

proof –

have *union-bound:*

$\bigwedge dg. \text{nnreal}$

$(\text{card } (\text{composition-fri-degree-union-bad-sets } s \text{ bad-sets } dg)) /$
 $\text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq$
composition-fri-error

proof –

fix *dg*

show *nnreal*

$(\text{card } (\text{composition-fri-degree-union-bad-sets } s \text{ bad-sets } dg)) /$
 $\text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq$
composition-fri-error

by (*rule*

composition-fri-degree-union-bad-sets-fraction-bound-if-unique-candidate
 $[OF \text{ unique}]$)

(*use bounded in*

$\langle \text{auto simp: composition-fri-bad-challenge-sets-bounded-def} \rangle$)

qed

show *?thesis*

by (*rule wp-composition-fri-challenge-set-hit-bound-via-degree-union*
 $[OF \text{ future union-bound}]$)

qed

lemma *wp-composition-fri-challenge-set-hit-bound-if-supported-root-candidate-unique:*

assumes *future: composition-fri-future-fresh s*

and *unique:*

$\bigwedge fr \text{ f-fri-roots f-final as dg composition-fri-roots.}$

$\exists$ *composition-table.*

composition-fri-supported-root-composition-table-candidates s fr
f-fri-roots f-final as dg composition-fri-roots $\subseteq$
 $\{ \text{composition-table} \}$

and *bounded: composition-fri-bad-challenge-sets-bounded bad-sets*

shows

wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s $\leq$
composition-fri-error

proof –

```

have union-bound:
   $\bigwedge fr\ f\text{-fri-roots}\ f\text{-final}\ \text{as}\ dg\ \text{composition-fri-roots}.$ 
  nnreal
    (card
      (composition-fri-supported-root-union-bad-sets s bad-sets fr
        f-fri-roots f-final as dg composition-fri-roots)) /
  nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤
  composition-fri-error
proof -
  fix fr f-fri-roots f-final as dg composition-fri-roots
  show nnreal
    (card
      (composition-fri-supported-root-union-bad-sets s bad-sets fr
        f-fri-roots f-final as dg composition-fri-roots)) /
  nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤
  composition-fri-error
  by (rule
    composition-fri-supported-root-union-bad-sets-fraction-bound-if-unique-candidate
    [OF unique])
    (use bounded in
      ‹auto simp: composition-fri-bad-challenge-sets-bounded-def›)
qed
show ?thesis
  by (rule
    wp-composition-fri-challenge-set-hit-bound-via-supported-root-union
    [OF future union-bound])
qed

lemma wp-composition-fri-challenge-set-hit-bound-if-no-supported-pairwise-merkle-bad:
  assumes future: composition-fri-future-fresh s
  and no-bad: ¬ composition-fri-supported-pairwise-merkle-bad s
  and bounded: composition-fri-bad-challenge-sets-bounded bad-sets
  shows
    wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s ≤
    composition-fri-error
proof (rule
  wp-composition-fri-challenge-set-hit-bound-if-supported-root-candidate-unique
  [OF future - bounded])
  fix fr f-fri-roots f-final as dg composition-fri-roots
  show
    ∃ composition-table.
      composition-fri-supported-root-composition-table-candidates s fr
        f-fri-roots f-final as dg composition-fri-roots ⊆
        {composition-table}
  by (rule
    composition-fri-supported-root-candidate-unique-if-no-global-pairwise-merkle-bad
    [OF no-bad])
qed

```


definition *trace-fri-opening-algebraic-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow \text{bool}$

where

$\text{trace-fri-opening-algebraic-cover } s \text{ bad-sets} \longleftrightarrow$
 $(\forall \text{ out trace-table composition-table as query-idxs opening-query-idxs}$
 $\text{ trace-roots trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final trace-round-layers composition-round-layers.}$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs} \longrightarrow$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs trace-final}$
 $\text{ dg composition-roots composition-bs composition-final opening-query-idxs}$
 $\text{ trace-round-layers composition-round-layers} \longrightarrow$
 $\neg \text{ trace-table-low-degree trace-table} \longrightarrow$
 $\text{trace-bs} \in \text{bad-sets trace-table})$

definition *composition-fri-opening-algebraic-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow \text{bool}$

where

$\text{composition-fri-opening-algebraic-cover } s \text{ bad-sets} \longleftrightarrow$
 $(\forall \text{ out trace-table composition-table as query-idxs opening-query-idxs}$
 $\text{ trace-roots trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final trace-round-layers composition-round-layers.}$
 $\text{accepted-with-bound-tables } s \text{ out trace-table composition-table as query-idxs} \longrightarrow$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs trace-final}$
 $\text{ dg composition-roots composition-bs composition-final opening-query-idxs}$
 $\text{ trace-round-layers composition-round-layers} \longrightarrow$
 $\text{trace-table-low-degree trace-table} \longrightarrow$
 $\neg \text{ composition-table-low-degree maxDegree composition-table} \longrightarrow$
 $\text{composition-bs} \in \text{bad-sets dg composition-table})$

definition *trace-fri-opening-reduction-assumption*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{trace-fri-opening-reduction-assumption } s \longleftrightarrow$
 $(\exists \text{ bad-sets.}$
 $\text{ trace-fri-opening-algebraic-cover } s \text{ bad-sets} \wedge$
 $\text{ trace-fri-bad-challenge-sets-bounded bad-sets})$

definition *composition-fri-opening-reduction-assumption*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

$\text{composition-fri-opening-reduction-assumption } s \longleftrightarrow$
 $(\exists \text{ bad-sets.}$
 $\text{ composition-fri-opening-algebraic-cover } s \text{ bad-sets} \wedge$
 $\text{ composition-fri-bad-challenge-sets-bounded bad-sets})$

lemma *trace-fri-opening-reductionI:*

assumes $\text{trace-fri-opening-algebraic-cover } s \text{ bad-sets}$

and *trace-fri-bad-challenge-sets-bounded bad-sets*
shows *trace-fri-opening-reduction-assumption s*
using *assms* **unfolding** *trace-fri-opening-reduction-assumption-def* **by** *blast*

lemma *trace-fri-opening-reductionE*:
assumes *trace-fri-opening-reduction-assumption s*
obtains *bad-sets*
where *trace-fri-opening-algebraic-cover s bad-sets*
and *trace-fri-bad-challenge-sets-bounded bad-sets*
using *assms* **unfolding** *trace-fri-opening-reduction-assumption-def* **by** *blast*

lemma *composition-fri-opening-reductionI*:
assumes *composition-fri-opening-algebraic-cover s bad-sets*
and *composition-fri-bad-challenge-sets-bounded bad-sets*
shows *composition-fri-opening-reduction-assumption s*
using *assms* **unfolding** *composition-fri-opening-reduction-assumption-def* **by** *blast*

lemma *composition-fri-opening-reductionE*:
assumes *composition-fri-opening-reduction-assumption s*
obtains *bad-sets*
where *composition-fri-opening-algebraic-cover s bad-sets*
and *composition-fri-bad-challenge-sets-bounded bad-sets*
using *assms* **unfolding** *composition-fri-opening-reduction-assumption-def*
by *blast*

lemma *trace-fri-multiround-bad-sets-bounded*:
assumes *frac*:
 $\bigwedge \text{trace-table.}$
 $\text{nnreal (card (trace-fri-multiround-bad-sets bad trace-table)) /}$
 $\text{nnreal (CARD('f) ^ ceil-log clength) } \leq \text{trace-fri-error}$
shows *trace-fri-bad-challenge-sets-bounded*
 $(\text{trace-fri-multiround-bad-sets bad})$
unfolding *trace-fri-bad-challenge-sets-bounded-def*
proof (*intro allI conjI*)
fix *trace-table*
show *trace-fri-multiround-bad-sets bad trace-table* $\subseteq$
 $\text{fri-challenge-space (ceil-log clength)}$
by (*rule trace-fri-multiround-bad-sets-subset*)
show $\text{nnreal (card (trace-fri-multiround-bad-sets bad trace-table)) /}$
 $\text{nnreal (CARD('f) ^ ceil-log clength) } \leq \text{trace-fri-error}$
by (*rule frac*)
qed

lemma *composition-fri-multiround-bad-sets-bounded*:
assumes *frac*:
 $\bigwedge dg \text{ composition-table.}$
 nnreal (card
 $(\text{composition-fri-multiround-bad-sets bad dg composition-table})) /$
 $\text{nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) } \leq$

```

      composition-fri-error
    shows composition-fri-bad-challenge-sets-bounded
      (composition-fri-multiround-bad-sets bad)
    unfolding composition-fri-bad-challenge-sets-bounded-def
  proof (intro allI conjI)
    fix dg composition-table
    show composition-fri-multiround-bad-sets bad dg composition-table  $\subseteq$ 
      fri-challenge-space (ceil-log (to-nat dg + 1))
    by (rule composition-fri-multiround-bad-sets-subset)
    show nnreal (card
      (composition-fri-multiround-bad-sets bad dg composition-table)) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))  $\leq$ 
      composition-fri-error
    by (rule frac)
  qed

```

```

lemma trace-fri-opening-reduction-from-multiround-bad-sets:
  assumes cover:
    trace-fri-opening-algebraic-cover s
    (trace-fri-multiround-bad-sets bad)
  and frac:
     $\bigwedge$  trace-table.
      nnreal (card (trace-fri-multiround-bad-sets bad trace-table)) /
      nnreal (CARD('f) ^ ceil-log clength)  $\leq$  trace-fri-error
  shows trace-fri-opening-reduction-assumption s
  by (rule trace-fri-opening-reductionI
    [OF cover trace-fri-multiround-bad-sets-bounded[OF frac]])

```

```

lemma composition-fri-opening-reduction-from-multiround-bad-sets:
  assumes cover:
    composition-fri-opening-algebraic-cover s
    (composition-fri-multiround-bad-sets bad)
  and frac:
     $\bigwedge$  dg composition-table.
      nnreal (card
        (composition-fri-multiround-bad-sets bad dg composition-table)) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))  $\leq$ 
      composition-fri-error
  shows composition-fri-opening-reduction-assumption s
  by (rule composition-fri-opening-reductionI
    [OF cover composition-fri-multiround-bad-sets-bounded[OF frac]])

```

end

end

```

theory Soundness-Reductions-Opening
  imports Soundness-Reductions-Partial

```

begin

FRI opening algebraic covers and final reduction bounds.

context *soundness*

begin

The multi-round bad-set constructors above isolate the proof-only part of the FRI reductions: once an algebraic cover and the corresponding error-fraction bound are available, the existing opening-reduction assumptions follow directly. The current transcript predicate records authenticated local openings, but it does not bind complete intermediate FRI layer tables. Thus the algebraic cover is still an external FRI/table-binding theorem target unless the protocol model is extended with such tables or an equivalent binding interface.

definition *trace-fri-bad-challenge-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow \text{bool}$

where

$\text{trace-fri-bad-challenge-cover } s \text{ bad-sets} \longleftrightarrow$
 $(\forall \text{ out} \in \text{set-dist } (\text{execute verify-monad } s).$
 $\text{trace-fri-bad } s \text{ out} \longrightarrow$
 $\text{trace-fri-challenge-set-hit } s \text{ bad-sets out})$

definition *composition-fri-bad-challenge-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow \text{bool}$

where

$\text{composition-fri-bad-challenge-cover } s \text{ bad-sets} \longleftrightarrow$
 $(\forall \text{ out} \in \text{set-dist } (\text{execute verify-monad } s).$
 $\text{composition-fri-bad } s \text{ out} \longrightarrow$
 $\text{composition-fri-challenge-set-hit } s \text{ bad-sets out})$

lemma *trace-fri-opening-algebraic-cover-imp-bad-challenge-cover:*

assumes *cover*: *trace-fri-opening-algebraic-cover* *s bad-sets*

shows *trace-fri-bad-challenge-cover* *s bad-sets*

unfolding *trace-fri-bad-challenge-cover-def*

proof (*intro ballI impI*)

fix *out* $:: (\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option}$

assume *outcome*: *out* $\in \text{set-dist } (\text{execute verify-monad } s)$

and *bad*: *trace-fri-bad* *s out*

then obtain *trace-table composition-table as query-idxs* **where** *bound*:

accepted-with-bound-tables *s out trace-table composition-table as query-idxs*

and *not-low*: $\neg \text{trace-table-low-degree } \text{trace-table}$

unfolding *trace-fri-bad-def* **by** *blast*

from *accepted-with-bound-tables-imp-accepted*[*OF bound*]

obtain *result final-state* **where** *out-eq*: *out* = *Some (result, final-state)*

unfolding *accepted-def* **by** (*cases out*) *auto*

have *verify-out*:

Some (result, final-state) $\in \text{set-dist } (\text{execute verify-monad } s)$

```

using outcome out-eq by simp
from verify-monad-accepted-fri-opening-transcript[OF verify-out]
obtain trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs' trace-round-layers composition-round-layers
where openings:
  accepted-fri-opening-transcript s (Some (result, final-state))
  trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs' trace-round-layers composition-round-layers
by blast
have trace-bs-bad: trace-bs ∈ bad-sets trace-table
using cover bound openings not-low out-eq
unfolding trace-fri-opening-algebraic-cover-def by blast
have challenges:
  accepted-fri-challenges s out trace-bs dg composition-bs
using accepted-fri-opening-transcript-challenges[OF openings] out-eq by simp
show trace-fri-challenge-set-hit s bad-sets out
unfolding trace-fri-challenge-set-hit-def
using bound challenges not-low trace-bs-bad by blast
qed

lemma composition-fri-opening-algebraic-cover-imp-bad-challenge-cover:
assumes cover: composition-fri-opening-algebraic-cover s bad-sets
shows composition-fri-bad-challenge-cover s bad-sets
unfolding composition-fri-bad-challenge-cover-def
proof (intro ballI impI)
fix out :: (unit list × ('f, 'a) protocol-channel-scheme) option
assume outcome: out ∈ set-dist (execute verify-monad s)
and bad: composition-fri-bad s out
then obtain trace-table composition-table as query-idxs where bound:
  accepted-with-bound-tables s out trace-table composition-table as query-idxs
and trace-low: trace-table-low-degree trace-table
and not-low: ¬ composition-table-low-degree maxDegree composition-table
unfolding composition-fri-bad-def by blast
from accepted-with-bound-tables-imp-accepted[OF bound]
obtain result final-state where out-eq: out = Some (result, final-state)
unfolding accepted-def by (cases out) auto
have verify-out:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
using outcome out-eq by simp
from verify-monad-accepted-fri-opening-transcript[OF verify-out]
obtain trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs' trace-round-layers composition-round-layers
where openings:
  accepted-fri-opening-transcript s (Some (result, final-state))
  trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs' trace-round-layers composition-round-layers
by blast
have composition-bs-bad: composition-bs ∈ bad-sets dg composition-table
using cover bound openings trace-low not-low out-eq

```

unfolding *composition-fri-opening-algebraic-cover-def* **by** *blast*
have *challenges*:
accepted-fri-challenges s out trace-bs dg composition-bs
using *accepted-fri-opening-transcript-challenges[OF openings] out-eq* **by** *simp*
show *composition-fri-challenge-set-hit s bad-sets out*
unfolding *composition-fri-challenge-set-hit-def*
using *bound challenges trace-low not-low composition-bs-bad* **by** *blast*
qed

definition *trace-fri-challenge-freshness-assumption*
 $:: (f, a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
trace-fri-challenge-freshness-assumption s $\longleftrightarrow$
 $(\forall \text{ bad-sets.}$
 $\text{trace-fri-bad-challenge-sets-bounded bad-sets} \longrightarrow$
 $\text{wp-event verify-monad } (\text{trace-fri-challenge-set-hit s bad-sets}) \text{ s} \leq$
 $\text{trace-fri-error})$

definition *composition-fri-challenge-freshness-assumption*
 $:: (f, a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
composition-fri-challenge-freshness-assumption s $\longleftrightarrow$
 $(\forall \text{ bad-sets.}$
 $\text{composition-fri-bad-challenge-sets-bounded bad-sets} \longrightarrow$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-challenge-set-hit s bad-sets}) \text{ s} \leq$
 $\text{composition-fri-error})$

definition *random-oracle-freshness-assumption*
 $:: (f, a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
random-oracle-freshness-assumption s $\longleftrightarrow$
 $\alpha\text{-challenge-freshness-assumption s} \wedge$
 $\text{query-index-freshness-assumption s} \wedge$
 $\text{trace-fri-challenge-freshness-assumption s} \wedge$
 $\text{composition-fri-challenge-freshness-assumption s}$

lemma *random-oracle-freshness-alpha*:
assumes *random-oracle-freshness-assumption s*
shows *alpha-challenge-freshness-assumption s*
using *assms* **unfolding** *random-oracle-freshness-assumption-def* **by** *simp*

lemma *random-oracle-freshness-query*:
assumes *random-oracle-freshness-assumption s*
shows *query-index-freshness-assumption s*
using *assms* **unfolding** *random-oracle-freshness-assumption-def* **by** *simp*

lemma *random-oracle-freshness-trace-fri*:
assumes *random-oracle-freshness-assumption s*

```

shows trace-fri-challenge-freshness-assumption s
using assms unfolding random-oracle-freshness-assumption-def by simp

lemma random-oracle-freshness-composition-fri:
  assumes random-oracle-freshness-assumption s
  shows composition-fri-challenge-freshness-assumption s
  using assms unfolding random-oracle-freshness-assumption-def by simp

lemma alpha-challenge-freshness-if-no-supported-pairwise-merkle-bad:
  assumes future: alpha-future-fresh s
  and no-bad:  $\neg$  alpha-supported-pairwise-merkle-bad s
  shows alpha-challenge-freshness-assumption s
  unfolding alpha-challenge-freshness-assumption-def
proof (intro allI impI)
  fix bad-sets :: 'f list  $\Rightarrow$  'f list set
  assume subset:
     $\forall$  trace-table. bad-sets trace-table  $\subseteq$  alpha-space
  and bounded:
     $\forall$  trace-table.
       $\text{nnreal} (\text{card} (\text{bad-sets trace-table})) /$ 
       $\text{nnreal} (\text{card alpha-space}) \leq \text{composition-error-bound}$ 
  show wp-event verify-monad (alpha-bad-set-hit s bad-sets) s  $\leq$ 
    composition-error-bound
proof (rule wp-alpha-bad-set-hit-bound-via-supported-header-union
  [OF future])
  fix trace-table
  show bad-sets trace-table  $\subseteq$  alpha-space
    using subset by blast
next
  fix fr f-fri-roots f-final
  have unique:
     $\exists$  trace-table.
      alpha-header-supported-trace-table-candidates s fr f-fri-roots
      f-final  $\subseteq$  {trace-table}
  by (rule alpha-header-supported-candidate-unique-if-no-global-pairwise-merkle-bad
  [OF no-bad])
  show
    nnreal
    (card
      (alpha-header-supported-union-bad-sets s bad-sets fr f-fri-roots
      f-final)) /
     $\text{nnreal} (\text{card alpha-space}) \leq \text{composition-error-bound}$ 
  by (rule alpha-header-supported-union-bad-sets-fraction-bound-if-unique-candidate
  [OF unique])
  (use subset bounded in blast)+
qed
qed

lemma trace-fri-challenge-freshness-if-no-supported-pairwise-merkle-bad:

```

assumes *future: trace-fri-future-fresh s*
and *no-bad: $\neg$ trace-fri-supported-pairwise-merkle-bad s*
shows *trace-fri-challenge-freshness-assumption s*
unfolding *trace-fri-challenge-freshness-assumption-def*
proof (*intro allI impI*)
fix *bad-sets :: 'f list $\Rightarrow$ 'f list set*
assume *bounded: trace-fri-bad-challenge-sets-bounded bad-sets*
show *wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s $\leq$*
trace-fri-error
by (*rule*
wp-trace-fri-challenge-set-hit-bound-if-no-supported-pairwise-merkle-bad
[OF future no-bad bounded])
qed

lemma *composition-fri-challenge-freshness-if-no-supported-pairwise-merkle-bad:*
assumes *future: composition-fri-future-fresh s*
and *no-bad: $\neg$ composition-fri-supported-pairwise-merkle-bad s*
shows *composition-fri-challenge-freshness-assumption s*
unfolding *composition-fri-challenge-freshness-assumption-def*
proof (*intro allI impI*)
fix *bad-sets :: 'f $\Rightarrow$ 'f list $\Rightarrow$ 'f list set*
assume *bounded: composition-fri-bad-challenge-sets-bounded bad-sets*
show
wp-event verify-monad
(composition-fri-challenge-set-hit s bad-sets) s $\leq$
composition-fri-error
by (*rule*
wp-composition-fri-challenge-set-hit-bound-if-no-supported-pairwise-merkle-bad
[OF future no-bad bounded])
qed

lemma *verifier-initial-alpha-challenge-freshness-if-no-supported-pairwise-merkle-bad:*
assumes *no-bad:*
 $\neg$ alpha-supported-pairwise-merkle-bad (verifier-initial-state tr)
shows *alpha-challenge-freshness-assumption (verifier-initial-state tr)*
by (*rule alpha-challenge-freshness-if-no-supported-pairwise-merkle-bad*)
(use no-bad in simp-all)

lemma *verifier-initial-trace-fri-challenge-freshness-if-no-supported-pairwise-merkle-bad:*
assumes *no-bad:*
 $\neg$ trace-fri-supported-pairwise-merkle-bad (verifier-initial-state tr)
shows
trace-fri-challenge-freshness-assumption (verifier-initial-state tr)
by (*rule trace-fri-challenge-freshness-if-no-supported-pairwise-merkle-bad*)
(use no-bad in simp-all)

lemma *verifier-initial-composition-fri-challenge-freshness-if-no-supported-pairwise-merkle-bad:*
assumes *no-bad:*
 $\neg$ composition-fri-supported-pairwise-merkle-bad

(verifier-initial-state tr)
shows
 composition-fri-challenge-freshness-assumption
 (verifier-initial-state tr)
by (rule
 composition-fri-challenge-freshness-if-no-supported-pairwise-merkle-bad)
 (use no-bad in simp-all)

lemma alpha-challenge-freshness-if-no-pairwise-side-bad:
 assumes future: alpha-future-fresh s
 and no-collision: $\neg$ supported-hash-output-collision-possible s
 and no-coupling: $\neg$ supported-pairwise-coupling-bad s
 shows alpha-challenge-freshness-assumption s
proof (rule alpha-challenge-freshness-if-no-supported-pairwise-merkle-bad
 [OF future])
 show $\neg$ alpha-supported-pairwise-merkle-bad s
 by (rule no-alpha-supported-pairwise-merkle-bad-if-no-pairwise-side-bad
 [OF no-collision no-coupling])
qed

lemma query-index-freshness-if-no-pairwise-side-bad:
 assumes future: query-future-fresh s
 and raw-bound: query-index-raw-preimage-bound
 and no-collision: $\neg$ supported-hash-output-collision-possible s
 and no-coupling: $\neg$ supported-pairwise-coupling-bad s
 shows query-index-freshness-assumption s
proof (rule query-index-freshness-if-no-query-supported-pairwise-merkle-bad
 [OF future raw-bound])
 show $\neg$ query-supported-pairwise-merkle-bad s
 by (rule no-query-supported-pairwise-merkle-bad-if-no-pairwise-side-bad
 [OF no-collision no-coupling])
qed

lemma trace-fri-challenge-freshness-if-no-pairwise-side-bad:
 assumes future: trace-fri-future-fresh s
 and no-collision: $\neg$ supported-hash-output-collision-possible s
 and no-coupling: $\neg$ supported-pairwise-coupling-bad s
 shows trace-fri-challenge-freshness-assumption s
proof (rule trace-fri-challenge-freshness-if-no-supported-pairwise-merkle-bad
 [OF future])
 show $\neg$ trace-fri-supported-pairwise-merkle-bad s
 by (rule
 no-trace-fri-supported-pairwise-merkle-bad-if-no-pairwise-side-bad
 [OF no-collision no-coupling])
qed

lemma composition-fri-challenge-freshness-if-no-pairwise-side-bad:
 assumes future: composition-fri-future-fresh s
 and no-collision: $\neg$ supported-hash-output-collision-possible s

and *no-coupling*: $\neg$ *supported-pairwise-coupling-bad* *s*
shows *composition-fri-challenge-freshness-assumption* *s*
proof (rule
 composition-fri-challenge-freshness-if-no-supported-pairwise-merkle-bad
 [*OF future*])
show $\neg$ *composition-fri-supported-pairwise-merkle-bad* *s*
 by (rule
 no-composition-fri-supported-pairwise-merkle-bad-if-no-pairwise-side-bad
 [*OF no-collision no-coupling*])
qed

lemma *random-oracle-freshness-if-no-pairwise-side-bad*:
assumes *alpha-future*: *alpha-future-fresh* *s*
and *query-future*: *query-future-fresh* *s*
and *trace-future*: *trace-fri-future-fresh* *s*
and *composition-future*: *composition-fri-future-fresh* *s*
and *raw-bound*: *query-index-raw-preimage-bound*
and *no-collision*: $\neg$ *supported-hash-output-collision-possible* *s*
and *no-coupling*: $\neg$ *supported-pairwise-coupling-bad* *s*
shows *random-oracle-freshness-assumption* *s*
unfolding *random-oracle-freshness-assumption-def*
proof (intro conjI)
show *alpha-challenge-freshness-assumption* *s*
 by (rule *alpha-challenge-freshness-if-no-pairwise-side-bad*
 [*OF alpha-future no-collision no-coupling*])
show *query-index-freshness-assumption* *s*
 by (rule *query-index-freshness-if-no-pairwise-side-bad*
 [*OF query-future raw-bound no-collision no-coupling*])
show *trace-fri-challenge-freshness-assumption* *s*
 by (rule *trace-fri-challenge-freshness-if-no-pairwise-side-bad*
 [*OF trace-future no-collision no-coupling*])
show *composition-fri-challenge-freshness-assumption* *s*
 by (rule *composition-fri-challenge-freshness-if-no-pairwise-side-bad*
 [*OF composition-future no-collision no-coupling*])
qed

lemma *query-bad-bound-if-no-pairwise-side-bad*:
assumes *future*: *query-future-fresh* *s*
and *raw-bound*: *query-index-raw-preimage-bound*
and *no-collision*: $\neg$ *supported-hash-output-collision-possible* *s*
and *no-coupling*: $\neg$ *supported-pairwise-coupling-bad* *s*
shows *wp-event verify-monad* (*query-bad* *s*) *s* $\leq$
 nnreal rounds * *query-error-bound*
proof –
have *query-fresh*: *query-index-freshness-assumption* *s*
 by (rule *query-index-freshness-if-no-pairwise-side-bad*
 [*OF future raw-bound no-collision no-coupling*])
show ?thesis
 by (rule *query-bad-bound-from-phase4B* [OF *query-fresh*])

qed

lemma *query-bad-bound-if-no-pairwise-side-event-on-support:*

assumes *future: query-future-fresh s*

and *raw-bound: query-index-raw-preimage-bound*

and *no-side:*

$\bigwedge out. out \in \text{set-dist } (\text{execute verify-monad } s) \implies$

$\neg \text{supported-pairwise-side-bad } s \text{ out}$

shows *wp-event verify-monad (query-bad s) s $\leq$*

*nnreal rounds * query-error-bound*

proof –

have *no-collision: $\neg \text{supported-hash-output-collision-possible } s$*

by (rule

no-supported-pairwise-side-bad-on-support-imp-no-hash-collision-possible

[OF *no-side*])

have *no-coupling: $\neg \text{supported-pairwise-coupling-bad } s$*

by (rule *no-supported-pairwise-side-bad-on-support-imp-no-coupling-bad*

[OF *no-side*])

show *?thesis*

by (rule *query-bad-bound-if-no-pairwise-side-bad*

[OF *future raw-bound no-collision no-coupling*])

qed

lemma *query-bad-bound-if-no-pairwise-side-event-on-support-from-sampler-wellformed:*

assumes *future: query-future-fresh s*

and *no-side:*

$\bigwedge out. out \in \text{set-dist } (\text{execute verify-monad } s) \implies$

$\neg \text{supported-pairwise-side-bad } s \text{ out}$

shows *wp-event verify-monad (query-bad s) s $\leq$*

*nnreal rounds * query-error-bound*

by (rule *query-bad-bound-if-no-pairwise-side-event-on-support*

[OF *future query-index-raw-preimage-bound-from-sampler-wellformed no-side*])

lemma *composition-bad-bound-if-no-pairwise-side-bad:*

assumes *wf: spec-degree-wellformed*

and *future: alpha-future-fresh s*

and *no-collision: $\neg \text{supported-hash-output-collision-possible } s$*

and *no-coupling: $\neg \text{supported-pairwise-coupling-bad } s$*

shows *wp-event verify-monad (composition-bad s) s $\leq$*

composition-error-bound

proof –

have *alpha-fresh: alpha-challenge-freshness-assumption s*

by (rule *alpha-challenge-freshness-if-no-pairwise-side-bad*

[OF *future no-collision no-coupling*])

show *?thesis*

by (rule *composition-bad-bound-from-phase4A*[OF *wf alpha-fresh*])

qed

lemma *composition-bad-bound-if-no-pairwise-side-event-on-support:*

assumes *wf: spec-degree-wellformed*

and *future: alpha-future-fresh s*

and *no-side:*

$\bigwedge out. out \in \text{set-dist } (\text{execute verify-monad } s) \implies$
 $\neg \text{supported-pairwise-side-bad } s \text{ out}$

shows *wp-event verify-monad (composition-bad s) s $\leq$*
composition-error-bound

proof —

have *no-collision: $\neg$ supported-hash-output-collision-possible s*

by (rule

no-supported-pairwise-side-bad-on-support-imp-no-hash-collision-possible
 $[OF \text{ no-side}]$)

have *no-coupling: $\neg$ supported-pairwise-coupling-bad s*

by (rule *no-supported-pairwise-side-bad-on-support-imp-no-coupling-bad*
 $[OF \text{ no-side}]$)

show *?thesis*

by (rule *composition-bad-bound-if-no-pairwise-side-bad*
 $[OF \text{ wf future no-collision no-coupling}]$)

qed

lemma *verifier-initial-random-oracle-freshness-if-uniform-sampler-and-no-pairwise-side-bad:*

— The range premise states that *to-nat* enumerates exactly the field indices used by the query sampler; the divisibility premise makes the modulo/index projection uniform over *query-sample-space*.

assumes *range-eq: range to-nat = $\{0..<size\}$*

and *dvd: query-sample-space-size dvd size*

and *no-collision:*

$\neg \text{supported-hash-output-collision-possible } (\text{verifier-initial-state } tr)$

and *no-coupling:*

$\neg \text{supported-pairwise-coupling-bad } (\text{verifier-initial-state } tr)$

shows *random-oracle-freshness-assumption (verifier-initial-state tr)*

proof —

have *raw-bound: query-index-raw-preimage-bound*

by (rule *query-index-raw-preimage-bound-if-uniform-range*
 $[OF \text{ range-eq dvd}]$)

show *?thesis*

by (rule *random-oracle-freshness-if-no-pairwise-side-bad*
 $[OF \text{ - - - raw-bound no-collision no-coupling}]$)

simp-all

qed

The remaining soundness interfaces have the following status without changing the protocol. The predicate *initial-merkle-binding-no-bad* is a cryptographic/model boundary excluding accepted executions whose local Merkle openings cannot be extended to full committed tables. The predicate *random-oracle-freshness-assumption* is a random-oracle model boundary: its component projections are deterministic, but deriving freshness itself would require assumptions about the hash oracle, and deriving domain separation internally would require protocol labels that are not present in the current formalization.

The two FRI reduction interfaces are proof obligations rather than protocol changes. Each packages an opening-transcript algebraic cover with a bounded bad-challenge-set statement. The lemmas below derive the verifier bad-event bounds directly from those opening-based interfaces.

lemma *trace-fri-bad-bound-from-opening-reduction*:
assumes *reduction*: *trace-fri-opening-reduction-assumption s*
and *fresh*: *trace-fri-challenge-freshness-assumption s*
shows *wp-event verify-monad (trace-fri-bad s) s* $\leq$ *trace-fri-error*
proof –
from *reduction* **obtain** *bad-sets* **where** *opening-cover*:
trace-fri-opening-algebraic-cover s bad-sets
and *bounded*:
trace-fri-bad-challenge-sets-bounded bad-sets
by (rule *trace-fri-opening-reductionE*)
have *cover*: *trace-fri-bad-challenge-cover s bad-sets*
by (rule *trace-fri-opening-algebraic-cover-imp-bad-challenge-cover*
[*OF opening-cover*])
have *wp-event verify-monad (trace-fri-bad s) s* $\leq$
wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s
by (rule *wp-event-mono-on-support*)
(use *cover* **in** $\langle$ *auto simp: trace-fri-bad-challenge-cover-def* $\rangle$)
also have ... $\leq$ *trace-fri-error*
using *fresh bounded*
unfolding *trace-fri-challenge-freshness-assumption-def* **by** *blast*
finally show *?thesis* .
qed

lemma *composition-fri-bad-bound-from-opening-reduction*:
assumes *reduction*: *composition-fri-opening-reduction-assumption s*
and *fresh*: *composition-fri-challenge-freshness-assumption s*
shows *wp-event verify-monad (composition-fri-bad s) s* $\leq$
composition-fri-error
proof –
from *reduction* **obtain** *bad-sets* **where** *opening-cover*:
composition-fri-opening-algebraic-cover s bad-sets
and *bounded*:
composition-fri-bad-challenge-sets-bounded bad-sets
by (rule *composition-fri-opening-reductionE*)
have *cover*: *composition-fri-bad-challenge-cover s bad-sets*
by (rule *composition-fri-opening-algebraic-cover-imp-bad-challenge-cover*
[*OF opening-cover*])
have *wp-event verify-monad (composition-fri-bad s) s* $\leq$
wp-event verify-monad
(composition-fri-challenge-set-hit s bad-sets) s
by (rule *wp-event-mono-on-support*)
(use *cover* **in** $\langle$ *auto simp: composition-fri-bad-challenge-cover-def* $\rangle$)
also have ... $\leq$ *composition-fri-error*
using *fresh bounded*
unfolding *composition-fri-challenge-freshness-assumption-def* **by** *blast*

finally show ?thesis .
qed

lemma *trace-fri-bad-with-partial-openings-bound-from-opening-reduction:*
assumes *reduction: trace-fri-opening-reduction-assumption s*
and *fresh: trace-fri-challenge-freshness-assumption s*
shows *wp-event verify-monad (trace-fri-bad-with-partial-openings s) s ≤ trace-fri-error*

proof –
have *wp-event verify-monad (trace-fri-bad-with-partial-openings s) s ≤ wp-event verify-monad (trace-fri-bad s) s*
by (rule *wp-trace-fri-bad-with-partial-openings-le-trace-fri-bad*)
also have *... ≤ trace-fri-error*
by (rule *trace-fri-bad-bound-from-opening-reduction* [OF *reduction fresh*])
finally show ?thesis .
qed

lemma *composition-fri-bad-with-partial-openings-bound-from-opening-reduction:*
assumes *reduction: composition-fri-opening-reduction-assumption s*
and *fresh: composition-fri-challenge-freshness-assumption s*
shows
wp-event verify-monad (composition-fri-bad-with-partial-openings s) s ≤ composition-fri-error

proof –
have *wp-event verify-monad (composition-fri-bad-with-partial-openings s) s ≤ wp-event verify-monad (composition-fri-bad s) s*
by (rule *wp-composition-fri-bad-with-partial-openings-le-composition-fri-bad*)
also have *... ≤ composition-fri-error*
by (rule *composition-fri-bad-bound-from-opening-reduction* [OF *reduction fresh*])
finally show ?thesis .
qed

lemma *trace-fri-with-tables-bad-sets-bounded-if-error-ge-one:*
assumes *error-ge-one: 1 ≤ trace-fri-error*
shows *trace-fri-bad-challenge-sets-bounded (trace-fri-with-tables-bad-sets s)*
unfolding *trace-fri-bad-challenge-sets-bounded-def*

proof (intro *allI conjI*)
fix *trace-table*
show *trace-fri-with-tables-bad-sets s trace-table ⊆ fri-challenge-space (ceil-log clength)*
by (rule *trace-fri-with-tables-bad-sets-subset*)
show *nnreal (card (trace-fri-with-tables-bad-sets s trace-table)) / nnreal (CARD('f) ^ ceil-log clength) ≤ trace-fri-error*

```

    by (rule order-trans
        [OF trace-fri-with-tables-bad-sets-fraction-le-one error-ge-one])
qed

lemma composition-fri-with-tables-bad-sets-bounded-if-error-ge-one:
  assumes error-ge-one:  $1 \leq \text{composition-fri-error}$ 
  shows composition-fri-bad-challenge-sets-bounded
    (composition-fri-with-tables-bad-sets s)
  unfolding composition-fri-bad-challenge-sets-bounded-def
proof (intro allI conjI)
  fix dg composition-table
  show composition-fri-with-tables-bad-sets s dg composition-table  $\subseteq$ 
    fri-challenge-space (ceil-log (to-nat dg + 1))
  by (rule composition-fri-with-tables-bad-sets-subset)
  show nnreal
    (card (composition-fri-with-tables-bad-sets s dg composition-table)) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))  $\leq$ 
    composition-fri-error
  by (rule order-trans
      [OF composition-fri-with-tables-bad-sets-fraction-le-one error-ge-one])
qed

lemma trace-fri-bad-with-tables-bound-from-cover:
  assumes cover:
    trace-fri-with-tables-opening-algebraic-cover s bad-sets
  and bounded: trace-fri-bad-challenge-sets-bounded bad-sets
  and fresh: trace-fri-challenge-freshness-assumption s
  shows wp-event verify-monad (trace-fri-bad-with-tables s) s  $\leq$ 
    trace-fri-error
proof -
  have challenge-cover:
    trace-fri-bad-with-tables-challenge-cover s bad-sets
  by (rule trace-fri-with-tables-opening-algebraic-cover-imp-bad-challenge-cover
      [OF cover])
  have wp-event verify-monad (trace-fri-bad-with-tables s) s  $\leq$ 
    wp-event verify-monad (trace-fri-challenge-set-hit s bad-sets) s
  by (rule wp-event-mono-on-support)
  (use challenge-cover in
    ⟨auto simp: trace-fri-bad-with-tables-challenge-cover-def⟩)
  also have ...  $\leq$  trace-fri-error
  using fresh bounded
  unfolding trace-fri-challenge-freshness-assumption-def by blast
  finally show ?thesis .
qed

lemma composition-fri-bad-with-tables-bound-from-cover:
  assumes cover:
    composition-fri-with-tables-opening-algebraic-cover s bad-sets
  and bounded: composition-fri-bad-challenge-sets-bounded bad-sets

```

and *fresh: composition-fri-challenge-freshness-assumption s*
shows *wp-event verify-monad (composition-fri-bad-with-tables s) s ≤ composition-fri-error*
proof –
have *challenge-cover:*
composition-fri-bad-with-tables-challenge-cover s bad-sets
by (rule
composition-fri-with-tables-opening-algebraic-cover-imp-bad-challenge-cover
[OF cover])
have *wp-event verify-monad (composition-fri-bad-with-tables s) s ≤ wp-event verify-monad (composition-fri-challenge-set-hit s bad-sets) s*
by (rule *wp-event-mono-on-support*)
(use challenge-cover in
⟨auto simp: composition-fri-bad-with-tables-challenge-cover-def⟩)
also have *... ≤ composition-fri-error*
using *fresh bounded*
unfolding *composition-fri-challenge-freshness-assumption-def* **by** *blast*
finally show *?thesis* .
qed

definition *trace-fri-with-tables-opening-reduction-assumption*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
trace-fri-with-tables-opening-reduction-assumption s $\longleftrightarrow$
 $(\exists \text{ bad-sets.}$
trace-fri-with-tables-opening-algebraic-cover s bad-sets $\wedge$
trace-fri-bad-challenge-sets-bounded bad-sets)

definition *composition-fri-with-tables-opening-reduction-assumption*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
composition-fri-with-tables-opening-reduction-assumption s $\longleftrightarrow$
 $(\exists \text{ bad-sets.}$
composition-fri-with-tables-opening-algebraic-cover s bad-sets $\wedge$
composition-fri-bad-challenge-sets-bounded bad-sets)

lemma *trace-fri-with-tables-opening-reductionE:*
assumes *trace-fri-with-tables-opening-reduction-assumption s*
obtains *bad-sets*
where *trace-fri-with-tables-opening-algebraic-cover s bad-sets*
and *trace-fri-bad-challenge-sets-bounded bad-sets*
using *assms*
unfolding *trace-fri-with-tables-opening-reduction-assumption-def* **by** *blast*

lemma *composition-fri-with-tables-opening-reductionE:*
assumes *composition-fri-with-tables-opening-reduction-assumption s*
obtains *bad-sets*
where *composition-fri-with-tables-opening-algebraic-cover s bad-sets*


```

    and composition-fri-bad-challenge-sets-bounded bad-sets
using assms
unfolding composition-fri-with-tables-opening-reduction-assumption-def
by blast

lemma trace-fri-bad-with-tables-bound-from-opening-reduction:
  assumes reduction:
    trace-fri-with-tables-opening-reduction-assumption s
  and fresh: trace-fri-challenge-freshness-assumption s
  shows wp-event verify-monad (trace-fri-bad-with-tables s) s ≤
    trace-fri-error
proof -
  from reduction obtain bad-sets where cover:
    trace-fri-with-tables-opening-algebraic-cover s bad-sets
  and bounded: trace-fri-bad-challenge-sets-bounded bad-sets
  by (rule trace-fri-with-tables-opening-reductionE)
  show ?thesis
  by (rule trace-fri-bad-with-tables-bound-from-cover
    [OF cover bounded fresh])
qed

lemma composition-fri-bad-with-tables-bound-from-opening-reduction:
  assumes reduction:
    composition-fri-with-tables-opening-reduction-assumption s
  and fresh: composition-fri-challenge-freshness-assumption s
  shows wp-event verify-monad (composition-fri-bad-with-tables s) s ≤
    composition-fri-error
proof -
  from reduction obtain bad-sets where cover:
    composition-fri-with-tables-opening-algebraic-cover s bad-sets
  and bounded: composition-fri-bad-challenge-sets-bounded bad-sets
  by (rule composition-fri-with-tables-opening-reductionE)
  show ?thesis
  by (rule composition-fri-bad-with-tables-bound-from-cover
    [OF cover bounded fresh])
qed

end

end

theory Soundness-Reductions
  imports Soundness-Reductions-Opening
begin

Compatibility umbrella for soundness reduction lemmas.

end

```

```

theory Soundness-Staged-Query-Prefix
  imports Soundness-Reductions Staged-Security-Experiment
begin

```

4.6 Query and Partial-Opening Reductions

Staged query-prefix replay and prefix receive decomposition lemmas.

```

context soundness
begin

```

The deterministic partition above reduces soundness to four probability bounds. The bounds themselves require additional cryptographic/algebraic interfaces: random-linear-combination soundness, query-sampling freshness, and FRI low-degree soundness. Those assumptions are kept in the separate bounded-soundness locale below, rather than hidden as unproved lemmas in this base locale.

```

lemma soundness-bound-component-accounting:
  soundness-bound =
    composition-error-bound + trace-fri-error + composition-fri-error +
    nnreal rounds * query-error-bound
unfolding soundness-bound-def by simp

```

```

lemma soundness-bound-with-merkle-component-accounting:
  soundness-bound-with-merkle =
    merkle-binding-error + composition-error-bound + trace-fri-error +
    composition-fri-error + nnreal rounds * query-error-bound
unfolding soundness-bound-with-merkle-def soundness-bound-def
by (simp add: algebra-simps)

```

```

lemma wp-event-finite-union-bound:
  fixes B :: 'i  $\Rightarrow$  prob
  assumes finite: finite I
  and bounds:  $\bigwedge i. i \in I \implies \text{wp-event } m (E\ i) \ s \leq B\ i$ 
  shows
    wp-event m ( $\lambda out. \exists i \in I. E\ i\ out$ ) s  $\leq (\sum i \in I. B\ i)$ 
  using finite bounds
proof (induction I)
  case empty
  show ?case
  unfolding wp-event-def wp-def dist-expect-def by simp
next
  case (insert i I)
  have split:
    wp-event m ( $\lambda out. \exists j \in \text{insert } i\ I. E\ j\ out$ ) s  $\leq$ 
      wp-event m (E i) s +
      wp-event m ( $\lambda out. \exists j \in I. E\ j\ out$ ) s
  proof -
  have wp-event m ( $\lambda out. \exists j \in \text{insert } i\ I. E\ j\ out$ ) s  $\leq$ 

```

```

      wp-event m ( $\lambda out. E\ i\ out \vee (\exists j \in I. E\ j\ out)$ ) s
    by (rule wp-event-mono) auto
  also have ...  $\leq$ 
    wp-event m ( $E\ i$ ) s +
    wp-event m ( $\lambda out. \exists j \in I. E\ j\ out$ ) s
  by (rule wp-event-union-bound)
  finally show ?thesis .
qed
have tail:
  wp-event m ( $\lambda out. \exists j \in I. E\ j\ out$ ) s  $\leq$  ( $\sum j \in I. B\ j$ )
  by (rule insert.IH) (use insert.premis in blast)
have wp-event m ( $E\ i$ ) s +
  wp-event m ( $\lambda out. \exists j \in I. E\ j\ out$ ) s  $\leq$ 
  B i + ( $\sum j \in I. B\ j$ )
  by (intro add-mono tail insert.premis) simp
also have ... = ( $\sum j \in insert\ i\ I. B\ j$ )
  using insert.hyps by simp
finally show ?case
  by (rule order-trans[OF split])
qed

lemma hash-extends-verifier-state-from-adversary-right:
  assumes ext:  $s \leq t$ 
  shows  $s \leq \text{verifier-state-from-adversary}\ t\ tr$ 
  using ext
  unfolding verifier-state-from-adversary-def less-eq-hash-ext-def
    less-eq-fmap-def
  by simp

lemma checked-staged-security-with-data-state-supported-output-local-side-bad-bound-controlled:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        supported-output-local-side-bad)
    adversary-initial-state  $\leq$ 
    hash-collision-budget-value 0
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget + verifier-hash-query-budget)
  unfolding supported-output-local-side-bad-iff-hash-map-output-collision-bad
  by (rule
    checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
    [OF wf controlled])

lemma checked-staged-security-with-data-state-supported-pairwise-side-bad-bound-controlled:
  fixes coupling-error :: prob
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A

```

```

and coupling-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      supported-pairwise-coupling-event)
  adversary-initial-state ≤ coupling-error
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      supported-pairwise-side-bad)
  adversary-initial-state ≤
  hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget + verifier-hash-query-budget) +
    coupling-error
proof –
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  have side-eq:
    ?E supported-pairwise-side-bad =
      (λout. ?E hash-map-output-collision-bad out ∨
        ?E supported-pairwise-coupling-event out)
  unfolding staged-security-with-data-state-verifier-event-def
    supported-pairwise-side-bad-def supported-pairwise-coupling-event-def
  by (rule ext) (auto split: option.splits prod.splits)
  have wp-event ?M (?E supported-pairwise-side-bad)
    adversary-initial-state ≤
    wp-event ?M (?E hash-map-output-collision-bad)
      adversary-initial-state +
    wp-event ?M (?E supported-pairwise-coupling-event)
      adversary-initial-state
  unfolding side-eq
  by (rule wp-event-union-bound)
  also have ... ≤
    hash-collision-budget-value 0
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget + verifier-hash-query-budget) +
    coupling-error
  by (intro add-mono coupling-bound
    checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
    [OF wf controlled])
  finally show ?thesis .
qed

lemma checked-staged-accepted-imp-transcript-pre-or-new-hit:
  assumes support:
    out ∈
      set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)

```

```

    and acc: accepted out
  shows
    staged-security-with-data-state-transcript-pre-hit out  $\vee$ 
    staged-security-with-data-state-transcript-new-hit out
  proof (cases out)
  case None
  then show ?thesis
    using acc unfolding accepted-def by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?tr = staged-proof-transcript data
  let ?s = verifier-state-from-adversary attacker-state ?tr
  have verifier:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)
  using checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
  by blast
  have split:
    hash-map-output-values ?s  $\cap$  set ?tr  $\neq$  {}  $\vee$ 
    hash-map-new-output-hit (set ?tr) ?s final-state
  by (rule verify-monad-trace-root-target-preexisting-or-new[OF verifier])
    simp
  then show ?thesis
    unfolding out-eq
    staged-security-with-data-state-transcript-pre-hit-def
    staged-security-with-data-state-transcript-new-hit-def
  by simp
qed

lemma checked-staged-acceptance-transcript-hit-union-bound:
  wp-event (checked-staged-security-experiment-with-data-state A) accepted
    adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?pre = staged-security-with-data-state-transcript-pre-hit
  let ?new = staged-security-with-data-state-transcript-new-hit
  have wp-event ?M accepted adversary-initial-state  $\leq$ 
    wp-event ?M ( $\lambda$ out. ?pre out  $\vee$  ?new out) adversary-initial-state
  by (rule wp-event-mono-on-support)
    (rule checked-staged-accepted-imp-transcript-pre-or-new-hit)

```

also have ... $\leq$
 $wp\text{-}event\ ?M\ ?pre\ adversary\text{-}initial\text{-}state +$
 $wp\text{-}event\ ?M\ ?new\ adversary\text{-}initial\text{-}state$
by (rule *wp-event-union-bound*)
finally show *?thesis* .
qed

definition *staged-security-with-data-state-query-partial-header-key-hit*
 $:: (((f\ staged\text{-}proof\text{-}data \times f\ protocol\text{-}channel) \times unit\ list) \times$
 $f\ protocol\text{-}channel) option \Rightarrow bool$
where
 $staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}query\text{-}partial\text{-}header\text{-}key\text{-}hit\ out \longleftrightarrow$
 $(case\ out\ of$
 $None \Rightarrow False$
 $| Some\ (((data, attacker\text{-}state), -), final\text{-}state) \Rightarrow$
 $(let\ s =$
 $verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data);$
 $B =$
 $query\text{-}header\text{-}supported\text{-}partial\text{-}union\text{-}good\text{-}sets\ s$
 $query\text{-}sampling\text{-}success\text{-}space$
 $(staged\text{-}trace\text{-}root\ data)$
 $(staged\text{-}trace\text{-}fri\text{-}roots\ data)$
 $(staged\text{-}trace\text{-}final\ data)$
 $(staged\text{-}alphas\ data)$
 $(staged\text{-}degree\ data)$
 $(staged\text{-}composition\text{-}fri\text{-}roots\ data)$
 $(staged\text{-}composition\text{-}final\ data)$
 in
 $(\exists\ i\ raw.\ i < rounds \wedge$
 $fmlookup\ (HashMap\ final\text{-}state)$
 $(QueryIndexChallenge\ i$
 $(state\text{-}after\text{-}query\text{-}chunks$
 $(staged\text{-}query\text{-}start\text{-}hash\ data)$
 $(staged\text{-}query\text{-}chunks\ data)\ i)) =$
 $Some\ raw \wedge$
 $index\ (to\text{-}nat\ raw) \in B)))$

definition *staged-security-with-data-state-query-bad-hit-at*
 $:: nat \Rightarrow$
 $((f\ staged\text{-}proof\text{-}data \times f\ protocol\text{-}channel) \times unit\ list) \times$
 $f\ protocol\text{-}channel) option \Rightarrow bool$
where
 $staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}query\text{-}bad\text{-}hit\text{-}at\ i\ out \longleftrightarrow$
 $(case\ out\ of$
 $None \Rightarrow False$
 $| Some\ (((data, attacker\text{-}state), result), final\text{-}state) \Rightarrow$
 $(let\ s =$
 $verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$

```

      (staged-proof-transcript data)
    in
      Some (result, final-state) ∈ set-dist (execute verify-monad s) ∧
      query-index-round-set-hit-at s query-sampling-success-space i
      (Some (result, final-state))))

lemma staged-security-with-data-state-query-bad-hit-imp-round-hit:
  assumes bad: staged-security-with-data-state-query-bad-hit out
  shows ∃ i ∈ {..rounds}.
    staged-security-with-data-state-query-bad-hit-at i out
proof (cases out)
  case None
  then show ?thesis
    using bad unfolding staged-security-with-data-state-query-bad-hit-def
    by simp
next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from bad[unfolded out-eq
    staged-security-with-data-state-query-bad-hit-def Let-def]
  have verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
    and query-bad: query-bad ?s (Some (result, final-state))
    by simp-all
  from query-bad-imp-query-index-round-set-hit[OF query-bad]
  obtain i where i-bound: i < rounds
    and hit-at:
      query-index-round-set-hit-at ?s query-sampling-success-space i
      (Some (result, final-state))
    unfolding query-index-round-set-hit-def by blast
  have staged-security-with-data-state-query-bad-hit-at i out
    unfolding out-eq staged-security-with-data-state-query-bad-hit-at-def
    Let-def
    using verifier hit-at by simp
  then show ?thesis
    using i-bound by blast
qed

lemma checked-staged-security-with-data-state-query-bad-hit-bound-from-rounds:
  assumes round-bound:
    ∧ i. i < rounds ⇒
      wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-query-bad-hit-at i)
      adversary-initial-state ≤ C i

```

shows

wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-query-bad-hit
adversary-initial-state $\leq (\sum i < \text{rounds}. C i)$

proof –

have *wp-event (checked-staged-security-experiment-with-data-state A)*
staged-security-with-data-state-query-bad-hit
adversary-initial-state $\leq$
wp-event (checked-staged-security-experiment-with-data-state A)
 $(\lambda out. \exists i \in \{..<\text{rounds}\}.$
staged-security-with-data-state-query-bad-hit-at i out)
adversary-initial-state
by (rule *wp-event-mono*)
(rule *staged-security-with-data-state-query-bad-hit-imp-round-hit*)
also have $\dots \leq (\sum i < \text{rounds}. C i)$
by (rule *wp-event-finite-UN-bound*)
(simp-all add: *round-bound*)

finally show ?thesis .

qed

lemma *checked-staged-security-with-data-state-query-bad-imp-partial-header-key-hit-or-collision-on-support:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

out $\in$
set-dist
(execute
(checked-staged-security-experiment-with-data-state A)
adversary-initial-state)

and *bad: staged-security-with-data-state-query-bad-hit out*

shows

staged-security-with-data-state-query-partial-header-key-hit out $\vee$
staged-security-with-data-state-verifier-event
hash-map-output-collision-bad out

proof (cases *out*)

case *None*

then show ?thesis

using *bad unfolding staged-security-with-data-state-query-bad-hit-def*

by *simp*

next

case (Some *result-pack*)

then obtain *data attacker-state result final-state* **where** *out-eq:*

out = Some (((data, attacker-state), result), final-state)

by (cases *result-pack*, auto *split: prod.splits*)

let ?s =

verifier-state-from-adversary attacker-state
(staged-proof-transcript data)

have *support-some:*

Some (((data, attacker-state), result), final-state) $\in$


```

    set-dist
    (execute
      (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
  using support unfolding out-eq .
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support-some]
have builder:
  Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
and verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast+
have staged-header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
  [OF wf controlled builder])
from bad[unfolded out-eq
  staged-security-with-data-state-query-bad-hit-def Let-def]
have query-bad: query-bad ?s (Some (result, final-state))
by simp
from query-bad-imp-query-index-round-set-hit[OF query-bad]
obtain i trace-table composition-table as query-idxs where
  bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
and i-bound: i < rounds
and query-hit:
  query-idxs ! i ∈
    query-sampling-success-space trace-table composition-table as
unfolding query-index-round-set-hit-def
  query-index-round-set-hit-at-def
by blast
have shape:
  accepted-transcript-shape ?s (Some (result, final-state))
  as query-idxs
using bound
unfolding accepted-with-bound-tables-def accepted-with-tables-def
by simp

```

```

from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support-some shape]
obtain raw-idxs where
  as-eq: as = staged-alphas data
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
  and lookup:
     $\bigwedge j. j < \text{rounds} \implies$ 
      fmlookup (HashMap final-state)
        (QueryIndexChallenge j
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data j))) =
        Some (raw-idxs ! j)
  by blast
show ?thesis
proof (cases hash-map-output-collision final-state)
  case True
  then have
    staged-security-with-data-state-verifier-event
    hash-map-output-collision-bad
    (Some (((data, attacker-state), result), final-state))
  unfolding staged-security-with-data-state-verifier-event-def
    hash-map-output-collision-bad-def accepted-def
  by simp
  then show ?thesis
    unfolding out-eq by simp
next
  case clean: False
  have idx-sample:
    query-idxs ! i  $\in$  query-sample-space
  proof –
    have len-query: length query-idxs = rounds
      using query-idxs-eq len-raw by simp
    have query-idxs ! i  $\in$  set query-idxs
      by (rule nth-mem) (use i-bound len-query in simp)
    then show ?thesis
      by (rule accepted-with-bound-tables-query-sample-space[OF bound])
qed
from bound obtain fr f-fri-roots f-final dg composition-fri-roots final
  rest where
  header':
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and trace-bind:
    merkle-root-binds-table fr trace-table final-state
  and comp-nonempty: composition-fri-roots  $\neq$  []
  and comp-bind:

```

```

merkle-root-binds-table (hd composition-fri-roots) composition-table
  final-state
unfolding accepted-with-bound-tables-def by blast
have eqs:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  composition-fri-roots = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header' staged-header]
by simp
have state:
  query-header-supported-table-candidate-state ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    trace-table composition-table final-state
unfolding query-header-supported-table-candidate-state-def
by (intro exI[of - result] exI[of - query-idxs]
    exI[of - List.concat (staged-query-chunks data)]])
    (use verifier bound staged-header trace-bind comp-nonempty comp-bind
    eqs in simp)
have candidate:
  (trace-table, composition-table) ∈
    query-header-supported-partial-table-candidates ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
using
  query-header-supported-table-candidate-state-imp-partial-candidate-state
    [OF state clean]
  query-header-supported-partial-table-candidates-iff-state
by blast
have raw-hit:
  index (to-nat (raw-idxs ! i)) ∈
    query-header-supported-partial-union-good-sets ?s
      query-sampling-success-space
      (staged-trace-root data)

```

```

    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  proof -
    have raw-idx-eq: index (to-nat (raw-idxs ! i)) = query-idxs ! i
      using query-idxs-eq len-raw i-bound by simp
    show ?thesis
      unfolding query-header-supported-partial-union-good-sets-def
        raw-idx-eq
      using idx-sample candidate query-hit as-eq by blast
  qed
  have lookup-i:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
    by (rule lookup[OF i-bound])
  show ?thesis
    unfolding out-eq
      staged-security-with-data-state-query-partial-header-key-hit-def
      Let-def
    using i-bound lookup-i raw-hit by auto
  qed
qed

lemma checked-staged-security-with-data-state-query-bad-bound-from-partial-header-key-hit-or-collision:
  fixes C K :: prob
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and partial-header-key-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-header-key-hit
      adversary-initial-state  $\leq C$ 
  and collision-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        hash-map-output-collision-bad)
      adversary-initial-state  $\leq K$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-bad-hit
      adversary-initial-state  $\leq C + K$ 
  proof -
    let ?M = checked-staged-security-experiment-with-data-state A

```

```

have wp-event ?M staged-security-with-data-state-query-bad-hit
  adversary-initial-state  $\leq$ 
wp-event ?M
  ( $\lambda$ out.
    staged-security-with-data-state-query-partial-header-key-hit out  $\vee$ 
    staged-security-with-data-state-verifier-event
      hash-map-output-collision-bad out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out  $\in$  set-dist (execute ?M adversary-initial-state)
and bad: staged-security-with-data-state-query-bad-hit out
show
  staged-security-with-data-state-query-partial-header-key-hit out  $\vee$ 
  staged-security-with-data-state-verifier-event
    hash-map-output-collision-bad out
by (rule
  checked-staged-security-with-data-state-query-bad-imp-partial-header-key-hit-or-collision-on-support
    [OF wf controlled support bad])
qed
also have ...  $\leq$ 
  wp-event ?M staged-security-with-data-state-query-partial-header-key-hit
    adversary-initial-state +
  wp-event ?M
    (staged-security-with-data-state-verifier-event
      hash-map-output-collision-bad)
    adversary-initial-state
by (rule wp-event-union-bound)
also have ...  $\leq C + K$ 
by (intro add-mono partial-header-key-bound collision-bound)
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-partial-header-key-hit-or-collision
fixes C K :: prob
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and partial-header-key-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-header-key-hit
    adversary-initial-state  $\leq C$ 
and collision-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      hash-map-output-collision-bad)
    adversary-initial-state  $\leq K$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)

```

(staged-security-with-data-state-verifier-event query-bad)
 adversary-initial-state $\leq C + K$
proof –
have wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event query-bad)
 adversary-initial-state $\leq$
 wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-query-bad-hit
 adversary-initial-state
by (rule checked-staged-security-with-data-state-query-bad-verifier-event-bound
 [OF order-refl])
also have ... $\leq C + K$
by (rule
 checked-staged-security-with-data-state-query-bad-bound-from-partial-header-key-hit-or-collision
 [OF wf controlled partial-header-key-bound collision-bound])
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-partial-header-key-hit-bound-from-index-set:

assumes cover:

$\bigwedge$ data attacker-state.
 query-header-supported-partial-union-good-sets
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 query-sampling-success-space
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) $\subseteq B$

and index-bound:

wp-event (checked-staged-security-experiment-with-data A)
 (staged-security-with-data-query-index-set-hit B)
 adversary-initial-state $\leq C$

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-query-partial-header-key-hit
 adversary-initial-state $\leq C$

proof –

let ?projected =

$\lambda out. \text{case out of None} \Rightarrow$
 staged-security-with-data-query-index-set-hit B None
 | Some (((data, -), result), t) $\Rightarrow$
 staged-security-with-data-query-index-set-hit B
 (Some ((data, result), t))

have event-le:

wp-event (checked-staged-security-experiment-with-data-state A)

```

    staged-security-with-data-state-query-partial-header-key-hit
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
    ?projected adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume hit:
    staged-security-with-data-state-query-partial-header-key-hit out
  show ?projected out
  proof (cases out)
    case None
    then show ?thesis
      using hit
      unfolding
        staged-security-with-data-state-query-partial-header-key-hit-def
      by simp
  next
    case (Some packed)
    then obtain data attacker-state result final-state where out-eq:
      out = Some (((data, attacker-state), result), final-state)
      by (cases packed, auto split: prod.splits)
    let ?s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    let ?B =
      query-header-supported-partial-union-good-sets ?s
      query-sampling-success-space
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
    from hit[unfolded out-eq
      staged-security-with-data-state-query-partial-header-key-hit-def
      Let-def]
    obtain i raw where i-bound: i < rounds
    and lookup:
      fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
        Some raw
    and raw-hit: index (to-nat raw) ∈ ?B
    by auto
    have raw-in-B: index (to-nat raw) ∈ B
    proof –

```

```

    have ?B  $\subseteq$  B
      by (rule cover)
    then show ?thesis
      using raw-hit by blast
  qed
show ?thesis
  unfolding out-eq staged-security-with-data-query-index-set-hit-def
  using i-bound lookup raw-in-B by auto
qed
qed
also have
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?projected adversary-initial-state =
  wp-event (checked-staged-security-experiment-with-data A)
    (staged-security-with-data-query-index-set-hit B)
    adversary-initial-state
  using checked-staged-security-experiment-with-data-event-from-data-state
    [of A staged-security-with-data-query-index-set-hit B]
  by simp
also have ...  $\leq$  C
  by (rule index-bound)
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-partial-header-key-hit-bound-from-index-set-controlled:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and cover:
     $\bigwedge$  data attacker-state.
      query-header-supported-partial-union-good-sets
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
      query-sampling-success-space
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)  $\subseteq$  B
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-header-key-hit
      adversary-initial-state  $\leq$ 
      staged-phase-target-error (query-index-raw-preimage B)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  by (rule
    checked-staged-security-with-data-state-partial-header-key-hit-bound-from-index-set

```


[OF cover])
(rule checked-staged-security-with-data-query-index-set-hit-bound
[OF wf controlled])

lemma *checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-partial-fixed-query-error-cover:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *cover:*

$\bigwedge$ *data attacker-state.*

query-header-supported-partial-union-good-sets

(*verifier-state-from-adversary attacker-state*

(*staged-proof-transcript data*))

query-sampling-success-space

(*staged-trace-root data*)

(*staged-trace-fri-roots data*)

(*staged-trace-final data*)

(*staged-alphas data*)

(*staged-degree data*)

(*staged-composition-fri-roots data*)

(*staged-composition-final data*) $\subseteq B$

and *raw-bound: query-index-raw-preimage-bound*

and *subset: $B \subseteq \text{query-sample-space}$*

and *frac:*

$\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{query-sample-space}) \leq$

query-error-bound

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(*staged-security-with-data-state-verifier-event query-bad*)

adversary-initial-state $\leq$

nnreal

(*staged-attacker-query-budget budgets* +

staged-challenge-query-budget) * *query-error-bound* +

hash-collision-budget-value 0

(*staged-attacker-query-budget budgets* +

staged-challenge-query-budget + *verifier-hash-query-budget*)

proof –

have *partial-key-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-query-partial-header-key-hit

adversary-initial-state $\leq$

nnreal

(*staged-attacker-query-budget budgets* +

staged-challenge-query-budget) * *query-error-bound*

proof –

have *wp-event (checked-staged-security-experiment-with-data-state A)*

staged-security-with-data-state-query-partial-header-key-hit

adversary-initial-state $\leq$

staged-phase-target-error (query-index-raw-preimage B)

(*staged-attacker-query-budget budgets* +

$staged\text{-}challenge\text{-}query\text{-}budget$)
by (rule
 $checked\text{-}staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}partial\text{-}header\text{-}key\text{-}hit\text{-}bound\text{-}from\text{-}index\text{-}set\text{-}controlled$
 $[OF\text{ }wf\text{ }controlled\text{ }cover]$)
also have $\dots \leq$
 $nnreal$
 $(staged\text{-}attacker\text{-}query\text{-}budget\text{ }budgets +$
 $staged\text{-}challenge\text{-}query\text{-}budget) * query\text{-}error\text{-}bound$
by (rule $staged\text{-}phase\text{-}query\text{-}index\text{-}target\text{-}error\text{-}query\text{-}bound$
 $[OF\text{ }raw\text{-}bound\text{ }subset\text{ }frac]$)
finally show $?thesis$.
qed
have $collision\text{-}bound$:
 $wp\text{-}event\text{ } (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\text{ } A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $hash\text{-}map\text{-}output\text{-}collision\text{-}bad)$
 $adversary\text{-}initial\text{-}state \leq$
 $hash\text{-}collision\text{-}budget\text{-}value\text{ } 0$
 $(staged\text{-}attacker\text{-}query\text{-}budget\text{ } budgets +$
 $staged\text{-}challenge\text{-}query\text{-}budget + verifier\text{-}hash\text{-}query\text{-}budget)$
by (rule
 $checked\text{-}staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}hash\text{-}map\text{-}output\text{-}collision\text{-}bad\text{-}bound\text{-}controlled$
 $[OF\text{ }wf\text{ }controlled]$)
show $?thesis$
by (rule
 $checked\text{-}staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}query\text{-}bad\text{-}verifier\text{-}event\text{-}bound\text{-}from\text{-}partial\text{-}header\text{-}key\text{-}hit\text{-}or\text{-}collision$
 $[OF\text{ }wf\text{ }controlled\text{ }partial\text{-}key\text{-}bound\text{ }collision\text{-}bound]$)
qed

lemma $staged\text{-}dynamic\text{-}partial\text{-}query\text{-}set\text{-}subset$:
 $query\text{-}header\text{-}supported\text{-}partial\text{-}union\text{-}good\text{-}sets$
 $(verifier\text{-}state\text{-}from\text{-}adversary\text{ } attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\text{ } data))$
 $query\text{-}sampling\text{-}success\text{-}space$
 $(staged\text{-}trace\text{-}root\text{ } data)$
 $(staged\text{-}trace\text{-}fri\text{-}roots\text{ } data)$
 $(staged\text{-}trace\text{-}final\text{ } data)$
 $(staged\text{-}alphas\text{ } data)$
 $(staged\text{-}degree\text{ } data)$
 $(staged\text{-}composition\text{-}fri\text{-}roots\text{ } data)$
 $(staged\text{-}composition\text{-}final\text{ } data) \subseteq query\text{-}sample\text{-}space$
by (rule $query\text{-}header\text{-}supported\text{-}partial\text{-}union\text{-}good\text{-}sets\text{-}subset$)

lemma $staged\text{-}dynamic\text{-}partial\text{-}query\text{-}set\text{-}fraction\text{-}bound\text{-}if\text{-}unique\text{-}candidate$:
assumes $unique$:
 $\exists trace\text{-}table\text{ } composition\text{-}table.$
 $query\text{-}header\text{-}supported\text{-}partial\text{-}table\text{-}candidates$
 $(verifier\text{-}state\text{-}from\text{-}adversary\text{ } attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\text{ } data))$

(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data) $\subseteq$
{(trace-table, composition-table)}
shows
nnreal
(card
(query-header-supported-partial-union-good-sets
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
query-sampling-success-space
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data))) /
nnreal (card query-sample-space) $\leq$ query-error-bound
by *(rule*
query-header-supported-partial-union-good-sets-fraction-bound-if-unique-candidate
[OF unique query-sampling-success-space-subset
query-sampling-success-space-fraction-bound-query-sample-space])

lemma *staged-dynamic-partial-query-set-target-error-bound-if-unique-candidate:*

assumes *raw-bound: query-index-raw-preimage-bound*

and *unique:*

$\exists$ trace-table composition-table.
query-header-supported-partial-table-candidates
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data) $\subseteq$
{(trace-table, composition-table)}

shows

staged-phase-target-error
(query-index-raw-preimage
(query-header-supported-partial-union-good-sets
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))

query-sampling-success-space
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data))) $q \leq$
 $\text{nnreal } q * \text{query-error-bound}$
by (*rule staged-phase-query-index-target-error-query-bound*
[OF raw-bound staged-dynamic-partial-query-set-subset
staged-dynamic-partial-query-set-fraction-bound-if-unique-candidate
[OF unique]])

lemma *checked-staged-query-prefix-recv-bad-index-imp-new-output-hit:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i \leq \text{rounds}$*
and *absent: query-index-raw-preimage-absent B s*
and *out:*
Some (raw, t) ∈
set-dist
(execute
(checked-staged-query-challenge-prefix-program A i $\gg$
*(λ -. *recv-query-index-challenge*)) s)*
and *bad: index (to-nat raw) ∈ B*
shows
hash-new-output-hit-event (query-index-raw-preimage B) s
(Some (raw, t))

proof –
obtain *prefix-out u where prefix:*
Some (prefix-out, u) ∈
set-dist
(execute (checked-staged-query-challenge-prefix-program A i) s)
and *recv:*
*Some (raw, t) ∈ set-dist (execute *recv-query-index-challenge* u)*
using *out by (auto elim!: set-dist-bindE)*
have *unchecked-prefix:*
Some
((spp-degree prefix-out, spp-composition-final prefix-out,
spp-query-chunks prefix-out), u) ∈
set-dist (execute (staged-query-challenge-prefix-program A i) s)
by (*rule checked-staged-query-challenge-prefix-outcome-imp-staged-query-prefix*
[OF prefix])
have *prefix-ext: $s \leq u$*
using *hash-target-program-staged-query-challenge-prefix-program*
[OF wf controlled i-bound, of query-index-raw-preimage B]
unchecked-prefix
unfolding *hash-target-program-def hash-extension-preserving-def*

```

    by blast
have recv-out:
  u ≤ t ∧
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter u) (PState u)) = Some raw
  using receive-query-index-challenge-outcome[OF recv] by blast
have raw-in: raw ∈ query-index-raw-preimage B
  using bad unfolding query-index-raw-preimage-def by simp
show ?thesis
proof (cases
  fmlookup (HashMap s)
    (QueryIndexChallenge (PQueryCounter u) (PState u)))
case None
then have hash-map-new-output-hit (query-index-raw-preimage B) s t
  unfolding hash-map-new-output-hit-def
  using recv-out raw-in by blast
then show ?thesis
  unfolding hash-new-output-hit-event-def by simp
next
case (Some old-raw)
have old-lookup-t:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter u) (PState u)) =
    Some old-raw
  by (rule hash-extension-lookup[OF Some])
  (rule hash-ext-trans[OF prefix-ext conjunct1[OF recv-out]])
then have old-eq: old-raw = raw
  using recv-out by simp
have old-raw ∉ query-index-raw-preimage B
  using absent Some
  unfolding query-index-raw-preimage-absent-def by blast
then show ?thesis
  using old-eq raw-in by simp
qed
qed

```

definition *checked-staged-query-prefix-with-state*

```

:: 'f staged-adversary ⇒ nat ⇒
  (('f staged-query-prefix-data × 'f protocol-channel),
   'f protocol-channel) state-monad

```

where

```

checked-staged-query-prefix-with-state A i =
  checked-staged-query-challenge-prefix-program A i ≫=
  (λprefix. get ≫= (λprefix-state.
    return (prefix, prefix-state)))

```

definition *checked-staged-query-prefix-receive-with-state*

```

:: 'f staged-adversary ⇒ nat ⇒
  (((f staged-query-prefix-data × 'f protocol-channel) × 'f),

```

```

    'f protocol-channel) state-monad
  where
    checked-staged-query-prefix-receive-with-state A i =
      checked-staged-query-prefix-with-state A i >>=
        (λprefix-pack. receive-query-index-challenge >>= (λraw.
          return (prefix-pack, raw)))

lemma checked-staged-query-program-decomp:
  assumes i-bound: i < n
  shows
    checked-staged-query-program A trace-roots composition-roots start n =
      checked-staged-query-program A trace-roots composition-roots start i >>=
        (λprefix-chunks.
          receive-query-index-challenge >>= (λraw.
            let idx = index (to-nat raw) in
            query-opening-stage A (start + i) raw >>= (λchunk.
              assert
                (verifier-query-round-chunk idx trace-roots composition-roots
                  chunk) >>= (λ-.
                    record-staged-messages chunk >>= (λ-.
                      checked-staged-query-program A trace-roots composition-roots
                        (Suc (start + i)) (n - Suc i) >>= (λsuffix-chunks.
                          return (prefix-chunks @ chunk # suffix-chunks)))))))
          using i-bound
proof (induction i arbitrary: start n)
  case 0
  then obtain m where n-eq: n = Suc m
  by (cases n) simp-all
  show ?case
  unfolding n-eq
  by (simp add: sm-bind-assoc Let-def)
next
  case (Suc i)
  then obtain m where n-eq: n = Suc m
  by (cases n) simp-all
  have i-lt-m: i < m
  using Suc.premis unfolding n-eq by simp
  have IH:
    checked-staged-query-program A trace-roots composition-roots
      (Suc start) m =
      checked-staged-query-program A trace-roots composition-roots
        (Suc start) i >>=
        (λprefix-chunks.
          receive-query-index-challenge >>= (λraw.
            let idx = index (to-nat raw) in
            query-opening-stage A (Suc start + i) raw >>= (λchunk.
              assert
                (verifier-query-round-chunk idx trace-roots composition-roots
                  chunk) >>= (λ-.

```

```

    record-staged-messages chunk  $\gg$  ( $\lambda$ -.
    checked-staged-query-program A trace-roots composition-roots
      (Suc (Suc start + i)) (m - Suc i)  $\gg$  ( $\lambda$ suffix-chunks.
      return (prefix-chunks @ chunk # suffix-chunks))))))
  by (rule Suc.IH[OF i-lt-m])
show ?case
  unfolding n-eq
  by (simp add: sm-bind-assoc Let-def IH)
qed

lemma get-bind-const:
  (get  $\gg$  ( $\lambda$ -. m)) = m
proof transfer
  fix m
  show dist-bind dist-get ( $\lambda$ -. m) = m
proof
  fix s
  show dist-bind dist-get ( $\lambda$ -. m) s = m s
  apply (rule dist-inject[THEN iffD1])
  unfolding dist-bind.rep-eq dist-get-def bind-cont-map-def
  apply (simp add: o-def map-fun-def dist-delta-dist)
  apply (subst map-bind-delta-left-state[where f= $\lambda$ s. Some (s, s)])
  by (auto split: option.splits prod.splits)
qed
qed

lemma get-bind-return-state-support:
  Some ((x, s), s)  $\in$ 
  set-dist (execute (get  $\gg$  ( $\lambda$ t. return (x, t))) s)
proof -
  have get-out: Some (s, s)  $\in$  set-dist (execute get s)
  unfolding set-dist-def get.rep-eq dist-get-def dist-delta-dist
    delta-map-def
  by simp
  have ret-out:
    Some ((x, s), s)  $\in$  set-dist (execute (return (x, s)) s)
  unfolding set-dist-def return.rep-eq dist-return-def dist-delta-dist
    delta-map-def
  by simp
  show ?thesis
  by (rule-tac x=s and t=s in set-dist-bindI)
    (rule get-out, rule ret-out)
qed

lemma get-bind-return-state-outcome:
  assumes outcome:
    Some ((x-out, captured), final)  $\in$ 
    set-dist (execute (get  $\gg$  ( $\lambda$ t. return (x, t))) s)
  shows x-out = x  $\wedge$  captured = s  $\wedge$  final = s

```

proof –
from *outcome* **obtain** $y\ u$ **where** *get-out*:
 Some $((y, u) \in \text{set-dist } (\text{execute } \text{get } s))$
and *ret-out*:
 Some $((x\text{-out}, \text{captured}), \text{final}) \in$
 $\text{set-dist } (\text{execute } (\text{return } (x, y))\ u)$
by $(\text{auto elim!}:\ \text{set-dist-bindE})$
have $y\text{-}u$: $y = s \wedge u = s$
using *get-out*
unfolding *set-dist-def get.rep-eq dist-get-def dist-delta-dist*
 delta-map-def
by *auto*
have *captured-final*: $x\text{-out} = x \wedge \text{captured} = y \wedge \text{final} = u$
using *ret-out*
unfolding *set-dist-def return.rep-eq dist-return-def dist-delta-dist*
 delta-map-def
by *auto*
show *?thesis*
using $y\text{-}u$ *captured-final* **by** *simp*
qed

lemma *checked-staged-query-prefix-with-state-outcome-state*:
assumes *outcome*:
 Some $((\text{prefix}, \text{prefix-state}), u) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-query-prefix-with-state } A\ i)\ s)$
shows $u = \text{prefix-state}$
proof –
show *?thesis*
using *outcome*
unfolding *checked-staged-query-prefix-with-state-def*
by $(\text{auto dest!}:\ \text{get-bind-return-state-outcome}$
 $\text{elim!}:\ \text{set-dist-bindE})$
qed

lemma *checked-staged-query-prefix-with-state-outcome-canonical*:
assumes *outcome*:
 Some $((\text{prefix}, \text{prefix-state}), u) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-query-prefix-with-state } A\ i)\ s)$
shows
 Some $((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-query-prefix-with-state } A\ i)\ s)$
using *outcome* *checked-staged-query-prefix-with-state-outcome-state*[*OF outcome*]
by *simp*

lemma *checked-staged-query-prefix-recv-with-state-prefix-support*:
assumes *outcome*:
 Some $((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-recv-with-state } A\ i)\ s)$

shows
Some ((prefix, prefix-state), prefix-state) ∈
set-dist (execute (checked-staged-query-prefix-with-state A i) s)
proof –
from outcome obtain u where prefix-u:
Some ((prefix, prefix-state), u) ∈
set-dist (execute (checked-staged-query-prefix-with-state A i) s)
unfolding *checked-staged-query-prefix-receive-with-state-def*
by (*auto elim!:* *set-dist-bindE*)
show *?thesis*
by (*rule checked-staged-query-prefix-with-state-outcome-canonical*
[OF prefix-u])
qed

lemma *checked-staged-query-prefix-receive-with-state-receive-support:*
assumes *outcome:*
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i) s)
shows
Some (raw, raw-state) ∈
set-dist (execute receive-query-index-challenge prefix-state)
proof –
from outcome obtain u where prefix-u:
Some ((prefix, prefix-state), u) ∈
set-dist (execute (checked-staged-query-prefix-with-state A i) s)
and *recv-u:*
Some (raw, raw-state) ∈
set-dist (execute receive-query-index-challenge u)
unfolding *checked-staged-query-prefix-receive-with-state-def*
by (*auto elim!:* *set-dist-bindE*)
have *u = prefix-state*
by (*rule checked-staged-query-prefix-with-state-outcome-state*
[OF prefix-u])
then show *?thesis*
using *recv-u* **by** *simp*
qed

lemma *checked-staged-query-challenge-prefix-program-outcomeE:*
assumes *outcome:*
Some (prefix, prefix-state) ∈
set-dist
(execute (checked-staged-query-challenge-prefix-program A i)
adversary-initial-state)
obtains *fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5 as s6 dg*
s7 s8 s9 composition-roots composition-bs s10 composition-final s11
query-start query-chunks **where**
Some (fr, s1) ∈
set-dist (execute (trace-root-stage A) adversary-initial-state)

```

Some ((), s2) ∈ set-dist (execute (record-staged-message fr) s1)
Some ((trace-roots, trace-bs), s3) ∈
  set-dist
    (execute
      (staged-trace-fri-program A 0 (ceil-log clength) []) s2)
Some (trace-final, s4) ∈
  set-dist (execute (trace-final-stage A trace-bs) s3)
Some ((), s5) ∈
  set-dist (execute (record-staged-message trace-final) s4)
Some (as, s6) ∈
  set-dist (execute (staged-alpha-program (length spec)) s5)
Some (dg, s7) ∈ set-dist (execute (degree-stage A as) s6)
Some ((), s8) ∈
  set-dist (execute (record-staged-message dg) s7)
Some ((), s9) ∈
  set-dist
    (execute
      (assert (ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)))
      s8)
Some ((composition-roots, composition-bs), s10) ∈
  set-dist
    (execute
      (staged-composition-fri-program A dg 0
        (ceil-log (to-nat dg + 1)) []) s9)
Some (composition-final, s11) ∈
  set-dist
    (execute (composition-final-stage A dg composition-bs) s10)
Some ((), query-start) ∈
  set-dist (execute (record-staged-message composition-final) s11)
Some (query-chunks, prefix-state) ∈
  set-dist
    (execute
      (checked-staged-query-program A trace-roots composition-roots
        0 i)
      query-start)
prefix =
  (|sqp-trace-root = fr,
   sqp-trace-fri-roots = trace-roots,
   sqp-trace-fri-challenges = trace-bs,
   sqp-trace-final = trace-final,
   sqp-alphas = as,
   sqp-degree = dg,
   sqp-composition-fri-roots = composition-roots,
   sqp-composition-fri-challenges = composition-bs,
   sqp-composition-final = composition-final,
   sqp-query-chunks = query-chunks|)
using outcome
unfolding checked-staged-query-challenge-prefix-program-def
  staged-alpha-prefix-program-def Let-def

```

by (auto elim!: set-dist-bindE intro: that split: prod.splits)

definition *checked-staged-after-query-prefix-receive*

```
:: 'f staged-adversary  $\Rightarrow$  nat  $\Rightarrow$ 
  (('f staged-query-prefix-data  $\times$  'f protocol-channel)  $\times$  'f)  $\Rightarrow$ 
  ('f staged-proof-data, 'f protocol-channel) state-monad
```

where

```
checked-staged-after-query-prefix-receive A i prefix-pack =
  (case prefix-pack of ((prefix, -), raw)  $\Rightarrow$ 
    do {
      let idx = index (to-nat raw);
      chunk  $\leftarrow$  query-opening-stage A i raw;
      assert
        (verifier-query-round-chunk idx
          (sqp-trace-fri-roots prefix)
          (sqp-composition-fri-roots prefix) chunk);
      record-staged-messages chunk;
      suffix-chunks  $\leftarrow$ 
        checked-staged-query-program A
          (sqp-trace-fri-roots prefix)
          (sqp-composition-fri-roots prefix)
          (Suc i) (rounds - Suc i);
      return
        (|staged-trace-root = sqp-trace-root prefix,
          staged-trace-fri-roots = sqp-trace-fri-roots prefix,
          staged-trace-fri-challenges =
            sqp-trace-fri-challenges prefix,
          staged-trace-final = sqp-trace-final prefix,
          staged-alphas = sqp-alphas prefix,
          staged-degree = sqp-degree prefix,
          staged-composition-fri-roots =
            sqp-composition-fri-roots prefix,
          staged-composition-fri-challenges =
            sqp-composition-fri-challenges prefix,
          staged-composition-final = sqp-composition-final prefix,
          staged-query-chunks =
            sqp-query-chunks prefix @ chunk # suffix-chunks|)
    })
```

definition *checked-staged-after-query-prefix-receive-with-verifier*

```
:: 'f staged-adversary  $\Rightarrow$  nat  $\Rightarrow$ 
  (('f staged-query-prefix-data  $\times$  'f protocol-channel)  $\times$  'f)  $\Rightarrow$ 
  (((('f staged-proof-data  $\times$  'f protocol-channel)  $\times$  unit list),
    'f protocol-channel) state-monad
```

where

```
checked-staged-after-query-prefix-receive-with-verifier A i prefix-pack =
  checked-staged-after-query-prefix-receive A i prefix-pack  $\gg=$ 
    ( $\lambda$ data.
      get  $\gg=$  ( $\lambda$ s.
```

```

    put
      (verifier-state-from-adversary s
       (staged-proof-transcript data)) >>=
    (λ-.
      verify-monad >>=
        (λresult. return ((data, s), result))))

```

lemma *checked-staged-after-query-prefix-receive-with-verifier-outcome:*

assumes *outcome:*

```

  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive-with-verifier A i
         prefix-pack)
        s)

```

shows

```

  Some (data, attacker-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive A i prefix-pack) s)
  Some (result, final-state) ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
         (staged-proof-transcript data)))

```

proof –

from *outcome[unfolded*

checked-staged-after-query-prefix-receive-with-verifier-def]

obtain *data' attacker-state' where data':*

```

  Some (data', attacker-state') ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive A i prefix-pack) s)

```

and *rest:*

```

  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (get >>=
          (λsa.
            put
              (verifier-state-from-adversary sa
               (staged-proof-transcript data')) >>=
            (λ-.
              verify-monad >>=
                (λr. return ((data', sa), r))))
          attacker-state'))

```

by (*rule set-dist-bindE*) (*rule that*)

from *rest obtain saved-state state-after-get where get-out:*

```

  Some (saved-state, state-after-get) ∈

```

```

    set-dist (execute get attacker-state')
and after-get:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (put
          (verifier-state-from-adversary saved-state
            (staged-proof-transcript data')) >>=
            (λ-.
              verify-monad >>=
                (λr. return ((data', saved-state), r))))
          state-after-get)
        by (rule set-dist-bindE) (rule that)
from get-out have saved-state-eq: saved-state = attacker-state'
    and state-after-get-eq: state-after-get = attacker-state'
    by simp-all
from after-get[unfolded saved-state-eq state-after-get-eq]
obtain u verifier-start where put-out:
  Some (u, verifier-start) ∈
    set-dist
      (execute
        (put
          (verifier-state-from-adversary attacker-state'
            (staged-proof-transcript data')))
          attacker-state')
    and after-put:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (verify-monad >>=
          (λr. return ((data', attacker-state'), r)))
          verifier-start)
        by (rule set-dist-bindE) (rule that)
from put-out have verifier-start-eq:
  verifier-start =
    verifier-state-from-adversary attacker-state'
    (staged-proof-transcript data')
    by simp
from after-put[unfolded verifier-start-eq]
obtain result' final-state' where verifier:
  Some (result', final-state') ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state'
          (staged-proof-transcript data')))
    and returned:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute (return ((data', attacker-state'), result')))

```

```

      final-state')
    by (elim set-dist-bindE)
  from returned have data-eq: data = data'
  and attacker-state-eq: attacker-state = attacker-state'
  and result-eq: result = result'
  and final-state-eq: final-state = final-state'
  by simp-all
show
  Some (data, attacker-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive A i prefix-pack) s)
  using data' unfolding data-eq attacker-state-eq by simp
show
  Some (result, final-state) ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
  using verifier
  unfolding data-eq attacker-state-eq result-eq final-state-eq
  by simp
qed

```

lemma *checked-staged-security-experiment-with-data-state-outcomeI*:

assumes *builder*:

```

  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)

```

and *verifier*:

```

  Some (result, final-state) ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))

```

shows

```

  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)

```

unfolding *checked-staged-security-experiment-with-data-state-def*

apply (rule-tac *x=data and t=attacker-state in set-dist-bindI*)

apply (rule *builder*)

apply (rule-tac *x=attacker-state and t=attacker-state in set-dist-bindI*)

apply *simp*

apply (rule-tac *x=()* **and**

t=verifier-state-from-adversary attacker-state

```

    (staged-proof-transcript data) in set-dist-bindI)
  apply simp
  apply (rule-tac x=result and t=final-state in set-dist-bindI)
  apply (rule verifier)
  apply simp
done

lemma checked-staged-transcript-program-query-prefix-receive-decomp:
  assumes i-bound: i < rounds
  shows
    checked-staged-transcript-program A =
      checked-staged-query-prefix-receive-with-state A i  $\gg$ 
        checked-staged-after-query-prefix-receive A i
  proof -
    have query-decomp:
       $\bigwedge$  trace-roots composition-roots.
      checked-staged-query-program A trace-roots composition-roots 0 rounds =
        checked-staged-query-program A trace-roots composition-roots 0 i  $\gg$ 
          ( $\lambda$  prefix-chunks.
            receive-query-index-challenge  $\gg$  ( $\lambda$  raw.
              let idx = index (to-nat raw) in
              query-opening-stage A (0 + i) raw  $\gg$  ( $\lambda$  chunk.
                assert
                  (verifier-query-round-chunk idx trace-roots composition-roots
                    chunk)  $\gg$  ( $\lambda$ -.
                    record-staged-messages chunk  $\gg$  ( $\lambda$ -.
                      checked-staged-query-program A trace-roots composition-roots
                        (Suc (0 + i)) (rounds - Suc i)  $\gg$  ( $\lambda$  suffix-chunks.
                          return (prefix-chunks @ chunk # suffix-chunks)))))))
            by (rule checked-staged-query-program-decomp[OF i-bound])
          )
    show ?thesis
    unfolding checked-staged-transcript-program-def
      checked-staged-query-prefix-receive-with-state-def
      checked-staged-query-prefix-with-state-def
      checked-staged-query-challenge-prefix-program-def
      staged-alpha-prefix-program-def
      checked-staged-after-query-prefix-receive-def
    apply (simp add: sm-bind-assoc Let-def split-def query-decomp
      get-bind-const)
    done
  qed

```

```

lemma checked-staged-transcript-program-query-prefix-receive-cont-support:
  assumes i-bound: i < rounds
  and outcome:
    Some (data, attacker-state)  $\in$ 
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)

```

obtains *prefix prefix-state raw raw-state* **where**
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
Some (data, attacker-state) ∈
set-dist
(execute
(checked-staged-after-query-prefix-receive A i
((prefix, prefix-state), raw))
raw-state)
proof –
have *decomp*:
checked-staged-transcript-program A =
checked-staged-query-prefix-receive-with-state A i >>=
checked-staged-after-query-prefix-receive A i
by (*rule checked-staged-transcript-program-query-prefix-receive-decomp*
[OF i-bound])
from *outcome[unfolded decomp]*
obtain *x raw-state* **where** *head*:
Some (x, raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *tail*:
Some (data, attacker-state) ∈
set-dist
(execute
(checked-staged-after-query-prefix-receive A i x)
raw-state)
by (*auto elim!:* *set-dist-bindE*)
obtain *prefix-pack raw* **where** *x-eq: x = (prefix-pack, raw)*
by (*cases x*) *simp*
obtain *prefix prefix-state* **where** *prefix-pack-eq*:
prefix-pack = (prefix, prefix-state)
by (*cases prefix-pack*) *simp*
show *?thesis*
proof (*rule that[of prefix prefix-state raw raw-state]*)
show *Some (((prefix, prefix-state), raw), raw-state) ∈*
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
using *head* **unfolding** *x-eq prefix-pack-eq* **by** *simp*
show *Some (data, attacker-state) ∈*
set-dist
(execute
(checked-staged-after-query-prefix-receive A i
((prefix, prefix-state), raw))
raw-state)

using *tail unfolding* *x-eq prefix-pack-eq* by *simp*
 qed
 qed
lemma *checked-staged-security-experiment-with-data-state-query-prefix-receive-decomp*:
 assumes *i-bound*: $i < \text{rounds}$
 shows
 checked-staged-security-experiment-with-data-state *A* =
 checked-staged-query-prefix-receive-with-state *A* *i* $\gg$
 checked-staged-after-query-prefix-receive-with-verifier *A* *i*
unfolding *checked-staged-security-experiment-with-data-state-def*
 checked-staged-transcript-program-query-prefix-receive-decomp[*OF i-bound*]
 checked-staged-after-query-prefix-receive-with-verifier-def
by (*simp add: sm-bind-assoc*)

definition *staged-query-prefix-start-hash*
 :: 'f staged-query-prefix-data $\Rightarrow$ 'f
where
 staged-query-prefix-start-hash *prefix* =
 concat
 (foldl concat
 (concat
 (foldl concat
 (concat
 (foldl concat
 (concat 0 (*sqp-trace-root* *prefix*))
 (*sqp-trace-fri-roots* *prefix*))
 (*sqp-trace-final* *prefix*))
 (*sqp-alphas* *prefix*))
 (*sqp-degree* *prefix*))
 (*sqp-composition-fri-roots* *prefix*))
 (*sqp-composition-final* *prefix*))

lemma *checked-staged-query-prefix-with-state-query-chunks-length*:
 assumes *wf*: *staged-budget-wellformed* *budgets*
 and *controlled*: *staged-adversary-controlled* *budgets* *A*
 and *i-bound*: $i < \text{rounds}$
 and *support*:
 Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
 set-dist
 (*execute* (*checked-staged-query-prefix-with-state* *A* *i*)
 adversary-initial-state)
shows *length* (*sqp-query-chunks* *prefix*) = *i*
proof –
from *support* **have** *prefix-out*:
 Some (*prefix*, *prefix-state*) $\in$
 set-dist
 (*execute* (*checked-staged-query-challenge-prefix-program* *A* *i*)
 adversary-initial-state)

```

unfolding checked-staged-query-prefix-with-state-def
by (auto elim!: set-dist-bindE)
from prefix-out obtain fr s1 s2 trace-roots trace-bs s3 trace-final s4
s5 as s6 dg s7 s8 s9 composition-roots composition-bs s10
composition-final s11 query-start query-chunks where
query-out:
Some (query-chunks, prefix-state) ∈
set-dist
(execute
(checked-staged-query-program A trace-roots composition-roots
0 i) s11)
and prefix-eq:
prefix =
(|spp-trace-root = fr,
spp-trace-fri-roots = trace-roots,
spp-trace-fri-challenges = trace-bs,
spp-trace-final = trace-final,
spp-alphas = as,
spp-degree = dg,
spp-composition-fri-roots = composition-roots,
spp-composition-fri-challenges = composition-bs,
spp-composition-final = composition-final,
spp-query-chunks = query-chunks|)
using prefix-out
by (elim checked-staged-query-challenge-prefix-program-outcomeE)
have bound: 0 + i ≤ length (query-opening-budgets budgets)
using wf i-bound unfolding staged-budget-wellformed-def by simp
from checked-staged-query-program-outcome-with-raws
[OF controlled bound query-out]
obtain raw-idxs query-idxs where len: length query-chunks = i
by blast
then show ?thesis
unfolding prefix-eq by simp
qed

```

```

lemma checked-staged-query-prefix-with-state-alignment:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and support:
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state)
shows
PQueryCounter prefix-state = i ∧
PState prefix-state =
state-after-query-chunks
(staged-query-prefix-start-hash prefix)

```

```

      (spp-query-chunks prefix) i ∧
      length (spp-query-chunks prefix) = i
proof –
from support have prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
      (execute (checked-staged-query-challenge-prefix-program A i)
        adversary-initial-state)
  unfolding checked-staged-query-prefix-with-state-def
  by (auto elim!: set-dist-bindE)
from prefix-out obtain fr s1 s2 trace-roots trace-bs s3 trace-final s4
  s5 as s6 dg s7 s8 s9 composition-roots composition-bs s10
  composition-final s11 query-start query-chunks where
  root-out:
    Some (fr, s1) ∈
      set-dist (execute (trace-root-stage A) adversary-initial-state)
  and record-root:
    Some ((), s2) ∈ set-dist (execute (record-staged-message fr) s1)
  and trace-out:
    Some ((trace-roots, trace-bs), s3) ∈
      set-dist
        (execute
          (staged-trace-fri-program A 0 (ceil-log clength) []) s2)
  and final-out:
    Some (trace-final, s4) ∈
      set-dist (execute (trace-final-stage A trace-bs) s3)
  and record-final:
    Some ((), s5) ∈
      set-dist (execute (record-staged-message trace-final) s4)
  and alpha-out:
    Some (as, s6) ∈
      set-dist (execute (staged-alpha-program (length spec)) s5)
  and degree-out:
    Some (dg, s7) ∈ set-dist (execute (degree-stage A as) s6)
  and record-degree:
    Some ((), s8) ∈ set-dist (execute (record-staged-message dg) s7)
  and assert-out:
    Some ((), s9) ∈
      set-dist
        (execute
          (assert
            (ceil-log (to-nat dg + 1) ≤
              ceil-log (maxDegree + 1))) s8)
  and composition-out:
    Some ((composition-roots, composition-bs), s10) ∈
      set-dist
        (execute
          (staged-composition-fri-program A dg 0
            (ceil-log (to-nat dg + 1)) []) s9)

```

```

and composition-final-out:
  Some (composition-final, s11) ∈
    set-dist
      (execute (composition-final-stage A dg composition-bs) s10)
and record-composition-final:
  Some ((), query-start) ∈
    set-dist (execute (record-staged-message composition-final) s11)
and query-out:
  Some (query-chunks, prefix-state) ∈
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          0 i)
        query-start)
and prefix-eq:
  prefix =
    (sqp-trace-root = fr,
     sqp-trace-fri-roots = trace-roots,
     sqp-trace-fri-challenges = trace-bs,
     sqp-trace-final = trace-final,
     sqp-alphas = as,
     sqp-degree = dg,
     sqp-composition-fri-roots = composition-roots,
     sqp-composition-fri-challenges = composition-bs,
     sqp-composition-final = composition-final,
     sqp-query-chunks = query-chunks)
using prefix-out
by (elim checked-staged-query-challenge-prefix-program-outcomeE)
have root-controlled:
  controlled-ro-program (trace-root-budget budgets) (trace-root-stage A)
  using controlled unfolding staged-adversary-controlled-def by simp
have trace-final-controlled:
  controlled-ro-program (trace-final-budget budgets)
    (trace-final-stage A trace-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have degree-controlled:
  controlled-ro-program (degree-budget budgets) (degree-stage A as)
  using controlled unfolding staged-adversary-controlled-def by blast
have composition-final-controlled:
  controlled-ro-program (composition-final-budget budgets)
    (composition-final-stage A dg composition-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have root-state: PState s1 = 0
  using controlled-stage-outcome-fields[OF root-controlled root-out]
  by simp
have s2-eq:
  s2 = s1 (
    PState := concat (PState s1) fr,
    PTranscript := PTranscript s1 @ [fr])

```

```

  by (rule record-staged-message-outcome[OF record-root])
have trace-bound:
  0 + ceil-log clength ≤ length (trace-fri-budgets budgets)
  using wf unfolding staged-budget-wellformed-def by simp
have trace-alignment:
  PState s3 = foldl concat (PState s2) trace-roots
  using staged-trace-fri-program-alignment
    [OF controlled trace-bound trace-out]
  by simp
have state-s3:
  PState s3 = foldl concat (concat 0 fr) trace-roots
  using root-state trace-alignment unfolding s2-eq by simp
have state-s4: PState s4 = PState s3
  using controlled-stage-outcome-fields
    [OF trace-final-controlled final-out]
  by simp
have s5-eq:
  s5 = s4(|
    PState := concat (PState s4) trace-final,
    PTranscript := PTranscript s4 @ [trace-final]|)
  by (rule record-staged-message-outcome[OF record-final])
have alpha-alignment:
  PState s6 = foldl concat (PState s5) as
  using staged-alpha-program-alignment[OF alpha-out] by simp
have state-s6:
  PState s6 =
    foldl concat
      (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
    as
  using state-s3 state-s4 alpha-alignment unfolding s5-eq by simp
have state-s7: PState s7 = PState s6
  using controlled-stage-outcome-fields[OF degree-controlled degree-out]
  by simp
have s8-eq:
  s8 = s7(|
    PState := concat (PState s7) dg,
    PTranscript := PTranscript s7 @ [dg]|)
  by (rule record-staged-message-outcome[OF record-degree])
have s9-eq: s9 = s8
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using assert-out unfolding assert-def
  by (cases
    ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
    (auto simp: throw-no-outcome)

```

```

have composition-bound:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
  using wf round-bound unfolding staged-budget-wellformed-def by simp
have composition-alignment:
  PState s10 = foldl concat (PState s9) composition-roots
  using staged-composition-fri-program-alignment
    [OF controlled composition-bound composition-out]
  by simp
have state-s10:
  PState s10 =
    foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
          as)
        dg)
      composition-roots
  using state-s6 state-s7 composition-alignment
  unfolding s8-eq s9-eq by simp
have state-s11: PState s11 = PState s10
  using controlled-stage-outcome-fields
    [OF composition-final-controlled composition-final-out]
  by simp
have query-start-eq:
  query-start = s11(
    PState := concat (PState s11) composition-final,
    PTranscript := PTranscript s11 @ [composition-final])
  by (rule record-staged-message-outcome[OF record-composition-final])
have start-state:
  PState query-start = staged-query-prefix-start-hash prefix
  unfolding prefix-eq staged-query-prefix-start-hash-def
  using state-s10 state-s11 unfolding query-start-eq by simp
have query-counter-start: PQueryCounter query-start = 0
proof –
  have root-counter:
    PQueryCounter s1 = PQueryCounter adversary-initial-state
    using controlled-stage-outcome-fields[OF root-controlled root-out]
    by simp
  have trace-counter:
    PQueryCounter s3 = PQueryCounter s2
    using staged-trace-fri-program-alignment
      [OF controlled trace-bound trace-out]
    by simp
  have final-counter: PQueryCounter s4 = PQueryCounter s3
    using controlled-stage-outcome-fields
      [OF trace-final-controlled final-out]
    by simp
  have alpha-counter: PQueryCounter s6 = PQueryCounter s5

```

```

    using staged-alpha-program-alignment[OF alpha-out] by simp
  have degree-counter: PQueryCounter s7 = PQueryCounter s6
    using controlled-stage-outcome-fields[OF degree-controlled degree-out]
    by simp
  have comp-counter: PQueryCounter s10 = PQueryCounter s9
    using staged-composition-fri-program-alignment
      [OF controlled composition-bound composition-out]
    by simp
  have comp-final-counter: PQueryCounter s11 = PQueryCounter s10
    using controlled-stage-outcome-fields
      [OF composition-final-controlled composition-final-out]
    by simp
  show ?thesis
    using root-counter trace-counter final-counter alpha-counter
      degree-counter comp-counter comp-final-counter
    unfolding s2-eq s5-eq s8-eq s9-eq query-start-eq by simp
qed
have query-bound:  $0 + i \leq \text{length } (\text{query-opening-budgets budgets})$ 
  using wf i-bound unfolding staged-budget-wellformed-def by simp
from checked-staged-query-program-outcome-with-raws
  [OF controlled query-bound query-out]
obtain raw-idxs query-idxs where
  len:  $\text{length } \text{query-chunks} = i$ 
  and state:
    PState prefix-state =
      state-after-query-chunks (PState query-start) query-chunks i
  and counter:
    PQueryCounter prefix-state = PQueryCounter query-start + i
  by blast
show ?thesis
  using start-state query-counter-start state counter len
  unfolding prefix-eq by simp
qed

lemma checked-staged-query-program-prefix-receive-support:
  assumes i-bound:  $i < n$ 
  and outcome:
    Some (chunks, t)  $\in$ 
      set-dist
        (execute
          (checked-staged-query-program A trace-roots composition-roots
            start n) s)
  obtains prefix-chunks prefix-state raw raw-state chunk chunk-state
    record-state suffix-chunks
  where
    Some (prefix-chunks, prefix-state)  $\in$ 
      set-dist
        (execute
          (checked-staged-query-program A trace-roots composition-roots

```

```

    start i) s)
Some (raw, raw-state) ∈
  set-dist (execute receive-query-index-challenge prefix-state)
Some (chunk, chunk-state) ∈
  set-dist
    (execute (query-opening-stage A (start + i) raw) raw-state)
verifier-query-round-chunk (index (to-nat raw))
  trace-roots composition-roots chunk
Some ((), record-state) ∈
  set-dist (execute (record-staged-messages chunk) chunk-state)
Some (suffix-chunks, t) ∈
  set-dist
    (execute
      (checked-staged-query-program A trace-roots composition-roots
        (Suc (start + i)) (n - Suc i) record-state)
      chunks = prefix-chunks @ chunk # suffix-chunks)
proof -
have decomp:
  checked-staged-query-program A trace-roots composition-roots start n =
  checked-staged-query-program A trace-roots composition-roots start i ≫=
  (λprefix-chunks.
    receive-query-index-challenge ≫= (λraw.
      let idx = index (to-nat raw) in
      query-opening-stage A (start + i) raw ≫= (λchunk.
        assert
          (verifier-query-round-chunk idx trace-roots composition-roots
            chunk) ≫= (λ-.
              record-staged-messages chunk ≫= (λ-.
                checked-staged-query-program A trace-roots composition-roots
                  (Suc (start + i)) (n - Suc i) ≫= (λsuffix-chunks.
                    return (prefix-chunks @ chunk # suffix-chunks)))))))
by (rule checked-staged-query-program-decomp[OF i-bound])
from outcome[unfolded decomp]
obtain prefix-chunks prefix-state raw raw-state chunk chunk-state
  record-state suffix-chunks where
  prefix-out:
    Some (prefix-chunks, prefix-state) ∈
      set-dist
        (execute
          (checked-staged-query-program A trace-roots composition-roots
            start i) s)
and raw-out:
  Some (raw, raw-state) ∈
    set-dist (execute receive-query-index-challenge prefix-state)
and chunk-out:
  Some (chunk, chunk-state) ∈
    set-dist
      (execute (query-opening-stage A (start + i) raw) raw-state)
and chunk-shape:

```


verifier-query-round-chunk (*index* (*to-nat raw*))
trace-roots composition-roots chunk
and *record-out*:
Some (*()*, *record-state*) $\in$
set-dist (*execute* (*record-staged-messages chunk*) *chunk-state*)
and *suffix-out*:
Some (*suffix-chunks*, *t*) $\in$
set-dist
(*execute*
(*checked-staged-query-program A trace-roots composition-roots*
(*Suc* (*start + i*)) (*n - Suc i*)) *record-state*)
and *chunks-eq*: *chunks* = *prefix-chunks @ chunk # suffix-chunks*
by (*auto simp: assert-def throw-no-outcome elim!: set-dist-bindE*
split: if-splits)
show *?thesis*
by (*rule that[OF prefix-out raw-out chunk-out chunk-shape record-out*
suffix-out chunks-eq])
qed

lemma *state-after-query-chunks-append-prefix*:
assumes *len-prefix*: *length prefix = i*
shows
state-after-query-chunks st (prefix @ suffix) i =
state-after-query-chunks st prefix i
using *len-prefix*
unfolding *state-after-query-chunks-def* **by** *simp*

lemma *checked-staged-query-program-prefix-receive-state-alignment*:
assumes *controlled*: *staged-adversary-controlled budgets A*
and *bound*: *start + n $\leq$ length (query-opening-budgets budgets)*
and *i-bound*: *i < n*
and *outcome*:
Some (*chunks*, *t*) $\in$
set-dist
(*execute*
(*checked-staged-query-program A trace-roots composition-roots*
start n) *s*)
obtains *prefix-chunks prefix-state raw raw-state chunk chunk-state*
record-state suffix-chunks
where
Some (*prefix-chunks*, *prefix-state*) $\in$
set-dist
(*execute*
(*checked-staged-query-program A trace-roots composition-roots*
start i) *s*)
Some (*raw*, *raw-state*) $\in$
set-dist (*execute receive-query-index-challenge prefix-state*)
Some (*chunk*, *chunk-state*) $\in$
set-dist

$(\text{execute } (\text{query-opening-stage } A \text{ (start + i) raw}) \text{ raw-state})$
 $\text{verifier-query-round-chunk } (\text{index } (\text{to-nat raw}))$
 $\text{trace-roots composition-roots chunk}$
 $\text{Some } ((), \text{record-state}) \in$
 $\text{set-dist } (\text{execute } (\text{record-staged-messages chunk}) \text{ chunk-state})$
 $\text{Some } (\text{suffix-chunks}, t) \in$
 set-dist
 $(\text{execute}$
 $(\text{checked-staged-query-program } A \text{ trace-roots composition-roots}$
 $(\text{Suc } (\text{start + i})) (n - \text{Suc } i)) \text{ record-state})$
 $\text{chunks} = \text{prefix-chunks} @ \text{chunk} \# \text{suffix-chunks}$
 $P\text{QueryCounter prefix-state} = P\text{QueryCounter } s + i$
 $P\text{State prefix-state} = \text{state-after-query-chunks } (P\text{State } s) \text{ chunks } i$

proof –

from $\text{checked-staged-query-program-prefix-receive-support}$
 $[\text{OF } i\text{-bound outcome}]$

obtain $\text{prefix-chunks prefix-state raw raw-state chunk chunk-state}$
 $\text{record-state suffix-chunks}$ **where**

prefix-out:
 $\text{Some } (\text{prefix-chunks}, \text{prefix-state}) \in$
 set-dist
 $(\text{execute}$
 $(\text{checked-staged-query-program } A \text{ trace-roots composition-roots}$
 $\text{start } i) s)$

and raw-out:
 $\text{Some } (\text{raw}, \text{raw-state}) \in$
 $\text{set-dist } (\text{execute } \text{receive-query-index-challenge } \text{prefix-state})$

and chunk-out:
 $\text{Some } (\text{chunk}, \text{chunk-state}) \in$
 set-dist
 $(\text{execute } (\text{query-opening-stage } A \text{ (start + i) raw}) \text{ raw-state})$

and chunk-shape:
 $\text{verifier-query-round-chunk } (\text{index } (\text{to-nat raw}))$
 $\text{trace-roots composition-roots chunk}$

and record-out:
 $\text{Some } ((), \text{record-state}) \in$
 $\text{set-dist } (\text{execute } (\text{record-staged-messages chunk}) \text{ chunk-state})$

and suffix-out:
 $\text{Some } (\text{suffix-chunks}, t) \in$
 set-dist
 $(\text{execute}$
 $(\text{checked-staged-query-program } A \text{ trace-roots composition-roots}$
 $(\text{Suc } (\text{start + i})) (n - \text{Suc } i)) \text{ record-state})$

and $\text{chunks-eq: chunks} = \text{prefix-chunks} @ \text{chunk} \# \text{suffix-chunks}$

by $(\text{rule } \text{checked-staged-query-program-prefix-receive-support}$
 $[\text{OF } i\text{-bound outcome}])$

have $\text{prefix-bound: start + i} \leq \text{length } (\text{query-opening-budgets budgets})$
using $\text{bound } i\text{-bound}$ **by** simp

from $\text{checked-staged-query-program-outcome-with-raws}$

$[OF \text{ controlled prefix-bound prefix-out}]$
obtain *raw-prefix query-prefix* **where**
len-prefix: length prefix-chunks = i
and *state-prefix:*
PState prefix-state =
state-after-query-chunks (PState s) prefix-chunks i
and *counter-prefix:*
PQueryCounter prefix-state = PQueryCounter s + i
by *blast*
have *state-full:*
PState prefix-state = state-after-query-chunks (PState s) chunks i
using *state-prefix len-prefix*
unfolding *chunks-eq*
by (*simp add: state-after-query-chunks-append-prefix*)
show *?thesis*
by (*rule that[OF prefix-out raw-out chunk-out chunk-shape record-out*
suffix-out chunks-eq counter-prefix state-full])
qed

lemma *checked-staged-transcript-program-query-prefix-receive-support:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *outcome:*
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
obtains *prefix prefix-state raw raw-state*
where
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
sqp-trace-root prefix = staged-trace-root data
sqp-trace-fri-roots prefix = staged-trace-fri-roots data
sqp-trace-fri-challenges prefix =
staged-trace-fri-challenges data
sqp-trace-final prefix = staged-trace-final data
sqp-alphas prefix = staged-alphas data
sqp-degree prefix = staged-degree data
sqp-composition-fri-roots prefix =
staged-composition-fri-roots data
sqp-composition-fri-challenges prefix =
staged-composition-fri-challenges data
sqp-composition-final prefix = staged-composition-final data
sqp-query-chunks prefix = take i (staged-query-chunks data)
PQueryCounter prefix-state = i
PState prefix-state =

```

    state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i
    prefix-state ≤ attacker-state
    raw-state ≤ attacker-state
    fmlookup (HashMap raw-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw
  proof —
  from outcome obtain fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5
    as s6 dg s7 s8 s9 composition-roots composition-bs s10
    composition-final s11 query-start query-chunks where
  root-out:
    Some (fr, s1) ∈
      set-dist (execute (trace-root-stage A) adversary-initial-state)
  and record-root:
    Some ((), s2) ∈ set-dist (execute (record-staged-message fr) s1)
  and trace-out:
    Some ((trace-roots, trace-bs), s3) ∈
      set-dist
        (execute
          (staged-trace-fri-program A 0 (ceil-log clength) []) s2)
  and final-out:
    Some (trace-final, s4) ∈
      set-dist (execute (trace-final-stage A trace-bs) s3)
  and record-final:
    Some ((), s5) ∈
      set-dist (execute (record-staged-message trace-final) s4)
  and alpha-out:
    Some (as, s6) ∈
      set-dist (execute (staged-alpha-program (length spec)) s5)
  and degree-out:
    Some (dg, s7) ∈ set-dist (execute (degree-stage A as) s6)
  and record-degree:
    Some ((), s8) ∈ set-dist (execute (record-staged-message dg) s7)
  and assert-out:
    Some ((), s9) ∈
      set-dist
        (execute
          (assert
            (ceil-log (to-nat dg + 1) ≤
              ceil-log (maxDegree + 1))) s8)
  and composition-out:
    Some ((composition-roots, composition-bs), s10) ∈
      set-dist
        (execute
          (staged-composition-fri-program A dg 0
            (ceil-log (to-nat dg + 1)) []) s9)

```

```

and composition-final-out:
  Some (composition-final, s11) ∈
    set-dist
      (execute (composition-final-stage A dg composition-bs) s10)
and record-composition-final:
  Some ((), query-start) ∈
    set-dist (execute (record-staged-message composition-final) s11)
and query-out:
  Some (query-chunks, attacker-state) ∈
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          0 rounds)
        query-start)
and data-eq:
  data =
    (staged-trace-root = fr,
     staged-trace-fri-roots = trace-roots,
     staged-trace-fri-challenges = trace-bs,
     staged-trace-final = trace-final,
     staged-alphas = as,
     staged-degree = dg,
     staged-composition-fri-roots = composition-roots,
     staged-composition-fri-challenges = composition-bs,
     staged-composition-final = composition-final,
     staged-query-chunks = query-chunks)
unfolding checked-staged-transcript-program-def Let-def
by (auto elim!: set-dist-bindE split: prod.splits)
have assert-out-Suc:
  Some ((), s9) ∈
    set-dist
      (execute
        (assert
          (ceil-log (Suc (to-nat dg)) ≤
            ceil-log (Suc maxDegree))) s8)
unfolding Suc-eq-plus1
by (rule assert-out)
have composition-out-Suc:
  Some ((composition-roots, composition-bs), s10) ∈
    set-dist
      (execute
        (staged-composition-fri-program A dg 0
          (ceil-log (Suc (to-nat dg))) []) s9)
unfolding Suc-eq-plus1
by (rule composition-out)
have query-bound: 0 + rounds ≤ length (query-opening-budgets budgets)
using wf unfolding staged-budget-wellformed-def by simp
from checked-staged-query-program-prefix-receive-state-alignment
  [OF controlled query-bound i-bound query-out]

```

obtain *prefix-chunks prefix-state raw raw-state chunk chunk-state*
record-state suffix-chunks **where**
prefix-query-out:
Some (prefix-chunks, prefix-state) ∈
set-dist
(execute
(checked-staged-query-program A trace-roots composition-roots
0 i) query-start)
and *raw-out:*
Some (raw, raw-state) ∈
set-dist (execute receive-query-index-challenge prefix-state)
and *chunk-out:*
Some (chunk, chunk-state) ∈
set-dist
(execute (query-opening-stage A (0 + i) raw) raw-state)
and *chunk-shape:*
verifier-query-round-chunk (index (to-nat raw))
trace-roots composition-roots chunk
and *record-out:*
Some ((), record-state) ∈
set-dist (execute (record-staged-messages chunk) chunk-state)
and *suffix-out:*
Some (suffix-chunks, attacker-state) ∈
set-dist
(execute
(checked-staged-query-program A trace-roots composition-roots
(Suc (0 + i)) (rounds - Suc i) record-state)
and *chunks-eq:* *query-chunks = prefix-chunks @ chunk # suffix-chunks*
and *counter-prefix:*
PQueryCounter prefix-state = PQueryCounter query-start + i
and *state-prefix:*
PState prefix-state =
state-after-query-chunks (PState query-start) query-chunks i
by (*rule checked-staged-query-program-prefix-receive-state-alignment*
[OF controlled query-bound i-bound query-out])
let *?prefix =*
(|sqp-trace-root = fr,
sqp-trace-fri-roots = trace-roots,
sqp-trace-fri-challenges = trace-bs,
sqp-trace-final = trace-final,
sqp-alphas = as,
sqp-degree = dg,
sqp-composition-fri-roots = composition-roots,
sqp-composition-fri-challenges = composition-bs,
sqp-composition-final = composition-final,
sqp-query-chunks = prefix-chunks|)
have *alpha-prefix-out:*
Some ((fr, trace-roots, trace-bs, trace-final), s5) ∈
set-dist

(*execute* (*staged-alpha-prefix-program* *A*) *adversary-initial-state*)

proof –

have *after-record-final*:

Some ((*fr*, *trace-roots*, *trace-bs*, *trace-final*), *s5*) ∈

set-dist

(*execute*

(*record-staged-message* *trace-final* $\gg$

(λ -. *return* (*fr*, *trace-roots*, *trace-bs*, *trace-final*)))

s4)

by (*rule-tac* *x=()* **and** *t=s5* **in** *set-dist-bindI*)

(*rule* *record-final*, *simp*)

have *after-final*:

Some ((*fr*, *trace-roots*, *trace-bs*, *trace-final*), *s5*) ∈

set-dist

(*execute*

(*trace-final-stage* *A* *trace-bs* $\gg$

(λ *trace-final*.

record-staged-message *trace-final* $\gg$

(λ -. *return* (*fr*, *trace-roots*, *trace-bs*, *trace-final*))))

s3)

by (*rule-tac* *x=trace-final* **and** *t=s4* **in** *set-dist-bindI*)

(*rule* *final-out*, *rule* *after-record-final*)

have *after-trace*:

Some ((*fr*, *trace-roots*, *trace-bs*, *trace-final*), *s5*) ∈

set-dist

(*execute*

(*staged-trace-fri-program* *A* 0 (*ceil-log* *clength*) $\square$ $\gg$

(λ (*trace-roots*, *trace-bs*).

trace-final-stage *A* *trace-bs* $\gg$

(λ *trace-final*.

record-staged-message *trace-final* $\gg$

(λ -. *return*

(*fr*, *trace-roots*, *trace-bs*, *trace-final*))))

s2)

apply (*rule-tac* *x=(trace-roots, trace-bs)* **and** *t=s3*

in *set-dist-bindI*)

apply (*rule* *trace-out*)

apply (*simp* *add: after-final*)

done

have *after-record-root*:

Some ((*fr*, *trace-roots*, *trace-bs*, *trace-final*), *s5*) ∈

set-dist

(*execute*

(*record-staged-message* *fr* $\gg$

(λ -. *staged-trace-fri-program* *A* 0 (*ceil-log* *clength*) $\square$ $\gg$

(λ (*trace-roots*, *trace-bs*).

trace-final-stage *A* *trace-bs* $\gg$

(λ *trace-final*.

record-staged-message *trace-final* $\gg$

```

      (λ-. return
        (fr, trace-roots, trace-bs, trace-final))))))
    s1)
  by (rule-tac x=() and t=s2 in set-dist-bindI)
    (rule record-root, rule after-trace)
show ?thesis
  unfolding staged-alpha-prefix-program-def
  by (rule-tac x=fr and t=s1 in set-dist-bindI)
    (rule root-out, rule after-record-root)
qed
have prefix-program-out:
  Some (?prefix, prefix-state) ∈
    set-dist
      (execute
        (checked-staged-query-challenge-prefix-program A i)
        adversary-initial-state)
proof -
  have after-prefix-query:
    Some (?prefix, prefix-state) ∈
      set-dist
        (execute
          (checked-staged-query-program A trace-roots composition-roots
            0 i >>=
              (λquery-chunks.
                return
                  (sqp-trace-root = fr,
                    sqp-trace-fri-roots = trace-roots,
                    sqp-trace-fri-challenges = trace-bs,
                    sqp-trace-final = trace-final,
                    sqp-alphas = as,
                    sqp-degree = dg,
                    sqp-composition-fri-roots = composition-roots,
                    sqp-composition-fri-challenges = composition-bs,
                    sqp-composition-final = composition-final,
                    sqp-query-chunks = query-chunks))))
          query-start)
  by (rule-tac x=prefix-chunks and t=prefix-state in set-dist-bindI)
    (rule prefix-query-out, simp)
have after-record-composition-final:
  Some (?prefix, prefix-state) ∈
    set-dist
      (execute
        (record-staged-message composition-final >>=
          (λ-.
            checked-staged-query-program A trace-roots composition-roots
              0 i >>=
                (λquery-chunks.
                  return
                    (sqp-trace-root = fr,

```



```

      sqp-trace-fri-roots = trace-roots,
      sqp-trace-fri-challenges = trace-bs,
      sqp-trace-final = trace-final,
      sqp-alphas = as,
      sqp-degree = dg,
      sqp-composition-fri-roots = composition-roots,
      sqp-composition-fri-challenges = composition-bs,
      sqp-composition-final = composition-final,
      sqp-query-chunks = query-chunks))))
    s11)
  by (rule-tac x=() and t=query-start in set-dist-bindI)
    (rule record-composition-final, rule after-prefix-query)
have after-composition-final:
  Some (?prefix, prefix-state) ∈
    set-dist
    (execute
      (composition-final-stage A dg composition-bs ≫=
        (λcomposition-final.
          record-staged-message composition-final ≫=
            (λ-.
              checked-staged-query-program A trace-roots
                composition-roots 0 i ≫=
                  (λquery-chunks.
                    return
                      (|sqp-trace-root = fr,
                        sqp-trace-fri-roots = trace-roots,
                        sqp-trace-fri-challenges = trace-bs,
                        sqp-trace-final = trace-final,
                        sqp-alphas = as,
                        sqp-degree = dg,
                        sqp-composition-fri-roots = composition-roots,
                        sqp-composition-fri-challenges = composition-bs,
                        sqp-composition-final = composition-final,
                        sqp-query-chunks = query-chunks|))))))
    s10)
  by (rule-tac x=composition-final and t=s11 in set-dist-bindI)
    (rule composition-final-out, rule after-record-composition-final)
have after-composition:
  Some (?prefix, prefix-state) ∈
    set-dist
    (execute
      (staged-composition-fri-program A dg 0
        (ceil-log (Suc (to-nat dg))) [] ≫=
          (λ(composition-roots, composition-bs).
            composition-final-stage A dg composition-bs ≫=
              (λcomposition-final.
                record-staged-message composition-final ≫=
                  (λ-.
                    checked-staged-query-program A trace-roots

```

```

      composition-roots 0 i >>=
    (λquery-chunks.
      return
        (sqp-trace-root = fr,
         sqp-trace-fri-roots = trace-roots,
         sqp-trace-fri-challenges = trace-bs,
         sqp-trace-final = trace-final,
         sqp-alphas = as,
         sqp-degree = dg,
         sqp-composition-fri-roots = composition-roots,
         sqp-composition-fri-challenges = composition-bs,
         sqp-composition-final = composition-final,
         sqp-query-chunks = query-chunks))))))
  s9)
apply (rule-tac x=(composition-roots, composition-bs) and t=s10
  in set-dist-bindI)
  apply (rule composition-out-Suc)
apply (simp add: after-composition-final)
done
have after-assert:
  Some (?prefix, prefix-state) ∈
    set-dist
    (execute
      (assert (ceil-log (Suc (to-nat dg)) ≤
        ceil-log (Suc maxDegree)) >>=
      (λ-.
        staged-composition-fri-program A dg 0
        (ceil-log (Suc (to-nat dg))) [] >>=
      (λ(composition-roots, composition-bs).
        composition-final-stage A dg composition-bs >>=
      (λcomposition-final.
        record-staged-message composition-final >>=
      (λ-.
        checked-staged-query-program A trace-roots
        composition-roots 0 i >>=
      (λquery-chunks.
        return
          (sqp-trace-root = fr,
           sqp-trace-fri-roots = trace-roots,
           sqp-trace-fri-challenges = trace-bs,
           sqp-trace-final = trace-final,
           sqp-alphas = as,
           sqp-degree = dg,
           sqp-composition-fri-roots = composition-roots,
           sqp-composition-fri-challenges = composition-bs,
           sqp-composition-final = composition-final,
           sqp-query-chunks = query-chunks))))))))))
    s8)
apply (rule-tac x=() and t=s9 in set-dist-bindI)

```

```

    apply (rule assert-out-Suc)
  apply (simp add: after-composition)
done
have after-record-degree:
  Some (?prefix, prefix-state) ∈
    set-dist
    (execute
      (record-staged-message dg ≫=
        (λ-.
          assert (ceil-log (Suc (to-nat dg)) ≤
            ceil-log (Suc maxDegree)) ≫=
          (λ-.
            staged-composition-fri-program A dg 0
              (ceil-log (Suc (to-nat dg))) [] ≫=
            (λ(composition-roots, composition-bs).
              composition-final-stage A dg composition-bs ≫=
            (λcomposition-final.
              record-staged-message composition-final ≫=
            (λ-.
              checked-staged-query-program A trace-roots
                composition-roots 0 i ≫=
            (λquery-chunks.
              return
                (λsqp-trace-root = fr,
                  sqp-trace-fri-roots = trace-roots,
                  sqp-trace-fri-challenges = trace-bs,
                  sqp-trace-final = trace-final,
                  sqp-alphas = as,
                  sqp-degree = dg,
                  sqp-composition-fri-roots = composition-roots,
                  sqp-composition-fri-challenges = composition-bs,
                  sqp-composition-final = composition-final,
                  sqp-query-chunks = query-chunks))))))))))
    s7)
  apply (rule-tac x=() and t=s8 in set-dist-bindI)
  apply (rule record-degree)
  apply (simp add: after-assert)
done
have after-degree:
  Some (?prefix, prefix-state) ∈
    set-dist
    (execute
      (degree-stage A as ≫=
        (λdg.
          record-staged-message dg ≫=
          (λ-.
            assert (ceil-log (Suc (to-nat dg)) ≤
              ceil-log (Suc maxDegree)) ≫=
          (λ-.

```

```

    staged-composition-fri-program A dg 0
      (ceil-log (Suc (to-nat dg))) [] >=>
    (λ(composition-roots, composition-bs).
      composition-final-stage A dg composition-bs >=>
      (λcomposition-final.
        record-staged-message composition-final >=>
        (λ-.
          checked-staged-query-program A trace-roots
            composition-roots 0 i >=>
          (λquery-chunks.
            return
              (|spp-trace-root = fr,
                spp-trace-fri-roots = trace-roots,
                spp-trace-fri-challenges = trace-bs,
                spp-trace-final = trace-final,
                spp-alphas = as,
                spp-degree = dg,
                spp-composition-fri-roots = composition-roots,
                spp-composition-fri-challenges = composition-bs,
                spp-composition-final = composition-final,
                spp-query-chunks = query-chunks|)))))))))
  s6)
apply (rule-tac x=dg and t=s7 in set-dist-bindI)
apply (rule degree-out)
apply (simp add: after-record-degree)
done
have after-alpha:
  Some (?prefix, prefix-state) ∈
    set-dist
    (execute
      (staged-alpha-program (length spec) >=>
        (λas.
          degree-stage A as >=>
          (λdg.
            record-staged-message dg >=>
            (λ-.
              assert (ceil-log (Suc (to-nat dg)) ≤
                ceil-log (Suc maxDegree)) >=>
              (λ-.
                staged-composition-fri-program A dg 0
                  (ceil-log (Suc (to-nat dg))) [] >=>
                (λ(composition-roots, composition-bs).
                  composition-final-stage A dg composition-bs >=>
                  (λcomposition-final.
                    record-staged-message composition-final >=>
                    (λ-.
                      checked-staged-query-program A trace-roots
                        composition-roots 0 i >=>
                      (λquery-chunks.

```

```

    return
    (|sqp-trace-root = fr,
     sqp-trace-fri-roots = trace-roots,
     sqp-trace-fri-challenges = trace-bs,
     sqp-trace-final = trace-final,
     sqp-alphas = as,
     sqp-degree = dg,
     sqp-composition-fri-roots = composition-roots,
     sqp-composition-fri-challenges = composition-bs,
     sqp-composition-final = composition-final,
     sqp-query-chunks = query-chunks|)))))))))
s5)
apply (rule-tac x=as and t=s6 in set-dist-bindI)
apply (rule alpha-out)
apply (simp add: after-degree)
done
show ?thesis
unfolding checked-staged-query-challenge-prefix-program-def Let-def
apply (rule-tac x=(fr, trace-roots, trace-bs, trace-final) and t=s5
in set-dist-bindI)
apply (rule alpha-prefix-out)
apply (simp add: after-alpha)
done
qed
have prefix-with-state-out:
  Some ((?prefix, prefix-state), prefix-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)
unfolding checked-staged-query-prefix-with-state-def
by (rule set-dist-bindI[where x=?prefix and t=prefix-state,
  OF prefix-program-out get-bind-return-state-support])
have prefix-receive-out:
  Some (((?prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
unfolding checked-staged-query-prefix-receive-with-state-def
by (intro set-dist-bindI[OF prefix-with-state-out]
  set-dist-bindI[OF raw-out])
  simp
have root-controlled:
  controlled-ro-program (trace-root-budget budgets) (trace-root-stage A)
using controlled unfolding staged-adversary-controlled-def by simp
have trace-final-controlled:
  controlled-ro-program (trace-final-budget budgets)
    (trace-final-stage A trace-bs)
using controlled unfolding staged-adversary-controlled-def by blast
have degree-controlled:

```

```

controlled-ro-program (degree-budget budgets) (degree-stage A as)
using controlled unfolding staged-adversary-controlled-def by blast
have composition-final-controlled:
  controlled-ro-program (composition-final-budget budgets)
    (composition-final-stage A dg composition-bs)
  using controlled unfolding staged-adversary-controlled-def by blast
have root-state: PState s1 = 0
  using controlled-stage-outcome-fields[OF root-controlled root-out]
  by simp
have s2-eq:
  s2 = s1(
    PState := concat (PState s1) fr,
    PTranscript := PTranscript s1 @ [fr])
  by (rule record-staged-message-outcome[OF record-root])
have trace-bound:
   $0 + \text{ceil-log } \text{clength} \leq \text{length } (\text{trace-fri-budgets } \text{budgets})$ 
  using wf unfolding staged-budget-wellformed-def by simp
have trace-alignment:
  PState s3 = foldl concat (PState s2) trace-roots
  using staged-trace-fri-program-alignment
    [OF controlled trace-bound trace-out]
  by simp
have state-s3:
  PState s3 = foldl concat (concat 0 fr) trace-roots
  using root-state trace-alignment unfolding s2-eq by simp
have state-s4: PState s4 = PState s3
  using controlled-stage-outcome-fields
    [OF trace-final-controlled final-out]
  by simp
have s5-eq:
  s5 = s4(
    PState := concat (PState s4) trace-final,
    PTranscript := PTranscript s4 @ [trace-final])
  by (rule record-staged-message-outcome[OF record-final])
have alpha-alignment:
  PState s6 = foldl concat (PState s5) as
  using staged-alpha-program-alignment[OF alpha-out] by simp
have state-s6:
  PState s6 =
    foldl concat
      (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
      as
  using state-s3 state-s4 alpha-alignment unfolding s5-eq by simp
have state-s7: PState s7 = PState s6
  using controlled-stage-outcome-fields[OF degree-controlled degree-out]
  by simp
have s8-eq:
  s8 = s7(
    PState := concat (PState s7) dg,

```

```

    PTranscript := PTranscript s7 @ [dg])
  by (rule record-staged-message-outcome[OF record-degree])
have s9-eq: s9 = s8
  using assert-out unfolding assert-def
  by (cases
      ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
      (auto simp: throw-no-outcome)
have round-bound:
  ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1)
  using assert-out unfolding assert-def
  by (cases
      ceil-log (to-nat dg + 1) ≤ ceil-log (maxDegree + 1))
      (auto simp: throw-no-outcome)
have composition-bound:
  0 + ceil-log (to-nat dg + 1) ≤
    length (composition-fri-budgets budgets)
  using wf round-bound unfolding staged-budget-wellformed-def by simp
have composition-alignment:
  PState s10 = foldl concat (PState s9) composition-roots
  using staged-composition-fri-program-alignment
    [OF controlled composition-bound composition-out]
  by simp
have state-s10:
  PState s10 =
    foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat 0 fr) trace-roots) trace-final)
          as)
        dg)
      composition-roots
  using state-s6 state-s7 composition-alignment
  unfolding s8-eq s9-eq by simp
have state-s11: PState s11 = PState s10
  using controlled-stage-outcome-fields
    [OF composition-final-controlled composition-final-out]
  by simp
have query-start-eq:
  query-start = s11(
    PState := concat (PState s11) composition-final,
    PTranscript := PTranscript s11 @ [composition-final])
  by (rule record-staged-message-outcome[OF record-composition-final])
have state-query-start:
  PState query-start =
    concat
      (foldl concat
        (concat
          (foldl concat
            (concat (foldl concat (concat 0 fr) trace-roots) trace-final)

```

```

      as)
      dg)
      composition-roots)
      composition-final
    using state-s10 state-s11 unfolding query-start-eq by simp
  have start-state:
    PState query-start = staged-query-start-hash data
  unfolding data-eq staged-query-start-hash-def
    staged-composition-fri-start-hash-def staged-trace-fri-start-hash-def
  using state-query-start by simp
  have query-counter-start: PQueryCounter query-start = 0
  proof -
    have root-fields:
      PQueryCounter s1 = PQueryCounter adversary-initial-state
      using controlled-stage-outcome-fields[OF root-controlled root-out]
      by simp
    have trace-counters:
      PQueryCounter s3 = PQueryCounter s2
      using staged-trace-fri-program-alignment
        [OF controlled trace-bound trace-out]
      by simp
    have final-counter: PQueryCounter s4 = PQueryCounter s3
      using controlled-stage-outcome-fields
        [OF trace-final-controlled final-out]
      by simp
    have alpha-counter: PQueryCounter s6 = PQueryCounter s5
      using staged-alpha-program-alignment[OF alpha-out] by simp
    have degree-counter: PQueryCounter s7 = PQueryCounter s6
      using controlled-stage-outcome-fields[OF degree-controlled degree-out]
      by simp
    have comp-counter: PQueryCounter s10 = PQueryCounter s9
      using staged-composition-fri-program-alignment
        [OF controlled composition-bound composition-out]
      by simp
    have comp-final-counter: PQueryCounter s11 = PQueryCounter s10
      using controlled-stage-outcome-fields
        [OF composition-final-controlled composition-final-out]
      by simp
    show ?thesis
      using root-fields trace-counters final-counter alpha-counter
        degree-counter comp-counter comp-final-counter
      unfolding s2-eq s5-eq s8-eq s9-eq query-start-eq by simp
  qed
  have prefix-counter: PQueryCounter prefix-state = i
    using counter-prefix query-counter-start by simp
  have prefix-state-eq:
    PState prefix-state =
      state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i

```



```

using state-prefix start-state unfolding data-eq by simp
have prefix-query-bound:  $0 + i \leq \text{length } (\text{query-opening-budgets budgets})$ 
using query-bound i-bound by simp
have prefix-len:  $\text{length prefix-chunks} = i$ 
proof –
  from checked-staged-query-program-outcome-with-raws
    [OF controlled prefix-query-bound prefix-query-out]
  show ?thesis
    by blast
qed
have prefix-chunks-eq:
  prefix-chunks = take i query-chunks
using chunks-eq prefix-len by simp
have receive-lookup:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
using receive-query-index-challenge-outcome[OF raw-out]
  prefix-counter prefix-state-eq
by simp
have stage-controlled:
  controlled-ro-program (query-opening-budgets budgets ! i)
    (query-opening-stage A i raw)
using controlled wf i-bound
unfolding staged-adversary-controlled-def staged-budget-wellformed-def
by simp
have raw-chunk-ext:  $\text{raw-state} \leq \text{chunk-state}$ 
proof –
  have chunk-out-i:
    Some (chunk, chunk-state) ∈
      set-dist (execute (query-opening-stage A i raw) raw-state)
using chunk-out by simp
show ?thesis
using controlled-ro-program-extension[OF stage-controlled] chunk-out-i
unfolding hash-extension-preserving-def by blast
qed
have prefix-raw-ext:  $\text{prefix-state} \leq \text{raw-state}$ 
by (rule receive-query-index-challenge-extends[OF raw-out])
have record-state-eq:
  record-state = chunk-state[
    PState := foldl concat (PState chunk-state) chunk,
    PTranscript := PTranscript chunk-state @ chunk]
by (rule record-staged-messages-outcome[OF record-out])
have chunk-record-ext:  $\text{chunk-state} \leq \text{record-state}$ 
unfolding record-state-eq less-eq-hash-ext-def less-eq-fmap-def by simp
have suffix-bound:
  Suc (0 + i) + (rounds – Suc i) ≤

```

```

    length (query-opening-budgets budgets)
  using query-bound i-bound by simp
have suffix-target:
  hash-target-program {}
  (sum-list
    (take (rounds - Suc i)
      (drop (Suc (0 + i)) (query-opening-budgets budgets))) +
    (rounds - Suc i))
  (checked-staged-query-program A trace-roots composition-roots
    (Suc (0 + i)) (rounds - Suc i))
  by (rule hash-target-program-checked-staged-query-program
    [OF controlled suffix-bound])
have suffix-ext: record-state ≤ attacker-state
  using hash-target-program-extension[OF suffix-target] suffix-out
  unfolding hash-extension-preserving-def by blast
have raw-ext: raw-state ≤ attacker-state
  by (rule hash-ext-trans[OF raw-chunk-ext])
  (rule hash-ext-trans[OF chunk-record-ext suffix-ext])
have prefix-ext: prefix-state ≤ attacker-state
  by (rule hash-ext-trans[OF prefix-raw-ext raw-ext])
show ?thesis
  by (rule that[OF prefix-receive-out])
  (use data-eq prefix-chunks-eq prefix-counter prefix-state-eq prefix-ext
    raw-ext receive-lookup in simp-all)
qed

```

```

lemma checked-staged-transcript-program-query-prefix-receive-final-lookup-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and outcome:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
  and final-lookup:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw
  obtains prefix prefix-state raw-state where
    Some (((prefix, prefix-state), raw), raw-state) ∈

```

```

    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  show ?thesis
proof (rule checked-staged-transcript-program-query-prefix-receive-support
  [OF wf controlled i-bound outcome])
  fix prefix prefix-state raw' raw-state
  assume prefix-receive:
    Some (((prefix, prefix-state), raw'), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
  and raw-ext: raw-state ≤ attacker-state
  and raw-lookup:
    fmlookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw'
  have raw-ext-s: raw-state ≤ ?s
  by (rule hash-extends-verifier-state-from-adversary-right[OF raw-ext])
  have s-ext-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
  have raw-ext-final: raw-state ≤ final-state
  by (rule hash-ext-trans[OF raw-ext-s s-ext-final])
  have final-lookup-raw':
    fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw'
  by (rule hash-extension-lookup[OF raw-lookup raw-ext-final])
  have raw-eq: raw' = raw
  using final-lookup final-lookup-raw' by simp
  have prefix-receive-raw:
    Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
  using prefix-receive raw-eq by simp
  show ?thesis
  by (rule that[OF prefix-receive-raw])
qed
qed

```

lemma *checked-staged-transcript-program-query-prefix-receive-final-lookup-support-with-alphas:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *outcome:*

Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)

and *verifier:*

Some (result, final-state) ∈
set-dist
(execute verify-monad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))))

and *final-lookup:*

fmlookup (HashMap final-state)
(QueryIndexChallenge i
(state-after-query-chunks
(staged-query-start-hash data) (staged-query-chunks data) i)) =
Some raw

obtains *prefix prefix-state raw-state where*

Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
sqp-alphas prefix = staged-alphas data

proof —

let *?s =*

verifier-state-from-adversary attacker-state
(staged-proof-transcript data)

show *?thesis*

proof (*rule checked-staged-transcript-program-query-prefix-receive-support*
[*OF wf controlled i-bound outcome*])

fix *prefix prefix-state raw' raw-state*

assume *prefix-receive:*

Some (((prefix, prefix-state), raw'), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)

and *alphas-eq: sqp-alphas prefix = staged-alphas data*

and *raw-ext: raw-state ≤ attacker-state*

and *raw-lookup:*

fmlookup (HashMap raw-state)
(QueryIndexChallenge i
(state-after-query-chunks
(staged-query-start-hash data) (staged-query-chunks data) i)) =
Some raw'

have *raw-ext-s: raw-state ≤ ?s*

```

    by (rule hash-extends-verifier-state-from-adversary-right[OF raw-ext])
  have s-ext-final: ?s ≤ final-state
    by (rule verify-monad-hash-extends[OF verifier])
  have raw-ext-final: raw-state ≤ final-state
    by (rule hash-ext-trans[OF raw-ext-s s-ext-final])
  have final-lookup-raw':
    fmllookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw'
    by (rule hash-extension-lookup[OF raw-lookup raw-ext-final])
  have raw-eq: raw' = raw
    using final-lookup final-lookup-raw' by simp
  have prefix-receive-raw:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
    using prefix-receive raw-eq by simp
  show ?thesis
    by (rule that[OF prefix-receive-raw alphas-eq])
qed
qed

end

end

```

```

theory Soundness-Staged-Query
imports Soundness-Staged-Query-Prefix
begin

```

Dynamic staged query target events and probability bounds.

```

context soundness
begin

```

```

definition checked-staged-query-prefix-dynamic-index-hit
:: ('f staged-query-prefix-data ⇒ 'f protocol-channel ⇒
  nat set) ⇒
  (((('f staged-query-prefix-data × 'f protocol-channel) × 'f) ×
    'f protocol-channel) option ⇒ bool
where
  checked-staged-query-prefix-dynamic-index-hit B out ⟷
    (case out of
      None ⇒ False
    | Some (((prefix, prefix-state), raw), -) ⇒
      index (to-nat raw) ∈ B prefix prefix-state)

```

definition *staged-partial-query-target*
 $:: 'f \text{ staged-proof-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow \text{nat set}$
where
 $\text{staged-partial-query-target data attacker-state} =$
 $\text{query-header-supported-partial-union-good-sets}$
 $\quad (\text{verifier-state-from-adversary attacker-state}$
 $\quad \quad (\text{staged-proof-transcript data}))$
 $\text{query-sampling-success-space}$
 $\quad (\text{staged-trace-root data})$
 $\quad (\text{staged-trace-fri-roots data})$
 $\quad (\text{staged-trace-final data})$
 $\quad (\text{staged-alphas data})$
 $\quad (\text{staged-degree data})$
 $\quad (\text{staged-composition-fri-roots data})$
 $\quad (\text{staged-composition-final data})$

definition *staged-query-prefix-matches-data*
 $:: \text{nat} \Rightarrow 'f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ staged-proof-data} \Rightarrow$
 bool
where
 $\text{staged-query-prefix-matches-data } i \text{ prefix data} \longleftrightarrow$
 $\text{sqp-trace-root prefix} = \text{staged-trace-root data} \wedge$
 $\text{sqp-trace-fri-roots prefix} = \text{staged-trace-fri-roots data} \wedge$
 $\text{sqp-trace-fri-challenges prefix} =$
 $\quad \text{staged-trace-fri-challenges data} \wedge$
 $\text{sqp-trace-final prefix} = \text{staged-trace-final data} \wedge$
 $\text{sqp-alphas prefix} = \text{staged-alphas data} \wedge$
 $\text{sqp-degree prefix} = \text{staged-degree data} \wedge$
 $\text{sqp-composition-fri-roots prefix} =$
 $\quad \text{staged-composition-fri-roots data} \wedge$
 $\text{sqp-composition-fri-challenges prefix} =$
 $\quad \text{staged-composition-fri-challenges data} \wedge$
 $\text{sqp-composition-final prefix} =$
 $\quad \text{staged-composition-final data} \wedge$
 $\text{sqp-query-chunks prefix} = \text{take } i \text{ (staged-query-chunks data)}$

lemma *staged-query-start-hash-of-prefix-completion:*
assumes *match:* $\text{staged-query-prefix-matches-data } i \text{ prefix data}$
shows $\text{staged-query-start-hash data} =$
 $\text{staged-query-prefix-start-hash prefix}$
using *match*
unfolding *staged-query-prefix-matches-data-def*
 $\text{staged-query-start-hash-def staged-composition-fri-start-hash-def}$
 $\text{staged-trace-fri-start-hash-def staged-query-prefix-start-hash-def}$
by *simp*

definition *staged-query-prefix-partial-query-target*
 $:: \text{nat} \Rightarrow 'f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow$

nat set
where
 $\text{staged-query-prefix-partial-query-target } i \text{ prefix prefix-state} =$
 $\{ \text{idx. } \exists \text{ data attacker-state.}$
 $\text{staged-query-prefix-matches-data } i \text{ prefix data } \wedge$
 $\text{prefix-state} \leq \text{attacker-state} \wedge$
 $\text{idx} \in \text{staged-partial-query-target data attacker-state} \}$

definition $\text{staged-query-prefix-committed-query-target}$
 $:: 'f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow \text{nat set}$
where
 $\text{staged-query-prefix-committed-query-target prefix prefix-state} =$
 $\text{query-header-committed-union-good-sets prefix-state}$
 $\text{query-sampling-success-space}$
 $(\text{sqp-trace-root prefix})$
 $(\text{sqp-alphas prefix})$
 $(\text{sqp-composition-fri-roots prefix})$

definition $\text{staged-query-prefix-length-committed-table-candidates}$
 $:: 'f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow$
 $('f \text{ list} \times 'f \text{ list}) \text{ set}$
where
 $\text{staged-query-prefix-length-committed-table-candidates prefix prefix-state} =$
 $\{ (\text{trace-table, composition-table}).$
 $\text{merkle-root-binds-table } (\text{sqp-trace-root prefix}) \text{ trace-table}$
 $\text{prefix-state} \wedge$
 $\text{sqp-composition-fri-roots prefix} \neq [] \wedge$
 $\text{merkle-root-binds-table } (\text{hd } (\text{sqp-composition-fri-roots prefix}))$
 $\text{composition-table prefix-state} \wedge$
 $\text{length trace-table} = \text{clength} * \text{scale} \wedge$
 $\text{length composition-table} = \text{clength} * \text{scale} \}$

definition $\text{staged-query-prefix-length-committed-query-target}$
 $:: 'f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow \text{nat set}$
where
 $\text{staged-query-prefix-length-committed-query-target prefix prefix-state} =$
 $\{ \text{idx} \in \text{query-sample-space.}$
 $\exists \text{ trace-table composition-table.}$
 $(\text{trace-table, composition-table}) \in$
 $\text{staged-query-prefix-length-committed-table-candidates prefix}$
 $\text{prefix-state} \wedge$
 $\text{idx} \in \text{query-sampling-success-space trace-table}$
 $\text{composition-table } (\text{sqp-alphas prefix}) \}$

definition $\text{staged-query-prefix-clean-length-committed-query-target}$
 $:: 'f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow \text{nat set}$
where
 $\text{staged-query-prefix-clean-length-committed-query-target prefix}$
 $\text{prefix-state} =$

(if hash-map-output-collision prefix-state then {}
 else staged-query-prefix-length-committed-query-target prefix
 prefix-state)

definition *checked-staged-query-prefix-hash-map-output-collision*
 $:: ((('f \text{ staged-query-prefix-data} \times 'f \text{ protocol-channel}) \times 'f) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-query-prefix-hash-map-output-collision out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((-, prefix-state), -), -) $\Rightarrow$
 hash-map-output-collision prefix-state)

definition *staged-query-prefix-committed-roots-clean*
 $:: 'f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow \text{bool}$

where

staged-query-prefix-committed-roots-clean prefix prefix-state $\longleftrightarrow$
 $\neg \text{merkle-root-binding-collision (sqp-trace-root prefix)}$
 prefix-state $\wedge$
 $(\forall \text{rt. sqp-composition-fri-roots prefix} \neq [] \longrightarrow$
 $\text{rt} = \text{hd (sqp-composition-fri-roots prefix)} \longrightarrow$
 $\neg \text{merkle-root-binding-collision rt prefix-state})$

definition *staged-query-prefix-clean-committed-query-target*
 $:: 'f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow \text{nat set}$

where

staged-query-prefix-clean-committed-query-target prefix prefix-state =
 (if staged-query-prefix-committed-roots-clean prefix prefix-state
 then staged-query-prefix-committed-query-target prefix prefix-state
 else {})

definition *checked-staged-query-prefix-committed-root-collision*
 $:: (((('f \text{ staged-query-prefix-data} \times 'f \text{ protocol-channel}) \times 'f) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-query-prefix-committed-root-collision out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((prefix, prefix-state), -), -) $\Rightarrow$
 $\neg \text{staged-query-prefix-committed-roots-clean prefix prefix-state})$

definition *staged-security-with-data-state-query-committed-prefix-hit*
 $:: 'f \text{ staged-adversary} \Rightarrow$
 $((('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

staged-security-with-data-state-query-committed-prefix-hit A out $\longleftrightarrow$
 (case out of

$None \Rightarrow False$
 $| Some (((data, attacker-state), -), -) \Rightarrow$
 $(\exists i \text{ prefix prefix-state raw raw-state.}$
 $i < rounds \wedge$
 $Some (data, attacker-state) \in$
 $set-dist$
 $(execute (checked-staged-transcript-program A)$
 $adversary-initial-state) \wedge$
 $Some (((prefix, prefix-state), raw), raw-state) \in$
 $set-dist$
 $(execute (checked-staged-query-prefix-receive-with-state A i)$
 $adversary-initial-state) \wedge$
 $checked-staged-query-prefix-dynamic-index-hit$
 $(\lambda prefix \text{ prefix-state.}$
 $staged-query-prefix-committed-query-target \text{ prefix prefix-state})$
 $(Some (((prefix, prefix-state), raw), raw-state))))$

definition *staged-security-with-data-state-query-length-committed-prefix-hit*

$:: 'f \text{ staged-adversary} \Rightarrow$
 $((('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \times unit \text{ list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow bool$

where

$staged-security-with-data-state-query-length-committed-prefix-hit A \text{ out} \longleftrightarrow$
 $(\text{case out of}$
 $None \Rightarrow False$
 $| Some (((data, attacker-state), -), -) \Rightarrow$
 $(\exists i \text{ prefix prefix-state raw raw-state.}$
 $i < rounds \wedge$
 $Some (data, attacker-state) \in$
 $set-dist$
 $(execute (checked-staged-transcript-program A)$
 $adversary-initial-state) \wedge$
 $Some (((prefix, prefix-state), raw), raw-state) \in$
 $set-dist$
 $(execute (checked-staged-query-prefix-receive-with-state A i)$
 $adversary-initial-state) \wedge$
 $checked-staged-query-prefix-dynamic-index-hit$
 $staged-query-prefix-length-committed-query-target$
 $(Some (((prefix, prefix-state), raw), raw-state))))$

definition *staged-security-with-data-state-query-tree-output-hit*

$:: 'f \text{ staged-adversary} \Rightarrow$
 $((('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \times unit \text{ list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow bool$

where

$staged-security-with-data-state-query-tree-output-hit A \text{ out} \longleftrightarrow$
 $(\text{case out of}$
 $None \Rightarrow False$
 $| Some (((data, attacker-state), result), final-state) \Rightarrow$

```

(let s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
in
  ( $\exists$  i prefix prefix-state raw raw-state trace-table
    composition-table as query-idxs trace-tree composition-tree.
    i < rounds  $\wedge$ 
    Some (data, attacker-state)  $\in$ 
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\wedge$ 
    Some (result, final-state)  $\in$ 
      set-dist (execute verify-monad s)  $\wedge$ 
      accepted-with-bound-tables s (Some (result, final-state))
      trace-table composition-table as query-idxs  $\wedge$ 
      fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some raw  $\wedge$ 
      index (to-nat raw)  $\in$ 
      query-sampling-success-space trace-table composition-table
      (staged-alphas data)  $\wedge$ 
      Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)  $\wedge$ 
      sqp-composition-fri-roots prefix  $\neq$  []  $\wedge$ 
      created-tree trace-table trace-tree final-state  $\wedge$ 
      sqp-trace-root prefix = value trace-tree  $\wedge$ 
      created-tree composition-table composition-tree final-state  $\wedge$ 
      hd (sqp-composition-fri-roots prefix) = value composition-tree  $\wedge$ 
      hash-map-new-output-hit
      (set-tree trace-tree  $\cup$  set-tree composition-tree)
      prefix-state final-state)))

```

lemma *staged-partial-query-target-subset-prefix-completion:*
assumes *match: staged-query-prefix-matches-data i prefix data*
and ext: *prefix-state $\leq$ attacker-state*
shows
 staged-partial-query-target data attacker-state $\subseteq$
 staged-query-prefix-partial-query-target i prefix prefix-state
unfolding *staged-query-prefix-partial-query-target-def*
using *match ext* **by** *blast*

lemma *staged-query-prefix-partial-query-target-subset:*
staged-query-prefix-partial-query-target i prefix prefix-state $\subseteq$
query-sample-space

unfolding *staged-query-prefix-partial-query-target-def*
staged-partial-query-target-def
using *query-header-supported-partial-union-good-sets-subset* **by** *blast*

lemma *staged-query-prefix-committed-query-target-subset*:
staged-query-prefix-committed-query-target prefix prefix-state $\subseteq$
query-sample-space
unfolding *staged-query-prefix-committed-query-target-def*
by (rule *query-header-committed-union-good-sets-subset*)

lemma *staged-query-prefix-committed-query-targetI*:
assumes *candidate*:
 (*trace-table*, *composition-table*) $\in$
query-header-committed-table-candidates prefix-state
 (*sqp-trace-root prefix*) (*sqp-composition-fri-roots prefix*)
and *hit*:
idx $\in$ *query-sampling-success-space trace-table composition-table*
 (*sqp-alphas prefix*)
shows *idx* $\in$ *staged-query-prefix-committed-query-target prefix prefix-state*
proof –
have *idx-space*: *idx* $\in$ *query-sample-space*
using *query-sampling-success-space-subset hit* **by** *blast*
show ?thesis
unfolding *staged-query-prefix-committed-query-target-def*
query-header-committed-union-good-sets-def
using *candidate hit idx-space* **by** *blast*
qed

lemma *staged-query-prefix-committed-query-target-fraction-bound-if-unique-candidate*:
assumes *unique*:
 $\exists$ *trace-table composition-table*.
query-header-committed-table-candidates prefix-state
 (*sqp-trace-root prefix*)
 (*sqp-composition-fri-roots prefix*) $\subseteq$
 {(*trace-table*, *composition-table*)}
shows
nnreal
 (card
 (*staged-query-prefix-committed-query-target prefix prefix-state*)) /
nnreal (card *query-sample-space*) $\leq$ *query-error-bound*
unfolding *staged-query-prefix-committed-query-target-def*
by (rule
query-header-committed-union-good-sets-fraction-bound-if-unique-candidate
 [OF *unique query-sampling-success-space-subset*
query-sampling-success-space-fraction-bound-query-sample-space])

lemma *staged-query-prefix-committed-query-target-fraction-bound-if-no-root-collisions*:
assumes *trace-clean*:
 $\neg$ *merkle-root-binding-collision* (*sqp-trace-root prefix*)

```

    prefix-state
  and comp-clean:
     $\wedge rt. \text{sqp-composition-fri-roots } prefix \neq [] \implies$ 
     $rt = hd (\text{sqp-composition-fri-roots } prefix) \implies$ 
     $\neg \text{merkle-root-binding-collision } rt \text{ } prefix\text{-state}$ 
  shows
    nnreal
    (card
      (staged-query-prefix-committed-query-target prefix prefix-state)) /
    nnreal (card query-sample-space)  $\leq$  query-error-bound
  proof -
    have unique:
       $\exists \text{trace-table composition-table.}$ 
      query-header-committed-table-candidates prefix-state
        (sqp-trace-root prefix)
        (sqp-composition-fri-roots prefix)  $\subseteq$ 
        {(trace-table, composition-table)}
    by (rule
      query-header-committed-table-candidates-subset-singleton-if-no-root-collisions
      [OF trace-clean comp-clean])
    show ?thesis
      by (rule
        staged-query-prefix-committed-query-target-fraction-bound-if-unique-candidate
        [OF unique])
  qed

lemma staged-query-prefix-length-committed-query-target-subset:
  staged-query-prefix-length-committed-query-target prefix prefix-state
     $\subseteq$  query-sample-space
  unfolding staged-query-prefix-length-committed-query-target-def by auto

lemma staged-query-prefix-length-committed-query-targetI:
  assumes candidate:
    (trace-table, composition-table)  $\in$ 
    staged-query-prefix-length-committed-table-candidates prefix prefix-state
  and hit:
    idx  $\in$  query-sampling-success-space trace-table composition-table
    (sqp-alphas prefix)
  shows
    idx  $\in$  staged-query-prefix-length-committed-query-target prefix
    prefix-state
  proof -
    have idx-space: idx  $\in$  query-sample-space
    using query-sampling-success-space-subset hit by blast
    show ?thesis
      unfolding staged-query-prefix-length-committed-query-target-def
      using candidate hit idx-space by blast
  qed

```

lemma *staged-query-prefix-clean-length-committed-query-target-subset:*
staged-query-prefix-clean-length-committed-query-target prefix prefix-state
 $\subseteq$ *query-sample-space*
unfolding *staged-query-prefix-clean-length-committed-query-target-def*
using *staged-query-prefix-length-committed-query-target-subset*
by *simp*

lemma *staged-query-prefix-length-committed-query-target-subset-committed:*
staged-query-prefix-length-committed-query-target prefix prefix-state
 $\subseteq$ *staged-query-prefix-committed-query-target prefix prefix-state*
unfolding *staged-query-prefix-length-committed-query-target-def*
staged-query-prefix-length-committed-table-candidates-def
staged-query-prefix-committed-query-target-def
query-header-committed-union-good-sets-def
query-header-committed-table-candidates-def
by *auto*

lemma *staged-query-prefix-length-committed-table-candidates-subset-singleton-if-no-hash-collision:*
assumes *clean: $\neg$ hash-map-output-collision prefix-state*
shows
 $\exists$ *trace-table composition-table.*
staged-query-prefix-length-committed-table-candidates prefix
prefix-state $\subseteq \{(trace-table, composition-table)\}$
proof (*cases*
staged-query-prefix-length-committed-table-candidates prefix
prefix-state = $\{\}$)
case *True*
then show *?thesis*
by (*intro exI[of -] exI[of -]*) *simp*
next
case *False*
then obtain *pair where pair:*
pair $\in$
staged-query-prefix-length-committed-table-candidates prefix
prefix-state
by *auto*
obtain *trace-table composition-table where pair-eq:*
pair = (trace-table, composition-table)
by (*cases pair*)
then have *candidate:*
(trace-table, composition-table) $\in$
staged-query-prefix-length-committed-table-candidates prefix
prefix-state
using *pair by simp*
have *trace-bind:*
merkle-root-binds-table (spp-trace-root prefix) trace-table
prefix-state
using *candidate*
unfolding *staged-query-prefix-length-committed-table-candidates-def*

```

  by simp
have comp-nonempty: sqp-composition-fri-roots prefix ≠ []
  using candidate
  unfolding staged-query-prefix-length-committed-table-candidates-def
  by simp
have comp-bind:
  merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
    composition-table prefix-state
  using candidate
  unfolding staged-query-prefix-length-committed-table-candidates-def
  by simp
have trace-len: length trace-table = clength * scale
  using candidate
  unfolding staged-query-prefix-length-committed-table-candidates-def
  by simp
have comp-len: length composition-table = clength * scale
  using candidate
  unfolding staged-query-prefix-length-committed-table-candidates-def
  by simp
have trace-nonempty: trace-table ≠ []
  using trace-len eval-domain-size-pos by auto
have comp-nonempty-table: composition-table ≠ []
  using comp-len eval-domain-size-pos by auto
have subset:
  staged-query-prefix-length-committed-table-candidates prefix
    prefix-state ⊆ {(trace-table, composition-table)}
proof
  fix pair
  assume pair-in:
    pair ∈
      staged-query-prefix-length-committed-table-candidates prefix
        prefix-state
  obtain trace-table' composition-table' where pair-eq:
    pair = (trace-table', composition-table')
  by (cases pair)
  have trace-bind':
    merkle-root-binds-table (sqp-trace-root prefix) trace-table'
      prefix-state
  using pair-in unfolding pair-eq
    staged-query-prefix-length-committed-table-candidates-def
  by simp
  have comp-bind':
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
      composition-table' prefix-state
  using pair-in unfolding pair-eq
    staged-query-prefix-length-committed-table-candidates-def
  by simp
  have trace-len':
    length trace-table' = clength * scale

```

```

using pair-in unfolding pair-eq
  staged-query-prefix-length-committed-table-candidates-def
by simp
have comp-len':
  length composition-table' = clength * scale
using pair-in unfolding pair-eq
  staged-query-prefix-length-committed-table-candidates-def
by simp
have trace-eq: trace-table' = trace-table
by (rule merkle-root-binds-same-length-tables-unique-if-no-hash-collision
  [OF trace-bind' trace-bind - clean])
  (simp add: trace-len' trace-len)
have comp-eq: composition-table' = composition-table
by (rule merkle-root-binds-same-length-tables-unique-if-no-hash-collision
  [OF comp-bind' comp-bind - clean])
  (simp add: comp-len' comp-len)
show pair ∈ {(trace-table, composition-table)}
  unfolding pair-eq trace-eq comp-eq by simp
qed
show ?thesis
  using subset by blast
qed

lemma staged-query-prefix-length-committed-query-target-fraction-bound-if-no-hash-collision:
assumes clean:  $\neg$  hash-map-output-collision prefix-state
shows
  nnreal
    (card
      (staged-query-prefix-length-committed-query-target prefix
        prefix-state)) /
    nnreal (card query-sample-space)  $\leq$  query-error-bound
proof –
from
  staged-query-prefix-length-committed-table-candidates-subset-singleton-if-no-hash-collision
  [OF clean, of prefix]
obtain trace-table composition-table where candidates:
  staged-query-prefix-length-committed-table-candidates prefix
  prefix-state  $\subseteq$  {(trace-table, composition-table)}
by blast
have union-subset:
  staged-query-prefix-length-committed-query-target prefix prefix-state
     $\subseteq$ 
  query-sampling-success-space trace-table composition-table
  (sqp-alphas prefix)
unfolding staged-query-prefix-length-committed-query-target-def
using candidates by auto
have finite-good:
  finite
    (query-sampling-success-space trace-table composition-table

```

```

    (sqp-alphas prefix))
  by (rule finite-subset[OF query-sampling-success-space-subset
    finite-query-sample-space])
have card-le:
  card
    (staged-query-prefix-length-committed-query-target prefix
      prefix-state) ≤
  card
    (query-sampling-success-space trace-table composition-table
      (sqp-alphas prefix))
  by (rule card-mono[OF finite-good union-subset])
have nnreal
  (card
    (staged-query-prefix-length-committed-query-target prefix
      prefix-state)) /
  nnreal (card query-sample-space) ≤
  nnreal
    (card
      (query-sampling-success-space trace-table composition-table
        (sqp-alphas prefix))) /
  nnreal (card query-sample-space)
  by (rule nnreal-nat-divide-right-mono[OF card-le])
also have ... ≤ query-error-bound
  by (rule query-sampling-success-space-fraction-bound-query-sample-space)
finally show ?thesis .
qed

lemma staged-query-prefix-clean-length-committed-query-target-fraction-bound:
  shows
    nnreal
      (card
        (staged-query-prefix-clean-length-committed-query-target prefix
          prefix-state)) /
      nnreal (card query-sample-space) ≤ query-error-bound
proof (cases hash-map-output-collision prefix-state)
case True
  then show ?thesis
    unfolding staged-query-prefix-clean-length-committed-query-target-def
    by simp
next
case False
have bound:
  nnreal
    (card
      (staged-query-prefix-length-committed-query-target prefix
        prefix-state)) /
    nnreal (card query-sample-space) ≤ query-error-bound
  by (rule
    staged-query-prefix-length-committed-query-target-fraction-bound-if-no-hash-collision

```



```

    [OF False])
  then show ?thesis
    using False
    unfolding staged-query-prefix-clean-length-committed-query-target-def
    by simp
qed

lemma checked-staged-query-prefix-length-committed-hit-imp-clean-hit-or-hash-collision:
  assumes hit:
    checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-length-committed-query-target out
  shows
    checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-clean-length-committed-query-target out ∨
    checked-staged-query-prefix-hash-map-output-collision out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding checked-staged-query-prefix-dynamic-index-hit-def by simp
next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state where out-eq:
    out = Some (((prefix, prefix-state), raw), raw-state)
    by (cases packed, auto split: prod.splits)
  show ?thesis
proof (cases hash-map-output-collision prefix-state)
  case True
  then have checked-staged-query-prefix-hash-map-output-collision out
    unfolding out-eq checked-staged-query-prefix-hash-map-output-collision-def
    by simp
  then show ?thesis by simp
next
  case False
  have
    checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-clean-length-committed-query-target out
    using hit False
    unfolding out-eq checked-staged-query-prefix-dynamic-index-hit-def
    staged-query-prefix-clean-length-committed-query-target-def
    by simp
  then show ?thesis by simp
qed
qed

lemma staged-query-prefix-clean-committed-query-target-subset:
  staged-query-prefix-clean-committed-query-target prefix prefix-state
  ⊆ query-sample-space
unfolding staged-query-prefix-clean-committed-query-target-def

```

```

using staged-query-prefix-committed-query-target-subset[of prefix prefix-state]
by simp

lemma staged-query-prefix-clean-committed-query-target-subset-committed:
  staged-query-prefix-clean-committed-query-target prefix prefix-state
     $\subseteq$  staged-query-prefix-committed-query-target prefix prefix-state
unfolding staged-query-prefix-clean-committed-query-target-def by auto

lemma staged-query-prefix-clean-committed-query-target-fraction-bound:
shows nnreal
  (card
    (staged-query-prefix-clean-committed-query-target prefix prefix-state)) /
    nnreal (card query-sample-space)  $\leq$  query-error-bound
proof (cases
  staged-query-prefix-committed-roots-clean prefix prefix-state)
case True
then have trace-clean:
   $\neg$  merkle-root-binding-collision (sqp-trace-root prefix)
    prefix-state
and comp-clean:
   $\bigwedge rt. \text{sqp-composition-fri-roots prefix} \neq [] \implies$ 
     $rt = \text{hd} (\text{sqp-composition-fri-roots prefix}) \implies$ 
     $\neg$  merkle-root-binding-collision rt prefix-state
unfolding staged-query-prefix-committed-roots-clean-def by blast+
show ?thesis
unfolding staged-query-prefix-clean-committed-query-target-def
using True
  staged-query-prefix-committed-query-target-fraction-bound-if-no-root-collisions
    [OF trace-clean comp-clean]
by simp
next
case False
then show ?thesis
unfolding staged-query-prefix-clean-committed-query-target-def by simp
qed

lemma checked-staged-query-prefix-committed-hit-imp-clean-hit-or-root-collision:
assumes hit:
  checked-staged-query-prefix-dynamic-index-hit
    ( $\lambda$ prefix prefix-state.
      staged-query-prefix-committed-query-target prefix prefix-state) out
shows
  checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-clean-committed-query-target out  $\vee$ 
    checked-staged-query-prefix-committed-root-collision out
proof (cases out)
case None
then show ?thesis
using hit

```

```

    unfolding checked-staged-query-prefix-dynamic-index-hit-def by simp
  next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state where out-eq:
    out = Some (((prefix, prefix-state), raw), raw-state)
    by (cases packed, auto split: prod.splits)
  show ?thesis
  proof (cases
    staged-query-prefix-committed-roots-clean prefix prefix-state)
  case True
  have clean-hit:
    checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-clean-committed-query-target out
  using hit True
  unfolding out-eq checked-staged-query-prefix-dynamic-index-hit-def
    staged-query-prefix-clean-committed-query-target-def
  by simp
  then show ?thesis by simp
  next
  case False
  then have
    checked-staged-query-prefix-committed-root-collision out
  unfolding out-eq checked-staged-query-prefix-committed-root-collision-def
  by simp
  then show ?thesis by simp
  qed
qed

```

```

lemma staged-partial-query-target-subset:
  staged-partial-query-target data attacker-state  $\subseteq$ 
  query-sample-space
unfolding staged-partial-query-target-def
by (rule query-header-supported-partial-union-good-sets-subset)

```

```

lemma staged-partial-query-target-fraction-bound-if-unique-candidate:
assumes unique:
   $\exists$  trace-table composition-table.
  query-header-supported-partial-table-candidates
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)  $\subseteq$ 
     $\{(trace-table, composition-table)\}$ 
shows

```

$$\text{nnreal } (\text{card } (\text{staged-partial-query-target data attacker-state})) /$$

$$\text{nnreal } (\text{card query-sample-space}) \leq \text{query-error-bound}$$

unfolding staged-partial-query-target-def

by (rule

query-header-supported-partial-union-good-sets-fraction-bound-if-unique-candidate

[OF unique query-sampling-success-space-subset

query-sampling-success-space-fraction-bound-query-sample-space])

definition checked-staged-query-prefix-fixed-transcript-target-hit

:: 'f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$

((('f staged-query-prefix-data $\times$ 'f protocol-channel) $\times$ 'f) $\times$

'f protocol-channel) option $\Rightarrow$ bool

where

checked-staged-query-prefix-fixed-transcript-target-hit data

attacker-state out $\longleftrightarrow$

checked-staged-query-prefix-dynamic-index-hit

(λ - -. staged-partial-query-target data attacker-state) out

lemma checked-staged-query-prefix-dynamic-index-hit-bound:

fixes C :: prob

assumes raw-bound: query-index-raw-preimage-bound

and prefix-bound:

$\bigwedge$ prefix prefix-state.

Some ((prefix, prefix-state), prefix-state) $\in$

set-dist

(execute (checked-staged-query-prefix-with-state A i) s) $\Rightarrow$

query-future-fresh prefix-state $\wedge$

B prefix prefix-state $\subseteq$ query-sample-space $\wedge$

nnreal (card (B prefix prefix-state)) /

nnreal (card query-sample-space) $\leq$ C

shows

wp-event (checked-staged-query-prefix-receive-with-state A i)

(checked-staged-query-prefix-dynamic-index-hit B) s $\leq$ C

unfolding checked-staged-query-prefix-receive-with-state-def

proof (rule wp-event-bind-bound-by-cont)

show $\neg$ checked-staged-query-prefix-dynamic-index-hit B None

unfolding checked-staged-query-prefix-dynamic-index-hit-def **by** simp

next

fix prefix-pack prefix-state

assume prefix-support:

Some (prefix-pack, prefix-state) $\in$

set-dist

(execute (checked-staged-query-prefix-with-state A i) s)

obtain prefix captured-state **where** prefix-pack-eq:

prefix-pack = (prefix, captured-state)

by (cases prefix-pack) simp

have captured-eq: captured-state = prefix-state

using checked-staged-query-prefix-with-state-outcome-state

[OF prefix-support[unfolded prefix-pack-eq]]

```

by simp
have support':
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-with-state A i) s)
using checked-staged-query-prefix-with-state-outcome-canonical
  [OF prefix-support[unfolded prefix-pack-eq]]
  captured-eq
by simp
from prefix-bound[OF support'] have future:
  query-future-fresh prefix-state
and subset:
  B prefix prefix-state ⊆ query-sample-space
and frac:
  nnreal (card (B prefix prefix-state)) /
    nnreal (card query-sample-space) ≤ C
by blast+
have receive-bound:
  wp-event
    (receive-query-index-challenge ≫= (λraw.
      return (prefix-pack, raw)))
    (checked-staged-query-prefix-dynamic-index-hit B) prefix-state ≤
    C
proof -
  let ?P =
    λout. case out of None ⇒ False
    | Some (raw, -) ⇒ index (to-nat raw) ∈ B prefix prefix-state
  have cont-eq:
    wp-event
      (receive-query-index-challenge ≫= (λraw.
        return (prefix-pack, raw)))
      (checked-staged-query-prefix-dynamic-index-hit B) prefix-state =
      wp-event receive-query-index-challenge ?P prefix-state
  unfolding wp-event-def
  apply (subst wp-bind)
  apply (rule arg-cong[where
    f = λQ. wp receive-query-index-challenge Q prefix-state])
  apply (rule ext)
  apply (simp add: wp-return prefix-pack-eq captured-eq
    checked-staged-query-prefix-dynamic-index-hit-def
    split: option.splits prod.splits)
  done
show ?thesis
  unfolding cont-eq
  by (rule order-trans
    [OF wp-receive-query-index-challenge-fresh-index-set-raw-bound
    [OF future raw-bound subset] frac])
qed
show wp-event

```

$(\text{receive-query-index-challenge} \gg (\lambda \text{raw}.$
 $\quad \text{return } (\text{prefix-pack}, \text{raw})))$
 $(\text{checked-staged-query-prefix-dynamic-index-hit } B) \text{ prefix-state} \leq$
 C
 by (rule receive-bound)
 qed

lemma *checked-staged-query-prefix-fixed-transcript-target-hit-bound:*
fixes $C :: \text{prob}$
assumes *raw-bound: query-index-raw-preimage-bound*
and *prefix-bound:*
 $\bigwedge \text{prefix prefix-state}.$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s) \implies$
 $\text{query-future-fresh prefix-state}$
and *subset: staged-partial-query-target data attacker-state $\subseteq$*
 $\text{query-sample-space}$
and *frac:*
 $\text{nnreal } (\text{card } (\text{staged-partial-query-target data attacker-state})) /$
 $\text{nnreal } (\text{card query-sample-space}) \leq C$
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-fixed-transcript-target-hit data}$
 $\text{attacker-state}) \ s \leq C$
unfolding *checked-staged-query-prefix-fixed-transcript-target-hit-def*
by (rule *checked-staged-query-prefix-dynamic-index-hit-bound*
 $[OF \text{ raw-bound}]$)
 (use *prefix-bound subset frac in auto*)

lemma *checked-staged-query-prefix-committed-target-hit-bound:*
assumes *raw-bound: query-index-raw-preimage-bound*
and *prefix-bound:*
 $\bigwedge \text{prefix prefix-state}.$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s) \implies$
 $\text{query-future-fresh prefix-state} \wedge$
 $\neg \text{merkle-root-binding-collision } (\text{sqp-trace-root prefix})$
 $\text{prefix-state} \wedge$
 $(\forall \text{rt. } \text{sqp-composition-fri-roots prefix} \neq [] \longrightarrow$
 $\text{rt} = \text{hd } (\text{sqp-composition-fri-roots prefix}) \longrightarrow$
 $\neg \text{merkle-root-binding-collision rt prefix-state})$
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $(\lambda \text{prefix prefix-state}.$
 $\text{staged-query-prefix-committed-query-target prefix prefix-state}))$
 $s \leq \text{query-error-bound}$

proof (*rule checked-staged-query-prefix-dynamic-index-hit-bound*
[OF raw-bound])
fix *prefix prefix-state*
assume *support*:
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i) s)
from *prefix-bound[OF support]*
have *future: query-future-fresh prefix-state*
and *trace-clean*:
 $\neg$ *merkle-root-binding-collision (sqp-trace-root prefix)*
prefix-state
and *comp-clean*:
 $\bigwedge rt. \text{sqp-composition-fri-roots } prefix \neq [] \implies$
 $rt = hd (\text{sqp-composition-fri-roots } prefix) \implies$
 $\neg \text{merkle-root-binding-collision } rt \text{ prefix-state}$
by *blast+*
show
query-future-fresh prefix-state $\wedge$
staged-query-prefix-committed-query-target prefix prefix-state
 $\subseteq$ *query-sample-space* $\wedge$
nnreal
(card
(staged-query-prefix-committed-query-target prefix prefix-state)) /
nnreal (card query-sample-space) ≤ query-error-bound
using *future*
staged-query-prefix-committed-query-target-subset[of prefix prefix-state]
staged-query-prefix-committed-query-target-fraction-bound-if-no-root-collisions
[OF trace-clean comp-clean]
by *blast*
qed

definition *checked-staged-query-prefix-dynamic-index-prehit*
 $:: ('f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow$
 $\text{nat set}) \Rightarrow$
 $((('f \text{ staged-query-prefix-data} \times 'f \text{ protocol-channel}) \times 'f) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-query-prefix-dynamic-index-prehit B out $\longleftrightarrow$
(case out of
None $\Rightarrow$ *False*
 $|$ *Some (((prefix, prefix-state), -), -)* $\Rightarrow$
 $(\exists \text{raw} \in \text{query-index-raw-preimage } (B \text{ prefix prefix-state}).$
 $\text{fmlookup } (\text{HashMap } \text{prefix-state})$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } \text{prefix-state})$
 $(P\text{State } \text{prefix-state})) = \text{Some raw}))$

definition *checked-staged-query-prefix-dynamic-index-prehit-relation*
 $:: 'f \text{ staged-adversary} \Rightarrow \text{nat} \Rightarrow 'f \text{ protocol-channel} \Rightarrow$

$(\text{'f staged-query-prefix-data} \Rightarrow \text{'f protocol-channel} \Rightarrow \text{nat set}) \Rightarrow$
 $\text{'f protocol-hash-input} \Rightarrow \text{'f} \Rightarrow \text{bool}$
where
checked-staged-query-prefix-dynamic-index-prehit-relation $A \ i \ s \ B \ x \ y$
 $\longleftrightarrow$
 $(\exists \text{ prefix prefix-state.}$
 $\text{Some } ((\text{prefix, prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s) \wedge$
 $x = \text{QueryIndexChallenge } (P\text{QueryCounter prefix-state})$
 $(P\text{State prefix-state}) \wedge$
 $y \in \text{query-index-raw-preimage } (B \ \text{prefix prefix-state}))$

lemma *checked-staged-query-prefix-dynamic-index-prehit-imp-relation-hit:*

assumes *empty*: $\text{HashMap } s = \text{fmempty}$
and *prefix-support*:
 $\text{Some } ((\text{prefix, prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s)$
and *receive-support*:
 $\text{Some } (((\text{prefix, prefix-state}), \text{raw}), \text{raw-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-receive-with-state } A \ i) \ s)$
and *prehit*:
 $\text{checked-staged-query-prefix-dynamic-index-prehit } B$
 $(\text{Some } (((\text{prefix, prefix-state}), \text{raw}), \text{raw-state})))$
shows
 $\text{hash-relation-hit-event}$
 $(\text{checked-staged-query-prefix-dynamic-index-prehit-relation } A \ i \ s \ B)$
 $s \ (\text{Some } (((\text{prefix, prefix-state}), \text{raw}), \text{raw-state})))$

proof —

from *prehit* **obtain** *old-raw* **where** *old-raw-in*:
 $\text{old-raw} \in \text{query-index-raw-preimage } (B \ \text{prefix prefix-state})$
and *lookup-prefix*:
 $\text{fmlookup } (\text{HashMap prefix-state})$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter prefix-state})$
 $(P\text{State prefix-state})) = \text{Some old-raw}$
unfolding *checked-staged-query-prefix-dynamic-index-prehit-def*
by *auto*
have *recv*:
 $\text{Some } (\text{raw, raw-state}) \in$
 $\text{set-dist } (\text{execute receive-query-index-challenge prefix-state})$

proof —

from *receive-support* **obtain** *u* **where**
prefix-u:
 $\text{Some } ((\text{prefix, prefix-state}), u) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s)$


```

and recv-u:
  Some (raw, raw-state)  $\in$ 
    set-dist (execute receive-query-index-challenge u)
unfolding checked-staged-query-prefix-receive-with-state-def
by (auto elim!: set-dist-bindE)
have u = prefix-state
by (rule checked-staged-query-prefix-with-state-outcome-state
    [OF prefix-u])
then show ?thesis
  using recv-u by simp
qed
have prefix-ext: prefix-state  $\leq$  raw-state
by (rule receive-query-index-challenge-extends[OF recv])
have lookup-raw-state:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge (PQueryCounter prefix-state)
      (PState prefix-state)) = Some old-raw
by (rule hash-extension-lookup[OF lookup-prefix prefix-ext])
have lookup-initial:
  fmlookup (HashMap s)
    (QueryIndexChallenge (PQueryCounter prefix-state)
      (PState prefix-state)) = None
using empty by simp
have rel:
  checked-staged-query-prefix-dynamic-index-prehit-relation A i s B
    (QueryIndexChallenge (PQueryCounter prefix-state)
      (PState prefix-state)) old-raw
unfolding
  checked-staged-query-prefix-dynamic-index-prehit-relation-def
using prefix-support old-raw-in by blast
have hit:
  hash-relation-hit
    (checked-staged-query-prefix-dynamic-index-prehit-relation A i s B)
    s raw-state
unfolding hash-relation-hit-def
by (intro exI[of -
    QueryIndexChallenge (PQueryCounter prefix-state)
      (PState prefix-state)
    exI[of - old-raw] conjI)
    (use lookup-initial lookup-raw-state rel in simp-all)
show ?thesis
unfolding hash-relation-hit-event-def using hit by simp
qed

lemma checked-staged-query-prefix-dynamic-index-prehit-bound-by-relation-hit:
assumes empty: HashMap s = fmempty
shows
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit B) s  $\leq$ 

```

```

    wp-event (checked-staged-query-prefix-receive-with-state A i)
      (hash-relation-hit-event
        (checked-staged-query-prefix-dynamic-index-prehit-relation A i s B)
          s) s
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i) s)
and prehit: checked-staged-query-prefix-dynamic-index-prehit B out
show
  hash-relation-hit-event
    (checked-staged-query-prefix-dynamic-index-prehit-relation A i s B)
    s out
proof (cases out)
case None
then show ?thesis
  using prehit
  unfolding checked-staged-query-prefix-dynamic-index-prehit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state where out-eq:
  out = Some (((prefix, prefix-state), raw), raw-state)
  by (cases packed, auto split: prod.splits)
have receive-support:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i) s)
  using support unfolding out-eq .
have prefix-support:
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-with-state A i) s)
proof –
  from receive-support obtain u where
    prefix-u:
      Some ((prefix, prefix-state), u) ∈
        set-dist
          (execute (checked-staged-query-prefix-with-state A i) s)
  unfolding checked-staged-query-prefix-receive-with-state-def
  by (auto elim!: set-dist-bindE)
show ?thesis
  by (rule checked-staged-query-prefix-with-state-outcome-canonical
    [OF prefix-u])
qed
show ?thesis
  unfolding out-eq

```

```

    by (rule checked-staged-query-prefix-dynamic-index-prehit-imp-relation-hit
        [OF empty prefix-support receive-support])
    (use prehit out-eq in simp)
qed
qed

lemma checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program:
  assumes empty: HashMap s = fmempty
  and relation-program:
    hash-relation-program
    (checked-staged-query-prefix-dynamic-index-prehit-relation A i s B)
    fiber-bound query-bound
    (checked-staged-query-prefix-receive-with-state A i)
  shows
    wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit B) s ≤
    staged-phase-relation-error fiber-bound query-bound
proof -
  have prehit-le-relation:
    wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit B) s ≤
    wp-event (checked-staged-query-prefix-receive-with-state A i)
    (hash-relation-hit-event
    (checked-staged-query-prefix-dynamic-index-prehit-relation A i s B)
    s) s
  by (rule
    checked-staged-query-prefix-dynamic-index-prehit-bound-by-relation-hit
    [OF empty])
  also have ... ≤ staged-phase-relation-error fiber-bound query-bound
  using relation-program
  unfolding hash-relation-program-def hash-relation-budget-def
    staged-phase-relation-error-def
  by blast
  finally show ?thesis .
qed

lemma hash-map-preserving-get:
  hash-map-preserving get
  unfolding hash-map-preserving-def by simp

lemma hash-relation-program-get:
  hash-relation-program R b 0 get
  by (rule hash-relation-program-zero)
  (rule hash-map-preserving-get)

lemma hash-relation-program-modify-preserves-hash-map:
  assumes bump:
     $\bigwedge s. \text{HashMap } (\text{bump } s) = \text{HashMap } s$ 
  shows hash-relation-program R b 0 (modify bump)

```

by (rule hash-relation-program-zero)
 (rule hash-map-preserving-modify[OF bump])

lemma hash-relation-program-receive-counted-tagged-random-field-element:

assumes bump:

$\bigwedge s. \text{HashMap } (\text{bump } s) = \text{HashMap } s$

and fibers: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$

shows

hash-relation-program $R \ b \ 1$

(protocol-receive-counted-tagged-random-field-element counter bump tag)

proof –

have modify-return:

hash-relation-program $R \ b \ (0 + 0)$

(modify bump $\gg=$ ($\lambda_. \text{return } r$))

for $r :: 'f$

by (rule hash-relation-program-bind)

(rule hash-relation-program-modify-preserves-hash-map[OF bump],

rule hash-relation-program-zero[OF hash-map-preserving-return])

have hash-tail:

hash-relation-program $R \ b \ (1 + (0 + 0))$

(hash (tag (counter s) (PState s)) $\gg=$

($\lambda r. \text{modify bump } \gg= (\lambda_. \text{return } r)$))

for s

by (rule hash-relation-program-bind)

(rule hash-relation-program-hash[OF fibers], rule modify-return)

have hash-relation-program $R \ b \ (0 + (1 + (0 + 0)))$

(get $\gg=$

($\lambda s.$

hash (tag (counter s) (PState s)) $\gg=$

($\lambda r. \text{modify bump } \gg= (\lambda_. \text{return } r)$)))

by (rule hash-relation-program-bind)

(rule hash-relation-program-get, rule hash-tail)

then show ?thesis

unfolding protocol-receive-counted-tagged-random-field-element-def

by simp

qed

lemma hash-relation-program-receive-query-index-challenge:

assumes fibers: $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$

shows hash-relation-program $R \ b \ 1$

receive-query-index-challenge

unfolding receive-query-index-challenge-def

by (rule hash-relation-program-receive-counted-tagged-random-field-element
 [OF - fibers])

simp

lemma hash-map-preserving-get-bind-return-pair:

hash-map-preserving (get $\gg= (\lambda s. \text{return } (x, s))$)

unfolding hash-map-preserving-def

```

by (auto elim!: set-dist-bindE)

lemma hash-map-preserving-imp-hash-range-budget-zero:
  fixes m :: ('r, ('f, 'a) protocol-channel-scheme) state-monad
  assumes preserving: hash-map-preserving m
  shows hash-range-budget 0 m
  unfolding hash-range-budget-def
proof (intro allI impI)
  fix s y t
  assume Some (y, t) ∈ set-dist (execute m s)
  then have HashMap t = HashMap s
    using preserving unfolding hash-map-preserving-def by blast
  then show
    card (hash-map-output-values t) ≤
      card (hash-map-output-values s) + 0
  by (simp add: hash-map-output-values-def)
qed

lemma hash-map-preserving-imp-hash-collision-budget-zero-scheme:
  fixes m :: ('r, ('f, 'a) protocol-channel-scheme) state-monad
  assumes preserving: hash-map-preserving m
  shows hash-collision-budget 0 m
  unfolding hash-collision-budget-def
proof (intro allI impI)
  fix s :: ('f, 'a) protocol-channel-scheme
  assume clean: ¬ hash-map-output-collision s
  have no-event:
     $\bigwedge \text{out. out} \in \text{set-dist (execute m s)} \implies$ 
     $\neg \text{hash-new-collision-event s out}$ 
  proof -
    fix out
    assume support: out ∈ set-dist (execute m s)
    show ¬ hash-new-collision-event s out
    proof (cases out)
      case None
      then show ?thesis
        unfolding hash-new-collision-event-def by simp
    next
      case (Some xt)
      then obtain x t where xt: xt = (x, t)
        by (cases xt) simp
      have same: HashMap t = HashMap s
        using preserving support Some xt
        unfolding hash-map-preserving-def by blast
      show ?thesis
        using clean same Some xt
        unfolding hash-new-collision-event-def
        hash-map-new-output-collision-def hash-map-output-collision-def
        by simp
    end
  end

```

```

qed
qed
have no-event-dom:
   $\bigwedge out. out \in dom (dist (execute\ m\ s)) \implies$ 
     $\neg hash\_new\_collision\_event\ s\ out$ 
  using no-event unfolding set-dist-def by blast
have wp-event m (hash-new-collision-event s) s = 0
  unfolding wp-event-def wp-def dist-expect-def
  by (intro sum.neutral) (simp add: no-event-dom)
then show
  wp-event m (hash-new-collision-event s) s  $\leq$ 
    hash-collision-budget-value
    (card (hash-map-output-values s)) 0
  unfolding hash-collision-budget-value-def by simp
qed

lemma checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound:
  assumes fibers:
     $\bigwedge x. card$ 
       $(\bigcup \{query\_index\_raw\_preimage\ (B\ prefix\ prefix\_state) \mid$ 
         $prefix\ prefix\_state.$ 
         $Some\ ((prefix,\ prefix\_state),\ prefix\_state) \in$ 
           $set\_dist$ 
           $(execute\ (checked\_staged\_query\_prefix\_with\_state\ A\ i)\ s) \wedge$ 
           $x = QueryIndexChallenge\ (PQueryCounter\ prefix\_state)$ 
           $(PState\ prefix\_state)\}) \leq b$ 
  shows
    card
       $\{y. checked\_staged\_query\_prefix\_dynamic\_index\_prehit\_relation$ 
         $A\ i\ s\ B\ x\ y\} \leq b$ 
  proof -
    let ?U =
       $\bigcup \{query\_index\_raw\_preimage\ (B\ prefix\ prefix\_state) \mid$ 
         $prefix\ prefix\_state.$ 
         $Some\ ((prefix,\ prefix\_state),\ prefix\_state) \in$ 
           $set\_dist$ 
           $(execute\ (checked\_staged\_query\_prefix\_with\_state\ A\ i)\ s) \wedge$ 
           $x = QueryIndexChallenge\ (PQueryCounter\ prefix\_state)$ 
           $(PState\ prefix\_state)\}$ 
    have subset:
       $\{y. checked\_staged\_query\_prefix\_dynamic\_index\_prehit\_relation$ 
         $A\ i\ s\ B\ x\ y\} \subseteq ?U$ 
    unfolding checked-staged-query-prefix-dynamic-index-prehit-relation-def
    by blast
    have finite-U: finite ?U
    by simp
    have card
       $\{y. checked\_staged\_query\_prefix\_dynamic\_index\_prehit\_relation$ 
         $A\ i\ s\ B\ x\ y\} \leq card\ ?U$ 

```

```

    by (rule card-mono[OF finite-U subset])
  also have ... ≤ b
    by (rule fibers)
  finally show ?thesis .
qed

lemma checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size:
  card
    {y. checked-staged-query-prefix-dynamic-index-prehit-relation
      A i s B x y} ≤ size
proof -
  have subset:
    {y. checked-staged-query-prefix-dynamic-index-prehit-relation
      A i s B x y} ⊆ (UNIV :: 'f set)
  by simp
  have finite-field: finite (UNIV :: 'f set)
  by simp
  have card
    {y. checked-staged-query-prefix-dynamic-index-prehit-relation
      A i s B x y} ≤ card (UNIV :: 'f set)
  by (rule card-mono[OF finite-field subset])
  also have ... = size
    using size-card by simp
  finally show ?thesis .
qed

lemma hash-relation-program-checked-staged-query-prefix-with-state:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound: i ≤ rounds
    and fibers:  $\bigwedge x. \text{card } \{y. R \ x \ y\} \leq b$ 
  shows
    hash-relation-program R b
      (staged-query-search-queries budgets i)
      (checked-staged-query-prefix-with-state A i)
  unfolding checked-staged-query-prefix-with-state-def
proof -
  have prefix:
    hash-relation-program R b
      (staged-query-search-queries budgets i)
      (checked-staged-query-challenge-prefix-program A i)
  by (rule hash-relation-program-checked-staged-query-challenge-prefix-program
    [OF wf controlled i-bound fibers])
  have tail:
     $\bigwedge \text{prefix}. \text{hash-relation-program } R \ b \ 0$ 
      (get  $\gg$  ( $\lambda \text{prefix-state}. \text{return } (\text{prefix}, \text{prefix-state})$ )))
  by (rule hash-relation-program-zero)
    (rule hash-map-preserving-get-bind-return-pair)
  have hash-relation-program R b

```

```

    (staged-query-search-queries budgets  $i + 0$ )
    (checked-staged-query-challenge-prefix-program  $A\ i \gg=$ 
      ( $\lambda$ prefix. get  $\gg=$  ( $\lambda$ prefix-state.
        return (prefix, prefix-state))))
  by (rule hash-relation-program-bind)
    (rule prefix, rule tail)
then show hash-relation-program  $R\ b$ 
  (staged-query-search-queries budgets  $i$ )
  (checked-staged-query-challenge-prefix-program  $A\ i \gg=$ 
    ( $\lambda$ prefix. get  $\gg=$  ( $\lambda$ prefix-state.
      return (prefix, prefix-state))))
  by simp
qed

lemma hash-relation-program-checked-staged-query-prefix-receive-with-state:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets  $A$ 
  and i-bound:  $i \leq \text{rounds}$ 
  and fibers:  $\bigwedge x. \text{card } \{y. R\ x\ y\} \leq b$ 
shows
  hash-relation-program  $R\ b$ 
    (staged-query-search-queries budgets  $i + 1$ )
    (checked-staged-query-prefix-receive-with-state  $A\ i$ )
unfolding checked-staged-query-prefix-receive-with-state-def
proof –
  have prefix:
    hash-relation-program  $R\ b$ 
      (staged-query-search-queries budgets  $i$ )
      (checked-staged-query-prefix-with-state  $A\ i$ )
  by (rule hash-relation-program-checked-staged-query-prefix-with-state
    [OF wf controlled i-bound fibers])
  have receive:
     $\bigwedge$ prefix-pack. hash-relation-program  $R\ b\ 1$ 
      (receive-query-index-challenge  $\gg=$  ( $\lambda$ raw.
        return (prefix-pack, raw)))
  proof –
    fix prefix-pack
    have hash-relation-program  $R\ b\ (1 + 0)$ 
      (receive-query-index-challenge  $\gg=$  ( $\lambda$ raw.
        return (prefix-pack, raw)))
    by (rule hash-relation-program-bind)
      (rule hash-relation-program-receive-query-index-challenge[OF fibers],
        rule hash-relation-program-zero[OF hash-map-preserving-return])
    then show hash-relation-program  $R\ b\ 1$ 
      (receive-query-index-challenge  $\gg=$  ( $\lambda$ raw.
        return (prefix-pack, raw)))
    by simp
  qed
show hash-relation-program  $R\ b$ 

```



```

    (staged-query-search-queries budgets  $i + 1$ )
    (checked-staged-query-prefix-with-state  $A\ i \gg=$ 
      ( $\lambda$ prefix-pack. receive-query-index-challenge  $\gg=$ 
        ( $\lambda$ raw. return (prefix-pack, raw))))
  by (rule hash-relation-program-bind)
    (rule prefix, rule receive)
qed

lemma hash-range-budget-checked-staged-query-prefix-with-state:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets  $A$ 
  and i-bound:  $i \leq \text{rounds}$ 
  shows
    hash-range-budget
      (staged-query-search-queries budgets  $i$ )
      (checked-staged-query-prefix-with-state  $A\ i$ )
  unfolding checked-staged-query-prefix-with-state-def
proof –
  have prefix:
    hash-range-budget
      (staged-query-search-queries budgets  $i$ )
      (checked-staged-query-challenge-prefix-program  $A\ i$ )
  by (rule hash-range-budget-checked-staged-query-challenge-prefix-program
    [OF wf controlled i-bound])
  have tail:
     $\wedge$ prefix. hash-range-budget 0
      (get  $\gg=$  ( $\lambda$ prefix-state. return (prefix, prefix-state)))
  by (rule hash-map-preserving-imp-hash-range-budget-zero)
    (rule hash-map-preserving-get-bind-return-pair)
  have hash-range-budget
    (staged-query-search-queries budgets  $i + 0$ )
    (checked-staged-query-challenge-prefix-program  $A\ i \gg=$ 
      ( $\lambda$ prefix. get  $\gg=$  ( $\lambda$ prefix-state.
        return (prefix, prefix-state))))
  by (rule hash-range-budget-bind)
    (rule prefix, rule tail)
  then show hash-range-budget
    (staged-query-search-queries budgets  $i$ )
    (checked-staged-query-challenge-prefix-program  $A\ i \gg=$ 
      ( $\lambda$ prefix. get  $\gg=$  ( $\lambda$ prefix-state.
        return (prefix, prefix-state))))
  by simp
qed

```

```

lemma hash-collision-budget-checked-staged-query-prefix-with-state:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets  $A$ 
  and i-bound:  $i \leq \text{rounds}$ 
  shows

```

```

    hash-collision-budget
      (staged-query-search-queries budgets i)
      (checked-staged-query-prefix-with-state A i)
  unfolding checked-staged-query-prefix-with-state-def
proof -
  have prefix-range:
    hash-range-budget
      (staged-query-search-queries budgets i)
      (checked-staged-query-challenge-prefix-program A i)
  by (rule hash-range-budget-checked-staged-query-challenge-prefix-program
      [OF wf controlled i-bound])
  have prefix-collision:
    hash-collision-budget
      (staged-query-search-queries budgets i)
      (checked-staged-query-challenge-prefix-program A i)
  by (rule hash-collision-budget-checked-staged-query-challenge-prefix-program
      [OF wf controlled i-bound])
  have tail-range:
     $\bigwedge$  prefix. hash-range-budget 0
      (get  $\gg$  ( $\lambda$ prefix-state. return (prefix, prefix-state)))
  by (rule hash-map-preserving-imp-hash-range-budget-zero)
      (rule hash-map-preserving-get-bind-return-pair)
  have tail-collision:
     $\bigwedge$  prefix. hash-collision-budget 0
      (get  $\gg$  ( $\lambda$ prefix-state. return (prefix, prefix-state)))
  by (rule hash-map-preserving-imp-hash-collision-budget-zero-scheme)
      (rule hash-map-preserving-get-bind-return-pair)
  have hash-collision-budget
    (staged-query-search-queries budgets i + 0)
    (checked-staged-query-challenge-prefix-program A i  $\gg$ 
      ( $\lambda$ prefix. get  $\gg$  ( $\lambda$ prefix-state.
        return (prefix, prefix-state)))))
  by (rule hash-collision-budget-bind)
      (rule prefix-range, rule prefix-collision, rule tail-range,
        rule tail-collision)
  then show hash-collision-budget
    (staged-query-search-queries budgets i)
    (checked-staged-query-challenge-prefix-program A i  $\gg$ 
      ( $\lambda$ prefix. get  $\gg$  ( $\lambda$ prefix-state.
        return (prefix, prefix-state)))))
  by simp
qed

lemma hash-range-budget-checked-staged-query-prefix-receive-with-state:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i  $\leq$  rounds
  shows
    hash-range-budget

```

```

    (staged-query-search-queries budgets  $i + 1$ )
    (checked-staged-query-prefix-receive-with-state  $A\ i$ )
unfolding checked-staged-query-prefix-receive-with-state-def
proof –
  have prefix:
    hash-range-budget
      (staged-query-search-queries budgets  $i$ )
      (checked-staged-query-prefix-with-state  $A\ i$ )
    by (rule hash-range-budget-checked-staged-query-prefix-with-state
      [OF wf controlled i-bound])
  have receive:
     $\bigwedge$ prefix-pack. hash-range-budget 1
      (receive-query-index-challenge  $\gg$  ( $\lambda$ raw.
        return (prefix-pack, raw)))
  proof –
    fix prefix-pack
    have hash-range-budget ( $1 + 0$ )
      (receive-query-index-challenge  $\gg$  ( $\lambda$ raw.
        return (prefix-pack, raw)))
    by (rule hash-range-budget-bind)
      (rule hash-range-budget-receive-query-index-challenge,
        rule hash-range-budget-return)
    then show hash-range-budget 1
      (receive-query-index-challenge  $\gg$  ( $\lambda$ raw.
        return (prefix-pack, raw)))
    by simp
  qed
show hash-range-budget
  (staged-query-search-queries budgets  $i + 1$ )
  (checked-staged-query-prefix-with-state  $A\ i$   $\gg$ 
    ( $\lambda$ prefix-pack. receive-query-index-challenge  $\gg$ 
      ( $\lambda$ raw. return (prefix-pack, raw)))))
  by (rule hash-range-budget-bind)
    (rule prefix, rule receive)
qed

lemma hash-collision-budget-checked-staged-query-prefix-receive-with-state:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and i-bound:  $i \leq \text{rounds}$ 
shows
  hash-collision-budget
    (staged-query-search-queries budgets  $i + 1$ )
    (checked-staged-query-prefix-receive-with-state  $A\ i$ )
unfolding checked-staged-query-prefix-receive-with-state-def
proof –
  have prefix-range:
    hash-range-budget
      (staged-query-search-queries budgets  $i$ )

```

(*checked-staged-query-prefix-with-state A i*)
by (rule *hash-range-budget-checked-staged-query-prefix-with-state*
 [*OF wf controlled i-bound*])
have *prefix-collision*:
hash-collision-budget
 (*staged-query-search-queries budgets i*)
 (*checked-staged-query-prefix-with-state A i*)
by (rule *hash-collision-budget-checked-staged-query-prefix-with-state*
 [*OF wf controlled i-bound*])
have *receive-range*:
 $\wedge$ *prefix-pack. hash-range-budget 1*
 (*receive-query-index-challenge* $\gg$ (λ *raw.*
return (prefix-pack, raw)))
proof –
fix *prefix-pack*
have *hash-range-budget (1 + 0)*
 (*receive-query-index-challenge* $\gg$ (λ *raw.*
return (prefix-pack, raw)))
by (rule *hash-range-budget-bind*)
 (rule *hash-range-budget-receive-query-index-challenge*,
 rule *hash-range-budget-return*)
then show *hash-range-budget 1*
 (*receive-query-index-challenge* $\gg$ (λ *raw.*
return (prefix-pack, raw)))
by simp
qed
have *receive-collision*:
 $\wedge$ *prefix-pack. hash-collision-budget 1*
 (*receive-query-index-challenge* $\gg$ (λ *raw.*
return (prefix-pack, raw)))
proof –
fix *prefix-pack*
have *hash-collision-budget (1 + 0)*
 (*receive-query-index-challenge* $\gg$ (λ *raw.*
return (prefix-pack, raw)))
by (rule *hash-collision-budget-bind*)
 (rule *hash-range-budget-receive-query-index-challenge*,
 rule *hash-collision-budget-receive-query-index-challenge*,
 rule *hash-range-budget-return*, rule *hash-collision-budget-return*)
then show *hash-collision-budget 1*
 (*receive-query-index-challenge* $\gg$ (λ *raw.*
return (prefix-pack, raw)))
by simp
qed
show *hash-collision-budget*
 (*staged-query-search-queries budgets i + 1*)
 (*checked-staged-query-prefix-with-state A i* $\gg$
 (λ *prefix-pack. receive-query-index-challenge* $\gg$
 (λ *raw. return (prefix-pack, raw)*)))

```

    by (rule hash-collision-budget-bind)
      (rule prefix-range, rule prefix-collision, rule receive-range,
       rule receive-collision)
qed

lemma hash-map-output-collision-mono:
  assumes collision: hash-map-output-collision s
    and ext:  $s \leq t$ 
  shows hash-map-output-collision t
proof -
  obtain  $x\ y\ z$  where neq:  $x \neq y$ 
    and x-lookup:  $\text{fmlookup } (\text{HashMap } s) \ x = \text{Some } z$ 
    and y-lookup:  $\text{fmlookup } (\text{HashMap } s) \ y = \text{Some } z$ 
    using collision unfolding hash-map-output-collision-def by blast
  have x-lookup-t:  $\text{fmlookup } (\text{HashMap } t) \ x = \text{Some } z$ 
    by (rule hash-extension-lookup[OF x-lookup ext])
  have y-lookup-t:  $\text{fmlookup } (\text{HashMap } t) \ y = \text{Some } z$ 
    by (rule hash-extension-lookup[OF y-lookup ext])
  show ?thesis
    by (rule hash-map-output-collisionI[OF neq x-lookup-t y-lookup-t])
qed

lemma checked-staged-query-prefix-hash-map-output-collision-imp-new-collision-on-support:
  assumes support:
    out  $\in$  set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
     adversary-initial-state)
    and collision:
      checked-staged-query-prefix-hash-map-output-collision out
  shows hash-new-collision-event adversary-initial-state out
proof (cases out)
  case None
  then show ?thesis
    using collision
    unfolding checked-staged-query-prefix-hash-map-output-collision-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state where out-eq:
    out = Some (((prefix, prefix-state), raw), raw-state)
    by (cases packed) auto
  have receive-support:
    Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
     adversary-initial-state)
    using support unfolding out-eq .
  have recv:
    Some (raw, raw-state)  $\in$ 

```

```

    set-dist (execute receive-query-index-challenge prefix-state)
  by (rule
    checked-staged-query-prefix-receive-with-state-receive-support
    [OF receive-support])
  have prefix-collision: hash-map-output-collision prefix-state
  using collision unfolding out-eq
    checked-staged-query-prefix-hash-map-output-collision-def by simp
  have raw-ext: prefix-state ≤ raw-state
  by (rule receive-query-index-challenge-extends[OF recv])
  have raw-collision: hash-map-output-collision raw-state
  by (rule hash-map-output-collision-mono[OF prefix-collision raw-ext])
  show ?thesis
  unfolding out-eq hash-new-collision-event-def
    hash-map-new-output-collision-def
  using raw-collision by simp
qed

lemma checked-staged-query-prefix-hash-map-output-collision-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i ≤ rounds
  shows
    wp-event (checked-staged-query-prefix-receive-with-state A i)
      checked-staged-query-prefix-hash-map-output-collision
      adversary-initial-state ≤
      hash-collision-budget-value 0
      (staged-query-search-queries budgets i + 1)
  proof -
    let ?M = checked-staged-query-prefix-receive-with-state A i
    let ?q = staged-query-search-queries budgets i + 1
    have event-le:
      wp-event ?M checked-staged-query-prefix-hash-map-output-collision
        adversary-initial-state ≤
      wp-event ?M (hash-new-collision-event adversary-initial-state)
        adversary-initial-state
    by (rule wp-event-mono-on-support)
    (rule
      checked-staged-query-prefix-hash-map-output-collision-imp-new-collision-on-support)
    have collision-budget:
      hash-collision-budget ?q ?M
    by (rule hash-collision-budget-checked-staged-query-prefix-receive-with-state
      [OF wf controlled i-bound])
    have new-collision-bound:
      wp-event ?M (hash-new-collision-event adversary-initial-state)
        adversary-initial-state ≤
      hash-collision-budget-value
        (card (hash-map-output-values adversary-initial-state)) ?q
    using collision-budget adversary-initial-state-no-output-collision
    unfolding hash-collision-budget-def by blast

```

have *wp-event* ?*M* *checked-staged-query-prefix-hash-map-output-collision*
adversary-initial-state $\leq$
hash-collision-budget-value
(*card* (*hash-map-output-values* *adversary-initial-state*)) ?*q*
by (*rule* *order-trans*[*OF* *event-le* *new-collision-bound*])
then show ?*thesis*
by *simp*
qed

lemma *checked-staged-query-prefix-partial-target-prehit-bound*:
assumes *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *i-bound*: $i \leq \text{rounds}$
shows
wp-event (*checked-staged-query-prefix-receive-with-state* *A* *i*)
(*checked-staged-query-prefix-dynamic-index-prehit*
(*staged-query-prefix-partial-query-target* *i*))
adversary-initial-state $\leq$
staged-phase-relation-error *size*
(*staged-query-search-queries* *budgets* $i + 1$)
proof (*rule* *checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program*)
show *HashMap* *adversary-initial-state* = *fmempty*
by (*simp* *add*: *adversary-initial-state-def*)
show *hash-relation-program*
(*checked-staged-query-prefix-dynamic-index-prehit-relation* *A* *i*
adversary-initial-state
(*staged-query-prefix-partial-query-target* *i*))
size (*staged-query-search-queries* *budgets* $i + 1$)
(*checked-staged-query-prefix-receive-with-state* *A* *i*)
by (*rule* *hash-relation-program-checked-staged-query-prefix-receive-with-state*
[*OF* *wf* *controlled* *i-bound*])
(*rule*
checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
qed

lemma *checked-staged-query-prefix-committed-target-prehit-bound*:
assumes *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *i-bound*: $i \leq \text{rounds}$
shows
wp-event (*checked-staged-query-prefix-receive-with-state* *A* *i*)
(*checked-staged-query-prefix-dynamic-index-prehit*
(λprefix *prefix-state*.
staged-query-prefix-committed-query-target *prefix* *prefix-state*))
adversary-initial-state $\leq$
staged-phase-relation-error *size*
(*staged-query-search-queries* *budgets* $i + 1$)
proof (*rule* *checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program*)
show *HashMap* *adversary-initial-state* = *fmempty*

by (simp add: adversary-initial-state-def)
 show hash-relation-program
 (checked-staged-query-prefix-dynamic-index-prehit-relation A i
 adversary-initial-state
 (λ prefix prefix-state.
 staged-query-prefix-committed-query-target prefix prefix-state))
 size (staged-query-search-queries budgets $i + 1$)
 (checked-staged-query-prefix-receive-with-state A i)
 by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state
 [OF wf controlled i-bound])
 (rule
 checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
 qed

lemma checked-staged-query-prefix-clean-committed-target-prehit-bound:
 assumes wf: staged-budget-wellformed budgets
 and controlled: staged-adversary-controlled budgets A
 and i-bound: $i \leq \text{rounds}$
 shows
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-prehit
 staged-query-prefix-clean-committed-query-target)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$)
proof (rule checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program)
 show HashMap adversary-initial-state = fmempty
 by (simp add: adversary-initial-state-def)
 show hash-relation-program
 (checked-staged-query-prefix-dynamic-index-prehit-relation A i
 adversary-initial-state
 staged-query-prefix-clean-committed-query-target)
 size (staged-query-search-queries budgets $i + 1$)
 (checked-staged-query-prefix-receive-with-state A i)
 by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state
 [OF wf controlled i-bound])
 (rule
 checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
 qed

lemma checked-staged-query-prefix-clean-length-committed-target-prehit-bound:
 assumes wf: staged-budget-wellformed budgets
 and controlled: staged-adversary-controlled budgets A
 and i-bound: $i \leq \text{rounds}$
 shows
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-prehit
 staged-query-prefix-clean-length-committed-query-target)
 adversary-initial-state $\leq$

staged-phase-relation-error size
(staged-query-search-queries budgets i + 1)
proof (rule *checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program*)
show *HashMap adversary-initial-state = fmempty*
by (simp add: *adversary-initial-state-def*)
show *hash-relation-program*
(checked-staged-query-prefix-dynamic-index-prehit-relation A i
adversary-initial-state
staged-query-prefix-clean-length-committed-query-target)
size (staged-query-search-queries budgets i + 1)
(checked-staged-query-prefix-receive-with-state A i)
by (rule *hash-relation-program-checked-staged-query-prefix-receive-with-state*
[OF wf controlled i-bound])
(rule
checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
qed

definition *checked-staged-query-prefix-dynamic-index-fresh-output-hit*
:: (*'f staged-query-prefix-data* $\Rightarrow$ *'f protocol-channel* $\Rightarrow$
nat set) $\Rightarrow$
((*'f staged-query-prefix-data* $\times$ *'f protocol-channel*) $\times$ *'f*) $\times$
'f protocol-channel) *option* $\Rightarrow$ *bool*
where
checked-staged-query-prefix-dynamic-index-fresh-output-hit B out $\longleftrightarrow$
(case out of
None $\Rightarrow$ *False*
| *Some (((prefix, prefix-state), raw), raw-state)* $\Rightarrow$
hash-new-output-hit-event
(query-index-raw-preimage (B prefix prefix-state))
prefix-state (Some (raw, raw-state)))

lemma *checked-staged-query-prefix-dynamic-index-hit-imp-prehit-or-fresh*:
assumes *absent*:
 $\neg$ *checked-staged-query-prefix-dynamic-index-prehit B*
(Some (((prefix, prefix-state), raw), raw-state))
and *prefix-support*:
Some ((prefix, prefix-state), prefix-state) $\in$
set-dist
(execute (checked-staged-query-prefix-with-state A i) s)
and *receive-support*:
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i) s)
and *hit*:
checked-staged-query-prefix-dynamic-index-hit B
(Some (((prefix, prefix-state), raw), raw-state))
shows
hash-new-output-hit-event
(query-index-raw-preimage (B prefix prefix-state)) prefix-state

```

      (Some (raw, raw-state))
proof –
  have raw-in:
    raw ∈ query-index-raw-preimage (B prefix prefix-state)
  using hit unfolding checked-staged-query-prefix-dynamic-index-hit-def
    query-index-raw-preimage-def
  by simp
  have no-lookup:
    fmlookup (HashMap prefix-state)
      (QueryIndexChallenge (PQueryCounter prefix-state)
        (PState prefix-state)) = None
  proof (cases
    fmlookup (HashMap prefix-state)
      (QueryIndexChallenge (PQueryCounter prefix-state)
        (PState prefix-state)))
  case None
  then show ?thesis .
next
  case (Some old-raw)
  have old-lookup-prefix:
    fmlookup (HashMap prefix-state)
      (QueryIndexChallenge (PQueryCounter prefix-state)
        (PState prefix-state)) = Some old-raw
  using Some by simp
  have old-in: old-raw ∈ query-index-raw-preimage
    (B prefix prefix-state)
  proof –
  have old-idx:
    index (to-nat old-raw) ∈ B prefix prefix-state
  proof –
  have raw-idx:
    index (to-nat raw) ∈ B prefix prefix-state
  using hit
  unfolding checked-staged-query-prefix-dynamic-index-hit-def
  by simp
  have recv:
    Some (raw, raw-state) ∈
      set-dist
        (execute receive-query-index-challenge prefix-state)
  by (rule
    checked-staged-query-prefix-receive-with-state-receive-support
      [OF receive-support])
  have recv-lookup:
    fmlookup (HashMap raw-state)
      (QueryIndexChallenge (PQueryCounter prefix-state)
        (PState prefix-state)) = Some raw
  using receive-query-index-challenge-outcome[OF recv] by blast
  have ext: prefix-state ≤ raw-state
  using receive-query-index-challenge-extends[OF recv] .

```

```

have old-lookup-raw-state:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge (PQueryCounter prefix-state)
      (PState prefix-state)) = Some old-raw
  by (rule hash-extension-lookup[OF old-lookup-prefix ext])
have old-raw-eq: old-raw = raw
  using recv-lookup old-lookup-raw-state by simp
show ?thesis
  using raw-idx unfolding old-raw-eq .
qed
show ?thesis
  using old-idx unfolding query-index-raw-preimage-def by simp
qed
have prehit:
  checked-staged-query-prefix-dynamic-index-prehit B
    (Some (((prefix, prefix-state), raw), raw-state))
  using old-lookup-prefix old-in
  by (auto simp: checked-staged-query-prefix-dynamic-index-prehit-def)
then show ?thesis
  using absent by contradiction
qed
from receive-support have recv:
  Some (raw, raw-state) ∈
    set-dist (execute receive-query-index-challenge prefix-state)
  by (rule
    checked-staged-query-prefix-recv-with-state-recv-support)
have recv-lookup:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge (PQueryCounter prefix-state)
      (PState prefix-state)) = Some raw
  using receive-query-index-challenge-outcome[OF recv] by blast
have new-hit:
  hash-map-new-output-hit
    (query-index-raw-preimage (B prefix prefix-state))
    prefix-state raw-state
  unfolding hash-map-new-output-hit-def
  using no-lookup recv-lookup raw-in by blast
show ?thesis
  unfolding hash-new-output-hit-event-def using new-hit by simp
qed

lemma checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-fresh:
  wp-event (checked-staged-query-prefix-recv-with-state A i)
    (checked-staged-query-prefix-dynamic-index-hit B) s ≤
  wp-event (checked-staged-query-prefix-recv-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit B) s +
  wp-event (checked-staged-query-prefix-recv-with-state A i)
    (checked-staged-query-prefix-dynamic-index-fresh-output-hit B) s
proof —

```

```

let ?M = checked-staged-query-prefix-receive-with-state A i
let ?hit = checked-staged-query-prefix-dynamic-index-hit B
let ?pre = checked-staged-query-prefix-dynamic-index-prehit B
let ?fresh = checked-staged-query-prefix-dynamic-index-fresh-output-hit B
have wp-event ?M ?hit s ≤ wp-event ?M (λout. ?pre out ∨ ?fresh out) s
proof (rule wp-event-mono-on-support)
  fix out
  assume support: out ∈ set-dist (execute ?M s)
  and hit: ?hit out
  show ?pre out ∨ ?fresh out
  proof (cases out)
    case None
    then show ?thesis
      using hit unfolding checked-staged-query-prefix-dynamic-index-hit-def
      by simp
  next
    case (Some packed)
    then obtain prefix prefix-state raw raw-state where out-eq:
      out = Some (((prefix, prefix-state), raw), raw-state)
      by (cases packed, auto split: prod.splits)
    show ?thesis
    proof (cases ?pre out)
      case True
      then show ?thesis by simp
    next
      case pre-absent: False
      have prefix-support:
        Some ((prefix, prefix-state), prefix-state) ∈
          set-dist
            (execute (checked-staged-query-prefix-with-state A i) s)
      by (rule
        checked-staged-query-prefix-receive-with-state-prefix-support
        [OF support[unfolded out-eq]])
      have fresh:
        hash-new-output-hit-event
          (query-index-raw-preimage (B prefix prefix-state))
          prefix-state (Some (raw, raw-state))
      by (rule checked-staged-query-prefix-dynamic-index-hit-imp-prehit-or-fresh
        [OF - prefix-support support[unfolded out-eq]
          hit[unfolded out-eq]])
      (use pre-absent out-eq in simp)
    then show ?thesis
      unfolding out-eq
      checked-staged-query-prefix-dynamic-index-fresh-output-hit-def
      by simp
  qed
qed
qed
also have ... ≤ wp-event ?M ?pre s + wp-event ?M ?fresh s

```

```

    by (rule wp-event-union-bound)
  finally show ?thesis .
qed

lemma wp-receive-query-index-challenge-new-output-hit-bound:
  fixes B :: 'f set
  and s :: ('f, 'a) protocol-channel-scheme
  shows
    wp-event receive-query-index-challenge
      (hash-new-output-hit-event B s) s ≤ nnreal (card B) / nnreal size
proof (cases
  fmlookup (HashMap s)
    (QueryIndexChallenge (PQueryCounter s) (PState s)))
case None
have event-le:
  wp-event receive-query-index-challenge
    (hash-new-output-hit-event B s) s ≤
  wp-event receive-query-index-challenge
    (λout. case out of None ⇒ False | Some (raw, -) ⇒ raw ∈ B) s
proof (rule wp-event-mono-on-support)
  fix out
  assume support: out ∈ set-dist (execute receive-query-index-challenge s)
  and hit: hash-new-output-hit-event B s out
  show (case out of None ⇒ False | Some (raw, -) ⇒ raw ∈ B)
proof (cases out)
  case None
  then show ?thesis
    using hit unfolding hash-new-output-hit-event-def by simp
next
  case (Some packed)
  then obtain raw t where packed-eq: packed = (raw, t)
  by (cases packed) simp
  have out':
    Some (raw, t) ∈
    set-dist (execute receive-query-index-challenge s)
  using support unfolding Some packed-eq .
  have lookup:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  using receive-query-index-challenge-outcome[OF out'] by blast
  have new-hit: hash-map-new-output-hit B s t
  using hit unfolding Some packed-eq hash-new-output-hit-event-def
  by simp
  then show ?thesis
proof -
  from new-hit obtain x y where none-x:
    fmlookup (HashMap s) x = None
  and some-x: fmlookup (HashMap t) x = Some y
  and y-in: y ∈ B

```

```

    unfolding hash-map-new-output-hit-def by blast
let  $?q = \text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)$ 
have  $\text{raw} \in B$ 
proof (cases  $x = ?q$ )
  case True
    then have  $\text{Some } y = \text{Some } \text{raw}$ 
      using some-x lookup by simp
    then show  $?thesis$ 
      using y-in by simp
  next
    case False
    have  $\text{fmlookup } (\text{HashMap } t) \ x = \text{fmlookup } (\text{HashMap } s) \ x$ 
      by (rule receive-query-index-challenge-preserves-other-lookup
        [OF out' False])
    then show  $?thesis$ 
      using none-x some-x by simp
  qed
then show  $?thesis$ 
  unfolding Some packed-eq by simp
qed
qed
qed
also have  $\dots = \text{nnreal } (\text{card } B) / \text{nnreal } \text{size}$ 
  by (rule wp-receive-query-index-challenge-fresh-set [OF None])
finally show  $?thesis$  .
next
case (Some old)
have old-lookup:
   $\text{fmlookup } (\text{HashMap } s)$ 
   $(\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) = \text{Some } \text{old}$ 
  using Some by simp
have no-new-on-support:
   $\bigwedge \text{out}. \text{out} \in \text{set-dist } (\text{execute } \text{receive-query-index-challenge } s) \implies$ 
   $\neg \text{hash-new-output-hit-event } B \ s \ \text{out}$ 
proof –
  fix out
  assume support:  $\text{out} \in \text{set-dist } (\text{execute } \text{receive-query-index-challenge } s)$ 
  show  $\neg \text{hash-new-output-hit-event } B \ s \ \text{out}$ 
  proof (cases out)
    case None
    then show  $?thesis$ 
      unfolding hash-new-output-hit-event-def by simp
  next
    case (Some packed)
    then obtain  $\text{raw } t$  where packed-eq:  $\text{packed} = (\text{raw}, t)$ 
    by (cases packed) simp
    have out':
       $\text{Some } (\text{raw}, t) \in$ 
       $\text{set-dist } (\text{execute } \text{receive-query-index-challenge } s)$ 

```

```

    using support unfolding Some packed-eq .
have ext:  $s \leq t$ 
  by (rule receive-query-index-challenge-extends[OF out'])
show ?thesis
proof
  assume bad: hash-new-output-hit-event B s out
  have bad':
     $\exists x y. \text{fmlookup } (\text{HashMap } s) \ x = \text{None} \wedge$ 
     $\text{fmlookup } (\text{HashMap } t) \ x = \text{Some } y \wedge y \in B$ 
  using bad
  unfolding Some packed-eq hash-new-output-hit-event-def
    hash-map-new-output-hit-def
  by simp
then obtain x y where none-x:
   $\text{fmlookup } (\text{HashMap } s) \ x = \text{None}$ 
  and some-x:  $\text{fmlookup } (\text{HashMap } t) \ x = \text{Some } y$ 
  and y-in:  $y \in B$ 
  by blast
let ?q = QueryIndexChallenge (PQueryCounter s) (PState s)
show False
proof (cases x = ?q)
  case True
  then show ?thesis
    using none-x old-lookup by simp
next
  case False
  have  $\text{fmlookup } (\text{HashMap } t) \ x = \text{fmlookup } (\text{HashMap } s) \ x$ 
    by (rule receive-query-index-challenge-preserves-other-lookup
      [OF out' False])
  then show ?thesis
    using none-x some-x by simp
qed
qed
qed
qed
have wp-event receive-query-index-challenge
  (hash-new-output-hit-event B s)  $s \leq$ 
  wp-event receive-query-index-challenge ( $\lambda \cdot$ . False) s
  by (rule wp-event-mono-on-support)
  (use no-new-on-support in blast)
also have ... = 0
  unfolding wp-event-def wp-def dist-expect-def by simp
also have  $0 \leq \text{nnreal } (\text{card } B) / \text{nnreal size}$ 
  by simp
finally show ?thesis .
qed

lemma checked-staged-query-prefix-dynamic-index-fresh-output-hit-bound:
  fixes C :: prob

```

assumes *raw-bound: query-index-raw-preimage-bound*
and *prefix-bound:*
 $\bigwedge \text{prefix prefix-state.}$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s) \implies$
 $B \ \text{prefix} \ \text{prefix-state} \subseteq \text{query-sample-space} \wedge$
 $\text{nnreal } (\text{card } (B \ \text{prefix} \ \text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq C$
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-recv-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-fresh-output-hit } B) \ s \leq C$
unfolding *checked-staged-query-prefix-recv-with-state-def*
proof (*rule wp-event-bind-bound-by-cont*)
show
 $\neg \text{checked-staged-query-prefix-dynamic-index-fresh-output-hit } B \ \text{None}$
unfolding *checked-staged-query-prefix-dynamic-index-fresh-output-hit-def*
by *simp*
next
fix *prefix-pack prefix-state*
assume *prefix-support:*
 $\text{Some } (\text{prefix-pack}, \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s)$
obtain *prefix captured-state* **where** *prefix-pack-eq:*
 $\text{prefix-pack} = (\text{prefix}, \text{captured-state})$
by (*cases prefix-pack*) *simp*
have *captured-eq: captured-state = prefix-state*
using *checked-staged-query-prefix-with-state-outcome-state*
 $[OF \ \text{prefix-support}[\text{unfolded } \text{prefix-pack-eq}]]$
by *simp*
have *support':*
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s)$
using *checked-staged-query-prefix-with-state-outcome-canonical*
 $[OF \ \text{prefix-support}[\text{unfolded } \text{prefix-pack-eq}]]$
 captured-eq
by *simp*
from *prefix-bound*[*OF support'*] **have** *subset:*
 $B \ \text{prefix} \ \text{prefix-state} \subseteq \text{query-sample-space}$
and *frac:*
 $\text{nnreal } (\text{card } (B \ \text{prefix} \ \text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq C$
by *blast+*
have *raw-frac:*
 $\text{nnreal } (\text{card } (\text{query-index-raw-preimage } (B \ \text{prefix} \ \text{prefix-state}))) /$
 $\text{nnreal } \text{size} \leq C$
proof –


```

have nnreal (card (query-index-raw-preimage (B prefix prefix-state))) /
  nnreal size ≤
  nnreal (card (B prefix prefix-state)) /
  nnreal (card query-sample-space)
using raw-bound subset unfolding query-index-raw-preimage-bound-def
by blast
also have ... ≤ C
by (rule frac)
finally show ?thesis .
qed
have cont-eq:
  wp-event
    (receive-query-index-challenge  $\gg$  (λraw.
      return (prefix-pack, raw)))
    (checked-staged-query-prefix-dynamic-index-fresh-output-hit B)
    prefix-state =
  wp-event receive-query-index-challenge
    (hash-new-output-hit-event
      (query-index-raw-preimage (B prefix prefix-state)) prefix-state)
    prefix-state
unfolding wp-event-def
apply (subst wp-bind)
apply (rule arg-cong[where
  f=λQ. wp receive-query-index-challenge Q prefix-state])
apply (rule ext)
apply (simp add: wp-return prefix-pack-eq captured-eq
  checked-staged-query-prefix-dynamic-index-fresh-output-hit-def
  hash-new-output-hit-event-def
  split: option.splits prod.splits)
done
show wp-event
  (receive-query-index-challenge  $\gg$  (λraw.
    return (prefix-pack, raw)))
  (checked-staged-query-prefix-dynamic-index-fresh-output-hit B)
  prefix-state ≤ C
unfolding cont-eq
by (rule order-trans[
  OF wp-receive-query-index-challenge-new-output-hit-bound raw-frac])
qed

lemma checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error:
fixes C :: prob
assumes raw-bound: query-index-raw-preimage-bound
and prefix-bound:
   $\bigwedge$  prefix prefix-state.
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i) s)  $\implies$ 
  B prefix prefix-state  $\subseteq$  query-sample-space  $\wedge$ 

```

$$\text{nnreal } (\text{card } (B \text{ prefix prefix-state})) / \text{nnreal } (\text{card query-sample-space}) \leq C$$
shows

$$\begin{aligned} & \text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (\text{checked-staged-query-prefix-dynamic-index-hit } B) \ s \leq \\ & \text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (\text{checked-staged-query-prefix-dynamic-index-prehit } B) \ s + C \end{aligned}$$
proof –
 have $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit } B) \ s \leq$
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-prehit } B) \ s +$
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-fresh-output-hit } B) \ s$
by (rule checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-fresh)
 also have $\dots \leq$
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-prehit } B) \ s + C$
by (intro add-mono order-refl
 checked-staged-query-prefix-dynamic-index-fresh-output-hit-bound
 [OF raw-bound prefix-bound])
 finally show ?thesis .

qed

lemma query-sample-space-fraction-le-one:
assumes subset: $B \subseteq \text{query-sample-space}$
shows $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card query-sample-space}) \leq (1::\text{prob})$
proof –
 have finite-B: finite B
by (rule finite-subset[OF subset finite-query-sample-space])
 have card-le: $\text{card } B \leq \text{card query-sample-space}$
by (rule card-mono[OF finite-query-sample-space subset])
 have $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card query-sample-space}) \leq$
 $\text{nnreal } (\text{card query-sample-space}) / \text{nnreal } (\text{card query-sample-space})$
by (rule nnreal-nat-divide-right-mono[OF card-le])
 also have $\dots = (1::\text{prob})$
by (rule nnreal-nat-divide-self) (simp add: query-sample-space-size-pos)
 finally show ?thesis .

qed

lemma checked-staged-query-prefix-partial-target-hit-bound-by-prehit-plus-one:
assumes raw-bound: query-index-raw-preimage-bound
shows

$$\begin{aligned} & \text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (\text{checked-staged-query-prefix-dynamic-index-hit} \\ & \quad \quad (\text{staged-query-prefix-partial-query-target } i)) \ s \leq \\ & \text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (\text{checked-staged-query-prefix-dynamic-index-prehit} \\ & \quad \quad (\text{staged-query-prefix-partial-query-target } i)) \ s + 1 \end{aligned}$$

```

proof (rule
  checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
  [OF raw-bound])
fix prefix prefix-state
assume support:
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist (execute (checked-staged-query-prefix-with-state A i) s)
have subset:
  staged-query-prefix-partial-query-target i prefix prefix-state ⊆
    query-sample-space
by (rule staged-query-prefix-partial-query-target-subset)
have frac:
  nnreal
    (card (staged-query-prefix-partial-query-target i prefix prefix-state)) /
    nnreal (card query-sample-space) ≤ (1::prob)
by (rule query-sample-space-fraction-le-one[OF subset])
show staged-query-prefix-partial-query-target i prefix prefix-state ⊆
  query-sample-space ∧
  nnreal
    (card (staged-query-prefix-partial-query-target i prefix prefix-state)) /
    nnreal (card query-sample-space) ≤ (1::prob)
using subset frac by blast
qed

```

lemma checked-staged-query-prefix-partial-target-hit-bound-by-relation-plus-one:

assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i \leq \text{rounds}$

shows

wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 (staged-query-prefix-partial-query-target i))
 adversary-initial-state ≤
 staged-phase-relation-error size
 (staged-query-search-queries budgets i + 1) + 1

proof –

have wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 (staged-query-prefix-partial-query-target i))
 adversary-initial-state ≤
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-prehit
 (staged-query-prefix-partial-query-target i))
 adversary-initial-state + 1
by (rule checked-staged-query-prefix-partial-target-hit-bound-by-prehit-plus-one
 [OF raw-bound])
also have ... ≤
 staged-phase-relation-error size

(staged-query-search-queries budgets $i + 1$) + 1
 by (intro add-mono order-refl
 checked-staged-query-prefix-partial-target-prehit-bound
 [OF wf controlled i-bound])
 finally show ?thesis .
 qed

lemma checked-staged-query-prefix-committed-target-hit-bound-by-relation-and-query-error:
assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i \leq \text{rounds}$
and prefix-clean:
 $\bigwedge \text{prefix prefix-state.}$
 Some ((prefix, prefix-state), prefix-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-with-state A i)
 adversary-initial-state) $\implies$
 $\neg$ merkle-root-binding-collision (sqp-trace-root prefix)
 prefix-state $\wedge$
 $(\forall \text{rt. sqp-composition-fri-roots prefix} \neq [] \longrightarrow$
 $\text{rt} = \text{hd (sqp-composition-fri-roots prefix)} \longrightarrow$
 $\neg \text{merkle-root-binding-collision rt prefix-state})$

shows
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 ($\lambda \text{prefix prefix-state.}$
 staged-query-prefix-committed-query-target prefix prefix-state))
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) + query-error-bound

proof –
have hit-bound:
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 ($\lambda \text{prefix prefix-state.}$
 staged-query-prefix-committed-query-target prefix prefix-state))
 adversary-initial-state $\leq$
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-prehit
 ($\lambda \text{prefix prefix-state.}$
 staged-query-prefix-committed-query-target prefix prefix-state))
 adversary-initial-state + query-error-bound

proof (rule checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
 [OF raw-bound])
fix prefix prefix-state
assume support:
 Some ((prefix, prefix-state), prefix-state) $\in$
 set-dist

(execute (checked-staged-query-prefix-with-state A i)
 adversary-initial-state)
from prefix-clean[OF support]
have trace-clean:
 $\neg$ merkle-root-binding-collision (sqp-trace-root prefix)
 prefix-state
and comp-clean:
 $\bigwedge rt. \text{sqp-composition-fri-roots prefix} \neq [] \implies$
 $rt = \text{hd} (\text{sqp-composition-fri-roots prefix}) \implies$
 $\neg$ merkle-root-binding-collision rt prefix-state
by blast+
show
 (λprefix prefix-state.
 staged-query-prefix-committed-query-target prefix prefix-state)
 prefix prefix-state $\subseteq$ query-sample-space $\wedge$
 nnreal
 (card
 ((λprefix prefix-state.
 staged-query-prefix-committed-query-target prefix prefix-state)
 prefix prefix-state)) /
 nnreal (card query-sample-space) $\leq$ query-error-bound
using
 staged-query-prefix-committed-query-target-subset[of prefix prefix-state]
 staged-query-prefix-committed-query-target-fraction-bound-if-no-root-collisions
 [OF trace-clean comp-clean]
by blast
qed
also have ... $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) + query-error-bound
by (intro add-mono order-refl
 checked-staged-query-prefix-committed-target-prehit-bound
 [OF wf controlled i-bound])
finally show ?thesis .
qed

lemma checked-staged-query-prefix-clean-committed-target-hit-bound:
assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i \leq$ rounds
shows
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-clean-committed-query-target)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) + query-error-bound
proof –

have *hit-bound*:

$$\begin{aligned} & wp\text{-}event \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state \ A \ i) \\ & \quad (checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit \\ & \quad \quad staged\text{-}query\text{-}prefix\text{-}clean\text{-}committed\text{-}query\text{-}target) \\ & \quad adversary\text{-}initial\text{-}state \leq \\ & wp\text{-}event \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state \ A \ i) \\ & \quad (checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}prehit \\ & \quad \quad staged\text{-}query\text{-}prefix\text{-}clean\text{-}committed\text{-}query\text{-}target) \\ & \quad adversary\text{-}initial\text{-}state + query\text{-}error\text{-}bound \end{aligned}$$

proof (rule
 $checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit\text{-}bound\text{-}by\text{-}prehit\text{-}and\text{-}query\text{-}error$
 $[OF \ raw\text{-}bound]$)
fix *prefix prefix-state*
assume *support*:
 $Some \ ((prefix, \ prefix\text{-}state), \ prefix\text{-}state) \in$
 $set\text{-}dist$
 $(execute \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}with\text{-}state \ A \ i)$
 $adversary\text{-}initial\text{-}state)$
have *subset*:
 $staged\text{-}query\text{-}prefix\text{-}clean\text{-}committed\text{-}query\text{-}target \ prefix \ prefix\text{-}state$
 $\subseteq query\text{-}sample\text{-}space$
by (rule $staged\text{-}query\text{-}prefix\text{-}clean\text{-}committed\text{-}query\text{-}target\text{-}subset$)
have *frac*:
 $nnreal$
 $(card$
 $(staged\text{-}query\text{-}prefix\text{-}clean\text{-}committed\text{-}query\text{-}target \ prefix$
 $prefix\text{-}state)) \ /$
 $nnreal \ (card \ query\text{-}sample\text{-}space) \leq query\text{-}error\text{-}bound$
by (rule $staged\text{-}query\text{-}prefix\text{-}clean\text{-}committed\text{-}query\text{-}target\text{-}fraction\text{-}bound$)
show $staged\text{-}query\text{-}prefix\text{-}clean\text{-}committed\text{-}query\text{-}target \ prefix \ prefix\text{-}state$
 $\subseteq query\text{-}sample\text{-}space \wedge$
 $nnreal$
 $(card$
 $(staged\text{-}query\text{-}prefix\text{-}clean\text{-}committed\text{-}query\text{-}target \ prefix$
 $prefix\text{-}state)) \ /$
 $nnreal \ (card \ query\text{-}sample\text{-}space) \leq query\text{-}error\text{-}bound$
using *subset frac by blast*
qed
also have $\dots \leq$
 $staged\text{-}phase\text{-}relation\text{-}error \ size$
 $(staged\text{-}query\text{-}search\text{-}queries \ budgets \ i + 1) + query\text{-}error\text{-}bound$
by (intro *add-mono order-refl*
 $checked\text{-}staged\text{-}query\text{-}prefix\text{-}clean\text{-}committed\text{-}target\text{-}prehit\text{-}bound$
 $[OF \ wf \ controlled \ i\text{-}bound]$)
finally show *?thesis* .
qed

lemma *checked-staged-query-prefix-committed-target-hit-bound-by-clean-and-collision*:
 $wp\text{-}event \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state \ A \ i)$

(checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-committed-query-target) $s \leq$
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-clean-committed-query-target) $s +$
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 checked-staged-query-prefix-committed-root-collision s
proof –
 let $?M =$ checked-staged-query-prefix-receive-with-state A i
 let $?hit =$
 checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-committed-query-target
 let $?clean =$
 checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-clean-committed-query-target
 let $?collision =$ checked-staged-query-prefix-committed-root-collision
 have wp-event $?M$ $?hit$ $s \leq$ wp-event $?M$ ($\lambda out. ?clean$ out $\vee$ $?collision$ out) s
 by (rule wp-event-mono)
 (rule checked-staged-query-prefix-committed-hit-imp-clean-hit-or-root-collision)
 also have $\dots \leq$ wp-event $?M$ $?clean$ $s +$ wp-event $?M$ $?collision$ s
 by (rule wp-event-union-bound)
 finally show $?thesis$.
qed

lemma checked-staged-query-prefix-committed-target-hit-bound-by-relation-query-error-and-collision:

assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i -bound: $i \leq$ rounds

shows

wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-committed-query-target)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
 query-error-bound +
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 checked-staged-query-prefix-committed-root-collision
 adversary-initial-state

proof –

have split:

wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-committed-query-target)
 adversary-initial-state $\leq$
 wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-clean-committed-query-target)

```

    adversary-initial-state +
    wp-event (checked-staged-query-prefix-receive-with-state A i)
    checked-staged-query-prefix-committed-root-collision
    adversary-initial-state
  by (rule
    checked-staged-query-prefix-committed-target-hit-bound-by-clean-and-collision)
have clean-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-hit
      staged-query-prefix-clean-committed-query-target)
  adversary-initial-state ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
  by (rule checked-staged-query-prefix-clean-committed-target-hit-bound
    [OF raw-bound wf controlled i-bound])
have tail-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-hit
      staged-query-prefix-clean-committed-query-target)
  adversary-initial-state +
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    checked-staged-query-prefix-committed-root-collision
  adversary-initial-state ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
  query-error-bound +
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    checked-staged-query-prefix-committed-root-collision
  adversary-initial-state
  using clean-bound by (simp add: add-mono)
show ?thesis
  by (rule order-trans[OF split tail-bound])
qed

```

lemma *checked-staged-query-prefix-clean-length-committed-target-hit-bound:*
assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i \leq \text{rounds}$

shows

```

  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-hit
      staged-query-prefix-clean-length-committed-query-target)
  adversary-initial-state ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound

```

proof —

have hit-bound:

```

  wp-event (checked-staged-query-prefix-receive-with-state A i)

```


$(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $\text{staged-query-prefix-clean-length-committed-query-target})$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-prehit}$
 $\text{staged-query-prefix-clean-length-committed-query-target})$
 $\text{adversary-initial-state} + \text{query-error-bound}$

proof (rule
 $\text{checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error}$
 $[\text{OF raw-bound}]$)

fix $\text{prefix prefix-state}$

assume support:
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state})$

have subset:
 $\text{staged-query-prefix-clean-length-committed-query-target prefix}$
 $\text{prefix-state} \subseteq \text{query-sample-space}$

by (rule $\text{staged-query-prefix-clean-length-committed-query-target-subset}$)

have frac:
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-clean-length-committed-query-target prefix}$
 $\text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$

by (rule $\text{staged-query-prefix-clean-length-committed-query-target-fraction-bound}$)

show $\text{staged-query-prefix-clean-length-committed-query-target prefix}$
 $\text{prefix-state} \subseteq \text{query-sample-space} \wedge$
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-clean-length-committed-query-target prefix}$
 $\text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$

using $\text{subset frac by blast}$

qed

also have $\dots \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) + \text{query-error-bound}$

by (intro $\text{add-mono order-refl}$
 $\text{checked-staged-query-prefix-clean-length-committed-target-prehit-bound}$
 $[\text{OF wf controlled } i\text{-bound}]$)

finally show $?thesis$.

qed

lemma $\text{checked-staged-query-prefix-length-committed-target-hit-bound-by-clean-and-collision:}$
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $\text{staged-query-prefix-length-committed-query-target}) \ s \leq$

$wp\text{-}event$ ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state$ A i)
 ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit$
 $staged\text{-}query\text{-}prefix\text{-}clean\text{-}length\text{-}committed\text{-}query\text{-}target$) s +
 $wp\text{-}event$ ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state$ A i)
 $checked\text{-}staged\text{-}query\text{-}prefix\text{-}hash\text{-}map\text{-}output\text{-}collision$ s
proof –
let $?M = checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state$ A i
let $?hit =$
 $checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit$
 $staged\text{-}query\text{-}prefix\text{-}length\text{-}committed\text{-}query\text{-}target$
let $?clean =$
 $checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit$
 $staged\text{-}query\text{-}prefix\text{-}clean\text{-}length\text{-}committed\text{-}query\text{-}target$
let $?collision = checked\text{-}staged\text{-}query\text{-}prefix\text{-}hash\text{-}map\text{-}output\text{-}collision$
have $wp\text{-}event$ $?M$ $?hit$ $s \leq wp\text{-}event$ $?M$ ($\lambda out. ?clean$ $out \vee ?collision$ out) s
by ($rule$ $wp\text{-}event\text{-}mono$)
 ($rule$
 $checked\text{-}staged\text{-}query\text{-}prefix\text{-}length\text{-}committed\text{-}hit\text{-}imp\text{-}clean\text{-}hit\text{-}or\text{-}hash\text{-}collision$)
also have $\dots \leq wp\text{-}event$ $?M$ $?clean$ $s + wp\text{-}event$ $?M$ $?collision$ s
by ($rule$ $wp\text{-}event\text{-}union\text{-}bound$)
finally show $?thesis$.
qed

lemma $checked\text{-}staged\text{-}query\text{-}prefix\text{-}length\text{-}committed\text{-}target\text{-}hit\text{-}bound\text{-}by\text{-}relation\text{-}query\text{-}error\text{-}and\text{-}collision$:

assumes $raw\text{-}bound$: $query\text{-}index\text{-}raw\text{-}preimage\text{-}bound$
and wf : $staged\text{-}budget\text{-}wellformed$ $budgets$
and $controlled$: $staged\text{-}adversary\text{-}controlled$ $budgets$ A
and $i\text{-}bound$: $i \leq rounds$

shows

$wp\text{-}event$ ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state$ A i)
 ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit$
 $staged\text{-}query\text{-}prefix\text{-}length\text{-}committed\text{-}query\text{-}target$)
 $adversary\text{-}initial\text{-}state \leq$
 $staged\text{-}phase\text{-}relation\text{-}error$ $size$
 ($staged\text{-}query\text{-}search\text{-}queries$ $budgets$ $i + 1$) +
 $query\text{-}error\text{-}bound$ +
 $wp\text{-}event$ ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state$ A i)
 $checked\text{-}staged\text{-}query\text{-}prefix\text{-}hash\text{-}map\text{-}output\text{-}collision$
 $adversary\text{-}initial\text{-}state$

proof –

have $split$:

$wp\text{-}event$ ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state$ A i)
 ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit$
 $staged\text{-}query\text{-}prefix\text{-}length\text{-}committed\text{-}query\text{-}target$)
 $adversary\text{-}initial\text{-}state \leq$
 $wp\text{-}event$ ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state$ A i)
 ($checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit$
 $staged\text{-}query\text{-}prefix\text{-}clean\text{-}length\text{-}committed\text{-}query\text{-}target$)
 $adversary\text{-}initial\text{-}state$ +

```

wp-event (checked-staged-query-prefix-receive-with-state A i)
checked-staged-query-prefix-hash-map-output-collision
adversary-initial-state
by (rule
  checked-staged-query-prefix-length-committed-target-hit-bound-by-clean-and-collision)
have clean-bound:
wp-event (checked-staged-query-prefix-receive-with-state A i)
  (checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-clean-length-committed-query-target)
adversary-initial-state ≤
staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) + query-error-bound
by (rule
  checked-staged-query-prefix-clean-length-committed-target-hit-bound
  [OF raw-bound wf controlled i-bound])
have tail-bound:
wp-event (checked-staged-query-prefix-receive-with-state A i)
  (checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-clean-length-committed-query-target)
adversary-initial-state +
wp-event (checked-staged-query-prefix-receive-with-state A i)
  checked-staged-query-prefix-hash-map-output-collision
adversary-initial-state ≤
staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
query-error-bound +
wp-event (checked-staged-query-prefix-receive-with-state A i)
  checked-staged-query-prefix-hash-map-output-collision
adversary-initial-state
using clean-bound by (simp add: add-mono)
show ?thesis
by (rule order-trans[OF split tail-bound])
qed

```

lemma *checked-staged-query-prefix-length-committed-target-hit-bound:*
assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i \leq \text{rounds}$

shows

```

wp-event (checked-staged-query-prefix-receive-with-state A i)
  (checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-length-committed-query-target)
adversary-initial-state ≤
staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
  (staged-query-search-queries budgets i + 1)

```

```

proof –
  have hit-bound:
    wp-event (checked-staged-query-prefix-receive-with-state A i)
      (checked-staged-query-prefix-dynamic-index-hit
        staged-query-prefix-length-committed-query-target)
    adversary-initial-state ≤
    staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
    query-error-bound +
    wp-event (checked-staged-query-prefix-receive-with-state A i)
      checked-staged-query-prefix-hash-map-output-collision
      adversary-initial-state
  by (rule
    checked-staged-query-prefix-length-committed-target-hit-bound-by-relation-query-error-and-collision
    [OF raw-bound wf controlled i-bound])
  have collision-bound:
    wp-event (checked-staged-query-prefix-receive-with-state A i)
      checked-staged-query-prefix-hash-map-output-collision
      adversary-initial-state ≤
      hash-collision-budget-value 0
      (staged-query-search-queries budgets i + 1)
  by (rule checked-staged-query-prefix-hash-map-output-collision-bound
    [OF wf controlled i-bound])
  have tail:
    staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
    query-error-bound +
    wp-event (checked-staged-query-prefix-receive-with-state A i)
      checked-staged-query-prefix-hash-map-output-collision
      adversary-initial-state ≤
    staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
    query-error-bound +
    hash-collision-budget-value 0
      (staged-query-search-queries budgets i + 1)
  using collision-bound by (intro add-mono order-refl)
  show ?thesis
  by (rule order-trans[OF hit-bound tail])
qed

end

end

```

```

theory Soundness-Staged-Transcript
  imports Soundness-Staged-Query
begin

```

Transcript-level staged query events and their transfer to verifier executions.

context *soundness*
begin

definition *staged-transcript-query-partial-header-key-hit*
 $:: ('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
staged-transcript-query-partial-header-key-hit out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ *False*
 | *Some (data, attacker-state)* $\Rightarrow$
 (let *s* =
 verifier-state-from-adversary attacker-state
 (*staged-proof-transcript data*);
 B =
 query-header-supported-partial-union-good-sets s
 query-sampling-success-space
 (*staged-trace-root data*)
 (*staged-trace-fri-roots data*)
 (*staged-trace-final data*)
 (*staged-alphas data*)
 (*staged-degree data*)
 (*staged-composition-fri-roots data*)
 (*staged-composition-final data*)
 in
 ($\exists i \text{ raw. } i < \text{rounds} \wedge$
 fmlookup (HashMap attacker-state)
 (*QueryIndexChallenge i*
 (*state-after-query-chunks*
 (*staged-query-start-hash data*)
 (*staged-query-chunks data*) i)) =
 Some raw $\wedge$
 index (to-nat raw) \in B)))

lemma *staged-transcript-query-partial-header-key-hit-alt:*
staged-transcript-query-partial-header-key-hit out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ *False*
 | *Some (data, attacker-state)* $\Rightarrow$
 ($\exists i \text{ raw. } i < \text{rounds} \wedge$
 fmlookup (HashMap attacker-state)
 (*QueryIndexChallenge i*
 (*state-after-query-chunks*
 (*staged-query-start-hash data*)
 (*staged-query-chunks data*) i)) =
 Some raw $\wedge$
 index (to-nat raw) \in
 staged-partial-query-target data attacker-state))
unfolding *staged-transcript-query-partial-header-key-hit-def*
staged-partial-query-target-def Let-def

by (simp split: option.splits prod.splits)

definition *staged-transcript-query-partial-header-key-hit-at*
 $:: \text{nat} \Rightarrow ('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

staged-transcript-query-partial-header-key-hit-at i $out \longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (data, attacker-state) $\Rightarrow$
 $(\exists \text{ raw. } i < \text{rounds} \wedge$
 fmlookup (HashMap attacker-state)
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data)
 (staged-query-chunks data) i)) =
 Some raw $\wedge$
 index (to-nat raw) $\in$
 staged-partial-query-target data attacker-state))

lemma *checked-staged-transcript-program-query-prefix-receive-lookup-support:*

assumes *wf*: staged-budget-wellformed budgets

and *controlled*: staged-adversary-controlled budgets A

and *i-bound*: $i < \text{rounds}$

and *outcome*:

Some (data, attacker-state) $\in$
 set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state)

obtains *prefix prefix-state raw raw-state*

where

Some (((prefix, prefix-state), raw), raw-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-receive-with-state A i)
 adversary-initial-state)
 $\text{raw-state} \leq \text{attacker-state}$
 fmlookup (HashMap raw-state)
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data) (staged-query-chunks data) i)) =
 Some raw

proof (rule checked-staged-transcript-program-query-prefix-receive-support
 [OF *wf controlled i-bound outcome*])

fix *prefix prefix-state raw raw-state*

assume *prefix-receive*:

Some (((prefix, prefix-state), raw), raw-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-receive-with-state A i)
 adversary-initial-state)

```

and raw-ext: raw-state  $\leq$  attacker-state
and lookup:
  fmllookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw
show ?thesis
  by (rule that[OF prefix-receive raw-ext lookup])
qed

lemma checked-staged-security-with-data-state-accepted-shape-query-prefix-receive-index:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((data, attacker-state), result), final-state)  $\in$ 
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
and shape:
  accepted-transcript-shape
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state)) as query-idxs
and i-bound: i < rounds
obtains prefix prefix-state raw raw-state
where
  Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  index (to-nat raw) = query-idxs ! i
proof -
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support]
have builder:
  Some (data, attacker-state)  $\in$ 
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and verifier:
  Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)
by blast+
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support shape]
obtain raw-idxs where

```

```

len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and lookup-final:
  ∧j. j < rounds ⇒
    fmllookup (HashMap final-state)
      (QueryIndexChallenge j
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) j)) =
      Some (raw-idxs ! j)
by blast
from checked-staged-transcript-program-query-prefix-receive-lookup-support
  [OF wf controlled i-bound builder]
obtain prefix prefix-state raw raw-state where
  prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
  and raw-ext: raw-state ≤ attacker-state
  and lookup-raw:
    fmllookup (HashMap raw-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw
  by blast
have attacker-ext-s: attacker-state ≤ ?s
proof -
  have attacker-state ≤ attacker-state
    unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  then show ?thesis
    by (rule hash-extends-verifier-state-from-adversary-right)
qed
have s-ext-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
have raw-ext-final: raw-state ≤ final-state
  by (rule hash-ext-trans[OF raw-ext])
  (rule hash-ext-trans[OF attacker-ext-s s-ext-final])
have lookup-final-raw:
  fmllookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
  by (rule hash-extension-lookup[OF lookup-raw raw-ext-final])
have raw-eq: raw-idxs ! i = raw
  using lookup-final[OF i-bound] lookup-final-raw by simp

```



```

have idx-eq: index (to-nat raw) = query-idxs ! i
  using query-idxs-eq len-row i-bound raw-eq by simp
show ?thesis
  by (rule that[OF prefix-receive idx-eq])
qed

lemma checked-staged-transcript-program-query-prefix-receive-match-support:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and outcome:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
obtains prefix prefix-state raw raw-state
where
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  sqp-trace-root prefix = staged-trace-root data
  sqp-trace-fri-roots prefix = staged-trace-fri-roots data
  sqp-trace-fri-challenges prefix =
    staged-trace-fri-challenges data
  sqp-trace-final prefix = staged-trace-final data
  sqp-alphas prefix = staged-alphas data
  sqp-degree prefix = staged-degree data
  sqp-composition-fri-roots prefix =
    staged-composition-fri-roots data
  sqp-composition-fri-challenges prefix =
    staged-composition-fri-challenges data
  sqp-composition-final prefix = staged-composition-final data
  sqp-query-chunks prefix = take i (staged-query-chunks data)
  prefix-state ≤ attacker-state
  raw-state ≤ attacker-state
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
proof (rule checked-staged-transcript-program-query-prefix-receive-support
  [OF wf controlled i-bound outcome])
fix prefix prefix-state raw raw-state
assume prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)

```

and *root-eq*: *sqp-trace-root prefix = staged-trace-root data*
and *trace-roots-eq*:
 sqp-trace-fri-roots prefix = staged-trace-fri-roots data
and *trace-bs-eq*:
 sqp-trace-fri-challenges prefix =
 staged-trace-fri-challenges data
and *trace-final-eq*: *sqp-trace-final prefix = staged-trace-final data*
and *alphas-eq*: *sqp-alphas prefix = staged-alphas data*
and *degree-eq*: *sqp-degree prefix = staged-degree data*
and *comp-roots-eq*:
 sqp-composition-fri-roots prefix =
 staged-composition-fri-roots data
and *comp-bs-eq*:
 sqp-composition-fri-challenges prefix =
 staged-composition-fri-challenges data
and *comp-final-eq*:
 sqp-composition-final prefix = staged-composition-final data
and *query-chunks-eq*:
 sqp-query-chunks prefix = take i (staged-query-chunks data)
and *prefix-ext*: *prefix-state ≤ attacker-state*
and *raw-ext*: *raw-state ≤ attacker-state*
and *lookup*:
 fmlookup (HashMap raw-state)
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data) (staged-query-chunks data) i)) =
 Some raw
show *?thesis*
by (*rule that*[*OF prefix-receive root-eq trace-roots-eq trace-bs-eq*
 trace-final-eq alphas-eq degree-eq comp-roots-eq comp-bs-eq
 comp-final-eq query-chunks-eq prefix-ext raw-ext lookup])
qed

lemma *checked-staged-transcript-query-hit-at-prefix-receive-support*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *outcome*:
 Some (data, attacker-state) ∈
 set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state)
and *hit*:
 staged-transcript-query-partial-header-key-hit-at i
 (Some (data, attacker-state))
shows
 $\exists$ *prefix prefix-state raw raw-state.*
 Some (((prefix, prefix-state), raw), raw-state) ∈
 set-dist
 (execute (checked-staged-query-prefix-receive-with-state A i)

```

      adversary-initial-state)  $\wedge$ 
      checked-staged-query-prefix-fixed-transcript-target-hit
      data attacker-state
      (Some (((prefix, prefix-state), raw), raw-state))
proof –
from hit obtain raw-hit where
  i-bound:  $i < \text{rounds}$ 
and lookup-attacker:
  fmlookup (HashMap attacker-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw-hit
and raw-hit-target:
  index (to-nat raw-hit)  $\in$  staged-partial-query-target data attacker-state
unfolding staged-transcript-query-partial-header-key-hit-at-def
by auto
from checked-staged-transcript-program-query-prefix-receive-lookup-support
  [OF wf controlled i-bound outcome]
obtain prefix prefix-state raw raw-state where
  prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
and raw-ext: raw-state  $\leq$  attacker-state
and lookup-raw:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
by (rule checked-staged-transcript-program-query-prefix-receive-lookup-support
  [OF wf controlled i-bound outcome])
have lookup-attacker-raw:
  fmlookup (HashMap attacker-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
by (rule hash-extension-lookup[OF lookup-raw raw-ext])
have raw-eq: raw = raw-hit
using lookup-attacker lookup-attacker-raw by simp
have fixed-hit:
  checked-staged-query-prefix-fixed-transcript-target-hit
  data attacker-state
  (Some (((prefix, prefix-state), raw), raw-state))
using raw-hit-target unfolding raw-eq
  checked-staged-query-prefix-fixed-transcript-target-hit-def

```

```

    checked-staged-query-prefix-dynamic-index-hit-def
  by simp
show ?thesis
  using prefix-receive fixed-hit by blast
qed

lemma checked-staged-transcript-query-hit-at-prefix-completion-receive-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and hit:
    staged-transcript-query-partial-header-key-hit-at i
      (Some (data, attacker-state))
  shows
    ∃ prefix prefix-state raw raw-state.
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state) ∧
        checked-staged-query-prefix-dynamic-index-hit
          (staged-query-prefix-partial-query-target i)
          (Some (((prefix, prefix-state), raw), raw-state))
  proof -
    from hit obtain raw-hit where
      i-bound: i < rounds
    and lookup-attacker:
      fmlookup (HashMap attacker-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data) (staged-query-chunks data) i)) =
        Some raw-hit
    and raw-hit-target:
      index (to-nat raw-hit) ∈ staged-partial-query-target data attacker-state
    unfolding staged-transcript-query-partial-header-key-hit-at-def
    by auto
  from checked-staged-transcript-program-query-prefix-receive-match-support
    [OF wf controlled i-bound outcome]
  obtain prefix prefix-state raw raw-state where
    prefix-receive:
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state)
  and root-eq: sqp-trace-root prefix = staged-trace-root data
  and trace-roots-eq:

```

```

    sqp-trace-fri-roots prefix = staged-trace-fri-roots data
and trace-bs-eq:
    sqp-trace-fri-challenges prefix =
        staged-trace-fri-challenges data
and trace-final-eq:
    sqp-trace-final prefix = staged-trace-final data
and alphas-eq: sqp-alphas prefix = staged-alphas data
and degree-eq: sqp-degree prefix = staged-degree data
and comp-roots-eq:
    sqp-composition-fri-roots prefix =
        staged-composition-fri-roots data
and comp-bs-eq:
    sqp-composition-fri-challenges prefix =
        staged-composition-fri-challenges data
and comp-final-eq:
    sqp-composition-final prefix = staged-composition-final data
and query-chunks-eq:
    sqp-query-chunks prefix = take i (staged-query-chunks data)
and prefix-ext: prefix-state  $\leq$  attacker-state
and raw-ext: raw-state  $\leq$  attacker-state
and lookup-raw:
    fmlookup (HashMap raw-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw
by (rule checked-staged-transcript-program-query-prefix-receive-match-support
      [OF wf controlled i-bound outcome])
have lookup-attacker-raw:
    fmlookup (HashMap attacker-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw
by (rule hash-extension-lookup[OF lookup-raw raw-ext])
have raw-eq: raw = raw-hit
using lookup-attacker lookup-attacker-raw by simp
have match:
    staged-query-prefix-matches-data i prefix data
unfolding staged-query-prefix-matches-data-def
using root-eq trace-roots-eq trace-bs-eq trace-final-eq alphas-eq
    degree-eq comp-roots-eq comp-bs-eq comp-final-eq query-chunks-eq
by simp
have prefix-target:
    index (to-nat raw)  $\in$ 
      staged-query-prefix-partial-query-target i prefix prefix-state
unfolding raw-eq
using raw-hit-target
    staged-partial-query-target-subset-prefix-completion[OF match prefix-ext]

```

```

    by blast
  have prefix-hit:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-partial-query-target i)
      (Some (((prefix, prefix-state), raw), raw-state))
  using prefix-target
  unfolding checked-staged-query-prefix-dynamic-index-hit-def
  by simp
  show ?thesis
    using prefix-receive prefix-hit by blast
qed

lemma checked-staged-after-query-prefix-receive-partial-hit-imp-prefix-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and head-support:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
  and cont-support:
    Some (data, attacker-state) ∈
      set-dist
        (execute
          (checked-staged-after-query-prefix-receive A i
            ((prefix, prefix-state), raw))
          raw-state)
  and hit:
    staged-transcript-query-partial-header-key-hit-at i
      (Some (data, attacker-state))
  shows
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-partial-query-target i)
      (Some (((prefix, prefix-state), raw), raw-state))
proof -
  have prefix-support:
    Some ((prefix, prefix-state), prefix-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-with-state A i)
          adversary-initial-state)
  by (rule checked-staged-query-prefix-receive-with-state-prefix-support
    [OF head-support])
  have recv:
    Some (raw, raw-state) ∈
      set-dist (execute receive-query-index-challenge prefix-state)
  by (rule checked-staged-query-prefix-receive-with-state-receive-support
    [OF head-support])
  from checked-staged-query-prefix-with-state-alignment

```

```

[OF wf controlled i-bound prefix-support]
have prefix-counter: PQueryCounter prefix-state = i
and prefix-state:
  PState prefix-state =
    state-after-query-chunks
      (staged-query-prefix-start-hash prefix)
      (sqp-query-chunks prefix) i
and prefix-len: length (sqp-query-chunks prefix) = i
by blast+
from cont-support obtain chunk chunk-state record-state suffix-chunks where
  chunk-out:
    Some (chunk, chunk-state) ∈
      set-dist (execute (query-opening-stage A i raw) raw-state)
and record-out:
    Some ((), record-state) ∈
      set-dist (execute (record-staged-messages chunk) chunk-state)
and suffix-out:
    Some (suffix-chunks, attacker-state) ∈
      set-dist
        (execute
          (checked-staged-query-program A
            (sqp-trace-fri-roots prefix)
            (sqp-composition-fri-roots prefix)
            (Suc i) (rounds - Suc i))
          record-state)
and data-eq:
  data =
    (|staged-trace-root = sqp-trace-root prefix,
      staged-trace-fri-roots = sqp-trace-fri-roots prefix,
      staged-trace-fri-challenges = sqp-trace-fri-challenges prefix,
      staged-trace-final = sqp-trace-final prefix,
      staged-alphas = sqp-alphas prefix,
      staged-degree = sqp-degree prefix,
      staged-composition-fri-roots = sqp-composition-fri-roots prefix,
      staged-composition-fri-challenges =
        sqp-composition-fri-challenges prefix,
      staged-composition-final = sqp-composition-final prefix,
      staged-query-chunks =
        sqp-query-chunks prefix @ chunk # suffix-chunks)
  unfolding checked-staged-after-query-prefix-receive-def Let-def
  by (auto elim!: set-dist-bindE simp: assert-def throw-no-outcome
    split: if-splits)
have match: staged-query-prefix-matches-data i prefix data
  unfolding staged-query-prefix-matches-data-def data-eq
  using prefix-len by simp
have start-hash:
  staged-query-start-hash data = staged-query-prefix-start-hash prefix
  by (rule staged-query-start-hash-of-prefix-completion[OF match])
have prefix-key-state:

```

```

PState prefix-state =
  state-after-query-chunks
    (staged-query-start-hash data) (staged-query-chunks data) i
using prefix-state prefix-len start-hash
unfolding data-eq
by (simp add: state-after-query-chunks-append-prefix)
have receive-lookup:
  fmllookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
using receive-query-index-challenge-outcome[OF recv]
  prefix-counter prefix-key-state
by simp
have prefix-raw-ext: prefix-state ≤ raw-state
  by (rule receive-query-index-challenge-extends[OF recv])
have stage-controlled:
  controlled-ro-program (query-opening-budgets budgets ! i)
    (query-opening-stage A i raw)
using controlled wf i-bound
unfolding staged-adversary-controlled-def staged-budget-wellformed-def
by simp
have raw-chunk-ext: raw-state ≤ chunk-state
using controlled-ro-program-extension[OF stage-controlled] chunk-out
unfolding hash-extension-preserving-def
by blast
have record-state-eq:
  record-state = chunk-state⟦
    PState := foldl concat (PState chunk-state) chunk,
    PTranscript := PTranscript chunk-state @ chunk⟧
by (rule record-staged-messages-outcome[OF record-out])
have chunk-record-ext: chunk-state ≤ record-state
unfolding record-state-eq less-eq-hash-ext-def less-eq-fmap-def
by simp
have suffix-bound:
  Suc i + (rounds - Suc i) ≤ length (query-opening-budgets budgets)
using wf i-bound unfolding staged-budget-wellformed-def by simp
have suffix-target:
  hash-target-program {}
    (sum-list
      (take (rounds - Suc i)
        (drop (Suc i) (query-opening-budgets budgets))) +
      (rounds - Suc i))
    (checked-staged-query-program A
      (sqp-trace-fri-roots prefix)
      (sqp-composition-fri-roots prefix)
      (Suc i) (rounds - Suc i))
by (rule hash-target-program-checked-staged-query-program

```



```

    [OF controlled suffix-bound])
  have suffix-ext: record-state  $\leq$  attacker-state
    using hash-target-program-extension[OF suffix-target] suffix-out
    unfolding hash-extension-preserving-def
    by blast
  have raw-ext: raw-state  $\leq$  attacker-state
    by (rule hash-ext-trans[OF raw-chunk-ext])
      (rule hash-ext-trans[OF chunk-record-ext suffix-ext])
  have prefix-ext: prefix-state  $\leq$  attacker-state
    by (rule hash-ext-trans[OF prefix-raw-ext raw-ext])
  from hit obtain raw-hit where
    lookup-attacker:
      fmlookup (HashMap attacker-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data) (staged-query-chunks data) i)) =
          Some raw-hit
  and raw-hit-target:
    index (to-nat raw-hit)  $\in$ 
      staged-partial-query-target data attacker-state
  unfolding staged-transcript-query-partial-header-key-hit-at-def
  by auto
  have lookup-attacker-raw:
    fmlookup (HashMap attacker-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
        Some raw
    by (rule hash-extension-lookup[OF receive-lookup raw-ext])
  have raw-eq: raw = raw-hit
    using lookup-attacker lookup-attacker-raw by simp
  have prefix-target:
    index (to-nat raw)  $\in$ 
      staged-query-prefix-partial-query-target i prefix prefix-state
  unfolding raw-eq
  using raw-hit-target
    staged-partial-query-target-subset-prefix-completion[OF match prefix-ext]
  by blast
  show ?thesis
    using prefix-target
    unfolding checked-staged-query-prefix-dynamic-index-hit-def
    by simp
qed

```

```

lemma checked-staged-transcript-program-query-prefix-receive-compact-support:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound:  $i < \text{rounds}$ 
    and outcome:

```

Some (*data*, *attacker-state*) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
obtains *prefix prefix-state raw raw-state*
where
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
sqp-trace-root prefix = staged-trace-root data
sqp-alphas prefix = staged-alphas data
sqp-composition-fri-roots prefix =
staged-composition-fri-roots data
prefix-state $\leq$ attacker-state
raw-state $\leq$ attacker-state
fmlookup (HashMap raw-state)
(QueryIndexChallenge i
(state-after-query-chunks
(staged-query-start-hash data) (staged-query-chunks data) i)) =
Some raw
proof (*rule checked-staged-transcript-program-query-prefix-receive-match-support*
[OF wf controlled i-bound outcome])
fix *prefix prefix-state raw raw-state*
assume *prefix-receive:*
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *root-eq: sqp-trace-root prefix = staged-trace-root data*
and *alphas-eq: sqp-alphas prefix = staged-alphas data*
and *comp-roots-eq:*
sqp-composition-fri-roots prefix =
staged-composition-fri-roots data
and *prefix-ext: prefix-state $\leq$ attacker-state*
and *raw-ext: raw-state $\leq$ attacker-state*
and *lookup:*
fmlookup (HashMap raw-state)
(QueryIndexChallenge i
(state-after-query-chunks
(staged-query-start-hash data) (staged-query-chunks data) i)) =
Some raw
show *?thesis*
by (*rule that[OF prefix-receive root-eq alphas-eq comp-roots-eq*
prefix-ext raw-ext lookup])
qed

lemma *checked-staged-after-query-prefix-receive-extends:*
assumes *wf: staged-budget-wellformed budgets*

and *controlled*: staged-adversary-controlled budgets A
and *i-bound*: $i < \text{rounds}$
and *data-out*:
 Some $(\text{data}, \text{attacker-state}) \in$
 set-dist
 (*execute*
 (*checked-staged-after-query-prefix-receive* A i
 $((\text{prefix}, \text{prefix-state}), \text{raw})$)
 raw-state)
shows $\text{raw-state} \leq \text{attacker-state}$
proof –
from *data-out* **obtain** *chunk chunk-state record-state suffix-chunks* **where**
 chunk-out:
 Some $(\text{chunk}, \text{chunk-state}) \in$
 set-dist (*execute* (*query-opening-stage* A i raw) raw-state)
and *record-out*:
 Some $((), \text{record-state}) \in$
 set-dist (*execute* (*record-staged-messages* chunk) chunk-state)
and *suffix-out*:
 Some $(\text{suffix-chunks}, \text{attacker-state}) \in$
 set-dist
 (*execute*
 (*checked-staged-query-program* A
 $(\text{sqp-trace-fri-roots } \text{prefix})$
 $(\text{sqp-composition-fri-roots } \text{prefix})$
 $(\text{Suc } i) (\text{rounds} - \text{Suc } i)$)
 record-state)
unfolding *checked-staged-after-query-prefix-receive-def Let-def*
by (*auto elim!*: *set-dist-bindE simp: assert-def throw-no-outcome*
 split: if-splits)
have *stage-controlled*:
 controlled-ro-program (*query-opening-budgets* $\text{budgets ! } i$)
 (*query-opening-stage* A i raw)
using *controlled wf i-bound*
unfolding *staged-adversary-controlled-def staged-budget-wellformed-def*
by *simp*
have *raw-chunk-ext*: $\text{raw-state} \leq \text{chunk-state}$
 using *controlled-ro-program-extension[OF stage-controlled]* *chunk-out*
unfolding *hash-extension-preserving-def*
by *blast*
have *record-state-eq*:
 $\text{record-state} = \text{chunk-state}(|$
 $\text{PState} := \text{foldl } \text{concat } (\text{PState } \text{chunk-state}) \text{ chunk},$
 $\text{PTranscript} := \text{PTranscript } \text{chunk-state} @ \text{chunk}|)$
by (*rule record-staged-messages-outcome[OF record-out]*)
have *chunk-record-ext*: $\text{chunk-state} \leq \text{record-state}$
unfolding *record-state-eq less-eq-hash-ext-def less-eq-fmap-def*
by *simp*
have *suffix-bound*:

```

    Suc i + (rounds - Suc i) ≤ length (query-opening-budgets budgets)
  using wf i-bound unfolding staged-budget-wellformed-def by simp
have suffix-target:
  hash-target-program {}
  (sum-list
    (take (rounds - Suc i)
      (drop (Suc i) (query-opening-budgets budgets))) +
    (rounds - Suc i))
  (checked-staged-query-program A
    (sqp-trace-fri-roots prefix)
    (sqp-composition-fri-roots prefix)
    (Suc i) (rounds - Suc i))
  by (rule hash-target-program-checked-staged-query-program
    [OF controlled suffix-bound])
have suffix-ext: record-state ≤ attacker-state
  using hash-target-program-extension[OF suffix-target] suffix-out
  unfolding hash-extension-preserving-def
  by blast
show ?thesis
  by (rule hash-ext-trans[OF raw-chunk-ext])
    (rule hash-ext-trans[OF chunk-record-ext suffix-ext])
qed

lemma checked-staged-after-query-prefix-receive-with-verifier-query-bad-at-imp-length-prefix-hit-or-tree-output:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and head-support:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
  and cont-support:
    out ∈
      set-dist
        (execute
          (checked-staged-after-query-prefix-receive-with-verifier A i
            ((prefix, prefix-state), raw))
          raw-state)
  and hit:
    staged-security-with-data-state-query-bad-hit-at i out
  shows
    checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-length-committed-query-target
    (Some (((prefix, prefix-state), raw), raw-state)) ∨
    staged-security-with-data-state-query-tree-output-hit A out
proof (cases out)
case None
then show ?thesis

```

```

    using hit unfolding staged-security-with-data-state-query-bad-hit-at-def
    by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have data-out':
  Some (data, attacker-state) ∈
  set-dist
  (execute
    (checked-staged-after-query-prefix-receive A i
      ((prefix, prefix-state), raw))
    raw-state)
  by (rule
    checked-staged-after-query-prefix-receive-with-verifier-outcome(1)
    [OF cont-support[unfolded out-eq]])
have verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by (rule
    checked-staged-after-query-prefix-receive-with-verifier-outcome(2)
    [OF cont-support[unfolded out-eq]])
from hit[unfolded out-eq]
  staged-security-with-data-state-query-bad-hit-at-def Let-def]
have hit-at:
  query-index-round-set-hit-at ?s query-sampling-success-space i
  (Some (result, final-state))
  by simp
from hit-at obtain trace-table composition-table as query-idxs where
  bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and query-hit:
    query-idxs ! i ∈
    query-sampling-success-space trace-table composition-table as
  unfolding query-index-round-set-hit-at-def by blast
have transcript-decomp:
  checked-staged-transcript-program A =
  checked-staged-query-prefix-receive-with-state A i ≫=
  checked-staged-after-query-prefix-receive A i
  by (rule checked-staged-transcript-program-query-prefix-receive-decomp
    [OF i-bound])
have builder:
  Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)

```

```

      adversary-initial-state)
unfolding transcript-decomp
by (rule-tac x=((prefix, prefix-state), raw) and t=raw-state
      in set-dist-bindI)
      (rule head-support, rule data-out')
have security-decomp:
  checked-staged-security-experiment-with-data-state A =
    checked-staged-query-prefix-receive-with-state A i  $\gg$ 
    checked-staged-after-query-prefix-receive-with-verifier A i
by (rule
      checked-staged-security-experiment-with-data-state-query-prefix-receive-decomp
      [OF i-bound])
have support-some:
  Some (((data, attacker-state), result), final-state)  $\in$ 
    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
unfolding security-decomp out-eq[symmetric]
by (rule-tac x=((prefix, prefix-state), raw) and t=raw-state
      in set-dist-bindI)
      (rule head-support, rule cont-support)
have shape:
  accepted-transcript-shape ?s (Some (result, final-state))
    as query-idxs
using bound
unfolding accepted-with-bound-tables-def accepted-with-tables-def
by simp
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support-some shape]
obtain final-raw-idxs where
  as-eq: as = staged-alphas data
and len-final-raw: length final-raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map ( $\lambda$ raw. index (to-nat raw)) final-raw-idxs
and lookup-final:
   $\bigwedge j. j < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge j
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) j)) =
      Some (final-raw-idxs ! j)
by blast
have prefix-support:
  Some ((prefix, prefix-state), prefix-state)  $\in$ 
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state)
by (rule checked-staged-query-prefix-receive-with-state-prefix-support

```

```

    [OF head-support])
have recv:
  Some (raw, raw-state) ∈
    set-dist (execute receive-query-index-challenge prefix-state)
  by (rule checked-staged-query-prefix-receive-with-state-receive-support
    [OF head-support])
from checked-staged-query-prefix-with-state-alignment
  [OF wf controlled i-bound prefix-support]
have prefix-counter: PQueryCounter prefix-state = i
and prefix-state-hash:
  PState prefix-state =
    state-after-query-chunks
      (staged-query-prefix-start-hash prefix)
      (sqp-query-chunks prefix) i
and prefix-len: length (sqp-query-chunks prefix) = i
by blast+
from data-out' obtain chunk chunk-state record-state suffix-chunks where
  chunk-out:
    Some (chunk, chunk-state) ∈
      set-dist (execute (query-opening-stage A i raw) raw-state)
and record-out:
  Some ((), record-state) ∈
    set-dist (execute (record-staged-messages chunk) chunk-state)
and suffix-out:
  Some (suffix-chunks, attacker-state) ∈
    set-dist
      (execute
        (checked-staged-query-program A
          (sqp-trace-fri-roots prefix)
          (sqp-composition-fri-roots prefix)
          (Suc i) (rounds - Suc i))
        record-state)
and data-eq:
  data =
    (|staged-trace-root = sqp-trace-root prefix,
      staged-trace-fri-roots = sqp-trace-fri-roots prefix,
      staged-trace-fri-challenges = sqp-trace-fri-challenges prefix,
      staged-trace-final = sqp-trace-final prefix,
      staged-alphas = sqp-alphas prefix,
      staged-degree = sqp-degree prefix,
      staged-composition-fri-roots = sqp-composition-fri-roots prefix,
      staged-composition-fri-challenges =
        sqp-composition-fri-challenges prefix,
      staged-composition-final = sqp-composition-final prefix,
      staged-query-chunks =
        sqp-query-chunks prefix @ chunk # suffix-chunks)
unfolding checked-staged-after-query-prefix-receive-def Let-def
by (auto elim!: set-dist-bindE simp: assert-def throw-no-outcome
  split: if-splits)

```

```

have match: staged-query-prefix-matches-data i prefix data
  unfolding staged-query-prefix-matches-data-def data-eq
  using prefix-len by simp
have start-hash:
  staged-query-start-hash data = staged-query-prefix-start-hash prefix
  by (rule staged-query-start-hash-of-prefix-completion[OF match])
have prefix-key-state:
  PState prefix-state =
    state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i
  using prefix-state-hash prefix-len start-hash
  unfolding data-eq
  by (simp add: state-after-query-chunks-append-prefix)
have receive-lookup:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
  using receive-query-index-challenge-outcome[OF recv]
  prefix-counter prefix-key-state
  by simp
have raw-ext-attacker: raw-state ≤ attacker-state
  by (rule checked-staged-after-query-prefix-receive-extends
    [OF wf controlled i-bound data-out])
have attacker-ext-s: attacker-state ≤ ?s
proof –
  have attacker-state ≤ attacker-state
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  then show ?thesis
  by (rule hash-extends-verifier-state-from-adversary-right)
qed
have s-ext-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
have raw-ext-final: raw-state ≤ final-state
  by (rule hash-ext-trans[OF raw-ext-attacker])
  (rule hash-ext-trans[OF attacker-ext-s s-ext-final])
have lookup-final-raw:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
  by (rule hash-extension-lookup[OF receive-lookup raw-ext-final])
have raw-eq: final-raw-idxs ! i = raw
  using lookup-final[OF i-bound] lookup-final-raw by simp
have raw-hit-actual:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table

```



```

    (staged-alphas data)
proof –
  have index (to-nat (final-row-idxs ! i)) = query-idxs ! i
    using query-idxs-eq len-final-row i-bound by simp
  then show ?thesis
    using query-hit as-eq raw-eq by simp
qed
have prefix-ext-raw: prefix-state ≤ raw-state
  by (rule receive-query-index-challenge-extends[OF recv])
have prefix-ext-final: prefix-state ≤ final-state
  by (rule hash-ext-trans[OF prefix-ext-raw raw-ext-final])
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
from accepted-with-bound-tables-initial-roots-pullback-or-new-tree-output-hit
  [OF prefix-ext-final bound]
obtain fr f-fri-roots f-final dg composition-fri-roots final rest
  trace-tree composition-tree where
  header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and comp-nonempty: composition-fri-roots ≠ []
  and trace-created:
    created-tree trace-table trace-tree final-state
  and trace-root: fr = value trace-tree
  and comp-created:
    created-tree composition-table composition-tree final-state
  and comp-root:
    hd composition-fri-roots = value composition-tree
  and split:
    (merkle-root-binds-table fr trace-table prefix-state ∧
      merkle-root-binds-table (hd composition-fri-roots)
        composition-table prefix-state) ∨
    hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
      prefix-state final-state
  by blast
have eqs:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧

```

```

f-final = staged-trace-final data ∧
as = staged-alphas data ∧
dg = staged-degree data ∧
composition-fri-roots = staged-composition-fri-roots data ∧
final = staged-composition-final data ∧
rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp
have trace-len: length trace-table = clength * scale
using bound unfolding accepted-with-bound-tables-def
accepted-with-tables-def by simp
have comp-len: length composition-table = clength * scale
using bound unfolding accepted-with-bound-tables-def
accepted-with-tables-def by simp
from split show ?thesis
proof
assume binds:
  merkle-root-binds-table fr trace-table prefix-state ∧
  merkle-root-binds-table (hd composition-fri-roots)
  composition-table prefix-state
have candidate:
  (trace-table, composition-table) ∈
    staged-query-prefix-length-committed-table-candidates prefix
    prefix-state
unfolding staged-query-prefix-length-committed-table-candidates-def
using binds comp-nonempty eqs data-eq trace-len comp-len by simp
have target:
  index (to-nat raw) ∈
    staged-query-prefix-length-committed-query-target prefix
    prefix-state
proof —
have raw-hit-prefix:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
    (sqp-alphas prefix)
using raw-hit-actual data-eq by simp
show ?thesis
by (rule staged-query-prefix-length-committed-query-targetI
  [OF candidate raw-hit-prefix])
qed
then show ?thesis
unfolding checked-staged-query-prefix-dynamic-index-hit-def
by simp
next
assume tree-hit:
  hash-map-new-output-hit
  (set-tree trace-tree ∪ set-tree composition-tree)
  prefix-state final-state
have comp-nonempty-prefix:

```

```

sqp-composition-fri-roots prefix ≠ []
using comp-nonempty eqs data-eq by simp
have trace-root-prefix:
  sqp-trace-root prefix = value trace-tree
  using trace-root eqs data-eq by simp
have comp-root-prefix:
  hd (sqp-composition-fri-roots prefix) = value composition-tree
  using comp-root eqs data-eq by simp
have tree-event-body:
  ∃ i prefix prefix-state raw raw-state trace-table
    composition-table as query-idxs trace-tree composition-tree.
    i < rounds ∧
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state) ∧
    Some (result, final-state) ∈
      set-dist (execute verify-monad ?s) ∧
      accepted-with-bound-tables ?s (Some (result, final-state))
      trace-table composition-table as query-idxs ∧
      fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some raw ∧
      index (to-nat raw) ∈
        query-sampling-success-space trace-table composition-table
        (staged-alphas data) ∧
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
          (execute
            (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state) ∧
          sqp-composition-fri-roots prefix ≠ [] ∧
          created-tree trace-table trace-tree final-state ∧
          sqp-trace-root prefix = value trace-tree ∧
          created-tree composition-table composition-tree final-state ∧
          hd (sqp-composition-fri-roots prefix) =
            value composition-tree ∧
          hash-map-new-output-hit
            (set-tree trace-tree ∪ set-tree composition-tree)
            prefix-state final-state
apply (rule-tac x=i in exI)
apply (rule-tac x=prefix in exI)
apply (rule-tac x=prefix-state in exI)
apply (rule-tac x=raw in exI)
apply (rule-tac x=raw-state in exI)
apply (rule-tac x=trace-table in exI)

```

```

apply (rule-tac x=composition-table in exI)
apply (rule-tac x=as in exI)
apply (rule-tac x=query-idxs in exI)
apply (rule-tac x=trace-tree in exI)
apply (rule-tac x=composition-tree in exI)
using i-bound builder verifier bound lookup-final-raw raw-hit-actual
  head-support comp-nonempty-prefix trace-created trace-root-prefix
  comp-created comp-root-prefix tree-hit
by blast
have tree-event:
  staged-security-with-data-state-query-tree-output-hit A out
using tree-event-body
unfolding out-eq
  staged-security-with-data-state-query-tree-output-hit-def Let-def
by simp
then show ?thesis by simp
qed
qed

```

lemma checked-staged-security-with-data-state-query-bad-hit-at-bound-by-length-prefix-and-tree:

```

assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i < \text{rounds}$ 

```

shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-query-bad-hit-at i)
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets  $i + 1$ ) +
  query-error-bound +
  hash-collision-budget-value 0
  (staged-query-search-queries budgets  $i + 1$ ) +
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-query-tree-output-hit A)
  adversary-initial-state

```

proof –

```

let ?M = checked-staged-security-experiment-with-data-state A
let ?Head = checked-staged-query-prefix-receive-with-state A i
let ?Cont =
  checked-staged-after-query-prefix-receive-with-verifier A i
let ?Prefix =
  checked-staged-query-prefix-dynamic-index-hit
  staged-query-prefix-length-committed-query-target
let ?Tree = staged-security-with-data-state-query-tree-output-hit A
let ?Bad = staged-security-with-data-state-query-bad-hit-at i
let ?BadNoTree =  $\lambda out. ?Bad\ out \wedge \neg ?Tree\ out$ 
have split:
  wp-event ?M ?Bad adversary-initial-state  $\leq$ 

```

```

wp-event ?M ?BadNoTree adversary-initial-state +
wp-event ?M ?Tree adversary-initial-state
proof –
  have wp-event ?M ?Bad adversary-initial-state ≤
    wp-event ?M (λout. ?BadNoTree out ∨ ?Tree out)
      adversary-initial-state
  by (rule wp-event-mono) blast
  also have ... ≤
    wp-event ?M ?BadNoTree adversary-initial-state +
    wp-event ?M ?Tree adversary-initial-state
  by (rule wp-event-union-bound)
  finally show ?thesis .
qed
have prefix-bound:
  wp-event ?Head ?Prefix adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound +
    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1)
  by (rule checked-staged-query-prefix-length-committed-target-hit-bound
    [OF raw-bound wf controlled])
    (use i-bound in simp)
have bad-no-tree-bound:
  wp-event ?M ?BadNoTree adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound +
    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1)
unfolding
  checked-staged-security-experiment-with-data-state-query-prefix-receive-decomp
  [OF i-bound]
proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
  show ?BadNoTree None ⇒ ?Prefix None
  unfolding staged-security-with-data-state-query-bad-hit-at-def
    checked-staged-query-prefix-dynamic-index-hit-def
  by simp
next
fix x t out
assume head:
  Some (x, t) ∈ set-dist (execute ?Head adversary-initial-state)
  and cont: out ∈ set-dist (execute (?Cont x) t)
  and bad-no-tree: ?BadNoTree out
show ?Prefix (Some (x, t))
proof (cases x)
  case x-eq: (Pair prefix-pack raw)
  then obtain prefix prefix-state where prefix-pack-eq:
    prefix-pack = (prefix, prefix-state)

```

```

    by (cases prefix-pack) simp
  have head':
    Some (((prefix, prefix-state), raw), t) ∈
      set-dist (execute ?Head adversary-initial-state)
    using head unfolding x-eq prefix-pack-eq .
  have cont':
    out ∈
      set-dist
        (execute
          (checked-staged-after-query-prefix-receive-with-verifier A i
            ((prefix, prefix-state), raw))
          t)
    using cont unfolding x-eq prefix-pack-eq .
  from checked-staged-after-query-prefix-receive-with-verifier-query-bad-at-imp-length-prefix-hit-or-tree-output
    [OF wf controlled i-bound head' cont']
  have ?Prefix (Some (((prefix, prefix-state), raw), t)) ∨
    ?Tree out
    using bad-no-tree by simp
  then show ?thesis
    using bad-no-tree unfolding x-eq prefix-pack-eq by simp
qed
qed
have wp-event ?M ?Bad adversary-initial-state ≤
  (staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound +
    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1)) +
  wp-event ?M ?Tree adversary-initial-state
  by (rule order-trans[OF split])
  (intro add-mono bad-no-tree-bound order-refl)
then show ?thesis
  by (simp add: algebra-simps)
qed

lemma checked-staged-security-with-data-state-query-bad-hit-at-no-tree-bound-by-length-prefix:
  assumes raw-bound: query-index-raw-preimage-bound
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (λout.
        staged-security-with-data-state-query-bad-hit-at i out ∧
        ¬ staged-security-with-data-state-query-tree-output-hit A out)
    adversary-initial-state ≤
      staged-phase-relation-error size
        (staged-query-search-queries budgets i + 1) +
        query-error-bound +

```

```

    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1)
proof –
  let ?Head = checked-staged-query-prefix-receive-with-state A i
  let ?Cont =
    checked-staged-after-query-prefix-receive-with-verifier A i
  let ?Prefix =
    checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-length-committed-query-target
  let ?Tree = staged-security-with-data-state-query-tree-output-hit A
  let ?Bad = staged-security-with-data-state-query-bad-hit-at i
  let ?BadNoTree =  $\lambda out. ?Bad\ out \wedge \neg ?Tree\ out$ 
  have prefix-bound:
    wp-event ?Head ?Prefix adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound +
    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1)
  by (rule checked-staged-query-prefix-length-committed-target-hit-bound
    [OF raw-bound wf controlled])
    (use i-bound in simp)
show ?thesis
unfolding
  checked-staged-security-experiment-with-data-state-query-prefix-receive-decomp
  [OF i-bound]
proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
show ?BadNoTree None  $\implies$  ?Prefix None
  unfolding staged-security-with-data-state-query-bad-hit-at-def
  checked-staged-query-prefix-dynamic-index-hit-def
  by simp
next
fix x t out
assume head:
  Some (x, t)  $\in$  set-dist (execute ?Head adversary-initial-state)
  and cont: out  $\in$  set-dist (execute (?Cont x) t)
  and bad-no-tree: ?BadNoTree out
show ?Prefix (Some (x, t))
proof (cases x)
  case x-eq: (Pair prefix-pack raw)
  then obtain prefix prefix-state where prefix-pack-eq:
    prefix-pack = (prefix, prefix-state)
  by (cases prefix-pack) simp
  have head':
    Some (((prefix, prefix-state), raw), t)  $\in$ 
    set-dist (execute ?Head adversary-initial-state)
  using head unfolding x-eq prefix-pack-eq .
  have cont':
    out  $\in$ 

```

```

    set-dist
    (execute
      (checked-staged-after-query-prefix-receive-with-verifier A i
        ((prefix, prefix-state), raw))
      t)
    using cont unfolding x-eq prefix-pack-eq .
  from checked-staged-after-query-prefix-receive-with-verifier-query-bad-at-imp-length-prefix-hit-or-tree-output
    [OF wf controlled i-bound head' cont]
  have ?Prefix (Some (((prefix, prefix-state), raw), t)) ∨
    ?Tree out
    using bad-no-tree by simp
  then show ?thesis
    using bad-no-tree unfolding x-eq prefix-pack-eq by simp
qed
qed
qed

```

lemma *checked-staged-security-with-data-state-query-bad-hit-bound-by-length-prefixes-and-tree:*

assumes raw-bound: *query-index-raw-preimage-bound*
and wf: *staged-budget-wellformed budgets*
and controlled: *staged-adversary-controlled budgets A*
shows

wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-query-bad-hit
adversary-initial-state ≤
 $(\sum i < \text{rounds.}$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1)) +
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-query-tree-output-hit A)
adversary-initial-state

proof –

let ?M = *checked-staged-security-experiment-with-data-state A*
let ?Tree = *staged-security-with-data-state-query-tree-output-hit A*
let ?NoTree =

λi out.
staged-security-with-data-state-query-bad-hit-at i out ∧
 $\neg$?Tree out

let ?E =

λi .
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1)

have *split:*


```

wp-event ?M staged-security-with-data-state-query-bad-hit
  adversary-initial-state ≤
wp-event ?M (λout. ∃ i ∈ {.. $\text{rounds}$ }. ?NoTree i out)
  adversary-initial-state +
wp-event ?M ?Tree adversary-initial-state
proof –
  have wp-event ?M staged-security-with-data-state-query-bad-hit
    adversary-initial-state ≤
wp-event ?M
  (λout. (∃ i ∈ {.. $\text{rounds}$ }. ?NoTree i out) ∨ ?Tree out)
    adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume bad: staged-security-with-data-state-query-bad-hit out
from staged-security-with-data-state-query-bad-hit-imp-round-hit[OF bad]
obtain i where i-bound: i ∈ {.. $\text{rounds}$ }
  and hit: staged-security-with-data-state-query-bad-hit-at i out
  by blast
show (∃ i ∈ {.. $\text{rounds}$ }. ?NoTree i out) ∨ ?Tree out
proof (cases ?Tree out)
  case True
  then show ?thesis by simp
next
  case False
  then show ?thesis
    using i-bound hit by blast
qed
qed
also have ... ≤
wp-event ?M (λout. ∃ i ∈ {.. $\text{rounds}$ }. ?NoTree i out)
  adversary-initial-state +
wp-event ?M ?Tree adversary-initial-state
by (rule wp-event-union-bound)
finally show ?thesis .
qed
have no-tree-bound:
wp-event ?M (λout. ∃ i ∈ {.. $\text{rounds}$ }. ?NoTree i out)
  adversary-initial-state ≤ (∑ i < rounds. ?E i)
proof (rule wp-event-finite-union-bound)
  show finite {.. $\text{rounds}$ }
  by simp
next
  fix i
  assume i-bound: i ∈ {.. $\text{rounds}$ }
  show wp-event ?M (?NoTree i) adversary-initial-state ≤ ?E i
  by (rule
    checked-staged-security-with-data-state-query-bad-hit-at-no-tree-bound-by-length-prefix
    [OF raw-bound wf controlled])
    (use i-bound in simp)

```

qed
have *wp-event ?M staged-security-with-data-state-query-bad-hit*
adversary-initial-state $\leq$
 $(\sum i < \text{rounds}. ?E i) + \text{wp-event ?M ?Tree adversary-initial-state}$
by (*rule order-trans[OF split]*)
(intro add-mono no-tree-bound order-refl)
then show *?thesis* .
qed

lemma *checked-staged-security-with-data-state-query-bad-hit-bound-by-length-prefixes-and-tree-from-budgets:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-query-bad-hit
adversary-initial-state $\leq$
 $(\sum i < \text{rounds}.$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1)) +
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-query-tree-output-hit A)
adversary-initial-state

proof –
have *raw-bound: query-index-raw-preimage-bound*
by (*rule query-index-raw-preimage-bound-from-sampler-wellformed*)
show *?thesis*
by (*rule*
checked-staged-security-with-data-state-query-bad-hit-bound-by-length-prefixes-and-tree
[OF raw-bound wf controlled])
qed

lemma *checked-staged-security-with-data-state-query-bad-verifier-event-bound-by-length-prefixes-and-tree:*
assumes *raw-bound: query-index-raw-preimage-bound*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event query-bad)
adversary-initial-state $\leq$
 $(\sum i < \text{rounds}.$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1)) +
wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-query-tree-output-hit A)
 adversary-initial-state
proof –
have wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event query-bad)
 adversary-initial-state $\leq$
 wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-query-bad-hit
 adversary-initial-state
by (rule checked-staged-security-with-data-state-query-bad-verifier-event-bound)
 simp
also have ... $\leq$
 ($\sum i < \text{rounds}$.
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
 query-error-bound +
 hash-collision-budget-value 0
 (staged-query-search-queries budgets $i + 1$)) +
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-query-tree-output-hit A)
 adversary-initial-state
by (rule
 checked-staged-security-with-data-state-query-bad-hit-bound-by-length-prefixes-and-tree
 [OF raw-bound wf controlled])
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-query-bad-verifier-event-bound-by-length-prefixes-and-tree-from-budget

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event query-bad)
 adversary-initial-state $\leq$
 ($\sum i < \text{rounds}$.
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
 query-error-bound +
 hash-collision-budget-value 0
 (staged-query-search-queries budgets $i + 1$)) +
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-query-tree-output-hit A)
 adversary-initial-state

proof –

have raw-bound: query-index-raw-preimage-bound

by (rule query-index-raw-preimage-bound-from-sampler-wellformed)

show ?thesis

by (rule

checked-staged-security-with-data-state-query-bad-verifier-event-bound-by-length-prefixes-and-tree

[OF raw-bound wf controlled])

qed

lemma *checked-staged-transcript-query-partial-header-key-hit-at-bound-from-prefix-receive:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *prefix-bound:*
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-partial-query-target i))
adversary-initial-state ≤ C
shows
wp-event (checked-staged-transcript-program A)
(staged-transcript-query-partial-header-key-hit-at i)
adversary-initial-state ≤ C
unfolding *checked-staged-transcript-program-query-prefix-receive-decomp*
[OF i-bound]
proof (rule *wp-event-bind-bound-by-head-event[OF prefix-bound]*)
show
staged-transcript-query-partial-header-key-hit-at i None ⇒
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-partial-query-target i) None
unfolding *staged-transcript-query-partial-header-key-hit-at-def*
checked-staged-query-prefix-dynamic-index-hit-def
by *simp*
next
fix *x t out*
assume *head:*
Some (x, t) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *cont:*
out ∈
set-dist
(execute (checked-staged-after-query-prefix-receive A i x) t)
and *hit: staged-transcript-query-partial-header-key-hit-at i out*
show
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-partial-query-target i) (Some (x, t))
proof (cases *x*)
case *x-eq: (Pair prefix-pack raw)*
then obtain *prefix prefix-state* **where** *prefix-pack-eq:*
prefix-pack = (prefix, prefix-state)
by (cases *prefix-pack*) *simp*
show *?thesis*
proof (cases *out*)
case *None*

```

then show ?thesis
  using hit
  unfolding staged-transcript-query-partial-header-key-hit-at-def
  by simp
next
case (Some packed)
then obtain data attacker-state where out-eq:
  out = Some (data, attacker-state)
  by (cases packed) simp
have head':
  Some (((prefix, prefix-state), raw), t) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  using head unfolding x-eq prefix-pack-eq .
have cont':
  Some (data, attacker-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive A i
          ((prefix, prefix-state), raw))
        t)
  using cont unfolding x-eq prefix-pack-eq out-eq .
have hit':
  staged-transcript-query-partial-header-key-hit-at i
    (Some (data, attacker-state))
  using hit unfolding out-eq .
have prefix-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-partial-query-target i)
    (Some (((prefix, prefix-state), raw), t))
  by (rule
    checked-staged-after-query-prefix-receive-partial-hit-imp-prefix-hit
    [OF wf controlled i-bound head' cont' hit'])
show ?thesis
  using prefix-hit unfolding x-eq prefix-pack-eq .
qed
qed
qed

```

lemma checked-staged-transcript-query-bound-table-hit-at-committed-prefix-receive-support:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and outcome:
 Some (data, attacker-state) ∈
 set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state)
and i-bound: i < rounds

and *lookup-attacker*:
fmlookup (*HashMap* *attacker-state*)
 (*QueryIndexChallenge* *i*
 (*state-after-query-chunks*
 (*staged-query-start-hash* *data*) (*staged-query-chunks* *data*) *i*)) =
Some raw-hit
and *raw-hit-target*:
index (*to-nat* *raw-hit*) ∈
query-sampling-success-space *trace-table* *composition-table*
 (*staged-alphas* *data*)
and *comp-nonempty*:
staged-composition-fri-roots *data* ≠ []
shows
 ∃ *prefix* *prefix-state* *raw* *raw-state*.
Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) ∈
set-dist
 (*execute* (*checked-staged-query-prefix-receive-with-state* *A* *i*)
adversary-initial-state) ∧
 (*merkle-root-binds-table* (*sqp-trace-root* *prefix*) *trace-table*
prefix-state →
merkle-root-binds-table (*hd* (*sqp-composition-fri-roots* *prefix*))
composition-table *prefix-state* →
checked-staged-query-prefix-dynamic-index-hit
 (λ*prefix* *prefix-state*.
staged-query-prefix-committed-query-target *prefix* *prefix-state*)
 (*Some* (((*prefix*, *prefix-state*), *raw*), *raw-state*)))
proof –
from *checked-staged-transcript-program-query-prefix-receive-compact-support*
 [*OF* *wf* *controlled* *i-bound* *outcome*]
obtain *prefix* *prefix-state* *raw* *raw-state* **where**
prefix-receive:
Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) ∈
set-dist
 (*execute* (*checked-staged-query-prefix-receive-with-state* *A* *i*)
adversary-initial-state)
and *root-eq*: *sqp-trace-root* *prefix* = *staged-trace-root* *data*
and *alphas-eq*: *sqp-alphas* *prefix* = *staged-alphas* *data*
and *comp-roots-eq*:
sqp-composition-fri-roots *prefix* =
staged-composition-fri-roots *data*
and *prefix-ext*: *prefix-state* ≤ *attacker-state*
and *raw-ext*: *raw-state* ≤ *attacker-state*
and *lookup-raw*:
fmlookup (*HashMap* *raw-state*)
 (*QueryIndexChallenge* *i*
 (*state-after-query-chunks*
 (*staged-query-start-hash* *data*) (*staged-query-chunks* *data*) *i*)) =
Some raw
by (*rule* *checked-staged-transcript-program-query-prefix-receive-compact-support*

```

    [OF wf controlled i-bound outcome])
have lookup-attacker-raw:
  fmlookup (HashMap attacker-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
  by (rule hash-extension-lookup[OF lookup-raw raw-ext])
have raw-eq: raw = raw-hit
  using lookup-attacker lookup-attacker-raw by simp
have comp-nonempty-prefix:
  sqp-composition-fri-roots prefix ≠ []
  using comp-nonempty comp-roots-eq by simp
have hit-if-bound:
  merkle-root-binds-table (sqp-trace-root prefix) trace-table
    prefix-state ⇒
  merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
    composition-table prefix-state ⇒
  checked-staged-query-prefix-dynamic-index-hit
    (λprefix prefix-state.
      staged-query-prefix-committed-query-target prefix prefix-state)
    (Some (((prefix, prefix-state), raw), raw-state))
proof –
  assume trace-bind:
    merkle-root-binds-table (sqp-trace-root prefix) trace-table
      prefix-state
  assume comp-bind:
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
      composition-table prefix-state
  have candidate:
    (trace-table, composition-table) ∈
      query-header-committed-table-candidates prefix-state
        (sqp-trace-root prefix)
        (sqp-composition-fri-roots prefix)
  unfolding query-header-committed-table-candidates-def
  using trace-bind comp-nonempty-prefix comp-bind by simp
have raw-target-prefix:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
      (sqp-alphas prefix)
  using raw-hit-target unfolding raw-eq alphas-eq .
have target:
  index (to-nat raw) ∈
    staged-query-prefix-committed-query-target prefix prefix-state
  by (rule staged-query-prefix-committed-query-targetI
    [OF candidate raw-target-prefix])
show ?thesis
  unfolding checked-staged-query-prefix-dynamic-index-hit-def
  using target by simp

```

```

qed
show ?thesis
  using prefix-receive hit-if-bound by blast
qed

lemma checked-staged-transcript-query-bound-table-hit-at-length-committed-prefix-receive-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and i-bound: i < rounds
  and lookup-attacker:
    fmlookup (HashMap attacker-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw-hit
  and raw-hit-target:
    index (to-nat raw-hit) ∈
      query-sampling-success-space trace-table composition-table
        (staged-alphas data)
  and comp-nonempty:
    staged-composition-fri-roots data ≠ []
  and trace-len: length trace-table = clength * scale
  and comp-len: length composition-table = clength * scale
  shows
    ∃ prefix prefix-state raw raw-state.
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state) ∧
        (merkle-root-binds-table (sqp-trace-root prefix) trace-table
          prefix-state →
            merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
              composition-table prefix-state →
                checked-staged-query-prefix-dynamic-index-hit
                  staged-query-prefix-length-committed-query-target
                    (Some (((prefix, prefix-state), raw), raw-state)))
  proof -
  from checked-staged-transcript-program-query-prefix-receive-compact-support
    [OF wf controlled i-bound outcome]
  obtain prefix prefix-state raw raw-state where
    prefix-receive:
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)

```



```

    adversary-initial-state)
  and root-eq: sqp-trace-root prefix = staged-trace-root data
  and alphas-eq: sqp-alphas prefix = staged-alphas data
  and comp-roots-eq:
    sqp-composition-fri-roots prefix =
      staged-composition-fri-roots data
  and prefix-ext: prefix-state ≤ attacker-state
  and raw-ext: raw-state ≤ attacker-state
  and lookup-raw:
    fmlookup (HashMap raw-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
        Some raw
  by (rule checked-staged-transcript-program-query-prefix-receive-compact-support
    [OF wf controlled i-bound outcome])
  have lookup-attacker-raw:
    fmlookup (HashMap attacker-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
        Some raw
  by (rule hash-extension-lookup[OF lookup-raw raw-ext])
  have raw-eq: raw = raw-hit
  using lookup-attacker lookup-attacker-raw by simp
  have comp-nonempty-prefix:
    sqp-composition-fri-roots prefix ≠ []
  using comp-nonempty comp-roots-eq by simp
  have hit-if-bound:
    merkle-root-binds-table (sqp-trace-root prefix) trace-table
      prefix-state ⇒
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
      composition-table prefix-state ⇒
    checked-staged-query-prefix-dynamic-index-hit
      staged-query-prefix-length-committed-query-target
      (Some (((prefix, prefix-state), raw), raw-state))
  proof -
    assume trace-bind:
      merkle-root-binds-table (sqp-trace-root prefix) trace-table
        prefix-state
    assume comp-bind:
      merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
        composition-table prefix-state
    have candidate:
      (trace-table, composition-table) ∈
        staged-query-prefix-length-committed-table-candidates prefix
        prefix-state
    unfolding staged-query-prefix-length-committed-table-candidates-def
    using trace-bind comp-nonempty-prefix comp-bind trace-len comp-len

```

```

    by simp
  have raw-target-prefix:
    index (to-nat raw) ∈
      query-sampling-success-space trace-table composition-table
        (sqp-alphas prefix)
    using raw-hit-target unfolding raw-eq alphas-eq .
  have target:
    index (to-nat raw) ∈
      staged-query-prefix-length-committed-query-target prefix
        prefix-state
    by (rule staged-query-prefix-length-committed-query-targetI
      [OF candidate raw-target-prefix])
  show ?thesis
    unfolding checked-staged-query-prefix-dynamic-index-hit-def
    using target by simp
qed
show ?thesis
  using prefix-receive hit-if-bound by blast
qed

lemma checked-staged-bound-tables-initial-roots-late-output-at-query-prefix-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
  and i-bound: i < rounds
  and bound:
    accepted-with-bound-tables
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result, final-state))
      trace-table composition-table as query-idxs
  shows
    ∃ prefix prefix-state raw raw-state.
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state) ∧
      sqp-composition-fri-roots prefix ≠ [] ∧
      merkle-root-binds-table (sqp-trace-root prefix) trace-table

```

```

    final-state  $\wedge$ 
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
    composition-table final-state  $\wedge$ 
    ((( $\exists$  input.
      fmlookup (HashMap prefix-state) input =
        Some (sqp-trace-root prefix))  $\wedge$ 
      ( $\exists$  input.
        fmlookup (HashMap prefix-state) input =
          Some (hd (sqp-composition-fri-roots prefix))))  $\vee$ 
      hash-map-new-output-hit
      {sqp-trace-root prefix, hd (sqp-composition-fri-roots prefix)}
      prefix-state final-state)
  proof -
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from checked-staged-transcript-program-query-prefix-receive-compact-support
    [OF wf controlled i-bound outcome]
  obtain prefix prefix-state raw raw-state where
    prefix-receive:
      Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state)
  and root-eq: sqp-trace-root prefix = staged-trace-root data
  and comp-roots-eq:
    sqp-composition-fri-roots prefix =
      staged-composition-fri-roots data
  and prefix-ext: prefix-state  $\leq$  attacker-state
  and raw-ext: raw-state  $\leq$  attacker-state
  and lookup-raw:
    fmlookup (HashMap raw-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw
  by (rule checked-staged-transcript-program-query-prefix-receive-compact-support
    [OF wf controlled i-bound outcome])
  have prefix-ext-s: prefix-state  $\leq$  ?s
  by (rule hash-extends-verifier-state-from-adversary-right
    [OF prefix-ext])
  have s-ext-final: ?s  $\leq$  final-state
  by (rule verify-monad-hash-extends[OF verifier])
  have prefix-ext-final: prefix-state  $\leq$  final-state
  by (rule hash-ext-trans[OF prefix-ext-s s-ext-final])
  have staged-header:
    verifier-header-transcript ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)

```

```

    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled outcome])
from accepted-with-bound-tables-initial-roots-late-output-preexisting-or-union-hit
  [OF prefix-ext-final bound]
obtain fr f-fri-roots f-final dg composition-fri-roots final rest where
  header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
and comp-nonempty: composition-fri-roots ≠ []
and trace-bind:
  merkle-root-binds-table fr trace-table final-state
and comp-bind:
  merkle-root-binds-table (hd composition-fri-roots) composition-table
  final-state
and late:
  ((∃ input. fmlookup (HashMap prefix-state) input = Some fr) ∧
   (∃ input.
    fmlookup (HashMap prefix-state) input =
      Some (hd composition-fri-roots))) ∨
  hash-map-new-output-hit {fr, hd composition-fri-roots}
  prefix-state final-state
by blast
have eqs:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  composition-fri-roots = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp
have comp-nonempty-prefix: sqp-composition-fri-roots prefix ≠ []
  using comp-nonempty comp-roots-eq eqs by simp
have trace-bind-prefix:
  merkle-root-binds-table (sqp-trace-root prefix) trace-table
  final-state
  using trace-bind root-eq eqs by simp
have comp-bind-prefix:
  merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
  composition-table final-state
  using comp-bind comp-roots-eq eqs by simp

```

```

have late-prefix:
  (( $\exists$  input.
    fmlookup (HashMap prefix-state) input =
      Some (sqp-trace-root prefix))  $\wedge$ 
    ( $\exists$  input.
      fmlookup (HashMap prefix-state) input =
        Some (hd (sqp-composition-fri-roots prefix))))  $\vee$ 
    hash-map-new-output-hit
    {sqp-trace-root prefix, hd (sqp-composition-fri-roots prefix)}
    prefix-state final-state
  using late root-eq comp-roots-eq eqs by simp
show ?thesis
  apply (rule-tac x=prefix in exI)
  apply (rule-tac x=prefix-state in exI)
  apply (rule-tac x=raw in exI)
  apply (rule-tac x=raw-state in exI)
  using prefix-receive comp-nonempty-prefix trace-bind-prefix
    comp-bind-prefix late-prefix
  by blast
qed

lemma checked-staged-bound-tables-initial-roots-pullback-at-query-prefix-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state)  $\in$ 
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:
    Some (result, final-state)  $\in$ 
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
  and i-bound:  $i < \text{rounds}$ 
  and bound:
    accepted-with-bound-tables
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result, final-state))
      trace-table composition-table as query-idxs
  and query-hit:
    query-idxs !  $i \in$ 
      query-sampling-success-space trace-table composition-table as
  shows
     $\exists$  prefix prefix-state raw raw-state trace-tree composition-tree.
    Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
      set-dist

```

```

    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state) ∧
  sqp-alphas prefix = staged-alphas data ∧
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw ∧
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
      (staged-alphas data) ∧
  sqp-composition-fri-roots prefix ≠ [] ∧
  created-tree trace-table trace-tree final-state ∧
  sqp-trace-root prefix = value trace-tree ∧
  created-tree composition-table composition-tree final-state ∧
  hd (sqp-composition-fri-roots prefix) = value composition-tree ∧
  ((merkle-root-binds-table (sqp-trace-root prefix) trace-table
    prefix-state ∧
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
      composition-table prefix-state) ∨
    hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
        prefix-state final-state)
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
from checked-staged-transcript-program-query-prefix-receive-compact-support
  [OF wf controlled i-bound outcome]
obtain prefix prefix-state raw raw-state where
  prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
and root-eq: sqp-trace-root prefix = staged-trace-root data
and alphas-eq: sqp-alphas prefix = staged-alphas data
and comp-roots-eq:
  sqp-composition-fri-roots prefix =
    staged-composition-fri-roots data
and prefix-ext: prefix-state ≤ attacker-state
and raw-ext: raw-state ≤ attacker-state
and lookup-raw:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
by (rule checked-staged-transcript-program-query-prefix-receive-compact-support

```

```

    [OF wf controlled i-bound outcome])
have prefix-ext-s: prefix-state ≤ ?s
  by (rule hash-extends-verifier-state-from-adversary-right
    [OF prefix-ext])
have s-ext-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
have prefix-ext-final: prefix-state ≤ final-state
  by (rule hash-ext-trans[OF prefix-ext-s s-ext-final])
have raw-ext-s: raw-state ≤ ?s
  by (rule hash-extends-verifier-state-from-adversary-right
    [OF raw-ext])
have raw-ext-final: raw-state ≤ final-state
  by (rule hash-ext-trans[OF raw-ext-s s-ext-final])
have lookup-final-raw:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
  by (rule hash-extension-lookup[OF lookup-raw raw-ext-final])
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled outcome])
from accepted-with-bound-tables-initial-roots-pullback-or-new-tree-output-hit
  [OF prefix-ext-final bound]
obtain fr f-fri-roots f-final dg composition-fri-roots final rest
  trace-tree composition-tree where
  header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
and comp-nonempty: composition-fri-roots ≠ []
and trace-created:
  created-tree trace-table trace-tree final-state
and trace-root: fr = value trace-tree
and comp-created:
  created-tree composition-table composition-tree final-state
and comp-root:
  hd composition-fri-roots = value composition-tree
and split:
  (merkle-root-binds-table fr trace-table prefix-state ∧

```

```

    merkle-root-binds-table (hd composition-fri-roots)
      composition-table prefix-state) ∨
  hash-map-new-output-hit
    (set-tree trace-tree ∪ set-tree composition-tree)
    prefix-state final-state
  by blast
have eqs:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  composition-fri-roots = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  rest = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header staged-header]
  by simp
have support-some:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  by (rule checked-staged-security-experiment-with-data-state-outcomeI
      [OF outcome verifier])
have shape:
  accepted-transcript-shape ?s (Some (result, final-state))
  as query-idxs
  using bound
  unfolding accepted-with-bound-tables-def accepted-with-tables-def
  by simp
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support-some shape]
obtain final-raw-idxs where
  as-eq: as = staged-alphas data
  and len-final-raw: length final-raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) final-raw-idxs
  and lookup-final:
    ∧j. j < rounds ⇒
      fmlookup (HashMap final-state)
        (QueryIndexChallenge j
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) j)) =
        Some (final-raw-idxs ! j)
  by blast
have raw-eq: final-raw-idxs ! i = raw
  using lookup-final[OF i-bound] lookup-final-raw by simp

```



```

have raw-hit:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
      (staged-alphas data)
proof -
  have index (to-nat (final-raw-idxs ! i)) = query-idxs ! i
    using query-idxs-eq len-final-raw i-bound by simp
  then show ?thesis
    using query-hit as-eq raw-eq by simp
qed
have comp-nonempty-prefix: sqp-composition-fri-roots prefix ≠ []
  using comp-nonempty comp-roots-eq eqs by simp
have trace-root-prefix:
  sqp-trace-root prefix = value trace-tree
  using root-eq trace-root eqs by simp
have comp-root-prefix:
  hd (sqp-composition-fri-roots prefix) = value composition-tree
  using comp-root comp-roots-eq eqs by simp
have split-prefix:
  ((merkle-root-binds-table (sqp-trace-root prefix) trace-table
    prefix-state ∧
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
      composition-table prefix-state) ∨
    hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
      prefix-state final-state)
  using split root-eq comp-roots-eq eqs by simp
show ?thesis
  apply (rule-tac x=prefix in exI)
  apply (rule-tac x=prefix-state in exI)
  apply (rule-tac x=raw in exI)
  apply (rule-tac x=raw-state in exI)
  apply (rule-tac x=trace-tree in exI)
  apply (rule-tac x=composition-tree in exI)
  using prefix-receive alphas-eq lookup-final-raw raw-hit
    comp-nonempty-prefix trace-created trace-root-prefix comp-created
    comp-root-prefix split-prefix
  by blast
qed

lemma checked-staged-transcript-query-bound-table-hit-or-tree-output-at-length-committed-prefix-receive-support:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and outcome:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and verifier:

```

Some (result, final-state) ∈
set-dist
(execute verify-monad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
and *i-bound: i < rounds*
and *bound:*
accepted-with-bound-tables
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(Some (result, final-state))
trace-table composition-table as query-idxs
and *query-hit:*
query-idxs ! i ∈
query-sampling-success-space trace-table composition-table as
shows
 $\exists$ *prefix prefix-state raw raw-state trace-tree composition-tree.*
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) ∧
fmlookup (HashMap final-state)
(QueryIndexChallenge i
(state-after-query-chunks
(staged-query-start-hash data) (staged-query-chunks data) i)) =
Some raw ∧
index (to-nat raw) ∈
query-sampling-success-space trace-table composition-table
(staged-alphas data) ∧
sqp-composition-fri-roots prefix ≠ [] ∧
created-tree trace-table trace-tree final-state ∧
sqp-trace-root prefix = value trace-tree ∧
created-tree composition-table composition-tree final-state ∧
hd (sqp-composition-fri-roots prefix) = value composition-tree ∧
(checked-staged-query-prefix-dynamic-index-hit
staged-query-prefix-length-committed-query-target
(Some (((prefix, prefix-state), raw), raw-state)) ∨
hash-map-new-output-hit
(set-tree trace-tree ∪ set-tree composition-tree)
prefix-state final-state)
proof –
from *checked-staged-bound-tables-initial-roots-pullback-at-query-prefix-support*
[OF wf controlled outcome verifier i-bound bound query-hit]
obtain *prefix prefix-state raw raw-state trace-tree composition-tree where*
prefix-receive:
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)

```

and alphas-eq: sqp-alphas prefix = staged-alphas data
and lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw
and raw-hit:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
      (staged-alphas data)
and comp-nonempty: sqp-composition-fri-roots prefix ≠ []
and trace-created: created-tree trace-table trace-tree final-state
and trace-root: sqp-trace-root prefix = value trace-tree
and comp-created:
  created-tree composition-table composition-tree final-state
and comp-root:
  hd (sqp-composition-fri-roots prefix) = value composition-tree
and split:
  (merkle-root-binds-table (sqp-trace-root prefix) trace-table
    prefix-state ∧
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
      composition-table prefix-state) ∨
    hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
      prefix-state final-state
by blast
have trace-len: length trace-table = clength * scale
using bound unfolding accepted-with-bound-tables-def
  accepted-with-tables-def by simp
have comp-len: length composition-table = clength * scale
using bound unfolding accepted-with-bound-tables-def
  accepted-with-tables-def by simp
have prefix-or-tree:
  checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-length-committed-query-target
    (Some (((prefix, prefix-state), raw), raw-state)) ∨
    hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
      prefix-state final-state
proof –
from split show ?thesis
proof
  assume binds:
    merkle-root-binds-table (sqp-trace-root prefix) trace-table
      prefix-state ∧
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
      composition-table prefix-state
  then have candidate:

```

```

    (trace-table, composition-table) ∈
      staged-query-prefix-length-committed-table-candidates prefix
      prefix-state
  unfolding staged-query-prefix-length-committed-table-candidates-def
  using comp-nonempty trace-len comp-len by simp
  have target:
    index (to-nat raw) ∈
      staged-query-prefix-length-committed-query-target prefix
      prefix-state
  proof –
    have raw-hit-prefix:
      index (to-nat raw) ∈
        query-sampling-success-space trace-table composition-table
        (sqp-alphas prefix)
      using raw-hit alphas-eq by simp
    show ?thesis
      by (rule staged-query-prefix-length-committed-query-targetI
        [OF candidate raw-hit-prefix])
  qed
  then show ?thesis
    unfolding checked-staged-query-prefix-dynamic-index-hit-def
    by simp
  next
    assume
      hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
      prefix-state final-state
    then show ?thesis by simp
  qed
  qed
  show ?thesis
    apply (rule-tac x=prefix in exI)
    apply (rule-tac x=prefix-state in exI)
    apply (rule-tac x=raw in exI)
    apply (rule-tac x=raw-state in exI)
    apply (rule-tac x=trace-tree in exI)
    apply (rule-tac x=composition-tree in exI)
    using prefix-receive lookup-final raw-hit comp-nonempty trace-created
      trace-root comp-created comp-root prefix-or-tree
    by blast
  qed

lemma checked-staged-transcript-query-bound-table-hit-or-tree-output-at-committed-prefix-receive-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state) ∈
      set-dist
      (execute (checked-staged-transcript-program A)

```

```

    adversary-initial-state)
and verifier:
  Some (result, final-state) ∈
    set-dist
    (execute verify-monad
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))))
and i-bound: i < rounds
and lookup-attacker:
  fmlookup (HashMap attacker-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw-hit
and raw-hit-target:
  index (to-nat raw-hit) ∈
    query-sampling-success-space trace-table composition-table
    (staged-alphas data)
and bound:
  accepted-with-bound-tables
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state))
    trace-table composition-table as query-idxs
shows
  ∃ prefix prefix-state raw raw-state trace-tree composition-tree.
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state) ∧
      fmlookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data) (staged-query-chunks data) i)) =
        Some raw ∧
      index (to-nat raw) ∈
        query-sampling-success-space trace-table composition-table
        (staged-alphas data) ∧
      sqp-composition-fri-roots prefix ≠ [] ∧
      created-tree trace-table trace-tree final-state ∧
      sqp-trace-root prefix = value trace-tree ∧
      created-tree composition-table composition-tree final-state ∧
      hd (sqp-composition-fri-roots prefix) = value composition-tree ∧
      (checked-staged-query-prefix-dynamic-index-hit
        (λprefix prefix-state.
          staged-query-prefix-committed-query-target prefix prefix-state)
        (Some (((prefix, prefix-state), raw), raw-state))) ∨
      hash-map-new-output-hit
        (set-tree trace-tree ∪ set-tree composition-tree)

```

```

    prefix-state final-state)
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from checked-staged-transcript-program-query-prefix-receive-compact-support
  [OF wf controlled i-bound outcome]
obtain prefix prefix-state raw raw-state where
  prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
and root-eq: sqp-trace-root prefix = staged-trace-root data
and alphas-eq: sqp-alphas prefix = staged-alphas data
and comp-roots-eq:
  sqp-composition-fri-roots prefix =
    staged-composition-fri-roots data
and prefix-ext: prefix-state ≤ attacker-state
and raw-ext: raw-state ≤ attacker-state
and lookup-raw:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
by (rule checked-staged-transcript-program-query-prefix-receive-compact-support
  [OF wf controlled i-bound outcome])
have lookup-attacker-raw:
  fmlookup (HashMap attacker-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
by (rule hash-extension-lookup[OF lookup-raw raw-ext])
have raw-eq: raw = raw-hit
using lookup-attacker lookup-attacker-raw by simp
have prefix-ext-s: prefix-state ≤ ?s
by (rule hash-extends-verifier-state-from-adversary-right
  [OF prefix-ext])
have s-ext-final: ?s ≤ final-state
by (rule verify-monad-hash-extends[OF verifier])
have prefix-ext-final: prefix-state ≤ final-state
by (rule hash-ext-trans[OF prefix-ext-s s-ext-final])
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)

```

```

    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled outcome])
from accepted-with-bound-tables-initial-roots-pullback-or-new-tree-output-hit
  [OF prefix-ext-final bound]
obtain fr f-fri-roots f-final dg composition-fri-roots final rest
  trace-tree composition-tree where
header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and comp-nonempty: composition-fri-roots  $\neq$  []
and trace-created:
  created-tree trace-table trace-tree final-state
and trace-root: fr = value trace-tree
and comp-created:
  created-tree composition-table composition-tree final-state
and comp-root:
  hd composition-fri-roots = value composition-tree
and split:
  (merkle-root-binds-table fr trace-table prefix-state  $\wedge$ 
    merkle-root-binds-table (hd composition-fri-roots)
      composition-table prefix-state)  $\vee$ 
  hash-map-new-output-hit
  (set-tree trace-tree  $\cup$  set-tree composition-tree)
  prefix-state final-state
by blast
have eqs:
  fr = staged-trace-root data  $\wedge$ 
  f-fri-roots = staged-trace-fri-roots data  $\wedge$ 
  f-final = staged-trace-final data  $\wedge$ 
  as = staged-alphas data  $\wedge$ 
  dg = staged-degree data  $\wedge$ 
  composition-fri-roots = staged-composition-fri-roots data  $\wedge$ 
  final = staged-composition-final data  $\wedge$ 
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp
have split-prefix:
  (merkle-root-binds-table (sqp-trace-root prefix) trace-table
    prefix-state  $\wedge$ 
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
      composition-table prefix-state)  $\vee$ 
  hash-map-new-output-hit
  (set-tree trace-tree  $\cup$  set-tree composition-tree)
  prefix-state final-state

```

```

using split root-eq comp-roots-eq eqs by simp
have comp-nonempty-prefix:
  sqp-composition-fri-roots prefix  $\neq$  []
using comp-nonempty comp-roots-eq eqs by simp
have raw-hit-actual:
  index (to-nat raw)  $\in$ 
    query-sampling-success-space trace-table composition-table
    (staged-alphas data)
using raw-hit-target unfolding raw-eq .
have lookup-initial-raw:
  fmllookup (HashMap ?s)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
using lookup-attacker-raw by simp
have lookup-final-raw:
  fmllookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
by (rule hash-extension-lookup[OF lookup-initial-raw s-ext-final])
have trace-root-prefix:
  sqp-trace-root prefix = value trace-tree
using root-eq trace-root eqs by simp
have comp-root-prefix:
  hd (sqp-composition-fri-roots prefix) = value composition-tree
using comp-root comp-roots-eq eqs by simp
have conclusion:
  checked-staged-query-prefix-dynamic-index-hit
    ( $\lambda$ prefix prefix-state.
      staged-query-prefix-committed-query-target prefix prefix-state)
    (Some (((prefix, prefix-state), raw), raw-state))  $\vee$ 
  hash-map-new-output-hit
    (set-tree trace-tree  $\cup$  set-tree composition-tree)
    prefix-state final-state
proof –
from split-prefix show ?thesis
proof
assume prefix-binds:
  merkle-root-binds-table (sqp-trace-root prefix) trace-table
  prefix-state  $\wedge$ 
  merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
    composition-table prefix-state
then have trace-bind:
  merkle-root-binds-table (sqp-trace-root prefix) trace-table
  prefix-state
and comp-bind:

```



```

    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
      composition-table prefix-state
  by simp-all
have candidate:
  (trace-table, composition-table) ∈
    query-header-committed-table-candidates prefix-state
      (sqp-trace-root prefix)
      (sqp-composition-fri-roots prefix)
  unfolding query-header-committed-table-candidates-def
  using trace-bind comp-nonempty-prefix comp-bind by simp
have raw-target-prefix:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
      (sqp-alphas prefix)
  using raw-hit-target unfolding raw-eq alphas-eq .
have target:
  index (to-nat raw) ∈
    staged-query-prefix-committed-query-target prefix prefix-state
  by (rule staged-query-prefix-committed-query-targetI
    [OF candidate raw-target-prefix])
then show ?thesis
  unfolding checked-staged-query-prefix-dynamic-index-hit-def by simp
next
  assume
    hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
      prefix-state final-state
  then show ?thesis by simp
qed
qed
show ?thesis
  apply (rule-tac x=prefix in exI)
  apply (rule-tac x=prefix-state in exI)
  apply (rule-tac x=raw in exI)
  apply (rule-tac x=raw-state in exI)
  apply (rule-tac x=trace-tree in exI)
  apply (rule-tac x=composition-tree in exI)
  using prefix-receive lookup-final-raw raw-hit-actual
    comp-nonempty-prefix trace-created trace-root-prefix comp-created
    comp-root-prefix conclusion
  by blast
qed

lemma checked-staged-security-with-data-state-query-bad-imp-committed-prefix-hit-or-tree-output-on-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    out ∈
      set-dist

```

```

      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
    and bad: staged-security-with-data-state-query-bad-hit out
  shows
    staged-security-with-data-state-query-committed-prefix-hit A out  $\vee$ 
    staged-security-with-data-state-query-tree-output-hit A out
proof (cases out)
  case None
  then show ?thesis
    using bad unfolding staged-security-with-data-state-query-bad-hit-def
    by simp
next
  case (Some result-pack)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases result-pack, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have support-some:
    Some (((data, attacker-state), result), final-state)  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
    using support unfolding out-eq .
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF support-some]
  have builder:
    Some (data, attacker-state)  $\in$ 
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
    and verifier:
      Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)
    by blast+
  from bad[unfolded out-eq
    staged-security-with-data-state-query-bad-hit-def Let-def]
  have query-bad: query-bad ?s (Some (result, final-state))
    by simp
  from query-bad-imp-query-index-round-set-hit[OF query-bad]
  obtain i trace-table composition-table as query-idxs where
    bound:
      accepted-with-bound-tables ?s (Some (result, final-state))
      trace-table composition-table as query-idxs
    and i-bound: i < rounds
    and query-hit:
      query-idxs ! i  $\in$ 

```

```

    query-sampling-success-space trace-table composition-table as
unfolding query-index-round-set-hit-def
    query-index-round-set-hit-at-def
by blast
have shape:
    accepted-transcript-shape ?s (Some (result, final-state))
    as query-idxs
using bound
unfolding accepted-with-bound-tables-def accepted-with-tables-def
by simp
from checked-staged-security-with-data-state-accepted-shape-query-keys
    [OF wf controlled support-some shape]
obtain final-raw-idxs where
    as-eq: as = staged-alphas data
    and len-final-raw: length final-raw-idxs = rounds
    and query-idxs-eq:
        query-idxs = map ( $\lambda$ raw. index (to-nat raw)) final-raw-idxs
    and lookup-final:
         $\bigwedge j. j < \text{rounds} \implies$ 
        fmlookup (HashMap final-state)
            (QueryIndexChallenge j
              (state-after-query-chunks
                (staged-query-start-hash data)
                (staged-query-chunks data) j)) =
        Some (final-raw-idxs ! j)
    by blast
from checked-staged-transcript-program-outcome-query-lookups
    [OF wf controlled builder]
obtain raw-idxs where
    len-raw: length raw-idxs = rounds
    and lookup-attacker-all:
         $\forall j < \text{rounds}.$ 
        fmlookup (HashMap attacker-state)
            (QueryIndexChallenge j
              (state-after-query-chunks
                (staged-query-start-hash data)
                (staged-query-chunks data) j)) =
        Some (raw-idxs ! j)
    by blast
have s-ext-final: ?s  $\leq$  final-state
    by (rule verify-monad-hash-extends[OF verifier])
have lookup-initial-i:
    fmlookup (HashMap ?s)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data) (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
    using lookup-attacker-all i-bound by simp
have lookup-final-i-from-initial:

```

```

fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i)) =
  Some (raw-idxs ! i)
  by (rule hash-extension-lookup[OF lookup-initial-i s-ext-final])
have raw-eq: final-raw-idxs ! i = raw-idxs ! i
  using lookup-final[OF i-bound] lookup-final-i-from-initial by simp
have raw-hit-target:
  index (to-nat (raw-idxs ! i)) ∈
    query-sampling-success-space trace-table composition-table
      (staged-alphas data)
proof -
  have index (to-nat (final-raw-idxs ! i)) = query-idxs ! i
    using query-idxs-eq len-final-raw i-bound by simp
  then show ?thesis
    using query-hit as-eq raw-eq by simp
qed
have lookup-attacker-i:
  fmlookup (HashMap attacker-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
  using lookup-attacker-all i-bound by simp
from
  checked-staged-transcript-query-bound-table-hit-or-tree-output-at-committed-prefix-receive-support
    [OF wf controlled builder verifier i-bound lookup-attacker-i
      raw-hit-target bound]
obtain prefix prefix-state raw raw-state trace-tree composition-tree where
  prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
and lookup-final-raw:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
and raw-hit-actual:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
      (staged-alphas data)
and comp-nonempty: sqp-composition-fri-roots prefix ≠ []
and trace-created:
  created-tree trace-table trace-tree final-state
and trace-root: sqp-trace-root prefix = value trace-tree

```

```

and comp-created:
  created-tree composition-table composition-tree final-state
and comp-root:
  hd (sqp-composition-fri-roots prefix) = value composition-tree
and split:
  checked-staged-query-prefix-dynamic-index-hit
    ( $\lambda$ prefix prefix-state.
      staged-query-prefix-committed-query-target prefix prefix-state)
    (Some (((prefix, prefix-state), raw), raw-state)))  $\vee$ 
  hash-map-new-output-hit
    (set-tree trace-tree  $\cup$  set-tree composition-tree)
    prefix-state final-state
by blast
from split show ?thesis
proof
  assume prefix-hit:
    checked-staged-query-prefix-dynamic-index-hit
      ( $\lambda$ prefix prefix-state.
        staged-query-prefix-committed-query-target prefix prefix-state)
      (Some (((prefix, prefix-state), raw), raw-state)))
  have committed-body:
     $\exists i$  prefix prefix-state raw raw-state.
       $i < \text{rounds} \wedge$ 
      Some (data, attacker-state)  $\in$ 
        set-dist
          (execute (checked-staged-transcript-program A)
            adversary-initial-state)  $\wedge$ 
          Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
            set-dist
              (execute
                (checked-staged-query-prefix-receive-with-state A i)
                adversary-initial-state)  $\wedge$ 
                checked-staged-query-prefix-dynamic-index-hit
                  staged-query-prefix-committed-query-target
                    (Some (((prefix, prefix-state), raw), raw-state)))
              )
          )
      apply (rule-tac x=i in exI)
      apply (rule-tac x=prefix in exI)
      apply (rule-tac x=prefix-state in exI)
      apply (rule-tac x=raw in exI)
      apply (rule-tac x=raw-state in exI)
      using i-bound builder prefix-receive prefix-hit
      by simp
  then show ?thesis
    unfolding out-eq
      staged-security-with-data-state-query-committed-prefix-hit-def
    by simp
next
  assume tree-hit:
    hash-map-new-output-hit

```

(set-tree trace-tree $\cup$ set-tree composition-tree)
 prefix-state final-state
have tree-body:
 $\exists i$ prefix prefix-state raw raw-state trace-table composition-table
 as query-idxs trace-tree composition-tree.
 $i < \text{rounds} \wedge$
 Some (data, attacker-state) $\in$
 set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\wedge$
 Some (result, final-state) $\in$ set-dist (execute verify-monad ?s) $\wedge$
 accepted-with-bound-tables ?s (Some (result, final-state))
 trace-table composition-table as query-idxs $\wedge$
 fmlookup (HashMap final-state)
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data)
 (staged-query-chunks data) i)) =
 Some raw $\wedge$
 index (to-nat raw) $\in$
 query-sampling-success-space trace-table composition-table
 (staged-alphas data) $\wedge$
 Some (((prefix, prefix-state), raw), raw-state) $\in$
 set-dist
 (execute
 (checked-staged-query-prefix-receive-with-state A i)
 adversary-initial-state) $\wedge$
 sqp-composition-fri-roots prefix $\neq [] \wedge$
 created-tree trace-table trace-tree final-state $\wedge$
 sqp-trace-root prefix = value trace-tree $\wedge$
 created-tree composition-table composition-tree final-state $\wedge$
 hd (sqp-composition-fri-roots prefix) =
 value composition-tree $\wedge$
 hash-map-new-output-hit
 (set-tree trace-tree $\cup$ set-tree composition-tree)
 prefix-state final-state
apply (rule-tac x=i in exI)
apply (rule-tac x=prefix in exI)
apply (rule-tac x=prefix-state in exI)
apply (rule-tac x=raw in exI)
apply (rule-tac x=raw-state in exI)
apply (rule-tac x=trace-table in exI)
apply (rule-tac x=composition-table in exI)
apply (rule-tac x=as in exI)
apply (rule-tac x=query-idxs in exI)
apply (rule-tac x=trace-tree in exI)
apply (rule-tac x=composition-tree in exI)
using i-bound builder verifier bound lookup-final-raw raw-hit-actual
 prefix-receive comp-nonempty trace-created trace-root comp-created

```

      comp-root tree-hit
    by blast
  then show ?thesis
    unfolding out-eq
      staged-security-with-data-state-query-tree-output-hit-def Let-def
    by simp
qed
qed

```

lemma *checked-staged-security-with-data-state-query-bad-imp-length-committed-prefix-hit-or-tree-output-on-support:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

out $\in$

set-dist

(*execute*

(*checked-staged-security-experiment-with-data-state A*)

adversary-initial-state)

and *bad: staged-security-with-data-state-query-bad-hit out*

shows

staged-security-with-data-state-query-length-committed-prefix-hit A out $\vee$

staged-security-with-data-state-query-tree-output-hit A out

proof (*cases out*)

case *None*

then show ?thesis

using *bad* **unfolding** *staged-security-with-data-state-query-bad-hit-def*

by *simp*

next

case (*Some result-pack*)

then obtain *data attacker-state result final-state* **where** *out-eq:*

out = *Some (((data, attacker-state), result), final-state)*

by (*cases result-pack, auto split: prod.splits*)

let ?s =

verifier-state-from-adversary attacker-state

(*staged-proof-transcript data*)

have *support-some:*

Some (((data, attacker-state), result), final-state) $\in$

set-dist

(*execute*

(*checked-staged-security-experiment-with-data-state A*)

adversary-initial-state)

using *support* **unfolding** *out-eq* .

from *checked-staged-security-experiment-with-data-state-outcomeE*

[*OF support-some*]

have *builder:*

Some (data, attacker-state) $\in$

set-dist

(*execute (checked-staged-transcript-program A*)

adversary-initial-state)

```

and verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast+
from bad[unfolded out-eq
  staged-security-with-data-state-query-bad-hit-def Let-def]
have query-bad: query-bad ?s (Some (result, final-state))
by simp
from query-bad-imp-query-index-round-set-hit[OF query-bad]
obtain i trace-table composition-table as query-idxs where
  bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
and i-bound: i < rounds
and query-hit:
  query-idxs ! i ∈
    query-sampling-success-space trace-table composition-table as
unfolding query-index-round-set-hit-def
  query-index-round-set-hit-at-def
by blast
from
  checked-staged-transcript-query-bound-table-hit-or-tree-output-at-length-committed-prefix-receive-support
  [OF wf controlled builder verifier i-bound bound query-hit]
obtain prefix prefix-state raw raw-state trace-tree composition-tree where
  prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
and lookup-final-raw:
  fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
and raw-hit-actual:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
    (staged-alphas data)
and comp-nonempty: sqp-composition-fri-roots prefix ≠ []
and trace-created:
  created-tree trace-table trace-tree final-state
and trace-root: sqp-trace-root prefix = value trace-tree
and comp-created:
  created-tree composition-table composition-tree final-state
and comp-root:
  hd (sqp-composition-fri-roots prefix) = value composition-tree
and split:
  checked-staged-query-prefix-dynamic-index-hit
  staged-query-prefix-length-committed-query-target

```



```

    (Some (((prefix, prefix-state), raw), raw-state)) ∨
    hash-map-new-output-hit
    (set-tree trace-tree ∪ set-tree composition-tree)
    prefix-state final-state
  by blast
from split show ?thesis
proof
  assume prefix-hit:
    checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-length-committed-query-target
    (Some (((prefix, prefix-state), raw), raw-state))
  have length-body:
    ∃ i prefix prefix-state raw raw-state.
    i < rounds ∧
    Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
    adversary-initial-state) ∧
    Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute
    (checked-staged-query-prefix-receive-with-state A i)
    adversary-initial-state) ∧
    checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-length-committed-query-target
    (Some (((prefix, prefix-state), raw), raw-state))
  apply (rule-tac x=i in exI)
  apply (rule-tac x=prefix in exI)
  apply (rule-tac x=prefix-state in exI)
  apply (rule-tac x=raw in exI)
  apply (rule-tac x=raw-state in exI)
  using i-bound builder prefix-receive prefix-hit
  by simp
then show ?thesis
  unfolding out-eq
  staged-security-with-data-state-query-length-committed-prefix-hit-def
  by simp
next
  assume tree-hit:
    hash-map-new-output-hit
    (set-tree trace-tree ∪ set-tree composition-tree)
    prefix-state final-state
  have tree-body:
    ∃ i prefix prefix-state raw raw-state trace-table composition-table
    as query-idxs trace-tree composition-tree.
    i < rounds ∧
    Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)

```

```

    adversary-initial-state) ∧
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s) ∧
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table as query-idxs ∧
  fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
  Some raw ∧
  index (to-nat raw) ∈
  query-sampling-success-space trace-table composition-table
  (staged-alphas data) ∧
  Some (((prefix, prefix-state), raw), raw-state) ∈
  set-dist
  (execute
    (checked-staged-query-prefix-receive-with-state A i)
    adversary-initial-state) ∧
  sqp-composition-fri-roots prefix ≠ [] ∧
  created-tree trace-table trace-tree final-state ∧
  sqp-trace-root prefix = value trace-tree ∧
  created-tree composition-table composition-tree final-state ∧
  hd (sqp-composition-fri-roots prefix) =
  value composition-tree ∧
  hash-map-new-output-hit
  (set-tree trace-tree ∪ set-tree composition-tree)
  prefix-state final-state
  apply (rule-tac x=i in exI)
  apply (rule-tac x=prefix in exI)
  apply (rule-tac x=prefix-state in exI)
  apply (rule-tac x=raw in exI)
  apply (rule-tac x=raw-state in exI)
  apply (rule-tac x=trace-table in exI)
  apply (rule-tac x=composition-table in exI)
  apply (rule-tac x=as in exI)
  apply (rule-tac x=query-idxs in exI)
  apply (rule-tac x=trace-tree in exI)
  apply (rule-tac x=composition-tree in exI)
  using i-bound builder verifier bound lookup-final-raw raw-hit-actual
  prefix-receive comp-nonempty trace-created trace-root comp-created
  comp-root tree-hit
  by blast
  then show ?thesis
  unfolding out-eq
  staged-security-with-data-state-query-tree-output-hit-def Let-def
  by simp
qed
qed

```

lemma *checked-staged-security-with-data-state-query-bad-bound-from-committed-prefix-hit-or-tree-output:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows

$$\begin{aligned} & wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A) \\ & \quad staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-bad}\text{-hit } adversary\text{-initial}\text{-state} \leq \\ & wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A) \\ & \quad (staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-committed}\text{-prefix}\text{-hit } A) \\ & \quad adversary\text{-initial}\text{-state} + \\ & wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A) \\ & \quad (staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-tree}\text{-output}\text{-hit } A) \\ & \quad adversary\text{-initial}\text{-state} \end{aligned}$$

proof –
let $?M = checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A$
let $?committed =$
 $staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-committed}\text{-prefix}\text{-hit } A$
let $?tree = staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-tree}\text{-output}\text{-hit } A$
have $wp\text{-event } ?M \text{ staged}\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-bad}\text{-hit}$
 $adversary\text{-initial}\text{-state} \leq$
 $wp\text{-event } ?M (\lambda out. ?committed \text{ out } \vee ?tree \text{ out})$
 $adversary\text{-initial}\text{-state}$
by (rule *wp-event-mono-on-support*)
(rule *checked-staged-security-with-data-state-query-bad-imp-committed-prefix-hit-or-tree-output-on-support*
[OF *wf controlled*])
also have $\dots \leq$
 $wp\text{-event } ?M ?committed \text{ adversary}\text{-initial}\text{-state} +$
 $wp\text{-event } ?M ?tree \text{ adversary}\text{-initial}\text{-state}$
by (rule *wp-event-union-bound*)
finally show $?thesis$.
qed

lemma *checked-staged-security-with-data-state-query-bad-bound-from-length-committed-prefix-hit-or-tree-output:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows

$$\begin{aligned} & wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A) \\ & \quad staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-bad}\text{-hit } adversary\text{-initial}\text{-state} \leq \\ & wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A) \\ & \quad (staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-length}\text{-committed}\text{-prefix}\text{-hit } A) \\ & \quad adversary\text{-initial}\text{-state} + \\ & wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A) \\ & \quad (staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-tree}\text{-output}\text{-hit } A) \\ & \quad adversary\text{-initial}\text{-state} \end{aligned}$$

proof –
let $?M = checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A$
let $?committed =$
 $staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-length}\text{-committed}\text{-prefix}\text{-hit } A$
let $?tree = staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-tree}\text{-output}\text{-hit } A$
have $wp\text{-event } ?M \text{ staged}\text{-security}\text{-with}\text{-data}\text{-state}\text{-query}\text{-bad}\text{-hit}$

```

    adversary-initial-state ≤
wp-event ?M (λout. ?committed out ∨ ?tree out)
    adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule
checked-staged-security-with-data-state-query-bad-imp-length-committed-prefix-hit-or-tree-output-on-support
[OF wf controlled])
also have ... ≤
  wp-event ?M ?committed adversary-initial-state +
  wp-event ?M ?tree adversary-initial-state
by (rule wp-event-union-bound)
finally show ?thesis .
qed

end

end

```

```

theory Soundness-Bound-Table-Partial-Openings
imports Soundness-Partial-Merkle
begin

```

Bridges from complete Merkle-bound table witnesses to partial authenticated openings. This proof layer is kept separate from `Soundness_Partial_Merkle` to avoid growing the base Merkle/event theory.

```

context soundness
begin

```

```

definition partial-composition-openings-inconsistent-bad
:: ('f, 'a) protocol-channel-scheme ⇒
  (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool
where
  partial-composition-openings-inconsistent-bad s out ⟷
  (∃ composition-root query-idxs composition-openings result final-state
    i opn opn'.
    out = Some (result, final-state) ∧
    accepted-with-partial-composition-openings s out composition-root
      query-idxs composition-openings ∧
    i < rounds ∧
    opn ∈ set (composition-openings ! i) ∧
    opn' ∈ set (composition-openings ! i) ∧
    opening-index opn = opening-index opn' ∧
    opening-value opn ≠ opening-value opn')

```

```

lemma partial-composition-openings-inconsistent-bad-imp-partial-merkle-inconsistency-bad:
  assumes bad: partial-composition-openings-inconsistent-bad s out
  shows partial-merkle-inconsistency-bad s out
proof –

```

```

obtain composition-root query-idxs composition-openings result final-state
  i opn opn' where
    out: out = Some (result, final-state)
and partial:
  accepted-with-partial-composition-openings s out composition-root
  query-idxs composition-openings
and i-bound: i < rounds
and opn-in: opn ∈ set (composition-openings ! i)
and opn'-in: opn' ∈ set (composition-openings ! i)
and same-index: opening-index opn = opening-index opn'
and diff-value: opening-value opn ≠ opening-value opn'
using bad
unfolding partial-composition-openings-inconsistent-bad-def by blast
have table:
  partial-authenticated-table composition-root (scale * clength)
  (composition-openings ! i) final-state
proof –
  obtain result' final-state' where
    partial-out: out = Some (result', final-state')
and tables:
     $\forall i < \text{rounds.}$ 
    partial-authenticated-table composition-root (scale * clength)
    (composition-openings ! i) final-state'
using partial
unfolding accepted-with-partial-composition-openings-def by auto
have final-state-eq: final-state' = final-state
using out partial-out by simp
show ?thesis
using tables i-bound unfolding final-state-eq by simp
qed
have opn-auth: authenticated-opening-in final-state opn
using table opn-in unfolding partial-authenticated-table-def by blast
have opn'-auth: authenticated-opening-in final-state opn'
using table opn'-in unfolding partial-authenticated-table-def by blast
have same-root: opening-root opn = opening-root opn'
proof –
  have root-opn: opening-root opn = composition-root
using table opn-in unfolding partial-authenticated-table-def by simp
have root-opn': opening-root opn' = composition-root
using table opn'-in unfolding partial-authenticated-table-def by simp
show ?thesis
using root-opn root-opn' by simp
qed
have same-length: opening-length opn = opening-length opn'
proof –
  have length-opn: opening-length opn = scale * clength
using table opn-in unfolding partial-authenticated-table-def by simp
have length-opn': opening-length opn' = scale * clength
using table opn'-in unfolding partial-authenticated-table-def by simp

```

```

show ?thesis
  using length-opn length-opn' by simp
qed
have accepted-out: accepted out
  using partial
  unfolding accepted-with-partial-composition-openings-def by simp
show ?thesis
  unfolding partial-merkle-inconsistency-bad-def
  using accepted-out out opn-auth opn'-auth same-root same-length
    same-index diff-value
  by blast
qed

lemma partial-composition-openings-inconsistent-bad-imp-hash-map-output-collision-bad:
  assumes partial-composition-openings-inconsistent-bad s out
  shows hash-map-output-collision-bad s out
  by (rule partial-merkle-inconsistency-bad-imp-hash-map-output-collision-bad)
  (rule
    partial-composition-openings-inconsistent-bad-imp-partial-merkle-inconsistency-bad
    [OF assms])

lemma partial-composition-openings-inconsistent-bad-mono-hash-map-output-collision-bad:
  wp-event verify-monad (partial-composition-openings-inconsistent-bad s) s
  ≤ wp-event verify-monad (hash-map-output-collision-bad s) s
  by (rule wp-event-mono)
  (rule partial-composition-openings-inconsistent-bad-imp-hash-map-output-collision-bad)

lemma partial-composition-openings-inconsistent-bad-bound-from-hash-map-output-collision:
  fixes H :: prob
  assumes wp-event verify-monad (hash-map-output-collision-bad s) s ≤ H
  shows
    wp-event verify-monad
      (partial-composition-openings-inconsistent-bad s) s ≤ H
  by (rule order-trans
    [OF
      partial-composition-openings-inconsistent-bad-mono-hash-map-output-collision-bad
      assms])

lemma table-agrees-with-authenticated-openings-from-maps:
  assumes len-table: length table = len
    and indices: map opening-index openings = idxs
    and value-map: map opening-value openings = map (!) table idxs
    and idx-bound:  $\bigwedge \text{idx}. \text{idx} \in \text{set } \text{idxs} \implies \text{idx} < \text{len}$ 
  shows table-agrees-with-authenticated-openings table len openings
proof –
  have opening-props:
     $\forall \text{opn} \in \text{set } \text{openings}.$ 
      opening-index opn < len  $\wedge$ 
      table ! opening-index opn = opening-value opn

```

```

proof
  fix opn
  assume opn-in: opn ∈ set openings
  then obtain k where k-bound: k < length openings
    and opn-eq: opn = openings ! k
    by (metis in-set-conv-nth)
  have idxs-len: length idxs = length openings
    using arg-cong[OF indices, of length] by simp
  have idx-eq: opening-index opn = idxs ! k
    using indices k-bound idxs-len opn-eq
    by (metis length-map nth-map)
  have value-eq: opening-value opn = table ! (idxs ! k)
  proof –
    have opening-value opn = map opening-value openings ! k
      using k-bound opn-eq by simp
    also have ... = map (! table) idxs ! k
      using value-map by simp
    also have ... = table ! (idxs ! k)
      using k-bound idxs-len by simp
    finally show ?thesis .
  qed
  have idx-in: idxs ! k ∈ set idxs
    by (rule nth-mem) (use k-bound idxs-len in simp)
  show opening-index opn < len ∧
    table ! opening-index opn = opening-value opn
    using idx-bound[OF idx-in] idx-eq value-eq by simp
  qed
  show ?thesis
    unfolding table-agrees-with-authenticated-openings-def
    using len-table opening-props by simp
qed

lemma accepted-with-bound-tables-partial-opening-witnesses:
  assumes bound:
    accepted-with-bound-tables s out trace-table composition-table as
      query-idxs
  obtains result final-state fr f-fri-roots f-final dg composition-fri-roots
    final rest trace-openings composition-openings where
    out = Some (result, final-state)
    verifier-header-transcript s fr f-fri-roots f-final as dg
      composition-fri-roots final rest
    composition-fri-roots ≠ []
    accepted-with-partial-trace-openings s out fr query-idxs
      trace-openings
    accepted-with-partial-composition-openings s out
      (hd composition-fri-roots) query-idxs composition-openings
    partial-trace-table-candidate trace-table trace-openings
    partial-composition-table-candidate composition-table
      composition-openings

```

```

merkle-root-binds-table fr trace-table final-state
merkle-root-binds-table (hd composition-fri-roots) composition-table
  final-state
proof –
from bound obtain result final-state fr f-fri-roots f-final dg
  composition-fri-roots final rest where out-eq:
    out = Some (result, final-state)
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
and comp-nonempty: composition-fri-roots ≠ []
and trace-bind:
  merkle-root-binds-table fr trace-table final-state
and comp-bind:
  merkle-root-binds-table (hd composition-fri-roots) composition-table
    final-state
unfolding accepted-with-bound-tables-def by blast
have accepted-out: accepted out
  by (rule accepted-with-bound-tables-imp-accepted[OF bound])
have trace-len-mult: length trace-table = scale * clength
  using accepted-with-bound-tables-shapes(1)[OF bound]
  by (simp add: mult.commute)
have comp-len-mult: length composition-table = scale * clength
  using accepted-with-bound-tables-shapes(2)[OF bound]
  by (simp add: mult.commute)
have query-len: length query-idxs = rounds
  by (rule accepted-with-bound-tables-shapes(4)[OF bound])
have trace-exists:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\exists \text{openings.}$ 
    map opening-index openings = powers-scaled (query-idxs ! i)  $\wedge$ 
    map opening-value openings =
      map ((! trace-table) (powers-scaled (query-idxs ! i)))  $\wedge$ 
    partial-authenticated-table fr (scale * clength) openings
      final-state
proof –
fix i
assume i-bound: i < rounds
have idx-in: query-idxs ! i  $\in$  set query-idxs
  by (rule nth-mem) (use i-bound query-len in simp)
have idx-sample: query-idxs ! i  $\in$  query-sample-space
  by (rule accepted-with-bound-tables-query-sample-space[OF bound idx-in])
have idxs-bound:
   $\bigwedge \text{idx. idx} \in \text{set (powers-scaled (query-idxs ! i))} \implies$ 
    idx < scale * clength
  using query-sample-space-powers-scaled-bound[OF idx-sample]
  by (simp add: mult.commute)
show
   $\exists \text{openings.}$ 

```



```

    map opening-index openings = powers-scaled (query-idxs ! i) ∧
    map opening-value openings =
      map (!) trace-table (powers-scaled (query-idxs ! i)) ∧
    partial-authenticated-table fr (scale * clength) openings
    final-state
proof (rule merkle-root-binds-table-authenticated-openings[
  OF trace-bind trace-len-mult idxs-bound])
fix openings
assume index-map:
  map opening-index openings = powers-scaled (query-idxs ! i)
and value-map:
  map opening-value openings =
    map (!) trace-table (powers-scaled (query-idxs ! i))
and partial:
  partial-authenticated-table fr (scale * clength) openings
  final-state
show ?thesis
  by (intro exI[of - openings] conjI index-map value-map partial)
qed
qed
let ?trace-opening = λi. SOME openings.
  map opening-index openings = powers-scaled (query-idxs ! i) ∧
  map opening-value openings =
    map (!) trace-table (powers-scaled (query-idxs ! i)) ∧
  partial-authenticated-table fr (scale * clength) openings final-state
define trace-openings where
  trace-openings = map ?trace-opening [0..have trace-openings-len: length trace-openings = rounds
  unfolding trace-openings-def by simp
have trace-openings-props:
  ∧i. i < rounds ⇒
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i) ∧
    map opening-value (trace-openings ! i) =
      map (!) trace-table (powers-scaled (query-idxs ! i)) ∧
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
proof –
  fix i
  assume i-bound: i < rounds
  have some-prop:
    map opening-index (?trace-opening i) = powers-scaled (query-idxs ! i) ∧
    map opening-value (?trace-opening i) =
      map (!) trace-table (powers-scaled (query-idxs ! i)) ∧
    partial-authenticated-table fr (scale * clength)
      (?trace-opening i) final-state
  by (rule someI-ex[OF trace-exists[OF i-bound]])
show
  map opening-index (trace-openings ! i) =

```

```

    powers-scaled (query-idxs ! i) ∧
    map opening-value (trace-openings ! i) =
      map (! trace-table) (powers-scaled (query-idxs ! i)) ∧
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
    using some-prop i-bound unfolding trace-openings-def by simp
qed
have comp-exists:
  ∧i. i < rounds ⇒
    ∃ openings.
      map opening-index openings =
        [query-idxs ! i,
         fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
      map opening-value openings =
        map (! composition-table)
          [query-idxs ! i,
           fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
      partial-authenticated-table (hd composition-fri-roots)
        (scale * clength) openings final-state
proof -
  fix i
  assume i-bound: i < rounds
  have idx-in: query-idxs ! i ∈ set query-idxs
    by (rule nth-mem) (use i-bound query-len in simp)
  have idx-bound: query-idxs ! i < scale * clength
    using accepted-with-bound-tables-shapes(5)[OF bound idx-in]
    by (simp add: mult.commute)
  have sibling-bound:
    fri-sibling-index (scale * clength) (query-idxs ! i) <
      scale * clength
    unfolding fri-sibling-index-def
    using eval-domain-size-pos by (simp add: mult.commute)
  have idxs-bound:
    ∧idx. idx ∈
      set [query-idxs ! i,
           fri-sibling-index (scale * clength) (query-idxs ! i)] ⇒
      idx < scale * clength
  proof -
    fix idx
    assume idx ∈
      set [query-idxs ! i,
           fri-sibling-index (scale * clength) (query-idxs ! i)]
    then have idx = query-idxs ! i ∨
      idx = fri-sibling-index (scale * clength) (query-idxs ! i)
      by simp
    then show idx < scale * clength
      using idx-bound sibling-bound by blast
  qed
show

```

```

∃ openings.
  map opening-index openings =
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
  map opening-value openings =
    map (!) composition-table
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
  partial-authenticated-table (hd composition-fri-roots)
  (scale * clength) openings final-state
proof (rule merkle-root-binds-table-authenticated-openings[
  OF comp-bind comp-len-mult idxs-bound])
fix openings
assume index-map:
  map opening-index openings =
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)]
and value-map:
  map opening-value openings =
    map (!) composition-table
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)]
and partial:
  partial-authenticated-table (hd composition-fri-roots)
  (scale * clength) openings final-state
show ?thesis
  by (intro exI[of - openings] conjI index-map value-map partial)
qed
qed
let ?composition-opening = λi. SOME openings.
  map opening-index openings =
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
  map opening-value openings =
    map (!) composition-table
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
  partial-authenticated-table (hd composition-fri-roots)
  (scale * clength) openings final-state
define composition-openings where
  composition-openings = map ?composition-opening [0..<rounds]
have comp-openings-len: length composition-openings = rounds
unfolding composition-openings-def by simp
have comp-openings-props:
  ∧i. i < rounds ⇒
    map opening-index (composition-openings ! i) =
      [query-idxs ! i,
       fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
    map opening-value (composition-openings ! i) =

```

```

    map (!) composition-table)
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
  partial-authenticated-table (hd composition-fri-roots)
  (scale * clength) (composition-openings ! i) final-state
proof -
fix i
assume i-bound: i < rounds
have some-prop:
  map opening-index (?composition-opening i) =
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
  map opening-value (?composition-opening i) =
    map (!) composition-table)
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
  partial-authenticated-table (hd composition-fri-roots)
  (scale * clength) (?composition-opening i) final-state
by (rule someI-ex[OF comp-exists[OF i-bound]])
show
  map opening-index (composition-openings ! i) =
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
  map opening-value (composition-openings ! i) =
    map (!) composition-table)
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
  partial-authenticated-table (hd composition-fri-roots)
  (scale * clength) (composition-openings ! i) final-state
using some-prop i-bound unfolding composition-openings-def by simp
qed
have trace-partial:
  accepted-with-partial-trace-openings s out fr query-idxs
  trace-openings
proof -
have indices:
  ∧ i. i < rounds ⇒
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
using trace-openings-props by simp
have partials:
  ∧ i. i < rounds ⇒
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) final-state
using trace-openings-props by simp
show ?thesis
unfolding accepted-with-partial-trace-openings-def
by (intro conjI exI[of - result] exI[of - final-state])
  (use accepted-out out-eq query-len trace-openings-len indices partials

```

```

      in simp-all)
qed
have comp-partial:
  accepted-with-partial-composition-openings s out
  (hd composition-fri-roots) query-idxs composition-openings
proof -
  have indices:
     $\bigwedge i. i < \text{rounds} \implies$ 
    map opening-index (composition-openings ! i) =
      [query-idxs ! i,
       fri-sibling-index (scale * clength) (query-idxs ! i)]
  using comp-openings-props by simp
  have partials:
     $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table (hd composition-fri-roots)
      (scale * clength) (composition-openings ! i) final-state
  using comp-openings-props by simp
  show ?thesis
  unfolding accepted-with-partial-composition-openings-def
  by (intro conjI exI[of - result] exI[of - final-state])
    (use accepted-out out-eq query-len comp-openings-len indices partials
     in simp-all)
qed
have trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
proof -
  have agrees:
     $\bigwedge i. i < \text{rounds} \implies$ 
    table-agrees-with-authenticated-openings trace-table
      (scale * clength) (trace-openings ! i)
  proof -
    fix i
    assume i-bound: i < rounds
    have idx-in: query-idxs ! i  $\in$  set query-idxs
      by (rule nth-mem) (use i-bound query-len in simp)
    have idx-sample: query-idxs ! i  $\in$  query-sample-space
      by (rule accepted-with-bound-tables-query-sample-space[
        OF bound idx-in])
    have idxs-bound:
       $\bigwedge \text{idx}. \text{idx} \in \text{set (powers-scaled (query-idxs ! i))} \implies$ 
      idx < scale * clength
    using query-sample-space-powers-scaled-bound[OF idx-sample]
    by (simp add: mult.commute)
  show
    table-agrees-with-authenticated-openings trace-table
      (scale * clength) (trace-openings ! i)
  proof -
    have index-map:
      map opening-index (trace-openings ! i) =

```

```

    powers-scaled (query-idxs ! i)
  using trace-openings-props[OF i-bound] by blast
have value-map:
  map opening-value (trace-openings ! i) =
    map (!) trace-table (powers-scaled (query-idxs ! i))
  using trace-openings-props[OF i-bound] by blast
show ?thesis
  by (rule table-agrees-with-authenticated-openings-from-maps[
    OF trace-len-mult index-map value-map idxs-bound])
qed
qed
show ?thesis
  unfolding partial-trace-table-candidate-def
  by (intro conjI allI impI)
    (use trace-len-mult trace-openings-len agrees in simp-all)
qed
have comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
proof -
  have agrees:
     $\bigwedge i. i < \text{rounds} \implies$ 
    table-agrees-with-authenticated-openings composition-table
    (scale * clength) (composition-openings ! i)
  proof -
    fix i
    assume i-bound: i < rounds
    have idx-in: query-idxs ! i  $\in$  set query-idxs
      by (rule nth-mem) (use i-bound query-len in simp)
    have idx-bound: query-idxs ! i < scale * clength
      using accepted-with-bound-tables-shapes(5)[OF bound idx-in]
      by (simp add: mult.commute)
    have sibling-bound:
      fri-sibling-index (scale * clength) (query-idxs ! i) <
        scale * clength
    unfolding fri-sibling-index-def
    using eval-domain-size-pos by (simp add: mult.commute)
    have idxs-bound:
       $\bigwedge idx. idx \in$ 
      set [query-idxs ! i,
        fri-sibling-index (scale * clength) (query-idxs ! i)]  $\implies$ 
      idx < scale * clength
    proof -
      fix idx
      assume idx  $\in$ 
      set [query-idxs ! i,
        fri-sibling-index (scale * clength) (query-idxs ! i)]
      then have idx = query-idxs ! i  $\vee$ 
        idx = fri-sibling-index (scale * clength) (query-idxs ! i)

```

```

    by simp
  then show  $idx < scale * clength$ 
    using  $idx-bound$   $sibling-bound$  by blast
qed
show
   $table-agrees-with-authenticated-openings\ composition-table$ 
     $(scale * clength) (composition-openings ! i)$ 
proof -
  have  $index-map$ :
     $map\ opening-index\ (composition-openings ! i) =$ 
       $[query-idxs ! i,$ 
         $fri-sibling-index\ (scale * clength) (query-idxs ! i)]$ 
    using  $comp-openings-props[OF\ i-bound]$  by blast
  have  $value-map$ :
     $map\ opening-value\ (composition-openings ! i) =$ 
       $map\ (!)\ composition-table$ 
       $[query-idxs ! i,$ 
         $fri-sibling-index\ (scale * clength) (query-idxs ! i)]$ 
    using  $comp-openings-props[OF\ i-bound]$  by blast
  show ?thesis
    by (rule  $table-agrees-with-authenticated-openings-from-maps[$ 
       $OF\ comp-len-mult\ index-map\ value-map\ idxs-bound]$ )
qed
qed
show ?thesis
  unfolding  $partial-composition-table-candidate-def$ 
  by (intro  $conjI$   $allI$   $impI$ )
    (use  $comp-len-mult\ comp-openings-len\ agrees$  in  $simp-all$ )
qed
show ?thesis
  by (rule  $that[OF\ out-eq\ header\ comp-nonempty\ trace-partial\ comp-partial$ 
     $trace-candidate\ comp-candidate\ trace-bind\ comp-bind]$ )
qed

lemma  $accepted-with-bound-tables-partial-opening-witnesses-Some$ :
  assumes  $bound$ :
     $accepted-with-bound-tables\ s\ (Some\ (result,\ final-state))$ 
     $trace-table\ composition-table\ as\ query-idxs$ 
  obtains  $fr\ f-fri-roots\ f-final\ dg\ composition-fri-roots\ final\ rest$ 
     $trace-openings\ composition-openings$  where
     $verifier-header-transcript\ s\ fr\ f-fri-roots\ f-final\ as\ dg$ 
     $composition-fri-roots\ final\ rest$ 
     $composition-fri-roots \neq []$ 
     $accepted-with-partial-trace-openings\ s\ (Some\ (result,\ final-state))$ 
     $fr\ query-idxs\ trace-openings$ 
     $accepted-with-partial-composition-openings\ s$ 
     $(Some\ (result,\ final-state))\ (hd\ composition-fri-roots)\ query-idxs$ 
     $composition-openings$ 
     $partial-trace-table-candidate\ trace-table\ trace-openings$ 

```

```

partial-composition-table-candidate composition-table
  composition-openings
merkle-root-binds-table fr trace-table final-state
merkle-root-binds-table (hd composition-fri-roots) composition-table
  final-state
proof -
show ?thesis
proof (rule accepted-with-bound-tables-partial-opening-witnesses[OF bound])
fix result' final-state' fr f-fri-roots f-final dg
  composition-fri-roots final rest trace-openings composition-openings
assume out-eq:
  Some (result, final-state) = Some (result', final-state')
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and comp-nonempty: composition-fri-roots ≠ []
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr query-idxs trace-openings
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots) query-idxs
  composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-bind:
  merkle-root-binds-table fr trace-table final-state'
and comp-bind:
  merkle-root-binds-table (hd composition-fri-roots) composition-table
  final-state'
have final-state-eq: final-state' = final-state
  using out-eq by simp
have trace-bind-outer:
  merkle-root-binds-table fr trace-table final-state
  using trace-bind final-state-eq by simp
have comp-bind-outer:
  merkle-root-binds-table (hd composition-fri-roots) composition-table
  final-state
  using comp-bind final-state-eq by simp
show ?thesis
  by (rule that[OF header comp-nonempty trace-partial comp-partial
    trace-candidate comp-candidate trace-bind-outer comp-bind-outer])
qed
qed
end

```


end

theory *Soundness-Staged-Partial-Query*
imports
 Soundness-Staged-Query-Prefix
 Soundness-Bound-Table-Partial-Openings
begin

Partial-opening query events for staged soundness.

This layer records query consistency directly from authenticated openings. It deliberately avoids the stronger, generally false claim that sampled openings determine unique complete trace and composition tables.

context *soundness*
begin

lemma *checked-staged-security-with-data-state-accepted-shape-query-chunks:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *outcome:*
 Some (((data, attacker-state), result), final-state) ∈
 set-dist
 (execute
 (staged-security-experiment-with-data-state A)
 adversary-initial-state)
and *shape:*
 accepted-transcript-shape
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (Some (result, final-state)) as query-idxs

shows

as = staged-alphas data ∧
(∃ raw-idxs.
 length raw-idxs = rounds ∧
 query-idxs = map (λraw. index (to-nat raw)) raw-idxs ∧
 (∀ i < rounds.
 verifier-query-round-chunk (query-idxs ! i)
 (staged-trace-fri-roots data)
 (staged-composition-fri-roots data)
 (staged-query-chunks data ! i)) ∧
 (∀ i < rounds.
 fmllookup (HashMap final-state)
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data)
 (staged-query-chunks data) i)) =
 Some (raw-idxs ! i) ∧
 *(∀ idx ∈ set query-idxs. idx < clength * scale))*

```

proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF outcome]
  have builder:
    Some (data, attacker-state) ∈
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  by blast
  have staged-header:
    verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
  from accepted-transcript-shape-query-chunksE[OF shape]
  obtain result' final-state' fr f-fri-roots f-final dg
    composition-fri-roots final rest raw-idxs query-chunks trailing
  where out-eq:
    Some (result, final-state) = Some (result', final-state')
  and header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-chunks: length query-chunks = rounds
  and concat-chunks: List.concat query-chunks @ trailing = rest
  and chunk-shape:
    ∧i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
      f-fri-roots composition-fri-roots (query-chunks ! i)
  and lookup:
    ∧i. i < rounds ⇒
      fmlookup (HashMap final-state')
      (QueryIndexChallenge (PQueryCounter ?s + i)
        (state-after-query-chunks
          (verifier-header-state ?s fr f-fri-roots f-final as dg
            composition-fri-roots final)
            query-chunks i)) =

```

```

    Some (raw-idxs ! i)
  and idx-bound:
     $\forall idx \in \text{set query-idxs}. idx < \text{clength} * \text{scale}$ 
  by blast
  have header-eq:
    fr = staged-trace-root data  $\wedge$ 
    f-fri-roots = staged-trace-fri-roots data  $\wedge$ 
    f-final = staged-trace-final data  $\wedge$ 
    as = staged-alphas data  $\wedge$ 
    dg = staged-degree data  $\wedge$ 
    composition-fri-roots = staged-composition-fri-roots data  $\wedge$ 
    final = staged-composition-final data  $\wedge$ 
    rest = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF staged-header header]
  by simp
  have final-state-eq: final-state' = final-state
  using out-eq by simp
  have header-state-eq:
    verifier-header-state ?s fr f-fri-roots f-final as dg
    composition-fri-roots final =
    staged-query-start-hash data
  using header-eq
  unfolding verifier-header-state-def verifier-header-messages-def
    staged-query-start-hash-def staged-composition-fri-start-hash-def
    staged-trace-fri-start-hash-def
  by simp
  from checked-staged-transcript-program-query-chunks-match-verifier-lengths
    [OF builder]
  obtain staged-query-idxs where
    staged-match:
      staged-query-chunks-match-verifier-lengths data staged-query-idxs
  by blast
  have query-idxs-len: length query-idxs = rounds
  using query-idxs-eq len-raw by simp
  have staged-match-query-idxs:
    staged-query-chunks-match-verifier-lengths data query-idxs
  by (rule staged-query-chunks-match-verifier-lengths-transfer
    [OF staged-match query-idxs-len])
  have parser-chunk-len:
     $\wedge i. i < \text{rounds} \implies$ 
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
      (staged-trace-fri-roots data)
      (staged-composition-fri-roots data)
  proof -
    fix i
    assume i-bound: i < rounds
    show
      length (query-chunks ! i) =

```

```

    verifier-query-round-transcript-length (query-idxs ! i)
      (staged-trace-fri-roots data)
      (staged-composition-fri-roots data)
  using verifier-query-round-chunk-length[OF chunk-shape[OF i-bound]]
    header-eq by simp
qed
have concat-staged:
  List.concat query-chunks @ trailing =
    List.concat (staged-query-chunks data)
  using concat-chunks header-eq by simp
have query-chunks-eq:
  query-chunks = staged-query-chunks data ∧ trailing = []
  by (rule query-chunks-eq-staged-if-matching-lengths
      [OF len-chunks concat-staged parser-chunk-len
        staged-match-query-idxs])
have chunk-shape-staged:
  ∧i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
      (staged-trace-fri-roots data)
      (staged-composition-fri-roots data)
      (staged-query-chunks data ! i)
  using chunk-shape header-eq query-chunks-eq by simp
have lookup-staged:
  ∧i. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
proof -
  fix i
  assume i-bound: i < rounds
  have lookup-i:
    fmlookup (HashMap final-state')
      (QueryIndexChallenge (PQueryCounter ?s + i)
        (state-after-query-chunks
          (verifier-header-state ?s fr f-fri-roots f-final as dg
            composition-fri-roots final)
          query-chunks i)) =
      Some (raw-idxs ! i)
  by (rule lookup[OF i-bound])
show
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)

```

```

    using lookup-i final-state-eq header-state-eq query-chunks-eq by simp
qed
show ?thesis
    using header-eq len-raw query-idxs-eq chunk-shape-staged lookup-staged
    idx-bound
    by blast
qed

lemma checked-staged-security-with-data-state-query-bad-round-chunkE:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  and bad:
    staged-security-with-data-state-query-bad-hit
    (Some (((data, attacker-state), result), final-state))
  obtains i trace-table composition-table query-idxs raw-idxs where
    i < rounds
    accepted-with-bound-tables
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result, final-state))
    trace-table composition-table (staged-alphas data) query-idxs
    query-idxs ! i ∈
      query-sampling-success-space trace-table composition-table
      (staged-alphas data)
    length raw-idxs = rounds
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
    verifier-query-round-chunk (query-idxs ! i)
      (staged-trace-fri-roots data)
      (staged-composition-fri-roots data)
      (staged-query-chunks data ! i)
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
  proof -
    let ?s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    from bad[unfolded staged-security-with-data-state-query-bad-hit-def
      Let-def]
    have query-bad: query-bad ?s (Some (result, final-state))
    by simp

```

```

from query-bad-imp-query-index-round-set-hit [OF query-bad]
obtain i trace-table composition-table as query-idxs where
  bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
and i-bound: i < rounds
and query-hit:
  query-idxs ! i ∈
    query-sampling-success-space trace-table composition-table as
unfolding query-index-round-set-hit-def
  query-index-round-set-hit-at-def
by blast
have shape:
  accepted-transcript-shape ?s (Some (result, final-state))
  as query-idxs
using bound
unfolding accepted-with-bound-tables-def accepted-with-tables-def
by simp
from checked-staged-security-with-data-state-accepted-shape-query-chunks
  [OF wf controlled support shape]
obtain raw-idxs where
  as-eq: as = staged-alphas data
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and chunk-shape:
   $\bigwedge j. j < rounds \implies$ 
    verifier-query-round-chunk (query-idxs ! j)
    (staged-trace-fri-roots data)
    (staged-composition-fri-roots data)
    (staged-query-chunks data ! j)
and lookup:
   $\bigwedge j. j < rounds \implies$ 
    fmlookup (HashMap final-state)
    (QueryIndexChallenge j
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data j))) =
      Some (raw-idxs ! j)
by blast
have bound-staged:
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table (staged-alphas data) query-idxs
using bound as-eq by simp
have query-hit-staged:
  query-idxs ! i ∈
    query-sampling-success-space trace-table composition-table
    (staged-alphas data)
using query-hit as-eq by simp

```

```

show ?thesis
  by (rule that[OF i-bound bound-staged query-hit-staged len-raw
    query-idxs-eq chunk-shape[OF i-bound] lookup[OF i-bound]])
qed

lemma ntimes-outcome-decomp-at:
  assumes outcome:
    Some (results, t) ∈ set-dist (execute (ntimes m n) s)
  and i-bound: i < n
  shows
    ∃ prefix x suffix s-i s-suc.
      Some (prefix, s-i) ∈ set-dist (execute (ntimes m i) s) ∧
      Some (x, s-suc) ∈ set-dist (execute m s-i) ∧
      Some (suffix, t) ∈
        set-dist (execute (ntimes m (n - Suc i)) s-suc) ∧
      results = prefix @ x # suffix
  using outcome i-bound
proof (induction n arbitrary: i s results t)
  case 0
  then show ?case by simp
next
  case (Suc n)
  from Suc.prem1(1) obtain x results' s1 where
    head: Some (x, s1) ∈ set-dist (execute m s)
  and tail:
    Some (results', t) ∈ set-dist (execute (ntimes m n) s1)
  and results-eq: results = x # results'
  by (auto elim!: set-dist-bindE)
show ?case
proof (cases i)
  case 0
  have prefix0:
    Some ([], s) ∈ set-dist (execute (ntimes m i) s)
  using 0 by simp
  have suffix0:
    Some (results', t) ∈
      set-dist (execute (ntimes m (Suc n - Suc i)) s1)
  using tail 0 by simp
  show ?thesis
  using prefix0 head suffix0 results-eq
  by (intro exI[of - []] exI[of - x] exI[of - results']
    exI[of - s] exI[of - s1]) simp
next
  case (Suc j)
  have j-bound: j < n
  using Suc.prem1(2) Suc by simp
  from Suc.IH[OF tail j-bound]
  obtain prefix y suffix s-j s-suc where
    prefix:

```

```

    Some (prefix, s-j) ∈ set-dist (execute (ntimes m j) s1)
and round: Some (y, s-suc) ∈ set-dist (execute m s-j)
and suffix:
    Some (suffix, t) ∈
      set-dist (execute (ntimes m (n - Suc j)) s-suc)
and results'-eq: results' = prefix @ y # suffix
by blast
have prefix-suc:
    Some (x # prefix, s-j) ∈
      set-dist (execute (ntimes m (Suc j)) s)
unfolding ntimes.simps
apply (rule set-dist-bindI[OF head])
apply (rule set-dist-bindI[OF prefix])
apply simp
done
have n-minus: Suc n - Suc (Suc j) = n - Suc j
by simp
have suffix-suc:
    Some (suffix, t) ∈
      set-dist (execute (ntimes m (Suc n - Suc i)) s-suc)
using suffix n-minus Suc by simp
show ?thesis
using prefix-suc round suffix-suc results-eq results'-eq Suc
by (intro exI[of - x # prefix] exI[of - y] exI[of - suffix]
      exI[of - s-j] exI[of - s-suc]) simp
qed
qed

```

definition *partial-query-round-consistent*

```

:: 'f authenticated-opening list list ⇒
  'f authenticated-opening list list ⇒ 'f list ⇒ nat ⇒
  nat ⇒ bool

```

where

```

partial-query-round-consistent trace-openings composition-openings as i idx
⟷
  i < rounds ∧
  idx ∈ query-sample-space ∧
  length trace-openings = rounds ∧
  length composition-openings = rounds ∧
  map opening-index (trace-openings ! i) = powers-scaled idx ∧
  map opening-index (composition-openings ! i) =
    [idx, fri-sibling-index (scale * clength) idx] ∧
  composition-openings ! i ≠ [] ∧
  opening-value ((composition-openings ! i) ! 0) =
    cp-eval as (map opening-value (trace-openings ! i))
    (h ^ idx * shift)

```

definition *partial-query-success-indices-at*

```

:: 'f authenticated-opening list list ⇒

```


$\text{'f authenticated-opening list list} \Rightarrow \text{'f list} \Rightarrow \text{nat} \Rightarrow \text{nat set}$

where

$\text{partial-query-success-indices-at trace-openings composition-openings as } i = \{idx \in \text{query-sample-space}.$
 $\text{partial-query-round-consistent trace-openings composition-openings as } i \text{ } idx\}$

lemma *partial-query-success-indices-at-subset:*

$\text{partial-query-success-indices-at trace-openings composition-openings as } i \subseteq \text{query-sample-space}$
unfolding *partial-query-success-indices-at-def* **by** *auto*

lemma *partial-query-success-indices-at-subset-query-sampling-success-space:*

assumes *trace-candidate:*

$\text{partial-trace-table-candidate trace-table trace-openings}$
and *comp-candidate:*
 $\text{partial-composition-table-candidate composition-table composition-openings}$
and *trace-low:* $\text{trace-table-low-degree trace-table}$
and *comp-low:* $\text{composition-table-low-degree maxDegree composition-table}$
and *not-all:*
 $\neg \text{all-queries-consistent trace-table composition-table as}$

shows

$\text{partial-query-success-indices-at trace-openings composition-openings as } i \subseteq \text{query-sampling-success-space trace-table composition-table as}$

proof

fix *idx*

assume *idx-in:*

$idx \in \text{partial-query-success-indices-at trace-openings composition-openings as } i$

have *round:*

$\text{partial-query-round-consistent trace-openings composition-openings as } i \text{ } idx$

using *idx-in* **unfolding** *partial-query-success-indices-at-def* **by** *simp*

have *i-bound:* $i < \text{rounds}$

using *round* **unfolding** *partial-query-round-consistent-def* **by** *simp*

have *idx-sample:* $idx \in \text{query-sample-space}$

using *round* **unfolding** *partial-query-round-consistent-def* **by** *simp*

have *trace-indices:*

$\text{map opening-index (trace-openings ! } i) = \text{powers-scaled } idx$
using *round* **unfolding** *partial-query-round-consistent-def* **by** *simp*

have *comp-indices:*

$\text{map opening-index (composition-openings ! } i) = [idx, \text{fri-sibling-index (scale * clength) } idx]$
using *round* **unfolding** *partial-query-round-consistent-def* **by** *simp*

have *comp-value-opened:*

$\text{opening-value ((composition-openings ! } i) ! 0) = \text{cp-eval as (map opening-value (trace-openings ! } i))$

```

    (h ^ idx * shift)
  using round unfolding partial-query-round-consistent-def by simp
have trace-len: length trace-table = scale * clength
  using trace-candidate unfolding partial-trace-table-candidate-def
  by simp
have comp-len: length composition-table = scale * clength
  using comp-candidate unfolding partial-composition-table-candidate-def
  by simp
have trace-values:
  map (! trace-table) (powers-scaled idx) =
    map opening-value (trace-openings ! i)
  by (rule partial-trace-table-candidate-query-values
      [OF trace-candidate i-bound trace-indices])
have comp-value:
  composition-table ! idx =
    opening-value ((composition-openings ! i) ! 0)
  by (rule partial-composition-table-candidate-query-value
      [OF comp-candidate i-bound comp-indices])
have idx-domain: idx < clength * scale
  by (rule query-sample-space-less-domain [OF idx-sample])
have powers-bound:
  ∀ j ∈ set (powers-scaled idx). j < length trace-table
  using query-sample-space-powers-scaled-bound [OF idx-sample] trace-len
  by (simp add: mult.commute)
have consistent:
  query-consistent-at trace-table composition-table as idx
  unfolding query-consistent-at-def
  using idx-domain trace-len comp-len powers-bound trace-values comp-value
  comp-value-opened
  by (simp add: mult.commute)
show idx ∈ query-sampling-success-space trace-table composition-table as
  unfolding query-sampling-success-space-def query-agreement-indices-def
  using idx-sample consistent trace-low comp-low not-all by simp
qed

```

```

lemma partial-query-round-consistent-imp-query-consistent-at:
  assumes trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  and round:
    partial-query-round-consistent trace-openings composition-openings
    as i idx
  shows query-consistent-at trace-table composition-table as idx
proof -
  have i-bound: i < rounds
    using round unfolding partial-query-round-consistent-def by simp
  have idx-sample: idx ∈ query-sample-space

```

```

using round unfolding partial-query-round-consistent-def by simp
have trace-indices:
  map opening-index (trace-openings ! i) = powers-scaled idx
  using round unfolding partial-query-round-consistent-def by simp
have comp-indices:
  map opening-index (composition-openings ! i) =
    [idx, fri-sibling-index (scale * clength) idx]
  using round unfolding partial-query-round-consistent-def by simp
have comp-value-opened:
  opening-value ((composition-openings ! i) ! 0) =
    cp-eval as (map opening-value (trace-openings ! i))
    (h ^ idx * shift)
  using round unfolding partial-query-round-consistent-def by simp
have trace-len: length trace-table = scale * clength
  using trace-candidate unfolding partial-trace-table-candidate-def
  by simp
have comp-len: length composition-table = scale * clength
  using comp-candidate unfolding partial-composition-table-candidate-def
  by simp
have trace-values:
  map (! trace-table) (powers-scaled idx) =
    map opening-value (trace-openings ! i)
  by (rule partial-trace-table-candidate-query-values
    [OF trace-candidate i-bound trace-indices])
have comp-value:
  composition-table ! idx =
    opening-value ((composition-openings ! i) ! 0)
  by (rule partial-composition-table-candidate-query-value
    [OF comp-candidate i-bound comp-indices])
have idx-domain: idx < clength * scale
  by (rule query-sample-space-less-domain[OF idx-sample])
have powers-bound:
   $\forall j \in \text{set } (\text{powers-scaled idx}). j < \text{length trace-table}$ 
  using query-sample-space-powers-scaled-bound[OF idx-sample] trace-len
  by (simp add: mult.commute)
show ?thesis
  unfolding query-consistent-at-def
  using idx-domain trace-len comp-len powers-bound trace-values comp-value
    comp-value-opened
  by (simp add: mult.commute)
qed

```

```

lemma partial-query-success-indices-at-fraction-bound-if-candidate-low-degree:
  assumes trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  and trace-low: trace-table-low-degree trace-table

```

and *comp-low*: *composition-table-low-degree maxDegree composition-table*
and *not-all*:
 $\neg$ *all-queries-consistent trace-table composition-table as*
shows
 nnreal
 (*card*
 (*partial-query-success-indices-at trace-openings composition-openings*
 as i)) /
 nnreal (card query-sample-space) ≤ query-error-bound
proof –
 have *subset*:
 partial-query-success-indices-at trace-openings composition-openings as i
 $\subseteq$ *query-sampling-success-space trace-table composition-table as*
 by (rule *partial-query-success-indices-at-subset-query-sampling-success-space*
 [*OF trace-candidate comp-candidate trace-low comp-low not-all*])
 have *finite-success*:
 finite (query-sampling-success-space trace-table composition-table as)
 by (rule *finite-subset*[*OF query-sampling-success-space-subset*
 finite-query-sample-space])
 have *card-le*:
 card
 (*partial-query-success-indices-at trace-openings composition-openings*
 as i) $\leq$
 card (query-sampling-success-space trace-table composition-table as)
 by (rule *card-mono*[*OF finite-success subset*])
 have *nnreal*
 (*card*
 (*partial-query-success-indices-at trace-openings*
 composition-openings as i)) /
 nnreal (card query-sample-space) ≤
 nnreal (card (query-sampling-success-space trace-table
 composition-table as)) /
 nnreal (card query-sample-space)
 by (rule *nnreal-nat-divide-right-mono*[*OF card-le*])
 also have ... $\leq$ *query-error-bound*
 by (rule *query-sampling-success-space-fraction-bound-query-sample-space*)
 finally show ?thesis .
qed

lemma *partial-query-round-consistentI*:
 assumes *i-bound*: $i < \text{rounds}$
 and *idx-sample*: $\text{idx} \in \text{query-sample-space}$
 and *trace-len*: $\text{length trace-openings} = \text{rounds}$
 and *comp-len*: $\text{length composition-openings} = \text{rounds}$
 and *trace-indices*:
 $\text{map opening-index (trace-openings ! i)} = \text{powers-scaled idx}$
 and *comp-indices*:
 $\text{map opening-index (composition-openings ! i)} =$
 $[\text{idx}, \text{fri-sibling-index (scale * clength) idx}]$

```

and comp-value:
  opening-value ((composition-openings ! i) ! 0) =
    cp-eval as (map opening-value (trace-openings ! i))
      (h ^ idx * shift)
shows
  partial-query-round-consistent trace-openings composition-openings as
    i idx
proof –
  have comp-round-nonempty: composition-openings ! i ≠ []
  proof –
    have length (map opening-index (composition-openings ! i)) =
      length [idx, fri-sibling-index (scale * clength) idx]
    using arg-cong[OF comp-indices, of length] by simp
    then have length (composition-openings ! i) = 2
    by simp
    then show ?thesis by auto
  qed
show ?thesis
  unfolding partial-query-round-consistent-def
  using i-bound idx-sample trace-len comp-len trace-indices comp-indices
    comp-round-nonempty comp-value
  by simp
qed

lemma partial-query-success-indices-atI:
  assumes partial-query-round-consistent trace-openings composition-openings
    as i idx
  shows
    idx ∈ partial-query-success-indices-at trace-openings
      composition-openings as i
  using assms unfolding partial-query-success-indices-at-def
    partial-query-round-consistent-def by simp

definition partial-query-openings-consistent
  :: 'f authenticated-opening list ⇒
    'f authenticated-opening list ⇒ 'f list ⇒ nat ⇒ bool
where
  partial-query-openings-consistent trace-openings composition-openings as idx
    ⇔
    idx ∈ query-sample-space ∧
    map opening-index trace-openings = powers-scaled idx ∧
    map opening-index composition-openings =
      [idx, fri-sibling-index (scale * clength) idx] ∧
    composition-openings ≠ [] ∧
    opening-value (composition-openings ! 0) =
      cp-eval as (map opening-value trace-openings) (h ^ idx * shift)

lemma partial-query-openings-consistentI:
  assumes idx-sample: idx ∈ query-sample-space

```

```

and trace-indices:
  map opening-index trace-openings = powers-scaled idx
and comp-indices:
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
and comp-value:
  opening-value (composition-openings ! 0) =
    cp-eval as (map opening-value trace-openings) (h ^ idx * shift)
shows
  partial-query-openings-consistent trace-openings composition-openings
    as idx
proof –
  have comp-nonempty: composition-openings ≠ []
  proof –
    have length (map opening-index composition-openings) =
      length [idx, fri-sibling-index (scale * clength) idx]
    using arg-cong[OF comp-indices, of length] by simp
    then have length composition-openings = 2
      by simp
    then show ?thesis by auto
  qed
  show ?thesis
    unfolding partial-query-openings-consistent-def
    using idx-sample trace-indices comp-indices comp-nonempty comp-value
    by simp
  qed

lemma partial-query-round-consistent-single-roundI:
  assumes i-bound: i < rounds
  and consistent:
    partial-query-openings-consistent trace-openings composition-openings
      as idx
  shows
    partial-query-round-consistent
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      as i idx
  proof –
    have idx-sample: idx ∈ query-sample-space
    using consistent unfolding partial-query-openings-consistent-def by simp
    have trace-indices:
      map opening-index trace-openings = powers-scaled idx
    using consistent unfolding partial-query-openings-consistent-def by simp
    have comp-indices:
      map opening-index composition-openings =
        [idx, fri-sibling-index (scale * clength) idx]
    using consistent unfolding partial-query-openings-consistent-def by simp
    have comp-value:
      opening-value (composition-openings ! 0) =

```

```

      cp-eval as (map opening-value trace-openings) (h ^ idx * shift)
    using consistent unfolding partial-query-openings-consistent-def by simp
  have comp-nonempty: composition-openings ≠ []
    using consistent unfolding partial-query-openings-consistent-def by simp
  have comp-round-value:
    opening-value
    (((replicate rounds []) [i := composition-openings]) ! i) ! 0) =
    cp-eval as
      (map opening-value
        (((replicate rounds []) [i := trace-openings]) ! i))
      (h ^ idx * shift)
    using i-bound comp-value by simp
  show ?thesis
    unfolding partial-query-round-consistent-def
    using i-bound idx-sample trace-indices comp-indices comp-nonempty
      comp-round-value
    by simp
qed

```

```

lemma powers-scaled-hd:
  hd (powers-scaled idx) = idx
  unfolding powers-scaled-def
  using powers-pos by (simp add: upt-conv-Cons)

```

```

lemma partial-query-openings-consistent-index-unique:
  assumes left:
    partial-query-openings-consistent trace-openings
    composition-openings as idx
  and right:
    partial-query-openings-consistent trace-openings
    composition-openings as idx'
  shows idx = idx'
proof -
  have powers-scaled idx = powers-scaled idx'
    using left right
    unfolding partial-query-openings-consistent-def by simp
  then have hd (powers-scaled idx) = hd (powers-scaled idx')
    by simp
  then show ?thesis
    by (simp add: powers-scaled-hd)
qed

```

```

lemma partial-query-openings-consistent-indices-card-le-one:
  card
  {idx ∈ query-sample-space.
    partial-query-openings-consistent trace-openings composition-openings
    as idx} ≤ 1
proof -
  let ?S =

```

```

    {idx ∈ query-sample-space.
      partial-query-openings-consistent trace-openings composition-openings
      as idx}
  have finite-S: finite ?S
  by simp
  have unique:  $\bigwedge idx\ idx'. idx \in ?S \implies idx' \in ?S \implies idx = idx'$ 
  by (rule partial-query-openings-consistent-index-unique) auto
  show ?thesis
  proof (cases  $\exists idx. idx \in ?S$ )
    case False
    have empty:  $?S = \{\}$ 
    proof (rule equals0I)
      fix idx
      assume  $idx \in ?S$ 
      then show False
      using False by blast
    qed
    show ?thesis
    by (subst empty, simp)
  next
  case True
  then obtain idx where idx-in:  $idx \in ?S$ 
  by blast
  have subset-single:  $?S \subseteq \{idx\}$ 
  using idx-in unique by blast
  have card ?S ≤ card {idx}
  by (rule card-mono) (use subset-single in simp-all)
  then show ?thesis
  by simp
qed
qed

lemma partial-query-openings-consistent-imp-round-consistent-list-update:
  assumes i-bound:  $i < rounds$ 
  and consistent:
    partial-query-openings-consistent trace-openings composition-openings
    as idx
  shows
    partial-query-round-consistent
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    as i idx
  proof -
    have idx-sample:  $idx \in query-sample-space$ 
    using consistent
    unfolding partial-query-openings-consistent-def by simp
    have trace-indices:
      map opening-index
      (((replicate rounds []) [i := trace-openings]) ! i) =

```



```

    powers-scaled idx
  using consistent i-bound
  unfolding partial-query-openings-consistent-def by simp
have comp-indices:
  map opening-index
  (((replicate rounds []) [i := composition-openings]) ! i) =
  [idx, fri-sibling-index (scale * clength) idx]
  using consistent i-bound
  unfolding partial-query-openings-consistent-def by simp
have comp-value:
  opening-value
  (((((replicate rounds []) [i := composition-openings]) ! i) ! 0) =
  cp-eval as
  (map opening-value
    (((replicate rounds []) [i := trace-openings]) ! i))
  (h ^ idx * shift)
  using consistent i-bound
  unfolding partial-query-openings-consistent-def by simp
show ?thesis
  by (rule partial-query-round-consistentI
    [OF i-bound idx-sample])
    (use i-bound trace-indices comp-indices comp-value in simp-all)
qed

lemma partial-query-openings-consistent-indices-subset-round-success-list-update:
  assumes i-bound: i < rounds
  shows
    {idx ∈ query-sample-space.
      partial-query-openings-consistent trace-openings composition-openings
      as idx}
    ⊆
    partial-query-success-indices-at
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      as i
  unfolding partial-query-success-indices-at-def
  using partial-query-openings-consistent-imp-round-consistent-list-update
    [OF i-bound]
  by blast

lemma partial-query-openings-consistent-indices-fraction-bound-if-candidate-low-degree:
  assumes i-bound: i < rounds
  and trace-candidate:
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  and comp-candidate:
    partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
  and trace-low: trace-table-low-degree trace-table

```

and *comp-low*: *composition-table-low-degree maxDegree composition-table*
and *not-all*:
 $\neg$ *all-queries-consistent trace-table composition-table as*
shows
nnreal
 $(\text{card } \{idx \in \text{query-sample-space.}$
 $\text{partial-query-openings-consistent trace-openings}$
 $\text{composition-openings as idx}\}) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
proof –
let *?trace-openings* = $(\text{replicate rounds } []) [i := \text{trace-openings}]$
let *?composition-openings* =
 $(\text{replicate rounds } []) [i := \text{composition-openings}]$
let *?flat* =
 $\{idx \in \text{query-sample-space.}$
 $\text{partial-query-openings-consistent trace-openings composition-openings}$
 $\text{as idx}\}$
let *?round* =
 $\text{partial-query-success-indices-at } ?\text{trace-openings}$
 $??\text{composition-openings as } i$
have *subset*: $?flat \subseteq ?round$
by (rule
 $\text{partial-query-openings-consistent-indices-subset-round-success-list-update}$
 $[OF \text{ i-bound}]$)
have *finite-round*: *finite ?round*
by (rule *finite-subset*[*OF partial-query-success-indices-at-subset*
 $\text{finite-query-sample-space}$])
have *card-le*: $\text{card } ?flat \leq \text{card } ?round$
by (rule *card-mono*[*OF finite-round subset*])
have *nnreal* $(\text{card } ?flat) / \text{nnreal } (\text{card } \text{query-sample-space}) \leq$
 $\text{nnreal } (\text{card } ?round) / \text{nnreal } (\text{card } \text{query-sample-space})$
by (rule *nnreal-nat-divide-right-mono*[*OF card-le*])
also have $\dots \leq \text{query-error-bound}$
by (rule *partial-query-success-indices-at-fraction-bound-if-candidate-low-degree*
 $[OF \text{ trace-candidate comp-candidate trace-low comp-low not-all}]$)
finally show *?thesis* .
qed

lemma *partial-trace-table-candidate-single-roundI*:
assumes *i-bound*: $i < \text{rounds}$
and *len-table*: $\text{length } \text{trace-table} = \text{scale} * \text{clength}$
and *agrees*:
 $\text{table-agrees-with-authenticated-openings trace-table}$
 $(\text{scale} * \text{clength}) \text{ trace-openings}$
shows
 $\text{partial-trace-table-candidate trace-table}$
 $((\text{replicate rounds } []) [i := \text{trace-openings}])$
unfolding *partial-trace-table-candidate-def*

```

proof (intro conjI allI impI)
  show  $\text{length } \text{trace-table} = \text{scale} * \text{clength}$ 
    by (rule len-table)
next
  show  $\text{length } ((\text{replicate } \text{rounds } []) [i := \text{trace-openings}]) = \text{rounds}$ 
    using i-bound by simp
next
  fix j
  assume j-bound:  $j < \text{rounds}$ 
  show
    table-agrees-with-authenticated-openings trace-table
      (scale * clength)
      (((replicate rounds []) [i := trace-openings]) ! j)
  proof (cases j = i)
    case True
    then show ?thesis
      using agrees i-bound by simp
  next
    case False
    then show ?thesis
      using j-bound len-table
      unfolding table-agrees-with-authenticated-openings-def
      by simp
  qed
qed

```

```

lemma partial-composition-table-candidate-single-roundI:
  assumes i-bound:  $i < \text{rounds}$ 
  and len-table:  $\text{length } \text{composition-table} = \text{scale} * \text{clength}$ 
  and agrees:
    table-agrees-with-authenticated-openings composition-table
      (scale * clength) composition-openings
  shows
    partial-composition-table-candidate composition-table
      ((replicate rounds []) [i := composition-openings])
  unfolding partial-composition-table-candidate-def
proof (intro conjI allI impI)
  show  $\text{length } \text{composition-table} = \text{scale} * \text{clength}$ 
    by (rule len-table)
next
  show  $\text{length } ((\text{replicate } \text{rounds } []) [i := \text{composition-openings}]) = \text{rounds}$ 
    using i-bound by simp
next
  fix j
  assume j-bound:  $j < \text{rounds}$ 
  show
    table-agrees-with-authenticated-openings composition-table
      (scale * clength)

```

```

      (((replicate rounds []) [i := composition-openings]) ! j)
proof (cases j = i)
  case True
  then show ?thesis
    using agrees i-bound by simp
next
  case False
  then show ?thesis
    using j-bound len-table
    unfolding table-agrees-with-authenticated-openings-def
    by simp
qed
qed

lemma partial-query-openings-consistent-indices-fraction-bound-if-tables-agree-low-degree:
  assumes i-bound: i < rounds
  and trace-len: length trace-table = scale * clength
  and comp-len: length composition-table = scale * clength
  and trace-agrees:
    table-agrees-with-authenticated-openings trace-table
      (scale * clength) trace-openings
  and comp-agrees:
    table-agrees-with-authenticated-openings composition-table
      (scale * clength) composition-openings
  and trace-low: trace-table-low-degree trace-table
  and comp-low: composition-table-low-degree maxDegree composition-table
  and not-all:
    ¬ all-queries-consistent trace-table composition-table as
  shows
    nnreal
      (card
        {idx ∈ query-sample-space.
          partial-query-openings-consistent trace-openings
            composition-openings as idx}) /
      nnreal (card query-sample-space) ≤ query-error-bound
proof –
  have trace-candidate:
    partial-trace-table-candidate trace-table
      ((replicate rounds []) [i := trace-openings])
  by (rule partial-trace-table-candidate-single-roundI
    [OF i-bound trace-len trace-agrees])
  have comp-candidate:
    partial-composition-table-candidate composition-table
      ((replicate rounds []) [i := composition-openings])
  by (rule partial-composition-table-candidate-single-roundI
    [OF i-bound comp-len comp-agrees])
  show ?thesis
  by (rule
    partial-query-openings-consistent-indices-fraction-bound-if-candidate-low-degree

```

```

      [OF i-bound trace-candidate comp-candidate trace-low comp-low
       not-all])
qed

lemma table-agrees-with-authenticated-openings-values:
  assumes agrees:
    table-agrees-with-authenticated-openings table len openings
  and indices: map opening-index openings = idxs
  shows map (!) table idxs = map opening-value openings
proof (rule nth-equalityI)
  show length (map (!) table idxs) = length (map opening-value openings)
  by (simp add: indices[symmetric])
next
  fix i
  assume i-bound: i < length (map (!) table idxs)
  have i-openings: i < length openings
    using i-bound by (simp add: indices[symmetric])
  have opn-in: openings ! i ∈ set openings
    by (rule nth-mem[OF i-openings])
  have table-value:
    table ! opening-index (openings ! i) = opening-value (openings ! i)
    using agrees opn-in
  unfolding table-agrees-with-authenticated-openings-def by auto
  have idxs ! i = opening-index (openings ! i)
    using i-bound by (simp add: indices[symmetric])
  then show map (!) table idxs ! i = map opening-value openings ! i
    using table-value i-bound i-openings by simp
qed

lemma partial-authenticated-table-agrees-with-bound-table-if-clean:
  assumes bind: merkle-root-binds-table rt table final-state
  and len-table: length table = scale * clength
  and partial:
    partial-authenticated-table rt (scale * clength) openings final-state
  and clean: ¬ hash-map-output-collision final-state
  shows
    table-agrees-with-authenticated-openings table (scale * clength)
    openings
proof -
  obtain n where len-pow: clength * scale = 2 ^ n
  using eval-domain-length-power by blast
  have table-pow: length table = 2 ^ n
  using len-table len-pow by (simp add: mult.commute)
  have opening-agrees:
    ∀ opn ∈ set openings.
      opening-index opn < scale * clength ∧
      table ! opening-index opn = opening-value opn
proof
  fix opn

```

```

assume opn-in: opn  $\in$  set openings
have root: opening-root opn = rt
  using partial opn-in unfolding partial-authenticated-table-def
  by blast
have len: opening-length opn = length table
  using partial opn-in len-table
  unfolding partial-authenticated-table-def by simp
have auth: authenticated-opening-in final-state opn
  using partial opn-in unfolding partial-authenticated-table-def
  by blast
have agreement:
  opening-index opn < length table  $\wedge$ 
  table ! opening-index opn = opening-value opn
  by (rule merkle-bound-table-agrees-with-authenticated-opening-if-clean
    [OF bind table-pow auth root len clean])
then show
  opening-index opn < scale * clength  $\wedge$ 
  table ! opening-index opn = opening-value opn
  using len-table by simp
qed
show ?thesis
  unfolding table-agrees-with-authenticated-openings-def
  using len-table opening-agrees by simp
qed

lemma partial-query-openings-consistent-from-sampling-success:
assumes trace-bind:
  merkle-root-binds-table fr trace-table final-state
and comp-bind:
  merkle-root-binds-table composition-root composition-table
  final-state
and trace-len: length trace-table = scale * clength
and comp-len: length composition-table = scale * clength
and trace-auth:
  partial-authenticated-table fr (scale * clength) trace-openings
  final-state
and comp-auth:
  partial-authenticated-table composition-root (scale * clength)
  composition-openings final-state
and clean:  $\neg$  hash-map-output-collision final-state
and trace-indices:
  map opening-index trace-openings = powers-scaled idx
and comp-indices:
  map opening-index composition-openings =
  [idx, fri-sibling-index (scale * clength) idx]
and success:
  idx  $\in$  query-sampling-success-space trace-table composition-table as
shows
  partial-query-openings-consistent trace-openings composition-openings

```

```

    as idx
proof –
  have idx-sample:  $idx \in \text{query-sample-space}$ 
    using success
    unfolding query-sampling-success-space-def query-agreement-indices-def
    by (auto split: if-splits)
  have consistent: query-consistent-at trace-table composition-table as idx
    using success
    unfolding query-sampling-success-space-def query-agreement-indices-def
    by (auto split: if-splits)
  have trace-agrees:
    table-agrees-with-authenticated-openings trace-table
      (scale * clength) trace-openings
    by (rule partial-authenticated-table-agrees-with-bound-table-if-clean
      [OF trace-bind trace-len trace-auth clean])
  have comp-agrees:
    table-agrees-with-authenticated-openings composition-table
      (scale * clength) composition-openings
    by (rule partial-authenticated-table-agrees-with-bound-table-if-clean
      [OF comp-bind comp-len comp-auth clean])
  have trace-values:
    map (! trace-table) (powers-scaled idx) =
      map opening-value trace-openings
    by (rule table-agrees-with-authenticated-openings-values
      [OF trace-agrees trace-indices])
  have comp-len-openings: length composition-openings = 2
    using arg-cong[OF comp-indices, of length] by simp
  have first-in: composition-openings ! 0 ∈ set composition-openings
    by (rule nth-mem) (use comp-len-openings in simp)
  have first-index: opening-index (composition-openings ! 0) = idx
proof –
  have map opening-index composition-openings ! 0 = idx
    using comp-indices by simp
  then show ?thesis
    using comp-len-openings by simp
qed
have composition-value:
  composition-table ! idx = opening-value (composition-openings ! 0)
    using comp-agrees first-in first-index
    unfolding table-agrees-with-authenticated-openings-def by auto
have opened-value:
  opening-value (composition-openings ! 0) =
    cp-eval as (map opening-value trace-openings) (h ^ idx * shift)
    using consistent trace-values composition-value
    unfolding query-consistent-at-def by simp
show ?thesis
  by (rule partial-query-openings-consistentI
    [OF idx-sample trace-indices comp-indices opened-value])
qed

```

lemma *partial-query-round-consistent-from-sampling-success*:

assumes *trace-partial*:

accepted-with-partial-trace-openings s out fr query-idxs
trace-openings

and *comp-partial*:

accepted-with-partial-composition-openings s out composition-root
query-idxs composition-openings

and *trace-candidate*:

partial-trace-table-candidate trace-table trace-openings

and *comp-candidate*:

partial-composition-table-candidate composition-table
composition-openings

and *i-bound*: $i < \text{rounds}$

and *success*:

query-idxs ! i $\in$
query-sampling-success-space trace-table composition-table as

shows

partial-query-round-consistent trace-openings composition-openings as i
(query-idxs ! i)

proof –

let $?idx = \text{query-idxs ! } i$

have *idx-sample*: $?idx \in \text{query-sample-space}$

using *success*

unfolding *query-sampling-success-space-def query-agreement-indices-def*

by (*auto split: if-splits*)

have *consistent*: *query-consistent-at trace-table composition-table as ?idx*

using *success*

unfolding *query-sampling-success-space-def query-agreement-indices-def*

by (*auto split: if-splits*)

have *trace-len*: $\text{length trace-openings} = \text{rounds}$

by (*rule accepted-with-partial-trace-openings-shapes(2)*
 $[\text{OF trace-partial}]$)

have *comp-len*: $\text{length composition-openings} = \text{rounds}$

by (*rule accepted-with-partial-composition-openings-shapes(2)*
 $[\text{OF comp-partial}]$)

have *trace-indices*:

map opening-index (trace-openings ! i) = powers-scaled ?idx

by (*rule accepted-with-partial-trace-openings-shapes(3)*
 $[\text{OF trace-partial } i\text{-bound}]$)

have *comp-indices*:

map opening-index (composition-openings ! i) =
 $[\text{?idx, fri-sibling-index (scale * clength) ?idx}]$

by (*rule accepted-with-partial-composition-openings-shapes(3)*
 $[\text{OF comp-partial } i\text{-bound}]$)

have *trace-values*:

map (! trace-table) (powers-scaled ?idx) =
map opening-value (trace-openings ! i)

by (*rule partial-trace-table-candidate-query-values*)

$[OF \text{ trace-candidate } i\text{-bound } \text{trace-indices}]$
have *composition-value*:
 $\text{composition-table } ! ?idx =$
 $\text{opening-value } ((\text{composition-openings } ! i) ! 0)$
by (rule *partial-composition-table-candidate-query-value*
 $[OF \text{ comp-candidate } i\text{-bound } \text{comp-indices}]$)
have *opened-value*:
 $\text{opening-value } ((\text{composition-openings } ! i) ! 0) =$
 $\text{cp-eval as } (\text{map opening-value } (\text{trace-openings } ! i))$
 $(h \wedge ?idx * \text{shift})$
using *consistent trace-values composition-value*
unfolding *query-consistent-at-def* **by** *simp*
show *?thesis*
by (rule *partial-query-round-consistentI*
 $[OF \text{ } i\text{-bound } idx\text{-sample } \text{trace-len } \text{comp-len } \text{trace-indices } \text{comp-indices}$
 $\text{opened-value}]$)
qed

definition *query-header-supported-partial-opening-witnesses*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow$
 $('f \text{ authenticated-opening list} \times$
 $'f \text{ authenticated-opening list}) \text{ set}$

where

$\text{query-header-supported-partial-opening-witnesses } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots final} =$
 $\{(\text{trace-openings}, \text{composition-openings}).$
 $\text{composition-fri-roots} \neq [] \wedge$
 $(\exists \text{ result final-state rest.}$
 $\text{Some } (\text{result}, \text{final-state}) \in \text{set-dist } (\text{execute verify-monad } s) \wedge$
 $\text{partial-authenticated-table fr } (\text{scale} * \text{clength}) \text{ trace-openings}$
 $\text{final-state} \wedge$
 $\text{partial-authenticated-table } (\text{hd composition-fri-roots})$
 $(\text{scale} * \text{clength}) \text{ composition-openings final-state} \wedge$
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final as dg}$
 $\text{composition-fri-roots final rest})\}$

lemma *query-header-supported-partial-opening-witnessesE*:

assumes

$(\text{trace-openings}, \text{composition-openings}) \in$
 $\text{query-header-supported-partial-opening-witnesses } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots final}$

obtains *result final-state rest* **where**

$\text{composition-fri-roots} \neq []$
 $\text{Some } (\text{result}, \text{final-state}) \in \text{set-dist } (\text{execute verify-monad } s)$
 $\text{partial-authenticated-table fr } (\text{scale} * \text{clength}) \text{ trace-openings}$
 final-state
 $\text{partial-authenticated-table } (\text{hd composition-fri-roots})$

$(scale * clength) \text{ composition-openings final-state}$
 $verifier\text{-}header\text{-}transcript \ s \ fr \ f\text{-}fri\text{-}roots \ f\text{-}final \ as \ dg$
 $composition\text{-}fri\text{-}roots \ final \ rest$
using *assms*
unfolding *query-header-supported-partial-opening-witnesses-def*
by *blast*

definition *query-header-supported-partial-opening-success-indices-at*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat} \Rightarrow \text{nat set}$

where

$query\text{-}header\text{-}supported\text{-}partial\text{-}opening\text{-}success\text{-}indices\text{-}at \ s \ fr$
 $f\text{-}fri\text{-}roots \ f\text{-}final \ as \ dg \ composition\text{-}fri\text{-}roots \ final \ i =$
 $\{idx \in query\text{-}sample\text{-}space.$
 $\exists \text{ trace-openings } composition\text{-}openings.$
 $(\text{trace-openings}, \text{composition-openings}) \in$
 $query\text{-}header\text{-}supported\text{-}partial\text{-}opening\text{-}witnesses \ s \ fr$
 $f\text{-}fri\text{-}roots \ f\text{-}final \ as \ dg \ composition\text{-}fri\text{-}roots \ final \ \wedge$
 $partial\text{-}query\text{-}openings\text{-}consistent \ \text{trace-openings } composition\text{-}openings$
 $as \ idx\}$

lemma *query-header-supported-partial-opening-success-indices-at-subset:*

$query\text{-}header\text{-}supported\text{-}partial\text{-}opening\text{-}success\text{-}indices\text{-}at \ s \ fr$
 $f\text{-}fri\text{-}roots \ f\text{-}final \ as \ dg \ composition\text{-}fri\text{-}roots \ final \ i$
 $\subseteq query\text{-}sample\text{-}space$

unfolding *query-header-supported-partial-opening-success-indices-at-def*
by *auto*

lemma *query-header-supported-partial-opening-success-indices-atE:*

assumes

$idx \in query\text{-}header\text{-}supported\text{-}partial\text{-}opening\text{-}success\text{-}indices\text{-}at \ s \ fr$
 $f\text{-}fri\text{-}roots \ f\text{-}final \ as \ dg \ composition\text{-}fri\text{-}roots \ final \ i$

obtains *trace-openings composition-openings* **where**

$idx \in query\text{-}sample\text{-}space$
 $(\text{trace-openings}, \text{composition-openings}) \in$
 $query\text{-}header\text{-}supported\text{-}partial\text{-}opening\text{-}witnesses \ s \ fr$
 $f\text{-}fri\text{-}roots \ f\text{-}final \ as \ dg \ composition\text{-}fri\text{-}roots \ final$
 $partial\text{-}query\text{-}openings\text{-}consistent \ \text{trace-openings } composition\text{-}openings$
 $as \ idx$

using *assms*

unfolding *query-header-supported-partial-opening-success-indices-at-def*
by *blast*

lemma *query-header-supported-partial-opening-success-indices-at-UN:*

$query\text{-}header\text{-}supported\text{-}partial\text{-}opening\text{-}success\text{-}indices\text{-}at \ s \ fr$
 $f\text{-}fri\text{-}roots \ f\text{-}final \ as \ dg \ composition\text{-}fri\text{-}roots \ final \ i =$
 $(\bigcup (\text{trace-openings}, \text{composition-openings}) \in$
 $query\text{-}header\text{-}supported\text{-}partial\text{-}opening\text{-}witnesses \ s \ fr$

f-fri-roots f-final as dg composition-fri-roots final.
{idx ∈ query-sample-space.
partial-query-openings-consistent trace-openings
composition-openings as idx})
unfolding *query-header-supported-partial-opening-success-indices-at-def*
by *auto*

lemma *query-header-supported-partial-opening-success-indices-at-subset-if-witnesses:*

assumes *cover:*

$\bigwedge$ *trace-openings composition-openings.*
(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses s fr
f-fri-roots f-final as dg composition-fri-roots final $\implies$
{idx ∈ query-sample-space.
partial-query-openings-consistent trace-openings
composition-openings as idx} $\subseteq B$

shows

query-header-supported-partial-opening-success-indices-at s fr
f-fri-roots f-final as dg composition-fri-roots final i $\subseteq B$

unfolding *query-header-supported-partial-opening-success-indices-at-UN*

using *cover* **by** *auto*

lemma *query-header-supported-partial-opening-success-indices-atI-from-bound-tables:*

assumes *outcome:*

Some (result, final-state) ∈ set-dist (execute verify-monad s)

and *header:*

verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest

and *comp-nonempty:* *composition-fri-roots $\neq []$*

and *trace-bind:*

merkle-root-binds-table fr trace-table final-state

and *comp-bind:*

merkle-root-binds-table (hd composition-fri-roots) composition-table
final-state

and *trace-len:* *length trace-table = scale * clength*

and *comp-len:* *length composition-table = scale * clength*

and *trace-auth:*

*partial-authenticated-table fr (scale * clength) trace-openings*
final-state

and *comp-auth:*

partial-authenticated-table (hd composition-fri-roots)
*(scale * clength) composition-openings final-state*

and *clean:* $\neg$ *hash-map-output-collision final-state*

and *trace-indices:*

map opening-index trace-openings = powers-scaled idx

and *comp-indices:*

map opening-index composition-openings =
*[idx, fri-sibling-index (scale * clength) idx]*

and *success:*

$idx \in \text{query-sampling-success-space trace-table composition-table as}$
shows
 $idx \in$
 $\text{query-header-supported-partial-opening-success-indices-at } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as } dg \text{ composition-fri-roots final } i$
proof –
have *consistent*:
 $\text{partial-query-openings-consistent trace-openings composition-openings}$
 $\text{as } idx$
by (rule *partial-query-openings-consistent-from-sampling-success*
 $[OF \text{ trace-bind comp-bind trace-len comp-len trace-auth comp-auth}$
 $\text{clean trace-indices comp-indices success}]$)
have *witness*:
 $(\text{trace-openings}, \text{composition-openings}) \in$
 $\text{query-header-supported-partial-opening-witnesses } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as } dg \text{ composition-fri-roots final}$
unfolding *query-header-supported-partial-opening-witnesses-def*
using *comp-nonempty outcome trace-auth comp-auth header*
by (auto split: prod.splits)
have *idx-sample*: $idx \in \text{query-sample-space}$
using *consistent*
unfolding *partial-query-openings-consistent-def* **by** *simp*
show *?thesis*
unfolding *query-header-supported-partial-opening-success-indices-at-def*
by (intro CollectI conjI exI[*of - trace-openings*]
 $\text{exI[of - composition-openings]}$)
 $(\text{use } idx\text{-sample witness consistent in simp-all})$
qed

lemma *query-header-supported-partial-opening-success-indices-atI-from-partials*:
assumes *outcome*:
 $\text{Some (result, final-state)} \in \text{set-dist (execute verify-monad } s)$
and *trace-partial*:
 $\text{accepted-with-partial-trace-openings } s$
 $(\text{Some (result, final-state)}) \text{ fr query-idxs trace-openings}$
and *comp-partial*:
 $\text{accepted-with-partial-composition-openings } s$
 $(\text{Some (result, final-state)}) \text{ composition-root query-idxs}$
 $\text{composition-openings}$
and *trace-candidate*:
 $\text{partial-trace-table-candidate trace-table trace-openings}$
and *comp-candidate*:
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings}$
and *header*:
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final as } dg$
 $\text{composition-fri-roots final rest}$
and *comp-nonempty*: $\text{composition-fri-roots} \neq []$
and *root-eq*: $\text{composition-root} = \text{hd composition-fri-roots}$

```

and i-bound:  $i < \text{rounds}$ 
and success:
  query-idxs !  $i \in$ 
    query-sampling-success-space trace-table composition-table as
shows
  query-idxs !  $i \in$ 
    query-header-supported-partial-opening-success-indices-at s fr
    f-fri-roots f-final as dg composition-fri-roots final i
proof –
  let ?trace-openings = trace-openings !  $i$ 
  let ?composition-openings = composition-openings !  $i$ 
  have consistent-round:
    partial-query-round-consistent trace-openings composition-openings as
     $i (query-idxs ! i)$ 
  by (rule partial-query-round-consistent-from-sampling-success
    [OF trace-partial comp-partial trace-candidate comp-candidate
    i-bound success])
  have consistent:
    partial-query-openings-consistent ?trace-openings
    ?composition-openings as (query-idxs ! i)
  unfolding partial-query-openings-consistent-def
  using consistent-round
  unfolding partial-query-round-consistent-def
  by simp
  have trace-table:
    partial-authenticated-table fr (scale * clength) ?trace-openings
    final-state
  using trace-partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
  have comp-table:
    partial-authenticated-table (hd composition-fri-roots)
     $(scale * clength) ?composition-openings final-state$ 
  using comp-partial i-bound root-eq
  unfolding accepted-with-partial-composition-openings-def by blast
  have witness:
     $(?trace-openings, ?composition-openings) \in$ 
    query-header-supported-partial-opening-witnesses s fr f-fri-roots
    f-final as dg composition-fri-roots final
  unfolding query-header-supported-partial-opening-witnesses-def
  using comp-nonempty outcome trace-table comp-table header
  by auto
  have idx-sample:
    query-idxs !  $i \in query-sample-space$ 
  using consistent unfolding partial-query-openings-consistent-def by simp
  show ?thesis
  unfolding query-header-supported-partial-opening-success-indices-at-def
  by (intro CollectI conjI exI [of - ?trace-openings]
    [exI [of - ?composition-openings]])
    (use idx-sample witness consistent in simp-all)

```

qed

lemma *partial-query-round-consistent-from-all-queries-consistent:*

assumes *trace-partial:*

accepted-with-partial-trace-openings s

(Some (result, final-state)) fr query-idxs trace-openings

and *comp-partial:*

accepted-with-partial-composition-openings s

(Some (result, final-state)) composition-root query-idxs

composition-openings

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *comp-candidate:*

partial-composition-table-candidate composition-table

composition-openings

and *i-bound: i < rounds*

and *idx-sample: query-idxs ! i ∈ query-sample-space*

and *all-queries:*

all-queries-consistent trace-table composition-table as

shows

partial-query-round-consistent trace-openings composition-openings as

i (query-idxs ! i)

proof —

let *?idx = query-idxs ! i*

have *trace-len: length trace-openings = rounds*

using *trace-partial* **unfolding** *accepted-with-partial-trace-openings-def*

by *blast*

have *comp-len: length composition-openings = rounds*

using *comp-partial*

unfolding *accepted-with-partial-composition-openings-def* **by** *blast*

have *trace-indices:*

map opening-index (trace-openings ! i) = powers-scaled ?idx

using *trace-partial i-bound*

unfolding *accepted-with-partial-trace-openings-def* **by** *blast*

have *comp-indices:*

map opening-index (composition-openings ! i) =

*[?idx, fri-sibling-index (scale * clength) ?idx]*

using *comp-partial i-bound*

unfolding *accepted-with-partial-composition-openings-def* **by** *blast*

have *qc:*

query-consistent-at trace-table composition-table as ?idx

using *all-queries idx-sample* **unfolding** *all-queries-consistent-def*

by *simp*

have *trace-values:*

map ((! trace-table) (powers-scaled ?idx)) =

map opening-value (trace-openings ! i)

by *(rule partial-trace-table-candidate-query-values*

[OF trace-candidate i-bound trace-indices])

have *comp-value:*

```

composition-table ! ?idx =
  opening-value ((composition-openings ! i) ! 0)
by (rule partial-composition-table-candidate-query-value
  [OF comp-candidate i-bound comp-indices])
have opened-value:
  opening-value ((composition-openings ! i) ! 0) =
    cp-eval as (map opening-value (trace-openings ! i))
    (h ^ ?idx * shift)
using qc trace-values comp-value unfolding query-consistent-at-def
by simp
show ?thesis
by (rule partial-query-round-consistentI
  [OF i-bound idx-sample trace-len comp-len trace-indices
    comp-indices opened-value])
qed

lemma query-header-supported-partial-opening-success-indices-atI-from-partials-consistent:
assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr query-idxs trace-openings
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) composition-root query-idxs
  composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and comp-nonempty: composition-fri-roots ≠ []
and root-eq: composition-root = hd composition-fri-roots
and i-bound: i < rounds
and idx-sample: query-idxs ! i ∈ query-sample-space
and all-queries:
  all-queries-consistent trace-table composition-table as
shows
  query-idxs ! i ∈
  query-header-supported-partial-opening-success-indices-at s fr
  f-fri-roots f-final as dg composition-fri-roots final i
proof –
  let ?trace-openings = trace-openings ! i
  let ?composition-openings = composition-openings ! i
  have consistent-round:
    partial-query-round-consistent trace-openings composition-openings as

```

```

    i (query-idxs ! i)
  by (rule partial-query-round-consistent-from-all-queries-consistent
    [OF trace-partial comp-partial trace-candidate comp-candidate
      i-bound idx-sample all-queries])
have consistent:
  partial-query-openings-consistent ?trace-openings
    ?composition-openings as (query-idxs ! i)
  unfolding partial-query-openings-consistent-def
  using consistent-round
  unfolding partial-query-round-consistent-def
  by simp
have trace-table:
  partial-authenticated-table fr (scale * clength) ?trace-openings
    final-state
  using trace-partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
have comp-table:
  partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) ?composition-openings final-state
  using comp-partial i-bound root-eq
  unfolding accepted-with-partial-composition-openings-def by blast
have witness:
  (?trace-openings, ?composition-openings) ∈
    query-header-supported-partial-opening-witnesses s fr f-fri-roots
    f-final as dg composition-fri-roots final
  unfolding query-header-supported-partial-opening-witnesses-def
  using comp-nonempty outcome trace-table comp-table header
  by auto
show ?thesis
  unfolding query-header-supported-partial-opening-success-indices-at-def
  by (intro CollectI conjI exI[of - ?trace-openings]
    exI[of - ?composition-openings])
    (use idx-sample witness consistent in simp-all)
qed

```

definition staged-security-with-data-state-query-partial-opening-hit-at

```

:: nat ⇒
  (((f staged-proof-data × f protocol-channel) × unit list) ×
    f protocol-channel) option ⇒ bool

```

where

```

staged-security-with-data-state-query-partial-opening-hit-at i out
  ⇔
  (case out of
    None ⇒ False
  | Some (((data, attacker-state), -), final-state) ⇒
    (let s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data);
      B =

```


query-header-supported-partial-opening-success-indices-at s
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
 i
 in
 (∃ raw.
 i < rounds ∧
 fmlookup (HashMap final-state)
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data)
 (staged-query-chunks data) i)) =
 Some raw ∧
 index (to-nat raw) ∈ B)))

definition *staged-security-with-data-state-query-partial-opening-hit*
 :: (((f staged-proof-data × f protocol-channel) × unit list) ×
 f protocol-channel) option ⇒ bool

where

staged-security-with-data-state-query-partial-opening-hit out $\longleftrightarrow$
 (∃ i < rounds.
 staged-security-with-data-state-query-partial-opening-hit-at i out)

lemma *staged-security-with-data-state-query-partial-opening-hit-atD:*

assumes

staged-security-with-data-state-query-partial-opening-hit-at i out

shows i < rounds

using *assms*

unfolding *staged-security-with-data-state-query-partial-opening-hit-at-def*

by (cases out; auto split: prod.splits)

lemma *staged-security-with-data-state-query-partial-opening-hit-atE:*

assumes

staged-security-with-data-state-query-partial-opening-hit-at i
 (Some (((data, attacker-state), result), final-state))

obtains raw **where**

i < rounds

fmlookup (HashMap final-state)

(QueryIndexChallenge i

(state-after-query-chunks

(staged-query-start-hash data)

(staged-query-chunks data) i)) =

Some raw

index (to-nat raw) ∈

```

    query-header-supported-partial-opening-success-indices-at
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
      i
  using assms
  unfolding staged-security-with-data-state-query-partial-opening-hit-at-def
    Let-def
  by auto

lemma staged-security-with-data-state-query-partial-opening-hitI:
  assumes i < rounds
    and staged-security-with-data-state-query-partial-opening-hit-at i out
  shows staged-security-with-data-state-query-partial-opening-hit out
  using assms
  unfolding staged-security-with-data-state-query-partial-opening-hit-def
  by blast

lemma staged-security-with-data-state-query-partial-opening-hit-imp-round-hit:
  assumes hit: staged-security-with-data-state-query-partial-opening-hit out
  shows  $\exists i \in \{..<rounds\}.$ 
    staged-security-with-data-state-query-partial-opening-hit-at i out
  using assms
  unfolding staged-security-with-data-state-query-partial-opening-hit-def
  by blast

lemma checked-staged-security-with-data-state-query-partial-opening-hit-bound-from-rounds:
  assumes round-bound:
     $\bigwedge i. i < rounds \implies$ 
      wp-event (checked-staged-security-experiment-with-data-state A)
        (staged-security-with-data-state-query-partial-opening-hit-at i)
        adversary-initial-state  $\leq C i$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-opening-hit
      adversary-initial-state  $\leq (\sum i < rounds. C i)$ 
proof -
  have wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-opening-hit
    adversary-initial-state  $\leq$ 
    wp-event (checked-staged-security-experiment-with-data-state A)
      ( $\lambda out. \exists i \in \{..<rounds\}.$ 
        staged-security-with-data-state-query-partial-opening-hit-at i out)

```

```

    adversary-initial-state
  by (rule wp-event-mono)
    (rule staged-security-with-data-state-query-partial-opening-hit-imp-round-hit)
also have ...  $\leq$  ( $\sum i < \text{rounds}. C\ i$ )
  by (rule wp-event-finite-UN-bound)
    (simp-all add: round-bound)
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-partial-opening-hit-bound-from-index-set:
  assumes cover:
     $\bigwedge \text{data attacker-state } i.$ 
     $i < \text{rounds} \implies$ 
    query-header-supported-partial-opening-success-indices-at
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
       $i \subseteq B$ 
  and index-bound:
    wp-event (checked-staged-security-experiment-with-data A)
      (staged-security-with-data-query-index-set-hit B)
      adversary-initial-state  $\leq C$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-opening-hit
      adversary-initial-state  $\leq C$ 
proof -
  let ?projected =
     $\lambda \text{out}. \text{case out of None} \implies$ 
    staged-security-with-data-query-index-set-hit B None
    | Some (((data, -), result), t)  $\implies$ 
    staged-security-with-data-query-index-set-hit B
      (Some ((data, result), t))
  have event-le:
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-opening-hit
      adversary-initial-state  $\leq$ 
    wp-event (checked-staged-security-experiment-with-data-state A)
      ?projected adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume hit:
    staged-security-with-data-state-query-partial-opening-hit out

```

```

show ?projected out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding staged-security-with-data-state-query-partial-opening-hit-def
      staged-security-with-data-state-query-partial-opening-hit-at-def
    by simp
next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  let ?B =
     $\lambda i.$  query-header-supported-partial-opening-success-indices-at ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
    i
  from hit[unfolded out-eq
    staged-security-with-data-state-query-partial-opening-hit-def
    staged-security-with-data-state-query-partial-opening-hit-at-def
    Let-def]
  obtain i raw where i-bound:  $i < \text{rounds}$ 
  and lookup:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some raw
  and raw-hit:  $\text{index } (\text{to-nat raw}) \in ?B \ i$ 
  by auto
  have raw-in-B:  $\text{index } (\text{to-nat raw}) \in B$ 
  proof –
    have ?B  $i \subseteq B$ 
    by (rule cover[OF i-bound])
    then show ?thesis
      using raw-hit by blast
  qed
show ?thesis
  unfolding out-eq staged-security-with-data-query-index-set-hit-def

```

```

    using i-bound lookup raw-in-B by auto
qed
qed
also have
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?projected adversary-initial-state =
  wp-event (checked-staged-security-experiment-with-data A)
    (staged-security-with-data-query-index-set-hit B)
    adversary-initial-state
  using checked-staged-security-experiment-with-data-event-from-data-state
    [of A staged-security-with-data-query-index-set-hit B]
  by simp
also have ...  $\leq C$ 
  by (rule index-bound)
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-partial-opening-hit-bound-from-index-set-controlled:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and cover:
     $\bigwedge$  data attacker-state i.
      i < rounds  $\implies$ 
      query-header-supported-partial-opening-success-indices-at
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)
      i  $\subseteq B$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-opening-hit
      adversary-initial-state  $\leq$ 
      staged-phase-target-error (query-index-raw-preimage B)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  proof (rule
    checked-staged-security-with-data-state-partial-opening-hit-bound-from-index-set)
  fix data attacker-state i
  assume i-bound: i < rounds
  show
    query-header-supported-partial-opening-success-indices-at
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))

```

```

    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
     $i \subseteq B$ 
  by (rule cover[OF i-bound])
next
show
  wp-event (checked-staged-security-experiment-with-data A)
    (staged-security-with-data-query-index-set-hit B)
    adversary-initial-state  $\leq$ 
    staged-phase-target-error (query-index-raw-preimage B)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  by (rule checked-staged-security-with-data-query-index-set-hit-bound
    [OF wf controlled])
qed

lemma checked-staged-security-with-data-state-partial-opening-hit-bound-from-fixed-query-error-cover:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and cover:
 $\bigwedge$ data attacker-state  $i$ .
   $i < \text{rounds} \implies$ 
  query-header-supported-partial-opening-success-indices-at
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
     $i \subseteq B$ 
and raw-bound: query-index-raw-preimage-bound
and subset:  $B \subseteq \text{query-sample-space}$ 
and frac:
   $\text{nnreal}(\text{card } B) / \text{nnreal}(\text{card } \text{query-sample-space}) \leq$ 
  query-error-bound
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-opening-hit
    adversary-initial-state  $\leq$ 
    nnreal
    (staged-attacker-query-budget budgets +

```

staged-challenge-query-budget) * *query-error-bound*

proof –

have *target*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

staged-security-with-data-state-query-partial-opening-hit

adversary-initial-state ≤

staged-phase-target-error (*query-index-raw-preimage B*)

 (*staged-attacker-query-budget budgets* +

staged-challenge-query-budget)

by (*rule*

checked-staged-security-with-data-state-partial-opening-hit-bound-from-index-set-controlled

 [*OF wf controlled cover*])

also have ... ≤

nnreal

 (*staged-attacker-query-budget budgets* +

staged-challenge-query-budget) * *query-error-bound*

by (*rule staged-phase-query-index-target-error-query-bound*

 [*OF raw-bound subset frac*])

finally show *?thesis* .

qed

lemma *checked-staged-security-with-data-state-query-bad-imp-partial-opening-hit-on-support*:

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *support*:

Some (((*data*, *attacker-state*), *result*), *final-state*) ∈

set-dist

 (*execute*

 (*checked-staged-security-experiment-with-data-state A*)

adversary-initial-state)

and *bad*:

staged-security-with-data-state-query-bad-hit

 (*Some* (((*data*, *attacker-state*), *result*), *final-state*))

shows

staged-security-with-data-state-query-partial-opening-hit

 (*Some* (((*data*, *attacker-state*), *result*), *final-state*))

proof –

let *?s* =

verifier-state-from-adversary attacker-state

 (*staged-proof-transcript data*)

from *checked-staged-security-experiment-with-data-state-outcomeE*

 [*OF support*]

obtain *builder* **where** *builder*:

Some (*data*, *attacker-state*) ∈

set-dist

 (*execute* (*checked-staged-transcript-program A*)

adversary-initial-state)

and *verifier*:

Some (*result*, *final-state*) ∈ *set-dist* (*execute verify-monad ?s*)

```

by blast
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
show ?thesis
proof (rule checked-staged-security-with-data-state-query-bad-round-chunkE
    [OF wf controlled support bad])
fix i trace-table composition-table query-idxs raw-idxs
assume i-bound: i < rounds
and bound:
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table (staged-alphas data) query-idxs
and success:
  query-idxs ! i ∈
    query-sampling-success-space trace-table composition-table
    (staged-alphas data)
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and query-chunk:
  verifier-query-round-chunk (query-idxs ! i)
    (staged-trace-fri-roots data)
    (staged-composition-fri-roots data)
    (staged-query-chunks data ! i)
and lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
show ?thesis
proof (rule accepted-with-bound-tables-partial-opening-witnesses-Some
    [OF bound])
fix fr f-fri-roots f-final dg composition-fri-roots final rest
  trace-openings composition-openings
assume header:
  verifier-header-transcript ?s fr f-fri-roots f-final
    (staged-alphas data) dg composition-fri-roots final rest
and comp-nonempty: composition-fri-roots ≠ []
and trace-partial:

```



```

accepted-with-partial-trace-openings ?s
  (Some (result, final-state)) fr query-idxs trace-openings
and comp-partial:
accepted-with-partial-composition-openings ?s
  (Some (result, final-state)) (hd composition-fri-roots)
  query-idxs composition-openings
and trace-candidate:
partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
partial-composition-table-candidate composition-table
  composition-openings
and trace-bind:
merkle-root-binds-table fr trace-table final-state
and comp-bind:
merkle-root-binds-table (hd composition-fri-roots)
  composition-table final-state
have header-eq:
fr = staged-trace-root data ∧
f-fri-roots = staged-trace-fri-roots data ∧
f-final = staged-trace-final data ∧
dg = staged-degree data ∧
composition-fri-roots = staged-composition-fri-roots data ∧
final = staged-composition-final data ∧
rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF staged-header header]
by simp
have idx-in-header-set:
query-idxs ! i ∈
  query-header-supported-partial-opening-success-indices-at ?s fr
  f-fri-roots f-final (staged-alphas data) dg
  composition-fri-roots final i
by (rule
  query-header-supported-partial-opening-success-indices-atI-from-partials
  [OF verifier trace-partial comp-partial trace-candidate
  comp-candidate header comp-nonempty refl i-bound success])
have idx-in-staged-set:
query-idxs ! i ∈
  query-header-supported-partial-opening-success-indices-at ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  i
using idx-in-header-set header-eq by simp
have raw-idx:
index (to-nat (raw-idxs ! i)) = query-idxs ! i

```

```

    using query-idxs-eq len-raw i-bound by simp
  have hit-at:
    staged-security-with-data-state-query-partial-opening-hit-at i
      (Some (((data, attacker-state), result), final-state))
  unfolding
    staged-security-with-data-state-query-partial-opening-hit-at-def
  using i-bound lookup idx-in-staged-set raw-idx
  by (simp add: Let-def)
show ?thesis
  by (rule staged-security-with-data-state-query-partial-opening-hitI
    [OF i-bound hit-at])
qed
qed
qed

lemma checked-staged-security-with-data-state-query-bad-bound-from-partial-opening-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-bad-hit
      adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-opening-hit
      adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  and bad: staged-security-with-data-state-query-bad-hit out
  show staged-security-with-data-state-query-partial-opening-hit out
proof (cases out)
  case None
  then show ?thesis
    using bad unfolding staged-security-with-data-state-query-bad-hit-def
    by simp
next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
  show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-data-state-query-bad-imp-partial-opening-hit-on-support

```

$[OF \text{ wf controlled}]$
 (use support bad out-eq in simp-all)
 qed
 qed

lemma *checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-partial-opening-hit*:
assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *partial-bound*:
 wp-event (checked-staged-security-experiment-with-data-state *A*)
 staged-security-with-data-state-query-partial-opening-hit
 adversary-initial-state $\leq Q$
shows
 wp-event (checked-staged-security-experiment-with-data-state *A*)
 (*staged-security-with-data-state-verifier-event query-bad*)
 adversary-initial-state $\leq Q$
proof –
have *query-hit-bound*:
 wp-event (checked-staged-security-experiment-with-data-state *A*)
 staged-security-with-data-state-query-bad-hit
 adversary-initial-state $\leq Q$
by (rule order.trans
 $[OF \text{ checked-staged-security-with-data-state-query-bad-bound-from-partial-opening-hit}$
 $[OF \text{ wf controlled}] \text{ partial-bound}]$)
show ?thesis
by (rule checked-staged-security-with-data-state-query-bad-verifier-event-bound
 $[OF \text{ query-hit-bound}]$)
 qed

lemma *checked-staged-security-with-data-state-query-bad-bound-from-partial-opening-fixed-query-error-cover*:
assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *cover*:
 $\bigwedge \text{data attacker-state } i.$
 $i < \text{rounds} \implies$
 query-header-supported-partial-opening-success-indices-at
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*))
 (*staged-trace-root data*)
 (*staged-trace-fri-roots data*)
 (*staged-trace-final data*)
 (*staged-alphas data*)
 (*staged-degree data*)
 (*staged-composition-fri-roots data*)
 (*staged-composition-final data*)
 $i \subseteq B$
and *raw-bound*: *query-index-raw-preimage-bound*
and *subset*: $B \subseteq \text{query-sample-space}$
and *frac*:

$$\text{nnreal } (\text{card } B) / \text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$$

shows

$$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \text{ staged-security-with-data-state-query-bad-hit adversary-initial-state} \leq \text{nnreal } ((\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget}) * \text{query-error-bound})$$

proof –

have $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \text{ staged-security-with-data-state-query-bad-hit adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \text{ staged-security-with-data-state-query-partial-opening-hit adversary-initial-state}$
by (rule
 $\text{checked-staged-security-with-data-state-query-bad-bound-from-partial-opening-hit}$
 $[\text{OF wf controlled}]$)

also have $\dots \leq$
 $\text{nnreal } ((\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget}) * \text{query-error-bound})$
by (rule
 $\text{checked-staged-security-with-data-state-partial-opening-hit-bound-from-fixed-query-error-cover}$
 $[\text{OF wf controlled cover raw-bound subset frac}]$)

finally show ?thesis .

qed

lemma *verifier-query-round-program-authenticated-initial-openings*:

fixes $s \ t :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes $\text{fl-eq: fl} = (b, \text{composition-root}) \# \text{fl-tail}$

and *outcome*:

$\text{Some } ((\), t) \in$
 set-dist
 $(\text{execute}$
 $(\text{verifier-query-round-program fr f-fl f-final as fl final}) \ s)$

obtains $\text{raw idx trace-leaves trace-openings composition-openings}$ **where**

$\text{idx} = \text{index } (\text{to-nat raw})$
 $\text{map opening-value trace-openings} = \text{trace-leaves}$
 $\text{map opening-index trace-openings} = \text{powers-scaled idx}$
 $\text{partial-authenticated-table fr } (\text{scale} * \text{clength}) \text{ trace-openings } t$
 $\text{partial-authenticated-table composition-root } (\text{scale} * \text{clength})$
 $\text{composition-openings } t$
 $\text{map opening-index composition-openings} =$
 $[\text{idx}, \text{fri-sibling-index } (\text{scale} * \text{clength}) \text{ idx}]$

proof –

from $\text{outcome}[\text{unfolded verifier-query-round-program-alt-def}]$

obtain raw s0 **where** *tail*:

```

Some ((), t) ∈
  set-dist
    (execute
      (verifier-query-round-after-index-program
        fr f-fl f-final as fl final raw) s0)
  by (auto elim!: set-dist-bindE)
show ?thesis
proof (rule verifier-query-round-after-index-authenticated-trace-openings
  [OF tail])
fix idx trace-leaves trace-openings
assume idx-eq: idx = index (to-nat raw)
and trace-values:
  map opening-value trace-openings = trace-leaves
and trace-indices:
  map opening-index trace-openings = powers-scaled idx
and trace-table:
  partial-authenticated-table fr (scale * clength) trace-openings t
show ?thesis
proof (rule
  verifier-query-round-after-index-authenticated-composition-openings
  [OF fl-eq tail])
fix idx' composition-openings
assume idx'-eq: idx' = index (to-nat raw)
and comp-table:
  partial-authenticated-table composition-root (scale * clength)
  composition-openings t
and comp-indices:
  map opening-index composition-openings =
    [idx', fri-sibling-index (scale * clength) idx']
have idx'-idx: idx' = idx
  using idx-eq idx'-eq by simp
show ?thesis
  by (rule that[OF idx-eq trace-values trace-indices trace-table
    comp-table])
    (use comp-indices idx'-idx in simp)
qed
qed
qed

lemma verifier-query-round-program-authenticated-initial-openings-with-lookup:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-eq: fl = (b, composition-root) # fl-tail
and outcome:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
obtains raw idx trace-leaves trace-openings composition-openings where
  idx = index (to-nat raw)

```

```

    map opening-value trace-openings = trace-leaves
    map opening-index trace-openings = powers-scaled idx
    partial-authenticated-table fr (scale * clength) trace-openings t
    partial-authenticated-table composition-root (scale * clength)
      composition-openings t
    map opening-index composition-openings =
      [idx, fri-sibling-index (scale * clength) idx]
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where rand:
  Some (raw, s0) ∈
    set-dist (execute receive-query-index-challenge s)
and tail:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
by (auto elim!: set-dist-bindE)
have lookup-s0:
  fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
using receive-query-index-challenge-outcome[OF rand] by simp
have lookup-t:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
using verifier-query-round-after-index-program-preserves-query-lookup
  [OF tail, of PQueryCounter s PState s] lookup-s0
by simp
show ?thesis
proof (rule verifier-query-round-after-index-authenticated-trace-openings
  [OF tail])
fix idx trace-leaves trace-openings
assume idx-eq: idx = index (to-nat raw)
and trace-values:
  map opening-value trace-openings = trace-leaves
and trace-indices:
  map opening-index trace-openings = powers-scaled idx
and trace-table:
  partial-authenticated-table fr (scale * clength) trace-openings t
show ?thesis
proof (rule
  verifier-query-round-after-index-authenticated-composition-openings
  [OF fl-eq tail])
fix idx' composition-openings
assume idx'-eq: idx' = index (to-nat raw)
and comp-table:

```

```

    partial-authenticated-table composition-root (scale * clength)
    composition-openings t
  and comp-indices:
    map opening-index composition-openings =
      [idx', fri-sibling-index (scale * clength) idx']
  have idx'-idx: idx' = idx
  using idx-eq idx'-eq by simp
  show ?thesis
  by (rule that[OF idx-eq trace-values trace-indices trace-table
    comp-table - lookup-t])
    (use comp-indices idx'-idx in simp)
qed
qed
qed

lemma verifier-query-round-program-authenticated-initial-openings-with-chunk:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes fl-eq: fl = (b, composition-root) # fl-tail
  and outcome:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  obtains raw idx chunk trace-leaves trace-openings composition-openings
  where
    idx = index (to-nat raw)
    verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
    PTranscript s = chunk @ PTranscript t
    PState t = foldl concat (PState s) chunk
    s ≤ t
    PQueryCounter t = Suc (PQueryCounter s)
    map opening-value trace-openings = trace-leaves
    map opening-index trace-openings = powers-scaled idx
    partial-authenticated-table fr (scale * clength) trace-openings t
    partial-authenticated-table composition-root (scale * clength)
      composition-openings t
    map opening-index composition-openings =
      [idx, fri-sibling-index (scale * clength) idx]
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof -
  show ?thesis
proof (rule verifier-query-round-program-outcome[OF outcome])
  fix raw0 idx0 chunk
  assume idx0-eq: idx0 = index (to-nat raw0)
  and chunk-shape:
    verifier-query-round-chunk idx0 (map snd f-fl) (map snd fl)
    chunk
  and transcript:

```

```

    PTranscript s = chunk @ PTranscript t
  and state:
    PState t = foldl concat (PState s) chunk
  and ext:  $s \leq t$ 
  and counter:
    PQueryCounter t = Suc (PQueryCounter s)
  and lookup0:
    fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw0
show ?thesis
proof (rule
  verifier-query-round-program-authenticated-initial-openings-with-lookup
  [OF fl-eq outcome])
fix raw idx trace-leaves trace-openings composition-openings
assume idx-eq:  $idx = index\ (to\ nat\ raw)$ 
  and trace-values:
    map opening-value trace-openings = trace-leaves
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and trace-table:
    partial-authenticated-table fr (scale * clength)
    trace-openings t
  and comp-table:
    partial-authenticated-table composition-root (scale * clength)
    composition-openings t
  and comp-indices:
    map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
  and lookup:
    fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
have raw-eq:  $raw = raw0$ 
  using lookup lookup0 by simp
have idx-idx0:  $idx = idx0$ 
  using idx-eq idx0-eq raw-eq by simp
show ?thesis
  by (rule that[OF idx-eq - transcript state ext counter
    trace-values trace-indices trace-table comp-table comp-indices
    lookup])
    (use chunk-shape idx-idx0 in simp)
qed
qed
qed

lemma ntimes-verifier-query-rounds-authenticated-initial-openings-at:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes fl-eq:  $fl = (b, composition-root) \# fl\text{-tail}$ 
  and outcome:
    Some (results, t)  $\in$ 

```



```

    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final) n)
        s)
  and i-bound:  $i < n$ 
obtains raw idx trace-leaves trace-openings composition-openings where
  idx = index (to-nat raw)
  map opening-value trace-openings = trace-leaves
  map opening-index trace-openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) trace-openings t
  partial-authenticated-table composition-root (scale * clength)
    composition-openings t
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
proof —
have
   $\exists$  raw idx trace-leaves trace-openings composition-openings.
  idx = index (to-nat raw)  $\wedge$ 
  map opening-value trace-openings = trace-leaves  $\wedge$ 
  map opening-index trace-openings = powers-scaled idx  $\wedge$ 
  partial-authenticated-table fr (scale * clength) trace-openings t  $\wedge$ 
  partial-authenticated-table composition-root (scale * clength)
    composition-openings t  $\wedge$ 
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
using outcome i-bound
proof (induction n arbitrary: i s results t)
case 0
then show ?case
  by simp
next
case (Suc n)
from Suc.prem(1) obtain u results' s1 where
  head:
    Some (u, s1)  $\in$ 
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results', t)  $\in$ 
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            n) s1)
  by (auto elim!: set-dist-bindE)
have u-eq: u = ()
  by (cases u) simp

```

```

show ?case
proof (cases i)
  case 0
  show ?thesis
  proof (rule verifier-query-round-program-authenticated-initial-openings
    [OF fl-eq head[unfolded u-eq]])
  fix raw idx trace-leaves trace-openings composition-openings
  assume idx-eq: idx = index (to-nat raw)
  and trace-values:
    map opening-value trace-openings = trace-leaves
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and trace-table-s1:
    partial-authenticated-table fr (scale * clength)
    trace-openings s1
  and comp-table-s1:
    partial-authenticated-table composition-root (scale * clength)
    composition-openings s1
  and comp-indices:
    map opening-index composition-openings =
      [idx, fri-sibling-index (scale * clength) idx]
  from ntimes-verifier-query-rounds-outcome[OF tail]
  obtain raw-idxs query-idxs query-chunks where
    s1-t: s1 ≤ t
  by (elim exE conjE)
  have trace-table-t:
    partial-authenticated-table fr (scale * clength)
    trace-openings t
  by (rule partial-authenticated-table-mono[OF trace-table-s1 s1-t])
  have comp-table-t:
    partial-authenticated-table composition-root (scale * clength)
    composition-openings t
  by (rule partial-authenticated-table-mono[OF comp-table-s1 s1-t])
  show ?thesis
  apply (rule exI[where x=raw])
  apply (rule exI[where x=idx])
  apply (rule exI[where x=trace-leaves])
  apply (rule exI[where x=trace-openings])
  apply (rule exI[where x=composition-openings])
  apply (intro conjI)
  apply (rule idx-eq)
  apply (rule trace-values)
  apply (rule trace-indices)
  apply (rule trace-table-t)
  apply (rule comp-table-t)
  apply (rule comp-indices)
  done
qed
next

```

```

    case (Suc j)
    have j-bound:  $j < n$ 
      using Suc.premis(2)  $\langle i = \text{Suc } j \rangle$  by simp
    from Suc.IH[OF tail j-bound] show ?thesis .
  qed
qed
then obtain raw idx trace-leaves trace-openings composition-openings where
  idx-eq:  $\text{idx} = \text{index } (\text{to-nat raw})$ 
  and trace-values:
    map opening-value trace-openings = trace-leaves
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and trace-table:
    partial-authenticated-table fr (scale * clength) trace-openings t
  and comp-table:
    partial-authenticated-table composition-root (scale * clength)
      composition-openings t
  and comp-indices:
    map opening-index composition-openings =
      [idx, fri-sibling-index (scale * clength) idx]
  by (elim exE conjE)
show ?thesis
  by (rule that[OF idx-eq trace-values trace-indices trace-table
    comp-table comp-indices])
qed

lemma accepted-with-bound-tables-partial-opening-header-witness-at:
  assumes outcome:
    Some (result, final-state)  $\in \text{set-dist } (\text{execute verify-monad } s)$ 
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and i-bound:  $i < \text{rounds}$ 
  obtains trace-openings composition-openings where
    (trace-openings, composition-openings)  $\in$ 
    query-header-supported-partial-opening-witnesses s fr f-fri-roots
    f-final as dg composition-fri-roots final
proof -
  show ?thesis
proof (rule accepted-with-bound-tables-partial-opening-witnesses-Some[OF bound])
  fix fr' f-fri-roots' f-final' dg' composition-fri-roots' final'
    rest' trace-openingss composition-openingss
  assume
  header':
    verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
    composition-fri-roots' final' rest'

```

```

and comp-nonempty': composition-fri-roots' ≠ []
and trace-partial:
  accepted-with-partial-trace-openings s (Some (result, final-state))
  fr' query-idxs trace-openingsss
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots')
  query-idxs composition-openingsss
have eqs:
  fr' = fr ∧
  f-fri-roots' = f-fri-roots ∧
  f-final' = f-final ∧
  dg' = dg ∧
  composition-fri-roots' = composition-fri-roots ∧
  final' = final ∧
  rest' = rest
  using verifier-header-transcript-unique[OF header' header] by simp
have trace-table-i:
  partial-authenticated-table fr (scale * clength)
  (trace-openingsss ! i) final-state
  using trace-partial i-bound eqs
  unfolding accepted-with-partial-trace-openings-def by blast
have comp-table-i:
  partial-authenticated-table (hd composition-fri-roots) (scale * clength)
  (composition-openingsss ! i) final-state
  using comp-partial i-bound eqs
  unfolding accepted-with-partial-composition-openings-def by simp
have witness:
  (trace-openingsss ! i, composition-openingsss ! i) ∈
  query-header-supported-partial-opening-witnesses s fr f-fri-roots
  f-final as dg composition-fri-roots final
  unfolding query-header-supported-partial-opening-witnesses-def
  using outcome header trace-table-i comp-table-i comp-nonempty' eqs
  by auto
show ?thesis
  by (rule that[OF witness])
qed
qed

```

```

lemma checked-actual-alpha-prefix-support-partial-opening-header-witness-at:
fixes prefix prefix-state data attacker-state result final-state
defines
  s ≡
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
assumes support:
  Some (((((prefix, prefix-state), data), attacker-state),
    result), final-state) ∈
  set-dist

```

```

      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
    and header:
      verifier-header-transcript s fr f-fri-roots f-final as dg
      composition-fri-roots final rest
    and bound:
      accepted-with-bound-tables s (Some (result, final-state))
      trace-table composition-table as query-idxs
    and i-bound: i < rounds
  obtains trace-openings composition-openings where
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses s fr f-fri-roots
      f-final as dg composition-fri-roots final
proof —
  have verify-out:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using support
  unfolding checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
    s-def
  by (auto elim!: set-dist-bindE split: prod.splits)
  show ?thesis
  by (rule accepted-with-bound-tables-partial-opening-header-witness-at
    [OF verify-out header bound i-bound that])
qed

lemma checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-header-witness-at:
  fixes prefix prefix-state data attacker-state result final-state
  defines
    s ≡
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  assumes support:
    Some (((((prefix, prefix-state), data), attacker-state),
      result), final-state) ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
      (Some (((((prefix, prefix-state), data), attacker-state),
        result), final-state))
  and i-bound: i < rounds
  obtains as dg composition-fri-roots final rest trace-table composition-table
    query-idxs trace-openings composition-openings where
    accepted-with-bound-tables s (Some (result, final-state))

```

```

    trace-table composition-table as query-idxs
  verifier-header-transcript s
    (staged-trace-root data) (staged-trace-fri-roots data)
    (staged-trace-final data) as dg composition-fri-roots final rest
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses s
    (staged-trace-root data) (staged-trace-fri-roots data)
    (staged-trace-final data) as dg composition-fri-roots final
proof –
  obtain trace-table composition-table as query-idxs dg
    composition-fri-roots final rest trace-tree composition-tree where
  verify-out:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and bound:
    accepted-with-bound-tables s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and header:
    verifier-header-transcript s
    (staged-trace-root data) (staged-trace-fri-roots data)
    (staged-trace-final data) as dg composition-fri-roots final rest
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit-def
    s-def Let-def
  by auto
  obtain trace-openings composition-openings where witness:
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final
  by (rule accepted-with-bound-tables-partial-opening-header-witness-at
    [OF verify-out header bound i-bound])
  show ?thesis
  by (rule that[OF bound header witness])
qed

definition
  checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit
  :: ((((((f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool
where
  checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit
  out ⇔
  (case out of
    None ⇒ False
  | Some (((((prefix, prefix-state), data), attacker-state),
    result), final-state) ⇒

```

```

    (let s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    in ∃ as dg composition-fri-roots final rest trace-table
      composition-table query-idxs trace-openings composition-openings.
      accepted-with-bound-tables s (Some (result, final-state))
      trace-table composition-table as query-idxs ∧
      verifier-header-transcript s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final rest ∧
      (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final))

lemma checked-actual-alpha-prefix-branch-subtree-verifier-imp-partial-opening-witness-hit-on-support:
  assumes support:
    out ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
    out
  shows
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit
    out
  proof (cases out)
  case None
  then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit-def
  by simp
  next
  case (Some result-pack)
  then obtain prefix prefix-state data attacker-state result final-state where
    out-eq:
      out = Some (((((prefix, prefix-state), data), attacker-state),
        result), final-state)
  by (cases result-pack, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have support-some:
    Some (((((prefix, prefix-state), data), attacker-state),
      result), final-state) ∈

```

```

    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
    using support unfolding out-eq .
  have hit-some:
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
      (Some (((((prefix, prefix-state), data), attacker-state),
        result), final-state))
    using hit unfolding out-eq .
  obtain as dg composition-fri-roots final rest trace-table composition-table
    query-idxs trace-openings composition-openings where
  bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and header:
    verifier-header-transcript ?s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final rest
  and witness:
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses ?s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final
  proof —
  obtain trace-table composition-table as query-idxs dg
    composition-fri-roots final rest trace-tree composition-tree where
  verify-out:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  and bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs
  and header:
    verifier-header-transcript ?s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final rest
  using hit-some
  unfolding
    checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit-def
    Let-def
  by auto
  obtain trace-openings composition-openings where witness:
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses ?s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final
  by (rule accepted-with-bound-tables-partial-opening-header-witness-at
    [OF verify-out header bound rounds-positive])

```



```

show ?thesis
  by (rule that[OF bound header witness])
qed
show ?thesis
  unfolding out-eg
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-def
    Let-def
  apply simp
  apply (rule exI[where x=as])
  apply (intro conjI)
  apply (rule exI[where x=trace-table])
  apply (rule exI[where x=composition-table])
  apply (rule exI[where x=query-idxs])
  apply (rule bound)
  apply (rule exI[where x=dg])
  apply (rule exI[where x=composition-fri-roots])
  apply (rule exI[where x=final])
  apply (intro conjI)
  apply (rule exI[where x=rest])
  apply (rule header)
  apply (rule exI[where x=trace-openings])
  apply (rule exI[where x=composition-openings])
  apply (rule witness)
  done
qed

end

end

```

```

theory Soundness-Staged-Partial-Query-Branch
imports Soundness-Staged-Partial-Query
begin

```

Branch-specific probability bridges for the partial-opening query layer. This keeps branch/tree-output integration material out of the size-sensitive base partial-query theory.

```

context soundness
begin

```

```

definition checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
  :: ((((((f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool
where
  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    out ⇔

```

```

(case out of
  None  $\Rightarrow$  False
| Some (x, t)  $\Rightarrow$ 
  staged-security-with-data-state-query-partial-opening-hit
    (Some (((snd (fst (fst x)), snd (fst x)), snd x), t)))

lemma checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hitE:
assumes hit:
  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    (Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state))
obtains i raw trace-openings composition-openings where
  i < rounds
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some raw
  index (to-nat raw)  $\in$  query-sample-space
  (trace-openings, composition-openings)  $\in$ 
    query-header-supported-partial-opening-witnesses
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))

proof –
have data-hit:
  staged-security-with-data-state-query-partial-opening-hit
    (Some (((data, attacker-state), result), final-state))
using hit
unfolding
  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-def
by simp
then obtain i where i-bound: i < rounds
and hit-at:
  staged-security-with-data-state-query-partial-opening-hit-at i
    (Some (((data, attacker-state), result), final-state))
unfolding staged-security-with-data-state-query-partial-opening-hit-def
by blast
from staged-security-with-data-state-query-partial-opening-hit-atE
  [OF hit-at]

```

```

obtain raw where lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some raw
and raw-hit:
  index (to-nat raw) ∈
    query-header-supported-partial-opening-success-indices-at
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)
        i
  by blast
from query-header-supported-partial-opening-success-indices-atE
  [OF raw-hit]
obtain trace-openings composition-openings where idx-sample:
  index (to-nat raw) ∈ query-sample-space
and witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)
and consistent:
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
  by blast
show ?thesis
  by (rule that[OF i-bound lookup idx-sample witness consistent])
qed

lemma checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-projection:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit

```

```

    adversary-initial-state =
wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-opening-hit
    adversary-initial-state
proof -
let ?project =
    λx. ((snd (fst (fst x)), snd (fst x)), snd x)
have
wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-opening-hit
    adversary-initial-state =
wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
      >>= (λx. return (?project x)))
    staged-security-with-data-state-query-partial-opening-hit
    adversary-initial-state
using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
by simp
also have ... =
wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (λout. case out of
      None ⇒ staged-security-with-data-state-query-partial-opening-hit None
    | Some (x, t) ⇒
      staged-security-with-data-state-query-partial-opening-hit
      (Some (?project x, t)))
    adversary-initial-state
by (rule wp-event-bind-return-map)
also have ... =
wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    adversary-initial-state
unfolding
    checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-def
    staged-security-with-data-state-query-partial-opening-hit-def
    staged-security-with-data-state-query-partial-opening-hit-at-def
by (rule arg-cong[where
    f = λQ. wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      Q adversary-initial-state])
    (rule ext, simp split: option.splits prod.splits)
finally show ?thesis
by simp
qed

```

lemma checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-bound-from-fixed-query-error-cover:

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *cover*:
 $\bigwedge$ *data* *attacker-state* *i*.
 $i < \text{rounds} \implies$
query-header-supported-partial-opening-success-indices-at
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)
 $i \subseteq B$
and *raw-bound*: *query-index-raw-preimage-bound*
and *subset*: $B \subseteq \text{query-sample-space}$
and *frac*:
 $\text{nnreal}(\text{card } B) / \text{nnreal}(\text{card } \text{query-sample-space}) \leq$
query-error-bound
shows
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
adversary-initial-state $\leq$
 nnreal
(staged-attacker-query-budget budgets +
*staged-challenge-query-budget) * query-error-bound*
proof –
have *data-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-query-partial-opening-hit
adversary-initial-state $\leq$
 nnreal
(staged-attacker-query-budget budgets +
*staged-challenge-query-budget) * query-error-bound*
by (rule
checked-staged-security-with-data-state-partial-opening-hit-bound-from-fixed-query-error-cover
 $[OF\ wf\ controlled\ cover\ raw-bound\ subset\ frac]$)
show ?thesis
unfolding
checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-projection
by (rule *data-bound*)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support-branch*:

assumes *support*:

Some (((((prefix, prefix-state), data), attacker-state), result),

```

    final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  shows
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  proof –
    let ?project =
      λx. ((snd (fst (fst x)), snd (fst x)), snd x)
  have projected:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
            A >> (λx. return (?project x)))
          adversary-initial-state)
  apply (rule set-dist-bindI[OF support])
  apply simp
  done
  show ?thesis
  using projected
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
  by simp
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-bound-tables-all-queries-imp-query-partial-opening-hit-on-sup*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *support:*
 Some (((((prefix, prefix-state), data), attacker-state), result),
 final-state) ∈
 set-dist
 (execute
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 adversary-initial-state)
and *bound:*
 accepted-with-bound-tables
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (Some (result, final-state)) trace-table composition-table as
 query-idxs
and *header:*

```

    verifier-header-transcript
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final rest
  and all-queries:
    all-queries-consistent trace-table composition-table as
  shows
    checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
      (Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state))
  proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support-branch
      [OF support])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  obtain builder where builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
  have staged-header:
    verifier-header-transcript ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
      (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
  have header-eq:
    as = staged-alphas data ∧
    dg = staged-degree data ∧
    composition-fri-roots = staged-composition-fri-roots data ∧

```

```

    final = staged-composition-final data  $\wedge$ 
    rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF staged-header header]
by simp
have shape:
    accepted-transcript-shape ?s (Some (result, final-state)) as query-idxs
using bound
unfolding accepted-with-bound-tables-def accepted-with-tables-def
by simp
from checked-staged-security-with-data-state-accepted-shape-query-chunks
    [OF wf controlled data-support shape]
obtain raw-idxs where
    as-eq: as = staged-alphas data
    and len-raw: length raw-idxs = rounds
    and query-idxs-eq:
        query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
    and lookup:
         $\bigwedge i. i < rounds \implies$ 
        fmllookup (HashMap final-state)
            (QueryIndexChallenge i
              (state-after-query-chunks
                (staged-query-start-hash data)
                (staged-query-chunks data) i)) =
            Some (raw-idxs ! i)
    by blast
have bound-staged:
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table (staged-alphas data) query-idxs
using bound as-eq by simp
have all-queries-staged:
    all-queries-consistent trace-table composition-table
    (staged-alphas data)
using all-queries as-eq by simp
have i-bound: 0 < rounds
    by (rule rounds-positive)
have idx-sample:
    query-idxs ! 0  $\in$  query-sample-space
proof –
    have len-query-idxs: length query-idxs = rounds
    by (rule accepted-with-bound-tables-shapes(4)[OF bound-staged])
    have query-idxs ! 0  $\in$  set query-idxs
    by (rule nth-mem) (use i-bound len-query-idxs in simp)
    then show ?thesis
    by (rule accepted-with-bound-tables-query-sample-space
        [OF bound-staged])
qed
show ?thesis
proof (rule accepted-with-bound-tables-partial-opening-witnesses-Some
    [OF bound-staged])

```



```

fix fr f-fri-roots f-final dg' composition-fri-roots' final'
  rest' trace-openings composition-openings
assume header':
  verifier-header-transcript ?s fr f-fri-roots f-final
    (staged-alphas data) dg' composition-fri-roots' final' rest'
and comp-nonempty: composition-fri-roots' ≠ []
and trace-partial:
  accepted-with-partial-trace-openings ?s
    (Some (result, final-state)) fr query-idxs trace-openings
and comp-partial:
  accepted-with-partial-composition-openings ?s
    (Some (result, final-state)) (hd composition-fri-roots')
    query-idxs composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
    composition-openings
have header-eq':
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  dg' = staged-degree data ∧
  composition-fri-roots' = staged-composition-fri-roots data ∧
  final' = staged-composition-final data ∧
  rest' = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF staged-header header']
by simp
have idx-in:
  query-idxs ! 0 ∈
    query-header-supported-partial-opening-success-indices-at ?s fr
      f-fri-roots f-final (staged-alphas data) dg'
      composition-fri-roots' final' 0
by (rule
  query-header-supported-partial-opening-success-indices-atI-from-partials-consistent
  [OF verifier trace-partial comp-partial trace-candidate
    comp-candidate header' comp-nonempty refl i-bound idx-sample
    all-queries-staged])
have idx-in-staged:
  query-idxs ! 0 ∈
    query-header-supported-partial-opening-success-indices-at ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
      0

```

```

    using idx-in header-eq' by simp
  have raw-idx:
    index (to-nat (raw-idxs ! 0)) = query-idxs ! 0
    using query-idxs-eq len-raw i-bound by simp
  have hit-at:
    staged-security-with-data-state-query-partial-opening-hit-at 0
      (Some (((data, attacker-state), result), final-state))
    unfolding staged-security-with-data-state-query-partial-opening-hit-at-def
    using i-bound lookup[OF i-bound] idx-in-staged raw-idx
    by (simp add: Let-def)
  have hit:
    staged-security-with-data-state-query-partial-opening-hit
      (Some (((data, attacker-state), result), final-state))
    by (rule staged-security-with-data-state-query-partial-opening-hitI
      [OF i-bound hit-at])
  show ?thesis
    unfolding
      checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-def
    using hit by simp
qed
qed

definition checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit
  :: (((((('f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool
where
  checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit
    out ⇔
    (case out of
      None ⇒ False
    | Some (((((prefix, prefix-state), data), attacker-state),
      result), final-state) ⇒
      (let s =
        verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)
        in ∃ as dg composition-fri-roots final rest trace-table
          composition-table query-idxs trace-openings composition-openings.
          accepted-with-bound-tables s (Some (result, final-state))
          trace-table composition-table as query-idxs ∧
          verifier-header-transcript s
            (staged-trace-root data) (staged-trace-fri-roots data)
            (staged-trace-final data) as dg composition-fri-roots final rest ∧
          (trace-openings, composition-openings) ∈
            query-header-supported-partial-opening-witnesses s
              (staged-trace-root data) (staged-trace-fri-roots data)
              (staged-trace-final data) as dg composition-fri-roots final ∧
          all-queries-consistent trace-table composition-table as))

```

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit-imp-query-hit-on-su*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

out $\in$

set-dist

(*execute*

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

adversary-initial-state)

and *hit:*

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit

out

shows

checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit

out

proof (*cases out*)

case *None*

then show *?thesis*

using *hit*

unfolding

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit-def

by *simp*

next

case (*Some result-pack*)

then obtain *prefix prefix-state data attacker-state result final-state* **where**

out-eq:

out = *Some* (((((*prefix, prefix-state*), *data*), *attacker-state*),

result), *final-state*)

by (*cases result-pack, auto split: prod.splits*)

let *?s* =

verifier-state-from-adversary attacker-state

(*staged-proof-transcript data*)

have *support-some:*

Some (((((*prefix, prefix-state*), *data*), *attacker-state*),

result), *final-state*) $\in$

set-dist

(*execute*

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

adversary-initial-state)

using *support* **unfolding** *out-eq* .

from *hit* **obtain** *as dg composition-fri-roots final rest trace-table*

composition-table query-idxs trace-openings composition-openings **where**

bound:

accepted-with-bound-tables ?s (*Some* (*result, final-state*))

trace-table composition-table as query-idxs

and *header:*

```

    verifier-header-transcript ?s
    (staged-trace-root data) (staged-trace-fri-roots data)
    (staged-trace-final data) as dg composition-fri-roots final rest
and all-queries:
    all-queries-consistent trace-table composition-table as
unfolding out-eq
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit-def
    Let-def
by auto
show ?thesis
    unfolding out-eq
    by (rule
        checked-staged-security-with-actual-alpha-prefix-bound-tables-all-queries-imp-query-partial-opening-hit-on-support
        [OF wf controlled support-some bound header all-queries])
qed

lemma checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit-bound-from-query-h
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and query-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    adversary-initial-state  $\leq$  Q
shows
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit
    adversary-initial-state  $\leq$  Q
proof –
have
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit
    adversary-initial-state  $\leq$ 
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    adversary-initial-state
by (rule wp-event-mono-on-support)
    (rule
        checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit-imp-query-hit-on-support
        [OF wf controlled])
then show ?thesis
    by (rule order-trans[OF - query-bound])
qed

definition checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit
    :: (((('f  $\times$  'f list  $\times$  'f list  $\times$  'f)  $\times$ 

```

$'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

$\text{checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit}$
 $\text{out} \longleftrightarrow$
 $\text{checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit}$
 $\text{out} \wedge$
 $\text{checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit}$
 out

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit-imp-query-hit:*

assumes

$\text{checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit}$
 out

shows *checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit*

out

using *assms*

unfolding

$\text{checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit-def}$

by *simp*

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit-bound-from-query-hit:*

assumes *query-bound:*

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit}$
 $\text{adversary-initial-state} \leq Q$

shows

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit}$
 $\text{adversary-initial-state} \leq Q$

by $(\text{rule } \text{order.trans}[OF - \text{query-bound}])$

$(\text{rule } \text{wp-event-mono},$
 rule

$\text{checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit-imp-query-hit})$

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit-bound-from-fixed-query-e*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *cover:*

$\bigwedge \text{data attacker-state } i.$

$i < \text{rounds} \implies$

$\text{query-header-supported-partial-opening-success-indices-at}$

$(\text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data}))$

$(\text{staged-trace-root data})$

$(\text{staged-trace-fri-roots data}))$

```

    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
     $i \subseteq B$ 
and raw-bound: query-index-raw-preimage-bound
and subset:  $B \subseteq \text{query-sample-space}$ 
and frac:
     $\text{nnreal} (\text{card } B) / \text{nnreal} (\text{card } \text{query-sample-space}) \leq$ 
    query-error-bound
shows
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit
    adversary-initial-state  $\leq$ 
    nnreal
    (staged-attacker-query-budget budgets +
    staged-challenge-query-budget) * query-error-bound
proof –
have query-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    adversary-initial-state  $\leq$ 
    nnreal
    (staged-attacker-query-budget budgets +
    staged-challenge-query-budget) * query-error-bound
by (rule
    checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-bound-from-fixed-query-error-cover
    [OF wf controlled cover raw-bound subset frac])
show ?thesis
by (rule
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit-bound-from-query-hit
    [OF query-bound])
qed

definition checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-query-hit
:: ((((((f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool
where
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-query-hit
    out  $\longleftrightarrow$ 
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit
    out ∧
    ¬ checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    out

```

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-imp-query-or-without-query:*

assumes

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit
out

shows

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit
out $\vee$
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-query-hit
out

using *assms*

unfolding

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit-def
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-query-hit-def
by *blast*

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-bound-from-query-and-without-*

assumes *query-bound:*

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit
adversary-initial-state $\leq Q$

and *without-bound:*

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-query-hit
adversary-initial-state $\leq W$

shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit
adversary-initial-state $\leq Q + W$

proof —

let *?M* =

checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

let *?Witness* =

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit

let *?Query* =

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit

let *?Without* =

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-query-hit

have *wp-event ?M ?Witness adversary-initial-state* $\leq$

wp-event ?M ($\lambda out. ?Query out \vee ?Without out$)
adversary-initial-state

by (*rule wp-event-mono*)

(*rule*

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-imp-query-or-without-query)

also have ... $\leq$
 $wp\text{-event } ?M \text{ ?Query adversary-initial-state} +$
 $wp\text{-event } ?M \text{ ?Without adversary-initial-state}$
by (rule *wp-event-union-bound*)
also have ... $\leq Q + W$
by (intro *add-mono query-bound without-bound*)
finally show ?thesis .
qed

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-partial-opening-witness-hit*:
assumes *witness-bound*:
 $wp\text{-event}$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state } A)$
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-partial}\text{-opening}\text{-witness}\text{-hit}$
 $adversary\text{-initial}\text{-state} \leq W$
shows
 $wp\text{-event}$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state } A)$
 $checked\text{-staged}\text{-security}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-tree}\text{-subtree}\text{-verifier}\text{-output}\text{-hit}$
 $adversary\text{-initial}\text{-state} \leq W$
proof –
have
 $wp\text{-event}$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state } A)$
 $checked\text{-staged}\text{-security}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-tree}\text{-subtree}\text{-verifier}\text{-output}\text{-hit}$
 $adversary\text{-initial}\text{-state} \leq$
 $wp\text{-event}$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state } A)$
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-partial}\text{-opening}\text{-witness}\text{-hit}$
 $adversary\text{-initial}\text{-state}$
by (rule *wp-event-mono-on-support*)
 $(rule$
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-imp}\text{-partial}\text{-opening}\text{-witness}\text{-hit}\text{-on}\text{-support})$
also have ... $\leq W$
by (rule *witness-bound*)
finally show ?thesis .
qed

definition *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit*
 $:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$
where
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-partial}\text{-opening}\text{-witness}\text{-without}\text{-all}\text{-queries}\text{-hit}$
 $out \longleftrightarrow$
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-partial}\text{-opening}\text{-witness}\text{-hit}$
 $out \wedge$
 $\neg checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-partial}\text{-opening}\text{-witness}\text{-all}\text{-queries}\text{-hit}$

out

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-imp-all-queries-or-without-all-q*

assumes

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit

out

shows

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit

out $\vee$

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit

out

using *assms*

unfolding

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit-def

by *blast*

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-bound-from-all-queries-and-wit*

assumes *all-queries-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit

adversary-initial-state $\leq Q$

and *without-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit

adversary-initial-state $\leq W$

shows

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit

adversary-initial-state $\leq Q + W$

proof –

let *?M* =

checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

let *?Witness* =

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit

let *?All* =

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit

let *?Without* =

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit

have *wp-event ?M ?Witness adversary-initial-state* $\leq$

wp-event ?M ($\lambda out. ?All\ out \vee ?Without\ out$)

adversary-initial-state

by (*rule wp-event-mono*)

(*rule*

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-imp-all-queries-or-without-all-quer

also have ... $\leq$
 $wp\text{-event } ?M \text{ ?All adversary-initial-state} +$
 $wp\text{-event } ?M \text{ ?Without adversary-initial-state}$
by (rule *wp-event-union-bound*)
also have ... $\leq Q + W$
by (intro *add-mono all-queries-bound without-bound*)
finally show ?thesis .
qed

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-all-queries-and-without-all-queries:*
assumes *all-queries-bound:*
 $wp\text{-event}$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state}$
 $A)$
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-partial}\text{-opening}\text{-witness}\text{-all}\text{-queries}\text{-hit}$
 $adversary\text{-initial}\text{-state} \leq Q$
and *without-bound:*
 $wp\text{-event}$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state}$
 $A)$
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-partial}\text{-opening}\text{-witness}\text{-without}\text{-all}\text{-queries}\text{-hit}$
 $adversary\text{-initial}\text{-state} \leq W$
shows
 $wp\text{-event}$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state } A)$
 $checked\text{-staged}\text{-security}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-tree}\text{-subtree}\text{-verifier}\text{-output}\text{-hit}$
 $adversary\text{-initial}\text{-state} \leq Q + W$
proof –
have *witness-bound:*
 $wp\text{-event}$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state } A)$
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-partial}\text{-opening}\text{-witness}\text{-hit}$
 $adversary\text{-initial}\text{-state} \leq Q + W$
by (rule
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-partial}\text{-opening}\text{-witness}\text{-hit}\text{-bound}\text{-from}\text{-all}\text{-queries}\text{-and}\text{-without}$
 $[OF \text{ all-queries-bound without-bound}])$
show ?thesis
by (rule
 $checked\text{-actual}\text{-alpha}\text{-prefix}\text{-branch}\text{-subtree}\text{-verifier}\text{-output}\text{-hit}\text{-bound}\text{-from}\text{-partial}\text{-opening}\text{-witness}\text{-hit}$
 $[OF \text{ witness-bound}])$
qed

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit-imp-fri-or-q*
assumes *support:*
 $out \in$
 $set\text{-dist}$
 $(execute$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state}$
 $A)$

```

      adversary-initial-state)
and hit:
  checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit
  out
shows
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  trace-fri-bad-with-partial-openings out  $\vee$ 
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-fri-bad-with-partial-openings out  $\vee$ 
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  query-bad out
proof (cases out)
  case None
  then show ?thesis
  using hit
  unfolding
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit-def
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-def
  by simp
next
  case (Some result-pack)
  then obtain prefix prefix-state data attacker-state result final-state where
    out-eq:
      out = Some (((((prefix, prefix-state), data), attacker-state),
        result), final-state)
    by (cases result-pack, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have support-some:
    Some (((((prefix, prefix-state), data), attacker-state),
      result), final-state)  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  using support unfolding out-eq .
  have witness-hit:
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit
    out
  using hit
  unfolding
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit-def
  by simp
  have not-all-hit:
     $\neg$ 
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit
    out

```

```

using hit
unfolding
  checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit-def
by simp
from witness-hit obtain as dg composition-fri-roots final rest trace-table
  composition-table query-idxs trace-openings composition-openings where
    bound:
      accepted-with-bound-tables ?s (Some (result, final-state))
      trace-table composition-table as query-idxs
and header:
      verifier-header-transcript ?s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final rest
and witness:
      (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses ?s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) as dg composition-fri-roots final
unfolding out-eq
  checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-def
  Let-def
by auto
have not-all-queries:
   $\neg$  all-queries-consistent trace-table composition-table as
proof
  assume all-queries:
    all-queries-consistent trace-table composition-table as
  have
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit
    out
  unfolding out-eq
  checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit-def
  Let-def
  apply simp
  apply (rule exI[where x=as])
  apply (rule exI[where x=dg])
  apply (rule exI[where x=composition-fri-roots])
  apply (rule exI[where x=final])
  apply (rule exI[where x=rest])
  apply (rule exI[where x=trace-table])
  apply (rule exI[where x=composition-table])
  apply (intro conjI)
    apply (rule exI[where x=query-idxs])
    apply (rule bound)
    apply (rule header)
  apply (rule exI[where x=trace-openings])
  apply (rule exI[where x=composition-openings])
  apply (rule witness)
  apply (rule all-queries)

```

```

    done
  then show False
    using not-all-hit by contradiction
qed
show ?thesis
proof (cases trace-table-low-degree trace-table)
  case False
  obtain fr f-fri-roots f-final dg' composition-fri-roots' final'
    rest' trace-openings' composition-openings' where
    trace-partial:
    accepted-with-partial-trace-openings ?s
    (Some (result, final-state)) fr query-idxs trace-openings'
  by (rule accepted-with-bound-tables-partial-opening-witnesses-Some
    [OF bound]) blast
  have trace-bad:
    trace-fri-bad-with-partial-openings ?s
    (Some (result, final-state))
  unfolding trace-fri-bad-with-partial-openings-def trace-fri-bad-def
  using bound False trace-partial by blast
  then show ?thesis
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    Let-def
    by simp
next
  case trace-low: True
  show ?thesis
  proof (cases
    composition-table-low-degree maxDegree composition-table)
    case False
    obtain fr f-fri-roots f-final dg' composition-fri-roots' final'
      rest' trace-openings' composition-openings' where
      trace-partial:
      accepted-with-partial-trace-openings ?s
      (Some (result, final-state)) fr query-idxs trace-openings'
    by (rule accepted-with-bound-tables-partial-opening-witnesses-Some
      [OF bound]) blast
    have comp-bad:
      composition-fri-bad-with-partial-openings ?s
      (Some (result, final-state))
    unfolding composition-fri-bad-with-partial-openings-def
      composition-fri-bad-def
    using bound trace-low False trace-partial by blast
    then show ?thesis
      unfolding out-eq
      checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      Let-def
      by simp
  next

```

```

case composition-low: True
have query-bad:
  query-bad ?s (Some (result, final-state))
  unfolding query-bad-def
  using bound trace-low composition-low not-all-queries by blast
then show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    Let-def
  by simp
qed
qed
qed

lemma checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit-bound-from
assumes trace-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      trace-fri-bad-with-partial-openings)
    adversary-initial-state  $\leq T$ 
and composition-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-fri-bad-with-partial-openings)
    adversary-initial-state  $\leq C$ 
and query-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      query-bad)
    adversary-initial-state  $\leq Q$ 
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit
    adversary-initial-state  $\leq T + C + Q$ 
proof –
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?Without =
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit
  let ?Trace =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    trace-fri-bad-with-partial-openings

```

```

let ?Composition =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-fri-bad-with-partial-openings
let ?Query =
  checked-staged-security-with-actual-alpha-prefix-verifier-event query-bad
have wp-event ?M ?Without adversary-initial-state ≤
  wp-event ?M (λout. ?Trace out ∨ (?Composition out ∨ ?Query out))
  adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule
    checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit-imp-fri-or-query)
also have ... ≤
  wp-event ?M ?Trace adversary-initial-state +
  wp-event ?M (λout. ?Composition out ∨ ?Query out)
  adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤
  wp-event ?M ?Trace adversary-initial-state +
  (wp-event ?M ?Composition adversary-initial-state +
  wp-event ?M ?Query adversary-initial-state)
by (intro add-mono order-refl wp-event-union-bound)
also have ... ≤ T + (C + Q)
by (intro add-mono trace-bound composition-bound query-bound)
also have ... = T + C + Q
by (simp add: add.assoc)
finally show ?thesis .
qed

```

```

lemma checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-query-hit-partial-fri-and-query:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and query-hit-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
  adversary-initial-state ≤ QH
and trace-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      trace-fri-bad-with-partial-openings)
  adversary-initial-state ≤ T
and composition-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event

```

composition-fri-bad-with-partial-openings
adversary-initial-state $\leq C$
and *query-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-verifier-event
query-bad)
adversary-initial-state $\leq Q$
shows
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
adversary-initial-state $\leq QH + (T + C + Q)$
proof –
have *all-queries-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit
adversary-initial-state $\leq QH$
by (rule
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-all-queries-hit-bound-from-query-hit
[OF wf controlled query-hit-bound])
have *without-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit
adversary-initial-state $\leq T + C + Q$
by (rule
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-all-queries-hit-bound-from-par
[OF trace-bound composition-bound query-bound])
show *?thesis*
by (rule
checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-all-queries-and-without-all-queries
[OF all-queries-bound without-bound])
qed

lemma *checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-partial-opening-witness-query-and-u*
assumes *query-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit
adversary-initial-state $\leq Q$
and *without-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-query-hit

$\text{adversary-initial-state} \leq W$
shows
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit}$
 $\text{adversary-initial-state} \leq Q + W$
proof –
have witness-bound :
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit}$
 $\text{adversary-initial-state} \leq Q + W$
by (rule
 $\text{checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit-bound-from-query-and-without-query}$
 $[OF \text{ query-bound without-bound}]$)
show $?thesis$
by (rule
 $\text{checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-partial-opening-witness-hit}$
 $[OF \text{ witness-bound}]$)
qed

lemma $\text{checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-partial-opening-fixed-query-cover-and}$
assumes $\text{wf: staged-budget-wellformed budgets}$
and $\text{controlled: staged-adversary-controlled budgets } A$
and cover:
 $\bigwedge \text{data attacker-state } i.$
 $i < \text{rounds} \implies$
 $\text{query-header-supported-partial-opening-success-indices-at}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{staged-trace-root data})$
 $(\text{staged-trace-fri-roots data})$
 $(\text{staged-trace-final data})$
 $(\text{staged-alphas data})$
 $(\text{staged-degree data})$
 $(\text{staged-composition-fri-roots data})$
 $(\text{staged-composition-final data})$
 $i \subseteq B$
and $\text{raw-bound: query-index-raw-preimage-bound}$
and $\text{subset: } B \subseteq \text{query-sample-space}$
and frac:
 $\text{nnreal (card } B) / \text{nnreal (card query-sample-space)} \leq$
 query-error-bound
and without-bound:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-without-query-hit}$
 $\text{adversary-initial-state} \leq W$

shows

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
adversary-initial-state $\leq$

nnreal

(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget) * *query-error-bound* + *W*

proof –

have *query-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit
adversary-initial-state $\leq$

nnreal

(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget) * *query-error-bound*

by (rule

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-query-hit-bound-from-fixed-query-error
[*OF wf controlled cover raw-bound subset frac*])

show *?thesis*

by (rule

checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-partial-opening-witness-query-and-without-bound
[*OF query-bound without-bound*])

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-branch-split-witness-and-transcript-budget*

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *root-pre-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-branch-transcript-root-preverifier-output-hit
adversary-initial-state $\leq$ *RP*

and *subtree-pre-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit
adversary-initial-state $\leq$ *SP*

and *subtree-verifier-witness-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-actual-alpha-prefix-branch-subtree-verifier-partial-opening-witness-hit
adversary-initial-state $\leq$ *WV*

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)

$(\text{staged-security-with-data-state-verifier-event composition-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(RP + \text{staged-concrete-transcript-target-error-bound} + SP + WV)$

proof –

have *subtree-verifier-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit}$
 $\text{adversary-initial-state} \leq WV$

by (rule
 $\text{checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-partial-opening-witness-hit}$
 $[OF \text{ subtree-verifier-witness-bound}]$)

show *?thesis*

by (rule
 $\text{checked-staged-security-with-data-state-composition-bad-bound-from-branch-split-side-events-and-transcript-budg}$
 $[OF \text{ wf controlled root-pre-bound subtree-pre-bound subtree-verifier-bound}]$)

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-branch-split-query-hit-partial-fri-and-que*

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *subtree-pre-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-preverifier-output-hit}$
 $\text{adversary-initial-state} \leq SP$

and *query-hit-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit}$
 $\text{adversary-initial-state} \leq QH$

and *trace-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-verifier-event trace-fri-bad-with-partial-openings})$
 $\text{adversary-initial-state} \leq T$

and *composition-bound*:

wp-event

```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-fri-bad-with-partial-openings)
    adversary-initial-state  $\leq C$ 
and query-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      query-bad)
    adversary-initial-state  $\leq Q$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event composition-bad)
    adversary-initial-state  $\leq$ 
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    (staged-phase-relation-error size
      (staged-attacker-query-budget budgets + staged-challenge-query-budget) +
      staged-concrete-transcript-target-error-bound + SP +
      (QH + (T + C + Q)))
proof –
  have subtree-verifier-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      checked-staged-security-with-actual-alpha-prefix-branch-tree-subtree-verifier-output-hit
      adversary-initial-state  $\leq QH + (T + C + Q)$ 
    by (rule
      checked-actual-alpha-prefix-branch-subtree-verifier-output-hit-bound-from-query-hit-partial-fri-and-query
      [OF wf controlled query-hit-bound trace-bound composition-bound
        query-bound])
  show ?thesis
    by (rule
      checked-staged-security-with-data-state-composition-bad-bound-from-branch-split-side-events-and-transcript-budg
      [OF wf controlled subtree-pre-bound subtree-verifier-bound])
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Partial
imports

```

begin

Partial-candidate composition reductions for the staged security experiment.

This layer avoids treating verifier-side Merkle subtree recomputation as a rare event. Instead, it lifts the verifier-local reduction from composition alpha badness to a partial-header candidate event plus hash collision.

context *soundness*

begin

definition *checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit*

$:: ('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $(((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
 $\text{bad-sets out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (x, \text{final-state}) \Rightarrow$
 $(\text{let packed} = \text{fst } (\text{fst } x);$
 $\text{data} = \text{snd packed};$
 $\text{attacker-state} = \text{snd } (\text{fst } x);$
 $\text{result} = \text{snd } x;$
 $s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
 $\text{in } \alpha\text{-header-list-set-hit } s$
 $(\alpha\text{-header-supported-partial-union-bad-sets } s \text{ bad-sets})$
 $(\text{Some } (\text{result}, \text{final-state}))))$

definition *checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit*

$:: ('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $(((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
 $\text{bad-sets out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (x, \text{final-state}) \Rightarrow$
 $(\text{let packed} = \text{fst } (\text{fst } x);$
 $\text{prefix-state} = \text{snd } (\text{fst packed});$
 $\text{data} = \text{snd packed};$

```

    attacker-state = snd (fst x);
    result = snd x;
    s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    in alpha-header-list-set-hit s
      (alpha-header-supported-partial-union-bad-sets s bad-sets)
      (Some (result, final-state)) ∧
      alpha-vector-prequeried-in-state prefix-state
      (staged-alphas data) (length spec)))

```

definition *checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit*

```

:: ('f list ⇒ 'f list set) ⇒
  ((((((('f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
  bad-sets out ⇔
checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
  bad-sets out ∧
(case out of
  None ⇒ False
| Some (x, -) ⇒
  (let packed = fst (fst x);
    prefix-state = snd (fst packed);
    data = snd packed
  in ¬ alpha-vector-prequeried-in-state prefix-state
    (staged-alphas data) (length spec)))

```

definition *checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*

```

:: ('f list ⇒ 'f list set) ⇒
  ((((((('f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
  bad-sets out ⇔
checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
  bad-sets out ∧
¬ checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  bad-sets out

```

definition *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift*

```

:: ((((((('f × 'f list × 'f list × 'f) ×
  'f protocol-channel) × 'f staged-proof-data) ×
  'f protocol-channel) × unit list) ×

```

'f protocol-channel) *option* $\Rightarrow$ *bool*
where
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
out $\longleftrightarrow$
checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-randomization-bad out $\wedge$
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space out

definition *staged-alpha-prefix-partial-header-prequery-hit*
 $:: 'f \text{ staged-adversary} \Rightarrow ('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool})$
where
staged-alpha-prefix-partial-header-prequery-hit A bad-sets out
 $\longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some (prefix, prefix-state)} \Rightarrow$
 $(\exists \text{ data attacker-state result final-state.}$
 $\text{Some } (((((\text{prefix, prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}) \in$
 set-dist
 $(\text{execute}$
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $\text{A})$
 $\text{adversary-initial-state}) \wedge$
 $\text{checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit}$
 bad-sets
 $(\text{Some } (((((\text{prefix, prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}))))))$

definition *staged-alpha-prefix-partial-header-prequery-relation*
 $:: 'f \text{ staged-adversary} \Rightarrow 'f \text{ protocol-channel} \Rightarrow$
 $('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $'f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}$
where
staged-alpha-prefix-partial-header-prequery-relation A s bad-sets x y
 $\longleftrightarrow$
 $(\exists \text{ prefix prefix-state data attacker-state result final-state i.}$
 $\text{Some } (((((\text{prefix, prefix-state}), \text{data}), \text{attacker-state}), \text{result}),$
 $\text{final-state}) \in$
 set-dist
 $(\text{execute}$
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $\text{A}) \text{ s}) \wedge$
 $\text{checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit}$
 bad-sets
 $(\text{Some } (((((\text{prefix, prefix-state}), \text{data}), \text{attacker-state}), \text{result}),$

$final-state)) \wedge$
 $i < length\ spec \wedge$
 $i < length\ (staged-alphas\ data) \wedge$
 $x = alpha-challenge-key-at\ prefix-state\ i\ (staged-alphas\ data) \wedge$
 $y = staged-alphas\ data\ !\ i)$

lemma *staged-alpha-prefix-partial-header-prequery-relation-fiber-bound-size:*

$card$
 $\{y. staged-alpha-prefix-partial-header-prequery-relation\ A\ s\ bad-sets\ x\ y\}$
 $\leq size$

proof –

have *subset:*

$\{y. staged-alpha-prefix-partial-header-prequery-relation\ A\ s\ bad-sets\ x\ y\}$
 $\subseteq (UNIV :: 'f\ set)$

by *simp*

have *finite-field:* $finite\ (UNIV :: 'f\ set)$

by *simp*

have *card*

$\{y. staged-alpha-prefix-partial-header-prequery-relation\ A\ s\ bad-sets\ x\ y\}$
 $\leq card\ (UNIV :: 'f\ set)$

by $(rule\ card-mono[OF\ finite-field\ subset])$

also have $... = size$

using *size-card* **by** *simp*

finally show *?thesis* .

qed

lemma *staged-alpha-prefix-partial-header-prequery-hit-imp-relation-hit:*

assumes *empty:* $HashMap\ s = fmempty$

and *support:*

$Some\ (prefix,\ prefix-state) \in$
 $set-dist\ (execute\ (staged-alpha-prefix-program\ A)\ s)$

and *hit:*

$staged-alpha-prefix-partial-header-prequery-hit\ A\ bad-sets$
 $(Some\ (prefix,\ prefix-state))$

and *init:* $s = adversary-initial-state$

shows

$hash-relation-hit-event$
 $(staged-alpha-prefix-partial-header-prequery-relation\ A\ s\ bad-sets)$
 $s\ (Some\ (prefix,\ prefix-state))$

proof –

from *hit* **obtain** *data attacker-state result final-state* **where** *full-support:*

$Some\ (((((prefix,\ prefix-state),\ data),\ attacker-state),\ result),$
 $final-state) \in$
 $set-dist$
 $(execute$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $adversary-initial-state)$

and *prehit:*


```

    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
    bad-sets
    (Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state))
  unfolding staged-alpha-prefix-partial-header-prequery-hit-def by auto
from prehit obtain i where
  i-bound:  $i < \text{length spec}$ 
  and i-len:  $i < \text{length (staged-alphas data)}$ 
  and lookup:
    fmlookup (HashMap prefix-state)
    (alpha-challenge-key-at prefix-state i (staged-alphas data)) =
    Some (staged-alphas data ! i)
  unfolding
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit-def
    alpha-vector-prequeried-in-state-def
  by (auto simp: Let-def)
have initial-none:
  fmlookup (HashMap s)
  (alpha-challenge-key-at prefix-state i (staged-alphas data)) = None
  using empty by simp
have relation:
  staged-alpha-prefix-partial-header-prequery-relation A s bad-sets
  (alpha-challenge-key-at prefix-state i (staged-alphas data))
  (staged-alphas data ! i)
  unfolding staged-alpha-prefix-partial-header-prequery-relation-def
  using full-support prehit i-bound i-len init by blast
have
  hash-relation-hit
  (staged-alpha-prefix-partial-header-prequery-relation A s bad-sets)
  s prefix-state
  unfolding hash-relation-hit-def
  by (intro exI[ $\text{of - alpha-challenge-key-at prefix-state i}$ 
    (staged-alphas data)]
    exI[ $\text{of - staged-alphas data ! i}$ ] conjI]
    (use initial-none lookup relation in simp-all))
then show ?thesis
  unfolding hash-relation-hit-event-def by simp
qed

lemma staged-alpha-prefix-partial-header-prequery-hit-bound-by-relation-hit:
  assumes init:  $s = \text{adversary-initial-state}$ 
  shows
    wp-event (staged-alpha-prefix-program A)
    (staged-alpha-prefix-partial-header-prequery-hit A bad-sets)  $s \leq$ 
    wp-event (staged-alpha-prefix-program A)
    (hash-relation-hit-event
    (staged-alpha-prefix-partial-header-prequery-relation A s bad-sets) s)
    s
proof (rule wp-event-mono-on-support)

```

```

fix out
assume support:
  out  $\in$  set-dist (execute (staged-alpha-prefix-program A) s)
and hit:
  staged-alpha-prefix-partial-header-prequery-hit A bad-sets out
show
  hash-relation-hit-event
    (staged-alpha-prefix-partial-header-prequery-relation A s bad-sets) s
  out
proof (cases out)
  case None
  then show ?thesis
    using hit unfolding staged-alpha-prefix-partial-header-prequery-hit-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state where out-eq:
    out = Some (prefix, prefix-state)
  by (cases packed) simp
show ?thesis
  unfolding out-eq
  by (rule staged-alpha-prefix-partial-header-prequery-hit-imp-relation-hit
    [OF - - init])
    (use support hit out-eq init in
       $\langle$ simp-all add: adversary-initial-state-def $\rangle$ )
qed
qed

lemma staged-alpha-prefix-partial-header-prequery-hit-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event (staged-alpha-prefix-program A)
    (staged-alpha-prefix-partial-header-prequery-hit A bad-sets)
    adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)
proof -
have prehit-le-relation:
  wp-event (staged-alpha-prefix-program A)
    (staged-alpha-prefix-partial-header-prequery-hit A bad-sets)
    adversary-initial-state  $\leq$ 
  wp-event (staged-alpha-prefix-program A)
    (hash-relation-hit-event
      (staged-alpha-prefix-partial-header-prequery-relation A
        adversary-initial-state bad-sets) adversary-initial-state)
    adversary-initial-state
by (rule staged-alpha-prefix-partial-header-prequery-hit-bound-by-relation-hit)
  simp

```

```

also have ... ≤
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0)
proof —
  have program:
    hash-relation-program
    (staged-alpha-prefix-partial-header-prequery-relation A
      adversary-initial-state bad-sets)
    size (staged-alpha-search-queries budgets 0)
    (staged-alpha-prefix-program A)
  by (rule hash-relation-program-staged-alpha-prefix-program
    [OF wf controlled])
    (rule
      staged-alpha-prefix-partial-header-prequery-relation-fiber-bound-size)
  show ?thesis
  using program
  unfolding hash-relation-program-def hash-relation-budget-def
    staged-phase-relation-error-def
  by blast
qed
finally show ?thesis .
qed

lemma checked-staged-security-experiment-with-actual-alpha-prefix-data-stateI:
  assumes prefix:
    Some (prefix, prefix-state) ∈
      set-dist
        (execute (staged-alpha-prefix-program A) adversary-initial-state)
  and after:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-after-alpha-prefix-program A prefix)
          prefix-state)
  and verifier:
    Some (result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
  shows
    Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state) ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
            A)
          adversary-initial-state)
proof —
  have transcript:

```

```

Some (((prefix, prefix-state), data), attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-with-alpha-prefix-program A)
    adversary-initial-state)
unfolding checked-staged-transcript-with-alpha-prefix-program-def
apply (rule-tac x=prefix and t=prefix-state in set-dist-bindI)
apply (rule prefix)
apply (rule-tac x=prefix-state and t=prefix-state in set-dist-bindI)
apply simp
apply (rule-tac x=data and t=attacker-state in set-dist-bindI)
apply (rule after)
apply simp
done
show ?thesis
unfolding
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
apply (rule-tac x=((prefix, prefix-state), data) and
  t=attacker-state in set-dist-bindI)
apply (rule transcript)
apply (rule-tac x=attacker-state and t=attacker-state in set-dist-bindI)
apply simp
apply (rule-tac x=() and
  t=verifier-state-from-adversary attacker-state
  (staged-proof-transcript data) in set-dist-bindI)
apply simp
apply (rule-tac x=result and t=final-state in set-dist-bindI)
apply (rule verifier)
apply simp
done
qed

lemma checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit-bound-from-prefix:
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
      bad-sets)
    adversary-initial-state ≤
  wp-event (staged-alpha-prefix-program A)
    (staged-alpha-prefix-partial-header-prequery-hit A bad-sets)
    adversary-initial-state
unfolding checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
  checked-staged-transcript-with-alpha-prefix-program-def
  sm-bind-assoc
apply (rule wp-event-bind-bound-by-head-event
  [where P = staged-alpha-prefix-partial-header-prequery-hit A bad-sets])
apply simp
apply (simp add:

```

```

    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit-def
    staged-alpha-prefix-partial-header-prequery-hit-def)
apply (auto simp:
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit-def
    staged-alpha-prefix-partial-header-prequery-hit-def Let-def
    elim!: set-dist-bindE
    intro!: set-dist-bindI
    checked-staged-security-experiment-with-actual-alpha-prefix-data-stateI
    split: option.splits prod.splits)
apply (rule-tac x=bi in exI)
apply (rule-tac x=bj in exI)
apply (rule-tac x=bk in exI)
apply (rule-tac x=x2a in exI)
apply (intro conjI)
    apply (rule checked-staged-security-experiment-with-actual-alpha-prefix-data-stateI)
        apply assumption
        apply assumption
        apply assumption
        apply assumption
apply assumption
done

lemma checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
     A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
     bad-sets)
    adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)
proof –
have
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
     bad-sets)
    adversary-initial-state  $\leq$ 
    wp-event (staged-alpha-prefix-program A)
    (staged-alpha-prefix-partial-header-prequery-hit A bad-sets)
    adversary-initial-state
by (rule
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit-bound-from-prefix)
also have ...  $\leq$ 
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)

```

```

    by (rule staged-alpha-prefix-partial-header-prequery-hit-bound
        [OF wf controlled])
  finally show ?thesis .
qed

```

Vector-valued, prefix-state-dependent accounting for future alpha challenge keys. The bad vector set may be selected from the concrete verifier/header state reached after the alpha prefix; the bound is derived from the existing relation-budget interface for the prefix program, not from a freshness assumption.

lemma *checked-staged-security-with-actual-alpha-prefix-vector-alpha-prequery-accounting:*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows

```

```

  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
      bad-sets)
  adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)

```

```

by (rule
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit-bound
    [OF wf controlled])

```

lemma *checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-imp-fresh-or-prequery:*

```

assumes
  checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
  bad-sets out
shows

```

```

  checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
  bad-sets out  $\vee$ 
  checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
  bad-sets out

```

```

using assms

```

```

unfolding

```

```

  checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-def
  checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-def
  checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit-def

```

```

by (auto simp: Let-def split: option.splits prod.splits)

```

lemma *checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-bound-from-fresh-and-prequery:*

```

assumes fresh-bound:

```

```

  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
      bad-sets)
  adversary-initial-state  $\leq$  F

```

and *prequery-bound*:

$$\begin{aligned} & wp\text{-}event \\ & \quad (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}data\text{-}state \\ & \quad \quad A) \\ & \quad (checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}partial\text{-}header\text{-}prequery\text{-}hit \\ & \quad \quad \quad bad\text{-}sets) \\ & \quad adversary\text{-}initial\text{-}state \leq P \end{aligned}$$
shows

$$\begin{aligned} & wp\text{-}event \\ & \quad (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}data\text{-}state \\ & \quad \quad A) \\ & \quad (checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}partial\text{-}header\text{-}bad\text{-}set\text{-}hit \\ & \quad \quad \quad bad\text{-}sets) \\ & \quad adversary\text{-}initial\text{-}state \leq F + P \end{aligned}$$
proof –
let $?M =$

$$checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}data\text{-}state \ A$$
let $?Bad =$

$$checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}partial\text{-}header\text{-}bad\text{-}set\text{-}hit \ bad\text{-}sets$$
let $?Fresh =$

$$checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}partial\text{-}header\text{-}fresh\text{-}hit \ bad\text{-}sets$$
let $?Pre =$

$$checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}partial\text{-}header\text{-}prequery\text{-}hit \ bad\text{-}sets$$
have $wp\text{-}event \ ?M \ ?Bad \ adversary\text{-}initial\text{-}state \leq$

$$wp\text{-}event \ ?M \ (\lambda out. \ ?Fresh \ out \ \vee \ ?Pre \ out) \ adversary\text{-}initial\text{-}state$$
by (rule *wp-event-mono*)

$$(rule \ checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}partial\text{-}header\text{-}bad\text{-}set\text{-}hit\text{-}imp\text{-}fresh\text{-}or\text{-}prequery)$$
also have $\dots \leq$

$$wp\text{-}event \ ?M \ ?Fresh \ adversary\text{-}initial\text{-}state \ +$$

$$wp\text{-}event \ ?M \ ?Pre \ adversary\text{-}initial\text{-}state$$
by (rule *wp-event-union-bound*)
also have $\dots \leq F + P$
by (intro *add-mono* *fresh-bound* *prequery-bound*)
finally show $?thesis$.
qed

lemma *checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-bound-from-fresh-and-budgets*:
assumes *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *fresh-bound*:

$$\begin{aligned} & wp\text{-}event \\ & \quad (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}data\text{-}state \\ & \quad \quad A) \\ & \quad (checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}\alpha\text{-}prefix\text{-}partial\text{-}header\text{-}fresh\text{-}hit \\ & \quad \quad \quad bad\text{-}sets) \end{aligned}$$

$adversary-initial-state \leq F$
shows
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state\ A)$
 $(checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit\ bad-sets)$
 $adversary-initial-state \leq$
 $F + staged-phase-relation-error\ size$
 $(staged-alpha-search-queries\ budgets\ 0)$
proof –
have $prequery-bound$:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state\ A)$
 $(checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit\ bad-sets)$
 $adversary-initial-state \leq$
 $staged-phase-relation-error\ size$
 $(staged-alpha-search-queries\ budgets\ 0)$
by (rule
 $checked-staged-security-with-actual-alpha-prefix-vector-alpha-prequery-accounting$
 $[OF\ wf\ controlled])$
show $?thesis$
by (rule
 $checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-bound-from-fresh-and-prequery$
 $[OF\ fresh-bound\ prequery-bound])$
qed

lemma $checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-imp-prefix-or-drift$:
assumes
 $checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit$
 $bad-sets\ out$
shows
 $checked-staged-security-with-actual-alpha-prefix-bad-set-hit\ bad-sets$
 $out \vee$
 $checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift$
 $bad-sets\ out$
using $assms$
unfolding
 $checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def$
by $blast$

lemma $checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-bound-from-prefix-and-drift$:
assumes $prefix-bound$:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state\ A)$
 $(checked-staged-security-with-actual-alpha-prefix-bad-set-hit\ bad-sets)$


```

    adversary-initial-state  $\leq P$ 
and drift-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
       A)
      (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
       bad-sets)
    adversary-initial-state  $\leq D$ 
shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
       A)
      (checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
       bad-sets)
    adversary-initial-state  $\leq P + D$ 
proof –
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?Fresh =
    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
    bad-sets
  let ?Prefix =
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit bad-sets
  let ?Drift =
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
    bad-sets
  have wp-event ?M ?Fresh adversary-initial-state  $\leq$ 
    wp-event ?M ( $\lambda$ out. ?Prefix out  $\vee$  ?Drift out)
    adversary-initial-state
  by (rule wp-event-mono)
    (rule
     checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-imp-prefix-or-drift)
  also have ...  $\leq$ 
    wp-event ?M ?Prefix adversary-initial-state +
    wp-event ?M ?Drift adversary-initial-state
  by (rule wp-event-union-bound)
  also have ...  $\leq P + D$ 
  by (intro add-mono prefix-bound drift-bound)
  finally show ?thesis .
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-imp-branch-tree-on-support:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  out  $\in$ 
  set-dist
  (execute
   (checked-staged-security-experiment-with-actual-alpha-prefix-data-state

```

```

      A)
      adversary-initial-state)
and drift:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
  out
shows
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
  out
proof —
have random-bad:
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-randomization-bad out
using drift
unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
by simp
have not-prefix:
  ¬ checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  composition-trace-bad-alpha-space out
using drift
unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def
by simp
have split:
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  composition-trace-bad-alpha-space out ∨
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
  out
by (rule
  checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-prefix-or-branch-tree-on-sup
  [OF wf controlled support random-bad])
then show ?thesis
  using not-prefix by blast
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-branch-tree:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and branch-tree-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
  A)
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
  adversary-initial-state ≤ T
shows
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
  A)

```

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq T$

proof –

let *?M* =
checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
have *wp-event ?M*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq$
wp-event ?M
checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
adversary-initial-state
by (rule *wp-event-mono-on-support*)
(rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-imp-branch-tree-on-support
[*OF wf controlled*])
then show *?thesis*
by (rule *order-trans[OF - branch-tree-bound]*)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-bound-from-prefix-drift-and-budgets*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *prefix-bound:*
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
bad-sets)
adversary-initial-state $\leq P$
and *drift-bound:*
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
bad-sets)
adversary-initial-state $\leq D$
shows
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit*
bad-sets)
adversary-initial-state $\leq$
P + D +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*)
proof –
have *fresh-bound:*
wp-event

```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
      bad-sets)
    adversary-initial-state  $\leq P + D$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-bound-from-prefix-and-drift
    [OF prefix-bound drift-bound])
  have
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
      bad-sets)
    adversary-initial-state  $\leq$ 
    (P + D) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-bound-from-fresh-and-budgets
    [OF wf controlled fresh-bound])
  then show ?thesis
  by (simp add: add.assoc)
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-partial-header-or-collision
  assumes support:
    out  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  and bad:
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-randomization-bad out
  shows
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
    composition-trace-bad-alpha-space out  $\vee$ 
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    hash-map-output-collision-bad out
proof (cases out)
case None
  then show ?thesis
  using bad
  unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
case (Some result-pack)
  then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:

```

```

    out =
      Some (((((prefix, prefix-state), data), attacker-state),
              result), final-state)
  by (cases result-pack, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have verify-out:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
using support
unfolding out-eq
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
by (auto elim!: set-dist-bindE split: prod.splits)
have random-bad:
  composition-randomization-bad ?s (Some (result, final-state))
using bad
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  Let-def
by simp
have alpha-bad:
  alpha-bad-set-hit ?s composition-trace-bad-alpha-space
  (Some (result, final-state))
by (rule composition-randomization-bad-imp-alpha-bad-set-hit
    [OF random-bad])
have split:
  alpha-header-list-set-hit ?s
  (alpha-header-supported-partial-union-bad-sets ?s
    composition-trace-bad-alpha-space)
  (Some (result, final-state)) ∨
  hash-map-output-collision-bad ?s (Some (result, final-state))
by (rule alpha-bad-set-hit-imp-partial-union-or-collision
    [OF verify-out alpha-bad
      composition-trace-bad-alpha-space-subset-alpha-space])
then show ?thesis
proof
  assume partial:
    alpha-header-list-set-hit ?s
    (alpha-header-supported-partial-union-bad-sets ?s
      composition-trace-bad-alpha-space)
    (Some (result, final-state))
  have
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
    composition-trace-bad-alpha-space out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-def
    Let-def
  using partial by simp
then show ?thesis by simp

```

```

next
  assume collision:
    hash-map-output-collision-bad ?s (Some (result, final-state))
  have
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    hash-map-output-collision-bad out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    Let-def
  using collision by simp
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-prefix-prequery-drift-or-collision:
  assumes support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
            A)
          adversary-initial-state)
  and bad:
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-randomization-bad out
  shows
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space out ∨
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
    composition-trace-bad-alpha-space out ∨
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
    out ∨
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    hash-map-output-collision-bad out
proof —
  have partial-or-collision:
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
    composition-trace-bad-alpha-space out ∨
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    hash-map-output-collision-bad out
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-partial-header-or-collision-or-collision-or-collision
    [OF support bad])
  then show ?thesis
proof
  assume partial:
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
    composition-trace-bad-alpha-space out
  have fresh-or-prequery:

```

```

    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
      composition-trace-bad-alpha-space out ∨
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
      composition-trace-bad-alpha-space out
  by (rule
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-imp-fresh-or-prequery
      [OF partial])
  then show ?thesis
proof
  assume fresh:
    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
      composition-trace-bad-alpha-space out
  have prefix-or-drift:
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
      composition-trace-bad-alpha-space out ∨
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
      composition-trace-bad-alpha-space out
  by (rule
    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-imp-prefix-or-drift
      [OF fresh])
  then show ?thesis
proof
  assume
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
      composition-trace-bad-alpha-space out
  then show ?thesis by simp
next
  assume drift:
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
      composition-trace-bad-alpha-space out
  have
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
      out
  unfolding
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
  using bad drift by simp
  then show ?thesis by simp
qed
next
  assume
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
      composition-trace-bad-alpha-space out
  then show ?thesis by simp
qed
next
  assume
    checked-staged-security-with-actual-alpha-prefix-verifier-event
      hash-map-output-collision-bad out
  then show ?thesis by simp

```

qed
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-bound-from-prefix-prequery-d*

assumes *prefix-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

(*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
 composition-trace-bad-alpha-space)

adversary-initial-state $\leq P$

and *prequery-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

(*checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit*
 composition-trace-bad-alpha-space)

adversary-initial-state $\leq Q$

and *drift-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq D$

and *collision-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
 hash-map-output-collision-bad)

adversary-initial-state $\leq C$

shows

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
 composition-randomization-bad)

adversary-initial-state $\leq P + Q + D + C$

proof –

let *?M* =

checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

let *?Prefix* =

checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space

let *?Prequery* =

checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
composition-trace-bad-alpha-space

let *?Drift* =

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift


```

let ?Collision =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  hash-map-output-collision-bad
let ?Random =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-randomization-bad
have wp-event ?M ?Random adversary-initial-state ≤
  wp-event ?M
  (λout. ?Prefix out ∨ ?Prequery out ∨ ?Drift out ∨
   ?Collision out)
  adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule
   checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-prefix-prequery-drift-or-collision)
also have ... ≤
  wp-event ?M ?Prefix adversary-initial-state +
  wp-event ?M ?Prequery adversary-initial-state +
  wp-event ?M ?Drift adversary-initial-state +
  wp-event ?M ?Collision adversary-initial-state
by (rule wp-event-union-bound4)
also have ... ≤ P + Q + D + C
by (intro add-mono prefix-bound prequery-bound drift-bound
  collision-bound)
finally show ?thesis .
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-bound-from-partial-header-and-partial-data-state
assumes partial-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  (checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
   composition-trace-bad-alpha-space)
  adversary-initial-state ≤ P
and collision-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
   hash-map-output-collision-bad)
  adversary-initial-state ≤ C
shows
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
   composition-randomization-bad)
  adversary-initial-state ≤ P + C
proof –

```

```

let ?M =
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let ?Partial =
  checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
  composition-trace-bad-alpha-space
let ?Collision =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  hash-map-output-collision-bad
let ?Random =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-randomization-bad
have wp-event ?M ?Random adversary-initial-state ≤
  wp-event ?M (λout. ?Partial out ∨ ?Collision out)
  adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule
    checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-partial-header-or-collision-o
also have ... ≤
  wp-event ?M ?Partial adversary-initial-state +
  wp-event ?M ?Collision adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤ P + C
by (intro add-mono partial-bound collision-bound)
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-partial-header-and-co
assumes partial-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
      composition-trace-bad-alpha-space)
  adversary-initial-state ≤ P
and collision-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      hash-map-output-collision-bad)
  adversary-initial-state ≤ C
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event composition-bad)
  adversary-initial-state ≤ P + C
proof –
let ?M =
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let ?degree =

```

```

    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-degree-bad
  let ?random =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-randomization-bad
  let ?bad =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-bad
  have projected:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event composition-bad)
    adversary-initial-state =
    wp-event ?M ?bad adversary-initial-state
  by (rule
    checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)
  have event-eq: ?bad = ( $\lambda$ out. ?degree out  $\vee$  ?random out)
  by (rule ext)
    (auto simp:
      checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      composition-bad-def split: option.splits prod.splits)
  have degree-false: ?degree = ( $\lambda$ out. False)
  by (rule ext)
    (auto simp:
      checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      composition-degree-bad-false[OF spec-degree-wellformed-from-spec-query-margin]
      split: option.splits prod.splits)
  have degree-bound:
    wp-event ?M ?degree adversary-initial-state  $\leq 0$ 
  unfolding degree-false wp-event-def wp-def dist-expect-def by simp
  have random-bound:
    wp-event ?M ?random adversary-initial-state  $\leq P + C$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-bound-from-partial-header-and-c
    [OF partial-bound collision-bound])
  have wp-event ?M ?bad adversary-initial-state  $\leq$ 
    wp-event ?M ?degree adversary-initial-state +
    wp-event ?M ?random adversary-initial-state
  unfolding event-eq by (rule wp-event-union-bound)
  also have ...  $\leq 0 + (P + C)$ 
  by (intro add-mono degree-bound random-bound)
  finally show ?thesis
  unfolding projected by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-hash-map-output-collision-bad-bound-controlled:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event

```

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 hash-map-output-collision-bad)
 adversary-initial-state $\leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)
proof –
 have data-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 hash-map-output-collision-bad)
 adversary-initial-state $\leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)
 by (rule
 checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
 [OF wf controlled])
 then show ?thesis
 using
 checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection
 [of A hash-map-output-collision-bad]
 by simp
qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-partial-header-and-co

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and partial-bound:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq P$

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event composition-bad)
 adversary-initial-state $\leq$
 P +
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)

proof –

have collision-bound:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event

```

    hash-map-output-collision-bad)
  adversary-initial-state  $\leq$ 
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
   staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
  checked-staged-security-with-actual-alpha-prefix-hash-map-output-collision-bad-bound-controlled
  [OF wf controlled])
show ?thesis
by (rule
  checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-partial-header-and-collision-bad-bound-controlled
  [OF partial-bound collision-bound])
qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-partial-header-candidate-drift-and-budget
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and drift-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
   composition-trace-bad-alpha-space)
  adversary-initial-state  $\leq$  D
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event composition-bad)
  adversary-initial-state  $\leq$ 
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  D +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
   staged-challenge-query-budget + verifier-hash-query-budget)
proof -
let ?Prefix =
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0)
have transcript-bound:
  wp-event (checked-staged-transcript-with-alpha-prefix-program A)
  (checked-staged-transcript-actual-alpha-prefix-bad-set-hit

```

```

    composition-trace-bad-alpha-space)
  adversary-initial-state  $\leq$  ?Prefix
by (rule
  checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
  [OF wf controlled])
have prefix-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space)
  adversary-initial-state  $\leq$  ?Prefix
by (rule
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
  [OF transcript-bound])
have partial-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
    composition-trace-bad-alpha-space)
  adversary-initial-state  $\leq$ 
  ?Prefix + D +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0)
by (rule
  checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-bound-from-prefix-drift-and-budgets
  [OF wf controlled prefix-bound drift-bound])
have
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event composition-bad)
  adversary-initial-state  $\leq$ 
  (?Prefix + D +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)) +
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
  checked-staged-security-with-data-state-composition-bad-bound-from-actual-alpha-prefix-partial-header-and-contr
  [OF wf controlled partial-bound])
then show ?thesis
  by (simp add: add.assoc)
qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-partial-prequery-branch-tree-and-budgets
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and branch-tree-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state

```

A)
checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
adversary-initial-state $\leq T$

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event composition-bad*)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
T +
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)

proof –

let *?M* =

checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let *?Prefix* =
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*)

let *?Pre* =

staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*)

let *?Coll* =

hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)

have *transcript-bound*:

wp-event (*checked-staged-transcript-with-alpha-prefix-program A*)
 (*checked-staged-transcript-actual-alpha-prefix-bad-set-hit*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$ *?Prefix*

by (*rule*

checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
 [*OF wf controlled*])

have *prefix-bound*:

wp-event *?M*
 (*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$ *?Prefix*

by (*rule*

checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript

```

    [OF transcript-bound])
have prequery-bound:
  wp-event ?M
  (checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
   composition-trace-bad-alpha-space)
  adversary-initial-state ≤ ?Pre
by (rule
  checked-staged-security-with-actual-alpha-prefix-vector-alpha-prequery-accounting
  [OF wf controlled])
have drift-bound:
  wp-event ?M
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
  adversary-initial-state ≤ T
by (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-branch-tree
  [OF wf controlled branch-tree-bound])
have collision-bound:
  wp-event ?M
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
   hash-map-output-collision-bad)
  adversary-initial-state ≤ ?Coll
by (rule
  checked-staged-security-with-actual-alpha-prefix-hash-map-output-collision-bad-bound-controlled
  [OF wf controlled])
let ?degree =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-degree-bad
let ?random =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-randomization-bad
let ?bad =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-bad
have projected:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event composition-bad)
  adversary-initial-state =
  wp-event ?M ?bad adversary-initial-state
by (rule
  checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)
have event-eq: ?bad = (λout. ?degree out ∨ ?random out)
by (rule ext)
  (auto simp:
  checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  composition-bad-def split: option.splits prod.splits)
have degree-false: ?degree = (λout. False)
by (rule ext)
  (auto simp:
  checked-staged-security-with-actual-alpha-prefix-verifier-event-def

```



```

    composition-degree-bad-false[OF spec-degree-wellformed-from-spec-query-margin]
    split: option.splits prod.splits)
  have degree-bound:
    wp-event ?M ?degree adversary-initial-state ≤ 0
    unfolding degree-false wp-event-def wp-def dist-expect-def by simp
  have random-bound:
    wp-event ?M ?random adversary-initial-state ≤
      ?Prefix + ?Pre + T + ?Coll
    by (rule
      checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-bound-from-prefix-prequery-drift-
      [OF prefix-bound prequery-bound drift-bound collision-bound])
  have wp-event ?M ?bad adversary-initial-state ≤
    wp-event ?M ?degree adversary-initial-state +
    wp-event ?M ?random adversary-initial-state
    unfolding event-eq by (rule wp-event-union-bound)
  also have ... ≤ 0 + (?Prefix + ?Pre + T + ?Coll)
    by (intro add-mono degree-bound random-bound)
  finally show ?thesis
    unfolding projected by (simp add: add.assoc)
qed

end

end

```

theory *Soundness-Staged-Candidate-Query*

imports

Soundness-Staged-Query

Soundness-Staged-Partial-Query

begin

Candidate-indexed dynamic query-prefix targets.

These targets are narrower than the old global partial-opening fixed cover: they are parameterized by one fixed trace/composition candidate pair and the authenticated openings extracted for that pair. This is the shape needed for the composition candidate-drift query case, where the target must be chosen before the query-index challenge is sampled.

context *soundness*

begin

definition *staged-query-prefix-candidate-opening-query-target*

```

:: 'f authenticated-opening list list ⇒
  'f authenticated-opening list list ⇒
  'f list ⇒ nat ⇒
  'f staged-query-prefix-data ⇒ 'f protocol-channel ⇒ nat set

```

where

```

staged-query-prefix-candidate-opening-query-target trace-openings
  composition-openings as i prefix prefix-state =
  partial-query-success-indices-at trace-openings composition-openings as i

```

definition *staged-query-prefix-candidate-pair-query-target*
 $:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow$
 $'f \text{ staged-query-prefix-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow \text{nat set}$

where

staged-query-prefix-candidate-pair-query-target trace-table
composition-table as prefix prefix-state =
query-sampling-success-space trace-table composition-table as

definition *staged-query-prefix-candidate-opening-query-target-from-prefix*

$:: 'f \text{ authenticated-opening list list} \Rightarrow$
 $'f \text{ authenticated-opening list list} \Rightarrow$
 $\text{nat} \Rightarrow 'f \text{ staged-query-prefix-data} \Rightarrow$
 $'f \text{ protocol-channel} \Rightarrow \text{nat set}$

where

staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i prefix prefix-state =
partial-query-success-indices-at trace-openings composition-openings
(sqp-alphas prefix) i

definition *staged-query-prefix-candidate-pair-query-target-from-prefix*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ staged-query-prefix-data} \Rightarrow$
 $'f \text{ protocol-channel} \Rightarrow \text{nat set}$

where

staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table prefix prefix-state =
query-sampling-success-space trace-table composition-table
(sqp-alphas prefix)

definition *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit*

where

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit A
trace-table composition-table out $\longleftrightarrow$
(case out of
None $\Rightarrow$ False
| Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
final-state) $\Rightarrow$
($\exists i$ prefix prefix-state raw raw-state.
i < rounds $\wedge$
Some (data, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) $\wedge$
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) $\wedge$
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target trace-table

composition-table (staged-alphas data))
(Some (((prefix, prefix-state), raw), raw-state))))

definition *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit*
where

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
trace-openings composition-openings as i out $\longleftrightarrow$
(case out of
None $\Rightarrow$ False
| Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
final-state) $\Rightarrow$
as = staged-alphas data $\wedge$
i < rounds $\wedge$
($\exists$ prefix prefix-state raw raw-state.
Some (data, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) $\wedge$
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) $\wedge$
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target
trace-openings composition-openings as i)
(Some (((prefix, prefix-state), raw), raw-state))))

lemma *staged-query-prefix-candidate-opening-query-target-subset:*
staged-query-prefix-candidate-opening-query-target trace-openings
composition-openings as i prefix prefix-state $\subseteq$ query-sample-space
unfolding *staged-query-prefix-candidate-opening-query-target-def*
by (rule *partial-query-success-indices-at-subset*)

lemma *staged-query-prefix-candidate-pair-query-target-subset:*
staged-query-prefix-candidate-pair-query-target trace-table
composition-table as prefix prefix-state $\subseteq$ query-sample-space
unfolding *staged-query-prefix-candidate-pair-query-target-def*
by (rule *query-sampling-success-space-subset*)

lemma *staged-query-prefix-candidate-opening-query-target-from-prefix-subset:*
staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i prefix prefix-state $\subseteq$
query-sample-space
unfolding *staged-query-prefix-candidate-opening-query-target-from-prefix-def*
by (rule *partial-query-success-indices-at-subset*)

lemma *staged-query-prefix-candidate-pair-query-target-from-prefix-subset:*
staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table prefix prefix-state $\subseteq$ query-sample-space

unfolding *staged-query-prefix-candidate-pair-query-target-from-prefix-def*
by (rule *query-sampling-success-space-subset*)

lemma *staged-query-prefix-candidate-opening-query-target-subset-candidate-pair:*

assumes *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *comp-candidate:*

partial-composition-table-candidate composition-table

composition-openings

and *trace-low: trace-table-low-degree trace-table*

and *comp-low: composition-table-low-degree maxDegree composition-table*

and *not-all:*

$\neg$ *all-queries-consistent trace-table composition-table as*

shows

staged-query-prefix-candidate-opening-query-target trace-openings

composition-openings as i prefix prefix-state $\subseteq$

staged-query-prefix-candidate-pair-query-target trace-table

composition-table as prefix prefix-state

unfolding *staged-query-prefix-candidate-opening-query-target-def*

staged-query-prefix-candidate-pair-query-target-def

by (rule *partial-query-success-indices-at-subset-query-sampling-success-space*

[*OF trace-candidate comp-candidate trace-low comp-low not-all*])

lemma *staged-query-prefix-candidate-opening-query-target-from-prefix-subset-candidate-pair-from-prefix:*

assumes *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *comp-candidate:*

partial-composition-table-candidate composition-table

composition-openings

and *trace-low: trace-table-low-degree trace-table*

and *comp-low: composition-table-low-degree maxDegree composition-table*

and *not-all:*

$\neg$ *all-queries-consistent trace-table composition-table*

(*sqp-alphas prefix*)

shows

staged-query-prefix-candidate-opening-query-target-from-prefix

trace-openings composition-openings i prefix prefix-state $\subseteq$

staged-query-prefix-candidate-pair-query-target-from-prefix trace-table

composition-table prefix prefix-state

unfolding *staged-query-prefix-candidate-opening-query-target-from-prefix-def*

staged-query-prefix-candidate-pair-query-target-from-prefix-def

by (rule *partial-query-success-indices-at-subset-query-sampling-success-space*

[*OF trace-candidate comp-candidate trace-low comp-low not-all*])

lemma *checked-staged-query-prefix-dynamic-index-hit-mono:*

assumes *subset: $\bigwedge$ prefix prefix-state. A prefix prefix-state $\subseteq$*

B prefix prefix-state

and *hit:*

checked-staged-query-prefix-dynamic-index-hit A

```

    (Some (((prefix, prefix-state), raw), raw-state))
  shows
    checked-staged-query-prefix-dynamic-index-hit B
    (Some (((prefix, prefix-state), raw), raw-state))
  using subset hit
  unfolding checked-staged-query-prefix-dynamic-index-hit-def by auto

lemma partial-trace-table-candidate-single-round-from-all-rounds:
  assumes cand: partial-trace-table-candidate trace-table trace-openingss
    and i-bound: i < rounds
  shows
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openingss ! i])
  proof -
    have len-table: length trace-table = scale * clength
      using cand unfolding partial-trace-table-candidate-def by simp
    have agrees:
      table-agrees-with-authenticated-openings trace-table
      (scale * clength) (trace-openingss ! i)
      using cand i-bound unfolding partial-trace-table-candidate-def by blast
    show ?thesis
      by (rule partial-trace-table-candidate-single-roundI
        [OF i-bound len-table agrees])
  qed

lemma partial-composition-table-candidate-single-round-from-all-rounds:
  assumes cand:
    partial-composition-table-candidate composition-table
    composition-openingss
    and i-bound: i < rounds
  shows
    partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openingss ! i])
  proof -
    have len-table: length composition-table = scale * clength
      using cand unfolding partial-composition-table-candidate-def by simp
    have agrees:
      table-agrees-with-authenticated-openings composition-table
      (scale * clength) (composition-openingss ! i)
      using cand i-bound unfolding partial-composition-table-candidate-def
      by blast
    show ?thesis
      by (rule partial-composition-table-candidate-single-roundI
        [OF i-bound len-table agrees])
  qed

lemma staged-query-prefix-candidate-opening-query-target-single-roundI:
  assumes i-bound: i < rounds
    and consistent:

```

partial-query-openings-consistent trace-openings composition-openings
as idx
shows
idx $\in$
staged-query-prefix-candidate-opening-query-target
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
as i prefix prefix-state
unfolding *staged-query-prefix-candidate-opening-query-target-def*
by (rule *partial-query-success-indices-atI*,
rule *partial-query-round-consistent-single-roundI*
[OF *i-bound consistent*])

lemma *checked-staged-query-prefix-candidate-opening-target-single-round-hitI*:
assumes *i-bound: i < rounds*
and *consistent*:
partial-query-openings-consistent trace-openings composition-openings
as (index (to-nat raw))
shows
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
as i)
(Some (((prefix, prefix-state), raw), raw-state))

proof –
have *index (to-nat raw)* $\in$
staged-query-prefix-candidate-opening-query-target
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
as i prefix prefix-state
by (rule *staged-query-prefix-candidate-opening-query-target-single-roundI*
[OF *i-bound consistent*])
then show *?thesis*
unfolding *checked-staged-query-prefix-dynamic-index-hit-def* **by** *simp*
qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hitI*:
assumes *as-eq: as = staged-alphas data*
and *i-bound: i < rounds*
and *builder*:
Some (data, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive*:
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)

adversary-initial-state)
and *target-hit*:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target
trace-openings composition-openings as i)
(Some (((prefix, prefix-state), raw), raw-state))
shows
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
trace-openings composition-openings as i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
final-state))
proof –
have *body*:
 $\exists$ *prefix prefix-state raw raw-state.*
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) ∧
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) ∧
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target
trace-openings composition-openings as i)
(Some (((prefix, prefix-state), raw), raw-state))
by (*intro exI[of - prefix] exI[of - prefix-state]*
exI[of - raw] exI[of - raw-state])
(use builder prefix-receive target-hit in simp)
show *?thesis*
unfolding
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-def
using *as-eq i-bound body by simp*
qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hitI-from-raw*:
assumes *as-eq: as = staged-alphas data*
and *i-bound: i < rounds*
and *builder*:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive*:
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *consistent*:

```

    partial-query-openings-consistent trace-openings composition-openings
    as (index (to-nat raw))
  shows
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    as i
    (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
    final-state))
  proof -
  have target-hit:
    checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-opening-query-target
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    as i)
    (Some ((prefix, prefix-state), raw), raw-state))
  by (rule checked-staged-query-prefix-candidate-opening-target-single-round-hitI
    [OF i-bound consistent])
  show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hitI
    [OF as-eq i-bound builder prefix-receive target-hit])
qed

lemma checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hitI-from-prefix-receive:
  assumes i-bound: i < rounds
  and builder:
    Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
    adversary-initial-state)
  and prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
    adversary-initial-state)
  and consistent:
    partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
  shows
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    (staged-alphas data) i
    (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
    final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hitI-from-raw

```


[*OF* refl *i*-bound builder prefix-receive consistent])

definition *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix*
where

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix *A*
trace-openings composition-openings i out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ *False*
 | *Some* (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*), *result*),
 final-state) $\Rightarrow$
 i < *rounds* $\wedge$
 ($\exists$ *prefix prefix-state raw raw-state*.
 Some (*data*, *attacker-state*) $\in$
 set-dist
 (*execute* (*checked-staged-transcript-program* *A*)

adversary-initial-state) $\wedge$

Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) $\in$
 set-dist

 (*execute* (*checked-staged-query-prefix-receive-with-state* *A i*)

adversary-initial-state) $\wedge$

checked-staged-query-prefix-dynamic-index-hit

 (*staged-query-prefix-candidate-opening-query-target-from-prefix*
 trace-openings composition-openings i)

 (*Some* (((*prefix*, *prefix-state*), *raw*), *raw-state*))))

definition *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix*
where

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix *A*
trace-table composition-table i out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ *False*
 | *Some* (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*), *result*),
 final-state) $\Rightarrow$
 i < *rounds* $\wedge$
 ($\exists$ *prefix prefix-state raw raw-state*.
 Some (*data*, *attacker-state*) $\in$
 set-dist
 (*execute* (*checked-staged-transcript-program* *A*)

adversary-initial-state) $\wedge$

Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) $\in$
 set-dist

 (*execute* (*checked-staged-query-prefix-receive-with-state* *A i*)

adversary-initial-state) $\wedge$

checked-staged-query-prefix-dynamic-index-hit

 (*staged-query-prefix-candidate-pair-query-target-from-prefix*
 trace-table composition-table)

 (*Some* (((*prefix*, *prefix-state*), *raw*), *raw-state*))))

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixI*:

assumes *i-bound*: $i < \text{rounds}$
and *builder*:
Some (*data*, *attacker-state*) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive*:
Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *target-hit*:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i)
*(Some (((*prefix*, *prefix-state*), *raw*), *raw-state*))*
shows
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix A
trace-openings composition-openings i
*(Some (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*), *result*),*
final-state))
proof —
have *body*:
 $\exists$ *prefix prefix-state raw raw-state*.
Some (*data*, *attacker-state*) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) \wedge
Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) \wedge
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i)
*(Some (((*prefix*, *prefix-state*), *raw*), *raw-state*))*
by (*intro exI[of - prefix] exI[of - prefix-state]*
exI[of - raw] exI[of - raw-state])
(use builder prefix-receive target-hit in simp)
show *?thesis*
unfolding
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix-def
using *i-bound body by simp*
qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixE*:
assumes *hit*:
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix A
trace-openings composition-openings i

$(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}), \text{result}), \text{final-state}))$
obtains $\text{prefix prefix-state raw raw-state}$ **where**
 $i < \text{rounds}$
 $\text{Some } (\text{data}, \text{attacker-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$
 $\text{Some } (((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $\text{adversary-initial-state})$
 $\text{checked-staged-query-prefix-dynamic-index-hit}$
 $(\text{staged-query-prefix-candidate-opening-query-target-from-prefix}$
 $\text{trace-openings composition-openings } i)$
 $(\text{Some } (((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}))$
using hit
proof –
have body:
 $i < \text{rounds} \wedge$
 $(\exists \text{prefix prefix-state raw raw-state.}$
 $\text{Some } (\text{data}, \text{attacker-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \wedge$
 $\text{Some } (((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $\text{adversary-initial-state}) \wedge$
 $\text{checked-staged-query-prefix-dynamic-index-hit}$
 $(\text{staged-query-prefix-candidate-opening-query-target-from-prefix}$
 $\text{trace-openings composition-openings } i)$
 $(\text{Some } (((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state})))$
using hit
unfolding
 $\text{checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix-def}$
by simp
then obtain $\text{prefix prefix-state raw raw-state}$ **where**
 $i\text{-bound: } i < \text{rounds}$
and builder:
 $\text{Some } (\text{data}, \text{attacker-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$
and prefix-receive:
 $\text{Some } (((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $\text{adversary-initial-state})$

```

and target-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-opening-query-target-from-prefix
      trace-openings composition-openings i)
    (Some (((prefix, prefix-state), raw), raw-state))
by blast
show ?thesis
  by (rule that[OF i-bound builder prefix-receive target-hit])
qed

lemma checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixI:
assumes i-bound: i < rounds
and builder:
  Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
and prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
and target-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target-from-prefix
      trace-table composition-table)
    (Some (((prefix, prefix-state), raw), raw-state))
shows
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix A
    trace-table composition-table i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
      final-state)))
proof –
have body:
   $\exists$  prefix prefix-state raw raw-state.
    Some (data, attacker-state) ∈
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\wedge$ 
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)  $\wedge$ 
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table)
      (Some (((prefix, prefix-state), raw), raw-state))
by (intro exI[of - prefix] exI[of - prefix-state]
  exI[of - raw] exI[of - raw-state])

```

```

    (use builder prefix-receive target-hit in simp)
show ?thesis
  unfolding
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-def
  using i-bound body by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE:
  assumes hit:
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix A
    trace-table composition-table i
    (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  obtains prefix prefix-state raw raw-state where
    i < rounds
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table)
      (Some (((prefix, prefix-state), raw), raw-state))
  using hit
proof —
  have body:
    i < rounds ∧
    (∃ prefix prefix-state raw raw-state.
      Some (data, attacker-state) ∈
        set-dist
          (execute (checked-staged-transcript-program A)
            adversary-initial-state) ∧
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state) ∧
      checked-staged-query-prefix-dynamic-index-hit
        (staged-query-prefix-candidate-pair-query-target-from-prefix
          trace-table composition-table)
        (Some (((prefix, prefix-state), raw), raw-state)))
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-def
  by simp
then obtain prefix prefix-state raw raw-state where

```

i-bound: $i < \text{rounds}$
and *builder:*
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive:*
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *target-hit:*
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state))
by *blast*
show *?thesis*
by *(rule that[OF i-bound builder prefix-receive target-hit])*
qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix-imp-candidate-pair-q*

assumes *hit:*

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix A
trace-openings composition-openings i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *comp-candidate:*

partial-composition-table-candidate composition-table
composition-openings

and *trace-low: trace-table-low-degree trace-table*

and *comp-low: composition-table-low-degree maxDegree composition-table*

and *not-all:*

$\bigwedge \text{prefix prefix-state.}$

Some ((prefix, prefix-state), prefix-state) ∈
set-dist

(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) ⇒

$\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)

shows

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix A
trace-table composition-table i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))

proof —

from *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixE*

$[OF \text{ hit}]$
obtain *prefix prefix-state raw raw-state* **where**
i-bound: $i < \text{rounds}$
and *builder:*
Some (data, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive:*
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *target-hit:*
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i)
(Some (((prefix, prefix-state), raw), raw-state))
by *blast*
have *prefix-support:*
Some ((prefix, prefix-state), prefix-state) $\in$
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state)
by *(rule checked-staged-query-prefix-receive-with-state-prefix-support*
 $[OF \text{ prefix-receive}]$
have *pair-target-hit:*
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table)
(Some (((prefix, prefix-state), raw), raw-state))
proof –
have *subset:*
staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i prefix prefix-state $\subseteq$
staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table prefix prefix-state
by *(rule*
staged-query-prefix-candidate-opening-query-target-from-prefix-subset-candidate-pair-from-prefix
 $[OF \text{ trace-candidate comp-candidate trace-low comp-low}$
 $\text{not-all}[OF \text{ prefix-support}]]$
show *?thesis*
using *target-hit subset*
unfolding *checked-staged-query-prefix-dynamic-index-hit-def* **by** *auto*
qed
show *?thesis*
by *(rule*
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixI
 $[OF \text{ i-bound builder prefix-receive pair-target-hit}]$

qed

lemma *checked-staged-query-prefix-candidate-opening-target-from-prefix-single-round-hitI:*

assumes *i-bound: $i < \text{rounds}$*

and *consistent:*

partial-query-openings-consistent trace-openings composition-openings
(sqp-alphas prefix) (index (to-nat raw))

shows

checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i)
(Some (((prefix, prefix-state), raw), raw-state))

proof –

have *index (to-nat raw) ∈*

staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i prefix prefix-state

unfolding *staged-query-prefix-candidate-opening-query-target-from-prefix-def*

by (*rule partial-query-success-indices-atI,*
rule partial-query-round-consistent-single-roundI
[OF i-bound consistent])

then show *?thesis*

unfolding *checked-staged-query-prefix-dynamic-index-hit-def* **by** *simp*

qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixI-from-prefix-receive:*

assumes *i-bound: $i < \text{rounds}$*

and *builder:*

Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)

and *prefix-receive:*

Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)

and *alphas-eq: sqp-alphas prefix = staged-alphas data*

and *consistent:*

partial-query-openings-consistent trace-openings composition-openings
(staged-alphas data) (index (to-nat raw))

shows

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix A
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i


```

    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
            final-state))
proof –
  have consistent-prefix:
    partial-query-openings-consistent trace-openings composition-openings
      (sqp-alphas prefix) (index (to-nat raw))
  using consistent alphas-eq by simp
  have target-hit:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-opening-query-target-from-prefix
        ((replicate rounds []) [i := trace-openings])
        ((replicate rounds []) [i := composition-openings])
        i)
      (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
    checked-staged-query-prefix-candidate-opening-target-from-prefix-single-round-hitI
    [OF i-bound consistent-prefix])
  show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixI
    [OF i-bound builder prefix-receive target-hit])
qed

```

lemma checked-staged-query-prefix-candidate-pair-target-hitI:

assumes hit:

index (to-nat raw) ∈

query-sampling-success-space trace-table composition-table as

shows

checked-staged-query-prefix-dynamic-index-hit

(staged-query-prefix-candidate-pair-query-target trace-table

composition-table as)

(Some (((prefix, prefix-state), raw), raw-state))

using hit

unfolding checked-staged-query-prefix-dynamic-index-hit-def

staged-query-prefix-candidate-pair-query-target-def

by simp

lemma checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hitI:

assumes i-bound: $i < \text{rounds}$

and builder:

Some (data, attacker-state) ∈

set-dist

(execute (checked-staged-transcript-program A)

adversary-initial-state)

and prefix-receive:

Some (((prefix, prefix-state), raw), raw-state) ∈

set-dist

(execute (checked-staged-query-prefix-receive-with-state A i)

adversary-initial-state)

and *target-hit*:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target trace-table
composition-table (staged-alphas data))
(Some (((prefix, prefix-state), raw), raw-state))
shows
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit A
trace-table composition-table
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
final-state))
proof –
have *body*:
 $\exists i$ *prefix prefix-state raw raw-state.*
i < rounds $\wedge$
Some (data, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) $\wedge$
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) $\wedge$
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target trace-table
composition-table (staged-alphas data))
(Some (((prefix, prefix-state), raw), raw-state))
by (*intro exI[of - i] exI[of - prefix] exI[of - prefix-state]*
exI[of - raw] exI[of - raw-state])
(use i-bound builder prefix-receive target-hit in simp)
show *?thesis*
unfolding
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-def
using *body by simp*
qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-imp-candidate-pair-query-hit*:
assumes *hit*:
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
trace-openings composition-openings as i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))
and *trace-candidate*:
partial-trace-table-candidate trace-table trace-openings
and *comp-candidate*:
partial-composition-table-candidate composition-table
composition-openings
and *trace-low*: *trace-table-low-degree trace-table*
and *comp-low*: *composition-table-low-degree maxDegree composition-table*
and *not-all*:

$\neg$ *all-queries-consistent trace-table composition-table as*
shows
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit A
trace-table composition-table
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)))
proof –
obtain *prefix prefix-state raw raw-state* **where**
as-eq: as = staged-alphas data
and *i-bound: i < rounds*
and *builder:*
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive:*
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *target-hit:*
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target
trace-openings composition-openings as i)
(Some (((prefix, prefix-state), raw), raw-state)))
using *hit*
unfolding
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-def
by *auto*
have *pair-target-hit:*
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target trace-table
composition-table as)
(Some (((prefix, prefix-state), raw), raw-state)))
by (rule *checked-staged-query-prefix-dynamic-index-hit-mono*
[*OF - target-hit*])
(rule
staged-query-prefix-candidate-opening-query-target-subset-candidate-pair
[*OF trace-candidate comp-candidate trace-low comp-low not-all*])
have *pair-target-hit':*
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target trace-table
composition-table (staged-alphas data))
(Some (((prefix, prefix-state), raw), raw-state)))
using *pair-target-hit as-eq* **by** *simp*
show *?thesis*
by (rule *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hitI*
[*OF i-bound builder prefix-receive pair-target-hit'*])
qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hitI-from-raw:*

assumes *i-bound*: $i < \text{rounds}$

and *builder*:

Some (*data*, *attacker-state*) $\in$
set-dist
 (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state)

and *prefix-receive*:

Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) $\in$
set-dist
 (*execute* (*checked-staged-query-prefix-receive-with-state* *A i*)
adversary-initial-state)

and *target-raw*:

index (*to-nat raw*) $\in$
query-sampling-success-space trace-table composition-table
 (*staged-alphas data*)

shows

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit A
trace-table composition-table
 (*Some* (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*), *result*),
final-state)))

proof —

have *target-hit*:

checked-staged-query-prefix-dynamic-index-hit
 (*staged-query-prefix-candidate-pair-query-target trace-table*
composition-table (staged-alphas data))
 (*Some* (((*prefix*, *prefix-state*), *raw*), *raw-state*)))

by (*rule checked-staged-query-prefix-candidate-pair-target-hitI*
 [*OF target-raw*])

show *?thesis*

by (*rule*
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hitI
 [*OF i-bound builder prefix-receive target-hit*])

qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hitE:*

assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit A
trace-table composition-table
 (*Some* (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*), *result*),
final-state)))

obtains *i prefix prefix-state raw raw-state* **where**

i < rounds

Some (*data*, *attacker-state*) $\in$
set-dist
 (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state)

Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) $\in$

```

    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-candidate-pair-query-target trace-table
    composition-table (staged-alphas data))
  (Some (((prefix, prefix-state), raw), raw-state))
using hit
proof –
have body:
  ∃ i prefix prefix-state raw raw-state.
  i < rounds ∧
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state) ∧
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state) ∧
checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-candidate-pair-query-target trace-table
    composition-table (staged-alphas data))
  (Some (((prefix, prefix-state), raw), raw-state))
using hit
unfolding
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-def
by simp
then obtain i prefix prefix-state raw raw-state where
  i-bound: i < rounds
and builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
and target-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target trace-table
      composition-table (staged-alphas data))
    (Some (((prefix, prefix-state), raw), raw-state))
by blast
show ?thesis
by (rule that[OF i-bound builder prefix-receive target-hit])
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-rawE*:

assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit A
trace-table composition-table
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
final-state)))

obtains *i prefix prefix-state raw raw-state* **where**

i < rounds

Some (data, attacker-state) ∈

set-dist

(execute (checked-staged-transcript-program A)
adversary-initial-state)

Some (((prefix, prefix-state), raw), raw-state) ∈

set-dist

(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)

index (to-nat raw) ∈

query-sampling-success-space trace-table composition-table
(staged-alphas data)

proof –

from *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hitE*

[OF hit]

obtain *i prefix prefix-state raw raw-state* **where**

i-bound: i < rounds

and *builder*:

Some (data, attacker-state) ∈

set-dist

(execute (checked-staged-transcript-program A)
adversary-initial-state)

and *prefix-receive*:

Some (((prefix, prefix-state), raw), raw-state) ∈

set-dist

(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)

and *target-hit*:

checked-staged-query-prefix-dynamic-index-hit

(staged-query-prefix-candidate-pair-query-target trace-table
composition-table (staged-alphas data))

(Some (((prefix, prefix-state), raw), raw-state))

by *blast*

have *raw-hit*:

index (to-nat raw) ∈

query-sampling-success-space trace-table composition-table
(staged-alphas data)

using *target-hit*

unfolding *checked-staged-query-prefix-dynamic-index-hit-def*

staged-query-prefix-candidate-pair-query-target-def

by *simp*

show ?thesis
by (rule that[OF i-bound builder prefix-receive raw-hit])
qed

lemma staged-query-prefix-candidate-pair-query-target-fraction-bound:
assumes trace-low: trace-table-low-degree trace-table
and comp-low: composition-table-low-degree maxDegree composition-table
and not-all:
 $\neg$ all-queries-consistent trace-table composition-table as
shows
nnreal
(card
(staged-query-prefix-candidate-pair-query-target trace-table
composition-table as prefix prefix-state)) /
nnreal (card query-sample-space) $\leq$ query-error-bound
unfolding staged-query-prefix-candidate-pair-query-target-def
by (rule query-sampling-success-space-fraction-bound-query-sample-space)

lemma staged-query-prefix-candidate-opening-query-target-fraction-bound:
assumes trace-candidate:
partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
partial-composition-table-candidate composition-table
composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-low: composition-table-low-degree maxDegree composition-table
and not-all:
 $\neg$ all-queries-consistent trace-table composition-table as
shows
nnreal
(card
(staged-query-prefix-candidate-opening-query-target trace-openings
composition-openings as i prefix prefix-state)) /
nnreal (card query-sample-space) $\leq$ query-error-bound
unfolding staged-query-prefix-candidate-opening-query-target-def
by (rule partial-query-success-indices-at-fraction-bound-if-candidate-low-degree
[OF trace-candidate comp-candidate trace-low comp-low not-all])

lemma staged-query-prefix-candidate-opening-query-target-from-prefix-fraction-bound:
assumes trace-candidate:
partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
partial-composition-table-candidate composition-table
composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-low: composition-table-low-degree maxDegree composition-table
and not-all:
 $\neg$ all-queries-consistent trace-table composition-table
(sqp-alphas prefix)

shows
nnreal
 (*card*
 (*staged-query-prefix-candidate-opening-query-target-from-prefix*
 trace-openings composition-openings i prefix prefix-state)) /
nnreal (card query-sample-space) ≤ query-error-bound
unfolding *staged-query-prefix-candidate-opening-query-target-from-prefix-def*
by (rule *partial-query-success-indices-at-fraction-bound-if-candidate-low-degree*
 [OF *trace-candidate comp-candidate trace-low comp-low not-all*])

lemma *staged-query-prefix-candidate-pair-query-target-from-prefix-fraction-bound:*

assumes *trace-low: trace-table-low-degree trace-table*
and *comp-low: composition-table-low-degree maxDegree composition-table*
and *not-all:*
 ¬ *all-queries-consistent trace-table composition-table*
 (*sqp-alphas prefix*)

shows
nnreal
 (*card*
 (*staged-query-prefix-candidate-pair-query-target-from-prefix*
 trace-table composition-table prefix prefix-state)) /
nnreal (card query-sample-space) ≤ query-error-bound
unfolding *staged-query-prefix-candidate-pair-query-target-from-prefix-def*
by (rule *query-sampling-success-space-fraction-bound-query-sample-space*)

lemma *checked-staged-query-prefix-candidate-opening-target-prehit-bound:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i ≤ rounds*

shows
wp-event (checked-staged-query-prefix-receive-with-state A i)
 (*checked-staged-query-prefix-dynamic-index-prehit*
 (*staged-query-prefix-candidate-opening-query-target*
 trace-openings composition-openings as i))
adversary-initial-state ≤
staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*)
proof (rule *checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program*)
show *HashMap adversary-initial-state = fmempty*
by (simp add: *adversary-initial-state-def*)
show *hash-relation-program*
 (*checked-staged-query-prefix-dynamic-index-prehit-relation A i*
 adversary-initial-state
 (*staged-query-prefix-candidate-opening-query-target*
 trace-openings composition-openings as i))
size (staged-query-search-queries budgets i + 1)
 (*checked-staged-query-prefix-receive-with-state A i*)
by (rule *hash-relation-program-checked-staged-query-prefix-receive-with-state*
 [OF *wf controlled i-bound*])

(rule
checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)

qed

lemma *checked-staged-query-prefix-candidate-opening-target-hit-bound*:
assumes *raw-bound*: *query-index-raw-preimage-bound*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: $i \leq \text{rounds}$
and *trace-candidate*:
partial-trace-table-candidate trace-table trace-openings
and *comp-candidate*:
partial-composition-table-candidate composition-table
composition-openings
and *trace-low*: *trace-table-low-degree trace-table*
and *comp-low*: *composition-table-low-degree maxDegree composition-table*
and *not-all*:
 $\neg \text{all-queries-consistent trace-table composition-table as}$
shows
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target
trace-openings composition-openings as i))
adversary-initial-state $\leq$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) + query-error-bound

proof –
have *hit-bound*:
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target
trace-openings composition-openings as i))
adversary-initial-state $\leq$
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-prehit
(staged-query-prefix-candidate-opening-query-target
trace-openings composition-openings as i))
adversary-initial-state + query-error-bound

proof (rule
checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
 $[OF \text{ raw-bound}]$)
fix *prefix prefix-state*
assume *support*:
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state)
have *subset*:
staged-query-prefix-candidate-opening-query-target trace-openings

```

    composition-openings as i prefix prefix-state  $\subseteq$ 
    query-sample-space
  by (rule staged-query-prefix-candidate-opening-query-target-subset)
have frac:
  nnreal
  (card
    (staged-query-prefix-candidate-opening-query-target
      trace-openings composition-openings as i prefix prefix-state)) /
  nnreal (card query-sample-space)  $\leq$  query-error-bound
  by (rule staged-query-prefix-candidate-opening-query-target-fraction-bound
    [OF trace-candidate comp-candidate trace-low comp-low not-all])
show
  staged-query-prefix-candidate-opening-query-target trace-openings
  composition-openings as i prefix prefix-state  $\subseteq$ 
  query-sample-space  $\wedge$ 
  nnreal
  (card
    (staged-query-prefix-candidate-opening-query-target
      trace-openings composition-openings as i prefix prefix-state)) /
  nnreal (card query-sample-space)  $\leq$  query-error-bound
  using subset frac by blast
qed
have prehit-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
  (checked-staged-query-prefix-dynamic-index-prehit
    (staged-query-prefix-candidate-opening-query-target
      trace-openings composition-openings as i))
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1)
  by (rule checked-staged-query-prefix-candidate-opening-target-prehit-bound
    [OF wf controlled i-bound])
have tail:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
  (checked-staged-query-prefix-dynamic-index-prehit
    (staged-query-prefix-candidate-opening-query-target
      trace-openings composition-openings as i))
  adversary-initial-state + query-error-bound  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) + query-error-bound
  by (intro add-mono prehit-bound order-refl)
show ?thesis
  by (rule order-trans[OF hit-bound tail])
qed

lemma checked-staged-query-prefix-candidate-opening-target-from-prefix-prehit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i  $\leq$  rounds

```

shows

wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-prehit
(staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i))
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1)

proof (rule checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program)

show *HashMap adversary-initial-state = fmempty*

by (simp add: adversary-initial-state-def)

show *hash-relation-program*

(checked-staged-query-prefix-dynamic-index-prehit-relation A i
adversary-initial-state
(staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i))
size (staged-query-search-queries budgets i + 1)
(checked-staged-query-prefix-receive-with-state A i)

by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state
 $[OF\ wf\ controlled\ i-bound]$)

(rule
checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)

qed

lemma *checked-staged-query-prefix-candidate-opening-target-from-prefix-hit-bound:*

assumes *raw-bound: query-index-raw-preimage-bound*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i ≤ rounds*

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *comp-candidate:*

partial-composition-table-candidate composition-table
composition-openings

and *trace-low: trace-table-low-degree trace-table*

and *comp-low: composition-table-low-degree maxDegree composition-table*

and *not-all:*

$\bigwedge prefix\ prefix-state.$

Some ((prefix, prefix-state), prefix-state) ∈
set-dist

(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) ⇒

$\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)

shows

wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i))

$adversary-initial-state \leq$
 $staged-phase-relation-error\ size$
 $(staged-query-search-queries\ budgets\ i + 1) + query-error-bound$

proof –

have *hit-bound*:

$wp-event\ (checked-staged-query-prefix-receive-with-state\ A\ i)$
 $(checked-staged-query-prefix-dynamic-index-hit$
 $(staged-query-prefix-candidate-opening-query-target-from-prefix$
 $trace-openings\ composition-openings\ i))$
 $adversary-initial-state \leq$
 $wp-event\ (checked-staged-query-prefix-receive-with-state\ A\ i)$
 $(checked-staged-query-prefix-dynamic-index-prehit$
 $(staged-query-prefix-candidate-opening-query-target-from-prefix$
 $trace-openings\ composition-openings\ i))$
 $adversary-initial-state + query-error-bound$

proof (rule

$checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error$
 $[OF\ raw-bound])$

fix *prefix prefix-state*

assume *support*:

$Some\ ((prefix,\ prefix-state),\ prefix-state) \in$
 $set-dist$
 $(execute\ (checked-staged-query-prefix-with-state\ A\ i)$
 $adversary-initial-state)$

have *subset*:

$staged-query-prefix-candidate-opening-query-target-from-prefix$
 $trace-openings\ composition-openings\ i\ prefix\ prefix-state \subseteq$
 $query-sample-space$

by (rule *staged-query-prefix-candidate-opening-query-target-from-prefix-subset*)

have *frac*:

$nnreal$
 $(card$
 $(staged-query-prefix-candidate-opening-query-target-from-prefix$
 $trace-openings\ composition-openings\ i\ prefix\ prefix-state)) /$
 $nnreal\ (card\ query-sample-space) \leq query-error-bound$

by (rule

$staged-query-prefix-candidate-opening-query-target-from-prefix-fraction-bound$
 $[OF\ trace-candidate\ comp-candidate\ trace-low\ comp-low\ not-all[OF\ support]])$

show

$staged-query-prefix-candidate-opening-query-target-from-prefix$
 $trace-openings\ composition-openings\ i\ prefix\ prefix-state \subseteq$
 $query-sample-space \wedge$
 $nnreal$
 $(card$
 $(staged-query-prefix-candidate-opening-query-target-from-prefix$
 $trace-openings\ composition-openings\ i\ prefix\ prefix-state)) /$
 $nnreal\ (card\ query-sample-space) \leq query-error-bound$

using *subset frac by blast*

qed

```

have prehit-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit
      (staged-query-prefix-candidate-opening-query-target-from-prefix
        trace-openings composition-openings i))
  adversary-initial-state ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1)
by (rule
  checked-staged-query-prefix-candidate-opening-target-from-prefix-prehit-bound
  [OF wf controlled i-bound])
have tail:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit
      (staged-query-prefix-candidate-opening-query-target-from-prefix
        trace-openings composition-openings i))
  adversary-initial-state + query-error-bound ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
by (intro add-mono prehit-bound order-refl)
show ?thesis
by (rule order-trans[OF hit-bound tail])
qed

lemma checked-staged-query-prefix-candidate-pair-target-prehit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i ≤ rounds
  shows
    wp-event (checked-staged-query-prefix-receive-with-state A i)
      (checked-staged-query-prefix-dynamic-index-prehit
        (staged-query-prefix-candidate-pair-query-target trace-table
          composition-table as))
    adversary-initial-state ≤
    staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1)
proof (rule checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program)
show HashMap adversary-initial-state = fmempty
by (simp add: adversary-initial-state-def)
show hash-relation-program
  (checked-staged-query-prefix-dynamic-index-prehit-relation A i
    adversary-initial-state
    (staged-query-prefix-candidate-pair-query-target trace-table
      composition-table as))
  size (staged-query-search-queries budgets i + 1)
  (checked-staged-query-prefix-receive-with-state A i)
by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state
  [OF wf controlled i-bound])
  (rule

```

checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)

qed

lemma *checked-staged-query-prefix-candidate-pair-target-hit-bound:*
assumes *raw-bound: query-index-raw-preimage-bound*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i \leq \text{rounds}$*
and *trace-low: trace-table-low-degree trace-table*
and *comp-low: composition-table-low-degree maxDegree composition-table*
and *not-all:*
 $\neg$ *all-queries-consistent trace-table composition-table as*
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $(\text{staged-query-prefix-candidate-pair-query-target trace-table}$
 $\text{composition-table as}))$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) + \text{query-error-bound}$

proof –
have *hit-bound:*
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $(\text{staged-query-prefix-candidate-pair-query-target trace-table}$
 $\text{composition-table as}))$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-prehit}$
 $(\text{staged-query-prefix-candidate-pair-query-target trace-table}$
 $\text{composition-table as}))$
 $\text{adversary-initial-state} + \text{query-error-bound}$

proof (rule
 $\text{checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error}$
 $[\text{OF raw-bound}]$)
fix *prefix prefix-state*
assume *support:*
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state})$

have *subset:*
 $\text{staged-query-prefix-candidate-pair-query-target trace-table}$
 $\text{composition-table as prefix prefix-state} \subseteq$
 $\text{query-sample-space}$
by (rule *staged-query-prefix-candidate-pair-query-target-subset*)

have *frac:*
 nnreal
 $(\text{card}$

```

      (staged-query-prefix-candidate-pair-query-target trace-table
        composition-table as prefix prefix-state)) /
      nnreal (card query-sample-space) ≤ query-error-bound
  by (rule staged-query-prefix-candidate-pair-query-target-fraction-bound
      [OF trace-low comp-low not-all])
show
  staged-query-prefix-candidate-pair-query-target trace-table
    composition-table as prefix prefix-state ⊆
    query-sample-space ∧
  nnreal
    (card
      (staged-query-prefix-candidate-pair-query-target trace-table
        composition-table as prefix prefix-state)) /
    nnreal (card query-sample-space) ≤ query-error-bound
  using subset frac by blast
qed
have prehit-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit
      (staged-query-prefix-candidate-pair-query-target trace-table
        composition-table as))
  adversary-initial-state ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1)
  by (rule checked-staged-query-prefix-candidate-pair-target-prehit-bound
      [OF wf controlled i-bound])
have tail:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit
      (staged-query-prefix-candidate-pair-query-target trace-table
        composition-table as))
  adversary-initial-state + query-error-bound ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
  by (intro add-mono prehit-bound order-refl)
show ?thesis
  by (rule order-trans[OF hit-bound tail])
qed

lemma checked-staged-query-prefix-candidate-pair-target-from-prefix-prehit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i ≤ rounds
  shows
    wp-event (checked-staged-query-prefix-receive-with-state A i)
      (checked-staged-query-prefix-dynamic-index-prehit
        (staged-query-prefix-candidate-pair-query-target-from-prefix
          trace-table composition-table))
    adversary-initial-state ≤

```

staged-phase-relation-error size
(staged-query-search-queries budgets $i + 1$)
proof (rule checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program)
show *HashMap adversary-initial-state = fmempty*
by (simp add: adversary-initial-state-def)
show *hash-relation-program*
(checked-staged-query-prefix-dynamic-index-prehit-relation A i
adversary-initial-state
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table))
size (staged-query-search-queries budgets $i + 1$)
(checked-staged-query-prefix-receive-with-state A i)
by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state
[*OF wf controlled i-bound*])
(rule
checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
qed

lemma *checked-staged-query-prefix-candidate-pair-target-from-prefix-hit-bound:*

assumes *raw-bound: query-index-raw-preimage-bound*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i \leq \text{rounds}$*
and *trace-low: trace-table-low-degree trace-table*
and *comp-low: composition-table-low-degree maxDegree composition-table*
and *not-all:*
 $\bigwedge \text{prefix prefix-state.}$
Some ((prefix, prefix-state), prefix-state) $\in$
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) $\implies$
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)

shows

wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table))
adversary-initial-state $\leq$
staged-phase-relation-error size
(staged-query-search-queries budgets $i + 1$) + query-error-bound

proof –

have *hit-bound:*

wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table))
adversary-initial-state $\leq$
wp-event (checked-staged-query-prefix-receive-with-state A i)


```

    (checked-staged-query-prefix-dynamic-index-prehit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table))
    adversary-initial-state + query-error-bound
proof (rule
  checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
  [OF raw-bound])
fix prefix prefix-state
assume support:
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state)
have subset:
  staged-query-prefix-candidate-pair-query-target-from-prefix
    trace-table composition-table prefix prefix-state ⊆
    query-sample-space
by (rule staged-query-prefix-candidate-pair-query-target-from-prefix-subset)
have frac:
  nnreal
    (card
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table prefix prefix-state)) /
    nnreal (card query-sample-space) ≤ query-error-bound
by (rule
  staged-query-prefix-candidate-pair-query-target-from-prefix-fraction-bound)
  (use trace-low comp-low not-all[OF support] in simp-all)
show
  staged-query-prefix-candidate-pair-query-target-from-prefix
    trace-table composition-table prefix prefix-state ⊆
    query-sample-space ∧
  nnreal
    (card
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table prefix prefix-state)) /
    nnreal (card query-sample-space) ≤ query-error-bound
using subset frac by blast
qed
have prehit-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table))
  adversary-initial-state ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1)
by (rule
  checked-staged-query-prefix-candidate-pair-target-from-prefix-prehit-bound
  [OF wf controlled i-bound])

```

have *tail*:
wp-event (*checked-staged-query-prefix-receive-with-state* *A* *i*)
 (*checked-staged-query-prefix-dynamic-index-prehit*
 (*staged-query-prefix-candidate-pair-query-target-from-prefix*
 trace-table composition-table))
 adversary-initial-state + *query-error-bound* ≤
 staged-phase-relation-error size
 (*staged-query-search-queries* *budgets* *i* + 1) + *query-error-bound*
by (*intro add-mono prehit-bound order-refl*)
show ?thesis
 by (*rule order-trans*[*OF hit-bound tail*])
qed

lemma *checked-staged-query-prefix-candidate-pair-target-from-prefix-hit-bound-from-budgets*:
assumes *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *i-bound*: *i* < *rounds*
and *trace-low*: *trace-table-low-degree* *trace-table*
and *comp-low*: *composition-table-low-degree* *maxDegree* *composition-table*
and *not-all*:
 $\bigwedge \text{prefix } \text{prefix-state.}$
 Some ((*prefix*, *prefix-state*), *prefix-state*) ∈
 set-dist
 (*execute* (*checked-staged-query-prefix-with-state* *A* *i*)
 adversary-initial-state) ⇒
 ¬ *all-queries-consistent* *trace-table composition-table*
 (*sqp-alphas* *prefix*)
shows
wp-event (*checked-staged-query-prefix-receive-with-state* *A* *i*)
 (*checked-staged-query-prefix-dynamic-index-hit*
 (*staged-query-prefix-candidate-pair-query-target-from-prefix*
 trace-table composition-table))
 adversary-initial-state ≤
 staged-phase-relation-error size
 (*staged-query-search-queries* *budgets* *i* + 1) + *query-error-bound*
proof –
have *raw-bound*: *query-index-raw-preimage-bound*
by (*rule query-index-raw-preimage-bound-from-sampler-wellformed*)
show ?thesis
by (*rule*
 checked-staged-query-prefix-candidate-pair-target-from-prefix-hit-bound
 [*OF raw-bound wf controlled - trace-low comp-low not-all*])
 (*use i-bound in simp*)
qed

end

end

theory *Staged-Security-Experiment-Composition-Query-Residual*

imports

Staged-Security-Experiment-Composition-Partial

Soundness-Staged-Partial-Query-Branch

Soundness-Staged-Candidate-Query

begin

Composition-specific residual split after the alpha-prequery accounting.

The broad branch-subtree witness event is not rare: honest verifier-side Merkle path re-computation can produce such witnesses. This layer therefore splits the actual remaining composition drift event itself into a part covered by the sampled partial-opening query event and a residual part that is still to be discharged by the Merkle/candidate proof.

context *soundness*

begin

The existing committed query-prefix targets are based on *query-sampling-success-space*. They are suitable for ordinary query inconsistency, but not directly for composition randomization witnesses: *composition-bad-context* already requires all sampled queries to be consistent. Hence the query-sampling success space of the actual composition witness is empty, and the composition-drift query case needs a narrower candidate-indexed target rather than the global partial-opening fixed cover.

lemma *composition-bad-context-query-sampling-success-space-empty:*

assumes

composition-bad-context s out trace-table composition-table as query-idxs f

shows *query-sampling-success-space trace-table composition-table as = {}*

proof –

have *all-queries: all-queries-consistent trace-table composition-table as*

using *assms unfolding composition-bad-context-def* **by** *blast*

then show *?thesis*

unfolding *query-sampling-success-space-def* **by** *simp*

qed

lemma *composition-randomization-bad-witness-query-sampling-success-space-empty:*

assumes *composition-randomization-bad s out*

obtains *trace-table composition-table as query-idxs f* **where**

composition-bad-context s out trace-table composition-table as query-idxs f

common-denominator-degree-bounds f as

random-combination-common-denominator-hides-violations f as

query-sampling-success-space trace-table composition-table as = {}

using *assms*

unfolding *composition-randomization-bad-def*

by *(metis composition-bad-context-query-sampling-success-space-empty)*

lemma *composition-trace-bad-alpha-spaceE:*

assumes *bad: as ∈ composition-trace-bad-alpha-space trace-table*

obtains *f* **where**

f = low-degree-trace-witness trace-table

```

    degree f < clength
    trace-table = map (poly f) eval-domain
    violated-constraints f ≠ {}
    as ∈ common-denominator-hiding-alpha-space f
    common-denominator-degree-bounds f as
    trace-table-low-degree trace-table
proof –
  let ?f = low-degree-trace-witness trace-table
  have deg: degree ?f < clength
  and trace-table: trace-table = map (poly ?f) eval-domain
  and violated: violated-constraints ?f ≠ {}
  and alpha: as ∈ common-denominator-hiding-alpha-space ?f
  and degree-bounds: common-denominator-degree-bounds ?f as
  using bad
  unfolding composition-trace-bad-alpha-space-def Let-def
  by (auto split: if-splits)
  have trace-low: trace-table-low-degree trace-table
  unfolding trace-table-low-degree-def
  using deg trace-table by blast
  show ?thesis
  by (rule that[OF refl deg trace-table violated alpha degree-bounds
    trace-low])
qed

lemma composition-trace-bad-alpha-space-trace-low-degree:
  assumes as ∈ composition-trace-bad-alpha-space trace-table
  shows trace-table-low-degree trace-table
  using assms by (elim composition-trace-bad-alpha-spaceE)

definition checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
  :: ((((((f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool
where
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    out ⇔
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
    out ∧
  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    out

definition checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
  :: ((((((f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool
where
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit

```

$out \longleftrightarrow$
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
 $out \wedge$
 $\neg$ *checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit*
 out

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-imp-drift:*

assumes

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
 out

shows

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
 out

using *assms*

unfolding

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
by *simp*

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-imp-not-query-*

assumes

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
 out

shows

$\neg$ *checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit*
 out

using *assms*

unfolding

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
by *simp*

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-imp-randomiza*

assumes

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
 out

shows

checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-randomization-bad out

using *assms*

unfolding

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def

by *simp*

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-imp-partial-hea*

assumes

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
 out

shows

checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift

```

      composition-trace-bad-alpha-space out
using assms
unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
by simp

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-imp-partial-head
assumes
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
  out
shows
  checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
  composition-trace-bad-alpha-space out
using assms
unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def
by simp

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-imp-not-prefix-head
assumes
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
  out
shows
  ¬ checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  composition-trace-bad-alpha-space out
using assms
unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def
by simp

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hitE:
assumes hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
  (Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state))
obtains
  alpha-header-list-set-hit
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (alpha-header-supported-partial-union-bad-sets
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    composition-trace-bad-alpha-space)
  (Some (result, final-state))

```

```

    ⊢ alpha-vector-prequeried-in-state prefix-state
      (staged-alphas data) (length spec)
    ⊢ checked-staged-security-with-actual-alpha-prefix-bad-set-hit
      composition-trace-bad-alpha-space
      (Some ((((prefix, prefix-state), data), attacker-state), result),
        final-state))
    ⊢ checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
      (Some ((((prefix, prefix-state), data), attacker-state), result),
        final-state))
    checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-randomization-bad
      (Some ((((prefix, prefix-state), data), attacker-state), result),
        final-state))
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def
    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-def
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-def
    Let-def
  by auto

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-imp-branch-tree
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    out ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
      out
  shows
    checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
      out
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-imp-branch-tree-on-support
      [OF wf controlled support])
    (rule
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-imp-drift
        [OF hit])

lemma checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support:
  assumes support:
    Some ((((prefix, prefix-state), data), attacker-state), result),

```

```

    final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  shows
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  proof -
    let ?project =
      λx. ((snd (fst (fst x)), snd (fst x)), snd x)
  have projected:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
            A >> (λx. return (?project x)))
          adversary-initial-state)
  apply (rule set-dist-bindI[OF support])
  apply simp
  done
  then show ?thesis
  using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
  by simp
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-query-partial-opening-hit*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

```

  Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)

```

and *bad:*

```

  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-randomization-bad
  (Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state))

```

shows

```

  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit

```



```

      (Some (((((prefix, prefix-state), data), attacker-state), result),
              final-state))
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
                 adversary-initial-state)
  by (rule
      checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
      [OF support])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  obtain builder verifier where
    builder:
      Some (data, attacker-state) ∈
        set-dist
          (execute (checked-staged-transcript-program A)
                   adversary-initial-state)
    and verifier:
      Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
  have staged-header:
    verifier-header-transcript ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
      (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled builder])
  from bad obtain trace-table composition-table as query-idxs f where
    ctx:
      composition-bad-context ?s (Some (result, final-state))
      trace-table composition-table as query-idxs f
  unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    composition-randomization-bad-def Let-def
  by auto
  have bound:
    accepted-with-bound-tables ?s (Some (result, final-state))
      trace-table composition-table as query-idxs
  using ctx unfolding composition-bad-context-def by blast
  have all-queries:

```

```

all-queries-consistent trace-table composition-table as
using ctx unfolding composition-bad-context-def by blast
have shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as query-idxs
using bound
unfolding accepted-with-bound-tables-def accepted-with-tables-def
by simp
from checked-staged-security-with-data-state-accepted-shape-query-chunks
  [OF wf controlled data-support shape]
obtain raw-idxs where
  as-eq: as = staged-alphas data
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and lookup:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
and idx-bound:
   $\forall \text{idx} \in \text{set } \text{query-idxs}. \text{idx} < \text{clength} * \text{scale}$ 
by blast
have bound-staged:
  accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table (staged-alphas data) query-idxs
using bound as-eq by simp
have all-queries-staged:
  all-queries-consistent trace-table composition-table
    (staged-alphas data)
using all-queries as-eq by simp
have i-bound:  $0 < \text{rounds}$ 
by (rule rounds-positive)
have idx-sample:
  query-idxs ! 0  $\in$  query-sample-space
proof –
  have len-query-idxs: length query-idxs = rounds
    by (rule accepted-with-bound-tables-shapes(4)[OF bound-staged])
  have query-idxs ! 0  $\in$  set query-idxs
    by (rule nth-mem) (use i-bound len-query-idxs in simp)
  then show ?thesis
    by (rule accepted-with-bound-tables-query-sample-space
      [OF bound-staged])
qed
show ?thesis
proof (rule accepted-with-bound-tables-partial-opening-witnesses-Some
  [OF bound-staged])

```

```

fix fr f-fri-roots f-final dg composition-fri-roots final rest
  trace-openings composition-openings
assume header:
  verifier-header-transcript ?s fr f-fri-roots f-final
    (staged-alphas data) dg composition-fri-roots final rest
and comp-nonempty: composition-fri-roots  $\neq$  []
and trace-partial:
  accepted-with-partial-trace-openings ?s
    (Some (result, final-state)) fr query-idxs trace-openings
and comp-partial:
  accepted-with-partial-composition-openings ?s
    (Some (result, final-state)) (hd composition-fri-roots)
    query-idxs composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
    composition-openings
have header-eq:
  fr = staged-trace-root data  $\wedge$ 
  f-fri-roots = staged-trace-fri-roots data  $\wedge$ 
  f-final = staged-trace-final data  $\wedge$ 
  dg = staged-degree data  $\wedge$ 
  composition-fri-roots = staged-composition-fri-roots data  $\wedge$ 
  final = staged-composition-final data  $\wedge$ 
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF staged-header header]
by simp
have idx-in:
  query-idxs ! 0  $\in$ 
    query-header-supported-partial-opening-success-indices-at ?s fr
      f-fri-roots f-final (staged-alphas data) dg
      composition-fri-roots final 0
by (rule
  query-header-supported-partial-opening-success-indices-atI-from-partials-consistent
  [OF verifier trace-partial comp-partial trace-candidate
    comp-candidate header comp-nonempty refl i-bound idx-sample
    all-queries-staged])
have idx-in-staged:
  query-idxs ! 0  $\in$ 
    query-header-supported-partial-opening-success-indices-at ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
      0

```

```

    using idx-in header-eq by simp
  have raw-idx:
    index (to-nat (raw-idxs ! 0)) = query-idxs ! 0
    using query-idxs-eq len-raw i-bound by simp
  have hit-at:
    staged-security-with-data-state-query-partial-opening-hit-at 0
      (Some (((data, attacker-state), result), final-state))
    unfolding staged-security-with-data-state-query-partial-opening-hit-at-def
    using i-bound lookup[OF i-bound] idx-in-staged raw-idx
    by (simp add: Let-def)
  have hit:
    staged-security-with-data-state-query-partial-opening-hit
      (Some (((data, attacker-state), result), final-state))
    by (rule staged-security-with-data-state-query-partial-opening-hitI
      [OF i-bound hit-at])
  show ?thesis
    unfolding
      checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-def
    using hit by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-false-on-support
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
            A)
          adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
      out
  shows False
proof (cases out)
case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
      checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    by simp
next
case (Some packed)
  then obtain prefix prefix-state data attacker-state result final-state where
    out-eq:

```

```

    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
            final-state)
  by (cases packed, auto split: prod.splits)
have support-some:
  Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  using support unfolding out-eq .
have random-bad:
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-randomization-bad
  (Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state))
  using hit unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
  by simp
have query-hit:
  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
  (Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state))
  by (rule
      checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-imp-query-partial-opening-hit-on
      [OF wf controlled support-some random-bad])
have not-query:
  ¬ checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
  (Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state))
  using hit unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
  by simp
show ?thesis
  using query-hit not-query by contradiction
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
    adversary-initial-state ≤ 0

```

```

proof —
  have
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
      adversary-initial-state ≤
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      (λ-. False)
      adversary-initial-state
  by (rule wp-event-mono-on-support)
    (use
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-false-on-support
        [OF wf controlled] in blast)
  also have ... = 0
    unfolding wp-event-def wp-def dist-expect-def by simp
  finally show ?thesis .
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-imp-query-or-without-query:*

assumes

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
out

shows

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
out ∨
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
out

using *assms*

unfolding

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-def
by *blast*

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-query-hit:*

assumes

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
out

shows

checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
out

using *assms*

unfolding

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
by *simp*

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-branch-tree-on-sup*

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *support*:
 out $\in$
 set-dist
 (*execute*
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 adversary-initial-state)
and *hit*:
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 out
shows
 checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
 out
proof –
 have *drift*:
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
 out
 using *hit*
 unfolding
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
 by *simp*
 show *?thesis*
 by (*rule*
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-imp-branch-tree-on-support
 [*OF wf controlled support drift*])
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-branch-tree*
assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *branch-tree-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
 adversary-initial-state $\leq T$
shows
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 adversary-initial-state $\leq T$
proof –
 have
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 adversary-initial-state $\leq$

```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-branch-tree-output-hit
  adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-branch-tree-on-support
    [OF wf controlled])
then show ?thesis
  by (rule order-trans[OF - branch-tree-bound])
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-driftE:*

assumes *hit:*

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
 (Some (((prefix, prefix-state), data), attacker-state), result),
 final-state))

obtains *s where*

s = verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)
composition-randomization-bad s (Some (result, final-state))
alpha-header-list-set-hit s
 (alpha-header-supported-partial-union-bad-sets s
 composition-trace-bad-alpha-space)
 (Some (result, final-state))
 $\neg$ staged-alphas data $\in$
 alpha-prefix-union-bad-sets prefix-state
 composition-trace-bad-alpha-space (fst prefix)

proof —

let *?s =*

verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)

have *random-bad:*

composition-randomization-bad ?s (Some (result, final-state))

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
checked-staged-security-with-actual-alpha-prefix-verifier-event-def
Let-def

by *simp*

have *header-hit:*

alpha-header-list-set-hit ?s
 (alpha-header-supported-partial-union-bad-sets ?s
 composition-trace-bad-alpha-space)
 (Some (result, final-state))

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def


```

    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-def
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-def
    Let-def
  by simp
have not-prefix:
  ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
using hit
unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit-def
  Let-def
  by simp
show ?thesis
  by (rule that[OF refl random-bad header-hit not-prefix])
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hitE:
  assumes hit:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
      (Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state))
  obtains s where
    s = verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    composition-randomization-bad s (Some (result, final-state))
    alpha-header-list-set-hit s
      (alpha-header-supported-partial-union-bad-sets s
        composition-trace-bad-alpha-space)
      (Some (result, final-state))
  ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    (Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state))
proof -
  have drift:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
      (Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state))
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
  by simp
have query-hit:
  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit

```

```

    (Some (((((prefix, prefix-state), data), attacker-state), result),
            final-state))
using hit
unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
by simp
from checked-staged-security-with-actual-alpha-prefix-composition-candidate-driftE
  [OF drift]
obtain s where
  s-def:
    s = verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
and random-bad:
    composition-randomization-bad s (Some (result, final-state))
and header-hit:
    alpha-header-list-set-hit s
      (alpha-header-supported-partial-union-bad-sets s
        composition-trace-bad-alpha-space)
      (Some (result, final-state))
and not-prefix:
    ¬ staged-alphas data ∈
      alpha-prefix-union-bad-sets prefix-state
        composition-trace-bad-alpha-space (fst prefix)
by blast
show ?thesis
by (rule that[OF s-def random-bad header-hit not-prefix query-hit])
qed

```

```

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-supportedE:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    (Some (((((prefix, prefix-state), data), attacker-state), result),
            final-state))
obtains s trace-table where
  s = verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  composition-randomization-bad s (Some (result, final-state))
  staged-alphas data ∈
    alpha-header-supported-partial-union-bad-sets s

```

```

    composition-trace-bad-alpha-space
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
  trace-table ∈
    alpha-header-supported-partial-trace-table-candidates s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
  staged-alphas data ∈ composition-trace-bad-alpha-space trace-table
  ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
  checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    (Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state)))
proof —
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hitE
    [OF hit]
  have random-bad:
    composition-randomization-bad ?s (Some (result, final-state))
  and header-list-hit:
    alpha-header-list-set-hit ?s
    (alpha-header-supported-partial-union-bad-sets ?s
    composition-trace-bad-alpha-space)
    (Some (result, final-state))
  and not-prefix:
    ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
  and query-hit:
    checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
    (Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state)))
  by blast+
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
    adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  obtain builder verifier where

```

```

builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
  [OF wf controlled builder])
from header-list-hit obtain as query-idxs fr f-fri-roots f-final dg
  composition-fri-roots final rest where
  shape: accepted-transcript-shape ?s (Some (result, final-state))
  as query-idxs
and header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and as-bad:
  as ∈ alpha-header-supported-partial-union-bad-sets ?s
  composition-trace-bad-alpha-space fr f-fri-roots f-final
unfolding alpha-header-list-set-hit-def by blast
have header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  composition-fri-roots = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp
have staged-bad:
  staged-alphas data ∈
    alpha-header-supported-partial-union-bad-sets ?s
    composition-trace-bad-alpha-space
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)

```

```

using as-bad header-eq by simp
from staged-bad obtain trace-table where trace-candidate:
  trace-table ∈
    alpha-header-supported-partial-trace-table-candidates ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
and trace-bad:
  staged-alphas data ∈ composition-trace-bad-alpha-space trace-table
unfolding alpha-header-supported-partial-union-bad-sets-def by blast
show ?thesis
by (rule that[OF refl random-bad staged-bad trace-candidate trace-bad
  not-prefix query-hit])
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-supported-openingsE:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some ((((prefix, prefix-state), data), attacker-state), result),
    final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    (Some ((((prefix, prefix-state), data), attacker-state), result),
      final-state))
obtains s trace-table i raw trace-openings composition-openings where
  s = verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  composition-randomization-bad s (Some (result, final-state))
  trace-table ∈
    alpha-header-supported-partial-trace-table-candidates s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
  staged-alphas data ∈ composition-trace-bad-alpha-space trace-table
  ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
  i < rounds
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data i))) =

```

Some raw
index (to-nat raw) ∈ query-sample-space
(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses s
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)
partial-query-openings-consistent trace-openings composition-openings
(staged-alphas data) (index (to-nat raw))
proof –
from *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-supportedE*
[OF wf controlled support hit]
obtain *s trace-table* **where**
s-def:
s = verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
and *random-bad:*
composition-randomization-bad s (Some (result, final-state))
and *trace-candidate:*
trace-table ∈
alpha-header-supported-partial-trace-table-candidates s
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
and *trace-bad:*
staged-alphas data ∈ composition-trace-bad-alpha-space trace-table
and *not-prefix:*
¬ staged-alphas data ∈
alpha-prefix-union-bad-sets prefix-state
composition-trace-bad-alpha-space (fst prefix)
and *query-hit:*
checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
(Some (((prefix, prefix-state), data), attacker-state), result),
final-state))
by *blast*
from *checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hitE*
[OF query-hit]
obtain *i raw trace-openings composition-openings* **where**
i-bound: i < rounds
and *lookup:*
fmlookup (HashMap final-state)
(QueryIndexChallenge i
(state-after-query-chunks
(staged-query-start-hash data)
(staged-query-chunks data) i)) =

Some raw
and *idx-sample: index (to-nat raw) ∈ query-sample-space*
and *witness:*
 (*trace-openings, composition-openings*) ∈
 query-header-supported-partial-opening-witnesses
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*))
 (*staged-trace-root data*)
 (*staged-trace-fri-roots data*)
 (*staged-trace-final data*)
 (*staged-alphas data*)
 (*staged-degree data*)
 (*staged-composition-fri-roots data*)
 (*staged-composition-final data*)
and *consistent:*
 partial-query-openings-consistent trace-openings composition-openings
 (*staged-alphas data*) (*index (to-nat raw)*)
by *blast*
have *witness-s:*
 (*trace-openings, composition-openings*) ∈
 query-header-supported-partial-opening-witnesses s
 (*staged-trace-root data*)
 (*staged-trace-fri-roots data*)
 (*staged-trace-final data*)
 (*staged-alphas data*)
 (*staged-degree data*)
 (*staged-composition-fri-roots data*)
 (*staged-composition-final data*)
using *witness s-def by simp*
show *?thesis*
by (*rule that[OF s-def random-bad trace-candidate trace-bad not-prefix*
 i-bound lookup idx-sample witness-s consistent])
qed

definition *checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit*
 $:: ('f \text{ staged-adversary} \Rightarrow$
 $(((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool})$
where
checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit
 $A \text{ out} \longleftrightarrow$
(case out of
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((((\alpha\text{-prefix}, \alpha\text{-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}) \Rightarrow$
 $(\exists i \text{ trace-openings composition-openings.}$
 $\text{checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit } A$

((replicate rounds []) [i := trace-openings])
 ((replicate rounds []) [i := composition-openings])
 (staged-alphas data) i
 (Some (((((alpha-prefix, alpha-prefix-state), data),
 attacker-state), result), final-state))))

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hitI:*

assumes *component:*

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
 ((replicate rounds []) [i := trace-openings])
 ((replicate rounds []) [i := composition-openings])
 (staged-alphas data) i
 (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
 final-state))

shows

checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit A
 (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
 final-state))

proof —

have

$\exists j$ *trace-openings'* *composition-openings'*.
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
 ((replicate rounds []) [j := trace-openings'])
 ((replicate rounds []) [j := composition-openings'])
 (staged-alphas data) j
 (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state), result),
 final-state))

by (intro *exI*[of - i] *exI*[of - trace-openings]
exI[of - composition-openings])

(rule *component*)

then show *?thesis*

unfolding

checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit-def

by *simp*

qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-hit:*

assumes *query-bound:*

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
adversary-initial-state $\leq Q$

shows

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq Q$

by (rule *order.trans*[OF - *query-bound*])

(rule *wp-event-mono*,

rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-query-hit)

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-query-and-without-q*

assumes *query-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq Q$

and *without-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state $\leq W$

shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq Q + W$

proof –

let *?M* =
checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let *?Drift* =
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
let *?Query* =
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
let *?Without* =
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
have *wp-event ?M ?Drift adversary-initial-state* $\leq$
wp-event ?M ($\lambda out. ?Query out \vee ?Without out$)
adversary-initial-state
by (*rule wp-event-mono*)
 (*rule*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-imp-query-or-without-query)
also have ... $\leq$
wp-event ?M ?Query adversary-initial-state +
wp-event ?M ?Without adversary-initial-state
by (*rule wp-event-union-bound*)
also have ... $\leq Q + W$
by (*intro add-mono query-bound without-bound*)
finally show *?thesis* .

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-composition-candidate-drift-and-budgets*:

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *drift-bound*:

$wp\text{-}event$
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}data\text{-}state$
 $A)$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}composition\text{-}candidate\text{-}drift$
 $adversary\text{-}initial\text{-}state \leq D$

shows

$wp\text{-}event (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event\text{-}composition\text{-}bad)$
 $adversary\text{-}initial\text{-}state \leq$
 $composition\text{-}error\text{-}bound +$
 $staged\text{-}phase\text{-}relation\text{-}error\text{-}size$
 $(staged\text{-}alpha\text{-}search\text{-}queries\text{-}budgets\text{-}0) +$
 $hash\text{-}collision\text{-}budget\text{-}value\text{-}0$
 $(staged\text{-}alpha\text{-}search\text{-}queries\text{-}budgets\text{-}0) +$
 $staged\text{-}phase\text{-}relation\text{-}error\text{-}size$
 $(staged\text{-}alpha\text{-}search\text{-}queries\text{-}budgets\text{-}0) +$
 $D +$
 $hash\text{-}collision\text{-}budget\text{-}value\text{-}0$
 $(staged\text{-}attacker\text{-}query\text{-}budget\text{-}budgets +$
 $staged\text{-}challenge\text{-}query\text{-}budget + verifier\text{-}hash\text{-}query\text{-}budget)$

proof –

let $?M =$
 $checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}data\text{-}state A$

let $?Prefix =$
 $composition\text{-}error\text{-}bound +$
 $staged\text{-}phase\text{-}relation\text{-}error\text{-}size$
 $(staged\text{-}alpha\text{-}search\text{-}queries\text{-}budgets\text{-}0) +$
 $hash\text{-}collision\text{-}budget\text{-}value\text{-}0$
 $(staged\text{-}alpha\text{-}search\text{-}queries\text{-}budgets\text{-}0)$

let $?Pre =$
 $staged\text{-}phase\text{-}relation\text{-}error\text{-}size$
 $(staged\text{-}alpha\text{-}search\text{-}queries\text{-}budgets\text{-}0)$

let $?Coll =$
 $hash\text{-}collision\text{-}budget\text{-}value\text{-}0$
 $(staged\text{-}attacker\text{-}query\text{-}budget\text{-}budgets +$
 $staged\text{-}challenge\text{-}query\text{-}budget + verifier\text{-}hash\text{-}query\text{-}budget)$

have $transcript\text{-}bound:$
 $wp\text{-}event (checked\text{-}staged\text{-}transcript\text{-}with\text{-}alpha\text{-}prefix\text{-}program A)$
 $(checked\text{-}staged\text{-}transcript\text{-}actual\text{-}alpha\text{-}prefix\text{-}bad\text{-}set\text{-}hit$
 $composition\text{-}trace\text{-}bad\text{-}alpha\text{-}space)$
 $adversary\text{-}initial\text{-}state \leq ?Prefix$

by (rule
 $checked\text{-}staged\text{-}transcript\text{-}with\text{-}alpha\text{-}prefix\text{-}actual\text{-}bad\text{-}set\text{-}hit\text{-}bound\text{-}from\text{-}budgets$
 $[OF\text{-}wf\text{-}controlled])$

have $prefix\text{-}bound:$
 $wp\text{-}event\text{-}?M$
 $(checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}bad\text{-}set\text{-}hit$
 $composition\text{-}trace\text{-}bad\text{-}alpha\text{-}space)$
 $adversary\text{-}initial\text{-}state \leq ?Prefix$

```

by (rule
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
  [OF transcript-bound])
have prequery-bound:
  wp-event ?M
  ( checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
    composition-trace-bad-alpha-space)
  adversary-initial-state  $\leq$  ?Pre
by (rule
  checked-staged-security-with-actual-alpha-prefix-vector-alpha-prequery-accounting
  [OF wf controlled])
have collision-bound:
  wp-event ?M
  ( checked-staged-security-with-actual-alpha-prefix-verifier-event
    hash-map-output-collision-bad)
  adversary-initial-state  $\leq$  ?Coll
by (rule
  checked-staged-security-with-actual-alpha-prefix-hash-map-output-collision-bad-bound-controlled
  [OF wf controlled])
let ?degree =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-degree-bad
let ?random =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-randomization-bad
let ?bad =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-bad
have projected:
  wp-event ( checked-staged-security-experiment-with-data-state A )
  ( staged-security-with-data-state-verifier-event composition-bad )
  adversary-initial-state =
  wp-event ?M ?bad adversary-initial-state
by (rule
  checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)
have event-eq: ?bad = ( $\lambda$ out. ?degree out  $\vee$  ?random out)
by (rule ext)
  (auto simp:
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    composition-bad-def split: option.splits prod.splits)
have degree-false: ?degree = ( $\lambda$ out. False)
by (rule ext)
  (auto simp:
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    composition-degree-bad-false[OF spec-degree-wellformed-from-spec-query-margin]
    split: option.splits prod.splits)
have degree-bound:
  wp-event ?M ?degree adversary-initial-state  $\leq$  0
unfolding degree-false wp-event-def wp-def dist-expect-def by simp

```

have *random-bound*:
 $wp\text{-}event\ ?M\ ?random\ adversary\text{-}initial\text{-}state \leq$
 $\ ?Prefix + ?Pre + D + ?Coll$
by (rule
checked-staged-security-with-actual-alpha-prefix-composition-randomization-bad-bound-from-prefix-prequery-drift
 $[OF\ prefix\text{-}bound\ prequery\text{-}bound\ drift\text{-}bound\ collision\text{-}bound]$)
have *wp-event ?M ?bad adversary-initial-state* $\leq$
 $wp\text{-}event\ ?M\ ?degree\ adversary\text{-}initial\text{-}state +$
 $wp\text{-}event\ ?M\ ?random\ adversary\text{-}initial\text{-}state$
unfolding *event-eq* **by** (rule *wp-event-union-bound*)
also have $\dots \leq 0 + (?Prefix + ?Pre + D + ?Coll)$
by (intro *add-mono degree-bound random-bound*)
finally show *?thesis*
unfolding *projected* **by** (simp *add: add.assoc*)
qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-partial-opening-hit-and-budgets*:

fixes *query-error :: prob*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *partial-opening-bound*:

$wp\text{-}event\ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A)$
 $\ staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}query\text{-}partial\text{-}opening\text{-}hit$
 $\ adversary\text{-}initial\text{-}state \leq query\text{-}error$

shows

$wp\text{-}event\ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A)$
 $\ (staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event\ composition\text{-}bad)$
 $\ adversary\text{-}initial\text{-}state \leq$
 $\ composition\text{-}error\text{-}bound +$
 $\ staged\text{-}phase\text{-}relation\text{-}error\ size$
 $\ (staged\text{-}alpha\text{-}search\text{-}queries\ budgets\ 0) +$
 $\ hash\text{-}collision\text{-}budget\text{-}value\ 0$
 $\ (staged\text{-}alpha\text{-}search\text{-}queries\ budgets\ 0) +$
 $\ staged\text{-}phase\text{-}relation\text{-}error\ size$
 $\ (staged\text{-}alpha\text{-}search\text{-}queries\ budgets\ 0) +$
 $\ (query\text{-}error + 0) +$
 $\ hash\text{-}collision\text{-}budget\text{-}value\ 0$
 $\ (staged\text{-}attacker\text{-}query\text{-}budget\ budgets +$
 $\ staged\text{-}challenge\text{-}query\text{-}budget + verifier\text{-}hash\text{-}query\text{-}budget)$

proof –

let *?M =*

checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

have *query-bound*:

$wp\text{-}event\ ?M$

$checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}query\text{-}partial\text{-}opening\text{-}hit$
 $\ adversary\text{-}initial\text{-}state \leq query\text{-}error$

unfolding

checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-projection

by (rule *partial-opening-bound*)

have *drift-query-bound*:
wp-event ?*M*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq$ *query-error*
by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-hit
[*OF query-bound*])
have *without-bound*:
wp-event ?*M*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state $\leq$ 0
by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-bound
[*OF wf controlled*])
have *drift-bound*:
wp-event ?*M*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq$ *query-error* + 0
by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-query-and-without-query
[*OF drift-query-bound without-bound*])
show ?*thesis*
by (rule
checked-staged-security-with-data-state-composition-bad-bound-from-composition-candidate-drift-and-budgets
[*OF wf controlled drift-bound*])
qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-partial-query-residual-and-budgets*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *cover*:
 $\bigwedge$ *data attacker-state i*.
i < *rounds* $\implies$
query-header-supported-partial-opening-success-indices-at
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))
(*staged-trace-root data*)
(*staged-trace-fri-roots data*)
(*staged-trace-final data*)
(*staged-alphas data*)
(*staged-degree data*)
(*staged-composition-fri-roots data*)
(*staged-composition-final data*)
i $\subseteq$ *B*
and *raw-bound*: *query-index-raw-preimage-bound*
and *subset*: *B* $\subseteq$ *query-sample-space*
and *frac*:
 $\text{nnreal}(\text{card } B) / \text{nnreal}(\text{card } \text{query-sample-space}) \leq$
query-error-bound

and *residual-bound*:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state $\leq R$

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event composition-bad)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
(nnreal
(staged-attacker-query-budget budgets +
*staged-challenge-query-budget) * query-error-bound + R)* +
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)

proof –

let *?M* =
checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

let *?QueryCost* =
nnreal
(staged-attacker-query-budget budgets +
*staged-challenge-query-budget) * query-error-bound*

have *query-bound*:

wp-event ?M
checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit
adversary-initial-state $\leq ?QueryCost$

by (rule
checked-staged-security-with-actual-alpha-prefix-query-partial-opening-hit-bound-from-fixed-query-error-cover
[OF wf controlled cover raw-bound subset frac])

have *drift-query-bound*:

wp-event ?M
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq ?QueryCost$

by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-hit
[OF query-bound])

have *drift-bound*:

wp-event ?M
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq ?QueryCost + R$

by (rule

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-query-and-without-query
 [OF *drift-query-bound residual-bound*])
have
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event composition-bad*)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 (?*QueryCost* + *R*) +
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)
by (rule
checked-staged-security-with-data-state-composition-bad-bound-from-composition-candidate-drift-and-budgets
 [OF *wf controlled drift-bound*])
then show ?thesis
by (simp add: add.assoc)
qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-partial-query-and-budgets:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *cover:*

$\bigwedge$ *data attacker-state i.*

i < *rounds* $\implies$

query-header-supported-partial-opening-success-indices-at

(*verifier-state-from-adversary attacker-state*

(*staged-proof-transcript data*))

(*staged-trace-root data*)

(*staged-trace-fri-roots data*)

(*staged-trace-final data*)

(*staged-alphas data*)

(*staged-degree data*)

(*staged-composition-fri-roots data*)

(*staged-composition-final data*)

i $\subseteq$ *B*

and *raw-bound: query-index-raw-preimage-bound*

and *subset: B* $\subseteq$ *query-sample-space*

and *frac:*

nnreal (*card B*) / *nnreal* (*card query-sample-space*) $\leq$

query-error-bound

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event composition-bad*)

```

    adversary-initial-state ≤
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    nnreal
    (staged-attacker-query-budget budgets +
    staged-challenge-query-budget) * query-error-bound +
    hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget)
proof —
have residual-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
  adversary-initial-state ≤ 0
by (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-bound
  [OF wf controlled])
have
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event composition-bad)
  adversary-initial-state ≤
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  (nnreal
  (staged-attacker-query-budget budgets +
  staged-challenge-query-budget) * query-error-bound + 0) +
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
  staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
  checked-staged-security-with-data-state-composition-bad-bound-from-partial-query-residual-and-budgets
  [OF wf controlled cover raw-bound subset frac residual-bound])
then show ?thesis
by simp
qed

end

```


end

theory *Soundness-Query-Opening-Extraction*
imports *Soundness-Partial-Merkle*
begin

Low-level verifier query extraction lemmas used by staged soundness.

This layer is deliberately kept below the staged event definitions: it only talks about verifier query rounds, authenticated openings, transcript chunks, and random-oracle lookups.

context *soundness*
begin

lemma *query-opening-fri-layer-opening-step-partial-authenticated-table-first-value:*

fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes $\text{len-pos}: 0 < \text{len}$
and $\text{idx-bound}: \text{idx} < \text{len}$
and *outcome:*
 $\text{Some } (out, t) \in$
 set-dist
 $(\text{execute } (\text{fri-layer-opening-step } (b, rt) (\text{idx}, x, \text{len}, pw))\ s)$
obtains *openings* **where**
 $\text{partial-authenticated-table } rt\ \text{len}\ \text{openings}\ t$
 $\text{map opening-index openings} = [\text{idx}, (\text{idx} + \text{len} \text{ div } 2) \bmod \text{len}]$
 $\text{length openings} = 2$
 $\text{opening-value } (\text{openings } !\ 0) = x$

proof –

from $\text{outcome}[\text{unfolded fri-layer-opening-step-unfold-finish}]$
obtain $xp\ s1\ xp\text{-path}\ s2\ xn\ s3\ xn\text{-path}\ s4$ **where**
 $\text{read-xp}: \text{Some } (xp, s1) \in \text{set-dist } (\text{execute read } s)$
and $\text{read-xp-path}:$
 $\text{Some } (xp\text{-path}, s2) \in$
 $\text{set-dist } (\text{execute } (\text{ntimes read } (\text{floor-log len}))\ s1)$
and $\text{read-xn}: \text{Some } (xn, s3) \in \text{set-dist } (\text{execute read } s2)$
and $\text{read-xn-path}:$
 $\text{Some } (xn\text{-path}, s4) \in$
 $\text{set-dist } (\text{execute } (\text{ntimes read } (\text{floor-log len}))\ s3)$
and *finish:*
 $\text{Some } (out, t) \in$
 set-dist
 $(\text{execute}$
 $(\text{fri-layer-opening-finish } b\ rt\ \text{idx}\ x\ \text{len}\ pw\ xp\ xp\text{-path}\ xn$
 $xn\text{-path})\ s4)$
by $(\text{auto elim!}: \text{set-dist-bindE})$
have $xp\text{-path-length}: \text{length } xp\text{-path} = \text{floor-log len}$
using $\text{ntimes-read-any-outcome}[OF\ \text{read-xp-path}]$ **by** *simp*
have $xn\text{-path-length}: \text{length } xn\text{-path} = \text{floor-log len}$
using $\text{ntimes-read-any-outcome}[OF\ \text{read-xn-path}]$ **by** *simp*

```

have sibling-bound: (idx + len div 2) mod len < len
  using len-pos by simp
have xp-x: xp = x
  using finish
  unfolding fri-layer-opening-finish-def assert-def
  by (auto simp: throw-no-outcome elim!: set-dist-bindE split: if-splits)
from fri-layer-opening-finish-authenticated-openings
  [OF idx-bound sibling-bound xp-path-length xn-path-length finish]
obtain xp-opening xn-opening where
  xp-eq:
    xp-opening =
      (|opening-root = rt,
        opening-length = len,
        opening-index = idx,
        opening-value = xp,
        opening-path = xp-path)
  and xn-eq:
    xn-opening =
      (|opening-root = rt,
        opening-length = len,
        opening-index = (idx + len div 2) mod len,
        opening-value = xn,
        opening-path = xn-path)
  and xp-auth: authenticated-opening-in t xp-opening
  and xn-auth: authenticated-opening-in t xn-opening
  by blast
let ?openings = [xp-opening, xn-opening]
have table: partial-authenticated-table rt len ?openings t
  unfolding partial-authenticated-table-def
  using xp-eq xn-eq xp-auth xn-auth by simp
have indices:
  map opening-index ?openings = [idx, (idx + len div 2) mod len]
  using xp-eq xn-eq by simp
have first-value: opening-value (?openings ! 0) = x
  using xp-eq xp-x by simp
show ?thesis
  by (rule that[OF table indices - first-value]) simp
qed

```

lemma *query-opening-ntimes-outcome-decomp-at*:

assumes *outcome*:

Some (*results*, *t*) ∈ *set-dist* (*execute* (*ntimes* *m n*) *s*)

and *i-bound*: *i* < *n*

shows

∃ *prefix x suffix s-i s-suc*.

Some (*prefix*, *s-i*) ∈ *set-dist* (*execute* (*ntimes* *m i*) *s*) ∧

Some (*x*, *s-suc*) ∈ *set-dist* (*execute* *m s-i*) ∧

Some (*suffix*, *t*) ∈

set-dist (*execute* (*ntimes* *m (n - Suc i)*) *s-suc*) ∧

```

    results = prefix @ x # suffix
using outcome i-bound
proof (induction n arbitrary: i s results t)
  case 0
  then show ?case by simp
next
  case (Suc n)
from Suc.premis(1) obtain x results' s1 where
    head: Some (x, s1) ∈ set-dist (execute m s)
    and tail:
      Some (results', t) ∈ set-dist (execute (ntimes m n) s1)
    and results-eq: results = x # results'
    by (auto elim!: set-dist-bindE)
show ?case
proof (cases i)
  case 0
  have prefix0:
    Some ([], s) ∈ set-dist (execute (ntimes m i) s)
    using 0 by simp
  have suffix0:
    Some (results', t) ∈
      set-dist (execute (ntimes m (Suc n - Suc i)) s1)
    using tail 0 by simp
  show ?thesis
    using prefix0 head suffix0 results-eq
    by (intro exI[of - []] exI[of - x] exI[of - results']
      exI[of - s] exI[of - s1]) simp
next
  case (Suc j)
  have j-bound: j < n
    using Suc.premis(2) Suc by simp
  from Suc.IH[OF tail j-bound]
  obtain prefix y suffix s-j s-suc where
    prefix:
      Some (prefix, s-j) ∈ set-dist (execute (ntimes m j) s1)
    and round: Some (y, s-suc) ∈ set-dist (execute m s-j)
    and suffix:
      Some (suffix, t) ∈
        set-dist (execute (ntimes m (n - Suc j)) s-suc)
    and results'-eq: results' = prefix @ y # suffix
    by blast
  have prefix-suc:
    Some (x # prefix, s-j) ∈
      set-dist (execute (ntimes m (Suc j)) s)
    unfolding ntimes.simps
    apply (rule set-dist-bindI[OF head])
    apply (rule set-dist-bindI[OF prefix])
    apply simp
  done

```

```

have n-minus:  $\text{Suc } n - \text{Suc } (\text{Suc } j) = n - \text{Suc } j$ 
  by simp
have suffix-suc:
  Some (suffix, t)  $\in$ 
    set-dist (execute (ntimes m ( $\text{Suc } n - \text{Suc } i$ )) s-suc)
  using suffix n-minus Suc by simp
show ?thesis
  using prefix-suc round suffix-suc results-eq results'-eq Suc
  by (intro exI[of - x # prefix] exI[of - y] exI[of - suffix]
    exI[of - s-j] exI[of - s-suc]) simp
qed
qed

lemma query-opening-verifier-query-round-program-authenticated-with-lookup:
fixes s t :: (f, 'a) protocol-channel-scheme
assumes fl-eq: fl = (b, composition-root) # fl-tail
and outcome:
  Some ((), t)  $\in$ 
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
obtains raw idx trace-leaves trace-openings composition-openings where
  idx = index (to-nat raw)
  map opening-value trace-openings = trace-leaves
  map opening-index trace-openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) trace-openings t
  partial-authenticated-table composition-root (scale * clength)
    composition-openings t
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof -
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where rand:
  Some (raw, s0)  $\in$ 
    set-dist (execute receive-query-index-challenge s)
and tail:
  Some ((), t)  $\in$ 
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
  by (auto elim!: set-dist-bindE)
have lookup-s0:
  fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  using receive-query-index-challenge-outcome[OF rand] by simp
have lookup-t:

```

```

    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  using verifier-query-round-after-index-program-preserves-query-lookup
    [OF tail, of PQueryCounter s PState s] lookup-s0
  by simp
show ?thesis
proof (rule verifier-query-round-after-index-authenticated-trace-openings
  [OF tail])
  fix idx trace-leaves trace-openings
  assume idx-eq: idx = index (to-nat raw)
  and trace-values:
    map opening-value trace-openings = trace-leaves
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and trace-table:
    partial-authenticated-table fr (scale * clength)
    trace-openings t
  show ?thesis
  proof (rule
    verifier-query-round-after-index-authenticated-composition-openings
    [OF fl-eq tail])
    fix idx' composition-openings
    assume idx'-eq: idx' = index (to-nat raw)
    and comp-table:
      partial-authenticated-table composition-root (scale * clength)
      composition-openings t
    and comp-indices:
      map opening-index composition-openings =
        [idx', fri-sibling-index (scale * clength) idx']
    have idx'-idx: idx' = idx
    using idx-eq idx'-eq by simp
    show ?thesis
    by (rule that[OF idx-eq trace-values trace-indices trace-table
      comp-table - lookup-t])
      (use comp-indices idx'-idx in simp)
  qed
qed
qed

lemma query-opening-verifier-query-round-program-authenticated-with-chunk:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes fl-eq: fl = (b, composition-root) # fl-tail
  and outcome:
    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  obtains raw idx chunk trace-leaves trace-openings composition-openings
  where

```

```

idx = index (to-nat raw)
verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
PTranscript s = chunk @ PTranscript t
PState t = foldl concat (PState s) chunk
s ≤ t
PQueryCounter t = Suc (PQueryCounter s)
map opening-value trace-openings = trace-leaves
map opening-index trace-openings = powers-scaled idx
partial-authenticated-table fr (scale * clength) trace-openings t
partial-authenticated-table composition-root (scale * clength)
  composition-openings t
map opening-index composition-openings =
  [idx, fri-sibling-index (scale * clength) idx]
fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof –
  show ?thesis
proof (rule verifier-query-round-program-outcome[OF outcome])
  fix raw0 idx0 chunk
  assume idx0-eq: idx0 = index (to-nat raw0)
  and chunk-shape:
    verifier-query-round-chunk idx0 (map snd f-fl) (map snd fl)
    chunk
  and transcript:
    PTranscript s = chunk @ PTranscript t
  and state:
    PState t = foldl concat (PState s) chunk
  and ext: s ≤ t
  and counter:
    PQueryCounter t = Suc (PQueryCounter s)
  and lookup0:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw0
  show ?thesis
proof (rule
  query-opening-verifier-query-round-program-authenticated-with-lookup
  [OF fl-eq outcome])
  fix raw idx trace-leaves trace-openings composition-openings
  assume idx-eq: idx = index (to-nat raw)
  and trace-values:
    map opening-value trace-openings = trace-leaves
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and trace-table:
    partial-authenticated-table fr (scale * clength)
    trace-openings t
  and comp-table:
    partial-authenticated-table composition-root (scale * clength)
    composition-openings t

```

```

and comp-indices:
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
and lookup:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
have raw-eq: raw = raw0
  using lookup lookup0 by simp
have idx-idx0: idx = idx0
  using idx-eq idx0-eq raw-eq by simp
show ?thesis
  by (rule that[OF idx-eq - transcript state ext counter
    trace-values trace-indices trace-table comp-table comp-indices
    lookup])
    (use chunk-shape idx-idx0 in simp)
qed
qed
qed

```

```

lemma query-opening-ntimes-prefix-state-alignment:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (prefix, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) i)
          s)
  obtains prefix-chunks where
    length prefix-chunks = i
    PState t = state-after-query-chunks (PState s) prefix-chunks i
    PQueryCounter t = PQueryCounter s + i
    s ≤ t

```

```

proof –
  from ntimes-verifier-query-rounds-outcome[OF outcome]
  obtain raw-idxs query-idxs prefix-chunks where
    len-chunks: length prefix-chunks = i
  and state:
    PState t = state-after-query-chunks (PState s) prefix-chunks i
  and ext: s ≤ t
  and counter: PQueryCounter t = PQueryCounter s + i
  by blast
  show ?thesis
  by (rule that[OF len-chunks state counter ext])
qed

```

```

lemma query-opening-ntimes-prefix-transcript-state-alignment:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:

```

Some (prefix, t) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final) i)
s)
obtains *prefix-query-idxs prefix-chunks where*
length prefix-query-idxs = i
length prefix-chunks = i
PTranscript s = List.concat prefix-chunks @ PTranscript t
PState t = state-after-query-chunks (PState s) prefix-chunks i
PQueryCounter t = PQueryCounter s + i
s ≤ t
 $\bigwedge j. j < i \implies$
verifier-query-round-chunk (prefix-query-idxs ! j)
(map snd f-fl) (map snd fl) (prefix-chunks ! j)
proof –
from *ntimes-verifier-query-rounds-outcome[OF outcome]*
obtain *raw-idxs query-idxs prefix-chunks where*
len-raw-idxs: length raw-idxs = i
and *query-idxs-eq:*
query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and
len-chunks: length prefix-chunks = i
and *transcript:*
PTranscript s = List.concat prefix-chunks @ PTranscript t
and *state:*
PState t = state-after-query-chunks (PState s) prefix-chunks i
and *ext: s ≤ t*
and *counter: PQueryCounter t = PQueryCounter s + i*
and *rounds:*
 $\bigwedge j. j < i \implies$
verifier-query-round-chunk (query-idxs ! j)
(map snd f-fl) (map snd fl) (prefix-chunks ! j)
by *blast*
have *len-idxs: length query-idxs = i*
using *len-raw-idxs query-idxs-eq by simp*
show *?thesis*
by *(rule that[OF len-idxs len-chunks transcript state counter ext])*
(use rounds in blast)
qed

lemma *query-opening-ntimes-suffix-extends:*
fixes *s t :: ('f, 'a) protocol-channel-scheme*
assumes *outcome:*
Some (suffix, t) ∈
set-dist
(execute
(ntimes


```

      (verifier-query-round-program fr f-fl f-final as fl final) n)
    s)
  shows  $s \leq t$ 
proof -
  from ntimes-verifier-query-rounds-outcome[OF outcome]
  show ?thesis
    by blast
qed

lemma query-opening-ntimes-suffix-transcript-state-alignment:
  fixes  $s\ t :: ('f, 'a) \text{protocol-channel-scheme}$ 
  assumes outcome:
    Some (suffix, t)  $\in$ 
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s)
  obtains suffix-query-idxs suffix-chunks where
    length suffix-query-idxs = n
    length suffix-chunks = n
    PTranscript s = List.concat suffix-chunks @ PTranscript t
    PState t = state-after-query-chunks (PState s) suffix-chunks n
    PQueryCounter t = PQueryCounter s + n
     $s \leq t$ 
     $\bigwedge j. j < n \implies$ 
      verifier-query-round-chunk (suffix-query-idxs ! j)
      (map snd f-fl) (map snd fl) (suffix-chunks ! j)
proof -
  from ntimes-verifier-query-rounds-outcome[OF outcome]
  obtain raw-idxs query-idxs suffix-chunks where
    len-raw-idxs: length raw-idxs = n
    and query-idxs-eq:
      query-idxs = map ( $\lambda \text{raw}. \text{index } (\text{to-nat raw})$ ) raw-idxs
    and len-chunks: length suffix-chunks = n
    and transcript:
      PTranscript s = List.concat suffix-chunks @ PTranscript t
    and state:
      PState t = state-after-query-chunks (PState s) suffix-chunks n
    and ext:  $s \leq t$ 
    and counter: PQueryCounter t = PQueryCounter s + n
    and rounds:
       $\bigwedge j. j < n \implies$ 
        verifier-query-round-chunk (query-idxs ! j)
        (map snd f-fl) (map snd fl) (suffix-chunks ! j)
    by blast
  have len-idxs: length query-idxs = n
    using len-raw-idxs query-idxs-eq by simp
  show ?thesis

```

by (rule that[OF len-idxs len-chunks transcript state counter ext])
 (use rounds in blast)

qed

lemma *query-opening-ntimes-authenticated-initial-openings-at:*
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes $fl\text{-eq}: fl = (b, \text{composition-root}) \# fl\text{-tail}$
and *outcome:*
Some (results, t) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final) n)
s)
and *i-bound: $i < n$*
obtains *raw idx prefix-chunks trace-openings composition-openings* **where**
idx = index (to-nat raw)
length prefix-chunks = i
map opening-index trace-openings = powers-scaled idx
*partial-authenticated-table fr (scale * clength) trace-openings t*
*partial-authenticated-table composition-root (scale * clength)*
composition-openings t
map opening-index composition-openings =
*[idx, fri-sibling-index (scale * clength) idx]*
fmlookup (HashMap t)
(QueryIndexChallenge (PQueryCounter s + i)
(state-after-query-chunks (PState s) prefix-chunks i)) = Some raw
proof –
from *query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]*
obtain *prefix round suffix s-i s-suc* **where**
prefix:
Some (prefix, s-i) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final) i)
s)
and *round:*
Some (round, s-suc) ∈
set-dist
(execute
(verifier-query-round-program fr f-fl f-final as fl final) s-i)
and *suffix:*
Some (suffix, t) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final)
(n – Suc i))

```

      s-suc)
  by blast
have round-unit: round = ()
  by (cases round) simp
show ?thesis
proof (rule
  query-opening-verifier-query-round-program-authenticated-with-chunk
  [OF fl-eq round[unfolded round-unit]])
fix raw idx chunk trace-leaves trace-openings composition-openings
assume idx-eq: idx = index (to-nat raw)
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
  chunk
and transcript: PTranscript s-i = chunk @ PTranscript s-suc
and state-suc: PState s-suc = foldl concat (PState s-i) chunk
and ext-round: s-i ≤ s-suc
and counter-suc: PQueryCounter s-suc = Suc (PQueryCounter s-i)
and trace-values: map opening-value trace-openings = trace-leaves
and trace-indices:
  map opening-index trace-openings = powers-scaled idx
and trace-table-suc:
  partial-authenticated-table fr (scale * clength)
  trace-openings s-suc
and comp-table-suc:
  partial-authenticated-table composition-root (scale * clength)
  composition-openings s-suc
and comp-indices:
  map opening-index composition-openings =
  [idx, fri-sibling-index (scale * clength) idx]
and lookup-suc:
  fmllookup (HashMap s-suc)
  (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
  Some raw
show ?thesis
proof (rule query-opening-ntimes-prefix-state-alignment[OF prefix])
fix prefix-chunks
assume len-chunks: length prefix-chunks = i
and state-i:
  PState s-i =
  state-after-query-chunks (PState s) prefix-chunks i
and counter-i: PQueryCounter s-i = PQueryCounter s + i
have suffix-ext: s-suc ≤ t
  by (rule query-opening-ntimes-suffix-extends[OF suffix])
have trace-table-t:
  partial-authenticated-table fr (scale * clength)
  trace-openings t
  by (rule
    partial-authenticated-table-mono[OF trace-table-suc
    suffix-ext])

```

```

have comp-table-t:
  partial-authenticated-table composition-root (scale * clength)
  composition-openings t
by (rule
  partial-authenticated-table-mono[OF comp-table-suc suffix-ext])
have lookup-t:
  fmllookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s + i)
  (state-after-query-chunks (PState s) prefix-chunks i)) =
  Some raw
using hash-extension-lookup[OF lookup-suc suffix-ext]
unfolding counter-i state-i .
show ?thesis
by (rule that[OF idx-eq len-chunks trace-indices trace-table-t
  comp-table-t comp-indices lookup-t])
qed
qed
qed
end
end

```

```

theory Soundness-Query-Opening-Consistency
imports
  Soundness-Query-Opening-Extraction
  Soundness-Staged-Partial-Query
begin

```

Verifier-local query-opening consistency facts. These lemmas package the composition check enforced by a successful query round together with the authenticated openings extracted from that round.

```

context soundness
begin

```

```

lemma verifier-query-round-after-index-authenticated-openings-consistent:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-eq: fl = (b, composition-root) # fl-tail
and outcome:
  Some ((), t) ∈
  set-dist
  (execute
  (verifier-query-round-after-index-program
  fr f-fl f-final as fl final raw) s)
obtains idx trace-openings composition-openings where
  idx = index (to-nat raw)
  map opening-index trace-openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) trace-openings t

```

```

partial-authenticated-table composition-root (scale * clength)
  composition-openings t
map opening-index composition-openings =
  [idx, fri-sibling-index (scale * clength) idx]
partial-query-openings-consistent trace-openings composition-openings
  as idx
proof –
let ?idx = index (to-nat raw)
from outcome obtain fv s1 f-out s2 s3 c-out s4 where
  query-decommit:
    Some (fv, s1) ∈
      set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
and trace-fri:
    Some (f-out, s2) ∈
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
and assert-trace:
    Some ((), s3) ∈
      set-dist
        (execute
          (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
    Some (c-out, s4) ∈
      set-dist (execute
        (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
          clength * scale, 1)
          (receive-query-commits fl)) s3)
and assert-comp:
    Some ((), t) ∈
      set-dist
        (execute
          (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
unfolding verifier-query-round-after-index-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have idx-sample: ?idx ∈ query-sample-space
using index-less-query-sample-space
unfolding query-sample-space-def by simp
from check-decommit-on-query-authenticated-openings
  [OF idx-sample query-decommit]
obtain trace-openings where
  trace-values: map opening-value trace-openings = fv
and trace-indices:
  map opening-index trace-openings = powers-scaled ?idx
and trace-table-s1:
  partial-authenticated-table fr (scale * clength) trace-openings s1
by blast
from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where s1-s2: s1 ≤ s2

```

```

    by blast
  have s3-eq:  $s3 = s2$ 
    using assert-trace unfolding assert-def
    by (cases case f-out of  $(-, f-x, -, -) \Rightarrow f-x = f-final$ )
      (auto simp: throw-no-outcome)
  have comp-steps:
    receive-query-commits fl =
      fri-layer-opening-step (b, composition-root) #
        receive-query-commits fl-tail
    unfolding fl-eq receive-query-commits-def fri-layer-opening-step-def
    by simp
  from comp-fri[unfolded comp-steps]
  obtain out1 s-head where
    head:
      Some (out1, s-head)  $\in$ 
        set-dist
          (execute
            (fri-layer-opening-step (b, composition-root)
              (?idx, cp-eval as fv ( $h \wedge ?idx * shift$ ),
                clength * scale, 1)) s3)
    and tail:
      Some (c-out, s4)  $\in$ 
        set-dist (execute (mfold out1 (receive-query-commits fl-tail))
          s-head)
    by (auto elim!: set-dist-bindE)
  have len-pos:  $0 < clength * scale$ 
    using eval-domain-nontrivial by linarith
  have idx-bound:  $?idx < clength * scale$ 
    by (rule index-less-domain)
  from query-opening-fri-layer-opening-step-partial-authenticated-table-first-value
    [OF len-pos idx-bound head]
  obtain composition-openings where
    comp-table-head:
      partial-authenticated-table composition-root (clength * scale)
        composition-openings s-head
    and comp-indices0:
      map opening-index composition-openings =
        [ $?idx$ ,  $(?idx + (clength * scale) \text{ div } 2) \bmod (clength * scale)$ ]
    and comp-len: length composition-openings = 2
    and comp-value:
      opening-value (composition-openings ! 0) =
        cp-eval as fv ( $h \wedge ?idx * shift$ )
    by blast
  from fri-layer-opening-step-outcome[OF head]
  have s3-s-head:  $s3 \leq s-head$ 
    by blast
  obtain i1 x1 len1 pw1 where out1-eq:  $out1 = (i1, x1, len1, pw1)$ 
    by (cases out1)
  from receive-query-commits-layers-outcome[OF tail[unfolded out1-eq]]

```

```

obtain tail-layer-chunks tail-chunk where s-head-s4: s-head ≤ s4
  by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of  $(-, x, -, -) \Rightarrow x = \text{final}$ )
    (auto simp: throw-no-outcome)
have s1-t: s1 ≤ t
  using s1-s2 s3-s-head s-head-s4 unfolding s3-eq t-eq
  by (meson hash-ext-trans)
have trace-table-t:
  partial-authenticated-table fr (scale * clength) trace-openings t
  by (rule partial-authenticated-table-mono[OF trace-table-s1 s1-t])
have comp-table-t:
  partial-authenticated-table composition-root (scale * clength)
    composition-openings t
  using partial-authenticated-table-mono[OF comp-table-head s-head-s4]
  unfolding t-eq by (simp add: mult.commute)
have comp-indices:
  map opening-index composition-openings =
    [?idx, fri-sibling-index (scale * clength) ?idx]
  using comp-indices0
  unfolding fri-sibling-index-def
  by (simp add: mult.commute)
have consistent:
  partial-query-openings-consistent trace-openings composition-openings
    as ?idx
  by (rule partial-query-openings-consistentI
    [OF idx-sample trace-indices comp-indices])
    (use comp-value trace-values in simp)
show ?thesis
  by (rule that[OF refl trace-indices trace-table-t comp-table-t
    comp-indices consistent])
qed

```

```

lemma verifier-query-round-after-index-empty-composition-consistent:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-empty: fl = []
and outcome:
  Some  $((\ ), t) \in$ 
    set-dist
    (execute
      (verifier-query-round-after-index-program
        fr f-fl f-final as fl final raw) s)
obtains idx trace-openings where
  idx = index (to-nat raw)
  map opening-index trace-openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) trace-openings t
  cp-eval as (map opening-value trace-openings) (h ^ idx * shift) =
    final

```

```

proof –
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
    query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
  and trace-fri:
    Some (f-out, s2) ∈
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
  and assert-trace:
    Some ((), s3) ∈
      set-dist
        (execute
          (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
  and comp-fri:
    Some (c-out, s4) ∈
      set-dist (execute
        (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
          clength * scale, 1)
          (receive-query-commits fl)) s3)
  and assert-comp:
    Some ((), t) ∈
      set-dist
        (execute
          (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
  unfolding verifier-query-round-after-index-program-def
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
  have idx-sample: ?idx ∈ query-sample-space
  using index-less-query-sample-space
  unfolding query-sample-space-def by simp
  from check-decommit-on-query-authenticated-openings
    [OF idx-sample query-decommit]
  obtain trace-openings where
    trace-values: map opening-value trace-openings = fv
  and trace-indices:
    map opening-index trace-openings = powers-scaled ?idx
  and trace-table-s1:
    partial-authenticated-table fr (scale * clength) trace-openings s1
  by blast
  from receive-query-commits-layers-outcome[OF trace-fri]
  obtain trace-layer-chunks trace-fri-chunk where s1-s2: s1 ≤ s2
  by blast
  have s3-eq: s3 = s2
  using assert-trace unfolding assert-def
  by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
    (auto simp: throw-no-outcome)
  have comp-out:

```



```

c-out =
  (?idx, cp-eval as fv (h ^ ?idx * shift), clength * scale, 1) ∧
  s4 = s3
using comp-fri fl-empty
unfolding receive-query-commits-def by simp
have final-eq:
  cp-eval as fv (h ^ ?idx * shift) = final
proof (rule ccontr)
  assume ¬ cp-eval as fv (h ^ ?idx * shift) = final
  then have Some ((), t) ∈ set-dist (execute throw s3)
    using assert-comp comp-out unfolding assert-def by simp
  then show False
    by (simp add: throw-no-outcome)
qed
have t-eq: t = s4
  using assert-comp comp-out final-eq unfolding assert-def by simp
have t-s2: t = s2
  using s3-eq t-eq comp-out by simp
have s1-t: s1 ≤ t
  using s1-s2 t-s2 by simp
have trace-table-t:
  partial-authenticated-table fr (scale * clength) trace-openings t
  by (rule partial-authenticated-table-mono[OF trace-table-s1 s1-t])
have final-eq':
  cp-eval as (map opening-value trace-openings) (h ^ ?idx * shift) =
    final
  using final-eq trace-values by simp
show ?thesis
  by (rule that[OF refl trace-indices trace-table-t final-eq'])
qed

lemma verifier-query-round-program-authenticated-openings-consistent-with-lookup:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-eq: fl = (b, composition-root) # fl-tail
and outcome:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
obtains raw idx trace-openings composition-openings where
  idx = index (to-nat raw)
  map opening-index trace-openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) trace-openings t
  partial-authenticated-table composition-root (scale * clength)
    composition-openings t
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
  partial-query-openings-consistent trace-openings composition-openings
    as idx

```

```

    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where rand:
  Some (raw, s0) ∈
    set-dist (execute receive-query-index-challenge s)
and tail:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
by (auto elim!: set-dist-bindE)
have lookup-s0:
  fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
using receive-query-index-challenge-outcome[OF rand] by simp
have lookup-t:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
using verifier-query-round-after-index-program-preserves-query-lookup
  [OF tail, of PQueryCounter s PState s] lookup-s0
by simp
show ?thesis
proof (rule verifier-query-round-after-index-authenticated-openings-consistent
  [OF fl-eq tail])
fix idx trace-openings composition-openings
assume idx-eq: idx = index (to-nat raw)
and trace-indices:
  map opening-index trace-openings = powers-scaled idx
and trace-table:
  partial-authenticated-table fr (scale * clength) trace-openings t
and comp-table:
  partial-authenticated-table composition-root (scale * clength)
    composition-openings t
and comp-indices:
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
and consistent:
  partial-query-openings-consistent trace-openings
    composition-openings as idx
show ?thesis
by (rule that[OF idx-eq trace-indices trace-table comp-table
  comp-indices consistent lookup-t])
qed
qed

```

lemma verifier-query-round-program-empty-composition-consistent-with-lookup:

```

fixes  $s\ t :: ('f, 'a) \text{ protocol-channel-scheme}$ 
assumes  $fl\text{-empty}$ :  $fl = []$ 
and  $outcome$ :
   $Some\ ((),\ t) \in$ 
     $set\text{-dist}$ 
     $(execute$ 
       $(verifier\text{-query-round-program}\ fr\ f\text{-}fl\ f\text{-}final\ as\ fl\ final)\ s)$ 
obtains  $raw\ idx\ trace\text{-openings}$  where
   $idx = index\ (to\text{-nat}\ raw)$ 
   $map\ opening\text{-index}\ trace\text{-openings} = powers\text{-scaled}\ idx$ 
   $partial\text{-authenticated-table}\ fr\ (scale * clength)\ trace\text{-openings}\ t$ 
   $cp\text{-eval}\ as\ (map\ opening\text{-value}\ trace\text{-openings})\ (h \wedge idx * shift) =$ 
     $final$ 
   $fmlookup\ (HashMap\ s)$ 
   $(QueryIndexChallenge\ (PQueryCounter\ s)\ (PState\ s)) = Some\ raw$ 
proof –
from  $outcome[unfolded\ verifier\text{-query-round-program-alt-def}]$ 
obtain  $raw\ s0$  where  $rand$ :
   $Some\ (raw,\ s0) \in$ 
     $set\text{-dist}\ (execute\ receive\text{-query-index-challenge}\ s)$ 
and  $tail$ :
   $Some\ ((),\ t) \in$ 
     $set\text{-dist}$ 
     $(execute$ 
       $(verifier\text{-query-round-after-index-program}$ 
         $fr\ f\text{-}fl\ f\text{-}final\ as\ fl\ final\ raw)\ s0)$ 
by  $(auto\ elim!:\ set\text{-dist}\text{-bindE})$ 
have  $lookup\text{-}s0$ :
   $fmlookup\ (HashMap\ s0)$ 
   $(QueryIndexChallenge\ (PQueryCounter\ s)\ (PState\ s)) = Some\ raw$ 
using  $receive\text{-query-index-challenge-outcome}[OF\ rand]$  by  $simp$ 
have  $lookup\text{-}t$ :
   $fmlookup\ (HashMap\ t)$ 
   $(QueryIndexChallenge\ (PQueryCounter\ s)\ (PState\ s)) = Some\ raw$ 
using  $verifier\text{-query-round-after-index-program-preserves-query-lookup}$ 
   $[OF\ tail,\ of\ PQueryCounter\ s\ PState\ s]\ lookup\text{-}s0$ 
by  $simp$ 
show  $?thesis$ 
proof  $(rule\ verifier\text{-query-round-after-index-empty-composition-consistent}$ 
   $[OF\ fl\text{-empty}\ tail])$ 
fix  $idx\ trace\text{-openings}$ 
assume  $idx\text{-eq}$ :  $idx = index\ (to\text{-nat}\ raw)$ 
and  $trace\text{-indices}$ :
   $map\ opening\text{-index}\ trace\text{-openings} = powers\text{-scaled}\ idx$ 
and  $trace\text{-table}$ :
   $partial\text{-authenticated-table}\ fr\ (scale * clength)\ trace\text{-openings}\ t$ 
and  $consistent$ :
   $cp\text{-eval}\ as\ (map\ opening\text{-value}\ trace\text{-openings})\ (h \wedge idx * shift) =$ 
     $final$ 

```

```

show ?thesis
  by (rule that[OF idx-eq trace-indices trace-table consistent
    lookup-t])
qed
qed

lemma ntimes-verifier-query-rounds-empty-composition-consistent:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-empty: fl = []
and outcome:
  Some (results, t) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final) n)
        s)
obtains raw-idxs query-idxs trace-openings where
  length raw-idxs = n
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  length trace-openings = n
   $\bigwedge i. i < n \implies$ 
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
   $\bigwedge i. i < n \implies$ 
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) t
   $\bigwedge i. i < n \implies$ 
    cp-eval as (map opening-value (trace-openings ! i))
      (h ^ (query-idxs ! i) * shift) = final
proof -
have
   $\exists$  raw-idxs query-idxs trace-openings.
    length raw-idxs = n ∧
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs ∧
    length trace-openings = n ∧
    ( $\forall i < n.$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)) ∧
    ( $\forall i < n.$ 
      partial-authenticated-table fr (scale * clength)
        (trace-openings ! i) t) ∧
    ( $\forall i < n.$ 
      cp-eval as (map opening-value (trace-openings ! i))
        (h ^ (query-idxs ! i) * shift) = final)
using outcome
proof (induction n arbitrary: s results t)
case 0
then show ?case
  by (intro exI[of - []] simp)

```

```

next
case (Suc n)
from Suc.premis obtain u results' s1 where
  head:
    Some (u, s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results', t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            n) s1)
  by (auto elim!: set-dist-bindE)
have u-eq: u = ()
  by (cases u) simp
from verifier-query-round-program-empty-composition-consistent-with-lookup
  [OF fl-empty head[unfolded u-eq]]
obtain raw idx openings where
  idx-eq: idx = index (to-nat raw)
  and idx-openings: map opening-index openings = powers-scaled idx
  and table-s1:
    partial-authenticated-table fr (scale * clength) openings s1
  and consistent-head:
    cp-eval as (map opening-value openings) (h ^ idx * shift) =
      final
  by blast
from Suc.IH[OF tail] obtain raw-tail idx-tail openings-tail where
  len-raw-tail: length raw-tail = n
  and idx-tail-eq:
    idx-tail = map (λraw. index (to-nat raw)) raw-tail
  and len-openings-tail: length openings-tail = n
  and tail-indices:
    ∀ i < n.
      map opening-index (openings-tail ! i) =
        powers-scaled (idx-tail ! i)
  and tail-tables:
    ∀ i < n.
      partial-authenticated-table fr (scale * clength)
        (openings-tail ! i) t
  and tail-consistent:
    ∀ i < n.
      cp-eval as (map opening-value (openings-tail ! i))
        (h ^ (idx-tail ! i) * shift) = final
  by blast
have s1-t: s1 ≤ t
  using ntimes-verifier-query-rounds-outcome[OF tail] by blast

```

```

have table-t:
  partial-authenticated-table fr (scale * clength) openings t
  by (rule partial-authenticated-table-mono[OF table-s1 s1-t])
let ?raws = raw # raw-tail
let ?idxs = idx # idx-tail
let ?openings = openings # openings-tail
have idxs-eq:
  ?idxs = map (λraw. index (to-nat raw)) ?raws
  using idx-eq idx-tail-eq by simp
have indices-all:
  ∀ i < Suc n.
    map opening-index (?openings ! i) =
      powers-scaled (?idxs ! i)
proof (intro allI impI)
  fix i
  assume i-bound: i < Suc n
  show map opening-index (?openings ! i) =
    powers-scaled (?idxs ! i)
  proof (cases i)
    case 0
    then show ?thesis
      using idx-openings by simp
  next
    case (Suc j)
    then show ?thesis
      using i-bound tail-indices by simp
  qed
qed
have tables-all:
  ∀ i < Suc n.
    partial-authenticated-table fr (scale * clength)
      (?openings ! i) t
proof (intro allI impI)
  fix i
  assume i-bound: i < Suc n
  show partial-authenticated-table fr (scale * clength)
    (?openings ! i) t
  proof (cases i)
    case 0
    then show ?thesis
      using table-t by simp
  next
    case (Suc j)
    then show ?thesis
      using i-bound tail-tables by simp
  qed
qed
have consistent-all:
  ∀ i < Suc n.

```

```

      cp-eval as (map opening-value (?openings ! i))
      (h ^ (?idxs ! i) * shift) = final
proof (intro allI impI)
  fix i
  assume i-bound: i < Suc n
  show cp-eval as (map opening-value (?openings ! i))
    (h ^ (?idxs ! i) * shift) = final
  proof (cases i)
    case 0
    then show ?thesis
    using consistent-head by simp
  next
    case (Suc j)
    then show ?thesis
    using i-bound tail-consistent by simp
  qed
qed
show ?case
  apply (intro exI[of - ?rows] exI[of - ?idxs]
    exI[of - ?openings] conjI)
  apply (simp add: len-row-tail)
  apply (rule idxs-eq)
  apply (simp add: len-openings-tail)
  apply (rule indices-all)
  apply (rule tables-all)
  apply (rule consistent-all)
  done
qed
then obtain raw-idxs query-idxs trace-openings where
  len-row: length raw-idxs = n
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-trace: length trace-openings = n
  and indices:
    ∀ i < n.
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
  and tables:
    ∀ i < n.
      partial-authenticated-table fr (scale * clength)
        (trace-openings ! i) t
  and consistent:
    ∀ i < n.
      cp-eval as (map opening-value (trace-openings ! i))
        (h ^ (query-idxs ! i) * shift) = final
  by blast
show ?thesis
  by (rule that[OF len-row query-idxs-eq len-trace])
    (use indices tables consistent in auto)

```

qed

lemma *ntimes-verifier-query-rounds-empty-composition-consistent-with-lookup*:

fixes $s\ t :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *fl-empty*: $fl = []$

and *outcome*:

Some $(results, t) \in$

set-dist

(execute

(ntimes

(verifier-query-round-program fr f-fl f-final as fl final) n)

s)

obtains *raw-idxs query-idxs query-chunks trace-openings* **where**

length raw-idxs = n

query-idxs = map ($\lambda raw. \text{index } (to\text{-nat } raw)$) raw-idxs

length query-chunks = n

length trace-openings = n

$\bigwedge i. i < n \implies$

fmlookup (HashMap t)

(QueryIndexChallenge (PQueryCounter s + i)

(state-after-query-chunks (PState s) query-chunks i)) =

Some (raw-idxs ! i)

$\bigwedge i. i < n \implies$

map opening-index (trace-openings ! i) =

powers-scaled (query-idxs ! i)

$\bigwedge i. i < n \implies$

*partial-authenticated-table fr (scale * clength)*

(trace-openings ! i) t

$\bigwedge i. i < n \implies$

cp-eval as (map opening-value (trace-openings ! i))

*(h ^ (query-idxs ! i) * shift) = final*

proof –

have

$\exists \text{raw-idxs query-idxs query-chunks trace-openings.}$

length raw-idxs = n $\wedge$

query-idxs = map ($\lambda raw. \text{index } (to\text{-nat } raw)$) raw-idxs $\wedge$

length query-chunks = n $\wedge$

length trace-openings = n $\wedge$

$(\forall i < n.$

fmlookup (HashMap t)

(QueryIndexChallenge (PQueryCounter s + i)

(state-after-query-chunks (PState s) query-chunks i)) =

Some (raw-idxs ! i)) $\wedge$

$(\forall i < n.$

map opening-index (trace-openings ! i) =

powers-scaled (query-idxs ! i)) $\wedge$

$(\forall i < n.$

*partial-authenticated-table fr (scale * clength)*

(trace-openings ! i) t) $\wedge$


```

  (∀ i < n.
    cp-eval as (map opening-value (trace-openings ! i))
      (h ^ (query-idxs ! i) * shift) = final)
using outcome
proof (induction n arbitrary: s results t)
  case 0
  then show ?case
    by (intro exI[of - []]) simp
next
  case (Suc n)
from Suc.premis obtain u results' s1 where
  head:
    Some (u, s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results', t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            n) s1)
    by (auto elim!: set-dist-bindE)
have u-eq: u = ()
  by (cases u) simp
from verifier-query-round-program-empty-composition-consistent-with-lookup
  [OF fl-empty head[unfolded u-eq]]
obtain raw idx openings where
  idx-eq: idx = index (to-nat raw)
  and idx-openings: map opening-index openings = powers-scaled idx
  and table-s1:
    partial-authenticated-table fr (scale * clength) openings s1
  and consistent-head:
    cp-eval as (map opening-value openings) (h ^ idx * shift) =
      final
  and lookup-s1:
    fmlookup (HashMap s1)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  by blast
from verifier-query-round-program-outcome[OF head[unfolded u-eq]]
obtain raw' idx' chunk where
  idx'-eq: idx' = index (to-nat raw')
  and st-s1: PState s1 = foldl concat (PState s) chunk
  and query-count-s1: PQueryCounter s1 = Suc (PQueryCounter s)
  and lookup-s1':
    fmlookup (HashMap s1)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw'
  by blast

```

```

have raw'-eq: raw' = raw
  using lookup-s1 lookup-s1' by simp
from Suc.IH[OF tail] obtain raw-tail idx-tail chunk-tail openings-tail
  where len-raw-tail: length raw-tail = n
  and idx-tail-eq:
    idx-tail = map ( $\lambda$ raw. index (to-nat raw)) raw-tail
  and len-chunk-tail: length chunk-tail = n
  and len-openings-tail: length openings-tail = n
  and lookup-tail:
     $\forall i < n.$ 
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s1 + i)
        (state-after-query-chunks (PState s1) chunk-tail i)) =
      Some (raw-tail ! i)
  and tail-indices:
     $\forall i < n.$ 
    map opening-index (openings-tail ! i) =
      powers-scaled (idx-tail ! i)
  and tail-tables:
     $\forall i < n.$ 
    partial-authenticated-table fr (scale * clength)
      (openings-tail ! i) t
  and tail-consistent:
     $\forall i < n.$ 
    cp-eval as (map opening-value (openings-tail ! i))
      (h ^ (idx-tail ! i) * shift) = final
  by blast
have s1-t: s1 ≤ t
  using ntimes-verifier-query-rounds-outcome[OF tail] by blast
have table-t:
  partial-authenticated-table fr (scale * clength) openings t
  by (rule partial-authenticated-table-mono[OF table-s1 s1-t])
have lookup-head-t:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  using hash-extension-lookup[OF lookup-s1 s1-t] .
let ?raws = raw # raw-tail
let ?idxs = idx # idx-tail
let ?chunks = chunk # chunk-tail
let ?openings = openings # openings-tail
have idxs-eq:
  ?idxs = map ( $\lambda$ raw. index (to-nat raw)) ?raws
  using idx-eq idx-tail-eq by simp
have state-shift:
   $\bigwedge i.$  state-after-query-chunks (PState s) ?chunks (Suc i) =
    state-after-query-chunks (PState s1) chunk-tail i
  using st-s1 unfolding state-after-query-chunks-def by simp
have lookup-all:
   $\forall i < \text{Suc } n.$ 

```

```

    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) ?chunks i)) =
      Some (?rows ! i)
  proof (intro allI impI)
    fix i
    assume i-bound: i < Suc n
    show fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) ?chunks i)) =
      Some (?rows ! i)
  proof (cases i)
    case 0
    then show ?thesis
      using lookup-head-t unfolding state-after-query-chunks-def by simp
  next
    case (Suc j)
    then show ?thesis
      using i-bound lookup-tail query-count-s1 state-shift[of j] by simp
  qed
qed
have indices-all:
  ∀ i < Suc n.
    map opening-index (?openings ! i) =
    powers-scaled (?idxs ! i)
  proof (intro allI impI)
    fix i
    assume i-bound: i < Suc n
    show map opening-index (?openings ! i) =
      powers-scaled (?idxs ! i)
  proof (cases i)
    case 0
    then show ?thesis
      using idx-openings by simp
  next
    case (Suc j)
    then show ?thesis
      using i-bound tail-indices by simp
  qed
qed
have tables-all:
  ∀ i < Suc n.
    partial-authenticated-table fr (scale * clength)
      (?openings ! i) t
  proof (intro allI impI)
    fix i
    assume i-bound: i < Suc n
    show partial-authenticated-table fr (scale * clength)
      (?openings ! i) t

```

```

proof (cases i)
  case 0
  then show ?thesis
    using table-t by simp
next
  case (Suc j)
  then show ?thesis
    using i-bound tail-tables by simp
qed
qed
have consistent-all:
   $\forall i < \text{Suc } n.$ 
  cp-eval as (map opening-value (?openings ! i))
  (h ^ (?idxs ! i) * shift) = final
proof (intro allI impI)
  fix i
  assume i-bound: i < Suc n
  show cp-eval as (map opening-value (?openings ! i))
  (h ^ (?idxs ! i) * shift) = final
  proof (cases i)
    case 0
    then show ?thesis
      using consistent-head by simp
  next
    case (Suc j)
    then show ?thesis
      using i-bound tail-consistent by simp
  qed
qed
show ?case
  by (intro exI[of - ?rows] exI[of - ?idxs] exI[of - ?chunks]
    exI[of - ?openings] conjI)
  (use len-row-tail idxs-eq len-chunk-tail len-openings-tail
    lookup-all indices-all tables-all consistent-all in simp-all)
qed
then obtain raw-idxs query-idxs query-chunks trace-openings where
  len-row: length raw-idxs = n
  and query-idxs-eq:
    query-idxs = map ( $\lambda \text{raw}. \text{index } (\text{to-nat } \text{raw})$ ) raw-idxs
  and len-chunks: length query-chunks = n
  and len-trace: length trace-openings = n
  and lookup:
     $\forall i < n.$ 
    fmllookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) query-chunks i)) =
      Some (raw-idxs ! i)
  and indices:
     $\forall i < n.$ 

```

```

    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
and tables:
  ∀ i < n.
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) t
and consistent:
  ∀ i < n.
    cp-eval as (map opening-value (trace-openings ! i))
      (h ^ (query-idxs ! i) * shift) = final
by blast
show ?thesis
by (rule that[OF len-row query-idxs-eq len-chunks len-trace])
    (use lookup indices tables consistent in auto)
qed

lemma ntimes-verifier-query-rounds-empty-composition-first-consistent:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-empty: fl = []
and outcome:
  Some (results, t) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        s)
obtains raw trace-openings where
  map opening-index trace-openings =
    powers-scaled (index (to-nat raw))
  partial-authenticated-table fr (scale * clength) trace-openings t
  cp-eval as (map opening-value trace-openings)
    (h ^ index (to-nat raw) * shift) = final
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof –
from outcome obtain u results' s1 where
  head:
    Some (u, s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
and tail:
    Some (results', t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            (rounds – Suc 0))
          s)

```

```

      s1)
    using rounds-positive
    by (cases rounds) (auto elim!: set-dist-bindE)
  have u-eq: u = ()
    by (cases u) simp
  from verifier-query-round-program-empty-composition-consistent-with-lookup
    [OF fl-empty head[unfolded u-eq]]
  obtain raw idx trace-openings where idx-eq:
    idx = index (to-nat raw)
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and trace-table-s1:
    partial-authenticated-table fr (scale * clength) trace-openings s1
  and consistent:
    cp-eval as (map opening-value trace-openings) (h ^ idx * shift) =
      final
  and lookup-s1:
    fmlookup (HashMap s1)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  by blast
  have s1-t: s1 ≤ t
    using ntimes-verifier-query-rounds-outcome[OF tail] by blast
  have trace-table-t:
    partial-authenticated-table fr (scale * clength) trace-openings t
  by (rule partial-authenticated-table-mono[OF trace-table-s1 s1-t])
  have lookup-t:
    fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  using hash-extension-lookup[OF lookup-s1 s1-t] .
  show ?thesis
  by (rule that[of trace-openings raw])
    (use idx-eq trace-indices trace-table-t consistent lookup-t in simp-all)
qed

```

```

lemma ntimes-verifier-query-rounds-authenticated-openings-consistent-at:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes fl-eq: fl = (b, composition-root) # fl-tail
  and outcome:
    Some (results, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s)
  and i-bound: i < n
  obtains raw idx prefix-chunks trace-openings composition-openings where
    idx = index (to-nat raw)
    length prefix-chunks = i
    map opening-index trace-openings = powers-scaled idx

```

```

partial-authenticated-table fr (scale * clength) trace-openings t
partial-authenticated-table composition-root (scale * clength)
  composition-openings t
map opening-index composition-openings =
  [idx, fri-sibling-index (scale * clength) idx]
partial-query-openings-consistent trace-openings composition-openings
  as idx
fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s + i)
    (state-after-query-chunks (PState s) prefix-chunks i)) = Some raw
proof –
from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
obtain prefix round suffix s-i s-suc where
  prefix:
    Some (prefix, s-i) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) i)
          s)
and round:
    Some (round, s-suc) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s-i)
and suffix:
    Some (suffix, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            (n – Suc i))
          s-suc)
by blast
have round-unit: round = ()
by (cases round) simp
show ?thesis
proof (rule
  verifier-query-round-program-authenticated-openings-consistent-with-lookup
    [OF fl-eq round[unfolded round-unit]])
fix raw idx trace-openings composition-openings
assume idx-eq: idx = index (to-nat raw)
and trace-indices:
  map opening-index trace-openings = powers-scaled idx
and trace-table-suc:
  partial-authenticated-table fr (scale * clength)
    trace-openings s-suc
and comp-table-suc:
  partial-authenticated-table composition-root (scale * clength)

```

```

    composition-openings s-suc
  and comp-indices:
    map opening-index composition-openings =
      [idx, fri-sibling-index (scale * clength) idx]
  and consistent:
    partial-query-openings-consistent trace-openings
      composition-openings as idx
  and lookup-suc:
    fmllookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
        Some raw
show ?thesis
proof (rule query-opening-ntimes-prefix-state-alignment[OF prefix])
  fix prefix-chunks
  assume len-chunks: length prefix-chunks = i
  and state-i:
    PState s-i =
      state-after-query-chunks (PState s) prefix-chunks i
  and counter-i: PQueryCounter s-i = PQueryCounter s + i
  have suffix-ext: s-suc ≤ t
    by (rule query-opening-ntimes-suffix-extends[OF suffix])
  have trace-table-t:
    partial-authenticated-table fr (scale * clength)
      trace-openings t
    by (rule
      partial-authenticated-table-mono[OF trace-table-suc suffix-ext])
  have comp-table-t:
    partial-authenticated-table composition-root (scale * clength)
      composition-openings t
    by (rule
      partial-authenticated-table-mono[OF comp-table-suc suffix-ext])
  have lookup-t:
    fmllookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) prefix-chunks i)) =
      Some raw
  using hash-extension-lookup[OF lookup-suc suffix-ext]
  unfolding counter-i state-i .
show ?thesis
  by (rule that[OF idx-eq len-chunks trace-indices trace-table-t
    comp-table-t comp-indices consistent lookup-t])
qed
qed
qed

lemma ntimes-verifier-query-rounds-empty-composition-consistent-at:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes fl-empty: fl = []
  and outcome:

```



```

    Some (results, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s)
  and i-bound: i < n
obtains raw idx prefix-chunks trace-openings where
  idx = index (to-nat raw)
  length prefix-chunks = i
  map opening-index trace-openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) trace-openings t
  cp-eval as (map opening-value trace-openings) (h ^ idx * shift) =
    final
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks (PState s) prefix-chunks i)) = Some raw
proof –
from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
obtain prefix round suffix s-i s-suc where
  prefix:
    Some (prefix, s-i) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) i)
          s)
  and round:
    Some (round, s-suc) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s-i)
  and suffix:
    Some (suffix, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
              (n – Suc i))
          s-suc)
  by blast
have round-unit: round = ()
  by (cases round) simp
show ?thesis
proof (rule
  verifier-query-round-program-empty-composition-consistent-with-lookup
    [OF fl-empty round[unfolded round-unit]])
  fix raw idx trace-openings
  assume idx-eq: idx = index (to-nat raw)

```

```

and trace-indices:
  map opening-index trace-openings = powers-scaled idx
and trace-table-suc:
  partial-authenticated-table fr (scale * clength)
  trace-openings s-suc
and consistent:
  cp-eval as (map opening-value trace-openings) (h ^ idx * shift) =
  final
and lookup-suc:
  fmlookup (HashMap s-suc)
  (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
  Some raw
show ?thesis
proof (rule query-opening-ntimes-prefix-state-alignment[OF prefix])
fix prefix-chunks
assume len-chunks: length prefix-chunks = i
and state-i:
  PState s-i =
  state-after-query-chunks (PState s) prefix-chunks i
and counter-i: PQueryCounter s-i = PQueryCounter s + i
have suffix-ext: s-suc ≤ t
by (rule query-opening-ntimes-suffix-extends[OF suffix])
have trace-table-t:
  partial-authenticated-table fr (scale * clength)
  trace-openings t
by (rule
  partial-authenticated-table-mono[OF trace-table-suc suffix-ext])
have lookup-t:
  fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s + i)
  (state-after-query-chunks (PState s) prefix-chunks i)) =
  Some raw
using hash-extension-lookup[OF lookup-suc suffix-ext]
unfolding counter-i state-i .
show ?thesis
by (rule that[OF idx-eq len-chunks trace-indices trace-table-t
  consistent lookup-t])
qed
qed
qed

```

```

lemma ntimes-verifier-query-rounds-selected-authenticated-chunks-at:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-eq: fl = (b, composition-root) # fl-tail
and outcome:
  Some (results, t) ∈
  set-dist
  (execute
  (ntimes

```

```

      (verifier-query-round-program fr f-fl f-final as fl final) n)
    s)
  and i-bound:  $i < n$ 
obtains raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
  suffix-chunks trace-openings composition-openings where
  idx = index (to-nat raw)
  length prefix-query-idxs = i
  length prefix-chunks = i
  length suffix-query-idxs =  $n - \text{Suc } i$ 
  length suffix-chunks =  $n - \text{Suc } i$ 
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
   $\bigwedge j. j < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! j)
    (map snd f-fl) (map snd fl) (prefix-chunks ! j)
   $\bigwedge j. j < n - \text{Suc } i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! j)
    (map snd f-fl) (map snd fl) (suffix-chunks ! j)
  PTranscript s =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @ PTranscript t
  map opening-index trace-openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) trace-openings t
  partial-authenticated-table composition-root (scale * clength)
    composition-openings t
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
  partial-query-openings-consistent trace-openings composition-openings
    as idx
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks (PState s) prefix-chunks i)) = Some raw
proof –
from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
obtain prefix round suffix s-i s-suc where
  prefix:
    Some (prefix, s-i)  $\in$ 
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) i)
          s)
  and round:
    Some (round, s-suc)  $\in$ 
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s-i)
  and suffix:
    Some (suffix, t)  $\in$ 
      set-dist
        (execute

```

```

      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        (n - Suc i))
      s-suc)
  by blast
have round-unit: round = ()
  by (cases round) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
  fix prefix-query-idxs prefix-chunks
  assume len-prefix: length prefix-query-idxs = i
  and len-prefix-chunks: length prefix-chunks = i
  and transcript-prefix:
    PTranscript s = List.concat prefix-chunks @ PTranscript s-i
  and state-i:
    PState s-i =
      state-after-query-chunks (PState s) prefix-chunks i
  and counter-i: PQueryCounter s-i = PQueryCounter s + i
  and prefix-ext: s ≤ s-i
  and prefix-rounds:
    ∧j. j < i ⇒
      verifier-query-round-chunk (prefix-query-idxs ! j)
      (map snd f-fl) (map snd fl) (prefix-chunks ! j)
  show ?thesis
proof (rule
  query-opening-verifier-query-round-program-authenticated-with-chunk
  [OF fl-eq round[unfolded round-unit]])
  fix raw idx chunk trace-leaves trace-openings composition-openings
  assume idx-eq: idx = index (to-nat raw)
  and chunk-shape:
    verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
    chunk
  and transcript-round:
    PTranscript s-i = chunk @ PTranscript s-suc
  and state-suc:
    PState s-suc = foldl concat (PState s-i) chunk
  and round-ext: s-i ≤ s-suc
  and counter-suc: PQueryCounter s-suc = Suc (PQueryCounter s-i)
  and trace-values:
    map opening-value trace-openings = trace-leaves
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and trace-table-suc:
    partial-authenticated-table fr (scale * clength)
    trace-openings s-suc
  and comp-table-suc:
    partial-authenticated-table composition-root (scale * clength)
    composition-openings s-suc

```

```

and comp-indices:
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
and lookup-suc:
  fmllookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
show ?thesis
proof (rule verifier-query-round-program-authenticated-openings-consistent-with-lookup
  [OF fl-eq round[unfolded round-unit]])
fix raw' idx' trace-openings' composition-openings'
assume idx'-eq: idx' = index (to-nat raw')
and trace-indices':
  map opening-index trace-openings' = powers-scaled idx'
and trace-table-suc':
  partial-authenticated-table fr (scale * clength)
    trace-openings' s-suc
and comp-table-suc':
  partial-authenticated-table composition-root (scale * clength)
    composition-openings' s-suc
and comp-indices':
  map opening-index composition-openings' =
    [idx', fri-sibling-index (scale * clength) idx']
and consistent':
  partial-query-openings-consistent trace-openings'
    composition-openings' as idx'
and lookup-suc':
  fmllookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw'
have raw'-eq: raw' = raw
  using lookup-suc lookup-suc' by simp
have idx'-idx: idx' = idx
  using idx'-eq idx-eq raw'-eq by simp
have suffix-ext: s-suc ≤ t
  by (rule query-opening-ntimes-suffix-extends[OF suffix])
have trace-table-t:
  partial-authenticated-table fr (scale * clength)
    trace-openings' t
  by (rule partial-authenticated-table-mono
    [OF trace-table-suc' suffix-ext])
have comp-table-t:
  partial-authenticated-table composition-root (scale * clength)
    composition-openings' t
  by (rule partial-authenticated-table-mono
    [OF comp-table-suc' suffix-ext])
have lookup-t:
  fmllookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + i)

```

```

      (state-after-query-chunks (PState s) prefix-chunks i)) =
      Some raw
    using hash-extension-lookup[OF lookup-suc suffix-ext]
    unfolding counter-i state-i .
  show ?thesis
  proof (rule query-opening-ntimes-suffix-transcript-state-alignment
    [OF suffix])
    fix suffix-query-idxs suffix-chunks
    assume len-suffix-idxs:
      length suffix-query-idxs = n - Suc i
    and len-suffix-chunks:
      length suffix-chunks = n - Suc i
    and transcript-suffix:
      PTranscript s-suc =
        List.concat suffix-chunks @ PTranscript t
    and suffix-state:
      PState t =
        state-after-query-chunks (PState s-suc) suffix-chunks
        (n - Suc i)
    and suffix-counter:
      PQueryCounter t = PQueryCounter s-suc + (n - Suc i)
    and suffix-ext2: s-suc ≤ t
    and suffix-rounds:
       $\bigwedge j. j < n - \text{Suc } i \implies$ 
        verifier-query-round-chunk (suffix-query-idxs ! j)
        (map snd f-fl) (map snd fl) (suffix-chunks ! j)
    have transcript-all:
      PTranscript s =
        List.concat (prefix-chunks @ chunk # suffix-chunks) @
        PTranscript t
    using transcript-prefix transcript-round transcript-suffix
    by simp
    show ?thesis
    by (rule that[OF idx-eq len-prefix len-prefix-chunks
      len-suffix-idxs len-suffix-chunks chunk-shape - -
      transcript-all - trace-table-t comp-table-t - - lookup-t])
      (use trace-indices' comp-indices' consistent' idx'-idx
        prefix-rounds suffix-rounds in simp-all)
  qed
qed
qed
qed
qed
qed

lemma ntimes-verifier-query-rounds-aligned-authenticated-openings-consistent:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes fl-eq: fl = (b, composition-root) # fl-tail
  and outcome:
    Some (results, t) ∈

```

```

    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        s)
  obtains query-idxs trace-openings composition-openings where
    length query-idxs = rounds
    length trace-openings = rounds
    length composition-openings = rounds
     $\bigwedge i. i < \text{rounds} \implies$ 
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
     $\bigwedge i. i < \text{rounds} \implies$ 
      partial-authenticated-table fr (scale * clength)
        (trace-openings ! i) t
     $\bigwedge i. i < \text{rounds} \implies$ 
      partial-authenticated-table composition-root (scale * clength)
        (composition-openings ! i) t
     $\bigwedge i. i < \text{rounds} \implies$ 
      map opening-index (composition-openings ! i) =
        [query-idxs ! i,
          fri-sibling-index (scale * clength) (query-idxs ! i)]
     $\bigwedge i. i < \text{rounds} \implies$ 
      partial-query-round-consistent trace-openings composition-openings as i
        (query-idxs ! i)
  proof (cases rounds = 0)
  case True
  show ?thesis
  by (rule that[of [] [] []]) (use True in simp-all)
next
case nonzero: False
let ?P =
   $\lambda i w. \text{map opening-index (fst (snd w))} = \text{powers-scaled (fst w)} \wedge$ 
   $\text{partial-authenticated-table fr (scale * clength) (fst (snd w)) } t \wedge$ 
   $\text{partial-authenticated-table composition-root (scale * clength)}$ 
   $(\text{snd (snd w)}) t \wedge$ 
   $\text{map opening-index (snd (snd w))} =$ 
   $[\text{fst w, fri-sibling-index (scale * clength) (fst w)}] \wedge$ 
   $\text{partial-query-openings-consistent (fst (snd w)) (snd (snd w))}$ 
   $\text{as (fst w)}$ 
let ?wit =  $\lambda i. \text{SOME } w. ?P i w$ 
let ?idx =  $\lambda i. \text{fst (?wit } i)$ 
let ?trace-openings =  $\lambda i. \text{fst (snd (?wit } i))$ 
let ?composition-openings =  $\lambda i. \text{snd (snd (?wit } i))$ 
let ?query-idxs = map ?idx [0..

```

```

 $\bigwedge i. i < \text{rounds} \implies$ 
   $\exists w. ?P\ i\ w$ 
proof –
  fix  $i$ 
  assume  $i\text{-bound}: i < \text{rounds}$ 
  show  $\exists w. ?P\ i\ w$ 
  proof (rule ntimes-verifier-query-rounds-selected-authenticated-chunks-at
    [OF fl-eq outcome i-bound])
    fix  $\text{raw}\ \text{idx}\ \text{prefix-query-idxs}\ \text{prefix-chunks}\ \text{chunk}\ \text{suffix-query-idxs}$ 
       $\text{suffix-chunks}\ \text{trace-openings}\ \text{composition-openings}$ 
    assume  $\text{idx-eq}: \text{idx} = \text{index}\ (\text{to-nat}\ \text{raw})$ 
    and  $\text{trace-indices}$ :
       $\text{map}\ \text{opening-index}\ \text{trace-openings} = \text{powers-scaled}\ \text{idx}$ 
    and  $\text{trace-table}$ :
       $\text{partial-authenticated-table}\ \text{fr}\ (\text{scale} * \text{clength})\ \text{trace-openings}\ t$ 
    and  $\text{comp-table}$ :
       $\text{partial-authenticated-table}\ \text{composition-root}\ (\text{scale} * \text{clength})$ 
         $\text{composition-openings}\ t$ 
    and  $\text{comp-indices}$ :
       $\text{map}\ \text{opening-index}\ \text{composition-openings} =$ 
         $[\text{idx}, \text{fri-sibling-index}\ (\text{scale} * \text{clength})\ \text{idx}]$ 
    and  $\text{consistent}$ :
       $\text{partial-query-openings-consistent}\ \text{trace-openings}$ 
         $\text{composition-openings}\ \text{as}\ \text{idx}$ 
    show  $?thesis$ 
    by (intro  $\text{exI}[\text{of} - (\text{idx}, \text{trace-openings}, \text{composition-openings})]$ 
       $\text{conjI}$ )
      (use  $\text{idx-eq}\ \text{trace-indices}\ \text{trace-table}\ \text{comp-table}\ \text{comp-indices}$ 
         $\text{consistent}$  in  $\text{simp-all}$ )

  qed
qed
have  $\text{round-props}$ :
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{map}\ \text{opening-index}\ (?trace\text{-tables}\ !\ i) =$ 
       $\text{powers-scaled}\ (?query\text{-idxs}\ !\ i) \wedge$ 
       $\text{partial-authenticated-table}\ \text{fr}\ (\text{scale} * \text{clength})$ 
         $(?trace\text{-tables}\ !\ i)\ t \wedge$ 
       $\text{partial-authenticated-table}\ \text{composition-root}\ (\text{scale} * \text{clength})$ 
         $(?comp\text{-tables}\ !\ i)\ t \wedge$ 
       $\text{map}\ \text{opening-index}\ (?comp\text{-tables}\ !\ i) =$ 
         $[\text{?query-idxs}\ !\ i,$ 
           $\text{fri-sibling-index}\ (\text{scale} * \text{clength})\ (?query\text{-idxs}\ !\ i)] \wedge$ 
       $\text{partial-query-round-consistent}\ ?trace\text{-tables}\ ?comp\text{-tables}\ \text{as}\ i$ 
         $(?query\text{-idxs}\ !\ i)$ 

  proof –
  fix  $i$ 
  assume  $i\text{-bound}: i < \text{rounds}$ 
  have  $\text{chosen}: ?P\ i\ (?wit\ i)$ 
  by (rule  $\text{someI-ex}[\text{OF}\ \text{ex-round}[\text{OF}\ i\text{-bound}]]$ )

```



```

have nths:
  ?query-idxs ! i = ?idx i
  ?trace-tables ! i = ?trace-openings i
  ?comp-tables ! i = ?composition-openings i
  using i-bound by simp-all
have lengths:
  length ?trace-tables = rounds
  length ?comp-tables = rounds
  by simp-all
show
  map opening-index (?trace-tables ! i) =
    powers-scaled (?query-idxs ! i) ∧
    partial-authenticated-table fr (scale * clength)
    (?trace-tables ! i) t ∧
    partial-authenticated-table composition-root (scale * clength)
    (?comp-tables ! i) t ∧
  map opening-index (?comp-tables ! i) =
    [?query-idxs ! i,
     fri-sibling-index (scale * clength) (?query-idxs ! i)] ∧
  partial-query-round-consistent ?trace-tables ?comp-tables as i
    (?query-idxs ! i)
  using chosen i-bound nths lengths
  unfolding partial-query-round-consistent-def
    partial-query-openings-consistent-def
  by simp
qed
show ?thesis
  by (rule that[of ?query-idxs ?trace-tables ?comp-tables])
    (use round-props in simp-all)
qed

end

end

```

```

theory Soundness-Partial-Initial-Aligned
  imports Soundness-Query-Opening-Consistency
begin

```

Aligned partial initial-opening evidence.

The predicate *accepted-with-partial-initial-openings* records trace and composition openings with separate query-index lists. For query soundness bounds this theory packages the stronger verifier-local fact that both opening families come from the same query-round execution and therefore use the same sampled query indices.

```

context soundness
begin

```

```

definition accepted-with-partial-initial-openings-aligned

```

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ authenticated-opening list list} \Rightarrow$
 $'f \text{ authenticated-opening list list} \Rightarrow \text{bool}$

where

$\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots final query-idxs trace-openings}$
 $\text{composition-openings} \longleftrightarrow$
 $\text{accepted out} \wedge$
 $\text{composition-fri-roots} \neq [] \wedge$
 $(\exists \text{ rest.}$
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final as dg}$
 $\text{composition-fri-roots final rest}) \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{accepted-with-partial-composition-openings } s \text{ out}$
 $(\text{hd composition-fri-roots}) \text{ query-idxs composition-openings}$

lemma *accepted-with-partial-initial-openings-aligned-imp-unaligned:*

assumes

$\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots final query-idxs trace-openings}$
 $\text{composition-openings}$

shows

$\text{accepted-with-partial-initial-openings } s \text{ out fr } f\text{-fri-roots } f\text{-final as dg}$
 $\text{composition-fri-roots final query-idxs trace-openings query-idxs}$
 $\text{composition-openings}$

using *assms*

unfolding *accepted-with-partial-initial-openings-aligned-def*

accepted-with-partial-initial-openings-def

by *blast*

lemma *accepted-with-partial-initial-openings-aligned-shapes:*

assumes

$\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots final query-idxs trace-openings}$
 $\text{composition-openings}$

shows $\text{composition-fri-roots} \neq []$

and $\exists \text{ rest. verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final as dg}$
 $\text{composition-fri-roots final rest}$

and $\text{accepted-with-partial-trace-openings } s \text{ out fr query-idxs}$
 trace-openings

and $\text{accepted-with-partial-composition-openings } s \text{ out}$
 $(\text{hd composition-fri-roots}) \text{ query-idxs composition-openings}$

using *assms*

unfolding *accepted-with-partial-initial-openings-aligned-def* **by** *blast+*

lemma *accepted-with-partial-initial-openings-aligned-candidates-if-no-partial-merkle-bad:*

assumes *partial:*

accepted-with-partial-initial-openings-aligned s
(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final query-idxs trace-openings
composition-openings

and *no-bad:*

$\neg$ *partial-merkle-inconsistency-bad s*
(Some (result, final-state))

shows $\exists$ *trace-table composition-table.*

partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings

proof –

have *unaligned:*

accepted-with-partial-initial-openings s
(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final query-idxs trace-openings query-idxs
composition-openings

by (rule *accepted-with-partial-initial-openings-aligned-imp-unaligned*
 $[OF\ partial]$)

show *?thesis*

by (rule
accepted-with-partial-initial-openings-candidates-if-no-partial-merkle-bad
 $[OF\ unaligned\ no-bad]$)

qed

lemma *verify-monad-accepted-with-partial-initial-openings-aligned:*

fixes *s :: ('f, 'a) protocol-channel-scheme*

assumes *outcome:*

Some (result, final-state) $\in$ set-dist (execute verify-monad s)

and *header0:*

verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest

and *comp-nonempty: composition-fri-roots $\neq []$*

obtains *query-idxs trace-openings composition-openings* **where**

accepted-with-partial-initial-openings-aligned s
(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final query-idxs trace-openings
composition-openings

$\wedge i. i < rounds \implies$

partial-query-round-consistent trace-openings composition-openings as i
(query-idxs ! i)

proof –

from *verify-monad-header-extraction* $[OF\ outcome]$

obtain *fr' f-fl f-final' as' dg' fl final' query-state* **where**

header:

verifier-header-transcript s fr' (map snd f-fl) f-final' as' dg'
(map snd fl) final' (PTranscript query-state)

```

and query-out:
  Some (result, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr' f-fl f-final' as' fl
            final') rounds)
        query-state)
by metis
have eqs:
  fr' = fr ∧
  map snd f-fl = f-fri-roots ∧
  f-final' = f-final ∧
  as' = as ∧
  dg' = dg ∧
  map snd fl = composition-fri-roots ∧
  final' = final ∧
  PTranscript query-state = rest
using verifier-header-transcript-unique[OF header header0] by simp
have fl-nonempty: fl ≠ []
using comp-nonempty eqs by (cases fl) simp-all
then obtain b composition-root fl-tail where fl-eq:
  fl = (b, composition-root) # fl-tail
by (cases fl) auto
have root-eq: composition-root = hd composition-fri-roots
proof –
  have composition-root # map snd fl-tail = composition-fri-roots
    using eqs unfolding fl-eq by simp
  then have hd composition-fri-roots = composition-root
    by (metis list.sel(1))
  then show ?thesis
    by simp
qed
obtain query-idxs trace-openings composition-openings where
  len-query-idxs: length query-idxs = rounds
  and len-trace: length trace-openings = rounds
  and len-comp: length composition-openings = rounds
  and trace-indices:
    ∧ i. i < rounds ⇒
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
  and trace-tables:
    ∧ i. i < rounds ⇒
      partial-authenticated-table fr' (scale * clength)
        (trace-openings ! i) final-state
  and comp-tables:
    ∧ i. i < rounds ⇒
      partial-authenticated-table composition-root (scale * clength)
        (composition-openings ! i) final-state

```

```

and comp-indices:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{map } \text{opening-index } (\text{composition-openings} ! i) =$ 
       $[\text{query-idxs} ! i,$ 
         $\text{fri-sibling-index } (\text{scale} * \text{clength}) (\text{query-idxs} ! i)]$ 
and consistent:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{partial-query-round-consistent } \text{trace-openings } \text{composition-openings}$ 
     $\text{as}' i (\text{query-idxs} ! i)$ 
proof (rule
  ntimes-verifier-query-rounds-aligned-authenticated-openings-consistent
  [OF fl-eq query-out])
fix query-idxs trace-openings composition-openings
assume len-query-idxs:  $\text{length } \text{query-idxs} = \text{rounds}$ 
and len-trace:  $\text{length } \text{trace-openings} = \text{rounds}$ 
and len-comp:  $\text{length } \text{composition-openings} = \text{rounds}$ 
and trace-indices:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{map } \text{opening-index } (\text{trace-openings} ! i) =$ 
     $\text{powers-scaled } (\text{query-idxs} ! i)$ 
and trace-tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{partial-authenticated-table } \text{fr}' (\text{scale} * \text{clength})$ 
     $(\text{trace-openings} ! i) \text{ final-state}$ 
and comp-tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{partial-authenticated-table } \text{composition-root } (\text{scale} * \text{clength})$ 
     $(\text{composition-openings} ! i) \text{ final-state}$ 
and comp-indices:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{map } \text{opening-index } (\text{composition-openings} ! i) =$ 
     $[\text{query-idxs} ! i,$ 
       $\text{fri-sibling-index } (\text{scale} * \text{clength}) (\text{query-idxs} ! i)]$ 
and consistent:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{partial-query-round-consistent } \text{trace-openings } \text{composition-openings}$ 
     $\text{as}' i (\text{query-idxs} ! i)$ 
show ?thesis
  by (rule that[OF len-query-idxs len-trace len-comp trace-indices
    trace-tables comp-tables comp-indices consistent])
qed
have consistent-as:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{partial-query-round-consistent } \text{trace-openings } \text{composition-openings as } i$ 
     $(\text{query-idxs} ! i)$ 
using consistent eqs by simp
have trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr query-idxs trace-openings

```

```

proof –
  have all-indices:
     $\forall i < \text{rounds}.$ 
       $\text{map } \text{opening-index } (\text{trace-openings } ! i) =$ 
         $\text{powers-scaled } (\text{query-idxs } ! i)$ 
    using trace-indices by blast
  have all-tables:
     $\forall i < \text{rounds}.$ 
       $\text{partial-authenticated-table } \text{fr } (\text{scale} * \text{clength})$ 
         $(\text{trace-openings } ! i) \text{ final-state}$ 
    using trace-tables eqs by simp
  show ?thesis
    unfolding accepted-with-partial-trace-openings-def accepted-def
    by (intro conjI exI[of - result] exI[of - final-state])
      (use len-query-idxs len-trace all-indices all-tables in simp-all)
  qed
have comp-partial:
  accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots) query-idxs
    composition-openings
proof –
  have all-indices:
     $\forall i < \text{rounds}.$ 
       $\text{map } \text{opening-index } (\text{composition-openings } ! i) =$ 
         $[\text{query-idxs } ! i,$ 
           $\text{fri-sibling-index } (\text{scale} * \text{clength}) (\text{query-idxs } ! i)]$ 
    using comp-indices by blast
  have all-tables:
     $\forall i < \text{rounds}.$ 
       $\text{partial-authenticated-table } (\text{hd } \text{composition-fri-roots})$ 
         $(\text{scale} * \text{clength}) (\text{composition-openings } ! i) \text{ final-state}$ 
    using comp-tables root-eq by simp
  show ?thesis
    unfolding accepted-with-partial-composition-openings-def accepted-def
    by (intro conjI exI[of - result] exI[of - final-state])
      (use len-query-idxs len-comp all-indices all-tables in simp-all)
  qed
have aligned:
  accepted-with-partial-initial-openings-aligned s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
proof –
  have accepted-out: accepted (Some (result, final-state))
    using trace-partial
    by (rule accepted-with-partial-trace-openings-imp-accepted)
  show ?thesis
  unfolding accepted-with-partial-initial-openings-aligned-def
  by (intro conjI exI[of - rest])

```

```

      (use accepted-out comp-nonempty header0 trace-partial comp-partial in
       simp-all)
    qed
  show ?thesis
  by (rule that[OF aligned consistent-as])
qed

end

end

```

```

theory Soundness-Aligned-Partial-Query
imports Soundness-Partial-Initial-Aligned
begin

```

Aligned partial-candidate query events.

This layer records the query-bad shape needed by the staged prefix-target proof: one shared verifier query-index list for trace and composition openings, together with the local composition equation checked in each query round. It is downstream of the staged query-opening extraction layer to avoid import cycles with the core deterministic reductions.

```

context soundness
begin

```

definition *accepted-with-partial-initial-openings-aligned-consistent*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$ 
  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$ 
  'f list  $\Rightarrow$  'f  $\Rightarrow$  nat list  $\Rightarrow$ 
  'f authenticated-opening list list  $\Rightarrow$ 
  'f authenticated-opening list list  $\Rightarrow$  bool

```

where

```

accepted-with-partial-initial-openings-aligned-consistent s out fr
  f-fri-roots f-final as dg composition-fri-roots final query-idxs
  trace-openings composition-openings  $\longleftrightarrow$ 
accepted-with-partial-initial-openings-aligned s out fr f-fri-roots
  f-final as dg composition-fri-roots final query-idxs trace-openings
  composition-openings  $\wedge$ 
( $\forall i < \text{rounds}$ .
  partial-query-round-consistent trace-openings composition-openings as i
  (query-idxs ! i))

```

definition *query-bad-with-aligned-partial-candidates*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool

```

where

```

query-bad-with-aligned-partial-candidates s out  $\longleftrightarrow$ 

```

$(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
 $\text{ query-idxs trace-openings composition-openings trace-table}$
 $\text{ composition-table.}$
 $\text{ accepted-with-partial-initial-openings-aligned-consistent } s \text{ out fr}$
 $\text{ f-fri-roots f-final as dg composition-fri-roots final query-idxs}$
 $\text{ trace-openings composition-openings } \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings } \wedge$
 $\text{ partial-composition-table-candidate composition-table}$
 $\text{ composition-openings } \wedge$
 $\text{ trace-table-low-degree trace-table } \wedge$
 $\text{ composition-table-low-degree maxDegree composition-table } \wedge$
 $\neg \text{ all-queries-consistent trace-table composition-table as})$

lemma *verify-monad-accepted-with-partial-initial-openings-aligned-consistent:*

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *outcome:*

$\text{Some (result, final-state)} \in \text{set-dist (execute verify-monad } s)$

and *header0:*

$\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final as dg}$
 $\text{ composition-fri-roots final rest}$

and *comp-nonempty:* $\text{composition-fri-roots} \neq []$

obtains $\text{query-idxs trace-openings composition-openings}$ **where**

$\text{accepted-with-partial-initial-openings-aligned-consistent } s$
 $(\text{Some (result, final-state)}) \text{ fr } f\text{-fri-roots } f\text{-final as dg}$
 $\text{ composition-fri-roots final query-idxs trace-openings}$
 $\text{ composition-openings}$

proof –

from *verify-monad-accepted-with-partial-initial-openings-aligned*

$[OF \text{ outcome header0 comp-nonempty}]$

obtain $\text{query-idxs trace-openings composition-openings}$ **where** *aligned:*

$\text{accepted-with-partial-initial-openings-aligned } s$
 $(\text{Some (result, final-state)}) \text{ fr } f\text{-fri-roots } f\text{-final as dg}$
 $\text{ composition-fri-roots final query-idxs trace-openings}$
 $\text{ composition-openings}$

and *consistent:*

$\bigwedge i. i < \text{rounds} \implies$
 $\text{partial-query-round-consistent trace-openings composition-openings as } i$
 $(\text{query-idxs ! } i)$

by *blast*

have *all-consistent:*

$\forall i < \text{rounds.}$
 $\text{partial-query-round-consistent trace-openings composition-openings as } i$
 $(\text{query-idxs ! } i)$

using *consistent by blast*

have *partial:*

$\text{accepted-with-partial-initial-openings-aligned-consistent } s$
 $(\text{Some (result, final-state)}) \text{ fr } f\text{-fri-roots } f\text{-final as dg}$
 $\text{ composition-fri-roots final query-idxs trace-openings}$
 $\text{ composition-openings}$

unfolding *accepted-with-partial-initial-openings-aligned-consistent-def*
using *aligned all-consistent* **by** *simp*
show *?thesis*
by (rule *that[OF partial]*)
qed

definition *soundness-bad-event-aligned-partial-candidate*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{soundness-bad-event-aligned-partial-candidate } s \text{ out} \longleftrightarrow$
 $\text{composition-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{trace-fri-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{composition-fri-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{query-bad-with-aligned-partial-candidates } s \text{ out}$

definition *soundness-bad-event-aligned-partial-candidate-with-partial-merkle*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{soundness-bad-event-aligned-partial-candidate-with-partial-merkle } s \text{ out} \longleftrightarrow$
 $\text{partial-merkle-inconsistency-bad } s \text{ out} \vee$
 $\text{soundness-bad-event-aligned-partial-candidate } s \text{ out}$

lemma *accepted-with-partial-initial-openings-aligned-consistent-imp-aligned:*
assumes
 $\text{accepted-with-partial-initial-openings-aligned-consistent } s \text{ out fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final query-idxs}$
 $\text{trace-openings composition-openings}$
shows
 $\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots final query-idxs trace-openings}$
 $\text{composition-openings}$
using *assms*
unfolding *accepted-with-partial-initial-openings-aligned-consistent-def*
by *simp*

lemma *accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned:*
assumes
 $\text{accepted-with-partial-initial-openings-aligned-consistent } s \text{ out fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final query-idxs}$
 $\text{trace-openings composition-openings}$
shows
 $\text{accepted-with-partial-initial-openings } s \text{ out fr } f\text{-fri-roots } f\text{-final as dg}$
 $\text{composition-fri-roots final query-idxs trace-openings query-idxs}$
 $\text{composition-openings}$
by (rule *accepted-with-partial-initial-openings-aligned-imp-unaligned*)
(rule *accepted-with-partial-initial-openings-aligned-consistent-imp-aligned*
[OF assms])

lemma *accepted-with-partial-initial-openings-aligned-consistent-success-index:*
assumes *partial:*
 accepted-with-partial-initial-openings-aligned-consistent s out fr
 f-fri-roots f-final as dg composition-fri-roots final query-idxs
 trace-openings composition-openings
and *i-bound: i < rounds*
shows
 query-idxs ! i ∈
 partial-query-success-indices-at trace-openings composition-openings
 as i
using *partial i-bound*
unfolding *accepted-with-partial-initial-openings-aligned-consistent-def*
 partial-query-success-indices-at-def partial-query-round-consistent-def
by *simp*

lemma *query-bad-with-aligned-partial-candidates-imp-query-bad-with-partial-candidates:*
assumes *query-bad-with-aligned-partial-candidates s out*
shows *query-bad-with-partial-candidates s out*
proof –
from *assms* **obtain** *fr f-fri-roots f-final as dg composition-fri-roots*
 final query-idxs trace-openings composition-openings trace-table
 composition-table **where** *partial:*
 accepted-with-partial-initial-openings-aligned-consistent s out fr
 f-fri-roots f-final as dg composition-fri-roots final query-idxs
 trace-openings composition-openings
and *trace-candidate:*
 partial-trace-table-candidate trace-table trace-openings
and *comp-candidate:*
 partial-composition-table-candidate composition-table
 composition-openings
and *trace-low: trace-table-low-degree trace-table*
and *comp-low:*
 composition-table-low-degree maxDegree composition-table
and *not-all:*
 $\neg$ *all-queries-consistent trace-table composition-table as*
unfolding *query-bad-with-aligned-partial-candidates-def* **by** *blast*
have *unaligned:*
 accepted-with-partial-initial-openings s out fr f-fri-roots f-final as dg
 composition-fri-roots final query-idxs trace-openings query-idxs
 composition-openings
by *(rule*
 accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned
 [OF partial])
show *?thesis*
unfolding *query-bad-with-partial-candidates-def*
using *unaligned trace-candidate comp-candidate trace-low comp-low not-all*
by *blast*
qed

lemma *soundness-bad-event-aligned-partial-candidate-imp-partial-candidate:*
assumes *soundness-bad-event-aligned-partial-candidate s out*
shows *soundness-bad-event-partial-candidate s out*
using *assms query-bad-with-aligned-partial-candidates-imp-query-bad-with-partial-candidates*
unfolding *soundness-bad-event-aligned-partial-candidate-def*
soundness-bad-event-partial-candidate-def
by *blast*

lemma *soundness-bad-event-aligned-partial-candidate-with-partial-merkle-imp-partial-candidate-with-partial-merkle:*
assumes
soundness-bad-event-aligned-partial-candidate-with-partial-merkle s out
shows *soundness-bad-event-partial-candidate-with-partial-merkle s out*
using *assms soundness-bad-event-aligned-partial-candidate-imp-partial-candidate*
unfolding *soundness-bad-event-aligned-partial-candidate-with-partial-merkle-def*
soundness-bad-event-partial-candidate-with-partial-merkle-def
by *blast*

lemma *accepted-aligned-partial-candidate-partition:*
assumes *false-statement: $\neg$ exists-valid-trace*
and *partial:*
accepted-with-partial-initial-openings-aligned-consistent s out fr
f-fri-roots f-final as dg composition-fri-roots final query-idxs
trace-openings composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *composition-candidate:*
partial-composition-table-candidate composition-table
composition-openings
shows *soundness-bad-event-aligned-partial-candidate s out*
proof (*cases trace-table-low-degree trace-table*)
case *False*
have *trace-fri-bad-with-partial-candidates s out*
unfolding *trace-fri-bad-with-partial-candidates-def*
using *accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned*
[OF partial]
trace-candidate False **by** *blast*
then show *?thesis*
unfolding *soundness-bad-event-aligned-partial-candidate-def* **by** *blast*

next
case *True*
note *trace-low = True*
show *?thesis*
proof (*cases composition-table-low-degree maxDegree composition-table*)
case *False*
have *composition-fri-bad-with-partial-candidates s out*
unfolding *composition-fri-bad-with-partial-candidates-def*
using *accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned*
[OF partial]

```

      trace-candidate composition-candidate trace-low False
    by blast
  then show ?thesis
    unfolding soundness-bad-event-aligned-partial-candidate-def by blast
next
case True
note composition-low = True
show ?thesis
proof (cases all-queries-consistent trace-table composition-table as)
  case False
  have query-bad-with-aligned-partial-candidates s out
    unfolding query-bad-with-aligned-partial-candidates-def
    using partial trace-candidate composition-candidate trace-low
      composition-low False
    by blast
  then show ?thesis
    unfolding soundness-bad-event-aligned-partial-candidate-def by blast
next
case True
note all-queries = True
from trace-low obtain f where
  deg-f: degree f < clength
  and trace-table-eq: trace-table = map (poly f) eval-domain
  unfolding trace-table-low-degree-def by blast
have violated: violated-constraints f ≠ {}
  by (rule false-statement-violated-constraints
    [OF false-statement deg-f])
have composition-bad-with-partial-candidates s out
  unfolding composition-bad-with-partial-candidates-def
  using accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned
    [OF partial]
    trace-candidate composition-candidate deg-f trace-table-eq
    violated composition-low all-queries
  by blast
then show ?thesis
  unfolding soundness-bad-event-aligned-partial-candidate-def by blast
qed
qed
qed

```

lemma *soundness-bad-event-aligned-partial-candidate-union-bound:*
fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$
assumes *comp-bound:*
 $\text{wp-event verify-monad}$
 $(\text{composition-bad-with-partial-candidates } s) \ s \leq$
 composition-error
and *trace-fri-bound:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-partial-candidates } s) \ s \leq$

```

    trace-fri-error'
  and comp-fri-bound:
    wp-event verify-monad
      (composition-fri-bad-with-partial-candidates s) s ≤
      composition-fri-error'
  and query-bound:
    wp-event verify-monad
      (query-bad-with-aligned-partial-candidates s) s ≤
      query-error
  shows
    wp-event verify-monad
      (soundness-bad-event-aligned-partial-candidate s) s ≤
      composition-error + trace-fri-error' + composition-fri-error' +
      query-error
  proof -
    have wp-event verify-monad
      (soundness-bad-event-aligned-partial-candidate s) s ≤
      wp-event verify-monad
        (composition-bad-with-partial-candidates s) s +
      wp-event verify-monad
        (trace-fri-bad-with-partial-candidates s) s +
      wp-event verify-monad
        (composition-fri-bad-with-partial-candidates s) s +
      wp-event verify-monad
        (query-bad-with-aligned-partial-candidates s) s
    unfolding soundness-bad-event-aligned-partial-candidate-def
    by (rule wp-event-union-bound4)
  also have ... ≤
    composition-error + trace-fri-error' + composition-fri-error' +
    query-error
  by (intro add-mono comp-bound trace-fri-bound comp-fri-bound
      query-bound)
  finally show ?thesis .
qed

lemma soundness-bad-event-aligned-partial-candidate-with-partial-merkle-union-bound:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤
    merkle-error
  and comp-bound:
    wp-event verify-monad
      (composition-bad-with-partial-candidates s) s ≤
      composition-error
  and trace-fri-bound:
    wp-event verify-monad
      (trace-fri-bad-with-partial-candidates s) s ≤
      trace-fri-error'
  and comp-fri-bound:

```

```

    wp-event verify-monad
      (composition-fri-bad-with-partial-candidates s) s ≤
      composition-fri-error'
  and query-bound:
    wp-event verify-monad
      (query-bad-with-aligned-partial-candidates s) s ≤
      query-error
  shows
    wp-event verify-monad
      (soundness-bad-event-aligned-partial-candidate-with-partial-merkle s) s
      ≤
      merkle-error + composition-error + trace-fri-error' +
      composition-fri-error' + query-error
  proof -
    have partial-bound:
      wp-event verify-monad
        (soundness-bad-event-aligned-partial-candidate s) s ≤
        composition-error + trace-fri-error' + composition-fri-error' +
        query-error
    by (rule soundness-bad-event-aligned-partial-candidate-union-bound
        [OF comp-bound trace-fri-bound comp-fri-bound query-bound])
    have wp-event verify-monad
      (soundness-bad-event-aligned-partial-candidate-with-partial-merkle s) s
      ≤
      wp-event verify-monad (partial-merkle-inconsistency-bad s) s +
      wp-event verify-monad
        (soundness-bad-event-aligned-partial-candidate s) s
    unfolding
      soundness-bad-event-aligned-partial-candidate-with-partial-merkle-def
    by (rule wp-event-union-bound)
    also have ... ≤
      merkle-error +
      (composition-error + trace-fri-error' + composition-fri-error' +
      query-error)
    by (intro add-mono merkle-bound partial-bound)
    finally show ?thesis
    by (simp add: algebra-simps)
  qed

  lemma checked-staged-security-with-data-state-aligned-partial-candidate-with-partial-merkle-union-bound:
    fixes merkle-error composition-error trace-fri-error'
      composition-fri-error' query-error :: prob
    assumes merkle-bound:
      wp-event (checked-staged-security-experiment-with-data-state A)
        (staged-security-with-data-state-verifier-event
          partial-merkle-inconsistency-bad)
        adversary-initial-state ≤ merkle-error
    and comp-bound:
      wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (staged-security-with-data-state-verifier-event
     composition-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-error
and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  trace-fri-error'
and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-fri-error'
and query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     query-bad-with-aligned-partial-candidates)
    adversary-initial-state  $\leq$  query-error
shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     soundness-bad-event-aligned-partial-candidate-with-partial-merkle)
    adversary-initial-state  $\leq$ 
    merkle-error + composition-error + trace-fri-error' +
    composition-fri-error' + query-error
proof –
    let ?M = checked-staged-security-experiment-with-data-state A
    let ?E = staged-security-with-data-state-verifier-event
    let ?merkle = ?E partial-merkle-inconsistency-bad
    let ?comp = ?E composition-bad-with-partial-candidates
    let ?trace = ?E trace-fri-bad-with-partial-candidates
    let ?comp-fri = ?E composition-fri-bad-with-partial-candidates
    let ?query = ?E query-bad-with-aligned-partial-candidates
    let ?partial = ?E soundness-bad-event-aligned-partial-candidate
    let ?with-merkle =
    ?E soundness-bad-event-aligned-partial-candidate-with-partial-merkle
    have partial-bound:
    wp-event ?M ?partial adversary-initial-state  $\leq$ 
    composition-error + trace-fri-error' + composition-fri-error' +
    query-error
    proof –
    have wp-event ?M ?partial adversary-initial-state  $\leq$ 
    wp-event ?M
    ( $\lambda$  out. ?comp out  $\vee$  ?trace out  $\vee$  ?comp-fri out  $\vee$ 
    ?query out)
    adversary-initial-state
    by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-verifier-event-def
    soundness-bad-event-aligned-partial-candidate-def

```

```

      split: option.splits prod.splits)
also have ... ≤
  wp-event ?M ?comp adversary-initial-state +
  wp-event ?M ?trace adversary-initial-state +
  wp-event ?M ?comp-fri adversary-initial-state +
  wp-event ?M ?query adversary-initial-state
  by (rule wp-event-union-bound4)
also have ... ≤
  composition-error + trace-fri-error' + composition-fri-error' +
  query-error
  by (intro add-mono comp-bound trace-fri-bound comp-fri-bound
    query-bound)
finally show ?thesis .
qed
have wp-event ?M ?with-merkle adversary-initial-state ≤
  wp-event ?M (λout. ?merkle out ∨ ?partial out)
  adversary-initial-state
  by (rule wp-event-mono)
  (auto simp: staged-security-with-data-state-verifier-event-def
    soundness-bad-event-aligned-partial-candidate-with-partial-merkle-def
    split: option.splits prod.splits)
also have ... ≤
  wp-event ?M ?merkle adversary-initial-state +
  wp-event ?M ?partial adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤
  merkle-error +
  (composition-error + trace-fri-error' + composition-fri-error' +
  query-error)
  by (intro add-mono merkle-bound partial-bound)
finally show ?thesis
  by (simp add: algebra-simps)
qed

lemma query-bad-with-aligned-partial-candidates-success-indexE:
assumes bad: query-bad-with-aligned-partial-candidates s out
and i-bound: i < rounds
obtains fr f-fri-roots f-final as dg composition-fri-roots final
  query-idxs trace-openings composition-openings trace-table
  composition-table where
  accepted-with-partial-initial-openings-aligned-consistent s out fr
  f-fri-roots f-final as dg composition-fri-roots final query-idxs
  trace-openings composition-openings
  partial-trace-table-candidate trace-table trace-openings
  partial-composition-table-candidate composition-table
  composition-openings
  trace-table-low-degree trace-table
  composition-table-low-degree maxDegree composition-table
  ¬ all-queries-consistent trace-table composition-table as

```


query-idxs ! i ∈
partial-query-success-indices-at trace-openings composition-openings
as i
proof –
from *bad* **obtain** *fr f-fri-roots f-final as dg composition-fri-roots*
final query-idxs trace-openings composition-openings trace-table
composition-table **where** *partial:*
accepted-with-partial-initial-openings-aligned-consistent s out fr
f-fri-roots f-final as dg composition-fri-roots final query-idxs
trace-openings composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *comp-candidate:*
partial-composition-table-candidate composition-table
composition-openings
and *trace-low: trace-table-low-degree trace-table*
and *comp-low:*
composition-table-low-degree maxDegree composition-table
and *not-all:*
 $\neg$ *all-queries-consistent trace-table composition-table as*
unfolding *query-bad-with-aligned-partial-candidates-def* **by** *blast*
have *success:*
query-idxs ! i ∈
partial-query-success-indices-at trace-openings composition-openings
as i
by (*rule*
accepted-with-partial-initial-openings-aligned-consistent-success-index
[OF partial i-bound])
show *?thesis*
by (*rule that[OF partial trace-candidate comp-candidate trace-low comp-low*
not-all success])
qed

lemma *query-bad-with-aligned-partial-candidates-target-fractionE:*
assumes *bad: query-bad-with-aligned-partial-candidates s out*
and *i-bound: i < rounds*
obtains *fr f-fri-roots f-final as dg composition-fri-roots final*
query-idxs trace-openings composition-openings trace-table
composition-table **where**
accepted-with-partial-initial-openings-aligned-consistent s out fr
f-fri-roots f-final as dg composition-fri-roots final query-idxs
trace-openings composition-openings
query-idxs ! i ∈
partial-query-success-indices-at trace-openings composition-openings
as i
nnreal
(card
(partial-query-success-indices-at trace-openings composition-openings
as i)) /

$nnreal \ (card \ query\ sample\ space) \leq \ query\ error\ bound$
proof –
from *query-bad-with-aligned-partial-candidates-success-indexE*
 $[OF \ bad \ i\ bound]$
obtain *fr f-fri-roots f-final as dg composition-fri-roots final*
query-idxs trace-openings composition-openings trace-table
composition-table **where** *partial:*
accepted-with-partial-initial-openings-aligned-consistent s out fr
f-fri-roots f-final as dg composition-fri-roots final query-idxs
trace-openings composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *comp-candidate:*
partial-composition-table-candidate composition-table
composition-openings
and *trace-low: trace-table-low-degree trace-table*
and *comp-low:*
composition-table-low-degree maxDegree composition-table
and *not-all:*
 $\neg \ all\ queries\ consistent \ trace\ table \ composition\ table \ as$
and *success:*
query-idxs ! i ∈
partial-query-success-indices-at trace-openings composition-openings
as i
by *blast*
have *fraction:*
 $nnreal$
 $(card$
 $(partial\ query\ success\ indices\ at \ trace\ openings \ composition\ openings$
 $as \ i)) \ /$
 $nnreal \ (card \ query\ sample\ space) \leq \ query\ error\ bound$
by $(rule \ partial\ query\ success\ indices\ at\ fraction\ bound\ if\ candidate\ low\ degree$
 $[OF \ trace\ candidate \ comp\ candidate \ trace\ low \ comp\ low \ not\ all])$
show *?thesis*
by $(rule \ that[OF \ partial \ success \ fraction])$
qed

lemma *query-bad-with-aligned-partial-candidates-miss-disagreements:*
assumes *bad: query-bad-with-aligned-partial-candidates s out*
shows $\exists \ trace\ table \ composition\ table \ as \ query\ idxs.$
query-samples-miss-disagreements trace-table composition-table as
query-idxs

proof –
from *bad* **obtain** *fr f-fri-roots f-final as dg composition-fri-roots*
final query-idxs trace-openings composition-openings trace-table
composition-table **where** *partial:*
accepted-with-partial-initial-openings-aligned-consistent s out fr
f-fri-roots f-final as dg composition-fri-roots final query-idxs
trace-openings composition-openings

```

and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-low:
  composition-table-low-degree maxDegree composition-table
and not-all:
   $\neg$  all-queries-consistent trace-table composition-table as
unfolding query-bad-with-aligned-partial-candidates-def by blast
have aligned:
  accepted-with-partial-initial-openings-aligned s out fr f-fri-roots
  f-final as dg composition-fri-roots final query-idxs trace-openings
  composition-openings
by (rule accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
  [OF partial])
have trace-partial:
  accepted-with-partial-trace-openings s out fr query-idxs
  trace-openings
using accepted-with-partial-initial-openings-aligned-shapes(3)
  [OF aligned] .
have len-query: length query-idxs = rounds
using accepted-with-partial-trace-openings-shapes(1) [OF trace-partial] .
have nonempty-disagreements:
  query-disagreement-indices trace-table composition-table as  $\neq \{\}$ 
using not-all
unfolding query-disagreement-indices-def bad-query-indices-def
  all-queries-consistent-def
by auto
have subset-agreement:
  set query-idxs  $\subseteq$ 
  query-agreement-indices trace-table composition-table as
proof
  fix idx
  assume idx-in: idx  $\in$  set query-idxs
  then obtain i where i-len: i < length query-idxs
    and idx-eq: query-idxs ! i = idx
    by (auto simp: in-set-conv-nth)
  have i-bound: i < rounds
    using i-len len-query by simp
  have success:
    query-idxs ! i  $\in$ 
    partial-query-success-indices-at trace-openings composition-openings
    as i
    by (rule
      accepted-with-partial-initial-openings-aligned-consistent-success-index
      [OF partial i-bound])
  have success-space:

```

```

    query-idxs ! i ∈
      query-sampling-success-space trace-table composition-table as
    using success
      partial-query-success-indices-at-subset-query-sampling-success-space
      [OF trace-candidate comp-candidate trace-low comp-low not-all,
       of i]
    by blast
  then show idx ∈
    query-agreement-indices trace-table composition-table as
  using idx-eq trace-low comp-low not-all
  unfolding query-sampling-success-space-def by simp
qed
have
  query-samples-miss-disagreements trace-table composition-table as
  query-idxs
  unfolding query-samples-miss-disagreements-def
  using nonempty-disagreements subset-agreement by simp
then show ?thesis by blast
qed

end

end

```

```

theory Soundness-Aligned-Partial-Query-Transcript
imports
  Soundness-Aligned-Partial-Query
  Soundness-Staged-Partial-Query
begin

```

Transcript-tied aligned partial-query events.

The aligned partial-opening predicate records that trace and composition openings use the same sampled query indices. For probabilistic query bounds we additionally need those indices to be the verifier-derived Fiat-Shamir query indices. This layer adds that transcript replay fact without changing the protocol model.

```

context soundness
begin

```

definition *accepted-with-partial-initial-openings-aligned-transcript-consistent*

```

:: ('f, 'a) protocol-channel-scheme ⇒
  (unit list × ('f, 'a) protocol-channel-scheme) option ⇒
  'f ⇒ 'f list ⇒ 'f ⇒ 'f list ⇒ 'f ⇒
  'f list ⇒ 'f ⇒ nat list ⇒
  'f authenticated-opening list list ⇒
  'f authenticated-opening list list ⇒ bool

```

where

```

  accepted-with-partial-initial-openings-aligned-transcript-consistent s out
  fr f-fri-roots f-final as dg composition-fri-roots final query-idxs

```

$\text{trace-openings composition-openings} \longleftrightarrow$
 $\text{accepted-with-partial-initial-openings-aligned-consistent } s \text{ out fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots final query-idxs}$
 $\text{trace-openings composition-openings} \wedge$
 $\text{accepted-transcript-shape } s \text{ out as query-idxs}$

definition *query-bad-with-aligned-transcript-partial-candidates*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{query-bad-with-aligned-transcript-partial-candidates } s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{query-idxs trace-openings composition-openings trace-table}$
 $\text{composition-table.}$
 $\text{accepted-with-partial-initial-openings-aligned-transcript-consistent } s$
 $\text{out fr f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{query-idxs trace-openings composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\text{composition-table-low-degree maxDegree composition-table} \wedge$
 $\neg \text{all-queries-consistent trace-table composition-table as})$

definition *soundness-bad-event-aligned-transcript-partial-candidate*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{soundness-bad-event-aligned-transcript-partial-candidate } s \text{ out} \longleftrightarrow$
 $\text{composition-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{trace-fri-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{composition-fri-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{query-bad-with-aligned-transcript-partial-candidates } s \text{ out}$

definition *soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle}$
 $s \text{ out} \longleftrightarrow$
 $\text{partial-merkle-inconsistency-bad } s \text{ out} \vee$
 $\text{soundness-bad-event-aligned-transcript-partial-candidate } s \text{ out}$

lemma *accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned:*

assumes

$\text{accepted-with-partial-initial-openings-aligned-transcript-consistent } s \text{ out}$
 $\text{fr f-fri-roots f-final as dg composition-fri-roots final query-idxs}$
 $\text{trace-openings composition-openings}$

shows

*accepted-with-partial-initial-openings-aligned-consistent s out fr
 f-fri-roots f-final as dg composition-fri-roots final query-idxs
 trace-openings composition-openings*
using *assms*
unfolding
accepted-with-partial-initial-openings-aligned-transcript-consistent-def
by *simp*

lemma *accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape:*
assumes
*accepted-with-partial-initial-openings-aligned-transcript-consistent s out
 fr f-fri-roots f-final as dg composition-fri-roots final query-idxs
 trace-openings composition-openings*
shows *accepted-transcript-shape s out as query-idxs*
using *assms*
unfolding
accepted-with-partial-initial-openings-aligned-transcript-consistent-def
by *simp*

lemma *query-bad-with-aligned-transcript-partial-candidates-imp-aligned:*
assumes *query-bad-with-aligned-transcript-partial-candidates s out*
shows *query-bad-with-aligned-partial-candidates s out*
proof —
from *assms* **obtain** *fr f-fri-roots f-final as dg composition-fri-roots
 final query-idxs trace-openings composition-openings trace-table
 composition-table* **where** *partial:*
*accepted-with-partial-initial-openings-aligned-transcript-consistent s
 out fr f-fri-roots f-final as dg composition-fri-roots final
 query-idxs trace-openings composition-openings*
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *comp-candidate:*
*partial-composition-table-candidate composition-table
 composition-openings*
and *trace-low: trace-table-low-degree trace-table*
and *comp-low:*
composition-table-low-degree maxDegree composition-table
and *not-all:*
 $\neg$ *all-queries-consistent trace-table composition-table as*
unfolding *query-bad-with-aligned-transcript-partial-candidates-def*
by *blast*
have *aligned:*
*accepted-with-partial-initial-openings-aligned-consistent s out fr
 f-fri-roots f-final as dg composition-fri-roots final query-idxs
 trace-openings composition-openings*
by *(rule*
accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
[OF partial])
show *?thesis*

unfolding *query-bad-with-aligned-partial-candidates-def*
using *aligned trace-candidate comp-candidate trace-low comp-low not-all*
by *blast*
qed

lemma *query-bad-with-aligned-transcript-partial-candidates-miss-disagreements:*
assumes *query-bad-with-aligned-transcript-partial-candidates s out*
shows $\exists$ *trace-table composition-table as query-idxs.*
query-samples-miss-disagreements trace-table composition-table as
query-idxs
by (*rule query-bad-with-aligned-partial-candidates-miss-disagreements*)
(*rule query-bad-with-aligned-transcript-partial-candidates-imp-aligned*
[*OF assms*])

lemma *soundness-bad-event-aligned-transcript-partial-candidate-imp-aligned:*
assumes *soundness-bad-event-aligned-transcript-partial-candidate s out*
shows *soundness-bad-event-aligned-partial-candidate s out*
using *assms query-bad-with-aligned-transcript-partial-candidates-imp-aligned*
unfolding *soundness-bad-event-aligned-transcript-partial-candidate-def*
soundness-bad-event-aligned-partial-candidate-def
by *blast*

lemma *soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-imp-aligned:*
assumes
soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
s out
shows
soundness-bad-event-aligned-partial-candidate-with-partial-merkle s out
using *assms soundness-bad-event-aligned-transcript-partial-candidate-imp-aligned*
unfolding
soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-def
soundness-bad-event-aligned-partial-candidate-with-partial-merkle-def
by *blast*

lemma *accepted-aligned-transcript-partial-candidate-partition:*
assumes *false-statement: $\neg$ exists-valid-trace*
and *partial:*
accepted-with-partial-initial-openings-aligned-transcript-consistent s
out fr f-fri-roots f-final as dg composition-fri-roots final
query-idxs trace-openings composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *composition-candidate:*
partial-composition-table-candidate composition-table
composition-openings
shows *soundness-bad-event-aligned-transcript-partial-candidate s out*
proof (*cases trace-table-low-degree trace-table*)
case *False*
have *aligned:*

```

    accepted-with-partial-initial-openings-aligned-consistent s out fr
      f-fri-roots f-final as dg composition-fri-roots final query-idxs
      trace-openings composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
  have trace-fri-bad-with-partial-candidates s out
    unfolding trace-fri-bad-with-partial-candidates-def
    using accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned
      [OF aligned]
    trace-candidate False by blast
  then show ?thesis
    unfolding soundness-bad-event-aligned-transcript-partial-candidate-def
    by blast
next
case True
note trace-low = True
show ?thesis
proof (cases composition-table-low-degree maxDegree composition-table)
  case False
  have aligned:
    accepted-with-partial-initial-openings-aligned-consistent s out fr
      f-fri-roots f-final as dg composition-fri-roots final query-idxs
      trace-openings composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
  have composition-fri-bad-with-partial-candidates s out
    unfolding composition-fri-bad-with-partial-candidates-def
    using accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned
      [OF aligned]
    trace-candidate composition-candidate trace-low False
  by blast
  then show ?thesis
    unfolding soundness-bad-event-aligned-transcript-partial-candidate-def
    by blast
next
case True
note composition-low = True
show ?thesis
proof (cases all-queries-consistent trace-table composition-table as)
  case False
  have query-bad-with-aligned-transcript-partial-candidates s out
    unfolding query-bad-with-aligned-transcript-partial-candidates-def
    using partial trace-candidate composition-candidate trace-low
      composition-low False
  by blast
  then show ?thesis
    unfolding soundness-bad-event-aligned-transcript-partial-candidate-def

```



```

    by blast
next
case True
note all-queries = True
from trace-low obtain f where
  deg-f: degree f < clength
  and trace-table-eq: trace-table = map (poly f) eval-domain
  unfolding trace-table-low-degree-def by blast
have violated: violated-constraints f ≠ {}
  by (rule false-statement-violated-constraints
      [OF false-statement deg-f])
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent s out fr
  f-fri-roots f-final as dg composition-fri-roots final query-idxs
  trace-openings composition-openings
  by (rule
      accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
      [OF partial])
have composition-bad-with-partial-candidates s out
  unfolding composition-bad-with-partial-candidates-def
  using accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned
  [OF aligned]
  trace-candidate composition-candidate deg-f trace-table-eq
  violated composition-low all-queries
  by blast
then show ?thesis
  unfolding soundness-bad-event-aligned-transcript-partial-candidate-def
  by blast
qed
qed
qed

```

lemma soundness-bad-event-aligned-transcript-partial-candidate-union-bound:

```

fixes s :: ('f, 'a) protocol-channel-scheme
assumes comp-bound:
  wp-event verify-monad
  (composition-bad-with-partial-candidates s) s ≤
  composition-error
and trace-fri-bound:
  wp-event verify-monad
  (trace-fri-bad-with-partial-candidates s) s ≤
  trace-fri-error'
and comp-fri-bound:
  wp-event verify-monad
  (composition-fri-bad-with-partial-candidates s) s ≤
  composition-fri-error'
and query-bound:
  wp-event verify-monad
  (query-bad-with-aligned-transcript-partial-candidates s) s ≤

```

query-error
shows
wp-event verify-monad
 (soundness-bad-event-aligned-transcript-partial-candidate s) $s \leq$
 composition-error + trace-fri-error' + composition-fri-error' +
query-error
proof –
have *wp-event verify-monad*
 (soundness-bad-event-aligned-transcript-partial-candidate s) $s \leq$
wp-event verify-monad
 (composition-bad-with-partial-candidates s) s +
wp-event verify-monad
 (trace-fri-bad-with-partial-candidates s) s +
wp-event verify-monad
 (composition-fri-bad-with-partial-candidates s) s +
wp-event verify-monad
 (query-bad-with-aligned-transcript-partial-candidates s) s
unfolding soundness-bad-event-aligned-transcript-partial-candidate-def
by (rule wp-event-union-bound4)
also have ... $\leq$
 composition-error + trace-fri-error' + composition-fri-error' +
query-error
by (intro add-mono comp-bound trace-fri-bound comp-fri-bound
 query-bound)
finally show ?thesis .
qed

lemma soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-union-bound:
fixes $s :: ('f, 'a)$ protocol-channel-scheme
assumes merkle-bound:
wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq$
 merkle-error
and comp-bound:
wp-event verify-monad
 (composition-bad-with-partial-candidates s) $s \leq$
 composition-error
and trace-fri-bound:
wp-event verify-monad
 (trace-fri-bad-with-partial-candidates s) $s \leq$
 trace-fri-error'
and comp-fri-bound:
wp-event verify-monad
 (composition-fri-bad-with-partial-candidates s) $s \leq$
 composition-fri-error'
and query-bound:
wp-event verify-monad
 (query-bad-with-aligned-transcript-partial-candidates s) $s \leq$
 query-error
shows

```

wp-event verify-monad
  (soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle s) s
≤
merkle-error + composition-error + trace-fri-error' +
composition-fri-error' + query-error
proof –
  have partial-bound:
    wp-event verify-monad
      (soundness-bad-event-aligned-transcript-partial-candidate s) s ≤
      composition-error + trace-fri-error' + composition-fri-error' +
      query-error
  by (rule
    soundness-bad-event-aligned-transcript-partial-candidate-union-bound
    [OF comp-bound trace-fri-bound comp-fri-bound query-bound])
  have wp-event verify-monad
    (soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle s) s
    ≤
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s +
    wp-event verify-monad
      (soundness-bad-event-aligned-transcript-partial-candidate s) s
  unfolding
    soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-def
  by (rule wp-event-union-bound)
  also have ... ≤
    merkle-error +
    (composition-error + trace-fri-error' + composition-fri-error' +
    query-error)
  by (intro add-mono merkle-bound partial-bound)
  finally show ?thesis
  by (simp add: algebra-simps)
qed

lemma checked-staged-security-with-data-state-aligned-transcript-partial-candidate-union-bound:
fixes composition-error trace-fri-error' composition-fri-error'
query-error :: prob
assumes comp-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-error
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ trace-fri-error'
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)

```

```

    adversary-initial-state ≤ composition-fri-error'
and query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state ≤ query-error
shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-aligned-transcript-partial-candidate)
    adversary-initial-state ≤
      composition-error + trace-fri-error' + composition-fri-error' +
      query-error
proof –
    let ?M = checked-staged-security-experiment-with-data-state A
    let ?E = staged-security-with-data-state-verifier-event
    let ?comp = ?E composition-bad-with-partial-candidates
    let ?trace = ?E trace-fri-bad-with-partial-candidates
    let ?comp-fri = ?E composition-fri-bad-with-partial-candidates
    let ?query = ?E query-bad-with-aligned-transcript-partial-candidates
    let ?partial = ?E soundness-bad-event-aligned-transcript-partial-candidate
    have wp-event ?M ?partial adversary-initial-state ≤
      wp-event ?M
        (λout. ?comp out ∨ ?trace out ∨ ?comp-fri out ∨
          ?query out)
      adversary-initial-state
    by (rule wp-event-mono)
      (auto simp: staged-security-with-data-state-verifier-event-def
        soundness-bad-event-aligned-transcript-partial-candidate-def
        split: option.splits prod.splits)
    also have ... ≤
      wp-event ?M ?comp adversary-initial-state +
      wp-event ?M ?trace adversary-initial-state +
      wp-event ?M ?comp-fri adversary-initial-state +
      wp-event ?M ?query adversary-initial-state
    by (rule wp-event-union-bound4)
    also have ... ≤
      composition-error + trace-fri-error' + composition-fri-error' +
      query-error
    by (intro add-mono comp-bound trace-fri-bound comp-fri-bound
      query-bound)
    finally show ?thesis .
qed

```

```

lemma checked-staged-security-with-data-state-aligned-transcript-partial-candidate-with-partial-merkle-component-un
fixes merkle-error composition-error trace-fri-error'
  composition-fri-error' query-error :: prob
assumes merkle-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (staged-security-with-data-state-verifier-event
     partial-merkle-inconsistency-bad)
    adversary-initial-state ≤ merkle-error
and comp-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-error
and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ trace-fri-error'
and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-fri-error'
and query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state ≤ query-error
shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle)
    adversary-initial-state ≤
    merkle-error + composition-error + trace-fri-error' +
    composition-fri-error' + query-error
proof –
have partial-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     soundness-bad-event-aligned-transcript-partial-candidate)
    adversary-initial-state ≤
    composition-error + trace-fri-error' + composition-fri-error' +
    query-error
by (rule
    checked-staged-security-with-data-state-aligned-transcript-partial-candidate-union-bound
    [OF comp-bound trace-fri-bound comp-fri-bound query-bound])
let ?M = checked-staged-security-experiment-with-data-state A
let ?E = staged-security-with-data-state-verifier-event
let ?merkle = ?E partial-merkle-inconsistency-bad
let ?partial = ?E soundness-bad-event-aligned-transcript-partial-candidate
let ?with-merkle =
    ?E soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
have wp-event ?M ?with-merkle adversary-initial-state ≤
    wp-event ?M (λout. ?merkle out ∨ ?partial out)

```

```

    adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-verifier-event-def
      soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-def
      split: option.splits prod.splits)
  also have ... ≤
    wp-event ?M ?merkle adversary-initial-state +
    wp-event ?M ?partial adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤
    merkle-error +
    (composition-error + trace-fri-error' + composition-fri-error' +
      query-error)
  by (intro add-mono merkle-bound partial-bound)
  finally show ?thesis
  by (simp add: algebra-simps)
qed

```

lemma *checked-staged-security-with-data-state-aligned-transcript-partial-candidate-with-partial-merkle-union-bound:*

fixes *merkle-error composition-error trace-fri-error'*
composition-fri-error' query-error :: prob

assumes *merkle-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad)
adversary-initial-state ≤ merkle-error

and *partial-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
soundness-bad-event-aligned-transcript-partial-candidate)
adversary-initial-state ≤
composition-error + trace-fri-error' + composition-fri-error' +
query-error

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle)
adversary-initial-state ≤
merkle-error + composition-error + trace-fri-error' +
composition-fri-error' + query-error

proof –

let *?M = checked-staged-security-experiment-with-data-state A*
let *?E = staged-security-with-data-state-verifier-event*
let *?merkle = ?E partial-merkle-inconsistency-bad*
let *?partial = ?E soundness-bad-event-aligned-transcript-partial-candidate*
have *wp-event ?M*
(?E soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle)
adversary-initial-state ≤
wp-event ?M (λout. ?merkle out ∨ ?partial out)

```

    adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-verifier-event-def
      soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-def
      split: option.splits prod.splits)
  also have ... ≤
    wp-event ?M ?merkle adversary-initial-state +
    wp-event ?M ?partial adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤
    merkle-error +
    (composition-error + trace-fri-error' + composition-fri-error' +
      query-error)
  by (intro add-mono merkle-bound partial-bound)
  finally show ?thesis
  by (simp add: algebra-simps)
qed

```

lemma *checked-staged-security-with-data-state-aligned-transcript-query-bad-imp-partial-opening-hit-on-support:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

Some (((data, attacker-state), result), final-state) ∈
set-dist
(execute
(ichecked-staged-security-experiment-with-data-state A)
adversary-initial-state)

and *bad:*

staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates
(Some (((data, attacker-state), result), final-state))

shows

staged-security-with-data-state-query-partial-opening-hit
(Some (((data, attacker-state), result), final-state))

proof —

let *?s =*

verifier-state-from-adversary attacker-state
(staged-proof-transcript data)

from *checked-staged-security-experiment-with-data-state-outcomeE*
[OF support]

obtain *builder where builder:*

Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)

and *verifier:*

Some (result, final-state) ∈ set-dist (execute verify-monad ?s)

by *blast*

have *staged-header:*

```

verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
  [OF wf controlled builder])
from bad obtain fr f-fri-roots f-final as dg composition-fri-roots
  final query-idxs trace-openings composition-openings trace-table
  composition-table where partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-low:
  composition-table-low-degree maxDegree composition-table
and not-all:
  ¬ all-queries-consistent trace-table composition-table as
unfolding staged-security-with-data-state-verifier-event-def
  query-bad-with-aligned-transcript-partial-candidates-def
by simp blast
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
by (rule
  accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
  [OF partial])
have shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
  query-idxs
by (rule
  accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape
  [OF partial])
from checked-staged-security-with-data-state-accepted-shape-query-chunks
  [OF wf controlled support shape]
obtain raw-idxs where
  as-eq: as = staged-alphas data

```



```

and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and lookup:
  ∀ i < rounds.
    fmllookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
by blast
have aligned0:
  accepted-with-partial-initial-openings-aligned ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
by (rule accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
  [OF aligned])
from accepted-with-partial-initial-openings-aligned-shapes[OF aligned0]
obtain rest where header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
by blast
have comp-nonempty: composition-fri-roots ≠ []
using accepted-with-partial-initial-openings-aligned-shapes(1)
  [OF aligned0] .
have trace-partial:
  accepted-with-partial-trace-openings ?s
    (Some (result, final-state)) fr query-idxs trace-openings
using accepted-with-partial-initial-openings-aligned-shapes(3)
  [OF aligned0] .
have comp-partial:
  accepted-with-partial-composition-openings ?s
    (Some (result, final-state)) (hd composition-fri-roots) query-idxs
    composition-openings
using accepted-with-partial-initial-openings-aligned-shapes(4)
  [OF aligned0] .
have header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  composition-fri-roots = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp

```

```

let ?i = 0
have i-bound: ?i < rounds
  using rounds-positive by simp
have round-consistent:
  partial-query-round-consistent trace-openings composition-openings as
    ?i (query-idxs ! ?i)
  using aligned i-bound
  unfolding accepted-with-partial-initial-openings-aligned-consistent-def
  by simp
have opening-consistent:
  partial-query-openings-consistent (trace-openings ! ?i)
    (composition-openings ! ?i) as (query-idxs ! ?i)
  using round-consistent
  unfolding partial-query-round-consistent-def
    partial-query-openings-consistent-def
  by simp
have idx-sample: query-idxs ! ?i ∈ query-sample-space
  using round-consistent
  unfolding partial-query-round-consistent-def by simp
have trace-table:
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! ?i) final-state
  using trace-partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
have comp-table:
  partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) (composition-openings ! ?i) final-state
  using comp-partial i-bound
  unfolding accepted-with-partial-composition-openings-def by blast
have witness:
  (trace-openings ! ?i, composition-openings ! ?i) ∈
    query-header-supported-partial-opening-witnesses ?s fr f-fri-roots
    f-final as dg composition-fri-roots final
  unfolding query-header-supported-partial-opening-witnesses-def
  using comp-nonempty verifier trace-table comp-table header by blast
have idx-in-header-set:
  query-idxs ! ?i ∈
    query-header-supported-partial-opening-success-indices-at ?s fr
    f-fri-roots f-final as dg composition-fri-roots final ?i
  unfolding query-header-supported-partial-opening-success-indices-at-def
  using idx-sample witness opening-consistent by blast
have idx-in-staged-set:
  query-idxs ! ?i ∈
    query-header-supported-partial-opening-success-indices-at ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)

```

```

      (staged-composition-fri-roots data)
      (staged-composition-final data)
      ?i
    using idx-in-header-set header-eq by simp
  have raw-idx:
    index (to-nat (raw-idxs ! ?i)) = query-idxs ! ?i
    using query-idxs-eq len-raw i-bound by simp
  have lookup-i:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge ?i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) ?i)) =
      Some (raw-idxs ! ?i)
    using lookup i-bound by simp
  have hit-at:
    staged-security-with-data-state-query-partial-opening-hit-at ?i
    (Some (((data, attacker-state), result), final-state))
  unfolding staged-security-with-data-state-query-partial-opening-hit-at-def
  using i-bound lookup-i idx-in-staged-set raw-idx
  by (simp add: Let-def)
show ?thesis
by (rule staged-security-with-data-state-query-partial-opening-hitI
  [OF i-bound hit-at])
qed

```

lemma *checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-partial-opening-hit:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-query-partial-opening-hit
adversary-initial-state

proof (rule *wp-event-mono-on-support*)

fix *out*

assume *support:*

out ∈
set-dist
(execute
(checked-staged-security-experiment-with-data-state A)
adversary-initial-state)

and *bad:*

staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates out

show *staged-security-with-data-state-query-partial-opening-hit out*

```

proof (cases out)
  case None
  then show ?thesis
    using bad
    unfolding staged-security-with-data-state-verifier-event-def by simp
next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  show ?thesis
    unfolding out-eq
    by (rule
      checked-staged-security-with-data-state-aligned-transcript-query-bad-imp-partial-opening-hit-on-support
      [OF wf controlled])
      (use support bad out-eq in simp-all)
qed
qed

lemma selected-query-openings-align-with-replayed-query-key:
  assumes i-bound: i < rounds
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and selected-idx: idx = index (to-nat raw)
  and selected-lookup:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge k prefix-state) = Some raw
  and replay-lookup:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge k prefix-state) = Some (raw-idxs ! i)
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and comp-indices:
    map opening-index composition-openings =
      [idx, fri-sibling-index (scale * clength) idx]
  and consistent:
    partial-query-openings-consistent trace-openings
      composition-openings as idx
shows
  idx = query-idxs ! i
  map opening-index trace-openings = powers-scaled (query-idxs ! i)
  map opening-index composition-openings =
    [query-idxs ! i,
      fri-sibling-index (scale * clength) (query-idxs ! i)]
  partial-query-openings-consistent trace-openings composition-openings
    as (query-idxs ! i)
proof –
  have raw-eq: raw = raw-idxs ! i

```

```

    using selected-lookup replay-lookup by simp
  have idx-eq:  $idx = query-idxs ! i$ 
    using selected-idx raw-eq query-idxs-eq len-raw i-bound by simp
  show  $idx = query-idxs ! i$ 
    by (rule idx-eq)
  show map opening-index trace-openings = powers-scaled (query-idxs ! i)
    using trace-indices idx-eq by simp
  show map opening-index composition-openings =
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)]
    using comp-indices idx-eq by simp
  show partial-query-openings-consistent trace-openings composition-openings
    as (query-idxs ! i)
    using consistent idx-eq by simp
qed

```

lemma *selected-query-openings-align-with-replayed-query-index:*

```

  assumes i-bound:  $i < rounds$ 
  and len-raw:  $length\ raw-idxs = rounds$ 
  and query-idxs-eq:
     $query-idxs = map (\lambda raw. index\ (to-nat\ raw))\ raw-idxs$ 
  and selected-idx:  $idx = index\ (to-nat\ raw)$ 
  and selected-lookup:
     $fmlookup\ (HashMap\ final-state)$ 
     $(QueryIndexChallenge\ i\ prefix-state) = Some\ raw$ 
  and replay-lookup:
     $fmlookup\ (HashMap\ final-state)$ 
     $(QueryIndexChallenge\ i\ prefix-state) = Some\ (raw-idxs ! i)$ 
  and trace-indices:
     $map\ opening-index\ trace-openings = powers-scaled\ idx$ 
  and comp-indices:
     $map\ opening-index\ composition-openings =$ 
    [ $idx$ ,  $fri-sibling-index\ (scale * clength)\ idx$ ]
  and consistent:
     $partial-query-openings-consistent\ trace-openings$ 
     $composition-openings\ as\ idx$ 
  shows
     $idx = query-idxs ! i$ 
     $map\ opening-index\ trace-openings = powers-scaled\ (query-idxs ! i)$ 
     $map\ opening-index\ composition-openings =$ 
    [ $query-idxs ! i$ ,
      $fri-sibling-index\ (scale * clength)\ (query-idxs ! i)$ ]
     $partial-query-openings-consistent\ trace-openings\ composition-openings$ 
     $as\ (query-idxs ! i)$ 
  by (rule selected-query-openings-align-with-replayed-query-key
    [OF i-bound len-raw query-idxs-eq selected-idx selected-lookup
     replay-lookup trace-indices comp-indices consistent])

```

lemma *ntimes-verifier-query-rounds-selected-authenticated-chunks-aligned-with-replay-at:*

```

assumes fl-eq: fl = (b, composition-root) # fl-tail
and outcome:
  Some (results, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
and i-bound: i < rounds
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and query-chunk:
  Λi. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and replay-lookup:
  Λi. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
obtains trace-openings composition-openings where
  map opening-index trace-openings = powers-scaled (query-idxs ! i)
  partial-authenticated-table fr (scale * clength) trace-openings
    final-state
  partial-authenticated-table composition-root (scale * clength)
    composition-openings final-state
  map opening-index composition-openings =
    [query-idxs ! i,
      fri-sibling-index (scale * clength) (query-idxs ! i)]
  partial-query-openings-consistent trace-openings composition-openings
    as (query-idxs ! i)
proof (rule ntimes-verifier-query-rounds-selected-authenticated-chunks-at
  [OF fl-eq outcome i-bound])
fix raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
  suffix-chunks trace-openings composition-openings
assume selected-idx: idx = index (to-nat raw)
and len-prefix-chunks: length prefix-chunks = i
and len-suffix-chunks: length suffix-chunks = rounds - Suc i
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
and prefix-rounds:

```

```

 $\bigwedge j. j < i \implies$ 
  verifier-query-round-chunk (prefix-query-idxs ! j)
  (map snd f-fl) (map snd fl) (prefix-chunks ! j)
and suffix-rounds:
 $\bigwedge j. j < \text{rounds} - \text{Suc } i \implies$ 
  verifier-query-round-chunk (suffix-query-idxs ! j)
  (map snd f-fl) (map snd fl) (suffix-chunks ! j)
and selected-transcript:
  PTranscript query-state =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @
      PTranscript final-state
and trace-indices:
  map opening-index trace-openings = powers-scaled idx
and trace-table:
  partial-authenticated-table fr (scale * clength) trace-openings
  final-state
and comp-table:
  partial-authenticated-table composition-root (scale * clength)
  composition-openings final-state
and comp-indices:
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
and consistent:
  partial-query-openings-consistent trace-openings
  composition-openings as idx
and selected-lookup:
  fmlookup (HashMap final-state)
  (QueryIndexChallenge (PQueryCounter query-state + i)
    (state-after-query-chunks (PState query-state) prefix-chunks i)) =
    Some raw
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks i-bound by simp
have selected-chunk-lens:
 $\bigwedge j. j < \text{rounds} \implies$ 
  length (?selected-chunks ! j) =
    verifier-query-round-transcript-length (query-idxs ! j)
    (map snd f-fl) (map snd fl)
proof –
  fix j
  assume j-bound: j < rounds
  show
    length (?selected-chunks ! j) =
      verifier-query-round-transcript-length (query-idxs ! j)
      (map snd f-fl) (map snd fl)
proof (cases j < i)
  case True
  have nth-eq: ?selected-chunks ! j = prefix-chunks ! j
    using True len-prefix-chunks by (simp add: nth-append)

```

```

have
  length (prefix-chunks ! j) =
    verifier-query-round-transcript-length
      (prefix-query-idxs ! j) (map snd f-fl) (map snd fl)
  by (rule verifier-query-round-chunk-length[OF prefix-rounds[OF True]])
then show ?thesis
  using nth-eq
  by (simp add: verifier-query-round-transcript-length-index-irrelevant)
next
case False
show ?thesis
proof (cases j = i)
  case True
  have nth-eq: ?selected-chunks ! j = chunk
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length chunk =
      verifier-query-round-transcript-length idx
        (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length[OF chunk-shape])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
  next
  case False
  let ?k = j - Suc i
  have ge: Suc i ≤ j
    using <¬ j < i> False by linarith
  have k-bound: ?k < rounds - Suc i
    using ge j-bound by linarith
  have nth-eq: ?selected-chunks ! j = suffix-chunks ! ?k
    using ge len-prefix-chunks by (simp add: nth-append)
  have
    length (suffix-chunks ! ?k) =
      verifier-query-round-transcript-length
        (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
      [OF suffix-rounds[OF k-bound]])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-lens:
   $\bigwedge j. j < rounds \implies$ 
  length (query-chunks ! j) =

```



```

    verifier-query-round-transcript-length (query-idxs ! j)
      (map snd f-fl) (map snd fl)
  using query-chunk verifier-query-round-chunk-length by blast
have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
  using selected-transcript transcript-query by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
     $\bigwedge j. j < \text{length } ?\text{selected-chunks} \implies$ 
       $\text{length } (?selected-chunks ! j) = \text{length } (query-chunks ! j)$ 
    using selected-chunk-lens query-chunk-lens len-selected-chunks by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks  $\wedge$  ([] :: 'f list) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
      [OF len-eq same-lens concat-eq-empty])
  then show ?thesis
    by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix[OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have selected-lookup-global:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw
  using selected-lookup prefix-state-eq by simp
have aligned:
  idx = query-idxs ! i
  map opening-index trace-openings = powers-scaled (query-idxs ! i)
  map opening-index composition-openings =
    [query-idxs ! i,
      fri-sibling-index (scale * clength) (query-idxs ! i)]
  partial-query-openings-consistent trace-openings composition-openings
    as (query-idxs ! i)
  by (rule selected-query-openings-align-with-replayed-query-key
    [OF i-bound len-raw query-idxs-eq selected-idx])

```

```

      selected-lookup-global replay-lookup[OF i-bound] trace-indices
      comp-indices consistent])+)
show ?thesis
by (rule that)
  (use aligned trace-table comp-table in simp-all)
qed

lemma ntimes-verifier-query-rounds-aligned-authenticated-openings-consistent-with-replay:
assumes fl-eq: fl = (b, composition-root) # fl-tail
and outcome:
  Some (results, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and query-chunk:
  ∧ i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
and replay-lookup:
  ∧ i. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
obtains trace-openings composition-openings where
  length trace-openings = rounds
  length composition-openings = rounds
  ∧ i. i < rounds ⇒
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
  ∧ i. i < rounds ⇒
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
  ∧ i. i < rounds ⇒
    partial-authenticated-table composition-root (scale * clength)
      (composition-openings ! i) final-state
  ∧ i. i < rounds ⇒
    map opening-index (composition-openings ! i) =

```

```

    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)]
  ∧ i. i < rounds ⇒
    partial-query-round-consistent trace-openings composition-openings as i
    (query-idxs ! i)
proof –
  let ?P =
    λ i w.
      map opening-index (fst w) = powers-scaled (query-idxs ! i) ∧
      partial-authenticated-table fr (scale * clength) (fst w)
      final-state ∧
      partial-authenticated-table composition-root (scale * clength)
      (snd w) final-state ∧
      map opening-index (snd w) =
        [query-idxs ! i,
         fri-sibling-index (scale * clength) (query-idxs ! i)] ∧
      query-idxs ! i ∈ query-sample-space ∧
      snd w ≠ [] ∧
      opening-value (snd w ! 0) =
        cp-eval as (map opening-value (fst w))
        (h ^ query-idxs ! i * shift)
have ex-round: ∧ i. i < rounds ⇒ ∃ w. ?P i w
proof –
  fix i
  assume i-bound: i < rounds
  show ∃ w. ?P i w
  proof (rule
    ntimes-verifier-query-rounds-selected-authenticated-chunks-aligned-with-replay-at
    [OF fl-eq outcome i-bound len-raw query-idxs-eq len-query-chunks
     transcript-query query-chunk replay-lookup])
  fix trace-openings composition-openings
  assume trace-indices:
    map opening-index trace-openings =
      powers-scaled (query-idxs ! i)
  and trace-table:
    partial-authenticated-table fr (scale * clength) trace-openings
    final-state
  and comp-table:
    partial-authenticated-table composition-root (scale * clength)
    composition-openings final-state
  and comp-indices:
    map opening-index composition-openings =
      [query-idxs ! i,
       fri-sibling-index (scale * clength) (query-idxs ! i)]
  and consistent:
    partial-query-openings-consistent trace-openings
    composition-openings as (query-idxs ! i)
have idx-sample: query-idxs ! i ∈ query-sample-space
  using consistent unfolding partial-query-openings-consistent-def

```

```

    by simp
  have comp-nonempty: composition-openings  $\neq$  []
    using consistent unfolding partial-query-openings-consistent-def
    by simp
  have comp-value:
    opening-value (composition-openings ! 0) =
      cp-eval as (map opening-value trace-openings)
        (h  $\wedge$  query-idxs ! i * shift)
    using consistent unfolding partial-query-openings-consistent-def
    by simp
  show ?thesis
    by (intro exI[of - (trace-openings, composition-openings)]
        conjI)
      (use trace-indices trace-table comp-table comp-indices consistent
        idx-sample comp-nonempty comp-value in simp-all)
qed
qed
let ?wit =  $\lambda i$ . SOME w. ?P i w
let ?trace-openings = map ( $\lambda i$ . fst (?wit i)) [0.. $\text{rounds}$ ]
let ?composition-openings = map ( $\lambda i$ . snd (?wit i)) [0.. $\text{rounds}$ ]
have round-props:  $\bigwedge i$ . i < rounds  $\implies$  ?P i (?wit i)
  by (rule someI-ex[OF ex-round])
have consistent:
   $\bigwedge i$ . i < rounds  $\implies$ 
    partial-query-round-consistent ?trace-openings ?composition-openings as
      i (query-idxs ! i)
proof -
  fix i
  assume i-bound: i < rounds
  have chosen: ?P i (?wit i)
    by (rule round-props[OF i-bound])
  have nth:
    ?trace-openings ! i = fst (?wit i)
    ?composition-openings ! i = snd (?wit i)
    using i-bound by simp-all
  have lengths:
    length ?trace-openings = rounds
    length ?composition-openings = rounds
    by simp-all
  have idx-sample: query-idxs ! i  $\in$  query-sample-space
    using chosen by blast
  have trace-indices0:
    map opening-index (fst (?wit i)) =
      powers-scaled (query-idxs ! i)
    using chosen by blast
  have comp-indices0:
    map opening-index (snd (?wit i)) =
      [query-idxs ! i,
        fri-sibling-index (scale * clength) (query-idxs ! i)]

```

```

using chosen by blast
have comp-value0:
  opening-value (snd (?wit i) ! 0) =
    cp-eval as (map opening-value (fst (?wit i)))
      (h ^ query-idxs ! i * shift)
using chosen by blast
have trace-indices:
  map opening-index (?trace-openings ! i) =
    powers-scaled (query-idxs ! i)
using trace-indices0 nths by simp
have comp-indices:
  map opening-index (?composition-openings ! i) =
    [query-idxs ! i,
      fri-sibling-index (scale * clength) (query-idxs ! i)]
using comp-indices0 nths by simp
have comp-value:
  opening-value ((?composition-openings ! i) ! 0) =
    cp-eval as (map opening-value (?trace-openings ! i))
      (h ^ query-idxs ! i * shift)
using comp-value0 nths by simp
show
  partial-query-round-consistent ?trace-openings ?composition-openings as
    i (query-idxs ! i)
by (rule partial-query-round-consistentI
  [OF i-bound idx-sample lengths trace-indices comp-indices comp-value])
qed
show ?thesis
proof (rule that[of ?trace-openings ?composition-openings])
  show length ?trace-openings = rounds
    by simp
  show length ?composition-openings = rounds
    by simp
  show  $\bigwedge i. i < \text{rounds} \implies$ 
    map opening-index (?trace-openings ! i) =
      powers-scaled (query-idxs ! i)
  proof -
    fix i
    assume i-bound: i < rounds
    have chosen: ?P i (?wit i)
      by (rule round-props[OF i-bound])
    have nth: ?trace-openings ! i = fst (?wit i)
      using i-bound by simp
    have trace-indices0:
      map opening-index (fst (?wit i)) =
        powers-scaled (query-idxs ! i)
      using chosen by blast
    show map opening-index (?trace-openings ! i) =
      powers-scaled (query-idxs ! i)
      using trace-indices0 nth by simp

```

```

qed
show  $\bigwedge i. i < \text{rounds} \implies$ 
  partial-authenticated-table fr (scale * clength)
  (?trace-openings ! i) final-state
proof -
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  have chosen: ?P i (?wit i)
    by (rule round-props[OF i-bound])
  have nth: ?trace-openings ! i = fst (?wit i)
    using i-bound by simp
  have trace-table0:
    partial-authenticated-table fr (scale * clength)
    (fst (?wit i)) final-state
    using chosen by blast
  show partial-authenticated-table fr (scale * clength)
    (?trace-openings ! i) final-state
    using trace-table0 nth by simp
qed
show  $\bigwedge i. i < \text{rounds} \implies$ 
  partial-authenticated-table composition-root (scale * clength)
  (?composition-openings ! i) final-state
proof -
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  have chosen: ?P i (?wit i)
    by (rule round-props[OF i-bound])
  have nth: ?composition-openings ! i = snd (?wit i)
    using i-bound by simp
  have comp-table0:
    partial-authenticated-table composition-root (scale * clength)
    (snd (?wit i)) final-state
    using chosen by blast
  show partial-authenticated-table composition-root (scale * clength)
    (?composition-openings ! i) final-state
    using comp-table0 nth by simp
qed
show  $\bigwedge i. i < \text{rounds} \implies$ 
  map opening-index (?composition-openings ! i) =
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)]
proof -
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  have chosen: ?P i (?wit i)
    by (rule round-props[OF i-bound])
  have nth: ?composition-openings ! i = snd (?wit i)
    using i-bound by simp
  have comp-indices0:

```

```

    map opening-index (snd (?wit i)) =
      [query-idxs ! i,
       fri-sibling-index (scale * clength) (query-idxs ! i)]
    using chosen by blast
  show map opening-index (?composition-openings ! i) =
    [query-idxs ! i,
     fri-sibling-index (scale * clength) (query-idxs ! i)]
    using comp-indices0 nth by simp
qed
show  $\bigwedge i. i < \text{rounds} \implies$ 
  partial-query-round-consistent ?trace-openings ?composition-openings as
    i (query-idxs ! i)
  by (rule consistent)
qed
qed

lemma query-rounds-replay-accepted-with-partial-initial-openings-aligned-consistent:
  assumes header0:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and comp-nonempty: composition-fri-roots  $\neq []$ 
  and fl-eq: fl = (b, composition-root) # fl-tail
  and f-roots: map snd f-fl = f-fri-roots
  and comp-roots: map snd fl = composition-fri-roots
  and query-out:
    Some (result, final-state)  $\in$ 
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            rounds)
          query-state)
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda \text{raw}. \text{index } (\text{to-nat } \text{raw})$ ) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
  and query-chunk:
     $\bigwedge i. i < \text{rounds} \implies$ 
      verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
  and replay-lookup:
     $\bigwedge i. i < \text{rounds} \implies$ 
      fmlookup (HashMap final-state)
        (QueryIndexChallenge (PQueryCounter query-state + i)
          (state-after-query-chunks (PState query-state) query-chunks i)) =
        Some (raw-idxs ! i)

```

```

obtains trace-openings composition-openings where
  accepted-with-partial-initial-openings-aligned-consistent s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
proof –
  have root-eq: composition-root = hd composition-fri-roots
  proof –
    have composition-root # map snd fl-tail = composition-fri-roots
      using comp-roots unfolding fl-eq by simp
    then have hd composition-fri-roots = composition-root
      by (metis list.sel(1))
    then show ?thesis by simp
  qed
show ?thesis
proof (rule
  ntimes-verifier-query-rounds-aligned-authenticated-openings-consistent-with-replay
  [OF fl-eq query-out len-raw query-idxs-eq len-query-chunks
   transcript-query query-chunk replay-lookup])
fix trace-openings composition-openings
assume len-trace: length trace-openings = rounds
and len-comp: length composition-openings = rounds
and trace-indices:
   $\bigwedge i. i < rounds \implies$ 
  map opening-index (trace-openings ! i) =
  powers-scaled (query-idxs ! i)
and trace-tables:
   $\bigwedge i. i < rounds \implies$ 
  partial-authenticated-table fr (scale * clength)
  (trace-openings ! i final-state)
and comp-tables:
   $\bigwedge i. i < rounds \implies$ 
  partial-authenticated-table composition-root (scale * clength)
  (composition-openings ! i final-state)
and comp-indices:
   $\bigwedge i. i < rounds \implies$ 
  map opening-index (composition-openings ! i) =
  [query-idxs ! i,
   fri-sibling-index (scale * clength) (query-idxs ! i)]
and consistent:
   $\bigwedge i. i < rounds \implies$ 
  partial-query-round-consistent trace-openings composition-openings
  as i (query-idxs ! i)
have len-query-idxs: length query-idxs = rounds
  using len-raw query-idxs-eq by simp
have trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr query-idxs trace-openings
proof –

```



```

have all-indices:
   $\forall i < \text{rounds}.$ 
     $\text{map } \text{opening-index } (\text{trace-openings } ! i) =$ 
       $\text{powers-scaled } (\text{query-idxs } ! i)$ 
    using trace-indices by blast
have all-tables:
   $\forall i < \text{rounds}.$ 
     $\text{partial-authenticated-table } \text{fr } (\text{scale} * \text{clength})$ 
       $(\text{trace-openings } ! i) \text{ final-state}$ 
    using trace-tables by blast
show ?thesis
  unfolding accepted-with-partial-trace-openings-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state])
    (use len-query-idxs len-trace all-indices all-tables in simp-all)
qed
have comp-partial:
  accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots) query-idxs
      composition-openings
proof –
  have all-indices:
     $\forall i < \text{rounds}.$ 
       $\text{map } \text{opening-index } (\text{composition-openings } ! i) =$ 
        [query-idxs ! i,
           $\text{fri-sibling-index } (\text{scale} * \text{clength}) (\text{query-idxs } ! i)$ ]
      using comp-indices by blast
  have all-tables:
     $\forall i < \text{rounds}.$ 
       $\text{partial-authenticated-table } (\text{hd } \text{composition-fri-roots})$ 
         $(\text{scale} * \text{clength}) (\text{composition-openings } ! i) \text{ final-state}$ 
      using comp-tables root-eq by simp
  show ?thesis
  unfolding accepted-with-partial-composition-openings-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state])
    (use len-query-idxs len-comp all-indices all-tables in simp-all)
qed
have aligned:
  accepted-with-partial-initial-openings-aligned s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
      composition-fri-roots final query-idxs trace-openings
      composition-openings
proof –
  have accepted-out: accepted (Some (result, final-state))
    using trace-partial
    by (rule accepted-with-partial-trace-openings-imp-accepted)
  show ?thesis
  unfolding accepted-with-partial-initial-openings-aligned-def
  by (intro conjI exI[of - rest])
    (use accepted-out comp-nonempty header0 trace-partial comp-partial)

```

```

      in simp-all)
qed
have all-consistent:
  ∀ i < rounds.
    partial-query-round-consistent trace-openings composition-openings as i
    (query-idxs ! i)
  using consistent by blast
have aligned-consistent:
  accepted-with-partial-initial-openings-aligned-consistent s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
  unfolding accepted-with-partial-initial-openings-aligned-consistent-def
  using aligned all-consistent by simp
show ?thesis
  by (rule that[OF aligned-consistent])
qed
qed

lemma query-rounds-replay-accepted-with-partial-initial-openings-aligned-transcript-consistent:
  assumes header0:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and comp-nonempty: composition-fri-roots ≠ []
  and fl-eq: fl = (b, composition-root) # fl-tail
  and f-roots: map snd f-fl = f-fri-roots
  and comp-roots: map snd fl = composition-fri-roots
  and query-state-transcript: PTranscript query-state = rest
  and query-state-state:
    PState query-state =
      verifier-header-state s fr f-fri-roots f-final as dg
      composition-fri-roots final
  and query-state-counter: PQueryCounter query-state = PQueryCounter s
  and query-out:
    Some (result, final-state) ∈
      set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
  and query-chunk:

```

$\bigwedge i. i < \text{rounds} \implies$
 $\text{verifier-query-round-chunk } (\text{query-idxs} ! i)$
 $(\text{map snd f-fl}) (\text{map snd fl}) (\text{query-chunks} ! i)$
and *replay-lookup*:
 $\bigwedge i. i < \text{rounds} \implies$
 $\text{fmlookup } (\text{HashMap } \text{final-state})$
 $(\text{QueryIndexChallenge } (\text{PQueryCounter } \text{query-state} + i)$
 $(\text{state-after-query-chunks } (\text{PState } \text{query-state}) \text{query-chunks } i)) =$
 $\text{Some } (\text{raw-idxs} ! i)$
obtains *trace-openings composition-openings* **where**
accepted-with-partial-initial-openings-aligned-transcript-consistent s
 $(\text{Some } (\text{result}, \text{final-state})) \text{ fr } f\text{-fri-roots } f\text{-final as } dg$
composition-fri-roots final query-idxs trace-openings
composition-openings
proof –
from *query-rounds-replay-accepted-with-partial-initial-openings-aligned-consistent*
 $[\text{OF header0 comp-nonempty fl-eq } f\text{-roots comp-roots query-out}$
 $\text{len-raw query-idxs-eq len-query-chunks transcript-query}$
 $\text{query-chunk replay-lookup}]$
obtain *trace-openings composition-openings* **where** *aligned*:
accepted-with-partial-initial-openings-aligned-consistent s
 $(\text{Some } (\text{result}, \text{final-state})) \text{ fr } f\text{-fri-roots } f\text{-final as } dg$
composition-fri-roots final query-idxs trace-openings
composition-openings
by *blast*
have *transcript-rest*:
 $\text{List.concat query-chunks @ PTranscript final-state} = \text{rest}$
using *transcript-query query-state-transcript* **by** *simp*
have *chunks*:
 $\forall i < \text{rounds.}$
 $\text{verifier-query-round-chunk } (\text{query-idxs} ! i)$
 $f\text{-fri-roots composition-fri-roots } (\text{query-chunks} ! i)$
using *query-chunk f-roots comp-roots* **by** *simp*
have *lookups*:
 $\forall i < \text{rounds.}$
 $\text{fmlookup } (\text{HashMap } \text{final-state})$
 $(\text{QueryIndexChallenge } (\text{PQueryCounter } s + i)$
 $(\text{state-after-query-chunks}$
 $(\text{verifier-header-state } s \text{ fr } f\text{-fri-roots } f\text{-final as } dg$
 $\text{composition-fri-roots final})$
 $\text{query-chunks } i)) =$
 $\text{Some } (\text{raw-idxs} ! i)$
using *replay-lookup query-state-state query-state-counter* **by** *simp*
have *idx-bounds*:
 $\forall \text{idx} \in \text{set query-idxs. idx} < \text{clength} * \text{scale}$
proof
fix *idx*
assume $\text{idx} \in \text{set query-idxs}$
then obtain *raw* **where** $\text{raw} \in \text{set raw-idxs}$

```

    and idx: idx = index (to-nat raw)
    unfolding query-idxs-eq by auto
have index (to-nat raw) < query-sample-space-size
  by (rule index-less-query-sample-space)
also have ... ≤ clength * scale
  by (rule query-sample-space-size-le-domain)
finally show idx < clength * scale
  using idx by simp
qed
have derived:
  verifier-query-indices-derived s (Some (result, final-state))
    (verifier-header-state s fr f-fri-roots f-final as dg
      composition-fri-roots final)
  rest f-fri-roots composition-fri-roots query-idxs
unfolding verifier-query-indices-derived-def
by (intro exI[of - result] exI[of - final-state]
    exI[of - raw-idxs] exI[of - query-chunks]
    exI[of - PTranscript final-state] conjI)
  (use len-raw query-idxs-eq len-query-chunks transcript-rest chunks
    lookups idx-bounds in auto)
have shape:
  accepted-transcript-shape s (Some (result, final-state)) as query-idxs
unfolding accepted-transcript-shape-def
by (intro exI[of - result] exI[of - final-state] exI[of - fr]
    exI[of - f-fri-roots] exI[of - f-final] exI[of - dg]
    exI[of - composition-fri-roots] exI[of - final]
    exI[of - rest] conjI)
  (use header0 derived in simp-all)
have tied:
  accepted-with-partial-initial-openings-aligned-transcript-consistent s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
unfolding
  accepted-with-partial-initial-openings-aligned-transcript-consistent-def
using aligned shape by simp
show ?thesis
  by (rule that[OF tied])
qed

end

end

```

```

theory Soundness-Aligned-Partial-Query-Prefix
imports
  Soundness-Aligned-Partial-Query-Transcript
  Soundness-Staged-Candidate-Query

```

begin

Bridge aligned partial-query evidence to prefix-local staged query targets.

The lemmas here are deterministic: they say that, once a verifier query index is replayed to a prefix receive and the prefix alphas agree with the accepted transcript, the aligned partial-candidate query evidence produces a hit in the existing prefix-local target. Probability bounds remain in the staged prefix-target layer.

context *soundness*

begin

definition *checked-staged-security-experiment-with-query-prefix-data-state*

where

checked-staged-security-experiment-with-query-prefix-data-state $A\ i =$
checked-staged-query-prefix-receive-with-state $A\ i \gg=$
 $(\lambda \text{prefix-out. get } \gg=$
 $(\lambda \text{prefix-state.}$
 $\text{checked-staged-after-query-prefix-receive-with-verifier } A\ i$
 $\text{prefix-out } \gg=$
 $(\lambda \text{out. return } ((\text{prefix-out}, \text{prefix-state}), \text{out}))))$

lemma *checked-staged-security-experiment-with-query-prefix-data-state-projection:*

assumes *i-bound*: $i < \text{rounds}$

shows

checked-staged-security-experiment-with-query-prefix-data-state $A\ i \gg=$
 $(\lambda \text{packed. return } (\text{snd } \text{packed})) =$
checked-staged-security-experiment-with-data-state A

unfolding

checked-staged-security-experiment-with-query-prefix-data-state-def
checked-staged-security-experiment-with-data-state-query-prefix-receive-decomp
 $[OF\ i\text{-bound}]$

by (*simp add: sm-bind-assoc get-bind-const*)

lemma *checked-staged-security-experiment-with-query-prefix-data-state-projection-event:*

assumes *i-bound*: $i < \text{rounds}$

shows

wp-event (*checked-staged-security-experiment-with-data-state* A) E
adversary-initial-state =
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A\ i)$
 $(\lambda \text{out. case out of}$
 $\text{None } \Rightarrow E\ \text{None}$
 $| \text{Some } (\text{packed}, t) \Rightarrow E\ (\text{Some } (\text{snd } \text{packed}, t)))$
adversary-initial-state

proof –

have

wp-event (*checked-staged-security-experiment-with-data-state* A) E
adversary-initial-state =
wp-event

```

    (checked-staged-security-experiment-with-query-prefix-data-state A i >>=
      (λpacked. return (snd packed)))
    E adversary-initial-state
  using
    checked-staged-security-experiment-with-query-prefix-data-state-projection
      [OF i-bound, of A]
  by simp
  also have ... =
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (λout. case out of
        None ⇒ E None
      | Some (packed, t) ⇒ E (Some (snd packed, t)))
      adversary-initial-state
    by (rule wp-event-bind-return-map)
  finally show ?thesis .
qed

lemma checked-staged-security-experiment-with-query-prefix-data-state-imp-data-state-support:
  assumes i-bound: i < rounds
  and outcome:
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
    ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
  shows
    Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
proof -
  from outcome obtain t where prefix-receive:
    Some (((prefix, prefix-state), raw), t) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  and t-eq: t = raw-state
  and cont:
    Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive-with-verifier A i
          ((prefix, prefix-state), raw))
        raw-state)
  unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
  by (auto elim!: set-dist-bindE split: prod.splits)

```

```

have data-out:
  Some (data, attacker-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive A i
          ((prefix, prefix-state), raw))
        raw-state)
  by (rule checked-staged-after-query-prefix-receive-with-verifier-outcome(1)
    [OF cont])
have verifier:
  Some (result, final-state) ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
  by (rule checked-staged-after-query-prefix-receive-with-verifier-outcome(2)
    [OF cont])
have builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  unfolding checked-staged-transcript-program-query-prefix-receive-decomp
    [OF i-bound]
  apply (rule-tac x=((prefix, prefix-state), raw) and
    t=raw-state in set-dist-bindI)
  apply (rule prefix-receive[unfolded t-eq])
  apply (rule data-out)
  done
show ?thesis
  by (rule checked-staged-security-experiment-with-data-state-outcomeI
    [OF builder verifier])
qed

lemma checked-staged-security-experiment-with-query-prefix-data-state-bound-by-prefix:
assumes prefix-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i) P
  adversary-initial-state ≤ B
and none-imp: Q None ⇒ P None
and cont-imp:
   $\bigwedge$ prefix-out t out.
  Some (prefix-out, t) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state) ⇒
  out ∈
    set-dist
      (execute
        (get  $\gg$ 

```

$(\lambda \text{prefix-state.}$
 $\quad \text{checked-staged-after-query-prefix-receive-with-verifier } A \ i$
 $\quad \text{prefix-out} \gg=$
 $\quad (\lambda \text{result. return } ((\text{prefix-out}, \text{prefix-state}), \text{result}))))$
 $t) \implies$
 $Q \text{ out} \implies P \text{ (Some (prefix-out}, t))$
shows
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $Q \text{ adversary-initial-state} \leq B$
unfolding $\text{checked-staged-security-experiment-with-query-prefix-data-state-def}$
by $(\text{rule wp-event-bind-bound-by-head-event}$
 $\quad [OF \text{ prefix-bound none-imp cont-imp}])$

definition $\text{checked-staged-security-with-query-prefix-dynamic-index-hit}$

where

$\text{checked-staged-security-with-query-prefix-dynamic-index-hit } T \text{ out} \longleftrightarrow$
 $(\text{case out of}$
 $\quad \text{None} \Rightarrow \text{False}$
 $\quad | \text{Some } (((\text{prefix-pack}, \text{raw}), \text{raw-state}), \text{result}), \text{final-state})) \Rightarrow$
 $\quad \text{checked-staged-query-prefix-dynamic-index-hit } T$
 $\quad (\text{Some } ((\text{prefix-pack}, \text{raw}), \text{raw-state})))$

definition $\text{checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit}$

where

$\text{checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit}$
 $\text{trace-table composition-table out} \longleftrightarrow$
 $\text{checked-staged-security-with-query-prefix-dynamic-index-hit}$
 $(\text{staged-query-prefix-candidate-pair-query-target-from-prefix}$
 $\text{trace-table composition-table}) \text{ out}$

lemma $\text{checked-staged-security-with-query-prefix-dynamic-index-hit-bound:}$

assumes prefix-bound:

$\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit } T)$
 $\text{adversary-initial-state} \leq B$

shows

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-dynamic-index-hit } T)$
 $\text{adversary-initial-state} \leq B$

proof $(\text{rule}$

$\text{checked-staged-security-experiment-with-query-prefix-data-state-bound-by-prefix}$
 $[OF \text{ prefix-bound}])$

show

$\text{checked-staged-security-with-query-prefix-dynamic-index-hit } T \text{ None} \implies$
 $\text{checked-staged-query-prefix-dynamic-index-hit } T \text{ None}$

unfolding $\text{checked-staged-security-with-query-prefix-dynamic-index-hit-def}$

by simp


```

next
  fix prefix-out t out
  assume head:
    Some (prefix-out, t) ∈
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  and cont:
    out ∈
      set-dist
      (execute
        (get  $\gg$ 
          ( $\lambda$ prefix-state.
            checked-staged-after-query-prefix-receive-with-verifier A i
              prefix-out  $\gg$ 
              ( $\lambda$ result. return ((prefix-out, prefix-state), result))))
        t)
  assume hit:
    checked-staged-security-with-query-prefix-dynamic-index-hit T out
  show
    checked-staged-query-prefix-dynamic-index-hit T
    (Some (prefix-out, t))
  using cont hit
  unfolding checked-staged-security-with-query-prefix-dynamic-index-hit-def
  by (auto elim!: set-dist-bindE split: option.splits prod.splits)
qed

```

lemma *checked-staged-query-prefix-dynamic-index-hit-bound-from-fraction:*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and target-bound:
   $\bigwedge$ prefix prefix-state.
    Some ((prefix, prefix-state), prefix-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)  $\implies$ 
    T prefix prefix-state  $\subseteq$  query-sample-space  $\wedge$ 
    nnreal (card (T prefix prefix-state)) /
      nnreal (card query-sample-space) ≤ C

```

shows

```

wp-event (checked-staged-query-prefix-receive-with-state A i)
  (checked-staged-query-prefix-dynamic-index-hit T)
adversary-initial-state  $\leq$ 
staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) + C

```

proof –

```

have raw-bound: query-index-raw-preimage-bound
by (rule query-index-raw-preimage-bound-from-sampler-wellformed)

```

have *hit-bound*:

$$\begin{aligned} & wp\text{-event } (checked\text{-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (checked\text{-staged-query-prefix-dynamic-index-hit } T) \\ & \quad adversary\text{-initial-state} \leq \\ & wp\text{-event } (checked\text{-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (checked\text{-staged-query-prefix-dynamic-index-prehit } T) \\ & \quad adversary\text{-initial-state} + C \end{aligned}$$
by (rule

$$checked\text{-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error}$$

$$[OF \ raw\text{-bound} \ target\text{-bound}])$$
have *prehit-bound*:

$$\begin{aligned} & wp\text{-event } (checked\text{-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (checked\text{-staged-query-prefix-dynamic-index-prehit } T) \\ & \quad adversary\text{-initial-state} \leq \\ & \quad staged\text{-phase-relation-error } size \\ & \quad (staged\text{-query-search-queries } budgets \ i + 1) \end{aligned}$$
proof (rule *checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program*)
show $HashMap \ adversary\text{-initial-state} = fmempty$
by (*simp add: adversary-initial-state-def*)
show *hash-relation-program*

$$\begin{aligned} & (checked\text{-staged-query-prefix-dynamic-index-prehit-relation } A \ i) \\ & \quad adversary\text{-initial-state } T) \\ & \quad size \ (staged\text{-query-search-queries } budgets \ i + 1) \\ & \quad (checked\text{-staged-query-prefix-receive-with-state } A \ i) \end{aligned}$$
by (rule *hash-relation-program-checked-staged-query-prefix-receive-with-state*

$$[OF \ wf \ controlled])$$

$$(use \ i\text{-bound} \ in \ simp\text{-all},$$

$$rule \ checked\text{-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size})$$
qed
show *?thesis*
by (rule *order-trans* $[OF \ hit\text{-bound}]$)

$$(intro \ add\text{-mono} \ prehit\text{-bound} \ order\text{-refl})$$
qed

lemma *checked-staged-security-with-query-prefix-dynamic-index-hit-bound-from-fraction*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *target-bound*:

$$\bigwedge prefix \ prefix\text{-state}.$$

$$Some \ ((prefix, \ prefix\text{-state}), \ prefix\text{-state}) \in$$

$$set\text{-dist}$$

$$(execute \ (checked\text{-staged-query-prefix-with-state } A \ i)$$

$$adversary\text{-initial-state}) \implies$$

$$T \ prefix \ prefix\text{-state} \subseteq query\text{-sample-space} \wedge$$

$$nnreal \ (card \ (T \ prefix \ prefix\text{-state})) \ /$$

$$nnreal \ (card \ query\text{-sample-space}) \leq C$$
shows

$$wp\text{-event}$$

(checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-dynamic-index-hit T)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) + C
by (rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound,
 rule checked-staged-query-prefix-dynamic-index-hit-bound-from-fraction
 [OF wf controlled i -bound target-bound])

lemma checked-staged-security-with-query-prefix-candidate-opening-target-from-prefix-hit-bound:

assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i -bound: $i \leq \text{rounds}$
and trace-candidate:
 partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
 partial-composition-table-candidate composition-table
 composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-low:
 composition-table-low-degree maxDegree composition-table
and not-all:
 $\bigwedge \text{prefix prefix-state.}$
 Some ((prefix, prefix-state), prefix-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-with-state A i)
 adversary-initial-state) $\implies$
 $\neg$ all-queries-consistent trace-table composition-table
 (sqp-alphas prefix)

shows

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-dynamic-index-hit
 (staged-query-prefix-candidate-opening-query-target-from-prefix
 trace-openings composition-openings i))
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) + query-error-bound
by (rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound,
 rule checked-staged-query-prefix-candidate-opening-target-from-prefix-hit-bound
 [OF raw-bound wf controlled i -bound trace-candidate comp-candidate
 trace-low comp-low not-all])

lemma checked-staged-security-with-query-prefix-candidate-opening-target-from-prefix-hit-bound-from-budgets:

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i -bound: $i < \text{rounds}$
and trace-candidate:

partial-trace-table-candidate trace-table trace-openings
and *comp-candidate:*
partial-composition-table-candidate composition-table
composition-openings
and *trace-low: trace-table-low-degree trace-table*
and *comp-low:*
composition-table-low-degree maxDegree composition-table
and *not-all:*
 $\bigwedge \text{prefix prefix-state.}$
Some ((prefix, prefix-state), prefix-state) $\in$
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) $\implies$
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)
shows
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i))
adversary-initial-state $\leq$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) + query-error-bound
proof –
have *raw-bound: query-index-raw-preimage-bound*
by *(rule query-index-raw-preimage-bound-from-sampler-wellformed)*
show *?thesis*
by *(rule*
checked-staged-security-with-query-prefix-candidate-opening-target-from-prefix-hit-bound
[OF raw-bound wf controlled - trace-candidate comp-candidate trace-low
comp-low not-all])
(use i-bound in simp)
qed

lemma *checked-staged-security-with-query-prefix-candidate-pair-target-from-prefix-hit-bound:*
assumes *raw-bound: query-index-raw-preimage-bound*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i $\leq$ rounds*
and *trace-low: trace-table-low-degree trace-table*
and *comp-low:*
composition-table-low-degree maxDegree composition-table
and *not-all:*
 $\bigwedge \text{prefix prefix-state.}$
Some ((prefix, prefix-state), prefix-state) $\in$
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) $\implies$

$\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)

shows

wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table))
adversary-initial-state $\leq$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) + query-error-bound

by *(rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound,*
rule checked-staged-query-prefix-candidate-pair-target-from-prefix-hit-bound
[OF raw-bound wf controlled i-bound trace-low comp-low not-all])

lemma *checked-staged-security-with-query-prefix-candidate-pair-target-from-prefix-hit-bound-from-budgets:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *trace-low: trace-table-low-degree trace-table*
and *comp-low:*
composition-table-low-degree maxDegree composition-table
and *not-all:*
 $\bigwedge_{\text{prefix}} \text{prefix-state.}$
Some ((prefix, prefix-state), prefix-state) $\in$
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) $\implies$
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)

shows

wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table))
adversary-initial-state $\leq$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) + query-error-bound

proof –

have *raw-bound: query-index-raw-preimage-bound*
by *(rule query-index-raw-preimage-bound-from-sampler-wellformed)*

show *?thesis*
by *(rule*
checked-staged-security-with-query-prefix-candidate-pair-target-from-prefix-hit-bound
[OF raw-bound wf controlled - trace-low comp-low not-all])
(use i-bound in simp)

qed

lemma *checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-bound-from-budgets:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *trace-low: trace-table-low-degree trace-table*

and *comp-low:*

composition-table-low-degree maxDegree composition-table

and *not-all:*

$\bigwedge \text{prefix prefix-state.}$

Some ((prefix, prefix-state), prefix-state) ∈

set-dist

(execute (checked-staged-query-prefix-with-state A i)

adversary-initial-state) ⇒

$\neg$ *all-queries-consistent trace-table composition-table*

(sqp-alphas prefix)

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit

trace-table composition-table)

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) + query-error-bound

unfolding

checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-def

by *(rule*

checked-staged-security-with-query-prefix-candidate-pair-target-from-prefix-hit-bound-from-budgets

[OF wf controlled i-bound trace-low comp-low not-all])

lemma *powers-scaled-eq-imp-eq:*

assumes *powers-scaled idx = powers-scaled idx'*

shows *idx = idx'*

proof *–*

have *left: powers-scaled idx ! 0 = idx*

unfolding *powers-scaled-def using powers-pos by simp*

have *right: powers-scaled idx' ! 0 = idx'*

unfolding *powers-scaled-def using powers-pos by simp*

show *?thesis*

using *assms left right by metis*

qed

lemma *partial-query-success-indices-at-card-le-one:*

card (partial-query-success-indices-at trace-openings

composition-openings as i) ≤ Suc 0

proof *–*

have *finite-target:*

finite (partial-query-success-indices-at trace-openings

composition-openings as i)

by *(rule finite-subset[OF partial-query-success-indices-at-subset*

```

    finite-query-sample-space])
have pairwise:
   $\forall idx \in \text{partial-query-success-indices-at trace-openings}$ 
    composition-openings as  $i$ .
   $\forall idx' \in \text{partial-query-success-indices-at trace-openings}$ 
    composition-openings as  $i$ .
     $idx = idx'$ 
proof (intro ballI)
  fix  $idx\ idx'$ 
  assume  $idx$ -in:
     $idx \in \text{partial-query-success-indices-at trace-openings}$ 
      composition-openings as  $i$ 
  and  $idx'$ -in:
     $idx' \in \text{partial-query-success-indices-at trace-openings}$ 
      composition-openings as  $i$ 
  have  $idx$ -indices:
    map opening-index (trace-openings !  $i$ ) = powers-scaled  $idx$ 
  using  $idx$ -in
  unfolding partial-query-success-indices-at-def
    partial-query-round-consistent-def
  by simp
  have  $idx'$ -indices:
    map opening-index (trace-openings !  $i$ ) = powers-scaled  $idx'$ 
  using  $idx'$ -in
  unfolding partial-query-success-indices-at-def
    partial-query-round-consistent-def
  by simp
  have powers-scaled  $idx$  = powers-scaled  $idx'$ 
  using  $idx$ -indices  $idx'$ -indices by simp
  then show  $idx = idx'$ 
  by (rule powers-scaled-eq-imp-eq)
qed
show ?thesis
  using card-le-Suc0-iff-eq[OF finite-target] pairwise by simp
qed

lemma prob-divide-right-mono:
  assumes  $(a::prob) \leq b$ 
  shows  $a / c \leq b / c$ 
  using assms
  apply transfer
  by (simp add: divide-right-mono)

lemma partial-query-success-indices-at-singleton-fraction-bound:
  nnreal
    (card
      (partial-query-success-indices-at trace-openings composition-openings
        as  $i$ )) /
  nnreal (card query-sample-space)  $\leq$ 

```

$(1::prob) / \text{nnreal } (\text{card } \text{query-sample-space})$
proof –
have *card-le*:
 nnreal
 $(\text{card}$
 $(\text{partial-query-success-indices-at } \text{trace-openings } \text{composition-openings}$
 $\text{as } i)) \leq (1::prob)$
using *partial-query-success-indices-at-card-le-one*
by *simp*
show *?thesis*
by (rule *prob-divide-right-mono*[*OF card-le*])
qed

lemma *staged-query-prefix-candidate-opening-query-target-from-prefix-singleton-fraction-bound*:
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-candidate-opening-query-target-from-prefix}$
 $\text{trace-openings } \text{composition-openings } i \text{ prefix prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq$
 $(1::prob) / \text{nnreal } (\text{card } \text{query-sample-space})$
unfolding *staged-query-prefix-candidate-opening-query-target-from-prefix-def*
by (rule *partial-query-success-indices-at-singleton-fraction-bound*)

lemma *checked-staged-query-prefix-candidate-opening-target-from-prefix-singleton-hit-bound*:
assumes *raw-bound*: *query-index-raw-preimage-bound*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: $i \leq \text{rounds}$
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $(\text{staged-query-prefix-candidate-opening-query-target-from-prefix}$
 $\text{trace-openings } \text{composition-openings } i))$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries } \text{budgets } i + 1) +$
 $(1::prob) / \text{nnreal } (\text{card } \text{query-sample-space})$

proof –
have *hit-bound*:
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $(\text{staged-query-prefix-candidate-opening-query-target-from-prefix}$
 $\text{trace-openings } \text{composition-openings } i))$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-prehit}$
 $(\text{staged-query-prefix-candidate-opening-query-target-from-prefix}$
 $\text{trace-openings } \text{composition-openings } i))$
 $\text{adversary-initial-state} +$


```

(1::prob) / nnreal (card query-sample-space)
proof (rule checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
  [OF raw-bound])
fix prefix prefix-state
assume
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state)
show
  staged-query-prefix-candidate-opening-query-target-from-prefix
    trace-openings composition-openings i prefix prefix-state ⊆
    query-sample-space ∧
  nnreal
    (card
      (staged-query-prefix-candidate-opening-query-target-from-prefix
        trace-openings composition-openings i prefix prefix-state)) /
  nnreal (card query-sample-space)
  ≤ (1::prob) / nnreal (card query-sample-space)
using
  staged-query-prefix-candidate-opening-query-target-from-prefix-subset
  staged-query-prefix-candidate-opening-query-target-from-prefix-singleton-fraction-bound
by blast
qed
have prehit-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit
      (staged-query-prefix-candidate-opening-query-target-from-prefix
        trace-openings composition-openings i))
  adversary-initial-state ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1)
by (rule checked-staged-query-prefix-candidate-opening-target-from-prefix-prehit-bound
  [OF wf controlled i-bound])
have tail:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit
      (staged-query-prefix-candidate-opening-query-target-from-prefix
        trace-openings composition-openings i))
  adversary-initial-state +
  (1::prob) / nnreal (card query-sample-space) ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
  (1::prob) / nnreal (card query-sample-space)
by (intro add-mono prehit-bound order-refl)
show ?thesis
by (rule order-trans[OF hit-bound tail])
qed

```

lemma *checked-staged-security-with-query-prefix-candidate-opening-target-from-prefix-singleton-hit-bound:*
assumes *raw-bound: query-index-raw-preimage-bound*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i \leq \text{rounds}$*

shows

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
(*checked-staged-security-with-query-prefix-dynamic-index-hit*
(*staged-query-prefix-candidate-opening-query-target-from-prefix*
trace-openings composition-openings i))

adversary-initial-state $\leq$

staged-phase-relation-error size

(*staged-query-search-queries budgets $i + 1$*) +

(*$1::\text{prob}$*) / *nnreal (card query-sample-space)*

by (*rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound,*
rule

checked-staged-query-prefix-candidate-opening-target-from-prefix-singleton-hit-bound
[*OF raw-bound wf controlled i-bound*])

lemma *checked-staged-security-with-query-prefix-candidate-opening-target-from-prefix-singleton-hit-bound-from-budget*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: $i < \text{rounds}$*

shows

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
(*checked-staged-security-with-query-prefix-dynamic-index-hit*
(*staged-query-prefix-candidate-opening-query-target-from-prefix*
trace-openings composition-openings i))

adversary-initial-state $\leq$

staged-phase-relation-error size

(*staged-query-search-queries budgets $i + 1$*) +

(*$1::\text{prob}$*) / *nnreal (card query-sample-space)*

proof –

have *raw-bound: query-index-raw-preimage-bound*

by (*rule query-index-raw-preimage-bound-from-sampler-wellformed*)

show *?thesis*

by (*rule*

checked-staged-security-with-query-prefix-candidate-opening-target-from-prefix-singleton-hit-bound
[*OF raw-bound wf controlled*])

(*use i-bound in simp*)

qed

definition *checked-staged-security-with-actual-query-prefix-candidate-opening-hit*

where

checked-staged-security-with-actual-query-prefix-candidate-opening-hit

trace-openings composition-openings i out $\longleftrightarrow$

checked-staged-security-with-query-prefix-dynamic-index-hit

(*staged-query-prefix-candidate-opening-query-target-from-prefix*
trace-openings composition-openings i) *out*

lemma *checked-staged-security-with-actual-query-prefix-candidate-opening-hitE*:

assumes *hit*:

checked-staged-security-with-actual-query-prefix-candidate-opening-hit
trace-openings composition-openings i
 (Some ((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*))

obtains

index (*to-nat raw*) ∈
staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i prefix prefix-state

using *hit*

unfolding

checked-staged-security-with-actual-query-prefix-candidate-opening-hit-def
checked-staged-security-with-query-prefix-dynamic-index-hit-def
checked-staged-query-prefix-dynamic-index-hit-def

by *simp*

lemma *checked-staged-security-with-actual-query-prefix-candidate-opening-hit-bound*:

assumes *raw-bound*: *query-index-raw-preimage-bound*

and *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *i-bound*: $i \leq \text{rounds}$

shows

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-actual-query-prefix-candidate-opening-hit*
trace-openings composition-openings i)
adversary-initial-state ≤
staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 ($1::\text{prob}$) / *nnreal* (*card query-sample-space*)

unfolding

checked-staged-security-with-actual-query-prefix-candidate-opening-hit-def

by (*rule*

checked-staged-security-with-query-prefix-candidate-opening-target-from-prefix-singleton-hit-bound
 [*OF raw-bound wf controlled i-bound*])

lemma *checked-staged-security-with-actual-query-prefix-candidate-opening-hit-bound-from-budgets*:

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *i-bound*: $i < \text{rounds}$

shows

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-actual-query-prefix-candidate-opening-hit*
trace-openings composition-openings i)

$\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $(1::\text{prob}) / \text{nnreal (card query-sample-space)}$

unfolding

$\text{checked-staged-security-with-actual-query-prefix-candidate-opening-hit-def}$
by (rule
 $\text{checked-staged-security-with-query-prefix-candidate-opening-target-from-prefix-singleton-hit-bound-from-budgets}$
 $[\text{OF wf controlled } i\text{-bound}]$)

lemma $\text{checked-staged-security-with-data-state-bound-by-actual-query-prefix-candidate-opening-hit}$:

assumes $\text{raw-bound: query-index-raw-preimage-bound}$
and $\text{wf: staged-budget-wellformed budgets}$
and $\text{controlled: staged-adversary-controlled budgets } A$
and $\text{i-bound: } i < \text{rounds}$
and event-imp:
 $\bigwedge \text{out.}$
 $(\text{case out of}$
 $\text{None} \Rightarrow E \text{ None}$
 $| \text{Some (packed, t)} \Rightarrow E (\text{Some (snd packed, t)})) \Rightarrow$
 $\text{checked-staged-security-with-actual-query-prefix-candidate-opening-hit}$
 $\text{trace-openings composition-openings } i \text{ out}$

shows

$\text{wp-event (checked-staged-security-experiment-with-data-state } A) E$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $(1::\text{prob}) / \text{nnreal (card query-sample-space)}$

proof –

have projection:
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A) E$
 $\text{adversary-initial-state} =$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\lambda \text{out. case out of}$
 $\text{None} \Rightarrow E \text{ None}$
 $| \text{Some (packed, t)} \Rightarrow E (\text{Some (snd packed, t)}))$
 $\text{adversary-initial-state}$

by (rule
 $\text{checked-staged-security-experiment-with-query-prefix-data-state-projection-event}$
 $[\text{OF } i\text{-bound}]$)

have lifted-bound:

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\lambda \text{out. case out of}$
 $\text{None} \Rightarrow E \text{ None}$
 $| \text{Some (packed, t)} \Rightarrow E (\text{Some (snd packed, t)}))$
 $\text{adversary-initial-state} \leq$
 wp-event

```

    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-actual-query-prefix-candidate-opening-hit
      trace-openings composition-openings i)
    adversary-initial-state
  by (rule wp-event-mono) (use event-imp in blast)
have fixed-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-actual-query-prefix-candidate-opening-hit
      trace-openings composition-openings i)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space)
  by (rule
    checked-staged-security-with-actual-query-prefix-candidate-opening-hit-bound
    [OF raw-bound wf controlled])
    (use i-bound in simp)
show ?thesis
  unfolding projection
  by (rule order-trans[OF lifted-bound fixed-bound])
qed

lemma checked-staged-security-with-data-state-bound-by-actual-query-prefix-candidate-opening-hit-from-budgets:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and event-imp:
     $\bigwedge out.$ 
    (case out of
      None  $\Rightarrow$  E None
      | Some (packed, t)  $\Rightarrow$  E (Some (snd packed, t)))  $\Rightarrow$ 
      checked-staged-security-with-actual-query-prefix-candidate-opening-hit
      trace-openings composition-openings i out
  shows
    wp-event (checked-staged-security-experiment-with-data-state A) E
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space)
proof -
  have raw-bound: query-index-raw-preimage-bound
  by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
show ?thesis
  by (rule
    checked-staged-security-with-data-state-bound-by-actual-query-prefix-candidate-opening-hit
    [OF raw-bound wf controlled i-bound event-imp])
qed

```

lemma *checked-staged-security-with-query-prefix-data-state-bound-by-actual-query-prefix-candidate-opening-hit*:
assumes *raw-bound*: *query-index-raw-preimage-bound*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: $i < \text{rounds}$
and *event-imp*:
 $\bigwedge \text{out}. E \text{ out} \implies$
 $\text{checked-staged-security-with-actual-query-prefix-candidate-opening-hit}$
 $\text{trace-openings composition-openings } i \text{ out}$
shows
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $E \text{ adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $(1::\text{prob}) / \text{nnreal (card query-sample-space)}$
proof –
have *event-le*:
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $E \text{ adversary-initial-state} \leq$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-actual-query-prefix-candidate-opening-hit}$
 $\text{trace-openings composition-openings } i)$
 $\text{adversary-initial-state}$
by (rule *wp-event-mono*) (rule *event-imp*)
have *fixed-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-actual-query-prefix-candidate-opening-hit}$
 $\text{trace-openings composition-openings } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $(1::\text{prob}) / \text{nnreal (card query-sample-space)}$
by (rule
 $\text{checked-staged-security-with-actual-query-prefix-candidate-opening-hit-bound}$
 $[\text{OF raw-bound wf controlled}]$)
 $(\text{use } i\text{-bound in simp})$
show *?thesis*
by (rule *order-trans*[*OF event-le fixed-bound*])
qed

lemma *checked-staged-security-with-query-prefix-data-state-bound-by-actual-query-prefix-candidate-opening-hit-from-*
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: $i < \text{rounds}$
and *event-imp*:

$\bigwedge out. E \text{ out} \implies$
checked-staged-security-with-actual-query-prefix-candidate-opening-hit
trace-openings composition-openings i out
shows
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
E adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
(1::prob) / nnreal (card query-sample-space)
proof –
have *raw-bound: query-index-raw-preimage-bound*
by *(rule query-index-raw-preimage-bound-from-sampler-wellformed)*
show *?thesis*
by *(rule*
checked-staged-security-with-query-prefix-data-state-bound-by-actual-query-prefix-candidate-opening-hit
[OF raw-bound wf controlled i-bound event-imp])
qed

lemma *checked-staged-query-prefix-with-state-alpha-prefix-support:*
assumes *outcome:*
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state)
obtains *alpha-prefix alpha-prefix-state where*
alpha-prefix =
(sqp-trace-root prefix, sqp-trace-fri-roots prefix,
sqp-trace-fri-challenges prefix, sqp-trace-final prefix)
Some (alpha-prefix, alpha-prefix-state) ∈
set-dist
(execute (staged-alpha-prefix-program A) adversary-initial-state)
proof –
from *outcome obtain prefix-out prefix-state' where prefix-program-out:*
Some (prefix-out, prefix-state') ∈
set-dist
(execute (checked-staged-query-challenge-prefix-program A i)
adversary-initial-state)
and *prefix-eq0: prefix-out = prefix*
unfolding *checked-staged-query-prefix-with-state-def*
by *(auto elim!: set-dist-bindE dest!: get-bind-return-state-outcome)*
obtain *fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5 as*
s6 dg s7 s8 s9 composition-roots composition-bs s10
composition-final s11 query-start query-chunks where
fr-out:
Some (fr, s1) ∈
set-dist (execute (trace-root-stage A) adversary-initial-state)
and *record-fr:*
Some ((), s2) ∈ set-dist (execute (record-staged-message fr) s1)

```

and trace-fri:
  Some ((trace-roots, trace-bs), s3) ∈
    set-dist
      (execute
        (staged-trace-fri-program A 0 (ceil-log clength) [])) s2)
and trace-final-out:
  Some (trace-final, s4) ∈
    set-dist (execute (trace-final-stage A trace-bs) s3)
and record-trace-final:
  Some ((), s5) ∈
    set-dist (execute (record-staged-message trace-final) s4)
and alpha-out:
  Some (as, s6) ∈
    set-dist (execute (staged-alpha-program (length spec)) s5)
and degree-out:
  Some (dg, s7) ∈ set-dist (execute (degree-stage A as) s6)
and record-degree:
  Some ((), s8) ∈
    set-dist (execute (record-staged-message dg) s7)
and assert-degree:
  Some ((), s9) ∈
    set-dist
      (execute
        (assert (ceil-log (to-nat dg + 1) ≤
          ceil-log (maxDegree + 1)))) s8)
and comp-fri:
  Some ((composition-roots, composition-bs), s10) ∈
    set-dist
      (execute
        (staged-composition-fri-program A dg 0
          (ceil-log (to-nat dg + 1)) [])) s9)
and comp-final:
  Some (composition-final, s11) ∈
    set-dist
      (execute (composition-final-stage A dg composition-bs) s10)
and record-comp-final:
  Some ((), query-start) ∈
    set-dist (execute (record-staged-message composition-final) s11)
and query-prefix:
  Some (query-chunks, prefix-state') ∈
    set-dist
      (execute
        (checked-staged-query-program A trace-roots composition-roots
          0 i) query-start)
and prefix-out-eq:
  prefix-out =
    (sqp-trace-root = fr,
     sqp-trace-fri-roots = trace-roots,
     sqp-trace-fri-challenges = trace-bs,

```



```

    sqp-trace-final = trace-final,
    sqp-alphas = as,
    sqp-degree = dg,
    sqp-composition-fri-roots = composition-roots,
    sqp-composition-fri-challenges = composition-bs,
    sqp-composition-final = composition-final,
    sqp-query-chunks = query-chunks)
  by (rule checked-staged-query-challenge-prefix-program-outcomeE[
    OF prefix-program-out])
have prefix-eq:
  prefix =
    (|sqp-trace-root = fr,
     sqp-trace-fri-roots = trace-roots,
     sqp-trace-fri-challenges = trace-bs,
     sqp-trace-final = trace-final,
     sqp-alphas = as,
     sqp-degree = dg,
     sqp-composition-fri-roots = composition-roots,
     sqp-composition-fri-challenges = composition-bs,
     sqp-composition-final = composition-final,
     sqp-query-chunks = query-chunks)
  using prefix-eq0 prefix-out-eq by simp
let ?alpha-prefix = (fr, trace-roots, trace-bs, trace-final)
have alpha-prefix-out:
  Some (?alpha-prefix, s5) ∈
    set-dist
    (execute (staged-alpha-prefix-program A) adversary-initial-state)
proof -
  have after-record-final:
    Some (?alpha-prefix, s5) ∈
      set-dist
      (execute
        (record-staged-message trace-final >>=
          (λ-. return (fr, trace-roots, trace-bs, trace-final))))
        s4)
  by (rule-tac x=() and t=s5 in set-dist-bindI)
    (rule record-trace-final, simp)
have after-final:
  Some (?alpha-prefix, s5) ∈
    set-dist
    (execute
      (trace-final-stage A trace-bs >>=
        (λtrace-final.
          record-staged-message trace-final >>=
            (λ-. return (fr, trace-roots, trace-bs, trace-final))))
        s3)
  by (rule-tac x=trace-final and t=s4 in set-dist-bindI)
    (rule trace-final-out, rule after-record-final)
have after-trace:

```

```

Some (?alpha-prefix, s5) ∈
  set-dist
  (execute
    (staged-trace-fri-program A 0 (ceil-log clength) [] ≫=
      (λ(trace-roots, trace-bs).
        trace-final-stage A trace-bs ≫=
        (λtrace-final.
          record-staged-message trace-final ≫=
          (λ-. return
            (fr, trace-roots, trace-bs, trace-final))))))
    s2)
apply (rule-tac x=(trace-roots, trace-bs) and t=s3
  in set-dist-bindI)
apply (rule trace-fri)
apply (simp add: after-final)
done
have after-record-root:
  Some (?alpha-prefix, s5) ∈
    set-dist
    (execute
      (record-staged-message fr ≫=
        (λ-. staged-trace-fri-program A 0 (ceil-log clength) [] ≫=
          (λ(trace-roots, trace-bs).
            trace-final-stage A trace-bs ≫=
            (λtrace-final.
              record-staged-message trace-final ≫=
              (λ-. return
                (fr, trace-roots, trace-bs, trace-final))))))
        s1)
    by (rule-tac x=() and t=s2 in set-dist-bindI)
      (rule record-fr, rule after-trace)
show ?thesis
unfolding staged-alpha-prefix-program-def
apply (rule-tac x=fr and t=s1 in set-dist-bindI)
apply (rule fr-out)
apply (rule after-record-root)
done
qed
show ?thesis
by (rule that[OF - alpha-prefix-out]) (simp add: prefix-eq)
qed

lemma checked-staged-security-with-query-prefix-data-state-alpha-prefix-support:
assumes outcome:
  Some (((((prefix, prefix-state), raw), raw-state), result), final-state)
  ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A i)

```

adversary-initial-state)
obtains *alpha-prefix alpha-prefix-state* **where**
alpha-prefix =
 (*sqp-trace-root prefix*, *sqp-trace-fri-roots prefix*,
sqp-trace-fri-challenges prefix, *sqp-trace-final prefix*)
Some (alpha-prefix, alpha-prefix-state) ∈
set-dist
(execute (staged-alpha-prefix-program A) adversary-initial-state)
proof –
from *outcome* **obtain** *t* **where** *prefix-receive*:
Some (((prefix, prefix-state), raw), t) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *t-eq: t = raw-state*
unfolding *checked-staged-security-experiment-with-query-prefix-data-state-def*
by (*auto elim!:* *set-dist-bindE*)
have *prefix-support*:
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state)
by (*rule checked-staged-query-prefix-receive-with-state-prefix-support*)
(use prefix-receive t-eq in simp)
show *?thesis*
by (*rule*
checked-staged-query-prefix-with-state-alpha-prefix-support
[OF prefix-support])
(rule that)
qed

lemma *checked-staged-security-with-query-prefix-data-state-alphas-eq*:
assumes *outcome*:
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)
∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)
shows *staged-alphas data = sqp-alphas prefix*
using *outcome*
unfolding
checked-staged-security-experiment-with-query-prefix-data-state-def
checked-staged-after-query-prefix-receive-with-verifier-def
checked-staged-after-query-prefix-receive-def
by (*auto elim!:* *set-dist-bindE split: prod.splits*)

definition *checked-staged-security-with-query-prefix-authenticated-opening-hit*

where

checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 staged-composition-fri-roots data $\neq \square$ $\wedge$
 partial-authenticated-table (staged-trace-root data)
 (scale * clength) trace-openings final-state $\wedge$
 partial-authenticated-table (hd (staged-composition-fri-roots data))
 (scale * clength) composition-openings final-state $\wedge$
 partial-query-openings-consistent trace-openings composition-openings
 (staged-alphas data) (index (to-nat raw)))

definition *checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit*
where

checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
trace-openings composition-openings i out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 staged-composition-fri-roots data $\neq \square$ $\wedge$
 partial-authenticated-table (staged-trace-root data)
 (scale * clength) trace-openings prefix-state $\wedge$
 partial-authenticated-table (hd (staged-composition-fri-roots data))
 (scale * clength) composition-openings prefix-state $\wedge$
 partial-query-openings-consistent trace-openings composition-openings
 (staged-alphas data) (index (to-nat raw)))

definition *checked-staged-security-with-query-prefix-opening-path-output-hit*
where

checked-staged-security-with-query-prefix-opening-path-output-hit
trace-openings composition-openings out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 ($\exists$ opening $\in$ set trace-openings $\cup$ set composition-openings.
 hash-map-new-output-hit
 (merkle-path-target-roots final-state
 (opening-root opening)
 (opening-length opening)
 (opening-index opening)
 (opening-value opening)
 (opening-path opening))
 prefix-state final-state))

lemma *checked-staged-security-with-query-prefix-opening-path-output-hitE:*
assumes *hit:*
checked-staged-security-with-query-prefix-opening-path-output-hit
trace-openings composition-openings out
obtains *prefix prefix-state raw raw-state data attacker-state result*
final-state opn **where**
out = Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)
opn ∈ set trace-openings ∪ set composition-openings
hash-map-new-output-hit
(merkle-path-target-roots final-state
(opening-root opn)
(opening-length opn)
(opening-index opn)
(opening-value opn)
(opening-path opn))
prefix-state final-state
finite
(merkle-path-target-roots final-state
(opening-root opn)
(opening-length opn)
(opening-index opn)
(opening-value opn)
(opening-path opn))
card
(merkle-path-target-roots final-state
(opening-root opn)
(opening-length opn)
(opening-index opn)
(opening-value opn)
(opening-path opn)) ≤ Suc (length (opening-path opn))
using *hit*
unfolding *checked-staged-security-with-query-prefix-opening-path-output-hit-def*
by (*cases out*)
(auto intro: that card-merkle-path-target-roots-le split: prod.splits)

lemma *checked-staged-security-experiment-with-query-prefix-data-state-prefix-extends-final:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *outcome:*
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)
∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)
shows *prefix-state ≤ final-state*

proof –

from *outcome* **obtain** *t* **where** *head-support*:

Some (((*prefix*, *prefix-state*), *raw*), *t*) ∈

set-dist

(*execute* (*checked-staged-query-prefix-receive-with-state* *A i*)

adversary-initial-state)

and *t-eq*: *t* = *raw-state*

and *cont-support*:

Some (((*data*, *attacker-state*), *result*), *final-state*) ∈

set-dist

(*execute*

(*checked-staged-after-query-prefix-receive-with-verifier* *A i*

((*prefix*, *prefix-state*), *raw*))

raw-state)

unfolding *checked-staged-security-experiment-with-query-prefix-data-state-def*

by (*auto elim!*: *set-dist-bindE* *split*: *prod.splits*)

have *head*:

Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) ∈

set-dist

(*execute* (*checked-staged-query-prefix-receive-with-state* *A i*)

adversary-initial-state)

using *head-support t-eq* **by** *simp*

have *recv*:

Some (*raw*, *raw-state*) ∈

set-dist (*execute receive-query-index-challenge prefix-state*)

by (*rule checked-staged-query-prefix-receive-with-state-receive-support*

[*OF head*])

have *prefix-ext-raw*: *prefix-state* ≤ *raw-state*

by (*rule receive-query-index-challenge-extends*[*OF recv*])

have *data-out*:

Some (*data*, *attacker-state*) ∈

set-dist

(*execute*

(*checked-staged-after-query-prefix-receive* *A i*

((*prefix*, *prefix-state*), *raw*))

raw-state)

by (*rule*

checked-staged-after-query-prefix-receive-with-verifier-outcome(1)

[*OF cont-support*])

have *verifier-out*:

Some (*result*, *final-state*) ∈

set-dist

(*execute verify-monad*

(*verifier-state-from-adversary attacker-state*

(*staged-proof-transcript data*)))

by (*rule*

checked-staged-after-query-prefix-receive-with-verifier-outcome(2)

[*OF cont-support*])

have *raw-ext-attacker*: *raw-state* ≤ *attacker-state*

```

    by (rule checked-staged-after-query-prefix-receive-extends
        [OF wf controlled i-bound data-out])
  have attacker-ext-verifier:
    attacker-state ≤
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  proof -
    have attacker-state ≤ attacker-state
      unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
    then show ?thesis
      by (rule hash-extends-verifier-state-from-adversary-right)
  qed
  have verifier-ext-final:
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data) ≤
      final-state
    by (rule verify-monad-hash-extends[OF verifier-out])
  have raw-ext-final: raw-state ≤ final-state
    by (rule hash-ext-trans[OF raw-ext-attacker])
      (rule hash-ext-trans[OF attacker-ext-verifier verifier-ext-final])
  show ?thesis
    by (rule hash-ext-trans[OF prefix-ext-raw raw-ext-final])
  qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-prefix-or-path-output-hit:
  assumes ext: prefix-state ≤ final-state
  and hit:
    checked-staged-security-with-query-prefix-authenticated-opening-hit
    trace-openings composition-openings i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  shows
    checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
    trace-openings composition-openings i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)) ∨
    checked-staged-security-with-query-prefix-opening-path-output-hit
    trace-openings composition-openings
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  proof -
    have comp-nonempty: staged-composition-fri-roots data ≠ []
    and trace-table:
      partial-authenticated-table (staged-trace-root data)
      (scale * clength) trace-openings final-state
    and comp-table:
      partial-authenticated-table (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings final-state
    and consistent:

```

```

    partial-query-openings-consistent trace-openings composition-openings
      (staged-alphas data) (index (to-nat raw))
using hit
unfolding checked-staged-security-with-query-prefix-authenticated-opening-hit-def
by simp-all
have trace-pullback:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings prefix-state ∨
  (∃ opn ∈ set trace-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots final-state (staged-trace-root data)
        (scale * clength)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn)) prefix-state final-state)
by (rule partial-authenticated-table-pullback-or-new-output-hit
  [OF ext trace-table])
then show ?thesis
proof
  assume trace-prefix:
    partial-authenticated-table (staged-trace-root data)
      (scale * clength) trace-openings prefix-state
  have comp-pullback:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings prefix-state ∨
    (∃ opn ∈ set composition-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots final-state
          (hd (staged-composition-fri-roots data))
          (scale * clength)
          (opening-index opn)
          (opening-value opn)
          (opening-path opn)) prefix-state final-state)
    by (rule partial-authenticated-table-pullback-or-new-output-hit
      [OF ext comp-table])
  then show ?thesis
proof
  assume comp-prefix:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings prefix-state
  then have
    checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
      trace-openings composition-openings i
      (Some (((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state))
  using comp-nonempty trace-prefix consistent
unfolding
  checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-def
by simp

```



```

then show ?thesis by simp
next
assume comp-path:
   $\exists \text{opn} \in \text{set composition-openings.}$ 
  hash-map-new-output-hit
    (merkle-path-target-roots final-state
      (hd (staged-composition-fri-roots data))
      (scale * clength)
      (opening-index opn)
      (opening-value opn)
      (opening-path opn)) prefix-state final-state
then obtain opn where opn-in: opn  $\in$  set composition-openings
and path-hit:
  hash-map-new-output-hit
    (merkle-path-target-roots final-state
      (hd (staged-composition-fri-roots data))
      (scale * clength)
      (opening-index opn)
      (opening-value opn)
      (opening-path opn)) prefix-state final-state
by blast
have root-eq:
  opening-root opn = hd (staged-composition-fri-roots data)
using comp-table opn-in
unfolding partial-authenticated-table-def by blast
have len-eq: opening-length opn = scale * clength
using comp-table opn-in
unfolding partial-authenticated-table-def by blast
have
  checked-staged-security-with-query-prefix-opening-path-output-hit
  trace-openings composition-openings
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
proof –
  have path-hit':
    hash-map-new-output-hit
      (merkle-path-target-roots final-state
        (opening-root opn)
        (opening-length opn)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn)) prefix-state final-state
    using path-hit root-eq len-eq by simp
  show ?thesis
  unfolding
    checked-staged-security-with-query-prefix-opening-path-output-hit-def
  using opn-in path-hit' by auto
qed
then show ?thesis by simp

```

```

qed
next
assume trace-path:
  ∃ opn ∈ set trace-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots final-state (staged-trace-root data)
        (scale * clength)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn)) prefix-state final-state
then obtain opn where opn-in: opn ∈ set trace-openings
and path-hit:
  hash-map-new-output-hit
    (merkle-path-target-roots final-state (staged-trace-root data)
      (scale * clength)
      (opening-index opn)
      (opening-value opn)
      (opening-path opn)) prefix-state final-state
  by blast
have root-eq: opening-root opn = staged-trace-root data
  using trace-table opn-in
  unfolding partial-authenticated-table-def by blast
have len-eq: opening-length opn = scale * clength
  using trace-table opn-in
  unfolding partial-authenticated-table-def by blast
have
  checked-staged-security-with-query-prefix-opening-path-output-hit
    trace-openings composition-openings
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
proof -
  have path-hit':
    hash-map-new-output-hit
      (merkle-path-target-roots final-state
        (opening-root opn)
        (opening-length opn)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn)) prefix-state final-state
    using path-hit root-eq len-eq by simp
  show ?thesis
    unfolding checked-staged-security-with-query-prefix-opening-path-output-hit-def
    using opn-in path-hit' by auto
qed
then show ?thesis by simp
qed
qed
lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-prefix-or-path-output-hit-on-support:

```

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i < \text{rounds}$ 
and support:
  out  $\in$ 
    set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
  trace-openings composition-openings i out
shows
  checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
  trace-openings composition-openings i out  $\vee$ 
  checked-staged-security-with-query-prefix-opening-path-output-hit
  trace-openings composition-openings out
proof (cases out)
  case None
  then show ?thesis
  using hit
  unfolding checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  by simp
next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
  by (cases packed, auto split: prod.splits)
  have ext: prefix-state  $\leq$  final-state
  by (rule
    checked-staged-security-experiment-with-query-prefix-data-state-prefix-extends-final
    [OF wf controlled i-bound])
    (use support out-eq in simp)
  show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-prefix-or-path-output-hit
    [OF ext])
    (use hit out-eq in simp)
qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-le-actual-query-prefix-hit:
assumes i-bound:  $i < \text{rounds}$ 
shows
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)

```

```

    (checked-staged-security-with-query-prefix-authenticated-opening-hit
      trace-openings composition-openings i)
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-actual-query-prefix-candidate-opening-hit
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
    trace-openings composition-openings i out
show
  checked-staged-security-with-actual-query-prefix-candidate-opening-hit
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings]) i out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
  by (cases packed, auto split: prod.splits)
have alphas-eq: staged-alphas data = sqp-alphas prefix
  by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq)
    (use support out-eq in simp)
have consistent:
  partial-query-openings-consistent trace-openings composition-openings
    (sqp-alphas prefix) (index (to-nat raw))
  using hit alphas-eq
  unfolding out-eq
    checked-staged-security-with-query-prefix-authenticated-opening-hit-def

```

```

    by simp
  have target-hit:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-opening-query-target-from-prefix
        ((replicate rounds []) [i := trace-openings])
        ((replicate rounds []) [i := composition-openings])
        i)
      (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
    checked-staged-query-prefix-candidate-opening-target-from-prefix-single-round-hitI)
    (use i-bound consistent in simp-all)
  show ?thesis
    unfolding out-eq
      checked-staged-security-with-actual-query-prefix-candidate-opening-hit-def
      checked-staged-security-with-query-prefix-dynamic-index-hit-def
    using target-hit by simp
qed
qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-bound:
  assumes raw-bound: query-index-raw-preimage-bound
    and wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound: i < rounds
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-authenticated-opening-hit
        trace-openings composition-openings i)
      adversary-initial-state ≤
      staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      (1::prob) / nnreal (card query-sample-space)
  proof -
    have actual-bound:
      wp-event
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        (checked-staged-security-with-actual-query-prefix-candidate-opening-hit
          ((replicate rounds []) [i := trace-openings])
          ((replicate rounds []) [i := composition-openings])
          i)
        adversary-initial-state ≤
        staged-phase-relation-error size
        (staged-query-search-queries budgets i + 1) +
        (1::prob) / nnreal (card query-sample-space)
    by (rule checked-staged-security-with-actual-query-prefix-candidate-opening-hit-bound
      [OF raw-bound wf controlled])
      (use i-bound in simp)
  show ?thesis

```

```

    by (rule order-trans[
      OF checked-staged-security-with-query-prefix-authenticated-opening-hit-le-actual-query-prefix-hit
        actual-bound])
      (use i-bound in simp)
qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-budgets:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound: i < rounds
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-authenticated-opening-hit
        trace-openings composition-openings i)
      adversary-initial-state ≤
      staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      (1::prob) / nnreal (card query-sample-space)
  proof -
    have raw-bound: query-index-raw-preimage-bound
    by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
    show ?thesis
    by (rule
      checked-staged-security-with-query-prefix-authenticated-opening-hit-bound
      [OF raw-bound wf controlled i-bound])
  qed

lemma checked-staged-security-with-query-prefix-data-state-bound-by-query-prefix-authenticated-opening-hit:
  assumes raw-bound: query-index-raw-preimage-bound
    and wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound: i < rounds
    and event-imp:
       $\bigwedge out. E \text{ out} \implies$ 
      checked-staged-security-with-query-prefix-authenticated-opening-hit
        trace-openings composition-openings i out
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      E adversary-initial-state ≤
      staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      (1::prob) / nnreal (card query-sample-space)
  proof -
    have event-le:
      wp-event
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        E adversary-initial-state ≤

```

```

wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-authenticated-opening-hit
    trace-openings composition-openings i)
  adversary-initial-state
by (rule wp-event-mono) (rule event-imp)
have fixed-bound:
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-authenticated-opening-hit
    trace-openings composition-openings i)
  adversary-initial-state ≤
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
  (1::prob) / nnreal (card query-sample-space)
by (rule checked-staged-security-with-query-prefix-authenticated-opening-hit-bound
  [OF raw-bound wf controlled i-bound])
show ?thesis
by (rule order-trans[OF event-le fixed-bound])
qed

```

lemma *checked-staged-security-with-query-prefix-data-state-bound-by-query-prefix-authenticated-opening-hit-from-budget*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *event-imp:*
 $\bigwedge out. E out \implies$
checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i out

shows

```

wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  E adversary-initial-state ≤
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
  (1::prob) / nnreal (card query-sample-space)

```

proof –

```

have raw-bound: query-index-raw-preimage-bound
by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
show ?thesis
by (rule
  checked-staged-security-with-query-prefix-data-state-bound-by-query-prefix-authenticated-opening-hit
  [OF raw-bound wf controlled i-bound event-imp])

```

qed

lemma *checked-staged-security-with-query-prefix-authenticated-opening-hit-le-actual-query-prefix-hit:*

assumes *i-bound: i < rounds*
shows
wp-event

```

    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
      trace-openings composition-openings i)
    adversary-initial-state ≤
wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-actual-query-prefix-candidate-opening-hit
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i)
    adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
    trace-openings composition-openings i out
show
  checked-staged-security-with-actual-query-prefix-candidate-opening-hit
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings]) i out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
  by (cases packed, auto split: prod.splits)
have alphas-eq: staged-alphas data = sqp-alphas prefix
  by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq)
    (use support out-eq in simp)
have consistent:
  partial-query-openings-consistent trace-openings composition-openings
    (sqp-alphas prefix) (index (to-nat raw))
  using hit alphas-eq
  unfolding out-eq

```



```

    checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-def
  by simp
have target-hit:
  checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-candidate-opening-query-target-from-prefix
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i)
  (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
    checked-staged-query-prefix-candidate-opening-target-from-prefix-single-round-hitI)
  (use i-bound consistent in simp-all)
show ?thesis
  unfolding out-eq
  checked-staged-security-with-actual-query-prefix-candidate-opening-hit-def
  checked-staged-security-with-query-prefix-dynamic-index-hit-def
  using target-hit by simp
qed
qed

lemma checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-bound:
  assumes raw-bound: query-index-raw-preimage-bound
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  shows
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
      trace-openings composition-openings i)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space)
proof -
  have actual-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-actual-query-prefix-candidate-opening-hit
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space)
  by (rule checked-staged-security-with-actual-query-prefix-candidate-opening-hit-bound
    [OF raw-bound wf controlled])
  (use i-bound in simp)

```

show *?thesis*
by (rule *order-trans*[
OF checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-le-actual-query-prefix-hit
actual-bound])
 (use *i-bound* **in** *simp*)
qed

lemma *checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-bound-from-budgets*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: *i < rounds*
shows
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit*
trace-openings composition-openings i)
adversary-initial-state $\leq$
staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 (*1::prob*) / *nnreal (card query-sample-space)*

proof –
have *raw-bound*: *query-index-raw-preimage-bound*
by (rule *query-index-raw-preimage-bound-from-sampler-wellformed*)
show *?thesis*
by (rule
checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-bound
 [*OF raw-bound wf controlled i-bound*])
qed

lemma *checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-prefix-and-path-output*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: *i < rounds*
and *prefix-bound*:
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit*
trace-openings composition-openings i)
adversary-initial-state $\leq P$
and *path-bound*:
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-opening-path-output-hit*
trace-openings composition-openings)
adversary-initial-state $\leq$ *path-error*
shows
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-authenticated-opening-hit*

$\text{trace-openings composition-openings } i)$
 $\text{adversary-initial-state} \leq P + \text{path-error}$
proof –
have *event-le*:
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-authenticated-opening-hit}$
 $\text{trace-openings composition-openings } i)$
 $\text{adversary-initial-state} \leq$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\lambda \text{out.}$
 $\text{checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit}$
 $\text{trace-openings composition-openings } i \ \text{out} \ \vee$
 $\text{checked-staged-security-with-query-prefix-opening-path-output-hit}$
 $\text{trace-openings composition-openings } \text{out})$
 $\text{adversary-initial-state}$
by (rule *wp-event-mono-on-support*)
 $(\text{rule}$
 $\text{checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-prefix-or-path-output-hit-on-support}$
 $[\text{OF } \text{wf controlled } i\text{-bound}])$
have *union-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\lambda \text{out.}$
 $\text{checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit}$
 $\text{trace-openings composition-openings } i \ \text{out} \ \vee$
 $\text{checked-staged-security-with-query-prefix-opening-path-output-hit}$
 $\text{trace-openings composition-openings } \text{out})$
 $\text{adversary-initial-state} \leq$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit}$
 $\text{trace-openings composition-openings } i)$
 $\text{adversary-initial-state} +$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-opening-path-output-hit}$
 $\text{trace-openings composition-openings})$
 $\text{adversary-initial-state}$
by (rule *wp-event-union-bound*)
have *sum-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit}$
 $\text{trace-openings composition-openings } i)$
 $\text{adversary-initial-state} +$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$

```

    (checked-staged-security-with-query-prefix-opening-path-output-hit
      trace-openings composition-openings)
    adversary-initial-state  $\leq P + \text{path-error}$ 
  by (intro add-mono prefix-bound path-bound)
show ?thesis
  by (rule order-trans[OF event-le])
    (rule order-trans[OF union-bound sum-bound])
qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-path-output:
assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and i-bound:  $i < \text{rounds}$ 
and path-bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state  $A\ i$ )
  (checked-staged-security-with-query-prefix-opening-path-output-hit
    trace-openings composition-openings)
  adversary-initial-state  $\leq \text{path-error}$ 
shows
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state  $A\ i$ )
  (checked-staged-security-with-query-prefix-authenticated-opening-hit
    trace-openings composition-openings  $i$ )
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets  $i + 1$ ) +
  (1::prob) / nnreal (card query-sample-space) + path-error
proof -
have prefix-bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state  $A\ i$ )
  (checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
    trace-openings composition-openings  $i$ )
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets  $i + 1$ ) +
  (1::prob) / nnreal (card query-sample-space)
by (rule checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-bound
  [OF raw-bound wf controlled i-bound])
show ?thesis
by (rule
  checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-prefix-and-path-output
  [OF wf controlled i-bound prefix-bound path-bound])
qed

lemma accepted-with-partial-initial-openings-aligned-consistent-prefix-target-hit:
assumes partial:

```

```

    accepted-with-partial-initial-openings-aligned-consistent s out fr
      f-fri-roots f-final as dg composition-fri-roots final query-idxs
      trace-openings composition-openings
  and i-bound: i < rounds
  and idx-eq: query-idxs ! i = index (to-nat raw)
  and alphas-eq: as = sqp-alphas prefix
shows
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-opening-query-target-from-prefix
      trace-openings composition-openings i)
    (Some (((prefix, prefix-state), raw), raw-state))
proof —
  have success:
    query-idxs ! i ∈
      partial-query-success-indices-at trace-openings composition-openings
      as i
  by (rule
    accepted-with-partial-initial-openings-aligned-consistent-success-index
    [OF partial i-bound])
  have index (to-nat raw) ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix
      trace-openings composition-openings i prefix prefix-state
  using success idx-eq alphas-eq
  unfolding staged-query-prefix-candidate-opening-query-target-from-prefix-def
  by simp
  then show ?thesis
    unfolding checked-staged-query-prefix-dynamic-index-hit-def by simp
qed

lemma query-bad-with-aligned-partial-candidates-prefix-target-hitE:
  assumes bad: query-bad-with-aligned-partial-candidates s out
  and i-bound: i < rounds
  and idx-eq: query-idxs ! i = index (to-nat raw)
  and alphas-eq: as = sqp-alphas prefix
  and partial:
    accepted-with-partial-initial-openings-aligned-consistent s out fr
      f-fri-roots f-final as dg composition-fri-roots final query-idxs
      trace-openings composition-openings
  and trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  and trace-low: trace-table-low-degree trace-table
  and comp-low:
    composition-table-low-degree maxDegree composition-table
  and not-all:
    ¬ all-queries-consistent trace-table composition-table as
  obtains

```

```

checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-candidate-opening-query-target-from-prefix
    trace-openings composition-openings i)
  (Some (((prefix, prefix-state), raw), raw-state))
nnreal
  (card
    (staged-query-prefix-candidate-opening-query-target-from-prefix
      trace-openings composition-openings i prefix prefix-state)) /
  nnreal (card query-sample-space) ≤ query-error-bound
proof –
  have hit:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-opening-query-target-from-prefix
        trace-openings composition-openings i)
      (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
    accepted-with-partial-initial-openings-aligned-consistent-prefix-target-hit
      [OF partial i-bound idx-eq alphas-eq])
  have not-all-prefix:
    ¬ all-queries-consistent trace-table composition-table
      (sqp-alphas prefix)
  using not-all alphas-eq by simp
  have fraction:
    nnreal
      (card
        (staged-query-prefix-candidate-opening-query-target-from-prefix
          trace-openings composition-openings i prefix prefix-state)) /
      nnreal (card query-sample-space) ≤ query-error-bound
  by (rule
    staged-query-prefix-candidate-opening-query-target-from-prefix-fraction-bound
      [OF trace-candidate comp-candidate trace-low comp-low not-all-prefix])
  show ?thesis
  by (rule that[OF hit fraction])
qed

```

lemma *accepted-with-partial-initial-openings-aligned-consistent-candidate-opening-hit-from-prefix:*

```

assumes partial:
  accepted-with-partial-initial-openings-aligned-consistent s out fr
  f-fri-roots f-final as dg composition-fri-roots final query-idxs
  trace-openings composition-openings
and i-bound: i < rounds
and idx-eq: query-idxs ! i = index (to-nat raw)
and alphas-eq: as = sqp-alphas prefix
and builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and prefix-receive:

```

Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
shows
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A trace-openings composition-openings i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))
proof –
have *target-hit*:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
trace-openings composition-openings i)
(Some (((prefix, prefix-state), raw), raw-state))
by (rule
accepted-with-partial-initial-openings-aligned-consistent-prefix-target-hit
[OF partial i-bound idx-eq alphas-eq])
show *?thesis*
by (rule
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixI
[OF i-bound builder prefix-receive target-hit])
qed

lemma *accepted-with-partial-initial-openings-aligned-transcript-consistent-candidate-opening-hit-from-prefix*:
assumes *partial*:
accepted-with-partial-initial-openings-aligned-transcript-consistent s
out fr f-fri-roots f-final as dg composition-fri-roots final
query-idxs trace-openings composition-openings
and *i-bound*: *i < rounds*
and *idx-eq*: *query-idxs ! i = index (to-nat raw)*
and *alphas-eq*: *as = sqp-alphas prefix*
and *builder*:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive*:
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
shows
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A trace-openings composition-openings i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))
proof –
have *aligned*:

```

    accepted-with-partial-initial-openings-aligned-consistent s out fr
      f-fri-roots f-final as dg composition-fri-roots final query-idxs
      trace-openings composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
  show ?thesis
  by (rule
    accepted-with-partial-initial-openings-aligned-consistent-candidate-opening-hit-from-prefix
    [OF aligned i-bound idx-eq alphas-eq builder prefix-receive])
qed

lemma query-bad-with-aligned-partial-candidates-candidate-opening-hit-from-prefixE:
  assumes bad: query-bad-with-aligned-partial-candidates s out
    and i-bound: i < rounds
    and idx-eq: query-idxs ! i = index (to-nat raw)
    and alphas-eq: as = sqp-alphas prefix
    and partial:
      accepted-with-partial-initial-openings-aligned-consistent s out fr
        f-fri-roots f-final as dg composition-fri-roots final query-idxs
        trace-openings composition-openings
  and builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
  obtains
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
      A trace-openings composition-openings i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))

proof -
  have hit:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
      A trace-openings composition-openings i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  by (rule
    accepted-with-partial-initial-openings-aligned-consistent-candidate-opening-hit-from-prefix
    [OF partial i-bound idx-eq alphas-eq builder prefix-receive])
  show ?thesis
  by (rule that[OF hit])
qed

```


lemma *query-bad-with-aligned-transcript-partial-candidates-candidate-opening-hit-from-prefixE*:
assumes *bad*: *query-bad-with-aligned-transcript-partial-candidates s out*
and *i-bound*: $i < \text{rounds}$
and *idx-eq*: *query-idxs ! i = index (to-nat raw)*
and *alphas-eq*: *as = sqp-alphas prefix*
and *partial*:
accepted-with-partial-initial-openings-aligned-transcript-consistent s
out fr f-fri-roots f-final as dg composition-fri-roots final
query-idxs trace-openings composition-openings
and *builder*:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive*:
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
obtains
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A trace-openings composition-openings i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)))
proof –
have *hit*:
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A trace-openings composition-openings i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)))
by (rule
accepted-with-partial-initial-openings-aligned-transcript-consistent-candidate-opening-hit-from-prefix
[OF partial i-bound idx-eq alphas-eq builder prefix-receive])
show *?thesis*
by (rule *that[OF hit]*)
qed

lemma *partial-candidate-query-witness-candidate-opening-hit-from-prefix*:
assumes *partial*:
accepted-with-partial-initial-openings s out fr f-fri-roots f-final as
dg composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings
and *trace-candidate*:
partial-trace-table-candidate trace-table trace-openings
and *comp-candidate*:
partial-composition-table-candidate composition-table
composition-openings
and *trace-low*: *trace-table-low-degree trace-table*
and *comp-low*:

composition-table-low-degree maxDegree composition-table
and *not-all:*
 $\neg$ *all-queries-consistent trace-table composition-table as*
and *i-bound: $i < \text{rounds}$*
and *alphas-eq: $as = \text{sqp-alphas prefix}$*
and *consistent:*
partial-query-openings-consistent (trace-openings ! i)
(composition-openings ! i) as (index (to-nat raw))
and *builder:*
Some (data, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive:*
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
shows
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A ((replicate rounds []) [i := trace-openings ! i])
((replicate rounds []) [i := composition-openings ! i]) i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)))
proof –
have *target-hit:*
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings ! i])
((replicate rounds []) [i := composition-openings ! i])
i)
(Some (((prefix, prefix-state), raw), raw-state)))
by (rule
checked-staged-query-prefix-candidate-opening-target-from-prefix-single-round-hitI)
(use i-bound consistent alphas-eq in simp-all)
show *?thesis*
by (rule
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixI
[OF i-bound builder prefix-receive target-hit])
qed

lemma *query-bad-with-partial-candidates-witness-candidate-opening-hit-from-prefix:*
assumes *bad: query-bad-with-partial-candidates s out*
and *partial:*
accepted-with-partial-initial-openings s out fr f-fri-roots f-final as
dg composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings

```

and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-low:
  composition-table-low-degree maxDegree composition-table
and not-all:
   $\neg$  all-queries-consistent trace-table composition-table as
and i-bound:  $i < \text{rounds}$ 
and alphas-eq:  $as = \text{sqp-alphas prefix}$ 
and consistent:
  partial-query-openings-consistent (trace-openings ! i)
  (composition-openings ! i) as (index (to-nat raw))
and builder:
  Some (data, attacker-state)  $\in$ 
  set-dist
  (execute (checked-staged-transcript-program A)
  adversary-initial-state)
and prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
  set-dist
  (execute (checked-staged-query-prefix-receive-with-state A i)
  adversary-initial-state)
shows
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
  A ((replicate rounds []) [i := trace-openings ! i])
  ((replicate rounds []) [i := composition-openings ! i]) i
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
  result), final-state))
by (rule partial-candidate-query-witness-candidate-opening-hit-from-prefix
  [OF partial trace-candidate comp-candidate trace-low comp-low not-all
  i-bound alphas-eq consistent builder prefix-receive])

end

end

```

```

theory Staged-Security-Experiment-Composition-Query-Bridge
imports
  Staged-Security-Experiment-Composition-Query-Residual
  Soundness-Query-Opening-Consistency
  Soundness-Bound-Table-Partial-Openings
  Soundness-Aligned-Partial-Query-Prefix
begin

```

Narrow bridge from the composition-drift query residual to the candidate-opening query target. This is kept separate from the residual split theory because the surrounding staged soundness layers are already heavy to consolidate.

context *soundness*
begin

lemma *partial-query-openings-consistent-synthetic*:
assumes *idx-sample*: $idx \in \text{query-sample-space}$
obtains *trace-openings composition-openings* **where**
partial-query-openings-consistent trace-openings composition-openings
as idx

proof –

let *?trace-openings* =
 map ($\lambda j.$
 ($\sqcup$ *opening-root* = 0,
 opening-length = *scale* * *clength*,
 opening-index = *j*,
 opening-value = 0,
 opening-path = [] $\sqcup$) (*powers-scaled idx*)
let *?composition-openings* =
 [($\sqcup$ *opening-root* = 0,
 opening-length = *scale* * *clength*,
 opening-index = *idx*,
 opening-value =
 cp-eval as (map *opening-value* *?trace-openings*) ($h \wedge idx * \text{shift}$),
 opening-path = [] $\sqcup$),
 ($\sqcup$ *opening-root* = 0,
 opening-length = *scale* * *clength*,
 opening-index = *fri-sibling-index* (*scale* * *clength*) *idx*,
 opening-value = 0,
 opening-path = [] $\sqcup$)]
have *trace-indices*:
 map *opening-index* *?trace-openings* = *powers-scaled idx*
by (*simp add: o-def*)
have *comp-indices*:
 map *opening-index* *?composition-openings* =
 [*idx*, *fri-sibling-index* (*scale* * *clength*) *idx*]
by *simp*
have *consistent*:
partial-query-openings-consistent ?trace-openings
?composition-openings as idx
by (*rule partial-query-openings-consistentI*
 [*OF idx-sample trace-indices comp-indices*])
simp
show *?thesis*
by (*rule that[OF consistent]*)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hitI-from-prefix-receive*:
assumes *i-bound*: $i < \text{rounds}$
and *builder*:
 Some (*data*, *attacker-state*) $\in$

```

    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  and prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
  shows
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit
    A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  proof -
  obtain trace-openings composition-openings where consistent:
    partial-query-openings-consistent trace-openings composition-openings
      (staged-alphas data) (index (to-nat raw))
  proof (rule partial-query-openings-consistent-synthetic)
  show index (to-nat raw) ∈ query-sample-space
    using index-less-query-sample-space
    unfolding query-sample-space-def by simp
  qed
  have component:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit
    A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    (staged-alphas data) i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hitI-from-prefix-receive
    [OF i-bound builder prefix-receive consistent])
  show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hitI
    [OF component])
  qed

definition checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
  :: ('f staged-adversary ⇒
    ((((((f × f list × f list × f) ×
      f protocol-channel) × f staged-proof-data) ×
      f protocol-channel) × unit list) ×
      f protocol-channel) option ⇒ bool)
  where
    checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
    A out ⟷
    (case out of

```

$None \Rightarrow False$
 $| \text{Some } (((((\alpha\text{-prefix}, \alpha\text{-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}) \Rightarrow$
 $\text{staged-composition-fri-roots data} \neq [] \wedge$
 $(\exists i \text{ trace-openings composition-openings.}$
 $\text{partial-authenticated-table (staged-trace-root data)}$
 $(\text{scale} * \text{clength}) \text{ trace-openings final-state} \wedge$
 $\text{partial-authenticated-table (hd (staged-composition-fri-roots data))}$
 $(\text{scale} * \text{clength}) \text{ composition-openings final-state} \wedge$
 $\text{checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit } A$
 $((\text{replicate rounds } []) [i := \text{trace-openings}])$
 $((\text{replicate rounds } []) [i := \text{composition-openings}])$
 $(\text{staged-alphas data}) i$
 $(\text{Some } (((((\alpha\text{-prefix}, \alpha\text{-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}))))$

lemma *checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hitI:*

assumes *comp-nonempty:* $\text{staged-composition-fri-roots data} \neq []$

and *trace-auth:*

$\text{partial-authenticated-table (staged-trace-root data)}$
 $(\text{scale} * \text{clength}) \text{ trace-openings final-state}$

and *comp-auth:*

$\text{partial-authenticated-table (hd (staged-composition-fri-roots data))}$
 $(\text{scale} * \text{clength}) \text{ composition-openings final-state}$

and *component:*

$\text{checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit } A$
 $((\text{replicate rounds } []) [i := \text{trace-openings}])$
 $((\text{replicate rounds } []) [i := \text{composition-openings}])$
 $(\text{staged-alphas data}) i$
 $(\text{Some } (((((\alpha\text{-prefix}, \alpha\text{-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state})))$

shows

$\text{checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit}$
 A
 $(\text{Some } (((((\alpha\text{-prefix}, \alpha\text{-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state})))$

proof –

have *witness:*

$\exists j \text{ trace-openings}' \text{ composition-openings'}.$
 $\text{partial-authenticated-table (staged-trace-root data)}$
 $(\text{scale} * \text{clength}) \text{ trace-openings}' \text{ final-state} \wedge$
 $\text{partial-authenticated-table (hd (staged-composition-fri-roots data))}$
 $(\text{scale} * \text{clength}) \text{ composition-openings}' \text{ final-state} \wedge$
 $\text{checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit } A$
 $((\text{replicate rounds } []) [j := \text{trace-openings}'])$
 $((\text{replicate rounds } []) [j := \text{composition-openings}'])$
 $(\text{staged-alphas data}) j$
 $(\text{Some } (((((\alpha\text{-prefix}, \alpha\text{-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state})))$

```

    by (intro exI[of - i] exI[of - trace-openings]
        exI[of - composition-openings] conjI)
        (use trace-auth comp-auth component in simp-all)
  show ?thesis
    unfolding
      checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-def
    using comp-nonempty witness by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-imp-compact:
  assumes hit:
    checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
    A out
  shows
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit
    A out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-def
    by simp
next
  case (Some packed)
  then obtain alpha-prefix alpha-prefix-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)
    by (cases packed, auto split: prod.splits)
  from hit[unfolded out-eq]
    checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-def]
  obtain i trace-openings composition-openings where component:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    (staged-alphas data) i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  by auto
  show ?thesis
    unfolding out-eq
    by (rule
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hitI
      [OF component])
qed

lemma checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-le-compact:

```

```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
    A)
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit
    A)
  adversary-initial-state
by (rule wp-event-mono)
(rule
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-imp-compact)

```

lemma *checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hitI-from-current-open*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and comp-nonempty: staged-composition-fri-roots data ≠ []
and i-bound: i < rounds
and builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and verifier:
  Some (result, final-state) ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))))
and final-lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and comp-auth:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings final-state
and consistent:
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
shows
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
    A
    (Some ((((((alpha-prefix, alpha-prefix-state), data), attacker-state),

```



```

    result), final-state))
proof –
from checked-staged-transcript-program-query-prefix-receive-final-lookup-support
  [OF wf controlled i-bound builder verifier final-lookup]
obtain prefix prefix-state raw-state where prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
by blast
have component:
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    (staged-alphas data) i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
by (rule
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hitI-from-prefix-receive
    [OF i-bound builder prefix-receive consistent])
show ?thesis
by (rule
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hitI
    [OF comp-nonempty trace-auth comp-auth component])
qed

definition checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
  :: ('f staged-adversary ⇒
    ((((((f × f list × f list × f) ×
      f protocol-channel) × f staged-proof-data) ×
      f protocol-channel) × unit list) ×
      f protocol-channel) option ⇒ bool)
where
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
    A out ⟷
    (case out of
      None ⇒ False
    | Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state) ⇒
      staged-composition-fri-roots data ≠ [] ∧
      (∃ i trace-openings composition-openings.
        partial-authenticated-table (staged-trace-root data)
          (scale * clength) trace-openings final-state ∧
        partial-authenticated-table (hd (staged-composition-fri-roots data))
          (scale * clength) composition-openings final-state ∧
        checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
          A
          ((replicate rounds []) [i := trace-openings])
          ((replicate rounds []) [i := composition-openings])
      )
    )

```

i
*(Some (((((alpha-prefix, alpha-prefix-state), data),
attacker-state), result), final-state))))*

lemma *checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefixI:*
assumes *comp-nonempty: staged-composition-fri-roots data* $\neq \square$
and *trace-auth:*
partial-authenticated-table (staged-trace-root data)
*(scale * clength) trace-openings final-state*
and *comp-auth:*
partial-authenticated-table (hd (staged-composition-fri-roots data))
*(scale * clength) composition-openings final-state*
and *component:*
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A
((replicate rounds $\square$) [i := trace-openings])
((replicate rounds $\square$) [i := composition-openings])
 i
*(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)))*

shows
checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
A
*(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)))*

proof –
have *witness:*
 $\exists j$ *trace-openings' composition-openings'.*
partial-authenticated-table (staged-trace-root data)
*(scale * clength) trace-openings' final-state* $\wedge$
partial-authenticated-table (hd (staged-composition-fri-roots data))
*(scale * clength) composition-openings' final-state* $\wedge$
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A
((replicate rounds $\square$) [j := trace-openings'])
((replicate rounds $\square$) [j := composition-openings'])
 j
*(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)))*
by (*intro exI[of - i] exI[of - trace-openings]*
exI[of - composition-openings] conjI)
(use trace-auth comp-auth component in simp-all)

show *?thesis*
unfolding
checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix-def
using *comp-nonempty witness by simp*
qed

lemma *checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefixI-from-*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and comp-nonempty: staged-composition-fri-roots data  $\neq []$ 
and i-bound: i < rounds
and builder:
  Some (data, attacker-state)  $\in$ 
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and verifier:
  Some (result, final-state)  $\in$ 
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
and final-lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and comp-auth:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings final-state
and consistent:
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
shows
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
    A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
proof –
from checked-staged-transcript-program-query-prefix-receive-final-lookup-support-with-alphas
  [OF wf controlled i-bound builder verifier final-lookup]
obtain prefix prefix-state raw-state where prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
and alphas-eq: sqp-alphas prefix = staged-alphas data
by blast
have component:
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])

```

```

i
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
by (rule
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixI-from-prefix-receive
  [OF i-bound builder prefix-receive alphas-eq consistent])
show ?thesis
by (rule
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefixI
  [OF comp-nonempty trace-auth comp-auth component])
qed

```

```

lemma selected-query-chunks-match-verifier-lengths:
assumes i-bound:  $i < \text{rounds}$ 
and len-prefix-chunks:  $\text{length prefix-chunks} = i$ 
and len-suffix-chunks:  $\text{length suffix-chunks} = \text{rounds} - \text{Suc } i$ 
and chunk-shape:
  verifier-query-round-chunk idx trace-roots composition-roots chunk
and prefix-rounds:
   $\bigwedge j. j < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! j)
    trace-roots composition-roots (prefix-chunks ! j)
and suffix-rounds:
   $\bigwedge j. j < \text{rounds} - \text{Suc } i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! j)
    trace-roots composition-roots (suffix-chunks ! j)
shows
   $\text{length (prefix-chunks @ chunk \# suffix-chunks)} = \text{rounds}$ 
   $\bigwedge j. j < \text{rounds} \implies$ 
     $\text{length ((prefix-chunks @ chunk \# suffix-chunks) ! j)} =$ 
     $\text{verifier-query-round-transcript-length (query-idxs ! j)}$ 
    trace-roots composition-roots
proof -
  show  $\text{length (prefix-chunks @ chunk \# suffix-chunks)} = \text{rounds}$ 
  using len-prefix-chunks len-suffix-chunks i-bound by simp
next
  fix j
  assume j-bound:  $j < \text{rounds}$ 
  let ?chunks =  $\text{prefix-chunks @ chunk \# suffix-chunks}$ 
  show
     $\text{length (?chunks ! j)} =$ 
     $\text{verifier-query-round-transcript-length (query-idxs ! j)}$ 
    trace-roots composition-roots
  proof (cases  $j < i$ )
  case True
  have nth-eq:  $?chunks ! j = \text{prefix-chunks ! } j$ 
  using True len-prefix-chunks by (simp add: nth-append)
  have
     $\text{length (prefix-chunks ! } j) =$ 

```

```

    verifier-query-round-transcript-length
      (prefix-query-idxs ! j) trace-roots composition-roots
  by (rule verifier-query-round-chunk-length[OF prefix-rounds[OF True]])
then show ?thesis
  using nth-eq
  by (simp add: verifier-query-round-transcript-length-index-irrelevant)
next
case False
note not-lt = False
show ?thesis
proof (cases j = i)
  case True
  have nth-eq: ?chunks ! j = chunk
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length chunk =
      verifier-query-round-transcript-length idx trace-roots
        composition-roots
    by (rule verifier-query-round-chunk-length[OF chunk-shape])
  then show ?thesis
    using nth-eq
    by (simp add: verifier-query-round-transcript-length-index-irrelevant)
next
case False
note not-eq = False
let ?k = j - Suc i
have ge: Suc i ≤ j
  using not-lt not-eq by linarith
have k-bound: ?k < rounds - Suc i
  using ge j-bound by linarith
have nth-eq: ?chunks ! j = suffix-chunks ! ?k
  using ge len-prefix-chunks by (simp add: nth-append)
have
  length (suffix-chunks ! ?k) =
    verifier-query-round-transcript-length
      (suffix-query-idxs ! ?k) trace-roots composition-roots
  by (rule verifier-query-round-chunk-length
    [OF suffix-rounds[OF k-bound]])
then show ?thesis
  using nth-eq
  by (simp add: verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed

```

lemma *selected-query-chunks-eq-staged:*
assumes *builder:*
Some (data, attacker-state) ∈
set-dist

```

      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
    and query-idxs-len: length query-idxs = rounds
    and chunks-len: length query-chunks = rounds
    and concat-eq:
      List.concat query-chunks @ trailing =
        List.concat (staged-query-chunks data)
    and parser-chunk-len:
       $\bigwedge j. j < rounds \implies$ 
        length (query-chunks ! j) =
          verifier-query-round-transcript-length (query-idxs ! j)
            (staged-trace-fri-roots data)
            (staged-composition-fri-roots data)
    shows query-chunks = staged-query-chunks data  $\wedge$  trailing = []
  proof -
    from checked-staged-transcript-program-query-chunks-match-verifier-lengths
      [OF builder]
    obtain staged-query-idxs where staged-match:
      staged-query-chunks-match-verifier-lengths data staged-query-idxs
    by blast
    have staged-match-query-idxs:
      staged-query-chunks-match-verifier-lengths data query-idxs
    by (rule staged-query-chunks-match-verifier-lengths-transfer
      [OF staged-match query-idxs-len])
    show ?thesis
    by (rule query-chunks-eq-staged-if-matching-lengths
      [OF chunks-len concat-eq parser-chunk-len
        staged-match-query-idxs])
  qed

lemma selected-query-prefix-state-eq-staged:
  assumes len-prefix-chunks: length prefix-chunks = i
  and chunks-eq:
    prefix-chunks @ chunk # suffix-chunks = staged-query-chunks data
  and query-state-hash: PState query-state = staged-query-start-hash data
  shows
    state-after-query-chunks (PState query-state) prefix-chunks i =
      state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i
  proof -
    have
      state-after-query-chunks (staged-query-start-hash data)
        (prefix-chunks @ chunk # suffix-chunks) i =
        state-after-query-chunks (staged-query-start-hash data)
          prefix-chunks i
    by (rule state-after-query-chunks-append-prefix [OF len-prefix-chunks])
    then show ?thesis
    using chunks-eq query-state-hash by simp
  qed

```

definition *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit*

```

:: ('f staged-adversary  $\Rightarrow$ 
  ((((((('f  $\times$  'f list  $\times$  'f list  $\times$  'f)  $\times$ 
    'f protocol-channel)  $\times$  'f staged-proof-data)  $\times$ 
    'f protocol-channel)  $\times$  unit list)  $\times$ 
    'f protocol-channel) option  $\Rightarrow$  bool)

```

where

```

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit
  A out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state)  $\Rightarrow$ 
    (let s =
      verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)
    in  $\exists i$  trace-openings composition-openings.
      (trace-openings, composition-openings)  $\in$ 
        query-header-supported-partial-opening-witnesses s
          (staged-trace-root data)
          (staged-trace-fri-roots data)
          (staged-trace-final data)
          (staged-alphas data)
          (staged-degree data)
          (staged-composition-fri-roots data)
          (staged-composition-final data)  $\wedge$ 
        checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
          ((replicate rounds []) [i := trace-openings])
          ((replicate rounds []) [i := composition-openings])
          (staged-alphas data) i
          (Some (((((alpha-prefix, alpha-prefix-state), data),
            attacker-state), result), final-state))))))

```

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hitI:*

assumes *witness:*

```

(trace-openings, composition-openings)  $\in$ 
  query-header-supported-partial-opening-witnesses
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)

```

and *component:*

```

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A

```

```

    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    (staged-alphas data) i
    (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  shows
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit
      A
      (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state))
  proof -
    have ex:
      ∃ j trace-openings' composition-openings'.
        (trace-openings', composition-openings') ∈
          query-header-supported-partial-opening-witnesses
            (verifier-state-from-adversary attacker-state
              (staged-proof-transcript data))
              (staged-trace-root data)
              (staged-trace-fri-roots data)
              (staged-trace-final data)
              (staged-alphas data)
              (staged-degree data)
              (staged-composition-fri-roots data)
              (staged-composition-final data) ∧
            checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
              ((replicate rounds []) [j := trace-openings'])
              ((replicate rounds []) [j := composition-openings'])
              (staged-alphas data) j
              (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
                result), final-state))
    by (intro exI[of - i] exI[of - trace-openings]
        exI[of - composition-openings] conjI)
        (use witness component in simp-all)
    show ?thesis
      unfolding
        checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-def
        Let-def
      using ex by simp
  qed

  lemma checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-imp-compa
    assumes hit:
      checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit
        A out
    shows
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit
        A out
  proof (cases out)
    case None

```



```

then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-def
  by simp
next
  case (Some packed)
  then obtain alpha-prefix alpha-prefix-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)
    by (cases packed, auto split: prod.splits)
  from hit[unfolded out-eq
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-def
    Let-def]
  obtain i trace-openings composition-openings where component:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    (staged-alphas data) i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  by auto
  show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hitI
    [OF component])
qed

```

```

lemma checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hitE:
assumes hit:
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit
  A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
obtains i trace-openings composition-openings where
  (trace-openings, composition-openings) ∈
  query-header-supported-partial-opening-witnesses
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)

```

```

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit
A
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
(staged-alphas data) i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))
proof –
from hit[unfolded
checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-def
Let-def]
obtain i trace-openings composition-openings where witness:
(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)
and component:
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit
A
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
(staged-alphas data) i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))
by auto
show ?thesis
by (rule that[OF witness component])
qed

lemma checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-le-compact
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit
A)
adversary-initial-state ≤
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit
A)
adversary-initial-state
by (rule wp-event-mono)
(rule

```

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-imp-compact)

definition *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*

$:: ('f \text{ staged-adversary} \Rightarrow$
 $(((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool})$

where

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix

$A \text{ out} \longleftrightarrow$

(*case out of*

$\text{None} \Rightarrow \text{False}$

| $\text{Some} (((((\alpha\text{-prefix}, \alpha\text{-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}) \Rightarrow$

(*let s =*

verifier-state-from-adversary attacker-state

(*staged-proof-transcript data*)

in $\exists i \text{ trace-openings composition-openings.}$

(*trace-openings, composition-openings*) $\in$

query-header-supported-partial-opening-witnesses s

(*staged-trace-root data*)

(*staged-trace-fri-roots data*)

(*staged-trace-final data*)

(*staged-alphas data*)

(*staged-degree data*)

(*staged-composition-fri-roots data*)

(*staged-composition-final data*) $\wedge$

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix

A

((*replicate rounds []*) [$i := \text{trace-openings}$])

((*replicate rounds []*) [$i := \text{composition-openings}$])

i

($\text{Some} (((((\alpha\text{-prefix}, \alpha\text{-prefix-state}), \text{data}),$
 $\text{attacker-state}), \text{result}), \text{final-state}))))$

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*

assumes *witness:*

(*trace-openings, composition-openings*) $\in$

query-header-supported-partial-opening-witnesses

(*verifier-state-from-adversary attacker-state*

(*staged-proof-transcript data*))

(*staged-trace-root data*)

(*staged-trace-fri-roots data*)

(*staged-trace-final data*)

(*staged-alphas data*)

(*staged-degree data*)

(*staged-composition-fri-roots data*)

(*staged-composition-final data*)

and component:

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix A

((replicate rounds []) [i := trace-openings])

((replicate rounds []) [i := composition-openings])

i

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))

shows

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix A

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))

proof –

have *ex*:

∃j trace-openings' composition-openings'.

(trace-openings', composition-openings') ∈

query-header-supported-partial-opening-witnesses

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data))

(staged-trace-root data)

(staged-trace-fri-roots data)

(staged-trace-final data)

(staged-alphas data)

(staged-degree data)

(staged-composition-fri-roots data) ∧

(staged-composition-final data) ∧

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix A

((replicate rounds []) [j := trace-openings'])

((replicate rounds []) [j := composition-openings'])

j

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))

by (intro *exI*[of - *i*] *exI*[of - trace-openings]

exI[of - composition-openings] *conjI*)

(use witness component **in** *simp-all*)

show *?thesis*

unfolding

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-def

Let-def

using *ex* **by** *simp*

qed

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*

assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix A

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))

obtains *i* trace-openings composition-openings **where**

```

    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
    A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
proof –
from hit[unfolded
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-def
  Let-def]
obtain i trace-openings composition-openings where witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  and component:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
    A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  by auto
show ?thesis
  by (rule that[OF witness component])
qed

lemma checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
assumes hit:

```

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
A
 (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))
obtains *i* trace-openings composition-openings query-prefix query-prefix-state
 raw raw-state **where**
 (trace-openings, composition-openings) ∈
 query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
i < rounds
 Some (data, attacker-state) ∈
 set-dist
 (execute (checked-staged-transcript-program *A*)
 adversary-initial-state)
 Some (((query-prefix, query-prefix-state), raw), raw-state) ∈
 set-dist
 (execute (checked-staged-query-prefix-receive-with-state *A* *i*)
 adversary-initial-state)
 checked-staged-query-prefix-dynamic-index-hit
 (staged-query-prefix-candidate-opening-query-target-from-prefix
 ((replicate rounds []) [i := trace-openings])
 ((replicate rounds []) [i := composition-openings])
 i)
 (Some (((query-prefix, query-prefix-state), raw), raw-state))
proof –
from *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*
 [OF hit]
obtain *i* trace-openings composition-openings **where** witness:
 (trace-openings, composition-openings) ∈
 query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
and component:
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix

```

A
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))
by blast
from checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixE
[OF component]
obtain query-prefix query-prefix-state raw raw-state where
i-bound: i < rounds
and builder:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and prefix-receive:
Some (((query-prefix, query-prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and target-hit:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i)
(Some (((query-prefix, query-prefix-state), raw), raw-state))
by blast
show ?thesis
by (rule that[OF witness i-bound builder prefix-receive target-hit])
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-header-authenticat
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
Some (((((prefix, prefix-state), data), attacker-state), result),
final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
adversary-initial-state)
and hit:
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
(Some (((((prefix, prefix-state), data), attacker-state), result),
final-state))
shows

```

```

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit
  A
  (Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state))
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  show ?thesis
  proof (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-supported-openingsE
    [OF wf controlled support hit])
  fix s trace-table i raw trace-openings composition-openings
  assume s-def: s = ?s
  and i-bound: i < rounds
  and final-lookup:
    fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
      Some raw
  and witness:
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  and consistent:
    partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
  from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
  obtain builder verifier where
    builder:
      Some (data, attacker-state) ∈
        set-dist

```



```

      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
    and verifier:
      Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
    by blast
  from checked-staged-transcript-program-query-prefix-receive-final-lookup-support
    [OF wf controlled i-bound builder verifier final-lookup]
  obtain query-prefix query-prefix-state raw-state where prefix-receive:
    Some (((query-prefix, query-prefix-state), raw), raw-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
    by blast
  have component:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit
      A
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      (staged-alphas data) i
      (Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state))
    by (rule
      checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hitI-from-prefix-receive
      [OF i-bound builder prefix-receive consistent])
  show ?thesis
    by (rule
      checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hitI
      [OF - component])
      (use witness s-def in simp)
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-header-authenticated
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit
      A)
    adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈

```

```

    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    out
show
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit
  A out
proof (cases out)
  case None
  then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
  case (Some packed)
  then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state)
  by (cases packed, auto split: prod.splits)
show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-header-authenticated-
    [OF wf controlled])
  (use support hit out-eq in simp-all)
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-header-authenticat
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state) ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    adversary-initial-state)
and hit:

```

```

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
  (Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state))
shows
checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
  A
  (Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state))
proof —
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-supported-openingsE
  [OF wf controlled support hit])
fix s trace-table i raw trace-openings composition-openings
assume s-def: s = ?s
and i-bound: i < rounds
and final-lookup:
fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
  Some raw
and witness:
(trace-openings, composition-openings) ∈
  query-header-supported-partial-opening-witnesses s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
and consistent:
partial-query-openings-consistent trace-openings composition-openings
  (staged-alphas data) (index (to-nat raw))
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
by (rule
  checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
  [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]

```

```

obtain builder verifier where
  builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
from checked-staged-transcript-program-query-prefix-receive-final-lookup-support-with-alphas
  [OF wf controlled i-bound builder verifier final-lookup]
obtain query-prefix query-prefix-state raw-state where prefix-receive:
  Some (((query-prefix, query-prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  and alphas-eq: sqp-alphas query-prefix = staged-alphas data
  by blast
have component:
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
    A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i
    (Some ((((prefix, prefix-state), data), attacker-state), result),
      final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixI-from-prefix-receive
      [OF i-bound builder prefix-receive alphas-eq consistent])
show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefixI
      [OF - component])
    (use witness s-def in simp)
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-header-authenticated
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
      A)

```

```

      adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    out
show
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
  A out
proof (cases out)
  case None
  then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
  case (Some packed)
  then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state)
  by (cases packed, auto split: prod.splits)
show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-header-authenticated
    [OF wf controlled])
  (use support hit out-eq in simp-all)
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-header-auth
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and header-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)

```

(*checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit*
A)
adversary-initial-state $\leq$ (*Q::prob*)
shows
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq$ *Q*
by (*rule order.trans*
 [*OF*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-header-authenticated-can
header-bound])
 (*use wf controlled in simp-all*)

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-header-authenticated-candidate-opening-query-hit*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *header-bound:*
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
 (*checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*
A)
adversary-initial-state $\leq$ (*Q::prob*)
shows
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq$ *Q*
by (*rule order.trans*
 [*OF*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-header-authenticated-can
header-bound])
 (*use wf controlled in simp-all*)

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-header-authenticated-candidate-opening-query-hit*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *header-bound:*
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
 (*checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit*
A)
adversary-initial-state $\leq$ (*Q::prob*)
and *without-bound:*
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state $\leq$ (*W::prob*)

shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq$ *Q* + *W*

proof (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-query-and-without-query)

show

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq$ *Q*

by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-header-authen
 [*OF wf controlled header-bound*])

show

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state $\leq$ *W*

by (rule *without-bound*)

qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-alpha-bad-trace-openings*

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *support*:

Some (((((*prefix*, *prefix-state*), *data*), *attacker-state*), *result*),
final-state) $\in$
set-dist
 (*execute*
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
adversary-initial-state)

and *hit*:

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 (*Some* (((((*prefix*, *prefix-state*), *data*), *attacker-state*), *result*),
final-state))

obtains *s* *trace-table witness-out query-idxs trace-openings*
witness-as witness-dg witness-composition-fri-roots witness-final
witness-rest **where**
s = *verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)
witness-out $\in$ *set-dist* (*execute verify-monad s*)
accepted-with-partial-trace-openings s witness-out
 (*staged-trace-root data*) *query-idxs trace-openings*
partial-trace-table-candidate trace-table trace-openings

```

verifier-header-transcript s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  witness-as witness-dg witness-composition-fri-roots witness-final
  witness-rest
witness-as = staged-alphas data
staged-alphas data ∈ composition-trace-bad-alpha-space trace-table
¬ staged-alphas data ∈
  alpha-prefix-union-bad-sets prefix-state
  composition-trace-bad-alpha-space (fst prefix)
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
obtain builder where
  builder:
    Some (data, attacker-state) ∈
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  by blast
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-supportedE
  [OF wf controlled support hit])
fix s trace-table

```



```

assume s-def:
  s = verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
and trace-candidate:
  trace-table ∈
    alpha-header-supported-partial-trace-table-candidates s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
and trace-bad:
  staged-alphas data ∈ composition-trace-bad-alpha-space trace-table
and not-prefix:
  ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
show ?thesis
proof (rule alpha-header-supported-partial-trace-table-candidatesE
  [OF trace-candidate])
fix witness-out query-idxs trace-openings witness-as witness-dg
  witness-composition-fri-roots witness-final witness-rest
assume witness-out:
  witness-out ∈ set-dist (execute verify-monad s)
and partial:
  accepted-with-partial-trace-openings s witness-out
    (staged-trace-root data) query-idxs trace-openings
and partial-candidate:
  partial-trace-table-candidate trace-table trace-openings
and header:
  verifier-header-transcript s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    witness-as witness-dg witness-composition-fri-roots
    witness-final witness-rest
have staged-header-s:
  verifier-header-transcript s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
using staged-header by (simp add: s-def[symmetric])
have as-eq: witness-as = staged-alphas data
using verifier-header-transcript-unique[OF staged-header-s header]
by simp
show ?thesis

```

by (rule that[*OF s-def witness-out partial partial-candidate*
header as-eq trace-bad not-prefix])
 qed
 qed
 qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-table-witnessE*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *support:*
Some (((((prefix, prefix-state), data), attacker-state), result),
final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
adversary-initial-state)
and *hit:*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
(Some (((((prefix, prefix-state), data), attacker-state), result),
final-state))
obtains *s trace-table composition-table query-idxs f bound-trace-openings*
bound-composition-openings **where**
s = verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
composition-bad-context s (Some (result, final-state)) trace-table
composition-table (staged-alphas data) query-idxs f
common-denominator-degree-bounds f (staged-alphas data)
random-combination-common-denominator-hides-violations f
(staged-alphas data)
partial-trace-table-candidate trace-table bound-trace-openings
partial-composition-table-candidate composition-table
bound-composition-openings
trace-table-low-degree trace-table
composition-table-low-degree maxDegree composition-table
all-queries-consistent trace-table composition-table
(staged-alphas data)
 $\neg$ *staged-alphas data ∈*
alpha-prefix-union-bad-sets prefix-state
composition-trace-bad-alpha-space (fst prefix)
proof –
from *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-supportedE*
[OF wf controlled support hit]
obtain *s trace-table* **where**
s-def:
s = verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
and *random-bad:*
composition-randomization-bad s (Some (result, final-state))

and *not-prefix*:
 $\neg$ *staged-alphas data* $\in$
 α -*prefix-union-bad-sets prefix-state*
 $\text{composition-trace-bad-alpha-space (fst prefix)}$
by *blast*
from *random-bad*
obtain *trace-table'* *composition-table as query-idxs f* **where** *bad-context*:
 $\text{composition-bad-context } s \text{ (Some (result, final-state)) } \text{trace-table'}$
 $\text{composition-table as query-idxs } f$
and *bounds: common-denominator-degree-bounds f as*
and *hides*:
 $\text{random-combination-common-denominator-hides-violations } f \text{ as}$
unfolding *composition-randomization-bad-def* **by** *blast*
have *bound-as*:
 $\text{accepted-with-bound-tables } s \text{ (Some (result, final-state)) } \text{trace-table'}$
 $\text{composition-table as query-idxs}$
using *bad-context unfolding composition-bad-context-def* **by** *blast*
from *bound-as obtain fr f-fri-roots f-final dg composition-fri-roots*
 final rest **where** *header-as*:
 $\text{verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final as } dg$
 $\text{composition-fri-roots final rest}$
unfolding *accepted-with-bound-tables-def* **by** *blast*
have *data-support*:
 $\text{Some } (((\text{data, attacker-state}), \text{result}), \text{final-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{adversary-initial-state})$
by (*rule*
 $\text{checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support}$
 $[\text{OF support}]$)
from *checked-staged-security-experiment-with-data-state-outcomeE*
 $[\text{OF data-support}]$
obtain *builder verifier* **where** *builder*:
 $\text{Some } (\text{data, attacker-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$
and *verifier*:
 $\text{Some } (\text{result, final-state}) \in$
 $\text{set-dist } (\text{execute } \text{verify-monad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
by *blast*
have *staged-header*:
 $\text{verifier-header-transcript } s$
 $(\text{staged-trace-root data})$
 $(\text{staged-trace-fri-roots data})$
 $(\text{staged-trace-final data})$
 $(\text{staged-alphas data})$

```

    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
unfolding s-def
by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
have as-eq: as = staged-alphas data
    using verifier-header-transcript-unique[OF header-as staged-header]
    by simp
have bad-context-staged:
    composition-bad-context s (Some (result, final-state)) trace-table'
    composition-table (staged-alphas data) query-idxs f
    using bad-context as-eq by simp
have bounds-staged:
    common-denominator-degree-bounds f (staged-alphas data)
    using bounds as-eq by simp
have hides-staged:
    random-combination-common-denominator-hides-violations f
    (staged-alphas data)
    using hides as-eq by simp
have bound:
    accepted-with-bound-tables s (Some (result, final-state)) trace-table'
    composition-table (staged-alphas data) query-idxs
    using bad-context-staged unfolding composition-bad-context-def by blast
from accepted-with-bound-tables-partial-opening-witnesses-Some[OF bound]
obtain fr f-fri-roots f-final dg composition-fri-roots final rest
    bound-trace-openings bound-composition-openings where
    trace-candidate:
    partial-trace-table-candidate trace-table' bound-trace-openings
    and comp-candidate:
    partial-composition-table-candidate composition-table
    bound-composition-openings
    by blast
have trace-low: trace-table-low-degree trace-table'
proof –
    have deg-f: degree f < clength
    and trace-table-eq: trace-table' = map (poly f) eval-domain
    using bad-context-staged unfolding composition-bad-context-def by blast+
    show ?thesis
    unfolding trace-table-low-degree-def
    using deg-f trace-table-eq by blast
qed
have comp-low:
    composition-table-low-degree maxDegree composition-table
    using bad-context-staged unfolding composition-bad-context-def by blast
have all-queries:
    all-queries-consistent trace-table' composition-table
    (staged-alphas data)

```

```

using bad-context-staged unfolding composition-bad-context-def by blast
show ?thesis
by (rule that[OF s-def bad-context-staged bounds-staged hides-staged
    trace-candidate comp-candidate trace-low comp-low all-queries
    not-prefix])
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-aligned-openingsE:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    (Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state))
obtains s trace-table composition-table query-idxs f i raw
  header-trace-openings header-composition-openings
  bound-trace-openings bound-composition-openings where
s = verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
composition-bad-context s (Some (result, final-state)) trace-table
  composition-table (staged-alphas data) query-idxs f
i < rounds
fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
    Some raw
index (to-nat raw) = query-idxs ! i
  (header-trace-openings, header-composition-openings) ∈
  query-header-supported-partial-opening-witnesses s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  partial-query-openings-consistent header-trace-openings
    header-composition-openings (staged-alphas data) (query-idxs ! i)
  partial-trace-table-candidate trace-table bound-trace-openings

```

```

partial-composition-table-candidate composition-table
  bound-composition-openings
all-queries-consistent trace-table composition-table
  (staged-alphas data)
¬ staged-alphas data ∈
  alpha-prefix-union-bad-sets prefix-state
  composition-trace-bad-alpha-space (fst prefix)
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  show ?thesis
  proof (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-supported-openingsE
    [OF wf controlled support hit])
  fix s' trace-table' i raw header-trace-openings
    header-composition-openings
  assume s'-def: s' = ?s
  and i-bound: i < rounds
  and final-lookup:
    fmllookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
      Some raw
  and header-witness:
    (header-trace-openings, header-composition-openings) ∈
    query-header-supported-partial-opening-witnesses s'
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and header-consistent:
    partial-query-openings-consistent header-trace-openings
    header-composition-openings (staged-alphas data)
    (index (to-nat raw))
  show ?thesis
  proof (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-table-witnessE
    [OF wf controlled support hit])
  fix s trace-table composition-table query-idxs f bound-trace-openings
    bound-composition-openings
  assume s-def: s = ?s
  and bad-context:
    composition-bad-context s (Some (result, final-state)) trace-table

```

```

    composition-table (staged-alphas data) query-idxs f
  and trace-candidate:
    partial-trace-table-candidate trace-table bound-trace-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
    bound-composition-openings
  and all-queries:
    all-queries-consistent trace-table composition-table
    (staged-alphas data)
  and not-prefix:
    ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
  set-dist
  (execute (checked-staged-security-experiment-with-data-state A)
    adversary-initial-state)
by (rule
  checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
  [OF support])
have bound:
  accepted-with-bound-tables s (Some (result, final-state)) trace-table
  composition-table (staged-alphas data) query-idxs
  using bad-context unfolding composition-bad-context-def by blast
have shape:
  accepted-transcript-shape s (Some (result, final-state))
  (staged-alphas data) query-idxs
  using bound unfolding accepted-with-bound-tables-def
  accepted-with-tables-def by simp
have shape-s:
  accepted-transcript-shape ?s (Some (result, final-state))
  (staged-alphas data) query-idxs
  using shape s-def by simp
from checked-staged-security-with-data-state-accepted-shape-query-chunks
  [OF wf controlled data-support shape-s]
obtain raw-idxs where
  len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and lookup:
     $\bigwedge j. j < \text{rounds} \implies$ 
    fmllookup (HashMap final-state)
    (QueryIndexChallenge j
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) j)) =
    Some (raw-idxs ! j)
by blast

```

```

have raw-eq: raw = raw-idxs ! i
  using final-lookup lookup[OF i-bound] by simp
have idx-eq: index (to-nat raw) = query-idxs ! i
  using raw-eq query-idxs-eq len-raw i-bound by simp
have header-witness-s:
  (header-trace-openings, header-composition-openings) ∈
    query-header-supported-partial-opening-witnesses s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  using header-witness s'-def s-def by simp
have header-consistent-idx:
  partial-query-openings-consistent header-trace-openings
    header-composition-openings (staged-alphas data) (query-idxs ! i)
  using header-consistent idx-eq by simp
  show ?thesis
    by (rule that[OF s-def bad-context i-bound final-lookup idx-eq
      header-witness-s header-consistent-idx trace-candidate
      comp-candidate all-queries not-prefix])
qed
qed
qed

```

```

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-authenticated-witnessE
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    (Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state))
obtains s trace-table composition-table query-idxs f i raw
  header-trace-openings header-composition-openings
  bound-trace-openings bound-composition-openings
  witness-result witness-final-state witness-rest where
  s = verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
  composition-bad-context s (Some (result, final-state)) trace-table

```



```

    composition-table (staged-alphas data) query-idxs f
  i < rounds
  index (to-nat raw) = query-idxs ! i
  Some (witness-result, witness-final-state) ∈
    set-dist (execute verify-monad s)
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) header-trace-openings witness-final-state
  partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) header-composition-openings witness-final-state
  verifier-header-transcript s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  witness-rest
  partial-query-openings-consistent header-trace-openings
    header-composition-openings (staged-alphas data) (query-idxs ! i)
  partial-trace-table-candidate trace-table bound-trace-openings
  partial-composition-table-candidate composition-table
    bound-composition-openings
  all-queries-consistent trace-table composition-table
    (staged-alphas data)
  ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
proof —
show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-aligned-openingsE
    [OF wf controlled support hit])
fix s trace-table composition-table query-idxs f i raw
  header-trace-openings header-composition-openings
  bound-trace-openings bound-composition-openings
assume s-def:
  s = verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
and bad-context:
  composition-bad-context s (Some (result, final-state))
  trace-table composition-table (staged-alphas data) query-idxs f
and i-bound: i < rounds
and idx-eq: index (to-nat raw) = query-idxs ! i
and header-witness:
  (header-trace-openings, header-composition-openings) ∈
  query-header-supported-partial-opening-witnesses s
    (staged-trace-root data)
    (staged-trace-fri-roots data)

```

```

    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and header-consistent:
    partial-query-openings-consistent header-trace-openings
    header-composition-openings (staged-alphas data) (query-idxs ! i)
  and trace-candidate:
    partial-trace-table-candidate trace-table bound-trace-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
    bound-composition-openings
  and all-queries:
    all-queries-consistent trace-table composition-table
    (staged-alphas data)
  and not-prefix:
    ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
show ?thesis
proof (rule query-header-supported-partial-opening-witnessesE
  [OF header-witness])
fix witness-result witness-final-state witness-rest
assume witness-out:
  Some (witness-result, witness-final-state) ∈
  set-dist (execute verify-monad s)
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
  (scale * clength) header-trace-openings witness-final-state
and comp-auth:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
  (scale * clength) header-composition-openings
  witness-final-state
and header:
  verifier-header-transcript s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  witness-rest
show ?thesis
by (rule that[OF s-def bad-context i-bound idx-eq witness-out
  trace-auth comp-auth header header-consistent trace-candidate
  comp-candidate all-queries not-prefix])
qed

```

qed
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-current-openingsE*:

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *support*:

Some (((((*prefix*, *prefix-state*), *data*), *attacker-state*), *result*),
final-state) ∈
set-dist
(*execute*
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
adversary-initial-state)

and *hit*:

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
(*Some* (((((*prefix*, *prefix-state*), *data*), *attacker-state*), *result*),
final-state))

obtains *s trace-table composition-table query-idxs f i raw*

trace-openings composition-openings

trace-openingss composition-openingss **where**

s = verifier-state-from-adversary attacker-state

(*staged-proof-transcript data*)

composition-bad-context s (Some (result, final-state)) trace-table

composition-table (staged-alphas data) query-idxs f

i < rounds

fmlookup (HashMap final-state)

(*QueryIndexChallenge i*

(*state-after-query-chunks*

(*staged-query-start-hash data*)

(*staged-query-chunks data*) *i*)) =

Some raw

index (to-nat raw) = query-idxs ! i

staged-composition-fri-roots data ≠ []

partial-authenticated-table (staged-trace-root data)

(*scale * clength*) *trace-openings final-state*

partial-authenticated-table (hd (staged-composition-fri-roots data))

(*scale * clength*) *composition-openings final-state*

partial-query-openings-consistent trace-openings composition-openings

(*staged-alphas data*) (*query-idxs ! i*)

partial-trace-table-candidate trace-table trace-openingss

partial-composition-table-candidate composition-table

composition-openingss

trace-openings = trace-openingss ! i

composition-openings = composition-openingss ! i

all-queries-consistent trace-table composition-table

(*staged-alphas data*)

¬ *staged-alphas data* ∈

alpha-prefix-union-bad-sets prefix-state

```

      composition-trace-bad-alpha-space (fst prefix)
proof –
  show ?thesis
proof (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-aligned-openingsE
    [OF wf controlled support hit])
fix s trace-table composition-table query-idxs f i raw
    header-trace-openings header-composition-openings
    bound-trace-openings bound-composition-openings
assume s-def:
  s = verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
and bad-context:
  composition-bad-context s (Some (result, final-state))
    trace-table composition-table (staged-alphas data) query-idxs f
and i-bound: i < rounds
and final-lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some raw
and idx-eq: index (to-nat raw) = query-idxs ! i
and all-queries:
  all-queries-consistent trace-table composition-table
    (staged-alphas data)
and not-prefix:
  ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space (fst prefix)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
obtain builder verifier where
  builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)

```

```

      adversary-initial-state)
and verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
have staged-header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
  [OF wf controlled builder])
have bound:
  accepted-with-bound-tables s (Some (result, final-state))
  trace-table composition-table (staged-alphas data) query-idxs
  using bad-context unfolding composition-bad-context-def by blast
show ?thesis
proof (rule accepted-with-bound-tables-partial-opening-witnesses-Some
  [OF bound])
fix fr f-fri-roots f-final dg composition-fri-roots final rest
  trace-openingss composition-openingss
assume header:
  verifier-header-transcript s fr f-fri-roots f-final
  (staged-alphas data) dg composition-fri-roots final rest
and comp-nonempty: composition-fri-roots ≠ []
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr query-idxs trace-openingss
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots)
  query-idxs composition-openingss
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openingss
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openingss
have header-s:
  verifier-header-transcript ?s fr f-fri-roots f-final
  (staged-alphas data) dg composition-fri-roots final rest
  using header s-def by simp
have header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧

```

```

    dg = staged-degree data ∧
    composition-fri-roots = staged-composition-fri-roots data ∧
    final = staged-composition-final data ∧
    rest = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF staged-header header-s]
  by simp
have query-len: length query-idxs = rounds
  by (rule accepted-with-bound-tables-shapes(4)[OF bound])
have idx-mem: query-idxs ! i ∈ set query-idxs
  by (rule nth-mem) (use i-bound query-len in simp)
have idx-sample: query-idxs ! i ∈ query-sample-space
  by (rule accepted-with-bound-tables-query-sample-space
      [OF bound idx-mem])
have consistent-round:
  partial-query-round-consistent trace-openingss
  composition-openingss (staged-alphas data) i (query-idxs ! i)
  by (rule partial-query-round-consistent-from-all-queries-consistent
      [OF trace-partial comp-partial trace-candidate comp-candidate
        i-bound idx-sample all-queries])
have consistent:
  partial-query-openings-consistent (trace-openingss ! i)
  (composition-openingss ! i) (staged-alphas data) (query-idxs ! i)
  using consistent-round
  unfolding partial-query-round-consistent-def
  partial-query-openings-consistent-def
  by simp
have trace-auth:
  partial-authenticated-table (staged-trace-root data)
  (scale * clength) (trace-openingss ! i) final-state
  using trace-partial header-eq i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
have comp-auth:
  partial-authenticated-table
  (hd (staged-composition-fri-roots data)) (scale * clength)
  (composition-openingss ! i) final-state
  using comp-partial header-eq i-bound
  unfolding accepted-with-partial-composition-openings-def by simp
have comp-nonempty-staged:
  staged-composition-fri-roots data ≠ []
  using comp-nonempty header-eq by simp
show ?thesis
  by (rule that[OF s-def bad-context i-bound final-lookup idx-eq
    comp-nonempty-staged trace-auth comp-auth consistent
    trace-candidate comp-candidate refl refl all-queries not-prefix])
qed
qed
qed

```

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-authenticated-candidate

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    (Some (((((prefix, prefix-state), data), attacker-state), result),
           final-state))
shows
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
    A
    (Some (((((prefix, prefix-state), data), attacker-state), result),
           final-state))
proof —
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
by (rule
  checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
obtain builder verifier where
  builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
and verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-current-openingsE
    [OF wf controlled support hit])
fix s trace-table composition-table query-idxs f i raw trace-openings
  composition-openings trace-openingsss composition-openingsss
assume i-bound: i < rounds

```

```

and final-lookup:
  fmllookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data i)) =
      Some raw
and idx-eq: index (to-nat raw) = query-idxs ! i
and comp-nonempty: staged-composition-fri-roots data ≠ []
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and comp-auth:
  partial-authenticated-table
    (hd (staged-composition-fri-roots data)) (scale * clength)
    composition-openings final-state
and consistent:
  partial-query-openings-consistent trace-openings
    composition-openings (staged-alphas data) (query-idxs ! i)
have consistent-raw:
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
  using consistent idx-eq by simp
show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hitI-from-current-openings
    [OF wf controlled comp-nonempty i-bound builder verifier
      final-lookup trace-auth comp-auth consistent-raw])
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-authenticated-candidate
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
      A)
    adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
    set-dist

```



```

      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
    and hit:
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
      out
  show
    checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
    A out
  proof (cases out)
  case None
  then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state)
  by (cases packed, auto split: prod.splits)
show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-authenticated-candid
    [OF wf controlled])
  (use support hit out-eq in simp-all)
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-authenticat
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and authenticated-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      (checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
        A)
      adversary-initial-state  $\leq$  (Q::prob)
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

```

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq Q$
by (*rule order.trans*
 $[OF$
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-authenticated-candidate-
authenticated-bound])
(use wf controlled in simp-all)

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-authenticated-candidate-*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *support:*
Some (((((prefix, prefix-state), data), attacker-state), result),
final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
adversary-initial-state)
and *hit:*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
(Some (((((prefix, prefix-state), data), attacker-state), result),
final-state))
shows
checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
A
(Some (((((prefix, prefix-state), data), attacker-state), result),
final-state))
proof –
let *?s =*
verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
have *data-support:*
Some (((data, attacker-state), result), final-state) ∈
set-dist
(execute (checked-staged-security-experiment-with-data-state A)
adversary-initial-state)
by (*rule*
checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
 $[OF support]$)
from *checked-staged-security-experiment-with-data-state-outcomeE*
 $[OF data-support]$
obtain *builder verifier where*
builder:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *verifier:*

```

    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-current-openingsE
  [OF wf controlled support hit])
fix s trace-table composition-table query-idxs f i raw trace-openings
  composition-openings trace-openingss composition-openingss
assume i-bound: i < rounds
and final-lookup:
  fmllookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some raw
and idx-eq: index (to-nat raw) = query-idxs ! i
and comp-nonempty: staged-composition-fri-roots data ≠ []
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and comp-auth:
  partial-authenticated-table
    (hd (staged-composition-fri-roots data)) (scale * clength)
    composition-openings final-state
and consistent:
  partial-query-openings-consistent trace-openings
    composition-openings (staged-alphas data) (query-idxs ! i)
have consistent-raw:
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
using consistent idx-eq by simp
show ?thesis
by (rule
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefixI-from-cu
  [OF wf controlled comp-nonempty i-bound builder verifier
    final-lookup trace-auth comp-auth consistent-raw])
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-authenticated-candid
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    adversary-initial-state ≤
    wp-event

```

```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
      A)
    adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
  out
show
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
  A out
proof (cases out)
case None
then show ?thesis
using hit
unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
  checked-staged-security-with-actual-alpha-prefix-verifier-event-def
by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state
where out-eq:
  out =
    Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state)
by (cases packed, auto split: prod.splits)
show ?thesis
unfolding out-eq
by (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-authenticated-candid
  [OF wf controlled])
  (use support hit out-eq in simp-all)
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-authenticat
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and authenticated-bound:

```

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
 (*checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix*
A)
adversary-initial-state $\leq$ (*Q::prob*)
shows
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq$ *Q*
by (*rule order.trans*
 [*OF*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-authenticated-candidate-
authenticated-bound])
 (use *wf controlled* **in** *simp-all*)

lemma *checked-staged-security-with-actual-alpha-prefix-aligned-transcript-query-bad-imp-candidate-opening-hit-from-*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *support:*
Some ((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*) $\in$
set-dist
 (execute
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
adversary-initial-state)
and *bad:*
checked-staged-security-with-actual-alpha-prefix-verifier-event
query-bad-with-aligned-transcript-partial-candidates
(Some ((((*alpha-prefix*, *alpha-prefix-state*), *data*),
attacker-state), *result*), *final-state*))
obtains *i trace-openings composition-openings* **where**
i < *rounds*
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A trace-openings composition-openings i
(Some ((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*))
proof –
let *?s* =
verifier-state-from-adversary attacker-state
 (*staged-proof-transcript data*)
have *data-support:*
Some (((*data*, *attacker-state*), *result*), *final-state*) $\in$
set-dist
 (execute (*checked-staged-security-experiment-with-data-state A*)
adversary-initial-state)
by (*rule*

```

    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
obtain builder verifier where
  builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
from bad obtain fr f-fri-roots f-final as dg composition-fri-roots
  final query-idxs trace-openings composition-openings trace-table
  composition-table where partial:
    accepted-with-partial-initial-openings-aligned-transcript-consistent
      ?s (Some (result, final-state)) fr f-fri-roots f-final as dg
      composition-fri-roots final query-idxs trace-openings
      composition-openings
  unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    query-bad-with-aligned-transcript-partial-candidates-def
  by auto
have shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
    query-idxs
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape
    [OF partial])
from checked-staged-security-with-data-state-accepted-shape-query-chunks
  [OF wf controlled data-support shape]
obtain raw-idxs where
  as-eq: as = staged-alphas data
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and lookup:
    ∀ i < rounds.
      fmlookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
  by blast
have zero-bound: 0 < rounds
  by (rule rounds-positive)
let ?raw = raw-idxs ! 0
have lookup0:

```

```

fmlookup (HashMap final-state)
  (QueryIndexChallenge 0
    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) 0)) =
  Some ?raw
using lookup zero-bound by simp
from checked-staged-transcript-program-query-prefix-receive-final-lookup-support-with-alphas
  [OF wf controlled zero-bound builder verifier lookup0]
obtain prefix prefix-state raw-state where prefix-receive:
  Some (((prefix, prefix-state), ?raw), raw-state) ∈
  set-dist
    (execute (checked-staged-query-prefix-receive-with-state A 0)
      adversary-initial-state)
and prefix-alphas: sqp-alphas prefix = staged-alphas data
by blast
have idx-eq: query-idxs ! 0 = index (to-nat ?raw)
  using query-idxs-eq len-raw zero-bound by simp
have alphas-eq: as = sqp-alphas prefix
  using as-eq prefix-alphas by simp
have hit:
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
  A trace-openings composition-openings 0
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
by (rule
  accepted-with-partial-initial-openings-aligned-transcript-consistent-candidate-opening-hit-from-prefix
  [OF partial zero-bound idx-eq alphas-eq builder prefix-receive])
show ?thesis
  by (rule that[OF zero-bound hit])
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-authenticated-and-wf
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and authenticated-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit
      A)
    adversary-initial-state ≤ (Q::prob)
and without-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
    adversary-initial-state ≤ (W::prob)
shows

```

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq Q + W$

proof (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-query-and-without-query)

show

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq Q$

by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-authenticated
 [OF wf controlled authenticated-bound])

show

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state $\leq W$

by (rule *without-bound*)

qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-authenticated-from-p*

assumes wf: *staged-budget-wellformed budgets*

and controlled: *staged-adversary-controlled budgets A*

and authenticated-bound:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix A*)
adversary-initial-state $\leq (Q::prob)$

and without-bound:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state $\leq (W::prob)$

shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq Q + W$

proof (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-query-and-without-query)

show

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit


```

    adversary-initial-state ≤ Q
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-authenticated-
    [OF wf controlled authenticated-bound])
  show
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
      adversary-initial-state ≤ W
  by (rule without-bound)
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-candidate-opening-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    (Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state))
shows
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit
    A
    (Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state))
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-supported-openingsE
    [OF wf controlled support hit])
  fix s trace-table i raw trace-openings composition-openings
  assume i-bound: i < rounds
  and final-lookup:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
    Some raw

```

and *consistent*:
partial-query-openings-consistent trace-openings composition-openings
(staged-alphas data) (index (to-nat raw))
have *data-support*:
Some (((data, attacker-state), result), final-state) ∈
set-dist
(execute (checked-staged-security-experiment-with-data-state A)
adversary-initial-state)
by *(rule*
checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
[OF support])
from *checked-staged-security-experiment-with-data-state-outcomeE*
[OF data-support]
obtain *builder verifier where*
builder:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *verifier:*
Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by *blast*
from *checked-staged-transcript-program-query-prefix-receive-final-lookup-support*
[OF wf controlled i-bound builder verifier final-lookup]
obtain *query-prefix query-prefix-state raw-state where prefix-receive:*
Some (((query-prefix, query-prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
by *blast*
have *component:*
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit
A
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
(staged-alphas data) i
(Some (((((prefix, prefix-state), data), attacker-state), result),
final-state))
by *(rule*
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hitI-from-prefix-receive
[OF i-bound builder prefix-receive consistent])
show *?thesis*
by *(rule*
checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hitI
[OF component])
qed
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-candidate-opening-q*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    adversary-initial-state  $\leq$ 
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit
        A)
      adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out  $\in$ 
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
  out
show
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-opening-query-hit
  A out
proof (cases out)
case None
then show ?thesis
using hit
unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
  checked-staged-security-with-actual-alpha-prefix-verifier-event-def
by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state
where out-eq:
  out =
    Some ((((((prefix, prefix-state), data), attacker-state), result),
      final-state)
by (cases packed, auto split: prod.splits)
show ?thesis
unfolding out-eq
by (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-candidate-opening-qu
  [OF wf controlled])

```

(use support hit out-eq in simp-all)

qed

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-authenticated-query-and-budgets:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *authenticated-bound:*

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state

A)

(checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit

A)

adversary-initial-state ≤ Q

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event composition-bad)

adversary-initial-state ≤

composition-error-bound +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +

hash-collision-budget-value 0

(staged-alpha-search-queries budgets 0) +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +

Q +

hash-collision-budget-value 0

(staged-attacker-query-budget budgets +

staged-challenge-query-budget + verifier-hash-query-budget)

proof –

have *without-bound:*

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state

A)

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit

adversary-initial-state ≤ 0

by (rule

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-bound

[OF wf controlled])

have *drift-bound:*

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift

adversary-initial-state ≤ Q + 0

by (rule

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-authenticated-and-without-bound

[OF wf controlled authenticated-bound without-bound])

have *comp-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event composition-bad)
 adversary-initial-state $\leq$
 composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (Q + 0) +
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
 checked-staged-security-with-data-state-composition-bad-bound-from-composition-candidate-drift-and-budgets
 [OF wf controlled drift-bound])
then show ?thesis
by simp
qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-header-authenticated-query-from-prefix-a
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and header-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
 A)
 adversary-initial-state $\leq$ Q
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event composition-bad)
 adversary-initial-state $\leq$
 composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 Q +
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)
proof –
have without-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state

A)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state ≤ 0
by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-bound
[OF wf controlled])
have *drift-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq Q + 0$
proof (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-query-and-without-query)
show
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq Q$
by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-header-authen
[OF wf controlled header-bound])
show
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state ≤ 0
by (rule *without-bound*)
qed
have *comp-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event composition-bad*)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets* 0) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets* 0) +
 (*Q* + 0) +
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)
by (rule
checked-staged-security-with-data-state-composition-bad-bound-from-composition-candidate-drift-and-budgets
[OF wf controlled drift-bound])
then show ?thesis

by simp
qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-authenticated-query-from-prefix-and-budg*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *authenticated-bound:*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

(*checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix*
A)

adversary-initial-state $\leq Q$

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event composition-bad*)

adversary-initial-state $\leq$

composition-error-bound +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

hash-collision-budget-value 0

(*staged-alpha-search-queries budgets 0*) +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

Q +

hash-collision-budget-value 0

(*staged-attacker-query-budget budgets* +

staged-challenge-query-budget + *verifier-hash-query-budget*)

proof –

have *without-bound:*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit

adversary-initial-state ≤ 0

by (rule

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-bound
[OF *wf controlled*])

have *drift-bound:*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift

adversary-initial-state $\leq Q + 0$

by (rule

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-authenticated-from-prefix
[OF *wf controlled authenticated-bound without-bound*])

have *comp-bound:*

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event composition-bad*)

```

    adversary-initial-state ≤
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    (Q + 0) +
    hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget)
  by (rule
    checked-staged-security-with-data-state-composition-bad-bound-from-composition-candidate-drift-and-budgets
    [OF wf controlled drift-bound])
  then show ?thesis
  by simp
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Query-Augmented
imports Staged-Security-Experiment-Composition-Query-Bridge
begin

```

Augmented query-prefix events for the composition query side.

These events live in the query-prefix augmented experiment, so the sampled query challenge is part of the event output. This is the correct setting for applying fixed-witness query-prefix bounds. The fixed-witness event below deliberately does not existentially choose openings after the query challenge; the remaining public-proof work is to derive such fixed witnesses from pre-query Merkle/candidate evidence or charge failures to existing side events.

```

context soundness
begin

```

lemma *checked-staged-security-with-data-state-query-prefix-continuationE:*

assumes *i-bound: i < rounds*

and *support:*

Some (((data, attacker-state), result), final-state) ∈
set-dist

(execute (checked-staged-security-experiment-with-data-state A)
adversary-initial-state)

obtains *prefix prefix-state raw raw-state* **where**

Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist


```

    (execute
      (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
  Some (((data, attacker-state), result), final-state) ∈
  set-dist
    (execute
      (checked-staged-after-query-prefix-receive-with-verifier A i
        ((prefix, prefix-state), raw))
      raw-state)
proof –
  have decomp:
    checked-staged-security-experiment-with-data-state A =
    checked-staged-query-prefix-receive-with-state A i  $\gg$ 
    checked-staged-after-query-prefix-receive-with-verifier A i
  by (rule
    checked-staged-security-experiment-with-data-state-query-prefix-receive-decomp
    [OF i-bound])
  from support[unfolded decomp]
  obtain x raw-state where prefix-receive:
    Some (x, raw-state) ∈
    set-dist
      (execute
        (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  and continuation:
    Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive-with-verifier A i x)
        raw-state)
  by (auto elim!: set-dist-bindE)
  obtain prefix-pack raw where x-eq: x = (prefix-pack, raw)
  by (cases x) simp
  obtain prefix prefix-state where prefix-pack-eq:
    prefix-pack = (prefix, prefix-state)
  by (cases prefix-pack) simp
  show ?thesis
  by (rule that[of prefix prefix-state raw raw-state])
    (use prefix-receive continuation x-eq prefix-pack-eq in simp-all)
qed

definition checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
where
  checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
  trace-openings composition-openings i out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)  $\Rightarrow$ 

```

staged-composition-fri-roots data $\neq \square$ $\wedge$
(trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data) $\wedge$
checked-staged-security-with-actual-query-prefix-candidate-opening-hit
((replicate rounds $\square$) [i := trace-openings])
((replicate rounds $\square$) [i := composition-openings])
i out

lemma *checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-imp-actual:*

assumes *hit:*

checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
trace-openings composition-openings i out

shows

checked-staged-security-with-actual-query-prefix-candidate-opening-hit
((replicate rounds $\square$) [i := trace-openings])
((replicate rounds $\square$) [i := composition-openings])
i out

using *hit*

unfolding

checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-def

by (*cases out*) (*auto split: prod.splits*)

lemma *checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitE:*

assumes *hit:*

checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
trace-openings composition-openings i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

obtains

staged-composition-fri-roots data $\neq \square$
(trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)

```

    (staged-composition-final data)
  index (to-nat raw) ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
    i prefix prefix-state
proof —
  have comp-nonempty:
    staged-composition-fri-roots data ≠ []
  and witness:
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)
  and actual:
    checked-staged-security-with-actual-query-prefix-candidate-opening-hit
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
    i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  using hit
  unfolding
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-def
  by simp-all
from checked-staged-security-with-actual-query-prefix-candidate-opening-hitE
  [OF actual]
have target:
  index (to-nat raw) ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
    i prefix prefix-state
  .
show ?thesis
  by (rule that[OF comp-nonempty witness target])
qed

lemma checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-prefix-facts:
assumes support:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈

```

```

    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
    trace-openings composition-openings i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  shows
    staged-alphas data = sqp-alphas prefix
    staged-trace-root data = sqp-trace-root prefix
    staged-composition-fri-roots data = sqp-composition-fri-roots prefix
    index (to-nat raw) ∈
      staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i prefix prefix-state
  proof —
    show alphas: staged-alphas data = sqp-alphas prefix
    by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq)
      (rule support)
    from hit show target:
    index (to-nat raw) ∈
      staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i prefix prefix-state
    by (rule
      checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitE)
    from support show staged-trace-root data = sqp-trace-root prefix
    unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
      checked-staged-after-query-prefix-receive-with-verifier-def
      checked-staged-after-query-prefix-receive-def
    by (auto elim!: set-dist-bindE split: prod.splits)
    from support show
      staged-composition-fri-roots data = sqp-composition-fri-roots prefix
    unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
      checked-staged-after-query-prefix-receive-with-verifier-def
      checked-staged-after-query-prefix-receive-def
    by (auto elim!: set-dist-bindE split: prod.splits)
  qed

```

The next wrapper lets downstream composition proofs conjoin their own side-condition with the fixed authenticated-opening query-prefix hit, without changing the probability bound. This is deliberately fixed-witness: the trace and composition openings are chosen before the query challenge is sampled.

definition *checked-staged-security-with-query-prefix-fixed-header-component*
where

checked-staged-security-with-query-prefix-fixed-header-component

$E \text{ trace-openings composition-openings } i \text{ out } \longleftrightarrow$

$E \text{ out } \wedge$

checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
trace-openings composition-openings i out

definition *checked-staged-security-with-query-prefix-fixed-current-authenticated-component*
where

checked-staged-security-with-query-prefix-fixed-current-authenticated-component

$E \text{ trace-openings composition-openings } i \text{ out } \longleftrightarrow$

checked-staged-security-with-query-prefix-fixed-header-component

$E \text{ trace-openings composition-openings } i \text{ out } \wedge$

checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i out

definition *checked-staged-security-with-query-prefix-header-witness-not-current*
where

checked-staged-security-with-query-prefix-header-witness-not-current

$E \text{ trace-openings composition-openings } i \text{ out } \longleftrightarrow$

checked-staged-security-with-query-prefix-fixed-header-component

$E \text{ trace-openings composition-openings } i \text{ out } \wedge$

$\neg$ *checked-staged-security-with-query-prefix-authenticated-opening-hit*
trace-openings composition-openings i out

definition *checked-staged-security-with-query-prefix-header-witness-path-output-hit*
where

checked-staged-security-with-query-prefix-header-witness-path-output-hit

$E \text{ trace-openings composition-openings } i \text{ out } \longleftrightarrow$

(case out of

$\text{None} \Rightarrow \text{False}$

| $\text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state}) \Rightarrow$
 $E \text{ out } \wedge$

$\text{staged-composition-fri-roots data} \neq [] \wedge$

$(\exists \text{ witness-result witness-state rest.}$

$\text{let } s = \text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data}) \text{ in}$

$\text{Some } (\text{witness-result}, \text{witness-state}) \in$

$\text{set-dist } (\text{execute verify-monad } s) \wedge$

$\text{verifier-header-transcript } s$

$(\text{staged-trace-root data})$

$(\text{staged-trace-fri-roots data})$

$(\text{staged-trace-final data})$

$(\text{staged-alphas data})$

$(\text{staged-degree data})$

$(\text{staged-composition-fri-roots data})$

$(\text{staged-composition-final data})$

$\text{rest} \wedge$

$(\exists \text{ opn} \in \text{set trace-openings.}$

hash-map-new-output-hit
 (merkle-path-target-roots witness-state
 (staged-trace-root data)
 (scale * clength)
 (opening-index opn)
 (opening-value opn)
 (opening-path opn))
 s witness-state) $\vee$
 ($\exists$ opn $\in$ set composition-openings.
hash-map-new-output-hit
 (merkle-path-target-roots witness-state
 (hd (staged-composition-fri-roots data))
 (scale * clength)
 (opening-index opn)
 (opening-value opn)
 (opening-path opn))
 s witness-state))))

definition *checked-staged-security-with-query-prefix-header-witness-transcript-hit*
where

checked-staged-security-with-query-prefix-header-witness-transcript-hit
 E out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 E out $\wedge$
 ($\exists$ witness-result witness-state.
 let s = verifier-state-from-adversary attacker-state
 (staged-proof-transcript data) in
 Some (witness-result, witness-state) $\in$
 set-dist (execute verify-monad s) $\wedge$
 (hash-map-output-values s $\cap$
 set (staged-proof-transcript data) $\neq$ {} $\vee$
 hash-map-new-output-hit (set (staged-proof-transcript data))
 s witness-state)))

definition *checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit*
where

checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
 E out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 E out $\wedge$
 hash-map-output-values
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\cap$

$\text{set } (\text{staged-proof-transcript data}) \neq \{\}$

definition *checked-staged-security-with-query-prefix-header-witness-transcript-new-hit*
where

checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
 $E \text{ out} \longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 $E \text{ out} \wedge$
 ($\exists$ witness-result witness-state.
 let $s = \text{verifier-state-from-adversary attacker-state}$
 (staged-proof-transcript data) in
 Some (witness-result, witness-state) $\in$
 set-dist (execute verify-monad s) $\wedge$
 hash-map-new-output-hit (set (staged-proof-transcript data))
 s witness-state))

lemma *checked-staged-security-with-query-prefix-header-witness-transcript-hit-imp-pre-or-new:*
assumes *hit:*

checked-staged-security-with-query-prefix-header-witness-transcript-hit
 $E \text{ out}$

shows

checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
 $E \text{ out} \vee$
checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
 $E \text{ out}$

using *hit*

unfolding

checked-staged-security-with-query-prefix-header-witness-transcript-hit-def
checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit-def
checked-staged-security-with-query-prefix-header-witness-transcript-new-hit-def

by (cases out) (fastforce split: prod.splits)+

lemma *checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit-imp-data-state-pre-hit:*
assumes *hit:*

checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
 E
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))

shows

staged-security-with-data-state-transcript-pre-hit
 (Some (((data, attacker-state), result), final-state))

using *hit*

unfolding

checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit-def
staged-security-with-data-state-transcript-pre-hit-def

by *simp*

lemma *checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit-bound-from-data-state-pre-hit:*

assumes *i-bound: $i < \text{rounds}$*

and *pre-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-transcript-pre-hit

adversary-initial-state $\leq P$

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit

E)

adversary-initial-state $\leq P$

proof –

have *projection:*

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-transcript-pre-hit

adversary-initial-state =

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

($\lambda \text{out. case out of}$

None $\Rightarrow$ staged-security-with-data-state-transcript-pre-hit None

| Some (packed, t) $\Rightarrow$

staged-security-with-data-state-transcript-pre-hit

(Some (snd packed, t)))

adversary-initial-state

by *(rule*

checked-staged-security-experiment-with-query-prefix-data-state-projection-event

[OF i-bound])

have *event-le:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit

E)

adversary-initial-state $\leq$

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

($\lambda \text{out. case out of}$

None $\Rightarrow$ staged-security-with-data-state-transcript-pre-hit None

| Some (packed, t) $\Rightarrow$

staged-security-with-data-state-transcript-pre-hit

(Some (snd packed, t)))

adversary-initial-state

proof *(rule wp-event-mono)*

fix *out*

assume *hit:*

checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit

E out

show


```

    (case out of
      None  $\Rightarrow$  staged-security-with-data-state-transcript-pre-hit None
    | Some (packed, t)  $\Rightarrow$ 
      staged-security-with-data-state-transcript-pre-hit
        (Some (snd packed, t)))
  proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit-def
    by simp
  next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  show ?thesis
    using
      checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit-imp-data-state-pre-hit
      [OF hit[unfolded out-eq]]
    unfolding out-eq by simp
  qed
qed
show ?thesis
  by (rule order-trans[OF event-le])
  (use pre-bound projection in simp)
qed

lemma checked-staged-security-with-query-prefix-header-witness-path-output-hitE:
  assumes hit:
    checked-staged-security-with-query-prefix-header-witness-path-output-hit
    E trace-openings composition-openings i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  and that:
     $\bigwedge$  witness-result witness-state rest.
    E (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))  $\Rightarrow$ 
    staged-composition-fri-roots data  $\neq$  []  $\Rightarrow$ 
    Some (witness-result, witness-state)  $\in$ 
      set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))  $\Rightarrow$ 
    verifier-header-transcript

```

```

    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    rest  $\Rightarrow$ 
  (( $\exists$  opn  $\in$  set trace-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots witness-state
        (staged-trace-root data)
        (scale * clength)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      witness-state)  $\vee$ 
    ( $\exists$  opn  $\in$  set composition-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state
          (hd (staged-composition-fri-roots data))
          (scale * clength)
          (opening-index opn)
          (opening-value opn)
          (opening-path opn))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        witness-state))  $\Rightarrow$ 
    thesis
  shows thesis
proof -
have E:
  E (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
and nonempty:
  staged-composition-fri-roots data  $\neq$  []
using hit
unfolding checked-staged-security-with-query-prefix-header-witness-path-output-hit-def
by simp-all
from hit obtain witness-result witness-state rest where
witness:
  Some (witness-result, witness-state)  $\in$ 
  set-dist
    (execute verify-monad
      (verifier-state-from-adversary attacker-state

```

```

      (staged-proof-transcript data)))
and header:
  verifier-header-transcript
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  rest
and path-hit:
  (( $\exists$  opn  $\in$  set trace-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots witness-state
        (staged-trace-root data)
        (scale * clength)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      witness-state)  $\vee$ 
    ( $\exists$  opn  $\in$  set composition-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state
          (hd (staged-composition-fri-roots data))
          (scale * clength)
          (opening-index opn)
          (opening-value opn)
          (opening-path opn))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        witness-state))
  unfolding checked-staged-security-with-query-prefix-header-witness-path-output-hit-def
  by (auto simp: Let-def; blast)
show ?thesis
  by (rule that[OF E nonempty witness header path-hit])
qed

lemma checked-staged-security-with-query-prefix-header-witness-path-output-imp-transcript-hit:
assumes hit:
  checked-staged-security-with-query-prefix-header-witness-path-output-hit
  E trace-openings composition-openings i out
shows
  checked-staged-security-with-query-prefix-header-witness-transcript-hit
  E out

```

```

proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding checked-staged-security-with-query-prefix-header-witness-path-output-hit-def
      checked-staged-security-with-query-prefix-header-witness-transcript-hit-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from hit[unfolded out-eq]
  obtain witness-result witness-state rest where
    E:
      E (Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state))
    and witness:
      Some (witness-result, witness-state) ∈
      set-dist (execute verify-monad ?s)
    by (rule
      checked-staged-security-with-query-prefix-header-witness-path-output-hitE)
  have tr-eq: PTranscript ?s = staged-proof-transcript data
    by simp
  have transcript-hit:
    hash-map-output-values ?s ∩ set (staged-proof-transcript data)
    ≠ {} ∨
    hash-map-new-output-hit (set (staged-proof-transcript data))
    ?s witness-state
    by (rule verify-monad-trace-root-target-preexisting-or-new
      [OF witness tr-eq])
  show ?thesis
    unfolding out-eq
      checked-staged-security-with-query-prefix-header-witness-transcript-hit-def
    using E witness transcript-hit by (auto simp: Let-def)
qed

lemma checked-staged-security-with-query-prefix-fixed-header-component-imp-header:
assumes hit:
  checked-staged-security-with-query-prefix-fixed-header-component
  E trace-openings composition-openings i out
shows
  checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit

```

trace-openings composition-openings i out
using *hit*
unfolding *checked-staged-security-with-query-prefix-fixed-header-component-def*
by *simp*

lemma *checked-staged-security-with-query-prefix-fixed-current-authenticated-component-imp-current:*
assumes *hit:*
checked-staged-security-with-query-prefix-fixed-current-authenticated-component
E trace-openings composition-openings i out
shows
checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i out
using *hit*
unfolding
checked-staged-security-with-query-prefix-fixed-current-authenticated-component-def
by *simp*

lemma *checked-staged-security-with-query-prefix-fixed-header-component-split-current:*
assumes *hit:*
checked-staged-security-with-query-prefix-fixed-header-component
E trace-openings composition-openings i out
shows
checked-staged-security-with-query-prefix-fixed-current-authenticated-component
E trace-openings composition-openings i out $\vee$
checked-staged-security-with-query-prefix-header-witness-not-current
E trace-openings composition-openings i out
using *hit*
unfolding
checked-staged-security-with-query-prefix-fixed-current-authenticated-component-def
checked-staged-security-with-query-prefix-header-witness-not-current-def
by *blast*

lemma *checked-staged-security-with-query-prefix-fixed-header-componentE:*
assumes *hit:*
checked-staged-security-with-query-prefix-fixed-header-component
E trace-openings composition-openings i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))
obtains
E (Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))
staged-composition-fri-roots data $\neq$ []
(trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)

```

    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  index (to-nat raw) ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i prefix prefix-state
proof —
  have E:
    E (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  and header:
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
      trace-openings composition-openings i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  using hit
  unfolding checked-staged-security-with-query-prefix-fixed-header-component-def
  by simp-all
from header show ?thesis
proof (rule
  checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitE)
assume comp-nonempty:
  staged-composition-fri-roots data ≠ []
and witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
and target:
  index (to-nat raw) ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i prefix prefix-state
  show ?thesis
  by (rule that[OF E comp-nonempty witness target])
qed
qed

```

lemma *checked-staged-security-with-query-prefix-fixed-header-component-supportE*:

assumes *support*:

Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) ∈
set-dist
 (execute
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 adversary-initial-state)

and *hit*:

checked-staged-security-with-query-prefix-fixed-header-component
E trace-openings composition-openings i
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))

obtains

E (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))
 Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
 (execute
 (checked-staged-query-prefix-receive-with-state A i)
 adversary-initial-state)
 Some (((data, attacker-state), result), final-state) ∈
set-dist
 (execute
 (checked-staged-after-query-prefix-receive-with-verifier A i
 ((prefix, prefix-state), raw))
 raw-state)
staged-alphas data = *sqp-alphas prefix*
staged-trace-root data = *sqp-trace-root prefix*
staged-composition-fri-roots data = *sqp-composition-fri-roots prefix*
staged-composition-fri-roots data ≠ []
 (trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
index (to-nat raw) ∈
staged-query-prefix-candidate-opening-query-target-from-prefix
 ((replicate rounds []) [i := trace-openings])
 ((replicate rounds []) [i := composition-openings])
 i prefix prefix-state

proof —

have *prefix-receive*:

```

Some (((prefix, prefix-state), raw), raw-state) ∈
  set-dist
  (execute
    (checked-staged-query-prefix-receive-with-state A i)
    adversary-initial-state)
and continuation:
Some (((data, attacker-state), result), final-state) ∈
  set-dist
  (execute
    (checked-staged-after-query-prefix-receive-with-verifier A i
      ((prefix, prefix-state), raw))
    raw-state)
using support
unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
by (auto elim!: set-dist-bindE)
have E:
  E (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
and header:
checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
  trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
using hit
unfolding checked-staged-security-with-query-prefix-fixed-header-component-def
by simp-all
from header show ?thesis
proof (rule
  checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitE)
assume comp-nonempty:
  staged-composition-fri-roots data ≠ []
and witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
and target:
  index (to-nat raw) ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i prefix prefix-state

```



```

have alphas:
  staged-alphas data = sqp-alphas prefix
by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq)
  (rule support)
have trace-root:
  staged-trace-root data = sqp-trace-root prefix
and comp-roots:
  staged-composition-fri-roots data = sqp-composition-fri-roots prefix
using support
unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
  checked-staged-after-query-prefix-receive-with-verifier-def
  checked-staged-after-query-prefix-receive-def
by (auto elim!: set-dist-bindE split: prod.splits)
show ?thesis
by (rule that[OF E prefix-receive continuation alphas trace-root
  comp-roots comp-nonempty witness target])
qed
qed

lemma checked-staged-security-with-query-prefix-header-witness-not-current-imp-path-output-on-support:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and support:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-header-witness-not-current
  E trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
shows
  checked-staged-security-with-query-prefix-header-witness-path-output-hit
  E trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
proof –
let ?out =
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have fixed:
  checked-staged-security-with-query-prefix-fixed-header-component

```

E trace-openings composition-openings *i* ?out
and not-current:
 $\neg$ checked-staged-security-with-query-prefix-authenticated-opening-hit
 trace-openings composition-openings *i* ?out
using hit
unfolding
 checked-staged-security-with-query-prefix-header-witness-not-current-def
by simp-all
have support-facts:
E ?out $\wedge$
 Some (((data, attacker-state), result), final-state) $\in$
 set-dist
 (execute
 (checked-staged-after-query-prefix-receive-with-verifier *A* *i*
 ((prefix, prefix-state), raw))
 raw-state) $\wedge$
 staged-alphas data = sqp-alphas prefix $\wedge$
 staged-composition-fri-roots data $\neq \square$ $\wedge$
 (trace-openings, composition-openings) $\in$
 query-header-supported-partial-opening-witnesses ?s
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) $\wedge$
 index (to-nat raw) $\in$
 staged-query-prefix-candidate-opening-query-target-from-prefix
 ((replicate rounds $\square$) [*i* := trace-openings])
 ((replicate rounds $\square$) [*i* := composition-openings])
i prefix prefix-state
proof (rule
 checked-staged-security-with-query-prefix-fixed-header-component-supportE
 [OF support fixed])
assume *E*: *E* ?out
and continuation:
 Some (((data, attacker-state), result), final-state) $\in$
 set-dist
 (execute
 (checked-staged-after-query-prefix-receive-with-verifier *A* *i*
 ((prefix, prefix-state), raw))
 raw-state)
and alphas-eq: staged-alphas data = sqp-alphas prefix
and comp-nonempty: staged-composition-fri-roots data $\neq \square$
and witness:
 (trace-openings, composition-openings) $\in$
 query-header-supported-partial-opening-witnesses ?s
 (staged-trace-root data)

```

    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and target:
  index (to-nat raw) ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
    i prefix prefix-state
  show ?thesis
    using E continuation alphas-eq comp-nonempty witness target by simp
qed
have E: E ?out
  using support-facts by blast
have continuation:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive-with-verifier A i
          ((prefix, prefix-state), raw))
        raw-state)
  using support-facts by blast
have alphas-eq: staged-alphas data = sqp-alphas prefix
  using support-facts by blast
have comp-nonempty: staged-composition-fri-roots data ≠ []
  using support-facts by blast
have witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  using support-facts by blast
have target:
  index (to-nat raw) ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
    i prefix prefix-state
  using support-facts by blast
from checked-staged-after-query-prefix-receive-with-verifier-outcome(2)
  [OF continuation]

```

```

have current-verifier:
  Some (result, final-state) ∈
    set-dist (execute verify-monad ?s)
  .
have s-ext-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF current-verifier])
from witness obtain witness-result witness-state rest where
  witness-out:
    Some (witness-result, witness-state) ∈
      set-dist (execute verify-monad ?s)
  and trace-table-witness:
    partial-authenticated-table (staged-trace-root data)
      (scale * clength) trace-openings witness-state
  and comp-table-witness:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings witness-state
  and header:
    verifier-header-transcript ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
      rest
  by (rule query-header-supported-partial-opening-witnessesE)
have s-ext-witness: ?s ≤ witness-state
  by (rule verify-monad-hash-extends[OF witness-out])
have consistent:
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
proof –
  have target-success:
    index (to-nat raw) ∈
      partial-query-success-indices-at
        ((replicate rounds []) [i := trace-openings])
        ((replicate rounds []) [i := composition-openings])
        (sqp-alphas prefix) i
  using target
  unfolding staged-query-prefix-candidate-opening-query-target-from-prefix-def
  by simp
then have round:
  partial-query-round-consistent
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    (sqp-alphas prefix) i (index (to-nat raw))
  unfolding partial-query-success-indices-at-def by simp
have trace-nth:

```

```

  ((replicate rounds []) [i := trace-openings]) ! i = trace-openings
  using i-bound by simp
have comp-nth:
  ((replicate rounds []) [i := composition-openings]) ! i =
    composition-openings
  using i-bound by simp
have idx-sample: index (to-nat raw) ∈ query-sample-space
  using round unfolding partial-query-round-consistent-def by simp
have trace-indices-full:
  map opening-index
    (((replicate rounds []) [i := trace-openings]) ! i) =
    powers-scaled (index (to-nat raw))
  using round unfolding partial-query-round-consistent-def by blast
have comp-indices-full:
  map opening-index
    (((replicate rounds []) [i := composition-openings]) ! i) =
    [index (to-nat raw),
     fri-sibling-index (scale * clength) (index (to-nat raw))]
  using round unfolding partial-query-round-consistent-def by blast
have comp-nonempty-full:
  ((replicate rounds []) [i := composition-openings]) ! i ≠ []
  using round unfolding partial-query-round-consistent-def by blast
have comp-value-full:
  opening-value
    (((replicate rounds []) [i := composition-openings]) ! i ! 0) =
    cp-eval (sqp-alphas prefix)
    (map opening-value
      (((replicate rounds []) [i := trace-openings]) ! i))
    (h ^ index (to-nat raw) * shift)
  using round unfolding partial-query-round-consistent-def by blast
have
  partial-query-openings-consistent trace-openings composition-openings
    (sqp-alphas prefix) (index (to-nat raw))
  unfolding partial-query-openings-consistent-def
  using idx-sample trace-indices-full comp-indices-full comp-nonempty-full
    comp-value-full trace-nth comp-nth
  by simp
then show ?thesis
  using alphas-eq by simp
qed
have trace-pullback:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings ?s ∨
  (∃ opn ∈ set trace-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots witness-state
        (staged-trace-root data)
        (scale * clength)
        (opening-index opn))

```

```

      (opening-value opn)
      (opening-path opn)) ?s witness-state)
  by (rule partial-authenticated-table-pullback-or-new-output-hit
      [OF s-ext-witness trace-table-witness])
  then show ?thesis
proof
  assume trace-table-s:
    partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings ?s
  have comp-pullback:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings ?s ∨
    (∃ opn ∈ set composition-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state
          (hd (staged-composition-fri-roots data))
            (scale * clength)
            (opening-index opn)
            (opening-value opn)
            (opening-path opn)) ?s witness-state)
      by (rule partial-authenticated-table-pullback-or-new-output-hit
          [OF s-ext-witness comp-table-witness])
  then show ?thesis
proof
  assume comp-table-s:
    partial-authenticated-table
      (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings ?s
  have trace-table-final:
    partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
  by (rule partial-authenticated-table-mono[OF trace-table-s s-ext-final])
  have comp-table-final:
    partial-authenticated-table
      (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings final-state
  by (rule partial-authenticated-table-mono[OF comp-table-s s-ext-final])
  have current:
    checked-staged-security-with-query-prefix-authenticated-opening-hit
    trace-openings composition-openings i ?out
  unfolding
    checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  using comp-nonempty trace-table-final comp-table-final consistent
  by simp
  then show ?thesis
    using not-current by contradiction
next
  assume comp-path:
    ∃ opn ∈ set composition-openings.

```

```

    hash-map-new-output-hit
      (merkle-path-target-roots witness-state
        (hd (staged-composition-fri-roots data)) (scale * clength)
        (opening-index opn) (opening-value opn) (opening-path opn))
      ?s witness-state
  have body:
    ∃ witness-result witness-state rest.
      let s = ?s in
      Some (witness-result, witness-state) ∈
        set-dist (execute verify-monad s) ∧
      verifier-header-transcript s
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)
        rest ∧
      ((∃ opn ∈ set trace-openings.
        hash-map-new-output-hit
          (merkle-path-target-roots witness-state
            (staged-trace-root data)
            (scale * clength)
            (opening-index opn)
            (opening-value opn)
            (opening-path opn))
          s witness-state) ∨
        (∃ opn ∈ set composition-openings.
          hash-map-new-output-hit
            (merkle-path-target-roots witness-state
              (hd (staged-composition-fri-roots data))
              (scale * clength)
              (opening-index opn)
              (opening-value opn)
              (opening-path opn))
            s witness-state))
      by (intro exI[of - witness-result] exI[of - witness-state]
        exI[of - rest])
        (simp add: witness-out header comp-path)
  show ?thesis
  unfolding
    checked-staged-security-with-query-prefix-header-witness-path-output-hit-def
  using E comp-nonempty body by simp
qed
next
assume trace-path:
  ∃ opn ∈ set trace-openings.
    hash-map-new-output-hit

```

```

    (merkle-path-target-roots witness-state
      (staged-trace-root data) (scale * clength)
      (opening-index opn) (opening-value opn) (opening-path opn))
    ?s witness-state
have body:
  ∃ witness-result witness-state rest.
    let s = ?s in
    Some (witness-result, witness-state) ∈
      set-dist (execute verify-monad s) ∧
    verifier-header-transcript s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
      rest ∧
    ((∃ opn ∈ set trace-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state
          (staged-trace-root data)
          (scale * clength)
          (opening-index opn)
          (opening-value opn)
          (opening-path opn))
        s witness-state) ∨
      (∃ opn ∈ set composition-openings.
        hash-map-new-output-hit
          (merkle-path-target-roots witness-state
            (hd (staged-composition-fri-roots data))
            (scale * clength)
            (opening-index opn)
            (opening-value opn)
            (opening-path opn))
          s witness-state))
    by (intro exI[of - witness-result] exI[of - witness-state]
      exI[of - rest])
      (simp add: witness-out header trace-path)
show ?thesis
unfolding
  checked-staged-security-with-query-prefix-header-witness-path-output-hit-def
using E comp-nonempty body by simp
qed
qed

lemma checked-staged-security-with-query-prefix-fixed-current-authenticated-component-imp-prefix-or-path-output-on
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A

```


and *i-bound*: $i < \text{rounds}$
and *support*:
 $\text{out} \in$
 set-dist
 (*execute*
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 adversary-initial-state)
and *hit*:
 checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 E *trace-openings* *composition-openings* i *out*
shows
 checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
 trace-openings *composition-openings* i *out* $\vee$
 checked-staged-security-with-query-prefix-opening-path-output-hit
 trace-openings *composition-openings* *out*
by (*rule*
 checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-prefix-or-path-output-hit-on-support
 [*OF* *wf* *controlled* *i-bound* *support*])
 (*rule*
 checked-staged-security-with-query-prefix-fixed-current-authenticated-component-imp-current
 [*OF* *hit*])

lemma *checked-staged-security-with-query-prefix-fixed-header-component-bound*:
assumes *raw-bound*: *query-index-raw-preimage-bound*
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* A
and *i-bound*: $i < \text{rounds}$
shows
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-fixed-header-component*
 E *trace-openings* *composition-openings* i)
 adversary-initial-state $\leq$
 staged-phase-relation-error *size*
 (*staged-query-search-queries* *budgets* $i + 1$) +
 $(1::\text{prob}) / \text{nnreal} (\text{card } \text{query-sample-space})$

proof –
have *event-le*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-fixed-header-component*
 E *trace-openings* *composition-openings* i)
 adversary-initial-state $\leq$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-actual-query-prefix-candidate-opening-hit*
 ((*replicate* *rounds* []) [$i := \text{trace-openings}$])
 ((*replicate* *rounds* []) [$i := \text{composition-openings}$])
 i)

```

    adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume
    checked-staged-security-with-query-prefix-fixed-header-component
    E trace-openings composition-openings i out
  then have
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
    trace-openings composition-openings i out
  by (rule
    checked-staged-security-with-query-prefix-fixed-header-component-imp-header)
  then show
    checked-staged-security-with-actual-query-prefix-candidate-opening-hit
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i out
  by (rule
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-imp-actual)
qed
have actual-bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-actual-query-prefix-candidate-opening-hit
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i)
  adversary-initial-state ≤
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
  (1::prob) / nnreal (card query-sample-space)
  by (rule checked-staged-security-with-actual-query-prefix-candidate-opening-hit-bound
    [OF raw-bound wf controlled])
  (use i-bound in simp)
  show ?thesis
  by (rule order-trans[OF event-le actual-bound])
qed

```

lemma checked-staged-security-with-query-prefix-fixed-header-component-bound-from-budgets:

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds

shows

```

wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-fixed-header-component
  E trace-openings composition-openings i)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +

```

```

      (1::prob) / nnreal (card query-sample-space)
proof –
  have raw-bound: query-index-raw-preimage-bound
    by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
  show ?thesis
    by (rule
      checked-staged-security-with-query-prefix-fixed-header-component-bound
      [OF raw-bound wf controlled i-bound])
qed

lemma checked-staged-security-with-query-prefix-fixed-current-authenticated-component-bound:
assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
shows
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-fixed-current-authenticated-component
      E trace-openings composition-openings i)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space)
proof –
  have event-le:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-fixed-current-authenticated-component
        E trace-openings composition-openings i)
      adversary-initial-state ≤
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-authenticated-opening-hit
        trace-openings composition-openings i)
      adversary-initial-state
    by (rule wp-event-mono)
      (rule
        checked-staged-security-with-query-prefix-fixed-current-authenticated-component-imp-current)
  have current-bound:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-authenticated-opening-hit
        trace-openings composition-openings i)
      adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space)
    by (rule checked-staged-security-with-query-prefix-authenticated-opening-hit-bound

```

```

    [OF raw-bound wf controlled i-bound])
  show ?thesis
    by (rule order-trans[OF event-le current-bound])
qed

lemma checked-staged-security-with-query-prefix-fixed-current-authenticated-component-bound-from-budgets:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound: i < rounds
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-fixed-current-authenticated-component
        E trace-openings composition-openings i)
      adversary-initial-state ≤
      staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      (1::prob) / nnreal (card query-sample-space)
  proof -
    have raw-bound: query-index-raw-preimage-bound
      by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
    show ?thesis
      by (rule
        checked-staged-security-with-query-prefix-fixed-current-authenticated-component-bound
        [OF raw-bound wf controlled i-bound])
  qed

lemma checked-staged-security-with-query-prefix-fixed-header-component-imp-current-or-witness-path-output-on-supp:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound: i < rounds
    and support:
      out ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
    and hit:
      checked-staged-security-with-query-prefix-fixed-header-component
      E trace-openings composition-openings i out
  shows
    checked-staged-security-with-query-prefix-fixed-current-authenticated-component
      E trace-openings composition-openings i out ∨
    checked-staged-security-with-query-prefix-header-witness-path-output-hit
      E trace-openings composition-openings i out
  proof (cases out)
    case None
    then have False
      using hit

```

```

unfolding
  checked-staged-security-with-query-prefix-fixed-header-component-def
  checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-def
by simp
then show ?thesis by simp
next
case (Some packed)
obtain prefix prefix-state raw raw-state data attacker-state result final-state
where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
using Some by (cases packed) (auto split: prod.splits)
from checked-staged-security-with-query-prefix-fixed-header-component-split-current
  [OF hit]
show ?thesis
proof
assume current:
  checked-staged-security-with-query-prefix-fixed-current-authenticated-component
  E trace-openings composition-openings i out
then show ?thesis by simp
next
assume not-current:
  checked-staged-security-with-query-prefix-header-witness-not-current
  E trace-openings composition-openings i out
have support':
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
using support out-eq by simp
have not-current':
  checked-staged-security-with-query-prefix-header-witness-not-current
  E trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
using not-current out-eq by simp
have path:
  checked-staged-security-with-query-prefix-header-witness-path-output-hit
  E trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
by (rule
    checked-staged-security-with-query-prefix-header-witness-not-current-imp-path-output-on-support
    [OF wf controlled i-bound support' not-current'])
then show ?thesis
using out-eq by simp

```

qed
qed

lemma *checked-staged-security-with-query-prefix-fixed-header-component-bound-from-current-and-witness-path-output*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *current-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

*(checked-staged-security-with-query-prefix-fixed-current-authenticated-component
E trace-openings composition-openings i)*

adversary-initial-state ≤ P

and *path-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

*(checked-staged-security-with-query-prefix-header-witness-path-output-hit
E trace-openings composition-openings i)*

adversary-initial-state ≤ Q

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

*(checked-staged-security-with-query-prefix-fixed-header-component
E trace-openings composition-openings i)*

adversary-initial-state ≤ P + Q

proof —

have *event-le:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

*(checked-staged-security-with-query-prefix-fixed-header-component
E trace-openings composition-openings i)*

adversary-initial-state ≤

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(λout.

checked-staged-security-with-query-prefix-fixed-current-authenticated-component

E trace-openings composition-openings i out ∨

checked-staged-security-with-query-prefix-header-witness-path-output-hit

E trace-openings composition-openings i out)

adversary-initial-state

by *(rule wp-event-mono-on-support)*

(rule

checked-staged-security-with-query-prefix-fixed-header-component-imp-current-or-witness-path-output-on-support
[OF wf controlled i-bound])

have *union-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(λout.

checked-staged-security-with-query-prefix-fixed-current-authenticated-component

$E \text{ trace-openings composition-openings } i \text{ out } \vee$
checked-staged-security-with-query-prefix-header-witness-path-output-hit
 $E \text{ trace-openings composition-openings } i \text{ out})$
adversary-initial-state $\leq$
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-fixed-current-authenticated-component}$
 $\quad E \text{ trace-openings composition-openings } i)$
adversary-initial-state $+$
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-header-witness-path-output-hit}$
 $\quad E \text{ trace-openings composition-openings } i)$
adversary-initial-state
by (rule *wp-event-union-bound*)
have *sum-bound*:
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-fixed-current-authenticated-component}$
 $\quad E \text{ trace-openings composition-openings } i)$
adversary-initial-state $+$
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-header-witness-path-output-hit}$
 $\quad E \text{ trace-openings composition-openings } i)$
adversary-initial-state $\leq P + Q$
by (intro *add-mono current-bound path-bound*)
show ?thesis
by (rule *order-trans*[*OF event-le*])
(rule *order-trans*[*OF union-bound sum-bound*])
qed

lemma *checked-staged-security-with-query-prefix-fixed-header-component-imp-current-or-witness-transcript-hit-on-su*

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *i-bound*: $i < \text{rounds}$

and *support*:

out $\in$

set-dist

(*execute*

$(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$

adversary-initial-state)

and *hit*:

checked-staged-security-with-query-prefix-fixed-header-component

$E \text{ trace-openings composition-openings } i \text{ out}$

shows

checked-staged-security-with-query-prefix-fixed-current-authenticated-component

$E \text{ trace-openings composition-openings } i \text{ out } \vee$

checked-staged-security-with-query-prefix-header-witness-transcript-hit

E out
proof –
have *split*:
checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 E trace-openings composition-openings i out $\vee$
checked-staged-security-with-query-prefix-header-witness-path-output-hit
 E trace-openings composition-openings i out
by (rule
checked-staged-security-with-query-prefix-fixed-header-component-imp-current-or-witness-path-output-on-support
 $[OF\ wf\ controlled\ i\text{-bound}\ support\ hit]$)
then show *?thesis*
proof
assume
checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 E trace-openings composition-openings i out
then show *?thesis* **by** *simp*
next
assume *path*:
checked-staged-security-with-query-prefix-header-witness-path-output-hit
 E trace-openings composition-openings i out
have
checked-staged-security-with-query-prefix-header-witness-transcript-hit
 E out
by (rule
checked-staged-security-with-query-prefix-header-witness-path-output-imp-transcript-hit
 $[OF\ path]$)
then show *?thesis* **by** *simp*
qed
qed

lemma *checked-staged-security-with-query-prefix-fixed-header-component-bound-from-current-and-witness-transcript-*
assumes *wf*: *staged-budget-wellformed* budgets
and *controlled*: *staged-adversary-controlled* budgets A
and *i-bound*: $i < rounds$
and *current-bound*:
wp-event
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}query\text{-}prefix\text{-}data\text{-}state\ A\ i)$
 $(checked\text{-}staged\text{-}security\text{-}with\text{-}query\text{-}prefix\text{-}fixed\text{-}current\text{-}authenticated\text{-}component$
 $E\ trace\text{-}openings\ composition\text{-}openings\ i)$
 $adversary\text{-}initial\text{-}state \leq P$
and *transcript-bound*:
wp-event
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}query\text{-}prefix\text{-}data\text{-}state\ A\ i)$
 $(checked\text{-}staged\text{-}security\text{-}with\text{-}query\text{-}prefix\text{-}header\text{-}witness\text{-}transcript\text{-}hit$
 $E)$
 $adversary\text{-}initial\text{-}state \leq Q$
shows
wp-event
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}query\text{-}prefix\text{-}data\text{-}state\ A\ i)$

(checked-staged-security-with-query-prefix-fixed-header-component
 E trace-openings composition-openings i)
 adversary-initial-state $\leq P + Q$

proof –

have *event-le*:

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state *A i*)
 (checked-staged-security-with-query-prefix-fixed-header-component
 E trace-openings composition-openings i)
 adversary-initial-state $\leq$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state *A i*)
 (λ out.
 checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 E trace-openings composition-openings i out $\vee$
 checked-staged-security-with-query-prefix-header-witness-transcript-hit
 E out)
 adversary-initial-state

by (rule *wp-event-mono-on-support*)
 (rule
 checked-staged-security-with-query-prefix-fixed-header-component-imp-current-or-witness-transcript-hit-on-support
 [*OF wf controlled i-bound*])

have *union-bound*:

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state *A i*)
 (λ out.
 checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 E trace-openings composition-openings i out $\vee$
 checked-staged-security-with-query-prefix-header-witness-transcript-hit
 E out)
 adversary-initial-state $\leq$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state *A i*)
 (checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 E trace-openings composition-openings i)
 adversary-initial-state +
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state *A i*)
 (checked-staged-security-with-query-prefix-header-witness-transcript-hit
 E)
 adversary-initial-state

by (rule *wp-event-union-bound*)

have *sum-bound*:

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state *A i*)
 (checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 E trace-openings composition-openings i)
 adversary-initial-state +
 wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-header-witness-transcript-hit
 E)
 $\text{adversary-initial-state} \leq P + Q$
 by (intro add-mono current-bound transcript-bound)
 show ?thesis
 by (rule order-trans[OF event-le])
 (rule order-trans[OF union-bound sum-bound])
 qed

lemma checked-staged-security-with-query-prefix-fixed-header-component-bound-from-current-pre-and-witness-new-hi

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and i-bound: $i < \text{rounds}$

and current-bound:

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 E trace-openings composition-openings i)

$\text{adversary-initial-state} \leq P$

and pre-bound:

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
 E)

$\text{adversary-initial-state} \leq R$

and new-bound:

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
 E)

$\text{adversary-initial-state} \leq Q$

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-fixed-header-component
 E trace-openings composition-openings i)

$\text{adversary-initial-state} \leq P + R + Q$

proof –

have transcript-bound:

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-header-witness-transcript-hit
 E)

$\text{adversary-initial-state} \leq R + Q$

proof –

have event-le:

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

```

    (checked-staged-security-with-query-prefix-header-witness-transcript-hit
      E)
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (λout.
    checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
      E out ∨
    checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
      E out)
  adversary-initial-state
by (rule wp-event-mono)
  (rule
    checked-staged-security-with-query-prefix-header-witness-transcript-hit-imp-pre-or-new)
have union-bound:
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (λout.
    checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
      E out ∨
    checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
      E out)
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
    E)
  adversary-initial-state +
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
    E)
  adversary-initial-state
by (rule wp-event-union-bound)
have sum-bound:
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
    E)
  adversary-initial-state +
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
    E)
  adversary-initial-state ≤ R + Q
by (intro add-mono pre-bound new-bound)
show ?thesis
by (rule order-trans[OF event-le])
  (rule order-trans[OF union-bound sum-bound])

```

qed
have *header-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
 (*checked-staged-security-with-query-prefix-fixed-header-component*
 E trace-openings composition-openings i)
 adversary-initial-state $\leq P + (R + Q)$
by (*rule*
 checked-staged-security-with-query-prefix-fixed-header-component-bound-from-current-and-witness-transcript-hit
 [*OF wf controlled i-bound current-bound transcript-bound*])
then show *?thesis*
 by (*simp add: algebra-simps*)
qed

lemma *checked-staged-security-with-query-prefix-fixed-header-component-bound-from-witness-path-output*:

assumes *raw-bound*: *query-index-raw-preimage-bound*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: *i < rounds*
and *path-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-header-witness-path-output-hit*
 E trace-openings composition-openings i)
 adversary-initial-state $\leq Q$

shows

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-fixed-header-component*
 E trace-openings composition-openings i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 (*1::prob*) / *nnreal (card query-sample-space)* + *Q*

proof —

have *current-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-fixed-current-authenticated-component*
 E trace-openings composition-openings i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 (*1::prob*) / *nnreal (card query-sample-space)*
by (*rule*
 checked-staged-security-with-query-prefix-fixed-current-authenticated-component-bound
 [*OF raw-bound wf controlled i-bound*])
show *?thesis*
by (*rule*

checked-staged-security-with-query-prefix-fixed-header-component-bound-from-current-and-witness-path-output
 [OF wf controlled i-bound current-bound path-bound])

qed

lemma *checked-staged-security-with-query-prefix-fixed-header-component-bound-from-witness-path-output-and-budget*
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i < \text{rounds}$
and path-bound:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-header-witness-path-output-hit*
 E trace-openings composition-openings i)
 adversary-initial-state $\leq Q$
shows
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-fixed-header-component*
 E trace-openings composition-openings i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets* i + 1) +
 (1::prob) / nnreal (card query-sample-space) + Q
proof –
have raw-bound: *query-index-raw-preimage-bound*
by (rule *query-index-raw-preimage-bound-from-sampler-wellformed*)
show ?thesis
by (rule
 checked-staged-security-with-query-prefix-fixed-header-component-bound-from-witness-path-output
 [OF raw-bound wf controlled i-bound path-bound])
qed

lemma *checked-staged-security-with-query-prefix-fixed-header-component-bound-from-witness-transcript-hit*
assumes raw-bound: *query-index-raw-preimage-bound*
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i < \text{rounds}$
and transcript-bound:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-header-witness-transcript-hit*
 E)
 adversary-initial-state $\leq Q$
shows
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-fixed-header-component*
 E trace-openings composition-openings i)
 adversary-initial-state $\leq$

staged-phase-relation-error size
(staged-query-search-queries budgets $i + 1$) +
(1::prob) / nnreal (card query-sample-space) + Q
proof –
have *current-bound*:
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-fixed-current-authenticated-component
E trace-openings composition-openings i)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets $i + 1$) +
(1::prob) / nnreal (card query-sample-space)
by (rule
checked-staged-security-with-query-prefix-fixed-current-authenticated-component-bound
[OF raw-bound wf controlled i-bound])
show *?thesis*
by (rule
checked-staged-security-with-query-prefix-fixed-header-component-bound-from-current-and-witness-transcript-hit
[OF wf controlled i-bound current-bound transcript-bound])
qed

lemma *checked-staged-security-with-query-prefix-fixed-header-component-bound-from-witness-transcript-hit-and-budget*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i < \text{rounds}$*
and *transcript-bound*:
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-header-witness-transcript-hit
E)
adversary-initial-state ≤ Q
shows
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-fixed-header-component
E trace-openings composition-openings i)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets $i + 1$) +
(1::prob) / nnreal (card query-sample-space) + Q
proof –
have *raw-bound: query-index-raw-preimage-bound*
by (rule *query-index-raw-preimage-bound-from-sampler-wellformed*)
show *?thesis*
by (rule
checked-staged-security-with-query-prefix-fixed-header-component-bound-from-witness-transcript-hit
[OF raw-bound wf controlled i-bound transcript-bound])
qed

lemma *checked-staged-security-with-query-prefix-fixed-header-component-bound-from-data-state-pre-and-witness-new*

assumes *raw-bound: query-index-raw-preimage-bound*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *pre-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-transcript-pre-hit

adversary-initial-state ≤ R

and *new-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-header-witness-transcript-new-hit E)

adversary-initial-state ≤ Q

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-fixed-header-component E trace-openings composition-openings i)

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) +

(1::prob) / nnreal (card query-sample-space) + R + Q

proof —

have *current-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-fixed-current-authenticated-component E trace-openings composition-openings i)

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) +

(1::prob) / nnreal (card query-sample-space)

by (rule

checked-staged-security-with-query-prefix-fixed-current-authenticated-component-bound
[OF raw-bound wf controlled i-bound])

have *pre-hit-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit E)

adversary-initial-state ≤ R

by (rule

checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit-bound-from-data-state-pre-hit
[OF i-bound pre-bound])

have *header-bound:*

wp-event

```

    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-fixed-header-component
      E trace-openings composition-openings i)
    adversary-initial-state ≤
    (staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      (1::prob) / nnreal (card query-sample-space)) + R + Q
  by (rule
    checked-staged-security-with-query-prefix-fixed-header-component-bound-from-current-pre-and-witness-new-hit
    [OF wf controlled i-bound current-bound pre-hit-bound new-bound])
  then show ?thesis
  by (simp add: algebra-simps)
qed

```

lemma *checked-staged-security-with-data-state-bound-by-query-prefix-fixed-header-component:*

```

assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and event-imp:
  ∧ out.
  (case out of
    None ⇒ E None
    | Some (packed, t) ⇒ E (Some (snd packed, t))) ⇒
    checked-staged-security-with-query-prefix-fixed-header-component
    C trace-openings composition-openings i out

```

shows

```

wp-event (checked-staged-security-experiment-with-data-state A) E
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
(1::prob) / nnreal (card query-sample-space)

```

proof –

have *projection:*

```

wp-event (checked-staged-security-experiment-with-data-state A) E
adversary-initial-state =
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(λ out. case out of
  None ⇒ E None
  | Some (packed, t) ⇒ E (Some (snd packed, t)))
adversary-initial-state
by (rule
  checked-staged-security-experiment-with-query-prefix-data-state-projection-event
  [OF i-bound])

```

have *lifted-bound:*

```

wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(λ out. case out of

```



```

    None  $\Rightarrow$  E None
  | Some (packed, t)  $\Rightarrow$  E (Some (snd packed, t)))
adversary-initial-state  $\leq$ 
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-fixed-header-component
    C trace-openings composition-openings i)
  adversary-initial-state
by (rule wp-event-mono) (use event-imp in blast)
have fixed-bound:
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-fixed-header-component
    C trace-openings composition-openings i)
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
  (1::prob) / nnreal (card query-sample-space)
by (rule checked-staged-security-with-query-prefix-fixed-header-component-bound
  [OF raw-bound wf controlled i-bound])
show ?thesis
  unfolding projection
  by (rule order-trans[OF lifted-bound fixed-bound])
qed

```

lemma *checked-staged-security-with-data-state-bound-by-query-prefix-fixed-header-component-from-budgets:*

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and i-bound: $i < \text{rounds}$

and event-imp:

$\bigwedge \text{out.}$

(case out of

None $\Rightarrow$ E None

| Some (packed, t) $\Rightarrow$ E (Some (snd packed, t))) $\Rightarrow$

checked-staged-security-with-query-prefix-fixed-header-component

C trace-openings composition-openings i out

shows

wp-event (checked-staged-security-experiment-with-data-state A) E

adversary-initial-state $\leq$

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) +

(1::prob) / nnreal (card query-sample-space)

proof –

have raw-bound: query-index-raw-preimage-bound

by (rule query-index-raw-preimage-bound-from-sampler-wellformed)

show ?thesis

by (rule

checked-staged-security-with-data-state-bound-by-query-prefix-fixed-header-component

[OF raw-bound wf controlled i-bound event-imp])

qed

lemma *checked-staged-security-with-data-state-bound-by-query-prefix-fixed-header-component-from-witness-path-output*

assumes *raw-bound: query-index-raw-preimage-bound*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *path-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-header-witness-path-output-hit
C trace-openings composition-openings i)

adversary-initial-state ≤ Q

and *event-imp:*

$\bigwedge out.$

(case out of

None ⇒ E None

| Some (packed, t) ⇒ E (Some (snd packed, t))) ⇒

checked-staged-security-with-query-prefix-fixed-header-component

C trace-openings composition-openings i out

shows

wp-event (checked-staged-security-experiment-with-data-state A) E

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) +

(1::prob) / nnreal (card query-sample-space) + Q

proof –

have *projection:*

wp-event (checked-staged-security-experiment-with-data-state A) E

adversary-initial-state =

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(λout. case out of

None ⇒ E None

| Some (packed, t) ⇒ E (Some (snd packed, t)))

adversary-initial-state

by *(rule*

checked-staged-security-experiment-with-query-prefix-data-state-projection-event

[OF i-bound])

have *lifted-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(λout. case out of

None ⇒ E None

| Some (packed, t) ⇒ E (Some (snd packed, t)))

adversary-initial-state ≤

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-fixed-header-component

C trace-openings composition-openings i)
 adversary-initial-state
by (rule *wp-event-mono*) (use *event-imp* in *blast*)
have *fixed-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-fixed-header-component*
 C trace-openings composition-openings i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets* $i + 1$) +
 ($1::\text{prob}$) / $\text{nnreal}(\text{card } \text{query-sample-space}) + Q$
by (rule
 checked-staged-security-with-query-prefix-fixed-header-component-bound-from-witness-path-output
 [*OF raw-bound wf controlled i-bound path-bound*])
show ?thesis
 unfolding *projection*
 by (rule *order-trans*[*OF lifted-bound fixed-bound*])
qed

lemma *checked-staged-security-with-data-state-bound-by-query-prefix-fixed-header-component-from-witness-path-output*
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets* A
and *i-bound*: $i < \text{rounds}$
and *path-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-header-witness-path-output-hit*
 C trace-openings composition-openings i)
 adversary-initial-state $\leq Q$
and *event-imp*:
 $\bigwedge \text{out.}$
 (case out of
 $\text{None} \Rightarrow E \text{ None}$
 | $\text{Some } (\text{packed}, t) \Rightarrow E (\text{Some } (\text{snd } \text{packed}, t)) \Rightarrow$
 checked-staged-security-with-query-prefix-fixed-header-component
 C trace-openings composition-openings i out
shows
 wp-event (*checked-staged-security-experiment-with-data-state* A) E
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets* $i + 1$) +
 ($1::\text{prob}$) / $\text{nnreal}(\text{card } \text{query-sample-space}) + Q$
proof –
have *raw-bound*: *query-index-raw-preimage-bound*
by (rule *query-index-raw-preimage-bound-from-sampler-wellformed*)
show ?thesis
by (rule
 checked-staged-security-with-data-state-bound-by-query-prefix-fixed-header-component-from-witness-path-output

[OF raw-bound wf controlled i-bound path-bound event-imp])

qed

lemma *checked-staged-security-with-actual-alpha-prefix-data-state-bound-by-query-prefix-fixed-header-component:*

assumes *raw-bound: query-index-raw-preimage-bound*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *event-imp:*

$\bigwedge out.$

 (case out of

 None $\Rightarrow D$ None

 | Some (packed, t) $\Rightarrow D$ (Some (snd packed, t))) $\Rightarrow$

 checked-staged-security-with-query-prefix-fixed-header-component

 C trace-openings composition-openings i out

shows

 wp-event

 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

 ($\lambda out.$ case out of

 None $\Rightarrow D$ None

 | Some (packed, t) $\Rightarrow$

 D (Some (((snd (fst (fst packed)), snd (fst packed)), snd packed),

 t)))

 adversary-initial-state $\leq$

 staged-phase-relation-error size

 (staged-query-search-queries budgets i + 1) +

 (1::prob) / nnreal (card query-sample-space)

proof –

 let ?project =

$\lambda x.$ ((snd (fst (fst x)), snd (fst x)), snd x)

have *projection:*

 wp-event

 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

 ($\lambda out.$ case out of

 None $\Rightarrow D$ None

 | Some (packed, t) $\Rightarrow D$ (Some (?project packed, t)))

 adversary-initial-state =

 wp-event

 (checked-staged-security-experiment-with-data-state A)

 D

 adversary-initial-state

proof –

have

 wp-event (checked-staged-security-experiment-with-data-state A)

 D adversary-initial-state =

 wp-event

 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

$\gg=$ ($\lambda x.$ return (?project x)))

 D adversary-initial-state

```

using
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
  [of A]
by simp
also have ... =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ( $\lambda out. \text{case out of}$ 
      None  $\Rightarrow D$  None
      | Some (packed, t)  $\Rightarrow D$  (Some (?project packed, t)))
    adversary-initial-state
  by (rule wp-event-bind-return-map)
finally show ?thesis
  by simp
qed
have data-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    D adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      (1::prob) / nnreal (card query-sample-space)
  by (rule
    checked-staged-security-with-data-state-bound-by-query-prefix-fixed-header-component
    [OF raw-bound wf controlled i-bound event-imp])
show ?thesis
  unfolding projection
  by (rule data-bound)
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-data-state-bound-by-query-prefix-fixed-header-component-from*

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *i-bound*: *i* < *rounds*

and *event-imp*:

$\bigwedge out.$

(*case out of*

None $\Rightarrow D$ *None*

| *Some* (*packed*, *t*) $\Rightarrow D$ (*Some* (*snd* *packed*, *t*))) $\implies$

checked-staged-security-with-query-prefix-fixed-header-component

C trace-openings composition-openings i out

shows

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)

($\lambda out. \text{case out of}$

None $\Rightarrow D$ *None*

| *Some* (*packed*, *t*) $\Rightarrow$

D (*Some* (((*snd* (*fst* (*fst* *packed*))), *snd* (*fst* *packed*))), *snd* *packed*),
t)))

adversary-initial-state $\leq$

```

    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space)
proof -
  have raw-bound: query-index-raw-preimage-bound
  by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
  show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-data-state-bound-by-query-prefix-fixed-header-component
    [OF raw-bound wf controlled i-bound event-imp])
qed

lemma checked-staged-security-with-actual-alpha-prefix-data-state-bound-by-query-prefix-fixed-header-component-from
assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and path-bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-header-witness-path-output-hit
    C trace-openings composition-openings i)
  adversary-initial-state ≤ Q
and event-imp:
  ∧out.
  (case out of
    None ⇒ D None
    | Some (packed, t) ⇒ D (Some (snd packed, t))) ⇒
    checked-staged-security-with-query-prefix-fixed-header-component
    C trace-openings composition-openings i out
shows
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (λout. case out of
    None ⇒ D None
    | Some (packed, t) ⇒
      D (Some (((snd (fst (fst packed)), snd (fst packed)), snd packed),
        t)))
  adversary-initial-state ≤
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
  (1::prob) / nnreal (card query-sample-space) + Q
proof -
  let ?project =
    λx. ((snd (fst (fst x)), snd (fst x)), snd x)
  have projection:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (λout. case out of

```

```

      None  $\Rightarrow$  D None
    | Some (packed, t)  $\Rightarrow$  D (Some (?project packed, t)))
  adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-data-state A)
  D
  adversary-initial-state
proof -
have
  wp-event (checked-staged-security-experiment-with-data-state A)
    D adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
       $\gg$  ( $\lambda x.$  return (?project x)))
    D adversary-initial-state
using
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
  [of A]
by simp
also have ... =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ( $\lambda out.$  case out of
      None  $\Rightarrow$  D None
    | Some (packed, t)  $\Rightarrow$  D (Some (?project packed, t)))
    adversary-initial-state
by (rule wp-event-bind-return-map)
finally show ?thesis
by simp
qed
have data-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    D adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
  (1::prob) / nnreal (card query-sample-space) + Q
by (rule
  checked-staged-security-with-data-state-bound-by-query-prefix-fixed-header-component-from-witness-path-output
  [OF raw-bound wf controlled i-bound path-bound event-imp])
show ?thesis
unfolding projection
by (rule data-bound)
qed

lemma checked-staged-security-with-actual-alpha-prefix-data-state-bound-by-query-prefix-fixed-header-component-from
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and path-bound:

```

```

wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-header-witness-path-output-hit
    C trace-openings composition-openings i)
  adversary-initial-state ≤ Q
and event-imp:
  ∧ out.
    (case out of
      None ⇒ D None
    | Some (packed, t) ⇒ D (Some (snd packed, t))) ⇒
      checked-staged-security-with-query-prefix-fixed-header-component
        C trace-openings composition-openings i out
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (λ out. case out of
      None ⇒ D None
    | Some (packed, t) ⇒
      D (Some (((snd (fst (fst packed)), snd (fst packed)), snd packed),
        t)))
    adversary-initial-state ≤
      staged-phase-relation-error size
        (staged-query-search-queries budgets i + 1) +
        (1::prob) / nnreal (card query-sample-space) + Q
proof -
  have raw-bound: query-index-raw-preimage-bound
  by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-data-state-bound-by-query-prefix-fixed-header-component-from-w
      [OF raw-bound wf controlled i-bound path-bound event-imp])
qed

lemma checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-bound:
  assumes raw-bound: query-index-raw-preimage-bound
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
        trace-openings composition-openings i)
      adversary-initial-state ≤
        staged-phase-relation-error size
          (staged-query-search-queries budgets i + 1) +
          (1::prob) / nnreal (card query-sample-space)
proof -
  have event-le:

```



```

wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
    trace-openings composition-openings i)
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-actual-query-prefix-candidate-opening-hit
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i)
  adversary-initial-state
by (rule wp-event-mono)
  (rule
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-imp-actual)
have actual-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-actual-query-prefix-candidate-opening-hit
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i)
    adversary-initial-state ≤
    staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      (1::prob) / nnreal (card query-sample-space)
  by (rule checked-staged-security-with-actual-query-prefix-candidate-opening-hit-bound
    [OF raw-bound wf controlled])
    (use i-bound in simp)
show ?thesis
  by (rule order-trans[OF event-le actual-bound])
qed

lemma checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-bound-from-budgets:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
        trace-openings composition-openings i)
      adversary-initial-state ≤
      staged-phase-relation-error size
        (staged-query-search-queries budgets i + 1) +
        (1::prob) / nnreal (card query-sample-space)
proof -
  have raw-bound: query-index-raw-preimage-bound
  by (rule query-index-raw-preimage-bound-from-sampler-wellformed)

```

```

show ?thesis
  by (rule
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-bound
    [OF raw-bound wf controlled i-bound])
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Query-Prefix-Witness
  imports Staged-Security-Experiment-Composition-Query-Augmented
begin

```

Witness-level bridge for the composition query-prefix path.

The existing query-prefix bounds are fixed-witness bounds: the trace and composition openings must already be selected before the query challenge is sampled. This layer records the exact local bridge from such fixed witness data to the augmented query-prefix component event. The remaining public soundness work is to derive these fixed witnesses from prefix Merkle/candidate evidence, or charge failures to existing Merkle/collision side events.

```

context soundness
begin

```

```

lemma partial-query-openings-consistent-canonical-trace-index:

```

```

  assumes consistent:

```

```

    partial-query-openings-consistent trace-openings composition-openings
    as idx

```

```

  shows trace-openings ≠ []

```

```

    and idx = opening-index (hd trace-openings)

```

```

proof –

```

```

  have trace-indices:

```

```

    map opening-index trace-openings = powers-scaled idx

```

```

    using consistent unfolding partial-query-openings-consistent-def by simp

```

```

  have powers-nonempty: powers-scaled idx ≠ []

```

```

    unfolding powers-scaled-def using powers-pos by simp

```

```

  then show trace-nonempty: trace-openings ≠ []

```

```

    using trace-indices by auto

```

```

  have hd (map opening-index trace-openings) = hd (powers-scaled idx)

```

```

    using trace-indices by simp

```

```

  then have opening-index (hd trace-openings) = hd (powers-scaled idx)

```

```

    using trace-nonempty by (simp add: hd-map)

```

```

  then show idx = opening-index (hd trace-openings)

```

```

    using powers-scaled-hd by simp

```

```

qed

```

```

lemma partial-query-success-indices-at-subset-canonical-trace-index:

```

$\text{partial-query-success-indices-at } \text{trace-openings } \text{composition-openings as } i$
 $\subseteq \{ \text{opening-index } (\text{hd } (\text{trace-openings } ! i)) \}$

proof

fix idx

assume idx-in :

$\text{idx} \in \text{partial-query-success-indices-at } \text{trace-openings}$
 $\text{composition-openings as } i$

have trace-indices :

$\text{map opening-index } (\text{trace-openings } ! i) = \text{powers-scaled idx}$

using idx-in

unfolding $\text{partial-query-success-indices-at-def}$
 $\text{partial-query-round-consistent-def}$

by simp

have trace-nonempty : $\text{trace-openings } ! i \neq []$

proof –

have $\text{powers-scaled idx} \neq []$

unfolding powers-scaled-def **using** powers-pos **by** simp

then show $?thesis$

using trace-indices **by** auto

qed

have $\text{hd } (\text{map opening-index } (\text{trace-openings } ! i)) = \text{hd } (\text{powers-scaled idx})$

using trace-indices **by** simp

then have $\text{opening-index } (\text{hd } (\text{trace-openings } ! i)) = \text{hd } (\text{powers-scaled idx})$

using trace-nonempty **by** $(\text{simp add: hd-map})$

then show $\text{idx} \in \{ \text{opening-index } (\text{hd } (\text{trace-openings } ! i)) \}$

using powers-scaled-hd **by** simp

qed

lemma $\text{staged-query-prefix-candidate-opening-query-target-from-prefix-subset-canonical-trace-index}$:

$\text{staged-query-prefix-candidate-opening-query-target-from-prefix}$
 $\text{trace-openings } \text{composition-openings } i \text{ prefix prefix-state} \subseteq$
 $\{ \text{opening-index } (\text{hd } (\text{trace-openings } ! i)) \}$

unfolding $\text{staged-query-prefix-candidate-opening-query-target-from-prefix-def}$

by $(\text{rule partial-query-success-indices-at-subset-canonical-trace-index})$

definition $\text{query-header-supported-partial-opening-canonical-indices}$

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat set}$

where

$\text{query-header-supported-partial-opening-canonical-indices } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg } \text{composition-fri-roots } \text{final} =$
 $\{ \text{opening-index } (\text{hd } \text{trace-openings}) \mid$
 $\text{trace-openings } \text{composition-openings.}$
 $(\text{trace-openings}, \text{composition-openings}) \in$
 $\text{query-header-supported-partial-opening-witnesses } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg } \text{composition-fri-roots } \text{final} \wedge$
 $\text{trace-openings} \neq [] \}$

definition *query-header-supported-partial-opening-canonical-query-indices*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat set}$

where

query-header-supported-partial-opening-canonical-query-indices s fr
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final} =$
query-header-supported-partial-opening-canonical-indices s fr
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final} \cap$
query-sample-space

lemma *query-header-supported-partial-opening-canonical-query-indices-subset:*

query-header-supported-partial-opening-canonical-query-indices s fr
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final} \subseteq$
query-sample-space

unfolding *query-header-supported-partial-opening-canonical-query-indices-def*
by *blast*

lemma *query-header-supported-partial-opening-canonical-query-indices-iff:*

$idx \in \text{query-header-supported-partial-opening-canonical-query-indices } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final} \longleftrightarrow$
 $idx \in \text{query-sample-space} \wedge$
 $idx \in \text{query-header-supported-partial-opening-canonical-indices } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$

unfolding *query-header-supported-partial-opening-canonical-query-indices-def*
by *blast*

lemma *finite-query-header-supported-partial-opening-canonical-query-indices[simp]:*

finite
 $(\text{query-header-supported-partial-opening-canonical-query-indices } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final})$
by $(\text{rule finite-subset}$
 $[OF \text{ query-header-supported-partial-opening-canonical-query-indices-subset}$
 $\text{finite-query-sample-space}])$

lemma *card-query-header-supported-partial-opening-canonical-query-indices-le:*

card
 $(\text{query-header-supported-partial-opening-canonical-query-indices } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final})$
 $\leq \text{card query-sample-space}$
by (rule card-mono)
 $(\text{rule finite-query-sample-space},$
 $\text{rule query-header-supported-partial-opening-canonical-query-indices-subset})$

lemma *query-header-supported-partial-opening-canonical-query-indices-eq-query-sample-space-if-all-indices-witnessed*

assumes *witnessed:*
 $\bigwedge idx. idx \in \text{query-sample-space} \implies$
 $\exists \text{ trace-openings composition-openings.}$
 $(\text{trace-openings}, \text{composition-openings}) \in$

```

      query-header-supported-partial-opening-witnesses s fr
      f-fri-roots f-final as dg composition-fri-roots final  $\wedge$ 
      trace-openings  $\neq \square$   $\wedge$ 
      opening-index (hd trace-openings) = idx
    shows
      query-header-supported-partial-opening-canonical-query-indices s fr
      f-fri-roots f-final as dg composition-fri-roots final =
      query-sample-space
  proof
    show
      query-header-supported-partial-opening-canonical-query-indices s fr
      f-fri-roots f-final as dg composition-fri-roots final  $\subseteq$ 
      query-sample-space
    by (rule query-header-supported-partial-opening-canonical-query-indices-subset)
  next
    show
      query-sample-space  $\subseteq$ 
      query-header-supported-partial-opening-canonical-query-indices s fr
      f-fri-roots f-final as dg composition-fri-roots final
    proof
      fix idx
      assume idx-sample: idx  $\in$  query-sample-space
      then obtain trace-openings composition-openings where witness:
        (trace-openings, composition-openings)  $\in$ 
        query-header-supported-partial-opening-witnesses s fr
        f-fri-roots f-final as dg composition-fri-roots final
      and nonempty: trace-openings  $\neq \square$ 
      and idx-eq: opening-index (hd trace-openings) = idx
      using witnessed by blast
    have idx-canonical:
      idx  $\in$ 
      query-header-supported-partial-opening-canonical-indices s fr
      f-fri-roots f-final as dg composition-fri-roots final
    unfolding query-header-supported-partial-opening-canonical-indices-def
    using witness nonempty idx-eq by blast
    show
      idx  $\in$ 
      query-header-supported-partial-opening-canonical-query-indices s fr
      f-fri-roots f-final as dg composition-fri-roots final
    unfolding query-header-supported-partial-opening-canonical-query-indices-def
    using idx-sample idx-canonical by simp
  qed
qed

lemma table-agrees-with-authenticated-openings-from-partial-authenticated-table-if-no-partial-merkle-bad:
  assumes table:
    partial-authenticated-table rt (scale * clength) openings final-state
  and outcome:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad s)

```

```

and no-bad:
   $\neg$  partial-merkle-inconsistency-bad s (Some (result, final-state))
shows  $\exists$  table.
  table-agrees-with-authenticated-openings table (scale * clength)
  openings
proof -
  have accepted-out: accepted (Some (result, final-state))
  unfolding accepted-def by simp
  have idx-bound:
     $\bigwedge \text{opn}. \text{opn} \in \text{set openings} \implies$ 
    opening-index opn < scale * clength
  proof -
    fix opn
    assume opn-in: opn  $\in$  set openings
    have auth: authenticated-opening-in final-state opn
    using table opn-in unfolding partial-authenticated-table-def by blast
    have len: opening-length opn = scale * clength
    using table opn-in unfolding partial-authenticated-table-def by blast
    show opening-index opn < scale * clength
    using auth len unfolding authenticated-opening-in-def by simp
  qed
  have consistent:
     $\bigwedge \text{opn opn}'. \text{opn} \in \text{set openings} \implies$ 
    opn'  $\in$  set openings  $\implies$ 
    opening-index opn = opening-index opn'  $\implies$ 
    opening-value opn = opening-value opn'
  proof (rule ccontr)
    fix opn opn'
    assume opn-in: opn  $\in$  set openings
    and opn'-in: opn'  $\in$  set openings
    and same-index: opening-index opn = opening-index opn'
    and neq: opening-value opn  $\neq$  opening-value opn'
    have auth: authenticated-opening-in final-state opn
    using table opn-in unfolding partial-authenticated-table-def by blast
    have auth': authenticated-opening-in final-state opn'
    using table opn'-in unfolding partial-authenticated-table-def by blast
    have root: opening-root opn = opening-root opn'
    using table opn-in opn'-in unfolding partial-authenticated-table-def
    by auto
    have len: opening-length opn = opening-length opn'
    using table opn-in opn'-in unfolding partial-authenticated-table-def
    by auto
    have bad:
      partial-merkle-inconsistency-bad s (Some (result, final-state))
      unfolding partial-merkle-inconsistency-bad-def
      by (intro conjI exI[of - result] exI[of - final-state]
        exI[of - opn] exI[of - opn'])
      (use accepted-out auth auth' root len same-index neq in simp-all)
    then show False

```

```

    using no-bad by contradiction
qed
show ?thesis
  by (rule table-agrees-with-authenticated-openings-consistent-exists
      [OF idx-bound consistent])
qed

lemma partial-trace-table-candidate-single-witness-if-no-partial-merkle-bad:
  assumes auth:
    partial-authenticated-table fr (scale * clength) trace-openings
    final-state
  and outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and no-bad:
    ¬ partial-merkle-inconsistency-bad s (Some (result, final-state))
  and i-bound: i < rounds
  shows ∃ trace-table.
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  proof -
    obtain trace-table where agrees:
      table-agrees-with-authenticated-openings trace-table (scale * clength)
      trace-openings
    using
      table-agrees-with-authenticated-openings-from-partial-authenticated-table-if-no-partial-merkle-bad
      [OF auth outcome no-bad]
    by blast
    have len: length trace-table = scale * clength
    using agrees unfolding table-agrees-with-authenticated-openings-def
    by simp
    have candidate:
      partial-trace-table-candidate trace-table
      ((replicate rounds []) [i := trace-openings])
    by (rule partial-trace-table-candidate-single-roundI
        [OF i-bound len agrees])
    then show ?thesis by blast
  qed

lemma partial-composition-table-candidate-single-witness-if-no-partial-merkle-bad:
  assumes auth:
    partial-authenticated-table composition-root (scale * clength)
    composition-openings final-state
  and outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and no-bad:
    ¬ partial-merkle-inconsistency-bad s (Some (result, final-state))
  and i-bound: i < rounds
  shows ∃ composition-table.
    partial-composition-table-candidate composition-table

```

```

      ((replicate rounds []) [i := composition-openings])
proof –
  obtain composition-table where agrees:
    table-agrees-with-authenticated-openings composition-table
      (scale * clength) composition-openings
  using
    table-agrees-with-authenticated-openings-from-partial-authenticated-table-if-no-partial-merkle-bad
      [OF auth outcome no-bad]
  by blast
  have len: length composition-table = scale * clength
  using agrees unfolding table-agrees-with-authenticated-openings-def
  by simp
  have candidate:
    partial-composition-table-candidate composition-table
      ((replicate rounds []) [i := composition-openings])
  by (rule partial-composition-table-candidate-single-roundI
      [OF i-bound len agrees])
  then show ?thesis by blast
qed

```

lemma *query-header-supported-partial-opening-witness-single-round-candidates-or-partial-merkle-bad*:

```

assumes witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses s fr f-fri-roots
    f-final as dg composition-fri-roots final
and i-bound: i < rounds
shows
  (∃ result final-state.
    partial-merkle-inconsistency-bad s (Some (result, final-state))) ∨
  (∃ trace-table composition-table.
    partial-trace-table-candidate trace-table
      ((replicate rounds []) [i := trace-openings]) ∧
    partial-composition-table-candidate composition-table
      ((replicate rounds []) [i := composition-openings]))

```

```

proof –
from query-header-supported-partial-opening-witnessesE[OF witness]
obtain result final-state rest where outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and trace-auth:
  partial-authenticated-table fr (scale * clength) trace-openings
    final-state
and comp-auth:
  partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) composition-openings final-state
by blast
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True

```



```

then show ?thesis by blast
next
case False
obtain trace-table where trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
using partial-trace-table-candidate-single-witness-if-no-partial-merkle-bad
  [OF trace-auth outcome False i-bound]
by blast
obtain composition-table where comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
using
  partial-composition-table-candidate-single-witness-if-no-partial-merkle-bad
  [OF comp-auth outcome False i-bound]
by blast
show ?thesis
using trace-candidate comp-candidate by blast
qed
qed

lemma query-header-supported-partial-opening-witness-success-indices-fraction-bound-if-low-degree-candidates:
assumes i-bound: i < rounds
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
and trace-low: trace-table-low-degree trace-table
and comp-low: composition-table-low-degree maxDegree composition-table
and not-all:
  ¬ all-queries-consistent trace-table composition-table as
shows
  nnreal
  (card
    {idx ∈ query-sample-space.
      partial-query-openings-consistent trace-openings
      composition-openings as idx}) /
  nnreal (card query-sample-space) ≤ query-error-bound
by (rule
  partial-query-openings-consistent-indices-fraction-bound-if-candidate-low-degree
  [OF i-bound trace-candidate comp-candidate trace-low comp-low not-all])

lemma accepted-with-partial-trace-openings-update-round:
assumes partial:
  accepted-with-partial-trace-openings s (Some (result, final-state))
  fr query-idxs trace-openings
and i-bound: i < rounds

```

```

and consistent:
  partial-query-openings-consistent trace-openings-i
    composition-openings-i as (query-idxs ! i)
and auth:
  partial-authenticated-table fr (scale * clength) trace-openings-i
    final-state
shows
  accepted-with-partial-trace-openings s (Some (result, final-state))
    fr query-idxs (trace-openings[i := trace-openings-i])
proof –
  have len-query: length query-idxs = rounds
    using accepted-with-partial-trace-openings-shapes(1)[OF partial] .
  have len-trace: length trace-openings = rounds
    using accepted-with-partial-trace-openings-shapes(2)[OF partial] .
  have trace-indices-i:
    map opening-index trace-openings-i = powers-scaled (query-idxs ! i)
    using consistent unfolding partial-query-openings-consistent-def by simp
  have indices:
     $\forall j < \text{rounds}.$ 
    map opening-index ((trace-openings[i := trace-openings-i]) ! j) =
      powers-scaled (query-idxs ! j)
  proof (intro allI impI)
    fix j
    assume j-bound: j < rounds
    show
      map opening-index ((trace-openings[i := trace-openings-i]) ! j) =
        powers-scaled (query-idxs ! j)
    proof (cases j = i)
      case True
        then show ?thesis
          using i-bound len-trace trace-indices-i by simp
      next
        case False
          then show ?thesis
            using accepted-with-partial-trace-openings-shapes(3)[OF partial j-bound]
              len-trace j-bound
            by simp
    qed
  qed
  have tables:
     $\forall j < \text{rounds}.$ 
    partial-authenticated-table fr (scale * clength)
      ((trace-openings[i := trace-openings-i]) ! j) final-state
  proof (intro allI impI)
    fix j
    assume j-bound: j < rounds
    show
      partial-authenticated-table fr (scale * clength)
        ((trace-openings[i := trace-openings-i]) ! j) final-state

```

```

proof (cases  $j = i$ )
  case True
  then show ?thesis
    using i-bound len-trace auth by simp
next
  case False
  then show ?thesis
    using partial j-bound len-trace
    unfolding accepted-with-partial-trace-openings-def
    by simp
qed
qed
show ?thesis
  unfolding accepted-with-partial-trace-openings-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state])
    (use len-query len-trace indices tables in simp-all)
qed

lemma accepted-with-partial-composition-openings-update-round:
assumes partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) composition-root query-idxs
  composition-openings
and i-bound:  $i < \text{rounds}$ 
and consistent:
  partial-query-openings-consistent trace-openings-i
  composition-openings-i as (query-idxs ! i)
and auth:
  partial-authenticated-table composition-root (scale * clength)
  composition-openings-i final-state
shows
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) composition-root query-idxs
  (composition-openings[i := composition-openings-i])
proof –
have len-query:  $\text{length } \text{query-idxs} = \text{rounds}$ 
  using accepted-with-partial-composition-openings-shapes(1)[OF partial] .
have len-comp:  $\text{length } \text{composition-openings} = \text{rounds}$ 
  using accepted-with-partial-composition-openings-shapes(2)[OF partial] .
have comp-indices-i:
  map opening-index composition-openings-i =
  [query-idxs ! i,
   fri-sibling-index (scale * clength) (query-idxs ! i)]
  using consistent unfolding partial-query-openings-consistent-def by simp
have indices:
   $\forall j < \text{rounds}.$ 
  map opening-index
  (((composition-openings[i := composition-openings-i]) ! j) =
   [query-idxs ! j,

```

```

      fri-sibling-index (scale * clength) (query-idxs ! j)]
proof (intro allI impI)
  fix j
  assume j-bound: j < rounds
  show
    map opening-index
      ((composition-openings[i := composition-openings-i]) ! j) =
      [query-idxs ! j,
       fri-sibling-index (scale * clength) (query-idxs ! j)]
proof (cases j = i)
  case True
  then show ?thesis
    using i-bound len-comp comp-indices-i by simp
next
  case False
  then show ?thesis
    using accepted-with-partial-composition-openings-shapes(3)
      [OF partial j-bound]
      len-comp j-bound
    by simp
qed
qed
have tables:
  ∀ j < rounds.
    partial-authenticated-table composition-root (scale * clength)
      ((composition-openings[i := composition-openings-i]) ! j)
      final-state
proof (intro allI impI)
  fix j
  assume j-bound: j < rounds
  show
    partial-authenticated-table composition-root (scale * clength)
      ((composition-openings[i := composition-openings-i]) ! j)
      final-state
proof (cases j = i)
  case True
  then show ?thesis
    using i-bound len-comp auth by simp
next
  case False
  then show ?thesis
    using partial j-bound len-comp
    unfolding accepted-with-partial-composition-openings-def
    by simp
qed
qed
show ?thesis
  unfolding accepted-with-partial-composition-openings-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state])

```

(use len-query len-comp indices tables in simp-all)

qed

lemma *accepted-with-partial-initial-openings-update-round*:

assumes *partial*:

accepted-with-partial-initial-openings s
 (Some (result, final-state)) fr *f-fri-roots f-final* as *dg*
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings

and *i-bound*: $i < \text{rounds}$

and *same-idx*: $\text{trace-query-idxs} ! i = \text{composition-query-idxs} ! i$

and *consistent*:

partial-query-openings-consistent trace-openings-i
composition-openings-i as (trace-query-idxs ! i)

and *trace-auth*:

partial-authenticated-table fr (scale * clength) *trace-openings-i*
final-state

and *comp-auth*:

partial-authenticated-table (hd *composition-fri-roots*)
 (scale * clength) *composition-openings-i final-state*

shows

accepted-with-partial-initial-openings s
 (Some (result, final-state)) fr *f-fri-roots f-final* as *dg*
composition-fri-roots final trace-query-idxs
 (trace-openings[i := trace-openings-i]) *composition-query-idxs*
 (composition-openings[i := composition-openings-i])

proof –

have *comp-nonempty*: $\text{composition-fri-roots} \neq []$

using *accepted-with-partial-initial-openings-shapes*(1)[OF *partial*] .

have *header*:

$\exists \text{rest. verifier-header-transcript } s \text{ fr } f\text{-fri-roots } f\text{-final as } dg$
composition-fri-roots final rest

using *accepted-with-partial-initial-openings-shapes*(2)[OF *partial*] .

have *trace-partial*:

accepted-with-partial-trace-openings s
 (Some (result, final-state)) fr *trace-query-idxs trace-openings*

using *accepted-with-partial-initial-openings-shapes*(3)[OF *partial*] .

have *comp-partial*:

accepted-with-partial-composition-openings s
 (Some (result, final-state)) (hd *composition-fri-roots*)
composition-query-idxs composition-openings

using *accepted-with-partial-initial-openings-shapes*(4)[OF *partial*] .

have *trace-updated*:

accepted-with-partial-trace-openings s
 (Some (result, final-state)) fr *trace-query-idxs*
 (trace-openings[i := trace-openings-i])

by (rule *accepted-with-partial-trace-openings-update-round*
 [OF *trace-partial i-bound consistent trace-auth*])

have *comp-consistent*:

```

    partial-query-openings-consistent trace-openings-i
    composition-openings-i as (composition-query-idxs ! i)
    using consistent same-idx by simp
  have comp-updated:
    accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots)
    composition-query-idxs
    (composition-openings[i := composition-openings-i])
  by (rule accepted-with-partial-composition-openings-update-round
    [OF comp-partial i-bound comp-consistent comp-auth])
  have accepted-out: accepted (Some (result, final-state))
  by (rule accepted-with-partial-trace-openings-imp-accepted
    [OF trace-updated])
  show ?thesis
    unfolding accepted-with-partial-initial-openings-def
    using accepted-out comp-nonempty header trace-updated comp-updated
    by simp
qed

lemma accepted-with-partial-initial-openings-update-round-candidates-if-no-partial-merkle-bad:
  assumes partial:
    accepted-with-partial-initial-openings s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings
  and no-bad:
    ¬ partial-merkle-inconsistency-bad s (Some (result, final-state))
  and i-bound: i < rounds
  and same-idx: trace-query-idxs ! i = composition-query-idxs ! i
  and consistent:
    partial-query-openings-consistent trace-openings-i
    composition-openings-i as (trace-query-idxs ! i)
  and trace-auth:
    partial-authenticated-table fr (scale * clength) trace-openings-i
    final-state
  and comp-auth:
    partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) composition-openings-i final-state
  shows
    ∃ trace-table composition-table.
    partial-trace-table-candidate trace-table
    (trace-openings[i := trace-openings-i]) ∧
    partial-composition-table-candidate composition-table
    (composition-openings[i := composition-openings-i])
  proof -
    have updated:
      accepted-with-partial-initial-openings s
      (Some (result, final-state)) fr f-fri-roots f-final as dg
      composition-fri-roots final trace-query-idxs

```

(trace-openings[i := trace-openings-i]) composition-query-idxs
 (composition-openings[i := composition-openings-i])
by (rule accepted-with-partial-initial-openings-update-round
 [OF partial i-bound same-idx consistent trace-auth comp-auth])
show ?thesis
by (rule accepted-with-partial-initial-openings-candidates-if-no-partial-merkle-bad
 [OF updated no-bad])
qed

lemma query-header-supported-partial-opening-success-indices-at-subset-canonical-indices:
 query-header-supported-partial-opening-success-indices-at s fr
 f-fri-roots f-final as dg composition-fri-roots final i $\subseteq$
 query-header-supported-partial-opening-canonical-indices s fr
 f-fri-roots f-final as dg composition-fri-roots final
proof
fix idx
assume idx-in:
 idx $\in$ query-header-supported-partial-opening-success-indices-at s fr
 f-fri-roots f-final as dg composition-fri-roots final i
then obtain trace-openings composition-openings **where** witness:
 (trace-openings, composition-openings) $\in$
 query-header-supported-partial-opening-witnesses s fr
 f-fri-roots f-final as dg composition-fri-roots final
and consistent:
 partial-query-openings-consistent trace-openings composition-openings
 as idx
by (rule query-header-supported-partial-opening-success-indices-atE)
have nonempty: trace-openings $\neq []$
and idx-eq: idx = opening-index (hd trace-openings)
by (rule partial-query-openings-consistent-canonical-trace-index
 [OF consistent])
show
 idx $\in$ query-header-supported-partial-opening-canonical-indices s fr
 f-fri-roots f-final as dg composition-fri-roots final
unfolding query-header-supported-partial-opening-canonical-indices-def
using witness nonempty idx-eq **by** blast
qed

lemma query-header-supported-partial-opening-success-indices-at-subset-canonical-query-indices:
 query-header-supported-partial-opening-success-indices-at s fr
 f-fri-roots f-final as dg composition-fri-roots final i $\subseteq$
 query-header-supported-partial-opening-canonical-query-indices s fr
 f-fri-roots f-final as dg composition-fri-roots final
proof
fix idx
assume idx-in:
 idx $\in$ query-header-supported-partial-opening-success-indices-at s fr
 f-fri-roots f-final as dg composition-fri-roots final i
have idx-canonical:

```

    idx ∈ query-header-supported-partial-opening-canonical-indices s fr
      f-fri-roots f-final as dg composition-fri-roots final
  using idx-in
    query-header-supported-partial-opening-success-indices-at-subset-canonical-indices
  by blast
have idx-sample: idx ∈ query-sample-space
  using idx-in query-header-supported-partial-opening-success-indices-at-subset
  by blast
show
  idx ∈ query-header-supported-partial-opening-canonical-query-indices s fr
    f-fri-roots f-final as dg composition-fri-roots final
  unfolding query-header-supported-partial-opening-canonical-query-indices-def
  using idx-canonical idx-sample by simp
qed

lemma query-header-supported-partial-opening-success-indices-at-fraction-bound-if-canonical-query-indices:
  assumes canonical-bound:
    nnreal
      (card
        (query-header-supported-partial-opening-canonical-query-indices
          s fr f-fri-roots f-final as dg composition-fri-roots final)) /
        nnreal (card query-sample-space) ≤ query-error-bound
  shows
    nnreal
      (card
        (query-header-supported-partial-opening-success-indices-at s fr
          f-fri-roots f-final as dg composition-fri-roots final i)) /
        nnreal (card query-sample-space) ≤ query-error-bound
proof –
  have subset:
    query-header-supported-partial-opening-success-indices-at s fr
      f-fri-roots f-final as dg composition-fri-roots final i ⊆
    query-header-supported-partial-opening-canonical-query-indices s fr
      f-fri-roots f-final as dg composition-fri-roots final
  by (rule
    query-header-supported-partial-opening-success-indices-at-subset-canonical-query-indices)
  have finite-canonical:
    finite
      (query-header-supported-partial-opening-canonical-query-indices s fr
        f-fri-roots f-final as dg composition-fri-roots final)
  by simp
  have card-le:
    card
      (query-header-supported-partial-opening-success-indices-at s fr
        f-fri-roots f-final as dg composition-fri-roots final i) ≤
    card
      (query-header-supported-partial-opening-canonical-query-indices s fr
        f-fri-roots f-final as dg composition-fri-roots final)
  by (rule card-mono[OF finite-canonical subset])

```


have

$$\frac{\text{nnreal} \left(\text{card} \left(\text{query-header-supported-partial-opening-success-indices-at } s \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final } i \right) \right)}{\text{nnreal} \left(\text{card query-sample-space} \right)} \leq$$

$$\frac{\text{nnreal} \left(\text{card} \left(\text{query-header-supported-partial-opening-canonical-query-indices } s \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final} \right) \right)}{\text{nnreal} \left(\text{card query-sample-space} \right)}$$
by (rule nnreal-nat-divide-right-mono[OF card-le])
also have $\dots \leq \text{query-error-bound}$
by (rule canonical-bound)
finally show ?thesis .
qed

lemma query-header-supported-partial-opening-success-indices-at-subset-if-canonical-indices:
assumes canonical-cover:

$$\text{query-header-supported-partial-opening-canonical-indices } s \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final} \subseteq B$$
shows

$$\text{query-header-supported-partial-opening-success-indices-at } s \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final } i \subseteq B$$
by (rule order-trans
[OF
query-header-supported-partial-opening-success-indices-at-subset-canonical-indices
canonical-cover])

lemma query-header-supported-partial-opening-success-indices-at-subset-if-canonical-query-indices:
assumes canonical-cover:

$$\text{query-header-supported-partial-opening-canonical-query-indices } s \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final} \subseteq B$$
shows

$$\text{query-header-supported-partial-opening-success-indices-at } s \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final } i \subseteq B$$
by (rule order-trans
[OF
query-header-supported-partial-opening-success-indices-at-subset-canonical-query-indices
canonical-cover])

lemma staged-query-header-supported-partial-opening-success-indices-at-subset-if-canonical-indices:
assumes canonical-cover:

$$\text{query-header-supported-partial-opening-canonical-indices}$$

$$(\text{verifier-state-from-adversary attacker-state}$$

$$(\text{staged-proof-transcript data}))$$

$$(\text{staged-trace-root data})$$

$$(\text{staged-trace-fri-roots data})$$

$$(\text{staged-trace-final data})$$

(staged-*alphas* data)
 (staged-*degree* data)
 (staged-*composition-fri-roots* data)
 (staged-*composition-final* data) $\subseteq B$

shows

query-header-supported-partial-opening-success-indices-at
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-*alphas* data)
 (staged-*degree* data)
 (staged-*composition-fri-roots* data)
 (staged-*composition-final* data)
 $i \subseteq B$

by (rule

query-header-supported-partial-opening-success-indices-at-subset-if-canonical-indices
 [OF canonical-cover])

lemma *staged-query-header-supported-partial-opening-success-indices-at-subset-if-canonical-query-indices:*

assumes canonical-cover:

query-header-supported-partial-opening-canonical-query-indices
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-*alphas* data)
 (staged-*degree* data)
 (staged-*composition-fri-roots* data)
 (staged-*composition-final* data) $\subseteq B$

shows

query-header-supported-partial-opening-success-indices-at
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-*alphas* data)
 (staged-*degree* data)
 (staged-*composition-fri-roots* data)
 (staged-*composition-final* data)
 $i \subseteq B$

by (rule

query-header-supported-partial-opening-success-indices-at-subset-if-canonical-query-indices
 [OF canonical-cover])

lemma *checked-staged-security-with-data-state-partial-opening-hit-bound-from-canonical-query-error-cover:*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and cover:
   $\bigwedge$  data attacker-state.
    query-header-supported-partial-opening-canonical-query-indices
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)  $\subseteq B$ 
and subset:  $B \subseteq \text{query-sample-space}$ 
and frac:
   $\text{nnreal} (\text{card } B) / \text{nnreal} (\text{card } \text{query-sample-space}) \leq$ 
    query-error-bound
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-opening-hit
    adversary-initial-state  $\leq$ 
      nnreal
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget) * query-error-bound
proof –
have raw-bound: query-index-raw-preimage-bound
  by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
show ?thesis
proof (rule
  checked-staged-security-with-data-state-partial-opening-hit-bound-from-fixed-query-error-cover
  [OF wf controlled - raw-bound subset frac])
fix data attacker-state i
assume i < rounds
show
  query-header-supported-partial-opening-success-indices-at
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    i  $\subseteq B$ 
by (rule
  staged-query-header-supported-partial-opening-success-indices-at-subset-if-canonical-query-indices
  [OF cover])

```

qed
qed

lemma *checked-staged-security-with-data-state-query-bad-bound-from-canonical-query-error-cover:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *cover:*

$\wedge$ *data attacker-state.*

query-header-supported-partial-opening-canonical-query-indices

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data))

(staged-trace-root data)

(staged-trace-fri-roots data)

(staged-trace-final data)

(staged-alphas data)

(staged-degree data)

(staged-composition-fri-roots data)

(staged-composition-final data) $\subseteq B$

and *subset: $B \subseteq \text{query-sample-space}$*

and *frac:*

nnreal (card B) / nnreal (card query-sample-space) $\leq$

query-error-bound

shows

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-query-bad-hit

adversary-initial-state $\leq$

nnreal

(staged-attacker-query-budget budgets +

*staged-challenge-query-budget) * query-error-bound*

proof –

have *partial-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-query-partial-opening-hit

adversary-initial-state $\leq$

nnreal

(staged-attacker-query-budget budgets +

*staged-challenge-query-budget) * query-error-bound*

by (rule

checked-staged-security-with-data-state-partial-opening-hit-bound-from-canonical-query-error-cover

[OF wf controlled cover subset frac])

show *?thesis*

by (rule *order.trans*

[OF checked-staged-security-with-data-state-query-bad-bound-from-partial-opening-hit

[OF wf controlled] partial-bound])

qed

lemma *checked-staged-query-prefix-candidate-opening-target-from-prefix-canonical-indexD:*

assumes *i-bound: $i < \text{rounds}$*

and *hit:*

```

checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-candidate-opening-query-target-from-prefix
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i)
  (Some (((prefix, prefix-state), raw), raw-state))
shows index (to-nat raw) = opening-index (hd trace-openings)
proof -
  have idx-in:
    index (to-nat raw) ∈
      staged-query-prefix-candidate-opening-query-target-from-prefix
        ((replicate rounds []) [i := trace-openings])
        ((replicate rounds []) [i := composition-openings])
        i prefix prefix-state
  using hit unfolding checked-staged-query-prefix-dynamic-index-hit-def
  by simp
  have singleton:
    staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i prefix prefix-state ⊆
      {opening-index
        (hd (((replicate rounds []) [i := trace-openings]) ! i))}
  by (rule
    staged-query-prefix-candidate-opening-query-target-from-prefix-subset-canonical-trace-index)
  have index (to-nat raw) =
    opening-index
      (hd (((replicate rounds []) [i := trace-openings]) ! i))
  using idx-in singleton by auto
  then show ?thesis
  using i-bound by simp
qed

```

lemma *checked-staged-security-with-query-prefix-fixed-header-component-canonical-indexD:*

assumes *i-bound: i < rounds*

and *hit:*

checked-staged-security-with-query-prefix-fixed-header-component
E trace-openings composition-openings i
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))

shows *index (to-nat raw) = opening-index (hd trace-openings)*

proof -

from *checked-staged-security-with-query-prefix-fixed-header-componentE[OF hit]*

have *target:*

index (to-nat raw) ∈
staged-query-prefix-candidate-opening-query-target-from-prefix
 ((replicate rounds []) [i := trace-openings])
 ((replicate rounds []) [i := composition-openings])
 i prefix prefix-state

```

  by blast
have singleton:
  staged-query-prefix-candidate-opening-query-target-from-prefix
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i prefix prefix-state  $\subseteq$ 
    {opening-index
      (hd (((replicate rounds []) [i := trace-openings]) ! i))}
  by (rule
    staged-query-prefix-candidate-opening-query-target-from-prefix-subset-canonical-trace-index)
have index (to-nat raw) =
  opening-index
    (hd (((replicate rounds []) [i := trace-openings]) ! i))
  using target singleton by auto
then show ?thesis
  using i-bound by simp
qed

lemma checked-staged-security-with-query-prefix-fixed-header-component-trace-openings-nonempty:
  assumes i-bound:  $i < \text{rounds}$ 
  and hit:
    checked-staged-security-with-query-prefix-fixed-header-component
      E trace-openings composition-openings i
      (Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state))
  shows trace-openings  $\neq []$ 
proof -
  from checked-staged-security-with-query-prefix-fixed-header-componentE[OF hit]
  have target:
    index (to-nat raw)  $\in$ 
      staged-query-prefix-candidate-opening-query-target-from-prefix
        ((replicate rounds []) [i := trace-openings])
        ((replicate rounds []) [i := composition-openings])
        i prefix prefix-state
  by blast
then have round:
    partial-query-round-consistent
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      (sqp-alphas prefix) i (index (to-nat raw))
  unfolding staged-query-prefix-candidate-opening-query-target-from-prefix-def
    partial-query-success-indices-at-def
  by simp
have trace-indices:
  map opening-index
    (((replicate rounds []) [i := trace-openings]) ! i) =
    powers-scaled (index (to-nat raw))
  using round unfolding partial-query-round-consistent-def by blast
have powers-scaled (index (to-nat raw))  $\neq []$ 

```

```

    unfolding powers-scaled-def using powers-pos by simp
  then have ((replicate rounds []) [i := trace-openings]) ! i ≠ []
    using trace-indices by auto
  then show ?thesis
    using i-bound by simp
qed

```

lemma *checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-canonical-query-indexD*

assumes *i-bound*: $i < \text{rounds}$

and *hit*:

checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
trace-openings composition-openings i
(Some (((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

shows

index (to-nat raw) ∈
query-header-supported-partial-opening-canonical-query-indices
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)

proof –

from *checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitE*
[OF hit]

have *witness*:

(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)

and *target*:

index (to-nat raw) ∈
staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i prefix prefix-state

by *blast+*

have *idx-sample*: *index (to-nat raw) ∈ query-sample-space*

```

using target staged-query-prefix-candidate-opening-query-target-from-prefix-subset
by blast
have round:
  partial-query-round-consistent
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    (sqp-alphas prefix) i (index (to-nat raw))
using target
unfolding staged-query-prefix-candidate-opening-query-target-from-prefix-def
  partial-query-success-indices-at-def
by simp
have trace-indices:
  map opening-index
    (((replicate rounds []) [i := trace-openings]) ! i) =
    powers-scaled (index (to-nat raw))
using round unfolding partial-query-round-consistent-def by blast
have trace-nonempty: trace-openings ≠ []
proof -
  have powers-scaled (index (to-nat raw)) ≠ []
    unfolding powers-scaled-def using powers-pos by simp
  then have ((replicate rounds []) [i := trace-openings]) ! i ≠ []
    using trace-indices by auto
  then show ?thesis
    using i-bound by simp
qed
have singleton:
  staged-query-prefix-candidate-opening-query-target-from-prefix
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i prefix prefix-state ⊆
    {opening-index
     (hd (((replicate rounds []) [i := trace-openings]) ! i))}
  by (rule
    staged-query-prefix-candidate-opening-query-target-from-prefix-subset-canonical-trace-index)
have idx-eq:
  index (to-nat raw) = opening-index (hd trace-openings)
  using target singleton i-bound by auto
have idx-canonical:
  index (to-nat raw) ∈
    query-header-supported-partial-opening-canonical-indices
      (verifier-state-from-adversary attacker-state
       (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)

```



```

unfolding query-header-supported-partial-opening-canonical-indices-def
using witness trace-nonempty idx-eq by blast
show ?thesis
unfolding query-header-supported-partial-opening-canonical-query-indices-def
using idx-sample idx-canonical by simp
qed

lemma checked-staged-security-with-query-prefix-fixed-header-component-canonical-query-indexD:
assumes i-bound:  $i < \text{rounds}$ 
and hit:
  checked-staged-security-with-query-prefix-fixed-header-component
  E trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
shows
  index (to-nat raw)  $\in$ 
  query-header-supported-partial-opening-canonical-query-indices
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
proof –
have header:
  checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
  trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
using hit
unfolding checked-staged-security-with-query-prefix-fixed-header-component-def
by simp
show ?thesis
by (rule
  checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-canonical-query-indexD
  [OF i-bound header])
qed

lemma checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitI-from-dynamic-hit:
assumes witness:
  (trace-openings, composition-openings)  $\in$ 
  query-header-supported-partial-opening-witnesses
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (staged-trace-root data)
  (staged-trace-fri-roots data)

```

```

    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and target-hit:
    checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i)
    (Some (((prefix, prefix-state), raw), raw-state))
  shows
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
    trace-openings composition-openings i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  proof -
    from query-header-supported-partial-opening-witnessesE[OF witness]
    have comp-nonempty: staged-composition-fri-roots data ≠ []
    by blast
    have actual:
      checked-staged-security-with-actual-query-prefix-candidate-opening-hit
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i
      (Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state))
    using target-hit
    unfolding checked-staged-security-with-actual-query-prefix-candidate-opening-hit-def
      checked-staged-security-with-query-prefix-dynamic-index-hit-def
    by simp
  show ?thesis
    unfolding
      checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-def
    using comp-nonempty witness actual by simp
qed

```

lemma *checked-staged-security-with-query-prefix-fixed-header-componentI-from-dynamic-hit:*

assumes *event:*

*E (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))*

and *witness:*

*(trace-openings, composition-openings) ∈
 query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)*

```

    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and target-hit:
    checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i)
    (Some (((prefix, prefix-state), raw), raw-state))
  shows
    checked-staged-security-with-query-prefix-fixed-header-component
      E trace-openings composition-openings i
      (Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state))
  unfolding checked-staged-security-with-query-prefix-fixed-header-component-def
  by (intro conjI event
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitI-from-dynamic-hit
      [OF witness target-hit])

lemma checked-staged-security-with-query-prefix-fixed-header-component-bound-from-budgets:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-fixed-header-component
        E trace-openings composition-openings i)
      adversary-initial-state ≤
      staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      (1::prob) / nnreal (card query-sample-space)
  proof –
  have raw-bound: query-index-raw-preimage-bound
  by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
  show ?thesis
  by (rule
    checked-staged-security-with-query-prefix-fixed-header-component-bound
      [OF raw-bound wf controlled i-bound])
  qed

lemma checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-bound-from-budgets:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  shows

```

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
 trace-openings composition-openings i)
adversary-initial-state $\leq$
staged-phase-relation-error size
 (staged-query-search-queries *budgets* $i + 1$) +
 (1::prob) / nnreal (card *query-sample-space*)

proof –

have *raw-bound: query-index-raw-preimage-bound*
by (rule *query-index-raw-preimage-bound-from-sampler-wellformed*)
show ?thesis
by (rule
 checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-bound
 [OF *raw-bound wf controlled i-bound*])

qed

lemma *checked-staged-query-prefix-candidate-opening-target-from-prefix-hit-bound-from-budgets:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *trace-candidate:*
 partial-trace-table-candidate trace-table trace-openings
and *comp-candidate:*
 partial-composition-table-candidate composition-table
 composition-openings
and *trace-low: trace-table-low-degree trace-table*
and *comp-low: composition-table-low-degree maxDegree composition-table*
and *not-all:*
 $\bigwedge \text{prefix } \text{prefix-state.}$
 Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
 set-dist
 (*execute* (*checked-staged-query-prefix-with-state* A i)
 adversary-initial-state) $\implies$
 $\neg$ *all-queries-consistent trace-table composition-table*
 (*sqp-alphas prefix*)

shows

wp-event (*checked-staged-query-prefix-receive-with-state* A i)
 (*checked-staged-query-prefix-dynamic-index-hit*
 (*staged-query-prefix-candidate-opening-query-target-from-prefix*
 trace-openings composition-openings i))
adversary-initial-state $\leq$
staged-phase-relation-error size
 (staged-query-search-queries *budgets* $i + 1$) + *query-error-bound*

proof –

have *raw-bound: query-index-raw-preimage-bound*
by (rule *query-index-raw-preimage-bound-from-sampler-wellformed*)
show ?thesis
by (rule

```

      checked-staged-query-prefix-candidate-opening-target-from-prefix-hit-bound
      [OF raw-bound wf controlled - trace-candidate comp-candidate
       trace-low comp-low not-all])
    (use i-bound in simp)
qed

end

end

```

```

theory Soundness-Partial-Merkle-Cross
  imports Soundness-Partial-Merkle
begin

```

Cross-state Merkle facts for partial authenticated openings.

The broad query-header candidate relation can be supported by different verifier executions and therefore different final hash-map states. These lemmas isolate the reusable binding fact available in the current model: two authenticated openings for the same root, length, and index cannot carry different values across compatible cleanly-merged states.

```

context soundness
begin

```

definition *partial-openings-cross-cover-all-indices*

```

:: 'f authenticated-opening list list  $\Rightarrow$ 
   'f authenticated-opening list list  $\Rightarrow$  bool

```

where

```

partial-openings-cross-cover-all-indices openings openings'  $\longleftrightarrow$ 
  ( $\forall j < \text{scale} * \text{clength}$ .
    $\exists i < \text{rounds}$ .  $\exists i' < \text{rounds}$ .
     $\exists \text{opn} \in \text{set } (\text{openings} ! i)$ .
     $\exists \text{opn}' \in \text{set } (\text{openings}' ! i')$ .
     $\text{opening-index } \text{opn} = j \wedge$ 
     $\text{opening-index } \text{opn}' = j$ )

```

lemma *partial-openings-cross-cover-all-indicesD*:

assumes

```

  partial-openings-cross-cover-all-indices openings openings'
  and  $j < \text{scale} * \text{clength}$ 

```

obtains $i \ i' \ \text{opn} \ \text{opn}'$ **where**

```

   $i < \text{rounds}$ 
   $i' < \text{rounds}$ 
   $\text{opn} \in \text{set } (\text{openings} ! i)$ 
   $\text{opn}' \in \text{set } (\text{openings}' ! i')$ 
   $\text{opening-index } \text{opn} = j$ 
   $\text{opening-index } \text{opn}' = j$ 

```

using *assms*

unfolding *partial-openings-cross-cover-all-indices-def* **by** *blast*

lemma *partial-openings-cross-cover-all-indices-sym*:

assumes *partial-openings-cross-cover-all-indices openings openings'*

shows *partial-openings-cross-cover-all-indices openings' openings*

using *assms*

unfolding *partial-openings-cross-cover-all-indices-def* **by** *blast*

lemma *not-partial-openings-cross-cover-all-indicesE*:

assumes $\neg$ *partial-openings-cross-cover-all-indices openings openings'*

obtains *j* **where**

j < *scale* * *clength*

$\bigwedge i \ i' \ \text{opn} \ \text{opn}'. \ i < \text{rounds} \implies i' < \text{rounds} \implies$

$\text{opn} \in \text{set} (\text{openings} ! i) \implies$

$\text{opn}' \in \text{set} (\text{openings}' ! i') \implies$

$\text{opening-index} \ \text{opn} = j \implies \text{opening-index} \ \text{opn}' \neq j$

using *assms*

unfolding *partial-openings-cross-cover-all-indices-def* **by** *blast*

lemma *partial-openings-cross-cover-all-indices-imp-left-cover*:

assumes *cover*:

partial-openings-cross-cover-all-indices openings openings'

and *j-bound*: *j* < *scale* * *clength*

shows $\exists i < \text{rounds}. \exists \text{opn} \in \text{set} (\text{openings} ! i).$

$\text{opening-index} \ \text{opn} = j$

using *partial-openings-cross-cover-all-indicesD[OF cover j-bound]* **by** *blast*

lemma *partial-openings-cross-cover-all-indices-imp-right-cover*:

assumes *cover*:

partial-openings-cross-cover-all-indices openings openings'

and *j-bound*: *j* < *scale* * *clength*

shows $\exists i < \text{rounds}. \exists \text{opn} \in \text{set} (\text{openings}' ! i).$

$\text{opening-index} \ \text{opn} = j$

using *partial-openings-cross-cover-all-indicesD[OF cover j-bound]* **by** *blast*

lemma *cross-sparse-openings-obtain-unopened-index*:

assumes *rounds-len*: *length openings* = *rounds*

and *sparse*: *length (List.concat openings)* < *scale* * *clength*

obtains *j* **where**

j < *scale* * *clength*

$\bigwedge i \ \text{opn}. \ i < \text{rounds} \implies \text{opn} \in \text{set} (\text{openings} ! i) \implies$

$\text{opening-index} \ \text{opn} \neq j$

proof –

let *?N* = *scale* * *clength*

let *?I* = *opening-index* ‘ *set (List.concat openings)*

have *finite-I*: *finite ?I*

by *simp*

have *card-I-le*: *card ?I* ≤ *length (List.concat openings)*

proof –

```

have card ?I ≤ card (set (List.concat openings))
  by (rule card-image-le) simp
also have ... ≤ length (List.concat openings)
  by (rule card-length)
finally show ?thesis .
qed
have not-all: ¬ {0.. $?N$ } ⊆ ?I
proof
  assume subset: {0.. $?N$ } ⊆ ?I
  have ?N = card {0.. $?N$ }
    by simp
  also have ... ≤ card ?I
    by (rule card-mono[OF finite-I subset])
  also have ... ≤ length (List.concat openings)
    by (rule card-I-le)
  finally show False
    using sparse by simp
qed
from not-all obtain j where j-in: j ∈ {0.. $?N$ } and j-notin: j ∉ ?I
  by blast
then have j-bound: j < ?N
  by simp
have unopened:
   $\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set } (\text{openings} ! i) \implies$ 
   $\text{opening-index } \text{opn} \neq j$ 
proof -
  fix i opn
  assume i-bound: i < rounds
    and opn-in: opn ∈ set (openings ! i)
  have openings ! i ∈ set openings
    using i-bound rounds-len by simp
  then have opn ∈ set (List.concat openings)
    using opn-in by auto
  then show opening-index opn ≠ j
    using j-notin by auto
qed
show ?thesis
  by (rule that[OF j-bound unopened])
qed

lemma partial-openings-cross-cover-all-indices-impossible-if-left-sparse:
  assumes rounds-len: length openings = rounds
    and sparse: length (List.concat openings) < scale * clength
  shows ¬ partial-openings-cross-cover-all-indices openings openings'
proof
  assume cover:
    partial-openings-cross-cover-all-indices openings openings'
  from cross-sparse-openings-obtain-unopened-index[OF rounds-len sparse]
  obtain j where j-bound: j < scale * clength

```

```

and unopened:
   $\bigwedge i \text{ } \textit{opn}. i < \textit{rounds} \implies \textit{opn} \in \textit{set} (\textit{openings} ! i) \implies$ 
     $\textit{opening-index } \textit{opn} \neq j$ 
by blast
from partial-openings-cross-cover-all-indices-imp-left-cover
  [OF cover j-bound]
obtain i opn where i-bound: i < rounds
  and opn-in: opn ∈ set (openings ! i)
  and opn-idx: opening-index opn = j
by blast
show False
  using unopened[OF i-bound opn-in] opn-idx by simp
qed

lemma partial-openings-cross-cover-all-indices-impossible-if-right-sparse:
  assumes rounds-len: length openings' = rounds
  and sparse: length (List.concat openings') < scale * clength
  shows  $\neg \textit{partial-openings-cross-cover-all-indices } \textit{openings } \textit{openings}'$ 
proof –
  have not-sym:
     $\neg \textit{partial-openings-cross-cover-all-indices } \textit{openings}' \textit{ openings}$ 
    by (rule partial-openings-cross-cover-all-indices-impossible-if-left-sparse
      [OF rounds-len sparse])
  show ?thesis
proof
  assume cover:
    partial-openings-cross-cover-all-indices openings openings'
  have partial-openings-cross-cover-all-indices openings' openings
    by (rule partial-openings-cross-cover-all-indices-sym[OF cover])
  then show False
    using not-sym by contradiction
qed
qed

lemma merkle-path-bound-merkle-mono:
  assumes bound: merkle-path-bound rt len idx v path s
  and ext: merkle-hash-extends s t
  shows merkle-path-bound rt len idx v path t
  using bound
proof (induction path arbitrary: rt len idx)
  case Nil
  then show ?case
    using ext unfolding merkle-hash-extends-def by simp
next
  case (Cons sibling path)
  show ?case
  proof (cases idx < len div 2)
  case True
  then obtain child where

```



```

  child:
    merkle-path-bound child (len div 2) idx v path s
  and lookup:
    fmlookup (HashMap s) (MerkleNode child sibling) = Some rt
  using Cons.premis by auto
have child-t:
  merkle-path-bound child (len div 2) idx v path t
  by (rule Cons.IH[OF child])
have lookup-t:
  fmlookup (HashMap t) (MerkleNode child sibling) = Some rt
  using ext lookup unfolding merkle-hash-extends-def by simp
show ?thesis
  using True child-t lookup-t
  by (auto intro!: exI[of - child])
next
case False
then obtain child where
  child:
    merkle-path-bound child (len div 2) (idx - len div 2)
    v path s
  and lookup:
    fmlookup (HashMap s) (MerkleNode sibling child) = Some rt
  using Cons.premis by auto
have child-t:
  merkle-path-bound child (len div 2) (idx - len div 2)
  v path t
  by (rule Cons.IH[OF child])
have lookup-t:
  fmlookup (HashMap t) (MerkleNode sibling child) = Some rt
  using ext lookup unfolding merkle-hash-extends-def by simp
show ?thesis
  using False child-t lookup-t
  by (auto intro!: exI[of - child])
qed
qed

```

```

lemma authenticated-opening-in-merkle-mono:
  assumes opn-auth: authenticated-opening-in s opn
  and ext: merkle-hash-extends s t
  shows authenticated-opening-in t opn
  using opn-auth merkle-path-bound-merkle-mono[OF - ext]
  unfolding authenticated-opening-in-def by blast

```

```

lemma inconsistent-authenticated-openings-imp-merkle-conflict-or-merge-collision:
  assumes first: authenticated-opening-in s opening
  and second: authenticated-opening-in t opening'
  and same-root: opening-root opening = opening-root opening'
  and same-length: opening-length opening = opening-length opening'
  and same-index: opening-index opening = opening-index opening'

```

```

    and different-value: opening-value opening  $\neq$  opening-value opening'
  shows
    merkle-hash-value-conflict s t  $\vee$ 
      hash-map-output-collision (merkle-hash-state-merge s t)
  proof (cases merkle-hash-value-conflict s t)
    case True
    then show ?thesis by simp
  next
    case no-conflict: False
    have compatible: merkle-hash-maps-compatible s t
      using no-conflict
      unfolding merkle-hash-maps-compatible-iff-no-value-conflict .
    let ?u = merkle-hash-state-merge s t
    have s-ext: merkle-hash-extends s ?u
      by (rule merkle-hash-state-merge-extends-left-if-compatible
        [OF compatible])
    have t-ext: merkle-hash-extends t ?u
      by (rule merkle-hash-state-merge-extends-right)
    have first-u: authenticated-opening-in ?u opening
      by (rule authenticated-opening-in-merkle-mono[OF first s-ext])
    have second-u: authenticated-opening-in ?u opening'
      by (rule authenticated-opening-in-merkle-mono[OF second t-ext])
    have collision: hash-map-output-collision ?u
      by (rule inconsistent-authenticated-openings-imp-hash-collision
        [OF first-u second-u same-root same-length same-index
          different-value])
    then show ?thesis by simp
  qed

lemma authenticated-openings-values-eq-if-no-merkle-conflict-clean-merge:
  assumes first: authenticated-opening-in s opening
    and second: authenticated-opening-in t opening'
    and same-root: opening-root opening = opening-root opening'
    and same-length: opening-length opening = opening-length opening'
    and same-index: opening-index opening = opening-index opening'
    and no-conflict:  $\neg$  merkle-hash-value-conflict s t
    and clean-merge:
       $\neg$  hash-map-output-collision (merkle-hash-state-merge s t)
  shows opening-value opening = opening-value opening'
  proof (rule ccontr)
    assume different:
      opening-value opening  $\neq$  opening-value opening'
    have
      merkle-hash-value-conflict s t  $\vee$ 
        hash-map-output-collision (merkle-hash-state-merge s t)
      by (rule
        inconsistent-authenticated-openings-imp-merkle-conflict-or-merge-collision
        [OF first second same-root same-length same-index different])
    then show False

```

using *no-conflict clean-merge* by *blast*
qed

lemma *partial-authenticated-tables-values-eq-if-no-merkle-conflict-clean-merge*:

assumes *table*:

partial-authenticated-table *rt len openings s*

and *table'*:

partial-authenticated-table *rt len openings' t*

and *opn-in*: *opn* ∈ *set openings*

and *opn'-in*: *opn'* ∈ *set openings'*

and *same-index*: *opening-index opn* = *opening-index opn'*

and *no-conflict*: $\neg$ *merkle-hash-value-conflict s t*

and *clean-merge*:

$\neg$ *hash-map-output-collision (merkle-hash-state-merge s t)*

shows *opening-value opn* = *opening-value opn'*

proof –

have *first*: *authenticated-opening-in s opn*

using *table opn-in unfolding partial-authenticated-table-def* by *blast*

have *second*: *authenticated-opening-in t opn'*

using *table' opn'-in unfolding partial-authenticated-table-def* by *blast*

have *root*: *opening-root opn* = *rt*

using *table opn-in unfolding partial-authenticated-table-def* by *blast*

have *root'*: *opening-root opn'* = *rt*

using *table' opn'-in unfolding partial-authenticated-table-def* by *blast*

have *same-root*: *opening-root opn* = *opening-root opn'*

using *root root'* by *simp*

have *len*: *opening-length opn* = *len*

using *table opn-in unfolding partial-authenticated-table-def* by *blast*

have *len'*: *opening-length opn'* = *len*

using *table' opn'-in unfolding partial-authenticated-table-def* by *blast*

have *same-length*: *opening-length opn* = *opening-length opn'*

using *len len'* by *simp*

show *?thesis*

by (rule *authenticated-openings-values-eq-if-no-merkle-conflict-clean-merge*
[OF *first second same-root same-length same-index no-conflict*
clean-merge])

qed

lemma *partial-trace-openings-values-eq-if-no-merkle-conflict-clean-merge*:

assumes *partial*:

accepted-with-partial-trace-openings s

(*Some (result, final-state)*) fr *query-idxs trace-openings*

and *partial'*:

accepted-with-partial-trace-openings s'

(*Some (result', final-state')*) fr *query-idxs' trace-openings'*

and *i-bound*: *i* < *rounds*

and *i'-bound*: *i'* < *rounds*

and *opn-in*: *opn* ∈ *set (trace-openings ! i)*

and *opn'-in*: *opn'* ∈ *set (trace-openings' ! i')*

```

and same-index: opening-index opn = opening-index opn'
and no-conflict:
  ¬ merkle-hash-value-conflict final-state final-state'
and clean-merge:
  ¬ hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
shows opening-value opn = opening-value opn'
proof –
  have table:
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
  using partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
  have table':
    partial-authenticated-table fr (scale * clength)
      (trace-openings' ! i') final-state'
  using partial' i'-bound
  unfolding accepted-with-partial-trace-openings-def by blast
  show ?thesis
  by (rule
    partial-authenticated-tables-values-eq-if-no-merkle-conflict-clean-merge
    [OF table table' opn-in opn'-in same-index no-conflict
      clean-merge])
qed

lemma partial-composition-openings-values-eq-if-no-merkle-conflict-clean-merge:
assumes partial:
  accepted-with-partial-composition-openings s
    (Some (result, final-state)) composition-root query-idxs
    composition-openings
and partial':
  accepted-with-partial-composition-openings s'
    (Some (result', final-state')) composition-root query-idxs'
    composition-openings'
and i-bound: i < rounds
and i'-bound: i' < rounds
and opn-in: opn ∈ set (composition-openings ! i)
and opn'-in: opn' ∈ set (composition-openings' ! i')
and same-index: opening-index opn = opening-index opn'
and no-conflict:
  ¬ merkle-hash-value-conflict final-state final-state'
and clean-merge:
  ¬ hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
shows opening-value opn = opening-value opn'
proof –
  have table:
    partial-authenticated-table composition-root (scale * clength)
      (composition-openings ! i) final-state

```

```

using partial i-bound
unfolding accepted-with-partial-composition-openings-def by blast
have table':
  partial-authenticated-table composition-root (scale * clength)
    (composition-openings' ! i') final-state'
using partial' i'-bound
unfolding accepted-with-partial-composition-openings-def by blast
show ?thesis
by (rule
  partial-authenticated-tables-values-eq-if-no-merkle-conflict-clean-merge
  [OF table table' opn-in opn'-in same-index no-conflict
    clean-merge])
qed

lemma partial-trace-table-candidates-eq-if-cross-openings-cover-clean-merge:
assumes partial:
  accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs trace-openings
and partial':
  accepted-with-partial-trace-openings s'
    (Some (result', final-state')) fr query-idxs' trace-openings'
and cand: partial-trace-table-candidate trace-table trace-openings
and cand':
  partial-trace-table-candidate trace-table' trace-openings'
and cover:
   $\bigwedge j. j < scale * clength \implies$ 
     $\exists i < rounds. \exists i' < rounds.$ 
     $\exists opn \in set (trace-openings ! i).$ 
     $\exists opn' \in set (trace-openings' ! i').$ 
     $opening-index\ opn = j \wedge$ 
     $opening-index\ opn' = j$ 
and no-conflict:
   $\neg merkle-hash-value-conflict\ final-state\ final-state'$ 
and clean-merge:
   $\neg hash-map-output-collision$ 
    (merkle-hash-state-merge final-state final-state')
shows trace-table = trace-table'
proof (rule nth-equalityI)
show length trace-table = length trace-table'
  using cand cand' unfolding partial-trace-table-candidate-def by simp
next
  fix j
  assume j-bound: j < length trace-table
  have len: length trace-table = scale * clength
    using cand unfolding partial-trace-table-candidate-def by simp
  from cover[OF j-bound[unfolded len]]
  obtain i i' opn opn' where
    i-bound: i < rounds
    and i'-bound: i' < rounds

```

```

and opn-in: opn ∈ set (trace-openings ! i)
and opn'-in: opn' ∈ set (trace-openings' ! i')
and opn-idx: opening-index opn = j
and opn'-idx: opening-index opn' = j
by blast
have trace-val: trace-table ! j = opening-value opn
  using partial-trace-table-candidateD(3)[OF cand i-bound opn-in]
    opn-idx by simp
have trace'-val: trace-table' ! j = opening-value opn'
  using partial-trace-table-candidateD(3)[OF cand' i'-bound opn'-in]
    opn'-idx by simp
have opening-values:
  opening-value opn = opening-value opn'
  by (rule partial-trace-openings-values-eq-if-no-merkle-conflict-clean-merge
    [OF partial partial' i-bound i'-bound opn-in opn'-in - no-conflict
      clean-merge])
    (use opn-idx opn'-idx in simp)
show trace-table ! j = trace-table' ! j
  using trace-val trace'-val opening-values by simp
qed

```

lemma *partial-trace-table-candidates-eq-if-cross-cover-clean-merge*:

```

assumes partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr query-idxs trace-openings
and partial':
  accepted-with-partial-trace-openings s'
  (Some (result', final-state')) fr query-idxs' trace-openings'
and cand: partial-trace-table-candidate trace-table trace-openings
and cand':
  partial-trace-table-candidate trace-table' trace-openings'
and cover:
  partial-openings-cross-cover-all-indices trace-openings
  trace-openings'
and no-conflict:
  ¬ merkle-hash-value-conflict final-state final-state'
and clean-merge:
  ¬ hash-map-output-collision
  (merkle-hash-state-merge final-state final-state')
shows trace-table = trace-table'

```

```

proof (rule
  partial-trace-table-candidates-eq-if-cross-openings-cover-clean-merge
  [OF partial partial' cand cand' - no-conflict clean-merge])
fix j
assume j < scale * clength
from partial-openings-cross-cover-all-indicesD[OF cover this]
obtain i i' opn opn' where
  i < rounds i' < rounds
  opn ∈ set (trace-openings ! i)

```

```

    opn' ∈ set (trace-openings' ! i')
    opening-index opn = j
    opening-index opn' = j
  by blast
then show
  ∃ i < rounds. ∃ i' < rounds.
    ∃ opn ∈ set (trace-openings ! i).
    ∃ opn' ∈ set (trace-openings' ! i').
      opening-index opn = j ∧ opening-index opn' = j
  by blast
qed

lemma partial-composition-table-candidates-eq-if-cross-openings-cover-clean-merge:
  assumes partial:
    accepted-with-partial-composition-openings s
    (Some (result, final-state)) composition-root query-idxs
    composition-openings
  and partial':
    accepted-with-partial-composition-openings s'
    (Some (result', final-state')) composition-root query-idxs'
    composition-openings'
  and cand:
    partial-composition-table-candidate composition-table
    composition-openings
  and cand':
    partial-composition-table-candidate composition-table'
    composition-openings'
  and cover:
    ∧ j. j < scale * clength ⇒
      ∃ i < rounds. ∃ i' < rounds.
        ∃ opn ∈ set (composition-openings ! i).
        ∃ opn' ∈ set (composition-openings' ! i').
          opening-index opn = j ∧
          opening-index opn' = j
  and no-conflict:
    ¬ merkle-hash-value-conflict final-state final-state'
  and clean-merge:
    ¬ hash-map-output-collision
    (merkle-hash-state-merge final-state final-state')
  shows composition-table = composition-table'
proof (rule nth-equalityI)
  show length composition-table = length composition-table'
  using cand cand'
  unfolding partial-composition-table-candidate-def by simp
next
fix j
assume j-bound: j < length composition-table
have len: length composition-table = scale * clength
  using cand unfolding partial-composition-table-candidate-def by simp

```

```

from cover[OF j-bound[unfolded len]]
obtain i i' opn opn' where
  i-bound: i < rounds
  and i'-bound: i' < rounds
  and opn-in: opn ∈ set (composition-openings ! i)
  and opn'-in: opn' ∈ set (composition-openings' ! i')
  and opn-idx: opening-index opn = j
  and opn'-idx: opening-index opn' = j
  by blast
have composition-val:
  composition-table ! j = opening-value opn
  using partial-composition-table-candidateD(3)[OF cand i-bound opn-in]
    opn-idx by simp
have composition'-val:
  composition-table' ! j = opening-value opn'
  using partial-composition-table-candidateD(3)[OF cand' i'-bound opn'-in]
    opn'-idx by simp
have opening-values:
  opening-value opn = opening-value opn'
  by (rule
    partial-composition-openings-values-eq-if-no-merkle-conflict-clean-merge
    [OF partial partial' i-bound i'-bound opn-in opn'-in - no-conflict
      clean-merge])
    (use opn-idx opn'-idx in simp)
show composition-table ! j = composition-table' ! j
  using composition-val composition'-val opening-values by simp
qed

```

lemma *partial-composition-table-candidates-eq-if-cross-cover-clean-merge*:

```

assumes partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) composition-root query-idxs
  composition-openings
and partial':
  accepted-with-partial-composition-openings s'
  (Some (result', final-state')) composition-root query-idxs'
  composition-openings'
and cand:
  partial-composition-table-candidate composition-table
  composition-openings
and cand':
  partial-composition-table-candidate composition-table'
  composition-openings'
and cover:
  partial-openings-cross-cover-all-indices composition-openings
  composition-openings'
and no-conflict:
  ¬ merkle-hash-value-conflict final-state final-state'
and clean-merge:

```



```

    ¬ hash-map-output-collision
      (merkle-hash-state-merge final-state final-state')
  shows composition-table = composition-table'
proof (rule
  partial-composition-table-candidates-eq-if-cross-openings-cover-clean-merge
    [OF partial partial' cand cand' - no-conflict clean-merge])
  fix j
  assume j < scale * clength
  from partial-openings-cross-cover-all-indicesD[OF cover this]
  obtain i i' opn opn' where
    i < rounds i' < rounds
    opn ∈ set (composition-openings ! i)
    opn' ∈ set (composition-openings' ! i')
    opening-index opn = j
    opening-index opn' = j
  by blast
  then show
    ∃ i < rounds. ∃ i' < rounds.
      ∃ opn ∈ set (composition-openings ! i).
        ∃ opn' ∈ set (composition-openings' ! i').
          opening-index opn = j ∧ opening-index opn' = j
  by blast
qed

end

end

```

```

theory Soundness-Query-Header-Cross-Candidate
imports
  Soundness-Reductions-Composition
  Soundness-Partial-Merkle-Cross
begin

```

Witnessed query-header partial candidates.

The public query-header candidate set hides the authenticated opening witnesses behind existentials. This layer exposes those witnesses so that the cross-state partial-Merkle binding lemmas can be applied before reducing back to the old candidate-set interface.

```

context soundness
begin

```

definition query-header-supported-partial-table-candidate-witness

```

:: ('f, 'a) protocol-channel-scheme ⇒
  'f ⇒ 'f list ⇒ 'f ⇒ 'f list ⇒ 'f ⇒
  'f list ⇒ 'f ⇒ 'f list ⇒ 'f list ⇒
  ('f, 'a) protocol-channel-scheme ⇒
  'f authenticated-opening list list ⇒
  'f authenticated-opening list list ⇒ bool

```

where

query-header-supported-partial-table-candidate-witness s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state trace-openings composition-openings
 $\longleftrightarrow$
composition-fri-roots $\neq [] \wedge$
 $(\exists \text{result trace-query-idxs composition-query-idxs rest.}$
Some (result, final-state) $\in$ set-dist (execute verify-monad s) $\wedge$
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr trace-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
accepted-with-partial-composition-openings s
(Some (result, final-state)) (hd composition-fri-roots)
composition-query-idxs composition-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest)

lemma *query-header-supported-partial-table-candidate-witness-imp-state:*

assumes

query-header-supported-partial-table-candidate-witness s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state trace-openings composition-openings

shows

query-header-supported-partial-table-candidate-state s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state

using *assms*

unfolding *query-header-supported-partial-table-candidate-witness-def*

query-header-supported-partial-table-candidate-state-def

by *blast*

lemma *query-header-supported-partial-table-candidate-state-obtain-witness:*

assumes

query-header-supported-partial-table-candidate-state s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state

obtains *trace-openings composition-openings* **where**

query-header-supported-partial-table-candidate-witness s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state trace-openings composition-openings

using *assms*

unfolding *query-header-supported-partial-table-candidate-witness-def*

query-header-supported-partial-table-candidate-state-def

by *blast*

lemma *query-header-supported-partial-table-candidate-witness-imp-candidate:*

assumes

query-header-supported-partial-table-candidate-witness s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state trace-openings composition-openings

shows

(trace-table, composition-table) ∈
query-header-supported-partial-table-candidates s fr f-fri-roots
f-final as dg composition-fri-roots final

using *query-header-supported-partial-table-candidate-witness-imp-state*
[OF *assms*]

query-header-supported-partial-table-candidates-iff-state

by *blast*

lemma *query-header-supported-partial-table-candidates-obtain-witness:*

assumes

(trace-table, composition-table) ∈
query-header-supported-partial-table-candidates s fr f-fri-roots
f-final as dg composition-fri-roots final

obtains *final-state trace-openings composition-openings* **where**

query-header-supported-partial-table-candidate-witness s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state trace-openings composition-openings

proof —

from *assms* **obtain** *final-state* **where** *state:*

query-header-supported-partial-table-candidate-state s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state

using *query-header-supported-partial-table-candidates-iff-state* **by** *blast*

from *query-header-supported-partial-table-candidate-state-obtain-witness*
[OF *state*]

obtain *trace-openings composition-openings* **where** *witness:*

query-header-supported-partial-table-candidate-witness s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state trace-openings composition-openings

by *blast*

show *?thesis*

by (*rule that*[OF *witness*])

qed

lemma *query-header-supported-partial-table-candidate-witness-shapes:*

assumes *witness:*

query-header-supported-partial-table-candidate-witness s fr f-fri-roots
f-final as dg composition-fri-roots final trace-table
composition-table final-state trace-openings composition-openings

shows

partial-trace-table-candidate trace-table trace-openings
partial-composition-table-candidate composition-table composition-openings

using *witness*

unfolding *query-header-supported-partial-table-candidate-witness-def*

by blast+

lemma *query-header-supported-partial-table-candidate-witness-eq-if-cross-cover-clean-merge:*

assumes *witness:*

query-header-supported-partial-table-candidate-witness s fr f-fri-roots

f-final as dg composition-fri-roots final trace-table

composition-table final-state trace-openings composition-openings

and *witness':*

query-header-supported-partial-table-candidate-witness s fr f-fri-roots

f-final as dg composition-fri-roots final trace-table'

composition-table' final-state' trace-openings' composition-openings'

and *trace-cover:*

partial-openings-cross-cover-all-indices trace-openings

trace-openings'

and *composition-cover:*

partial-openings-cross-cover-all-indices composition-openings

composition-openings'

and *no-conflict:*

$\neg$ *merkle-hash-value-conflict final-state final-state'*

and *clean-merge:*

$\neg$ *hash-map-output-collision*

(merkle-hash-state-merge final-state final-state')

shows *trace-table = trace-table' $\wedge$ composition-table = composition-table'*

proof –

from *witness* **obtain** *result trace-query-idxs composition-query-idxs rest* **where**

trace-partial:

accepted-with-partial-trace-openings s

(Some (result, final-state)) fr trace-query-idxs trace-openings

and *comp-partial:*

accepted-with-partial-composition-openings s

(Some (result, final-state)) (hd composition-fri-roots)

composition-query-idxs composition-openings

and *trace-cand:*

partial-trace-table-candidate trace-table trace-openings

and *comp-cand:*

partial-composition-table-candidate composition-table

composition-openings

unfolding *query-header-supported-partial-table-candidate-witness-def*

by *blast*

from *witness'* **obtain** *result' trace-query-idxs' composition-query-idxs' rest'*

where *trace-partial':*

accepted-with-partial-trace-openings s

(Some (result', final-state')) fr trace-query-idxs'

trace-openings'

and *comp-partial':*

accepted-with-partial-composition-openings s

(Some (result', final-state')) (hd composition-fri-roots)

composition-query-idxs' composition-openings'

and *trace-cand':*

```

    partial-trace-table-candidate trace-table' trace-openings'
  and comp-cand':
    partial-composition-table-candidate composition-table'
    composition-openings'
  unfolding query-header-supported-partial-table-candidate-witness-def
  by blast
  have trace-eq: trace-table = trace-table'
  by (rule partial-trace-table-candidates-eq-if-cross-cover-clean-merge
    [OF trace-partial trace-partial' trace-cand trace-cand' trace-cover
      no-conflict clean-merge])
  have comp-eq: composition-table = composition-table'
  by (rule partial-composition-table-candidates-eq-if-cross-cover-clean-merge
    [OF comp-partial comp-partial' comp-cand comp-cand'
      composition-cover no-conflict clean-merge])
  show ?thesis
  using trace-eq comp-eq by simp
qed

lemma query-header-supported-partial-table-candidates-subset-singleton-if-pairwise-cross-cover-clean-merge:
  assumes pairwise:
     $\bigwedge$  trace-table composition-table final-state trace-openings
      composition-openings trace-table' composition-table' final-state'
      trace-openings' composition-openings'.
    query-header-supported-partial-table-candidate-witness s fr
      f-fri-roots f-final as dg composition-fri-roots final trace-table
      composition-table final-state trace-openings composition-openings
   $\implies$ 
    query-header-supported-partial-table-candidate-witness s fr
      f-fri-roots f-final as dg composition-fri-roots final trace-table'
      composition-table' final-state' trace-openings' composition-openings'
   $\implies$ 
    partial-openings-cross-cover-all-indices trace-openings
      trace-openings'  $\wedge$ 
    partial-openings-cross-cover-all-indices composition-openings
      composition-openings'  $\wedge$ 
     $\neg$  merkle-hash-value-conflict final-state final-state'  $\wedge$ 
     $\neg$  hash-map-output-collision
      (merkle-hash-state-merge final-state final-state')
  shows
     $\exists$  trace-table composition-table.
    query-header-supported-partial-table-candidates s fr f-fri-roots
      f-final as dg composition-fri-roots final  $\subseteq$ 
      {(trace-table, composition-table)}
  proof (cases
    query-header-supported-partial-table-candidates s fr f-fri-roots
      f-final as dg composition-fri-roots final = {})
  case True
  then show ?thesis
  by (intro exI[of -] exI[of -]) simp

```

```

next
  case False
  then obtain pair0 where pair0-in:
    pair0 ∈
      query-header-supported-partial-table-candidates s fr f-fri-roots
      f-final as dg composition-fri-roots final
    by blast
  then obtain trace-table0 composition-table0 where pair0-eq:
    pair0 = (trace-table0, composition-table0)
    by (cases pair0)
  have cand0:
    (trace-table0, composition-table0) ∈
      query-header-supported-partial-table-candidates s fr f-fri-roots
      f-final as dg composition-fri-roots final
    using pair0-in unfolding pair0-eq .
  from query-header-supported-partial-table-candidates-obtain-witness[OF cand0]
  obtain final-state0 trace-openings0 composition-openings0 where witness0:
    query-header-supported-partial-table-candidate-witness s fr f-fri-roots
    f-final as dg composition-fri-roots final trace-table0
    composition-table0 final-state0 trace-openings0
    composition-openings0
  by blast
  have subset:
    query-header-supported-partial-table-candidates s fr f-fri-roots
    f-final as dg composition-fri-roots final ⊆
      {(trace-table0, composition-table0)}
  proof
    fix pair
    assume pair-in:
      pair ∈
        query-header-supported-partial-table-candidates s fr f-fri-roots
        f-final as dg composition-fri-roots final
    obtain trace-table composition-table where pair-eq:
      pair = (trace-table, composition-table)
      by (cases pair)
    from query-header-supported-partial-table-candidates-obtain-witness
      [OF pair-in[unfolded pair-eq]]
    obtain final-state trace-openings composition-openings where witness:
      query-header-supported-partial-table-candidate-witness s fr
      f-fri-roots f-final as dg composition-fri-roots final trace-table
      composition-table final-state trace-openings composition-openings
    by blast
  have pairwise-facts:
    partial-openings-cross-cover-all-indices trace-openings
    trace-openings0
    partial-openings-cross-cover-all-indices composition-openings
    composition-openings0
    ¬ merkle-hash-value-conflict final-state final-state0
    ¬ hash-map-output-collision

```

```

    (merkle-hash-state-merge final-state final-state0)
  using pairwise[OF witness witness0] by simp-all
have eq:
  trace-table = trace-table0  $\wedge$  composition-table = composition-table0
  by (rule
    query-header-supported-partial-table-candidate-witness-eq-if-cross-cover-clean-merge
    [OF witness witness0 pairwise-facts])
show pair  $\in$  {(trace-table0, composition-table0)}
  using pair-eq eq by simp
qed
then show ?thesis
  by blast
qed

lemma query-header-supported-partial-table-candidates-not-singleton-imp-cross-cover-or-merkle-bad:
  assumes not-unique:
     $\neg$  ( $\exists$  trace-table composition-table.
      query-header-supported-partial-table-candidates s fr f-fri-roots
        f-final as dg composition-fri-roots final  $\subseteq$ 
        {(trace-table, composition-table)}))
  shows
     $\exists$  trace-table composition-table final-state trace-openings
      composition-openings trace-table' composition-table' final-state'
      trace-openings' composition-openings'.
      query-header-supported-partial-table-candidate-witness s fr
        f-fri-roots f-final as dg composition-fri-roots final trace-table
        composition-table final-state trace-openings composition-openings  $\wedge$ 
      query-header-supported-partial-table-candidate-witness s fr
        f-fri-roots f-final as dg composition-fri-roots final trace-table'
        composition-table' final-state' trace-openings'
        composition-openings'  $\wedge$ 
      (trace-table  $\neq$  trace-table'  $\vee$ 
        composition-table  $\neq$  composition-table')  $\wedge$ 
      ( $\neg$  partial-openings-cross-cover-all-indices trace-openings
        trace-openings'  $\vee$ 
         $\neg$  partial-openings-cross-cover-all-indices composition-openings
        composition-openings'  $\vee$ 
        merkle-hash-value-conflict final-state final-state'  $\vee$ 
        hash-map-output-collision
        (merkle-hash-state-merge final-state final-state'))
  proof -
    let ?C =
      query-header-supported-partial-table-candidates s fr f-fri-roots
        f-final as dg composition-fri-roots final
    have nonempty: ?C  $\neq$  {}
      using not-unique by auto
    then obtain pair where pair-in: pair  $\in$  ?C
      by blast
    obtain trace-table composition-table where pair-eq:

```

pair = (*trace-table*, *composition-table*)
 by (*cases pair*)
have *cand*:
 (*trace-table*, *composition-table*) ∈ ?*C*
 using *pair-in* **unfolding** *pair-eq* .
have *not-subset*: ¬ ?*C* ⊆ {(*trace-table*, *composition-table*)}
 using *not-unique* **by** *blast*
then obtain *pair'* **where** *pair'-in*: *pair'* ∈ ?*C*
 and *pair'-not*: *pair'* ∉ {(*trace-table*, *composition-table*)}
 by *auto*
obtain *trace-table'* *composition-table'* **where** *pair'-eq*:
 pair' = (*trace-table'*, *composition-table'*)
 by (*cases pair'*)
have *cand'*:
 (*trace-table'*, *composition-table'*) ∈ ?*C*
 using *pair'-in* **unfolding** *pair'-eq* .
have *distinct*:
 trace-table ≠ *trace-table'* ∨
 composition-table ≠ *composition-table'*
 using *pair'-not* **unfolding** *pair'-eq* **by** *auto*
from *query-header-supported-partial-table-candidates-obtain-witness*[*OF cand*]
obtain *final-state* *trace-openings* *composition-openings* **where** *witness*:
 query-header-supported-partial-table-candidate-witness s fr
 f-fri-roots f-final as dg composition-fri-roots final trace-table
 composition-table final-state trace-openings composition-openings
 by *blast*
from *query-header-supported-partial-table-candidates-obtain-witness*[*OF cand'*]
obtain *final-state'* *trace-openings'* *composition-openings'* **where** *witness'*:
 query-header-supported-partial-table-candidate-witness s fr
 f-fri-roots f-final as dg composition-fri-roots final trace-table'
 composition-table' final-state' trace-openings'
 composition-openings'
 by *blast*
have *side*:
 ¬ *partial-openings-cross-cover-all-indices trace-openings*
 trace-openings' ∨
 ¬ *partial-openings-cross-cover-all-indices composition-openings*
 composition-openings' ∨
 merkle-hash-value-conflict final-state final-state' ∨
 hash-map-output-collision
 (*merkle-hash-state-merge final-state final-state'*)
proof (*rule ccontr*)
assume *no-side*:
 ¬ (¬ *partial-openings-cross-cover-all-indices trace-openings*
 trace-openings' ∨
 ¬ *partial-openings-cross-cover-all-indices composition-openings*
 composition-openings' ∨
 merkle-hash-value-conflict final-state final-state' ∨
 hash-map-output-collision)


```

    (merkle-hash-state-merge final-state final-state'))
have eq:
  trace-table = trace-table' ∧
  composition-table = composition-table'
by (rule
  query-header-supported-partial-table-candidate-witness-eq-if-cross-cover-clean-merge
  [OF witness witness'])
  (use no-side in auto)
then show False
  using distinct by simp
qed
show ?thesis
  using witness witness' distinct side by blast
qed

```

definition query-header-supported-partial-pairwise-cross-or-merkle-bad

```

:: ('f, 'a) protocol-channel-scheme ⇒
  'f ⇒ 'f list ⇒ 'f ⇒ 'f list ⇒ 'f ⇒
  'f list ⇒ 'f ⇒ bool

```

where

```

query-header-supported-partial-pairwise-cross-or-merkle-bad s fr
  f-fri-roots f-final as dg composition-fri-roots final ←→
  (∃ trace-table composition-table final-state trace-openings
    composition-openings trace-table' composition-table' final-state'
    trace-openings' composition-openings'.
  query-header-supported-partial-table-candidate-witness s fr
    f-fri-roots f-final as dg composition-fri-roots final trace-table
    composition-table final-state trace-openings composition-openings ∧
  query-header-supported-partial-table-candidate-witness s fr
    f-fri-roots f-final as dg composition-fri-roots final trace-table'
    composition-table' final-state' trace-openings'
    composition-openings' ∧
  (trace-table ≠ trace-table' ∨
    composition-table ≠ composition-table') ∧
  (¬ partial-openings-cross-cover-all-indices trace-openings
    trace-openings' ∨
    ¬ partial-openings-cross-cover-all-indices composition-openings
    composition-openings' ∨
    merkle-hash-value-conflict final-state final-state' ∨
    hash-map-output-collision
    (merkle-hash-state-merge final-state final-state'))))

```

definition query-supported-partial-pairwise-cross-or-merkle-bad

```

:: ('f, 'a) protocol-channel-scheme ⇒ bool

```

where

```

query-supported-partial-pairwise-cross-or-merkle-bad s ←→
  (∃ fr f-fri-roots f-final as dg composition-fri-roots final.
  query-header-supported-partial-pairwise-cross-or-merkle-bad s fr
    f-fri-roots f-final as dg composition-fri-roots final)

```

lemma *query-header-supported-partial-pairwise-cross-or-merkle-bad-if-not-singleton:*
assumes *not-unique:*
 $\neg (\exists \text{ trace-table } \text{composition-table.}$
 $\text{query-header-supported-partial-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots final } \subseteq$
 $\{(\text{trace-table}, \text{composition-table})\})$
shows
 $\text{query-header-supported-partial-pairwise-cross-or-merkle-bad } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
unfolding *query-header-supported-partial-pairwise-cross-or-merkle-bad-def*
using
 $\text{query-header-supported-partial-table-candidates-not-singleton-imp-cross-cover-or-merkle-bad}$
 $[OF \text{ not-unique}]$
by *blast*

lemma *query-header-supported-partial-table-candidates-subset-singleton-if-no-cross-or-merkle-bad:*
assumes *no-bad:*
 $\neg \text{query-header-supported-partial-pairwise-cross-or-merkle-bad } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
shows
 $\exists \text{ trace-table } \text{composition-table.}$
 $\text{query-header-supported-partial-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots final } \subseteq$
 $\{(\text{trace-table}, \text{composition-table})\}$
proof (rule *ccontr*)
assume *not-unique:*
 $\neg (\exists \text{ trace-table } \text{composition-table.}$
 $\text{query-header-supported-partial-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg composition-fri-roots final } \subseteq$
 $\{(\text{trace-table}, \text{composition-table})\})$
then have
 $\text{query-header-supported-partial-pairwise-cross-or-merkle-bad } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
unfolding *query-header-supported-partial-pairwise-cross-or-merkle-bad-def*
using
 $\text{query-header-supported-partial-table-candidates-not-singleton-imp-cross-cover-or-merkle-bad}$
by *blast*
then show *False*
using *no-bad by contradiction*
qed

lemma *query-supported-partial-pairwise-cross-or-merkle-badD:*
assumes
 $\text{query-supported-partial-pairwise-cross-or-merkle-bad } s$
obtains *fr f-fri-roots f-final as dg composition-fri-roots final* **where**
 $\text{query-header-supported-partial-pairwise-cross-or-merkle-bad } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
using *assms*

unfolding *query-supported-partial-pairwise-cross-or-merkle-bad-def* **by** *blast*

lemma *query-header-supported-partial-table-candidates-subset-singleton-if-no-global-cross-or-merkle-bad*:

assumes *no-bad*:

$\neg$ *query-supported-partial-pairwise-cross-or-merkle-bad* *s*

shows

$\exists$ *trace-table composition-table.*
query-header-supported-partial-table-candidates *s* *fr* *f-fri-roots*
f-final *as* *dg composition-fri-roots final* $\subseteq$
 $\{(trace-table, composition-table)\}$

proof (rule

query-header-supported-partial-table-candidates-subset-singleton-if-no-cross-or-merkle-bad)

show

$\neg$ *query-header-supported-partial-pairwise-cross-or-merkle-bad* *s* *fr*
f-fri-roots *f-final* *as* *dg composition-fri-roots final*

using *no-bad*

unfolding *query-supported-partial-pairwise-cross-or-merkle-bad-def* **by** *blast*

qed

lemma *query-header-supported-partial-union-good-sets-fraction-bound-if-no-cross-or-merkle-bad*:

assumes *no-bad*:

$\neg$ *query-header-supported-partial-pairwise-cross-or-merkle-bad* *s* *fr*
f-fri-roots *f-final* *as* *dg composition-fri-roots final*

and *subset*:

$\bigwedge$ *trace-table composition-table* *as.*
good-sets *trace-table composition-table* *as* $\subseteq$ *query-sample-space*

and *bounded*:

$\bigwedge$ *trace-table composition-table* *as.*
 $nnreal (card (good-sets trace-table composition-table as)) /$
 $nnreal (card query-sample-space) \leq query-error-bound$

shows

nnreal
 $(card$
 $(query-header-supported-partial-union-good-sets s good-sets fr$
 $f-fri-roots f-final as dg composition-fri-roots final)) /$
 $nnreal (card query-sample-space) \leq query-error-bound$

proof –

have *unique*:

$\exists$ *trace-table composition-table.*
query-header-supported-partial-table-candidates *s* *fr* *f-fri-roots*
f-final *as* *dg composition-fri-roots final* $\subseteq$
 $\{(trace-table, composition-table)\}$

by (rule

query-header-supported-partial-table-candidates-subset-singleton-if-no-cross-or-merkle-bad
 $[OF no-bad]$)

show *?thesis*

by (rule

query-header-supported-partial-union-good-sets-fraction-bound-if-unique-candidate
 $[OF unique subset bounded]$)

qed

lemma *query-header-supported-partial-union-good-sets-fraction-bound-if-no-global-cross-or-merkle-bad:*

assumes *no-bad:*

$\neg$ *query-supported-partial-pairwise-cross-or-merkle-bad s*

and *subset:*

$\bigwedge$ *trace-table composition-table as.*

good-sets trace-table composition-table as $\subseteq$ *query-sample-space*

and *bounded:*

$\bigwedge$ *trace-table composition-table as.*

$\text{nnreal } (\text{card } (\text{good-sets trace-table composition-table as})) /$

$\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$

shows

nnreal

$(\text{card}$

$(\text{query-header-supported-partial-union-good-sets } s \text{ good-sets fr}$

$\text{f-fri-roots f-final as dg composition-fri-roots final})) /$

$\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$

proof (rule

query-header-supported-partial-union-good-sets-fraction-bound-if-no-cross-or-merkle-bad

[*OF - subset bounded*])

show

$\neg$ *query-header-supported-partial-pairwise-cross-or-merkle-bad s fr*

f-fri-roots f-final as dg composition-fri-roots final

using *no-bad*

unfolding *query-supported-partial-pairwise-cross-or-merkle-bad-def* **by** *blast*

qed

lemma *wp-query-index-round-set-hit-bound-if-no-global-cross-or-merkle-bad:*

fixes *H :: prob*

assumes *future: query-future-fresh s*

and *raw-bound: query-index-raw-preimage-bound*

and *no-bad:*

$\neg$ *query-supported-partial-pairwise-cross-or-merkle-bad s*

and *subset:*

$\bigwedge$ *trace-table composition-table as.*

good-sets trace-table composition-table as $\subseteq$ *query-sample-space*

and *bounded:*

$\bigwedge$ *trace-table composition-table as.*

$\text{nnreal } (\text{card } (\text{good-sets trace-table composition-table as})) /$

$\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$

and *collision-bound:*

wp-event verify-monad (hash-map-output-collision-bad s) s $\leq H$

shows

wp-event verify-monad (query-index-round-set-hit s good-sets) s $\leq$

$\text{nnreal rounds} * \text{query-error-bound} + H$

proof (rule

wp-query-index-round-set-hit-bound-via-supported-partial-union-or-collision

[*OF future raw-bound - collision-bound*])

```

fix fr f-fri-roots f-final as dg composition-fri-roots final
show
  nnreal
    (card
      (query-header-supported-partial-union-good-sets s good-sets fr
        f-fri-roots f-final as dg composition-fri-roots final)) /
    nnreal (card query-sample-space) ≤ query-error-bound
by (rule
  query-header-supported-partial-union-good-sets-fraction-bound-if-no-global-cross-or-merkle-bad
    [OF no-bad subset bounded])
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Query-Header-Rounds
imports
  Staged-Security-Experiment-Composition-Query-Prefix-Witness
  Soundness-Query-Header-Cross-Candidate
begin

```

Per-round decomposition for the header-authenticated composition query event.

The global event existentially ranges over the query round and authenticated opening witnesses. This layer peels off the finite round existential. The remaining witness selection problem is deliberately left explicit: later proofs must either choose prefix-fixed witnesses before the query challenge or charge failures to Merkle/collision side events.

```

context soundness
begin

```

```

definition staged-query-prefix-header-supported-query-target
  :: 'f staged-query-prefix-data ⇒ 'f protocol-channel ⇒ nat set
where
  staged-query-prefix-header-supported-query-target prefix prefix-state =
    query-header-supported-partial-union-good-sets prefix-state
      query-sampling-success-space
      (sqp-trace-root prefix)
      (sqp-trace-fri-roots prefix)
      (sqp-trace-final prefix)
      (sqp-alphas prefix)
      (sqp-degree prefix)
      (sqp-composition-fri-roots prefix)
      (sqp-composition-final prefix)

```

```

lemma staged-query-prefix-header-supported-query-target-subset:
  staged-query-prefix-header-supported-query-target prefix prefix-state
    ⊆ query-sample-space
unfolding staged-query-prefix-header-supported-query-target-def

```

by (rule query-header-supported-partial-union-good-sets-subset)

lemma staged-query-prefix-header-supported-query-target-fraction-bound-if-no-cross-or-merkle-bad:

assumes no-bad:

¬ query-supported-partial-pairwise-cross-or-merkle-bad prefix-state

shows

nnreal

(card

(staged-query-prefix-header-supported-query-target prefix
prefix-state)) /

nnreal (card query-sample-space) ≤ query-error-bound

unfolding staged-query-prefix-header-supported-query-target-def

by (rule

query-header-supported-partial-union-good-sets-fraction-bound-if-no-global-cross-or-merkle-bad

[OF no-bad query-sampling-success-space-subset

query-sampling-success-space-fraction-bound-query-sample-space])

lemma checked-staged-query-prefix-header-supported-query-target-prehit-bound:

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and i-bound: $i \leq \text{rounds}$

shows

wp-event (checked-staged-query-prefix-receive-with-state A i)

(checked-staged-query-prefix-dynamic-index-prehit
staged-query-prefix-header-supported-query-target)

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1)

proof (rule checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program)

show HashMap adversary-initial-state = fmempty

by (simp add: adversary-initial-state-def)

show hash-relation-program

(checked-staged-query-prefix-dynamic-index-prehit-relation A i
adversary-initial-state

staged-query-prefix-header-supported-query-target)

size (staged-query-search-queries budgets i + 1)

(checked-staged-query-prefix-receive-with-state A i)

by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state

[OF wf controlled i-bound])

(rule

checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)

qed

lemma checked-staged-query-prefix-header-supported-query-target-hit-bound-from-budgets:

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and i-bound: $i < \text{rounds}$

and no-bad:

$\bigwedge \text{prefix prefix-state.}$

Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) ⇒
 $\neg$ *query-supported-partial-pairwise-cross-or-merkle-bad*
prefix-state
shows
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
staged-query-prefix-header-supported-query-target)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) + query-error-bound
proof –
have *raw-bound: query-index-raw-preimage-bound*
by *(rule query-index-raw-preimage-bound-from-sampler-wellformed)*
have *hit-bound:*
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
staged-query-prefix-header-supported-query-target)
adversary-initial-state ≤
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-prehit
staged-query-prefix-header-supported-query-target)
adversary-initial-state + query-error-bound
proof *(rule*
checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
[OF raw-bound])
fix *prefix prefix-state*
assume *support:*
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state)
show
staged-query-prefix-header-supported-query-target prefix prefix-state
 $\subseteq$ *query-sample-space* $\wedge$
nnreal
(card
(staged-query-prefix-header-supported-query-target prefix
prefix-state)) /
nnreal (card query-sample-space) ≤ query-error-bound
using
staged-query-prefix-header-supported-query-target-subset
[of prefix prefix-state]
staged-query-prefix-header-supported-query-target-fraction-bound-if-no-cross-or-merkle-bad
[of prefix-state prefix, OF no-bad[OF support]]
by *blast*
qed

have *prehit-bound*:
wp-event (*checked-staged-query-prefix-receive-with-state* *A i*)
(checked-staged-query-prefix-dynamic-index-prehit
staged-query-prefix-header-supported-query-target)
adversary-initial-state $\leq$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1)
by (*rule*
checked-staged-query-prefix-header-supported-query-target-prehit-bound
[OF wf controlled])
(use i-bound in simp)
show *?thesis*
by (*rule order-trans[OF hit-bound]*)
(intro add-mono prehit-bound order-refl)
qed

definition *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at*

$:: \text{nat} \Rightarrow 'f \text{ staged-adversary} \Rightarrow$
 $(((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
i A out $\longleftrightarrow$
(case out of
None $\Rightarrow$ *False*
| Some $(((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}) \Rightarrow$
(let *s =*
verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
in $\exists \text{trace-openings composition-openings.}$
 $(\text{trace-openings}, \text{composition-openings}) \in$
query-header-supported-partial-opening-witnesses *s*
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data) $\wedge$
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A
 $((\text{replicate rounds } [] [i := \text{trace-openings}])$
 $((\text{replicate rounds } [] [i := \text{composition-openings}])$
 i
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}),$
 $\text{attacker-state}), \text{result}), \text{final-state}))))))$

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*

assumes *witness*:

(*trace-openings*, *composition-openings*) ∈
query-header-supported-partial-opening-witnesses
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*))
(*staged-trace-root* *data*)
(*staged-trace-fri-roots* *data*)
(*staged-trace-final* *data*)
(*staged-alphas* *data*)
(*staged-degree* *data*)
(*staged-composition-fri-roots* *data*)
(*staged-composition-final* *data*)

and *component*:

checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix *A*
((*replicate* *rounds* []) [*i* := *trace-openings*])
((*replicate* *rounds* []) [*i* := *composition-openings*])
i
(*Some* ((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*))

shows

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
i *A*
(*Some* ((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*))

unfolding

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at-def
Let-def

using *witness component*

by (*auto split: option.splits prod.splits*)

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*

assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
A out

shows

∃ *i* < *rounds*.
checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
i *A out*

proof (*cases out*)

case *None*

then show *?thesis*

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-def
by *simp*

next

case (*Some packed*)

```

then obtain alpha-prefix alpha-prefix-state data attacker-state result
final-state where out-eq:
  out =
    Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state)
  by (cases packed) (auto split: prod.splits)
from checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
  [OF hit[unfolded out-eq]]
obtain i trace-openings composition-openings where witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and component:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  by blast
have i-bound: i < rounds
  using component
  unfolding
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix-def
  by (simp split: option.splits prod.splits)
have round-hit:
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
  i A out
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
    [OF witness component])
show ?thesis
  using i-bound round-hit by blast
qed

```

```

lemma checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
assumes hit:
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
  i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),

```

```

    result), final-state))
obtains trace-openings composition-openings where
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A
  ((replicate rounds []) [i := trace-openings])
  ((replicate rounds []) [i := composition-openings])
  i
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
using hit
unfolding
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at-de
  Let-def
by (auto split: option.splits prod.splits)

lemma checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
assumes hit:
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
  i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
and i-bound: i < rounds
shows
  (∃ result' final-state'.
    partial-merkle-inconsistency-bad
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result', final-state'))) ∨
  (∃ trace-openings composition-openings trace-table composition-table.
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
          (staged-trace-root data)
          (staged-trace-fri-roots data)
          (staged-trace-final data)
          (staged-alphas data)
          (staged-degree data))

```

```

      (staged-composition-fri-roots data)
      (staged-composition-final data) ∧
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A
  ((replicate rounds []) [i := trace-openings])
  ((replicate rounds []) [i := composition-openings])
i
  (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state)) ∧
partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings]) ∧
partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings]))
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-atE
    [OF hit]
  obtain trace-openings composition-openings where witness:
    (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and component:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
    A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i
    (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  by blast
from query-header-supported-partial-opening-witness-single-round-candidates-or-partial-merkle-bad
  [OF witness i-bound]
show ?thesis
  using witness component by blast
qed

lemma checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
assumes hit:
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
  i A

```

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))
and *i-bound*: $i < \text{rounds}$
and *trace-low*:
 $\bigwedge \text{trace-openings composition-openings trace-table}$
 $\text{composition-table.}$
 (trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) $\implies$
partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings]) $\implies$
partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings]) $\implies$
trace-table-low-degree trace-table
and *comp-low*:
 $\bigwedge \text{trace-openings composition-openings trace-table}$
 $\text{composition-table.}$
 (trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) $\implies$
partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings]) $\implies$
partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings]) $\implies$
composition-table-low-degree maxDegree composition-table
and *not-all*:
 $\bigwedge \text{trace-openings composition-openings trace-table}$
 $\text{composition-table prefix prefix-state.}$
 (trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)

```

      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)  $\implies$ 
partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])  $\implies$ 
partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])  $\implies$ 
Some ((prefix, prefix-state), prefix-state)  $\in$ 
  set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state)  $\implies$ 
 $\neg$  all-queries-consistent trace-table composition-table
  (sqp-alphas prefix)
shows
  ( $\exists$  result' final-state'.
    partial-merkle-inconsistency-bad
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result', final-state')))  $\vee$ 
  ( $\exists$  trace-table composition-table.
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
      A trace-table composition-table i
      (Some (((alpha-prefix, alpha-prefix-state), data),
        attacker-state), result), final-state)))
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at-ca
    [OF hit i-bound]
  show ?thesis
  proof
    assume
       $\exists$  result' final-state'.
        partial-merkle-inconsistency-bad ?s (Some (result', final-state'))
    then show ?thesis by simp
  next
    assume candidates:
       $\exists$  trace-openings composition-openings trace-table
        composition-table.
      (trace-openings, composition-openings)  $\in$ 
        query-header-supported-partial-opening-witnesses ?s
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)

```

```

    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)  $\wedge$ 
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A
  ((replicate rounds []) [i := trace-openings])
  ((replicate rounds []) [i := composition-openings])
  i
  (Some (((((alpha-prefix, alpha-prefix-state), data),
    attacker-state), result), final-state))  $\wedge$ 
partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])  $\wedge$ 
partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
then obtain trace-openings composition-openings trace-table
composition-table where witness:
  (trace-openings, composition-openings)  $\in$ 
  query-header-supported-partial-opening-witnesses ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
and opening-hit:
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
  A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i
    (Some (((((alpha-prefix, alpha-prefix-state), data),
      attacker-state), result), final-state))
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
by blast
have pair-hit:
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
  A trace-table composition-table i
  (Some (((((alpha-prefix, alpha-prefix-state), data),
    attacker-state), result), final-state))
by (rule
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix-imp-candidate-pair-que
  [OF opening-hit trace-candidate comp-candidate

```

```

      trace-low[OF witness trace-candidate comp-candidate]
      comp-low[OF witness trace-candidate comp-candidate]])
    (rule not-all[OF witness trace-candidate comp-candidate])
  then show ?thesis by blast
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-implies
fixes Q :: nat  $\Rightarrow$  prob
assumes round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
        i A)
      adversary-initial-state  $\leq Q\ i$ 
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
      A)
    adversary-initial-state  $\leq (\sum i < \text{rounds}. Q\ i)$ 
proof -
  have event-le:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
        A)
      adversary-initial-state  $\leq$ 
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      ( $\lambda \text{out}. \exists i \in \{.. < \text{rounds}\}. \text{checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at } i\ A\ \text{out}$ )
      adversary-initial-state
  by (rule wp-event-mono)
  (use
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-implies
    in blast)
  have union-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      ( $\lambda \text{out}. \exists i \in \{.. < \text{rounds}\}. \text{checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at } i\ A\ \text{out}$ )
      adversary-initial-state  $\leq$ 
      ( $\sum i < \text{rounds}. Q\ i$ )
  by (rule wp-event-finite-UN-bound) (simp-all add: round-bound)
show ?thesis

```


by (rule order-trans[OF event-le union-bound])
 qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-header-authenticated-query-from-prefix-round*
 fixes $Q :: \text{nat} \Rightarrow \text{prob}$
 assumes *wf*: *staged-budget-wellformed budgets*
 and *controlled*: *staged-adversary-controlled budgets A*
 and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-round } i A)$
 $\text{adversary-initial-state} \leq Q i$
 shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event composition-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\sum i < \text{rounds}. Q i) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget})$
proof –
 have *header-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-round } A)$
 $\text{adversary-initial-state} \leq (\sum i < \text{rounds}. Q i)$
 by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-round}$
 $[OF \text{ round-bound}]])$
 show ?thesis
 by (rule
 $\text{checked-staged-security-with-data-state-composition-bad-bound-from-header-authenticated-query-from-prefix-and-round}$
 $[OF \text{ wf controlled header-bound}])$
 qed

end

end

theory *Staged-Security-Experiment-Composition-Query-Classification*
 imports *Staged-Security-Experiment-Composition-Query-Prefix-Witness*
begin

Local classification lemmas for current-final-state partial-opening evidence. These lemmas deliberately live downstream of the fixed-witness bridge to avoid growing the size-sensitive prefix-witness theory.

context *soundness*
begin

lemma *trace-fri-bad-with-updated-partial-candidatesI*:
 assumes *updated*:
 accepted-with-partial-initial-openings s
 (Some (result, final-state)) fr f-fri-roots f-final as dg
 composition-fri-roots final trace-query-idxs trace-openings
 composition-query-idxs composition-openings
 and *trace-candidate*:
 partial-trace-table-candidate trace-table trace-openings
 and *not-low*: $\neg$ *trace-table-low-degree trace-table*
shows *trace-fri-bad-with-partial-candidates s*
 (Some (result, final-state))
using *assms*
unfolding *trace-fri-bad-with-partial-candidates-def* **by** *blast*

lemma *composition-fri-bad-with-updated-partial-candidatesI*:
 assumes *updated*:
 accepted-with-partial-initial-openings s
 (Some (result, final-state)) fr f-fri-roots f-final as dg
 composition-fri-roots final trace-query-idxs trace-openings
 composition-query-idxs composition-openings
 and *trace-candidate*:
 partial-trace-table-candidate trace-table trace-openings
 and *comp-candidate*:
 partial-composition-table-candidate composition-table
 composition-openings
 and *trace-low*: *trace-table-low-degree trace-table*
 and *not-comp-low*:
 $\neg$ *composition-table-low-degree maxDegree composition-table*
shows *composition-fri-bad-with-partial-candidates s*
 (Some (result, final-state))
using *assms*
unfolding *composition-fri-bad-with-partial-candidates-def* **by** *blast*

lemma *query-bad-with-updated-partial-candidatesI*:
 assumes *updated*:
 accepted-with-partial-initial-openings s

(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *comp-candidate:*
partial-composition-table-candidate composition-table
composition-openings
and *trace-low:* *trace-table-low-degree trace-table*
and *comp-low:*
composition-table-low-degree maxDegree composition-table
and *not-all:* $\neg$ *all-queries-consistent trace-table composition-table as*
shows *query-bad-with-partial-candidates s (Some (result, final-state))*
using *assms*
unfolding *query-bad-with-partial-candidates-def* **by** *blast*

lemma *accepted-with-partial-initial-openings-update-round-candidate-classification:*

assumes *partial:*
accepted-with-partial-initial-openings s
(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings
and *no-bad:*
 $\neg$ *partial-merkle-inconsistency-bad s (Some (result, final-state))*
and *i-bound:* $i < \text{rounds}$
and *same-idx:* *trace-query-idxs ! i = composition-query-idxs ! i*
and *consistent:*
partial-query-openings-consistent trace-openings-i
composition-openings-i as (trace-query-idxs ! i)
and *trace-auth:*
*partial-authenticated-table fr (scale * clength) trace-openings-i*
final-state
and *comp-auth:*
partial-authenticated-table (hd composition-fri-roots)
*(scale * clength) composition-openings-i final-state*
shows
trace-fri-bad-with-partial-candidates s (Some (result, final-state)) $\vee$
composition-fri-bad-with-partial-candidates s
(Some (result, final-state)) $\vee$
query-bad-with-partial-candidates s (Some (result, final-state)) $\vee$
 $(\exists \text{ trace-table composition-table.}$
partial-trace-table-candidate trace-table
 $(\text{trace-openings}[i := \text{trace-openings-}i]) \wedge$
partial-composition-table-candidate composition-table
 $(\text{composition-openings}[i := \text{composition-openings-}i]) \wedge$
trace-table-low-degree trace-table $\wedge$
composition-table-low-degree maxDegree composition-table $\wedge$
all-queries-consistent trace-table composition-table as)

proof —

```

let ?trace-openings = trace-openings[i := trace-openings-i]
let ?composition-openings =
  composition-openings[i := composition-openings-i]
have updated:
  accepted-with-partial-initial-openings s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs ?trace-openings
  composition-query-idxs ?composition-openings
by (rule accepted-with-partial-initial-openings-update-round
  [OF partial i-bound same-idx consistent trace-auth comp-auth])
obtain trace-table composition-table where trace-candidate:
  partial-trace-table-candidate trace-table ?trace-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  ?composition-openings
using accepted-with-partial-initial-openings-candidates-if-no-partial-merkle-bad
  [OF updated no-bad]
by blast
show ?thesis
proof (cases trace-table-low-degree trace-table)
case False
then have trace-fri-bad-with-partial-candidates s
  (Some (result, final-state))
by (rule trace-fri-bad-with-updated-partial-candidatesI
  [OF updated trace-candidate])
then show ?thesis by simp
next
case trace-low: True
show ?thesis
proof (cases composition-table-low-degree maxDegree composition-table)
case False
then have composition-fri-bad-with-partial-candidates s
  (Some (result, final-state))
by (rule composition-fri-bad-with-updated-partial-candidatesI
  [OF updated trace-candidate comp-candidate trace-low])
then show ?thesis by simp
next
case comp-low: True
show ?thesis
proof (cases all-queries-consistent trace-table composition-table as)
case False
then have query-bad-with-partial-candidates s
  (Some (result, final-state))
by (rule query-bad-with-updated-partial-candidatesI
  [OF updated trace-candidate comp-candidate trace-low comp-low])
then show ?thesis by simp
next
case True
then show ?thesis

```

```

      using trace-candidate comp-candidate trace-low comp-low by blast
    qed
  qed
qed
qed

```

lemma *accepted-with-partial-initial-openings-aligned-consistent-update-round-candidate-classification:*

assumes *partial:*

accepted-with-partial-initial-openings-aligned-consistent s
(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final query-idxs trace-openings
composition-openings

and *no-bad:*

$\neg$ *partial-merkle-inconsistency-bad s (Some (result, final-state))*

and *i-bound: i < rounds*

and *consistent:*

partial-query-openings-consistent trace-openings-i
composition-openings-i as (query-idxs ! i)

and *trace-auth:*

*partial-authenticated-table fr (scale * clength) trace-openings-i*
final-state

and *comp-auth:*

partial-authenticated-table (hd composition-fri-roots)
*(scale * clength) composition-openings-i final-state*

shows

trace-fri-bad-with-partial-candidates s (Some (result, final-state)) $\vee$
composition-fri-bad-with-partial-candidates s
(Some (result, final-state)) $\vee$
query-bad-with-partial-candidates s (Some (result, final-state)) $\vee$
 $(\exists$ *trace-table composition-table.*
partial-trace-table-candidate trace-table
 $(\text{trace-openings}[i := \text{trace-openings-}i]) \wedge$
partial-composition-table-candidate composition-table
 $(\text{composition-openings}[i := \text{composition-openings-}i]) \wedge$
trace-table-low-degree trace-table $\wedge$
composition-table-low-degree maxDegree composition-table $\wedge$
all-queries-consistent trace-table composition-table as)

proof —

have *aligned:*

accepted-with-partial-initial-openings-aligned s
(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final query-idxs trace-openings
composition-openings

using *partial*

unfolding *accepted-with-partial-initial-openings-aligned-consistent-def*

by *simp*

have *unaligned:*

accepted-with-partial-initial-openings s
(Some (result, final-state)) fr f-fri-roots f-final as dg

```

    composition-fri-roots final query-idxs trace-openings query-idxs
    composition-openings
  by (rule accepted-with-partial-initial-openings-aligned-imp-unaligned
    [OF aligned])
show ?thesis
  by (rule
    accepted-with-partial-initial-openings-update-round-candidate-classification
    [OF unaligned no-bad i-bound - consistent trace-auth comp-auth])
    simp
qed

lemma accepted-with-partial-initial-openings-aligned-transcript-consistent-update-round-candidate-classification:
  assumes partial:
    accepted-with-partial-initial-openings-aligned-transcript-consistent s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
  and no-bad:
    ¬ partial-merkle-inconsistency-bad s (Some (result, final-state))
  and i-bound: i < rounds
  and consistent:
    partial-query-openings-consistent trace-openings-i
    composition-openings-i as (query-idxs ! i)
  and trace-auth:
    partial-authenticated-table fr (scale * clength) trace-openings-i
    final-state
  and comp-auth:
    partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) composition-openings-i final-state
  shows
    trace-fri-bad-with-partial-candidates s (Some (result, final-state)) ∨
    composition-fri-bad-with-partial-candidates s
    (Some (result, final-state)) ∨
    query-bad-with-partial-candidates s (Some (result, final-state)) ∨
    (∃ trace-table composition-table.
      partial-trace-table-candidate trace-table
        (trace-openings[i := trace-openings-i]) ∧
      partial-composition-table-candidate composition-table
        (composition-openings[i := composition-openings-i]) ∧
      trace-table-low-degree trace-table ∧
      composition-table-low-degree maxDegree composition-table ∧
      all-queries-consistent trace-table composition-table as)
  proof -
    have aligned:
      accepted-with-partial-initial-openings-aligned-consistent s
      (Some (result, final-state)) fr f-fri-roots f-final as dg
      composition-fri-roots final query-idxs trace-openings
      composition-openings
    by (rule

```

```

      accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
      [OF partial])
  show ?thesis
  by (rule
      accepted-with-partial-initial-openings-aligned-consistent-update-round-candidate-classification
      [OF aligned no-bad i-bound consistent trace-auth comp-auth])
qed
end
end
end

```

```

theory Staged-Security-Experiment-Composition-Query-Current
imports Staged-Security-Experiment-Composition-Query-Classification
begin

```

Current-execution projection for the composition query branch.

The broader header-supported query witnesses are existential over possible verifier executions. This layer only reasons about openings authenticated in the actual sampled verifier final state, so the sampled query lookup is available and the event projects to the existing data-state partial-opening query event.

```

context soundness
begin

```

lemma *checked-staged-security-experiment-with-query-prefix-data-state-raw-final-lookup:*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and outcome:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    adversary-initial-state)

```

shows

```

  fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i)) =
  Some raw

```

proof –

```

have head:
  Some (((prefix, prefix-state), raw), raw-state) ∈
  set-dist
  (execute (checked-staged-query-prefix-receive-with-state A i)
    adversary-initial-state)
and cont:

```

```

Some (((data, attacker-state), result), final-state) ∈
  set-dist
    (execute
      (checked-staged-after-query-prefix-receive-with-verifier A i
        ((prefix, prefix-state), raw))
      raw-state)
using outcome
unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
by (auto elim!: set-dist-bindE)
have recv:
  Some (raw, raw-state) ∈
    set-dist (execute receive-query-index-challenge prefix-state)
by (rule checked-staged-query-prefix-receive-with-state-receive-support
      [OF head])
have raw-lookup:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge (PQueryCounter prefix-state) (PState prefix-state)) =
    Some raw
using receive-query-index-challenge-outcome[OF recv] by blast
have data-out:
  Some (data, attacker-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive A i
          ((prefix, prefix-state), raw))
        raw-state)
by (rule
      checked-staged-after-query-prefix-receive-with-verifier-outcome(1)
      [OF cont])
have verifier-out:
  Some (result, final-state) ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
by (rule
      checked-staged-after-query-prefix-receive-with-verifier-outcome(2)
      [OF cont])
have raw-ext-attacker: raw-state ≤ attacker-state
by (rule checked-staged-after-query-prefix-receive-extends
      [OF wf controlled i-bound data-out])
have attacker-ext-verifier:
  attacker-state ≤
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
proof –
have attacker-state ≤ attacker-state
unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
then show ?thesis

```



```

    by (rule hash-extends-verifier-state-from-adversary-right)
qed
have verifier-ext-final:
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data) ≤
    final-state
  by (rule verify-monad-hash-extends[OF verifier-out])
have raw-ext-final: raw-state ≤ final-state
  by (rule hash-ext-trans[OF raw-ext-attacker])
    (rule hash-ext-trans[OF attacker-ext-verifier verifier-ext-final])
have prefix-support:
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state)
  by (rule checked-staged-query-prefix-receive-with-state-prefix-support
    [OF head])
have prefix-alignment:
  PQueryCounter prefix-state = i ∧
  PState prefix-state =
    state-after-query-chunks
    (staged-query-prefix-start-hash prefix)
    (sqp-query-chunks prefix) i ∧
    length (sqp-query-chunks prefix) = i
  by (rule checked-staged-query-prefix-with-state-alignment
    [OF wf controlled i-bound prefix-support])
from cont obtain chunk chunk-state assert-state record-state suffix-chunks
where data-eq:
  data =
    (|staged-trace-root = sqp-trace-root prefix,
    staged-trace-fri-roots = sqp-trace-fri-roots prefix,
    staged-trace-fri-challenges = sqp-trace-fri-challenges prefix,
    staged-trace-final = sqp-trace-final prefix,
    staged-alphas = sqp-alphas prefix,
    staged-degree = sqp-degree prefix,
    staged-composition-fri-roots = sqp-composition-fri-roots prefix,
    staged-composition-fri-challenges =
      sqp-composition-fri-challenges prefix,
    staged-composition-final = sqp-composition-final prefix,
    staged-query-chunks = sqp-query-chunks prefix @ chunk # suffix-chunks|)
unfolding checked-staged-after-query-prefix-receive-with-verifier-def
  checked-staged-after-query-prefix-receive-def
  by (auto elim!: set-dist-bindE split: prod.splits)
have start-hash-eq:
  staged-query-start-hash data = staged-query-prefix-start-hash prefix
unfolding data-eq staged-query-start-hash-def
  staged-query-prefix-start-hash-def
  staged-composition-fri-start-hash-def
  staged-trace-fri-start-hash-def

```

```

    by simp
  have chunks-state-eq:
    state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i =
    state-after-query-chunks
      (staged-query-prefix-start-hash prefix) (sqp-query-chunks prefix) i
  proof -
    have len-prefix: length (sqp-query-chunks prefix) = i
      using prefix-alignment by simp
    have
      state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i =
      state-after-query-chunks
        (staged-query-prefix-start-hash prefix)
        (sqp-query-chunks prefix @ chunk # suffix-chunks) i
      using start-hash-eq unfolding data-eq by simp
    also have ... =
      state-after-query-chunks
        (staged-query-prefix-start-hash prefix) (sqp-query-chunks prefix) i
      by (rule state-after-query-chunks-append-prefix[OF len-prefix])
    finally show ?thesis .
  qed
  have fields:
    PQueryCounter prefix-state = i
    PState prefix-state =
      state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i
    using prefix-alignment chunks-state-eq by simp-all
  have raw-lookup':
    fmlookup (HashMap raw-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
    using raw-lookup fields by simp
  show ?thesis
    by (rule hash-extension-lookup[OF raw-lookup' raw-ext-final])
  qed

definition checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
where
  checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
    i out  $\longleftrightarrow$ 
    ( $\exists$  trace-openings composition-openings.
      checked-staged-security-with-query-prefix-authenticated-opening-hit
        trace-openings composition-openings i out)

definition checked-staged-security-with-query-prefix-current-query-partial-opening-hit
where

```

checked-staged-security-with-query-prefix-current-query-partial-opening-hit
 $out \longleftrightarrow$
 $(\exists i < rounds.$
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
 $i \ out)$

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-atI:*

assumes

checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i out

shows

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
 $i \ out$

using *assms*

unfolding

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-def
by *blast*

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hitI:*

assumes $i < rounds$

and

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
 $i \ out$

shows

checked-staged-security-with-query-prefix-current-query-partial-opening-hit
 out

using *assms*

unfolding

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-def
by *blast*

definition *staged-query-prefix-prefix-authenticated-opening-target*

$:: 'f \ staged\text{-}query\text{-}prefix\text{-}data \Rightarrow 'f \ protocol\text{-}channel \Rightarrow nat \ set$

where

staged-query-prefix-prefix-authenticated-opening-target prefix prefix-state =
 $\{idx \in query\text{-}sample\text{-}space.$
 $\exists (trace\text{-}openings :: 'f \ authenticated\text{-}opening \ list)$
 $(composition\text{-}openings :: 'f \ authenticated\text{-}opening \ list).$
 $sqp\text{-}composition\text{-}fri\text{-}roots \ prefix \neq [] \wedge$
 $partial\text{-}authenticated\text{-}table \ (sqp\text{-}trace\text{-}root \ prefix)$
 $(scale * clength) \ trace\text{-}openings \ prefix\text{-}state \wedge$
 $partial\text{-}authenticated\text{-}table \ (hd \ (sqp\text{-}composition\text{-}fri\text{-}roots \ prefix))$
 $(scale * clength) \ composition\text{-}openings \ prefix\text{-}state \wedge$
 $partial\text{-}query\text{-}openings\text{-}consistent \ trace\text{-}openings \ composition\text{-}openings$
 $(sqp\text{-}alphas \ prefix) \ idx\}$

definition *staged-query-prefix-prefix-authenticated-trace-indices*

$:: 'f \ staged\text{-}query\text{-}prefix\text{-}data \Rightarrow 'f \ protocol\text{-}channel \Rightarrow nat \ set$

where

staged-query-prefix-prefix-authenticated-trace-indices *prefix prefix-state* =
 $\{idx \in \text{query-sample-space}.$
 $\exists \text{opn} :: 'f \text{ authenticated-opening}.$
 $\text{authenticated-opening-in } \text{prefix-state } \text{opn} \wedge$
 $\text{opening-root } \text{opn} = \text{sqp-trace-root } \text{prefix} \wedge$
 $\text{opening-length } \text{opn} = \text{scale} * \text{clength} \wedge$
 $\text{opening-index } \text{opn} = \text{idx}\}$

lemma *staged-query-prefix-prefix-authenticated-opening-target-subset*:
staged-query-prefix-prefix-authenticated-opening-target *prefix prefix-state*
 $\subseteq \text{query-sample-space}$
unfolding *staged-query-prefix-prefix-authenticated-opening-target-def*
by *blast*

lemma *staged-query-prefix-prefix-authenticated-opening-target-eq-query-sample-space-if-all-indices-witnessed*:
assumes *witnessed*:

$\bigwedge idx. idx \in \text{query-sample-space} \implies$
 $\exists \text{trace-openings } \text{composition-openings}.$
 $\text{sqp-composition-fri-roots } \text{prefix} \neq [] \wedge$
 $\text{partial-authenticated-table } (\text{sqp-trace-root } \text{prefix})$
 $(\text{scale} * \text{clength}) \text{ trace-openings } \text{prefix-state} \wedge$
 $\text{partial-authenticated-table}$
 $(\text{hd } (\text{sqp-composition-fri-roots } \text{prefix}))$
 $(\text{scale} * \text{clength}) \text{ composition-openings } \text{prefix-state} \wedge$
 $\text{partial-query-openings-consistent } \text{trace-openings}$
 $\text{composition-openings } (\text{sqp-alphas } \text{prefix}) \text{ idx}$

shows

staged-query-prefix-prefix-authenticated-opening-target *prefix*
prefix-state = *query-sample-space*

proof

show

staged-query-prefix-prefix-authenticated-opening-target *prefix*
prefix-state $\subseteq \text{query-sample-space}$

by (rule *staged-query-prefix-prefix-authenticated-opening-target-subset*)

next

show

query-sample-space $\subseteq$
staged-query-prefix-prefix-authenticated-opening-target *prefix*
prefix-state

proof

fix *idx*

assume *idx-sample*: *idx* $\in \text{query-sample-space}$

then obtain *trace-openings composition-openings* **where**

$\text{sqp-composition-fri-roots } \text{prefix} \neq []$
 $\text{partial-authenticated-table } (\text{sqp-trace-root } \text{prefix})$
 $(\text{scale} * \text{clength}) \text{ trace-openings } \text{prefix-state}$
 $\text{partial-authenticated-table}$
 $(\text{hd } (\text{sqp-composition-fri-roots } \text{prefix}))$
 $(\text{scale} * \text{clength}) \text{ composition-openings } \text{prefix-state}$

```

    partial-query-openings-consistent trace-openings composition-openings
      (sqp-alphas prefix) idx
    using witnessed by blast
  then show
    idx ∈
      staged-query-prefix-prefix-authenticated-opening-target prefix
      prefix-state
    unfolding staged-query-prefix-prefix-authenticated-opening-target-def
    using idx-sample by blast
qed
qed

lemma staged-query-prefix-prefix-authenticated-opening-target-subset-trace-indices:
  staged-query-prefix-prefix-authenticated-opening-target prefix prefix-state
  ⊆
  staged-query-prefix-prefix-authenticated-trace-indices prefix prefix-state
proof
  fix idx
  assume idx-in:
    idx ∈
      staged-query-prefix-prefix-authenticated-opening-target prefix
      prefix-state
  then obtain trace-openings composition-openings where
    idx-sample: idx ∈ query-sample-space
    and trace-table:
      partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) trace-openings prefix-state
    and consistent:
      partial-query-openings-consistent trace-openings
      composition-openings (sqp-alphas prefix) idx
    unfolding staged-query-prefix-prefix-authenticated-opening-target-def
    by blast
  have trace-indices:
    map opening-index trace-openings = powers-scaled idx
    using consistent unfolding partial-query-openings-consistent-def
    by simp
  have trace-nonempty: trace-openings ≠ []
proof -
  have powers-scaled idx ≠ []
    unfolding powers-scaled-def using powers-pos by simp
  then show ?thesis
    using trace-indices by auto
qed
have hd-idx: opening-index (hd trace-openings) = idx
proof -
  have hd (map opening-index trace-openings) = hd (powers-scaled idx)
    using trace-indices by simp
  then have opening-index (hd trace-openings) = hd (powers-scaled idx)
    using trace-nonempty by (simp add: hd-map)

```

```

then show ?thesis
  using powers-scaled-hd by simp
qed
have hd-in: hd trace-openings ∈ set trace-openings
  using trace-nonempty by simp
have auth:
  authenticated-opening-in prefix-state (hd trace-openings)
  using trace-table hd-in unfolding partial-authenticated-table-def
  by blast
have root:
  opening-root (hd trace-openings) = sqp-trace-root prefix
  using trace-table hd-in unfolding partial-authenticated-table-def
  by blast
have len:
  opening-length (hd trace-openings) = scale * clength
  using trace-table hd-in unfolding partial-authenticated-table-def
  by blast
show
  idx ∈
    staged-query-prefix-prefix-authenticated-trace-indices prefix
    prefix-state
  unfolding staged-query-prefix-prefix-authenticated-trace-indices-def
  by (intro CollectI conjI exI[of - hd trace-openings])
    (use idx-sample auth root len hd-idx in simp-all)
qed

lemma staged-query-prefix-prefix-authenticated-trace-indices-subset:
  staged-query-prefix-prefix-authenticated-trace-indices prefix prefix-state
  ⊆ query-sample-space
  unfolding staged-query-prefix-prefix-authenticated-trace-indices-def
  by blast

lemma staged-query-prefix-prefix-authenticated-trace-indices-eq-query-sample-space-if-all-indices-witnessed:
  assumes witnessed:
     $\bigwedge idx. idx \in \text{query-sample-space} \implies$ 
     $\exists \text{opn} :: 'f \text{ authenticated-opening.}$ 
    authenticated-opening-in prefix-state opn  $\wedge$ 
    opening-root opn = sqp-trace-root prefix  $\wedge$ 
    opening-length opn = scale * clength  $\wedge$ 
    opening-index opn = idx
  shows
    staged-query-prefix-prefix-authenticated-trace-indices prefix
    prefix-state = query-sample-space
proof
  show
    staged-query-prefix-prefix-authenticated-trace-indices prefix
    prefix-state ⊆ query-sample-space
  by (rule staged-query-prefix-prefix-authenticated-trace-indices-subset)
next

```

```

show
  query-sample-space  $\subseteq$ 
    staged-query-prefix-prefix-authenticated-trace-indices prefix
    prefix-state
unfolding staged-query-prefix-prefix-authenticated-trace-indices-def
using witnessed by blast
qed

lemma finite-staged-query-prefix-prefix-authenticated-trace-indices[simp]:
  finite
    (staged-query-prefix-prefix-authenticated-trace-indices prefix
     prefix-state)
by (rule finite-subset
      [OF staged-query-prefix-prefix-authenticated-trace-indices-subset
         finite-query-sample-space])

lemma staged-query-prefix-prefix-authenticated-opening-target-fraction-bound-if-trace-indices:
assumes trace-frac:
  nnreal
    (card
      (staged-query-prefix-prefix-authenticated-trace-indices prefix
       prefix-state)) /
  nnreal (card query-sample-space)  $\leq$  query-error-bound
shows
  nnreal
    (card
      (staged-query-prefix-prefix-authenticated-opening-target prefix
       prefix-state)) /
  nnreal (card query-sample-space)  $\leq$  query-error-bound
proof –
have card-le:
  card
    (staged-query-prefix-prefix-authenticated-opening-target prefix
     prefix-state)  $\leq$ 
  card
    (staged-query-prefix-prefix-authenticated-trace-indices prefix
     prefix-state)
by (rule card-mono)
    (simp,
     rule staged-query-prefix-prefix-authenticated-opening-target-subset-trace-indices)
have
  nnreal
    (card
      (staged-query-prefix-prefix-authenticated-opening-target prefix
       prefix-state)) /
  nnreal (card query-sample-space)  $\leq$ 
  nnreal
    (card
      (staged-query-prefix-prefix-authenticated-trace-indices prefix

```

```

      prefix-state)) /
    nreal (card query-sample-space)
  by (rule nreal-nat-divide-right-mono[OF card-le])
also have ... ≤ query-error-bound
  by (rule trace-frac)
finally show ?thesis .
qed

```

lemma *checked-staged-query-prefix-prefix-authenticated-target-hit-bound-from-trace-indices:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *trace-frac:*

$\bigwedge \text{prefix prefix-state.}$

Some ((prefix, prefix-state), prefix-state) ∈

set-dist

(execute (checked-staged-query-prefix-with-state A i)

adversary-initial-state) ⇒

nreal

(card

(staged-query-prefix-prefix-authenticated-trace-indices prefix

prefix-state)) /

nreal (card query-sample-space) ≤ query-error-bound

shows

wp-event (checked-staged-query-prefix-receive-with-state A i)

(checked-staged-query-prefix-dynamic-index-hit

staged-query-prefix-prefix-authenticated-opening-target)

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) + query-error-bound

proof —

have *raw-bound: query-index-raw-preimage-bound*

by (rule *query-index-raw-preimage-bound-from-sampler-wellformed*)

have *hit-bound:*

wp-event (checked-staged-query-prefix-receive-with-state A i)

(checked-staged-query-prefix-dynamic-index-hit

staged-query-prefix-prefix-authenticated-opening-target)

adversary-initial-state ≤

wp-event (checked-staged-query-prefix-receive-with-state A i)

(checked-staged-query-prefix-dynamic-index-prehit

staged-query-prefix-prefix-authenticated-opening-target)

adversary-initial-state + query-error-bound

proof (rule

checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error

[OF raw-bound])

fix *prefix prefix-state*

assume *support:*

Some ((prefix, prefix-state), prefix-state) ∈

set-dist


```

      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)
have subset:
  staged-query-prefix-prefix-authenticated-opening-target prefix
    prefix-state  $\subseteq$  query-sample-space
by (rule staged-query-prefix-prefix-authenticated-opening-target-subset)
have frac:
  nnreal
    (card
      (staged-query-prefix-prefix-authenticated-opening-target prefix
        prefix-state)) /
    nnreal (card query-sample-space)  $\leq$  query-error-bound
by (rule
  staged-query-prefix-prefix-authenticated-opening-target-fraction-bound-if-trace-indices)
  (rule trace-frac[OF support])
show
  staged-query-prefix-prefix-authenticated-opening-target prefix
    prefix-state  $\subseteq$  query-sample-space  $\wedge$ 
  nnreal
    (card
      (staged-query-prefix-prefix-authenticated-opening-target prefix
        prefix-state)) /
    nnreal (card query-sample-space)  $\leq$  query-error-bound
using subset frac by blast
qed
have prehit-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit
      staged-query-prefix-prefix-authenticated-opening-target)
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1)
proof (rule
  checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program)
show HashMap adversary-initial-state = fmempty
by (simp add: adversary-initial-state-def)
show hash-relation-program
  (checked-staged-query-prefix-dynamic-index-prehit-relation A i
    adversary-initial-state
      staged-query-prefix-prefix-authenticated-opening-target)
  size (staged-query-search-queries budgets i + 1)
  (checked-staged-query-prefix-receive-with-state A i)
by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state
  [OF wf controlled])
  (use i-bound in simp-all,
    rule checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
qed
show ?thesis
by (rule order-trans[OF hit-bound])

```

(intro add-mono prehit-bound order-refl)

qed

definition *checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at*
where
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
i out $\longleftrightarrow$
 $(\exists (trace-openings :: 'f authenticated-opening list)$
 $(composition-openings :: 'f authenticated-opening list).$
checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
trace-openings composition-openings i out)

definition *checked-staged-security-with-query-prefix-current-path-output-hit-at*
where
checked-staged-security-with-query-prefix-current-path-output-hit-at
 $(i::nat) out \longleftrightarrow$
 $(\exists (trace-openings :: 'f authenticated-opening list)$
 $(composition-openings :: 'f authenticated-opening list).$
checked-staged-security-with-query-prefix-opening-path-output-hit
trace-openings composition-openings out)

lemma *checked-staged-security-with-query-prefix-current-path-output-hit-at-imp-lower:*
assumes *hit:*
checked-staged-security-with-query-prefix-current-path-output-hit-at
i out
shows
 $\exists (trace-openings :: 'f authenticated-opening list)$
 $(composition-openings :: 'f authenticated-opening list).$
checked-staged-security-with-query-prefix-opening-path-output-hit
trace-openings composition-openings out
using *hit*
unfolding
checked-staged-security-with-query-prefix-current-path-output-hit-at-def
by *assumption*

lemma *checked-staged-security-with-query-prefix-current-path-output-hit-atE:*
assumes *hit:*
checked-staged-security-with-query-prefix-current-path-output-hit-at
i out
obtains *prefix prefix-state raw raw-state data attacker-state result*
final-state rt len idx val pth **where**
out = Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)
hash-map-new-output-hit
(merkle-path-target-roots final-state rt len idx val pth)
prefix-state final-state
finite (merkle-path-target-roots final-state rt len idx val pth)
card (merkle-path-target-roots final-state rt len idx val pth) $\leq$
Suc (length pth)

```

proof –
from hit have lower-ex:
   $\exists$ (trace-openings :: 'f authenticated-opening list)
    (composition-openings :: 'f authenticated-opening list).
  checked-staged-security-with-query-prefix-opening-path-output-hit
    trace-openings composition-openings out
unfolding
  checked-staged-security-with-query-prefix-current-path-output-hit-at-def
by assumption
then show ?thesis
proof
  fix trace-openings
  assume composition-ex:
     $\exists$  composition-openings.
    checked-staged-security-with-query-prefix-opening-path-output-hit
      trace-openings composition-openings out
  then show ?thesis
  proof
    fix composition-openings
    assume lower:
      checked-staged-security-with-query-prefix-opening-path-output-hit
        trace-openings composition-openings out
    then show ?thesis
    proof (rule checked-staged-security-with-query-prefix-opening-path-output-hitE)
      fix prefix prefix-state raw raw-state data attacker-state result
        final-state opn
      assume out-eq:
        out = Some ((((prefix, prefix-state), raw), raw-state),
          ((data, attacker-state), result)), final-state)
      assume new-output:
        hash-map-new-output-hit
          (merkle-path-target-roots final-state
            (opening-root opn)
            (opening-length opn)
            (opening-index opn)
            (opening-value opn)
            (opening-path opn))
          prefix-state final-state
      assume finite-roots:
        finite
          (merkle-path-target-roots final-state
            (opening-root opn)
            (opening-length opn)
            (opening-index opn)
            (opening-value opn)
            (opening-path opn))
      assume card-roots:
        card
          (merkle-path-target-roots final-state

```

```

      (opening-root opn)
      (opening-length opn)
      (opening-index opn)
      (opening-value opn)
      (opening-path opn)) ≤
    Suc (length (opening-path opn))
  show thesis
  by (rule that
      [of prefix prefix-state raw raw-state data attacker-state result
        final-state opening-root opn opening-length opn
        opening-index opn opening-value opn opening-path opn])
      (rule out-eq, rule new-output, rule finite-roots, rule card-roots))
qed
qed
qed
qed

lemma checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-imp-dynamic-target:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-query-prefix-data-state A i)
          adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
      i out
  shows
    checked-staged-security-with-query-prefix-dynamic-index-hit
      staged-query-prefix-prefix-authenticated-opening-target out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-def
    checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
  by (cases packed, auto split: prod.splits)

```

```

have support-out:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
using support out-eq by simp
have head:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
and cont:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
    (execute
      (checked-staged-after-query-prefix-receive-with-verifier A i
        ((prefix, prefix-state), raw))
      raw-state)
using support-out
unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
by (auto elim!: set-dist-bindE)
have prefix-support:
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state)
by (rule checked-staged-query-prefix-receive-with-state-prefix-support
  [OF head])
have prefix-alignment:
  PQueryCounter prefix-state = i ∧
  PState prefix-state =
    state-after-query-chunks
    (staged-query-prefix-start-hash prefix)
    (sqp-query-chunks prefix) i ∧
  length (sqp-query-chunks prefix) = i
by (rule checked-staged-query-prefix-with-state-alignment
  [OF wf controlled i-bound prefix-support])
from cont obtain chunk chunk-state assert-state record-state suffix-chunks
where data-eq:
  data =
    (|staged-trace-root = sqp-trace-root prefix,
      staged-trace-fri-roots = sqp-trace-fri-roots prefix,
      staged-trace-fri-challenges = sqp-trace-fri-challenges prefix,
      staged-trace-final = sqp-trace-final prefix,
      staged-alphas = sqp-alphas prefix,
      staged-degree = sqp-degree prefix,
      staged-composition-fri-roots = sqp-composition-fri-roots prefix,

```

```

    staged-composition-fri-challenges =
      sqp-composition-fri-challenges prefix,
    staged-composition-final = sqp-composition-final prefix,
    staged-query-chunks = sqp-query-chunks prefix @ chunk # suffix-chunks)
unfolding checked-staged-after-query-prefix-receive-with-verifier-def
  checked-staged-after-query-prefix-receive-def
by (auto elim!: set-dist-bindE split: prod.splits)
from hit obtain trace-openings composition-openings where prefix-hit:
  checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
  trace-openings composition-openings i out
unfolding
  checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-def
by blast
have idx-in:
  index (to-nat raw) ∈
    staged-query-prefix-prefix-authenticated-opening-target prefix prefix-state
using prefix-hit index-less-query-sample-space
unfolding out-eq
  checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-def
  staged-query-prefix-prefix-authenticated-opening-target-def
  query-sample-space-def
  data-eq
by auto
show ?thesis
unfolding out-eq
  checked-staged-security-with-query-prefix-dynamic-index-hit-def
  checked-staged-query-prefix-dynamic-index-hit-def
using idx-in by simp
qed

lemma checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-le-dynamic-target:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
shows
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
    adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-dynamic-index-hit
      staged-query-prefix-prefix-authenticated-opening-target)
    adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-imp-dynamic-target
    [OF wf controlled i-bound])

```

lemma *checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target:*

assumes *wf*: staged-budget-wellformed budgets

and *controlled*: staged-adversary-controlled budgets *A*

and *i-bound*: $i < \text{rounds}$

and *prefix-bound*:

wp-event (*checked-staged-query-prefix-receive-with-state* *A i*)

(*checked-staged-query-prefix-dynamic-index-hit*

staged-query-prefix-prefix-authenticated-opening-target)

adversary-initial-state $\leq B$

shows

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)

(*checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i*)

adversary-initial-state $\leq B$

proof –

have *lifted*:

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)

(*checked-staged-security-with-query-prefix-dynamic-index-hit*

staged-query-prefix-prefix-authenticated-opening-target)

adversary-initial-state $\leq B$

by (rule *checked-staged-security-with-query-prefix-dynamic-index-hit-bound*

[*OF prefix-bound*])

show *?thesis*

by (rule *order-trans*[

OF checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-le-dynamic-target

lifted])

(use *wf controlled i-bound in simp-all*)

qed

lemma *checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-trace-indices:*

assumes *wf*: staged-budget-wellformed budgets

and *controlled*: staged-adversary-controlled budgets *A*

and *i-bound*: $i < \text{rounds}$

and *trace-frac*:

$\bigwedge \text{prefix } \text{prefix-state}.$

Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$

set-dist

(*execute* (*checked-staged-query-prefix-with-state* *A i*)

adversary-initial-state) $\implies$

nnreal

(*card*

(*staged-query-prefix-prefix-authenticated-trace-indices prefix*

prefix-state)) /

nnreal (*card query-sample-space*) $\leq \text{query-error-bound}$

shows

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)

(*checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i*)

$adversary-initial-state \leq$
 $staged-phase-relation-error\ size$
 $(staged-query-search-queries\ budgets\ i + 1) + query-error-bound$

proof (rule
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target
[OF wf controlled i-bound])

show
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
staged-query-prefix-prefix-authenticated-opening-target)
 $adversary-initial-state \leq$
 $staged-phase-relation-error\ size$
 $(staged-query-search-queries\ budgets\ i + 1) + query-error-bound$

by (rule
checked-staged-query-prefix-prefix-authenticated-target-hit-bound-from-trace-indices
[OF wf controlled i-bound trace-fac])

qed

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-prefix-or-path-on-support:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *support:*
out ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)
and *hit:*
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
i out

shows
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
i out ∨
checked-staged-security-with-query-prefix-current-path-output-hit-at
i out

proof —
from *hit* **obtain** *trace-openings composition-openings* **where** *current:*
checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i out

unfolding
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-def

by *blast*

have *split:*
checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
trace-openings composition-openings i out ∨
checked-staged-security-with-query-prefix-opening-path-output-hit
trace-openings composition-openings out

by (rule

checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-prefix-or-path-output-hit-on-support
 [OF wf controlled i-bound support current])
then show ?thesis
unfolding
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-def
checked-staged-security-with-query-prefix-current-path-output-hit-at-def
by blast
qed

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-path*:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i < \text{rounds}$
and prefix-bound:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at* i)
 adversary-initial-state $\leq P$
and path-bound:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-current-path-output-hit-at* i)
 adversary-initial-state $\leq Q$
shows
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at* i)
 adversary-initial-state $\leq P + Q$

proof –
have event-le:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at* i)
 adversary-initial-state $\leq$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 ($\lambda \text{out.}$
 checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
 i out $\vee$
 checked-staged-security-with-query-prefix-current-path-output-hit-at
 i out)
 adversary-initial-state
by (rule wp-event-mono-on-support)
 (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-prefix-or-path-on-support
 [OF wf controlled i-bound])
have union-bound:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)

```

(λout.
  checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
    i out ∨
  checked-staged-security-with-query-prefix-current-path-output-hit-at
    i out)
adversary-initial-state ≤
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
adversary-initial-state +
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-path-output-hit-at i)
adversary-initial-state
by (rule wp-event-union-bound)
have sum-bound:
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
adversary-initial-state +
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-path-output-hit-at i)
adversary-initial-state ≤ P + Q
by (intro add-mono prefix-bound path-bound)
show ?thesis
by (rule order-trans[OF event-le])
(rule order-trans[OF union-bound sum-bound])
qed

```

definition staged-security-with-data-state-current-query-partial-opening-hit-at

```

:: nat ⇒
  (((f staged-proof-data × f protocol-channel) × unit list) ×
   f protocol-channel) option ⇒ bool

```

where

```

staged-security-with-data-state-current-query-partial-opening-hit-at i out
↔
(case out of
  None ⇒ False
| Some (((data, attacker-state), result), final-state) ⇒
  (let s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  in
    (∃ raw trace-openings composition-openings rest.
      i < rounds ∧
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks

```

$(\text{staged-query-start-hash } \text{data})$
 $(\text{staged-query-chunks } \text{data}) \ i)) =$
 $\text{Some } \text{raw} \wedge$
 $\text{staged-composition-fri-roots } \text{data} \neq [] \wedge$
 $\text{Some } (\text{result}, \text{final-state}) \in$
 $\text{set-dist } (\text{execute } \text{verify-monad } s) \wedge$
 $\text{partial-authenticated-table } (\text{staged-trace-root } \text{data})$
 $(\text{scale} * \text{clength}) \text{ trace-openings } \text{final-state} \wedge$
 $\text{partial-authenticated-table}$
 $(\text{hd } (\text{staged-composition-fri-roots } \text{data}))$
 $(\text{scale} * \text{clength}) \text{ composition-openings } \text{final-state} \wedge$
 $\text{verifier-header-transcript } s$
 $(\text{staged-trace-root } \text{data})$
 $(\text{staged-trace-fri-roots } \text{data})$
 $(\text{staged-trace-final } \text{data})$
 $(\text{staged-alphas } \text{data})$
 $(\text{staged-degree } \text{data})$
 $(\text{staged-composition-fri-roots } \text{data})$
 $(\text{staged-composition-final } \text{data})$
 $\text{rest} \wedge$
 $\text{partial-query-openings-consistent } \text{trace-openings}$
 $\text{composition-openings } (\text{staged-alphas } \text{data})$
 $(\text{index } (\text{to-nat } \text{raw}))))$

definition *staged-security-with-data-state-current-query-partial-opening-hit*

$:: (((\text{'f staged-proof-data} \times \text{'f protocol-channel}) \times \text{unit list}) \times$
 $\text{'f protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

staged-security-with-data-state-current-query-partial-opening-hit out

$\longleftrightarrow$

$(\exists i < \text{rounds}.$

$\text{staged-security-with-data-state-current-query-partial-opening-hit-at}$
 $i \text{ out})$

lemma *staged-security-with-data-state-current-query-partial-opening-hit-at-aligned-transcript-classification-on-support*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

$\text{Some } (((\text{data}, \text{attacker-state}), \text{result}), \text{final-state}) \in$

set-dist

$(\text{execute}$

$(\text{checked-staged-security-experiment-with-data-state } A)$

$\text{adversary-initial-state})$

and *partial:*

accepted-with-partial-initial-openings-aligned-transcript-consistent

$(\text{verifier-state-from-adversary } \text{attacker-state}$

$(\text{staged-proof-transcript } \text{data}))$

$(\text{Some } (\text{result}, \text{final-state})) \text{ fr } \text{f-fri-roots } \text{f-final as dg}$

$\text{composition-fri-roots } \text{final } \text{query-idxs } \text{trace-openings}$

```

    composition-openings
and no-bad:
  ¬ partial-merkle-inconsistency-bad
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))
  (Some (result, final-state))
and hit:
  staged-security-with-data-state-current-query-partial-opening-hit-at i
  (Some (((data, attacker-state), result), final-state))
shows
  trace-fri-bad-with-partial-candidates
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))
  (Some (result, final-state)) ∨
  composition-fri-bad-with-partial-candidates
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))
  (Some (result, final-state)) ∨
  query-bad-with-partial-candidates
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))
  (Some (result, final-state)) ∨
  (∃ trace-table composition-table trace-openings-i composition-openings-i.
   partial-trace-table-candidate trace-table
    (trace-openings[i := trace-openings-i]) ∧
   partial-composition-table-candidate composition-table
    (composition-openings[i := composition-openings-i]) ∧
   trace-table-low-degree trace-table ∧
   composition-table-low-degree maxDegree composition-table ∧
   all-queries-consistent trace-table composition-table as)
proof —
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support]
  obtain builder where builder:
    Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
     adversary-initial-state)
  by blast
  have staged-header:
    verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)

```

```

    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled builder])
have shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
    query-idxs
  by (rule
      accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape
      [OF partial])
from checked-staged-security-with-data-state-accepted-shape-query-chunks
  [OF wf controlled support shape]
obtain raw-idxs where as-eq: as = staged-alphas data
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and lookup:
  ∀ j < rounds.
    fmllookup (HashMap final-state)
      (QueryIndexChallenge j
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) j)) =
      Some (raw-idxs ! j)
  by blast
from partial have aligned:
  accepted-with-partial-initial-openings-aligned-consistent ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
  by (rule
      accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned)
have aligned0:
  accepted-with-partial-initial-openings-aligned ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
  by (rule
      accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
      [OF aligned])
from accepted-with-partial-initial-openings-aligned-shapes[OF aligned0]
obtain rest where header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
  by blast
have header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧

```

```

f-final = staged-trace-final data ∧
as = staged-alphas data ∧
dg = staged-degree data ∧
composition-fri-roots = staged-composition-fri-roots data ∧
final = staged-composition-final data ∧
rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp
from hit obtain raw trace-openings-i composition-openings-i rest' where
i-bound: i < rounds
and final-lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some raw
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings-i final-state
and comp-auth:
  partial-authenticated-table
    (hd (staged-composition-fri-roots data)) (scale * clength)
    composition-openings-i final-state
and consistent-raw:
  partial-query-openings-consistent trace-openings-i
    composition-openings-i (staged-alphas data) (index (to-nat raw))
unfolding staged-security-with-data-state-current-query-partial-opening-hit-at-def
  Let-def
by auto
have raw-eq: raw = raw-idxs ! i
  using final-lookup lookup i-bound by simp
have idx-eq: index (to-nat raw) = query-idxs ! i
  using raw-eq query-idxs-eq len-raw i-bound by simp
have consistent:
  partial-query-openings-consistent trace-openings-i
    composition-openings-i as (query-idxs ! i)
  using consistent-raw idx-eq header-eq by simp
have trace-auth':
  partial-authenticated-table fr (scale * clength) trace-openings-i
    final-state
  using trace-auth header-eq by simp
have comp-auth':
  partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) composition-openings-i final-state
  using comp-auth header-eq by simp
have classified:
  trace-fri-bad-with-partial-candidates ?s
    (Some (result, final-state)) ∨

```

```

composition-fri-bad-with-partial-candidates ?s
  (Some (result, final-state)) ∨
query-bad-with-partial-candidates ?s (Some (result, final-state)) ∨
(∃ trace-table composition-table.
  partial-trace-table-candidate trace-table
    (trace-openings[i := trace-openings-i]) ∧
  partial-composition-table-candidate composition-table
    (composition-openings[i := composition-openings-i]) ∧
  trace-table-low-degree trace-table ∧
  composition-table-low-degree maxDegree composition-table ∧
  all-queries-consistent trace-table composition-table as)
by (rule
  accepted-with-partial-initial-openings-aligned-transcript-consistent-update-round-candidate-classification
  [OF partial no-bad i-bound consistent trace-auth' comp-auth'])
then show ?thesis
  by blast
qed

lemma staged-security-with-data-state-current-query-partial-opening-hit-at-imp-query-prefix-current-on-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and support:
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
  and hit:
    staged-security-with-data-state-current-query-partial-opening-hit-at i
      (Some (((data, attacker-state), result), final-state))
  shows
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
      i
      (Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state))
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have hit':
    ∃ raw' trace-openings composition-openings rest.
      i < rounds ∧
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =

```

```

Some raw' ∧
staged-composition-fri-roots data ≠ [] ∧
Some (result, final-state) ∈ set-dist (execute verify-monad ?s) ∧
partial-authenticated-table (staged-trace-root data)
  (scale * clength) trace-openings final-state ∧
partial-authenticated-table
  (hd (staged-composition-fri-roots data)) (scale * clength)
  composition-openings final-state ∧
verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  rest ∧
partial-query-openings-consistent trace-openings
  composition-openings (staged-alphas data) (index (to-nat raw'))
using hit
unfolding staged-security-with-data-state-current-query-partial-opening-hit-at-def
  Let-def
by simp
from hit'
obtain raw' trace-openings composition-openings rest where
  final-lookup':
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some raw'
and comp-nonempty: staged-composition-fri-roots data ≠ []
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and comp-auth:
  partial-authenticated-table
    (hd (staged-composition-fri-roots data)) (scale * clength)
    composition-openings final-state
and consistent':
  partial-query-openings-consistent trace-openings
    composition-openings (staged-alphas data) (index (to-nat raw'))
by blast
have final-lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =

```


Some raw
 by (rule
 checked-staged-security-experiment-with-query-prefix-data-state-raw-final-lookup
 [OF wf controlled i-bound support])
 have raw-eq: raw' = raw
 using final-lookup' final-lookup by simp
 have current:
 checked-staged-security-with-query-prefix-authenticated-opening-hit
 trace-openings composition-openings i
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))
 unfolding checked-staged-security-with-query-prefix-authenticated-opening-hit-def
 using comp-nonempty trace-auth comp-auth consistent' raw-eq by simp
 show ?thesis
 by (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-atI
 [OF current])
 qed

lemma checked-staged-security-with-data-state-current-query-partial-opening-hit-at-le-query-prefix-current:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
shows
 wp-event
 (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-query-partial-opening-hit-at i)
 adversary-initial-state ≤
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state

proof –
 let ?projected =
 λout. case out of
 None ⇒
 staged-security-with-data-state-current-query-partial-opening-hit-at
 i None
 | Some (packed, t) ⇒
 staged-security-with-data-state-current-query-partial-opening-hit-at
 i (Some (snd packed, t))
 have projection:
 wp-event
 (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-query-partial-opening-hit-at i)
 adversary-initial-state =
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 ?projected

```

    adversary-initial-state
  by (rule
    checked-staged-security-experiment-with-query-prefix-data-state-projection-event
    [OF i-bound])
  have event-le:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    ?projected
    adversary-initial-state ≤
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
  and hit: ?projected out
  show
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
    i out
  proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      staged-security-with-data-state-current-query-partial-opening-hit-at-def
    by simp
  next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
  show ?thesis
    unfolding out-eq
  by (rule
    staged-security-with-data-state-current-query-partial-opening-hit-at-imp-query-prefix-current-on-support
    [OF wf controlled i-bound])
    (use support hit out-eq in simp-all)
  qed
  qed
  show ?thesis

```

unfolding *projection*
 by (rule *event-le*)
 qed

lemma *checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds*:

assumes *round-bound*:

$\bigwedge i. i < \text{rounds} \implies$

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i*)

adversary-initial-state $\leq C i$

and *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

shows

wp-event

(*checked-staged-security-experiment-with-data-state A*)

staged-security-with-data-state-current-query-partial-opening-hit

adversary-initial-state $\leq (\sum i < \text{rounds}. C i)$

proof –

have *round-le*:

$\bigwedge i. i < \text{rounds} \implies$

wp-event

(*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-current-query-partial-opening-hit-at i*)

adversary-initial-state $\leq C i$

proof –

fix *i*

assume *i-bound*: $i < \text{rounds}$

show

wp-event

(*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-current-query-partial-opening-hit-at i*)

adversary-initial-state $\leq C i$

by (rule *order-trans*

[*OF*

checked-staged-security-with-data-state-current-query-partial-opening-hit-at-le-query-prefix-current
round-bound[*OF i-bound*]])

(use *wf controlled i-bound* **in** *simp-all*)

qed

have *event-le*:

wp-event

(*checked-staged-security-experiment-with-data-state A*)

staged-security-with-data-state-current-query-partial-opening-hit

adversary-initial-state $\leq$

wp-event

(*checked-staged-security-experiment-with-data-state A*)

($\lambda \text{out}. \exists i \in \{..<\text{rounds}\}.$

staged-security-with-data-state-current-query-partial-opening-hit-at

i out)

```

    adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-current-query-partial-opening-hit-def)
also have ... ≤ (∑ i<rounds. C i)
  by (rule wp-event-finite-UN-bound)
    (simp-all add: round-le)
finally show ?thesis .
qed

lemma staged-security-with-data-state-current-query-partial-opening-hit-at-imp-partial-opening-hit-at:
  assumes hit:
    staged-security-with-data-state-current-query-partial-opening-hit-at i out
  shows
    staged-security-with-data-state-query-partial-opening-hit-at i out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    staged-security-with-data-state-current-query-partial-opening-hit-at-def
    staged-security-with-data-state-query-partial-opening-hit-at-def
  by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have hit':
  ∃ raw trace-openings composition-openings rest.
    i < rounds ∧
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some raw ∧
    staged-composition-fri-roots data ≠ [] ∧
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s) ∧
    partial-authenticated-table (staged-trace-root data)
      (scale * clength) trace-openings final-state ∧
    partial-authenticated-table
      (hd (staged-composition-fri-roots data)) (scale * clength)
      composition-openings final-state ∧
    verifier-header-transcript ?s
      (staged-trace-root data)

```

```

    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    rest ∧
  partial-query-openings-consistent trace-openings
    composition-openings (staged-alphas data) (index (to-nat raw))
using hit
unfolding out-eq
  staged-security-with-data-state-current-query-partial-opening-hit-at-def
  Let-def
by simp
from hit'
obtain raw trace-openings composition-openings rest where
  i-bound:  $i < \text{rounds}$ 
and lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some raw
and comp-nonempty: staged-composition-fri-roots data  $\neq []$ 
and verifier:
  Some (result, final-state)  $\in \text{set-dist } (\text{execute verify-monad } ?s)$ 
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and comp-auth:
  partial-authenticated-table
    (hd (staged-composition-fri-roots data)) (scale * clength)
    composition-openings final-state
and header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    rest
and consistent:
  partial-query-openings-consistent trace-openings
    composition-openings (staged-alphas data) (index (to-nat raw))
by blast
have idx-sample:  $\text{index } (\text{to-nat raw}) \in \text{query-sample-space}$ 

```

```

using consistent
unfolding partial-query-openings-consistent-def by simp
have witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
unfolding query-header-supported-partial-opening-witnesses-def
using comp-nonempty verifier trace-auth comp-auth header
by blast
have success:
  index (to-nat raw) ∈
    query-header-supported-partial-opening-success-indices-at ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    i
unfolding query-header-supported-partial-opening-success-indices-at-def
by (intro CollectI conjI exI[of - trace-openings]
    exI[of - composition-openings])
    (use idx-sample witness consistent in simp-all)
show ?thesis
unfolding out-eq staged-security-with-data-state-query-partial-opening-hit-at-def
  Let-def
using i-bound lookup success by simp
qed

lemma staged-security-with-data-state-current-query-partial-opening-hit-imp-partial-opening-hit:
assumes hit:
  staged-security-with-data-state-current-query-partial-opening-hit out
shows staged-security-with-data-state-query-partial-opening-hit out
proof –
from hit obtain i where i-bound: i < rounds
and round-hit:
  staged-security-with-data-state-current-query-partial-opening-hit-at
    i out
unfolding staged-security-with-data-state-current-query-partial-opening-hit-def
by blast
have
  staged-security-with-data-state-query-partial-opening-hit-at i out

```

by (rule
 staged-security-with-data-state-current-query-partial-opening-hit-at-imp-partial-opening-hit-at
 [*OF round-hit*])
 then show ?thesis
 by (rule *staged-security-with-data-state-query-partial-opening-hitI*
 [*OF i-bound*])
 qed

lemma *staged-security-with-data-state-current-query-partial-opening-hitI*:
 assumes *i < rounds*
 and
 staged-security-with-data-state-current-query-partial-opening-hit-at
 i out
 shows
 staged-security-with-data-state-current-query-partial-opening-hit out
 using *assms*
 unfolding *staged-security-with-data-state-current-query-partial-opening-hit-def*
 by blast

lemma *checked-staged-security-with-data-state-aligned-transcript-query-bad-imp-current-partial-opening-hit-on-suppo*:
 assumes *wf: staged-budget-wellformed budgets*
 and *controlled: staged-adversary-controlled budgets A*
 and *support:*
 Some (((data, attacker-state), result), final-state) ∈
 set-dist
 (execute
 (checked-staged-security-experiment-with-data-state A)
 adversary-initial-state)
 and *bad:*
 staged-security-with-data-state-verifier-event
 query-bad-with-aligned-transcript-partial-candidates
 (Some (((data, attacker-state), result), final-state))
 shows
 staged-security-with-data-state-current-query-partial-opening-hit
 (Some (((data, attacker-state), result), final-state))
proof –
 let ?s =
 verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)
from *checked-staged-security-experiment-with-data-state-outcomeE*
 [*OF support*]
obtain *builder where builder:*
 Some (data, attacker-state) ∈
 set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state)
 and *verifier:*
 Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
 by blast

```

have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled builder])
from bad obtain fr f-fri-roots f-final as dg composition-fri-roots
  final query-idxs trace-openings composition-openings trace-table
  composition-table where partial:
    accepted-with-partial-initial-openings-aligned-transcript-consistent ?s
      (Some (result, final-state)) fr f-fri-roots f-final as dg
      composition-fri-roots final query-idxs trace-openings
      composition-openings
unfolding staged-security-with-data-state-verifier-event-def
  query-bad-with-aligned-transcript-partial-candidates-def
by simp blast
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
have shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
    query-idxs
by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape
    [OF partial])
from checked-staged-security-with-data-state-accepted-shape-query-chunks
  [OF wf controlled support shape]
obtain raw-idxs where
  as-eq: as = staged-alphas data
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
  and lookup:
     $\forall i < \text{rounds}.$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)

```



```

      (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
  by blast
have aligned0:
  accepted-with-partial-initial-openings-aligned ?s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
    [OF aligned])
from accepted-with-partial-initial-openings-aligned-shapes[OF aligned0]
obtain rest where header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
  by blast
have comp-nonempty: composition-fri-roots ≠ []
  using accepted-with-partial-initial-openings-aligned-shapes(1)
  [OF aligned0] .
have trace-partial:
  accepted-with-partial-trace-openings ?s
  (Some (result, final-state)) fr query-idxs trace-openings
  using accepted-with-partial-initial-openings-aligned-shapes(3)
  [OF aligned0] .
have comp-partial:
  accepted-with-partial-composition-openings ?s
  (Some (result, final-state)) (hd composition-fri-roots) query-idxs
  composition-openings
  using accepted-with-partial-initial-openings-aligned-shapes(4)
  [OF aligned0] .
have header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  composition-fri-roots = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  rest = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header staged-header]
  by simp
let ?i = 0
have i-bound: ?i < rounds
  using rounds-positive by simp
have round-consistent:
  partial-query-round-consistent trace-openings composition-openings as
  ?i (query-idxs ! ?i)
  using aligned i-bound
unfolding accepted-with-partial-initial-openings-aligned-consistent-def

```

```

by simp
have opening-consistent:
  partial-query-openings-consistent (trace-openings ! ?i)
    (composition-openings ! ?i) as (query-idxs ! ?i)
using round-consistent
unfolding partial-query-round-consistent-def
  partial-query-openings-consistent-def
by simp
have trace-table:
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! ?i) final-state
using trace-partial i-bound
unfolding accepted-with-partial-trace-openings-def by blast
have comp-table:
  partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) (composition-openings ! ?i) final-state
using comp-partial i-bound
unfolding accepted-with-partial-composition-openings-def by blast
have raw-idx:
  index (to-nat (raw-idxs ! ?i)) = query-idxs ! ?i
using query-idxs-eq len-raw i-bound by simp
have lookup-i:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge ?i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) ?i)) =
    Some (raw-idxs ! ?i)
using lookup i-bound by simp
have exists-hit:
  ∃ raw trace-openings' composition-openings' rest'.
    ?i < rounds ∧
    fmlookup (HashMap final-state)
      (QueryIndexChallenge ?i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) ?i)) =
      Some raw ∧
    staged-composition-fri-roots data ≠ [] ∧
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s) ∧
    partial-authenticated-table (staged-trace-root data)
      (scale * clength) trace-openings' final-state ∧
    partial-authenticated-table (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings' final-state ∧
    verifier-header-transcript ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)

```

```

    (staged-composition-final data)
    rest' ∧
    partial-query-openings-consistent trace-openings'
    composition-openings' (staged-alphas data) (index (to-nat raw))
  apply (rule exI[where x = raw-idxs ! ?i])
  apply (rule exI[where x = trace-openings ! ?i])
  apply (rule exI[where x = composition-openings ! ?i])
  apply (rule exI[where x = List.concat (staged-query-chunks data)])
  using i-bound lookup-i comp-nonempty verifier trace-table comp-table
    opening-consistent raw-idx header-eq staged-header
  by simp
have hit-at:
  staged-security-with-data-state-current-query-partial-opening-hit-at ?i
  (Some (((data, attacker-state), result), final-state))
unfolding
  staged-security-with-data-state-current-query-partial-opening-hit-at-def
  Let-def
  using exists-hit by simp
show ?thesis
  by (rule staged-security-with-data-state-current-query-partial-opening-hitI
      [OF i-bound hit-at])
qed

lemma checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
  and bad:
    staged-security-with-data-state-verifier-event
    query-bad-with-aligned-transcript-partial-candidates out
  show staged-security-with-data-state-current-query-partial-opening-hit out
proof (cases out)
  case None
  then show ?thesis

```

```

    using bad
    unfolding staged-security-with-data-state-verifier-event-def
      query-bad-with-aligned-transcript-partial-candidates-def
    by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eg:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
show ?thesis
  unfolding out-eg
  by (rule
    checked-staged-security-with-data-state-aligned-transcript-query-bad-imp-current-partial-opening-hit-on-support
      [OF wf controlled])
    (use support bad out-eg in simp-all)
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-data-state-current-
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
      (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state))
  shows
    staged-security-with-data-state-current-query-partial-opening-hit
      (Some (((data, attacker-state), result), final-state))
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
      [OF support])
  from checked-staged-security-experiment-with-data-state-outcomeE

```

```

    [OF data-support]
obtain builder verifier where
  builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
have header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-current-openingsE
    [OF wf controlled support hit])
  fix s trace-table composition-table query-idxs f i raw trace-openings
    composition-openings trace-openingss composition-openingsss
  assume s-def:
    s = verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  and i-bound: i < rounds
  and final-lookup:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some raw
  and idx-eq: index (to-nat raw) = query-idxs ! i
  and comp-nonempty: staged-composition-fri-roots data ≠ []
  and trace-auth:
    partial-authenticated-table (staged-trace-root data)
      (scale * clength) trace-openings final-state
  and comp-auth:
    partial-authenticated-table
      (hd (staged-composition-fri-roots data)) (scale * clength)
      composition-openings final-state
  and consistent:

```

```

    partial-query-openings-consistent trace-openings
    composition-openings (staged-alphas data) (query-idxs ! i)
have consistent-raw:
    partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
    using consistent idx-eq by simp
have round-hit:
    staged-security-with-data-state-current-query-partial-opening-hit-at i
    (Some (((data, attacker-state), result), final-state))
unfolding
    staged-security-with-data-state-current-query-partial-opening-hit-at-def
    Let-def
apply simp
by (intro exI[of - raw] exI[of - trace-openings]
    exI[of - composition-openings]
    exI[of - List.concat (staged-query-chunks data)] conjI)
    (use i-bound final-lookup comp-nonempty verifier trace-auth
    comp-auth header consistent-raw in simp-all)
show
    staged-security-with-data-state-current-query-partial-opening-hit
    (Some (((data, attacker-state), result), final-state))
by (rule
    staged-security-with-data-state-current-query-partial-opening-hitI
    [OF i-bound round-hit])
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-data-state-current-po
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    adversary-initial-state ≤
    wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state
proof -
let ?project =
    λx. ((snd (fst (fst x)), snd (fst x)), snd x)
let ?projected =
    λout. case out of
    None ⇒ False
    | Some (packed, t) ⇒
    staged-security-with-data-state-current-query-partial-opening-hit
    (Some (?project packed, t))
have event-le:

```

```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  ?projected adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
  out
show ?projected out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
case (Some packed)
then obtain alpha-prefix alpha-prefix-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((alpha-prefix, alpha-prefix-state), data),
      attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
have partial-hit:
  staged-security-with-data-state-current-query-partial-opening-hit
  (Some (((data, attacker-state), result), final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-data-state-current-p
    [OF wf controlled])
  (use support hit out-eq in simp-all)
show ?thesis
  unfolding out-eq
  using partial-hit by simp
qed
qed

```

```

have projection:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state
proof –
  have
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-current-query-partial-opening-hit
      adversary-initial-state =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
         $\gg (\lambda x. \text{return } (?project\ x))$ )
      staged-security-with-data-state-current-query-partial-opening-hit
      adversary-initial-state
  using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
  by simp
also have ... =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state
  by (subst wp-event-bind-return-map)
    (simp add:
      staged-security-with-data-state-current-query-partial-opening-hit-def
      staged-security-with-data-state-current-query-partial-opening-hit-at-def)
  finally show ?thesis
    by simp
qed
show ?thesis
  by (rule order-trans[OF event-le])
    (simp add: projection)
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-data-state-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and partial-bound:
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state  $\leq (Q::prob)$ 
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

```


checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq Q$
by (rule *order.trans*
 $[OF$
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-data-state-current-parti
partial-bound])
 (use *wf controlled* **in** *simp-all*)

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix*
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i*)
adversary-initial-state $\leq C i$
shows
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state $\leq (\sum i < \text{rounds}. C i)$

proof –
have *current-bound*:
wp-event
 (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state $\leq (\sum i < \text{rounds}. C i)$
by (rule
checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
 $[OF \text{ round-bound } wf \text{ controlled}]$)
show *?thesis*
by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-data-state-curr
 $[OF \text{ wf controlled } \text{current-bound}]$)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-prefix-target*
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *prefix-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event (*checked-staged-query-prefix-receive-with-state A i*)
 (*checked-staged-query-prefix-dynamic-index-hit*
staged-query-prefix-prefix-authenticated-opening-target)
adversary-initial-state $\leq P i$
and *path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-path-output-hit-at i)
 adversary-initial-state $\leq Q$ i

shows

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 adversary-initial-state $\leq (\sum i < \text{rounds}. P$ $i + Q$ $i)$

proof (rule

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix-curr
 [OF wf controlled])

fix i

assume i -bound: $i < \text{rounds}$

have prefix-hit-bound:

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
 adversary-initial-state $\leq P$ i

by (rule

checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target
 [OF wf controlled i -bound prefix-bound[OF i -bound]])

show

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state $\leq P$ $i + Q$ i

by (rule

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-path
 [OF wf controlled i -bound prefix-hit-bound path-bound[OF i -bound]])

qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-trace-indic

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and trace-frac:

$\bigwedge i$ prefix prefix-state.
 $i < \text{rounds} \implies$
 Some ((prefix, prefix-state), prefix-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-with-state A i)
 adversary-initial-state) $\implies$

nnreal

(card
 (staged-query-prefix-prefix-authenticated-trace-indices prefix
 prefix-state)) /
 nnreal (card query-sample-space) $\leq$ query-error-bound

and path-bound:

$\bigwedge i. i < \text{rounds} \implies$
 wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-path-output-hit-at i)
adversary-initial-state ≤ Q i

shows

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
adversary-initial-state ≤
(∑ i < rounds.
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound + Q i)

proof (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix-curr
[OF wf controlled])

fix *i*

assume *i-bound: i < rounds*

have *prefix-hit-bound:*

wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) + query-error-bound

by (rule
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-trace-indices
[OF wf controlled i-bound])
(use trace-frac[OF i-bound] in blast)

show

wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound + Q i

by (rule
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-path
[OF wf controlled i-bound prefix-hit-bound path-bound[OF i-bound]])

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-trace-indices-path-and-budgets:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *trace-frac:*

$\bigwedge i \text{ prefix prefix-state.}$
 $i < \text{rounds} \implies$
Some ((prefix, prefix-state), prefix-state) ∈
set-dist

$(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \text{ adversary-initial-state}) \implies$
 nnreal
 $(\text{card } (\text{staged-query-prefix-prefix-authenticated-trace-indices } \text{prefix} \text{ prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
and path-bound :
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq Q \ i$
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event } \text{composition-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries } \text{budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries } \text{budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries } \text{budgets } 0) +$
 $(\sum i < \text{rounds}.$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries } \text{budgets } i + 1) +$
 $\text{query-error-bound} + Q \ i) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget } \text{budgets } +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget})$
proof –
let $?Q =$
 $(\sum i < \text{rounds}.$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries } \text{budgets } i + 1) +$
 $\text{query-error-bound} + Q \ i)$
have query-bound :
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit}$
 $\text{adversary-initial-state} \leq ?Q$
by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-trace-indices-a}$
 $[\text{OF } \text{wf controlled trace-frac } \text{path-bound}])$
have without-bound :
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
adversary-initial-state ≤ 0
by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-bound
[OF wf controlled])
have *drift-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
adversary-initial-state $\leq ?Q + 0$
by (rule
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-query-and-without-query
[OF query-bound without-bound])
have *comp-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event composition-bad*)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
(*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +
(*?Q + 0*) +
hash-collision-budget-value 0
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
checked-staged-security-with-data-state-composition-bad-bound-from-composition-candidate-drift-and-budgets
[OF wf controlled drift-bound])
then show *?thesis*
by *simp*
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-data-state-partial-*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *support*:
Some (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*) $\in$
set-dist
(*execute*
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
adversary-initial-state)
and *hit*:
checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
(*Some* (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),

```

    result), final-state))
shows
  staged-security-with-data-state-query-partial-opening-hit
    (Some (((data, attacker-state), result), final-state))
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
      [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
obtain builder verifier where
  builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
have header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-current-openingsE
    [OF wf controlled support hit])
  fix s trace-table composition-table query-idxs f i raw trace-openings
    composition-openings trace-openingss composition-openingss
  assume s-def:
    s = verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  and i-bound: i < rounds

```

```

and final-lookup:
  fmllookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data i)) =
      Some raw
and idx-eq: index (to-nat raw) = query-idxs ! i
and comp-nonempty: staged-composition-fri-roots data  $\neq$  []
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and comp-auth:
  partial-authenticated-table
    (hd (staged-composition-fri-roots data)) (scale * clength)
    composition-openings final-state
and consistent:
  partial-query-openings-consistent trace-openings
    composition-openings (staged-alphas data) (query-idxs ! i)
have consistent-raw:
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
using consistent idx-eq by simp
have idx-sample: index (to-nat raw)  $\in$  query-sample-space
using consistent-raw
unfolding partial-query-openings-consistent-def by simp
have witness:
  (trace-openings, composition-openings)  $\in$ 
    query-header-supported-partial-opening-witnesses ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
unfolding query-header-supported-partial-opening-witnesses-def
using comp-nonempty verifier trace-auth comp-auth header
by auto
have success:
  index (to-nat raw)  $\in$ 
    query-header-supported-partial-opening-success-indices-at ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)

```

```

      i
unfolding query-header-supported-partial-opening-success-indices-at-def
by (intro CollectI conjI exI[of - trace-openings]
      exI[of - composition-openings])
      (use idx-sample witness consistent-row in simp-all)
have round-hit:
  staged-security-with-data-state-query-partial-opening-hit-at i
  (Some (((data, attacker-state), result), final-state))
unfolding staged-security-with-data-state-query-partial-opening-hit-at-def
  Let-def
using i-bound final-lookup success by simp
show
  staged-security-with-data-state-query-partial-opening-hit
  (Some (((data, attacker-state), result), final-state))
by (rule staged-security-with-data-state-query-partial-opening-hitI
      [OF i-bound round-hit])
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-data-state-partial-op
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
  adversary-initial-state  $\leq$ 
  wp-event
  (checked-staged-security-experiment-with-data-state A)
  staged-security-with-data-state-query-partial-opening-hit
  adversary-initial-state

proof -
let ?project =
   $\lambda x. ((snd (fst (fst x)), snd (fst x)), snd x)$ 
let ?projected =
   $\lambda out. case\ out\ of$ 
    None  $\Rightarrow$  False
  | Some (packed, t)  $\Rightarrow$ 
    staged-security-with-data-state-query-partial-opening-hit
    (Some (?project packed, t))
have event-le:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
  adversary-initial-state  $\leq$ 
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  ?projected adversary-initial-state
proof (rule wp-event-mono-on-support)

```



```

fix out
assume support:
  out ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    out
show ?projected out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
      checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    by simp
next
  case (Some packed)
  then obtain alpha-prefix alpha-prefix-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data),
        attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  have partial-hit:
    staged-security-with-data-state-query-partial-opening-hit
      (Some (((data, attacker-state), result), final-state))
    by (rule
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-data-state-partial-op
        [OF wf controlled])
      (use support hit out-eq in simp-all)
  show ?thesis
    unfolding out-eq
    using partial-hit by simp
qed
qed
have projection:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-opening-hit
    adversary-initial-state

```

```

proof –
  have
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-opening-hit
      adversary-initial-state =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
         $\gg$  ( $\lambda x.$  return (?project x)))
      staged-security-with-data-state-query-partial-opening-hit
      adversary-initial-state
  using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
  by simp
also have ... =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state
  by (subst wp-event-bind-return-map)
    (simp add: staged-security-with-data-state-query-partial-opening-hit-def
      staged-security-with-data-state-query-partial-opening-hit-at-def)
finally show ?thesis
  by simp
qed
show ?thesis
  by (rule order-trans[OF event-le])
    (simp add: projection)
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-data-state-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and partial-bound:
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-opening-hit
    adversary-initial-state  $\leq$  (Q::prob)
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    adversary-initial-state  $\leq$  Q
by (rule order.trans
  [OF
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-data-state-partial-opening-
    partial-bound])
  (use wf controlled in simp-all)

end

```

end

theory *Staged-Security-Experiment-Composition-Query-Binding-Gap*
imports

Staged-Security-Experiment-Composition-Query-Header-Rounds

Staged-Security-Experiment-Composition-Query-Current

begin

Candidate-binding gap for the query-prefix composition route.

The dynamic header-supported query target is small only for candidates that are already visible in the prefix oracle state. This layer isolates the remaining proof obligation: if a later authenticated candidate pair is not in that prefix candidate set, the gap must be charged to Merkle/collision side events in the next proof layer.

context *soundness*

begin

lemma *checked-staged-query-prefix-candidate-pair-hit-imp-header-supported-target-hit:*

assumes *candidate:*

(*trace-table*, *composition-table*) $\in$
query-header-supported-partial-table-candidates prefix-state
(*sqp-trace-root prefix*)
(*sqp-trace-fri-roots prefix*)
(*sqp-trace-final prefix*)
(*sqp-alphas prefix*)
(*sqp-degree prefix*)
(*sqp-composition-fri-roots prefix*)
(*sqp-composition-final prefix*)

and *hit:*

checked-staged-query-prefix-dynamic-index-hit
(*staged-query-prefix-candidate-pair-query-target-from-prefix*
trace-table composition-table)
(*Some (((prefix, prefix-state), raw), raw-state)*)

shows

checked-staged-query-prefix-dynamic-index-hit
staged-query-prefix-header-supported-query-target
(*Some (((prefix, prefix-state), raw), raw-state)*)

proof –

have *subset:*

staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table prefix prefix-state $\subseteq$
staged-query-prefix-header-supported-query-target prefix prefix-state

unfolding

staged-query-prefix-candidate-pair-query-target-from-prefix-def
staged-query-prefix-header-supported-query-target-def
query-header-supported-partial-union-good-sets-def

using *candidate query-sampling-success-space-subset* **by** *auto*

show *?thesis*

using *hit subset*
unfolding *checked-staged-query-prefix-dynamic-index-hit-def*
by *auto*
qed

definition *query-header-supported-single-round-partial-table-candidates*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat} \Rightarrow ('f \text{ list} \times 'f \text{ list}) \text{ set}$

where

query-header-supported-single-round-partial-table-candidates s fr
f-fri-roots f-final as dg composition-fri-roots final i =
 $\{(trace\text{-}table, composition\text{-}table).$
 $composition\text{-}fri\text{-}roots \neq [] \wedge$
 $(\exists trace\text{-}openings composition\text{-}openings \text{ rest.}$
 $partial\text{-}authenticated\text{-}table \text{ fr } (scale * clength)$
 $trace\text{-}openings \text{ s} \wedge$
 $partial\text{-}authenticated\text{-}table (hd \text{ composition-fri-roots})$
 $(scale * clength) \text{ composition-openings s} \wedge$
 $partial\text{-}trace\text{-}table\text{-}candidate \text{ trace-table}$
 $((replicate \text{ rounds } []) [i := trace\text{-}openings]) \wedge$
 $partial\text{-}composition\text{-}table\text{-}candidate \text{ composition-table}$
 $((replicate \text{ rounds } []) [i := composition\text{-}openings]) \wedge$
 $verifier\text{-}header\text{-}transcript \text{ s fr f-fri-roots f-final as dg}$
 $composition\text{-}fri\text{-}roots \text{ final rest})\}$

lemma *query-header-supported-single-round-partial-table-candidatesI:*

assumes *comp-nonempty: composition-fri-roots $\neq []$*

and *trace-auth:*

$partial\text{-}authenticated\text{-}table \text{ fr } (scale * clength)$
 $trace\text{-}openings \text{ s}$

and *comp-auth:*

$partial\text{-}authenticated\text{-}table (hd \text{ composition-fri-roots})$
 $(scale * clength) \text{ composition-openings s}$

and *trace-candidate:*

$partial\text{-}trace\text{-}table\text{-}candidate \text{ trace-table}$
 $((replicate \text{ rounds } []) [i := trace\text{-}openings])$

and *comp-candidate:*

$partial\text{-}composition\text{-}table\text{-}candidate \text{ composition-table}$
 $((replicate \text{ rounds } []) [i := composition\text{-}openings])$

and *header:*

$verifier\text{-}header\text{-}transcript \text{ s fr f-fri-roots f-final as dg}$
 $composition\text{-}fri\text{-}roots \text{ final rest}$

shows

$(trace\text{-}table, composition\text{-}table) \in$
 $query\text{-}header\text{-}supported\text{-}single\text{-}round\text{-}partial\text{-}table\text{-}candidates \text{ s fr}$
 $f\text{-}fri\text{-}roots \text{ f-final as dg composition-fri-roots final i}$

unfolding *query-header-supported-single-round-partial-table-candidates-def*

using *assms by blast*

lemma *query-header-supported-single-round-partial-table-candidate-notin-contradicts:*

assumes *notin:*

(*trace-table*, *composition-table*) $\notin$
query-header-supported-single-round-partial-table-candidates *s* *fr*
f-fri-roots *f-final* *as dg composition-fri-roots final i*

and *comp-nonempty:* *composition-fri-roots* $\neq \square$

and *trace-auth:*

partial-authenticated-table *fr* (*scale* * *clength*)
trace-openings *s*

and *comp-auth:*

partial-authenticated-table (*hd composition-fri-roots*)
(*scale* * *clength*) *composition-openings* *s*

and *trace-candidate:*

partial-trace-table-candidate *trace-table*
((*replicate rounds* $\square$) [*i* := *trace-openings*])

and *comp-candidate:*

partial-composition-table-candidate *composition-table*
((*replicate rounds* $\square$) [*i* := *composition-openings*])

and *header:*

verifier-header-transcript *s* *fr f-fri-roots f-final as dg*
composition-fri-roots final rest

shows *False*

using *notin*

query-header-supported-single-round-partial-table-candidatesI
[*OF comp-nonempty trace-auth comp-auth trace-candidate*
comp-candidate header]

by *contradiction*

definition *staged-query-prefix-single-round-header-supported-query-target*

$:: \text{nat} \Rightarrow 'f \text{ staged-query-prefix-data} \Rightarrow$

$'f \text{ protocol-channel} \Rightarrow \text{nat set}$

where

staged-query-prefix-single-round-header-supported-query-target *i* *prefix*

prefix-state =

{*idx*.

$\exists$ *trace-table* *composition-table*.

(*trace-table*, *composition-table*) $\in$

query-header-supported-single-round-partial-table-candidates

prefix-state

(*sqp-trace-root* *prefix*)

(*sqp-trace-fri-roots* *prefix*)

(*sqp-trace-final* *prefix*)

(*sqp-alphas* *prefix*)

(*sqp-degree* *prefix*)

(*sqp-composition-fri-roots* *prefix*)

(*sqp-composition-final* *prefix*)

i $\wedge$

idx $\in$

staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table prefix prefix-state}

lemma *checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit:*

assumes *candidate:*

(trace-table, composition-table) ∈
query-header-supported-single-round-partial-table-candidates
prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)
i

and *hit:*

checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state))

shows

checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-single-round-header-supported-query-target i)
(Some (((prefix, prefix-state), raw), raw-state))

proof —

have *idx-in:*

index (to-nat raw) ∈
staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table prefix prefix-state

using *hit unfolding checked-staged-query-prefix-dynamic-index-hit-def*

by *simp*

have

index (to-nat raw) ∈
staged-query-prefix-single-round-header-supported-query-target i
prefix prefix-state

unfolding *staged-query-prefix-single-round-header-supported-query-target-def*

by *(intro CollectI exI[of - trace-table]*

exI[of - composition-table])

(use candidate idx-in in simp)

then show *?thesis*

unfolding *checked-staged-query-prefix-dynamic-index-hit-def*

by *simp*

qed

lemma *query-header-supported-partial-table-candidatesI-from-partials:*

assumes *comp-nonempty: composition-fri-roots ≠ []*

and *outcome:*

Some (result, final-state) ∈ set-dist (execute verify-monad s)

```

and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots)
  composition-query-idxs composition-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
shows
  (trace-table, composition-table) ∈
  query-header-supported-partial-table-candidates s fr f-fri-roots
  f-final as dg composition-fri-roots final
proof —
have body:
  composition-fri-roots ≠ [] ∧
  (∃ out trace-query-idxs composition-query-idxs trace-openings
   composition-openings rest.
   out ∈ set-dist (execute verify-monad s) ∧
   accepted-with-partial-trace-openings s out fr trace-query-idxs
   trace-openings ∧
   partial-trace-table-candidate trace-table trace-openings ∧
   accepted-with-partial-composition-openings s out
   (hd composition-fri-roots) composition-query-idxs
   composition-openings ∧
   partial-composition-table-candidate composition-table
   composition-openings ∧
   verifier-header-transcript s fr f-fri-roots f-final as dg
   composition-fri-roots final rest)
using comp-nonempty outcome trace-partial trace-candidate comp-partial
  comp-candidate header
by blast
show ?thesis
unfolding query-header-supported-partial-table-candidates-def
using body by simp
qed

lemma query-header-supported-partial-table-candidate-notin-contradicts-partials:
assumes notin:
  (trace-table, composition-table) ∉
  query-header-supported-partial-table-candidates s fr f-fri-roots
  f-final as dg composition-fri-roots final
and comp-nonempty: composition-fri-roots ≠ []

```

and *outcome*:
Some (result, final-state) ∈ set-dist (execute verify-monad s)
and *trace-partial*:
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr trace-query-idxs trace-openings
and *trace-candidate*:
partial-trace-table-candidate trace-table trace-openings
and *comp-partial*:
accepted-with-partial-composition-openings s
(Some (result, final-state)) (hd composition-fri-roots)
composition-query-idxs composition-openings
and *comp-candidate*:
partial-composition-table-candidate composition-table
composition-openings
and *header*:
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
shows *False*
using *notin*
query-header-supported-partial-table-candidatesI-from-partials
[OF comp-nonempty outcome trace-partial trace-candidate comp-partial
comp-candidate header]
by *contradiction*

lemma *query-header-supported-partial-table-candidate-notin-no-accepted-partials*:

assumes *notin*:
(trace-table, composition-table) ∉
query-header-supported-partial-table-candidates s fr f-fri-roots
f-final as dg composition-fri-roots final
and *comp-nonempty*: *composition-fri-roots ≠ []*
and *trace-candidate*:
partial-trace-table-candidate trace-table trace-openings
and *comp-candidate*:
partial-composition-table-candidate composition-table
composition-openings
shows
 $\neg (\exists \text{result final-state trace-query-idxs composition-query-idxs rest.}$
Some (result, final-state) ∈ set-dist (execute verify-monad s) ∧
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr trace-query-idxs trace-openings ∧
accepted-with-partial-composition-openings s
(Some (result, final-state)) (hd composition-fri-roots)
composition-query-idxs composition-openings ∧
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest)

proof

assume
 $\exists \text{result final-state trace-query-idxs composition-query-idxs rest.}$
Some (result, final-state) ∈ set-dist (execute verify-monad s) ∧

accepted-with-partial-trace-openings s
(Some (result, final-state)) fr trace-query-idxs trace-openings $\wedge$
accepted-with-partial-composition-openings s
(Some (result, final-state)) (hd composition-fri-roots)
composition-query-idxs composition-openings $\wedge$
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
then obtain *result final-state trace-query-idxs composition-query-idxs rest*
where *outcome:*
Some (result, final-state) $\in$ set-dist (execute verify-monad s)
and *trace-partial:*
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr trace-query-idxs trace-openings
and *comp-partial:*
accepted-with-partial-composition-openings s
(Some (result, final-state)) (hd composition-fri-roots)
composition-query-idxs composition-openings
and *header:*
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
by *blast*
show *False*
by (*rule query-header-supported-partial-table-candidate-notin-contradicts-partials*
[OF notin comp-nonempty outcome trace-partial trace-candidate
comp-partial comp-candidate header])
qed

lemma *accepted-with-partial-trace-openings-single-round-update-other-empty:*

assumes *partial:*

accepted-with-partial-trace-openings s out fr query-idxs
((replicate rounds []) [i := trace-openings])

and *j-bound: $j < \text{rounds}$*

and *j-ne: $j \neq i$*

shows *powers-scaled (query-idxs ! j) = []*

proof —

have *indices:*

map opening-index (((replicate rounds []) [i := trace-openings]) ! j) =
powers-scaled (query-idxs ! j)

using *accepted-with-partial-trace-openings-shapes(3)[OF partial j-bound]*

.

have *((replicate rounds []) [i := trace-openings]) ! j = []*

using *j-bound j-ne* **by** *simp*

then show *?thesis*

using *indices* **by** *simp*

qed

lemma *accepted-with-partial-composition-openings-single-round-update-other-false:*

assumes *partial:*

accepted-with-partial-composition-openings s out composition-root

```

      query-idxs ((replicate rounds []) [i := composition-openings])
    and j-bound: j < rounds
    and j-ne: j ≠ i
  shows False
proof -
  have indices:
    map opening-index (((replicate rounds []) [i := composition-openings]) ! j) =
      [query-idxs ! j, fri-sibling-index (scale * clength) (query-idxs ! j)]
  using
    accepted-with-partial-composition-openings-shapes(3)
    [OF partial j-bound]
  .
  have ((replicate rounds []) [i := composition-openings]) ! j = []
    using j-bound j-ne by simp
  then show False
    using indices by simp
qed

lemma query-header-supported-partial-opening-witnesses-pullback-or-path-output:
  assumes witness:
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses s fr f-fri-roots
      f-final as dg composition-fri-roots final
  shows
    (partial-authenticated-table fr (scale * clength) trace-openings s ∧
      partial-authenticated-table (hd composition-fri-roots) (scale * clength)
        composition-openings s) ∨
    (∃ witness-state. ∃ opn ∈ set trace-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state fr (scale * clength)
          (opening-index opn) (opening-value opn) (opening-path opn))
        s witness-state) ∨
    (∃ witness-state. ∃ opn ∈ set composition-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state (hd composition-fri-roots)
          (scale * clength) (opening-index opn) (opening-value opn)
          (opening-path opn))
        s witness-state)
proof -
  from witness obtain result witness-state rest where outcome:
    Some (result, witness-state) ∈ set-dist (execute verify-monad s)
  and trace-table:
    partial-authenticated-table fr (scale * clength) trace-openings
      witness-state
  and comp-table:
    partial-authenticated-table (hd composition-fri-roots)
      (scale * clength) composition-openings witness-state
  by (elim query-header-supported-partial-opening-witnessesE)
  have ext: s ≤ witness-state

```

```

    by (rule verify-monad-hash-extends[OF outcome])
  have trace-pull:
    partial-authenticated-table fr (scale * clength) trace-openings s ∨
    (∃ opn ∈ set trace-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state fr (scale * clength)
          (opening-index opn) (opening-value opn) (opening-path opn))
        s witness-state)
    by (rule partial-authenticated-table-pullback-or-new-output-hit
      [OF ext trace-table])
  have comp-pull:
    partial-authenticated-table (hd composition-fri-roots) (scale * clength)
      composition-openings s ∨
    (∃ opn ∈ set composition-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state (hd composition-fri-roots)
          (scale * clength) (opening-index opn) (opening-value opn)
          (opening-path opn))
        s witness-state)
    by (rule partial-authenticated-table-pullback-or-new-output-hit
      [OF ext comp-table])
  from trace-pull comp-pull show ?thesis by blast
qed

```

lemma *query-header-supported-single-round-candidate-from-witness-or-path-output:*

```

  assumes witness:
    (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses s fr f-fri-roots
    f-final as dg composition-fri-roots final
  and trace-candidate:
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  and comp-candidate:
    partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
  shows
    (trace-table, composition-table) ∈
    query-header-supported-single-round-partial-table-candidates s fr
    f-fri-roots f-final as dg composition-fri-roots final i ∨
    (∃ witness-state. ∃ opn ∈ set trace-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state fr (scale * clength)
          (opening-index opn) (opening-value opn) (opening-path opn))
        s witness-state) ∨
    (∃ witness-state. ∃ opn ∈ set composition-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state (hd composition-fri-roots)
          (scale * clength) (opening-index opn) (opening-value opn)
          (opening-path opn))

```

```

      s witness-state)
proof –
  from witness obtain result witness-state rest where comp-nonempty:
    composition-fri-roots ≠ []
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  by (elim query-header-supported-partial-opening-witnessesE)
have pull:
  (partial-authenticated-table fr (scale * clength) trace-openings s ∧
   partial-authenticated-table (hd composition-fri-roots) (scale * clength)
    composition-openings s) ∨
  (∃ witness-state. ∃ opn ∈ set trace-openings.
   hash-map-new-output-hit
    (merkle-path-target-roots witness-state fr (scale * clength)
     (opening-index opn) (opening-value opn) (opening-path opn))
    s witness-state) ∨
  (∃ witness-state. ∃ opn ∈ set composition-openings.
   hash-map-new-output-hit
    (merkle-path-target-roots witness-state (hd composition-fri-roots)
     (scale * clength) (opening-index opn) (opening-value opn)
```

(*opening-path opn*))

```

    s witness-state)
  by (rule query-header-supported-partial-opening-witnesses-pullback-or-path-output
    [OF witness])
then show ?thesis
proof
  assume auths:
    partial-authenticated-table fr (scale * clength) trace-openings s ∧
    partial-authenticated-table (hd composition-fri-roots) (scale * clength)
      composition-openings s
  have candidate:
    (trace-table, composition-table) ∈
      query-header-supported-single-round-partial-table-candidates s fr
        f-fri-roots f-final as dg composition-fri-roots final i
  by (rule query-header-supported-single-round-partial-table-candidatesI
    [OF comp-nonempty - - trace-candidate comp-candidate header])
    (use auths in simp-all)
  then show ?thesis by simp
next
  assume
    (∃ witness-state. ∃ opn ∈ set trace-openings.
     hash-map-new-output-hit
      (merkle-path-target-roots witness-state fr (scale * clength)
       (opening-index opn) (opening-value opn) (opening-path opn))
      s witness-state) ∨
    (∃ witness-state. ∃ opn ∈ set composition-openings.
     hash-map-new-output-hit
      (merkle-path-target-roots witness-state (hd composition-fri-roots)
```

```

      (scale * clength) (opening-index opn) (opening-value opn)
      (opening-path opn))
    s witness-state)
  then show ?thesis by simp
qed
qed

lemma query-header-supported-single-round-candidate-notin-imp-path-output:
  assumes witness:
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses s fr f-fri-roots
      f-final as dg composition-fri-roots final
  and trace-candidate:
    partial-trace-table-candidate trace-table
      ((replicate rounds []) [i := trace-openings])
  and comp-candidate:
    partial-composition-table-candidate composition-table
      ((replicate rounds []) [i := composition-openings])
  and notin:
    (trace-table, composition-table) ∉
      query-header-supported-single-round-partial-table-candidates s fr
      f-fri-roots f-final as dg composition-fri-roots final i
  shows
    (∃ witness-state. ∃ opn ∈ set trace-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state fr (scale * clength)
          (opening-index opn) (opening-value opn) (opening-path opn))
        s witness-state) ∨
    (∃ witness-state. ∃ opn ∈ set composition-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots witness-state (hd composition-fri-roots)
          (scale * clength) (opening-index opn) (opening-value opn)
          (opening-path opn))
        s witness-state)
  using query-header-supported-single-round-candidate-from-witness-or-path-output
    [OF witness trace-candidate comp-candidate] notin
  by blast

```

definition *checked-staged-security-with-query-prefix-header-supported-query-target-hit-at*
where

```

checked-staged-security-with-query-prefix-header-supported-query-target-hit-at
  i out ⟷
  (case out of
    None ⇒ False
  | Some (((((prefix, prefix-state), raw), raw-state), continuation),
    final-state) ⇒
    checked-staged-query-prefix-dynamic-index-hit
      staged-query-prefix-header-supported-query-target
      (Some (((prefix, prefix-state), raw), raw-state)))

```

definition *checked-staged-security-with-query-prefix-candidate-binding-gap-at*
where

checked-staged-security-with-query-prefix-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
 trace-openings composition-openings i out $\wedge$
 partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings]) $\wedge$
 partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings]) $\wedge$
 (trace-table, composition-table) $\notin$
 query-header-supported-partial-table-candidates prefix-state
 (sqp-trace-root prefix)
 (sqp-trace-fri-roots prefix)
 (sqp-trace-final prefix)
 (sqp-alphas prefix)
 (sqp-degree prefix)
 (sqp-composition-fri-roots prefix)
 (sqp-composition-final prefix))

definition *checked-staged-security-with-query-prefix-candidate-binding-gap-side*
where

checked-staged-security-with-query-prefix-candidate-binding-gap-side
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings]) $\wedge$
 partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings]) $\wedge$
 (trace-table, composition-table) $\notin$
 query-header-supported-partial-table-candidates prefix-state
 (sqp-trace-root prefix)
 (sqp-trace-fri-roots prefix)
 (sqp-trace-final prefix)
 (sqp-alphas prefix)
 (sqp-degree prefix)
 (sqp-composition-fri-roots prefix)
 (sqp-composition-final prefix))

definition *checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at*
where

checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-candidate-binding-gap-side
trace-openings composition-openings trace-table composition-table i out $\wedge$
checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
trace-openings composition-openings i out

lemma *checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-imp-current-prefix-authenticated:*

assumes *gap:*

checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out

shows

checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
i out

using *gap*

unfolding

checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at-def
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-def

by *blast*

lemma *checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-bound-from-dynamic-target:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *prefix-bound:*

wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
staged-query-prefix-prefix-authenticated-opening-target)
adversary-initial-state $\leq$ B

shows

wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq$ B

proof –

have *event-le:*

wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq$
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
i)

adversary-initial-state
by (rule *wp-event-mono*)
 (rule
checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-imp-current-prefix-authenticated)
have *current-bound*:
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at*
i)
adversary-initial-state $\leq B$
by (rule
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target
 [*OF wf controlled i-bound prefix-bound*])
show *?thesis*
by (rule *order-trans*[*OF event-le current-bound*])
qed

lemma *checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-bound-from-trace-indices*:

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *i-bound*: *i < rounds*

and *trace-frac*:

$\bigwedge \text{prefix } \text{prefix-state.}$

Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$

set-dist

(*execute* (*checked-staged-query-prefix-with-state A i*)

adversary-initial-state) $\implies$

nnreal

(*card*

(*staged-query-prefix-prefix-authenticated-trace-indices prefix*
prefix-state)) /

nnreal (*card query-sample-space*) $\leq \text{query-error-bound}$

shows

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at*

trace-openings composition-openings trace-table composition-table i)

adversary-initial-state $\leq$

staged-phase-relation-error size

(*staged-query-search-queries budgets i + 1*) + *query-error-bound*

proof –

have *current-bound*:

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at*

i)

adversary-initial-state $\leq$

staged-phase-relation-error size

(*staged-query-search-queries budgets i + 1*) + *query-error-bound*


```

by (rule
  checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-trace-indices
  [OF wf controlled i-bound trace-frac])
have event-le:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at
    trace-openings composition-openings trace-table composition-table i)
  adversary-initial-state ≤
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
    i)
  adversary-initial-state
by (rule wp-event-mono)
  (rule
    checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-imp-current-prefix-authenticated)
show ?thesis
  by (rule order-trans[OF event-le current-bound])
qed

lemma checked-staged-security-with-query-prefix-candidate-binding-gap-iff-fixed-header-component:
  checked-staged-security-with-query-prefix-candidate-binding-gap-at
    trace-openings composition-openings trace-table composition-table i out
  ↔
  checked-staged-security-with-query-prefix-fixed-header-component
    (checked-staged-security-with-query-prefix-candidate-binding-gap-side
      trace-openings composition-openings trace-table composition-table i)
    trace-openings composition-openings i out
unfolding
  checked-staged-security-with-query-prefix-candidate-binding-gap-at-def
  checked-staged-security-with-query-prefix-candidate-binding-gap-side-def
  checked-staged-security-with-query-prefix-fixed-header-component-def
by (cases out) (auto split: prod.splits)

lemma checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-imp-header-target-or-bir
assumes hit:
  checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
  trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
and trace-low: trace-table-low-degree trace-table
and comp-low: composition-table-low-degree maxDegree composition-table

```

and *not-all*:
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)

shows
checked-staged-security-with-query-prefix-header-supported-query-target-hit-at
i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)) $\vee$
checked-staged-security-with-query-prefix-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

proof –
from *hit* **have** *opening-target*:
index (to-nat raw) $\in$
staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i prefix prefix-state
by (rule
checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitE)

have *opening-subset*:
staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i prefix prefix-state $\subseteq$
staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table prefix prefix-state
by (rule
staged-query-prefix-candidate-opening-query-target-from-prefix-subset-candidate-pair-from-prefix
[OF trace-candidate comp-candidate trace-low comp-low not-all])

have *pair-target-hit*:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state))

using *opening-target opening-subset*
unfolding *checked-staged-query-prefix-dynamic-index-hit-def* **by** *auto*

show *?thesis*

proof (cases
(trace-table, composition-table) $\in$
query-header-supported-partial-table-candidates prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix))

```

case True
have header-target:
  checked-staged-query-prefix-dynamic-index-hit
  staged-query-prefix-header-supported-query-target
  (Some (((prefix, prefix-state), raw), raw-state))
by (rule
  checked-staged-query-prefix-candidate-pair-hit-imp-header-supported-target-hit
  [OF True pair-target-hit])
have
  checked-staged-security-with-query-prefix-header-supported-query-target-hit-at
  i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-header-supported-query-target-hit-at-def
using header-target by simp
then show ?thesis by simp
next
case False
have
  checked-staged-security-with-query-prefix-candidate-binding-gap-at
  trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-candidate-binding-gap-at-def
using hit trace-candidate comp-candidate False by simp
then show ?thesis by simp
qed
qed

```

```

lemma checked-staged-security-with-query-prefix-candidate-binding-gap-imp-header-authenticated:
assumes gap:
  checked-staged-security-with-query-prefix-candidate-binding-gap-at
  trace-openings composition-openings trace-table composition-table i out
shows
  checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
  trace-openings composition-openings i out
using gap
unfolding checked-staged-security-with-query-prefix-candidate-binding-gap-at-def
by (cases out) (auto split: prod.splits)

```

```

lemma checked-staged-security-with-query-prefix-candidate-binding-gap-current-imp-prefix-gap-or-opening-path-output:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and support:
  out ∈
  set-dist

```

```

    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state  $A$   $i$ )
      adversary-initial-state)
  and current:
    checked-staged-security-with-query-prefix-fixed-current-authenticated-component
      (checked-staged-security-with-query-prefix-candidate-binding-gap-side
        trace-openings composition-openings trace-table composition-table  $i$ )
      trace-openings composition-openings  $i$  out
  shows
    checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at
      trace-openings composition-openings trace-table composition-table  $i$  out  $\vee$ 
    checked-staged-security-with-query-prefix-opening-path-output-hit
      trace-openings composition-openings out
  proof —
  have side:
    checked-staged-security-with-query-prefix-candidate-binding-gap-side
      trace-openings composition-openings trace-table composition-table  $i$  out
  and current-hit:
    checked-staged-security-with-query-prefix-authenticated-opening-hit
      trace-openings composition-openings  $i$  out
  using current
  unfolding
    checked-staged-security-with-query-prefix-fixed-current-authenticated-component-def
    checked-staged-security-with-query-prefix-fixed-header-component-def
  by simp-all
  have split:
    checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
      trace-openings composition-openings  $i$  out  $\vee$ 
    checked-staged-security-with-query-prefix-opening-path-output-hit
      trace-openings composition-openings out
  by (rule
    checked-staged-security-with-query-prefix-fixed-current-authenticated-component-imp-prefix-or-path-output-on-sup
    [OF wf controlled i-bound support current])
  then show ?thesis
  proof
  assume prefix:
    checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
      trace-openings composition-openings  $i$  out
  have
    checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at
      trace-openings composition-openings trace-table composition-table  $i$  out
  unfolding
    checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at-def
  using side prefix by simp
  then show ?thesis by simp
  next
  assume
    checked-staged-security-with-query-prefix-opening-path-output-hit
      trace-openings composition-openings out

```

then show *?thesis* by simp
 qed
 qed

lemma *checked-staged-security-with-query-prefix-candidate-binding-gap-imp-current-or-path-output-on-support:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *support:*

out ∈

set-dist

(*execute*

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

adversary-initial-state)

and *gap:*

checked-staged-security-with-query-prefix-candidate-binding-gap-at

trace-openings composition-openings trace-table composition-table i out

shows

checked-staged-security-with-query-prefix-fixed-current-authenticated-component

(*checked-staged-security-with-query-prefix-candidate-binding-gap-side*

trace-openings composition-openings trace-table composition-table i)

trace-openings composition-openings i out ∨

checked-staged-security-with-query-prefix-header-witness-path-output-hit

(*checked-staged-security-with-query-prefix-candidate-binding-gap-side*

trace-openings composition-openings trace-table composition-table i)

trace-openings composition-openings i out

proof —

have *fixed:*

checked-staged-security-with-query-prefix-fixed-header-component

(*checked-staged-security-with-query-prefix-candidate-binding-gap-side*

trace-openings composition-openings trace-table composition-table i)

trace-openings composition-openings i out

using *gap*

unfolding

checked-staged-security-with-query-prefix-candidate-binding-gap-iff-fixed-header-component

.

show *?thesis*

by (rule

checked-staged-security-with-query-prefix-fixed-header-component-imp-current-or-witness-path-output-on-support

[*OF wf controlled i-bound support fixed*])

qed

lemma *checked-staged-security-with-query-prefix-candidate-binding-gap-imp-prefix-gap-or-path-outputs-on-support:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *support:*

out ∈

set-dist

```

      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state  $A$   $i$ )
        adversary-initial-state)
    and gap:
      checked-staged-security-with-query-prefix-candidate-binding-gap-at
        trace-openings composition-openings trace-table composition-table  $i$  out
    shows
      checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at
        trace-openings composition-openings trace-table composition-table  $i$  out  $\vee$ 
      checked-staged-security-with-query-prefix-opening-path-output-hit
        trace-openings composition-openings out  $\vee$ 
      checked-staged-security-with-query-prefix-header-witness-path-output-hit
        (checked-staged-security-with-query-prefix-candidate-binding-gap-side
          trace-openings composition-openings trace-table composition-table  $i$ )
          trace-openings composition-openings  $i$  out
    proof —
    have split:
      checked-staged-security-with-query-prefix-fixed-current-authenticated-component
        (checked-staged-security-with-query-prefix-candidate-binding-gap-side
          trace-openings composition-openings trace-table composition-table  $i$ )
          trace-openings composition-openings  $i$  out  $\vee$ 
      checked-staged-security-with-query-prefix-header-witness-path-output-hit
        (checked-staged-security-with-query-prefix-candidate-binding-gap-side
          trace-openings composition-openings trace-table composition-table  $i$ )
          trace-openings composition-openings  $i$  out
    by (rule
      checked-staged-security-with-query-prefix-candidate-binding-gap-imp-current-or-path-output-on-support
      [OF wf controlled  $i$ -bound support gap])
    then show ?thesis
    proof
    assume current:
      checked-staged-security-with-query-prefix-fixed-current-authenticated-component
        (checked-staged-security-with-query-prefix-candidate-binding-gap-side
          trace-openings composition-openings trace-table composition-table  $i$ )
          trace-openings composition-openings  $i$  out
    have
      checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at
        trace-openings composition-openings trace-table composition-table  $i$  out  $\vee$ 
      checked-staged-security-with-query-prefix-opening-path-output-hit
        trace-openings composition-openings out
    by (rule
      checked-staged-security-with-query-prefix-candidate-binding-gap-current-imp-prefix-gap-or-opening-path-output-on-support
      [OF wf controlled  $i$ -bound support current])
    then show ?thesis by blast
  next
  assume
    checked-staged-security-with-query-prefix-header-witness-path-output-hit
      (checked-staged-security-with-query-prefix-candidate-binding-gap-side
        trace-openings composition-openings trace-table composition-table  $i$ )

```

$\text{trace-openings composition-openings } i \text{ out}$
then show $?thesis$ **by** simp
qed
qed

lemma *checked-staged-security-with-query-prefix-candidate-binding-gap-bound-from-prefix-target-and-path-outputs:*
assumes $\text{wf: staged-budget-wellformed budgets}$
and $\text{controlled: staged-adversary-controlled budgets } A$
and $\text{i-bound: } i < \text{rounds}$
and prefix-bound:
 $\text{wp-event (checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $\text{staged-query-prefix-prefix-authenticated-opening-target})$
 $\text{adversary-initial-state} \leq P$
and $\text{opening-path-bound:}$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-opening-path-output-hit}$
 $\text{trace-openings composition-openings})$
 $\text{adversary-initial-state} \leq Q$
and $\text{witness-path-bound:}$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-header-witness-path-output-hit}$
 $(\text{checked-staged-security-with-query-prefix-candidate-binding-gap-side}$
 $\text{trace-openings composition-openings trace-table composition-table } i)$
 $\text{trace-openings composition-openings } i)$
 $\text{adversary-initial-state} \leq R$
shows
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-candidate-binding-gap-at}$
 $\text{trace-openings composition-openings trace-table composition-table } i)$
 $\text{adversary-initial-state} \leq P + Q + R$
proof –
let $?m =$
 $\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i$
let $?prefix =$
 $\text{checked-staged-security-with-query-prefix-prefix-candidate-binding-gap-at}$
 $\text{trace-openings composition-openings trace-table composition-table } i$
let $?open-path =$
 $\text{checked-staged-security-with-query-prefix-opening-path-output-hit}$
 $\text{trace-openings composition-openings}$
let $?witness-path =$
 $\text{checked-staged-security-with-query-prefix-header-witness-path-output-hit}$
 $(\text{checked-staged-security-with-query-prefix-candidate-binding-gap-side}$
 $\text{trace-openings composition-openings trace-table composition-table } i)$
 $\text{trace-openings composition-openings } i$
have event-le:

$wp\text{-event } ?m$
 $(checked\text{-staged}\text{-security}\text{-with}\text{-query}\text{-prefix}\text{-candidate}\text{-binding}\text{-gap}\text{-at}$
 $\quad trace\text{-openings } composition\text{-openings } trace\text{-table } composition\text{-table } i)$
 $adversary\text{-initial}\text{-state} \leq$
 $wp\text{-event } ?m$
 $(\lambda out. ?prefix \text{ out} \vee ?open\text{-path } out \vee ?witness\text{-path } out)$
 $adversary\text{-initial}\text{-state}$
by (rule $wp\text{-event}\text{-mono}\text{-on}\text{-support}$)
 $(rule$
 $checked\text{-staged}\text{-security}\text{-with}\text{-query}\text{-prefix}\text{-candidate}\text{-binding}\text{-gap}\text{-imp}\text{-prefix}\text{-gap}\text{-or}\text{-path}\text{-outputs}\text{-on}\text{-support}$
 $\quad [OF \text{ wf controlled } i\text{-bound}])$
have $union\text{-le}$:
 $wp\text{-event } ?m$
 $(\lambda out. ?prefix \text{ out} \vee ?open\text{-path } out \vee ?witness\text{-path } out)$
 $adversary\text{-initial}\text{-state} \leq$
 $wp\text{-event } ?m \text{ ?prefix } adversary\text{-initial}\text{-state} +$
 $(wp\text{-event } ?m \text{ ?open}\text{-path } adversary\text{-initial}\text{-state} +$
 $\quad wp\text{-event } ?m \text{ ?witness}\text{-path } adversary\text{-initial}\text{-state})$
proof –
have
 $wp\text{-event } ?m$
 $(\lambda out. ?prefix \text{ out} \vee ?open\text{-path } out \vee ?witness\text{-path } out)$
 $adversary\text{-initial}\text{-state} \leq$
 $wp\text{-event } ?m \text{ ?prefix } adversary\text{-initial}\text{-state} +$
 $wp\text{-event } ?m$
 $(\lambda out. ?open\text{-path } out \vee ?witness\text{-path } out)$
 $adversary\text{-initial}\text{-state}$
by (rule $wp\text{-event}\text{-union}\text{-bound}$)
also have
 $\dots \leq$
 $wp\text{-event } ?m \text{ ?prefix } adversary\text{-initial}\text{-state} +$
 $(wp\text{-event } ?m \text{ ?open}\text{-path } adversary\text{-initial}\text{-state} +$
 $\quad wp\text{-event } ?m \text{ ?witness}\text{-path } adversary\text{-initial}\text{-state})$
by (intro $add\text{-mono } order\text{-refl } wp\text{-event}\text{-union}\text{-bound}$)
finally show $?thesis$.
qed
have $prefix\text{-gap}\text{-bound}$:
 $wp\text{-event } ?m \text{ ?prefix } adversary\text{-initial}\text{-state} \leq P$
by (rule
 $checked\text{-staged}\text{-security}\text{-with}\text{-query}\text{-prefix}\text{-prefix}\text{-candidate}\text{-binding}\text{-gap}\text{-bound}\text{-from}\text{-dynamic}\text{-target}$
 $\quad [OF \text{ wf controlled } i\text{-bound } prefix\text{-bound}])$
have $sum\text{-bound}$:
 $wp\text{-event } ?m \text{ ?prefix } adversary\text{-initial}\text{-state} +$
 $(wp\text{-event } ?m \text{ ?open}\text{-path } adversary\text{-initial}\text{-state} +$
 $\quad wp\text{-event } ?m \text{ ?witness}\text{-path } adversary\text{-initial}\text{-state}) \leq P + Q + R$
proof –
have $path\text{-sum}\text{-bound}$:
 $wp\text{-event } ?m \text{ ?open}\text{-path } adversary\text{-initial}\text{-state} +$
 $wp\text{-event } ?m \text{ ?witness}\text{-path } adversary\text{-initial}\text{-state} \leq Q + R$


```

    by (rule add-mono[OF opening-path-bound witness-path-bound])
  have
    wp-event ?m ?prefix adversary-initial-state +
      (wp-event ?m ?open-path adversary-initial-state +
        wp-event ?m ?witness-path adversary-initial-state) ≤ P + (Q + R)
    by (rule add-mono[OF prefix-gap-bound path-sum-bound])
  then show ?thesis
    by (simp add: add.assoc)
qed
show ?thesis
  by (rule order-trans[OF event-le order-trans[OF union-le sum-bound]])
qed

lemma checked-staged-security-with-query-prefix-candidate-binding-gap-bound-from-budgets:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound: i < rounds
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-candidate-binding-gap-at
        trace-openings composition-openings trace-table composition-table i)
      adversary-initial-state ≤
      staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      (1::prob) / nnreal (card query-sample-space)
proof -
  have event-le:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-candidate-binding-gap-at
        trace-openings composition-openings trace-table composition-table i)
      adversary-initial-state ≤
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
        trace-openings composition-openings i)
      adversary-initial-state
  by (rule wp-event-mono)
  (rule
    checked-staged-security-with-query-prefix-candidate-binding-gap-imp-header-authenticated)
  have header-bound:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
        trace-openings composition-openings i)
      adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +

```

```

    (1::prob) / nnreal (card query-sample-space)
  by (rule
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-bound-from-budgets
    [OF wf controlled i-bound])
  show ?thesis
  by (rule order-trans[OF event-le header-bound])
qed

```

definition *checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at*

```

:: nat ⇒ 'f staged-adversary ⇒
  ((((((f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at
  i A out ⟷
  (case out of
    None ⇒ False
  | Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state) ⇒
    (∃ prefix prefix-state raw raw-state.
      Some (data, attacker-state) ∈
        set-dist
          (execute (checked-staged-transcript-program A)
            adversary-initial-state) ∧
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state) ∧
      checked-staged-query-prefix-dynamic-index-hit
        staged-query-prefix-header-supported-query-target
        (Some (((prefix, prefix-state), raw), raw-state))))))

```

definition *checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at*

```

:: nat ⇒ 'f staged-adversary ⇒
  ((((((f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
  i A out ⟷
  (case out of
    None ⇒ False
  | Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state) ⇒
    (∃ prefix prefix-state raw raw-state.
      Some (data, attacker-state) ∈

```

$set-dist$
 $(execute (checked-staged-transcript-program A)$
 $adversary-initial-state) \wedge$
 $Some (((prefix, prefix-state), raw), raw-state) \in$
 $set-dist$
 $(execute (checked-staged-query-prefix-receive-with-state A i)$
 $adversary-initial-state) \wedge$
 $checked-staged-query-prefix-dynamic-index-hit$
 $(staged-query-prefix-single-round-header-supported-query-target i)$
 $(Some (((prefix, prefix-state), raw), raw-state))))$

The preceding actual-alpha event is a diagnostic compatibility predicate: it existentially ranges over query-prefix branches. It should not be bounded directly from query-prefix target budgets unless a separate projection lemma is proved. Bounded public-route arguments should keep the event at the query-prefix data-state layer, or use an explicitly aligned actual-alpha predicate whose query-prefix output is fixed by the run.

definition *checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-at*
 $:: nat \Rightarrow 'f staged-adversary \Rightarrow$
 $(((((('f \times 'f list \times 'f list \times 'f) \times$
 $'f protocol-channel) \times 'f staged-proof-data) \times$
 $'f protocol-channel) \times unit list) \times$
 $'f protocol-channel) option \Rightarrow bool$

where

$checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-at$
 $i A out \longleftrightarrow$
 $(case out of$
 $None \Rightarrow False$
 $| Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),$
 $result), final-state) \Rightarrow$
 $(\exists trace-table composition-table prefix prefix-state raw raw-state.$
 $Some (data, attacker-state) \in$
 $set-dist$
 $(execute (checked-staged-transcript-program A)$
 $adversary-initial-state) \wedge$
 $Some (((prefix, prefix-state), raw), raw-state) \in$
 $set-dist$
 $(execute (checked-staged-query-prefix-receive-with-state A i)$
 $adversary-initial-state) \wedge$
 $checked-staged-query-prefix-dynamic-index-hit$
 $(staged-query-prefix-candidate-pair-query-target-from-prefix$
 $trace-table composition-table)$
 $(Some (((prefix, prefix-state), raw), raw-state)) \wedge$
 $(trace-table, composition-table) \notin$
 $query-header-supported-partial-table-candidates prefix-state$
 $(sqp-trace-root prefix)$
 $(sqp-trace-fri-roots prefix)$
 $(sqp-trace-final prefix)$
 $(sqp-alphas prefix)$
 $(sqp-degree prefix)$

(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)))

definition *checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at*

:: nat $\Rightarrow$ 'f staged-adversary $\Rightarrow$
((((('f $\times$ 'f list $\times$ 'f list $\times$ 'f) $\times$
'f protocol-channel) $\times$ 'f staged-proof-data) $\times$
'f protocol-channel) $\times$ unit list) $\times$
'f protocol-channel) option $\Rightarrow$ bool

where

checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at

i A out $\longleftrightarrow$

(case out of

None $\Rightarrow$ False

| Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state) $\Rightarrow$

($\exists$ trace-table composition-table prefix prefix-state raw raw-state.

Some (data, attacker-state) $\in$

set-dist

(execute (checked-staged-transcript-program A)

adversary-initial-state) $\wedge$

Some (((prefix, prefix-state), raw), raw-state) $\in$

set-dist

(execute (checked-staged-query-prefix-receive-with-state A i)

adversary-initial-state) $\wedge$

checked-staged-query-prefix-dynamic-index-hit

(staged-query-prefix-candidate-pair-query-target-from-prefix

trace-table composition-table)

(Some (((prefix, prefix-state), raw), raw-state)) $\wedge$

(trace-table, composition-table) $\notin$

query-header-supported-single-round-partial-table-candidates

prefix-state

(sqp-trace-root prefix)

(sqp-trace-fri-roots prefix)

(sqp-trace-final prefix)

(sqp-alphas prefix)

(sqp-degree prefix)

(sqp-composition-fri-roots prefix)

(sqp-composition-final prefix)

i))

definition *checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-a*

:: nat $\Rightarrow$ 'f staged-adversary $\Rightarrow$
((((('f $\times$ 'f list $\times$ 'f list $\times$ 'f) $\times$
'f protocol-channel) $\times$ 'f staged-proof-data) $\times$
'f protocol-channel) $\times$ unit list) $\times$
'f protocol-channel) option $\Rightarrow$ bool

where

checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at

$i \text{ } A \text{ } \text{out} \longleftrightarrow$
(case out of
None $\Rightarrow$ False
| Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state) $\Rightarrow$
($\exists$ trace-openings composition-openings trace-table composition-table
prefix prefix-state raw raw-state.
Some (data, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) $\wedge$
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) $\wedge$
(trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data) $\wedge$
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings]) $\wedge$
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings]) $\wedge$
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state)) $\wedge$
(trace-table, composition-table) $\notin$
query-header-supported-single-round-partial-table-candidates
prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)
i))

This witness variant has the same diagnostic status: the authenticated openings are carried, but the query-prefix branch is still existential in the actual-alpha output. Use the same-run query-prefix witness-gap predicates for probability bounds.

```

lemma checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-imp-
assumes gap:
  checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
    i A out
shows
  checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at
    i A out
proof (cases out)
case None
then show ?thesis
  using gap
  unfolding
    checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at-def
  by simp
next
case (Some packed)
then obtain alpha-prefix alpha-prefix-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)
  by (cases packed) (auto split: prod.splits)
from gap obtain trace-openings composition-openings trace-table
  composition-table prefix prefix-state raw raw-state where builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
  and target-hit:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table)
      (Some (((prefix, prefix-state), raw), raw-state))
  and notin:
    (trace-table, composition-table)  $\notin$ 
      query-header-supported-single-round-partial-table-candidates
        prefix-state
        (sqp-trace-root prefix)
        (sqp-trace-fri-roots prefix)
        (sqp-trace-final prefix)
        (sqp-alphas prefix)
        (sqp-degree prefix)
        (sqp-composition-fri-roots prefix)
        (sqp-composition-final prefix)

```

i
unfolding *out-eq*
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at-def
by *auto*
have *body*:
 $\exists$ *trace-table composition-table prefix prefix-state raw raw-state*.
Some (data, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) $\wedge$
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) $\wedge$
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state)) $\wedge$
(trace-table, composition-table) $\notin$
query-header-supported-single-round-partial-table-candidates
prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)
 i
by (*intro exI[of - trace-table] exI[of - composition-table]*
exI[of - prefix] exI[of - prefix-state]
exI[of - raw] exI[of - raw-state])
(use builder prefix-receive target-hit notin in simp)
show *?thesis*
unfolding *out-eq*
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at-def
using *body by simp*
qed

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witnessE*:
assumes *gap*:
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
 i *A*
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)))
obtains *trace-openings composition-openings trace-table composition-table*
prefix prefix-state raw raw-state **where**
Some (data, attacker-state) $\in$
set-dist

(execute (checked-staged-transcript-program A)
adversary-initial-state)
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings])
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings])
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state))
(trace-table, composition-table) ∉
query-header-supported-single-round-partial-table-candidates
prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)
i
proof –
have *body*:
 $\exists$ *trace-openings composition-openings trace-table composition-table*
prefix prefix-state raw raw-state.
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) ∧
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) ∧
(trace-openings, composition-openings) ∈

query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) $\wedge$
partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings]) $\wedge$
partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings]) $\wedge$
checked-staged-query-prefix-dynamic-index-hit
 (staged-query-prefix-candidate-pair-query-target-from-prefix
 trace-table composition-table)
 (Some (((prefix, prefix-state), raw), raw-state)) $\wedge$
 (trace-table, composition-table) $\notin$
query-header-supported-single-round-partial-table-candidates
 prefix-state
 (sqp-trace-root prefix)
 (sqp-trace-fri-roots prefix)
 (sqp-trace-final prefix)
 (sqp-alphas prefix)
 (sqp-degree prefix)
 (sqp-composition-fri-roots prefix)
 (sqp-composition-final prefix)
 i
using gap
unfolding
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at-def
by simp
then obtain trace-openings composition-openings trace-table composition-table
 prefix prefix-state raw raw-state **where**
 Some (data, attacker-state) $\in$
 set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state)
 Some (((prefix, prefix-state), raw), raw-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-receive-with-state A i)
 adversary-initial-state)
 (trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)

```

    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target-from-prefix
     trace-table composition-table)
    (Some (((prefix, prefix-state), raw), raw-state))
  (trace-table, composition-table) ∉
  query-header-supported-single-round-partial-table-candidates
    prefix-state
    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)
    (sqp-composition-final prefix)
  i
  by blast
  then show ?thesis
  by (rule that)
qed

```

definition *checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at*

```

:: nat ⇒ 'f staged-adversary ⇒
  ((((((('f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at
  i A out ⟷
  (case out of
    None ⇒ False
  | Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state) ⇒
    (∃ trace-openings composition-openings trace-table composition-table
     prefix prefix-state raw raw-state.
     Some (data, attacker-state) ∈
       set-dist
         (execute (checked-staged-transcript-program A)
          adversary-initial-state) ∧
     Some (((prefix, prefix-state), raw), raw-state) ∈

```

set-dist
 (execute (checked-staged-query-prefix-receive-with-state *A i*)
 adversary-initial-state) ∧
 (trace-openings, composition-openings) ∈
 query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) ∧
 partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings]) ∧
 partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings]) ∧
 checked-staged-query-prefix-dynamic-index-hit
 (staged-query-prefix-candidate-pair-query-target-from-prefix
 trace-table composition-table)
 (Some (((prefix, prefix-state), raw), raw-state)) ∧
 (trace-table, composition-table) ∉
 query-header-supported-partial-table-candidates prefix-state
 (sqp-trace-root prefix)
 (sqp-trace-fri-roots prefix)
 (sqp-trace-final prefix)
 (sqp-alphas prefix)
 (sqp-degree prefix)
 (sqp-composition-fri-roots prefix)
 (sqp-composition-final prefix)))

lemma checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-imp-gap:

assumes gap:

checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at
i A out

shows

checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-at
i A out

using gap

proof (cases out)

case None

then show ?thesis

using gap

unfolding

checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at-def

by simp

next

case (Some packed)

then obtain *alpha-prefix alpha-prefix-state data attacker-state result*
final-state **where** *out-eq*:
out =
Some (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*)
by (*cases packed*) (*auto split: prod.splits*)
have *gap-body*:
 $\exists$ *trace-openings composition-openings trace-table composition-table*
prefix prefix-state raw raw-state.
Some (*data*, *attacker-state*) $\in$
set-dist
(*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\wedge$
Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) $\in$
set-dist
(*execute* (*checked-staged-query-prefix-receive-with-state* *A i*)
adversary-initial-state) $\wedge$
(*trace-openings*, *composition-openings*) $\in$
query-header-supported-partial-opening-witnesses
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*))
(*staged-trace-root* *data*)
(*staged-trace-fri-roots* *data*)
(*staged-trace-final* *data*)
(*staged-alphas* *data*)
(*staged-degree* *data*)
(*staged-composition-fri-roots* *data*)
(*staged-composition-final* *data*) $\wedge$
partial-trace-table-candidate trace-table
((*replicate rounds* []) [*i := trace-openings*]) $\wedge$
partial-composition-table-candidate composition-table
((*replicate rounds* []) [*i := composition-openings*]) $\wedge$
checked-staged-query-prefix-dynamic-index-hit
(*staged-query-prefix-candidate-pair-query-target-from-prefix*
trace-table composition-table)
(*Some* (((*prefix*, *prefix-state*), *raw*), *raw-state*)) $\wedge$
(*trace-table*, *composition-table*) $\notin$
query-header-supported-partial-table-candidates prefix-state
(*sqp-trace-root* *prefix*)
(*sqp-trace-fri-roots* *prefix*)
(*sqp-trace-final* *prefix*)
(*sqp-alphas* *prefix*)
(*sqp-degree* *prefix*)
(*sqp-composition-fri-roots* *prefix*)
(*sqp-composition-final* *prefix*)
using *gap*
unfolding *out-eq*
checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at-def
by *simp*

from *gap-body*
obtain *trace-openings composition-openings trace-table composition-table*
prefix prefix-state raw raw-state **where** *builder*:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *prefix-receive*:
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *target-hit*:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state))
and *notin*:
(trace-table, composition-table) ∉
query-header-supported-partial-table-candidates prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)
by *blast*
have *body*:
 $\exists$ *trace-table composition-table prefix prefix-state raw raw-state.*
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) ∧
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) ∧
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state)) ∧
(trace-table, composition-table) ∉
query-header-supported-partial-table-candidates prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)

```

      (sqp-composition-fri-roots prefix)
      (sqp-composition-final prefix)
    by (intro exI[of - trace-table] exI[of - composition-table]
        exI[of - prefix] exI[of - prefix-state]
        exI[of - raw] exI[of - raw-state])
      (use builder prefix-receive target-hit notin in simp)
  show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-at-def
  using body by simp
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witnessE*:

assumes *gap*:

checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at
i A

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))

obtains *trace-openings composition-openings trace-table composition-table*

prefix prefix-state raw raw-state **where**

Some (data, attacker-state) ∈

set-dist

(execute (checked-staged-transcript-program A)
 adversary-initial-state)

Some (((prefix, prefix-state), raw), raw-state) ∈

set-dist

(execute (checked-staged-query-prefix-receive-with-state A i)
 adversary-initial-state)

(trace-openings, composition-openings) ∈

query-header-supported-partial-opening-witnesses

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data))

(staged-trace-root data)

(staged-trace-fri-roots data)

(staged-trace-final data)

(staged-alphas data)

(staged-degree data)

(staged-composition-fri-roots data)

(staged-composition-final data)

partial-trace-table-candidate trace-table

((replicate rounds []) [i := trace-openings])

partial-composition-table-candidate composition-table

((replicate rounds []) [i := composition-openings])

checked-staged-query-prefix-dynamic-index-hit

(staged-query-prefix-candidate-pair-query-target-from-prefix
 trace-table composition-table)

(Some (((prefix, prefix-state), raw), raw-state))

(trace-table, composition-table) ∉

query-header-supported-partial-table-candidates prefix-state

```

    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)
    (sqp-composition-final prefix)
proof –
  have body:
    ∃ trace-openings composition-openings trace-table composition-table
      prefix prefix-state raw raw-state.
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state) ∧
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state) ∧
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data) ∧
    partial-trace-table-candidate trace-table
      ((replicate rounds []) [i := trace-openings]) ∧
    partial-composition-table-candidate composition-table
      ((replicate rounds []) [i := composition-openings]) ∧
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table)
      (Some (((prefix, prefix-state), raw), raw-state)) ∧
    (trace-table, composition-table) ∉
      query-header-supported-partial-table-candidates prefix-state
        (sqp-trace-root prefix)
        (sqp-trace-fri-roots prefix)
        (sqp-trace-final prefix)
        (sqp-alphas prefix)
        (sqp-degree prefix)
        (sqp-composition-fri-roots prefix)
        (sqp-composition-final prefix)
  using gap
  unfolding

```

checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at-def
by simp
then obtain *trace-openings composition-openings trace-table composition-table*
prefix prefix-state raw raw-state where
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings])
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings])
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state))
(trace-table, composition-table) ∉
query-header-supported-partial-table-candidates prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)
by blast
then show *?thesis*
by *(rule that)*
qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-imp-single-round-header*
assumes *hit:*
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
A trace-table composition-table i

$(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}), \text{result}), \text{final-state})))$
shows
checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
i A
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}), \text{result}), \text{final-state}))) \vee$
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at
i A
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}), \text{result}), \text{final-state})))$
proof –
from
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
 $[\text{OF hit}]$
obtain *prefix prefix-state raw raw-state* **where**
builder:
 $\text{Some } (\text{data}, \text{attacker-state}) \in$
set-dist
 $(\text{execute } (\text{checked-staged-transcript-program } A)$
adversary-initial-state)
and *prefix-receive:*
 $\text{Some } (((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}) \in$
set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
adversary-initial-state)
and *target-hit:*
checked-staged-query-prefix-dynamic-index-hit
 $(\text{staged-query-prefix-candidate-pair-query-target-from-prefix}$
trace-table composition-table)
 $(\text{Some } (((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state})))$
by (rule
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
 $[\text{OF hit}]$)
show *?thesis*
proof (cases
 $(\text{trace-table}, \text{composition-table}) \in$
query-header-supported-single-round-partial-table-candidates
prefix-state
 $(\text{sqp-trace-root } \text{prefix})$
 $(\text{sqp-trace-fri-roots } \text{prefix})$
 $(\text{sqp-trace-final } \text{prefix})$
 $(\text{sqp-alphas } \text{prefix})$
 $(\text{sqp-degree } \text{prefix})$
 $(\text{sqp-composition-fri-roots } \text{prefix})$
 $(\text{sqp-composition-final } \text{prefix})$
i)
case *True*
have *single-round-hit:*

```

checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-single-round-header-supported-query-target i)
  (Some (((prefix, prefix-state), raw), raw-state))
by (rule
  checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit
  [OF True target-hit])
have
checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
  i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
proof –
  have inner:
     $\exists$  prefix prefix-state raw raw-state.
    Some (data, attacker-state)  $\in$ 
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\wedge$ 
    Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)  $\wedge$ 
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-single-round-header-supported-query-target i)
      (Some (((prefix, prefix-state), raw), raw-state))
    by (intro exI[of - prefix] exI[of - prefix-state]
      exI[of - raw] exI[of - raw-state])
      (use builder prefix-receive single-round-hit in simp)
  show ?thesis
    unfolding
    checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at-def
    using inner by simp
  qed
  then show ?thesis by simp
next
case False
have
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at
  i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
proof –
  have inner:
     $\exists$  trace-table composition-table prefix prefix-state raw raw-state.
    Some (data, attacker-state)  $\in$ 
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\wedge$ 
    Some (((prefix, prefix-state), raw), raw-state)  $\in$ 

```

```

    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state) ∧
checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-candidate-pair-query-target-from-prefix
    trace-table composition-table)
  (Some (((prefix, prefix-state), raw), raw-state)) ∧
(trace-table, composition-table) ∉
query-header-supported-single-round-partial-table-candidates
  prefix-state
  (sqp-trace-root prefix)
  (sqp-trace-fri-roots prefix)
  (sqp-trace-final prefix)
  (sqp-alphas prefix)
  (sqp-degree prefix)
  (sqp-composition-fri-roots prefix)
  (sqp-composition-final prefix)
  i
by (intro exI[of - trace-table] exI[of - composition-table]
  exI[of - prefix] exI[of - prefix-state]
  exI[of - raw] exI[of - raw-state])
  (use builder prefix-receive target-hit False in simp)
show ?thesis
  unfolding
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at-def
  using inner by simp
qed
then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-with-witness-imp-single-
assumes hit:
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
  A trace-table composition-table i
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
and witness:
  (trace-openings, composition-openings) ∈
  query-header-supported-partial-opening-witnesses
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)

```

```

and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
shows
  checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
    i A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state)) ∨
  checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
    i A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
proof –
from
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
    [OF hit]
obtain prefix prefix-state raw raw-state where
  builder:
    Some (data, attacker-state) ∈
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
and target-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target-from-prefix
      trace-table composition-table)
    (Some (((prefix, prefix-state), raw), raw-state))
by (rule
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
    [OF hit])
show ?thesis
proof (cases
  (trace-table, composition-table) ∈
    query-header-supported-single-round-partial-table-candidates
    prefix-state
    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)

```

```

      (sqp-composition-final prefix)
    i)
  case True
  have single-round-hit:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-single-round-header-supported-query-target i)
      (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
      checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit
      [OF True target-hit])
  have
    checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
      i A
      (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state))
  proof -
    have inner:
      ∃ prefix prefix-state raw raw-state.
        Some (data, attacker-state) ∈
          set-dist
            (execute (checked-staged-transcript-program A)
              adversary-initial-state) ∧
        Some (((prefix, prefix-state), raw), raw-state) ∈
          set-dist
            (execute (checked-staged-query-prefix-receive-with-state A i)
              adversary-initial-state) ∧
        checked-staged-query-prefix-dynamic-index-hit
          (staged-query-prefix-single-round-header-supported-query-target i)
          (Some (((prefix, prefix-state), raw), raw-state))
    by (intro exI[of - prefix] exI[of - prefix-state]
        exI[of - raw] exI[of - raw-state])
        (use builder prefix-receive single-round-hit in simp)
    show ?thesis
      unfolding
        checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at-def
      using inner by simp
  qed
  then show ?thesis by simp
next
case False
have
  checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
    i A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  proof -
    have inner:
      ∃ trace-openings composition-openings trace-table composition-table
        prefix prefix-state raw raw-state.

```

```

Some (data, attacker-state) ∈
  set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state) ∧
Some (((prefix, prefix-state), raw), raw-state) ∈
  set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state) ∧
(trace-openings, composition-openings) ∈
  query-header-supported-partial-opening-witnesses
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data) ∧
partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings]) ∧
partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings]) ∧
checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-candidate-pair-query-target-from-prefix
    trace-table composition-table)
  (Some (((prefix, prefix-state), raw), raw-state)) ∧
(trace-table, composition-table) ∉
  query-header-supported-single-round-partial-table-candidates
    prefix-state
    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)
    (sqp-composition-final prefix)
i
by (intro exI[of - trace-openings] exI[of - composition-openings]
  exI[of - trace-table] exI[of - composition-table]
  exI[of - prefix] exI[of - prefix-state]
  exI[of - raw] exI[of - raw-state])
  (use builder prefix-receive witness trace-candidate comp-candidate
    target-hit False in simp)
show ?thesis
unfolding
  checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at-def
using inner by simp
qed

```

then show *?thesis* by simp
 qed
 qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-with-witness-imp-header*

assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix

A trace-table composition-table *i*

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))

and *witness*:

(trace-openings, composition-openings) ∈
 query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)

and *trace-candidate*:

partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings])

and *comp-candidate*:

partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings])

shows

checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at
i A

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state)) ∨

checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at
i A

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))

proof —

from

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
 [OF *hit*]

obtain *prefix prefix-state raw raw-state* **where**

builder:

Some (data, attacker-state) ∈
 set-dist
 (execute (checked-staged-transcript-program *A*)
 adversary-initial-state)

and *prefix-receive*:

Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) ∈

```

    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  and target-hit:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table)
      (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
      [OF hit])
show ?thesis
proof (cases
  (trace-table, composition-table) ∈
    query-header-supported-partial-table-candidates prefix-state
      (sqp-trace-root prefix)
      (sqp-trace-fri-roots prefix)
      (sqp-trace-final prefix)
      (sqp-alphas prefix)
      (sqp-degree prefix)
      (sqp-composition-fri-roots prefix)
      (sqp-composition-final prefix))
case True
have header-hit:
  checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-header-supported-query-target
      (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
    checked-staged-query-prefix-candidate-pair-hit-imp-header-supported-target-hit
      [OF True target-hit])
have
  checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at
    i A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
proof -
  have inner:
    ∃ prefix prefix-state raw raw-state.
      Some (data, attacker-state) ∈
        set-dist
          (execute (checked-staged-transcript-program A)
            adversary-initial-state) ∧
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state) ∧
    checked-staged-query-prefix-dynamic-index-hit
      staged-query-prefix-header-supported-query-target
        (Some (((prefix, prefix-state), raw), raw-state))

```



```

    by (intro exI[of - prefix] exI[of - prefix-state]
        exI[of - raw] exI[of - raw-state])
      (use builder prefix-receive header-hit in simp)
  show ?thesis
    unfolding
      checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at-def
    using inner by simp
qed
then show ?thesis by simp
next
case False
have
  checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at
  i A
  (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
proof -
  have inner:
     $\exists$  trace-openings composition-openings trace-table composition-table
      prefix prefix-state raw raw-state.
    Some (data, attacker-state)  $\in$ 
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)  $\wedge$ 
    Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)  $\wedge$ 
    (trace-openings, composition-openings)  $\in$ 
      query-header-supported-partial-opening-witnesses
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)  $\wedge$ 
    partial-trace-table-candidate trace-table
      ((replicate rounds []) [i := trace-openings])  $\wedge$ 
    partial-composition-table-candidate composition-table
      ((replicate rounds []) [i := composition-openings])  $\wedge$ 
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table)
      (Some (((prefix, prefix-state), raw), raw-state))  $\wedge$ 
    (trace-table, composition-table)  $\notin$ 
      query-header-supported-partial-table-candidates prefix-state

```

```

      (sqp-trace-root prefix)
      (sqp-trace-fri-roots prefix)
      (sqp-trace-final prefix)
      (sqp-alphas prefix)
      (sqp-degree prefix)
      (sqp-composition-fri-roots prefix)
      (sqp-composition-final prefix)
    by (intro exI[of - trace-openings] exI[of - composition-openings]
        exI[of - trace-table] exI[of - composition-table]
        exI[of - prefix] exI[of - prefix-state]
        exI[of - raw] exI[of - raw-state])
      (use builder prefix-receive witness trace-candidate comp-candidate
        target-hit False in simp)
  show ?thesis
    unfolding
      checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at-def
    using inner by simp
qed
then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
  assumes hit:
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
      i A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  and i-bound: i < rounds
  and trace-low:
     $\bigwedge$  trace-openings composition-openings trace-table
      composition-table.
    (trace-openings, composition-openings)  $\in$ 
      query-header-supported-partial-opening-witnesses
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)  $\implies$ 
      partial-trace-table-candidate trace-table
        ((replicate rounds []) [i := trace-openings])  $\implies$ 
      partial-composition-table-candidate composition-table
        ((replicate rounds []) [i := composition-openings])  $\implies$ 
      trace-table-low-degree trace-table
  and comp-low:

```

$\wedge$ *trace-openings composition-openings trace-table*
composition-table.
(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data) ⇒
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings]) ⇒
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings]) ⇒
composition-table-low-degree maxDegree composition-table
and not-all:

$\wedge$ *trace-openings composition-openings trace-table*
composition-table prefix prefix-state.
(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data) ⇒
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings]) ⇒
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings]) ⇒
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) ⇒
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)

shows

$(\exists \text{result}' \text{ final-state'}. \text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{Some (result}', \text{final-state'}))) \vee$
 $\text{checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at}$

```

i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state)) ∨
checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at
i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at-ca
    [OF hit i-bound]
  show ?thesis
  proof
    assume
      ∃ result' final-state'.
      partial-merkle-inconsistency-bad ?s (Some (result', final-state'))
    then show ?thesis by simp
  next
    assume candidates:
      ∃ trace-openings composition-openings trace-table
        composition-table.
      (trace-openings, composition-openings) ∈
        query-header-supported-partial-opening-witnesses ?s
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data) ∧
checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i
    (Some (((((alpha-prefix, alpha-prefix-state), data),
      attacker-state), result), final-state)) ∧
partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings]) ∧
partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
  then obtain trace-openings composition-openings trace-table
    composition-table where witness:
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses ?s
      (staged-trace-root data)

```

```

    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and opening-hit:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
    A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i
    (Some (((alpha-prefix, alpha-prefix-state), data),
      attacker-state), result), final-state))
  and trace-candidate:
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  and comp-candidate:
    partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
  by blast
have pair-hit:
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
  A trace-table composition-table i
  (Some (((alpha-prefix, alpha-prefix-state), data),
    attacker-state), result), final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix-imp-candidate-pair-que
    [OF opening-hit trace-candidate comp-candidate
      trace-low[OF witness trace-candidate comp-candidate]
      comp-low[OF witness trace-candidate comp-candidate]])
    (rule not-all[OF witness trace-candidate comp-candidate])
have
  checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at
  i A
  (Some (((alpha-prefix, alpha-prefix-state), data),
    attacker-state), result), final-state)) ∨
checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-witness-at
i A
  (Some (((alpha-prefix, alpha-prefix-state), data),
    attacker-state), result), final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-with-witness-imp-header-ta
    [OF pair-hit witness trace-candidate comp-candidate])
then show ?thesis by simp
qed
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*

assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
i A

(*Some* ((((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*)))

and *i-bound*: *i* < *rounds*

and *trace-low*:

$\bigwedge$ *trace-openings composition-openings trace-table*
composition-table.

(*trace-openings*, *composition-openings*) $\in$
query-header-supported-partial-opening-witnesses
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))
(*staged-trace-root data*)
(*staged-trace-fri-roots data*)
(*staged-trace-final data*)
(*staged-alphas data*)
(*staged-degree data*)
(*staged-composition-fri-roots data*)
(*staged-composition-final data*) $\implies$

partial-trace-table-candidate trace-table

((*replicate rounds* []) [*i := trace-openings*]) $\implies$

partial-composition-table-candidate composition-table

((*replicate rounds* []) [*i := composition-openings*]) $\implies$

trace-table-low-degree trace-table

and *comp-low*:

$\bigwedge$ *trace-openings composition-openings trace-table*
composition-table.

(*trace-openings*, *composition-openings*) $\in$
query-header-supported-partial-opening-witnesses
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))
(*staged-trace-root data*)
(*staged-trace-fri-roots data*)
(*staged-trace-final data*)
(*staged-alphas data*)
(*staged-degree data*)
(*staged-composition-fri-roots data*)
(*staged-composition-final data*) $\implies$

partial-trace-table-candidate trace-table

((*replicate rounds* []) [*i := trace-openings*]) $\implies$

partial-composition-table-candidate composition-table

((*replicate rounds* []) [*i := composition-openings*]) $\implies$

composition-table-low-degree maxDegree composition-table

and *not-all*:

$\bigwedge$ *trace-openings composition-openings trace-table*
composition-table prefix prefix-state.

(*trace-openings*, *composition-openings*) $\in$
query-header-supported-partial-opening-witnesses

```

    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)  $\implies$ 
    partial-trace-table-candidate trace-table
      ((replicate rounds []) [i := trace-openings])  $\implies$ 
    partial-composition-table-candidate composition-table
      ((replicate rounds []) [i := composition-openings])  $\implies$ 
    Some ((prefix, prefix-state), prefix-state)  $\in$ 
      set-dist
        (execute (checked-staged-query-prefix-with-state A i)
          adversary-initial-state)  $\implies$ 
     $\neg$  all-queries-consistent trace-table composition-table
      (sqp-alphas prefix)
  shows
    ( $\exists$  result' final-state'.
      partial-merkle-inconsistency-bad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        (Some (result', final-state')))  $\vee$ 
    checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
      i A
      (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)))  $\vee$ 
    checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
      i A
      (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)))
  proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  from
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at-ca
      [OF hit i-bound]
  show ?thesis
  proof
    assume
       $\exists$  result' final-state'.
        partial-merkle-inconsistency-bad ?s (Some (result', final-state'))
    then show ?thesis by simp
  next
    assume candidates:
       $\exists$  trace-openings composition-openings trace-table

```

composition-table.
 (trace-openings, composition-openings) ∈
 query-header-supported-partial-opening-witnesses ?s
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) ∧
 checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
 A
 ((replicate rounds []) [i := trace-openings])
 ((replicate rounds []) [i := composition-openings])
 i
 (Some (((alpha-prefix, alpha-prefix-state), data),
 attacker-state), result), final-state)) ∧
 partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings]) ∧
 partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings])
then obtain trace-openings composition-openings trace-table
 composition-table **where** witness:
 (trace-openings, composition-openings) ∈
 query-header-supported-partial-opening-witnesses ?s
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
and opening-hit:
 checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
 A
 ((replicate rounds []) [i := trace-openings])
 ((replicate rounds []) [i := composition-openings])
 i
 (Some (((alpha-prefix, alpha-prefix-state), data),
 attacker-state), result), final-state))
and trace-candidate:
 partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings])
and comp-candidate:
 partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings])
by blast
have pair-hit:
 checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix


```

    A trace-table composition-table i
    (Some (((alpha-prefix, alpha-prefix-state), data),
      attacker-state), result), final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix-imp-candidate-pair-que
      [OF opening-hit trace-candidate comp-candidate
        trace-low[OF witness trace-candidate comp-candidate]
        comp-low[OF witness trace-candidate comp-candidate]])
    (rule not-all[OF witness trace-candidate comp-candidate])
  have
    checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
      i A
      (Some (((alpha-prefix, alpha-prefix-state), data),
        attacker-state), result), final-state)) ∨
    checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
      i A
      (Some (((alpha-prefix, alpha-prefix-state), data),
        attacker-state), result), final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-with-witness-imp-single-ron
      [OF pair-hit witness trace-candidate comp-candidate])
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-imp-header-target-or-bin
  assumes hit:
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
      A trace-table composition-table i
      (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state))
  shows
    checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at
      i A
      (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)) ∨
    checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-at
      i A
      (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state))
proof -
  from
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
      [OF hit]
  obtain prefix prefix-state raw raw-state where
    i-bound: i < rounds
  and builder:
    Some (data, attacker-state) ∈
      set-dist

```

```

      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
and target-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target-from-prefix
      trace-table composition-table)
    (Some (((prefix, prefix-state), raw), raw-state))
by (rule
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
  [OF hit])
show ?thesis
proof (cases
  (trace-table, composition-table) ∈
    query-header-supported-partial-table-candidates prefix-state
    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)
    (sqp-composition-final prefix))
  case True
  have header-hit:
    checked-staged-query-prefix-dynamic-index-hit
      staged-query-prefix-header-supported-query-target
      (Some (((prefix, prefix-state), raw), raw-state))
    by (rule
      checked-staged-query-prefix-candidate-pair-hit-imp-header-supported-target-hit
      [OF True target-hit])
  have
    checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at
    i A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  proof –
  have inner:
    ∃ prefix prefix-state raw raw-state.
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state) ∧
    checked-staged-query-prefix-dynamic-index-hit
      staged-query-prefix-header-supported-query-target
      (Some (((prefix, prefix-state), raw), raw-state))

```

```

    by (intro exI[of - prefix] exI[of - prefix-state]
        exI[of - raw] exI[of - raw-state])
      (use prefix-receive header-hit in simp)
  show ?thesis
    unfolding
      checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at-def
    using builder inner by simp
qed
then show ?thesis by simp
next
case False
have
  checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-at
  i A
  (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
proof -
  have inner:
     $\exists$  trace-table composition-table prefix prefix-state raw raw-state.
    Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)  $\wedge$ 
    checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target-from-prefix
      trace-table composition-table)
    (Some (((prefix, prefix-state), raw), raw-state))  $\wedge$ 
    (trace-table, composition-table)  $\notin$ 
    query-header-supported-partial-table-candidates prefix-state
    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)
    (sqp-composition-final prefix)
  by (intro exI[of - trace-table] exI[of - composition-table]
      exI[of - prefix] exI[of - prefix-state]
      exI[of - raw] exI[of - raw-state])
    (use prefix-receive target-hit False in simp)
  show ?thesis
    unfolding
      checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-at-def
    using builder inner by simp
qed
then show ?thesis by simp
qed
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*

assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at

i A

(*Some* ((((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*)))

and *i-bound*: $i < \text{rounds}$

and *trace-low*:

$\bigwedge \text{trace-openings composition-openings trace-table}$
composition-table.

(*trace-openings*, *composition-openings*) $\in$
query-header-supported-partial-opening-witnesses

(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))

(*staged-trace-root data*)

(*staged-trace-fri-roots data*)

(*staged-trace-final data*)

(*staged-alphas data*)

(*staged-degree data*)

(*staged-composition-fri-roots data*)

(*staged-composition-final data*) $\implies$

partial-trace-table-candidate trace-table

((*replicate rounds []*) [*i := trace-openings*]) $\implies$

partial-composition-table-candidate composition-table

((*replicate rounds []*) [*i := composition-openings*]) $\implies$

trace-table-low-degree trace-table

and *comp-low*:

$\bigwedge \text{trace-openings composition-openings trace-table}$
composition-table.

(*trace-openings*, *composition-openings*) $\in$
query-header-supported-partial-opening-witnesses

(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))

(*staged-trace-root data*)

(*staged-trace-fri-roots data*)

(*staged-trace-final data*)

(*staged-alphas data*)

(*staged-degree data*)

(*staged-composition-fri-roots data*)

(*staged-composition-final data*) $\implies$

partial-trace-table-candidate trace-table

((*replicate rounds []*) [*i := trace-openings*]) $\implies$

partial-composition-table-candidate composition-table

((*replicate rounds []*) [*i := composition-openings*]) $\implies$

composition-table-low-degree maxDegree composition-table

and *not-all*:

$\bigwedge \text{trace-openings composition-openings trace-table}$
composition-table prefix prefix-state.

(*trace-openings*, *composition-openings*) $\in$

query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) $\implies$
partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings]) $\implies$
partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings]) $\implies$
 Some ((prefix, prefix-state), prefix-state) $\in$
set-dist
 (execute (checked-staged-query-prefix-with-state A i)
 adversary-initial-state) $\implies$
 $\neg$ all-queries-consistent trace-table composition-table
 (sqp-alphas prefix)

shows

($\exists$ result' final-state'.
partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (Some (result', final-state'))) $\vee$
checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at
 i A
 (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))) $\vee$
checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-at
 i A
 (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state)))

proof —

have base:

($\exists$ result' final-state'.
partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (Some (result', final-state'))) $\vee$
 ($\exists$ trace-table composition-table.
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
 A trace-table composition-table i
 (Some (((((alpha-prefix, alpha-prefix-state), data),
 attacker-state), result), final-state)))

by (rule

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at-
 [OF hit i-bound trace-low comp-low not-all])

```

then show ?thesis
proof
  assume
     $\exists \text{result}' \text{ final-state}'.$ 
    partial-merkle-inconsistency-bad
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result', final-state'))
  then show ?thesis by simp
next
  assume
     $\exists \text{trace-table composition-table}.$ 
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
    A trace-table composition-table i
    (Some (((((alpha-prefix, alpha-prefix-state), data),
      attacker-state), result), final-state)))
  then obtain trace-table composition-table where pair-hit:
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
    A trace-table composition-table i
    (Some (((((alpha-prefix, alpha-prefix-state), data),
      attacker-state), result), final-state)))
  by blast
  have
    checked-staged-security-with-actual-alpha-prefix-header-supported-query-target-hit-at
    i A
    (Some (((((alpha-prefix, alpha-prefix-state), data),
      attacker-state), result), final-state)))  $\vee$ 
    checked-staged-security-with-actual-alpha-prefix-query-prefix-candidate-binding-gap-at
    i A
    (Some (((((alpha-prefix, alpha-prefix-state), data),
      attacker-state), result), final-state)))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-imp-header-target-or-binding-gap
    [OF pair-hit])
  then show ?thesis by simp
qed
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Query-Single-Round
  imports Staged-Security-Experiment-Composition-Query-Binding-Gap
begin

```

Same-run single-round candidate binding for the composition query branch.

The actual-alpha diagnostic events in the previous layer intentionally record enough data

to relate a candidate pair to a query-prefix target, but they are too broad for Merkle path-output accounting because the query-prefix branch is existential. This layer states the single-round binding gap directly over the query-prefix data-state experiment, where prefix state, sampled query, verifier continuation, and authenticated openings are from the same run.

context *soundness*
begin

definition *checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at*
where

checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
 $i \text{ out} \longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((prefix, prefix-state), raw), raw-state), continuation),
 final-state) $\Rightarrow$
 checked-staged-query-prefix-dynamic-index-hit
 (staged-query-prefix-single-round-header-supported-query-target i)
 (Some (((prefix, prefix-state), raw), raw-state)))

lemma *checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at-iff-dynamic:*

checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
 $i \text{ out} \longleftrightarrow$
checked-staged-security-with-query-prefix-dynamic-index-hit
 (staged-query-prefix-single-round-header-supported-query-target i) out

unfolding

checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at-def
checked-staged-security-with-query-prefix-dynamic-index-hit-def
by (cases out) (auto split: prod.splits)

lemma *checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at-bound:*

assumes *prefix-bound:*

wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 (staged-query-prefix-single-round-header-supported-query-target i))
adversary-initial-state $\leq B$

shows

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
 i)
adversary-initial-state $\leq B$

unfolding

checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at-iff-dynamic
by (rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound
 [OF prefix-bound])

lemma *staged-query-prefix-candidate-pair-query-target-from-prefix-empty-if-all-queries-consistent:*

assumes *all-queries*:

all-queries-consistent trace-table composition-table (sqp-alphas prefix)

shows

*staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table prefix prefix-state = {}*

using *all-queries*

unfolding *staged-query-prefix-candidate-pair-query-target-from-prefix-def*

query-sampling-success-space-def

by *simp*

lemma *checked-staged-query-prefix-candidate-pair-hit-from-prefix-false-if-all-queries-consistent*:

assumes *all-queries*:

all-queries-consistent trace-table composition-table (sqp-alphas prefix)

shows

$\neg$ *checked-staged-query-prefix-dynamic-index-hit*
*(staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table)*
(Some (((prefix, prefix-state), raw), raw-state))

using

staged-query-prefix-candidate-pair-query-target-from-prefix-empty-if-all-queries-consistent
[OF all-queries, of prefix-state]

unfolding *checked-staged-query-prefix-dynamic-index-hit-def*

by *simp*

definition *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*

where

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side

trace-openings composition-openings trace-table composition-table i out

$\longleftrightarrow$

(case out of

None $\Rightarrow$ False

| *Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) $\Rightarrow$*

partial-trace-table-candidate trace-table

((replicate rounds []) [i := trace-openings]) $\wedge$

partial-composition-table-candidate composition-table

((replicate rounds []) [i := composition-openings]) $\wedge$

(trace-table, composition-table) $\notin$

query-header-supported-single-round-partial-table-candidates

prefix-state

(sqp-trace-root prefix)

(sqp-trace-fri-roots prefix)

(sqp-trace-final prefix)

(sqp-alphas prefix)

(sqp-degree prefix)

(sqp-composition-fri-roots prefix)

(sqp-composition-final prefix)

i)

definition *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at*
where

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
(case out of
None $\Rightarrow$ False
| Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) $\Rightarrow$
checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
trace-openings composition-openings i out $\wedge$
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings]) $\wedge$
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings]) $\wedge$
(trace-table, composition-table) $\notin$
query-header-supported-single-round-partial-table-candidates
prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)
i)

definition *checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at*
where

checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
trace-openings composition-openings trace-table composition-table i out $\wedge$
checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
trace-openings composition-openings i out

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-iff-fixed-header-component:*

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-fixed-header-component
(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
trace-openings composition-openings trace-table composition-table i)
trace-openings composition-openings i out

unfolding

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at-def
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side-def
checked-staged-security-with-query-prefix-fixed-header-component-def

by (cases out) (auto split: prod.splits)

lemma *checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-imp-current-prefix-authen*

assumes gap:

checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out

shows

checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
i out

using gap

unfolding

checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at-def
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-def

by blast

lemma *checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-bound-from-dynamic-targ*

assumes wf: *staged-budget-wellformed* budgets

and controlled: *staged-adversary-controlled* budgets A

and i-bound: $i < \text{rounds}$

and prefix-bound:

wp-event (*checked-staged-query-prefix-receive-with-state* A i)
(*checked-staged-query-prefix-dynamic-index-hit*
staged-query-prefix-prefix-authenticated-opening-target)
adversary-initial-state $\leq$ B

shows

wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* A i)
(*checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at*
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq$ B

proof —

have event-le:

wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* A i)
(*checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at*
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq$

wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* A i)
(*checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at*
i)
adversary-initial-state

by (rule wp-event-mono)

(rule

checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-imp-current-prefix-authen

have current-bound:

wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* A i)
(*checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at*

i)
adversary-initial-state $\leq B$
by (rule
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target
[OF wf controlled i-bound prefix-bound])
show ?thesis
by (rule *order-trans*[*OF event-le current-bound*])
qed

lemma *checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-imp-single-round-header*
assumes *hit*:
checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
trace-openings composition-openings i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))
and *trace-candidate*:
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings])
and *comp-candidate*:
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings])
and *trace-low*: *trace-table-low-degree trace-table*
and *comp-low*: *composition-table-low-degree maxDegree composition-table*
and *not-all*:
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)
shows
checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)) $\vee$
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

proof –
from *hit* **have** *opening-target*:
index (to-nat raw) $\in$
staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i prefix prefix-state
by (rule
checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitE)
have *opening-subset*:
staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i prefix prefix-state $\subseteq$

```

    staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
    composition-table prefix prefix-state
  by (rule
    staged-query-prefix-candidate-opening-query-target-from-prefix-subset-candidate-pair-from-prefix
    [OF trace-candidate comp-candidate trace-low comp-low not-all])
  have pair-target-hit:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table)
      (Some (((prefix, prefix-state), raw), raw-state))
  using opening-target opening-subset
  unfolding checked-staged-query-prefix-dynamic-index-hit-def by auto
  show ?thesis
  proof (cases
    (trace-table, composition-table) ∈
    query-header-supported-single-round-partial-table-candidates
    prefix-state
    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)
    (sqp-composition-final prefix)
    i)
  case True
  have target:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-single-round-header-supported-query-target i)
      (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
    checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit
    [OF True pair-target-hit])
  have
    checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
    i
    (Some ((((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)))
  unfolding
    checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at-def
  using target by simp
  then show ?thesis by simp
next
case False
have
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
  trace-openings composition-openings trace-table composition-table i
  (Some ((((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)))

```

```

    unfolding
      checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at-def
    using hit trace-candidate comp-candidate False by simp
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-current-imp-prefix-gap-or-openings
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-query-prefix-data-state A i)
          adversary-initial-state)
  and current:
    checked-staged-security-with-query-prefix-fixed-current-authenticated-component
      (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
        trace-openings composition-openings trace-table composition-table i)
    trace-openings composition-openings i out
  shows
    checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at
      trace-openings composition-openings trace-table composition-table i out ∨
    checked-staged-security-with-query-prefix-opening-path-output-hit
      trace-openings composition-openings out
  proof —
    have side:
      checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
        trace-openings composition-openings trace-table composition-table i out
    and current-hit:
      checked-staged-security-with-query-prefix-authenticated-opening-hit
        trace-openings composition-openings i out
    using current
    unfolding
      checked-staged-security-with-query-prefix-fixed-current-authenticated-component-def
      checked-staged-security-with-query-prefix-fixed-header-component-def
    by simp-all
    have split:
      checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
        trace-openings composition-openings i out ∨
      checked-staged-security-with-query-prefix-opening-path-output-hit
        trace-openings composition-openings out
    by (rule
      checked-staged-security-with-query-prefix-fixed-current-authenticated-component-imp-prefix-or-path-output-on-sup
        [OF wf controlled i-bound support current])
    then show ?thesis
  proof

```

```

assume prefix:
  checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit
  trace-openings composition-openings i out
have
  checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at
  trace-openings composition-openings trace-table composition-table i out
unfolding
  checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at-def
using side prefix by simp
then show ?thesis by simp
next
assume
  checked-staged-security-with-query-prefix-opening-path-output-hit
  trace-openings composition-openings out
then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-imp-current-or-path-output-on-s
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and support:
  out ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    adversary-initial-state)
and gap:
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
  trace-openings composition-openings trace-table composition-table i out
shows
  checked-staged-security-with-query-prefix-fixed-current-authenticated-component
  (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
    trace-openings composition-openings trace-table composition-table i)
  trace-openings composition-openings i out ∨
  checked-staged-security-with-query-prefix-header-witness-path-output-hit
  (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
    trace-openings composition-openings trace-table composition-table i)
  trace-openings composition-openings i out
proof –
have fixed:
  checked-staged-security-with-query-prefix-fixed-header-component
  (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
    trace-openings composition-openings trace-table composition-table i)
  trace-openings composition-openings i out
using gap
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-iff-fixed-header-component

```

.
show ?thesis
 by (rule
 checked-staged-security-with-query-prefix-fixed-header-component-imp-current-or-witness-path-output-on-support
 [*OF wf controlled i-bound support fixed*])
qed

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-imp-prefix-gap-or-path-outputs-c*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *support:*
 out ∈
 set-dist
 (*execute*
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 adversary-initial-state)
and *gap:*
 checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
 trace-openings composition-openings trace-table composition-table i out
shows
 checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at
 trace-openings composition-openings trace-table composition-table i out ∨
 checked-staged-security-with-query-prefix-opening-path-output-hit
 trace-openings composition-openings out ∨
 checked-staged-security-with-query-prefix-header-witness-path-output-hit
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
 trace-openings composition-openings trace-table composition-table i)
 trace-openings composition-openings i out
proof –
have *split:*
 checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
 trace-openings composition-openings trace-table composition-table i)
 trace-openings composition-openings i out ∨
 checked-staged-security-with-query-prefix-header-witness-path-output-hit
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
 trace-openings composition-openings trace-table composition-table i)
 trace-openings composition-openings i out
by (rule
 checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-imp-current-or-path-output-on-supp
 [*OF wf controlled i-bound support gap*])
then show ?thesis
proof
assume *current:*
 checked-staged-security-with-query-prefix-fixed-current-authenticated-component
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
 trace-openings composition-openings trace-table composition-table i)
 trace-openings composition-openings i out

have
checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out $\vee$
checked-staged-security-with-query-prefix-opening-path-output-hit
trace-openings composition-openings out
by (rule
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-current-imp-prefix-gap-or-opening
 $[OF\ wf\ controlled\ i\text{-bound}\ support\ current]$)
then show *?thesis* **by** *blast*
next
assume
checked-staged-security-with-query-prefix-header-witness-path-output-hit
(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
trace-openings composition-openings trace-table composition-table i)
trace-openings composition-openings i out
then show *?thesis* **by** *simp*
qed
qed

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-bound-from-prefix-target-and-pa*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *prefix-bound:*
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
staged-query-prefix-prefix-authenticated-opening-target)
adversary-initial-state $\leq P$
and *opening-path-bound:*
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-opening-path-output-hit
trace-openings composition-openings)
adversary-initial-state $\leq Q$
and *witness-path-bound:*
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-header-witness-path-output-hit
(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
trace-openings composition-openings trace-table composition-table i)
trace-openings composition-openings i)
adversary-initial-state $\leq R$
shows
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq P + Q + R$
proof –


```

let ?m =
  checked-staged-security-experiment-with-query-prefix-data-state A i
let ?prefix =
  checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-at
    trace-openings composition-openings trace-table composition-table i
let ?open-path =
  checked-staged-security-with-query-prefix-opening-path-output-hit
    trace-openings composition-openings
let ?witness-path =
  checked-staged-security-with-query-prefix-header-witness-path-output-hit
    (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
      trace-openings composition-openings trace-table composition-table i)
    trace-openings composition-openings i
have event-le:
  wp-event ?m
    (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
      trace-openings composition-openings trace-table composition-table i)
    adversary-initial-state ≤
  wp-event ?m
    (λout. ?prefix out ∨ ?open-path out ∨ ?witness-path out)
    adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule
    checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-imp-prefix-gap-or-path-outputs-on-s
      [OF wf controlled i-bound])
have union-le:
  wp-event ?m
    (λout. ?prefix out ∨ ?open-path out ∨ ?witness-path out)
    adversary-initial-state ≤
  wp-event ?m ?prefix adversary-initial-state +
  (wp-event ?m ?open-path adversary-initial-state +
    wp-event ?m ?witness-path adversary-initial-state)
proof –
  have
    wp-event ?m
      (λout. ?prefix out ∨ ?open-path out ∨ ?witness-path out)
      adversary-initial-state ≤
    wp-event ?m ?prefix adversary-initial-state +
    wp-event ?m
      (λout. ?open-path out ∨ ?witness-path out)
      adversary-initial-state
    by (rule wp-event-union-bound)
  also have
    ... ≤
    wp-event ?m ?prefix adversary-initial-state +
    (wp-event ?m ?open-path adversary-initial-state +
      wp-event ?m ?witness-path adversary-initial-state)
    by (intro add-mono order-refl wp-event-union-bound)
  finally show ?thesis .

```

qed
have *prefix-gap-bound*:
 wp-event ?*m* ?*prefix* *adversary-initial-state* $\leq P$
 by (rule
 checked-staged-security-with-query-prefix-prefix-single-round-candidate-binding-gap-bound-from-dynamic-target
 [*OF wf controlled i-bound prefix-bound*])
have *sum-bound*:
 wp-event ?*m* ?*prefix* *adversary-initial-state* +
 (*wp-event* ?*m* ?*open-path* *adversary-initial-state* +
 wp-event ?*m* ?*witness-path* *adversary-initial-state*) $\leq P + Q + R$
proof –
 have *path-sum-bound*:
 wp-event ?*m* ?*open-path* *adversary-initial-state* +
 wp-event ?*m* ?*witness-path* *adversary-initial-state* $\leq Q + R$
 by (rule *add-mono*[*OF opening-path-bound witness-path-bound*])
 have
 wp-event ?*m* ?*prefix* *adversary-initial-state* +
 (*wp-event* ?*m* ?*open-path* *adversary-initial-state* +
 wp-event ?*m* ?*witness-path* *adversary-initial-state*) $\leq P + (Q + R)$
 by (rule *add-mono*[*OF prefix-gap-bound path-sum-bound*])
 then show ?*thesis*
 by (*simp* *add*: *add.assoc*)
qed
show ?*thesis*
 by (rule *order-trans*[*OF event-le order-trans*[*OF union-le sum-bound*]])
qed

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-bound-from-witness-path-output*
assumes *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *i-bound*: *i* < *rounds*
and *path-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
 (*checked-staged-security-with-query-prefix-header-witness-path-output-hit*
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
 trace-openings composition-openings trace-table composition-table i)
 trace-openings composition-openings i)
 adversary-initial-state $\leq Q$
shows
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at*
 trace-openings composition-openings trace-table composition-table i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries* *budgets i + 1*) +
 (*1::prob*) / *nnreal* (*card query-sample-space*) + *Q*
proof –

have *event-eq*:
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i =
checked-staged-security-with-query-prefix-fixed-header-component
(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
trace-openings composition-openings trace-table composition-table i)
trace-openings composition-openings i
by (rule *ext*)
(rule
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-iff-fixed-header-component)
show *?thesis*
unfolding *event-eq*
by (rule
checked-staged-security-with-query-prefix-fixed-header-component-bound-from-witness-path-output-and-budgets
[*OF wf controlled i-bound path-bound*])
qed

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-bound-from-budgets*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
shows
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
(1::prob) / nnreal (card query-sample-space)
proof –
have *event-eq*:
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i =
checked-staged-security-with-query-prefix-fixed-header-component
(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
trace-openings composition-openings trace-table composition-table i)
trace-openings composition-openings i
by (rule *ext*)
(rule
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-iff-fixed-header-component)
show *?thesis*
unfolding *event-eq*
by (rule
checked-staged-security-with-query-prefix-fixed-header-component-bound-from-budgets
[*OF wf controlled i-bound*])
qed

lemma *checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-bound-from-target-and-f*

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *i-bound*: $i < \text{rounds}$
and *trace-candidate*:
 partial-trace-table-candidate *trace-table*
 ((replicate rounds []) [$i := \text{trace-openings}$])
and *comp-candidate*:
 partial-composition-table-candidate *composition-table*
 ((replicate rounds []) [$i := \text{composition-openings}$])
and *trace-low*: *trace-table-low-degree* *trace-table*
and *comp-low*: *composition-table-low-degree* *maxDegree* *composition-table*
and *not-all*:
 $\bigwedge \text{prefix prefix-state raw raw-state data attacker-state result}$
 final-state.
 Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\in$
 set-dist
 (*execute*
 (*checked-staged-security-experiment-with-query-prefix-data-state*
 A i)
 adversary-initial-state) $\implies$
 $\neg$ *all-queries-consistent* *trace-table* *composition-table*
 (*sqp-alphas* *prefix*)
and *target-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
 (*checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at*
 i)
 adversary-initial-state $\leq T$
shows
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
 (*checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit*
 trace-openings composition-openings i)
 adversary-initial-state $\leq$
 T +
 (*staged-phase-relation-error* *size*
 (*staged-query-search-queries* *budgets i + 1*) +
 ($1::\text{prob}$) / *nnreal* (*card query-sample-space*))
proof –
let *?m* =
 checked-staged-security-experiment-with-query-prefix-data-state *A i*
let *?target* =
 checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
 i
let *?gap* =
 checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
 trace-openings composition-openings trace-table composition-table i
have *event-le*:

```

wp-event ?m
  (checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
   trace-openings composition-openings i)
  adversary-initial-state ≤
wp-event ?m (λout. ?target out ∨ ?gap out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈ set-dist (execute ?m adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
    trace-openings composition-openings i out
  show ?target out ∨ ?gap out
  proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-def
    by simp
  next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state
    result final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  have not-all':
    ¬ all-queries-consistent trace-table composition-table
      (sqp-alphas prefix)
    by (rule not-all)
      (use support out-eq in simp)
  show ?thesis
    unfolding out-eq
    by (rule
      checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-imp-single-round-header-
      [OF - trace-candidate comp-candidate trace-low comp-low not-all'])
      (use hit out-eq in simp)
  qed
qed
have union-le:
  wp-event ?m (λout. ?target out ∨ ?gap out)
    adversary-initial-state ≤
  wp-event ?m ?target adversary-initial-state +
  wp-event ?m ?gap adversary-initial-state
  by (rule wp-event-union-bound)
have gap-bound:

```

```

wp-event ?m ?gap adversary-initial-state ≤
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
  (1::prob) / nnreal (card query-sample-space)
by (rule
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-bound-from-budgets
  [OF wf controlled i-bound])
have sum-bound:
  wp-event ?m ?target adversary-initial-state +
  wp-event ?m ?gap adversary-initial-state ≤
  T +
  (staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space))
by (rule add-mono[OF target-bound gap-bound])
show ?thesis
by (rule order-trans[OF event-le order-trans[OF union-le sum-bound]])
qed

lemma checked-staged-security-with-query-prefix-verifier-unsupported-witness-gap-imp-header-witness-path-output:
  assumes witness:
    (trace-openings, composition-openings) ∈
      query-header-supported-partial-opening-witnesses
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  and trace-candidate:
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  and comp-candidate:
    partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
  and notin-prefix:
    (trace-table, composition-table) ∉
      query-header-supported-single-round-partial-table-candidates
      prefix-state
      (sqp-trace-root prefix)
      (sqp-trace-fri-roots prefix)
      (sqp-trace-final prefix)
      (sqp-alphas prefix)
      (sqp-degree prefix)
      (sqp-composition-fri-roots prefix)
      (sqp-composition-final prefix)

```

i
and *notin-verifier*:
 (trace-table, composition-table) $\notin$
 query-header-supported-single-round-partial-table-candidates
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
 i
shows
 checked-staged-security-with-query-prefix-header-witness-path-output-hit
 (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
 trace-openings composition-openings trace-table composition-table i)
 trace-openings composition-openings i
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))
proof –
let $?s =$
 verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)
from query-header-supported-partial-opening-witnessesE[OF witness]
obtain witness-result witness-state rest **where** comp-nonempty:
 staged-composition-fri-roots data $\neq []$
and outcome:
 Some (witness-result, witness-state) $\in$
 set-dist (execute verify-monad $?s$)
and trace-auth:
 partial-authenticated-table (staged-trace-root data)
 (scale * clength) trace-openings witness-state
and comp-auth:
 partial-authenticated-table (hd (staged-composition-fri-roots data))
 (scale * clength) composition-openings witness-state
and header:
 verifier-header-transcript $?s$
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
 rest
by blast
have ext: $?s \leq$ witness-state

```

by (rule verify-monad-hash-extends[OF outcome])
have trace-pull:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings ?s ∨
  (∃ opn ∈ set trace-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots witness-state
        (staged-trace-root data) (scale * clength)
        (opening-index opn) (opening-value opn) (opening-path opn))
        ?s witness-state)
  by (rule partial-authenticated-table-pullback-or-new-output-hit
    [OF ext trace-auth])
have comp-pull:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings ?s ∨
  (∃ opn ∈ set composition-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots witness-state
        (hd (staged-composition-fri-roots data)) (scale * clength)
        (opening-index opn) (opening-value opn) (opening-path opn))
        ?s witness-state)
  by (rule partial-authenticated-table-pullback-or-new-output-hit
    [OF ext comp-auth])
have path:
  (∃ opn ∈ set trace-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots witness-state
        (staged-trace-root data) (scale * clength)
        (opening-index opn) (opening-value opn) (opening-path opn))
        ?s witness-state) ∨
  (∃ opn ∈ set composition-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots witness-state
        (hd (staged-composition-fri-roots data)) (scale * clength)
        (opening-index opn) (opening-value opn) (opening-path opn))
        ?s witness-state)
proof (rule ccontr)
assume not-path: ¬ ?thesis
have trace-on-s:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings ?s
using trace-pull not-path by blast
have comp-on-s:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings ?s
using comp-pull not-path by blast
show False
by (rule
  query-header-supported-single-round-partial-table-candidate-notin-contradicts

```



```

      [OF notin-verifier comp-nonempty trace-on-s comp-on-s
        trace-candidate comp-candidate header]])
qed
have side:
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
    trace-openings composition-openings trace-table composition-table i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side-def
using trace-candidate comp-candidate notin-prefix by simp
show ?thesis
unfolding
  checked-staged-security-with-query-prefix-header-witness-path-output-hit-def
using side comp-nonempty outcome header path
apply (simp add: Let-def)
apply (intro exI[of - witness-result] exI[of - witness-state]
  exI[of - rest])
apply auto
done
qed

lemma checked-staged-security-with-query-prefix-witness-gap-side-imp-verifier-supported-or-header-witness-path-outp
assumes witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
and side:
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
    trace-openings composition-openings trace-table composition-table i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
shows
  (trace-table, composition-table) ∈
    query-header-supported-single-round-partial-table-candidates
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)

```

```

    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    i ∨
checked-staged-security-with-query-prefix-header-witness-path-output-hit
(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
  trace-openings composition-openings trace-table composition-table i)
trace-openings composition-openings i
(Some (((((prefix, prefix-state), raw), raw-state),
  ((data, attacker-state), result)), final-state))
proof (cases
  (trace-table, composition-table) ∈
  query-header-supported-single-round-partial-table-candidates
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  i)
case True
then show ?thesis by simp
next
case notin-verifier: False
from side have trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
and notin-prefix:
  (trace-table, composition-table) ∉
  query-header-supported-single-round-partial-table-candidates
  prefix-state
  (sqp-trace-root prefix)
  (sqp-trace-fri-roots prefix)
  (sqp-trace-final prefix)
  (sqp-alphas prefix)
  (sqp-degree prefix)
  (sqp-composition-fri-roots prefix)
  (sqp-composition-final prefix)
  i
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side-def
by simp-all
have

```

checked-staged-security-with-query-prefix-header-witness-path-output-hit
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
 trace-openings composition-openings trace-table composition-table i)
trace-openings composition-openings i
 (*Some* ((((*prefix, prefix-state*), *raw*), *raw-state*),
 ((*data, attacker-state*), *result*)), *final-state*))
 by (rule
 checked-staged-security-with-query-prefix-verifier-unsupported-witness-gap-imp-header-witness-path-output
 [*OF witness trace-candidate comp-candidate notin-prefix notin-verifier*])
 then show ?thesis by simp
 qed

definition *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side*
where

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | *Some* ((((*prefix, prefix-state*), *raw*), *raw-state*),
 ((*data, attacker-state*), *result*)), *final-state*) $\Rightarrow$
 checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
 trace-openings composition-openings trace-table composition-table i
 out $\wedge$
 (*trace-openings, composition-openings*) $\in$
 query-header-supported-partial-opening-witnesses
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*))
 (*staged-trace-root data*)
 (*staged-trace-fri-roots data*)
 (*staged-trace-final data*)
 (*staged-alphas data*)
 (*staged-degree data*)
 (*staged-composition-fri-roots data*)
 (*staged-composition-final data*))

definition *checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at*
where

checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-dynamic-index-hit
 (*staged-query-prefix-candidate-pair-query-target-from-prefix*
 trace-table composition-table) out $\wedge$
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
trace-openings composition-openings trace-table composition-table i out

lemma *checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-imp-dynamic-hit:*
assumes gap:

checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
trace-openings composition-openings trace-table composition-table i out
shows
checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table) out
using *gap*
unfolding
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-def
by *simp*

lemma *checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-imp-witness-side:*
assumes *gap:*
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
trace-openings composition-openings trace-table composition-table i out
shows
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
trace-openings composition-openings trace-table composition-table i out
using *gap*
unfolding
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-def
by *simp*

definition *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at*
where
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out $\wedge$
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
trace-openings composition-openings trace-table composition-table i out

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-imp-gap:*
assumes *hit:*
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at
trace-openings composition-openings trace-table composition-table i out
shows
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i out
using *hit*
unfolding
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-def
by *simp*

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-imp-witness-side:*
assumes *hit:*
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at
trace-openings composition-openings trace-table composition-table i out

shows

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
trace-openings composition-openings trace-table composition-table i out

using *hit*

unfolding

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-def

by *simp*

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-bound-from-budgets:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at
trace-openings composition-openings trace-table composition-table i)

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) +

(1::prob) / nnreal (card query-sample-space)

proof —

have *event-le:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at
trace-openings composition-openings trace-table composition-table i)

adversary-initial-state ≤

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i)

adversary-initial-state

by *(rule wp-event-mono)*

(rule

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-imp-gap)

have *gap-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
trace-openings composition-openings trace-table composition-table i)

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) +

(1::prob) / nnreal (card query-sample-space)

by *(rule*

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-bound-from-budgets
[OF wf controlled i-bound])

show *?thesis*

by (rule order-trans[*OF event-le gap-bound*])
qed

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-bound-from-witness-side*
assumes *side-bound*:

wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
(*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side*
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq S$

shows

wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
(*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at*
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq S$

by (rule order-trans[*OF - side-bound*])

(rule *wp-event-mono*,
rule *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-imp-witness-side*)

definition *checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side*
where

checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side
trace-openings composition-openings trace-table composition-table i out
 $\longleftrightarrow$
(*case out of*
None $\Rightarrow$ *False*
| *Some* ((((*prefix*, *prefix-state*), *raw*), *raw-state*),
((*data*, *attacker-state*), *result*)), *final-state*) $\Rightarrow$
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
trace-openings composition-openings trace-table composition-table i
out $\wedge$
(*trace-table*, *composition-table*) $\in$
query-header-supported-single-round-partial-table-candidates
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))
(*staged-trace-root data*)
(*staged-trace-fri-roots data*)
(*staged-trace-final data*)
(*staged-alphas data*)
(*staged-degree data*)
(*staged-composition-fri-roots data*)
(*staged-composition-final data*)
i)

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-imp-supported-or-p*
assumes *hit*:

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
trace-openings composition-openings trace-table composition-table i out

```

shows
  checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side
    trace-openings composition-openings trace-table composition-table i out  $\vee$ 
  checked-staged-security-with-query-prefix-header-witness-path-output-hit
    (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
      trace-openings composition-openings trace-table composition-table i)
    trace-openings composition-openings i out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
from hit have side:
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
    trace-openings composition-openings trace-table composition-table i
    out
and witness:
  (trace-openings, composition-openings)  $\in$ 
    query-header-supported-partial-opening-witnesses
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  unfolding out-eq
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-def
  by simp-all
have split:
  (trace-table, composition-table)  $\in$ 
    query-header-supported-single-round-partial-table-candidates
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)

```

```

    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    i ∨
checked-staged-security-with-query-prefix-header-witness-path-output-hit
  (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
    trace-openings composition-openings trace-table composition-table i)
  trace-openings composition-openings i out
unfolding out-eq
by (rule
  checked-staged-security-with-query-prefix-witness-gap-side-imp-verifier-supported-or-header-witness-path-output
    [OF witness side[unfolded out-eq]])
then show ?thesis
proof
  assume supported:
    (trace-table, composition-table) ∈
      query-header-supported-single-round-partial-table-candidates
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        (staged-trace-root data)
        (staged-trace-fri-roots data)
        (staged-trace-final data)
        (staged-alphas data)
        (staged-degree data)
        (staged-composition-fri-roots data)
        (staged-composition-final data)
        i
  have
    checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side
      trace-openings composition-openings trace-table composition-table i out
  unfolding out-eq
  checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side-def
  using hit[unfolded out-eq] supported by simp
  then show ?thesis by simp
next
  assume
    checked-staged-security-with-query-prefix-header-witness-path-output-hit
      (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
        trace-openings composition-openings trace-table composition-table i)
      trace-openings composition-openings i out
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-bound-from-support
  assumes supported-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)

```



```

    (checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side
      trace-openings composition-openings trace-table composition-table i)
    adversary-initial-state  $\leq S$ 
and path-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-header-witness-path-output-hit
      (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
        trace-openings composition-openings trace-table composition-table i)
      trace-openings composition-openings i)
    adversary-initial-state  $\leq P$ 
shows
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
      trace-openings composition-openings trace-table composition-table i)
    adversary-initial-state  $\leq S + P$ 
proof –
  let ?m =
    checked-staged-security-experiment-with-query-prefix-data-state A i
  let ?W =
    checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
      trace-openings composition-openings trace-table composition-table i
  let ?S =
    checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side
      trace-openings composition-openings trace-table composition-table i
  let ?P =
    checked-staged-security-with-query-prefix-header-witness-path-output-hit
      (checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
        trace-openings composition-openings trace-table composition-table i)
      trace-openings composition-openings i
  have event-le:
    wp-event ?m ?W adversary-initial-state  $\leq$ 
    wp-event ?m ( $\lambda$ out. ?S out  $\vee$  ?P out) adversary-initial-state
  by (rule wp-event-mono)
    (rule
      checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-imp-supported-or-path-
    )
  have union-le:
    wp-event ?m ( $\lambda$ out. ?S out  $\vee$  ?P out) adversary-initial-state  $\leq$ 
    wp-event ?m ?S adversary-initial-state +
    wp-event ?m ?P adversary-initial-state
  by (rule wp-event-union-bound)
  have sum-le:
    wp-event ?m ?S adversary-initial-state +
    wp-event ?m ?P adversary-initial-state  $\leq S + P$ 
  by (rule add-mono[OF supported-bound path-bound])
  show ?thesis
  by (rule order-trans[OF event-le order-trans[OF union-le sum-le]])
qed

```

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-bound-from-support*

assumes *supported-bound*:

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side*
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq S$

and *transcript-bound*:

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-header-witness-transcript-hit*
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
trace-openings composition-openings trace-table composition-table i))
adversary-initial-state $\leq P$

shows

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side*
trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq S + P$

proof —

let *?m* =

checked-staged-security-experiment-with-query-prefix-data-state A i

let *?side* =

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
trace-openings composition-openings trace-table composition-table i

have *path-bound*:

wp-event ?m
 (*checked-staged-security-with-query-prefix-header-witness-path-output-hit*
?side trace-openings composition-openings i)
adversary-initial-state $\leq P$

proof —

have

wp-event ?m
 (*checked-staged-security-with-query-prefix-header-witness-path-output-hit*
?side trace-openings composition-openings i)
adversary-initial-state $\leq$

wp-event ?m

(*checked-staged-security-with-query-prefix-header-witness-transcript-hit*
?side)

adversary-initial-state

by (rule *wp-event-mono*)

(rule

checked-staged-security-with-query-prefix-header-witness-path-output-imp-transcript-hit)

then show *?thesis*

by (rule *order-trans*[*OF* - *transcript-bound*])

qed

show *?thesis*

by (rule
 checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-bound-from-supported-
 [*OF supported-bound path-bound*])
 qed

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-bound-from-supported-*
assumes *supported-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side*
 trace-openings composition-openings trace-table composition-table i)
 adversary-initial-state $\leq S$
and *pre-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit*
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
 trace-openings composition-openings trace-table composition-table i))
 adversary-initial-state $\leq R$
and *new-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-header-witness-transcript-new-hit*
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
 trace-openings composition-openings trace-table composition-table i))
 adversary-initial-state $\leq Q$
shows
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side*
 trace-openings composition-openings trace-table composition-table i)
 adversary-initial-state $\leq S + (R + Q)$
proof –
 let *?m* =
 checked-staged-security-experiment-with-query-prefix-data-state A i
 let *?side* =
 checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side
 trace-openings composition-openings trace-table composition-table i
 have *transcript-bound*:
 wp-event ?m
 (*checked-staged-security-with-query-prefix-header-witness-transcript-hit*
 ?side)
 adversary-initial-state $\leq R + Q$
 proof –
 have *event-le*:
 wp-event ?m
 (*checked-staged-security-with-query-prefix-header-witness-transcript-hit*
 ?side)
 adversary-initial-state $\leq$

```

wp-event ?m
(λout.
  checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
    ?side out ∨
  checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
    ?side out)
adversary-initial-state
by (rule wp-event-mono)
(rule
  checked-staged-security-with-query-prefix-header-witness-transcript-hit-imp-pre-or-new)
have union-le:
wp-event ?m
(λout.
  checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
    ?side out ∨
  checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
    ?side out)
adversary-initial-state ≤
wp-event ?m
(checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
  ?side)
adversary-initial-state +
wp-event ?m
(checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
  ?side)
adversary-initial-state
by (rule wp-event-union-bound)
have sum-le:
wp-event ?m
(checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit
  ?side)
adversary-initial-state +
wp-event ?m
(checked-staged-security-with-query-prefix-header-witness-transcript-new-hit
  ?side)
adversary-initial-state ≤ R + Q
by (rule add-mono[OF pre-bound new-bound])
show ?thesis
by (rule order-trans[OF event-le order-trans[OF union-le sum-le]])
qed
have bound:
wp-event ?m
(checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
  trace-openings composition-openings trace-table composition-table i)
adversary-initial-state ≤ S + (R + Q)
by (rule
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-bound-from-supported-
    [OF supported-bound transcript-bound])
then show ?thesis .

```

qed

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-bound-from-supported-bound*

assumes *supported-bound*:

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-single-round-verifier-supported-gap-witness-side*
trace-openings composition-openings trace-table composition-table i)

adversary-initial-state $\leq S$

and *pre-bound*:

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-header-witness-transcript-pre-hit*
(*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
trace-openings composition-openings trace-table composition-table i))

adversary-initial-state $\leq R$

and *new-bound*:

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-header-witness-transcript-new-hit*
(*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side*
trace-openings composition-openings trace-table composition-table i))

adversary-initial-state $\leq Q$

shows

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at*
trace-openings composition-openings trace-table composition-table i)

adversary-initial-state $\leq S + (R + Q)$

proof –

have *side-bound*:

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side*
trace-openings composition-openings trace-table composition-table i)

adversary-initial-state $\leq S + (R + Q)$

by (rule

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-bound-from-supported-bound
[OF *supported-bound pre-bound new-bound*])

show *?thesis*

by (rule

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-bound-from-witness-side
[OF *side-bound*])

qed

definition *checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-exists*

where

checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-exists

out $\longleftrightarrow$

(case out of
 None $\Rightarrow$ False
 | Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state) $\Rightarrow$
 ($\exists$ result' final-state'.
 partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (Some (result', final-state'))))

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at*
assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at

i A

(Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))

and *i-bound*: $i < \text{rounds}$

and *trace-low*:

$\bigwedge$ *trace-openings composition-openings trace-table*
composition-table.

(trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data) $\Rightarrow$

partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings]) $\Rightarrow$
partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings]) $\Rightarrow$
trace-table-low-degree trace-table

and *comp-low*:

$\bigwedge$ *trace-openings composition-openings trace-table*
composition-table.

(trace-openings, composition-openings) $\in$
query-header-supported-partial-opening-witnesses
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)

$(\text{staged-composition-final data}) \Rightarrow$
 $\text{partial-trace-table-candidate trace-table}$
 $((\text{replicate rounds } []) [i := \text{trace-openings}]) \Rightarrow$
 $\text{partial-composition-table-candidate composition-table}$
 $((\text{replicate rounds } []) [i := \text{composition-openings}]) \Rightarrow$
 $\text{composition-table-low-degree maxDegree composition-table}$

and not-all:

$\wedge \text{trace-openings composition-openings trace-table}$
 $\text{composition-table prefix prefix-state.}$
 $(\text{trace-openings, composition-openings}) \in$
 $\text{query-header-supported-partial-opening-witnesses}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{staged-trace-root data})$
 $(\text{staged-trace-fri-roots data})$
 $(\text{staged-trace-final data})$
 $(\text{staged-alphas data})$
 $(\text{staged-degree data})$
 $(\text{staged-composition-fri-roots data})$
 $(\text{staged-composition-final data}) \Rightarrow$
 $\text{partial-trace-table-candidate trace-table}$
 $((\text{replicate rounds } []) [i := \text{trace-openings}]) \Rightarrow$
 $\text{partial-composition-table-candidate composition-table}$
 $((\text{replicate rounds } []) [i := \text{composition-openings}]) \Rightarrow$
 $\text{Some } ((\text{prefix, prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \Rightarrow$
 $\neg \text{all-queries-consistent trace-table composition-table}$
 $(\text{sqp-alphas prefix})$

shows

$\text{checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-exists}$
 $(\text{Some } (((((\text{alpha-prefix, alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state})) \vee$
 $\text{checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at}$
 $i \ A$
 $(\text{Some } (((((\text{alpha-prefix, alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state})) \vee$
 $\text{checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at}$
 $i \ A$
 $(\text{Some } (((((\text{alpha-prefix, alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}))$

using

$\text{checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at-im}$
 $[\text{OF hit } i\text{-bound trace-low comp-low not-all}]$

unfolding

$\text{checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-exists-def}$

by simp

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at*
assumes *split*:

$\bigwedge out.$

$out \in$

$set-dist$

$(execute$

$(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$

$A)$

$adversary-initial-state) \implies$

$checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at$

$i A out \implies$

$checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-exists$

$out \vee$

$checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at$

$i A out \vee$

$checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at$

$i A out$

and *partial-merkle-bound*:

$wp-event$

$(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$

$A)$

$checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-exists$

$adversary-initial-state \leq P$

and *target-bound*:

$wp-event$

$(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$

$A)$

$(checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at$

$i A)$

$adversary-initial-state \leq Q$

and *gap-bound*:

$wp-event$

$(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$

$A)$

$(checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at$

$i A)$

$adversary-initial-state \leq R$

shows

$wp-event$

$(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)$

$(checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at$

$i A)$

$adversary-initial-state \leq P + Q + R$

proof –

let $?m =$

$checked-staged-security-experiment-with-actual-alpha-prefix-data-state A$

let $?pm =$

$checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-exists$

let $?target =$


```

checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
  i A
let ?gap =
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
  i A
have event-le:
  wp-event ?m
  (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-at
    i A)
  adversary-initial-state ≤
  wp-event ?m (λout. ?pm out ∨ ?target out ∨ ?gap out)
  adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule split)
have union-le:
  wp-event ?m (λout. ?pm out ∨ ?target out ∨ ?gap out)
  adversary-initial-state ≤
  wp-event ?m ?pm adversary-initial-state +
  (wp-event ?m ?target adversary-initial-state +
  wp-event ?m ?gap adversary-initial-state)
proof -
  have
    wp-event ?m (λout. ?pm out ∨ ?target out ∨ ?gap out)
    adversary-initial-state ≤
    wp-event ?m ?pm adversary-initial-state +
    wp-event ?m (λout. ?target out ∨ ?gap out)
    adversary-initial-state
  by (rule wp-event-union-bound)
  also have
    ... ≤
    wp-event ?m ?pm adversary-initial-state +
    (wp-event ?m ?target adversary-initial-state +
    wp-event ?m ?gap adversary-initial-state)
  by (intro add-mono order-refl wp-event-union-bound)
  finally show ?thesis .
qed
have sum-bound:
  wp-event ?m ?pm adversary-initial-state +
  (wp-event ?m ?target adversary-initial-state +
  wp-event ?m ?gap adversary-initial-state) ≤ P + Q + R
proof -
  have
    wp-event ?m ?target adversary-initial-state +
    wp-event ?m ?gap adversary-initial-state ≤ Q + R
  by (rule add-mono[OF target-bound gap-bound])
  then have
    wp-event ?m ?pm adversary-initial-state +
    (wp-event ?m ?target adversary-initial-state +
    wp-event ?m ?gap adversary-initial-state) ≤ P + (Q + R)

```

```

    by (rule add-mono[OF partial-merkle-bound])
  then show ?thesis
    by (simp add: add.assoc)
qed
show ?thesis
  by (rule order-trans[OF event-le order-trans[OF union-le sum-bound]])
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Query-Actual-Run
imports Staged-Security-Experiment-Composition-Query-Single-Round
begin

```

Same-run authenticated-opening split for the composition query branch.

This layer keeps the verifier continuation from the staged experiment fixed, so partial-Merkle inconsistency is expressed as the actual verifier event for the sampled run.

```

context soundness
begin

```

```

definition checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad
where

```

```

  checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad
    out  $\longleftrightarrow$ 
    checked-staged-security-with-actual-alpha-prefix-verifier-event
      partial-merkle-inconsistency-bad out

```

```

lemma checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix-imp-a

```

```

assumes support:

```

```

  Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state)  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)

```

```

and hit:

```

```

  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
    A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))

```

```

and trace-low:

```

```

   $\bigwedge i$  trace-openings composition-openings trace-table
    composition-table.
     $i < \text{rounds} \implies$ 
    partial-trace-table-candidate trace-table

```

$((\text{replicate rounds } []) [i := \text{trace-openings}]) \Rightarrow$
 $\text{partial-composition-table-candidate composition-table}$
 $((\text{replicate rounds } []) [i := \text{composition-openings}]) \Rightarrow$
 $\text{trace-table-low-degree trace-table}$
and comp-low:
 $\bigwedge i \text{ trace-openings composition-openings trace-table}$
 $\text{composition-table.}$
 $i < \text{rounds} \Rightarrow$
 $\text{partial-trace-table-candidate trace-table}$
 $((\text{replicate rounds } []) [i := \text{trace-openings}]) \Rightarrow$
 $\text{partial-composition-table-candidate composition-table}$
 $((\text{replicate rounds } []) [i := \text{composition-openings}]) \Rightarrow$
 $\text{composition-table-low-degree maxDegree composition-table}$
and not-all:
 $\bigwedge i \text{ trace-openings composition-openings trace-table}$
 $\text{composition-table prefix prefix-state.}$
 $i < \text{rounds} \Rightarrow$
 $\text{partial-trace-table-candidate trace-table}$
 $((\text{replicate rounds } []) [i := \text{trace-openings}]) \Rightarrow$
 $\text{partial-composition-table-candidate composition-table}$
 $((\text{replicate rounds } []) [i := \text{composition-openings}]) \Rightarrow$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \Rightarrow$
 $\neg \text{all-queries-consistent trace-table composition-table}$
 $(\text{sqp-alphas prefix})$
shows
 $\text{checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad}$
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}))) \vee$
 $(\exists i \text{ trace-table composition-table.}$
 $\text{checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix}$
 $A \text{ trace-table composition-table } i$
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}))))$
proof –
let ?out =
 $\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state})$
let ?vout = $\text{Some } (\text{result}, \text{final-state})$
let ?s =
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
have data-support:
 $\text{Some } (((\text{data}, \text{attacker-state}), \text{result}), \text{final-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{adversary-initial-state})$

```

    by (rule
      checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
      [OF support])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  obtain builder verifier where
    verifier:
      Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
    by blast
  from hit obtain i trace-openings composition-openings where
    comp-nonempty: staged-composition-fri-roots data ≠ []
  and trace-auth:
    partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
  and comp-auth:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings final-state
  and component:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix
    A
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i ?out
  unfolding
    checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix-def
  by auto
  have i-bound: i < rounds
  using component
  unfolding
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix-def
  by (simp split: option.splits prod.splits)
  show ?thesis
  proof (cases partial-merkle-inconsistency-bad ?s ?vout)
  case True
  then show ?thesis
    unfolding
      checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad-def
      checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    by (simp add: Let-def)
  next
  case no-bad: False
  obtain trace-table where trace-candidate:
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  using
    partial-trace-table-candidate-single-witness-if-no-partial-merkle-bad
    [OF trace-auth verifier no-bad i-bound]
  by blast
  obtain composition-table where comp-candidate:

```

```

partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
using
  partial-composition-table-candidate-single-witness-if-no-partial-merkle-bad
  [OF comp-auth verifier no-bad i-bound]
by blast
have pair-hit:
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
  A trace-table composition-table i ?out
by (rule
  checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix-imp-candidate-pair-que
  [OF component trace-candidate comp-candidate
  trace-low[OF i-bound trace-candidate comp-candidate]
  comp-low[OF i-bound trace-candidate comp-candidate]])
  (rule not-all[OF i-bound trace-candidate comp-candidate])
then show ?thesis
  using i-bound by blast
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix-imp-ac
assumes support:
  Some ((((alpha-prefix, alpha-prefix-state), data), attacker-state),
  result), final-state) ∈
  set-dist
  (execute
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
  A)
  adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
  A
  (Some ((((alpha-prefix, alpha-prefix-state), data), attacker-state),
  result), final-state))
and trace-low:
   $\bigwedge i$  trace-openings composition-openings trace-table
  composition-table.
  i < rounds  $\implies$ 
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])  $\implies$ 
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])  $\implies$ 
  trace-table-low-degree trace-table
and comp-low:
   $\bigwedge i$  trace-openings composition-openings trace-table
  composition-table.
  i < rounds  $\implies$ 
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])  $\implies$ 

```

partial-composition-table-candidate composition-table
 $((\text{replicate rounds } []) [i := \text{composition-openings}]) \implies$
composition-table-low-degree maxDegree composition-table
and not-all:
 $\bigwedge i \text{ trace-openings composition-openings trace-table}$
composition-table prefix prefix-state.
 $i < \text{rounds} \implies$
partial-trace-table-candidate trace-table
 $((\text{replicate rounds } []) [i := \text{trace-openings}]) \implies$
partial-composition-table-candidate composition-table
 $((\text{replicate rounds } []) [i := \text{composition-openings}]) \implies$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
adversary-initial-state) \implies
 $\neg \text{all-queries-consistent trace-table composition-table}$
 $(\text{sqp-alphas prefix})$
shows
checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state})) \vee$
 $(\exists i < \text{rounds}.$
checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
 $i \ A$
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state})) \vee$
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at
 $i \ A$
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}))))$
proof –
have split:
checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state})) \vee$
 $(\exists i \text{ trace-table composition-table.}$
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
 $A \text{ trace-table composition-table } i$
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}))))$
by (rule
checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix-imp-actua
 $[\text{OF support hit trace-low comp-low not-all}]$
then show ?thesis
proof
assume
checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad
 $(\text{Some } (((((\text{alpha-prefix}, \text{alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}))$

```

then show ?thesis by simp
next
assume
   $\exists i$  trace-table composition-table.
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
  A trace-table composition-table i
  (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
then obtain i trace-table composition-table where pair-hit:
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
  A trace-table composition-table i
  (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
by blast
have i-bound:  $i < \text{rounds}$ 
by (rule
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
  [OF pair-hit])
have
  checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
  i A
  (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))  $\vee$ 
  checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at
  i A
  (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
by (rule
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-imp-single-round-header-target-hit-atE
  [OF pair-hit])
then show ?thesis
  using i-bound by blast
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix-bound-
fixes PM :: prob and T G :: nat  $\Rightarrow$  prob
assumes split:
   $\bigwedge out.$ 
  out  $\in$ 
  set-dist
  (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    adversary-initial-state)  $\Rightarrow$ 
  checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
  A out  $\Rightarrow$ 
  checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad
  out  $\vee$ 

```

$(\exists i < \text{rounds}.$
checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
 $i \ A \ \text{out} \vee$
checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at
 $i \ A \ \text{out})$
and *partial-merkle-bound*:
wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad
 $\text{adversary-initial-state} \leq PM$
and *target-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at}$
 $i \ A)$
 $\text{adversary-initial-state} \leq T \ i$
and *gap-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at}$
 $i \ A)$
 $\text{adversary-initial-state} \leq G \ i$
shows
wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix}$
 $A)$
 $\text{adversary-initial-state} \leq$
 $PM + (\sum i < \text{rounds}. T \ i + G \ i)$
proof –
let $?M =$
 $\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A$
let $?auth =$
 $\text{checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix}$
 A
let $?pm =$
 $\text{checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad}$
let $?round =$
 $\lambda i \ \text{out}.$
 $\text{checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at}$
 $i \ A \ \text{out} \vee$
 $\text{checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-at}$
 $i \ A \ \text{out}$
have *event-le*:


```

wp-event ?M ?auth adversary-initial-state ≤
wp-event ?M (λout. ?pm out ∨ (∃ i ∈ {..

```

lemma checked-staged-security-with-data-state-composition-bad-bound-from-actual-single-round-components-and-bud.

fixes $PM :: \text{prob}$ **and** $T\ G :: \text{nat} \Rightarrow \text{prob}$
assumes wf : *staged-budget-wellformed budgets*
and $controlled$: *staged-adversary-controlled budgets A*
and $split$:
 $\bigwedge out.$
 $out \in$
 $set\text{-}dist$
 $(execute$
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}data\text{-}state$
 $A)$
 $adversary\text{-}initial\text{-}state) \Rightarrow$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}authenticated\text{-}candidate\text{-}opening\text{-}query\text{-}hit\text{-}from\text{-}prefix$
 $A\ out \Rightarrow$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}partial\text{-}merkle\text{-}inconsistency\text{-}bad$
 $out \vee$
 $(\exists i < rounds.$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}single\text{-}round\text{-}header\text{-}supported\text{-}query\text{-}target\text{-}hit\text{-}at$
 $i\ A\ out \vee$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}query\text{-}prefix\text{-}single\text{-}round\text{-}candidate\text{-}binding\text{-}gap\text{-}at$
 $i\ A\ out)$
and $partial\text{-}merkle\text{-}bound$:
 $wp\text{-}event$
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}data\text{-}state$
 $A)$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}partial\text{-}merkle\text{-}inconsistency\text{-}bad$
 $adversary\text{-}initial\text{-}state \leq PM$
and $target\text{-}bound$:
 $\bigwedge i. i < rounds \Rightarrow$
 $wp\text{-}event$
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}data\text{-}state$
 $A)$
 $(checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}single\text{-}round\text{-}header\text{-}supported\text{-}query\text{-}target\text{-}hit\text{-}at$
 $i\ A)$
 $adversary\text{-}initial\text{-}state \leq T\ i$
and $gap\text{-}bound$:
 $\bigwedge i. i < rounds \Rightarrow$
 $wp\text{-}event$
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}data\text{-}state$
 $A)$
 $(checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}query\text{-}prefix\text{-}single\text{-}round\text{-}candidate\text{-}binding\text{-}gap\text{-}at$
 $i\ A)$
 $adversary\text{-}initial\text{-}state \leq G\ i$
shows
 $wp\text{-}event\ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event\ composition\text{-}bad)$
 $adversary\text{-}initial\text{-}state \leq$
 $composition\text{-}error\text{-}bound +$
 $staged\text{-}phase\text{-}relation\text{-}error\ size$
 $(staged\text{-}alpha\text{-}search\text{-}queries\ budgets\ 0) +$

```

    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    (PM + ( $\sum i < \text{rounds. } T i + G i$ )) +
    hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget)
proof –
  have authenticated-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix
    A)
    adversary-initial-state  $\leq PM + (\sum i < \text{rounds. } T i + G i)$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-authenticated-candidate-opening-query-hit-from-prefix-bound-from
    [OF split partial-merkle-bound target-bound gap-bound])
  show ?thesis
  by (rule
    checked-staged-security-with-data-state-composition-bad-bound-from-authenticated-query-from-prefix-and-budgets
    [OF wf controlled authenticated-bound])
qed

end

end

```

```

theory Soundness-Staged-Bounds
imports
  Soundness-Staged-Transcript
  Soundness-Staged-Partial-Query
  Soundness-Staged-Partial-Query-Branch
  Staged-Security-Experiment-Composition-Partial
  Staged-Security-Experiment-Composition-Query-Residual
  Staged-Security-Experiment-Composition-Query-Actual-Run
begin

```

Staged soundness bad-event bounds and accepted-execution partitions.

```

context soundness
begin

```

```

lemma checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-query-bad-rounds:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event (checked-staged-security-experiment-with-data-state A)

```

$(\text{staged-security-with-data-state-query-bad-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$

shows

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event query-bad})$
 $\text{adversary-initial-state} \leq (\sum i < \text{rounds. } C \ i)$

proof –

have $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event query-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-query-bad-hit}$
 $\text{adversary-initial-state}$
by (rule $\text{checked-staged-security-with-data-state-query-bad-verifier-event-bound}$)
 simp

also have $\dots \leq (\sum i < \text{rounds. } C \ i)$

by (rule $\text{checked-staged-security-with-data-state-query-bad-hit-bound-from-rounds}$)
(rule round-bound)

finally show $?thesis$.

qed

lemma $\text{checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-committed-prefix-hit-or-tree-output}$

fixes $\text{committed-error tree-error} :: \text{prob}$

assumes $\text{wf: staged-budget-wellformed budgets}$

and $\text{controlled: staged-adversary-controlled budgets } A$

and committed-bound:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-query-committed-prefix-hit } A)$
 $\text{adversary-initial-state} \leq \text{committed-error}$

and tree-bound:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-query-tree-output-hit } A)$
 $\text{adversary-initial-state} \leq \text{tree-error}$

shows

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event query-bad})$
 $\text{adversary-initial-state} \leq \text{committed-error} + \text{tree-error}$

proof –

have $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event query-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-query-bad-hit}$
 $\text{adversary-initial-state}$
by (rule $\text{checked-staged-security-with-data-state-query-bad-verifier-event-bound}$)
 simp

also have $\dots \leq$

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$

(staged-security-with-data-state-query-committed-prefix-hit A)
 adversary-initial-state +
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-query-tree-output-hit A)
 adversary-initial-state
 by (rule
 checked-staged-security-with-data-state-query-bad-bound-from-committed-prefix-hit-or-tree-output
 [OF wf controlled])
 also have ... $\leq$ committed-error + tree-error
 by (intro add-mono committed-bound tree-bound)
 finally show ?thesis .
 qed

lemma checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-length-committed-prefix-hit-or-tree-output

fixes committed-error tree-error :: prob

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and committed-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-query-length-committed-prefix-hit A)
 adversary-initial-state $\leq$ committed-error

and tree-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-query-tree-output-hit A)
 adversary-initial-state $\leq$ tree-error

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event query-bad)
 adversary-initial-state $\leq$ committed-error + tree-error

proof –

have wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event query-bad)
 adversary-initial-state $\leq$

wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-query-bad-hit
 adversary-initial-state

by (rule checked-staged-security-with-data-state-query-bad-verifier-event-bound)
 simp

also have ... $\leq$

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-query-length-committed-prefix-hit A)
 adversary-initial-state +
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-query-tree-output-hit A)
 adversary-initial-state

by (rule

checked-staged-security-with-data-state-query-bad-bound-from-length-committed-prefix-hit-or-tree-output
 [OF wf controlled])

also have ... $\leq$ committed-error + tree-error

```

    by (intro add-mono committed-bound tree-bound)
  finally show ?thesis .
qed

lemma staged-security-with-data-state-query-tree-output-hit-imp-partial-header-key-hit-or-collision:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and tree-hit:
      staged-security-with-data-state-query-tree-output-hit A out
  shows
    staged-security-with-data-state-query-partial-header-key-hit out  $\vee$ 
    staged-security-with-data-state-verifier-event
    hash-map-output-collision-bad out
proof (cases out)
  case None
  then show ?thesis
    using tree-hit
    unfolding staged-security-with-data-state-query-tree-output-hit-def
    by simp
next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have tree-body:
     $\exists i$  prefix prefix-state raw raw-state trace-table composition-table
      as query-idxs trace-tree composition-tree.
     $i < \text{rounds} \wedge$ 
    Some (data, attacker-state)  $\in$ 
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)  $\wedge$ 
    Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)  $\wedge$ 
    accepted-with-bound-tables ?s (Some (result, final-state))
    trace-table composition-table as query-idxs  $\wedge$ 
    fmllookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw  $\wedge$ 
    index (to-nat raw)  $\in$ 
      query-sampling-success-space trace-table composition-table
      (staged-alphas data)  $\wedge$ 
    Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)

```

```

    adversary-initial-state) ∧
    sqp-composition-fri-roots prefix ≠ [] ∧
    created-tree trace-table trace-tree final-state ∧
    sqp-trace-root prefix = value trace-tree ∧
    created-tree composition-table composition-tree final-state ∧
    hd (sqp-composition-fri-roots prefix) = value composition-tree ∧
    hash-map-new-output-hit
      (set-tree trace-tree ∪ set-tree composition-tree)
    prefix-state final-state
  using tree-hit
  unfolding out-eq
    staged-security-with-data-state-query-tree-output-hit-def Let-def
  by simp
from tree-body
obtain i prefix prefix-state raw raw-state trace-table composition-table
  as query-idxs trace-tree composition-tree where
  i-bound: i < rounds
and builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and verifier:
  Some (result, final-state) ∈
    set-dist (execute verify-monad ?s)
and bound:
  accepted-with-bound-tables ?s (Some (result, final-state))
  trace-table composition-table as query-idxs
and lookup-final-raw:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
and raw-hit:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
      (staged-alphas data)
  by blast
have accepted-out: accepted (Some (result, final-state))
  unfolding accepted-def by simp
show ?thesis
proof (cases hash-map-output-collision final-state)
case True
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
    hash-map-output-collision-bad-def
  using accepted-out by simp
next

```

```

case clean: False
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
from bound obtain fr f-fri-roots f-final dg composition-fri-roots final
  rest where
    header:
      verifier-header-transcript ?s fr f-fri-roots f-final as dg
        composition-fri-roots final rest
    and trace-bind:
      merkle-root-binds-table fr trace-table final-state
    and comp-nonempty: composition-fri-roots  $\neq$  []
    and comp-bind:
      merkle-root-binds-table (hd composition-fri-roots)
        composition-table final-state
    unfolding accepted-with-bound-tables-def by blast
have eqs:
  fr = staged-trace-root data  $\wedge$ 
  f-fri-roots = staged-trace-fri-roots data  $\wedge$ 
  f-final = staged-trace-final data  $\wedge$ 
  as = staged-alphas data  $\wedge$ 
  dg = staged-degree data  $\wedge$ 
  composition-fri-roots = staged-composition-fri-roots data  $\wedge$ 
  final = staged-composition-final data  $\wedge$ 
  rest = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header staged-header]
  by simp
have state:
  query-header-supported-table-candidate-state ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    trace-table composition-table final-state
  unfolding query-header-supported-table-candidate-state-def
  by (intro exI[of - result] exI[of - query-idxs]
    exI[of - List.concat (staged-query-chunks data)])

```



```

    (use verifier bound staged-header trace-bind comp-nonempty comp-bind
      eqs in simp)
  have candidate:
    (trace-table, composition-table) ∈
      query-header-supported-partial-table-candidates ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  using
    query-header-supported-table-candidate-state-imp-partial-candidate-state
    [OF state clean]
    query-header-supported-partial-table-candidates-iff-state
  by blast
  have raw-partial-hit:
    index (to-nat raw) ∈
      query-header-supported-partial-union-good-sets ?s
      query-sampling-success-space
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  proof —
    have raw-in-sample:
      index (to-nat raw) ∈ query-sample-space
    using raw-hit query-sampling-success-space-subset by blast
  show ?thesis
  unfolding query-header-supported-partial-union-good-sets-def
  using candidate raw-hit raw-in-sample by blast
qed
then show ?thesis
  unfolding out-eq
  staged-security-with-data-state-query-partial-header-key-hit-def
  Let-def
  using i-bound lookup-final-raw by auto
qed
qed

lemma staged-security-with-data-state-query-tree-output-hit-bound-from-partial-header-key-hit-or-collision:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (staged-security-with-data-state-query-tree-output-hit A)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-header-key-hit
    adversary-initial-state +
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     hash-map-output-collision-bad)
    adversary-initial-state
proof –
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?partial = staged-security-with-data-state-query-partial-header-key-hit
  let ?collision =
    staged-security-with-data-state-verifier-event
    hash-map-output-collision-bad
  have wp-event ?M (staged-security-with-data-state-query-tree-output-hit A)
    adversary-initial-state ≤
  wp-event ?M (λout. ?partial out ∨ ?collision out)
    adversary-initial-state
  by (rule wp-event-mono)
    (rule
     staged-security-with-data-state-query-tree-output-hit-imp-partial-header-key-hit-or-collision
     [OF wf controlled])
  also have ... ≤
    wp-event ?M ?partial adversary-initial-state +
    wp-event ?M ?collision adversary-initial-state
  by (rule wp-event-union-bound)
  finally show ?thesis .
qed

```

```

lemma checked-staged-security-with-data-state-query-bad-verifier-event-bound-by-length-prefixes-partial-header-and-c
fixes partial-error collision-error :: prob
assumes raw-bound: query-index-raw-preimage-bound
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and partial-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  staged-security-with-data-state-query-partial-header-key-hit
  adversary-initial-state ≤ partial-error
and collision-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   hash-map-output-collision-bad)
  adversary-initial-state ≤ collision-error
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event query-bad)
  adversary-initial-state ≤
  (∑ i < rounds.

```

```

    staged-phase-relation-error size
      (staged-query-search-queries budgets  $i + 1$ ) +
      query-error-bound +
      hash-collision-budget-value 0
      (staged-query-search-queries budgets  $i + 1$ )) +
    (partial-error + collision-error)
  proof -
    let ?M = checked-staged-security-experiment-with-data-state A
    let ?Tree = staged-security-with-data-state-query-tree-output-hit A
    let ?PrefixSum =
      ( $\sum i < \text{rounds}$ .
        staged-phase-relation-error size
          (staged-query-search-queries budgets  $i + 1$ ) +
          query-error-bound +
          hash-collision-budget-value 0
          (staged-query-search-queries budgets  $i + 1$ ))
    have tree-bound:
      wp-event ?M ?Tree adversary-initial-state  $\leq$ 
        partial-error + collision-error
    proof -
      have wp-event ?M ?Tree adversary-initial-state  $\leq$ 
        wp-event ?M staged-security-with-data-state-query-partial-header-key-hit
          adversary-initial-state +
        wp-event ?M
          (staged-security-with-data-state-verifier-event
            hash-map-output-collision-bad)
          adversary-initial-state
      by (rule
        staged-security-with-data-state-query-tree-output-hit-bound-from-partial-header-key-hit-or-collision
          [OF wf controlled])
      also have ...  $\leq$  partial-error + collision-error
      by (intro add-mono partial-bound collision-bound)
      finally show ?thesis .
    qed
  have wp-event ?M
    (staged-security-with-data-state-verifier-event query-bad)
    adversary-initial-state  $\leq$ 
    ?PrefixSum + wp-event ?M ?Tree adversary-initial-state
  by (rule
    checked-staged-security-with-data-state-query-bad-verifier-event-bound-by-length-prefixes-and-tree
      [OF raw-bound wf controlled])
  also have ...  $\leq$  ?PrefixSum + (partial-error + collision-error)
  by (intro add-mono order-refl tree-bound)
  finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-query-bad-verifier-event-bound-by-length-prefixes-partial-header-and-c
  assumes raw-bound: query-index-raw-preimage-bound
  and wf: staged-budget-wellformed budgets

```

and *controlled: staged-adversary-controlled budgets A*
and *partial-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-query-partial-header-key-hit
adversary-initial-state ≤ partial-error
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event query-bad)
adversary-initial-state ≤
(∑ i < rounds.
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1)) +
(partial-error +
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget))
proof –
have *collision-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
hash-map-output-collision-bad)
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
[OF wf controlled])
show *?thesis*
by (rule
checked-staged-security-with-data-state-query-bad-verifier-event-bound-by-length-prefixes-partial-header-and-collision-bound-controlled
[OF raw-bound wf controlled partial-bound collision-bound])
qed

lemma *checked-staged-security-with-data-state-query-bad-verifier-event-bound-by-length-prefixes-partial-header-and-collision-bound-controlled*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *partial-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-query-partial-header-key-hit
adversary-initial-state ≤ partial-error
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event query-bad)
adversary-initial-state ≤
(∑ i < rounds.

```

    staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) +
      query-error-bound +
      hash-collision-budget-value 0
      (staged-query-search-queries budgets i + 1)) +
    (partial-error +
      hash-collision-budget-value 0
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget + verifier-hash-query-budget))
proof –
  have raw-bound: query-index-raw-preimage-bound
  by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
  show ?thesis
  by (rule
    checked-staged-security-with-data-state-query-bad-verifier-event-bound-by-length-prefixes-partial-header-and-control
    [OF raw-bound wf controlled partial-bound])
qed

lemma checked-staged-security-with-data-state-partial-trace-openings-inconsistent-bad-bound-controlled:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        partial-trace-openings-inconsistent-bad)
    adversary-initial-state ≤
    hash-collision-budget-value 0
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget + verifier-hash-query-budget)
proof –
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      partial-trace-openings-inconsistent-bad)
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      hash-map-output-collision-bad)
  adversary-initial-state
  by (rule wp-event-mono)
  (auto simp: staged-security-with-data-state-verifier-event-def
    intro: partial-trace-openings-inconsistent-bad-imp-hash-map-output-collision-bad
    split: option.splits prod.splits)
  also have ... ≤
    hash-collision-budget-value 0
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget + verifier-hash-query-budget)
  by (rule
    checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
    [OF wf controlled])

```

finally show ?thesis .
qed

lemma *checked-staged-security-with-data-state-partial-composition-openings-inconsistent-bad-bound-controlled:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
partial-composition-openings-inconsistent-bad)
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)

proof –

have *wp-event (checked-staged-security-experiment-with-data-state A)*

(staged-security-with-data-state-verifier-event
partial-composition-openings-inconsistent-bad)

adversary-initial-state ≤

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
hash-map-output-collision-bad)
adversary-initial-state

by *(rule wp-event-mono)*

(auto simp: staged-security-with-data-state-verifier-event-def
intro:

partial-composition-openings-inconsistent-bad-imp-hash-map-output-collision-bad

split: option.splits prod.splits)

also have *... ≤*

hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)

by *(rule*

checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
[OF wf controlled])

finally show ?thesis .

qed

lemma *checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad)
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)

proof –

have *wp-event* (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
hash-map-output-collision-bad)
adversary-initial-state
by (*rule wp-event-mono*)
 (*auto simp: staged-security-with-data-state-verifier-event-def*
intro: partial-merkle-inconsistency-bad-imp-hash-map-output-collision-bad
split: option.splits prod.splits)
also have ... $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)
by (*rule*
checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
[OF wf controlled])
finally show ?thesis .
qed

lemma *checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad-bound-controlled:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad
adversary-initial-state $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)

proof –

have *event-eq:*

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad
adversary-initial-state =
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
partial-merkle-inconsistency-bad)
adversary-initial-state

unfolding

checked-staged-security-with-actual-alpha-prefix-partial-merkle-inconsistency-bad-def

using

checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection
[of A partial-merkle-inconsistency-bad]

```

    by simp
  show ?thesis
    unfolding event-eq
    by (rule
      checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
      [OF wf controlled])
qed

lemma checked-staged-security-with-data-state-supported-output-local-side-bad-bound-controlled:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      supported-output-local-side-bad)
  adversary-initial-state ≤
  hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget + verifier-hash-query-budget)
proof -
have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    supported-output-local-side-bad)
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      hash-map-output-collision-bad)
  adversary-initial-state
by (rule wp-event-mono)
  (auto simp: staged-security-with-data-state-verifier-event-def
    supported-output-local-side-bad-iff-hash-map-output-collision-bad
    split: option.splits prod.splits)
also have ... ≤
  hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
  checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
  [OF wf controlled])
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-supported-pairwise-side-bad-bound-from-coupling:
fixes coupling-error :: prob
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and coupling-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event

```



```

      supported-pairwise-coupling-event)
    adversary-initial-state ≤ coupling-error
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      supported-pairwise-side-bad)
  adversary-initial-state ≤
  hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget + verifier-hash-query-budget) +
    coupling-error
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  have event-eq:
    ?E supported-pairwise-side-bad =
      (λout. ?E hash-map-output-collision-bad out ∨
        ?E supported-pairwise-coupling-event out)
  by (rule ext)
    (auto simp: staged-security-with-data-state-verifier-event-def
      supported-pairwise-side-bad-def
      split: option.splits prod.splits)
  have wp-event ?M (?E supported-pairwise-side-bad)
    adversary-initial-state ≤
    wp-event ?M (?E hash-map-output-collision-bad)
      adversary-initial-state +
    wp-event ?M (?E supported-pairwise-coupling-event)
      adversary-initial-state
  unfolding event-eq by (rule wp-event-union-bound)
  also have ... ≤
    hash-collision-budget-value 0
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget + verifier-hash-query-budget) +
    coupling-error
  by (intro add-mono coupling-bound
    checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
    [OF wf controlled])
  finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-committed-prefix-partial-header-a
  fixes committed-error partial-error collision-error :: prob
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and committed-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-query-committed-prefix-hit A)
      adversary-initial-state ≤ committed-error
  and partial-bound:

```

```

    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-header-key-hit
      adversary-initial-state  $\leq$  partial-error
  and collision-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        hash-map-output-collision-bad)
      adversary-initial-state  $\leq$  collision-error
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event query-bad)
      adversary-initial-state  $\leq$ 
        committed-error + (partial-error + collision-error)
  proof -
    let ?M = checked-staged-security-experiment-with-data-state A
    have tree-bound:
      wp-event ?M (staged-security-with-data-state-query-tree-output-hit A)
      adversary-initial-state  $\leq$  partial-error + collision-error
    proof -
      have wp-event ?M
        (staged-security-with-data-state-query-tree-output-hit A)
        adversary-initial-state  $\leq$ 
          wp-event ?M staged-security-with-data-state-query-partial-header-key-hit
            adversary-initial-state +
          wp-event ?M
            (staged-security-with-data-state-verifier-event
              hash-map-output-collision-bad)
            adversary-initial-state
      by (rule
        staged-security-with-data-state-query-tree-output-hit-bound-from-partial-header-key-hit-or-collision
          [OF wf controlled])
      also have ...  $\leq$  partial-error + collision-error
      by (intro add-mono partial-bound collision-bound)
      finally show ?thesis .
    qed
  show ?thesis
  by (rule
    checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-committed-prefix-hit-or-tree-output
      [OF wf controlled committed-bound tree-bound])
  qed

lemma checked-staged-security-with-data-state-query-bad-verifier-event-bound-from-partial-opening-fixed-query-error-
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and cover:
     $\bigwedge$ data attacker-state i.
      i < rounds  $\implies$ 
        query-header-supported-partial-opening-success-indices-at
          (verifier-state-from-adversary attacker-state

```

(staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
 $i \subseteq B$
and raw-bound: query-index-raw-preimage-bound
and subset: $B \subseteq \text{query-sample-space}$
and frac:
 $\text{nnreal} (\text{card } B) / \text{nnreal} (\text{card } \text{query-sample-space}) \leq$
 query-error-bound
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event } \text{query-bad})$
 $\text{adversary-initial-state} \leq$
 nnreal
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget}) * \text{query-error-bound}$
by (rule checked-staged-security-with-data-state-query-bad-verifier-event-bound)
 (rule
 $\text{checked-staged-security-with-data-state-query-bad-bound-from-partial-opening-fixed-query-error-cover}$
 $[\text{OF wf controlled cover raw-bound subset frac}]$)

lemma staged-transcript-query-partial-header-key-hit-iff-round-hit:

staged-transcript-query-partial-header-key-hit out $\longleftrightarrow$

$(\exists i \in \{..<\text{rounds}\}.$

$\text{staged-transcript-query-partial-header-key-hit-at } i \text{ out})$

unfolding staged-transcript-query-partial-header-key-hit-alt

$\text{staged-transcript-query-partial-header-key-hit-at-def}$

by (auto split: option.splits prod.splits)

lemma checked-staged-transcript-query-partial-header-key-hit-bound-from-rounds:

assumes round-bound:

$\bigwedge i. i < \text{rounds} \implies$

$\text{wp-event } (\text{checked-staged-transcript-program } A)$

$(\text{staged-transcript-query-partial-header-key-hit-at } i)$

$\text{adversary-initial-state} \leq C \ i$

shows

$\text{wp-event } (\text{checked-staged-transcript-program } A)$

$\text{staged-transcript-query-partial-header-key-hit}$

$\text{adversary-initial-state} \leq (\sum i < \text{rounds}. C \ i)$

proof –

have event-eq:

$\text{staged-transcript-query-partial-header-key-hit} =$

$(\lambda \text{out}. \exists i \in \{..<\text{rounds}\}.$

$\text{staged-transcript-query-partial-header-key-hit-at } i \text{ out})$

```

    by (rule ext)
      (simp add:
        staged-transcript-query-partial-header-key-hit-iff-round-hit)
  show ?thesis
    unfolding event-eq
    by (rule wp-event-finite-union-bound)
      (simp-all add: round-bound)
qed

lemma checked-staged-security-with-data-state-partial-header-key-hit-bound-from-dynamic-transcript:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and transcript-bound:
    wp-event (checked-staged-transcript-program A)
      staged-transcript-query-partial-header-key-hit
      adversary-initial-state  $\leq C$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-query-partial-header-key-hit
      adversary-initial-state  $\leq C$ 
  unfolding checked-staged-security-experiment-with-data-state-def
proof (rule wp-event-bind-bound-by-head-event[OF transcript-bound])
  show
    staged-security-with-data-state-query-partial-header-key-hit None  $\implies$ 
      staged-transcript-query-partial-header-key-hit None
  unfolding staged-security-with-data-state-query-partial-header-key-hit-def
    staged-transcript-query-partial-header-key-hit-def
  by simp
next
  fix data attacker-state out
  assume builder:
    Some (data, attacker-state)  $\in$ 
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and cont:
    out  $\in$ 
      set-dist
        (execute
          (get  $\gg=$ 
            ( $\lambda s$ . put
              (verifier-state-from-adversary s
                (staged-proof-transcript data))  $\gg=$ 
              ( $\lambda$ -. verify-monad  $\gg=$ 
                ( $\lambda$ result. return ((data, s), result))))))
          attacker-state)
  and hit:
    staged-security-with-data-state-query-partial-header-key-hit out
  from checked-staged-transcript-program-outcome-query-lookups

```

```

[OF wf controlled builder]
obtain raw-idxs query-idxs where
  len-raw: length raw-idxs = rounds
and lookup-attacker:
  ∀ j < rounds.
    fmlookup (HashMap attacker-state)
      (QueryIndexChallenge j
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) j)) =
        Some (raw-idxs ! j)
by blast
from hit obtain data' attacker-state' result final-state i raw where
  out-eq:
    out = Some (((data', attacker-state'), result), final-state)
and i-bound: i < rounds
and lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data')
        (staged-query-chunks data') i)) =
      Some raw
and raw-hit:
  index (to-nat raw) ∈
    query-header-supported-partial-union-good-sets
      (verifier-state-from-adversary attacker-state'
        (staged-proof-transcript data'))
    query-sampling-success-space
      (staged-trace-root data')
      (staged-trace-fri-roots data')
      (staged-trace-final data')
      (staged-alphas data')
      (staged-degree data')
      (staged-composition-fri-roots data')
      (staged-composition-final data')
unfolding staged-security-with-data-state-query-partial-header-key-hit-def
by (auto split: option.splits prod.splits)
from cont[unfolded out-eq] obtain verifier-state verifier-result where
  verifier:
    Some (verifier-result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
and data'-eq: data' = data
and attacker-state'-eq: attacker-state' = attacker-state
by (auto elim!: set-dist-bindE)
have ext:

```

```

    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data) ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
have lookup-initial:
  fmlookup
    (HashMap
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
  using lookup-attacker i-bound by simp
have lookup-final-from-initial:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
  by (rule hash-extension-lookup[OF lookup-initial ext])
have raw-eq: raw = raw-idxs ! i
  using lookup-final lookup-final-from-initial data'-eq by simp
have lookup-attacker-i:
  fmlookup (HashMap attacker-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
  using lookup-attacker i-bound by simp
have raw-idxs-hit:
  index (to-nat (raw-idxs ! i)) ∈
    query-header-supported-partial-union-good-sets
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
    query-sampling-success-space
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  using raw-hit raw-eq data'-eq attacker-state'-eq by simp
show staged-transcript-query-partial-header-key-hit
  (Some (data, attacker-state))
unfolding staged-transcript-query-partial-header-key-hit-def Let-def

```

using *i-bound lookup-attacker-i raw-idxs-hit* by auto
qed

definition *checked-staged-event-soundness-bound*

$:: prob \Rightarrow prob \Rightarrow prob \Rightarrow prob$

where

checked-staged-event-soundness-bound merkle-error composition-error
query-error =
merkle-error + *composition-error* + *trace-fri-error* +
composition-fri-error + *query-error*

lemma *soundness-bad-event-union-bound:*

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *comp-bound:*

wp-event verify-monad (composition-bad s) s $\leq$ *composition-error-bound*

and *trace-fri-bound:*

wp-event verify-monad (trace-fri-bad s) s $\leq$ *trace-fri-error*

and *comp-fri-bound:*

wp-event verify-monad (composition-fri-bad s) s $\leq$ *composition-fri-error*

and *query-bound:*

wp-event verify-monad (query-bad s) s $\leq$ *nnreal rounds* * *query-error-bound*

shows *wp-event verify-monad (soundness-bad-event s) s* $\leq$ *soundness-bound*

proof –

have *wp-event verify-monad (soundness-bad-event s) s* $\leq$

wp-event verify-monad (composition-bad s) s +

wp-event verify-monad (trace-fri-bad s) s +

wp-event verify-monad (composition-fri-bad s) s +

wp-event verify-monad (query-bad s) s

unfolding *soundness-bad-event-def*

by (rule *wp-event-union-bound4*)

also have ... $\leq$

composition-error-bound + *trace-fri-error* + *composition-fri-error* +

nnreal rounds * *query-error-bound*

by (intro *add-mono comp-bound trace-fri-bound comp-fri-bound query-bound*)

also have ... = *soundness-bound*

by (*simp add: soundness-bound-component-accounting*)

finally show *?thesis* .

qed

lemma *soundness-bad-event-table-aware-union-bound:*

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *comp-bound:*

wp-event verify-monad (composition-bad s) s $\leq$ *composition-error-bound*

and *trace-fri-bound:*

wp-event verify-monad (trace-fri-bad-with-tables s) s $\leq$

trace-fri-error

and *comp-fri-bound:*

wp-event verify-monad (composition-fri-bad-with-tables s) s $\leq$

composition-fri-error

and *query-bound*:
 $wp\text{-event verify-monad (query-bad } s) s \leq nnreal\ rounds * query\text{-error-bound}$
shows
 $wp\text{-event verify-monad (soundness-bad-event-table-aware } s) s \leq$
 $soundness\text{-bound}$
proof –
have $wp\text{-event verify-monad (soundness-bad-event-table-aware } s) s \leq$
 $wp\text{-event verify-monad (composition-bad } s) s +$
 $wp\text{-event verify-monad (trace-fri-bad-with-tables } s) s +$
 $wp\text{-event verify-monad (composition-fri-bad-with-tables } s) s +$
 $wp\text{-event verify-monad (query-bad } s) s$
unfolding *soundness-bad-event-table-aware-def*
by (*rule wp-event-union-bound4*)
also have $\dots \leq$
 $composition\text{-error-bound} + trace\text{-fri-error} + composition\text{-fri-error} +$
 $nnreal\ rounds * query\text{-error-bound}$
by (*intro add-mono comp-bound trace-fri-bound comp-fri-bound query-bound*)
also have $\dots = soundness\text{-bound}$
by (*simp add: soundness-bound-component-accounting*)
finally show *?thesis* .

qed

lemma *soundness-bad-event-partial-opening-union-bound*:
fixes $s :: ('f, 'a)\ protocol\ channel\ scheme$
assumes *comp-bound*:
 $wp\text{-event verify-monad (composition-bad } s) s \leq composition\text{-error-bound}$
and *trace-fri-bound*:
 $wp\text{-event verify-monad (trace-fri-bad-with-partial-openings } s) s \leq$
 $trace\text{-fri-error}$
and *comp-fri-bound*:
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-bad-with-partial-openings } s) s \leq$
 $composition\text{-fri-error}$
and *query-bound*:
 $wp\text{-event verify-monad (query-bad } s) s \leq nnreal\ rounds * query\text{-error-bound}$
shows
 $wp\text{-event verify-monad (soundness-bad-event-partial-opening } s) s \leq$
 $soundness\text{-bound}$
proof –
have $wp\text{-event verify-monad (soundness-bad-event-partial-opening } s) s \leq$
 $wp\text{-event verify-monad (composition-bad } s) s +$
 $wp\text{-event verify-monad (trace-fri-bad-with-partial-openings } s) s +$
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-bad-with-partial-openings } s) s +$
 $wp\text{-event verify-monad (query-bad } s) s$
unfolding *soundness-bad-event-partial-opening-def*
by (*rule wp-event-union-bound4*)
also have $\dots \leq$
 $composition\text{-error-bound} + trace\text{-fri-error} + composition\text{-fri-error} +$

$nnreal\ rounds * query\ error\ bound$
by (intro add-mono comp-bound trace-fri-bound comp-fri-bound query-bound)
also have ... = soundness-bound
by (simp add: soundness-bound-component-accounting)
finally show ?thesis .
qed

lemma soundness-bad-event-table-aware-with-merkle-union-bound:

fixes $s :: ('f, 'a)\ protocol\ channel\ scheme$

assumes merkle-bound:

$wp\ event\ verify\ monad\ (fri\ merkle\ binding\ bad\ s)\ s \leq$
 $merkle\ binding\ error$

and comp-bound:

$wp\ event\ verify\ monad\ (composition\ bad\ s)\ s \leq$
 $composition\ error\ bound$

and trace-fri-bound:

$wp\ event\ verify\ monad\ (trace\ fri\ bad\ with\ tables\ s)\ s \leq$
 $trace\ fri\ error$

and comp-fri-bound:

$wp\ event\ verify\ monad\ (composition\ fri\ bad\ with\ tables\ s)\ s \leq$
 $composition\ fri\ error$

and query-bound:

$wp\ event\ verify\ monad\ (query\ bad\ s)\ s \leq$
 $nnreal\ rounds * query\ error\ bound$

shows

$wp\ event\ verify\ monad$
 $(soundness\ bad\ event\ table\ aware\ with\ merkle\ s)\ s \leq$
 $soundness\ bound\ with\ merkle$

proof –

have table-bound:

$wp\ event\ verify\ monad\ (soundness\ bad\ event\ table\ aware\ s)\ s \leq$
 $soundness\ bound$

by (rule soundness-bad-event-table-aware-union-bound
 $[OF\ comp\ bound\ trace\ fri\ bound\ comp\ fri\ bound\ query\ bound])$

have $wp\ event\ verify\ monad$

$(soundness\ bad\ event\ table\ aware\ with\ merkle\ s)\ s \leq$
 $wp\ event\ verify\ monad\ (fri\ merkle\ binding\ bad\ s)\ s +$
 $wp\ event\ verify\ monad\ (soundness\ bad\ event\ table\ aware\ s)\ s$

unfolding soundness-bad-event-table-aware-with-merkle-def

by (rule wp-event-union-bound)

also have ... $\leq merkle\ binding\ error + soundness\ bound$

by (intro add-mono merkle-bound table-bound)

also have ... = soundness-bound-with-merkle

unfolding soundness-bound-with-merkle-def **by** (simp add: algebra-simps)

finally show ?thesis .

qed

lemma soundness-bad-event-partial-opening-with-merkle-union-bound:

fixes $s :: ('f, 'a)\ protocol\ channel\ scheme$

assumes *merkle-bound*:
 $wp\text{-event verify-monad (fri-merkle-binding-bad } s) s \leq merkle\text{-binding-error}$
and *comp-bound*:
 $wp\text{-event verify-monad (composition-bad } s) s \leq composition\text{-error-bound}$
and *trace-fri-bound*:
 $wp\text{-event verify-monad (trace-fri-bad-with-partial-openings } s) s \leq trace\text{-fri-error}$
and *comp-fri-bound*:
 $wp\text{-event verify-monad (composition-fri-bad-with-partial-openings } s) s \leq composition\text{-fri-error}$
and *query-bound*:
 $wp\text{-event verify-monad (query-bad } s) s \leq nnreal\ rounds * query\text{-error-bound}$
shows
 $wp\text{-event verify-monad (soundness-bad-event-partial-opening-with-merkle } s) s \leq soundness\text{-bound-with-merkle}$
proof –
have *partial-bound*:
 $wp\text{-event verify-monad (soundness-bad-event-partial-opening } s) s \leq soundness\text{-bound}$
by (rule *soundness-bad-event-partial-opening-union-bound* [OF *comp-bound trace-fri-bound comp-fri-bound query-bound*])
have *wp-event verify-monad*
 $(soundness\text{-bad-event-partial-opening-with-merkle } s) s \leq wp\text{-event verify-monad (fri-merkle-binding-bad } s) s + wp\text{-event verify-monad (soundness-bad-event-partial-opening } s) s$
unfolding *soundness-bad-event-partial-opening-with-merkle-def*
by (rule *wp-event-union-bound*)
also have $\dots \leq merkle\text{-binding-error} + soundness\text{-bound}$
by (intro *add-mono merkle-bound partial-bound*)
also have $\dots = soundness\text{-bound-with-merkle}$
unfolding *soundness-bound-with-merkle-def* **by** (*simp add: algebra-simps*)
finally show *?thesis* .
qed

lemma *soundness-bad-event-partial-opening-with-initial-merkle-union-bound*:

fixes $s :: ('f, 'a)\ \text{protocol-channel-scheme}$

assumes *merkle-bound*:

$wp\text{-event verify-monad (initial-merkle-binding-bad } s) s \leq merkle\text{-binding-error}$

and *comp-bound*:

$wp\text{-event verify-monad (composition-bad } s) s \leq composition\text{-error-bound}$

and *trace-fri-bound*:

$wp\text{-event verify-monad (trace-fri-bad-with-partial-openings } s) s \leq$

trace-fri-error
and *comp-fri-bound*:
wp-event verify-monad
(composition-fri-bad-with-partial-openings s) s ≤
composition-fri-error
and *query-bound*:
wp-event verify-monad (query-bad s) s ≤
*nnreal rounds * query-error-bound*
shows
wp-event verify-monad
(soundness-bad-event-partial-opening-with-initial-merkle s) s ≤
soundness-bound-with-merkle
proof –
have *partial-bound*:
wp-event verify-monad (soundness-bad-event-partial-opening s) s ≤
soundness-bound
by (rule *soundness-bad-event-partial-opening-union-bound*
[*OF comp-bound trace-fri-bound comp-fri-bound query-bound*])
have *wp-event verify-monad*
(soundness-bad-event-partial-opening-with-initial-merkle s) s ≤
wp-event verify-monad (initial-merkle-binding-bad s) s +
wp-event verify-monad (soundness-bad-event-partial-opening s) s
unfolding *soundness-bad-event-partial-opening-with-initial-merkle-def*
by (rule *wp-event-union-bound*)
also have ... ≤ *merkle-binding-error + soundness-bound*
by (intro *add-mono merkle-bound partial-bound*)
also have ... = *soundness-bound-with-merkle*
unfolding *soundness-bound-with-merkle-def* **by** (*simp add: algebra-simps*)
finally show ?thesis .
qed

lemma *soundness-bad-event-partial-opening-with-partial-merkle-union-bound*:
fixes *s :: ('f, 'a) protocol-channel-scheme*
assumes *merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤
merkle-binding-error
and *comp-bound*:
wp-event verify-monad (composition-bad s) s ≤
composition-error-bound
and *trace-fri-bound*:
wp-event verify-monad (trace-fri-bad-with-partial-openings s) s ≤
trace-fri-error
and *comp-fri-bound*:
wp-event verify-monad
(composition-fri-bad-with-partial-openings s) s ≤
composition-fri-error
and *query-bound*:
wp-event verify-monad (query-bad s) s ≤
*nnreal rounds * query-error-bound*

shows
 $wp\text{-event verify-monad}$
 $(\text{soundness-bad-event-partial-opening-with-partial-merkle } s) \ s \leq$
 $\text{soundness-bound-with-merkle}$
proof –
have partial-bound :
 $wp\text{-event verify-monad } (\text{soundness-bad-event-partial-opening } s) \ s \leq$
 soundness-bound
by (rule $\text{soundness-bad-event-partial-opening-union-bound}$
 $[OF \text{ comp-bound trace-fri-bound comp-fri-bound query-bound}]$)
have $wp\text{-event verify-monad}$
 $(\text{soundness-bad-event-partial-opening-with-partial-merkle } s) \ s \leq$
 $wp\text{-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s +$
 $wp\text{-event verify-monad } (\text{soundness-bad-event-partial-opening } s) \ s$
unfolding $\text{soundness-bad-event-partial-opening-with-partial-merkle-def}$
by (rule $\text{wp-event-union-bound}$)
also have $\dots \leq \text{merkle-binding-error} + \text{soundness-bound}$
by (intro $\text{add-mono merkle-bound partial-bound}$)
also have $\dots = \text{soundness-bound-with-merkle}$
unfolding $\text{soundness-bound-with-merkle-def}$ **by** (simp $\text{add: algebra-simps}$)
finally show $?thesis$.
qed

lemma $\text{soundness-bad-event-partial-candidate-union-bound}$:
fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$
assumes comp-bound :
 $wp\text{-event verify-monad}$
 $(\text{composition-bad-with-partial-candidates } s) \ s \leq$
 $\text{composition-error-bound}$
and trace-fri-bound :
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-bad-with-partial-candidates } s) \ s \leq$
 trace-fri-error
and comp-fri-bound :
 $wp\text{-event verify-monad}$
 $(\text{composition-fri-bad-with-partial-candidates } s) \ s \leq$
 $\text{composition-fri-error}$
and query-bound :
 $wp\text{-event verify-monad}$
 $(\text{query-bad-with-partial-candidates } s) \ s \leq$
 $nnreal \text{ rounds} * \text{query-error-bound}$
shows
 $wp\text{-event verify-monad } (\text{soundness-bad-event-partial-candidate } s) \ s \leq$
 soundness-bound
proof –
have $wp\text{-event verify-monad } (\text{soundness-bad-event-partial-candidate } s) \ s \leq$
 $wp\text{-event verify-monad}$
 $(\text{composition-bad-with-partial-candidates } s) \ s +$
 $wp\text{-event verify-monad}$

```

      (trace-fri-bad-with-partial-candidates s) s +
    wp-event verify-monad
      (composition-fri-bad-with-partial-candidates s) s +
    wp-event verify-monad (query-bad-with-partial-candidates s) s
  unfolding soundness-bad-event-partial-candidate-def
  by (rule wp-event-union-bound4)
also have ... ≤ composition-error-bound + trace-fri-error +
  composition-fri-error + nnreal rounds * query-error-bound
  by (intro add-mono comp-bound trace-fri-bound comp-fri-bound
    query-bound)
also have ... = soundness-bound
  unfolding soundness-bound-def by (simp add: algebra-simps)
finally show ?thesis .
qed

lemma soundness-bad-event-partial-candidate-with-partial-merkle-union-bound:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤
      merkle-binding-error
  and comp-bound:
    wp-event verify-monad
      (composition-bad-with-partial-candidates s) s ≤
        composition-error-bound
  and trace-fri-bound:
    wp-event verify-monad
      (trace-fri-bad-with-partial-candidates s) s ≤
        trace-fri-error
  and comp-fri-bound:
    wp-event verify-monad
      (composition-fri-bad-with-partial-candidates s) s ≤
        composition-fri-error
  and query-bound:
    wp-event verify-monad
      (query-bad-with-partial-candidates s) s ≤
        nnreal rounds * query-error-bound
  shows
    wp-event verify-monad
      (soundness-bad-event-partial-candidate-with-partial-merkle s) s ≤
        soundness-bound-with-merkle
proof -
  have partial-bound:
    wp-event verify-monad (soundness-bad-event-partial-candidate s) s ≤
      soundness-bound
  by (rule soundness-bad-event-partial-candidate-union-bound
    [OF comp-bound trace-fri-bound comp-fri-bound query-bound])
  have wp-event verify-monad
    (soundness-bad-event-partial-candidate-with-partial-merkle s) s ≤
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s +

```

```

    wp-event verify-monad (soundness-bad-event-partial-candidate s) s
  unfolding soundness-bad-event-partial-candidate-with-partial-merkle-def
  by (rule wp-event-union-bound)
also have ... ≤ merkle-binding-error + soundness-bound
  by (intro add-mono merkle-bound partial-bound)
also have ... = soundness-bound-with-merkle
  unfolding soundness-bound-with-merkle-def by (simp add: algebra-simps)
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-partial-candidate-with-partial-merkle-union-bound:
  fixes merkle-error composition-error trace-fri-error'
  composition-fri-error' query-error :: prob
  assumes merkle-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        partial-merkle-inconsistency-bad)
    adversary-initial-state ≤ merkle-error
  and comp-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-error
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ trace-fri-error'
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-fri-error'
  and query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        query-bad-with-partial-candidates)
    adversary-initial-state ≤ query-error
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-partial-candidate-with-partial-merkle)
    adversary-initial-state ≤
      merkle-error + composition-error + trace-fri-error' +
      composition-fri-error' + query-error
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  let ?merkle = ?E partial-merkle-inconsistency-bad

```

```

let ?comp = ?E composition-bad-with-partial-candidates
let ?trace = ?E trace-fri-bad-with-partial-candidates
let ?comp-fri = ?E composition-fri-bad-with-partial-candidates
let ?query = ?E query-bad-with-partial-candidates
let ?partial = ?E soundness-bad-event-partial-candidate
let ?with-merkle =
  ?E soundness-bad-event-partial-candidate-with-partial-merkle
have partial-bound:
  wp-event ?M ?partial adversary-initial-state ≤
    composition-error + trace-fri-error' + composition-fri-error' +
    query-error
proof -
  have wp-event ?M ?partial adversary-initial-state ≤
    wp-event ?M
      (λout. ?comp out ∨ ?trace out ∨ ?comp-fri out ∨
        ?query out)
      adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-verifier-event-def
      soundness-bad-event-partial-candidate-def
      split: option.splits prod.splits)
  also have ... ≤
    wp-event ?M ?comp adversary-initial-state +
    wp-event ?M ?trace adversary-initial-state +
    wp-event ?M ?comp-fri adversary-initial-state +
    wp-event ?M ?query adversary-initial-state
  by (rule wp-event-union-bound4)
  also have ... ≤
    composition-error + trace-fri-error' + composition-fri-error' +
    query-error
  by (intro add-mono comp-bound trace-fri-bound comp-fri-bound
    query-bound)
  finally show ?thesis .
qed
have wp-event ?M ?with-merkle adversary-initial-state ≤
  wp-event ?M (λout. ?merkle out ∨ ?partial out)
    adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-verifier-event-def
      soundness-bad-event-partial-candidate-with-partial-merkle-def
      split: option.splits prod.splits)
  also have ... ≤
    wp-event ?M ?merkle adversary-initial-state +
    wp-event ?M ?partial adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤
    merkle-error +
    (composition-error + trace-fri-error' + composition-fri-error' +
      query-error)

```

```

    by (intro add-mono merkle-bound partial-bound)
  finally show ?thesis
    by (simp add: algebra-simps)
qed

lemma soundness-bad-event-partial-candidate-empty-header-union-bound:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes comp-bound:
    wp-event verify-monad
      (composition-bad-with-empty-composition-header-candidates s) s ≤
      composition-error-bound
  and trace-fri-bound:
    wp-event verify-monad
      (trace-fri-bad-with-empty-composition-header-candidates s) s ≤
      trace-fri-error
  and query-bound:
    wp-event verify-monad
      (query-bad-with-empty-composition-header-candidates s) s ≤
      nnreal rounds * query-error-bound
  shows
    wp-event verify-monad
      (soundness-bad-event-partial-candidate-empty-header s) s ≤
      composition-error-bound + trace-fri-error +
      nnreal rounds * query-error-bound
proof -
  have wp-event verify-monad
    (soundness-bad-event-partial-candidate-empty-header s) s ≤
    wp-event verify-monad
      (λout.
        composition-bad-with-empty-composition-header-candidates s out ∨
        trace-fri-bad-with-empty-composition-header-candidates s out ∨
        query-bad-with-empty-composition-header-candidates s out ∨
        False) s
  unfolding soundness-bad-event-partial-candidate-empty-header-def
  by simp
  also have ... ≤
    wp-event verify-monad
      (composition-bad-with-empty-composition-header-candidates s) s +
    wp-event verify-monad
      (trace-fri-bad-with-empty-composition-header-candidates s) s +
    wp-event verify-monad
      (query-bad-with-empty-composition-header-candidates s) s +
    wp-event verify-monad (λ-. False) s
  by (rule wp-event-union-bound4)
  also have ... =
    wp-event verify-monad
      (composition-bad-with-empty-composition-header-candidates s) s +
    wp-event verify-monad
      (trace-fri-bad-with-empty-composition-header-candidates s) s +

```


$wp\text{-}event\ verify\text{-}monad$
 $(query\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates\ s)\ s$
unfolding $wp\text{-}event\text{-}def\ wp\text{-}def\ dist\text{-}expect\text{-}def$ **by** $simp$
also have $\dots \leq$
 $composition\text{-}error\text{-}bound + trace\text{-}fri\text{-}error +$
 $nnreal\ rounds * query\text{-}error\text{-}bound$
by $(intro\ add\text{-}mono\ comp\text{-}bound\ trace\text{-}fri\text{-}bound\ query\text{-}bound)$
finally show $?thesis$.
qed

lemma $composition\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates\text{-}union\text{-}bound$:
fixes $s :: ('f, 'a)\ protocol\text{-}channel\text{-}scheme$
assumes $false\text{-}statement: \neg\ exists\text{-}valid\text{-}trace$
and $degree\text{-}bound$:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}degree\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates\ s)\ s$
 $\leq degree\text{-}error$
and $randomization\text{-}bound$:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}randomization\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates$
 $s)\ s \leq randomization\text{-}error$
shows
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates\ s)\ s \leq$
 $degree\text{-}error + randomization\text{-}error$
proof –
have $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates\ s)\ s \leq$
 $wp\text{-}event\ verify\text{-}monad$
 $(\lambda out.$
 $composition\text{-}degree\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates$
 $s\ out \vee$
 $composition\text{-}randomization\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates$
 $s\ out)\ s$
by $(rule\ wp\text{-}event\text{-}mono)$
 $(rule\ composition\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates\text{-}split$
 $[OF\ false\text{-}statement])$
also have $\dots \leq$
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}degree\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates\ s)\ s +$
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}randomization\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates$
 $s)\ s$
by $(rule\ wp\text{-}event\text{-}union\text{-}bound)$
also have $\dots \leq degree\text{-}error + randomization\text{-}error$
by $(intro\ add\text{-}mono\ degree\text{-}bound\ randomization\text{-}bound)$
finally show $?thesis$.
qed

lemma *soundness-bad-event-partial-candidate-empty-header-split-union-bound*:

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *false-statement*: $\neg \text{exists-valid-trace}$

and *degree-bound*:

wp-event verify-monad
 $(\text{composition-degree-bad-with-empty-composition-header-candidates } s) \ s \leq \text{degree-error}$

and *randomization-bound*:

wp-event verify-monad
 $(\text{composition-randomization-bad-with-empty-composition-header-candidates } s) \ s \leq \text{randomization-error}$

and *trace-fri-bound*:

wp-event verify-monad
 $(\text{trace-fri-bad-with-empty-composition-header-candidates } s) \ s \leq \text{trace-fri-error}$

and *query-bound*:

wp-event verify-monad
 $(\text{query-bad-with-empty-composition-header-candidates } s) \ s \leq \text{nnreal rounds} * \text{query-error-bound}$

shows

wp-event verify-monad
 $(\text{soundness-bad-event-partial-candidate-empty-header } s) \ s \leq \text{degree-error} + \text{randomization-error} + \text{trace-fri-error} + \text{nnreal rounds} * \text{query-error-bound}$

proof –

have *wp-event verify-monad*
 $(\text{soundness-bad-event-partial-candidate-empty-header } s) \ s \leq$
wp-event verify-monad
 $(\lambda \text{out.}$
 $\text{composition-degree-bad-with-empty-composition-header-candidates } s \ \text{out} \ \vee$
 $\text{composition-randomization-bad-with-empty-composition-header-candidates } s \ \text{out} \ \vee$
 $\text{trace-fri-bad-with-empty-composition-header-candidates } s \ \text{out} \ \vee$
 $\text{query-bad-with-empty-composition-header-candidates } s \ \text{out}) \ s$

proof (rule *wp-event-mono*)

fix *out*

assume *empty*:

soundness-bad-event-partial-candidate-empty-header $s \ \text{out}$

then consider

$(\text{comp}) \ \text{composition-bad-with-empty-composition-header-candidates } s \ \text{out}$
 $| \ (\text{trace}) \ \text{trace-fri-bad-with-empty-composition-header-candidates } s \ \text{out}$
 $| \ (\text{query}) \ \text{query-bad-with-empty-composition-header-candidates } s \ \text{out}$

unfolding *soundness-bad-event-partial-candidate-empty-header-def*
by *blast*

then show

composition-degree-bad-with-empty-composition-header-candidates
 $s \ \text{out} \ \vee$
composition-randomization-bad-with-empty-composition-header-candidates

```

      s out  $\vee$ 
      trace-fri-bad-with-empty-composition-header-candidates s out  $\vee$ 
      query-bad-with-empty-composition-header-candidates s out
proof cases
  case comp
  then show ?thesis
    using composition-bad-with-empty-composition-header-candidates-split
      [OF false-statement comp]
    by blast
  qed blast+
qed
also have ...  $\leq$ 
  wp-event verify-monad
    (composition-degree-bad-with-empty-composition-header-candidates s) s +
  wp-event verify-monad
    (composition-randomization-bad-with-empty-composition-header-candidates
      s) s +
  wp-event verify-monad
    (trace-fri-bad-with-empty-composition-header-candidates s) s +
  wp-event verify-monad
    (query-bad-with-empty-composition-header-candidates s) s
  by (rule wp-event-union-bound4)
also have ...  $\leq$ 
  degree-error + randomization-error + trace-fri-error +
  nnreal rounds * query-error-bound
  by (intro add-mono degree-bound randomization-bound trace-fri-bound
    query-bound)
finally show ?thesis .
qed

lemma soundness-bad-event-partial-candidate-empty-header-reduced-union-bound:
fixes s :: ('f, 'a) protocol-channel-scheme
assumes false-statement:  $\neg$  exists-valid-trace
and randomization-bound:
  wp-event verify-monad
    (composition-randomization-bad-with-empty-composition-header-candidates
      s) s  $\leq$  composition-error-bound
and trace-fri-bound:
  wp-event verify-monad
    (trace-fri-bad-with-empty-composition-header-candidates s) s  $\leq$ 
    trace-fri-error
and query-bound:
  wp-event verify-monad
    (query-bad-with-empty-composition-header-candidates s) s  $\leq$ 
    nnreal rounds * query-error-bound
shows
  wp-event verify-monad
    (soundness-bad-event-partial-candidate-empty-header s) s  $\leq$ 
    composition-error-bound + trace-fri-error +

```

```

      nnreal rounds * query-error-bound
proof –
  have split-bound:
    wp-event verify-monad
      (soundness-bad-event-partial-candidate-empty-header s) s ≤
      0 + composition-error-bound + trace-fri-error +
      nnreal rounds * query-error-bound
  by (rule soundness-bad-event-partial-candidate-empty-header-split-union-bound
    [OF false-statement
      composition-degree-bad-with-empty-composition-header-candidates-bound
      randomization-bound trace-fri-bound query-bound])
  then show ?thesis by simp
qed

lemma soundness-bad-event-partial-candidate-empty-header-bound-from-alpha-hit:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes false-statement: ¬ exists-valid-trace
  and alpha-bound:
    wp-event verify-monad
      (composition-alpha-bad-set-hit-with-empty-composition-header-candidates
        s) s ≤ composition-error-bound
  and trace-fri-bound:
    wp-event verify-monad
      (trace-fri-bad-with-empty-composition-header-candidates s) s ≤
      trace-fri-error
  and query-bound:
    wp-event verify-monad
      (query-bad-with-empty-composition-header-candidates s) s ≤
      nnreal rounds * query-error-bound
  shows
    wp-event verify-monad
      (soundness-bad-event-partial-candidate-empty-header s) s ≤
      composition-error-bound + trace-fri-error +
      nnreal rounds * query-error-bound
proof –
  have randomization-bound:
    wp-event verify-monad
      (composition-randomization-bad-with-empty-composition-header-candidates
        s) s ≤ composition-error-bound
  proof –
    have wp-event verify-monad
      (composition-randomization-bad-with-empty-composition-header-candidates
        s) s ≤
      wp-event verify-monad
        (composition-alpha-bad-set-hit-with-empty-composition-header-candidates
          s) s
    by (rule
      wp-composition-randomization-bad-with-empty-composition-header-candidates-le-alpha-hit)
    also have ... ≤ composition-error-bound

```

```

    by (rule alpha-bound)
  finally show ?thesis .
qed
show ?thesis
  by (rule soundness-bad-event-partial-candidate-empty-header-reduced-union-bound
    [OF false-statement randomization-bound trace-fri-bound query-bound])
qed

lemma soundness-bad-event-partial-candidate-empty-header-bound-from-partial-opening-alpha-union-hit:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes false-statement:  $\neg$  exists-valid-trace
  and alpha-bound:
    wp-event verify-monad
    (composition-alpha-partial-opening-union-hit-with-empty-header s) s  $\leq$ 
    composition-error-bound
  and trace-fri-bound:
    wp-event verify-monad
    (trace-fri-bad-with-empty-composition-header-candidates s) s  $\leq$ 
    trace-fri-error
  and query-bound:
    wp-event verify-monad
    (query-bad-with-empty-composition-header-candidates s) s  $\leq$ 
    nnreal rounds * query-error-bound
  shows
    wp-event verify-monad
    (soundness-bad-event-partial-candidate-empty-header s) s  $\leq$ 
    composition-error-bound + trace-fri-error +
    nnreal rounds * query-error-bound
proof -
  have alpha-hit-bound:
    wp-event verify-monad
    (composition-alpha-bad-set-hit-with-empty-composition-header-candidates
      s) s  $\leq$  composition-error-bound
  proof -
    have wp-event verify-monad
      (composition-alpha-bad-set-hit-with-empty-composition-header-candidates
        s) s  $\leq$ 
      wp-event verify-monad
      (composition-alpha-partial-opening-union-hit-with-empty-header s) s
    by (rule
      wp-composition-alpha-bad-set-hit-with-empty-header-le-partial-opening-union-hit)
    also have ...  $\leq$  composition-error-bound
    by (rule alpha-bound)
  finally show ?thesis .
qed
show ?thesis
  by (rule soundness-bad-event-partial-candidate-empty-header-bound-from-alpha-hit
    [OF false-statement alpha-hit-bound trace-fri-bound query-bound])
qed

```

lemma *soundness-bad-event-partial-candidate-empty-header-bound-from-alpha-header-partial-union-hit:*

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *false-statement*: $\neg \text{exists-valid-trace}$

and *alpha-bound*:

wp-event verify-monad
 (*alpha-header-list-set-hit* s
 (*alpha-header-supported-partial-union-bad-sets* s
composition-trace-bad-alpha-space)) $s \leq$
composition-error-bound

and *trace-fri-bound*:

wp-event verify-monad
 (*trace-fri-bad-with-empty-composition-header-candidates* s) $s \leq$
trace-fri-error

and *query-bound*:

wp-event verify-monad
 (*query-bad-with-empty-composition-header-candidates* s) $s \leq$
 $\text{nnreal rounds} * \text{query-error-bound}$

shows

wp-event verify-monad
 (*soundness-bad-event-partial-candidate-empty-header* s) $s \leq$
composition-error-bound + *trace-fri-error* +
 $\text{nnreal rounds} * \text{query-error-bound}$

proof –

have *partial-alpha-bound*:

wp-event verify-monad
 (*composition-alpha-partial-opening-union-hit-with-empty-header* s) $s \leq$
composition-error-bound

proof –

have *wp-event verify-monad*

(*composition-alpha-partial-opening-union-hit-with-empty-header* s) s
 $\leq$

wp-event verify-monad
 (*alpha-header-list-set-hit* s
 (*alpha-header-supported-partial-union-bad-sets* s
composition-trace-bad-alpha-space)) s

by (*rule*

wp-composition-alpha-partial-opening-union-hit-with-empty-header-le-alpha-header-partial-union)

also have $\dots \leq \text{composition-error-bound}$

by (*rule alpha-bound*)

finally show *?thesis* .

qed

show *?thesis*

by (*rule*

soundness-bad-event-partial-candidate-empty-header-bound-from-partial-opening-alpha-union-hit
 [*OF false-statement partial-alpha-bound trace-fri-bound query-bound*])

qed

lemma *checked-staged-security-with-actual-alpha-prefix-partial-candidate-randomization-imp-partial-header-bad-set-h*

```

assumes support:
  out ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-randomization-bad-with-partial-candidates out
shows
  checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
    composition-trace-bad-alpha-space out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
case (Some result-pack)
then obtain prefix prefix-state data attacker-state result final-state
where out-eq:
  out =
    Some (((((prefix, prefix-state), data), attacker-state),
      result), final-state)
  by (cases result-pack, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
using support
unfolding out-eq
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
by (auto elim!: set-dist-bindE split: prod.splits)
have local-random:
  composition-randomization-bad-with-partial-candidates ?s
    (Some (result, final-state))
using hit
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  Let-def
by simp
have local-hit:
  composition-alpha-bad-set-hit-with-partial-candidates ?s
    (Some (result, final-state))
by (rule composition-randomization-bad-with-partial-candidates-imp-alpha-hit
  [OF local-random])

```

```

have partial-header:
  alpha-header-list-set-hit ?s
    (alpha-header-supported-partial-union-bad-sets ?s
      composition-trace-bad-alpha-space)
    (Some (result, final-state))
  by (rule
    composition-alpha-bad-set-hit-with-partial-candidates-imp-alpha-header-partial-union-on-support
    [OF verifier local-hit])
show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-def
    Let-def
  using partial-header by simp
qed

```

```

lemma checked-staged-security-with-actual-alpha-prefix-partial-candidate-randomization-le-partial-header-bad-set-hit-on:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-randomization-bad-with-partial-candidates)
    adversary-initial-state  $\leq$ 
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
      composition-trace-bad-alpha-space)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
  (rule
    checked-staged-security-with-actual-alpha-prefix-partial-candidate-randomization-imp-partial-header-bad-set-hit-on)

```

```

lemma checked-staged-security-with-actual-alpha-prefix-partial-candidate-composition-randomization-bound-from-pre:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and prefix-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
        composition-trace-bad-alpha-space)
      adversary-initial-state  $\leq$  P
  and drift-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
        composition-trace-bad-alpha-space)
      adversary-initial-state  $\leq$  D
  shows
    wp-event

```


(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 composition-randomization-bad-with-partial-candidates)
 adversary-initial-state $\leq$
 $P + D +$
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)
proof –
have partial-header-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$
 $P + D +$
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)
by (rule
 checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-bound-from-prefix-drift-and-budgets
 [OF wf controlled prefix-bound drift-bound])
have wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 composition-randomization-bad-with-partial-candidates)
 adversary-initial-state $\leq$
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
 composition-trace-bad-alpha-space)
 adversary-initial-state
by (rule
 checked-staged-security-with-actual-alpha-prefix-partial-candidate-randomization-le-partial-header-bad-set-hit)
also have ... $\leq$
 $P + D +$
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)
by (rule partial-header-bound)
finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-partial-candidate-composition-randomization-bound-from-prefix-drift-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and prefix-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
 composition-trace-bad-alpha-space)

$adversary-initial-state \leq P$
and *drift-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift$
 $composition-trace-bad-alpha-space)$
 $adversary-initial-state \leq D$
shows
 $wp-event (checked-staged-security-experiment-with-data-state A)$
 $(staged-security-with-data-state-verifier-event$
 $composition-randomization-bad-with-partial-candidates)$
 $adversary-initial-state \leq$
 $P + D +$
 $staged-phase-relation-error\ size$
 $(staged-alpha-search-queries\ budgets\ 0)$
proof –
have *projected*:
 $wp-event (checked-staged-security-experiment-with-data-state A)$
 $(staged-security-with-data-state-verifier-event$
 $composition-randomization-bad-with-partial-candidates)$
 $adversary-initial-state =$
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)$
 $(checked-staged-security-with-actual-alpha-prefix-verifier-event$
 $composition-randomization-bad-with-partial-candidates)$
 $adversary-initial-state$
by (rule
 $checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)$
show *?thesis*
unfolding *projected*
by (rule
 $checked-staged-security-with-actual-alpha-prefix-partial-candidate-composition-randomization-bound-from-prefix-$
 $[OF\ wf\ controlled\ prefix-bound\ drift-bound])$
qed

lemma *checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-prefix-drift-and-budgets*
assumes *false-statement*: $\neg\ exists-valid-trace$
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *prefix-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $(checked-staged-security-with-actual-alpha-prefix-bad-set-hit$
 $composition-trace-bad-alpha-space)$
 $adversary-initial-state \leq P$
and *drift-bound*:
 $wp-event$

```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
      composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq D$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-bad-with-partial-candidates)
    adversary-initial-state  $\leq$ 
      P + D +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0)
  proof -
    let ?M = checked-staged-security-experiment-with-data-state A
    let ?E = staged-security-with-data-state-verifier-event
    let ?degree = ?E composition-degree-bad-with-partial-candidates
    let ?random = ?E composition-randomization-bad-with-partial-candidates
    let ?bad = ?E composition-bad-with-partial-candidates
    have degree-false: ?degree = ( $\lambda$ out. False)
    by (rule ext)
      (auto simp: staged-security-with-data-state-verifier-event-def
        composition-degree-bad-with-partial-candidates-false
        split: option.splits prod.splits)
    have degree-bound:
      wp-event ?M ?degree adversary-initial-state  $\leq 0$ 
    unfolding degree-false wp-event-def wp-def dist-expect-def by simp
    have random-bound:
      wp-event ?M ?random adversary-initial-state  $\leq$ 
        P + D +
        staged-phase-relation-error size
        (staged-alpha-search-queries budgets 0)
    by (rule
      checked-staged-security-with-data-state-partial-candidate-composition-randomization-bound-from-prefix-drift-and-
        [OF wf controlled prefix-bound drift-bound])
    have wp-event ?M ?bad adversary-initial-state  $\leq$ 
      wp-event ?M ( $\lambda$ out. ?degree out  $\vee$  ?random out)
      adversary-initial-state
    by (rule wp-event-mono)
      (auto simp: staged-security-with-data-state-verifier-event-def
        dest: composition-bad-with-partial-candidates-split
        [OF false-statement]
        split: option.splits prod.splits)
    also have ...  $\leq$ 
      wp-event ?M ?degree adversary-initial-state +
      wp-event ?M ?random adversary-initial-state
    by (rule wp-event-union-bound)
    also have ...  $\leq$ 
      0 + (P + D +

```

staged-phase-relation-error size
(staged-alpha-search-queries budgets 0))
by (*intro add-mono degree-bound random-bound*)
finally show *?thesis*
by *simp*
qed

lemma *checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-drift-and-budgets:*

assumes *false-statement: $\neg$ exists-valid-trace*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *drift-bound:*

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)

(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space)

adversary-initial-state $\leq D$

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event
composition-bad-with-partial-candidates)

adversary-initial-state $\leq$

composition-error-bound +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +

hash-collision-budget-value 0

(staged-alpha-search-queries budgets 0) +

D +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0)

proof $-$

let *?P =*

composition-error-bound +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +

hash-collision-budget-value 0

(staged-alpha-search-queries budgets 0)

have *transcript-bound:*

wp-event (checked-staged-transcript-with-alpha-prefix-program A)

(checked-staged-transcript-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)

adversary-initial-state $\leq ?P$

by (*rule*

checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets

[OF wf controlled])

have *prefix-bound:*

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state

```

    A)
    (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
     composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq$  ?P
  by (rule
      checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
      [OF transcript-bound])
  have
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-bad-with-partial-candidates)
    adversary-initial-state  $\leq$ 
    ?P + D +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)
  by (rule
      checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-prefix-drift-and-budgets
      [OF false-statement wf controlled prefix-bound drift-bound])
  then show ?thesis
  by (simp add: add.assoc)
qed

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-partial-alpha-hit-imp-partial-header-bad-set-hit
  assumes support:
    out  $\in$ 
    set-dist
    (execute
     (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
     adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-alpha-partial-opening-union-hit-with-empty-header out
  shows
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
    composition-trace-bad-alpha-space out
proof (cases out)
case None
  then show ?thesis
  using hit
  unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
case (Some result-pack)
  then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state),
              result), final-state)

```

```

    by (cases result-pack, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  using support
  unfolding out-eq
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
  by (auto elim!: set-dist-bindE split: prod.splits)
  have local-hit:
    composition-alpha-partial-opening-union-hit-with-empty-header ?s
      (Some (result, final-state))
  using hit
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    Let-def
  by simp
  have partial-header:
    alpha-header-list-set-hit ?s
      (alpha-header-supported-partial-union-bad-sets ?s
        composition-trace-bad-alpha-space)
      (Some (result, final-state))
  by (rule
    composition-alpha-partial-opening-union-hit-with-empty-header-imp-alpha-header-partial-union-on-support
      [OF verifier local-hit])
  show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-def
    Let-def
  using partial-header by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-partial-alpha-hit-le-partial-header-bad-set-hit:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-partial-opening-union-hit-with-empty-header)
  adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
      composition-trace-bad-alpha-space)
  adversary-initial-state
  by (rule wp-event-mono-on-support)
  (rule
    checked-staged-security-with-actual-alpha-prefix-empty-header-partial-alpha-hit-imp-partial-header-bad-set-hit-on-s)

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-partial-alpha-hit-bound-from-prefix-drift-and-b

```

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
 (*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq P$
and *drift-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq D$
shows
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
 (*checked-staged-security-with-actual-alpha-prefix-verifier-event*
 composition-alpha-partial-opening-union-hit-with-empty-header)
 adversary-initial-state $\leq$
 $P + D +$
 staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0)
proof –
 have *partial-header-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$
 $P + D +$
 staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0)
 by (rule
 checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-bound-from-prefix-drift-and-budgets
 [*OF wf controlled prefix-bound drift-bound*])
 have *wp-event*
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
 (*checked-staged-security-with-actual-alpha-prefix-verifier-event*
 composition-alpha-partial-opening-union-hit-with-empty-header)
 adversary-initial-state $\leq$
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit*
 composition-trace-bad-alpha-space)
 adversary-initial-state
 by (rule

checked-staged-security-with-actual-alpha-prefix-empty-header-partial-alpha-hit-le-partial-header-bad-set-hit)
also have ... $\leq$
 $P + D +$
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)
by (rule *partial-header-bound*)
finally show ?thesis .
qed

lemma *checked-staged-security-with-data-state-empty-header-partial-alpha-hit-bound-from-prefix-drift-and-budgets*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *prefix-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state $\leq P$
and *drift-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space)
adversary-initial-state $\leq D$
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-alpha-partial-opening-union-hit-with-empty-header)
adversary-initial-state $\leq$
 $P + D +$
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)
proof –
have *projected*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-alpha-partial-opening-union-hit-with-empty-header)
adversary-initial-state =
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-partial-opening-union-hit-with-empty-header)
adversary-initial-state
by (rule
checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)
show ?thesis
unfolding *projected*

by (rule
 checked-staged-security-with-actual-alpha-prefix-empty-header-partial-alpha-hit-bound-from-prefix-drift-and-budg
 [*OF wf controlled prefix-bound drift-bound*])
 qed

lemma *checked-staged-security-with-data-state-empty-header-composition-randomization-bound-from-prefix-drift-and*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *prefix-bound:*
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq P$
and *drift-bound:*
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq D$
shows
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 composition-randomization-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$
 P + D +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*)
proof –
have *alpha-bound:*
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 composition-alpha-partial-opening-union-hit-with-empty-header)
 adversary-initial-state $\leq$
 P + D +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*)
by (rule
 checked-staged-security-with-data-state-empty-header-partial-alpha-hit-bound-from-prefix-drift-and-budg
 [*OF wf controlled prefix-bound drift-bound*])
have *wp-event* (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 composition-randomization-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 composition-alpha-partial-opening-union-hit-with-empty-header)

adversary-initial-state
by (rule *wp-event-mono*)
 (auto simp: *staged-security-with-data-state-verifier-event-def*
 intro:
 composition-alpha-bad-set-hit-with-empty-header-imp-partial-opening-union-hit
 composition-randomization-bad-with-empty-composition-header-candidates-imp-alpha-hit
 split: *option.splits prod.splits*)
also have ... $\leq$
 $P + D +$
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*)
by (rule *alpha-bound*)
finally show ?thesis .
qed

lemma *checked-staged-security-with-data-state-empty-header-bad-event-bound-from-prefix-drift-trace-fri-and-query:*

assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq P$
and *drift-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq D$
and *trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq \text{trace-fri-error}$
and *query-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq \text{empty-query-error}$
shows
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 soundness-bad-event-partial-candidate-empty-header)
 adversary-initial-state $\leq$
 $P + D +$

```

    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    trace-fri-error + empty-query-error
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  let ?degree =
    ?E composition-degree-bad-with-empty-composition-header-candidates
  let ?random =
    ?E composition-randomization-bad-with-empty-composition-header-candidates
  let ?trace =
    ?E trace-fri-bad-with-empty-composition-header-candidates
  let ?query =
    ?E query-bad-with-empty-composition-header-candidates
  let ?bad = ?E soundness-bad-event-partial-candidate-empty-header
  have degree-false: ?degree = ( $\lambda$ out. False)
    by (rule ext)
      (auto simp: staged-security-with-data-state-verifier-event-def
        composition-degree-bad-with-empty-composition-header-candidates-false
        split: option.splits prod.splits)
  have degree-bound:
    wp-event ?M ?degree adversary-initial-state  $\leq$  0
    unfolding degree-false wp-event-def wp-def dist-expect-def by simp
  have random-bound:
    wp-event ?M ?random adversary-initial-state  $\leq$ 
      P + D +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0)
    by (rule
      checked-staged-security-with-data-state-empty-header-composition-randomization-bound-from-prefix-drift-and-buo
      [OF wf controlled prefix-bound drift-bound])
  have wp-event ?M ?bad adversary-initial-state  $\leq$ 
    wp-event ?M
      ( $\lambda$ out. ?degree out  $\vee$  ?random out  $\vee$  ?trace out  $\vee$ 
        ?query out)
      adversary-initial-state
  proof (rule wp-event-mono)
    fix out
    assume bad: ?bad out
    show ?degree out  $\vee$  ?random out  $\vee$  ?trace out  $\vee$  ?query out
    proof (cases out)
      case None
      then show ?thesis
        using bad
        unfolding staged-security-with-data-state-verifier-event-def
        by simp
    next
      case (Some packed)
      then obtain data attacker-state result final-state where out-eq:

```

```

    out = Some (((data, attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have local-bad:
  soundness-bad-event-partial-candidate-empty-header ?s
  (Some (result, final-state))
using bad
unfolding out-eq staged-security-with-data-state-verifier-event-def
by simp
then consider
  (comp)
  composition-bad-with-empty-composition-header-candidates ?s
  (Some (result, final-state))
| (trace)
  trace-fri-bad-with-empty-composition-header-candidates ?s
  (Some (result, final-state))
| (query)
  query-bad-with-empty-composition-header-candidates ?s
  (Some (result, final-state))
unfolding soundness-bad-event-partial-candidate-empty-header-def
by blast
then show ?thesis
proof cases
  case comp
  then have comp-split:
    composition-degree-bad-with-empty-composition-header-candidates ?s
    (Some (result, final-state)) ∨
    composition-randomization-bad-with-empty-composition-header-candidates
    ?s (Some (result, final-state))
  by (rule
    composition-bad-with-empty-composition-header-candidates-split
    [OF false-statement])
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    using comp-split by (auto split: option.splits prod.splits)
next
  case trace
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
next
  case query
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
qed
qed

```

qed
also have ... $\leq$
 $wp\text{-event } ?M \text{ ?degree adversary-initial-state} +$
 $wp\text{-event } ?M \text{ ?random adversary-initial-state} +$
 $wp\text{-event } ?M \text{ ?trace adversary-initial-state} +$
 $wp\text{-event } ?M \text{ ?query adversary-initial-state}$
by (rule wp-event-union-bound4)
also have ... $\leq$
 $0 +$
 $(P + D +$
 $\text{staged-phase-relation-error size}$
 $\text{(staged-alpha-search-queries budgets 0)}) +$
 $\text{trace-fri-error} + \text{empty-query-error}$
by (intro add-mono degree-bound random-bound trace-fri-bound query-bound)
finally show ?thesis
by (simp add: algebra-simps)
qed

lemma checked-staged-security-with-data-state-empty-header-bad-event-bound-from-drift-trace-fri-and-query:
assumes false-statement: $\neg \text{exists-valid-trace}$
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and drift-bound:
 $wp\text{-event}$
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq D$
and trace-fri-bound:
 $wp\text{-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and query-bound:
 $wp\text{-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{nnreal rounds} * \text{query-error-bound}$
shows
 $wp\text{-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{soundness-bad-event-partial-candidate-empty-header})$
 $\text{adversary-initial-state} \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $\text{(staged-alpha-search-queries budgets 0)} +$
 $\text{hash-collision-budget-value } 0$
 $\text{(staged-alpha-search-queries budgets 0)} +$

```

    D +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    trace-fri-error + nnreal rounds * query-error-bound
proof -
  let ?P =
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0)
  have transcript-bound:
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
    (checked-staged-transcript-actual-alpha-prefix-bad-set-hit
     composition-trace-bad-alpha-space)
    adversary-initial-state ≤ ?P
  by (rule
    checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
    [OF wf controlled])
  have prefix-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
     A)
    (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
     composition-trace-bad-alpha-space)
    adversary-initial-state ≤ ?P
  by (rule
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
    [OF transcript-bound])
  have
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     soundness-bad-event-partial-candidate-empty-header)
    adversary-initial-state ≤
    ?P + D +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    trace-fri-error + nnreal rounds * query-error-bound
  by (rule
    checked-staged-security-with-data-state-empty-header-bad-event-bound-from-prefix-drift-trace-fri-and-query
    [OF false-statement wf controlled prefix-bound drift-bound
     trace-fri-bound query-bound])
  then show ?thesis
  by (simp add: add.assoc)
qed

lemma soundness-bad-event-table-aware-with-pairwise-side-union-bound:
  fixes s :: ('f, 'a) protocol-channel-scheme
  and pairwise-side-error :: prob

```

assumes *side-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (supported\text{-pairwise}\text{-side}\text{-bad}\ s)\ s \leq pairwise\text{-side}\text{-error}$
and *merkle-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (fri\text{-merkle}\text{-binding}\text{-bad}\ s)\ s \leq merkle\text{-binding}\text{-error}$
and *comp-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (composition\text{-bad}\ s)\ s \leq composition\text{-error}\text{-bound}$
and *trace-fri-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (trace\text{-fri}\text{-bad}\text{-with}\text{-tables}\ s)\ s \leq trace\text{-fri}\text{-error}$
and *comp-fri-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (composition\text{-fri}\text{-bad}\text{-with}\text{-tables}\ s)\ s \leq composition\text{-fri}\text{-error}$
and *query-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (query\text{-bad}\ s)\ s \leq nreal\ rounds * query\text{-error}\text{-bound}$
shows
 $wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad}\text{-event}\text{-table}\text{-aware}\text{-with}\text{-pairwise}\text{-side}\ s)\ s \leq pairwise\text{-side}\text{-error} + soundness\text{-bound}\text{-with}\text{-merkle}$
proof –
have *merkle-table-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad}\text{-event}\text{-table}\text{-aware}\text{-with}\text{-merkle}\ s)\ s \leq soundness\text{-bound}\text{-with}\text{-merkle}$
by (rule *soundness-bad-event-table-aware-with-merkle-union-bound* [OF *merkle-bound comp-bound trace-fri-bound comp-fri-bound query-bound*])
have *wp-event verify-monad*
 $(soundness\text{-bad}\text{-event}\text{-table}\text{-aware}\text{-with}\text{-pairwise}\text{-side}\ s)\ s \leq wp\text{-event}\ verify\text{-monad}\ (supported\text{-pairwise}\text{-side}\text{-bad}\ s)\ s + wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad}\text{-event}\text{-table}\text{-aware}\text{-with}\text{-merkle}\ s)\ s$
unfolding *soundness-bad-event-table-aware-with-pairwise-side-def*
by (rule *wp-event-union-bound*)
also have $\dots \leq pairwise\text{-side}\text{-error} + soundness\text{-bound}\text{-with}\text{-merkle}$
by (intro *add-mono side-bound merkle-table-bound*)
finally show *?thesis* .
qed

lemma *soundness-bad-event-table-aware-with-output-local-side-union-bound*:
fixes $s :: ('f, 'a)\ protocol\text{-channel}\text{-scheme}$
and *output-local-side-error* :: *prob*
assumes *side-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (supported\text{-output}\text{-local}\text{-side}\text{-bad}\ s)\ s \leq output\text{-local}\text{-side}\text{-error}$
and *merkle-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (fri\text{-merkle}\text{-binding}\text{-bad}\ s)\ s \leq merkle\text{-binding}\text{-error}$

and *comp-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (composition\text{-bad}\ s)\ s \leq composition\text{-error}\text{-bound}$
and *trace-fri-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (trace\text{-fri}\text{-bad}\text{-with}\text{-tables}\ s)\ s \leq trace\text{-fri}\text{-error}$
and *comp-fri-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (composition\text{-fri}\text{-bad}\text{-with}\text{-tables}\ s)\ s \leq composition\text{-fri}\text{-error}$
and *query-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (query\text{-bad}\ s)\ s \leq nnreal\ rounds * query\text{-error}\text{-bound}$
shows
 $wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad}\text{-event}\text{-table}\text{-aware}\text{-with}\text{-output}\text{-local}\text{-side}\ s)\ s \leq output\text{-local}\text{-side}\text{-error} + soundness\text{-bound}\text{-with}\text{-merkle}$
proof –
have *merkle-table-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad}\text{-event}\text{-table}\text{-aware}\text{-with}\text{-merkle}\ s)\ s \leq soundness\text{-bound}\text{-with}\text{-merkle}$
by (rule *soundness-bad-event-table-aware-with-merkle-union-bound* [OF *merkle-bound comp-bound trace-fri-bound comp-fri-bound query-bound*])
have *wp-event verify-monad*
 $(soundness\text{-bad}\text{-event}\text{-table}\text{-aware}\text{-with}\text{-output}\text{-local}\text{-side}\ s)\ s \leq wp\text{-event}\ verify\text{-monad}\ (supported\text{-output}\text{-local}\text{-side}\text{-bad}\ s)\ s + wp\text{-event}\ verify\text{-monad}\ (soundness\text{-bad}\text{-event}\text{-table}\text{-aware}\text{-with}\text{-merkle}\ s)\ s$
unfolding *soundness-bad-event-table-aware-with-output-local-side-def*
by (rule *wp-event-union-bound*)
also have ... $\leq output\text{-local}\text{-side}\text{-error} + soundness\text{-bound}\text{-with}\text{-merkle}$
by (intro *add-mono side-bound merkle-table-bound*)
finally show ?thesis .
qed

lemma *soundness-from-bad-event-partition*:
fixes $s :: ('f, 'a)\ protocol\text{-channel}\text{-scheme}$
assumes *false-statement*: $\neg exists\text{-valid}\text{-trace}$
and *partition*:
 $\bigwedge out :: (unit\ list \times ('f, 'a)\ protocol\text{-channel}\text{-scheme})\ option.$
 $out \in set\text{-dist}\ (execute\ verify\text{-monad}\ s) \implies accepted\ out \implies soundness\text{-bad}\text{-event}\ s\ out$
and *comp-bound*: $wp\text{-event}\ verify\text{-monad}\ (composition\text{-bad}\ s)\ s \leq composition\text{-error}\text{-bound}$
and *trace-fri-bound*: $wp\text{-event}\ verify\text{-monad}\ (trace\text{-fri}\text{-bad}\ s)\ s \leq trace\text{-fri}\text{-error}$
and *comp-fri-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (composition\text{-fri}\text{-bad}\ s)\ s \leq composition\text{-fri}\text{-error}$
and *query-bound*:
 $wp\text{-event}\ verify\text{-monad}\ (query\text{-bad}\ s)\ s \leq nnreal\ rounds * query\text{-error}\text{-bound}$

shows *wp-event verify-monad accepted s ≤ soundness-bound*
proof –
have *bad-event-bound*:
wp-event verify-monad (soundness-bad-event s) s ≤ soundness-bound
by (rule *soundness-bad-event-union-bound*
 $[OF\ comp-bound\ trace-fri-bound\ comp-fri-bound\ query-bound]$)
have *accepted-subset*:
 $\bigwedge out :: (unit\ list \times ('f, 'a)\ protocol-channel-scheme)\ option.$
 $out \in set-dist\ (execute\ verify-monad\ s) \implies accepted\ out \implies$
 $soundness-bad-event\ s\ out$
proof –
fix *out* :: $(unit\ list \times ('f, 'a)\ protocol-channel-scheme)\ option$
assume *supp*: $out \in set-dist\ (execute\ verify-monad\ s)$
assume *accepted out*
then show *soundness-bad-event s out*
by (rule *partition[OF supp]*)
qed
have *wp-event verify-monad accepted s ≤*
wp-event verify-monad (soundness-bad-event s) s
by (rule *wp-event-mono-on-support*) (rule *accepted-subset*)
also have $\dots \leq soundness-bound$
by (rule *bad-event-bound*)
finally show *?thesis* .
qed

lemma *soundness-from-table-aware-bad-event-partition*:
fixes *s* :: $('f, 'a)\ protocol-channel-scheme$
assumes *false-statement*: $\neg exists-valid-trace$
and *partition*:
 $\bigwedge out :: (unit\ list \times ('f, 'a)\ protocol-channel-scheme)\ option.$
 $out \in set-dist\ (execute\ verify-monad\ s) \implies$
 $accepted\ out \implies$
 $soundness-bad-event-table-aware-with-merkle\ s\ out$
and *merkle-bound*:
 $wp-event\ verify-monad\ (fri-merkle-binding-bad\ s)\ s \leq$
 $merkle-binding-error$
and *comp-bound*:
 $wp-event\ verify-monad\ (composition-bad\ s)\ s \leq$
 $composition-error-bound$
and *trace-fri-bound*:
 $wp-event\ verify-monad\ (trace-fri-bad-with-tables\ s)\ s \leq$
 $trace-fri-error$
and *comp-fri-bound*:
 $wp-event\ verify-monad\ (composition-fri-bad-with-tables\ s)\ s \leq$
 $composition-fri-error$
and *query-bound*:
 $wp-event\ verify-monad\ (query-bad\ s)\ s \leq$
 $nnreal\ rounds * query-error-bound$
shows *wp-event verify-monad accepted s ≤ soundness-bound-with-merkle*

proof –
have *bad-event-bound*:
 wp-event verify-monad
 (*soundness-bad-event-table-aware-with-merkle s*) $s \leq$
 soundness-bound-with-merkle
by (*rule soundness-bad-event-table-aware-with-merkle-union-bound*
 [*OF merkle-bound comp-bound trace-fri-bound comp-fri-bound*
 query-bound])
have *accepted-subset*:
 $\bigwedge out :: (unit\ list \times ('f, 'a)\ protocol-channel-scheme)\ option.$
 $out \in set-dist\ (execute\ verify-monad\ s) \implies accepted\ out \implies$
 $soundness-bad-event-table-aware-with-merkle\ s\ out$
proof –
fix *out* :: (*unit list* $\times$ (*'f*, *'a*) *protocol-channel-scheme*) *option*
assume *supp*: *out* $\in$ *set-dist* (*execute verify-monad s*)
assume *accepted out*
then show *soundness-bad-event-table-aware-with-merkle s out*
by (*rule partition*[*OF supp*])
qed
have *wp-event verify-monad accepted s* $\leq$
 wp-event verify-monad
 (*soundness-bad-event-table-aware-with-merkle s*) *s*
by (*rule wp-event-mono-on-support*) (*rule accepted-subset*)
also have ... $\leq$ *soundness-bound-with-merkle*
by (*rule bad-event-bound*)
finally show ?thesis .
qed

lemma *soundness-from-partial-opening-bad-event-partition*:
fixes *s* :: (*'f*, *'a*) *protocol-channel-scheme*
and *B* :: *prob*
assumes *partition*:
 $\bigwedge out :: (unit\ list \times ('f, 'a)\ protocol-channel-scheme)\ option.$
 $out \in set-dist\ (execute\ verify-monad\ s) \implies$
 $accepted\ out \implies$
 $soundness-bad-event-partial-opening-with-merkle\ s\ out$
and *bad-event-bound*:
 wp-event verify-monad
 (*soundness-bad-event-partial-opening-with-merkle s*) $s \leq B$
shows *wp-event verify-monad accepted s* $\leq B$
proof –
have *wp-event verify-monad accepted s* $\leq$
 wp-event verify-monad
 (*soundness-bad-event-partial-opening-with-merkle s*) *s*
by (*rule wp-event-mono-on-support*) (*rule partition*)
also have ... $\leq B$
by (*rule bad-event-bound*)
finally show ?thesis .
qed

lemma *soundness-from-table-aware-bad-event-partition-with-pairwise-side:*
fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$
and $\text{pairwise-side-error} :: \text{prob}$
assumes $\text{false-statement}: \neg \text{exists-valid-trace}$
and partition :
 $\bigwedge \text{out} :: (\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option}.$
 $\text{out} \in \text{set-dist} (\text{execute verify-monad } s) \implies$
 $\text{accepted out} \implies$
 $\text{soundness-bad-event-table-aware-with-pairwise-side } s \text{ out}$
and side-bound :
 $\text{wp-event verify-monad} (\text{supported-pairwise-side-bad } s) \ s \leq$
 $\text{pairwise-side-error}$
and merkle-bound :
 $\text{wp-event verify-monad} (\text{fri-merkle-binding-bad } s) \ s \leq$
 $\text{merkle-binding-error}$
and comp-bound :
 $\text{wp-event verify-monad} (\text{composition-bad } s) \ s \leq$
 $\text{composition-error-bound}$
and trace-fri-bound :
 $\text{wp-event verify-monad} (\text{trace-fri-bad-with-tables } s) \ s \leq$
 trace-fri-error
and comp-fri-bound :
 $\text{wp-event verify-monad} (\text{composition-fri-bad-with-tables } s) \ s \leq$
 $\text{composition-fri-error}$
and query-bound :
 $\text{wp-event verify-monad} (\text{query-bad } s) \ s \leq$
 $\text{nnreal rounds} * \text{query-error-bound}$
shows
 $\text{wp-event verify-monad accepted } s \leq$
 $\text{pairwise-side-error} + \text{soundness-bound-with-merkle}$
proof –
have bad-event-bound :
 $\text{wp-event verify-monad}$
 $(\text{soundness-bad-event-table-aware-with-pairwise-side } s) \ s \leq$
 $\text{pairwise-side-error} + \text{soundness-bound-with-merkle}$
by (rule $\text{soundness-bad-event-table-aware-with-pairwise-side-union-bound}$
 $[\text{OF side-bound merkle-bound comp-bound trace-fri-bound comp-fri-bound}$
 $\text{query-bound}]$)
have accepted-subset :
 $\bigwedge \text{out} :: (\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option}.$
 $\text{out} \in \text{set-dist} (\text{execute verify-monad } s) \implies \text{accepted out} \implies$
 $\text{soundness-bad-event-table-aware-with-pairwise-side } s \text{ out}$
proof –
fix $\text{out} :: (\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option}$
assume $\text{supp}: \text{out} \in \text{set-dist} (\text{execute verify-monad } s)$
assume accepted out
then show $\text{soundness-bad-event-table-aware-with-pairwise-side } s \text{ out}$
by (rule $\text{partition}[\text{OF supp}]$)

qed
have *wp-event verify-monad accepted s* $\leq$
 wp-event verify-monad
 (*soundness-bad-event-table-aware-with-pairwise-side s*) *s*
 by (*rule wp-event-mono-on-support*) (*rule accepted-subset*)
also have ... $\leq$ *pairwise-side-error* + *soundness-bound-with-merkle*
 by (*rule bad-event-bound*)
finally show ?thesis .
qed

lemma *soundness-from-table-aware-bad-event-partition-with-output-local-side*:
fixes *s* :: ('f, 'a) *protocol-channel-scheme*
and *output-local-side-error* :: *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *partition*:
 $\bigwedge$ *out* :: (unit list $\times$ ('f, 'a) *protocol-channel-scheme*) *option*.
 out $\in$ *set-dist* (*execute verify-monad s*) $\implies$
 accepted out $\implies$
 soundness-bad-event-table-aware-with-output-local-side s out
and *side-bound*:
 wp-event verify-monad (*supported-output-local-side-bad s*) *s* $\leq$
 output-local-side-error
and *merkle-bound*:
 wp-event verify-monad (*fri-merkle-binding-bad s*) *s* $\leq$
 merkle-binding-error
and *comp-bound*:
 wp-event verify-monad (*composition-bad s*) *s* $\leq$
 composition-error-bound
and *trace-fri-bound*:
 wp-event verify-monad (*trace-fri-bad-with-tables s*) *s* $\leq$
 trace-fri-error
and *comp-fri-bound*:
 wp-event verify-monad (*composition-fri-bad-with-tables s*) *s* $\leq$
 composition-fri-error
and *query-bound*:
 wp-event verify-monad (*query-bad s*) *s* $\leq$
 nnreal rounds * *query-error-bound*
shows
 wp-event verify-monad accepted s $\leq$
 output-local-side-error + *soundness-bound-with-merkle*
proof –
have *bad-event-bound*:
 wp-event verify-monad
 (*soundness-bad-event-table-aware-with-output-local-side s*) *s* $\leq$
 output-local-side-error + *soundness-bound-with-merkle*
by (*rule*
 soundness-bad-event-table-aware-with-output-local-side-union-bound
 [*OF side-bound merkle-bound comp-bound trace-fri-bound comp-fri-bound*
 query-bound])

have *wp-event verify-monad accepted s* $\leq$
 wp-event verify-monad
 (*soundness-bad-event-table-aware-with-output-local-side s*) *s*
by (*rule wp-event-mono-on-support*) (*rule partition*)
also have ... $\leq$
 output-local-side-error + *soundness-bound-with-merkle*
by (*rule bad-event-bound*)
finally show *?thesis* .
qed

lemma *accepted-execution-partition-obligation:*
fixes *s* :: ('f, 'a) *protocol-channel-scheme*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *bind*: *initial-merkle-binding-no-bad s*
shows
 $\bigwedge out :: (unit\ list \times ('f, 'a)\ protocol-channel-scheme)\ option.$
 $out \in set-dist\ (execute\ verify-monad\ s) \implies$
 $accepted\ out \implies$
 $soundness-bad-event\ s\ out$
proof –
fix *out* :: (unit list $\times$ ('f, 'a) protocol-channel-scheme) option
assume *outcome*: *out* $\in$ *set-dist* (*execute verify-monad s*)
and *acc*: *accepted out*
from *acc* **obtain** *result final-state* **where** *out-eq*: *out* = *Some (result, final-state)*
unfolding *accepted-def* **by** (*cases out*) *auto*
have *verify-out*:
 $Some\ (result,\ final-state) \in set-dist\ (execute\ verify-monad\ s)$
using *outcome out-eq* **by** *simp*
from *verify-monad-merkle-bound-witnesses*[*OF bind verify-out*]
obtain *trace-table composition-table* **as** *query-idxs* **where** *bound*:
 $accepted-with-bound-tables\ s\ (Some\ (result,\ final-state))$
 $trace-table\ composition-table\ as\ query-idxs$
by *blast*
have *composition-bad s* (*Some (result, final-state)*) $\vee$
 $trace-fri-bad\ s\ (Some\ (result,\ final-state)) \vee$
 $composition-fri-bad\ s\ (Some\ (result,\ final-state)) \vee$
 $query-bad\ s\ (Some\ (result,\ final-state))$
by (*rule accepted-bound-table-partition*[*OF false-statement bound*])
then show *soundness-bad-event s out*
unfolding *soundness-bad-event-def* **using** *out-eq* **by** *simp*
qed

lemma *accepted-execution-partition-obligation-partial-opening:*
fixes *s* :: ('f, 'a) *protocol-channel-scheme*
assumes *false-statement*: $\neg$ *exists-valid-trace*
shows
 $\bigwedge out :: (unit\ list \times ('f, 'a)\ protocol-channel-scheme)\ option.$
 $out \in set-dist\ (execute\ verify-monad\ s) \implies$
 $accepted\ out \implies$

```

    soundness-bad-event-partial-opening-with-merkle s out
  by (rule accepted-partition-soundness-bad-event-partial-opening-with-merkle
    [OF false-statement])

```

end

end

```

theory Soundness-Staged-Partial-Merkle
  imports Soundness-Staged-Bounds
begin

```

Aggregator for staged partial-Merkle soundness infrastructure. The old public-path compatibility wrappers based on partial-Merkle event-bound premises were removed before publication; downstream active-route theories import this file only for the underlying staged bounds.

end

```

theory Soundness-Partial-Candidate-Blocker
  imports Soundness-Partial-Merkle
begin

```

Diagnostic lemmas for the remaining query-bound route.

Partial authenticated openings constrain only the opened indices. Without a separate Merkle-binding/completion theorem, the corresponding partial table candidate sets are not singleton in general: unopened positions can be changed while preserving all partial-opening evidence.

```

context soundness
begin

```

lemma *table-agrees-with-authenticated-openings-update-unopened:*

assumes *agrees:*

table-agrees-with-authenticated-openings table len openings

and *unopened:*

$\bigwedge \text{opn. } \text{opn} \in \text{set openings} \implies \text{opening-index } \text{opn} \neq j$

shows

table-agrees-with-authenticated-openings (table[j := v]) len openings

proof –

have *len:* $\text{len} = \text{length table}$

using *agrees* **unfolding** *table-agrees-with-authenticated-openings-def*

by *simp*

have *vals:*

$\bigwedge \text{opn. } \text{opn} \in \text{set openings} \implies$

$\text{opening-index } \text{opn} < \text{length table} \wedge$

$\text{table ! opening-index } \text{opn} = \text{opening-value } \text{opn}$

using *agrees* **unfolding** *table-agrees-with-authenticated-openings-def*

```

    by blast
  have updated-values:
     $\bigwedge \text{opn}. \text{opn} \in \text{set openings} \implies$ 
       $\text{opening-index } \text{opn} < \text{length } (\text{table}[j := v]) \wedge$ 
       $(\text{table}[j := v]) ! \text{opening-index } \text{opn} = \text{opening-value } \text{opn}$ 
  proof -
    fix opn
    assume opn-in:  $\text{opn} \in \text{set openings}$ 
    have idx-bound:  $\text{opening-index } \text{opn} < \text{length } \text{table}$ 
      using vals[OF opn-in] by simp
    have idx-ne:  $\text{opening-index } \text{opn} \neq j$ 
      by (rule unopened[OF opn-in])
    have  $(\text{table}[j := v]) ! \text{opening-index } \text{opn} =$ 
       $\text{table} ! \text{opening-index } \text{opn}$ 
      using idx-bound idx-ne by simp
    then show
       $\text{opening-index } \text{opn} < \text{length } (\text{table}[j := v]) \wedge$ 
       $(\text{table}[j := v]) ! \text{opening-index } \text{opn} = \text{opening-value } \text{opn}$ 
      using vals[OF opn-in] by simp
  qed
  show ?thesis
    unfolding table-agrees-with-authenticated-openings-def
    using len updated-values by simp
qed

lemma table-agrees-with-authenticated-openings-eq-if-all-indices-opened:
  assumes agrees:
     $\text{table-agrees-with-authenticated-openings } \text{table } \text{len } \text{openings}$ 
  and agrees':
     $\text{table-agrees-with-authenticated-openings } \text{table}' \text{ len } \text{openings}$ 
  and cover:
     $\bigwedge j. j < \text{len} \implies \exists \text{opn} \in \text{set openings}. \text{opening-index } \text{opn} = j$ 
  shows  $\text{table} = \text{table}'$ 
proof (rule nth-equalityI)
  show  $\text{length } \text{table} = \text{length } \text{table}'$ 
    using agrees agrees'
    unfolding table-agrees-with-authenticated-openings-def by simp
next
  fix j
  assume j-bound:  $j < \text{length } \text{table}$ 
  have len-eq:  $\text{length } \text{table} = \text{len}$ 
    using agrees unfolding table-agrees-with-authenticated-openings-def
    by simp
  then obtain opn where opn-in:  $\text{opn} \in \text{set openings}$ 
    and opn-idx:  $\text{opening-index } \text{opn} = j$ 
    using cover j-bound by blast
  have table-val:  $\text{table} ! j = \text{opening-value } \text{opn}$ 
    using agrees opn-in opn-idx

```

```

    unfolding table-agrees-with-authenticated-openings-def by blast
  have table'-val: table' ! j = opening-value opn
    using agrees' opn-in opn-idx
    unfolding table-agrees-with-authenticated-openings-def by blast
  show table ! j = table' ! j
    using table-val table'-val by simp
qed

lemma partial-trace-table-candidate-update-unopened:
  assumes cand: partial-trace-table-candidate trace-table trace-openings
    and unopened:
       $\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set } (\text{trace-openings} ! i) \implies$ 
        opening-index opn  $\neq j$ 
  shows
    partial-trace-table-candidate (trace-table[j := v]) trace-openings
proof -
  have len: length (trace-table[j := v]) = scale * clength
    using cand unfolding partial-trace-table-candidate-def by simp
  have rounds-len: length trace-openings = rounds
    using cand unfolding partial-trace-table-candidate-def by simp
  have agrees:
     $\bigwedge i. i < \text{rounds} \implies$ 
      table-agrees-with-authenticated-openings (trace-table[j := v])
        (scale * clength) (trace-openings ! i)
  proof -
    fix i
    assume i-bound: i < rounds
    have old:
      table-agrees-with-authenticated-openings trace-table
        (scale * clength) (trace-openings ! i)
      using cand i-bound unfolding partial-trace-table-candidate-def by blast
    show
      table-agrees-with-authenticated-openings (trace-table[j := v])
        (scale * clength) (trace-openings ! i)
      by (rule table-agrees-with-authenticated-openings-update-unopened
        [OF old])
        (use unopened[OF i-bound] in blast)
  qed
  show ?thesis
    unfolding partial-trace-table-candidate-def
    using len rounds-len agrees by simp
qed

```

```

lemma partial-trace-table-candidates-not-singleton-if-unopened:
  assumes cand: partial-trace-table-candidate trace-table trace-openings
    and j-bound: j < scale * clength
    and unopened:
       $\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set } (\text{trace-openings} ! i) \implies$ 
        opening-index opn  $\neq j$ 

```


shows
 $\neg (\exists \text{table. } \text{partial-trace-table-candidates } \text{trace-openings} \subseteq \{\text{table}\})$

proof
assume *unique*:
 $\exists \text{table. } \text{partial-trace-table-candidates } \text{trace-openings} \subseteq \{\text{table}\}$
let $? \text{trace-table}' = \text{trace-table}[j := \text{trace-table} ! j + 1]$
have *cand-in*:
 $\text{trace-table} \in \text{partial-trace-table-candidates } \text{trace-openings}$
using *cand* **unfolding** *partial-trace-table-candidates-def* **by** *simp*
have *cand'-in*:
 $? \text{trace-table}' \in \text{partial-trace-table-candidates } \text{trace-openings}$
using *partial-trace-table-candidate-update-unopened* [*OF cand unopened*]
unfolding *partial-trace-table-candidates-def* **by** *simp*
from *unique* **obtain** *table* **where** *subset*:
 $\text{partial-trace-table-candidates } \text{trace-openings} \subseteq \{\text{table}\}$
by *blast*
have *trace-eq*: $\text{trace-table} = \text{table}$
using *subset cand-in* **by** *blast*
have *trace'-eq*: $? \text{trace-table}' = \text{table}$
using *subset cand'-in* **by** *blast*
have *len*: $\text{length } \text{trace-table} = \text{scale} * \text{clength}$
using *cand* **unfolding** *partial-trace-table-candidate-def* **by** *simp*
have *changed*: $? \text{trace-table}' ! j \neq \text{trace-table} ! j$
using *j-bound len* **by** *simp*
show *False*
using *changed trace-eq trace'-eq* **by** *simp*
qed

lemma *partial-trace-table-candidate-eq-if-all-indices-opened*:
assumes *cand*: $\text{partial-trace-table-candidate } \text{trace-table } \text{trace-openings}$
and *cand'*:
 $\text{partial-trace-table-candidate } \text{trace-table}' \text{ trace-openings}$
and *cover*:
 $\bigwedge j. j < \text{scale} * \text{clength} \implies$
 $\exists i < \text{rounds}. \exists \text{opn} \in \text{set } (\text{trace-openings} ! i).$
 $\text{opening-index } \text{opn} = j$
shows $\text{trace-table} = \text{trace-table}'$

proof (*rule nth-equalityI*)
show $\text{length } \text{trace-table} = \text{length } \text{trace-table}'$
using *cand cand'* **unfolding** *partial-trace-table-candidate-def* **by** *simp*
next
fix *j*
assume *j-bound*: $j < \text{length } \text{trace-table}$
have *len*: $\text{length } \text{trace-table} = \text{scale} * \text{clength}$
using *cand* **unfolding** *partial-trace-table-candidate-def* **by** *simp*
from *cover* [*OF j-bound* [*unfolded len*]]
obtain *i opn* **where** *i-bound*: $i < \text{rounds}$
and *opn-in*: $\text{opn} \in \text{set } (\text{trace-openings} ! i)$

```

    and opn-idx: opening-index opn = j
    by blast
  have table-val: trace-table ! j = opening-value opn
    using partial-trace-table-candidateD(3)[OF cand i-bound opn-in]
    opn-idx by simp
  have table'-val: trace-table' ! j = opening-value opn
    using partial-trace-table-candidateD(3)[OF cand' i-bound opn-in]
    opn-idx by simp
  show trace-table ! j = trace-table' ! j
    using table-val table'-val by simp
qed

```

lemma *partial-trace-table-candidates-subset-singleton-if-all-indices-opened:*

assumes *cand: partial-trace-table-candidate trace-table trace-openings*
and *cover:*

$\bigwedge j. j < \text{scale} * \text{clength} \implies$
 $\exists i < \text{rounds}. \exists \text{opn} \in \text{set } (\text{trace-openings} ! i).$
 $\text{opening-index } \text{opn} = j$

shows *partial-trace-table-candidates trace-openings $\subseteq \{\text{trace-table}\}$*

proof

fix *trace-table'*

assume *trace-table' $\in$ partial-trace-table-candidates trace-openings*

then have *cand':*

partial-trace-table-candidate trace-table' trace-openings

unfolding *partial-trace-table-candidates-def* **by** *simp*

have *trace-table' = trace-table*

using *partial-trace-table-candidate-eq-if-all-indices-opened*
 $[OF \text{ cand' cand cover}]$.

then show *trace-table' $\in \{\text{trace-table}\}$*

by *simp*

qed

lemma *partial-composition-table-candidate-update-unopened:*

assumes *cand:*

partial-composition-table-candidate composition-table
composition-openings

and *unopened:*

$\bigwedge i \text{ opn}. i < \text{rounds} \implies$
 $\text{opn} \in \text{set } (\text{composition-openings} ! i) \implies$
 $\text{opening-index } \text{opn} \neq j$

shows

partial-composition-table-candidate
(composition-table[j := v]) composition-openings

proof –

have *len: length (composition-table[j := v]) = scale * clength*

using *cand* **unfolding** *partial-composition-table-candidate-def* **by** *simp*

have *rounds-len: length composition-openings = rounds*

using *cand* **unfolding** *partial-composition-table-candidate-def* **by** *simp*

have *agrees:*

```

 $\bigwedge i. i < \text{rounds} \implies$ 
  table-agrees-with-authenticated-openings
    (composition-table[j := v]) (scale * clength)
    (composition-openings ! i)
proof –
  fix i
  assume i-bound: i < rounds
  have old:
    table-agrees-with-authenticated-openings composition-table
      (scale * clength) (composition-openings ! i)
  using cand i-bound
  unfolding partial-composition-table-candidate-def by blast
  show
    table-agrees-with-authenticated-openings
      (composition-table[j := v]) (scale * clength)
      (composition-openings ! i)
  by (rule table-agrees-with-authenticated-openings-update-unopened
    [OF old])
    (use unopened[OF i-bound] in blast)
  qed
  show ?thesis
    unfolding partial-composition-table-candidate-def
    using len rounds-len agrees by simp
  qed

lemma partial-composition-table-candidates-not-singleton-if-unopened:
  assumes cand:
    partial-composition-table-candidate composition-table
      composition-openings
  and j-bound: j < scale * clength
  and unopened:
     $\bigwedge i \text{ opn. } i < \text{rounds} \implies$ 
      opn  $\in \text{set } (\text{composition-openings} ! i) \implies$ 
      opening-index opn  $\neq j$ 
  shows
     $\neg (\exists \text{ table. } \text{partial-composition-table-candidates } \text{composition-openings} \subseteq \{\text{table}\})$ 
proof
  assume unique:
     $\exists \text{ table. } \text{partial-composition-table-candidates } \text{composition-openings} \subseteq \{\text{table}\}$ 
  let ?composition-table' =
    composition-table[j := composition-table ! j + 1]
  have cand-in:
    composition-table  $\in$ 
      partial-composition-table-candidates composition-openings
  using cand unfolding partial-composition-table-candidates-def by simp
  have cand'-in:

```

```

    ?composition-table' ∈
      partial-composition-table-candidates composition-openings
    using partial-composition-table-candidate-update-unopened[OF cand unopened]
    unfolding partial-composition-table-candidates-def by simp
  from unique obtain table where subset:
    partial-composition-table-candidates composition-openings ⊆
      {table}
  by blast
  have comp-eq: composition-table = table
    using subset cand-in by blast
  have comp'-eq: ?composition-table' = table
    using subset cand'-in by blast
  have len: length composition-table = scale * clength
    using cand unfolding partial-composition-table-candidate-def by simp
  have changed: ?composition-table' ! j ≠ composition-table ! j
    using j-bound len by simp
  show False
    using changed comp-eq comp'-eq by simp
qed

```

lemma *partial-composition-table-candidate-eq-if-all-indices-opened:*

```

  assumes cand:
    partial-composition-table-candidate composition-table
    composition-openings
  and cand':
    partial-composition-table-candidate composition-table'
    composition-openings
  and cover:
     $\bigwedge j. j < \text{scale} * \text{clength} \implies$ 
       $\exists i < \text{rounds}. \exists \text{opn} \in \text{set } (\text{composition-openings } ! i).$ 
       $\text{opening-index } \text{opn} = j$ 
  shows composition-table = composition-table'
proof (rule nth-equalityI)
  show length composition-table = length composition-table'
    using cand cand'
    unfolding partial-composition-table-candidate-def by simp
next
  fix j
  assume j-bound: j < length composition-table
  have len: length composition-table = scale * clength
    using cand unfolding partial-composition-table-candidate-def by simp
  from cover[OF j-bound[unfolded len]]
  obtain i opn where i-bound: i < rounds
    and opn-in: opn ∈ set (composition-openings ! i)
    and opn-idx: opening-index opn = j
  by blast
  have table-val: composition-table ! j = opening-value opn
    using partial-composition-table-candidateD(3)[OF cand i-bound opn-in]
    opn-idx by simp

```

```

have table'-val: composition-table' ! j = opening-value opn
  using partial-composition-table-candidateD(3)[OF cand' i-bound opn-in]
    opn-idx by simp
show composition-table ! j = composition-table' ! j
  using table-val table'-val by simp
qed

lemma partial-composition-table-candidates-subset-singleton-if-all-indices-opened:
  assumes cand:
    partial-composition-table-candidate composition-table
    composition-openings
  and cover:
     $\bigwedge j. j < \text{scale} * \text{clength} \implies$ 
     $\exists i < \text{rounds}. \exists \text{opn} \in \text{set } (\text{composition-openings } ! i).$ 
    opening-index opn = j
  shows
    partial-composition-table-candidates composition-openings  $\subseteq$ 
    {composition-table}
proof
  fix composition-table'
  assume
    composition-table'  $\in$ 
    partial-composition-table-candidates composition-openings
  then have cand':
    partial-composition-table-candidate composition-table'
    composition-openings
  unfolding partial-composition-table-candidates-def by simp
  have composition-table' = composition-table
  using partial-composition-table-candidate-eq-if-all-indices-opened
    [OF cand' cand cover] .
  then show composition-table'  $\in$  {composition-table}
    by simp
qed

end

end

```

```

theory Soundness-Query-Partial-Candidate-Blocker
imports
  Soundness-Query-Header-Cross-Candidate
  Soundness-Partial-Candidate-Blocker
begin

```

Exact query-header version of the partial-candidate non-uniqueness diagnostic. These lemmas show why the old singleton-candidate route cannot be applied to the broad partial-opening candidate relation.

```

context soundness

```

begin

lemma *sparse-openings-obtain-unopened-index:*

assumes *rounds-len:* $\text{length openings} = \text{rounds}$

and *sparse:* $\text{length (List.concat openings)} < \text{scale} * \text{clength}$

obtains *j* **where**

$j < \text{scale} * \text{clength}$

$\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set (openings ! } i) \implies$
 $\text{opening-index opn} \neq j$

proof –

let $?N = \text{scale} * \text{clength}$

let $?I = \text{opening-index ' set (List.concat openings)}$

have *finite-I:* *finite* $?I$

by *simp*

have *card-I-le:* $\text{card } ?I \leq \text{length (List.concat openings)}$

proof –

have $\text{card } ?I \leq \text{card (set (List.concat openings))}$

by (*rule card-image-le*) *simp*

also have $\dots \leq \text{length (List.concat openings)}$

by (*rule card-length*)

finally show *?thesis* .

qed

have *not-all:* $\neg \{0..<?N\} \subseteq ?I$

proof

assume *subset:* $\{0..<?N\} \subseteq ?I$

have $?N = \text{card } \{0..<?N\}$

by *simp*

also have $\dots \leq \text{card } ?I$

by (*rule card-mono[OF finite-I subset]*)

also have $\dots \leq \text{length (List.concat openings)}$

by (*rule card-I-le*)

finally show *False*

using *sparse* **by** *simp*

qed

from *not-all* **obtain** *j* **where** *j-in:* $j \in \{0..<?N\}$ **and** *j-notin:* $j \notin ?I$

by *blast*

then have *j-bound:* $j < ?N$

by *simp*

have *unopened:*

$\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set (openings ! } i) \implies$
 $\text{opening-index opn} \neq j$

proof –

fix *i opn*

assume *i-bound:* $i < \text{rounds}$

and *opn-in:* $\text{opn} \in \text{set (openings ! } i)$

have $\text{openings ! } i \in \text{set openings}$

using *i-bound rounds-len* **by** *simp*

then have $\text{opn} \in \text{set (List.concat openings)}$

using *opn-in* **by** *auto*

```

    then show opening-index opn  $\neq$  j
      using j-notin by auto
  qed
  show ?thesis
    by (rule that[OF j-bound unopened])
  qed

lemma accepted-with-partial-trace-openings-concat-length:
  assumes partial:
    accepted-with-partial-trace-openings s out fr query-idxs trace-openings
  shows length (List.concat trace-openings) = rounds * powers
proof -
  have len: length trace-openings = rounds
    by (rule accepted-with-partial-trace-openings-shapes(2)[OF partial])
  have entry-len:
     $\bigwedge xs. xs \in \text{set trace-openings} \implies \text{length } xs = \text{powers}$ 
  proof -
    fix xs
    assume xs-in:  $xs \in \text{set trace-openings}$ 
    then obtain i where i-bound:  $i < \text{length trace-openings}$ 
      and xs-eq:  $xs = \text{trace-openings} ! i$ 
      by (auto simp: in-set-conv-nth)
    have idxs:
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
      by (rule accepted-with-partial-trace-openings-shapes(3)[OF partial])
        (use i-bound len in simp)
    have len-xs: length (map opening-index xs) =
      length (powers-scaled (query-idxs ! i))
      using idxs xs-eq by simp
    show length xs = powers
      using len-xs unfolding powers-scaled-def by simp
  qed
  have map-len: map length trace-openings = replicate rounds powers
  proof (rule nth-equalityI)
    show length (map length trace-openings) =
      length (replicate rounds powers)
      using len by simp
  next
    fix i
    assume i-bound:  $i < \text{length (map length trace-openings)}$ 
    have trace-openings ! i  $\in \text{set trace-openings}$ 
      by (rule nth-mem) (use i-bound in simp)
    then show
      map length trace-openings ! i = replicate rounds powers ! i
      using entry-len[of trace-openings ! i] i-bound len by simp
  qed
  show ?thesis
  proof -

```

```

have length (List.concat trace-openings) =
  sum-list (map length trace-openings)
  by (induction trace-openings) simp-all
also have ... = sum-list (replicate rounds powers)
  using map-len by simp
also have ... = rounds * powers
  by (induction rounds) simp-all
finally show ?thesis .
qed
qed

lemma accepted-with-partial-composition-openings-concat-length:
  assumes partial:
    accepted-with-partial-composition-openings s out composition-root
    query-idxs composition-openings
  shows length (List.concat composition-openings) = rounds * 2
proof -
  have len: length composition-openings = rounds
    by (rule accepted-with-partial-composition-openings-shapes(2)
      [OF partial])
  have entry-len:
     $\bigwedge xs. xs \in \text{set } \text{composition-openings} \implies \text{length } xs = 2$ 
  proof -
    fix xs
    assume xs-in:  $xs \in \text{set } \text{composition-openings}$ 
    then obtain i where i-bound:  $i < \text{length } \text{composition-openings}$ 
      and xs-eq:  $xs = \text{composition-openings} ! i$ 
      by (auto simp: in-set-conv-nth)
    have idxs:
      map opening-index (composition-openings ! i) =
        [query-idxs ! i,
         fri-sibling-index (scale * clength) (query-idxs ! i)]
      by (rule accepted-with-partial-composition-openings-shapes(3)
        [OF partial])
        (use i-bound len in simp)
    have len-xs: length (map opening-index xs) = length
      [query-idxs ! i,
       fri-sibling-index (scale * clength) (query-idxs ! i)]
      using idxs xs-eq by simp
    show length xs = 2
      using len-xs by simp
  qed
  have map-len: map length composition-openings = replicate rounds 2
  proof (rule nth-equalityI)
    show length (map length composition-openings) =
      length (replicate rounds 2)
      using len by simp
  next
    fix i

```



```

assume i-bound:  $i < \text{length } (\text{map length composition-openings})$ 
have composition-openings !  $i \in \text{set composition-openings}$ 
  by (rule nth-mem) (use i-bound in simp)
then show
   $\text{map length composition-openings ! } i = \text{replicate rounds } 2 ! i$ 
  using entry-len[of composition-openings !  $i$ ] i-bound len by simp
qed
show ?thesis
proof –
  have  $\text{length } (\text{List.concat composition-openings}) =$ 
     $\text{sum-list } (\text{map length composition-openings})$ 
    by (induction composition-openings) simp-all
  also have  $\dots = \text{sum-list } (\text{replicate rounds } 2)$ 
    using map-len by simp
  also have  $\dots = \text{rounds} * 2$ 
    by (induction rounds) simp-all
  finally show ?thesis .
qed
qed

lemma query-header-supported-partial-table-candidates-update-trace-unopened:
assumes comp-nonempty:  $\text{composition-fri-roots} \neq []$ 
and outcome:
   $\text{Some } (\text{result}, \text{final-state}) \in \text{set-dist } (\text{execute verify-monad } s)$ 
and trace-partial:
   $\text{accepted-with-partial-trace-openings } s$ 
   $(\text{Some } (\text{result}, \text{final-state})) \text{ fr trace-query-idxs trace-openings}$ 
and trace-candidate:
   $\text{partial-trace-table-candidate trace-table trace-openings}$ 
and comp-partial:
   $\text{accepted-with-partial-composition-openings } s$ 
   $(\text{Some } (\text{result}, \text{final-state})) (\text{hd composition-fri-roots})$ 
   $\text{composition-query-idxs composition-openings}$ 
and comp-candidate:
   $\text{partial-composition-table-candidate composition-table}$ 
   $\text{composition-openings}$ 
and header:
   $\text{verifier-header-transcript } s \text{ fr f-fri-roots f-final as dg}$ 
   $\text{composition-fri-roots final rest}$ 
and unopened:
   $\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set } (\text{trace-openings ! } i) \implies$ 
   $\text{opening-index opn} \neq j$ 
shows
   $(\text{trace-table}[j := v], \text{composition-table}) \in$ 
   $\text{query-header-supported-partial-table-candidates } s \text{ fr f-fri-roots}$ 
   $\text{f-final as dg composition-fri-roots final}$ 
proof –
  have trace-candidate':
   $\text{partial-trace-table-candidate } (\text{trace-table}[j := v]) \text{ trace-openings}$ 

```

```

    by (rule partial-trace-table-candidate-update-unopened
        [OF trace-candidate unopened])
  show ?thesis
    unfolding query-header-supported-partial-table-candidates-def
    using comp-nonempty outcome trace-partial trace-candidate'
        comp-partial comp-candidate header
    by blast
qed

lemma query-header-supported-partial-table-candidates-not-singleton-if-trace-unopened:
  assumes comp-nonempty: composition-fri-roots  $\neq$  []
  and outcome:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
  and trace-partial:
    accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr trace-query-idxs trace-openings
  and trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and comp-partial:
    accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots)
    composition-query-idxs composition-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and j-bound:  $j < \text{scale} * \text{clength}$ 
  and unopened:
     $\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set } (\text{trace-openings} ! i) \implies$ 
    opening-index opn  $\neq j$ 
  shows
     $\neg (\exists \text{ trace-table0 } \text{composition-table0.}$ 
    query-header-supported-partial-table-candidates s fr f-fri-roots
    f-final as dg composition-fri-roots final  $\subseteq$ 
     $\{(\text{trace-table0}, \text{composition-table0})\})$ 
  proof
    assume unique:
       $\exists \text{ trace-table0 } \text{composition-table0.}$ 
      query-header-supported-partial-table-candidates s fr f-fri-roots
      f-final as dg composition-fri-roots final  $\subseteq$ 
       $\{(\text{trace-table0}, \text{composition-table0})\}$ 
    let ?trace-table' = trace-table[j := trace-table ! j + 1]
    have candidate:
      (trace-table, composition-table)  $\in$ 
      query-header-supported-partial-table-candidates s fr f-fri-roots
      f-final as dg composition-fri-roots final
    unfolding query-header-supported-partial-table-candidates-def

```

```

using comp-nonempty outcome trace-partial trace-candidate
      comp-partial comp-candidate header
by blast
have candidate':
  (?trace-table', composition-table) ∈
    query-header-supported-partial-table-candidates s fr f-fri-roots
    f-final as dg composition-fri-roots final
by (rule
      query-header-supported-partial-table-candidates-update-trace-unopened
      [OF comp-nonempty outcome trace-partial trace-candidate comp-partial
        comp-candidate header unopened])
from unique obtain trace-table0 composition-table0 where subset:
  query-header-supported-partial-table-candidates s fr f-fri-roots
  f-final as dg composition-fri-roots final ⊆
  {(trace-table0, composition-table0)}
by blast
have pair-eq:
  (trace-table, composition-table) = (trace-table0, composition-table0)
using subset candidate by blast
have pair'-eq:
  (?trace-table', composition-table) = (trace-table0, composition-table0)
using subset candidate' by blast
have len: length trace-table = scale * clength
using trace-candidate unfolding partial-trace-table-candidate-def by simp
have changed: ?trace-table' ! j ≠ trace-table ! j
using j-bound len by simp
show False
using pair-eq pair'-eq changed by simp
qed

lemma query-header-supported-partial-table-candidates-not-singleton-if-trace-sparse:
assumes comp-nonempty: composition-fri-roots ≠ []
and outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots)
  composition-query-idxs composition-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest

```

```

    and sparse: length (List.concat trace-openings) < scale * clength
  shows
    ¬ (∃ trace-table0 composition-table0.
      query-header-supported-partial-table-candidates s fr f-fri-roots
        f-final as dg composition-fri-roots final ⊆
        {(trace-table0, composition-table0)})
  proof -
    have rounds-len: length trace-openings = rounds
    using trace-candidate unfolding partial-trace-table-candidate-def by simp
  obtain j where j-bound: j < scale * clength
  and unopened:
    ∧ i opn. i < rounds ⇒ opn ∈ set (trace-openings ! i) ⇒
      opening-index opn ≠ j
  using sparse-openings-obtain-unopened-index[OF rounds-len sparse]
  by blast
  show ?thesis
  by (rule
    query-header-supported-partial-table-candidates-not-singleton-if-trace-unopened
    [OF comp-nonempty outcome trace-partial trace-candidate comp-partial
      comp-candidate header j-bound unopened])
qed

lemma query-header-supported-partial-pairwise-cross-or-merkle-bad-if-trace-sparse:
  assumes comp-nonempty: composition-fri-roots ≠ []
  and outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and trace-partial:
    accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr trace-query-idxs trace-openings
  and trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and comp-partial:
    accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots)
    composition-query-idxs composition-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and sparse: length (List.concat trace-openings) < scale * clength
  shows
    query-header-supported-partial-pairwise-cross-or-merkle-bad s fr
      f-fri-roots f-final as dg composition-fri-roots final
  proof -
    have not-unique:
      ¬ (∃ trace-table0 composition-table0.
        query-header-supported-partial-table-candidates s fr f-fri-roots

```

```

    f-final as dg composition-fri-roots final  $\subseteq$ 
    {(trace-table0, composition-table0)})
  by (rule
    query-header-supported-partial-table-candidates-not-singleton-if-trace-sparse
    [OF comp-nonempty outcome trace-partial trace-candidate comp-partial
      comp-candidate header sparse])
  show ?thesis
  by (rule
    query-header-supported-partial-pairwise-cross-or-merkle-bad-if-not-singleton
    [OF not-unique])
qed

lemma query-header-supported-partial-table-candidates-not-singleton-if-trace-query-sparse:
assumes comp-nonempty: composition-fri-roots  $\neq$  []
and outcome:
  Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots)
  composition-query-idxs composition-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and sparse: rounds * powers < scale * clength
shows
   $\neg (\exists \text{ trace-table0 composition-table0.}$ 
    query-header-supported-partial-table-candidates s fr f-fri-roots
    f-final as dg composition-fri-roots final  $\subseteq$ 
    {(trace-table0, composition-table0)})
proof –
have concat-sparse:
  length (List.concat trace-openings) < scale * clength
using accepted-with-partial-trace-openings-concat-length [OF trace-partial]
  sparse by simp
show ?thesis
  by (rule
    query-header-supported-partial-table-candidates-not-singleton-if-trace-sparse
    [OF comp-nonempty outcome trace-partial trace-candidate comp-partial
      comp-candidate header concat-sparse])
qed

```

lemma *query-header-supported-partial-pairwise-cross-or-merkle-bad-if-trace-query-sparse:*

assumes *comp-nonempty: composition-fri-roots $\neq \square$*

and *outcome:*

Some (result, final-state) $\in$ set-dist (execute verify-monad s)

and *trace-partial:*

accepted-with-partial-trace-openings s

(Some (result, final-state)) fr trace-query-idxs trace-openings

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *comp-partial:*

accepted-with-partial-composition-openings s

(Some (result, final-state)) (hd composition-fri-roots)

composition-query-idxs composition-openings

and *comp-candidate:*

partial-composition-table-candidate composition-table

composition-openings

and *header:*

verifier-header-transcript s fr f-fri-roots f-final as dg

composition-fri-roots final rest

and *sparse: rounds * powers < scale * clength*

shows

query-header-supported-partial-pairwise-cross-or-merkle-bad s fr

f-fri-roots f-final as dg composition-fri-roots final

proof —

have *not-unique:*

$\neg (\exists \text{ trace-table0 } \text{composition-table0}.$

query-header-supported-partial-table-candidates s fr f-fri-roots

f-final as dg composition-fri-roots final $\subseteq$

\{(trace-table0, composition-table0)\})

by (rule

query-header-supported-partial-table-candidates-not-singleton-if-trace-query-sparse

[OF comp-nonempty outcome trace-partial trace-candidate comp-partial

comp-candidate header sparse])

show *?thesis*

by (rule

query-header-supported-partial-pairwise-cross-or-merkle-bad-if-not-singleton

[OF not-unique])

qed

lemma *query-header-supported-partial-table-candidates-update-composition-unopened:*

assumes *comp-nonempty: composition-fri-roots $\neq \square$*

and *outcome:*

Some (result, final-state) $\in$ set-dist (execute verify-monad s)

and *trace-partial:*

accepted-with-partial-trace-openings s

(Some (result, final-state)) fr trace-query-idxs trace-openings

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *comp-partial:*

```

    accepted-with-partial-composition-openings s
      (Some (result, final-state)) (hd composition-fri-roots)
      composition-query-idxs composition-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
      composition-openings
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
      composition-fri-roots final rest
  and unopened:
     $\bigwedge i \text{ opn. } i < \text{rounds} \implies$ 
      opn  $\in$  set (composition-openings ! i)  $\implies$ 
      opening-index opn  $\neq$  j
  shows
    (trace-table, composition-table[j := v])  $\in$ 
      query-header-supported-partial-table-candidates s fr f-fri-roots
      f-final as dg composition-fri-roots final
  proof -
  have comp-candidate':
    partial-composition-table-candidate (composition-table[j := v])
      composition-openings
  by (rule partial-composition-table-candidate-update-unopened
      [OF comp-candidate unopened])
  show ?thesis
  unfolding query-header-supported-partial-table-candidates-def
  using comp-nonempty outcome trace-partial trace-candidate
    comp-partial comp-candidate' header
  by blast
qed

```

lemma query-header-supported-partial-table-candidates-not-singleton-if-composition-unopened:

```

  assumes comp-nonempty: composition-fri-roots  $\neq$  []
  and outcome:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
  and trace-partial:
    accepted-with-partial-trace-openings s
      (Some (result, final-state)) fr trace-query-idxs trace-openings
  and trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and comp-partial:
    accepted-with-partial-composition-openings s
      (Some (result, final-state)) (hd composition-fri-roots)
      composition-query-idxs composition-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
      composition-openings
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
      composition-fri-roots final rest

```

and *j-bound*: $j < \text{scale} * \text{clength}$
and *unopened*:
 $\bigwedge i \text{ opn. } i < \text{rounds} \implies$
 $\text{opn} \in \text{set } (\text{composition-openings} ! i) \implies$
 $\text{opening-index } \text{opn} \neq j$
shows
 $\neg (\exists \text{ trace-table0 } \text{composition-table0}.$
 $\text{query-header-supported-partial-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg } \text{composition-fri-roots } \text{final} \subseteq$
 $\{(\text{trace-table0}, \text{composition-table0})\})$
proof
assume *unique*:
 $\exists \text{ trace-table0 } \text{composition-table0}.$
 $\text{query-header-supported-partial-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg } \text{composition-fri-roots } \text{final} \subseteq$
 $\{(\text{trace-table0}, \text{composition-table0})\}$
let *?composition-table'* =
 $\text{composition-table}[j := \text{composition-table} ! j + 1]$
have *candidate*:
 $(\text{trace-table}, \text{composition-table}) \in$
 $\text{query-header-supported-partial-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg } \text{composition-fri-roots } \text{final}$
unfolding *query-header-supported-partial-table-candidates-def*
using *comp-nonempty outcome trace-partial trace-candidate*
 $\text{comp-partial comp-candidate header}$
by *blast*
have *candidate'*:
 $(\text{trace-table}, ?\text{composition-table}') \in$
 $\text{query-header-supported-partial-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg } \text{composition-fri-roots } \text{final}$
by (*rule*
 $\text{query-header-supported-partial-table-candidates-update-composition-unopened}$
 $[\text{OF comp-nonempty outcome trace-partial trace-candidate comp-partial}$
 $\text{comp-candidate header unopened}]$)
from *unique* **obtain** *trace-table0 composition-table0* **where** *subset*:
 $\text{query-header-supported-partial-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final as dg } \text{composition-fri-roots } \text{final} \subseteq$
 $\{(\text{trace-table0}, \text{composition-table0})\}$
by *blast*
have *pair-eq*:
 $(\text{trace-table}, \text{composition-table}) = (\text{trace-table0}, \text{composition-table0})$
using *subset candidate* **by** *blast*
have *pair'-eq*:
 $(\text{trace-table}, ?\text{composition-table}') =$
 $(\text{trace-table0}, \text{composition-table0})$
using *subset candidate'* **by** *blast*
have *len*: $\text{length } \text{composition-table} = \text{scale} * \text{clength}$
using *comp-candidate*
unfolding *partial-composition-table-candidate-def* **by** *simp*


```

have changed: ?composition-table' ! j ≠ composition-table ! j
  using j-bound len by simp
show False
  using pair-eq pair'-eq changed by simp
qed

lemma query-header-supported-partial-table-candidates-not-singleton-if-composition-sparse:
assumes comp-nonempty: composition-fri-roots ≠ []
and outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots)
  composition-query-idxs composition-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and sparse:
  length (List.concat composition-openings) < scale * clength
shows
  ¬ (∃ trace-table0 composition-table0.
    query-header-supported-partial-table-candidates s fr f-fri-roots
    f-final as dg composition-fri-roots final ⊆
    {(trace-table0, composition-table0)})
proof –
  have rounds-len: length composition-openings = rounds
  using comp-candidate
  unfolding partial-composition-table-candidate-def by simp
obtain j where j-bound: j < scale * clength
and unopened:
  ∧ i opn. i < rounds ⇒
    opn ∈ set (composition-openings ! i) ⇒
    opening-index opn ≠ j
  using sparse-openings-obtain-unopened-index[OF rounds-len sparse]
  by blast
show ?thesis
  by (rule
    query-header-supported-partial-table-candidates-not-singleton-if-composition-unopened
    [OF comp-nonempty outcome trace-partial trace-candidate comp-partial
      comp-candidate header j-bound unopened])
qed

```

lemma *query-header-supported-partial-pairwise-cross-or-merkle-bad-if-composition-sparse:*

assumes *comp-nonempty: composition-fri-roots $\neq \square$*

and *outcome:*

Some (result, final-state) $\in$ set-dist (execute verify-monad s)

and *trace-partial:*

accepted-with-partial-trace-openings s

(Some (result, final-state)) fr trace-query-idxs trace-openings

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *comp-partial:*

accepted-with-partial-composition-openings s

(Some (result, final-state)) (hd composition-fri-roots)

composition-query-idxs composition-openings

and *comp-candidate:*

partial-composition-table-candidate composition-table

composition-openings

and *header:*

verifier-header-transcript s fr f-fri-roots f-final as dg

composition-fri-roots final rest

and *sparse:*

*length (List.concat composition-openings) $<$ scale * clength*

shows

query-header-supported-partial-pairwise-cross-or-merkle-bad s fr

f-fri-roots f-final as dg composition-fri-roots final

proof —

have *not-unique:*

$\neg (\exists \text{ trace-table0 } \text{composition-table0}.$

query-header-supported-partial-table-candidates s fr f-fri-roots

f-final as dg composition-fri-roots final $\subseteq$

\{(trace-table0, composition-table0)\})

by (rule

query-header-supported-partial-table-candidates-not-singleton-if-composition-sparse

[OF comp-nonempty outcome trace-partial trace-candidate comp-partial

comp-candidate header sparse])

show *?thesis*

by (rule

query-header-supported-partial-pairwise-cross-or-merkle-bad-if-not-singleton

[OF not-unique])

qed

lemma *query-header-supported-partial-table-candidates-not-singleton-if-composition-query-sparse:*

assumes *comp-nonempty: composition-fri-roots $\neq \square$*

and *outcome:*

Some (result, final-state) $\in$ set-dist (execute verify-monad s)

and *trace-partial:*

accepted-with-partial-trace-openings s

(Some (result, final-state)) fr trace-query-idxs trace-openings

and *trace-candidate:*

```

    partial-trace-table-candidate trace-table trace-openings
  and comp-partial:
    accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-fri-roots)
    composition-query-idxs composition-openings
  and comp-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and sparse: rounds * 2 < scale * clength
shows
  ¬ (∃ trace-table0 composition-table0.
    query-header-supported-partial-table-candidates s fr f-fri-roots
    f-final as dg composition-fri-roots final ⊆
    {(trace-table0, composition-table0)})
proof -
  have concat-sparse:
    length (List.concat composition-openings) < scale * clength
  using accepted-with-partial-composition-openings-concat-length
  [OF comp-partial] sparse by simp
show ?thesis
  by (rule
    query-header-supported-partial-table-candidates-not-singleton-if-composition-sparse
    [OF comp-nonempty outcome trace-partial trace-candidate comp-partial
    comp-candidate header concat-sparse])
qed

lemma query-header-supported-partial-pairwise-cross-or-merkle-bad-if-composition-query-sparse:
assumes comp-nonempty: composition-fri-roots ≠ []
and outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-partial:
  accepted-with-partial-composition-openings s
  (Some (result, final-state)) (hd composition-fri-roots)
  composition-query-idxs composition-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and sparse: rounds * 2 < scale * clength

```

```

shows
  query-header-supported-partial-pairwise-cross-or-merkle-bad s fr
    f-fri-roots f-final as dg composition-fri-roots final
proof –
  have not-unique:
    ¬ (∃ trace-table0 composition-table0.
      query-header-supported-partial-table-candidates s fr f-fri-roots
        f-final as dg composition-fri-roots final ⊆
        {(trace-table0, composition-table0)})
  by (rule
    query-header-supported-partial-table-candidates-not-singleton-if-composition-query-sparse
    [OF comp-nonempty outcome trace-partial trace-candidate comp-partial
      comp-candidate header sparse])
  show ?thesis
  by (rule
    query-header-supported-partial-pairwise-cross-or-merkle-bad-if-not-singleton
    [OF not-unique])
qed

end

end

```

theory Soundness-Alpha-Header-Candidate-Blocker

imports

Soundness-Query-Partial-Candidate-Blocker

Staged-Security-Experiment-Composition

begin

Diagnostic lemmas for the alpha-header partial-candidate route.

The staged composition proof uses *alpha-header-supported-partial-trace-table-candidates*. This candidate set is trace-only, but it has the same sparse-opening obstruction as the query-header partial-candidate set: sampled authenticated openings do not determine unopened trace-table positions.

context soundness

begin

definition alpha-header-partial-candidate-not-prefix-bound

∷ 'f protocol-channel ⇒ ('f, 'a) protocol-channel-scheme ⇒
 'f ⇒ 'f list ⇒ 'f ⇒ bool

where

alpha-header-partial-candidate-not-prefix-bound prefix-state s fr
 f-fri-roots f-final $\longleftrightarrow$
 (∃ trace-table ∈
 alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
 f-final.
 trace-table ∉ alpha-prefix-trace-table-candidates prefix-state fr)

lemma *alpha-header-partial-candidate-not-prefix-bound-iff-not-subset:*
alpha-header-partial-candidate-not-prefix-bound prefix-state s fr
f-fri-roots f-final $\longleftrightarrow$
 $\neg$ *alpha-header-supported-partial-trace-table-candidates s fr*
f-fri-roots f-final $\subseteq$
alpha-prefix-trace-table-candidates prefix-state fr
unfolding *alpha-header-partial-candidate-not-prefix-bound-def* **by** *blast*

lemma *alpha-header-supported-partial-union-bad-sets-subset-prefix-union-if-bound:*
assumes *bound:*
 $\neg$ *alpha-header-partial-candidate-not-prefix-bound prefix-state s fr*
f-fri-roots f-final
shows
alpha-header-supported-partial-union-bad-sets s bad-sets fr f-fri-roots
f-final $\subseteq$ alpha-prefix-union-bad-sets prefix-state bad-sets fr
using *bound*
unfolding *alpha-header-partial-candidate-not-prefix-bound-def*
alpha-header-supported-partial-union-bad-sets-def
alpha-prefix-union-bad-sets-def
by *blast*

lemma *alpha-header-partial-union-hit-not-prefix-hit-imp-not-prefix-bound:*
assumes *hit:*
as $\in$
alpha-header-supported-partial-union-bad-sets s bad-sets fr
f-fri-roots f-final
and *not-prefix:*
as $\notin$ alpha-prefix-union-bad-sets prefix-state bad-sets fr
shows
alpha-header-partial-candidate-not-prefix-bound prefix-state s fr
f-fri-roots f-final
proof (rule *ccontr*)
assume
 $\neg$ *alpha-header-partial-candidate-not-prefix-bound prefix-state s fr*
f-fri-roots f-final
then have
alpha-header-supported-partial-union-bad-sets s bad-sets fr
f-fri-roots f-final $\subseteq$
alpha-prefix-union-bad-sets prefix-state bad-sets fr
by (rule *alpha-header-supported-partial-union-bad-sets-subset-prefix-union-if-bound*)
then show *False*
using *hit not-prefix* **by** *blast*
qed

lemma *alpha-header-supported-partial-trace-table-candidates-update-unopened:*
assumes *outcome:*
Some (result, final-state) $\in$ set-dist (execute verify-monad s)
and *partial:*
accepted-with-partial-trace-openings s

```

    (Some (result, final-state)) fr query-idxs trace-openings
and candidate:
    partial-trace-table-candidate trace-table trace-openings
and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
and unopened:
     $\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set } (\text{trace-openings} ! i) \implies$ 
    opening-index opn  $\neq j$ 
shows
    trace-table[j := v]  $\in$ 
    alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
    f-final
proof –
    have candidate':
    partial-trace-table-candidate (trace-table[j := v]) trace-openings
    by (rule partial-trace-table-candidate-update-unopened
    [OF candidate unopened])
    show ?thesis
    unfolding alpha-header-supported-partial-trace-table-candidates-def
    using outcome partial candidate' header by blast
qed

lemma alpha-header-supported-partial-trace-table-candidates-not-singleton-if-trace-unopened:
assumes outcome:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
and partial:
    accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs trace-openings
and candidate:
    partial-trace-table-candidate trace-table trace-openings
and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
and j-bound: j < scale * clength
and unopened:
     $\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set } (\text{trace-openings} ! i) \implies$ 
    opening-index opn  $\neq j$ 
shows
     $\neg (\exists \text{ trace-table0.}$ 
    alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
    f-final  $\subseteq \{\text{trace-table0}\})$ 
proof
assume unique:
     $\exists \text{ trace-table0.}$ 
    alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
    f-final  $\subseteq \{\text{trace-table0}\}$ 
let ?trace-table' = trace-table[j := trace-table ! j + 1]
have candidate-in:

```

```

    trace-table ∈
      alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
        f-final
  unfolding alpha-header-supported-partial-trace-table-candidates-def
  using outcome partial candidate header by blast
have candidate'-in:
  ?trace-table' ∈
    alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
      f-final
  by (rule
    alpha-header-supported-partial-trace-table-candidates-update-unopened
    [OF outcome partial candidate header unopened])
from unique obtain trace-table0 where subset:
  alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
    f-final ⊆ {trace-table0}
  by blast
have trace-eq: trace-table = trace-table0
  using subset candidate-in by blast
have trace'-eq: ?trace-table' = trace-table0
  using subset candidate'-in by blast
have len: length trace-table = scale * clength
  using candidate unfolding partial-trace-table-candidate-def by simp
have changed: ?trace-table' ! j ≠ trace-table ! j
  using j-bound len by simp
show False
  using changed trace-eq trace'-eq by simp
qed

lemma alpha-header-supported-partial-trace-table-candidates-not-singleton-if-trace-sparse:
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and partial:
    accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs trace-openings
  and candidate:
    partial-trace-table-candidate trace-table trace-openings
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and sparse: length (List.concat trace-openings) < scale * clength
  shows
    ¬ (∃ trace-table0.
      alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
        f-final ⊆ {trace-table0})
proof -
  have rounds-len: length trace-openings = rounds
    using candidate unfolding partial-trace-table-candidate-def by simp
  obtain j where j-bound: j < scale * clength
    and unopened:

```

```

 $\bigwedge i \text{ opn. } i < \text{rounds} \implies \text{opn} \in \text{set } (\text{trace-openings } ! i) \implies$ 
  opening-index opn  $\neq j$ 
using sparse-openings-obtain-unopened-index[OF rounds-len sparse]
by blast
show ?thesis
by (rule
  alpha-header-supported-partial-trace-table-candidates-not-singleton-if-trace-unopened
  [OF outcome partial candidate header j-bound unopened])
qed

end

end

```

```

theory Soundness-Alpha-Header-Cross-Candidate
imports
  Soundness-Alpha-Header-Candidate-Blocker
  Soundness-Partial-Merkle-Cross
begin

```

Witnessed alpha-header partial trace candidates.

This layer keeps verifier outcomes and authenticated opening witnesses explicit. It proves the trace-only analogue of the query-header cross-candidate decomposition: if witnessed candidates are not singleton, then either pairwise opening coverage fails or a Merkle/collision side condition occurs.

```

context soundness
begin

```

```

definition alpha-header-supported-partial-trace-table-candidate-witness
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list  $\Rightarrow$ 
    ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    'f authenticated-opening list list  $\Rightarrow$  bool

```

```

where
  alpha-header-supported-partial-trace-table-candidate-witness s fr
    f-fri-roots f-final trace-table final-state trace-openings  $\longleftrightarrow$ 
    ( $\exists$  result query-idxs as dg composition-fri-roots final rest.
      Some (result, final-state)  $\in$  set-dist (execute verify-monad s)  $\wedge$ 
      accepted-with-partial-trace-openings s
      (Some (result, final-state)) fr query-idxs trace-openings  $\wedge$ 
      partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
      verifier-header-transcript s fr f-fri-roots f-final as dg
      composition-fri-roots final rest)

```

```

definition alpha-header-supported-witnessed-partial-trace-table-candidates
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list set
where

```


$\text{alpha-header-supported-witnessed-partial-trace-table-candidates } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final} =$
 $\{ \text{trace-table.}$
 $\exists \text{final-state trace-openings.}$
 $\text{alpha-header-supported-partial-trace-table-candidate-witness } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final trace-table final-state trace-openings} \}$

definition $\text{alpha-header-supported-witnessed-partial-union-bad-sets}$

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list set}$

where

$\text{alpha-header-supported-witnessed-partial-union-bad-sets } s \text{ bad-sets fr}$
 $f\text{-fri-roots } f\text{-final} =$
 $\{ as \in \text{alpha-space.}$
 $\exists \text{trace-table} \in$
 $\text{alpha-header-supported-witnessed-partial-trace-table-candidates } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final.}$
 $as \in \text{bad-sets trace-table} \}$

lemma $\text{alpha-header-supported-partial-trace-table-candidate-witness-imp-candidate:}$

assumes

$\text{alpha-header-supported-partial-trace-table-candidate-witness } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final trace-table final-state trace-openings}$

shows

$\text{trace-table} \in$
 $\text{alpha-header-supported-partial-trace-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final}$

using assms

unfolding $\text{alpha-header-supported-partial-trace-table-candidate-witness-def}$

$\text{alpha-header-supported-partial-trace-table-candidates-def}$

by blast

lemma $\text{alpha-header-supported-witnessed-candidates-subset-candidates:}$

$\text{alpha-header-supported-witnessed-partial-trace-table-candidates } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final} \subseteq$
 $\text{alpha-header-supported-partial-trace-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final}$

unfolding $\text{alpha-header-supported-witnessed-partial-trace-table-candidates-def}$

using $\text{alpha-header-supported-partial-trace-table-candidate-witness-imp-candidate}$

by blast

lemma $\text{alpha-header-supported-partial-trace-table-candidates-subset-witnessed-candidates:}$

$\text{alpha-header-supported-partial-trace-table-candidates } s \text{ fr } f\text{-fri-roots}$
 $f\text{-final} \subseteq$
 $\text{alpha-header-supported-witnessed-partial-trace-table-candidates } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final}$

unfolding $\text{alpha-header-supported-partial-trace-table-candidates-def}$

$\text{alpha-header-supported-witnessed-partial-trace-table-candidates-def}$

alpha-header-supported-partial-trace-table-candidate-witness-def
accepted-with-partial-trace-openings-def
by *blast*

lemma *alpha-header-supported-witnessed-candidates-eq-candidates:*
alpha-header-supported-witnessed-partial-trace-table-candidates s fr
f-fri-roots f-final =
alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
f-final
using *alpha-header-supported-witnessed-candidates-subset-candidates*
alpha-header-supported-partial-trace-table-candidates-subset-witnessed-candidates
by *blast*

lemma *alpha-header-supported-witnessed-union-bad-sets-subset-union-bad-sets:*
alpha-header-supported-witnessed-partial-union-bad-sets s bad-sets fr
f-fri-roots f-final $\subseteq$
alpha-header-supported-partial-union-bad-sets s bad-sets fr f-fri-roots
f-final
unfolding *alpha-header-supported-witnessed-partial-union-bad-sets-def*
alpha-header-supported-partial-union-bad-sets-def
using *alpha-header-supported-witnessed-candidates-subset-candidates*
by *blast*

lemma *alpha-header-supported-union-bad-sets-subset-witnessed-union-bad-sets:*
alpha-header-supported-partial-union-bad-sets s bad-sets fr f-fri-roots
f-final $\subseteq$
alpha-header-supported-witnessed-partial-union-bad-sets s bad-sets fr
f-fri-roots f-final
unfolding *alpha-header-supported-witnessed-partial-union-bad-sets-def*
alpha-header-supported-partial-union-bad-sets-def
using *alpha-header-supported-partial-trace-table-candidates-subset-witnessed-candidates*
by *blast*

lemma *alpha-header-supported-witnessed-union-bad-sets-eq-union-bad-sets:*
alpha-header-supported-witnessed-partial-union-bad-sets s bad-sets fr
f-fri-roots f-final =
alpha-header-supported-partial-union-bad-sets s bad-sets fr f-fri-roots
f-final
using *alpha-header-supported-witnessed-union-bad-sets-subset-union-bad-sets*
alpha-header-supported-union-bad-sets-subset-witnessed-union-bad-sets
by *blast*

lemma *alpha-header-partial-candidate-not-prefix-bound-obtain-witness:*
assumes
alpha-header-partial-candidate-not-prefix-bound prefix-state s fr
f-fri-roots f-final
obtains *trace-table final-state trace-openings where*
alpha-header-supported-partial-trace-table-candidate-witness s fr
f-fri-roots f-final trace-table final-state trace-openings

trace-table $\notin$ *alpha-prefix-trace-table-candidates* *prefix-state* *fr*
using *assms*
unfolding *alpha-header-partial-candidate-not-prefix-bound-def*
using *alpha-header-supported-partial-trace-table-candidates-subset-witnessed-candidates*
unfolding *alpha-header-supported-witnessed-partial-trace-table-candidates-def*
by *blast*

lemma *alpha-header-supported-witnessed-partial-trace-table-candidates-obtain-witness:*

assumes
trace-table $\in$
alpha-header-supported-witnessed-partial-trace-table-candidates *s* *fr*
f-fri-roots *f-final*
obtains *final-state* *trace-openings* **where**
alpha-header-supported-partial-trace-table-candidate-witness *s* *fr*
f-fri-roots *f-final* *trace-table* *final-state* *trace-openings*
using *assms*
unfolding *alpha-header-supported-witnessed-partial-trace-table-candidates-def*
by *blast*

lemma *alpha-header-supported-partial-trace-table-candidate-witness-shape:*

assumes
alpha-header-supported-partial-trace-table-candidate-witness *s* *fr*
f-fri-roots *f-final* *trace-table* *final-state* *trace-openings*
shows *partial-trace-table-candidate* *trace-table* *trace-openings*
using *assms*
unfolding *alpha-header-supported-partial-trace-table-candidate-witness-def*
by *blast*

lemma *alpha-header-supported-partial-trace-table-candidate-witness-eq-if-cross-cover-clean-merge:*

assumes *witness:*
alpha-header-supported-partial-trace-table-candidate-witness *s* *fr*
f-fri-roots *f-final* *trace-table* *final-state* *trace-openings*
and *witness':*
alpha-header-supported-partial-trace-table-candidate-witness *s* *fr*
f-fri-roots *f-final* *trace-table'* *final-state'* *trace-openings'*
and *trace-cover:*
partial-openings-cross-cover-all-indices *trace-openings*
trace-openings'
and *no-conflict:*
 $\neg$ *merkle-hash-value-conflict* *final-state* *final-state'*
and *clean-merge:*
 $\neg$ *hash-map-output-collision*
(merkle-hash-state-merge *final-state* *final-state')*
shows *trace-table* = *trace-table'*

proof –

from *witness* **obtain** *result* *query-idxs* *as* *dg* *composition-fri-roots* *final* *rest*
where *trace-partial:*
accepted-with-partial-trace-openings *s*
(Some (result, final-state)) *fr* *query-idxs* *trace-openings*

```

and trace-cand:
  partial-trace-table-candidate trace-table trace-openings
unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
by blast
from witness' obtain result' query-idxs' as' dg' composition-fri-roots'
  final' rest' where trace-partial':
    accepted-with-partial-trace-openings s
    (Some (result', final-state')) fr query-idxs' trace-openings'
and trace-cand':
  partial-trace-table-candidate trace-table' trace-openings'
unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
by blast
show ?thesis
by (rule partial-trace-table-candidates-eq-if-cross-cover-clean-merge
  [OF trace-partial trace-partial' trace-cand trace-cand' trace-cover
    no-conflict clean-merge])
qed

```

lemma *alpha-header-supported-witnessed-partial-trace-table-candidates-subset-singleton-if-pairwise-cross-cover-clean-*

assumes *pairwise*:

```

   $\bigwedge \text{trace-table final-state trace-openings trace-table' final-state'}$ 
   $\text{trace-openings'}$ .
  alpha-header-supported-partial-trace-table-candidate-witness s fr
  f-fri-roots f-final trace-table final-state trace-openings
 $\implies$ 
  alpha-header-supported-partial-trace-table-candidate-witness s fr
  f-fri-roots f-final trace-table' final-state' trace-openings'
 $\implies$ 
  partial-openings-cross-cover-all-indices trace-openings
  trace-openings'  $\wedge$ 
   $\neg \text{merkle-hash-value-conflict final-state final-state'}$   $\wedge$ 
   $\neg \text{hash-map-output-collision}$ 
  (merkle-hash-state-merge final-state final-state')

```

shows

```

 $\exists \text{trace-table}$ .
  alpha-header-supported-witnessed-partial-trace-table-candidates s fr
  f-fri-roots f-final  $\subseteq \{\text{trace-table}\}$ 

```

proof (*cases*

```

  alpha-header-supported-witnessed-partial-trace-table-candidates s fr
  f-fri-roots f-final = \{\})

```

case *True*

then show *?thesis*

```

  by (intro exI[of - \{\}] simp)

```

next

case *False*

then obtain *trace-table0 where cand0*:

```

  trace-table0  $\in$ 
  alpha-header-supported-witnessed-partial-trace-table-candidates s fr
  f-fri-roots f-final

```

```

    by blast
  from alpha-header-supported-witnessed-partial-trace-table-candidates-obtain-witness
    [OF cand0]
  obtain final-state0 trace-openings0 where witness0:
    alpha-header-supported-partial-trace-table-candidate-witness s fr
    f-fri-roots f-final trace-table0 final-state0 trace-openings0
  by blast
  have subset:
    alpha-header-supported-witnessed-partial-trace-table-candidates s fr
    f-fri-roots f-final  $\subseteq$  {trace-table0}
  proof
    fix trace-table
    assume cand:
      trace-table  $\in$ 
      alpha-header-supported-witnessed-partial-trace-table-candidates s fr
      f-fri-roots f-final
    from alpha-header-supported-witnessed-partial-trace-table-candidates-obtain-witness
      [OF cand]
    obtain final-state trace-openings where witness:
      alpha-header-supported-partial-trace-table-candidate-witness s fr
      f-fri-roots f-final trace-table final-state trace-openings
    by blast
    have pairwise-facts:
      partial-openings-cross-cover-all-indices trace-openings
      trace-openings0
       $\neg$  merkle-hash-value-conflict final-state final-state0
       $\neg$  hash-map-output-collision
      (merkle-hash-state-merge final-state final-state0)
    using pairwise[OF witness witness0] by simp-all
    have trace-table = trace-table0
    by (rule
      alpha-header-supported-partial-trace-table-candidate-witness-eq-if-cross-cover-clean-merge
      [OF witness witness0 pairwise-facts])
    then show trace-table  $\in$  {trace-table0}
    by simp
  qed
  then show ?thesis
  by blast
qed

```

```

lemma alpha-header-supported-witnessed-partial-trace-table-candidates-not-singleton-imp-cross-cover-or-merkle-bad:
  assumes not-unique:
     $\neg (\exists \text{trace-table.}$ 
      alpha-header-supported-witnessed-partial-trace-table-candidates s fr
      f-fri-roots f-final  $\subseteq$  {trace-table})
  shows
     $\exists \text{trace-table final-state trace-openings trace-table' final-state'}$ 
    trace-openings'.
    alpha-header-supported-partial-trace-table-candidate-witness s fr

```

```

    f-fri-roots f-final trace-table final-state trace-openings  $\wedge$ 
    alpha-header-supported-partial-trace-table-candidate-witness s fr
    f-fri-roots f-final trace-table' final-state' trace-openings'  $\wedge$ 
    trace-table  $\neq$  trace-table'  $\wedge$ 
    ( $\neg$  partial-openings-cross-cover-all-indices trace-openings
      trace-openings'  $\vee$ 
      merkle-hash-value-conflict final-state final-state'  $\vee$ 
      hash-map-output-collision
      (merkle-hash-state-merge final-state final-state'))
  proof -
  let ?C =
    alpha-header-supported-witnessed-partial-trace-table-candidates s fr
    f-fri-roots f-final
  have nonempty: ?C  $\neq$  {}
  using not-unique by auto
  then obtain trace-table where cand: trace-table  $\in$  ?C
  by blast
  have not-subset:  $\neg$  ?C  $\subseteq$  {trace-table}
  using not-unique by blast
  then obtain trace-table' where cand': trace-table'  $\in$  ?C
  and distinct: trace-table  $\neq$  trace-table'
  by auto
  from alpha-header-supported-witnessed-partial-trace-table-candidates-obtain-witness
  [OF cand]
  obtain final-state trace-openings where witness:
    alpha-header-supported-partial-trace-table-candidate-witness s fr
    f-fri-roots f-final trace-table final-state trace-openings
  by blast
  from alpha-header-supported-witnessed-partial-trace-table-candidates-obtain-witness
  [OF cand']
  obtain final-state' trace-openings' where witness':
    alpha-header-supported-partial-trace-table-candidate-witness s fr
    f-fri-roots f-final trace-table' final-state' trace-openings'
  by blast
  have side:
     $\neg$  partial-openings-cross-cover-all-indices trace-openings
      trace-openings'  $\vee$ 
      merkle-hash-value-conflict final-state final-state'  $\vee$ 
      hash-map-output-collision
      (merkle-hash-state-merge final-state final-state')
  proof (rule ccontr)
  assume no-side:
     $\neg$  ( $\neg$  partial-openings-cross-cover-all-indices trace-openings
      trace-openings'  $\vee$ 
      merkle-hash-value-conflict final-state final-state'  $\vee$ 
      hash-map-output-collision
      (merkle-hash-state-merge final-state final-state'))
  have trace-table = trace-table'
  by (rule

```

```

      alpha-header-supported-partial-trace-table-candidate-witness-eq-if-cross-cover-clean-merge
      [OF witness witness'])
    (use no-side in auto)
  then show False
    using distinct by simp
qed
show ?thesis
  using witness witness' distinct side by blast
qed

```

definition *alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  bool

```

where

```

alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad
  s fr f-fri-roots f-final  $\longleftrightarrow$ 
  ( $\exists$  trace-table final-state trace-openings trace-table' final-state'
    trace-openings'.
    alpha-header-supported-partial-trace-table-candidate-witness s fr
      f-fri-roots f-final trace-table final-state trace-openings  $\wedge$ 
    alpha-header-supported-partial-trace-table-candidate-witness s fr
      f-fri-roots f-final trace-table' final-state' trace-openings'  $\wedge$ 
    trace-table  $\neq$  trace-table'  $\wedge$ 
    ( $\neg$  partial-openings-cross-cover-all-indices trace-openings
      trace-openings'  $\vee$ 
    merkle-hash-value-conflict final-state final-state'  $\vee$ 
    hash-map-output-collision
    (merkle-hash-state-merge final-state final-state'))))

```

definition *alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$  bool

```

where

```

alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad s
 $\longleftrightarrow$ 
  ( $\exists$  fr f-fri-roots f-final.
    alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad
      s fr f-fri-roots f-final)

```

lemma *alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-if-not-singleton:*

assumes *not-unique:*

```

 $\neg$  ( $\exists$  trace-table.
  alpha-header-supported-witnessed-partial-trace-table-candidates s fr
    f-fri-roots f-final  $\subseteq$  {trace-table})

```

shows

```

alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad
  s fr f-fri-roots f-final

```

unfolding

```

alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-def

```

using

alpha-header-supported-witnessed-partial-trace-table-candidates-not-singleton-imp-cross-cover-or-merkle-bad
 [OF not-unique]
 by blast

lemma *alpha-header-supported-witnessed-partial-trace-table-candidates-subset-singleton-if-no-cross-or-merkle-bad*:
assumes *no-bad*:
 $\neg$ *alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad*
 s fr *f-fri-roots* *f-final*
shows
 $\exists$ *trace-table*.
 alpha-header-supported-witnessed-partial-trace-table-candidates *s* fr
 f-fri-roots *f-final* $\subseteq$ {*trace-table*}
proof (rule *ccontr*)
assume *not-unique*:
 $\neg$ ($\exists$ *trace-table*.
 alpha-header-supported-witnessed-partial-trace-table-candidates *s* fr
 f-fri-roots *f-final* $\subseteq$ {*trace-table*})
then have
 alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad
 s fr *f-fri-roots* *f-final*
by (rule
 alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-if-not-singleton)
then show *False*
using *no-bad* **by** *contradiction*
qed

lemma *alpha-header-supported-partial-trace-table-candidates-subset-singleton-if-no-witnessed-cross-or-merkle-bad*:
assumes *no-bad*:
 $\neg$ *alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad*
 s fr *f-fri-roots* *f-final*
shows
 $\exists$ *trace-table*.
 alpha-header-supported-partial-trace-table-candidates *s* fr *f-fri-roots*
 f-final $\subseteq$ {*trace-table*}
proof –
have
 $\exists$ *trace-table*.
 alpha-header-supported-witnessed-partial-trace-table-candidates *s* fr
 f-fri-roots *f-final* $\subseteq$ {*trace-table*}
by (rule
 alpha-header-supported-witnessed-partial-trace-table-candidates-subset-singleton-if-no-cross-or-merkle-bad
 [OF *no-bad*])
moreover have
 alpha-header-supported-partial-trace-table-candidates *s* fr *f-fri-roots*
 f-final =
 alpha-header-supported-witnessed-partial-trace-table-candidates *s* fr
 f-fri-roots *f-final*
using *alpha-header-supported-witnessed-candidates-eq-candidates*
 [*of s fr f-fri-roots f-final*]

by *simp*
 then show ?thesis
 using *calculation* by *simp*
 qed

lemma *alpha-header-supported-partial-trace-table-candidates-subset-singleton-if-no-global-witnessed-cross-or-merkle-bad*

assumes *no-bad*:

$\neg$ *alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad*
s

shows

$\exists$ *trace-table*.

alpha-header-supported-partial-trace-table-candidates s fr f-fri-roots
f-final $\subseteq \{\text{trace-table}\}$

proof (rule

alpha-header-supported-partial-trace-table-candidates-subset-singleton-if-no-witnessed-cross-or-merkle-bad)

show

$\neg$ *alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad*
s fr f-fri-roots f-final

using *no-bad*

unfolding

alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-def

by *blast*

qed

lemma *alpha-header-supported-witnessed-partial-union-bad-sets-fraction-bound-if-no-cross-or-merkle-bad*:

fixes *C* :: *prob*

assumes *no-bad*:

$\neg$ *alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad*
s fr f-fri-roots f-final

and *subset*: $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$

and *bounded*:

$\bigwedge \text{trace-table.}$
 $\text{nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)} \leq$
C

shows

nnreal
 $(\text{card}$
 $(\text{alpha-header-supported-witnessed-partial-union-bad-sets } s \text{ bad-sets}$
 $\text{fr f-fri-roots f-final})) /$
 $\text{nnreal (card alpha-space)} \leq C$

proof –

have *unique*:

$\exists$ *trace-table*.

alpha-header-supported-witnessed-partial-trace-table-candidates s fr
f-fri-roots f-final $\subseteq \{\text{trace-table}\}$

by (rule

alpha-header-supported-witnessed-partial-trace-table-candidates-subset-singleton-if-no-cross-or-merkle-bad
 $[OF \text{ no-bad}]$

from *unique* **obtain** *trace-table* **where** *candidates*:

```

    alpha-header-supported-witnessed-partial-trace-table-candidates s fr
      f-fri-roots f-final  $\subseteq \{\text{trace-table}\}$ 
  by blast
have union-subset:
  alpha-header-supported-witnessed-partial-union-bad-sets s bad-sets fr
    f-fri-roots f-final  $\subseteq \text{bad-sets trace-table}$ 
  unfolding alpha-header-supported-witnessed-partial-union-bad-sets-def
  using candidates by auto
have finite-bad: finite (bad-sets trace-table)
  by (rule finite-subset[OF subset finite-alpha-space])
have card-le:
  card
    (alpha-header-supported-witnessed-partial-union-bad-sets s bad-sets fr
      f-fri-roots f-final)  $\leq$  card (bad-sets trace-table)
  by (rule card-mono[OF finite-bad union-subset])
have nnreal
  (card
    (alpha-header-supported-witnessed-partial-union-bad-sets s bad-sets
      fr f-fri-roots f-final)) /
  nnreal (card alpha-space)  $\leq$ 
  nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)
  by (rule nnreal-nat-divide-right-mono[OF card-le])
also have ...  $\leq C$ 
  by (rule bounded)
finally show ?thesis .
qed

lemma alpha-header-supported-partial-union-bad-sets-fraction-bound-if-no-witnessed-cross-or-merkle-bad:
  fixes C :: prob
  assumes no-bad:
     $\neg$  alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad
      s fr f-fri-roots f-final
  and subset:  $\bigwedge \text{trace-table}. \text{bad-sets trace-table} \subseteq \text{alpha-space}$ 
  and bounded:
     $\bigwedge \text{trace-table}. \text{nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)} \leq C$ 
  shows
    nnreal
      (card
        (alpha-header-supported-partial-union-bad-sets s bad-sets fr
          f-fri-roots f-final)) /
      nnreal (card alpha-space)  $\leq C$ 
  using
    alpha-header-supported-witnessed-partial-union-bad-sets-fraction-bound-if-no-cross-or-merkle-bad
      [OF no-bad subset bounded]
  by (simp add: alpha-header-supported-witnessed-union-bad-sets-eq-union-bad-sets)

lemma alpha-header-supported-witnessed-partial-union-bad-sets-fraction-bound-if-no-global-cross-or-merkle-bad:

```

```

fixes  $C :: \text{prob}$ 
assumes  $\text{no-bad}$ :
   $\neg \text{alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad}$ 
   $s$ 
and  $\text{subset}$ :  $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$ 
and  $\text{bounded}$ :
   $\bigwedge \text{trace-table.}$ 
   $\text{nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)} \leq$ 
   $C$ 
shows
   $\text{nnreal}$ 
   $(\text{card}$ 
     $(\text{alpha-header-supported-witnessed-partial-union-bad-sets } s \text{ bad-sets}$ 
       $\text{fr } f\text{-fri-roots } f\text{-final})) /$ 
     $\text{nnreal (card alpha-space)} \leq C$ 
proof –
  have  $\text{no-header-bad}$ :
     $\neg \text{alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad}$ 
     $s \text{ fr } f\text{-fri-roots } f\text{-final}$ 
  using  $\text{no-bad}$ 
  unfolding
     $\text{alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-def}$ 
  by  $\text{blast}$ 
show  $?thesis$ 
  by  $(\text{rule}$ 
     $\text{alpha-header-supported-witnessed-partial-union-bad-sets-fraction-bound-if-no-cross-or-merkle-bad}$ 
     $[\text{OF no-header-bad subset bounded}])$ 
qed

lemma  $\text{alpha-header-supported-partial-union-bad-sets-fraction-bound-if-no-global-witnessed-cross-or-merkle-bad}$ :
  fixes  $C :: \text{prob}$ 
  assumes  $\text{no-bad}$ :
     $\neg \text{alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad}$ 
     $s$ 
  and  $\text{subset}$ :  $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$ 
  and  $\text{bounded}$ :
     $\bigwedge \text{trace-table.}$ 
     $\text{nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)} \leq$ 
     $C$ 
  shows
     $\text{nnreal}$ 
     $(\text{card}$ 
       $(\text{alpha-header-supported-partial-union-bad-sets } s \text{ bad-sets fr}$ 
         $f\text{-fri-roots } f\text{-final})) /$ 
       $\text{nnreal (card alpha-space)} \leq C$ 
  using
     $\text{alpha-header-supported-witnessed-partial-union-bad-sets-fraction-bound-if-no-global-cross-or-merkle-bad}$ 
     $[\text{OF no-bad subset bounded}]$ 
  by  $(\text{simp add: alpha-header-supported-witnessed-union-bad-sets-eq-union-bad-sets})$ 

```

lemma *wp-alpha-bad-set-hit-bound-if-no-global-witnessed-cross-or-merkle-bad:*

fixes $C\ H :: \text{prob}$

assumes *future: alpha-future-fresh s*

and *no-bad:*

$\neg$ *alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad*
 s

and *subset:* $\bigwedge \text{trace-table. bad-sets trace-table} \subseteq \text{alpha-space}$

and *bounded:*

$\bigwedge \text{trace-table.}$
 $\text{nnreal (card (bad-sets trace-table)) / nnreal (card alpha-space)} \leq$
 C

and *collision-bound:*

$\text{wp-event verify-monad (hash-map-output-collision-bad } s) s \leq H$

shows $\text{wp-event verify-monad (alpha-bad-set-hit } s \text{ bad-sets)} s \leq C + H$

proof –

have *unique:*

$\bigwedge \text{fr f-fri-roots f-final.}$

$\exists \text{ trace-table.}$

$\text{alpha-header-supported-partial-trace-table-candidates } s \text{ fr}$

$\text{f-fri-roots f-final} \subseteq \{\text{trace-table}\}$

by (rule

$\text{alpha-header-supported-partial-trace-table-candidates-subset-singleton-if-no-global-witnessed-cross-or-merkle-bad}$
[OF no-bad])

show *?thesis*

by (rule

$\text{wp-alpha-bad-set-hit-bound-if-supported-partial-header-candidate-unique-or-collision}$
[OF future unique subset bounded collision-bound])

qed

lemma *composition-randomization-bad-bound-if-no-global-witnessed-cross-or-merkle-bad:*

fixes $H :: \text{prob}$

assumes *future: alpha-future-fresh s*

and *no-bad:*

$\neg$ *alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad*
 s

and *collision-bound:*

$\text{wp-event verify-monad (hash-map-output-collision-bad } s) s \leq H$

shows

$\text{wp-event verify-monad (composition-randomization-bad } s) s \leq$
 $\text{composition-error-bound} + H$

proof –

have *unique:*

$\bigwedge \text{fr f-fri-roots f-final.}$

$\exists \text{ trace-table.}$

$\text{alpha-header-supported-partial-trace-table-candidates } s \text{ fr}$

$\text{f-fri-roots f-final} \subseteq \{\text{trace-table}\}$

by (rule

$\text{alpha-header-supported-partial-trace-table-candidates-subset-singleton-if-no-global-witnessed-cross-or-merkle-bad}$

```

    [OF no-bad])
  show ?thesis
  by (rule
    composition-randomization-bad-bound-if-alpha-header-supported-partial-candidate-unique-or-collision
    [OF future unique collision-bound])
qed

lemma alpha-header-witness-candidate-in-prefix-if-pullback-clean-cover:
  assumes prefix-candidate:
    prefix-trace-table ∈
      alpha-prefix-trace-table-candidates prefix-state fr
  and witness:
    alpha-header-supported-partial-trace-table-candidate-witness s fr
    f-fri-roots f-final trace-table final-state trace-openings
  and pullback:
     $\bigwedge i. i < \text{rounds} \implies$ 
      partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) prefix-state
  and cover:
     $\bigwedge j. j < \text{scale} * \text{clength} \implies$ 
       $\exists i < \text{rounds}. \exists \text{opn} \in \text{set } (\text{trace-openings} ! i).$ 
      opening-index opn = j
  and clean-prefix:  $\neg \text{hash-map-output-collision prefix-state}$ 
  shows
    trace-table ∈
      alpha-prefix-trace-table-candidates prefix-state fr
  proof -
    have prefix-len:
      length prefix-trace-table = clength * scale
    using prefix-candidate
    unfolding alpha-prefix-trace-table-candidates-def by simp
    have prefix-bind:
      merkle-root-binds-table fr prefix-trace-table prefix-state
    using prefix-candidate
    unfolding alpha-prefix-trace-table-candidates-def by simp
    from witness obtain result query-idxs as dg composition-fri-roots final rest
    where trace-cand:
      partial-trace-table-candidate trace-table trace-openings
    unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
    by blast
    have prefix-trace-cand:
      partial-trace-table-candidate prefix-trace-table trace-openings
    proof -
      obtain n where len-pow: clength * scale =  $2^n$ 
      using eval-domain-length-power by blast
      have table-pow: length prefix-trace-table =  $2^n$ 
      using prefix-len len-pow by (simp add: mult.commute)
      have rounds-len: length trace-openings = rounds
      using trace-cand unfolding partial-trace-table-candidate-def by simp

```

```

have agrees:
   $\bigwedge i. i < \text{rounds} \implies$ 
     $\text{table-agrees-with-authenticated-openings } \text{prefix-trace-table}$ 
       $(\text{scale} * \text{clength}) (\text{trace-openings} ! i)$ 
proof –
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  have table-i:
     $\text{partial-authenticated-table } \text{fr } (\text{scale} * \text{clength})$ 
       $(\text{trace-openings} ! i) \text{ prefix-state}$ 
    by (rule pullback[OF i-bound])
  have opening-agrees:
     $\forall \text{opn} \in \text{set } (\text{trace-openings} ! i).$ 
       $\text{opening-index } \text{opn} < \text{scale} * \text{clength} \wedge$ 
       $\text{prefix-trace-table} ! \text{opening-index } \text{opn} = \text{opening-value } \text{opn}$ 
  proof
    fix opn
    assume opn-in:  $\text{opn} \in \text{set } (\text{trace-openings} ! i)$ 
    have root:  $\text{opening-root } \text{opn} = \text{fr}$ 
      using table-i opn-in unfolding partial-authenticated-table-def
      by blast
    have len:  $\text{opening-length } \text{opn} = \text{length } \text{prefix-trace-table}$ 
      using table-i opn-in prefix-len
      unfolding partial-authenticated-table-def by simp
    have auth:  $\text{authenticated-opening-in } \text{prefix-state } \text{opn}$ 
      using table-i opn-in unfolding partial-authenticated-table-def
      by blast
    have agreement:
       $\text{opening-index } \text{opn} < \text{length } \text{prefix-trace-table} \wedge$ 
       $\text{prefix-trace-table} ! \text{opening-index } \text{opn} = \text{opening-value } \text{opn}$ 
      by (rule merkle-bound-table-agrees-with-authenticated-opening-if-clean
        [OF prefix-bind table-pow auth root len clean-prefix])
    then show
       $\text{opening-index } \text{opn} < \text{scale} * \text{clength} \wedge$ 
       $\text{prefix-trace-table} ! \text{opening-index } \text{opn} = \text{opening-value } \text{opn}$ 
      using prefix-len by (simp add: mult.commute)
    qed
  show
     $\text{table-agrees-with-authenticated-openings } \text{prefix-trace-table}$ 
       $(\text{scale} * \text{clength}) (\text{trace-openings} ! i)$ 
    unfolding table-agrees-with-authenticated-openings-def
    using prefix-len opening-agrees by simp
  qed
show ?thesis
  unfolding partial-trace-table-candidate-def
  using prefix-len rounds-len agrees by simp
qed
have trace-eq:  $\text{trace-table} = \text{prefix-trace-table}$ 
  by (rule partial-trace-table-candidate-eq-if-all-indices-opened)

```

$[OF \text{ trace-cand prefix-trace-cand cover}]$
show *?thesis*
using *prefix-candidate unfolding trace-eq* .
qed

lemma *alpha-header-not-prefix-witness-imp-no-prefix-candidate-or-pullback-gap:*
assumes *not-prefix:*
 $\text{trace-table} \notin \text{alpha-prefix-trace-table-candidates prefix-state fr}$
and *witness:*
 $\text{alpha-header-supported-partial-trace-table-candidate-witness } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final trace-table final-state trace-openings}$
and *clean-prefix:* $\neg \text{hash-map-output-collision prefix-state}$
and *pullback:*
 $\bigwedge i. i < \text{rounds} \implies$
 $\text{partial-authenticated-table fr (scale * clength)}$
 $(\text{trace-openings ! } i) \text{ prefix-state}$
and *cover:*
 $\bigwedge j. j < \text{scale * clength} \implies$
 $\exists i < \text{rounds}. \exists \text{opn} \in \text{set } (\text{trace-openings ! } i).$
 $\text{opening-index opn} = j$
shows
 $\text{alpha-prefix-trace-table-candidates prefix-state fr} = \{\}$
proof (rule *ccontr*)
assume $\text{alpha-prefix-trace-table-candidates prefix-state fr} \neq \{\}$
then obtain *prefix-trace-table* **where** *prefix-candidate:*
 $\text{prefix-trace-table} \in$
 $\text{alpha-prefix-trace-table-candidates prefix-state fr}$
by *blast*
have
 $\text{trace-table} \in$
 $\text{alpha-prefix-trace-table-candidates prefix-state fr}$
by (rule *alpha-header-witness-candidate-in-prefix-if-pullback-clean-cover*
 $[OF \text{ prefix-candidate witness pullback cover clean-prefix}]$)
then show *False*
using *not-prefix* **by** *contradiction*
qed

lemma *alpha-header-witness-opening-pullback-or-new-output-hit:*
assumes *ext:* $\text{prefix-state} \leq \text{final-state}$
and *witness:*
 $\text{alpha-header-supported-partial-trace-table-candidate-witness } s \text{ fr}$
 $f\text{-fri-roots } f\text{-final trace-table final-state trace-openings}$
and *i-bound:* $i < \text{rounds}$
and *opn-in:* $\text{opn} \in \text{set } (\text{trace-openings ! } i)$
shows
 $\text{authenticated-opening-in prefix-state opn} \vee$
 $\text{hash-map-new-output-hit}$
 $(\text{merkle-path-target-roots final-state fr (scale * clength)})$
 $(\text{opening-index opn}) (\text{opening-value opn}) (\text{opening-path opn})$

prefix-state final-state
proof –
from *witness* **obtain** *result query-idxs as dg composition-fri-roots final rest*
where *partial:*
accepted-with-partial-trace-openings s
(Some (result, final-state)) fr query-idxs trace-openings
unfolding *alpha-header-supported-partial-trace-table-candidate-witness-def*
by *blast*
have *table-i:*
*partial-authenticated-table fr (scale * clength)*
(trace-openings ! i) final-state
using *partial i-bound*
unfolding *accepted-with-partial-trace-openings-def* **by** *blast*
show *?thesis*
by *(rule partial-authenticated-table-opening-pullback-or-new-output-hit*
[OF ext table-i opn-in])
qed

lemma *alpha-header-witness-openings-pullback-or-new-output-hit:*
assumes *ext: prefix-state $\leq$ final-state*
and *witness:*
alpha-header-supported-partial-trace-table-candidate-witness s fr
f-fri-roots f-final trace-table final-state trace-openings
shows
 $(\forall i < \text{rounds}.$
*partial-authenticated-table fr (scale * clength)*
(trace-openings ! i) prefix-state) $\vee$
 $(\exists i < \text{rounds}.$ $\exists \text{opn} \in \text{set } (\text{trace-openings ! } i).$
hash-map-new-output-hit
*(merkle-path-target-roots final-state fr (scale * clength)*
(opening-index opn) (opening-value opn) (opening-path opn))
prefix-state final-state)
proof *(cases*
 $\exists i < \text{rounds}.$ $\exists \text{opn} \in \text{set } (\text{trace-openings ! } i).$
hash-map-new-output-hit
*(merkle-path-target-roots final-state fr (scale * clength)*
(opening-index opn) (opening-value opn) (opening-path opn))
prefix-state final-state)
case *True*
then show *?thesis* **by** *simp*
next
case *False*
have *pullback-all:*
 $\forall i < \text{rounds}.$
*partial-authenticated-table fr (scale * clength)*
(trace-openings ! i) prefix-state
proof *(intro allI impI)*
fix *i*
assume *i-bound: i < rounds*


```

from witness obtain result query-idxs as dg composition-fri-roots final rest
  where partial:
    accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs trace-openings
  unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
  by blast
have table-i:
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) final-state
  using partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
show
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) prefix-state
  unfolding partial-authenticated-table-def
proof
  fix opn
  assume opn-in: opn ∈ set (trace-openings ! i)
  have root-len:
    opening-root opn = fr ∧ opening-length opn = scale * clength
  using table-i opn-in unfolding partial-authenticated-table-def
  by blast
  have pullback:
    authenticated-opening-in prefix-state opn ∨
    hash-map-new-output-hit
      (merkle-path-target-roots final-state fr (scale * clength)
        (opening-index opn) (opening-value opn) (opening-path opn))
    prefix-state final-state
  by (rule alpha-header-witness-opening-pullback-or-new-output-hit
    [OF ext witness i-bound opn-in])
  have auth-prefix: authenticated-opening-in prefix-state opn
  using pullback False i-bound opn-in by blast
  show
    opening-root opn = fr ∧
    opening-length opn = scale * clength ∧
    authenticated-opening-in prefix-state opn
  using root-len auth-prefix by simp
  qed
  qed
  then show ?thesis by simp
  qed
end
end

```

```

theory Staged-Security-Experiment-Alpha-Header-Binding
imports

```

begin

Staged wrapper for the alpha-header candidate-binding diagnostic.

This does not bound the event. It isolates the remaining proof obligation: a partial-header alpha hit that is not already a prefix-fixed alpha hit must use a header-supported partial trace candidate that is not bound by the prefix Merkle evidence.

context *soundness*

begin

definition *checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*

$:: ('f \text{ list} \Rightarrow 'f \text{ list set}) \Rightarrow$
 $(((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound

bad-sets out $\longleftrightarrow$

(*case out of*

None $\Rightarrow$ *False*

| *Some* (*x*, *final-state*) $\Rightarrow$

(*let packed* = *fst* (*fst x*);

prefix = *fst* (*fst packed*);

prefix-state = *snd* (*fst packed*);

data = *snd packed*;

attacker-state = *snd* (*fst x*);

result = *snd x*;

s =

verifier-state-from-adversary attacker-state

(*staged-proof-transcript data*)

in $\exists$ *as query-idxs fr f-fri-roots f-final dg composition-fri-roots*
final rest.

accepted-transcript-shape s (*Some* (*result*, *final-state*))

as query-idxs $\wedge$

verifier-header-transcript s fr f-fri-roots f-final as dg

composition-fri-roots final rest $\wedge$

as $\in$

alpha-header-supported-partial-union-bad-sets s bad-sets fr

f-fri-roots f-final $\wedge$

as $\notin$ *alpha-prefix-union-bad-sets prefix-state bad-sets fr* $\wedge$

alpha-header-partial-candidate-not-prefix-bound prefix-state s
fr f-fri-roots f-final)

lemma *checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-boundI*:

assumes *s-eq*:

s =

verifier-state-from-adversary attacker-state

```

      (staged-proof-transcript data)
and shape:
  accepted-transcript-shape s (Some (result, final-state))
    as query-idxs
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
and hit:
  as ∈
    alpha-header-supported-partial-union-bad-sets s bad-sets fr
      f-fri-roots f-final
and not-prefix:
  as ∉ alpha-prefix-union-bad-sets prefix-state bad-sets fr
and not-bound:
  alpha-header-partial-candidate-not-prefix-bound prefix-state s fr
    f-fri-roots f-final
shows
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
    bad-sets
    (Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state))
proof —
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
have shape':
  accepted-transcript-shape ?s (Some (result, final-state))
    as query-idxs
using shape by (simp add: s-eq)
have header':
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
using header by (simp add: s-eq)
have hit':
  as ∈
    alpha-header-supported-partial-union-bad-sets ?s bad-sets fr
      f-fri-roots f-final
using hit by (simp add: s-eq)
have not-bound':
  alpha-header-partial-candidate-not-prefix-bound prefix-state ?s fr
    f-fri-roots f-final
using not-bound by (simp add: s-eq)
show ?thesis
unfolding
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound-def
  Let-def
apply simp
apply (rule exI[where x=as])
apply (intro conjI)

```

```

  apply (rule exI[where x=query-idxs])
  apply (rule shape')
  apply (rule exI[where x=fr])
  apply (rule exI[where x=f-fri-roots])
  apply (rule exI[where x=f-final])
  apply (intro conjI)
    apply (rule exI[where x=dg])
    apply (rule exI[where x=composition-fri-roots])
    apply (rule exI[where x=final])
    apply (rule exI[where x=rest])
    apply (rule header')
    apply (rule hit')
    apply (rule not-prefix)
    apply (rule not-bound')
  done
qed

lemma checked-staged-security-with-actual-alpha-prefix-support-prefix-extends-verifier-state:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    Some (((prefix, prefix-state), data), attacker-state), result),
    final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  shows
    prefix-state ≤
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
proof -
  from support obtain trans-out where trans-out:
    Some (((prefix, prefix-state), data), attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-with-alpha-prefix-program A)
      adversary-initial-state)
  unfolding
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
  by (auto elim!: set-dist-bindE split: prod.splits)
  from checked-staged-transcript-with-alpha-prefix-program-support
  [OF wf controlled trans-out]
  have prefix-ext: prefix-state ≤ attacker-state
  by blast
  show ?thesis
  by (rule
    alpha-prefix-hash-extends-verifier-state-from-adversary-right
    [OF prefix-ext])

```

qed

lemma *checked-staged-security-with-actual-alpha-prefix-support-prefix-extends-verifier-final:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

*Some (((((prefix, prefix-state), data), attacker-state), result),
final-state) ∈*

set-dist

(execute

*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)*

adversary-initial-state)

and *verifier-out:*

Some (result', witness-final-state) ∈

set-dist

(execute verify-monad

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

shows *prefix-state ≤ witness-final-state*

proof –

have *prefix-s:*

prefix-state ≤

verifier-state-from-adversary attacker-state

(staged-proof-transcript data)

by *(rule*

checked-staged-security-with-actual-alpha-prefix-support-prefix-extends-verifier-state

[OF wf controlled support])

have *s-final:*

verifier-state-from-adversary attacker-state

(staged-proof-transcript data) ≤

witness-final-state

by *(rule verify-monad-hash-extends[OF verifier-out])*

show *?thesis*

by *(rule hash-ext-trans[OF prefix-s s-final])*

qed

lemma *checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-obtain-witness-final-ext:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

*Some (((((prefix, prefix-state), data), attacker-state), result),
final-state) ∈*

set-dist

(execute

*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)*

adversary-initial-state)

and *bound:*

```

    alpha-header-partial-candidate-not-prefix-bound prefix-state
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
    fr f-fri-roots f-final
obtains trace-table witness-final-state trace-openings where
  alpha-header-supported-partial-trace-table-candidate-witness
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  fr f-fri-roots f-final trace-table witness-final-state
    trace-openings
  trace-table  $\notin$  alpha-prefix-trace-table-candidates prefix-state fr
  prefix-state  $\leq$  witness-final-state
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
from alpha-header-partial-candidate-not-prefix-bound-obtain-witness
  [OF bound]
obtain trace-table witness-final-state trace-openings where witness:
  alpha-header-supported-partial-trace-table-candidate-witness ?s fr
  f-fri-roots f-final trace-table witness-final-state
    trace-openings
and not-prefix:
  trace-table  $\notin$  alpha-prefix-trace-table-candidates prefix-state fr
by blast
from witness obtain witness-result query-idxs as dg composition-fri-roots
  final rest where verifier-out:
  Some (witness-result, witness-final-state)  $\in$ 
    set-dist (execute verify-monad ?s)
unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
by blast
have prefix-final: prefix-state  $\leq$  witness-final-state
by (rule
  checked-staged-security-with-actual-alpha-prefix-support-prefix-extends-verifier-final
  [OF wf controlled support verifier-out])
show ?thesis
by (rule that[OF witness not-prefix prefix-final])
qed

```

```

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-decompose:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state)  $\in$ 
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)

```

```

      adversary-initial-state)
and bound:
  alpha-header-partial-candidate-not-prefix-bound prefix-state
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  fr f-fri-roots f-final
shows
  hash-map-output-collision prefix-state ∨
  (∃ trace-table witness-final-state trace-openings.
    alpha-header-supported-partial-trace-table-candidate-witness
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
    fr f-fri-roots f-final trace-table witness-final-state
    trace-openings ∧
    trace-table ∉
      alpha-prefix-trace-table-candidates prefix-state fr ∧
    prefix-state ≤ witness-final-state ∧
    ((∃ j < scale * clength.
      ∀ i opn. i < rounds →
        opn ∈ set (trace-openings ! i) →
        opening-index opn ≠ j) ∨
      alpha-prefix-trace-table-candidates prefix-state fr = {} ∨
      (∃ i < rounds. ∃ opn ∈ set (trace-openings ! i).
        hash-map-new-output-hit
          (merkle-path-target-roots witness-final-state fr
            (scale * clength) (opening-index opn)
            (opening-value opn) (opening-path opn))
          prefix-state witness-final-state))))
proof —
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-obtain-witness-final-ext
  [OF wf controlled support bound]
obtain trace-table witness-final-state trace-openings where witness:
  alpha-header-supported-partial-trace-table-candidate-witness ?s fr
  f-fri-roots f-final trace-table witness-final-state
  trace-openings
and not-prefix:
  trace-table ∉
    alpha-prefix-trace-table-candidates prefix-state fr
and prefix-final: prefix-state ≤ witness-final-state
by blast
show ?thesis
proof (cases hash-map-output-collision prefix-state)
case True
then show ?thesis by simp
next
case clean-prefix: False

```

```

show ?thesis
proof (cases
   $\forall j < \text{scale} * \text{clength}.$ 
   $\exists i < \text{rounds}. \exists \text{opn} \in \text{set} (\text{trace-openings} ! i).$ 
   $\text{opening-index } \text{opn} = j$ )
case False
then have uncovered:
   $\exists j < \text{scale} * \text{clength}.$ 
   $\forall i \text{ opn}. i < \text{rounds} \longrightarrow$ 
   $\text{opn} \in \text{set} (\text{trace-openings} ! i) \longrightarrow$ 
   $\text{opening-index } \text{opn} \neq j$ 
by blast
show ?thesis
using witness not-prefix prefix-final uncovered by blast
next
case True
have pullback-or-new:
   $(\forall i < \text{rounds}.$ 
     $\text{partial-authenticated-table fr } (\text{scale} * \text{clength})$ 
     $(\text{trace-openings} ! i) \text{ prefix-state}) \vee$ 
     $(\exists i < \text{rounds}. \exists \text{opn} \in \text{set} (\text{trace-openings} ! i).$ 
     $\text{hash-map-new-output-hit}$ 
     $(\text{merkle-path-target-roots witness-final-state fr}$ 
     $(\text{scale} * \text{clength}) (\text{opening-index } \text{opn})$ 
     $(\text{opening-value } \text{opn}) (\text{opening-path } \text{opn}))$ 
     $\text{prefix-state witness-final-state})$ 
by (rule alpha-header-witness-openings-pullback-or-new-output-hit
    [OF prefix-final witness])
from pullback-or-new show ?thesis
proof
assume pullback-all:
   $\forall i < \text{rounds}.$ 
   $\text{partial-authenticated-table fr } (\text{scale} * \text{clength})$ 
   $(\text{trace-openings} ! i) \text{ prefix-state}$ 
have no-prefix-candidate:
   $\text{alpha-prefix-trace-table-candidates prefix-state fr} = \{\}$ 
proof (rule
   $\text{alpha-header-not-prefix-witness-imp-no-prefix-candidate-or-pullback-gap}$ 
  [OF not-prefix witness clean-prefix])
fix i
assume  $i < \text{rounds}$ 
then show
   $\text{partial-authenticated-table fr } (\text{scale} * \text{clength})$ 
   $(\text{trace-openings} ! i) \text{ prefix-state}$ 
using pullback-all by blast
next
fix j
assume  $j < \text{scale} * \text{clength}$ 
then show

```



```

     $\exists i < \text{rounds}. \exists \text{opn} \in \text{set } (\text{trace-openings } ! i).$ 
    opening-index opn = j
    using True by blast
qed
show ?thesis
    using witness not-prefix prefix-final no-prefix-candidate by blast
next
assume
     $\exists i < \text{rounds}. \exists \text{opn} \in \text{set } (\text{trace-openings } ! i).$ 
    hash-map-new-output-hit
    (merkle-path-target-roots witness-final-state fr
      (scale * clength) (opening-index opn)
      (opening-value opn) (opening-path opn))
    prefix-state witness-final-state
    then show ?thesis
    using witness not-prefix prefix-final by blast
qed
qed
qed
qed

definition checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
:: ((((((f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool
where
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision out
 $\longleftrightarrow$ 
(case out of
  None ⇒ False
| Some (x, -) ⇒
  (let packed = fst (fst x);
    prefix-state = snd (fst packed)
  in hash-map-output-collision prefix-state))

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
adversary-initial-state ≤
hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0)
proof –
have projection:
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

```

```

checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
adversary-initial-state ≤
wp-event
(checked-staged-transcript-with-alpha-prefix-program A)
checked-staged-transcript-actual-alpha-prefix-collision-hit
adversary-initial-state
unfolding
checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
proof (rule wp-event-bind-bound-by-head-event)
show
wp-event (checked-staged-transcript-with-alpha-prefix-program A)
checked-staged-transcript-actual-alpha-prefix-collision-hit
adversary-initial-state ≤
wp-event (checked-staged-transcript-with-alpha-prefix-program A)
checked-staged-transcript-actual-alpha-prefix-collision-hit
adversary-initial-state
by simp
next
show
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
None ⇒
checked-staged-transcript-actual-alpha-prefix-collision-hit None
unfolding
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-def
checked-staged-transcript-actual-alpha-prefix-collision-hit-def
by simp
next
fix packed attacker-state out
assume cont:
out ∈
set-dist
(execute
(get >>=
(λs. put
(verifier-state-from-adversary s
(staged-proof-transcript (snd packed)))) >>=
(λ-. verify-monad >>=
(λresult. return ((packed, s), result))))))
attacker-state)
and hit:
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
out
obtain prefix prefix-state data where packed-eq:
packed = ((prefix, prefix-state), data)
by (cases packed, auto split: prod.splits)
have hash-map-output-collision prefix-state
using cont hit
unfolding
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-def

```

```

    packed-eq
  by (auto elim!: set-dist-bindE split: option.splits prod.splits)
then show
  checked-staged-transcript-actual-alpha-prefix-collision-hit
    (Some (packed, attacker-state))
  unfolding checked-staged-transcript-actual-alpha-prefix-collision-hit-def
    packed-eq
  by simp
qed
have transcript-bound:
  wp-event
    (checked-staged-transcript-with-alpha-prefix-program A)
    checked-staged-transcript-actual-alpha-prefix-collision-hit
    adversary-initial-state ≤
  wp-event (staged-alpha-prefix-program A)
    (hash-new-collision-event adversary-initial-state)
    adversary-initial-state
  by (rule checked-staged-transcript-with-alpha-prefix-collision-bound)
have collision-budget:
  hash-collision-budget (staged-alpha-search-queries budgets 0)
    (staged-alpha-prefix-program A)
  by (rule hash-collision-budget-staged-alpha-prefix-program
    [OF wf controlled])
have collision-bound:
  wp-event (staged-alpha-prefix-program A)
    (hash-new-collision-event adversary-initial-state)
    adversary-initial-state ≤
  hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0)
proof -
  have
    wp-event (staged-alpha-prefix-program A)
      (hash-new-collision-event adversary-initial-state)
      adversary-initial-state ≤
      hash-collision-budget-value
        (card (hash-map-output-values adversary-initial-state))
        (staged-alpha-search-queries budgets 0)
    using collision-budget adversary-initial-state-no-output-collision
    unfolding hash-collision-budget-def by blast
  also have ... =
    hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0)
  by simp
  finally show ?thesis .
qed
show ?thesis
  by (rule order-trans[OF projection])
  (rule order-trans[OF transcript-bound collision-bound])
qed

definition checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index

```

$$\begin{aligned} &:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times \\ &\quad 'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times \\ &\quad 'f \text{ protocol-channel}) \times \text{unit list}) \times \\ &\quad 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool} \end{aligned}$$

where

$$\begin{aligned} &\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index} \\ &\quad \text{out} \longleftrightarrow \\ &\quad (\text{case out of} \\ &\quad \quad \text{None} \Rightarrow \text{False} \\ &\quad | \text{Some } (x, -) \Rightarrow \\ &\quad \quad (\text{let packed} = \text{fst } (\text{fst } x); \\ &\quad \quad \quad \text{prefix-state} = \text{snd } (\text{fst } \text{packed}); \\ &\quad \quad \quad \text{data} = \text{snd } \text{packed}; \\ &\quad \quad \quad \text{attacker-state} = \text{snd } (\text{fst } x); \\ &\quad \quad \quad s = \\ &\quad \quad \quad \text{verifier-state-from-adversary attacker-state} \\ &\quad \quad \quad \quad (\text{staged-proof-transcript data}) \\ &\quad \text{in } \exists \text{fr } f\text{-fri-roots } f\text{-final trace-table witness-final-state} \\ &\quad \quad \text{trace-openings.} \\ &\quad \quad \text{alpha-header-supported-partial-trace-table-candidate-witness} \\ &\quad \quad \quad s \text{ fr } f\text{-fri-roots } f\text{-final trace-table witness-final-state} \\ &\quad \quad \quad \text{trace-openings} \wedge \\ &\quad \quad \text{trace-table} \notin \\ &\quad \quad \quad \text{alpha-prefix-trace-table-candidates prefix-state fr} \wedge \\ &\quad \quad \text{prefix-state} \leq \text{witness-final-state} \wedge \\ &\quad \quad (\exists j < \text{scale} * \text{clength.} \\ &\quad \quad \quad \forall i \text{ opn. } i < \text{rounds} \longrightarrow \\ &\quad \quad \quad \text{opn} \in \text{set } (\text{trace-openings ! } i) \longrightarrow \\ &\quad \quad \quad \text{opening-index opn} \neq j))) \end{aligned}$$

definition *checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate*

$$\begin{aligned} &:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times \\ &\quad 'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times \\ &\quad 'f \text{ protocol-channel}) \times \text{unit list}) \times \\ &\quad 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool} \end{aligned}$$

where

$$\begin{aligned} &\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate} \\ &\quad \text{out} \longleftrightarrow \\ &\quad (\text{case out of} \\ &\quad \quad \text{None} \Rightarrow \text{False} \\ &\quad | \text{Some } (x, -) \Rightarrow \\ &\quad \quad (\text{let packed} = \text{fst } (\text{fst } x); \\ &\quad \quad \quad \text{prefix-state} = \text{snd } (\text{fst } \text{packed}); \\ &\quad \quad \quad \text{data} = \text{snd } \text{packed}; \\ &\quad \quad \quad \text{attacker-state} = \text{snd } (\text{fst } x); \\ &\quad \quad \quad s = \\ &\quad \quad \quad \text{verifier-state-from-adversary attacker-state} \\ &\quad \quad \quad \quad (\text{staged-proof-transcript data}) \\ &\quad \text{in } \exists \text{fr } f\text{-fri-roots } f\text{-final trace-table witness-final-state} \end{aligned}$$

trace-openings.
alpha-header-supported-partial-trace-table-candidate-witness
s fr f-fri-roots f-final trace-table witness-final-state
trace-openings $\wedge$
trace-table $\notin$
alpha-prefix-trace-table-candidates prefix-state fr $\wedge$
prefix-state $\leq$ *witness-final-state* $\wedge$
alpha-prefix-trace-table-candidates prefix-state fr = $\{\}$

definition *checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output*

$:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
out $\longleftrightarrow$
(case out of
None $\Rightarrow$ *False*
| Some (x, -) $\Rightarrow$
 $(\text{let packed} = \text{fst } (\text{fst } x);$
 $\text{prefix-state} = \text{snd } (\text{fst } \text{packed});$
 $\text{data} = \text{snd } \text{packed};$
 $\text{attacker-state} = \text{snd } (\text{fst } x);$
 $s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
in $\exists \text{fr f-fri-roots f-final trace-table witness-final-state}$
trace-openings.
alpha-header-supported-partial-trace-table-candidate-witness
s fr f-fri-roots f-final trace-table witness-final-state
trace-openings $\wedge$
trace-table $\notin$
alpha-prefix-trace-table-candidates prefix-state fr $\wedge$
prefix-state $\leq$ *witness-final-state* $\wedge$
 $(\exists i < \text{rounds}. \exists \text{opn} \in \text{set } (\text{trace-openings } ! i).$
 $\text{hash-map-new-output-hit}$
 $(\text{merkle-path-target-roots witness-final-state fr}$
 $(\text{scale} * \text{clength}) (\text{opening-index } \text{opn})$
 $(\text{opening-value } \text{opn}) (\text{opening-path } \text{opn}))$
 $\text{prefix-state witness-final-state}))$

definition *checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad*

$:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad

```

    out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some (x, -)  $\Rightarrow$ 
    (let packed = fst (fst x);
      data = snd packed;
      attacker-state = snd (fst x);
      s =
        verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)
      in alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad
        s))

```

definition *checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit*

```

:: ((((((f  $\times$  'f list  $\times$  'f list  $\times$  'f)  $\times$ 
  'f protocol-channel)  $\times$  'f staged-proof-data)  $\times$ 
  'f protocol-channel)  $\times$  unit list)  $\times$ 
  'f protocol-channel) option  $\Rightarrow$  bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
  out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some (x, -)  $\Rightarrow$ 
    (let packed = fst (fst x);
      data = snd packed;
      attacker-state = snd (fst x);
      s =
        verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)
      in  $\exists$  witness-result witness-state.
        Some (witness-result, witness-state)  $\in$ 
          set-dist (execute verify-monad s)  $\wedge$ 
          (hash-map-output-values s  $\cap$ 
            set (staged-proof-transcript data)  $\neq$   $\{\}$   $\vee$ 
            hash-map-new-output-hit (set (staged-proof-transcript data))
              s witness-state)))

```

definition *checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit*

```

:: ((((((f  $\times$  'f list  $\times$  'f list  $\times$  'f)  $\times$ 
  'f protocol-channel)  $\times$  'f staged-proof-data)  $\times$ 
  'f protocol-channel)  $\times$  unit list)  $\times$ 
  'f protocol-channel) option  $\Rightarrow$  bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
  out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some (x, -)  $\Rightarrow$ 

```

```

    (let packed = fst (fst x);
      data = snd packed;
      attacker-state = snd (fst x);
      s =
        verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)
    in hash-map-output-values s ∩
      set (staged-proof-transcript data) ≠ {}))

```

definition *checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit*

```

:: ((((((f × 'f list × 'f list × 'f) ×
  'f protocol-channel) × 'f staged-proof-data) ×
  'f protocol-channel) × unit list) ×
  'f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
  out ⇔
    (case out of
      None ⇒ False
    | Some (x, -) ⇒
      (let packed = fst (fst x);
        data = snd packed;
        attacker-state = snd (fst x);
        s =
          verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)
      in ∃ witness-result witness-state.
        Some (witness-result, witness-state) ∈
          set-dist (execute verify-monad s) ∧
          hash-map-new-output-hit (set (staged-proof-transcript data))
            s witness-state))

```

lemma *checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-imp-pre-or-new:*

assumes

```

checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
  out

```

shows

```

checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
  out ∨
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
  out

```

using *assms*

unfolding

```

checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-def
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-def
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit-def
by (auto split: option.splits prod.splits)

```

lemma *checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-imp-witness-transcript-hit:*

```

assumes
  checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
  out
shows
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
  out
proof (cases out)
case None
then show ?thesis
  using assms
  unfolding
    checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-def
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state
where out-eq:
  out =
    Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from assms obtain fr f-fri-roots f-final trace-table witness-final-state
  trace-openings where witness:
  alpha-header-supported-partial-trace-table-candidate-witness ?s fr
  f-fri-roots f-final trace-table witness-final-state trace-openings
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-def
  Let-def
by (auto split: option.splits prod.splits)
from witness obtain witness-result query-idxs as dg composition-fri-roots
  final rest where outcome:
  Some (witness-result, witness-final-state) ∈
  set-dist (execute verify-monad ?s)
unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
by blast
have tr-eq: PTranscript ?s = staged-proof-transcript data
by simp
have transcript-hit:
  hash-map-output-values ?s ∩ set (staged-proof-transcript data)
  ≠ {} ∨
  hash-map-new-output-hit (set (staged-proof-transcript data))
  ?s witness-final-state
by (rule verify-monad-trace-root-target-preexisting-or-new
  [OF outcome tr-eq])
show ?thesis

```



```

unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-def
  Let-def
using outcome transcript-hit
by (auto split: option.splits prod.splits)
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-le-witness-transcript-hit:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
  adversary-initial-state  $\leq$ 
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
  adversary-initial-state
by (rule wp-event-mono)
  (rule
    checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-imp-witness-transcript-hit)

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-imp-witness-transcript-hit:
assumes
  checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
  out
shows
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
  out
proof (cases out)
case None
then show ?thesis
using assms
unfolding
  checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-def
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-def
by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state
where out-eq:
  out =
    Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state)
by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from assms obtain fr f-fri-roots f-final trace-table witness-final-state
  trace-openings where witness:
  alpha-header-supported-partial-trace-table-candidate-witness ?s fr

```

```

    f-fri-roots f-final trace-table witness-final-state trace-openings
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-def
    Let-def
  by (auto split: option.splits prod.splits)
from witness obtain witness-result query-idxs as dg composition-fri-roots
  final rest where outcome:
    Some (witness-result, witness-final-state) ∈
      set-dist (execute verify-monad ?s)
  unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
  by blast
have tr-eq: PTranscript ?s = staged-proof-transcript data
  by simp
have transcript-hit:
  hash-map-output-values ?s ∩ set (staged-proof-transcript data)
    ≠ {} ∨
  hash-map-new-output-hit (set (staged-proof-transcript data))
    ?s witness-final-state
  by (rule verify-monad-trace-root-target-preexisting-or-new
    [OF outcome tr-eq])
show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-def
    Let-def
  using outcome transcript-hit
  by (auto split: option.splits prod.splits)
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-le-witness-transcript-hit:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
    adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
    adversary-initial-state
  by (rule wp-event-mono)
  (rule
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-imp-witness-transcript-hit)

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-imp-witnessed-cross-or-merkle-bad:
  assumes
    checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
    out
  shows
    checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
    out
proof (cases out)

```

```

case None
then show ?thesis
  using assms
  unfolding
    checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from assms obtain fr f-fri-roots f-final trace-table witness-final-state
  trace-openings j where witness:
    alpha-header-supported-partial-trace-table-candidate-witness ?s fr
    f-fri-roots f-final trace-table witness-final-state trace-openings
and j-bound: j < scale * clength
and unopened:
   $\bigwedge i \text{ opn. } i < \text{rounds} \implies$ 
   $\text{opn} \in \text{set } (\text{trace-openings } ! i) \implies$ 
   $\text{opening-index opn} \neq j$ 
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-def
  Let-def
by (auto split: option.splits prod.splits)
from witness obtain result' query-idxs as dg composition-fri-roots final rest
where outcome:
  Some (result', witness-final-state) ∈ set-dist (execute verify-monad ?s)
and partial:
  accepted-with-partial-trace-openings ?s
  (Some (result', witness-final-state)) fr query-idxs trace-openings
and candidate: partial-trace-table-candidate trace-table trace-openings
and header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
by blast
have not-singleton:
   $\neg (\exists \text{ trace-table0.}$ 
  alpha-header-supported-partial-trace-table-candidates ?s fr f-fri-roots
  f-final ⊆ {trace-table0})
by (rule
  alpha-header-supported-partial-trace-table-candidates-not-singleton-if-trace-unopened
  [OF outcome partial candidate header j-bound unopened])

```

```

have not-singleton-witnessed:
  ¬ (∃ trace-table0.
    alpha-header-supported-witnessed-partial-trace-table-candidates ?s fr
      f-fri-roots f-final ⊆ {trace-table0})
  using not-singleton
  by (simp add: alpha-header-supported-witnessed-candidates-eq-candidates)
have
  alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad
    ?s fr f-fri-roots f-final
  by (rule
    alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-if-not-singleton
      [OF not-singleton-witnessed])
then have
  alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad ?s
  unfolding alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-def
  by blast
then show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad-def
    Let-def
  by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-witnessed-cross-or-merkle-bad:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
    adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
    adversary-initial-state
  by (rule wp-event-mono)
  (rule
    checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-imp-witnessed-cross-or-merkle-bad)

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-imp-decomposition-on-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
            A)
          adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
    bad-sets out

```

```

shows
  checked-staged-security-with-actual-alpha-prefix-not-prefix-collision out ∨
  checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
    out ∨
  checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
    out ∨
  checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
    out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state data attacker-state result final-state
    where out-eq:
      out =
        Some (((((prefix, prefix-state), data), attacker-state), result),
          final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have hit-body:
    ∃ as query-idxs fr f-fri-roots f-final dg composition-fri-roots
      final rest.
      accepted-transcript-shape ?s (Some (result, final-state))
        as query-idxs ∧
      verifier-header-transcript ?s fr f-fri-roots f-final as dg
        composition-fri-roots final rest ∧
      as ∈
        alpha-header-supported-partial-union-bad-sets ?s bad-sets fr
          f-fri-roots f-final ∧
      as ∉ alpha-prefix-union-bad-sets prefix-state bad-sets fr ∧
      alpha-header-partial-candidate-not-prefix-bound prefix-state ?s fr
        f-fri-roots f-final
    using hit
    unfolding out-eq
      checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound-def
      Let-def
    by simp
  from hit-body obtain as query-idxs fr f-fri-roots f-final dg
    composition-fri-roots final rest where bound:
      alpha-header-partial-candidate-not-prefix-bound prefix-state ?s fr
        f-fri-roots f-final
  by blast

```

```

have support-some:
  Some (((((prefix, prefix-state), data), attacker-state), result),
    final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
using support out-eq by simp
from
  checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-decompose
  [OF wf controlled support-some bound]
show ?thesis
proof
  assume collision: hash-map-output-collision prefix-state
  have
    checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
    out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-def
    Let-def
  using collision by simp
  then show ?thesis by simp
next
  assume witness-cases:
    ∃ trace-table witness-final-state trace-openings.
      alpha-header-supported-partial-trace-table-candidate-witness ?s fr
      f-fri-roots f-final trace-table witness-final-state
      trace-openings ∧
      trace-table ∉
        alpha-prefix-trace-table-candidates prefix-state fr ∧
      prefix-state ≤ witness-final-state ∧
      ((∃ j < scale * clength.
        ∀ i opn.
          i < rounds ⟶
          opn ∈ set (trace-openings ! i) ⟶
          opening-index opn ≠ j) ∨
        alpha-prefix-trace-table-candidates prefix-state fr = {} ∨
        (∃ i < rounds. ∃ opn ∈ set (trace-openings ! i).
          hash-map-new-output-hit
            (merkle-path-target-roots witness-final-state fr
              (scale * clength) (opening-index opn)
              (opening-value opn) (opening-path opn))
            prefix-state witness-final-state))
  from witness-cases obtain trace-table witness-final-state trace-openings
  where witness:
    alpha-header-supported-partial-trace-table-candidate-witness ?s fr
    f-fri-roots f-final trace-table witness-final-state
    trace-openings

```

```

and not-prefix:
  trace-table  $\notin$ 
    alpha-prefix-trace-table-candidates prefix-state fr
and ext: prefix-state  $\leq$  witness-final-state
and cases:
  ( $\exists j < \text{scale} * \text{clength}$ .
     $\forall i \text{ opn.}$ 
       $i < \text{rounds} \longrightarrow$ 
         $\text{opn} \in \text{set } (\text{trace-openings} ! i) \longrightarrow$ 
           $\text{opening-index } \text{opn} \neq j) \vee$ 
    alpha-prefix-trace-table-candidates prefix-state fr = {}  $\vee$ 
    ( $\exists i < \text{rounds. } \exists \text{opn} \in \text{set } (\text{trace-openings} ! i).$ 
      hash-map-new-output-hit
      (merkle-path-target-roots witness-final-state fr
        ( $\text{scale} * \text{clength}$ ) (opening-index opn)
        (opening-value opn) (opening-path opn))
      prefix-state witness-final-state)
by blast
from cases show ?thesis
proof
assume uncovered:
  ( $\exists j < \text{scale} * \text{clength}$ .
     $\forall i \text{ opn.}$ 
       $i < \text{rounds} \longrightarrow$ 
         $\text{opn} \in \text{set } (\text{trace-openings} ! i) \longrightarrow$ 
           $\text{opening-index } \text{opn} \neq j$ 
have body:
    ( $\exists \text{fr}' \text{ f-fri-roots}' \text{ f-final}' \text{ trace-table}'$ 
      witness-final-state' trace-openings'.
      alpha-header-supported-partial-trace-table-candidate-witness ?s
      fr' f-fri-roots' f-final' trace-table'
      witness-final-state' trace-openings'  $\wedge$ 
      trace-table'  $\notin$ 
      alpha-prefix-trace-table-candidates prefix-state fr'  $\wedge$ 
      prefix-state  $\leq$  witness-final-state'  $\wedge$ 
      ( $\exists j < \text{scale} * \text{clength}$ .
         $\forall i \text{ opn. } i < \text{rounds} \longrightarrow$ 
           $\text{opn} \in \text{set } (\text{trace-openings}' ! i) \longrightarrow$ 
             $\text{opening-index } \text{opn} \neq j$ )
using witness not-prefix ext uncovered by blast
have
  checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
  out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-def
  Let-def
using body by (simp split: prod.splits)
then show ?thesis by simp
next

```

```

assume rest:
  alpha-prefix-trace-table-candidates prefix-state fr = {} ∨
  (∃ i < rounds. ∃ opn ∈ set (trace-openings ! i).
    hash-map-new-output-hit
      (merkle-path-target-roots witness-final-state fr
        (scale * clength) (opening-index opn)
        (opening-value opn) (opening-path opn))
      prefix-state witness-final-state)
then show ?thesis
proof
  assume no-prefix:
    alpha-prefix-trace-table-candidates prefix-state fr = {}
  have body:
    ∃ fr' f-fri-roots' f-final' trace-table'
      witness-final-state' trace-openings'.
    alpha-header-supported-partial-trace-table-candidate-witness ?s
      fr' f-fri-roots' f-final' trace-table'
      witness-final-state' trace-openings' ∧
    trace-table' ∉
      alpha-prefix-trace-table-candidates prefix-state fr' ∧
    prefix-state ≤ witness-final-state' ∧
    alpha-prefix-trace-table-candidates prefix-state fr' = {}
  using witness not-prefix ext no-prefix by blast
  have
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
      out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-def
    Let-def
  using body by (simp split: prod.splits)
  then show ?thesis by simp
next
  assume path:
    ∃ i < rounds. ∃ opn ∈ set (trace-openings ! i).
    hash-map-new-output-hit
      (merkle-path-target-roots witness-final-state fr
        (scale * clength) (opening-index opn)
        (opening-value opn) (opening-path opn))
    prefix-state witness-final-state
  have body:
    ∃ fr' f-fri-roots' f-final' trace-table'
      witness-final-state' trace-openings'.
    alpha-header-supported-partial-trace-table-candidate-witness ?s
      fr' f-fri-roots' f-final' trace-table'
      witness-final-state' trace-openings' ∧
    trace-table' ∉
      alpha-prefix-trace-table-candidates prefix-state fr' ∧
    prefix-state ≤ witness-final-state' ∧
    (∃ i < rounds. ∃ opn ∈ set (trace-openings' ! i)).

```



```

    hash-map-new-output-hit
    (merkle-path-target-roots witness-final-state' fr'
      (scale * clength) (opening-index opn)
      (opening-value opn) (opening-path opn))
    prefix-state witness-final-state')
  using witness not-prefix ext path by blast
have
  checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
  out
  unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-def
  Let-def
  using body by (simp split: prod.splits)
  then show ?thesis by simp
qed
qed
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound:
  fixes C U N P :: prob
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and collision-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
    adversary-initial-state ≤ C
  and uncovered-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
    adversary-initial-state ≤ U
  and no-prefix-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
    adversary-initial-state ≤ N
  and path-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
    adversary-initial-state ≤ P
shows
  wp-event

```

```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
      bad-sets)
    adversary-initial-state  $\leq C + U + N + P$ 
proof –
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?B =
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
      bad-sets
  let ?C = checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
  let ?U =
    checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
  let ?N =
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
  let ?P = checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
  have wp-event ?M ?B adversary-initial-state  $\leq$ 
    wp-event ?M ( $\lambda$ out. ?C out  $\vee$  ?U out  $\vee$  ?N out  $\vee$  ?P out)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (rule
      checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-imp-decomposition-on-support
      [OF wf controlled])
  also have ...  $\leq$ 
    wp-event ?M ?C adversary-initial-state +
    wp-event ?M ?U adversary-initial-state +
    wp-event ?M ?N adversary-initial-state +
    wp-event ?M ?P adversary-initial-state
  by (rule wp-event-union-bound4)
  also have ...  $\leq C + U + N + P$ 
  by (intro add-mono collision-bound uncovered-bound no-prefix-bound
    path-bound)
  finally show ?thesis .
qed

lemma checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-imp-not-prefix-bound-on-support
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  out  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
and drift:
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
    bad-sets out

```

```

shows
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
  bad-sets out
proof (cases out)
case None
then show ?thesis
  using drift
  unfolding
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def
    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-def
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-def
  by simp
next
case (Some packed-out)
then obtain prefix prefix-state data attacker-state result final-state
where out-eq:
  out =
    Some (((((prefix, prefix-state), data), attacker-state), result),
          final-state)
  by (cases packed-out, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have partial-hit:
  checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
  bad-sets out
using drift
unfolding
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def
  checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-def
by simp
have not-prefix-event:
  ¬ checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  bad-sets out
using drift
unfolding
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-def
by simp
have builder:
  Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)
    adversary-initial-state)
proof —
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
  set-dist
  (execute (checked-staged-security-experiment-with-data-state A)
    adversary-initial-state)

```

```

    by (rule
        checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
        [OF support[unfolded out-eq]])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  show ?thesis
    by blast
qed
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled builder])
have prefix-root-eq: staged-trace-root data = fst prefix
  using support out-eq
  unfolding checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
    checked-staged-transcript-with-alpha-prefix-program-def
    checked-staged-after-alpha-prefix-program-def Let-def
  by (auto elim!: set-dist-bindE split: prod.splits)
have alpha-header-hit:
  alpha-header-list-set-hit ?s
    (alpha-header-supported-partial-union-bad-sets ?s bad-sets)
    (Some (result, final-state))
  using partial-hit unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-def
  by (simp add: Let-def)
from alpha-header-hit obtain as query-idxs fr f-fri-roots f-final dg
  composition-fri-roots final rest where shape:
    accepted-transcript-shape ?s (Some (result, final-state)) as query-idxs
and header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and hit:
  as ∈
    alpha-header-supported-partial-union-bad-sets ?s bad-sets fr
    f-fri-roots f-final
  unfolding alpha-header-list-set-hit-def by blast
have as-eq: as = staged-alphas data
  using verifier-header-transcript-unique[OF staged-header header]
  by simp
have fr-eq: fr = staged-trace-root data
  using verifier-header-transcript-unique[OF header staged-header]

```

by *simp*
 have *not-prefix*:
 $as \notin \text{alpha-prefix-union-bad-sets prefix-state bad-sets fr}$
 proof
 assume *as-prefix*:
 $as \in \text{alpha-prefix-union-bad-sets prefix-state bad-sets fr}$
 have
 $\text{staged-alphas data} \in$
 $\text{alpha-prefix-union-bad-sets prefix-state bad-sets (fst prefix)}$
 using *as-prefix as-eq fr-eq prefix-root-eq* by *simp*
 then show *False*
 using *not-prefix-event unfolding out-eq*
 checked-staged-security-with-actual-alpha-prefix-bad-set-hit-def
 by *simp*
 qed
 have *not-bound*:
 $\text{alpha-header-partial-candidate-not-prefix-bound prefix-state ?s fr}$
 $\text{f-fri-roots f-final}$
 by (rule *alpha-header-partial-union-hit-not-prefix-hit-imp-not-prefix-bound*
 [*OF hit not-prefix*])
 show *?thesis*
 unfolding *out-eq*
 by (rule
 checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-boundI
 [*OF refl shape header hit not-prefix not-bound*])
 qed

lemma *checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-le-not-prefix-bound*:
 assumes *wf*: *staged-budget-wellformed budgets*
 and *controlled*: *staged-adversary-controlled budgets A*
 shows
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
 bad-sets)
 $\text{adversary-initial-state} \leq$
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
 bad-sets)
 $\text{adversary-initial-state}$
 by (rule *wp-event-mono-on-support*)
 (rule
 checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-imp-not-prefix-bound-on-support
 [*OF wf controlled*])

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-candidate-not-prefix-and-budgets*:
 assumes *wf*: *staged-budget-wellformed budgets*
 and *controlled*: *staged-adversary-controlled budgets A*

and *not-prefix-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)

adversary-initial-state $\leq D$

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event composition-bad*)

adversary-initial-state $\leq$

composition-error-bound +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

hash-collision-budget-value 0

(*staged-alpha-search-queries budgets 0*) +

D +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

hash-collision-budget-value 0

(*staged-attacker-query-budget budgets* +

staged-challenge-query-budget + *verifier-hash-query-budget*)

proof –

have *drift-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
composition-trace-bad-alpha-space)

adversary-initial-state $\leq D$

proof –

have *wp-event*

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
composition-trace-bad-alpha-space)

adversary-initial-state $\leq$

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)

adversary-initial-state

by (*rule*

checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-le-not-prefix-bound
[*OF wf controlled*])

also have ... $\leq D$

by (*rule not-prefix-bound*)

finally show *?thesis* .

qed
 show ?thesis
 by (rule
 checked-staged-security-with-data-state-composition-bad-bound-from-partial-header-candidate-drift-and-budgets
 [OF wf controlled drift-bound])
 qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-not-prefix-components-and-budgets*:
 fixes $C\ U\ N\ P :: \text{prob}$
 assumes *wf*: staged-budget-wellformed budgets
 and *controlled*: staged-adversary-controlled budgets A
 and *collision-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
 adversary-initial-state $\leq C$
 and *uncovered-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
 adversary-initial-state $\leq U$
 and *no-prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 adversary-initial-state $\leq N$
 and *path-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
 adversary-initial-state $\leq P$
 shows
 wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event composition-bad*)
 adversary-initial-state $\leq$
 composition-error-bound +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets* 0) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets* 0) +
 ($C + U + N + P$) +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets* 0) +
 hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +

$\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}$)

proof –

have *not-prefix-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)
 $\text{adversary-initial-state} \leq C + U + N + P$

by (*rule*
checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound
 [OF *wf controlled collision-bound uncovered-bound no-prefix-bound*
path-bound])

show *?thesis*

by (*rule*
checked-staged-security-with-data-state-composition-bad-bound-from-candidate-not-prefix-and-budgets
 [OF *wf controlled not-prefix-bound*])

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-not-prefix-noncollision-components-and-*
fixes $U\ N\ P :: \text{prob}$

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets* A

and *uncovered-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
 $\text{adversary-initial-state} \leq U$

and *no-prefix-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 $\text{adversary-initial-state} \leq N$

and *path-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
 $\text{adversary-initial-state} \leq P$

shows

wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event composition-bad*)
 $\text{adversary-initial-state} \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 (*staged-alpha-search-queries budgets* 0) +
 $\text{hash-collision-budget-value } 0$

(staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + U + N + P) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)

proof (rule
 checked-staged-security-with-data-state-composition-bad-bound-from-not-prefix-components-and-budgets
 [OF wf controlled - uncovered-bound no-prefix-bound path-bound])

show
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
 adversary-initial-state ≤
 hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0)

by (rule
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
 [OF wf controlled])

qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-cross-no-prefix-path-components-and-budgets

fixes X N P :: prob

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and cross-bound:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
 adversary-initial-state ≤ X

and no-prefix-bound:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 adversary-initial-state ≤ N

and path-bound:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
 adversary-initial-state ≤ P

shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event composition-bad)
 adversary-initial-state ≤

composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + X + N + P) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)
proof (rule
checked-staged-security-with-data-state-composition-bad-bound-from-not-prefix-noncollision-components-and-budget
[OF wf controlled - no-prefix-bound path-bound])
show
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
adversary-initial-state ≤ X
by (rule *order-trans*
[OF
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-witnessed-cross-or-merkle-bad
cross-bound])
qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-cross-no-prefix-transcript-components-and*
fixes *X N T :: prob*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *cross-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
adversary-initial-state ≤ X
and *no-prefix-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
adversary-initial-state ≤ N
and *transcript-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
adversary-initial-state ≤ T

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event composition-bad)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + $X + N + T$) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)

proof (rule

checked-staged-security-with-data-state-composition-bad-bound-from-cross-no-prefix-path-components-and-budgets
 $[OF \text{ wf controlled cross-bound no-prefix-bound}]$)

show

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
adversary-initial-state $\leq T$
by (rule *order-trans*
 $[OF$
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-le-witness-transcript-hit
transcript-bound])

qed

lemma *checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new:*

assumes *pre-bound:*

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
adversary-initial-state $\leq P$

and *new-bound:*

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq Q$

shows

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit

$adversary-initial-state \leq P + Q$
proof –
have *event-le*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
adversary-initial-state $\leq$
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 ($\lambda out.$
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
 out $\vee$
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
 out)
adversary-initial-state
by (rule *wp-event-mono*)
 (rule
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-imp-pre-or-new)
have *union-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 ($\lambda out.$
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
 out $\vee$
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
 out)
adversary-initial-state $\leq$
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
adversary-initial-state +
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state
by (rule *wp-event-union-bound*)
have *sum-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
adversary-initial-state +
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

```

    A)
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
    adversary-initial-state  $\leq P + Q$ 
  by (intro add-mono pre-bound new-bound)
show ?thesis
  by (rule order-trans[OF event-le])
    (rule order-trans[OF union-bound sum-bound])
qed

lemma checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-imp-data-state-pre-hit:
  assumes
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
    (Some (((prefix, prefix-state), data), attacker-state), result),
    final-state))
  shows
    staged-security-with-data-state-transcript-pre-hit
    (Some (((data, attacker-state), result), final-state))
  using assms
  unfolding
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-def
    staged-security-with-data-state-transcript-pre-hit-def
  by simp

lemma checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit:
  assumes pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state  $\leq P$ 
  shows
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
    adversary-initial-state  $\leq P$ 
proof -
  let ?project =
     $\lambda x. ((snd (fst (fst x)), snd (fst x)), snd x)$ 
  have projection:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state =
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ( $\lambda out. case out of$ 
      None  $\Rightarrow$  staged-security-with-data-state-transcript-pre-hit None
    | Some (packed, t)  $\Rightarrow$ 
      staged-security-with-data-state-transcript-pre-hit
      (Some (?project packed, t)))
    adversary-initial-state
proof -

```

```

have
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
       $\gg$  ( $\lambda x.$  return (?project x)))
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state
using
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
  [of A]
by simp
also have ... =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ( $\lambda out.$  case out of
      None  $\Rightarrow$  staged-security-with-data-state-transcript-pre-hit None
    | Some (packed, t)  $\Rightarrow$ 
      staged-security-with-data-state-transcript-pre-hit
      (Some (?project packed, t)))
    adversary-initial-state
by (rule wp-event-bind-return-map)
finally show ?thesis .
qed
have event-le:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
    adversary-initial-state  $\leq$ 
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ( $\lambda out.$  case out of
      None  $\Rightarrow$  staged-security-with-data-state-transcript-pre-hit None
    | Some (packed, t)  $\Rightarrow$ 
      staged-security-with-data-state-transcript-pre-hit
      (Some (?project packed, t)))
    adversary-initial-state
proof (rule wp-event-mono)
fix out
assume hit:
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
  out
show
  (case out of
    None  $\Rightarrow$  staged-security-with-data-state-transcript-pre-hit None
  | Some (packed, t)  $\Rightarrow$ 
    staged-security-with-data-state-transcript-pre-hit
    (Some (?project packed, t)))

```

```

proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state)
    by (cases packed) (auto split: prod.splits)
  show ?thesis
    using
      checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-imp-data-state-pre-hit
      [OF hit[unfolded out-eq]]
    unfolding out-eq by simp
  qed
qed
show ?thesis
  by (rule order-trans[OF event-le])
    (use pre-bound projection in simp)
qed

definition checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
  :: (((((('f × 'f list × 'f list × 'f) ×
    'f protocol-channel) × 'f staged-proof-data) ×
    'f protocol-channel) × unit list) ×
    'f protocol-channel) option ⇒ bool
where
  checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
    out ⇔
    (case out of
      None ⇒ False
    | Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state) ⇒
      staged-security-with-data-state-transcript-pre-hit
        (Some (((data, attacker-state), result), final-state)) ∨
      staged-security-with-data-state-transcript-new-hit
        (Some (((data, attacker-state), result), final-state))))

lemma checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit-bound-from-data-state:
  assumes pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state ≤ P

```

```

and new-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state  $\leq R$ 
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
    adversary-initial-state  $\leq P + R$ 
proof –
  let ?project =
     $\lambda x. ((\text{snd } (\text{fst } (\text{fst } x)), \text{snd } (\text{fst } x)), \text{snd } x)$ 
  let ?E =
     $\lambda out.$ 
      staged-security-with-data-state-transcript-pre-hit out  $\vee$ 
      staged-security-with-data-state-transcript-new-hit out
  let ?projected =
     $\lambda out. \text{case } out \text{ of}$ 
      None  $\Rightarrow ?E \text{ None}$ 
      | Some (packed, t)  $\Rightarrow ?E (\text{Some } (?project \text{ packed}, t))$ 
  have projected-eq:
    ?projected =
      checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit-def
    staged-security-with-data-state-transcript-pre-hit-def
    staged-security-with-data-state-transcript-new-hit-def
  by (rule ext) (auto split: option.splits prod.splits)
  have projection:
    wp-event (checked-staged-security-experiment-with-data-state A) ?E
    adversary-initial-state =
      wp-event
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
        checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
        adversary-initial-state
  proof –
  have
    wp-event (checked-staged-security-experiment-with-data-state A) ?E
    adversary-initial-state =
      wp-event
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
           $\gg (\lambda x. \text{return } (?project x))$ )
        ?E adversary-initial-state
  using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
  by simp
  also have ... =
    wp-event

```



```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state
  by (rule wp-event-bind-return-map)
  finally show ?thesis
    unfolding projected-eq .
qed
have union-bound:
  wp-event (checked-staged-security-experiment-with-data-state A) ?E
    adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state
  by (rule wp-event-union-bound)
have
  wp-event (checked-staged-security-experiment-with-data-state A) ?E
    adversary-initial-state ≤ P + R
  by (rule order-trans[OF union-bound])
    (intro add-mono pre-bound new-bound)
then show ?thesis
  unfolding projection .
qed

lemma checked-staged-security-with-actual-alpha-prefix-some-support-imp-sampled-transcript-hit:
  assumes support:
    Some (((((prefix, prefix-state), data), attacker-state), result),
      final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
  shows
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
      (Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state))
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  by (rule

```

```

    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
have verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
have transcript-split:
    hash-map-output-values ?s ∩ set (staged-proof-transcript data)
    ≠ {} ∨
    hash-map-new-output-hit (set (staged-proof-transcript data))
    ?s final-state
by (rule verify-monad-trace-root-target-preexisting-or-new[OF verifier])
    simp
show ?thesis
unfolding
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit-def
    staged-security-with-data-state-transcript-pre-hit-def
    staged-security-with-data-state-transcript-new-hit-def
using transcript-split by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-bound-from-sampled-transcript
assumes pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state ≤ P
and new-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state ≤ R
shows
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
    adversary-initial-state ≤ P + R
proof –
have event-le:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
    adversary-initial-state ≤
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
    adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:

```

```

    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
          adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
    out
show
  checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
  out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state)
  by (cases packed) (auto split: prod.splits)
show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-some-support-imp-sampled-transcript-hit)
    (rule support[unfolded out-eq])
qed
qed
have transcript-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
    adversary-initial-state ≤ P + R
  by (rule
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit-bound-from-data-state
    [OF pre-bound new-bound])
show ?thesis
  by (rule order-trans[OF event-le transcript-bound])
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-bound-from-sampled-transcript:
  assumes pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit

```

```

    adversary-initial-state  $\leq P$ 
and new-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-transcript-new-hit
      adversary-initial-state  $\leq R$ 
shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
      adversary-initial-state  $\leq P + R$ 
proof –
have event-le:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
      adversary-initial-state  $\leq$ 
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
      adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
    out  $\in$ 
      set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
        adversary-initial-state)
and hit:
    checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
    out
show
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
    out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state
where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state)
by (cases packed) (auto split: prod.splits)

```

```

show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-some-support-imp-sampled-transcript-hit)
    (rule support[unfolded out-eq])
qed
qed
have transcript-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
    adversary-initial-state  $\leq P + R$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit-bound-from-data-state
    [OF pre-bound new-bound])
show ?thesis
  by (rule order-trans[OF event-le transcript-bound])
qed

lemma checked-staged-security-with-data-state-transcript-new-hit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state  $\leq$  staged-concrete-transcript-target-error-bound
proof –
  let ?Builder = checked-staged-transcript-program A
  let ?Tail =
     $\lambda$ data s.
      wp-event verify-monad
        (hash-new-output-hit-event (set (staged-proof-transcript data))
          (verifier-state-from-adversary s (staged-proof-transcript data)))
        (verifier-state-from-adversary s (staged-proof-transcript data))
  have new-none[simp]:  $\neg$  staged-security-with-data-state-transcript-new-hit None
  unfolding staged-security-with-data-state-transcript-new-hit-def by simp
  have return-eq:
    wp (return ((data, s), result))
      ( $\lambda$ out.
        if staged-security-with-data-state-transcript-new-hit out then 1
        else 0)
    final-state =
      (if hash-map-new-output-hit (set (staged-proof-transcript data))
        (verifier-state-from-adversary s (staged-proof-transcript data))
        final-state then 1 else 0)
  for data s result final-state
  unfolding staged-security-with-data-state-transcript-new-hit-def
  by (simp add: wpsimps)
  have verify-post-eq:

```

```

( $\lambda r$ . case  $r$  of
  None  $\Rightarrow$  0
  | Some (result, final-state)  $\Rightarrow$ 
    wp (return ((data, s), result))
      ( $\lambda out$ .
        if staged-security-with-data-state-transcript-new-hit out
        then 1 else 0)
    final-state) =
( $\lambda r$ .
  if hash-new-output-hit-event (set (staged-proof-transcript data))
  (verifier-state-from-adversary s (staged-proof-transcript data))  $r$ 
  then 1 else 0)
for data s
proof (rule ext)
fix  $r$ 
show (case  $r$  of
  None  $\Rightarrow$  0
  | Some (result, final-state)  $\Rightarrow$ 
    wp (return ((data, s), result))
      ( $\lambda out$ .
        if staged-security-with-data-state-transcript-new-hit out
        then 1 else 0)
    final-state) =
  (if hash-new-output-hit-event (set (staged-proof-transcript data))
  (verifier-state-from-adversary s (staged-proof-transcript data))  $r$ 
  then 1 else 0)
proof (cases  $r$ )
case None
then show ?thesis
  unfolding hash-new-output-hit-event-def by simp
next
case (Some packed)
then obtain result final-state where
  packed-eq: packed = (result, final-state)
by (cases packed) simp
show ?thesis
  using return-eq[of data s result final-state]
  unfolding Some packed-eq hash-new-output-hit-event-def
  by simp
qed
qed
have tail-eq:
  wp
  (get  $\gg=$  ( $\lambda s$ .
    put (verifier-state-from-adversary s
      (staged-proof-transcript data))  $\gg=$ 
    ( $\lambda$ -. verify-monad  $\gg=$ 
      ( $\lambda result$ . return ((data, s), result))))))
  ( $\lambda out$ .

```

```

    if staged-security-with-data-state-transcript-new-hit out then 1
    else 0)
  s =
  ?Tail data s
for data s
unfolding wp-event-def
by (simp add: wpsimps verify-post-eq)
have post-eq:
  (λout. case out of
    None ⇒ 0
    | Some (x, y) ⇒
      wp
      (get >>= (λs.
        put (verifier-state-from-adversary s
          (staged-proof-transcript x)) >>=
          (λ-. verify-monad >>=
            (λresult. return ((x, s), result))))))
      (λout.
        if staged-security-with-data-state-transcript-new-hit out
        then 1 else 0)
      y) =
  (λout. case out of
    None ⇒ 0
    | Some (data, s) ⇒ ?Tail data s)
by (rule ext) (simp add: tail-eq split: option.splits prod.splits)
have post-tail-unfold:
  (λout. case out of
    None ⇒ 0
    | Some (data, s) ⇒ ?Tail data s) =
  (λout. case out of
    None ⇒ 0
    | Some (data, s) ⇒
      wp verify-monad
      (λx. if hash-new-output-hit-event
        (set (staged-proof-transcript data))
        (verifier-state-from-adversary s
          (staged-proof-transcript data))
        x then 1 else 0)
      (verifier-state-from-adversary s
        (staged-proof-transcript data)))
by (rule ext) (simp add: wp-event-def split: option.splits prod.splits)
have event-eq:
  wp-event (checked-staged-security-experiment-with-data-state A)
  staged-security-with-data-state-transcript-new-hit
  adversary-initial-state =
  wp ?Builder
  (λout. case out of
    None ⇒ 0
    | Some (data, s) ⇒ ?Tail data s)

```

```

    adversary-initial-state
unfolding checked-staged-security-experiment-with-data-state-def
    wp-event-def
by (simp add: wpsimps post-eq post-tail-unfold
    split: prod.splits option.splits)
have tail-bound:
 $\bigwedge out. out \in \text{set-dist } (\text{execute } ?Builder \text{ adversary-initial-state}) \implies$ 
    (case out of None  $\Rightarrow 0$  | Some (data, s)  $\Rightarrow ?Tail \text{ data } s$ )
     $\leq \text{staged-concrete-transcript-target-error-bound}$ 
proof -
    fix out
    assume support:
        out  $\in \text{set-dist } (\text{execute } ?Builder \text{ adversary-initial-state})$ 
    show (case out of None  $\Rightarrow 0$  | Some (data, s)  $\Rightarrow ?Tail \text{ data } s$ )
         $\leq \text{staged-concrete-transcript-target-error-bound}$ 
    proof (cases out)
    case None
    then show ?thesis
        unfolding staged-concrete-transcript-target-error-bound-def by simp
    next
    case (Some packed)
    then obtain data s where packed-eq: packed = (data, s)
        by (cases packed) simp
    have outcome:
        Some (data, s)  $\in$ 
            set-dist (execute ?Builder adversary-initial-state)
        using support Some packed-eq by simp
    have verifier-bound:
        ?Tail data s  $\leq$ 
            concrete-transcript-target-error (staged-proof-transcript data)
        using wp-verify-monad-hash-new-output-hit-bound
            [of set (staged-proof-transcript data)
            verifier-state-from-adversary s
            (staged-proof-transcript data)]
        unfolding concrete-transcript-target-error-def
            hash-target-budget-value-def
        by (simp add: mult.commute)
    have target-bound:
        concrete-transcript-target-error (staged-proof-transcript data)
             $\leq \text{staged-concrete-transcript-target-error-bound}$ 
    proof -
        have len:
            length (staged-proof-transcript data)  $\leq$ 
                staged-proof-transcript-length-bound
            by (rule
                checked-staged-transcript-program-staged-proof-transcript-length-bound
                [OF wf controlled outcome])
        show ?thesis
            unfolding staged-concrete-transcript-target-error-bound-def

```



```

      by (rule concrete-transcript-target-error-bound-from-length[OF len])
    qed
  show ?thesis
    using Some packed-eq order-trans[OF verifier-bound target-bound]
    by simp
  qed
  qed
  show ?thesis
    unfolding event-eq
    by (rule wp-le-const-on-support[OF tail-bound])
  qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-cross-no-prefix-transcript-split-componen
fixes X N P Q :: prob
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and cross-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
  adversary-initial-state  $\leq$  X
and no-prefix-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
  adversary-initial-state  $\leq$  N
and pre-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
  adversary-initial-state  $\leq$  P
and new-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
  adversary-initial-state  $\leq$  Q
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event composition-bad)
  adversary-initial-state  $\leq$ 
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +

```

(hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + X + N + (P + Q)) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)

proof (rule
 checked-staged-security-with-data-state-composition-bad-bound-from-cross-no-prefix-transcript-components-and-bud
 [OF wf controlled cross-bound no-prefix-bound])

show
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
 adversary-initial-state $\leq P + Q$

by (rule
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new
 [OF pre-bound new-bound])

qed

lemma checked-staged-security-with-data-state-composition-bad-bound-from-cross-no-prefix-data-pre-and-witness-new
 fixes X N P Q :: prob
 assumes wf: staged-budget-wellformed budgets
 and controlled: staged-adversary-controlled budgets A
 and cross-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
 adversary-initial-state $\leq X$
 and no-prefix-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 adversary-initial-state $\leq N$
 and data-pre-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-transcript-pre-hit
 adversary-initial-state $\leq P$
 and new-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
 adversary-initial-state $\leq Q$

shows
 wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event composition-bad)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + X + N + (P + Q)) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)

proof (rule
checked-staged-security-with-data-state-composition-bad-bound-from-cross-no-prefix-transcript-split-components-and
[OF wf controlled cross-bound no-prefix-bound - new-bound])

show
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
adversary-initial-state $\leq P$

by (rule
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
[OF data-pre-bound])

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-cross-and-transcript-components-and-bu*
fixes *X T :: prob*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *cross-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
adversary-initial-state $\leq X$
and *transcript-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
adversary-initial-state $\leq T$

shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event composition-bad)
adversary-initial-state $\leq$
composition-error-bound +

staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
 (*hash-collision-budget-value 0*
 (*staged-alpha-search-queries budgets 0*) + $X + T + T$) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)

proof (rule
checked-staged-security-with-data-state-composition-bad-bound-from-cross-no-prefix-transcript-components-and-bud
 [*OF wf controlled cross-bound - transcript-bound*])

show
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
adversary-initial-state $\leq T$
by (rule *order-trans*
 [*OF*
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-le-witness-transcript-hit
transcript-bound])

qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-cross-data-pre-and-witness-new-compone*
fixes $X P Q :: prob$
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *cross-bound:*
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
adversary-initial-state $\leq X$
and *data-pre-bound:*
wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$
and *new-bound:*
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq Q$
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event composition-bad*)

```

    adversary-initial-state ≤
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    (hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) + X + (P + Q) + (P + Q)) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget)
proof (rule
    checked-staged-security-with-data-state-composition-bad-bound-from-cross-and-transcript-components-and-budgets
    [OF wf controlled cross-bound])
show
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
    adversary-initial-state ≤ P + Q
proof (rule
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new
    [OF - new-bound])
show
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
    adversary-initial-state ≤ P
by (rule
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
    [OF data-pre-bound])
qed
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Query-Current-Alignment
imports Staged-Security-Experiment-Composition-Query-Current
begin

```

Query-prefix alignment facts for the current-final-state composition-query route. This theory is intentionally narrow so that the already large current-query projection theory does not grow past the project size threshold.

context *soundness*
begin

lemma *checked-staged-security-experiment-with-query-prefix-data-state-outcomeI*:

assumes *prefix-receive*:

Some (((*prefix*, *prefix-state*), *raw*), *raw-state*) ∈
set-dist
(*execute* (*checked-staged-query-prefix-receive-with-state* *A i*)
adversary-initial-state)

and *after*:

Some (*data*, *attacker-state*) ∈
set-dist
(*execute*
(*checked-staged-after-query-prefix-receive* *A i*
((*prefix*, *prefix-state*), *raw*))
raw-state)

and *verifier*:

Some (*result*, *final-state*) ∈
set-dist
(*execute* *verify-monad*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*))))

shows

Some ((((((*prefix*, *prefix-state*), *raw*), *raw-state*),
((*data*, *attacker-state*), *result*)), *final-state*) ∈
set-dist
(*execute*
(*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
adversary-initial-state)

proof –

have *with-verifier*:

Some (((*data*, *attacker-state*), *result*), *final-state*) ∈
set-dist
(*execute*
(*checked-staged-after-query-prefix-receive-with-verifier* *A i*
((*prefix*, *prefix-state*), *raw*))
raw-state)

unfolding *checked-staged-after-query-prefix-receive-with-verifier-def*

apply (*rule-tac* *x=**data* **and** *t=**attacker-state* **in** *set-dist-bindI*)

apply (*rule* *after*)

apply (*rule-tac* *x=**attacker-state* **and** *t=**attacker-state* **in** *set-dist-bindI*)

apply *simp*

apply (*rule-tac* *x=*() **and**

*t=**verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*) **in** *set-dist-bindI*)

apply *simp*

apply (*rule-tac* *x=**result* **and** *t=**final-state* **in** *set-dist-bindI*)

apply (*rule* *verifier*)

apply *simp*

```

done
show ?thesis
  unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
  apply (rule set-dist-bindI[OF prefix-receive])
  apply (rule-tac x=raw-state and t=raw-state in set-dist-bindI)
  apply simp
  apply (rule-tac x=((data, attacker-state), result) and
    t=final-state in set-dist-bindI)
  apply (rule with-verifier)
  apply simp
done
qed

```

lemma *checked-staged-after-query-prefix-receive-alphas:*

assumes *after:*

Some (data, attacker-state) ∈
set-dist
(execute
(checked-staged-after-query-prefix-receive A i
((prefix, prefix-state), raw))
raw-state)

shows *staged-alphas data = sqp-alphas prefix*

using *after*

unfolding *checked-staged-after-query-prefix-receive-def Let-def*

by (*auto elim!: set-dist-bindE split: prod.splits*)

lemma *checked-staged-security-with-actual-alpha-prefix-support-query-prefix-data-stateE:*

assumes *i-bound: i < rounds*

and *support:*

Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
adversary-initial-state)

obtains *prefix prefix-state raw raw-state where*

Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)

proof –

let *?s =*

verifier-state-from-adversary attacker-state
(staged-proof-transcript data)

have *data-support:*

Some (((data, attacker-state), result), final-state) ∈

```

    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  obtain builder verifier where builder:
    Some (data, attacker-state) ∈
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈
      set-dist (execute verify-monad ?s)
  by blast
  from checked-staged-transcript-program-query-prefix-receive-cont-support
    [OF i-bound builder]
  obtain prefix prefix-state raw raw-state where prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  and after:
    Some (data, attacker-state) ∈
      set-dist
      (execute
        (checked-staged-after-query-prefix-receive A i
          ((prefix, prefix-state), raw))
        raw-state)
  by blast
  have qprefix-support:
    Some ((((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state) ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
  by (rule checked-staged-security-experiment-with-query-prefix-data-state-outcomeI
    [OF prefix-receive after verifier])
  show ?thesis
  by (rule that[OF qprefix-support])
qed

lemma checked-staged-security-with-actual-alpha-prefix-support-query-prefix-data-state-alphasE:
  assumes i-bound: i < rounds
  and support:
    Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),

```



```

    result), final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
obtains prefix prefix-state raw raw-state where
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
  sqp-alphas prefix = staged-alphas data
proof —
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
obtain builder verifier where builder:
  Some (data, attacker-state) ∈
  set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈
    set-dist (execute verify-monad ?s)
  by blast
from checked-staged-transcript-program-query-prefix-receive-cont-support
  [OF i-bound builder]
obtain prefix prefix-state raw raw-state where prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state) ∈
  set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
  and after:
    Some (data, attacker-state) ∈
    set-dist
      (execute
        (checked-staged-after-query-prefix-receive A i

```

```

      ((prefix, prefix-state), raw))
    raw-state)
  by blast
have qprefix-support:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
  by (rule checked-staged-security-experiment-with-query-prefix-data-state-outcomeI
    [OF prefix-receive after verifier])
have alphas-eq: sqp-alphas prefix = staged-alphas data
  using checked-staged-after-query-prefix-receive-alphas[OF after]
  by simp
show ?thesis
  by (rule that[OF qprefix-support alphas-eq])
qed

lemma checked-staged-security-with-actual-alpha-prefix-candidate-pair-success-query-prefix-hitE:
assumes i-bound: i < rounds
and support:
  Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
and success:
   $\bigwedge$  prefix prefix-state raw raw-state.
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state
        A i)
      adversary-initial-state)  $\implies$ 
  sqp-alphas prefix = staged-alphas data  $\implies$ 
  index (to-nat raw) ∈
  query-sampling-success-space trace-table composition-table
  (staged-alphas data)
obtains prefix prefix-state raw raw-state where
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)

```

checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

proof –

from *checked-staged-security-with-actual-alpha-prefix-support-query-prefix-data-state-alphasE*
[OF i-bound support]

obtain *prefix prefix-state raw raw-state* **where** *qprefix-support*:
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)

and *alphas-eq: sqp-alphas prefix = staged-alphas data*
by *blast*

have *hit*:
checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

using *success[OF qprefix-support alphas-eq] alphas-eq*

unfolding
checked-staged-security-with-query-prefix-dynamic-index-hit-def
checked-staged-query-prefix-dynamic-index-hit-def
staged-query-prefix-candidate-pair-query-target-from-prefix-def

by *simp*

show *?thesis*
by *(rule that[OF qprefix-support hit])*

qed

lemma *checked-staged-actual-alpha-prefix-current-query-partial-opening-hit-at-imp-query-prefix-current-on-support*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *support*:
Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
adversary-initial-state)

and *hit*:
staged-security-with-data-state-current-query-partial-opening-hit-at i
(Some (((data, attacker-state), result), final-state))

obtains *prefix prefix-state raw raw-state* **where**

```

    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
    i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have data-support:
    Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
obtain builder verifier where builder:
    Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
have decomp:
    checked-staged-transcript-program A =
    checked-staged-query-prefix-receive-with-state A i ≫=
    checked-staged-after-query-prefix-receive A i
  by (rule checked-staged-transcript-program-query-prefix-receive-decomp
    [OF i-bound])
from builder[unfolded decomp]
obtain prefix-out raw-state where prefix-receive:
    Some (prefix-out, raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  and after:
    Some (data, attacker-state) ∈
    set-dist
      (execute

```

```

      (checked-staged-after-query-prefix-receive A i prefix-out)
      raw-state)
  by (elim set-dist-bindE)
obtain prefix prefix-state raw where prefix-out-eq:
  prefix-out = ((prefix, prefix-state), raw)
  by (cases prefix-out) auto
have qprefix-support:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    adversary-initial-state)
  by (rule checked-staged-security-experiment-with-query-prefix-data-state-outcomeI)
  (use prefix-receive after verifier prefix-out-eq in simp-all)
have qprefix-hit:
  checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
  i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  by (rule
    staged-security-with-data-state-current-query-partial-opening-hit-at-imp-query-prefix-current-on-support
    [OF wf controlled i-bound qprefix-support hit])
show ?thesis
  by (rule that[OF qprefix-support qprefix-hit])
qed

```

definition checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at

```

:: nat ⇒
  ((((((f × f list × f list × f) ×
    f protocol-channel) × f staged-proof-data) ×
    f protocol-channel) × unit list) ×
    f protocol-channel) option ⇒ bool

```

where

```

checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at
i out ⇔
(case out of
  None ⇒ False
| Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
  result), final-state) ⇒
  staged-security-with-data-state-current-query-partial-opening-hit-at i
  (Some (((data, attacker-state), result), final-state)))

```

definition checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit

```

:: ((((((f × f list × f list × f) ×
  f protocol-channel) × f staged-proof-data) ×
  f protocol-channel) × unit list) ×
  f protocol-channel) option ⇒ bool

```

where

checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
 $out \longleftrightarrow$
 $(\exists i \in \{..<rounds\}.$
checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at
i out)

lemma *checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-imp-query-prefix-current*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *support:*

out $\in$

set-dist

(*execute*

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

adversary-initial-state)

and *hit:*

checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at

i out

shows

$\exists$ *prefix prefix-state raw raw-state data attacker-state result final-state.*

Some (((((*prefix, prefix-state*), *raw*), *raw-state*),

((*data, attacker-state*), *result*)), *final-state*) $\in$

set-dist

(*execute*

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

adversary-initial-state) $\wedge$

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at

i

(*Some* (((((*prefix, prefix-state*), *raw*), *raw-state*),

((*data, attacker-state*), *result*)), *final-state*))

proof (*cases out*)

case *None*

then show *?thesis*

using *hit*

unfolding

checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-def

by *simp*

next

case (*Some packed*)

then obtain *alpha-prefix alpha-prefix-state data attacker-state result*

final-state **where** *out-eq:*

out =

Some (((((*alpha-prefix, alpha-prefix-state*), *data*), *attacker-state*),

result), *final-state*)

by (*cases packed*) (*auto split: prod.splits*)

have *data-hit:*

staged-security-with-data-state-current-query-partial-opening-hit-at i

```

    (Some (((data, attacker-state), result), final-state))
using hit
unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-def
by simp
have support-some:
    Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state) ∈
    set-dist
    (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
    adversary-initial-state)
using support out-eq by simp
from checked-staged-actual-alpha-prefix-current-query-partial-opening-hit-at-imp-query-prefix-current-on-support
    [OF wf controlled i-bound support-some data-hit]
obtain prefix prefix-state raw raw-state where qsupport:
    Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
    set-dist
    (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    adversary-initial-state)
and qhit:
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
    i
    (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
by blast
show ?thesis
    using qsupport qhit by blast
qed

lemma checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-bound-from-query-prefix-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i < \text{rounds}$ 
and query-prefix-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq Q$ 
shows
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq Q$ 
proof –
    let ?project =

```

```

λx. ((snd (fst (fst x)), snd (fst x)), snd x)
have projection:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at i)
  adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-query-partial-opening-hit-at i)
  adversary-initial-state
proof —
  have
    wp-event
      (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-current-query-partial-opening-hit-at i)
    adversary-initial-state =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
        >>= (λx. return (?project x)))
      (staged-security-with-data-state-current-query-partial-opening-hit-at i)
    adversary-initial-state
  using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
  by simp
also have ... =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at i)
  adversary-initial-state
proof —
  have pred-eq:
    (λout. case out of
      None ⇒
        staged-security-with-data-state-current-query-partial-opening-hit-at
          i None
      | Some (x, t) ⇒
        staged-security-with-data-state-current-query-partial-opening-hit-at
          i (Some (?project x, t))) =
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at
      i
  by (rule ext)
  (simp add:
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-def
    staged-security-with-data-state-current-query-partial-opening-hit-at-def
    split: option.splits prod.splits)
show ?thesis
  by (subst wp-event-bind-return-map) (simp add: pred-eq)
qed

```



```

finally show ?thesis
  by simp
qed
have data-bound:
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq$  Q
  by (rule order-trans
    [OF
      checked-staged-security-with-data-state-current-query-partial-opening-hit-at-le-query-prefix-current
      query-prefix-bound])
    (use wf controlled i-bound in simp-all)
show ?thesis
  unfolding projection by (rule data-bound)
qed

lemma checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-bound-from-query-prefix-current
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq$  C i
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
    adversary-initial-state  $\leq$  ( $\sum i < \text{rounds}. C i$ )
proof –
have round-le:
   $\bigwedge i. i < \text{rounds} \implies$ 
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq$  C i
  by (rule
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-bound-from-query-prefix-current
    [OF wf controlled - round-bound])
have
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
    adversary-initial-state  $\leq$ 
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ( $\lambda \text{out}. \exists i \in \{.. < \text{rounds}\}.$ 

```

```

      checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at
      i out)
    adversary-initial-state
  by (rule wp-event-mono)
    (simp add:
      checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-def)
  also have ... ≤ (∑ i<rounds. C i)
    by (rule wp-event-finite-UN-bound)
      (simp-all add: round-le)
  finally show ?thesis .
qed

lemma checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-projection:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
  adversary-initial-state =
  wp-event
  (checked-staged-security-experiment-with-data-state A)
  staged-security-with-data-state-current-query-partial-opening-hit
  adversary-initial-state
proof -
  let ?project =
    λx. ((snd (fst (fst x))), snd (fst x)), snd x)
  have
    wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state =
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
      ≫ (λx. return (?project x)))
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state
  using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
  by simp
  also have ... =
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
    adversary-initial-state
proof -
  have pred-eq:
    (λout. case out of
      None ⇒
        staged-security-with-data-state-current-query-partial-opening-hit
        None

```

```

| Some (x, t) ⇒
  staged-security-with-data-state-current-query-partial-opening-hit
  (Some (?project x, t))) =
checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
by (rule ext)
  (auto simp:
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-def
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-def
    staged-security-with-data-state-current-query-partial-opening-hit-def
    staged-security-with-data-state-current-query-partial-opening-hit-at-def
    split: option.splits prod.splits)
show ?thesis
  by (subst wp-event-bind-return-map) (simp add: pred-eq)
qed
finally show ?thesis
  by simp
qed

lemma checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-actual-alpha-current-qu
assumes bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
  adversary-initial-state ≤ Q
shows
  wp-event
  (checked-staged-security-experiment-with-data-state A)
  staged-security-with-data-state-current-query-partial-opening-hit
  adversary-initial-state ≤ Q
using bound
unfolding
  checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-projection
by simp

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-current-query-part
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  out ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
  out
shows
  checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit

```

```

    out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
      checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-def
      checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-def
    by simp
next
  case (Some packed)
  then obtain alpha-prefix alpha-prefix-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)
    by (cases packed) (auto split: prod.splits)
  have support-some:
    Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
    using support out-eq by simp
  have hit-some:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
      (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state))
    using hit out-eq by simp
  have data-hit:
    staged-security-with-data-state-current-query-partial-opening-hit
      (Some (((data, attacker-state), result), final-state))
    by (rule
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-data-state-current-par
      [OF wf controlled support-some hit-some])
  then obtain i where i-bound: i < rounds
    and round-hit:
      staged-security-with-data-state-current-query-partial-opening-hit-at i
        (Some (((data, attacker-state), result), final-state))
    unfolding staged-security-with-data-state-current-query-partial-opening-hit-def
    by blast
  have actual-round:
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at
      i

```

```

    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  using round-hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-def
  by simp
show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-def
  using i-bound actual-round by blast
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-actual-alpha-prefix-composition-candidate-drift-query-hit
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and current-query-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
      adversary-initial-state ≤ Q
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
      adversary-initial-state ≤ Q
  by (rule order-trans[
    OF - current-query-bound])
    (rule wp-event-mono-on-support,
    rule checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-current-query-partial-opening-hit-at-aligned-transcript-consistent
    [OF wf controlled])

lemma checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-aligned-transcript-consistent
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
  and partial:
    accepted-with-partial-initial-openings-aligned-transcript-consistent
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings

```

```

and no-bad:
   $\neg$  partial-merkle-inconsistency-bad
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state))
and hit:
  checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at
    i
    (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state)
shows
  trace-fri-bad-with-partial-candidates
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state))  $\vee$ 
  composition-fri-bad-with-partial-candidates
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state))  $\vee$ 
  query-bad-with-partial-candidates
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state))  $\vee$ 
  ( $\exists$  trace-table composition-table trace-openings-i
    composition-openings-i.
    partial-trace-table-candidate trace-table
      (trace-openings[i := trace-openings-i])  $\wedge$ 
    partial-composition-table-candidate composition-table
      (composition-openings[i := composition-openings-i])  $\wedge$ 
    trace-table-low-degree trace-table  $\wedge$ 
    composition-table-low-degree maxDegree composition-table  $\wedge$ 
    all-queries-consistent trace-table composition-table as)
proof –
  have data-support:
    Some (((data, attacker-state), result), final-state)  $\in$ 
      set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support])
  have data-hit:
    staged-security-with-data-state-current-query-partial-opening-hit-at i
      (Some (((data, attacker-state), result), final-state))
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-def
  by simp

```

show *?thesis*
by (rule
staged-security-with-data-state-current-query-partial-opening-hit-at-aligned-transcript-classification-on-support
[OF wf controlled data-support partial no-bad data-hit])
qed

lemma *checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-aligned-transcript-classification-on-support*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *support:*
Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
adversary-initial-state)
and *partial:*
accepted-with-partial-initial-openings-aligned-transcript-consistent
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final query-idxs trace-openings
composition-openings
and *no-bad:*
 $\neg$ *partial-merkle-inconsistency-bad*
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(Some (result, final-state))
and *hit:*
checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))
shows
trace-fri-bad-with-partial-candidates
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(Some (result, final-state)) ∨
composition-fri-bad-with-partial-candidates
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(Some (result, final-state)) ∨
query-bad-with-partial-candidates
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(Some (result, final-state)) ∨
 $(\exists i$ *trace-table composition-table trace-openings-i*
composition-openings-i.
 $i < \text{rounds} \wedge$

```

    partial-trace-table-candidate trace-table
      (trace-openings[i := trace-openings-i]) ∧
    partial-composition-table-candidate composition-table
      (composition-openings[i := composition-openings-i]) ∧
    trace-table-low-degree trace-table ∧
    composition-table-low-degree maxDegree composition-table ∧
    all-queries-consistent trace-table composition-table as)
proof –
  from hit obtain i where i-bound: i < rounds
  and round-hit:
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at
      i
      (Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state))
  unfolding
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-def
  by blast
have classified:
  trace-fri-bad-with-partial-candidates
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state)) ∨
  composition-fri-bad-with-partial-candidates
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state)) ∨
  query-bad-with-partial-candidates
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state)) ∨
  (∃ trace-table composition-table trace-openings-i
    composition-openings-i.
    partial-trace-table-candidate trace-table
      (trace-openings[i := trace-openings-i]) ∧
    partial-composition-table-candidate composition-table
      (composition-openings[i := composition-openings-i]) ∧
    trace-table-low-degree trace-table ∧
    composition-table-low-degree maxDegree composition-table ∧
    all-queries-consistent trace-table composition-table as)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-at-aligned-transcript-classification
    [OF wf controlled support partial no-bad round-hit])
  then show ?thesis
  using i-bound by blast
qed

lemma checked-staged-actual-alpha-prefix-query-prefix-receive-alignment-from-final-lookup:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A

```



```

and i-bound:  $i < \text{rounds}$ 
and support:
  Some ((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state)  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
and final-lookup:
  fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
obtains prefix prefix-state raw-state where
  Some (((prefix, prefix-state), raw), raw-state)  $\in$ 
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
  sqp-alphas prefix = staged-alphas data
  sqp-query-chunks prefix = take i (staged-query-chunks data)
  PQueryCounter prefix-state = i
  PState prefix-state =
    state-after-query-chunks
    (staged-query-start-hash data) (staged-query-chunks data) i
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have data-support:
    Some (((data, attacker-state), result), final-state)  $\in$ 
      set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
    by (rule
      checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
      [OF support])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  obtain builder verifier where builder:
    Some (data, attacker-state)  $\in$ 
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
    and verifier:
      Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)
  by blast

```

```

show ?thesis
proof (rule checked-staged-transcript-program-query-prefix-receive-support
  [OF wf controlled i-bound builder])
fix prefix prefix-state raw' raw-state
assume prefix-receive:
  Some (((prefix, prefix-state), raw'), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
and alphas-eq: sqp-alphas prefix = staged-alphas data
and chunks-eq:
  sqp-query-chunks prefix = take i (staged-query-chunks data)
and counter-eq: PQueryCounter prefix-state = i
and state-eq:
  PState prefix-state =
    state-after-query-chunks
    (staged-query-start-hash data) (staged-query-chunks data) i
and raw-ext: raw-state ≤ attacker-state
and raw-lookup:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw'
have raw-ext-s: raw-state ≤ ?s
  by (rule hash-extends-verifier-state-from-adversary-right[OF raw-ext])
have s-ext-final: ?s ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
have raw-ext-final: raw-state ≤ final-state
  by (rule hash-ext-trans[OF raw-ext-s s-ext-final])
have final-lookup-raw':
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw'
  by (rule hash-extension-lookup[OF raw-lookup raw-ext-final])
have raw-eq: raw' = raw
  using final-lookup final-lookup-raw' by simp
have prefix-receive-raw:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
  using prefix-receive raw-eq by simp
show ?thesis
  by (rule that[OF prefix-receive-raw alphas-eq chunks-eq counter-eq
    state-eq])
qed

```

qed

end

end

theory *Soundness-Conceptual-Table*
 imports *Soundness-Partial-Merkle*
begin

Conceptual Merkle tables.

This layer deliberately does not reconstruct a complete *created-tree* witness from sampled openings. Instead it records the value that is uniquely authenticated for a root, length, and index in a hash-map state. Values at unauthenticated indices are arbitrary Hilbert-choice defaults; later proofs must only use them after showing agreement with authenticated openings or charging disagreement to a Merkle/hash side event.

The intended prefix-fixed use is to build these tables from the query-prefix oracle state, before the query-index challenge is sampled. Later openings checked by the verifier are handled in a downstream layer via pullback or controlled collision/inconsistency events.

context *soundness*
begin

definition *authenticated-value-at*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $'f \Rightarrow \text{nat} \Rightarrow \text{nat} \Rightarrow 'f \Rightarrow \text{bool}$

where

$\text{authenticated-value-at } s \ r \ \text{len } i \ v \longleftrightarrow$
 $(\exists \text{ opn} :: 'f \text{ authenticated-opening.}$
 $\text{authenticated-opening-in } s \ \text{opn} \wedge$
 $\text{opening-root } \text{opn} = r \wedge$
 $\text{opening-length } \text{opn} = \text{len} \wedge$
 $\text{opening-index } \text{opn} = i \wedge$
 $\text{opening-value } \text{opn} = v)$

lemma *authenticated-value-atI:*

assumes *authenticated-opening-in* $s \ \text{opn}$
 and $\text{opening-root } \text{opn} = r$
 and $\text{opening-length } \text{opn} = \text{len}$
 and $\text{opening-index } \text{opn} = i$
 and $\text{opening-value } \text{opn} = v$
 shows *authenticated-value-at* $s \ r \ \text{len } i \ v$
 using *assms unfolding authenticated-value-at-def* **by** *blast*

lemma *authenticated-value-atE:*

assumes *authenticated-value-at* $s \ r \ \text{len } i \ v$
 obtains *opn* **where**
 authenticated-opening-in $s \ \text{opn}$

opening-root *opn* = *r*
opening-length *opn* = *len*
opening-index *opn* = *i*
opening-value *opn* = *v*
using *assms* **unfolding** *authenticated-value-at-def* **by** *blast*

lemma *authenticated-value-at-unique-if-clean*:
assumes *clean*: $\neg$ *hash-map-output-collision* *s*
and *v*: *authenticated-value-at* *s* *r* *len* *i* *v*
and *v'*: *authenticated-value-at* *s* *r* *len* *i* *v'*
shows *v* = *v'*
proof –
from *authenticated-value-atE*[*OF v*] **obtain** *opn* **where**
auth: *authenticated-opening-in* *s* *opn*
and *root*: *opening-root* *opn* = *r*
and *len-eq*: *opening-length* *opn* = *len*
and *idx*: *opening-index* *opn* = *i*
and *val*: *opening-value* *opn* = *v*
by *blast*
from *authenticated-value-atE*[*OF v'*] **obtain** *opn'* **where**
auth': *authenticated-opening-in* *s* *opn'*
and *root'*: *opening-root* *opn'* = *r*
and *len-eq'*: *opening-length* *opn'* = *len*
and *idx'*: *opening-index* *opn'* = *i*
and *val'*: *opening-value* *opn'* = *v'*
by *blast*
show *?thesis*
proof (*rule ccontr*)
assume *v* $\neq$ *v'*
then have *diff*: *opening-value* *opn* $\neq$ *opening-value* *opn'*
using *val* *val'* **by** *simp*
have *collision*: *hash-map-output-collision* *s*
by (*rule inconsistent-authenticated-openings-imp-hash-collision*
[*OF auth auth'*])
(*use root root' len-eq len-eq' idx idx' diff in simp-all*)
then show *False*
using *clean* **by** *contradiction*
qed
qed

lemma *authenticated-value-at-mono*:
assumes *v*: *authenticated-value-at* *s* *r* *len* *i* *x*
and *ext*: *s* $\leq$ *t*
shows *authenticated-value-at* *t* *r* *len* *i* *x*
proof –

from *authenticated-value-atE*[*OF v*] **obtain** *opn* **where**
auth: *authenticated-opening-in* *s* *opn*
and *root*: *opening-root* *opn* = *r*
and *len-eq*: *opening-length* *opn* = *len*

```

    and idx: opening-index opn = i
    and val: opening-value opn = x
    by blast
  have authenticated-opening-in t opn
    by (rule authenticated-opening-in-mono[OF auth ext])
  then show ?thesis
    by (rule authenticated-value-atI[OF - root len-eq idx val])
qed

definition conceptual-opening-value
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$  'f  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  'f
where
  conceptual-opening-value s r len i =
    (THE v. authenticated-value-at s r len i v)

lemma conceptual-opening-value-eq-if-authenticated:
  assumes clean:  $\neg$  hash-map-output-collision s
    and v: authenticated-value-at s r len i x
  shows conceptual-opening-value s r len i = x
  unfolding conceptual-opening-value-def
proof (rule the-equality)
  show authenticated-value-at s r len i x
    by (rule v)
next
  fix y
  assume authenticated-value-at s r len i y
  then show y = x
    using authenticated-value-at-unique-if-clean[OF clean - v] by blast
qed

definition conceptual-table
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$  'f  $\Rightarrow$  nat  $\Rightarrow$  'f list
where
  conceptual-table s r len =
    map (conceptual-opening-value s r len) [0.. $\text{len}$ ]

definition conceptual-opening-value-with-default
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$  'f  $\Rightarrow$  nat  $\Rightarrow$ 
    'f  $\Rightarrow$  nat  $\Rightarrow$  'f
where
  conceptual-opening-value-with-default s r len dflt i =
    (if  $\exists v$ . authenticated-value-at s r len i v
     then conceptual-opening-value s r len i
     else dflt)

definition conceptual-table-with-default
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$  'f  $\Rightarrow$  nat  $\Rightarrow$ 
    'f list  $\Rightarrow$  'f list
where

```

conceptual-table-with-default *s r len default-table* =
 map ($\lambda i.$ *conceptual-opening-value-with-default* *s r len*
 (*default-table* ! *i*) *i*) [0..*len*]

lemma *length-conceptual-table*[*simp*]:
length (*conceptual-table* *s r len*) = *len*
unfolding *conceptual-table-def* **by** *simp*

lemma *length-conceptual-table-with-default*[*simp*]:
length (*conceptual-table-with-default* *s r len default-table*) = *len*
unfolding *conceptual-table-with-default-def* **by** *simp*

lemma *conceptual-table-nth*:
assumes *i* < *len*
shows *conceptual-table* *s r len* ! *i* = *conceptual-opening-value* *s r len* *i*
using *assms* **unfolding** *conceptual-table-def* **by** *simp*

lemma *conceptual-table-with-default-nth*:
assumes *i* < *len*
shows
conceptual-table-with-default *s r len default-table* ! *i* =
conceptual-opening-value-with-default *s r len* (*default-table* ! *i*) *i*
using *assms* **unfolding** *conceptual-table-with-default-def* **by** *simp*

lemma *conceptual-opening-value-with-default-eq-if-authenticated*:
assumes *clean*: $\neg$ *hash-map-output-collision* *s*
and *v*: *authenticated-value-at* *s r len* *i* *x*
shows *conceptual-opening-value-with-default* *s r len* *dft* *i* = *x*
proof –
have *ex*: $\exists v.$ *authenticated-value-at* *s r len* *i* *v*
using *v* **by** *blast*
show ?*thesis*
using *ex* *conceptual-opening-value-eq-if-authenticated*[*OF clean v*]
unfolding *conceptual-opening-value-with-default-def* **by** *simp*
qed

lemma *conceptual-opening-value-with-default-eq-default-if-not-authenticated*:
assumes $\neg (\exists v.$ *authenticated-value-at* *s r len* *i* *v*)
shows *conceptual-opening-value-with-default* *s r len* *dft* *i* = *dft*
using *assms* **unfolding** *conceptual-opening-value-with-default-def*
by (*auto split: if-splits*)

lemma *conceptual-table-with-default-eq-if-agrees-with-authenticated-values*:
assumes *clean*: $\neg$ *hash-map-output-collision* *s*
and *len-eq*: *length* *default-table* = *len*
and *agrees*:
 $\bigwedge i v. i < len \implies \text{authenticated-value-at } s \text{ } r \text{ } len \text{ } i \text{ } v \implies$
 $\text{default-table } ! \text{ } i = v$
shows *conceptual-table-with-default* *s r len* *default-table* = *default-table*

```

proof (rule nth-equalityI)
  show length (conceptual-table-with-default s r len default-table) =
    length default-table
  using len-eq by simp
next
  fix i
  assume i-bound:
    i < length (conceptual-table-with-default s r len default-table)
  then have i-len: i < len
    by simp
  show conceptual-table-with-default s r len default-table ! i =
    default-table ! i
  proof (cases  $\exists v. \text{authenticated-value-at } s \ r \ \text{len } i \ v$ )
    case True
      then obtain v where v: authenticated-value-at s r len i v
      by blast
      have
        conceptual-table-with-default s r len default-table ! i =
          conceptual-opening-value-with-default s r len (default-table ! i) i
        by (rule conceptual-table-with-default-nth[OF i-len])
      also have ... = v
        by (rule conceptual-opening-value-with-default-eq-if-authenticated
          [OF clean v])
      also have ... = default-table ! i
        using agrees[OF i-len v] by simp
      finally show ?thesis .
    case False
      have
        conceptual-table-with-default s r len default-table ! i =
          conceptual-opening-value-with-default s r len (default-table ! i) i
        by (rule conceptual-table-with-default-nth[OF i-len])
      also have ... = default-table ! i
        by (rule
          conceptual-opening-value-with-default-eq-default-if-not-authenticated
          [OF False])
      finally show ?thesis .
  qed
qed

lemma conceptual-table-agrees-with-authenticated-opening:
  assumes clean:  $\neg \text{hash-map-output-collision } s$ 
  and auth: authenticated-opening-in s opn
  and root: opening-root opn = r
  and len-eq: opening-length opn = len
  and idx-bound: opening-index opn < len
  shows
    conceptual-table s r len ! opening-index opn = opening-value opn
proof –

```

```

have v:
  authenticated-value-at s r len (opening-index opn)
    (opening-value opn)
  by (rule authenticated-value-atI[OF auth root len-eq refl refl])
have conceptual-table s r len ! opening-index opn =
  conceptual-opening-value s r len (opening-index opn)
  by (rule conceptual-table-nth[OF idx-bound])
also have ... = opening-value opn
  by (rule conceptual-opening-value-eq-if-authenticated[OF clean v])
finally show ?thesis .
qed

lemma conceptual-table-agrees-with-partial-authenticated-table:
  assumes clean:  $\neg$  hash-map-output-collision s
    and table: partial-authenticated-table r len openings s
    and opn-in:  $\text{opn} \in \text{set openings}$ 
  shows
    conceptual-table s r len ! opening-index opn = opening-value opn
proof -
  have auth: authenticated-opening-in s opn
    using table opn-in unfolding partial-authenticated-table-def by blast
  have root: opening-root opn = r
    using table opn-in unfolding partial-authenticated-table-def by blast
  have len-eq: opening-length opn = len
    using table opn-in unfolding partial-authenticated-table-def by blast
  have idx-bound: opening-index opn < len
    using auth len-eq unfolding authenticated-opening-in-def by simp
  show ?thesis
    by (rule conceptual-table-agrees-with-authenticated-opening
        [OF clean auth root len-eq idx-bound])
qed

end

end

```

```

theory Soundness-Conceptual-Query-Prefix
  imports Soundness-Conceptual-Table
begin

```

Prefix-fixed conceptual tables.

These tables are built from a staged query-prefix state, i.e. after the transcript prefix and earlier query rounds have been processed, but before the next query-index challenge is sampled. This is the state selection needed for later dynamic query-target bounds.

```

context soundness
begin

```

```

definition query-prefix-trace-conceptual-table

```


$:: 'f \text{ staged-query-prefix-data} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list}$

where

$\text{query-prefix-trace-conceptual-table prefix } s =$
 $\text{conceptual-table } s \text{ (sqp-trace-root prefix) (scale * clength)}$

definition *query-prefix-composition-conceptual-table*

$:: 'f \text{ staged-query-prefix-data} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list}$

where

$\text{query-prefix-composition-conceptual-table prefix } s =$
 $\text{conceptual-table } s \text{ (hd (sqp-composition-fri-roots prefix))}$
 $(\text{scale} * \text{clength})$

definition *query-prefix-trace-conceptual-table-with-default*

$:: 'f \text{ staged-query-prefix-data} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list}$

where

$\text{query-prefix-trace-conceptual-table-with-default prefix } s \text{ default-table} =$
 $\text{conceptual-table-with-default } s \text{ (sqp-trace-root prefix)}$
 $(\text{scale} * \text{clength}) \text{ default-table}$

definition *query-prefix-composition-conceptual-table-with-default*

$:: 'f \text{ staged-query-prefix-data} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list}$

where

$\text{query-prefix-composition-conceptual-table-with-default prefix } s$
 $\text{default-table} =$
 $\text{conceptual-table-with-default } s \text{ (hd (sqp-composition-fri-roots prefix))}$
 $(\text{scale} * \text{clength}) \text{ default-table}$

lemma *length-query-prefix-trace-conceptual-table[simp]:*

$\text{length (query-prefix-trace-conceptual-table prefix } s) = \text{scale} * \text{clength}$
unfolding *query-prefix-trace-conceptual-table-def* **by** *simp*

lemma *length-query-prefix-composition-conceptual-table[simp]:*

$\text{length (query-prefix-composition-conceptual-table prefix } s) =$
 $\text{scale} * \text{clength}$
unfolding *query-prefix-composition-conceptual-table-def* **by** *simp*

lemma *length-query-prefix-trace-conceptual-table-with-default[simp]:*

length
 $(\text{query-prefix-trace-conceptual-table-with-default prefix } s$
 $\text{default-table}) = \text{scale} * \text{clength}$
unfolding *query-prefix-trace-conceptual-table-with-default-def* **by** *simp*

lemma *length-query-prefix-composition-conceptual-table-with-default[simp]:*

length
 $(\text{query-prefix-composition-conceptual-table-with-default prefix } s$

$\text{default-table}) = \text{scale} * \text{clength}$
unfolding *query-prefix-composition-conceptual-table-with-default-def*
by *simp*

lemma *query-prefix-trace-conceptual-table-agrees-with-authenticated-opening:*

assumes *clean*: $\neg \text{hash-map-output-collision } s$
and *auth*: *authenticated-opening-in* *s* *opn*
and *root*: *opening-root* *opn* = *sqp-trace-root* *prefix*
and *len-eq*: *opening-length* *opn* = *scale* * *clength*
and *idx-bound*: *opening-index* *opn* < *scale* * *clength*

shows

query-prefix-trace-conceptual-table *prefix* *s* ! *opening-index* *opn* =
opening-value *opn*

unfolding *query-prefix-trace-conceptual-table-def*

by (*rule conceptual-table-agrees-with-authenticated-opening*
[*OF clean auth root len-eq idx-bound*])

lemma *query-prefix-composition-conceptual-table-agrees-with-authenticated-opening:*

assumes *clean*: $\neg \text{hash-map-output-collision } s$
and *auth*: *authenticated-opening-in* *s* *opn*
and *roots-nonempty*: *sqp-composition-fri-roots* *prefix* $\neq []$
and *root*: *opening-root* *opn* = *hd* (*sqp-composition-fri-roots* *prefix*)
and *len-eq*: *opening-length* *opn* = *scale* * *clength*
and *idx-bound*: *opening-index* *opn* < *scale* * *clength*

shows

query-prefix-composition-conceptual-table *prefix* *s* !
opening-index *opn* = *opening-value* *opn*

unfolding *query-prefix-composition-conceptual-table-def*

by (*rule conceptual-table-agrees-with-authenticated-opening*
[*OF clean auth root len-eq idx-bound*])

lemma

query-prefix-trace-conceptual-table-with-default-agrees-with-authenticated-opening:

assumes *clean*: $\neg \text{hash-map-output-collision } s$
and *auth*: *authenticated-opening-in* *s* *opn*
and *root*: *opening-root* *opn* = *sqp-trace-root* *prefix*
and *len-eq*: *opening-length* *opn* = *scale* * *clength*
and *idx-bound*: *opening-index* *opn* < *scale* * *clength*

shows

query-prefix-trace-conceptual-table-with-default *prefix* *s*
default-table ! *opening-index* *opn* =
opening-value *opn*

proof –

have *v*:

authenticated-value-at *s* (*sqp-trace-root* *prefix*) (*scale* * *clength*)
(*opening-index* *opn*) (*opening-value* *opn*)

by (*rule authenticated-value-atI*[*OF auth root len-eq refl refl*])

have

query-prefix-trace-conceptual-table-with-default *prefix* *s*

```

    default-table ! opening-index opn =
    conceptual-opening-value-with-default s (sqp-trace-root prefix)
      (scale * clength) (default-table ! opening-index opn)
      (opening-index opn)
  unfolding query-prefix-trace-conceptual-table-with-default-def
  by (rule conceptual-table-with-default-nth[OF idx-bound])
also have ... = opening-value opn
  by (rule conceptual-opening-value-with-default-eq-if-authenticated
      [OF clean v])
finally show ?thesis .
qed

lemma
  query-prefix-composition-conceptual-table-with-default-agrees-with-authenticated-opening:
  assumes clean:  $\neg$  hash-map-output-collision s
    and auth: authenticated-opening-in s opn
    and root: opening-root opn = hd (sqp-composition-fri-roots prefix)
    and len-eq: opening-length opn = scale * clength
    and idx-bound: opening-index opn < scale * clength
  shows
    query-prefix-composition-conceptual-table-with-default prefix s
      default-table ! opening-index opn =
      opening-value opn
  proof -
    have v:
      authenticated-value-at s (hd (sqp-composition-fri-roots prefix))
        (scale * clength) (opening-index opn) (opening-value opn)
    by (rule authenticated-value-atI[OF auth root len-eq refl refl])
    have
      query-prefix-composition-conceptual-table-with-default prefix s
        default-table ! opening-index opn =
        conceptual-opening-value-with-default s
          (hd (sqp-composition-fri-roots prefix)) (scale * clength)
          (default-table ! opening-index opn) (opening-index opn)
    unfolding query-prefix-composition-conceptual-table-with-default-def
    by (rule conceptual-table-with-default-nth[OF idx-bound])
    also have ... = opening-value opn
      by (rule conceptual-opening-value-with-default-eq-if-authenticated
          [OF clean v])
    finally show ?thesis .
  qed

lemma query-prefix-trace-conceptual-table-agrees-with-partial-authenticated-table:
  assumes clean:  $\neg$  hash-map-output-collision s
    and table:
      partial-authenticated-table (sqp-trace-root prefix)
        (scale * clength) openings s
    and opn-in: opn  $\in$  set openings
  shows

```

query-prefix-trace-conceptual-table prefix s ! opening-index opn =
opening-value opn
unfolding *query-prefix-trace-conceptual-table-def*
by (*rule conceptual-table-agrees-with-partial-authenticated-table*
[OF clean table opn-in])

lemma

query-prefix-composition-conceptual-table-agrees-with-partial-authenticated-table:
assumes *clean: $\neg$ hash-map-output-collision s*
and *table:*
partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
*(scale * clength) openings s*
and *opn-in: opn $\in$ set openings*
shows
query-prefix-composition-conceptual-table prefix s ! opening-index opn =
opening-value opn
unfolding *query-prefix-composition-conceptual-table-def*
by (*rule conceptual-table-agrees-with-partial-authenticated-table*
[OF clean table opn-in])

lemma

query-prefix-trace-conceptual-table-with-default-agrees-with-partial-authenticated-table:
assumes *clean: $\neg$ hash-map-output-collision s*
and *table:*
partial-authenticated-table (sqp-trace-root prefix)
*(scale * clength) openings s*
and *opn-in: opn $\in$ set openings*
shows
query-prefix-trace-conceptual-table-with-default prefix s
default-table ! opening-index opn =
opening-value opn

proof —

have *auth: authenticated-opening-in s opn*
using *table opn-in* **unfolding** *partial-authenticated-table-def* **by** *blast*
have *root: opening-root opn = sqp-trace-root prefix*
using *table opn-in* **unfolding** *partial-authenticated-table-def* **by** *blast*
have *len-eq: opening-length opn = scale * clength*
using *table opn-in* **unfolding** *partial-authenticated-table-def* **by** *blast*
have *idx-bound: opening-index opn < scale * clength*
using *auth len-eq* **unfolding** *authenticated-opening-in-def* **by** *simp*
show *?thesis*
by (*rule*
query-prefix-trace-conceptual-table-with-default-agrees-with-authenticated-opening
[OF clean auth root len-eq idx-bound])

qed

lemma

query-prefix-composition-conceptual-table-with-default-agrees-with-partial-authenticated-table:
assumes *clean: $\neg$ hash-map-output-collision s*

and *table*:
partial-authenticated-table (*hd* (*sqp-composition-fri-roots prefix*))
*(scale * clength) openings s*
and *opn-in*: *opn* $\in$ *set openings*
shows
query-prefix-composition-conceptual-table-with-default prefix s
default-table ! opening-index opn =
opening-value opn
proof –
have *auth*: *authenticated-opening-in s opn*
using *table opn-in unfolding partial-authenticated-table-def by blast*
have *root*: *opening-root opn = hd (sqp-composition-fri-roots prefix)*
using *table opn-in unfolding partial-authenticated-table-def by blast*
have *len-eq*: *opening-length opn = scale * clength*
using *table opn-in unfolding partial-authenticated-table-def by blast*
have *idx-bound*: *opening-index opn < scale * clength*
using *auth len-eq unfolding authenticated-opening-in-def by simp*
show *?thesis*
by (*rule*
query-prefix-composition-conceptual-table-with-default-agrees-with-authenticated-opening
[OF clean auth root len-eq idx-bound])
qed

lemma *query-prefix-trace-conceptual-table-with-default-eq-default-if-agrees-with-authenticated-values*:

assumes *clean*: $\neg$ *hash-map-output-collision s*
and *len-eq*: *length default-table = scale * clength*
and *agrees*:
 $\bigwedge i v. i < scale * clength \implies$
authenticated-value-at s (sqp-trace-root prefix)
*(scale * clength) i v \implies*
default-table ! i = v

shows

query-prefix-trace-conceptual-table-with-default prefix s
default-table = default-table
unfolding *query-prefix-trace-conceptual-table-with-default-def*
by (*rule conceptual-table-with-default-eq-if-agrees-with-authenticated-values*
[OF clean len-eq agrees])

lemma *query-prefix-composition-conceptual-table-with-default-eq-default-if-agrees-with-authenticated-values*:

assumes *clean*: $\neg$ *hash-map-output-collision s*
and *len-eq*: *length default-table = scale * clength*
and *agrees*:
 $\bigwedge i v. i < scale * clength \implies$
authenticated-value-at s (hd (sqp-composition-fri-roots prefix))
*(scale * clength) i v \implies*
default-table ! i = v

shows

query-prefix-composition-conceptual-table-with-default prefix s
default-table = default-table

```

unfolding query-prefix-composition-conceptual-table-with-default-def
by (rule conceptual-table-with-default-eq-if-agrees-with-authenticated-values
    [OF clean len-eq agrees])

lemma query-prefix-trace-conceptual-table-partial-trace-candidate:
assumes clean:  $\neg$  hash-map-output-collision s
and len: length trace-openings = rounds
and tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) (trace-openings ! i) s
shows
  partial-trace-table-candidate
    (query-prefix-trace-conceptual-table prefix s) trace-openings
unfolding partial-trace-table-candidate-def
proof (intro conjI allI impI)
show length (query-prefix-trace-conceptual-table prefix s) =
  scale * clength
by simp
next
show length trace-openings = rounds
by (rule len)
next
fix i
assume i-bound:  $i < \text{rounds}$ 
show
  table-agrees-with-authenticated-openings
    (query-prefix-trace-conceptual-table prefix s) (scale * clength)
      (trace-openings ! i)
unfolding table-agrees-with-authenticated-openings-def
proof (intro conjI ballI)
show length (query-prefix-trace-conceptual-table prefix s) =
  scale * clength
by simp
next
fix opn
assume opn-in:  $\text{opn} \in \text{set } (\text{trace-openings} ! i)$ 
have table: partial-authenticated-table (sqp-trace-root prefix)
  (scale * clength) (trace-openings ! i) s
by (rule tables[OF i-bound])
show opening-index opn < scale * clength
using table opn-in
unfolding partial-authenticated-table-def authenticated-opening-in-def
by auto
next
fix opn
assume opn-in:  $\text{opn} \in \text{set } (\text{trace-openings} ! i)$ 
have table: partial-authenticated-table (sqp-trace-root prefix)
  (scale * clength) (trace-openings ! i) s

```

```

    by (rule tables[OF i-bound])
  show
    query-prefix-trace-conceptual-table prefix s ! opening-index opn =
      opening-value opn
  by (rule
    query-prefix-trace-conceptual-table-agrees-with-partial-authenticated-table
    [OF clean table opn-in])
qed
qed

lemma query-prefix-trace-conceptual-table-with-default-partial-trace-candidate:
  assumes clean:  $\neg$  hash-map-output-collision s
  and len: length trace-openings = rounds
  and tables:
     $\bigwedge i. i < \text{rounds} \implies$ 
      partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) (trace-openings ! i) s
  shows
    partial-trace-table-candidate
      (query-prefix-trace-conceptual-table-with-default prefix s
        default-table) trace-openings
  unfolding partial-trace-table-candidate-def
proof (intro conjI allI impI)
  show
    length
      (query-prefix-trace-conceptual-table-with-default prefix s
        default-table) =
      scale * clength
  by simp
next
  show length trace-openings = rounds
  by (rule len)
next
  fix i
  assume i-bound: i < rounds
  show
    table-agrees-with-authenticated-openings
      (query-prefix-trace-conceptual-table-with-default prefix s
        default-table)
      (scale * clength) (trace-openings ! i)
  unfolding table-agrees-with-authenticated-openings-def
proof (intro conjI ballI)
  show
    length
      (query-prefix-trace-conceptual-table-with-default prefix s
        default-table) =
      scale * clength
  by simp
next

```

```

fix opn
  assume opn-in: opn  $\in$  set (trace-openings ! i)
  have table: partial-authenticated-table (sqp-trace-root prefix)
    (scale * clength) (trace-openings ! i) s
    by (rule tables[OF i-bound])
  show opening-index opn < scale * clength
    using table opn-in
    unfolding partial-authenticated-table-def authenticated-opening-in-def
    by auto
next
  fix opn
  assume opn-in: opn  $\in$  set (trace-openings ! i)
  have table: partial-authenticated-table (sqp-trace-root prefix)
    (scale * clength) (trace-openings ! i) s
    by (rule tables[OF i-bound])
  show
    query-prefix-trace-conceptual-table-with-default prefix s
    default-table ! opening-index opn =
    opening-value opn
    by (rule
      query-prefix-trace-conceptual-table-with-default-agrees-with-partial-authenticated-table
      [OF clean table opn-in])
  qed
qed

lemma query-prefix-composition-conceptual-table-partial-composition-candidate:
  assumes clean:  $\neg$  hash-map-output-collision s
  and len: length composition-openings = rounds
  and tables:
     $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
    (scale * clength) (composition-openings ! i) s
  shows
    partial-composition-table-candidate
    (query-prefix-composition-conceptual-table prefix s)
    composition-openings
  unfolding partial-composition-table-candidate-def
proof (intro conjI allI impI)
  show
    length (query-prefix-composition-conceptual-table prefix s) =
    scale * clength
    by simp
next
  show length composition-openings = rounds
    by (rule len)
next
  fix i
  assume i-bound: i < rounds
  show

```



```

table-agrees-with-authenticated-openings
  (query-prefix-composition-conceptual-table prefix s)
  (scale * clength) (composition-openings ! i)
unfolding table-agrees-with-authenticated-openings-def
proof (intro conjI ballI)
show
  length (query-prefix-composition-conceptual-table prefix s) =
    scale * clength
  by simp
next
fix opn
assume opn-in: opn ∈ set (composition-openings ! i)
have table:
  partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
    (scale * clength) (composition-openings ! i) s
  by (rule tables[OF i-bound])
show opening-index opn < scale * clength
  using table opn-in
  unfolding partial-authenticated-table-def authenticated-opening-in-def
  by auto
next
fix opn
assume opn-in: opn ∈ set (composition-openings ! i)
have table:
  partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
    (scale * clength) (composition-openings ! i) s
  by (rule tables[OF i-bound])
show
  query-prefix-composition-conceptual-table prefix s !
    opening-index opn = opening-value opn
  by (rule
    query-prefix-composition-conceptual-table-agrees-with-partial-authenticated-table
    [OF clean table opn-in])
qed
qed

lemma
query-prefix-composition-conceptual-table-with-default-partial-composition-candidate:
assumes clean:  $\neg$  hash-map-output-collision s
and len: length composition-openings = rounds
and tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
      (scale * clength) (composition-openings ! i) s
shows
  partial-composition-table-candidate
    (query-prefix-composition-conceptual-table-with-default prefix s
      default-table)
    composition-openings

```

```

unfolding partial-composition-table-candidate-def
proof (intro conjI allI impI)
  show
    length
      (query-prefix-composition-conceptual-table-with-default prefix s
        default-table) =
        scale * clength
    by simp
next
  show length composition-openings = rounds
    by (rule len)
next
  fix i
  assume i-bound: i < rounds
  show
    table-agrees-with-authenticated-openings
      (query-prefix-composition-conceptual-table-with-default prefix s
        default-table)
      (scale * clength) (composition-openings ! i)
    unfolding table-agrees-with-authenticated-openings-def
  proof (intro conjI ballI)
    show
      length
        (query-prefix-composition-conceptual-table-with-default prefix s
          default-table) =
          scale * clength
      by simp
    next
      fix opn
      assume opn-in: opn ∈ set (composition-openings ! i)
      have table:
        partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
          (scale * clength) (composition-openings ! i) s
        by (rule tables[OF i-bound])
      show opening-index opn < scale * clength
        using table opn-in
        unfolding partial-authenticated-table-def authenticated-opening-in-def
        by auto
    next
      fix opn
      assume opn-in: opn ∈ set (composition-openings ! i)
      have table:
        partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
          (scale * clength) (composition-openings ! i) s
        by (rule tables[OF i-bound])
      show
        query-prefix-composition-conceptual-table-with-default prefix s
          default-table ! opening-index opn =
          opening-value opn

```

```

    by (rule
      query-prefix-composition-conceptual-table-with-default-agrees-with-partial-authenticated-table
      [OF clean table opn-in])
  qed
qed
end
end

```

```

theory Soundness-Conceptual-Query-Target
imports
  Soundness-Conceptual-Query-Prefix
  Soundness-Aligned-Partial-Query-Prefix
begin

```

Prefix-fixed query targets built from conceptual Merkle tables.

This layer deliberately uses only the query-prefix state. It therefore supports the dynamic query-index probability bound, but it does not claim that openings first authenticated after the query challenge agree with these tables. That later-opening bridge is a separate proof obligation.

```

context soundness
begin

```

```

definition staged-query-prefix-conceptual-query-target
  :: 'f staged-query-prefix-data  $\Rightarrow$  'f protocol-channel  $\Rightarrow$  nat set
where
  staged-query-prefix-conceptual-query-target prefix prefix-state =
    query-sampling-success-space
      (query-prefix-trace-conceptual-table prefix prefix-state)
      (query-prefix-composition-conceptual-table prefix prefix-state)
      (sqp-alphas prefix)

```

```

definition staged-query-prefix-conceptual-default-query-target
  :: 'f list  $\Rightarrow$  'f list  $\Rightarrow$ 
    'f staged-query-prefix-data  $\Rightarrow$  'f protocol-channel  $\Rightarrow$  nat set
where
  staged-query-prefix-conceptual-default-query-target trace-default
    composition-default prefix prefix-state =
    query-sampling-success-space
      (query-prefix-trace-conceptual-table-with-default prefix prefix-state
        trace-default)
      (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-default)
      (sqp-alphas prefix)

```

```

definition staged-query-prefix-conceptual-empty-query-target
  :: 'f staged-query-prefix-data  $\Rightarrow$  'f protocol-channel  $\Rightarrow$  nat set

```

where

staged-query-prefix-conceptual-empty-query-target prefix prefix-state =
query-sampling-success-space
(query-prefix-trace-conceptual-table prefix prefix-state)
*(replicate (scale * clength) (sqp-composition-final prefix))*
(sqp-alphas prefix)

lemma *staged-query-prefix-conceptual-query-target-subset:*

staged-query-prefix-conceptual-query-target prefix prefix-state $\subseteq$
query-sample-space

unfolding *staged-query-prefix-conceptual-query-target-def*
by (rule *query-sampling-success-space-subset*)

lemma *staged-query-prefix-conceptual-default-query-target-subset:*

staged-query-prefix-conceptual-default-query-target trace-default
composition-default prefix prefix-state $\subseteq$
query-sample-space

unfolding *staged-query-prefix-conceptual-default-query-target-def*
by (rule *query-sampling-success-space-subset*)

lemma *staged-query-prefix-conceptual-empty-query-target-subset:*

staged-query-prefix-conceptual-empty-query-target prefix prefix-state
 $\subseteq$ *query-sample-space*

unfolding *staged-query-prefix-conceptual-empty-query-target-def*
by (rule *query-sampling-success-space-subset*)

lemma *staged-query-prefix-conceptual-query-target-fraction-bound:*

nnreal
(card
(staged-query-prefix-conceptual-query-target prefix prefix-state)) /
nnreal (card query-sample-space) ≤ query-error-bound
unfolding *staged-query-prefix-conceptual-query-target-def*
by (rule *query-sampling-success-space-fraction-bound-query-sample-space*)

lemma *staged-query-prefix-conceptual-default-query-target-fraction-bound:*

nnreal
(card
(staged-query-prefix-conceptual-default-query-target trace-default
composition-default prefix prefix-state)) /
nnreal (card query-sample-space) ≤ query-error-bound
unfolding *staged-query-prefix-conceptual-default-query-target-def*
by (rule *query-sampling-success-space-fraction-bound-query-sample-space*)

lemma *staged-query-prefix-conceptual-empty-query-target-fraction-bound:*

nnreal
(card
(staged-query-prefix-conceptual-empty-query-target prefix
prefix-state)) /
nnreal (card query-sample-space) ≤ query-error-bound

unfolding *staged-query-prefix-conceptual-empty-query-target-def*
by (*rule query-sampling-success-space-fraction-bound-query-sample-space*)

lemma *query-prefix-trace-conceptual-table-with-default-eq-if-agrees:*

assumes *clean*: $\neg$ *hash-map-output-collision prefix-state*

and *len*: *length trace-default* = *scale* * *clength*

and *agrees*:

$\bigwedge i v. i < \text{scale} * \text{clength} \implies$
authenticated-value-at prefix-state (sqp-trace-root prefix)
*(scale * clength) i v* $\implies$
trace-default ! i = v

shows

query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default = trace-default

unfolding *query-prefix-trace-conceptual-table-with-default-def*

by (*rule conceptual-table-with-default-eq-if-agrees-with-authenticated-values*
[*OF clean len agrees*])

lemma *query-prefix-composition-conceptual-table-with-default-eq-if-agrees:*

assumes *clean*: $\neg$ *hash-map-output-collision prefix-state*

and *len*: *length composition-default* = *scale* * *clength*

and *agrees*:

$\bigwedge i v. i < \text{scale} * \text{clength} \implies$
authenticated-value-at prefix-state
*(hd (sqp-composition-fri-roots prefix)) (scale * clength) i v* $\implies$
composition-default ! i = v

shows

query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default = composition-default

unfolding *query-prefix-composition-conceptual-table-with-default-def*

by (*rule conceptual-table-with-default-eq-if-agrees-with-authenticated-values*
[*OF clean len agrees*])

lemma

staged-query-prefix-candidate-pair-query-target-from-prefix-subset-conceptual-default-if-agrees:

assumes *clean*: $\neg$ *hash-map-output-collision prefix-state*

and *trace-len*: *length trace-table* = *scale* * *clength*

and *composition-len*: *length composition-table* = *scale* * *clength*

and *trace-agrees*:

$\bigwedge i v. i < \text{scale} * \text{clength} \implies$
authenticated-value-at prefix-state (sqp-trace-root prefix)
*(scale * clength) i v* $\implies$
trace-table ! i = v

and *composition-agrees*:

$\bigwedge i v. i < \text{scale} * \text{clength} \implies$
authenticated-value-at prefix-state
*(hd (sqp-composition-fri-roots prefix)) (scale * clength) i v* $\implies$
composition-table ! i = v

shows

staged-query-prefix-candidate-pair-query-target-from-prefix *trace-table*
composition-table *prefix* *prefix-state* $\subseteq$
staged-query-prefix-conceptual-default-query-target *trace-table*
composition-table *prefix* *prefix-state*
proof –
have *trace-eq*:
query-prefix-trace-conceptual-table-with-default *prefix* *prefix-state*
trace-table = *trace-table*
by (rule *query-prefix-trace-conceptual-table-with-default-eq-if-agrees*
[*OF clean trace-len trace-agrees*])
have *composition-eq*:
query-prefix-composition-conceptual-table-with-default *prefix*
prefix-state *composition-table* = *composition-table*
by (rule
query-prefix-composition-conceptual-table-with-default-eq-if-agrees
[*OF clean composition-len composition-agrees*])
show ?thesis
unfolding *staged-query-prefix-candidate-pair-query-target-from-prefix-def*
staged-query-prefix-conceptual-default-query-target-def
using *trace-eq* *composition-eq* **by** *simp*
qed

definition *query-prefix-trace-default-agrees-with-authenticated-values*
 $:: 'f \text{ list} \Rightarrow 'f \text{ staged-query-prefix-data} \Rightarrow$
 $'f \text{ protocol-channel} \Rightarrow \text{bool}$

where
query-prefix-trace-default-agrees-with-authenticated-values *trace-default*
prefix *prefix-state* $\longleftrightarrow$
 $(\forall i \ v. \ i < \text{scale} * \text{clength} \longrightarrow$
authenticated-value-at *prefix-state* (*sqp-trace-root* *prefix*)
 $(\text{scale} * \text{clength}) \ i \ v \longrightarrow$
 $\text{trace-default} \ ! \ i = v)$

definition *query-prefix-composition-default-agrees-with-authenticated-values*
 $:: 'f \text{ list} \Rightarrow 'f \text{ staged-query-prefix-data} \Rightarrow$
 $'f \text{ protocol-channel} \Rightarrow \text{bool}$

where
query-prefix-composition-default-agrees-with-authenticated-values
composition-default *prefix* *prefix-state* $\longleftrightarrow$
 $(\forall i \ v. \ i < \text{scale} * \text{clength} \longrightarrow$
authenticated-value-at *prefix-state*
 $(\text{hd} (\text{sqp-composition-fri-roots } \text{prefix})) (\text{scale} * \text{clength}) \ i \ v \longrightarrow$
 $\text{composition-default} \ ! \ i = v)$

definition *query-prefix-candidate-defaults-agree-with-authenticated-values*
 $:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ staged-query-prefix-data} \Rightarrow$
 $'f \text{ protocol-channel} \Rightarrow \text{bool}$

where
query-prefix-candidate-defaults-agree-with-authenticated-values

$\text{trace-default } \text{composition-default } \text{prefix } \text{prefix-state} \longleftrightarrow$
 $\text{query-prefix-trace-default-agrees-with-authenticated-values } \text{trace-default}$
 $\text{prefix } \text{prefix-state} \wedge$
 $\text{query-prefix-composition-default-agrees-with-authenticated-values}$
 $\text{composition-default } \text{prefix } \text{prefix-state}$

lemma *query-prefix-trace-default-agrees-with-authenticated-valuesD:*

assumes

$\text{query-prefix-trace-default-agrees-with-authenticated-values } \text{trace-default}$
 $\text{prefix } \text{prefix-state}$

and $i < \text{scale} * \text{clength}$

and

$\text{authenticated-value-at } \text{prefix-state } (\text{sqp-trace-root } \text{prefix})$
 $(\text{scale} * \text{clength}) \ i \ v$

shows $\text{trace-default} \ ! \ i = v$

using *assms*

unfolding *query-prefix-trace-default-agrees-with-authenticated-values-def*

by *blast*

lemma *query-prefix-composition-default-agrees-with-authenticated-valuesD:*

assumes

$\text{query-prefix-composition-default-agrees-with-authenticated-values}$
 $\text{composition-default } \text{prefix } \text{prefix-state}$

and $i < \text{scale} * \text{clength}$

and

$\text{authenticated-value-at } \text{prefix-state}$
 $(\text{hd } (\text{sqp-composition-fri-roots } \text{prefix})) \ (\text{scale} * \text{clength}) \ i \ v$

shows $\text{composition-default} \ ! \ i = v$

using *assms*

unfolding

query-prefix-composition-default-agrees-with-authenticated-values-def

by *blast*

lemma

staged-query-prefix-candidate-pair-query-target-from-prefix-subset-conceptual-default-if-prefix-agrees:

assumes *clean:* $\neg \text{hash-map-output-collision } \text{prefix-state}$

and *trace-len:* $\text{length } \text{trace-table} = \text{scale} * \text{clength}$

and *composition-len:* $\text{length } \text{composition-table} = \text{scale} * \text{clength}$

and *agrees:*

$\text{query-prefix-candidate-defaults-agree-with-authenticated-values}$
 $\text{trace-table } \text{composition-table } \text{prefix } \text{prefix-state}$

shows

$\text{staged-query-prefix-candidate-pair-query-target-from-prefix } \text{trace-table}$
 $\text{composition-table } \text{prefix } \text{prefix-state} \subseteq$
 $\text{staged-query-prefix-conceptual-default-query-target } \text{trace-table}$
 $\text{composition-table } \text{prefix } \text{prefix-state}$

proof (rule

$\text{staged-query-prefix-candidate-pair-query-target-from-prefix-subset-conceptual-default-if-agrees}$
 $[\text{OF } \text{clean } \text{trace-len } \text{composition-len}])$

```

fix  $i\ v$ 
assume  $i\text{-bound}: i < \text{scale} * \text{clength}$ 
and  $\text{auth}$ :
   $\text{authenticated-value-at prefix-state (sqp-trace-root prefix)}$ 
   $(\text{scale} * \text{clength})\ i\ v$ 
have  $\text{trace-agrees}$ :
   $\text{query-prefix-trace-default-agrees-with-authenticated-values trace-table}$ 
   $\text{prefix prefix-state}$ 
using  $\text{agrees}$ 
unfolding
   $\text{query-prefix-candidate-defaults-agree-with-authenticated-values-def}$ 
by  $\text{simp}$ 
show  $\text{trace-table} !\ i = v$ 
by ( $\text{rule}$ 
   $\text{query-prefix-trace-default-agrees-with-authenticated-valuesD}$ 
   $[\text{OF trace-agrees } i\text{-bound } \text{auth}]$ )
next
fix  $i\ v$ 
assume  $i\text{-bound}: i < \text{scale} * \text{clength}$ 
and  $\text{auth}$ :
   $\text{authenticated-value-at prefix-state}$ 
   $(\text{hd (sqp-composition-fri-roots prefix)}) (\text{scale} * \text{clength})\ i\ v$ 
have  $\text{composition-agrees}$ :
   $\text{query-prefix-composition-default-agrees-with-authenticated-values}$ 
   $\text{composition-table prefix prefix-state}$ 
using  $\text{agrees}$ 
unfolding
   $\text{query-prefix-candidate-defaults-agree-with-authenticated-values-def}$ 
by  $\text{simp}$ 
show  $\text{composition-table} !\ i = v$ 
by ( $\text{rule}$ 
   $\text{query-prefix-composition-default-agrees-with-authenticated-valuesD}$ 
   $[\text{OF composition-agrees } i\text{-bound } \text{auth}]$ )
qed

```

definition $\text{query-prefix-candidate-defaults-agree-at-query}$

$:: 'f\ \text{list} \Rightarrow 'f\ \text{list} \Rightarrow \text{nat} \Rightarrow$
 $'f\ \text{staged-query-prefix-data} \Rightarrow 'f\ \text{protocol-channel} \Rightarrow \text{bool}$

where

$\text{query-prefix-candidate-defaults-agree-at-query trace-table}$
 $\text{composition-table idx prefix prefix-state} \longleftrightarrow$
 $(\forall j \in \text{set (powers-scaled idx)}).$
 $\text{query-prefix-trace-conceptual-table-with-default prefix prefix-state}$
 $\text{trace-table} !\ j = \text{trace-table} !\ j) \wedge$
 $\text{query-prefix-composition-conceptual-table-with-default prefix prefix-state}$
 $\text{composition-table} !\ \text{idx} = \text{composition-table} !\ \text{idx}$

lemma $\text{query-prefix-candidate-pair-query-hit-imp-conceptual-default-if-agrees-at-query}$:

assumes hit :


```

  idx ∈
    staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
    composition-table prefix prefix-state
and trace-len: length trace-table = scale * clength
and composition-len: length composition-table = scale * clength
and agrees:
  query-prefix-candidate-defaults-agree-at-query trace-table
  composition-table idx prefix prefix-state
and default-trace-low:
  trace-table-low-degree
  (query-prefix-trace-conceptual-table-with-default prefix prefix-state
   trace-table)
and default-composition-low:
  composition-table-low-degree maxDegree
  (query-prefix-composition-conceptual-table-with-default prefix
   prefix-state composition-table)
and default-not-all:
  ¬ all-queries-consistent
  (query-prefix-trace-conceptual-table-with-default prefix prefix-state
   trace-table)
  (query-prefix-composition-conceptual-table-with-default prefix
   prefix-state composition-table)
  (sqp-alphas prefix)
shows
  idx ∈
    staged-query-prefix-conceptual-default-query-target trace-table
    composition-table prefix prefix-state
proof —
  let ?trace-default =
    query-prefix-trace-conceptual-table-with-default prefix prefix-state
    trace-table
  let ?composition-default =
    query-prefix-composition-conceptual-table-with-default prefix
    prefix-state composition-table
  have trace-len-eq: length trace-table = length ?trace-default
    using trace-len by simp
  have composition-len-eq:
    length composition-table = length ?composition-default
    using composition-len by simp
  have trace-values:
    map ((! trace-table) (powers-scaled idx)) =
      map ((! ?trace-default) (powers-scaled idx))
  proof (rule nth-equalityI)
  show
    length (map ((! trace-table) (powers-scaled idx))) =
      length (map ((! ?trace-default) (powers-scaled idx)))
    by simp
next
  fix k

```

```

assume k-bound:
  k < length (map (!) trace-table) (powers-scaled idx)
then have ps-bound: k < length (powers-scaled idx)
  by simp
have ps-mem: powers-scaled idx ! k ∈ set (powers-scaled idx)
  by (rule nth-mem[OF ps-bound])
have eq:
  ?trace-default ! (powers-scaled idx ! k) =
    trace-table ! (powers-scaled idx ! k)
  using agrees ps-mem
  unfolding query-prefix-candidate-defaults-agree-at-query-def
  by blast
show
  map (!) trace-table (powers-scaled idx) ! k =
    map (!) ?trace-default (powers-scaled idx) ! k
  using ps-bound eq by simp
qed
have composition-value:
  composition-table ! idx = ?composition-default ! idx
  using agrees
  unfolding query-prefix-candidate-defaults-agree-at-query-def
  by simp
have qc-iff:
  query-consistent-at trace-table composition-table (sqp-alphas prefix)
    idx  $\longleftrightarrow$ 
    query-consistent-at ?trace-default ?composition-default
      (sqp-alphas prefix) idx
  by (rule query-consistent-at-cong-opened-values
    [OF trace-len-eq composition-len-eq trace-values composition-value])
have qc:
  query-consistent-at trace-table composition-table (sqp-alphas prefix)
    idx
  using hit
  unfolding staged-query-prefix-candidate-pair-query-target-from-prefix-def
    query-sampling-success-space-def query-agreement-indices-def
  by (cases
    trace-table-low-degree trace-table ∧
    composition-table-low-degree maxDegree composition-table ∧
    ¬ all-queries-consistent trace-table composition-table
      (sqp-alphas prefix))
    auto
have query-idx: idx ∈ query-sample-space
  using hit query-sampling-success-space-subset
  unfolding staged-query-prefix-candidate-pair-query-target-from-prefix-def
  by blast
have default-qc:
  query-consistent-at ?trace-default ?composition-default
    (sqp-alphas prefix) idx
  using qc qc-iff by simp

```

```

show ?thesis
using query-idx default-qc default-trace-low default-composition-low
      default-not-all
unfolding staged-query-prefix-candidate-pair-query-target-from-prefix-def
      staged-query-prefix-conceptual-default-query-target-def
      query-sampling-success-space-def query-agreement-indices-def
by auto
qed

lemma query-prefix-candidate-defaults-agree-at-query-from-prefix-authenticated-openings:
assumes clean:  $\neg$  hash-map-output-collision prefix-state
and i-bound:  $i < \text{rounds}$ 
and trace-prefix:
  partial-authenticated-table (spp-trace-root prefix)
    (scale * clength) trace-openings prefix-state
and composition-prefix:
  partial-authenticated-table (hd (spp-composition-fri-roots prefix))
    (scale * clength) composition-openings prefix-state
and consistent:
  partial-query-openings-consistent trace-openings composition-openings
    (spp-alphas prefix) idx
and trace-candidate:
  partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
and composition-candidate:
  partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
shows
  query-prefix-candidate-defaults-agree-at-query trace-table
    composition-table idx prefix prefix-state
proof –
let ?trace-openingsss = (replicate rounds []) [i := trace-openings]
let ?composition-openingsss =
  (replicate rounds []) [i := composition-openings]
have trace-agrees:
  table-agrees-with-authenticated-openings trace-table
    (scale * clength) trace-openings
proof –
have all:
   $\forall ia < \text{rounds}.$ 
    table-agrees-with-authenticated-openings trace-table
      (scale * clength) (?trace-openingsss ! ia)
using trace-candidate
unfolding partial-trace-table-candidate-def by blast
have at-i:
  table-agrees-with-authenticated-openings trace-table
    (scale * clength) (?trace-openingsss ! i)
using all i-bound by blast
have ?trace-openingsss ! i = trace-openings

```

```

    using i-bound by simp
  then show ?thesis
    using at-i by simp
qed
have composition-agrees:
  table-agrees-with-authenticated-openings composition-table
    (scale * clength) composition-openings
proof -
  have all:
     $\forall ia < rounds.$ 
    table-agrees-with-authenticated-openings composition-table
      (scale * clength) (?composition-openings ! ia)
    using composition-candidate
    unfolding partial-composition-table-candidate-def by blast
  have at-i:
    table-agrees-with-authenticated-openings composition-table
      (scale * clength) (?composition-openings ! i)
    using all i-bound by blast
  have ?composition-openings ! i = composition-openings
    using i-bound by simp
  then show ?thesis
    using at-i by simp
qed
have trace-indices:
  map opening-index trace-openings = powers-scaled idx
  using consistent unfolding partial-query-openings-consistent-def by simp
have composition-indices:
  map opening-index composition-openings =
    [idx, fri-sibling-index (scale * clength) idx]
  using consistent unfolding partial-query-openings-consistent-def by simp
have composition-nonempty: composition-openings  $\neq$  []
  using consistent unfolding partial-query-openings-consistent-def by simp
have trace-point:
   $\bigwedge j. j \in \text{set (powers-scaled idx)} \implies$ 
  query-prefix-trace-conceptual-table-with-default prefix prefix-state
    trace-table ! j =
  trace-table ! j
proof -
  fix j
  assume j-in:  $j \in \text{set (powers-scaled idx)}$ 
  then obtain opn where opn-in:  $opn \in \text{set trace-openings}$ 
    and opn-idx: opening-index opn = j
    using trace-indices
    by (metis (mono-tags, lifting) in-set-conv-nth length-map nth-map)
  have conceptual:
    query-prefix-trace-conceptual-table-with-default prefix prefix-state
      trace-table ! opening-index opn =
    opening-value opn
  by (rule

```

```

      query-prefix-trace-conceptual-table-with-default-agrees-with-partial-authenticated-table
      [OF clean trace-prefix opn-in])
have table:
  trace-table ! opening-index opn = opening-value opn
  using trace-agrees opn-in
  unfolding table-agrees-with-authenticated-openings-def by blast
show
  query-prefix-trace-conceptual-table-with-default prefix prefix-state
  trace-table ! j =
  trace-table ! j
  using conceptual table opn-idx by simp
qed
have composition-point:
  query-prefix-composition-conceptual-table-with-default prefix
  prefix-state composition-table ! idx =
  composition-table ! idx
proof –
  have opn0-in: composition-openings ! 0 ∈ set composition-openings
  using composition-nonempty by simp
  have opn0-idx: opening-index (composition-openings ! 0) = idx
  using composition-indices composition-nonempty
  by (cases composition-openings) simp-all
  have conceptual:
  query-prefix-composition-conceptual-table-with-default prefix
  prefix-state composition-table ! opening-index (composition-openings ! 0) =
  opening-value (composition-openings ! 0)
  by (rule
  query-prefix-composition-conceptual-table-with-default-agrees-with-partial-authenticated-table
  [OF clean composition-prefix opn0-in])
  have table:
  composition-table ! opening-index (composition-openings ! 0) =
  opening-value (composition-openings ! 0)
  using composition-agrees opn0-in
  unfolding table-agrees-with-authenticated-openings-def by blast
  show ?thesis
  using conceptual table opn0-idx by simp
qed
show ?thesis
  unfolding query-prefix-candidate-defaults-agree-at-query-def
  using trace-point composition-point by blast
qed

lemma query-prefix-current-openings-imp-conceptual-default-query-target:
assumes clean: ¬ hash-map-output-collision prefix-state
and i-bound: i < rounds
and trace-prefix:
  partial-authenticated-table (sqp-trace-root prefix)
  (scale * clength) trace-openings prefix-state
and composition-prefix:

```

```

    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
      (scale * clength) composition-openings prefix-state
  and consistent:
    partial-query-openings-consistent trace-openings composition-openings
      (sqp-alphas prefix) idx
  and trace-candidate:
    partial-trace-table-candidate trace-table
      ((replicate rounds []) [i := trace-openings])
  and composition-candidate:
    partial-composition-table-candidate composition-table
      ((replicate rounds []) [i := composition-openings])
  and default-trace-low:
    trace-table-low-degree
      (query-prefix-trace-conceptual-table-with-default prefix prefix-state
        trace-table)
  and default-composition-low:
    composition-table-low-degree maxDegree
      (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-table)
  and default-not-all:
    ¬ all-queries-consistent
      (query-prefix-trace-conceptual-table-with-default prefix prefix-state
        trace-table)
      (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-table)
      (sqp-alphas prefix)
  shows
    idx ∈
      staged-query-prefix-conceptual-default-query-target trace-table
        composition-table prefix prefix-state
  proof -
  let ?trace-openings = (replicate rounds []) [i := trace-openings]
  let ?composition-openings =
    (replicate rounds []) [i := composition-openings]
  let ?trace-default =
    query-prefix-trace-conceptual-table-with-default prefix prefix-state
      trace-table
  let ?composition-default =
    query-prefix-composition-conceptual-table-with-default prefix
      prefix-state composition-table
  have trace-len: length trace-table = scale * clength
    using trace-candidate unfolding partial-trace-table-candidate-def
    by simp
  have composition-len: length composition-table = scale * clength
    using composition-candidate
    unfolding partial-composition-table-candidate-def by simp
  have idx-sample: idx ∈ query-sample-space
    using consistent unfolding partial-query-openings-consistent-def by simp
  have round:

```

```

partial-query-round-consistent ?trace-openingsss ?composition-openingsss
  (sqp-alphas prefix) i idx
using i-bound consistent
unfolding partial-query-round-consistent-def
  partial-query-openings-consistent-def
by simp
have candidate-consistent:
  query-consistent-at trace-table composition-table (sqp-alphas prefix)
    idx
by (rule partial-query-round-consistent-imp-query-consistent-at
  [OF trace-candidate composition-candidate round])
have agrees:
  query-prefix-candidate-defaults-agree-at-query trace-table
    composition-table idx prefix prefix-state
by (rule
  query-prefix-candidate-defaults-agree-at-query-from-prefix-authenticated-openings
  [OF clean i-bound trace-prefix composition-prefix consistent
    trace-candidate composition-candidate])
have trace-len-eq: length trace-table = length ?trace-default
  using trace-len by simp
have composition-len-eq:
  length composition-table = length ?composition-default
  using composition-len by simp
have trace-values-eq:
  map (! trace-table) (powers-scaled idx) =
    map (! ?trace-default) (powers-scaled idx)
proof (rule nth-equalityI)
  show
    length (map (! trace-table) (powers-scaled idx)) =
      length (map (! ?trace-default) (powers-scaled idx))
  by simp
next
fix k
assume k-bound:
  k < length (map (! trace-table) (powers-scaled idx))
then have ps-bound: k < length (powers-scaled idx)
  by simp
have ps-mem: powers-scaled idx ! k ∈ set (powers-scaled idx)
  by (rule nth-mem[OF ps-bound])
have eq:
  ?trace-default ! (powers-scaled idx ! k) =
    trace-table ! (powers-scaled idx ! k)
  using agrees ps-mem
unfolding query-prefix-candidate-defaults-agree-at-query-def
by blast
show
  map (! trace-table) (powers-scaled idx) ! k =
    map (! ?trace-default) (powers-scaled idx) ! k
  using ps-bound eq by simp

```

```

qed
have composition-value-eq:
  composition-table ! idx = ?composition-default ! idx
  using agrees
  unfolding query-prefix-candidate-defaults-agree-at-query-def by simp
have qc-iff:
  query-consistent-at trace-table composition-table (sqp-alphas prefix)
    idx  $\longleftrightarrow$ 
  query-consistent-at ?trace-default ?composition-default
    (sqp-alphas prefix) idx
  by (rule query-consistent-at-cong-opened-values
    [OF trace-len-eq composition-len-eq trace-values-eq
      composition-value-eq])
have default-consistent:
  query-consistent-at ?trace-default ?composition-default
    (sqp-alphas prefix) idx
  using candidate-consistent qc-iff by simp
show ?thesis
  unfolding staged-query-prefix-conceptual-default-query-target-def
    query-sampling-success-space-def query-agreement-indices-def
  using idx-sample default-consistent default-trace-low
    default-composition-low default-not-all by simp
qed

lemma trace-table-agrees-with-prefix-authenticated-values-if-final-merkle-bound:
  assumes ext: prefix-state  $\leq$  final-state
  and clean:  $\neg$  hash-map-output-collision final-state
  and bind:
    merkle-root-binds-table (sqp-trace-root prefix) trace-table
    final-state
  and len: length trace-table = scale * clength
  and i-bound: i < scale * clength
  and auth:
    authenticated-value-at prefix-state (sqp-trace-root prefix)
    (scale * clength) i v
  shows trace-table ! i = v
proof -
  from authenticated-value-atE[OF auth] obtain opn where
    opn-auth-prefix: authenticated-opening-in prefix-state opn
  and root: opening-root opn = sqp-trace-root prefix
  and opn-len: opening-length opn = scale * clength
  and idx: opening-index opn = i
  and val: opening-value opn = v
  by blast
  have opn-auth-final: authenticated-opening-in final-state opn
  by (rule authenticated-opening-in-mono[OF opn-auth-prefix ext])
  obtain n where len-pow: length trace-table =  $2^n$ 
  using len eval-domain-length-power by (metis mult.commute)
  have agrees:

```



```

opening-index opn < length trace-table ∧
  trace-table ! opening-index opn = opening-value opn
by (rule merkle-bound-table-agrees-with-authenticated-opening-if-clean
  [OF bind len-pow opn-auth-final root - clean])
  (use len opn-len in simp)
then show ?thesis
  using idx val by simp
qed

```

lemma *composition-table-agrees-with-prefix-authenticated-values-if-final-merkle-bound:*

```

assumes ext: prefix-state ≤ final-state
and clean: ¬ hash-map-output-collision final-state
and bind:
  merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
    composition-table final-state
and len: length composition-table = scale * clength
and i-bound: i < scale * clength
and auth:
  authenticated-value-at prefix-state
    (hd (sqp-composition-fri-roots prefix)) (scale * clength) i v
shows composition-table ! i = v
proof -
from authenticated-value-atE[OF auth] obtain opn where
  opn-auth-prefix: authenticated-opening-in prefix-state opn
and root:
  opening-root opn = hd (sqp-composition-fri-roots prefix)
and opn-len: opening-length opn = scale * clength
and idx: opening-index opn = i
and val: opening-value opn = v
by blast
have opn-auth-final: authenticated-opening-in final-state opn
  by (rule authenticated-opening-in-mono[OF opn-auth-prefix ext])
obtain n where len-pow: length composition-table = 2 ^ n
  using len eval-domain-length-power by (metis mult.commute)
have agrees:
  opening-index opn < length composition-table ∧
    composition-table ! opening-index opn = opening-value opn
by (rule merkle-bound-table-agrees-with-authenticated-opening-if-clean
  [OF bind len-pow opn-auth-final root - clean])
  (use len opn-len in simp)
then show ?thesis
  using idx val by simp
qed

```

lemma

staged-query-prefix-candidate-pair-query-target-from-prefix-subset-conceptual-default-if-final-merkle-bound:

```

assumes prefix-clean: ¬ hash-map-output-collision prefix-state
and final-clean: ¬ hash-map-output-collision final-state
and ext: prefix-state ≤ final-state

```

and *trace-bind*:
 merkle-root-binds-table (*sqp-trace-root prefix*) *trace-table*
 final-state
and *composition-bind*:
 merkle-root-binds-table (*hd (sqp-composition-fri-roots prefix)*)
 composition-table final-state
and *trace-len*: *length trace-table = scale * clength*
and *composition-len*: *length composition-table = scale * clength*
shows
 staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
 composition-table prefix prefix-state $\subseteq$
 staged-query-prefix-conceptual-default-query-target trace-table
 composition-table prefix prefix-state
proof (*rule*
 staged-query-prefix-candidate-pair-query-target-from-prefix-subset-conceptual-default-if-agrees
 [*OF prefix-clean trace-len composition-len*])
fix *i v*
assume *i-bound*: *i < scale * clength*
and *auth*:
 authenticated-value-at prefix-state (sqp-trace-root prefix)
 (*scale * clength*) *i v*
show *trace-table ! i = v*
by (*rule*
 trace-table-agrees-with-prefix-authenticated-values-if-final-merkle-bound
 [*OF ext final-clean trace-bind trace-len i-bound auth*])
next
fix *i v*
assume *i-bound*: *i < scale * clength*
and *auth*:
 authenticated-value-at prefix-state
 (*hd (sqp-composition-fri-roots prefix)*) (*scale * clength*) *i v*
show *composition-table ! i = v*
by (*rule*
 composition-table-agrees-with-prefix-authenticated-values-if-final-merkle-bound
 [*OF ext final-clean composition-bind composition-len i-bound auth*])
qed

lemma
 staged-query-prefix-candidate-opening-query-target-from-prefix-subset-conceptual:
assumes *clean*: $\neg$ *hash-map-output-collision prefix-state*
and *trace-len*: *length trace-openings = rounds*
and *composition-len*: *length composition-openings = rounds*
and *trace-tables*:
 $\bigwedge j. j < rounds \implies$
 partial-authenticated-table (sqp-trace-root prefix)
 (*scale * clength*) (*trace-openings ! j*) *prefix-state*
and *composition-tables*:
 $\bigwedge j. j < rounds \implies$
 partial-authenticated-table (hd (sqp-composition-fri-roots prefix))

```

    (scale * clength) (composition-openings ! j) prefix-state
and trace-low:
  trace-table-low-degree
    (query-prefix-trace-conceptual-table prefix prefix-state)
and composition-low:
  composition-table-low-degree maxDegree
    (query-prefix-composition-conceptual-table prefix prefix-state)
and not-all:
  ¬ all-queries-consistent
    (query-prefix-trace-conceptual-table prefix prefix-state)
    (query-prefix-composition-conceptual-table prefix prefix-state)
    (sqp-alphas prefix)
shows
  staged-query-prefix-candidate-opening-query-target-from-prefix
    trace-openings composition-openings i prefix prefix-state ⊆
    staged-query-prefix-conceptual-query-target prefix prefix-state
proof –
  have trace-candidate:
    partial-trace-table-candidate
      (query-prefix-trace-conceptual-table prefix prefix-state)
      trace-openings
  by (rule query-prefix-trace-conceptual-table-partial-trace-candidate
    [OF clean trace-len trace-tables])
  have composition-candidate:
    partial-composition-table-candidate
      (query-prefix-composition-conceptual-table prefix prefix-state)
      composition-openings
  by (rule
    query-prefix-composition-conceptual-table-partial-composition-candidate
    [OF clean composition-len composition-tables])
  show ?thesis
  unfolding staged-query-prefix-candidate-opening-query-target-from-prefix-def
    staged-query-prefix-conceptual-query-target-def
  by (rule partial-query-success-indices-at-subset-query-sampling-success-space
    [OF trace-candidate composition-candidate trace-low composition-low
    not-all])
qed

```

```

lemma
  staged-query-prefix-candidate-opening-query-target-from-prefix-subset-conceptual-default:
assumes clean: ¬ hash-map-output-collision prefix-state
and trace-len: length trace-openings = rounds
and composition-len: length composition-openings = rounds
and trace-tables:
  ∧j. j < rounds ⇒
    partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) (trace-openings ! j) prefix-state
and composition-tables:
  ∧j. j < rounds ⇒

```

```

    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
      (scale * clength) (composition-openings ! j) prefix-state
and trace-low:
  trace-table-low-degree
    (query-prefix-trace-conceptual-table-with-default prefix
      prefix-state trace-default)
and composition-low:
  composition-table-low-degree maxDegree
    (query-prefix-composition-conceptual-table-with-default prefix
      prefix-state composition-default)
and not-all:
  ¬ all-queries-consistent
    (query-prefix-trace-conceptual-table-with-default prefix
      prefix-state trace-default)
    (query-prefix-composition-conceptual-table-with-default prefix
      prefix-state composition-default)
    (sqp-alphas prefix)
shows
  staged-query-prefix-candidate-opening-query-target-from-prefix
    trace-openings composition-openings i prefix prefix-state ⊆
  staged-query-prefix-conceptual-default-query-target trace-default
    composition-default prefix prefix-state
proof —
  have trace-candidate:
    partial-trace-table-candidate
      (query-prefix-trace-conceptual-table-with-default prefix prefix-state
        trace-default)
    trace-openings
  by (rule
    query-prefix-trace-conceptual-table-with-default-partial-trace-candidate
    [OF clean trace-len trace-tables])
  have composition-candidate:
    partial-composition-table-candidate
      (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-default)
    composition-openings
  by (rule
    query-prefix-composition-conceptual-table-with-default-partial-composition-candidate
    [OF clean composition-len composition-tables])
  show ?thesis
  unfolding staged-query-prefix-candidate-opening-query-target-from-prefix-def
    staged-query-prefix-conceptual-default-query-target-def
  by (rule partial-query-success-indices-at-subset-query-sampling-success-space
    [OF trace-candidate composition-candidate trace-low composition-low
      not-all])
qed

lemma checked-staged-query-prefix-conceptual-query-target-hit-bound:
  fixes C :: prob

```

assumes *raw-bound: query-index-raw-preimage-bound*
and *prefix-bound:*
 $\bigwedge \text{prefix } \text{prefix-state}.$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s) \implies$
 $\text{query-future-fresh } \text{prefix-state}$
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $\text{staged-query-prefix-conceptual-query-target}) \ s \leq$
 query-error-bound
proof (*rule checked-staged-query-prefix-dynamic-index-hit-bound[OF raw-bound]*)
fix *prefix prefix-state*
assume *support:*
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s)$
have *future: query-future-fresh prefix-state*
by (*rule prefix-bound[OF support]*)
show
 $\text{query-future-fresh } \text{prefix-state} \wedge$
 $\text{staged-query-prefix-conceptual-query-target } \text{prefix } \text{prefix-state}$
 $\subseteq \text{query-sample-space} \wedge$
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-conceptual-query-target } \text{prefix } \text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
using *future*
 $\text{staged-query-prefix-conceptual-query-target-subset[of prefix prefix-state]}$
 $\text{staged-query-prefix-conceptual-query-target-fraction-bound}$
by *blast*
qed

lemma *checked-staged-query-prefix-conceptual-query-target-hit-bound-from-sampler:*

assumes *prefix-bound:*
 $\bigwedge \text{prefix } \text{prefix-state}.$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s) \implies$
 $\text{query-future-fresh } \text{prefix-state}$

shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $\text{staged-query-prefix-conceptual-query-target}) \ s \leq$
 query-error-bound
by (*rule checked-staged-query-prefix-conceptual-query-target-hit-bound*
 $[\text{OF query-index-raw-preimage-bound-from-sampler-wellformed}$
 $\text{prefix-bound}]$)

lemma *checked-staged-query-prefix-conceptual-default-query-target-hit-bound:*
fixes $C :: \text{prob}$
assumes *raw-bound: query-index-raw-preimage-bound*
and *prefix-bound:*
 $\bigwedge \text{prefix prefix-state.}$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s) \implies$
 $\text{query-future-fresh prefix-state}$
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $(\text{staged-query-prefix-conceptual-default-query-target trace-default}$
 $\text{composition-default})) \ s \leq$
 query-error-bound
proof (rule *checked-staged-query-prefix-dynamic-index-hit-bound*[OF *raw-bound*])
fix $\text{prefix prefix-state}$
assume *support:*
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s)$
have *future: query-future-fresh prefix-state*
by (rule *prefix-bound*[OF *support*])
show
 $\text{query-future-fresh prefix-state} \wedge$
 $\text{staged-query-prefix-conceptual-default-query-target trace-default}$
 $\text{composition-default prefix prefix-state} \subseteq \text{query-sample-space} \wedge$
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-conceptual-default-query-target trace-default}$
 $\text{composition-default prefix prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
using *future*
 $\text{staged-query-prefix-conceptual-default-query-target-subset}$
 $\text{staged-query-prefix-conceptual-default-query-target-fraction-bound}$
by *blast*
qed

lemma *checked-staged-query-prefix-conceptual-default-query-target-hit-bound-from-sampler:*
assumes *prefix-bound:*
 $\bigwedge \text{prefix prefix-state.}$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s) \implies$
 $\text{query-future-fresh prefix-state}$
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$

$(\text{staged-query-prefix-conceptual-default-query-target } \text{trace-default} \\ \text{composition-default})) \ s \leq$
 query-error-bound
by (rule
 $\text{checked-staged-query-prefix-conceptual-default-query-target-hit-bound}$
 $[OF \ \text{query-index-raw-preimage-bound-from-sampler-wellformed}$
 $\text{prefix-bound}]$)

lemma *checked-staged-query-prefix-conceptual-empty-query-target-hit-bound:*
fixes $C :: \text{prob}$
assumes $\text{raw-bound: query-index-raw-preimage-bound}$
and prefix-bound:
 $\bigwedge \text{prefix prefix-state.}$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s) \implies$
 $\text{query-future-fresh prefix-state}$
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $\text{staged-query-prefix-conceptual-empty-query-target}) \ s \leq$
 query-error-bound
proof (rule $\text{checked-staged-query-prefix-dynamic-index-hit-bound}[OF \ \text{raw-bound}]$)
fix $\text{prefix prefix-state}$
assume support:
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i) \ s)$
have $\text{future: query-future-fresh prefix-state}$
by (rule $\text{prefix-bound}[OF \ \text{support}]$)
show
 $\text{query-future-fresh prefix-state} \wedge$
 $\text{staged-query-prefix-conceptual-empty-query-target prefix prefix-state}$
 $\subseteq \text{query-sample-space} \wedge$
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-conceptual-empty-query-target prefix}$
 $\text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
using future
 $\text{staged-query-prefix-conceptual-empty-query-target-subset}$
 $\text{staged-query-prefix-conceptual-empty-query-target-fraction-bound}$
by blast
qed

lemma *checked-staged-query-prefix-conceptual-empty-query-target-hit-bound-from-sampler:*
assumes prefix-bound:
 $\bigwedge \text{prefix prefix-state.}$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$

$set-dist$
 $(execute (checked-staged-query-prefix-with-state A i) s) \implies$
 $query-future-fresh\ prefix-state$
shows
 $wp-event (checked-staged-query-prefix-receive-with-state A i)$
 $(checked-staged-query-prefix-dynamic-index-hit$
 $staged-query-prefix-conceptual-empty-query-target) s \leq$
 $query-error-bound$
by (rule checked-staged-query-prefix-conceptual-empty-query-target-hit-bound
 $[OF query-index-raw-preimage-bound-from-sampler-wellformed$
 $prefix-bound]$)

lemma *checked-staged-query-prefix-conceptual-query-target-prehit-bound:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i \leq rounds$*
shows
 $wp-event (checked-staged-query-prefix-receive-with-state A i)$
 $(checked-staged-query-prefix-dynamic-index-prehit$
 $staged-query-prefix-conceptual-query-target)$
 $adversary-initial-state \leq$
 $staged-phase-relation-error\ size$
 $(staged-query-search-queries\ budgets\ i + 1)$
proof (rule checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program)
show $HashMap\ adversary-initial-state = fmempty$
by (simp add: adversary-initial-state-def)
show *hash-relation-program*
 $(checked-staged-query-prefix-dynamic-index-prehit-relation A i$
 $adversary-initial-state\ staged-query-prefix-conceptual-query-target)$
 $size (staged-query-search-queries\ budgets\ i + 1)$
 $(checked-staged-query-prefix-receive-with-state A i)$
by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state
 $[OF wf\ controlled\ i-bound]$)
(rule
 $checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)$
qed

lemma *checked-staged-query-prefix-conceptual-default-query-target-prehit-bound:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i \leq rounds$*
shows
 $wp-event (checked-staged-query-prefix-receive-with-state A i)$
 $(checked-staged-query-prefix-dynamic-index-prehit$
 $(staged-query-prefix-conceptual-default-query-target\ trace-default$
 $composition-default))$
 $adversary-initial-state \leq$
 $staged-phase-relation-error\ size$
 $(staged-query-search-queries\ budgets\ i + 1)$

proof (rule *checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program*)
show *HashMap adversary-initial-state = fmempty*
by (simp add: *adversary-initial-state-def*)
show *hash-relation-program*
(checked-staged-query-prefix-dynamic-index-prehit-relation *A i*
adversary-initial-state
(staged-query-prefix-conceptual-default-query-target *trace-default*
composition-default))
size (staged-query-search-queries *budgets i + 1*)
(checked-staged-query-prefix-receive-with-state *A i*)
by (rule *hash-relation-program-checked-staged-query-prefix-receive-with-state*
[*OF wf controlled i-bound*])
(rule
checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
qed

lemma *checked-staged-query-prefix-conceptual-empty-query-target-prehit-bound*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i ≤ rounds*
shows
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-prehit
staged-query-prefix-conceptual-empty-query-target)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries *budgets i + 1*)
proof (rule *checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program*)
show *HashMap adversary-initial-state = fmempty*
by (simp add: *adversary-initial-state-def*)
show *hash-relation-program*
(checked-staged-query-prefix-dynamic-index-prehit-relation *A i*
adversary-initial-state
staged-query-prefix-conceptual-empty-query-target)
size (staged-query-search-queries *budgets i + 1*)
(checked-staged-query-prefix-receive-with-state *A i*)
by (rule *hash-relation-program-checked-staged-query-prefix-receive-with-state*
[*OF wf controlled i-bound*])
(rule
checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
qed

lemma *checked-staged-query-prefix-conceptual-query-target-hit-bound-from-budgets*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
shows
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit

```

    staged-query-prefix-conceptual-query-target)
  adversary-initial-state ≤
  staged-phase-relation-error size
  (staged-query-search-queries budgets  $i + 1$ ) + query-error-bound
proof –
  have raw-bound: query-index-raw-preimage-bound
  by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
  have hit-bound:
    wp-event (checked-staged-query-prefix-receive-with-state  $A\ i$ )
      (checked-staged-query-prefix-dynamic-index-hit
        staged-query-prefix-conceptual-query-target)
    adversary-initial-state ≤
    wp-event (checked-staged-query-prefix-receive-with-state  $A\ i$ )
      (checked-staged-query-prefix-dynamic-index-prehit
        staged-query-prefix-conceptual-query-target)
    adversary-initial-state + query-error-bound
proof (rule
  checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
  [OF raw-bound])
fix prefix prefix-state
assume
  Some ((prefix, prefix-state), prefix-state) ∈
  set-dist
  (execute (checked-staged-query-prefix-with-state  $A\ i$ )
    adversary-initial-state)
show
  staged-query-prefix-conceptual-query-target prefix prefix-state
  ⊆ query-sample-space ∧
  nnreal
  (card
    (staged-query-prefix-conceptual-query-target prefix
      prefix-state)) /
  nnreal (card query-sample-space) ≤ query-error-bound
using staged-query-prefix-conceptual-query-target-subset
  staged-query-prefix-conceptual-query-target-fraction-bound
by blast
qed
have prehit-bound:
  wp-event (checked-staged-query-prefix-receive-with-state  $A\ i$ )
    (checked-staged-query-prefix-dynamic-index-prehit
      staged-query-prefix-conceptual-query-target)
  adversary-initial-state ≤
  staged-phase-relation-error size
  (staged-query-search-queries budgets  $i + 1$ )
by (rule checked-staged-query-prefix-conceptual-query-target-prehit-bound
  [OF wf controlled])
  (use i-bound in simp)
show ?thesis
by (rule order-trans[OF hit-bound])

```

(intro add-mono prehit-bound order-refl)

qed

lemma *checked-staged-query-prefix-conceptual-default-query-target-hit-bound-from-budgets:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
shows

$$\begin{aligned} & wp\text{-}event \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state \ A \ i) \\ & \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit \\ & \ \ (staged\text{-}query\text{-}prefix\text{-}conceptual\text{-}default\text{-}query\text{-}target \ trace\text{-}default \\ & \ \ composition\text{-}default)) \\ & \ adversary\text{-}initial\text{-}state \leq \\ & \ staged\text{-}phase\text{-}relation\text{-}error \ size \\ & \ (staged\text{-}query\text{-}search\text{-}queries \ budgets \ i + 1) + query\text{-}error\text{-}bound \end{aligned}$$

proof –
have *raw-bound: query-index-raw-preimage-bound*
by (rule *query-index-raw-preimage-bound-from-sampler-wellformed*)
have *hit-bound:*

$$\begin{aligned} & wp\text{-}event \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state \ A \ i) \\ & \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}hit \\ & \ \ (staged\text{-}query\text{-}prefix\text{-}conceptual\text{-}default\text{-}query\text{-}target \ trace\text{-}default \\ & \ \ composition\text{-}default)) \\ & \ adversary\text{-}initial\text{-}state \leq \\ & wp\text{-}event \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}receive\text{-}with\text{-}state \ A \ i) \\ & \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}dynamic\text{-}index\text{-}prehit \\ & \ \ (staged\text{-}query\text{-}prefix\text{-}conceptual\text{-}default\text{-}query\text{-}target \ trace\text{-}default \\ & \ \ composition\text{-}default)) \\ & \ adversary\text{-}initial\text{-}state + query\text{-}error\text{-}bound \end{aligned}$$

proof (rule
checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
[OF *raw-bound*])
fix *prefix prefix-state*
assume

$$\begin{aligned} & Some \ ((prefix, \ prefix\text{-}state), \ prefix\text{-}state) \in \\ & \ set\text{-}dist \\ & \ (execute \ (checked\text{-}staged\text{-}query\text{-}prefix\text{-}with\text{-}state \ A \ i) \\ & \ \ adversary\text{-}initial\text{-}state) \end{aligned}$$

show

$$\begin{aligned} & staged\text{-}query\text{-}prefix\text{-}conceptual\text{-}default\text{-}query\text{-}target \ trace\text{-}default \\ & \ composition\text{-}default \ prefix \ prefix\text{-}state \subseteq query\text{-}sample\text{-}space \wedge \\ & \ nnreal \\ & \ (card \\ & \ \ (staged\text{-}query\text{-}prefix\text{-}conceptual\text{-}default\text{-}query\text{-}target \ trace\text{-}default \\ & \ \ composition\text{-}default \ prefix \ prefix\text{-}state)) / \\ & \ nnreal \ (card \ query\text{-}sample\text{-}space) \leq query\text{-}error\text{-}bound \end{aligned}$$

using *staged-query-prefix-conceptual-default-query-target-subset*
staged-query-prefix-conceptual-default-query-target-fraction-bound
by *blast*

qed
have *prehit-bound*:
 wp-event (*checked-staged-query-prefix-receive-with-state A i*)
 (*checked-staged-query-prefix-dynamic-index-prehit*
 (*staged-query-prefix-conceptual-default-query-target trace-default*
 composition-default))
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*)
by (*rule*
 checked-staged-query-prefix-conceptual-default-query-target-prehit-bound
 [*OF wf controlled*])
 (*use i-bound in simp*)
show *?thesis*
 by (*rule order-trans[OF hit-bound]*)
 (*intro add-mono prehit-bound order-refl*)
qed

lemma *checked-staged-query-prefix-conceptual-empty-query-target-hit-bound-from-budgets*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: *i < rounds*
shows
 wp-event (*checked-staged-query-prefix-receive-with-state A i*)
 (*checked-staged-query-prefix-dynamic-index-hit*
 staged-query-prefix-conceptual-empty-query-target)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) + *query-error-bound*
proof –
have *raw-bound*: *query-index-raw-preimage-bound*
by (*rule query-index-raw-preimage-bound-from-sampler-wellformed*)
have *hit-bound*:
 wp-event (*checked-staged-query-prefix-receive-with-state A i*)
 (*checked-staged-query-prefix-dynamic-index-hit*
 staged-query-prefix-conceptual-empty-query-target)
 adversary-initial-state $\leq$
 wp-event (*checked-staged-query-prefix-receive-with-state A i*)
 (*checked-staged-query-prefix-dynamic-index-prehit*
 staged-query-prefix-conceptual-empty-query-target)
 adversary-initial-state + *query-error-bound*
proof (*rule*
 checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
 [*OF raw-bound*])
fix *prefix prefix-state*
assume
 Some ((prefix, prefix-state), prefix-state) ∈
 set-dist
 (*execute* (*checked-staged-query-prefix-with-state A i*)

$adversary-initial-state)$
show
 $staged-query-prefix-conceptual-empty-query-target\ prefix\ prefix-state$
 $\subseteq query-sample-space \wedge$
 $nnreal$
 $(card$
 $(staged-query-prefix-conceptual-empty-query-target\ prefix$
 $prefix-state)) /$
 $nnreal (card\ query-sample-space) \leq query-error-bound$
using $staged-query-prefix-conceptual-empty-query-target-subset$
 $staged-query-prefix-conceptual-empty-query-target-fraction-bound$
by *blast*
qed
have *prehit-bound*:
 $wp-event (checked-staged-query-prefix-receive-with-state\ A\ i)$
 $(checked-staged-query-prefix-dynamic-index-prehit$
 $staged-query-prefix-conceptual-empty-query-target)$
 $adversary-initial-state \leq$
 $staged-phase-relation-error\ size$
 $(staged-query-search-queries\ budgets\ i + 1)$
by (rule
 $checked-staged-query-prefix-conceptual-empty-query-target-prehit-bound$
 $[OF\ wf\ controlled])$
 $(use\ i-bound\ in\ simp)$
show *?thesis*
by (rule $order-trans[OF\ hit-bound]$)
 $(intro\ add-mono\ prehit-bound\ order-refl)$
qed

definition *checked-staged-security-with-query-prefix-conceptual-target-hit*
where
 $checked-staged-security-with-query-prefix-conceptual-target-hit\ out \longleftrightarrow$
 $checked-staged-security-with-query-prefix-dynamic-index-hit$
 $staged-query-prefix-conceptual-query-target\ out$

definition *checked-staged-security-with-query-prefix-conceptual-default-target-hit*
where
 $checked-staged-security-with-query-prefix-conceptual-default-target-hit$
 $trace-default\ composition-default\ out \longleftrightarrow$
 $checked-staged-security-with-query-prefix-dynamic-index-hit$
 $(staged-query-prefix-conceptual-default-query-target\ trace-default$
 $composition-default)\ out$

definition *checked-staged-security-with-query-prefix-conceptual-empty-target-hit*
where
 $checked-staged-security-with-query-prefix-conceptual-empty-target-hit\ out$
 $\longleftrightarrow$
 $checked-staged-security-with-query-prefix-dynamic-index-hit$
 $staged-query-prefix-conceptual-empty-query-target\ out$

lemma *checked-staged-security-with-query-prefix-conceptual-target-hit-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

checked-staged-security-with-query-prefix-conceptual-target-hit

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) + query-error-bound

unfolding *checked-staged-security-with-query-prefix-conceptual-target-hit-def*

by *(rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound)*

(rule checked-staged-query-prefix-conceptual-query-target-hit-bound-from-budgets

[OF wf controlled i-bound])

lemma *checked-staged-security-with-query-prefix-conceptual-default-target-hit-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-conceptual-default-target-hit

trace-default composition-default)

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) + query-error-bound

unfolding

checked-staged-security-with-query-prefix-conceptual-default-target-hit-def

by *(rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound)*

(rule

checked-staged-query-prefix-conceptual-default-query-target-hit-bound-from-budgets

[OF wf controlled i-bound])

lemma *checked-staged-security-with-query-prefix-conceptual-empty-target-hit-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

checked-staged-security-with-query-prefix-conceptual-empty-target-hit

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) + query-error-bound

unfolding

checked-staged-security-with-query-prefix-conceptual-empty-target-hit-def

by (rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound)
 (rule
 checked-staged-query-prefix-conceptual-empty-query-target-hit-bound-from-budgets
 [OF wf controlled i-bound])

end

end

theory Soundness-Conceptual-Query-Current
imports
 Soundness-Conceptual-Query-Target
 Staged-Security-Experiment-Composition-Query-Current
begin

Connect the existing prefix-authenticated query target to the conceptual prefix tables.
 This is a deterministic bridge: it only uses openings that are already authenticated in the
 query-prefix state, so the resulting target remains fixed before the query-index challenge
 is sampled.

context soundness
begin

definition staged-query-prefix-prefix-authenticated-empty-opening-target
 :: 'f staged-query-prefix-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ nat set

where

 staged-query-prefix-prefix-authenticated-empty-opening-target
 prefix prefix-state =
 {idx $\in$ query-sample-space.
 $\exists$ (trace-openings :: 'f authenticated-opening list).
 partial-authenticated-table (sqp-trace-root prefix)
 (scale * clength) trace-openings prefix-state $\wedge$
 map opening-index trace-openings = powers-scaled idx $\wedge$
 cp-eval (sqp-alphas prefix) (map opening-value trace-openings)
 (h $\wedge$ idx * shift) = sqp-composition-final prefix}

lemma staged-query-prefix-prefix-authenticated-empty-opening-target-subset:
 staged-query-prefix-prefix-authenticated-empty-opening-target
 prefix prefix-state $\subseteq$ query-sample-space
unfolding staged-query-prefix-prefix-authenticated-empty-opening-target-def
by blast

definition checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit
where

 checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit
 trace-openings i out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),

$((data, attacker-state), result)), final-state) \Rightarrow$
 $partial-authenticated-table (spp-trace-root prefix)$
 $(scale * clength) trace-openings prefix-state \wedge$
 $map opening-index trace-openings = powers-scaled (index (to-nat raw)) \wedge$
 $cp-eval (spp-alphas prefix) (map opening-value trace-openings)$
 $(h \wedge index (to-nat raw) * shift) =$
 $spp-composition-final prefix)$

definition *checked-staged-security-with-query-prefix-empty-authenticated-opening-hit*
where

$checked-staged-security-with-query-prefix-empty-authenticated-opening-hit$
 $trace-openings i out \longleftrightarrow$
 $(case out of$
 $None \Rightarrow False$
 $| Some (((((prefix, prefix-state), raw), raw-state),$
 $((data, attacker-state), result)), final-state) \Rightarrow$
 $partial-authenticated-table (spp-trace-root prefix)$
 $(scale * clength) trace-openings final-state \wedge$
 $map opening-index trace-openings = powers-scaled (index (to-nat raw)) \wedge$
 $cp-eval (spp-alphas prefix) (map opening-value trace-openings)$
 $(h \wedge index (to-nat raw) * shift) =$
 $spp-composition-final prefix)$

definition *checked-staged-security-with-query-prefix-empty-trace-path-output-hit*
where

$checked-staged-security-with-query-prefix-empty-trace-path-output-hit$
 $trace-openings out \longleftrightarrow$
 $(case out of$
 $None \Rightarrow False$
 $| Some (((((prefix, prefix-state), raw), raw-state),$
 $((data, attacker-state), result)), final-state) \Rightarrow$
 $(\exists opening \in set trace-openings.$
 $hash-map-new-output-hit$
 $(merkle-path-target-roots final-state$
 $(spp-trace-root prefix)$
 $(scale * clength)$
 $(opening-index opening)$
 $(opening-value opening)$
 $(opening-path opening))$
 $prefix-state final-state))$

definition *checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at*
where

$checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at$
 $i out \longleftrightarrow$
 $(\exists (trace-openings :: 'f authenticated-opening list).$
 $checked-staged-security-with-query-prefix-authenticated-empty-opening-hit$
 $trace-openings i out)$

definition *checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at*
where

checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at
 $i \text{ out} \longleftrightarrow$
 $(\exists (trace\text{-}openings :: 'f \text{ authenticated-opening list}).$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}query\text{-}prefix\text{-}empty\text{-}authenticated\text{-}opening\text{-}hit$
 $trace\text{-}openings \ i \ out)$

definition *checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at*
where

checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at
 $i \text{ out} \longleftrightarrow$
 $(\exists (trace\text{-}openings :: 'f \text{ authenticated-opening list}).$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}query\text{-}prefix\text{-}empty\text{-}trace\text{-}path\text{-}output\text{-}hit$
 $trace\text{-}openings \ out)$

definition *checked-staged-security-with-query-prefix-conceptual-empty-context-residual-at*
where

checked-staged-security-with-query-prefix-conceptual-empty-context-residual-at
 $out \longleftrightarrow$
 $(case \ out \ of$
 $\quad None \Rightarrow False$
 $\mid Some \ (((((prefix, \ prefix\text{-}state), \ raw), \ raw\text{-}state),$
 $\quad ((data, \ attacker\text{-}state), \ result)), \ final\text{-}state) \Rightarrow$
 $\quad hash\text{-}map\text{-}output\text{-}collision \ prefix\text{-}state \ \vee$
 $\quad \neg \ trace\text{-}table\text{-}low\text{-}degree$
 $\quad (query\text{-}prefix\text{-}trace\text{-}conceptual\text{-}table \ prefix \ prefix\text{-}state) \ \vee$
 $\quad all\text{-}queries\text{-}consistent$
 $\quad (query\text{-}prefix\text{-}trace\text{-}conceptual\text{-}table \ prefix \ prefix\text{-}state)$
 $\quad (replicate \ (scale * \ clength) \ (sqp\text{-}composition\text{-}final \ prefix))$
 $\quad (sqp\text{-}alphas \ prefix))$

definition *checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at*
where

checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
 $(i :: nat) \ out \longleftrightarrow$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}query\text{-}prefix\text{-}current\text{-}empty\text{-}prefix\text{-}authenticated\text{-}hit\text{-}at$
 $i \ out \wedge$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}query\text{-}prefix\text{-}conceptual\text{-}empty\text{-}context\text{-}residual\text{-}at$
 out

lemma *checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at-imp-dynamic-target:*

assumes *hit:*

checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at
 $i \ out$

shows

checked-staged-security-with-query-prefix-dynamic-index-hit
 $staged\text{-}query\text{-}prefix\text{-}authenticated\text{-}empty\text{-}opening\text{-}target \ out$

proof (*cases out*)

```

case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at-def
    checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
from hit obtain trace-openings where prefix-hit:
  checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit
  trace-openings i out
unfolding
  checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at-def
by blast
have idx-in:
  index (to-nat raw) ∈
    staged-query-prefix-prefix-authenticated-empty-opening-target
    prefix prefix-state
using prefix-hit index-less-query-sample-space
unfolding out-eq
  checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit-def
  staged-query-prefix-prefix-authenticated-empty-opening-target-def
  query-sample-space-def
by auto
show ?thesis
unfolding out-eq
  checked-staged-security-with-query-prefix-dynamic-index-hit-def
  checked-staged-query-prefix-dynamic-index-hit-def
using idx-in by simp
qed

```

```

lemma checked-staged-security-with-query-prefix-empty-authenticated-opening-hit-imp-prefix-or-path-output-hit:
assumes ext: prefix-state ≤ final-state
and hit:
  checked-staged-security-with-query-prefix-empty-authenticated-opening-hit
  trace-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)))
shows
  checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit
  trace-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),

```

```

      ((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-empty-trace-path-output-hit
  trace-openings
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
proof —
have trace-table:
  partial-authenticated-table (sqp-trace-root prefix)
    (scale * clength) trace-openings final-state
and trace-indices:
  map opening-index trace-openings = powers-scaled (index (to-nat raw))
and consistent:
  cp-eval (sqp-alphas prefix) (map opening-value trace-openings)
    (h ^ index (to-nat raw) * shift) =
    sqp-composition-final prefix
using hit
unfolding
  checked-staged-security-with-query-prefix-empty-authenticated-opening-hit-def
by simp-all
have trace-pullback:
  partial-authenticated-table (sqp-trace-root prefix)
    (scale * clength) trace-openings prefix-state ∨
  (∃ opn ∈ set trace-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots final-state (sqp-trace-root prefix)
        (scale * clength)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn)) prefix-state final-state)
  by (rule partial-authenticated-table-pullback-or-new-output-hit
    [OF ext trace-table])
then show ?thesis
proof
  assume trace-prefix:
    partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) trace-openings prefix-state
  then have
    checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit
      trace-openings i
      (Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state))
  using trace-indices consistent
  unfolding
    checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit-def
  by simp
then show ?thesis by simp
next
  assume
    ∃ opn ∈ set trace-openings.

```

```

    hash-map-new-output-hit
      (merkle-path-target-roots final-state (sqp-trace-root prefix)
        (scale * clength)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn)) prefix-state final-state
  then have
    checked-staged-security-with-query-prefix-empty-trace-path-output-hit
      trace-openings
      (Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state))
  unfolding
    checked-staged-security-with-query-prefix-empty-trace-path-output-hit-def
  by simp
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-query-prefix-empty-authenticated-opening-hit-imp-prefix-or-path-output-hit-on-sv
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-query-prefix-data-state A i)
          adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-empty-authenticated-opening-hit
      trace-openings i out
  shows
    checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit
      trace-openings i out ∨
    checked-staged-security-with-query-prefix-empty-trace-path-output-hit
      trace-openings out
proof (cases out)
case None
  then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-empty-authenticated-opening-hit-def
  by simp
next
case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),

```

```

      ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  have ext: prefix-state  $\leq$  final-state
  by (rule
      checked-staged-security-experiment-with-query-prefix-data-state-prefix-extends-final
      [OF wf controlled i-bound])
      (use support out-eq in simp)
  show ?thesis
  unfolding out-eq
  by (rule
      checked-staged-security-with-query-prefix-empty-authenticated-opening-hit-imp-prefix-or-path-output-hit
      [OF ext])
      (use hit out-eq in simp)
qed

lemma checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at-imp-prefix-or-path-output-hit-on-
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and support:
    out  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at
    i out
  shows
    checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at
    i out  $\vee$ 
    checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at
    i out
proof -
  from hit obtain trace-openings where trace-hit:
    checked-staged-security-with-query-prefix-empty-authenticated-opening-hit
    trace-openings i out
  unfolding
    checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at-def
  by blast
  have split:
    checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit
    trace-openings i out  $\vee$ 
    checked-staged-security-with-query-prefix-empty-trace-path-output-hit
    trace-openings out
  by (rule
      checked-staged-security-with-query-prefix-empty-authenticated-opening-hit-imp-prefix-or-path-output-hit-on-suppo
      [OF wf controlled i-bound support trace-hit])
  then show ?thesis

```

unfolding
checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at-def
checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at-def
by *blast*
qed

lemma *low-degree-all-queries-consistent-imp-composition-trace-bad-alpha-space:*

assumes *false-statement: $\neg$ exists-valid-trace*
and *len-as: length as = length spec*
and *trace-low: trace-table-low-degree trace-table*
and *composition-low:*
composition-table-low-degree maxDegree composition-table
and *all-queries:*
all-queries-consistent trace-table composition-table as
shows *as $\in$ composition-trace-bad-alpha-space trace-table*

proof –

from *trace-low* **obtain** *f* **where** *deg-f: degree f < clength*
and *trace-table: trace-table = map (poly f) eval-domain*
unfolding *trace-table-low-degree-def* **by** *blast*
have *bounds: common-denominator-degree-bounds f as*
by (*rule common-denominator-degree-bounds-from-spec-degree-wellformed*
[OF spec-degree-wellformed-from-spec-query-margin deg-f])
have *hides:*
random-combination-common-denominator-hides-violations f as
by (*rule random-combination-common-denominator-hides-from-queries*
[OF len-as false-statement deg-f trace-table composition-low
all-queries bounds])
have *violated: violated-constraints f $\neq$ {}*
by (*rule false-statement-violated-constraints*
[OF false-statement deg-f])
have *as-common: as $\in$ common-denominator-hiding-alpha-space f*
using *len-as hides*
unfolding *common-denominator-hiding-alpha-space-def alpha-space-def*
by *simp*
have *witness-eq: low-degree-trace-witness trace-table = f*
by (*rule low-degree-trace-witness-eq [OF deg-f trace-table]*)
show *?thesis*
using *deg-f trace-table violated bounds as-common*
unfolding *composition-trace-bad-alpha-space-def witness-eq*
by *simp*

qed

lemma *staged-query-prefix-prefix-authenticated-empty-opening-target-subset-conceptual-empty:*

assumes *clean: $\neg$ hash-map-output-collision prefix-state*
and *trace-low:*
trace-table-low-degree
(query-prefix-trace-conceptual-table prefix prefix-state)
and *not-all:*
 $\neg$ all-queries-consistent

```

    (query-prefix-trace-conceptual-table prefix prefix-state)
    (replicate (scale * clength) (sqp-composition-final prefix))
    (sqp-alphas prefix)
  shows
    staged-query-prefix-prefix-authenticated-empty-opening-target
      prefix prefix-state  $\subseteq$ 
      staged-query-prefix-conceptual-empty-query-target prefix prefix-state
proof
  fix idx
  assume idx-in:
    idx  $\in$ 
      staged-query-prefix-prefix-authenticated-empty-opening-target
      prefix prefix-state
  then obtain trace-openings where
    idx-sample: idx  $\in$  query-sample-space
    and trace-table:
      partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) trace-openings prefix-state
    and trace-indices:
      map opening-index trace-openings = powers-scaled idx
    and final-eq:
      cp-eval (sqp-alphas prefix) (map opening-value trace-openings)
      (h  $\wedge$  idx * shift) = sqp-composition-final prefix
  unfolding staged-query-prefix-prefix-authenticated-empty-opening-target-def
  by blast
let ?trace-table =
  query-prefix-trace-conceptual-table prefix prefix-state
let ?composition-table =
  replicate (scale * clength) (sqp-composition-final prefix)
have idx-bound: idx < clength * scale
  by (rule query-sample-space-less-domain[OF idx-sample])
have trace-values:
  map (!) ?trace-table (powers-scaled idx) =
    map opening-value trace-openings
proof (rule nth-equalityI)
  show
    length (map (!) ?trace-table (powers-scaled idx)) =
      length (map opening-value trace-openings)
    using trace-indices by (metis length-map)
next
  fix k
  assume k-bound:
    k < length (map (!) ?trace-table (powers-scaled idx))
  then have ps-bound: k < length (powers-scaled idx)
    by simp
  have opn-bound: k < length trace-openings
    using trace-indices ps-bound by (metis length-map)
  have opn-in: trace-openings ! k  $\in$  set trace-openings
    by (rule nth-mem[OF opn-bound])

```

```

have idx-eq:
  opening-index (trace-openings ! k) = powers-scaled idx ! k
  using trace-indices ps-bound opn-bound
  by (metis nth-map)
have conceptual:
  ?trace-table ! opening-index (trace-openings ! k) =
    opening-value (trace-openings ! k)
  by (rule
    query-prefix-trace-conceptual-table-agrees-with-partial-authenticated-table
    [OF clean trace-table opn-in])
show
  map (!) ?trace-table (powers-scaled idx) ! k =
    map opening-value trace-openings ! k
  using ps-bound opn-bound idx-eq conceptual by simp
qed
have query-consistent:
  query-consistent-at ?trace-table ?composition-table
    (sqp-alphas prefix) idx
proof –
  have composition-value:
    ?composition-table ! idx = sqp-composition-final prefix
    using idx-bound by (simp add: mult.commute)
  have cp-eq:
    cp-eval (sqp-alphas prefix)
      (map (!) ?trace-table (powers-scaled idx))
      (h ^ idx * shift) =
      sqp-composition-final prefix
    using trace-values final-eq by simp
  show ?thesis
    unfolding query-consistent-at-def
    using idx-bound query-sample-space-powers-scaled-bound [OF idx-sample]
      composition-value cp-eq by (simp add: mult.commute)
qed
show
  idx ∈
    staged-query-prefix-conceptual-empty-query-target prefix prefix-state
  unfolding staged-query-prefix-conceptual-empty-query-target-def
    query-sampling-success-space-def query-agreement-indices-def
  using idx-sample trace-low constant-composition-table-low-degree not-all
    query-consistent by simp
qed

lemma checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-imp-conceptual-empty-or-residual-hit
assumes hit:
  checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at
    i out
shows
  checked-staged-security-with-query-prefix-conceptual-empty-target-hit out ∨
  checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at

```



```

      i out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at-def
      checked-staged-security-with-query-prefix-prefix-authenticated-empty-opening-hit-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  show ?thesis
  proof (cases
     $\neg$  hash-map-output-collision prefix-state  $\wedge$ 
    trace-table-low-degree
    (query-prefix-trace-conceptual-table prefix prefix-state)  $\wedge$ 
     $\neg$  all-queries-consistent
    (query-prefix-trace-conceptual-table prefix prefix-state)
    (replicate (scale * clength) (sqp-composition-final prefix))
    (sqp-alphas prefix))
  case False
  then have
    checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
    i out
  unfolding
    checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at-def
    checked-staged-security-with-query-prefix-conceptual-empty-context-residual-at-def
    out-eq
  using hit out-eq by simp
  then show ?thesis by simp
next
  case True
  then have clean:
     $\neg$  hash-map-output-collision prefix-state
  and trace-low:
    trace-table-low-degree
    (query-prefix-trace-conceptual-table prefix prefix-state)
  and not-all:
     $\neg$  all-queries-consistent
    (query-prefix-trace-conceptual-table prefix prefix-state)
    (replicate (scale * clength) (sqp-composition-final prefix))
    (sqp-alphas prefix)
  by blast+

```

```

have prefix-hit:
  checked-staged-security-with-query-prefix-dynamic-index-hit
    staged-query-prefix-authenticated-empty-opening-target out
by (rule
  checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at-imp-dynamic-target
    [OF hit])
have subset:
  staged-query-prefix-authenticated-empty-opening-target
    prefix prefix-state  $\subseteq$ 
  staged-query-prefix-conceptual-empty-query-target prefix prefix-state
by (rule
  staged-query-prefix-authenticated-empty-opening-target-subset-conceptual-empty
    [OF clean trace-low not-all])
have conceptual-hit:
  checked-staged-security-with-query-prefix-conceptual-empty-target-hit
    out
using prefix-hit subset
unfolding out-eq
  checked-staged-security-with-query-prefix-conceptual-empty-target-hit-def
  checked-staged-security-with-query-prefix-dynamic-index-hit-def
  checked-staged-query-prefix-dynamic-index-hit-def
by auto
then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at-bound-from-conceptual-em
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and residual-bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
    i)
  adversary-initial-state  $\leq R$ 
shows
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at
    i)
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
  query-error-bound + R
proof –
let ?M = checked-staged-security-experiment-with-query-prefix-data-state A i
let ?Conceptual =
  checked-staged-security-with-query-prefix-conceptual-empty-target-hit

```

```

let ?Residual =
  checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
  i
have event-le:
  wp-event ?M
    (checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at
     i)
    adversary-initial-state ≤
    wp-event ?M (λout. ?Conceptual out ∨ ?Residual out)
    adversary-initial-state
by (rule wp-event-mono)
    (rule
     checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-imp-conceptual-empty-or-residual)
have conceptual-bound:
  wp-event ?M ?Conceptual adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
by (rule
    checked-staged-security-with-query-prefix-conceptual-empty-target-hit-bound
    [OF wf controlled i-bound])
have union-bound:
  wp-event ?M (λout. ?Conceptual out ∨ ?Residual out)
    adversary-initial-state ≤
    wp-event ?M ?Conceptual adversary-initial-state +
    wp-event ?M ?Residual adversary-initial-state
by (rule wp-event-union-bound)
have sum-bound:
  wp-event ?M ?Conceptual adversary-initial-state +
  wp-event ?M ?Residual adversary-initial-state ≤
    (staged-phase-relation-error size
     (staged-query-search-queries budgets i + 1) + query-error-bound) +
    R
by (intro add-mono conceptual-bound residual-bound)
show ?thesis
by (rule order-trans[OF event-le])
    (rule order-trans[OF union-bound sum-bound])
qed

```

lemma checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at-bound-from-prefix-and-path:

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and prefix-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at
     i)
    adversary-initial-state ≤ P
and path-bound:

```

```

    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at
        i)
      adversary-initial-state  $\leq T$ 
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at
        i)
      adversary-initial-state  $\leq P + T$ 
  proof -
    let ?M = checked-staged-security-experiment-with-query-prefix-data-state A i
    let ?Prefix =
      checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at
        i
    let ?Path =
      checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at
        i
    have event-le:
      wp-event ?M
        (checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at
          i)
        adversary-initial-state  $\leq$ 
        wp-event ?M ( $\lambda out. ?Prefix\ out \vee ?Path\ out$ )
        adversary-initial-state
    by (rule wp-event-mono-on-support)
      (rule
        checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at-imp-prefix-or-path-output-hit-on-sup
        [OF wf controlled i-bound])
    have union-bound:
      wp-event ?M ( $\lambda out. ?Prefix\ out \vee ?Path\ out$ )
        adversary-initial-state  $\leq$ 
        wp-event ?M ?Prefix adversary-initial-state +
        wp-event ?M ?Path adversary-initial-state
    by (rule wp-event-union-bound)
    have sum-bound:
      wp-event ?M ?Prefix adversary-initial-state +
      wp-event ?M ?Path adversary-initial-state  $\leq P + T$ 
    by (intro add-mono prefix-bound path-bound)
    show ?thesis
      by (rule order-trans[OF event-le])
        (rule order-trans[OF union-bound sum-bound])
  qed

  lemma staged-query-prefix-prefix-authenticated-opening-target-subset-conceptual:
    assumes clean:  $\neg$  hash-map-output-collision prefix-state
    and trace-low:
      trace-table-low-degree

```

```

    (query-prefix-trace-conceptual-table prefix prefix-state)
and composition-low:
    composition-table-low-degree maxDegree
    (query-prefix-composition-conceptual-table prefix prefix-state)
and not-all:
    ¬ all-queries-consistent
    (query-prefix-trace-conceptual-table prefix prefix-state)
    (query-prefix-composition-conceptual-table prefix prefix-state)
    (sqp-alphas prefix)
shows
    staged-query-prefix-prefix-authenticated-opening-target prefix
    prefix-state ⊆
    staged-query-prefix-conceptual-query-target prefix prefix-state
proof
fix idx
assume idx-in:
    idx ∈ staged-query-prefix-prefix-authenticated-opening-target prefix
    prefix-state
then obtain trace-openings composition-openings where
    trace-table:
    partial-authenticated-table (sqp-trace-root prefix)
    (scale * clength) trace-openings prefix-state
and composition-table:
    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
    (scale * clength) composition-openings prefix-state
and consistent:
    partial-query-openings-consistent trace-openings composition-openings
    (sqp-alphas prefix) idx
unfolding staged-query-prefix-prefix-authenticated-opening-target-def
by blast
let ?trace-openings =
    (replicate rounds []) [0 := trace-openings]
let ?composition-openings =
    (replicate rounds []) [0 := composition-openings]
have zero-bound: 0 < rounds
by (rule rounds-positive)
have trace-len: length ?trace-openings = rounds
by simp
have composition-len: length ?composition-openings = rounds
by simp
have trace-tables:
    ∧j. j < rounds ⇒
    partial-authenticated-table (sqp-trace-root prefix)
    (scale * clength) (?trace-openings ! j) prefix-state
proof –
fix j
assume j-bound: j < rounds
show
    partial-authenticated-table (sqp-trace-root prefix)

```

```

    (scale * clength) (?trace-openings ! j) prefix-state
proof (cases j = 0)
  case True
    then show ?thesis
      using trace-table zero-bound by simp
  next
    case False
    then have ?trace-openings ! j = []
      using j-bound by simp
    then show ?thesis
      unfolding partial-authenticated-table-def by simp
  qed
qed
have composition-tables:
   $\bigwedge j. j < \text{rounds} \implies$ 
    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
    (scale * clength) (?composition-openings ! j) prefix-state
proof –
  fix j
  assume j-bound: j < rounds
  show
    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
    (scale * clength) (?composition-openings ! j) prefix-state
proof (cases j = 0)
  case True
    then show ?thesis
      using composition-table zero-bound by simp
  next
    case False
    then have ?composition-openings ! j = []
      using j-bound by simp
    then show ?thesis
      unfolding partial-authenticated-table-def by simp
  qed
qed
have opening-target:
  idx  $\in$ 
    staged-query-prefix-candidate-opening-query-target-from-prefix
    ?trace-openings ?composition-openings 0 prefix prefix-state
unfolding staged-query-prefix-candidate-opening-query-target-from-prefix-def
by (rule partial-query-success-indices-atI,
    rule partial-query-round-consistent-single-roundI
    [OF zero-bound consistent])
have target-subset:
  staged-query-prefix-candidate-opening-query-target-from-prefix
  ?trace-openings ?composition-openings 0 prefix prefix-state  $\subseteq$ 
  staged-query-prefix-conceptual-query-target prefix prefix-state
by (rule
  staged-query-prefix-candidate-opening-query-target-from-prefix-subset-conceptual

```

```

    [OF clean trace-len composition-len trace-tables composition-tables
      trace-low composition-low not-all])
show
  idx ∈ staged-query-prefix-conceptual-query-target prefix prefix-state
using opening-target target-subset by blast
qed

lemma staged-query-prefix-prefix-authenticated-opening-target-fraction-bound-if-conceptual:
assumes clean:  $\neg$  hash-map-output-collision prefix-state
and trace-low:
  trace-table-low-degree
  (query-prefix-trace-conceptual-table prefix prefix-state)
and composition-low:
  composition-table-low-degree maxDegree
  (query-prefix-composition-conceptual-table prefix prefix-state)
and not-all:
   $\neg$  all-queries-consistent
  (query-prefix-trace-conceptual-table prefix prefix-state)
  (query-prefix-composition-conceptual-table prefix prefix-state)
  (sqp-alphas prefix)
shows
  nnreal
  (card
    (staged-query-prefix-prefix-authenticated-opening-target prefix
      prefix-state)) /
  nnreal (card query-sample-space)  $\leq$  query-error-bound
proof –
have subset:
  staged-query-prefix-prefix-authenticated-opening-target prefix
  prefix-state  $\subseteq$ 
  staged-query-prefix-conceptual-query-target prefix prefix-state
by (rule
  staged-query-prefix-prefix-authenticated-opening-target-subset-conceptual
  [OF clean trace-low composition-low not-all])
have finite-conceptual:
  finite (staged-query-prefix-conceptual-query-target prefix prefix-state)
by (rule finite-subset
  [OF staged-query-prefix-conceptual-query-target-subset
  finite-query-sample-space])
have card-le:
  card
  (staged-query-prefix-prefix-authenticated-opening-target prefix
    prefix-state)  $\leq$ 
  card (staged-query-prefix-conceptual-query-target prefix prefix-state)
by (rule card-mono[OF finite-conceptual subset])
have frac-le:
  nnreal
  (card
    (staged-query-prefix-prefix-authenticated-opening-target prefix

```

```

    prefix-state)) /
  nnreal (card query-sample-space) ≤
  nnreal
    (card
      (staged-query-prefix-conceptual-query-target prefix prefix-state)) /
    nnreal (card query-sample-space)
  by (rule nnreal-nat-divide-right-mono[OF card-le])
show ?thesis
  by (rule order-trans[OF frac-le
    staged-query-prefix-conceptual-query-target-fraction-bound])
qed

lemma staged-query-prefix-prefix-authenticated-opening-target-subset-conceptual-default:
  assumes clean: ¬ hash-map-output-collision prefix-state
  and trace-low:
    trace-table-low-degree
    (query-prefix-trace-conceptual-table-with-default prefix
      prefix-state trace-default)
  and composition-low:
    composition-table-low-degree maxDegree
    (query-prefix-composition-conceptual-table-with-default prefix
      prefix-state composition-default)
  and not-all:
    ¬ all-queries-consistent
    (query-prefix-trace-conceptual-table-with-default prefix
      prefix-state trace-default)
    (query-prefix-composition-conceptual-table-with-default prefix
      prefix-state composition-default)
    (sqp-alphas prefix)
  shows
    staged-query-prefix-prefix-authenticated-opening-target prefix
      prefix-state ⊆
    staged-query-prefix-conceptual-default-query-target trace-default
      composition-default prefix prefix-state
proof
  fix idx
  assume idx-in:
    idx ∈ staged-query-prefix-prefix-authenticated-opening-target prefix
      prefix-state
  then obtain trace-openings composition-openings where
    trace-table:
      partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) trace-openings prefix-state
  and composition-table:
    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
    (scale * clength) composition-openings prefix-state
  and consistent:
    partial-query-openings-consistent trace-openings composition-openings
    (sqp-alphas prefix) idx

```



```

unfolding staged-query-prefix-prefix-authenticated-opening-target-def
by blast
let ?trace-openings =
  (replicate rounds []) [0 := trace-openings]
let ?composition-openings =
  (replicate rounds []) [0 := composition-openings]
have zero-bound: 0 < rounds
  by (rule rounds-positive)
have trace-len: length ?trace-openings = rounds
  by simp
have composition-len: length ?composition-openings = rounds
  by simp
have trace-tables:
   $\bigwedge j. j < rounds \implies$ 
    partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) (?trace-openings ! j) prefix-state
proof -
  fix j
  assume j-bound: j < rounds
  show
    partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) (?trace-openings ! j) prefix-state
proof (cases j = 0)
  case True
  then show ?thesis
    using trace-table zero-bound by simp
next
  case False
  then have ?trace-openings ! j = []
    using j-bound by simp
  then show ?thesis
    unfolding partial-authenticated-table-def by simp
qed
qed
have composition-tables:
   $\bigwedge j. j < rounds \implies$ 
    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
      (scale * clength) (?composition-openings ! j) prefix-state
proof -
  fix j
  assume j-bound: j < rounds
  show
    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
      (scale * clength) (?composition-openings ! j) prefix-state
proof (cases j = 0)
  case True
  then show ?thesis
    using composition-table zero-bound by simp
next

```

```

case False
then have ?composition-openings ! j = []
  using j-bound by simp
then show ?thesis
  unfolding partial-authenticated-table-def by simp
qed
qed
have opening-target:
  idx ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix
    ?trace-openings ?composition-openings 0 prefix prefix-state
unfolding staged-query-prefix-candidate-opening-query-target-from-prefix-def
by (rule partial-query-success-indices-atI,
  rule partial-query-round-consistent-single-roundI
  [OF zero-bound consistent])
have target-subset:
  staged-query-prefix-candidate-opening-query-target-from-prefix
  ?trace-openings ?composition-openings 0 prefix prefix-state ⊆
  staged-query-prefix-conceptual-default-query-target trace-default
  composition-default prefix prefix-state
by (rule
  staged-query-prefix-candidate-opening-query-target-from-prefix-subset-conceptual-default
  [OF clean trace-len composition-len trace-tables composition-tables
  trace-low composition-low not-all])
show
  idx ∈
    staged-query-prefix-conceptual-default-query-target trace-default
    composition-default prefix prefix-state
  using opening-target target-subset by blast
qed

definition checked-staged-security-with-query-prefix-conceptual-context-residual-at
where
  checked-staged-security-with-query-prefix-conceptual-context-residual-at
  out  $\longleftrightarrow$ 
  (case out of
  None  $\Rightarrow$  False
  | Some ((((((prefix, prefix-state), raw), raw-state),
  ((data, attacker-state), result)), final-state)  $\Rightarrow$ 
  hash-map-output-collision prefix-state  $\vee$ 
   $\neg$  trace-table-low-degree
  (query-prefix-trace-conceptual-table prefix prefix-state)  $\vee$ 
   $\neg$  composition-table-low-degree maxDegree
  (query-prefix-composition-conceptual-table prefix prefix-state)  $\vee$ 
  all-queries-consistent
  (query-prefix-trace-conceptual-table prefix prefix-state)
  (query-prefix-composition-conceptual-table prefix prefix-state)
  (sqp-alphas prefix))

```

definition *checked-staged-security-with-query-prefix-conceptual-default-context-residual-at*
where

checked-staged-security-with-query-prefix-conceptual-default-context-residual-at
 $\text{trace-default composition-default out} \longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 hash-map-output-collision prefix-state $\vee$
 $\neg$ trace-table-low-degree
 (query-prefix-trace-conceptual-table-with-default prefix
 prefix-state trace-default) $\vee$
 $\neg$ composition-table-low-degree maxDegree
 (query-prefix-composition-conceptual-table-with-default prefix
 prefix-state composition-default) $\vee$
 all-queries-consistent
 (query-prefix-trace-conceptual-table-with-default prefix
 prefix-state trace-default)
 (query-prefix-composition-conceptual-table-with-default prefix
 prefix-state composition-default)
 (sq α -alphas prefix))

lemma *checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-imp-conceptual-or-residual-on-suppo*

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and i-bound: $i < \text{rounds}$

and support:

 out $\in$
 set-dist
 (execute
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 adversary-initial-state)

and hit:

 checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
 i out

shows

 checked-staged-security-with-query-prefix-conceptual-target-hit out $\vee$
 checked-staged-security-with-query-prefix-conceptual-context-residual-at
 out

proof (cases out)

case None

then show ?thesis

using hit

unfolding

 checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-def
 checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-def

by simp

next

case (Some packed)

```

then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
show ?thesis
proof (cases
   $\neg$  hash-map-output-collision prefix-state  $\wedge$ 
  trace-table-low-degree
  (query-prefix-trace-conceptual-table prefix prefix-state)  $\wedge$ 
  composition-table-low-degree maxDegree
  (query-prefix-composition-conceptual-table prefix prefix-state)  $\wedge$ 
   $\neg$  all-queries-consistent
  (query-prefix-trace-conceptual-table prefix prefix-state)
  (query-prefix-composition-conceptual-table prefix prefix-state)
  (sqp-alphas prefix))
case False
then have
  checked-staged-security-with-query-prefix-conceptual-context-residual-at
    out
  unfolding out-eq
    checked-staged-security-with-query-prefix-conceptual-context-residual-at-def
  by simp
then show ?thesis by simp
next
case True
then have clean:
   $\neg$  hash-map-output-collision prefix-state
and trace-low:
  trace-table-low-degree
  (query-prefix-trace-conceptual-table prefix prefix-state)
and composition-low:
  composition-table-low-degree maxDegree
  (query-prefix-composition-conceptual-table prefix prefix-state)
and not-all:
   $\neg$  all-queries-consistent
  (query-prefix-trace-conceptual-table prefix prefix-state)
  (query-prefix-composition-conceptual-table prefix prefix-state)
  (sqp-alphas prefix)
by blast+
have prefix-hit:
  checked-staged-security-with-query-prefix-dynamic-index-hit
  staged-query-prefix-prefix-authenticated-opening-target out
by (rule
  checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-imp-dynamic-target
  [OF wf controlled i-bound support hit])
have subset:
  staged-query-prefix-prefix-authenticated-opening-target prefix

```

```

    prefix-state  $\subseteq$ 
    staged-query-prefix-conceptual-query-target prefix prefix-state
  by (rule
    staged-query-prefix-prefix-authenticated-opening-target-subset-conceptual
    [OF clean trace-low composition-low not-all])
  have conceptual-hit:
    checked-staged-security-with-query-prefix-conceptual-target-hit out
  using prefix-hit subset
  unfolding out-eq
    checked-staged-security-with-query-prefix-conceptual-target-hit-def
    checked-staged-security-with-query-prefix-dynamic-index-hit-def
    checked-staged-query-prefix-dynamic-index-hit-def
  by auto
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-conceptual-residual:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound:  $i < \text{rounds}$ 
  and residual-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    checked-staged-security-with-query-prefix-conceptual-context-residual-at
    adversary-initial-state  $\leq R$ 
  shows
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
      i)
    adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-query-search-queries budgets  $i + 1$ ) +
    query-error-bound + R
proof -
  let ?M = checked-staged-security-experiment-with-query-prefix-data-state A i
  let ?Conceptual =
    checked-staged-security-with-query-prefix-conceptual-target-hit
  let ?Residual =
    checked-staged-security-with-query-prefix-conceptual-context-residual-at
  have event-le:
    wp-event ?M
    (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
      i)
    adversary-initial-state  $\leq$ 
    wp-event ?M ( $\lambda \text{out. ?Conceptual out} \vee \text{?Residual out}$ )
    adversary-initial-state
  by (rule wp-event-mono-on-support)

```

```

(rule
  checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-imp-conceptual-or-residual-on-support
  [OF wf controlled i-bound])
have conceptual-bound:
  wp-event ?M ?Conceptual adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
by (rule
  checked-staged-security-with-query-prefix-conceptual-target-hit-bound
  [OF wf controlled i-bound])
have union-bound:
  wp-event ?M (λout. ?Conceptual out ∨ ?Residual out)
    adversary-initial-state ≤
  wp-event ?M ?Conceptual adversary-initial-state +
  wp-event ?M ?Residual adversary-initial-state
by (rule wp-event-union-bound)
have sum-bound:
  wp-event ?M ?Conceptual adversary-initial-state +
  wp-event ?M ?Residual adversary-initial-state ≤
    (staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound) +
  R
by (intro add-mono conceptual-bound residual-bound)
show ?thesis
by (rule order-trans[OF event-le])
  (rule order-trans[OF union-bound sum-bound])
qed

lemma checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-imp-conceptual-default-or-residual-on-support
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and support:
    out ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
      i out
  shows
    checked-staged-security-with-query-prefix-conceptual-default-target-hit
      trace-default composition-default out ∨
    checked-staged-security-with-query-prefix-conceptual-default-context-residual-at
      trace-default composition-default out
proof (cases out)
case None
then show ?thesis

```

```

using hit
unfolding
  checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-def
  checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-def
by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
out =
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)
by (cases packed) (auto split: prod.splits)
show ?thesis
proof (cases
  ¬ hash-map-output-collision prefix-state ∧
  trace-table-low-degree
  (query-prefix-trace-conceptual-table-with-default prefix
    prefix-state trace-default) ∧
  composition-table-low-degree maxDegree
  (query-prefix-composition-conceptual-table-with-default prefix
    prefix-state composition-default) ∧
  ¬ all-queries-consistent
  (query-prefix-trace-conceptual-table-with-default prefix
    prefix-state trace-default)
  (query-prefix-composition-conceptual-table-with-default prefix
    prefix-state composition-default)
  (sqp-alphas prefix))
case False
then have
  checked-staged-security-with-query-prefix-conceptual-default-context-residual-at
  trace-default composition-default out
unfolding out-eq
  checked-staged-security-with-query-prefix-conceptual-default-context-residual-at-def
by simp
then show ?thesis by simp
next
case True
then have clean:
  ¬ hash-map-output-collision prefix-state
and trace-low:
  trace-table-low-degree
  (query-prefix-trace-conceptual-table-with-default prefix
    prefix-state trace-default)
and composition-low:
  composition-table-low-degree maxDegree
  (query-prefix-composition-conceptual-table-with-default prefix
    prefix-state composition-default)
and not-all:

```

```

  ¬ all-queries-consistent
    (query-prefix-trace-conceptual-table-with-default prefix
      prefix-state trace-default)
    (query-prefix-composition-conceptual-table-with-default prefix
      prefix-state composition-default)
    (sqp-alphas prefix)
  by blast+
have prefix-hit:
  checked-staged-security-with-query-prefix-dynamic-index-hit
    staged-query-prefix-prefix-authenticated-opening-target out
  by (rule
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-imp-dynamic-target
      [OF wf controlled i-bound support hit])
have subset:
  staged-query-prefix-prefix-authenticated-opening-target prefix
    prefix-state ⊆
  staged-query-prefix-conceptual-default-query-target trace-default
    composition-default prefix prefix-state
  by (rule
    staged-query-prefix-prefix-authenticated-opening-target-subset-conceptual-default
      [OF clean trace-low composition-low not-all])
have conceptual-hit:
  checked-staged-security-with-query-prefix-conceptual-default-target-hit
    trace-default composition-default out
  using prefix-hit subset
  unfolding out-eq
    checked-staged-security-with-query-prefix-conceptual-default-target-hit-def
    checked-staged-security-with-query-prefix-dynamic-index-hit-def
    checked-staged-query-prefix-dynamic-index-hit-def
  by auto
then show ?thesis by simp
qed
qed

```

lemma *checked-staged-query-prefix-prefix-authenticated-target-hit-bound-from-conceptual:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *conceptual-context:*
 $\bigwedge \text{prefix prefix-state.}$
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) ⇒
 $\neg \text{hash-map-output-collision prefix-state} \wedge$
trace-table-low-degree
(query-prefix-trace-conceptual-table prefix prefix-state) ∧
composition-table-low-degree maxDegree
(query-prefix-composition-conceptual-table prefix prefix-state) ∧

$\neg$ *all-queries-consistent*
 (query-prefix-trace-conceptual-table prefix prefix-state)
 (query-prefix-composition-conceptual-table prefix prefix-state)
 (sqp-alphas prefix)

shows

wp-event (checked-staged-query-prefix-receive-with-state *A i*)
 (checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-prefix-authenticated-opening-target)
adversary-initial-state $\leq$
staged-phase-relation-error size
 (staged-query-search-queries budgets *i + 1*) + *query-error-bound*

proof –

have *raw-bound*: query-index-raw-preimage-bound
by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
have *hit-bound*:
wp-event (checked-staged-query-prefix-receive-with-state *A i*)
 (checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-prefix-authenticated-opening-target)
adversary-initial-state $\leq$
wp-event (checked-staged-query-prefix-receive-with-state *A i*)
 (checked-staged-query-prefix-dynamic-index-prehit
 staged-query-prefix-prefix-authenticated-opening-target)
adversary-initial-state + *query-error-bound*

proof (rule
 checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
 [OF *raw-bound*])

fix *prefix prefix-state*

assume *support*:
 Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
set-dist
 (execute (checked-staged-query-prefix-with-state *A i*)
adversary-initial-state)

have *ctx*:
 $\neg$ *hash-map-output-collision prefix-state* $\wedge$
trace-table-low-degree
 (query-prefix-trace-conceptual-table prefix *prefix-state*) $\wedge$
composition-table-low-degree maxDegree
 (query-prefix-composition-conceptual-table prefix *prefix-state*) $\wedge$
 $\neg$ *all-queries-consistent*
 (query-prefix-trace-conceptual-table prefix *prefix-state*)
 (query-prefix-composition-conceptual-table prefix *prefix-state*)
 (sqp-alphas prefix)

by (rule conceptual-context[OF *support*])

have *subset*:
staged-query-prefix-prefix-authenticated-opening-target prefix
prefix-state $\subseteq$ *query-sample-space*

by (rule staged-query-prefix-prefix-authenticated-opening-target-subset)

have *frac*:
nnreal

```

    (card
      (staged-query-prefix-prefix-authenticated-opening-target prefix
        prefix-state)) /
    nnreal (card query-sample-space) ≤ query-error-bound
using ctx
by (intro
      staged-query-prefix-prefix-authenticated-opening-target-fraction-bound-if-conceptual)
    blast+
show
  staged-query-prefix-prefix-authenticated-opening-target prefix
    prefix-state ⊆ query-sample-space ∧
  nnreal
    (card
      (staged-query-prefix-prefix-authenticated-opening-target prefix
        prefix-state)) /
    nnreal (card query-sample-space) ≤ query-error-bound
using subset frac by blast
qed
have prehit-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit
      staged-query-prefix-prefix-authenticated-opening-target)
  adversary-initial-state ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1)
proof (rule
  checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program)
show HashMap adversary-initial-state = fmempty
  by (simp add: adversary-initial-state-def)
show hash-relation-program
  (checked-staged-query-prefix-dynamic-index-prehit-relation A i
    adversary-initial-state
      staged-query-prefix-prefix-authenticated-opening-target)
  size (staged-query-search-queries budgets i + 1)
  (checked-staged-query-prefix-receive-with-state A i)
by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state
  [OF wf controlled])
  (use i-bound in simp-all,
    rule checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
qed
show ?thesis
  by (rule order-trans[OF hit-bound])
  (intro add-mono prehit-bound order-refl)
qed

end

end

```

theory *Staged-Security-Experiment-Composition-Query-Current-Path*
imports *Staged-Security-Experiment-Composition-Query-Current*
begin

Structured current-query Merkle path-output events.

The broad *checked-staged-security-with-query-prefix-current-path-output-hit-at* event intentionally forgets why the path was relevant. This layer keeps the authenticated-opening root, length, and final-state authentication evidence, so later Merkle/candidate arguments can distinguish ordinary verifier hashing from candidate drift.

context *soundness*
begin

definition *checked-staged-security-with-query-prefix-transcript-hit*
where

checked-staged-security-with-query-prefix-transcript-hit out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ *False*
 | *Some* ((((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) $\Rightarrow$
 hash-map-output-values
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\cap$
 set (*staged-proof-transcript data*) $\neq \{\}$ $\vee$
 hash-map-new-output-hit (*set* (*staged-proof-transcript data*))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*))
 final-state)

definition *checked-staged-security-with-query-prefix-current-trace-path-output-hit-at*
where

checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
i out $\longleftrightarrow$
 (case out of
 None $\Rightarrow$ *False*
 | *Some* ((((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) $\Rightarrow$
 ($\exists$ *opn* :: 'f *authenticated-opening*.
 opening-root opn = *staged-trace-root data* $\wedge$
 opening-length opn = *scale* * *clength* $\wedge$
 authenticated-opening-in final-state opn $\wedge$
 hash-map-new-output-hit
 (*merkle-path-target-roots final-state*
 (*staged-trace-root data*)
 (*scale* * *clength*)
 (*opening-index opn*)
 (*opening-value opn*)
 (*opening-path opn*)
 prefix-state final-state)

definition *checked-staged-security-with-query-prefix-current-composition-path-output-hit-at*
where

checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
 $i \text{ out} \longleftrightarrow$
(case out of
None $\Rightarrow$ False
| Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) $\Rightarrow$
staged-composition-fri-roots data $\neq []$ $\wedge$
 $(\exists \text{ opn} :: \text{'f authenticated-opening.}$
opening-root opn = hd (staged-composition-fri-roots data) $\wedge$
opening-length opn = scale * clength $\wedge$
authenticated-opening-in final-state opn $\wedge$
hash-map-new-output-hit
(merkle-path-target-roots final-state
(hd (staged-composition-fri-roots data))
(scale * clength)
(opening-index opn)
(opening-value opn)
(opening-path opn))
prefix-state final-state))

lemma *checked-staged-security-with-query-prefix-structured-path-hit-imp-transcript-hit-on-support:*

assumes *i-bound: i < rounds*

and *support:*

out $\in$
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)

and *hit:*

checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
i out $\vee$
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
i out

shows *checked-staged-security-with-query-prefix-transcript-hit out*

proof (cases out)

case None

then show ?thesis

using hit

unfolding

checked-staged-security-with-query-prefix-current-trace-path-output-hit-at-def
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at-def

by simp

next

case (Some packed)

then obtain prefix prefix-state raw raw-state data attacker-state result

final-state **where** out-eq:

```

    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  have mapped-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-query-prefix-data-state A i ≫≫
            (λpacked. return (snd packed)))
          adversary-initial-state)
    unfolding out-eq
    by (rule set-dist-bindI[OF support[unfolded out-eq]]) simp
  have projection:
    checked-staged-security-experiment-with-query-prefix-data-state A i ≫≫
      (λpacked. return (snd packed)) =
      checked-staged-security-experiment-with-data-state A
    by (rule checked-staged-security-experiment-with-query-prefix-data-state-projection
      [OF i-bound])
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
    using mapped-support unfolding projection .
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  have verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
    by blast
  have transcript-split:
    hash-map-output-values ?s ∩ set (staged-proof-transcript data)
      ≠ {} ∨
    hash-map-new-output-hit (set (staged-proof-transcript data))
      ?s final-state
    by (rule verify-monad-trace-root-target-preexisting-or-new[OF verifier])
      simp
  show ?thesis
    unfolding out-eq checked-staged-security-with-query-prefix-transcript-hit-def
    using transcript-split by simp
qed

```

```

lemma checked-staged-security-with-query-prefix-transcript-hit-bound-from-data-state:
  assumes i-bound: i < rounds
  and pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state  $\leq P$ 
and new-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state  $\leq R$ 
shows
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    checked-staged-security-with-query-prefix-transcript-hit
    adversary-initial-state  $\leq P + R$ 
proof –
    let ?E =
    λout.
    staged-security-with-data-state-transcript-pre-hit out ∨
    staged-security-with-data-state-transcript-new-hit out
    let ?projected =
    λout. case out of
    None  $\Rightarrow$  ?E None
    | Some (packed, t)  $\Rightarrow$  ?E (Some (snd packed, t))
    have projected-eq:
    ?projected = checked-staged-security-with-query-prefix-transcript-hit
    unfolding
    checked-staged-security-with-query-prefix-transcript-hit-def
    staged-security-with-data-state-transcript-pre-hit-def
    staged-security-with-data-state-transcript-new-hit-def
    by (rule ext) (auto split: option.splits prod.splits)
    have projection:
    wp-event (checked-staged-security-experiment-with-data-state A)
    ?E
    adversary-initial-state =
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    checked-staged-security-with-query-prefix-transcript-hit
    adversary-initial-state
    using
    checked-staged-security-experiment-with-query-prefix-data-state-projection-event
    [OF i-bound, of A ?E]
    unfolding projected-eq .
    have union-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    ?E
    adversary-initial-state  $\leq$ 
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state +
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state

```

```

    by (rule wp-event-union-bound)
  have wp-event (checked-staged-security-experiment-with-data-state A)
    ?E
    adversary-initial-state  $\leq P + R$ 
  by (rule order-trans[OF union-bound])
    (intro add-mono pre-bound new-bound)
  then show ?thesis
    unfolding projection .
qed

lemma checked-staged-security-with-query-prefix-current-trace-path-output-hit-at-bound-from-transcript:
  assumes i-bound:  $i < \text{rounds}$ 
  and pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state  $\leq P$ 
  and new-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state  $\leq R$ 
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
      adversary-initial-state  $\leq P + R$ 
proof -
  have event-le:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
      adversary-initial-state  $\leq$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      checked-staged-security-with-query-prefix-transcript-hit
      adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support:
      out  $\in$ 
      set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
    and hit:
      checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
      i out
  show checked-staged-security-with-query-prefix-transcript-hit out
  by (rule
    checked-staged-security-with-query-prefix-structured-path-hit-imp-transcript-hit-on-support

```

```

      [OF i-bound support])
    (use hit in simp)
qed
have transcript-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    checked-staged-security-with-query-prefix-transcript-hit
    adversary-initial-state ≤ P + R
  by (rule
      checked-staged-security-with-query-prefix-transcript-hit-bound-from-data-state
      [OF i-bound pre-bound new-bound])
show ?thesis
  by (rule order-trans[OF event-le transcript-bound])
qed

lemma checked-staged-security-with-query-prefix-current-composition-path-output-hit-at-bound-from-transcript:
  assumes i-bound: i < rounds
  and pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state ≤ P
  and new-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state ≤ R
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
      adversary-initial-state ≤ P + R
proof -
  have event-le:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
      adversary-initial-state ≤
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      checked-staged-security-with-query-prefix-transcript-hit
      adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support:
      out ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
    and hit:

```



```

    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
    i out
show checked-staged-security-with-query-prefix-transcript-hit out
  by (rule
    checked-staged-security-with-query-prefix-structured-path-hit-imp-transcript-hit-on-support
    [OF i-bound support])
    (use hit in simp)
qed
have transcript-bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  checked-staged-security-with-query-prefix-transcript-hit
  adversary-initial-state  $\leq P + R$ 
  by (rule
    checked-staged-security-with-query-prefix-transcript-hit-bound-from-data-state
    [OF i-bound pre-bound new-bound])
show ?thesis
  by (rule order-trans[OF event-le transcript-bound])
qed

lemma checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-prefix-or-structured-path-or
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i < \text{rounds}$ 
and support:
  out  $\in$ 
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
  i out
shows
  checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
  i out  $\vee$ 
  checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
  i out  $\vee$ 
  checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
  i out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-def
    checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  by simp
next
case (Some packed)

```

```

then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
from hit obtain trace-openings composition-openings where current:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
    trace-openings composition-openings i out
unfolding checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-def
by blast
have ext: prefix-state ≤ final-state
using support unfolding out-eq
by (rule
  checked-staged-security-experiment-with-query-prefix-data-state-prefix-extends-final
  [OF wf controlled i-bound])
have comp-nonempty: staged-composition-fri-roots data ≠ []
and trace-table:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and comp-table:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings final-state
and consistent:
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
using current unfolding out-eq
  checked-staged-security-with-query-prefix-authenticated-opening-hit-def
by simp-all
have trace-pullback:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings prefix-state ∨
    ( $\exists \text{opn} \in \text{set } \text{trace-openings}.$ 
      hash-map-new-output-hit
        (merkle-path-target-roots final-state (staged-trace-root data)
          (scale * clength)
          (opening-index opn)
          (opening-value opn)
          (opening-path opn)) prefix-state final-state)
by (rule partial-authenticated-table-pullback-or-new-output-hit
  [OF ext trace-table])
then show ?thesis
proof
assume trace-prefix:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings prefix-state
have comp-pullback:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings prefix-state ∨

```

```

( $\exists$  opn  $\in$  set composition-openings.
  hash-map-new-output-hit
    (merkle-path-target-roots final-state
      (hd (staged-composition-fri-roots data))
      (scale * clength)
      (opening-index opn)
      (opening-value opn)
      (opening-path opn)) prefix-state final-state)
  by (rule partial-authenticated-table-pullback-or-new-output-hit
    [OF ext comp-table])
then show ?thesis
proof
  assume comp-prefix:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings prefix-state
  have
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
      i out
  unfolding out-eq
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-def
    checked-staged-security-with-query-prefix-prefix-authenticated-opening-hit-def
  using comp-nonempty trace-prefix comp-prefix consistent by auto
  then show ?thesis by simp
next
  assume comp-path:
     $\exists$  opn  $\in$  set composition-openings.
    hash-map-new-output-hit
      (merkle-path-target-roots final-state
        (hd (staged-composition-fri-roots data))
        (scale * clength)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn)) prefix-state final-state
  then obtain opn where opn-in: opn  $\in$  set composition-openings
  and path-hit:
    hash-map-new-output-hit
      (merkle-path-target-roots final-state
        (hd (staged-composition-fri-roots data))
        (scale * clength)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn)) prefix-state final-state
  by blast
  have auth: authenticated-opening-in final-state opn
  using comp-table opn-in
  unfolding partial-authenticated-table-def by blast
  have root-eq:
    opening-root opn = hd (staged-composition-fri-roots data)
  using comp-table opn-in

```

```

    unfolding partial-authenticated-table-def by blast
  have len-eq: opening-length opn = scale * clength
    using comp-table opn-in
    unfolding partial-authenticated-table-def by blast
  have
    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
      i out
    unfolding out-eq
      checked-staged-security-with-query-prefix-current-composition-path-output-hit-at-def
    using comp-nonempty auth root-eq len-eq path-hit by auto
  then show ?thesis by simp
qed
next
assume trace-path:
   $\exists$  opn  $\in$  set trace-openings.
  hash-map-new-output-hit
    (merkle-path-target-roots final-state (staged-trace-root data)
      (scale * clength)
      (opening-index opn)
      (opening-value opn)
      (opening-path opn)) prefix-state final-state
  then obtain opn where opn-in: opn  $\in$  set trace-openings
  and path-hit:
    hash-map-new-output-hit
      (merkle-path-target-roots final-state (staged-trace-root data)
        (scale * clength)
        (opening-index opn)
        (opening-value opn)
        (opening-path opn)) prefix-state final-state
    by blast
  have auth: authenticated-opening-in final-state opn
    using trace-table opn-in
    unfolding partial-authenticated-table-def by blast
  have root-eq: opening-root opn = staged-trace-root data
    using trace-table opn-in
    unfolding partial-authenticated-table-def by blast
  have len-eq: opening-length opn = scale * clength
    using trace-table opn-in
    unfolding partial-authenticated-table-def by blast
  have
    checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
      i out
    unfolding out-eq
      checked-staged-security-with-query-prefix-current-trace-path-output-hit-at-def
    using auth root-eq len-eq path-hit by auto
  then show ?thesis by simp
qed
qed

```

lemma *checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-candidate-pair-or-structured-path-on-*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *support:*
Some ((((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)
and *hit:*
checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i
(Some ((((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))
and *trace-candidate:*
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings])
and *comp-candidate:*
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings])
and *trace-low: trace-table-low-degree trace-table*
and *comp-low: composition-table-low-degree maxDegree composition-table*
and *not-all:*
¬ all-queries-consistent trace-table composition-table
(sqp-alphas prefix)
shows
checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
trace-table composition-table
(Some ((((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
i
(Some ((((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
i
(Some ((((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))
proof –
let *?out =*
Some ((((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)
have *ext: prefix-state ≤ final-state*
by (*rule*
checked-staged-security-experiment-with-query-prefix-data-state-prefix-extends-final
[OF wf controlled i-bound support])
have *alphas-eq: staged-alphas data = sqp-alphas prefix*

```

by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq
  [OF support])
have comp-nonempty: staged-composition-fri-roots data ≠ []
and trace-table-final:
  partial-authenticated-table (staged-trace-root data)
  (scale * clength) trace-openings final-state
and comp-table-final:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
  (scale * clength) composition-openings final-state
and consistent:
  partial-query-openings-consistent trace-openings composition-openings
  (staged-alphas data) (index (to-nat raw))
using hit
unfolding checked-staged-security-with-query-prefix-authenticated-opening-hit-def
by simp-all
have trace-pullback:
  partial-authenticated-table (staged-trace-root data)
  (scale * clength) trace-openings prefix-state ∨
  (∃ opn ∈ set trace-openings.
    hash-map-new-output-hit
    (merkle-path-target-roots final-state (staged-trace-root data)
      (scale * clength) (opening-index opn) (opening-value opn)
      (opening-path opn)) prefix-state final-state)
  by (rule partial-authenticated-table-pullback-or-new-output-hit
    [OF ext trace-table-final])
then show ?thesis
proof
  assume trace-prefix:
    partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings prefix-state
  have comp-pullback:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings prefix-state ∨
    (∃ opn ∈ set composition-openings.
      hash-map-new-output-hit
      (merkle-path-target-roots final-state
        (hd (staged-composition-fri-roots data))
        (scale * clength) (opening-index opn) (opening-value opn)
        (opening-path opn)) prefix-state final-state)
    by (rule partial-authenticated-table-pullback-or-new-output-hit
      [OF ext comp-table-final])
  then show ?thesis
proof
  assume comp-prefix:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) composition-openings prefix-state
  have opening-target:
    index (to-nat raw) ∈
    staged-query-prefix-candidate-opening-query-target-from-prefix

```

```

    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i prefix prefix-state
unfolding staged-query-prefix-candidate-opening-query-target-from-prefix-def
by (rule partial-query-success-indices-atI,
      rule partial-query-round-consistent-single-roundI
      [OF i-bound])
    (use consistent alphas-eq in simp)
have opening-subset:
  staged-query-prefix-candidate-opening-query-target-from-prefix
    ((replicate rounds []) [i := trace-openings])
    ((replicate rounds []) [i := composition-openings])
    i prefix prefix-state  $\subseteq$ 
  staged-query-prefix-candidate-pair-query-target-from-prefix
    trace-table composition-table prefix prefix-state
by (rule
      staged-query-prefix-candidate-opening-query-target-from-prefix-subset-candidate-pair-from-prefix
      [OF trace-candidate comp-candidate trace-low comp-low not-all])
have pair-hit:
  checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
    trace-table composition-table ?out
unfolding
  checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-def
  checked-staged-security-with-query-prefix-dynamic-index-hit-def
  checked-staged-query-prefix-dynamic-index-hit-def
using opening-target opening-subset by auto
then show ?thesis by simp
next
assume comp-path:
   $\exists \text{opn} \in \text{set composition-openings}.$ 
  hash-map-new-output-hit
    (merkle-path-target-roots final-state
      (hd (staged-composition-fri-roots data))
      (scale * clength) (opening-index opn) (opening-value opn)
      (opening-path opn)) prefix-state final-state
then obtain opn where opn-in:  $\text{opn} \in \text{set composition-openings}$ 
and path-hit:
  hash-map-new-output-hit
    (merkle-path-target-roots final-state
      (hd (staged-composition-fri-roots data))
      (scale * clength) (opening-index opn) (opening-value opn)
      (opening-path opn)) prefix-state final-state
by blast
have auth: authenticated-opening-in final-state opn
using comp-table-final opn-in
unfolding partial-authenticated-table-def by blast
have root-eq:
  opening-root opn = hd (staged-composition-fri-roots data)
using comp-table-final opn-in

```

```

    unfolding partial-authenticated-table-def by blast
  have len-eq: opening-length opn = scale * clength
    using comp-table-final opn-in
    unfolding partial-authenticated-table-def by blast
  have
    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
      i ?out
    unfolding
      checked-staged-security-with-query-prefix-current-composition-path-output-hit-at-def
    using comp-nonempty auth root-eq len-eq path-hit by auto
  then show ?thesis by simp
qed
next
assume trace-path:
   $\exists$  opn  $\in$  set trace-openings.
  hash-map-new-output-hit
    (merkle-path-target-roots final-state (staged-trace-root data)
      (scale * clength) (opening-index opn) (opening-value opn)
      (opening-path opn)) prefix-state final-state
  then obtain opn where opn-in: opn  $\in$  set trace-openings
  and path-hit:
    hash-map-new-output-hit
      (merkle-path-target-roots final-state (staged-trace-root data)
        (scale * clength) (opening-index opn) (opening-value opn)
        (opening-path opn)) prefix-state final-state
  by blast
  have auth: authenticated-opening-in final-state opn
    using trace-table-final opn-in
    unfolding partial-authenticated-table-def by blast
  have root-eq: opening-root opn = staged-trace-root data
    using trace-table-final opn-in
    unfolding partial-authenticated-table-def by blast
  have len-eq: opening-length opn = scale * clength
    using trace-table-final opn-in
    unfolding partial-authenticated-table-def by blast
  have
    checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
      i ?out
    unfolding
      checked-staged-security-with-query-prefix-current-trace-path-output-hit-at-def
    using auth root-eq len-eq path-hit by auto
  then show ?thesis by simp
qed
qed

definition staged-query-prefix-inconsistent-candidate-pair-query-target
where
  staged-query-prefix-inconsistent-candidate-pair-query-target
    trace-table composition-table prefix prefix-state =

```



```

    (if all-queries-consistent trace-table composition-table
      (sqp-alphas prefix)
      then {}
      else staged-query-prefix-candidate-pair-query-target-from-prefix
           trace-table composition-table prefix prefix-state)

lemma staged-query-prefix-inconsistent-candidate-pair-query-target-bound:
assumes trace-low: trace-table-low-degree trace-table
and comp-low:
  composition-table-low-degree maxDegree composition-table
shows
  staged-query-prefix-inconsistent-candidate-pair-query-target
    trace-table composition-table prefix prefix-state
     $\subseteq$  query-sample-space  $\wedge$ 
    nnreal
      (card
        (staged-query-prefix-inconsistent-candidate-pair-query-target
          trace-table composition-table prefix prefix-state)) /
      nnreal (card query-sample-space)  $\leq$  query-error-bound
proof (cases
  all-queries-consistent trace-table composition-table (sqp-alphas prefix))
case True
then show ?thesis
  unfolding staged-query-prefix-inconsistent-candidate-pair-query-target-def
  by simp
next
case False
have subset:
  staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
    composition-table prefix prefix-state  $\subseteq$  query-sample-space
by (rule staged-query-prefix-candidate-pair-query-target-from-prefix-subset)
have frac:
  nnreal
    (card
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table prefix prefix-state)) /
    nnreal (card query-sample-space)  $\leq$  query-error-bound
by (rule
  staged-query-prefix-candidate-pair-query-target-from-prefix-fraction-bound
  [OF trace-low comp-low False])
show ?thesis
  unfolding staged-query-prefix-inconsistent-candidate-pair-query-target-def
  using False subset frac by simp
qed

lemma checked-staged-security-with-query-prefix-inconsistent-candidate-pair-hit-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i < \text{rounds}$ 

```

and *trace-low*: *trace-table-low-degree trace-table*
and *comp-low*:
composition-table-low-degree maxDegree composition-table
shows
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-inconsistent-candidate-pair-query-target
trace-table composition-table))
adversary-initial-state $\leq$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) + query-error-bound
by (*rule*
checked-staged-security-with-query-prefix-dynamic-index-hit-bound-from-fraction
[OF wf controlled i-bound])
(use
staged-query-prefix-inconsistent-candidate-pair-query-target-bound
[OF trace-low comp-low] in blast)

lemma *checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-imp-inconsistent-target*:

assumes *hit*:
checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
trace-table composition-table
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))
and *not-all*:
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)

shows

checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-inconsistent-candidate-pair-query-target
trace-table composition-table)
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

using *hit not-all*

unfolding

checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-def
checked-staged-security-with-query-prefix-dynamic-index-hit-def
checked-staged-query-prefix-dynamic-index-hit-def
staged-query-prefix-inconsistent-candidate-pair-query-target-def

by *simp*

lemma *checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-candidate-pair-and-structured*

assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: *i < rounds*
and *trace-candidate*:
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings])

and *comp-candidate*:
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings])
and *trace-low*: *trace-table-low-degree trace-table*
and *comp-low*: *composition-table-low-degree maxDegree composition-table*
and *not-all*:
 $\bigwedge \text{prefix prefix-state raw raw-state data attacker-state result}$
final-state.
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state
A i)
adversary-initial-state) ⇒
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)
and *pair-bound*:
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
trace-table composition-table)
adversary-initial-state ≤ P
and *trace-path-bound*:
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
i)
adversary-initial-state ≤ T
and *composition-path-bound*:
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
i)
adversary-initial-state ≤ C
shows
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i)
adversary-initial-state ≤ P + T + C
proof –
let *?M* =
checked-staged-security-experiment-with-query-prefix-data-state A i
let *?Pair* =
checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
trace-table composition-table
let *?Trace* =
checked-staged-security-with-query-prefix-current-trace-path-output-hit-at

```

i
let ?Comp =
  checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
  i
have event-le:
  wp-event ?M
    (checked-staged-security-with-query-prefix-authenticated-opening-hit
     trace-openings composition-openings i)
  adversary-initial-state ≤
  wp-event ?M (λout. ?Pair out ∨ ?Trace out ∨ ?Comp out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈ set-dist (execute ?M adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
  trace-openings composition-openings i out
show ?Pair out ∨ ?Trace out ∨ ?Comp out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
             ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
have support-some:
  Some (((((prefix, prefix-state), raw), raw-state),
           ((data, attacker-state), result)), final-state) ∈
  set-dist (execute ?M adversary-initial-state)
using support out-eq by simp
show ?thesis
unfolding out-eq
by (rule
  checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-candidate-pair-or-structured-path-on-s
  [OF wf controlled i-bound support-some])
  (use hit[unfolded out-eq] trace-candidate comp-candidate
   trace-low comp-low not-all[OF support-some] in simp-all)
qed
qed
have union-le:

```

```

wp-event ?M ( $\lambda out. ?Pair\ out \vee ?Trace\ out \vee ?Comp\ out$ )
  adversary-initial-state  $\leq$ 
wp-event ?M ?Pair adversary-initial-state +
  (wp-event ?M ?Trace adversary-initial-state +
    wp-event ?M ?Comp adversary-initial-state)
proof –
  have
    wp-event ?M ( $\lambda out. ?Pair\ out \vee ?Trace\ out \vee ?Comp\ out$ )
      adversary-initial-state  $\leq$ 
      wp-event ?M ?Pair adversary-initial-state +
      wp-event ?M ( $\lambda out. ?Trace\ out \vee ?Comp\ out$ )
        adversary-initial-state
    by (rule wp-event-union-bound)
  also have
    ...  $\leq$ 
    wp-event ?M ?Pair adversary-initial-state +
    (wp-event ?M ?Trace adversary-initial-state +
      wp-event ?M ?Comp adversary-initial-state)
    by (intro add-mono order-refl wp-event-union-bound)
  finally show ?thesis .
qed
have sum-bound:
  wp-event ?M ?Pair adversary-initial-state +
    (wp-event ?M ?Trace adversary-initial-state +
      wp-event ?M ?Comp adversary-initial-state)  $\leq P + T + C$ 
proof –
  have path-sum:
    wp-event ?M ?Trace adversary-initial-state +
    wp-event ?M ?Comp adversary-initial-state  $\leq T + C$ 
    by (rule add-mono[OF trace-path-bound composition-path-bound])
  have
    wp-event ?M ?Pair adversary-initial-state +
    (wp-event ?M ?Trace adversary-initial-state +
      wp-event ?M ?Comp adversary-initial-state)  $\leq P + (T + C)$ 
    by (rule add-mono[OF pair-bound path-sum])
  then show ?thesis
    by (simp add: add.assoc)
qed
show ?thesis
  by (rule order-trans[OF event-le order-trans[OF union-le sum-bound]])
qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-inconsistent-candidate-pair-an
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i < rounds$ 
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])

```

and *comp-candidate*:
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings])
and *trace-low*: *trace-table-low-degree trace-table*
and *comp-low*: *composition-table-low-degree maxDegree composition-table*
and *not-all-on-hit*:
 $\bigwedge$ *prefix prefix-state raw raw-state data attacker-state result*
final-state.
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state
A i)
adversary-initial-state) ⇒
checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)) ⇒
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)
and *trace-path-bound*:
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
i)
adversary-initial-state ≤ T
and *composition-path-bound*:
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
i)
adversary-initial-state ≤ C
shows
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound + T + C
proof –
let *?M* =
checked-staged-security-experiment-with-query-prefix-data-state A i
let *?Pair* =
checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-inconsistent-candidate-pair-query-target
trace-table composition-table)

```

let ?Trace =
  checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
  i
let ?Comp =
  checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
  i
have event-le:
  wp-event ?M
  (checked-staged-security-with-query-prefix-authenticated-opening-hit
   trace-openings composition-openings i)
  adversary-initial-state ≤
  wp-event ?M (λout. ?Pair out ∨ ?Trace out ∨ ?Comp out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈ set-dist (execute ?M adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
  trace-openings composition-openings i out
show ?Pair out ∨ ?Trace out ∨ ?Comp out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
have support-some:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist (execute ?M adversary-initial-state)
  using support out-eq by simp
have not-all:
  ¬ all-queries-consistent trace-table composition-table
  (sqp-alphas prefix)
  by (rule not-all-on-hit[OF support-some])
  (use hit out-eq in simp)
have split:
  checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
  trace-table composition-table

```

```

    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
  i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
  i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
by (rule
checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-candidate-pair-or-structured-path-on-s
  [OF wf controlled i-bound support-some])
  (use hit[unfolded out-eq] trace-candidate comp-candidate
    trace-low comp-low not-all in simp-all)
then show ?thesis
proof
  assume pair:
    checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
    trace-table composition-table
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  have ?Pair
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  by (rule
checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-imp-inconsistent-target
  [OF pair not-all])
  then show ?thesis
    unfolding out-eq by simp
next
  assume
    checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
    i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
    i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  then show ?thesis
    unfolding out-eq by simp
qed
qed
qed
have pair-bound:
  wp-event ?M ?Pair adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound

```


by (rule
 checked-staged-security-with-query-prefix-inconsistent-candidate-pair-hit-bound
 [*OF wf controlled i-bound trace-low comp-low*])

have *union-le*:
 $wp\text{-}event\ ?M\ (\lambda out. ?Pair\ out \vee ?Trace\ out \vee ?Comp\ out)$
 $adversary\text{-}initial\text{-}state \leq$
 $wp\text{-}event\ ?M\ ?Pair\ adversary\text{-}initial\text{-}state +$
 $(wp\text{-}event\ ?M\ ?Trace\ adversary\text{-}initial\text{-}state +$
 $wp\text{-}event\ ?M\ ?Comp\ adversary\text{-}initial\text{-}state)$

proof –
 have
 $wp\text{-}event\ ?M\ (\lambda out. ?Pair\ out \vee ?Trace\ out \vee ?Comp\ out)$
 $adversary\text{-}initial\text{-}state \leq$
 $wp\text{-}event\ ?M\ ?Pair\ adversary\text{-}initial\text{-}state +$
 $wp\text{-}event\ ?M\ (\lambda out. ?Trace\ out \vee ?Comp\ out)$
 $adversary\text{-}initial\text{-}state$
 by (rule *wp-event-union-bound*)

also have
 $\dots \leq$
 $wp\text{-}event\ ?M\ ?Pair\ adversary\text{-}initial\text{-}state +$
 $(wp\text{-}event\ ?M\ ?Trace\ adversary\text{-}initial\text{-}state +$
 $wp\text{-}event\ ?M\ ?Comp\ adversary\text{-}initial\text{-}state)$
 by (intro *add-mono order-refl wp-event-union-bound*)

finally show *?thesis* .

qed

have *sum-bound*:
 $wp\text{-}event\ ?M\ ?Pair\ adversary\text{-}initial\text{-}state +$
 $(wp\text{-}event\ ?M\ ?Trace\ adversary\text{-}initial\text{-}state +$
 $wp\text{-}event\ ?M\ ?Comp\ adversary\text{-}initial\text{-}state) \leq$
 $(staged\text{-}phase\text{-}relation\text{-}error\ size$
 $(staged\text{-}query\text{-}search\text{-}queries\ budgets\ i + 1) + query\text{-}error\text{-}bound) +$
 $(T + C)$
 by (intro *add-mono pair-bound add-mono trace-path-bound*
 composition-path-bound)

have *rhs-eq*:
 $(staged\text{-}phase\text{-}relation\text{-}error\ size$
 $(staged\text{-}query\text{-}search\text{-}queries\ budgets\ i + 1) + query\text{-}error\text{-}bound) +$
 $(T + C) =$
 $staged\text{-}phase\text{-}relation\text{-}error\ size$
 $(staged\text{-}query\text{-}search\text{-}queries\ budgets\ i + 1) +$
 $query\text{-}error\text{-}bound + T + C$
 by (*simp add: add.assoc*)

have *bound*:
 $wp\text{-}event\ ?M$
 $(checked\text{-}staged\text{-}security\text{-}with\text{-}query\text{-}prefix\text{-}authenticated\text{-}opening\text{-}hit$
 $trace\text{-}openings\ composition\text{-}openings\ i)$
 $adversary\text{-}initial\text{-}state \leq$
 $(staged\text{-}phase\text{-}relation\text{-}error\ size$
 $(staged\text{-}query\text{-}search\text{-}queries\ budgets\ i + 1) + query\text{-}error\text{-}bound) +$

```

    (T + C)
  by (rule order-trans[OF event-le order-trans[OF union-le sum-bound]])
show ?thesis
  using bound rhs-eq by simp
qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-candidate-pair-budgets-and-st
assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and trace-candidate:
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  and comp-candidate:
    partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
  and trace-low: trace-table-low-degree trace-table
  and comp-low: composition-table-low-degree maxDegree composition-table
  and not-all:
     $\bigwedge \text{prefix prefix-state.}$ 
    Some ((prefix, prefix-state), prefix-state)  $\in$ 
      set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)  $\implies$ 
       $\neg$  all-queries-consistent trace-table composition-table
      (sqp-alphas prefix)
  and trace-path-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
      i)
    adversary-initial-state  $\leq T$ 
  and composition-path-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
      i)
    adversary-initial-state  $\leq C$ 
shows
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-authenticated-opening-hit
    trace-openings composition-openings i)
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
  query-error-bound + T + C
proof -
  have pair-bound:

```

```

wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
    trace-table composition-table)
  adversary-initial-state ≤
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) + query-error-bound
by (rule
  checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-bound-from-budgets
  [OF wf controlled i-bound trace-low comp-low not-all])
have not-all-support:
  ∧ prefix prefix-state raw raw-state data attacker-state result
  final-state.
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state
      A i)
    adversary-initial-state) ⇒
  ¬ all-queries-consistent trace-table composition-table
  (sqp-alphas prefix)
proof -
  fix prefix prefix-state raw raw-state data attacker-state result final-state
  assume support:
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state
        A i)
      adversary-initial-state)
  have head:
    Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A i)
      adversary-initial-state)
  using support
  unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
  by (auto elim!: set-dist-bindE split: prod.splits)
  have prefix-support:
    Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state)
  by (rule checked-staged-query-prefix-receive-with-state-prefix-support
    [OF head])
  show
    ¬ all-queries-consistent trace-table composition-table

```

```

      (sqp-alphas prefix)
    by (rule not-all[OF prefix-support])
qed
show ?thesis
by (rule
  checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-candidate-pair-and-structured-pa
  [OF wf controlled i-bound trace-candidate comp-candidate trace-low
    comp-low not-all-support pair-bound trace-path-bound
    composition-path-bound])
qed

lemma checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-structur
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and prefix-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
    adversary-initial-state ≤ P
and trace-path-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
    adversary-initial-state ≤ T
and composition-path-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
    adversary-initial-state ≤ C
shows
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state ≤ P + T + C
proof -
let ?M = checked-staged-security-experiment-with-query-prefix-data-state A i
let ?Prefix =
  checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i
let ?Trace =
  checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i
let ?Composition =
  checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i
have event-le:
  wp-event ?M
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state ≤
  wp-event ?M (λout. ?Prefix out ∨ ?Trace out ∨ ?Composition out)
    adversary-initial-state

```

by (rule *wp-event-mono-on-support*)
 (rule
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-prefix-or-structured-path-on-su
[OF wf controlled i-bound])
have *first-union*:
wp-event ?M (λout. ?Prefix out ∨ ?Trace out ∨ ?Composition out)
adversary-initial-state ≤
wp-event ?M ?Prefix adversary-initial-state +
wp-event ?M (λout. ?Trace out ∨ ?Composition out)
adversary-initial-state
by (rule *wp-event-union-bound*)
have *second-union*:
wp-event ?M (λout. ?Trace out ∨ ?Composition out)
adversary-initial-state ≤
wp-event ?M ?Trace adversary-initial-state +
wp-event ?M ?Composition adversary-initial-state
by (rule *wp-event-union-bound*)
have *path-sum-bound*:
wp-event ?M (λout. ?Trace out ∨ ?Composition out)
adversary-initial-state ≤ T + C
by (rule *order-trans[OF second-union]*)
 (intro *add-mono trace-path-bound composition-path-bound*)
have *sum-bound*:
wp-event ?M ?Prefix adversary-initial-state +
wp-event ?M (λout. ?Trace out ∨ ?Composition out)
adversary-initial-state ≤ P + (T + C)
by (intro *add-mono prefix-bound path-sum-bound*)
have *union-sum-bound*:
wp-event ?M (λout. ?Prefix out ∨ ?Trace out ∨ ?Composition out)
adversary-initial-state ≤ P + T + C
by (rule *order-trans[OF first-union]*)
 (use *sum-bound* in ⟨*simp add: add.assoc*⟩)
show *?thesis*
by (rule *order-trans[OF event-le]*)
 (rule *union-sum-bound*)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-prefix-target*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *prefix-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event (checked-staged-query-prefix-receive-with-state A i)
 (checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-prefix-authenticated-opening-target)
adversary-initial-state ≤ P i
and *trace-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 adversary-initial-state $\leq T$ i
and composition-path-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 adversary-initial-state $\leq C$ i
shows
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 adversary-initial-state $\leq (\sum_{i < \text{rounds}} P$ $i + T$ $i + C$ $i)$
proof (rule
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix-curr
 [OF wf controlled])
fix i
assume i -bound: $i < \text{rounds}$
have prefix-hit-bound:
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
 adversary-initial-state $\leq P$ i
by (rule
 checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target
 [OF wf controlled i -bound prefix-bound[OF i -bound]])
show
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state $\leq P$ $i + T$ $i + C$ i
by (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-structured-p
 [OF wf controlled i -bound prefix-hit-bound trace-path-bound[OF i -bound]
 composition-path-bound[OF i -bound]])
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-trace-indic
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fac:
 $\bigwedge i$ prefix prefix-state.
 $i < \text{rounds} \implies$
 Some ((prefix, prefix-state), prefix-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-with-state A i)
 adversary-initial-state) $\implies$
 nnreal

```

    (card
      (staged-query-prefix-prefix-authenticated-trace-indices prefix
        prefix-state)) /
    nnreal (card query-sample-space) ≤ query-error-bound
and trace-path-bound:
  ∧ i. i < rounds ⇒
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
      adversary-initial-state ≤ T i
and composition-path-bound:
  ∧ i. i < rounds ⇒
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
      adversary-initial-state ≤ C i
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    adversary-initial-state ≤
      (∑ i < rounds.
        staged-phase-relation-error size
          (staged-query-search-queries budgets i + 1) +
          query-error-bound + T i + C i)
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-prefix-target-and-
  [OF wf controlled - trace-path-bound composition-path-bound])
fix i
assume i-bound: i < rounds
show wp-event (checked-staged-query-prefix-receive-with-state A i)
  (checked-staged-query-prefix-dynamic-index-hit
    staged-query-prefix-prefix-authenticated-opening-target)
  adversary-initial-state ≤
    staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) + query-error-bound
proof (rule
  checked-staged-query-prefix-prefix-authenticated-target-hit-bound-from-trace-indices
  [OF wf controlled i-bound])
fix prefix prefix-state
assume support:
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)
show
  nnreal
    (card
      (staged-query-prefix-prefix-authenticated-trace-indices prefix

```

$\text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
by (rule trace-frac[OF i-bound support])
qed
qed

lemma *checked-staged-security-with-data-state-composition-bad-bound-from-trace-indices-structured-paths-and-budget*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-frac:*
 $\bigwedge i \text{ prefix prefix-state.}$
 $i < \text{rounds} \implies$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \implies$
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-prefix-authenticated-trace-indices prefix}$
 $\text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
and *trace-path-bound:*
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T \ i$
and *composition-path-bound:*
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event composition-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\sum i < \text{rounds.}$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} + T \ i + C \ i) +$
 $\text{hash-collision-budget-value } 0$

(staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)
proof –
 let ?Q =
 (∑ i < rounds.
 staged-phase-relation-error size
 (staged-query-search-queries budgets i + 1) +
 query-error-bound + T i + C i)
have query-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 adversary-initial-state ≤ ?Q
by (rule
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-trace-indices-a
 [OF wf controlled trace-frac trace-path-bound composition-path-bound])
have without-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit
 adversary-initial-state ≤ 0
by (rule
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-without-query-hit-bound
 [OF wf controlled])
have drift-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift
 adversary-initial-state ≤ ?Q + 0
by (rule
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-bound-from-query-and-without-query
 [OF query-bound without-bound])
have comp-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event composition-bad)
 adversary-initial-state ≤
 composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (?Q + 0) +
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
 checked-staged-security-with-data-state-composition-bad-bound-from-composition-candidate-drift-and-budgets

```

    [OF wf controlled drift-bound])
  then show ?thesis
    by simp
qed

end

end

```

```

theory Soundness-Conceptual-Query-Empty-Current
imports
  Soundness-Query-Opening-Consistency
  Soundness-Conceptual-Query-Current
  Staged-Security-Experiment-Composition-Query-Current-Path
  Soundness-Staged-Bounds
begin

```

Empty-composition current-query projection.

This layer connects the verifier-local empty-composition query extraction to the prefix-fixed conceptual empty query target infrastructure. It is kept separate from the non-empty current-query layer to avoid growing the already size-sensitive staged/query theories.

```

context soundness
begin

```

definition staged-security-with-data-state-current-empty-query-opening-hit-at

```

:: nat  $\Rightarrow$ 
  (((f staged-proof-data  $\times$  'f protocol-channel)  $\times$  unit list)  $\times$ 
   'f protocol-channel) option  $\Rightarrow$  bool

```

where

```

staged-security-with-data-state-current-empty-query-opening-hit-at i out
 $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some (((data, attacker-state), result), final-state)  $\Rightarrow$ 
    (let s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    in
      ( $\exists$  raw trace-openings rest.
        i < rounds  $\wedge$ 
        fmlookup (HashMap final-state)
          (QueryIndexChallenge i
            (state-after-query-chunks
              (staged-query-start-hash data)
              (staged-query-chunks data) i)) =
          Some raw  $\wedge$ 
        staged-composition-fri-roots data = []  $\wedge$ 

```

Some (*result*, *final-state*) ∈
set-dist (*execute verify-monad s*) ∧
partial-authenticated-table (*staged-trace-root data*)
*(scale * clength) trace-openings final-state* ∧
verifier-header-transcript s
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
 []
(staged-composition-final data)
rest ∧
map opening-index trace-openings =
powers-scaled (index (to-nat raw)) ∧
cp-eval (staged-alphas data) (map opening-value trace-openings)
*(h ^ index (to-nat raw) * shift)* =
staged-composition-final data)))

definition *staged-security-with-data-state-current-empty-query-opening-hit*
 :: (((*f staged-proof-data* × *f protocol-channel*) × *unit list*) ×
f protocol-channel) *option* ⇒ *bool*

where

staged-security-with-data-state-current-empty-query-opening-hit out
 ⇔
 (∃ *i* < *rounds*.
staged-security-with-data-state-current-empty-query-opening-hit-at
i out)

lemma *staged-security-with-data-state-current-empty-query-opening-hitI*:
assumes *i* < *rounds*
and *staged-security-with-data-state-current-empty-query-opening-hit-at*
i out
shows *staged-security-with-data-state-current-empty-query-opening-hit out*
using *assms*
unfolding *staged-security-with-data-state-current-empty-query-opening-hit-def*
by *blast*

lemma *staged-security-with-data-state-current-empty-query-opening-hit-atI*:
assumes *i-bound: i* < *rounds*
and *lookup:*
fmlookup (HashMap final-state)
(QueryIndexChallenge i
(state-after-query-chunks
(staged-query-start-hash data)
(staged-query-chunks data) i)) =
Some raw
and *comp-empty: staged-composition-fri-roots data* = []
and *verifier:*

```

Some (result, final-state) ∈
  set-dist
    (execute verify-monad
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))))
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and header:
  verifier-header-transcript
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
  []
  (staged-composition-final data)
  rest
and trace-indices:
  map opening-index trace-openings =
    powers-scaled (index (to-nat raw))
and consistent:
  cp-eval (staged-alphas data) (map opening-value trace-openings)
    (h ^ index (to-nat raw) * shift) =
      staged-composition-final data
shows
  staged-security-with-data-state-current-empty-query-opening-hit-at i
    (Some (((data, attacker-state), result), final-state))
unfolding staged-security-with-data-state-current-empty-query-opening-hit-at-def
  Let-def
apply simp
by (intro exI[of - raw] exI[of - trace-openings] exI[of - rest] conjI)
  (use assms in simp-all)

lemma empty-query-bad-header-alignment-on-support:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
and bad:
  staged-security-with-data-state-verifier-event
    query-bad-with-empty-composition-header-candidates
    (Some (((data, attacker-state), result), final-state))

```

```

obtains fr f-fri-roots f-final as dg final rest where
  verifier-header-transcript
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  fr f-fri-roots f-final as dg [] final rest
  fr = staged-trace-root data
  f-fri-roots = staged-trace-fri-roots data
  f-final = staged-trace-final data
  as = staged-alphas data
  dg = staged-degree data
  staged-composition-fri-roots data = []
  final = staged-composition-final data
  rest = List.concat (staged-query-chunks data)
  verifier-header-state
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  fr f-fri-roots f-final as dg [] final =
    staged-query-start-hash data
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF support]
  obtain builder where builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  by blast
  have staged-header:
    verifier-header-transcript ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
      (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
  have bad-verifier:
    query-bad-with-empty-composition-header-candidates ?s
      (Some (result, final-state))
  using bad
  unfolding staged-security-with-data-state-verifier-event-def
  by simp
  from bad-verifier obtain fr f-fri-roots f-final as dg final

```

```

    trace-query-idxs trace-openings trace-table composition-table
where partial:
    accepted-with-empty-composition-header-candidates ?s
      (Some (result, final-state)) fr f-fri-roots f-final as dg final
      trace-query-idxs trace-openings trace-table composition-table
    unfolding query-bad-with-empty-composition-header-candidates-def
    by blast
from partial obtain rest where header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg []
    final rest
    unfolding accepted-with-empty-composition-header-candidates-def
    by blast
have header-eq:
    fr = staged-trace-root data ∧
    f-fri-roots = staged-trace-fri-roots data ∧
    f-final = staged-trace-final data ∧
    as = staged-alphas data ∧
    dg = staged-degree data ∧
    [] = staged-composition-fri-roots data ∧
    final = staged-composition-final data ∧
    rest = List.concat (staged-query-chunks data)
    using verifier-header-transcript-unique[OF header staged-header]
    by simp
have header-state:
    verifier-header-state ?s fr f-fri-roots f-final as dg [] final =
      staged-query-start-hash data
    using header-eq
    unfolding verifier-header-state-def verifier-header-messages-def
      staged-query-start-hash-def staged-composition-fri-start-hash-def
      staged-trace-fri-start-hash-def
    by simp
show ?thesis
    by (rule that[OF header])
      (use header-eq header-state in simp-all)
qed

```

```

lemma empty-query-bad-query-rounds-on-support:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
and bad:
  staged-security-with-data-state-verifier-event
    query-bad-with-empty-composition-header-candidates
    (Some (((data, attacker-state), result), final-state))

```

obtains *f-fl query-state* **where**
PState query-state = *staged-query-start-hash data*
PQueryCounter query-state = 0
Some (result, final-state) ∈
set-dist
 (execute
 (ntimes
 (verifier-query-round-program
 (staged-trace-root data)
f-fl
 (staged-trace-final data)
 (staged-alphas data)
 []
 (staged-composition-final data))
 rounds)
query-state)

proof –
let *?s* =
verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)
from *checked-staged-security-experiment-with-data-state-outcomeE*
 [OF support]
have *verifier*:
Some (result, final-state) ∈ *set-dist (execute verify-monad ?s)*
by *blast*
from *empty-query-bad-header-alignment-on-support*
 [OF *wf controlled support bad*]
obtain *fr f-fri-roots f-final as dg final rest* **where** *header*:
verifier-header-transcript ?s fr f-fri-roots f-final as dg []
final rest
and *fr-eq*: *fr* = *staged-trace-root data*
and *f-final-eq*: *f-final* = *staged-trace-final data*
and *as-eq*: *as* = *staged-alphas data*
and *final-eq*: *final* = *staged-composition-final data*
and *header-state*:
verifier-header-state ?s fr f-fri-roots f-final as dg [] final =
staged-query-start-hash data
by *blast*
from *verify-monad-supplied-empty-header-query-rounds*[OF *verifier header*]
obtain *f-fl query-state* **where**
query-state-state:
PState query-state =
verifier-header-state ?s fr f-fri-roots f-final as dg [] final
and *query-counter*: *PQueryCounter query-state* = *PQueryCounter ?s*
and *query-out*:
Some (result, final-state) ∈
set-dist
 (execute
 (ntimes

```

      (verifier-query-round-program fr f-fl f-final as [] final)
      rounds)
    query-state)
  by blast
show ?thesis
  by (rule that[of query-state f-fl])
    (use query-state-state query-counter query-out header-state fr-eq
      f-final-eq as-eq final-eq in simp-all)
qed

lemma staged-security-with-data-state-current-empty-query-opening-hit-at-imp-query-prefix-current-on-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and support:
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
  and hit:
    staged-security-with-data-state-current-empty-query-opening-hit-at i
    (Some (((data, attacker-state), result), final-state))
  shows
    checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at
    i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have hit':
    ∃ raw' trace-openings rest.
      i < rounds ∧
      fmlookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some raw' ∧
      staged-composition-fri-roots data = [] ∧
      Some (result, final-state) ∈ set-dist (execute verify-monad ?s) ∧
      partial-authenticated-table (staged-trace-root data)
        (scale * clength) trace-openings final-state ∧
      verifier-header-transcript ?s
        (staged-trace-root data)
        (staged-trace-fri-roots data)

```



```

    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    []
    (staged-composition-final data)
    rest ∧
    map opening-index trace-openings =
      powers-scaled (index (to-nat raw')) ∧
    cp-eval (staged-alphas data) (map opening-value trace-openings)
    (h ^ index (to-nat raw') * shift) =
      staged-composition-final data
using hit
unfolding
  staged-security-with-data-state-current-empty-query-opening-hit-at-def
  Let-def
by simp
from hit' obtain raw' trace-openings rest where
  final-lookup':
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some raw'
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and trace-indices:
  map opening-index trace-openings =
    powers-scaled (index (to-nat raw'))
and consistent:
  cp-eval (staged-alphas data) (map opening-value trace-openings)
    (h ^ index (to-nat raw') * shift) =
      staged-composition-final data
by blast
have final-lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
by (rule
  checked-staged-security-experiment-with-query-prefix-data-state-raw-final-lookup
  [OF wf controlled i-bound support])
have raw-eq: raw' = raw
using final-lookup' final-lookup by simp
from support obtain chunk chunk-state assert-state record-state suffix-chunks
where data-eq:
  data =

```

```

(staged-trace-root = sqp-trace-root prefix,
 staged-trace-fri-roots = sqp-trace-fri-roots prefix,
 staged-trace-fri-challenges = sqp-trace-fri-challenges prefix,
 staged-trace-final = sqp-trace-final prefix,
 staged-alphas = sqp-alphas prefix,
 staged-degree = sqp-degree prefix,
 staged-composition-fri-roots = sqp-composition-fri-roots prefix,
 staged-composition-fri-challenges =
   sqp-composition-fri-challenges prefix,
 staged-composition-final = sqp-composition-final prefix,
 staged-query-chunks = sqp-query-chunks prefix @ chunk # suffix-chunks)
unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
checked-staged-after-query-prefix-receive-with-verifier-def
checked-staged-after-query-prefix-receive-def
by (auto elim!: set-dist-bindE split: prod.splits)
have current:
checked-staged-security-with-query-prefix-empty-authenticated-opening-hit
trace-openings i
(Some ((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state)
unfolding
checked-staged-security-with-query-prefix-empty-authenticated-opening-hit-def
using trace-auth trace-indices consistent raw-eq data-eq by simp
show ?thesis
unfolding
checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at-def
using current by blast
qed

lemma checked-staged-security-with-data-state-current-empty-query-opening-hit-at-le-query-prefix-current:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
shows
wp-event
(checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at i)
adversary-initial-state ≤
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at i)
adversary-initial-state
proof –
let ?projected =
λout. case out of
  None ⇒
    staged-security-with-data-state-current-empty-query-opening-hit-at
      i None
  | Some (packed, t) ⇒

```

```

    staged-security-with-data-state-current-empty-query-opening-hit-at
    i (Some (snd packed, t))
have projection:
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-empty-query-opening-hit-at i)
    adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    ?projected
    adversary-initial-state
by (rule
    checked-staged-security-experiment-with-query-prefix-data-state-projection-event
    [OF i-bound])
have event-le:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    ?projected
    adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at i)
    adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
and hit: ?projected out
show
  checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at
  i out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    staged-security-with-data-state-current-empty-query-opening-hit-at-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)

```

```

    by (cases packed) (auto split: prod.splits)
show ?thesis
  unfolding out-eq
  by (rule
    staged-security-with-data-state-current-empty-query-opening-hit-at-imp-query-prefix-current-on-support
      [OF wf controlled i-bound])
    (use support hit out-eq in simp-all)
qed
qed
show ?thesis
  unfolding projection
  by (rule event-le)
qed

lemma empty-query-bad-imp-current-empty-opening-hit-at-zero-on-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  and bad:
    staged-security-with-data-state-verifier-event
      query-bad-with-empty-composition-header-candidates
      (Some (((data, attacker-state), result), final-state))
  shows
    staged-security-with-data-state-current-empty-query-opening-hit-at 0
      (Some (((data, attacker-state), result), final-state))
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF support]
  have verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
  from empty-query-bad-header-alignment-on-support
    [OF wf controlled support bad]
  obtain rest where header:
    verifier-header-transcript ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
  []

```

```

    (staged-composition-final data)
  rest
and comp-empty:
    staged-composition-fri-roots data = []
by blast
from empty-query-bad-query-rounds-on-support
  [OF wf controlled support bad]
obtain f-fl query-state where query-state-state:
  PState query-state = staged-query-start-hash data
and query-counter:
  PQueryCounter query-state = 0
and query-out:
  Some (result, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program
            (staged-trace-root data)
            f-fl
            (staged-trace-final data)
            (staged-alphas data)
            []
            (staged-composition-final data))
          rounds)
        query-state)
    by blast
from ntimes-verifier-query-rounds-empty-composition-first-consistent
  [OF refl query-out]
obtain raw trace-openings where trace-indices:
  map opening-index trace-openings =
    powers-scaled (index (to-nat raw))
and trace-table:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and consistent:
  cp-eval (staged-alphas data) (map opening-value trace-openings)
    (h ^ index (to-nat raw) * shift) =
    staged-composition-final data
and lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state) (PState query-state)) =
    Some raw
by blast
have zero-bound: 0 < rounds
using rounds-positive by simp
have lookup0:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge 0
      (state-after-query-chunks

```

```

      (staged-query-start-hash data)
      (staged-query-chunks data) 0)) =
    Some raw
  using lookup query-state-state query-counter
  unfolding state-after-query-chunks-def by simp
show ?thesis
  by (rule
      staged-security-with-data-state-current-empty-query-opening-hit-atI
      [OF zero-bound lookup0 comp-empty verifier trace-table header
        trace-indices consistent])
qed

lemma checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event
      (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        query-bad-with-empty-composition-header-candidates)
      adversary-initial-state ≤
    wp-event
      (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
      adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  and bad:
    staged-security-with-data-state-verifier-event
      query-bad-with-empty-composition-header-candidates out
  show
    staged-security-with-data-state-current-empty-query-opening-hit-at 0 out
proof (cases out)
  case None
  then show ?thesis
    using bad
    unfolding staged-security-with-data-state-verifier-event-def
      query-bad-with-empty-composition-header-candidates-def
    by simp
next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)

```

```

    by (cases packed) (auto split: prod.splits)
show ?thesis
  unfolding out-eq
  by (rule empty-query-bad-imp-current-empty-opening-hit-at-zero-on-support
      [OF wf controlled])
      (use support bad out-eq in simp-all)
qed
qed

lemma checked-staged-security-with-data-state-current-empty-query-opening-hit-at-bound-from-prefix-and-path:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound:  $i < \text{rounds}$ 
  and prefix-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at
      i)
    adversary-initial-state  $\leq P$ 
  and path-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at
      i)
    adversary-initial-state  $\leq T$ 
  shows
    wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-empty-query-opening-hit-at i)
    adversary-initial-state  $\leq P + T$ 
proof -
  have current-le-prefix:
    wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-empty-query-opening-hit-at i)
    adversary-initial-state  $\leq$ 
  and
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at i)
    adversary-initial-state
  by (rule
    checked-staged-security-with-data-state-current-empty-query-opening-hit-at-le-query-prefix-current
    [OF wf controlled i-bound])
  have prefix-bound':
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at i)
    adversary-initial-state  $\leq P + T$ 
  by (rule

```

checked-staged-security-with-query-prefix-current-empty-authenticated-hit-at-bound-from-prefix-and-path
 [OF wf controlled i-bound prefix-bound path-bound])
show ?thesis
 by (rule order-trans[OF current-le-prefix prefix-bound])
qed

lemma *checked-staged-security-with-data-state-current-empty-query-opening-hit-at-bound-from-conceptual-empty-residual-hit-at*
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i < \text{rounds}$
and residual-bound:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at*
 i)
 adversary-initial-state $\leq R$
and path-bound:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at*
 i)
 adversary-initial-state $\leq T$
shows
 wp-event
 (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-current-empty-query-opening-hit-at* i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
 query-error-bound + R + T
proof –
have prefix-bound:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at*
 i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
 query-error-bound + R
by (rule
 checked-staged-security-with-query-prefix-current-empty-prefix-authenticated-hit-at-bound-from-conceptual-empty-residual-hit-at
 [OF wf controlled i-bound residual-bound])
have current-bound:
 wp-event
 (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-current-empty-query-opening-hit-at* i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size


```

    (staged-query-search-queries budgets  $i + 1$ ) +
    query-error-bound +  $R$ ) +  $T$ 
  by (rule
    checked-staged-security-with-data-state-current-empty-query-opening-hit-at-bound-from-prefix-and-path
    [OF wf controlled i-bound prefix-bound path-bound])
  then show ?thesis
    by (simp add: algebra-simps)
qed

lemma checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at-bound-from-transcript:
  assumes i-bound:  $i < \text{rounds}$ 
  and pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state  $\leq P$ 
  and new-bound:
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state  $\leq R$ 
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state  $A$   $i$ )
      (checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at
         $i$ )
      adversary-initial-state  $\leq P + R$ 
proof -
  have event-le:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state  $A$   $i$ )
      (checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at
         $i$ )
      adversary-initial-state  $\leq$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state  $A$   $i$ )
      checked-staged-security-with-query-prefix-transcript-hit
      adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support:
      out  $\in$ 
      set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state  $A$   $i$ )
        adversary-initial-state)
    and hit:
      checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at
       $i$  out
    show checked-staged-security-with-query-prefix-transcript-hit out
  proof (cases out)

```

```

case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at-def
    checked-staged-security-with-query-prefix-empty-trace-path-output-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
have mapped-support:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i  $\gg$ 
        ( $\lambda$ packed. return (snd packed))))
      adversary-initial-state)
    unfolding out-eq
    by (rule set-dist-bindI[OF support[unfolded out-eq]]) simp
have projection:
  checked-staged-security-experiment-with-query-prefix-data-state A i  $\gg$ 
    ( $\lambda$ packed. return (snd packed)) =
    checked-staged-security-experiment-with-data-state A
    by (rule checked-staged-security-experiment-with-query-prefix-data-state-projection
      [OF i-bound])
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
    using mapped-support unfolding projection .
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
have verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
have transcript-split:
  hash-map-output-values ?s ∩ set (staged-proof-transcript data)
     $\neq \{\}$   $\vee$ 
  hash-map-new-output-hit (set (staged-proof-transcript data))
    ?s final-state

```

```

    by (rule verify-monad-trace-root-target-preexisting-or-new[OF verifier])
      simp
  show ?thesis
    unfolding out-eq
      checked-staged-security-with-query-prefix-transcript-hit-def
    using transcript-split by simp
qed
qed
have transcript-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    checked-staged-security-with-query-prefix-transcript-hit
    adversary-initial-state  $\leq P + R$ 
  by (rule
      checked-staged-security-with-query-prefix-transcript-hit-bound-from-data-state
      [OF i-bound pre-bound new-bound])
show ?thesis
  by (rule order-trans[OF event-le transcript-bound])
qed

lemma checked-staged-security-with-data-state-current-empty-query-opening-hit-at-bound-from-conceptual-empty-residual-hit-at
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound:  $i < \text{rounds}$ 
  and residual-bound:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at i)
      adversary-initial-state  $\leq E$ 
  and pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-transcript-pre-hit
      adversary-initial-state  $\leq P$ 
  and new-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-transcript-new-hit
      adversary-initial-state  $\leq R$ 
  shows
    wp-event
      (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-current-empty-query-opening-hit-at i)
      adversary-initial-state  $\leq$ 
      staged-phase-relation-error size
      (staged-query-search-queries budgets  $i + 1$ ) +
      query-error-bound +  $E + (P + R)$ 
proof -
  have path-bound:
    wp-event

```

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at*
i)
 $\text{adversary-initial-state} \leq P + R$
by (rule
checked-staged-security-with-query-prefix-current-empty-trace-path-output-hit-at-bound-from-transcript
[OF i-bound pre-bound new-bound])
show ?thesis
by (rule
checked-staged-security-with-data-state-current-empty-query-opening-hit-at-bound-from-conceptual-empty-residual
[OF wf controlled i-bound residual-bound path-bound])
qed

lemma *checked-staged-security-with-data-state-empty-header-bad-event-bound-from-prefix-drift-trace-fri-and-current-*
assumes *false-statement: $\neg$ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *prefix-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
 $\text{adversary-initial-state} \leq P$
and *drift-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space)
 $\text{adversary-initial-state} \leq D$
and *trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and *current-empty-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
soundness-bad-event-partial-candidate-empty-header)
 $\text{adversary-initial-state} \leq$
 $P + D +$
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + empty-query-error

```

proof –
  have query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        query-bad-with-empty-composition-header-candidates)
    adversary-initial-state ≤ empty-query-error
  by (rule order-trans[
    OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero[
      OF wf controlled] current-empty-bound])
  show ?thesis
  by (rule
    checked-staged-security-with-data-state-empty-header-bad-event-bound-from-prefix-drift-trace-fri-and-query
      [OF false-statement wf controlled prefix-bound drift-bound
        trace-fri-bound query-bound])
qed

end

end

```

```

theory Soundness-Staged-Aligned-Partial-Merkle
  imports
    Soundness-Staged-Partial-Merkle
    Soundness-Aligned-Partial-Query-Transcript
    Staged-Security-Experiment-Alpha-Header-Binding
    Staged-Security-Experiment-Composition-Query-Current-Alignment
    Soundness-Conceptual-Query-Empty-Current
  begin

```

Staged partial-Merkle packaging that uses aligned nonempty-header query evidence. This downstream layer avoids importing verifier-query extraction machinery into the upstream deterministic reductions.

```

context soundness
begin

```

```

lemma partial-merkle-inconsistency-bad-imp-aligned-partial-candidate-with-partial-merkle:
  assumes partial-merkle-inconsistency-bad s out
  shows
    soundness-bad-event-aligned-partial-candidate-with-partial-merkle s out
  using assms
  unfolding
    soundness-bad-event-aligned-partial-candidate-with-partial-merkle-def
  by simp

```

```

lemma soundness-bad-event-aligned-partial-candidate-imp-with-partial-merkle:
  assumes soundness-bad-event-aligned-partial-candidate s out
  shows
    soundness-bad-event-aligned-partial-candidate-with-partial-merkle s out

```

```

using assms
unfolding
  soundness-bad-event-aligned-partial-candidate-with-partial-merkle-def
by simp

lemma accepted-partition-soundness-bad-event-aligned-partial-candidate-with-partial-merkle-if-header-nonempty:
assumes false-statement:  $\neg$  exists-valid-trace
and supp: out  $\in$  set-dist (execute verify-monad s)
and acc: accepted out
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and comp-nonempty: composition-fri-roots  $\neq$  []
shows
  soundness-bad-event-aligned-partial-candidate-with-partial-merkle s out
proof -
from acc obtain result final-state where out-eq:
  out = Some (result, final-state)
unfolding accepted-def by (cases out) auto
from
  verify-monad-accepted-with-partial-initial-openings-aligned-consistent
  [OF supp[unfolded out-eq] header comp-nonempty]
obtain query-idxs trace-openings composition-openings where partial:
  accepted-with-partial-initial-openings-aligned-consistent s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
by blast
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then show ?thesis
unfolding out-eq
by (rule
  partial-merkle-inconsistency-bad-imp-aligned-partial-candidate-with-partial-merkle)
next
case False
have aligned:
  accepted-with-partial-initial-openings-aligned s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
by (rule
  accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
  [OF partial])
obtain trace-table composition-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and composition-candidate:

```

```

partial-composition-table-candidate composition-table
composition-openings
using
  accepted-with-partial-initial-openings-aligned-candidates-if-no-partial-merkle-bad
  [OF aligned False]
by blast
have bad:
  soundness-bad-event-aligned-partial-candidate s
  (Some (result, final-state))
by (rule accepted-aligned-partial-candidate-partition
  [OF false-statement partial trace-candidate composition-candidate])
then show ?thesis
  unfolding out-eq
  by (rule soundness-bad-event-aligned-partial-candidate-imp-with-partial-merkle)
qed
qed

```

lemma *accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-if-header-non*

assumes *false-statement*: $\neg \text{exists-valid-trace}$

and *supp*: $\text{out} \in \text{set-dist } (\text{execute verify-monad } s)$

and *acc*: *accepted out*

and *header*:

verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest

and *comp-nonempty*: *composition-fri-roots* $\neq []$

shows

soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
s out

proof —

from *acc* **obtain** *result final-state* **where** *out-eq*:

out = *Some (result, final-state)*

unfolding *accepted-def* **by** (*cases out*) *auto*

from *verify-monad-supplied-header-transcript-shape*

[OF *supp*[*unfolded out-eq*] *header*]

obtain *query-idxs raw-idxs query-chunks* **where** *shape*:

accepted-transcript-shape s (Some (result, final-state)) as query-idxs

and *len-raw*: *length raw-idxs* = *rounds*

and *query-idxs-eq*:

query-idxs = *map* ($\lambda \text{raw. index } (\text{to-nat raw})$) *raw-idxs*

and *len-query-chunks*: *length query-chunks* = *rounds*

and *transcript-rest*:

List.concat query-chunks @ PTranscript final-state = *rest*

and *query-chunk-header*:

$\bigwedge i. i < \text{rounds} \implies$

verifier-query-round-chunk (query-idxs ! i)

f-fri-roots composition-fri-roots (query-chunks ! i)

and *replay-lookup-header*:

$\bigwedge i. i < \text{rounds} \implies$

fmlookup (HashMap final-state)

```

    (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks
        (verifier-header-state s fr f-fri-roots f-final as dg
          composition-fri-roots final)
        query-chunks i)) =
    Some (raw-idxs ! i)
  by blast
from verify-monad-supplied-header-query-rounds
  [OF supp[unfolded out-eq] header]
obtain f-fl fl query-state where f-roots:
  map snd f-fl = f-fri-roots
and comp-roots: map snd fl = composition-fri-roots
and query-transcript-state: PTranscript query-state = rest
and query-hash-state:
  PState query-state =
    verifier-header-state s fr f-fri-roots f-final as dg
    composition-fri-roots final
and query-counter: PQueryCounter query-state = PQueryCounter s
and query-out:
  Some (result, final-state) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
      query-state)
  by blast
have fl-nonempty: fl ≠ []
  using comp-nonempty comp-roots by (cases fl) simp-all
then obtain b composition-root fl-tail where fl-eq:
  fl = (b, composition-root) # fl-tail
  by (cases fl) auto
have transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
  using query-transcript-state transcript-rest by simp
have query-chunk:
  ∧i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
  using query-chunk-header f-roots comp-roots by simp
have replay-lookup:
  ∧i. i < rounds ⇒
    fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some (raw-idxs ! i)
  using replay-lookup-header query-hash-state query-counter by simp
from

```



```

query-rounds-replay-accepted-with-partial-initial-openings-aligned-transcript-consistent
[OF header comp-nonempty fl-eq f-roots comp-roots
 query-transcript-state query-hash-state query-counter query-out
 len-raw query-idxs-eq len-query-chunks transcript-query query-chunk
 replay-lookup]
obtain trace-openings composition-openings where partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
by blast
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then show ?thesis
  unfolding out-eq
  soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-def
  by simp
next
case False
have aligned0:
  accepted-with-partial-initial-openings-aligned s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
  using partial
  by (blast dest:
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    accepted-with-partial-initial-openings-aligned-consistent-imp-aligned)
obtain trace-table composition-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and composition-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
using
  accepted-with-partial-initial-openings-aligned-candidates-if-no-partial-merkle-bad
  [OF aligned0 False]
by blast
have bad:
  soundness-bad-event-aligned-transcript-partial-candidate s
  (Some (result, final-state))
by (rule accepted-aligned-transcript-partial-candidate-partition
  [OF false-statement partial trace-candidate composition-candidate])
then show ?thesis
  unfolding out-eq
  by (simp add:
    soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-def)
qed

```

qed

lemma *accepted-partition-soundness-bad-event-aligned-partial-candidate-with-partial-merkle-or-empty-header:*
assumes *false-statement:* $\neg \text{exists-valid-trace}$
and *supp:* $\text{out} \in \text{set-dist } (\text{execute verify-monad } s)$
and *acc:* *accepted out*
and *header:*
 verifier-header-transcript s fr f-fri-roots f-final as dg
 composition-fri-roots final rest
shows
 soundness-bad-event-aligned-partial-candidate-with-partial-merkle s out $\vee$
 soundness-bad-event-partial-candidate-empty-header s out
proof (*cases composition-fri-roots = []*)
case *False*
then have *comp-nonempty:* *composition-fri-roots $\neq []$*
by *simp*
have *soundness-bad-event-aligned-partial-candidate-with-partial-merkle s*
 out
by (*rule*
 accepted-partition-soundness-bad-event-aligned-partial-candidate-with-partial-merkle-if-header-nonempty
 [*OF false-statement supp acc header comp-nonempty*])
then show *?thesis* **by** *simp*
next
case *True*
note *comp-empty = True*
from *acc* **obtain** *result final-state* **where** *out-eq:*
 out = Some (result, final-state)
unfolding *accepted-def* **by** (*cases out*) *auto*
show *?thesis*
proof (*cases*
 partial-merkle-inconsistency-bad s (Some (result, final-state)))
case *True*
then have
 soundness-bad-event-aligned-partial-candidate-with-partial-merkle s out
unfolding *out-eq*
by (*rule*
 partial-merkle-inconsistency-bad-imp-aligned-partial-candidate-with-partial-merkle)
then show *?thesis* **by** *simp*
next
case *False*
from *verify-monad-accepted-with-partial-trace-openings-for-header*
 [*OF supp[unfolded out-eq] header*]
obtain *trace-query-idxs trace-openings* **where** *trace-partial:*
 accepted-with-partial-trace-openings s
 (*Some (result, final-state)*) *fr trace-query-idxs trace-openings*
by *blast*
obtain *trace-table* **where** *trace-candidate:*
 partial-trace-table-candidate trace-table trace-openings
using *accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad*

```

    [OF trace-partial False]
  by blast
have accepted-out: accepted (Some (result, final-state))
  by (rule accepted-with-partial-trace-openings-imp-accepted
    [OF trace-partial])
let ?composition-table = replicate (scale * clength) final
have empty-partial:
  accepted-with-empty-composition-header-candidates s
    (Some (result, final-state)) fr f-fri-roots f-final as dg final
    trace-query-idxs trace-openings trace-table ?composition-table
  unfolding accepted-with-empty-composition-header-candidates-def
  by (intro conjI exI[of - rest])
    (use accepted-out header comp-empty trace-partial trace-candidate
      in simp-all)
have soundness-bad-event-partial-candidate-empty-header s
  (Some (result, final-state))
  by (rule accepted-empty-composition-header-candidate-partition
    [OF false-statement empty-partial])
then show ?thesis
  unfolding out-eq by simp
qed
qed

lemma accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-or-empty-header
  assumes false-statement:  $\neg$  exists-valid-trace
  and supp:  $out \in \text{set-dist } (\text{execute verify-monad } s)$ 
  and acc: accepted out
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  shows
    soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
      s out  $\vee$ 
    soundness-bad-event-partial-candidate-empty-header s out
proof (cases composition-fri-roots = [])
case False
then have comp-nonempty: composition-fri-roots  $\neq$  []
  by simp
have
  soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
    s out
  by (rule
    accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-if-header-nonempty
    [OF false-statement supp acc header comp-nonempty])
then show ?thesis by simp
next
case True
note comp-empty = True
from acc obtain result final-state where out-eq:

```

```

    out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then have
  soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
    s out
  unfolding out-eq
    soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-def
  by simp
then show ?thesis by simp
next
case False
from verify-monad-accepted-with-partial-trace-openings-for-header
  [OF supp[unfolded out-eq] header]
obtain trace-query-idxs trace-openings where trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
by blast
obtain trace-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
using accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
  [OF trace-partial False]
by blast
have accepted-out: accepted (Some (result, final-state))
  by (rule accepted-with-partial-trace-openings-imp-accepted
    [OF trace-partial])
let ?composition-table = replicate (scale * clength) final
have empty-partial:
  accepted-with-empty-composition-header-candidates s
  (Some (result, final-state)) fr f-fri-roots f-final as dg final
  trace-query-idxs trace-openings trace-table ?composition-table
unfolding accepted-with-empty-composition-header-candidates-def
by (intro conjI exI[of - rest])
  (use accepted-out header comp-empty trace-partial trace-candidate
    in simp-all)
have soundness-bad-event-partial-candidate-empty-header s
  (Some (result, final-state))
  by (rule accepted-empty-composition-header-candidate-partition
    [OF false-statement empty-partial])
then show ?thesis
  unfolding out-eq by simp
qed
qed

lemma checked-staged-soundness-from-aligned-partial-candidate-and-empty-header-bounds:
  fixes partial-candidate-error empty-header-error :: prob

```

```

assumes false-statement:  $\neg$  exists-valid-trace
and partial-candidate-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      soundness-bad-event-aligned-partial-candidate-with-partial-merkle)
    adversary-initial-state  $\leq$  partial-candidate-error
and empty-header-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      soundness-bad-event-partial-candidate-empty-header)
    adversary-initial-state  $\leq$  empty-header-error
shows checked-staged-adversary-acceptance-probability A  $\leq$ 
  partial-candidate-error + empty-header-error
proof –
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  let ?partial =
    ?E soundness-bad-event-aligned-partial-candidate-with-partial-merkle
  let ?empty = ?E soundness-bad-event-partial-candidate-empty-header
  let ?bad =  $\lambda out. ?partial\ out \vee ?empty\ out$ 
  have accepted-bad:
    wp-event ?M accepted adversary-initial-state  $\leq$ 
      wp-event ?M ?bad adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support: out  $\in$  set-dist (execute ?M adversary-initial-state)
  and acc: accepted out
  show ?bad out
  proof (cases out)
  case None
  then show ?thesis
    using acc unfolding accepted-def by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have verifier:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)
  using checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
  by blast
  have verifier-acc: accepted (Some (result, final-state))
  unfolding accepted-def by simp
  from verify-monad-accepted-transcript-shape[OF verifier]
  obtain alphas query-idxs where shape:

```

```

    accepted-transcript-shape ?s (Some (result, final-state))
    alphas query-idxs
  by blast
from shape obtain result' final-state' fr f-fri-roots f-final dg
    composition-fri-roots final rest where
  header:
    verifier-header-transcript ?s fr f-fri-roots f-final alphas dg
    composition-fri-roots final rest
  by (elim accepted-transcript-shape-header-query-extraction)
have local-bad:
  soundness-bad-event-aligned-partial-candidate-with-partial-merkle ?s
    (Some (result, final-state))  $\vee$ 
  soundness-bad-event-partial-candidate-empty-header ?s
    (Some (result, final-state))
  by (rule
    accepted-partition-soundness-bad-event-aligned-partial-candidate-with-partial-merkle-or-empty-header
    [OF false-statement verifier verifier-acc header])
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
have bad-bound:
  wp-event ?M ?bad adversary-initial-state  $\leq$ 
    partial-candidate-error + empty-header-error
proof -
  have wp-event ?M ?bad adversary-initial-state  $\leq$ 
    wp-event ?M ?partial adversary-initial-state +
    wp-event ?M ?empty adversary-initial-state
  by (rule wp-event-union-bound)
  also have ...  $\leq$  partial-candidate-error + empty-header-error
  by (intro add-mono partial-candidate-bound empty-header-bound)
  finally show ?thesis .
qed
have staged-bound:
  wp-event (checked-staged-security-experiment A) accepted
    adversary-initial-state  $\leq$ 
    partial-candidate-error + empty-header-error
  unfolding checked-staged-security-experiment-acceptance-with-data-state
  by (rule order-trans[OF accepted-bad bad-bound])
show ?thesis
proof -
  have
    wp-event (checked-staged-security-experiment A)
      ( $\lambda$ out.  $\neg$  Option.is-none out) adversary-initial-state =
    wp-event (checked-staged-security-experiment A) accepted
      adversary-initial-state
  unfolding wp-event-def accepted-def
  by (rule arg-cong[where

```

```

    f = λQ. wp (checked-staged-security-experiment A) Q
      adversary-initial-state])
  (rule ext, simp)
then show ?thesis
  unfolding checked-staged-adversary-acceptance-probability-def
  using staged-bound by simp
qed
qed

lemma checked-staged-soundness-from-aligned-transcript-partial-candidate-and-empty-header-bounds:
  fixes partial-candidate-error empty-header-error :: prob
  assumes false-statement: ¬ exists-valid-trace
  and partial-candidate-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle)
        adversary-initial-state ≤ partial-candidate-error
  and empty-header-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-partial-candidate-empty-header)
        adversary-initial-state ≤ empty-header-error
  shows checked-staged-adversary-acceptance-probability A ≤
    partial-candidate-error + empty-header-error
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  let ?partial =
    ?E soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
  let ?empty = ?E soundness-bad-event-partial-candidate-empty-header
  let ?bad = λout. ?partial out ∨ ?empty out
  have accepted-bad:
    wp-event ?M accepted adversary-initial-state ≤
      wp-event ?M ?bad adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support: out ∈ set-dist (execute ?M adversary-initial-state)
    and acc: accepted out
    show ?bad out
    proof (cases out)
      case None
      then show ?thesis
        using acc unfolding accepted-def by simp
    next
      case (Some packed)
      then obtain data attacker-state result final-state where out-eq:
        out = Some (((data, attacker-state), result), final-state)
        by (cases packed, auto split: prod.splits)
      let ?s =

```

```

    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  using checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
  by blast
  have verifier-acc: accepted (Some (result, final-state))
    unfolding accepted-def by simp
  from verify-monad-accepted-transcript-shape[OF verifier]
  obtain alphas query-idxs where shape:
    accepted-transcript-shape ?s (Some (result, final-state))
    alphas query-idxs
  by blast
  from shape obtain result' final-state' fr f-fri-roots f-final dg
    composition-fri-roots final rest where
    header:
      verifier-header-transcript ?s fr f-fri-roots f-final alphas dg
      composition-fri-roots final rest
  by (elim accepted-transcript-shape-header-query-extraction)
  have local-bad:
    soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
      ?s (Some (result, final-state)) ∨
    soundness-bad-event-partial-candidate-empty-header ?s
      (Some (result, final-state))
  by (rule
    accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-or-empty-header)
    [OF false-statement verifier verifier-acc header])
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
qed
qed
have bad-bound:
  wp-event ?M ?bad adversary-initial-state ≤
    partial-candidate-error + empty-header-error
proof -
  have wp-event ?M ?bad adversary-initial-state ≤
    wp-event ?M ?partial adversary-initial-state +
    wp-event ?M ?empty adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤ partial-candidate-error + empty-header-error
  by (intro add-mono partial-candidate-bound empty-header-bound)
  finally show ?thesis .
qed
have staged-bound:
  wp-event (checked-staged-security-experiment A) accepted
    adversary-initial-state ≤
    partial-candidate-error + empty-header-error

```



```

unfolding checked-staged-security-experiment-acceptance-with-data-state
by (rule order-trans[OF accepted-bad bad-bound])
show ?thesis
proof –
  have
    wp-event (checked-staged-security-experiment A)
      ( $\lambda out. \neg Option.is\_none\ out$ ) adversary-initial-state =
      wp-event (checked-staged-security-experiment A) accepted
      adversary-initial-state
    unfolding wp-event-def accepted-def
    by (rule arg-cong[where
       $f = \lambda Q. wp\ (checked\_staged\_security\_experiment\ A)\ Q$ 
      adversary-initial-state])
      (rule ext, simp)
    then show ?thesis
    unfolding checked-staged-adversary-acceptance-probability-def
    using staged-bound by simp
  qed
qed

lemma checked-staged-soundness-from-aligned-partial-candidate-components-and-empty-header-bound:
fixes merkle-error composition-error trace-fri-error'
  composition-fri-error' query-error empty-header-error :: prob
assumes false-statement:  $\neg exists\_valid\_trace$ 
and merkle-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      partial-merkle-inconsistency-bad)
    adversary-initial-state  $\leq$  merkle-error
and comp-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-error
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  trace-fri-error'
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-fri-error'
and query-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      query-bad-with-aligned-partial-candidates)
    adversary-initial-state  $\leq$  query-error

```

and *empty-header-bound*:
 $wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $\quad soundness\text{-}bad\text{-}event\text{-}partial\text{-}candidate\text{-}empty\text{-}header)$
 $adversary\text{-}initial\text{-}state \leq empty\text{-}header\text{-}error$
shows *checked-staged-adversary-acceptance-probability* $A \leq$
 $merkle\text{-}error + composition\text{-}error + trace\text{-}fri\text{-}error' +$
 $composition\text{-}fri\text{-}error' + query\text{-}error + empty\text{-}header\text{-}error$
proof –
have *partial-candidate-bound*:
 $wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $\quad soundness\text{-}bad\text{-}event\text{-}aligned\text{-}partial\text{-}candidate\text{-}with\text{-}partial\text{-}merkle)$
 $adversary\text{-}initial\text{-}state \leq$
 $merkle\text{-}error + composition\text{-}error + trace\text{-}fri\text{-}error' +$
 $composition\text{-}fri\text{-}error' + query\text{-}error$
by (rule
 $checked\text{-}staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}aligned\text{-}partial\text{-}candidate\text{-}with\text{-}partial\text{-}merkle\text{-}union\text{-}bound$
 $\ [OF \ merkle\text{-}bound \ comp\text{-}bound \ trace\text{-}fri\text{-}bound \ comp\text{-}fri\text{-}bound$
 $\quad query\text{-}bound])$
have *bound*:
 $checked\text{-}staged\text{-}adversary\text{-}acceptance\text{-}probability \ A \leq$
 $(merkle\text{-}error + composition\text{-}error + trace\text{-}fri\text{-}error' +$
 $\quad composition\text{-}fri\text{-}error' + query\text{-}error) + empty\text{-}header\text{-}error$
by (rule
 $checked\text{-}staged\text{-}soundness\text{-}from\text{-}aligned\text{-}partial\text{-}candidate\text{-}and\text{-}empty\text{-}header\text{-}bounds$
 $\ [OF \ false\text{-}statement \ partial\text{-}candidate\text{-}bound \ empty\text{-}header\text{-}bound])$
then show *?thesis*
by (*simp add: algebra-simps*)
qed

lemma *checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-empty-header-bound*:
fixes *merkle-error composition-error trace-fri-error'*
 $composition\text{-}fri\text{-}error' \ query\text{-}error \ empty\text{-}header\text{-}error :: prob$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *merkle-bound*:
 $wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $\quad partial\text{-}merkle\text{-}inconsistency\text{-}bad)$
 $adversary\text{-}initial\text{-}state \leq merkle\text{-}error$
and *comp-bound*:
 $wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $\quad composition\text{-}bad\text{-}with\text{-}partial\text{-}candidates)$
 $adversary\text{-}initial\text{-}state \leq composition\text{-}error$
and *trace-fri-bound*:
 $wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $\quad trace\text{-}fri\text{-}bad\text{-}with\text{-}partial\text{-}candidates)$

$\text{adversary-initial-state} \leq \text{trace-fri-error}'$
and *comp-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and *query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-aligned-transcript-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{query-error}$
and *empty-header-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{soundness-bad-event-partial-candidate-empty-header})$
 $\text{adversary-initial-state} \leq \text{empty-header-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{merkle-error} + \text{composition-error} + \text{trace-fri-error}' +$
 $\text{composition-fri-error}' + \text{query-error} + \text{empty-header-error}$
proof –
have *partial-candidate-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle})$
 $\text{adversary-initial-state} \leq$
 $\text{merkle-error} + \text{composition-error} + \text{trace-fri-error}' +$
 $\text{composition-fri-error}' + \text{query-error}$
by (rule
 $\text{checked-staged-security-with-data-state-aligned-transcript-partial-candidate-with-partial-merkle-component-union}$
 $[\text{OF merkle-bound comp-bound trace-fri-bound comp-fri-bound}$
 $\text{query-bound}])$
have *bound*:
 $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $(\text{merkle-error} + \text{composition-error} + \text{trace-fri-error}' +$
 $\text{composition-fri-error}' + \text{query-error}) + \text{empty-header-error}$
by (rule
 $\text{checked-staged-soundness-from-aligned-transcript-partial-candidate-and-empty-header-bounds}$
 $[\text{OF false-statement partial-candidate-bound empty-header-bound}])$
then show *?thesis*
by (*simp add: algebra-simps*)
qed

lemma *checked-staged-soundness-from-aligned-partial-candidate-components-and-empty-header-components*:
fixes $\text{trace-fri-error}'$ $\text{composition-fri-error}'$ query-error P $D :: \text{prob}$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* A
and *trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$

*(staged-security-with-data-state-verifier-event
 trace-fri-bad-with-partial-candidates)*
adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
 composition-fri-bad-with-partial-candidates)*
adversary-initial-state $\leq$ *composition-fri-error'*
and *query-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
 query-bad-with-aligned-partial-candidates)*
adversary-initial-state $\leq$ *query-error*
and *prefix-bound:*
wp-event
*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)*
*(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
 composition-trace-bad-alpha-space)*
adversary-initial-state $\leq$ *P*
and *drift-bound:*
wp-event
*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)*
*(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
 composition-trace-bad-alpha-space)*
adversary-initial-state $\leq$ *D*
and *empty-trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)*
adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
 query-bad-with-empty-composition-header-candidates)*
adversary-initial-state $\leq$ *nnreal rounds * query-error-bound*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(P + D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
query-error +
(P + D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +

$\text{trace-fri-error} + \text{nnreal rounds} * \text{query-error-bound}$
proof –
have *comp-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{composition-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq$
 $P + D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)$
by (rule
 $\text{checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-prefix-drift-and-budgets}$
 $[\text{OF false-statement wf controlled prefix-bound drift-bound}])$
have *merkle-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{partial-merkle-inconsistency-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\quad \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget})$
by (rule
 $\text{checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled}$
 $[\text{OF wf controlled}])$
have *empty-header-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{soundness-bad-event-partial-candidate-empty-header})$
 $\text{adversary-initial-state} \leq$
 $P + D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{trace-fri-error} + \text{nnreal rounds} * \text{query-error-bound}$
by (rule
 $\text{checked-staged-security-with-data-state-empty-header-bad-event-bound-from-prefix-drift-trace-fri-and-query}$
 $[\text{OF false-statement wf controlled prefix-bound drift-bound}$
 $\quad \text{empty-trace-fri-bound empty-query-bound}])$
have *bound*:
 $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\quad \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(P + D +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $\text{query-error} +$
 $(P + D +$
 $\quad \text{staged-phase-relation-error size}$

(staged-alpha-search-queries budgets 0) +
 trace-fri-error + nnreal rounds * query-error-bound)
by (rule
 checked-staged-soundness-from-aligned-partial-candidate-components-and-empty-header-bound
 [OF false-statement merkle-bound comp-bound trace-fri-bound
 comp-fri-bound query-bound empty-header-bound])
then show ?thesis .
qed

lemma checked-staged-soundness-from-aligned-partial-candidate-components-and-current-empty-query:
fixes trace-fri-error' composition-fri-error' query-error P D
 empty-query-error :: prob
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ trace-fri-error'
and comp-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ composition-fri-error'
and query-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 query-bad-with-aligned-partial-candidates)
 adversary-initial-state $\leq$ query-error
and prefix-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$ P
and drift-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$ D
and empty-trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ trace-fri-error

and *current-empty-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-current-empty-query-opening-hit-at 0*)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
(*P* + *D* +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*)) +
trace-fri-error' + *composition-fri-error'* +
query-error +
(*P* + *D* +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +
trace-fri-error + *empty-query-error*)
proof –
have *comp-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
composition-bad-with-partial-candidates)
adversary-initial-state $\leq$
P + *D* +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*)
by (*rule*
checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-prefix-drift-and-budgets
[*OF false-statement wf controlled prefix-bound drift-bound*])
have *merkle-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$
hash-collision-budget-value 0
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)
by (*rule*
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
[*OF wf controlled*])
have *empty-header-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state $\leq$
P + *D* +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +
trace-fri-error + *empty-query-error*

by (rule
checked-staged-security-with-data-state-empty-header-bad-event-bound-from-prefix-drift-trace-fri-and-current-empty-trace-fri-bound current-empty-bound)
have bound:
checked-staged-adversary-acceptance-probability $A \leq$
hash-collision-budget-value 0
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
($P + D +$
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0)) +
trace-fri-error' + *composition-fri-error'* +
query-error +
($P + D +$
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
trace-fri-error + *empty-query-error*)
by (rule
checked-staged-soundness-from-aligned-partial-candidate-components-and-empty-header-bound
[*OF* *false-statement* *merkle-bound* *comp-bound* *trace-fri-bound*
comp-fri-bound *query-bound* *empty-header-bound*])
then show ?thesis .
qed

lemma *checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-empty-header-components*
fixes *trace-fri-error'* *composition-fri-error'* *query-error* P $D :: \text{prob}$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* A
and *trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-fri-error'*
and *query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq$ *query-error*
and *prefix-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

*(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)*
adversary-initial-state $\leq P$
and *drift-bound*:
wp-event
*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)*
*(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space)*
adversary-initial-state $\leq D$
and *empty-trace-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)*
adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)*
adversary-initial-state $\leq$ *nnreal rounds * query-error-bound*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
*(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +*
(P + D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
query-error +
(P + D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
*trace-fri-error + nnreal rounds * query-error-bound)*
proof –
have *comp-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
composition-bad-with-partial-candidates)*
adversary-initial-state $\leq$
P + D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)
by (rule
checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-prefix-drift-and-budgets
[OF false-statement wf controlled prefix-bound drift-bound])
have *merkle-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad)*

adversary-initial-state $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* +
 staged-challenge-query-budget + *verifier-hash-query-budget*)
by (rule
 checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
 [*OF wf controlled*])
have *empty-header-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
 soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state $\leq$
P + *D* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
trace-fri-error + *nnreal* *rounds* * *query-error-bound*
by (rule
 checked-staged-security-with-data-state-empty-header-bad-event-bound-from-prefix-drift-trace-fri-and-query
 [*OF false-statement wf controlled prefix-bound drift-bound*
 empty-trace-fri-bound empty-query-bound])
have *bound*:
checked-staged-adversary-acceptance-probability *A* $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* +
 staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*P* + *D* +
 staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0)) +
trace-fri-error' + *composition-fri-error'* +
query-error +
 (*P* + *D* +
 staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
 trace-fri-error + *nnreal* *rounds* * *query-error-bound*)
by (rule
 checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-empty-header-bound
 [*OF false-statement merkle-bound comp-bound trace-fri-bound*
 comp-fri-bound query-bound empty-header-bound])
then show ?thesis .
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-partial-opening-hit*:
fixes *trace-fri-error'* *composition-fri-error'* *query-error* *P* *D* :: *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*

$\text{trace-fri-bad-with-partial-candidates}$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}'$
and *comp-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and *partial-opening-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-query-partial-opening-hit}$
 $\text{adversary-initial-state} \leq \text{query-error}$
and *prefix-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-bad-set-hit}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq P$
and *drift-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq D$
and *empty-trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and *empty-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{nnreal rounds} * \text{query-error-bound}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(P + D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $\text{query-error} +$
 $(P + D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{trace-fri-error} + \text{nnreal rounds} * \text{query-error-bound})$
proof –

have *query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq$ *query-error*
by (*rule order-trans*[
OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-partial-opening-hit
partial-opening-bound])
(*use wf controlled in simp-all*)
show *?thesis*
by (*rule*
checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-empty-header-components
[*OF false-statement wf controlled trace-fri-bound comp-fri-bound*
query-bound prefix-bound drift-bound empty-trace-fri-bound
empty-query-bound])
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit*:
fixes *trace-fri-error'* *composition-fri-error'* *query-error P D :: prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-fri-error'*
and *current-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state $\leq$ *query-error*
and *prefix-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$ *P*
and *drift-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
composition-trace-bad-alpha-space)

$adversary-initial-state \leq D$
and *empty-trace-fri-bound*:
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $trace-fri-bad-with-empty-composition-header-candidates)$
 $adversary-initial-state \leq trace-fri-error$
and *empty-query-bound*:
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $query-bad-with-empty-composition-header-candidates)$
 $adversary-initial-state \leq nnreal \ rounds * query-error-bound$
shows *checked-staged-adversary-acceptance-probability* $A \leq$
 $hash-collision-budget-value \ 0$
 $(staged-attacker-query-budget \ budgets +$
 $staged-challenge-query-budget + verifier-hash-query-budget) +$
 $(P + D +$
 $staged-phase-relation-error \ size$
 $(staged-alpha-search-queries \ budgets \ 0)) +$
 $trace-fri-error' + composition-fri-error' +$
 $query-error +$
 $(P + D +$
 $staged-phase-relation-error \ size$
 $(staged-alpha-search-queries \ budgets \ 0) +$
 $trace-fri-error + nnreal \ rounds * query-error-bound)$
proof –
have *query-bound*:
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $query-bad-with-aligned-transcript-partial-candidates)$
 $adversary-initial-state \leq query-error$
by (rule *order-trans*[
 $OF \ checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit$
 $current-query-bound]$)
 $(use \ wf \ controlled \ in \ simp-all)$
show *?thesis*
by (rule
 $checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-empty-header-components$
 $[OF \ false-statement \ wf \ controlled \ trace-fri-bound \ comp-fri-bound$
 $query-bound \ prefix-bound \ drift-bound \ empty-trace-fri-bound$
 $empty-query-bound]$)
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds*:
fixes $trace-fri-error' \ composition-fri-error' \ P \ D :: prob$
and $C :: nat \Rightarrow prob$
assumes *false-statement*: $\neg \ exists-valid-trace$
and *wf*: *staged-budget-wellformed* $budgets$
and *controlled*: *staged-adversary-controlled* $budgets \ A$
and *trace-fri-bound*:

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
 $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-fri-error'*
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 $wp\text{-}event$
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at* i)
adversary-initial-state $\leq C$ i
and *prefix-bound*:
 $wp\text{-}event$
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq P$
and *drift-bound*:
 $wp\text{-}event$
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq D$
and *empty-trace-fri-bound*:
 $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound*:
 $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *nnreal rounds* $\ast$ *query-error-bound*
shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* $+$
staged-challenge-query-budget $+$ *verifier-hash-query-budget*) $+$
 $(P + D +$
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* $0)) +$
trace-fri-error' $+$ *composition-fri-error'* $+$
 $(\sum i < \text{rounds}. C$ $i) +$
 $(P + D +$

staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
*trace-fri-error + nnreal rounds * query-error-bound)*
proof –
have *current-query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state $\leq (\sum i < \text{rounds. } C \ i)$
by (rule
checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
[OF round-bound wf controlled])
show *?thesis*
by (rule
checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit
[OF false-statement wf controlled trace-fri-bound comp-fri-bound
current-query-bound prefix-bound drift-bound empty-trace-fri-bound
empty-query-bound])
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-partial-opening-hit-from-drift*:
fixes *trace-fri-error' composition-fri-error' query-error D :: prob*
assumes *false-statement: $\neg$ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ trace-fri-error'
and *comp-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ composition-fri-error'
and *partial-opening-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-query-partial-opening-hit
adversary-initial-state $\leq$ query-error
and *drift-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$ D
and *empty-trace-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)

$\text{adversary-initial-state} \leq \text{trace-fri-error}$
and *empty-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{nnreal rounds} * \text{query-error-bound}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $\text{query-error} +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{trace-fri-error} + \text{nnreal rounds} * \text{query-error-bound})$
proof –
have *comp-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)$
by (rule
 $\text{checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-drift-and-budgets}$
 $[\text{OF false-statement wf controlled drift-bound}])$
have *query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$

query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq$ *query-error*
by (rule *order-trans*[
OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-partial-opening-hit
partial-opening-bound])
 (use *wf* controlled **in** *simp-all*)
have *merkle-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)
by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
 [*OF wf* controlled])
have *empty-header-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
D +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
trace-fri-error + *nnreal rounds* * *query-error-bound*
by (rule
checked-staged-security-with-data-state-empty-header-bad-event-bound-from-drift-trace-fri-and-query
 [*OF false-statement wf* controlled *drift-bound empty-trace-fri-bound*
empty-query-bound])
show *?thesis*
by (rule
checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-empty-header-bound
 [*OF false-statement merkle-bound comp-bound trace-fri-bound*
comp-fri-bound query-bound empty-header-bound])
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit-from-drift*:
fixes *trace-fri-error'* *composition-fri-error'* *query-error D* :: *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)

(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-fri-error'*
and *current-query-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state $\leq$ *query-error*
and *drift-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$ *D*
and *empty-trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *nnreal rounds * query-error-bound*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
query-error +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
D +

staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
*trace-fri-error + nnreal rounds * query-error-bound)*

proof –

have *comp-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-bad-with-partial-candidates)
adversary-initial-state ≤
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)

by (rule
checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-drift-and-budgets
[OF false-statement wf controlled drift-bound])

have *query-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state ≤ query-error

by (rule *order-trans*[
OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit
current-query-bound])
(use wf controlled in simp-all)

have *merkle-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad)
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)

by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
[OF wf controlled])

have *empty-header-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state ≤
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0

```

    (staged-alpha-search-queries budgets 0) +
    D +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    trace-fri-error + nnreal rounds * query-error-bound
  by (rule
    checked-staged-security-with-data-state-empty-header-bad-event-bound-from-drift-trace-fri-and-query
    [OF false-statement wf controlled drift-bound empty-trace-fri-bound
      empty-query-bound])
  show ?thesis
  by (rule
    checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-empty-header-bound
    [OF false-statement merkle-bound comp-bound trace-fri-bound
      comp-fri-bound query-bound empty-header-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-partial-opening-hit-from-not-prefix:
  fixes trace-fri-error' composition-fri-error' query-error N :: prob
  assumes false-statement:  $\neg$  exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  trace-fri-error'
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-fri-error'
  and partial-opening-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-query-partial-opening-hit
    adversary-initial-state  $\leq$  query-error
  and not-prefix-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
      composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq$  N
  and empty-trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state  $\leq$  trace-fri-error
  and empty-query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

*(staged-security-with-data-state-verifier-event
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state ≤ nnreal rounds * query-error-bound*
shows *checked-staged-adversary-acceptance-probability A ≤
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 query-error +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 trace-fri-error + nnreal rounds * query-error-bound)*

proof –

have *drift-bound:*

wp-event
*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
 composition-trace-bad-alpha-space)
 adversary-initial-state ≤ N*

proof –

have *wp-event*
*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
 composition-trace-bad-alpha-space)
 adversary-initial-state ≤*

wp-event
*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
 composition-trace-bad-alpha-space)
 adversary-initial-state*

by (rule
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-le-not-prefix-bound

```

    [OF wf controlled])
  also have ...  $\leq N$ 
    by (rule not-prefix-bound)
  finally show ?thesis .
qed
show ?thesis
  by (rule
    checked-staged-soundness-from-aligned-transcript-components-and-partial-opening-hit-from-drift
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
      partial-opening-bound drift-bound empty-trace-fri-bound
      empty-query-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit-from-not-prefix-bound
  fixes trace-fri-error' composition-fri-error' query-error N :: prob
  assumes false-statement:  $\neg$  exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  trace-fri-error'
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-fri-error'
  and current-query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state  $\leq$  query-error
  and not-prefix-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
      composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq N$ 
  and empty-trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state  $\leq$  trace-fri-error
  and empty-query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      query-bad-with-empty-composition-header-candidates)
    adversary-initial-state  $\leq$  nnreal rounds * query-error-bound

```

shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* budgets +
 staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
 staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries* budgets 0) +
 N +
 staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0)) +
trace-fri-error' + *composition-fri-error'* +
query-error +
 (*composition-error-bound* +
 staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries* budgets 0) +
 N +
 staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0) +
 trace-fri-error + *nnreal* rounds * *query-error-bound*)

proof –

have *drift-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
 composition-trace-bad-alpha-space)
adversary-initial-state $\leq N$

proof –

have *wp-event*

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
 composition-trace-bad-alpha-space)
adversary-initial-state $\leq$

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
 composition-trace-bad-alpha-space)
adversary-initial-state

by (rule

checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-le-not-prefix-bound
 [OF wf controlled])

also have ... $\leq N$

by (rule *not-prefix-bound*)

```

    finally show ?thesis .
qed
show ?thesis
by (rule
  checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit-from-drift
  [OF false-statement wf controlled trace-fri-bound comp-fri-bound
    current-query-bound drift-bound empty-trace-fri-bound
    empty-query-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-not-prefix
fixes trace-fri-error' composition-fri-error' N :: prob
and C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-partial-candidates)
  adversary-initial-state  $\leq$  trace-fri-error'
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-fri-bad-with-partial-candidates)
  adversary-initial-state  $\leq$  composition-fri-error'
and round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
  adversary-initial-state  $\leq$  C i
and not-prefix-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
  (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
    composition-trace-bad-alpha-space)
  adversary-initial-state  $\leq$  N
and empty-trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state  $\leq$  trace-fri-error
and empty-query-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    query-bad-with-empty-composition-header-candidates)
  adversary-initial-state  $\leq$  nnreal rounds * query-error-bound

```


shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
N +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0)) +
trace-fri-error' + *composition-fri-error'* +
 ($\sum i < \text{rounds}. C\ i$) +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
N +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
trace-fri-error + *nnreal rounds* * *query-error-bound*)
proof –
have *current-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state $\leq$ ($\sum i < \text{rounds}. C\ i$)
by (rule
checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
 [*OF round-bound wf controlled*])
show *?thesis*
by (rule
checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit-from-not-prefix
 [*OF false-statement wf controlled trace-fri-bound comp-fri-bound*
current-query-bound not-prefix-bound empty-trace-fri-bound
empty-query-bound])
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-prefix-target-path-bounds-from-not-prefix*:
fixes *trace-fri-error'* *composition-fri-error'* $N :: \text{prob}$
and $P\ Q :: \text{nat} \Rightarrow \text{prob}$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* A
and *trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)

$\text{adversary-initial-state} \leq \text{trace-fri-error}'$
and *comp-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and *prefix-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i)$
 $(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $\text{staged-query-prefix-authenticated-opening-target})$
 $\text{adversary-initial-state} \leq P \ i$
and *path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq Q \ i$
and *not-prefix-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and *empty-trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and *empty-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{nnreal rounds} * \text{query-error-bound}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $N +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum i < \text{rounds}. P \ i + Q \ i) +$

```

    (composition-error-bound +
     staged-phase-relation-error size
     (staged-alpha-search-queries budgets 0) +
     hash-collision-budget-value 0
     (staged-alpha-search-queries budgets 0) +
     N +
     staged-phase-relation-error size
     (staged-alpha-search-queries budgets 0) +
     trace-fri-error + nnreal rounds * query-error-bound)
  proof -
  have round-bound:
     $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq P\ i + Q\ i$ 
  proof -
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  have prefix-hit-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
    adversary-initial-state  $\leq P\ i$ 
  by (rule
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target
    [OF wf controlled i-bound prefix-bound[OF i-bound]])
  show
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq P\ i + Q\ i$ 
  by (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-path
    [OF wf controlled i-bound prefix-hit-bound path-bound[OF i-bound]])
  qed
  show ?thesis
  by (rule
    checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-not-prefix
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
     round-bound not-prefix-bound empty-trace-fri-bound
     empty-query-bound])
  qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-not-prefix:
  fixes trace-fri-error' composition-fri-error' N :: prob
  and Q :: nat  $\Rightarrow$  prob
  assumes false-statement:  $\neg \text{exists-valid-trace}$ 
  and wf: staged-budget-wellformed budgets

```

and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state ≤ trace-fri-error'
and *comp-fri-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (*staged-security-with-data-state-verifier-event*
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state ≤ composition-fri-error'
and *trace-frac:*
 $\bigwedge i$ *prefix prefix-state.*
 $i < \text{rounds} \implies$
 Some ((prefix, prefix-state), prefix-state) ∈
 set-dist
 (*execute (checked-staged-query-prefix-with-state A i)*
 adversary-initial-state) ⇒
 nnreal
 (*card*
 (*staged-query-prefix-prefix-authenticated-trace-indices prefix*
 prefix-state)) /
 nnreal (card query-sample-space) ≤ query-error-bound
and *path-bound:*
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-path-output-hit-at i*)
 adversary-initial-state ≤ Q i
and *not-prefix-bound:*
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
 composition-trace-bad-alpha-space)
 adversary-initial-state ≤ N
and *empty-trace-fri-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state ≤ trace-fri-error
and *empty-query-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (*staged-security-with-data-state-verifier-event*
 query-bad-with-empty-composition-header-candidates)
 *adversary-initial-state ≤ nnreal rounds * query-error-bound*
shows *checked-staged-adversary-acceptance-probability A ≤*
 hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets +*

```

    staged-challenge-query-budget + verifier-hash-query-budget) +
  (composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    N +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)) +
  trace-fri-error' + composition-fri-error' +
  (∑ i < rounds.
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound + Q i) +
  (composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    N +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    trace-fri-error + nnreal rounds * query-error-bound)
proof -
  have round-bound:
    ∧ i. i < rounds ⇒
      wp-event
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
        adversary-initial-state ≤
      staged-phase-relation-error size
        (staged-query-search-queries budgets i + 1) +
        query-error-bound + Q i
proof -
  fix i
  assume i-bound: i < rounds
  have prefix-hit-bound:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
      adversary-initial-state ≤
      staged-phase-relation-error size
        (staged-query-search-queries budgets i + 1) + query-error-bound
  by (rule
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-trace-indices
    [OF wf controlled i-bound])
    (use trace-frac[OF i-bound] in blast)
  show
    wp-event

```

```

    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound + Q i
  by (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-path
    [OF wf controlled i-bound prefix-hit-bound path-bound[OF i-bound]])
qed
show ?thesis
  by (rule
    checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-not-prefix
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
      round-bound not-prefix-bound empty-trace-fri-bound
      empty-query-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-not-prefix-co
  fixes trace-fri-error' composition-fri-error' C U M R :: prob
  and Q :: nat ⇒ prob
  assumes false-statement: ¬ exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ trace-fri-error'
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-fri-error'
  and trace-frac:
    ∧ i prefix prefix-state.
    i < rounds ⇒
    Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state) ⇒
    nnreal
    (card
      (staged-query-prefix-prefix-authenticated-trace-indices prefix
        prefix-state)) /
    nnreal (card query-sample-space) ≤ query-error-bound
  and path-bound:
    ∧ i. i < rounds ⇒
    wp-event

```

(checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-path-output-hit-at i)
 adversary-initial-state $\leq Q$ i
and collision-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
 adversary-initial-state $\leq C$
and uncovered-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
 adversary-initial-state $\leq U$
and no-prefix-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 adversary-initial-state $\leq M$
and not-prefix-path-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
 adversary-initial-state $\leq R$
and empty-trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ trace-fri-error
and empty-query-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ nnreal rounds * query-error-bound
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 (C + U + M + R) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +

$\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum i < \text{rounds}.$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-query-search-queries budgets } i + 1) +$
 $\quad \text{query-error-bound} + Q \ i) +$
 $(\text{composition-error-bound} +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad \text{hash-collision-budget-value } 0$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad (C + U + M + R) +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad \text{trace-fri-error} + \text{nnreal rounds} * \text{query-error-bound})$

proof –

have *not-prefix-bound*:

wp-event

$(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $\quad A)$

$(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound}$
 $\quad \text{composition-trace-bad-alpha-space})$

$\text{adversary-initial-state} \leq C + U + M + R$

by (rule

$\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound}$
 $[\text{OF } \text{wf controlled collision-bound uncovered-bound no-prefix-bound}$
 $\quad \text{not-prefix-path-bound}]$)

show *?thesis*

by (rule

$\text{checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-not-prefix-no}$
 $[\text{OF } \text{false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\quad \text{trace-frac path-bound not-prefix-bound empty-trace-fri-bound}$
 $\quad \text{empty-query-bound}]$)

qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-not-prefix-no*

fixes $\text{trace-fri-error}' \text{ composition-fri-error}' U M R :: \text{prob}$

and $Q :: \text{nat} \Rightarrow \text{prob}$

assumes *false-statement*: $\neg \text{exists-valid-trace}$

and *wf*: *staged-budget-wellformed* *budgets*

and *controlled*: *staged-adversary-controlled* *budgets* *A*

and *trace-fri-bound*:

wp-event $(\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}'$

and *comp-fri-bound*:

wp-event $(\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{composition-fri-bad-with-partial-candidates})$

$adversary-initial-state \leq composition-fri-error'$
and *trace-frac*:
 $\bigwedge i \text{ prefix prefix-state.}$
 $i < rounds \implies$
 $Some ((prefix, prefix-state), prefix-state) \in$
 $set-dist$
 $(execute (checked-staged-query-prefix-with-state A i)$
 $adversary-initial-state) \implies$
 $nnreal$
 $(card$
 $(staged-query-prefix-prefix-authenticated-trace-indices prefix$
 $prefix-state)) /$
 $nnreal (card query-sample-space) \leq query-error-bound$
and *path-bound*:
 $\bigwedge i. i < rounds \implies$
 $wp-event$
 $(checked-staged-security-experiment-with-query-prefix-data-state A i)$
 $(checked-staged-security-with-query-prefix-current-path-output-hit-at i)$
 $adversary-initial-state \leq Q i$
and *uncovered-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index$
 $adversary-initial-state \leq U$
and *no-prefix-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate$
 $adversary-initial-state \leq M$
and *not-prefix-path-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output$
 $adversary-initial-state \leq R$
and *empty-trace-fri-bound*:
 $wp-event (checked-staged-security-experiment-with-data-state A)$
 $(staged-security-with-data-state-verifier-event$
 $trace-fri-bad-with-empty-composition-header-candidates)$
 $adversary-initial-state \leq trace-fri-error$
and *empty-query-bound*:
 $wp-event (checked-staged-security-experiment-with-data-state A)$
 $(staged-security-with-data-state-verifier-event$
 $query-bad-with-empty-composition-header-candidates)$
 $adversary-initial-state \leq nnreal rounds * query-error-bound$
shows $checked-staged-adversary-acceptance-probability A \leq$
 $hash-collision-budget-value 0$

(staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + U + M + R) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 (∑ i < rounds.
 staged-phase-relation-error size
 (staged-query-search-queries budgets i + 1) +
 query-error-bound + Q i) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + U + M + R) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 trace-fri-error + nnreal rounds * query-error-bound)

proof (rule
 checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-not-prefix-compon
 [OF false-statement wf controlled trace-fri-bound comp-fri-bound
 trace-frac path-bound - uncovered-bound no-prefix-bound
 not-prefix-path-bound empty-trace-fri-bound empty-query-bound])

show
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
 adversary-initial-state ≤
 hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0)

by (rule
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
 [OF wf controlled])

qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-cross-no-pre
fixes trace-fri-error' composition-fri-error' X M R :: prob
and Q :: nat ⇒ prob
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A

and *trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}'$

and *comp-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$

and *trace-frac*:
 $\bigwedge i \text{ prefix prefix-state.}$
 $i < \text{rounds} \implies$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \implies$
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-prefix-authenticated-trace-indices prefix}$
 $\text{prefix-state})) /$
 $\text{nnreal } (\text{card query-sample-space}) \leq \text{query-error-bound}$

and *path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq Q \ i$

and *cross-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad}$
 $\text{adversary-initial-state} \leq X$

and *no-prefix-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate}$
 $\text{adversary-initial-state} \leq M$

and *not-prefix-path-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output}$
 $\text{adversary-initial-state} \leq R$

and *empty-trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$

trace-fri-bad-with-empty-composition-header-candidates
adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *nnreal rounds* * *query-error-bound*
shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
(composition-error-bound +
staged-phase-relation-error *size*
(staged-alpha-search-queries *budgets* 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries *budgets* 0) +
(hash-collision-budget-value 0
(staged-alpha-search-queries *budgets* 0) + *X* + *M* + *R*) +
staged-phase-relation-error *size*
(staged-alpha-search-queries *budgets* 0)) +
trace-fri-error' + *composition-fri-error'* +
 $(\sum i < \text{rounds}.$
staged-phase-relation-error *size*
(staged-query-search-queries *budgets* *i* + 1) +
query-error-bound + *Q* *i*) +
(composition-error-bound +
staged-phase-relation-error *size*
(staged-alpha-search-queries *budgets* 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries *budgets* 0) +
(hash-collision-budget-value 0
(staged-alpha-search-queries *budgets* 0) + *X* + *M* + *R*) +
staged-phase-relation-error *size*
(staged-alpha-search-queries *budgets* 0) +
trace-fri-error + *nnreal rounds* * *query-error-bound*)
proof (*rule*
checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-not-prefix-noncoll
[*OF* *false-statement* *wf* *controlled* *trace-fri-bound* *comp-fri-bound*
trace-frac *path-bound* - *no-prefix-bound* *not-prefix-path-bound*
empty-trace-fri-bound *empty-query-bound*])
show
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
adversary-initial-state $\leq$ *X*
by (*rule* *order-trans*
[*OF*
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-witnessed-cross-or-merkle-bad

cross-bound])
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-cross-no-pre*

fixes *trace-fri-error'* *composition-fri-error'* *X M P R* :: *prob*

and *Q* :: *nat* $\Rightarrow$ *prob*

assumes *false-statement*: $\neg$ *exists-valid-trace*

and *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

trace-fri-bad-with-partial-candidates)

adversary-initial-state $\leq$ *trace-fri-error'*

and *comp-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

composition-fri-bad-with-partial-candidates)

adversary-initial-state $\leq$ *composition-fri-error'*

and *trace-frac*:

$\bigwedge i$ *prefix prefix-state*.

i < *rounds* $\Rightarrow$

Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$

set-dist

(*execute* (*checked-staged-query-prefix-with-state A i*)

adversary-initial-state) $\Rightarrow$

nnreal

(*card*

(*staged-query-prefix-prefix-authenticated-trace-indices prefix*

prefix-state)) /

nnreal (*card query-sample-space*) $\leq$ *query-error-bound*

and *path-bound*:

$\bigwedge i$. *i* < *rounds* $\Rightarrow$

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-current-path-output-hit-at i*)

adversary-initial-state $\leq$ *Q i*

and *cross-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad

adversary-initial-state $\leq$ *X*

and *no-prefix-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate

adversary-initial-state $\leq$ *M*

and *transcript-pre-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit

adversary-initial-state $\leq P$

and *transcript-new-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit

adversary-initial-state $\leq R$

and *empty-trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

trace-fri-bad-with-empty-composition-header-candidates)

adversary-initial-state $\leq$ *trace-fri-error*

and *empty-query-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

query-bad-with-empty-composition-header-candidates)

adversary-initial-state $\leq$ *nnreal rounds* * *query-error-bound*

shows *checked-staged-adversary-acceptance-probability A* $\leq$

hash-collision-budget-value 0

(*staged-attacker-query-budget budgets* +

staged-challenge-query-budget + *verifier-hash-query-budget*) +

(*composition-error-bound* +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

hash-collision-budget-value 0

(*staged-alpha-search-queries budgets 0*) +

(*hash-collision-budget-value 0*

(*staged-alpha-search-queries budgets 0*) + *X* + *M* + (*P* + *R*)) +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*)) +

trace-fri-error' + *composition-fri-error'* +

($\sum i < \text{rounds}$.

staged-phase-relation-error size

(*staged-query-search-queries budgets i* + 1) +

query-error-bound + *Q i*) +

(*composition-error-bound* +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

hash-collision-budget-value 0

(*staged-alpha-search-queries budgets 0*) +

(*hash-collision-budget-value 0*

(*staged-alpha-search-queries budgets 0*) + *X* + *M* + (*P* + *R*)) +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

$\text{trace-fri-error} + \text{nnreal rounds} * \text{query-error-bound}$
proof (rule
checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-cross-no-prefix-pat
 $[OF \text{ false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\text{trace-frac path-bound cross-bound no-prefix-bound} - \text{empty-trace-fri-bound}$
 $\text{empty-query-bound}]$)
show
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output}$
 $\text{adversary-initial-state} \leq P + R$
by (rule *order-trans*
 $[OF$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-le-witness-transcript-hit}]$)
(rule
 $\text{checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new}$
 $[OF \text{ transcript-pre-bound transcript-new-bound}]$)
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-cross-no-pre*
fixes $\text{trace-fri-error}' \text{ composition-fri-error}' X M P R :: \text{prob}$
and $Q :: \text{nat} \Rightarrow \text{prob}$
assumes *false-statement: $\neg \text{exists-valid-trace}$*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound:*
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}'$
and *comp-fri-bound:*
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and *trace-frac:*
 $\bigwedge i \text{ prefix prefix-state.}$
 $i < \text{rounds} \implies$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \implies$
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-prefix-authenticated-trace-indices prefix}$
 $\text{prefix-state})) /$
 $\text{nnreal } (\text{card query-sample-space}) \leq \text{query-error-bound}$
and *path-bound:*

$\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq Q \ i$
and *cross-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad}$
 $\text{adversary-initial-state} \leq X$
and *no-prefix-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate}$
 $\text{adversary-initial-state} \leq M$
and *data-pre-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-pre-hit}$
 $\text{adversary-initial-state} \leq P$
and *transcript-new-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit}$
 $\text{adversary-initial-state} \leq R$
and *empty-trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and *empty-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{nnreal rounds} * \text{query-error-bound}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) + X + M + (P + R)) +$
 $\text{staged-phase-relation-error size}$

$(\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum i < \text{rounds}.$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-query-search-queries budgets } i + 1) +$
 $\quad \text{query-error-bound} + Q \ i) +$
 $(\text{composition-error-bound} +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad \text{hash-collision-budget-value } 0$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad (\text{hash-collision-budget-value } 0$
 $\quad (\text{staged-alpha-search-queries budgets } 0) + X + M + (P + R)) +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad \text{trace-fri-error} + \text{nnreal rounds} * \text{query-error-bound})$

proof (rule
checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-cross-no-prefix-tra
 $[OF \text{ false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\text{trace-frac path-bound cross-bound no-prefix-bound} - \text{transcript-new-bound}$
 $\text{empty-trace-fri-bound empty-query-bound}]$)

show
wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit}$
 $\text{adversary-initial-state} \leq P$

by (rule
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
 $[OF \text{ data-pre-bound}]$)

qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-cross-data-p*
fixes $\text{trace-fri-error}' \text{ composition-fri-error}' X P R :: \text{prob}$
and $Q :: \text{nat} \Rightarrow \text{prob}$
assumes *false-statement: $\neg \text{exists-valid-trace}$*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound:*
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}'$
and *comp-fri-bound:*
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and *trace-frac:*

$\bigwedge i$ *prefix prefix-state*.
 $i < \text{rounds} \implies$
Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
set-dist
(*execute* (*checked-staged-query-prefix-with-state* *A i*)
adversary-initial-state) $\implies$
nnreal
(*card*
(*staged-query-prefix-prefix-authenticated-trace-indices* *prefix*
prefix-state)) /
nnreal (*card* *query-sample-space*) $\leq$ *query-error-bound*
and *path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
(*checked-staged-security-with-query-prefix-current-path-output-hit-at* *i*)
adversary-initial-state $\leq$ *Q i*
and *cross-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
adversary-initial-state $\leq$ *X*
and *data-pre-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq$ *P*
and *transcript-new-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq$ *R*
and *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *nnreal* *rounds* * *query-error-bound*
shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
hash-collision-budget-value *0*
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
(*composition-error-bound* +
staged-phase-relation-error *size*)

(staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + X + (P + R) + (P + R)) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 ($\sum i < \text{rounds}$.
 staged-phase-relation-error size
 (staged-query-search-queries budgets i + 1) +
 query-error-bound + Q i) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + X + (P + R) + (P + R)) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 trace-fri-error + nnreal rounds * query-error-bound)

proof (rule
 checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-cross-no-prefix-dat
 [OF false-statement wf controlled trace-fri-bound comp-fri-bound
 trace-frac path-bound cross-bound - data-pre-bound transcript-new-bound
 empty-trace-fri-bound empty-query-bound])

show
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 adversary-initial-state $\leq P + R$

proof (rule order-trans
 [OF
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-le-witness-transcript-hit])

have pre-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
 adversary-initial-state $\leq P$

by (rule
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
 [OF data-pre-bound])

show
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)

```

    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
    adversary-initial-state  $\leq P + R$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new
    [OF pre-bound transcript-new-bound])
qed
qed
end
end

```

```

theory Soundness-Staged-Aligned-Current-Query
  imports Soundness-Staged-Aligned-Partial-Merkle
begin

```

Thin wrappers that expose the actual-alpha current-query event in the aligned partial-candidate soundness path. The underlying query reduction still lives in the partial-Merkle layer; this theory only uses the projection proved in the current-query alignment layer.

```

context soundness
begin

```

```

lemma checked-staged-soundness-from-aligned-transcript-components-and-actual-alpha-current-partial-opening-hit-fr
  fixes trace-fri-error' composition-fri-error' query-error  $N :: \text{prob}$ 
  assumes false-statement:  $\neg \text{exists-valid-trace}$ 
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets  $A$ 
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-partial-candidates)
      adversary-initial-state  $\leq \text{trace-fri-error}'$ 
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-partial-candidates)
      adversary-initial-state  $\leq \text{composition-fri-error}'$ 
  and current-query-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
         $A$ )
      checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
      adversary-initial-state  $\leq \text{query-error}$ 
  and not-prefix-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
         $A$ )

```

(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq N$
and empty-trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ trace-fri-error
and empty-query-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ nnreal rounds * query-error-bound
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 query-error +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 trace-fri-error + nnreal rounds * query-error-bound)

proof –

have data-current-query-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-current-query-partial-opening-hit
 adversary-initial-state $\leq$ query-error
by (rule
 checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-actual-alpha-current-query
 [OF current-query-bound])

show ?thesis

by (rule
 checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit-from-not-prefix
 [OF false-statement wf controlled trace-fri-bound comp-fri-bound
 data-current-query-bound not-prefix-bound empty-trace-fri-bound])

empty-query-bound])

qed

theorem *stark-soundness-from-aligned-partial-actual-alpha-current-query:*
fixes *trace-fri-error'* *composition-fri-error'* *query-error* $N :: \text{prob}$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* A
and *trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *composition-fri-error'*
and *current-query-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
 adversary-initial-state $\leq$ *query-error*
and *not-prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$ N
and *empty-trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ *nnreal rounds* * *query-error-bound*
shows *checked-staged-adversary-acceptance-probability* $A \leq$
 hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* +
 staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
 staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
 hash-collision-budget-value 0)

(staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 query-error +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 trace-fri-error + nnreal rounds * query-error-bound)
by (rule
 checked-staged-soundness-from-aligned-transcript-components-and-actual-alpha-current-partial-opening-hit-from-n
 [OF false-statement wf controlled trace-fri-bound comp-fri-bound
 current-query-bound not-prefix-bound empty-trace-fri-bound
 empty-query-bound])

theorem stark-soundness-from-aligned-partial-query-prefix-current-rounds:
fixes trace-fri-error' composition-fri-error' N :: prob
and C :: nat $\Rightarrow$ prob
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ trace-fri-error'
and comp-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ composition-fri-error'
and round-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state $\leq$ C i
and not-prefix-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
 composition-trace-bad-alpha-space)

adversary-initial-state $\leq N$
and *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *nnreal rounds* * *query-error-bound*
shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
(*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
(*staged-alpha-search-queries* *budgets* 0) +
 N +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0)) +
trace-fri-error' + *composition-fri-error'* +
($\sum i < \text{rounds. } C \ i$) +
(*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
(*staged-alpha-search-queries* *budgets* 0) +
 N +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
trace-fri-error + *nnreal rounds* * *query-error-bound*)
proof –
have *current-query-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)
checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit
adversary-initial-state $\leq$ ($\sum i < \text{rounds. } C \ i$)
by (*rule*
checked-staged-security-with-actual-alpha-prefix-current-query-partial-opening-hit-bound-from-query-prefix-current
[*OF wf controlled round-bound*])
show *?thesis*
by (*rule*
stark-soundness-from-aligned-partial-actual-alpha-current-query
[*OF false-statement wf controlled trace-fri-bound comp-fri-bound*
current-query-bound not-prefix-bound empty-trace-fri-bound
empty-query-bound])

qed

theorem *stark-soundness-from-aligned-partial-query-prefix-current-prefix-path:*
fixes *trace-fri-error'* *composition-fri-error'* *N* :: *prob*
and *P* *Q* :: *nat* $\Rightarrow$ *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *composition-fri-error'*
and *prefix-bound*:
 $\bigwedge i. i < \text{rounds} \Rightarrow$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A* *i*)
 (*checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at* *i*)
 adversary-initial-state $\leq$ *P* *i*
and *path-bound*:
 $\bigwedge i. i < \text{rounds} \Rightarrow$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A* *i*)
 (*checked-staged-security-with-query-prefix-current-path-output-hit-at* *i*)
 adversary-initial-state $\leq$ *Q* *i*
and *not-prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$ *N*
and *empty-trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ *nnreal* *rounds* * *query-error-bound*
shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
 hash-collision-budget-value *0*

(staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 ($\sum i < \text{rounds}. P\ i + Q\ i$) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 trace-fri-error + nnreal rounds * query-error-bound)

proof (rule stark-soundness-from-aligned-partial-query-prefix-current-rounds
 [OF false-statement wf controlled trace-fri-bound comp-fri-bound])
fix i
assume i-bound: i < rounds
show
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state $\leq P\ i + Q\ i$
by (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-path
 [OF wf controlled i-bound prefix-bound[OF i-bound]
 path-bound[OF i-bound]])
qed (rule not-prefix-bound empty-trace-fri-bound empty-query-bound)+

theorem stark-soundness-from-aligned-partial-query-prefix-dynamic-target-path:
fixes trace-fri-error' composition-fri-error' N :: prob
and P Q :: nat $\Rightarrow$ prob
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ trace-fri-error'
and comp-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)

*(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)*
adversary-initial-state $\leq$ *composition-fri-error'*
and *prefix-target-bound:*
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
(checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
staged-query-prefix-authenticated-opening-target)
adversary-initial-state $\leq P\ i$
and *path-bound:*
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-path-output-hit-at i)
adversary-initial-state $\leq Q\ i$
and *not-prefix-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state $\leq N$
and *empty-trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq \text{trace-fri-error}$
and *empty-query-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq \text{nnreal rounds} * \text{query-error-bound}$
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) $+$
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) $+$
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0)) $+$
N +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) $+$
trace-fri-error' + composition-fri-error' +
 $(\sum i < \text{rounds}. P\ i + Q\ i)$ $+$
(composition-error-bound +
staged-phase-relation-error size

```

      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
      N +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      trace-fri-error + nnreal rounds * query-error-bound)
proof -
  have prefix-bound:
     $\bigwedge i. i < \text{rounds} \implies$ 
      wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
      adversary-initial-state  $\leq P$  i
  by (rule
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target
    [OF wf controlled - prefix-target-bound])
  show ?thesis
  by (rule stark-soundness-from-aligned-partial-query-prefix-current-prefix-path
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
      prefix-bound path-bound not-prefix-bound empty-trace-fri-bound
      empty-query-bound])
qed

end

end

```

```

theory Soundness-Conceptual-Query-Current-Path
imports
  Soundness-Conceptual-Query-Current
  Staged-Security-Experiment-Composition-Query-Current-Path
begin

```

Integration of conceptual prefix-query targets with the existing structured current-query path split.

```

context soundness
begin

```

```

definition checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
where
  checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
    i out  $\longleftrightarrow$ 
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
      i out  $\wedge$ 
    checked-staged-security-with-query-prefix-conceptual-context-residual-at
      out

```

definition *checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at*
where

checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
 $\text{trace-default } \text{composition-default } i \text{ out} \longleftrightarrow$
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
 $i \text{ out} \wedge$
checked-staged-security-with-query-prefix-conceptual-default-context-residual-at
 $\text{trace-default } \text{composition-default } \text{out}$

definition *checked-staged-security-with-query-prefix-candidate-pair-default-agreement-gap*
where

checked-staged-security-with-query-prefix-candidate-pair-default-agreement-gap
 $\text{trace-table } \text{composition-table } \text{out} \longleftrightarrow$
(case out of
 $\text{None} \Rightarrow \text{False}$
| $\text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state}) \Rightarrow$
 $\neg \text{query-prefix-candidate-defaults-agree-with-authenticated-values}$
 $\text{trace-table } \text{composition-table } \text{prefix } \text{prefix-state})$

definition *checked-staged-security-with-query-prefix-candidate-pair-sampled-default-agreement-gap*
where

checked-staged-security-with-query-prefix-candidate-pair-sampled-default-agreement-gap
 $\text{trace-table } \text{composition-table } \text{out} \longleftrightarrow$
(case out of
 $\text{None} \Rightarrow \text{False}$
| $\text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state}) \Rightarrow$
 $\neg \text{query-prefix-candidate-defaults-agree-at-query } \text{trace-table}$
 $\text{composition-table } (\text{index } (\text{to-nat } \text{raw})) \text{ prefix } \text{prefix-state})$

definition *checked-staged-security-with-query-prefix-authenticated-candidate-hit-at*
where

checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
 $\text{trace-table } \text{composition-table } i \text{ out} \longleftrightarrow$
 $(\exists \text{trace-openings } \text{composition-openings}.$
checked-staged-security-with-query-prefix-authenticated-opening-hit
 $\text{trace-openings } \text{composition-openings } i \text{ out} \wedge$
partial-trace-table-candidate trace-table
 $((\text{replicate } \text{rounds } []) [i := \text{trace-openings}]) \wedge$
partial-composition-table-candidate composition-table
 $((\text{replicate } \text{rounds } []) [i := \text{composition-openings}]))$

lemma *checked-staged-security-with-query-prefix-candidate-pair-hit-imp-conceptual-default-or-agreement-gap:*
assumes *hit:*

checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
 $\text{trace-table } \text{composition-table}$
 $(\text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state}))$

```

and clean:  $\neg$  hash-map-output-collision prefix-state
and trace-len: length trace-table = scale * clength
and composition-len: length composition-table = scale * clength
shows
  checked-staged-security-with-query-prefix-conceptual-default-target-hit
    trace-table composition-table
    (Some ((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)  $\vee$ 
  checked-staged-security-with-query-prefix-candidate-pair-default-agreement-gap
    trace-table composition-table
    (Some ((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
proof (cases
  query-prefix-candidate-defaults-agree-with-authenticated-values
    trace-table composition-table prefix prefix-state)
case True
have subset:
  staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
    composition-table prefix prefix-state  $\subseteq$ 
  staged-query-prefix-conceptual-default-query-target trace-table
    composition-table prefix prefix-state
by (rule
  staged-query-prefix-candidate-pair-query-target-from-prefix-subset-conceptual-default-if-prefix-agrees
    [OF clean trace-len composition-len True])
have
  checked-staged-security-with-query-prefix-conceptual-default-target-hit
    trace-table composition-table
    (Some ((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
using hit subset
unfolding
  checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-def
  checked-staged-security-with-query-prefix-conceptual-default-target-hit-def
  checked-staged-security-with-query-prefix-dynamic-index-hit-def
  checked-staged-query-prefix-dynamic-index-hit-def
by auto
then show ?thesis by simp
next
case False
then have
  checked-staged-security-with-query-prefix-candidate-pair-default-agreement-gap
    trace-table composition-table
    (Some ((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
unfolding
  checked-staged-security-with-query-prefix-candidate-pair-default-agreement-gap-def
by simp
then show ?thesis by simp
qed

```

lemma *checked-staged-security-with-query-prefix-candidate-pair-hit-imp-conceptual-default-or-sampled-agreement-gap*

assumes *hit*:

checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
trace-table composition-table

*(Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))*

and *trace-len*: *length trace-table = scale * clength*

and *composition-len*: *length composition-table = scale * clength*

and *default-trace-low*:

trace-table-low-degree

*(query-prefix-trace-conceptual-table-with-default prefix prefix-state
 trace-table)*

and *default-composition-low*:

composition-table-low-degree maxDegree

*(query-prefix-composition-conceptual-table-with-default prefix
 prefix-state composition-table)*

and *default-not-all*:

$\neg$ *all-queries-consistent*

*(query-prefix-trace-conceptual-table-with-default prefix prefix-state
 trace-table)*

*(query-prefix-composition-conceptual-table-with-default prefix
 prefix-state composition-table)*

(sqp-alphas prefix)

shows

checked-staged-security-with-query-prefix-conceptual-default-target-hit
trace-table composition-table

*(Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state)) $\vee$*

checked-staged-security-with-query-prefix-candidate-pair-sampled-default-agreement-gap
trace-table composition-table

*(Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))*

proof (*cases*

query-prefix-candidate-defaults-agree-at-query trace-table
composition-table (index (to-nat raw)) prefix prefix-state)

case *True*

have *raw-hit*:

index (to-nat raw) $\in$

staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table prefix prefix-state

using *hit*

unfolding

checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-def

checked-staged-security-with-query-prefix-dynamic-index-hit-def

checked-staged-query-prefix-dynamic-index-hit-def

by *simp*

have *default-raw-hit*:

index (to-nat raw) $\in$

```

    staged-query-prefix-conceptual-default-query-target trace-table
    composition-table prefix prefix-state
  by (rule
    query-prefix-candidate-pair-query-hit-imp-conceptual-default-if-agrees-at-query
    [OF raw-hit trace-len composition-len True default-trace-low
    default-composition-low default-not-all])
  have
    checked-staged-security-with-query-prefix-conceptual-default-target-hit
    trace-table composition-table
    (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  unfolding
    checked-staged-security-with-query-prefix-conceptual-default-target-hit-def
    checked-staged-security-with-query-prefix-dynamic-index-hit-def
    checked-staged-query-prefix-dynamic-index-hit-def
  using default-raw-hit by simp
  then show ?thesis by simp
next
  case False
  then have
    checked-staged-security-with-query-prefix-candidate-pair-sampled-default-agreement-gap
    trace-table composition-table
    (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  unfolding
    checked-staged-security-with-query-prefix-candidate-pair-sampled-default-agreement-gap-def
  by simp
  then show ?thesis by simp
qed

lemma checked-staged-security-with-query-prefix-candidate-pair-hit-imp-conceptual-default-if-final-merkle-bound:
  assumes hit:
    checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
    trace-table composition-table
    (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  and prefix-clean:  $\neg$  hash-map-output-collision prefix-state
  and final-clean:  $\neg$  hash-map-output-collision final-state
  and ext:  $\text{prefix-state} \leq \text{final-state}$ 
  and trace-bind:
    merkle-root-binds-table (sqp-trace-root prefix) trace-table
    final-state
  and composition-bind:
    merkle-root-binds-table (hd (sqp-composition-fri-roots prefix))
    composition-table final-state
  and trace-len:  $\text{length trace-table} = \text{scale} * \text{clength}$ 
  and composition-len:  $\text{length composition-table} = \text{scale} * \text{clength}$ 
  shows
    checked-staged-security-with-query-prefix-conceptual-default-target-hit

```



```

    trace-table composition-table
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
proof –
  have subset:
    staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
    composition-table prefix prefix-state  $\subseteq$ 
    staged-query-prefix-conceptual-default-query-target trace-table
    composition-table prefix prefix-state
  by (rule
    staged-query-prefix-candidate-pair-query-target-from-prefix-subset-conceptual-default-if-final-merkle-bound
    [OF prefix-clean final-clean ext trace-bind composition-bind trace-len
      composition-len])
  show ?thesis
  using hit subset
  unfolding
    checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-def
    checked-staged-security-with-query-prefix-conceptual-default-target-hit-def
    checked-staged-security-with-query-prefix-dynamic-index-hit-def
    checked-staged-query-prefix-dynamic-index-hit-def
  by auto
qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-conceptual-default-or-residual-or-stru
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i < \text{rounds}$ 
and support:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
  trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
shows
  checked-staged-security-with-query-prefix-conceptual-default-target-hit
  trace-table composition-table
  (Some (((((prefix, prefix-state), raw), raw-state),

```

```

      ((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
trace-table composition-table i
      (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
i
      (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
i
      (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
proof –
  let ?out =
    Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)
  have current-hit:
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
    i ?out
  by (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-atI
    [OF hit])
  show ?thesis
  proof (cases
    ¬ hash-map-output-collision prefix-state ∧
    trace-table-low-degree
    (query-prefix-trace-conceptual-table-with-default prefix
    prefix-state trace-table) ∧
    composition-table-low-degree maxDegree
    (query-prefix-composition-conceptual-table-with-default prefix
    prefix-state composition-table) ∧
    ¬ all-queries-consistent
    (query-prefix-trace-conceptual-table-with-default prefix
    prefix-state trace-table)
    (query-prefix-composition-conceptual-table-with-default prefix
    prefix-state composition-table)
    (sqp-alphas prefix))
  case False
  then have
    checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
    trace-table composition-table i ?out
  unfolding
    checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-def
    checked-staged-security-with-query-prefix-conceptual-default-context-residual-at-def
  using current-hit by simp
  then show ?thesis by simp
next
  case True

```

```

then have clean:
   $\neg$  hash-map-output-collision prefix-state
and default-trace-low:
  trace-table-low-degree
    (query-prefix-trace-conceptual-table-with-default prefix
     prefix-state trace-table)
and default-composition-low:
  composition-table-low-degree maxDegree
    (query-prefix-composition-conceptual-table-with-default prefix
     prefix-state composition-table)
and default-not-all:
   $\neg$  all-queries-consistent
    (query-prefix-trace-conceptual-table-with-default prefix
     prefix-state trace-table)
    (query-prefix-composition-conceptual-table-with-default prefix
     prefix-state composition-table)
    (sqp-alphas prefix)
by blast+
have ext: prefix-state  $\leq$  final-state
by (rule
  checked-staged-security-experiment-with-query-prefix-data-state-prefix-extends-final
  [OF wf controlled i-bound support])
have alphas-eq: staged-alphas data = sqp-alphas prefix
by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq
  [OF support])
have trace-root-eq: staged-trace-root data = sqp-trace-root prefix
using support
unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
  checked-staged-after-query-prefix-receive-with-verifier-def
  checked-staged-after-query-prefix-receive-def
by (auto elim!: set-dist-bindE split: prod.splits)
have composition-roots-eq:
  staged-composition-fri-roots data = sqp-composition-fri-roots prefix
using support
unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
  checked-staged-after-query-prefix-receive-with-verifier-def
  checked-staged-after-query-prefix-receive-def
by (auto elim!: set-dist-bindE split: prod.splits)
have comp-nonempty: staged-composition-fri-roots data  $\neq$  []
and trace-table-final:
  partial-authenticated-table (staged-trace-root data)
  (scale * clength) trace-openings final-state
and comp-table-final:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
  (scale * clength) composition-openings final-state
and consistent:
  partial-query-openings-consistent trace-openings composition-openings
  (staged-alphas data) (index (to-nat raw))
using hit

```

```

unfolding checked-staged-security-with-query-prefix-authenticated-opening-hit-def
by simp-all
have trace-pullback:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings prefix-state  $\vee$ 
    ( $\exists$  opn  $\in$  set trace-openings.
      hash-map-new-output-hit
        (merkle-path-target-roots final-state (staged-trace-root data)
          (scale * clength) (opening-index opn) (opening-value opn)
          (opening-path opn)) prefix-state final-state)
  by (rule partial-authenticated-table-pullback-or-new-output-hit
    [OF ext trace-table-final])
then show ?thesis
proof
  assume trace-prefix-staged:
    partial-authenticated-table (staged-trace-root data)
      (scale * clength) trace-openings prefix-state
  have comp-pullback:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings prefix-state  $\vee$ 
      ( $\exists$  opn  $\in$  set composition-openings.
        hash-map-new-output-hit
          (merkle-path-target-roots final-state
            (hd (staged-composition-fri-roots data))
            (scale * clength) (opening-index opn) (opening-value opn)
            (opening-path opn)) prefix-state final-state)
    by (rule partial-authenticated-table-pullback-or-new-output-hit
      [OF ext comp-table-final])
  then show ?thesis
proof
  assume comp-prefix-staged:
    partial-authenticated-table (hd (staged-composition-fri-roots data))
      (scale * clength) composition-openings prefix-state
  have trace-prefix:
    partial-authenticated-table (sqp-trace-root prefix)
      (scale * clength) trace-openings prefix-state
    using trace-prefix-staged trace-root-eq by simp
  have composition-prefix:
    partial-authenticated-table (hd (sqp-composition-fri-roots prefix))
      (scale * clength) composition-openings prefix-state
    using comp-prefix-staged composition-roots-eq by simp
  have consistent-prefix:
    partial-query-openings-consistent trace-openings
      composition-openings (sqp-alphas prefix) (index (to-nat raw))
    using consistent alphas-eq by simp
  have target:
    index (to-nat raw)  $\in$ 
      staged-query-prefix-conceptual-default-query-target trace-table
      composition-table prefix prefix-state

```

```

    by (rule
      query-prefix-current-openings-imp-conceptual-default-query-target
      [OF clean i-bound trace-prefix composition-prefix
        consistent-prefix trace-candidate comp-candidate
        default-trace-low default-composition-low default-not-all])
  have
    checked-staged-security-with-query-prefix-conceptual-default-target-hit
    trace-table composition-table ?out
  unfolding
    checked-staged-security-with-query-prefix-conceptual-default-target-hit-def
    checked-staged-security-with-query-prefix-dynamic-index-hit-def
    checked-staged-query-prefix-dynamic-index-hit-def
  using target by simp
  then show ?thesis by simp
next
  assume comp-path:
     $\exists \text{opn} \in \text{set composition-openings.}$ 
    hash-map-new-output-hit
    (merkle-path-target-roots final-state
      (hd (staged-composition-fri-roots data))
      (scale * clength) (opening-index opn) (opening-value opn)
      (opening-path opn)) prefix-state final-state
  then obtain opn where opn-in:  $\text{opn} \in \text{set composition-openings}$ 
  and path-hit:
    hash-map-new-output-hit
    (merkle-path-target-roots final-state
      (hd (staged-composition-fri-roots data))
      (scale * clength) (opening-index opn) (opening-value opn)
      (opening-path opn)) prefix-state final-state
  by blast
  have auth: authenticated-opening-in final-state opn
  using comp-table-final opn-in
  unfolding partial-authenticated-table-def by blast
  have root-eq:
    opening-root opn = hd (staged-composition-fri-roots data)
  using comp-table-final opn-in
  unfolding partial-authenticated-table-def by blast
  have len-eq: opening-length opn = scale * clength
  using comp-table-final opn-in
  unfolding partial-authenticated-table-def by blast
  have
    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
    i ?out
  unfolding
    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at-def
  using comp-nonempty auth root-eq len-eq path-hit by auto
  then show ?thesis by simp
qed
next

```

```

assume trace-path:
   $\exists \text{opn} \in \text{set trace-openings.}$ 
  hash-map-new-output-hit
    (merkle-path-target-roots final-state (staged-trace-root data)
     (scale * clength) (opening-index opn) (opening-value opn)
     (opening-path opn)) prefix-state final-state
then obtain opn where opn-in: opn  $\in \text{set trace-openings}$ 
and path-hit:
  hash-map-new-output-hit
    (merkle-path-target-roots final-state (staged-trace-root data)
     (scale * clength) (opening-index opn) (opening-value opn)
     (opening-path opn)) prefix-state final-state
by blast
have auth: authenticated-opening-in final-state opn
using trace-table-final opn-in
unfolding partial-authenticated-table-def by blast
have root-eq: opening-root opn = staged-trace-root data
using trace-table-final opn-in
unfolding partial-authenticated-table-def by blast
have len-eq: opening-length opn = scale * clength
using trace-table-final opn-in
unfolding partial-authenticated-table-def by blast
have
  checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
    i ?out
unfolding
  checked-staged-security-with-query-prefix-current-trace-path-output-hit-at-def
using auth root-eq len-eq path-hit by auto
then show ?thesis by simp
qed
qed
qed

```

```

lemma checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-imp-conceptual-default-or-residual-or
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and support:
  out  $\in$ 
    set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
    trace-table composition-table i out
shows
  checked-staged-security-with-query-prefix-conceptual-default-target-hit
    trace-table composition-table out  $\vee$ 

```

```

checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
  trace-table composition-table i out  $\vee$ 
checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
  i out  $\vee$ 
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
  i out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-def
    checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
from hit[unfolded out-eq]
  checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-def]
obtain trace-openings composition-openings where opening-hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
  trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
by blast
have support':
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)  $\in$ 
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    adversary-initial-state)
  using support unfolding out-eq .
show ?thesis
unfolding out-eq
by (rule
  checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-conceptual-default-or-residual-or-structu
  [OF wf controlled i-bound support' opening-hit trace-candidate

```

comp-candidate])
qed

lemma *checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-imp-current:*

assumes

checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
i out

shows

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
i out

using *assms*

unfolding

checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-def
by *simp*

lemma *checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-imp-residual:*

assumes

checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
i out

shows

checked-staged-security-with-query-prefix-conceptual-context-residual-at
out

using *assms*

unfolding

checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-def
by *simp*

definition *checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at*

where

checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at
i out $\longleftrightarrow$
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
i out $\wedge$
(case *out* of
 None $\Rightarrow$ False
 | Some (((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) $\Rightarrow$
 hash-map-output-collision prefix-state)

definition *checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at*

where

checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
i out $\longleftrightarrow$
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
i out $\wedge$
(case *out* of
 None $\Rightarrow$ False
 | Some (((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) $\Rightarrow$

$\neg$ *trace-table-low-degree*
 (*query-prefix-trace-conceptual-table* *prefix* *prefix-state*) $\vee$
 $\neg$ *composition-table-low-degree* *maxDegree*
 (*query-prefix-composition-conceptual-table* *prefix* *prefix-state*) $\vee$
all-queries-consistent
 (*query-prefix-trace-conceptual-table* *prefix* *prefix-state*)
 (*query-prefix-composition-conceptual-table* *prefix* *prefix-state*)
 (*sqp-alphas* *prefix*)

definition *checked-staged-security-with-query-prefix-conceptual-trace-low-degree-bad-at*
where

checked-staged-security-with-query-prefix-conceptual-trace-low-degree-bad-at
i out $\longleftrightarrow$
checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
i out $\wedge$
 (case out of
 None $\Rightarrow$ *False*
 | *Some* ((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) $\Rightarrow$
 $\neg$ *trace-table-low-degree*
 (*query-prefix-trace-conceptual-table* *prefix* *prefix-state*))

definition *checked-staged-security-with-query-prefix-conceptual-composition-low-degree-bad-at*
where

checked-staged-security-with-query-prefix-conceptual-composition-low-degree-bad-at
i out $\longleftrightarrow$
checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
i out $\wedge$
 (case out of
 None $\Rightarrow$ *False*
 | *Some* ((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) $\Rightarrow$
 $\neg$ *composition-table-low-degree* *maxDegree*
 (*query-prefix-composition-conceptual-table* *prefix* *prefix-state*))

definition *checked-staged-security-with-query-prefix-conceptual-alpha-bad-set-hit-at*
where

checked-staged-security-with-query-prefix-conceptual-alpha-bad-set-hit-at
i out $\longleftrightarrow$
checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
i out $\wedge$
 (case out of
 None $\Rightarrow$ *False*
 | *Some* ((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) $\Rightarrow$
 sqp-alphas *prefix* $\in$
 composition-trace-bad-alpha-space
 (*query-prefix-trace-conceptual-table* *prefix* *prefix-state*))

lemma *checked-staged-query-prefix-with-state-alphas-length*:

assumes *support*:

Some ((prefix, prefix-state), prefix-state) ∈ set-dist (execute (checked-staged-query-prefix-with-state A i) adversary-initial-state)

shows *length (sqp-alphas prefix) = length spec*

proof –

from *support* **have** *prefix-out*:

Some (prefix, prefix-state) ∈ set-dist (execute (checked-staged-query-challenge-prefix-program A i) adversary-initial-state)

unfolding *checked-staged-query-prefix-with-state-def*

by (*auto elim!: set-dist-bindE*)

from *prefix-out* **obtain** *fr s1 s2 trace-roots trace-bs s3 trace-final s4 s5 as s6 dg s7 s8 s9 composition-roots composition-bs s10 composition-final s11 query-start query-chunks* **where**

alpha-out:

Some (as, s6) ∈ set-dist (execute (staged-alpha-program (length spec)) s5)

and *prefix-eq*:

prefix =

(|sqp-trace-root = fr,

sqp-trace-fri-roots = trace-roots,

sqp-trace-fri-challenges = trace-bs,

sqp-trace-final = trace-final,

sqp-alphas = as,

sqp-degree = dg,

sqp-composition-fri-roots = composition-roots,

sqp-composition-fri-challenges = composition-bs,

sqp-composition-final = composition-final,

sqp-query-chunks = query-chunks)

using *prefix-out*

by (*elim checked-staged-query-challenge-prefix-program-outcomeE*)

have *length as = length spec*

using *staged-alpha-program-alignment[OF alpha-out]* **by** *simp*

then show *?thesis*

unfolding *prefix-eq* **by** *simp*

qed

lemma *checked-staged-security-with-query-prefix-data-state-prefix-support*:

assumes *support*:

Some (((((prefix, prefix-state), raw), raw-state), ((data, attacker-state), result)), final-state) ∈ set-dist (execute (checked-staged-security-experiment-with-query-prefix-data-state A i) adversary-initial-state)

shows
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state)

proof –
from *support* **have** *receive-support*:
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
unfolding *checked-staged-security-experiment-with-query-prefix-data-state-def*
by (*auto elim!*: *set-dist-bindE split: prod.splits*)
then show *?thesis*
by (*rule checked-staged-query-prefix-receive-with-state-prefix-support*)
qed

lemma *checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-cases-on-support*:
assumes *false-statement: ¬ exists-valid-trace*
and *support*:
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)
and *hit*:
checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

shows
 $\neg$ *trace-table-low-degree*
(query-prefix-trace-conceptual-table prefix prefix-state) ∨
 $\neg$ *composition-table-low-degree maxDegree*
(query-prefix-composition-conceptual-table prefix prefix-state) ∨
sqp-alphas prefix ∈
composition-trace-bad-alpha-space
(query-prefix-trace-conceptual-table prefix prefix-state)

proof –
let *?trace = query-prefix-trace-conceptual-table prefix prefix-state*
let *?composition =*
query-prefix-composition-conceptual-table prefix prefix-state
from *hit* **have** *residual*:
 $\neg$ *trace-table-low-degree ?trace ∨*
 $\neg$ *composition-table-low-degree maxDegree ?composition ∨*
all-queries-consistent ?trace ?composition (sqp-alphas prefix)
unfolding
checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-def

```

    by simp
show ?thesis
proof (cases trace-table-low-degree ?trace)
  case False
  then show ?thesis by simp
next
case trace-low: True
show ?thesis
proof (cases composition-table-low-degree maxDegree ?composition)
  case False
  then show ?thesis by simp
next
case composition-low: True
from residual trace-low composition-low have all-queries:
  all-queries-consistent ?trace ?composition (sqp-alphas prefix)
  by simp
have prefix-support:
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)
  by (rule
    checked-staged-security-with-query-prefix-data-state-prefix-support
    [OF support])
have alpha-len: length (sqp-alphas prefix) = length spec
  by (rule checked-staged-query-prefix-with-state-alphas-length
    [OF prefix-support])
have
  sqp-alphas prefix ∈
    composition-trace-bad-alpha-space ?trace
  by (rule low-degree-all-queries-consistent-imp-composition-trace-bad-alpha-space
    [OF false-statement alpha-len trace-low composition-low
      all-queries])
  then show ?thesis by simp
qed
qed
qed

lemma checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-split-on-support:
  assumes false-statement:  $\neg$  exists-valid-trace
  and support:
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state) ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-query-prefix-data-state A i)
          adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at

```

i
 $(\text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state}))$

shows

$\text{checked-staged-security-with-query-prefix-conceptual-trace-low-degree-bad-at}$
 i
 $(\text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state})) \vee$
 $\text{checked-staged-security-with-query-prefix-conceptual-composition-low-degree-bad-at}$
 i
 $(\text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state})) \vee$
 $\text{checked-staged-security-with-query-prefix-conceptual-alpha-bad-set-hit-at}$
 i
 $(\text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state}))$

using

$\text{checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-cases-on-support}$
 $[OF \text{ false-statement support hit}] \text{ hit}$

unfolding

$\text{checked-staged-security-with-query-prefix-conceptual-trace-low-degree-bad-at-def}$
 $\text{checked-staged-security-with-query-prefix-conceptual-composition-low-degree-bad-at-def}$
 $\text{checked-staged-security-with-query-prefix-conceptual-alpha-bad-set-hit-at-def}$

by simp

lemma $\text{checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-bound-from-split:}$

assumes $\text{false-statement: } \neg \text{ exists-valid-trace}$

and trace-bound:

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-conceptual-trace-low-degree-bad-at}$
 $i)$
 $\text{adversary-initial-state} \leq T$

and $\text{composition-bound:}$

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-conceptual-composition-low-degree-bad-at}$
 $i)$
 $\text{adversary-initial-state} \leq C$

and alpha-bound:

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-conceptual-alpha-bad-set-hit-at}$
 $i)$
 $\text{adversary-initial-state} \leq P$

shows

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at}$

```

    i)
    adversary-initial-state  $\leq T + C + P$ 
proof –
  let ?M = checked-staged-security-experiment-with-query-prefix-data-state A i
  let ?Trace =
    checked-staged-security-with-query-prefix-conceptual-trace-low-degree-bad-at
    i
  let ?Composition =
    checked-staged-security-with-query-prefix-conceptual-composition-low-degree-bad-at
    i
  let ?Alpha =
    checked-staged-security-with-query-prefix-conceptual-alpha-bad-set-hit-at
    i
  have event-le:
    wp-event ?M
    (checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
     i)
    adversary-initial-state  $\leq$ 
    wp-event ?M ( $\lambda out. ?Trace out \vee ?Composition out \vee ?Alpha out$ )
    adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out  $\in$  set-dist (execute ?M adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
    i out
  show ?Trace out  $\vee ?Composition out \vee ?Alpha out$ 
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
    checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-def
    by simp
  next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  have support':
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)  $\in$ 
    set-dist (execute ?M adversary-initial-state)
    using support unfolding out-eq .
  have hit':

```

```

checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
  i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  using hit unfolding out-eq .
show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-split-on-support
      [OF false-statement support' hit'])
qed
qed
also have ... ≤
  wp-event ?M ?Trace adversary-initial-state +
  wp-event ?M ?Composition adversary-initial-state +
  wp-event ?M ?Alpha adversary-initial-state
proof -
  have ... ≤
    wp-event ?M ?Trace
      adversary-initial-state +
    wp-event ?M (λout. ?Composition out ∨ ?Alpha out)
      adversary-initial-state
  by (rule wp-event-union-bound
    [where P = ?Trace
      and Q = λout. ?Composition out ∨ ?Alpha out])
  also have ... ≤
    wp-event ?M ?Trace adversary-initial-state +
    (wp-event ?M ?Composition adversary-initial-state +
      wp-event ?M ?Alpha adversary-initial-state)
  by (intro add-mono order-refl wp-event-union-bound)
  finally show ?thesis
    by (simp add: add.assoc)
qed
also have ... ≤ T + C + P
  by (intro add-mono trace-bound composition-bound alpha-bound)
  finally show ?thesis .
qed

lemma checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-split:
  assumes
    checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
      i out
  shows
    checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at
      i out ∨
    checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
      i out
  using assms
  unfolding

```

checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-def
checked-staged-security-with-query-prefix-conceptual-context-residual-at-def
checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at-def
checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-def
by (cases out) (auto split: prod.splits)

lemma *checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

shows

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at*
i)

adversary-initial-state ≤

hash-collision-budget-value 0

(*staged-query-search-queries budgets i + 1*)

proof —

have *i-le: i ≤ rounds*

using *i-bound* **by** *simp*

have *prefix-bound:*

wp-event (checked-staged-query-prefix-receive-with-state A i)

checked-staged-query-prefix-hash-map-output-collision

adversary-initial-state ≤

hash-collision-budget-value 0

(*staged-query-search-queries budgets i + 1*)

by (rule *checked-staged-query-prefix-hash-map-output-collision-bound*

[*OF wf controlled i-le*])

show *?thesis*

proof (rule

checked-staged-security-experiment-with-query-prefix-data-state-bound-by-prefix

[*OF prefix-bound*])

show

checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at
i None ⇒

checked-staged-query-prefix-hash-map-output-collision None

unfolding

checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at-def

checked-staged-query-prefix-hash-map-output-collision-def

by *simp*

next

fix *prefix-out t out*

assume

Some (prefix-out, t) ∈

set-dist

(*execute (checked-staged-query-prefix-receive-with-state A i)*

adversary-initial-state)

and *cont:*


```

    out ∈
      set-dist
        (execute
          (get ≫=
            (λprefix-state.
              checked-staged-after-query-prefix-receive-with-verifier A i
                prefix-out ≫=
                  (λresult. return ((prefix-out, prefix-state), result))))
            t)
    and event:
      checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at
        i out
  show
    checked-staged-query-prefix-hash-map-output-collision
      (Some (prefix-out, t))
  using cont event
  unfolding
    checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at-def
    checked-staged-query-prefix-hash-map-output-collision-def
  by (auto elim!: set-dist-bindE split: option.splits prod.splits)
qed
qed

lemma checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-imp-prefix-collision-on-sup
  assumes default-context:
    ∧prefix prefix-state.
      Some ((prefix, prefix-state), prefix-state) ∈
        set-dist
          (execute (checked-staged-query-prefix-with-state A i)
            adversary-initial-state) ⇒
      trace-table-low-degree
        (query-prefix-trace-conceptual-table-with-default prefix
          prefix-state trace-default) ∧
      composition-table-low-degree maxDegree
        (query-prefix-composition-conceptual-table-with-default prefix
          prefix-state composition-default) ∧
    ¬ all-queries-consistent
      (query-prefix-trace-conceptual-table-with-default prefix
        prefix-state trace-default)
      (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-default)
      (sqp-alphas prefix)
  and support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-query-prefix-data-state A i)
            adversary-initial-state)
  and hit:

```

```

    checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
    trace-default composition-default i out
shows
    checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at
    i out
proof (cases out)
case None
then show ?thesis
    using hit
    unfolding
        checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-def
        checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-def
        checked-staged-security-with-query-prefix-authenticated-opening-hit-def
    by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
        Some (((((prefix, prefix-state), raw), raw-state),
            ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
have support':
    Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state) ∈
    set-dist
    (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
    using support unfolding out-eq .
have prefix-support:
    Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)
    by (rule
        checked-staged-security-with-query-prefix-data-state-prefix-support
        [OF support'])
have ctxt-fact:
    trace-table-low-degree
    (query-prefix-trace-conceptual-table-with-default prefix prefix-state
        trace-default) ∧
    composition-table-low-degree maxDegree
    (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-default) ∧
    ¬ all-queries-consistent
    (query-prefix-trace-conceptual-table-with-default prefix prefix-state
        trace-default)
    (query-prefix-composition-conceptual-table-with-default prefix

```

```

    prefix-state composition-default)
  (sqp-alphas prefix)
  by (rule default-context[OF prefix-support])
from hit ctxt-fact have collision: hash-map-output-collision prefix-state
unfolding out-eq
  checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-def
  checked-staged-security-with-query-prefix-conceptual-default-context-residual-at-def
by auto
have current-hit:
  checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
  i out
using hit
unfolding
  checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-def
by simp
show ?thesis
unfolding
  checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at-def
using current-hit collision out-eq by simp
qed

lemma checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-context:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and default-context:
     $\bigwedge \text{prefix prefix-state.}$ 
    Some ((prefix, prefix-state), prefix-state)  $\in$ 
      set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)  $\implies$ 
      trace-table-low-degree
      (query-prefix-trace-conceptual-table-with-default prefix
        prefix-state trace-default)  $\wedge$ 
      composition-table-low-degree maxDegree
      (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-default)  $\wedge$ 
       $\neg$  all-queries-consistent
      (query-prefix-trace-conceptual-table-with-default prefix
        prefix-state trace-default)
      (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-default)
      (sqp-alphas prefix)
  shows
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
      trace-default composition-default i)
    adversary-initial-state  $\leq$ 

```

hash-collision-budget-value 0
(staged-query-search-queries budgets $i + 1$)
proof –
let $?M = \text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i$
have *event-le:*
wp-event ?M
(checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
trace-default composition-default i)
adversary-initial-state $\leq$
wp-event ?M
(checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at
 i)
adversary-initial-state
by (rule *wp-event-mono-on-support*)
(rule
checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-imp-prefix-collision-on-support
[*OF default-context*])
also have ... $\leq$
hash-collision-budget-value 0
(staged-query-search-queries budgets $i + 1$)
by (rule
checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at-bound
[*OF wf controlled i-bound*])
finally show *?thesis* .
qed

lemma *checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-bound-from-split:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: $i < \text{rounds}$*

and *algebraic-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at

i)

adversary-initial-state $\leq R$

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at

i)

adversary-initial-state $\leq$

hash-collision-budget-value 0

(staged-query-search-queries budgets $i + 1$) + R

proof –

let $?M = \text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i$

let $?Collision =$

checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at

i

```

let ?Algebraic =
  checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
  i
have event-le:
  wp-event ?M
    (checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at i)
    adversary-initial-state ≤
    wp-event ?M (λout. ?Collision out ∨ ?Algebraic out)
    adversary-initial-state
by (rule wp-event-mono)
    (rule
      checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-split)
have union-bound:
  wp-event ?M (λout. ?Collision out ∨ ?Algebraic out)
    adversary-initial-state ≤
    wp-event ?M ?Collision adversary-initial-state +
    wp-event ?M ?Algebraic adversary-initial-state
by (rule wp-event-union-bound)
have collision-bound:
  wp-event ?M ?Collision adversary-initial-state ≤
    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1)
by (rule
    checked-staged-security-with-query-prefix-conceptual-context-prefix-collision-hit-at-bound
    [OF wf controlled i-bound])
have sum-bound:
  wp-event ?M ?Collision adversary-initial-state +
    wp-event ?M ?Algebraic adversary-initial-state ≤
    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1) + R
by (intro add-mono collision-bound algebraic-bound)
show ?thesis
by (rule order-trans[OF event-le order-trans[OF union-bound sum-bound]])
qed

```

lemma checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-bound-from-current:

assumes current-bound:

```

  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
      i)
    adversary-initial-state ≤ C

```

shows

```

  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
      i)
    adversary-initial-state ≤ C

```

by (rule order-trans[OF - current-bound])

(rule *wp-event-mono*,
 rule *checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-imp-current*)

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-conceptual-or-residual-or-s*

assumes *wf*: *staged-budget-wellformed* budgets

and *controlled*: *staged-adversary-controlled* budgets *A*

and *i-bound*: *i* < *rounds*

and *support*:

out ∈

set-dist

(*execute*

(*checked-staged-security-experiment-with-query-prefix-data-state* *A* *i*)

adversary-initial-state)

and *hit*:

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at

i out

shows

checked-staged-security-with-query-prefix-conceptual-target-hit out ∨

checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at

i out ∨

checked-staged-security-with-query-prefix-current-trace-path-output-hit-at

i out ∨

checked-staged-security-with-query-prefix-current-composition-path-output-hit-at

i out

proof —

have *split*:

checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at

i out ∨

checked-staged-security-with-query-prefix-current-trace-path-output-hit-at

i out ∨

checked-staged-security-with-query-prefix-current-composition-path-output-hit-at

i out

by (rule

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-prefix-or-structured-path-on-su

[*OF wf controlled i-bound support hit*])

then show *?thesis*

proof

assume *prefix*:

checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at

i out

have

checked-staged-security-with-query-prefix-conceptual-target-hit out ∨

checked-staged-security-with-query-prefix-conceptual-context-residual-at

out

by (rule

checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-imp-conceptual-or-residual-on-support

[*OF wf controlled i-bound support prefix*])

then show *?thesis*

unfolding

```

    checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-def
using hit by blast
next
assume
    checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
    i out  $\vee$ 
    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
    i out
then show ?thesis by blast
qed
qed

lemma checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-conceptual-default-or-resid
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and support:
    out  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
and hit:
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
    i out
shows
    checked-staged-security-with-query-prefix-conceptual-default-target-hit
    trace-default composition-default out  $\vee$ 
    checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
    trace-default composition-default i out  $\vee$ 
    checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
    i out  $\vee$ 
    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
    i out
proof —
have split:
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
    i out  $\vee$ 
    checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
    i out  $\vee$ 
    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
    i out
by (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-prefix-or-structured-path-on-su
    [OF wf controlled i-bound support hit])
then show ?thesis
proof
assume prefix:
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at

```

i out
have
checked-staged-security-with-query-prefix-conceptual-default-target-hit
trace-default composition-default out $\vee$
checked-staged-security-with-query-prefix-conceptual-default-context-residual-at
trace-default composition-default out
by (*rule*
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-imp-conceptual-default-or-residual-on-
[OF wf controlled i-bound support prefix])
then show *?thesis*
unfolding
checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-def
using *hit* **by** *blast*
next
assume
checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
i out $\vee$
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
i out
then show *?thesis* **by** *blast*
qed
qed

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-residual*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i < \text{rounds}$*
and *residual-bound:*
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
i)
adversary-initial-state $\leq R$
and *trace-path-bound:*
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
adversary-initial-state $\leq T$
and *composition-path-bound:*
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
adversary-initial-state $\leq C$
shows
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
adversary-initial-state $\leq$
staged-phase-relation-error size

$(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} + R + T + C$
proof –
let $?M = \text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i$
let $?Conceptual =$
 $\text{checked-staged-security-with-query-prefix-conceptual-target-hit}$
let $?Residual =$
 $\text{checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at } i$
let $?Trace =$
 $\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i$
let $?Composition =$
 $\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i$
have *event-le*:
 $\text{wp-event } ?M$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } ?M$
 $(\lambda \text{out. } ?Conceptual \text{ out} \vee ?Residual \text{ out} \vee ?Trace \text{ out} \vee$
 $?Composition \text{ out})$
 $\text{adversary-initial-state}$
by (rule *wp-event-mono-on-support*)
(rule
 $\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-conceptual-or-residual-or-stru}$
 $[OF \text{ wf controlled } i\text{-bound}]$)
have *conceptual-bound*:
 $\text{wp-event } ?M \ ?Conceptual \text{ adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) + \text{query-error-bound}$
by (rule *checked-staged-security-with-query-prefix-conceptual-target-hit-bound*
 $[OF \text{ wf controlled } i\text{-bound}]$)
have *union-bound*:
 $\text{wp-event } ?M$
 $(\lambda \text{out. } ?Conceptual \text{ out} \vee ?Residual \text{ out} \vee ?Trace \text{ out} \vee$
 $?Composition \text{ out})$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } ?M \ ?Conceptual \text{ adversary-initial-state} +$
 $\text{wp-event } ?M \ ?Residual \text{ adversary-initial-state} +$
 $\text{wp-event } ?M \ ?Trace \text{ adversary-initial-state} +$
 $\text{wp-event } ?M \ ?Composition \text{ adversary-initial-state}$
by (rule *wp-event-union-bound4*)
have *sum-bound*:
 $\text{wp-event } ?M \ ?Conceptual \text{ adversary-initial-state} +$
 $\text{wp-event } ?M \ ?Residual \text{ adversary-initial-state} +$
 $\text{wp-event } ?M \ ?Trace \text{ adversary-initial-state} +$
 $\text{wp-event } ?M \ ?Composition \text{ adversary-initial-state} \leq$
 $(\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) + \text{query-error-bound}) +$
 $R + T + C$

```

    by (intro add-mono conceptual-bound residual-bound trace-path-bound
        composition-path-bound)
  have event-bound:
    wp-event ?M
    ( $\lambda out. ?Conceptual\ out \vee ?Residual\ out \vee ?Trace\ out \vee$ 
       $?Composition\ out$ )
    adversary-initial-state  $\leq$ 
    (staged-phase-relation-error size
      (staged-query-search-queries budgets  $i + 1$ ) + query-error-bound) +
     $R + T + C$ 
    by (rule order-trans[OF union-bound sum-bound])
  show ?thesis
    by (rule order-trans[OF event-le])
      (use event-bound in  $\langle simp\ add: algebra-simps \rangle$ )
qed

lemma checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-
assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound:  $i < rounds$ 
  and residual-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
      trace-default composition-default i)
    adversary-initial-state  $\leq R$ 
  and trace-path-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
    adversary-initial-state  $\leq T$ 
  and composition-path-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
    adversary-initial-state  $\leq C$ 
shows
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets  $i + 1$ ) +
  query-error-bound +  $R + T + C$ 
proof -
  let ?M = checked-staged-security-experiment-with-query-prefix-data-state A i
  let ?Conceptual =
    checked-staged-security-with-query-prefix-conceptual-default-target-hit
    trace-default composition-default

```

```

let ?Residual =
  checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
    trace-default composition-default i
let ?Trace =
  checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i
let ?Composition =
  checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i
have event-le:
  wp-event ?M
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq$ 
  wp-event ?M
    ( $\lambda$ out. ?Conceptual out  $\vee$  ?Residual out  $\vee$  ?Trace out  $\vee$ 
      ?Composition out)
    adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-imp-conceptual-default-or-residual-
      [OF wf controlled i-bound])
have conceptual-bound:
  wp-event ?M ?Conceptual adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
by (rule
    checked-staged-security-with-query-prefix-conceptual-default-target-hit-bound
      [OF wf controlled i-bound])
have union-bound:
  wp-event ?M
    ( $\lambda$ out. ?Conceptual out  $\vee$  ?Residual out  $\vee$  ?Trace out  $\vee$ 
      ?Composition out)
    adversary-initial-state  $\leq$ 
  wp-event ?M ?Conceptual adversary-initial-state +
  wp-event ?M ?Residual adversary-initial-state +
  wp-event ?M ?Trace adversary-initial-state +
  wp-event ?M ?Composition adversary-initial-state
by (rule wp-event-union-bound4)
have sum-bound:
  wp-event ?M ?Conceptual adversary-initial-state +
  wp-event ?M ?Residual adversary-initial-state +
  wp-event ?M ?Trace adversary-initial-state +
  wp-event ?M ?Composition adversary-initial-state  $\leq$ 
    (staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) + query-error-bound) +
    R + T + C
by (intro add-mono conceptual-bound residual-bound trace-path-bound
    composition-path-bound)
have event-bound:
  wp-event ?M
    ( $\lambda$ out. ?Conceptual out  $\vee$  ?Residual out  $\vee$  ?Trace out  $\vee$ 

```

?Composition out)
adversary-initial-state $\leq$
(staged-phase-relation-error size
(staged-query-search-queries budgets $i + 1$) + query-error-bound) +
R + T + C
by (*rule order-trans*[*OF union-bound sum-bound*])
show *?thesis*
by (*rule order-trans*[*OF event-le*])
(use event-bound in ‹simp add: algebra-simps›)
qed

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: $i < \text{rounds}$*

and *default-context:*

$\bigwedge \text{prefix prefix-state.}$

Some ((prefix, prefix-state), prefix-state) $\in$
set-dist

(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) $\implies$

trace-table-low-degree

(query-prefix-trace-conceptual-table-with-default prefix
prefix-state trace-default) $\wedge$

composition-table-low-degree maxDegree

(query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default) $\wedge$

$\neg$ all-queries-consistent

(query-prefix-trace-conceptual-table-with-default prefix
prefix-state trace-default)

(query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default)

(sqp-alphas prefix)

and *trace-path-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)

adversary-initial-state $\leq T$

and *composition-path-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)

adversary-initial-state $\leq C$

shows

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)

adversary-initial-state $\leq$

staged-phase-relation-error size

(staged-query-search-queries budgets $i + 1$) +
 query-error-bound +
 hash-collision-budget-value 0
 (staged-query-search-queries budgets $i + 1$) + $T + C$

proof –

have residual-bound:

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
 trace-default composition-default i)

adversary-initial-state $\leq$

hash-collision-budget-value 0

(staged-query-search-queries budgets $i + 1$)

by (rule

checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-context
 [OF wf controlled i -bound default-context])

have bound:

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)

adversary-initial-state $\leq$

staged-phase-relation-error size

(staged-query-search-queries budgets $i + 1$) +

query-error-bound +

hash-collision-budget-value 0

(staged-query-search-queries budgets $i + 1$) + $T + C$

by (rule

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-residual-hit-at
 [OF wf controlled i -bound residual-bound trace-path-bound
 composition-path-bound])

show ?thesis

using bound **by** simp

qed

lemma checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-bound-from-conceptual-default-residual-hit-at

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and i -bound: $i < \text{rounds}$

and residual-bound:

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
 trace-table composition-table i)

adversary-initial-state $\leq R$

and trace-path-bound:

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)

adversary-initial-state $\leq T$

and *composition-path-bound*:

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i*)
adversary-initial-state $\leq C$

shows

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-authenticated-candidate-hit-at*
 trace-table composition-table i)
adversary-initial-state $\leq$
staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
query-error-bound + R + T + C

proof –

let *?M* = *checked-staged-security-experiment-with-query-prefix-data-state A i*

let *?Conceptual* =
checked-staged-security-with-query-prefix-conceptual-default-target-hit
trace-table composition-table

let *?Residual* =
checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
trace-table composition-table i

let *?Trace* =
checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i

let *?Composition* =
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i

have *event-le*:

wp-event ?M
 (*checked-staged-security-with-query-prefix-authenticated-candidate-hit-at*
 trace-table composition-table i)
adversary-initial-state $\leq$
wp-event ?M
 ($\lambda out. ?Conceptual\ out \vee ?Residual\ out \vee ?Trace\ out \vee$
 $?Composition\ out$)
adversary-initial-state

by (*rule wp-event-mono-on-support*)
 (*rule*
checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-imp-conceptual-default-or-residual-or-str
 [*OF wf controlled i-bound*])

have *conceptual-bound*:

wp-event ?M ?Conceptual adversary-initial-state $\leq$
staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) + *query-error-bound*

by (*rule*
checked-staged-security-with-query-prefix-conceptual-default-target-hit-bound
 [*OF wf controlled i-bound*])

have *union-bound*:

wp-event ?M
 ($\lambda out. ?Conceptual\ out \vee ?Residual\ out \vee ?Trace\ out \vee$

$?Composition\ out)$
 $adversary-initial-state \leq$
 $wp-event\ ?M\ ?Conceptual\ adversary-initial-state +$
 $wp-event\ ?M\ ?Residual\ adversary-initial-state +$
 $wp-event\ ?M\ ?Trace\ adversary-initial-state +$
 $wp-event\ ?M\ ?Composition\ adversary-initial-state$
by (rule *wp-event-union-bound4*)
have *sum-bound*:
 $wp-event\ ?M\ ?Conceptual\ adversary-initial-state +$
 $wp-event\ ?M\ ?Residual\ adversary-initial-state +$
 $wp-event\ ?M\ ?Trace\ adversary-initial-state +$
 $wp-event\ ?M\ ?Composition\ adversary-initial-state \leq$
 $(staged-phase-relation-error\ size$
 $(staged-query-search-queries\ budgets\ i + 1) + query-error-bound) +$
 $R + T + C$
by (intro *add-mono conceptual-bound residual-bound trace-path-bound*
composition-path-bound)
have *event-bound*:
 $wp-event\ ?M$
 $(\lambda out. ?Conceptual\ out \vee ?Residual\ out \vee ?Trace\ out \vee$
 $?Composition\ out)$
 $adversary-initial-state \leq$
 $(staged-phase-relation-error\ size$
 $(staged-query-search-queries\ budgets\ i + 1) + query-error-bound) +$
 $R + T + C$
by (rule *order-trans[OF union-bound sum-bound]*)
show *?thesis*
by (rule *order-trans[OF event-le]*)
 $(use\ event-bound\ in\ \langle simp\ add: algebra-simps \rangle)$
qed

lemma *checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-bound-from-conceptual-default-context*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *default-context:*

$\bigwedge prefix\ prefix-state.$

$Some\ ((prefix, prefix-state), prefix-state) \in$

set-dist

$(execute\ (checked-staged-query-prefix-with-state\ A\ i)$

$adversary-initial-state) \implies$

trace-table-low-degree

$(query-prefix-trace-conceptual-table-with-default\ prefix$

$prefix-state\ trace-table) \wedge$

composition-table-low-degree maxDegree

$(query-prefix-composition-conceptual-table-with-default\ prefix$

$prefix-state\ composition-table) \wedge$

$\neg\ all-queries-consistent$

$(query-prefix-trace-conceptual-table-with-default\ prefix$

```

    prefix-state trace-table)
  (query-prefix-composition-conceptual-table-with-default prefix
    prefix-state composition-table)
  (sqp-alphas prefix)
and trace-path-bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
  adversary-initial-state  $\leq T$ 
and composition-path-bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
  adversary-initial-state  $\leq C$ 
shows
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
    trace-table composition-table i)
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets  $i + 1$ ) +
  query-error-bound +
  hash-collision-budget-value 0
  (staged-query-search-queries budgets  $i + 1$ ) +  $T + C$ 
proof –
have residual-bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
    trace-table composition-table i)
  adversary-initial-state  $\leq$ 
  hash-collision-budget-value 0
  (staged-query-search-queries budgets  $i + 1$ )
by (rule
  checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-context
  [OF wf controlled i-bound default-context])
have bound:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
    trace-table composition-table i)
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
  (staged-query-search-queries budgets  $i + 1$ ) +
  query-error-bound +
  hash-collision-budget-value 0
  (staged-query-search-queries budgets  $i + 1$ ) +  $T + C$ 
by (rule

```


checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-bound-from-conceptual-default-residual-a
 [*OF wf controlled i-bound residual-bound trace-path-bound*
composition-path-bound])
show ?thesis
 using bound by simp
qed

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-algebraic*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *algebraic-residual-bound:*
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at*
 i)
 adversary-initial-state ≤ R
and *trace-path-bound:*
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i*)
 adversary-initial-state ≤ T
and *composition-path-bound:*
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i*)
 adversary-initial-state ≤ C
shows
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i*)
 adversary-initial-state ≤
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 query-error-bound +
 hash-collision-budget-value 0
 (*staged-query-search-queries budgets i + 1*) +
 R + T + C
proof –
have *residual-bound:*
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at*
 i)
 adversary-initial-state ≤
 hash-collision-budget-value 0
 (*staged-query-search-queries budgets i + 1*) + *R*
by (rule
 checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-bound-from-split

[OF wf controlled *i*-bound algebraic-residual-bound])
have bound:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i*)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 query-error-bound +
 (*hash-collision-budget-value 0*
 (*staged-query-search-queries budgets i + 1*) + *R*) + *T* + *C*
by (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-residual-and
 [OF wf controlled *i*-bound residual-bound trace-path-bound
 composition-path-bound])
show ?thesis
 using bound **by** (simp add: algebra-simps)
qed

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-residual-and*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: *i* < *rounds*
and *trace-low-degree-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-conceptual-trace-low-degree-bad-at*
 i)
 adversary-initial-state $\leq$ *LT*
and *composition-low-degree-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-conceptual-composition-low-degree-bad-at*
 i)
 adversary-initial-state $\leq$ *LC*
and *alpha-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-conceptual-alpha-bad-set-hit-at*
 i)
 adversary-initial-state $\leq$ *A'*
and *trace-path-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i*)
 adversary-initial-state $\leq$ *T*
and *composition-path-bound*:
 wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 adversary-initial-state $\leq C$

shows

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
 query-error-bound +
 hash-collision-budget-value 0
 (staged-query-search-queries budgets $i + 1$) +
 ($LT + LC + A'$) + $T + C$

proof –

have residual-bound:

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
 i)
 adversary-initial-state $\leq LT + LC + A'$

by (rule
 checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-bound-from-split
 [OF false-statement trace-low-degree-bound
 composition-low-degree-bound alpha-bound])

show ?thesis

by (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-algebraic-re
 [OF wf controlled i -bound residual-bound trace-path-bound
 composition-path-bound])

qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-conceptual-

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and residual-bound:

$\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
 i)
 adversary-initial-state $\leq R$ i

and trace-path-bound:

$\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 adversary-initial-state $\leq T$ i

and composition-path-bound:

$\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
shows
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit}$
 $\text{adversary-initial-state} \leq$
 $(\sum_{i < \text{rounds}}.$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} + R \ i + T \ i + C \ i)$
proof (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix-curr}$
 $[\text{OF wf controlled}])$
fix i
assume $i\text{-bound}: i < \text{rounds}$
show
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} + R \ i + T \ i + C \ i$
by (rule
 $\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-residual-and}$
 $[\text{OF wf controlled } i\text{-bound residual-bound}[\text{OF } i\text{-bound}]$
 $\text{trace-path-bound}[\text{OF } i\text{-bound}] \text{ composition-path-bound}[\text{OF } i\text{-bound}]])$
qed
lemma $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-conceptual-}$
assumes $\text{wf}: \text{staged-budget-wellformed budgets}$
and $\text{controlled}: \text{staged-adversary-controlled budgets } A$
and $\text{residual-bound}:$
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at}$
 $\text{trace-default composition-default } i)$
 $\text{adversary-initial-state} \leq R \ i$
and $\text{trace-path-bound}:$
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T \ i$

and *composition-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
shows
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit}$
 $\text{adversary-initial-state} \leq$
 $(\sum i < \text{rounds}.$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} + R \ i + T \ i + C \ i)$
proof (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix-curr}$
 $[\text{OF wf controlled}])$
fix i
assume $i\text{-bound}: i < \text{rounds}$
show
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} + R \ i + T \ i + C \ i$
by (rule
 $\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-resi}$
 $[\text{OF wf controlled } i\text{-bound residual-bound}[\text{OF } i\text{-bound}]$
 $\text{trace-path-bound}[\text{OF } i\text{-bound}] \text{ composition-path-bound}[\text{OF } i\text{-bound}]])$
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-conceptual-*
assumes $\text{wf}: \text{staged-budget-wellformed budgets}$
and $\text{controlled}: \text{staged-adversary-controlled budgets } A$
and *default-context*:
 $\bigwedge i \text{ prefix prefix-state}. i < \text{rounds} \implies$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \implies$
 $\text{trace-table-low-degree}$
 $(\text{query-prefix-trace-conceptual-table-with-default prefix}$
 $\text{prefix-state trace-default}) \wedge$
 $\text{composition-table-low-degree maxDegree}$
 $(\text{query-prefix-composition-conceptual-table-with-default prefix}$
 $\text{prefix-state composition-default}) \wedge$

$\neg$ *all-queries-consistent*
 (query-prefix-trace-conceptual-table-with-default prefix
 prefix-state trace-default)
 (query-prefix-composition-conceptual-table-with-default prefix
 prefix-state composition-default)
 (sqp-alphas prefix)

and *trace-path-bound*:

$\bigwedge i. i < \text{rounds} \implies$
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 adversary-initial-state $\leq T i$

and *composition-path-bound*:

$\bigwedge i. i < \text{rounds} \implies$
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 adversary-initial-state $\leq C i$

shows

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 adversary-initial-state $\leq$
 ($\sum i < \text{rounds}.$
 staged-phase-relation-error size
 (staged-query-search-queries budgets i + 1) +
 query-error-bound +
 hash-collision-budget-value 0
 (staged-query-search-queries budgets i + 1) +
 T i + C i)

proof (rule
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix-curr
 [OF wf controlled])

fix i

assume *i-bound*: i < rounds

show

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets i + 1) +
 query-error-bound +
 hash-collision-budget-value 0
 (staged-query-search-queries budgets i + 1) +
 T i + C i

by (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-cont
 [OF wf controlled i-bound - trace-path-bound[OF i-bound])

$\text{composition-path-bound}[OF\ i\text{-bound}]]])$
 $(\text{rule default-context}[OF\ i\text{-bound}])$

qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-conceptual-*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *algebraic-residual-bound:*

$\bigwedge i. i < \text{rounds} \implies$
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A\ i)$
 $(\text{checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at } i)$
 $\text{adversary-initial-state} \leq R\ i$

and *trace-path-bound:*

$\bigwedge i. i < \text{rounds} \implies$
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A\ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T\ i$

and *composition-path-bound:*

$\bigwedge i. i < \text{rounds} \implies$
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A\ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C\ i$

shows

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit}$
 $\text{adversary-initial-state} \leq$
 $(\sum i < \text{rounds}.$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $R\ i + T\ i + C\ i)$

proof (rule

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix-curr

$[OF\ wf\ \text{controlled}]])$

fix *i*

assume *i-bound: i < rounds*

show

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A\ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$

(staged-query-search-queries budgets $i + 1$) +
 query-error-bound +
 hash-collision-budget-value 0
 (staged-query-search-queries budgets $i + 1$) +
 $R\ i + T\ i + C\ i$
 by (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-algebraic-re
 [OF wf controlled i-bound algebraic-residual-bound[OF i-bound]
 trace-path-bound[OF i-bound] composition-path-bound[OF i-bound]])
 qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-conceptual-
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-low-degree-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-conceptual-trace-low-degree-bad-at
 i)
 adversary-initial-state $\leq LT\ i$
and composition-low-degree-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-conceptual-composition-low-degree-bad-at
 i)
 adversary-initial-state $\leq LC\ i$
and alpha-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-conceptual-alpha-bad-set-hit-at
 i)
 adversary-initial-state $\leq P\ i$
and trace-path-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 adversary-initial-state $\leq T\ i$
and composition-path-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 adversary-initial-state $\leq C\ i$
shows

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 adversary-initial-state $\leq$
 ($\sum i < \text{rounds}$.
 staged-phase-relation-error size
 (*staged-query-search-queries budgets* $i + 1$) +
 query-error-bound +
 hash-collision-budget-value 0
 (*staged-query-search-queries budgets* $i + 1$) +
 ($LT\ i + LC\ i + P\ i$) + $T\ i + C\ i$)

proof (rule
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix-curr
 [*OF wf controlled*])
fix i
assume $i\text{-bound}$: $i < \text{rounds}$
show
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* $A\ i$)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at* i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets* $i + 1$) +
 query-error-bound +
 hash-collision-budget-value 0
 (*staged-query-search-queries budgets* $i + 1$) +
 ($LT\ i + LC\ i + P\ i$) + $T\ i + C\ i$

by (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-residual-spl
 [*OF false-statement wf controlled i-bound*
 trace-low-degree-bound[*OF i-bound*]
 composition-low-degree-bound[*OF i-bound*]
 alpha-bound[*OF i-bound*]
 trace-path-bound[*OF i-bound*]
 composition-path-bound[*OF i-bound*]])

qed

lemma *checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-trivial-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* $A\ i$)
 (*checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at*
 i)
 adversary-initial-state ≤ 1
by (rule *wp-event-le-1*)

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-trivial-r*
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets* A
and $i\text{-bound}$: $i < \text{rounds}$

and *trace-path-bound*:
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 $\text{adversary-initial-state} \leq T$
and *composition-path-bound*:
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 $\text{adversary-initial-state} \leq C$
shows
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 (staged-query-search-queries budgets $i + 1$) +
 $\text{query-error-bound} + 1 + T + C$
by (rule
 checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-residual-and-
 [OF wf controlled i -bound
 checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-trivial-bound
 trace-path-bound composition-path-bound])

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-conceptual-*
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and *trace-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 $\text{adversary-initial-state} \leq T\ i$
and *composition-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 $\text{adversary-initial-state} \leq C\ i$
shows
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 $\text{adversary-initial-state} \leq$
 ($\sum i < \text{rounds}.$
 $\text{staged-phase-relation-error size}$
 (staged-query-search-queries budgets $i + 1$) +
 $\text{query-error-bound} + 1 + T\ i + C\ i$)

proof (rule

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-conceptual-residual
 [OF wf controlled - trace-path-bound composition-path-bound])
fix i
assume $i < \text{rounds}$
show
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at* i)
 adversary-initial-state ≤ 1
 by (rule
 checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-trivial-bound)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-conceptual-*
assumes wf: *staged-budget-wellformed* budgets
and controlled: *staged-adversary-controlled* budgets A
and *conceptual-context*:
 $\bigwedge i$ *prefix* *prefix-state*.
 $i < \text{rounds} \implies$
 Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
 set-dist
 (*execute* (*checked-staged-query-prefix-with-state* A i)
 adversary-initial-state) $\implies$
 $\neg$ *hash-map-output-collision* *prefix-state* $\wedge$
 trace-table-low-degree
 (*query-prefix-trace-conceptual-table* *prefix* *prefix-state*) $\wedge$
 composition-table-low-degree *maxDegree*
 (*query-prefix-composition-conceptual-table* *prefix* *prefix-state*) $\wedge$
 $\neg$ *all-queries-consistent*
 (*query-prefix-trace-conceptual-table* *prefix* *prefix-state*)
 (*query-prefix-composition-conceptual-table* *prefix* *prefix-state*)
 (*sqp-alphas* *prefix*)
and *trace-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-current-trace-path-output-hit-at* i)
 adversary-initial-state $\leq T$ i
and *composition-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* A i)
 (*checked-staged-security-with-query-prefix-current-composition-path-output-hit-at* i)
 adversary-initial-state $\leq C$ i
shows
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)
 checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
 adversary-initial-state $\leq$

```

    ( $\sum_{i < \text{rounds}}$ .
      staged-phase-relation-error size
      (staged-query-search-queries budgets  $i + 1$ ) +
      query-error-bound +  $T i + C i$ )
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-prefix-target-and-
  [OF wf controlled - trace-path-bound composition-path-bound])
fix  $i$ 
assume  $i\text{-bound}$ :  $i < \text{rounds}$ 
show
  wp-event (checked-staged-query-prefix-receive-with-state  $A i$ )
    (checked-staged-query-prefix-dynamic-index-hit
      staged-query-prefix-prefix-authenticated-opening-target)
  adversary-initial-state  $\leq$ 
  staged-phase-relation-error size
    (staged-query-search-queries budgets  $i + 1$ ) + query-error-bound
by (rule
  checked-staged-query-prefix-prefix-authenticated-target-hit-bound-from-conceptual
  [OF wf controlled  $i\text{-bound}$ ])
  (rule conceptual-context[OF  $i\text{-bound}$ ])
qed

end

end

```

theory *Soundness-Conceptual-Query-Current-Candidate*

imports *Soundness-Conceptual-Query-Current-Path*

begin

Fixed-candidate projection for current-query openings.

This layer keeps the trace/composition candidate tables as fixed parameters while projecting the data-state current-opening event to the corresponding query-prefix event. It is the form needed by the conceptual/default query target bound: the target is fixed before the query-index challenge.

context *soundness*

begin

definition *staged-security-with-data-state-current-query-partial-candidate-hit-at*

where

```

  staged-security-with-data-state-current-query-partial-candidate-hit-at
    trace-table composition-table  $i$  out  $\longleftrightarrow$ 
  (case out of
    None  $\Rightarrow$  False
  | Some (((data, attacker-state), result), final-state)  $\Rightarrow$ 
    (let  $s =$ 
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)

```

in $\exists$ *raw* *trace-openings* *composition-openings* *rest*.
i < *rounds* $\wedge$
fmlookup (*HashMap* *final-state*)
 (*QueryIndexChallenge* *i*
 (*state-after-query-chunks*
 (*staged-query-start-hash* *data*)
 (*staged-query-chunks* *data*) *i*)) =
 Some *raw* $\wedge$
 staged-composition-fri-roots *data* $\neq$ [] $\wedge$
 Some (*result*, *final-state*) $\in$ *set-dist* (*execute* *verify-monad* *s*) $\wedge$
 partial-authenticated-table (*staged-trace-root* *data*)
 (*scale* * *clength*) *trace-openings* *final-state* $\wedge$
 partial-authenticated-table
 (*hd* (*staged-composition-fri-roots* *data*)) (*scale* * *clength*)
 composition-openings *final-state* $\wedge$
 verifier-header-transcript *s*
 (*staged-trace-root* *data*)
 (*staged-trace-fri-roots* *data*)
 (*staged-trace-final* *data*)
 (*staged-alphas* *data*)
 (*staged-degree* *data*)
 (*staged-composition-fri-roots* *data*)
 (*staged-composition-final* *data*)
 rest $\wedge$
 partial-query-openings-consistent *trace-openings*
 composition-openings (*staged-alphas* *data*) (*index* (*to-nat* *raw*)) $\wedge$
 partial-trace-table-candidate *trace-table*
 ((*replicate* *rounds* []) [*i* := *trace-openings*]) $\wedge$
 partial-composition-table-candidate *composition-table*
 ((*replicate* *rounds* []) [*i* := *composition-openings*]))

definition *staged-security-with-data-state-current-query-some-partial-candidate-hit*
where

staged-security-with-data-state-current-query-some-partial-candidate-hit
out $\longleftrightarrow$
($\exists$ *trace-table* *composition-table* *i*.
i < *rounds* $\wedge$
staged-security-with-data-state-current-query-partial-candidate-hit-at
trace-table *composition-table* *i* *out*)

lemma *staged-security-with-data-state-current-query-partial-candidate-hit-at-imp-query-prefix-authenticated-candidate*

assumes *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *i-bound*: *i* < *rounds*
and *support*:
Some (((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) $\in$
set-dist
(*execute*

```

      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
and hit:
  staged-security-with-data-state-current-query-partial-candidate-hit-at
  trace-table composition-table i
  (Some (((data, attacker-state), result), final-state))
shows
  checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
  trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from hit obtain raw' trace-openings composition-openings rest where
  final-lookup':
    fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw'
and comp-nonempty: staged-composition-fri-roots data ≠ []
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
  (scale * clength) trace-openings final-state
and comp-auth:
  partial-authenticated-table
  (hd (staged-composition-fri-roots data)) (scale * clength)
  composition-openings final-state
and consistent':
  partial-query-openings-consistent trace-openings
  composition-openings (staged-alphas data) (index (to-nat raw'))
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
unfolding
  staged-security-with-data-state-current-query-partial-candidate-hit-at-def
  Let-def
by auto
have final-lookup:
  fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i)) =
  Some raw

```

```

by (rule
  checked-staged-security-experiment-with-query-prefix-data-state-raw-final-lookup
  [OF wf controlled i-bound support])
have raw-eq: raw' = raw
  using final-lookup' final-lookup by simp
have opening-hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
  trace-openings composition-openings i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  unfolding checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  using comp-nonempty trace-auth comp-auth consistent' raw-eq by simp
show ?thesis
  unfolding checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-def
  by (intro exI[of - trace-openings] exI[of - composition-openings]
    conjI opening-hit trace-candidate comp-candidate)
qed

```

lemma *staged-security-with-data-state-current-query-partial-candidate-hit-at-le-query-prefix-authenticated-candidate:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*

shows

wp-event
 (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-current-query-partial-candidate-hit-at*
trace-table composition-table i)
adversary-initial-state ≤
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-authenticated-candidate-hit-at*
trace-table composition-table i)
adversary-initial-state

proof —

let *?projected* =
 λ*out. case out of*
None ⇒
staged-security-with-data-state-current-query-partial-candidate-hit-at
trace-table composition-table i None
 | *Some (packed, t)* ⇒
staged-security-with-data-state-current-query-partial-candidate-hit-at
trace-table composition-table i (Some (snd packed, t))

have *projection:*

wp-event
 (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-current-query-partial-candidate-hit-at*
trace-table composition-table i)
adversary-initial-state =
wp-event

```

    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    ?projected
    adversary-initial-state
  by (rule
    checked-staged-security-experiment-with-query-prefix-data-state-projection-event
    [OF i-bound])
have event-le:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  ?projected
  adversary-initial-state ≤
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
    trace-table composition-table i)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    adversary-initial-state)
  and hit: ?projected out
show
  checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
  trace-table composition-table i out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    staged-security-with-data-state-current-query-partial-candidate-hit-at-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
show ?thesis
  unfolding out-eq
  by (rule
    staged-security-with-data-state-current-query-partial-candidate-hit-at-imp-query-prefix-authenticated-candidate
    [OF wf controlled i-bound])
  (use support hit out-eq in simp-all)

```



```

qed
qed
show ?thesis
  unfolding projection
  by (rule event-le)
qed

```

lemma *staged-security-with-data-state-current-query-partial-candidate-hit-at-bound-from-conceptual-default:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *residual-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

*(checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
trace-table composition-table i)*

adversary-initial-state ≤ R

and *trace-path-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)

adversary-initial-state ≤ T

and *composition-path-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)

adversary-initial-state ≤ C

shows

wp-event

(checked-staged-security-experiment-with-data-state A)

*(staged-security-with-data-state-current-query-partial-candidate-hit-at
trace-table composition-table i)*

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) +

query-error-bound + R + T + C

proof —

have *prefix-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

*(checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
trace-table composition-table i)*

adversary-initial-state ≤

staged-phase-relation-error size

(staged-query-search-queries budgets i + 1) +

query-error-bound + R + T + C

by (rule

checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-bound-from-conceptual-default-residual-a

[OF wf controlled i-bound residual-bound trace-path-bound

```

      composition-path-bound])
show ?thesis
by (rule order-trans
  [OF
    staged-security-with-data-state-current-query-partial-candidate-hit-at-le-query-prefix-authenticated-candidate
    prefix-bound])
  (use wf controlled i-bound in simp-all)
qed

lemma staged-security-with-data-state-current-query-partial-candidate-hit-at-bound-from-conceptual-default-context:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and default-context:
     $\bigwedge \text{prefix prefix-state.}$ 
    Some ((prefix, prefix-state), prefix-state)  $\in$ 
      set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)  $\implies$ 
      trace-table-low-degree
      (query-prefix-trace-conceptual-table-with-default prefix
        prefix-state trace-table)  $\wedge$ 
      composition-table-low-degree maxDegree
      (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-table)  $\wedge$ 
       $\neg$  all-queries-consistent
      (query-prefix-trace-conceptual-table-with-default prefix
        prefix-state trace-table)
      (query-prefix-composition-conceptual-table-with-default prefix
        prefix-state composition-table)
      (sqp-alphas prefix)
  and trace-path-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
    adversary-initial-state  $\leq T$ 
  and composition-path-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
    adversary-initial-state  $\leq C$ 
  shows
    wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-query-partial-candidate-hit-at
      trace-table composition-table i)
    adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +

```

query-error-bound +
hash-collision-budget-value 0
(*staged-query-search-queries* budgets *i* + 1) + *T* + *C*

proof –

have *prefix-bound*:

wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* *A* *i*)
(*checked-staged-security-with-query-prefix-authenticated-candidate-hit-at*
trace-table composition-table *i*)
adversary-initial-state ≤
staged-phase-relation-error size
(*staged-query-search-queries* budgets *i* + 1) +
query-error-bound +
hash-collision-budget-value 0
(*staged-query-search-queries* budgets *i* + 1) + *T* + *C*

by (rule
checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-bound-from-conceptual-default-context-an
[*OF wf controlled i-bound default-context trace-path-bound*
composition-path-bound])

show ?thesis

by (rule *order-trans*
[*OF*
staged-security-with-data-state-current-query-partial-candidate-hit-at-le-query-prefix-authenticated-candidate
prefix-bound])
(*use wf controlled i-bound in simp-all*)

qed

lemma *checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-data-state-current-*

assumes *wf*: *staged-budget-wellformed* budgets

and *controlled*: *staged-adversary-controlled* budgets *A*

and *support*:

Some (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*) ∈
set-dist
(*execute*
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
adversary-initial-state)

and *hit*:

checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
(*Some* (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*))

obtains *trace-table composition-table* *i* **where**
i < *rounds*
staged-security-with-data-state-current-query-partial-candidate-hit-at
trace-table composition-table *i*
(*Some* (((*data*, *attacker-state*), *result*), *final-state*))

proof –

let ?s =

```

    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
    by (rule
      checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
      [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
obtain builder verifier where
  builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
    and verifier:
      Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
    by blast
have header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
    by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled builder])
show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-current-openingsE
  [OF wf controlled support hit])
fix s trace-table composition-table query-idxs f i raw trace-openings
  composition-openings trace-openingsss composition-openingsss
assume s-def:
  s = verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
    and bad-context:
      composition-bad-context s (Some (result, final-state))
      trace-table composition-table (staged-alphas data) query-idxs f
    and i-bound: i < rounds
    and final-lookup:
      fmlookup (HashMap final-state)
      (QueryIndexChallenge i

```

```

    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
  Some raw
and idx-eq: index (to-nat raw) = query-idxs ! i
and comp-nonempty: staged-composition-fri-roots data ≠ []
and trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) trace-openings final-state
and comp-auth:
  partial-authenticated-table
    (hd (staged-composition-fri-roots data)) (scale * clength)
    composition-openings final-state
and consistent:
  partial-query-openings-consistent trace-openings
    composition-openings (staged-alphas data) (query-idxs ! i)
and trace-candidate-full:
  partial-trace-table-candidate trace-table trace-openings
and comp-candidate-full:
  partial-composition-table-candidate composition-table
    composition-openings
and trace-openings-eq: trace-openings = trace-openings ! i
and comp-openings-eq: composition-openings = composition-openings ! i
and all-queries:
  all-queries-consistent trace-table composition-table
    (staged-alphas data)
and not-alpha-prefix:
  ¬ staged-alphas data ∈
    alpha-prefix-union-bad-sets alpha-prefix-state
    composition-trace-bad-alpha-space (fst alpha-prefix)
have consistent-raw:
  partial-query-openings-consistent trace-openings composition-openings
    (staged-alphas data) (index (to-nat raw))
using consistent idx-eq by simp
have trace-candidate:
  partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
by (rule partial-trace-table-candidate-single-roundI)
    (use i-bound trace-candidate-full trace-openings-eq in
      ⟨auto simp: partial-trace-table-candidate-def⟩)
have comp-candidate:
  partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
by (rule partial-composition-table-candidate-single-roundI)
    (use i-bound comp-candidate-full comp-openings-eq in
      ⟨auto simp: partial-composition-table-candidate-def⟩)
have round-hit:
  staged-security-with-data-state-current-query-partial-candidate-hit-at
    trace-table composition-table i

```

```

    (Some (((data, attacker-state), result), final-state))
unfolding
    staged-security-with-data-state-current-query-partial-candidate-hit-at-def
    Let-def
apply simp
by (intro exI[of - raw] exI[of - trace-openings]
    exI[of - composition-openings]
    exI[of - List.concat (staged-query-chunks data)] conjI)
    (use i-bound final-lookup comp-nonempty verifier trace-auth
    comp-auth header consistent-raw trace-candidate comp-candidate
    in simp-all)
show ?thesis
by (rule that[OF i-bound round-hit])
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-data-state-current-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
    Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state) ∈
    set-dist
    (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
    adversary-initial-state)
and hit:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
shows
    staged-security-with-data-state-current-query-some-partial-candidate-hit
    (Some (((data, attacker-state), result), final-state))
proof –
from
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-data-state-current-partia
    [OF wf controlled support hit]
obtain trace-table composition-table i where
    i-bound: i < rounds
and round-hit:
    staged-security-with-data-state-current-query-partial-candidate-hit-at
    trace-table composition-table i
    (Some (((data, attacker-state), result), final-state))
by blast
show ?thesis
unfolding
    staged-security-with-data-state-current-query-some-partial-candidate-hit-def
by (intro exI[of - trace-table] exI[of - composition-table])

```

```

    exI[of - i] conjI i-bound round-hit)
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-le-data-state-current-so
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-some-partial-candidate-hit
    adversary-initial-state
proof -
  let ?project =
    λx. ((snd (fst (fst x)), snd (fst x)), snd x)
  let ?projected =
    λout. case out of
      None ⇒ False
    | Some (packed, t) ⇒
      staged-security-with-data-state-current-query-some-partial-candidate-hit
      (Some (?project packed, t))
  have event-le:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
      adversary-initial-state ≤
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      ?projected adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit
    out
  show ?projected out
  proof (cases out)
  case None
  then show ?thesis
  using hit

```

```

unfolding
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-def
  checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-def
  checked-staged-security-with-actual-alpha-prefix-verifier-event-def
by simp
next
case (Some packed)
then obtain alpha-prefix alpha-prefix-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data),
        attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
have candidate-hit:
  staged-security-with-data-state-current-query-some-partial-candidate-hit
    (Some (((data, attacker-state), result), final-state))
    by (rule
      checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-imp-data-state-current-s
        [OF wf controlled])
      (use support hit out-eq in simp-all)
show ?thesis
  unfolding out-eq
  using candidate-hit by simp
qed
qed
have projection:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-some-partial-candidate-hit
    adversary-initial-state
proof –
have
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-some-partial-candidate-hit
    adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
       $\gg$  ( $\lambda x.$  return (?project x)))
    staged-security-with-data-state-current-query-some-partial-candidate-hit
    adversary-initial-state
using
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
  [of A]
by simp
also have ... =
  wp-event

```



```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state
  by (subst wp-event-bind-return-map)
    (simp add:
      staged-security-with-data-state-current-query-some-partial-candidate-hit-def
      staged-security-with-data-state-current-query-partial-candidate-hit-at-def)
  finally show ?thesis
    by simp
qed
show ?thesis
  by (rule order-trans[OF event-le])
    (simp add: projection)
qed
end
end
end

```

```

theory Soundness-Staged-Aligned-Current-Query-Path
imports
  Soundness-Staged-Aligned-Current-Query
  Soundness-Conceptual-Query-Current-Candidate
begin

```

Current-query wrappers that use structured Merkle path-output events.

This layer keeps trace and composition authenticated-opening path outputs separate, which is the form needed for Merkle and candidate uniqueness arguments in the staged soundness proof.

```

context soundness
begin

```

```

theorem stark-soundness-from-aligned-partial-query-prefix-current-prefix-structured-paths:
fixes trace-fri-error' composition-fri-error' N :: prob
and P T C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  trace-fri-error'
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-fri-error'
and prefix-bound:

```

$\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at } i)$
 $\text{adversary-initial-state} \leq P \ i$
and trace-path-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T \ i$
and $\text{composition-path-bound:}$
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and not-prefix-bound:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and $\text{empty-trace-fri-bound:}$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and $\text{empty-query-bound:}$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{nnreal rounds} * \text{query-error-bound}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $N +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum i < \text{rounds}. P \ i + T \ i + C \ i) +$
 $(\text{composition-error-bound} +$

```

    staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
    N +
    staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
    trace-fri-error + nreal rounds * query-error-bound)
proof (rule stark-soundness-from-aligned-partial-query-prefix-current-rounds
  [OF false-statement wf controlled trace-fri-bound comp-fri-bound])
fix i
assume i-bound: i < rounds
show
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state ≤ P i + T i + C i
by (rule
  checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-structured-p
    [OF wf controlled i-bound prefix-bound[OF i-bound]
      trace-path-bound[OF i-bound] composition-path-bound[OF i-bound]])
qed (rule not-prefix-bound empty-trace-fri-bound empty-query-bound)+

theorem stark-soundness-from-aligned-partial-query-prefix-dynamic-target-structured-paths:
fixes trace-fri-error' composition-fri-error' N :: prob
and P T C :: nat ⇒ prob
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ trace-fri-error'
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-fri-error'
and prefix-target-bound:
  ∧ i. i < rounds ⇒
    wp-event
      (checked-staged-query-prefix-receive-with-state A i)
      (checked-staged-query-prefix-dynamic-index-hit
        staged-query-prefix-prefix-authenticated-opening-target)
      adversary-initial-state ≤ P i
and trace-path-bound:
  ∧ i. i < rounds ⇒
    wp-event

```

(checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 adversary-initial-state $\leq T$ i
and composition-path-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 adversary-initial-state $\leq C$ i
and not-prefix-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq N$
and empty-trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq \text{trace-fri-error}$
and empty-query-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq \text{nnreal rounds} * \text{query-error-bound}$
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 ($\sum i < \text{rounds}. P$ i + T i + C i) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 trace-fri-error + nnreal rounds * query-error-bound)

proof –
have *prefix-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at } i)$
 $\text{adversary-initial-state} \leq P \ i$
by (rule
 $\text{checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target}$
 $[OF \ \text{wf controlled} - \text{prefix-target-bound}])$
show *?thesis*
by (rule
 $\text{stark-soundness-from-aligned-partial-query-prefix-current-prefix-structured-paths}$
 $[OF \ \text{false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\text{prefix-bound trace-path-bound composition-path-bound not-prefix-bound}$
 $\text{empty-trace-fri-bound empty-query-bound}])$
qed

theorem *stark-soundness-from-aligned-partial-query-prefix-trace-index-structured-paths*:

fixes *trace-fri-error'* *composition-fri-error'* $N :: \text{prob}$
and $T \ C :: \text{nat} \Rightarrow \text{prob}$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}'$
and *comp-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and *trace-frac*:
 $\bigwedge i \text{ prefix prefix-state.}$
 $i < \text{rounds} \implies$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \implies$
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-prefix-authenticated-trace-indices prefix}$
 $\text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
and *trace-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 adversary-initial-state $\leq T$ i
and composition-path-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 adversary-initial-state $\leq C$ i
and not-prefix-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq N$
and empty-trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq \text{trace-fri-error}$
and empty-query-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq \text{nnreal rounds} * \text{query-error-bound}$
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 ($\sum i < \text{rounds}.$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
 query-error-bound + T i + C i) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +

staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
*trace-fri-error + nnreal rounds * query-error-bound)*

proof (rule
stark-soundness-from-aligned-partial-query-prefix-dynamic-target-structured-paths
[OF false-statement wf controlled trace-fri-bound comp-fri-bound - trace-path-bound
composition-path-bound not-prefix-bound empty-trace-fri-bound empty-query-bound])
fix *i*
assume *i-bound: i < rounds*
show
wp-event (checked-staged-query-prefix-receive-with-state A i)
(checked-staged-query-prefix-dynamic-index-hit
staged-query-prefix-prefix-authenticated-opening-target)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) + query-error-bound
by (rule
checked-staged-query-prefix-prefix-authenticated-target-hit-bound-from-trace-indices
[OF wf controlled i-bound])
(use trace-frac[OF i-bound] in blast)
qed

theorem *stark-soundness-from-aligned-partial-query-prefix-conceptual-structured-paths:*
fixes *trace-fri-error' composition-fri-error' N :: prob*
and *T C :: nat ⇒ prob*
assumes *false-statement: ¬ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state ≤ trace-fri-error'
and *comp-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state ≤ composition-fri-error'
and *conceptual-context:*
 $\bigwedge i$ *prefix prefix-state.*
i < rounds ⇒
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) ⇒
 $\neg$ *hash-map-output-collision prefix-state ∧*
trace-table-low-degree
(query-prefix-trace-conceptual-table prefix prefix-state) ∧
composition-table-low-degree maxDegree

$(\text{query-prefix-composition-conceptual-table } \text{prefix } \text{prefix-state}) \wedge$
 $\neg \text{all-queries-consistent}$
 $(\text{query-prefix-trace-conceptual-table } \text{prefix } \text{prefix-state})$
 $(\text{query-prefix-composition-conceptual-table } \text{prefix } \text{prefix-state})$
 $(\text{sqp-alphas } \text{prefix})$
and *trace-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T \ i$
and *composition-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *not-prefix-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound } \text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and *empty-trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event } \text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and *empty-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event } \text{query-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{nnreal rounds} * \text{query-error-bound}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $N +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum i < \text{rounds.}$
 $\text{staged-phase-relation-error size}$

(staged-query-search-queries budgets $i + 1$) +
 query-error-bound + $T i + C i$) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 trace-fri-error + $nnreal \text{ rounds} * \text{query-error-bound}$)

proof (rule
 stark-soundness-from-aligned-partial-query-prefix-dynamic-target-structured-paths
 [OF false-statement wf controlled trace-fri-bound comp-fri-bound - trace-path-bound
 composition-path-bound not-prefix-bound empty-trace-fri-bound empty-query-bound])

fix i
assume $i\text{-bound}$: $i < \text{rounds}$
show
 wp-event (checked-staged-query-prefix-receive-with-state $A i$)
 (checked-staged-query-prefix-dynamic-index-hit
 staged-query-prefix-prefix-authenticated-opening-target)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) + query-error-bound

by (rule
 checked-staged-query-prefix-prefix-authenticated-target-hit-bound-from-conceptual
 [OF wf controlled $i\text{-bound}$])
 (rule conceptual-context[OF $i\text{-bound}$])

qed

theorem stark-soundness-from-aligned-partial-query-prefix-conceptual-default-structured-paths:
fixes trace-fri-error' composition-fri-error' $N :: \text{prob}$
and $T C :: \text{nat} \Rightarrow \text{prob}$
and trace-default composition-default :: 'f list
assumes false-statement: $\neg \text{exists-valid-trace}$
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ trace-fri-error'
and comp-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ composition-fri-error'
and default-context:
 $\bigwedge i$ prefix prefix-state.

$i < \text{rounds} \implies$
Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
set-dist
 (*execute* (*checked-staged-query-prefix-with-state* *A i*)
adversary-initial-state) $\implies$
trace-table-low-degree
 (*query-prefix-trace-conceptual-table-with-default* *prefix*
prefix-state trace-default) $\wedge$
composition-table-low-degree maxDegree
 (*query-prefix-composition-conceptual-table-with-default* *prefix*
prefix-state composition-default) $\wedge$
 $\neg$ *all-queries-consistent*
 (*query-prefix-trace-conceptual-table-with-default* *prefix*
prefix-state trace-default)
 (*query-prefix-composition-conceptual-table-with-default* *prefix*
prefix-state composition-default)
 (*sqp-alphas* *prefix*)
and *trace-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
 (*checked-staged-security-with-query-prefix-current-trace-path-output-hit-at* *i*)
adversary-initial-state $\leq T i$
and *composition-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
 (*checked-staged-security-with-query-prefix-current-composition-path-output-hit-at* *i*)
adversary-initial-state $\leq C i$
and *not-prefix-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq N$
and *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq \text{trace-fri-error}$
and *empty-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq \text{nnreal rounds} * \text{query-error-bound}$
shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
hash-collision-budget-value *0*
 (*staged-attacker-query-budget* *budgets* +

staged-challenge-query-budget + *verifier-hash-query-budget*) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
N +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + *composition-fri-error'* +
 $(\sum i < \text{rounds}.$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
T i + C i) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
N +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + *nnreal rounds * query-error-bound*)
proof (*rule stark-soundness-from-aligned-partial-query-prefix-current-rounds*
[*OF false-statement wf controlled trace-fri-bound comp-fri-bound - not-prefix-bound*
empty-trace-fri-bound empty-query-bound])
fix *i*
assume *i-bound: i < rounds*
show
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
T i + C i
by (*rule*
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-cont
[*OF wf controlled i-bound - trace-path-bound[OF i-bound]*
composition-path-bound[OF i-bound]])
(rule default-context[OF i-bound])
qed

end

end

theory *Soundness-Conceptual-Query-Current-Alpha-Clean*
 imports *Soundness-Staged-Aligned-Current-Query-Path*
begin

Alpha-clean wrappers for the default-backed conceptual current-query route.

This layer derives the default-context non-consistency premise from the false statement, low degree of the fixed default tables, and the fact that the sampled alpha vector is not in the corresponding composition bad set.

context *soundness*
begin

lemma *checked-staged-security-with-query-prefix-conceptual-default-context-from-alpha-clean:*

assumes *false-statement*: $\neg \text{exists-valid-trace}$
 and *alphas-length*: $\text{length } (\text{sqp-alphas prefix}) = \text{length spec}$
 and *trace-low*:
 trace-table-low-degree
 (*query-prefix-trace-conceptual-table-with-default prefix prefix-state*
 trace-default)
 and *composition-low*:
 composition-table-low-degree maxDegree
 (*query-prefix-composition-conceptual-table-with-default prefix*
 prefix-state composition-default)
 and *alpha-clean*:
 sqp-alphas prefix $\notin$ *composition-trace-bad-alpha-space*
 (*query-prefix-trace-conceptual-table-with-default prefix prefix-state*
 trace-default)

shows

trace-table-low-degree
 (*query-prefix-trace-conceptual-table-with-default prefix prefix-state*
 trace-default) $\wedge$
 composition-table-low-degree maxDegree
 (*query-prefix-composition-conceptual-table-with-default prefix*
 prefix-state composition-default) $\wedge$
 $\neg \text{all-queries-consistent}$
 (*query-prefix-trace-conceptual-table-with-default prefix prefix-state*
 trace-default)
 (*query-prefix-composition-conceptual-table-with-default prefix*
 prefix-state composition-default)
 (*sqp-alphas prefix*)

proof (*intro conjI trace-low composition-low notI*)

assume *all-queries*:
 all-queries-consistent
 (*query-prefix-trace-conceptual-table-with-default prefix prefix-state*
 trace-default)

```

    (query-prefix-composition-conceptual-table-with-default prefix
     prefix-state composition-default)
    (sqp-alphas prefix)
have alpha-bad:
  sqp-alphas prefix ∈ composition-trace-bad-alpha-space
  (query-prefix-trace-conceptual-table-with-default prefix prefix-state
   trace-default)
by (rule low-degree-all-queries-consistent-imp-composition-trace-bad-alpha-space
     [OF false-statement alphas-length trace-low composition-low all-queries])
show False
  using alpha-clean alpha-bad by simp
qed

lemma checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-alpha-clean:
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound:  $i < \text{rounds}$ 
and default-alpha-clean:
   $\bigwedge \text{prefix prefix-state.}$ 
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
     adversary-initial-state)  $\implies$ 
  trace-table-low-degree
  (query-prefix-trace-conceptual-table-with-default prefix
   prefix-state trace-default)  $\wedge$ 
  composition-table-low-degree maxDegree
  (query-prefix-composition-conceptual-table-with-default prefix
   prefix-state composition-default)  $\wedge$ 
  sqp-alphas prefix  $\notin$  composition-trace-bad-alpha-space
  (query-prefix-trace-conceptual-table-with-default prefix
   prefix-state trace-default)
shows
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
   trace-default composition-default i)
  adversary-initial-state  $\leq$ 
  hash-collision-budget-value 0
  (staged-query-search-queries budgets i + 1)
proof (rule
  checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-context
  [OF wf controlled i-bound])
fix prefix prefix-state
assume prefix-support:
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)

```

adversary-initial-state)
from *default-alpha-clean*[*OF prefix-support*] **have** *trace-low*:
trace-table-low-degree
(query-prefix-trace-conceptual-table-with-default prefix
prefix-state trace-default)
and *composition-low*:
composition-table-low-degree maxDegree
(query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default)
and *alpha-clean*:
sqp-alphas prefix $\notin$ *composition-trace-bad-alpha-space*
(query-prefix-trace-conceptual-table-with-default prefix
prefix-state trace-default)
by *blast+*
have *alphas-length*: *length (sqp-alphas prefix) = length spec*
by (*rule checked-staged-query-prefix-with-state-alphas-length*
[*OF prefix-support*])
show
trace-table-low-degree
(query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default) $\wedge$
composition-table-low-degree maxDegree
(query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default) $\wedge$
 $\neg$ *all-queries-consistent*
(query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default)
(query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default)
(sqp-alphas prefix)
by (*rule*
checked-staged-security-with-query-prefix-conceptual-default-context-from-alpha-clean
[*OF false-statement alphas-length trace-low composition-low alpha-clean*])
qed

lemma *query-prefix-default-alpha-clean-from-alpha-prefix-union-clean*:
assumes *trace-eq*:
query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default = trace-default
and *trace-candidate*:
trace-default $\in$
alpha-prefix-trace-table-candidates prefix-state (sqp-trace-root prefix)
and *union-clean*:
sqp-alphas prefix $\notin$
alpha-prefix-union-bad-sets prefix-state composition-trace-bad-alpha-space
(sqp-trace-root prefix)
shows
sqp-alphas prefix $\notin$ *composition-trace-bad-alpha-space*
(query-prefix-trace-conceptual-table-with-default prefix prefix-state

trace-default)

proof

assume *alpha-bad*:

sqp-alphas prefix $\in$ *composition-trace-bad-alpha-space*
 (*query-prefix-trace-conceptual-table-with-default prefix prefix-state*
trace-default)

have *alpha-space*:

sqp-alphas prefix $\in$ *alpha-space*
using *alpha-bad composition-trace-bad-alpha-space-subset-alpha-space*
by *blast*

have

sqp-alphas prefix $\in$
alpha-prefix-union-bad-sets prefix-state composition-trace-bad-alpha-space
(sqp-trace-root prefix)

unfolding *alpha-prefix-union-bad-sets-def*
using *trace-candidate alpha-bad trace-eq alpha-space* **by** *auto*

then show *False*
using *union-clean* **by** *simp*

qed

lemma *checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-alpha-prefix-u*

assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: $i < \text{rounds}$
and *default-prefix-clean*:

$\bigwedge \text{prefix prefix-state.}$
Some ((prefix, prefix-state), prefix-state) $\in$
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) $\implies$
query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default = trace-default $\wedge$
query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default = composition-default $\wedge$
trace-table-low-degree trace-default $\wedge$
composition-table-low-degree maxDegree composition-default $\wedge$
trace-default $\in$
alpha-prefix-trace-table-candidates prefix-state
(sqp-trace-root prefix) $\wedge$
sqp-alphas prefix $\notin$
alpha-prefix-union-bad-sets prefix-state
composition-trace-bad-alpha-space (sqp-trace-root prefix)

shows

wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
trace-default composition-default i)
adversary-initial-state $\leq$

hash-collision-budget-value 0
(staged-query-search-queries budgets $i + 1$)
proof (rule
checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-alpha-clean
[OF false-statement wf controlled i-bound])
fix *prefix prefix-state*
assume *prefix-support*:
Some ((prefix, prefix-state), prefix-state) ∈
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state)
from *default-prefix-clean[OF prefix-support]* **have** *trace-eq*:
query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default = trace-default
and *composition-eq*:
query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default = composition-default
and *trace-low*: *trace-table-low-degree trace-default*
and *composition-low*:
composition-table-low-degree maxDegree composition-default
and *trace-candidate*:
trace-default ∈
alpha-prefix-trace-table-candidates prefix-state
(sqp-trace-root prefix)
and *union-clean*:
sqp-alphas prefix ∉
alpha-prefix-union-bad-sets prefix-state
composition-trace-bad-alpha-space (sqp-trace-root prefix)
by *blast+*
have *alpha-clean*:
sqp-alphas prefix ∉ composition-trace-bad-alpha-space
(query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default)
by (rule *query-prefix-default-alpha-clean-from-alpha-prefix-union-clean*
[OF trace-eq trace-candidate union-clean])
show
trace-table-low-degree
(query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default) ∧
composition-table-low-degree maxDegree
(query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default) ∧
sqp-alphas prefix ∉ composition-trace-bad-alpha-space
(query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default)
using *trace-eq composition-eq trace-low composition-low alpha-clean* **by** *simp*
qed

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-*

assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *i-bound*: $i < \text{rounds}$
and *default-alpha-clean*:
 $\bigwedge \text{prefix prefix-state.}$
Some $((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
set-dist
 $(\text{execute}(\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \implies$
trace-table-low-degree
 $(\text{query-prefix-trace-conceptual-table-with-default prefix}$
 $\text{prefix-state trace-default}) \wedge$
composition-table-low-degree maxDegree
 $(\text{query-prefix-composition-conceptual-table-with-default prefix}$
 $\text{prefix-state composition-default}) \wedge$
 $\text{sqp-alphas prefix} \notin \text{composition-trace-bad-alpha-space}$
 $(\text{query-prefix-trace-conceptual-table-with-default prefix}$
 $\text{prefix-state trace-default})$
and *trace-path-bound*:
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T$
and *composition-path-bound*:
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C$
shows
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-query-search-queries budgets } i + 1) + T + C$
proof –
have *residual-bound*:
wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at}$
 $\text{trace-default composition-default } i)$
 $\text{adversary-initial-state} \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-query-search-queries budgets } i + 1)$
by (rule

checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-alpha-clean
 [OF *false-statement* wf *controlled* *i-bound* *default-alpha-clean*])

have *bound*:

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at* *i*)
adversary-initial-state $\leq$
staged-phase-relation-error *size*
 (*staged-query-search-queries* *budgets* *i + 1*) +
query-error-bound +
hash-collision-budget-value 0
 (*staged-query-search-queries* *budgets* *i + 1*) + *T* + *C*

by (rule
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-residual-hit-at-bound
 [OF wf *controlled* *i-bound* *residual-bound* *trace-path-bound*
composition-path-bound])

show ?thesis

using *bound* **by** *simp*

qed

lemma *checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-residual-hit-at-bound*

assumes *false-statement*: $\neg$ *exists-valid-trace*

and *wf*: *staged-budget-wellformed* *budgets*

and *controlled*: *staged-adversary-controlled* *budgets* *A*

and *i-bound*: *i* < *rounds*

and *default-prefix-clean*:

$\bigwedge$ *prefix* *prefix-state*.
 Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
set-dist
 (*execute* (*checked-staged-query-prefix-with-state* *A i*)
adversary-initial-state) $\implies$
query-prefix-trace-conceptual-table-with-default *prefix* *prefix-state*
trace-default = *trace-default* $\wedge$
query-prefix-composition-conceptual-table-with-default *prefix*
prefix-state *composition-default* = *composition-default* $\wedge$
trace-table-low-degree *trace-default* $\wedge$
composition-table-low-degree *maxDegree* *composition-default* $\wedge$
trace-default $\in$
alpha-prefix-trace-table-candidates *prefix-state*
 (*sqp-trace-root* *prefix*) $\wedge$
sqp-alphas *prefix* $\notin$
alpha-prefix-union-bad-sets *prefix-state*
composition-trace-bad-alpha-space (*sqp-trace-root* *prefix*)

and *trace-path-bound*:

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
 (*checked-staged-security-with-query-prefix-current-trace-path-output-hit-at* *i*)
adversary-initial-state $\leq$ *T*

and *composition-path-bound*:

```

    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
      adversary-initial-state ≤ C
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
      adversary-initial-state ≤
        staged-phase-relation-error size
          (staged-query-search-queries budgets i + 1) +
        query-error-bound +
        hash-collision-budget-value 0
          (staged-query-search-queries budgets i + 1) + T + C
  proof -
    have residual-bound:
      wp-event
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        (checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at
          trace-default composition-default i)
        adversary-initial-state ≤
          hash-collision-budget-value 0
            (staged-query-search-queries budgets i + 1)
      by (rule
        checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-alpha-prefix-union
          [OF false-statement wf controlled i-bound default-prefix-clean])
    have bound:
      wp-event
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
        adversary-initial-state ≤
          staged-phase-relation-error size
            (staged-query-search-queries budgets i + 1) +
          query-error-bound +
          hash-collision-budget-value 0
            (staged-query-search-queries budgets i + 1) + T + C
      by (rule
        checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-residual
          [OF wf controlled i-bound residual-bound trace-path-bound
            composition-path-bound])
    show ?thesis
      using bound by simp
  qed

  lemma checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-bound-from-conceptual-default-alpha-
    assumes false-statement: ¬ exists-valid-trace
    and wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and i-bound: i < rounds

```

and *default-alpha-clean*:
 $\bigwedge \text{prefix } \text{prefix-state}.$
Some $((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \implies$
 $\text{trace-table-low-degree}$
 $(\text{query-prefix-trace-conceptual-table-with-default } \text{prefix}$
 $\text{prefix-state } \text{trace-table}) \wedge$
 $\text{composition-table-low-degree } \text{maxDegree}$
 $(\text{query-prefix-composition-conceptual-table-with-default } \text{prefix}$
 $\text{prefix-state } \text{composition-table}) \wedge$
 $\text{sqp-alphas } \text{prefix} \notin \text{composition-trace-bad-alpha-space}$
 $(\text{query-prefix-trace-conceptual-table-with-default } \text{prefix}$
 $\text{prefix-state } \text{trace-table})$
and *trace-path-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T$
and *composition-path-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C$
shows
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-authenticated-candidate-hit-at}$
 $\text{trace-table } \text{composition-table } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-query-search-queries budgets } i + 1) + T + C$
proof –
have *residual-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at}$
 $\text{trace-table } \text{composition-table } i)$
 $\text{adversary-initial-state} \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-query-search-queries budgets } i + 1)$
by (rule
 $\text{checked-staged-security-with-query-prefix-conceptual-default-context-residual-hit-at-bound-from-alpha-clean}$
 $[\text{OF false-statement wf controlled } i\text{-bound default-alpha-clean}])$
have *bound*:

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
 trace-table composition-table i)
adversary-initial-state $\leq$
staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
query-error-bound +
hash-collision-budget-value 0
 (staged-query-search-queries budgets $i + 1$) + $T + C$
by (rule
 checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-bound-from-conceptual-default-residual-a
 [*OF wf controlled i-bound residual-bound trace-path-bound*
 composition-path-bound])
show ?thesis
 using bound **by** simp
qed

lemma staged-security-with-data-state-current-query-partial-candidate-hit-at-bound-from-conceptual-default-alpha-clean
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: $i < \text{rounds}$
and default-alpha-clean:
 $\bigwedge \text{prefix prefix-state.}$
 Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
 set-dist
 (execute (checked-staged-query-prefix-with-state $A\ i$)
 adversary-initial-state) $\implies$
 trace-table-low-degree
 (query-prefix-trace-conceptual-table-with-default *prefix*
 prefix-state trace-table) $\wedge$
 composition-table-low-degree maxDegree
 (query-prefix-composition-conceptual-table-with-default *prefix*
 prefix-state composition-table) $\wedge$
 spp-alphas *prefix* $\notin$ *composition-trace-bad-alpha-space*
 (query-prefix-trace-conceptual-table-with-default *prefix*
 prefix-state trace-table)
and *trace-path-bound*:
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 adversary-initial-state $\leq T$
and *composition-path-bound*:
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 adversary-initial-state $\leq C$
shows

wp-event
 (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-query-partial-candidate-hit-at
 trace-table composition-table i)
adversary-initial-state $\leq$
staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
query-error-bound +
hash-collision-budget-value 0
 (staged-query-search-queries budgets $i + 1$) + $T + C$
proof –
have *prefix-bound*:
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-authenticated-candidate-hit-at
 trace-table composition-table i)
adversary-initial-state $\leq$
staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
query-error-bound +
hash-collision-budget-value 0
 (staged-query-search-queries budgets $i + 1$) + $T + C$
by (rule
 checked-staged-security-with-query-prefix-authenticated-candidate-hit-at-bound-from-conceptual-default-alpha-clean
 [OF false-statement wf controlled i -bound default-alpha-clean
 trace-path-bound composition-path-bound])
show ?thesis
by (rule order-trans
 [OF
 staged-security-with-data-state-current-query-partial-candidate-hit-at-le-query-prefix-authenticated-candidate
 prefix-bound])
 (use wf controlled i -bound **in** simp-all)
qed

lemma checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-conceptual-
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and default-alpha-clean:
 $\bigwedge i$ prefix prefix-state. $i < \text{rounds} \implies$
 Some ((prefix, prefix-state), prefix-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-with-state A i)
 adversary-initial-state) $\implies$
trace-table-low-degree
 (query-prefix-trace-conceptual-table-with-default prefix
 prefix-state trace-default) $\wedge$
composition-table-low-degree maxDegree
 (query-prefix-composition-conceptual-table-with-default prefix

$\text{prefix-state composition-default}) \wedge$
 $\text{sqp-alphas prefix} \notin \text{composition-trace-bad-alpha-space}$
 $(\text{query-prefix-trace-conceptual-table-with-default prefix}$
 $\text{prefix-state trace-default})$
and trace-path-bound :
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T \ i$
and $\text{composition-path-bound}$:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
shows
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit}$
 $\text{adversary-initial-state} \leq$
 $(\sum i < \text{rounds}.$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $T \ i + C \ i)$
proof (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-composition-candidate-drift-query-hit-bound-from-query-prefix-curr}$
 $[\text{OF wf controlled}])$
fix i
assume $i\text{-bound}: i < \text{rounds}$
show
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $T \ i + C \ i$
by (rule
 $\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-alpha}$
 $[\text{OF false-statement wf controlled } i\text{-bound} - \text{trace-path-bound}[\text{OF } i\text{-bound}]$
 $\text{composition-path-bound}[\text{OF } i\text{-bound}]])$
 $(\text{rule default-alpha-clean}[\text{OF } i\text{-bound}])$

qed

theorem *stark-soundness-from-aligned-partial-query-prefix-conceptual-default-alpha-clean-structured-paths:*

fixes *trace-fri-error'* *composition-fri-error'* *N* :: *prob*
and *T C* :: *nat* $\Rightarrow$ *prob*
and *trace-default* *composition-default* :: '*f list*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets A*
and *trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-fri-error'*
and *default-alpha-clean*:
 $\bigwedge i$ *prefix* *prefix-state*.
i < *rounds* $\Rightarrow$
Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
set-dist
(*execute* (*checked-staged-query-prefix-with-state A i*)
adversary-initial-state) $\Rightarrow$
trace-table-low-degree
(*query-prefix-trace-conceptual-table-with-default prefix*
prefix-state trace-default) $\wedge$
composition-table-low-degree maxDegree
(*query-prefix-composition-conceptual-table-with-default prefix*
prefix-state composition-default) $\wedge$
sqp-alphas prefix $\notin$ *composition-trace-bad-alpha-space*
(*query-prefix-trace-conceptual-table-with-default prefix*
prefix-state trace-default)
and *trace-path-bound*:
 $\bigwedge i$. *i* < *rounds* $\Rightarrow$
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
(*checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i*)
adversary-initial-state $\leq$ *T i*
and *composition-path-bound*:
 $\bigwedge i$. *i* < *rounds* $\Rightarrow$
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
(*checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i*)
adversary-initial-state $\leq$ *C i*
and *not-prefix-bound*:
wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state $\leq N$
and *empty-trace-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
and *empty-query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *nnreal rounds * query-error-bound*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
N +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
($\sum i < \text{rounds}$.
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
T i + C i) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
N +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
*trace-fri-error + nnreal rounds * query-error-bound)*
proof (*rule stark-soundness-from-aligned-partial-query-prefix-current-rounds*
[OF false-statement wf controlled trace-fri-bound comp-fri-bound - not-prefix-bound
empty-trace-fri-bound empty-query-bound])
fix *i*
assume *i-bound: i < rounds*

show

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i*)
adversary-initial-state $\leq$
staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
query-error-bound +
hash-collision-budget-value 0
 (*staged-query-search-queries budgets i + 1*) +
T i + C i

by (*rule*

checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-alpha-prefix-union-clean-structured-path
 [*OF false-statement wf controlled i-bound - trace-path-bound[OF i-bound]*
composition-path-bound[OF i-bound]])
 (*rule default-alpha-clean[OF i-bound]*)

qed

theorem *stark-soundness-from-aligned-partial-query-prefix-conceptual-default-alpha-prefix-union-clean-structured-path*

fixes *trace-fri-error' composition-fri-error' N :: prob*

and *T C :: nat $\Rightarrow$ prob*

and *trace-default composition-default :: 'f list*

assumes *false-statement: $\neg$ exists-valid-trace*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *trace-fri-bound:*

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error'*

and *comp-fri-bound:*

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-fri-error'*

and *default-prefix-clean:*

$\bigwedge i$ *prefix prefix-state.*

i < rounds $\implies$

Some ((prefix, prefix-state), prefix-state) $\in$

set-dist

(*execute* (*checked-staged-query-prefix-with-state A i*)

adversary-initial-state) $\implies$

query-prefix-trace-conceptual-table-with-default prefix prefix-state

trace-default = *trace-default* $\wedge$

query-prefix-composition-conceptual-table-with-default prefix

prefix-state composition-default = *composition-default* $\wedge$

trace-table-low-degree trace-default $\wedge$

composition-table-low-degree maxDegree composition-default $\wedge$

trace-default $\in$

$\alpha\text{-prefix-trace-table-candidates } \text{prefix-state}$
 $(\text{sqp-trace-root } \text{prefix}) \wedge$
 $\text{sqp-alphas } \text{prefix} \notin$
 $\alpha\text{-prefix-union-bad-sets } \text{prefix-state}$
 $\text{composition-trace-bad-alpha-space } (\text{sqp-trace-root } \text{prefix})$
and trace-path-bound :
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T \ i$
and $\text{composition-path-bound}$:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and not-prefix-bound :
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound } \text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and $\text{empty-trace-fri-bound}$:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event } \text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and empty-query-bound :
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event } \text{query-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{nnreal rounds} * \text{query-error-bound}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $N +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum i < \text{rounds}.$
 $\text{staged-phase-relation-error size}$

```

      (staged-query-search-queries budgets  $i + 1$ ) +
      query-error-bound +
      hash-collision-budget-value 0
      (staged-query-search-queries budgets  $i + 1$ ) +
       $T\ i + C\ i$ ) +
    (composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
       $N$  +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      trace-fri-error + nreal rounds * query-error-bound)
  proof (rule stark-soundness-from-aligned-partial-query-prefix-current-rounds
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound - not-prefix-bound
      empty-trace-fri-bound empty-query-bound])
  fix  $i$ 
  assume  $i$ -bound:  $i < \text{rounds}$ 
  show
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state  $A\ i$ )
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at  $i$ )
    adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-query-search-queries budgets  $i + 1$ ) +
    query-error-bound +
    hash-collision-budget-value 0
    (staged-query-search-queries budgets  $i + 1$ ) +
     $T\ i + C\ i$ 
  by (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-alpha-clean
    [OF false-statement wf controlled  $i$ -bound - trace-path-bound[OF  $i$ -bound]
      composition-path-bound[OF  $i$ -bound]])
    (rule default-prefix-clean[OF  $i$ -bound])
  qed

end

end

theory Soundness-Staged-Aligned-Current-Empty
imports
  Soundness-Conceptual-Query-Current-Alpha-Clean
  Soundness-Staged-Aligned-Partial-Merkle
begin

```

Packaging for the aligned-transcript partial-candidate path with the current-empty query

bound. This combines the current-query non-empty branch with the empty-composition branch introduced in the current-empty query layer, using a refined empty-header query bound in this downstream route.

context *soundness*
begin

definition *checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift*

$$\begin{aligned} &:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times \\ &\quad 'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times \\ &\quad 'f \text{ protocol-channel}) \times \text{unit list}) \times \\ &\quad 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool} \end{aligned}$$

where

checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
 $out \longleftrightarrow$
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space out $\wedge$
(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
composition-randomization-bad-with-partial-candidates out $\vee$
checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-partial-opening-union-hit-with-empty-header out)

lemma *checked-staged-security-with-actual-alpha-prefix-partial-candidate-randomization-imp-prefix-prequery-or-relevant*

assumes *support*:

out $\in$
set-dist
(*execute*
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
adversary-initial-state)

and *bad*:

checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-randomization-bad-with-partial-candidates out

shows

checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space out $\vee$
checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
composition-trace-bad-alpha-space out $\vee$
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
out

proof —

have *partial*:

checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
composition-trace-bad-alpha-space out

by (*rule*

checked-staged-security-with-actual-alpha-prefix-partial-candidate-randomization-imp-partial-header-bad-set-hit-o
 $[OF \text{ support bad}]$)

have *fresh-or-prequery*:

checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
composition-trace-bad-alpha-space out $\vee$

```

    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
    composition-trace-bad-alpha-space out
  by (rule
    checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-imp-fresh-or-prequery
    [OF partial])
  then show ?thesis
proof
  assume fresh:
    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
    composition-trace-bad-alpha-space out
  have prefix-or-drift:
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space out ∨
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
    composition-trace-bad-alpha-space out
  by (rule
    checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-imp-prefix-or-drift
    [OF fresh])
  then show ?thesis
proof
  assume
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space out
  then show ?thesis by simp
next
  assume drift:
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
    composition-trace-bad-alpha-space out
  have
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
    out
  unfolding
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-def
  using drift bad by simp
  then show ?thesis by simp
qed
next
  assume
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
    composition-trace-bad-alpha-space out
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-imp-prefix-prequery-or-relevant-drift-
  assumes support:
    out ∈
    set-dist
    (execute

```

```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    adversary-initial-state)
and hit:
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-alpha-partial-opening-union-hit-with-empty-header out
shows
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  composition-trace-bad-alpha-space out ∨
  checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
  composition-trace-bad-alpha-space out ∨
  checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
  out
proof –
have partial:
  checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
  composition-trace-bad-alpha-space out
by (rule
  checked-staged-security-with-actual-alpha-prefix-empty-header-partial-alpha-hit-imp-partial-header-bad-set-hit-on-
  [OF support hit])
have fresh-or-prequery:
  checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
  composition-trace-bad-alpha-space out ∨
  checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
  composition-trace-bad-alpha-space out
by (rule
  checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-imp-fresh-or-prequery
  [OF partial])
then show ?thesis
proof
assume fresh:
  checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
  composition-trace-bad-alpha-space out
have prefix-or-drift:
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  composition-trace-bad-alpha-space out ∨
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
  composition-trace-bad-alpha-space out
by (rule
  checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-imp-prefix-or-drift
  [OF fresh])
then show ?thesis
proof
assume
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  composition-trace-bad-alpha-space out
then show ?thesis by simp
next
assume drift:

```

```

    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
    composition-trace-bad-alpha-space out
  have
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
    out
  unfolding
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-def
  using drift hit by simp
  then show ?thesis by simp
qed
next
assume
  checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
  composition-trace-bad-alpha-space out
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-partial-candidate-randomization-bound-from-relevant-drift-and-
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and drift-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
      adversary-initial-state  $\leq D$ 
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        composition-randomization-bad-with-partial-candidates)
      adversary-initial-state  $\leq$ 
      composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      D
proof -
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?Prefix =
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space
  let ?Prequery =
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit

```



```

    composition-trace-bad-alpha-space
let ?Drift =
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
let ?Random =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-randomization-bad-with-partial-candidates
let ?P =
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0)
let ?Q =
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0)
have transcript-bound:
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
    (checked-staged-transcript-actual-alpha-prefix-bad-set-hit
     composition-trace-bad-alpha-space)
    adversary-initial-state ≤ ?P
by (rule
    checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
    [OF wf controlled])
have prefix-bound:
    wp-event ?M ?Prefix adversary-initial-state ≤ ?P
by (rule
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
    [OF transcript-bound])
have prequery-bound:
    wp-event ?M ?Prequery adversary-initial-state ≤ ?Q
by (rule
    checked-staged-security-with-actual-alpha-prefix-vector-alpha-prequery-accounting
    [OF wf controlled])
have wp-event ?M ?Random adversary-initial-state ≤
    wp-event ?M
    (λout. ?Prefix out ∨ ?Prequery out ∨ ?Drift out ∨ False)
    adversary-initial-state
by (rule wp-event-mono-on-support)
    (use
    checked-staged-security-with-actual-alpha-prefix-partial-candidate-randomization-imp-prefix-prequery-or-relevant-
    [of - A] in blast)
also have ... ≤
    wp-event ?M ?Prefix adversary-initial-state +
    wp-event ?M ?Prequery adversary-initial-state +
    wp-event ?M ?Drift adversary-initial-state +
    wp-event ?M (λ-. False) adversary-initial-state
by (rule wp-event-union-bound4)
also have ... ≤ ?P + ?Q + D + 0
by (intro add-mono prefix-bound prequery-bound drift-bound)

```

```

    (simp add: wp-event-def wp-def dist-expect-def)
  finally show ?thesis
    by (simp add: algebra-simps)
qed

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-bound-from-relevant-drift-and-budget.
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and drift-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
    adversary-initial-state  $\leq$  D
  shows
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-partial-opening-union-hit-with-empty-header)
    adversary-initial-state  $\leq$ 
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D
proof -
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?Prefix =
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space
  let ?Prequery =
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
    composition-trace-bad-alpha-space
  let ?Drift =
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
  let ?Hit =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-alpha-partial-opening-union-hit-with-empty-header
  let ?P =
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0)
  let ?Q =

```

```

    staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0)
have transcript-bound:
  wp-event (checked-staged-transcript-with-alpha-prefix-program A)
    (checked-staged-transcript-actual-alpha-prefix-bad-set-hit
      composition-trace-bad-alpha-space)
    adversary-initial-state ≤ ?P
  by (rule
    checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
    [OF wf controlled])
have prefix-bound:
  wp-event ?M ?Prefix adversary-initial-state ≤ ?P
  by (rule
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
    [OF transcript-bound])
have prequery-bound:
  wp-event ?M ?Prequery adversary-initial-state ≤ ?Q
  by (rule
    checked-staged-security-with-actual-alpha-prefix-vector-alpha-prequery-accounting
    [OF wf controlled])
have wp-event ?M ?Hit adversary-initial-state ≤
  wp-event ?M
    (λout. ?Prefix out ∨ ?Prequery out ∨ ?Drift out ∨ False)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (use
      checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-imp-prefix-prequery-or-relevant-drift-on-s
      [of - A] in blast)
also have ... ≤
  wp-event ?M ?Prefix adversary-initial-state +
  wp-event ?M ?Prequery adversary-initial-state +
  wp-event ?M ?Drift adversary-initial-state +
  wp-event ?M (λ-. False) adversary-initial-state
  by (rule wp-event-union-bound4)
also have ... ≤ ?P + ?Q + D + 0
  by (intro add-mono prefix-bound prequery-bound drift-bound)
    (simp add: wp-event-def wp-def dist-expect-def)
finally show ?thesis
  by (simp add: algebra-simps)
qed

lemma checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-relevant-drift-and-budg
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and drift-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)

```

checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq D$

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
composition-bad-with-partial-candidates)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
D

proof –

let *?M* = *checked-staged-security-experiment-with-data-state A*
let *?E* = *staged-security-with-data-state-verifier-event*
let *?degree* = *?E composition-degree-bad-with-partial-candidates*
let *?random* = *?E composition-randomization-bad-with-partial-candidates*
let *?bad* = *?E composition-bad-with-partial-candidates*
have *degree-false*: *?degree* = (λ out. *False*)
by (*rule ext*)
 (*auto simp*: *staged-security-with-data-state-verifier-event-def*
composition-degree-bad-with-partial-candidates-false
split: *option.splits prod.splits*)

have *degree-bound*:

wp-event ?M ?degree adversary-initial-state ≤ 0

unfolding *degree-false wp-event-def wp-def dist-expect-def* **by** *simp*

have *projected*:

wp-event ?M ?random adversary-initial-state =
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-verifier-event*
composition-randomization-bad-with-partial-candidates)
adversary-initial-state
by (*rule*
checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)

have *random-bound*:

wp-event ?M ?random adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
D

unfolding *projected*

by (rule
checked-staged-security-with-actual-alpha-prefix-partial-candidate-randomization-bound-from-relevant-drift-and-b
 [OF wf controlled drift-bound])
have *wp-event* ?M ?bad *adversary-initial-state* $\leq$
wp-event ?M (λ out. ?degree out $\vee$?random out)
adversary-initial-state
by (rule *wp-event-mono*)
 (auto simp: *staged-security-with-data-state-verifier-event-def*
dest: composition-bad-with-partial-candidates-split
 [OF false-statement]
split: option.splits prod.splits)
also have ... $\leq$
wp-event ?M ?degree *adversary-initial-state* +
wp-event ?M ?random *adversary-initial-state*
by (rule *wp-event-union-bound*)
also have ... $\leq$
 0 +
 (composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets* 0) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets* 0) +
 D)
by (intro add-mono degree-bound random-bound)
finally show ?thesis
by simp
qed

lemma *checked-staged-security-with-data-state-empty-header-composition-randomization-bound-from-relevant-drift-a*

assumes wf: *staged-budget-wellformed budgets*

and controlled: *staged-adversary-controlled budgets A*

and drift-bound:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$ D

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*
composition-randomization-bad-with-empty-composition-header-candidates)

adversary-initial-state $\leq$

composition-error-bound +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets* 0) +

hash-collision-budget-value 0

(*staged-alpha-search-queries budgets* 0) +

```

    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D
proof –
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?random =
    staged-security-with-data-state-verifier-event
    composition-randomization-bad-with-empty-composition-header-candidates
  let ?alpha =
    staged-security-with-data-state-verifier-event
    composition-alpha-partial-opening-union-hit-with-empty-header
  have alpha-projected:
    wp-event ?M ?alpha adversary-initial-state =
      wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        composition-alpha-partial-opening-union-hit-with-empty-header)
      adversary-initial-state
  by (rule
    checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)
  have alpha-bound:
    wp-event ?M ?alpha adversary-initial-state ≤
      composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      D
  unfolding alpha-projected
  by (rule
    checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-bound-from-relevant-drift-and-budgets
    [OF wf controlled drift-bound])
  have wp-event ?M ?random adversary-initial-state ≤
    wp-event ?M ?alpha adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro:
        composition-alpha-bad-set-hit-with-empty-header-imp-partial-opening-union-hit
        composition-randomization-bad-with-empty-composition-header-candidates-imp-alpha-hit
        split: option.splits prod.splits)
  also have ... ≤
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size

```

```

      (staged-alpha-search-queries budgets 0) +
    D
  by (rule alpha-bound)
  finally show ?thesis .
qed

lemma checked-staged-security-with-data-state-empty-header-bad-event-bound-from-relevant-drift-trace-fri-and-query
  fixes trace-fri-error' :: prob
  assumes false-statement:  $\neg$  exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and drift-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
      adversary-initial-state  $\leq$  D
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-empty-composition-header-candidates)
      adversary-initial-state  $\leq$  trace-fri-error'
  and query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        query-bad-with-empty-composition-header-candidates)
      adversary-initial-state  $\leq$  empty-query-error
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-partial-candidate-empty-header)
      adversary-initial-state  $\leq$ 
        composition-error-bound +
        staged-phase-relation-error size
        (staged-alpha-search-queries budgets 0) +
        hash-collision-budget-value 0
        (staged-alpha-search-queries budgets 0) +
        staged-phase-relation-error size
        (staged-alpha-search-queries budgets 0) +
        D + trace-fri-error' + empty-query-error
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  let ?degree =
    ?E composition-degree-bad-with-empty-composition-header-candidates
  let ?random =
    ?E composition-randomization-bad-with-empty-composition-header-candidates
  let ?trace =
    ?E trace-fri-bad-with-empty-composition-header-candidates

```

```

let ?query =
  ?E query-bad-with-empty-composition-header-candidates
let ?bad = ?E soundness-bad-event-partial-candidate-empty-header
have degree-false: ?degree = ( $\lambda$ out. False)
  by (rule ext)
    (auto simp: staged-security-with-data-state-verifier-event-def
      composition-degree-bad-with-empty-composition-header-candidates-false
      split: option.splits prod.splits)
have degree-bound:
  wp-event ?M ?degree adversary-initial-state  $\leq$  0
  unfolding degree-false wp-event-def wp-def dist-expect-def by simp
have random-bound:
  wp-event ?M ?random adversary-initial-state  $\leq$ 
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D
  by (rule
    checked-staged-security-with-data-state-empty-header-composition-randomization-bound-from-relevant-drift-and-l
    [OF wf controlled drift-bound])
have wp-event ?M ?bad adversary-initial-state  $\leq$ 
  wp-event ?M
    ( $\lambda$ out. ?degree out  $\vee$  ?random out  $\vee$  ?trace out  $\vee$ 
      ?query out)
    adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume bad: ?bad out
  show ?degree out  $\vee$  ?random out  $\vee$  ?trace out  $\vee$  ?query out
  proof (cases out)
    case None
    then show ?thesis
      using bad
      unfolding staged-security-with-data-state-verifier-event-def
      by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have local-bad:
    soundness-bad-event-partial-candidate-empty-header ?s

```



```

    (Some (result, final-state))
using bad
unfolding out-eq staged-security-with-data-state-verifier-event-def
by simp
then consider
  (comp)
    composition-bad-with-empty-composition-header-candidates ?s
    (Some (result, final-state))
  | (trace)
    trace-fri-bad-with-empty-composition-header-candidates ?s
    (Some (result, final-state))
  | (query)
    query-bad-with-empty-composition-header-candidates ?s
    (Some (result, final-state))
unfolding soundness-bad-event-partial-candidate-empty-header-def
by blast
then show ?thesis
proof cases
  case comp
    then have comp-split:
      composition-degree-bad-with-empty-composition-header-candidates ?s
      (Some (result, final-state))  $\vee$ 
      composition-randomization-bad-with-empty-composition-header-candidates
      ?s (Some (result, final-state))
    by (rule
      composition-bad-with-empty-composition-header-candidates-split
      [OF false-statement])
    then show ?thesis
      unfolding out-eq staged-security-with-data-state-verifier-event-def
      using comp-split by (auto split: option.splits prod.splits)
  next
    case trace
      then show ?thesis
        unfolding out-eq staged-security-with-data-state-verifier-event-def
        by simp
  next
    case query
      then show ?thesis
        unfolding out-eq staged-security-with-data-state-verifier-event-def
        by simp
  qed
qed
qed
also have ...  $\leq$ 
  wp-event ?M ?degree adversary-initial-state +
  wp-event ?M ?random adversary-initial-state +
  wp-event ?M ?trace adversary-initial-state +
  wp-event ?M ?query adversary-initial-state
by (rule wp-event-union-bound4)

```

also have ... $\leq$
 $0 +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D) +$
 $\text{trace-fri-error}' + \text{empty-query-error}$
by (intro add-mono degree-bound random-bound trace-fri-bound query-bound)
finally show ?thesis
by (simp add: algebra-simps)
qed

lemma checked-staged-soundness-from-aligned-partial-candidate-components-and-current-empty-query-from-relevant-
fixes trace-fri-error' composition-fri-error' query-error D
empty-query-error :: prob
assumes false-statement: $\neg \text{exists-valid-trace}$
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ trace-fri-error'
and comp-fri-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ composition-fri-error'
and query-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-partial-candidates)
adversary-initial-state $\leq$ query-error
and drift-bound:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$ D
and empty-trace-fri-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ trace-fri-error
and current-empty-bound:

$$\begin{aligned} & \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\ & \quad (\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0) \\ & \quad \text{adversary-initial-state} \leq \text{empty-query-error} \\ \text{shows } & \text{checked-staged-adversary-acceptance-probability } A \leq \\ & \text{hash-collision-budget-value } 0 \\ & \quad (\text{staged-attacker-query-budget budgets} + \\ & \quad \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) + \\ & (\text{composition-error-bound} + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \text{hash-collision-budget-value } 0 \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad D) + \\ & \text{trace-fri-error}' + \text{composition-fri-error}' + \\ & \text{query-error} + \\ & (\text{composition-error-bound} + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \text{hash-collision-budget-value } 0 \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad D + \text{trace-fri-error} + \text{empty-query-error}) \\ \text{proof } & - \\ \text{have } & \text{comp-bound:} \\ & \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\ & \quad (\text{staged-security-with-data-state-verifier-event} \\ & \quad \text{composition-bad-with-partial-candidates}) \\ & \quad \text{adversary-initial-state} \leq \\ & \quad \text{composition-error-bound} + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad \text{hash-collision-budget-value } 0 \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad D \\ \text{by } & (\text{rule} \\ & \quad \text{checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-relevant-drift-and-budgets} \\ & \quad [\text{OF false-statement wf controlled drift-bound}]) \\ \text{have } & \text{merkle-bound:} \\ & \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\ & \quad (\text{staged-security-with-data-state-verifier-event} \\ & \quad \text{partial-merkle-inconsistency-bad}) \\ & \quad \text{adversary-initial-state} \leq \\ & \quad \text{hash-collision-budget-value } 0 \\ & \quad (\text{staged-attacker-query-budget budgets} +
\end{aligned}$$

staged-challenge-query-budget + verifier-hash-query-budget)

by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
[OF wf controlled])

have *empty-query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤ empty-query-error

by (rule *order-trans*[
OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero
OF wf controlled] *current-empty-bound*])

have *empty-header-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state ≤
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D + trace-fri-error + empty-query-error

by (rule
checked-staged-security-with-data-state-empty-header-bad-event-bound-from-relevant-drift-trace-fri-and-query
[OF false-statement wf controlled drift-bound empty-trace-fri-bound
empty-query-bound])

show *?thesis*

by (rule
checked-staged-soundness-from-aligned-partial-candidate-components-and-empty-header-bound
[OF false-statement merkle-bound comp-bound trace-fri-bound
comp-fri-bound query-bound empty-header-bound])

qed

lemma *checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query*:
fixes *trace-fri-error' composition-fri-error' query-error P D*
empty-query-error :: prob
assumes *false-statement: ¬ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state ≤ trace-fri-error'
and *comp-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)

*(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)*
adversary-initial-state $\leq$ *composition-fri-error'*
and *query-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates)*
adversary-initial-state $\leq$ *query-error*
and *prefix-bound:*
wp-event
*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)*
*(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)*
adversary-initial-state $\leq$ *P*
and *drift-bound:*
wp-event
*(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)*
*(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space)*
adversary-initial-state $\leq$ *D*
and *empty-trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)*
adversary-initial-state $\leq$ *trace-fri-error*
and *current-empty-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(P + D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
query-error +
(P + D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + empty-query-error)
proof –
have *comp-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
*(staged-security-with-data-state-verifier-event
composition-bad-with-partial-candidates)*

adversary-initial-state $\leq$
P + *D* +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)
by (rule
checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-prefix-drift-and-budgets
[OF false-statement wf controlled prefix-bound drift-bound])
have merkle-bound:
wp-event (checked-staged-security-experiment-with-data-state *A*)
(staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
[OF wf controlled])
have empty-header-bound:
wp-event (checked-staged-security-experiment-with-data-state *A*)
(staged-security-with-data-state-verifier-event
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state $\leq$
P + *D* +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + empty-query-error
by (rule
checked-staged-security-with-data-state-empty-header-bad-event-bound-from-prefix-drift-trace-fri-and-current-emp
[OF false-statement wf controlled prefix-bound drift-bound
empty-trace-fri-bound current-empty-bound])
have bound:
checked-staged-adversary-acceptance-probability *A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(*P* + *D* +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
query-error +
(*P* + *D* +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + empty-query-error)
by (rule
checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-empty-header-bound
[OF false-statement merkle-bound comp-bound trace-fri-bound
comp-fri-bound query-bound empty-header-bound])

then show ?thesis .
qed

lemma *checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from*

fixes *trace-fri-error'* *composition-fri-error'* *query-error D*

empty-query-error :: prob

assumes *false-statement*: $\neg$ *exists-valid-trace*

and *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

trace-fri-bad-with-partial-candidates)

adversary-initial-state $\leq$ *trace-fri-error'*

and *comp-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

composition-fri-bad-with-partial-candidates)

adversary-initial-state $\leq$ *composition-fri-error'*

and *query-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

query-bad-with-aligned-transcript-partial-candidates)

adversary-initial-state $\leq$ *query-error*

and *drift-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*

composition-trace-bad-alpha-space)

adversary-initial-state $\leq$ *D*

and *empty-trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

trace-fri-bad-with-empty-composition-header-candidates)

adversary-initial-state $\leq$ *trace-fri-error*

and *current-empty-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-current-empty-query-opening-hit-at 0*)

adversary-initial-state $\leq$ *empty-query-error*

shows *checked-staged-adversary-acceptance-probability A* $\leq$

hash-collision-budget-value 0

(*staged-attacker-query-budget budgets* +

staged-challenge-query-budget + *verifier-hash-query-budget*) +

(*composition-error-bound* +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

hash-collision-budget-value 0

(*staged-alpha-search-queries budgets 0*) +

$D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $\text{query-error} +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{trace-fri-error} + \text{empty-query-error})$

proof –

let $?P =$

$\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0)$

have *transcript-bound*:

$\text{wp-event } (\text{checked-staged-transcript-with-alpha-prefix-program } A)$
 $(\text{checked-staged-transcript-actual-alpha-prefix-bad-set-hit}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq ?P$

by (rule

$\text{checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets}$
 $[\text{OF wf controlled}])$

have *prefix-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-bad-set-hit}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq ?P$

by (rule

$\text{checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript}$
 $[\text{OF transcript-bound}])$

show *?thesis*

by (rule

$\text{checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query}$
 $[\text{OF false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\text{query-bound prefix-bound drift-bound empty-trace-fri-bound}$
 $\text{current-empty-bound}])$

qed

lemma *checked-staged-soundness-from-aligned-partial-candidate-components-and-current-empty-query-from-drift*:

fixes $\text{trace-fri-error}' \text{ composition-fri-error}' \text{ query-error } D$

empty-query-error :: prob
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *composition-fri-error'*
and *query-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 query-bad-with-aligned-partial-candidates)
 adversary-initial-state $\leq$ *query-error*
and *drift-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$ *D*
and *empty-trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ *trace-fri-error*
and *current-empty-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-current-empty-query-opening-hit-at 0*)
 adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
 hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
 staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
 D +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*)) +
 trace-fri-error' + composition-fri-error' +
 query-error +

```

    (composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
      D +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      trace-fri-error + empty-query-error)
  proof -
    let ?P =
      composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0)
    have transcript-bound:
      wp-event (checked-staged-transcript-with-alpha-prefix-program A)
      (checked-staged-transcript-actual-alpha-prefix-bad-set-hit
        composition-trace-bad-alpha-space)
      adversary-initial-state ≤ ?P
    by (rule
      checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
      [OF wf controlled])
    have prefix-bound:
      wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
        composition-trace-bad-alpha-space)
      adversary-initial-state ≤ ?P
    by (rule
      checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
      [OF transcript-bound])
    show ?thesis
    by (rule
      checked-staged-soundness-from-aligned-partial-candidate-components-and-current-empty-query
      [OF false-statement wf controlled trace-fri-bound comp-fri-bound
        query-bound prefix-bound drift-bound empty-trace-fri-bound
        current-empty-bound])
  qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit-from-drift-curr
  fixes trace-fri-error' composition-fri-error' query-error D
  empty-query-error :: prob
  assumes false-statement: ¬ exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:

```

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
 $adversary\text{-}initial\text{-}state \leq trace\text{-}fri\text{-}error'$

and *comp-fri-bound*:

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
 $adversary\text{-}initial\text{-}state \leq composition\text{-}fri\text{-}error'$

and *current-query-bound*:

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
staged-security-with-data-state-current-query-partial-opening-hit
 $adversary\text{-}initial\text{-}state \leq query\text{-}error$

and *drift-bound*:

$wp\text{-}event$
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
composition-trace-bad-alpha-space)
 $adversary\text{-}initial\text{-}state \leq D$

and *empty-trace-fri-bound*:

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
 $adversary\text{-}initial\text{-}state \leq trace\text{-}fri\text{-}error$

and *current-empty-bound*:

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-current-empty-query-opening-hit-at* 0)
 $adversary\text{-}initial\text{-}state \leq empty\text{-}query\text{-}error$

shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0)) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
 D +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0)) +
 $trace\text{-}fri\text{-}error' + composition\text{-}fri\text{-}error' +$
 $query\text{-}error +$
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0)) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
 $D +$

staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + empty-query-error)

proof –

have *query-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state ≤ query-error

by (rule *order-trans*[
OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit
current-query-bound])
(use wf controlled in simp-all)

show *?thesis*

by (rule
checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from-d
[OF false-statement wf controlled trace-fri-bound comp-fri-bound
query-bound drift-bound empty-trace-fri-bound current-empty-bound])

qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit-from-not-prefix*

fixes *trace-fri-error' composition-fri-error' query-error N*
empty-query-error :: prob

assumes *false-statement: ¬ exists-valid-trace*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *trace-fri-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state ≤ trace-fri-error'

and *comp-fri-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state ≤ composition-fri-error'

and *current-query-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state ≤ query-error

and *not-prefix-bound*:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state ≤ N

and *empty-trace-fri-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
and *current-empty-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
N +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
query-error +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
N +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + empty-query-error)
proof –
have *query-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq$ *query-error*
by (rule *order-trans*[
OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit
current-query-bound])
(use wf controlled in simp-all)
have *drift-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$
N
proof –

have *wp-event*
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
 composition-trace-bad-alpha-space)
 adversary-initial-state
by (*rule*
 checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-le-not-prefix-bound
 [*OF wf controlled*])
also have ... $\leq N$
 by (*rule not-prefix-bound*)
finally show *?thesis* .
qed
have *prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$
 composition-error-bound +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*)
proof –
 let *?P* =
 composition-error-bound +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*)
have *transcript-bound*:
 wp-event (*checked-staged-transcript-with-alpha-prefix-program A*)
 (*checked-staged-transcript-actual-alpha-prefix-bad-set-hit*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$ *?P*
by (*rule*
 checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
 [*OF wf controlled*])
show *?thesis*
 by (*rule*
 checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript

```

    [OF transcript-bound])
qed
show ?thesis
by (rule
  checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query
  [OF false-statement wf controlled trace-fri-bound comp-fri-bound
    query-bound prefix-bound drift-bound empty-trace-fri-bound
    current-empty-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-drift-cur
fixes trace-fri-error' composition-fri-error' D empty-query-error :: prob
and C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  trace-fri-error'
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-fri-error'
and round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
      adversary-initial-state  $\leq$  C i
and drift-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
      composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq$  D
and empty-trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state  $\leq$  trace-fri-error
and current-empty-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
    adversary-initial-state  $\leq$  empty-query-error
shows checked-staged-adversary-acceptance-probability A  $\leq$ 

```

hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
($\sum i < \text{rounds}. C\ i$) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
D +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + empty-query-error)

proof –

have *current-query-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state $\leq (\sum i < \text{rounds}. C\ i)$

by (rule
checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
[OF round-bound wf controlled])

show *?thesis*

by (rule
checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit-from-drift-current
[OF false-statement wf controlled trace-fri-bound comp-fri-bound
current-query-bound drift-bound empty-trace-fri-bound
current-empty-bound])

qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-not-prefi*

fixes *trace-fri-error' composition-fri-error' N empty-query-error :: prob*

and *C :: nat $\Rightarrow$ prob*

assumes *false-statement: $\neg$ exists-valid-trace*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *trace-fri-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ trace-fri-error'

and *comp-fri-bound*:
 $wp\text{-event}$ (*checked-staged-security-experiment-with-data-state* A)
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 $wp\text{-event}$
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *not-prefix-bound*:
 $wp\text{-event}$
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and *empty-trace-fri-bound*:
 $wp\text{-event}$ (*checked-staged-security-experiment-with-data-state* A)
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and *current-empty-bound*:
 $wp\text{-event}$ (*checked-staged-security-experiment-with-data-state* A)
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows *checked-staged-adversary-acceptance-probability* $A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $N +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum i < \text{rounds}. C \ i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $N +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$

$\text{trace-fri-error} + \text{empty-query-error}$)
proof –
have *current-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-current-query-partial-opening-hit}$
 $\text{adversary-initial-state} \leq (\sum_{i < \text{rounds}} C \ i)$
by (rule
 $\text{checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds}$
 $[OF \ \text{round-bound} \ \text{wf controlled}]$)
show ?thesis
by (rule
 $\text{checked-staged-soundness-from-aligned-transcript-components-and-current-partial-opening-hit-from-not-prefix-curre}$
 $[OF \ \text{false-statement} \ \text{wf controlled} \ \text{trace-fri-bound} \ \text{comp-fri-bound}$
 $\text{current-query-bound} \ \text{not-prefix-bound} \ \text{empty-trace-fri-bound}$
 $\text{current-empty-bound}]$)
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-not-prefix-cu*
fixes $\text{trace-fri-error}' \ \text{composition-fri-error}' \ N \ \text{empty-query-error} :: \text{prob}$
and $Q :: \text{nat} \Rightarrow \text{prob}$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed* budgets
and *controlled*: *staged-adversary-controlled* budgets A
and *trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}'$
and *comp-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and *trace-frac*:
 $\bigwedge i \ \text{prefix} \ \text{prefix-state}.$
 $i < \text{rounds} \implies$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\text{adversary-initial-state}) \implies$
 nnreal
 $(\text{card}$
 $(\text{staged-query-prefix-prefix-authenticated-trace-indices } \text{prefix}$
 $\text{prefix-state})) /$
 $\text{nnreal } (\text{card } \text{query-sample-space}) \leq \text{query-error-bound}$
and *path-bound*:
 $\bigwedge i. \ i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$

(checked-staged-security-with-query-prefix-current-path-output-hit-at i)
 adversary-initial-state $\leq Q\ i$
and not-prefix-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq N$
and empty-trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ trace-fri-error
and current-empty-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 adversary-initial-state $\leq$ empty-query-error
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 ($\sum i < \text{rounds}$.
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
 query-error-bound + $Q\ i$) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 trace-fri-error + empty-query-error)
proof –
have round-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)

```

    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound + Q i
proof —
  fix i
  assume i-bound: i < rounds
  have prefix-hit-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
  by (rule
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-trace-indices
    [OF wf controlled i-bound])
    (use trace-frac[OF i-bound] in blast)
  show
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound + Q i
  by (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-path
    [OF wf controlled i-bound prefix-hit-bound path-bound[OF i-bound]])
  qed
show ?thesis
  by (rule
    checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-not-prefix-c
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
    round-bound not-prefix-bound empty-trace-fri-bound
    current-empty-bound])
  qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-trace-index-structured-path-bounds-from-n
  fixes trace-fri-error' composition-fri-error' N empty-query-error :: prob
  and T C :: nat ⇒ prob
  assumes false-statement: ¬ exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-partial-candidates)

```

$\text{adversary-initial-state} \leq \text{trace-fri-error}'$
and *comp-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and *trace-frac*:
 $\bigwedge i \text{ prefix prefix-state.}$
 $i < \text{rounds} \implies$
 $\text{Some } ((\text{prefix}, \text{prefix-state}), \text{prefix-state}) \in$
 $\quad \text{set-dist}$
 $\quad (\text{execute } (\text{checked-staged-query-prefix-with-state } A \ i)$
 $\quad \quad \text{adversary-initial-state}) \implies$
 nnreal
 $\quad (\text{card}$
 $\quad \quad (\text{staged-query-prefix-prefix-authenticated-trace-indices prefix}$
 $\quad \quad \quad \text{prefix-state})) /$
 $\quad \text{nnreal } (\text{card query-sample-space}) \leq \text{query-error-bound}$
and *trace-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T \ i$
and *composition-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *not-prefix-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $\quad A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound}$
 $\quad \text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and *empty-trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and *current-empty-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets } +$

staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (composition-error-bound +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
 N +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*)) +
trace-fri-error' + *composition-fri-error'* +
 (∑ *i* < *rounds*.
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 query-error-bound + *T i* + *C i*) +
 (composition-error-bound +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
 N +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 trace-fri-error + *empty-query-error*)
proof –
 have *round-bound*:
 ∧ *i. i < rounds* ⇒
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i*)
 adversary-initial-state ≤
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 query-error-bound + *T i* + *C i*
 proof –
 fix *i*
 assume *i-bound*: *i < rounds*
 have *prefix-hit-bound*:
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at i*)
 adversary-initial-state ≤
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) + *query-error-bound*
 by (rule
 checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-trace-indices
 [*OF wf controlled i-bound*])
 (use *trace-frac[OF i-bound]* **in** *blast*)
 show
 wp-event

```

    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound + T i + C i
  by (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-structured-
    [OF wf controlled i-bound prefix-hit-bound trace-path-bound[OF i-bound]
    composition-path-bound[OF i-bound]])
qed
show ?thesis
  by (rule
    checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-not-prefix-cv
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
    round-bound not-prefix-bound empty-trace-fri-bound
    current-empty-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-conceptual-structured-paths-from-not-prefix-cv
  fixes trace-fri-error' composition-fri-error' N empty-query-error :: prob
  and T C :: nat ⇒ prob
  assumes false-statement: ¬ exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ trace-fri-error'
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    composition-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-fri-error'
  and conceptual-context:
    ∧ i prefix prefix-state.
    i < rounds ⇒
    Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
    adversary-initial-state) ⇒
    ¬ hash-map-output-collision prefix-state ∧
    trace-table-low-degree
    (query-prefix-trace-conceptual-table prefix prefix-state) ∧
    composition-table-low-degree maxDegree
    (query-prefix-composition-conceptual-table prefix prefix-state) ∧
    ¬ all-queries-consistent
    (query-prefix-trace-conceptual-table prefix prefix-state)

```

(query-prefix-composition-conceptual-table prefix prefix-state)
 (sqp-alphas prefix)
and trace-path-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i)
 adversary-initial-state $\leq T i$
and composition-path-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i)
 adversary-initial-state $\leq C i$
and not-prefix-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq N$
and empty-trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq \text{trace-fri-error}$
and current-empty-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 adversary-initial-state $\leq \text{empty-query-error}$
shows checked-staged-adversary-acceptance-probability A $\leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 N +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0)) +
 trace-fri-error' + composition-fri-error' +
 ($\sum i < \text{rounds}.$
 staged-phase-relation-error size
 (staged-query-search-queries budgets i + 1) +
 query-error-bound + T i + C i) +
 (composition-error-bound +
 staged-phase-relation-error size

$(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $N +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{trace-fri-error} + \text{empty-query-error})$

proof –

have *round-bound*:

$\bigwedge i. i < \text{rounds} \implies$

wp-event

$(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} + T \ i + C \ i$

proof –

fix *i*

assume *i-bound*: $i < \text{rounds}$

have *prefix-target-bound*:

wp-event $(\text{checked-staged-query-prefix-receive-with-state } A \ i)$

$(\text{checked-staged-query-prefix-dynamic-index-hit}$
 $\text{staged-query-prefix-prefix-authenticated-opening-target})$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) + \text{query-error-bound}$

by (rule

$\text{checked-staged-query-prefix-prefix-authenticated-target-hit-bound-from-conceptual}$
 $[\text{OF wf controlled } i\text{-bound}])$
 $(\text{rule conceptual-context}[\text{OF } i\text{-bound}])$

have *prefix-bound*:

wp-event

$(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) + \text{query-error-bound}$

by (rule

$\text{checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at-bound-from-dynamic-target}$
 $[\text{OF wf controlled } i\text{-bound } \text{prefix-target-bound}])$

show

wp-event

$(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1) +$
 $\text{query-error-bound} + T \ i + C \ i$

```

    by (rule
      checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-structured-
        [OF wf controlled i-bound prefix-bound trace-path-bound [OF i-bound]
          composition-path-bound [OF i-bound]])
  qed
show ?thesis
  by (rule
    checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-not-prefix-cv
      [OF false-statement wf controlled trace-fri-bound comp-fri-bound
        round-bound not-prefix-bound empty-trace-fri-bound
        current-empty-bound])
  qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-cross-data-p
  fixes trace-fri-error' composition-fri-error' X P R empty-query-error :: prob
  and Q :: nat  $\Rightarrow$  prob
  assumes false-statement:  $\neg$  exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-partial-candidates)
      adversary-initial-state  $\leq$  trace-fri-error'
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-partial-candidates)
      adversary-initial-state  $\leq$  composition-fri-error'
  and trace-frac:
     $\bigwedge i$  prefix prefix-state.
     $i < \text{rounds} \implies$ 
    Some ((prefix, prefix-state), prefix-state)  $\in$ 
      set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)  $\implies$ 
    nnreal
      (card
        (staged-query-prefix-prefix-authenticated-trace-indices prefix
          prefix-state)) /
      nnreal (card query-sample-space)  $\leq$  query-error-bound
  and path-bound:
     $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-path-output-hit-at i)
      adversary-initial-state  $\leq$  Q i
  and cross-bound:
    wp-event

```

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
adversary-initial-state $\leq X$
and *data-pre-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$
and *transcript-new-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq R$
and *empty-trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq \text{trace-fri-error}$
and *current-empty-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq \text{empty-query-error}$
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + X + (P + R) + (P + R)) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
 $(\sum_{i < \text{rounds.}}$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound + Q i) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + X + (P + R) + (P + R)) +
staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +
 trace-fri-error + empty-query-error)

proof –

have *transcript-pre-bound*:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
 adversary-initial-state $\leq P$

by (rule
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
 [OF data-pre-bound])

have *witness-hit-bound*:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
 adversary-initial-state $\leq P + R$

by (rule
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new
 [OF transcript-pre-bound transcript-new-bound])

have *no-prefix-bound*:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 adversary-initial-state $\leq P + R$

by (rule *order-trans*
 [OF
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-le-witness-transcript-hit
 witness-hit-bound])

have *path-output-bound*:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
 adversary-initial-state $\leq P + R$

by (rule *order-trans*
 [OF
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-le-witness-transcript-hit
 witness-hit-bound])

have *uncovered-bound*:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
 adversary-initial-state $\leq X$

by (rule *order-trans*
 [OF
 checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-witnessed-cross-or-merkle-bad
 cross-bound])

have *collision-bound*:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
adversary-initial-state $\leq$
hash-collision-budget-value 0 (*staged-alpha-search-queries* budgets 0)
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
[*OF wf controlled*])
have *not-prefix-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)
(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$
hash-collision-budget-value 0 (*staged-alpha-search-queries* budgets 0) +
X + (P + R) + (P + R)
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound
[*OF wf controlled collision-bound uncovered-bound no-prefix-bound*
path-output-bound])
show ?thesis
by (rule
checked-staged-soundness-from-aligned-transcript-components-and-trace-index-path-bounds-from-not-prefix-current-bound
[*OF false-statement wf controlled trace-fri-bound comp-fri-bound*
trace-frac path-bound not-prefix-bound empty-trace-fri-bound
current-empty-bound])
qed

lemma *checked-staged-soundness-from-aligned-transcript-components-and-trace-index-structured-path-bounds-from-current-bound*
fixes *trace-fri-error'* *composition-fri-error'* X P R *empty-query-error* :: prob
and T C :: nat $\Rightarrow$ prob
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed* budgets
and *controlled*: *staged-adversary-controlled* budgets A
and *trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-fri-error'*
and *trace-frac*:
 $\bigwedge i$ *prefix* *prefix-state*.
i < rounds $\Rightarrow$
Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
set-dist
(*execute* (*checked-staged-query-prefix-with-state* A i)
adversary-initial-state) $\Rightarrow$

$nnreal$
 $(card$
 $(staged-query-prefix-prefix-authenticated-trace-indices\ prefix$
 $prefix-state)) /$
 $nnreal (card\ query-sample-space) \leq query-error-bound$
and *trace-path-bound*:
 $\bigwedge i. i < rounds \implies$
 $wp-event$
 $(checked-staged-security-experiment-with-query-prefix-data-state\ A\ i)$
 $(checked-staged-security-with-query-prefix-current-trace-path-output-hit-at\ i)$
 $adversary-initial-state \leq T\ i$
and *composition-path-bound*:
 $\bigwedge i. i < rounds \implies$
 $wp-event$
 $(checked-staged-security-experiment-with-query-prefix-data-state\ A\ i)$
 $(checked-staged-security-with-query-prefix-current-composition-path-output-hit-at\ i)$
 $adversary-initial-state \leq C\ i$
and *cross-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad$
 $adversary-initial-state \leq X$
and *data-pre-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $staged-security-with-data-state-transcript-pre-hit$
 $adversary-initial-state \leq P$
and *transcript-new-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state$
 $A)$
 $checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit$
 $adversary-initial-state \leq R$
and *empty-trace-fri-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $trace-fri-bad-with-empty-composition-header-candidates)$
 $adversary-initial-state \leq trace-fri-error$
and *current-empty-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-current-empty-query-opening-hit-at\ 0)$
 $adversary-initial-state \leq empty-query-error$
shows $checked-staged-adversary-acceptance-probability\ A \leq$
 $hash-collision-budget-value\ 0$
 $(staged-attacker-query-budget\ budgets +$
 $staged-challenge-query-budget + verifier-hash-query-budget) +$
 $(composition-error-bound +$
 $staged-phase-relation-error\ size$
 $(staged-alpha-search-queries\ budgets\ 0) +$

hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + X + (P + R) + (P + R)) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)) +
trace-fri-error' + composition-fri-error' +
 $(\sum i < \text{rounds}.$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound + T i + C i) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + X + (P + R) + (P + R)) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + empty-query-error)

proof –

have *transcript-pre-bound*:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
adversary-initial-state $\leq P$

by (rule
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
 [OF *data-pre-bound*])

have *witness-hit-bound*:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
adversary-initial-state $\leq P + R$

by (rule
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new
 [OF *transcript-pre-bound transcript-new-bound*])

have *no-prefix-bound*:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
adversary-initial-state $\leq P + R$

by (rule *order-trans*
 [OF
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-le-witness-transcript-hit
witness-hit-bound])

have *path-output-bound*:

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
adversary-initial-state $\leq P + R$
by (*rule order-trans*
[OF
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-le-witness-transcript-hit
witness-hit-bound])
have *uncovered-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
adversary-initial-state $\leq X$
by (*rule order-trans*
[OF
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-witnessed-cross-or-merkle-bad
cross-bound])
have *collision-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
adversary-initial-state $\leq$
hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0)
by (*rule*
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
[OF wf controlled])
have *not-prefix-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$
hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0) +
X + (P + R) + (P + R)
by (*rule*
checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound
[OF wf controlled collision-bound uncovered-bound no-prefix-bound
path-output-bound])
show *?thesis*
by (*rule*
checked-staged-soundness-from-aligned-transcript-components-and-trace-index-structured-path-bounds-from-not-p
[OF false-statement wf controlled trace-fri-bound comp-fri-bound
trace-frac trace-path-bound composition-path-bound not-prefix-bound
empty-trace-fri-bound current-empty-bound])
qed

theorem *checked-staged-soundness-from-aligned-transcript-conceptual-default-alpha-prefix-union-clean-structured-pa*
fixes *trace-fri-error' composition-fri-error' N empty-query-error :: prob*
and *T C :: nat $\Rightarrow$ prob*
and *trace-default composition-default :: 'f list*

assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *composition-fri-error'*
and *default-prefix-clean*:
 $\bigwedge i$ *prefix prefix-state*.
 i < *rounds* $\implies$
 Some ((*prefix, prefix-state*), *prefix-state*) $\in$
 set-dist
 (*execute* (*checked-staged-query-prefix-with-state A i*)
 adversary-initial-state) $\implies$
 query-prefix-trace-conceptual-table-with-default prefix prefix-state
 trace-default = *trace-default* $\wedge$
 query-prefix-composition-conceptual-table-with-default prefix
 prefix-state composition-default = *composition-default* $\wedge$
 trace-table-low-degree trace-default $\wedge$
 composition-table-low-degree maxDegree composition-default $\wedge$
 trace-default $\in$
 alpha-prefix-trace-table-candidates prefix-state
 (*sqp-trace-root prefix*) $\wedge$
 sqp-alphas prefix $\notin$
 alpha-prefix-union-bad-sets prefix-state
 composition-trace-bad-alpha-space (sqp-trace-root prefix)
and *trace-path-bound*:
 $\bigwedge i$. *i* < *rounds* $\implies$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-trace-path-output-hit-at i*)
 adversary-initial-state $\leq$ *T i*
and *composition-path-bound*:
 $\bigwedge i$. *i* < *rounds* $\implies$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-composition-path-output-hit-at i*)
 adversary-initial-state $\leq$ *C i*
and *not-prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*)

composition-trace-bad-alpha-space)
adversary-initial-state $\leq N$
and *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
and *current-empty-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-current-empty-query-opening-hit-at* *0*)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
hash-collision-budget-value *0*
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
(*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* *0*) +
hash-collision-budget-value *0*
(*staged-alpha-search-queries* *budgets* *0*) +
N +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* *0*)) +
trace-fri-error' + *composition-fri-error'* +
($\sum i < \text{rounds}$.
staged-phase-relation-error *size*
(*staged-query-search-queries* *budgets* *i* + *1*) +
query-error-bound +
hash-collision-budget-value *0*
(*staged-query-search-queries* *budgets* *i* + *1*) +
T i + *C i*) +
(*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* *0*) +
hash-collision-budget-value *0*
(*staged-alpha-search-queries* *budgets* *0*) +
N +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* *0*) +
trace-fri-error + *empty-query-error*)
proof (*rule*
checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-not-prefix-curr
[*OF false-statement wf controlled trace-fri-bound comp-fri-bound - not-prefix-bound*
empty-trace-fri-bound current-empty-bound])
fix *i*
assume *i-bound*: *i* < *rounds*
show
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)

(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
adversary-initial-state $\leq$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
T i + C i
by (rule
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-alpha
[OF false-statement wf controlled i-bound - trace-path-bound[OF i-bound]
composition-path-bound[OF i-bound]]])
(rule default-prefix-clean[OF i-bound])
qed

theorem *checked-staged-soundness-from-aligned-transcript-conceptual-default-alpha-prefix-union-clean-structured-pat*
fixes *trace-fri-error' composition-fri-error' D empty-query-error :: prob*
and *T C :: nat $\Rightarrow$ prob*
and *trace-default composition-default :: 'f list*
assumes *false-statement: $\neg$ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ trace-fri-error'
and *comp-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ composition-fri-error'
and *default-prefix-clean:*
 $\bigwedge i$ *prefix prefix-state.*
i < rounds $\implies$
Some ((prefix, prefix-state), prefix-state) $\in$
set-dist
(execute (checked-staged-query-prefix-with-state A i)
adversary-initial-state) $\implies$
query-prefix-trace-conceptual-table-with-default prefix prefix-state
trace-default = trace-default $\wedge$
query-prefix-composition-conceptual-table-with-default prefix
prefix-state composition-default = composition-default $\wedge$
trace-table-low-degree trace-default $\wedge$
composition-table-low-degree maxDegree composition-default $\wedge$
trace-default $\in$
alpha-prefix-trace-table-candidates prefix-state
(sqp-trace-root prefix) $\wedge$
sqp-alphas prefix $\notin$

$\alpha\text{-prefix-union-bad-sets } \text{prefix-state}$
 $\text{composition-trace-bad-alpha-space } (\text{sqp-trace-root } \text{prefix})$
and trace-path-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq T \ i$
and $\text{composition-path-bound:}$
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and drift-bound:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift } \text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq D$
and $\text{empty-trace-fri-bound:}$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event } \text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and $\text{current-empty-bound:}$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget } \text{budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries } \text{budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries } \text{budgets } 0) +$
 $D +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries } \text{budgets } 0)) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum_{i < \text{rounds}}.$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries } \text{budgets } i + 1) +$
 $\text{query-error-bound} +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-query-search-queries } \text{budgets } i + 1) +$

```

     $T\ i + C\ i) +$ 
    (composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
      D +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      trace-fri-error + empty-query-error)
  proof (rule
    checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-drift-current-e
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound - drift-bound
      empty-trace-fri-bound current-empty-bound])
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  show
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-query-search-queries budgets  $i + 1$ ) +
    query-error-bound +
    hash-collision-budget-value 0
    (staged-query-search-queries budgets  $i + 1$ ) +
     $T\ i + C\ i$ 
  by (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-alpha
    [OF false-statement wf controlled i-bound - trace-path-bound[OF i-bound]
      composition-path-bound[OF i-bound]])
    (rule default-prefix-clean[OF i-bound])
qed

end

end

```

```

theory Soundness-Alpha-Fixed-Candidate
  imports Soundness-Conceptual-Query-Current-Alpha-Clean
begin

```

Fixed-trace-table alpha targets.

This layer records the sound probability component that is available when a trace table has already been fixed before the alpha challenge is sampled. It deliberately does not claim that sampled Merkle openings determine such a table.

```

context soundness
begin

```

definition *staged-alpha-prefix-fixed-trace-prequery-hit*

$:: 'f \text{ list} \Rightarrow$
 $((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

staged-alpha-prefix-fixed-trace-prequery-hit *trace-table out* $\longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (-, \text{prefix-state}) \Rightarrow$
 $(\exists as \in \text{composition-trace-bad-alpha-space } \text{trace-table}.$
 $\text{alpha-vector-prequeried-in-state } \text{prefix-state as}$
 $(\text{length spec})))$

definition *staged-alpha-prefix-fixed-trace-prequery-relation*

$:: 'f \text{ staged-adversary} \Rightarrow 'f \text{ protocol-channel} \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}$

where

staged-alpha-prefix-fixed-trace-prequery-relation *A s trace-table x y*
 $\longleftrightarrow$
 $(\exists \text{prefix } \text{prefix-state as } i.$
 $\text{Some } (\text{prefix}, \text{prefix-state}) \in$
 $\text{set-dist } (\text{execute } (\text{staged-alpha-prefix-program } A) s) \wedge$
 $as \in \text{composition-trace-bad-alpha-space } \text{trace-table} \wedge$
 $i < \text{length spec} \wedge$
 $i < \text{length as} \wedge$
 $x = \text{alpha-challenge-key-at } \text{prefix-state } i \text{ as} \wedge$
 $y = \text{as} ! i)$

definition *checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried*

$:: 'f \text{ list} \Rightarrow$
 $(((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow$
 bool

where

checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried
trace-table out $\longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((\text{prefix}, \text{prefix-state}), \text{data}), -) \Rightarrow$
 $\text{staged-alphas } \text{data} \in \text{composition-trace-bad-alpha-space } \text{trace-table} \wedge$
 $\neg \text{alpha-vector-prequeried-in-state } \text{prefix-state}$
 $(\text{staged-alphas } \text{data}) (\text{length spec}))$

definition *checked-staged-transcript-fixed-trace-alpha-prequery-hit*

$:: 'f \text{ list} \Rightarrow$
 $(((((('f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$

```

      'f protocol-channel) option ⇒
    bool
  where
    checked-staged-transcript-fixed-trace-alpha-prequery-hit trace-table out
    ↔
    (case out of
      None ⇒ False
    | Some (((prefix, prefix-state), data), -) ⇒
      staged-alphas data ∈ composition-trace-bad-alpha-space trace-table ∧
      alpha-vector-prequeried-in-state prefix-state
      (staged-alphas data) (length spec))

definition checked-staged-transcript-fixed-trace-alpha-bad
  :: 'f list ⇒
    (((('f × 'f list × 'f list × 'f) ×
      'f protocol-channel) × 'f staged-proof-data) ×
      'f protocol-channel) option ⇒
    bool
  where
    checked-staged-transcript-fixed-trace-alpha-bad trace-table out
    ↔
    (case out of
      None ⇒ False
    | Some (((prefix, prefix-state), data), -) ⇒
      staged-alphas data ∈ composition-trace-bad-alpha-space trace-table)

lemma staged-alpha-prefix-fixed-trace-prequery-hit-imp-relation-hit:
  assumes empty: HashMap s = fmempty
  and support:
    Some (prefix, prefix-state) ∈
      set-dist (execute (staged-alpha-prefix-program A) s)
  and hit:
    staged-alpha-prefix-fixed-trace-prequery-hit trace-table
      (Some (prefix, prefix-state))
  shows
    hash-relation-hit-event
      (staged-alpha-prefix-fixed-trace-prequery-relation A s trace-table)
      s (Some (prefix, prefix-state))
proof -
  from hit obtain as i where member:
    as ∈ composition-trace-bad-alpha-space trace-table
  and i-bound: i < length spec
  and i-len: i < length as
  and lookup:
    fmllookup (HashMap prefix-state)
      (alpha-challenge-key-at prefix-state i as) = Some (as ! i)
  unfolding staged-alpha-prefix-fixed-trace-prequery-hit-def
    alpha-vector-prequeried-in-state-def
  by auto

```

```

have initial-none:
  fmlookup (HashMap s) (alpha-challenge-key-at prefix-state i as) =
    None
using empty by simp
have relation:
  staged-alpha-prefix-fixed-trace-prequery-relation A s trace-table
    (alpha-challenge-key-at prefix-state i as) (as ! i)
unfolding staged-alpha-prefix-fixed-trace-prequery-relation-def
using support member i-bound i-len by blast
have hit-relation:
  hash-relation-hit
    (staged-alpha-prefix-fixed-trace-prequery-relation A s trace-table)
      s prefix-state
unfolding hash-relation-hit-def
using initial-none lookup relation by blast
show ?thesis
unfolding hash-relation-hit-event-def using hit-relation by simp
qed

lemma staged-alpha-prefix-fixed-trace-prequery-hit-bound-by-relation-hit:
assumes empty: HashMap s = fmempty
shows
  wp-event (staged-alpha-prefix-program A)
    (staged-alpha-prefix-fixed-trace-prequery-hit trace-table) s ≤
  wp-event (staged-alpha-prefix-program A)
    (hash-relation-hit-event
      (staged-alpha-prefix-fixed-trace-prequery-relation A s trace-table)
      s) s
proof (rule wp-event-mono-on-support)
fix out
assume support: out ∈ set-dist (execute (staged-alpha-prefix-program A) s)
and hit: staged-alpha-prefix-fixed-trace-prequery-hit trace-table out
show
  hash-relation-hit-event
    (staged-alpha-prefix-fixed-trace-prequery-relation A s trace-table)
      s out
proof (cases out)
case None
then show ?thesis
using hit unfolding staged-alpha-prefix-fixed-trace-prequery-hit-def
by simp
next
case (Some result)
then obtain prefix prefix-state where out-eq:
  out = Some (prefix, prefix-state)
by (cases result) auto
show ?thesis
unfolding out-eq
by (rule staged-alpha-prefix-fixed-trace-prequery-hit-imp-relation-hit)

```



```

      [OF empty])
    (use support hit out-eq in simp-all)
  qed
qed

lemma staged-alpha-prefix-fixed-trace-prequery-relation-fiber-bound-size:
  card
    {y. staged-alpha-prefix-fixed-trace-prequery-relation A s trace-table x y}
  ≤ size
proof -
  have subset:
    {y. staged-alpha-prefix-fixed-trace-prequery-relation A s trace-table x y}
    ⊆ (UNIV :: 'f set)
  by simp
  have finite-field: finite (UNIV :: 'f set)
  by simp
  have card
    {y. staged-alpha-prefix-fixed-trace-prequery-relation A s trace-table x y}
    ≤ card (UNIV :: 'f set)
  by (rule card-mono[OF finite-field subset])
  also have ... = size
  using size-card by simp
  finally show ?thesis .
qed

lemma staged-alpha-prefix-fixed-trace-prequery-hit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (staged-alpha-prefix-program A)
      (staged-alpha-prefix-fixed-trace-prequery-hit trace-table)
      adversary-initial-state ≤
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0)
proof -
  have prehit-le-relation:
    wp-event (staged-alpha-prefix-program A)
      (staged-alpha-prefix-fixed-trace-prequery-hit trace-table)
      adversary-initial-state ≤
    wp-event (staged-alpha-prefix-program A)
      (hash-relation-hit-event
        (staged-alpha-prefix-fixed-trace-prequery-relation A
          adversary-initial-state trace-table)
        adversary-initial-state)
      adversary-initial-state
  by (rule staged-alpha-prefix-fixed-trace-prequery-hit-bound-by-relation-hit)
    (simp add: adversary-initial-state-def)
  also have ... ≤
    staged-phase-relation-error size

```

```

      (staged-alpha-search-queries budgets 0)
proof -
  have program:
    hash-relation-program
      (staged-alpha-prefix-fixed-trace-prequery-relation A
        adversary-initial-state trace-table)
      size (staged-alpha-search-queries budgets 0)
      (staged-alpha-prefix-program A)
  by (rule hash-relation-program-staged-alpha-prefix-program
    [OF wf controlled])
      (rule staged-alpha-prefix-fixed-trace-prequery-relation-fiber-bound-size)
show ?thesis
  using program
  unfolding hash-relation-program-def hash-relation-budget-def
    staged-phase-relation-error-def
  by blast
qed
finally show ?thesis .
qed

lemma checked-staged-after-alpha-prefix-fixed-trace-alpha-bad-excluding-prequeried-bound:
shows
  wp-event (checked-staged-after-alpha-prefix-program A prefix)
    (λout. case out of
      None ⇒ False
    | Some (data, -) ⇒
      staged-alphas data ∈
        composition-trace-bad-alpha-space trace-table ∧
        ¬ alpha-vector-prequeried-in-state s
        (staged-alphas data) (length spec))
    s ≤ composition-error-bound
proof -
let ?B = composition-trace-bad-alpha-space trace-table
have bound:
  wp-event (checked-staged-after-alpha-prefix-program A prefix)
    (λout. case out of
      None ⇒ False
    | Some (data, -) ⇒
      staged-alphas data ∈ ?B ∧
      ¬ alpha-vector-prequeried-in-state s
      (staged-alphas data) (length spec))
    s ≤ nnreal (card ?B) / nnreal (CARD('f) ^ length spec)
  by (rule
    checked-staged-after-alpha-prefix-program-alpha-list-bound-excluding-prequeried
    [OF composition-trace-bad-alpha-space-subset-alpha-space])
also have ... ≤ composition-error-bound
using composition-trace-bad-alpha-space-fraction-bound-alpha-space
  [of trace-table]
by simp

```

```

finally show ?thesis .
qed

lemma checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried-bound:
shows
  wp-event (checked-staged-transcript-with-alpha-prefix-program A)
    (checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried
      trace-table)
  adversary-initial-state ≤ composition-error-bound
unfolding checked-staged-transcript-with-alpha-prefix-program-def
proof (rule wp-event-bind-bound-by-cont)
show
  ¬ checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried
    trace-table None
unfolding
  checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried-def
by simp
next
fix prefix prefix-state
assume prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
    (execute (staged-alpha-prefix-program A) adversary-initial-state)
let ?Q =
  checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried
    trace-table
have get-eq:
  wp-event
    (get ≥=
      (λprefix-state'.
        checked-staged-after-alpha-prefix-program A prefix ≥=
          (λdata. return ((prefix, prefix-state'), data))))
  ?Q prefix-state =
  wp-event
    (checked-staged-after-alpha-prefix-program A prefix ≥=
      (λdata. return ((prefix, prefix-state), data)))
  ?Q
  prefix-state
unfolding wp-event-def by (simp add: wpsimps)
have return-eq:
  wp-event
    (checked-staged-after-alpha-prefix-program A prefix ≥=
      (λdata. return ((prefix, prefix-state), data)))
  ?Q prefix-state =
  wp-event (checked-staged-after-alpha-prefix-program A prefix)
    (λout. case out of
      None ⇒ False
    | Some (data, -) ⇒
      staged-alphas data ∈

```

```

      composition-trace-bad-alpha-space trace-table  $\wedge$ 
     $\neg$  alpha-vector-prequeried-in-state prefix-state
      (staged-alphas data) (length spec))
    prefix-state
  apply (subst wp-event-bind-return-map)
  unfolding
    checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried-def
  by simp
show
  wp-event
    (get  $\gg$ 
      ( $\lambda$ prefix-state.
        checked-staged-after-alpha-prefix-program A prefix  $\gg$ 
          ( $\lambda$ data. return ((prefix, prefix-state), data))))
    ?Q prefix-state  $\leq$  composition-error-bound
  unfolding get-eq return-eq
  by (rule
    checked-staged-after-alpha-prefix-fixed-trace-alpha-bad-excluding-prequeried-bound)
qed

lemma checked-staged-transcript-fixed-trace-alpha-prequery-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
      (checked-staged-transcript-fixed-trace-alpha-prequery-hit trace-table)
      adversary-initial-state  $\leq$ 
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0)
  unfolding checked-staged-transcript-with-alpha-prefix-program-def
proof (rule wp-event-bind-bound-by-head-event)
  show
    wp-event (staged-alpha-prefix-program A)
      (staged-alpha-prefix-fixed-trace-prequery-hit trace-table)
      adversary-initial-state
     $\leq$  staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0)
  by (rule staged-alpha-prefix-fixed-trace-prequery-hit-bound
    [OF wf controlled])
next
  show
    checked-staged-transcript-fixed-trace-alpha-prequery-hit trace-table
    None  $\implies$ 
    staged-alpha-prefix-fixed-trace-prequery-hit trace-table None
  unfolding checked-staged-transcript-fixed-trace-alpha-prequery-hit-def
    staged-alpha-prefix-fixed-trace-prequery-hit-def
  by simp
next
  fix prefix prefix-state out

```

```

assume prefix-out:
  Some (prefix, prefix-state) ∈
    set-dist
      (execute (staged-alpha-prefix-program A) adversary-initial-state)
and cont:
  out ∈
    set-dist
      (execute
        (get  $\gg$ 
          ( $\lambda$ prefix-state.
            checked-staged-after-alpha-prefix-program A prefix  $\gg$ 
              ( $\lambda$ data. return ((prefix, prefix-state), data))))
          prefix-state)
and hit:
  checked-staged-transcript-fixed-trace-alpha-prequery-hit trace-table
    out
from hit cont show
  staged-alpha-prefix-fixed-trace-prequery-hit trace-table
    (Some (prefix, prefix-state))
unfolding checked-staged-transcript-fixed-trace-alpha-prequery-hit-def
  staged-alpha-prefix-fixed-trace-prequery-hit-def
by (auto elim!: set-dist-bindE split: option.splits prod.splits)
qed

```

```

lemma checked-staged-transcript-fixed-trace-alpha-bad-split:
assumes hit:
  checked-staged-transcript-fixed-trace-alpha-bad trace-table out
shows
  checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried
    trace-table out  $\vee$ 
    checked-staged-transcript-fixed-trace-alpha-prequery-hit trace-table out
using hit
unfolding checked-staged-transcript-fixed-trace-alpha-bad-def
  checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried-def
  checked-staged-transcript-fixed-trace-alpha-prequery-hit-def
by (auto split: option.splits prod.splits)

```

```

lemma checked-staged-transcript-fixed-trace-alpha-bad-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event (checked-staged-transcript-with-alpha-prefix-program A)
    (checked-staged-transcript-fixed-trace-alpha-bad trace-table)
    adversary-initial-state  $\leq$ 
      composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0)
proof –
  let ?M = checked-staged-transcript-with-alpha-prefix-program A

```

```

let ?Fresh =
  checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried
  trace-table
let ?Pre =
  checked-staged-transcript-fixed-trace-alpha-prequery-hit trace-table
have event-le:
  wp-event ?M
    (checked-staged-transcript-fixed-trace-alpha-bad trace-table)
    adversary-initial-state ≤
    wp-event ?M (λout. ?Fresh out ∨ ?Pre out)
    adversary-initial-state
  by (rule wp-event-mono) (rule checked-staged-transcript-fixed-trace-alpha-bad-split)
also have ... ≤
  wp-event ?M ?Fresh adversary-initial-state +
  wp-event ?M ?Pre adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0)
  by (intro add-mono
    checked-staged-transcript-fixed-trace-alpha-bad-excluding-prequeried-bound
    checked-staged-transcript-fixed-trace-alpha-prequery-bound
    wf controlled)
  finally show ?thesis .
qed

end

end

```

```

theory Staged-Security-Experiment-Composition-Query-Canonical
  imports Staged-Security-Experiment-Composition-Query-Classification
begin

```

Canonical-index projections for the header-authenticated composition query bridge. These lemmas do not bound the broad existential event by themselves; they expose the canonical query-index cover that a later Merkle/candidate binding argument has to control.

In particular, the projection from the actual-alpha-prefix event to a query-prefix hit is support-level evidence. It is not by itself a probability reduction to the separately sampled query-prefix program: the authenticated opening witness is still selected existentially from verifier-reachable evidence. A sound probability bound needs either a cover fixed before the query challenge is sampled, or a Merkle/collision/partial-opening side event accounting for failure to select such a witness.

```

context soundness
begin

```

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*

assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix

A

(*Some* (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*))

obtains *i trace-openings composition-openings query-prefix query-prefix-state*

raw raw-state where

(*trace-openings*, *composition-openings*) ∈

query-header-supported-partial-opening-witnesses

(*verifier-state-from-adversary attacker-state*

(*staged-proof-transcript data*))

(*staged-trace-root data*)

(*staged-trace-fri-roots data*)

(*staged-trace-final data*)

(*staged-alphas data*)

(*staged-degree data*)

(*staged-composition-fri-roots data*)

(*staged-composition-final data*)

i < rounds

Some (data, attacker-state) ∈

set-dist

(*execute (checked-staged-transcript-program A)*

adversary-initial-state)

Some (((query-prefix, query-prefix-state), raw), raw-state) ∈

set-dist

(*execute (checked-staged-query-prefix-receive-with-state A i)*

adversary-initial-state)

index (to-nat raw) ∈

query-header-supported-partial-opening-canonical-query-indices

(*verifier-state-from-adversary attacker-state*

(*staged-proof-transcript data*))

(*staged-trace-root data*)

(*staged-trace-fri-roots data*)

(*staged-trace-final data*)

(*staged-alphas data*)

(*staged-degree data*)

(*staged-composition-fri-roots data*)

(*staged-composition-final data*)

proof (*rule*

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-query

[*OF hit*])

fix *i trace-openings composition-openings query-prefix query-prefix-state*

raw raw-state

assume *witness*:

(*trace-openings*, *composition-openings*) ∈

query-header-supported-partial-opening-witnesses

(*verifier-state-from-adversary attacker-state*

(*staged-proof-transcript data*))

```

    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
and i-bound:  $i < \text{rounds}$ 
and builder:
    Some (data, attacker-state)  $\in$ 
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and prefix-receive:
    Some (((query-prefix, query-prefix-state), raw), raw-state)  $\in$ 
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
and target-hit:
    checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i)
    (Some (((query-prefix, query-prefix-state), raw), raw-state))
have header-hit:
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
    trace-openings composition-openings i
    (Some (((((query-prefix, query-prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
by (rule
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitI-from-dynamic-hit
    [OF witness target-hit])
have canonical:
    index (to-nat raw)  $\in$ 
    query-header-supported-partial-opening-canonical-query-indices
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
by (rule
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-canonical-query-indexD
    [OF i-bound header-hit])
show ?thesis

```


by (rule that[OF witness i-bound builder prefix-receive canonical])
qed

lemma *checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*

assumes *hit*:

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix

A

(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))

and *cover*:

query-header-supported-partial-opening-canonical-query-indices

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data))

(staged-trace-root data)

(staged-trace-fri-roots data)

(staged-trace-final data)

(staged-alphas data)

(staged-degree data)

(staged-composition-fri-roots data)

(staged-composition-final data) $\subseteq B$

obtains *i query-prefix query-prefix-state raw raw-state* **where**

i < rounds

Some (data, attacker-state) $\in$

set-dist

(execute (checked-staged-transcript-program A)

adversary-initial-state)

Some (((query-prefix, query-prefix-state), raw), raw-state) $\in$

set-dist

(execute (checked-staged-query-prefix-receive-with-state A i)

adversary-initial-state)

checked-staged-query-prefix-dynamic-index-hit

(λ - . *B*)

(Some (((query-prefix, query-prefix-state), raw), raw-state))

proof (rule

checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix-canon

[OF hit])

fix *i trace-openings composition-openings query-prefix query-prefix-state raw*

raw-state

assume *i-bound*: *i < rounds*

and *builder*:

Some (data, attacker-state) $\in$

set-dist

(execute (checked-staged-transcript-program A)

adversary-initial-state)

and *prefix-receive*:

Some (((query-prefix, query-prefix-state), raw), raw-state) $\in$

set-dist

(execute (checked-staged-query-prefix-receive-with-state A i)

adversary-initial-state)

and *canonical*:
index (*to-nat raw*) $\in$
query-header-supported-partial-opening-canonical-query-indices
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)
have *hit-B*:
checked-staged-query-prefix-dynamic-index-hit
*(λ - -. *B*)*
(Some (((query-prefix, query-prefix-state), raw), raw-state))
using *canonical cover*
unfolding *checked-staged-query-prefix-dynamic-index-hit-def*
by *auto*
show *?thesis*
by (*rule that*[*OF i-bound builder prefix-receive hit-B*])
qed

lemma *checked-staged-query-prefix-constant-index-hit-bound-from-budgets*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
and *subset: B $\subseteq$ query-sample-space*
and *frac*:

$$\text{nnreal (card B) / nnreal (card query-sample-space)} \leq$$

$$\text{query-error-bound}$$
shows

$$\text{wp-event (checked-staged-query-prefix-receive-with-state A i)}$$

$$\text{(checked-staged-query-prefix-dynamic-index-hit (λ - -. B))}$$

$$\text{adversary-initial-state} \leq$$

$$\text{staged-phase-relation-error size}$$

$$\text{(staged-query-search-queries budgets i + 1) + query-error-bound}$$
proof –
have *raw-bound: query-index-raw-preimage-bound*
by (*rule query-index-raw-preimage-bound-from-sampler-wellformed*)
have *hit-bound*:

$$\text{wp-event (checked-staged-query-prefix-receive-with-state A i)}$$

$$\text{(checked-staged-query-prefix-dynamic-index-hit (λ - -. B))}$$

$$\text{adversary-initial-state} \leq$$

$$\text{wp-event (checked-staged-query-prefix-receive-with-state A i)}$$

$$\text{(checked-staged-query-prefix-dynamic-index-prehit (λ - -. B))}$$

$$\text{adversary-initial-state} + \text{query-error-bound}$$
proof (*rule*
checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error

```

    [OF raw-bound])
  fix prefix prefix-state
  assume
    Some ((prefix, prefix-state), prefix-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)
  show
    (λ- -. B) prefix prefix-state ⊆ query-sample-space ∧
      nnreal (card ((λ- -. B) prefix prefix-state)) /
      nnreal (card query-sample-space) ≤ query-error-bound
  using subset frac by simp
qed
have prehit-bound:
  wp-event (checked-staged-query-prefix-receive-with-state A i)
    (checked-staged-query-prefix-dynamic-index-prehit (λ- -. B))
    adversary-initial-state ≤
      staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1)
proof (rule
  checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program)
show HashMap adversary-initial-state = fmempty
  by (simp add: adversary-initial-state-def)
show hash-relation-program
  (checked-staged-query-prefix-dynamic-index-prehit-relation A i
    adversary-initial-state (λ- -. B))
  size (staged-query-search-queries budgets i + 1)
  (checked-staged-query-prefix-receive-with-state A i)
  by (rule hash-relation-program-checked-staged-query-prefix-receive-with-state
    [OF wf controlled])
    (use i-bound in simp-all,
      rule checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size)
qed
show ?thesis
  by (rule order-trans[OF hit-bound])
    (intro add-mono prehit-bound order-refl)
qed

lemma checked-staged-security-with-query-prefix-constant-index-hit-bound-from-budgets:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and subset: B ⊆ query-sample-space
  and frac:
    nnreal (card B) / nnreal (card query-sample-space) ≤
      query-error-bound
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)

```

```

    (checked-staged-security-with-query-prefix-dynamic-index-hit
     ( $\lambda$ - .  $B$ ))
    adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-query-search-queries budgets  $i + 1$ ) + query-error-bound
  by (rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound,
      rule checked-staged-query-prefix-constant-index-hit-bound-from-budgets
      [OF wf controlled i-bound subset frac])

end

end

```

theory *Soundness*

imports

Soundness-Staged-Aligned-Current-Query-Path
Soundness-Staged-Aligned-Current-Empty
Soundness-Alpha-Fixed-Candidate
Staged-Security-Experiment-Composition-Query-Header-Rounds
Staged-Security-Experiment-Composition-Query-Canonical
Staged-Security-Experiment-Composition-Query-Current

begin

Aggregator for the soundness infrastructure used by the active public endpoint. The public soundness theorem is exported by `Soundness_FRI_Trace_Restricted_Query_Endpoint.thy`.

end

theory *Staged-Security-Experiment-Composition-Query-Single-Round-Witness-Bridge*

imports

Staged-Security-Experiment-Composition-Query-Single-Round
Staged-Security-Experiment-Composition-Query-Current-Alignment
Staged-Security-Experiment-Composition-Query-Current-Path
Staged-Security-Experiment-Alpha-Header-Binding

begin

Bridge from the broad actual-alpha diagnostic witness event to a fixed query-prefix candidate-pair witness event.

The bridge is deterministic: it extracts the concrete prefix branch and candidate pair already present in the diagnostic event. It does not bound the diagnostic event probabilistically; later proof layers must still avoid treating the broad existential diagnostic event as a final public-path event.

context *soundness*

begin

definition *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at*
where

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
trace-openings composition-openings trace-table composition-table i A out
 $\longleftrightarrow$
(case out of
None $\Rightarrow$ False
| Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state) $\Rightarrow$
($\exists$ prefix prefix-state raw raw-state.
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) $\in$
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state
A i)
adversary-initial-state) $\wedge$
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
trace-openings composition-openings trace-table composition-table i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))))

definition *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at*
where

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
i A out $\longleftrightarrow$
($\exists$ trace-openings composition-openings trace-table composition-table.
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
trace-openings composition-openings trace-table composition-table i A
out)

definition *checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at*
where

checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at
i out $\longleftrightarrow$
($\exists$ trace-openings composition-openings trace-table composition-table.
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
trace-openings composition-openings trace-table composition-table i
out)

definition *staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at*
where

staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
trace-openings composition-openings trace-table composition-table i A out
 $\longleftrightarrow$
(case out of
None $\Rightarrow$ False
| Some (((data, attacker-state), result), final-state) $\Rightarrow$
($\exists$ prefix prefix-state raw raw-state.
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) $\in$

set-dist
 (execute
 (checked-staged-security-experiment-with-query-prefix-data-state
 A i)
 adversary-initial-state) $\wedge$
 checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
 trace-openings composition-openings trace-table composition-table i
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))))

definition *staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at*
where

staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
 i A out $\longleftrightarrow$
 ($\exists$ trace-openings composition-openings trace-table composition-table.
 staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
 trace-openings composition-openings trace-table composition-table i A
 out)

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-pr*
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
 trace-openings composition-openings trace-table composition-table i A out
 $\longleftrightarrow$

(case out of
 None $\Rightarrow$ False
 | Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state) $\Rightarrow$
 staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
 trace-openings composition-openings trace-table composition-table i A
 (Some (((data, attacker-state), result), final-state))))

unfolding

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-def
 staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-def
by (auto split: option.splits prod.splits)

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-wp*
wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 (checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
 trace-openings composition-openings trace-table composition-table i A)
 adversary-initial-state =
wp-event
 (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
 trace-openings composition-openings trace-table composition-table i A)
 adversary-initial-state

proof –

let ?project =
 $\lambda x. ((snd (fst (fst x)), snd (fst x)), snd x)$

```

have
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
      trace-openings composition-openings trace-table composition-table i A)
  adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
      ≫ (λx. return (?project x)))
    (staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
      trace-openings composition-openings trace-table composition-table i A)
  adversary-initial-state
using
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
  [of A]
by simp
also have ... =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
      trace-openings composition-openings trace-table composition-table i A)
  adversary-initial-state
proof -
  have pred-eq:
    (λout. case out of
      None ⇒
        staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
          trace-openings composition-openings trace-table composition-table i A
      Some (x, t) ⇒
        staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
          trace-openings composition-openings trace-table composition-table i A
        (Some (?project x, t))) =
    checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
      trace-openings composition-openings trace-table composition-table i A
  by (rule ext)
  (simp add:
    checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-proje
    staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-def
    split: option.splits prod.splits)
  show ?thesis
  by (subst wp-event-bind-return-map) (simp add: pred-eq)
qed
finally show ?thesis
  by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-bor
  assumes bound:

```

wp-event
 (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at*
 trace-openings composition-openings trace-table composition-table i A)
adversary-initial-state $\leq B$

shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at*
 trace-openings composition-openings trace-table composition-table i A)
adversary-initial-state $\leq B$

using *bound*

unfolding

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-wp-project

by *simp*

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists*
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
i A out
 $\longleftrightarrow$
 (*case out of*
 None $\Rightarrow$ *False*
 | *Some* ((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
 result), *final-state*) $\Rightarrow$
 staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
 i A (Some (((data, attacker-state), result), final-state)))

unfolding

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-def
staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-def

by (*auto simp*;
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-projecti
split: option.splits prod.splits)

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists*
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at*
 i A)
adversary-initial-state =
wp-event
 (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at*
 i A)
adversary-initial-state

proof –

let *?project* =
 $\lambda x. ((snd (fst (fst x)), snd (fst x)), snd x)$

have

wp-event


```

    (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
    i A)
  adversary-initial-state =
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
    >>= (λx. return (?project x)))
  (staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
    i A)
  adversary-initial-state
using
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
  [of A]
by simp
also have ... =
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
    i A)
  adversary-initial-state
proof -
have pred-eq:
  (λout. case out of
    None =>
      staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
        i A None
  | Some (x, t) =>
      staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
        i A (Some (?project x, t))) =
  checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
    i A
by (rule ext)
  (simp add:
    checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
    staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-def
    staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-def
    split: option.splits prod.splits)
show ?thesis
by (subst wp-event-bind-return-map) (simp add: pred-eq)
qed
finally show ?thesis
by simp
qed

lemma checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at-le-same-run-data-
assumes i-bound: i < rounds
shows
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)

```

```

    (checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at
     i)
    adversary-initial-state ≤
    wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
     i A)
    adversary-initial-state
proof –
  let ?M =
    checked-staged-security-experiment-with-query-prefix-data-state A i
  let ?Gap =
    checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at
    i
  let ?DataGap =
    staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
    i A
  let ?projected =
    λout. case out of
      None ⇒ ?DataGap None
    | Some (packed, t) ⇒ ?DataGap (Some (snd packed, t))
  have projection:
    wp-event (checked-staged-security-experiment-with-data-state A)
    ?DataGap adversary-initial-state =
    wp-event ?M ?projected adversary-initial-state
  by (rule checked-staged-security-experiment-with-query-prefix-data-state-projection-event
    [OF i-bound])
  have event-le:
    wp-event ?M ?Gap adversary-initial-state ≤
    wp-event ?M ?projected adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support: out ∈ set-dist (execute ?M adversary-initial-state)
  and gap: ?Gap out
  show ?projected out
  proof (cases out)
  case None
  then show ?thesis
    using gap
    unfolding
    checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at-def
    checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-def
    checked-staged-security-with-query-prefix-dynamic-index-hit-def
    by simp
  next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =

```

```

    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
from gap obtain trace-openings composition-openings trace-table
  composition-table where fixed-gap:
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
    trace-openings composition-openings trace-table composition-table i
  out
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at-def
by blast
have qsupport:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
    set-dist (execute ?M adversary-initial-state)
using support out-eq by simp
have fixed-gap-out:
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
    trace-openings composition-openings trace-table composition-table i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
using fixed-gap out-eq by simp
have data-gap:
  staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
    trace-openings composition-openings trace-table composition-table i A
    (Some (((data, attacker-state), result), final-state))
unfolding
  staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-def
using qsupport fixed-gap-out by auto
have projected-gap:
  ?DataGap (Some (((data, attacker-state), result), final-state))
unfolding
  staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-def
using data-gap by blast
show ?thesis
unfolding out-eq
using projected-gap by simp
qed
qed
show ?thesis
unfolding projection
by (rule event-le)
qed

lemma checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-atE:
assumes gap:
  checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
    trace-openings composition-openings trace-table composition-table i A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),

```

result), *final-state*))
obtains *prefix prefix-state raw raw-state* **where**
Some (((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) ∈
set-dist
 (*execute*
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
adversary-initial-state)
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
trace-openings composition-openings trace-table composition-table i
 (*Some* (((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*))
using *gap*
unfolding
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-def
by (*auto split: prod.splits*)

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-atI*:
assumes *support*:
Some (((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*) ∈
set-dist
 (*execute*
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
adversary-initial-state)
and *gap*:
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
trace-openings composition-openings trace-table composition-table i
 (*Some* (((((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*)), *final-state*))
shows
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
trace-openings composition-openings trace-table composition-table i A
 (*Some* (((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*))
unfolding
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-def
using *support gap* **by** *auto*

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists*:
assumes *gap*:
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
trace-openings composition-openings trace-table composition-table i A out
shows
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
i A out
unfolding
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-def
using *gap* **by** *blast*

lemma *checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-atI*:

assumes *gap*:

checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
trace-openings composition-openings trace-table composition-table i out

shows

checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at
i out

unfolding

checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at-def

using *gap* **by** *blast*

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists*:

assumes *gap*:

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
i A out

shows

checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
out

proof (*cases out*)

case *None*

then show *?thesis*

using *gap*

unfolding

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-d

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at-def

by *simp*

next

case (*Some packed*)

then obtain *alpha-prefix alpha-prefix-state data attacker-state result*

final-state where out-eq:

out =

Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)

by (*cases packed, auto split: prod.splits*)

from *gap* **obtain** *trace-openings composition-openings trace-table*

composition-table prefix prefix-state raw raw-state where support:

Some (((((prefix, prefix-state), raw), raw-state),

((data, attacker-state), result)), final-state) ∈

set-dist

(execute

(checked-staged-security-experiment-with-query-prefix-data-state
A i)

adversary-initial-state)

and *fixed-gap*:

checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at

trace-openings composition-openings trace-table composition-table i

(Some (((((prefix, prefix-state), raw), raw-state),

((data, attacker-state), result)), final-state))

```

unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-d
by (auto elim!:
  checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-atE)
have witness-side:
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
  trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
by (rule
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-imp-witness-side
  [OF fixed-gap])
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from witness-side have witness:
  (trace-openings, composition-openings) ∈
  query-header-supported-partial-opening-witnesses ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-def
by simp
from query-header-supported-partial-opening-witnessesE[OF witness]
obtain witness-result witness-state rest where outcome:
  Some (witness-result, witness-state) ∈
  set-dist (execute verify-monad ?s)
by blast
have tr-eq: PTranscript ?s = staged-proof-transcript data
by simp
have transcript-hit:
  hash-map-output-values ?s ∩ set (staged-proof-transcript data)
  ≠ {} ∨
  hash-map-new-output-hit (set (staged-proof-transcript data))
  ?s witness-state
by (rule verify-monad-trace-root-target-preexisting-or-new
  [OF outcome tr-eq])
show ?thesis
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-def
  Let-def
using outcome transcript-hit
by (auto split: prod.splits)
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)

(*checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at*
 i A)

adversary-initial-state $\leq$

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit

adversary-initial-state

by (*rule wp-event-mono*)

(*rule*

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-i

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists*

assumes *pre-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit

adversary-initial-state $\leq P$

and *new-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit

adversary-initial-state $\leq Q$

shows

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)

(*checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at*
 i A)

adversary-initial-state $\leq P + Q$

by (*rule order-trans*

[*OF*

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-

checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new

[*OF pre-bound new-bound*]])

lemma *checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists*

assumes *data-pre-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* A)

staged-security-with-data-state-transcript-pre-hit

adversary-initial-state $\leq P$

and *new-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq Q$

shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)
 (*checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at*
i A)
adversary-initial-state $\leq P + Q$

proof –

have *pre-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
adversary-initial-state $\leq P$

by (rule
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
 [OF *data-pre-bound*])

show *?thesis*

by (rule
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at
 [OF *pre-bound new-bound*])

qed

lemma *staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-bound-from-d*

assumes *data-pre-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* A)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$

and *new-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq Q$

shows

wp-event
 (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at*
i A)
adversary-initial-state $\leq P + Q$

using

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-bo
 [OF *data-pre-bound new-bound*, of i]

unfolding

checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-w
by *simp*

lemma *checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at-bound-from-data-p*

assumes *i-bound*: $i < \text{rounds}$

and *data-pre-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$
and *new-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq Q$
shows
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
(*checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at i*)
adversary-initial-state $\leq P + Q$
by (*rule order-trans*[
OF checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at-le-same-run-data-
OF i-bound]
staged-security-with-data-state-query-prefix-fixed-candidate-pair-witness-gap-same-run-exists-at-bound-from-data-
OF data-pre-bound new-bound]])
lemma *checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-with-witness-imp-single-*
assumes *hit*:
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
A trace-table composition-table i
(*Some* (((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*))
and *witness*:
(*trace-openings*, *composition-openings*) $\in$
query-header-supported-partial-opening-witnesses
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))
(*staged-trace-root data*)
(*staged-trace-fri-roots data*)
(*staged-trace-final data*)
(*staged-alphas data*)
(*staged-degree data*)
(*staged-composition-fri-roots data*)
(*staged-composition-final data*)
and *trace-candidate*:
partial-trace-table-candidate trace-table
((*replicate rounds []*) [*i := trace-openings*])
and *comp-candidate*:
partial-composition-table-candidate composition-table
((*replicate rounds []*) [*i := composition-openings*])
and *same-run-support*:
 $\bigwedge$ *prefix prefix-state raw raw-state.*
Some (*data*, *attacker-state*) $\in$

set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
 (execute (checked-staged-query-prefix-receive-with-state A i)
 adversary-initial-state) $\implies$
checked-staged-query-prefix-dynamic-index-hit
 (staged-query-prefix-candidate-pair-query-target-from-prefix
 trace-table composition-table)
 (Some (((prefix, prefix-state), raw), raw-state)) $\implies$
 Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\in$
set-dist
 (execute
 (checked-staged-security-experiment-with-query-prefix-data-state
 A i)
 adversary-initial-state)

shows

checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
 i A
 (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state)) $\vee$
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
 trace-openings composition-openings trace-table composition-table i A
 (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))

proof —

from

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE
 [OF hit]

obtain prefix prefix-state raw raw-state **where**

builder:

Some (data, attacker-state) $\in$
set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state)

and *prefix-receive:*

Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
 (execute (checked-staged-query-prefix-receive-with-state A i)
 adversary-initial-state)

and *target-hit:*

checked-staged-query-prefix-dynamic-index-hit
 (staged-query-prefix-candidate-pair-query-target-from-prefix
 trace-table composition-table)
 (Some (((prefix, prefix-state), raw), raw-state))

by (rule

checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixE

```

[OF hit])
have support:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state
        A i)
      adversary-initial-state)
  by (rule same-run-support[OF builder prefix-receive target-hit])
show ?thesis
proof (cases
  (trace-table, composition-table) ∈
  query-header-supported-single-round-partial-table-candidates
  prefix-state
  (sqp-trace-root prefix)
  (sqp-trace-fri-roots prefix)
  (sqp-trace-final prefix)
  (sqp-alphas prefix)
  (sqp-degree prefix)
  (sqp-composition-fri-roots prefix)
  (sqp-composition-final prefix)
  i)
case True
have single-round-hit:
  checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-single-round-header-supported-query-target i)
  (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
    checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit
    [OF True target-hit])
have
  checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
  i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
proof –
  have inner:
    ∃ prefix prefix-state raw raw-state.
    Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state) ∧
    Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state) ∧
    checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-single-round-header-supported-query-target i)

```

```

      (Some (((prefix, prefix-state), raw), raw-state))
    by (intro exI[of - prefix] exI[of - prefix-state]
        exI[of - raw] exI[of - raw-state])
      (use builder prefix-receive single-round-hit in simp)
  show ?thesis
    unfolding
      checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at-def
    using inner by simp
qed
then show ?thesis by simp
next
case False
have success:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
      (sqp-alphas prefix)
  using target-hit
  unfolding
    checked-staged-query-prefix-dynamic-index-hit-def
    staged-query-prefix-candidate-pair-query-target-from-prefix-def
  by simp
have fixed-gap:
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
    trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
proof -
  have dyn:
    checked-staged-security-with-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table)
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  unfolding
    checked-staged-security-with-query-prefix-dynamic-index-hit-def
    checked-staged-query-prefix-dynamic-index-hit-def
    staged-query-prefix-candidate-pair-query-target-from-prefix-def
  using success by simp
  have side:
    checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
      trace-openings composition-openings trace-table composition-table i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  unfolding
    checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-def
    checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side-def
  using witness trace-candidate comp-candidate False by simp
show ?thesis
  unfolding

```

```

    checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-def
  using dyn side by simp
qed
have
  checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
    trace-openings composition-openings trace-table composition-table i A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-atI
    [OF support fixed-gap])
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-candidate-pair-success-with-witness-imp-single-round-header-
  assumes i-bound: i < rounds
  and support:
    Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
  and success:
    ∧ prefix prefix-state raw raw-state.
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state) ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state
          A i)
        adversary-initial-state) ⇒
    sqp-alphas prefix = staged-alphas data ⇒
    index (to-nat raw) ∈
      query-sampling-success-space trace-table composition-table
        (staged-alphas data)
  and witness:
    (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)

```

```

    (staged-composition-final data)
and trace-candidate:
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
and comp-candidate:
    partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
shows
checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
  i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state)) ∨
checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
  trace-openings composition-openings trace-table composition-table i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
proof (rule
  checked-staged-security-with-actual-alpha-prefix-candidate-pair-success-query-prefix-hitE
  [OF i-bound support success])
fix prefix prefix-state raw raw-state
assume qsupport:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    adversary-initial-state)
and qhit:
checked-staged-security-with-query-prefix-dynamic-index-hit
  (staged-query-prefix-candidate-pair-query-target-from-prefix
    trace-table composition-table)
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
  set-dist
  (execute (checked-staged-security-experiment-with-data-state A)
    adversary-initial-state)
by (rule
  checked-staged-security-experiment-with-query-prefix-data-state-imp-data-state-support
  [OF i-bound qsupport])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
obtain builder where builder:
  Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)
    adversary-initial-state)
by blast

```

```

have prefix-receive:
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
using qsupport
unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
by (auto elim!: set-dist-bindE split: prod.splits)
have raw-pair-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target-from-prefix
      trace-table composition-table)
    (Some (((prefix, prefix-state), raw), raw-state))
using qhit
unfolding checked-staged-security-with-query-prefix-dynamic-index-hit-def
by simp
show ?thesis
proof (cases
  (trace-table, composition-table) ∈
    query-header-supported-single-round-partial-table-candidates
      prefix-state
      (sqp-trace-root prefix)
      (sqp-trace-fri-roots prefix)
      (sqp-trace-final prefix)
      (sqp-alphas prefix)
      (sqp-degree prefix)
      (sqp-composition-fri-roots prefix)
      (sqp-composition-final prefix)
      i)
  case True
have single-round-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-single-round-header-supported-query-target i)
    (Some (((prefix, prefix-state), raw), raw-state))
by (rule
    checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit
    [OF True raw-pair-hit])
have
  checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
    i A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
proof –
  have inner:
    ∃ prefix prefix-state raw raw-state.
      Some (data, attacker-state) ∈
        set-dist
          (execute (checked-staged-transcript-program A)
            adversary-initial-state) ∧

```

```

    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state) ∧
      checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-single-round-header-supported-query-target i)
      (Some (((prefix, prefix-state), raw), raw-state))
  by (intro exI[of - prefix] exI[of - prefix-state]
    exI[of - raw] exI[of - raw-state])
    (use builder prefix-receive single-round-hit in simp)
show ?thesis
  unfolding
    checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at-def
  using inner by simp
qed
then show ?thesis by simp
next
case False
have side:
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
  trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  unfolding
    checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-def
    checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side-def
  using witness trace-candidate comp-candidate False by simp
have fixed-gap:
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
  trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  unfolding
    checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-def
  using qhit side by simp
have
  checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-at
  trace-openings composition-openings trace-table composition-table i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
  by (rule
    checked-staged-security-with-actual-alpha-prefix-query-prefix-fixed-candidate-pair-witness-gap-same-run-atI
    [OF qsupport fixed-gap])
then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-imp-single-round-header
  assumes hit:

```


checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
trace-openings composition-openings i
((Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))
and *trace-candidate:*
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings])
and *comp-candidate:*
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings])
and *trace-low: trace-table-low-degree trace-table*
and *comp-low: composition-table-low-degree maxDegree composition-table*
and *not-all:*
¬ all-queries-consistent trace-table composition-table
(sqp-alphas prefix)
shows
checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
i
((Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state)) ∨
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at
trace-openings composition-openings trace-table composition-table i
((Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))
proof –
from *hit* **obtain** *comp-nonempty witness opening-target*
where *comp-nonempty:*
staged-composition-fri-roots data ≠ []
and *witness:*
(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)
and *opening-target:*
index (to-nat raw) ∈
staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [i := trace-openings])
((replicate rounds []) [i := composition-openings])
i prefix prefix-state
by *(rule*
checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hitE)
have *opening-subset:*

```

    staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [i := trace-openings])
      ((replicate rounds []) [i := composition-openings])
      i prefix prefix-state  $\subseteq$ 
      staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
      composition-table prefix prefix-state
  by (rule
    staged-query-prefix-candidate-opening-query-target-from-prefix-subset-candidate-pair-from-prefix
    [OF trace-candidate comp-candidate trace-low comp-low not-all])
  have pair-target-hit:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table)
      (Some (((prefix, prefix-state), raw), raw-state))
  using opening-target opening-subset
  unfolding checked-staged-query-prefix-dynamic-index-hit-def by auto
  show ?thesis
  proof (cases
    (trace-table, composition-table)  $\in$ 
    query-header-supported-single-round-partial-table-candidates
    prefix-state
    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)
    (sqp-composition-final prefix)
    i)
  case True
  have target:
    checked-staged-query-prefix-dynamic-index-hit
      (staged-query-prefix-single-round-header-supported-query-target i)
      (Some (((prefix, prefix-state), raw), raw-state))
  by (rule
    checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit
    [OF True pair-target-hit])
  have
    checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
      i
      (Some ((((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)))
  unfolding
    checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at-def
  using target by simp
  then show ?thesis by simp
next
case False
have gap:

```

```

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at
  trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-at-def
using hit trace-candidate comp-candidate False by simp
have side:
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
    trace-openings composition-openings trace-table composition-table i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-def
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side-def
using witness trace-candidate comp-candidate False by simp
have
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at
    trace-openings composition-openings trace-table composition-table i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-def
using gap side by simp
then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-bound-from-target-and-f
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
and trace-low: trace-table-low-degree trace-table
and comp-low: composition-table-low-degree maxDegree composition-table
and not-all:
   $\bigwedge$  prefix prefix-state raw raw-state data attacker-state result
    final-state.
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state
        A i)

```

```

      adversary-initial-state)  $\implies$ 
 $\neg$  all-queries-consistent trace-table composition-table
      (sqp-alphas prefix)
and target-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
      i)
    adversary-initial-state  $\leq$  T
shows
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
      trace-openings composition-openings i)
    adversary-initial-state  $\leq$ 
      T +
      (staged-phase-relation-error size
        (staged-query-search-queries budgets i + 1) +
        (1::prob) / nnreal (card query-sample-space))
proof –
  let ?m =
    checked-staged-security-experiment-with-query-prefix-data-state A i
  let ?target =
    checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
      i
  let ?gap =
    checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at
      trace-openings composition-openings trace-table composition-table i
  have event-le:
    wp-event ?m
      (checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
        trace-openings composition-openings i)
      adversary-initial-state  $\leq$ 
      wp-event ?m ( $\lambda$ out. ?target out  $\vee$  ?gap out)
      adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out  $\in$  set-dist (execute ?m adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit
      trace-openings composition-openings i out
  show ?target out  $\vee$  ?gap out
  proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-def

```

```

    by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state
  result final-state where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
have not-all':
  ¬ all-queries-consistent trace-table composition-table
    (sqp-alphas prefix)
  by (rule not-all)
    (use support out-eq in simp)
show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-query-prefix-header-authenticated-candidate-opening-hit-imp-single-round-header-
      [OF - trace-candidate comp-candidate trace-low comp-low not-all'])
    (use hit out-eq in simp)
qed
qed
have union-le:
  wp-event ?m (λout. ?target out ∨ ?gap out)
  adversary-initial-state ≤
  wp-event ?m ?target adversary-initial-state +
  wp-event ?m ?gap adversary-initial-state
  by (rule wp-event-union-bound)
have gap-bound:
  wp-event ?m ?gap adversary-initial-state ≤
  staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space)
  by (rule
    checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at-bound-from-budgets
      [OF wf controlled i-bound])
have sum-bound:
  wp-event ?m ?target adversary-initial-state +
  wp-event ?m ?gap adversary-initial-state ≤
  T +
  (staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    (1::prob) / nnreal (card query-sample-space))
  by (rule add-mono[OF target-bound gap-bound])
show ?thesis
  by (rule order-trans[OF event-le order-trans[OF union-le sum-bound]])
qed

```

lemma *checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-atI:*

```

assumes success:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
    (sqp-alphas prefix)
and witness:
  (trace-openings, composition-openings) ∈
    query-header-supported-partial-opening-witnesses
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [i := composition-openings])
and notin:
  (trace-table, composition-table) ∉
    query-header-supported-single-round-partial-table-candidates
    prefix-state
    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)
    (sqp-composition-final prefix)
  i
shows
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
  trace-openings composition-openings trace-table composition-table i
  (Some ((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
proof –
have dyn:
  checked-staged-security-with-query-prefix-dynamic-index-hit
  (staged-query-prefix-candidate-pair-query-target-from-prefix
    trace-table composition-table)
  (Some ((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
using success
unfolding
  checked-staged-security-with-query-prefix-dynamic-index-hit-def

```

```

checked-staged-query-prefix-dynamic-index-hit-def
staged-query-prefix-candidate-pair-query-target-from-prefix-def
by simp
have side:
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
  trace-openings composition-openings trace-table composition-table i
  (Some (((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
unfolding
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-def
checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side-def
using witness trace-candidate comp-candidate notin by simp
show ?thesis
unfolding
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-def
using dyn side by simp
qed

lemma checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-bound-from-budgets:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and trace-low: trace-table-low-degree trace-table
  and comp-low:
    composition-table-low-degree maxDegree composition-table
  and not-all:
     $\bigwedge \text{prefix prefix-state.}$ 
    Some ((prefix, prefix-state), prefix-state)  $\in$ 
      set-dist
      (execute (checked-staged-query-prefix-with-state A i)
        adversary-initial-state)  $\implies$ 
     $\neg$  all-queries-consistent trace-table composition-table
      (sqp-alphas prefix)
  shows
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
        trace-openings composition-openings trace-table composition-table i)
      adversary-initial-state  $\leq$ 
      staged-phase-relation-error size
      (staged-query-search-queries budgets i + 1) + query-error-bound
  proof -
    have event-le:
      wp-event
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        (checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
          trace-openings composition-openings trace-table composition-table i)
        adversary-initial-state  $\leq$ 
      wp-event

```

```

    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table))
    adversary-initial-state
  by (rule wp-event-mono)
    (rule
      checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-imp-dynamic-hit)
  have target-bound:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-dynamic-index-hit
      (staged-query-prefix-candidate-pair-query-target-from-prefix
        trace-table composition-table))
    adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
  by (rule
    checked-staged-security-with-query-prefix-candidate-pair-target-from-prefix-hit-bound-from-budgets
      [OF wf controlled i-bound trace-low comp-low not-all])
  show ?thesis
  by (rule order-trans[OF event-le target-bound])
qed

lemma checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witnessE-can
assumes gap:
  checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
    i A
  (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
obtains trace-openings composition-openings trace-table composition-table
  prefix prefix-state raw raw-state where
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
    trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
proof -
  from gap show ?thesis
proof (rule
  checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witnessE)
fix trace-openings composition-openings trace-table composition-table

```


prefix prefix-state raw raw-state
assume *builder*:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
assume *prefix-receive*:
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
assume *witness*:
(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)
(staged-trace-final data)
(staged-alphas data)
(staged-degree data)
(staged-composition-fri-roots data)
(staged-composition-final data)
assume *trace-candidate*:
partial-trace-table-candidate trace-table
((replicate rounds []) [i := trace-openings])
assume *comp-candidate*:
partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings])
assume *target-hit*:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix
trace-table composition-table)
(Some (((prefix, prefix-state), raw), raw-state))
assume *notin*:
(trace-table, composition-table) ∉
query-header-supported-single-round-partial-table-candidates
prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)
i
have *dyn*:
checked-staged-security-with-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix

```

    trace-table composition-table)
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
unfolding checked-staged-security-with-query-prefix-dynamic-index-hit-def
using target-hit by simp
have side:
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side
  trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-side-def
  checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-side-def
using witness trace-candidate comp-candidate notin by simp
have fixed-gap:
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
  trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-def
using dyn side by simp
show ?thesis
by (rule that[OF builder prefix-receive fixed-gap])
qed
qed

lemma checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-imp-single-round-header-target-or-fix
assumes pair-hit:
  checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
  trace-table composition-table
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
and witness:
  (trace-openings, composition-openings) ∈
  query-header-supported-partial-opening-witnesses
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
and trace-candidate:
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [i := trace-openings])
and comp-candidate:

```

```

    partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
shows
  checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
  i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)) ∨
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
  trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
proof -
  let ?out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  have raw-hit:
    checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target-from-prefix
      trace-table composition-table)
    (Some (((prefix, prefix-state), raw), raw-state))
  using pair-hit
  unfolding
    checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-def
    checked-staged-security-with-query-prefix-dynamic-index-hit-def
  by simp
show ?thesis
proof (cases
  (trace-table, composition-table) ∈
  query-header-supported-single-round-partial-table-candidates
  prefix-state
  (sqp-trace-root prefix)
  (sqp-trace-fri-roots prefix)
  (sqp-trace-final prefix)
  (sqp-alphas prefix)
  (sqp-degree prefix)
  (sqp-composition-fri-roots prefix)
  (sqp-composition-final prefix)
  i)
case True
have target:
  checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-single-round-header-supported-query-target i)
  (Some (((prefix, prefix-state), raw), raw-state))
by (rule
  checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit
  [OF True raw-hit])
have
  checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
  i ?out

```

```

unfolding
  checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at-def
using target by simp
then show ?thesis by simp
next
case False
have success:
  index (to-nat raw) ∈
    query-sampling-success-space trace-table composition-table
    (sqp-alphas prefix)
using raw-hit
unfolding
  checked-staged-query-prefix-dynamic-index-hit-def
  staged-query-prefix-candidate-pair-query-target-from-prefix-def
by simp
have
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
    trace-openings composition-openings trace-table composition-table i
    ?out
by (rule
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-atI
  [OF success witness trace-candidate comp-candidate False])
then show ?thesis by simp
qed
qed

```

lemma *checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-single-round-header-target-or-fixed-w*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *i-bound: i < rounds*

and *support:*

Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)

and *hit:*

checked-staged-security-with-query-prefix-authenticated-opening-hit
trace-openings composition-openings i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

and *witness:*

(trace-openings, composition-openings) ∈
query-header-supported-partial-opening-witnesses
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
(staged-trace-root data)
(staged-trace-fri-roots data)

```

    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and trace-candidate:
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  and comp-candidate:
    partial-composition-table-candidate composition-table
    ((replicate rounds []) [i := composition-openings])
  and trace-low: trace-table-low-degree trace-table
  and comp-low: composition-table-low-degree maxDegree composition-table
  and not-all:
    ¬ all-queries-consistent trace-table composition-table
    (sqp-alphas prefix)
shows
  checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
  i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)) ∨
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
  trace-openings composition-openings trace-table composition-table i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)) ∨
  checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
  i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)) ∨
  checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
  i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
proof —
  let ?out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  have split:
    checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
    trace-table composition-table ?out ∨
    checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
    i ?out ∨
    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
    i ?out
  by (rule
    checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-candidate-pair-or-structured-path-on-sup
    [OF wf controlled i-bound support hit trace-candidate
      comp-candidate trace-low comp-low not-all])
  then show ?thesis

```

```

proof
  assume pair:
    checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit
    trace-table composition-table ?out
  from
    checked-staged-security-with-query-prefix-candidate-pair-from-prefix-hit-imp-single-round-header-target-or-fixed-witness
    [OF pair witness trace-candidate comp-candidate]
  show ?thesis
    by blast
  next
    assume
      checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
      i ?out  $\vee$ 
      checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
      i ?out
    then show ?thesis by blast
  qed
qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-single-round-target-fixed-witness
  fixes H G T C :: prob
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and witness:
     $\bigwedge \text{prefix prefix-state raw raw-state data attacker-state result}$ 
    final-state.
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)  $\in$ 
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state
        A i)
      adversary-initial-state)  $\implies$ 
    (trace-openings, composition-openings)  $\in$ 
    query-header-supported-partial-opening-witnesses
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  and trace-candidate:
    partial-trace-table-candidate trace-table
    ((replicate rounds []) [i := trace-openings])
  and comp-candidate:

```

partial-composition-table-candidate composition-table
 ((*replicate rounds* []) [*i* := *composition-openings*])
and *trace-low: trace-table-low-degree trace-table*
and *comp-low: composition-table-low-degree maxDegree composition-table*
and *not-all:*
 $\bigwedge$ *prefix prefix-state raw raw-state data attacker-state result*
 final-state.
 Some ((((*prefix, prefix-state*), *raw*), *raw-state*),
 ((*data, attacker-state*), *result*)), *final-state*) $\in$
 set-dist
 (*execute*
 (*checked-staged-security-experiment-with-query-prefix-data-state*
 A i)
 adversary-initial-state) $\implies$
 $\neg$ *all-queries-consistent trace-table composition-table*
 (*sqp-alphas prefix*)
and *target-bound:*
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at*
 i)
 adversary-initial-state $\leq H$
and *fixed-gap-bound:*
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at*
 trace-openings composition-openings trace-table composition-table i)
 adversary-initial-state $\leq G$
and *trace-path-bound:*
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-trace-path-output-hit-at*
 i)
 adversary-initial-state $\leq T$
and *composition-path-bound:*
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-composition-path-output-hit-at*
 i)
 adversary-initial-state $\leq C$
shows
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-authenticated-opening-hit*
 trace-openings composition-openings i)
 adversary-initial-state $\leq H + G + T + C$
proof –
let *?M* =
 checked-staged-security-experiment-with-query-prefix-data-state A i

```

let ?Target =
  checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
  i
let ?Gap =
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
  trace-openings composition-openings trace-table composition-table i
let ?Trace =
  checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
  i
let ?Comp =
  checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
  i
have event-le:
  wp-event ?M
  (checked-staged-security-with-query-prefix-authenticated-opening-hit
   trace-openings composition-openings i)
  adversary-initial-state ≤
  wp-event ?M
  (λout. ?Target out ∨ ?Gap out ∨ ?Trace out ∨ ?Comp out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈ set-dist (execute ?M adversary-initial-state)
and hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
  trace-openings composition-openings i out
show ?Target out ∨ ?Gap out ∨ ?Trace out ∨ ?Comp out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
             ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
have support-some:
  Some (((((prefix, prefix-state), raw), raw-state),
           ((data, attacker-state), result)), final-state) ∈
  set-dist (execute ?M adversary-initial-state)
  using support out-eq by simp
show ?thesis

```



```

unfolding out-eq
by (rule
  checked-staged-security-with-query-prefix-authenticated-opening-hit-imp-single-round-header-target-or-fixed-wit
  [ OF wf controlled i-bound support-some - witness[OF support-some]
    trace-candidate comp-candidate trace-low comp-low
    not-all[OF support-some] ])
  (use hit out-eq in simp)
qed
qed
have union-bound:
  wp-event ?M
  ( $\lambda out. ?Target\ out \vee ?Gap\ out \vee ?Trace\ out \vee ?Comp\ out$ )
  adversary-initial-state  $\leq$ 
  wp-event ?M ?Target adversary-initial-state +
  wp-event ?M ?Gap adversary-initial-state +
  wp-event ?M ?Trace adversary-initial-state +
  wp-event ?M ?Comp adversary-initial-state
by (rule wp-event-union-bound4)
have component-bound:
  wp-event ?M ?Target adversary-initial-state +
  wp-event ?M ?Gap adversary-initial-state +
  wp-event ?M ?Trace adversary-initial-state +
  wp-event ?M ?Comp adversary-initial-state  $\leq$ 
  H + G + T + C
by (intro add-mono target-bound fixed-gap-bound trace-path-bound
  composition-path-bound)
show ?thesis
by (rule order-trans[OF event-le order-trans[OF union-bound
  component-bound]])
qed

lemma checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-single-round-target-budgets-an
fixes H T C :: prob
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and i-bound: i < rounds
and witness:
   $\bigwedge prefix\ prefix-state\ raw\ raw-state\ data\ attacker-state\ result$ 
  final-state.
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)  $\in$ 
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state
      A i)
    adversary-initial-state)  $\implies$ 
  (trace-openings, composition-openings)  $\in$ 
  query-header-supported-partial-opening-witnesses
  (verifier-state-from-adversary attacker-state

```

(staged-proof-transcript data))
 (staged-trace-root data)
 (staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (staged-composition-final data)
and trace-candidate:
 partial-trace-table-candidate trace-table
 ((replicate rounds []) [i := trace-openings])
and comp-candidate:
 partial-composition-table-candidate composition-table
 ((replicate rounds []) [i := composition-openings])
and trace-low: trace-table-low-degree trace-table
and comp-low: composition-table-low-degree maxDegree composition-table
and not-all:
 $\bigwedge$ prefix prefix-state.
 Some ((prefix, prefix-state), prefix-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-with-state A i)
 adversary-initial-state) $\implies$
 $\neg$ all-queries-consistent trace-table composition-table
 (sqp-alphas prefix)
and target-bound:
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
 i)
 adversary-initial-state $\leq H$
and trace-path-bound:
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
 i)
 adversary-initial-state $\leq T$
and composition-path-bound:
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
 i)
 adversary-initial-state $\leq C$
shows
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-authenticated-opening-hit
 trace-openings composition-openings i)
 adversary-initial-state $\leq$
 H +

(staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) + query-error-bound) +
 $T + C$
proof –
have *fixed-gap-bound*:
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state $A\ i$)
 (checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
 trace-openings composition-openings trace-table composition-table i)
adversary-initial-state $\leq$
 staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) + query-error-bound
by (rule
 checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at-bound-from-budgets
 [OF wf controlled i -bound trace-low comp-low])
 (use not-all in blast)
have *not-all-on-data-support*:
 $\bigwedge$ prefix prefix-state raw raw-state data attacker-state result
 final-state.
 Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\in$
 set-dist
 (execute
 (checked-staged-security-experiment-with-query-prefix-data-state
 $A\ i$)
 adversary-initial-state) $\implies$
 $\neg$ all-queries-consistent trace-table composition-table
 (sqp-alphas prefix)
proof –
fix prefix prefix-state raw raw-state data attacker-state result
 final-state
assume *support*:
 Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\in$
 set-dist
 (execute
 (checked-staged-security-experiment-with-query-prefix-data-state
 $A\ i$)
 adversary-initial-state)
have *receive-support*:
 Some (((prefix, prefix-state), raw), raw-state) $\in$
 set-dist
 (execute (checked-staged-query-prefix-receive-with-state $A\ i$)
 adversary-initial-state)
using *support*
unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
by (auto elim!: set-dist-bindE split: prod.splits)
have *prefix-support*:
 Some ((prefix, prefix-state), prefix-state) $\in$

```

    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state)
  by (rule checked-staged-query-prefix-receive-with-state-prefix-support
    [OF receive-support])
show  $\neg$  all-queries-consistent trace-table composition-table
  (sqp-alphas prefix)
  by (rule not-all[OF prefix-support])
qed
show ?thesis
  by (rule
    checked-staged-security-with-query-prefix-authenticated-opening-hit-bound-from-single-round-target-fixed-witness-
    [OF wf controlled i-bound witness trace-candidate comp-candidate
      trace-low comp-low not-all-on-data-support target-bound fixed-gap-bound
      trace-path-bound composition-path-bound])
qed
end
end
end

```

theory *Staged-Security-Experiment-Composition-Query-Aligned-Witness-Bridge*
imports
Staged-Security-Experiment-Composition-Query-Current-Alignment
Staged-Security-Experiment-Composition-Query-Single-Round-Witness-Bridge
begin

Same-run bridge from actual-alpha support to the fixed query-prefix candidate-pair witness-gap event.

This layer avoids the broad diagnostic event: the sampled query index, prefix state, verifier output, and witness side facts all live on the same query-prefix data-state output.

context *soundness*
begin

lemma *checked-staged-security-with-query-prefix-data-state-accepted-shape-query-index:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i < \text{rounds}$*
and *support:*
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) $\in$
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)
and *shape:*
accepted-transcript-shape
(verifier-state-from-adversary attacker-state

```

      (staged-proof-transcript data))
    (Some (result, final-state)) as query-idxs
  shows index (to-nat raw) = query-idxs ! i
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  by (rule
    checked-staged-security-experiment-with-query-prefix-data-state-imp-data-state-support
    [OF i-bound support])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  obtain verifier where verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
  from checked-staged-security-with-data-state-accepted-shape-query-chunks
    [OF wf controlled data-support shape]
  obtain raw-idxs where len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and lookup-final:
    ∀ j < rounds.
      fmlookup (HashMap final-state)
        (QueryIndexChallenge j
          (state-after-query-chunks
            (staged-query-start-hash data) (staged-query-chunks data) j)) =
        Some (raw-idxs ! j)
  by blast
  from support obtain t where prefix-receive:
    Some (((prefix, prefix-state), raw), t) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
  and t-eq: t = raw-state
  and cont:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute
          (checked-staged-after-query-prefix-receive-with-verifier A i
            ((prefix, prefix-state), raw))
          raw-state)
  unfolding checked-staged-security-experiment-with-query-prefix-data-state-def
  by (auto elim!: set-dist-bindE split: prod.splits)
  have data-out:

```

Some (*data*, *attacker-state*) $\in$
set-dist
(execute
(checked-staged-after-query-prefix-receive *A i*
((prefix, prefix-state), raw))
raw-state)
by (*rule checked-staged-after-query-prefix-receive-with-verifier-outcome*(1)
[OF cont])
have *prefix-support*:
Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
set-dist
(execute (checked-staged-query-prefix-with-state *A i*)
adversary-initial-state)
by (*rule checked-staged-query-prefix-receive-with-state-prefix-support*)
(use prefix-receive t-eq in simp)
have *recv*:
Some (*raw*, *raw-state*) $\in$
set-dist (execute receive-query-index-challenge prefix-state)
by (*rule checked-staged-query-prefix-receive-with-state-receive-support*)
(use prefix-receive t-eq in simp)
from *checked-staged-query-prefix-with-state-alignment*
[OF wf controlled i-bound prefix-support]
have *prefix-counter*: *PQueryCounter prefix-state = i*
and *prefix-state-hash*:
PState prefix-state =
state-after-query-chunks
(staged-query-prefix-start-hash prefix)
(sqp-query-chunks prefix) i
and *prefix-len*: *length (sqp-query-chunks prefix) = i*
by *blast+*
from *data-out* **obtain** *chunk chunk-state record-state suffix-chunks* **where**
chunk-out:
Some (*chunk*, *chunk-state*) $\in$
set-dist (execute (query-opening-stage *A i raw) raw-state)*
and *record-out*:
Some (*()*, *record-state*) $\in$
set-dist (execute (record-staged-messages *chunk) chunk-state)*
and *suffix-out*:
Some (*suffix-chunks*, *attacker-state*) $\in$
set-dist
(execute
(checked-staged-query-program *A*
(sqp-trace-fri-roots prefix)
(sqp-composition-fri-roots prefix)
(Suc i) (rounds - Suc i))
record-state)
and *data-eq*:
data =
(!staged-trace-root = sqp-trace-root prefix,

```

    staged-trace-fri-roots = sqp-trace-fri-roots prefix,
    staged-trace-fri-challenges = sqp-trace-fri-challenges prefix,
    staged-trace-final = sqp-trace-final prefix,
    staged-alphas = sqp-alphas prefix,
    staged-degree = sqp-degree prefix,
    staged-composition-fri-roots = sqp-composition-fri-roots prefix,
    staged-composition-fri-challenges =
      sqp-composition-fri-challenges prefix,
    staged-composition-final = sqp-composition-final prefix,
    staged-query-chunks =
      sqp-query-chunks prefix @ chunk # suffix-chunks)
unfolding checked-staged-after-query-prefix-receive-def Let-def
by (auto elim!: set-dist-bindE simp: assert-def throw-no-outcome
    split: if-splits)
have match: staged-query-prefix-matches-data i prefix data
unfolding staged-query-prefix-matches-data-def data-eq
using prefix-len by simp
have start-hash:
  staged-query-start-hash data = staged-query-prefix-start-hash prefix
by (rule staged-query-start-hash-of-prefix-completion[OF match])
have prefix-key-state:
  PState prefix-state =
    state-after-query-chunks
    (staged-query-start-hash data) (staged-query-chunks data) i
using prefix-state-hash prefix-len start-hash
unfolding data-eq
by (simp add: state-after-query-chunks-append-prefix)
have receive-lookup:
  fmlookup (HashMap raw-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
using receive-query-index-challenge-outcome[OF recv]
    prefix-counter prefix-key-state
by simp
have raw-ext-attacker: raw-state ≤ attacker-state
by (rule checked-staged-after-query-prefix-receive-extends
  [OF wf controlled i-bound data-out])
have attacker-ext-s: attacker-state ≤ ?s
proof -
  have attacker-state ≤ attacker-state
  unfolding less-eq-hash-ext-def less-eq-fmap-def by simp
  then show ?thesis
  by (rule hash-extends-verifier-state-from-adversary-right)
qed
have s-ext-final: ?s ≤ final-state
by (rule verify-monad-hash-extends[OF verifier])
have raw-ext-final: raw-state ≤ final-state

```

```

    by (rule hash-ext-trans[OF raw-ext-attacker])
      (rule hash-ext-trans[OF attacker-ext-s s-ext-final])
  have lookup-final-raw:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
      Some raw
    by (rule hash-extension-lookup[OF receive-lookup raw-ext-final])
  have raw-eq: raw-idxs ! i = raw
    using lookup-final[rule-format, OF i-bound] lookup-final-raw by simp
  show ?thesis
    using query-idxs-eq len-raw i-bound raw-eq by simp
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-aligned-transcript-query-bad-imp-header-authenticated-candidate*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

*Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state) ∈
 set-dist
 (execute
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 adversary-initial-state)*

and *bad:*

*checked-staged-security-with-actual-alpha-prefix-verifier-event
 query-bad-with-aligned-transcript-partial-candidates
 (Some (((((alpha-prefix, alpha-prefix-state), data),
 attacker-state), result), final-state))*

shows

*checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
 A
 (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state))*

proof —

let *?s =*

*verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)*

have *data-support:*

*Some (((data, attacker-state), result), final-state) ∈
 set-dist
 (execute (checked-staged-security-experiment-with-data-state A)
 adversary-initial-state)*

by (rule

checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
 [OF support])

from *checked-staged-security-experiment-with-data-state-outcomeE*


```

    [OF data-support]
obtain builder verifier where builder:
    Some (data, attacker-state) ∈
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
  [OF wf controlled builder])
from bad obtain fr f-fri-roots f-final as dg composition-fri-roots
  final query-idxs trace-openings composition-openings trace-table
  composition-table where partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent
    ?s (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  query-bad-with-aligned-transcript-partial-candidates-def
by auto
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
by (rule
  accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
  [OF partial])
have shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
    query-idxs
by (rule
  accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape
  [OF partial])
from checked-staged-security-with-data-state-accepted-shape-query-chunks
  [OF wf controlled data-support shape]
obtain raw-idxs where as-eq: as = staged-alphas data
and len-raw: length raw-idxs = rounds

```

```

and query-idxs-eq:
  query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and lookup:
   $\forall i < \text{rounds}$ .
    fmllookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
by blast
have aligned0:
  accepted-with-partial-initial-openings-aligned ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
by (rule accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
  [OF aligned])
from accepted-with-partial-initial-openings-aligned-shapes[OF aligned0]
obtain rest where header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
by blast
have comp-nonempty: composition-fri-roots  $\neq$  []
using accepted-with-partial-initial-openings-aligned-shapes(1)
  [OF aligned0] .
have trace-partial:
  accepted-with-partial-trace-openings ?s
    (Some (result, final-state)) fr query-idxs trace-openings
using accepted-with-partial-initial-openings-aligned-shapes(3)
  [OF aligned0] .
have comp-partial:
  accepted-with-partial-composition-openings ?s
    (Some (result, final-state)) (hd composition-fri-roots) query-idxs
    composition-openings
using accepted-with-partial-initial-openings-aligned-shapes(4)
  [OF aligned0] .
have header-eq:
  fr = staged-trace-root data  $\wedge$ 
  f-fri-roots = staged-trace-fri-roots data  $\wedge$ 
  f-final = staged-trace-final data  $\wedge$ 
  as = staged-alphas data  $\wedge$ 
  dg = staged-degree data  $\wedge$ 
  composition-fri-roots = staged-composition-fri-roots data  $\wedge$ 
  final = staged-composition-final data  $\wedge$ 
  rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp
let ?i = 0

```

```

have i-bound: ?i < rounds
  using rounds-positive by simp
have round-consistent:
  partial-query-round-consistent trace-openings composition-openings as
    ?i (query-idxs ! ?i)
  using aligned i-bound
  unfolding accepted-with-partial-initial-openings-aligned-consistent-def
  by simp
have opening-consistent:
  partial-query-openings-consistent (trace-openings ! ?i)
    (composition-openings ! ?i) (staged-alphas data)
    (index (to-nat (raw-idxs ! ?i)))
  using round-consistent query-idxs-eq len-raw i-bound header-eq
  unfolding partial-query-round-consistent-def
    partial-query-openings-consistent-def
  by simp
have trace-table-auth:
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! ?i) final-state
  using trace-partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
have comp-table-auth:
  partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) (composition-openings ! ?i) final-state
  using comp-partial i-bound
  unfolding accepted-with-partial-composition-openings-def by blast
have witness-header:
  (trace-openings ! ?i, composition-openings ! ?i) ∈
    query-header-supported-partial-opening-witnesses ?s fr f-fri-roots
    f-final as dg composition-fri-roots final
  unfolding query-header-supported-partial-opening-witnesses-def
  using comp-nonempty verifier trace-table-auth comp-table-auth header
  by blast
have witness:
  (trace-openings ! ?i, composition-openings ! ?i) ∈
    query-header-supported-partial-opening-witnesses ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
  using witness-header header-eq by simp
have lookup0:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge ?i
      (state-after-query-chunks
        (staged-query-start-hash data))

```

```

    (staged-query-chunks data) ?i)) =
    Some (raw-idxs ! ?i)
using lookup i-bound by simp
from checked-staged-transcript-program-query-prefix-receive-final-lookup-support-with-alphas
    [OF wf controlled i-bound builder verifier lookup0]
obtain prefix prefix-state raw-state where prefix-receive:
    Some (((prefix, prefix-state), raw-idxs ! ?i), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A ?i)
    adversary-initial-state)
and alphas-eq: sqp-alphas prefix = staged-alphas data
by blast
have component:
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefix A
    ((replicate rounds []) [?i := trace-openings ! ?i])
    ((replicate rounds []) [?i := composition-openings ! ?i])
    ?i
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state))
by (rule
    checked-staged-security-with-actual-alpha-prefix-candidate-opening-query-hit-from-prefixI-from-prefix-receive
    [OF i-bound builder prefix-receive alphas-eq opening-consistent])
show ?thesis
by (rule
    checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefixI
    [OF witness component])
qed

lemma checked-staged-security-with-actual-alpha-prefix-aligned-transcript-query-bad-imp-single-round-header-target-c
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
    Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state) ∈
    set-dist
    (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
    adversary-initial-state)
and bad:
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    query-bad-with-aligned-transcript-partial-candidates
    (Some (((((alpha-prefix, alpha-prefix-state), data),
    attacker-state), result), final-state))
shows
    checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
    0 A
    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state)) ∨

```

```

checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
0 A
(Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state))
proof –
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
  set-dist
  (execute (checked-staged-security-experiment-with-data-state A)
  adversary-initial-state)
by (rule
  checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
  [OF support])
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF data-support]
obtain builder verifier where builder:
  Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)
  adversary-initial-state)
and verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
have staged-header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
  [OF wf controlled builder])
from bad obtain fr f-fri-roots f-final as dg composition-fri-roots
  final query-idxs trace-openings composition-openings trace-table
  composition-table where partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent
  ?s (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
  partial-composition-table-candidate composition-table

```

```

    composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-low:
    composition-table-low-degree maxDegree composition-table
and not-all:
     $\neg$  all-queries-consistent trace-table composition-table as
unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    query-bad-with-aligned-transcript-partial-candidates-def
by auto
have aligned:
    accepted-with-partial-initial-openings-aligned-consistent ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
have shape:
    accepted-transcript-shape ?s (Some (result, final-state)) as
    query-idxs
by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape
    [OF partial])
from checked-staged-security-with-data-state-accepted-shape-query-chunks
    [OF wf controlled data-support shape]
obtain raw-idxs where as-eq: as = staged-alphas data
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
    query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and lookup:
     $\forall i < \text{rounds}.$ 
    fmlookup (HashMap final-state)
    (QueryIndexChallenge i
    (state-after-query-chunks
    (staged-query-start-hash data)
    (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
by blast
have aligned0:
    accepted-with-partial-initial-openings-aligned ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
by (rule accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
    [OF aligned])
from accepted-with-partial-initial-openings-aligned-shapes[OF aligned0]
obtain rest where header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest

```

```

by blast
have comp-nonempty: composition-fri-roots  $\neq$  []
  using accepted-with-partial-initial-openings-aligned-shapes(1)
  [OF aligned0] .
have trace-partial:
  accepted-with-partial-trace-openings ?s
    (Some (result, final-state)) fr query-idxs trace-openings
  using accepted-with-partial-initial-openings-aligned-shapes(3)
  [OF aligned0] .
have comp-partial:
  accepted-with-partial-composition-openings ?s
    (Some (result, final-state)) (hd composition-fri-roots) query-idxs
    composition-openings
  using accepted-with-partial-initial-openings-aligned-shapes(4)
  [OF aligned0] .
have header-eq:
  fr = staged-trace-root data  $\wedge$ 
  f-fri-roots = staged-trace-fri-roots data  $\wedge$ 
  f-final = staged-trace-final data  $\wedge$ 
  as = staged-alphas data  $\wedge$ 
  dg = staged-degree data  $\wedge$ 
  composition-fri-roots = staged-composition-fri-roots data  $\wedge$ 
  final = staged-composition-final data  $\wedge$ 
  rest = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header staged-header]
  by simp
let ?i = 0
have i-bound: ?i < rounds
  using rounds-positive by simp
have round-consistent:
  partial-query-round-consistent trace-openings composition-openings as
    ?i (query-idxs ! ?i)
  using aligned i-bound
  unfolding accepted-with-partial-initial-openings-aligned-consistent-def
  by simp
have opening-consistent:
  partial-query-openings-consistent (trace-openings ! ?i)
    (composition-openings ! ?i) (staged-alphas data)
    (index (to-nat (raw-idxs ! ?i)))
  using round-consistent query-idxs-eq len-raw i-bound header-eq
  unfolding partial-query-round-consistent-def
    partial-query-openings-consistent-def
  by simp
have trace-table-auth:
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! ?i) final-state
  using trace-partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
have comp-table-auth:

```

```

partial-authenticated-table (hd composition-fri-roots)
  (scale * clength) (composition-openings ! ?i) final-state
using comp-partial i-bound
unfolding accepted-with-partial-composition-openings-def by blast
have witness-header:
  (trace-openings ! ?i, composition-openings ! ?i) ∈
    query-header-supported-partial-opening-witnesses ?s fr f-fri-roots
    f-final as dg composition-fri-roots final
unfolding query-header-supported-partial-opening-witnesses-def
using comp-nonempty verifier trace-table-auth comp-table-auth header
by blast
have witness:
  (trace-openings ! ?i, composition-openings ! ?i) ∈
    query-header-supported-partial-opening-witnesses ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
using witness-header header-eq by simp
have lookup0:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge ?i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) ?i)) =
    Some (raw-idxs ! ?i)
using lookup i-bound by simp
from checked-staged-transcript-program-query-prefix-receive-final-lookup-support-with-alphas
  [OF wf controlled i-bound builder verifier lookup0]
obtain prefix prefix-state raw-state where prefix-receive:
  Some (((prefix, prefix-state), raw-idxs ! ?i), raw-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-receive-with-state A ?i)
      adversary-initial-state)
and alphas-eq: sqp-alphas prefix = staged-alphas data
by blast
have trace-candidate-single:
  partial-trace-table-candidate trace-table
    ((replicate rounds []) [?i := trace-openings ! ?i])
by (rule partial-trace-table-candidate-single-round-from-all-rounds
    [OF trace-candidate i-bound])
have comp-candidate-single:
  partial-composition-table-candidate composition-table
    ((replicate rounds []) [?i := composition-openings ! ?i])
by (rule partial-composition-table-candidate-single-round-from-all-rounds
    [OF comp-candidate i-bound])

```


have *opening-target*:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [?i := trace-openings ! ?i])
((replicate rounds []) [?i := composition-openings ! ?i])
?i)
(Some (((prefix, prefix-state), raw-idxs ! ?i), raw-state))
by (rule *checked-staged-query-prefix-candidate-opening-target-from-prefix-single-round-hitI*
[OF i-bound])
(use opening-consistent alphas-eq in simp)
have *not-all-prefix*:
 $\neg$ *all-queries-consistent trace-table composition-table*
(sqp-alphas prefix)
using *not-all header-eq alphas-eq* **by** *simp*
have *opening-subset*:
staged-query-prefix-candidate-opening-query-target-from-prefix
((replicate rounds []) [?i := trace-openings ! ?i])
((replicate rounds []) [?i := composition-openings ! ?i])
?i prefix prefix-state $\subseteq$
staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table prefix prefix-state
by (rule
staged-query-prefix-candidate-opening-query-target-from-prefix-subset-candidate-pair-from-prefix
[OF trace-candidate-single comp-candidate-single trace-low comp-low
not-all-prefix])
have *pair-target*:
checked-staged-query-prefix-dynamic-index-hit
(staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
composition-table)
(Some (((prefix, prefix-state), raw-idxs ! ?i), raw-state))
using *opening-target opening-subset*
unfolding *checked-staged-query-prefix-dynamic-index-hit-def* **by** *auto*
have *pair-hit*:
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix
A trace-table composition-table ?i
(Some ((((((alpha-prefix, alpha-prefix-state), data), attacker-state),
result), final-state)))
by (rule
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefixI
[OF i-bound builder prefix-receive pair-target])
show *?thesis*
by (rule
checked-staged-security-with-actual-alpha-prefix-candidate-pair-query-hit-from-prefix-with-witness-imp-single-round
[OF pair-hit witness trace-candidate-single comp-candidate-single])
qed

definition *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-exists-at*
where

checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-exists-at

$i \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-openings composition-openings trace-table composition-table.}$
 $\text{checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-at}$
 $\text{trace-openings composition-openings trace-table composition-table } i \text{ out})$

definition *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-current-exists-at*
where

$\text{checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-current-exists-at}$
 $i \text{ out} \longleftrightarrow$
 $\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at}$
 $i \text{ out} \wedge$
 $\text{checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-exists-at}$
 $i \text{ out}$

definition *checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at*
where

$\text{checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at}$
 $i \text{ out} \longleftrightarrow$
 $\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at}$
 $i \text{ out} \wedge$
 $\text{checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at}$
 $i \text{ out}$

lemma *checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-current-exists-bound-from*
assumes *current-bound:*

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at}$
 $i)$
 $\text{adversary-initial-state} \leq C$

shows

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-current-exists-at}$
 $i)$
 $\text{adversary-initial-state} \leq C$

by (*rule order-trans[OF - current-bound]*,
rule wp-event-mono)

$(\text{simp add:}$
 $\text{checked-staged-security-with-query-prefix-single-round-candidate-binding-gap-witness-current-exists-at-def})$

lemma *checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-bound-from*
assumes *current-bound:*

wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at}$
 $i)$
 $\text{adversary-initial-state} \leq C$

shows

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at
 i)
 $\text{adversary-initial-state} \leq C$
by (rule order-trans[OF - current-bound],
 rule wp-event-mono)
 (simp add:
 checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at-def)

lemma *checked-staged-security-with-query-prefix-data-state-aligned-transcript-query-bad-imp-single-round-component*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\in$
set-dist
 (execute
 (checked-staged-security-experiment-with-query-prefix-data-state A 0)
 adversary-initial-state)

and *bad:*

staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates
 (Some (((data, attacker-state), result), final-state))

shows

checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at
 0
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state)) $\vee$
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at
 0
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))

proof –

let $?s =$

verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)

let $?i = 0$

have *i-bound: $?i < \text{rounds}$*

using *rounds-positive* **by** *simp*

have *data-support:*

Some (((data, attacker-state), result), final-state) $\in$
set-dist
 (execute (checked-staged-security-experiment-with-data-state A)
 adversary-initial-state)

by (rule

checked-staged-security-experiment-with-query-prefix-data-state-imp-data-state-support
 [OF *i-bound support*])

from *checked-staged-security-experiment-with-data-state-outcomeE*

```

[OF data-support]
obtain builder verifier where builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
  [OF wf controlled builder])
from bad obtain fr f-fri-roots f-final as dg composition-fri-roots
  final query-idxs trace-openings composition-openings trace-table
  composition-table where partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent
    ?s (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-low:
  composition-table-low-degree maxDegree composition-table
and not-all:
  ¬ all-queries-consistent trace-table composition-table as
unfolding staged-security-with-data-state-verifier-event-def
  query-bad-with-aligned-transcript-partial-candidates-def
by auto
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
by (rule
  accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
  [OF partial])

```

```

have shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
    query-idxs
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape
    [OF partial])
have index-eq: index (to-nat raw) = query-idxs ! ?i
  by (rule
    checked-staged-security-with-query-prefix-data-state-accepted-shape-query-index
    [OF wf controlled i-bound support shape])
have aligned0:
  accepted-with-partial-initial-openings-aligned ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
  by (rule accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
    [OF aligned])
from accepted-with-partial-initial-openings-aligned-shapes[OF aligned0]
obtain rest where header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  by blast
have comp-nonempty: composition-fri-roots ≠ []
  using accepted-with-partial-initial-openings-aligned-shapes(1)
    [OF aligned0] .
have trace-partial:
  accepted-with-partial-trace-openings ?s
    (Some (result, final-state)) fr query-idxs trace-openings
  using accepted-with-partial-initial-openings-aligned-shapes(3)
    [OF aligned0] .
have comp-partial:
  accepted-with-partial-composition-openings ?s
    (Some (result, final-state)) (hd composition-fri-roots) query-idxs
    composition-openings
  using accepted-with-partial-initial-openings-aligned-shapes(4)
    [OF aligned0] .
have header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  composition-fri-roots = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  rest = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header staged-header]
  by simp
have alphas-eq: staged-alphas data = sqp-alphas prefix
  by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq)

```

```

    (rule support)
have round-consistent:
  partial-query-round-consistent trace-openings composition-openings as
    ?i (query-idxs ! ?i)
  using aligned i-bound
  unfolding accepted-with-partial-initial-openings-aligned-consistent-def
  by simp
have opening-consistent:
  partial-query-openings-consistent (trace-openings ! ?i)
    (composition-openings ! ?i) (sqp-alphas prefix)
    (index (to-nat raw))
  using round-consistent index-eq header-eq alphas-eq
  unfolding partial-query-round-consistent-def
    partial-query-openings-consistent-def
  by simp
have trace-table-auth:
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! ?i) final-state
  using trace-partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
have comp-table-auth:
  partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) (composition-openings ! ?i) final-state
  using comp-partial i-bound
  unfolding accepted-with-partial-composition-openings-def by blast
have witness-header:
  (trace-openings ! ?i, composition-openings ! ?i) ∈
    query-header-supported-partial-opening-witnesses ?s fr f-fri-roots
    f-final as dg composition-fri-roots final
  unfolding query-header-supported-partial-opening-witnesses-def
  using comp-nonempty verifier trace-table-auth comp-table-auth header
  by blast
have witness:
  (trace-openings ! ?i, composition-openings ! ?i) ∈
    query-header-supported-partial-opening-witnesses ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
  using witness-header header-eq by simp
have trace-candidate-single:
  partial-trace-table-candidate trace-table
    ((replicate rounds []) [?i := trace-openings ! ?i])
  by (rule partial-trace-table-candidate-single-round-from-all-rounds
    [OF trace-candidate i-bound])
have comp-candidate-single:

```

```

partial-composition-table-candidate composition-table
  ((replicate rounds []) [?i := composition-openings ! ?i])
by (rule partial-composition-table-candidate-single-round-from-all-rounds
  [OF comp-candidate i-bound])
have opening-target:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-opening-query-target-from-prefix
      ((replicate rounds []) [?i := trace-openings ! ?i])
      ((replicate rounds []) [?i := composition-openings ! ?i])
      ?i)
    (Some (((prefix, prefix-state), raw), raw-state))
by (rule checked-staged-query-prefix-candidate-opening-target-from-prefix-single-round-hitI
  [OF i-bound opening-consistent])
have not-all-prefix:
  ¬ all-queries-consistent trace-table composition-table
    (sqp-alphas prefix)
  using not-all header-eq alphas-eq by simp
have opening-subset:
  staged-query-prefix-candidate-opening-query-target-from-prefix
    ((replicate rounds []) [?i := trace-openings ! ?i])
    ((replicate rounds []) [?i := composition-openings ! ?i])
    ?i prefix prefix-state ⊆
  staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
    composition-table prefix prefix-state
by (rule
  staged-query-prefix-candidate-opening-query-target-from-prefix-subset-candidate-pair-from-prefix
  [OF trace-candidate-single comp-candidate-single trace-low comp-low
    not-all-prefix])
have pair-target:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target-from-prefix trace-table
      composition-table)
    (Some (((prefix, prefix-state), raw), raw-state))
  using opening-target opening-subset
  unfolding checked-staged-query-prefix-dynamic-index-hit-def by auto
have auth-hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
    (trace-openings ! ?i) (composition-openings ! ?i) ?i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  unfolding
    checked-staged-security-with-query-prefix-authenticated-opening-hit-def
  using comp-nonempty trace-table-auth comp-table-auth opening-consistent
    header-eq alphas-eq
  by simp
show ?thesis
proof (cases
  (trace-table, composition-table) ∈
  query-header-supported-single-round-partial-table-candidates

```

```

    prefix-state
    (sqp-trace-root prefix)
    (sqp-trace-fri-roots prefix)
    (sqp-trace-final prefix)
    (sqp-alphas prefix)
    (sqp-degree prefix)
    (sqp-composition-fri-roots prefix)
    (sqp-composition-final prefix)
    ?i)
case True
have target-hit:
  checked-staged-query-prefix-dynamic-index-hit
  (staged-query-prefix-single-round-header-supported-query-target ?i)
  (Some (((prefix, prefix-state), raw), raw-state))
by (rule
  checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit
  [OF True pair-target])
have
checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at
  ?i
  (Some ((((((prefix, prefix-state), raw), raw-state),
  ((data, attacker-state), result)), final-state))
unfolding
checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at-def
checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at-def
using target-hit
  checked-staged-security-with-query-prefix-current-query-partial-opening-hit-atI
  [OF auth-hit]
by simp
then show ?thesis by simp
next
case False
have success-prefix:
  index (to-nat raw) ∈
  query-sampling-success-space trace-table composition-table
  (sqp-alphas prefix)
using pair-target
unfolding
  checked-staged-query-prefix-dynamic-index-hit-def
  staged-query-prefix-candidate-pair-query-target-from-prefix-def
by simp
have gap-witness:
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
  (trace-openings ! ?i) (composition-openings ! ?i)
  trace-table composition-table ?i
  (Some ((((((prefix, prefix-state), raw), raw-state),
  ((data, attacker-state), result)), final-state))
by (rule
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-atI

```



```

    [OF success-prefix witness trace-candidate-single
      comp-candidate-single False])
have gap-exists:
  checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at
    ?i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  by (rule
    checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-atI
      [OF gap-witness])
  then show ?thesis
    using gap-exists by simp
qed
qed

lemma checked-staged-security-with-query-prefix-data-state-aligned-transcript-query-bad-extract-round0-authenticated
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A 0)
    adversary-initial-state)
and bad:
  staged-security-with-data-state-verifier-event
  query-bad-with-aligned-transcript-partial-candidates
  (Some (((data, attacker-state), result), final-state))
obtains trace-opening composition-opening trace-table composition-table where
  checked-staged-security-with-query-prefix-authenticated-opening-hit
  trace-opening composition-opening 0
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  (trace-opening, composition-opening) ∈
  query-header-supported-partial-opening-witnesses
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)
  (staged-composition-fri-roots data)
  (staged-composition-final data)
  partial-trace-table-candidate trace-table
  ((replicate rounds []) [0 := trace-opening])
  partial-composition-table-candidate composition-table
  ((replicate rounds []) [0 := composition-opening])

```

```

    trace-table-low-degree trace-table
    composition-table-low-degree maxDegree composition-table
    ¬ all-queries-consistent trace-table composition-table
    (sqp-alphas prefix)
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  let ?i = 0
  have i-bound: ?i < rounds
    using rounds-positive by simp
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
    by (rule
      checked-staged-security-experiment-with-query-prefix-data-state-imp-data-state-support
      [OF i-bound support])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  obtain builder verifier where builder:
    Some (data, attacker-state) ∈
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
  have staged-header:
    verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
  from bad obtain fr f-fri-roots f-final as dg composition-fri-roots
    final query-idxs trace-openings composition-openings trace-table
    composition-table where partial:
    accepted-with-partial-initial-openings-aligned-transcript-consistent
    ?s (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
  and trace-candidate:

```

```

    partial-trace-table-candidate trace-table trace-openings
and comp-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-low:
    composition-table-low-degree maxDegree composition-table
and not-all:
     $\neg$  all-queries-consistent trace-table composition-table as
unfolding staged-security-with-data-state-verifier-event-def
    query-bad-with-aligned-transcript-partial-candidates-def
by auto
have aligned:
    accepted-with-partial-initial-openings-aligned-consistent ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
have aligned0:
    accepted-with-partial-initial-openings-aligned ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
by (rule accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
    [OF aligned])
from accepted-with-partial-initial-openings-aligned-shapes[OF aligned0]
obtain rest where header:
    verifier-header-transcript ?s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
by blast
have comp-nonempty: composition-fri-roots  $\neq$  []
using accepted-with-partial-initial-openings-aligned-shapes(1)
    [OF aligned0] .
have trace-partial:
    accepted-with-partial-trace-openings ?s
    (Some (result, final-state)) fr query-idxs trace-openings
using accepted-with-partial-initial-openings-aligned-shapes(3)
    [OF aligned0] .
have comp-partial:
    accepted-with-partial-composition-openings ?s
    (Some (result, final-state)) (hd composition-fri-roots) query-idxs
    composition-openings
using accepted-with-partial-initial-openings-aligned-shapes(4)
    [OF aligned0] .
have header-eq:
    fr = staged-trace-root data  $\wedge$ 
    f-fri-roots = staged-trace-fri-roots data  $\wedge$ 

```

```

f-final = staged-trace-final data ∧
as = staged-alphas data ∧
dg = staged-degree data ∧
composition-fri-roots = staged-composition-fri-roots data ∧
final = staged-composition-final data ∧
rest = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header staged-header]
by simp
have alphas-eq: staged-alphas data = sqp-alphas prefix
by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq)
    (rule support)
have shape:
    accepted-transcript-shape ?s (Some (result, final-state)) as
    query-idxs
by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape
    [OF partial])
have index-eq: index (to-nat raw) = query-idxs ! ?i
by (rule
    checked-staged-security-with-query-prefix-data-state-accepted-shape-query-index
    [OF wf controlled i-bound support shape])
have round-consistent:
    partial-query-round-consistent trace-openings composition-openings as
    ?i (query-idxs ! ?i)
using aligned i-bound
unfolding accepted-with-partial-initial-openings-aligned-consistent-def
by simp
have opening-consistent:
    partial-query-openings-consistent (trace-openings ! ?i)
    (composition-openings ! ?i) (sqp-alphas prefix)
    (index (to-nat raw))
using round-consistent index-eq header-eq alphas-eq
unfolding partial-query-round-consistent-def
    partial-query-openings-consistent-def
by simp
have trace-table-auth:
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! ?i) final-state
using trace-partial i-bound
unfolding accepted-with-partial-trace-openings-def by blast
have comp-table-auth:
    partial-authenticated-table (hd composition-fri-roots)
    (scale * clength) (composition-openings ! ?i) final-state
using comp-partial i-bound
unfolding accepted-with-partial-composition-openings-def by blast
have witness-header:
    (trace-openings ! ?i, composition-openings ! ?i) ∈
    query-header-supported-partial-opening-witnesses ?s fr f-fri-roots
    f-final as dg composition-fri-roots final

```

```

unfolding query-header-supported-partial-opening-witnesses-def
using comp-nonempty verifier trace-table-auth comp-table-auth header
by blast
have witness:
  (trace-openings ! ?i, composition-openings ! ?i) ∈
    query-header-supported-partial-opening-witnesses ?s
      (staged-trace-root data)
      (staged-trace-fri-roots data)
      (staged-trace-final data)
      (staged-alphas data)
      (staged-degree data)
      (staged-composition-fri-roots data)
      (staged-composition-final data)
using witness-header header-eq by simp
have trace-candidate-single:
  partial-trace-table-candidate trace-table
    ((replicate rounds []) [?i := trace-openings ! ?i])
by (rule partial-trace-table-candidate-single-round-from-all-rounds
    [OF trace-candidate i-bound])
have comp-candidate-single:
  partial-composition-table-candidate composition-table
    ((replicate rounds []) [?i := composition-openings ! ?i])
by (rule partial-composition-table-candidate-single-round-from-all-rounds
    [OF comp-candidate i-bound])
have auth-hit:
  checked-staged-security-with-query-prefix-authenticated-opening-hit
    (trace-openings ! ?i) (composition-openings ! ?i) ?i
    (Some ((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-authenticated-opening-hit-def
using comp-nonempty trace-table-auth comp-table-auth opening-consistent
  header-eq alphas-eq
by simp
have not-all-prefix:
  ¬ all-queries-consistent trace-table composition-table
    (sqp-alphas prefix)
using not-all header-eq alphas-eq by simp
show ?thesis
by (rule that[OF auth-hit witness trace-candidate-single
  comp-candidate-single trace-low comp-low not-all-prefix])
qed

lemma checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-query-prefix-single-round-consistent
fixes T Gap :: prob
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and target-bound:
  wp-event

```

```

    (checked-staged-security-experiment-with-query-prefix-data-state A 0)
  (checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at
    0)
  adversary-initial-state ≤ T
and gap-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A 0)
  (checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at
    0)
  adversary-initial-state ≤ Gap
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      query-bad-with-aligned-transcript-partial-candidates)
  adversary-initial-state ≤ T + Gap
proof –
  let ?i = 0
  let ?m =
    checked-staged-security-experiment-with-query-prefix-data-state A ?i
  let ?query =
    staged-security-with-data-state-verifier-event
      query-bad-with-aligned-transcript-partial-candidates
  let ?projected =
    λout. case out of
      None ⇒ ?query None
    | Some (packed, t) ⇒ ?query (Some (snd packed, t))
  let ?target =
    checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at
      ?i
  let ?gap =
    checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at
      ?i
  have i-bound: ?i < rounds
    using rounds-positive by simp
  have projection:
    wp-event (checked-staged-security-experiment-with-data-state A)
      ?query adversary-initial-state =
    wp-event ?m ?projected adversary-initial-state
  by (rule
    checked-staged-security-experiment-with-query-prefix-data-state-projection-event
    [OF i-bound])
  have event-le:
    wp-event ?m ?projected adversary-initial-state ≤
    wp-event ?m (λout. ?target out ∨ ?gap out)
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support: out ∈ set-dist (execute ?m adversary-initial-state)
  and projected: ?projected out

```

```

show ?target out  $\vee$  ?gap out
proof (cases out)
  case None
  then show ?thesis
    using projected
    unfolding staged-security-with-data-state-verifier-event-def by simp
next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state
    result final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  have bad:
    staged-security-with-data-state-verifier-event
      query-bad-with-aligned-transcript-partial-candidates
        (Some (((data, attacker-state), result), final-state))
    using projected unfolding out-eq by simp
  show ?thesis
    unfolding out-eq
    by (rule
      checked-staged-security-with-query-prefix-data-state-aligned-transcript-query-bad-imp-single-round-components
        [OF wf controlled])
      (use support bad out-eq in simp-all)
  qed
qed
have union-bound:
  wp-event ?m ( $\lambda$ out. ?target out  $\vee$  ?gap out)
    adversary-initial-state  $\leq$ 
  wp-event ?m ?target adversary-initial-state +
  wp-event ?m ?gap adversary-initial-state
  by (rule wp-event-union-bound)
have sum-bound:
  wp-event ?m ?target adversary-initial-state +
  wp-event ?m ?gap adversary-initial-state  $\leq T + \text{Gap}$ 
  by (rule add-mono[OF target-bound gap-bound])
show ?thesis
  unfolding projection
  by (rule order-trans[OF event-le order-trans[OF union-bound sum-bound]])
qed

lemma checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-query-prefix-single-round-target
  fixes T Gap :: prob
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and target-bound:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A 0)

```

(*checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at*
0)
adversary-initial-state $\leq T$
and *gap-bound*:
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A 0*)
 (*checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at*
0)
adversary-initial-state $\leq \text{Gap}$
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq T + \text{Gap}$
by (*rule*
checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-query-prefix-single-round-compon
[OF wf controlled target-bound gap-bound])

lemma *checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-query-prefix-single-round-tar*
fixes *T P Q :: prob*
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *target-bound*:
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A 0*)
 (*checked-staged-security-with-query-prefix-single-round-header-supported-query-target-current-hit-at*
0)
adversary-initial-state $\leq T$
and *data-pre-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$
and *new-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq Q$
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq T + (P + Q)$
proof –
have *i-bound*: *0 < rounds*
using *rounds-positive* **by** *simp*
have *gap-bound*:
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A 0*)


```

    (checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at
      0)
    adversary-initial-state  $\leq P + Q$ 
  by (rule
    checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-exists-at-bound-from-data-pre-
      [OF i-bound data-pre-bound new-bound])
  show ?thesis
  by (rule
    checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-query-prefix-single-round-target-
      [OF wf controlled target-bound gap-bound])
qed

lemma checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-header-authenticated-candida
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and header-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
    (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
      A)
    adversary-initial-state  $\leq Q$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state  $\leq Q$ 
proof -
  let ?project =
     $\lambda x. ((snd (fst (fst x)), snd (fst x)), snd x)$ 
  let ?query =
    staged-security-with-data-state-verifier-event
      query-bad-with-aligned-transcript-partial-candidates
  let ?projected =
     $\lambda out. case\ out\ of$ 
      None  $\Rightarrow$  False
    | Some (packed, t)  $\Rightarrow$  ?query (Some (?project packed, t))
  have projection:
    wp-event (checked-staged-security-experiment-with-data-state A)
    ?query adversary-initial-state =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state
proof -
  have
    wp-event (checked-staged-security-experiment-with-data-state A)
    ?query adversary-initial-state =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

```

```

    >>= (λx. return (?project x)))
  ?query adversary-initial-state
using
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
  [of A]
by simp
also have ... =
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  ?projected adversary-initial-state
by (subst wp-event-bind-return-map)
  (simp add: staged-security-with-data-state-verifier-event-def)
finally show ?thesis
  by simp
qed
have event-le:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  ?projected adversary-initial-state ≤
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
  A)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    adversary-initial-state)
and projected: ?projected out
show
  checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
  A out
proof (cases out)
case None
then show ?thesis
  using projected by simp
next
case (Some packed)
then obtain alpha-prefix alpha-prefix-state data attacker-state
  result final-state where out-eq:
  out =
  Some (((((alpha-prefix, alpha-prefix-state), data),
    attacker-state), result), final-state)
by (cases packed, auto split: prod.splits)

```

```

have bad:
  checked-staged-security-with-actual-alpha-prefix-verifier-event
    query-bad-with-aligned-transcript-partial-candidates out
  using projected unfolding out-eq
  by (simp add:
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    staged-security-with-data-state-verifier-event-def Let-def)
show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-aligned-transcript-query-bad-imp-header-authenticated-candidates
    [OF wf controlled])
    (use support bad out-eq in simp-all)
qed
qed
show ?thesis
  unfolding projection
  by (rule order-trans[OF event-le header-bound])
qed

lemma checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-single-round-components:
fixes T Gap :: prob
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and target-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
      0 A)
    adversary-initial-state ≤ T
and gap-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
      0 A)
    adversary-initial-state ≤ Gap
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state ≤ T + Gap
proof –
  let ?project =
     $\lambda x. ((snd (fst (fst x)), snd (fst x)), snd x)$ 
  let ?query =
    staged-security-with-data-state-verifier-event
      query-bad-with-aligned-transcript-partial-candidates

```

```

let ?projected =
  λout. case out of
    None ⇒ False
  | Some (packed, t) ⇒ ?query (Some (?project packed, t))
let ?target =
  checked-staged-security-with-actual-alpha-prefix-single-round-header-supported-query-target-hit-at
    0 A
let ?gap =
  checked-staged-security-with-actual-alpha-prefix-query-prefix-single-round-candidate-binding-gap-witness-at
    0 A
have projection:
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?query adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state
proof –
  have
    wp-event (checked-staged-security-experiment-with-data-state A)
      ?query adversary-initial-state =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
        ≫ (λx. return (?project x)))
      ?query adversary-initial-state
    using
      checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
      [of A]
    by simp
  also have ... =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      ?projected adversary-initial-state
    by (subst wp-event-bind-return-map)
      (simp add: staged-security-with-data-state-verifier-event-def)
  finally show ?thesis
    by simp
qed
have event-le:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (λout. ?target out ∨ ?gap out)
    adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈

```

```

    set-dist
    (execute
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
        A)
      adversary-initial-state)
  and projected: ?projected out
show ?target out  $\vee$  ?gap out
proof (cases out)
  case None
  then show ?thesis
    using projected by simp
next
  case (Some packed)
  then obtain alpha-prefix alpha-prefix-state data attacker-state
    result final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data),
        attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  have bad:
    checked-staged-security-with-actual-alpha-prefix-verifier-event
      query-bad-with-aligned-transcript-partial-candidates out
    using projected unfolding out-eq
    by (simp add:
      checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      staged-security-with-data-state-verifier-event-def Let-def)
  show ?thesis
    unfolding out-eq
    by (rule
      checked-staged-security-with-actual-alpha-prefix-aligned-transcript-query-bad-imp-single-round-header-target-on
      [OF wf controlled])
    (use support bad out-eq in simp-all)
qed
qed
have union-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ( $\lambda$ out. ?target out  $\vee$  ?gap out)
    adversary-initial-state  $\leq$ 
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?target adversary-initial-state +
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?gap adversary-initial-state
  by (rule wp-event-union-bound)
have sum-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

```

$?target \text{ adversary-initial-state} +$
 $wp\text{-event}$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state } A)$
 $?gap \text{ adversary-initial-state} \leq T + Gap$
by (rule add-mono[$OF \text{ target-bound gap-bound}$])
show $?thesis$
unfolding $projection$
by (rule order-trans[$OF \text{ event-le order-trans}[OF \text{ union-bound sum-bound}]$])
qed

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-pair-success-witness-gapE*:

assumes $i\text{-bound}: i < rounds$

and *support*:

$Some \ (((((\alpha\text{-prefix}, \alpha\text{-prefix}\text{-state}), \text{data}), \text{attacker}\text{-state}),$
 $\text{result}), \text{final}\text{-state}) \in$
 $set\text{-dist}$
 $(execute$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-actual}\text{-alpha}\text{-prefix}\text{-data}\text{-state}$
 $A)$
 $\text{adversary}\text{-initial}\text{-state})$

and *success*:

$\bigwedge \text{prefix prefix}\text{-state raw raw}\text{-state}.$
 $Some \ (((((\text{prefix}, \text{prefix}\text{-state}), \text{raw}), \text{raw}\text{-state}),$
 $((\text{data}, \text{attacker}\text{-state}), \text{result})), \text{final}\text{-state}) \in$
 $set\text{-dist}$
 $(execute$
 $(checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-query}\text{-prefix}\text{-data}\text{-state}$
 $A \ i)$
 $\text{adversary}\text{-initial}\text{-state}) \implies$
 $sqp\text{-alphas prefix} = staged\text{-alphas data} \implies$
 $index \ (to\text{-nat raw}) \in$
 $query\text{-sampling}\text{-success}\text{-space trace}\text{-table composition}\text{-table}$
 $(staged\text{-alphas data})$

and *witness*:

$(\text{trace}\text{-openings}, \text{composition}\text{-openings}) \in$
 $query\text{-header}\text{-supported}\text{-partial}\text{-opening}\text{-witnesses}$
 $(\text{verifier}\text{-state}\text{-from}\text{-adversary } \text{attacker}\text{-state}$
 $(\text{staged}\text{-proof}\text{-transcript } \text{data}))$
 $(\text{staged}\text{-trace}\text{-root } \text{data})$
 $(\text{staged}\text{-trace}\text{-fri}\text{-roots } \text{data})$
 $(\text{staged}\text{-trace}\text{-final } \text{data})$
 $(\text{staged}\text{-alphas } \text{data})$
 $(\text{staged}\text{-degree } \text{data})$
 $(\text{staged}\text{-composition}\text{-fri}\text{-roots } \text{data})$
 $(\text{staged}\text{-composition}\text{-final } \text{data})$

and *trace-candidate*:

$partial\text{-trace}\text{-table}\text{-candidate trace}\text{-table}$
 $((\text{replicate } rounds \ \square) \ [i := \text{trace}\text{-openings}])$

and *comp-candidate*:

partial-composition-table-candidate composition-table
((replicate rounds []) [i := composition-openings])
and *notin:*
 $\bigwedge \text{prefix prefix-state raw raw-state.}$
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state
A i)
adversary-initial-state) ⇒
sqp-alphas prefix = staged-alphas data ⇒
(trace-table, composition-table) ∉
query-header-supported-single-round-partial-table-candidates
prefix-state
(sqp-trace-root prefix)
(sqp-trace-fri-roots prefix)
(sqp-trace-final prefix)
(sqp-alphas prefix)
(sqp-degree prefix)
(sqp-composition-fri-roots prefix)
(sqp-composition-final prefix)
i
obtains *prefix prefix-state raw raw-state where*
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)
checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
trace-openings composition-openings trace-table composition-table i
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))
proof –
from *checked-staged-security-with-actual-alpha-prefix-support-query-prefix-data-state-alphasE*
[OF i-bound support]
obtain *prefix prefix-state raw raw-state where* *qsupport:*
Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)
and *alphas-eq:* *sqp-alphas prefix = staged-alphas data*
by *blast*
have *success-prefix:*
index (to-nat raw) ∈
query-sampling-success-space trace-table composition-table

```

      (sqp-alphas prefix)
    using success[OF qsupport alphas-eq] alphas-eq by simp
  have notin-prefix:
    (trace-table, composition-table)  $\notin$ 
      query-header-supported-single-round-partial-table-candidates
        prefix-state
        (sqp-trace-root prefix)
        (sqp-trace-fri-roots prefix)
        (sqp-trace-final prefix)
        (sqp-alphas prefix)
        (sqp-degree prefix)
        (sqp-composition-fri-roots prefix)
        (sqp-composition-final prefix)
    i
  by (rule notin[OF qsupport alphas-eq])
  have fixed-gap:
    checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-at
      trace-openings composition-openings trace-table composition-table i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  by (rule checked-staged-security-with-query-prefix-single-round-candidate-pair-witness-gap-atI
    [OF success-prefix witness trace-candidate comp-candidate notin-prefix])
  show ?thesis
  by (rule that[OF qsupport fixed-gap])
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-candidate-pair-success-header-targetE:*

assumes *i-bound: $i < \text{rounds}$*

and *support:*

*Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
 result), final-state) $\in$
 set-dist
 (execute
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 adversary-initial-state)*

and *success:*

$\bigwedge$ *prefix prefix-state raw raw-state.
 Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\in$
 set-dist
 (execute
 (checked-staged-security-experiment-with-query-prefix-data-state
 A i)
 adversary-initial-state) $\implies$
 sqp-alphas prefix = staged-alphas data $\implies$
 index (to-nat raw) $\in$
 query-sampling-success-space trace-table composition-table
 (staged-alphas data)*

and *supported*:

$\bigwedge \text{prefix prefix-state raw raw-state.}$
Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\in$
set-dist
 (execute
 (changed-staged-security-experiment-with-query-prefix-data-state
 A i)
 adversary-initial-state) $\implies$
sqp-alphas prefix = staged-alphas data $\implies$
 (trace-table, composition-table) $\in$
query-header-supported-single-round-partial-table-candidates
prefix-state
 (sqp-trace-root prefix)
 (sqp-trace-fri-roots prefix)
 (sqp-trace-final prefix)
 (sqp-alphas prefix)
 (sqp-degree prefix)
 (sqp-composition-fri-roots prefix)
 (sqp-composition-final prefix)
i

obtains *prefix prefix-state raw raw-state where*
Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\in$
set-dist
 (execute
 (changed-staged-security-experiment-with-query-prefix-data-state *A i*)
 adversary-initial-state)
checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
i
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))

proof –

from *checked-staged-security-with-actual-alpha-prefix-support-query-prefix-data-state-alphasE*
 [*OF i-bound support*]

obtain *prefix prefix-state raw raw-state where qsupport*:
Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\in$
set-dist
 (execute
 (changed-staged-security-experiment-with-query-prefix-data-state *A i*)
 adversary-initial-state)

and *alphas-eq: sqp-alphas prefix = staged-alphas data*
by *blast*

have *success-prefix*:
index (to-nat raw) $\in$
query-sampling-success-space trace-table composition-table
 (sqp-alphas prefix)

using *success[OF qsupport alphas-eq] alphas-eq by simp*

```

have pair-target-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-candidate-pair-query-target-from-prefix
      trace-table composition-table)
    (Some (((prefix, prefix-state), raw), raw-state))
using success-prefix
unfolding
  checked-staged-query-prefix-dynamic-index-hit-def
  staged-query-prefix-candidate-pair-query-target-from-prefix-def
by simp
have supported-prefix:
  (trace-table, composition-table) ∈
    query-header-supported-single-round-partial-table-candidates
      prefix-state
      (sqp-trace-root prefix)
      (sqp-trace-fri-roots prefix)
      (sqp-trace-final prefix)
      (sqp-alphas prefix)
      (sqp-degree prefix)
      (sqp-composition-fri-roots prefix)
      (sqp-composition-final prefix)
    i
by (rule supported[OF qsupport alphas-eq])
have target-hit:
  checked-staged-query-prefix-dynamic-index-hit
    (staged-query-prefix-single-round-header-supported-query-target i)
    (Some (((prefix, prefix-state), raw), raw-state))
by (rule
    checked-staged-query-prefix-candidate-pair-hit-imp-single-round-header-target-hit
    [OF supported-prefix pair-target-hit])
have query-prefix-hit:
  checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at
    i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
unfolding
  checked-staged-security-with-query-prefix-single-round-header-supported-query-target-hit-at-def
using target-hit by simp
show ?thesis
by (rule that[OF qsupport query-prefix-hit])
qed

end

end

```

```

theory Soundness-Staged-Relevant-Drift
imports

```

Soundness-Staged-Aligned-Current-Empty
Staged-Security-Experiment-Composition-Query-Aligned-Witness-Bridge
begin

4.7 Relevant Drift and Reachable FRI Evidence

Small downstream layer for the refined relevant-drift event. The purpose is to keep diagnostic and bridge lemmas out of the already large current-empty packaging theory.

context *soundness*
begin

definition *composition-randomization-bad-with-aligned-transcript-partial-candidates*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-randomization-bad-with-aligned-transcript-partial-candidates
 $s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
 $\text{ query-idxs trace-openings composition-openings trace-table}$
 $\text{ composition-table } f.$
 $\text{ accepted-with-partial-initial-openings-aligned-transcript-consistent } s$
 $\text{ out fr } f\text{-fri-roots } f\text{-final as dg composition-fri-roots final}$
 $\text{ query-idxs trace-openings composition-openings} \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{ partial-composition-table-candidate composition-table}$
 $\text{ composition-openings} \wedge$
 $\text{ degree } f < \text{ clength} \wedge$
 $\text{ trace-table} = \text{ map } (\text{poly } f) \text{ eval-domain} \wedge$
 $\text{ violated-constraints } f \neq \{\}$
 $\text{ composition-table-low-degree maxDegree composition-table} \wedge$
 $\text{ all-queries-consistent trace-table composition-table as} \wedge$
 $\text{ common-denominator-degree-bounds } f \text{ as} \wedge$
 $\text{ random-combination-common-denominator-hides-violations } f \text{ as})$

definition *soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization
 $s \text{ out} \longleftrightarrow$
 $\text{ composition-randomization-bad-with-aligned-transcript-partial-candidates}$
 $s \text{ out} \vee$
 $\text{ trace-fri-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{ composition-fri-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{ query-bad-with-aligned-transcript-partial-candidates } s \text{ out}$

definition *soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
 s out $\longleftrightarrow$
 partial-merkle-inconsistency-bad s out $\vee$
 soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization
 s out

lemma *accepted-aligned-transcript-partial-candidate-partition-aligned-randomization:*

assumes *false-statement:* $\neg$ *exists-valid-trace*

and *partial:*

accepted-with-partial-initial-openings-aligned-transcript-consistent s
 out fr f-fri-roots f-final as dg composition-fri-roots final
 query-idxs trace-openings composition-openings

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *composition-candidate:*

partial-composition-table-candidate composition-table
 composition-openings

shows

trace-fri-bad-with-partial-candidates s out $\vee$
 composition-fri-bad-with-partial-candidates s out $\vee$
 query-bad-with-aligned-transcript-partial-candidates s out $\vee$
 composition-randomization-bad-with-aligned-transcript-partial-candidates
 s out

proof (*cases trace-table-low-degree trace-table*)

case *False*

have *aligned:*

accepted-with-partial-initial-openings-aligned-consistent s out fr
 f-fri-roots f-final as dg composition-fri-roots final query-idxs
 trace-openings composition-openings

by (*rule*

accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
 [*OF partial*])

have *trace-fri-bad-with-partial-candidates s out*

unfolding *trace-fri-bad-with-partial-candidates-def*

using *accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned*
 [*OF aligned*]

trace-candidate False **by** *blast*

then show *?thesis* **by** *blast*

next

case *True*

note *trace-low = True*

show *?thesis*

proof (*cases composition-table-low-degree maxDegree composition-table*)

case *False*

have *aligned:*

accepted-with-partial-initial-openings-aligned-consistent s out fr
 f-fri-roots f-final as dg composition-fri-roots final query-idxs
 trace-openings composition-openings

```

    by (rule
        accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
        [OF partial])
  have composition-fri-bad-with-partial-candidates s out
    unfolding composition-fri-bad-with-partial-candidates-def
    using accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned
        [OF aligned]
        trace-candidate composition-candidate trace-low False
    by blast
  then show ?thesis by blast
next
case True
note composition-low = True
show ?thesis
proof (cases all-queries-consistent trace-table composition-table as)
  case False
  have query-bad-with-aligned-transcript-partial-candidates s out
    unfolding query-bad-with-aligned-transcript-partial-candidates-def
    using partial trace-candidate composition-candidate trace-low
        composition-low False
    by blast
  then show ?thesis by blast
next
case True
note all-queries = True
from trace-low obtain f where
  deg-f: degree f < clength
  and trace-table-eq: trace-table = map (poly f) eval-domain
  unfolding trace-table-low-degree-def by blast
have violated: violated-constraints f ≠ {}
  by (rule false-statement-violated-constraints
      [OF false-statement deg-f])
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent s out fr
    f-fri-roots f-final as dg composition-fri-roots final query-idxs
    trace-openings composition-openings
  by (rule
      accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
      [OF partial])
have unaligned:
  accepted-with-partial-initial-openings s out fr f-fri-roots f-final
    as dg composition-fri-roots final query-idxs trace-openings
    query-idxs composition-openings
  by (rule
      accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned
      [OF aligned])
have len-as: length as = length spec
  by (rule accepted-with-partial-initial-openings-alpha-length
      [OF unaligned])

```

```

have bounds: common-denominator-degree-bounds f as
  by (rule common-denominator-degree-bounds-from-spec-degree-wellformed
    [OF spec-degree-wellformed-from-spec-query-margin deg-f])
have hides:
  random-combination-common-denominator-hides-violations f as
  by (rule random-combination-common-denominator-hides-from-queries
    [OF len-as false-statement deg-f trace-table-eq composition-low
      all-queries bounds])
have
  composition-randomization-bad-with-aligned-transcript-partial-candidates
  s out
  unfolding
    composition-randomization-bad-with-aligned-transcript-partial-candidates-def
  using partial trace-candidate composition-candidate deg-f
    trace-table-eq violated composition-low all-queries bounds hides
  by blast
  then show ?thesis by blast
qed
qed
qed

lemma accepted-aligned-transcript-partial-candidate-partition-with-aligned-randomization-event:
assumes false-statement:  $\neg$  exists-valid-trace
and partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent s
  out fr f-fri-roots f-final as dg composition-fri-roots final
  query-idxs trace-openings composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and composition-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
shows
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization
  s out
using
  accepted-aligned-transcript-partial-candidate-partition-aligned-randomization
  [OF false-statement partial trace-candidate composition-candidate]
unfolding
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-def
by blast

lemma checked-staged-security-with-data-state-aligned-transcript-partial-candidate-aligned-randomization-union-bou
fixes composition-error trace-fri-error' composition-fri-error'
  query-error :: prob
assumes comp-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-randomization-bad-with-aligned-transcript-partial-candidates)

```

```

    adversary-initial-state ≤ composition-error
and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ trace-fri-error'
and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-fri-error'
and query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state ≤ query-error
shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization)
    adversary-initial-state ≤
      composition-error + trace-fri-error' + composition-fri-error' +
      query-error
proof –
    let ?M = checked-staged-security-experiment-with-data-state A
    let ?E = staged-security-with-data-state-verifier-event
    let ?comp =
      ?E composition-randomization-bad-with-aligned-transcript-partial-candidates
    let ?trace = ?E trace-fri-bad-with-partial-candidates
    let ?comp-fri = ?E composition-fri-bad-with-partial-candidates
    let ?query = ?E query-bad-with-aligned-transcript-partial-candidates
    let ?partial =
      ?E soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization
    have wp-event ?M ?partial adversary-initial-state ≤
      wp-event ?M
        (λout. ?comp out ∨ ?trace out ∨ ?comp-fri out ∨
          ?query out)
      adversary-initial-state
    by (rule wp-event-mono)
      (auto simp: staged-security-with-data-state-verifier-event-def
        soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-def
        split: option.splits prod.splits)
    also have ... ≤
      wp-event ?M ?comp adversary-initial-state +
      wp-event ?M ?trace adversary-initial-state +
      wp-event ?M ?comp-fri adversary-initial-state +
      wp-event ?M ?query adversary-initial-state
    by (rule wp-event-union-bound4)
    also have ... ≤

```

```

    composition-error + trace-fri-error' + composition-fri-error' +
    query-error
  by (intro add-mono comp-bound trace-fri-bound comp-fri-bound
      query-bound)
  finally show ?thesis .
qed

```

lemma *checked-staged-security-with-data-state-aligned-transcript-partial-candidate-aligned-randomization-with-merkle*

fixes *merkle-error composition-error trace-fri-error'*

composition-fri-error' query-error :: prob

assumes *merkle-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

partial-merkle-inconsistency-bad)

adversary-initial-state ≤ merkle-error

and *comp-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

composition-randomization-bad-with-aligned-transcript-partial-candidates)

adversary-initial-state ≤ composition-error

and *trace-fri-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

trace-fri-bad-with-partial-candidates)

adversary-initial-state ≤ trace-fri-error'

and *comp-fri-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

composition-fri-bad-with-partial-candidates)

adversary-initial-state ≤ composition-fri-error'

and *query-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

query-bad-with-aligned-transcript-partial-candidates)

adversary-initial-state ≤ query-error

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle)

adversary-initial-state ≤

merkle-error + composition-error + trace-fri-error' +

composition-fri-error' + query-error

proof –

have *partial-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization)

adversary-initial-state ≤

composition-error + trace-fri-error' + composition-fri-error' +


```

    query-error
  by (rule
    checked-staged-security-with-data-state-aligned-transcript-partial-candidate-aligned-randomization-union-bound
    [OF comp-bound trace-fri-bound comp-fri-bound query-bound])
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        partial-merkle-inconsistency-bad)
      adversary-initial-state +
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization)
      adversary-initial-state
  proof -
  have event-le:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
      soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle)
      adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (λout.
        staged-security-with-data-state-verifier-event
        partial-merkle-inconsistency-bad out ∨
        staged-security-with-data-state-verifier-event
        soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization
        out)
      adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-verifier-event-def
    soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle-def
    split: option.splits prod.splits)
  also have ... ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        partial-merkle-inconsistency-bad)
      adversary-initial-state +
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization)
      adversary-initial-state
  by (rule wp-event-union-bound)
  finally show ?thesis .
qed
also have ... ≤
  merkle-error +
  (composition-error + trace-fri-error' + composition-fri-error' +

```

```

    query-error)
  by (intro add-mono merkle-bound partial-bound)
finally show ?thesis
  by (simp add: algebra-simps)
qed

lemma accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
  assumes false-statement:  $\neg$  exists-valid-trace
  and supp:  $out \in \text{set-dist } (\text{execute verify-monad } s)$ 
  and acc: accepted out
  and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
  and comp-nonempty: composition-fri-roots  $\neq []$ 
shows
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
    s out
proof -
  from acc obtain result final-state where out-eq:
    out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto
  from verify-monad-supplied-header-transcript-shape
    [OF supp[unfolded out-eq] header]
  obtain query-idxs raw-idxs query-chunks where shape:
    accepted-transcript-shape s (Some (result, final-state)) as query-idxs
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda raw. \text{index } (to\text{-nat } raw)$ ) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-rest:
    List.concat query-chunks @ PTranscript final-state = rest
  and query-chunk-header:
     $\bigwedge i. i < \text{rounds} \implies$ 
      verifier-query-round-chunk (query-idxs ! i)
      f-fri-roots composition-fri-roots (query-chunks ! i)
  and replay-lookup-header:
     $\bigwedge i. i < \text{rounds} \implies$ 
      fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks
          (verifier-header-state s fr f-fri-roots f-final as dg
            composition-fri-roots final)
            query-chunks i)) =
      Some (raw-idxs ! i)
  by blast
  from verify-monad-supplied-header-query-rounds
    [OF supp[unfolded out-eq] header]
  obtain f-fl fl query-state where f-roots:
    map snd f-fl = f-fri-roots

```

```

and comp-roots: map snd fl = composition-fri-roots
and query-transcript-state: PTranscript query-state = rest
and query-hash-state:
  PState query-state =
    verifier-header-state s fr f-fri-roots f-final as dg
    composition-fri-roots final
and query-counter: PQueryCounter query-state = PQueryCounter s
and query-out:
  Some (result, final-state) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
      query-state)
by blast
have fl-nonempty: fl ≠ []
  using comp-nonempty comp-roots by (cases fl) simp-all
then obtain b composition-root fl-tail where fl-eq:
  fl = (b, composition-root) # fl-tail
  by (cases fl) auto
have transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
  using query-transcript-state transcript-rest by simp
have query-chunk:
   $\bigwedge i. i < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
  using query-chunk-header f-roots comp-roots by simp
have replay-lookup:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some (raw-idxs ! i)
  using replay-lookup-header query-hash-state query-counter by simp
from
  query-rounds-replay-accepted-with-partial-initial-openings-aligned-transcript-consistent
  [OF header comp-nonempty fl-eq f-roots comp-roots
    query-transcript-state query-hash-state query-counter query-out
    len-raw query-idxs-eq len-query-chunks transcript-query query-chunk
    replay-lookup]
obtain trace-openings composition-openings where partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
by blast

```

```

show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then show ?thesis
  unfolding out-eq
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle-def
  by simp
next
case False
have aligned0:
  accepted-with-partial-initial-openings-aligned s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
  using partial
  by (blast dest:
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    accepted-with-partial-initial-openings-aligned-consistent-imp-aligned)
obtain trace-table composition-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and composition-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
using
  accepted-with-partial-initial-openings-aligned-candidates-if-no-partial-merkle-bad
  [OF aligned0 False]
by blast
have bad:
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization
  s (Some (result, final-state))
  by (rule
    accepted-aligned-transcript-partial-candidate-partition-with-aligned-randomization-event
    [OF false-statement partial trace-candidate composition-candidate])
then show ?thesis
  unfolding out-eq
  by (simp add:
    soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle-def)
qed
qed

lemma accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-n
assumes false-statement:  $\neg$  exists-valid-trace
and supp:  $out \in \text{set-dist } (\text{execute verify-monad } s)$ 
and acc: accepted out
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
shows

```

```

    soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
      s out ∨
    soundness-bad-event-partial-candidate-empty-header s out
proof (cases composition-fri-roots = [])
case False
then have comp-nonempty: composition-fri-roots ≠ []
  by simp
have
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
    s out
  by (rule
    accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
      [OF false-statement supp acc header comp-nonempty])
then show ?thesis by simp
next
case True
note comp-empty = True
from acc obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then have
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
    s out
  unfolding out-eq
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle-def
  by simp
then show ?thesis by simp
next
case False
from verify-monad-accepted-with-partial-trace-openings-for-header
  [OF supp[unfolded out-eq] header]
obtain trace-query-idxs trace-openings where trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
by blast
obtain trace-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
using accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
  [OF trace-partial False]
by blast
have accepted-out: accepted (Some (result, final-state))
  by (rule accepted-with-partial-trace-openings-imp-accepted
    [OF trace-partial])
let ?composition-table = replicate (scale * clength) final
have empty-partial:

```

```

    accepted-with-empty-composition-header-candidates s
      (Some (result, final-state)) fr f-fri-roots f-final as dg final
      trace-query-idxs trace-openings trace-table ?composition-table
  unfolding accepted-with-empty-composition-header-candidates-def
  by (intro conjI exI[of - rest])
    (use accepted-out header comp-empty trace-partial trace-candidate
      in simp-all)
  have soundness-bad-event-partial-candidate-empty-header s
    (Some (result, final-state))
  by (rule accepted-empty-composition-header-candidate-partition
    [OF false-statement empty-partial])
  then show ?thesis
    unfolding out-eq by simp
qed
qed

lemma checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-and-empty-header-
  fixes partial-candidate-error empty-header-error :: prob
  assumes false-statement:  $\neg$  exists-valid-trace
  and partial-candidate-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle)
        adversary-initial-state  $\leq$  partial-candidate-error
  and empty-header-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-partial-candidate-empty-header)
        adversary-initial-state  $\leq$  empty-header-error
  shows checked-staged-adversary-acceptance-probability A  $\leq$ 
    partial-candidate-error + empty-header-error
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  let ?partial =
    ?E soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
  let ?empty = ?E soundness-bad-event-partial-candidate-empty-header
  let ?bad =  $\lambda$ out. ?partial out  $\vee$  ?empty out
  have accepted-bad:
    wp-event ?M accepted adversary-initial-state  $\leq$ 
      wp-event ?M ?bad adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support: out  $\in$  set-dist (execute ?M adversary-initial-state)
    and acc: accepted out
    show ?bad out
    proof (cases out)
      case None
      then show ?thesis

```

```

    using acc unfolding accepted-def by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  using checked-staged-security-experiment-with-data-state-outcomeE
  [OF support[unfolded out-eq]]
  by blast
have verifier-acc: accepted (Some (result, final-state))
  unfolding accepted-def by simp
from verify-monad-accepted-transcript-shape[OF verifier]
obtain alphas query-idxs where shape:
  accepted-transcript-shape ?s (Some (result, final-state))
  alphas query-idxs
  by blast
from shape obtain result' final-state' fr f-fri-roots f-final dg
  composition-fri-roots final rest where
  header:
    verifier-header-transcript ?s fr f-fri-roots f-final alphas dg
    composition-fri-roots final rest
  by (elim accepted-transcript-shape-header-query-extraction)
have local-bad:
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
    ?s (Some (result, final-state)) ∨
  soundness-bad-event-partial-candidate-empty-header ?s
    (Some (result, final-state))
  by (rule
    accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
    [OF false-statement verifier verifier-acc header])
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
have bad-bound:
  wp-event ?M ?bad adversary-initial-state ≤
  partial-candidate-error + empty-header-error
proof -
  have wp-event ?M ?bad adversary-initial-state ≤
    wp-event ?M ?partial adversary-initial-state +
    wp-event ?M ?empty adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤ partial-candidate-error + empty-header-error

```

```

    by (intro add-mono partial-candidate-bound empty-header-bound)
  finally show ?thesis .
qed
have staged-bound:
  wp-event (checked-staged-security-experiment A) accepted
  adversary-initial-state ≤
  partial-candidate-error + empty-header-error
  unfolding checked-staged-security-experiment-acceptance-with-data-state
  by (rule order-trans[OF accepted-bad bad-bound])
show ?thesis
proof -
  have
    wp-event (checked-staged-security-experiment A)
    (λout. ¬ Option.is-none out) adversary-initial-state =
    wp-event (checked-staged-security-experiment A) accepted
    adversary-initial-state
  unfolding wp-event-def accepted-def
  by (rule arg-cong[where
    f=λQ. wp (checked-staged-security-experiment A) Q
    adversary-initial-state])
    (rule ext, simp)
  then show ?thesis
    unfolding checked-staged-adversary-acceptance-probability-def
    using staged-bound by simp
qed
qed

lemma checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-components-and-en
fixes merkle-error composition-error trace-fri-error'
  composition-fri-error' query-error empty-header-error :: prob
assumes false-statement: ¬ exists-valid-trace
and merkle-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
  partial-merkle-inconsistency-bad)
  adversary-initial-state ≤ merkle-error
and comp-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
  composition-randomization-bad-with-aligned-transcript-partial-candidates)
  adversary-initial-state ≤ composition-error
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
  trace-fri-bad-with-partial-candidates)
  adversary-initial-state ≤ trace-fri-error'
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event

```


composition-fri-bad-with-partial-candidates
adversary-initial-state $\leq$ *composition-fri-error'*
and *query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq$ *query-error*
and *empty-header-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state $\leq$ *empty-header-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
merkle-error + *composition-error* + *trace-fri-error'* +
composition-fri-error' + *query-error* + *empty-header-error*
proof –
have *partial-candidate-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle)
adversary-initial-state $\leq$
merkle-error + *composition-error* + *trace-fri-error'* +
composition-fri-error' + *query-error*
by (rule
checked-staged-security-with-data-state-aligned-transcript-partial-candidate-aligned-randomization-with-merkle-u
[*OF merkle-bound comp-bound trace-fri-bound comp-fri-bound*
query-bound])
have *bound*:
checked-staged-adversary-acceptance-probability A $\leq$
(*merkle-error* + *composition-error* + *trace-fri-error'* +
composition-fri-error' + *query-error*) + *empty-header-error*
by (rule
checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-and-empty-header-bou
[*OF false-statement partial-candidate-bound empty-header-bound*])
then show *?thesis*
by (*simp add: algebra-simps*)
qed

lemma *accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-current-query-on-support*:

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support*:

Some (((*data*, *attacker-state*), *result*), *final-state*) $\in$
set-dist

(*execute*

(*checked-staged-security-experiment-with-data-state A*)

adversary-initial-state)

and *partial*:

accepted-with-partial-initial-openings-aligned-transcript-consistent

```

    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
  shows
    staged-security-with-data-state-current-query-partial-opening-hit
    (Some (((data, attacker-state), result), final-state))
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support]
obtain builder verifier where builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
and verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
by (rule checked-staged-transcript-program-outcome-header-transcript
  [OF wf controlled builder])
have shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
    query-idxs
by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-shape
    [OF partial])
from checked-staged-security-with-data-state-accepted-shape-query-chunks
  [OF wf controlled support shape]
obtain raw-idxs where as-eq: as = staged-alphas data
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and lookup:
  ∀ i < rounds.
    fmlookup (HashMap final-state)

```

```

    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
  by blast
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent ?s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
have aligned0:
  accepted-with-partial-initial-openings-aligned ?s
  (Some (result, final-state)) fr f-fri-roots f-final as dg
  composition-fri-roots final query-idxs trace-openings
  composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
    [OF aligned])
from accepted-with-partial-initial-openings-aligned-shapes[OF aligned0]
obtain rest where header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
  by blast
have comp-nonempty: composition-fri-roots ≠ []
  using accepted-with-partial-initial-openings-aligned-shapes(1)
  [OF aligned0] .
have trace-partial:
  accepted-with-partial-trace-openings ?s
  (Some (result, final-state)) fr query-idxs trace-openings
  using accepted-with-partial-initial-openings-aligned-shapes(3)
  [OF aligned0] .
have comp-partial:
  accepted-with-partial-composition-openings ?s
  (Some (result, final-state)) (hd composition-fri-roots) query-idxs
  composition-openings
  using accepted-with-partial-initial-openings-aligned-shapes(4)
  [OF aligned0] .
have header-eq:
  fr = staged-trace-root data ∧
  f-fri-roots = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  composition-fri-roots = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧

```

```

    rest = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header staged-header]
  by simp
let ?i = 0
have i-bound: ?i < rounds
  using rounds-positive by simp
have round-consistent:
  partial-query-round-consistent trace-openings composition-openings as
    ?i (query-idxs ! ?i)
  using aligned i-bound
  unfolding accepted-with-partial-initial-openings-aligned-consistent-def
  by simp
have opening-consistent:
  partial-query-openings-consistent (trace-openings ! ?i)
    (composition-openings ! ?i) as (query-idxs ! ?i)
  using round-consistent
  unfolding partial-query-round-consistent-def
    partial-query-openings-consistent-def
  by simp
have trace-auth:
  partial-authenticated-table (staged-trace-root data)
    (scale * clength) (trace-openings ! ?i) final-state
  using trace-partial i-bound header-eq
  unfolding accepted-with-partial-trace-openings-def by blast
have comp-auth:
  partial-authenticated-table (hd (staged-composition-fri-roots data))
    (scale * clength) (composition-openings ! ?i) final-state
  using comp-partial i-bound header-eq
  unfolding accepted-with-partial-composition-openings-def by blast
have final-lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge ?i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) ?i)) =
    Some (raw-idxs ! ?i)
  using lookup i-bound by simp
have idx-raw:
  query-idxs ! ?i = index (to-nat (raw-idxs ! ?i))
  using query-idxs-eq len-raw i-bound by simp
have hit-at:
  staged-security-with-data-state-current-query-partial-opening-hit-at ?i
    (Some (((data, attacker-state), result), final-state))
  unfolding staged-security-with-data-state-current-query-partial-opening-hit-at-def
    Let-def
  apply simp
by (intro exI[of - raw-idxs ! ?i]
  exI[of - trace-openings ! ?i]
  exI[of - composition-openings ! ?i] exI[of - rest] conjI)

```

```

    (use i-bound final-lookup comp-nonempty verifier trace-auth comp-auth
      staged-header opening-consistent idx-raw header-eq as-eq in simp-all)
  show ?thesis
  by (rule staged-security-with-data-state-current-query-partial-opening-hitI
      [OF i-bound hit-at])
qed

lemma checked-staged-security-with-data-state-aligned-partial-candidate-randomization-imp-current-query-on-support
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  and bad:
    staged-security-with-data-state-verifier-event
    composition-randomization-bad-with-aligned-transcript-partial-candidates
    out
  shows staged-security-with-data-state-current-query-partial-opening-hit out
proof (cases out)
case None
then show ?thesis
  using bad
  unfolding staged-security-with-data-state-verifier-event-def
    composition-randomization-bad-with-aligned-transcript-partial-candidates-def
  by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have bad-some:
  composition-randomization-bad-with-aligned-transcript-partial-candidates
  ?s (Some (result, final-state))
using bad
unfolding out-eq staged-security-with-data-state-verifier-event-def
  Let-def
by simp
obtain fr f-fri-roots f-final as dg composition-fri-roots final
  query-idxs trace-openings composition-openings trace-table
  composition-table f
where partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent
  ?s (Some (result, final-state)) fr f-fri-roots f-final as dg

```

```

    composition-fri-roots final query-idxs trace-openings
    composition-openings
using bad-some
unfolding
    composition-randomization-bad-with-aligned-transcript-partial-candidates-def
by blast
show ?thesis
unfolding out-eq
by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-current-query-on-support
    [OF wf controlled support[unfolded out-eq] partial])
qed

lemma checked-staged-security-with-data-state-aligned-partial-candidate-randomization-le-current-query:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
    wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    composition-randomization-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state  $\leq$ 
    wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state
by (rule wp-event-mono-on-support)
    (rule
    checked-staged-security-with-data-state-aligned-partial-candidate-randomization-imp-current-query-on-support
    [OF wf controlled])

lemma checked-staged-security-with-data-state-aligned-partial-candidate-randomization-bound-from-query-prefix-curr
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and round-bound:
     $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
    adversary-initial-state  $\leq C i$ 
shows
    wp-event
    (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    composition-randomization-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state  $\leq (\sum i < \text{rounds}. C i)$ 
proof –
have current-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state  $\leq (\sum i < \text{rounds}. C\ i)$ 
  by (rule
    checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
    [OF round-bound wf controlled])
  show ?thesis
  by (rule order-trans[
    OF
    checked-staged-security-with-data-state-aligned-partial-candidate-randomization-le-current-query
    current-bound])
    (use wf controlled in simp-all)
qed

lemma checked-staged-security-with-data-state-aligned-randomization-or-query-bad-le-current-query:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event
  (checked-staged-security-experiment-with-data-state A)
  ( $\lambda$ out.
    staged-security-with-data-state-verifier-event
    composition-randomization-bad-with-aligned-transcript-partial-candidates
    out  $\vee$ 
    staged-security-with-data-state-verifier-event
    query-bad-with-aligned-transcript-partial-candidates out)
  adversary-initial-state  $\leq$ 
  wp-event
  (checked-staged-security-experiment-with-data-state A)
  staged-security-with-data-state-current-query-partial-opening-hit
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out  $\in$ 
  set-dist
  (execute
    (checked-staged-security-experiment-with-data-state A)
    adversary-initial-state)
and bad:
  staged-security-with-data-state-verifier-event
  composition-randomization-bad-with-aligned-transcript-partial-candidates
  out  $\vee$ 
  staged-security-with-data-state-verifier-event
  query-bad-with-aligned-transcript-partial-candidates out
show
  staged-security-with-data-state-current-query-partial-opening-hit out
proof (cases
  staged-security-with-data-state-verifier-event
  composition-randomization-bad-with-aligned-transcript-partial-candidates

```

```

      out)
    case True
    then show ?thesis
    by (rule
      checked-staged-security-with-data-state-aligned-partial-candidate-randomization-imp-current-query-on-support
      [OF wf controlled support])
  next
  case False
  then have query-bad:
    staged-security-with-data-state-verifier-event
    query-bad-with-aligned-transcript-partial-candidates out
  using bad by simp
  show ?thesis
  proof (cases out)
  case None
  then show ?thesis
    using query-bad
    unfolding staged-security-with-data-state-verifier-event-def
    query-bad-with-aligned-transcript-partial-candidates-def
    by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
  by (cases packed, auto split: prod.splits)
  show ?thesis
    unfolding out-eq
    by (rule
      checked-staged-security-with-data-state-aligned-transcript-query-bad-imp-current-partial-opening-hit-on-support
      [OF wf controlled support[unfolded out-eq]
      query-bad[unfolded out-eq]])
qed
qed
qed

```

lemma *checked-staged-security-with-data-state-aligned-transcript-partial-candidate-aligned-randomization-with-merkle*
fixes merkle-error current-query-error trace-fri-error'
 composition-fri-error' :: prob
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and merkle-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 partial-merkle-inconsistency-bad)
 adversary-initial-state $\leq$ merkle-error
and current-query-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-current-query-partial-opening-hit
 adversary-initial-state $\leq$ current-query-error


```

and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  trace-fri-error'
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-fri-error'
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle)
    adversary-initial-state  $\leq$ 
      merkle-error + current-query-error + trace-fri-error' +
      composition-fri-error'
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  let ?merkle = ?E partial-merkle-inconsistency-bad
  let ?comp =
    ?E composition-randomization-bad-with-aligned-transcript-partial-candidates
  let ?trace = ?E trace-fri-bad-with-partial-candidates
  let ?comp-fri = ?E composition-fri-bad-with-partial-candidates
  let ?query = ?E query-bad-with-aligned-transcript-partial-candidates
  let ?current =
    staged-security-with-data-state-current-query-partial-opening-hit
  let ?combined =  $\lambda$ out. ?comp out  $\vee$  ?query out
  let ?partial =
    ?E soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
  have event-le:
    wp-event ?M ?partial adversary-initial-state  $\leq$ 
      wp-event ?M
        ( $\lambda$ out. ?merkle out  $\vee$  ?combined out  $\vee$  ?trace out  $\vee$ 
          ?comp-fri out)
        adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-verifier-event-def
      soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle-def
      soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-def
      split: option.splits prod.splits)
  have union-bound:
    wp-event ?M
      ( $\lambda$ out. ?merkle out  $\vee$  ?combined out  $\vee$  ?trace out  $\vee$ 
        ?comp-fri out)
      adversary-initial-state  $\leq$ 
      wp-event ?M ?merkle adversary-initial-state +
      wp-event ?M ?combined adversary-initial-state +

```

```

    wp-event ?M ?trace adversary-initial-state +
    wp-event ?M ?comp-fri adversary-initial-state
  by (rule wp-event-union-bound4)
have combined-bound:
  wp-event ?M ?combined adversary-initial-state ≤
  wp-event ?M ?current adversary-initial-state
by (rule
  checked-staged-security-with-data-state-aligned-randomization-or-query-bad-le-current-query
  [OF wf controlled])
have
  wp-event ?M ?partial adversary-initial-state ≤
  wp-event ?M ?merkle adversary-initial-state +
  wp-event ?M ?current adversary-initial-state +
  wp-event ?M ?trace adversary-initial-state +
  wp-event ?M ?comp-fri adversary-initial-state
by (rule order-trans[OF event-le])
  (rule order-trans[OF union-bound],
  intro add-mono order-refl combined-bound)
also have ... ≤
  merkle-error + current-query-error + trace-fri-error' +
  composition-fri-error'
by (intro add-mono merkle-bound current-query-bound trace-fri-bound
  comp-fri-bound)
finally show ?thesis .
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-aligned-partial-candidate-randomization-imp-current-query-o*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

```

  out ∈
  set-dist
  (execute
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    adversary-initial-state)

```

and *bad:*

```

  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-randomization-bad-with-aligned-transcript-partial-candidates
  out

```

shows

```

  (case out of
    None ⇒ False
  | Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state) ⇒
    staged-security-with-data-state-current-query-partial-opening-hit
    (Some (((data, attacker-state), result), final-state)))

```

proof (cases out)

case *None*

```

then show ?thesis
  using bad
  unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    composition-randomization-bad-with-aligned-transcript-partial-candidates-def
  by simp
next
case (Some packed)
then obtain alpha-prefix alpha-prefix-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have bad-some:
  composition-randomization-bad-with-aligned-transcript-partial-candidates
    ?s (Some (result, final-state))
  using bad
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    Let-def
  by simp
obtain fr f-fri-roots f-final as dg composition-fri-roots final
  query-idxs trace-openings composition-openings trace-table
  composition-table f
where partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent
    ?s (Some (result, final-state)) fr f-fri-roots f-final as dg
    composition-fri-roots final query-idxs trace-openings
    composition-openings
  using bad-some
  unfolding
    composition-randomization-bad-with-aligned-transcript-partial-candidates-def
  by blast
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
      [OF support[unfolded out-eq]])
have data-hit:
  staged-security-with-data-state-current-query-partial-opening-hit
    (Some (((data, attacker-state), result), final-state))
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-current-query-on-support

```

```

    [OF wf controlled data-support partial])
  then show ?thesis
    unfolding out-eq by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-aligned-partial-candidate-randomization-le-current-query:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        composition-randomization-bad-with-aligned-transcript-partial-candidates)
      adversary-initial-state ≤
    wp-event
      (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-current-query-partial-opening-hit
      adversary-initial-state
  proof -
    let ?project =
      λx. ((snd (fst (fst x)), snd (fst x)), snd x)
    let ?projected =
      λout. case out of
        None ⇒ False
      | Some (packed, t) ⇒
        staged-security-with-data-state-current-query-partial-opening-hit
          (Some (?project packed, t))
    have event-le:
      wp-event
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
        (checked-staged-security-with-actual-alpha-prefix-verifier-event
          composition-randomization-bad-with-aligned-transcript-partial-candidates)
        adversary-initial-state ≤
      wp-event
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
        ?projected adversary-initial-state
    proof (rule wp-event-mono-on-support)
      fix out
      assume support:
        out ∈
          set-dist
            (execute
              (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
                A)
              adversary-initial-state)
      and bad:
        checked-staged-security-with-actual-alpha-prefix-verifier-event
          composition-randomization-bad-with-aligned-transcript-partial-candidates
        out
    qed
  qed

```

```

show ?projected out
proof (cases out)
  case None
  then show ?thesis
    using bad
    unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      composition-randomization-bad-with-aligned-transcript-partial-candidates-def
    by simp
next
  case (Some packed)
  then obtain alpha-prefix alpha-prefix-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data),
        attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  have hit:
    staged-security-with-data-state-current-query-partial-opening-hit
      (Some (((data, attacker-state), result), final-state))
    using
      checked-staged-security-with-actual-alpha-prefix-aligned-partial-candidate-randomization-imp-current-query-on-s
        [OF wf controlled support bad]
    unfolding out-eq by simp
  show ?thesis
    unfolding out-eq using hit by simp
qed
qed
have projection:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    ?projected adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state
proof –
have
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state =
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
       $\gg$  (λx. return (?project x)))
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state
  using
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
    [of A]
  by simp

```

also have ... =
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
?projected adversary-initial-state
by (*subst wp-event-bind-return-map*)
(simp add:
staged-security-with-data-state-current-query-partial-opening-hit-def
staged-security-with-data-state-current-query-partial-opening-hit-at-def)
finally show *?thesis*
by *simp*
qed
show *?thesis*
by (*rule order-trans[OF event-le]*)
(simp add: projection)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-aligned-partial-candidate-randomization-bound-from-query-pr*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *round-bound:*
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
adversary-initial-state $\leq C i$
shows
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-randomization-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq (\sum i < \text{rounds}. C i)$
proof –
have *current-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state $\leq (\sum i < \text{rounds}. C i)$
by (*rule*
checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
[OF round-bound wf controlled])
show *?thesis*
by (*rule order-trans[*
OF
checked-staged-security-with-actual-alpha-prefix-aligned-partial-candidate-randomization-le-current-query
current-bound])
(use wf controlled in simp-all)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-driftE:*
assumes

checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
out
obtains
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space out
checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-randomization-bad-with-partial-candidates out
| *checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
composition-trace-bad-alpha-space out
checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-partial-opening-union-hit-with-empty-header out
using *assms*
unfolding
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-def
by *blast*

lemma *checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-le-branches:*
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
composition-randomization-bad-with-partial-candidates)
adversary-initial-state +
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
composition-alpha-partial-opening-union-hit-with-empty-header)
adversary-initial-state

proof —
have *event-le:*
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
($\lambda out.$
checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-randomization-bad-with-partial-candidates out $\vee$
checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-partial-opening-union-hit-with-empty-header out)
adversary-initial-state
by (*rule wp-event-mono*)
(*auto simp:*
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-def)
have *union-le:*

```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (λout.
    checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-randomization-bad-with-partial-candidates out ∨
    checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-partial-opening-union-hit-with-empty-header out)
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-randomization-bad-with-partial-candidates)
  adversary-initial-state +
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-alpha-partial-opening-union-hit-with-empty-header)
  adversary-initial-state
  by (rule wp-event-union-bound)
show ?thesis
  by (rule order-trans[OF event-le union-le])
qed

lemma checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-branch-bound:
  assumes random-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        composition-randomization-bad-with-partial-candidates)
      adversary-initial-state ≤ R
  and empty-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        composition-alpha-partial-opening-union-hit-with-empty-header)
      adversary-initial-state ≤ E
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
      adversary-initial-state ≤ R + E
  by (rule order-trans[
    OF
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-le-branches])
  (intro add-mono random-bound empty-bound)

lemma checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-le-not-prefix-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A

```


shows
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)
adversary-initial-state
proof –
have *event-le*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
composition-trace-bad-alpha-space)
adversary-initial-state
by (rule *wp-event-mono*)
 (auto simp:
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-def)
have *drift-le*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)
adversary-initial-state
by (rule
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift-le-not-prefix-bound
[*OF wf controlled*])
show ?thesis
by (rule *order-trans*[*OF event-le drift-le*])
qed

lemma *checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-not-prefix*
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *collision-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
adversary-initial-state $\leq C$
and *uncovered-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
adversary-initial-state $\leq U$
and *no-prefix-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
adversary-initial-state $\leq N$
and *path-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
adversary-initial-state $\leq P$
shows
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq C + U + N + P$
proof –
have *not-prefix-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state $\leq C + U + N + P$
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound
[OF wf controlled collision-bound uncovered-bound no-prefix-bound
path-bound])
have *relevant-le*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state
by (rule
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-le-not-prefix-bound
[OF wf controlled])

show *?thesis*
by (rule order-trans[*OF relevant-le not-prefix-bound*])
qed

lemma *checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-cross-and-*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *cross-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
adversary-initial-state ≤ X
and *witness-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
adversary-initial-state ≤ W
shows
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + X + W + W
proof –
have *collision-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
adversary-initial-state ≤
hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0)
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
[*OF wf controlled*])
have *uncovered-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
adversary-initial-state ≤ X
by (rule order-trans
[*OF*
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-witnessed-cross-or-merkle-bad
cross-bound])
have *no-prefix-bound:*
wp-event

```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
    adversary-initial-state  $\leq W$ 
  by (rule order-trans
    [OF
      checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-le-witness-transcript-hit
      witness-bound])
  have path-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
    adversary-initial-state  $\leq W$ 
  by (rule order-trans
    [OF
      checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-le-witness-transcript-hit
      witness-bound])
  show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-not-prefix-com
    [OF wf controlled collision-bound uncovered-bound no-prefix-bound
      path-bound])
qed

lemma checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-cross-data
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and cross-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
    adversary-initial-state  $\leq X$ 
  and data-pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state  $\leq P$ 
  and transcript-new-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
    adversary-initial-state  $\leq R$ 
  shows
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
    adversary-initial-state  $\leq$ 

```

hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + X + (P + R) + (P + R)
proof –
have *transcript-pre-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
adversary-initial-state ≤ P
by (rule
checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
[OF data-pre-bound])
have *witness-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
adversary-initial-state ≤ P + R
by (rule
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new
[OF transcript-pre-bound transcript-new-bound])
show *?thesis*
by (rule
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-cross-and-witness
[OF wf controlled cross-bound witness-bound])
qed

lemma *checked-staged-soundness-from-aligned-partial-candidate-components-and-current-empty-query-from-cross-data*
fixes *trace-fri-error' composition-fri-error' query-error X P R*
empty-query-error :: prob
assumes *false-statement: ¬ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state ≤ trace-fri-error'
and *comp-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state ≤ composition-fri-error'
and *query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-partial-candidates)
adversary-initial-state ≤ query-error
and *cross-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state

A)
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
adversary-initial-state $\leq X$
and *data-pre-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$
and *transcript-new-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq R$
and *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
and *current-empty-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-current-empty-query-opening-hit-at* 0)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability* A $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* budgets +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* budgets 0) +
staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0) +
 (*hash-collision-budget-value* 0
 (*staged-alpha-search-queries* budgets 0) + X + (P + R) + (P + R))) +
trace-fri-error' + *composition-fri-error'* +
query-error +
 (*composition-error-bound* +
staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* budgets 0) +
staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0) +
 (*hash-collision-budget-value* 0
 (*staged-alpha-search-queries* budgets 0) + X + (P + R) + (P + R)) +
trace-fri-error + *empty-query-error*)
proof –
let ?D =

hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + X + (P + R) + (P + R)
have *drift-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state ≤ ?D
by (rule
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-cross-data-pre
[OF wf controlled cross-bound data-pre-bound transcript-new-bound])
show *?thesis*
by (rule
checked-staged-soundness-from-aligned-partial-candidate-components-and-current-empty-query-from-relevant-dri
[OF false-statement wf controlled trace-fri-bound comp-fri-bound
query-bound drift-bound empty-trace-fri-bound current-empty-bound])
qed

lemma *checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from*
fixes *trace-fri-error' composition-fri-error' query-error D*
empty-query-error :: prob
assumes *false-statement: ¬ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state ≤ trace-fri-error'
and *comp-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state ≤ composition-fri-error'
and *query-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state ≤ query-error
and *drift-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state ≤ D
and *empty-trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤ trace-fri-error

and *current-empty-bound*:

$$\begin{aligned} & \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\ & \quad (\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0) \\ & \quad \text{adversary-initial-state} \leq \text{empty-query-error} \end{aligned}$$
shows *checked-staged-adversary-acceptance-probability* $A \leq$

$$\begin{aligned} & \text{hash-collision-budget-value } 0 \\ & \quad (\text{staged-attacker-query-budget budgets} + \\ & \quad \quad \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) + \\ & (\text{composition-error-bound} + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \text{hash-collision-budget-value } 0 \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad D) + \\ & \text{trace-fri-error}' + \text{composition-fri-error}' + \\ & \text{query-error} + \\ & (\text{composition-error-bound} + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \text{hash-collision-budget-value } 0 \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad D + \text{trace-fri-error} + \text{empty-query-error}) \end{aligned}$$
proof –
have *comp-bound*:

$$\begin{aligned} & \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\ & \quad (\text{staged-security-with-data-state-verifier-event} \\ & \quad \quad \text{composition-bad-with-partial-candidates}) \\ & \quad \text{adversary-initial-state} \leq \\ & \quad \text{composition-error-bound} + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \text{hash-collision-budget-value } 0 \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad D \end{aligned}$$
by (rule

$$\begin{aligned} & \text{checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-relevant-drift-and-budgets} \\ & \quad [\text{OF false-statement wf controlled drift-bound}]) \end{aligned}$$
have *merkle-bound*:

$$\begin{aligned} & \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\ & \quad (\text{staged-security-with-data-state-verifier-event} \\ & \quad \quad \text{partial-merkle-inconsistency-bad}) \\ & \quad \text{adversary-initial-state} \leq \\ & \quad \text{hash-collision-budget-value } 0 \end{aligned}$$

(staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
 checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
 [OF wf controlled])
have empty-query-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ empty-query-error
by (rule order-trans[
 OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero[
 OF wf controlled] current-empty-bound])
have empty-header-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 soundness-bad-event-partial-candidate-empty-header)
 adversary-initial-state $\leq$
 composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 D + trace-fri-error + empty-query-error
by (rule
 checked-staged-security-with-data-state-empty-header-bad-event-bound-from-relevant-drift-trace-fri-and-query
 [OF false-statement wf controlled drift-bound empty-trace-fri-bound
 empty-query-bound])
show ?thesis
by (rule
 checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-empty-header-bound
 [OF false-statement merkle-bound comp-bound trace-fri-bound
 comp-fri-bound query-bound empty-header-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-components-and-current-query-prefix-rounds-from-relevant-
fixes trace-fri-error' composition-fri-error' D empty-query-error :: prob
and C :: nat $\Rightarrow$ prob
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ trace-fri-error'
and comp-fri-bound:

$wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $\ (staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $\ \ \ composition\text{-}fri\text{-}bad\text{-}with\text{-}partial\text{-}candidates)$
 $\ \ \ adversary\text{-}initial\text{-}state \leq \ composition\text{-}fri\text{-}error'$
and $round\text{-}bound$:
 $\bigwedge i. i < rounds \implies$
 $\ \ \ wp\text{-}event$
 $\ \ \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}query\text{-}prefix\text{-}data\text{-}state \ A \ i)$
 $\ \ \ (checked\text{-}staged\text{-}security\text{-}with\text{-}query\text{-}prefix\text{-}current\text{-}query\text{-}partial\text{-}opening\text{-}hit\text{-}at \ i)$
 $\ \ \ adversary\text{-}initial\text{-}state \leq \ C \ i$
and $drift\text{-}bound$:
 $\ \ \ wp\text{-}event$
 $\ \ \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}data\text{-}state$
 $\ \ \ \ \ A)$
 $\ \ \ checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}relevant\text{-}partial\text{-}header\text{-}candidate\text{-}drift$
 $\ \ \ adversary\text{-}initial\text{-}state \leq \ D$
and $empty\text{-}trace\text{-}fri\text{-}bound$:
 $\ \ \ wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $\ \ \ (staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $\ \ \ \ \ trace\text{-}fri\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates)$
 $\ \ \ adversary\text{-}initial\text{-}state \leq \ trace\text{-}fri\text{-}error$
and $current\text{-}empty\text{-}bound$:
 $\ \ \ wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $\ \ \ (staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}current\text{-}empty\text{-}query\text{-}opening\text{-}hit\text{-}at \ 0)$
 $\ \ \ adversary\text{-}initial\text{-}state \leq \ empty\text{-}query\text{-}error$
shows $checked\text{-}staged\text{-}adversary\text{-}acceptance\text{-}probability \ A \leq$
 $\ hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $\ \ (staged\text{-}attacker\text{-}query\text{-}budget \ budgets \ +$
 $\ \ \ staged\text{-}challenge\text{-}query\text{-}budget \ + \ verifier\text{-}hash\text{-}query\text{-}budget) \ +$
 $\ (composition\text{-}error\text{-}bound \ +$
 $\ \ staged\text{-}phase\text{-}relation\text{-}error \ size$
 $\ \ \ (staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) \ +$
 $\ hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $\ \ \ (staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) \ +$
 $\ \ staged\text{-}phase\text{-}relation\text{-}error \ size$
 $\ \ \ (staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) \ +$
 $\ \ D) \ +$
 $\ trace\text{-}fri\text{-}error' \ + \ composition\text{-}fri\text{-}error' \ +$
 $\ (\sum i < rounds. \ C \ i) \ +$
 $\ (composition\text{-}error\text{-}bound \ +$
 $\ \ staged\text{-}phase\text{-}relation\text{-}error \ size$
 $\ \ \ (staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) \ +$
 $\ hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $\ \ \ (staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) \ +$
 $\ \ staged\text{-}phase\text{-}relation\text{-}error \ size$
 $\ \ \ (staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) \ +$
 $\ \ D \ + \ trace\text{-}fri\text{-}error \ + \ empty\text{-}query\text{-}error)$
proof –
have $current\text{-}query\text{-}bound$:

```

wp-event (checked-staged-security-experiment-with-data-state A)
  staged-security-with-data-state-current-query-partial-opening-hit
  adversary-initial-state  $\leq (\sum i < \text{rounds}. C\ i)$ 
by (rule
  checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
  [OF round-bound wf controlled])
have query-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    query-bad-with-aligned-transcript-partial-candidates)
  adversary-initial-state  $\leq (\sum i < \text{rounds}. C\ i)$ 
by (rule order-trans[
  OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit
  current-query-bound])
  (use wf controlled in simp-all)
show ?thesis
by (rule
  checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from-re
  [OF false-statement wf controlled trace-fri-bound comp-fri-bound
    query-bound drift-bound empty-trace-fri-bound current-empty-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix-
fixes trace-fri-error' composition-fri-error' D empty-query-error :: prob
and C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-partial-candidates)
  adversary-initial-state  $\leq$  trace-fri-error'
and comp-fri-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-fri-bad-with-partial-candidates)
  adversary-initial-state  $\leq$  composition-fri-error'
and round-bound:
 $\bigwedge i. i < \text{rounds} \implies$ 
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
  adversary-initial-state  $\leq C\ i$ 
and drift-bound:
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
  checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift

```

$\text{adversary-initial-state} \leq D$
and *empty-trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and *current-empty-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows *checked-staged-adversary-acceptance-probability* $A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds}. C i) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum i < \text{rounds}. C i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D + \text{trace-fri-error} + \text{empty-query-error})$
proof –
have *comp-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-randomization-bad-with-aligned-transcript-partial-candidates})$
 $\text{adversary-initial-state} \leq (\sum i < \text{rounds}. C i)$
by (rule
 $\text{checked-staged-security-with-data-state-aligned-partial-candidate-randomization-bound-from-query-prefix-current-}$
 $[\text{OF wf controlled round-bound}])$
have *current-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-current-query-partial-opening-hit}$
 $\text{adversary-initial-state} \leq (\sum i < \text{rounds}. C i)$
by (rule
 $\text{checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds}$
 $[\text{OF round-bound wf controlled}])$
have *query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-aligned-transcript-partial-candidates})$
 $\text{adversary-initial-state} \leq (\sum i < \text{rounds}. C i)$
by (rule *order-trans*[
 $\text{OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit}$
 $\text{current-query-bound}])$

(use *wf* controlled in *simp-all*)

have *merkle-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)

by (*rule*
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
 [*OF wf controlled*])

have *empty-query-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *empty-query-error*

by (*rule order-trans*[
OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero
OF wf controlled] *current-empty-bound*])

have *empty-header-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
D + *trace-fri-error* + *empty-query-error*

by (*rule*
checked-staged-security-with-data-state-empty-header-bad-event-bound-from-relevant-drift-trace-fri-and-query
 [*OF false-statement wf controlled drift-bound empty-trace-fri-bound*
empty-query-bound])

have *bound*:

checked-staged-adversary-acceptance-probability A $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 ($\sum i < \text{rounds. } C \ i$) +
trace-fri-error' + *composition-fri-error'* +
 ($\sum i < \text{rounds. } C \ i$) +
 (*composition-error-bound* +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0)

(staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 D + trace-fri-error + empty-query-error)
 by (rule
 checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-components-and-empty-
 [OF false-statement merkle-bound comp-bound trace-fri-bound
 comp-fri-bound query-bound empty-header-bound])
 then show ?thesis .
 qed

lemma checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-structured-query-fr
fixes trace-fri-error' composition-fri-error' D empty-query-error
 header-query-error :: prob
and C :: nat $\Rightarrow$ prob
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ trace-fri-error'
and comp-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ composition-fri-error'
and round-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state $\leq$ C i
and header-query-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix
 A)
 adversary-initial-state $\leq$ header-query-error
and drift-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
 adversary-initial-state $\leq$ D
and empty-trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
and *current-empty-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
($\sum i < \text{rounds. } C i$) +
trace-fri-error' + composition-fri-error' +
header-query-error +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D + trace-fri-error + empty-query-error)
proof –
have *comp-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-randomization-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq$ *($\sum i < \text{rounds. } C i$)*
by (rule
checked-staged-security-with-data-state-aligned-partial-candidate-randomization-bound-from-query-prefix-current-
[OF wf controlled round-bound])
have *query-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq$ *header-query-error*
by (rule
checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-header-authenticated-candidate-
[OF wf controlled header-query-bound])
have *merkle-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled

$[OF \text{ wf controlled}]$
have *empty-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *empty-query-error*
by (*rule order-trans*[
OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero
OF wf controlled] *current-empty-bound*])
have *empty-header-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D + trace-fri-error + empty-query-error
by (*rule*
checked-staged-security-with-data-state-empty-header-bad-event-bound-from-relevant-drift-trace-fri-and-query
 $[OF \text{ false-statement wf controlled drift-bound empty-trace-fri-bound}$
 $\text{empty-query-bound}]$)
have *bound*:
checked-staged-adversary-acceptance-probability A $\leq$
(hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
 $(\sum i < \text{rounds}. C \ i) + \text{trace-fri-error}' +$
composition-fri-error' + header-query-error) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D + trace-fri-error + empty-query-error)
by (*rule*
checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-components-and-empty
 $[OF \text{ false-statement merkle-bound comp-bound trace-fri-bound}$
 $\text{comp-fri-bound query-bound empty-header-bound}]$)
then show *?thesis*
by (*simp add: algebra-simps*)
qed

lemma *checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix*
fixes *trace-fri-error'* *composition-fri-error'* *D* *empty-query-error* :: *prob*
and *C* :: *nat* $\Rightarrow$ *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*
and *trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *composition-fri-error'*
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \Rightarrow$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* *A* *i*)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at* *i*)
 adversary-initial-state $\leq$ *C* *i*
and *drift-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
 adversary-initial-state $\leq$ *D*
and *empty-trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ *trace-fri-error*
and *current-empty-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-current-empty-query-opening-hit-at* *0*)
 adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
 hash-collision-budget-value *0*
 (*staged-attacker-query-budget* *budgets* +
 staged-challenge-query-budget + *verifier-hash-query-budget*) +
 ($\sum i < \text{rounds}. C\ i$) +
 trace-fri-error' + *composition-fri-error'* +
 (*composition-error-bound* +
 staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* *0*) +
 hash-collision-budget-value *0*
 (*staged-alpha-search-queries* *budgets* *0*) +
 staged-phase-relation-error *size*)

$(\text{staged-alpha-search-queries budgets } 0) +$
 $D + \text{trace-fri-error} + \text{empty-query-error})$
proof –
have *current-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-current-query-partial-opening-hit}$
 $\text{adversary-initial-state} \leq (\sum_{i < \text{rounds.}} C \ i)$
by (rule
 $\text{checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds}$
 $[\text{OF round-bound wf controlled}])$
have *merkle-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{partial-merkle-inconsistency-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget})$
by (rule
 $\text{checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled}$
 $[\text{OF wf controlled}])$
have *partial-candidate-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle})$
 $\text{adversary-initial-state} \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum_{i < \text{rounds.}} C \ i) + \text{trace-fri-error}' + \text{composition-fri-error}'$
by (rule
 $\text{checked-staged-security-with-data-state-aligned-transcript-partial-candidate-aligned-randomization-with-merkle-c}$
 $[\text{OF wf controlled merkle-bound current-query-bound trace-fri-bound}$
 $\text{comp-fri-bound}])$
have *empty-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
by (rule *order-trans*[
 $\text{OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero}$
 $\text{OF wf controlled}] \text{ current-empty-bound}])$
have *empty-header-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{soundness-bad-event-partial-candidate-empty-header})$
 $\text{adversary-initial-state} \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$

```

    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D + trace-fri-error + empty-query-error
  by (rule
    checked-staged-security-with-data-state-empty-header-bad-event-bound-from-relevant-drift-trace-fri-and-query
    [OF false-statement wf controlled drift-bound empty-trace-fri-bound
      empty-query-bound])
  have bound:
    checked-staged-adversary-acceptance-probability A ≤
    (hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget + verifier-hash-query-budget) +
    (∑ i<rounds. C i) + trace-fri-error' +
    composition-fri-error') +
    (composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D + trace-fri-error + empty-query-error)
  by (rule
    checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-and-empty-header-bou
    [OF false-statement partial-candidate-bound empty-header-bound])
  then show ?thesis
  by (simp add: algebra-simps)
qed

lemma checked-staged-soundness-from-aligned-transcript-conceptual-default-alpha-prefix-union-clean-structured-paths
  fixes trace-fri-error' composition-fri-error' D empty-query-error :: prob
  and T C :: nat ⇒ prob
  and trace-default composition-default :: 'f list
  assumes false-statement: ¬ exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ trace-fri-error'
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-fri-error'

```

and *default-prefix-clean*:
 $\bigwedge i$ *prefix* *prefix-state*.
 $i < \text{rounds} \implies$
Some ((*prefix*, *prefix-state*), *prefix-state*) $\in$
set-dist
(*execute* (*checked-staged-query-prefix-with-state* *A* *i*)
adversary-initial-state) $\implies$
query-prefix-trace-conceptual-table-with-default *prefix* *prefix-state*
trace-default = *trace-default* $\wedge$
query-prefix-composition-conceptual-table-with-default *prefix*
prefix-state *composition-default* = *composition-default* $\wedge$
trace-table-low-degree *trace-default* $\wedge$
composition-table-low-degree *maxDegree* *composition-default* $\wedge$
trace-default $\in$
alpha-prefix-trace-table-candidates *prefix-state*
(*sqp-trace-root* *prefix*) $\wedge$
sqp-alphas *prefix* $\notin$
alpha-prefix-union-bad-sets *prefix-state*
composition-trace-bad-alpha-space (*sqp-trace-root* *prefix*)

and *trace-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* *A* *i*)
(*checked-staged-security-with-query-prefix-current-trace-path-output-hit-at* *i*)
adversary-initial-state $\leq T$ *i*

and *composition-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* *A* *i*)
(*checked-staged-security-with-query-prefix-current-composition-path-output-hit-at* *i*)
adversary-initial-state $\leq C$ *i*

and *drift-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq D$

and *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*

and *current-empty-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-current-empty-query-opening-hit-at* *0*)
adversary-initial-state $\leq$ *empty-query-error*

shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
hash-collision-budget-value *0*
(*staged-attacker-query-budget* *budgets* +

staged-challenge-query-budget + verifier-hash-query-budget) +
 $(\sum i < \text{rounds}.$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
T i + C i) +
trace-fri-error' + composition-fri-error' +
 $(\sum i < \text{rounds}.$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
T i + C i) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D + trace-fri-error + empty-query-error)
proof (rule
checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix-round
[OF false-statement wf controlled trace-fri-bound comp-fri-bound -
drift-bound empty-trace-fri-bound current-empty-bound])
fix *i*
assume *i-bound: i < rounds*
show
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
adversary-initial-state ≤
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
T i + C i
by (rule
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-default-alpha
[OF false-statement wf controlled i-bound - trace-path-bound[OF i-bound]
composition-path-bound[OF i-bound]])
(rule default-prefix-clean[OF i-bound])
qed
end

end

theory *Soundness-Staged-Empty-Relevant-Drift*
imports *Soundness-Staged-Relevant-Drift*
begin

Empty-header-specific relevant-drift packaging.

This theory is kept downstream of the broader relevant-drift layer so that the latter remains focused on reusable drift decompositions.

context *soundness*
begin

definition *checked-staged-security-with-actual-alpha-prefix-witnessed-sparse-cross-bad*

$$\begin{aligned} &:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times \\ &\quad 'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times \\ &\quad 'f \text{ protocol-channel}) \times \text{unit list}) \times \\ &\quad 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool} \end{aligned}$$

where

checked-staged-security-with-actual-alpha-prefix-witnessed-sparse-cross-bad

$$\begin{aligned} &\text{out} \longleftrightarrow \\ &(\text{case out of} \\ &\quad \text{None} \Rightarrow \text{False} \\ &| \text{Some } (x, -) \Rightarrow \\ &\quad (\text{let packed} = \text{fst } (\text{fst } x); \\ &\quad \quad \text{data} = \text{snd packed}; \\ &\quad \quad \text{attacker-state} = \text{snd } (\text{fst } x); \\ &\quad \quad s = \\ &\quad \quad \quad \text{verifier-state-from-adversary attacker-state} \\ &\quad \quad \quad (\text{staged-proof-transcript data}) \\ &\quad \text{in } \exists \text{ fr f-fri-roots f-final trace-table final-state} \\ &\quad \quad \text{trace-openings trace-table' final-state' trace-openings'}. \\ &\quad \quad \text{alpha-header-supported-partial-trace-table-candidate-witness} \\ &\quad \quad \quad s \text{ fr f-fri-roots f-final trace-table final-state} \\ &\quad \quad \quad \text{trace-openings} \wedge \\ &\quad \quad \text{alpha-header-supported-partial-trace-table-candidate-witness} \\ &\quad \quad \quad s \text{ fr f-fri-roots f-final trace-table' final-state'} \\ &\quad \quad \quad \text{trace-openings'} \wedge \\ &\quad \quad \text{trace-table} \neq \text{trace-table'} \wedge \\ &\quad \quad \neg \text{partial-openings-cross-cover-all-indices trace-openings} \\ &\quad \quad \quad \text{trace-openings'})) \end{aligned}$$

definition *checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad*

$$\begin{aligned} &:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times \\ &\quad 'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times \\ &\quad 'f \text{ protocol-channel}) \times \text{unit list}) \times \\ &\quad 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool} \end{aligned}$$

where

checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad

```

out  $\longleftrightarrow$ 
(case out of
  None  $\Rightarrow$  False
| Some (x, -)  $\Rightarrow$ 
  (let packed = fst (fst x);
    data = snd packed;
    attacker-state = snd (fst x);
    s =
      verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)
  in  $\exists$  fr f-fri-roots f-final trace-table final-state
    trace-openings trace-table' final-state' trace-openings'.
    alpha-header-supported-partial-trace-table-candidate-witness
      s fr f-fri-roots f-final trace-table final-state
      trace-openings  $\wedge$ 
    alpha-header-supported-partial-trace-table-candidate-witness
      s fr f-fri-roots f-final trace-table' final-state'
      trace-openings'  $\wedge$ 
    trace-table  $\neq$  trace-table'  $\wedge$ 
    (merkle-hash-value-conflict final-state final-state'  $\vee$ 
      hash-map-output-collision
        (merkle-hash-state-merge final-state final-state'))))

```

definition *checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad*

```

:: ((((((f  $\times$  'f list  $\times$  'f list  $\times$  'f)  $\times$ 
  'f protocol-channel)  $\times$  'f staged-proof-data)  $\times$ 
  'f protocol-channel)  $\times$  unit list)  $\times$ 
  'f protocol-channel) option  $\Rightarrow$  bool

```

where

checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad

```

out  $\longleftrightarrow$ 
(case out of
  None  $\Rightarrow$  False
| Some (x, -)  $\Rightarrow$ 
  (let packed = fst (fst x);
    data = snd packed;
    attacker-state = snd (fst x);
    s =
      verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)
  in  $\exists$  fr f-fri-roots f-final trace-table final-state
    trace-openings.
    alpha-header-supported-partial-trace-table-candidate-witness
      s fr f-fri-roots f-final trace-table final-state
      trace-openings  $\wedge$ 
    hash-map-output-collision final-state))

```

definition *checked-staged-security-with-actual-alpha-prefix-witnessed-coupling-bad*

```

:: ((((((f × 'f list × 'f list × 'f) ×
      'f protocol-channel) × 'f staged-proof-data) ×
      'f protocol-channel) × unit list) ×
      'f protocol-channel) option ⇒ bool
where
  checked-staged-security-with-actual-alpha-prefix-witnessed-coupling-bad
    out ⇔
    (case out of
      None ⇒ False
    | Some (x, -) ⇒
      (let packed = fst (fst x);
        data = snd packed;
        attacker-state = snd (fst x);
        s =
          verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)
        in ∃ fr f-fri-roots f-final trace-table final-state
          trace-openings trace-table' final-state' trace-openings'.
          alpha-header-supported-partial-trace-table-candidate-witness
            s fr f-fri-roots f-final trace-table final-state
              trace-openings ∧
          alpha-header-supported-partial-trace-table-candidate-witness
            s fr f-fri-roots f-final trace-table' final-state'
              trace-openings' ∧
          trace-table ≠ trace-table' ∧
          merkle-hash-pairwise-coupling-bad final-state final-state'))

lemma checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad-split:
assumes
  checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad
  out
shows
  checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad
  out ∨
  checked-staged-security-with-actual-alpha-prefix-witnessed-coupling-bad
  out
proof (cases out)
case None
then show ?thesis
using assms
unfolding
  checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad-def
by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state0
where out-eq:
  out =
    Some (((((prefix, prefix-state), data), attacker-state), result),

```



```

    final-state0)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from assms obtain fr f-fri-roots f-final trace-table final-state
  trace-openings trace-table' final-state' trace-openings' where
witness:
  alpha-header-supported-partial-trace-table-candidate-witness ?s fr
  f-fri-roots f-final trace-table final-state trace-openings
and witness':
  alpha-header-supported-partial-trace-table-candidate-witness ?s fr
  f-fri-roots f-final trace-table' final-state' trace-openings'
and distinct: trace-table  $\neq$  trace-table'
and pair-bad:
  merkle-hash-value-conflict final-state final-state'  $\vee$ 
  hash-map-output-collision
  (merkle-hash-state-merge final-state final-state')
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad-def
  Let-def
by auto
have split:
  hash-map-output-collision final-state  $\vee$ 
  hash-map-output-collision final-state'  $\vee$ 
  merkle-hash-pairwise-coupling-bad final-state final-state'
by (rule merkle-hash-pairwise-bad-imp-local-collision-or-coupling-bad
  [OF pair-bad])
then show ?thesis
proof
  assume collision: hash-map-output-collision final-state
  have
    checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad
    out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad-def
    Let-def
  apply simp
  apply (intro exI conjI)
  apply (rule witness)
  apply (rule collision)
  done
  then show ?thesis by simp
next
  assume rest:
    hash-map-output-collision final-state'  $\vee$ 
    merkle-hash-pairwise-coupling-bad final-state final-state'
  then show ?thesis
proof

```

```

assume collision': hash-map-output-collision final-state'
have
  checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad
  out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad-def
  Let-def
  apply simp
  apply (intro exI conjI)
    apply (rule witness')
    apply (rule collision')
  done
then show ?thesis by simp
next
assume coupling:
  merkle-hash-pairwise-coupling-bad final-state final-state'
have
  checked-staged-security-with-actual-alpha-prefix-witnessed-coupling-bad
  out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-witnessed-coupling-bad-def
  Let-def
  apply simp
  apply (intro exI conjI)
    apply (rule witness)
    apply (rule witness')
    apply (rule distinct)
    apply (rule coupling)
  done
then show ?thesis by simp
qed
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad-le-local-or-coupling:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad
  adversary-initial-state  $\leq$ 
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad
  adversary-initial-state +
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-witnessed-coupling-bad
  adversary-initial-state
proof –
  let ?M =

```

```

    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?H =
    checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad
  let ?L =
    checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad
  let ?C =
    checked-staged-security-with-actual-alpha-prefix-witnessed-coupling-bad
  have wp-event ?M ?H adversary-initial-state ≤
    wp-event ?M (λout. ?L out ∨ ?C out) adversary-initial-state
  by (rule wp-event-mono)
    (rule
      checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad-split)
  also have ... ≤
    wp-event ?M ?L adversary-initial-state +
    wp-event ?M ?C adversary-initial-state
  by (rule wp-event-union-bound)
  finally show ?thesis .
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad-split:*

assumes

checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
out

shows

checked-staged-security-with-actual-alpha-prefix-witnessed-sparse-cross-bad
out ∨
checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad
out

using *assms*

unfolding

checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad-def
checked-staged-security-with-actual-alpha-prefix-witnessed-sparse-cross-bad-def
checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad-def
alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-def
alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-def
Let-def

by (*auto split: option.splits prod.splits; blast*)

lemma *checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad-le-sparse-or-merkle-side:*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
adversary-initial-state ≤

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-witnessed-sparse-cross-bad
adversary-initial-state +

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)

checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad
adversary-initial-state
proof –
let ?M =
checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let ?W =
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
let ?S =
checked-staged-security-with-actual-alpha-prefix-witnessed-sparse-cross-bad
let ?H =
checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad
have *wp-event ?M ?W adversary-initial-state* ≤
wp-event ?M (λout. ?S out ∨ ?H out) adversary-initial-state
by (rule *wp-event-mono*)
(rule
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad-split)
also have ... ≤
wp-event ?M ?S adversary-initial-state +
wp-event ?M ?H adversary-initial-state
by (rule *wp-event-union-bound*)
finally show ?thesis .
qed

definition *empty-header-partial-trace-candidate-nonunique*
:: (f, 'a) protocol-channel-scheme ⇒
(unit list × (f, 'a) protocol-channel-scheme) option ⇒ bool
where
empty-header-partial-trace-candidate-nonunique s out ⇔
(∃ fr f-fri-roots f-final as dg final trace-query-idxs
trace-openings trace-table composition-table.
accepted-with-empty-composition-header-candidates s out fr
f-fri-roots f-final as dg final trace-query-idxs trace-openings
trace-table composition-table ∧
¬ (∃ trace-table0.
partial-trace-table-candidates trace-openings ⊆
{trace-table0}))

lemma *composition-alpha-partial-opening-union-hit-with-empty-header-split*:
assumes *hit*:
composition-alpha-partial-opening-union-hit-with-empty-header s out
shows
composition-alpha-bad-set-hit-with-empty-composition-header-candidates
s out ∨
empty-header-partial-trace-candidate-nonunique s out
proof –

obtain fr f-fri-roots f-final as dg final trace-query-idxs
trace-openings trace-table composition-table **where** *partial*:
accepted-with-empty-composition-header-candidates s out fr
f-fri-roots f-final as dg final trace-query-idxs trace-openings

```

    trace-table composition-table
  and as-union:
    as ∈ partial-trace-opening-alpha-union-bad-sets trace-openings
  using hit
  unfolding composition-alpha-partial-opening-union-hit-with-empty-header-def
  by blast
from as-union obtain trace-table' where candidate':
  trace-table' ∈ partial-trace-table-candidates trace-openings
  and as-bad': as ∈ composition-trace-bad-alpha-space trace-table'
  unfolding partial-trace-opening-alpha-union-bad-sets-def by blast
have candidate:
  trace-table ∈ partial-trace-table-candidates trace-openings
  using partial
  unfolding accepted-with-empty-composition-header-candidates-def
  partial-trace-table-candidates-def
  by simp
show ?thesis
proof (cases
  ∃ trace-table0.
    partial-trace-table-candidates trace-openings ⊆
      {trace-table0})
case True
then obtain trace-table0 where subset:
  partial-trace-table-candidates trace-openings ⊆
    {trace-table0}
  by blast
have trace-eq: trace-table = trace-table0
  using subset candidate by blast
have trace'-eq: trace-table' = trace-table0
  using subset candidate' by blast
have as ∈ composition-trace-bad-alpha-space trace-table
  using as-bad' trace-eq trace'-eq by simp
then have
  composition-alpha-bad-set-hit-with-empty-composition-header-candidates
    s out
  unfolding
    composition-alpha-bad-set-hit-with-empty-composition-header-candidates-def
  using partial by blast
then show ?thesis by simp
next
case False
have empty-header-partial-trace-candidate-nonunique s out
  unfolding empty-header-partial-trace-candidate-nonunique-def
  using partial False by blast
then show ?thesis by simp
qed
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-nonunique-imp-witnessed-cross-or-merkle-on-s*

```

assumes support:
  out ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
and nonunique:
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  empty-header-partial-trace-candidate-nonunique out
shows
  checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
  out
proof (cases out)
case None
then show ?thesis
  using nonunique
  unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
case (Some result-pack)
then obtain prefix prefix-state data attacker-state result final-state
where out-eq:
  out =
    Some (((((prefix, prefix-state), data), attacker-state),
      result), final-state)
  by (cases result-pack, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have verifier:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  using support
  unfolding out-eq
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
  by (auto elim!: set-dist-bindE split: prod.splits)
have local-nonunique:
  empty-header-partial-trace-candidate-nonunique ?s
  (Some (result, final-state))
  using nonunique
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    Let-def
  by simp
then obtain fr f-fri-roots f-final as dg final trace-query-idxs
  trace-openings trace-table composition-table where empty:
  accepted-with-empty-composition-header-candidates ?s
  (Some (result, final-state)) fr f-fri-roots f-final as dg final
  trace-query-idxs trace-openings trace-table composition-table

```

```

and not-unique-partial:
   $\neg (\exists \text{ trace-table0.}$ 
     $\text{partial-trace-table-candidates } \text{trace-openings} \subseteq$ 
     $\{\text{trace-table0}\})$ 
unfolding empty-header-partial-trace-candidate-nonunique-def
by blast
from empty obtain rest where header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg []
  final rest
and trace-partial:
  accepted-with-partial-trace-openings ?s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
unfolding accepted-with-empty-composition-header-candidates-def
by blast
have partial-subset:
   $\text{partial-trace-table-candidates } \text{trace-openings} \subseteq$ 
   $\text{alpha-header-supported-partial-trace-table-candidates } ?s \text{ fr } f\text{-fri-roots}$ 
   $f\text{-final}$ 
proof
  fix trace-table'
  assume
     $\text{trace-table}' \in \text{partial-trace-table-candidates } \text{trace-openings}$ 
  then have candidate':
     $\text{partial-trace-table-candidate } \text{trace-table}' \text{ trace-openings}$ 
  unfolding partial-trace-table-candidates-def by simp
show
   $\text{trace-table}' \in$ 
   $\text{alpha-header-supported-partial-trace-table-candidates } ?s \text{ fr}$ 
   $f\text{-fri-roots } f\text{-final}$ 
unfolding alpha-header-supported-partial-trace-table-candidates-def
by (intro CollectI exI[of - Some (result, final-state)])
  exI[of - trace-query-idxs] exI[of - trace-openings]
  exI[of - as] exI[of - dg] exI[of - []]
  exI[of - final] exI[of - rest] conjI)
  (use verifier trace-partial candidate' header in simp-all)
qed
have not-unique-header:
   $\neg (\exists \text{ trace-table0.}$ 
     $\text{alpha-header-supported-partial-trace-table-candidates } ?s \text{ fr}$ 
     $f\text{-fri-roots } f\text{-final} \subseteq \{\text{trace-table0}\})$ 
proof
  assume
     $\exists \text{ trace-table0.}$ 
     $\text{alpha-header-supported-partial-trace-table-candidates } ?s \text{ fr}$ 
     $f\text{-fri-roots } f\text{-final} \subseteq \{\text{trace-table0}\}$ 
  then obtain trace-table0 where header-subset:
     $\text{alpha-header-supported-partial-trace-table-candidates } ?s \text{ fr}$ 
     $f\text{-fri-roots } f\text{-final} \subseteq \{\text{trace-table0}\}$ 
  by blast

```

```

have
  partial-trace-table-candidates trace-openings  $\subseteq$ 
    {trace-table0}
  using partial-subset header-subset by blast
then show False
  using not-unique-partial by blast
qed
have not-unique-witnessed:
   $\neg (\exists \text{ trace-table0.}$ 
    alpha-header-supported-witnessed-partial-trace-table-candidates ?s fr
      f-fri-roots f-final  $\subseteq \{\text{trace-table0}\})$ 
  using not-unique-header
  by (simp add: alpha-header-supported-witnessed-candidates-eq-candidates)
have
  alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad
    ?s fr f-fri-roots f-final
  by (rule
    alpha-header-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-if-not-singleton
      [OF not-unique-witnessed])
then have
  alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad ?s
unfolding
  alpha-supported-witnessed-partial-trace-pairwise-cross-or-merkle-bad-def
by blast
then show ?thesis
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad-def
  Let-def
by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-nonunique-le-witnessed-cross-or-merkle:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      empty-header-partial-trace-candidate-nonunique)
  adversary-initial-state  $\leq$ 
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
    adversary-initial-state
by (rule wp-event-mono-on-support)
  (rule
    checked-staged-security-with-actual-alpha-prefix-empty-header-nonunique-imp-witnessed-cross-or-merkle-on-support)

definition checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
  :: (((('f  $\times$  'f list  $\times$  'f list  $\times$  'f)  $\times$ 
    'f protocol-channel)  $\times$  'f staged-proof-data)  $\times$ 
    'f protocol-channel)  $\times$  unit list)  $\times$ 

```



```

      'f protocol-channel) option  $\Rightarrow$  bool
where
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
    out  $\longleftrightarrow$ 
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
      composition-trace-bad-alpha-space out  $\wedge$ 
    checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-partial-opening-union-hit-with-empty-header out

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-le-fixed-or-nonunique:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
    adversary-initial-state  $\leq$ 
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
    adversary-initial-state +
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      empty-header-partial-trace-candidate-nonunique)
    adversary-initial-state
proof -
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?fixed =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-bad-set-hit-with-empty-composition-header-candidates
  let ?nonunique =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
      empty-header-partial-trace-candidate-nonunique
  have event-le:
    wp-event ?M
      checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
      adversary-initial-state  $\leq$ 
    wp-event ?M ( $\lambda$ out. ?fixed out  $\vee$  ?nonunique out)
      adversary-initial-state
  by (rule wp-event-mono)
    (auto simp:
      checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-def
      checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      split: option.splits prod.splits
      dest:
        composition-alpha-partial-opening-union-hit-with-empty-header-split)
  also have ...  $\leq$ 
    wp-event ?M ?fixed adversary-initial-state +
    wp-event ?M ?nonunique adversary-initial-state

```

by (rule wp-event-union-bound)
 finally show ?thesis .
 qed

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-le-fixed-or-witnessed-cross:*

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
 adversary-initial-state $\leq$
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
 adversary-initial-state +
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
 adversary-initial-state

proof –

let ?M =
 checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
 let ?fixed =
 checked-staged-security-with-actual-alpha-prefix-verifier-event
 composition-alpha-bad-set-hit-with-empty-composition-header-candidates
 let ?nonunique =
 checked-staged-security-with-actual-alpha-prefix-verifier-event
 empty-header-partial-trace-candidate-nonunique

have base:

wp-event ?M
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
 adversary-initial-state $\leq$
wp-event ?M ?fixed adversary-initial-state +
wp-event ?M ?nonunique adversary-initial-state
 by (rule
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-le-fixed-or-nonunique)

have nonunique-bound:

wp-event ?M ?nonunique adversary-initial-state $\leq$
wp-event ?M
 checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
 adversary-initial-state

by (rule
 checked-staged-security-with-actual-alpha-prefix-empty-header-nonunique-le-witnessed-cross-or-merkle)

show ?thesis

by (rule order-trans[OF base])
 (intro add-mono order-refl nonunique-bound)

qed

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-bound-from-fixed-and-witnessed-cross:*

assumes fixed-bound:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
 adversary-initial-state $\leq F$
and cross-bound:
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
 adversary-initial-state $\leq X$
shows
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state *A*)
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
 adversary-initial-state $\leq F + X$
by (rule order-trans[
 OF
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-le-fixed-or-witnessed-cross])
 (intro add-mono fixed-bound cross-bound)

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-bound-from-fixed-and-nonunique:
assumes fixed-bound:
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
 adversary-initial-state $\leq F$
and nonunique-bound:
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 empty-header-partial-trace-candidate-nonunique)
 adversary-initial-state $\leq N$
shows
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state *A*)
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
 adversary-initial-state $\leq F + N$
by (rule order-trans[
 OF
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-le-fixed-or-nonunique])
 (intro add-mono fixed-bound nonunique-bound)

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-le-relevant-drift:
wp-event

```

    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
    adversary-initial-state ≤
wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
    adversary-initial-state
by (rule wp-event-mono)
    (auto simp:
      checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-def
      checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-def)

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-bound-from-relevant-drift:
assumes drift-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
    adversary-initial-state ≤ D
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
    adversary-initial-state ≤ D
by (rule order-trans[
  OF
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-le-relevant-drift
    drift-bound])

lemma checked-staged-security-with-actual-alpha-prefix-fixed-empty-alpha-hit-imp-prefix-or-not-prefix-on-support:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  out ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
and fixed-hit:
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-alpha-bad-set-hit-with-empty-composition-header-candidates
  out
shows
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
  composition-trace-bad-alpha-space out ∨
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
  composition-trace-bad-alpha-space out
proof (cases out)

```

```

case None
then show ?thesis
  using fixed-hit
  unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
case (Some result-pack)
then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state),
        result), final-state)
  by (cases result-pack, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have verify-out:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  using support
  unfolding out-eq
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
  by (auto elim!: set-dist-bindE split: prod.splits)
have local-fixed:
composition-alpha-bad-set-hit-with-empty-composition-header-candidates ?s
  (Some (result, final-state))
using fixed-hit
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  Let-def
by simp
then obtain fr f-fri-roots f-final as dg final trace-query-idxs
  trace-openings trace-table composition-table rest where empty:
accepted-with-empty-composition-header-candidates ?s
  (Some (result, final-state)) fr f-fri-roots f-final as dg final
  trace-query-idxs trace-openings trace-table composition-table
and header:
  verifier-header-transcript ?s fr f-fri-roots f-final as dg []
  final rest
and trace-partial:
  accepted-with-partial-trace-openings ?s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and as-bad:
  as ∈ composition-trace-bad-alpha-space trace-table
unfolding
  composition-alpha-bad-set-hit-with-empty-composition-header-candidates-def
  accepted-with-empty-composition-header-candidates-def
by blast

```

```

from verify-monad-accepted-transcript-shape[OF verify-out]
obtain as' query-idxs where shape':
  accepted-transcript-shape ?s (Some (result, final-state)) as'
    query-idxs
by blast
from shape' obtain result' final-state' fr' f-fri-roots' f-final' dg'
  composition-fri-roots' final' rest' where out-shape:
    Some (result, final-state) = Some (result', final-state')
and header':
  verifier-header-transcript ?s fr' f-fri-roots' f-final' as' dg'
    composition-fri-roots' final' rest'
by (elim accepted-transcript-shape-header-query-extraction)
have as'-eq: as' = as
  using verifier-header-transcript-unique[OF header' header] by simp
have shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
    query-idxs
  using shape' as'-eq by simp
have header-union:
  as ∈
    alpha-header-supported-partial-union-bad-sets ?s
      composition-trace-bad-alpha-space fr f-fri-roots f-final
proof —
  have candidate:
    trace-table ∈
      alpha-header-supported-partial-trace-table-candidates ?s fr
        f-fri-roots f-final
  unfolding alpha-header-supported-partial-trace-table-candidates-def
  by (intro CollectI exI[of - Some (result, final-state)]
    exI[of - trace-query-idxs] exI[of - trace-openings]
    exI[of - as] exI[of - dg] exI[of - []]
    exI[of - final] exI[of - rest] conjI)
    (use verify-out trace-partial trace-candidate header in simp-all)
  show ?thesis
  unfolding alpha-header-supported-partial-union-bad-sets-def
  using candidate as-bad
    composition-trace-bad-alpha-space-subset-alpha-space[of trace-table]
  by auto
qed
have builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
proof —
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)

```

```

      adversary-initial-state)
    by (rule
      checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
      [OF support[unfolded out-eq]])
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  show ?thesis
    by blast
qed
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
have prefix-root-eq: staged-trace-root data = fst prefix
  using support out-eq
  unfolding checked-staged-security-experiment-with-actual-alpha-prefix-data-state-def
    checked-staged-transcript-with-alpha-prefix-program-def
    checked-staged-after-alpha-prefix-program-def Let-def
  by (auto elim!: set-dist-bindE split: prod.splits)
have as-eq: as = staged-alphas data
  using verifier-header-transcript-unique[OF header staged-header]
  by simp
have fr-eq: fr = staged-trace-root data
  using verifier-header-transcript-unique[OF header staged-header]
  by simp
show ?thesis
proof (cases
  as ∈ alpha-prefix-union-bad-sets prefix-state
    composition-trace-bad-alpha-space fr)
case True
have
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit-def
    Let-def
  using True as-eq fr-eq prefix-root-eq by simp
then show ?thesis by simp
next
case not-prefix: False
have not-bound:

```

```

    alpha-header-partial-candidate-not-prefix-bound prefix-state ?s fr
    f-fri-roots f-final
  by (rule alpha-header-partial-union-hit-not-prefix-hit-imp-not-prefix-bound
      [OF header-union not-prefix])
  have
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
      composition-trace-bad-alpha-space out
  unfolding out-eq
  by (rule
      checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-boundI
      [OF refl shape header header-union not-prefix not-bound])
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-fixed-empty-alpha-hit-le-prefix-or-not-prefix:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
      adversary-initial-state ≤
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
        composition-trace-bad-alpha-space)
      adversary-initial-state +
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
        composition-trace-bad-alpha-space)
      adversary-initial-state
proof -
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?Fixed =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-alpha-bad-set-hit-with-empty-composition-header-candidates
  let ?Prefix =
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space
  let ?NotPrefix =
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
    composition-trace-bad-alpha-space
  have event-le:
    wp-event ?M ?Fixed adversary-initial-state ≤
    wp-event ?M (λout. ?Prefix out ∨ ?NotPrefix out)

```


adversary-initial-state
by (rule *wp-event-mono-on-support*)
(rule
checked-staged-security-with-actual-alpha-prefix-fixed-empty-alpha-hit-imp-prefix-or-not-prefix-on-support
[*OF wf controlled*])
also have ... $\leq$
wp-event ?M ?Prefix adversary-initial-state +
wp-event ?M ?NotPrefix adversary-initial-state
by (rule *wp-event-union-bound*)
finally show ?thesis .
qed

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-imp-prefix-prequery-or-empty-relevant*
assumes *support*:
out $\in$
set-dist
(execute
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
adversary-initial-state)
and *hit*:
checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-partial-opening-union-hit-with-empty-header out
shows
checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space out $\vee$
checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
composition-trace-bad-alpha-space out $\vee$
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
out
proof –
have *partial*:
checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit
composition-trace-bad-alpha-space out
by (rule
checked-staged-security-with-actual-alpha-prefix-empty-header-partial-alpha-hit-imp-partial-header-bad-set-hit-on-
[*OF support hit*])
have *fresh-or-prequery*:
checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
composition-trace-bad-alpha-space out $\vee$
checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
composition-trace-bad-alpha-space out
by (rule
checked-staged-security-with-actual-alpha-prefix-partial-header-bad-set-hit-imp-fresh-or-prequery
[*OF partial*])
then show ?thesis
proof
assume *fresh*:
checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit

```

    composition-trace-bad-alpha-space out
have prefix-or-drift:
  checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space out  $\vee$ 
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
    composition-trace-bad-alpha-space out
by (rule
  checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-imp-prefix-or-drift
    [OF fresh])
then show ?thesis
proof
  assume
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space out
  then show ?thesis by simp
next
  assume drift:
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
    composition-trace-bad-alpha-space out
  have
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
    out
  unfolding
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-def
  using drift hit by simp
  then show ?thesis by simp
qed
next
  assume
    checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
    composition-trace-bad-alpha-space out
  then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-bound-from-empty-relevant-drift-and-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and drift-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
  adversary-initial-state  $\leq D$ 
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-partial-opening-union-hit-with-empty-header)

```

$$\begin{aligned}
& \text{adversary-initial-state} \leq \\
& \text{composition-error-bound} + \\
& \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \text{hash-collision-budget-value } 0 \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& D
\end{aligned}$$

proof –

let $?M =$
checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

let $?Prefix =$
checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space

let $?Prequery =$
checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
composition-trace-bad-alpha-space

let $?Drift =$
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift

let $?Hit =$
checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-partial-opening-union-hit-with-empty-header

let $?P =$
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0)

let $?Q =$
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0)

have *transcript-bound*:
wp-event (checked-staged-transcript-with-alpha-prefix-program A)
(checked-staged-transcript-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state ≤ ?P

by (rule
checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
[OF wf controlled])

have *prefix-bound*:
wp-event ?M ?Prefix adversary-initial-state ≤ ?P

by (rule
checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
[OF transcript-bound])

have *prequery-bound*:
wp-event ?M ?Prequery adversary-initial-state ≤ ?Q

by (rule
checked-staged-security-with-actual-alpha-prefix-vector-alpha-prequery-accounting

```

    [OF wf controlled])
have wp-event ?M ?Hit adversary-initial-state ≤
  wp-event ?M
  (λout. ?Prefix out ∨ ?Prequery out ∨ ?Drift out ∨ False)
  adversary-initial-state
by (rule wp-event-mono-on-support)
  (use
    checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-imp-prefix-prequery-or-empty-relevant-drift
    [of - A] in blast)
also have ... ≤
  wp-event ?M ?Prefix adversary-initial-state +
  wp-event ?M ?Prequery adversary-initial-state +
  wp-event ?M ?Drift adversary-initial-state +
  wp-event ?M (λ-. False) adversary-initial-state
by (rule wp-event-union-bound4)
also have ... ≤ ?P + ?Q + D + 0
by (intro add-mono prefix-bound prequery-bound drift-bound)
  (simp add: wp-event-def wp-def dist-expect-def)
finally show ?thesis
by (simp add: algebra-simps)
qed

lemma checked-staged-security-with-data-state-empty-header-composition-randomization-bound-from-empty-relevant-drift
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and drift-bound:
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
  adversary-initial-state ≤ D
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-randomization-bad-with-empty-composition-header-candidates)
  adversary-initial-state ≤
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  D
proof -
let ?M = checked-staged-security-experiment-with-data-state A
let ?random =
  staged-security-with-data-state-verifier-event
  composition-randomization-bad-with-empty-composition-header-candidates

```

```

let ?alpha =
  staged-security-with-data-state-verifier-event
  composition-alpha-partial-opening-union-hit-with-empty-header
have alpha-projected:
  wp-event ?M ?alpha adversary-initial-state =
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-partial-opening-union-hit-with-empty-header)
    adversary-initial-state
  by (rule
    checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)
have alpha-bound:
  wp-event ?M ?alpha adversary-initial-state ≤
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D
  unfolding alpha-projected
  by (rule
    checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-bound-from-empty-relevant-drift-and-bud
    [OF wf controlled drift-bound])
have wp-event ?M ?random adversary-initial-state ≤
  wp-event ?M ?alpha adversary-initial-state
  by (rule wp-event-mono)
  (auto simp: staged-security-with-data-state-verifier-event-def
    intro:
      composition-alpha-bad-set-hit-with-empty-header-imp-partial-opening-union-hit
      composition-randomization-bad-with-empty-composition-header-candidates-imp-alpha-hit
      split: option.splits prod.splits)
also have ... ≤
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  D
  by (rule alpha-bound)
finally show ?thesis .
qed

```

lemma checked-staged-security-with-data-state-empty-header-bad-event-bound-from-empty-relevant-drift-trace-fri-and
assumes false-statement: $\neg$ exists-valid-trace

and *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *drift-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
 adversary-initial-state $\leq D$
and *trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ *trace-fri-error*
and *query-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 query-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ *empty-query-error*
shows
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 soundness-bad-event-partial-candidate-empty-header)
 adversary-initial-state $\leq$
 composition-error-bound +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 D + trace-fri-error + empty-query-error
proof –
let *?M* = *checked-staged-security-experiment-with-data-state A*
let *?E* = *staged-security-with-data-state-verifier-event*
let *?degree* =
 ?E composition-degree-bad-with-empty-composition-header-candidates
let *?random* =
 ?E composition-randomization-bad-with-empty-composition-header-candidates
let *?trace* =
 ?E trace-fri-bad-with-empty-composition-header-candidates
let *?query* =
 ?E query-bad-with-empty-composition-header-candidates
let *?bad* = *?E soundness-bad-event-partial-candidate-empty-header*
have *degree-false*: *?degree* = (λ out. *False*)
 by (*rule ext*)
 (*auto simp*: *staged-security-with-data-state-verifier-event-def*
 composition-degree-bad-with-empty-composition-header-candidates-false
 split: *option.splits prod.splits*)
have *degree-bound*:

```

wp-event ?M ?degree adversary-initial-state ≤ 0
unfolding degree-false wp-event-def wp-def dist-expect-def by simp
have random-bound:
wp-event ?M ?random adversary-initial-state ≤
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  D
by (rule
  checked-staged-security-with-data-state-empty-header-composition-randomization-bound-from-empty-relevant-dri
  [OF wf controlled drift-bound])
have wp-event ?M ?bad adversary-initial-state ≤
  wp-event ?M
  (λout. ?degree out ∨ ?random out ∨ ?trace out ∨
    ?query out)
  adversary-initial-state
proof (rule wp-event-mono)
fix out
assume bad: ?bad out
show ?degree out ∨ ?random out ∨ ?trace out ∨ ?query out
proof (cases out)
case None
then show ?thesis
using bad
unfolding staged-security-with-data-state-verifier-event-def
by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have local-bad:
  soundness-bad-event-partial-candidate-empty-header ?s
  (Some (result, final-state))
using bad
unfolding out-eq staged-security-with-data-state-verifier-event-def
by simp
then consider
  (comp)
  composition-bad-with-empty-composition-header-candidates ?s
  (Some (result, final-state))
  | (trace)

```

```

    trace-fri-bad-with-empty-composition-header-candidates ?s
    (Some (result, final-state))
  | (query)
    query-bad-with-empty-composition-header-candidates ?s
    (Some (result, final-state))
  unfolding soundness-bad-event-partial-candidate-empty-header-def
  by blast
then show ?thesis
proof cases
case comp
then have comp-split:
  composition-degree-bad-with-empty-composition-header-candidates ?s
  (Some (result, final-state)) ∨
  composition-randomization-bad-with-empty-composition-header-candidates
  ?s (Some (result, final-state))
by (rule
  composition-bad-with-empty-composition-header-candidates-split
  [OF false-statement])
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  using comp-split by (auto split: option.splits prod.splits)
next
case trace
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
next
case query
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
qed
also have ... ≤
  wp-event ?M ?degree adversary-initial-state +
  wp-event ?M ?random adversary-initial-state +
  wp-event ?M ?trace adversary-initial-state +
  wp-event ?M ?query adversary-initial-state
by (rule wp-event-union-bound4)
also have ... ≤
  0 +
  (composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +

```


$D) +$
 $\text{trace-fri-error} + \text{empty-query-error}$
by (intro add-mono degree-bound random-bound trace-fri-bound query-bound)
finally show ?thesis
by (simp add: algebra-simps)
qed

lemma checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix
fixes trace-fri-error' composition-fri-error' D empty-query-error :: prob
and C :: nat $\Rightarrow$ prob
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}'$
and comp-fri-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$
and round-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and drift-bound:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift}$
 $\text{adversary-initial-state} \leq D$
and empty-trace-fri-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
and current-empty-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows checked-staged-adversary-acceptance-probability A $\leq$
hash-collision-budget-value 0
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds}. C \ i) +$

$\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum_{i < \text{rounds}} C\ i) +$
 $(\text{composition-error-bound} +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad \text{hash-collision-budget-value } 0$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $D + \text{trace-fri-error} + \text{empty-query-error})$
proof –
have *comp-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{composition-randomization-bad-with-aligned-transcript-partial-candidates})$
 $\text{adversary-initial-state} \leq (\sum_{i < \text{rounds}} C\ i)$
by (rule
 $\text{checked-staged-security-with-data-state-aligned-partial-candidate-randomization-bound-from-query-prefix-current-}$
 $\quad [\text{OF wf controlled round-bound}])$
have *current-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-current-query-partial-opening-hit}$
 $\text{adversary-initial-state} \leq (\sum_{i < \text{rounds}} C\ i)$
by (rule
 $\text{checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds}$
 $\quad [\text{OF round-bound wf controlled}])$
have *query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{query-bad-with-aligned-transcript-partial-candidates})$
 $\text{adversary-initial-state} \leq (\sum_{i < \text{rounds}} C\ i)$
by (rule *order-trans*[
 $\text{OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit}$
 $\quad \text{current-query-bound}])$
 $(\text{use wf controlled in simp-all})$
have *merkle-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\quad \text{partial-merkle-inconsistency-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets } +$
 $\quad \text{staged-challenge-query-budget } + \text{verifier-hash-query-budget})$
by (rule
 $\text{checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled}$
 $\quad [\text{OF wf controlled}])$
have *empty-query-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$

```

    query-bad-with-empty-composition-header-candidates)
  adversary-initial-state ≤ empty-query-error
by (rule order-trans[
  OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero[
    OF wf controlled] current-empty-bound])
have empty-header-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      soundness-bad-event-partial-candidate-empty-header)
  adversary-initial-state ≤
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  D + trace-fri-error + empty-query-error
by (rule
  checked-staged-security-with-data-state-empty-header-bad-event-bound-from-empty-relevant-drift-trace-fri-and-qu
  [OF false-statement wf controlled drift-bound empty-trace-fri-bound
    empty-query-bound])
have bound:
  checked-staged-adversary-acceptance-probability A ≤
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget) +
  (∑ i<rounds. C i) +
  trace-fri-error' + composition-fri-error' +
  (∑ i<rounds. C i) +
  (composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D + trace-fri-error + empty-query-error)
by (rule
  checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-components-and-empt
  [OF false-statement merkle-bound comp-bound trace-fri-bound
    comp-fri-bound query-bound empty-header-bound])
then show ?thesis .
qed

lemma checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-structured-query-fr
  fixes trace-fri-error' composition-fri-error' D empty-query-error
  header-query-error :: prob
  and C :: nat ⇒ prob

```

assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ *composition-fri-error'*
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i*)
 adversary-initial-state $\leq C i$
and *header-query-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 (*checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix*
 A)
 adversary-initial-state $\leq$ *header-query-error*
and *drift-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
 adversary-initial-state $\leq D$
and *empty-trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ *trace-fri-error*
and *current-empty-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-current-empty-query-opening-hit-at 0*)
 adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
 hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
 staged-challenge-query-budget + *verifier-hash-query-budget*) +
 ($\sum i < \text{rounds}. C i$) +
 trace-fri-error' + *composition-fri-error'* +
 header-query-error +
 (*composition-error-bound* +

staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D + trace-fri-error + empty-query-error)
proof –
have *comp-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-randomization-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state ≤ (∑ i<rounds. C i)
by (rule
checked-staged-security-with-data-state-aligned-partial-candidate-randomization-bound-from-query-prefix-current-
[OF wf controlled round-bound])
have *query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state ≤ header-query-error
by (rule
checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-header-authenticated-candidate-
[OF wf controlled header-query-bound])
have *merkle-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad)
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
[OF wf controlled])
have *empty-query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤ empty-query-error
by (rule *order-trans*[
OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero
*OF wf controlled] *current-empty-bound*])
have *empty-header-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state ≤
*composition-error-bound +**

staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
D + trace-fri-error + empty-query-error
by (rule
checked-staged-security-with-data-state-empty-header-bad-event-bound-from-empty-relevant-drift-trace-fri-and-qu
 [*OF false-statement wf controlled drift-bound empty-trace-fri-bound*
empty-query-bound])
have bound:
checked-staged-adversary-acceptance-probability A ≤
 (*hash-collision-budget-value 0*
 (*staged-attacker-query-budget budgets +*
staged-challenge-query-budget + verifier-hash-query-budget) +
 ($\sum i < \text{rounds}. C\ i$) + *trace-fri-error' +*
composition-fri-error' + header-query-error) +
 (*composition-error-bound +*
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
D + trace-fri-error + empty-query-error)
by (rule
checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-components-and-empt
 [*OF false-statement merkle-bound comp-bound trace-fri-bound*
comp-fri-bound query-bound empty-header-bound])
then show ?thesis
by (simp add: algebra-simps)
qed

lemma *checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix*
fixes *trace-fri-error' composition-fri-error' D empty-query-error :: prob*
and *C :: nat ⇒ prob*
assumes *false-statement: ¬ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state ≤ trace-fri-error'
and *comp-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)

$adversary-initial-state \leq composition-fri-error'$
and *round-bound*:
 $\bigwedge i. i < rounds \implies$
 $wp-event$
 $(checked-staged-security-experiment-with-query-prefix-data-state\ A\ i)$
 $(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at\ i)$
 $adversary-initial-state \leq C\ i$
and *drift-bound*:
 $wp-event$
 $(checked-staged-security-experiment-with-actual-alpha-prefix-data-state\ A)$
 $checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift$
 $adversary-initial-state \leq D$
and *empty-trace-fri-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $\quad trace-fri-bad-with-empty-composition-header-candidates)$
 $adversary-initial-state \leq trace-fri-error$
and *current-empty-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-current-empty-query-opening-hit-at\ 0)$
 $adversary-initial-state \leq empty-query-error$
shows $checked-staged-adversary-acceptance-probability\ A \leq$
 $hash-collision-budget-value\ 0$
 $(staged-attacker-query-budget\ budgets +$
 $\quad staged-challenge-query-budget + verifier-hash-query-budget) +$
 $(\sum i < rounds. C\ i) +$
 $trace-fri-error' + composition-fri-error' +$
 $(composition-error-bound +$
 $\quad staged-phase-relation-error\ size$
 $\quad (staged-alpha-search-queries\ budgets\ 0) +$
 $hash-collision-budget-value\ 0$
 $\quad (staged-alpha-search-queries\ budgets\ 0) +$
 $\quad staged-phase-relation-error\ size$
 $\quad (staged-alpha-search-queries\ budgets\ 0) +$
 $\quad D + trace-fri-error + empty-query-error)$
proof –
have *current-query-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $staged-security-with-data-state-current-query-partial-opening-hit$
 $adversary-initial-state \leq (\sum i < rounds. C\ i)$
by (rule
 $checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds$
 $[OF\ round-bound\ wf\ controlled])$
have *merkle-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $\quad partial-merkle-inconsistency-bad)$
 $adversary-initial-state \leq$

hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
[OF wf controlled])
have *empty-query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤ empty-query-error
by (rule *order-trans*[
OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero
OF wf controlled] current-empty-bound])
have *empty-header-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state ≤
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D + trace-fri-error + empty-query-error
by (rule
checked-staged-security-with-data-state-empty-header-bad-event-bound-from-empty-relevant-drift-trace-fri-and-qu
[OF false-statement wf controlled drift-bound empty-trace-fri-bound
empty-query-bound])
have *partial-candidate-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle)
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
($\sum i < \text{rounds. } C\ i$) + trace-fri-error' + composition-fri-error'
by (rule
checked-staged-security-with-data-state-aligned-transcript-partial-candidate-aligned-randomization-with-merkle-c
[OF wf controlled merkle-bound current-query-bound trace-fri-bound
comp-fri-bound])
have *bound*:
checked-staged-adversary-acceptance-probability A ≤
(hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +


```

    ( $\sum_{i < \text{rounds}} C\ i$ ) + trace-fri-error' +
    composition-fri-error') +
    (composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D + trace-fri-error + empty-query-error)
  by (rule
    checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-and-empty-header-bound
    [OF false-statement partial-candidate-bound empty-header-bound])
  then show ?thesis
  by (simp add: algebra-simps)
qed

end

end

```

```

theory Soundness-Relevant-Drift-Public
imports
  Soundness
  Soundness-Staged-Empty-Relevant-Drift
begin

```

Public migration wrappers for the refined relevant-drift route.

These theorems sit downstream of *Soundness* so the main public compatibility layer does not keep growing. The route below derives the aligned composition-randomization branch and aligned query branch from the same per-round current-query prefix bounds. It exposes only the empty-header-specific relevant-drift event and keeps the partial-candidate FRI premises internal to this route.

```

context soundness
begin

```

```

theorem stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query:
  fixes trace-fri-error' composition-fri-error' D empty-query-error :: prob
  and C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-partial-candidates)
  adversary-initial-state  $\leq$  trace-fri-error'

```

and *comp-fri-bound*:
 $wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $\quad composition\text{-}fri\text{-}bad\text{-}with\text{-}partial\text{-}candidates)$
 $adversary\text{-}initial\text{-}state \leq composition\text{-}fri\text{-}error'$
and *round-bound*:
 $\bigwedge i. i < rounds \implies$
 $wp\text{-}event$
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}query\text{-}prefix\text{-}data\text{-}state \ A \ i)$
 $(checked\text{-}staged\text{-}security\text{-}with\text{-}query\text{-}prefix\text{-}current\text{-}query\text{-}partial\text{-}opening\text{-}hit\text{-}at \ i)$
 $adversary\text{-}initial\text{-}state \leq C \ i$
and *drift-bound*:
 $wp\text{-}event$
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}data\text{-}state$
 $\quad A)$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}empty\text{-}header\text{-}relevant\text{-}drift$
 $adversary\text{-}initial\text{-}state \leq D$
and *empty-trace-fri-bound*:
 $wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $\quad trace\text{-}fri\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates)$
 $adversary\text{-}initial\text{-}state \leq trace\text{-}fri\text{-}error$
and *current-empty-bound*:
 $wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}current\text{-}empty\text{-}query\text{-}opening\text{-}hit\text{-}at \ 0)$
 $adversary\text{-}initial\text{-}state \leq empty\text{-}query\text{-}error$
shows $checked\text{-}staged\text{-}adversary\text{-}acceptance\text{-}probability \ A \leq$
 $hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $(staged\text{-}attacker\text{-}query\text{-}budget \ budgets +$
 $\quad staged\text{-}challenge\text{-}query\text{-}budget + verifier\text{-}hash\text{-}query\text{-}budget) +$
 $(\sum i < rounds. C \ i) +$
 $trace\text{-}fri\text{-}error' + composition\text{-}fri\text{-}error' +$
 $(\sum i < rounds. C \ i) +$
 $(composition\text{-}error\text{-}bound +$
 $\quad staged\text{-}phase\text{-}relation\text{-}error \ size$
 $\quad (staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) +$
 $hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $\quad (staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) +$
 $D +$
 $\quad staged\text{-}phase\text{-}relation\text{-}error \ size$
 $\quad (staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) +$
 $\quad trace\text{-}fri\text{-}error + empty\text{-}query\text{-}error)$
proof –
have *bound*:
 $checked\text{-}staged\text{-}adversary\text{-}acceptance\text{-}probability \ A \leq$
 $hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $(staged\text{-}attacker\text{-}query\text{-}budget \ budgets +$
 $\quad staged\text{-}challenge\text{-}query\text{-}budget + verifier\text{-}hash\text{-}query\text{-}budget) +$
 $(\sum i < rounds. C \ i) +$

```

    trace-fri-error' + composition-fri-error' +
    (∑ i < rounds. C i) +
    (composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D + trace-fri-error + empty-query-error)
  by (rule
    checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix-ro
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
    round-bound drift-bound empty-trace-fri-bound current-empty-bound])
  then show ?thesis
  by (simp add: algebra-simps)
qed

```

theorem stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-single-query

fixes trace-fri-error' composition-fri-error' D empty-query-error :: prob

and C :: nat $\Rightarrow$ prob

assumes false-statement: $\neg$ exists-valid-trace

and wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and trace-fri-bound:

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

trace-fri-bad-with-partial-candidates)

adversary-initial-state $\leq$ trace-fri-error'

and comp-fri-bound:

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

composition-fri-bad-with-partial-candidates)

adversary-initial-state $\leq$ composition-fri-error'

and round-bound:

$\bigwedge i. i < \text{rounds} \implies$

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)

adversary-initial-state $\leq$ C i

and drift-bound:

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state

A)

checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift

adversary-initial-state $\leq$ D

and empty-trace-fri-bound:

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

```

    trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state ≤ trace-fri-error
and current-empty-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
    adversary-initial-state ≤ empty-query-error
shows checked-staged-adversary-acceptance-probability A ≤
  hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget + verifier-hash-query-budget) +
  (∑ i<rounds. C i) +
  trace-fri-error' + composition-fri-error' +
  (composition-error-bound +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
   (staged-alpha-search-queries budgets 0) +
  D +
  staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
  trace-fri-error + empty-query-error)
proof –
  have bound:
    checked-staged-adversary-acceptance-probability A ≤
    hash-collision-budget-value 0
      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget + verifier-hash-query-budget) +
    (∑ i<rounds. C i) +
    trace-fri-error' + composition-fri-error' +
    (composition-error-bound +
     staged-phase-relation-error size
     (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
     (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
     (staged-alpha-search-queries budgets 0) +
    D + trace-fri-error + empty-query-error)
  by (rule
    checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix-ro
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
     round-bound drift-bound empty-trace-fri-bound current-empty-bound])
  then show ?thesis
  by (simp add: algebra-simps)
qed

end

end

```

```

theory Soundness-Relevant-Drift-Reachable-FRI
  imports Soundness-Relevant-Drift-Public
begin

```

Reachable partial-candidate FRI interface for the refined relevant-drift migration path. This layer does not change the compatibility predicates *trace-fri-bad-with-partial-openings* and *composition-fri-bad-with-partial-openings*. Instead it records the partial-candidate FRI obligations used by the aligned partial-opening route: the verifier has authenticated sampled openings, and the FRI low-degree interface rules out a low-degree violation for any candidate table compatible with those openings.

```

context soundness
begin

```

```

definition trace-fri-bad-with-reachable-partial-candidate
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool

```

where

```

  trace-fri-bad-with-reachable-partial-candidate s out  $\longleftrightarrow$ 
    ( $\exists$  fr query-idxs trace-openings trace-table.
      accepted-with-partial-trace-openings s out fr query-idxs
        trace-openings  $\wedge$ 
      partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
       $\neg$  trace-table-low-degree trace-table)

```

```

definition composition-fri-bad-with-reachable-partial-candidate
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool

```

where

```

  composition-fri-bad-with-reachable-partial-candidate s out  $\longleftrightarrow$ 
    ( $\exists$  fr f-fri-roots f-final as dg composition-fri-roots final
      trace-query-idxs trace-openings composition-query-idxs
      composition-openings trace-table composition-table.
      accepted-with-partial-initial-openings s out fr f-fri-roots f-final
        as dg composition-fri-roots final trace-query-idxs trace-openings
          composition-query-idxs composition-openings  $\wedge$ 
      partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
      partial-composition-table-candidate composition-table
        composition-openings  $\wedge$ 
      trace-table-low-degree trace-table  $\wedge$ 
       $\neg$  composition-table-low-degree maxDegree composition-table)

```

```

definition trace-fri-reachable-partial-candidate-reduction-assumption
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$  bool

```

where

```

  trace-fri-reachable-partial-candidate-reduction-assumption s  $\longleftrightarrow$ 
    wp-event verify-monad
      (trace-fri-bad-with-reachable-partial-candidate s) s  $\leq$ 

```

trace-fri-error

definition *composition-fri-reachable-partial-candidate-reduction-assumption*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

composition-fri-reachable-partial-candidate-reduction-assumption *s*

$\longleftrightarrow$

wp-event verify-monad

$(\text{composition-fri-bad-with-reachable-partial-candidate } s) \leq$

composition-fri-error

lemma *trace-fri-bad-with-partial-candidates-imp-reachable-partial-candidate:*

assumes *trace-fri-bad-with-partial-candidates* *s out*

shows *trace-fri-bad-with-reachable-partial-candidate* *s out*

using *assms*

unfolding *trace-fri-bad-with-partial-candidates-def*

trace-fri-bad-with-reachable-partial-candidate-def

accepted-with-partial-initial-openings-def

by *blast*

lemma *composition-fri-bad-with-partial-candidates-imp-reachable-partial-candidate:*

assumes *composition-fri-bad-with-partial-candidates* *s out*

shows *composition-fri-bad-with-reachable-partial-candidate* *s out*

using *assms*

unfolding *composition-fri-bad-with-partial-candidates-def*

composition-fri-bad-with-reachable-partial-candidate-def

by *blast*

lemma *trace-fri-bad-with-empty-composition-header-candidates-imp-reachable-partial-candidate:*

assumes *trace-fri-bad-with-empty-composition-header-candidates* *s out*

shows *trace-fri-bad-with-reachable-partial-candidate* *s out*

using *assms*

unfolding *trace-fri-bad-with-empty-composition-header-candidates-def*

accepted-with-empty-composition-header-candidates-def

trace-fri-bad-with-reachable-partial-candidate-def

by *blast*

lemma *trace-fri-bad-with-partial-candidates-not-None:*

$\neg \text{trace-fri-bad-with-partial-candidates } s \text{ None}$

unfolding *trace-fri-bad-with-partial-candidates-def*

accepted-with-partial-initial-openings-def *accepted-def*

by *simp*

lemma *composition-fri-bad-with-partial-candidates-not-None:*

$\neg \text{composition-fri-bad-with-partial-candidates } s \text{ None}$

unfolding *composition-fri-bad-with-partial-candidates-def*

accepted-with-partial-initial-openings-def *accepted-def*

by *simp*

lemma *trace-fri-bad-with-empty-composition-header-candidates-not-None*:
 $\neg \text{trace-fri-bad-with-empty-composition-header-candidates } s \text{ None}$
unfolding *trace-fri-bad-with-empty-composition-header-candidates-def*
accepted-with-empty-composition-header-candidates-def *accepted-def*
by *simp*

lemma *trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction*:
assumes
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
shows
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
proof (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
show $\bigwedge s. \neg \text{trace-fri-bad-with-partial-candidates } s \text{ None}$
by (rule *trace-fri-bad-with-partial-candidates-not-None*)
next
fix *data attacker-state*
assume *builder*:
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$
let $?s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
have *reduction*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-reachable-partial-candidate } ?s) \text{ } ?s \leq$
 trace-fri-error
using *assms*[*OF* *builder*]
unfolding *trace-fri-reachable-partial-candidate-reduction-assumption-def*
by *simp*
show $\text{wp-event verify-monad (trace-fri-bad-with-partial-candidates } ?s) \text{ } ?s$
 $\leq \text{trace-fri-error}$
by (rule *order-trans*
[*OF* - *reduction*])
(rule *wp-event-mono*,
rule *trace-fri-bad-with-partial-candidates-imp-reachable-partial-candidate*)
qed

lemma *composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction*:
assumes

```

 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\impl$ 
    composition-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
  adversary-initial-state  $\leq$  composition-fri-error
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\wedge s. \neg$  composition-fri-bad-with-partial-candidates s None
  by (rule composition-fri-bad-with-partial-candidates-not-None)
next
fix data attacker-state
assume builder:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have reduction:
  wp-event verify-mono
    (composition-fri-bad-with-reachable-partial-candidate ?s) ?s  $\leq$ 
    composition-fri-error
using assms[OF builder]
unfolding
  composition-fri-reachable-partial-candidate-reduction-assumption-def
by simp
show wp-event verify-mono
  (composition-fri-bad-with-partial-candidates ?s) ?s  $\leq$ 
  composition-fri-error
by (rule order-trans
  [OF - reduction])
  (rule wp-event-mono,
    rule
      composition-fri-bad-with-partial-candidates-imp-reachable-partial-candidate)
qed

lemma trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction:
assumes
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\impl$ 
    trace-fri-reachable-partial-candidate-reduction-assumption

```



```

    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       trace-fri-bad-with-empty-composition-header-candidates)
      adversary-initial-state ≤ trace-fri-error
  proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
    show  $\bigwedge s. \neg \text{trace-fri-bad-with-empty-composition-header-candidates } s \text{ None}$ 
      by (rule trace-fri-bad-with-empty-composition-header-candidates-not-None)
  next
    fix data attacker-state
    assume builder:
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state)
    let ?s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    have reduction:
      wp-event verify-monad
        (trace-fri-bad-with-reachable-partial-candidate ?s) ?s ≤
          trace-fri-error
    using assms[OF builder]
    unfolding trace-fri-reachable-partial-candidate-reduction-assumption-def
    by simp
    show wp-event verify-monad
      (trace-fri-bad-with-empty-composition-header-candidates ?s) ?s ≤
        trace-fri-error
    by (rule order-trans
      [OF - reduction])
      (rule wp-event-mono,
       rule
        trace-fri-bad-with-empty-composition-header-candidates-imp-reachable-partial-candidate)
  qed

```

definition *empty-header-low-degree-trace-candidate-nonunique*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

empty-header-low-degree-trace-candidate-nonunique $s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr f-fri-roots f-final as dg final trace-query-idxs}$
 $\text{trace-openings trace-table composition-table trace-table'.$
 $\text{accepted-with-empty-composition-header-candidates } s \text{ out fr}$
 $\text{f-fri-roots f-final as dg final trace-query-idxs trace-openings}$
 $\text{trace-table composition-table} \wedge$
 $\text{partial-trace-table-candidate trace-table' trace-openings} \wedge$
 $\text{trace-table} \neq \text{trace-table'} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$

trace-table-low-degree trace-table')

definition *empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit s out

$\longleftrightarrow$

$(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{ trace-openings trace-table composition-table trace-table' } i.$
 $\text{ accepted-with-empty-composition-header-candidates s out fr}$
 $\text{ f-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{ trace-table composition-table} \wedge$
 $\text{ partial-trace-table-candidate trace-table' trace-openings} \wedge$
 $\text{ trace-table} \neq \text{ trace-table' } \wedge$
 $\text{ trace-table-low-degree trace-table} \wedge$
 $\text{ trace-table-low-degree trace-table' } \wedge$
 $i < \text{ rounds} \wedge$
 $\text{ map } (!) \text{ trace-table) (powers-scaled (trace-query-idxs ! i)) =}$
 $\text{ map } (!) \text{ trace-table') (powers-scaled (trace-query-idxs ! i))})$

definition *empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit

s out $\longleftrightarrow$

$(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{ trace-openings trace-table composition-table trace-table' } i.$
 $\text{ accepted-with-empty-composition-header-candidates s out fr}$
 $\text{ f-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{ trace-table composition-table} \wedge$
 $\text{ accepted-transcript-shape s out as trace-query-idxs} \wedge$
 $\text{ partial-trace-table-candidate trace-table' trace-openings} \wedge$
 $\text{ trace-table} \neq \text{ trace-table' } \wedge$
 $\text{ trace-table-low-degree trace-table} \wedge$
 $\text{ trace-table-low-degree trace-table' } \wedge$
 $i < \text{ rounds} \wedge$
 $\text{ map } (!) \text{ trace-table) (powers-scaled (trace-query-idxs ! i)) =}$
 $\text{ map } (!) \text{ trace-table') (powers-scaled (trace-query-idxs ! i))})$

lemma *empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-imp-query-agreement:*

assumes

empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
s out

shows

empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit
s out

using *assms*

unfolding

empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit-def
empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit-def

by *blast*

definition *trace-table-agreement-indices* :: 'f list $\Rightarrow$ 'f list $\Rightarrow$ nat set

where

trace-table-agreement-indices trace-table trace-table' =
 {*idx* $\in$ *query-sample-space*.
 map (! *trace-table*) (*powers-scaled idx*) =
 map (! *trace-table'*) (*powers-scaled idx*)}

lemma *empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreementE*:

assumes

empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
s out

obtains *fr f-fri-roots f-final as dg final trace-query-idxs*

trace-openings trace-table composition-table trace-table' i

where

accepted-with-empty-composition-header-candidates s out fr
f-fri-roots f-final as dg final trace-query-idxs trace-openings
trace-table composition-table
accepted-transcript-shape s out as trace-query-idxs
partial-trace-table-candidate trace-table' trace-openings
trace-table $\neq$ trace-table'
trace-table-low-degree trace-table
trace-table-low-degree trace-table'
i < rounds
trace-query-idxs ! i $\in$
trace-table-agreement-indices trace-table trace-table'

proof –

obtain *fr f-fri-roots f-final as dg final trace-query-idxs*

trace-openings trace-table composition-table trace-table' i

where *empty*:

accepted-with-empty-composition-header-candidates s out fr
f-fri-roots f-final as dg final trace-query-idxs trace-openings
trace-table composition-table

and *shape*: *accepted-transcript-shape s out as trace-query-idxs*

and *cand'*: *partial-trace-table-candidate trace-table' trace-openings*

and *distinct*: *trace-table $\neq$ trace-table'*

and *trace-low*: *trace-table-low-degree trace-table*

and *trace'-low*: *trace-table-low-degree trace-table'*

and *i-bound*: *i < rounds*

and *agree*:

map (! *trace-table*) (*powers-scaled (trace-query-idxs ! i)*) =
 map (! *trace-table'*) (*powers-scaled (trace-query-idxs ! i)*)

using *assms*

unfolding

empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit-def

```

  by blast
have len-query: length trace-query-idxs = rounds
  using accepted-transcript-shape-query-shapes(1)[OF shape] .
have idx-mem: trace-query-idxs ! i ∈ set trace-query-idxs
  by (rule nth-mem) (use i-bound len-query in simp)
have idx-sample: trace-query-idxs ! i ∈ query-sample-space
  by (rule accepted-transcript-shape-query-sample-space[OF shape idx-mem])
have idx-agree:
  trace-query-idxs ! i ∈
    trace-table-agreement-indices trace-table trace-table'
  using idx-sample agree unfolding trace-table-agreement-indices-def
  by simp
show ?thesis
  by (rule that[OF empty shape cand' distinct trace-low trace'-low
    i-bound idx-agree])
qed

lemma trace-table-agreement-indices-card-bound:
  assumes trace-low: trace-table-low-degree trace-table
    and trace'-low: trace-table-low-degree trace-table'
    and distinct: trace-table ≠ trace-table'
  shows card (trace-table-agreement-indices trace-table trace-table') <
    clength
proof -
  from trace-low obtain f where deg-f: degree f < clength
    and trace-table: trace-table = map (poly f) eval-domain
    unfolding trace-table-low-degree-def by blast
  from trace'-low obtain f' where deg-f': degree f' < clength
    and trace-table': trace-table' = map (poly f') eval-domain
    unfolding trace-table-low-degree-def by blast
  let ?A = trace-table-agreement-indices trace-table trace-table'
  let ?R = f - f'
  have A-subset: ?A ⊆ query-sample-space
    unfolding trace-table-agreement-indices-def by auto
  have finite-A: finite ?A
    unfolding trace-table-agreement-indices-def query-sample-space-def by simp
  have inj: inj-on (λidx. h ^ idx * shift) ?A
  proof (intro inj-onI)
    fix i j
    assume i-in: i ∈ ?A
      and j-in: j ∈ ?A
      and eq: h ^ i * shift = h ^ j * shift
    have i-bound: i < clength * scale
      by (rule query-sample-space-less-domain[OF set-mp[OF A-subset i-in]])
    have j-bound: j < clength * scale
      by (rule query-sample-space-less-domain[OF set-mp[OF A-subset j-in]])
    have h ^ i = h ^ j
      using eq shift-nonzero by simp
    then show i = j
  end

```

```

    by (rule h-power-inj-on-eval-domain[OF i-bound j-bound])
qed
have residual-nonzero: ?R ≠ 0
proof
  assume ?R = 0
  then have f = f'
    by simp
  then show False
    using distinct trace-table trace-table' by simp
qed
have image-subset:
  (λidx. h ^ idx * shift) ' ?A ⊆ {x. poly ?R x = 0}
proof
  fix x
  assume x-in: x ∈ (λidx. h ^ idx * shift) ' ?A
  then obtain idx where idx-in: idx ∈ ?A
    and x-eq: x = h ^ idx * shift
    by blast
  have idx-sample: idx ∈ query-sample-space
    by (rule set-mp[OF A-subset idx-in])
  have idx-bound: idx < clength * scale
    by (rule query-sample-space-less-domain[OF idx-sample])
  have agree:
    map (!) trace-table (powers-scaled idx) =
    map (!) trace-table' (powers-scaled idx)
    using idx-in unfolding trace-table-agreement-indices-def by simp
  have ps0: powers-scaled idx ! 0 = idx
    using powers-pos unfolding powers-scaled-def by simp
  have len-ps: 0 < length (powers-scaled idx)
    using powers-pos unfolding powers-scaled-def by simp
  have value-eq: trace-table ! idx = trace-table' ! idx
    using arg-cong[OF agree, of λxs. xs ! 0] ps0 len-ps
    by simp
  have left:
    trace-table ! idx = poly f (h ^ idx * shift)
    using trace-table eval-domain-nth[OF idx-bound] idx-bound
    eval-domain-length by simp
  have right:
    trace-table' ! idx = poly f' (h ^ idx * shift)
    using trace-table' eval-domain-nth[OF idx-bound] idx-bound
    eval-domain-length by simp
  have poly f (h ^ idx * shift) = poly f' (h ^ idx * shift)
    using value-eq left right by simp
  then show x ∈ {x. poly ?R x = 0}
    using x-eq by simp
qed
have card ?A = card ((λidx. h ^ idx * shift) ' ?A)
  by (simp add: card-image inj)
also have ... ≤ card {x. poly ?R x = 0}

```

```

    by (rule card-mono[OF poly-roots-finite[OF residual-nonzero]
      image-subset])
  also have ...  $\leq$  degree ?R
    by (rule card-poly-roots-bound[OF residual-nonzero])
  also have ...  $<$  clength
    using deg-f deg-f' by (simp add: degree-diff-less)
  finally show ?thesis .
qed

lemma trace-table-agreement-indices-fraction-bound:
  assumes trace-low: trace-table-low-degree trace-table
    and trace'-low: trace-table-low-degree trace-table'
    and distinct: trace-table  $\neq$  trace-table'
  shows nnreal
    (card (trace-table-agreement-indices trace-table trace-table')) /
    nnreal (card query-sample-space)  $\leq$ 
    nnreal clength / nnreal (card query-sample-space)
proof -
  have card-le:
    card (trace-table-agreement-indices trace-table trace-table')  $\leq$ 
    clength
  using trace-table-agreement-indices-card-bound
    [OF trace-low trace'-low distinct]
  by simp
  show ?thesis
    by (rule nnreal-nat-divide-right-mono[OF card-le])
qed

lemma checked-staged-query-prefix-trace-pair-agreement-hit-bound-by-prehit:
  assumes trace-low: trace-table-low-degree trace-table
    and trace'-low: trace-table-low-degree trace-table'
    and distinct: trace-table  $\neq$  trace-table'
  shows
    wp-event (checked-staged-query-prefix-receive-with-state A i)
      (checked-staged-query-prefix-dynamic-index-hit
        ( $\lambda$ - -. trace-table-agreement-indices trace-table trace-table'))
      adversary-initial-state  $\leq$ 
    wp-event (checked-staged-query-prefix-receive-with-state A i)
      (checked-staged-query-prefix-dynamic-index-prehit
        ( $\lambda$ - -. trace-table-agreement-indices trace-table trace-table'))
      adversary-initial-state +
    nnreal clength / nnreal (card query-sample-space)
proof -
  have raw-bound: query-index-raw-preimage-bound
    by (rule query-index-raw-preimage-bound-from-sampler-wellformed)
  show ?thesis
  proof (rule
    checked-staged-query-prefix-dynamic-index-hit-bound-by-prehit-and-query-error
      [OF raw-bound])

```

```

fix prefix prefix-state
assume
  Some ((prefix, prefix-state), prefix-state) ∈
    set-dist
    (execute (checked-staged-query-prefix-with-state A i)
      adversary-initial-state)
show
  trace-table-agreement-indices trace-table trace-table'
    ⊆ query-sample-space ∧
  nnreal
    (card (trace-table-agreement-indices trace-table trace-table')) /
  nnreal (card query-sample-space) ≤
  nnreal clength / nnreal (card query-sample-space)
using trace-table-agreement-indices-fraction-bound
  [OF trace-low trace'-low distinct]
unfolding trace-table-agreement-indices-def by auto
qed
qed

definition checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-from-prefix
  :: 'f staged-adversary ⇒ 'f list ⇒ 'f list ⇒ nat ⇒
    ((((((('f × 'f list × 'f list × 'f) ×
      'f protocol-channel) × 'f staged-proof-data) ×
      'f protocol-channel) × unit list) ×
      'f protocol-channel) option ⇒ bool
where
  checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-from-prefix
    A trace-table trace-table' i out ⟷
  (case out of
    None ⇒ False
  | Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
    result), final-state) ⇒
    (∃ prefix prefix-state raw raw-state.
      Some (data, attacker-state) ∈
        set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state) ∧
      Some (((prefix, prefix-state), raw), raw-state) ∈
        set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state) ∧
      checked-staged-query-prefix-dynamic-index-hit
        (λ- -. trace-table-agreement-indices trace-table trace-table')
        (Some (((prefix, prefix-state), raw), raw-state))))))

lemma checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-from-prefixE:
assumes
  checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-from-prefix
  A trace-table trace-table' i

```

```

    (Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
      result), final-state))
obtains prefix prefix-state raw raw-state where
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  Some (((prefix, prefix-state), raw), raw-state) ∈
    set-dist
      (execute (checked-staged-query-prefix-receive-with-state A i)
        adversary-initial-state)
  checked-staged-query-prefix-dynamic-index-hit
    (λ- -. trace-table-agreement-indices trace-table trace-table')
    (Some (((prefix, prefix-state), raw), raw-state))
using assms
proof –
obtain prefix prefix-state raw raw-state where
  builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and prefix-receive:
    Some (((prefix, prefix-state), raw), raw-state) ∈
      set-dist
        (execute (checked-staged-query-prefix-receive-with-state A i)
          adversary-initial-state)
  and hit:
    checked-staged-query-prefix-dynamic-index-hit
      (λ- -. trace-table-agreement-indices trace-table trace-table')
      (Some (((prefix, prefix-state), raw), raw-state))
  using assms
  unfolding
    checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-from-prefix-def
  by simp blast
  show ?thesis
  by (rule that[OF builder prefix-receive hit])
qed

definition checked-staged-security-with-actual-alpha-prefix-empty-header-low-degree-prefix-agreement-hit
  :: 'f staged-adversary ⇒
    ((((((f × 'f list × 'f list × 'f) ×
      'f protocol-channel) × 'f staged-proof-data) ×
      'f protocol-channel) × unit list) ×
      'f protocol-channel) option ⇒ bool
where
  checked-staged-security-with-actual-alpha-prefix-empty-header-low-degree-prefix-agreement-hit
    A out ⟷
    (case out of

```


None $\Rightarrow$ *False*
| *Some* ((((*alpha-prefix*, *alpha-prefix-state*), *data*), *attacker-state*),
result), *final-state*) $\Rightarrow$
(let *s* =
verifier-state-from-adversary attacker-state
(*staged-proof-transcript data*)
in
($\exists$ fr *f-fri-roots f-final* as dg *final trace-query-idxs*
trace-openings trace-table composition-table trace-table' i
prefix prefix-state raw raw-state.
accepted-with-empty-composition-header-candidates s
(*Some (result, final-state)*) fr *f-fri-roots f-final* as dg
final trace-query-idxs trace-openings trace-table
composition-table $\wedge$
accepted-transcript-shape s (Some (result, final-state)) as
trace-query-idxs $\wedge$
partial-trace-table-candidate trace-table' trace-openings $\wedge$
trace-table $\neq$ *trace-table'* $\wedge$
trace-table-low-degree trace-table $\wedge$
trace-table-low-degree trace-table' $\wedge$
i < *rounds* $\wedge$
Some (data, attacker-state) $\in$
set-dist
(*execute (checked-staged-transcript-program A)*
adversary-initial-state) $\wedge$
Some (((prefix, prefix-state), raw), raw-state) $\in$
set-dist
(*execute (checked-staged-query-prefix-receive-with-state A i)*
adversary-initial-state) $\wedge$
index (to-nat raw) $\in$
trace-table-agreement-indices trace-table trace-table')))

lemma *empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-imp-prefix-agreement-hit-on-*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *support:*

out $\in$
set-dist
(*execute*
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
adversary-initial-state)

and *hit:*

checked-staged-security-with-actual-alpha-prefix-verifier-event
empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
out

shows

checked-staged-security-with-actual-alpha-prefix-empty-header-low-degree-prefix-agreement-hit
A out

```

proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      checked-staged-security-with-actual-alpha-prefix-empty-header-low-degree-prefix-agreement-hit-def
    by simp
next
  case (Some packed)
  then obtain alpha-prefix alpha-prefix-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have local-hit:
    empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
    ?s (Some (result, final-state))
  using hit
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
from empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreementE
  [OF local-hit]
obtain fr f-fri-roots f-final as dg final trace-query-idxs
  trace-openings trace-table composition-table trace-table' i where
  empty:
    accepted-with-empty-composition-header-candidates ?s
    (Some (result, final-state)) fr f-fri-roots f-final as dg final
    trace-query-idxs trace-openings trace-table composition-table
  and shape:
    accepted-transcript-shape ?s (Some (result, final-state)) as
    trace-query-idxs
  and cand': partial-trace-table-candidate trace-table' trace-openings
  and distinct: trace-table ≠ trace-table'
  and trace-low: trace-table-low-degree trace-table
  and trace'-low: trace-table-low-degree trace-table'
  and i-bound: i < rounds
  and idx-agree:
    trace-query-idxs ! i ∈
    trace-table-agreement-indices trace-table trace-table'
  by blast
have data-support:
  Some (((data, attacker-state), result), final-state) ∈
  set-dist
  (execute (checked-staged-security-experiment-with-data-state A))

```

adversary-initial-state)
by (rule
checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
[OF support[unfolded out-eq]])
from
checked-staged-security-with-data-state-accepted-shape-query-prefix-receive-index
[OF wf controlled data-support shape i-bound]
obtain *prefix prefix-state raw raw-state* **where**
prefix-receive:
Some (((prefix, prefix-state), raw), raw-state) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state)
and *raw-idx: index (to-nat raw) = trace-query-idxs ! i*
by *blast*
have *raw-agree:*
index (to-nat raw) ∈
trace-table-agreement-indices trace-table trace-table'
using *raw-idx idx-agree* **by** *simp*
have *builder:*
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
proof –
from *checked-staged-security-experiment-with-data-state-outcomeE*
[OF data-support]
show *?thesis* **by** *blast*
qed
show *?thesis*
unfolding *out-eq*
checked-staged-security-with-actual-alpha-prefix-empty-header-low-degree-prefix-agreement-hit-def
apply (*simp add: Let-def*)
by (*intro exI[of - fr] exI[of - f-fri-roots]*
exI[of - f-final] exI[of - as] exI[of - dg]
exI[of - final] exI[of - trace-query-idxs]
exI[of - trace-openings] exI[of - trace-table]
exI[of - composition-table] exI[of - trace-table']
exI[of - i] exI[of - prefix] exI[of - prefix-state]
exI[of - raw] exI[of - raw-state] conjI empty shape cand'
distinct trace-low trace'-low i-bound builder prefix-receive
raw-agree)
qed

lemma *empty-header-low-degree-trace-candidate-nonunique-imp-query-agreement-hit:*
assumes *empty-header-low-degree-trace-candidate-nonunique s out*
shows
empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit
s out

```

proof –
  obtain fr f-fri-roots f-final as dg final trace-query-idxs
    trace-openings trace-table composition-table trace-table' where empty:
    accepted-with-empty-composition-header-candidates s out fr
    f-fri-roots f-final as dg final trace-query-idxs trace-openings
    trace-table composition-table
  and cand': partial-trace-table-candidate trace-table' trace-openings
  and distinct: trace-table ≠ trace-table'
  and trace-low: trace-table-low-degree trace-table
  and trace'-low: trace-table-low-degree trace-table'
  using assms
  unfolding empty-header-low-degree-trace-candidate-nonunique-def by blast
have partial:
  accepted-with-partial-trace-openings s out fr trace-query-idxs
  trace-openings
  using empty unfolding accepted-with-empty-composition-header-candidates-def
by blast
have cand:
  partial-trace-table-candidate trace-table trace-openings
  using empty unfolding accepted-with-empty-composition-header-candidates-def
by blast
have idxs:
  map opening-index (trace-openings ! 0) =
  powers-scaled (trace-query-idxs ! 0)
  using accepted-with-partial-trace-openings-shapes(3)
  [OF partial rounds-positive] .
have vals:
  map (! trace-table) (powers-scaled (trace-query-idxs ! 0)) =
  map (! trace-table') (powers-scaled (trace-query-idxs ! 0))
proof –
  have left:
    map (! trace-table) (powers-scaled (trace-query-idxs ! 0)) =
    map opening-value (trace-openings ! 0)
    by (rule partial-trace-table-candidate-query-values
      [OF cand rounds-positive idxs])
  have right:
    map (! trace-table') (powers-scaled (trace-query-idxs ! 0)) =
    map opening-value (trace-openings ! 0)
    by (rule partial-trace-table-candidate-query-values
      [OF cand' rounds-positive idxs])
  show ?thesis
    using left right by simp
qed
show ?thesis
  unfolding
    empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit-def
  using empty cand' distinct trace-low trace'-low rounds-positive vals
by blast
qed

```

lemma *empty-header-low-degree-trace-candidate-nonunique-le-query-agreement-hit:*

```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-low-degree-trace-candidate-nonunique)
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit)
  adversary-initial-state
by (rule wp-event-mono)
  (auto simp:
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    split: option.splits prod.splits
    dest:
      empty-header-low-degree-trace-candidate-nonunique-imp-query-agreement-hit)

```

lemma *empty-header-partial-trace-candidate-nonunique-imp-trace-fri-or-low-degree-nonunique:*

assumes *empty-header-partial-trace-candidate-nonunique s out*
shows

trace-fri-bad-with-empty-composition-header-candidates s out ∨
empty-header-low-degree-trace-candidate-nonunique s out

proof —

obtain *fr f-fri-roots f-final as dg final trace-query-idxs trace-openings*
trace-table composition-table **where** *empty:*
accepted-with-empty-composition-header-candidates s out fr
f-fri-roots f-final as dg final trace-query-idxs trace-openings
trace-table composition-table

and *not-singleton:*

$\neg (\exists \text{ trace-table0.}$
partial-trace-table-candidates trace-openings $\subseteq$
 $\{\text{trace-table0}\})$

using *assms*

unfolding *empty-header-partial-trace-candidate-nonunique-def* **by** *blast*

have *trace-candidate:*

trace-table $\in$ *partial-trace-table-candidates trace-openings*

using *empty*

unfolding *accepted-with-empty-composition-header-candidates-def*
partial-trace-table-candidates-def

by *simp*

obtain *trace-table'* **where** *trace-candidate':*

trace-table' $\in$ *partial-trace-table-candidates trace-openings*

and *distinct:* *trace-table* $\neq$ *trace-table'*

proof —

have $\neg$ *partial-trace-table-candidates trace-openings* $\subseteq$
 $\{\text{trace-table}\}$

using *not-singleton* **by** *blast*

```

then obtain trace-table' where
  trace-table'  $\in$  partial-trace-table-candidates trace-openings
  trace-table'  $\notin$  {trace-table}
  by blast
then show ?thesis
  using that by blast
qed
show ?thesis
proof (cases trace-table-low-degree trace-table)
  case False
  then have trace-fri-bad-with-empty-composition-header-candidates s out
    unfolding trace-fri-bad-with-empty-composition-header-candidates-def
    using empty by blast
  then show ?thesis by simp
next
  case trace-low: True
  show ?thesis
  proof (cases trace-table-low-degree trace-table')
    case False
    have empty':
      accepted-with-empty-composition-header-candidates s out fr
      f-fri-roots f-final as dg final trace-query-idxs trace-openings
      trace-table' composition-table
      using empty trace-candidate'
      unfolding accepted-with-empty-composition-header-candidates-def
      partial-trace-table-candidates-def
      by blast
    then have
      trace-fri-bad-with-empty-composition-header-candidates s out
      unfolding trace-fri-bad-with-empty-composition-header-candidates-def
      using False by blast
    then show ?thesis by simp
  next
  case trace'-low: True
  have empty-header-low-degree-trace-candidate-nonunique s out
    unfolding empty-header-low-degree-trace-candidate-nonunique-def
    using empty trace-candidate' distinct trace-low trace'-low
    unfolding partial-trace-table-candidates-def
    by blast
  then show ?thesis by simp
  qed
  qed
qed

definition empty-header-partial-trace-candidate-nonunique-transcript
  :: (f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  (f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  empty-header-partial-trace-candidate-nonunique-transcript s out  $\longleftrightarrow$ 

```

$(\exists \text{fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{trace-openings trace-table composition-table.}$
 $\text{accepted-with-empty-composition-header-candidates s out fr}$
 $f\text{-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{trace-table composition-table} \wedge$
 $\text{accepted-transcript-shape s out as trace-query-idxs} \wedge$
 $\neg (\exists \text{trace-table0.}$
 $\text{partial-trace-table-candidates trace-openings} \subseteq$
 $\{\text{trace-table0}\}))$

lemma *empty-header-partial-trace-candidate-nonunique-transcript-imp-trace-fri-or-low-degree-transcript-query-agree*

assumes *empty-header-partial-trace-candidate-nonunique-transcript s out*

shows

$\text{trace-fri-bad-with-empty-composition-header-candidates s out} \vee$
 $\text{empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit}$
 $s \text{ out}$

proof –

obtain *fr f-fri-roots f-final as dg final trace-query-idxs trace-openings*
trace-table composition-table **where** *empty:*

$\text{accepted-with-empty-composition-header-candidates s out fr}$
 $f\text{-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{trace-table composition-table}$

and *shape: accepted-transcript-shape s out as trace-query-idxs*

and *not-singleton:*

$\neg (\exists \text{trace-table0.}$
 $\text{partial-trace-table-candidates trace-openings} \subseteq$
 $\{\text{trace-table0}\})$

using *assms*

unfolding *empty-header-partial-trace-candidate-nonunique-transcript-def*

by *blast*

have *trace-candidate:*

$\text{trace-table} \in \text{partial-trace-table-candidates trace-openings}$

using *empty*

unfolding *accepted-with-empty-composition-header-candidates-def*

partial-trace-table-candidates-def

by *simp*

obtain *trace-table'* **where** *trace-candidate':*

$\text{trace-table}' \in \text{partial-trace-table-candidates trace-openings}$

and *distinct: trace-table $\neq$ trace-table'*

proof –

have $\neg \text{partial-trace-table-candidates trace-openings} \subseteq$
 $\{\text{trace-table}\}$

using *not-singleton* **by** *blast*

then obtain *trace-table'* **where**

$\text{trace-table}' \in \text{partial-trace-table-candidates trace-openings}$

$\text{trace-table}' \notin \{\text{trace-table}\}$

by *blast*

then show *?thesis*

using *that* **by** *blast*

```

qed
show ?thesis
proof (cases trace-table-low-degree trace-table)
  case False
  then have trace-fri-bad-with-empty-composition-header-candidates s out
    unfolding trace-fri-bad-with-empty-composition-header-candidates-def
    using empty by blast
  then show ?thesis by simp
next
case trace-low: True
show ?thesis
proof (cases trace-table-low-degree trace-table')
  case False
  have empty':
    accepted-with-empty-composition-header-candidates s out fr
      f-fri-roots f-final as dg final trace-query-idxs trace-openings
      trace-table' composition-table
  using empty trace-candidate'
  unfolding accepted-with-empty-composition-header-candidates-def
    partial-trace-table-candidates-def
  by blast
  then have
    trace-fri-bad-with-empty-composition-header-candidates s out
  unfolding trace-fri-bad-with-empty-composition-header-candidates-def
  using False by blast
  then show ?thesis by simp
next
case trace'-low: True
have idxs:
  map opening-index (trace-openings ! 0) =
    powers-scaled (trace-query-idxs ! 0)
proof -
  have partial:
    accepted-with-partial-trace-openings s out fr trace-query-idxs
      trace-openings
  using empty
  unfolding accepted-with-empty-composition-header-candidates-def
  by blast
  show ?thesis
    using accepted-with-partial-trace-openings-shapes(3)
      [OF partial rounds-positive] .
qed
have vals:
  map ((! trace-table) (powers-scaled (trace-query-idxs ! 0))) =
    map ((! trace-table') (powers-scaled (trace-query-idxs ! 0)))
proof -
  have cand:
    partial-trace-table-candidate trace-table trace-openings
  using empty

```



```

    unfolding accepted-with-empty-composition-header-candidates-def
    by blast
have cand':
  partial-trace-table-candidate trace-table' trace-openings
  using trace-candidate'
  unfolding partial-trace-table-candidates-def by simp
have left:
  map ((! trace-table) (powers-scaled (trace-query-idxs ! 0))) =
    map opening-value (trace-openings ! 0)
  by (rule partial-trace-table-candidate-query-values
    [OF cand rounds-positive idxs])
have right:
  map ((! trace-table') (powers-scaled (trace-query-idxs ! 0))) =
    map opening-value (trace-openings ! 0)
  by (rule partial-trace-table-candidate-query-values
    [OF cand' rounds-positive idxs])
show ?thesis
  using left right by simp
qed
have
  empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
  s out
  unfolding
    empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit-def
  using empty shape trace-candidate' distinct trace-low trace'-low
    rounds-positive vals
  unfolding partial-trace-table-candidates-def
  by blast
then show ?thesis by simp
qed
qed
qed
qed

lemma empty-header-partial-trace-candidate-nonunique-transcript-le-trace-fri-or-low-degree-transcript-query-agreement
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-partial-trace-candidate-nonunique-transcript)
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
    trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state +
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit)
  adversary-initial-state

```

```

proof –
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?N =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-partial-trace-candidate-nonunique-transcript
  let ?T =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    trace-fri-bad-with-empty-composition-header-candidates
  let ?L =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
  have wp-event ?M ?N adversary-initial-state ≤
    wp-event ?M (λout. ?T out ∨ ?L out) adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      split: option.splits prod.splits
      dest:
        empty-header-partial-trace-candidate-nonunique-transcript-imp-trace-fri-or-low-degree-transcript-query-agreement-hit)
  also have ... ≤
    wp-event ?M ?T adversary-initial-state +
    wp-event ?M ?L adversary-initial-state
  by (rule wp-event-union-bound)
  finally show ?thesis .
qed

```

definition *composition-alpha-partial-opening-union-hit-with-empty-header-transcript*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-alpha-partial-opening-union-hit-with-empty-header-transcript
 $s \text{ out} \longleftrightarrow$
 $(\exists fr \text{ f-fri-roots } f\text{-final as } dg \text{ final } \text{trace-query-idxs}$
 $\text{trace-openings } \text{trace-table } \text{composition-table.}$
 $\text{accepted-with-empty-composition-header-candidates } s \text{ out } fr$
 $f\text{-fri-roots } f\text{-final as } dg \text{ final } \text{trace-query-idxs } \text{trace-openings}$
 $\text{trace-table } \text{composition-table} \wedge$
 $\text{accepted-transcript-shape } s \text{ out as } \text{trace-query-idxs} \wedge$
 $as \in \text{partial-trace-opening-alpha-union-bad-sets } \text{trace-openings})$

lemma *composition-alpha-partial-opening-union-hit-with-empty-header-transcript-imp-union-hit:*

assumes

composition-alpha-partial-opening-union-hit-with-empty-header-transcript
 $s \text{ out}$

shows

composition-alpha-partial-opening-union-hit-with-empty-header $s \text{ out}$

using *assms*

unfolding

composition-alpha-partial-opening-union-hit-with-empty-header-transcript-def

composition-alpha-partial-opening-union-hit-with-empty-header-def
by blast

definition *composition-bad-with-empty-composition-header-candidates-transcript*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-bad-with-empty-composition-header-candidates-transcript
 $s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{ trace-openings trace-table composition-table } f.$
 $\text{ accepted-with-empty-composition-header-candidates } s \text{ out fr}$
 $f\text{-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{ trace-table composition-table} \wedge$
 $\text{ accepted-transcript-shape } s \text{ out as trace-query-idxs} \wedge$
 $\text{ degree } f < \text{ clength} \wedge$
 $\text{ trace-table} = \text{ map } (\text{poly } f) \text{ eval-domain} \wedge$
 $\text{ violated-constraints } f \neq \{\}$
 $\text{ composition-table-low-degree maxDegree composition-table} \wedge$
 $\text{ all-queries-consistent trace-table composition-table as})$

definition *composition-degree-bad-with-empty-composition-header-candidates-transcript*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-degree-bad-with-empty-composition-header-candidates-transcript
 $s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{ trace-openings trace-table composition-table } f.$
 $\text{ accepted-with-empty-composition-header-candidates } s \text{ out fr}$
 $f\text{-fri-roots } f\text{-final as dg final trace-query-idxs trace-openings}$
 $\text{ trace-table composition-table} \wedge$
 $\text{ accepted-transcript-shape } s \text{ out as trace-query-idxs} \wedge$
 $\text{ degree } f < \text{ clength} \wedge$
 $\text{ trace-table} = \text{ map } (\text{poly } f) \text{ eval-domain} \wedge$
 $\text{ violated-constraints } f \neq \{\}$
 $\text{ composition-table-low-degree maxDegree composition-table} \wedge$
 $\text{ all-queries-consistent trace-table composition-table as} \wedge$
 $\neg \text{ common-denominator-degree-bounds } f \text{ as})$

definition *composition-randomization-bad-with-empty-composition-header-candidates-transcript*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-randomization-bad-with-empty-composition-header-candidates-transcript
 $s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr } f\text{-fri-roots } f\text{-final as dg final trace-query-idxs}$
 $\text{ trace-openings trace-table composition-table } f.$
 $\text{ accepted-with-empty-composition-header-candidates } s \text{ out fr}$

f-fri-roots f-final as dg final trace-query-idxs trace-openings
trace-table composition-table $\wedge$
accepted-transcript-shape s out as trace-query-idxs $\wedge$
degree f < clength $\wedge$
trace-table = map (poly f) eval-domain $\wedge$
violated-constraints f $\neq \{\}$ $\wedge$
composition-table-low-degree maxDegree composition-table $\wedge$
all-queries-consistent trace-table composition-table as $\wedge$
common-denominator-degree-bounds f as $\wedge$
random-combination-common-denominator-hides-violations f as)

lemma *composition-bad-with-empty-composition-header-candidates-transcript-split:*

assumes *false-statement: $\neg$ exists-valid-trace*

and *bad:*

composition-bad-with-empty-composition-header-candidates-transcript
s out

shows

composition-degree-bad-with-empty-composition-header-candidates-transcript
s out $\vee$

composition-randomization-bad-with-empty-composition-header-candidates-transcript
s out

proof —

obtain *fr f-fri-roots f-final as dg final trace-query-idxs*

trace-openings trace-table composition-table f **where** *partial:*

accepted-with-empty-composition-header-candidates s out fr
f-fri-roots f-final as dg final trace-query-idxs trace-openings
trace-table composition-table

and *shape: accepted-transcript-shape s out as trace-query-idxs*

and *deg-f: degree f < clength*

and *trace-table: trace-table = map (poly f) eval-domain*

and *violated: violated-constraints f $\neq \{\}$*

and *composition-low:*

composition-table-low-degree maxDegree composition-table

and *all-queries:*

all-queries-consistent trace-table composition-table as

using *bad*

unfolding

composition-bad-with-empty-composition-header-candidates-transcript-def

by *blast*

show *?thesis*

proof (*cases common-denominator-degree-bounds f as*)

case *False*

then have

composition-degree-bad-with-empty-composition-header-candidates-transcript
s out

unfolding

composition-degree-bad-with-empty-composition-header-candidates-transcript-def

using *partial shape deg-f trace-table violated composition-low*

all-queries

```

    by blast
  then show ?thesis by simp
next
case True
note bounds = True
have len-as: length as = length spec
proof -
  obtain result final-state fr' f-fri-roots' f-final' dg'
    composition-fri-roots' final' rest' where
    verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
    composition-fri-roots' final' rest'
  using shape
  by (elim accepted-transcript-shape-header-query-extraction)
  then show ?thesis
  by (rule verifier-header-transcript-shapes(3))
qed
have hides:
  random-combination-common-denominator-hides-violations f as
  by (rule random-combination-common-denominator-hides-from-queries
    [OF len-as false-statement deg-f trace-table composition-low
    all-queries bounds])
have
  composition-randomization-bad-with-empty-composition-header-candidates-transcript
  s out
  unfolding
    composition-randomization-bad-with-empty-composition-header-candidates-transcript-def
  using partial shape deg-f trace-table violated composition-low
  all-queries bounds hides
  by blast
  then show ?thesis by simp
qed
qed

lemma composition-degree-bad-with-empty-composition-header-candidates-transcript-imp-degree-bad:
  assumes
    composition-degree-bad-with-empty-composition-header-candidates-transcript
    s out
  shows
    composition-degree-bad-with-empty-composition-header-candidates s out
  using assms
  unfolding
    composition-degree-bad-with-empty-composition-header-candidates-transcript-def
    composition-degree-bad-with-empty-composition-header-candidates-def
  by blast

lemma composition-degree-bad-with-empty-composition-header-candidates-transcript-false:
  ¬ composition-degree-bad-with-empty-composition-header-candidates-transcript
  s out
  using

```

composition-degree-bad-with-empty-composition-header-candidates-transcript-imp-degree-bad
composition-degree-bad-with-empty-composition-header-candidates-false
by *blast*

lemma *composition-degree-bad-with-empty-composition-header-candidates-transcript-bound:*

wp-event verify-monad

(*composition-degree-bad-with-empty-composition-header-candidates-transcript*
s) *s* ≤
0

proof —

have *composition-degree-bad-with-empty-composition-header-candidates-transcript*

s =

(*λout :: (unit list × ('f, 'a) protocol-channel-scheme) option.*
False)

by (*rule ext*)

(*simp add:*

composition-degree-bad-with-empty-composition-header-candidates-transcript-false)

then show *?thesis*

unfolding *wp-event-def wp-def dist-expect-def* **by** *simp*

qed

lemma *composition-randomization-bad-with-empty-composition-header-candidates-transcript-imp-alpha-union-hit:*

assumes *bad:*

composition-randomization-bad-with-empty-composition-header-candidates-transcript

s out

shows

composition-alpha-partial-opening-union-hit-with-empty-header-transcript

s out

proof —

obtain *fr f-fri-roots f-final as dg final trace-query-idxs*

trace-openings trace-table composition-table f **where** *partial:*

accepted-with-empty-composition-header-candidates s out fr

f-fri-roots f-final as dg final trace-query-idxs trace-openings

trace-table composition-table

and *shape: accepted-transcript-shape s out as trace-query-idxs*

and *deg-f: degree f < clength*

and *trace-table: trace-table = map (poly f) eval-domain*

and *violated: violated-constraints f ≠ {}*

and *bounds: common-denominator-degree-bounds f as*

and *hides:*

random-combination-common-denominator-hides-violations f as

using *bad*

unfolding

composition-randomization-bad-with-empty-composition-header-candidates-transcript-def

by *blast*

have *as-in:*

as ∈ common-denominator-hiding-alpha-space f

proof —

obtain *result final-state fr' f-fri-roots' f-final' dg'*

```

    composition-fri-roots' final' rest' where header:
    verifier-header-transcript s fr' f-fri-roots' f-final' as dg'
    composition-fri-roots' final' rest'
using shape
by (elim accepted-transcript-shape-header-query-extraction)
have len-as: length as = length spec
    by (rule verifier-header-transcript-shapes(3)[OF header])
show ?thesis
    using len-as hides
    unfolding common-denominator-hiding-alpha-space-def alpha-space-def
    by simp
qed
have candidate:
    trace-table ∈ partial-trace-table-candidates trace-openings
    using partial
    unfolding accepted-with-empty-composition-header-candidates-def
    partial-trace-table-candidates-def
    by simp
have witness-eq: low-degree-trace-witness trace-table = f
    by (rule low-degree-trace-witness-eq[OF deg-f trace-table])
have as-bad: as ∈ composition-trace-bad-alpha-space trace-table
    using deg-f trace-table violated bounds as-in
    unfolding composition-trace-bad-alpha-space-def witness-eq
    by simp
have as ∈ partial-trace-opening-alpha-union-bad-sets trace-openings
    unfolding partial-trace-opening-alpha-union-bad-sets-def
    using candidate as-bad by blast
then show ?thesis
    unfolding
    composition-alpha-partial-opening-union-hit-with-empty-header-transcript-def
    using partial shape by blast
qed

lemma composition-alpha-partial-opening-union-hit-with-empty-header-transcript-split:
assumes hit:
    composition-alpha-partial-opening-union-hit-with-empty-header-transcript
    s out
shows
    composition-alpha-bad-set-hit-with-empty-composition-header-candidates
    s out ∨
    empty-header-partial-trace-candidate-nonunique-transcript s out
proof –
obtain fr f-fri-roots f-final as dg final trace-query-idxs
    trace-openings trace-table composition-table where partial:
    accepted-with-empty-composition-header-candidates s out fr
    f-fri-roots f-final as dg final trace-query-idxs trace-openings
    trace-table composition-table
and shape: accepted-transcript-shape s out as trace-query-idxs
and as-union:

```

```

    as ∈ partial-trace-opening-alpha-union-bad-sets trace-openings
using hit
unfolding
    composition-alpha-partial-opening-union-hit-with-empty-header-transcript-def
by blast
from as-union obtain trace-table' where candidate':
    trace-table' ∈ partial-trace-table-candidates trace-openings
and as-bad': as ∈ composition-trace-bad-alpha-space trace-table'
unfolding partial-trace-opening-alpha-union-bad-sets-def by blast
have candidate:
    trace-table ∈ partial-trace-table-candidates trace-openings
using partial
unfolding accepted-with-empty-composition-header-candidates-def
    partial-trace-table-candidates-def
by simp
show ?thesis
proof (cases
    ∃ trace-table0.
        partial-trace-table-candidates trace-openings ⊆
        {trace-table0})
case True
then obtain trace-table0 where subset:
    partial-trace-table-candidates trace-openings ⊆
    {trace-table0}
by blast
have trace-eq: trace-table = trace-table0
using subset candidate by blast
have trace'-eq: trace-table' = trace-table0
using subset candidate' by blast
have as ∈ composition-trace-bad-alpha-space trace-table
using as-bad' trace-eq trace'-eq by simp
then have
    composition-alpha-bad-set-hit-with-empty-composition-header-candidates
    s out
unfolding
    composition-alpha-bad-set-hit-with-empty-composition-header-candidates-def
using partial by blast
then show ?thesis by simp
next
case False
have empty-header-partial-trace-candidate-nonunique-transcript s out
unfolding empty-header-partial-trace-candidate-nonunique-transcript-def
using partial shape False by blast
then show ?thesis by simp
qed
qed

definition checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
:: ((((((f × f list × f list × f) ×
```



```

      'f protocol-channel) × 'f staged-proof-data) ×
      'f protocol-channel) × unit list) ×
      'f protocol-channel) option ⇒ bool
where
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
    out ↔
  checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
    composition-trace-bad-alpha-space out ∧
  checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-alpha-partial-opening-union-hit-with-empty-header-transcript
    out

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-le-fixed-or-nonunique-
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
    composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
  adversary-initial-state +
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-partial-trace-candidate-nonunique-transcript)
  adversary-initial-state
proof —
let ?M =
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let ?fixed =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  composition-alpha-bad-set-hit-with-empty-composition-header-candidates
let ?nonunique =
  checked-staged-security-with-actual-alpha-prefix-verifier-event
  empty-header-partial-trace-candidate-nonunique-transcript
have wp-event ?M
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
  adversary-initial-state ≤
wp-event ?M (λout. ?fixed out ∨ ?nonunique out)
  adversary-initial-state
by (rule wp-event-mono)
  (auto simp:
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-def
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    split: option.splits prod.splits
    dest:
      composition-alpha-partial-opening-union-hit-with-empty-header-transcript-split)
also have ... ≤

```

```

    wp-event ?M ?fixed adversary-initial-state +
    wp-event ?M ?nonunique adversary-initial-state
  by (rule wp-event-union-bound)
  finally show ?thesis .
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-fixed-and-*
assumes *fixed-bound:*

```

  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
   composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
  adversary-initial-state ≤ F

```

and *nonunique-bound:*

```

  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  (checked-staged-security-with-actual-alpha-prefix-verifier-event
   empty-header-partial-trace-candidate-nonunique-transcript)
  adversary-initial-state ≤ N

```

shows

```

  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
  adversary-initial-state ≤ F + N

```

by (rule order-trans[

```

  OF
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-le-fixed-or-nonunique-tran
  (intro add-mono fixed-bound nonunique-bound)

```

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-le-relevant-drift:*

```

  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
  adversary-initial-state ≤
  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
  adversary-initial-state
  by (rule wp-event-mono)
  (auto simp:
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-def
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-def
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    split: option.splits prod.splits
    intro:
      composition-alpha-partial-opening-union-hit-with-empty-header-transcript-imp-union-hit)

```

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-relevant-d*

assumes *drift-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)

checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift

adversary-initial-state $\leq D$

shows

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)

checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript

adversary-initial-state $\leq D$

by (*rule order-trans*[

OF

checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-le-relevant-drift

drift-bound])

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-le-fixed-or-trace-fri-or-*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)

checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript

adversary-initial-state $\leq$

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)

(*checked-staged-security-with-actual-alpha-prefix-verifier-event*

composition-alpha-bad-set-hit-with-empty-composition-header-candidates)

adversary-initial-state +

(*wp-event*

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)

(*checked-staged-security-with-actual-alpha-prefix-verifier-event*

trace-fri-bad-with-empty-composition-header-candidates)

adversary-initial-state +

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)

(*checked-staged-security-with-actual-alpha-prefix-verifier-event*

empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit)

adversary-initial-state)

proof –

let *?M* =

checked-staged-security-experiment-with-actual-alpha-prefix-data-state *A*

let *?fixed* =

checked-staged-security-with-actual-alpha-prefix-verifier-event

composition-alpha-bad-set-hit-with-empty-composition-header-candidates

let *?nonunique* =

checked-staged-security-with-actual-alpha-prefix-verifier-event

empty-header-partial-trace-candidate-nonunique-transcript

let *?trace-fri* =

checked-staged-security-with-actual-alpha-prefix-verifier-event

trace-fri-bad-with-empty-composition-header-candidates

let *?low-degree* =
checked-staged-security-with-actual-alpha-prefix-verifier-event
empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
have *split*:
wp-event ?M
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
adversary-initial-state $\leq$
wp-event ?M ?fixed adversary-initial-state +
wp-event ?M ?nonunique adversary-initial-state
by (rule
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-le-fixed-or-nonunique-tran
have *nonunique*:
wp-event ?M ?nonunique adversary-initial-state $\leq$
wp-event ?M ?trace-fri adversary-initial-state +
wp-event ?M ?low-degree adversary-initial-state
by (rule
empty-header-partial-trace-candidate-nonunique-transcript-le-trace-fri-or-low-degree-transcript-query-agreement)
show *?thesis*
by (rule *order-trans[OF split]*)
(intro *add-mono order-refl nonunique*)
qed

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-fixed-trace*
assumes *fixed-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state $\leq F$
and *trace-fri-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq T$
and *low-degree-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit)
adversary-initial-state $\leq L$
shows
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
adversary-initial-state $\leq F + (T + L)$

```

by (rule order-trans[
  OF
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-le-fixed-or-trace-fri-or-low
  (intro add-mono fixed-bound add-mono trace-fri-bound low-degree-bound)

lemma empty-header-partial-trace-candidate-nonunique-le-trace-fri-or-low-degree-nonunique:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      empty-header-partial-trace-candidate-nonunique)
  adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state +
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      empty-header-low-degree-trace-candidate-nonunique)
  adversary-initial-state
proof -
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?N =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-partial-trace-candidate-nonunique
  let ?T =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    trace-fri-bad-with-empty-composition-header-candidates
  let ?L =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-low-degree-trace-candidate-nonunique
  have wp-event ?M ?N adversary-initial-state ≤
    wp-event ?M (λout. ?T out ∨ ?L out) adversary-initial-state
  by (rule wp-event-mono)
  (auto simp: checked-staged-security-with-actual-alpha-prefix-verifier-event-def
    split: option.splits prod.splits
    dest:
      empty-header-partial-trace-candidate-nonunique-imp-trace-fri-or-low-degree-nonunique)
  also have ... ≤
    wp-event ?M ?T adversary-initial-state +
    wp-event ?M ?L adversary-initial-state
  by (rule wp-event-union-bound)
  finally show ?thesis .
qed

theorem stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-f
  fixes D empty-query-error :: prob

```

and $C :: \text{nat} \Rightarrow \text{prob}$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *composition-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state A } i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *drift-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift}$
 $\text{adversary-initial-state} \leq D$
and *current-empty-bound*:
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows *checked-staged-adversary-acceptance-probability A* $\leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds. } C \ i) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum i < \text{rounds. } C \ i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D +$

```

    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    trace-fri-error + empty-query-error)
proof –
  have trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ trace-fri-error
  by (rule
    trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction
    [OF trace-fri-reduction])
  have comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-partial-candidates)
    adversary-initial-state ≤ composition-fri-error
  by (rule
    composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction
    [OF composition-fri-reduction])
  have empty-trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state ≤ trace-fri-error
  by (rule
    trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
    [OF trace-fri-reduction])
  show ?thesis
  by (rule
    stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
      round-bound drift-bound empty-trace-fri-bound current-empty-bound])
qed

theorem stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-f
fixes F X empty-query-error :: prob
and C :: nat ⇒ prob
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-reduction:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
    trace-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))

```

and *composition-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)

adversary-initial-state) $\implies$
composition-fri-reachable-partial-candidate-reduction-assumption
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*))

and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* *A i*)
(*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at* *i*)
adversary-initial-state $\leq C i$

and *fixed-empty-alpha-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-verifier-event*
composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state $\leq F$

and *cross-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
adversary-initial-state $\leq X$

and *current-empty-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-current-empty-query-opening-hit-at* *0*)
adversary-initial-state $\leq \text{empty-query-error}$

shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
hash-collision-budget-value *0*
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 $(\sum i < \text{rounds}. C i) +$
trace-fri-error + *composition-fri-error* +
 $(\sum i < \text{rounds}. C i) +$
(*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* *0*) +
hash-collision-budget-value *0*
(*staged-alpha-search-queries* *budgets* *0*) +
(*F* + *X*) +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* *0*) +
trace-fri-error + *empty-query-error*)

proof –
have *drift-bound*:


```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
  adversary-initial-state  $\leq F + X$ 
by (rule
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-bound-from-fixed-and-witnessed-cross-bound
  [OF fixed-empty-alpha-bound cross-bound])
show ?thesis
by (rule
  stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-reduction
  [OF false-statement wf controlled trace-fri-reduction
    composition-fri-reduction round-bound drift-bound
    current-empty-bound])
qed

theorem stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-reduction
fixes F N empty-query-error :: prob
and C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-reduction:
   $\bigwedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    trace-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
and composition-fri-reduction:
   $\bigwedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    composition-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
and round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
      adversary-initial-state  $\leq C i$ 
and fixed-empty-alpha-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-bad-set-hit-with-empty-composition-header-candidates)

```

```

    adversary-initial-state  $\leq F$ 
and nonunique-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
     A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
     empty-header-partial-trace-candidate-nonunique)
    adversary-initial-state  $\leq N$ 
and current-empty-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
    adversary-initial-state  $\leq$  empty-query-error
shows checked-staged-adversary-acceptance-probability A  $\leq$ 
  hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget + verifier-hash-query-budget) +
  ( $\sum i < \text{rounds}. C i$ ) +
  trace-fri-error + composition-fri-error +
  ( $\sum i < \text{rounds}. C i$ ) +
  (composition-error-bound +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
   hash-collision-budget-value 0
   (staged-alpha-search-queries budgets 0) +
   (F + N) +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
   trace-fri-error + empty-query-error)
proof –
  have drift-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift
      adversary-initial-state  $\leq F + N$ 
    by (rule
      checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-bound-from-fixed-and-nonunique
      [OF fixed-empty-alpha-bound nonunique-bound])
  show ?thesis
    by (rule
      stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri
      [OF false-statement wf controlled trace-fri-reduction
       composition-fri-reduction round-bound drift-bound
       current-empty-bound])
qed

theorem stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri
fixes F L empty-query-error :: prob
and C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace

```

and *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets A
and *trace-fri-reduction*:
 $\bigwedge$ *data* *attacker-state*.
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary } \text{attacker-state}$
 $(\text{staged-proof-transcript } data))$
and *composition-fri-reduction*:
 $\bigwedge$ *data* *attacker-state*.
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary } \text{attacker-state}$
 $(\text{staged-proof-transcript } data))$
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *fixed-empty-alpha-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-verifier-event}$
 $\text{composition-alpha-bad-set-hit-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq F$
and *low-degree-nonunique-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-verifier-event}$
 $\text{empty-header-low-degree-trace-candidate-nonunique})$
 $\text{adversary-initial-state} \leq L$
and *current-empty-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget } \text{budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds}. C \ i) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum i < \text{rounds}. C \ i) +$

```

    (composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
      (F + (trace-fri-error + L)) +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      trace-fri-error + empty-query-error)
  proof -
    let ?M =
      checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
    let ?T =
      checked-staged-security-with-actual-alpha-prefix-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates
    let ?L =
      checked-staged-security-with-actual-alpha-prefix-verifier-event
      empty-header-low-degree-trace-candidate-nonunique
  have trace-fri-data-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state ≤ trace-fri-error
  by (rule
      trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
      [OF trace-fri-reduction])
  have trace-fri-prefix-bound:
    wp-event ?M ?T adversary-initial-state ≤ trace-fri-error
  proof -
    have projected:
      wp-event (checked-staged-security-experiment-with-data-state A)
        (staged-security-with-data-state-verifier-event
          trace-fri-bad-with-empty-composition-header-candidates)
      adversary-initial-state =
      wp-event ?M ?T adversary-initial-state
    by (rule
        checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection)
    show ?thesis
      using projected trace-fri-data-bound by simp
  qed
  have nonunique-bound:
    wp-event ?M
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        empty-header-partial-trace-candidate-nonunique)
    adversary-initial-state ≤ trace-fri-error + L
  proof (rule order-trans)
    show wp-event ?M
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        empty-header-partial-trace-candidate-nonunique)

```

```

    adversary-initial-state ≤
wp-event ?M ?T adversary-initial-state +
wp-event ?M ?L adversary-initial-state
  by (rule
    empty-header-partial-trace-candidate-nonunique-le-trace-fri-or-low-degree-nonunique)
show ... ≤ trace-fri-error + L
  by (intro add-mono trace-fri-prefix-bound low-degree-nonunique-bound)
qed
show ?thesis
  by (rule
    stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-spli
    [OF false-statement wf controlled trace-fri-reduction
      composition-fri-reduction round-bound fixed-empty-alpha-bound
      nonunique-bound current-empty-bound])
qed

theorem stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-spli
fixes F L empty-query-error :: prob
and C :: nat ⇒ prob
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-reduction:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      trace-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
and composition-fri-reduction:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      composition-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
and round-bound:
  ∧ i. i < rounds ⇒
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
      adversary-initial-state ≤ C i
and fixed-empty-alpha-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event

```

composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state $\leq F$
and *low-degree-query-agreement-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-verifier-event
empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit)
adversary-initial-state $\leq L$
and *current-empty-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
($\sum i < \text{rounds}. C i$ +
trace-fri-error + composition-fri-error +
($\sum i < \text{rounds}. C i$ +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(F + (trace-fri-error + L)) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + empty-query-error)
proof –
have *low-degree-nonunique-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-verifier-event
empty-header-low-degree-trace-candidate-nonunique)
adversary-initial-state $\leq L$
by *(rule order-trans*
[OF empty-header-low-degree-trace-candidate-nonunique-le-query-agreement-hit
low-degree-query-agreement-bound])
show *?thesis*
by *(rule*
stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-split
[OF false-statement wf controlled trace-fri-reduction
composition-fri-reduction round-bound fixed-empty-alpha-bound
low-degree-nonunique-bound current-empty-bound])
qed
theorem *stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-split*
fixes *X N empty-query-error :: prob*

and $C :: \text{nat} \Rightarrow \text{prob}$
assumes $\text{false-statement}: \neg \text{exists-valid-trace}$
and $\text{wf}: \text{staged-budget-wellformed budgets}$
and $\text{controlled}: \text{staged-adversary-controlled budgets } A$
and $\text{trace-fri-reduction}$:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and $\text{composition-fri-reduction}$:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and round-bound :
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and not-prefix-bound :
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and cross-bound :
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad}$
 $\text{adversary-initial-state} \leq X$
and $\text{current-empty-bound}$:
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds. } C \ i) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$

$(\sum_{i < \text{rounds. } C} i) +$
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
 $((\text{composition-error-bound} +$
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + N) + X) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
trace-fri-error + *empty-query-error*)

proof –

let $?M =$
checked-staged-security-experiment-with-actual-alpha-prefix-data-state A

let $?P =$
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0)

have *transcript-bound*:
wp-event (checked-staged-transcript-with-alpha-prefix-program A)
(checked-staged-transcript-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state $\leq ?P$

by (rule
checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
 $[OF \text{ wf controlled}]$)

have *prefix-bound*:
wp-event ?M
(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state $\leq ?P$

by (rule
checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
 $[OF \text{ transcript-bound}]$)

have *fixed-empty-alpha-bound*:
wp-event ?M
(checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state $\leq ?P + N$

proof (rule *order-trans*)

show *wp-event ?M*
(checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state $\leq$


```

wp-event ?M
  (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
    composition-trace-bad-alpha-space)
  adversary-initial-state +
wp-event ?M
  (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
    composition-trace-bad-alpha-space)
  adversary-initial-state
by (rule
  checked-staged-security-with-actual-alpha-prefix-fixed-empty-alpha-hit-le-prefix-or-not-prefix
  [OF wf controlled])
show ... ≤ ?P + N
  by (intro add-mono prefix-bound not-prefix-bound)
qed
show ?thesis
  by (rule
    stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-spli
    [OF false-statement wf controlled trace-fri-reduction
      composition-fri-reduction round-bound fixed-empty-alpha-bound
      cross-bound current-empty-bound])
qed

theorem stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-spli
fixes X P R empty-query-error :: prob
and C :: nat ⇒ prob
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-reduction:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      trace-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
and composition-fri-reduction:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      composition-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
and round-bound:
  ∧ i. i < rounds ⇒
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)

```

$\text{adversary-initial-state} \leq C \ i$
and *cross-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad}$
 $\text{adversary-initial-state} \leq X$
and *data-pre-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-pre-hit}$
 $\text{adversary-initial-state} \leq P$
and *sampled-transcript-new-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-new-hit}$
 $\text{adversary-initial-state} \leq R$
and *current-empty-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds}. C \ i) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum i < \text{rounds}. C \ i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $((\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $X + (P + R) + (P + R))) + X) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{trace-fri-error} + \text{empty-query-error})$
proof –
let $?M =$
 $\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A$
let $?N =$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $X + (P + R) + (P + R)$

have *no-prefix-bound*:
wp-event ?*M*
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
adversary-initial-state $\leq P + R$
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-bound-from-sampled-transcript
[*OF data-pre-bound sampled-transcript-new-bound*])
have *path-bound*:
wp-event ?*M*
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
adversary-initial-state $\leq P + R$
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-bound-from-sampled-transcript
[*OF data-pre-bound sampled-transcript-new-bound*])
have *collision-bound*:
wp-event ?*M*
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
adversary-initial-state $\leq$
hash-collision-budget-value 0 (*staged-alpha-search-queries* budgets 0)
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
[*OF wf controlled*])
have *uncovered-bound*:
wp-event ?*M*
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
adversary-initial-state $\leq X$
by (rule *order-trans*
[*OF*
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-witnessed-cross-or-merkle-bad
cross-bound])
have *not-prefix-bound*:
wp-event ?*M*
(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq ?N$
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound
[*OF wf controlled collision-bound uncovered-bound no-prefix-bound*
path-bound])
show ?*thesis*
by (rule
stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-spli
[*OF false-statement wf controlled trace-fri-reduction*
composition-fri-reduction round-bound not-prefix-bound cross-bound
current-empty-bound])
qed

theorem *stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-spli*
fixes *X P empty-query-error* :: *prob*

and $C :: \text{nat} \Rightarrow \text{prob}$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *composition-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state A } i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *cross-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad}$
 $\text{adversary-initial-state} \leq X$
and *data-pre-bound*:
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $\text{staged-security-with-data-state-transcript-pre-hit}$
 $\text{adversary-initial-state} \leq P$
and *current-empty-bound*:
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows *checked-staged-adversary-acceptance-probability A* $\leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds. } C \ i) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum i < \text{rounds. } C \ i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size})$

```

    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    ((composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    (hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    X + (P + staged-concrete-transcript-target-error-bound) +
    (P + staged-concrete-transcript-target-error-bound))) + X) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    trace-fri-error + empty-query-error)
proof -
  have sampled-transcript-new-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state ≤ staged-concrete-transcript-target-error-bound
  by (rule checked-staged-security-with-data-state-transcript-new-hit-bound
    [OF wf controlled])
  show ?thesis
  by (rule
    stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-spli
    [OF false-statement wf controlled trace-fri-reduction
    composition-fri-reduction round-bound cross-bound data-pre-bound
    sampled-transcript-new-bound current-empty-bound])
qed

theorem stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-spli
fixes S H P empty-query-error :: prob
and C :: nat ⇒ prob
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-reduction:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state) ⇒
    trace-fri-reachable-partial-candidate-reduction-assumption
    (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
and composition-fri-reduction:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state) ⇒

```

composition-fri-reachable-partial-candidate-reduction-assumption
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
and round-bound:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state $\leq C$ i
and sparse-bound:
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witnessed-sparse-cross-bad
 adversary-initial-state $\leq S$
and merkle-side-bound:
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad
 adversary-initial-state $\leq H$
and data-pre-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-transcript-pre-hit
 adversary-initial-state $\leq P$
and current-empty-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 adversary-initial-state $\leq \text{empty-query-error}$
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 ($\sum i < \text{rounds}. C$ i) +
 trace-fri-error + composition-fri-error +
 ($\sum i < \text{rounds}. C$ i) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 ((composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 ($S + H$) + ($P + \text{staged-concrete-transcript-target-error-bound}$) +

$(P + \text{staged-concrete-transcript-target-error-bound})) + (S + H)) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{trace-fri-error} + \text{empty-query-error})$

proof –

have *cross-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad}$
 $\text{adversary-initial-state} \leq S + H$

proof (*rule order-trans*)

show *wp-event*
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad}$
 $\text{adversary-initial-state} \leq$
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-sparse-cross-bad}$
 $\text{adversary-initial-state} +$
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad}$
 $\text{adversary-initial-state}$
by (*rule*
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad-le-sparse-or-merkle-side}$)

show $\dots \leq S + H$
by (*intro add-mono sparse-bound merkle-side-bound*)

qed

show *?thesis*
by (*rule*
 $\text{stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-split}$
 $[\text{OF false-statement wf controlled trace-fri-reduction}$
 $\text{composition-fri-reduction round-bound cross-bound data-pre-bound}$
 $\text{current-empty-bound}]$)

qed

theorem *stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-split*
fixes $S L K P \text{ empty-query-error} :: \text{prob}$
and $C :: \text{nat} \Rightarrow \text{prob}$
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* A
and *trace-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$

$\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \\ \text{adversary-initial-state}) \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and $\text{composition-fri-reduction:}$
 $\bigwedge \text{data attacker-state.}$
 $\text{Some } (\text{data, attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \\ \text{adversary-initial-state}) \implies$
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and round-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and sparse-bound:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-sparse-cross-bad}$
 $\text{adversary-initial-state} \leq S$
and $\text{local-collision-bound:}$
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad}$
 $\text{adversary-initial-state} \leq L$
and coupling-bound:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-coupling-bad}$
 $\text{adversary-initial-state} \leq K$
and data-pre-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-pre-hit}$
 $\text{adversary-initial-state} \leq P$
and $\text{current-empty-bound:}$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$

$(\sum_{i < \text{rounds.}} C\ i) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum_{i < \text{rounds.}} C\ i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $((\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(S + (L + K)) + (P + \text{staged-concrete-transcript-target-error-bound}) +$
 $(P + \text{staged-concrete-transcript-target-error-bound}))) +$
 $(S + (L + K))) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{trace-fri-error} + \text{empty-query-error})$

proof –

have *merkle-side-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad}$
 $\text{adversary-initial-state} \leq L + K$

proof (rule *order-trans*)

show *wp-event*
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad}$
 $\text{adversary-initial-state} \leq$
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-local-collision-bad}$
 $\text{adversary-initial-state} +$
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-coupling-bad}$
 $\text{adversary-initial-state}$
by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-witnessed-merkle-side-bad-le-local-or-coupling})$

show $\dots \leq L + K$
by (*intro add-mono local-collision-bound coupling-bound*)

qed

```

show ?thesis
by (rule
  stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable-fri-split
  [OF false-statement wf controlled trace-fri-reduction
    composition-fri-reduction round-bound sparse-bound
    merkle-side-bound data-pre-bound current-empty-bound])
qed

end

```

These wrappers expose the broad witnessed cross/Merkle diagnostic split used during development of the reachable-FRI route. They remain available only as internal plumbing; the public route should use aligned transcript-indexed wrappers that do not expose these diagnostic events.

```

hide-fact
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
  soundness.stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-reachable
end

```

```

theory Soundness-FRI-Partial-Evidence
imports Soundness-Relevant-Drift-Reachable-FRI
begin

```

Normalized partial-opening FRI evidence predicates for the reachable-state FRI obligations.

This theory is intentionally narrow. It does not change the public soundness theorem path; it only factors the remaining reachable FRI obligations into explicit evidence predicates that later proofs can target.

```

context soundness
begin

```

```

definition trace-fri-partial-candidate-evidence
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$ 
    'f  $\Rightarrow$  nat list  $\Rightarrow$  'f authenticated-opening list list  $\Rightarrow$ 
    'f list  $\Rightarrow$  bool

```

```

where
  trace-fri-partial-candidate-evidence s out fr query-idxs

```

$\text{trace-openings trace-table} \longleftrightarrow$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table}$

definition *composition-fri-partial-candidate-evidence*

$:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow$
 $\text{'f} \Rightarrow \text{'f list} \Rightarrow \text{'f} \Rightarrow \text{'f list} \Rightarrow \text{'f} \Rightarrow$
 $\text{'f list} \Rightarrow \text{'f} \Rightarrow \text{nat list} \Rightarrow$
 $\text{'f authenticated-opening list list} \Rightarrow \text{nat list} \Rightarrow$
 $\text{'f authenticated-opening list list} \Rightarrow \text{'f list} \Rightarrow$
 $\text{'f list} \Rightarrow \text{bool}$

where

$\text{composition-fri-partial-candidate-evidence } s \text{ out fr f-fri-roots}$
 $\text{f-final as dg composition-fri-roots final trace-query-idxs}$
 $\text{trace-openings composition-query-idxs composition-openings trace-table}$
 $\text{composition-table} \longleftrightarrow$
 $\text{accepted-with-partial-initial-openings } s \text{ out fr f-fri-roots f-final}$
 $\text{as dg composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{composition-query-idxs composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\neg \text{composition-table-low-degree maxDegree composition-table}$

lemma *trace-fri-bad-with-reachable-partial-candidate-iff-evidence:*

$\text{trace-fri-bad-with-reachable-partial-candidate } s \text{ out} \longleftrightarrow$
 $(\exists \text{ fr query-idxs trace-openings trace-table.}$
 $\text{trace-fri-partial-candidate-evidence } s \text{ out fr query-idxs}$
 $\text{trace-openings trace-table})$

unfolding *trace-fri-bad-with-reachable-partial-candidate-def*

trace-fri-partial-candidate-evidence-def

by *blast*

lemma *composition-fri-bad-with-reachable-partial-candidate-iff-evidence:*

$\text{composition-fri-bad-with-reachable-partial-candidate } s \text{ out} \longleftrightarrow$

$(\exists \text{ fr f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table.}$
 $\text{composition-fri-partial-candidate-evidence } s \text{ out fr f-fri-roots}$
 $\text{f-final as dg composition-fri-roots final trace-query-idxs}$
 $\text{trace-openings composition-query-idxs composition-openings}$
 $\text{trace-table composition-table})$

unfolding *composition-fri-bad-with-reachable-partial-candidate-def*

composition-fri-partial-candidate-evidence-def

by *blast*

lemma *trace-fri-partial-candidate-evidenceI*:
assumes
 accepted-with-partial-trace-openings s out fr query-idxs
 trace-openings
 partial-trace-table-candidate trace-table trace-openings
 $\neg$ *trace-table-low-degree trace-table*
shows
 trace-fri-partial-candidate-evidence s out fr query-idxs
 trace-openings trace-table
using *assms* **unfolding** *trace-fri-partial-candidate-evidence-def* **by** *blast*

lemma *trace-fri-partial-candidate-evidenceD*:
assumes
 trace-fri-partial-candidate-evidence s out fr query-idxs
 trace-openings trace-table
shows
 accepted-with-partial-trace-openings s out fr query-idxs
 trace-openings
 partial-trace-table-candidate trace-table trace-openings
 $\neg$ *trace-table-low-degree trace-table*
using *assms* **unfolding** *trace-fri-partial-candidate-evidence-def* **by** *blast+*

lemma *trace-fri-bad-with-reachable-partial-candidateE*:
assumes *trace-fri-bad-with-reachable-partial-candidate s out*
obtains *fr query-idxs trace-openings trace-table* **where**
 trace-fri-partial-candidate-evidence s out fr query-idxs
 trace-openings trace-table
using *assms*
unfolding *trace-fri-bad-with-reachable-partial-candidate-iff-evidence*
by *blast*

lemma *composition-fri-partial-candidate-evidenceI*:
assumes
 accepted-with-partial-initial-openings s out fr f-fri-roots f-final
 as dg composition-fri-roots final trace-query-idxs trace-openings
 composition-query-idxs composition-openings
 partial-trace-table-candidate trace-table trace-openings
 partial-composition-table-candidate composition-table
 composition-openings
 trace-table-low-degree trace-table
 $\neg$ *composition-table-low-degree maxDegree composition-table*
shows
 composition-fri-partial-candidate-evidence s out fr f-fri-roots
 f-final as dg composition-fri-roots final trace-query-idxs
 trace-openings composition-query-idxs composition-openings
 trace-table composition-table
using *assms*
unfolding *composition-fri-partial-candidate-evidence-def* **by** *blast*

lemma *composition-fri-partial-candidate-evidenceD:*

assumes

*composition-fri-partial-candidate-evidence s out fr f-fri-roots
f-final as dg composition-fri-roots final trace-query-idxs
trace-openings composition-query-idxs composition-openings trace-table
composition-table*

shows

*accepted-with-partial-initial-openings s out fr f-fri-roots f-final
as dg composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings
partial-trace-table-candidate trace-table trace-openings
partial-composition-table-candidate composition-table
composition-openings
trace-table-low-degree trace-table
 $\neg$ composition-table-low-degree maxDegree composition-table*

using *assms*

unfolding *composition-fri-partial-candidate-evidence-def* **by** *blast+*

lemma *composition-fri-bad-with-reachable-partial-candidateE:*

assumes *composition-fri-bad-with-reachable-partial-candidate s out*

obtains *fr f-fri-roots f-final as dg composition-fri-roots final*

*trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table where
composition-fri-partial-candidate-evidence s out fr f-fri-roots
f-final as dg composition-fri-roots final trace-query-idxs
trace-openings composition-query-idxs composition-openings
trace-table composition-table*

using *assms*

unfolding *composition-fri-bad-with-reachable-partial-candidate-iff-evidence*
by *blast*

lemma *trace-fri-reachable-partial-candidate-reductionD:*

assumes *trace-fri-reachable-partial-candidate-reduction-assumption s*

shows

*wp-event verify-monad
(trace-fri-bad-with-reachable-partial-candidate s) s $\leq$
trace-fri-error*

using *assms*

unfolding *trace-fri-reachable-partial-candidate-reduction-assumption-def*
by *simp*

lemma *composition-fri-reachable-partial-candidate-reductionD:*

assumes *composition-fri-reachable-partial-candidate-reduction-assumption s*

shows

*wp-event verify-monad
(composition-fri-bad-with-reachable-partial-candidate s) s $\leq$
composition-fri-error*

using *assms*

unfolding *composition-fri-reachable-partial-candidate-reduction-assumption-def*
by *simp*

definition *trace-fri-partial-candidate-opening-evidence*

```

:: ('f, 'a) protocol-channel-scheme =>
  (unit list × ('f, 'a) protocol-channel-scheme) option =>
  'f list => 'f list => 'f => 'f =>
  'f list => 'f list => 'f => nat list =>
  'f list list list => 'f list list list =>
  'f => nat list => 'f authenticated-opening list list =>
  'f list => bool

```

where

```

trace-fri-partial-candidate-opening-evidence s out trace-roots
  trace-bs trace-final dg composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr candidate-query-idxs trace-openings
  trace-table <=>
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers ∧
trace-fri-partial-candidate-evidence s out fr candidate-query-idxs
  trace-openings trace-table

```

definition *composition-fri-partial-candidate-opening-evidence*

```

:: ('f, 'a) protocol-channel-scheme =>
  (unit list × ('f, 'a) protocol-channel-scheme) option =>
  'f list => 'f list => 'f => 'f =>
  'f list => 'f list => 'f => nat list =>
  'f list list list => 'f list list list =>
  'f => 'f list => 'f => 'f list => 'f =>
  'f list => 'f => nat list =>
  'f authenticated-opening list list => nat list =>
  'f authenticated-opening list list => 'f list =>
  'f list => bool

```

where

```

composition-fri-partial-candidate-opening-evidence s out trace-roots
  trace-bs trace-final fri-dg opening-composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs trace-openings
  composition-query-idxs composition-openings trace-table
  composition-table <=>
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  fri-dg opening-composition-roots composition-bs composition-final
  fri-query-idxs trace-round-layers composition-round-layers ∧
composition-fri-partial-candidate-evidence s out fr f-fri-roots
  f-final as dg composition-fri-roots final trace-query-idxs
  trace-openings composition-query-idxs composition-openings
  trace-table composition-table

```

lemma *trace-fri-partial-candidate-opening-evidenceD:*

assumes

trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table

shows

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
trace-fri-partial-candidate-evidence s out fr candidate-query-idxs
trace-openings trace-table

using *assms*

unfolding *trace-fri-partial-candidate-opening-evidence-def* **by** *blast+*

lemma *composition-fri-partial-candidate-opening-evidenceD:*

assumes

composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table

shows

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
composition-fri-partial-candidate-evidence s out fr f-fri-roots
f-final as dg composition-fri-roots final trace-query-idxs
trace-openings composition-query-idxs composition-openings
trace-table composition-table

using *assms*

unfolding *composition-fri-partial-candidate-opening-evidence-def* **by** *blast+*

lemma *verify-monad-trace-fri-bad-extracts-opening-evidence:*

assumes *outcome:*

Some (result, final-state) ∈ set-dist (execute verify-monad s)

and *bad:*

trace-fri-bad-with-reachable-partial-candidate s
(Some (result, final-state))

obtains *trace-roots trace-bs trace-final dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table

where

```

    trace-fri-partial-candidate-opening-evidence s
      (Some (result, final-state)) trace-roots trace-bs trace-final dg
      composition-roots composition-bs composition-final fri-query-idxs
      trace-round-layers composition-round-layers fr candidate-query-idxs
      trace-openings trace-table
  proof -
    from verify-monad-accepted-fri-opening-transcript[OF outcome]
    obtain trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final fri-query-idxs trace-round-layers
      composition-round-layers where fri-opening:
      accepted-fri-opening-transcript s (Some (result, final-state))
      trace-roots trace-bs trace-final dg composition-roots composition-bs
      composition-final fri-query-idxs trace-round-layers
      composition-round-layers
    by blast
    from trace-fri-bad-with-reachable-partial-candidateE[OF bad]
    obtain fr candidate-query-idxs trace-openings trace-table where evidence:
      trace-fri-partial-candidate-evidence s
      (Some (result, final-state)) fr candidate-query-idxs trace-openings
      trace-table
    by blast
    have combined:
      trace-fri-partial-candidate-opening-evidence s
      (Some (result, final-state)) trace-roots trace-bs trace-final dg
      composition-roots composition-bs composition-final fri-query-idxs
      trace-round-layers composition-round-layers fr candidate-query-idxs
      trace-openings trace-table
    unfolding trace-fri-partial-candidate-opening-evidence-def
    using fri-opening evidence by simp
    show ?thesis
      by (rule that[OF combined])
  qed

lemma verify-monad-composition-fri-bad-extracts-opening-evidence:
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and bad:
    composition-fri-bad-with-reachable-partial-candidate s
    (Some (result, final-state))
  obtains trace-roots trace-bs trace-final fri-dg opening-composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table
  where
    composition-fri-partial-candidate-opening-evidence s
      (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
      opening-composition-roots composition-bs composition-final

```


fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table
proof –
from *verify-monad-accepted-fri-opening-transcript*[*OF outcome*]
obtain *trace-roots trace-bs trace-final fri-dg opening-composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers **where** *fri-opening:*
accepted-fri-opening-transcript s (Some (result, final-state))
trace-roots trace-bs trace-final fri-dg opening-composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers
by *blast*
from *composition-fri-bad-with-reachable-partial-candidateE*[*OF bad*]
obtain *fr f-fri-roots f-final as dg composition-fri-roots final*
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table **where** *evidence:*
composition-fri-partial-candidate-evidence s
(Some (result, final-state)) fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
by *blast*
have *combined:*
composition-fri-partial-candidate-opening-evidence s
(Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table
unfolding *composition-fri-partial-candidate-opening-evidence-def*
using *fri-opening evidence* **by** *simp*
show *?thesis*
by *(rule that[OF combined])*
qed

lemma *verify-monad-composition-fri-bad-extracts-opening-evidence-with-degree-bound:*

assumes *outcome:*

Some (result, final-state) ∈ set-dist (execute verify-monad s)

and *bad:*

composition-fri-bad-with-reachable-partial-candidate s

(Some (result, final-state))

obtains *trace-roots trace-bs trace-final fri-dg opening-composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr f-fri-roots f-final as dg

composition-fri-roots final trace-query-idxs trace-openings

composition-query-idxs composition-openings trace-table

```

    composition-table
where
    composition-fri-partial-candidate-opening-evidence s
      (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
      opening-composition-roots composition-bs composition-final
      fri-query-idxs trace-round-layers composition-round-layers fr
      f-fri-roots f-final as dg composition-fri-roots final
      trace-query-idxs trace-openings composition-query-idxs
      composition-openings trace-table composition-table
    and to-nat fri-dg ≤ maxDegree
proof (rule verify-monad-composition-fri-bad-extracts-opening-evidence
  [OF outcome bad])
fix trace-roots trace-bs trace-final fri-dg opening-composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs trace-openings
  composition-query-idxs composition-openings trace-table composition-table
assume evidence:
  composition-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
    opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers fr
    f-fri-roots f-final as dg composition-fri-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table
have fri-opening:
  accepted-fri-opening-transcript s (Some (result, final-state))
    trace-roots trace-bs trace-final fri-dg opening-composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers
using composition-fri-partial-candidate-opening-evidenceD(1)
  [OF evidence] .
have challenges:
  accepted-fri-challenges s (Some (result, final-state))
    trace-bs fri-dg composition-bs
by (rule accepted-fri-opening-transcript-challenges[OF fri-opening])
have degree-bound: to-nat fri-dg ≤ maxDegree
by (rule verify-monad-accepted-fri-challenges-degree-bound
  [OF outcome challenges])
show ?thesis
by (rule that[OF evidence degree-bound])
qed

lemma verifier-query-round-after-index-final-checks:
assumes outcome:
  Some ((), t) ∈
    set-dist
    (execute
      (verifier-query-round-after-index-program

```

```

      fr f-fl f-final as fl final raw) s)
obtains idx fv s1 f-i f-len f-pow s2 c-i c-len c-pow s4 where
  idx = index (to-nat raw)
  Some (fv, s1) ∈
    set-dist (execute (mmap (check-decommit-on-query fr idx)) s)
  Some ((f-i, f-final, f-len, f-pow), s2) ∈
    set-dist
      (execute
        (mfold (idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
  Some ((c-i, final, c-len, c-pow), s4) ∈
    set-dist
      (execute
        (mfold (idx, cp-eval as fv (h ^ idx * shift),
          clength * scale, 1)
          (receive-query-commits fl)) s2)
  t = s4
proof -
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
    query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
    and trace-fri:
      Some (f-out, s2) ∈
        set-dist (execute
          (mfold (?idx, hd fv, clength * scale, 1)
            (receive-query-commits f-fl)) s1)
    and assert-trace:
      Some ((), s3) ∈
        set-dist
          (execute
            (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
    and comp-fri:
      Some (c-out, s4) ∈
        set-dist (execute
          (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
            clength * scale, 1)
            (receive-query-commits fl)) s3)
    and assert-comp:
      Some ((), t) ∈
        set-dist
          (execute
            (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
  unfolding verifier-query-round-after-index-program-def
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
obtain f-i f-x f-len f-pow where f-out-eq:
  f-out = (f-i, f-x, f-len, f-pow)
  by (cases f-out) auto

```

```

have f-x-eq: f-x = f-final
  using assert-trace unfolding assert-def f-out-eq
  by (cases f-x = f-final) (auto simp: throw-no-outcome)
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def f-out-eq f-x-eq by simp
obtain c-i c-x c-len c-pow where c-out-eq:
  c-out = (c-i, c-x, c-len, c-pow)
  by (cases c-out) auto
have c-x-eq: c-x = final
  using assert-comp unfolding assert-def c-out-eq
  by (cases c-x = final) (auto simp: throw-no-outcome)
have t-eq: t = s4
  using assert-comp unfolding assert-def c-out-eq c-x-eq by simp
have trace-fri':
  Some ((f-i, f-final, f-len, f-pow), s2) ∈
    set-dist
    (execute
      (mfold (?idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl)) s1)
  using trace-fri f-out-eq f-x-eq by simp
have comp-fri':
  Some ((c-i, final, c-len, c-pow), s4) ∈
    set-dist
    (execute
      (mfold (?idx, cp-eval as fv (h  $\wedge$  ?idx * shift),
        clength * scale, 1)
        (receive-query-commits fl)) s2)
  using comp-fri c-out-eq c-x-eq s3-eq by simp
show ?thesis
  by (rule that[OF refl query-decommit trace-fri' comp-fri' t-eq])
qed

```

```

lemma verifier-query-round-program-final-checks:
assumes outcome:
  Some ((), t)  $\in$ 
    set-dist
    (execute
      (verifier-query-round-program fr f-fl f-final as fl final) s)
obtains raw idx fv s0 s1 f-i f-len f-pow s2 c-i c-len c-pow s4
where
  Some (raw, s0)  $\in$  set-dist (execute receive-query-index-challenge s)
  idx = index (to-nat raw)
  Some (fv, s1)  $\in$ 
    set-dist (execute (mmap (check-decommit-on-query fr idx)) s0)
  Some ((f-i, f-final, f-len, f-pow), s2)  $\in$ 
    set-dist
    (execute
      (mfold (idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl)) s1)

```

```

    Some ((c-i, final, c-len, c-pow), s4) ∈
      set-dist
        (execute
          (mfold (idx, cp-eval as fv (h ^ idx * shift),
            clength * scale, 1)
            (receive-query-commits fl)) s2)
    t = s4
proof -
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where
  raw:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s)
and after:
    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s0)
  by (auto elim!: set-dist-bindE)
from verifier-query-round-after-index-final-checks[OF after]
obtain idx fv s1 f-i f-len f-pow s2 c-i c-len c-pow s4 where
  idx-eq: idx = index (to-nat raw)
and query-decommit:
    Some (fv, s1) ∈
      set-dist (execute (mmap (check-decommit-on-query fr idx)) s0)
and trace-fri:
    Some ((f-i, f-final, f-len, f-pow), s2) ∈
      set-dist
        (execute
          (mfold (idx, hd fv, clength * scale, 1)
            (receive-query-commits f-fl)) s1)
and comp-fri:
    Some ((c-i, final, c-len, c-pow), s4) ∈
      set-dist
        (execute
          (mfold (idx, cp-eval as fv (h ^ idx * shift),
            clength * scale, 1)
            (receive-query-commits fl)) s2)
and t-eq: t = s4
by blast
show ?thesis
by (rule that[OF raw idx-eq query-decommit trace-fri comp-fri t-eq])
qed

end

end

```

```

theory Soundness-FRI-Layer-Evidence
  imports Soundness-FRI-Partial-Evidence
begin

```

Partial FRI layer evidence extracted from the verifier transcript.

This theory only packages facts already exposed by the trace and composition layer-opening lemmas for *accepted-fri-opening-transcript*. It does not reconstruct complete FRI tables from sampled openings.

```

context soundness
begin

```

```

definition fri-layer-step-evidence
  :: 'f  $\Rightarrow$  'f  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$  nat  $\Rightarrow$ 
    nat  $\Rightarrow$  'f  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f list  $\Rightarrow$ 
    nat  $\Rightarrow$  'f  $\Rightarrow$  'f list  $\Rightarrow$  bool

```

where

```

  fri-layer-step-evidence rt ch len idx pw sibling-idx
    xp xp-path xn xn-path next-idx next-value chunk  $\longleftrightarrow$ 
    sibling-idx = fri-sibling-index len idx  $\wedge$ 
    next-idx = idx mod (len div 2)  $\wedge$ 
    next-value =
      fri-fold-value ch xp xn
      (fri-fold-denominator ((h  $\wedge$  idx) * shift) pw)  $\wedge$ 
    fri-layer-opening-chunk len xp xp-path xn xn-path chunk

```

```

definition fri-round-layer-evidence
  :: 'f list  $\Rightarrow$  'f list  $\Rightarrow$  nat list  $\Rightarrow$ 
    nat  $\Rightarrow$  nat  $\Rightarrow$  'f list list list  $\Rightarrow$  bool

```

where

```

  fri-round-layer-evidence roots challenges query-idxs round-idx
    layer-idx round-layers  $\longleftrightarrow$ 
    round-idx < length query-idxs  $\wedge$ 
    layer-idx < length roots  $\wedge$ 
    length challenges = length roots  $\wedge$ 
    length round-layers = length query-idxs  $\wedge$ 
    ( $\exists$  xp xp-path xn xn-path.
      fri-layer-step-evidence
        (roots ! layer-idx)
        (challenges ! layer-idx)
        (fri-layer-lengths (length roots) (clength * scale) ! layer-idx)
        (fri-layer-indices (length roots) (query-idxs ! round-idx)
          (clength * scale) ! layer-idx)
        (2  $\wedge$  layer-idx)
        (fri-sibling-index
          (fri-layer-lengths (length roots) (clength * scale) ! layer-idx)
          (fri-layer-indices (length roots) (query-idxs ! round-idx)
            (clength * scale) ! layer-idx))
        xp xp-path xn xn-path
    )

```

$$\begin{aligned}
& ((\text{fri-layer-indices } (\text{length roots}) (\text{query-idxs ! round-idx}) \\
& \quad (\text{clength * scale}) ! \text{layer-idx}) \bmod \\
& \quad ((\text{fri-layer-lengths } (\text{length roots}) (\text{clength * scale}) ! \text{layer-idx}) \\
& \quad \text{div } 2)) \\
& (\text{fri-fold-value } (\text{challenges ! layer-idx}) \text{ xp xn} \\
& \quad (\text{fri-fold-denominator} \\
& \quad ((h \wedge \\
& \quad \quad (\text{fri-layer-indices } (\text{length roots}) (\text{query-idxs ! round-idx}) \\
& \quad \quad (\text{clength * scale}) ! \text{layer-idx})) * \text{shift}) \\
& \quad (2 \wedge \text{layer-idx}))) \\
& (\text{round-layers ! round-idx ! layer-idx}))
\end{aligned}$$

definition *fri-all-round-layer-evidence*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow \text{bool}$

where

fri-all-round-layer-evidence roots challenges query-idxs round-layers
 $\longleftrightarrow$
 $\text{length challenges} = \text{length roots} \wedge$
 $\text{length round-layers} = \text{length query-idxs} \wedge$
 $(\forall \text{round-idx} < \text{length query-idxs}.$
 $\quad \forall \text{layer-idx} < \text{length roots}.$
 $\quad \text{fri-round-layer-evidence roots challenges query-idxs round-idx}$
 $\quad \text{layer-idx round-layers})$

definition *fri-evidence-layer-len* $:: 'f \text{ list} \Rightarrow \text{nat} \Rightarrow \text{nat}$

where

fri-evidence-layer-len roots layer-idx =
 $\text{fri-layer-lengths } (\text{length roots}) (\text{clength * scale}) ! \text{layer-idx}$

definition *fri-evidence-layer-idx*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow \text{nat} \Rightarrow \text{nat} \Rightarrow \text{nat}$

where

fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
 $\text{fri-layer-indices } (\text{length roots}) (\text{query-idxs ! round-idx})$
 $(\text{clength * scale}) ! \text{layer-idx}$

definition *fri-evidence-next-idx*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow \text{nat} \Rightarrow \text{nat} \Rightarrow \text{nat}$

where

fri-evidence-next-idx roots query-idxs round-idx layer-idx =
 $\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} \bmod$
 $(\text{fri-evidence-layer-len roots layer-idx} \text{ div } 2)$

definition *fri-evidence-next-value*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow \text{nat} \Rightarrow \text{nat} \Rightarrow$
 $'f \Rightarrow 'f \Rightarrow 'f$

where

fri-evidence-next-value roots challenges query-idxs round-idx layer-idx

$xp\ xn =$
 $fri\text{-}fold\text{-}value\ (challenges\ !\ layer\text{-}idx)\ xp\ xn$
 $(fri\text{-}fold\text{-}denominator$
 $((h\ \wedge\ fri\text{-}evidence\text{-}layer\text{-}idx\ roots\ query\text{-}idxs\ round\text{-}idx\ layer\text{-}idx)\ * shift)$
 $(2\ \wedge\ layer\text{-}idx))$

definition *fri-sampled-table-fold*

$:: 'f \Rightarrow nat \Rightarrow nat \Rightarrow nat \Rightarrow 'f\ list \Rightarrow$
 $'f\ list \Rightarrow 'f\ list \Rightarrow 'f \Rightarrow bool$

where

$fri\text{-}sampled\text{-}table\text{-}fold\ ch\ len\ idx\ pw\ fri\text{-}dom\ layer\ chunk\ next\text{-}value$
 $\longleftrightarrow$
 $(\exists xp\ xp\text{-}path\ xn\ xn\text{-}path.$
 $fri\text{-}layer\text{-}opening\text{-}chunk\ len\ xp\ xp\text{-}path\ xn\ xn\text{-}path\ chunk\ \wedge$
 $fri\text{-}opening\text{-}matches\text{-}table\ len\ idx\ layer\ xp\ xn\ \wedge$
 $fri\text{-}dom\ !\ idx = (h\ \wedge\ idx\ * shift)\ \wedge\ pw\ \wedge$
 $next\text{-}value = fri\text{-}table\text{-}fold\text{-}value\ ch\ layer\ fri\text{-}dom\ len\ pw\ idx)$

definition *generic-fri-round-forced-next-value*

$:: 'f\ list \Rightarrow 'f\ list \Rightarrow nat\ list \Rightarrow$
 $'f\ list\ list\ list \Rightarrow nat \Rightarrow nat \Rightarrow 'f \Rightarrow bool$

where

$generic\text{-}fri\text{-}round\text{-}forced\text{-}next\text{-}value\ roots\ challenges\ query\text{-}idxs$
 $round\text{-}layers\ round\text{-}idx\ layer\text{-}idx\ v \longleftrightarrow$
 $(\exists xp\ xp\text{-}path\ xn\ xn\text{-}path.$
 $fri\text{-}layer\text{-}step\text{-}evidence$
 $(roots\ !\ layer\text{-}idx)$
 $(challenges\ !\ layer\text{-}idx)$
 $(fri\text{-}evidence\text{-}layer\text{-}len\ roots\ layer\text{-}idx)$
 $(fri\text{-}evidence\text{-}layer\text{-}idx\ roots\ query\text{-}idxs\ round\text{-}idx\ layer\text{-}idx)$
 $(2\ \wedge\ layer\text{-}idx)$
 $(fri\text{-}sibling\text{-}index\ (fri\text{-}evidence\text{-}layer\text{-}len\ roots\ layer\text{-}idx)$
 $(fri\text{-}evidence\text{-}layer\text{-}idx\ roots\ query\text{-}idxs\ round\text{-}idx\ layer\text{-}idx))$
 $xp\ xp\text{-}path\ xn\ xn\text{-}path$
 $(fri\text{-}evidence\text{-}next\text{-}idx\ roots\ query\text{-}idxs\ round\text{-}idx\ layer\text{-}idx)$
 v
 $(round\text{-}layers\ !\ round\text{-}idx\ !\ layer\text{-}idx))$

lemma *fri-layer-step-evidenceI:*

assumes $sibling\text{-}idx = fri\text{-}sibling\text{-}index\ len\ idx$
and $next\text{-}idx = idx\ mod\ (len\ div\ 2)$
and $next\text{-}value =$
 $fri\text{-}fold\text{-}value\ ch\ xp\ xn$
 $(fri\text{-}fold\text{-}denominator\ ((h\ \wedge\ idx)\ * shift)\ pw)$
and $fri\text{-}layer\text{-}opening\text{-}chunk\ len\ xp\ xp\text{-}path\ xn\ xn\text{-}path\ chunk$
shows $fri\text{-}layer\text{-}step\text{-}evidence\ rt\ ch\ len\ idx\ pw\ sibling\text{-}idx$
 $xp\ xp\text{-}path\ xn\ xn\text{-}path\ next\text{-}idx\ next\text{-}value\ chunk$
using *assms unfolding fri-layer-step-evidence-def by simp*

lemma *fri-layer-step-evidenceD*:
assumes *fri-layer-step-evidence* *rt ch len idx pw sibling-idx*
xp xp-path xn xn-path next-idx next-value chunk
shows *sibling-idx = fri-sibling-index len idx*
and *next-idx = idx mod (len div 2)*
and *next-value =*
fri-fold-value ch xp xn
*(fri-fold-denominator ((h ^ idx) * shift) pw)*
and *fri-layer-opening-chunk len xp xp-path xn xn-path chunk*
using *assms unfolding fri-layer-step-evidence-def* **by** *simp-all*

lemma *fri-layer-opening-chunk-values-unique*:
assumes *first*:
fri-layer-opening-chunk len xp xp-path xn xn-path chunk
and *second*:
fri-layer-opening-chunk len yp yp-path yn yn-path chunk
shows *xp = yp*
and *xn = yn*
proof –
have *chunk-eq*:
 $[xp] @ xp\text{-}path @ [xn] @ xn\text{-}path =$
 $[yp] @ yp\text{-}path @ [yn] @ yn\text{-}path$
using *first second unfolding fri-layer-opening-chunk-def* **by** *simp*
then show *xp = yp*
by *simp*
have *len-eq*: *length xp-path = length yp-path*
using *first second unfolding fri-layer-opening-chunk-def* **by** *simp*
have *left*:
 $([xp] @ xp\text{-}path @ [xn] @ xn\text{-}path) ! \text{Suc } (\text{length } xp\text{-}path) = xn$
by *simp*
have *right*:
 $([yp] @ yp\text{-}path @ [yn] @ yn\text{-}path) ! \text{Suc } (\text{length } xp\text{-}path) = yn$
using *len-eq* **by** *simp*
show *xn = yn*
using *chunk-eq left right* **by** *metis*
qed

lemma *fri-round-layer-evidenceE*:
assumes *evidence*:
fri-round-layer-evidence roots challenges query-idxs round-idx
layer-idx round-layers
obtains *xp xp-path xn xn-path*
where *round-idx < length query-idxs*
and *layer-idx < length roots*
and *length challenges = length roots*
and *length round-layers = length query-idxs*
and *fri-layer-step-evidence*
(roots ! layer-idx)

```

(challenges ! layer-idx)
(fri-layer-lengths (length roots) (clength * scale) ! layer-idx)
(fri-layer-indices (length roots) (query-idxs ! round-idx)
  (clength * scale) ! layer-idx)
(2 ^ layer-idx)
(fri-sibling-index
  (fri-layer-lengths (length roots) (clength * scale) ! layer-idx)
  (fri-layer-indices (length roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx))
xp xp-path xn xn-path
((fri-layer-indices (length roots) (query-idxs ! round-idx)
  (clength * scale) ! layer-idx) mod
  ((fri-layer-lengths (length roots) (clength * scale) ! layer-idx)
    div 2))
(fri-fold-value (challenges ! layer-idx) xp xn
  (fri-fold-denominator
    ((h ^
      (fri-layer-indices (length roots) (query-idxs ! round-idx)
        (clength * scale) ! layer-idx)) * shift)
      (2 ^ layer-idx))))
(round-layers ! round-idx ! layer-idx)
using evidence unfolding fri-round-layer-evidence-def by blast

```

lemma *fri-round-layer-evidence-compactE*:

assumes *evidence*:

fri-round-layer-evidence roots challenges query-idxs round-idx
 layer-idx round-layers

obtains *xp xp-path xn xn-path*

where *round-idx < length query-idxs*

and *layer-idx < length roots*

and *length challenges = length roots*

and *length round-layers = length query-idxs*

and *fri-layer-step-evidence*

(roots ! layer-idx)

(challenges ! layer-idx)

(fri-evidence-layer-len roots layer-idx)

(fri-evidence-layer-idx roots query-idxs round-idx layer-idx)

(2 ^ layer-idx)

(fri-sibling-index (fri-evidence-layer-len roots layer-idx)

(fri-evidence-layer-idx roots query-idxs round-idx layer-idx))

xp xp-path xn xn-path

(fri-evidence-next-idx roots query-idxs round-idx layer-idx)

(fri-evidence-next-value roots challenges query-idxs round-idx
 layer-idx xp xn)

(round-layers ! round-idx ! layer-idx)

proof –

from *fri-round-layer-evidenceE[OF evidence]*

obtain *xp xp-path xn xn-path* **where**

round-bound: round-idx < length query-idxs

```

and layer-bound: layer-idx < length roots
and challenges-len: length challenges = length roots
and layers-len: length round-layers = length query-idxs
and step:
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-layer-lengths (length roots) (clength * scale) ! layer-idx)
    (fri-layer-indices (length roots) (query-idxs ! round-idx)
      (clength * scale) ! layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index
      (fri-layer-lengths (length roots) (clength * scale) ! layer-idx)
      (fri-layer-indices (length roots) (query-idxs ! round-idx)
        (clength * scale) ! layer-idx))
    xp xp-path xn xn-path
    ((fri-layer-indices (length roots) (query-idxs ! round-idx)
      (clength * scale) ! layer-idx) mod
      ((fri-layer-lengths (length roots) (clength * scale) ! layer-idx)
        div 2))
    (fri-fold-value (challenges ! layer-idx) xp xn
      (fri-fold-denominator
        ((h ^
          (fri-layer-indices (length roots) (query-idxs ! round-idx)
            (clength * scale) ! layer-idx)) * shift)
          (2 ^ layer-idx)))
      (round-layers ! round-idx ! layer-idx))
  by blast
have compact-step:
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
    xp xp-path xn xn-path
    (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx
      layer-idx xp xn)
    (round-layers ! round-idx ! layer-idx)
  using step
unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
  fri-evidence-next-idx-def fri-evidence-next-value-def
by simp
show ?thesis
by (rule that[OF round-bound layer-bound challenges-len layers-len
  compact-step])

```

qed

lemma *fri-all-round-layer-evidenceD*:

assumes *evidence*:

fri-all-round-layer-evidence roots challenges query-idxs round-layers

and *round-bound*: *round-idx* < *length query-idxs*

and *layer-bound*: *layer-idx* < *length roots*

shows *fri-round-layer-evidence roots challenges query-idxs round-idx layer-idx round-layers*

using *assms* **unfolding** *fri-all-round-layer-evidence-def* **by** *blast*

lemma *fri-layer-step-evidence-from-outcome*:

assumes *outcome*:

*Some ((next-idx, next-value, next-len, next-pw), t) ∈
set-dist (execute (fri-layer-opening-step (ch, rt)
(idx, x, len, pw)) s)*

obtains *xp xp-path xn xn-path chunk*

where *fri-layer-step-evidence rt ch len idx pw*

(fri-sibling-index len idx) xp xp-path xn xn-path

next-idx next-value chunk

and *chunk* = [*xp*] @ *xp-path* @ [*xn*] @ *xn-path*

and *x* = *xp*

and *next-len* = *len div 2*

and *next-pw* = *pw + pw*

proof –

from *fri-layer-opening-step-outcome[OF outcome]*

obtain *xp xp-path xn xn-path x'* **where**

out-eq:

*(next-idx, next-value, next-len, next-pw) =
(idx mod (len div 2), x', len div 2, pw + pw)*

and *xp-eq*: *xp* = *x*

and *x'-eq*:

*x' = fri-fold-value ch xp xn
(fri-fold-denominator ((h ^ idx) * shift) pw)*

and *chunk*:

*fri-layer-opening-chunk len xp xp-path xn xn-path
([xp] @ xp-path @ [xn] @ xn-path)*

by *blast*

let *?chunk* = [*xp*] @ *xp-path* @ [*xn*] @ *xn-path*

have *step*:

fri-layer-step-evidence rt ch len idx pw

(fri-sibling-index len idx) xp xp-path xn xn-path next-idx

next-value ?chunk

using *out-eq x'-eq chunk* **unfolding** *fri-layer-step-evidence-def* **by** *simp*

show *?thesis*

by (*rule that[OF step refl]*)

(*use out-eq xp-eq in simp-all*)

qed

lemma *fri-layer-step-evidence-fold-table-if-opening-matches*:
assumes *evidence*:
fri-layer-step-evidence *rt ch len idx pw sibling-idx*
xp xp-path xn xn-path next-idx next-value chunk
and *dom-i*: *fri-dom ! idx = (h ^ idx * shift) ^ pw*
and *match*: *fri-opening-matches-table len idx layer xp xn*
shows *next-value = fri-table-fold-value ch layer fri-dom len pw idx*
proof –
have *next-value-eq*:
next-value =
fri-fold-value ch xp xn
*(fri-fold-denominator ((h ^ idx) * shift) pw)*
using *fri-layer-step-evidenceD(3)[OF evidence]* .
also have ... = *fri-table-fold-value ch layer fri-dom len pw idx*
by (*rule fri-fold-value-eq-table-fold-value[OF dom-i match]*)
finally show ?thesis .
qed

lemma *fri-layer-step-evidence-fold-table-if-matching-chunk*:
assumes *evidence*:
fri-layer-step-evidence *rt ch len idx pw sibling-idx*
xp xp-path xn xn-path next-idx next-value chunk
and *dom-i*: *fri-dom ! idx = (h ^ idx * shift) ^ pw*
and *chunk-match*:
fri-layer-opening-chunk len yp yp-path yn yn-path chunk
and *table-match*: *fri-opening-matches-table len idx layer yp yn*
shows *next-value = fri-table-fold-value ch layer fri-dom len pw idx*
proof –
have *step-chunk*: *fri-layer-opening-chunk len xp xp-path xn xn-path chunk*
by (*rule fri-layer-step-evidenceD(4)[OF evidence]*)
have *xp-eq*: *xp = yp*
by (*rule fri-layer-opening-chunk-values-unique(1)*
[OF step-chunk chunk-match])
have *xn-eq*: *xn = yn*
by (*rule fri-layer-opening-chunk-values-unique(2)*
[OF step-chunk chunk-match])
have *table-match'*:
fri-opening-matches-table len idx layer xp xn
using *table-match xp-eq xn-eq by simp*
show ?thesis
by (*rule fri-layer-step-evidence-fold-table-if-opening-matches*
[OF evidence dom-i table-match'])
qed

lemma *fri-sampled-table-foldI*:
assumes
fri-layer-opening-chunk len xp xp-path xn xn-path chunk
and *fri-opening-matches-table len idx layer xp xn*
and *fri-dom ! idx = (h ^ idx * shift) ^ pw*

and *next-value* = *fri-table-fold-value* *ch* *layer* *fri-dom* *len* *pw* *idx*
shows *fri-sampled-table-fold* *ch* *len* *idx* *pw* *fri-dom* *layer* *chunk*
next-value
using *assms* **unfolding** *fri-sampled-table-fold-def* **by** *blast*

lemma *fri-sampled-table-foldE*:

assumes *fri-sampled-table-fold* *ch* *len* *idx* *pw* *fri-dom* *layer* *chunk*
next-value
obtains *xp* *xp-path* *xn* *xn-path*
where *fri-layer-opening-chunk* *len* *xp* *xp-path* *xn* *xn-path* *chunk*
and *fri-opening-matches-table* *len* *idx* *layer* *xp* *xn*
and *fri-dom* ! *idx* = (*h* ^ *idx* * *shift*) ^ *pw*
and *next-value* = *fri-table-fold-value* *ch* *layer* *fri-dom* *len* *pw* *idx*
using *assms* **unfolding** *fri-sampled-table-fold-def* **by** *blast*

lemma *fri-layer-step-evidence-sampled-table-fold-if-matching-chunk*:

assumes *evidence*:
fri-layer-step-evidence *rt* *ch* *len* *idx* *pw* *sibling-idx*
xp *xp-path* *xn* *xn-path* *next-idx* *next-value* *chunk*
and *dom-i*: *fri-dom* ! *idx* = (*h* ^ *idx* * *shift*) ^ *pw*
and *chunk-match*:
fri-layer-opening-chunk *len* *yp* *yp-path* *yn* *yn-path* *chunk*
and *table-match*: *fri-opening-matches-table* *len* *idx* *layer* *yp* *yn*
shows *fri-sampled-table-fold* *ch* *len* *idx* *pw* *fri-dom* *layer* *chunk*
next-value

proof —

have *fold-eq*:
next-value = *fri-table-fold-value* *ch* *layer* *fri-dom* *len* *pw* *idx*
by (rule *fri-layer-step-evidence-fold-table-if-matching-chunk*
[*OF evidence dom-i chunk-match table-match*])
show ?*thesis*
by (rule *fri-sampled-table-foldI*
[*OF chunk-match table-match dom-i fold-eq*])

qed

lemma *fri-layer-opening-matches-bound-table-compactE*:

assumes *match*:
fri-layer-opening-matches-bound-table *query-idx* *layer-idx* *tables*
layer-chunks
and *tables-len*: *length* *tables* = *length* *roots*
obtains *xp* *xp-path* *xn* *xn-path*
where
fri-layer-opening-chunk (*fri-evidence-layer-len* *roots* *layer-idx*)
xp *xp-path* *xn* *xn-path* (*layer-chunks* ! *layer-idx*)
and *fri-opening-matches-table*
(*fri-evidence-layer-len* *roots* *layer-idx*)
(*fri-layer-indices* (*length* *roots*) *query-idx* (*clength* * *scale*) !
layer-idx)
(*tables* ! *layer-idx*) *xp* *xn*

proof –
 let ?len = fri-evidence-layer-len roots layer-idx
 let ?idx =
 fri-layer-indices (length roots) query-idx (clength * scale) !
 layer-idx
from match **obtain** xp xp-path xn xn-path **where**
 chunk:
 fri-layer-opening-chunk ?len xp xp-path xn xn-path
 (layer-chunks ! layer-idx)
and table-match:
 fri-opening-matches-table ?len ?idx (tables ! layer-idx) xp xn
unfolding fri-layer-opening-matches-bound-table-def
by (auto simp: Let-def tables-len fri-evidence-layer-len-def)
show ?thesis
by (rule that[OF chunk table-match])
qed

lemma fri-all-round-openings-match-tablesD:
assumes all:
 fri-all-round-openings-match-tables tables query-idxs round-layers
and round-bound: round-idx < length query-idxs
and layer-bound: layer-idx < length tables
shows fri-layer-opening-matches-bound-table
 (query-idxs ! round-idx) layer-idx tables
 (round-layers ! round-idx)
using assms
unfolding fri-all-round-openings-match-tables-def
 fri-round-openings-match-tables-def
by blast

lemma accepted-fri-opening-transcript-trace-round-layer-evidence:
assumes fri-openings:
 accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
 dg composition-roots composition-bs composition-final query-idxs
 trace-round-layers composition-round-layers
and round-bound: round-idx < rounds
and layer-bound: layer-idx < length trace-roots
shows fri-round-layer-evidence trace-roots trace-bs query-idxs
 round-idx layer-idx trace-round-layers

proof –
have shapes:
 length trace-bs = length trace-roots
 length trace-round-layers = rounds
 length query-idxs = rounds
using accepted-fri-opening-transcript-shapes[OF fri-openings] **by** blast+
from accepted-fri-opening-transcript-trace-layer-opening
 [OF fri-openings round-bound layer-bound]
obtain root challenge len idx sibling-idx xp xp-path xn xn-path **where**
 root-eq: root = trace-roots ! layer-idx

```

and challenge-eq: challenge = trace-bs ! layer-idx
and len-eq:
  len =
    fri-layer-lengths (length trace-roots) (clength * scale) ! layer-idx
and idx-eq:
  idx =
    fri-layer-indices (length trace-roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx
and sibling-eq: sibling-idx = fri-sibling-index len idx
and chunk:
  fri-layer-opening-chunk len xp xp-path xn xn-path
  (trace-round-layers ! round-idx ! layer-idx)
by blast
have step:
  fri-layer-step-evidence (trace-roots ! layer-idx)
  (trace-bs ! layer-idx)
  (fri-layer-lengths (length trace-roots) (clength * scale) ! layer-idx)
  (fri-layer-indices (length trace-roots) (query-idxs ! round-idx)
  (clength * scale) ! layer-idx)
  (2 ^ layer-idx)
  (fri-sibling-index
  (fri-layer-lengths (length trace-roots) (clength * scale) ! layer-idx)
  (fri-layer-indices (length trace-roots) (query-idxs ! round-idx)
  (clength * scale) ! layer-idx))
  xp xp-path xn xn-path
  ((fri-layer-indices (length trace-roots) (query-idxs ! round-idx)
  (clength * scale) ! layer-idx) mod
  ((fri-layer-lengths (length trace-roots) (clength * scale) ! layer-idx)
  div 2))
  (fri-fold-value (trace-bs ! layer-idx) xp xn
  (fri-fold-denominator
  ((h ^
  (fri-layer-indices (length trace-roots) (query-idxs ! round-idx)
  (clength * scale) ! layer-idx)) * shift)
  (2 ^ layer-idx)))
  (trace-round-layers ! round-idx ! layer-idx)
using sibling-eq chunk unfolding fri-layer-step-evidence-def
by (simp add: len-eq idx-eq)
show ?thesis
unfolding fri-round-layer-evidence-def
using round-bound layer-bound shapes step
by auto
qed

```

```

lemma accepted-fri-opening-transcript-composition-round-layer-evidence:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers

```



```

and round-bound: round-idx < rounds
and layer-bound: layer-idx < length composition-roots
shows fri-round-layer-evidence composition-roots composition-bs query-idxs
  round-idx layer-idx composition-round-layers
proof –
  have shapes:
    length composition-bs = length composition-roots
    length composition-round-layers = rounds
    length query-idxs = rounds
    using accepted-fri-opening-transcript-shapes[OF fri-openings] by blast+
from accepted-fri-opening-transcript-composition-layer-opening
  [OF fri-openings round-bound layer-bound]
obtain root challenge len idx sibling-idx xp xp-path xn xn-path where
  root-eq: root = composition-roots ! layer-idx
and challenge-eq: challenge = composition-bs ! layer-idx
and len-eq:
  len =
    fri-layer-lengths (length composition-roots) (clength * scale) !
    layer-idx
and idx-eq:
  idx =
    fri-layer-indices (length composition-roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx
and sibling-eq: sibling-idx = fri-sibling-index len idx
and chunk:
  fri-layer-opening-chunk len xp xp-path xn xn-path
  (composition-round-layers ! round-idx ! layer-idx)
by blast
have step:
  fri-layer-step-evidence (composition-roots ! layer-idx)
  (composition-bs ! layer-idx)
  (fri-layer-lengths (length composition-roots) (clength * scale) !
    layer-idx)
  (fri-layer-indices (length composition-roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx)
  (2 ^ layer-idx)
  (fri-sibling-index
    (fri-layer-lengths (length composition-roots) (clength * scale) !
      layer-idx)
    (fri-layer-indices (length composition-roots) (query-idxs ! round-idx)
      (clength * scale) ! layer-idx))
  xp xp-path xn xn-path
  ((fri-layer-indices (length composition-roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx) mod
    ((fri-layer-lengths (length composition-roots) (clength * scale) !
      layer-idx) div 2))
  (fri-fold-value (composition-bs ! layer-idx) xp xn
    (fri-fold-denominator
      ((h ^

```

```

      (fri-layer-indices (length composition-roots)
        (query-idxs ! round-idx) (clength * scale) ! layer-idx)) *
      shift)
    (2 ^ layer-idx)))
  (composition-round-layers ! round-idx ! layer-idx)
using sibling-eq chunk unfolding fri-layer-step-evidence-def
by (simp add: len-eq idx-eq)
show ?thesis
  unfolding fri-round-layer-evidence-def
  using round-bound layer-bound shapes step
  by auto
qed

lemma accepted-fri-opening-transcript-trace-all-round-layer-evidence:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
shows fri-all-round-layer-evidence trace-roots trace-bs query-idxs
  trace-round-layers
proof -
have shapes:
  length trace-bs = length trace-roots
  length trace-round-layers = rounds
  length query-idxs = rounds
  using accepted-fri-opening-transcript-shapes[OF fri-openings] by blast+
show ?thesis
  unfolding fri-all-round-layer-evidence-def
  using shapes
  by (intro conjI allI impI)
  (auto intro:
    accepted-fri-opening-transcript-trace-round-layer-evidence
    [OF fri-openings])
qed

lemma accepted-fri-opening-transcript-composition-all-round-layer-evidence:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
shows fri-all-round-layer-evidence composition-roots composition-bs
  query-idxs composition-round-layers
proof -
have shapes:
  length composition-bs = length composition-roots
  length composition-round-layers = rounds
  length query-idxs = rounds
  using accepted-fri-opening-transcript-shapes[OF fri-openings] by blast+
show ?thesis

```

```

unfolding fri-all-round-layer-evidence-def
using shapes
by (intro conjI allI impI)
  (auto intro:
    accepted-fri-opening-transcript-composition-round-layer-evidence
    [OF fri-openings])
qed

lemma ntimes-verifier-query-round-program-final-checks-at:
assumes outcome:
  Some (results, t) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final) n)
      s)
and i-bound: i < n
obtains prefix suffix s-i s-suc raw idx fv s0 s1
  f-i f-len f-pow s2 c-i c-len c-pow s4
where
  Some (prefix, s-i) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final) i)
      s)
  Some (( ), s-suc) ∈
    set-dist
    (execute
      (verifier-query-round-program fr f-fl f-final as fl final) s-i)
  Some (suffix, t) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        (n - Suc i))
      s-suc)
  results = prefix @ ( ) # suffix
  Some (raw, s0) ∈ set-dist (execute receive-query-index-challenge s-i)
  idx = index (to-nat raw)
  Some (fv, s1) ∈
    set-dist (execute (mmap (check-decommit-on-query fr idx)) s0)
  Some ((f-i, f-final, f-len, f-pow), s2) ∈
    set-dist
    (execute
      (mfold (idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl)) s1)
  Some ((c-i, final, c-len, c-pow), s4) ∈
    set-dist

```

```

      (execute
        (mfold (idx, cp-eval as fv (h ^ idx * shift),
          clength * scale, 1)
          (receive-query-commits fl)) s2)
    s-suc = s4
proof –
  let ?round = verifier-query-round-program fr f-fl f-final as fl final
  from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
  obtain prefix x suffix s-i s-suc where
    prefix:
      Some (prefix, s-i) ∈ set-dist (execute (ntimes ?round i) s)
  and round:
      Some (x, s-suc) ∈ set-dist (execute ?round s-i)
  and suffix:
      Some (suffix, t) ∈
        set-dist (execute (ntimes ?round (n – Suc i)) s-suc)
  and results-eq: results = prefix @ x # suffix
  by blast
  have x-eq: x = ()
  by (cases x) simp
  have round-unit:
      Some ((), s-suc) ∈ set-dist (execute ?round s-i)
  using round x-eq by simp
  show ?thesis
  proof (rule verifier-query-round-program-final-checks[OF round-unit])
  fix raw idx fv s0 s1 f-i f-len f-pow s2 c-i c-len c-pow s4
  assume raw:
      Some (raw, s0) ∈
        set-dist (execute receive-query-index-challenge s-i)
  and idx-eq: idx = index (to-nat raw)
  and query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr idx)) s0)
  and trace-final:
      Some ((f-i, f-final, f-len, f-pow), s2) ∈
        set-dist
          (execute
            (mfold (idx, hd fv, clength * scale, 1)
              (receive-query-commits f-fl)) s1)
  and composition-final:
      Some ((c-i, final, c-len, c-pow), s4) ∈
        set-dist
          (execute
            (mfold (idx, cp-eval as fv (h ^ idx * shift),
              clength * scale, 1)
              (receive-query-commits fl)) s2)
  and s-suc-eq: s-suc = s4
  have results-unit: results = prefix @ () # suffix
  using results-eq x-eq by simp

```

```

show ?thesis
  by (rule that[OF prefix round-unit suffix results-unit raw idx-eq
    query-decommit trace-final composition-final s-suc-eq])
qed
qed

end

end

```

```

theory Soundness-FRI-Generic-Low-Degree
  imports Soundness-FRI-Layer-Evidence
begin

```

Generic FRI low-degree proof interface.

This theory factors the trace/composition-independent bad-challenge-list layer. It deliberately stays at the generic FRI level: no trace-specific or composition-specific candidate predicate is mentioned here.

```

context soundness
begin

```

```

definition generic-fri-partial-evidence
  :: ('f list  $\Rightarrow$  bool)  $\Rightarrow$  'f list  $\Rightarrow$  nat  $\Rightarrow$ 
    'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  nat list  $\Rightarrow$ 
    'f list list list  $\Rightarrow$  bool

```

```

where
  generic-fri-partial-evidence low-degree candidate-table degree-bound
    roots challenges final-value query-idxs round-layers  $\longleftrightarrow$ 
    length roots = fri-round-count-for-degree-bound degree-bound  $\wedge$ 
    length challenges = length roots  $\wedge$ 
    fri-all-round-layer-evidence roots challenges query-idxs round-layers

```

```

definition generic-fri-bad-challenge-lists
  :: nat  $\Rightarrow$  (nat  $\Rightarrow$  'f list  $\Rightarrow$  'f set)  $\Rightarrow$  'f list set
where
  generic-fri-bad-challenge-lists n bad =
    fri-multiround-bad-challenge-lists n bad

```

```

definition generic-fri-round-bad-challenge-lists
  :: nat  $\Rightarrow$  (nat  $\Rightarrow$  'f list  $\Rightarrow$  'f set)  $\Rightarrow$  nat  $\Rightarrow$ 
    'f list set

```

```

where
  generic-fri-round-bad-challenge-lists n bad i =
    {bs  $\in$  fri-challenge-space n. i < n  $\wedge$  bs ! i  $\in$  bad i (take i bs)}

```

```

definition generic-fri-round-bad-witnesses
  :: nat  $\Rightarrow$  (nat  $\Rightarrow$  'f list  $\Rightarrow$  'f set)  $\Rightarrow$  nat  $\Rightarrow$ 
    ('f list  $\times$  'f  $\times$  'f list) set

```

where

generic-fri-round-bad-witnesses n *bad* $i =$
 $\{(prefix, b, suffix).$
 $i < n \wedge$
 $length\ prefix = i \wedge$
 $b \in bad\ i\ prefix \wedge$
 $length\ suffix = n - Suc\ i\}$

definition *generic-fri-bad-candidate*

$:: ('f\ list \Rightarrow bool) \Rightarrow 'f\ list \Rightarrow nat \Rightarrow$
 $'f\ list \Rightarrow 'f\ list \Rightarrow 'f \Rightarrow nat\ list \Rightarrow$
 $'f\ list\ list\ list \Rightarrow$
 $(nat \Rightarrow 'f\ list \Rightarrow 'f\ set) \Rightarrow bool$

where

generic-fri-bad-candidate *low-degree candidate-table degree-bound*
 $roots\ challenges\ final-value\ query-idxs\ round-layers\ bad \longleftrightarrow$
generic-fri-partial-evidence *low-degree candidate-table degree-bound*
 $roots\ challenges\ final-value\ query-idxs\ round-layers \wedge$
 $\neg\ low-degree\ candidate-table \wedge$
 $challenges \in generic-fri-bad-challenge-lists\ (length\ challenges)\ bad$

definition *generic-fri-proximity-evidence*

$:: nat \Rightarrow nat \Rightarrow nat\ list \Rightarrow nat\ list \Rightarrow$
 $'f\ list\ list \Rightarrow 'f\ list\ list \Rightarrow 'f\ list \Rightarrow$
 $(nat \Rightarrow 'f\ list \Rightarrow 'f\ set) \Rightarrow bool$

where

generic-fri-proximity-evidence $n\ d\ lens\ pws\ doms\ layers\ challenges\ bad$
 $\longleftrightarrow$
 $fri-multiround-proximity-steps\ n\ d\ lens\ pws\ doms\ layers\ bad \wedge$
 $challenges \in fri-challenge-space\ n \wedge$
 $(\forall i < n.$
 $fri-algebraic-layer-relation\ (lens\ !\ i)\ (pws\ !\ i)$
 $(challenges\ !\ i)\ (doms\ !\ i)\ (layers\ !\ i)$
 $(doms\ !\ Suc\ i)\ (layers\ !\ Suc\ i)) \wedge$
 $\neg\ fri-table-low-degree-on\ d\ (doms\ !\ 0)\ (layers\ !\ 0) \wedge$
 $fri-table-low-degree-on\ (fri-degree-after\ n\ d)\ (doms\ !\ n)$
 $(layers\ !\ n)$

definition *generic-fri-canonical-layer-chain-evidence*

$:: ('f\ list \Rightarrow bool) \Rightarrow 'f\ list \Rightarrow nat \Rightarrow$
 $'f\ list \Rightarrow 'f\ list \Rightarrow 'f \Rightarrow nat\ list \Rightarrow$
 $'f\ list\ list\ list \Rightarrow 'f\ list\ list \Rightarrow 'f\ list\ list \Rightarrow$
 $(nat \Rightarrow 'f\ list \Rightarrow 'f\ set) \Rightarrow bool$

where

generic-fri-canonical-layer-chain-evidence *low-degree candidate-table*
 $degree-bound\ roots\ challenges\ final-value\ query-idxs\ round-layers$
 $doms\ layers\ bad \longleftrightarrow$
generic-fri-partial-evidence *low-degree candidate-table degree-bound*
 $roots\ challenges\ final-value\ query-idxs\ round-layers \wedge$

$\text{doms} ! 0 = \text{eval-domain} \wedge$
 $\text{layers} ! 0 = \text{candidate-table} \wedge$
 $\text{fri-final-constant-consistent} (\text{layers} ! \text{length challenges}) \text{final-value} \wedge$
 $\text{generic-fri-proximity-evidence} (\text{length challenges}) \text{degree-bound}$
 $(\text{fri-layer-lengths} (\text{length challenges}) (\text{clength} * \text{scale}))$
 $(\text{map } (\lambda i. 2 \wedge i) [0..<\text{length challenges}])$
 $\text{doms layers challenges bad}$

definition *generic-fri-sampled-layer-chain-evidence*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

$\text{generic-fri-sampled-layer-chain-evidence low-degree candidate-table}$
 $\text{degree-bound roots challenges final-value query-idxs round-layers}$
 $\text{doms layers} \longleftrightarrow$
 $\text{generic-fri-partial-evidence low-degree candidate-table degree-bound}$
 $\text{roots challenges final-value query-idxs round-layers} \wedge$
 $\text{length doms} = \text{Suc} (\text{length challenges}) \wedge$
 $\text{length layers} = \text{Suc} (\text{length challenges}) \wedge$
 $\text{doms} ! 0 = \text{eval-domain} \wedge$
 $\text{layers} ! 0 = \text{candidate-table} \wedge$
 $\text{fri-final-constant-consistent} (\text{layers} ! \text{length challenges}) \text{final-value} \wedge$
 $(\forall \text{round-idx} < \text{length query-idxs}.$
 $\forall \text{layer-idx} < \text{length challenges}.$
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} <$
 $\text{length} (\text{layers} ! \text{Suc layer-idx}) \wedge$
 $\text{fri-sampled-table-fold}$
 $(\text{challenges} ! \text{layer-idx})$
 $(\text{fri-evidence-layer-len roots layer-idx})$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx})$
 $(2 \wedge \text{layer-idx})$
 $(\text{doms} ! \text{layer-idx})$
 $(\text{layers} ! \text{layer-idx})$
 $(\text{round-layers} ! \text{round-idx} ! \text{layer-idx})$
 $(\text{layers} ! \text{Suc layer-idx} !$
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx}))$

definition *fri-sampled-domains-align*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

$\text{fri-sampled-domains-align roots query-idxs doms} \longleftrightarrow$
 $(\forall \text{round-idx} < \text{length query-idxs}.$
 $\forall \text{layer-idx} < \text{length roots}.$
 $\text{doms} ! \text{layer-idx} !$
 $\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} =$
 $(h \wedge \text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} *$
 $\text{shift}) \wedge (2 \wedge \text{layer-idx}))$

definition *fri-sampled-next-domains-align*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

fri-sampled-next-domains-align roots query-idxs doms $\longleftrightarrow$
 $(\forall \text{round-idx} < \text{length query-idxs}.$
 $\quad \forall \text{layer-idx} < \text{length roots}.$
 $\quad \text{doms ! layer-idx !}$
 $\quad \text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} =$
 $\quad (\text{h} \wedge \text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} *$
 $\quad \text{shift}) \wedge (2 \wedge \text{layer-idx}))$

definition *fri-sampled-indices-in-fold-range*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow \text{bool}$

where

fri-sampled-indices-in-fold-range roots query-idxs $\longleftrightarrow$
 $(\forall \text{round-idx} < \text{length query-idxs}.$
 $\quad \forall \text{layer-idx} < \text{length roots}.$
 $\quad \text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} <$
 $\quad \text{fri-evidence-layer-len roots layer-idx div } 2)$

definition *fri-sampled-layer-covers-fold-indices*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow \text{nat} \Rightarrow \text{bool}$

where

fri-sampled-layer-covers-fold-indices roots query-idxs layer-idx
 $\longleftrightarrow$
 $(\forall \text{idx} < \text{fri-evidence-layer-len roots layer-idx div } 2.$
 $\quad \exists \text{round-idx} < \text{length query-idxs}.$
 $\quad \text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} = \text{idx})$

definition *fri-sampled-layers-cover-fold-indices*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow \text{bool}$

where

fri-sampled-layers-cover-fold-indices roots query-idxs $\longleftrightarrow$
 $(\forall \text{layer-idx} < \text{length roots}.$
 $\quad \text{fri-sampled-layer-covers-fold-indices roots query-idxs layer-idx})$

definition *generic-fri-sampled-layer-chain-full-cover*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow$
 $\quad 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat list} \Rightarrow$
 $\quad 'f \text{ list list list} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list list} \Rightarrow$
 $\quad (\text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow \text{bool}$

where

generic-fri-sampled-layer-chain-full-cover low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers bad $\longleftrightarrow$
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers $\wedge$
fri-multiround-proximity-steps (length challenges) degree-bound

$(\text{fri-layer-lengths } (\text{length challenges}) (\text{clength} * \text{scale}))$
 $(\text{map } (\lambda i. 2 \wedge i) [0..<\text{length challenges}])$
 $\text{doms layers bad} \wedge$
 $\text{fri-sampled-layers-cover-fold-indices roots query-idxs} \wedge$
 $(\forall \text{layer-idx} < \text{length challenges.}$
 $2 \text{ dvd fri-evidence-layer-len roots layer-idx}) \wedge$
 $(\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} * 2 \wedge \text{layer-idx} =$
 $\text{clength} * \text{scale}) \wedge$
 $(\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} \leq$
 $\text{length } (\text{doms ! layer-idx})) \wedge$
 $(\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} \leq$
 $\text{length } (\text{layers ! layer-idx})) \wedge$
 $(\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx div } 2 \leq$
 $\text{length } (\text{doms ! Suc layer-idx})) \wedge$
 $(\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx div } 2 \leq$
 $\text{length } (\text{layers ! Suc layer-idx})) \wedge$
 $(\forall \text{layer-idx} < \text{length challenges.}$
 $\forall \text{idx} < \text{fri-evidence-layer-len roots layer-idx div } 2.$
 $\text{doms ! Suc layer-idx ! idx} =$
 $(\text{doms ! layer-idx ! idx})^2) \wedge$
 $(\forall \text{layer-idx} < \text{length challenges.}$
 $\forall \text{idx} < \text{fri-evidence-layer-len roots layer-idx div } 2.$
 $\text{doms ! layer-idx ! idx} = (\text{h} \wedge \text{idx} * \text{shift}) \wedge (2 \wedge \text{layer-idx})) \wedge$
 $\neg \text{fri-table-low-degree-on degree-bound eval-domain candidate-table} \wedge$
 $\text{fri-table-low-degree-on}$
 $(\text{fri-degree-after } (\text{length challenges}) \text{ degree-bound})$
 $(\text{doms ! length challenges}) (\text{layers ! length challenges})$

lemma *generic-fri-partial-evidence-shapes:*

assumes *generic-fri-partial-evidence low-degree candidate-table degree-bound*
roots challenges final-value query-idxs round-layers
shows *length roots = fri-round-count-for-degree-bound degree-bound*
and *length challenges = length roots*
and *fri-all-round-layer-evidence roots challenges query-idxs round-layers*
using *assms unfolding generic-fri-partial-evidence-def by simp-all*

lemma *generic-fri-sampled-layer-chain-evidenceD:*

assumes
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers
shows
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers

$\text{length doms} = \text{Suc} (\text{length challenges})$
 $\text{length layers} = \text{Suc} (\text{length challenges})$
 $\text{doms} ! 0 = \text{eval-domain}$
 $\text{layers} ! 0 = \text{candidate-table}$
 $\text{fri-final-constant-consistent} (\text{layers} ! \text{length challenges}) \text{ final-value}$
using *assms*
unfolding *generic-fri-sampled-layer-chain-evidence-def* **by** *simp-all*

lemma *generic-fri-sampled-layer-chain-evidence-sample:*

assumes *chain:*
 $\text{generic-fri-sampled-layer-chain-evidence low-degree candidate-table}$
 $\text{degree-bound roots challenges final-value query-idxs round-layers}$
 doms layers
and *round-bound:* $\text{round-idx} < \text{length query-idxs}$
and *layer-bound:* $\text{layer-idx} < \text{length challenges}$
shows
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} <$
 $\text{length} (\text{layers} ! \text{Suc layer-idx})$
 $\text{fri-sampled-table-fold}$
 $(\text{challenges} ! \text{layer-idx})$
 $(\text{fri-evidence-layer-len roots layer-idx})$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx})$
 $(2 \wedge \text{layer-idx})$
 $(\text{doms} ! \text{layer-idx})$
 $(\text{layers} ! \text{layer-idx})$
 $(\text{round-layers} ! \text{round-idx} ! \text{layer-idx})$
 $(\text{layers} ! \text{Suc layer-idx} !$
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx})$
using *chain round-bound layer-bound*
unfolding *generic-fri-sampled-layer-chain-evidence-def* **by** *simp-all*

lemma *fri-sampled-domains-alignD:*

assumes *fri-sampled-domains-align roots query-idxs doms*
and $\text{round-idx} < \text{length query-idxs}$
and $\text{layer-idx} < \text{length roots}$
shows
 $\text{doms} ! \text{layer-idx} !$
 $\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} =$
 $(h \wedge \text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} *$
 $\text{shift}) \wedge (2 \wedge \text{layer-idx})$
using *assms* **unfolding** *fri-sampled-domains-align-def* **by** *blast*

lemma *fri-sampled-next-domains-alignD:*

assumes *fri-sampled-next-domains-align roots query-idxs doms*
and $\text{round-idx} < \text{length query-idxs}$
and $\text{layer-idx} < \text{length roots}$
shows
 $\text{doms} ! \text{layer-idx} !$
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} =$

$(h \wedge \text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} * \text{shift}) \wedge (2 \wedge \text{layer-idx})$
using *assms* **unfolding** *fri-sampled-next-domains-align-def* **by** *blast*

lemma *fri-sampled-indices-in-fold-rangeD*:
assumes *fri-sampled-indices-in-fold-range roots query-idxs*
and *round-idx < length query-idxs*
and *layer-idx < length roots*
shows
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} < \text{fri-evidence-layer-len roots layer-idx div } 2$
using *assms* **unfolding** *fri-sampled-indices-in-fold-range-def* **by** *blast*

lemma *fri-sampled-layers-cover-fold-indicesD*:
assumes *fri-sampled-layers-cover-fold-indices roots query-idxs*
and *layer-idx < length roots*
and *idx < fri-evidence-layer-len roots layer-idx div 2*
obtains *round-idx*
where *round-idx < length query-idxs*
and $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} = \text{idx}$
using *assms*
unfolding *fri-sampled-layers-cover-fold-indices-def*
fri-sampled-layer-covers-fold-indices-def
by *blast*

lemma *fri-sampled-layer-covers-fold-indices-card-le-rounds*:
assumes *cover: fri-sampled-layer-covers-fold-indices roots query-idxs layer-idx*
shows $\text{fri-evidence-layer-len roots layer-idx div } 2 \leq \text{length query-idxs}$

proof –
let $?N = \text{fri-evidence-layer-len roots layer-idx div } 2$
have *subset*:
 $\{..< ?N\} \subseteq$
 $(\lambda \text{round-idx.}$
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx})$ ‘
 $\{..< \text{length query-idxs}\}$
proof
fix *idx*
assume $\text{idx} \in \{..< ?N\}$
then have *idx-bound*: $\text{idx} < ?N$
by *simp*
then obtain *round-idx* **where** *round-bound*: $\text{round-idx} < \text{length query-idxs}$
and *idx-eq*:
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} = \text{idx}$
using *cover*
unfolding *fri-sampled-layer-covers-fold-indices-def* **by** *blast*
then show $\text{idx} \in$
 $(\lambda \text{round-idx.}$
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx})$ ‘
 $\{..< \text{length query-idxs}\}$

```

    by blast
qed
have ?N = card {.. $?N$ }
  by simp
also have ...  $\leq$ 
  card (( $\lambda$ round-idx.
    fri-evidence-next-idx roots query-idxs round-idx layer-idx) ‘
    {.. $\text{length query-idxs}$ })
  by (rule card-mono) (simp-all add: subset)
also have ...  $\leq$  card {.. $\text{length query-idxs}$ }
  by (rule card-image-le) simp
also have ... = length query-idxs
  by simp
finally show ?thesis .
qed

lemma fri-sampled-layers-cover-fold-indices-card-le-rounds:
  assumes cover: fri-sampled-layers-cover-fold-indices roots query-idxs
    and layer-bound: layer-idx < length roots
  shows fri-evidence-layer-len roots layer-idx div 2  $\leq$  length query-idxs
proof -
  have layer-cover:
    fri-sampled-layer-covers-fold-indices roots query-idxs layer-idx
  using cover layer-bound
  unfolding fri-sampled-layers-cover-fold-indices-def by blast
  show ?thesis
    by (rule fri-sampled-layer-covers-fold-indices-card-le-rounds
      [OF layer-cover])
qed

lemma fri-sampled-layers-not-cover-if-layer-exceeds-rounds:
  assumes layer-bound: layer-idx < length roots
    and too-few:
      length query-idxs < fri-evidence-layer-len roots layer-idx div 2
  shows  $\neg$  fri-sampled-layers-cover-fold-indices roots query-idxs
proof
  assume cover: fri-sampled-layers-cover-fold-indices roots query-idxs
  have fri-evidence-layer-len roots layer-idx div 2  $\leq$  length query-idxs
    by (rule fri-sampled-layers-cover-fold-indices-card-le-rounds
      [OF cover layer-bound])
  then show False
    using too-few by simp
qed

lemma generic-fri-proximity-evidenceD:
  assumes generic-fri-proximity-evidence n d lens pws doms layers
    challenges bad
  shows
    fri-multiround-proximity-steps n d lens pws doms layers bad

```

challenges $\in$ *fri-challenge-space* *n*
 $\neg$ *fri-table-low-degree-on* *d* (*doms* ! 0) (*layers* ! 0)
fri-table-low-degree-on (*fri-degree-after* *n* *d*) (*doms* ! *n*)
(*layers* ! *n*)
using *assms* **unfolding** *generic-fri-proximity-evidence-def* **by** *simp-all*

lemma *generic-fri-proximity-evidence-relation*:
assumes *proximity*:
generic-fri-proximity-evidence *n* *d* *lens* *pws* *doms* *layers*
challenges *bad*
and *i-bound*: *i* < *n*
shows
fri-algebraic-layer-relation (*lens* ! *i*) (*pws* ! *i*)
(*challenges* ! *i*) (*doms* ! *i*) (*layers* ! *i*)
(*doms* ! *Suc* *i*) (*layers* ! *Suc* *i*)
using *proximity* *i-bound*
unfolding *generic-fri-proximity-evidence-def* **by** *blast*

lemma *generic-fri-canonical-layer-chain-proximity*:
assumes *chain*:
generic-fri-canonical-layer-chain-evidence *low-degree* *candidate-table*
degree-bound *roots* *challenges* *final-value* *query-idxs* *round-layers*
doms *layers* *bad*
shows
generic-fri-proximity-evidence (*length* *challenges*) *degree-bound*
(*fri-layer-lengths* (*length* *challenges*) (*clength* * *scale*))
(*map* ($\lambda i. 2^i$) [0..*length* *challenges*])
doms *layers* *challenges* *bad*
using *chain* **unfolding** *generic-fri-canonical-layer-chain-evidence-def*
by *simp*

lemma *generic-fri-canonical-layer-chain-relation*:
assumes *chain*:
generic-fri-canonical-layer-chain-evidence *low-degree* *candidate-table*
degree-bound *roots* *challenges* *final-value* *query-idxs* *round-layers*
doms *layers* *bad*
and *i-bound*: *i* < *length* *challenges*
shows
fri-algebraic-layer-relation
(*fri-layer-lengths* (*length* *challenges*) (*clength* * *scale*) ! *i*)
(2^i)
(*challenges* ! *i*) (*doms* ! *i*) (*layers* ! *i*)
(*doms* ! *Suc* *i*) (*layers* ! *Suc* *i*)

proof –
have *proximity*:
generic-fri-proximity-evidence (*length* *challenges*) *degree-bound*
(*fri-layer-lengths* (*length* *challenges*) (*clength* * *scale*))
(*map* ($\lambda i. 2^i$) [0..*length* *challenges*])
doms *layers* *challenges* *bad*

```

  by (rule generic-fri-canonical-layer-chain-proximity[OF chain])
have rel:
  fri-algebraic-layer-relation
    (fri-layer-lengths (length challenges) (clength * scale) ! i)
    ((map (λi. 2 ^ i) [0..

```

```

lemma generic-fri-canonical-layer-chain-lengths:
  assumes chain:
    generic-fri-canonical-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers bad
  shows length doms = Suc (length challenges)
    and length layers = Suc (length challenges)
proof -
  have proximity:
    generic-fri-proximity-evidence (length challenges) degree-bound
    (fri-layer-lengths (length challenges) (clength * scale))
    (map (λi. 2 ^ i) [0..

```

```

lemma fri-algebraic-layer-relation-next-index-bound:
  assumes relation:
    fri-algebraic-layer-relation len pw ch fri-dom layer next-dom
    next-layer
  and idx-bound: idx < len div 2
  shows idx < length next-layer
proof -
  have len div 2 ≤ length next-layer
    using relation unfolding fri-algebraic-layer-relation-def by simp
  then show ?thesis
    using idx-bound by linarith

```

qed

lemma *fri-algebraic-layer-relation-sampled-fold-eq*:
assumes *relation*:
fri-algebraic-layer-relation len pw ch fri-dom layer next-dom
next-layer
and *idx-bound*: $idx < len \text{ div } 2$
shows *next-layer ! idx =*
fri-table-fold-value ch layer fri-dom len pw idx
by (rule *fri-algebraic-layer-relationD*(2)[*OF relation idx-bound*])

lemma *fri-half-dvd-len-if-even*:
fixes *len* :: nat
assumes *even-len*: $2 \text{ dvd } len$
shows $len \text{ div } 2 \text{ dvd } len$
proof (cases $len = 0$)
case *True*
then show ?thesis **by** *simp*
next
case *False*
then have $len = 2 * (len \text{ div } 2)$
using *even-len* **by** *simp*
then have $len = (len \text{ div } 2) * 2$
by *simp*
then show ?thesis
unfolding *dvd-def* **by** *blast*
qed

lemma *fri-sibling-index-mod-half*:
fixes *len idx* :: nat
assumes *even-len*: $2 \text{ dvd } len$
shows *fri-sibling-index len idx mod (len div 2) =*
idx mod (len div 2)
proof –
have *dvd-half*: $len \text{ div } 2 \text{ dvd } len$
by (rule *fri-half-dvd-len-if-even*[*OF even-len*])
have *fri-sibling-index len idx mod (len div 2) =*
((idx + len div 2) mod len) mod (len div 2)
unfolding *fri-sibling-index-def* **by** *simp*
also have $\dots = (idx + len \text{ div } 2) \text{ mod } (len \text{ div } 2)$
using *mod-mod-cancel*[*OF dvd-half, of idx + len div 2*] **by** *simp*
also have $\dots = idx \text{ mod } (len \text{ div } 2)$
by *simp*
finally show ?thesis .
qed

lemma *fri-fold-value-swap-neg-denominator*:
fri-fold-value b xp xn (– denom) =
fri-fold-value b xn xp denom

```

proof (cases denom = 0)
  case True
  then show ?thesis
    unfolding fri-fold-value-def by (simp add: algebra-simps)
next
  case False
  then show ?thesis
    unfolding fri-fold-value-def by (simp add: field-simps)
qed

```

lemma soundness-h-power-mod-eval-domain:

$h \wedge (k \bmod (\text{clength} * \text{scale})) = h \wedge k$

```

proof -
  let ?N = clength * scale
  have order:  $h \wedge ?N = 1$ 
    using h-order .
  have k-eq:  $k = (k \text{ div } ?N) * ?N + k \bmod ?N$ 
    using div-mult-mod-eq[of k ?N] by (simp add: mult.commute)
  have period:  $h \wedge ((k \text{ div } ?N) * ?N) = 1$ 
proof -
  have  $h \wedge ((k \text{ div } ?N) * ?N) = h \wedge (?N * (k \text{ div } ?N))$ 
    by (simp add: mult.commute)
  also have ... =  $(h \wedge ?N) \wedge (k \text{ div } ?N)$ 
    by (simp only: power-mult)
  also have ... = 1
    using order by simp
  finally show ?thesis .
qed
have  $h \wedge k = h \wedge ((k \text{ div } ?N) * ?N + k \bmod ?N)$ 
  using k-eq by simp
also have ... =  $h \wedge ((k \text{ div } ?N) * ?N) * h \wedge (k \bmod ?N)$ 
  by (simp add: power-add)
also have ... =  $h \wedge (k \bmod ?N)$ 
  using period by simp
finally show ?thesis
  by simp
qed

```

lemma soundness-h-half-shift:

$h \wedge (k + (\text{clength} * \text{scale}) \text{ div } 2) = -(h \wedge k)$

using h-half-order

by (simp add: power-add algebra-simps)

lemma soundness-mod-mult-right-factor-nat:

fixes a b c :: nat

assumes b-pos: $0 < b$

and c-pos: $0 < c$

shows $((a \bmod b) * c) \bmod (b * c) = (a * c) \bmod (b * c)$

proof -


```

have decomp:  $a = a \text{ div } b * b + a \text{ mod } b$ 
  using div-mult-mod-eq[of a b] by simp
have r-less:  $(a \text{ mod } b) * c < b * c$ 
  using b-pos c-pos mod-less-divisor[OF b-pos]
  by (simp add: mult-less-mono2)
have r-le:  $(a \text{ mod } b) * c \leq a * c$ 
proof -
  have  $a \text{ mod } b \leq a$ 
    using decomp by linarith
  then show ?thesis
    by simp
qed
have dvd-diff:  $b * c \text{ dvd } a * c - (a \text{ mod } b) * c$ 
proof -
  have ac-decomp:  $a * c = (a \text{ div } b) * (b * c) + (a \text{ mod } b) * c$ 
  proof -
    have  $a * c = (a \text{ div } b * b + a \text{ mod } b) * c$ 
      using decomp by simp
    also have  $\dots = (a \text{ div } b * b) * c + (a \text{ mod } b) * c$ 
      by (simp only: distrib-right)
    also have  $\dots = (a \text{ div } b) * (b * c) + (a \text{ mod } b) * c$ 
      by (simp add: algebra-simps)
    finally show ?thesis .
  qed
  have  $a * c - (a \text{ mod } b) * c = (a \text{ div } b) * (b * c)$ 
    using ac-decomp by simp
  then show ?thesis by simp
qed
have  $(a * c) \text{ mod } (b * c) = (a \text{ mod } b) * c$ 
  by (rule mod-nat-eqI[OF r-less r-le dvd-diff])
then show ?thesis by simp
qed

lemma fri-layer-lengths-nth-div:
  assumes j-bound:  $j < n$ 
  shows fri-layer-lengths n len ! j = len div 2 ^ j
  using j-bound
proof (induction n arbitrary: len j)
  case 0
  then show ?case by simp
next
  case (Suc n)
  then show ?case
  proof (cases j)
    case 0
    then show ?thesis by simp
  next
    case (Suc k)
    then have k-bound:  $k < n$ 

```

```

    using Suc.premis by simp
  have fri-layer-lengths (Suc n) len ! j =
    fri-layer-lengths n (len div 2) ! k
    using Suc by simp
  also have ... = (len div 2) div 2 ^ k
    by (rule Suc.IH[OF k-bound])
  also have ... = len div 2 ^ j
    unfolding Suc by (simp add: div-mult2-eq mult.commute)
  finally show ?thesis .
qed
qed

```

```

lemma fri-layer-lengths-round-product-if-dvd:
  assumes j-bound: j < n
    and dvd: 2 ^ j dvd len
  shows fri-layer-lengths n len ! j * 2 ^ j = len
proof -
  have fri-layer-lengths n len ! j = len div 2 ^ j
    by (rule fri-layer-lengths-nth-div[OF j-bound])
  then show ?thesis
    using dvd by (simp add: dvd-div-mult-self)
qed

```

```

lemma fri-layer-lengths-even-if-dvd:
  assumes j-bound: j < n
    and dvd: 2 ^ Suc j dvd len
  shows 2 dvd fri-layer-lengths n len ! j
proof -
  obtain k where len-eq: len = 2 ^ Suc j * k
    using dvd by blast
  have fri-layer-lengths n len ! j = len div 2 ^ j
    by (rule fri-layer-lengths-nth-div[OF j-bound])
  also have ... = 2 * k
    using len-eq by simp
  finally show ?thesis by simp
qed

```

```

lemma fri-layer-lengths-Suc-nth:
  assumes Suc j < n
  shows fri-layer-lengths n len ! Suc j =
    fri-layer-lengths n len ! j div 2
proof -
  have j-bound: j < n
    using assms by simp
  have fri-layer-lengths n len ! Suc j = len div 2 ^ Suc j
    by (rule fri-layer-lengths-nth-div[OF assms])
  also have ... = (len div 2 ^ j) div 2
    by (metis div-mult2-eq mult.commute power-Suc)
  also have ... = fri-layer-lengths n len ! j div 2

```

```

    using fri-layer-lengths-nth-div[OF j-bound, of len] by simp
  finally show ?thesis .
qed

lemma generic-fri-canonical-domain-power:
  assumes chain:
    generic-fri-canonical-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers bad
  and roots-len: length challenges = length roots
  and layer-bound: layer-idx < length challenges
  and idx-bound: idx < fri-evidence-layer-len roots layer-idx
  shows doms ! layer-idx ! idx = (h ^ idx * shift) ^ (2 ^ layer-idx)
  using layer-bound idx-bound
proof (induction layer-idx arbitrary: idx)
  case 0
  note zero = 0
  have dom0: doms ! 0 = eval-domain
  using chain unfolding generic-fri-canonical-layer-chain-evidence-def
  by simp
  have len0: fri-evidence-layer-len roots 0 = clength * scale
  proof (cases roots)
    case Nil
    then show ?thesis
      using zero roots-len unfolding fri-evidence-layer-len-def by simp
  next
    case (Cons r rs)
    then show ?thesis
      unfolding fri-evidence-layer-len-def by simp
  qed
  show ?case
  using eval-domain-nth[of idx] zero len0
  unfolding dom0 by simp
next
  case (Suc j)
  have j-bound: j < length challenges
  using Suc.prem1 by simp
  have len-eq-j:
    fri-layer-lengths (length challenges) (clength * scale) ! j =
    fri-evidence-layer-len roots j
  using roots-len unfolding fri-evidence-layer-len-def by simp
  have len-eq-suc:
    fri-layer-lengths (length challenges) (clength * scale) ! Suc j =
    fri-evidence-layer-len roots (Suc j)
  using roots-len unfolding fri-evidence-layer-len-def by simp
  have suc-len:
    fri-evidence-layer-len roots (Suc j) =
    fri-evidence-layer-len roots j div 2
  using fri-layer-lengths-Suc-nth[OF Suc.prem1, of clength * scale]

```

```

    len-eq-j len-eq-suc by simp
have idx-half:  $idx < \text{fri-evidence-layer-len roots } j \text{ div } 2$ 
    using Suc.premis(2) suc-len by simp
have idx-prev:  $idx < \text{fri-evidence-layer-len roots } j$ 
    using idx-half by simp
have rel:
  fri-algebraic-layer-relation
    (fri-evidence-layer-len roots j)
    ( $2^j$ )
    (challenges ! j) (doms ! j) (layers ! j)
    (doms ! Suc j) (layers ! Suc j)
    using generic-fri-canonical-layer-chain-relation[OF chain j-bound]
    len-eq-j by simp
have next-dom:
  doms ! Suc j ! idx = (doms ! j ! idx)2
  by (rule fri-algebraic-layer-relationD(1)[OF rel idx-half])
have prev-dom:
  doms ! j ! idx = ( $h^{\wedge} idx * \text{shift}$ )( $2^j$ )
  by (rule Suc.IH[OF j-bound idx-prev])
let ?x =  $h^{\wedge} idx * \text{shift}$ 
have pow-step:  $(?x^{\wedge} (2^j))^2 = ?x^{\wedge} (2^{\wedge} \text{Suc } j)$ 
proof -
  have  $(?x^{\wedge} (2^j))^2 = ?x^{\wedge} ((2^j) * 2)$ 
    by (simp add: power-mult)
  also have ... =  $?x^{\wedge} (2^{\wedge} \text{Suc } j)$ 
    by (simp add: mult.commute)
  finally show ?thesis .
qed
show ?case
  using next-dom prev-dom pow-step by simp
qed

lemma fri-sibling-domain-round:
assumes len-pos:  $0 < \text{len}$ 
and even-len:  $2 \text{ dvd } \text{len}$ 
and round:  $\text{len} * \text{pw} = \text{clength} * \text{scale}$ 
shows
   $((h^{\wedge} ((i + \text{len div } 2) \bmod \text{len})) * \text{shift})^{\wedge} \text{pw} =$ 
   $- (((h^{\wedge} i) * \text{shift})^{\wedge} \text{pw})$ 
proof -
let ?N =  $\text{clength} * \text{scale}$ 
have pw-pos:  $0 < \text{pw}$ 
  using len-pos round eval-domain-size-pos by (cases pw) simp-all
have half-round:  $(\text{len div } 2) * \text{pw} = ?N \text{ div } 2$ 
proof -
from even-len obtain k where len-eq:  $\text{len} = 2 * k$ 
  by blast
then have ?N =  $2 * (k * \text{pw})$ 
  using round by simp

```

```

then show ?thesis
  using len-eq by simp
qed
have exp-mod:
  
$$(((i + \text{len div } 2) \bmod \text{len}) * pw) \bmod ?N =$$


$$(i * pw + ?N \text{ div } 2) \bmod ?N$$

proof -
  have 
$$(((i + \text{len div } 2) \bmod \text{len}) * pw) \bmod ?N =$$


$$((i + \text{len div } 2) * pw) \bmod ?N$$

  using soundness-mod-mult-right-factor-nat
  [OF len-pos pw-pos, of i + len div 2] round by simp
  also have ... =  $(i * pw + ?N \text{ div } 2) \bmod ?N$ 
  using half-round by (simp add: algebra-simps)
  finally show ?thesis .
qed
have h-exp:

$$h \wedge (((i + \text{len div } 2) \bmod \text{len}) * pw) = -(h \wedge (i * pw))$$

proof -
  have 
$$h \wedge (((i + \text{len div } 2) \bmod \text{len}) * pw) =$$


$$h \wedge (((i + \text{len div } 2) \bmod \text{len}) * pw) \bmod ?N$$

  using soundness-h-power-mod-eval-domain
  [of  $((i + \text{len div } 2) \bmod \text{len}) * pw]$ 
  by simp
  also have ... =  $h \wedge (i * pw + ?N \text{ div } 2) \bmod ?N$ 
  using exp-mod by simp
  also have ... =  $h \wedge (i * pw + ?N \text{ div } 2)$ 
  using soundness-h-power-mod-eval-domain[of i * pw + ?N div 2]
  by simp
  also have ... =  $-(h \wedge (i * pw))$ 
  by (rule soundness-h-half-shift)
  finally show ?thesis .
qed
show ?thesis
  using h-exp by (simp add: power-mult power-mult-distrib)
qed

lemma fri-sibling-fold-denominator:
  assumes len-pos:  $0 < \text{len}$ 
  and even-len:  $2 \text{ dvd } \text{len}$ 
  and round:  $\text{len} * pw = \text{clength} * \text{scale}$ 
  shows

$$\text{fri-fold-denominator } ((h \wedge \text{fri-sibling-index } \text{len } i * \text{shift}) \wedge pw) \text{ } 1 =$$


$$- \text{fri-fold-denominator } ((h \wedge i * \text{shift}) \wedge pw) \text{ } 1$$

  using fri-sibling-domain-round[OF len-pos even-len round, of i]
  unfolding fri-sibling-index-def fri-fold-denominator-def
  by simp

lemma fri-sibling-index-first-half:
  fixes len idx :: nat

```

```

assumes even-len: 2 dvd len
and idx-bound: idx < len div 2
shows fri-sibling-index len idx = idx + len div 2
proof -
  have len-eq: len = 2 * (len div 2)
    using even-len by simp
  have idx + len div 2 < len
    using idx-bound len-eq by linarith
  then show ?thesis
    unfolding fri-sibling-index-def by simp
qed

lemma fri-sibling-index-involution-first-half:
  fixes len idx :: nat
  assumes even-len: 2 dvd len
  and idx-bound: idx < len div 2
  shows fri-sibling-index len (fri-sibling-index len idx) = idx
proof -
  have len-eq: len = 2 * (len div 2)
    using even-len by simp
  have sib-eq: fri-sibling-index len idx = idx + len div 2
    by (rule fri-sibling-index-first-half[OF even-len idx-bound])
  have idx-len: idx < len
    using idx-bound len-eq by linarith
  have sum-eq: idx + len div 2 + len div 2 = idx + len
    using len-eq by linarith
  have fri-sibling-index len (fri-sibling-index len idx) =
    ((idx + len div 2) mod len + len div 2) mod len
    unfolding sib-eq fri-sibling-index-def by simp
  also have ... = (idx + len div 2 + len div 2) mod len
    by (simp add: mod-add-left-eq)
  also have ... = (idx + len) mod len
    by (simp only: sum-eq)
  also have ... = idx
    using idx-len by simp
  finally show ?thesis .
qed

lemma fri-sibling-index-mod-half-second-half:
  fixes len raw :: nat
  assumes even-len: 2 dvd len
  and raw-bound: raw < len
  and second-half: ¬ raw < len div 2
  shows raw = fri-sibling-index len (raw mod (len div 2))
proof -
  let ?half = len div 2
  have len-eq: len = 2 * ?half
    using even-len by simp
  have half-pos: 0 < ?half

```

```

    using raw-bound second-half len-eq by linarith
have half-le-raw: ?half ≤ raw
    using second-half by simp
define k where k = raw - ?half
have raw-eq: raw = ?half + k
    using half-le-raw unfolding k-def by simp
have k-bound: k < ?half
    using raw-bound len-eq raw-eq by linarith
have raw-mod: raw mod ?half = k
    using raw-eq k-bound half-pos by simp
have fri-sibling-index len (raw mod ?half) =
  (k + ?half) mod len
    unfolding raw-mod fri-sibling-index-def by simp
also have ... = raw
    using raw-eq raw-bound by simp
finally show ?thesis by simp
qed

lemma fri-table-fold-value-sibling-first-half:
  assumes even-len: 2 dvd len
    and idx-bound: idx < len div 2
    and round: len * pw = clength * scale
    and dom-idx: fri-dom ! idx = (h ^ idx * shift) ^ pw
    and dom-sib:
      fri-dom ! fri-sibling-index len idx =
        (h ^ fri-sibling-index len idx * shift) ^ pw
  shows
    fri-table-fold-value ch layer fri-dom len pw
      (fri-sibling-index len idx) =
        fri-table-fold-value ch layer fri-dom len pw idx
proof -
  have len-pos: 0 < len
    using idx-bound by simp
  have sib-sib: fri-sibling-index len (fri-sibling-index len idx) = idx
    by (rule fri-sibling-index-involution-first-half
      [OF even-len idx-bound])
  have denom-sib:
    fri-fold-denominator
      ((h ^ fri-sibling-index len idx * shift) ^ pw) 1 =
      - fri-fold-denominator ((h ^ idx * shift) ^ pw) 1
    by (rule fri-sibling-fold-denominator
      [OF len-pos even-len round])
  have fri-table-fold-value ch layer fri-dom len pw
    (fri-sibling-index len idx) =
    fri-fold-value ch
      (layer ! fri-sibling-index len idx)
      (layer ! idx)
      (- fri-fold-denominator ((h ^ idx * shift) ^ pw) 1)
    unfolding fri-table-fold-value-def

```

```

    using sib-sib dom-sib denom-sib by simp
  also have ... =
    fri-fold-value ch
      (layer ! idx)
      (layer ! fri-sibling-index len idx)
      (fri-fold-denominator ((h ^ idx * shift) ^ pw) 1)
    by (rule fri-fold-value-swap-neg-denominator)
  also have ... = fri-table-fold-value ch layer fri-dom len pw idx
    unfolding fri-table-fold-value-def using dom-idx by simp
  finally show ?thesis .
qed

lemma fri-sampled-table-fold-from-algebraic-relation:
  assumes relation:
    fri-algebraic-layer-relation len pw ch fri-dom layer next-dom
    next-layer
  and idx-bound: idx < len div 2
  and chunk:
    fri-layer-opening-chunk len xp xp-path xn xn-path chunk
  and match: fri-opening-matches-table len idx layer xp xn
  and dom-idx: fri-dom ! idx = (h ^ idx * shift) ^ pw
  shows idx < length next-layer
    and fri-sampled-table-fold ch len idx pw fri-dom layer chunk
      (next-layer ! idx)
  proof -
    show idx-len: idx < length next-layer
      by (rule fri-algebraic-layer-relation-next-index-bound
        [OF relation idx-bound])
    have fold-eq:
      next-layer ! idx = fri-table-fold-value ch layer fri-dom len pw idx
      by (rule fri-algebraic-layer-relation-sampled-fold-eq
        [OF relation idx-bound])
    show fri-sampled-table-fold ch len idx pw fri-dom layer chunk
      (next-layer ! idx)
      by (rule fri-sampled-table-foldI[OF chunk match dom-idx fold-eq])
  qed

```

```

lemma fri-sampled-table-fold-from-algebraic-relation-sibling:
  assumes relation:
    fri-algebraic-layer-relation len pw ch fri-dom layer next-dom
    next-layer
  and even-len: 2 dvd len
  and round: len * pw = clength * scale
  and idx-bound: idx < len div 2
  and raw-eq: raw = fri-sibling-index len idx
  and chunk:
    fri-layer-opening-chunk len xp xp-path xn xn-path chunk
  and match: fri-opening-matches-table len raw layer xp xn
  and dom-raw: fri-dom ! raw = (h ^ raw * shift) ^ pw

```



```

    and dom-idx: fri-dom ! idx = (h ^ idx * shift) ^ pw
  shows idx < length next-layer
    and fri-sampled-table-fold ch len raw pw fri-dom layer chunk
      (next-layer ! idx)
proof -
  show idx-len: idx < length next-layer
    by (rule fri-algebraic-layer-relation-next-index-bound
        [OF relation idx-bound])
  have fold-idx:
    next-layer ! idx = fri-table-fold-value ch layer fri-dom len pw idx
  by (rule fri-algebraic-layer-relation-sampled-fold-eq
      [OF relation idx-bound])
  have fold-row:
    fri-table-fold-value ch layer fri-dom len pw raw =
    fri-table-fold-value ch layer fri-dom len pw idx
  unfolding raw-eq
  by (rule fri-table-fold-value-sibling-first-half
      [OF even-len idx-bound round dom-idx])
    (use dom-row raw-eq in simp)
  have fold-eq:
    next-layer ! idx = fri-table-fold-value ch layer fri-dom len pw raw
  using fold-idx fold-row by simp
  show fri-sampled-table-fold ch len raw pw fri-dom layer chunk
    (next-layer ! idx)
  by (rule fri-sampled-table-foldI[OF chunk match dom-row fold-eq])
qed

```

lemma *fri-sampled-table-fold-from-algebraic-relation-mod-index:*

```

  assumes relation:
    fri-algebraic-layer-relation len pw ch fri-dom layer next-dom
      next-layer
  and even-len: 2 dvd len
  and round: len * pw = clength * scale
  and chunk:
    fri-layer-opening-chunk len xp xp-path xn xn-path chunk
  and match: fri-opening-matches-table len raw layer xp xn
  and dom-row: fri-dom ! raw = (h ^ raw * shift) ^ pw
  and dom-norm:
    fri-dom ! (raw mod (len div 2)) =
    (h ^ (raw mod (len div 2)) * shift) ^ pw
  shows raw mod (len div 2) < length next-layer
    and fri-sampled-table-fold ch len raw pw fri-dom layer chunk
      (next-layer ! (raw mod (len div 2)))

```

```

proof -
  have raw-bound: raw < len
  by (rule fri-opening-matches-tableD(1)[OF match])
  show idx-len: raw mod (len div 2) < length next-layer
  proof (cases raw < len div 2)
    case True

```

```

then have raw-mod: raw mod (len div 2) = raw
  by simp
show ?thesis
  using fri-sampled-table-fold-from-algebraic-relation(1)
  [OF relation True chunk match dom-raw]
  unfolding raw-mod .
next
case False
let ?idx = raw mod (len div 2)
have half-pos: 0 < len div 2
  using raw-bound False even-len by (cases len) auto
have idx-bound: ?idx < len div 2
  using half-pos by simp
show ?thesis
  by (rule fri-sampled-table-fold-from-algebraic-relation-sibling(1)
    [OF relation even-len round idx-bound - chunk match dom-raw
      dom-norm])
    (rule fri-sibling-index-mod-half-second-half
      [OF even-len raw-bound False])
qed
show fri-sampled-table-fold ch len raw pw fri-dom layer chunk
  (next-layer ! (raw mod (len div 2)))
proof (cases raw < len div 2)
case True
then have raw-mod: raw mod (len div 2) = raw
  by simp
show ?thesis
  using fri-sampled-table-fold-from-algebraic-relation(2)
  [OF relation True chunk match dom-raw]
  unfolding raw-mod .
next
case False
let ?idx = raw mod (len div 2)
have half-pos: 0 < len div 2
  using raw-bound False even-len by (cases len) auto
have idx-bound: ?idx < len div 2
  using half-pos by simp
show ?thesis
  by (rule fri-sampled-table-fold-from-algebraic-relation-sibling(2)
    [OF relation even-len round idx-bound - chunk match dom-raw
      dom-norm])
    (rule fri-sibling-index-mod-half-second-half
      [OF even-len raw-bound False])
qed
qed

lemma fri-sampled-table-fold-eq-at-mod-index:
assumes fold:
  fri-sampled-table-fold ch len raw pw fri-dom layer chunk next-value

```

```

and even-len: 2 dvd len
and round: len * pw = clength * scale
and dom-norm:
  fri-dom ! (raw mod (len div 2)) =
    (h ^ (raw mod (len div 2)) * shift) ^ pw
shows next-value =
  fri-table-fold-value ch layer fri-dom len pw (raw mod (len div 2))
proof -
from fold obtain xp xp-path xn xn-path where
  match: fri-opening-matches-table len raw layer xp xn
  and dom-row: fri-dom ! raw = (h ^ raw * shift) ^ pw
  and next-eq: next-value = fri-table-fold-value ch layer fri-dom len pw raw
  by (elim fri-sampled-table-foldE)
have raw-bound: raw < len
  by (rule fri-opening-matches-tableD(1)[OF match])
show ?thesis
proof (cases raw < len div 2)
  case True
  then have raw mod (len div 2) = raw
  by simp
  then show ?thesis
  using next-eq by simp
next
  case False
  let ?idx = raw mod (len div 2)
  have half-pos: 0 < len div 2
  using raw-bound False even-len by (cases len) auto
  have idx-bound: ?idx < len div 2
  using half-pos by simp
  have raw-eq: raw = fri-sibling-index len ?idx
  by (rule fri-sibling-index-mod-half-second-half
    [OF even-len raw-bound False])
  have dom-sib:
    fri-dom ! fri-sibling-index len ?idx =
      (h ^ fri-sibling-index len ?idx * shift) ^ pw
  using dom-row raw-eq by simp
  have fold-sib:
    fri-table-fold-value ch layer fri-dom len pw
      (fri-sibling-index len ?idx) =
      fri-table-fold-value ch layer fri-dom len pw ?idx
  by (rule fri-table-fold-value-sibling-first-half
    [OF even-len idx-bound round dom-norm dom-sib])
  have fri-table-fold-value ch layer fri-dom len pw raw =
    fri-table-fold-value ch layer fri-dom len pw ?idx
  using raw-eq fold-sib by simp
  then show ?thesis
  using next-eq by simp
qed
qed

```

lemma *fri-algebraic-layer-relation-from-sampled-coverage*:

assumes *even-len*: $2 \text{ dvd } \text{len}$

and *round*: $\text{len} * \text{pw} = \text{clength} * \text{scale}$

and *len-pos*: $0 < \text{len}$

and *dom-len*: $\text{len} \leq \text{length } \text{fri-dom}$

and *layer-len*: $\text{len} \leq \text{length } \text{layer}$

and *next-dom-len*: $\text{len} \text{ div } 2 \leq \text{length } \text{next-dom}$

and *next-layer-len*: $\text{len} \text{ div } 2 \leq \text{length } \text{next-layer}$

and *next-dom*:

$\bigwedge \text{idx}. \text{idx} < \text{len} \text{ div } 2 \implies$
 $\text{next-dom} ! \text{idx} = (\text{fri-dom} ! \text{idx})^2$

and *dom-norm*:

$\bigwedge \text{idx}. \text{idx} < \text{len} \text{ div } 2 \implies$
 $\text{fri-dom} ! \text{idx} = (h \wedge \text{idx} * \text{shift}) \wedge \text{pw}$

and *sampled-cover*:

$\bigwedge \text{idx}. \text{idx} < \text{len} \text{ div } 2 \implies$
 $\exists \text{raw chunk}.$
 $\text{raw} \bmod (\text{len} \text{ div } 2) = \text{idx} \wedge$
 $\text{fri-sampled-table-fold } \text{ch } \text{len } \text{raw } \text{pw } \text{fri-dom } \text{layer } \text{chunk}$
 $(\text{next-layer} ! \text{idx})$

shows

fri-algebraic-layer-relation $\text{len } \text{pw } \text{ch } \text{fri-dom } \text{layer } \text{next-dom}$
 next-layer

unfolding *fri-algebraic-layer-relation-def*

proof (*intro conjI allI impI*)

show $0 < \text{len}$

by (*rule len-pos*)

show $\text{len} \leq \text{length } \text{fri-dom}$

by (*rule dom-len*)

show $\text{len} \leq \text{length } \text{layer}$

by (*rule layer-len*)

show $\text{len} \text{ div } 2 \leq \text{length } \text{next-dom}$

by (*rule next-dom-len*)

show $\text{len} \text{ div } 2 \leq \text{length } \text{next-layer}$

by (*rule next-layer-len*)

fix *idx*

assume *idx-bound*: $\text{idx} < \text{len} \text{ div } 2$

show $\text{next-dom} ! \text{idx} = (\text{fri-dom} ! \text{idx})^2$

by (*rule next-dom*[*OF idx-bound*])

obtain *raw chunk* **where** *raw-mod*: $\text{raw} \bmod (\text{len} \text{ div } 2) = \text{idx}$

and *fold*:

fri-sampled-table-fold $\text{ch } \text{len } \text{raw } \text{pw } \text{fri-dom } \text{layer } \text{chunk}$
 $(\text{next-layer} ! \text{idx})$

using *sampled-cover*[*OF idx-bound*] **by** *blast*

have *fold-eq*:

$\text{next-layer} ! \text{idx} =$
 $\text{fri-table-fold-value } \text{ch } \text{layer } \text{fri-dom } \text{len } \text{pw } (\text{raw} \bmod (\text{len} \text{ div } 2))$

by (*rule fri-sampled-table-fold-eq-at-mod-index*)

```

    [OF fold even-len round])
  (use dom-norm idx-bound raw-mod in simp)
then show next-layer ! idx =
  fri-table-fold-value ch layer fri-dom len pw idx
using raw-mod by simp
qed

lemma generic-fri-sampled-layer-chain-bad-challenge-list-from-coverage:
  assumes chain:
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers
  and steps:
    fri-multiround-proximity-steps (length challenges) degree-bound
    (fri-layer-lengths (length challenges) (clength * scale))
    (map ( $\lambda i. 2^i$ ) [0.. $\text{length challenges}$ ])
    doms layers bad
  and covers: fri-sampled-layers-cover-fold-indices roots query-idxs
  and even-len:
     $\bigwedge \text{layer-idx}. \text{layer-idx} < \text{length challenges} \implies$ 
    2 dvd fri-evidence-layer-len roots layer-idx
  and round-product:
     $\bigwedge \text{layer-idx}. \text{layer-idx} < \text{length challenges} \implies$ 
    fri-evidence-layer-len roots layer-idx *  $2^{\text{layer-idx}}$  =
    clength * scale
  and dom-len:
     $\bigwedge \text{layer-idx}. \text{layer-idx} < \text{length challenges} \implies$ 
    fri-evidence-layer-len roots layer-idx  $\leq \text{length (doms ! layer-idx)}$ 
  and layer-len:
     $\bigwedge \text{layer-idx}. \text{layer-idx} < \text{length challenges} \implies$ 
    fri-evidence-layer-len roots layer-idx  $\leq$ 
    length (layers ! layer-idx)
  and next-dom-len:
     $\bigwedge \text{layer-idx}. \text{layer-idx} < \text{length challenges} \implies$ 
    fri-evidence-layer-len roots layer-idx div 2  $\leq$ 
    length (doms ! Suc layer-idx)
  and next-layer-len:
     $\bigwedge \text{layer-idx}. \text{layer-idx} < \text{length challenges} \implies$ 
    fri-evidence-layer-len roots layer-idx div 2  $\leq$ 
    length (layers ! Suc layer-idx)
  and next-dom:
     $\bigwedge \text{layer-idx idx}.$ 
    layer-idx < length challenges  $\implies$ 
    idx < fri-evidence-layer-len roots layer-idx div 2  $\implies$ 
    doms ! Suc layer-idx ! idx =
    (doms ! layer-idx ! idx)2
  and dom-norm:
     $\bigwedge \text{layer-idx idx}.$ 
    layer-idx < length challenges  $\implies$ 

```

```

    idx < fri-evidence-layer-len roots layer-idx div 2  $\implies$ 
    doms ! layer-idx ! idx = (h ^ idx * shift) ^ (2 ^ layer-idx)
and start-not-low:
  ¬ fri-table-low-degree-on degree-bound eval-domain candidate-table
and final-low:
    fri-table-low-degree-on
      (fri-degree-after (length challenges) degree-bound)
      (doms ! length challenges) (layers ! length challenges)
shows challenges ∈
  generic-fri-bad-challenge-lists (length challenges) bad
proof –
  have partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
      roots challenges final-value query-idxs round-layers
    by (rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])
  have roots-len: length challenges = length roots
    by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
  have challenges-space: challenges ∈ fri-challenge-space (length challenges)
    unfolding fri-challenge-space-def by simp
  have dom0: doms ! 0 = eval-domain
    by (rule generic-fri-sampled-layer-chain-evidenceD(4)[OF chain])
  have layer0: layers ! 0 = candidate-table
    by (rule generic-fri-sampled-layer-chain-evidenceD(5)[OF chain])
  have relations:
    ∧ i. i < length challenges  $\implies$ 
      fri-algebraic-layer-relation
        (fri-layer-lengths (length challenges) (clength * scale) ! i)
        ((map (λ i. 2 ^ i) [0..<length challenges]) ! i)
        (challenges ! i) (doms ! i) (layers ! i)
        (doms ! Suc i) (layers ! Suc i)
  proof –
    fix i
    assume i-bound: i < length challenges
    have i-root-bound: i < length roots
      using i-bound roots-len by simp
    have len-eq:
      fri-layer-lengths (length challenges) (clength * scale) ! i =
      fri-evidence-layer-len roots i
      using roots-len unfolding fri-evidence-layer-len-def by simp
    have pw-eq:
      (map (λ i. 2 ^ i) [0..<length challenges]) ! i = 2 ^ i
      using i-bound by simp
    have sampled-cover:
      ∧ idx. idx < fri-evidence-layer-len roots i div 2  $\implies$ 
      ∃ raw chunk.
        raw mod (fri-evidence-layer-len roots i div 2) = idx ∧
        fri-sampled-table-fold (challenges ! i)
          (fri-evidence-layer-len roots i) raw (2 ^ i)
          (doms ! i) (layers ! i) chunk

```

```

      (layers ! Suc i ! idx)
proof -
  fix idx
  assume idx-bound: idx < fri-evidence-layer-len roots i div 2
  obtain round-idx where round-bound: round-idx < length query-idxs
  and idx-eq:
    fri-evidence-next-idx roots query-idxs round-idx i = idx
  by (rule fri-sampled-layers-cover-fold-indicesD
    [OF covers i-root-bound idx-bound])
  have sample:
    fri-sampled-table-fold (challenges ! i)
      (fri-evidence-layer-len roots i)
      (fri-evidence-layer-idx roots query-idxs round-idx i)
      (2 ^ i) (doms ! i) (layers ! i)
      (round-layers ! round-idx ! i)
      (layers ! Suc i !
        fri-evidence-next-idx roots query-idxs round-idx i)
  by (rule generic-fri-sampled-layer-chain-evidence-sample(2)
    [OF chain round-bound i-bound])
  show  $\exists$  raw chunk.
    raw mod (fri-evidence-layer-len roots i div 2) = idx  $\wedge$ 
    fri-sampled-table-fold (challenges ! i)
      (fri-evidence-layer-len roots i) raw (2 ^ i)
      (doms ! i) (layers ! i) chunk
      (layers ! Suc i ! idx)
  using sample idx-eq
  unfolding fri-evidence-next-idx-def by blast
qed
have len-pos: 0 < fri-evidence-layer-len roots i
  using round-product[OF i-bound] eval-domain-size-pos
  by (cases fri-evidence-layer-len roots i) simp-all
have rel:
  fri-algebraic-layer-relation
    (fri-evidence-layer-len roots i) (2 ^ i)
    (challenges ! i) (doms ! i) (layers ! i)
    (doms ! Suc i) (layers ! Suc i)
  by (rule fri-algebraic-layer-relation-from-sampled-coverage
    [OF even-len[OF i-bound] round-product[OF i-bound] len-pos
      dom-len[OF i-bound] layer-len[OF i-bound]
      next-dom-len[OF i-bound] next-layer-len[OF i-bound]])
    (use next-dom[OF i-bound] dom-norm[OF i-bound] sampled-cover
      in simp-all)
then show
  fri-algebraic-layer-relation
    (fri-layer-lengths (length challenges) (clength * scale) ! i)
    ((map ( $\lambda i$ . 2 ^ i) [0..<length challenges]) ! i)
    (challenges ! i) (doms ! i) (layers ! i)
    (doms ! Suc i) (layers ! Suc i)
  using rel by (simp add: len-eq pw-eq)

```

```

qed
show ?thesis
  unfolding generic-fri-bad-challenge-lists-def
  by (rule fri-multi-round-proximity-skeleton
      [OF steps challenges-space relations])
      (use start-not-low final-low dom0 layer0 in simp-all)
qed

lemma generic-fri-sampled-layer-chain-full-cover-bad-challenge-list:
  assumes
    generic-fri-sampled-layer-chain-full-cover low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers bad
  shows challenges ∈
    generic-fri-bad-challenge-lists (length challenges) bad
proof -
  have chain:
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers
  and steps:
    fri-multi-round-proximity-steps (length challenges) degree-bound
    (fri-layer-lengths (length challenges) (clength * scale))
    (map (λi. 2 ^ i) [0..<length challenges])
    doms layers bad
  and covers: fri-sampled-layers-cover-fold-indices roots query-idxs
  and even-len:
    ∧ layer-idx. layer-idx < length challenges ⇒
      2 dvd fri-evidence-layer-len roots layer-idx
  and round-product:
    ∧ layer-idx. layer-idx < length challenges ⇒
      fri-evidence-layer-len roots layer-idx * 2 ^ layer-idx =
        clength * scale
  and dom-len:
    ∧ layer-idx. layer-idx < length challenges ⇒
      fri-evidence-layer-len roots layer-idx ≤
        length (doms ! layer-idx)
  and layer-len:
    ∧ layer-idx. layer-idx < length challenges ⇒
      fri-evidence-layer-len roots layer-idx ≤
        length (layers ! layer-idx)
  and next-dom-len:
    ∧ layer-idx. layer-idx < length challenges ⇒
      fri-evidence-layer-len roots layer-idx div 2 ≤
        length (doms ! Suc layer-idx)
  and next-layer-len:
    ∧ layer-idx. layer-idx < length challenges ⇒
      fri-evidence-layer-len roots layer-idx div 2 ≤
        length (layers ! Suc layer-idx)

```



```

and next-dom:
   $\bigwedge \text{layer-idx } idx.
    \text{layer-idx} < \text{length challenges} \implies
    idx < \text{fri-evidence-layer-len roots layer-idx div } 2 \implies
    \text{doms ! Suc layer-idx ! idx} =
    (\text{doms ! layer-idx ! idx})^2
and dom-norm:
   $\bigwedge \text{layer-idx } idx.
    \text{layer-idx} < \text{length challenges} \implies
    idx < \text{fri-evidence-layer-len roots layer-idx div } 2 \implies
    \text{doms ! layer-idx ! idx} = (h \wedge idx * \text{shift}) \wedge (2 \wedge \text{layer-idx})
and start-not-low:
   $\neg \text{fri-table-low-degree-on degree-bound eval-domain candidate-table}$ 
and final-low:
  fri-table-low-degree-on
    (fri-degree-after (length challenges) degree-bound)
    (doms ! length challenges) (layers ! length challenges)
using assms
unfolding generic-fri-sampled-layer-chain-full-cover-def
by blast+
show ?thesis
by (rule generic-fri-sampled-layer-chain-bad-challenge-list-from-coverage
  [OF chain steps covers even-len round-product dom-len layer-len
   next-dom-len next-layer-len next-dom dom-norm start-not-low
   final-low])
qed

lemma generic-fri-bad-candidate-from-sampled-layer-chain-full-cover:
assumes cover:
  generic-fri-sampled-layer-chain-full-cover low-degree candidate-table
  degree-bound roots challenges final-value query-idxs round-layers
  doms layers bad
and not-low:  $\neg \text{low-degree candidate-table}$ 
shows
  generic-fri-bad-candidate low-degree candidate-table degree-bound
  roots challenges final-value query-idxs round-layers bad
proof –
have sampled:
  generic-fri-sampled-layer-chain-evidence low-degree candidate-table
  degree-bound roots challenges final-value query-idxs round-layers
  doms layers
using cover
unfolding generic-fri-sampled-layer-chain-full-cover-def by blast
have partial:
  generic-fri-partial-evidence low-degree candidate-table degree-bound
  roots challenges final-value query-idxs round-layers
by (rule generic-fri-sampled-layer-chain-evidenceD(1)[OF sampled])
have bad:
  challenges  $\in$  generic-fri-bad-challenge-lists (length challenges) bad$$ 
```

```

    by (rule generic-fri-sampled-layer-chain-full-cover-bad-challenge-list
        [OF cover])
  show ?thesis
    unfolding generic-fri-bad-candidate-def
    using partial-not-low-bad by simp
qed

lemma generic-fri-canonical-opening-match-sample:
  assumes openings:
    fri-all-round-openings-match-tables
    (take (length challenges) layers) query-idxs round-layers
  and layers-len: length layers = Suc (length challenges)
  and roots-len: length challenges = length roots
  and round-bound: round-idx < length query-idxs
  and layer-bound: layer-idx < length challenges
  obtains xp xp-path xn xn-path
  where
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
      xp xp-path xn xn-path (round-layers ! round-idx ! layer-idx)
  and fri-opening-matches-table
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (layers ! layer-idx) xp xn
proof -
  have layer-bound-take:
    layer-idx < length (take (length challenges) layers)
  using layer-bound layers-len by simp
  have openings-match:
    fri-layer-opening-matches-bound-table (query-idxs ! round-idx)
      layer-idx (take (length challenges) layers)
      (round-layers ! round-idx)
  by (rule fri-all-round-openings-match-tablesD
      [OF openings round-bound layer-bound-take])
  have tables-len:
    length (take (length challenges) layers) = length roots
  using layers-len roots-len by simp
  from fri-layer-opening-matches-bound-table-compactE
    [OF openings-match tables-len]
  obtain xp xp-path xn xn-path where
    chunk:
      fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
        xp xp-path xn xn-path (round-layers ! round-idx ! layer-idx)
  and table-match-take:
    fri-opening-matches-table
      (fri-evidence-layer-len roots layer-idx)
      (fri-layer-indices (length roots) (query-idxs ! round-idx)
        (clength * scale) ! layer-idx)
      (take (length challenges) layers ! layer-idx) xp xn
  by blast

```

```

have idx-eq:
  fri-layer-indices (length roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx =
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx
  unfolding fri-evidence-layer-idx-def by simp
have table-eq:
  take (length challenges) layers ! layer-idx = layers ! layer-idx
  using layer-bound layers-len by simp
have table-match:
  fri-opening-matches-table
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (layers ! layer-idx) xp xn
  using table-match-take idx-eq table-eq by simp
show ?thesis
  by (rule that[OF chunk table-match])
qed

lemma generic-fri-canonical-sampled-fold-at-first-half:
assumes chain:
  generic-fri-canonical-layer-chain-evidence low-degree candidate-table
  degree-bound roots challenges final-value query-idxs round-layers
  doms layers bad
and openings:
  fri-all-round-openings-match-tables
    (take (length challenges) layers) query-idxs round-layers
and domains: fri-sampled-domains-align roots query-idxs doms
and idx-bound:
  fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
  fri-evidence-layer-len roots layer-idx div 2
and round-bound: round-idx < length query-idxs
and layer-bound: layer-idx < length challenges
shows
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
  length (layers ! Suc layer-idx)
  fri-sampled-table-fold
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (doms ! layer-idx)
    (layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (layers ! Suc layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx)
proof –
have partial:
  generic-fri-partial-evidence low-degree candidate-table degree-bound
  roots challenges final-value query-idxs round-layers

```

```

using chain unfolding generic-fri-canonical-layer-chain-evidence-def
by simp
have roots-len: length challenges = length roots
  by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
have layers-len: length layers = Suc (length challenges)
  by (rule generic-fri-canonical-layer-chain-lengths(2)[OF chain])
have len-eq:
  fri-layer-lengths (length challenges) (clength * scale) ! layer-idx =
    fri-evidence-layer-len roots layer-idx
  using roots-len unfolding fri-evidence-layer-len-def by simp
have relation:
  fri-algebraic-layer-relation
    (fri-evidence-layer-len roots layer-idx)
    (2 ^ layer-idx)
    (challenges ! layer-idx) (doms ! layer-idx) (layers ! layer-idx)
    (doms ! Suc layer-idx) (layers ! Suc layer-idx)
  using generic-fri-canonical-layer-chain-relation
    [OF chain layer-bound] len-eq by simp
have dom-idx:
  doms ! layer-idx !
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
    (h ^ fri-evidence-layer-idx roots query-idxs round-idx layer-idx *
      shift) ^ (2 ^ layer-idx)
  by (rule fri-sampled-domains-alignD[OF domains round-bound])
    (use layer-bound roots-len in simp)
obtain xp xp-path xn xn-path where
  chunk:
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
      xp xp-path xn xn-path (round-layers ! round-idx ! layer-idx)
  and match:
    fri-opening-matches-table
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      (layers ! layer-idx) xp xn
  by (rule generic-fri-canonical-opening-match-sample
    [OF openings layers-len roots-len round-bound layer-bound])
have next-idx-eq:
  fri-evidence-next-idx roots query-idxs round-idx layer-idx =
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx
  using idx-bound unfolding fri-evidence-next-idx-def by simp
have idx-len-orig:
  fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
    length (layers ! Suc layer-idx)
  by (rule fri-sampled-table-fold-from-algebraic-relation(1)
    [OF relation idx-bound chunk match dom-idx])
then show
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (layers ! Suc layer-idx)
  using next-idx-eq by simp

```

have *fold-orig*:
fri-sampled-table-fold
 (*challenges* ! *layer-idx*)
 (*fri-evidence-layer-len* *roots* *layer-idx*)
 (*fri-evidence-layer-idx* *roots* *query-idxs* *round-idx* *layer-idx*)
 ($2 \wedge \text{layer-idx}$)
 (*doms* ! *layer-idx*)
 (*layers* ! *layer-idx*)
 (*round-layers* ! *round-idx* ! *layer-idx*)
 (*layers* ! *Suc* *layer-idx* !
 fri-evidence-layer-idx *roots* *query-idxs* *round-idx* *layer-idx*)
by (*rule* *fri-sampled-table-fold-from-algebraic-relation*
 [*OF* *relation* *idx-bound* *chunk* *match* *dom-idx*])
then show
fri-sampled-table-fold
 (*challenges* ! *layer-idx*)
 (*fri-evidence-layer-len* *roots* *layer-idx*)
 (*fri-evidence-layer-idx* *roots* *query-idxs* *round-idx* *layer-idx*)
 ($2 \wedge \text{layer-idx}$)
 (*doms* ! *layer-idx*)
 (*layers* ! *layer-idx*)
 (*round-layers* ! *round-idx* ! *layer-idx*)
 (*layers* ! *Suc* *layer-idx* !
 fri-evidence-next-idx *roots* *query-idxs* *round-idx* *layer-idx*)
using *next-idx-eq* **by** *simp*
qed

lemma *generic-fri-canonical-sampled-fold-at-any-index*:
assumes *chain*:
 generic-fri-canonical-layer-chain-evidence *low-degree* *candidate-table*
 degree-bound *roots* *challenges* *final-value* *query-idxs* *round-layers*
 doms *layers* *bad*
and *openings*:
 fri-all-round-openings-match-tables
 (*take* (*length* *challenges*) *layers*) *query-idxs* *round-layers*
and *domains*: *fri-sampled-domains-align* *roots* *query-idxs* *doms*
and *even-len*: $2 \text{ dvd } \text{fri-evidence-layer-len}$ *roots* *layer-idx*
and *round*:
 fri-evidence-layer-len *roots* *layer-idx* * $2 \wedge \text{layer-idx} =$
 length * *scale*
and *dom-norm*:
 doms ! *layer-idx* !
 fri-evidence-next-idx *roots* *query-idxs* *round-idx* *layer-idx* =
 ($h \wedge \text{fri-evidence-next-idx}$ *roots* *query-idxs* *round-idx* *layer-idx* *
 shift) $\wedge (2 \wedge \text{layer-idx})$
and *round-bound*: *round-idx* < *length* *query-idxs*
and *layer-bound*: *layer-idx* < *length* *challenges*
shows
 fri-evidence-next-idx *roots* *query-idxs* *round-idx* *layer-idx* <

```

    length (layers ! Suc layer-idx)
  fri-sampled-table-fold
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (doms ! layer-idx)
    (layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (layers ! Suc layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx)
proof -
  have partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
      roots challenges final-value query-idxs round-layers
  using chain unfolding generic-fri-canonical-layer-chain-evidence-def
  by simp
  have roots-len: length challenges = length roots
  by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
  have layers-len: length layers = Suc (length challenges)
  by (rule generic-fri-canonical-layer-chain-lengths(2)[OF chain])
  have len-eq:
    fri-layer-lengths (length challenges) (clength * scale) ! layer-idx =
      fri-evidence-layer-len roots layer-idx
  using roots-len unfolding fri-evidence-layer-len-def by simp
  have relation:
    fri-algebraic-layer-relation
      (fri-evidence-layer-len roots layer-idx)
      (2 ^ layer-idx)
      (challenges ! layer-idx) (doms ! layer-idx) (layers ! layer-idx)
      (doms ! Suc layer-idx) (layers ! Suc layer-idx)
  using generic-fri-canonical-layer-chain-relation
    [OF chain layer-bound] len-eq by simp
  have dom-raw:
    doms ! layer-idx !
      fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
      (h ^ fri-evidence-layer-idx roots query-idxs round-idx layer-idx *
        shift) ^ (2 ^ layer-idx)
  by (rule fri-sampled-domains-alignD[OF domains round-bound])
    (use layer-bound roots-len in simp)
  obtain xp xp-path xn xn-path where
    chunk:
      fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
      xp xp-path xn xn-path (round-layers ! round-idx ! layer-idx)
  and match:
    fri-opening-matches-table
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      (layers ! layer-idx) xp xn

```

```

  by (rule generic-fri-canonical-opening-match-sample
      [OF openings layers-len roots-len round-bound layer-bound])
show
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (layers ! Suc layer-idx)
  unfolding fri-evidence-next-idx-def
  by (rule fri-sampled-table-fold-from-algebraic-relation-mod-index(1)
      [OF relation even-len round chunk match dom-row])
      (use dom-norm fri-evidence-next-idx-def in simp)
show
  fri-sampled-table-fold
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (doms ! layer-idx)
    (layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (layers ! Suc layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  unfolding fri-evidence-next-idx-def
  by (rule fri-sampled-table-fold-from-algebraic-relation-mod-index(2)
      [OF relation even-len round chunk match dom-row])
      (use dom-norm fri-evidence-next-idx-def in simp)
qed

```

lemma *generic-fri-canonical-sampled-raw-domain-power:*

```

  assumes chain:
    generic-fri-canonical-layer-chain-evidence low-degree candidate-table
      degree-bound roots challenges final-value query-idxs round-layers
      doms layers bad
  and openings:
    fri-all-round-openings-match-tables
      (take (length challenges) layers) query-idxs round-layers
  and round-bound: round-idx < length query-idxs
  and layer-bound: layer-idx < length challenges
  shows
    doms ! layer-idx !
      fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
      (h ^ fri-evidence-layer-idx roots query-idxs round-idx layer-idx *
        shift) ^ (2 ^ layer-idx)

```

proof –

```

  have partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
      roots challenges final-value query-idxs round-layers
  using chain unfolding generic-fri-canonical-layer-chain-evidence-def
  by simp
  have roots-len: length challenges = length roots
  by (rule generic-fri-partial-evidence-shapes(2)[OF partial])

```

```

have layers-len: length layers = Suc (length challenges)
  by (rule generic-fri-canonical-layer-chain-lengths(2)[OF chain])
obtain xp xp-path xn xn-path where match:
  fri-opening-matches-table
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (layers ! layer-idx) xp xn
  by (rule generic-fri-canonical-opening-match-sample
    [OF openings layers-len roots-len round-bound layer-bound])
have idx-bound:
  fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
    fri-evidence-layer-len roots layer-idx
  by (rule fri-opening-matches-tableD(1)[OF match])
show ?thesis
  by (rule generic-fri-canonical-domain-power
    [OF chain roots-len layer-bound idx-bound])
qed

lemma generic-fri-canonical-layer-chain-as-sampled:
assumes chain:
  generic-fri-canonical-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers bad
and openings:
  fri-all-round-openings-match-tables
    (take (length challenges) layers) query-idxs round-layers
and domains: fri-sampled-domains-align roots query-idxs doms
and next-domains: fri-sampled-next-domains-align roots query-idxs doms
and even-len:
   $\bigwedge \text{layer-idx. layer-idx} < \text{length challenges} \implies$ 
    2 dvd fri-evidence-layer-len roots layer-idx
and round:
   $\bigwedge \text{layer-idx. layer-idx} < \text{length challenges} \implies$ 
    fri-evidence-layer-len roots layer-idx * 2 ^ layer-idx =
      clength * scale
shows
  generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers
proof –
have partial:
  generic-fri-partial-evidence low-degree candidate-table degree-bound
    roots challenges final-value query-idxs round-layers
using chain unfolding generic-fri-canonical-layer-chain-evidence-def
by simp
have roots-len: length challenges = length roots
  by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
have doms-len: length doms = Suc (length challenges)
  by (rule generic-fri-canonical-layer-chain-lengths(1)[OF chain])

```



```

have layers-len: length layers = Suc (length challenges)
  by (rule generic-fri-canonical-layer-chain-lengths(2)[OF chain])
have dom0: doms ! 0 = eval-domain
  using chain unfolding generic-fri-canonical-layer-chain-evidence-def
  by simp
have layer0: layers ! 0 = candidate-table
  using chain unfolding generic-fri-canonical-layer-chain-evidence-def
  by simp
have final:
  fri-final-constant-consistent (layers ! length challenges) final-value
  using chain unfolding generic-fri-canonical-layer-chain-evidence-def
  by simp
have sampled:
   $\bigwedge \text{round-idx layer-idx.}$ 
    round-idx < length query-idxs  $\implies$ 
    layer-idx < length challenges  $\implies$ 
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
      length (layers ! Suc layer-idx)  $\wedge$ 
    fri-sampled-table-fold
      (challenges ! layer-idx)
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      (2 ^ layer-idx)
      (doms ! layer-idx)
      (layers ! layer-idx)
      (round-layers ! round-idx ! layer-idx)
      (layers ! Suc layer-idx !
        fri-evidence-next-idx roots query-idxs round-idx layer-idx)
proof –
  fix round-idx layer-idx
  assume round-bound: round-idx < length query-idxs
  and layer-bound: layer-idx < length challenges
  have layer-bound-roots: layer-idx < length roots
  using layer-bound roots-len by simp
  have dom-norm:
    doms ! layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx =
      (h ^ fri-evidence-next-idx roots query-idxs round-idx layer-idx *
        shift) ^ (2 ^ layer-idx)
  by (rule fri-sampled-next-domains-alignD
    [OF next-domains round-bound layer-bound-roots])
show
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (layers ! Suc layer-idx)  $\wedge$ 
  fri-sampled-table-fold
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)

```

```

    (doms ! layer-idx)
    (layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (layers ! Suc layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  by (intro conjI)
    (rule generic-fri-canonical-sampled-fold-at-any-index
      [OF chain openings domains even-len[OF layer-bound]
        round[OF layer-bound] dom-norm round-bound layer-bound])+
qed
show ?thesis
  unfolding generic-fri-sampled-layer-chain-evidence-def
  using partial doms-len layers-len dom0 layer0 final sampled by blast
qed

```

lemma *generic-fri-bad-challenge-lists-subset:*
generic-fri-bad-challenge-lists n bad $\subseteq$ fri-challenge-space n
unfolding *generic-fri-bad-challenge-lists-def*
by (rule *fri-multiround-bad-challenge-lists-subset*)

lemma *generic-fri-bad-challenge-lists-round-union:*
generic-fri-bad-challenge-lists n bad =
($\bigcup i \in \{..<n\}. \text{generic-fri-round-bad-challenge-lists } n \text{ bad } i$)
unfolding *generic-fri-bad-challenge-lists-def*
fri-multiround-bad-challenge-lists-def
generic-fri-round-bad-challenge-lists-def
by *auto*

lemma *generic-fri-round-bad-challenge-lists-image-subset:*
generic-fri-round-bad-challenge-lists n bad i $\subseteq$
($\lambda(\text{prefix}, b, \text{suffix}). \text{prefix} @ b \# \text{suffix}$) ‘
generic-fri-round-bad-witnesses n bad i
proof
fix *bs*
assume *bs-in: bs $\in$ generic-fri-round-bad-challenge-lists n bad i*
then have *bs-space: bs $\in$ fri-challenge-space n*
and *i-bound: i < n*
and *hit: bs ! i $\in$ bad i (take i bs)*
unfolding *generic-fri-round-bad-challenge-lists-def* **by** *simp-all*
have *len-bs: length bs = n*
using *bs-space* **unfolding** *fri-challenge-space-def* **by** *simp*
let *?prefix = take i bs*
let *?b = bs ! i*
let *?suffix = drop (Suc i) bs*
have *witness:*
(?prefix, ?b, ?suffix) $\in$ generic-fri-round-bad-witnesses n bad i
unfolding *generic-fri-round-bad-witnesses-def*
using *i-bound hit len-bs* **by** *simp*
have *bs-eq: bs = ?prefix @ ?b # ?suffix*

```

    using i-bound len-bs by (simp add: id-take-nth-drop)
  show bs ∈ (λ(prefix, b, suffix). prefix @ b # suffix) ‘
    generic-fri-round-bad-witnesses n bad i
  using witness bs-eq by force
qed

lemma generic-fri-round-bad-witnesses-subset:
  generic-fri-round-bad-witnesses n bad i ⊆
    {(prefix, b, suffix).
      length prefix = i ∧
      b ∈ bad i prefix ∧
      length suffix = n - Suc i}
  unfolding generic-fri-round-bad-witnesses-def by auto

lemma generic-fri-bad-challenge-lists-card-bound-by-rounds:
  card (generic-fri-bad-challenge-lists n bad) ≤
    (∑ i ∈ {.. $n$ }. card (generic-fri-round-bad-challenge-lists n bad i))
  unfolding generic-fri-bad-challenge-lists-round-union
  by (rule card-UN-le) simp

lemma generic-fri-bad-challenge-lists-bounded-byI:
  assumes card-bound: card (generic-fri-bad-challenge-lists n bad) ≤ bd
  shows fri-bad-challenge-lists-bounded-by n
    (generic-fri-bad-challenge-lists n bad) bd
  unfolding fri-bad-challenge-lists-bounded-by-def
  using generic-fri-bad-challenge-lists-subset card-bound by simp

lemma generic-fri-round-bad-witnesses-finite:
  finite (generic-fri-round-bad-witnesses n bad i)
proof -
  let ?P = {prefix :: 'f list. length prefix = i}
  let ?S = {suffix :: 'f list. length suffix = n - Suc i}
  have witness-eq:
    generic-fri-round-bad-witnesses n bad i =
      (if i < n then (SIGMA prefix: ?P. bad i prefix × ?S) else {})
  unfolding generic-fri-round-bad-witnesses-def by auto
  show ?thesis
proof (cases i < n)
  case True
  have finite-P: finite ?P
    by (rule finite-length-lists-UNIV)
  have finite-S: finite ?S
    by (rule finite-length-lists-UNIV)
  have finite-sigma: finite (SIGMA prefix: ?P. bad i prefix × ?S)
    by (rule finite-SigmaI[OF finite-P]) (use finite-S in simp)
  then show ?thesis
    using True witness-eq by simp
next
  case False

```

```

    then show ?thesis
    using witness-eq by simp
qed
qed

lemma generic-fri-round-bad-witnesses-card-bound:
  assumes local-bound:
     $\bigwedge \text{prefix}. \text{length prefix} = i \implies \text{card} (\text{bad } i \text{ prefix}) \leq \text{bd}$ 
  shows  $\text{card} (\text{generic-fri-round-bad-witnesses } n \text{ bad } i) \leq$ 
     $\text{CARD}('f) \wedge i * \text{bd} * \text{CARD}('f) \wedge (n - \text{Suc } i)$ 
  proof -
    let ?P = {prefix :: 'f list. length prefix = i}
    let ?S = {suffix :: 'f list. length suffix = n - Suc i}
    have witness-eq:
      generic-fri-round-bad-witnesses n bad i =
        (if i < n then (SIGMA prefix: ?P. bad i prefix  $\times$  ?S) else {})
    unfolding generic-fri-round-bad-witnesses-def by auto
    show ?thesis
  proof (cases i < n)
    case False
    then show ?thesis
    using witness-eq by simp
  next
    case True
    have finite-P: finite ?P
    by (rule finite-length-lists-UNIV)
    have finite-S: finite ?S
    by (rule finite-length-lists-UNIV)
    have finite-products:  $\forall \text{prefix} \in ?P. \text{finite} (\text{bad } i \text{ prefix} \times ?S)$ 
    using finite-S by simp
    have card-witness:
       $\text{card} (\text{generic-fri-round-bad-witnesses } n \text{ bad } i) =$ 
       $(\sum \text{prefix} \in ?P. \text{card} (\text{bad } i \text{ prefix} \times ?S))$ 
    proof -
      have  $\text{card} (\text{SIGMA prefix: ?P. bad } i \text{ prefix} \times ?S) =$ 
       $(\sum \text{prefix} \in ?P. \text{card} (\text{bad } i \text{ prefix} \times ?S))$ 
      by (rule card-SigmaI[OF finite-P finite-products])
      then show ?thesis
      using True witness-eq by simp
    qed
    also have ...  $\leq (\sum \text{prefix} \in ?P. \text{card} (\text{bad } i \text{ prefix}) * \text{CARD}('f) \wedge (n - \text{Suc } i))$ 
    proof (rule sum-mono)
      fix prefix
      assume prefix  $\in ?P$ 
      have  $\text{card} (\text{bad } i \text{ prefix} \times ?S) =$ 
       $\text{card} (\text{bad } i \text{ prefix}) * \text{card } ?S$ 
      by (simp add: card-cartesian-product)
      also have ... =

```

```

      card (bad i prefix) * CARD('f) ^ (n - Suc i)
    by (simp add: card-length-lists-UNIV)
  finally show card (bad i prefix × ?S) ≤
    card (bad i prefix) * CARD('f) ^ (n - Suc i)
  by simp
qed
also have ... ≤ (∑ prefix ∈ ?P.
  bd * CARD('f) ^ (n - Suc i))
proof (rule sum-mono)
  fix prefix
  assume prefix-in: prefix ∈ ?P
  have card (bad i prefix) ≤ bd
    by (rule local-bound) (use prefix-in in simp)
  then show card (bad i prefix) * CARD('f) ^ (n - Suc i)
    ≤ bd * CARD('f) ^ (n - Suc i)
  by simp
qed
also have ... = card ?P * (bd * CARD('f) ^ (n - Suc i))
  by simp
also have ... = CARD('f) ^ i * bd * CARD('f) ^ (n - Suc i)
  by (simp add: card-length-lists-UNIV mult.assoc)
finally show ?thesis
  using card-witness by simp
qed
qed

```

lemma *generic-fri-round-bad-challenge-lists-card-bound*:

assumes *local-bound*:

$\bigwedge \text{prefix}. \text{length prefix} = i \implies \text{card (bad i prefix)} \leq \text{bd}$

shows $\text{card (generic-fri-round-bad-challenge-lists n bad i)} \leq$
 $\text{CARD('f) ^ i * bd * CARD('f) ^ (n - Suc i)}$

proof —

have *image-subset*:

$\text{generic-fri-round-bad-challenge-lists n bad i} \subseteq$
 $(\lambda(\text{prefix}, b, \text{suffix}). \text{prefix} @ b \# \text{suffix}) \text{ `}$
 $\text{generic-fri-round-bad-witnesses n bad i}$

by (rule *generic-fri-round-bad-challenge-lists-image-subset*)

have *finite-image*:

$\text{finite } ((\lambda(\text{prefix}, b, \text{suffix}). \text{prefix} @ b \# \text{suffix}) \text{ `}$
 $\text{generic-fri-round-bad-witnesses n bad i})$

by (rule *finite-imageI*)

(rule *generic-fri-round-bad-witnesses-finite*)

have $\text{card (generic-fri-round-bad-challenge-lists n bad i)} \leq$

$\text{card } ((\lambda(\text{prefix}, b, \text{suffix}). \text{prefix} @ b \# \text{suffix}) \text{ `}$
 $\text{generic-fri-round-bad-witnesses n bad i})$

by (rule *card-mono[OF finite-image image-subset]*)

also have $\dots \leq \text{card (generic-fri-round-bad-witnesses n bad i)}$

by (rule *card-image-le*)

(rule *generic-fri-round-bad-witnesses-finite*)

also have ... $\leq \text{CARD}(f) \wedge i * \text{bd} * \text{CARD}(f) \wedge (n - \text{Suc } i)$
 by (rule generic-fri-round-bad-witnesses-card-bound)
 (rule local-bound)
 finally show ?thesis .
 qed

lemma generic-fri-bad-challenge-lists-card-bound-from-round-bounds:
 assumes local-bound:
 $\bigwedge i \text{ prefix. } i < n \implies \text{length prefix} = i \implies$
 $\text{card } (\text{bad } i \text{ prefix}) \leq \text{bd } i$
 shows $\text{card } (\text{generic-fri-bad-challenge-lists } n \text{ bad}) \leq$
 $(\sum i \in \{..<n\}. \text{CARD}(f) \wedge i * \text{bd } i * \text{CARD}(f) \wedge (n - \text{Suc } i))$
proof -
 have $\text{card } (\text{generic-fri-bad-challenge-lists } n \text{ bad}) \leq$
 $(\sum i \in \{..<n\}. \text{card } (\text{generic-fri-round-bad-challenge-lists } n \text{ bad } i))$
 by (rule generic-fri-bad-challenge-lists-card-bound-by-rounds)
 also have ... $\leq$
 $(\sum i \in \{..<n\}. \text{CARD}(f) \wedge i * \text{bd } i * \text{CARD}(f) \wedge (n - \text{Suc } i))$
proof (rule sum-mono)
 fix i
 assume i-in: $i \in \{..<n\}$
 show $\text{card } (\text{generic-fri-round-bad-challenge-lists } n \text{ bad } i) \leq$
 $\text{CARD}(f) \wedge i * \text{bd } i * \text{CARD}(f) \wedge (n - \text{Suc } i)$
 by (rule generic-fri-round-bad-challenge-lists-card-bound)
 (rule local-bound[of i], use i-in in simp-all)
 qed
 finally show ?thesis .
 qed

lemma generic-fri-bad-challenge-lists-bounded-by-from-round-bounds:
 assumes local-bound:
 $\bigwedge i \text{ prefix. } i < n \implies \text{length prefix} = i \implies$
 $\text{card } (\text{bad } i \text{ prefix}) \leq \text{bd } i$
 shows fri-bad-challenge-lists-bounded-by n
 $(\text{generic-fri-bad-challenge-lists } n \text{ bad})$
 $(\sum i \in \{..<n\}. \text{CARD}(f) \wedge i * \text{bd } i * \text{CARD}(f) \wedge (n - \text{Suc } i))$
 by (rule generic-fri-bad-challenge-lists-bounded-byI)
 (rule generic-fri-bad-challenge-lists-card-bound-from-round-bounds
 [OF local-bound])

lemma generic-fri-bad-challenge-lists-card-bound:
 $\text{card } (\text{generic-fri-bad-challenge-lists } n \text{ bad}) \leq \text{CARD}(f) \wedge n$
unfolding generic-fri-bad-challenge-lists-def
 by (rule fri-multiround-bad-challenge-lists-card-bound)

lemma generic-fri-bad-challenge-lists-bounded-by:
 fri-bad-challenge-lists-bounded-by n
 $(\text{generic-fri-bad-challenge-lists } n \text{ bad}) (\text{CARD}(f) \wedge n)$
unfolding generic-fri-bad-challenge-lists-def

by (rule fri-multi-round-bad-challenge-lists-bounded-by)

lemma *generic-fri-low-degree-bad-challenge-listI*:

assumes *steps*:

fri-multi-round-proximity-steps *n d lens pws doms layers bad*

and *challenges*: *bs* $\in$ *fri-challenge-space* *n*

and *relations*:

$\bigwedge i. i < n \implies$

fri-algebraic-layer-relation (*lens* ! *i*) (*pws* ! *i*) (*bs* ! *i*)

(*doms* ! *i*) (*layers* ! *i*) (*doms* ! *Suc i*) (*layers* ! *Suc i*)

and *start-not-low*:

$\neg$ *fri-table-low-degree-on* *d* (*doms* ! 0) (*layers* ! 0)

and *final-low*:

fri-table-low-degree-on (*fri-degree-after* *n d*) (*doms* ! *n*)

(*layers* ! *n*)

shows *bs* $\in$ *generic-fri-bad-challenge-lists* *n bad*

unfolding *generic-fri-bad-challenge-lists-def*

by (rule *fri-multi-round-proximity-skeleton*

[*OF steps challenges relations start-not-low final-low*])

lemma *generic-fri-proximity-evidence-bad-challenge-list*:

assumes *generic-fri-proximity-evidence* *n d lens pws doms layers bs bad*

shows *bs* $\in$ *generic-fri-bad-challenge-lists* *n bad*

proof –

have *steps*:

fri-multi-round-proximity-steps *n d lens pws doms layers bad*

and *challenges*: *bs* $\in$ *fri-challenge-space* *n*

and *relations*:

$\bigwedge i. i < n \implies$

fri-algebraic-layer-relation (*lens* ! *i*) (*pws* ! *i*) (*bs* ! *i*)

(*doms* ! *i*) (*layers* ! *i*) (*doms* ! *Suc i*) (*layers* ! *Suc i*)

and *start-not-low*:

$\neg$ *fri-table-low-degree-on* *d* (*doms* ! 0) (*layers* ! 0)

and *final-low*:

fri-table-low-degree-on (*fri-degree-after* *n d*) (*doms* ! *n*)

(*layers* ! *n*)

using *assms* **unfolding** *generic-fri-proximity-evidence-def* **by** *blast+*

show *?thesis*

by (rule *generic-fri-low-degree-bad-challenge-listI*

[*OF steps challenges relations start-not-low final-low*])

qed

lemma *generic-fri-canonical-layer-chain-bad-challenge-list*:

assumes

generic-fri-canonical-layer-chain-evidence *low-degree* *candidate-table*

degree-bound *roots* *challenges* *final-value* *query-idxs* *round-layers*

doms *layers* *bad*

shows *challenges* $\in$

generic-fri-bad-challenge-lists (*length* *challenges*) *bad*

proof –
have *generic-fri-proximity-evidence* (*length challenges*) *degree-bound*
(*fri-layer-lengths* (*length challenges*) (*clength * scale*))
(*map* ($\lambda i. 2^i$) [*0..<length challenges*])
doms layers challenges bad
using *assms*
unfolding *generic-fri-canonical-layer-chain-evidence-def* **by** *simp*
then show *?thesis*
by (*rule generic-fri-proximity-evidence-bad-challenge-list*)
qed

lemma *generic-fri-canonical-layer-chain-evidenceD*:
assumes
generic-fri-canonical-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers bad
shows
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers
doms ! 0 = eval-domain
layers ! 0 = candidate-table
fri-final-constant-consistent (layers ! length challenges) final-value
generic-fri-proximity-evidence (length challenges) degree-bound
(*fri-layer-lengths* (*length challenges*) (*clength * scale*))
(*map* ($\lambda i. 2^i$) [*0..<length challenges*])
doms layers challenges bad
using *assms* **unfolding** *generic-fri-canonical-layer-chain-evidence-def*
by *simp-all*

lemma *generic-fri-bad-candidate-from-canonical-layer-chain*:
assumes *chain*:
generic-fri-canonical-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers bad
and *not-low*: $\neg$ *low-degree candidate-table*
shows
generic-fri-bad-candidate low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers bad
proof –
have *partial*:
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers
by (*rule generic-fri-canonical-layer-chain-evidenceD(1)[OF chain]*)
have *bad*:
challenges $\in$
generic-fri-bad-challenge-lists (length challenges) bad
by (*rule generic-fri-canonical-layer-chain-bad-challenge-list[OF chain]*)
show *?thesis*
unfolding *generic-fri-bad-candidate-def*

using *partial not-low bad* by *simp*
qed

lemma *generic-fri-bad-challenge-lists-mono*:
assumes *mono*:
 $\bigwedge i \text{ prefix. } i < n \implies \text{length prefix} = i \implies$
 $\text{bad } i \text{ prefix} \subseteq \text{bad}' i \text{ prefix}$
shows *generic-fri-bad-challenge-lists* *n bad* $\subseteq$
generic-fri-bad-challenge-lists *n bad'*

proof
fix *bs*
assume *bs-in*: *bs* $\in$ *generic-fri-bad-challenge-lists* *n bad*
then have *space*: *bs* $\in$ *fri-challenge-space* *n*
unfolding *generic-fri-bad-challenge-lists-def*
using *fri-multiround-bad-challenge-lists-subset* by *blast*
from *bs-in* **obtain** *i* **where** *i-bound*: *i* $<$ *n*
and *hit*: *bs* ! *i* $\in$ *bad* *i* (take *i* *bs*)
unfolding *generic-fri-bad-challenge-lists-def*
fri-multiround-bad-challenge-lists-def
by *blast*
have *len-bs*: *length* *bs* = *n*
using *space* **unfolding** *fri-challenge-space-def* by *simp*
have *len-take*: *length* (take *i* *bs*) = *i*
using *i-bound* *len-bs* by *simp*
have *bs* ! *i* $\in$ *bad'* *i* (take *i* *bs*)
using *hit* *mono*[*OF* *i-bound* *len-take*] by *blast*
then show *bs* $\in$ *generic-fri-bad-challenge-lists* *n bad'*
using *space* *i-bound*
unfolding *generic-fri-bad-challenge-lists-def*
fri-multiround-bad-challenge-lists-def
by *blast*
qed

lemma *generic-fri-bad-candidateD*:
assumes *generic-fri-bad-candidate* *low-degree* *candidate-table* *degree-bound*
roots *challenges* *final-value* *query-idxs* *round-layers* *bad*
shows *generic-fri-partial-evidence* *low-degree* *candidate-table* *degree-bound*
roots *challenges* *final-value* *query-idxs* *round-layers*
and $\neg$ *low-degree* *candidate-table*
and *challenges* $\in$ *generic-fri-bad-challenge-lists* (*length* *challenges*) *bad*
using *assms* **unfolding** *generic-fri-bad-candidate-def* by *simp-all*

lemma *generic-fri-bad-candidate-challenges-in-space*:
assumes *bad-candidate*:
generic-fri-bad-candidate *low-degree* *candidate-table* *degree-bound*
roots *challenges* *final-value* *query-idxs* *round-layers* *bad*
shows *challenges* $\in$ *fri-challenge-space* (*length* *challenges*)
proof –
have *challenges* $\in$ *generic-fri-bad-challenge-lists* (*length* *challenges*) *bad*

```

    using generic-fri-bad-candidateD(3)[OF bad-candidate] .
  then show ?thesis
    using generic-fri-bad-challenge-lists-subset[of length challenges bad]
    by blast
qed

lemma generic-fri-bad-candidate-challenges-length:
  assumes bad-candidate:
    generic-fri-bad-candidate low-degree candidate-table degree-bound
    roots challenges final-value query-idxs round-layers bad
  shows length challenges = fri-round-count-for-degree-bound degree-bound
proof -
  have evidence:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
    roots challenges final-value query-idxs round-layers
  using generic-fri-bad-candidateD(1)[OF bad-candidate] .
  then show ?thesis
    using generic-fri-partial-evidence-shapes(1,2)[OF evidence] by simp
qed

lemma trace-table-low-degree-iff-fri-table-low-degree-on-eval-domain:
  trace-table-low-degree trace-table  $\longleftrightarrow$ 
  fri-table-low-degree-on (clength - 1) eval-domain trace-table
proof
  assume trace-table-low-degree trace-table
  then obtain f where deg-f: degree f < clength
    and table: trace-table = map (poly f) eval-domain
    unfolding trace-table-low-degree-def by blast
  have degree f  $\leq$  clength - 1
    using deg-f by simp
  then show fri-table-low-degree-on (clength - 1) eval-domain trace-table
    unfolding fri-table-low-degree-on-def
    using table by blast
next
  assume fri-table-low-degree-on (clength - 1) eval-domain trace-table
  then obtain p where deg-p: degree p  $\leq$  clength - 1
    and table: trace-table = map (poly p) eval-domain
    unfolding fri-table-low-degree-on-def by blast
  have degree p < clength
    using deg-p clength-pos by simp
  then show trace-table-low-degree trace-table
    unfolding trace-table-low-degree-def
    using table by blast
qed

lemma composition-table-low-degree-iff-fri-table-low-degree-on-eval-domain:
  composition-table-low-degree d composition-table  $\longleftrightarrow$ 
  fri-table-low-degree-on d eval-domain composition-table
  unfolding composition-table-low-degree-def fri-table-low-degree-on-def

```

by *simp*

lemma *composition-table-low-degree-mono:*

assumes $d \leq d'$

and *composition-table-low-degree d composition-table*

shows *composition-table-low-degree d' composition-table*

proof –

from *assms(2)* **obtain** *q* **where** *deg-q: degree q ≤ d*

and *table: composition-table = map (poly q) eval-domain*

unfolding *composition-table-low-degree-def* **by** *blast*

have *degree q ≤ d'*

using *deg-q assms(1)* **by** *simp*

then show *?thesis*

unfolding *composition-table-low-degree-def*

using *table* **by** *blast*

qed

lemma *composition-table-not-low-degree-mono:*

assumes $d \leq d'$

and $\neg$ *composition-table-low-degree d' composition-table*

shows $\neg$ *composition-table-low-degree d composition-table*

using *composition-table-low-degree-mono[OF assms(1)] assms(2)* **by** *blast*

lemma *trace-table-not-low-degree-iff-not-fri-table-low-degree-on-eval-domain:*

$\neg$ *trace-table-low-degree trace-table* $\longleftrightarrow$

$\neg$ *fri-table-low-degree-on (clength - 1) eval-domain trace-table*

by (*simp add: trace-table-low-degree-iff-fri-table-low-degree-on-eval-domain*)

lemma *composition-table-not-low-degree-iff-not-fri-table-low-degree-on-eval-domain:*

$\neg$ *composition-table-low-degree d composition-table* $\longleftrightarrow$

$\neg$ *fri-table-low-degree-on d eval-domain composition-table*

by (*simp add:*

composition-table-low-degree-iff-fri-table-low-degree-on-eval-domain)

lemma *accepted-fri-opening-transcript-trace-generic-partial-evidence:*

assumes *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final

dg composition-roots composition-bs composition-final query-idxs

trace-round-layers composition-round-layers

shows

generic-fri-partial-evidence trace-table-low-degree trace-table

(clength - 1) trace-roots trace-bs trace-final query-idxs

trace-round-layers

proof –

have *count:*

length trace-roots = fri-round-count-for-degree-bound (clength - 1)

using *accepted-fri-opening-transcript-algebraic-round-counts(1)*

[OF fri-openings]

by (*simp add: trace-fri-algebraic-round-count-def*)

have *len-bs*: *length trace-bs = length trace-roots*
using *accepted-fri-opening-transcript-shapes*[*OF fri-openings*] **by** *blast*
have *layers*:
fri-all-round-layer-evidence trace-roots trace-bs query-idxs
trace-round-layers
by (*rule accepted-fri-opening-transcript-trace-all-round-layer-evidence*
[OF fri-openings])
show *?thesis*
unfolding *generic-fri-partial-evidence-def*
using *count len-bs layers* **by** *simp*
qed

lemma *accepted-fri-opening-transcript-composition-generic-partial-evidence*:
assumes *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers
shows
generic-fri-partial-evidence (composition-table-low-degree (to-nat dg))
composition-table (to-nat dg) composition-roots composition-bs
composition-final query-idxs composition-round-layers
proof —
have *count*:
length composition-roots = fri-round-count-for-degree-bound (to-nat dg)
using *accepted-fri-opening-transcript-algebraic-round-counts*(3)
[OF fri-openings]
by (*simp add: composition-fri-algebraic-round-count-def*)
have *len-bs*: *length composition-bs = length composition-roots*
using *accepted-fri-opening-transcript-shapes*[*OF fri-openings*] **by** *blast*
have *layers*:
fri-all-round-layer-evidence composition-roots composition-bs query-idxs
composition-round-layers
by (*rule*
accepted-fri-opening-transcript-composition-all-round-layer-evidence
[OF fri-openings])
show *?thesis*
unfolding *generic-fri-partial-evidence-def*
using *count len-bs layers* **by** *simp*
qed

lemma *trace-fri-partial-candidate-opening-evidence-generic-partial*:
assumes *evidence*:
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
shows
generic-fri-partial-evidence trace-table-low-degree trace-table

$(clength - 1)$ *trace-roots trace-bs trace-final fri-query-idxs*
trace-round-layers
proof –
have *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
using *trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence]* .
show *?thesis*
by (rule *accepted-fri-opening-transcript-trace-generic-partial-evidence*
[OF fri-openings])
qed

lemma *composition-fri-partial-candidate-opening-evidence-generic-partial*:
assumes *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
shows
generic-fri-partial-evidence
(composition-table-low-degree (to-nat fri-dg))
composition-table (to-nat fri-dg) opening-composition-roots
composition-bs composition-final fri-query-idxs
composition-round-layers

proof –
have *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
using *composition-fri-partial-candidate-opening-evidenceD(1)*
[OF evidence] .
show *?thesis*
by (rule
accepted-fri-opening-transcript-composition-generic-partial-evidence
[OF fri-openings])
qed

lemma *trace-fri-partial-candidate-opening-evidence-not-low-on-eval-domain*:
assumes *evidence*:
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table

shows $\neg$ *fri-table-low-degree-on* (*clength* - 1) *eval-domain trace-table*
proof -
have *partial*:
trace-fri-partial-candidate-evidence s out fr candidate-query-idxs
trace-openings trace-table
using *trace-fri-partial-candidate-opening-evidenceD(2)[OF evidence]* .
have $\neg$ *trace-table-low-degree trace-table*
using *trace-fri-partial-candidate-evidenceD(3)[OF partial]* .
then show ?thesis
using *trace-table-not-low-degree-iff-not-fri-table-low-degree-on-eval-domain*
by *simp*
qed

lemma *composition-fri-partial-candidate-opening-evidence-not-low-declared*:
assumes *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *degree-bound*: *to-nat fri-dg* $\leq$ *maxDegree*
shows $\neg$ *composition-table-low-degree* (*to-nat fri-dg*) *composition-table*
proof -
have *partial*:
composition-fri-partial-candidate-evidence s out fr f-fri-roots
f-final as dg composition-fri-roots final trace-query-idxs
trace-openings composition-query-idxs composition-openings
trace-table composition-table
by (*rule composition-fri-partial-candidate-opening-evidenceD(2)*
[OF evidence])
have *not-low-max*:
 $\neg$ *composition-table-low-degree maxDegree composition-table*
by (*rule composition-fri-partial-candidate-evidenceD(5)[OF partial]*)
show ?thesis
by (*rule composition-table-not-low-degree-mono*
[OF degree-bound not-low-max])
qed

lemma *composition-fri-partial-candidate-opening-evidence-not-low-declared-on-eval-domain*:
assumes *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table

and *degree-bound*: $\text{to-nat } \text{fri-dg} \leq \text{maxDegree}$
shows $\neg \text{fri-table-low-degree-on } (\text{to-nat } \text{fri-dg}) \text{ eval-domain}$
composition-table
proof –
have $\neg \text{composition-table-low-degree } (\text{to-nat } \text{fri-dg}) \text{ composition-table}$
by (rule *composition-fri-partial-candidate-opening-evidence-not-low-declared*
 $[\text{OF evidence degree-bound}]$)
then show *?thesis*
using *composition-table-not-low-degree-iff-not-fri-table-low-degree-on-eval-domain*
by *simp*
qed

lemma *trace-fri-partial-candidate-opening-evidence-canonical-bad-candidate*:

assumes *evidence*:
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *chain*:
generic-fri-canonical-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers bad
shows
generic-fri-bad-candidate trace-table-low-degree trace-table
(clength - 1) trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers bad

proof –
have $\neg \text{trace-table-low-degree } \text{trace-table}$
using *trace-fri-partial-candidate-opening-evidenceD(2)[OF evidence]*
trace-fri-partial-candidate-evidenceD(3)
by *blast*
then show *?thesis*
by (rule *generic-fri-bad-candidate-from-canonical-layer-chain*
 $[\text{OF chain}]$)
qed

lemma *composition-fri-partial-candidate-opening-evidence-canonical-bad-candidate*:

assumes *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *degree-bound*: $\text{to-nat } \text{fri-dg} \leq \text{maxDegree}$
and *chain*:
generic-fri-canonical-layer-chain-evidence

```

    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers bad
shows
  generic-fri-bad-candidate
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers bad
proof –
  have  $\neg$  composition-table-low-degree (to-nat fri-dg) composition-table
    by (rule composition-fri-partial-candidate-opening-evidence-not-low-declared
      [OF evidence degree-bound])
  then show ?thesis
    by (rule generic-fri-bad-candidate-from-canonical-layer-chain
      [OF chain])
qed

lemma trace-fri-multiround-bad-sets-as-generic:
  trace-fri-multiround-bad-sets bad trace-table =
    generic-fri-bad-challenge-lists trace-fri-algebraic-round-count
      (bad trace-table)
unfolding trace-fri-multiround-bad-sets-def
  generic-fri-bad-challenge-lists-def
by simp

lemma composition-fri-multiround-bad-sets-as-generic:
  composition-fri-multiround-bad-sets bad dg composition-table =
    generic-fri-bad-challenge-lists
      (composition-fri-algebraic-round-count dg)
      (bad dg composition-table)
unfolding composition-fri-multiround-bad-sets-def
  generic-fri-bad-challenge-lists-def
by simp

end

end

```

```

theory Soundness-FRI-Internal-Bounds
  imports Soundness-FRI-Generic-Low-Degree
begin

```

Internal concrete FRI challenge-list bounds.

These lemmas deliberately avoid the symbolic parameters *trace-fri-error* and *composition-fri-error*. They expose the concrete bad-set fractions obtained from the generic FRI bad challenge-list interface. Later public-route work can either use these concrete terms directly, or relate the symbolic error parameters to these terms via an explicitly chosen FRI soundness interface.

context *soundness*
begin

definition *fri-bad-challenge-round-mass*

$:: \text{nat} \Rightarrow (\text{nat} \Rightarrow \text{nat}) \Rightarrow \text{nat}$

where

$\text{fri-bad-challenge-round-mass } n \text{ bd} =$
 $(\sum i \in \{..<n\}. \text{CARD}('f) \wedge i * \text{bd } i * \text{CARD}('f) \wedge (n - \text{Suc } i))$

definition *fri-bad-challenge-round-fraction*

$:: \text{nat} \Rightarrow (\text{nat} \Rightarrow \text{nat}) \Rightarrow \text{prob}$

where

$\text{fri-bad-challenge-round-fraction } n \text{ bd} =$
 $\text{nnreal } (\text{fri-bad-challenge-round-mass } n \text{ bd}) / \text{nnreal } (\text{CARD}('f) \wedge n)$

definition *fri-one-step-disagreement-round-mass*

$:: \text{nat} \Rightarrow (\text{nat} \Rightarrow \text{nat}) \Rightarrow \text{nat}$

where

$\text{fri-one-step-disagreement-round-mass } n \text{ len} =$
 $\text{fri-bad-challenge-round-mass } n (\lambda i. \text{len } i \text{ div } 2)$

definition *fri-one-step-disagreement-round-fraction*

$:: \text{nat} \Rightarrow (\text{nat} \Rightarrow \text{nat}) \Rightarrow \text{prob}$

where

$\text{fri-one-step-disagreement-round-fraction } n \text{ len} =$
 $\text{fri-bad-challenge-round-fraction } n (\lambda i. \text{len } i \text{ div } 2)$

lemma *fri-one-step-disagreement-round-fraction-unfold:*

$\text{fri-one-step-disagreement-round-fraction } n \text{ len} =$
 nnreal
 $(\sum i \in \{..<n\}. \text{CARD}('f) \wedge i * (\text{len } i \text{ div } 2) * \text{CARD}('f) \wedge (n - \text{Suc } i)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge n)$

unfolding *fri-one-step-disagreement-round-fraction-def*

fri-bad-challenge-round-fraction-def

fri-bad-challenge-round-mass-def

by *simp*

lemma *fri-one-step-disagreement-round-fraction-refl-bound:*

nnreal
 $(\sum i \in \{..<n\}. \text{CARD}('f) \wedge i * (\text{len } i \text{ div } 2) * \text{CARD}('f) \wedge (n - \text{Suc } i)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge n)$
 $\leq \text{fri-one-step-disagreement-round-fraction } n \text{ len}$
by (*simp add: fri-one-step-disagreement-round-fraction-unfold*)

definition *trace-fri-canonical-layer-len* $:: \text{nat} \Rightarrow \text{nat}$

where

$\text{trace-fri-canonical-layer-len } i =$

$fri\text{-}layer\text{-}lengths\ (fri\text{-}round\text{-}count\text{-}for\text{-}degree\text{-}bound\ (clength - 1))$
 $(clength * scale) ! i$

definition $composition\text{-}fri\text{-}canonical\text{-}layer\text{-}len :: 'f \Rightarrow nat \Rightarrow nat$
where

$composition\text{-}fri\text{-}canonical\text{-}layer\text{-}len\ dg\ i =$
 $fri\text{-}layer\text{-}lengths\ (fri\text{-}round\text{-}count\text{-}for\text{-}degree\text{-}bound\ (to\text{-}nat\ dg))$
 $(clength * scale) ! i$

definition $concrete\text{-}trace\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction :: prob$
where

$concrete\text{-}trace\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction =$
 $fri\text{-}one\text{-}step\text{-}disagreement\text{-}round\text{-}fraction$
 $(fri\text{-}round\text{-}count\text{-}for\text{-}degree\text{-}bound\ (clength - 1))$
 $trace\text{-}fri\text{-}canonical\text{-}layer\text{-}len$

definition $concrete\text{-}composition\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction :: prob$
where

$concrete\text{-}composition\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction =$
 Max
 $((\lambda dg :: 'f.$
 $fri\text{-}one\text{-}step\text{-}disagreement\text{-}round\text{-}fraction$
 $(fri\text{-}round\text{-}count\text{-}for\text{-}degree\text{-}bound\ (to\text{-}nat\ dg))$
 $(composition\text{-}fri\text{-}canonical\text{-}layer\text{-}len\ dg))\ 'UNIV)$

lemma $concrete\text{-}trace\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction\text{-}eq:$

$fri\text{-}one\text{-}step\text{-}disagreement\text{-}round\text{-}fraction$
 $(fri\text{-}round\text{-}count\text{-}for\text{-}degree\text{-}bound\ (clength - 1))$
 $trace\text{-}fri\text{-}canonical\text{-}layer\text{-}len$
 $= concrete\text{-}trace\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction$
by ($simp\ add: concrete\text{-}trace\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction\text{-}def$)

lemma $concrete\text{-}composition\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction\text{-}ge:$

$fri\text{-}one\text{-}step\text{-}disagreement\text{-}round\text{-}fraction$
 $(fri\text{-}round\text{-}count\text{-}for\text{-}degree\text{-}bound\ (to\text{-}nat\ dg))$
 $(composition\text{-}fri\text{-}canonical\text{-}layer\text{-}len\ dg)$
 $\leq concrete\text{-}composition\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction$
unfolding $concrete\text{-}composition\text{-}fri\text{-}bad\text{-}challenge\text{-}fraction\text{-}def$
by ($rule\ Max\text{-}ge$) $simp\text{-}all$

lemma $fri\text{-}bad\text{-}challenge\text{-}round\text{-}fraction\text{-}bound\text{-}from\text{-}round\text{-}bounds:$

assumes $local\text{-}bound:$
 $\bigwedge i\ prefix. i < n \implies length\ prefix = i \implies$
 $card\ (bad\ i\ prefix) \leq bd\ i$
shows $nnreal\ (card\ (generic\text{-}fri\text{-}bad\text{-}challenge\text{-}lists\ n\ bad)) /$
 $nnreal\ (CARD('f) ^ n)$
 $\leq fri\text{-}bad\text{-}challenge\text{-}round\text{-}fraction\ n\ bd$
unfolding $fri\text{-}bad\text{-}challenge\text{-}round\text{-}fraction\text{-}def$
 $fri\text{-}bad\text{-}challenge\text{-}round\text{-}mass\text{-}def$

by (rule nnreal-nat-divide-right-mono)
 (rule generic-fri-bad-challenge-lists-card-bound-from-round-bounds
 [OF local-bound])

lemma generic-fri-bad-challenge-lists-card-bound-from-one-step-disagreement-subset:

assumes local-subset:

$\bigwedge i \text{ prefix}. i < n \implies \text{length prefix} = i \implies$
 $\text{bad } i \text{ prefix} \subseteq$
 $\text{fri-one-step-disagreement-challenges}$
 $(\text{len } i) (\text{pw } i) (\text{layer } i \text{ prefix}) (\text{claimed } i \text{ prefix}) (\text{domains } i \text{ prefix})$

shows card (generic-fri-bad-challenge-lists n bad) $\leq$

$(\sum i \in \{..<n\}.$
 $\text{CARD}('f) \wedge i * (\text{len } i \text{ div } 2) * \text{CARD}('f) \wedge (n - \text{Suc } i))$

proof (rule generic-fri-bad-challenge-lists-card-bound-from-round-bounds)

fix $i :: \text{nat}$ **and** $\text{prefix} :: 'f \text{ list}$

assume $i\text{-bound}: i < n$ **and** $\text{prefix-len}: \text{length prefix} = i$

have card (bad $i \text{ prefix}$) $\leq$

card
 (fri-one-step-disagreement-challenges
 (len i) (pw i) (layer $i \text{ prefix}$) (claimed $i \text{ prefix}$) (domains $i \text{ prefix}$))
by (rule card-mono) (simp-all add: local-subset[OF $i\text{-bound}$ prefix-len])

also have ... $\leq \text{len } i \text{ div } 2$

by (rule fri-one-step-disagreement-challenges-card-bound-local)

finally show card (bad $i \text{ prefix}$) $\leq \text{len } i \text{ div } 2$.

qed

lemma generic-fri-bad-challenge-lists-bounded-by-from-one-step-disagreement-subset:

assumes local-subset:

$\bigwedge i \text{ prefix}. i < n \implies \text{length prefix} = i \implies$
 $\text{bad } i \text{ prefix} \subseteq$
 $\text{fri-one-step-disagreement-challenges}$
 $(\text{len } i) (\text{pw } i) (\text{layer } i \text{ prefix}) (\text{claimed } i \text{ prefix}) (\text{domains } i \text{ prefix})$

shows fri-bad-challenge-lists-bounded-by n

(generic-fri-bad-challenge-lists n bad)

$(\sum i \in \{..<n\}.$
 $\text{CARD}('f) \wedge i * (\text{len } i \text{ div } 2) * \text{CARD}('f) \wedge (n - \text{Suc } i))$

by (rule generic-fri-bad-challenge-lists-bounded-byI)

(rule generic-fri-bad-challenge-lists-card-bound-from-one-step-disagreement-subset
 [OF local-subset])

lemma fri-bad-challenge-round-fraction-bound-from-one-step-disagreement-subset:

assumes local-subset:

$\bigwedge i \text{ prefix}. i < n \implies \text{length prefix} = i \implies$
 $\text{bad } i \text{ prefix} \subseteq$
 $\text{fri-one-step-disagreement-challenges}$
 $(\text{len } i) (\text{pw } i) (\text{layer } i \text{ prefix}) (\text{claimed } i \text{ prefix}) (\text{domains } i \text{ prefix})$

shows nnreal (card (generic-fri-bad-challenge-lists n bad)) /

nnreal (CARD('f) $\wedge n$)

$\leq \text{fri-one-step-disagreement-round-fraction } n \text{ len}$

unfolding *fri-one-step-disagreement-round-fraction-def*
fri-bad-challenge-round-fraction-def
fri-one-step-disagreement-round-mass-def
fri-bad-challenge-round-mass-def
by (rule *nnreal-nat-divide-right-mono*)
(rule *generic-fri-bad-challenge-lists-card-bound-from-one-step-disagreement-subset*
[*OF local-subset*])

lemma *wp-verify-monad-trace-generic-fri-bad-challenge-list-bound:*

assumes *future: trace-fri-future-fresh s*

shows

wp-event verify-monad
(*trace-fri-challenge-list-set-hit s*
(*generic-fri-bad-challenge-lists trace-fri-algebraic-round-count bad*))
 s
 $\leq$
nnreal
(*card*
(*generic-fri-bad-challenge-lists trace-fri-algebraic-round-count*
bad)) /
nnreal (*CARD('f) ^ ceil-log clength*)

proof (rule *wp-verify-monad-trace-fri-challenge-list-set-bound[OF future]*)

show *generic-fri-bad-challenge-lists trace-fri-algebraic-round-count bad*
 $\subseteq$ *fri-challenge-space (ceil-log clength)*

using *generic-fri-bad-challenge-lists-subset*

by (*simp add: trace-fri-algebraic-round-count-eq*)

qed

lemma *wp-verify-monad-trace-generic-fri-bad-challenge-list-bound-from-card:*

assumes *future: trace-fri-future-fresh s*

and *card-bound:*

card (generic-fri-bad-challenge-lists trace-fri-algebraic-round-count
bad) ≤ bd

shows

wp-event verify-monad
(*trace-fri-challenge-list-set-hit s*
(*generic-fri-bad-challenge-lists trace-fri-algebraic-round-count bad*))
 s
 $\leq$ *nnreal* *bd* / *nnreal* (*CARD('f) ^ ceil-log clength*)

proof –

have *wp-event verify-monad*

(*trace-fri-challenge-list-set-hit s*
(*generic-fri-bad-challenge-lists trace-fri-algebraic-round-count bad*))

s

$\leq$

nnreal
(*card*
(*generic-fri-bad-challenge-lists trace-fri-algebraic-round-count*
bad)) /

$\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
by (rule *wp-verify-monad-trace-generic-fri-bad-challenge-list-bound*
 $[\text{OF future}]$)
also have $\dots \leq \text{nnreal } bd / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
by (rule *nnreal-nat-divide-right-mono* $[\text{OF card-bound}]$)
finally show *?thesis* .
qed

lemma *wp-verify-monad-trace-generic-fri-bad-challenge-list-bound-from-rounds*:
assumes *future: trace-fri-future-fresh s*
and *local-bound*:
 $\bigwedge i \text{ prefix. } i < \text{trace-fri-algebraic-round-count} \implies$
 $\text{length prefix} = i \implies \text{card } (\text{bad } i \text{ prefix}) \leq bd \ i$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-challenge-list-set-hit } s$
 $(\text{generic-fri-bad-challenge-lists trace-fri-algebraic-round-count bad}))$
 s
 $\leq$
 nnreal
 $(\sum i \in \{..<\text{trace-fri-algebraic-round-count}\}.$
 $\text{CARD}('f) \wedge i * bd \ i *$
 $\text{CARD}('f) \wedge (\text{trace-fri-algebraic-round-count} - \text{Suc } i)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
by (rule *wp-verify-monad-trace-generic-fri-bad-challenge-list-bound-from-card*
 $[\text{OF future}]$)
(rule *generic-fri-bad-challenge-lists-card-bound-from-round-bounds*
 $[\text{OF local-bound}]$)

lemma *wp-verify-monad-composition-generic-fri-bad-challenge-list-bound*:
assumes *future: composition-fri-future-fresh s*
and *bound*:
 $\bigwedge dg.$
 nnreal
 $(\text{card}$
 $(\text{generic-fri-bad-challenge-lists}$
 $(\text{composition-fri-algebraic-round-count } dg) (\text{bad } dg))) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-challenge-list-set-hit } s$
 $(\lambda dg. \text{generic-fri-bad-challenge-lists}$
 $(\text{composition-fri-algebraic-round-count } dg) (\text{bad } dg)))$
 $s \leq C$
proof (rule *wp-verify-monad-composition-fri-challenge-list-set-bound*
 $[\text{OF future}]$)
fix *dg*
show *generic-fri-bad-challenge-lists*
 $(\text{composition-fri-algebraic-round-count } dg) (\text{bad } dg)$

```

    ⊆ fri-challenge-space (ceil-log (to-nat dg + 1))
  using generic-fri-bad-challenge-lists-subset
  by (simp add: composition-fri-algebraic-round-count-eq)
show nnreal
  (card
    (generic-fri-bad-challenge-lists
      (composition-fri-algebraic-round-count dg) (bad dg))) /
  nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C
  by (rule bound)
qed

lemma wp-verify-monad-composition-generic-fri-bad-challenge-list-bound-from-card:
  assumes future: composition-fri-future-fresh s
  and card-bound:
    ∧ dg. card
      (generic-fri-bad-challenge-lists
        (composition-fri-algebraic-round-count dg) (bad dg)) ≤ bd dg
  and fraction-bound:
    ∧ dg. nnreal (bd dg) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C
  shows
    wp-event verify-monad
      (composition-fri-challenge-list-set-hit s
        (λ dg. generic-fri-bad-challenge-lists
          (composition-fri-algebraic-round-count dg) (bad dg)))
      s ≤ C
proof (rule wp-verify-monad-composition-generic-fri-bad-challenge-list-bound
  [OF future])
  fix dg
  have nnreal
    (card
      (generic-fri-bad-challenge-lists
        (composition-fri-algebraic-round-count dg) (bad dg))) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
  ≤ nnreal (bd dg) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
  by (rule nnreal-nat-divide-right-mono[OF card-bound])
  also have ... ≤ C
  by (rule fraction-bound)
  finally show nnreal
    (card
      (generic-fri-bad-challenge-lists
        (composition-fri-algebraic-round-count dg) (bad dg))) /
    nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C .
qed

lemma wp-verify-monad-composition-generic-fri-bad-challenge-list-bound-from-rounds:
  assumes future: composition-fri-future-fresh s
  and local-bound:

```

```

 $\bigwedge dg \ i \text{ prefix. } i < \text{composition-fri-algebraic-round-count } dg \implies$ 
 $\text{length prefix} = i \implies \text{card } (\text{bad } dg \ i \text{ prefix}) \leq \text{bd } dg \ i$ 
and fraction-bound:
 $\bigwedge dg. \text{nnreal}$ 
 $(\sum i \in \{..<\text{composition-fri-algebraic-round-count } dg\}.$ 
 $\text{CARD}(f) \wedge i * \text{bd } dg \ i *$ 
 $\text{CARD}(f) \wedge (\text{composition-fri-algebraic-round-count } dg - \text{Suc } i)) /$ 
 $\text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$ 
shows
 $\text{wp-event verify-monad}$ 
 $(\text{composition-fri-challenge-list-set-hit } s$ 
 $(\lambda dg. \text{generic-fri-bad-challenge-lists}$ 
 $(\text{composition-fri-algebraic-round-count } dg) (\text{bad } dg)))$ 
 $s \leq C$ 
proof (rule
 $\text{wp-verify-monad-composition-generic-fri-bad-challenge-list-bound-from-card}$ 
 $[\text{OF future - fraction-bound}]$ )
fix  $dg$ 
show  $\text{card}$ 
 $(\text{generic-fri-bad-challenge-lists}$ 
 $(\text{composition-fri-algebraic-round-count } dg) (\text{bad } dg))$ 
 $\leq$ 
 $(\sum i \in \{..<\text{composition-fri-algebraic-round-count } dg\}.$ 
 $\text{CARD}(f) \wedge i * \text{bd } dg \ i *$ 
 $\text{CARD}(f) \wedge (\text{composition-fri-algebraic-round-count } dg - \text{Suc } i))$ 
by (rule generic-fri-bad-challenge-lists-card-bound-from-round-bounds)
 $(\text{rule local-bound}[of - dg], \text{simp-all})$ 
qed

end

end

```

```

theory Soundness-FRI-Refined-Reachable
imports Soundness-FRI-Internal-Bounds
begin

```

Refined reachable FRI layer-chain targets.

Broad reachable partial-candidate events quantify full base tables compatible with sampled initial openings. Sampled base openings alone do not determine the committed FRI layer chain, so this theory introduces the narrower algebraic cover used by the next proof layer.

Its witnesses include the accepted FRI opening transcript and a canonical FRI layer chain. The cover is internal proof infrastructure; it does not change the protocol, verifier, Merkle interface, or public theorem assumptions.

```

context soundness
begin

```

definition *trace-fri-bad-with-refined-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-bad-with-refined-layer-chain } s \text{ bad out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-canonical-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers (bad trace-table))}$

definition *composition-fri-bad-with-refined-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-bad-with-refined-layer-chain } s \text{ bad out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{opening-composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table doms layers.}$
 $\text{composition-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{composition-query-idxs composition-openings trace-table}$
 $\text{composition-table} \wedge$
 $\text{generic-fri-canonical-layer-chain-evidence}$
 $(\text{composition-table-low-degree (to-nat fri-dg)}) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs composition-round-layers doms layers}$
 $(\text{bad fri-dg composition-table}))$

definition *trace-fri-reachable-missing-refined-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-reachable-missing-refined-layer-chain } s \text{ bad out} \longleftrightarrow$
 $\text{trace-fri-bad-with-reachable-partial-candidate } s \text{ out} \wedge$
 $\neg \text{trace-fri-bad-with-refined-layer-chain } s \text{ bad out}$

definition *composition-fri-reachable-missing-refined-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-reachable-missing-refined-layer-chain } s \text{ bad out}$
 $\longleftrightarrow$
 $\text{composition-fri-bad-with-reachable-partial-candidate } s \text{ out} \wedge$
 $\neg \text{composition-fri-bad-with-refined-layer-chain } s \text{ bad out}$

definition *trace-fri-bad-with-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-bad-with-sampled-layer-chain } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers})$

definition *composition-fri-bad-with-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-bad-with-sampled-layer-chain } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{opening-composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table doms layers.}$
 $\text{composition-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{composition-fri-roots final trace-query-idxs trace-openings})$

composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers)

definition *trace-fri-reachable-missing-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-reachable-missing-sampled-layer-chain s out $\longleftrightarrow$
trace-fri-bad-with-reachable-partial-candidate s out $\wedge$
 $\neg \text{trace-fri-bad-with-sampled-layer-chain } s \text{ out}$

definition *composition-fri-reachable-missing-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-reachable-missing-sampled-layer-chain s out $\longleftrightarrow$
composition-fri-bad-with-reachable-partial-candidate s out $\wedge$
 $\neg \text{composition-fri-bad-with-sampled-layer-chain } s \text{ out}$

definition *trace-fri-sampled-layer-chain-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow \text{bool}$

where

trace-fri-sampled-layer-chain-cover s bad $\longleftrightarrow$
 $(\forall \text{ out trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
trace-fri-partial-candidate-opening-evidence s out trace-roots
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \longrightarrow$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers} \longrightarrow$
 $\text{trace-bs} \in \text{trace-fri-multiround-bad-sets bad trace-table})$

definition *composition-fri-sampled-layer-chain-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow \text{bool}$

where

composition-fri-sampled-layer-chain-cover s bad $\longleftrightarrow$
 $(\forall \text{ out trace-roots trace-bs trace-final fri-dg}$
 $\text{opening-composition-roots composition-bs composition-final}$

fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\longrightarrow$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
 $\longrightarrow$
composition-bs $\in$
composition-fri-multi-round-bad-sets bad fri-dg composition-table)

definition *trace-fri-bad-with-full-cover-layer-chain*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \text{ list} \Rightarrow \text{nat} \Rightarrow f \text{ list} \Rightarrow f \text{ set}) \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-bad-with-full-cover-layer-chain s bad out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table doms layers.
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
generic-fri-sampled-layer-chain-full-cover trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers (bad trace-table))

definition *composition-fri-bad-with-full-cover-layer-chain*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \Rightarrow f \text{ list} \Rightarrow \text{nat} \Rightarrow f \text{ list} \Rightarrow f \text{ set}) \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-bad-with-full-cover-layer-chain s bad out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs

composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-full-cover
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
(bad fri-dg composition-table))

definition *trace-fri-sampled-missing-full-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-sampled-missing-full-cover s bad out $\longleftrightarrow$
trace-fri-bad-with-sampled-layer-chain s out $\wedge$
 $\neg \text{trace-fri-bad-with-full-cover-layer-chain s bad out}$

definition *composition-fri-sampled-missing-full-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-missing-full-cover s bad out $\longleftrightarrow$
composition-fri-bad-with-sampled-layer-chain s out $\wedge$
 $\neg \text{composition-fri-bad-with-full-cover-layer-chain s bad out}$

lemma *trace-fri-sampled-layer-chain-cover-from-full-cover:*

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

fixes $\text{bad} :: 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}$

assumes *full-cover:*

$\wedge \text{out trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$

trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\implies$

generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers $\implies$

generic-fri-sampled-layer-chain-full-cover trace-table-low-degree

```

    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers (bad trace-table)
shows trace-fri-sampled-layer-chain-cover s bad
proof (unfold trace-fri-sampled-layer-chain-cover-def, intro allI impI)
fix out trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table doms layers
assume evidence:
    trace-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
assume chain:
    generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
have full:
    generic-fri-sampled-layer-chain-full-cover trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers (bad trace-table)
by (rule full-cover[OF evidence chain])
have not-low:  $\neg$  trace-table-low-degree trace-table
using trace-fri-partial-candidate-opening-evidenceD(2)[OF evidence]
    trace-fri-partial-candidate-evidenceD(3)
by blast
have bad-candidate:
    generic-fri-bad-candidate trace-table-low-degree trace-table
    (clength - 1) trace-roots trace-bs trace-final fri-query-idxs
    trace-round-layers (bad trace-table)
by (rule generic-fri-bad-candidate-from-sampled-layer-chain-full-cover
    [OF full not-low])
have bad-generic:
    trace-bs  $\in$  generic-fri-bad-challenge-lists (length trace-bs)
    (bad trace-table)
by (rule generic-fri-bad-candidateD(3)[OF bad-candidate])
have partial:
    generic-fri-partial-evidence trace-table-low-degree trace-table
    (clength - 1) trace-roots trace-bs trace-final fri-query-idxs
    trace-round-layers
by (rule generic-fri-bad-candidateD(1)[OF bad-candidate])
have len-trace:
    length trace-bs = trace-fri-algebraic-round-count
using generic-fri-partial-evidence-shapes(1,2)[OF partial]
by (simp add: trace-fri-algebraic-round-count-def)
show trace-bs  $\in$  trace-fri-multiround-bad-sets bad trace-table
using bad-generic len-trace
by (simp add: trace-fri-multiround-bad-sets-as-generic)

```

qed

lemma *composition-fri-sampled-layer-chain-cover-from-full-cover:*

fixes *s* :: ('f, 'a) *protocol-channel-scheme*

fixes *bad* :: 'f $\Rightarrow$ 'f list $\Rightarrow$ nat $\Rightarrow$ 'f list $\Rightarrow$ 'f set

assumes *full-cover:*

$\wedge$ *out trace-roots trace-bs trace-final fri-dg*

opening-composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers fr

f-fri-roots f-final as dg composition-fri-roots final

trace-query-idxs trace-openings composition-query-idxs

composition-openings trace-table composition-table doms layers.

composition-fri-partial-candidate-opening-evidence s out trace-roots

trace-bs trace-final fri-dg opening-composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr f-fri-roots f-final as dg

composition-fri-roots final trace-query-idxs trace-openings

composition-query-idxs composition-openings trace-table

composition-table $\Rightarrow$

generic-fri-sampled-layer-chain-evidence

(composition-table-low-degree (to-nat fri-dg)) composition-table

(to-nat fri-dg) opening-composition-roots composition-bs

composition-final fri-query-idxs composition-round-layers doms

layers $\Rightarrow$

generic-fri-sampled-layer-chain-full-cover

(composition-table-low-degree (to-nat fri-dg)) composition-table

(to-nat fri-dg) opening-composition-roots composition-bs

composition-final fri-query-idxs composition-round-layers doms

layers (bad fri-dg composition-table)

shows *composition-fri-sampled-layer-chain-cover s bad*

proof (*unfold composition-fri-sampled-layer-chain-cover-def,*

intro allI impI)

fix *out trace-roots trace-bs trace-final fri-dg*

opening-composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers fr

f-fri-roots f-final as dg composition-fri-roots final

trace-query-idxs trace-openings composition-query-idxs

composition-openings trace-table composition-table doms layers

assume *evidence:*

composition-fri-partial-candidate-opening-evidence s out trace-roots

trace-bs trace-final fri-dg opening-composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr f-fri-roots f-final as dg

composition-fri-roots final trace-query-idxs trace-openings

composition-query-idxs composition-openings trace-table

composition-table

assume *chain:*

generic-fri-sampled-layer-chain-evidence

(composition-table-low-degree (to-nat fri-dg)) composition-table

```

    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms
    layers
  have full:
    generic-fri-sampled-layer-chain-full-cover
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms
    layers (bad fri-dg composition-table)
  by (rule full-cover[OF evidence chain])
  have not-low-on:
    ¬ fri-table-low-degree-on (to-nat fri-dg) eval-domain
    composition-table
  using full unfolding generic-fri-sampled-layer-chain-full-cover-def
  by blast
  have not-low:
    ¬ composition-table-low-degree (to-nat fri-dg) composition-table
  using not-low-on
    composition-table-not-low-degree-iff-not-fri-table-low-degree-on-eval-domain
  by simp
  have bad-candidate:
    generic-fri-bad-candidate
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
    (bad fri-dg composition-table)
  by (rule generic-fri-bad-candidate-from-sampled-layer-chain-full-cover
    [OF full not-low])
  have bad-generic:
    composition-bs ∈ generic-fri-bad-challenge-lists
    (length composition-bs) (bad fri-dg composition-table)
  by (rule generic-fri-bad-candidateD(3)[OF bad-candidate])
  have partial:
    generic-fri-partial-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
  by (rule generic-fri-bad-candidateD(1)[OF bad-candidate])
  have len-composition:
    length composition-bs = composition-fri-algebraic-round-count fri-dg
  using generic-fri-partial-evidence-shapes(1,2)[OF partial]
  by (simp add: composition-fri-algebraic-round-count-def)
  show composition-bs ∈
    composition-fri-multiround-bad-sets bad fri-dg composition-table
  using bad-generic len-composition
  by (simp add: composition-fri-multiround-bad-sets-as-generic)
qed

```

lemma *trace-fri-bad-with-full-cover-layer-chain-imp-list-hit:*

assumes *full*:
trace-fri-bad-with-full-cover-layer-chain s bad out
and *envelope*:
 $\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
shows *trace-fri-challenge-list-set-hit s B out*
proof –
from *full* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table doms layers
where *evidence*:
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *full-cover*:
generic-fri-sampled-layer-chain-full-cover trace-table-low-degree
trace-table (clength – 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers (bad trace-table)
unfolding *trace-fri-bad-with-full-cover-layer-chain-def* **by** *blast*
have *bad-generic*:
trace-bs $\in$ generic-fri-bad-challenge-lists (length trace-bs)
(bad trace-table)
proof –
have *not-low*: $\neg \text{trace-table-low-degree trace-table}$
using *trace-fri-partial-candidate-opening-evidenceD(2)[OF evidence]*
trace-fri-partial-candidate-evidenceD(3)
by *blast*
have *bad-candidate*:
generic-fri-bad-candidate trace-table-low-degree trace-table
(clength – 1) trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers (bad trace-table)
by (rule *generic-fri-bad-candidate-from-sampled-layer-chain-full-cover*
 $[\text{OF full-cover not-low}]$)
show *?thesis*
by (rule *generic-fri-bad-candidateD(3)[OF bad-candidate]*)
qed
have *partial*:
generic-fri-partial-evidence trace-table-low-degree trace-table
(clength – 1) trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers
proof –
have *not-low*: $\neg \text{trace-table-low-degree trace-table}$
using *trace-fri-partial-candidate-opening-evidenceD(2)[OF evidence]*
trace-fri-partial-candidate-evidenceD(3)
by *blast*
have *bad-candidate*:
generic-fri-bad-candidate trace-table-low-degree trace-table


```

    (clength - 1) trace-roots trace-bs trace-final fri-query-idxs
    trace-round-layers (bad trace-table)
  by (rule generic-fri-bad-candidate-from-sampled-layer-chain-full-cover
    [OF full-cover not-low])
show ?thesis
  by (rule generic-fri-bad-candidateD(1)[OF bad-candidate])
qed
have len-trace:
  length trace-bs = trace-fri-algebraic-round-count
  using generic-fri-partial-evidence-shapes(1,2)[OF partial]
  by (simp add: trace-fri-algebraic-round-count-def)
have bad-hit: trace-bs ∈ trace-fri-multiround-bad-sets bad trace-table
  using bad-generic len-trace
  by (simp add: trace-fri-multiround-bad-sets-as-generic)
have challenges:
  accepted-fri-challenges s out trace-bs dg composition-bs
  using trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence]
  accepted-fri-opening-transcript-challenges by blast
have trace-bs ∈ B
  using bad-hit envelope[of trace-table] by blast
then show ?thesis
  unfolding trace-fri-challenge-list-set-hit-def
  using challenges by blast
qed

```

lemma *composition-fri-bad-with-full-cover-layer-chain-imp-list-hit:*
assumes *full:*
composition-fri-bad-with-full-cover-layer-chain s bad out
and *envelope:*
 $\bigwedge dg$ *composition-table.*
composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B dg$
shows *composition-fri-challenge-list-set-hit s B out*
proof –
from *full* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers
where *evidence:*
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *full-cover:*

```

generic-fri-sampled-layer-chain-full-cover
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) opening-composition-roots composition-bs
  composition-final fri-query-idxs composition-round-layers doms layers
  (bad fri-dg composition-table)
unfolding composition-fri-bad-with-full-cover-layer-chain-def
by auto
have bad-generic:
  composition-bs ∈ generic-fri-bad-challenge-lists
  (length composition-bs) (bad fri-dg composition-table)
proof –
have not-low-on:
  ¬ fri-table-low-degree-on (to-nat fri-dg) eval-domain
  composition-table
using full-cover
unfolding generic-fri-sampled-layer-chain-full-cover-def by blast
have not-low:
  ¬ composition-table-low-degree (to-nat fri-dg) composition-table
using not-low-on
  composition-table-not-low-degree-iff-not-fri-table-low-degree-on-eval-domain
by simp
have bad-candidate:
  generic-fri-bad-candidate
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) opening-composition-roots composition-bs
  composition-final fri-query-idxs composition-round-layers
  (bad fri-dg composition-table)
by (rule generic-fri-bad-candidate-from-sampled-layer-chain-full-cover
  [OF full-cover not-low])
show ?thesis
by (rule generic-fri-bad-candidateD(3)[OF bad-candidate])
qed
have partial:
  generic-fri-partial-evidence
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) opening-composition-roots composition-bs
  composition-final fri-query-idxs composition-round-layers
proof –
have not-low-on:
  ¬ fri-table-low-degree-on (to-nat fri-dg) eval-domain
  composition-table
using full-cover
unfolding generic-fri-sampled-layer-chain-full-cover-def by blast
have not-low:
  ¬ composition-table-low-degree (to-nat fri-dg) composition-table
using not-low-on
  composition-table-not-low-degree-iff-not-fri-table-low-degree-on-eval-domain
by simp
have bad-candidate:

```

```

generic-fri-bad-candidate
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) opening-composition-roots composition-bs
  composition-final fri-query-idxs composition-round-layers
  (bad fri-dg composition-table)
  by (rule generic-fri-bad-candidate-from-sampled-layer-chain-full-cover
      [OF full-cover not-low])
show ?thesis
  by (rule generic-fri-bad-candidateD(1)[OF bad-candidate])
qed
have len-composition:
  length composition-bs = composition-fri-algebraic-round-count fri-dg
  using generic-fri-partial-evidence-shapes(1,2)[OF partial]
  by (simp add: composition-fri-algebraic-round-count-def)
have bad-hit:
  composition-bs ∈
    composition-fri-multiround-bad-sets bad fri-dg composition-table
  using bad-generic len-composition
  by (simp add: composition-fri-multiround-bad-sets-as-generic)
have challenges:
  accepted-fri-challenges s out trace-bs fri-dg composition-bs
  using composition-fri-partial-candidate-opening-evidenceD(1)[OF evidence]
  accepted-fri-opening-transcript-challenges by blast
have composition-bs ∈ B fri-dg
  using bad-hit envelope[of fri-dg composition-table] by blast
then show ?thesis
  unfolding composition-fri-challenge-list-set-hit-def
  using challenges by blast
qed

lemma trace-fri-sampled-imp-full-cover-or-missing:
  assumes trace-fri-bad-with-sampled-layer-chain s out
  shows
    trace-fri-bad-with-full-cover-layer-chain s bad out ∨
    trace-fri-sampled-missing-full-cover s bad out
  using assms unfolding trace-fri-sampled-missing-full-cover-def by blast

lemma composition-fri-sampled-imp-full-cover-or-missing:
  assumes composition-fri-bad-with-sampled-layer-chain s out
  shows
    composition-fri-bad-with-full-cover-layer-chain s bad out ∨
    composition-fri-sampled-missing-full-cover s bad out
  using assms unfolding composition-fri-sampled-missing-full-cover-def
  by blast

lemma wp-trace-fri-bad-with-full-cover-layer-chain-bound-from-envelope:
  assumes future: trace-fri-future-fresh s
  and envelope:
     $\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$ 

```

and *subset*: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and *bound*:
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength}) \leq C$
shows
 $\text{wp-event verify-monad } (\text{trace-fri-bad-with-full-cover-layer-chain } s \text{ bad}) s \leq C$
proof –
have *wp-event verify-monad*
 $(\text{trace-fri-bad-with-full-cover-layer-chain } s \text{ bad}) s \leq$
 $\text{wp-event verify-monad } (\text{trace-fri-challenge-list-set-hit } s B) s$
by (*rule wp-event-mono*)
 $(\text{rule trace-fri-bad-with-full-cover-layer-chain-imp-list-hit } [OF - \text{envelope}])$
also have $\dots \leq \text{nnreal } (\text{card } B) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
by (*rule wp-verify-monad-trace-fri-challenge-list-set-bound*
 $[OF \text{ future subset}])$
also have $\dots \leq C$
by (*rule bound*)
finally show *?thesis* .
qed

lemma *wp-composition-fri-bad-with-full-cover-layer-chain-bound-from-envelope*:
assumes *future: composition-fri-future-fresh s*
and *envelope*:
 $\bigwedge dg \text{ composition-table.}$
 $\text{composition-fri-multi-round-bad-sets bad dg composition-table}$
 $\subseteq B dg$
and *subset*:
 $\bigwedge dg. B dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and *bound*:
 $\bigwedge dg. \text{nnreal } (\text{card } (B dg)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$
shows
 $\text{wp-event verify-monad } (\text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad}) s \leq C$
proof –
have *wp-event verify-monad*
 $(\text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad}) s \leq$
 $\text{wp-event verify-monad } (\text{composition-fri-challenge-list-set-hit } s B) s$
by (*rule wp-event-mono*)
 $(\text{rule composition-fri-bad-with-full-cover-layer-chain-imp-list-hit } [OF - \text{envelope}])$
also have $\dots \leq C$
by (*rule wp-verify-monad-composition-fri-challenge-list-set-bound*
 $[OF \text{ future subset bound}])$
finally show *?thesis* .
qed

lemma *wp-trace-fri-sampled-bound-from-full-cover-and-missing:*

assumes *full-bound:*

wp-event verify-monad

$(\text{trace-fri-bad-with-full-cover-layer-chain } s \text{ bad}) \ s \leq R$

and *missing-bound:*

wp-event verify-monad

$(\text{trace-fri-sampled-missing-full-cover } s \text{ bad}) \ s \leq M$

shows

wp-event verify-monad

$(\text{trace-fri-bad-with-sampled-layer-chain } s) \ s \leq R + M$

proof –

have *wp-event verify-monad*

$(\text{trace-fri-bad-with-sampled-layer-chain } s) \ s \leq$

wp-event verify-monad

$(\lambda \text{out. trace-fri-bad-with-full-cover-layer-chain } s \text{ bad out} \vee$
 $\text{trace-fri-sampled-missing-full-cover } s \text{ bad out}) \ s$

by (rule *wp-event-mono*)

(rule *trace-fri-sampled-imp-full-cover-or-missing*)

also have $\dots \leq$

wp-event verify-monad

$(\text{trace-fri-bad-with-full-cover-layer-chain } s \text{ bad}) \ s +$

wp-event verify-monad

$(\text{trace-fri-sampled-missing-full-cover } s \text{ bad}) \ s$

by (rule *wp-event-union-bound*)

also have $\dots \leq R + M$

by (intro *add-mono full-bound missing-bound*)

finally show *?thesis* .

qed

lemma *wp-composition-fri-sampled-bound-from-full-cover-and-missing:*

assumes *full-bound:*

wp-event verify-monad

$(\text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad}) \ s \leq R$

and *missing-bound:*

wp-event verify-monad

$(\text{composition-fri-sampled-missing-full-cover } s \text{ bad}) \ s \leq M$

shows

wp-event verify-monad

$(\text{composition-fri-bad-with-sampled-layer-chain } s) \ s \leq R + M$

proof –

have *wp-event verify-monad*

$(\text{composition-fri-bad-with-sampled-layer-chain } s) \ s \leq$

wp-event verify-monad

$(\lambda \text{out. composition-fri-bad-with-full-cover-layer-chain } s \text{ bad out} \vee$
 $\text{composition-fri-sampled-missing-full-cover } s \text{ bad out}) \ s$

by (rule *wp-event-mono*)

(rule *composition-fri-sampled-imp-full-cover-or-missing*)

also have $\dots \leq$

wp-event verify-monad

(composition-fri-bad-with-full-cover-layer-chain s bad) s +
 wp-event verify-monad
 (composition-fri-sampled-missing-full-cover s bad) s
 by (rule wp-event-union-bound)
 also have ... $\leq R + M$
 by (intro add-mono full-bound missing-bound)
 finally show ?thesis .
 qed

definition trace-fri-canonical-layer-chain-cover

:: ('f, 'a) protocol-channel-scheme $\Rightarrow$
 ('f list $\Rightarrow$ nat $\Rightarrow$ 'f list $\Rightarrow$ 'f set) $\Rightarrow$ bool

where

trace-fri-canonical-layer-chain-cover s bad $\longleftrightarrow$
 ($\forall$ out trace-roots trace-bs trace-final dg composition-roots
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table doms layers.
 trace-fri-partial-candidate-opening-evidence s out trace-roots
 trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table $\longrightarrow$
 generic-fri-canonical-layer-chain-evidence trace-table-low-degree
 trace-table (clength - 1) trace-roots trace-bs trace-final
 fri-query-idxs trace-round-layers doms layers (bad trace-table)
 $\longrightarrow$
 trace-bs $\in$ trace-fri-multiround-bad-sets bad trace-table)

definition composition-fri-canonical-layer-chain-cover

:: ('f, 'a) protocol-channel-scheme $\Rightarrow$
 ('f $\Rightarrow$ 'f list $\Rightarrow$ nat $\Rightarrow$ 'f list $\Rightarrow$ 'f set) $\Rightarrow$ bool

where

composition-fri-canonical-layer-chain-cover s bad $\longleftrightarrow$
 ($\forall$ out trace-roots trace-bs trace-final fri-dg
 opening-composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers fr
 f-fri-roots f-final as dg composition-fri-roots final
 trace-query-idxs trace-openings composition-query-idxs
 composition-openings trace-table composition-table doms layers.
 composition-fri-partial-candidate-opening-evidence s out trace-roots
 trace-bs trace-final fri-dg opening-composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr f-fri-roots f-final as dg
 composition-fri-roots final trace-query-idxs trace-openings
 composition-query-idxs composition-openings trace-table
 composition-table $\longrightarrow$
 generic-fri-canonical-layer-chain-evidence
 (composition-table-low-degree (to-nat fri-dg)) composition-table)

(to-nat fri-dg) opening-composition-roots composition-bs
 composition-final fri-query-idxs composition-round-layers doms layers
 (bad fri-dg composition-table) $\longrightarrow$
 composition-bs $\in$
 composition-fri-multiround-bad-sets bad fri-dg composition-table)

lemma trace-fri-canonical-layer-chain-cover-generic:

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

fixes $bad :: 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}$

shows trace-fri-canonical-layer-chain-cover $s \text{ bad}$

proof (unfold trace-fri-canonical-layer-chain-cover-def, intro allI impI)

fix out trace-roots trace-bs trace-final dg composition-roots
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table doms layers

assume evidence:

trace-fri-partial-candidate-opening-evidence $s \text{ out trace-roots}$
 trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table

assume chain:

generic-fri-canonical-layer-chain-evidence trace-table-low-degree
 trace-table (clength - 1) trace-roots trace-bs trace-final
 fri-query-idxs trace-round-layers doms layers (bad trace-table)

have bad-generic:

trace-bs $\in$ generic-fri-bad-challenge-lists (length trace-bs)
 (bad trace-table)

by (rule generic-fri-canonical-layer-chain-bad-challenge-list[OF chain])

have partial:

generic-fri-partial-evidence trace-table-low-degree trace-table
 (clength - 1) trace-roots trace-bs trace-final fri-query-idxs
 trace-round-layers

by (rule generic-fri-canonical-layer-chain-evidenceD(1)[OF chain])

have len-trace:

length trace-bs = trace-fri-algebraic-round-count
using generic-fri-partial-evidence-shapes(1,2)[OF partial]
by (simp add: trace-fri-algebraic-round-count-def)

show trace-bs $\in$ trace-fri-multiround-bad-sets bad trace-table

using bad-generic len-trace

by (simp add: trace-fri-multiround-bad-sets-as-generic)

qed

lemma composition-fri-canonical-layer-chain-cover-generic:

fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$

fixes $bad :: 'f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}$

shows composition-fri-canonical-layer-chain-cover $s \text{ bad}$

proof (unfold composition-fri-canonical-layer-chain-cover-def,
 intro allI impI)

fix *out trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers
assume *evidence:*
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
assume *chain:*
generic-fri-canonical-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
(bad fri-dg composition-table)
have *bad-generic:*
composition-bs ∈ generic-fri-bad-challenge-lists
(length composition-bs) (bad fri-dg composition-table)
by *(rule generic-fri-canonical-layer-chain-bad-challenge-list[OF chain])*
have *partial:*
generic-fri-partial-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers
by *(rule generic-fri-canonical-layer-chain-evidenceD(1)[OF chain])*
have *len-composition:*
length composition-bs = composition-fri-algebraic-round-count fri-dg
using *generic-fri-partial-evidence-shapes(1,2)[OF partial]*
by *(simp add: composition-fri-algebraic-round-count-def)*
show *composition-bs ∈*
composition-fri-multiround-bad-sets bad fri-dg composition-table
using *bad-generic len-composition*
by *(simp add: composition-fri-multiround-bad-sets-as-generic)*
qed

lemma *trace-fri-bad-with-refined-layer-chain-imp-list-hit:*
assumes *refined:*
trace-fri-bad-with-refined-layer-chain s bad out
and *envelope:*
 $\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
shows *trace-fri-challenge-list-set-hit s B out*
proof —
from *refined* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr candidate-query-idxs trace-openings
trace-table doms layers
where *evidence:*
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *chain:*
generic-fri-canonical-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers (bad trace-table)
unfolding *trace-fri-bad-with-refined-layer-chain-def* **by** *blast*
have *cover: trace-fri-canonical-layer-chain-cover s bad*
by *(rule trace-fri-canonical-layer-chain-cover-generic)*
have *bad-hit: trace-bs ∈ trace-fri-multiround-bad-sets bad trace-table*
using *cover evidence chain*
unfolding *trace-fri-canonical-layer-chain-cover-def* **by** *blast*
have *challenges:*
accepted-fri-challenges s out trace-bs dg composition-bs
using *trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence]*
accepted-fri-opening-transcript-challenges **by** *blast*
have *trace-bs ∈ B*
using *bad-hit envelope[of trace-table]* **by** *blast*
then show *?thesis*
unfolding *trace-fri-challenge-list-set-hit-def*
using *challenges* **by** *blast*
qed

lemma *composition-fri-bad-with-refined-layer-chain-imp-list-hit:*
assumes *refined:*
composition-fri-bad-with-refined-layer-chain s bad out
and *envelope:*
 $\bigwedge dg$ *composition-table.*
composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B$ *dg*
shows *composition-fri-challenge-list-set-hit s B out*
proof –
from *refined* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers
where *evidence:*
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg

```

    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table
and chain:
    generic-fri-canonical-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers
    (bad fri-dg composition-table)
unfolding composition-fri-bad-with-refined-layer-chain-def
by auto
have bad-hit:
    composition-bs ∈
    composition-fri-multiround-bad-sets bad fri-dg composition-table
proof –
have bad-generic:
    composition-bs ∈ generic-fri-bad-challenge-lists
    (length composition-bs) (bad fri-dg composition-table)
by (rule generic-fri-canonical-layer-chain-bad-challenge-list
    [OF chain])
have partial:
    generic-fri-partial-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
by (rule generic-fri-canonical-layer-chain-evidenceD(1)[OF chain])
have len-composition:
    length composition-bs = composition-fri-algebraic-round-count fri-dg
using generic-fri-partial-evidence-shapes(1,2)[OF partial]
by (simp add: composition-fri-algebraic-round-count-def)
show ?thesis
using bad-generic len-composition
by (simp add: composition-fri-multiround-bad-sets-as-generic)
qed
have challenges:
    accepted-fri-challenges s out trace-bs fri-dg composition-bs
using composition-fri-partial-candidate-opening-evidenceD(1)[OF evidence]
    accepted-fri-opening-transcript-challenges by blast
have composition-bs ∈ B fri-dg
using bad-hit envelope[of fri-dg composition-table] by blast
then show ?thesis
unfolding composition-fri-challenge-list-set-hit-def
using challenges by blast
qed

lemma wp-trace-fri-bad-with-refined-layer-chain-bound-from-envelope:
assumes future: trace-fri-future-fresh s
and envelope:
     $\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$ 

```

and *subset*: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and *bound*:
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength}) \leq C$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-refined-layer-chain } s \text{ bad}) s \leq C$
proof –
have $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-refined-layer-chain } s \text{ bad}) s \leq$
 $\text{wp-event verify-monad } (\text{trace-fri-challenge-list-set-hit } s B) s$
by $(\text{rule wp-event-mono})$
 $(\text{rule trace-fri-bad-with-refined-layer-chain-imp-list-hit}$
 $[\text{OF - envelope}])$
also have $\dots \leq \text{nnreal } (\text{card } B) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
by $(\text{rule wp-verify-monad-trace-fri-challenge-list-set-bound}$
 $[\text{OF future subset}])$
also have $\dots \leq C$
by (rule bound)
finally show *?thesis* .
qed

lemma *wp-composition-fri-bad-with-refined-layer-chain-bound-from-envelope*:
assumes *future: composition-fri-future-fresh s*
and *envelope*:
 $\bigwedge dg \text{ composition-table.}$
 $\text{composition-fri-multi-round-bad-sets bad dg composition-table}$
 $\subseteq B dg$
and *subset*:
 $\bigwedge dg. B dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and *bound*:
 $\bigwedge dg. \text{nnreal } (\text{card } (B dg)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-refined-layer-chain } s \text{ bad}) s \leq C$
proof –
have $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-refined-layer-chain } s \text{ bad}) s \leq$
 $\text{wp-event verify-monad } (\text{composition-fri-challenge-list-set-hit } s B) s$
by $(\text{rule wp-event-mono})$
 $(\text{rule composition-fri-bad-with-refined-layer-chain-imp-list-hit}$
 $[\text{OF - envelope}])$
also have $\dots \leq C$
by $(\text{rule wp-verify-monad-composition-fri-challenge-list-set-bound}$
 $[\text{OF future subset bound}])$
finally show *?thesis* .
qed

lemma *trace-fri-bad-with-sampled-layer-chain-imp-list-hit*:
assumes *sampled*:
 trace-fri-bad-with-sampled-layer-chain s out
and *cover*: *trace-fri-sampled-layer-chain-cover s bad*
and *envelope*:
 $\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
shows *trace-fri-challenge-list-set-hit s B out*
proof –
from *sampled* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table doms layers
where *evidence*:
 trace-fri-partial-candidate-opening-evidence s out trace-roots
 trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table
and *chain*:
 generic-fri-sampled-layer-chain-evidence trace-table-low-degree
 trace-table (clength – 1) trace-roots trace-bs trace-final
 fri-query-idxs trace-round-layers doms layers
unfolding *trace-fri-bad-with-sampled-layer-chain-def* **by** *blast*
have *bad-hit*: *trace-bs* $\in$ *trace-fri-multiround-bad-sets bad trace-table*
using *cover evidence chain*
unfolding *trace-fri-sampled-layer-chain-cover-def* **by** *blast*
have *challenges*:
 accepted-fri-challenges s out trace-bs dg composition-bs
using *trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence]*
 accepted-fri-opening-transcript-challenges **by** *blast*
have *trace-bs* $\in$ *B*
using *bad-hit envelope[of trace-table]* **by** *blast*
then show *?thesis*
 unfolding *trace-fri-challenge-list-set-hit-def*
 using *challenges* **by** *blast*
qed

lemma *composition-fri-bad-with-sampled-layer-chain-imp-list-hit*:
assumes *sampled*:
 composition-fri-bad-with-sampled-layer-chain s out
and *cover*: *composition-fri-sampled-layer-chain-cover s bad*
and *envelope*:
 $\bigwedge dg \text{ composition-table. composition-fri-multiround-bad-sets bad dg composition-table} \subseteq B dg$
shows *composition-fri-challenge-list-set-hit s B out*
proof –
from *sampled* **obtain** *trace-roots trace-bs trace-final fri-dg*
 opening-composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers
where *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *chain*:
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
unfolding *composition-fri-bad-with-sampled-layer-chain-def* **by** *auto*
have *bad-hit*:
composition-bs $\in$
composition-fri-multiround-bad-sets bad fri-dg composition-table
using *cover evidence chain*
unfolding *composition-fri-sampled-layer-chain-cover-def* **by** *metis*
have *challenges*:
accepted-fri-challenges s out trace-bs fri-dg composition-bs
using *composition-fri-partial-candidate-opening-evidenceD(1)[OF evidence]*
accepted-fri-opening-transcript-challenges **by** *blast*
have *composition-bs* $\in B$ *fri-dg*
using *bad-hit envelope[of fri-dg composition-table]* **by** *blast*
then show *?thesis*
unfolding *composition-fri-challenge-list-set-hit-def*
using *challenges* **by** *blast*
qed

lemma *wp-trace-fri-bad-with-sampled-layer-chain-bound-from-envelope*:
assumes *future: trace-fri-future-fresh s*
and *cover: trace-fri-sampled-layer-chain-cover s bad*
and *envelope*:
 $\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
and *subset*: $B \subseteq \text{fri-challenge-space (ceil-log clength)}$
and *bound*:
 $\text{nnreal (card } B) / \text{nnreal (CARD('f) } \wedge \text{ ceil-log clength)} \leq C$
shows
wp-event verify-monad
 $(\text{trace-fri-bad-with-sampled-layer-chain } s) s \leq C$
proof –
have *wp-event verify-monad*
 $(\text{trace-fri-bad-with-sampled-layer-chain } s) s \leq$
wp-event verify-monad (trace-fri-challenge-list-set-hit s B) s

by (rule wp-event-mono)
 (rule trace-fri-bad-with-sampled-layer-chain-imp-list-hit
 [OF - cover envelope])
 also have ... $\leq$ nreal (card B) /
 nreal (CARD('f) $\wedge$ ceil-log clength)
 by (rule wp-verify-monad-trace-fri-challenge-list-set-bound
 [OF future subset])
 also have ... $\leq$ C
 by (rule bound)
 finally show ?thesis .
 qed

lemma wp-composition-fri-bad-with-sampled-layer-chain-bound-from-envelope:

assumes future: composition-fri-future-fresh s
 and cover: composition-fri-sampled-layer-chain-cover s bad
 and envelope:
 $\wedge$ dg composition-table.
 composition-fri-multi-round-bad-sets bad dg composition-table
 $\subseteq B$ dg
 and subset:
 $\wedge$ dg. B dg $\subseteq$ fri-challenge-space (ceil-log (to-nat dg + 1))
 and bound:
 $\wedge$ dg. nreal (card (B dg)) /
 nreal (CARD('f) $\wedge$ ceil-log (to-nat dg + 1)) $\leq$ C

shows

wp-event verify-monad
 (composition-fri-bad-with-sampled-layer-chain s) s $\leq$ C

proof –

have wp-event verify-monad
 (composition-fri-bad-with-sampled-layer-chain s) s $\leq$
 wp-event verify-monad (composition-fri-challenge-list-set-hit s B) s
 by (rule wp-event-mono)
 (rule composition-fri-bad-with-sampled-layer-chain-imp-list-hit
 [OF - cover envelope])
 also have ... $\leq$ C
 by (rule wp-verify-monad-composition-fri-challenge-list-set-bound
 [OF future subset bound])
 finally show ?thesis .

qed

lemma trace-fri-reachable-imp-refined-or-missing-layer-chain:

assumes trace-fri-bad-with-reachable-partial-candidate s out

shows

trace-fri-bad-with-refined-layer-chain s bad out $\vee$
 trace-fri-reachable-missing-refined-layer-chain s bad out

using assms

unfolding trace-fri-reachable-missing-refined-layer-chain-def by blast

lemma composition-fri-reachable-imp-refined-or-missing-layer-chain:

assumes *composition-fri-bad-with-reachable-partial-candidate s out*
shows
composition-fri-bad-with-refined-layer-chain s bad out $\vee$
composition-fri-reachable-missing-refined-layer-chain s bad out
using *assms*
unfolding *composition-fri-reachable-missing-refined-layer-chain-def* **by** *blast*

lemma *trace-fri-reachable-imp-sampled-or-missing-layer-chain:*
assumes *trace-fri-bad-with-reachable-partial-candidate s out*
shows
trace-fri-bad-with-sampled-layer-chain s out $\vee$
trace-fri-reachable-missing-sampled-layer-chain s out
using *assms*
unfolding *trace-fri-reachable-missing-sampled-layer-chain-def* **by** *blast*

lemma *composition-fri-reachable-imp-sampled-or-missing-layer-chain:*
assumes *composition-fri-bad-with-reachable-partial-candidate s out*
shows
composition-fri-bad-with-sampled-layer-chain s out $\vee$
composition-fri-reachable-missing-sampled-layer-chain s out
using *assms*
unfolding *composition-fri-reachable-missing-sampled-layer-chain-def*
by *blast*

lemma *verify-monad-trace-fri-missing-extracts-opening-evidence:*
assumes *outcome:*
Some (result, final-state) $\in$ set-dist (execute verify-monad s)
and *missing:*
trace-fri-reachable-missing-refined-layer-chain s bad
(Some (result, final-state))
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
where
trace-fri-partial-candidate-opening-evidence s
(Some (result, final-state)) trace-roots trace-bs trace-final dg
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr candidate-query-idxs
trace-openings trace-table
 $\wedge$ *doms layers. $\neg$*
generic-fri-canonical-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers (bad trace-table)

proof –
have *reachable:*
trace-fri-bad-with-reachable-partial-candidate s
(Some (result, final-state))
and *not-refined:*

```

    ¬ trace-fri-bad-with-refined-layer-chain s bad
      (Some (result, final-state))
  using missing
  unfolding trace-fri-reachable-missing-refined-layer-chain-def
  by blast+
from verify-monad-trace-fri-bad-extracts-opening-evidence
  [OF outcome reachable]
obtain trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr candidate-query-idxs trace-openings
  trace-table
where evidence:
  trace-fri-partial-candidate-opening-evidence s
  (Some (result, final-state)) trace-roots trace-bs trace-final dg
  composition-roots composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers fr candidate-query-idxs
  trace-openings trace-table
by blast
have no-chain:
  ∧ doms layers. ¬
    generic-fri-canonical-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers (bad trace-table)
proof
  fix doms layers
  assume chain:
    generic-fri-canonical-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers (bad trace-table)
  have trace-fri-bad-with-refined-layer-chain s bad
    (Some (result, final-state))
  unfolding trace-fri-bad-with-refined-layer-chain-def
  using evidence chain by blast
  then show False
    using not-refined by contradiction
qed
show ?thesis
  by (rule that[OF evidence no-chain])
qed

```

lemma *verify-monad-composition-fri-missing-extracts-opening-evidence:*
assumes *outcome:*
Some (result, final-state) ∈ set-dist (execute verify-monad s)
and *missing:*
composition-fri-reachable-missing-refined-layer-chain s bad
(Some (result, final-state))
obtains *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr


```

    f-fri-roots f-final as dg composition-fri-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table
where
    composition-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
    opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers fr
    f-fri-roots f-final as dg composition-fri-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table
    to-nat fri-dg ≤ maxDegree
     $\bigwedge$  doms layers.  $\neg$ 
    generic-fri-canonical-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers
    (bad fri-dg composition-table)
proof –
have reachable:
    composition-fri-bad-with-reachable-partial-candidate s
    (Some (result, final-state))
and not-refined:
     $\neg$  composition-fri-bad-with-refined-layer-chain s bad
    (Some (result, final-state))
using missing
unfolding composition-fri-reachable-missing-refined-layer-chain-def
by blast+
show ?thesis
proof (rule
    verify-monad-composition-fri-bad-extracts-opening-evidence-with-degree-bound
    [OF outcome reachable])
fix trace-roots trace-bs trace-final fri-dg
    opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers fr
    f-fri-roots f-final as dg composition-fri-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table
assume evidence:
    composition-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
    opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers fr
    f-fri-roots f-final as dg composition-fri-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table
assume degree-bound: to-nat fri-dg ≤ maxDegree
have no-chain:
     $\bigwedge$  doms layers.  $\neg$ 

```

```

    generic-fri-canonical-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms
    layers (bad fri-dg composition-table)
  proof
    fix doms layers
    assume chain:
      generic-fri-canonical-layer-chain-evidence
      (composition-table-low-degree (to-nat fri-dg)) composition-table
      (to-nat fri-dg) opening-composition-roots composition-bs
      composition-final fri-query-idxs composition-round-layers doms
      layers (bad fri-dg composition-table)
    have composition-fri-bad-with-refined-layer-chain s bad
      (Some (result, final-state))
    using evidence chain
    unfolding composition-fri-bad-with-refined-layer-chain-def
    by metis
    then show False
      using not-refined by contradiction
  qed
  show thesis
    by (rule that[OF evidence degree-bound no-chain])
  qed
qed

lemma verify-monad-trace-fri-missing-sampled-extracts-opening-evidence:
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and missing:
    trace-fri-reachable-missing-sampled-layer-chain s
    (Some (result, final-state))
  obtains trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
  where
    trace-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr candidate-query-idxs
    trace-openings trace-table
    ∧ doms layers. ¬
      generic-fri-sampled-layer-chain-evidence trace-table-low-degree
      trace-table (clength - 1) trace-roots trace-bs trace-final
      fri-query-idxs trace-round-layers doms layers
  proof -
    have reachable:
      trace-fri-bad-with-reachable-partial-candidate s

```

```

    (Some (result, final-state))
and not-sampled:
  ¬ trace-fri-bad-with-sampled-layer-chain s
    (Some (result, final-state))
using missing
unfolding trace-fri-reachable-missing-sampled-layer-chain-def
by blast+
from verify-monad-trace-fri-bad-extracts-opening-evidence
  [OF outcome reachable]
obtain trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr candidate-query-idxs trace-openings
  trace-table
where evidence:
  trace-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr candidate-query-idxs
    trace-openings trace-table
by blast
have no-sampled:
  ∧ doms layers. ¬
    generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
proof
  fix doms layers
  assume chain:
    generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
  have trace-fri-bad-with-sampled-layer-chain s
    (Some (result, final-state))
    unfolding trace-fri-bad-with-sampled-layer-chain-def
    using evidence chain by blast
  then show False
    using not-sampled by contradiction
qed
show ?thesis
  by (rule that[OF evidence no-sampled])
qed

lemma verify-monad-composition-fri-missing-sampled-extracts-opening-evidence:
assumes outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
and missing:
  composition-fri-reachable-missing-sampled-layer-chain s
    (Some (result, final-state))
obtains trace-roots trace-bs trace-final fri-dg

```

```

opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table
where
composition-fri-partial-candidate-opening-evidence s
  (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table
to-nat fri-dg ≤ maxDegree
 $\bigwedge$  doms layers.  $\neg$ 
  generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers
proof –
have reachable:
composition-fri-bad-with-reachable-partial-candidate s
  (Some (result, final-state))
and not-sampled:
 $\neg$  composition-fri-bad-with-sampled-layer-chain s
  (Some (result, final-state))
using missing
unfolding composition-fri-reachable-missing-sampled-layer-chain-def
by blast+
show ?thesis
proof (rule
  verify-monad-composition-fri-bad-extracts-opening-evidence-with-degree-bound
    [OF outcome reachable])
fix trace-roots trace-bs trace-final fri-dg
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table
assume evidence:
composition-fri-partial-candidate-opening-evidence s
  (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table
assume degree-bound: to-nat fri-dg ≤ maxDegree
have no-sampled:

```

```

 $\wedge$  doms layers.  $\neg$ 
  generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms
    layers
proof
  fix doms layers
  assume chain:
    generic-fri-sampled-layer-chain-evidence
      (composition-table-low-degree (to-nat fri-dg)) composition-table
      (to-nat fri-dg) opening-composition-roots composition-bs
      composition-final fri-query-idxs composition-round-layers doms
      layers
  have composition-fri-bad-with-sampled-layer-chain s
    (Some (result, final-state))
  using evidence chain
  unfolding composition-fri-bad-with-sampled-layer-chain-def
  by metis
  then show False
    using not-sampled by contradiction
qed
show thesis
  by (rule that[OF evidence degree-bound no-sampled])
qed
qed

lemma verify-monad-trace-fri-missing-full-cover-extracts-sampled-evidence:
assumes outcome:
  Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
and missing:
  trace-fri-sampled-missing-full-cover s bad
  (Some (result, final-state))
obtains trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr candidate-query-idxs trace-openings
  trace-table doms layers
where
  trace-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr candidate-query-idxs
    trace-openings trace-table
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
   $\neg$  generic-fri-sampled-layer-chain-full-cover trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers (bad trace-table)

```

```

proof –
  have sampld:
    trace-fri-bad-with-sampled-layer-chain s
      (Some (result, final-state))
  and not-full:
     $\neg$  trace-fri-bad-with-full-cover-layer-chain s bad
      (Some (result, final-state))
  using missing
  unfolding trace-fri-sampled-missing-full-cover-def by blast+
from sampld obtain trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr candidate-query-idxs trace-openings
  trace-table doms layers
where evidence:
  trace-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr candidate-query-idxs
    trace-openings trace-table
and chain:
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength – 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
  unfolding trace-fri-bad-with-sampled-layer-chain-def by blast
have no-full:
   $\neg$  generic-fri-sampled-layer-chain-full-cover trace-table-low-degree
    trace-table (clength – 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers (bad trace-table)
proof
  assume full:
    generic-fri-sampled-layer-chain-full-cover trace-table-low-degree
      trace-table (clength – 1) trace-roots trace-bs trace-final
      fri-query-idxs trace-round-layers doms layers (bad trace-table)
  have trace-fri-bad-with-full-cover-layer-chain s bad
    (Some (result, final-state))
  unfolding trace-fri-bad-with-full-cover-layer-chain-def
  using evidence full by blast
  then show False
    using not-full by contradiction
qed
show ?thesis
  by (rule that[OF evidence chain no-full])
qed

lemma verify-monad-composition-fri-missing-full-cover-extracts-sampled-evidence:
assumes outcome:
  Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
and missing:
  composition-fri-sampled-missing-full-cover s bad

```

(Some (result, final-state))

obtains trace-roots trace-bs trace-final fri-dg
 opening-composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers fr
 f-fri-roots f-final as dg composition-fri-roots final
 trace-query-idxs trace-openings composition-query-idxs
 composition-openings trace-table composition-table doms layers

where
 composition-fri-partial-candidate-opening-evidence s
 (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
 opening-composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers fr
 f-fri-roots f-final as dg composition-fri-roots final
 trace-query-idxs trace-openings composition-query-idxs
 composition-openings trace-table composition-table
 generic-fri-sampled-layer-chain-evidence
 (composition-table-low-degree (to-nat fri-dg)) composition-table
 (to-nat fri-dg) opening-composition-roots composition-bs
 composition-final fri-query-idxs composition-round-layers doms layers
 $\neg$ generic-fri-sampled-layer-chain-full-cover
 (composition-table-low-degree (to-nat fri-dg)) composition-table
 (to-nat fri-dg) opening-composition-roots composition-bs
 composition-final fri-query-idxs composition-round-layers doms layers
 (bad fri-dg composition-table)

proof –
have sampled:
 composition-fri-bad-with-sampled-layer-chain s
 (Some (result, final-state))
and not-full:
 $\neg$ composition-fri-bad-with-full-cover-layer-chain s bad
 (Some (result, final-state))
using missing
unfolding composition-fri-sampled-missing-full-cover-def **by** blast+
from sampled **obtain** trace-roots trace-bs trace-final fri-dg
 opening-composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers fr
 f-fri-roots f-final as dg composition-fri-roots final
 trace-query-idxs trace-openings composition-query-idxs
 composition-openings trace-table composition-table doms layers

where evidence:
 composition-fri-partial-candidate-opening-evidence s
 (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
 opening-composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers fr
 f-fri-roots f-final as dg composition-fri-roots final
 trace-query-idxs trace-openings composition-query-idxs
 composition-openings trace-table composition-table

and chain:
 generic-fri-sampled-layer-chain-evidence

```

    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers
unfolding composition-fri-bad-with-sampled-layer-chain-def by auto
have no-full:
  ¬ generic-fri-sampled-layer-chain-full-cover
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers
    (bad fri-dg composition-table)
proof
  assume full:
    generic-fri-sampled-layer-chain-full-cover
      (composition-table-low-degree (to-nat fri-dg)) composition-table
      (to-nat fri-dg) opening-composition-roots composition-bs
      composition-final fri-query-idxs composition-round-layers doms layers
      (bad fri-dg composition-table)
  have composition-fri-bad-with-full-cover-layer-chain s bad
    (Some (result, final-state))
  unfolding composition-fri-bad-with-full-cover-layer-chain-def
  using evidence full by metis
  then show False
    using not-full by contradiction
qed
show ?thesis
  by (rule that[OF evidence chain no-full])
qed

```

```

lemma wp-trace-fri-reachable-bound-from-refined-and-missing:
  assumes refined-bound:
    wp-event verify-monad
      (trace-fri-bad-with-refined-layer-chain s bad) s ≤ R
  and missing-bound:
    wp-event verify-monad
      (trace-fri-reachable-missing-refined-layer-chain s bad) s ≤ M
  shows
    wp-event verify-monad
      (trace-fri-bad-with-reachable-partial-candidate s) s ≤ R + M
proof −
  have wp-event verify-monad
    (trace-fri-bad-with-reachable-partial-candidate s) s ≤
  wp-event verify-monad
    (λout. trace-fri-bad-with-refined-layer-chain s bad out ∨
      trace-fri-reachable-missing-refined-layer-chain s bad out) s
  by (rule wp-event-mono)
    (rule trace-fri-reachable-imp-refined-or-missing-layer-chain)
  also have ... ≤
    wp-event verify-monad
      (trace-fri-bad-with-refined-layer-chain s bad) s +

```


wp-event verify-monad
 (trace-fri-reachable-missing-refined-layer-chain s bad) s
 by (rule wp-event-union-bound)
 also have $\dots \leq R + M$
 by (intro add-mono refined-bound missing-bound)
 finally show ?thesis .
 qed

lemma wp-composition-fri-reachable-bound-from-refined-and-missing:
 assumes refined-bound:
 wp-event verify-monad
 (composition-fri-bad-with-refined-layer-chain s bad) $s \leq R$
 and missing-bound:
 wp-event verify-monad
 (composition-fri-reachable-missing-refined-layer-chain s bad) $s \leq M$
 shows
 wp-event verify-monad
 (composition-fri-bad-with-reachable-partial-candidate s) $s \leq R + M$
proof –
 have *wp-event verify-monad*
 (composition-fri-bad-with-reachable-partial-candidate s) $s \leq$
 wp-event verify-monad
 ($\lambda out. \text{composition-fri-bad-with-refined-layer-chain } s \text{ bad } out \vee$
 $\text{composition-fri-reachable-missing-refined-layer-chain } s \text{ bad } out$) s
 by (rule wp-event-mono)
 (rule composition-fri-reachable-imp-refined-or-missing-layer-chain)
 also have $\dots \leq$
 wp-event verify-monad
 (composition-fri-bad-with-refined-layer-chain s bad) $s +$
 wp-event verify-monad
 (composition-fri-reachable-missing-refined-layer-chain s bad) s
 by (rule wp-event-union-bound)
 also have $\dots \leq R + M$
 by (intro add-mono refined-bound missing-bound)
 finally show ?thesis .
 qed

lemma wp-trace-fri-reachable-bound-from-sampled-and-missing:
 assumes sampled-bound:
 wp-event verify-monad
 (trace-fri-bad-with-sampled-layer-chain s) $s \leq R$
 and missing-bound:
 wp-event verify-monad
 (trace-fri-reachable-missing-sampled-layer-chain s) $s \leq M$
 shows
 wp-event verify-monad
 (trace-fri-bad-with-reachable-partial-candidate s) $s \leq R + M$
proof –
 have *wp-event verify-monad*

(trace-fri-bad-with-reachable-partial-candidate s) s ≤
 wp-event verify-monad
 (λout. trace-fri-bad-with-sampled-layer-chain s out ∨
 trace-fri-reachable-missing-sampled-layer-chain s out) s
 by (rule wp-event-mono)
 (rule trace-fri-reachable-imp-sampled-or-missing-layer-chain)
 also have ... ≤
 wp-event verify-monad
 (trace-fri-bad-with-sampled-layer-chain s) s +
 wp-event verify-monad
 (trace-fri-reachable-missing-sampled-layer-chain s) s
 by (rule wp-event-union-bound)
 also have ... ≤ R + M
 by (intro add-mono sampled-bound missing-bound)
 finally show ?thesis .
 qed

lemma wp-composition-fri-reachable-bound-from-sampled-and-missing:
 assumes sampled-bound:
 wp-event verify-monad
 (composition-fri-bad-with-sampled-layer-chain s) s ≤ R
 and missing-bound:
 wp-event verify-monad
 (composition-fri-reachable-missing-sampled-layer-chain s) s ≤ M
 shows
 wp-event verify-monad
 (composition-fri-bad-with-reachable-partial-candidate s) s ≤ R + M
proof –
 have wp-event verify-monad
 (composition-fri-bad-with-reachable-partial-candidate s) s ≤
 wp-event verify-monad
 (λout. composition-fri-bad-with-sampled-layer-chain s out ∨
 composition-fri-reachable-missing-sampled-layer-chain s out) s
 by (rule wp-event-mono)
 (rule composition-fri-reachable-imp-sampled-or-missing-layer-chain)
 also have ... ≤
 wp-event verify-monad
 (composition-fri-bad-with-sampled-layer-chain s) s +
 wp-event verify-monad
 (composition-fri-reachable-missing-sampled-layer-chain s) s
 by (rule wp-event-union-bound)
 also have ... ≤ R + M
 by (intro add-mono sampled-bound missing-bound)
 finally show ?thesis .
 qed

lemma trace-fri-reachable-partial-candidate-reduction-from-refined-and-missing:
 assumes refined-bound:
 wp-event verify-monad

(trace-fri-bad-with-refined-layer-chain s bad) $s \leq R$
and missing-bound:
 wp-event verify-monad
 (trace-fri-reachable-missing-refined-layer-chain s bad) $s \leq M$
and total-bound: $R + M \leq \text{trace-fri-error}$
shows trace-fri-reachable-partial-candidate-reduction-assumption s
proof –
have old-bound:
 wp-event verify-monad
 (trace-fri-bad-with-reachable-partial-candidate s) $s \leq R + M$
by (rule wp-trace-fri-reachable-bound-from-refined-and-missing
 [OF refined-bound missing-bound])
show ?thesis
unfolding trace-fri-reachable-partial-candidate-reduction-assumption-def
by (rule order-trans[OF old-bound total-bound])
qed

lemma composition-fri-reachable-partial-candidate-reduction-from-refined-and-missing:
assumes refined-bound:
 wp-event verify-monad
 (composition-fri-bad-with-refined-layer-chain s bad) $s \leq R$
and missing-bound:
 wp-event verify-monad
 (composition-fri-reachable-missing-refined-layer-chain s bad) $s \leq M$
and total-bound: $R + M \leq \text{composition-fri-error}$
shows composition-fri-reachable-partial-candidate-reduction-assumption s
proof –
have old-bound:
 wp-event verify-monad
 (composition-fri-bad-with-reachable-partial-candidate s) $s \leq R + M$
by (rule wp-composition-fri-reachable-bound-from-refined-and-missing
 [OF refined-bound missing-bound])
show ?thesis
unfolding
 composition-fri-reachable-partial-candidate-reduction-assumption-def
by (rule order-trans[OF old-bound total-bound])
qed

lemma trace-fri-reachable-partial-candidate-reduction-from-sampled-and-missing:
assumes sampled-bound:
 wp-event verify-monad
 (trace-fri-bad-with-sampled-layer-chain s) $s \leq R$
and missing-bound:
 wp-event verify-monad
 (trace-fri-reachable-missing-sampled-layer-chain s) $s \leq M$
and total-bound: $R + M \leq \text{trace-fri-error}$
shows trace-fri-reachable-partial-candidate-reduction-assumption s
proof –
have old-bound:

```

wp-event verify-monad
  (trace-fri-bad-with-reachable-partial-candidate s) s ≤ R + M
by (rule wp-trace-fri-reachable-bound-from-sampled-and-missing
    [OF sampled-bound missing-bound])
show ?thesis
  unfolding trace-fri-reachable-partial-candidate-reduction-assumption-def
  by (rule order-trans[OF old-bound total-bound])
qed

lemma composition-fri-reachable-partial-candidate-reduction-from-sampled-and-missing:
  assumes sampled-bound:
    wp-event verify-monad
      (composition-fri-bad-with-sampled-layer-chain s) s ≤ R
  and missing-bound:
    wp-event verify-monad
      (composition-fri-reachable-missing-sampled-layer-chain s) s ≤ M
  and total-bound: R + M ≤ composition-fri-error
  shows composition-fri-reachable-partial-candidate-reduction-assumption s
proof -
  have old-bound:
    wp-event verify-monad
      (composition-fri-bad-with-reachable-partial-candidate s) s ≤ R + M
  by (rule wp-composition-fri-reachable-bound-from-sampled-and-missing
      [OF sampled-bound missing-bound])
  show ?thesis
    unfolding
      composition-fri-reachable-partial-candidate-reduction-assumption-def
    by (rule order-trans[OF old-bound total-bound])
qed

lemma trace-fri-bad-with-partial-candidates-bound-from-refined-and-missing:
  fixes R M :: prob
  assumes refined-bound:
    ∧ data attacker-state.
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)
            adversary-initial-state) ⇒
      wp-event verify-monad
        (trace-fri-bad-with-refined-layer-chain
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)) bad)
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)) ≤ R
  and missing-bound:
    ∧ data attacker-state.
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)
            adversary-initial-state) ⇒
      wp-event verify-monad

```

$(\text{trace-fri-reachable-missing-refined-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad})$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq M$
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq R + M$
proof (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
show $\bigwedge s. \neg \text{trace-fri-bad-with-partial-candidates } s \text{ None}$
by (rule *trace-fri-bad-with-partial-candidates-not-None*)
next
fix *data attacker-state*
assume *builder*:
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$
let $?s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
have *old-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-reachable-partial-candidate } ?s) ?s \leq R + M$
by (rule *wp-trace-fri-reachable-bound-from-refined-and-missing*
 $[\text{OF refined-bound}[\text{OF builder}] \text{missing-bound}[\text{OF builder}]]$)
show $\text{wp-event verify-monad } (\text{trace-fri-bad-with-partial-candidates } ?s) ?s$
 $\leq R + M$
by (rule *order-trans*[*OF* - *old-bound*])
 $(\text{rule wp-event-mono},$
 $\text{rule trace-fri-bad-with-partial-candidates-imp-reachable-partial-candidate})$
qed

lemma *composition-fri-bad-with-partial-candidates-bound-from-refined-and-missing*:
fixes $R \ M :: \text{prob}$
assumes *refined-bound*:
 $\bigwedge data \text{ attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-refined-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad})$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq R$
and *missing-bound*:
 $\bigwedge data \text{ attacker-state.}$

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(composition-fri-reachable-missing-refined-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) bad)
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ M
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state ≤ R + M
proof (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
show $\bigwedge s. \neg \text{composition-fri-bad-with-partial-candidates } s \text{ None}$
by (rule *composition-fri-bad-with-partial-candidates-not-None*)
next
fix *data attacker-state*
assume *builder:*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)
let *?s =*
verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
have *old-bound:*
wp-event verify-monad
(composition-fri-bad-with-reachable-partial-candidate ?s) ?s ≤
R + M
by (rule *wp-composition-fri-reachable-bound-from-refined-and-missing*
 $[OF \text{ refined-bound}[OF \text{ builder}] \text{ missing-bound}[OF \text{ builder}]]$)
show *wp-event verify-monad*
(composition-fri-bad-with-partial-candidates ?s) ?s ≤ R + M
by (rule *order-trans[OF - old-bound]*)
(rule *wp-event-mono,*
rule
composition-fri-bad-with-partial-candidates-imp-reachable-partial-candidate)
qed

lemma *trace-fri-bad-with-empty-header-candidates-bound-from-refined-and-missing:*
fixes *R M :: prob*
assumes *refined-bound:*
 $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-bad-with-refined-layer-chain

```

    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad)
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ R
and missing-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (trace-fri-reachable-missing-refined-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad)
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ M
shows
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state ≤ R + M
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\wedge s. \neg \text{trace-fri-bad-with-empty-composition-header-candidates } s \text{ None}$ 
  by (rule trace-fri-bad-with-empty-composition-header-candidates-not-None)
next
  fix data attacker-state
  assume builder:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have old-bound:
    wp-event verify-monad
      (trace-fri-bad-with-reachable-partial-candidate ?s) ?s ≤ R + M
  by (rule wp-trace-fri-reachable-bound-from-refined-and-missing
    [OF refined-bound[OF builder] missing-bound[OF builder]])
  show wp-event verify-monad
    (trace-fri-bad-with-empty-composition-header-candidates ?s) ?s ≤
      R + M
  by (rule order-trans[OF - old-bound])
    (rule wp-event-mono,
      rule
        trace-fri-bad-with-empty-composition-header-candidates-imp-reachable-partial-candidate)
qed

lemma trace-fri-bad-with-partial-candidates-bound-from-sampled-and-missing:
  fixes R M :: prob
  assumes sampled-bound:

```

$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-bad-with-sampled-layer-chain
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq R$
and missing-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-reachable-missing-sampled-layer-chain
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq M$
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq R + M$
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show $\wedge s. \neg \text{trace-fri-bad-with-partial-candidates } s \text{ None}$
by (rule trace-fri-bad-with-partial-candidates-not-None)
next
fix data attacker-state
assume builder:
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state)
let ?s =
 verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)
have old-bound:
 wp-event verify-monad
 (trace-fri-bad-with-reachable-partial-candidate ?s) ?s $\leq R + M$
by (rule wp-trace-fri-reachable-bound-from-sampled-and-missing
 [OF sampled-bound[OF builder] missing-bound[OF builder]])
show wp-event verify-monad (trace-fri-bad-with-partial-candidates ?s) ?s
 $\leq R + M$
by (rule order-trans[OF - old-bound])
 (rule wp-event-mono,
 rule trace-fri-bad-with-partial-candidates-imp-reachable-partial-candidate)
qed

lemma *composition-fri-bad-with-partial-candidates-bound-from-sampled-and-missing:*

fixes $R\ M :: \text{prob}$

assumes *sampled-bound:*

$\bigwedge \text{data attacker-state.}$

$\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$

$\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq R$

and *missing-bound:*

$\bigwedge \text{data attacker-state.}$

$\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$

$\text{wp-event verify-monad}$
 $(\text{composition-fri-reachable-missing-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq M$

shows

$\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq R + M$

proof (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)

show $\bigwedge s. \neg \text{composition-fri-bad-with-partial-candidates } s \text{ None}$

by (rule *composition-fri-bad-with-partial-candidates-not-None*)

next

fix *data attacker-state*

assume *builder:*

$\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$

let $?s =$

$\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$

have *old-bound:*

$\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-reachable-partial-candidate } ?s) ?s \leq$
 $R + M$

by (rule *wp-composition-fri-reachable-bound-from-sampled-and-missing*
 $[\text{OF sampled-bound}[\text{OF builder}] \text{ missing-bound}[\text{OF builder}]]$)

show $\text{wp-event verify-monad}$

```

    (composition-fri-bad-with-partial-candidates ?s) ?s ≤ R + M
  by (rule order-trans[OF - old-bound])
    (rule wp-event-mono,
     rule
       composition-fri-bad-with-partial-candidates-imp-reachable-partial-candidate)
qed

lemma trace-fri-bad-with-empty-header-candidates-bound-from-sampled-and-missing:
  fixes R M :: prob
  assumes sampled-bound:
    ∧data attacker-state.
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state) ⇒
        wp-event verify-monad
          (trace-fri-bad-with-sampled-layer-chain
            (verifier-state-from-adversary attacker-state
              (staged-proof-transcript data)))
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)) ≤ R
  and missing-bound:
    ∧data attacker-state.
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state) ⇒
        wp-event verify-monad
          (trace-fri-reachable-missing-sampled-layer-chain
            (verifier-state-from-adversary attacker-state
              (staged-proof-transcript data)))
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)) ≤ M
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-empty-composition-header-candidates)
      adversary-initial-state ≤ R + M
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show ∧s. ¬ trace-fri-bad-with-empty-composition-header-candidates s None
  by (rule trace-fri-bad-with-empty-composition-header-candidates-not-None)
next
  fix data attacker-state
  assume builder:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have old-bound:

```

```

wp-event verify-monad
  (trace-fri-bad-with-reachable-partial-candidate ?s) ?s ≤ R + M
by (rule wp-trace-fri-reachable-bound-from-sampled-and-missing
    [OF sampled-bound[OF builder] missing-bound[OF builder]])
show wp-event verify-monad
  (trace-fri-bad-with-empty-composition-header-candidates ?s) ?s ≤
    R + M
by (rule order-trans[OF - old-bound])
  (rule wp-event-mono,
    rule
      trace-fri-bad-with-empty-composition-header-candidates-imp-reachable-partial-candidate)
qed

```

lemma *trace-fri-bad-with-partial-candidates-bound-from-full-cover-and-missing:*

fixes *R MF MS* :: *prob*

assumes *full-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒

wp-event verify-monad
(trace-fri-bad-with-full-cover-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) bad)
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ R

and *missing-full-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒

wp-event verify-monad
(trace-fri-sampled-missing-full-cover
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) bad)
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ MF

and *missing-sampled-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒

wp-event verify-monad
(trace-fri-reachable-missing-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ MS

shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-partial-candidates)
  adversary-initial-state  $\leq (R + MF) + MS$ 
proof (rule trace-fri-bad-with-partial-candidates-bound-from-sampled-and-missing)
fix data attacker-state
assume builder:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
  (trace-fri-bad-with-sampled-layer-chain ?s) ?s  $\leq R + MF$ 
by (rule wp-trace-fri-sampled-bound-from-full-cover-and-missing
  [OF full-bound[OF builder] missing-full-bound[OF builder]])
next
fix data attacker-state
assume builder:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (trace-fri-reachable-missing-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq MS$ 
by (rule missing-sampled-bound[OF builder])
qed

lemma composition-fri-bad-with-partial-candidates-bound-from-full-cover-and-missing:
fixes R MF MS :: prob
assumes full-bound:
   $\wedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
  wp-event verify-monad
    (composition-fri-bad-with-full-cover-layer-chain
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) bad)
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))  $\leq R$ 
and missing-full-bound:
   $\wedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)

```

```

    adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (composition-fri-sampled-missing-full-cover
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad)
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq MF$ 
and missing-sampled-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (composition-fri-reachable-missing-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq MS$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq (R + MF) + MS$ 
proof (rule
  composition-fri-bad-with-partial-candidates-bound-from-sampled-and-missing)
fix data attacker-state
assume builder:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
  (composition-fri-bad-with-sampled-layer-chain ?s) ?s  $\leq R + MF$ 
by (rule wp-composition-fri-sampled-bound-from-full-cover-and-missing
  [OF full-bound[OF builder] missing-full-bound[OF builder]])
next
fix data attacker-state
assume builder:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (composition-fri-reachable-missing-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq MS$ 

```

```

    by (rule missing-sampled-bound[OF builder])
qed

lemma trace-fri-bad-with-empty-header-candidates-bound-from-full-cover-and-missing:
  fixes R MF MS :: prob
  assumes full-bound:
     $\bigwedge \text{data attacker-state.}$ 
     $\text{Some (data, attacker-state)} \in$ 
     $\text{set-dist (execute (checked-staged-transcript-program A)}$ 
     $\text{adversary-initial-state)} \implies$ 
    wp-event verify-monad
    (trace-fri-bad-with-full-cover-layer-chain
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad)
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))  $\leq R$ 
  and missing-full-bound:
     $\bigwedge \text{data attacker-state.}$ 
     $\text{Some (data, attacker-state)} \in$ 
     $\text{set-dist (execute (checked-staged-transcript-program A)}$ 
     $\text{adversary-initial-state)} \implies$ 
    wp-event verify-monad
    (trace-fri-sampled-missing-full-cover
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad)
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))  $\leq MF$ 
  and missing-sampled-bound:
     $\bigwedge \text{data attacker-state.}$ 
     $\text{Some (data, attacker-state)} \in$ 
     $\text{set-dist (execute (checked-staged-transcript-program A)}$ 
     $\text{adversary-initial-state)} \implies$ 
    wp-event verify-monad
    (trace-fri-reachable-missing-sampled-layer-chain
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))  $\leq MS$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state  $\leq (R + MF) + MS$ 
proof (rule
  trace-fri-bad-with-empty-header-candidates-bound-from-sampled-and-missing)
fix data attacker-state
assume builder:
   $\text{Some (data, attacker-state)} \in$ 
   $\text{set-dist (execute (checked-staged-transcript-program A)}$ 

```

```

    adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  show wp-event verify-monad
    (trace-fri-bad-with-sampled-layer-chain ?s) ?s ≤ R + MF
  by (rule wp-trace-fri-sampled-bound-from-full-cover-and-missing
    [OF full-bound[OF builder] missing-full-bound[OF builder]])
next
fix data attacker-state
assume builder:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (trace-fri-reachable-missing-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ MS
by (rule missing-sampled-bound[OF builder])
qed

end

end

```

```

theory Soundness-FRI-Full-Cover-Reasons
  imports Soundness-FRI-Refined-Reachable
begin

```

4.8 FRI Cover and Verifier-Tied Reductions

Diagnostic split for missing full sampled FRI cover.

The predicates below do not add assumptions. They expose which clause of *generic-fri-sampled-layer-chain-j* must fail when verifier sampled layer-chain evidence exists but the corresponding full-cover witness is absent. Later proof layers should bound these failures by existing Merkle/hash/proximity side events.

```

context soundness
begin

```

definition *generic-fri-sampled-full-cover-failure*

```

:: ('f list ⇒ bool) ⇒ 'f list ⇒ nat ⇒
  'f list ⇒ 'f list ⇒ 'f ⇒ nat list ⇒
  'f list list list ⇒ 'f list list ⇒ 'f list list ⇒
  (nat ⇒ 'f list ⇒ 'f set) ⇒ bool

```

where

generic-fri-sampled-full-cover-failure low-degree candidate-table
 degree-bound roots challenges final-value query-idxs round-layers
 doms layers bad $\longleftrightarrow$
 $\neg$ fri-multiround-proximity-steps (length challenges) degree-bound
 (fri-layer-lengths (length challenges) (clength * scale))
 (map ($\lambda i. 2^i$) [0..
 doms layers bad $\vee$
 $\neg$ fri-sampled-layers-cover-fold-indices roots query-idxs $\vee$
 $\neg$ ($\forall$ layer-idx < length challenges.
 2 dvd fri-evidence-layer-len roots layer-idx) $\vee$
 $\neg$ ($\forall$ layer-idx < length challenges.
 fri-evidence-layer-len roots layer-idx * $2^{\text{layer-idx}}$ =
 clength * scale) $\vee$
 $\neg$ ($\forall$ layer-idx < length challenges.
 fri-evidence-layer-len roots layer-idx $\leq$
 length (doms ! layer-idx)) $\vee$
 $\neg$ ($\forall$ layer-idx < length challenges.
 fri-evidence-layer-len roots layer-idx $\leq$
 length (layers ! layer-idx)) $\vee$
 $\neg$ ($\forall$ layer-idx < length challenges.
 fri-evidence-layer-len roots layer-idx div 2 $\leq$
 length (doms ! Suc layer-idx)) $\vee$
 $\neg$ ($\forall$ layer-idx < length challenges.
 fri-evidence-layer-len roots layer-idx div 2 $\leq$
 length (layers ! Suc layer-idx)) $\vee$
 $\neg$ ($\forall$ layer-idx < length challenges.
 $\forall$ idx < fri-evidence-layer-len roots layer-idx div 2.
 doms ! Suc layer-idx ! idx =
 (doms ! layer-idx ! idx)²) $\vee$
 $\neg$ ($\forall$ layer-idx < length challenges.
 $\forall$ idx < fri-evidence-layer-len roots layer-idx div 2.
 doms ! layer-idx ! idx = (h ^ idx * shift) ^ ($2^{\text{layer-idx}}$)) $\vee$
 fri-table-low-degree-on degree-bound eval-domain candidate-table $\vee$
 $\neg$ fri-table-low-degree-on
 (fri-degree-after (length challenges) degree-bound)
 (doms ! length challenges) (layers ! length challenges)

definition *generic-fri-full-cover-proximity-failure*

$:: \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list list} \Rightarrow$
 (nat $\Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}$) $\Rightarrow$ bool

where

generic-fri-full-cover-proximity-failure degree-bound challenges
 doms layers bad $\longleftrightarrow$
 $\neg$ fri-multiround-proximity-steps (length challenges) degree-bound
 (fri-layer-lengths (length challenges) (clength * scale))
 (map ($\lambda i. 2^i$) [0..
 doms layers bad

definition *generic-fri-full-cover-index-failure*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow \text{bool}$

where

$\text{generic-fri-full-cover-index-failure roots query-idxs} \longleftrightarrow$
 $\neg \text{fri-sampled-layers-cover-fold-indices roots query-idxs}$

definition *generic-fri-full-cover-shape-failure*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

$\text{generic-fri-full-cover-shape-failure roots challenges doms layers} \longleftrightarrow$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$
 $2 \text{ dvd fri-evidence-layer-len roots layer-idx}) \vee$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} * 2 \wedge \text{layer-idx} =$
 $\text{clength} * \text{scale}) \vee$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} \leq$
 $\text{length} (\text{doms ! layer-idx})) \vee$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} \leq$
 $\text{length} (\text{layers ! layer-idx})) \vee$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} \text{ div } 2 \leq$
 $\text{length} (\text{doms ! Suc layer-idx})) \vee$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} \text{ div } 2 \leq$
 $\text{length} (\text{layers ! Suc layer-idx})) \vee$
 $\text{length} (\text{doms ! length challenges}) \neq$
 $\text{length} (\text{layers ! length challenges})$

definition *generic-fri-full-cover-arithmetic-shape-failure*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{bool}$

where

$\text{generic-fri-full-cover-arithmetic-shape-failure roots challenges}$
 $\longleftrightarrow$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$
 $2 \text{ dvd fri-evidence-layer-len roots layer-idx}) \vee$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} * 2 \wedge \text{layer-idx} =$
 $\text{clength} * \text{scale})$

definition *generic-fri-full-cover-witness-length-failure*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

$\text{generic-fri-full-cover-witness-length-failure roots challenges doms layers}$
 $\longleftrightarrow$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} \leq$
 $\text{length} (\text{doms ! layer-idx})) \vee$
 $\neg (\forall \text{layer-idx} < \text{length challenges.}$

$fri\text{-}evidence\text{-}layer\text{-}len\ roots\ layer\text{-}idx \leq$
 $length\ (layers\ !\ layer\text{-}idx)) \vee$
 $\neg (\forall\ layer\text{-}idx < length\ challenges.$
 $fri\text{-}evidence\text{-}layer\text{-}len\ roots\ layer\text{-}idx\ div\ 2 \leq$
 $length\ (doms\ !\ Suc\ layer\text{-}idx)) \vee$
 $\neg (\forall\ layer\text{-}idx < length\ challenges.$
 $fri\text{-}evidence\text{-}layer\text{-}len\ roots\ layer\text{-}idx\ div\ 2 \leq$
 $length\ (layers\ !\ Suc\ layer\text{-}idx)) \vee$
 $length\ (doms\ !\ length\ challenges) \neq$
 $length\ (layers\ !\ length\ challenges)$

lemma *generic-fri-full-cover-shape-failure-split:*
assumes *generic-fri-full-cover-shape-failure roots challenges doms layers*
shows
generic-fri-full-cover-arithmetic-shape-failure roots challenges $\vee$
generic-fri-full-cover-witness-length-failure roots challenges
doms layers
using *assms*
unfolding *generic-fri-full-cover-shape-failure-def*
generic-fri-full-cover-arithmetic-shape-failure-def
generic-fri-full-cover-witness-length-failure-def
by *blast*

lemma *ceil-log-Suc-le-power-local:*
assumes $d < 2^N$
shows $ceil\text{-}log\ (Suc\ d) \leq N$
proof (*cases* $d = 0$)
case *True*
then show *?thesis*
unfolding *ceil-log-def* **by** *simp*
next
case *False*
then have *d-pos: 0 < d*
by *simp*
have $2^{\text{floor-log}\ d} \leq d$
by (*rule floor-log-exp2-le[OF d-pos]*)
also have $\dots < 2^N$
using *assms* .
finally have $\text{floor-log}\ d < N$
by (*subst (asm) power-strict-increasing-iff*) *simp-all*
then show *?thesis*
using *False* **unfolding** *ceil-log-def* **by** *simp*
qed

lemma *ceil-log-le-power-bound:*
assumes $n \leq 2^N$
shows $ceil\text{-}log\ n \leq N$
proof (*cases* n)
case *0*

```

then show ?thesis
  unfolding ceil-log-def by simp
next
case (Suc d)
then have  $d < 2 \wedge N$ 
  using assms by simp
then show ?thesis
  unfolding Suc by (rule ceil-log-Suc-le-power-local)
qed

lemma generic-fri-full-cover-arithmetic-shape-failure-false:
  assumes challenges-len: length challenges = length roots
    and eval-power: clength * scale =  $2 \wedge N$ 
    and rounds-bound: length challenges  $\leq N$ 
  shows  $\neg$  generic-fri-full-cover-arithmetic-shape-failure roots challenges
proof -
  have even:
     $\bigwedge \text{layer-idx}. \text{layer-idx} < \text{length challenges} \implies$ 
      2 dvd fri-evidence-layer-len roots layer-idx
  proof -
    fix layer-idx
    assume layer-bound: layer-idx < length challenges
    have layer-root-bound: layer-idx < length roots
      using layer-bound challenges-len by simp
    have suc-le: Suc layer-idx  $\leq N$ 
      using layer-bound rounds-bound by simp
    have  $(2::\text{nat}) \wedge \text{Suc layer-idx} \text{ dvd } (2::\text{nat}) \wedge N$ 
      by (rule le-imp-power-dvd[OF suc-le])
    then have dvd-eval:  $2 \wedge \text{Suc layer-idx} \text{ dvd } \text{clength} * \text{scale}$ 
      using eval-power by simp
    show 2 dvd fri-evidence-layer-len roots layer-idx
      unfolding fri-evidence-layer-len-def
      by (rule fri-layer-lengths-even-if-dvd[OF layer-root-bound dvd-eval])
  qed
  have product:
     $\bigwedge \text{layer-idx}. \text{layer-idx} < \text{length challenges} \implies$ 
      fri-evidence-layer-len roots layer-idx *  $2 \wedge \text{layer-idx} =$ 
        clength * scale
  proof -
    fix layer-idx
    assume layer-bound: layer-idx < length challenges
    have layer-root-bound: layer-idx < length roots
      using layer-bound challenges-len by simp
    have le-N: layer-idx  $\leq N$ 
      using layer-bound rounds-bound by simp
    have  $(2::\text{nat}) \wedge \text{layer-idx} \text{ dvd } (2::\text{nat}) \wedge N$ 
      by (rule le-imp-power-dvd[OF le-N])
    then have dvd-eval:  $2 \wedge \text{layer-idx} \text{ dvd } \text{clength} * \text{scale}$ 
      using eval-power by simp
  qed

```

```

show fri-evidence-layer-len roots layer-idx * 2 ^ layer-idx =
  clength * scale
unfolding fri-evidence-layer-len-def
by (rule fri-layer-lengths-round-product-if-dvd
  [OF layer-root-bound dvd-eval])
qed
show ?thesis
unfolding generic-fri-full-cover-arithmetic-shape-failure-def
using even product by simp
qed

definition generic-fri-full-cover-domain-failure
  :: 'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f list list  $\Rightarrow$  bool
where
  generic-fri-full-cover-domain-failure roots challenges doms  $\longleftrightarrow$ 
     $\neg (\forall \text{layer-idx} < \text{length challenges}.$ 
       $\forall \text{idx} < \text{fri-evidence-layer-len roots layer-idx div } 2.$ 
         $\text{doms ! Suc layer-idx ! idx} =$ 
           $(\text{doms ! layer-idx ! idx})^2 \vee$ 
         $\neg (\forall \text{layer-idx} < \text{length challenges}.$ 
           $\forall \text{idx} < \text{fri-evidence-layer-len roots layer-idx div } 2.$ 
             $\text{doms ! layer-idx ! idx} = (h \wedge \text{idx} * \text{shift}) \wedge (2 \wedge \text{layer-idx}))$ 
      )

definition generic-fri-full-cover-start-low-failure
  :: nat  $\Rightarrow$  'f list  $\Rightarrow$  bool
where
  generic-fri-full-cover-start-low-failure degree-bound candidate-table
     $\longleftrightarrow \text{fri-table-low-degree-on degree-bound eval-domain candidate-table}$ 

definition generic-fri-full-cover-final-low-failure
  :: nat  $\Rightarrow$  'f list  $\Rightarrow$  'f list list  $\Rightarrow$  'f list list  $\Rightarrow$  bool
where
  generic-fri-full-cover-final-low-failure degree-bound challenges
    doms layers  $\longleftrightarrow$ 
       $\neg \text{fri-table-low-degree-on}$ 
        (fri-degree-after (length challenges) degree-bound)
        (doms ! length challenges) (layers ! length challenges)

definition generic-fri-full-cover-final-length-failure
  :: 'f list  $\Rightarrow$  'f list list  $\Rightarrow$  'f list list  $\Rightarrow$  bool
where
  generic-fri-full-cover-final-length-failure challenges doms layers
     $\longleftrightarrow$ 
      length (doms ! length challenges)  $\neq$ 
        length (layers ! length challenges)

lemma fri-final-constant-consistent-low-degree-if-lengths:
  assumes final: fri-final-constant-consistent table final-value
  and len: length table = length fri-dom

```

```

  shows fri-table-low-degree-on d fri-dom table
proof -
  have table: table = map (poly [:final-value:]) fri-dom
proof (rule nth-equalityI)
  show length table = length (map (poly [:final-value:]) fri-dom)
    using len by simp
next
  fix i
  assume i-bound: i < length table
  have table ! i = final-value
    by (rule fri-final-constant-consistent-nth[OF final i-bound])
  then show table ! i = map (poly [:final-value:]) fri-dom ! i
    using i-bound len by simp
qed
show ?thesis
  unfolding fri-table-low-degree-on-def
  using table by (intro exI[where x=[:final-value:]]) simp
qed

lemma generic-fri-final-low-failure-imp-length-failure:
  assumes chain:
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers
  and final-low:
    generic-fri-full-cover-final-low-failure degree-bound challenges
    doms layers
  shows generic-fri-full-cover-final-length-failure challenges doms layers
proof (unfold generic-fri-full-cover-final-length-failure-def, rule notI)
  assume len:
    length (doms ! length challenges) =
    length (layers ! length challenges)
  have final:
    fri-final-constant-consistent (layers ! length challenges) final-value
    by (rule generic-fri-sampled-layer-chain-evidenceD(6)[OF chain])
  have low:
    fri-table-low-degree-on
      (fri-degree-after (length challenges) degree-bound)
      (doms ! length challenges) (layers ! length challenges)
    by (rule fri-final-constant-consistent-low-degree-if-lengths[OF final])
      (rule len[symmetric])
  show False
    using final-low low
    unfolding generic-fri-full-cover-final-low-failure-def by contradiction
qed

```

definition *generic-fri-sampled-full-cover-failure-case*

```

:: ('f list  $\Rightarrow$  bool)  $\Rightarrow$  'f list  $\Rightarrow$  nat  $\Rightarrow$ 
  'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  nat list  $\Rightarrow$ 

```

'f list list list $\Rightarrow$ 'f list list $\Rightarrow$ 'f list list $\Rightarrow$
 (nat $\Rightarrow$ 'f list $\Rightarrow$ 'f set) $\Rightarrow$ bool

where

generic-fri-sampled-full-cover-failure-case low-degree candidate-table
 degree-bound roots challenges final-value query-idxs round-layers
 doms layers bad $\longleftrightarrow$
 generic-fri-full-cover-proximity-failure degree-bound challenges
 doms layers bad $\vee$
 generic-fri-full-cover-index-failure roots query-idxs $\vee$
 generic-fri-full-cover-shape-failure roots challenges doms layers $\vee$
 generic-fri-full-cover-domain-failure roots challenges doms $\vee$
 generic-fri-full-cover-start-low-failure degree-bound candidate-table $\vee$
 generic-fri-full-cover-final-low-failure degree-bound challenges
 doms layers

lemma generic-fri-sampled-full-cover-failure-iff-case:

generic-fri-sampled-full-cover-failure low-degree candidate-table
 degree-bound roots challenges final-value query-idxs round-layers
 doms layers bad $\longleftrightarrow$
 generic-fri-sampled-full-cover-failure-case low-degree candidate-table
 degree-bound roots challenges final-value query-idxs round-layers
 doms layers bad

unfolding generic-fri-sampled-full-cover-failure-def

generic-fri-sampled-full-cover-failure-case-def
 generic-fri-full-cover-proximity-failure-def
 generic-fri-full-cover-index-failure-def
 generic-fri-full-cover-shape-failure-def
 generic-fri-full-cover-domain-failure-def
 generic-fri-full-cover-start-low-failure-def
 generic-fri-full-cover-final-low-failure-def
 fri-table-low-degree-on-def

by auto

lemma generic-fri-sampled-full-cover-failure-cases:

assumes

generic-fri-sampled-full-cover-failure low-degree candidate-table
 degree-bound roots challenges final-value query-idxs round-layers
 doms layers bad

shows

generic-fri-full-cover-proximity-failure degree-bound challenges
 doms layers bad $\vee$
 generic-fri-full-cover-index-failure roots query-idxs $\vee$
 generic-fri-full-cover-shape-failure roots challenges doms layers $\vee$
 generic-fri-full-cover-domain-failure roots challenges doms $\vee$
 generic-fri-full-cover-start-low-failure degree-bound candidate-table $\vee$
 generic-fri-full-cover-final-low-failure degree-bound challenges
 doms layers

using assms

unfolding generic-fri-sampled-full-cover-failure-iff-case

generic-fri-sampled-full-cover-failure-case-def
by *blast*

definition *trace-fri-sampled-full-cover-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-sampled-full-cover-failure s bad out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{ trace-table doms layers.}$
trace-fri-partial-candidate-opening-evidence s out trace-roots
 $\text{ trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{ trace-table} \wedge$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
 $\text{ trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{ fri-query-idxs trace-round-layers doms layers} \wedge$
generic-fri-sampled-full-cover-failure trace-table-low-degree
 $\text{ trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{ fri-query-idxs trace-round-layers doms layers (bad trace-table)})$

definition *trace-fri-sampled-full-cover-failure-case*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-sampled-full-cover-failure-case s bad out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{ trace-table doms layers.}$
trace-fri-partial-candidate-opening-evidence s out trace-roots
 $\text{ trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{ trace-table} \wedge$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
 $\text{ trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{ fri-query-idxs trace-round-layers doms layers} \wedge$
generic-fri-sampled-full-cover-failure-case trace-table-low-degree
 $\text{ trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{ fri-query-idxs trace-round-layers doms layers (bad trace-table)})$

definition *trace-fri-sampled-full-cover-proximity-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(\text{'f list} \Rightarrow \text{nat} \Rightarrow \text{'f list} \Rightarrow \text{'f set}) \Rightarrow$
 $(\text{unit list} \times (\text{'f, 'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers} \wedge$
 $\text{generic-fri-full-cover-proximity-failure (clength - 1) trace-bs}$
 $\text{doms layers (bad trace-table)})$

definition *trace-fri-sampled-full-cover-index-failure*

$:: (\text{'f, 'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f, 'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-sampled-full-cover-index-failure } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers} \wedge$
 $\text{generic-fri-full-cover-index-failure trace-roots fri-query-idxs})$

definition *trace-fri-sampled-full-cover-shape-failure*

$:: (\text{'f, 'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f, 'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-sampled-full-cover-shape-failure } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$

composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers $\wedge$
generic-fri-full-cover-shape-failure trace-roots trace-bs
doms layers)

definition *trace-fri-sampled-full-cover-arithmetic-shape-failure*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-sampled-full-cover-arithmetic-shape-failure s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{ trace-table doms layers.}$
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers $\wedge$
generic-fri-full-cover-arithmetic-shape-failure trace-roots trace-bs)

definition *trace-fri-sampled-full-cover-witness-length-failure*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-sampled-full-cover-witness-length-failure s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{ trace-table doms layers.}$
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers $\wedge$
generic-fri-full-cover-witness-length-failure trace-roots trace-bs
doms layers)

definition *trace-fri-sampled-full-cover-domain-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-sampled-full-cover-domain-failure } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers} \wedge$
 $\text{generic-fri-full-cover-domain-failure trace-roots trace-bs doms})$

definition *trace-fri-sampled-full-cover-start-low-failure*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-sampled-full-cover-start-low-failure } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers} \wedge$
 $\text{generic-fri-full-cover-start-low-failure (clength - 1) trace-table})$

definition *trace-fri-sampled-full-cover-final-low-failure*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-sampled-full-cover-final-low-failure } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$

composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers $\wedge$
generic-fri-full-cover-final-low-failure (clength - 1) trace-bs
doms layers)

definition *trace-fri-sampled-full-cover-final-length-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-sampled-full-cover-final-length-failure s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{ trace-table doms layers.}$
trace-fri-partial-candidate-opening-evidence s out trace-roots
 $\text{ trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{ trace-table} \wedge$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
 $\text{ trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{ fri-query-idxs trace-round-layers doms layers} \wedge$
generic-fri-full-cover-final-length-failure trace-bs doms layers)

definition *composition-fri-sampled-full-cover-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(('f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-failure s bad out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{ opening-composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{ f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{ trace-query-idxs trace-openings composition-query-idxs}$
 $\text{ composition-openings trace-table composition-table doms layers.}$
composition-fri-partial-candidate-opening-evidence s out trace-roots
 $\text{ trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{ composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{ composition-query-idxs composition-openings trace-table}$
 $\text{ composition-table} \wedge$
generic-fri-sampled-layer-chain-evidence
 $(\text{composition-table-low-degree (to-nat fri-dg)}) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$

composition-final fri-query-idxs composition-round-layers doms layers
 $\wedge$
generic-fri-sampled-full-cover-failure
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
(bad fri-dg composition-table))

definition *composition-fri-sampled-full-cover-failure-case*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-failure-case s bad out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{ opening-composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{ f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{ trace-query-idxs trace-openings composition-query-idxs}$
 $\text{ composition-openings trace-table composition-table doms layers.}$
composition-fri-partial-candidate-opening-evidence s out trace-roots
 $\text{ trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{ composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{ composition-query-idxs composition-openings trace-table}$
 $\text{ composition-table} \wedge$
generic-fri-sampled-layer-chain-evidence
 $(\text{composition-table-low-degree (to-nat fri-dg)}) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs composition-round-layers doms layers}$
 $\wedge$
generic-fri-sampled-full-cover-failure-case
 $(\text{composition-table-low-degree (to-nat fri-dg)}) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs composition-round-layers doms layers}$
 $(\text{bad fri-dg composition-table}))$

definition *composition-fri-sampled-full-cover-proximity-failure*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-proximity-failure s bad out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{ opening-composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{ f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{ trace-query-idxs trace-openings composition-query-idxs}$

composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
 $\wedge$
generic-fri-full-cover-proximity-failure (to-nat fri-dg)
composition-bs doms layers (bad fri-dg composition-table))

definition *composition-fri-sampled-full-cover-index-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-index-failure s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
 $\wedge$
generic-fri-full-cover-index-failure opening-composition-roots
fri-query-idxs)

definition *composition-fri-sampled-full-cover-shape-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-shape-failure s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
opening-composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
 $\wedge$
generic-fri-full-cover-shape-failure opening-composition-roots
composition-bs doms layers)

definition *composition-fri-sampled-full-cover-arithmetic-shape-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-arithmetic-shape-failure s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
 $\wedge$
generic-fri-full-cover-arithmetic-shape-failure
opening-composition-roots composition-bs)

definition *composition-fri-sampled-full-cover-witness-length-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-witness-length-failure *s out* $\longleftrightarrow$
 $(\exists$ *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence *s out trace-roots*
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
 $\wedge$
generic-fri-full-cover-witness-length-failure
opening-composition-roots composition-bs doms layers)

definition *composition-fri-sampled-full-cover-domain-failure*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-domain-failure *s out* $\longleftrightarrow$
 $(\exists$ *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence *s out trace-roots*
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
 $\wedge$
generic-fri-full-cover-domain-failure *opening-composition-roots*
composition-bs doms)

definition *composition-fri-sampled-full-cover-start-low-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-start-low-failure $s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{ opening-composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{ f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{ trace-query-idxs trace-openings composition-query-idxs}$
 $\text{ composition-openings trace-table composition-table doms layers.}$
 $\text{ composition-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{ trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{ composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{ composition-query-idxs composition-openings trace-table}$
 $\text{ composition-table} \wedge$
 $\text{ generic-fri-sampled-layer-chain-evidence}$
 $(\text{ composition-table-low-degree } (\text{ to-nat fri-dg})) \text{ composition-table}$
 $(\text{ to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs composition-round-layers doms layers}$
 $\wedge$
 $\text{ generic-fri-full-cover-start-low-failure } (\text{ to-nat fri-dg})$
 $\text{ composition-table})$

definition *composition-fri-sampled-full-cover-final-low-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-final-low-failure $s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{ opening-composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{ f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{ trace-query-idxs trace-openings composition-query-idxs}$
 $\text{ composition-openings trace-table composition-table doms layers.}$
 $\text{ composition-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{ trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{ composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{ composition-query-idxs composition-openings trace-table}$
 $\text{ composition-table} \wedge$
 $\text{ generic-fri-sampled-layer-chain-evidence}$
 $(\text{ composition-table-low-degree } (\text{ to-nat fri-dg})) \text{ composition-table}$
 $(\text{ to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs composition-round-layers doms layers}$

$\wedge$
generic-fri-full-cover-final-low-failure (*to-nat fri-dg*)
composition-bs doms layers)

definition *composition-fri-sampled-full-cover-final-length-failure*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-final-length-failure *s out* $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{ opening-composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{ f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{ trace-query-idxs trace-openings composition-query-idxs}$
 $\text{ composition-openings trace-table composition-table doms layers.}$
 $\text{ composition-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{ trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{ composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{ composition-query-idxs composition-openings trace-table}$
 $\text{ composition-table } \wedge$
 $\text{ generic-fri-sampled-layer-chain-evidence}$
 $\text{ (composition-table-low-degree (to-nat fri-dg)) composition-table}$
 $\text{ (to-nat fri-dg) opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs composition-round-layers doms layers}$
 $\wedge$
 $\text{ generic-fri-full-cover-final-length-failure composition-bs doms}$
 $\text{ layers})$

lemma *trace-fri-sampled-full-cover-failure-iff-case*:
 $\text{ trace-fri-sampled-full-cover-failure } s \text{ bad out} \longleftrightarrow$
 $\text{ trace-fri-sampled-full-cover-failure-case } s \text{ bad out}$
unfolding *trace-fri-sampled-full-cover-failure-def*
trace-fri-sampled-full-cover-failure-case-def
by (*simp only: generic-fri-sampled-full-cover-failure-iff-case*)

lemma *composition-fri-sampled-full-cover-failure-iff-case*:
 $\text{ composition-fri-sampled-full-cover-failure } s \text{ bad out} \longleftrightarrow$
 $\text{ composition-fri-sampled-full-cover-failure-case } s \text{ bad out}$
unfolding *composition-fri-sampled-full-cover-failure-def*
composition-fri-sampled-full-cover-failure-case-def
by (*simp only: generic-fri-sampled-full-cover-failure-iff-case*)

lemma *trace-fri-sampled-full-cover-failure-case-split*:
assumes *trace-fri-sampled-full-cover-failure-case* *s bad out*
shows
 $\text{ trace-fri-sampled-full-cover-proximity-failure } s \text{ bad out} \vee$
 $\text{ trace-fri-sampled-full-cover-index-failure } s \text{ out} \vee$

trace-fri-sampled-full-cover-shape-failure s out $\vee$
trace-fri-sampled-full-cover-domain-failure s out $\vee$
trace-fri-sampled-full-cover-start-low-failure s out $\vee$
trace-fri-sampled-full-cover-final-low-failure s out
using *assms*
unfolding *trace-fri-sampled-full-cover-failure-case-def*
trace-fri-sampled-full-cover-proximity-failure-def
trace-fri-sampled-full-cover-index-failure-def
trace-fri-sampled-full-cover-shape-failure-def
trace-fri-sampled-full-cover-domain-failure-def
trace-fri-sampled-full-cover-start-low-failure-def
trace-fri-sampled-full-cover-final-low-failure-def
generic-fri-sampled-full-cover-failure-case-def
by *meson*

lemma *composition-fri-sampled-full-cover-failure-case-split:*
assumes *composition-fri-sampled-full-cover-failure-case s bad out*
shows
composition-fri-sampled-full-cover-proximity-failure s bad out $\vee$
composition-fri-sampled-full-cover-index-failure s out $\vee$
composition-fri-sampled-full-cover-shape-failure s out $\vee$
composition-fri-sampled-full-cover-domain-failure s out $\vee$
composition-fri-sampled-full-cover-start-low-failure s out $\vee$
composition-fri-sampled-full-cover-final-low-failure s out
using *assms*
unfolding *composition-fri-sampled-full-cover-failure-case-def*
composition-fri-sampled-full-cover-proximity-failure-def
composition-fri-sampled-full-cover-index-failure-def
composition-fri-sampled-full-cover-shape-failure-def
composition-fri-sampled-full-cover-domain-failure-def
composition-fri-sampled-full-cover-start-low-failure-def
composition-fri-sampled-full-cover-final-low-failure-def
generic-fri-sampled-full-cover-failure-case-def
by *meson*

lemma *trace-fri-sampled-full-cover-shape-failure-split:*
assumes *trace-fri-sampled-full-cover-shape-failure s out*
shows
trace-fri-sampled-full-cover-arithmetic-shape-failure s out $\vee$
trace-fri-sampled-full-cover-witness-length-failure s out
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table doms layers
where *evidence:*
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers

composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and chain:
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers
and shape:
generic-fri-full-cover-shape-failure trace-roots trace-bs
doms layers
unfolding *trace-fri-sampled-full-cover-shape-failure-def* **by** *blast*
have *split:*
generic-fri-full-cover-arithmetic-shape-failure trace-roots trace-bs $\vee$
generic-fri-full-cover-witness-length-failure trace-roots trace-bs
doms layers
by (rule *generic-fri-full-cover-shape-failure-split*[*OF shape*])
then show *?thesis*
unfolding *trace-fri-sampled-full-cover-arithmetic-shape-failure-def*
trace-fri-sampled-full-cover-witness-length-failure-def
using *evidence chain* **by** *meson*
qed

lemma *composition-fri-sampled-full-cover-shape-failure-split:*
assumes *composition-fri-sampled-full-cover-shape-failure s out*
shows
composition-fri-sampled-full-cover-arithmetic-shape-failure s out $\vee$
composition-fri-sampled-full-cover-witness-length-failure s out
proof —
from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers
where *evidence:*
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and chain:
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
and shape:
generic-fri-full-cover-shape-failure opening-composition-roots
composition-bs doms layers

unfolding *composition-fri-sampled-full-cover-shape-failure-def*
by *meson*
have *split*:
generic-fri-full-cover-arithmetic-shape-failure
opening-composition-roots composition-bs $\vee$
generic-fri-full-cover-witness-length-failure
opening-composition-roots composition-bs doms layers
by (rule *generic-fri-full-cover-shape-failure-split*[*OF shape*])
then show *?thesis*
unfolding
composition-fri-sampled-full-cover-arithmetic-shape-failure-def
composition-fri-sampled-full-cover-witness-length-failure-def
using *evidence chain* **by** *meson*
qed

lemma *wp-trace-fri-sampled-full-cover-shape-bound-from-split*:
assumes *arithmetic*:
wp-event verify-monad
(trace-fri-sampled-full-cover-arithmetic-shape-failure s) s $\leq$ A
and *witness*:
wp-event verify-monad
(trace-fri-sampled-full-cover-witness-length-failure s) s $\leq$ W
shows
wp-event verify-monad
(trace-fri-sampled-full-cover-shape-failure s) s $\leq$ A + W
proof –
have *wp-event verify-monad*
(trace-fri-sampled-full-cover-shape-failure s) s
 $\leq$ *wp-event verify-monad*
(λ out.
trace-fri-sampled-full-cover-arithmetic-shape-failure s out $\vee$
trace-fri-sampled-full-cover-witness-length-failure s out) s
by (rule *wp-event-mono*)
(rule *trace-fri-sampled-full-cover-shape-failure-split*)
also have $\dots \leq A + W$
by (rule *order-trans*[*OF wp-event-union-bound*
add-mono[*OF arithmetic witness*]])
finally show *?thesis* .
qed

lemma *wp-composition-fri-sampled-full-cover-shape-bound-from-split*:
assumes *arithmetic*:
wp-event verify-monad
(composition-fri-sampled-full-cover-arithmetic-shape-failure s) s
 $\leq A$
and *witness*:
wp-event verify-monad
(composition-fri-sampled-full-cover-witness-length-failure s) s
 $\leq W$

shows
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-sampled-full-cover-shape-failure } s) \ s \leq A + W$
proof –
have $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-sampled-full-cover-shape-failure } s) \ s$
 $\leq wp\text{-event } verify\text{-monad}$
 $(\lambda out.$
 $composition\text{-fri-sampled-full-cover-arithmetic-shape-failure } s \ out \ \vee$
 $composition\text{-fri-sampled-full-cover-witness-length-failure } s \ out) \ s$
by $(rule \ wp\text{-event-mono})$
 $(rule \ composition\text{-fri-sampled-full-cover-shape-failure-split})$
also have $\dots \leq A + W$
by $(rule \ order\text{-trans}[OF \ wp\text{-event-union-bound}$
 $add\text{-mono}[OF \ arithmetic \ witness]])$
finally show $?thesis$.
qed

lemma $wp\text{-event-six-union-bound}$:
assumes $A: wp\text{-event } m \ A \ s \leq a$
and $B: wp\text{-event } m \ B \ s \leq b$
and $C: wp\text{-event } m \ C \ s \leq c$
and $D: wp\text{-event } m \ D \ s \leq d$
and $E: wp\text{-event } m \ E \ s \leq e$
and $F: wp\text{-event } m \ F \ s \leq f$
shows $wp\text{-event } m$
 $(\lambda x. A \ x \ \vee \ B \ x \ \vee \ C \ x \ \vee \ D \ x \ \vee \ E \ x \ \vee \ F \ x) \ s$
 $\leq a + b + c + d + e + f$
proof –
have EF :
 $wp\text{-event } m \ (\lambda x. E \ x \ \vee \ F \ x) \ s \leq e + f$
by $(rule \ order\text{-trans}[OF \ wp\text{-event-union-bound} \ add\text{-mono}[OF \ E \ F]])$
have DEF :
 $wp\text{-event } m \ (\lambda x. D \ x \ \vee \ E \ x \ \vee \ F \ x) \ s \leq d + e + f$
proof –
have $wp\text{-event } m \ (\lambda x. D \ x \ \vee \ E \ x \ \vee \ F \ x) \ s$
 $\leq wp\text{-event } m \ D \ s + wp\text{-event } m \ (\lambda x. E \ x \ \vee \ F \ x) \ s$
by $(rule \ wp\text{-event-union-bound})$
also have $\dots \leq d + (e + f)$
by $(rule \ add\text{-mono}[OF \ D \ EF])$
finally show $?thesis$
by $(simp \ add: \ add.assoc)$
qed
have $CDEF$:
 $wp\text{-event } m \ (\lambda x. C \ x \ \vee \ D \ x \ \vee \ E \ x \ \vee \ F \ x) \ s$
 $\leq c + d + e + f$
proof –
have $wp\text{-event } m \ (\lambda x. C \ x \ \vee \ D \ x \ \vee \ E \ x \ \vee \ F \ x) \ s$
 $\leq wp\text{-event } m \ C \ s + wp\text{-event } m \ (\lambda x. D \ x \ \vee \ E \ x \ \vee \ F \ x) \ s$

```

    by (rule wp-event-union-bound)
  also have ... ≤ c + (d + e + f)
    by (rule add-mono[OF C DEF])
  finally show ?thesis
    by (simp add: add.assoc)
qed
have BCDEF:
  wp-event m (λx. B x ∨ C x ∨ D x ∨ E x ∨ F x) s
    ≤ b + c + d + e + f
proof -
  have wp-event m (λx. B x ∨ C x ∨ D x ∨ E x ∨ F x) s
    ≤ wp-event m B s +
      wp-event m (λx. C x ∨ D x ∨ E x ∨ F x) s
    by (rule wp-event-union-bound)
  also have ... ≤ b + (c + d + e + f)
    by (rule add-mono[OF B CDEF])
  finally show ?thesis
    by (simp add: add.assoc)
qed
have wp-event m
  (λx. A x ∨ B x ∨ C x ∨ D x ∨ E x ∨ F x) s
    ≤ wp-event m A s +
      wp-event m (λx. B x ∨ C x ∨ D x ∨ E x ∨ F x) s
    by (rule wp-event-union-bound)
  also have ... ≤ a + (b + c + d + e + f)
    by (rule add-mono[OF A BCDEF])
  finally show ?thesis
    by (simp add: add.assoc)
qed

lemma wp-trace-fri-sampled-full-cover-failure-case-union-bound:
  assumes proximity:
    wp-event verify-monad
      (trace-fri-sampled-full-cover-proximity-failure s bad) s ≤ P
  and index:
    wp-event verify-monad
      (trace-fri-sampled-full-cover-index-failure s) s ≤ I
  and shape:
    wp-event verify-monad
      (trace-fri-sampled-full-cover-shape-failure s) s ≤ Sh
  and domain:
    wp-event verify-monad
      (trace-fri-sampled-full-cover-domain-failure s) s ≤ D
  and start-low:
    wp-event verify-monad
      (trace-fri-sampled-full-cover-start-low-failure s) s ≤ SL
  and final-low:
    wp-event verify-monad
      (trace-fri-sampled-full-cover-final-low-failure s) s ≤ FL

```

shows

$$\begin{aligned} & \text{wp-event verify-monad} \\ & (\text{trace-fri-sampled-full-cover-failure-case } s \text{ bad}) \ s \\ & \leq P + I + Sh + D + SL + FL \end{aligned}$$

proof –

have *step1*:

$$\begin{aligned} & \text{wp-event verify-monad} \\ & (\text{trace-fri-sampled-full-cover-failure-case } s \text{ bad}) \ s \\ & \leq \\ & \text{wp-event verify-monad} \\ & (\lambda \text{out. trace-fri-sampled-full-cover-proximity-failure } s \text{ bad out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-index-failure } s \text{ out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-shape-failure } s \text{ out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-domain-failure } s \text{ out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-start-low-failure } s \text{ out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-final-low-failure } s \text{ out}) \ s \\ & \text{by (rule wp-event-mono)} \\ & (\text{rule trace-fri-sampled-full-cover-failure-case-split}) \end{aligned}$$

have *step2*:

$$\begin{aligned} & \text{wp-event verify-monad} \\ & (\lambda \text{out. trace-fri-sampled-full-cover-proximity-failure } s \text{ bad out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-index-failure } s \text{ out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-shape-failure } s \text{ out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-domain-failure } s \text{ out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-start-low-failure } s \text{ out } \vee \\ & \quad \text{trace-fri-sampled-full-cover-final-low-failure } s \text{ out}) \ s \\ & \leq P + I + Sh + D + SL + FL \\ & \text{by (rule wp-event-six-union-bound} \\ & \quad [\text{OF proximity index shape domain start-low final-low}]) \end{aligned}$$

show *?thesis*

$$\text{by (rule order-trans[OF step1 step2])}$$

qed

lemma *wp-composition-fri-sampled-full-cover-failure-case-union-bound*:
assumes *proximity*:

$$\text{wp-event verify-monad} \\ (\text{composition-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$$
and *index*:

$$\text{wp-event verify-monad} \\ (\text{composition-fri-sampled-full-cover-index-failure } s) \ s \leq I$$
and *shape*:

$$\text{wp-event verify-monad} \\ (\text{composition-fri-sampled-full-cover-shape-failure } s) \ s \leq Sh$$
and *domain*:

$$\text{wp-event verify-monad} \\ (\text{composition-fri-sampled-full-cover-domain-failure } s) \ s \leq D$$
and *start-low*:

$$\text{wp-event verify-monad} \\ (\text{composition-fri-sampled-full-cover-start-low-failure } s) \ s \leq SL$$

and *final-low*:
wp-event verify-monad
 (composition-fri-sampled-full-cover-final-low-failure *s*) *s* ≤ *FL*
shows
wp-event verify-monad
 (composition-fri-sampled-full-cover-failure-case *s bad*) *s*
 ≤ *P* + *I* + *Sh* + *D* + *SL* + *FL*
proof –
have *step1*:
wp-event verify-monad
 (composition-fri-sampled-full-cover-failure-case *s bad*) *s*
 ≤
wp-event verify-monad
 (λ*out*.
 composition-fri-sampled-full-cover-proximity-failure *s bad out* ∨
 composition-fri-sampled-full-cover-index-failure *s out* ∨
 composition-fri-sampled-full-cover-shape-failure *s out* ∨
 composition-fri-sampled-full-cover-domain-failure *s out* ∨
 composition-fri-sampled-full-cover-start-low-failure *s out* ∨
 composition-fri-sampled-full-cover-final-low-failure *s out*) *s*
by (rule *wp-event-mono*)
 (rule *composition-fri-sampled-full-cover-failure-case-split*)
have *step2*:
wp-event verify-monad
 (λ*out*.
 composition-fri-sampled-full-cover-proximity-failure *s bad out* ∨
 composition-fri-sampled-full-cover-index-failure *s out* ∨
 composition-fri-sampled-full-cover-shape-failure *s out* ∨
 composition-fri-sampled-full-cover-domain-failure *s out* ∨
 composition-fri-sampled-full-cover-start-low-failure *s out* ∨
 composition-fri-sampled-full-cover-final-low-failure *s out*) *s*
 ≤ *P* + *I* + *Sh* + *D* + *SL* + *FL*
by (rule *wp-event-six-union-bound*
 [*OF proximity index shape domain start-low final-low*])
show *?thesis*
by (rule *order-trans*[*OF step1 step2*])
qed

lemma *trace-fri-sampled-full-cover-start-low-failure-false*:

¬ *trace-fri-sampled-full-cover-start-low-failure s out*

proof

assume *failure*: *trace-fri-sampled-full-cover-start-low-failure s out*

then obtain *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table doms layers

where *evidence*:

trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs


```

    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
and start-low:
    generic-fri-full-cover-start-low-failure (clength - 1)
    trace-table
unfolding trace-fri-sampled-full-cover-start-low-failure-def by blast
have not-low:
  ¬ fri-table-low-degree-on (clength - 1) eval-domain trace-table
by (rule trace-fri-partial-candidate-opening-evidence-not-low-on-eval-domain
    [OF evidence])
have low:
    fri-table-low-degree-on (clength - 1) eval-domain trace-table
using start-low
unfolding generic-fri-full-cover-start-low-failure-def .
show False
using not-low low by contradiction
qed

lemma wp-trace-fri-sampled-full-cover-start-low-failure-zero:
  wp-event verify-monad
  (trace-fri-sampled-full-cover-start-low-failure s) s ≤ 0
proof -
  have wp-event verify-monad
    (trace-fri-sampled-full-cover-start-low-failure s) s
    ≤ wp-event verify-monad (λ-. False) s
  by (rule wp-event-mono)
    (simp add: trace-fri-sampled-full-cover-start-low-failure-false)
  also have ... = 0
  unfolding wp-event-def wp-def dist-expect-def by simp
  finally show ?thesis .
qed

lemma composition-fri-sampled-full-cover-start-low-failure-false-on-support:
assumes outcome:
  out ∈ set-dist (execute verify-monad s)
and failure:
  composition-fri-sampled-full-cover-start-low-failure s out
shows False
proof -
from failure obtain trace-roots trace-bs trace-final fri-dg
  opening-composition-roots composition-bs composition-final
  fri-query-idxs trace-round-layers composition-round-layers fr
  f-fri-roots f-final as dg composition-fri-roots final
  trace-query-idxs trace-openings composition-query-idxs
  composition-openings trace-table composition-table doms layers
where evidence:
  composition-fri-partial-candidate-opening-evidence s out trace-roots
  trace-bs trace-final fri-dg opening-composition-roots composition-bs

```

```

    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table
  and chain:
    generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers
  and start-low:
    generic-fri-full-cover-start-low-failure (to-nat fri-dg)
    composition-table
  unfolding composition-fri-sampled-full-cover-start-low-failure-def
  by meson
  have fri-opening:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    fri-dg opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  by (rule composition-fri-partial-candidate-opening-evidenceD(1)
    [OF evidence])
  from fri-opening obtain result final-state where
    out-eq: out = Some (result, final-state)
  unfolding accepted-fri-opening-transcript-def by blast
  have outcome':
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using outcome out-eq by simp
  have challenges:
    accepted-fri-challenges s (Some (result, final-state)) trace-bs
    fri-dg composition-bs
  using accepted-fri-opening-transcript-challenges[OF fri-opening]
    out-eq by simp
  have degree-bound: to-nat fri-dg ≤ maxDegree
  by (rule verify-monad-accepted-fri-challenges-degree-bound
    [OF outcome' challenges])
  have not-low:
    ¬ fri-table-low-degree-on (to-nat fri-dg) eval-domain
    composition-table
  by (rule
    composition-fri-partial-candidate-opening-evidence-not-low-declared-on-eval-domain
    [OF evidence degree-bound])
  have low:
    fri-table-low-degree-on (to-nat fri-dg) eval-domain composition-table
  using start-low
  unfolding generic-fri-full-cover-start-low-failure-def .
  show False
  using not-low low by contradiction
qed

```

lemma *wp-composition-fri-sampled-full-cover-start-low-failure-zero*:
wp-event verify-monad
(composition-fri-sampled-full-cover-start-low-failure s) s ≤ 0
proof –
have *wp-event verify-monad*
(composition-fri-sampled-full-cover-start-low-failure s) s
 $\leq$ *wp-event verify-monad* $(\lambda\cdot. \text{False})$ *s*
by *(rule wp-event-mono-on-support)*
(rule composition-fri-sampled-full-cover-start-low-failure-false-on-support)
also have ... = 0
unfolding *wp-event-def wp-def dist-expect-def* **by** *simp*
finally show *?thesis* .
qed

lemma *trace-fri-sampled-full-cover-arithmetic-shape-failure-false*:
assumes *trace-fri-sampled-full-cover-arithmetic-shape-failure s out*
shows *False*
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table doms layers
where *chain*:
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength – 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers
and *arithmetic*:
generic-fri-full-cover-arithmetic-shape-failure trace-roots trace-bs
unfolding *trace-fri-sampled-full-cover-arithmetic-shape-failure-def*
by *meson*
have *partial*:
generic-fri-partial-evidence trace-table-low-degree trace-table
(clength – 1) trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers
by *(rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])*
have *challenges-len*: *length trace-bs = length trace-roots*
using *generic-fri-partial-evidence-shapes(2)[OF partial]* .
have *rounds-eq*: *length trace-bs = ceil-log clength*
using *generic-fri-partial-evidence-shapes(1,2)[OF partial]*
unfolding *fri-round-count-for-degree-bound-def*
using *clength-pos* **by** *simp*
obtain *N* **where** *eval-power*: *clength * scale = 2 ^ N*
using *eval-domain-length-power* **by** *blast*
have *clength-le-eval*: *clength ≤ clength * scale*
using *scale-pos* **by** *simp*
have *rounds-bound*: *length trace-bs ≤ N*
unfolding *rounds-eq*
by *(rule ceil-log-le-power-bound)*
(use clength-le-eval eval-power in simp)

```

have  $\neg$  generic-fri-full-cover-arithmetic-shape-failure
  trace-roots trace-bs
by (rule generic-fri-full-cover-arithmetic-shape-failure-false
  [OF challenges-len eval-power rounds-bound])
then show False
  using arithmetic by contradiction
qed

lemma wp-trace-fri-sampled-full-cover-arithmetic-shape-failure-zero:
  wp-event verify-monad
  (trace-fri-sampled-full-cover-arithmetic-shape-failure s) s  $\leq$  0
proof –
  have wp-event verify-monad
    (trace-fri-sampled-full-cover-arithmetic-shape-failure s) s
     $\leq$  wp-event verify-monad ( $\lambda$ -. False) s
  by (rule wp-event-mono)
    (rule trace-fri-sampled-full-cover-arithmetic-shape-failure-false)
  also have ... = 0
  unfolding wp-event-def wp-def dist-expect-def by simp
  finally show ?thesis .
qed

lemma composition-fri-sampled-full-cover-arithmetic-shape-failure-false-on-support:
  assumes outcome:
    out  $\in$  set-dist (execute verify-monad s)
  and failure:
    composition-fri-sampled-full-cover-arithmetic-shape-failure s out
  shows False
proof –
  from failure obtain trace-roots trace-bs trace-final fri-dg
    opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers fr
    f-fri-roots f-final as dg composition-fri-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table doms layers
  where evidence:
    composition-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final fri-dg opening-composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table
  and chain:
    generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers
  and arithmetic:

```

```

    generic-fri-full-cover-arithmetic-shape-failure
    opening-composition-roots composition-bs
unfolding composition-fri-sampled-full-cover-arithmetic-shape-failure-def
by meson
have partial:
    generic-fri-partial-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
    by (rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])
have challenges-len:
    length composition-bs = length opening-composition-roots
    using generic-fri-partial-evidence-shapes(2)[OF partial] .
have rounds-eq:
    length composition-bs = ceil-log (to-nat fri-dg + 1)
    using generic-fri-partial-evidence-shapes(1,2)[OF partial]
    unfolding fri-round-count-for-degree-bound-def by simp
have fri-opening:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    fri-dg opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
    by (rule composition-fri-partial-candidate-opening-evidenceD(1)
        [OF evidence])
from fri-opening obtain result final-state where
    out-eq: out = Some (result, final-state)
    unfolding accepted-fri-opening-transcript-def by blast
have outcome':
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
    using outcome out-eq by simp
have challenges:
    accepted-fri-challenges s (Some (result, final-state)) trace-bs
    fri-dg composition-bs
    using accepted-fri-opening-transcript-challenges[OF fri-opening]
    out-eq by simp
have degree-bound: to-nat fri-dg ≤ maxDegree
    by (rule verify-monad-accepted-fri-challenges-degree-bound
        [OF outcome' challenges])
have degree-eval-bound: to-nat fri-dg + 1 ≤ clength * scale
    using degree-bound maxDegree-less-eval-domain by linarith
obtain N where eval-power: clength * scale = 2 ^ N
    using eval-domain-length-power by blast
have rounds-bound: length composition-bs ≤ N
    unfolding rounds-eq
    by (rule ceil-log-le-power-bound)
    (use degree-eval-bound eval-power in simp)
have ¬ generic-fri-full-cover-arithmetic-shape-failure
    opening-composition-roots composition-bs
    by (rule generic-fri-full-cover-arithmetic-shape-failure-false
        [OF challenges-len eval-power rounds-bound])

```

then show *False*
using *arithmetic* **by** *contradiction*
qed

lemma *wp-composition-fri-sampled-full-cover-arithmetic-shape-failure-zero:*

wp-event verify-monad

(composition-fri-sampled-full-cover-arithmetic-shape-failure s) s ≤ 0

proof –

have *wp-event verify-monad*

(composition-fri-sampled-full-cover-arithmetic-shape-failure s) s

≤ wp-event verify-monad (λ-. False) s

by *(rule wp-event-mono-on-support)*

(rule

composition-fri-sampled-full-cover-arithmetic-shape-failure-false-on-support)

also have *... = 0*

unfolding *wp-event-def wp-def dist-expect-def* **by** *simp*

finally show *?thesis* .

qed

lemma *wp-trace-fri-sampled-full-cover-shape-bound-from-witness-length:*

assumes *witness:*

wp-event verify-monad

(trace-fri-sampled-full-cover-witness-length-failure s) s ≤ W

shows

wp-event verify-monad

(trace-fri-sampled-full-cover-shape-failure s) s ≤ W

proof –

have *wp-event verify-monad*

(trace-fri-sampled-full-cover-shape-failure s) s ≤ 0 + W

by *(rule wp-trace-fri-sampled-full-cover-shape-bound-from-split*

[OF wp-trace-fri-sampled-full-cover-arithmetic-shape-failure-zero
witness])

then show *?thesis*

by *simp*

qed

lemma *wp-composition-fri-sampled-full-cover-shape-bound-from-witness-length:*

assumes *witness:*

wp-event verify-monad

(composition-fri-sampled-full-cover-witness-length-failure s) s

≤ W

shows

wp-event verify-monad

(composition-fri-sampled-full-cover-shape-failure s) s ≤ W

proof –

have *wp-event verify-monad*

(composition-fri-sampled-full-cover-shape-failure s) s ≤ 0 + W

by *(rule wp-composition-fri-sampled-full-cover-shape-bound-from-split*

[OF

$wp_composition_fri_sampled_full_cover_arithmetic_shape_failure_zero$
 $witness])$
then show $?thesis$
by $simp$
qed

lemma $trace_fri_sampled_full_cover_final_low_imp_length$:
assumes $trace_fri_sampled_full_cover_final_low_failure\ s\ out$
shows $trace_fri_sampled_full_cover_final_length_failure\ s\ out$
proof –
from $assms$ **obtain** $trace_roots\ trace_bs\ trace_final\ dg\ composition_roots$
 $composition_bs\ composition_final\ fri_query_idxs\ trace_round_layers$
 $composition_round_layers\ fr\ candidate_query_idxs\ trace_openings$
 $trace_table\ doms\ layers$
where $evidence$:
 $trace_fri_partial_candidate_opening_evidence\ s\ out\ trace_roots$
 $trace_bs\ trace_final\ dg\ composition_roots\ composition_bs$
 $composition_final\ fri_query_idxs\ trace_round_layers$
 $composition_round_layers\ fr\ candidate_query_idxs\ trace_openings$
 $trace_table$
and $chain$:
 $generic_fri_sampled_layer_chain_evidence\ trace_table_low_degree$
 $trace_table\ (clength - 1)\ trace_roots\ trace_bs\ trace_final$
 $fri_query_idxs\ trace_round_layers\ doms\ layers$
and $final_low$:
 $generic_fri_full_cover_final_low_failure\ (clength - 1)\ trace_bs$
 $doms\ layers$
unfolding $trace_fri_sampled_full_cover_final_low_failure_def$ **by** $blast$
have $length_failure$:
 $generic_fri_full_cover_final_length_failure\ trace_bs\ doms\ layers$
by $(rule\ generic_fri_final_low_failure_imp_length_failure$
 $[OF\ chain\ final_low])$
show $?thesis$
unfolding $trace_fri_sampled_full_cover_final_length_failure_def$
using $evidence\ chain\ length_failure$ **by** $blast$
qed

lemma $composition_fri_sampled_full_cover_final_low_imp_length$:
assumes $composition_fri_sampled_full_cover_final_low_failure\ s\ out$
shows $composition_fri_sampled_full_cover_final_length_failure\ s\ out$
proof –
from $assms$ **obtain** $trace_roots\ trace_bs\ trace_final\ fri_dg$
 $opening_composition_roots\ composition_bs\ composition_final$
 $fri_query_idxs\ trace_round_layers\ composition_round_layers\ fr$
 $f_fri_roots\ f_final\ as\ dg\ composition_fri_roots\ final$
 $trace_query_idxs\ trace_openings\ composition_query_idxs$
 $composition_openings\ trace_table\ composition_table\ doms\ layers$
where $evidence$:
 $composition_fri_partial_candidate_opening_evidence\ s\ out\ trace_roots$

trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and chain:
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
and final-low:
generic-fri-full-cover-final-low-failure (to-nat fri-dg)
composition-bs doms layers
unfolding *composition-fri-sampled-full-cover-final-low-failure-def*
by *meson*
have *length-failure:*
generic-fri-full-cover-final-length-failure composition-bs doms layers
by (rule *generic-fri-final-low-failure-imp-length-failure*
[*OF chain final-low*])
show *?thesis*
unfolding *composition-fri-sampled-full-cover-final-length-failure-def*
using *evidence chain length-failure* **by** *meson*
qed

lemma *wp-trace-fri-sampled-full-cover-final-low-bound-from-length:*
assumes *length-bound:*
wp-event verify-monad
(trace-fri-sampled-full-cover-final-length-failure s) s ≤ L
shows
wp-event verify-monad
(trace-fri-sampled-full-cover-final-low-failure s) s ≤ L
by (rule *order-trans[OF - length-bound]*)
(rule *wp-event-mono,*
rule trace-fri-sampled-full-cover-final-low-imp-length)

lemma *wp-composition-fri-sampled-full-cover-final-low-bound-from-length:*
assumes *length-bound:*
wp-event verify-monad
(composition-fri-sampled-full-cover-final-length-failure s) s ≤ L
shows
wp-event verify-monad
(composition-fri-sampled-full-cover-final-low-failure s) s ≤ L
by (rule *order-trans[OF - length-bound]*)
(rule *wp-event-mono,*
rule composition-fri-sampled-full-cover-final-low-imp-length)

lemma *trace-fri-sampled-full-cover-final-length-imp-shape:*
assumes *trace-fri-sampled-full-cover-final-length-failure s out*

shows *trace-fri-sampled-full-cover-shape-failure s out*
using *assms*
unfolding *trace-fri-sampled-full-cover-final-length-failure-def*
trace-fri-sampled-full-cover-shape-failure-def
generic-fri-full-cover-final-length-failure-def
generic-fri-full-cover-shape-failure-def
by *blast*

lemma *composition-fri-sampled-full-cover-final-length-imp-shape:*
assumes *composition-fri-sampled-full-cover-final-length-failure s out*
shows *composition-fri-sampled-full-cover-shape-failure s out*
using *assms*
unfolding *composition-fri-sampled-full-cover-final-length-failure-def*
composition-fri-sampled-full-cover-shape-failure-def
generic-fri-full-cover-final-length-failure-def
generic-fri-full-cover-shape-failure-def
by *meson*

lemma *trace-fri-sampled-full-cover-final-low-imp-shape:*
assumes *trace-fri-sampled-full-cover-final-low-failure s out*
shows *trace-fri-sampled-full-cover-shape-failure s out*
using *trace-fri-sampled-full-cover-final-low-imp-length[OF assms]*
trace-fri-sampled-full-cover-final-length-imp-shape
by *blast*

lemma *composition-fri-sampled-full-cover-final-low-imp-shape:*
assumes *composition-fri-sampled-full-cover-final-low-failure s out*
shows *composition-fri-sampled-full-cover-shape-failure s out*
using *composition-fri-sampled-full-cover-final-low-imp-length[OF assms]*
composition-fri-sampled-full-cover-final-length-imp-shape
by *blast*

lemma *trace-fri-sampled-full-cover-failure-case-split-no-final:*
assumes *trace-fri-sampled-full-cover-failure-case s bad out*
shows
trace-fri-sampled-full-cover-proximity-failure s bad out $\vee$
trace-fri-sampled-full-cover-index-failure s out $\vee$
trace-fri-sampled-full-cover-shape-failure s out $\vee$
trace-fri-sampled-full-cover-domain-failure s out $\vee$
trace-fri-sampled-full-cover-start-low-failure s out
using *trace-fri-sampled-full-cover-failure-case-split[OF assms]*
trace-fri-sampled-full-cover-final-low-imp-shape
by *blast*

lemma *composition-fri-sampled-full-cover-failure-case-split-no-final:*
assumes *composition-fri-sampled-full-cover-failure-case s bad out*
shows
composition-fri-sampled-full-cover-proximity-failure s bad out $\vee$
composition-fri-sampled-full-cover-index-failure s out $\vee$

composition-fri-sampled-full-cover-shape-failure s out $\vee$
composition-fri-sampled-full-cover-domain-failure s out $\vee$
composition-fri-sampled-full-cover-start-low-failure s out
using *composition-fri-sampled-full-cover-failure-case-split*[*OF assms*]
composition-fri-sampled-full-cover-final-low-imp-shape
by *blast*

lemma *wp-trace-fri-sampled-full-cover-failure-case-union-bound-no-final*:

assumes *proximity*:

wp-event verify-monad
(trace-fri-sampled-full-cover-proximity-failure s bad) s $\leq P$

and *index*:

wp-event verify-monad
(trace-fri-sampled-full-cover-index-failure s) s $\leq I$

and *shape*:

wp-event verify-monad
(trace-fri-sampled-full-cover-shape-failure s) s $\leq Sh$

and *domain*:

wp-event verify-monad
(trace-fri-sampled-full-cover-domain-failure s) s $\leq D$

and *start-low*:

wp-event verify-monad
(trace-fri-sampled-full-cover-start-low-failure s) s $\leq SL$

shows

wp-event verify-monad
(trace-fri-sampled-full-cover-failure-case s bad) s
 $\leq P + I + Sh + D + SL$

proof –

let *?A* = $\lambda out. trace-fri-sampled-full-cover-proximity-failure s bad out$

let *?B* = $\lambda out. trace-fri-sampled-full-cover-index-failure s out$

let *?C* = $\lambda out. trace-fri-sampled-full-cover-shape-failure s out$

let *?D* = $\lambda out. trace-fri-sampled-full-cover-domain-failure s out$

let *?E* = $\lambda out. trace-fri-sampled-full-cover-start-low-failure s out$

have *step1*:

wp-event verify-monad
(trace-fri-sampled-full-cover-failure-case s bad) s
 $\leq wp-event verify-monad$
 $(\lambda out. ?A out \vee ?B out \vee ?C out \vee ?D out \vee ?E out) s$

by (*rule wp-event-mono*)

(*rule trace-fri-sampled-full-cover-failure-case-split-no-final*)

have *four*:

wp-event verify-monad
 $(\lambda out. ?A out \vee ?B out \vee ?C out \vee ?D out) s$
 $\leq P + I + Sh + D$

by (*rule order-trans*[*OF wp-event-union-bound4*

add-mono[*OF add-mono*[*OF add-mono*[*OF proximity index*] *shape*]
domain]])

have *five*:

wp-event verify-monad

```

    (λout. ?A out ∨ ?B out ∨ ?C out ∨ ?D out ∨ ?E out) s
    ≤ P + I + Sh + D + SL
proof –
  have wp-event verify-monad
    (λout. ?A out ∨ ?B out ∨ ?C out ∨ ?D out ∨ ?E out) s
    ≤ wp-event verify-monad
      (λout. ?A out ∨ ?B out ∨ ?C out ∨ ?D out) s +
      wp-event verify-monad ?E s
    by (rule order-trans[OF - wp-event-union-bound])
      (rule wp-event-mono, blast)
  also have ... ≤ (P + I + Sh + D) + SL
    by (rule add-mono[OF four start-low])
  finally show ?thesis
    by (simp add: add.assoc)
qed
show ?thesis
  by (rule order-trans[OF step1 five])
qed

lemma wp-composition-fri-sampled-full-cover-failure-case-union-bound-no-final:
assumes proximity:
  wp-event verify-monad
    (composition-fri-sampled-full-cover-proximity-failure s bad) s ≤ P
and index:
  wp-event verify-monad
    (composition-fri-sampled-full-cover-index-failure s) s ≤ I
and shape:
  wp-event verify-monad
    (composition-fri-sampled-full-cover-shape-failure s) s ≤ Sh
and domain:
  wp-event verify-monad
    (composition-fri-sampled-full-cover-domain-failure s) s ≤ D
and start-low:
  wp-event verify-monad
    (composition-fri-sampled-full-cover-start-low-failure s) s ≤ SL
shows
  wp-event verify-monad
    (composition-fri-sampled-full-cover-failure-case s bad) s
    ≤ P + I + Sh + D + SL
proof –
  let ?A =
    λout. composition-fri-sampled-full-cover-proximity-failure s bad out
  let ?B = λout. composition-fri-sampled-full-cover-index-failure s out
  let ?C = λout. composition-fri-sampled-full-cover-shape-failure s out
  let ?D = λout. composition-fri-sampled-full-cover-domain-failure s out
  let ?E = λout. composition-fri-sampled-full-cover-start-low-failure s out
  have step1:
    wp-event verify-monad
      (composition-fri-sampled-full-cover-failure-case s bad) s

```

```

    ≤ wp-event verify-monad
      (λout. ?A out ∨ ?B out ∨ ?C out ∨ ?D out ∨ ?E out) s
  by (rule wp-event-mono)
    (rule composition-fri-sampled-full-cover-failure-case-split-no-final)
have four:
  wp-event verify-monad
    (λout. ?A out ∨ ?B out ∨ ?C out ∨ ?D out) s
    ≤ P + I + Sh + D
  by (rule order-trans[OF wp-event-union-bound4
    add-mono[OF add-mono[OF add-mono[OF proximity index] shape]
      domain]])
have five:
  wp-event verify-monad
    (λout. ?A out ∨ ?B out ∨ ?C out ∨ ?D out ∨ ?E out) s
    ≤ P + I + Sh + D + SL
proof -
  have wp-event verify-monad
    (λout. ?A out ∨ ?B out ∨ ?C out ∨ ?D out ∨ ?E out) s
    ≤ wp-event verify-monad
      (λout. ?A out ∨ ?B out ∨ ?C out ∨ ?D out) s +
      wp-event verify-monad ?E s
    by (rule order-trans[OF - wp-event-union-bound])
      (rule wp-event-mono, blast)
  also have ... ≤ (P + I + Sh + D) + SL
    by (rule add-mono[OF four start-low])
  finally show ?thesis
    by (simp add: add.assoc)
qed
show ?thesis
  by (rule order-trans[OF step1 five])
qed

lemma generic-fri-sampled-layer-chain-full-cover-or-failure:
  assumes
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers
  shows
    generic-fri-sampled-layer-chain-full-cover low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers bad ∨
    generic-fri-sampled-full-cover-failure low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers bad
  using assms
  unfolding generic-fri-sampled-layer-chain-full-cover-def
    generic-fri-sampled-full-cover-failure-def
  by blast

```

lemma *generic-fri-sampled-layer-chain-missing-full-cover-failure:*
assumes *chain:*
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers
and *no-full:*
 $\neg$ *generic-fri-sampled-layer-chain-full-cover low-degree*
candidate-table degree-bound roots challenges final-value query-idxs
round-layers doms layers bad
shows
generic-fri-sampled-full-cover-failure low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers bad
using *generic-fri-sampled-layer-chain-full-cover-or-failure*[*OF chain*]
no-full **by** *blast*

lemma *trace-fri-sampled-missing-full-cover-imp-failure:*
assumes *trace-fri-sampled-missing-full-cover s bad out*
shows *trace-fri-sampled-full-cover-failure s bad out*
proof —
have *sampled: trace-fri-bad-with-sampled-layer-chain s out*
and *not-full: $\neg$ trace-fri-bad-with-full-cover-layer-chain s bad out*
using *assms unfolding trace-fri-sampled-missing-full-cover-def* **by** *blast+*
from *sampled* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table doms layers
where *evidence:*
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *chain:*
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers
unfolding *trace-fri-bad-with-sampled-layer-chain-def* **by** *blast*
have *no-full:*
 $\neg$ *generic-fri-sampled-layer-chain-full-cover trace-table-low-degree*
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers (bad trace-table)
proof
assume *full:*
generic-fri-sampled-layer-chain-full-cover trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers (bad trace-table)
have *trace-fri-bad-with-full-cover-layer-chain s bad out*
unfolding *trace-fri-bad-with-full-cover-layer-chain-def*

```

    using evidence full by blast
  then show False
    using not-full by contradiction
qed
have failure:
  generic-fri-sampled-full-cover-failure trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers (bad trace-table)
  by (rule generic-fri-sampled-layer-chain-missing-full-cover-failure
    [OF chain no-full])
show ?thesis
  unfolding trace-fri-sampled-full-cover-failure-def
  using evidence chain failure by blast
qed

lemma composition-fri-sampled-missing-full-cover-imp-failure:
  assumes composition-fri-sampled-missing-full-cover s bad out
  shows composition-fri-sampled-full-cover-failure s bad out
proof -
  have sampled: composition-fri-bad-with-sampled-layer-chain s out
  and not-full:
    ¬ composition-fri-bad-with-full-cover-layer-chain s bad out
  using assms unfolding composition-fri-sampled-missing-full-cover-def
  by blast+
  from sampled obtain trace-roots trace-bs trace-final fri-dg
    opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers fr
    f-fri-roots f-final as dg composition-fri-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table doms layers
  where evidence:
    composition-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final fri-dg opening-composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table
  and chain:
    generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers
  unfolding composition-fri-bad-with-sampled-layer-chain-def by auto
  have no-full:
    ¬ generic-fri-sampled-layer-chain-full-cover
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms layers

```

(*bad fri-dg composition-table*)

proof

assume *full*:

generic-fri-sampled-layer-chain-full-cover
 (*composition-table-low-degree (to-nat fri-dg)*) *composition-table*
 (*to-nat fri-dg*) *opening-composition-roots composition-bs*
composition-final fri-query-idxs composition-round-layers doms
layers (bad fri-dg composition-table)

have *composition-fri-bad-with-full-cover-layer-chain s bad out*

unfolding *composition-fri-bad-with-full-cover-layer-chain-def*

using *evidence full* **by** *metis*

then show *False*

using *not-full* **by** *contradiction*

qed

have *failure*:

generic-fri-sampled-full-cover-failure
 (*composition-table-low-degree (to-nat fri-dg)*) *composition-table*
 (*to-nat fri-dg*) *opening-composition-roots composition-bs*
composition-final fri-query-idxs composition-round-layers doms layers
 (*bad fri-dg composition-table*)

by (*rule generic-fri-sampled-layer-chain-missing-full-cover-failure*
 [*OF chain no-full*])

show *?thesis*

unfolding *composition-fri-sampled-full-cover-failure-def*

using *evidence chain failure* **by** *metis*

qed

lemma *wp-trace-fri-sampled-missing-full-cover-bound-from-failure*:

assumes *failure-bound*:

wp-event verify-monad
 (*trace-fri-sampled-full-cover-failure s bad*) $s \leq M$

shows

wp-event verify-monad
 (*trace-fri-sampled-missing-full-cover s bad*) $s \leq M$

by (*rule order-trans[OF - failure-bound]*)
 (*rule wp-event-mono,*
rule trace-fri-sampled-missing-full-cover-imp-failure)

lemma *wp-trace-fri-sampled-missing-full-cover-bound-from-failure-case*:

assumes *failure-bound*:

wp-event verify-monad
 (*trace-fri-sampled-full-cover-failure-case s bad*) $s \leq M$

shows

wp-event verify-monad
 (*trace-fri-sampled-missing-full-cover s bad*) $s \leq M$

by (*rule wp-trace-fri-sampled-missing-full-cover-bound-from-failure*)
 (*rule order-trans[OF - failure-bound],*
rule wp-event-mono,
simp add: trace-fri-sampled-full-cover-failure-iff-case)

lemma *wp-trace-fri-sampled-missing-full-cover-bound-from-failure-subcases:*

assumes *proximity:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$

and *index:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$

and *shape:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-shape-failure } s) \ s \leq Sh$

and *domain:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-domain-failure } s) \ s \leq D$

and *start-low:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-start-low-failure } s) \ s \leq SL$

and *final-low:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-final-low-failure } s) \ s \leq FL$

shows

wp-event verify-monad

$(\text{trace-fri-sampled-missing-full-cover } s \text{ bad}) \ s$
 $\leq P + I + Sh + D + SL + FL$

by (*rule wp-trace-fri-sampled-missing-full-cover-bound-from-failure-case*)

(*rule wp-trace-fri-sampled-full-cover-failure-case-union-bound*
 $[OF \text{ proximity index shape domain start-low final-low}]$)

lemma *wp-trace-fri-sampled-missing-full-cover-bound-from-five-subcases:*

assumes *proximity:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$

and *index:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$

and *shape:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-shape-failure } s) \ s \leq Sh$

and *domain:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-domain-failure } s) \ s \leq D$

and *final-low:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-final-low-failure } s) \ s \leq FL$

shows

wp-event verify-monad

$(\text{trace-fri-sampled-missing-full-cover } s \text{ bad}) \ s$
 $\leq P + I + Sh + D + FL$

proof –

have *six*:
 wp-event verify-monad
 (trace-fri-sampled-missing-full-cover s bad) s
 $\leq P + I + Sh + D + 0 + FL$
by (rule wp-trace-fri-sampled-missing-full-cover-bound-from-failure-subcases
 [OF proximity index shape domain
 wp-trace-fri-sampled-full-cover-start-low-failure-zero final-low])
then show ?thesis
 by simp
qed

lemma wp-trace-fri-sampled-missing-full-cover-bound-from-length-subcases:

assumes proximity:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-proximity-failure s bad) s $\leq P$
and index:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-index-failure s) s $\leq I$
and shape:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-shape-failure s) s $\leq Sh$
and domain:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-domain-failure s) s $\leq D$
and final-length:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-final-length-failure s) s $\leq FL$
shows
 wp-event verify-monad
 (trace-fri-sampled-missing-full-cover s bad) s
 $\leq P + I + Sh + D + FL$
by (rule wp-trace-fri-sampled-missing-full-cover-bound-from-five-subcases
 [OF proximity index shape domain])
 (rule wp-trace-fri-sampled-full-cover-final-low-bound-from-length
 [OF final-length])

lemma wp-trace-fri-sampled-missing-full-cover-bound-from-shape-subcases:

assumes proximity:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-proximity-failure s bad) s $\leq P$
and index:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-index-failure s) s $\leq I$
and shape:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-shape-failure s) s $\leq Sh$
and domain:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-domain-failure s) s $\leq D$

shows

$$\begin{aligned} & \text{wp-event verify-monad} \\ & (\text{trace-fri-sampled-missing-full-cover } s \text{ bad}) \ s \\ & \leq P + I + Sh + D \end{aligned}$$
proof –
have *case-bound*:

$$\begin{aligned} & \text{wp-event verify-monad} \\ & (\text{trace-fri-sampled-full-cover-failure-case } s \text{ bad}) \ s \\ & \leq P + I + Sh + D + 0 \end{aligned}$$
by (rule *wp-trace-fri-sampled-full-cover-failure-case-union-bound-no-final*
 $[OF \text{ proximity index shape domain}$
 $\text{wp-trace-fri-sampled-full-cover-start-low-failure-zero}]$)
have *wp-event verify-monad*

$$\begin{aligned} & (\text{trace-fri-sampled-missing-full-cover } s \text{ bad}) \ s \\ & \leq P + I + Sh + D + 0 \end{aligned}$$
by (rule *wp-trace-fri-sampled-missing-full-cover-bound-from-failure-case*
 $[OF \text{ case-bound}]$)
then show *?thesis*
by *simp*
qed

lemma *wp-trace-fri-sampled-missing-full-cover-bound-from-split-shape-subcases*:
assumes *proximity*:

$$\text{wp-event verify-monad} \quad (\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$$
and *index*:

$$\text{wp-event verify-monad} \quad (\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$$
and *arithmetic-shape*:

$$\text{wp-event verify-monad} \quad (\text{trace-fri-sampled-full-cover-arithmetic-shape-failure } s) \ s \leq A$$
and *witness-length*:

$$\text{wp-event verify-monad} \quad (\text{trace-fri-sampled-full-cover-witness-length-failure } s) \ s \leq W$$
and *domain*:

$$\text{wp-event verify-monad} \quad (\text{trace-fri-sampled-full-cover-domain-failure } s) \ s \leq D$$
shows

$$\begin{aligned} & \text{wp-event verify-monad} \\ & (\text{trace-fri-sampled-missing-full-cover } s \text{ bad}) \ s \\ & \leq P + I + (A + W) + D \end{aligned}$$
by (rule *wp-trace-fri-sampled-missing-full-cover-bound-from-shape-subcases*
 $[OF \text{ proximity index - domain}]$)
(rule *wp-trace-fri-sampled-full-cover-shape-bound-from-split*
 $[OF \text{ arithmetic-shape witness-length}]$)

lemma *wp-trace-fri-sampled-missing-full-cover-bound-from-witness-length-subcases*:
assumes *proximity*:

$$\text{wp-event verify-monad}$$

(trace-fri-sampled-full-cover-proximity-failure s bad) $s \leq P$
and index:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-index-failure s) $s \leq I$
and witness-length:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-witness-length-failure s) $s \leq W$
and domain:
 wp-event verify-monad
 (trace-fri-sampled-full-cover-domain-failure s) $s \leq D$
shows
 wp-event verify-monad
 (trace-fri-sampled-missing-full-cover s bad) s
 $\leq P + I + W + D$
proof –
have wp-event verify-monad
 (trace-fri-sampled-missing-full-cover s bad) s
 $\leq P + I + (0 + W) + D$
by (rule
 wp-trace-fri-sampled-missing-full-cover-bound-from-split-shape-subcases
 [OF proximity index
 wp-trace-fri-sampled-full-cover-arithmetic-shape-failure-zero
 witness-length domain])
then show ?thesis
by simp
qed

lemma wp-composition-fri-sampled-missing-full-cover-bound-from-failure:

assumes failure-bound:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-failure s bad) $s \leq M$
shows
 wp-event verify-monad
 (composition-fri-sampled-missing-full-cover s bad) $s \leq M$
by (rule order-trans[OF - failure-bound])
 (rule wp-event-mono,
 rule composition-fri-sampled-missing-full-cover-imp-failure)

lemma wp-composition-fri-sampled-missing-full-cover-bound-from-failure-case:

assumes failure-bound:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-failure-case s bad) $s \leq M$
shows
 wp-event verify-monad
 (composition-fri-sampled-missing-full-cover s bad) $s \leq M$
by (rule wp-composition-fri-sampled-missing-full-cover-bound-from-failure)
 (rule order-trans[OF - failure-bound],
 rule wp-event-mono,
 simp add: composition-fri-sampled-full-cover-failure-iff-case)

lemma *wp-composition-fri-sampled-missing-full-cover-bound-from-failure-subcases:*

assumes *proximity:*

wp-event verify-monad

(composition-fri-sampled-full-cover-proximity-failure s bad) s ≤ P

and *index:*

wp-event verify-monad

(composition-fri-sampled-full-cover-index-failure s) s ≤ I

and *shape:*

wp-event verify-monad

(composition-fri-sampled-full-cover-shape-failure s) s ≤ Sh

and *domain:*

wp-event verify-monad

(composition-fri-sampled-full-cover-domain-failure s) s ≤ D

and *start-low:*

wp-event verify-monad

(composition-fri-sampled-full-cover-start-low-failure s) s ≤ SL

and *final-low:*

wp-event verify-monad

(composition-fri-sampled-full-cover-final-low-failure s) s ≤ FL

shows

wp-event verify-monad

(composition-fri-sampled-missing-full-cover s bad) s
 $\leq P + I + Sh + D + SL + FL$

by (*rule wp-composition-fri-sampled-missing-full-cover-bound-from-failure-case*)

(*rule wp-composition-fri-sampled-full-cover-failure-case-union-bound*
 $[OF\ proximity\ index\ shape\ domain\ start-low\ final-low]$)

lemma *wp-composition-fri-sampled-missing-full-cover-bound-from-five-subcases:*

assumes *proximity:*

wp-event verify-monad

(composition-fri-sampled-full-cover-proximity-failure s bad) s ≤ P

and *index:*

wp-event verify-monad

(composition-fri-sampled-full-cover-index-failure s) s ≤ I

and *shape:*

wp-event verify-monad

(composition-fri-sampled-full-cover-shape-failure s) s ≤ Sh

and *domain:*

wp-event verify-monad

(composition-fri-sampled-full-cover-domain-failure s) s ≤ D

and *final-low:*

wp-event verify-monad

(composition-fri-sampled-full-cover-final-low-failure s) s ≤ FL

shows

wp-event verify-monad

(composition-fri-sampled-missing-full-cover s bad) s
 $\leq P + I + Sh + D + FL$

proof –

have *six*:
 wp-event verify-monad
 (composition-fri-sampled-missing-full-cover s bad) s
 $\leq P + I + Sh + D + 0 + FL$
by (rule
 wp-composition-fri-sampled-missing-full-cover-bound-from-failure-subcases
 [OF proximity index shape domain
 wp-composition-fri-sampled-full-cover-start-low-failure-zero
 final-low])
then show ?thesis
by simp
qed

lemma wp-composition-fri-sampled-missing-full-cover-bound-from-length-subcases:
assumes proximity:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-proximity-failure s bad) s $\leq P$
and index:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-index-failure s) s $\leq I$
and shape:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-shape-failure s) s $\leq Sh$
and domain:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-domain-failure s) s $\leq D$
and final-length:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-final-length-failure s) s $\leq FL$
shows
 wp-event verify-monad
 (composition-fri-sampled-missing-full-cover s bad) s
 $\leq P + I + Sh + D + FL$
by (rule
 wp-composition-fri-sampled-missing-full-cover-bound-from-five-subcases
 [OF proximity index shape domain])
 (rule wp-composition-fri-sampled-full-cover-final-low-bound-from-length
 [OF final-length])

lemma wp-composition-fri-sampled-missing-full-cover-bound-from-shape-subcases:
assumes proximity:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-proximity-failure s bad) s $\leq P$
and index:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-index-failure s) s $\leq I$
and shape:
 wp-event verify-monad
 (composition-fri-sampled-full-cover-shape-failure s) s $\leq Sh$

and domain:
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-full-cover-domain-failure } s) \ s \leq D$
shows
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-missing-full-cover } s \text{ bad}) \ s$
 $\leq P + I + Sh + D$
proof –
have case-bound:
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-full-cover-failure-case } s \text{ bad}) \ s$
 $\leq P + I + Sh + D + 0$
by (rule
 $wp\text{-composition-fri-sampled-full-cover-failure-case-union-bound-no-final}$
 $[OF \text{ proximity index shape domain}$
 $wp\text{-composition-fri-sampled-full-cover-start-low-failure-zero}]$)
have $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-missing-full-cover } s \text{ bad}) \ s$
 $\leq P + I + Sh + D + 0$
by (rule
 $wp\text{-composition-fri-sampled-missing-full-cover-bound-from-failure-case}$
 $[OF \text{ case-bound}]$)
then show $?thesis$
by $simp$
qed

lemma
 $wp\text{-composition-fri-sampled-missing-full-cover-bound-from-split-shape-subcases:}$
assumes $proximity:$
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$
and $index:$
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-full-cover-index-failure } s) \ s \leq I$
and $arithmetic\text{-shape:}$
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-full-cover-arithmetic-shape-failure } s) \ s$
 $\leq A$
and $witness\text{-length:}$
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-full-cover-witness-length-failure } s) \ s \leq W$
and $domain:$
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-full-cover-domain-failure } s) \ s \leq D$
shows
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-missing-full-cover } s \text{ bad}) \ s$
 $\leq P + I + (A + W) + D$
by (rule

wp-composition-fri-sampled-missing-full-cover-bound-from-shape-subcases
 [OF proximity index - domain])
 (rule *wp-composition-fri-sampled-full-cover-shape-bound-from-split*
 [OF arithmetic-shape witness-length])

lemma

wp-composition-fri-sampled-missing-full-cover-bound-from-witness-length-subcases:

assumes *proximity:*

wp-event verify-monad

(*composition-fri-sampled-full-cover-proximity-failure s bad*) $s \leq P$

and *index:*

wp-event verify-monad

(*composition-fri-sampled-full-cover-index-failure s*) $s \leq I$

and *witness-length:*

wp-event verify-monad

(*composition-fri-sampled-full-cover-witness-length-failure s*) $s \leq W$

and *domain:*

wp-event verify-monad

(*composition-fri-sampled-full-cover-domain-failure s*) $s \leq D$

shows

wp-event verify-monad

(*composition-fri-sampled-missing-full-cover s bad*) s
 $\leq P + I + W + D$

proof –

have *wp-event verify-monad*

(*composition-fri-sampled-missing-full-cover s bad*) s
 $\leq P + I + (0 + W) + D$

by (rule

wp-composition-fri-sampled-missing-full-cover-bound-from-split-shape-subcases
 [OF proximity index
wp-composition-fri-sampled-full-cover-arithmetic-shape-failure-zero
witness-length domain])

then show *?thesis*

by *simp*

qed

end

end

theory *Soundness-FRI-Full-Cover-Lengths*

imports *Soundness-FRI-Full-Cover-Reasons*

begin

Length residuals for sampled full-cover FRI witnesses.

Arithmetic shape failures are impossible, and table-layer length failures follow from sampled fold coverage. This theory factors the remaining witness-length residual into domain-length/final-length obligations plus the already-classified sampled-index cover-

age failure.

context *soundness*
begin

definition *generic-fri-full-cover-domain-length-failure*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

generic-fri-full-cover-domain-length-failure roots challenges doms layers

$\longleftrightarrow$

$\neg (\forall \text{ layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx} \leq$
 $\text{length (doms ! layer-idx)}) \vee$
 $\neg (\forall \text{ layer-idx} < \text{length challenges.}$
 $\text{fri-evidence-layer-len roots layer-idx div } 2 \leq$
 $\text{length (doms ! Suc layer-idx)}) \vee$
 $\text{length (doms ! length challenges)} \neq$
 $\text{length (layers ! length challenges)}$

definition *trace-fri-sampled-full-cover-domain-length-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-sampled-full-cover-domain-length-failure s out $\longleftrightarrow$

$(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence s out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers} \wedge$
 $\text{generic-fri-full-cover-domain-length-failure trace-roots trace-bs}$
 $\text{doms layers})$

definition *composition-fri-sampled-full-cover-domain-length-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-domain-length-failure s out

$\longleftrightarrow$

$(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{opening-composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$

composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
 $\wedge$
generic-fri-full-cover-domain-length-failure
opening-composition-roots composition-bs doms layers)

lemma *generic-fri-sampled-layer-chain-next-layer-lengths-if-cover:*

assumes *chain:*

generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers

and *covers:* *fri-sampled-layers-cover-fold-indices roots query-idxs*

and *layer-bound:* *layer-idx < length challenges*

shows

fri-evidence-layer-len roots layer-idx div 2 $\leq$
length (layers ! Suc layer-idx)

proof (rule ccontr)

assume *not-len:*

$\neg$ *fri-evidence-layer-len roots layer-idx div 2 $\leq$*
length (layers ! Suc layer-idx)

let *?idx = length (layers ! Suc layer-idx)*

have *idx-bound:*

?idx < fri-evidence-layer-len roots layer-idx div 2

using *not-len by simp*

have *partial:*

generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers

by (rule *generic-fri-sampled-layer-chain-evidenceD(1)[OF chain]*)

have *roots-len:* *length challenges = length roots*

by (rule *generic-fri-partial-evidence-shapes(2)[OF partial]*)

obtain *round-idx where* *round-bound: round-idx < length query-idxs*

and *next-eq:*

fri-evidence-next-idx roots query-idxs round-idx layer-idx = ?idx

by (rule *fri-sampled-layers-cover-fold-indicesD*
 $[OF covers - idx-bound]$)

(use *layer-bound roots-len in simp*)

have *?idx < length (layers ! Suc layer-idx)*

using *generic-fri-sampled-layer-chain-evidence-sample(1)*
 $[OF chain round-bound layer-bound]$

```

    next-eq by simp
  then show False by simp
qed

lemma generic-fri-sampled-layer-chain-layer-lengths-if-cover:
  assumes chain:
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers
  and covers: fri-sampled-layers-cover-fold-indices roots query-idxs
  and base-len: length candidate-table = clength * scale
  and layer-bound: layer-idx < length challenges
  shows fri-evidence-layer-len roots layer-idx ≤
    length (layers ! layer-idx)
proof (cases layer-idx)
  case 0
  have partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
    roots challenges final-value query-idxs round-layers
    by (rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])
  have roots-len: length challenges = length roots
    by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
  have roots-nonempty: 0 < length roots
    using layer-bound roots-len unfolding 0 by simp
  then obtain n where roots-len-suc: length roots = Suc n
    by (cases length roots) auto
  have len0: fri-evidence-layer-len roots 0 = clength * scale
    using roots-len-suc unfolding 0 fri-evidence-layer-len-def by simp
  have layer0: layers ! 0 = candidate-table
    by (rule generic-fri-sampled-layer-chain-evidenceD(5)[OF chain])
  show ?thesis
    using len0 layer0 base-len unfolding 0 by simp
next
  case (Suc j)
  have partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
    roots challenges final-value query-idxs round-layers
    by (rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])
  have roots-len: length challenges = length roots
    by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
  have j-bound: j < length challenges
    using layer-bound unfolding Suc by simp
  have len-suc:
    fri-evidence-layer-len roots (Suc j) =
      fri-evidence-layer-len roots j div 2
    using fri-layer-lengths-Suc-nth[of j length roots clength * scale]
    layer-bound roots-len
    unfolding fri-evidence-layer-len-def Suc by simp
  have next-len:

```

```

    fri-evidence-layer-len roots  $j \text{ div } 2 \leq \text{length } (\text{layers } ! \text{ Suc } j)$ 
  by (rule generic-fri-sampled-layer-chain-next-layer-lengths-if-cover
      [OF chain covers j-bound])
  show ?thesis
    using len-suc next-len unfolding Suc by simp
qed

lemma generic-fri-witness-length-failure-imp-domain-length-if-cover:
  assumes chain:
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
      degree-bound roots challenges final-value query-idxs round-layers
      doms layers
  and no-index:
     $\neg$  generic-fri-full-cover-index-failure roots query-idxs
  and base-len:  $\text{length candidate-table} = \text{clength} * \text{scale}$ 
  and witness:
    generic-fri-full-cover-witness-length-failure roots challenges
      doms layers
  shows generic-fri-full-cover-domain-length-failure roots challenges
    doms layers
proof -
  have covers: fri-sampled-layers-cover-fold-indices roots query-idxs
    using no-index unfolding generic-fri-full-cover-index-failure-def by simp
  have layer-len:
     $\bigwedge \text{layer-idx. layer-idx} < \text{length challenges} \implies$ 
    fri-evidence-layer-len roots layer-idx  $\leq$ 
     $\text{length } (\text{layers } ! \text{ layer-idx})$ 
  by (rule generic-fri-sampled-layer-chain-layer-lengths-if-cover
      [OF chain covers base-len])
  have next-layer-len:
     $\bigwedge \text{layer-idx. layer-idx} < \text{length challenges} \implies$ 
    fri-evidence-layer-len roots layer-idx  $\text{div } 2 \leq$ 
     $\text{length } (\text{layers } ! \text{ Suc layer-idx})$ 
  by (rule generic-fri-sampled-layer-chain-next-layer-lengths-if-cover
      [OF chain covers])
  show ?thesis
    using witness layer-len next-layer-len
    unfolding generic-fri-full-cover-witness-length-failure-def
      generic-fri-full-cover-domain-length-failure-def
    by blast
qed

lemma trace-fri-sampled-full-cover-witness-length-imp-index-or-domain-length:
  assumes witness:
    trace-fri-sampled-full-cover-witness-length-failure s out
  shows
    trace-fri-sampled-full-cover-index-failure s out  $\vee$ 
    trace-fri-sampled-full-cover-domain-length-failure s out
proof -

```

```

from witness obtain trace-roots trace-bs trace-final dg composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table doms layers
where evidence:
  trace-fri-partial-candidate-opening-evidence s out trace-roots
  trace-bs trace-final dg composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr candidate-query-idxs trace-openings
  trace-table
and chain:
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
  trace-table (clength - 1) trace-roots trace-bs trace-final
  fri-query-idxs trace-round-layers doms layers
and generic-witness:
  generic-fri-full-cover-witness-length-failure trace-roots trace-bs
  doms layers
unfolding trace-fri-sampled-full-cover-witness-length-failure-def
by blast
show ?thesis
proof (cases
  generic-fri-full-cover-index-failure trace-roots fri-query-idxs)
case True
then show ?thesis
  unfolding trace-fri-sampled-full-cover-index-failure-def
  using evidence chain by blast
next
case False
have partial:
  trace-fri-partial-candidate-evidence s out fr candidate-query-idxs
  trace-openings trace-table
by (rule trace-fri-partial-candidate-opening-evidenceD(2)
  [OF evidence])
have cand:
  partial-trace-table-candidate trace-table trace-openings
by (rule trace-fri-partial-candidate-evidenceD(2)[OF partial])
have base-len: length trace-table = clength * scale
using cand unfolding partial-trace-table-candidate-def by simp
have domain-length:
  generic-fri-full-cover-domain-length-failure trace-roots trace-bs
  doms layers
by (rule generic-fri-witness-length-failure-imp-domain-length-if-cover
  [OF chain False base-len generic-witness])
then show ?thesis
  unfolding trace-fri-sampled-full-cover-domain-length-failure-def
  using evidence chain by blast
qed
qed

```

lemma *composition-fri-sampled-full-cover-witness-length-imp-index-or-domain-length:*

assumes *witness:*

composition-fri-sampled-full-cover-witness-length-failure s out

shows

composition-fri-sampled-full-cover-index-failure s out $\vee$

composition-fri-sampled-full-cover-domain-length-failure s out

proof —

from *witness* **obtain** *trace-roots trace-bs trace-final fri-dg*

opening-composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers fr

f-fri-roots f-final as dg composition-fri-roots final

trace-query-idxs trace-openings composition-query-idxs

composition-openings trace-table composition-table doms layers

where *evidence:*

composition-fri-partial-candidate-opening-evidence s out trace-roots

trace-bs trace-final fri-dg opening-composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr f-fri-roots f-final as dg

composition-fri-roots final trace-query-idxs trace-openings

composition-query-idxs composition-openings trace-table

composition-table

and *chain:*

generic-fri-sampled-layer-chain-evidence

(composition-table-low-degree (to-nat fri-dg)) composition-table

(to-nat fri-dg) opening-composition-roots composition-bs

composition-final fri-query-idxs composition-round-layers doms layers

and *generic-witness:*

generic-fri-full-cover-witness-length-failure

opening-composition-roots composition-bs doms layers

unfolding *composition-fri-sampled-full-cover-witness-length-failure-def*

by *meson*

show *?thesis*

proof (*cases*

generic-fri-full-cover-index-failure opening-composition-roots

fri-query-idxs)

case *True*

then show *?thesis*

unfolding *composition-fri-sampled-full-cover-index-failure-def*

using *evidence chain* **by** *meson*

next

case *False*

have *partial:*

composition-fri-partial-candidate-evidence s out fr f-fri-roots

f-final as dg composition-fri-roots final trace-query-idxs

trace-openings composition-query-idxs composition-openings

trace-table composition-table

by (*rule composition-fri-partial-candidate-opening-evidenceD(2)*

[OF evidence])

have *cand:*

```

    partial-composition-table-candidate composition-table
    composition-openings
  by (rule composition-fri-partial-candidate-evidenceD(3)[OF partial])
have base-len: length composition-table = clength * scale
  using cand unfolding partial-composition-table-candidate-def by simp
have domain-length:
  generic-fri-full-cover-domain-length-failure
  opening-composition-roots composition-bs doms layers
  by (rule generic-fri-witness-length-failure-imp-domain-length-if-cover
    [OF chain False base-len generic-witness])
then show ?thesis
  unfolding composition-fri-sampled-full-cover-domain-length-failure-def
  using evidence chain by meson
qed
qed

lemma wp-trace-fri-sampled-full-cover-witness-length-bound-from-index-or-domain-length:
  assumes index:
    wp-event verify-monad
    (trace-fri-sampled-full-cover-index-failure s) s ≤ I
  and domain-length:
    wp-event verify-monad
    (trace-fri-sampled-full-cover-domain-length-failure s) s ≤ DL
  shows
    wp-event verify-monad
    (trace-fri-sampled-full-cover-witness-length-failure s) s
    ≤ I + DL
proof -
  have wp-event verify-monad
    (trace-fri-sampled-full-cover-witness-length-failure s) s
    ≤ wp-event verify-monad
    (λout. trace-fri-sampled-full-cover-index-failure s out ∨
      trace-fri-sampled-full-cover-domain-length-failure s out) s
  by (rule wp-event-mono)
    (rule trace-fri-sampled-full-cover-witness-length-imp-index-or-domain-length)
  also have ... ≤ I + DL
  by (rule order-trans[OF wp-event-union-bound
    add-mono[OF index domain-length]])
  finally show ?thesis .
qed

lemma wp-composition-fri-sampled-full-cover-witness-length-bound-from-index-or-domain-length:
  assumes index:
    wp-event verify-monad
    (composition-fri-sampled-full-cover-index-failure s) s ≤ I
  and domain-length:
    wp-event verify-monad
    (composition-fri-sampled-full-cover-domain-length-failure s) s ≤ DL
  shows

```

$wp\text{-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-witness-length-failure } s) \ s$
 $\leq I + DL$
proof –
have $wp\text{-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-witness-length-failure } s) \ s$
 $\leq wp\text{-event verify-monad}$
 $(\lambda out. \text{composition-fri-sampled-full-cover-index-failure } s \ out \ \vee$
 $\text{composition-fri-sampled-full-cover-domain-length-failure } s \ out) \ s$
by $(\text{rule } wp\text{-event-mono})$
 $(\text{rule}$
 $\text{composition-fri-sampled-full-cover-witness-length-imp-index-or-domain-length})$
also have $\dots \leq I + DL$
by $(\text{rule } \text{order-trans}[OF \ wp\text{-event-union-bound}$
 $\text{add-mono}[OF \ \text{index domain-length}]])$
finally show $?thesis$.
qed

lemma $wp\text{-trace-fri-sampled-missing-full-cover-bound-from-length-split}$:

assumes $proximity$:

$wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-proximity-failure } s \ \text{bad}) \ s \leq P$

and $index$:

$wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$

and $domain\text{-length}$:

$wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-domain-length-failure } s) \ s \leq DL$

and $domain$:

$wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-domain-failure } s) \ s \leq D$

shows

$wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-missing-full-cover } s \ \text{bad}) \ s$
 $\leq P + I + (I + DL) + D$

by $(\text{rule } wp\text{-trace-fri-sampled-missing-full-cover-bound-from-witness-length-subcases}$
 $[OF \ proximity \ index$
 $wp\text{-trace-fri-sampled-full-cover-witness-length-bound-from-index-or-domain-length}$
 $domain])$
 $(\text{rule } index, \text{ rule } domain\text{-length})$

lemma $wp\text{-composition-fri-sampled-missing-full-cover-bound-from-length-split}$:

assumes $proximity$:

$wp\text{-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-proximity-failure } s \ \text{bad}) \ s \leq P$

and $index$:

$wp\text{-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-index-failure } s) \ s \leq I$

and $domain\text{-length}$:

```

wp-event verify-monad
  (composition-fri-sampled-full-cover-domain-length-failure s) s ≤ DL
and domain:
wp-event verify-monad
  (composition-fri-sampled-full-cover-domain-failure s) s ≤ D
shows
wp-event verify-monad
  (composition-fri-sampled-missing-full-cover s bad) s
  ≤ P + I + (I + DL) + D
by (rule
  wp-composition-fri-sampled-missing-full-cover-bound-from-witness-length-subcases
  [OF proximity index
    wp-composition-fri-sampled-full-cover-witness-length-bound-from-index-or-domain-length
    domain])
  (rule index, rule domain-length)

```

end

end

theory *Soundness-FRI-Full-Cover-Domains*
imports *Soundness-FRI-Full-Cover-Lengths*
begin

Domain residuals for sampled full-cover FRI witnesses.

This theory keeps the remaining domain issue explicitly verifier-sampled: a global full-cover domain failure is routed to missing sampled-index coverage or to sampled domain facts that still need to be bounded/extracted.

context *soundness*
begin

definition *fri-canonical-domain-at* :: nat ⇒ 'f list

where

```

fri-canonical-domain-at layer-idx =
  map (λidx. (h ^ idx * shift) ^ (2 ^ layer-idx))
  [0.. $((clength * scale) \text{ div } 2 ^ \text{layer-idx})$ ]

```

definition *fri-canonical-domains* :: nat ⇒ 'f list list

where

```

fri-canonical-domains n =
  map fri-canonical-domain-at [0.. $Suc\ n$ ]

```

definition *generic-fri-canonical-sampled-layer-chain-evidence*

```

:: ('f list ⇒ bool) ⇒ 'f list ⇒ nat ⇒
  'f list ⇒ 'f list ⇒ 'f ⇒ nat list ⇒
  'f list list list ⇒ 'f list list ⇒ bool

```

where

```

generic-fri-canonical-sampled-layer-chain-evidence low-degree

```


candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers $\longleftrightarrow$
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges)) layers

definition *trace-fri-bad-with-canonical-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-bad-with-canonical-sampled-layer-chain s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table layers.}$
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
generic-fri-canonical-sampled-layer-chain-evidence
trace-table-low-degree trace-table (clength - 1) trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers layers)

definition *composition-fri-bad-with-canonical-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-bad-with-canonical-sampled-layer-chain s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{opening-composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table layers.}$
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-canonical-sampled-layer-chain-evidence
 $(\text{composition-table-low-degree (to-nat fri-dg)}) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs composition-round-layers layers)$

definition *trace-fri-sampled-missing-canonical-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-sampled-missing-canonical-sampled-layer-chain } s \text{ out}$
 $\longleftrightarrow$
 $\text{trace-fri-bad-with-sampled-layer-chain } s \text{ out} \wedge$
 $\neg \text{trace-fri-bad-with-canonical-sampled-layer-chain } s \text{ out}$

definition *composition-fri-sampled-missing-canonical-sampled-layer-chain*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{composition-fri-sampled-missing-canonical-sampled-layer-chain } s \text{ out}$
 $\longleftrightarrow$
 $\text{composition-fri-bad-with-sampled-layer-chain } s \text{ out} \wedge$
 $\neg \text{composition-fri-bad-with-canonical-sampled-layer-chain } s \text{ out}$

definition *generic-fri-canonical-raw-index-failure*
 $:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow \text{bool}$
where
 $\text{generic-fri-canonical-raw-index-failure roots challenges query-idxs}$
 $\longleftrightarrow$
 $(\exists \text{round-idx} < \text{length query-idxs}.$
 $\exists \text{layer-idx} < \text{length challenges}.$
 $\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} \geq$
 $\text{length (fri-canonical-domain-at layer-idx)})$

definition *trace-fri-canonical-raw-index-failure*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-canonical-raw-index-failure } s \text{ out} \longleftrightarrow$
 $(\exists \text{trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table doms layers}.$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers} \wedge$
 $\text{generic-fri-canonical-raw-index-failure trace-roots trace-bs}$
 $\text{fri-query-idxs})$

definition *composition-fri-canonical-raw-index-failure*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{composition-fri-canonical-raw-index-failure } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{ opening-composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{ f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{ trace-query-idxs trace-openings composition-query-idxs}$
 $\text{ composition-openings trace-table composition-table doms layers.}$
 $\text{composition-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{ trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{ composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{ composition-query-idxs composition-openings trace-table}$
 $\text{ composition-table} \wedge$
 $\text{generic-fri-sampled-layer-chain-evidence}$
 $(\text{composition-table-low-degree } (\text{to-nat fri-dg})) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs composition-round-layers doms layers}$
 $\wedge$
 $\text{generic-fri-canonical-raw-index-failure opening-composition-roots}$
 $\text{ composition-bs fri-query-idxs})$

definition *fri-sampled-successor-domains-square*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

$\text{fri-sampled-successor-domains-square roots query-idxs doms} \longleftrightarrow$
 $(\forall \text{ round-idx} < \text{length query-idxs.}$
 $\forall \text{ layer-idx} < \text{length roots.}$
 $\text{doms ! Suc layer-idx !}$
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx} =$
 $(\text{doms ! layer-idx !}$
 $\text{fri-evidence-next-idx roots query-idxs round-idx layer-idx})^2)$

definition *generic-fri-full-cover-sampled-domain-failure*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

$\text{generic-fri-full-cover-sampled-domain-failure roots query-idxs doms}$
 $\longleftrightarrow$
 $\neg \text{fri-sampled-next-domains-align roots query-idxs doms} \vee$
 $\neg \text{fri-sampled-successor-domains-square roots query-idxs doms}$

definition *trace-fri-sampled-full-cover-sampled-domain-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-sampled-full-cover-sampled-domain-failure } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$

composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table doms layers.
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers $\wedge$
generic-fri-full-cover-sampled-domain-failure trace-roots
fri-query-idxs doms)

definition *composition-fri-sampled-full-cover-sampled-domain-failure*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-full-cover-sampled-domain-failure s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers.
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
 $\wedge$
generic-fri-full-cover-sampled-domain-failure
opening-composition-roots fri-query-idxs doms)

lemma *fri-sampled-successor-domains-squareD:*

assumes *fri-sampled-successor-domains-square roots query-idxs doms*
and *round-idx < length query-idxs*
and *layer-idx < length roots*

shows

doms ! Suc layer-idx !
fri-evidence-next-idx roots query-idxs round-idx layer-idx =
(doms ! layer-idx !

fri-evidence-next-idx roots query-idxs round-idx layer-idx)²
using *assms* **unfolding** *fri-sampled-successor-domains-square-def* **by** *blast*

lemma *fri-canonical-domain-at-length*:
 $\text{length } (\text{fri-canonical-domain-at } \text{layer-idx}) =$
 $(\text{clength} * \text{scale}) \text{ div } 2 \wedge \text{layer-idx}$
unfolding *fri-canonical-domain-at-def* **by** *simp*

lemma *fri-canonical-domain-at-nth*:
assumes $\text{idx} < \text{length } (\text{fri-canonical-domain-at } \text{layer-idx})$
shows $\text{fri-canonical-domain-at } \text{layer-idx} ! \text{idx} =$
 $(h \wedge \text{idx} * \text{shift}) \wedge (2 \wedge \text{layer-idx})$
using *assms* **unfolding** *fri-canonical-domain-at-def* **by** *simp*

lemma *fri-canonical-domains-length*:
 $\text{length } (\text{fri-canonical-domains } n) = \text{Suc } n$
unfolding *fri-canonical-domains-def* **by** *simp*

lemma *fri-canonical-domains-nth*:
assumes $\text{layer-idx} \leq n$
shows $\text{fri-canonical-domains } n ! \text{layer-idx} =$
 $\text{fri-canonical-domain-at } \text{layer-idx}$
proof –
show *?thesis*
proof (*cases* $\text{layer-idx} < n$)
case *True*
have *in-left*:
 $\text{layer-idx} < \text{length } (\text{map } \text{fri-canonical-domain-at } [0..<n])$
using *True* **by** *simp*
have *append-nth*:
 $(\text{map } \text{fri-canonical-domain-at } [0..<n] @$
 $[\text{fri-canonical-domain-at } n] ! \text{layer-idx} =$
 $(\text{if } \text{layer-idx} < \text{length } (\text{map } \text{fri-canonical-domain-at } [0..<n])$
 $\text{then } \text{map } \text{fri-canonical-domain-at } [0..<n] ! \text{layer-idx}$
 $\text{else } [\text{fri-canonical-domain-at } n] !$
 $(\text{layer-idx} - \text{length } (\text{map } \text{fri-canonical-domain-at } [0..<n])))$
by (*rule* *nth-append*)
have $\text{fri-canonical-domains } n ! \text{layer-idx} =$
 $\text{map } \text{fri-canonical-domain-at } [0..<n] ! \text{layer-idx}$
unfolding *fri-canonical-domains-def*
using *append-nth in-left* **by** *simp*
also have $\dots = \text{fri-canonical-domain-at } \text{layer-idx}$
using *True* **by** *simp*
finally show *?thesis* .
next
case *False*
then have $\text{layer-idx} = n$
using *assms* **by** *simp*
then have *eq-n*: $\text{layer-idx} = n$.

```

have len-left: length (map fri-canonical-domain-at [0.. $n$ ]) =  $n$ 
  by simp
have append-nth:
  (map fri-canonical-domain-at [0.. $n$ ] @
    [fri-canonical-domain-at  $n$ ]) !  $n$  =
    (if  $n < \text{length (map fri-canonical-domain-at [0.. $n$ ])}$ 
      then map fri-canonical-domain-at [0.. $n$ ] !  $n$ 
      else [fri-canonical-domain-at  $n$ ] !
        ( $n - \text{length (map fri-canonical-domain-at [0.. $n$ ])}$ ))
  by (rule nth-append)
show ?thesis
  unfolding fri-canonical-domains-def
  using eq-n len-left append-nth by simp
qed
qed

lemma fri-canonical-domain-at-0:
  fri-canonical-domain-at 0 = eval-domain
proof (rule nth-equalityI)
  show length (fri-canonical-domain-at 0) = length eval-domain
  by (simp add: fri-canonical-domain-at-length eval-domain-length)
next
  fix  $i$ 
  assume i-bound:  $i < \text{length (fri-canonical-domain-at 0)}$ 
  then have i-eval:  $i < \text{clength} * \text{scale}$ 
  by (simp add: fri-canonical-domain-at-length)
  show fri-canonical-domain-at 0 !  $i$  = eval-domain !  $i$ 
  using fri-canonical-domain-at-nth[OF i-bound] eval-domain-nth[OF i-eval]
  by simp
qed

lemma fri-canonical-domains-0:
  0 < length (fri-canonical-domains  $n$ )
  by (simp add: fri-canonical-domains-length)

lemma fri-canonical-domains-first:
  fri-canonical-domains  $n$  ! 0 = eval-domain
  using fri-canonical-domains-nth[of 0  $n$ ] fri-canonical-domain-at-0
  by simp

lemma fri-sampled-table-fold-to-canonical-domain:
  assumes fold:
    fri-sampled-table-fold  $ch$   $len$   $raw$  ( $2 \wedge \text{layer-idx}$ ) fri-dom  $layer$   $chunk$ 
    next-value
  and raw-bound:  $raw < \text{length (fri-canonical-domain-at layer-idx)}$ 
  shows
    fri-sampled-table-fold  $ch$   $len$   $raw$  ( $2 \wedge \text{layer-idx}$ )
    (fri-canonical-domain-at  $layer-idx$ )  $layer$   $chunk$  next-value
proof —

```

```

from fold obtain xp xp-path xn xn-path where
  chunk: fri-layer-opening-chunk len xp xp-path xn xn-path chunk
and match: fri-opening-matches-table len raw layer xp xn
and dom-row:
  fri-dom ! raw = (h ^ raw * shift) ^ (2 ^ layer-idx)
and next-eq:
  next-value =
    fri-table-fold-value ch layer fri-dom len (2 ^ layer-idx) raw
by (elim fri-sampled-table-foldE)
have canonical-row:
  fri-canonical-domain-at layer-idx ! raw =
    (h ^ raw * shift) ^ (2 ^ layer-idx)
by (rule fri-canonical-domain-at-nth[OF raw-bound])
have fold-eq:
  next-value =
    fri-table-fold-value ch layer (fri-canonical-domain-at layer-idx)
      len (2 ^ layer-idx) raw
using next-eq dom-row canonical-row
unfolding fri-table-fold-value-def by simp
show ?thesis
by (rule fri-sampled-table-foldI
  [OF chunk match canonical-row fold-eq])
qed

lemma generic-fri-sampled-layer-chain-to-canonical-domains:
assumes chain:
  generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers
and raw-bounds:
   $\bigwedge \text{round-idx layer-idx.}$ 
     $\text{round-idx} < \text{length query-idxs} \implies$ 
     $\text{layer-idx} < \text{length challenges} \implies$ 
     $\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} <$ 
     $\text{length (fri-canonical-domain-at layer-idx)}$ 
shows
  generic-fri-canonical-sampled-layer-chain-evidence low-degree
    candidate-table degree-bound roots challenges final-value query-idxs
    round-layers layers
proof –
have partial:
  generic-fri-partial-evidence low-degree candidate-table degree-bound
    roots challenges final-value query-idxs round-layers
by (rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])
have len-layers: length layers = Suc (length challenges)
by (rule generic-fri-sampled-layer-chain-evidenceD(3)[OF chain])
have layer0: layers ! 0 = candidate-table
by (rule generic-fri-sampled-layer-chain-evidenceD(5)[OF chain])
have final:

```

```

  fri-final-constant-consistent (layers ! length challenges)
    final-value
  by (rule generic-fri-sampled-layer-chain-evidenceD(6)[OF chain])
have canonical-chain:
  generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    (fri-canonical-domains (length challenges)) layers
  unfolding generic-fri-sampled-layer-chain-evidence-def
proof (intro conjI allI impI)
  show
    generic-fri-partial-evidence low-degree candidate-table degree-bound
      roots challenges final-value query-idxs round-layers
    by (rule partial)
next
  show length (fri-canonical-domains (length challenges)) =
    Suc (length challenges)
    by (rule fri-canonical-domains-length)
next
  show length layers = Suc (length challenges)
    by (rule len-layers)
next
  show fri-canonical-domains (length challenges) ! 0 = eval-domain
    by (rule fri-canonical-domains-first)
next
  show layers ! 0 = candidate-table
    by (rule layer0)
next
  show
    fri-final-constant-consistent (layers ! length challenges)
      final-value
    by (rule final)
next
  fix round-idx layer-idx
  assume round-bound: round-idx < length query-idxs
  assume layer-bound: layer-idx < length challenges
  show
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
      length (layers ! Suc layer-idx)
    by (rule generic-fri-sampled-layer-chain-evidence-sample(1)
      [OF chain round-bound layer-bound])
next
  fix round-idx layer-idx
  assume round-bound: round-idx < length query-idxs
  assume layer-bound: layer-idx < length challenges
  have old-fold:
    fri-sampled-table-fold (challenges ! layer-idx)
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      (2 ^ layer-idx)

```



```

    (doms ! layer-idx)
    (layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (layers ! Suc layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  by (rule generic-fri-sampled-layer-chain-evidence-sample(2)
    [OF chain round-bound layer-bound])
have canonical-at:
  fri-sampled-table-fold (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-canonical-domain-at layer-idx)
    (layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (layers ! Suc layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  by (rule fri-sampled-table-fold-to-canonical-domain
    [OF old-fold raw-bounds[OF round-bound layer-bound]])
have dom-nth:
  fri-canonical-domains (length challenges) ! layer-idx =
    fri-canonical-domain-at layer-idx
  by (rule fri-canonical-domains-nth) (use layer-bound in simp)
show
  fri-sampled-table-fold (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-canonical-domains (length challenges) ! layer-idx)
    (layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (layers ! Suc layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  using canonical-at dom-nth by simp
qed
show ?thesis
  using canonical-chain
  unfolding generic-fri-canonical-sampled-layer-chain-evidence-def
  by simp
qed

lemma generic-fri-canonical-sampled-layer-chain-evidenceD:
  assumes
    generic-fri-canonical-sampled-layer-chain-evidence low-degree
    candidate-table degree-bound roots challenges final-value query-idxs
    round-layers layers
  shows
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers

```

```

    (fri-canonical-domains (length challenges)) layers
using assms
unfolding generic-fri-canonical-sampled-layer-chain-evidence-def by simp

lemma generic-fri-canonical-sampled-layer-chain-evidenceI:
assumes
  generic-fri-sampled-layer-chain-evidence low-degree candidate-table
  degree-bound roots challenges final-value query-idxs round-layers
  (fri-canonical-domains (length challenges)) layers
shows
  generic-fri-canonical-sampled-layer-chain-evidence low-degree
  candidate-table degree-bound roots challenges final-value query-idxs
  round-layers layers
using assms
unfolding generic-fri-canonical-sampled-layer-chain-evidence-def by simp

lemma generic-fri-sampled-missing-canonical-imp-raw-index-failure:
assumes chain:
  generic-fri-sampled-layer-chain-evidence low-degree candidate-table
  degree-bound roots challenges final-value query-idxs round-layers
  doms layers
and missing:
   $\neg$  generic-fri-canonical-sampled-layer-chain-evidence low-degree
  candidate-table degree-bound roots challenges final-value query-idxs
  round-layers layers
shows generic-fri-canonical-raw-index-failure roots challenges query-idxs
proof (rule ccontr)
assume not-failure:
   $\neg$  generic-fri-canonical-raw-index-failure roots challenges
  query-idxs
have raw-bounds:
   $\bigwedge$  round-idx layer-idx.
    round-idx < length query-idxs  $\implies$ 
    layer-idx < length challenges  $\implies$ 
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
    length (fri-canonical-domain-at layer-idx)
using not-failure
unfolding generic-fri-canonical-raw-index-failure-def by force
have canonical:
  generic-fri-canonical-sampled-layer-chain-evidence low-degree
  candidate-table degree-bound roots challenges final-value query-idxs
  round-layers layers
by (rule generic-fri-sampled-layer-chain-to-canonical-domains
  [OF chain raw-bounds])
show False
using missing canonical by contradiction
qed

lemma trace-fri-missing-canonical-sampled-imp-raw-index-failure:

```

assumes *missing*:
trace-fri-sampled-missing-canonical-sampled-layer-chain s out
shows *trace-fri-canonical-raw-index-failure s out*
proof –
from *missing* **have** *sampled*:
trace-fri-bad-with-sampled-layer-chain s out
and *not-canonical*:
 $\neg$ *trace-fri-bad-with-canonical-sampled-layer-chain s out*
unfolding *trace-fri-sampled-missing-canonical-sampled-layer-chain-def*
by *blast+*
from *sampled* **obtain** *trace-roots trace-bs trace-final dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr candidate-query-idxs
trace-openings trace-table doms layers
where *evidence*:
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *chain*:
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength – 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers
unfolding *trace-fri-bad-with-sampled-layer-chain-def* **by** *blast*
have *generic-missing*:
 $\neg$ *generic-fri-canonical-sampled-layer-chain-evidence*
trace-table-low-degree trace-table (clength – 1) trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers layers
proof
assume *canonical*:
generic-fri-canonical-sampled-layer-chain-evidence
trace-table-low-degree trace-table (clength – 1) trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers layers
then have *trace-fri-bad-with-canonical-sampled-layer-chain s out*
unfolding *trace-fri-bad-with-canonical-sampled-layer-chain-def*
using *evidence* **by** *blast*
then show *False*
using *not-canonical* **by** *contradiction*
qed
have *raw-failure*:
generic-fri-canonical-raw-index-failure trace-roots trace-bs
fri-query-idxs
by (*rule generic-fri-sampled-missing-canonical-imp-raw-index-failure*
[*OF chain generic-missing*])
show *?thesis*
unfolding *trace-fri-canonical-raw-index-failure-def*
using *evidence chain raw-failure* **by** *blast*
qed

lemma *composition-fri-missing-canonical-sampled-imp-raw-index-failure:*
assumes *missing:*
composition-fri-sampled-missing-canonical-sampled-layer-chain s out
shows *composition-fri-canonical-raw-index-failure s out*
proof –
from *missing* **have** *sampled:*
composition-fri-bad-with-sampled-layer-chain s out
and *not-canonical:*
 $\neg$ *composition-fri-bad-with-canonical-sampled-layer-chain s out*
unfolding *composition-fri-sampled-missing-canonical-sampled-layer-chain-def*
by *blast+*
from *sampled* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers
where *evidence:*
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *chain:*
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
unfolding *composition-fri-bad-with-sampled-layer-chain-def* **by** *auto*
have *generic-missing:*
 $\neg$ *generic-fri-canonical-sampled-layer-chain-evidence*
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers layers
proof
assume *canonical:*
generic-fri-canonical-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers layers
then **have** *composition-fri-bad-with-canonical-sampled-layer-chain s out*
unfolding *composition-fri-bad-with-canonical-sampled-layer-chain-def*
using *evidence* **by** *meson*
then **show** *False*
using *not-canonical* **by** *contradiction*
qed

have *raw-failure*:
generic-fri-canonical-raw-index-failure opening-composition-roots
composition-bs fri-query-idxs
by (rule *generic-fri-sampled-missing-canonical-imp-raw-index-failure*
[OF chain generic-missing])
show *?thesis*
unfolding *composition-fri-canonical-raw-index-failure-def*
using *evidence chain raw-failure* **by** *meson*
qed

lemma *wp-trace-fri-missing-canonical-sampled-bound-from-raw-index*:
assumes *raw-index*:
wp-event verify-monad (trace-fri-canonical-raw-index-failure s) s
 $\leq R$
shows
wp-event verify-monad
(trace-fri-sampled-missing-canonical-sampled-layer-chain s) s $\leq R$
proof –
have *wp-event verify-monad*
(trace-fri-sampled-missing-canonical-sampled-layer-chain s) s
 $\leq$ *wp-event verify-monad*
(trace-fri-canonical-raw-index-failure s) s
by (rule *wp-event-mono*)
(rule *trace-fri-missing-canonical-sampled-imp-raw-index-failure*)
also have $\dots \leq R$
by (rule *raw-index*)
finally show *?thesis* .
qed

lemma *wp-composition-fri-missing-canonical-sampled-bound-from-raw-index*:
assumes *raw-index*:
wp-event verify-monad (composition-fri-canonical-raw-index-failure s) s
 $\leq R$
shows
wp-event verify-monad
(composition-fri-sampled-missing-canonical-sampled-layer-chain s) s
 $\leq R$
proof –
have *wp-event verify-monad*
(composition-fri-sampled-missing-canonical-sampled-layer-chain s) s
 $\leq$ *wp-event verify-monad*
(composition-fri-canonical-raw-index-failure s) s
by (rule *wp-event-mono*)
(rule *composition-fri-missing-canonical-sampled-imp-raw-index-failure*)
also have $\dots \leq R$
by (rule *raw-index*)
finally show *?thesis* .
qed

lemma *generic-fri-sampled-layer-chain-no-canonical-raw-index-failure:*
assumes *chain:*
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers
shows $\neg$ *generic-fri-canonical-raw-index-failure roots challenges*
query-idxs
proof
assume *failure:*
generic-fri-canonical-raw-index-failure roots challenges query-idxs
then obtain *round-idx layer-idx* **where** *round-bound:*
round-idx < length query-idxs
and *layer-bound:* *layer-idx < length challenges*
and *raw-ge:*
fri-evidence-layer-idx roots query-idxs round-idx layer-idx $\geq$
length (fri-canonical-domain-at layer-idx)
unfolding *generic-fri-canonical-raw-index-failure-def* **by** *blast*
have *partial:*
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers
by (rule *generic-fri-sampled-layer-chain-evidenceD(1)[OF chain]*)
have *roots-challenges:* *length challenges = length roots*
by (rule *generic-fri-partial-evidence-shapes(2)[OF partial]*)
have *roots-bound:* *layer-idx < length roots*
using *layer-bound roots-challenges* **by** *simp*
have *fold:*
fri-sampled-table-fold (challenges ! layer-idx)
(fri-evidence-layer-len roots layer-idx)
(fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
(2 ^ layer-idx)
(doms ! layer-idx)
(layers ! layer-idx)
(round-layers ! round-idx ! layer-idx)
(layers ! Suc layer-idx !
fri-evidence-next-idx roots query-idxs round-idx layer-idx)
by (rule *generic-fri-sampled-layer-chain-evidence-sample(2)*
[OF chain round-bound layer-bound])
from *fold* **obtain** *xp xp-path xn xn-path* **where**
match:
fri-opening-matches-table (fri-evidence-layer-len roots layer-idx)
(fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
(layers ! layer-idx) xp xn
by (elim *fri-sampled-table-foldE*)
have *raw-lt-len:*
fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
fri-evidence-layer-len roots layer-idx
by (rule *fri-opening-matches-tableD(1)[OF match]*)
have *len-eq:*
fri-evidence-layer-len roots layer-idx =

length (fri-canonical-domain-at layer-idx)
unfolding fri-evidence-layer-len-def fri-canonical-domain-at-length
using fri-layer-lengths-nth-div[OF roots-bound, of clength * scale]
by simp
show False
using raw-lt-len raw-ge **unfolding** len-eq **by** simp
qed

lemma trace-fri-canonical-raw-index-failure-false:

¬ trace-fri-canonical-raw-index-failure s out

proof

assume raw: trace-fri-canonical-raw-index-failure s out

then obtain trace-roots trace-bs trace-final dg composition-roots
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table doms layers

where chain:

generic-fri-sampled-layer-chain-evidence trace-table-low-degree
 trace-table (clength - 1) trace-roots trace-bs trace-final
 fri-query-idxs trace-round-layers doms layers

and failure:

generic-fri-canonical-raw-index-failure trace-roots trace-bs
 fri-query-idxs

unfolding trace-fri-canonical-raw-index-failure-def **by** blast

have ¬ generic-fri-canonical-raw-index-failure trace-roots trace-bs
 fri-query-idxs

by (rule generic-fri-sampled-layer-chain-no-canonical-raw-index-failure
 [OF chain])

then show False

using failure **by** contradiction

qed

lemma composition-fri-canonical-raw-index-failure-false:

¬ composition-fri-canonical-raw-index-failure s out

proof

assume raw: composition-fri-canonical-raw-index-failure s out

then obtain trace-roots trace-bs trace-final fri-dg
 opening-composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers fr
 f-fri-roots f-final as dg composition-fri-roots final
 trace-query-idxs trace-openings composition-query-idxs
 composition-openings trace-table composition-table doms layers

where chain:

generic-fri-sampled-layer-chain-evidence
 (composition-table-low-degree (to-nat fri-dg)) composition-table
 (to-nat fri-dg) opening-composition-roots composition-bs
 composition-final fri-query-idxs composition-round-layers doms layers

and failure:

generic-fri-canonical-raw-index-failure opening-composition-roots

```

    composition-bs fri-query-idxs
  unfolding composition-fri-canonical-raw-index-failure-def by meson
  have  $\neg$  generic-fri-canonical-raw-index-failure
    opening-composition-roots composition-bs fri-query-idxs
  by (rule generic-fri-sampled-layer-chain-no-canonical-raw-index-failure
    [OF chain])
  then show False
    using failure by contradiction
qed

lemma wp-trace-fri-canonical-raw-index-failure-zero:
  wp-event verify-monad (trace-fri-canonical-raw-index-failure s) s  $\leq$  0
proof -
  have wp-event verify-monad (trace-fri-canonical-raw-index-failure s) s
     $\leq$  wp-event verify-monad ( $\lambda$ -. False) s
  by (rule wp-event-mono)
    (simp add: trace-fri-canonical-raw-index-failure-false)
  also have ... = 0
    unfolding wp-event-def wp-def dist-expect-def by simp
  finally show ?thesis .
qed

lemma wp-composition-fri-canonical-raw-index-failure-zero:
  wp-event verify-monad
    (composition-fri-canonical-raw-index-failure s) s  $\leq$  0
proof -
  have wp-event verify-monad
    (composition-fri-canonical-raw-index-failure s) s
     $\leq$  wp-event verify-monad ( $\lambda$ -. False) s
  by (rule wp-event-mono)
    (simp add: composition-fri-canonical-raw-index-failure-false)
  also have ... = 0
    unfolding wp-event-def wp-def dist-expect-def by simp
  finally show ?thesis .
qed

lemma wp-trace-fri-missing-canonical-sampled-zero:
  wp-event verify-monad
    (trace-fri-sampled-missing-canonical-sampled-layer-chain s) s  $\leq$  0
  by (rule wp-trace-fri-missing-canonical-sampled-bound-from-raw-index)
    (rule wp-trace-fri-canonical-raw-index-failure-zero)

lemma wp-composition-fri-missing-canonical-sampled-zero:
  wp-event verify-monad
    (composition-fri-sampled-missing-canonical-sampled-layer-chain s) s
     $\leq$  0
  by (rule wp-composition-fri-missing-canonical-sampled-bound-from-raw-index)
    (rule wp-composition-fri-canonical-raw-index-failure-zero)

```


lemma *trace-fri-bad-with-canonical-sampled-layer-chain-imp-sampled:*
assumes *trace-fri-bad-with-canonical-sampled-layer-chain s out*
shows *trace-fri-bad-with-sampled-layer-chain s out*
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table layers
where *evidence:*
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *chain:*
generic-fri-canonical-sampled-layer-chain-evidence
trace-table-low-degree trace-table (clength – 1) trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers layers
unfolding *trace-fri-bad-with-canonical-sampled-layer-chain-def* **by** *blast*
have *sampled:*
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength – 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers
(fri-canonical-domains (length trace-bs)) layers
by *(rule generic-fri-canonical-sampled-layer-chain-evidenceD[OF chain])*
show *?thesis*
unfolding *trace-fri-bad-with-sampled-layer-chain-def*
using *evidence sampled* **by** *blast*
qed

lemma *composition-fri-bad-with-canonical-sampled-layer-chain-imp-sampled:*
assumes *composition-fri-bad-with-canonical-sampled-layer-chain s out*
shows *composition-fri-bad-with-sampled-layer-chain s out*
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table layers
where *evidence:*
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *chain:*

```

    generic-fri-canonical-sampled-layer-chain-evidence
      (composition-table-low-degree (to-nat fri-dg)) composition-table
      (to-nat fri-dg) opening-composition-roots composition-bs
      composition-final fri-query-idxs composition-round-layers layers
  unfolding composition-fri-bad-with-canonical-sampled-layer-chain-def
  by meson
have sampled:
  generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
    (fri-canonical-domains (length composition-bs)) layers
  by (rule generic-fri-canonical-sampled-layer-chain-evidenceD[OF chain])
show ?thesis
  unfolding composition-fri-bad-with-sampled-layer-chain-def
  using evidence sampled by meson
qed

lemma trace-fri-sampled-layer-chain-canonical-or-missing:
  assumes trace-fri-bad-with-sampled-layer-chain s out
  shows
    trace-fri-bad-with-canonical-sampled-layer-chain s out  $\vee$ 
    trace-fri-sampled-missing-canonical-sampled-layer-chain s out
  using assms
  unfolding trace-fri-sampled-missing-canonical-sampled-layer-chain-def
  by blast

lemma composition-fri-sampled-layer-chain-canonical-or-missing:
  assumes composition-fri-bad-with-sampled-layer-chain s out
  shows
    composition-fri-bad-with-canonical-sampled-layer-chain s out  $\vee$ 
    composition-fri-sampled-missing-canonical-sampled-layer-chain s out
  using assms
  unfolding composition-fri-sampled-missing-canonical-sampled-layer-chain-def
  by blast

lemma wp-trace-fri-sampled-bound-from-canonical-and-missing:
  assumes canonical:
    wp-event verify-monad
      (trace-fri-bad-with-canonical-sampled-layer-chain s)  $s \leq C$ 
  and missing:
    wp-event verify-monad
      (trace-fri-sampled-missing-canonical-sampled-layer-chain s)  $s \leq M$ 
  shows
    wp-event verify-monad (trace-fri-bad-with-sampled-layer-chain s)  $s$ 
     $\leq C + M$ 
proof -
  have wp-event verify-monad (trace-fri-bad-with-sampled-layer-chain s)  $s$ 

```

$\leq$ *wp-event verify-monad*
 $(\lambda out. \text{trace-fri-bad-with-canonical-sampled-layer-chain } s \text{ out} \vee$
 $\text{trace-fri-sampled-missing-canonical-sampled-layer-chain } s \text{ out}) \ s$
by (*rule wp-event-mono*)
 $(\text{rule trace-fri-sampled-layer-chain-canonical-or-missing})$
also have $\dots \leq C + M$
by (*rule order-trans[OF wp-event-union-bound*
 $\text{add-mono[OF canonical missing]]$)
finally show *?thesis* .
qed

lemma *wp-composition-fri-sampled-bound-from-canonical-and-missing:*
assumes *canonical:*
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-canonical-sampled-layer-chain } s) \ s \leq C$
and *missing:*
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-missing-canonical-sampled-layer-chain } s) \ s$
 $\leq M$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-sampled-layer-chain } s) \ s \leq C + M$
proof –
have *wp-event verify-monad*
 $(\text{composition-fri-bad-with-sampled-layer-chain } s) \ s$
 $\leq$ *wp-event verify-monad*
 $(\lambda out.$
 $\text{composition-fri-bad-with-canonical-sampled-layer-chain } s \text{ out} \vee$
 $\text{composition-fri-sampled-missing-canonical-sampled-layer-chain } s \text{ out})$
 s
by (*rule wp-event-mono*)
 $(\text{rule composition-fri-sampled-layer-chain-canonical-or-missing})$
also have $\dots \leq C + M$
by (*rule order-trans[OF wp-event-union-bound*
 $\text{add-mono[OF canonical missing]]$)
finally show *?thesis* .
qed

lemma *wp-trace-fri-sampled-bound-from-canonical:*
assumes *canonical:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-canonical-sampled-layer-chain } s) \ s \leq C$
shows
 $\text{wp-event verify-monad } (\text{trace-fri-bad-with-sampled-layer-chain } s) \ s$
 $\leq C$
proof –
have $\text{wp-event verify-monad } (\text{trace-fri-bad-with-sampled-layer-chain } s) \ s$
 $\leq C + 0$
by (*rule wp-trace-fri-sampled-bound-from-canonical-and-missing*)

$[OF \text{ canonical } wp\text{-trace-fri-missing-canonical-sampled-zero}])$
then show $?thesis$ **by** $simp$
qed

lemma *wp-composition-fri-sampled-bound-from-canonical*:
assumes *canonical*:
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-bad-with-canonical-sampled-layer-chain } s) \ s \leq C$
shows
 $wp\text{-event } verify\text{-monad } (composition\text{-fri-bad-with-sampled-layer-chain } s) \ s$
 $\leq C$
proof –
have $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-bad-with-sampled-layer-chain } s) \ s \leq C + 0$
by (rule *wp-composition-fri-sampled-bound-from-canonical-and-missing*
 $[OF \text{ canonical } wp\text{-composition-fri-missing-canonical-sampled-zero}])$
then show $?thesis$ **by** $simp$
qed

lemma *wp-trace-fri-canonical-sampled-bound-from-sampled*:
assumes *sampling*:
 $wp\text{-event } verify\text{-monad } (trace\text{-fri-bad-with-sampled-layer-chain } s) \ s$
 $\leq R$
shows
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-bad-with-canonical-sampled-layer-chain } s) \ s \leq R$
proof –
have $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-bad-with-canonical-sampled-layer-chain } s) \ s \leq$
 $wp\text{-event } verify\text{-monad } (trace\text{-fri-bad-with-sampled-layer-chain } s) \ s$
by (rule *wp-event-mono*)
 $(rule \text{ trace-fri-bad-with-canonical-sampled-layer-chain-imp-sampled})$
then show $?thesis$
by (rule *order-trans*[*OF* - *sampling*])
qed

lemma *wp-composition-fri-canonical-sampled-bound-from-sampled*:
assumes *sampling*:
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-bad-with-sampled-layer-chain } s) \ s \leq R$
shows
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-bad-with-canonical-sampled-layer-chain } s) \ s \leq R$
proof –
have $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-bad-with-canonical-sampled-layer-chain } s) \ s \leq$
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-bad-with-sampled-layer-chain } s) \ s$
by (rule *wp-event-mono*)

(rule composition-fri-bad-with-canonical-sampled-layer-chain-imp-sampled)
then show ?thesis
 by (rule order-trans[OF - sampled])
qed

lemma wp-trace-fri-canonical-sampled-bound-from-full-cover-and-missing:
assumes full-bound:
 wp-event verify-monad
 (trace-fri-bad-with-full-cover-layer-chain s bad) $s \leq R$
and missing-bound:
 wp-event verify-monad
 (trace-fri-sampled-missing-full-cover s bad) $s \leq M$
shows
 wp-event verify-monad
 (trace-fri-bad-with-canonical-sampled-layer-chain s) $s \leq R + M$
by (rule wp-trace-fri-canonical-sampled-bound-from-sampled)
 (rule wp-trace-fri-sampled-bound-from-full-cover-and-missing
 [OF full-bound missing-bound])

lemma wp-composition-fri-canonical-sampled-bound-from-full-cover-and-missing:
assumes full-bound:
 wp-event verify-monad
 (composition-fri-bad-with-full-cover-layer-chain s bad) $s \leq R$
and missing-bound:
 wp-event verify-monad
 (composition-fri-sampled-missing-full-cover s bad) $s \leq M$
shows
 wp-event verify-monad
 (composition-fri-bad-with-canonical-sampled-layer-chain s) $s \leq R + M$
by (rule wp-composition-fri-canonical-sampled-bound-from-sampled)
 (rule wp-composition-fri-sampled-bound-from-full-cover-and-missing
 [OF full-bound missing-bound])

lemma verify-monad-trace-fri-missing-canonical-sampled-extracts-evidence:
assumes outcome:
 Some (result, final-state) $\in$ set-dist (execute verify-monad s)
and missing:
 trace-fri-sampled-missing-canonical-sampled-layer-chain s
 (Some (result, final-state))
obtains trace-roots trace-bs trace-final dg composition-roots
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table doms layers
where
 trace-fri-partial-candidate-opening-evidence s
 (Some (result, final-state)) trace-roots trace-bs trace-final dg
 composition-roots composition-bs composition-final fri-query-idxs
 trace-round-layers composition-round-layers fr candidate-query-idxs

```

    trace-openings trace-table
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
   $\wedge$ canonical-layers.  $\neg$ 
    generic-fri-canonical-sampled-layer-chain-evidence
      trace-table-low-degree trace-table (clength - 1) trace-roots
      trace-bs trace-final fri-query-idxs trace-round-layers
      canonical-layers
proof -
have sampled:
  trace-fri-bad-with-sampled-layer-chain s
    (Some (result, final-state))
and not-canonical:
   $\neg$  trace-fri-bad-with-canonical-sampled-layer-chain s
    (Some (result, final-state))
using missing
unfolding trace-fri-sampled-missing-canonical-sampled-layer-chain-def
by blast+
from sampled obtain trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr candidate-query-idxs trace-openings
  trace-table doms layers
where evidence:
  trace-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr candidate-query-idxs
    trace-openings trace-table
and chain:
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
unfolding trace-fri-bad-with-sampled-layer-chain-def by blast
have no-canonical:
   $\wedge$ canonical-layers.  $\neg$ 
    generic-fri-canonical-sampled-layer-chain-evidence
      trace-table-low-degree trace-table (clength - 1) trace-roots
      trace-bs trace-final fri-query-idxs trace-round-layers
      canonical-layers
proof
fix canonical-layers
assume canonical:
  generic-fri-canonical-sampled-layer-chain-evidence
    trace-table-low-degree trace-table (clength - 1) trace-roots
    trace-bs trace-final fri-query-idxs trace-round-layers
    canonical-layers
have trace-fri-bad-with-canonical-sampled-layer-chain s
  (Some (result, final-state))

```

```

    unfolding trace-fri-bad-with-canonical-sampled-layer-chain-def
    using evidence canonical by blast
  then show False
    using not-canonical by contradiction
qed
show ?thesis
  by (rule that[OF evidence chain no-canonical])
qed

lemma verify-monad-composition-fri-missing-canonical-sampled-extracts-evidence:
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and missing:
    composition-fri-sampled-missing-canonical-sampled-layer-chain s
    (Some (result, final-state))
  obtains trace-roots trace-bs trace-final fri-dg
    opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers fr
    f-fri-roots f-final as dg composition-fri-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table doms layers
  where
    composition-fri-partial-candidate-opening-evidence s
      (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
      opening-composition-roots composition-bs composition-final
      fri-query-idxs trace-round-layers composition-round-layers fr
      f-fri-roots f-final as dg composition-fri-roots final
      trace-query-idxs trace-openings composition-query-idxs
      composition-openings trace-table composition-table
    generic-fri-sampled-layer-chain-evidence
      (composition-table-low-degree (to-nat fri-dg)) composition-table
      (to-nat fri-dg) opening-composition-roots composition-bs
      composition-final fri-query-idxs composition-round-layers doms layers
    ∧ canonical-layers. ¬
    generic-fri-canonical-sampled-layer-chain-evidence
      (composition-table-low-degree (to-nat fri-dg)) composition-table
      (to-nat fri-dg) opening-composition-roots composition-bs
      composition-final fri-query-idxs composition-round-layers
      canonical-layers
  proof -
  have sampled:
    composition-fri-bad-with-sampled-layer-chain s
      (Some (result, final-state))
  and not-canonical:
    ¬ composition-fri-bad-with-canonical-sampled-layer-chain s
      (Some (result, final-state))
  using missing
  unfolding
    composition-fri-sampled-missing-canonical-sampled-layer-chain-def

```

by blast+
from sampled **obtain** trace-roots trace-bs trace-final fri-dg
 opening-composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers fr
 f-fri-roots f-final as dg composition-fri-roots final
 trace-query-idxs trace-openings composition-query-idxs
 composition-openings trace-table composition-table doms layers
where evidence:
 composition-fri-partial-candidate-opening-evidence s
 (Some (result, final-state)) trace-roots trace-bs trace-final fri-dg
 opening-composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers fr
 f-fri-roots f-final as dg composition-fri-roots final
 trace-query-idxs trace-openings composition-query-idxs
 composition-openings trace-table composition-table
and chain:
 generic-fri-sampled-layer-chain-evidence
 (composition-table-low-degree (to-nat fri-dg)) composition-table
 (to-nat fri-dg) opening-composition-roots composition-bs
 composition-final fri-query-idxs composition-round-layers doms layers
unfolding composition-fri-bad-with-sampled-layer-chain-def **by** meson
have no-canonical:
 $\bigwedge$ canonical-layers. $\neg$
 generic-fri-canonical-sampled-layer-chain-evidence
 (composition-table-low-degree (to-nat fri-dg)) composition-table
 (to-nat fri-dg) opening-composition-roots composition-bs
 composition-final fri-query-idxs composition-round-layers
 canonical-layers
proof
fix canonical-layers
assume canonical:
 generic-fri-canonical-sampled-layer-chain-evidence
 (composition-table-low-degree (to-nat fri-dg)) composition-table
 (to-nat fri-dg) opening-composition-roots composition-bs
 composition-final fri-query-idxs composition-round-layers
 canonical-layers
have composition-fri-bad-with-canonical-sampled-layer-chain s
 (Some (result, final-state))
unfolding composition-fri-bad-with-canonical-sampled-layer-chain-def
using evidence canonical **by** meson
then show False
using not-canonical **by** contradiction
qed
show ?thesis
by (rule that[OF evidence chain no-canonical])
qed

lemma fri-canonical-domain-successor-square:
assumes next-bound:


```

    idx < length (fri-canonical-domain-at (Suc layer-idx))
  and cur-bound:
    idx < length (fri-canonical-domain-at layer-idx)
  shows fri-canonical-domain-at (Suc layer-idx) ! idx =
    (fri-canonical-domain-at layer-idx ! idx)2
proof -
  have next-val:
    fri-canonical-domain-at (Suc layer-idx) ! idx =
      (h ^ idx * shift) ^ (2 ^ Suc layer-idx)
  by (rule fri-canonical-domain-at-nth[OF next-bound])
  have cur-val:
    fri-canonical-domain-at layer-idx ! idx =
      (h ^ idx * shift) ^ (2 ^ layer-idx)
  by (rule fri-canonical-domain-at-nth[OF cur-bound])
  have square:
    (h ^ idx * shift) ^ (2 ^ Suc layer-idx) =
      ((h ^ idx * shift) ^ (2 ^ layer-idx))2
  by (metis mult.commute power2-eq-square power-Suc power-mult)
  show ?thesis
    using next-val cur-val square by simp
qed

```

```

lemma fri-canonical-domains-successor-square:
  assumes layer-bound: Suc layer-idx ≤ n
  and idx-bound:
    idx < length (fri-canonical-domains n ! Suc layer-idx)
  and cur-idx-bound:
    idx < length (fri-canonical-domains n ! layer-idx)
  shows fri-canonical-domains n ! Suc layer-idx ! idx =
    (fri-canonical-domains n ! layer-idx ! idx)2
proof -
  have cur-bound: layer-idx ≤ n
  using layer-bound by simp
  have next-dom:
    fri-canonical-domains n ! Suc layer-idx =
      fri-canonical-domain-at (Suc layer-idx)
  by (rule fri-canonical-domains-nth[OF layer-bound])
  have cur-dom:
    fri-canonical-domains n ! layer-idx =
      fri-canonical-domain-at layer-idx
  by (rule fri-canonical-domains-nth[OF cur-bound])
  show ?thesis
    using fri-canonical-domain-successor-square
      [of idx layer-idx] idx-bound cur-idx-bound
    unfolding next-dom cur-dom by simp
qed

```

```

lemma fri-canonical-domains-sampled-next-align:
  assumes roots-bound: length roots ≤ n

```

and *idx-bound*:
 $\bigwedge \text{round-idx } \text{layer-idx}.$
 $\text{round-idx} < \text{length } \text{query-idxs} \implies$
 $\text{layer-idx} < \text{length } \text{roots} \implies$
 $\text{fri-evidence-next-idx } \text{roots } \text{query-idxs } \text{round-idx } \text{layer-idx} <$
 $\text{length } (\text{fri-canonical-domain-at } \text{layer-idx})$
shows
 $\text{fri-sampled-next-domains-align } \text{roots } \text{query-idxs}$
 $(\text{fri-canonical-domains } n)$
unfolding *fri-sampled-next-domains-align-def*
proof (*intro allI impI*)
fix *round-idx layer-idx*
assume *round-bound*: $\text{round-idx} < \text{length } \text{query-idxs}$
and *layer-bound*: $\text{layer-idx} < \text{length } \text{roots}$
have *layer-le*: $\text{layer-idx} \leq n$
using *layer-bound roots-bound* **by** *simp*
have *dom-eq*:
 $\text{fri-canonical-domains } n ! \text{layer-idx} =$
 $\text{fri-canonical-domain-at } \text{layer-idx}$
by (*rule fri-canonical-domains-nth[OF layer-le]*)
have *idx*:
 $\text{fri-evidence-next-idx } \text{roots } \text{query-idxs } \text{round-idx } \text{layer-idx} <$
 $\text{length } (\text{fri-canonical-domain-at } \text{layer-idx})$
by (*rule idx-bound[OF round-bound layer-bound]*)
show
 $\text{fri-canonical-domains } n ! \text{layer-idx} !$
 $\text{fri-evidence-next-idx } \text{roots } \text{query-idxs } \text{round-idx } \text{layer-idx} =$
 $(h \wedge \text{fri-evidence-next-idx } \text{roots } \text{query-idxs } \text{round-idx } \text{layer-idx} *$
 $\text{shift}) \wedge (2 \wedge \text{layer-idx})$
unfolding *dom-eq*
by (*rule fri-canonical-domain-at-nth[OF idx]*)
qed

lemma *fri-canonical-domains-sampled-successor-square*:
assumes *roots-bound*: $\text{length } \text{roots} \leq n$
and *cur-idx-bound*:
 $\bigwedge \text{round-idx } \text{layer-idx}.$
 $\text{round-idx} < \text{length } \text{query-idxs} \implies$
 $\text{layer-idx} < \text{length } \text{roots} \implies$
 $\text{fri-evidence-next-idx } \text{roots } \text{query-idxs } \text{round-idx } \text{layer-idx} <$
 $\text{length } (\text{fri-canonical-domains } n ! \text{layer-idx})$
and *next-idx-bound*:
 $\bigwedge \text{round-idx } \text{layer-idx}.$
 $\text{round-idx} < \text{length } \text{query-idxs} \implies$
 $\text{layer-idx} < \text{length } \text{roots} \implies$
 $\text{fri-evidence-next-idx } \text{roots } \text{query-idxs } \text{round-idx } \text{layer-idx} <$
 $\text{length } (\text{fri-canonical-domains } n ! \text{Suc } \text{layer-idx})$
shows
 $\text{fri-sampled-successor-domains-square } \text{roots } \text{query-idxs}$

```

    (fri-canonical-domains n)
  unfolding fri-sampled-successor-domains-square-def
proof (intro allI impI)
  fix round-idx layer-idx
  assume round-bound: round-idx < length query-idxs
    and layer-bound: layer-idx < length roots
  have layer-le: Suc layer-idx ≤ n
    using layer-bound roots-bound by simp
  show
    fri-canonical-domains n ! Suc layer-idx !
    fri-evidence-next-idx roots query-idxs round-idx layer-idx =
    (fri-canonical-domains n ! layer-idx !
    fri-evidence-next-idx roots query-idxs round-idx layer-idx)2
  by (rule fri-canonical-domains-successor-square[OF layer-le])
    (rule next-idx-bound[OF round-bound layer-bound],
    rule cur-idx-bound[OF round-bound layer-bound])
qed

lemma fri-canonical-domains-no-sampled-domain-failure:
  assumes roots-bound: length roots ≤ n
    and canonical-next-bound:
     $\bigwedge \text{round-idx layer-idx.}$ 
    round-idx < length query-idxs  $\implies$ 
    layer-idx < length roots  $\implies$ 
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (fri-canonical-domain-at layer-idx)
    and canonical-successor-bound:
     $\bigwedge \text{round-idx layer-idx.}$ 
    round-idx < length query-idxs  $\implies$ 
    layer-idx < length roots  $\implies$ 
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (fri-canonical-domains n ! Suc layer-idx)
  shows  $\neg$  generic-fri-full-cover-sampled-domain-failure roots query-idxs
    (fri-canonical-domains n)
proof -
  have next-align:
    fri-sampled-next-domains-align roots query-idxs
    (fri-canonical-domains n)
  by (rule fri-canonical-domains-sampled-next-align
    [OF roots-bound canonical-next-bound])
  have cur-bound:
     $\bigwedge \text{round-idx layer-idx.}$ 
    round-idx < length query-idxs  $\implies$ 
    layer-idx < length roots  $\implies$ 
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (fri-canonical-domains n ! layer-idx)
  proof -
    fix round-idx layer-idx
    assume round-bound: round-idx < length query-idxs

```

```

    and layer-bound: layer-idx < length roots
  have layer-le: layer-idx ≤ n
    using layer-bound roots-bound by simp
  have dom-eq:
    fri-canonical-domains n ! layer-idx =
      fri-canonical-domain-at layer-idx
    by (rule fri-canonical-domains-nth[OF layer-le])
  show
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
      length (fri-canonical-domains n ! layer-idx)
    unfolding dom-eq
    by (rule canonical-next-bound[OF round-bound layer-bound])
qed
have succ-square:
  fri-sampled-successor-domains-square roots query-idxs
    (fri-canonical-domains n)
  by (rule fri-canonical-domains-sampled-successor-square
    [OF roots-bound cur-bound canonical-successor-bound])
show ?thesis
  unfolding generic-fri-full-cover-sampled-domain-failure-def
  using next-align succ-square by simp
qed

lemma generic-fri-full-cover-domain-failure-imp-index-or-sampled-domain:
  assumes roots-len: length challenges = length roots
    and domain: generic-fri-full-cover-domain-failure roots challenges doms
  shows
    generic-fri-full-cover-index-failure roots query-idxs ∨
    generic-fri-full-cover-sampled-domain-failure roots query-idxs doms
proof (cases generic-fri-full-cover-index-failure roots query-idxs)
  case True
    then show ?thesis by simp
next
  case False
  have covers: fri-sampled-layers-cover-fold-indices roots query-idxs
    using False unfolding generic-fri-full-cover-index-failure-def by simp
  show ?thesis
  proof (cases
    generic-fri-full-cover-sampled-domain-failure roots query-idxs doms)
    case True
      then show ?thesis by simp
  next
    case no-sampled: False
    have next-align: fri-sampled-next-domains-align roots query-idxs doms
      and succ-square:
        fri-sampled-successor-domains-square roots query-idxs doms
      using no-sampled
      unfolding generic-fri-full-cover-sampled-domain-failure-def
      by simp-all
  end
end

```

```

have no-square-fail:
   $\forall$  layer-idx < length challenges.
     $\forall$  idx < fri-evidence-layer-len roots layer-idx div 2.
      doms ! Suc layer-idx ! idx =
        (doms ! layer-idx ! idx)2
proof (intro allI impI)
  fix layer-idx idx
  assume layer-bound: layer-idx < length challenges
    and idx-bound: idx < fri-evidence-layer-len roots layer-idx div 2
  have layer-root-bound: layer-idx < length roots
    using layer-bound roots-len by simp
  obtain round-idx where round-bound: round-idx < length query-idxs
    and idx-eq:
      fri-evidence-next-idx roots query-idxs round-idx layer-idx = idx
  by (rule fri-sampled-layers-cover-fold-indicesD
    [OF covers layer-root-bound idx-bound])
  show doms ! Suc layer-idx ! idx =
    (doms ! layer-idx ! idx)2
    using fri-sampled-successor-domains-squareD
      [OF succ-square round-bound layer-root-bound] idx-eq
    by simp
qed
have no-norm-fail:
   $\forall$  layer-idx < length challenges.
     $\forall$  idx < fri-evidence-layer-len roots layer-idx div 2.
      doms ! layer-idx ! idx =
        (h ^ idx * shift) ^ (2 ^ layer-idx)
proof (intro allI impI)
  fix layer-idx idx
  assume layer-bound: layer-idx < length challenges
    and idx-bound: idx < fri-evidence-layer-len roots layer-idx div 2
  have layer-root-bound: layer-idx < length roots
    using layer-bound roots-len by simp
  obtain round-idx where round-bound: round-idx < length query-idxs
    and idx-eq:
      fri-evidence-next-idx roots query-idxs round-idx layer-idx = idx
  by (rule fri-sampled-layers-cover-fold-indicesD
    [OF covers layer-root-bound idx-bound])
  show doms ! layer-idx ! idx =
    (h ^ idx * shift) ^ (2 ^ layer-idx)
    using fri-sampled-next-domains-alignD
      [OF next-align round-bound layer-root-bound] idx-eq
    by simp
qed
have False
  using domain no-square-fail no-norm-fail
  unfolding generic-fri-full-cover-domain-failure-def by simp
then show ?thesis by simp
qed

```

qed

lemma *trace-fri-sampled-full-cover-domain-imp-index-or-sampled-domain:*

assumes *domain: trace-fri-sampled-full-cover-domain-failure s out*

shows

trace-fri-sampled-full-cover-index-failure s out $\vee$

trace-fri-sampled-full-cover-sampled-domain-failure s out

proof —

from *domain* **obtain** *trace-roots trace-bs trace-final dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr candidate-query-idxs trace-openings

trace-table doms layers

where *evidence:*

trace-fri-partial-candidate-opening-evidence s out trace-roots

trace-bs trace-final dg composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr candidate-query-idxs trace-openings

trace-table

and *chain:*

generic-fri-sampled-layer-chain-evidence trace-table-low-degree

trace-table (clength - 1) trace-roots trace-bs trace-final

fri-query-idxs trace-round-layers doms layers

and *generic-domain:*

generic-fri-full-cover-domain-failure trace-roots trace-bs doms

unfolding *trace-fri-sampled-full-cover-domain-failure-def* **by** *blast*

have *partial:*

generic-fri-partial-evidence trace-table-low-degree trace-table

(clength - 1) trace-roots trace-bs trace-final fri-query-idxs

trace-round-layers

by *(rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])*

have *roots-len: length trace-bs = length trace-roots*

by *(rule generic-fri-partial-evidence-shapes(2)[OF partial])*

have *split:*

generic-fri-full-cover-index-failure trace-roots fri-query-idxs $\vee$

generic-fri-full-cover-sampled-domain-failure trace-roots

fri-query-idxs doms

by *(rule generic-fri-full-cover-domain-failure-imp-index-or-sampled-domain*
[OF roots-len generic-domain])

then show *?thesis*

proof

assume *generic-fri-full-cover-index-failure trace-roots fri-query-idxs*

then show *?thesis*

unfolding *trace-fri-sampled-full-cover-index-failure-def*

using *evidence chain* **by** *blast*

next

assume

generic-fri-full-cover-sampled-domain-failure trace-roots

fri-query-idxs doms

then show *?thesis*

unfolding *trace-fri-sampled-full-cover-sampled-domain-failure-def*
using *evidence chain* **by** *blast*
qed
qed

lemma *composition-fri-sampled-full-cover-domain-imp-index-or-sampled-domain:*
assumes *domain: composition-fri-sampled-full-cover-domain-failure s out*
shows
composition-fri-sampled-full-cover-index-failure s out $\vee$
composition-fri-sampled-full-cover-sampled-domain-failure s out

proof –
from *domain* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table doms layers
where *evidence:*
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *chain:*
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers
and *generic-domain:*
generic-fri-full-cover-domain-failure opening-composition-roots
composition-bs doms
unfolding *composition-fri-sampled-full-cover-domain-failure-def*
by *meson*
have *partial:*
generic-fri-partial-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers
by *(rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])*
have *roots-len: length composition-bs = length opening-composition-roots*
by *(rule generic-fri-partial-evidence-shapes(2)[OF partial])*
have *split:*
generic-fri-full-cover-index-failure opening-composition-roots
fri-query-idxs $\vee$
generic-fri-full-cover-sampled-domain-failure
opening-composition-roots fri-query-idxs doms
by *(rule generic-fri-full-cover-domain-failure-imp-index-or-sampled-domain*

```

    [OF roots-len generic-domain])
  then show ?thesis
proof
  assume
    generic-fri-full-cover-index-failure opening-composition-roots
    fri-query-idxs
  then show ?thesis
    unfolding composition-fri-sampled-full-cover-index-failure-def
    using evidence chain by meson
next
  assume
    generic-fri-full-cover-sampled-domain-failure
    opening-composition-roots fri-query-idxs doms
  then show ?thesis
    unfolding composition-fri-sampled-full-cover-sampled-domain-failure-def
    using evidence chain by meson
qed
qed

lemma wp-trace-fri-sampled-full-cover-domain-bound-from-index-or-sampled-domain:
  assumes index:
    wp-event verify-monad
    (trace-fri-sampled-full-cover-index-failure s) s ≤ I
  and sampled-domain:
    wp-event verify-monad
    (trace-fri-sampled-full-cover-sampled-domain-failure s) s ≤ SD
  shows
    wp-event verify-monad
    (trace-fri-sampled-full-cover-domain-failure s) s ≤ I + SD
proof -
  have wp-event verify-monad
    (trace-fri-sampled-full-cover-domain-failure s) s
  ≤ wp-event verify-monad
    (λout. trace-fri-sampled-full-cover-index-failure s out ∨
    trace-fri-sampled-full-cover-sampled-domain-failure s out) s
  by (rule wp-event-mono)
    (rule trace-fri-sampled-full-cover-domain-imp-index-or-sampled-domain)
  also have ... ≤ I + SD
  by (rule order-trans[OF wp-event-union-bound
    add-mono[OF index sampled-domain]])
  finally show ?thesis .
qed

lemma
  wp-composition-fri-sampled-full-cover-domain-bound-from-index-or-sampled-domain:
  assumes index:
    wp-event verify-monad
    (composition-fri-sampled-full-cover-index-failure s) s ≤ I
  and sampled-domain:

```


$wp\text{-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-sampled-domain-failure } s) \ s$
 $\leq SD$
shows
 $wp\text{-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-domain-failure } s) \ s \leq I + SD$
proof –
have $wp\text{-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-domain-failure } s) \ s$
 $\leq wp\text{-event verify-monad}$
 $(\lambda out. \text{composition-fri-sampled-full-cover-index-failure } s \ out \ \vee$
 $\text{composition-fri-sampled-full-cover-sampled-domain-failure } s \ out) \ s$
by (rule $wp\text{-event-mono}$)
(rule
 $\text{composition-fri-sampled-full-cover-domain-imp-index-or-sampled-domain}$)
also have $\dots \leq I + SD$
by (rule $\text{order-trans}[OF \ wp\text{-event-union-bound}$
 $\text{add-mono}[OF \ \text{index sampled-domain}]])$
finally show $?thesis$.
qed

lemma $wp\text{-trace-fri-sampled-missing-full-cover-bound-from-domain-split}$:
assumes $proximity$:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-proximity-failure } s \ bad) \ s \leq P$
and $index$:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$
and $domain\text{-length}$:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-domain-length-failure } s) \ s \leq DL$
and $sampled\text{-domain}$:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-sampled-domain-failure } s) \ s \leq SD$
shows
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-missing-full-cover } s \ bad) \ s$
 $\leq P + I + (I + DL) + (I + SD)$
by (rule $wp\text{-trace-fri-sampled-missing-full-cover-bound-from-length-split}$
 $[OF \ proximity \ index \ domain\text{-length}$
 $wp\text{-trace-fri-sampled-full-cover-domain-bound-from-index-or-sampled-domain}]$)
(rule $index$, rule $sampled\text{-domain}$)

lemma $wp\text{-composition-fri-sampled-missing-full-cover-bound-from-domain-split}$:
assumes $proximity$:
 $wp\text{-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-proximity-failure } s \ bad) \ s \leq P$
and $index$:
 $wp\text{-event verify-monad}$

```

    (composition-fri-sampled-full-cover-index-failure s) s ≤ I
and domain-length:
  wp-event verify-monad
    (composition-fri-sampled-full-cover-domain-length-failure s) s ≤ DL
and sampled-domain:
  wp-event verify-monad
    (composition-fri-sampled-full-cover-sampled-domain-failure s) s
    ≤ SD
shows
  wp-event verify-monad
    (composition-fri-sampled-missing-full-cover s bad) s
    ≤ P + I + (I + DL) + (I + SD)
by (rule
  wp-composition-fri-sampled-missing-full-cover-bound-from-length-split
  [OF proximity index domain-length
    wp-composition-fri-sampled-full-cover-domain-bound-from-index-or-sampled-domain])
  (rule index, rule sampled-domain)

end

end

```

```

theory Soundness-FRI-Sampled-Obstructions
imports Soundness-FRI-Full-Cover-Domains
begin

```

Explicit obstruction events for the branch where an accepting verifier execution reaches the broad reachable FRI bad event, but no sampled FRI layer-chain witness exists. These predicates are proof-routing devices only. They do not reconstruct complete FRI tables from sampled openings; they record that the verifier's partial FRI evidence cannot be organized into the sampled layer-chain object required by the generic low-degree reduction.

```

context soundness
begin

```

```

definition generic-fri-sampled-next-value-conflict

```

```

  :: 'f list ⇒ 'f list ⇒ nat list ⇒
    'f list list list ⇒ bool

```

```

where

```

```

  generic-fri-sampled-next-value-conflict roots challenges query-idxs
    round-layers ⟷
  (∃ round-idx round-idx' layer-idx v v'.
    round-idx < length query-idxs ∧
    round-idx' < length query-idxs ∧
    layer-idx < length challenges ∧
    fri-evidence-next-idx roots query-idxs round-idx layer-idx =
    fri-evidence-next-idx roots query-idxs round-idx' layer-idx ∧

```

generic-fri-round-forced-next-value roots challenges query-idxs
round-layers round-idx layer-idx v $\wedge$
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers round-idx' layer-idx v' $\wedge$
v $\neq$ *v'*)

definition *generic-fri-sampled-final-value-conflict*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow \text{bool}$

where

generic-fri-sampled-final-value-conflict roots challenges final-value
query-idxs round-layers $\longleftrightarrow$
 $(\exists \text{ round-idx } v.$
 $0 < \text{length challenges} \wedge$
 $\text{round-idx} < \text{length query-idxs} \wedge$
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers round-idx (length challenges - 1) v $\wedge$
v $\neq$ *final-value*)

definition *generic-fri-sampled-base-opening-conflict*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow \text{bool}$

where

generic-fri-sampled-base-opening-conflict candidate-table roots challenges
query-idxs round-layers $\longleftrightarrow$
 $(\exists \text{ round-idx } xp \text{ xp-path } xn \text{ xn-path}.$
 $0 < \text{length challenges} \wedge$
 $\text{round-idx} < \text{length query-idxs} \wedge$
fri-layer-step-evidence
 $(\text{roots} ! 0)$
 $(\text{challenges} ! 0)$
 $(\text{fri-evidence-layer-len roots } 0)$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx } 0)$
 1
 $(\text{fri-sibling-index } (\text{fri-evidence-layer-len roots } 0)$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx } 0))$
 $xp \text{ xp-path } xn \text{ xn-path}$
 $(\text{fri-evidence-next-idx roots query-idxs round-idx } 0)$
 $(\text{fri-evidence-next-value roots challenges query-idxs round-idx } 0$
 $xp \text{ xn})$
 $(\text{round-layers} ! \text{round-idx} ! 0) \wedge$
 $\neg \text{fri-opening-matches-table } (\text{fri-evidence-layer-len roots } 0)$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx } 0)$
 $\text{candidate-table } xp \text{ xn})$

definition *generic-fri-sampled-successor-opening-conflict*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow \text{bool}$

where

generic-fri-sampled-successor-opening-conflict roots challenges query-idxs
 round-layers $\longleftrightarrow$
 $(\exists \text{ source-round target-round layer-idx } v \text{ } xp \text{ } xp\text{-path } xn \text{ } xn\text{-path}.$
 $\text{source-round} < \text{length query-idxs} \wedge$
 $\text{target-round} < \text{length query-idxs} \wedge$
 $\text{Suc layer-idx} < \text{length challenges} \wedge$
generic-fri-round-forced-next-value roots challenges query-idxs
 $\text{round-layers source-round layer-idx } v \wedge$
fri-layer-step-evidence
 $(\text{roots} ! \text{Suc layer-idx})$
 $(\text{challenges} ! \text{Suc layer-idx})$
 $(\text{fri-evidence-layer-len roots (Suc layer-idx)})$
 $(\text{fri-evidence-layer-idx roots query-idxs target-round}$
 $\text{(Suc layer-idx)})$
 $(2 \wedge \text{Suc layer-idx})$
 $(\text{fri-sibling-index (fri-evidence-layer-len roots (Suc layer-idx)})$
 $\text{(fri-evidence-layer-idx roots query-idxs target-round}$
 $\text{(Suc layer-idx))})$
 $xp \text{ } xp\text{-path } xn \text{ } xn\text{-path}$
 $(\text{fri-evidence-next-idx roots query-idxs target-round}$
 $\text{(Suc layer-idx)})$
 $(\text{fri-evidence-next-value roots challenges query-idxs target-round}$
 $\text{(Suc layer-idx) } xp \text{ } xn)$
 $(\text{round-layers} ! \text{target-round} ! \text{Suc layer-idx}) \wedge$
 $((\text{fri-evidence-next-idx roots query-idxs source-round layer-idx} =$
 $\text{fri-evidence-layer-idx roots query-idxs target-round}$
 $\text{(Suc layer-idx)}) \wedge$
 $v \neq xp) \vee$
 $(\text{fri-evidence-next-idx roots query-idxs source-round layer-idx} =$
 $\text{fri-sibling-index (fri-evidence-layer-len roots (Suc layer-idx)})$
 $\text{(fri-evidence-layer-idx roots query-idxs target-round}$
 $\text{(Suc layer-idx)}) \wedge$
 $v \neq xn)))$

definition *generic-fri-sampled-same-layer-opening-conflict*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow \text{bool}$

where

generic-fri-sampled-same-layer-opening-conflict roots challenges
 query-idxs round-layers $\longleftrightarrow$
 $(\exists \text{ round-idx round-idx' layer-idx}$
 $xp \text{ } xp\text{-path } xn \text{ } xn\text{-path } yp \text{ } yp\text{-path } yn \text{ } yn\text{-path}.$
 $\text{round-idx} < \text{length query-idxs} \wedge$
 $\text{round-idx'} < \text{length query-idxs} \wedge$
 $\text{layer-idx} < \text{length challenges} \wedge$
fri-layer-step-evidence
 $(\text{roots} ! \text{layer-idx})$
 $(\text{challenges} ! \text{layer-idx})$
 $(\text{fri-evidence-layer-len roots layer-idx})$

(fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
 (2 ^ layer-idx)
 (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
 xp xp-path xn xn-path
 (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
 (fri-evidence-next-value roots challenges query-idxs round-idx
 layer-idx xp xn)
 (round-layers ! round-idx ! layer-idx) ^
 fri-layer-step-evidence
 (roots ! layer-idx)
 (challenges ! layer-idx)
 (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
 (2 ^ layer-idx)
 (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
 yp yp-path yn yn-path
 (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
 (fri-evidence-next-value roots challenges query-idxs round-idx'
 layer-idx yp yn)
 (round-layers ! round-idx' ! layer-idx) ^
 ((fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
 fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ^
 xp ≠ yp) ∨
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
 fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ^
 xp ≠ yn) ∨
 (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
 fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ^
 xn ≠ yp) ∨
 (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
 fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ^
 xn ≠ yn)))

lemma *generic-fri-sampled-next-value-conflictE*:

assumes

*generic-fri-sampled-next-value-conflict roots challenges query-idxs
 round-layers*

obtains *round-idx round-idx' layer-idx v v'*

where

round-idx < length query-idxs

round-idx' < length query-idxs

layer-idx < length challenges

fri-evidence-next-idx roots query-idxs round-idx layer-idx =

fri-evidence-next-idx roots query-idxs round-idx' layer-idx
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers round-idx layer-idx v
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers round-idx' layer-idx v'
v ≠ v'
using *assms unfolding generic-fri-sampled-next-value-conflict-def*
by *blast*

lemma *generic-fri-sampled-final-value-conflictE:*

assumes
generic-fri-sampled-final-value-conflict roots challenges final-value
query-idxs round-layers
obtains *round-idx v*
where
0 < length challenges
round-idx < length query-idxs
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers round-idx (length challenges - 1) v
v ≠ final-value
using *assms unfolding generic-fri-sampled-final-value-conflict-def*
by *blast*

lemma *generic-fri-sampled-successor-opening-conflictE:*

assumes
generic-fri-sampled-successor-opening-conflict roots challenges
query-idxs round-layers
obtains *source-round target-round layer-idx v xp xp-path xn xn-path*
where
source-round < length query-idxs
target-round < length query-idxs
Suc layer-idx < length challenges
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers source-round layer-idx v
fri-layer-step-evidence
(roots ! Suc layer-idx)
(challenges ! Suc layer-idx)
(fri-evidence-layer-len roots (Suc layer-idx))
(fri-evidence-layer-idx roots query-idxs target-round
(Suc layer-idx))
(2 ^ Suc layer-idx)
(fri-sibling-index (fri-evidence-layer-len roots (Suc layer-idx))
(fri-evidence-layer-idx roots query-idxs target-round
(Suc layer-idx)))
xp xp-path xn xn-path
(fri-evidence-next-idx roots query-idxs target-round
(Suc layer-idx))
(fri-evidence-next-value roots challenges query-idxs target-round
(Suc layer-idx) xp xn)

$(\text{round-layers} \text{ ! } \text{target-round} \text{ ! } \text{Suc layer-idx})$
 $((\text{fri-evidence-next-idx roots query-idxs source-round layer-idx} =$
 $\quad \text{fri-evidence-layer-idx roots query-idxs target-round}$
 $\quad (\text{Suc layer-idx}) \wedge$
 $\quad v \neq xp) \vee$
 $(\text{fri-evidence-next-idx roots query-idxs source-round layer-idx} =$
 $\quad \text{fri-sibling-index (fri-evidence-layer-len roots (Suc layer-idx))}$
 $\quad (\text{fri-evidence-layer-idx roots query-idxs target-round}$
 $\quad (\text{Suc layer-idx})) \wedge$
 $\quad v \neq xn))$
using *assms unfolding generic-fri-sampled-successor-opening-conflict-def*
by *blast*

definition *generic-fri-sampled-layer-assignment-obstruction*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow$
 $\quad 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat list} \Rightarrow$
 $\quad 'f \text{ list list list} \Rightarrow \text{bool}$

where

generic-fri-sampled-layer-assignment-obstruction low-degree
 $\text{candidate-table degree-bound roots challenges final-value query-idxs}$
 $\text{round-layers} \longleftrightarrow$
generic-fri-partial-evidence low-degree candidate-table degree-bound
 $\text{roots challenges final-value query-idxs round-layers} \wedge$
 $(\forall \text{ doms layers. } \neg$
 $\text{generic-fri-sampled-layer-chain-evidence low-degree candidate-table}$
 $\text{degree-bound roots challenges final-value query-idxs round-layers}$
 $\text{doms layers})$

definition *generic-fri-sampled-assignment-conflict*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $\quad \text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow \text{bool}$

where

generic-fri-sampled-assignment-conflict candidate-table roots challenges
 $\text{final-value query-idxs round-layers} \longleftrightarrow$
generic-fri-sampled-base-opening-conflict candidate-table roots
 $\text{challenges query-idxs round-layers} \vee$
generic-fri-sampled-next-value-conflict roots challenges query-idxs
 $\text{round-layers} \vee$
generic-fri-sampled-successor-opening-conflict roots challenges
 $\text{query-idxs round-layers} \vee$
generic-fri-sampled-same-layer-opening-conflict roots challenges
 $\text{query-idxs round-layers} \vee$
generic-fri-sampled-final-value-conflict roots challenges final-value
 $\text{query-idxs round-layers}$

definition *generic-fri-sampled-layer-value-constraint*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $\quad \text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow \text{nat} \Rightarrow \text{nat} \Rightarrow$
 $\quad 'f \Rightarrow \text{bool}$

where

generic-fri-sampled-layer-value-constraint candidate-table roots
challenges final-value query-idxs round-layers layer-idx idx v
 $\longleftrightarrow$
 (if *layer-idx* = 0 then
 idx < *length candidate-table* $\wedge$ *v* = *candidate-table* ! *idx*
 else if *layer-idx* = *length challenges* then
 v = *final-value*
 else
 ($\exists$ *round-idx*.
 round-idx < *length query-idxs* $\wedge$
 generic-fri-round-forced-next-value roots challenges query-idxs
 round-layers round-idx (layer-idx - 1) v $\wedge$
 fri-evidence-next-idx roots query-idxs round-idx
 (*layer-idx* - 1) = *idx*) $\vee$
 ($\exists$ *round-idx xp xp-path xn xn-path*.
 round-idx < *length query-idxs* $\wedge$
 layer-idx < *length challenges* $\wedge$
 fri-layer-step-evidence
 (*roots* ! *layer-idx*)
 (*challenges* ! *layer-idx*)
 (*fri-evidence-layer-len roots layer-idx*)
 (*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*)
 ($2 \wedge \text{layer-idx}$)
 (*fri-sibling-index (fri-evidence-layer-len roots layer-idx)*
 (*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*))
 xp xp-path xn xn-path
 (*fri-evidence-next-idx roots query-idxs round-idx layer-idx*)
 (*fri-evidence-next-value roots challenges query-idxs round-idx*
 layer-idx xp xn)
 (*round-layers* ! *round-idx* ! *layer-idx*) $\wedge$
 ((*idx* = *fri-evidence-layer-idx roots query-idxs round-idx*
 layer-idx $\wedge$ *v* = *xp*) $\vee$
 (*idx* = *fri-sibling-index (fri-evidence-layer-len roots layer-idx)*
 (*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*) $\wedge$
 v = *xn*))))))

definition *generic-fri-sampled-layer-value*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $\text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow \text{nat} \Rightarrow \text{nat} \Rightarrow 'f$

where

generic-fri-sampled-layer-value candidate-table roots challenges
final-value query-idxs round-layers layer-idx idx =
 (if $\exists v.$ *generic-fri-sampled-layer-value-constraint candidate-table*
 roots challenges final-value query-idxs round-layers layer-idx idx v
 then *SOME v. generic-fri-sampled-layer-value-constraint*
 candidate-table roots challenges final-value query-idxs round-layers
 layer-idx idx v
 else *final-value*)

definition *generic-fri-sampled-assignment-layer*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $\text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list}$

where

generic-fri-sampled-assignment-layer candidate-table roots challenges
 $\text{final-value query-idxs round-layers layer-idx} =$
 (if $\text{layer-idx} = 0$ then *candidate-table*
 else $\text{map (generic-fri-sampled-layer-value candidate-table roots}$
 $\text{challenges final-value query-idxs round-layers layer-idx)}$
 $[0..<\text{clength} * \text{scale}]$)

definition *generic-fri-sampled-assignment-layers*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $\text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow 'f \text{ list list}$

where

generic-fri-sampled-assignment-layers candidate-table roots challenges
 $\text{final-value query-idxs round-layers} =$
 $\text{map (generic-fri-sampled-assignment-layer candidate-table roots}$
 $\text{challenges final-value query-idxs round-layers)}$
 $[0..<\text{Suc (length challenges)}]$

lemma *generic-fri-sampled-assignment-layer-length-nonzero[simp]:*

assumes $\text{layer-idx} \neq 0$

shows $\text{length (generic-fri-sampled-assignment-layer candidate-table roots}$
 $\text{challenges final-value query-idxs round-layers layer-idx}) =$
 $\text{clength} * \text{scale}$

using *assms* **unfolding** *generic-fri-sampled-assignment-layer-def* **by** *simp*

lemma *generic-fri-sampled-assignment-layer-zero[simp]:*

generic-fri-sampled-assignment-layer candidate-table roots challenges
 $\text{final-value query-idxs round-layers } 0 = \text{candidate-table}$

unfolding *generic-fri-sampled-assignment-layer-def* **by** *simp*

lemma *generic-fri-sampled-assignment-layers-length[simp]:*

$\text{length (generic-fri-sampled-assignment-layers candidate-table roots}$
 $\text{challenges final-value query-idxs round-layers}) =$
 $\text{Suc (length challenges)}$

unfolding *generic-fri-sampled-assignment-layers-def* **by** *simp*

lemma *generic-fri-sampled-assignment-layers-nth:*

assumes $\text{layer-idx} \leq \text{length challenges}$

shows

generic-fri-sampled-assignment-layers candidate-table roots challenges
 $\text{final-value query-idxs round-layers ! layer-idx} =$
generic-fri-sampled-assignment-layer candidate-table roots challenges
 $\text{final-value query-idxs round-layers layer-idx}$

proof (cases $\text{layer-idx} < \text{length challenges}$)

case *True*

```

then show ?thesis
  unfolding generic-fri-sampled-assignment-layers-def
  by (simp add: nth-append)
next
case False
then have layer-eq: layer-idx = length challenges
  using assms by simp
then show ?thesis
  unfolding generic-fri-sampled-assignment-layers-def
  by (simp add: nth-append)
qed

lemma generic-fri-sampled-assignment-layers-base[simp]:
  generic-fri-sampled-assignment-layers candidate-table roots challenges
  final-value query-idxs round-layers ! 0 = candidate-table
using generic-fri-sampled-assignment-layers-nth[of 0 challenges
  candidate-table roots final-value query-idxs round-layers]
by simp

lemma generic-fri-sampled-layer-value-unique:
  assumes constraint:
    generic-fri-sampled-layer-value-constraint candidate-table roots
    challenges final-value query-idxs round-layers layer-idx idx v
  and unique:
     $\bigwedge v'. \text{generic-fri-sampled-layer-value-constraint candidate-table}$ 
     $\text{roots challenges final-value query-idxs round-layers layer-idx idx } v'$ 
     $\implies v' = v$ 
  shows
    generic-fri-sampled-layer-value candidate-table roots challenges
    final-value query-idxs round-layers layer-idx idx = v
proof -
  have ex:  $\exists v. \text{generic-fri-sampled-layer-value-constraint}$ 
    candidate-table roots challenges final-value query-idxs round-layers
    layer-idx idx v
  using constraint by blast
  have some-eq:
    (SOME v. generic-fri-sampled-layer-value-constraint candidate-table
      roots challenges final-value query-idxs round-layers layer-idx idx v) =
      v
  by (rule some-equality) (use constraint unique in blast)+
  show ?thesis
    unfolding generic-fri-sampled-layer-value-def
    using ex some-eq by simp
qed

lemma generic-fri-sampled-assignment-layer-nth-unique:
  assumes idx-bound: idx < clength * scale
  and layer-nonzero: layer-idx  $\neq$  0
  and constraint:

```

generic-fri-sampled-layer-value-constraint candidate-table roots
challenges final-value query-idxs round-layers layer-idx idx v
and *unique:*
 $\bigwedge v'. \text{generic-fri-sampled-layer-value-constraint candidate-table}$
 $\text{roots challenges final-value query-idxs round-layers layer-idx idx } v'$
 $\implies v' = v$
shows
generic-fri-sampled-assignment-layer candidate-table roots challenges
final-value query-idxs round-layers layer-idx ! idx = v
unfolding *generic-fri-sampled-assignment-layer-def*
using *idx-bound layer-nonzero*
by (*simp add: generic-fri-sampled-layer-value-unique[OF constraint unique]*)

lemma *generic-fri-sampled-final-assignment-layer-consistent:*
assumes *challenges-nonempty: 0 < length challenges*
shows
fri-final-constant-consistent
(generic-fri-sampled-assignment-layers candidate-table roots challenges
final-value query-idxs round-layers ! length challenges)
final-value
proof –
let *?layer =*
generic-fri-sampled-assignment-layers candidate-table roots challenges
final-value query-idxs round-layers ! length challenges
have *layer-eq:*
?layer =
generic-fri-sampled-assignment-layer candidate-table roots challenges
final-value query-idxs round-layers (length challenges)
by (*rule generic-fri-sampled-assignment-layers-nth*) *simp*
have *layer-len: length ?layer = clength * scale*
using *challenges-nonempty layer-eq* **by** *simp*
have *nth-final:*
 $\bigwedge \text{idx. idx} < \text{length } ?\text{layer} \implies ?\text{layer} ! \text{idx} = \text{final-value}$
proof –
fix *idx*
assume *idx-bound-layer: idx < length ?layer*
then have *idx-bound: idx < clength * scale*
using *layer-len* **by** *simp*
have *constraint:*
generic-fri-sampled-layer-value-constraint candidate-table roots
challenges final-value query-idxs round-layers
(length challenges) idx final-value
using *challenges-nonempty*
unfolding *generic-fri-sampled-layer-value-constraint-def* **by** *simp*
have *unique:*
 $\bigwedge v'. \text{generic-fri-sampled-layer-value-constraint candidate-table}$
 $\text{roots challenges final-value query-idxs round-layers}$
 $(\text{length challenges}) \text{ idx } v' \implies v' = \text{final-value}$
using *challenges-nonempty*

```

unfolding generic-fri-sampled-layer-value-constraint-def by simp
show ?layer ! idx = final-value
unfolding layer-eq
by (rule generic-fri-sampled-assignment-layer-nth-unique
      [OF idx-bound - constraint unique])
      (use challenges-nonempty in simp)
qed
show ?thesis
unfolding fri-final-constant-consistent-def
using nth-final by (auto simp: in-set-conv-nth)
qed

```

definition trace-fri-sampled-layer-assignment-obstruction

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-sampled-layer-assignment-obstruction s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 trace-table.
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-layer-assignment-obstruction}$
 $\text{trace-table-low-degree trace-table (clength - 1) trace-roots}$
 $\text{trace-bs trace-final fri-query-idxs trace-round-layers})$

definition composition-fri-sampled-layer-assignment-obstruction

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-layer-assignment-obstruction s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{opening-composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table.}$
 $\text{composition-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{composition-query-idxs composition-openings trace-table}$
 $\text{composition-table} \wedge$

generic-fri-sampled-layer-assignment-obstruction
 (composition-table-low-degree (to-nat fri-dg)) composition-table
 (to-nat fri-dg) opening-composition-roots composition-bs
 composition-final fri-query-idxs composition-round-layers)

definition *trace-fri-sampled-assignment-conflict*

:: ('f, 'a) protocol-channel-scheme $\Rightarrow$
 (unit list $\times$ ('f, 'a) protocol-channel-scheme) option $\Rightarrow$ bool

where

trace-fri-sampled-assignment-conflict s out $\longleftrightarrow$
 ($\exists$ trace-roots trace-bs trace-final dg composition-roots
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table.
trace-fri-partial-candidate-opening-evidence s out trace-roots
 trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table $\wedge$
generic-fri-sampled-assignment-conflict trace-table trace-roots
 trace-bs trace-final fri-query-idxs trace-round-layers)

definition *composition-fri-sampled-assignment-conflict*

:: ('f, 'a) protocol-channel-scheme $\Rightarrow$
 (unit list $\times$ ('f, 'a) protocol-channel-scheme) option $\Rightarrow$ bool

where

composition-fri-sampled-assignment-conflict s out $\longleftrightarrow$
 ($\exists$ trace-roots trace-bs trace-final fri-dg
 opening-composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers fr
 f-fri-roots f-final as dg composition-fri-roots final
 trace-query-idxs trace-openings composition-query-idxs
 composition-openings trace-table composition-table.
composition-fri-partial-candidate-opening-evidence s out trace-roots
 trace-bs trace-final fri-dg opening-composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr f-fri-roots f-final as dg
 composition-fri-roots final trace-query-idxs trace-openings
 composition-query-idxs composition-openings trace-table
 composition-table $\wedge$
generic-fri-sampled-assignment-conflict composition-table
 opening-composition-roots composition-bs composition-final
 fri-query-idxs composition-round-layers)

lemma *generic-fri-sampled-layer-assignment-obstructionD:*

assumes

generic-fri-sampled-layer-assignment-obstruction low-degree
 candidate-table degree-bound roots challenges final-value query-idxs
 round-layers

shows

*generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers*

$\wedge \text{doms layers. } \neg$

*generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers*

using *assms*

unfolding *generic-fri-sampled-layer-assignment-obstruction-def*

by *simp-all*

lemma *trace-fri-sampled-layer-assignment-obstructionE:*

assumes

trace-fri-sampled-layer-assignment-obstruction s out

obtains *trace-roots trace-bs trace-final dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr candidate-query-idxs trace-openings

trace-table

where

trace-fri-partial-candidate-opening-evidence s out trace-roots

trace-bs trace-final dg composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr candidate-query-idxs trace-openings

trace-table

generic-fri-sampled-layer-assignment-obstruction

trace-table-low-degree trace-table (clength - 1) trace-roots

trace-bs trace-final fri-query-idxs trace-round-layers

using *assms*

unfolding *trace-fri-sampled-layer-assignment-obstruction-def*

apply *(elim exE conjE)*

apply *(rule that)*

apply *assumption*

apply *assumption*

done

lemma *composition-fri-sampled-layer-assignment-obstructionE:*

assumes

composition-fri-sampled-layer-assignment-obstruction s out

obtains *trace-roots trace-bs trace-final fri-dg*

opening-composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers fr

f-fri-roots f-final as dg composition-fri-roots final

trace-query-idxs trace-openings composition-query-idxs

composition-openings trace-table composition-table

where

composition-fri-partial-candidate-opening-evidence s out trace-roots

trace-bs trace-final fri-dg opening-composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr f-fri-roots f-final as dg

```

    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table
    generic-fri-sampled-layer-assignment-obstruction
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
using assms
unfolding composition-fri-sampled-layer-assignment-obstruction-def
apply (elim exE conjE)
apply (rule that)
apply assumption
apply assumption
done

```

lemma *generic-fri-no-next-value-conflict-unique*:

assumes *no-conflict*:

- $\neg$ *generic-fri-sampled-next-value-conflict roots challenges*
- query-idxs round-layers*
- and** *round-bound*: *round-idx* < *length query-idxs*
- and** *round-bound'*: *round-idx'* < *length query-idxs*
- and** *layer-bound*: *layer-idx* < *length challenges*
- and** *same-idx*:
- fri-evidence-next-idx roots query-idxs round-idx layer-idx* =
- fri-evidence-next-idx roots query-idxs round-idx' layer-idx*
- and** *forced*:
- generic-fri-round-forced-next-value roots challenges query-idxs*
- round-layers round-idx layer-idx v*
- and** *forced'*:
- generic-fri-round-forced-next-value roots challenges query-idxs*
- round-layers round-idx' layer-idx v'*

shows *v* = *v'*

using *assms*

unfolding *generic-fri-sampled-next-value-conflict-def*

by *blast*

lemma *generic-fri-no-final-value-conflict-forced-eq*:

assumes *no-conflict*:

- $\neg$ *generic-fri-sampled-final-value-conflict roots challenges*
- final-value query-idxs round-layers*
- and** *challenges-nonempty*: *0* < *length challenges*
- and** *round-bound*: *round-idx* < *length query-idxs*
- and** *forced*:
- generic-fri-round-forced-next-value roots challenges query-idxs*
- round-layers round-idx (length challenges - 1) v*

shows *v* = *final-value*

using *assms*

unfolding *generic-fri-sampled-final-value-conflict-def*

by *blast*

lemma *generic-fri-no-assignment-conflict-no-final:*
assumes
 $\neg$ *generic-fri-sampled-assignment-conflict candidate-table roots*
challenges final-value query-idxs round-layers
shows
 $\neg$ *generic-fri-sampled-final-value-conflict roots challenges*
final-value query-idxs round-layers
using *assms unfolding generic-fri-sampled-assignment-conflict-def* **by** *simp*

lemma *generic-fri-no-assignment-conflict-no-next:*
assumes
 $\neg$ *generic-fri-sampled-assignment-conflict candidate-table roots*
challenges final-value query-idxs round-layers
shows
 $\neg$ *generic-fri-sampled-next-value-conflict roots challenges*
query-idxs round-layers
using *assms unfolding generic-fri-sampled-assignment-conflict-def* **by** *simp*

lemma *generic-fri-sampled-final-layer-forced-value:*
assumes *no-conflict:*
 $\neg$ *generic-fri-sampled-assignment-conflict candidate-table roots*
challenges final-value query-idxs round-layers
and *challenges-nonempty:* $0 < \text{length challenges}$
and *round-bound:* $\text{round-idx} < \text{length query-idxs}$
and *idx-bound:*
fri-evidence-next-idx roots query-idxs round-idx
 $(\text{length challenges} - 1) < \text{clength} * \text{scale}$
and *forced:*
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers round-idx (length challenges - 1) v
shows
generic-fri-sampled-assignment-layers candidate-table roots challenges
final-value query-idxs round-layers ! length challenges !
fri-evidence-next-idx roots query-idxs round-idx
 $(\text{length challenges} - 1) = v$
proof –
have *v-final:* $v = \text{final-value}$
by (*rule generic-fri-no-final-value-conflict-forced-eq*
 $[OF \text{ generic-fri-no-assignment-conflict-no-final } [OF \text{ no-conflict}]$
 $\text{challenges-nonempty round-bound forced}]$)
let *?idx =*
fri-evidence-next-idx roots query-idxs round-idx
 $(\text{length challenges} - 1)$
have *layer-eq:*
generic-fri-sampled-assignment-layers candidate-table roots challenges
final-value query-idxs round-layers ! length challenges =
generic-fri-sampled-assignment-layer candidate-table roots challenges
final-value query-idxs round-layers (length challenges)


```

    by (rule generic-fri-sampled-assignment-layers-nth) simp
  have constraint:
    generic-fri-sampled-layer-value-constraint candidate-table roots
      challenges final-value query-idxs round-layers
      (length challenges) ?idx final-value
    using challenges-nonempty
    unfolding generic-fri-sampled-layer-value-constraint-def by simp
  have unique:
     $\bigwedge v'. \text{generic-fri-sampled-layer-value-constraint candidate-table roots}$ 
      challenges final-value query-idxs round-layers
      (length challenges) ?idx  $v' \implies v' = \text{final-value}$ 
    using challenges-nonempty
    unfolding generic-fri-sampled-layer-value-constraint-def by simp
  have nth-final:
    generic-fri-sampled-assignment-layer candidate-table roots challenges
      final-value query-idxs round-layers (length challenges) ! ?idx =
      final-value
    by (rule generic-fri-sampled-assignment-layer-nth-unique
        [OF idx-bound - constraint unique])
      (use challenges-nonempty in simp)
  show ?thesis
    using nth-final v-final unfolding layer-eq by simp
qed

```

```

lemma generic-fri-no-base-opening-conflict-matches-step:
  assumes no-conflict:
     $\neg \text{generic-fri-sampled-base-opening-conflict candidate-table roots}$ 
      challenges query-idxs round-layers
  and challenges-nonempty:  $0 < \text{length challenges}$ 
  and round-bound:  $\text{round-idx} < \text{length query-idxs}$ 
  and step:
    fri-layer-step-evidence
      (roots ! 0)
      (challenges ! 0)
      (fri-evidence-layer-len roots 0)
      (fri-evidence-layer-idx roots query-idxs round-idx 0)
      1
      (fri-sibling-index (fri-evidence-layer-len roots 0)
        (fri-evidence-layer-idx roots query-idxs round-idx 0))
      xp xp-path xn xn-path
      (fri-evidence-next-idx roots query-idxs round-idx 0)
      (fri-evidence-next-value roots challenges query-idxs round-idx 0
        xp xn)
      (round-layers ! round-idx ! 0)
  shows
    fri-opening-matches-table (fri-evidence-layer-len roots 0)
      (fri-evidence-layer-idx roots query-idxs round-idx 0)
      candidate-table xp xn
  using assms

```

unfolding *generic-fri-sampled-base-opening-conflict-def*
by *blast*

lemma *generic-fri-no-assignment-conflict-no-base:*

assumes

¬ *generic-fri-sampled-assignment-conflict candidate-table roots*
challenges final-value query-idxs round-layers

shows

¬ *generic-fri-sampled-base-opening-conflict candidate-table roots*
challenges query-idxs round-layers

using *assms unfolding generic-fri-sampled-assignment-conflict-def* **by** *simp*

lemma *generic-fri-sampled-assignment-base-matches-step:*

assumes *no-conflict:*

¬ *generic-fri-sampled-assignment-conflict candidate-table roots*
challenges final-value query-idxs round-layers

and *challenges-nonempty: 0 < length challenges*

and *round-bound: round-idx < length query-idxs*

and *step:*

fri-layer-step-evidence
(roots ! 0)
(challenges ! 0)
(fri-evidence-layer-len roots 0)
(fri-evidence-layer-idx roots query-idxs round-idx 0)
1
(fri-sibling-index (fri-evidence-layer-len roots 0)
(fri-evidence-layer-idx roots query-idxs round-idx 0))
xp xp-path xn xn-path
(fri-evidence-next-idx roots query-idxs round-idx 0)
(fri-evidence-next-value roots challenges query-idxs round-idx 0
xp xn)
(round-layers ! round-idx ! 0)

shows

fri-opening-matches-table (fri-evidence-layer-len roots 0)
(fri-evidence-layer-idx roots query-idxs round-idx 0)
(generic-fri-sampled-assignment-layers candidate-table roots
challenges final-value query-idxs round-layers ! 0) xp xn

proof —

have *base-match:*

fri-opening-matches-table (fri-evidence-layer-len roots 0)
(fri-evidence-layer-idx roots query-idxs round-idx 0)
candidate-table xp xn

by (*rule generic-fri-no-base-opening-conflict-matches-step*
[*OF generic-fri-no-assignment-conflict-no-base[OF no-conflict]*
challenges-nonempty round-bound step])

show *?thesis*

using *base-match* **by** *simp*

qed

lemma *generic-fri-no-successor-opening-conflict-matches-left*:
assumes *no-conflict*:
 $\neg$ *generic-fri-sampled-successor-opening-conflict roots challenges*
query-idxs round-layers
and *source-bound*: *source-round* < *length query-idxs*
and *target-bound*: *target-round* < *length query-idxs*
and *layer-bound*: *Suc layer-idx* < *length challenges*
and *forced*:
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers source-round layer-idx v
and *step*:
fri-layer-step-evidence
(*roots* ! *Suc layer-idx*)
(*challenges* ! *Suc layer-idx*)
(*fri-evidence-layer-len roots* (*Suc layer-idx*))
(*fri-evidence-layer-idx roots query-idxs target-round*
(*Suc layer-idx*))
($2 \wedge$ *Suc layer-idx*)
(*fri-sibling-index* (*fri-evidence-layer-len roots* (*Suc layer-idx*))
(*fri-evidence-layer-idx roots query-idxs target-round*
(*Suc layer-idx*)))
xp xp-path xn xn-path
(*fri-evidence-next-idx roots query-idxs target-round*
(*Suc layer-idx*))
(*fri-evidence-next-value roots challenges query-idxs target-round*
(*Suc layer-idx xp xn*)
(*round-layers* ! *target-round* ! *Suc layer-idx*)
and *same-idx*:
fri-evidence-next-idx roots query-idxs source-round layer-idx =
fri-evidence-layer-idx roots query-idxs target-round
(*Suc layer-idx*)
shows *v = xp*
using *assms*
unfolding *generic-fri-sampled-successor-opening-conflict-def*
by *blast*

lemma *generic-fri-no-successor-opening-conflict-matches-right*:
assumes *no-conflict*:
 $\neg$ *generic-fri-sampled-successor-opening-conflict roots challenges*
query-idxs round-layers
and *source-bound*: *source-round* < *length query-idxs*
and *target-bound*: *target-round* < *length query-idxs*
and *layer-bound*: *Suc layer-idx* < *length challenges*
and *forced*:
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers source-round layer-idx v
and *step*:
fri-layer-step-evidence
(*roots* ! *Suc layer-idx*)

```

(challenges ! Suc layer-idx)
(fri-evidence-layer-len roots (Suc layer-idx))
(fri-evidence-layer-idx roots query-idxs target-round
 (Suc layer-idx))
(2 ^ Suc layer-idx)
(fri-sibling-index (fri-evidence-layer-len roots (Suc layer-idx))
 (fri-evidence-layer-idx roots query-idxs target-round
 (Suc layer-idx)))
xp xp-path xn xn-path
(fri-evidence-next-idx roots query-idxs target-round
 (Suc layer-idx))
(fri-evidence-next-value roots challenges query-idxs target-round
 (Suc layer-idx) xp xn)
(round-layers ! target-round ! Suc layer-idx)
and same-idx:
  fri-evidence-next-idx roots query-idxs source-round layer-idx =
  fri-sibling-index (fri-evidence-layer-len roots (Suc layer-idx))
  (fri-evidence-layer-idx roots query-idxs target-round
  (Suc layer-idx))
shows v = xn
using assms
unfolding generic-fri-sampled-successor-opening-conflict-def
by blast

```

lemma generic-fri-no-same-layer-opening-conflict-left-left:

```

assumes no-conflict:
  ¬ generic-fri-sampled-same-layer-opening-conflict roots challenges
  query-idxs round-layers
and round-bound: round-idx < length query-idxs
and round-bound': round-idx' < length query-idxs
and layer-bound: layer-idx < length challenges
and step:
  fri-layer-step-evidence
  (roots ! layer-idx)
  (challenges ! layer-idx)
  (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
  (2 ^ layer-idx)
  (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
  xp xp-path xn xn-path
  (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  (fri-evidence-next-value roots challenges query-idxs round-idx
  layer-idx xp xn)
  (round-layers ! round-idx ! layer-idx)
and step':
  fri-layer-step-evidence
  (roots ! layer-idx)
  (challenges ! layer-idx)

```

```

(fri-evidence-layer-len roots layer-idx)
(fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
(2 ^ layer-idx)
(fri-sibling-index (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
yp yp-path yn yn-path
(fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
(fri-evidence-next-value roots challenges query-idxs round-idx'
  layer-idx yp yn)
(round-layers ! round-idx' ! layer-idx)
and same-idx:
  fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
  fri-evidence-layer-idx roots query-idxs round-idx' layer-idx
shows xp = yp
using assms
unfolding generic-fri-sampled-same-layer-opening-conflict-def
by blast

```

lemma generic-fri-no-same-layer-opening-conflict-left-right:

```

assumes no-conflict:
  ¬ generic-fri-sampled-same-layer-opening-conflict roots challenges
  query-idxs round-layers
and round-bound: round-idx < length query-idxs
and round-bound': round-idx' < length query-idxs
and layer-bound: layer-idx < length challenges
and step:
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
    xp xp-path xn xn-path
    (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx
      layer-idx xp xn)
    (round-layers ! round-idx ! layer-idx)
and step':
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
    yp yp-path yn yn-path

```

```

    (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx'
      layer-idx yp yn)
    (round-layers ! round-idx' ! layer-idx)
and same-idx:
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
shows xp = yn
using assms
unfolding generic-fri-sampled-same-layer-opening-conflict-def
by blast

```

lemma *generic-fri-no-same-layer-opening-conflict-right-left:*

assumes *no-conflict:*

```

  ¬ generic-fri-sampled-same-layer-opening-conflict roots challenges
    query-idxs round-layers
and round-bound: round-idx < length query-idxs
and round-bound': round-idx' < length query-idxs
and layer-bound: layer-idx < length challenges
and step:
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
    xp xp-path xn xn-path
    (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx
      layer-idx xp xn)
    (round-layers ! round-idx ! layer-idx)
and step':
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
    yp yp-path yn yn-path
    (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx'
      layer-idx yp yn)
    (round-layers ! round-idx' ! layer-idx)
and same-idx:

```

fri-sibling-index (*fri-evidence-layer-len* *roots* *layer-idx*)
 (*fri-evidence-layer-idx* *roots* *query-idxs* *round-idx* *layer-idx*) =
fri-evidence-layer-idx *roots* *query-idxs* *round-idx'* *layer-idx*
shows *xn* = *yp*
using *assms*
unfolding *generic-fri-sampled-same-layer-opening-conflict-def*
by *blast*

lemma *generic-fri-no-same-layer-opening-conflict-right-right*:
assumes *no-conflict*:
 $\neg$ *generic-fri-sampled-same-layer-opening-conflict* *roots* *challenges*
query-idxs *round-layers*
and *round-bound*: *round-idx* < *length* *query-idxs*
and *round-bound'*: *round-idx'* < *length* *query-idxs*
and *layer-bound*: *layer-idx* < *length* *challenges*
and *step*:
fri-layer-step-evidence
 (*roots* ! *layer-idx*)
 (*challenges* ! *layer-idx*)
 (*fri-evidence-layer-len* *roots* *layer-idx*)
 (*fri-evidence-layer-idx* *roots* *query-idxs* *round-idx* *layer-idx*)
 ($2 \wedge$ *layer-idx*)
 (*fri-sibling-index* (*fri-evidence-layer-len* *roots* *layer-idx*)
 (*fri-evidence-layer-idx* *roots* *query-idxs* *round-idx* *layer-idx*))
xp xp-path xn xn-path
 (*fri-evidence-next-idx* *roots* *query-idxs* *round-idx* *layer-idx*)
 (*fri-evidence-next-value* *roots* *challenges* *query-idxs* *round-idx*
layer-idx xp xn)
 (*round-layers* ! *round-idx* ! *layer-idx*)
and *step'*:
fri-layer-step-evidence
 (*roots* ! *layer-idx*)
 (*challenges* ! *layer-idx*)
 (*fri-evidence-layer-len* *roots* *layer-idx*)
 (*fri-evidence-layer-idx* *roots* *query-idxs* *round-idx'* *layer-idx*)
 ($2 \wedge$ *layer-idx*)
 (*fri-sibling-index* (*fri-evidence-layer-len* *roots* *layer-idx*)
 (*fri-evidence-layer-idx* *roots* *query-idxs* *round-idx'* *layer-idx*))
yp yp-path yn yn-path
 (*fri-evidence-next-idx* *roots* *query-idxs* *round-idx'* *layer-idx*)
 (*fri-evidence-next-value* *roots* *challenges* *query-idxs* *round-idx'*
layer-idx yp yn)
 (*round-layers* ! *round-idx'* ! *layer-idx*)
and *same-idx*:
fri-sibling-index (*fri-evidence-layer-len* *roots* *layer-idx*)
 (*fri-evidence-layer-idx* *roots* *query-idxs* *round-idx* *layer-idx*) =
fri-sibling-index (*fri-evidence-layer-len* *roots* *layer-idx*)
 (*fri-evidence-layer-idx* *roots* *query-idxs* *round-idx'* *layer-idx*)
shows *xn* = *yn*

```

using assms
unfolding generic-fri-sampled-same-layer-opening-conflict-def
by blast

lemma generic-fri-no-assignment-conflict-no-successor:
  assumes
     $\neg$  generic-fri-sampled-assignment-conflict candidate-table roots
    challenges final-value query-idxs round-layers
  shows
     $\neg$  generic-fri-sampled-successor-opening-conflict roots challenges
    query-idxs round-layers
  using assms unfolding generic-fri-sampled-assignment-conflict-def by simp

lemma generic-fri-no-assignment-conflict-no-same:
  assumes
     $\neg$  generic-fri-sampled-assignment-conflict candidate-table roots
    challenges final-value query-idxs round-layers
  shows
     $\neg$  generic-fri-sampled-same-layer-opening-conflict roots challenges
    query-idxs round-layers
  using assms unfolding generic-fri-sampled-assignment-conflict-def by simp

lemma generic-fri-sampled-layer-value-constraint-current-left:
  assumes layer-nonzero: 0 < layer-idx
  and layer-bound: layer-idx < length challenges
  and round-bound: round-idx < length query-idxs
  and step:
    fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    ( $2 \wedge \text{layer-idx}$ )
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
     (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
    xp xp-path xn xn-path
    (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx
     layer-idx xp xn)
    (round-layers ! round-idx ! layer-idx)
  shows
    generic-fri-sampled-layer-value-constraint candidate-table roots
    challenges final-value query-idxs round-layers layer-idx
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) xp
proof –
  have nz: layer-idx  $\neq$  0
  using layer-nonzero by simp
  have not-final: layer-idx  $\neq$  length challenges
  using layer-nonzero layer-bound by simp-all

```



```

show ?thesis
  unfolding generic-fri-sampled-layer-value-constraint-def
  apply (simp add: nz not-final)
  apply (rule disjI2)
  apply (rule exI[of - round-idx])
  apply (intro conjI)
    apply (rule round-bound)
    apply (rule layer-bound)
  apply (rule exI[of - xp])
  apply (rule exI[of - xp-path])
  apply (rule exI[of - xn])
  apply (intro conjI)
    apply (rule exI[of - xn-path])
    apply (rule step)
  apply simp
done
qed

lemma generic-fri-sampled-layer-value-constraint-current-right:
  assumes layer-nonzero: 0 < layer-idx
  and layer-bound: layer-idx < length challenges
  and round-bound: round-idx < length query-idxs
  and step:
    fri-layer-step-evidence
      (roots ! layer-idx)
      (challenges ! layer-idx)
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      (2 ^ layer-idx)
      (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
      xp xp-path xn xn-path
      (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
      (fri-evidence-next-value roots challenges query-idxs round-idx
        layer-idx xp xn)
      (round-layers ! round-idx ! layer-idx)
  shows
    generic-fri-sampled-layer-value-constraint candidate-table roots
      challenges final-value query-idxs round-layers layer-idx
      (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)) xn
proof -
  have nz: layer-idx ≠ 0
  using layer-nonzero by simp
  have not-final: layer-idx ≠ length challenges
  using layer-nonzero layer-bound by simp-all
  show ?thesis
  unfolding generic-fri-sampled-layer-value-constraint-def
  apply (simp add: nz not-final)

```

```

apply (rule disjI2)
apply (rule exI[of - round-idx])
apply (intro conjI)
  apply (rule round-bound)
  apply (rule layer-bound)
apply (rule exI[of - xp])
apply (rule exI[of - xp-path])
apply (rule exI[of - xn])
apply (intro conjI)
  apply (rule exI[of - xn-path])
  apply (rule step)
  apply simp
done
qed

lemma generic-fri-sampled-assignment-current-left-matches-step:
assumes no-conflict:
   $\neg$  generic-fri-sampled-assignment-conflict candidate-table roots
    challenges final-value query-idxs round-layers
and layer-nonzero:  $0 < \text{layer-idx}$ 
and layer-bound:  $\text{layer-idx} < \text{length challenges}$ 
and round-bound:  $\text{round-idx} < \text{length query-idxs}$ 
and idx-bound:
  fri-evidence-layer-idx roots query-idxs round-idx layer-idx  $<$ 
     $\text{length} * \text{scale}$ 
and step:
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    ( $2 \wedge \text{layer-idx}$ )
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
    xp xp-path xn xn-path
    (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx
      layer-idx xp xn)
    (round-layers ! round-idx ! layer-idx)
shows
  generic-fri-sampled-assignment-layers candidate-table roots challenges
    final-value query-idxs round-layers ! layer-idx !
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx = xp
proof –
let ?raw = fri-evidence-layer-idx roots query-idxs round-idx layer-idx
have layer-eq:
  generic-fri-sampled-assignment-layers candidate-table roots challenges
    final-value query-idxs round-layers ! layer-idx =
    generic-fri-sampled-assignment-layer candidate-table roots challenges

```

```

    final-value query-idxs round-layers layer-idx
  by (rule generic-fri-sampled-assignment-layers-nth)
    (use layer-bound in simp)
have constraint:
  generic-fri-sampled-layer-value-constraint candidate-table roots
  challenges final-value query-idxs round-layers layer-idx ?raw xp
  by (rule generic-fri-sampled-layer-value-constraint-current-left
    [OF layer-nonzero layer-bound round-bound step])
have unique:
   $\wedge v'. \text{generic-fri-sampled-layer-value-constraint candidate-table roots}$ 
  challenges final-value query-idxs round-layers layer-idx ?raw v'
   $\implies v' = xp$ 
proof -
  fix v'
  assume c:
    generic-fri-sampled-layer-value-constraint candidate-table roots
    challenges final-value query-idxs round-layers layer-idx ?raw v'
  have pred-or-current:
    ( $\exists$  source-round.
      source-round < length query-idxs  $\wedge$ 
      generic-fri-round-forced-next-value roots challenges query-idxs
      round-layers source-round (layer-idx - 1) v'  $\wedge$ 
      fri-evidence-next-idx roots query-idxs source-round
      (layer-idx - 1) = ?raw)  $\vee$ 
    ( $\exists$  round-idx' yp yp-path yn yn-path.
      round-idx' < length query-idxs  $\wedge$ 
      layer-idx < length challenges  $\wedge$ 
      fri-layer-step-evidence
        (roots ! layer-idx)
        (challenges ! layer-idx)
        (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
        (2  $\wedge$  layer-idx)
        (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
          (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
      yp yp-path yn yn-path
      (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
      (fri-evidence-next-value roots challenges query-idxs round-idx'
        layer-idx yp yn)
      (round-layers ! round-idx' ! layer-idx)  $\wedge$ 
      ((?raw = fri-evidence-layer-idx roots query-idxs round-idx'
        layer-idx  $\wedge$  v' = yp)  $\vee$ 
      (?raw = fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)  $\wedge$ 
        v' = yn)))
  using c layer-nonzero layer-bound
  unfolding generic-fri-sampled-layer-value-constraint-def
  by auto
then show v' = xp

```

```

proof
  assume
     $\exists \text{source-round.}$ 
     $\text{source-round} < \text{length query-idxs} \wedge$ 
     $\text{generic-fri-round-forced-next-value roots challenges query-idxs}$ 
     $\text{round-layers source-round (layer-idx - 1) } v' \wedge$ 
     $\text{fri-evidence-next-idx roots query-idxs source-round}$ 
     $(\text{layer-idx} - 1) = ?\text{raw}$ 
  then obtain  $\text{source-round}$  where  $\text{source-bound:}$ 
     $\text{source-round} < \text{length query-idxs}$ 
  and forced:
     $\text{generic-fri-round-forced-next-value roots challenges query-idxs}$ 
     $\text{round-layers source-round (layer-idx - 1) } v'$ 
  and same-idx:
     $\text{fri-evidence-next-idx roots query-idxs source-round}$ 
     $(\text{layer-idx} - 1) = ?\text{raw}$ 
  by blast
have  $\text{suc: Suc (layer-idx - 1) = layer-idx}$ 
  using layer-nonzero by simp
have  $v' = xp$ 
  using generic-fri-no-successor-opening-conflict-matches-left
   $[OF \text{ generic-fri-no-assignment-conflict-no-successor}[OF \text{ no-conflict}]$ 
     $\text{source-bound round-bound, of layer-idx - 1 } v'$ 
     $xp \text{ } xp\text{-path } xn \text{ } xn\text{-path}]$ 
     $\text{layer-bound forced step same-idx suc}$  by simp
  then show ?thesis .
next
  assume
     $\exists \text{round-idx' yp yp-path yn yn-path.}$ 
     $\text{round-idx'} < \text{length query-idxs} \wedge$ 
     $\text{layer-idx} < \text{length challenges} \wedge$ 
     $\text{fri-layer-step-evidence}$ 
     $(\text{roots ! layer-idx})$ 
     $(\text{challenges ! layer-idx})$ 
     $(\text{fri-evidence-layer-len roots layer-idx})$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx})$ 
     $(2 \wedge \text{layer-idx})$ 
     $(\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx}))$ 
     $\text{yp yp-path yn yn-path}$ 
     $(\text{fri-evidence-next-idx roots query-idxs round-idx' layer-idx})$ 
     $(\text{fri-evidence-next-value roots challenges query-idxs round-idx'}$ 
     $\text{layer-idx yp yn})$ 
     $(\text{round-layers ! round-idx' ! layer-idx}) \wedge$ 
     $((?raw = \text{fri-evidence-layer-idx roots query-idxs round-idx'}$ 
     $\text{layer-idx} \wedge v' = \text{yp}) \vee$ 
     $(?raw = \text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx}) \wedge$ 
     $v' = \text{yn}))$ 

```

```

then obtain round-idx' yp yp-path yn yn-path where round-bound':
  round-idx' < length query-idxs
and step':
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
  yp yp-path yn yn-path
    (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx'
      layer-idx yp yn)
    (round-layers ! round-idx' ! layer-idx)
and value-case:
  (?raw = fri-evidence-layer-idx roots query-idxs round-idx'
    layer-idx  $\wedge$  v' = yp)  $\vee$ 
  (?raw = fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)  $\wedge$ 
    v' = yn)
by blast
from value-case show ?thesis
proof
  assume raw-case:
    ?raw = fri-evidence-layer-idx roots query-idxs round-idx'
      layer-idx  $\wedge$  v' = yp
  have xp = yp
  by (rule generic-fri-no-same-layer-opening-conflict-left-left
    [OF generic-fri-no-assignment-conflict-no-same[OF no-conflict]
      round-bound round-bound' layer-bound step step'])
    (use raw-case in simp)
  then show ?thesis
    using raw-case by simp
next
  assume sibling-case:
    ?raw = fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)  $\wedge$ 
      v' = yn
  have xp = yn
  by (rule generic-fri-no-same-layer-opening-conflict-left-right
    [OF generic-fri-no-assignment-conflict-no-same[OF no-conflict]
      round-bound round-bound' layer-bound step step'])
    (use sibling-case in simp)
  then show ?thesis
    using sibling-case by simp
qed
qed

```

```

qed
have nth-row:
  generic-fri-sampled-assignment-layer candidate-table roots challenges
  final-value query-idxs round-layers layer-idx ! ?raw = xp
by (rule generic-fri-sampled-assignment-layer-nth-unique
  [OF idx-bound - constraint unique])
  (use layer-nonzero in simp)
show ?thesis
  using nth-row unfolding layer-eq by simp
qed

lemma generic-fri-sampled-assignment-current-right-matches-step:
  assumes no-conflict:
    ¬ generic-fri-sampled-assignment-conflict candidate-table roots
    challenges final-value query-idxs round-layers
  and layer-nonzero: 0 < layer-idx
  and layer-bound: layer-idx < length challenges
  and round-bound: round-idx < length query-idxs
  and idx-bound:
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) <
    length * scale
  and step:
    fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
    xp xp-path xn xn-path
    (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx
      layer-idx xp xn)
    (round-layers ! round-idx ! layer-idx)
  shows
    generic-fri-sampled-assignment-layers candidate-table roots challenges
    final-value query-idxs round-layers ! layer-idx !
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) = xn
proof -
  let ?sib =
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
  have layer-eq:
    generic-fri-sampled-assignment-layers candidate-table roots challenges
    final-value query-idxs round-layers ! layer-idx =
    generic-fri-sampled-assignment-layer candidate-table roots challenges

```

```

    final-value query-idxs round-layers layer-idx
  by (rule generic-fri-sampled-assignment-layers-nth)
    (use layer-bound in simp)
have constraint:
  generic-fri-sampled-layer-value-constraint candidate-table roots
  challenges final-value query-idxs round-layers layer-idx ?sib xn
  by (rule generic-fri-sampled-layer-value-constraint-current-right
    [OF layer-nonzero layer-bound round-bound step])
have unique:
   $\wedge v'. \text{generic-fri-sampled-layer-value-constraint candidate-table roots}$ 
  challenges final-value query-idxs round-layers layer-idx ?sib  $v'$ 
   $\implies v' = xn$ 
proof -
  fix  $v'$ 
  assume c:
    generic-fri-sampled-layer-value-constraint candidate-table roots
    challenges final-value query-idxs round-layers layer-idx ?sib  $v'$ 
  have pred-or-current:
    ( $\exists$  source-round.
      source-round < length query-idxs  $\wedge$ 
      generic-fri-round-forced-next-value roots challenges query-idxs
      round-layers source-round (layer-idx - 1)  $v'$   $\wedge$ 
      fri-evidence-next-idx roots query-idxs source-round
      (layer-idx - 1) = ?sib)  $\vee$ 
    ( $\exists$  round-idx' yp yp-path yn yn-path.
      round-idx' < length query-idxs  $\wedge$ 
      layer-idx < length challenges  $\wedge$ 
      fri-layer-step-evidence
        (roots ! layer-idx)
        (challenges ! layer-idx)
        (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
        (2  $\wedge$  layer-idx)
        (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
          (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
      yp yp-path yn yn-path
      (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
      (fri-evidence-next-value roots challenges query-idxs round-idx'
        layer-idx yp yn)
      (round-layers ! round-idx' ! layer-idx)  $\wedge$ 
      ((?sib = fri-evidence-layer-idx roots query-idxs round-idx'
        layer-idx  $\wedge$   $v' = yp$ )  $\vee$ 
      (?sib = fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)  $\wedge$ 
         $v' = yn$ )))
  using c layer-nonzero layer-bound
  unfolding generic-fri-sampled-layer-value-constraint-def
  by auto
then show  $v' = xn$ 

```

```

proof
  assume
     $\exists$  source-round.
      source-round < length query-idxs  $\wedge$ 
      generic-fri-round-forced-next-value roots challenges query-idxs
      round-layers source-round (layer-idx - 1) v'  $\wedge$ 
      fri-evidence-next-idx roots query-idxs source-round
      (layer-idx - 1) = ?sib
  then obtain source-round where source-bound:
    source-round < length query-idxs
  and forced:
    generic-fri-round-forced-next-value roots challenges query-idxs
    round-layers source-round (layer-idx - 1) v'
  and same-idx:
    fri-evidence-next-idx roots query-idxs source-round
    (layer-idx - 1) = ?sib
  by blast
  have suc: Suc (layer-idx - 1) = layer-idx
  using layer-nonzero by simp
  have v' = xn
  using generic-fri-no-successor-opening-conflict-matches-right
    [OF generic-fri-no-assignment-conflict-no-successor [OF no-conflict]
      source-bound round-bound, of layer-idx - 1 v'
      xp xp-path xn xn-path]
  layer-bound forced step same-idx suc by simp
  then show ?thesis .
next
  assume
     $\exists$  round-idx' yp yp-path yn yn-path.
      round-idx' < length query-idxs  $\wedge$ 
      layer-idx < length challenges  $\wedge$ 
      fri-layer-step-evidence
        (roots ! layer-idx)
        (challenges ! layer-idx)
        (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
        ( $2 \wedge \text{layer-idx}$ )
        (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
          (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
      yp yp-path yn yn-path
      (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
      (fri-evidence-next-value roots challenges query-idxs round-idx'
        layer-idx yp yn)
      (round-layers ! round-idx' ! layer-idx)  $\wedge$ 
      ((? sib = fri-evidence-layer-idx roots query-idxs round-idx'
        layer-idx  $\wedge$  v' = yp)  $\vee$ 
        (? sib = fri-sibling-index (fri-evidence-layer-len roots layer-idx)
          (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)  $\wedge$ 
          v' = yn))

```



```

then obtain round-idx' yp yp-path yn yn-path where round-bound':
  round-idx' < length query-idxs
and step':
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
  yp yp-path yn yn-path
    (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx'
      layer-idx yp yn)
    (round-layers ! round-idx' ! layer-idx)
and value-case:
  (?sib = fri-evidence-layer-idx roots query-idxs round-idx'
    layer-idx ∧ v' = yp) ∨
  (?sib = fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
    v' = yn)
by blast
from value-case show ?thesis
proof
  assume raw-case:
    ?sib = fri-evidence-layer-idx roots query-idxs round-idx'
      layer-idx ∧ v' = yp
  have xn = yp
    by (rule generic-fri-no-same-layer-opening-conflict-right-left
      [OF generic-fri-no-assignment-conflict-no-same[OF no-conflict]
        round-bound round-bound' layer-bound step step'])
    (use raw-case in simp)
  then show ?thesis
    using raw-case by simp
next
  assume sibling-case:
    ?sib = fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
      v' = yn
  have xn = yn
    by (rule generic-fri-no-same-layer-opening-conflict-right-right
      [OF generic-fri-no-assignment-conflict-no-same[OF no-conflict]
        round-bound round-bound' layer-bound step step'])
    (use sibling-case in simp)
  then show ?thesis
    using sibling-case by simp
qed
qed

```

```

qed
have nth-sib:
  generic-fri-sampled-assignment-layer candidate-table roots challenges
  final-value query-idxs round-layers layer-idx ! ?sib = xn
by (rule generic-fri-sampled-assignment-layer-nth-unique
    [OF idx-bound - constraint unique])
  (use layer-nonzero in simp)
show ?thesis
  using nth-sib unfolding layer-eq by simp
qed

```

```

lemma generic-fri-sampled-assignment-current-matches-step:
  assumes no-conflict:
    ¬ generic-fri-sampled-assignment-conflict candidate-table roots
      challenges final-value query-idxs round-layers
  and layer-nonzero: 0 < layer-idx
  and layer-bound: layer-idx < length challenges
  and round-bound: round-idx < length query-idxs
  and raw-bound:
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
      fri-evidence-layer-len roots layer-idx
  and len-bound:
    fri-evidence-layer-len roots layer-idx ≤ clength * scale
  and sibling-bound:
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) <
        clength * scale
  and step:
    fri-layer-step-evidence
      (roots ! layer-idx)
      (challenges ! layer-idx)
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      (2 ^ layer-idx)
      (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
      xp xp-path xn xn-path
      (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
      (fri-evidence-next-value roots challenges query-idxs round-idx
        layer-idx xp xn)
      (round-layers ! round-idx ! layer-idx)
  shows
    fri-opening-matches-table (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      (generic-fri-sampled-assignment-layers candidate-table roots
        challenges final-value query-idxs round-layers ! layer-idx) xp xn
proof -
  let ?len = fri-evidence-layer-len roots layer-idx
  let ?raw = fri-evidence-layer-idx roots query-idxs round-idx layer-idx

```

```

let ?sib = fri-sibling-index ?len ?raw
let ?layer =
  generic-fri-sampled-assignment-layers candidate-table roots challenges
  final-value query-idxs round-layers ! layer-idx
have raw-bound-assignment: ?raw < clength * scale
  using raw-bound len-bound by simp
have layer-eq:
  ?layer =
    generic-fri-sampled-assignment-layer candidate-table roots challenges
    final-value query-idxs round-layers layer-idx
  by (rule generic-fri-sampled-assignment-layers-nth)
    (use layer-bound in simp)
have layer-len: length ?layer = clength * scale
  using layer-eq layer-nonzero by simp
have left:
  ?layer ! ?raw = xp
  by (rule generic-fri-sampled-assignment-current-left-matches-step
    [OF no-conflict layer-nonzero layer-bound round-bound
      raw-bound-assignment step])
have right:
  ?layer ! ?sib = xn
  by (rule generic-fri-sampled-assignment-current-right-matches-step
    [OF no-conflict layer-nonzero layer-bound round-bound
      sibling-bound step])
show ?thesis
  unfolding fri-opening-matches-table-def
  using raw-bound len-bound layer-len left right by simp
qed

lemma generic-fri-sampled-assignment-successor-forced-value:
assumes no-conflict:
  ¬ generic-fri-sampled-assignment-conflict candidate-table roots
  challenges final-value query-idxs round-layers
and layer-bound: Suc layer-idx < length challenges
and round-bound: round-idx < length query-idxs
and next-bound:
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
  clength * scale
and forced:
  generic-fri-round-forced-next-value roots challenges query-idxs
  round-layers round-idx layer-idx v
shows
  generic-fri-sampled-assignment-layers candidate-table roots challenges
  final-value query-idxs round-layers ! Suc layer-idx !
  fri-evidence-next-idx roots query-idxs round-idx layer-idx = v
proof –
let ?next = fri-evidence-next-idx roots query-idxs round-idx layer-idx
have suc-nonzero: Suc layer-idx ≠ 0
  by simp

```

```

have layer-eq:
  generic-fri-sampled-assignment-layers candidate-table roots challenges
    final-value query-idxs round-layers ! Suc layer-idx =
  generic-fri-sampled-assignment-layer candidate-table roots challenges
    final-value query-idxs round-layers (Suc layer-idx)
by (rule generic-fri-sampled-assignment-layers-nth)
  (use layer-bound in simp)
have constraint:
  generic-fri-sampled-layer-value-constraint candidate-table roots
    challenges final-value query-idxs round-layers (Suc layer-idx)
    ?next v
using layer-bound round-bound forced
unfolding generic-fri-sampled-layer-value-constraint-def
by auto
have unique:
   $\bigwedge v'. \text{generic-fri-sampled-layer-value-constraint candidate-table roots}$ 
    challenges final-value query-idxs round-layers (Suc layer-idx)
    ?next v'  $\implies v' = v$ 
proof -
fix v'
assume c:
  generic-fri-sampled-layer-value-constraint candidate-table roots
    challenges final-value query-idxs round-layers (Suc layer-idx)
    ?next v'
have pred-or-current:
  ( $\exists$  source-round.
    source-round < length query-idxs  $\wedge$ 
    generic-fri-round-forced-next-value roots challenges query-idxs
      round-layers source-round layer-idx v'  $\wedge$ 
    fri-evidence-next-idx roots query-idxs source-round layer-idx =
      ?next)  $\vee$ 
  ( $\exists$  target-round xp xp-path xn xn-path.
    target-round < length query-idxs  $\wedge$ 
    Suc layer-idx < length challenges  $\wedge$ 
    fri-layer-step-evidence
      (roots ! Suc layer-idx)
      (challenges ! Suc layer-idx)
      (fri-evidence-layer-len roots (Suc layer-idx))
      (fri-evidence-layer-idx roots query-idxs target-round
        (Suc layer-idx))
      (2  $\wedge$  Suc layer-idx)
      (fri-sibling-index (fri-evidence-layer-len roots (Suc layer-idx))
        (fri-evidence-layer-idx roots query-idxs target-round
          (Suc layer-idx)))
    xp xp-path xn xn-path
    (fri-evidence-next-idx roots query-idxs target-round
      (Suc layer-idx))
    (fri-evidence-next-value roots challenges query-idxs target-round
      (Suc layer-idx) xp xn))

```

```

    (round-layers ! target-round ! Suc layer-idx) ∧
    ((?next = fri-evidence-layer-idx roots query-idxs target-round
      (Suc layer-idx) ∧ v' = xp) ∨
      (?next = fri-sibling-index
        (fri-evidence-layer-len roots (Suc layer-idx))
        (fri-evidence-layer-idx roots query-idxs target-round
          (Suc layer-idx)) ∧
          v' = xn)))
  using c layer-bound
  unfolding generic-fri-sampled-layer-value-constraint-def
  by auto
then show v' = v
proof
  assume
    ∃ source-round.
      source-round < length query-idxs ∧
      generic-fri-round-forced-next-value roots challenges query-idxs
        round-layers source-round layer-idx v' ∧
      fri-evidence-next-idx roots query-idxs source-round layer-idx =
        ?next
  then obtain source-round where source-bound:
    source-round < length query-idxs
  and forced':
    generic-fri-round-forced-next-value roots challenges query-idxs
      round-layers source-round layer-idx v'
  and same-idx:
    fri-evidence-next-idx roots query-idxs source-round layer-idx =
      ?next
  by blast
  have v = v'
  proof -
    have layer-bound0: layer-idx < length challenges
      using layer-bound by simp
    show ?thesis
    by (rule generic-fri-no-next-value-conflict-unique
      [OF generic-fri-no-assignment-conflict-no-next[OF no-conflict]
        round-bound source-bound layer-bound0 - forced forced'])
      (use same-idx in simp)
  qed
  then show ?thesis by simp
next
  assume
    ∃ target-round xp xp-path xn xn-path.
      target-round < length query-idxs ∧
      Suc layer-idx < length challenges ∧
      fri-layer-step-evidence
        (roots ! Suc layer-idx)
        (challenges ! Suc layer-idx)
        (fri-evidence-layer-len roots (Suc layer-idx))

```

```

(fri-evidence-layer-idx roots query-idxs target-round
 (Suc layer-idx))
(2 ^ Suc layer-idx)
(fri-sibling-index (fri-evidence-layer-len roots (Suc layer-idx))
 (fri-evidence-layer-idx roots query-idxs target-round
  (Suc layer-idx)))
xp xp-path xn xn-path
(fri-evidence-next-idx roots query-idxs target-round
 (Suc layer-idx))
(fri-evidence-next-value roots challenges query-idxs target-round
 (Suc layer-idx) xp xn)
(round-layers ! target-round ! Suc layer-idx) ∧
((?next = fri-evidence-layer-idx roots query-idxs target-round
 (Suc layer-idx) ∧ v' = xp) ∨
 (?next = fri-sibling-index
 (fri-evidence-layer-len roots (Suc layer-idx))
 (fri-evidence-layer-idx roots query-idxs target-round
  (Suc layer-idx)) ∧
 v' = xn))
then obtain target-round xp xp-path xn xn-path where target-bound:
  target-round < length query-idxs
and step:
  fri-layer-step-evidence
  (roots ! Suc layer-idx)
  (challenges ! Suc layer-idx)
  (fri-evidence-layer-len roots (Suc layer-idx))
  (fri-evidence-layer-idx roots query-idxs target-round
   (Suc layer-idx))
  (2 ^ Suc layer-idx)
  (fri-sibling-index (fri-evidence-layer-len roots (Suc layer-idx))
   (fri-evidence-layer-idx roots query-idxs target-round
    (Suc layer-idx)))
  xp xp-path xn xn-path
  (fri-evidence-next-idx roots query-idxs target-round
   (Suc layer-idx))
  (fri-evidence-next-value roots challenges query-idxs target-round
   (Suc layer-idx) xp xn)
  (round-layers ! target-round ! Suc layer-idx)
and value-case:
  (?next = fri-evidence-layer-idx roots query-idxs target-round
   (Suc layer-idx) ∧ v' = xp) ∨
  (?next = fri-sibling-index
   (fri-evidence-layer-len roots (Suc layer-idx))
   (fri-evidence-layer-idx roots query-idxs target-round
    (Suc layer-idx)) ∧
   v' = xn)
by blast
from value-case show ?thesis
proof

```

```

assume raw-case:
  ?next = fri-evidence-layer-idx roots query-idxs target-round
  (Suc layer-idx)  $\wedge$   $v' = xp$ 
have  $v = xp$ 
  by (rule generic-fri-no-successor-opening-conflict-matches-left
    [OF generic-fri-no-assignment-conflict-no-successor
      [OF no-conflict] round-bound target-bound layer-bound
        forced step])
    (use raw-case in simp)
then show ?thesis
  using raw-case by simp
next
assume sibling-case:
  ?next = fri-sibling-index
  (fri-evidence-layer-len roots (Suc layer-idx))
  (fri-evidence-layer-idx roots query-idxs target-round
    (Suc layer-idx))  $\wedge$ 
   $v' = xn$ 
have  $v = xn$ 
  by (rule generic-fri-no-successor-opening-conflict-matches-right
    [OF generic-fri-no-assignment-conflict-no-successor
      [OF no-conflict] round-bound target-bound layer-bound
        forced step])
    (use sibling-case in simp)
then show ?thesis
  using sibling-case by simp
qed
qed
qed
have nth-next:
  generic-fri-sampled-assignment-layer candidate-table roots challenges
  final-value query-idxs round-layers (Suc layer-idx) ! ?next = v
  by (rule generic-fri-sampled-assignment-layer-nth-unique
    [OF next-bound suc-nonzero constraint unique])
show ?thesis
  using nth-next unfolding layer-eq by simp
qed

lemma accepted-fri-opening-transcript-query-idx-bound:
assumes
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final dg composition-roots composition-bs composition-final
  query-idxs trace-round-layers composition-round-layers
and  $idx \in \text{set query-idxs}$ 
shows  $idx < \text{clength} * \text{scale}$ 
using assms
unfolding accepted-fri-opening-transcript-def by blast

lemma trace-fri-partial-candidate-opening-evidence-query-idx-bound:

```

assumes
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and $idx \in \text{set } \text{fri-query-idxs}$
shows $idx < \text{clength} * \text{scale}$
proof –
have *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
by (rule *trace-fri-partial-candidate-opening-evidenceD(1)*[*OF assms(1)*])
show *?thesis*
by (rule *accepted-fri-opening-transcript-query-idx-bound*
[*OF fri-openings assms(2)*])
qed

lemma *composition-fri-partial-candidate-opening-evidence-query-idx-bound*:

assumes
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and $idx \in \text{set } \text{fri-query-idxs}$
shows $idx < \text{clength} * \text{scale}$
proof –
have *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
by (rule *composition-fri-partial-candidate-opening-evidenceD(1)*
[*OF assms(1)*])
show *?thesis*
by (rule *accepted-fri-opening-transcript-query-idx-bound*
[*OF fri-openings assms(2)*])
qed

lemma *fri-layer-indices-nth-bound*:

assumes *query-bound: query-idx < len*
and *layer-bound: layer-idx < n*
and *eval-power: len = 2 ^ N*
and *rounds-bound: n ≤ N*
shows


```

    fri-layer-indices n query-idx len ! layer-idx <
    fri-layer-lengths n len ! layer-idx
    using layer-bound query-bound rounds-bound eval-power
proof (induction layer-idx arbitrary: n query-idx len N)
  case 0
  then show ?case
    by (cases n) simp-all
next
  case (Suc layer-idx)
  then obtain n' where n-eq: n = Suc n'
    by (cases n) auto
  have layer-bound': layer-idx < n'
    using Suc.premis(1) unfolding n-eq by simp
  have len-even: 2 dvd len
    using Suc.premis(3,4) n-eq Suc.premis(1)
    by (cases N) simp-all
  have len-half-power:
     $\exists N'. \text{len div } 2 = 2 \wedge N' \wedge n' \leq N'$ 
proof (cases N)
  case 0
  then show ?thesis
    using Suc.premis(3) n-eq by simp
next
  case (Suc N')
  have len div 2 =  $2 \wedge N'$ 
    using Suc.premis(4) unfolding Suc by simp
  moreover have  $n' \leq N'$ 
    using Suc.premis(3) unfolding n-eq Suc by simp
  ultimately show ?thesis by blast
qed
  then obtain N' where len-half-eq: len div 2 =  $2 \wedge N'$ 
    and rounds-bound':  $n' \leq N'$ 
    by blast
  have query-bound':
    query-idx mod (len div 2) < len div 2
    using len-half-eq by simp
  have ih:
    fri-layer-indices n' (query-idx mod (len div 2)) (len div 2) !
    layer-idx <
    fri-layer-lengths n' (len div 2) ! layer-idx
    by (rule Suc.IH[OF layer-bound' query-bound' rounds-bound'
      len-half-eq])
  show ?case
    using ih unfolding n-eq by simp
qed

lemma fri-evidence-layer-idx-bound-from-query-bound:
  assumes partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound

```

roots challenges final-value query-idxs round-layers
and *query-bound: query-idx < clength * scale*
and *round-idx-eq: query-idx = query-idxs ! round-idx*
and *layer-bound: layer-idx < length challenges*
and *eval-power: clength * scale = 2 ^ N*
and *rounds-bound: length challenges ≤ N*
shows
fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
fri-evidence-layer-len roots layer-idx
proof –
have *roots-len: length challenges = length roots*
by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
show ?thesis
unfolding *fri-evidence-layer-idx-def fri-evidence-layer-len-def*
using *fri-layer-indices-nth-bound*
[OF query-bound, of layer-idx length roots N]
layer-bound roots-len eval-power rounds-bound round-idx-eq
by *simp*
qed

lemma *fri-evidence-layer-len-le-eval-domain:*
assumes *partial:*
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers
and *layer-bound: layer-idx < length challenges*
and *eval-power: clength * scale = 2 ^ N*
shows *fri-evidence-layer-len roots layer-idx ≤ clength * scale*
proof –
have *roots-len: length challenges = length roots*
by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
have *roots-bound: layer-idx < length roots*
using *layer-bound roots-len* **by** *simp*
have *len-eq:*
fri-evidence-layer-len roots layer-idx =
*(clength * scale) div 2 ^ layer-idx*
unfolding *fri-evidence-layer-len-def*
by (rule *fri-layer-lengths-nth-div*[OF roots-bound])
show ?thesis
unfolding *len-eq* **by** *simp*
qed

lemma *fri-evidence-sibling-index-bound-from-raw:*
assumes *raw-bound:*
fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
fri-evidence-layer-len roots layer-idx
and *len-bound:*
*fri-evidence-layer-len roots layer-idx ≤ clength * scale*
shows
fri-sibling-index (fri-evidence-layer-len roots layer-idx)

```

      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) <
      clength * scale
proof –
  have len-pos:  $0 < \text{fri-evidence-layer-len roots layer-idx}$ 
    using raw-bound by simp
  have sibling-lt:
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) <
      fri-evidence-layer-len roots layer-idx
    unfolding fri-sibling-index-def
    using len-pos by simp
  then show ?thesis
    using len-bound by linarith
qed

```

lemma *fri-evidence-next-idx-bound-from-raw*:

```

assumes raw-bound:
  fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
  fri-evidence-layer-len roots layer-idx
and len-bound:
  fri-evidence-layer-len roots layer-idx ≤ clength * scale
shows
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
  clength * scale

```

```

proof –
  have len-pos:  $0 < \text{fri-evidence-layer-len roots layer-idx}$ 
    using raw-bound by simp
  have half-le:  $\text{fri-evidence-layer-len roots layer-idx div } 2 \leq$ 
     $\text{clength} * \text{scale}$ 
    using len-bound by simp
  show ?thesis
proof (cases fri-evidence-layer-len roots layer-idx div 2 = 0)
  case True
    then show ?thesis
      unfolding fri-evidence-next-idx-def
      using raw-bound len-bound by simp
  next
    case False
    then have half-pos:
       $0 < \text{fri-evidence-layer-len roots layer-idx div } 2$ 
      by simp
    have mod-lt:
      fri-evidence-layer-idx roots query-idxs round-idx layer-idx mod
      (fri-evidence-layer-len roots layer-idx div 2) <
      fri-evidence-layer-len roots layer-idx div 2
      by (rule mod-less-divisor[OF half-pos])
    then show ?thesis
      unfolding fri-evidence-next-idx-def
      using half-le by linarith

```

qed
qed

lemma *trace-fri-reachable-missing-sampled-layer-chain-None*[simp]:
 $\neg$ *trace-fri-reachable-missing-sampled-layer-chain* *s* *None*
 unfolding *trace-fri-reachable-missing-sampled-layer-chain-def*
 trace-fri-bad-with-reachable-partial-candidate-def
 accepted-with-partial-trace-openings-def
 by *simp*

lemma *composition-fri-reachable-missing-sampled-layer-chain-None*[simp]:
 $\neg$ *composition-fri-reachable-missing-sampled-layer-chain* *s* *None*
 unfolding *composition-fri-reachable-missing-sampled-layer-chain-def*
 composition-fri-bad-with-reachable-partial-candidate-def
 accepted-with-partial-initial-openings-def
 accepted-with-partial-trace-openings-def
 by *simp*

lemma *trace-fri-sampled-layer-assignment-obstruction-None*[simp]:
 $\neg$ *trace-fri-sampled-layer-assignment-obstruction* *s* *None*
 unfolding *trace-fri-sampled-layer-assignment-obstruction-def*
 trace-fri-partial-candidate-opening-evidence-def
 accepted-fri-opening-transcript-def
 by *simp*

lemma *composition-fri-sampled-layer-assignment-obstruction-None*[simp]:
 $\neg$ *composition-fri-sampled-layer-assignment-obstruction* *s* *None*
 unfolding *composition-fri-sampled-layer-assignment-obstruction-def*
 composition-fri-partial-candidate-opening-evidence-def
 accepted-fri-opening-transcript-def
 by *simp*

lemma *trace-fri-sampled-assignment-conflict-None*[simp]:
 $\neg$ *trace-fri-sampled-assignment-conflict* *s* *None*
 unfolding *trace-fri-sampled-assignment-conflict-def*
 trace-fri-partial-candidate-opening-evidence-def
 accepted-fri-opening-transcript-def
 by *simp*

lemma *composition-fri-sampled-assignment-conflict-None*[simp]:
 $\neg$ *composition-fri-sampled-assignment-conflict* *s* *None*
 unfolding *composition-fri-sampled-assignment-conflict-def*
 composition-fri-partial-candidate-opening-evidence-def
 accepted-fri-opening-transcript-def
 by *simp*

lemma *wp-trace-fri-assignment-obstruction-bound-from-conflict*:
 assumes *support-imp*:
 $\bigwedge out. out \in \text{set-dist } (\text{execute verify-monad } s) \implies$

$\text{trace-fri-sampled-layer-assignment-obstruction } s \text{ out} \implies$
 $\text{trace-fri-sampled-assignment-conflict } s \text{ out}$
and *conflict-bound*:
 $\text{wp-event verify-monad } (\text{trace-fri-sampled-assignment-conflict } s) \ s$
 $\leq C$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-layer-assignment-obstruction } s) \ s \leq C$
proof –
have $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-layer-assignment-obstruction } s) \ s \leq$
 $\text{wp-event verify-monad } (\text{trace-fri-sampled-assignment-conflict } s) \ s$
by (rule *wp-event-mono-on-support*) (use *support-imp* **in** *blast*)
also have $\dots \leq C$
by (rule *conflict-bound*)
finally show *?thesis* .
qed

lemma *wp-composition-fri-assignment-obstruction-bound-from-conflict*:
assumes *support-imp*:
 $\bigwedge \text{out. out} \in \text{set-dist } (\text{execute verify-monad } s) \implies$
 $\text{composition-fri-sampled-layer-assignment-obstruction } s \text{ out} \implies$
 $\text{composition-fri-sampled-assignment-conflict } s \text{ out}$
and *conflict-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-assignment-conflict } s) \ s \leq C$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-layer-assignment-obstruction } s) \ s \leq C$
proof –
have $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-layer-assignment-obstruction } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-assignment-conflict } s) \ s$
by (rule *wp-event-mono-on-support*) (use *support-imp* **in** *blast*)
also have $\dots \leq C$
by (rule *conflict-bound*)
finally show *?thesis* .
qed

lemma *trace-fri-reachable-missing-sampled-imp-assignment-obstruction-on-support*:
assumes *support*:
 $\text{out} \in \text{set-dist } (\text{execute verify-monad } s)$
and *missing*:
 $\text{trace-fri-reachable-missing-sampled-layer-chain } s \text{ out}$
shows $\text{trace-fri-sampled-layer-assignment-obstruction } s \text{ out}$
proof (cases *out*)
case *None*
then show *?thesis*

```

using missing
unfolding trace-fri-reachable-missing-sampled-layer-chain-def
           trace-fri-bad-with-reachable-partial-candidate-def
           accepted-with-partial-trace-openings-def
by simp
next
case (Some pair)
then obtain result final-state where pair: pair = (result, final-state)
by (cases pair) simp
have outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
using support Some pair by simp
show ?thesis
proof (rule verify-monad-trace-fri-missing-sampled-extracts-opening-evidence
  [OF outcome])
show
  trace-fri-reachable-missing-sampled-layer-chain s
  (Some (result, final-state))
using missing Some pair by simp
next
fix trace-roots trace-bs trace-final dg composition-roots
     composition-bs composition-final fri-query-idxs trace-round-layers
     composition-round-layers fr candidate-query-idxs trace-openings
     trace-table
assume evidence:
  trace-fri-partial-candidate-opening-evidence s
  (Some (result, final-state)) trace-roots trace-bs trace-final dg
  composition-roots composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers fr candidate-query-idxs
  trace-openings trace-table
assume no-chain:
   $\bigwedge \text{doms layers. } \neg$ 
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
  trace-table (clength - 1) trace-roots trace-bs trace-final
  fri-query-idxs trace-round-layers doms layers
have partial:
  generic-fri-partial-evidence trace-table-low-degree trace-table
  (clength - 1) trace-roots trace-bs trace-final fri-query-idxs
  trace-round-layers
by (rule trace-fri-partial-candidate-opening-evidence-generic-partial
  [OF evidence])
have obstruction:
  generic-fri-sampled-layer-assignment-obstruction
  trace-table-low-degree trace-table (clength - 1) trace-roots
  trace-bs trace-final fri-query-idxs trace-round-layers
unfolding generic-fri-sampled-layer-assignment-obstruction-def
using partial no-chain by blast
show trace-fri-sampled-layer-assignment-obstruction s out
unfolding trace-fri-sampled-layer-assignment-obstruction-def

```

```

    using evidence obstruction Some pair by blast
qed
qed

lemma composition-fri-reachable-missing-sampled-imp-assignment-obstruction-on-support:
  assumes support:
    out ∈ set-dist (execute verify-monad s)
  and missing:
    composition-fri-reachable-missing-sampled-layer-chain s out
  shows composition-fri-sampled-layer-assignment-obstruction s out
proof (cases out)
case None
then show ?thesis
  unfolding composition-fri-sampled-layer-assignment-obstruction-def
  using missing
  unfolding composition-fri-reachable-missing-sampled-layer-chain-def
    composition-fri-bad-with-reachable-partial-candidate-def
    accepted-with-partial-initial-openings-def
    accepted-with-partial-trace-openings-def
  by simp
next
case (Some pair)
then obtain result final-state where pair: pair = (result, final-state)
  by (cases pair) simp
have outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using support Some pair by simp
show ?thesis
proof (rule
  verify-monad-composition-fri-missing-sampled-extracts-opening-evidence
  [OF outcome])
  show
    composition-fri-reachable-missing-sampled-layer-chain s
      (Some (result, final-state))
    using missing Some pair by simp
next
fix trace-roots trace-bs trace-final fri-dg opening-composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs trace-openings
  composition-query-idxs composition-openings trace-table
  composition-table
assume evidence:
  composition-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final
  fri-dg opening-composition-roots composition-bs composition-final
  fri-query-idxs trace-round-layers composition-round-layers fr
  f-fri-roots f-final as dg composition-fri-roots final
  trace-query-idxs trace-openings composition-query-idxs

```

```

    composition-openings trace-table composition-table
assume degree-bound: to-nat fri-dg  $\leq$  maxDegree
assume no-chain:
   $\bigwedge$  doms layers.  $\neg$ 
    generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg))
    composition-table (to-nat fri-dg) opening-composition-roots
    composition-bs composition-final fri-query-idxs
    composition-round-layers doms layers
have partial:
  generic-fri-partial-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
  by (rule composition-fri-partial-candidate-opening-evidence-generic-partial
    [OF evidence])
have obstruction:
  generic-fri-sampled-layer-assignment-obstruction
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
  unfolding generic-fri-sampled-layer-assignment-obstruction-def
  using partial no-chain by blast
show composition-fri-sampled-layer-assignment-obstruction s out
  unfolding composition-fri-sampled-layer-assignment-obstruction-def
  using evidence obstruction Some pair degree-bound by meson
qed
qed

lemma wp-trace-fri-reachable-missing-sampled-bound-from-assignment-obstruction:
assumes obstruction:
  wp-event verify-monad
  (trace-fri-sampled-layer-assignment-obstruction s)  $s \leq M$ 
shows
  wp-event verify-monad
  (trace-fri-reachable-missing-sampled-layer-chain s)  $s \leq M$ 
proof –
have wp-event verify-monad
  (trace-fri-reachable-missing-sampled-layer-chain s) s
 $\leq$  wp-event verify-monad
  (trace-fri-sampled-layer-assignment-obstruction s) s
by (rule wp-event-mono-on-support)
  (rule trace-fri-reachable-missing-sampled-imp-assignment-obstruction-on-support)
also have ...  $\leq M$ 
by (rule obstruction)
finally show ?thesis .
qed

lemma wp-composition-fri-reachable-missing-sampled-bound-from-assignment-obstruction:

```


assumes *obstruction*:
 wp-event verify-monad
 (*composition-fri-sampled-layer-assignment-obstruction s*) $s \leq M$
shows
 wp-event verify-monad
 (*composition-fri-reachable-missing-sampled-layer-chain s*) $s \leq M$
proof –
 have *wp-event verify-monad*
 (*composition-fri-reachable-missing-sampled-layer-chain s*) s
 $\leq$ *wp-event verify-monad*
 (*composition-fri-sampled-layer-assignment-obstruction s*) s
 by (rule *wp-event-mono-on-support*)
 (rule *composition-fri-reachable-missing-sampled-imp-assignment-obstruction-on-support*)
 also have $\dots \leq M$
 by (rule *obstruction*)
 finally show *?thesis* .
qed

lemma *trace-fri-reachable-partial-candidate-reduction-from-sampled-and-assignment-obstruction*:

assumes *sampled-bound*:
 wp-event verify-monad
 (*trace-fri-bad-with-sampled-layer-chain s*) $s \leq R$
and *obstruction-bound*:
 wp-event verify-monad
 (*trace-fri-sampled-layer-assignment-obstruction s*) $s \leq M$
and *total-bound*: $R + M \leq \text{trace-fri-error}$
shows *trace-fri-reachable-partial-candidate-reduction-assumption s*
proof (rule *trace-fri-reachable-partial-candidate-reduction-from-sampled-and-missing*)
 show *wp-event verify-monad*
 (*trace-fri-bad-with-sampled-layer-chain s*) $s \leq R$
 by (rule *sampled-bound*)
 next
 show *wp-event verify-monad*
 (*trace-fri-reachable-missing-sampled-layer-chain s*) $s \leq M$
 by (rule *wp-trace-fri-reachable-missing-sampled-bound-from-assignment-obstruction*
 [*OF obstruction-bound*])
 next
 show $R + M \leq \text{trace-fri-error}$
 by (rule *total-bound*)
qed

lemma *composition-fri-reachable-partial-candidate-reduction-from-sampled-and-assignment-obstruction*:

assumes *sampled-bound*:
 wp-event verify-monad
 (*composition-fri-bad-with-sampled-layer-chain s*) $s \leq R$
and *obstruction-bound*:
 wp-event verify-monad
 (*composition-fri-sampled-layer-assignment-obstruction s*) $s \leq M$
and *total-bound*: $R + M \leq \text{composition-fri-error}$

shows *composition-fri-reachable-partial-candidate-reduction-assumption s*
proof (rule
 composition-fri-reachable-partial-candidate-reduction-from-sampled-and-missing)
show *wp-event verify-monad*
 (*composition-fri-bad-with-sampled-layer-chain s*) $s \leq R$
 by (rule *sampled-bound*)
next
show *wp-event verify-monad*
 (*composition-fri-reachable-missing-sampled-layer-chain s*) $s \leq M$
 by (rule
 wp-composition-fri-reachable-missing-sampled-bound-from-assignment-obstruction
 [*OF obstruction-bound*])
next
show $R + M \leq \text{composition-fri-error}$
 by (rule *total-bound*)
qed

lemma *trace-fri-reachable-partial-candidate-reduction-from-sampled-and-assignment-conflict:*
assumes *sampled-bound:*
 wp-event verify-monad
 (*trace-fri-bad-with-sampled-layer-chain s*) $s \leq R$
and *support-imp:*
 $\bigwedge \text{out. out} \in \text{set-dist (execute verify-monad s)} \implies$
 trace-fri-sampled-layer-assignment-obstruction s out $\implies$
 trace-fri-sampled-assignment-conflict s out
and *conflict-bound:*
 wp-event verify-monad
 (*trace-fri-sampled-assignment-conflict s*) $s \leq M$
and *total-bound:* $R + M \leq \text{trace-fri-error}$
shows *trace-fri-reachable-partial-candidate-reduction-assumption s*
proof (rule
 trace-fri-reachable-partial-candidate-reduction-from-sampled-and-assignment-obstruction)
show *wp-event verify-monad*
 (*trace-fri-bad-with-sampled-layer-chain s*) $s \leq R$
 by (rule *sampled-bound*)
next
show *wp-event verify-monad*
 (*trace-fri-sampled-layer-assignment-obstruction s*) $s \leq M$
 by (rule *wp-trace-fri-assignment-obstruction-bound-from-conflict*
 [*OF support-imp conflict-bound*])
next
show $R + M \leq \text{trace-fri-error}$
 by (rule *total-bound*)
qed

lemma *composition-fri-reachable-partial-candidate-reduction-from-sampled-and-assignment-conflict:*
assumes *sampled-bound:*
 wp-event verify-monad
 (*composition-fri-bad-with-sampled-layer-chain s*) $s \leq R$

and *support-imp*:
 $\bigwedge \text{out. out} \in \text{set-dist } (\text{execute verify-monad } s) \implies$
 $\text{composition-fri-sampled-layer-assignment-obstruction } s \text{ out} \implies$
 $\text{composition-fri-sampled-assignment-conflict } s \text{ out}$
and *conflict-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-assignment-conflict } s) \ s \leq M$
and *total-bound*: $R + M \leq \text{composition-fri-error}$
shows *composition-fri-reachable-partial-candidate-reduction-assumption* s
proof (rule
 $\text{composition-fri-reachable-partial-candidate-reduction-from-sampled-and-assignment-obstruction}$)
show *wp-event verify-monad*
 $(\text{composition-fri-bad-with-sampled-layer-chain } s) \ s \leq R$
by (rule *sampled-bound*)
next
show *wp-event verify-monad*
 $(\text{composition-fri-sampled-layer-assignment-obstruction } s) \ s \leq M$
by (rule *wp-composition-fri-assignment-obstruction-bound-from-conflict*
 $[\text{OF support-imp conflict-bound}]$)
next
show $R + M \leq \text{composition-fri-error}$
by (rule *total-bound*)
qed

lemma *trace-fri-bad-with-partial-candidates-bound-from-sampled-and-assignment-obstruction*:

fixes $R \ M :: \text{prob}$

assumes *sampled-bound*:

$\bigwedge \text{data attacker-state.}$

$\text{Some } (\text{data, attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$

$\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq R$

and *obstruction-bound*:

$\bigwedge \text{data attacker-state.}$

$\text{Some } (\text{data, attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$

$\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-layer-assignment-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq M$

shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-partial-candidates)
  adversary-initial-state  $\leq R + M$ 
proof (rule trace-fri-bad-with-partial-candidates-bound-from-sampled-and-missing)
fix data attacker-state
assume builder:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (trace-fri-bad-with-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq R$ 
by (rule sampled-bound[OF builder])
next
fix data attacker-state
assume builder:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (trace-fri-reachable-missing-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq M$ 
by (rule
  wp-trace-fri-reachable-missing-sampled-bound-from-assignment-obstruction)
  (rule obstruction-bound[OF builder])
qed

lemma composition-fri-bad-with-partial-candidates-bound-from-sampled-and-assignment-obstruction:
fixes R M :: prob
assumes sampled-bound:
   $\bigwedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
  wp-event verify-monad
  (composition-fri-bad-with-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq R$ 
and obstruction-bound:
   $\bigwedge$  data attacker-state.

```

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-sampled-layer-assignment-obstruction*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq M$

shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq R + M$

proof (*rule*
composition-fri-bad-with-partial-candidates-bound-from-sampled-and-missing)

fix *data attacker-state*

assume *builder*:
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state)

show *wp-event verify-monad*
(*composition-fri-bad-with-sampled-layer-chain*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq R$

by (*rule sampled-bound[OF builder]*)

next

fix *data attacker-state*

assume *builder*:
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state)

show *wp-event verify-monad*
(*composition-fri-reachable-missing-sampled-layer-chain*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq M$

by (*rule*
wp-composition-fri-reachable-missing-sampled-bound-from-assignment-obstruction)
(*rule obstruction-bound[OF builder]*)

qed

lemma *trace-fri-bad-with-partial-candidates-bound-from-sampled-and-assignment-conflict*:
fixes *R M* :: *prob*
assumes *sampled-bound*:
 $\bigwedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-bad-with-sampled-layer-chain*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq R$
and *support-imp*:
 $\wedge$ *data attacker-state out.*
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
out $\in$ *set-dist*
(*execute verify-monad*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))) $\implies$
trace-fri-sampled-layer-assignment-obstruction
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) *out* $\implies$
trace-fri-sampled-assignment-conflict
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) *out*
and *conflict-bound*:
 $\wedge$ *data attacker-state.*
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-sampled-assignment-conflict*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq M$
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq R + M$
proof (*rule*
trace-fri-bad-with-partial-candidates-bound-from-sampled-and-assignment-obstruction)
fix *data attacker-state*
assume *builder*:
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state)
show *wp-event verify-monad*
(*trace-fri-bad-with-sampled-layer-chain*

```

    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ R
  by (rule sampled-bound[OF builder])
next
fix data attacker-state
assume builder:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (trace-fri-sampled-layer-assignment-obstruction
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ M
  by (rule wp-trace-fri-assignment-obstruction-bound-from-conflict)
    (use builder support-imp conflict-bound in blast)+
qed

lemma composition-fri-bad-with-partial-candidates-bound-from-sampled-and-assignment-conflict:
  fixes R M :: prob
  assumes sampled-bound:
    ∧data attacker-state.
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state) ⇒
      wp-event verify-monad
        (composition-fri-bad-with-sampled-layer-chain
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)) ≤ R
  and support-imp:
    ∧data attacker-state out.
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state) ⇒
      out ∈ set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))) ⇒
      composition-fri-sampled-layer-assignment-obstruction
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)) out ⇒
      composition-fri-sampled-assignment-conflict
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)) out

```

```

and conflict-bound:
 $\wedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    wp-event verify-monad
      (composition-fri-sampled-assignment-conflict
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))  $\leq M$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq R + M$ 
proof (rule
  composition-fri-bad-with-partial-candidates-bound-from-sampled-and-assignment-obstruction)
fix data attacker-state
assume builder:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (composition-fri-bad-with-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq R$ 
by (rule sampled-bound[OF builder])
next
fix data attacker-state
assume builder:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (composition-fri-sampled-layer-assignment-obstruction
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq M$ 
by (rule wp-composition-fri-assignment-obstruction-bound-from-conflict)
  (use builder support-imp conflict-bound in blast) +
qed

end

end

```



```

theory Soundness-FRI-Sampled-Assignment-Chain
  imports Soundness-FRI-Sampled-Obstructions
begin

```

Construction of sampled FRI layer-chain witnesses from verifier-local sampled assignment layers. This theory is kept separate from the sampled obstruction-routing layer to keep that layer below the project size threshold.

```

context soundness
begin

```

```

lemma ceil-log-le-power:
  assumes le-power:  $m \leq (2::nat) \wedge N$ 
  shows ceil-log  $m \leq N$ 
proof (cases  $m \leq 1$ )
  case True
  then show ?thesis
    unfolding ceil-log-def by simp
next
  case False
  then have m-gt:  $1 < m$ 
    by simp
  then have m-pred-pos:  $0 < m - 1$ 
    by simp
  have m-pred-lt:  $m - 1 < (2::nat) \wedge N$ 
    using le-power m-gt by linarith
  have floor-less: floor-log  $(m - 1) < N$ 
proof (rule ccontr)
  assume  $\neg$  floor-log  $(m - 1) < N$ 
  then have N-le:  $N \leq \text{floor-log } (m - 1)$ 
    by simp
  then have  $(2::nat) \wedge N \leq 2 \wedge \text{floor-log } (m - 1)$ 
    by simp
  also have  $\dots \leq m - 1$ 
    by (rule floor-log-exp2-le[OF m-pred-pos])
  finally show False
    using m-pred-lt by simp
qed
then show ?thesis
  unfolding ceil-log-def using m-gt by simp
qed

```

```

lemma generic-fri-sampled-assignment-fold-at-step:
  assumes partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
    roots challenges final-value query-idxs round-layers
  and no-conflict:
     $\neg$  generic-fri-sampled-assignment-conflict candidate-table roots

```

```

    challenges final-value query-idxs round-layers
and round-bound: round-idx < length query-idxs
and layer-bound: layer-idx < length challenges
and raw-bound:
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
    fri-evidence-layer-len roots layer-idx
and len-bound:
    fri-evidence-layer-len roots layer-idx ≤ clength * scale
and step:
    fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
    xp xp-path xn xn-path
    (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx
    layer-idx xp xn)
    (round-layers ! round-idx ! layer-idx)
shows
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (generic-fri-sampled-assignment-layers candidate-table roots
    challenges final-value query-idxs round-layers ! Suc layer-idx)
    fri-sampled-table-fold
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-canonical-domains (length challenges) ! layer-idx)
    (generic-fri-sampled-assignment-layers candidate-table roots
    challenges final-value query-idxs round-layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (generic-fri-sampled-assignment-layers candidate-table roots
    challenges final-value query-idxs round-layers ! Suc layer-idx !
    fri-evidence-next-idx roots query-idxs round-idx layer-idx)
proof –
    let ?layers =
        generic-fri-sampled-assignment-layers candidate-table roots challenges
        final-value query-idxs round-layers
    let ?len = fri-evidence-layer-len roots layer-idx
    let ?raw = fri-evidence-layer-idx roots query-idxs round-idx layer-idx
    let ?next = fri-evidence-next-idx roots query-idxs round-idx layer-idx
    let ?next-value =
        fri-evidence-next-value roots challenges query-idxs round-idx layer-idx
        xp xn
    have roots-len: length challenges = length roots

```

```

    by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
  have sibling-bound:
    fri-sibling-index ?len ?raw < clength * scale
    by (rule fri-evidence-sibling-index-bound-from-row
        [OF raw-bound len-bound])
  have next-bound: ?next < clength * scale
    by (rule fri-evidence-next-idx-bound-from-row[OF raw-bound len-bound])
  have layer-next-len:
    length (?layers ! Suc layer-idx) = clength * scale
  proof -
    have Suc layer-idx ≤ length challenges
      using layer-bound by simp
    then have ?layers ! Suc layer-idx =
      generic-fri-sampled-assignment-layer candidate-table roots challenges
      final-value query-idxs round-layers (Suc layer-idx)
      by (rule generic-fri-sampled-assignment-layers-nth)
    then show ?thesis by simp
  qed
  then show next-bound-layer:
    ?next < length (?layers ! Suc layer-idx)
    using next-bound by simp
  have current-match:
    fri-opening-matches-table ?len ?raw (?layers ! layer-idx) xp xn
  proof (cases layer-idx = 0)
    case True
    have challenges-nonempty: 0 < length challenges
      using layer-bound by (cases challenges) auto
    have base-match:
      fri-opening-matches-table (fri-evidence-layer-len roots 0)
      (fri-evidence-layer-idx roots query-idxs round-idx 0)
      (?layers ! 0) xp xn
      by (rule generic-fri-sampled-assignment-base-matches-step
          [OF no-conflict challenges-nonempty round-bound])
      (use step True in simp)
    show ?thesis
      using base-match True by simp
  next
    case False
    then have layer-nonzero: 0 < layer-idx
      by simp
    show ?thesis
      by (rule generic-fri-sampled-assignment-current-matches-step
          [OF no-conflict layer-nonzero layer-bound round-bound raw-bound
              len-bound sibling-bound step])
  qed
  have forced:
    generic-fri-round-forced-next-value roots challenges query-idxs
    round-layers round-idx layer-idx ?next-value
  unfolding generic-fri-round-forced-next-value-def

```

```

using step by blast
have next-value:
  ?layers ! Suc layer-idx ! ?next = ?next-value
proof (cases Suc layer-idx < length challenges)
  case True
  show ?thesis
    by (rule generic-fri-sampled-assignment-successor-forced-value
      [OF no-conflict True round-bound next-bound forced])
next
  case False
  have suc-eq: Suc layer-idx = length challenges
    using False layer-bound by simp
  have challenges-nonempty: 0 < length challenges
    using layer-bound by (cases challenges) auto
  have layer-eq: layer-idx = length challenges - 1
    using suc-eq by simp
  have forced-last:
    generic-fri-round-forced-next-value roots challenges query-idxs
      round-layers round-idx (length challenges - 1) ?next-value
    using forced layer-eq by simp
  have next-bound-last:
    fri-evidence-next-idx roots query-idxs round-idx
      (length challenges - 1) < clength * scale
    using next-bound layer-eq by simp
  have final-next:
    ?layers ! length challenges !
    fri-evidence-next-idx roots query-idxs round-idx
      (length challenges - 1) = ?next-value
    apply (rule generic-fri-sampled-final-layer-forced-value)
    using no-conflict challenges-nonempty round-bound next-bound-last forced-last
    by simp-all
  show ?thesis
    using final-next layer-eq suc-eq by simp
qed
have chunk:
  fri-layer-opening-chunk ?len xp xp-path xn xn-path
    (round-layers ! round-idx ! layer-idx)
  by (rule fri-layer-step-evidenceD(4)[OF step])
have canonical-layer:
  fri-canonical-domains (length challenges) ! layer-idx =
    fri-canonical-domain-at layer-idx
  by (rule fri-canonical-domains-nth) (use layer-bound in simp)
have canonical-len:
  length (fri-canonical-domain-at layer-idx) = ?len
  unfolding fri-canonical-domain-at-length fri-evidence-layer-len-def
  using roots-len fri-layer-lengths-nth-div
    [of layer-idx length roots clength * scale] layer-bound
  by simp
have dom-raw:

```

```

    fri-canonical-domains (length challenges) ! layer-idx ! ?raw =
      (h ^ ?raw * shift) ^ (2 ^ layer-idx)
  unfolding canonical-layer
  by (rule fri-canonical-domain-at-nth)
    (use raw-bound canonical-len in simp)
have fold:
  fri-sampled-table-fold
    (challenges ! layer-idx) ?len ?raw (2 ^ layer-idx)
    (fri-canonical-domains (length challenges) ! layer-idx)
    (?layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    ?next-value
  by (rule fri-layer-step-evidence-sampled-table-fold-if-matching-chunk
      [OF step dom-row chunk current-match])
then show
  fri-sampled-table-fold
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-canonical-domains (length challenges) ! layer-idx)
    (?layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (?layers ! Suc layer-idx ! ?next)
  using next-value by simp
qed

lemma generic-fri-sampled-assignment-layer-chain-evidence:
  assumes partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
      roots challenges final-value query-idxs round-layers
  and no-conflict:
    ¬ generic-fri-sampled-assignment-conflict candidate-table roots
      challenges final-value query-idxs round-layers
  and challenges-nonempty: 0 < length challenges
  and query-bounds:
    ∧ round-idx. round-idx < length query-idxs ⇒
      query-idxs ! round-idx < clength * scale
  and eval-power: clength * scale = 2 ^ N
  and rounds-bound: length challenges ≤ N
  shows
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
      degree-bound roots challenges final-value query-idxs round-layers
    (fri-canonical-domains (length challenges))
    (generic-fri-sampled-assignment-layers candidate-table roots
      challenges final-value query-idxs round-layers)
proof -
  let ?doms = fri-canonical-domains (length challenges)
  let ?layers =

```

```

generic-fri-sampled-assignment-layers candidate-table roots challenges
  final-value query-idxs round-layers
have doms-len: length ?doms = Suc (length challenges)
  by (rule fri-canonical-domains-length)
have layers-len: length ?layers = Suc (length challenges)
  by (rule generic-fri-sampled-assignment-layers-length)
have doms0: ?doms ! 0 = eval-domain
  by (rule fri-canonical-domains-first)
have layers0: ?layers ! 0 = candidate-table
  by simp
have final:
  fri-final-constant-consistent (?layers ! length challenges)
  final-value
  by (rule generic-fri-sampled-final-assignment-layer-consistent
    [OF challenges-nonempty])
have samples:
   $\bigwedge$ round-idx layer-idx.
    round-idx < length query-idxs  $\implies$ 
    layer-idx < length challenges  $\implies$ 
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
      length (?layers ! Suc layer-idx)  $\wedge$ 
    fri-sampled-table-fold
      (challenges ! layer-idx)
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      ( $2^{\text{layer-idx}}$ )
      (?doms ! layer-idx)
      (?layers ! layer-idx)
      (round-layers ! round-idx ! layer-idx)
      (?layers ! Suc layer-idx !
        fri-evidence-next-idx roots query-idxs round-idx layer-idx)
proof –
  fix round-idx layer-idx
  assume round-bound: round-idx < length query-idxs
  and layer-bound: layer-idx < length challenges
  have roots-len: length challenges = length roots
  by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
  have all-layers:
    fri-all-round-layer-evidence roots challenges query-idxs round-layers
    by (rule generic-fri-partial-evidence-shapes(3)[OF partial])
  have layer-root-bound: layer-idx < length roots
  using layer-bound roots-len by simp
  have round-layer:
    fri-round-layer-evidence roots challenges query-idxs round-idx
    layer-idx round-layers
    by (rule fri-all-round-layer-evidenceD
      [OF all-layers round-bound layer-root-bound])
  then obtain xp xp-path xn xn-path where step:
    fri-layer-step-evidence

```

```

    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
    xp xp-path xn xn-path
    (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx
      layer-idx xp xn)
    (round-layers ! round-idx ! layer-idx)
  by (elim fri-round-layer-evidence-compactE)
have raw-bound:
  fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
    fri-evidence-layer-len roots layer-idx
  by (rule fri-evidence-layer-idx-bound-from-query-bound
    [OF partial query-bounds[OF round-bound] refl layer-bound
      eval-power rounds-bound])
have len-bound:
  fri-evidence-layer-len roots layer-idx ≤ clength * scale
  by (rule fri-evidence-layer-len-le-eval-domain
    [OF partial layer-bound eval-power])
show
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (?layers ! Suc layer-idx) ∧
  fri-sampled-table-fold
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (?doms ! layer-idx)
    (?layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (?layers ! Suc layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  by (rule conjI)
  (rule generic-fri-sampled-assignment-fold-at-step
    [OF partial no-conflict round-bound layer-bound raw-bound
      len-bound step])+
qed
show ?thesis
  unfolding generic-fri-sampled-layer-chain-evidence-def
  using partial doms-len layers-len doms0 layers0 final samples
  by blast
qed

lemma generic-fri-sampled-assignment-layer-chain-evidence-zero-round:
  assumes partial:

```

```

generic-fri-partial-evidence low-degree candidate-table degree-bound
  roots challenges final-value query-idxs round-layers
and challenges-empty: length challenges = 0
and final:
  fri-final-constant-consistent candidate-table final-value
shows
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
  degree-bound roots challenges final-value query-idxs round-layers
  (fri-canonical-domains (length challenges))
  (generic-fri-sampled-assignment-layers candidate-table roots
    challenges final-value query-idxs round-layers)
proof –
let ?doms = fri-canonical-domains (length challenges)
let ?layers =
  generic-fri-sampled-assignment-layers candidate-table roots challenges
  final-value query-idxs round-layers
have doms-len: length ?doms = Suc (length challenges)
  by (rule fri-canonical-domains-length)
have layers-len: length ?layers = Suc (length challenges)
  by (rule generic-fri-sampled-assignment-layers-length)
have doms0: ?doms ! 0 = eval-domain
  by (rule fri-canonical-domains-first)
have layers0: ?layers ! 0 = candidate-table
  by simp
have final-layer:
  fri-final-constant-consistent (?layers ! length challenges)
  final-value
  using challenges-empty final by simp
have samples:
   $\bigwedge \text{round-idx layer-idx.}$ 
  round-idx < length query-idxs  $\implies$ 
  layer-idx < length challenges  $\implies$ 
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (?layers ! Suc layer-idx)  $\wedge$ 
  fri-sampled-table-fold
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (?doms ! layer-idx)
    (?layers ! layer-idx)
    (round-layers ! round-idx ! layer-idx)
    (?layers ! Suc layer-idx !
      fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  using challenges-empty by simp
show ?thesis
unfolding generic-fri-sampled-layer-chain-evidence-def
using partial doms-len layers-len doms0 layers0 final-layer samples
by blast

```


qed

lemma *generic-fri-sampled-layer-assignment-obstruction-imp-conflict:*

assumes *obstruction:*

*generic-fri-sampled-layer-assignment-obstruction low-degree
candidate-table degree-bound roots challenges final-value query-idxs
round-layers*

and *challenges-nonempty: 0 < length challenges*

and *query-bounds:*

$\bigwedge \text{round-idx. round-idx} < \text{length query-idxs} \implies$
 $\text{query-idxs} ! \text{round-idx} < \text{clength} * \text{scale}$

and *eval-power: clength * scale = 2 ^ N*

and *rounds-bound: length challenges ≤ N*

shows

*generic-fri-sampled-assignment-conflict candidate-table roots
challenges final-value query-idxs round-layers*

proof (rule *ccontr*)

assume *no-conflict:*

$\neg$ *generic-fri-sampled-assignment-conflict candidate-table roots
challenges final-value query-idxs round-layers*

have *partial:*

*generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers*

by (rule *generic-fri-sampled-layer-assignment-obstructionD(1)*
[OF *obstruction*])

have *no-chain:*

$\bigwedge \text{doms layers. } \neg$
*generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
doms layers*

by (rule *generic-fri-sampled-layer-assignment-obstructionD(2)*
[OF *obstruction*])

have *chain:*

*generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges))
(generic-fri-sampled-assignment-layers candidate-table roots
challenges final-value query-idxs round-layers)*

by (rule *generic-fri-sampled-assignment-layer-chain-evidence*
[OF *partial no-conflict challenges-nonempty query-bounds
eval-power rounds-bound*])

show *False*

using *no-chain[of fri-canonical-domains (length challenges)
generic-fri-sampled-assignment-layers candidate-table roots
challenges final-value query-idxs round-layers]* *chain*

by *blast*

qed

lemma *generic-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final:*

```

assumes obstruction:
  generic-fri-sampled-layer-assignment-obstruction low-degree
  candidate-table degree-bound roots challenges final-value query-idxs
  round-layers
and query-bounds:
   $\bigwedge \text{round-idx. round-idx} < \text{length query-idxs} \implies$ 
     $\text{query-idxs} ! \text{round-idx} < \text{clength} * \text{scale}$ 
and eval-power:  $\text{clength} * \text{scale} = 2^N$ 
and rounds-bound:  $\text{length challenges} \leq N$ 
shows
  generic-fri-sampled-assignment-conflict candidate-table roots
  challenges final-value query-idxs round-layers  $\vee$ 
   $\text{length challenges} = 0 \wedge$ 
   $\neg \text{fri-final-constant-consistent candidate-table final-value}$ 
proof (cases  $0 < \text{length challenges}$ )
  case True
  then show ?thesis
  by (intro disjI1
    generic-fri-sampled-layer-assignment-obstruction-imp-conflict
    [OF obstruction - query-bounds eval-power rounds-bound])
next
  case False
  then have challenges-empty:  $\text{length challenges} = 0$ 
  by simp
  have partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
    roots challenges final-value query-idxs round-layers
  by (rule generic-fri-sampled-layer-assignment-obstructionD(1)
    [OF obstruction])
  have no-chain:
     $\bigwedge \text{doms layers. } \neg$ 
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    doms layers
  by (rule generic-fri-sampled-layer-assignment-obstructionD(2)
    [OF obstruction])
  show ?thesis
  proof (cases fri-final-constant-consistent candidate-table final-value)
  case True
  have chain:
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    (fri-canonical-domains (length challenges))
    (generic-fri-sampled-assignment-layers candidate-table roots
    challenges final-value query-idxs round-layers)
  by (rule generic-fri-sampled-assignment-layer-chain-evidence-zero-round
    [OF partial challenges-empty True])
  then show ?thesis
  using no-chain by blast

```

```

next
  case False
  then show ?thesis
    using challenges-empty by blast
qed
qed

```

definition *trace-fri-zero-round-final-obstruction*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool

```

where

```

trace-fri-zero-round-final-obstruction s out  $\longleftrightarrow$ 
  ( $\exists$  trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table.
    trace-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table  $\wedge$ 
    length trace-bs = 0  $\wedge$ 
     $\neg$  fri-final-constant-consistent trace-table trace-final)

```

definition *composition-fri-zero-round-final-obstruction*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool

```

where

```

composition-fri-zero-round-final-obstruction s out  $\longleftrightarrow$ 
  ( $\exists$  trace-roots trace-bs trace-final fri-dg
    opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers fr
    f-fri-roots f-final as dg composition-fri-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table.
    composition-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final fri-dg opening-composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table  $\wedge$ 
    length composition-bs = 0  $\wedge$ 
     $\neg$  fri-final-constant-consistent composition-table composition-final)

```

lemma *trace-fri-sampled-layer-assignment-obstruction-imp-conflict-if-nonempty*:

assumes *obstruction*: *trace-fri-sampled-layer-assignment-obstruction* *s out*

and *challenges-nonempty*:

$\wedge$ *trace-roots* *trace-bs* *trace-final* *dg* *composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table.
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\implies$
 $0 < \text{length trace-bs}$
shows *trace-fri-sampled-assignment-conflict s out*
proof (rule *trace-fri-sampled-layer-assignment-obstructionE*[*OF obstruction*])
fix *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
assume *evidence:*
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
assume *obstruction-generic:*
generic-fri-sampled-layer-assignment-obstruction
trace-table-low-degree trace-table ($\text{clength} - 1$) trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers
obtain *N* **where** *eval-power: $\text{clength} * \text{scale} = 2^N$*
using *eval-domain-length-power* **by** *blast*
have *trace-rounds-le: $\text{length trace-bs} \leq N$*
proof –
have *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
by (rule *trace-fri-partial-candidate-opening-evidenceD*(1)
[*OF evidence*])
have *len: $\text{length trace-bs} = \text{ceil-log clength}$*
using *accepted-fri-opening-transcript-shapes*(1,2)[*OF fri-openings*]
by *simp*
have *$\text{clength} \leq \text{length} * \text{scale}$*
using *scale-pos* **by** *simp*
also have *$\dots = 2^N$*
by (rule *eval-power*)
finally have *$\text{ceil-log clength} \leq N$*
by (rule *ceil-log-le-power*)
then show *?thesis*
using *len* **by** *simp*
qed
have *query-bounds:*
 $\bigwedge \text{round-idx. round-idx} < \text{length fri-query-idxs} \implies$

```

    fri-query-idxs ! round-idx < clength * scale
proof -
  fix round-idx
  assume round-bound: round-idx < length fri-query-idxs
  have fri-query-idxs ! round-idx ∈ set fri-query-idxs
    using round-bound by simp
  then show fri-query-idxs ! round-idx < clength * scale
    by (rule trace-fri-partial-candidate-opening-evidence-query-idx-bound
      [OF evidence])
qed
have conflict:
  generic-fri-sampled-assignment-conflict trace-table trace-roots
    trace-bs trace-final fri-query-idxs trace-round-layers
  by (rule generic-fri-sampled-layer-assignment-obstruction-imp-conflict
    [OF obstruction-generic challenges-nonempty [OF evidence]
      query-bounds eval-power trace-rounds-le])
show ?thesis
  unfolding trace-fri-sampled-assignment-conflict-def
  using evidence conflict by blast
qed

lemma composition-fri-sampled-layer-assignment-obstruction-imp-conflict-if-nonempty:
assumes support:
  out ∈ set-dist (execute verify-monad s)
and obstruction:
  composition-fri-sampled-layer-assignment-obstruction s out
and challenges-nonempty:
  ∧ trace-roots trace-bs trace-final fri-dg opening-composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table.
  composition-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final fri-dg opening-composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table ⇒
  0 < length composition-bs
shows composition-fri-sampled-assignment-conflict s out
proof (rule composition-fri-sampled-layer-assignment-obstructionE
  [OF obstruction])
fix trace-roots trace-bs trace-final fri-dg opening-composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs trace-openings
  composition-query-idxs composition-openings trace-table composition-table

```

```

assume evidence:
  composition-fri-partial-candidate-opening-evidence s out trace-roots
  trace-bs trace-final fri-dg opening-composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs trace-openings
  composition-query-idxs composition-openings trace-table
  composition-table
assume obstruction-generic:
  generic-fri-sampled-layer-assignment-obstruction
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) opening-composition-roots composition-bs
  composition-final fri-query-idxs composition-round-layers
obtain N where eval-power:  $\text{length} * \text{scale} = 2^N$ 
  using eval-domain-length-power by blast
have fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final fri-dg opening-composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers
  by (rule composition-fri-partial-candidate-opening-evidenceD(1)
    [OF evidence])
obtain result final-state where out-eq: out = Some (result, final-state)
  using fri-openings unfolding accepted-fri-opening-transcript-def by blast
have outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using support out-eq by simp
have challenges:
  accepted-fri-challenges s (Some (result, final-state))
  trace-bs fri-dg composition-bs
  using accepted-fri-opening-transcript-challenges[OF fri-openings]
  out-eq by simp
have degree-bound: to-nat fri-dg ≤ maxDegree
  by (rule verify-monad-accepted-fri-challenges-degree-bound
    [OF outcome challenges])
have composition-rounds-le: length composition-bs ≤ N
proof –
  have len: length composition-bs = ceil-log (to-nat fri-dg + 1)
    using accepted-fri-opening-transcript-shapes(3,4)[OF fri-openings]
    by simp
  have to-nat fri-dg + 1 ≤ length * scale
    using degree-bound maxDegree-less-eval-domain by linarith
  also have ... =  $2^N$ 
    by (rule eval-power)
  finally have ceil-log (to-nat fri-dg + 1) ≤ N
    by (rule ceil-log-le-power)
  then show ?thesis
    using len by simp
qed

```

```

have query-bounds:
   $\bigwedge \text{round-idx}. \text{round-idx} < \text{length fri-query-idxs} \implies$ 
     $\text{fri-query-idxs} ! \text{round-idx} < \text{clength} * \text{scale}$ 
proof –
  fix round-idx
  assume round-bound: round-idx < length fri-query-idxs
  have fri-query-idxs ! round-idx  $\in$  set fri-query-idxs
    using round-bound by simp
  then show fri-query-idxs ! round-idx < clength * scale
    by (rule composition-fri-partial-candidate-opening-evidence-query-idx-bound
      [OF evidence])
qed
have conflict:
  generic-fri-sampled-assignment-conflict composition-table
    opening-composition-roots composition-bs composition-final
    fri-query-idxs composition-round-layers
  by (rule generic-fri-sampled-layer-assignment-obstruction-imp-conflict
    [OF obstruction-generic challenges-nonempty[OF evidence]
      query-bounds eval-power composition-rounds-le])
show ?thesis
  unfolding composition-fri-sampled-assignment-conflict-def
  apply (intro exI conjI)
  apply (rule evidence)
  apply (rule conflict)
done
qed

lemma trace-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final:
  assumes obstruction: trace-fri-sampled-layer-assignment-obstruction s out
  shows
    trace-fri-sampled-assignment-conflict s out  $\vee$ 
    trace-fri-zero-round-final-obstruction s out
proof (rule trace-fri-sampled-layer-assignment-obstructionE[OF obstruction])
  fix trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
  assume evidence:
    trace-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
  assume obstruction-generic:
    generic-fri-sampled-layer-assignment-obstruction
    trace-table-low-degree trace-table (clength – 1) trace-roots
    trace-bs trace-final fri-query-idxs trace-round-layers
  obtain N where eval-power: clength * scale =  $2^N$ 
  using eval-domain-length-power by blast

```

```

have trace-rounds-le: length trace-bs ≤ N
proof -
  have fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  by (rule trace-fri-partial-candidate-opening-evidenceD(1)
    [OF evidence])
  have len: length trace-bs = ceil-log clength
    using accepted-fri-opening-transcript-shapes(1,2)[OF fri-openings]
  by simp
  have clength ≤ length * scale
    using scale-pos by simp
  also have ... = 2 ^ N
    by (rule eval-power)
  finally have ceil-log clength ≤ N
    by (rule ceil-log-le-power)
  then show ?thesis
    using len by simp
qed
have query-bounds:
  ∧ round-idx. round-idx < length fri-query-idxs ⇒
  fri-query-idxs ! round-idx < clength * scale
proof -
  fix round-idx
  assume round-bound: round-idx < length fri-query-idxs
  have fri-query-idxs ! round-idx ∈ set fri-query-idxs
    using round-bound by simp
  then show fri-query-idxs ! round-idx < clength * scale
    by (rule trace-fri-partial-candidate-opening-evidence-query-idx-bound
      [OF evidence])
qed
have generic:
  generic-fri-sampled-assignment-conflict trace-table trace-roots
  trace-bs trace-final fri-query-idxs trace-round-layers ∨
  length trace-bs = 0 ∧
  ¬ fri-final-constant-consistent trace-table trace-final
  by (rule
    generic-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final
    [OF obstruction-generic query-bounds eval-power trace-rounds-le])
then show ?thesis
proof
  assume conflict:
    generic-fri-sampled-assignment-conflict trace-table trace-roots
    trace-bs trace-final fri-query-idxs trace-round-layers
  then show ?thesis
    unfolding trace-fri-sampled-assignment-conflict-def
    using evidence by blast
next

```



```

assume zero:
  length trace-bs = 0 ∧
  ¬ fri-final-constant-consistent trace-table trace-final
then show ?thesis
  unfolding trace-fri-zero-round-final-obstruction-def
  using evidence by blast
qed
qed

lemma composition-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final:
assumes support:
  out ∈ set-dist (execute verify-monad s)
and obstruction:
  composition-fri-sampled-layer-assignment-obstruction s out
shows
  composition-fri-sampled-assignment-conflict s out ∨
  composition-fri-zero-round-final-obstruction s out
proof (rule composition-fri-sampled-layer-assignment-obstructionE
  [OF obstruction])
fix trace-roots trace-bs trace-final fri-dg opening-composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs trace-openings
  composition-query-idxs composition-openings trace-table composition-table
assume evidence:
  composition-fri-partial-candidate-opening-evidence s out trace-roots
  trace-bs trace-final fri-dg opening-composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs trace-openings
  composition-query-idxs composition-openings trace-table
  composition-table
assume obstruction-generic:
  generic-fri-sampled-layer-assignment-obstruction
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) opening-composition-roots composition-bs
  composition-final fri-query-idxs composition-round-layers
obtain N where eval-power: clength * scale = 2 ^ N
using eval-domain-length-power by blast
have fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final fri-dg opening-composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers
by (rule composition-fri-partial-candidate-opening-evidenceD(1)
  [OF evidence])
obtain result final-state where out-eq: out = Some (result, final-state)
using fri-openings unfolding accepted-fri-opening-transcript-def by blast
have outcome:

```

```

    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using support out-eq by simp
have challenges:
  accepted-fri-challenges s (Some (result, final-state))
  trace-bs fri-dg composition-bs
using accepted-fri-opening-transcript-challenges[OF fri-openings]
  out-eq by simp
have degree-bound: to-nat fri-dg ≤ maxDegree
  by (rule verify-monad-accepted-fri-challenges-degree-bound
    [OF outcome challenges])
have composition-rounds-le: length composition-bs ≤ N
proof -
  have len: length composition-bs = ceil-log (to-nat fri-dg + 1)
    using accepted-fri-opening-transcript-shapes(3,4)[OF fri-openings]
    by simp
  have to-nat fri-dg + 1 ≤ clength * scale
    using degree-bound maxDegree-less-eval-domain by linarith
  also have ... = 2 ^ N
    by (rule eval-power)
  finally have ceil-log (to-nat fri-dg + 1) ≤ N
    by (rule ceil-log-le-power)
  then show ?thesis
    using len by simp
qed
have query-bounds:
  ∧ round-idx. round-idx < length fri-query-idxs ⇒
  fri-query-idxs ! round-idx < clength * scale
proof -
  fix round-idx
  assume round-bound: round-idx < length fri-query-idxs
  have fri-query-idxs ! round-idx ∈ set fri-query-idxs
    using round-bound by simp
  then show fri-query-idxs ! round-idx < clength * scale
    by (rule composition-fri-partial-candidate-opening-evidence-query-idx-bound
      [OF evidence])
qed
have generic:
  generic-fri-sampled-assignment-conflict composition-table
  opening-composition-roots composition-bs composition-final
  fri-query-idxs composition-round-layers ∨
  length composition-bs = 0 ∧
  ¬ fri-final-constant-consistent composition-table
  composition-final
  by (rule
    generic-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final
    [OF obstruction-generic query-bounds eval-power
      composition-rounds-le])
then show ?thesis
proof

```

```

assume conflict:
  generic-fri-sampled-assignment-conflict composition-table
  opening-composition-roots composition-bs composition-final
  fri-query-idxs composition-round-layers
then show ?thesis
  unfolding composition-fri-sampled-assignment-conflict-def
  apply (intro disjI1 exI conjI)
  apply (rule evidence)
  apply assumption
  done
next
assume zero:
  length composition-bs = 0  $\wedge$ 
   $\neg$  fri-final-constant-consistent composition-table
  composition-final
then show ?thesis
  unfolding composition-fri-zero-round-final-obstruction-def
  apply (intro disjI2 exI conjI)
  apply (rule evidence)
  using zero
  apply simp-all
  done
qed
qed

end

end

```

```

theory Soundness-FRI-Zero-Round-Target
  imports Soundness-FRI-Sampled-Assignment-Chain
begin

```

Query-style target events for the zero-round FRI branch.

If no FRI fold round is present, sampled fold equations cannot produce a successor layer. Acceptance can only test that sampled base-layer entries agree with the final value. This theory factors the corresponding target set without changing the protocol or the public soundness interface.

```

context soundness
begin

```

```

definition fri-final-value-agreement-set :: 'f list  $\Rightarrow$  'f  $\Rightarrow$  nat set
where
  fri-final-value-agreement-set table final =
    {idx  $\in$  query-sample-space. idx < length table  $\wedge$  table ! idx = final}

```

```

definition fri-zero-round-final-value-target-hit
  :: 'f list  $\Rightarrow$  'f  $\Rightarrow$  nat list  $\Rightarrow$  bool

```

where

fri-zero-round-final-value-target-hit table final query-idxs
 $\longleftrightarrow$
 $\neg$ *fri-final-constant-consistent table final* $\wedge$
set query-idxs $\subseteq$ *fri-final-value-agreement-set table final*

definition *trace-zero-round-final-checked-candidate*

$:: (f, a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow$
 $f \Rightarrow \text{nat list} \Rightarrow f \text{ authenticated-opening list list} \Rightarrow$
 $f \text{ list} \Rightarrow f \Rightarrow \text{bool}$

where

trace-zero-round-final-checked-candidate s out fr query-idxs
trace-openings trace-table final $\longleftrightarrow$
accepted-with-partial-trace-openings s out fr query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $(\forall \text{idx} \in \text{set query-idxs}. \text{idx} \in \text{query-sample-space}) \wedge$
 $(\forall i < \text{rounds}. \text{opening-value} (\text{hd} (\text{trace-openings} ! i)) = \text{final})$

definition *trace-fri-zero-round-checked-final-obstruction*

$:: (f, a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-zero-round-checked-final-obstruction s out $\longleftrightarrow$
 $(\exists \text{trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr trace-openings trace-table.}$
accepted-fri-opening-transcript s out trace-roots trace-bs
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{length trace-bs} = 0 \wedge$
trace-zero-round-final-checked-candidate s out fr fri-query-idxs
trace-openings trace-table trace-final $\wedge$
 $\neg \text{fri-final-constant-consistent trace-table trace-final})$

lemma *fri-final-value-agreement-set-subset:*

fri-final-value-agreement-set table final $\subseteq$ *query-sample-space*
unfolding *fri-final-value-agreement-set-def* **by** *blast*

lemma *finite-fri-final-value-agreement-set[simp]:*

finite (fri-final-value-agreement-set table final)
by (*rule finite-subset[OF fri-final-value-agreement-set-subset]*)
simp

lemma *fri-final-value-agreement-setD:*

assumes *idx* $\in$ *fri-final-value-agreement-set table final*
shows *idx* $\in$ *query-sample-space*
and *idx* $<$ *length table*

and $table ! idx = final$
using *assms* **unfolding** *fri-final-value-agreement-set-def* **by** *blast+*

lemma *fri-zero-round-final-value-target-hitD*:
assumes *fri-zero-round-final-value-target-hit table final query-idxs*
shows $\neg fri\text{-}final\text{-}constant\text{-}consistent\ table\ final$
and $\bigwedge idx. idx \in set\ query\text{-}idxs \implies$
 $idx \in fri\text{-}final\text{-}value\text{-}agreement\text{-}set\ table\ final$
using *assms*
unfolding *fri-zero-round-final-value-target-hit-def* **by** *blast+*

lemma *powers-scaled-hd*:
 $hd\ (powers\text{-}scaled\ idx) = idx$
unfolding *powers-scaled-def*
using *powers-pos* **by** (*simp add: upt-conv-Cons*)

lemma *trace-zero-round-final-checked-candidate-query-agrees*:
assumes *checked*:
 $trace\text{-}zero\text{-}round\text{-}final\text{-}checked\text{-}candidate\ s\ out\ fr\ query\text{-}idxs$
 $trace\text{-}openings\ trace\text{-}table\ final$
and *i-bound*: $i < rounds$
shows
 $query\text{-}idxs ! i \in fri\text{-}final\text{-}value\text{-}agreement\text{-}set\ trace\text{-}table\ final$

proof –
have *partial*:
 $accepted\text{-}with\text{-}partial\text{-}trace\text{-}openings\ s\ out\ fr\ query\text{-}idxs\ trace\text{-}openings$
using *checked* **unfolding** *trace-zero-round-final-checked-candidate-def*
by *simp*
have *cand*:
 $partial\text{-}trace\text{-}table\text{-}candidate\ trace\text{-}table\ trace\text{-}openings$
using *checked* **unfolding** *trace-zero-round-final-checked-candidate-def*
by *simp*
have *query-sample*:
 $query\text{-}idxs ! i \in query\text{-}sample\text{-}space$
using *checked i-bound accepted-with-partial-trace-openings-shapes(1)*
 $[OF\ partial]$
unfolding *trace-zero-round-final-checked-candidate-def*
by *simp*
have *idxs*:
 $map\ opening\text{-}index\ (trace\text{-}openings ! i) =$
 $powers\text{-}scaled\ (query\text{-}idxs ! i)$
by (*rule accepted-with-partial-trace-openings-shapes(3)*
 $[OF\ partial\ i\text{-}bound]$)
have *openings-nonempty*: $trace\text{-}openings ! i \neq []$
proof –
have $map\ opening\text{-}index\ (trace\text{-}openings ! i) \neq []$
using *idxs powers-pos* **unfolding** *powers-scaled-def* **by** *simp*
then show *?thesis* **by** *auto*
qed

```

have hd-index:
  opening-index (hd (trace-openings ! i)) = query-idxs ! i
proof -
  have opening-index (hd (trace-openings ! i)) =
    hd (map opening-index (trace-openings ! i))
    using openings-nonempty by (cases trace-openings ! i) auto
  also have ... = hd (powers-scaled (query-idxs ! i))
    using idxs by simp
  also have ... = query-idxs ! i
    by (rule powers-scaled-hd)
  finally show ?thesis .
qed
have hd-in: hd (trace-openings ! i) ∈ set (trace-openings ! i)
  using openings-nonempty by simp
have table-value:
  trace-table ! (query-idxs ! i) =
    opening-value (hd (trace-openings ! i))
  using partial-trace-table-candidateD(3)[OF cand i-bound hd-in]
    hd-index by simp
have final:
  opening-value (hd (trace-openings ! i)) = final
  using checked i-bound
  unfolding trace-zero-round-final-checked-candidate-def by simp
have table-len:
  length trace-table = scale * clength
  using cand unfolding partial-trace-table-candidate-def by blast
have idx-bound: query-idxs ! i < length trace-table
  using partial-trace-table-candidateD(2)[OF cand i-bound hd-in]
    hd-index table-len by simp
show ?thesis
  unfolding fri-final-value-agreement-set-def
  using query-sample idx-bound table-value final by simp
qed

lemma trace-zero-round-final-checked-candidate-target-hit:
  assumes checked:
    trace-zero-round-final-checked-candidate s out fr query-idxs
    trace-openings trace-table final
  and not-final:
    ¬ fri-final-constant-consistent trace-table final
  shows
    fri-zero-round-final-value-target-hit trace-table final query-idxs
proof -
  have partial:
    accepted-with-partial-trace-openings s out fr query-idxs trace-openings
    using checked unfolding trace-zero-round-final-checked-candidate-def
    by simp
  have len-query: length query-idxs = rounds
    by (rule accepted-with-partial-trace-openings-shapes(1)[OF partial])

```

```

have subset:
  set query-idxs  $\subseteq$  fri-final-value-agreement-set trace-table final
proof
  fix idx
  assume idx-in: idx  $\in$  set query-idxs
  then obtain i where i-bound: i < length query-idxs
    and idx-eq: idx = query-idxs ! i
    by (auto simp: in-set-conv-nth)
  then have i < rounds
    using len-query by simp
  then show idx  $\in$  fri-final-value-agreement-set trace-table final
    using idx-eq
    by (simp add:
      trace-zero-round-final-checked-candidate-query-agrees[OF checked])
qed
show ?thesis
  unfolding fri-zero-round-final-value-target-hit-def
  using not-final subset by simp
qed

```

```

lemma trace-fri-zero-round-checked-final-obstructionE:
  assumes trace-fri-zero-round-checked-final-obstruction s out
  obtains trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr trace-openings trace-table
  where
    accepted-fri-opening-transcript s out trace-roots trace-bs
      trace-final dg composition-roots composition-bs composition-final
      fri-query-idxs trace-round-layers composition-round-layers
    length trace-bs = 0
    trace-zero-round-final-checked-candidate s out fr fri-query-idxs
      trace-openings trace-table trace-final
     $\neg$  fri-final-constant-consistent trace-table trace-final
  using assms
  unfolding trace-fri-zero-round-checked-final-obstruction-def
  by blast

```

```

lemma trace-fri-zero-round-checked-final-obstruction-target-hit:
  assumes trace-fri-zero-round-checked-final-obstruction s out
  obtains trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr trace-openings trace-table
  where
    accepted-fri-opening-transcript s out trace-roots trace-bs
      trace-final dg composition-roots composition-bs composition-final
      fri-query-idxs trace-round-layers composition-round-layers
    length trace-bs = 0
    trace-zero-round-final-checked-candidate s out fr fri-query-idxs
      trace-openings trace-table trace-final

```

$\neg$ *fri-final-constant-consistent* *trace-table* *trace-final*
fri-zero-round-final-value-target-hit *trace-table* *trace-final*
fri-query-idxs
proof (rule *trace-fri-zero-round-checked-final-obstructionE*[*OF assms*])
fix *trace-roots* *trace-bs* *trace-final* *dg* *composition-roots* *composition-bs*
composition-final *fri-query-idxs* *trace-round-layers*
composition-round-layers *fr* *trace-openings* *trace-table*
assume *fri-openings*:
accepted-fri-opening-transcript *s* *out* *trace-roots* *trace-bs*
trace-final *dg* *composition-roots* *composition-bs* *composition-final*
fri-query-idxs *trace-round-layers* *composition-round-layers*
and *rounds-empty*: *length* *trace-bs* = 0
and *checked*:
trace-zero-round-final-checked-candidate *s* *out* *fr* *fri-query-idxs*
trace-openings *trace-table* *trace-final*
and *not-final*:
 $\neg$ *fri-final-constant-consistent* *trace-table* *trace-final*
have *target*:
fri-zero-round-final-value-target-hit *trace-table* *trace-final*
fri-query-idxs
by (rule *trace-zero-round-final-checked-candidate-target-hit*
[*OF checked not-final*])
show ?thesis
by (rule *that*[*OF fri-openings rounds-empty checked not-final target*])
qed

lemma *fri-final-value-agreement-set-card-le*:
card (*fri-final-value-agreement-set* *table* *final*) $\leq$
query-sample-space-size
proof –
have *card* (*fri-final-value-agreement-set* *table* *final*) $\leq$
card *query-sample-space*
by (rule *card-mono*[*OF finite-query-sample-space*
fri-final-value-agreement-set-subset])
then show ?thesis **by** *simp*
qed

lemma *fri-final-value-agreement-set-card-lt-if-query-sample-disagrees*:
assumes *idx-sample*: *idx* $\in$ *query-sample-space*
and *idx-bound*: *idx* < *length* *table*
and *neg*: *table* ! *idx* $\neq$ *final*
shows *card* (*fri-final-value-agreement-set* *table* *final*) <
query-sample-space-size
proof –
have *proper*:
fri-final-value-agreement-set *table* *final* $\subset$ *query-sample-space*
proof
show *fri-final-value-agreement-set* *table* *final* $\subseteq$ *query-sample-space*
by (rule *fri-final-value-agreement-set-subset*)


```

next
show fri-final-value-agreement-set table final  $\neq$  query-sample-space
proof
  assume eq: fri-final-value-agreement-set table final =
    query-sample-space
  then have idx  $\in$  fri-final-value-agreement-set table final
    using idx-sample by simp
  then show False
    using fri-final-value-agreement-setD(3) neq by blast
qed
qed
have card (fri-final-value-agreement-set table final) <
  card query-sample-space
  by (rule psubset-card-mono[OF finite-query-sample-space proper])
then show ?thesis by simp
qed

lemma fri-final-value-agreement-set-all-query-sample-iff:
  assumes table-covers-query-sample:
     $\bigwedge \text{idx}. \text{idx} \in \text{query-sample-space} \implies \text{idx} < \text{length table}$ 
  shows
    fri-final-value-agreement-set table final = query-sample-space
     $\longleftrightarrow$ 
     $(\forall \text{idx} \in \text{query-sample-space}. \text{table} ! \text{idx} = \text{final})$ 
  unfolding fri-final-value-agreement-set-def
  using table-covers-query-sample by blast

lemma fri-final-constant-consistent-if-query-sample-is-whole-table:
  assumes table-len: length table = query-sample-space-size
    and all-query:  $\forall \text{idx} \in \text{query-sample-space}. \text{table} ! \text{idx} = \text{final}$ 
  shows fri-final-constant-consistent table final
proof -
  have all-idx:  $\bigwedge \text{idx}. \text{idx} < \text{length table} \implies \text{table} ! \text{idx} = \text{final}$ 
proof -
  fix idx
  assume idx-bound: idx < length table
  then have idx  $\in$  query-sample-space
    using table-len unfolding query-sample-space-def by simp
  then show table ! idx = final
    using all-query by simp
qed
show ?thesis
  unfolding fri-final-constant-consistent-def
  using all-idx by (auto simp: in-set-conv-nth)
qed

lemma fri-final-value-agreement-set-card-lt-if-not-constant-whole-sample:
  assumes table-len: length table = query-sample-space-size
    and not-final:  $\neg$  fri-final-constant-consistent table final

```

```

shows card (fri-final-value-agreement-set table final) <
  query-sample-space-size
proof (rule ccontr)
assume  $\neg$  card (fri-final-value-agreement-set table final) <
  query-sample-space-size
then have card-eq:
  card (fri-final-value-agreement-set table final) =
    query-sample-space-size
using fri-final-value-agreement-set-card-le[of table final] by linarith
have set-eq:
  fri-final-value-agreement-set table final = query-sample-space
by (rule card-subset-eq[OF finite-query-sample-space
  fri-final-value-agreement-set-subset])
  (use card-eq in simp)
have all-query:
   $\forall idx \in \text{query-sample-space}. \text{table } ! \text{idx} = \text{final}$ 
using set-eq fri-final-value-agreement-setD(3) by metis
have fri-final-constant-consistent table final
by (rule fri-final-constant-consistent-if-query-sample-is-whole-table
  [OF table-len all-query])
then show False
using not-final by simp
qed

```

```

lemma trace-zero-round-table-len-equals-query-sample-space:
assumes trace-rounds-empty: length trace-bs = 0
and fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final dg composition-roots composition-bs composition-final
  fri-query-idxs trace-round-layers composition-round-layers
and cand: partial-trace-table-candidate trace-table trace-openings
shows length trace-table = query-sample-space-size
proof –
have len-trace-bs: length trace-bs = ceil-log clength
using accepted-fri-opening-transcript-shapes(1,2)[OF fri-openings]
by simp
have ceil-zero: ceil-log clength = 0
using trace-rounds-empty len-trace-bs by simp
have clength-le-one: clength  $\leq 1$ 
using ceil-zero unfolding ceil-log-def
by (cases clength  $\leq 1$ ) simp-all
have clength-eq: clength = 1
using clength-le-one clength-pos by simp
have powers-eq: powers = 1
using powers-pos powers-le-clength clength-eq by simp
have table-len: length trace-table = scale * clength
using cand unfolding partial-trace-table-candidate-def by blast
show ?thesis
unfolding query-sample-space-size-def

```

using *table-len* *clength-eq* *powers-eq* by *simp*
 qed

lemma *trace-fri-zero-round-checked-final-obstruction-agreement-set-card-lt*:
 assumes *obstruction*:
 trace-fri-zero-round-checked-final-obstruction *s* *out*
 obtains *trace-roots* *trace-bs* *trace-final* *dg* *composition-roots*
 composition-bs *composition-final* *fri-query-idxs* *trace-round-layers*
 composition-round-layers *fr* *trace-openings* *trace-table*
 where
 accepted-fri-opening-transcript *s* *out* *trace-roots* *trace-bs*
 trace-final *dg* *composition-roots* *composition-bs* *composition-final*
 fri-query-idxs *trace-round-layers* *composition-round-layers*
 length *trace-bs* = 0
 trace-zero-round-final-checked-candidate *s* *out* *fr* *fri-query-idxs*
 trace-openings *trace-table* *trace-final*
 $\neg$ *fri-final-constant-consistent* *trace-table* *trace-final*
 fri-zero-round-final-value-target-hit *trace-table* *trace-final*
 fri-query-idxs
 card (*fri-final-value-agreement-set* *trace-table* *trace-final*) <
 query-sample-space-size
proof (rule *trace-fri-zero-round-checked-final-obstruction-target-hit*
 [OF *obstruction*])
 fix *trace-roots* *trace-bs* *trace-final* *dg* *composition-roots* *composition-bs*
 composition-final *fri-query-idxs* *trace-round-layers*
 composition-round-layers *fr* *trace-openings* *trace-table*
 assume *fri-openings*:
 accepted-fri-opening-transcript *s* *out* *trace-roots* *trace-bs*
 trace-final *dg* *composition-roots* *composition-bs* *composition-final*
 fri-query-idxs *trace-round-layers* *composition-round-layers*
 and *rounds-empty*: *length* *trace-bs* = 0
 and *checked*:
 trace-zero-round-final-checked-candidate *s* *out* *fr* *fri-query-idxs*
 trace-openings *trace-table* *trace-final*
 and *not-final*:
 $\neg$ *fri-final-constant-consistent* *trace-table* *trace-final*
 and *target*:
 fri-zero-round-final-value-target-hit *trace-table* *trace-final*
 fri-query-idxs
 have *cand*:
 partial-trace-table-candidate *trace-table* *trace-openings*
 using **checked** **unfolding** *trace-zero-round-final-checked-candidate-def*
 by *simp*
 have *table-len*:
 length *trace-table* = *query-sample-space-size*
 by (rule *trace-zero-round-table-len-equals-query-sample-space*
 [OF *rounds-empty* *fri-openings* *cand*])
 have *card-lt*:
 card (*fri-final-value-agreement-set* *trace-table* *trace-final*) <

query-sample-space-size
by (rule *fri-final-value-agreement-set-card-lt-if-not-constant-whole-sample*
[OF table-len not-final])
show *?thesis*
by (rule *that[OF fri-openings rounds-empty checked not-final target*
card-lt])
qed

lemma *trace-fri-zero-round-checked-final-obstruction-first-query-target:*
assumes *obstruction:*
trace-fri-zero-round-checked-final-obstruction s out
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr trace-openings trace-table
where
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
length trace-bs = 0
trace-zero-round-final-checked-candidate s out fr fri-query-idxs
trace-openings trace-table trace-final
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
fri-query-idxs ! 0 $\in$
fri-final-value-agreement-set trace-table trace-final
card (fri-final-value-agreement-set trace-table trace-final) <
query-sample-space-size
proof (rule *trace-fri-zero-round-checked-final-obstruction-agreement-set-card-lt*
[OF obstruction])
fix *trace-roots trace-bs trace-final dg composition-roots composition-bs*
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr trace-openings trace-table
assume *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *rounds-empty: length trace-bs = 0*
and *checked:*
trace-zero-round-final-checked-candidate s out fr fri-query-idxs
trace-openings trace-table trace-final
and *not-final:*
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
and *target:*
fri-zero-round-final-value-target-hit trace-table trace-final
fri-query-idxs
and *card-lt:*
card (fri-final-value-agreement-set trace-table trace-final) <
query-sample-space-size
have *first-in:*
fri-query-idxs ! 0 $\in$

```

    fri-final-value-agreement-set trace-table trace-final
proof –
  have partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  using checked unfolding trace-zero-round-final-checked-candidate-def
  by simp
  have length fri-query-idxs = rounds
    by (rule accepted-with-partial-trace-openings-shapes(1)[OF partial])
  then have fri-query-idxs ! 0 ∈ set fri-query-idxs
    using rounds-positive by (simp add: nth-mem)
  then show ?thesis
    by (rule fri-zero-round-final-value-target-hitD(2)[OF target])
qed
show ?thesis
  by (rule that[OF fri-openings rounds-empty checked not-final first-in
    card-lt])
qed

lemma trace-zero-round-checked-candidate-not-trace-low-degree:
  assumes rounds-empty: length trace-bs = 0
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and checked:
    trace-zero-round-final-checked-candidate s out fr fri-query-idxs
    trace-openings trace-table trace-final
  and not-final:
    ¬ fri-final-constant-consistent trace-table trace-final
  shows ¬ trace-table-low-degree trace-table
proof
  assume low: trace-table-low-degree trace-table
  have len-trace-bs: length trace-bs = ceil-log clength
    using accepted-fri-opening-transcript-shapes(1,2)[OF fri-openings]
    by simp
  have ceil-zero: ceil-log clength = 0
    using rounds-empty len-trace-bs by simp
  have clength-le-one: clength ≤ 1
    using ceil-zero unfolding ceil-log-def
    by (cases clength ≤ 1) simp-all
  have clength-eq: clength = 1
    using clength-le-one clength-pos by simp
  obtain f where deg-f: degree f < clength
    and trace-table: trace-table = map (poly f) eval-domain
    using low unfolding trace-table-low-degree-def by blast
  have deg0: degree f = 0
    using deg-f clength-eq by simp
  obtain a where f-eq: f = [:a:]

```

```

using degree0-coeffs[OF deg0] by blast
have table-const:
   $\bigwedge y. y \in \text{set trace-table} \implies y = a$ 
  using trace-table f-eq by auto
have sample-agrees:
  fri-query-idxs ! 0  $\in$ 
    fri-final-value-agreement-set trace-table trace-final
  by (rule trace-zero-round-final-checked-candidate-query-agrees
    [OF checked rounds-positive])
have idx-bound: fri-query-idxs ! 0 < length trace-table
  using fri-final-value-agreement-setD(2)[OF sample-agrees] .
have table-at-final:
  trace-table ! (fri-query-idxs ! 0) = trace-final
  using fri-final-value-agreement-setD(3)[OF sample-agrees] .
have trace-table ! (fri-query-idxs ! 0)  $\in$  set trace-table
  using idx-bound by simp
then have a-eq: a = trace-final
  using table-const table-at-final by simp
have fri-final-constant-consistent trace-table trace-final
  unfolding fri-final-constant-consistent-def
  using table-const a-eq by auto
then show False
  using not-final by contradiction
qed

```

lemma trace-fri-zero-round-checked-final-obstruction-not-trace-low-degree:

assumes obstruction:

trace-fri-zero-round-checked-final-obstruction s out

obtains trace-roots trace-bs trace-final dg composition-roots

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr trace-openings trace-table

where

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final dg composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers

length trace-bs = 0

trace-zero-round-final-checked-candidate s out fr fri-query-idxs

trace-openings trace-table trace-final

$\neg$ fri-final-constant-consistent trace-table trace-final

$\neg$ trace-table-low-degree trace-table

proof (rule trace-fri-zero-round-checked-final-obstructionE[*OF obstruction*])

fix trace-roots trace-bs trace-final dg composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr trace-openings trace-table

assume fri-openings:

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final dg composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers

and rounds-empty: length trace-bs = 0

and *checked*:
trace-zero-round-final-checked-candidate s out fr fri-query-idxs
trace-openings trace-table trace-final
and *not-final*:
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
have *not-low*:
 $\neg$ *trace-table-low-degree trace-table*
by (rule *trace-zero-round-checked-candidate-not-trace-low-degree*
[*OF rounds-empty fri-openings checked not-final*])
show *?thesis*
by (rule *that[OF fri-openings rounds-empty checked not-final not-low]*)
qed

definition *trace-fri-zero-round-checked-first-query-target-hit*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
trace-fri-zero-round-checked-first-query-target-hit s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr trace-openings trace-table.}$
accepted-fri-opening-transcript s out trace-roots trace-bs
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{length trace-bs} = 0 \wedge$
trace-zero-round-final-checked-candidate s out fr fri-query-idxs
trace-openings trace-table trace-final $\wedge$
 $\neg \text{fri-final-constant-consistent trace-table trace-final} \wedge$
 $\text{fri-query-idxs} \neq 0 \in$
 $\text{fri-final-value-agreement-set trace-table trace-final} \wedge$
 $\text{card}(\text{fri-final-value-agreement-set trace-table trace-final}) <$
 $\text{query-sample-space-size})$

lemma *trace-fri-zero-round-checked-final-obstruction-imp-first-query-target-hit*:

assumes *trace-fri-zero-round-checked-final-obstruction s out*

shows *trace-fri-zero-round-checked-first-query-target-hit s out*

proof (rule *trace-fri-zero-round-checked-final-obstruction-first-query-target*
[*OF assms*])

fix *trace-roots trace-bs trace-final dg composition-roots composition-bs*
 $\text{composition-final fri-query-idxs trace-round-layers composition-round-layers}$
 $\text{fr trace-openings trace-table}$

assume *fri-openings*:

accepted-fri-opening-transcript s out trace-roots trace-bs
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers}$

and *rounds-empty*: $\text{length trace-bs} = 0$

and *checked*:

trace-zero-round-final-checked-candidate s out fr fri-query-idxs
trace-openings trace-table trace-final

and *not-final*:
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
and *first-hit*:
fri-query-idxs ! 0 $\in$
fri-final-value-agreement-set trace-table trace-final
and *card-lt*:
card (fri-final-value-agreement-set trace-table trace-final) <
query-sample-space-size
show ?thesis
unfolding *trace-fri-zero-round-checked-first-query-target-hit-def*
by (*intro exI conjI*)
(rule *fri-openings*, rule *rounds-empty*, rule *checked*, rule *not-final*,
rule *first-hit*, rule *card-lt*)
qed

lemma *wp-trace-fri-zero-round-checked-final-obstruction-bound-from-first-query-target-hit*:
assumes *target-bound*:
wp-event verify-monad
(*trace-fri-zero-round-checked-first-query-target-hit s*) $s \leq T$
shows
wp-event verify-monad
(*trace-fri-zero-round-checked-final-obstruction s*) $s \leq T$
proof –
have *wp-event verify-monad*
(*trace-fri-zero-round-checked-final-obstruction s*) $s \leq$
wp-event verify-monad
(*trace-fri-zero-round-checked-first-query-target-hit s*) s
by (rule *wp-event-mono*)
(rule
trace-fri-zero-round-checked-final-obstruction-imp-first-query-target-hit)
then show ?thesis
by (rule *order-trans*[*OF - target-bound*])
qed

lemma *trace-fri-zero-round-final-obstruction-agreement-set-card-lt*:
assumes *obstruction*: *trace-fri-zero-round-final-obstruction s out*
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
where
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
length trace-bs = 0
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
card (fri-final-value-agreement-set trace-table trace-final) <

query-sample-space-size

proof –

from *obstruction* **obtain** *trace-roots trace-bs trace-final dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr candidate-query-idxs
trace-openings trace-table **where** *evidence:*
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *rounds-empty: length trace-bs = 0*
and *not-final:*
 ¬ *fri-final-constant-consistent trace-table trace-final*
unfolding *trace-fri-zero-round-final-obstruction-def* **by** *blast*
have *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
by (*rule trace-fri-partial-candidate-opening-evidenceD(1)*
 [*OF evidence*])
have *partial:*
trace-fri-partial-candidate-evidence s out fr candidate-query-idxs
trace-openings trace-table
by (*rule trace-fri-partial-candidate-opening-evidenceD(2)*
 [*OF evidence*])
have *cand:*
partial-trace-table-candidate trace-table trace-openings
by (*rule trace-fri-partial-candidate-evidenceD(2)[OF partial]*)
have *table-len:*
length trace-table = query-sample-space-size
by (*rule trace-zero-round-table-len-equals-query-sample-space*
 [*OF rounds-empty fri-openings cand*])
have *card-lt:*
card (fri-final-value-agreement-set trace-table trace-final) <
query-sample-space-size
by (*rule fri-final-value-agreement-set-card-lt-if-not-constant-whole-sample*
 [*OF table-len not-final*])
show *?thesis*
by (*rule that[OF evidence rounds-empty not-final card-lt]*)
qed

lemma *trace-fri-zero-round-final-obstructionE:*
assumes *trace-fri-zero-round-final-obstruction s out*
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
where

trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
length trace-bs = 0
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
using *assms*
unfolding *trace-fri-zero-round-final-obstruction-def* **by** *blast*

definition *trace-fri-zero-round-candidate-small-agreement-set*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where

trace-fri-zero-round-candidate-small-agreement-set s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 trace-table.
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
length trace-bs = 0 $\wedge$
 $\neg$ *fri-final-constant-consistent trace-table trace-final* $\wedge$
 $\text{card (fri-final-value-agreement-set trace-table trace-final)} <$
 $\text{query-sample-space-size}$)

definition *trace-fri-zero-round-candidate-first-query-target-hit*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where

trace-fri-zero-round-candidate-first-query-target-hit s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr candidate-query-idxs trace-openings}$
 trace-table.
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
length trace-bs = 0 $\wedge$
 $\neg$ *fri-final-constant-consistent trace-table trace-final* $\wedge$
 $\text{candidate-query-idxs} \neq 0 \in$
 $\text{fri-final-value-agreement-set trace-table trace-final} \wedge$
 $\text{card (fri-final-value-agreement-set trace-table trace-final)} <$
 $\text{query-sample-space-size}$)

definition *trace-fri-zero-round-candidate-first-query-alignment-gap*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-zero-round-candidate-first-query-alignment-gap } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 trace-table.
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{length trace-bs} = 0 \wedge$
 $\neg \text{fri-final-constant-consistent trace-table trace-final} \wedge$
 $\text{candidate-query-idxs} ! 0 \in$
 $\text{fri-final-value-agreement-set trace-table trace-final} \wedge$
 $\text{fri-query-idxs} ! 0 \notin$
 $\text{fri-final-value-agreement-set trace-table trace-final} \wedge$
 $\text{card} (\text{fri-final-value-agreement-set trace-table trace-final}) <$
 $\text{query-sample-space-size})$

definition *trace-fri-zero-round-candidate-first-query-value-gap*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-zero-round-candidate-first-query-value-gap } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 trace-table.
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{length trace-bs} = 0 \wedge$
 $\neg \text{fri-final-constant-consistent trace-table trace-final} \wedge$
 $\text{candidate-query-idxs} ! 0 \notin$
 $\text{fri-final-value-agreement-set trace-table trace-final} \wedge$
 $\text{card} (\text{fri-final-value-agreement-set trace-table trace-final}) <$
 $\text{query-sample-space-size})$

lemma *trace-fri-zero-round-candidate-small-agreement-setE:*
assumes *trace-fri-zero-round-candidate-small-agreement-set* $s \text{ out}$
obtains $\text{trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$

composition-round-layers fr candidate-query-idxs trace-openings
trace-table
where
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
length trace-bs = 0
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
card (fri-final-value-agreement-set trace-table trace-final) <
query-sample-space-size
using *assms*
unfolding *trace-fri-zero-round-candidate-small-agreement-set-def*
by *blast*

lemma *trace-fri-zero-round-candidate-first-query-target-hitE:*
assumes *trace-fri-zero-round-candidate-first-query-target-hit s out*
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
where
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
length trace-bs = 0
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
candidate-query-idxs ! 0 $\in$
fri-final-value-agreement-set trace-table trace-final
card (fri-final-value-agreement-set trace-table trace-final) <
query-sample-space-size
using *assms*
unfolding *trace-fri-zero-round-candidate-first-query-target-hit-def*
by *blast*

lemma *trace-fri-zero-round-candidate-first-query-alignment-gapE:*
assumes *trace-fri-zero-round-candidate-first-query-alignment-gap s out*
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
where
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings

trace-table
length trace-bs = 0
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
candidate-query-idxs ! 0 ∈
fri-final-value-agreement-set trace-table trace-final
fri-query-idxs ! 0 ∉
fri-final-value-agreement-set trace-table trace-final
card (fri-final-value-agreement-set trace-table trace-final) <
query-sample-space-size
using *assms*
unfolding *trace-fri-zero-round-candidate-first-query-alignment-gap-def*
by *blast*

lemma *trace-fri-zero-round-candidate-first-query-value-gapE*:
assumes *trace-fri-zero-round-candidate-first-query-value-gap s out*
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
where
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
length trace-bs = 0
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
candidate-query-idxs ! 0 ∉
fri-final-value-agreement-set trace-table trace-final
card (fri-final-value-agreement-set trace-table trace-final) <
query-sample-space-size
using *assms*
unfolding *trace-fri-zero-round-candidate-first-query-value-gap-def*
by *blast*

lemma *trace-fri-zero-round-candidate-first-query-target-hit-imp-small*:
assumes *trace-fri-zero-round-candidate-first-query-target-hit s out*
shows *trace-fri-zero-round-candidate-small-agreement-set s out*
proof (*rule trace-fri-zero-round-candidate-first-query-target-hitE[OF assms]*)
fix *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
assume *evidence*:
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table

and *rounds-empty*: $\text{length } \text{trace-bs} = 0$
and *not-final*:
 $\neg \text{fri-final-constant-consistent } \text{trace-table } \text{trace-final}$
and *card-lt*:
 $\text{card } (\text{fri-final-value-agreement-set } \text{trace-table } \text{trace-final}) < \text{query-sample-space-size}$
show *?thesis*
unfolding *trace-fri-zero-round-candidate-small-agreement-set-def*
by (*intro exI conjI*)
(rule *evidence*, rule *rounds-empty*, rule *not-final*, rule *card-lt*)
qed

lemma *trace-fri-zero-round-checked-first-query-target-hit-imp-candidate*:
assumes *trace-fri-zero-round-checked-first-query-target-hit s out*
shows *trace-fri-zero-round-candidate-first-query-target-hit s out*
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots composition-bs composition-final fri-query-idxs trace-round-layers composition-round-layers fr trace-openings trace-table*
where *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final dg composition-roots composition-bs composition-final fri-query-idxs trace-round-layers composition-round-layers
and *rounds-empty*: $\text{length } \text{trace-bs} = 0$
and *checked*:
trace-zero-round-final-checked-candidate s out fr fri-query-idxs trace-openings trace-table trace-final
and *not-final*:
 $\neg \text{fri-final-constant-consistent } \text{trace-table } \text{trace-final}$
and *first-hit*:
fri-query-idxs ! 0 ∈ fri-final-value-agreement-set trace-table trace-final
and *card-lt*:
 $\text{card } (\text{fri-final-value-agreement-set } \text{trace-table } \text{trace-final}) < \text{query-sample-space-size}$
unfolding *trace-fri-zero-round-checked-first-query-target-hit-def*
by *blast*
have *partial*:
accepted-with-partial-trace-openings s out fr fri-query-idxs trace-openings
using *checked* **unfolding** *trace-zero-round-final-checked-candidate-def*
by *simp*
have *candidate*:
partial-trace-table-candidate trace-table trace-openings
using *checked* **unfolding** *trace-zero-round-final-checked-candidate-def*
by *simp*
have *not-low*:
 $\neg \text{trace-table-low-degree } \text{trace-table}$
by (rule *trace-zero-round-checked-candidate-not-trace-low-degree*)

[OF rounds-empty fri-openings checked not-final])
have evidence:
 trace-fri-partial-candidate-opening-evidence s out trace-roots
 trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr fri-query-idxs trace-openings trace-table
unfolding trace-fri-partial-candidate-opening-evidence-def
 trace-fri-partial-candidate-evidence-def
by (intro conjI fri-openings partial candidate not-low)
show ?thesis
unfolding trace-fri-zero-round-candidate-first-query-target-hit-def
by (intro exI conjI)
 (rule evidence, rule rounds-empty, rule not-final, rule first-hit,
 rule card-lt)
qed

lemma trace-fri-zero-round-final-obstruction-imp-candidate-small-agreement-set:
assumes trace-fri-zero-round-final-obstruction s out
shows trace-fri-zero-round-candidate-small-agreement-set s out
proof (rule trace-fri-zero-round-final-obstruction-agreement-set-card-lt
 [OF assms])
fix trace-roots trace-bs trace-final dg composition-roots
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table
assume evidence:
 trace-fri-partial-candidate-opening-evidence s out trace-roots
 trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table
and rounds-empty: length trace-bs = 0
and not-final:
 ¬ fri-final-constant-consistent trace-table trace-final
and card-lt:
 card (fri-final-value-agreement-set trace-table trace-final) <
 query-sample-space-size
show ?thesis
unfolding trace-fri-zero-round-candidate-small-agreement-set-def
by (intro exI conjI)
 (rule evidence, rule rounds-empty, rule not-final, rule card-lt)
qed

lemma wp-trace-fri-zero-round-final-obstruction-bound-from-candidate-small-agreement-set:
assumes small-bound:
 wp-event verify-monad
 (trace-fri-zero-round-candidate-small-agreement-set s) s ≤ T
shows
 wp-event verify-monad

```

      (trace-fri-zero-round-final-obstruction s) s ≤ T
proof –
  have wp-event verify-monad
    (trace-fri-zero-round-final-obstruction s) s ≤
      wp-event verify-monad
        (trace-fri-zero-round-candidate-small-agreement-set s) s
  by (rule wp-event-mono)
    (rule
      trace-fri-zero-round-final-obstruction-imp-candidate-small-agreement-set)
  then show ?thesis
    by (rule order-trans[OF - small-bound])
qed

end

end

```

```

theory Soundness-Query-Transcript-Openings
imports Soundness-Query-Opening-Consistency
begin

```

Combined query-round extraction lemmas.

The existing execution layer extracts transcript chunks and authenticated Merkle openings separately. This layer keeps the witnesses coupled to the same verifier query-round execution, so downstream transcript-indexed reductions can use verifier-derived query indices.

```

context soundness
begin

```

```

lemma verifier-query-round-program-outcome-with-authenticated-trace-openings:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
obtains raw idx chunk openings where
  idx = index (to-nat raw)
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
  PTranscript s = chunk @ PTranscript t
  PState t = foldl concat (PState s) chunk
  s ≤ t
  PQueryCounter t = Suc (PQueryCounter s)
  fmlookup (HashMap t) (QueryIndexChallenge (PQueryCounter s) (PState s)) =
    Some raw
  length openings = length (powers-scaled idx)
  map opening-index openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) openings t

```



```

proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where
  rand:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s)
  and tail:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s0)
  by (auto elim! set-dist-bindE)
let ?idx = index (to-nat raw)
from tail obtain fv s1 f-out s2 s3 c-out s4 where
  query-decommit:
    Some (fv, s1) ∈
      set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s0)
  and trace-fri:
    Some (f-out, s2) ∈
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
  and assert-trace:
    Some ((), s3) ∈
      set-dist
        (execute
          (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
  and comp-fri:
    Some (c-out, s4) ∈
      set-dist (execute
        (mfold
          (?idx, cp-eval as fv (h ^ ?idx * shift), clength * scale, 1)
            (receive-query-commits fl)) s3)
  and assert-comp:
    Some ((), t) ∈
      set-dist
        (execute (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
  unfolding verifier-query-round-after-index-program-def
  by (auto simp: Let-def split: prod.splits elim! set-dist-bindE)
have rand-res:
  s ≤ s0 ∧
    PState s0 = PState s ∧
    PTranscript s0 = PTranscript s ∧
    fmlookup (HashMap s0)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  using receive-query-index-challenge-outcome[OF rand] .
have rand-counter:
  PQueryCounter s0 = Suc (PQueryCounter s)

```

```

using receive-query-index-challenge-counter-outcome[OF rand] by simp
from check-decommit-on-query-outcome[OF query-decommit]
obtain query-paths query-chunk where
  query-chunk-shape:
    query-decommitment-transcript ?idx fv query-paths query-chunk
  and tr-s0: PTranscript s0 = query-chunk @ PTranscript s1
  and st-s1: PState s1 = foldl concat (PState s0) query-chunk
  and ext-s0-s1: s0 ≤ s1
  and query-count-s1: PQueryCounter s1 = PQueryCounter s0
by blast
from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where
  trace-layers:
    fri-layers-transcript (length f-fl) (clength * scale)
    trace-layer-chunks trace-fri-chunk
  and tr-s1: PTranscript s1 = trace-fri-chunk @ PTranscript s2
  and st-s2: PState s2 = foldl concat (PState s1) trace-fri-chunk
  and ext-s1-s2: s1 ≤ s2
  and query-count-s2: PQueryCounter s2 = PQueryCounter s1
by blast
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def
  by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
    (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  comp-layers:
    fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
  and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
  and st-s4: PState s4 = foldl concat (PState s3) comp-fri-chunk
  and ext-s3-s4: s3 ≤ s4
  and query-count-s4: PQueryCounter s4 = PQueryCounter s3
by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -) ⇒ x = final)
    (auto simp: throw-no-outcome)
let ?chunk = query-chunk @ trace-fri-chunk @ comp-fri-chunk
have round-chunk:
  verifier-query-round-chunk ?idx (map snd f-fl) (map snd fl) ?chunk
  unfolding verifier-query-round-chunk-def
  apply (rule exI[where x=query-chunk])
  apply (rule exI[where x=trace-fri-chunk])
  apply (rule exI[where x=comp-fri-chunk])
  apply (rule exI[where x=fv])
  apply (rule exI[where x=query-paths])
  apply (rule exI[where x=trace-layer-chunks])
  apply (rule exI[where x=comp-layer-chunks])

```

```

    using query-chunk-shape trace-layers comp-layers
    by simp
have tr-t: PTranscript s = ?chunk @ PTranscript t
    using rand-res tr-s0 tr-s1 tr-s3 s3-eq t-eq by simp
have st-t: PState t = foldl concat (PState s) ?chunk
    using rand-res st-s1 st-s2 st-s4 s3-eq t-eq by simp
have ext-s0-t: s0 ≤ t
    using ext-s0-s1 ext-s1-s2 ext-s3-s4 t-eq unfolding s3-eq
    by (meson hash-ext-trans)
have ext-s-t: s ≤ t
    using rand-res ext-s0-t by (meson hash-ext-trans)
have query-count-t: PQueryCounter t = Suc (PQueryCounter s)
    using rand-counter query-count-s1 query-count-s2 query-count-s4
    unfolding s3-eq t-eq by simp
have lookup-t:
    fmllookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof -
    let ?key = QueryIndexChallenge (PQueryCounter s) (PState s)
    have fmllookup (HashMap s0) ?key = fmllookup (HashMap t) ?key
        using ext-s0-t rand-res
        unfolding less-eq-hash-ext-def less-eq-fmap-def
        by (metis option.distinct(1))
    then show ?thesis
        using rand-res by simp
qed
from verifier-query-round-after-index-authenticated-trace-openings[OF tail]
obtain idx' leaves openings where
    idx'-eq: idx' = ?idx
    and len-openings: length openings = length (powers-scaled idx')
    and idx-openings: map opening-index openings = powers-scaled idx'
    and table-t:
        partial-authenticated-table fr (scale * clength) openings t
    by blast
show ?thesis
proof (rule that)
    show ?idx = index (to-nat raw)
        by simp
    show verifier-query-round-chunk ?idx (map snd f-fl) (map snd fl)
        ?chunk
        by (rule round-chunk)
    show PTranscript s = ?chunk @ PTranscript t
        by (rule tr-t)
    show PState t = foldl concat (PState s) ?chunk
        by (rule st-t)
    show s ≤ t
        by (rule ext-s-t)
    show PQueryCounter t = Suc (PQueryCounter s)
        by (rule query-count-t)

```

```

show fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  by (rule lookup-t)
show length openings = length (powers-scaled ?idx)
  using idx'-eq len-openings by simp
show map opening-index openings = powers-scaled ?idx
  using idx'-eq idx-openings by simp
show partial-authenticated-table fr (scale * clength) openings t
  by (rule table-t)
qed
qed

lemma ntimes-verifier-query-rounds-selected-trace-openings-aligned-with-replay-at:
fixes query-state final-state :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (results, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
and i-bound: i < rounds
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and query-chunk:
  ∧i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and replay-lookup:
  ∧i. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
obtains openings where
  map opening-index openings = powers-scaled (query-idxs ! i)
  partial-authenticated-table fr (scale * clength) openings final-state
proof -
let ?round = verifier-query-round-program fr f-fl f-final as fl final
from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
obtain prefix x suffix s-i s-suc where
  prefix:

```

```

    Some (prefix, s-i) ∈
      set-dist (execute (ntimes ?round i) query-state)
and round:
    Some (x, s-suc) ∈ set-dist (execute ?round s-i)
and suffix:
    Some (suffix, final-state) ∈
      set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
by blast
have x-eq: x = ()
by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
fix prefix-query-idxs prefix-chunks
assume len-prefix-chunks: length prefix-chunks = i
and transcript-prefix:
  PTranscript query-state =
    List.concat prefix-chunks @ PTranscript s-i
and state-i:
  PState s-i =
    state-after-query-chunks (PState query-state) prefix-chunks i
and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
and prefix-rounds:
   $\bigwedge k. k < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
show ?thesis
proof (rule
  verifier-query-round-program-outcome-with-authenticated-trace-openings
  [OF round[unfolded x-eq]])
fix raw idx chunk openings
assume idx-eq: idx = index (to-nat raw)
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
    chunk
and transcript-round: PTranscript s-i = chunk @ PTranscript s-suc
and lookup-suc:
  fmlookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
and opening-indices: map opening-index openings = powers-scaled idx
and auth-table-suc:
  partial-authenticated-table fr (scale * clength) openings s-suc
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-chunks:
  length suffix-chunks = rounds - Suc i

```

```

and transcript-suffix:
  PTranscript s-suc =
    List.concat suffix-chunks @ PTranscript final-state
and suffix-ext: s-suc ≤ final-state
and suffix-rounds:
   $\bigwedge k. k < \text{rounds} - \text{Suc } i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks i-bound by simp
have selected-chunk-lens:
   $\bigwedge k. k < \text{rounds} \implies$ 
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
proof –
  fix k
  assume k-bound: k < rounds
  show
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
proof (cases k < i)
  case True
  have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length (prefix-chunks ! k) =
      verifier-query-round-transcript-length
        (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
      [OF prefix-rounds[OF True]])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
next
  case False
  show ?thesis
  proof (cases k = i)
  case True
  have nth-eq: ?selected-chunks ! k = chunk
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length chunk =
      verifier-query-round-transcript-length idx
        (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length[OF chunk-shape])

```

```

then show ?thesis
  using nth-eq
  by (simp add:
    verifier-query-round-transcript-length-index-irrelevant)
next
  case False
  let ?k = k - Suc i
  have ge: Suc i ≤ k
    using ⟨¬ k < i⟩ False by linarith
  have k-bound': ?k < rounds - Suc i
    using ge k-bound by linarith
  have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k
    using ge len-prefix-chunks by (simp add: nth-append)
  have
    length (suffix-chunks ! ?k) =
      verifier-query-round-transcript-length
        (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
      [OF suffix-rounds[OF k-bound']])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-lens:
  ∧k. k < rounds ⇒
    length (query-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  using query-chunk verifier-query-round-chunk-length by blast
have transcript-all:
  PTranscript query-state =
    List.concat ?selected-chunks @ PTranscript final-state
  using transcript-prefix transcript-round transcript-suffix by simp
have concat-eq:
  List.concat ?selected-chunks = List.concat query-chunks
  using transcript-all transcript-query by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
    ∧k. k < length ?selected-chunks ⇒
      length (?selected-chunks ! k) = length (query-chunks ! k)
    using selected-chunk-lens query-chunk-lens len-selected-chunks
    by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp

```

```

have len-eq: length ?selected-chunks = length query-chunks
  using len-selected-chunks len-query-chunks by simp
have ?selected-chunks = query-chunks  $\wedge$  ( $\square :: 'f \text{ list}$ ) =  $\square$ 
  by (rule concat-append-eq-concat-same-chunk-lengths
    [OF len-eq same-lens concat-eq-empty])
then show ?thesis
  by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state)
      ?selected-chunks i =
        state-after-query-chunks (PState query-state)
          prefix-chunks i
  by (rule state-after-query-chunks-append-prefix
    [OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks
        (PState query-state) query-chunks i)) =
    Some raw
  using hash-extension-lookup[OF lookup-suc suffix-ext]
  unfolding counter-i state-i prefix-state-eq .
have raw-eq: raw = raw-idxs ! i
  using lookup-final replay-lookup[OF i-bound] by simp
have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-eq query-idxs-eq len-raw i-bound by simp
have auth-table-final:
  partial-authenticated-table fr (scale * clength) openings
    final-state
  by (rule partial-authenticated-table-mono
    [OF auth-table-suc suffix-ext])
show ?thesis
  by (rule that[OF - auth-table-final])
    (use opening-indices idx-eq-query in simp)
qed
qed
qed
qed

lemma verifier-query-round-program-outcome-with-zero-trace-final-opening:
  fixes s t :: ('f, 'a) protocol-channel-scheme

```



```

assumes f-fl-empty: f-fl = []
and outcome:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
obtains raw idx chunk openings where
  idx = index (to-nat raw)
  verifier-query-round-chunk idx [] (map snd fl) chunk
  PTranscript s = chunk @ PTranscript t
  PState t = foldl concat (PState s) chunk
  s ≤ t
  PQueryCounter t = Suc (PQueryCounter s)
  fmlookup (HashMap t) (QueryIndexChallenge (PQueryCounter s) (PState s)) =
    Some raw
  map opening-index openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) openings t
  opening-value (hd openings) = f-final
proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where
  rand:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s)
and tail:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
by (auto elim!: set-dist-bindE)
let ?idx = index (to-nat raw)
from tail obtain fv s1 f-out s2 s3 c-out s4 where
  query-decommit:
    Some (fv, s1) ∈
      set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s0)
and trace-fri:
  Some (f-out, s2) ∈
    set-dist (execute
      (mfold (?idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl)) s1)
and assert-trace:
  Some ((), s3) ∈
    set-dist
      (execute
        (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
  Some (c-out, s4) ∈
    set-dist (execute

```

```

    (mfold
      (?idx, cp-eval as fv (h ^ ?idx * shift), clength * scale, 1)
      (receive-query-commits fl)) s3)
and assert-comp:
  Some (( ), t) ∈
    set-dist
      (execute (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
unfolding verifier-query-round-after-index-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have rand-res:
  s ≤ s0 ∧
  PState s0 = PState s ∧
  PTranscript s0 = PTranscript s ∧
  fmlookup (HashMap s0)
  (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
using receive-query-index-challenge-outcome[OF rand] .
have rand-counter:
  PQueryCounter s0 = Suc (PQueryCounter s)
using receive-query-index-challenge-counter-outcome[OF rand] by simp
from check-decommit-on-query-outcome[OF query-decommit]
obtain query-paths query-chunk where
  query-chunk-shape:
    query-decommitment-transcript ?idx fv query-paths query-chunk
and tr-s0: PTranscript s0 = query-chunk @ PTranscript s1
and st-s1: PState s1 = foldl concat (PState s0) query-chunk
and ext-s0-s1: s0 ≤ s1
and query-count-s1: PQueryCounter s1 = PQueryCounter s0
by blast
have trace-out:
  f-out = (?idx, hd fv, clength * scale, 1)
and s2-eq: s2 = s1
using trace-fri f-fl-empty
unfolding receive-query-commits-def by simp-all
have hd-fv: hd fv = f-final
using assert-trace unfolding assert-def trace-out
by (cases hd fv = f-final) (auto simp: throw-no-outcome)
have s3-eq: s3 = s2
using assert-trace unfolding assert-def trace-out hd-fv by simp
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  comp-layers:
    fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
and st-s4: PState s4 = foldl concat (PState s3) comp-fri-chunk
and ext-s3-s4: s3 ≤ s4
and query-count-s4: PQueryCounter s4 = PQueryCounter s3
by blast
have t-eq: t = s4

```

```

using assert-comp unfolding assert-def
by (cases case c-out of (-, x, -, -)  $\Rightarrow$   $x = \text{final}$ )
  (auto simp: throw-no-outcome)
have trace-layers-empty:
  fri-layers-transcript 0 (clength * scale) [] []
  unfolding fri-layers-transcript-def by simp
let ?chunk = query-chunk @ [] @ comp-fri-chunk
have round-chunk:
  verifier-query-round-chunk ?idx [] (map snd fl) ?chunk
  unfolding verifier-query-round-chunk-def
  apply (rule exI[where  $x = \text{query-chunk}$ ])
  apply (rule exI[where  $x = []$ ])
  apply (rule exI[where  $x = \text{comp-fri-chunk}$ ])
  apply (rule exI[where  $x = \text{fv}$ ])
  apply (rule exI[where  $x = \text{query-paths}$ ])
  apply (rule exI[where  $x = []$ ])
  apply (rule exI[where  $x = \text{comp-layer-chunks}$ ])
  using query-chunk-shape trace-layers-empty comp-layers
  by simp
have tr-t: PTranscript  $s = ?chunk @ PTranscript\ t$ 
  using rand-res tr-s0 tr-s3 s3-eq s2-eq t-eq by simp
have st-t: PState  $t = \text{foldl concat} (PState\ s)\ ?chunk$ 
  using rand-res st-s1 st-s4 s3-eq s2-eq t-eq by simp
have ext-s0-t:  $s0 \leq t$ 
  using ext-s0-s1 ext-s3-s4 t-eq unfolding s3-eq s2-eq
  by (meson hash-ext-trans)
have ext-s-t:  $s \leq t$ 
  using rand-res ext-s0-t by (meson hash-ext-trans)
have query-count-t: PQueryCounter  $t = \text{Suc} (PQueryCounter\ s)$ 
  using rand-counter query-count-s1 query-count-s4
  unfolding s3-eq s2-eq t-eq by simp
have lookup-t:
  fmlookup (HashMap  $t$ )
    (QueryIndexChallenge (PQueryCounter  $s$ ) (PState  $s$ )) = Some raw
  using hash-extension-lookup[OF conjunct2[OF conjunct2[OF conjunct2[OF rand-res]]] ext-s0-t]
    rand-res
  by simp
have idx-in:  $?idx \in \text{query-sample-space}$ 
  using index-less-query-sample-space
  unfolding query-sample-space-def by simp
from check-decommit-on-query-authenticated-openings
  [OF idx-in query-decommit]
obtain openings where values-openings:
  map opening-value openings = fv
  and idx-openings: map opening-index openings = powers-scaled ?idx
  and table-s1:
    partial-authenticated-table fr (scale * clength) openings s1
  by blast
have s1-t:  $s1 \leq t$ 

```

```

    using ext-s3-s4 unfolding s3-eq s2-eq t-eq .
have table-t:
  partial-authenticated-table fr (scale * clength) openings t
  by (rule partial-authenticated-table-mono[OF table-s1 s1-t])
have openings-nonempty: openings ≠ []
proof -
  have map opening-index openings ≠ []
    using idx-openings powers-pos unfolding powers-scaled-def by simp
  then show ?thesis by auto
qed
have hd-value: opening-value (hd openings) = hd fv
  using values-openings openings-nonempty
  by (cases openings; cases fv) simp-all
show ?thesis
proof (rule that)
  show ?idx = index (to-nat raw)
    by simp
  show verifier-query-round-chunk ?idx [] (map snd fl) ?chunk
    by (rule round-chunk)
  show PTranscript s = ?chunk @ PTranscript t
    by (rule tr-t)
  show PState t = foldl concat (PState s) ?chunk
    by (rule st-t)
  show s ≤ t
    by (rule ext-s-t)
  show PQueryCounter t = Suc (PQueryCounter s)
    by (rule query-count-t)
  show fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
    by (rule lookup-t)
  show map opening-index openings = powers-scaled ?idx
    by (rule idx-openings)
  show partial-authenticated-table fr (scale * clength) openings t
    by (rule table-t)
  show opening-value (hd openings) = f-final
    using hd-value hd-fv by simp
qed
qed

lemma ntimes-verifier-query-rounds-selected-empty-trace-chunks-at:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes fl-empty: fl = []
  and outcome:
    Some (results, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s)

```

```

and i-bound:  $i < n$ 
obtain raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
  suffix-chunks trace-openings where
  idx = index (to-nat raw)
  length prefix-query-idxs = i
  length prefix-chunks = i
  length suffix-query-idxs =  $n - \text{Suc } i$ 
  length suffix-chunks =  $n - \text{Suc } i$ 
  verifier-query-round-chunk idx (map snd f-fl) [] chunk
   $\bigwedge j. j < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! j)
      (map snd f-fl) [] (prefix-chunks ! j)
   $\bigwedge j. j < n - \text{Suc } i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! j)
      (map snd f-fl) [] (suffix-chunks ! j)
  PTranscript s =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @ PTranscript t
  map opening-index trace-openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) trace-openings t
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks (PState s) prefix-chunks i)) = Some raw
proof –
from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
obtain prefix round suffix s-i s-suc where
  prefix:
    Some (prefix, s-i) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) i)
          s)
  and round:
    Some (round, s-suc) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s-i)
  and suffix:
    Some (suffix, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
              ( $n - \text{Suc } i$ ))
          s-suc)
  by blast
have round-unit: round = ()
  by (cases round) simp
show ?thesis

```

```

proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
fix prefix-query-idxs prefix-chunks
assume len-prefix: length prefix-query-idxs = i
  and len-prefix-chunks: length prefix-chunks = i
  and transcript-prefix:
    PTranscript s = List.concat prefix-chunks @ PTranscript s-i
  and state-i:
    PState s-i =
      state-after-query-chunks (PState s) prefix-chunks i
  and counter-i: PQueryCounter s-i = PQueryCounter s + i
  and prefix-ext: s ≤ s-i
  and prefix-rounds:
     $\bigwedge j. j < i \implies$ 
      verifier-query-round-chunk (prefix-query-idxs ! j)
        (map snd f-fl) (map snd fl) (prefix-chunks ! j)
show ?thesis
proof (rule
  verifier-query-round-program-outcome-with-authenticated-trace-openings
  [OF round[unfolded round-unit]])
fix raw idx chunk trace-openings
assume idx-eq: idx = index (to-nat raw)
  and chunk-shape:
    verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
      chunk
  and transcript-round:
    PTranscript s-i = chunk @ PTranscript s-suc
  and state-suc:
    PState s-suc = foldl concat (PState s-i) chunk
  and round-ext: s-i ≤ s-suc
  and counter-suc: PQueryCounter s-suc = Suc (PQueryCounter s-i)
  and lookup-suc:
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
      Some raw
  and len-trace:
    length trace-openings = length (powers-scaled idx)
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and trace-table-suc:
    partial-authenticated-table fr (scale * clength)
      trace-openings s-suc
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-idxs:
  length suffix-query-idxs = n - Suc i
  and len-suffix-chunks:

```

```

    length suffix-chunks = n - Suc i
  and transcript-suffix:
    PTranscript s-suc =
      List.concat suffix-chunks @ PTranscript t
  and suffix-state:
    PState t =
      state-after-query-chunks (PState s-suc) suffix-chunks
      (n - Suc i)
  and suffix-counter:
    PQueryCounter t = PQueryCounter s-suc + (n - Suc i)
  and suffix-ext: s-suc ≤ t
  and suffix-rounds:
    ∧j. j < n - Suc i ⇒
      verifier-query-round-chunk (suffix-query-idxs ! j)
      (map snd f-fl) (map snd fl) (suffix-chunks ! j)
  have transcript-all:
    PTranscript s =
      List.concat (prefix-chunks @ chunk # suffix-chunks) @
      PTranscript t
    using transcript-prefix transcript-round transcript-suffix
    by simp
  have trace-table-t:
    partial-authenticated-table fr (scale * clength)
    trace-openings t
    by (rule partial-authenticated-table-mono
      [OF trace-table-suc suffix-ext])
  have lookup-t:
    fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks (PState s) prefix-chunks i)) =
    Some raw
    using hash-extension-lookup[OF lookup-suc suffix-ext]
    unfolding counter-i state-i .
  show ?thesis
    by (rule that[OF idx-eq len-prefix len-prefix-chunks
      len-suffix-idxs len-suffix-chunks - - transcript-all
      trace-indices trace-table-t lookup-t])
      (use chunk-shape prefix-rounds suffix-rounds fl-empty in simp-all)
qed
qed
qed
qed

lemma ntimes-verifier-query-rounds-selected-empty-trace-chunks-aligned-with-replay-at:
  assumes outcome:
    Some (results, final-state) ∈
      set-dist
      (execute
        (ntimes

```

```

      (verifier-query-round-program fr f-fl f-final as [] final)
      rounds)
    query-state)
  and i-bound:  $i < \text{rounds}$ 
  and len-raw:  $\text{length raw-idxs} = \text{rounds}$ 
  and query-idxs-eq:
    query-idxs = map ( $\lambda \text{raw. index (to-nat raw)}$ ) raw-idxs
  and len-query-chunks:  $\text{length query-chunks} = \text{rounds}$ 
  and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
  and query-chunk:
     $\bigwedge i. i < \text{rounds} \implies$ 
      verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) [] (query-chunks ! i)
  and replay-lookup:
     $\bigwedge i. i < \text{rounds} \implies$ 
      fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks
      (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
  obtains trace-openings where
    map opening-index trace-openings = powers-scaled (query-idxs ! i)
    partial-authenticated-table fr (scale * clength) trace-openings
    final-state
  proof (rule ntimes-verifier-query-rounds-selected-empty-trace-chunks-at
    [OF refl outcome i-bound])
  fix raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
    suffix-chunks trace-openings
  assume selected-idx:  $\text{idx} = \text{index (to-nat raw)}$ 
  and len-prefix-chunks:  $\text{length prefix-chunks} = i$ 
  and len-suffix-chunks:  $\text{length suffix-chunks} = \text{rounds} - \text{Suc } i$ 
  and chunk-shape:
    verifier-query-round-chunk idx (map snd f-fl) [] chunk
  and prefix-rounds:
     $\bigwedge j. j < i \implies$ 
      verifier-query-round-chunk (prefix-query-idxs ! j)
      (map snd f-fl) [] (prefix-chunks ! j)
  and suffix-rounds:
     $\bigwedge j. j < \text{rounds} - \text{Suc } i \implies$ 
      verifier-query-round-chunk (suffix-query-idxs ! j)
      (map snd f-fl) [] (suffix-chunks ! j)
  and selected-transcript:
    PTranscript query-state =
      List.concat (prefix-chunks @ chunk # suffix-chunks) @
      PTranscript final-state
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx

```



```

and trace-table:
  partial-authenticated-table fr (scale * clength) trace-openings
  final-state
and selected-lookup:
  fmlookup (HashMap final-state)
  (QueryIndexChallenge (PQueryCounter query-state + i)
    (state-after-query-chunks (PState query-state) prefix-chunks i)) =
    Some raw
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks i-bound by simp
have selected-chunk-lens:
   $\bigwedge j. j < \text{rounds} \implies$ 
    length (?selected-chunks ! j) =
      verifier-query-round-transcript-length (query-idxs ! j)
      (map snd f-fl) []
proof -
  fix j
  assume j-bound: j < rounds
  show
    length (?selected-chunks ! j) =
      verifier-query-round-transcript-length (query-idxs ! j)
      (map snd f-fl) []
proof (cases j < i)
  case True
  have nth-eq: ?selected-chunks ! j = prefix-chunks ! j
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length (prefix-chunks ! j) =
      verifier-query-round-transcript-length
        (prefix-query-idxs ! j) (map snd f-fl) []
    by (rule verifier-query-round-chunk-length[OF prefix-rounds[OF True]])
  then show ?thesis
    using nth-eq
    by (simp add: verifier-query-round-transcript-length-index-irrelevant)
next
  case False
  show ?thesis
  proof (cases j = i)
  case True
  have nth-eq: ?selected-chunks ! j = chunk
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length chunk =
      verifier-query-round-transcript-length idx (map snd f-fl) []
    by (rule verifier-query-round-chunk-length[OF chunk-shape])
  then show ?thesis
    using nth-eq
    by (simp add:

```

```

      verifier-query-round-transcript-length-index-irrelevant)
next
  case False
  let ?k = j - Suc i
  have ge: Suc i ≤ j
    using ⟨¬ j < i⟩ False by linarith
  have k-bound: ?k < rounds - Suc i
    using ge j-bound by linarith
  have nth-eq: ?selected-chunks ! j = suffix-chunks ! ?k
    using ge len-prefix-chunks by (simp add: nth-append)
  have
    length (suffix-chunks ! ?k) =
      verifier-query-round-transcript-length
        (suffix-query-idxs ! ?k) (map snd f-fl) []
    by (rule verifier-query-round-chunk-length
        [OF suffix-rounds [OF k-bound]])
  then show ?thesis
    using nth-eq
    by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-lens:
  ∧j. j < rounds ⇒
    length (query-chunks ! j) =
      verifier-query-round-transcript-length (query-idxs ! j)
        (map snd f-fl) []
    using query-chunk verifier-query-round-chunk-length by blast
have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
  using selected-transcript transcript-query by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
    ∧j. j < length ?selected-chunks ⇒
      length (?selected-chunks ! j) = length (query-chunks ! j)
    using selected-chunk-lens query-chunk-lens len-selected-chunks by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks ∧ ([] :: 'f list) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
        [OF len-eq same-lens concat-eq-empty])
  then show ?thesis
    by simp
qed
have prefix-state-eq:

```

```

state-after-query-chunks (PState query-state) prefix-chunks i =
  state-after-query-chunks (PState query-state) query-chunks i
proof —
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix[OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have selected-lookup-global:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw
  using selected-lookup prefix-state-eq by simp
have raw-eq: raw = raw-idxs ! i
  using selected-lookup-global replay-lookup[OF i-bound] by simp
have idx-eq: idx = query-idxs ! i
  using selected-idx raw-eq query-idxs-eq len-raw i-bound by simp
show ?thesis
  by (rule that)
  (use idx-eq trace-indices trace-table in simp-all)
qed

lemma ntimes-verifier-query-rounds-empty-trace-openings-aligned-with-replay:
  assumes outcome:
    Some (results, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as [] final)
            rounds)
          query-state)
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
  and query-chunk:
    ∧ i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
        (map snd f-fl) [] (query-chunks ! i)
  and replay-lookup:
    ∧ i. i < rounds ⇒
      fmlookup (HashMap final-state)
        (QueryIndexChallenge (PQueryCounter query-state + i)

```

```

      (state-after-query-chunks
       (PState query-state) query-chunks i)) =
    Some (raw-idxs ! i)
obtains trace-openings where
  length trace-openings = rounds
   $\bigwedge i. i < \text{rounds} \implies$ 
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
proof -
  let ?P =
     $\lambda i$  openings.
      map opening-index openings = powers-scaled (query-idxs ! i)  $\wedge$ 
      partial-authenticated-table fr (scale * clength) openings final-state
  have ex-round:  $\bigwedge i. i < \text{rounds} \implies \exists \text{ openings}. ?P i \text{ openings}$ 
proof -
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  show  $\exists \text{ openings}. ?P i \text{ openings}$ 
  proof (rule
    ntimes-verifier-query-rounds-selected-empty-trace-chunks-aligned-with-replay-at
    [OF outcome i-bound len-raw query-idxs-eq len-query-chunks
     transcript-query query-chunk replay-lookup])
  fix trace-openings
  assume
    map opening-index trace-openings =
      powers-scaled (query-idxs ! i)
      partial-authenticated-table fr (scale * clength)
      trace-openings final-state
  then show ?thesis
  by blast
qed
qed
let ?w =  $\lambda i. \text{SOME openings}. ?P i \text{ openings}$ 
let ?trace-openings = map ?w [0.. $\text{rounds}$ ]
have len-trace: length ?trace-openings = rounds
  by simp
have props:
   $\bigwedge i. i < \text{rounds} \implies ?P i (\text{?trace-openings ! } i)$ 
proof -
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  have ?P i (?w i)
    by (rule someI-ex[OF ex-round[OF i-bound]])
  then show ?P i (?trace-openings ! i)
    using i-bound by simp
qed

```

```

show ?thesis
  by (rule that[OF len-trace])
    (use props in simp-all)
qed

lemma ntimes-verifier-query-rounds-selected-zero-trace-final-opening-at:
  assumes outcome:
    Some (results, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr [] f-final as fl final)
            rounds)
          query-state)
  and i-bound: i < rounds
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
  and query-chunk:
    ∧ i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
        [] (map snd fl) (query-chunks ! i)
  and replay-lookup:
    ∧ i. i < rounds ⇒
      fmlookup (HashMap final-state)
        (QueryIndexChallenge (PQueryCounter query-state + i)
          (state-after-query-chunks
            (PState query-state) query-chunks i)) =
        Some (raw-idxs ! i)
  obtains trace-openings where
    map opening-index trace-openings = powers-scaled (query-idxs ! i)
    partial-authenticated-table fr (scale * clength) trace-openings
      final-state
    opening-value (hd trace-openings) = f-final
proof –
  from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
  obtain prefix round suffix s-i s-suc where
    prefix:
      Some (prefix, s-i) ∈
        set-dist
          (execute
            (ntimes
              (verifier-query-round-program fr [] f-final as fl final) i)
            query-state)
  and round:

```

```

    Some (round, s-suc) ∈
      set-dist
        (execute
          (verifier-query-round-program fr [] f-final as fl final) s-i)
  and suffix:
    Some (suffix, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr [] f-final as fl final)
            (rounds - Suc i))
          s-suc)
  by blast
have round-unit: round = ()
  by (cases round) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
  fix prefix-query-idxs prefix-chunks
  assume len-prefix: length prefix-query-idxs = i
  and len-prefix-chunks: length prefix-chunks = i
  and transcript-prefix:
    PTranscript query-state = List.concat prefix-chunks @ PTranscript s-i
  and state-i:
    PState s-i =
      state-after-query-chunks (PState query-state) prefix-chunks i
  and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
  and prefix-ext: query-state ≤ s-i
  and prefix-rounds:
    ∧j. j < i ⇒
      verifier-query-round-chunk (prefix-query-idxs ! j)
      (map snd []) (map snd fl) (prefix-chunks ! j)
  show ?thesis
proof (rule
  verifier-query-round-program-outcome-with-zero-trace-final-opening
  [OF refl round[unfolded round-unit]])
  fix raw idx chunk trace-openings
  assume selected-idx: idx = index (to-nat raw)
  and chunk-shape:
    verifier-query-round-chunk idx [] (map snd fl) chunk
  and transcript-round:
    PTranscript s-i = chunk @ PTranscript s-suc
  and state-suc:
    PState s-suc = foldl concat (PState s-i) chunk
  and round-ext: s-i ≤ s-suc
  and counter-suc: PQueryCounter s-suc = Suc (PQueryCounter s-i)
  and lookup-suc:
    fmlookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =

```

```

    Some raw
  and trace-indices:
    map opening-index trace-openings = powers-scaled idx
  and trace-table-suc:
    partial-authenticated-table fr (scale * clength)
    trace-openings s-suc
  and trace-value:
    opening-value (hd trace-openings) = f-final
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-idxs:
  length suffix-query-idxs = rounds - Suc i
and len-suffix-chunks:
  length suffix-chunks = rounds - Suc i
and transcript-suffix:
  PTranscript s-suc =
    List.concat suffix-chunks @ PTranscript final-state
and suffix-state:
  PState final-state =
    state-after-query-chunks (PState s-suc) suffix-chunks
    (rounds - Suc i)
and suffix-counter:
  PQueryCounter final-state =
    PQueryCounter s-suc + (rounds - Suc i)
and suffix-ext: s-suc ≤ final-state
and suffix-rounds:
  ∧j. j < rounds - Suc i ⇒
    verifier-query-round-chunk (suffix-query-idxs ! j)
    (map snd []) (map snd fl) (suffix-chunks ! j)
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have selected-transcript:
  PTranscript query-state =
    List.concat ?selected-chunks @ PTranscript final-state
  using transcript-prefix transcript-round transcript-suffix
  by simp
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks i-bound by simp
have selected-chunk-lens:
  ∧j. j < rounds ⇒
    length (?selected-chunks ! j) =
      verifier-query-round-transcript-length (query-idxs ! j)
      [] (map snd fl)
proof -
fix j
assume j-bound: j < rounds
show
  length (?selected-chunks ! j) =

```

```

    verifier-query-round-transcript-length (query-idxs ! j)
    [] (map snd fl)
  proof (cases j < i)
  case True
  have nth-eq: ?selected-chunks ! j = prefix-chunks ! j
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length (prefix-chunks ! j) =
      verifier-query-round-transcript-length
        (prefix-query-idxs ! j) [] (map snd fl)
  using verifier-query-round-chunk-length
    [OF prefix-rounds[OF True]]
  by simp
  then show ?thesis
    using nth-eq
  by (simp add:
    verifier-query-round-transcript-length-index-irrelevant)
next
case False
show ?thesis
proof (cases j = i)
case True
have nth-eq: ?selected-chunks ! j = chunk
  using True len-prefix-chunks by (simp add: nth-append)
have
  length chunk =
    verifier-query-round-transcript-length idx [] (map snd fl)
  by (rule verifier-query-round-chunk-length[OF chunk-shape])
then show ?thesis
  using nth-eq
  by (simp add:
    verifier-query-round-transcript-length-index-irrelevant)
next
case False
let ?k = j - Suc i
have ge: Suc i ≤ j
  using <¬ j < i> False by linarith
have k-bound: ?k < rounds - Suc i
  using ge j-bound by linarith
have nth-eq: ?selected-chunks ! j = suffix-chunks ! ?k
  using ge len-prefix-chunks by (simp add: nth-append)
have
  length (suffix-chunks ! ?k) =
    verifier-query-round-transcript-length
      (suffix-query-idxs ! ?k) [] (map snd fl)
  using verifier-query-round-chunk-length
    [OF suffix-rounds[OF k-bound]]
  by simp
then show ?thesis

```



```

    using nth-eq
    by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
  qed
qed
qed
have query-chunk-lens:
   $\bigwedge j. j < \text{rounds} \implies$ 
    length (query-chunks ! j) =
      verifier-query-round-transcript-length (query-idxs ! j)
    [] (map snd fl)
  using query-chunk verifier-query-round-chunk-length by blast
have concat-eq:
  List.concat ?selected-chunks = List.concat query-chunks
  using selected-transcript transcript-query by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
     $\bigwedge j. j < \text{length } ?\text{selected-chunks} \implies$ 
      length (?selected-chunks ! j) = length (query-chunks ! j)
    using selected-chunk-lens query-chunk-lens len-selected-chunks
    by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks  $\wedge$  ([] :: 'f list) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
        [OF len-eq same-lens concat-eq-empty])
  then show ?thesis
    by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix
        [OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have selected-lookup-global:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state)

```

```

      query-chunks i)) =
    Some raw
    using hash-extension-lookup[OF lookup-suc suffix-ext]
    unfolding counter-i state-i prefix-state-eq .
  have raw-eq: raw = raw-idxs ! i
    using selected-lookup-global replay-lookup[OF i-bound] by simp
  have idx-eq: idx = query-idxs ! i
    using selected-idx raw-eq query-idxs-eq len-raw i-bound by simp
  have trace-table:
    partial-authenticated-table fr (scale * clength) trace-openings
    final-state
    by (rule partial-authenticated-table-mono
      [OF trace-table-suc suffix-ext])
  show ?thesis
    by (rule that)
      (use idx-eq trace-indices trace-table trace-value in simp-all)
qed
qed
qed
qed

lemma ntimes-verifier-query-rounds-zero-trace-final-openings-aligned-with-replay:
  assumes outcome:
    Some (results, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr [] f-final as fl final)
            rounds)
          query-state)
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
  and query-chunk:
    ∧ i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
      [] (map snd fl) (query-chunks ! i)
  and replay-lookup:
    ∧ i. i < rounds ⇒
      fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
  obtains trace-openings where

```

```

length trace-openings = rounds
 $\wedge i. i < rounds \implies$ 
  map opening-index (trace-openings ! i) =
    powers-scaled (query-idxs ! i)
 $\wedge i. i < rounds \implies$ 
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) final-state
 $\wedge i. i < rounds \implies$ 
  opening-value (hd (trace-openings ! i)) = f-final
proof -
let ?P =
   $\lambda i$  openings.
    map opening-index openings = powers-scaled (query-idxs ! i)  $\wedge$ 
    partial-authenticated-table fr (scale * clength) openings final-state  $\wedge$ 
    opening-value (hd openings) = f-final
have ex-round:  $\wedge i. i < rounds \implies \exists$  openings. ?P i openings
proof -
  fix i
  assume i-bound:  $i < rounds$ 
  show  $\exists$  openings. ?P i openings
  proof (rule
    ntimes-verifier-query-rounds-selected-zero-trace-final-opening-at
    [OF outcome i-bound len-raw query-idxs-eq len-query-chunks
      transcript-query query-chunk replay-lookup])
  fix trace-openings
  assume
    map opening-index trace-openings =
      powers-scaled (query-idxs ! i)
      partial-authenticated-table fr (scale * clength)
        trace-openings final-state
    opening-value (hd trace-openings) = f-final
  then show ?thesis
  by blast
  qed
qed
let ?w =  $\lambda i. \text{SOME openings. ?P i openings}$ 
let ?trace-openings = map ?w [0.. $rounds$ ]
have len-trace: length ?trace-openings = rounds
  by simp
have props:
   $\wedge i. i < rounds \implies ?P i$  (?trace-openings ! i)
proof -
  fix i
  assume i-bound:  $i < rounds$ 
  have ?P i (?w i)
    by (rule someI-ex[OF ex-round[OF i-bound]])
  then show ?P i (?trace-openings ! i)
    using i-bound by simp
  qed

```

```

show ?thesis
  by (rule that[OF len-trace])
    (use props in simp-all)
qed

lemma verify-monad-supplied-empty-header-partial-trace-openings:
  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
  and header0:
    verifier-header-transcript s fr f-fri-roots f-final as dg []
    final rest
  obtains query-idxs trace-openings where
    accepted-transcript-shape s (Some (result, final-state)) as
    query-idxs
    accepted-with-partial-trace-openings s
    (Some (result, final-state)) fr query-idxs trace-openings
proof –
  obtain f-fl query-state where f-roots:
    map snd f-fl = f-fri-roots
  and query-rest: PTranscript query-state = rest
  and query-state-start:
    PState query-state =
      verifier-header-state s fr f-fri-roots f-final as dg [] final
  and query-count:
    PQueryCounter query-state = PQueryCounter s
  and query-out:
    Some (result, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as [] final)
            rounds)
          query-state)
  by (rule verify-monad-supplied-empty-header-query-rounds
    [OF outcome header0])
show ?thesis
proof (rule verify-monad-supplied-header-transcript-shape
  [OF outcome header0])
  fix query-idxs raw-idxs query-chunks
  assume shape:
    accepted-transcript-shape s (Some (result, final-state)) as
    query-idxs
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    List.concat query-chunks @ PTranscript final-state = rest

```

```

and query-chunk:
   $\bigwedge i. i < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! i)
    f-fri-roots [] (query-chunks ! i)
and replay-lookup:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks
          (verifier-header-state s fr f-fri-roots f-final as dg []
            final)
            query-chunks i)) =
      Some (raw-idxs ! i)
have transcript-query-state:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
using transcript-query query-rest by simp
have query-chunk-empty:
   $\bigwedge i. i < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! i)
    (map snd f-fl) [] (query-chunks ! i)
using query-chunk f-roots by simp
have replay-lookup-query-state:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
using replay-lookup query-state-start query-count by simp
show ?thesis
proof (rule
  ntimes-verifier-query-rounds-empty-trace-openings-aligned-with-replay
  [OF query-out len-raw query-idxs-eq len-query-chunks
    transcript-query-state query-chunk-empty
    replay-lookup-query-state])
fix trace-openings
assume len-trace: length trace-openings = rounds
and trace-indices:
   $\bigwedge i. i < \text{rounds} \implies$ 
    map opening-index (trace-openings ! i) =
    powers-scaled (query-idxs ! i)
and trace-tables:
   $\bigwedge i. i < \text{rounds} \implies$ 
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) final-state
have len-query-idxs: length query-idxs = rounds
using len-raw query-idxs-eq by simp
have partial:

```

```

    accepted-with-partial-trace-openings s
      (Some (result, final-state)) fr query-idxs trace-openings
  unfolding accepted-with-partial-trace-openings-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state])
      (use len-query-idxs len-trace trace-indices trace-tables in auto)
  show ?thesis
    by (rule that[OF shape partial])
qed
qed
qed
end
end
end

```

```

theory Soundness-FRI-Verifier-Tied
imports
  Soundness-FRI-Zero-Round-Target
  Soundness-Query-Transcript-Openings
begin

```

Verifier-tied FRI partial-candidate events.

Broad reachable partial-candidate events allow the candidate table to be compatible with sampled authenticated openings whose indices are not tied to the FRI opening transcript. This theory introduces narrower internal events where the candidate openings use the same query indices as the verifier's FRI transcript. The protocol and public soundness statement are unchanged.

```

context soundness
begin

```

```

lemma partial-authenticated-tables-values-unique-if-clean:
  assumes table:
    partial-authenticated-table rt len openings s
  and table':
    partial-authenticated-table rt len openings' s
  and clean:  $\neg$  hash-map-output-collision s
  and first:  $\text{opening} \in \text{set openings}$ 
  and second:  $\text{opening}' \in \text{set openings}'$ 
  and same-index:  $\text{opening-index opening} = \text{opening-index opening}'$ 
  shows  $\text{opening-value opening} = \text{opening-value opening}'$ 
proof -
  have combined:
    partial-authenticated-table rt len (openings @ openings') s
  using table table'
  unfolding partial-authenticated-table-def by auto
  show ?thesis
    by (rule partial-authenticated-table-values-unique-if-clean[

```

```

    OF combined clean])
    (use first second same-index in auto)
qed

lemma verifier-query-round-after-index-zero-trace-final-opening:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes empty-trace: f-fl = []
  and outcome:
    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s)
  obtains idx openings where
    idx = index (to-nat raw)
    and map opening-index openings = powers-scaled idx
    and partial-authenticated-table fr (scale * clength) openings t
    and opening-value (hd openings) = f-final
proof -
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
    query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
    and trace-fri:
      Some (f-out, s2) ∈
        set-dist
          (execute
            (mfold (?idx, hd fv, clength * scale, 1)
              (receive-query-commits f-fl)) s1)
    and assert-trace:
      Some (( ), s3) ∈
        set-dist
          (execute
            (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
    and comp-fri:
      Some (c-out, s4) ∈
        set-dist (execute
          (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
            clength * scale, 1)
            (receive-query-commits fl)) s3)
    and assert-comp:
      Some (( ), t) ∈
        set-dist
          (execute
            (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
  unfolding verifier-query-round-after-index-program-def
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have trace-out:

```

```

  f-out = (?idx, hd fv, clength * scale, 1)
  and s2-eq: s2 = s1
  using trace-fri empty-trace
  unfolding receive-query-commits-def by simp-all
have hd-fv: hd fv = f-final
  using assert-trace unfolding assert-def trace-out
  by (cases hd fv = f-final) (auto simp: throw-no-outcome)
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def trace-out hd-fv by simp
have idx-in: ?idx ∈ query-sample-space
  using index-less-query-sample-space
  unfolding query-sample-space-def by simp
from check-decommit-on-query-authenticated-openings
  [OF idx-in query-decommit]
obtain openings where len-openings:
  length openings = length (powers-scaled ?idx)
  and values-openings: map opening-value openings = fv
  and idx-openings: map opening-index openings = powers-scaled ?idx
  and table-s1:
    partial-authenticated-table fr (scale * clength) openings s1
  by blast
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where s3-s4: s3 ≤ s4
  by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -) ⇒ x = final)
    (auto simp: throw-no-outcome)
have s1-t: s1 ≤ t
  using s3-s4 unfolding s3-eq s2-eq t-eq .
have table-t:
  partial-authenticated-table fr (scale * clength) openings t
  by (rule partial-authenticated-table-mono[OF table-s1 s1-t])
have openings-nonempty: openings ≠ []
  using len-openings powers-pos unfolding powers-scaled-def by auto
have hd-value: opening-value (hd openings) = hd fv
  using values-openings openings-nonempty
  by (cases openings; cases fv) simp-all
show ?thesis
  by (rule that[OF refl idx-openings table-t])
    (simp add: hd-value hd-fv)
qed

```

```

lemma verifier-query-round-program-zero-trace-final-opening:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes empty-trace: f-fl = []
  and outcome:
    Some ((), t) ∈
      set-dist

```



```

      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
shows  $\exists$  raw idx openings.
  idx = index (to-nat raw)  $\wedge$ 
  map opening-index openings = powers-scaled idx  $\wedge$ 
  partial-authenticated-table fr (scale * clength) openings t  $\wedge$ 
  opening-value (hd openings) = f-final
proof -
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where
  Some (raw, s0)  $\in$  set-dist (execute receive-query-index-challenge s)
and after:
  Some ((), t)  $\in$ 
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
by (auto elim!: set-dist-bindE)
from verifier-query-round-after-index-zero-trace-final-opening
  [OF empty-trace after]
show ?thesis
by blast
qed

lemma trace-zero-round-checked-candidate-from-verified-openings-if-no-merkle:
fixes final-state :: ('f, 'a) protocol-channel-scheme
assumes partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr query-idxs trace-openings
and candidate:
  partial-trace-table-candidate trace-table trace-openings
and no-merkle:
   $\neg$  partial-merkle-inconsistency-bad s (Some (result, final-state))
and query-sample:
   $\bigwedge$  idx. idx  $\in$  set query-idxs  $\implies$  idx  $\in$  query-sample-space
and verifier-openings:
   $\bigwedge$  i. i < rounds  $\implies$ 
     $\exists$  openings.
      map opening-index openings = powers-scaled (query-idxs ! i)  $\wedge$ 
      partial-authenticated-table fr (scale * clength) openings
      final-state  $\wedge$ 
      opening-value (hd openings) = final
shows
  trace-zero-round-final-checked-candidate s
  (Some (result, final-state)) fr query-idxs trace-openings
  trace-table final
proof -
have final-values:
   $\bigwedge$  i. i < rounds  $\implies$ 

```

```

    opening-value (hd (trace-openings ! i)) = final
proof –
  fix i
  assume i-bound: i < rounds
  have candidate-indices:
    map opening-index (trace-openings ! i) =
      powers-scaled (query-idxs ! i)
  by (rule accepted-with-partial-trace-openings-shapes(3)
    [OF partial i-bound])
  have candidate-table:
    partial-authenticated-table fr (scale * clength)
      (trace-openings ! i) final-state
  using partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
from verifier-openings[OF i-bound]
obtain openings where verified-indices:
  map opening-index openings = powers-scaled (query-idxs ! i)
and verified-table:
  partial-authenticated-table fr (scale * clength) openings
    final-state
and verified-value: opening-value (hd openings) = final
by blast
have candidate-nonempty: trace-openings ! i ≠ []
proof –
  have map opening-index (trace-openings ! i) ≠ []
  using candidate-indices powers-pos unfolding powers-scaled-def by simp
  then show ?thesis by auto
qed
have verified-nonempty: openings ≠ []
proof –
  have map opening-index openings ≠ []
  using verified-indices powers-pos unfolding powers-scaled-def by simp
  then show ?thesis by auto
qed
let ?candidate = hd (trace-openings ! i)
let ?verified = hd openings
have candidate-in: ?candidate ∈ set (trace-openings ! i)
  using candidate-nonempty by simp
have verified-in: ?verified ∈ set openings
  using verified-nonempty by simp
have same-index:
  opening-index ?candidate = opening-index ?verified
proof –
  have opening-index ?candidate =
    hd (map opening-index (trace-openings ! i))
  using candidate-nonempty by (cases trace-openings ! i) auto
  also have ... = hd (powers-scaled (query-idxs ! i))
  using candidate-indices by simp
  also have ... = hd (map opening-index openings)

```

```

    using verified-indices by simp
  also have ... = opening-index ?verified
    using verified-nonempty by (cases openings) auto
  finally show ?thesis .
qed
have candidate-auth:
  authenticated-opening-in final-state ?candidate
  using candidate-table candidate-in
  unfolding partial-authenticated-table-def by blast
have verified-auth:
  authenticated-opening-in final-state ?verified
  using verified-table verified-in
  unfolding partial-authenticated-table-def by blast
have same-root:
  opening-root ?candidate = opening-root ?verified
  using candidate-table verified-table candidate-in verified-in
  unfolding partial-authenticated-table-def by auto
have same-length:
  opening-length ?candidate = opening-length ?verified
  using candidate-table verified-table candidate-in verified-in
  unfolding partial-authenticated-table-def by auto
show opening-value ?candidate = final
proof (rule ccontr)
  assume neg-final: opening-value ?candidate  $\neq$  final
  have diff:
    opening-value ?candidate  $\neq$  opening-value ?verified
    using neg-final verified-value by simp
  have partial-merkle-inconsistency-bad s
    (Some (result, final-state))
    unfolding partial-merkle-inconsistency-bad-def
    by (intro conjI exI)
    (use partial candidate-auth verified-auth same-root same-length
      same-index diff in
       $\langle$ auto simp: accepted-with-partial-trace-openings-def $\rangle$ )
  then show False
    using no-merkle by contradiction
qed
qed
show ?thesis
  unfolding trace-zero-round-final-checked-candidate-def
  using partial candidate query-sample final-values by blast
qed

definition trace-fri-bad-with-verifier-tied-partial-candidate
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  trace-fri-bad-with-verifier-tied-partial-candidate s out  $\longleftrightarrow$ 
    ( $\exists$  trace-roots trace-bs trace-final dg composition-roots

```

composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr trace-openings trace-table.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ trace-table-low-degree trace-table)

definition *trace-fri-bad-with-header-tied-partial-candidate*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-bad-with-header-tied-partial-candidate s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers $\wedge$
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ trace-table-low-degree trace-table)

lemma *trace-fri-bad-with-header-tied-imp-verifier-tied:*

assumes *trace-fri-bad-with-header-tied-partial-candidate s out*

shows *trace-fri-bad-with-verifier-tied-partial-candidate s out*

using *assms*

unfolding *trace-fri-bad-with-header-tied-partial-candidate-def*

trace-fri-bad-with-verifier-tied-partial-candidate-def

by *blast*

lemma *accepted-fri-opening-transcript-query-idx-in-sample-space:*

assumes *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final dg composition-roots composition-bs composition-final

query-idxs trace-round-layers composition-round-layers

and *idx-in: idx $\in$ set query-idxs*

shows *idx $\in$ query-sample-space*

using *fri-openings idx-in index-less-query-sample-space*

unfolding *accepted-fri-opening-transcript-def query-sample-space-def* **by** *auto*

definition *trace-fri-header-tied-zero-round-final-obstruction*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-zero-round-final-obstruction s out
 $\longleftrightarrow$
($\exists$ *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers $\wedge$
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ *trace-table-low-degree trace-table* $\wedge$
length trace-bs = 0 $\wedge$
 $\neg$ *fri-final-constant-consistent trace-table trace-final*)

lemma *trace-fri-header-tied-zero-round-final-obstruction-imp-checked-or-merkle:*

assumes *obstruction:*

trace-fri-header-tied-zero-round-final-obstruction s out

shows

trace-fri-zero-round-checked-final-obstruction s out $\vee$
partial-merkle-inconsistency-bad s out

proof —

from *obstruction* **obtain** *trace-roots trace-bs trace-final dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as rest trace-openings
trace-table **where** *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *header:*
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *candidate:*
partial-trace-table-candidate trace-table trace-openings
and *not-low:* $\neg$ *trace-table-low-degree trace-table*
and *zero-rounds:* *length trace-bs = 0*
and *not-final:*
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
unfolding *trace-fri-header-tied-zero-round-final-obstruction-def*
by *blast*
from *fri-openings* **obtain** *result final-state fr0 f-fl as0 fl query-state*
raw-idxs query-chunks **where** *out-eq:*
out = Some (result, final-state)

```

and trace-roots-eq: trace-roots = map snd f-fl
and trace-bs-eq: trace-bs = map fst f-fl
and composition-roots-eq: composition-roots = map snd fl
and composition-bs-eq: composition-bs = map fst fl
and header0:
  verifier-header-transcript s fr0 trace-roots trace-final as0 dg
  composition-roots composition-final (PTranscript query-state)
and query-state-eq:
  PState query-state =
    verifier-header-state s fr0 trace-roots trace-final as0 dg
    composition-roots composition-final
and query-counter-eq:
  PQueryCounter query-state = PQueryCounter s
and query-out:
  Some (result, final-state) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr0 f-fl trace-final as0 fl
          composition-final)
        rounds)
      query-state)
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  fri-query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and query-chunk:
  ∧i. i < rounds ⇒
    verifier-query-round-chunk (fri-query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
and replay-lookup:
  ∧i. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
unfolding accepted-fri-opening-transcript-def by blast
have f-fl-empty: f-fl = []
using zero-rounds trace-bs-eq by simp
have trace-roots-empty: trace-roots = []
using trace-roots-eq f-fl-empty by simp
have header-eqs:
  fr0 = fr ∧
  as0 = as ∧
  PTranscript query-state = rest
using verifier-header-transcript-unique[OF header0 header] by simp

```

```

have query-out-empty:
  Some (result, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr [] trace-final as fl
            composition-final)
          rounds)
        query-state)
using query-out f-fl-empty header-egs by simp
have query-chunk-empty:
  ∧i. i < rounds ⇒
    verifier-query-round-chunk (fri-query-idxs ! i)
      [] (map snd fl) (query-chunks ! i)
using query-chunk trace-roots-empty composition-roots-eq by simp
show ?thesis
proof (cases partial-merkle-inconsistency-bad s out)
  case True
    then show ?thesis by simp
next
  case False
    have no-merkle:
      ¬ partial-merkle-inconsistency-bad s
        (Some (result, final-state))
    using False out-eq by simp
    have verifier-openings:
      ∧i. i < rounds ⇒
        ∃ openings.
          map opening-index openings =
            powers-scaled (fri-query-idxs ! i) ∧
            partial-authenticated-table fr (scale * clength) openings
              final-state ∧
            opening-value (hd openings) = trace-final
    proof –
      fix i
      assume i-bound: i < rounds
      show
        ∃ openings.
          map opening-index openings =
            powers-scaled (fri-query-idxs ! i) ∧
            partial-authenticated-table fr (scale * clength) openings
              final-state ∧
            opening-value (hd openings) = trace-final
    proof (rule
      ntimes-verifier-query-rounds-selected-zero-trace-final-opening-at
      [OF query-out-empty i-bound len-raw query-idxs-eq len-query-chunks
        transcript-query query-chunk-empty replay-lookup])
    fix verified-openings
    assume

```

```

    map opening-index verified-openings =
      powers-scaled (fri-query-idxs ! i)
    partial-authenticated-table fr (scale * clength)
      verified-openings final-state
    opening-value (hd verified-openings) = trace-final
  then show
    ∃ openings.
      map opening-index openings =
        powers-scaled (fri-query-idxs ! i) ∧
        partial-authenticated-table fr (scale * clength) openings
        final-state ∧
        opening-value (hd openings) = trace-final
  by blast
qed
qed
have query-sample:
  ∧idx. idx ∈ set fri-query-idxs ⇒ idx ∈ query-sample-space
  by (rule accepted-fri-opening-transcript-query-idx-in-sample-space
    [OF fri-openings])
have partial-some:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr fri-query-idxs trace-openings
  using partial out-eq by simp
have checked:
  trace-zero-round-final-checked-candidate s
  (Some (result, final-state)) fr fri-query-idxs trace-openings
  trace-table trace-final
  by (rule trace-zero-round-checked-candidate-from-verified-openings-if-no-merkle
    [OF partial-some candidate no-merkle query-sample verifier-openings])
have trace-fri-zero-round-checked-final-obstruction s out
  unfolding trace-fri-zero-round-checked-final-obstruction-def out-eq
  by (intro exI conjI)
  (use fri-openings[unfolded out-eq] zero-rounds checked not-final
    in simp-all)
  then show ?thesis by simp
qed
qed

definition composition-fri-bad-with-verifier-tied-partial-candidate
  :: ('f, 'a) protocol-channel-scheme ⇒
    (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool
where
  composition-fri-bad-with-verifier-tied-partial-candidate s out
  ⇔
  (∃ trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings composition-openings
    trace-table composition-table.
    accepted-fri-opening-transcript s out trace-roots trace-bs

```


trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table*)

definition *trace-fri-bad-with-verifier-tied-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-bad-with-verifier-tied-sampled-layer-chain s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr trace-openings trace-table doms layers.}$
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ *trace-table-low-degree trace-table* $\wedge$
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers)

definition *composition-fri-bad-with-verifier-tied-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-bad-with-verifier-tied-sampled-layer-chain s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table doms layers.}$
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$

partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers doms layers)

definition *trace-fri-verifier-tied-missing-sampled-layer-chain*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-verifier-tied-missing-sampled-layer-chain s out $\longleftrightarrow$
trace-fri-bad-with-verifier-tied-partial-candidate s out $\wedge$
 $\neg$ *trace-fri-bad-with-verifier-tied-sampled-layer-chain s out*

definition *composition-fri-verifier-tied-missing-sampled-layer-chain*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-missing-sampled-layer-chain s out
 $\longleftrightarrow$
composition-fri-bad-with-verifier-tied-partial-candidate s out $\wedge$
 $\neg$ *composition-fri-bad-with-verifier-tied-sampled-layer-chain s out*

definition *trace-fri-verifier-tied-reduction*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

trace-fri-verifier-tied-reduction s $\longleftrightarrow$
wp-event verify-monad
(trace-fri-bad-with-verifier-tied-partial-candidate s) s $\leq$
trace-fri-error

definition *composition-fri-verifier-tied-reduction*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-reduction s $\longleftrightarrow$
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-partial-candidate s) s $\leq$
composition-fri-error

definition *soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate
s out $\longleftrightarrow$
composition-bad-with-partial-candidates s out $\vee$

trace-fri-bad-with-header-tied-partial-candidate s out $\vee$
composition-fri-bad-with-verifier-tied-partial-candidate s out $\vee$
query-bad-with-aligned-transcript-partial-candidates s out

definition *soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle
 $s \text{ out} \longleftrightarrow$
partial-merkle-inconsistency-bad s out $\vee$
soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate
 $s \text{ out}$

lemma *accepted-fri-opening-transcript-obtains-aligned-partial-openings:*

assumes *fri-openings:*

accepted-fri-opening-transcript s (Some (result, final-state))
trace-roots trace-bs trace-final dg composition-roots composition-bs
composition-final query-idxs trace-round-layers
composition-round-layers

and *comp-nonempty:* *composition-roots* $\neq []$

obtains *fr as rest trace-openings composition-openings* **where**

verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
accepted-with-partial-initial-openings-aligned-transcript-consistent s
(Some (result, final-state)) fr trace-roots trace-final as dg
composition-roots composition-final query-idxs trace-openings
composition-openings

proof –

from *fri-openings* **obtain** *result' final-state'* *fr f-fl as fl query-state*
raw-idxs query-chunks **where** *out-eq:*

Some (result, final-state) = Some (result', final-state')

and *trace-roots-eq:* *trace-roots = map snd f-fl*

and *comp-roots-eq:* *composition-roots = map snd fl*

and *header:*

verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final (PTranscript query-state)

and *query-state-state:*

PState query-state =
verifier-header-state s fr trace-roots trace-final as dg
composition-roots composition-final

and *query-state-counter:*

PQueryCounter query-state = PQueryCounter s

and *query-out:*

Some (result', final-state') $\in$
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl trace-final as fl

```

      composition-final)
    rounds)
  query-state)
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state'
and query-chunk:
  ∧i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
and replay-lookup:
  ∧i. i < rounds ⇒
    fmllookup (HashMap final-state')
    (QueryIndexChallenge
      (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
  unfolding accepted-fri-opening-transcript-def by blast
obtain b composition-root fl-tail where fl-eq:
  fl = (b, composition-root) # fl-tail
  using comp-nonempty comp-roots-eq by (cases fl) auto
have f-roots: map snd f-fl = trace-roots
  using trace-roots-eq by simp
have comp-roots: map snd fl = composition-roots
  using comp-roots-eq by simp
have result-eq: result' = result
  and final-state-eq: final-state' = final-state
  using out-eq by simp-all
have query-out':
  Some (result, final-state) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr f-fl trace-final as fl
          composition-final)
        rounds)
      query-state)
  using query-out result-eq final-state-eq by simp
have transcript-query':
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
  using transcript-query final-state-eq by simp
have query-chunk':
  ∧i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)

```

```

    (map snd f-fl) (map snd fl) (query-chunks ! i)
  using query-chunk trace-roots-eq comp-roots-eq by simp
have replay-lookup':
   $\wedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge
        (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
  using replay-lookup final-state-eq by simp
from
  query-rounds-replay-accepted-with-partial-initial-openings-aligned-transcript-consistent
  [OF header comp-nonempty fl-eq f-roots comp-roots refl
   query-state-state query-state-counter query-out' len-raw query-idxs-eq
   len-query-chunks transcript-query' query-chunk' replay-lookup']
obtain trace-openings composition-openings where partial:
  accepted-with-partial-initial-openings-aligned-transcript-consistent s
  (Some (result, final-state)) fr trace-roots trace-final as dg
  composition-roots composition-final query-idxs trace-openings
  composition-openings
by blast
show ?thesis
by (rule that[OF header partial])
qed

```

lemma *accepted-fri-opening-transcript-obtains-aligned-partial-candidates-or-partial-merkle:*

assumes *fri-openings:*

accepted-fri-opening-transcript s (Some (result, final-state))
trace-roots trace-bs trace-final dg composition-roots composition-bs
composition-final query-idxs trace-round-layers
composition-round-layers

and *comp-nonempty: composition-roots $\neq []$*

shows

partial-merkle-inconsistency-bad s (Some (result, final-state)) $\vee$
($\exists$ fr as rest trace-openings composition-openings trace-table
composition-table.
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest $\wedge$
accepted-with-partial-initial-openings-aligned s
(Some (result, final-state)) fr trace-roots trace-final as dg
composition-roots composition-final query-idxs trace-openings
composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings))

proof (cases

partial-merkle-inconsistency-bad s (Some (result, final-state)))

case *True*

then show *?thesis* **by** *simp*

next
case *no-merkle*: *False*
from *accepted-fri-opening-transcript-obtains-aligned-partial-openings*
[OF fri-openings comp-nonempty]
obtain *fr as rest trace-openings composition-openings* **where** *header:*
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
and *partial:*
accepted-with-partial-initial-openings-aligned-transcript-consistent s
(Some (result, final-state)) fr trace-roots trace-final as dg
composition-roots composition-final query-idxs trace-openings
composition-openings
by *blast*
have *aligned:*
accepted-with-partial-initial-openings-aligned s
(Some (result, final-state)) fr trace-roots trace-final as dg
composition-roots composition-final query-idxs trace-openings
composition-openings
using *partial*
by (*blast dest:*
accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
accepted-with-partial-initial-openings-aligned-consistent-imp-aligned)
obtain *trace-table composition-table* **where** *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *composition-candidate:*
partial-composition-table-candidate composition-table
composition-openings
using
accepted-with-partial-initial-openings-aligned-candidates-if-no-partial-merkle-bad
[OF aligned no-merkle]
by *blast*
show *?thesis*
by (*intro disjI2 exI conjI*)
(rule header, rule aligned, rule trace-candidate,
rule composition-candidate)
qed

lemma *accepted-verifier-tied-aligned-transcript-partial-candidate-partition:*
assumes *false-statement: $\neg$ exists-valid-trace*
and *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final query-idxs trace-round-layers
composition-round-layers
and *partial:*
accepted-with-partial-initial-openings-aligned-transcript-consistent s
out fr trace-roots trace-final as fri-dg composition-roots
composition-final
query-idxs trace-openings composition-openings

```

and header:
  verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and composition-candidate:
  partial-composition-table-candidate composition-table
    composition-openings
shows
  soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate
    s out
proof (cases trace-table-low-degree trace-table)
case False
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent s out fr
    trace-roots trace-final as fri-dg composition-roots composition-final query-idxs
    trace-openings composition-openings
by (rule
  accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
have aligned0:
  accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final query-idxs
    trace-openings composition-openings
by (rule accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
  [OF aligned])
have trace-partial:
  accepted-with-partial-trace-openings s out fr query-idxs
    trace-openings
by (rule accepted-with-partial-initial-openings-aligned-shapes(3)
  [OF aligned0])
have trace-fri-bad-with-header-tied-partial-candidate s out
  unfolding trace-fri-bad-with-header-tied-partial-candidate-def
by (intro exI conjI)
  (rule fri-openings, rule header, rule trace-partial,
    rule trace-candidate, rule False)
then show ?thesis
  unfolding
    soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-def
  by blast
next
case True
note trace-low = True
show ?thesis
proof (cases composition-table-low-degree maxDegree composition-table)
case False
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent s out fr
    trace-roots trace-final as fri-dg composition-roots composition-final query-idxs

```

```

    trace-openings composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
have aligned0:
  accepted-with-partial-initial-openings-aligned s out fr trace-roots
  trace-final as fri-dg composition-roots composition-final query-idxs
  trace-openings composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
    [OF aligned])
have composition-fri-bad-with-verifier-tied-partial-candidate s out
  unfolding composition-fri-bad-with-verifier-tied-partial-candidate-def
  by (intro exI conjI)
    (rule fri-openings, rule aligned0, rule trace-candidate,
     rule composition-candidate, rule trace-low, rule False)
then show ?thesis
  unfolding
    soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-def
  by blast
next
case True
note composition-low = True
show ?thesis
proof (cases all-queries-consistent trace-table composition-table as)
  case False
  have query-bad-with-aligned-transcript-partial-candidates s out
    unfolding query-bad-with-aligned-transcript-partial-candidates-def
    using partial trace-candidate composition-candidate trace-low
    composition-low False
  by blast
  then show ?thesis
    unfolding
      soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-def
    by blast
next
case True
note all-queries = True
from trace-low obtain f where
  deg-f: degree f < clength
  and trace-table-eq: trace-table = map (poly f) eval-domain
  unfolding trace-table-low-degree-def by blast
have violated: violated-constraints f ≠ {}
  by (rule false-statement-violated-constraints
    [OF false-statement deg-f])
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent s out fr
  trace-roots trace-final as fri-dg composition-roots composition-final query-idxs
  trace-openings composition-openings

```



```

    by (rule
      accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
      [OF partial])
  have composition-bad-with-partial-candidates s out
    unfolding composition-bad-with-partial-candidates-def
    using accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned
      [OF aligned]
      trace-candidate composition-candidate deg-f trace-table-eq
      violated composition-low all-queries
    by blast
  then show ?thesis
    unfolding
      soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-def
    by blast
qed
qed
qed

lemma accepted-fri-opening-transcript-verifier-tied-partition-with-partial-merkle:
  assumes false-statement:  $\neg$  exists-valid-trace
  and fri-openings:
    accepted-fri-opening-transcript s (Some (result, final-state))
    trace-roots trace-bs trace-final dg composition-roots composition-bs
    composition-final query-idxs trace-round-layers
    composition-round-layers
  and comp-nonempty: composition-roots  $\neq$  []
  shows
    soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle
      s (Some (result, final-state))
proof -
  from accepted-fri-opening-transcript-obtains-aligned-partial-openings
    [OF fri-openings comp-nonempty]
  obtain fr as rest trace-openings composition-openings where header:
    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-initial-openings-aligned-transcript-consistent s
      (Some (result, final-state)) fr trace-roots trace-final as dg
    composition-roots composition-final query-idxs trace-openings
    composition-openings
  by blast
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then show ?thesis
  unfolding
    soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle-def
  by simp

```

```

next
case False
have aligned:
  accepted-with-partial-initial-openings-aligned s
  (Some (result, final-state)) fr trace-roots trace-final as dg
  composition-roots composition-final query-idxs trace-openings
  composition-openings
using partial
by (blast dest:
  accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
  accepted-with-partial-initial-openings-aligned-consistent-imp-aligned)
from accepted-with-partial-initial-openings-aligned-candidates-if-no-partial-merkle-bad
  [OF aligned False]
obtain trace-table composition-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and composition-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
by blast
have bad:
  soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate
  s (Some (result, final-state))
by (rule accepted-verifier-tied-aligned-transcript-partial-candidate-partition
  [OF false-statement fri-openings partial header trace-candidate
  composition-candidate])
then show ?thesis
  unfolding
  soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle-def
  by simp
qed
qed

lemma accepted-fri-opening-transcript-headerE:
  assumes fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers
  obtains fr as rest where
    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
  using fri-openings
  unfolding accepted-fri-opening-transcript-def
  by blast

lemma accepted-partition-soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle-o
  assumes false-statement:  $\neg$  exists-valid-trace
  and supp:  $out \in \text{set-dist } (\text{execute verify-monad } s)$ 
  and acc:  $\text{accepted out}$ 
  shows

```

```

    soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle
      s out ∨
    soundness-bad-event-partial-candidate-empty-header s out
proof –
from acc obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto
from verify-monad-accepted-fri-opening-transcript[OF supp[unfolded out-eq]]
obtain trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final query-idxs trace-round-layers
  composition-round-layers
where fri-openings:
  accepted-fri-opening-transcript s (Some (result, final-state))
  trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs trace-round-layers
  composition-round-layers
by blast
show ?thesis
proof (cases composition-roots = [])
case False
have
  soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle
    s (Some (result, final-state))
by (rule
  accepted-fri-opening-transcript-verifier-tied-partition-with-partial-merkle
    [OF false-statement fri-openings False])
then show ?thesis
  unfolding out-eq by simp
next
case True
note comp-empty = True
from accepted-fri-opening-transcript-headerE[OF fri-openings]
obtain fr as rest where header:
  verifier-header-transcript s fr trace-roots trace-final as dg
  composition-roots composition-final rest
by blast
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then have
  soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle
    s (Some (result, final-state))
unfolding
  soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle-def
by simp
then show ?thesis
  unfolding out-eq by simp
next

```

```

case False
from verify-monad-accepted-with-partial-trace-openings-for-header
  [OF supp[unfolded out-eq] header]
obtain trace-query-idxs trace-openings where trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr trace-query-idxs trace-openings
by blast
obtain trace-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
using
  accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
  [OF trace-partial False]
by blast
have accepted-out: accepted (Some (result, final-state))
by (rule accepted-with-partial-trace-openings-imp-accepted
  [OF trace-partial])
let ?composition-table = replicate (scale * clength) composition-final
have empty-partial:
  accepted-with-empty-composition-header-candidates s
  (Some (result, final-state)) fr trace-roots trace-final as dg
  composition-final trace-query-idxs trace-openings trace-table
  ?composition-table
unfolding accepted-with-empty-composition-header-candidates-def
by (intro conjI exI[of - rest])
  (use accepted-out header comp-empty trace-partial trace-candidate
  in simp-all)
have soundness-bad-event-partial-candidate-empty-header s
  (Some (result, final-state))
by (rule accepted-empty-composition-header-candidate-partition
  [OF false-statement empty-partial])
then show ?thesis
  unfolding out-eq by simp
qed
qed
qed

lemma soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-union-bound:
assumes comp-bound:
  wp-event verify-monad
  (composition-bad-with-partial-candidates s)  $s \leq$ 
  composition-error
and trace-fri-bound:
  wp-event verify-monad
  (trace-fri-bad-with-header-tied-partial-candidate s)  $s \leq$ 
  trace-fri-error'
and comp-fri-bound:
  wp-event verify-monad
  (composition-fri-bad-with-verifier-tied-partial-candidate s)  $s \leq$ 
  composition-fri-error'

```

and *query-bound*:

$$\text{wp-event verify-monad } (\text{query-bad-with-aligned-transcript-partial-candidates } s) \leq \text{query-error}$$
shows

$$\begin{aligned} &\text{wp-event verify-monad } (\text{soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate } s) \leq \\ &\text{composition-error} + \text{trace-fri-error}' + \text{composition-fri-error}' + \text{query-error} \end{aligned}$$
proof –
have *wp-event verify-monad*

$$(\text{soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate } s) \leq$$

$$\text{wp-event verify-monad } (\text{composition-bad-with-partial-candidates } s) \leq$$

$$\text{wp-event verify-monad } (\text{trace-fri-bad-with-header-tied-partial-candidate } s) \leq$$

$$\text{wp-event verify-monad } (\text{composition-fri-bad-with-verifier-tied-partial-candidate } s) \leq$$

$$\text{wp-event verify-monad } (\text{query-bad-with-aligned-transcript-partial-candidates } s) \leq$$
unfolding

$$\text{soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-def}$$
by (*rule wp-event-union-bound4*)
also have ... $\leq$

$$\text{composition-error} + \text{trace-fri-error}' + \text{composition-fri-error}' + \text{query-error}$$
by (*intro add-mono comp-bound trace-fri-bound comp-fri-bound query-bound*)
finally show *?thesis* .
qed

lemma *soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle-union-bound*:

assumes *merkle-bound*:

$$\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \leq \text{merkle-error}$$
and *comp-bound*:

$$\text{wp-event verify-monad } (\text{composition-bad-with-partial-candidates } s) \leq \text{composition-error}$$
and *trace-fri-bound*:

$$\text{wp-event verify-monad } (\text{trace-fri-bad-with-header-tied-partial-candidate } s) \leq \text{trace-fri-error}'$$
and *comp-fri-bound*:

$$\text{wp-event verify-monad } (\text{composition-fri-bad-with-verifier-tied-partial-candidate } s) \leq \text{composition-fri-error}'$$

and *query-bound*:

$$\text{wp-event verify-monad } (\text{query-bad-with-aligned-transcript-partial-candidates } s) \leq \text{query-error}$$
shows

$$\begin{aligned} &\text{wp-event verify-monad } (\text{soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle } s) \leq \\ &\text{merkle-error} + \text{composition-error} + \text{trace-fri-error}' + \\ &\text{composition-fri-error}' + \text{query-error} \end{aligned}$$
proof –
have *partial-bound*:

$$\begin{aligned} &\text{wp-event verify-monad } (\text{soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate } s) \leq \\ &\text{composition-error} + \text{trace-fri-error}' + \text{composition-fri-error}' + \\ &\text{query-error} \end{aligned}$$
by (rule

$$\text{soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-union-bound } [OF \text{ comp-bound trace-fri-bound comp-fri-bound query-bound}])$$
have *wp-event verify-monad*

$$\begin{aligned} &(\text{soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle } s) \leq \\ &\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \leq \\ &\text{wp-event verify-monad } (\text{soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate } s) \leq \end{aligned}$$
unfolding

$$\text{soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle-def}$$
by (rule *wp-event-union-bound*)
also have ... $\leq$

$$\text{merkle-error} + (\text{composition-error} + \text{trace-fri-error}' + \text{composition-fri-error}' + \text{query-error})$$
by (intro *add-mono merkle-bound partial-bound*)
finally show *?thesis*
by (*simp add: algebra-simps*)
qed

lemma *checked-staged-security-with-data-state-verifier-tied-aligned-transcript-partial-candidate-union-bound*:

assumes *comp-bound*:

$$\begin{aligned} &\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\ &(\text{staged-security-with-data-state-verifier-event } \text{composition-bad-with-partial-candidates}) \\ &\text{adversary-initial-state} \leq \text{composition-error} \end{aligned}$$

and *trace-fri-bound*:

$$\begin{aligned} &\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\ &(\text{staged-security-with-data-state-verifier-event } \text{trace-fri-bad-with-header-tied-partial-candidate}) \end{aligned}$$

```

    adversary-initial-state ≤ trace-fri-error'
and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-verifier-tied-partial-candidate)
    adversary-initial-state ≤ composition-fri-error'
and query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state ≤ query-error
shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate)
    adversary-initial-state ≤
      composition-error + trace-fri-error' + composition-fri-error' +
      query-error
proof –
    let ?M = checked-staged-security-experiment-with-data-state A
    let ?E = staged-security-with-data-state-verifier-event
    let ?comp = ?E composition-bad-with-partial-candidates
    let ?trace = ?E trace-fri-bad-with-header-tied-partial-candidate
    let ?comp-fri =
      ?E composition-fri-bad-with-verifier-tied-partial-candidate
    let ?query = ?E query-bad-with-aligned-transcript-partial-candidates
    let ?partial =
      ?E soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate
    have wp-event ?M ?partial adversary-initial-state ≤
      wp-event ?M
        (λout. ?comp out ∨ ?trace out ∨ ?comp-fri out ∨
          ?query out)
      adversary-initial-state
    by (rule wp-event-mono)
      (auto simp: staged-security-with-data-state-verifier-event-def
        soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-def
        split: option.splits prod.splits)
    also have ... ≤
      wp-event ?M ?comp adversary-initial-state +
      wp-event ?M ?trace adversary-initial-state +
      wp-event ?M ?comp-fri adversary-initial-state +
      wp-event ?M ?query adversary-initial-state
    by (rule wp-event-union-bound4)
    also have ... ≤
      composition-error + trace-fri-error' + composition-fri-error' +
      query-error
    by (intro add-mono comp-bound trace-fri-bound comp-fri-bound
      query-bound)
    finally show ?thesis .

```

qed

lemma *checked-staged-security-with-data-state-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle-u*

assumes *merkle-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$ *merkle-error*

and *comp-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
composition-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-error*

and *trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state $\leq$ *trace-fri-error'*

and *comp-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state $\leq$ *composition-fri-error'*

and *query-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq$ *query-error*

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle)
adversary-initial-state $\leq$
merkle-error + *composition-error* + *trace-fri-error'* +
composition-fri-error' + *query-error*

proof –

have *partial-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate)
adversary-initial-state $\leq$
composition-error + *trace-fri-error'* + *composition-fri-error'* +
query-error

by (rule

checked-staged-security-with-data-state-verifier-tied-aligned-transcript-partial-candidate-union-bound
[*OF comp-bound trace-fri-bound comp-fri-bound query-bound*])

let *?M* = *checked-staged-security-experiment-with-data-state A*

let *?E* = *staged-security-with-data-state-verifier-event*

let *?merkle* = *?E partial-merkle-inconsistency-bad*


```

let ?partial =
  ?E soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate
let ?with-merkle =
  ?E soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle
have wp-event ?M ?with-merkle adversary-initial-state ≤
  wp-event ?M (λout. ?merkle out ∨ ?partial out)
  adversary-initial-state
by (rule wp-event-mono)
  (auto simp: staged-security-with-data-state-verifier-event-def
  soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle-def
  split: option.splits prod.splits)
also have ... ≤
  wp-event ?M ?merkle adversary-initial-state +
  wp-event ?M ?partial adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤
  merkle-error +
  (composition-error + trace-fri-error' + composition-fri-error' +
  query-error)
by (intro add-mono merkle-bound partial-bound)
finally show ?thesis
by (simp add: algebra-simps)
qed

```

lemma checked-staged-soundness-from-verifier-tied-aligned-transcript-partial-candidate-and-empty-header-bounds:

fixes partial-candidate-error empty-header-error :: prob

assumes false-statement: $\neg$ exists-valid-trace

and partial-candidate-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle)
 adversary-initial-state ≤ partial-candidate-error

and empty-header-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 soundness-bad-event-partial-candidate-empty-header)
 adversary-initial-state ≤ empty-header-error

shows checked-staged-adversary-acceptance-probability A ≤
 partial-candidate-error + empty-header-error

proof –

let ?M = checked-staged-security-experiment-with-data-state A

let ?E = staged-security-with-data-state-verifier-event

let ?partial =

?E soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle

let ?empty = ?E soundness-bad-event-partial-candidate-empty-header

let ?bad = λout. ?partial out ∨ ?empty out

have accepted-bad:

wp-event ?M accepted adversary-initial-state ≤
 wp-event ?M ?bad adversary-initial-state

```

proof (rule wp-event-mono-on-support)
  fix out
  assume support: out  $\in$  set-dist (execute ?M adversary-initial-state)
    and acc: accepted out
  show ?bad out
  proof (cases out)
    case None
    then show ?thesis
      using acc unfolding accepted-def by simp
  next
    case (Some packed)
    then obtain data attacker-state result final-state where out-eq:
      out = Some (((data, attacker-state), result), final-state)
      by (cases packed, auto split: prod.splits)
    let ?s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    have verifier:
      Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)
      using checked-staged-security-experiment-with-data-state-outcomeE
      [OF support[unfolded out-eq]]
      by blast
    have verifier-acc: accepted (Some (result, final-state))
      unfolding accepted-def by simp
    have local-bad:
      soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle
      ?s (Some (result, final-state))  $\vee$ 
      soundness-bad-event-partial-candidate-empty-header ?s
      (Some (result, final-state))
      by (rule
        accepted-partition-soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate-with-partial-merkle-or-
        [OF false-statement verifier verifier-acc])
    then show ?thesis
      unfolding out-eq staged-security-with-data-state-verifier-event-def
      by simp
  qed
qed
have bad-bound:
  wp-event ?M ?bad adversary-initial-state  $\leq$ 
  partial-candidate-error + empty-header-error
proof -
  have wp-event ?M ?bad adversary-initial-state  $\leq$ 
  wp-event ?M ?partial adversary-initial-state +
  wp-event ?M ?empty adversary-initial-state
  by (rule wp-event-union-bound)
  also have ...  $\leq$  partial-candidate-error + empty-header-error
  by (intro add-mono partial-candidate-bound empty-header-bound)
  finally show ?thesis .
qed

```

```

have staged-bound:
  wp-event (checked-staged-security-experiment A) accepted
    adversary-initial-state  $\leq$ 
      partial-candidate-error + empty-header-error
  unfolding checked-staged-security-experiment-acceptance-with-data-state
  by (rule order-trans[OF accepted-bad bad-bound])
show ?thesis
proof -
  have
    wp-event (checked-staged-security-experiment A)
      ( $\lambda$ out.  $\neg$  Option.is-none out) adversary-initial-state =
      wp-event (checked-staged-security-experiment A) accepted
        adversary-initial-state
  unfolding wp-event-def accepted-def
  by (rule arg-cong[where
    f= $\lambda$ Q. wp (checked-staged-security-experiment A) Q
    adversary-initial-state])
    (rule ext, simp)
  then show ?thesis
    unfolding checked-staged-adversary-acceptance-probability-def
    using staged-bound by simp
qed
qed

```

lemma checked-staged-soundness-from-verifier-tied-aligned-transcript-partial-candidate-components-and-empty-head

fixes merkle-error composition-error trace-fri-error'
 composition-fri-error' query-error empty-header-error :: prob

assumes false-statement: $\neg$ exists-valid-trace

and merkle-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 partial-merkle-inconsistency-bad)
 adversary-initial-state $\leq$ merkle-error

and comp-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-bad-with-partial-candidates)
 adversary-initial-state $\leq$ composition-error

and trace-fri-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-partial-candidate)
 adversary-initial-state $\leq$ trace-fri-error'

and comp-fri-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state $\leq$ composition-fri-error'

and query-bound:

$wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $query\text{-}bad\text{-}with\text{-}aligned\text{-}transcript\text{-}partial\text{-}candidates$)
 $adversary\text{-}initial\text{-}state \leq query\text{-}error$
and $empty\text{-}header\text{-}bound$:
 $wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $soundness\text{-}bad\text{-}event\text{-}partial\text{-}candidate\text{-}empty\text{-}header$)
 $adversary\text{-}initial\text{-}state \leq empty\text{-}header\text{-}error$
shows $checked\text{-}staged\text{-}adversary\text{-}acceptance\text{-}probability$ $A \leq$
 $merkle\text{-}error + composition\text{-}error + trace\text{-}fri\text{-}error' +$
 $composition\text{-}fri\text{-}error' + query\text{-}error + empty\text{-}header\text{-}error$
proof –
have $partial\text{-}candidate\text{-}bound$:
 $wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $soundness\text{-}bad\text{-}event\text{-}verifier\text{-}tied\text{-}aligned\text{-}transcript\text{-}partial\text{-}candidate\text{-}with\text{-}partial\text{-}merkle$)
 $adversary\text{-}initial\text{-}state \leq$
 $merkle\text{-}error + composition\text{-}error + trace\text{-}fri\text{-}error' +$
 $composition\text{-}fri\text{-}error' + query\text{-}error$
by ($rule$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}tied\text{-}aligned\text{-}transcript\text{-}partial\text{-}candidate\text{-}with\text{-}partial\text{-}merkle\text{-}union$
 $[OF\ merkle\text{-}bound\ comp\text{-}bound\ trace\text{-}fri\text{-}bound\ comp\text{-}fri\text{-}bound$
 $query\text{-}bound])$)
have $bound$:
 $checked\text{-}staged\text{-}adversary\text{-}acceptance\text{-}probability$ $A \leq$
 $(merkle\text{-}error + composition\text{-}error + trace\text{-}fri\text{-}error' +$
 $composition\text{-}fri\text{-}error' + query\text{-}error) + empty\text{-}header\text{-}error$
by ($rule$
 $checked\text{-}staged\text{-}soundness\text{-}from\text{-}verifier\text{-}tied\text{-}aligned\text{-}transcript\text{-}partial\text{-}candidate\text{-}and\text{-}empty\text{-}header\text{-}bounds$
 $[OF\ false\text{-}statement\ partial\text{-}candidate\text{-}bound\ empty\text{-}header\text{-}bound])$)
then show $?thesis$
by ($simp\ add$: $algebra\text{-}simps$)
qed

lemma $trace\text{-}fri\text{-}bad\text{-}with\text{-}verifier\text{-}tied\text{-}partial\text{-}candidateE$:
assumes
 $trace\text{-}fri\text{-}bad\text{-}with\text{-}verifier\text{-}tied\text{-}partial\text{-}candidate$ $s\ out$
obtains $trace\text{-}roots\ trace\text{-}bs\ trace\text{-}final\ dg\ composition\text{-}roots$
 $composition\text{-}bs\ composition\text{-}final\ fri\text{-}query\text{-}idxs\ trace\text{-}round\text{-}layers$
 $composition\text{-}round\text{-}layers\ fr\ trace\text{-}openings\ trace\text{-}table$
where
 $accepted\text{-}fri\text{-}opening\text{-}transcript$ $s\ out\ trace\text{-}roots\ trace\text{-}bs$
 $trace\text{-}final\ dg\ composition\text{-}roots\ composition\text{-}bs\ composition\text{-}final$
 $fri\text{-}query\text{-}idxs\ trace\text{-}round\text{-}layers\ composition\text{-}round\text{-}layers$
 $accepted\text{-}with\text{-}partial\text{-}trace\text{-}openings$ $s\ out\ fr\ fri\text{-}query\text{-}idxs$
 $trace\text{-}openings$
 $partial\text{-}trace\text{-}table\text{-}candidate\ trace\text{-}table\ trace\text{-}openings$
 $\neg\ trace\text{-}table\text{-}low\text{-}degree\ trace\text{-}table$

```

using assms
unfolding trace-fri-bad-with-verifier-tied-partial-candidate-def
by blast

lemma composition-fri-bad-with-verifier-tied-partial-candidateE:
  assumes
    composition-fri-bad-with-verifier-tied-partial-candidate s out
  obtains trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings composition-openings
    trace-table composition-table
  where
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final fri-query-idxs
    trace-openings composition-openings
    partial-trace-table-candidate trace-table trace-openings
    partial-composition-table-candidate composition-table
    composition-openings
    trace-table-low-degree trace-table
     $\neg$  composition-table-low-degree maxDegree composition-table
proof –
  obtain trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings composition-openings
    trace-table composition-table
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
  and aligned:
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final fri-query-idxs
    trace-openings composition-openings
  and trace-cand:
    partial-trace-table-candidate trace-table trace-openings
  and comp-cand:
    partial-composition-table-candidate composition-table
    composition-openings
  and trace-low:
    trace-table-low-degree trace-table
  and comp-not-low:
     $\neg$  composition-table-low-degree maxDegree composition-table
using assms
unfolding composition-fri-bad-with-verifier-tied-partial-candidate-def

```

by metis
 show ?thesis
 by (rule that[*OF fri-openings aligned trace-cand comp-cand trace-low comp-not-low*])
 qed

lemma *trace-fri-bad-with-verifier-tied-imp-reachable-partial-candidate:*
assumes
 trace-fri-bad-with-verifier-tied-partial-candidate s out
shows *trace-fri-bad-with-reachable-partial-candidate s out*
proof (rule *trace-fri-bad-with-verifier-tied-partial-candidateE[OF assms]*)
fix *trace-roots trace-bs trace-final dg composition-roots*
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr trace-openings trace-table
assume *partial:*
 accepted-with-partial-trace-openings s out fr fri-query-idxs
 trace-openings
and *cand:*
 partial-trace-table-candidate trace-table trace-openings
and *not-low:* $\neg$ *trace-table-low-degree trace-table*
show ?thesis
unfolding *trace-fri-bad-with-reachable-partial-candidate-def*
by (intro *exI conjI*; (rule *partial* | rule *cand* | rule *not-low*))
 qed

lemma *composition-fri-bad-with-verifier-tied-imp-reachable-partial-candidate:*
assumes
 composition-fri-bad-with-verifier-tied-partial-candidate s out
shows *composition-fri-bad-with-reachable-partial-candidate s out*
proof (rule
 composition-fri-bad-with-verifier-tied-partial-candidateE[OF assms])
fix *trace-roots trace-bs trace-final fri-dg composition-roots*
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr as trace-openings composition-openings
 trace-table composition-table
assume *aligned:*
 accepted-with-partial-initial-openings-aligned s out fr trace-roots
 trace-final as fri-dg composition-roots composition-final fri-query-idxs
 trace-openings composition-openings
and *trace-cand:*
 partial-trace-table-candidate trace-table trace-openings
and *comp-cand:*
 partial-composition-table-candidate composition-table
 composition-openings
and *trace-low:* *trace-table-low-degree trace-table*
and *comp-not-low:*
 $\neg$ *composition-table-low-degree maxDegree composition-table*
have *unaligned:*
 accepted-with-partial-initial-openings s out fr trace-roots

```

    trace-final as fri-dg composition-roots composition-final
    fri-query-idxs trace-openings fri-query-idxs composition-openings
  by (rule accepted-with-partial-initial-openings-aligned-imp-unaligned
      [OF aligned])
  show ?thesis
  unfolding composition-fri-bad-with-reachable-partial-candidate-def
  by (intro exI conjI; (rule unaligned | rule trace-cand |
      rule comp-cand | rule trace-low | rule comp-not-low))
qed

lemma trace-fri-bad-with-verifier-tied-partial-candidate-None[simp]:
  ¬ trace-fri-bad-with-verifier-tied-partial-candidate s None
  unfolding trace-fri-bad-with-verifier-tied-partial-candidate-def
    accepted-fri-opening-transcript-def
  by simp

lemma composition-fri-bad-with-verifier-tied-partial-candidate-None[simp]:
  ¬ composition-fri-bad-with-verifier-tied-partial-candidate s None
  unfolding composition-fri-bad-with-verifier-tied-partial-candidate-def
    accepted-fri-opening-transcript-def
  by simp

lemma trace-fri-verifier-tied-imp-sampled-or-missing-layer-chain:
  assumes trace-fri-bad-with-verifier-tied-partial-candidate s out
  shows
    trace-fri-bad-with-verifier-tied-sampled-layer-chain s out ∨
    trace-fri-verifier-tied-missing-sampled-layer-chain s out
  using assms
  unfolding trace-fri-verifier-tied-missing-sampled-layer-chain-def
  by blast

lemma composition-fri-verifier-tied-imp-sampled-or-missing-layer-chain:
  assumes composition-fri-bad-with-verifier-tied-partial-candidate s out
  shows
    composition-fri-bad-with-verifier-tied-sampled-layer-chain s out ∨
    composition-fri-verifier-tied-missing-sampled-layer-chain s out
  using assms
  unfolding composition-fri-verifier-tied-missing-sampled-layer-chain-def
  by blast

lemma wp-trace-fri-verifier-tied-bound-from-sampled-and-missing:
  assumes sampled-bound:
    wp-event verify-monad
      (trace-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R
  and missing-bound:
    wp-event verify-monad
      (trace-fri-verifier-tied-missing-sampled-layer-chain s) s ≤ M
  shows
    wp-event verify-monad

```

$(\text{trace-fri-bad-with-verifier-tied-partial-candidate } s) \ s \leq R + M$
proof –
have *wp-event verify-monad*
 $(\text{trace-fri-bad-with-verifier-tied-partial-candidate } s) \ s \leq$
wp-event verify-monad
 $(\lambda \text{out. trace-fri-bad-with-verifier-tied-sampled-layer-chain } s \ \text{out} \ \vee$
 $\text{trace-fri-verifier-tied-missing-sampled-layer-chain } s \ \text{out}) \ s$
by (rule *wp-event-mono*)
(rule *trace-fri-verifier-tied-imp-sampled-or-missing-layer-chain*)
also have $\dots \leq$
wp-event verify-monad
 $(\text{trace-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s +$
wp-event verify-monad
 $(\text{trace-fri-verifier-tied-missing-sampled-layer-chain } s) \ s$
by (rule *wp-event-union-bound*)
also have $\dots \leq R + M$
by (rule *add-mono[OF sampled-bound missing-bound]*)
finally show *?thesis* .
qed

lemma *wp-composition-fri-verifier-tied-bound-from-sampled-and-missing:*

assumes *sampled-bound:*

wp-event verify-monad
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s \leq R$

and *missing-bound:*

wp-event verify-monad
 $(\text{composition-fri-verifier-tied-missing-sampled-layer-chain } s) \ s \leq M$

shows

wp-event verify-monad
 $(\text{composition-fri-bad-with-verifier-tied-partial-candidate } s) \ s \leq$
 $R + M$

proof –

have *wp-event verify-monad*

$(\text{composition-fri-bad-with-verifier-tied-partial-candidate } s) \ s \leq$

wp-event verify-monad

$(\lambda \text{out.}$
 $\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain } s \ \text{out} \ \vee$
 $\text{composition-fri-verifier-tied-missing-sampled-layer-chain } s \ \text{out}) \ s$

by (rule *wp-event-mono*)

(rule *composition-fri-verifier-tied-imp-sampled-or-missing-layer-chain*)

also have $\dots \leq$

wp-event verify-monad

$(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s +$

wp-event verify-monad

$(\text{composition-fri-verifier-tied-missing-sampled-layer-chain } s) \ s$

by (rule *wp-event-union-bound*)

also have $\dots \leq R + M$

by (rule *add-mono[OF sampled-bound missing-bound]*)

finally show *?thesis* .

qed

definition *trace-fri-verifier-tied-sampled-layer-assignment-obstruction*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-verifier-tied-sampled-layer-assignment-obstruction s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr trace-openings trace-table.}$
 $\text{ accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final dg composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{ accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{ trace-table-low-degree trace-table} \wedge$
 $\text{ generic-fri-sampled-layer-assignment-obstruction}$
 $\text{ trace-table-low-degree trace-table (clength - 1) trace-roots}$
 $\text{ trace-bs trace-final fri-query-idxs trace-round-layers})$

definition *trace-fri-verifier-tied-sampled-assignment-conflict*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-verifier-tied-sampled-assignment-conflict s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr trace-openings trace-table.}$
 $\text{ accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final dg composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{ accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{ trace-table-low-degree trace-table} \wedge$
 $\text{ generic-fri-sampled-assignment-conflict trace-table trace-roots}$
 $\text{ trace-bs trace-final fri-query-idxs trace-round-layers})$

definition *trace-fri-verifier-tied-zero-round-final-obstruction*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-verifier-tied-zero-round-final-obstruction s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$

composition-round-layers fr trace-openings trace-table.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ *trace-table-low-degree trace-table $\wedge$*
length trace-bs = 0 $\wedge$
 $\neg$ *fri-final-constant-consistent trace-table trace-final*)

definition *composition-fri-verifier-tied-sampled-layer-assignment-obstruction*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-sampled-layer-assignment-obstruction s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table.}$
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table $\wedge$*
generic-fri-sampled-layer-assignment-obstruction
 $(\text{composition-table-low-degree } (\text{to-nat fri-dg})) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs composition-round-layers})$

definition *composition-fri-verifier-tied-sampled-assignment-conflict*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-sampled-assignment-conflict s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table.}$
accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
generic-fri-sampled-assignment-conflict composition-table
composition-roots composition-bs composition-final fri-query-idxs
composition-round-layers)

definition *composition-fri-verifier-tied-zero-round-final-obstruction*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-zero-round-final-obstruction s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table.}$
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
 $\text{length composition-bs} = 0$ $\wedge$
 $\neg$ *fri-final-constant-consistent composition-table*
composition-final)

lemma *trace-fri-header-tied-zero-round-final-obstruction-imp-verifier-tied:*

assumes *trace-fri-header-tied-zero-round-final-obstruction s out*

shows *trace-fri-verifier-tied-zero-round-final-obstruction s out*

using *assms*

unfolding *trace-fri-header-tied-zero-round-final-obstruction-def*

trace-fri-verifier-tied-zero-round-final-obstruction-def

by *blast*

lemma *composition-fri-verifier-tied-zero-round-final-obstruction-false:*
 $\neg$ *composition-fri-verifier-tied-zero-round-final-obstruction s out*
proof
assume *obstruction:*
composition-fri-verifier-tied-zero-round-final-obstruction s out
then obtain *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table
where *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *aligned:*
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings
and *zero: length composition-bs = 0*
unfolding *composition-fri-verifier-tied-zero-round-final-obstruction-def*
by *blast*
have *nonempty: composition-roots $\neq$ []*
by (*rule accepted-with-partial-initial-openings-aligned-shapes(1)*
[OF aligned])
have *composition-roots = []*
using *accepted-fri-opening-transcript-shapes(4)[OF fri-openings] zero*
by *simp*
then show *False*
using *nonempty by simp*
qed

lemma *wp-composition-fri-verifier-tied-zero-round-final-obstruction-zero:*
wp-event verify-monad
(composition-fri-verifier-tied-zero-round-final-obstruction s) s = 0
unfolding *wp-event-def wp-def dist-expect-def*
by (*simp add:*
composition-fri-verifier-tied-zero-round-final-obstruction-false)

lemma *trace-fri-verifier-tied-assignment-obstruction-imp-conflict-or-zero:*
assumes *obstruction:*
trace-fri-verifier-tied-sampled-layer-assignment-obstruction s out
shows
trace-fri-verifier-tied-sampled-assignment-conflict s out $\vee$
trace-fri-verifier-tied-zero-round-final-obstruction s out
proof –
obtain *trace-roots trace-bs' trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings trace-table
where *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs'
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *cand: partial-trace-table-candidate trace-table trace-openings*
and *not-low: $\neg$ trace-table-low-degree trace-table*
and *generic:*
generic-fri-sampled-layer-assignment-obstruction
trace-table-low-degree trace-table (clength - 1) trace-roots
trace-bs' trace-final fri-query-idxs trace-round-layers
using *obstruction*
unfolding *trace-fri-verifier-tied-sampled-layer-assignment-obstruction-def*
by *blast*
obtain *N where eval-power: clength * scale = 2^N*
using *eval-domain-length-power* **by** *blast*
have *trace-rounds-le: length trace-bs' $\leq N$*
proof –
have *len: length trace-bs' = ceil-log clength*
using *accepted-fri-opening-transcript-shapes(1,2)[OF fri-openings]*
by *simp*
have *clength $\leq$ clength * scale*
using *scale-pos* **by** *simp*
also have *... = 2^N*
by *(rule eval-power)*
finally have *ceil-log clength $\leq N$*
by *(rule ceil-log-le-power)*
then show *?thesis*
using *len* **by** *simp*
qed
have *query-bounds':*
 $\bigwedge \text{round-idx. round-idx} < \text{length fri-query-idxs} \implies$
 $\text{fri-query-idxs} ! \text{round-idx} < \text{clength} * \text{scale}$
proof –
fix *round-idx*
assume *round-bound: round-idx < length fri-query-idxs*
have *fri-query-idxs ! round-idx $\in$ set fri-query-idxs*
using *round-bound* **by** *simp*
then show *fri-query-idxs ! round-idx < clength * scale*
by *(rule accepted-fri-opening-transcript-query-idx-bound*
[OF fri-openings])
qed
have *split:*
generic-fri-sampled-assignment-conflict trace-table trace-roots
trace-bs' trace-final fri-query-idxs trace-round-layers $\vee$
length trace-bs' = 0 $\wedge$
 $\neg$ *fri-final-constant-consistent trace-table trace-final*
by *(rule*

```

    generic-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final
    [OF generic query-bounds' eval-power trace-rounds-le])
  then show ?thesis
proof
  assume conflict:
    generic-fri-sampled-assignment-conflict trace-table trace-roots
    trace-bs' trace-final fri-query-idxs trace-round-layers
  then show ?thesis
    unfolding trace-fri-verifier-tied-sampled-assignment-conflict-def
    by (intro disjI1 exI conjI)
      (rule fri-openings, rule partial, rule cand, rule not-low,
       rule conflict)
next
  assume zero:
    length trace-bs' = 0 ∧
    ¬ fri-final-constant-consistent trace-table trace-final
  then show ?thesis
    unfolding trace-fri-verifier-tied-zero-round-final-obstruction-def
    by (intro disjI2 exI conjI)
      (rule fri-openings, rule partial, rule cand, rule not-low,
       simp-all)
qed
qed

lemma composition-fri-verifier-tied-assignment-obstruction-imp-conflict-or-zero-on-support:
  assumes support: out ∈ set-dist (execute verify-monad s)
  and obstruction:
    composition-fri-verifier-tied-sampled-layer-assignment-obstruction
    s out
  shows
    composition-fri-verifier-tied-sampled-assignment-conflict s out ∨
    composition-fri-verifier-tied-zero-round-final-obstruction s out
proof -
  obtain trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings composition-openings
    trace-table composition-table
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
  and aligned:
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final fri-query-idxs
    trace-openings composition-openings
  and trace-cand:
    partial-trace-table-candidate trace-table trace-openings
  and comp-cand:

```

```

    partial-composition-table-candidate composition-table
    composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-not-low:
  ¬ composition-table-low-degree maxDegree composition-table
and generic:
  generic-fri-sampled-layer-assignment-obstruction
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) composition-roots composition-bs
  composition-final fri-query-idxs composition-round-layers
using obstruction
unfolding
  composition-fri-verifier-tied-sampled-layer-assignment-obstruction-def
by metis
obtain  $N$  where eval-power:  $\text{clength} * \text{scale} = 2^N$ 
  using eval-domain-length-power by blast
obtain result final-state where out-eq:  $\text{out} = \text{Some}(\text{result}, \text{final-state})$ 
  using fri-openings unfolding accepted-fri-opening-transcript-def by blast
have outcome:
   $\text{Some}(\text{result}, \text{final-state}) \in \text{set-dist}(\text{execute verify-monad } s)$ 
  using support out-eq by simp
have challenges:
  accepted-fri-challenges  $s(\text{Some}(\text{result}, \text{final-state}))$ 
  trace-bs fri-dg composition-bs
  using accepted-fri-opening-transcript-challenges[OF fri-openings]
  out-eq by simp
have degree-bound:  $\text{to-nat fri-dg} \leq \text{maxDegree}$ 
  by (rule verify-monad-accepted-fri-challenges-degree-bound
    [OF outcome challenges])
have composition-rounds-le:  $\text{length composition-bs} \leq N$ 
proof –
  have len:  $\text{length composition-bs} = \text{ceil-log}(\text{to-nat fri-dg} + 1)$ 
    using accepted-fri-opening-transcript-shapes(3,4)[OF fri-openings]
    by simp
  have  $\text{to-nat fri-dg} + 1 \leq \text{clength} * \text{scale}$ 
    using degree-bound maxDegree-less-eval-domain by linarith
  also have  $\dots = 2^N$ 
    by (rule eval-power)
  finally have  $\text{ceil-log}(\text{to-nat fri-dg} + 1) \leq N$ 
    by (rule ceil-log-le-power)
  then show ?thesis
    using len by simp
qed
have query-bounds:
   $\bigwedge \text{round-idx}. \text{round-idx} < \text{length fri-query-idxs} \implies$ 
   $\text{fri-query-idxs} ! \text{round-idx} < \text{clength} * \text{scale}$ 
proof –
  fix round-idx
  assume round-bound:  $\text{round-idx} < \text{length fri-query-idxs}$ 

```

```

have fri-query-idxs ! round-idx ∈ set fri-query-idxs
  using round-bound by simp
then show fri-query-idxs ! round-idx < clength * scale
  by (rule accepted-fri-opening-transcript-query-idx-bound
    [OF fri-openings])
qed
have split:
  generic-fri-sampled-assignment-conflict composition-table
    composition-roots composition-bs composition-final fri-query-idxs
    composition-round-layers  $\vee$ 
    length composition-bs = 0  $\wedge$ 
     $\neg$  fri-final-constant-consistent composition-table
      composition-final
  by (rule
    generic-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final
    [OF generic query-bounds eval-power composition-rounds-le])
then show ?thesis
proof
  assume conflict:
    generic-fri-sampled-assignment-conflict composition-table
      composition-roots composition-bs composition-final fri-query-idxs
      composition-round-layers
  then show ?thesis
    unfolding
      composition-fri-verifier-tied-sampled-assignment-conflict-def
    by (intro disjI1 exI conjI)
      (rule fri-openings, rule aligned, rule trace-cand, rule comp-cand,
        rule trace-low, rule comp-not-low, rule conflict)
next
  assume zero:
    length composition-bs = 0  $\wedge$ 
     $\neg$  fri-final-constant-consistent composition-table
      composition-final
  then show ?thesis
    unfolding
      composition-fri-verifier-tied-zero-round-final-obstruction-def
    by (intro disjI2 exI conjI)
      (rule fri-openings, rule aligned, rule trace-cand, rule comp-cand,
        rule trace-low, rule comp-not-low, simp-all)
qed
qed

lemma trace-fri-verifier-tied-missing-sampled-imp-assignment-obstruction:
  assumes missing:
    trace-fri-verifier-tied-missing-sampled-layer-chain s out
  shows
    trace-fri-verifier-tied-sampled-layer-assignment-obstruction s out
proof —
  have bad:

```



```

trace-fri-bad-with-verifier-tied-partial-candidate s out
and not-sampled:
   $\neg$  trace-fri-bad-with-verifier-tied-sampled-layer-chain s out
using missing
unfolding trace-fri-verifier-tied-missing-sampled-layer-chain-def
by blast+
from trace-fri-bad-with-verifier-tied-partial-candidateE[OF bad]
obtain trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr trace-openings trace-table
where fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final dg composition-roots composition-bs composition-final
  fri-query-idxs trace-round-layers composition-round-layers
and partial:
  accepted-with-partial-trace-openings s out fr fri-query-idxs
  trace-openings
and cand: partial-trace-table-candidate trace-table trace-openings
and not-low:  $\neg$  trace-table-low-degree trace-table
by blast
have partial-generic:
  generic-fri-partial-evidence trace-table-low-degree trace-table
  (clength - 1) trace-roots trace-bs trace-final fri-query-idxs
  trace-round-layers
by (rule accepted-fri-opening-transcript-trace-generic-partial-evidence
  [OF fri-openings])
have no-chain:
   $\bigwedge$  doms layers. \neg
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
  trace-table (clength - 1) trace-roots trace-bs trace-final
  fri-query-idxs trace-round-layers doms layers
proof
  fix doms layers
  assume chain:
    generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
  have trace-fri-bad-with-verifier-tied-sampled-layer-chain s out
  unfolding trace-fri-bad-with-verifier-tied-sampled-layer-chain-def
  by (intro exI conjI)
    (rule fri-openings, rule partial, rule cand, rule not-low,
    rule chain)
  then show False
  using not-sampled by contradiction
qed
have obstruction:
  generic-fri-sampled-layer-assignment-obstruction
  trace-table-low-degree trace-table (clength - 1) trace-roots
  trace-bs trace-final fri-query-idxs trace-round-layers

```

```

unfolding generic-fri-sampled-layer-assignment-obstruction-def
using partial-generic no-chain by blast
show ?thesis
unfolding trace-fri-verifier-tied-sampled-layer-assignment-obstruction-def
by (intro exI conjI)
  (rule fri-openings, rule partial, rule cand, rule not-low,
   rule obstruction)
qed

lemma composition-fri-verifier-tied-missing-sampled-imp-assignment-obstruction:
assumes missing:
  composition-fri-verifier-tied-missing-sampled-layer-chain s out
shows
  composition-fri-verifier-tied-sampled-layer-assignment-obstruction s out
proof –
have bad:
  composition-fri-bad-with-verifier-tied-partial-candidate s out
and not-sampled:
  ¬ composition-fri-bad-with-verifier-tied-sampled-layer-chain s out
using missing
unfolding composition-fri-verifier-tied-missing-sampled-layer-chain-def
by blast+
from composition-fri-bad-with-verifier-tied-partial-candidateE[OF bad]
obtain trace-roots trace-bs trace-final fri-dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr as trace-openings composition-openings
  trace-table composition-table
where fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final fri-dg composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers
and aligned:
  accepted-with-partial-initial-openings-aligned s out fr trace-roots
  trace-final as fri-dg composition-roots composition-final fri-query-idxs
  trace-openings composition-openings
and trace-cand:
  partial-trace-table-candidate trace-table trace-openings
and comp-cand:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-not-low:
  ¬ composition-table-low-degree maxDegree composition-table
by metis
have partial-generic:
  generic-fri-partial-evidence
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) composition-roots composition-bs composition-final

```

```

    fri-query-idxs composition-round-layers
  by (rule
    accepted-fri-opening-transcript-composition-generic-partial-evidence
    [OF fri-openings])
  have no-chain:
     $\bigwedge \text{doms layers. } \neg$ 
    generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers doms
    layers
  proof
    fix doms layers
    assume chain:
      generic-fri-sampled-layer-chain-evidence
      (composition-table-low-degree (to-nat fri-dg)) composition-table
      (to-nat fri-dg) composition-roots composition-bs
      composition-final fri-query-idxs composition-round-layers doms
      layers
    have composition-fri-bad-with-verifier-tied-sampled-layer-chain s out
    unfolding composition-fri-bad-with-verifier-tied-sampled-layer-chain-def
    by (intro exI conjI)
      (rule fri-openings, rule aligned, rule trace-cand, rule comp-cand,
        rule trace-low, rule comp-not-low, rule chain)
    then show False
    using not-sampled by contradiction
  qed
  have obstruction:
    generic-fri-sampled-layer-assignment-obstruction
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) composition-roots composition-bs composition-final
    fri-query-idxs composition-round-layers
    unfolding generic-fri-sampled-layer-assignment-obstruction-def
    using partial-generic no-chain by blast
  show ?thesis
  unfolding
    composition-fri-verifier-tied-sampled-layer-assignment-obstruction-def
  by (intro exI conjI)
    (rule fri-openings, rule aligned, rule trace-cand, rule comp-cand,
      rule trace-low, rule comp-not-low, rule obstruction)
qed

lemma wp-trace-fri-verifier-tied-missing-sampled-bound-from-assignment-obstruction:
  assumes obstruction-bound:
    wp-event verify-monad
    (trace-fri-verifier-tied-sampled-layer-assignment-obstruction s) s
     $\leq M$ 
  shows
    wp-event verify-monad

```

$(\text{trace-fri-verifier-tied-missing-sampled-layer-chain } s) \ s \leq M$
proof –
have $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-missing-sampled-layer-chain } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-sampled-layer-assignment-obstruction } s) \ s$
by $(\text{rule wp-event-mono})$
 $(\text{rule trace-fri-verifier-tied-missing-sampled-imp-assignment-obstruction})$
also have $\dots \leq M$
by $(\text{rule obstruction-bound})$
finally show $?thesis$.
qed

lemma $\text{wp-composition-fri-verifier-tied-missing-sampled-bound-from-assignment-obstruction}$:
assumes obstruction-bound :
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-layer-assignment-obstruction } s) \ s$
 $\leq M$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-missing-sampled-layer-chain } s) \ s \leq M$
proof –
have $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-missing-sampled-layer-chain } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-layer-assignment-obstruction } s) \ s$
by $(\text{rule wp-event-mono})$
 $(\text{rule}$
 $\text{composition-fri-verifier-tied-missing-sampled-imp-assignment-obstruction})$
also have $\dots \leq M$
by $(\text{rule obstruction-bound})$
finally show $?thesis$.
qed

lemma $\text{wp-trace-fri-verifier-tied-bound-from-sampled-and-assignment-obstruction}$:
assumes sampler-bound :
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s \leq R$
and obstruction-bound :
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-sampled-layer-assignment-obstruction } s) \ s$
 $\leq M$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-verifier-tied-partial-candidate } s) \ s \leq R + M$
by $(\text{rule wp-trace-fri-verifier-tied-bound-from-sampled-and-missing}$
 $[\text{OF sampler-bound}$
 $\text{wp-trace-fri-verifier-tied-missing-sampled-bound-from-assignment-obstruction}$
 $[\text{OF obstruction-bound}]])$

lemma *wp-composition-fri-verifier-tied-bound-from-sampled-and-assignment-obstruction:*
assumes *sampled-bound:*
 wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R
and *obstruction-bound:*
 wp-event verify-monad
 (composition-fri-verifier-tied-sampled-layer-assignment-obstruction s) s
 ≤ M
shows
 wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-partial-candidate s) s ≤
 R + M
by (*rule wp-composition-fri-verifier-tied-bound-from-sampled-and-missing*
 [*OF sampled-bound*
 wp-composition-fri-verifier-tied-missing-sampled-bound-from-assignment-obstruction
 [*OF obstruction-bound*]])

lemma *trace-fri-bad-with-verifier-tied-partial-candidates-staged-bound:*
assumes *verifier-bound:*
 $\bigwedge \text{data attacker-state.}$
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-bad-with-verifier-tied-partial-candidate
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ C
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state ≤ C
by (*rule checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
 (*simp-all add: verifier-bound*)

lemma *composition-fri-bad-with-verifier-tied-partial-candidates-staged-bound:*
assumes *verifier-bound:*
 $\bigwedge \text{data attacker-state.}$
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-partial-candidate
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state

$(\text{staged-proof-transcript data}) \leq C$
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq C$
by $(\text{rule checked-staged-security-with-data-state-verifier-event-bound-from-cont})$
 $(\text{simp-all add: verifier-bound})$

lemma $\text{wp-trace-fri-verifier-tied-assignment-obstruction-bound-from-conflict-and-zero:}$
assumes conflict-bound:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-sampled-assignment-conflict } s) s \leq C$
and zero-bound:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-zero-round-final-obstruction } s) s \leq Z$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-sampled-layer-assignment-obstruction } s) s$
 $\leq C + Z$
proof –
have $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-sampled-layer-assignment-obstruction } s) s$
 $\leq \text{wp-event verify-monad}$
 $(\lambda \text{out. trace-fri-verifier-tied-sampled-assignment-conflict } s \text{ out} \vee$
 $\text{trace-fri-verifier-tied-zero-round-final-obstruction } s \text{ out}) s$
by $(\text{rule wp-event-mono})$
 $(\text{rule trace-fri-verifier-tied-assignment-obstruction-imp-conflict-or-zero})$
also have $\dots \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-sampled-assignment-conflict } s) s +$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-zero-round-final-obstruction } s) s$
by $(\text{rule wp-event-union-bound})$
also have $\dots \leq C + Z$
by $(\text{rule add-mono}[OF \text{ conflict-bound zero-bound}])$
finally show $?thesis$.
qed

lemma $\text{wp-composition-fri-verifier-tied-assignment-obstruction-bound-from-conflict-and-zero:}$
assumes conflict-bound:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-assignment-conflict } s) s \leq C$
and zero-bound:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-zero-round-final-obstruction } s) s \leq Z$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-layer-assignment-obstruction } s) s$

$\leq C + Z$
proof –
have *wp-event verify-monad*
 $(\text{composition-fri-verifier-tied-sampled-layer-assignment-obstruction } s) \ s$
 $\leq \text{wp-event verify-monad}$
 $(\lambda \text{out. composition-fri-verifier-tied-sampled-assignment-conflict } s \ \text{out} \ \vee$
 $\text{composition-fri-verifier-tied-zero-round-final-obstruction } s \ \text{out}) \ s$
by (rule *wp-event-mono-on-support*)
(rule
 $\text{composition-fri-verifier-tied-assignment-obstruction-imp-conflict-or-zero-on-support}$)
also have ... $\leq$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-assignment-conflict } s) \ s +$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-zero-round-final-obstruction } s) \ s$
by (rule *wp-event-union-bound*)
also have ... $\leq C + Z$
by (rule *add-mono[OF conflict-bound zero-bound]*)
finally show ?thesis .
qed

lemma *wp-composition-fri-verifier-tied-assignment-obstruction-bound-from-conflict:*

assumes *conflict-bound:*

wp-event verify-monad

$(\text{composition-fri-verifier-tied-sampled-assignment-conflict } s) \ s \leq C$

shows

wp-event verify-monad

$(\text{composition-fri-verifier-tied-sampled-layer-assignment-obstruction } s) \ s$

$\leq C$

proof –

have *zero-bound:*

wp-event verify-monad

$(\text{composition-fri-verifier-tied-zero-round-final-obstruction } s) \ s \leq 0$

by (subst *wp-composition-fri-verifier-tied-zero-round-final-obstruction-zero*)

simp

have *wp-event verify-monad*

$(\text{composition-fri-verifier-tied-sampled-layer-assignment-obstruction } s) \ s$

$\leq C + 0$

by (rule

wp-composition-fri-verifier-tied-assignment-obstruction-bound-from-conflict-and-zero

[*OF conflict-bound zero-bound*])

then show ?thesis

by *simp*

qed

lemma *wp-trace-fri-verifier-tied-bound-from-sampled-conflict-and-zero:*

assumes *sampled-bound:*

wp-event verify-monad

$(\text{trace-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s \leq R$

and *conflict-bound*:
wp-event verify-monad
 (trace-fri-verifier-tied-sampled-assignment-conflict s) $s \leq C$
and *zero-bound*:
wp-event verify-monad
 (trace-fri-verifier-tied-zero-round-final-obstruction s) $s \leq Z$
shows
wp-event verify-monad
 (trace-fri-bad-with-verifier-tied-partial-candidate s) s
 $\leq R + (C + Z)$
by (rule
wp-trace-fri-verifier-tied-bound-from-sampled-and-assignment-obstruction
 [OF *sampled-bound*
wp-trace-fri-verifier-tied-assignment-obstruction-bound-from-conflict-and-zero
 [OF *conflict-bound zero-bound*]])

lemma *wp-composition-fri-verifier-tied-bound-from-sampled-conflict-and-zero*:

assumes *sampled-bound*:
wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-sampled-layer-chain s) $s \leq R$
and *conflict-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict s) $s \leq C$
and *zero-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-zero-round-final-obstruction s) $s \leq Z$
shows
wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-partial-candidate s) s
 $\leq R + (C + Z)$
by (rule
wp-composition-fri-verifier-tied-bound-from-sampled-and-assignment-obstruction
 [OF *sampled-bound*
wp-composition-fri-verifier-tied-assignment-obstruction-bound-from-conflict-and-zero
 [OF *conflict-bound zero-bound*]])

lemma *wp-composition-fri-verifier-tied-bound-from-sampled-conflict*:

assumes *sampled-bound*:
wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-sampled-layer-chain s) $s \leq R$
and *conflict-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict s) $s \leq C$
shows
wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-partial-candidate s) s
 $\leq R + C$
by (rule
wp-composition-fri-verifier-tied-bound-from-sampled-and-assignment-obstruction

[*OF sampled-bound*
wp-composition-fri-verifier-tied-assignment-obstruction-bound-from-conflict
[OF conflict-bound]])

lemma *trace-fri-verifier-tied-reductionD*:
assumes *trace-fri-verifier-tied-reduction s*
shows
wp-event verify-monad
(trace-fri-bad-with-verifier-tied-partial-candidate s) s ≤
trace-fri-error
using *assms unfolding trace-fri-verifier-tied-reduction-def* **by** *simp*

lemma *composition-fri-verifier-tied-reductionD*:
assumes *composition-fri-verifier-tied-reduction s*
shows
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-partial-candidate s) s ≤
composition-fri-error
using *assms unfolding composition-fri-verifier-tied-reduction-def* **by** *simp*

lemma *trace-fri-verifier-tied-reduction-from-sampled-conflict-and-zero*:
assumes *sampled-bound*:
wp-event verify-monad
(trace-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R
and *conflict-bound*:
wp-event verify-monad
(trace-fri-verifier-tied-sampled-assignment-conflict s) s ≤ C
and *zero-bound*:
wp-event verify-monad
(trace-fri-verifier-tied-zero-round-final-obstruction s) s ≤ Z
and *total-bound*: $R + (C + Z) \leq \text{trace-fri-error}$
shows *trace-fri-verifier-tied-reduction s*
unfolding *trace-fri-verifier-tied-reduction-def*
by (*rule order-trans*[
OF wp-trace-fri-verifier-tied-bound-from-sampled-conflict-and-zero
[OF sampled-bound conflict-bound zero-bound] total-bound])

lemma *composition-fri-verifier-tied-reduction-from-sampled-conflict-and-zero*:
assumes *sampled-bound*:
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R
and *conflict-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-sampled-assignment-conflict s) s ≤ C
and *zero-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-zero-round-final-obstruction s) s ≤ Z
and *total-bound*: $R + (C + Z) \leq \text{composition-fri-error}$
shows *composition-fri-verifier-tied-reduction s*

unfolding *composition-fri-verifier-tied-reduction-def*
by (rule order-trans[
OF wp-composition-fri-verifier-tied-bound-from-sampled-conflict-and-zero
[OF sampled-bound conflict-bound zero-bound] total-bound])

lemma *composition-fri-verifier-tied-reduction-from-sampled-conflict:*

assumes *sampled-bound:*

wp-event verify-monad

(*composition-fri-bad-with-verifier-tied-sampled-layer-chain s*) $s \leq R$

and *conflict-bound:*

wp-event verify-monad

(*composition-fri-verifier-tied-sampled-assignment-conflict s*) $s \leq C$

and *total-bound:* $R + C \leq \text{composition-fri-error}$

shows *composition-fri-verifier-tied-reduction s*

unfolding *composition-fri-verifier-tied-reduction-def*

by (rule order-trans[

OF wp-composition-fri-verifier-tied-bound-from-sampled-conflict

[OF sampled-bound conflict-bound] total-bound])

lemma *trace-fri-zero-round-checked-final-obstruction-imp-verifier-tied:*

assumes

trace-fri-zero-round-checked-final-obstruction s out

shows *trace-fri-bad-with-verifier-tied-partial-candidate s out*

proof (rule *trace-fri-zero-round-checked-final-obstruction-not-trace-low-degree*
[OF assms])

fix *trace-roots trace-bs trace-final dg composition-roots composition-bs*

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr trace-openings trace-table

assume *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final dg composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers

and *checked:*

trace-zero-round-final-checked-candidate s out fr fri-query-idxs

trace-openings trace-table trace-final

and *not-low:* $\neg \text{trace-table-low-degree trace-table}$

have *partial:*

accepted-with-partial-trace-openings s out fr fri-query-idxs

trace-openings

using *checked* **unfolding** *trace-zero-round-final-checked-candidate-def*

by *simp*

have *cand:*

partial-trace-table-candidate trace-table trace-openings

using *checked* **unfolding** *trace-zero-round-final-checked-candidate-def*

by *simp*

show *?thesis*

unfolding *trace-fri-bad-with-verifier-tied-partial-candidate-def*

by (*intro exI conjI*)

(rule *fri-openings*, rule *partial*, rule *cand*, rule *not-low*)

qed

lemma *trace-fri-zero-round-checked-final-obstruction-mono-verifier-tied:*
 $wp\text{-}event\ verify\text{-}monad\ (trace\text{-}fri\text{-}zero\text{-}round\text{-}checked\text{-}final\text{-}obstruction\ s)\ s \leq$
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}bad\text{-}with\text{-}verifier\text{-}tied\text{-}partial\text{-}candidate\ s)\ s$
by (rule *wp-event-mono*)
(rule *trace-fri-zero-round-checked-final-obstruction-imp-verifier-tied*)

end

end

theory *Soundness-Relevant-Drift-Query-Agreement*
imports *Soundness-Relevant-Drift-Reachable-FRI*
begin

Fixed trace-pair query-agreement bounds for the empty-header relevant-drift residual. This layer keeps the candidate pair fixed before the query challenge is sampled. It therefore avoids the invalid broad fixed-cover route where partial openings selected after the query determine the target.

context *soundness*
begin

definition *staged-security-with-data-state-trace-pair-agreement-hit-at*
 $:: 'f\ list \Rightarrow 'f\ list \Rightarrow nat \Rightarrow$
 $((('f\ staged\text{-}proof\text{-}data \times 'f\ protocol\text{-}channel) \times unit\ list) \times$
 $'f\ protocol\text{-}channel)\ option \Rightarrow bool$

where

staged-security-with-data-state-trace-pair-agreement-hit-at
 $trace\text{-}table\ trace\text{-}table'\ i\ out \longleftrightarrow$
(case out of
 $None \Rightarrow False$
| $Some\ (((data, attacker\text{-}state), result), final\text{-}state) \Rightarrow$
 $(\exists\ as\ query\text{-}idxs.$
 $accepted\text{-}transcript\text{-}shape$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data))$
 $(Some\ (result, final\text{-}state))\ as\ query\text{-}idxs \wedge$
 $i < rounds \wedge$
 $query\text{-}idxs\ !\ i \in$
 $trace\text{-}table\text{-}agreement\text{-}indices\ trace\text{-}table\ trace\text{-}table'))$

definition *checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at*
 $:: 'f\ list \Rightarrow 'f\ list \Rightarrow nat \Rightarrow$
 $(((((('f \times 'f\ list \times 'f\ list \times 'f) \times$
 $'f\ protocol\text{-}channel) \times 'f\ staged\text{-}proof\text{-}data) \times$
 $'f\ protocol\text{-}channel) \times unit\ list) \times$

$\text{'f protocol-channel) option} \Rightarrow \text{bool}$
where
checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
 $\text{trace-table trace-table' } i \text{ out} \longleftrightarrow$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((\text{alpha-prefix, alpha-prefix-state}), \text{data}), \text{attacker-state}),$
 $\text{result}), \text{final-state}) \Rightarrow$
 $\text{staged-security-with-data-state-trace-pair-agreement-hit-at}$
 $\text{trace-table trace-table' } i$
 $(\text{Some } (((\text{data, attacker-state}), \text{result}), \text{final-state}))))$
definition *checked-staged-security-with-query-prefix-trace-pair-agreement-hit-at*
 $:: \text{'f list} \Rightarrow \text{'f list} \Rightarrow \text{nat} \Rightarrow$
 $(((((\text{'f staged-query-prefix-data} \times \text{'f protocol-channel}) \times$
 $\text{'f}) \times \text{'f protocol-channel}) \times$
 $((\text{'f staged-proof-data} \times \text{'f protocol-channel}) \times \text{unit list})) \times$
 $\text{'f protocol-channel) option}) \Rightarrow \text{bool}$
where
checked-staged-security-with-query-prefix-trace-pair-agreement-hit-at
 $\text{trace-table trace-table' } i \text{ out} \longleftrightarrow$
checked-staged-security-with-query-prefix-dynamic-index-hit
 $(\lambda \text{ -. trace-table-agreement-indices trace-table trace-table'}) \text{ out}$
lemma *checked-staged-query-prefix-trace-pair-agreement-prehit-bound:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: i < rounds*
shows
 $\text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \text{ } i)$
 $(\text{checked-staged-query-prefix-dynamic-index-prehit}$
 $(\lambda \text{ -. trace-table-agreement-indices trace-table trace-table'}))$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-query-search-queries budgets } i + 1)$
proof $(\text{rule checked-staged-query-prefix-dynamic-index-prehit-bound-from-relation-program})$
show $\text{HashMap adversary-initial-state} = \text{fmempty}$
by $(\text{simp add: adversary-initial-state-def})$
show *hash-relation-program*
 $(\text{checked-staged-query-prefix-dynamic-index-prehit-relation } A \text{ } i$
 $\text{adversary-initial-state}$
 $(\lambda \text{ -. trace-table-agreement-indices trace-table trace-table'}))$
 $\text{size } (\text{staged-query-search-queries budgets } i + 1)$
 $(\text{checked-staged-query-prefix-receive-with-state } A \text{ } i)$
by $(\text{rule hash-relation-program-checked-staged-query-prefix-receive-with-state}$
 $[\text{OF wf controlled less-imp-le}[\text{OF i-bound}]])$
 $(\text{rule checked-staged-query-prefix-dynamic-index-prehit-relation-fiber-bound-size})$
qed

lemma *checked-staged-query-prefix-trace-pair-agreement-hit-bound:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i < \text{rounds}$*
and *trace-low: trace-table-low-degree trace-table*
and *trace'-low: trace-table-low-degree trace-table'*
and *distinct: $\text{trace-table} \neq \text{trace-table}'$*
shows

$$\begin{aligned} & \text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (\text{checked-staged-query-prefix-dynamic-index-hit} \\ & \quad \quad (\lambda - . \text{trace-table-agreement-indices trace-table trace-table}')) \\ & \text{adversary-initial-state} \leq \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-query-search-queries budgets } i + 1) + \\ & \quad \text{nnreal clength} / \text{nnreal (card query-sample-space)} \end{aligned}$$

proof –
have *hit:*

$$\begin{aligned} & \text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (\text{checked-staged-query-prefix-dynamic-index-hit} \\ & \quad \quad (\lambda - . \text{trace-table-agreement-indices trace-table trace-table}')) \\ & \text{adversary-initial-state} \leq \\ & \text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (\text{checked-staged-query-prefix-dynamic-index-prehit} \\ & \quad \quad (\lambda - . \text{trace-table-agreement-indices trace-table trace-table}')) \\ & \text{adversary-initial-state} + \\ & \quad \text{nnreal clength} / \text{nnreal (card query-sample-space)} \end{aligned}$$

by (rule *checked-staged-query-prefix-trace-pair-agreement-hit-bound-by-prehit*
 $[\text{OF trace-low trace'-low distinct}]$)
have *pre:*

$$\begin{aligned} & \text{wp-event } (\text{checked-staged-query-prefix-receive-with-state } A \ i) \\ & \quad (\text{checked-staged-query-prefix-dynamic-index-prehit} \\ & \quad \quad (\lambda - . \text{trace-table-agreement-indices trace-table trace-table}')) \\ & \text{adversary-initial-state} \leq \\ & \quad \text{staged-phase-relation-error size} \\ & \quad (\text{staged-query-search-queries budgets } i + 1) \end{aligned}$$

by (rule *checked-staged-query-prefix-trace-pair-agreement-prehit-bound*
 $[\text{OF wf controlled i-bound}]$)
show *?thesis*
by (rule *order-trans* $[\text{OF hit}]$) (intro *add-mono pre order-refl*)
qed

lemma *checked-staged-security-with-query-prefix-trace-pair-agreement-hit-bound:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i < \text{rounds}$*
and *trace-low: trace-table-low-degree trace-table*
and *trace'-low: trace-table-low-degree trace-table'*
and *distinct: $\text{trace-table} \neq \text{trace-table}'$*
shows

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-trace-pair-agreement-hit-at
 trace-table trace-table' i)
adversary-initial-state $\leq$
staged-phase-relation-error size
 (staged-query-search-queries budgets $i + 1$) +
 nnreal clength / nnreal (card query-sample-space)
unfolding checked-staged-security-with-query-prefix-trace-pair-agreement-hit-at-def
by (rule checked-staged-security-with-query-prefix-dynamic-index-hit-bound)
 (rule checked-staged-query-prefix-trace-pair-agreement-hit-bound
 [*OF wf controlled i-bound trace-low trace'-low distinct*])

lemma staged-security-with-data-state-trace-pair-agreement-hit-at-le-query-prefix:
assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets A
and *i-bound*: $i < \text{rounds}$
shows
wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-trace-pair-agreement-hit-at
 trace-table trace-table' i)
adversary-initial-state $\leq$
wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-trace-pair-agreement-hit-at
 trace-table trace-table' i)
adversary-initial-state

proof –
let $?E =$
 staged-security-with-data-state-trace-pair-agreement-hit-at
 trace-table trace-table' i
let $?Q =$
 checked-staged-security-with-query-prefix-trace-pair-agreement-hit-at
 trace-table trace-table' i
have *proj*:
 wp-event (checked-staged-security-experiment-with-data-state A) $?E$
 adversary-initial-state =
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 ($\lambda \text{out. case out of}$
 None $\Rightarrow ?E$ None
 | Some (packed, t) $\Rightarrow ?E$ (Some (snd packed, t)))
 adversary-initial-state
by (rule checked-staged-security-experiment-with-query-prefix-data-state-projection-event
 [*OF i-bound*])
show $?thesis$
unfolding *proj*
proof (rule *wp-event-mono-on-support*)
fix *out*

```

assume support:
  out ∈
    set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state
          A i)
        adversary-initial-state)
  and hit:
    (case out of None ⇒ ?E None
     | Some (packed, t) ⇒ ?E (Some (snd packed, t)))
show ?Q out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding staged-security-with-data-state-trace-pair-agreement-hit-at-def
      checked-staged-security-with-query-prefix-trace-pair-agreement-hit-at-def
      checked-staged-security-with-query-prefix-dynamic-index-hit-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state
    result final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        (((data, attacker-state), result)), final-state)
    by (cases packed, auto split: prod.splits)
  have support':
    Some (((((prefix, prefix-state), raw), raw-state),
      (((data, attacker-state), result)), final-state)
    ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-query-prefix-data-state
            A i)
          adversary-initial-state)
    using support unfolding out-eq .
  have data-support:
    Some (((data, attacker-state), result), final-state) ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
    by (rule
      checked-staged-security-experiment-with-query-prefix-data-state-imp-data-state-support
      [OF i-bound support'])
  have data-hit:
    staged-security-with-data-state-trace-pair-agreement-hit-at
      trace-table trace-table' i
    (Some (((data, attacker-state), result), final-state))

```

```

    using hit unfolding out-eq by simp
  obtain as query-idxs where shape:
    accepted-transcript-shape
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result, final-state)) as query-idxs
  and idx-agree:
    query-idxs ! i ∈
      trace-table-agreement-indices trace-table trace-table'
  using data-hit unfolding
    staged-security-with-data-state-trace-pair-agreement-hit-at-def
  by auto
  from checked-staged-security-with-data-state-accepted-shape-query-keys
    [OF wf controlled data-support shape]
  obtain raw-idxs where len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and lookup-final:
    ∧j. j < rounds ⇒
      fmllookup (HashMap final-state)
        (QueryIndexChallenge j
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) j)) =
        Some (raw-idxs ! j)
  by blast
  have final-lookup:
    fmllookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data) (staged-query-chunks data) i)) =
    Some raw
  by (rule
    checked-staged-security-experiment-with-query-prefix-data-state-raw-final-lookup
    [OF wf controlled i-bound support'])
  have raw-eq: raw-idxs ! i = raw
  using lookup-final[OF i-bound] final-lookup by simp
  have raw-agree:
    index (to-nat raw) ∈
      trace-table-agreement-indices trace-table trace-table'
  using idx-agree query-idxs-eq len-raw i-bound raw-eq by simp
  show ?thesis
  unfolding out-eq
    checked-staged-security-with-query-prefix-trace-pair-agreement-hit-at-def
    checked-staged-security-with-query-prefix-dynamic-index-hit-def
    checked-staged-query-prefix-dynamic-index-hit-def
  using raw-agree by simp
qed
qed

```


qed

lemma *staged-security-with-data-state-trace-pair-agreement-hit-at-bound:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *i-bound: $i < \text{rounds}$*
and *trace-low: trace-table-low-degree trace-table*
and *trace'-low: trace-table-low-degree trace-table'*
and *distinct: $\text{trace-table} \neq \text{trace-table}'$*

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-trace-pair-agreement-hit-at
trace-table trace-table' i)
adversary-initial-state $\leq$
staged-phase-relation-error size
(staged-query-search-queries budgets $i + 1$) +
nnreal clength / nnreal (card query-sample-space))

by (rule order-trans[
OF staged-security-with-data-state-trace-pair-agreement-hit-at-le-query-prefix
checked-staged-security-with-query-prefix-trace-pair-agreement-hit-bound])
 (use assms in simp-all)

lemma *checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at-projection:*

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
trace-table trace-table' i)
adversary-initial-state =
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-trace-pair-agreement-hit-at
trace-table trace-table' i)
adversary-initial-state

proof –

let *?project =*
 $\lambda x. ((\text{snd } (\text{fst } (\text{fst } x)), \text{snd } (\text{fst } x)), \text{snd } x)$

have

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-trace-pair-agreement-hit-at
trace-table trace-table' i)
adversary-initial-state =
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
 $\gg (\lambda x. \text{return } (?project\ x))$
(staged-security-with-data-state-trace-pair-agreement-hit-at
trace-table trace-table' i)
adversary-initial-state

using

checked-staged-security-experiment-with-actual-alpha-prefix-data-state-projection
[of A]

```

    by simp
  also have ... =
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
        trace-table trace-table' i)
      adversary-initial-state
  proof -
    have pred-eq:
      (λout. case out of
        None ⇒
          staged-security-with-data-state-trace-pair-agreement-hit-at
            trace-table trace-table' i None
        | Some (x, t) ⇒
          staged-security-with-data-state-trace-pair-agreement-hit-at
            trace-table trace-table' i (Some (?project x, t))) =
        checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
          trace-table trace-table' i
      by (rule ext)
      (auto simp:
        staged-security-with-data-state-trace-pair-agreement-hit-at-def
        checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at-def
        split: option.splits prod.splits)
    show ?thesis
      by (subst wp-event-bind-return-map) (simp add: pred-eq)
  qed
  finally show ?thesis
    by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound: i < rounds
  and trace-low: trace-table-low-degree trace-table
  and trace'-low: trace-table-low-degree trace-table'
  and distinct: trace-table ≠ trace-table'
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      (checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
        trace-table trace-table' i)
      adversary-initial-state ≤
      staged-phase-relation-error size
        (staged-query-search-queries budgets i + 1) +
        nnreal clength / nnreal (card query-sample-space)
  unfolding
    checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at-projection
  by (rule staged-security-with-data-state-trace-pair-agreement-hit-at-bound)

```

[*OF wf controlled i-bound trace-low trace'-low distinct*])

definition *low-degree-trace-pair-space* :: ('f list × 'f list) set
where

low-degree-trace-pair-space =
 {(*trace-table*, *trace-table'*).
 trace-table-low-degree trace-table ∧
 trace-table-low-degree trace-table' ∧
 trace-table ≠ *trace-table'*}

lemma *trace-table-low-degree-length*:
assumes *trace-table-low-degree trace-table*
shows *length trace-table* = *clength * scale*
using *assms unfolding trace-table-low-degree-def*
by (*auto simp: eval-domain-length*)

lemma *finite-low-degree-trace-pair-space[simp]*:
finite low-degree-trace-pair-space

proof —

have *subset*:
low-degree-trace-pair-space ⊆
 {*xs. length xs* = *clength * scale*} ×
 {*xs. length xs* = *clength * scale*}
unfolding *low-degree-trace-pair-space-def*
by (*auto dest: trace-table-low-degree-length*)
show ?*thesis*
by (*rule finite-subset[OF subset]*)
 (*simp add: finite-length-lists-UNIV*)

qed

definition *low-degree-trace-pair-agreement-round-error*

where

low-degree-trace-pair-agreement-round-error budgets i =
 (∑ (*trace-table*, *trace-table'*) ∈ *low-degree-trace-pair-space*.
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 nnreal clength / nnreal (card query-sample-space))

definition *low-degree-trace-pair-agreement-error*

where

low-degree-trace-pair-agreement-error budgets =
 (∑ *x* ∈ *low-degree-trace-pair-space* × {..*rounds*}.
 case x of (pair, i) ⇒
 staged-phase-relation-error size
 (*staged-query-search-queries budgets i + 1*) +
 nnreal clength / nnreal (card query-sample-space))

lemma *empty-header-low-degree-transcript-query-agreement-imp-pair-hit*:
assumes

```

checked-staged-security-with-actual-alpha-prefix-verifier-event
empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
out
shows
   $\exists \text{pair} \in \text{low-degree-trace-pair-space}.$ 
   $\exists i \in \{..<\text{rounds}\}.$ 
    checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
      (fst pair) (snd pair) i out
proof (cases out)
case None
then show ?thesis
  using assms
  unfolding checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
next
case (Some packed)
then obtain alpha-prefix alpha-prefix-state data attacker-state result
  final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have hit:
  empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
    ?s (Some (result, final-state))
  using assms
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-verifier-event-def
  by simp
then obtain fr f-fri-roots f-final as dg final trace-query-idxs
  trace-openings trace-table composition-table trace-table' i where
  shape:
    accepted-transcript-shape ?s (Some (result, final-state)) as
      trace-query-idxs
  and distinct: trace-table  $\neq$  trace-table'
  and trace-low: trace-table-low-degree trace-table
  and trace'-low: trace-table-low-degree trace-table'
  and i-bound:  $i < \text{rounds}$ 
  and idx-agree:
    trace-query-idxs !  $i \in$ 
      trace-table-agreement-indices trace-table trace-table'
  by (rule
    empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreementE)
have pair-in:
  (trace-table, trace-table')  $\in$  low-degree-trace-pair-space
  unfolding low-degree-trace-pair-space-def

```

```

using trace-low trace'-low distinct by simp
have pair-hit:
  checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
    trace-table trace-table' i out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at-def
    staged-security-with-data-state-trace-pair-agreement-hit-at-def
using shape i-bound idx-agree by auto
show ?thesis
by (intro bexI[of - (trace-table, trace-table')])
    (bexI[of - i])
    (use pair-in i-bound pair-hit in simp-all)
qed

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-low-degree-transcript-query-agreement-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit)
    adversary-initial-state  $\leq$ 
    low-degree-trace-pair-agreement-error budgets
proof -
let ?M =
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let ?pair-hit =
   $\lambda$ (pair, i) out.
    checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
      (fst pair) (snd pair) i out
let ?A = low-degree-trace-pair-space  $\times$  {..rounds}
have event-le:
  wp-event ?M
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit)
    adversary-initial-state  $\leq$ 
wp-event ?M ( $\lambda$ out.  $\exists x \in ?A. ?pair-hit x out$ )
    adversary-initial-state
by (rule wp-event-mono)
    (auto dest:
      empty-header-low-degree-transcript-query-agreement-imp-pair-hit)
have finite-A: finite ?A
by simp
have union-bound:
  wp-event ?M ( $\lambda$ out.  $\exists x \in ?A. ?pair-hit x out$ )
    adversary-initial-state  $\leq$ 
    ( $\sum x \in ?A.$ 
      case x of (pair, i)  $\Rightarrow$ 

```

```

    staged-phase-relation-error size
    (staged-query-search-queries budgets  $i + 1$ ) +
    nnreal clength / nnreal (card query-sample-space))
proof (rule wp-event-finite-UN-bound[OF finite-A])
  fix x
  assume x-in:  $x \in ?A$ 
  then obtain trace-table trace-table' i where x-eq:
     $x = ((\text{trace-table}, \text{trace-table}'), i)$ 
  and pair-in:
     $(\text{trace-table}, \text{trace-table}') \in \text{low-degree-trace-pair-space}$ 
  and i-bound:  $i < \text{rounds}$ 
  by (cases x, auto)
  have trace-low: trace-table-low-degree trace-table
    using pair-in unfolding low-degree-trace-pair-space-def by auto
  have trace'-low: trace-table-low-degree trace-table'
    using pair-in unfolding low-degree-trace-pair-space-def by auto
  have distinct:  $\text{trace-table} \neq \text{trace-table}'$ 
    using pair-in unfolding low-degree-trace-pair-space-def by auto
  have bound:
    wp-event ?M
    (checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
      trace-table trace-table' i)
    adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-query-search-queries budgets  $i + 1$ ) +
    nnreal clength / nnreal (card query-sample-space)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at-bound
    [OF wf controlled i-bound trace-low trace'-low distinct])
  show wp-event ?M (?pair-hit x) adversary-initial-state
     $\leq$  (case x of (pair, i)  $\Rightarrow$ 
      staged-phase-relation-error size
      (staged-query-search-queries budgets  $i + 1$ ) +
      nnreal clength / nnreal (card query-sample-space))
    unfolding x-eq
    using bound by simp
qed
have sum-eq:
   $(\sum x \in ?A. \text{case } x \text{ of } (\text{pair}, i) \Rightarrow \text{staged-phase-relation-error size} \text{ (staged-query-search-queries budgets } i + 1 \text{) + nnreal clength / nnreal (card query-sample-space)}) =$ 
  low-degree-trace-pair-agreement-error budgets
unfolding low-degree-trace-pair-agreement-error-def
  low-degree-trace-pair-agreement-round-error-def
by (simp add: sum.cartesian-product case-prod-unfold)
show ?thesis
proof (rule order-trans[OF event-le])

```

```

    show wp-event ?M (λout. ∃ x ∈ ?A. ?pair-hit x out)
      adversary-initial-state ≤
      low-degree-trace-pair-agreement-error budgets
    using union-bound sum-eq by simp
  qed
qed
end
end

```

```

theory Soundness-Relevant-Drift-Transcript-Empty
imports
  Soundness-Relevant-Drift-Query-Agreement
  Soundness-Query-Transcript-Openings
begin

```

Transcript-indexed empty-header packaging.

This layer avoids the broad empty-header composition residual whose authenticated-opening witnesses carry existential query indices. The composition branch instead carries the verifier transcript shape, so later query-agreement bounds can use verifier-derived query indices.

```

context soundness
begin

```

```

definition soundness-bad-event-partial-candidate-empty-header-transcript
  :: ('f, 'a) protocol-channel-scheme ⇒
    (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool

```

where

```

  soundness-bad-event-partial-candidate-empty-header-transcript s out
    ⇔
    composition-bad-with-empty-composition-header-candidates-transcript
      s out ∨
    trace-fri-bad-with-empty-composition-header-candidates s out ∨
    query-bad-with-empty-composition-header-candidates s out

```

lemma accepted-empty-composition-header-candidate-partition-transcript:

assumes false-statement: $\neg$ exists-valid-trace

and partial:

```

  accepted-with-empty-composition-header-candidates s out fr
  f-fri-roots f-final as dg final trace-query-idxs trace-openings
  trace-table composition-table

```

and shape: accepted-transcript-shape s out as trace-query-idxs

shows soundness-bad-event-partial-candidate-empty-header-transcript s out

proof (cases trace-table-low-degree trace-table)

case False

have trace-fri-bad-with-empty-composition-header-candidates s out

unfolding trace-fri-bad-with-empty-composition-header-candidates-def

```

    using partial False by blast
  then show ?thesis
    unfolding soundness-bad-event-partial-candidate-empty-header-transcript-def
    by blast
next
case True
note trace-low = True
have composition-low:
  composition-table-low-degree maxDegree composition-table
proof -
  have composition-table = replicate (scale * clength) final
    using partial
    unfolding accepted-with-empty-composition-header-candidates-def by simp
  then show ?thesis
    using constant-composition-table-low-degree by simp
qed
show ?thesis
proof (cases all-queries-consistent trace-table composition-table as)
  case False
  have query-bad-with-empty-composition-header-candidates s out
    unfolding query-bad-with-empty-composition-header-candidates-def
    using partial trace-low composition-low False by blast
  then show ?thesis
    unfolding soundness-bad-event-partial-candidate-empty-header-transcript-def
    by blast
next
case True
note all-queries = True
from trace-low obtain f where
  deg-f: degree f < clength
  and trace-table-eq: trace-table = map (poly f) eval-domain
  unfolding trace-table-low-degree-def by blast
have violated: violated-constraints f ≠ {}
  by (rule false-statement-violated-constraints
    [OF false-statement deg-f])
have composition-bad-with-empty-composition-header-candidates-transcript
  s out
  unfolding composition-bad-with-empty-composition-header-candidates-transcript-def
  using partial shape deg-f trace-table-eq violated composition-low
  all-queries
  by blast
then show ?thesis
  unfolding soundness-bad-event-partial-candidate-empty-header-transcript-def
  by blast
qed
qed

lemma accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-or-empty-head
  assumes false-statement: ¬ exists-valid-trace

```



```

and supp: out ∈ set-dist (execute verify-monad s)
and acc: accepted out
and shape: accepted-transcript-shape s out as query-idxs
and header:
  verifier-header-transcript s fr f-fri-roots f-final as dg
  composition-fri-roots final rest
and trace-partial-if-empty:
  composition-fri-roots = [] ⇒
    accepted-with-partial-trace-openings s out fr query-idxs
    trace-openings
shows
  soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
    s out ∨
  soundness-bad-event-partial-candidate-empty-header-transcript s out
proof (cases composition-fri-roots = [])
case False
then have comp-nonempty: composition-fri-roots ≠ []
  by simp
have
  soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
    s out
  by (rule
    accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-if-header-nonempty
    [OF false-statement supp acc header comp-nonempty])
then show ?thesis by simp
next
case True
note comp-empty = True
from acc obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto
have trace-partial:
  accepted-with-partial-trace-openings s (Some (result, final-state))
  fr query-idxs trace-openings
  using trace-partial-if-empty[OF comp-empty] unfolding out-eq .
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then have
  soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
    s out
  unfolding out-eq
  soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-def
  by simp
then show ?thesis by simp
next
case False
obtain trace-table where trace-candidate:

```

```

    partial-trace-table-candidate trace-table trace-openings
using accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
    [OF trace-partial False]
by blast
have accepted-out: accepted (Some (result, final-state))
    by (rule accepted-with-partial-trace-openings-imp-accepted
        [OF trace-partial])
let ?composition-table = replicate (scale * clength) final
have empty-partial:
    accepted-with-empty-composition-header-candidates s
    (Some (result, final-state)) fr f-fri-roots f-final as dg final
    query-idxs trace-openings trace-table ?composition-table
unfolding accepted-with-empty-composition-header-candidates-def
by (intro conjI exI[of - rest])
    (use accepted-out header comp-empty trace-partial trace-candidate
        in simp-all)
have soundness-bad-event-partial-candidate-empty-header-transcript s
    (Some (result, final-state))
    by (rule accepted-empty-composition-header-candidate-partition-transcript
        [OF false-statement empty-partial shape[unfolded out-eq]])
then show ?thesis
    unfolding out-eq by simp
qed
qed

lemma accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-or-empty-header
assumes false-statement:  $\neg$  exists-valid-trace
and supp:  $out \in \text{set-dist } (\text{execute verify-monad } s)$ 
and acc: accepted out
and header:
    verifier-header-transcript s fr f-fri-roots f-final as dg
    composition-fri-roots final rest
shows
    soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
    s out  $\vee$ 
    soundness-bad-event-partial-candidate-empty-header-transcript s out
proof (cases composition-fri-roots = [])
case False
then have comp-nonempty: composition-fri-roots  $\neq []$ 
by simp
have
    soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
    s out
by (rule
    accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-if-header-nonempty
    [OF false-statement supp acc header comp-nonempty])
then show ?thesis by simp
next
case True

```

```

note comp-empty = True
from acc obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto
have header-empty:
  verifier-header-transcript s fr f-fri-roots f-final as dg []
  final rest
  using header comp-empty by simp
from verify-monad-supplied-empty-header-partial-trace-openings
  [OF supp[unfolded out-eq] header-empty]
obtain query-idxs trace-openings where shape:
  accepted-transcript-shape s (Some (result, final-state)) as
  query-idxs
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr query-idxs trace-openings
by blast
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then have
  soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
  s out
  unfolding out-eq
  soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-def
  by simp
then show ?thesis by simp
next
case False
obtain trace-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
  using accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
  [OF trace-partial False]
  by blast
have accepted-out: accepted (Some (result, final-state))
  by (rule accepted-with-partial-trace-openings-imp-accepted
  [OF trace-partial])
let ?composition-table = replicate (scale * clength) final
have empty-partial:
  accepted-with-empty-composition-header-candidates s
  (Some (result, final-state)) fr f-fri-roots f-final as dg final
  query-idxs trace-openings trace-table ?composition-table
  unfolding accepted-with-empty-composition-header-candidates-def
  by (intro conjI exI[of - rest])
  (use accepted-out header-empty trace-partial trace-candidate
  in simp-all)
have soundness-bad-event-partial-candidate-empty-header-transcript s
  (Some (result, final-state))

```


$[OF \text{ partial}]$)
then show $?thesis$
proof
assume *fresh*:
checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit
composition-trace-bad-alpha-space out
have *prefix-or-drift*:
checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space out $\vee$
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space out
by (*rule*
checked-staged-security-with-actual-alpha-prefix-partial-header-fresh-hit-imp-prefix-or-drift
 $[OF \text{ fresh}]$)
then show $?thesis$
proof
assume
checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space out
then show $?thesis$ **by** *simp*
next
assume *drift*:
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-drift
composition-trace-bad-alpha-space out
have
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
out
unfolding
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-def
using *drift hit* **by** *simp*
then show $?thesis$ **by** *simp*
qed
next
assume
checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
composition-trace-bad-alpha-space out
then show $?thesis$ **by** *simp*
qed
qed

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-transcript-bound-from-empty-relevant*
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *drift-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
adversary-initial-state $\leq D$

shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-verifier-event*
 composition-alpha-partial-opening-union-hit-with-empty-header-transcript)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
D

proof –

let *?M* =
 checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let *?Prefix* =
 checked-staged-security-with-actual-alpha-prefix-bad-set-hit
 composition-trace-bad-alpha-space
let *?Prequery* =
 checked-staged-security-with-actual-alpha-prefix-partial-header-prequery-hit
 composition-trace-bad-alpha-space
let *?Drift* =
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
let *?Hit* =
 checked-staged-security-with-actual-alpha-prefix-verifier-event
 composition-alpha-partial-opening-union-hit-with-empty-header-transcript
let *?P* =
 composition-error-bound +
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*)
let *?Q* =
 staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*)
have *transcript-bound*:
 wp-event (*checked-staged-transcript-with-alpha-prefix-program A*)
 (*checked-staged-transcript-actual-alpha-prefix-bad-set-hit*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$ *?P*
by (*rule*
 checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
 [*OF wf controlled*])
have *prefix-bound*:
 wp-event ?M ?Prefix adversary-initial-state $\leq$ *?P*
by (*rule*
 checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript

```

    [OF transcript-bound])
have prequery-bound:
  wp-event ?M ?Prequery adversary-initial-state ≤ ?Q
  by (rule
    checked-staged-security-with-actual-alpha-prefix-vector-alpha-prequery-accounting
    [OF wf controlled])
have event-le:
  wp-event ?M ?Hit adversary-initial-state ≤
  wp-event ?M
  (λout. ?Prefix out ∨ ?Prequery out ∨ ?Drift out ∨ False)
  adversary-initial-state
  by (rule wp-event-mono-on-support)
  (use
    checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-transcript-imp-prefix-prequery-or-empty-
    [of - A] in blast)
also have ... ≤
  wp-event ?M ?Prefix adversary-initial-state +
  wp-event ?M ?Prequery adversary-initial-state +
  wp-event ?M ?Drift adversary-initial-state +
  wp-event ?M (λ-. False) adversary-initial-state
  by (rule wp-event-union-bound4)
also have ... ≤ ?P + ?Q + D + 0
  by (intro add-mono prefix-bound prequery-bound drift-bound)
  (simp add: wp-event-def wp-def dist-expect-def)
finally show ?thesis
  by (simp add: algebra-simps)
qed

```

lemma *checked-staged-security-with-data-state-empty-header-composition-randomization-transcript-bound-from-empty*

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and drift-bound:

```

  wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
    A)
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
  adversary-initial-state ≤ D

```

shows

```

  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-randomization-bad-with-empty-composition-header-candidates-transcript)
  adversary-initial-state ≤
  composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +

```

D
proof –
let $?M = \text{checked-staged-security-experiment-with-data-state } A$
let $?random =$
 $\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-randomization-bad-with-empty-composition-header-candidates-transcript}$
let $?alpha =$
 $\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-alpha-partial-opening-union-hit-with-empty-header-transcript}$
have alpha-projected:
 $\text{wp-event } ?M \text{ } ?alpha \text{ adversary-initial-state} =$
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-verifier-event}$
 $\text{composition-alpha-partial-opening-union-hit-with-empty-header-transcript})$
 $\text{adversary-initial-state}$
by (rule
 $\text{checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection})$
have alpha-bound:
 $\text{wp-event } ?M \text{ } ?alpha \text{ adversary-initial-state} \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 D
unfolding alpha-projected
by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-empty-header-alpha-hit-transcript-bound-from-empty-relevant-drift}$
 $[\text{OF wf controlled drift-bound}])$
have $\text{wp-event } ?M \text{ } ?random \text{ adversary-initial-state} \leq$
 $\text{wp-event } ?M \text{ } ?alpha \text{ adversary-initial-state}$
by (rule wp-event-mono)
 $(\text{auto simp: staged-security-with-data-state-verifier-event-def}$
 intro:
 $\text{composition-randomization-bad-with-empty-composition-header-candidates-transcript-imp-alpha-union-hit}$
 $\text{split: option.splits prod.splits})$
also have $\dots \leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 D
by (rule alpha-bound)

finally show *?thesis* .
qed

lemma *checked-staged-security-with-data-state-empty-header-bad-event-transcript-bound-from-empty-relevant-drift-tr*

fixes *empty-trace-fri-error* :: *prob*

assumes *false-statement*: $\neg$ *exists-valid-trace*

and *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *drift-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
adversary-initial-state $\leq D$

and *trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)

adversary-initial-state $\leq$ *empty-trace-fri-error*

and *query-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*
query-bad-with-empty-composition-header-candidates)

adversary-initial-state $\leq$ *empty-query-error*

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*
soundness-bad-event-partial-candidate-empty-header-transcript)

adversary-initial-state $\leq$

composition-error-bound +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

hash-collision-budget-value 0

(*staged-alpha-search-queries budgets 0*) +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

D + *empty-trace-fri-error* + *empty-query-error*

proof –

let *?M* = *checked-staged-security-experiment-with-data-state A*

let *?E* = *staged-security-with-data-state-verifier-event*

let *?degree* =

?E composition-degree-bad-with-empty-composition-header-candidates-transcript

let *?random* =

?E composition-randomization-bad-with-empty-composition-header-candidates-transcript

let *?trace* =

?E trace-fri-bad-with-empty-composition-header-candidates

let *?query* =

?E query-bad-with-empty-composition-header-candidates

let *?bad* =

```

?E soundness-bad-event-partial-candidate-empty-header-transcript
have degree-false: ?degree = ( $\lambda$ out. False)
by (rule ext)
  (auto simp: staged-security-with-data-state-verifier-event-def
    composition-degree-bad-with-empty-composition-header-candidates-transcript-false
    split: option.splits prod.splits)
have degree-bound:
  wp-event ?M ?degree adversary-initial-state  $\leq 0$ 
  unfolding degree-false wp-event-def wp-def dist-expect-def by simp
have random-bound:
  wp-event ?M ?random adversary-initial-state  $\leq$ 
    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    D
by (rule
  checked-staged-security-with-data-state-empty-header-composition-randomization-transcript-bound-from-empty-re
  [OF wf controlled drift-bound])
have wp-event ?M ?bad adversary-initial-state  $\leq$ 
  wp-event ?M
  ( $\lambda$ out. ?degree out  $\vee$  ?random out  $\vee$  ?trace out  $\vee$ 
    ?query out)
  adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume bad: ?bad out
  show ?degree out  $\vee$  ?random out  $\vee$  ?trace out  $\vee$  ?query out
  proof (cases out)
    case None
    then show ?thesis
      using bad
      unfolding staged-security-with-data-state-verifier-event-def
      by simp
  next
    case (Some packed)
    then obtain data attacker-state result final-state where out-eq:
      out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
    let ?s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    have local-bad:
      soundness-bad-event-partial-candidate-empty-header-transcript ?s
      (Some (result, final-state))
    using bad

```

```

unfolding out-eq staged-security-with-data-state-verifier-event-def
by simp
then consider
  (comp)
    composition-bad-with-empty-composition-header-candidates-transcript ?s
    (Some (result, final-state))
  | (trace)
    trace-fri-bad-with-empty-composition-header-candidates ?s
    (Some (result, final-state))
  | (query)
    query-bad-with-empty-composition-header-candidates ?s
    (Some (result, final-state))
unfolding
  soundness-bad-event-partial-candidate-empty-header-transcript-def
by blast
then show ?thesis
proof cases
case comp
then have comp-split:
  composition-degree-bad-with-empty-composition-header-candidates-transcript
    ?s (Some (result, final-state)) ∨
  composition-randomization-bad-with-empty-composition-header-candidates-transcript
    ?s (Some (result, final-state))
by (rule
  composition-bad-with-empty-composition-header-candidates-transcript-split
  [OF false-statement])
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  using comp-split by (auto split: option.splits prod.splits)
next
case trace
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
next
case query
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
qed
also have ... ≤
  wp-event ?M ?degree adversary-initial-state +
  wp-event ?M ?random adversary-initial-state +
  wp-event ?M ?trace adversary-initial-state +
  wp-event ?M ?query adversary-initial-state
by (rule wp-event-union-bound4)
also have ... ≤

```

```

0 +
(composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 D) +
empty-trace-fri-error + empty-query-error
by (intro add-mono degree-bound random-bound trace-fri-bound query-bound)
finally show ?thesis
by (simp add: algebra-simps)
qed

lemma checked-staged-soundness-from-aligned-transcript-partial-candidate-and-empty-header-transcript-bounds:
fixes partial-candidate-error empty-header-error :: prob
assumes false-statement:  $\neg$  exists-valid-trace
and partial-candidate-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
 soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle)
adversary-initial-state  $\leq$  partial-candidate-error
and empty-header-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
 soundness-bad-event-partial-candidate-empty-header-transcript)
adversary-initial-state  $\leq$  empty-header-error
shows checked-staged-adversary-acceptance-probability A  $\leq$ 
partial-candidate-error + empty-header-error
proof -
let ?M = checked-staged-security-experiment-with-data-state A
let ?E = staged-security-with-data-state-verifier-event
let ?partial =
?E soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
let ?empty = ?E soundness-bad-event-partial-candidate-empty-header-transcript
let ?bad =  $\lambda$ out. ?partial out  $\vee$  ?empty out
have accepted-bad:
wp-event ?M accepted adversary-initial-state  $\leq$ 
wp-event ?M ?bad adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support: out  $\in$  set-dist (execute ?M adversary-initial-state)
and acc: accepted out
show ?bad out
proof (cases out)
case None
then show ?thesis
using acc unfolding accepted-def by simp

```

```

next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  using checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
  by blast
  have verifier-acc: accepted (Some (result, final-state))
    unfolding accepted-def by simp
  from verify-monad-accepted-transcript-shape[OF verifier]
  obtain alphas query-idxs where shape:
    accepted-transcript-shape ?s (Some (result, final-state))
    alphas query-idxs
  by blast
  from shape obtain result' final-state' fr f-fri-roots f-final dg
    composition-fri-roots final rest where
    header:
      verifier-header-transcript ?s fr f-fri-roots f-final alphas dg
      composition-fri-roots final rest
  by (elim accepted-transcript-shape-header-query-extraction)
  have local-bad:
    soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle
      ?s (Some (result, final-state)) ∨
    soundness-bad-event-partial-candidate-empty-header-transcript ?s
      (Some (result, final-state))
  by (rule
    accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle-or-empty-header)
    [OF false-statement verifier verifier-acc header])
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
qed
qed
have bad-bound:
  wp-event ?M ?bad adversary-initial-state ≤
  partial-candidate-error + empty-header-error
proof -
  have wp-event ?M ?bad adversary-initial-state ≤
    wp-event ?M ?partial adversary-initial-state +
    wp-event ?M ?empty adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤ partial-candidate-error + empty-header-error
  by (intro add-mono partial-candidate-bound empty-header-bound)

```

```

    finally show ?thesis .
qed
have staged-bound:
  wp-event (checked-staged-security-experiment A) accepted
    adversary-initial-state ≤
      partial-candidate-error + empty-header-error
  unfolding checked-staged-security-experiment-acceptance-with-data-state
  by (rule order-trans[OF accepted-bad bad-bound])
show ?thesis
proof -
  have
    wp-event (checked-staged-security-experiment A)
      (λout. ¬ Option.is-none out) adversary-initial-state =
    wp-event (checked-staged-security-experiment A) accepted
      adversary-initial-state
  unfolding wp-event-def accepted-def
  by (rule arg-cong[where
    f=λQ. wp (checked-staged-security-experiment A) Q
    adversary-initial-state])
    (rule ext, simp)
  then show ?thesis
    unfolding checked-staged-adversary-acceptance-probability-def
    using staged-bound by simp
qed
qed

```

```

lemma checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from
  fixes trace-fri-error' composition-fri-error' query-error D E
  empty-trace-fri-error
  empty-query-error :: prob
  assumes false-statement: ¬ exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-partial-candidates)
      adversary-initial-state ≤ trace-fri-error'
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-partial-candidates)
      adversary-initial-state ≤ composition-fri-error'
  and query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        query-bad-with-aligned-transcript-partial-candidates)
      adversary-initial-state ≤ query-error
  and drift-bound:

```

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq D$
and *empty-drift-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
adversary-initial-state $\leq E$
and *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *empty-trace-fri-error*
and *current-empty-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-current-empty-query-opening-hit-at 0*)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
D) +
trace-fri-error' + composition-fri-error' +
query-error +
 (*composition-error-bound* +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
E + empty-trace-fri-error + empty-query-error)
proof –
have *comp-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
composition-bad-with-partial-candidates)
adversary-initial-state $\leq$
composition-error-bound +

staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D
by (rule
checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-relevant-drift-and-budgets
[OF false-statement wf controlled drift-bound])
have *merkle-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad)
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget)
by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
[OF wf controlled])
have *empty-query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤ empty-query-error
by (rule *order-trans*[
OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero
OF wf controlled] current-empty-bound])
have *empty-header-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
soundness-bad-event-partial-candidate-empty-header-transcript)
adversary-initial-state ≤
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
E + empty-trace-fri-error + empty-query-error
by (rule
checked-staged-security-with-data-state-empty-header-bad-event-transcript-bound-from-empty-relevant-drift-trans
[OF false-statement wf controlled empty-drift-bound
empty-trace-fri-bound empty-query-bound])
have *partial-candidate-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event


```

    soundness-bad-event-aligned-transcript-partial-candidate-with-partial-merkle)
  adversary-initial-state ≤
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
   staged-challenge-query-budget + verifier-hash-query-budget) +
  (composition-error-bound +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
   hash-collision-budget-value 0
   (staged-alpha-search-queries budgets 0) +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
   D) +
  trace-fri-error' + composition-fri-error' + query-error
by (rule
  checked-staged-security-with-data-state-aligned-transcript-partial-candidate-with-partial-merkle-component-union
  [OF merkle-bound comp-bound trace-fri-bound comp-fri-bound
   query-bound])
have bound:
  checked-staged-adversary-acceptance-probability A ≤
  (hash-collision-budget-value 0
   (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget) +
  (composition-error-bound +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
   hash-collision-budget-value 0
   (staged-alpha-search-queries budgets 0) +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
   D) +
  trace-fri-error' + composition-fri-error' + query-error) +
  (composition-error-bound +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
  hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  E + empty-trace-fri-error + empty-query-error)
by (rule
  checked-staged-soundness-from-aligned-transcript-partial-candidate-and-empty-header-transcript-bounds
  [OF false-statement partial-candidate-bound empty-header-bound])
then show ?thesis
by (simp add: algebra-simps)
qed

lemma checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from
  fixes trace-fri-error' composition-fri-error' query-error D

```

empty-trace-fri-error empty-query-error :: prob
assumes *false-statement: $\neg$ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ trace-fri-error'
and *comp-fri-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-partial-candidates)
 adversary-initial-state $\leq$ composition-fri-error'
and *query-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 query-bad-with-aligned-transcript-partial-candidates)
 adversary-initial-state $\leq$ query-error
and *drift-bound:*
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
 adversary-initial-state $\leq$ D
and *empty-trace-fri-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state $\leq$ empty-trace-fri-error
and *current-empty-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 adversary-initial-state $\leq$ empty-query-error
shows *checked-staged-adversary-acceptance-probability A $\leq$*
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 D) +
 trace-fri-error' + composition-fri-error' +
 query-error +
 (composition-error-bound +

```

    staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
    D + empty-trace-fri-error + empty-query-error)
proof -
  have empty-drift-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
      adversary-initial-state  $\leq D$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-relevant-drift-
    [OF drift-bound])
  show ?thesis
  by (rule
    checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from-re-
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
      query-bound drift-bound empty-drift-bound empty-trace-fri-bound
      current-empty-bound])
qed

lemma checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-reachable-
fixes F L :: prob
assumes trace-fri-reduction:
   $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      trace-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
and fixed-empty-alpha-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
    adversary-initial-state  $\leq F$ 
and low-degree-query-agreement-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit)
    adversary-initial-state  $\leq L$ 
shows

```

```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
  adversary-initial-state  $\leq F + (\text{trace-fri-error} + L)$ 
proof -
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?trace =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    trace-fri-bad-with-empty-composition-header-candidates
  let ?low-degree-transcript =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit
  let ?low-degree =
    checked-staged-security-with-actual-alpha-prefix-verifier-event
    empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit
  have trace-data-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state  $\leq \text{trace-fri-error}$ 
  by (rule
      trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
      [OF trace-fri-reduction])
  have trace-bound:
    wp-event ?M ?trace adversary-initial-state  $\leq \text{trace-fri-error}$ 
  using trace-data-bound
    checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection
    [of A trace-fri-bad-with-empty-composition-header-candidates]
  by simp
  have low-degree-transcript-bound:
    wp-event ?M ?low-degree-transcript adversary-initial-state  $\leq L$ 
  proof (rule order-trans)
    show wp-event ?M ?low-degree-transcript adversary-initial-state  $\leq$ 
      wp-event ?M ?low-degree adversary-initial-state
    by (rule wp-event-mono)
    (auto simp: checked-staged-security-with-actual-alpha-prefix-verifier-event-def
      intro:
        empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-imp-query-agreement
        split: option.splits prod.splits)
    show wp-event ?M ?low-degree adversary-initial-state  $\leq L$ 
    by (rule low-degree-query-agreement-bound)
  qed
qed
show ?thesis
  by (rule
      checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-fixed-trace-fri-
      [OF fixed-empty-alpha-bound trace-bound low-degree-transcript-bound])
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-reachable-*
fixes $F :: \text{prob}$
assumes wf : *staged-budget-wellformed budgets*
and $controlled$: *staged-adversary-controlled budgets A*
and $trace\text{-}fri\text{-}reduction$:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and $fixed\text{-}empty\text{-}alpha\text{-}bound$:
 $wp\text{-}event$
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-verifier-event}$
 $\text{composition-alpha-bad-set-hit-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq F$
shows
 $wp\text{-}event$
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state A})$
 $\text{checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript}$
 $\text{adversary-initial-state} \leq$
 $F + (\text{trace-fri-error} + \text{low-degree-trace-pair-agreement-error budgets})$
proof –
let $?M =$
 $\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state A}$
have $trace\text{-}data\text{-}bound$:
 $wp\text{-}event (\text{checked-staged-security-experiment-with-data-state A})$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
by (rule
 $\text{trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction}$
 $[\text{OF trace-fri-reduction}])$
have $trace\text{-}bound$:
 $wp\text{-}event ?M$
 $(\text{checked-staged-security-with-actual-alpha-prefix-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
using $trace\text{-}data\text{-}bound$
 $\text{checked-staged-security-with-data-state-verifier-event-actual-alpha-prefix-projection}$
 $[\text{of A trace-fri-bad-with-empty-composition-header-candidates}]$
by $simp$
have $low\text{-}degree\text{-}bound$:
 $wp\text{-}event ?M$
 $(\text{checked-staged-security-with-actual-alpha-prefix-verifier-event}$
 $\text{empty-header-low-degree-trace-candidate-nonunique-transcript-query-agreement-hit})$

```

    adversary-initial-state  $\leq$ 
    low-degree-trace-pair-agreement-error budgets
  by (rule
    checked-staged-security-with-actual-alpha-prefix-empty-header-low-degree-transcript-query-agreement-bound
    [OF wf controlled])
  show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-fixed-trace-fri-
    [OF fixed-empty-alpha-bound trace-bound low-degree-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from
  fixes trace-fri-error' composition-fri-error' query-error D F L
  empty-query-error :: prob
  assumes false-statement:  $\neg$  exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  trace-fri-error'
  and comp-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-fri-error'
  and query-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state  $\leq$  query-error
  and drift-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
    adversary-initial-state  $\leq$  D
  and trace-fri-reduction:
     $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
    trace-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
  and fixed-empty-alpha-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state

```

A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
 adversary-initial-state $\leq F$
and low-degree-query-agreement-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit)
 adversary-initial-state $\leq L$
and current-empty-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 adversary-initial-state $\leq$ empty-query-error
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 D) +
 trace-fri-error' + composition-fri-error' +
 query-error +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (F + (trace-fri-error + L)) + trace-fri-error +
 empty-query-error)
proof –
have empty-drift-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
 adversary-initial-state $\leq F + (\text{trace-fri-error} + L)$
by (rule
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-reachable-trac
 [OF trace-fri-reduction fixed-empty-alpha-bound
 low-degree-query-agreement-bound])
have empty-trace-fri-bound:

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state  $\leq$  trace-fri-error
by (rule
  trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
  [OF trace-fri-reduction])
show ?thesis
by (rule
  checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from-re
  [OF false-statement wf controlled trace-fri-bound comp-fri-bound
   query-bound drift-bound empty-drift-bound empty-trace-fri-bound
   current-empty-bound])
qed

lemma checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from-
fixes trace-fri-error' composition-fri-error' query-error D F
  empty-query-error :: prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  trace-fri-error'
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-fri-bad-with-partial-candidates)
    adversary-initial-state  $\leq$  composition-fri-error'
and query-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state  $\leq$  query-error
and drift-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
     A)
    checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
    adversary-initial-state  $\leq$  D
and trace-fri-reduction:
   $\bigwedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    trace-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state)

```


(staged-proof-transcript data))
and fixed-empty-alpha-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
 adversary-initial-state $\leq F$
and current-empty-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 adversary-initial-state $\leq$ empty-query-error
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 D) +
 trace-fri-error' + composition-fri-error' +
 query-error +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (F + (trace-fri-error +
 low-degree-trace-pair-agreement-error budgets)) +
 trace-fri-error + empty-query-error)
proof –
have empty-drift-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
 adversary-initial-state $\leq$
 F + (trace-fri-error + low-degree-trace-pair-agreement-error budgets)
by (rule
 checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-reachable-trac
 [OF wf controlled trace-fri-reduction fixed-empty-alpha-bound])
have empty-trace-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event

```

    trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state ≤ trace-fri-error
by (rule
  trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
  [OF trace-fri-reduction])
show ?thesis
by (rule
  checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from-re
  [OF false-statement wf controlled trace-fri-bound comp-fri-bound
    query-bound drift-bound empty-drift-bound empty-trace-fri-bound
    current-empty-bound])
qed

theorem stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-
fixes  $D F L$  empty-query-error :: prob
and  $C :: \text{nat} \Rightarrow \text{prob}$ 
assumes false-statement:  $\neg \text{exists-valid-trace}$ 
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and trace-fri-reduction:
   $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program  $A$ )
      adversary-initial-state)  $\implies$ 
    trace-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
and composition-fri-reduction:
   $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program  $A$ )
      adversary-initial-state)  $\implies$ 
    composition-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
and round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state  $A$   $i$ )
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at  $i$ )
      adversary-initial-state ≤  $C$   $i$ 
and drift-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
         $A$ )
      checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
      adversary-initial-state ≤  $D$ 
and fixed-empty-alpha-bound:
    wp-event

```

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-verifier-event composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state $\leq F$
and *low-degree-query-agreement-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-verifier-event empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit)
adversary-initial-state $\leq L$
and *current-empty-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ empty-query-error
shows *checked-staged-adversary-acceptance-probability $A \leq$*
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D) +
trace-fri-error + composition-fri-error +
($\sum_{i < \text{rounds}} C i$) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
($F + (\text{trace-fri-error} + L)$) + trace-fri-error +
empty-query-error)
proof –
have *trace-fri-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ trace-fri-error
by *(rule*
trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction
[OF trace-fri-reduction])
have *comp-fri-bound:*

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   composition-fri-bad-with-partial-candidates)
  adversary-initial-state ≤ composition-fri-error
by (rule
   composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction
   [OF composition-fri-reduction])
have current-query-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-current-query-partial-opening-hit
    adversary-initial-state ≤ (∑ i<rounds. C i)
  by (rule
      checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
      [OF round-bound wf controlled])
have query-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     query-bad-with-aligned-transcript-partial-candidates)
    adversary-initial-state ≤ (∑ i<rounds. C i)
  by (rule order-trans[
      OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit
      current-query-bound])
    (use wf controlled in simp-all)
show ?thesis
  by (rule
      checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from-re
      [OF false-statement wf controlled trace-fri-bound comp-fri-bound
       query-bound drift-bound trace-fri-reduction fixed-empty-alpha-bound
       low-degree-query-agreement-bound current-empty-bound])
qed

theorem stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-r
  fixes D F empty-query-error :: prob
  and C :: nat ⇒ prob
  assumes false-statement: ¬ exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-reduction:
    ∧ data attacker-state.
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state) ⇒
        trace-fri-reachable-partial-candidate-reduction-assumption
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))
  and composition-fri-reduction:
    ∧ data attacker-state.
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)

```

$\text{adversary-initial-state}) \implies$
composition-fri-reachable-partial-candidate-reduction-assumption
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *drift-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift}$
 $\text{adversary-initial-state} \leq D$
and *fixed-empty-alpha-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-verifier-event}$
 $\text{composition-alpha-bad-set-hit-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq F$
and *current-empty-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum i < \text{rounds}. C \ i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $F) + (\text{trace-fri-error} +$

$low-degree-trace-pair-agreement-error\ budgets)) +$
 $trace-fri-error + empty-query-error)$
proof –
have *trace-fri-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $\quad trace-fri-bad-with-partial-candidates)$
 $adversary-initial-state \leq trace-fri-error$
by (rule
 $trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction$
 $[OF\ trace-fri-reduction])$
have *comp-fri-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $\quad composition-fri-bad-with-partial-candidates)$
 $adversary-initial-state \leq composition-fri-error$
by (rule
 $composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction$
 $[OF\ composition-fri-reduction])$
have *current-query-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $staged-security-with-data-state-current-query-partial-opening-hit$
 $adversary-initial-state \leq (\sum i < rounds.\ C\ i)$
by (rule
 $checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds$
 $[OF\ round-bound\ wf\ controlled])$
have *query-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $\quad query-bad-with-aligned-transcript-partial-candidates)$
 $adversary-initial-state \leq (\sum i < rounds.\ C\ i)$
by (rule *order-trans*[
 $OF\ checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit$
 $\quad current-query-bound])$
 $(use\ wf\ controlled\ in\ simp-all)$
show *?thesis*
by (rule
 $checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from-re$
 $[OF\ false-statement\ wf\ controlled\ trace-fri-bound\ comp-fri-bound$
 $\quad query-bound\ drift-bound\ trace-fri-reduction\ fixed-empty-alpha-bound$
 $\quad current-empty-bound])$
qed

theorem *stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-r*
fixes $D\ empty-query-error :: prob$
and $C :: nat \Rightarrow prob$
assumes *false-statement*: $\neg\ exists-valid-trace$
and *wf*: *staged-budget-wellformed* $budgets$
and *controlled*: *staged-adversary-controlled* $budgets\ A$

and *trace-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *composition-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *drift-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift}$
 $\text{adversary-initial-state} \leq D$
and *current-empty-bound*:
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum i < \text{rounds. } C \ i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$

hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D + trace-fri-error + empty-query-error)
proof –
have *trace-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state ≤ trace-fri-error
by (rule
trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction
[OF trace-fri-reduction])
have *comp-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state ≤ composition-fri-error
by (rule
composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction
[OF composition-fri-reduction])
have *current-query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state ≤ (∑ i < rounds. C i)
by (rule
checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
[OF round-bound wf controlled])
have *query-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
query-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state ≤ (∑ i < rounds. C i)
by (rule *order-trans*[
OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit
current-query-bound])
(use wf controlled in simp-all)
have *empty-trace-fri-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤ trace-fri-error
by (rule
trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
[OF trace-fri-reduction])
show *?thesis*
by (rule
checked-staged-soundness-from-aligned-transcript-partial-candidate-components-and-current-empty-query-from-re

$[OF \text{ false-statement } wf \text{ controlled trace-fri-bound comp-fri-bound}$
 $\text{query-bound drift-bound empty-trace-fri-bound current-empty-bound}]$)

qed

theorem *stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-n*
fixes $D \ N \ L \ \text{empty-query-error} :: \text{prob}$
and $C :: \text{nat} \Rightarrow \text{prob}$
assumes *false-statement: $\neg$ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-reduction:*
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *composition-fri-reduction:*
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *round-bound:*
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state A } i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *drift-bound:*
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift}$
 $\text{adversary-initial-state} \leq D$
and *not-prefix-bound:*
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and *low-degree-query-agreement-bound:*
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$

(checked-staged-security-with-actual-alpha-prefix-verifier-event
empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit)
adversary-initial-state $\leq L$
and *current-empty-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
D) +
trace-fri-error + composition-fri-error +
($\sum i < \text{rounds}. C i$) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
((composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) + N) +
(trace-fri-error + L)) +
trace-fri-error + empty-query-error)
proof –
let *?M* =
checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let *?P* =
composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0)
have *transcript-bound:*
wp-event (checked-staged-transcript-with-alpha-prefix-program A)
(checked-staged-transcript-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$ *?P*

```

    by (rule
      checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
      [OF wf controlled])
  have prefix-bound:
    wp-event ?M
    (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
      composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq$  ?P
  by (rule
    checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
    [OF transcript-bound])
  have fixed-empty-alpha-bound:
    wp-event ?M
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
    adversary-initial-state  $\leq$  ?P + N
  proof (rule order-trans)
    show wp-event ?M
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
      adversary-initial-state  $\leq$ 
      wp-event ?M
      (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
        composition-trace-bad-alpha-space)
      adversary-initial-state +
      wp-event ?M
      (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
        composition-trace-bad-alpha-space)
      adversary-initial-state
    by (rule
      checked-staged-security-with-actual-alpha-prefix-fixed-empty-alpha-hit-le-prefix-or-not-prefix
      [OF wf controlled])
    show ...  $\leq$  ?P + N
      by (intro add-mono prefix-bound not-prefix-bound)
  qed
  show ?thesis
    by (rule
      stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-reach
      [OF false-statement wf controlled trace-fri-reduction
        composition-fri-reduction round-bound drift-bound
        fixed-empty-alpha-bound low-degree-query-agreement-bound
        current-empty-bound])
  qed

theorem stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-r
fixes D N empty-query-error :: prob
and C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets

```

and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-reduction:*
 $\bigwedge$ *data attacker-state.*
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *composition-fri-reduction:*
 $\bigwedge$ *data attacker-state.*
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *round-bound:*
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *drift-bound:*
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift}$
 $\text{adversary-initial-state} \leq D$
and *not-prefix-bound:*
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and *current-empty-bound:*
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$

staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
D) +
trace-fri-error + composition-fri-error +
 ($\sum_{i < \text{rounds. } C} i$) +
 (*composition-error-bound +*
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
 ((*composition-error-bound +*
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) + *N*) +
 (*trace-fri-error + low-degree-trace-pair-agreement-error budgets*)) +
trace-fri-error + empty-query-error)
proof –
let ?*M* =
checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let ?*P* =
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*)
have *transcript-bound*:
wp-event (checked-staged-transcript-with-alpha-prefix-program A)
 (*checked-staged-transcript-actual-alpha-prefix-bad-set-hit*
composition-trace-bad-alpha-space)
adversary-initial-state ≤ ?*P*
by (rule
checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
 [*OF wf controlled*])
have *prefix-bound*:
wp-event ?M
 (*checked-staged-security-with-actual-alpha-prefix-bad-set-hit*
composition-trace-bad-alpha-space)
adversary-initial-state ≤ ?*P*
by (rule
checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
 [*OF transcript-bound*])
have *fixed-empty-alpha-bound*:
wp-event ?M
 (*checked-staged-security-with-actual-alpha-prefix-verifier-event*
composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state ≤ ?*P* + *N*

proof (rule *order-trans*)
show *wp-event ?M*
(checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state $\leq$
wp-event ?M
(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state +
wp-event ?M
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state
by (rule
checked-staged-security-with-actual-alpha-prefix-fixed-empty-alpha-hit-le-prefix-or-not-prefix
[OF wf controlled])
show $\dots \leq ?P + N$
by (intro *add-mono prefix-bound not-prefix-bound*)
qed
show *?thesis*
by (rule
stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-reach
[OF false-statement wf controlled trace-fri-reduction
composition-fri-reduction round-bound drift-bound
fixed-empty-alpha-bound current-empty-bound])
qed

theorem *stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-j*
fixes *D N empty-query-error :: prob*
and *C :: nat $\Rightarrow$ prob*
assumes *false-statement: $\neg$ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-reduction:*
 $\bigwedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
trace-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *composition-fri-reduction:*
 $\bigwedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
composition-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))

and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *drift-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift}$
 $\text{adversary-initial-state} \leq D$
and *not-prefix-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound}$
 $\text{composition-trace-bad-alpha-space})$
 $\text{adversary-initial-state} \leq N$
and *current-empty-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum i < \text{rounds}. C \ i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $D + \text{trace-fri-error} + \text{empty-query-error})$
by (rule
 $\text{stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-relevant}$
 $[OF \text{ false-statement } wf \text{ controlled trace-fri-reduction}$
 $\text{composition-fri-reduction round-bound drift-bound}$

current-empty-bound])

theorem *stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-n*

fixes $X P R L$ *empty-query-error* :: *prob*

and $C :: \text{nat} \Rightarrow \text{prob}$

assumes *false-statement*: $\neg \text{exists-valid-trace}$

and *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *trace-fri-reduction*:

$\bigwedge \text{data attacker-state.}$

Some (*data, attacker-state*) $\in$

set-dist (*execute* (*checked-staged-transcript-program A*)

adversary-initial-state) $\implies$

trace-fri-reachable-partial-candidate-reduction-assumption

(*verifier-state-from-adversary attacker-state*

(*staged-proof-transcript data*))

and *composition-fri-reduction*:

$\bigwedge \text{data attacker-state.}$

Some (*data, attacker-state*) $\in$

set-dist (*execute* (*checked-staged-transcript-program A*)

adversary-initial-state) $\implies$

composition-fri-reachable-partial-candidate-reduction-assumption

(*verifier-state-from-adversary attacker-state*

(*staged-proof-transcript data*))

and *round-bound*:

$\bigwedge i. i < \text{rounds} \implies$

wp-event

(*checked-staged-security-experiment-with-query-prefix-data-state A i*)

(*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i*)

adversary-initial-state $\leq C i$

and *cross-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad

adversary-initial-state $\leq X$

and *data-pre-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

staged-security-with-data-state-transcript-pre-hit

adversary-initial-state $\leq P$

and *witness-new-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit

adversary-initial-state $\leq R$

and *low-degree-query-agreement-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)
 (checked-staged-security-with-actual-alpha-prefix-verifier-event
 empty-header-low-degree-trace-candidate-nonunique-query-agreement-hit)
 adversary-initial-state $\leq L$
and current-empty-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 adversary-initial-state $\leq$ empty-query-error
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + $X + (P + R) + (P + R)$)) +
 trace-fri-error + composition-fri-error +
 ($\sum_{i < \text{rounds}} C i$) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 ((composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + $X + (P + R) + (P + R)$)) +
 (trace-fri-error + L)) +
 trace-fri-error + empty-query-error)
proof –
let ?D =
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + $X + (P + R) + (P + R)$
have drift-bound:
 wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
 adversary-initial-state $\leq$?D
by (rule

checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-cross-data-pre
 [OF wf controlled cross-bound data-pre-bound witness-new-bound])
have *transcript-pre-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
 adversary-initial-state $\leq P$
by (rule
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
 [OF *data-pre-bound*])
have *witness-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
 adversary-initial-state $\leq P + R$
by (rule
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new
 [OF *transcript-pre-bound witness-new-bound*])
have *no-prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 adversary-initial-state $\leq P + R$
by (rule *order-trans*
 [OF
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-le-witness-transcript-hit
 witness-bound])
have *path-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
 adversary-initial-state $\leq P + R$
by (rule *order-trans*
 [OF
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-le-witness-transcript-hit
 witness-bound])
have *collision-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
 adversary-initial-state $\leq$
 hash-collision-budget-value 0 (*staged-alpha-search-queries budgets 0*)
by (rule
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
 [OF wf controlled])
have *uncovered-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*

A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
adversary-initial-state $\leq X$
by (rule *order-trans*
 [OF
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-witnessed-cross-or-merkle-bad
cross-bound])
have *not-prefix-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq ?D$
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound
 [OF *wf controlled collision-bound uncovered-bound no-prefix-bound*
path-bound])
show *?thesis*
by (rule
stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-reach
 [OF *false-statement wf controlled trace-fri-reduction*
composition-fri-reduction round-bound drift-bound not-prefix-bound
low-degree-query-agreement-bound current-empty-bound])
qed

theorem *stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-r*
fixes *X P R empty-query-error* :: *prob*
and *C* :: *nat* $\Rightarrow$ *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
trace-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *composition-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
composition-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \Rightarrow$

wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
 adversary-initial-state $\leq C$ i
and cross-bound:
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witnessed-cross-or-merkle-bad
 adversary-initial-state $\leq X$
and data-pre-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-transcript-pre-hit
 adversary-initial-state $\leq P$
and witness-new-bound:
wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
 A)
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
 adversary-initial-state $\leq R$
and current-empty-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 adversary-initial-state $\leq$ empty-query-error
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + $X + (P + R) + (P + R)$)) +
 trace-fri-error + composition-fri-error +
 ($\sum i < \text{rounds. } C$ i) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 ((composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +

$\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) + X + (P + R) + (P + R))) +$
 $(\text{trace-fri-error} + \text{low-degree-trace-pair-agreement-error budgets})) +$
 $\text{trace-fri-error} + \text{empty-query-error})$

proof –

let $?D =$

$\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) + X + (P + R) + (P + R)$

have *drift-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift}$
 $\text{adversary-initial-state} \leq ?D$

by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-cross-data-pre}$
 $[OF \text{ wf controlled cross-bound data-pre-bound witness-new-bound}]$)

have *transcript-pre-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit}$
 $\text{adversary-initial-state} \leq P$

by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit}$
 $[OF \text{ data-pre-bound}]$)

have *witness-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit}$
 $\text{adversary-initial-state} \leq P + R$

by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new}$
 $[OF \text{ transcript-pre-bound witness-new-bound}]$)

have *no-prefix-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate}$
 $\text{adversary-initial-state} \leq P + R$

by (rule *order-trans*
 $[OF$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-le-witness-transcript-hit}$
 $\text{witness-bound}]$)

have *path-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output}$
 $\text{adversary-initial-state} \leq P + R$

by (rule *order-trans*

```

    [OF
      checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-le-witness-transcript-hit
      witness-bound])
have collision-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
    adversary-initial-state  $\leq$ 
    hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
    [OF wf controlled])
have uncovered-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
    adversary-initial-state  $\leq X$ 
  by (rule order-trans
    [OF
      checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-witnessed-cross-or-merkle-bad
      cross-bound])
have not-prefix-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
      composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq ?D$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound
    [OF wf controlled collision-bound uncovered-bound no-prefix-bound
      path-bound])
show ?thesis
  by (rule
    stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query-reaching
    [OF false-statement wf controlled trace-fri-reduction
      composition-fri-reduction round-bound drift-bound not-prefix-bound
      current-empty-bound])
qed

```

end

These transcript-empty wrappers expose the broad witnessed cross/Merkle diagnostic event directly. They are retained only for internal comparison with the sampled-transcript route and are not part of the exported public path.

hide-fact

soundness.stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query
soundness.stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query
soundness.stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query
soundness.stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query
soundness.stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query
soundness.stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query
soundness.stark-soundness-from-aligned-transcript-partial-components-current-query-prefix-and-current-empty-query

end

theory *Soundness-Relevant-Drift-Transcript-Public*
imports *Soundness-Relevant-Drift-Transcript-Empty*
begin

Public-facing transcript-indexed packaging for the current relevant-drift proof route. This layer keeps new wrappers out of the larger transcript-empty theory and avoids the diagnostic broad witnessed cross/Merkle route for the nonempty partial-candidate branch.

context *soundness*
begin

lemma *accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle*
assumes *false-statement: $\neg$ exists-valid-trace*
and *supp: out $\in$ set-dist (execute verify-monad s)*
and *acc: accepted out*
and *header:*
verifier-header-transcript s fr f-fri-roots f-final as dg
composition-fri-roots final rest
shows
soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
s out $\vee$
soundness-bad-event-partial-candidate-empty-header-transcript s out
proof (*cases composition-fri-roots = []*)
case *False*
then have *comp-nonempty: composition-fri-roots $\neq$ []*
by *simp*
have
soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
s out
by (*rule*
accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
[OF false-statement supp acc header comp-nonempty])
then show *?thesis* **by** *simp*
next
case *True*
note *comp-empty = True*
from *acc* **obtain** *result final-state* **where** *out-eq:*

```

    out = Some (result, final-state)
  unfolding accepted-def by (cases out) auto
have header-empty:
  verifier-header-transcript s fr f-fri-roots f-final as dg []
  final rest
  using header comp-empty by simp
from verify-monad-supplied-empty-header-partial-trace-openings
  [OF supp[unfolded out-eq] header-empty]
obtain query-idxs trace-openings where shape:
  accepted-transcript-shape s (Some (result, final-state)) as
    query-idxs
and trace-partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr query-idxs trace-openings
by blast
show ?thesis
proof (cases
  partial-merkle-inconsistency-bad s (Some (result, final-state)))
case True
then have
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
    s out
  unfolding out-eq
  soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle-def
  by simp
then show ?thesis by simp
next
case False
obtain trace-table where trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
  using accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
    [OF trace-partial False]
  by blast
have accepted-out: accepted (Some (result, final-state))
  by (rule accepted-with-partial-trace-openings-imp-accepted
    [OF trace-partial])
let ?composition-table = replicate (scale * clength) final
have empty-partial:
  accepted-with-empty-composition-header-candidates s
    (Some (result, final-state)) fr f-fri-roots f-final as dg final
  query-idxs trace-openings trace-table ?composition-table
  unfolding accepted-with-empty-composition-header-candidates-def
  by (intro conjI exI [of - rest])
    (use accepted-out header-empty trace-partial trace-candidate
      in simp-all)
have soundness-bad-event-partial-candidate-empty-header-transcript s
  (Some (result, final-state))
  by (rule accepted-empty-composition-header-candidate-partition-transcript
    [OF false-statement empty-partial shape])

```



```

    then show ?thesis
      unfolding out-eq by simp
    qed
  qed

lemma checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-and-empty-header-
  fixes partial-candidate-error empty-header-error :: prob
  assumes false-statement:  $\neg$  exists-valid-trace
  and partial-candidate-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle)
        adversary-initial-state  $\leq$  partial-candidate-error
  and empty-header-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        soundness-bad-event-partial-candidate-empty-header-transcript)
        adversary-initial-state  $\leq$  empty-header-error
  shows checked-staged-adversary-acceptance-probability A  $\leq$ 
    partial-candidate-error + empty-header-error
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?E = staged-security-with-data-state-verifier-event
  let ?partial =
    ?E soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
  let ?empty = ?E soundness-bad-event-partial-candidate-empty-header-transcript
  let ?bad =  $\lambda$ out. ?partial out  $\vee$  ?empty out
  have accepted-bad:
    wp-event ?M accepted adversary-initial-state  $\leq$ 
      wp-event ?M ?bad adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support: out  $\in$  set-dist (execute ?M adversary-initial-state)
    and acc: accepted out
    show ?bad out
    proof (cases out)
      case None
      then show ?thesis
        using acc unfolding accepted-def by simp
    next
      case (Some packed)
      then obtain data attacker-state result final-state where out-eq:
        out = Some (((data, attacker-state), result), final-state)
        by (cases packed, auto split: prod.splits)
      let ?s =
        verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)
      have verifier:
        Some (result, final-state)  $\in$  set-dist (execute verify-monad ?s)
    
```

```

using checked-staged-security-experiment-with-data-state-outcomeE
  [OF support[unfolded out-eq]]
by blast
have verifier-acc: accepted (Some (result, final-state))
  unfolding accepted-def by simp
from verify-monad-accepted-transcript-shape[OF verifier]
obtain alphas query-idxs where shape:
  accepted-transcript-shape ?s (Some (result, final-state))
  alphas query-idxs
by blast
from shape obtain result' final-state' fr f-fri-roots f-final dg
  composition-fri-roots final rest where
  header:
    verifier-header-transcript ?s fr f-fri-roots f-final alphas dg
    composition-fri-roots final rest
  by (elim accepted-transcript-shape-header-query-extraction)
have local-bad:
soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle
  ?s (Some (result, final-state)) ∨
soundness-bad-event-partial-candidate-empty-header-transcript ?s
  (Some (result, final-state))
by (rule
  accepted-partition-soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-m
  [OF false-statement verifier verifier-acc header])
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
have bad-bound:
wp-event ?M ?bad adversary-initial-state ≤
partial-candidate-error + empty-header-error
proof –
  have wp-event ?M ?bad adversary-initial-state ≤
    wp-event ?M ?partial adversary-initial-state +
    wp-event ?M ?empty adversary-initial-state
    by (rule wp-event-union-bound)
  also have ... ≤ partial-candidate-error + empty-header-error
    by (intro add-mono partial-candidate-bound empty-header-bound)
  finally show ?thesis .
qed
have staged-bound:
wp-event (checked-staged-security-experiment A) accepted
  adversary-initial-state ≤
  partial-candidate-error + empty-header-error
unfolding checked-staged-security-experiment-acceptance-with-data-state
by (rule order-trans[OF accepted-bad bad-bound])
show ?thesis
proof –

```

```

have
  wp-event (checked-staged-security-experiment A)
    ( $\lambda out. \neg Option.is\_none\ out$ ) adversary-initial-state =
  wp-event (checked-staged-security-experiment A) accepted
    adversary-initial-state
unfolding wp-event-def accepted-def
by (rule arg-cong[where
  f= $\lambda Q. wp\ (checked\_staged\_security\_experiment\ A)\ Q$ 
  adversary-initial-state])
  (rule ext, simp)
then show ?thesis
  unfolding checked-staged-adversary-acceptance-probability-def
  using staged-bound by simp
qed
qed

theorem stark-soundness-from-transcript-indexed-partial-opening-route:
fixes F empty-query-error :: prob
and C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-reduction:
   $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      trace-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
and composition-fri-reduction:
   $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      composition-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
and round-bound:
   $\bigwedge i. i < rounds \implies$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
      adversary-initial-state  $\leq C\ i$ 
and fixed-empty-alpha-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event

```

composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state $\leq F$
and *current-empty-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-current-empty-query-opening-hit-at* 0)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 $(\sum_{i < \text{rounds. } C} i)$ +
trace-fri-error + *composition-fri-error* +
 $(\sum_{i < \text{rounds. } C} i)$ +
(*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
(*staged-alpha-search-queries* *budgets* 0) +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
($F + (\text{trace-fri-error} +$
low-degree-trace-pair-agreement-error *budgets*)) +
trace-fri-error + *empty-query-error*)
proof –
have *merkle-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$
hash-collision-budget-value 0
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*)
by (*rule*
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
 $[OF \text{ wf controlled}]$)
have *comp-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
composition-randomization-bad-with-aligned-transcript-partial-candidates)
adversary-initial-state $\leq (\sum_{i < \text{rounds. } C} i)$
by (*rule*
checked-staged-security-with-data-state-aligned-partial-candidate-randomization-bound-from-query-prefix-current
 $[OF \text{ wf controlled round-bound}]$)
have *trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
by (*rule*

$\text{trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction}$
 $[OF \text{ trace-fri-reduction}]])$

have *comp-fri-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}$

by (rule
 $\text{composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction}$
 $[OF \text{ composition-fri-reduction}]])$

have *current-query-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-current-query-partial-opening-hit}$
 $\text{adversary-initial-state} \leq (\sum i < \text{rounds. } C \ i)$

by (rule
 $\text{checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds}$
 $[OF \text{ round-bound wf controlled}]])$

have *query-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-aligned-transcript-partial-candidates})$
 $\text{adversary-initial-state} \leq (\sum i < \text{rounds. } C \ i)$

by (rule *order-trans*[
 $OF \text{ checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit}$
 $\text{current-query-bound}]])$
 $(\text{use wf controlled in simp-all})$

have *partial-candidate-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle})$
 $\text{adversary-initial-state} \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds. } C \ i) + \text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum i < \text{rounds. } C \ i)$

by (rule
 $\text{checked-staged-security-with-data-state-aligned-transcript-partial-candidate-aligned-randomization-with-merkle-un}$
 $[OF \text{ merkle-bound comp-bound trace-fri-bound comp-fri-bound}$
 $\text{query-bound}]])$

have *empty-drift-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript}$
 $\text{adversary-initial-state} \leq$
 $F + (\text{trace-fri-error} + \text{low-degree-trace-pair-agreement-error budgets})$

by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-reachable-trac}$
 $[OF \text{ wf controlled trace-fri-reduction fixed-empty-alpha-bound}]])$

have *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
by (*rule*
trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
[*OF trace-fri-reduction*])
have *empty-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *empty-query-error*
by (*rule order-trans*[
OF checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero
OF wf controlled] *current-empty-bound*])
have *empty-header-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
soundness-bad-event-partial-candidate-empty-header-transcript)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
(*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +
(*F* + (*trace-fri-error* +
low-degree-trace-pair-agreement-error budgets)) +
trace-fri-error + *empty-query-error*
by (*rule*
checked-staged-security-with-data-state-empty-header-bad-event-transcript-bound-from-empty-relevant-drift-trans
[*OF false-statement wf controlled empty-drift-bound*
empty-trace-fri-bound empty-query-bound])
have *bound*:
checked-staged-adversary-acceptance-probability A $\leq$
(*hash-collision-budget-value 0*
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
($\sum i < \text{rounds. } C \ i$) + *trace-fri-error* + *composition-fri-error* +
($\sum i < \text{rounds. } C \ i$)) +
(*composition-error-bound* +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
(*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +

```

    (F + (trace-fri-error +
      low-degree-trace-pair-agreement-error budgets)) +
    trace-fri-error + empty-query-error)
  by (rule
    checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-and-empty-header-tran
    [OF false-statement partial-candidate-bound empty-header-bound])
  then show ?thesis
    by (simp add: algebra-simps)
qed

```

theorem *stark-soundness-from-transcript-indexed-partial-opening-route-single-query-charge:*

```

fixes F empty-query-error :: prob
and C :: nat  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-reduction:
   $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      trace-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
and composition-fri-reduction:
   $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      composition-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
and round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
      adversary-initial-state  $\leq$  C i
and fixed-empty-alpha-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
    adversary-initial-state  $\leq$  F
and current-empty-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
    adversary-initial-state  $\leq$  empty-query-error

```

shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* budgets +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 ($\sum i < \text{rounds}. C\ i$) +
trace-fri-error + *composition-fri-error* +
 (*composition-error-bound* +
staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* budgets 0) +
staged-phase-relation-error size
 (*staged-alpha-search-queries* budgets 0) +
 ($F + (\text{trace-fri-error} +$
low-degree-trace-pair-agreement-error budgets)) +
trace-fri-error + *empty-query-error*)

proof –

have *merkle-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* budgets +
staged-challenge-query-budget + *verifier-hash-query-budget*)
by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
 [OF *wf controlled*])

have *trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
by (rule
trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction
 [OF *trace-fri-reduction*])

have *comp-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-fri-error*
by (rule
composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction
 [OF *composition-fri-reduction*])

have *current-query-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* A)
staged-security-with-data-state-current-query-partial-opening-hit
adversary-initial-state $\leq$ ($\sum i < \text{rounds}. C\ i$)
by (rule

checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds
 [OF round-bound wf controlled])
have *partial-candidate-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
soundness-bad-event-aligned-transcript-partial-candidate-with-aligned-randomization-and-merkle)
adversary-initial-state $\leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 ($\sum i < \text{rounds. } C \ i$) + *trace-fri-error* + *composition-fri-error*
by (rule
checked-staged-security-with-data-state-aligned-transcript-partial-candidate-aligned-randomization-with-merkle-c
 [OF wf controlled merkle-bound current-query-bound trace-fri-bound
comp-fri-bound])
have *empty-drift-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript
adversary-initial-state $\leq$
F + (*trace-fri-error* + *low-degree-trace-pair-agreement-error budgets*)
by (rule
checked-staged-security-with-actual-alpha-prefix-empty-header-relevant-drift-transcript-bound-from-reachable-trac
 [OF wf controlled trace-fri-reduction fixed-empty-alpha-bound])
have *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*
by (rule
trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
 [OF trace-fri-reduction])
have *empty-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *empty-query-error*
by (rule *order-trans*[
 OF *checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero*[
 OF wf controlled] *current-empty-bound*])
have *empty-header-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
soundness-bad-event-partial-candidate-empty-header-transcript)
adversary-initial-state $\leq$
composition-error-bound +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0

(staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (F + (trace-fri-error +
 low-degree-trace-pair-agreement-error budgets)) +
 trace-fri-error + empty-query-error
by (rule
 checked-staged-security-with-data-state-empty-header-bad-event-transcript-bound-from-empty-relevant-drift-trans
 [OF false-statement wf controlled empty-drift-bound
 empty-trace-fri-bound empty-query-bound])
have bound:
 checked-staged-adversary-acceptance-probability A ≤
 (hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (∑ i<rounds. C i) + trace-fri-error + composition-fri-error) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (F + (trace-fri-error +
 low-degree-trace-pair-agreement-error budgets)) +
 trace-fri-error + empty-query-error)
by (rule
 checked-staged-soundness-from-aligned-transcript-partial-candidate-aligned-randomization-and-empty-header-tran
 [OF false-statement partial-candidate-bound empty-header-bound])
then show ?thesis
by (simp add: algebra-simps)
qed

theorem stark-soundness-from-transcript-indexed-partial-opening-route-from-not-prefix:

fixes N empty-query-error :: prob
and C :: nat ⇒ prob
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-reduction:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 trace-fri-reachable-partial-candidate-reduction-assumption
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
and composition-fri-reduction:
 ∧ data attacker-state.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
composition-fri-reachable-partial-candidate-reduction-assumption
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*))
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* *A* *i*)
(*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at* *i*)
adversary-initial-state $\leq C$ *i*
and *not-prefix-bound*:
wp-event
(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)
(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq N$
and *current-empty-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-current-empty-query-opening-hit-at* *0*)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
hash-collision-budget-value *0*
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 $(\sum i < \text{rounds}. C\ i)$ +
trace-fri-error + *composition-fri-error* +
 $(\sum i < \text{rounds}. C\ i)$ +
(*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* *0*) +
hash-collision-budget-value *0*
(*staged-alpha-search-queries* *budgets* *0*) +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* *0*) +
((*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* *0*) +
hash-collision-budget-value *0*
(*staged-alpha-search-queries* *budgets* *0*) + *N*) +
(*trace-fri-error* +
low-degree-trace-pair-agreement-error *budgets*)) +
trace-fri-error + *empty-query-error*)
proof –
let *?M* =
checked-staged-security-experiment-with-actual-alpha-prefix-data-state *A*
let *?P* =

composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0)
have *transcript-bound*:
wp-event (checked-staged-transcript-with-alpha-prefix-program A)
(checked-staged-transcript-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state ≤ ?P
by (rule
checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
[OF wf controlled])
have *prefix-bound*:
wp-event ?M
(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state ≤ ?P
by (rule
checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
[OF transcript-bound])
have *fixed-empty-alpha-bound*:
wp-event ?M
(checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state ≤ ?P + N
proof (rule *order-trans*)
show *wp-event ?M*
(checked-staged-security-with-actual-alpha-prefix-verifier-event
composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
adversary-initial-state ≤
wp-event ?M
(checked-staged-security-with-actual-alpha-prefix-bad-set-hit
composition-trace-bad-alpha-space)
adversary-initial-state +
wp-event ?M
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state
by (rule
checked-staged-security-with-actual-alpha-prefix-fixed-empty-alpha-hit-le-prefix-or-not-prefix
[OF wf controlled])
show ... ≤ *?P + N*
by (intro *add-mono prefix-bound not-prefix-bound*)
qed
show *?thesis*
by (rule
stark-soundness-from-transcript-indexed-partial-opening-route
[OF false-statement wf controlled trace-fri-reduction])

*composition-fri-reduction round-bound fixed-empty-alpha-bound
current-empty-bound*])
qed

theorem *stark-soundness-from-transcript-indexed-partial-opening-route-from-not-prefix-single-query-charge:*
fixes *N empty-query-error :: prob*
and *C :: nat $\Rightarrow$ prob*
assumes *false-statement: $\neg$ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-fri-reduction:*
 $\bigwedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
trace-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *composition-fri-reduction:*
 $\bigwedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
composition-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *round-bound:*
 $\bigwedge i. i < \text{rounds} \Rightarrow$
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
adversary-initial-state $\leq$ C i
and *not-prefix-bound:*
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$ N
and *current-empty-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ empty-query-error
shows *checked-staged-adversary-acceptance-probability A $\leq$*
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
($\sum i < \text{rounds}. C i$) +
trace-fri-error + composition-fri-error +

```

    (composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      ((composition-error-bound +
        staged-phase-relation-error size
        (staged-alpha-search-queries budgets 0) +
        hash-collision-budget-value 0
        (staged-alpha-search-queries budgets 0) + N) +
        (trace-fri-error +
          low-degree-trace-pair-agreement-error budgets)) +
        trace-fri-error + empty-query-error)
  proof -
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?P =
    composition-error-bound +
      staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0)
  have transcript-bound:
    wp-event (checked-staged-transcript-with-alpha-prefix-program A)
      (checked-staged-transcript-actual-alpha-prefix-bad-set-hit
        composition-trace-bad-alpha-space)
    adversary-initial-state ≤ ?P
  by (rule
      checked-staged-transcript-with-alpha-prefix-actual-bad-set-hit-bound-from-budgets
      [OF wf controlled])
  have prefix-bound:
    wp-event ?M
      (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
        composition-trace-bad-alpha-space)
    adversary-initial-state ≤ ?P
  by (rule
      checked-staged-security-with-actual-alpha-prefix-bad-set-hit-bound-from-transcript
      [OF transcript-bound])
  have fixed-empty-alpha-bound:
    wp-event ?M
      (checked-staged-security-with-actual-alpha-prefix-verifier-event
        composition-alpha-bad-set-hit-with-empty-composition-header-candidates)
    adversary-initial-state ≤ ?P + N
  proof (rule order-trans)
  show wp-event ?M
    (checked-staged-security-with-actual-alpha-prefix-verifier-event
      composition-alpha-bad-set-hit-with-empty-composition-header-candidates)

```

```

adversary-initial-state ≤
wp-event ?M
  (checked-staged-security-with-actual-alpha-prefix-bad-set-hit
   composition-trace-bad-alpha-space)
adversary-initial-state +
wp-event ?M
  (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
   composition-trace-bad-alpha-space)
adversary-initial-state
by (rule
   checked-staged-security-with-actual-alpha-prefix-fixed-empty-alpha-hit-le-prefix-or-not-prefix
   [OF wf controlled])
show ... ≤ ?P + N
by (intro add-mono prefix-bound not-prefix-bound)
qed
show ?thesis
by (rule
   stark-soundness-from-transcript-indexed-partial-opening-route-single-query-charge
   [OF false-statement wf controlled trace-fri-reduction
    composition-fri-reduction round-bound fixed-empty-alpha-bound
    current-empty-bound])
qed

theorem stark-soundness-from-transcript-indexed-partial-opening-route-from-uncovered-and-transcript:
fixes U P empty-query-error :: prob
and C :: nat ⇒ prob
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-reduction:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                    adversary-initial-state) ⇒
      trace-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
         (staged-proof-transcript data))
and composition-fri-reduction:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                    adversary-initial-state) ⇒
      composition-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
         (staged-proof-transcript data))
and round-bound:
  ∧ i. i < rounds ⇒
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)

```

$(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *uncovered-bound*:
 wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index}$
 $\text{adversary-initial-state} \leq U$
and *data-pre-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-pre-hit}$
 $\text{adversary-initial-state} \leq P$
and *current-empty-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows *checked-staged-adversary-acceptance-probability* $A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds}. C \ i) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum i < \text{rounds}. C \ i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $((\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) + U +$
 $(P + \text{staged-concrete-transcript-target-error-bound}) +$
 $(P + \text{staged-concrete-transcript-target-error-bound}))) +$
 $(\text{trace-fri-error} +$
 $\text{low-degree-trace-pair-agreement-error budgets})) +$
 $\text{trace-fri-error} + \text{empty-query-error})$
proof –
have *transcript-new-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-new-hit}$
 $\text{adversary-initial-state} \leq \text{staged-concrete-transcript-target-error-bound}$
by (rule
 $\text{checked-staged-security-with-data-state-transcript-new-hit-bound}$

[OF wf controlled])
have *no-prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 adversary-initial-state $\leq$
 $P + \text{staged-concrete-transcript-target-error-bound}$
by (rule
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-bound-from-sampled-transcript
 [OF *data-pre-bound transcript-new-bound*])
have *path-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
 adversary-initial-state $\leq$
 $P + \text{staged-concrete-transcript-target-error-bound}$
by (rule
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-bound-from-sampled-transcript
 [OF *data-pre-bound transcript-new-bound*])
have *collision-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
 adversary-initial-state $\leq$
 hash-collision-budget-value 0 (*staged-alpha-search-queries budgets 0*)
by (rule
 checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
 [OF *wf controlled*])
have *not-prefix-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
 composition-trace-bad-alpha-space)
 adversary-initial-state $\leq$
 hash-collision-budget-value 0 (*staged-alpha-search-queries budgets 0*) +
 $U + (P + \text{staged-concrete-transcript-target-error-bound}) +$
 $(P + \text{staged-concrete-transcript-target-error-bound})$
by (rule
 checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound
 [OF *wf controlled collision-bound uncovered-bound no-prefix-bound*
 path-bound])
show *?thesis*
by (rule
 stark-soundness-from-transcript-indexed-partial-opening-route-from-not-prefix
 [OF *false-statement wf controlled trace-fri-reduction*
 composition-fri-reduction round-bound not-prefix-bound
 current-empty-bound])
qed

theorem *stark-soundness-from-transcript-indexed-partial-opening-route-from-uncovered-and-transcript-single-query-c*

fixes $U\ P$ *empty-query-error* :: *prob*

and C :: *nat* $\Rightarrow$ *prob*

assumes *false-statement*: $\neg$ *exists-valid-trace*

and *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *trace-fri-reduction*:

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

trace-fri-reachable-partial-candidate-reduction-assumption

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data))

and *composition-fri-reduction*:

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

composition-fri-reachable-partial-candidate-reduction-assumption

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data))

and *round-bound*:

$\bigwedge i. i < \text{rounds} \Rightarrow$

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)

adversary-initial-state $\leq C\ i$

and *uncovered-bound*:

wp-event

(checked-staged-security-experiment-with-actual-alpha-prefix-data-state

A)

checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index

adversary-initial-state $\leq U$

and *data-pre-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-transcript-pre-hit

adversary-initial-state $\leq P$

and *current-empty-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-current-empty-query-opening-hit-at 0)

adversary-initial-state $\leq$ empty-query-error

shows *checked-staged-adversary-acceptance-probability A $\leq$*

hash-collision-budget-value 0

(staged-attacker-query-budget budgets +

staged-challenge-query-budget + verifier-hash-query-budget) +

($\sum i < \text{rounds}. C\ i$) +

trace-fri-error + composition-fri-error +

(composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 ((composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 (hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) + U +
 (P + staged-concrete-transcript-target-error-bound) +
 (P + staged-concrete-transcript-target-error-bound))) +
 (trace-fri-error +
 low-degree-trace-pair-agreement-error budgets)) +
 trace-fri-error + empty-query-error)

proof –

have transcript-new-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-transcript-new-hit
 adversary-initial-state $\leq$ staged-concrete-transcript-target-error-bound

by (rule
 checked-staged-security-with-data-state-transcript-new-hit-bound
 [OF wf controlled])

have no-prefix-bound:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
 adversary-initial-state $\leq$
 P + staged-concrete-transcript-target-error-bound

by (rule
 checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-bound-from-sampled-transcript
 [OF data-pre-bound transcript-new-bound])

have path-bound:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
 adversary-initial-state $\leq$
 P + staged-concrete-transcript-target-error-bound

by (rule
 checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-bound-from-sampled-transcript
 [OF data-pre-bound transcript-new-bound])

have collision-bound:

wp-event
 (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)

```

    checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
    adversary-initial-state  $\leq$ 
    hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
    [OF wf controlled])
  have not-prefix-bound:
    wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
      composition-trace-bad-alpha-space)
    adversary-initial-state  $\leq$ 
    hash-collision-budget-value 0 (staged-alpha-search-queries budgets 0) +
    U + (P + staged-concrete-transcript-target-error-bound) +
    (P + staged-concrete-transcript-target-error-bound)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-union-bound
    [OF wf controlled collision-bound uncovered-bound no-prefix-bound
      path-bound])
  show ?thesis
  by (rule
    stark-soundness-from-transcript-indexed-partial-opening-route-from-not-prefix-single-query-charge
    [OF false-statement wf controlled trace-fri-reduction
      composition-fri-reduction round-bound not-prefix-bound
      current-empty-bound])
qed

end

end

```

```

theory Soundness-Not-Prefix-Partial-Opening
  imports Soundness-Relevant-Drift-Transcript-Public
begin

```

Same-run partial-opening packaging for the remaining not-prefix residual.

The residual is not a probability-small sparse-cover event by itself. Instead, it records the precise verifier evidence still carried by the current route: an accepting verifier run, authenticated partial trace openings from that same run, and a trace-domain index left unopened by those sampled openings. This layer gives that evidence a non-diagnostic name and keeps the broad witnessed cross/Merkle classifier out of the checked wrapper.

```

context soundness
begin

```

```

definition partial-trace-openings-leave-domain-gap
  :: 'f authenticated-opening list list  $\Rightarrow$  bool
where

```

partial-trace-openings-leave-domain-gap *trace-openings* $\longleftrightarrow$
 $(\exists j < \text{scale} * \text{clength}.$
 $\forall i \text{ opn. } i < \text{rounds} \longrightarrow$
 $\text{opn} \in \text{set } (\text{trace-openings} ! i) \longrightarrow$
 $\text{opening-index } \text{opn} \neq j)$

definition *checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap*
 $:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap
 $\text{out} \longleftrightarrow$
 $(\text{case } \text{out} \text{ of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (x, -) \Rightarrow$
 $(\text{let } \text{packed} = \text{fst } (\text{fst } x);$
 $\text{prefix-state} = \text{snd } (\text{fst } \text{packed});$
 $\text{data} = \text{snd } \text{packed};$
 $\text{attacker-state} = \text{snd } (\text{fst } x);$
 $s =$
 $\text{verifier-state-from-adversary } \text{attacker-state}$
 $(\text{staged-proof-transcript } \text{data})$
 $\text{in } \exists \text{fr } f\text{-fri-roots } f\text{-final } \text{trace-table } \text{witness-final-state}$
 trace-openings.
 $\text{alpha-header-supported-partial-trace-table-candidate-witness}$
 $s \text{ fr } f\text{-fri-roots } f\text{-final } \text{trace-table } \text{witness-final-state}$
 $\text{trace-openings} \wedge$
 $\text{trace-table} \notin$
 $\text{alpha-prefix-trace-table-candidates } \text{prefix-state } \text{fr} \wedge$
 $\text{prefix-state} \leq \text{witness-final-state} \wedge$
 $\text{partial-trace-openings-leave-domain-gap } \text{trace-openings}))$

lemma *checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap-iff-uncovered-index:*
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap}$
 $\text{out} \longleftrightarrow$
 $\text{checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index}$
 out

unfolding

checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap-def
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-def
partial-trace-openings-leave-domain-gap-def
by (*auto split: option.splits prod.splits*)

definition *checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement*
 $:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$

$\text{'f protocol-channel) option} \Rightarrow \text{bool}$
where
checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement
 $\text{out} \longleftrightarrow$
(case out of
 $\text{None} \Rightarrow \text{False}$
| $\text{Some } (x, -) \Rightarrow$
 $(\text{let packed} = \text{fst } (\text{fst } x);$
 $\text{prefix-state} = \text{snd } (\text{fst } \text{packed});$
 $\text{data} = \text{snd } \text{packed};$
 $\text{attacker-state} = \text{snd } (\text{fst } x);$
 $s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
in $\exists \text{fr f-fri-roots f-final trace-table prefix-trace-table}$
 $\text{witness-final-state trace-openings result query-idxs } i.$
 $\text{alpha-header-supported-partial-trace-table-candidate-witness}$
 $s \text{ fr f-fri-roots f-final trace-table witness-final-state}$
 $\text{trace-openings} \wedge$
 $\text{trace-table} \notin$
 $\text{alpha-prefix-trace-table-candidates prefix-state fr} \wedge$
 $\text{prefix-trace-table} \in$
 $\text{alpha-prefix-trace-table-candidates prefix-state fr} \wedge$
 $\text{prefix-state} \leq \text{witness-final-state} \wedge$
 $\text{accepted-with-partial-trace-openings } s$
 $(\text{Some } (\text{result}, \text{witness-final-state})) \text{ fr query-idxs}$
 $\text{trace-openings} \wedge$
 $i < \text{rounds} \wedge$
 $\text{query-idxs} ! i \in$
 $\text{trace-table-agreement-indices trace-table}$
 $\text{prefix-trace-table}))$

definition *checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-sampled-agreement*
 $:: \text{'f staged-adversary} \Rightarrow$
 $(((((\text{'f} \times \text{'f list} \times \text{'f list} \times \text{'f}) \times$
 $\text{'f protocol-channel}) \times \text{'f staged-proof-data}) \times$
 $\text{'f protocol-channel}) \times \text{unit list}) \times$
 $\text{'f protocol-channel) option} \Rightarrow \text{bool}$

where
checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-sampled-agreement
 $A \text{ out} \longleftrightarrow$
(case out of
 $\text{None} \Rightarrow \text{False}$
| $\text{Some } (x, -) \Rightarrow$
 $(\text{let packed} = \text{fst } (\text{fst } x);$
 $\text{prefix-state} = \text{snd } (\text{fst } \text{packed});$
 $\text{data} = \text{snd } \text{packed};$
 $\text{attacker-state} = \text{snd } (\text{fst } x);$
 $s =$

verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
in $\exists$ *fr f-fri-roots f-final trace-table prefix-trace-table*
witness-final-state trace-openings result query-idxs as i
prefix prefix-state' raw raw-state.
alpha-header-supported-partial-trace-table-candidate-witness
s fr f-fri-roots f-final trace-table witness-final-state
trace-openings $\wedge$
trace-table $\notin$
alpha-prefix-trace-table-candidates prefix-state fr $\wedge$
prefix-trace-table $\in$
alpha-prefix-trace-table-candidates prefix-state fr $\wedge$
trace-table-low-degree trace-table $\wedge$
trace-table-low-degree prefix-trace-table $\wedge$
trace-table $\neq$ *prefix-trace-table* $\wedge$
accepted-with-partial-trace-openings s
(Some (result, witness-final-state)) fr query-idxs
trace-openings $\wedge$
accepted-transcript-shape s (Some (result, witness-final-state))
as query-idxs $\wedge$
i < rounds $\wedge$
Some (data, attacker-state) \in
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state) \wedge
Some (((prefix, prefix-state'), raw), raw-state) \in
set-dist
(execute (checked-staged-query-prefix-receive-with-state A i)
adversary-initial-state) \wedge
index (to-nat raw) \in
trace-table-agreement-indices trace-table
prefix-trace-table))

definition *checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement*

$:: ((((((f \times 'f \text{ list} \times 'f \text{ list} \times 'f) \times$
 $'f \text{ protocol-channel}) \times 'f \text{ staged-proof-data}) \times$
 $'f \text{ protocol-channel}) \times \text{unit list}) \times$
 $'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement
out $\longleftrightarrow$
(case out of
None $\Rightarrow$ *False*
 $|$ *Some (((((alpha-prefix, prefix-state), data), attacker-state),*
result), final-state) \Rightarrow
(let s =
verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
in $\exists$ *fr f-fri-roots f-final trace-table prefix-trace-table*

witness-final-state trace-openings as query-idxs i.
alpha-header-supported-partial-trace-table-candidate-witness
s fr f-fri-roots f-final trace-table witness-final-state
trace-openings $\wedge$
trace-table $\notin$
alpha-prefix-trace-table-candidates prefix-state fr $\wedge$
prefix-trace-table $\in$
alpha-prefix-trace-table-candidates prefix-state fr $\wedge$
trace-table-low-degree trace-table $\wedge$
trace-table-low-degree prefix-trace-table $\wedge$
trace-table $\neq$ prefix-trace-table $\wedge$
accepted-transcript-shape s (Some (result, final-state))
as query-idxs $\wedge$
i < rounds $\wedge$
query-idxs ! i $\in$
trace-table-agreement-indices trace-table
prefix-trace-table))

lemma *checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-same-run-partial-opening-gap:*

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
adversary-initial-state $\leq$
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap
adversary-initial-state
by (rule wp-event-mono)
(simp add:
checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap-iff-uncovered-index)

lemma *checked-staged-security-with-actual-alpha-prefix-same-run-gap-imp-witness-transcript-hit:*

assumes
checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap
out
shows
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
out
proof (cases out)
case None
then show ?thesis
using *assms*
unfolding
checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap-def
by *simp*
next
case (Some packed)
then obtain *prefix prefix-state data attacker-state result final-state*
where *out-eq:*


```

    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
            final-state)
  by (cases packed, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from assms obtain fr f-fri-roots f-final trace-table
  witness-final-state trace-openings where witness:
  alpha-header-supported-partial-trace-table-candidate-witness ?s fr
  f-fri-roots f-final trace-table witness-final-state trace-openings
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap-def
  Let-def
  by (auto split: prod.splits)
from witness obtain result' query-idxs as dg composition-fri-roots final
  rest where outcome:
  Some (result', witness-final-state) ∈
    set-dist (execute verify-monad ?s)
unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
  by blast
have tr-eq: PTranscript ?s = staged-proof-transcript data
  by simp
have transcript-hit:
  hash-map-output-values ?s ∩ set (staged-proof-transcript data)
    ≠ {} ∨
  hash-map-new-output-hit (set (staged-proof-transcript data))
    ?s witness-final-state
  by (rule verify-monad-trace-root-target-preexisting-or-new
      [OF outcome tr-eq])
show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-def
    Let-def
  using outcome transcript-hit
  by (auto split: prod.splits)
qed

```

lemma *checked-staged-security-with-actual-alpha-prefix-same-run-gap-le-witness-transcript-hit:*

```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
  adversary-initial-state
by (rule wp-event-mono)
  (rule

```

checked-staged-security-with-actual-alpha-prefix-same-run-gap-imp-witness-transcript-hit)

lemma *checked-staged-security-with-actual-alpha-prefix-same-run-gap-bound-from-pre-and-new:*

assumes *pre-bound:*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit

adversary-initial-state $\leq P$

and *new-bound:*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit

adversary-initial-state $\leq Q$

shows

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap

adversary-initial-state $\leq P + Q$

by (*rule order-trans*

[*OF*

checked-staged-security-with-actual-alpha-prefix-same-run-gap-le-witness-transcript-hit

checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new

[*OF pre-bound new-bound*]])

lemma *checked-staged-security-with-actual-alpha-prefix-same-run-gap-bound-from-data-pre-and-new:*

assumes *data-pre-bound:*

wp-event (*checked-staged-security-experiment-with-data-state* A)

staged-security-with-data-state-transcript-pre-hit

adversary-initial-state $\leq P$

and *new-bound:*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit

adversary-initial-state $\leq Q$

shows

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
A)

checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap

adversary-initial-state $\leq P + Q$

proof –

have *pre-bound:*

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* A)

checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit

```

    adversary-initial-state  $\leq P$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit-bound-from-data-state-pre-hit
    [OF data-pre-bound])
  show ?thesis
  by (rule
    checked-staged-security-with-actual-alpha-prefix-same-run-gap-bound-from-pre-and-new
    [OF pre-bound new-bound])
qed

lemma checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit-bound-from-sampled-transcript:
  assumes pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state  $\leq P$ 
  and new-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-new-hit
    adversary-initial-state  $\leq R$ 
  shows
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
      adversary-initial-state  $\leq P + R$ 
proof -
  have event-le:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
      adversary-initial-state  $\leq$ 
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
      checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
      adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support:
      out  $\in$ 
      set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
        adversary-initial-state)
    and hit:
      checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
      out
    show
      checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
      out
  proof (cases out)

```

```

case None
then show ?thesis
  using hit
  unfolding
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state)
  by (cases packed) (auto split: prod.splits)
show ?thesis
  unfolding out-eq
  by (rule
    checked-staged-security-with-actual-alpha-prefix-some-support-imp-sampled-transcript-hit
    (rule support[unfolded out-eq])
  )
qed
qed
have sampled-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
    adversary-initial-state  $\leq P + R$ 
  by (rule
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit-bound-from-data-state
    [OF pre-bound new-bound])
show ?thesis
  by (rule order-trans[OF event-le sampled-bound])
qed

lemma checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-same-run
fixes G P R :: prob
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and gap-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap
    adversary-initial-state  $\leq G$ 
and data-pre-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state  $\leq P$ 
and transcript-new-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)

```

staged-security-with-data-state-transcript-new-hit
adversary-initial-state $\leq R$

shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$
hash-collision-budget-value 0
 (*staged-alpha-search-queries* budgets 0) + *G* + (*P* + *R*) + (*P* + *R*)

proof –

have *collision-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
adversary-initial-state $\leq$
hash-collision-budget-value 0 (*staged-alpha-search-queries* budgets 0)

by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-bound
 [OF *wf controlled*])

have *uncovered-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index
adversary-initial-state $\leq G$

by (rule *order-trans*
 [OF
checked-staged-security-with-actual-alpha-prefix-not-prefix-uncovered-index-le-same-run-partial-opening-gap
gap-bound])

have *no-prefix-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
adversary-initial-state $\leq P + R$

by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-bound-from-sampled-transcript
 [OF *data-pre-bound transcript-new-bound*])

have *path-bound*:

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state* *A*)
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
adversary-initial-state $\leq P + R$

by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-bound-from-sampled-transcript
 [OF *data-pre-bound transcript-new-bound*])

show ?thesis

by (rule
 checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-not-prefix-com
 [*OF wf controlled collision-bound uncovered-bound no-prefix-bound*
 path-bound])
 qed

lemma *checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-same-run*
 fixes $P\ Q\ R :: \text{prob}$
 assumes *wf*: *staged-budget-wellformed budgets*
 and *controlled*: *staged-adversary-controlled budgets A*
 and *data-pre-bound*:
 wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-transcript-pre-hit
 adversary-initial-state $\leq P$
 and *witness-new-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
 adversary-initial-state $\leq Q$
 and *transcript-new-bound*:
 wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-transcript-new-hit
 adversary-initial-state $\leq R$
 shows
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
 adversary-initial-state $\leq$
 hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) + ($P + Q$) + ($P + R$) +
 ($P + R$)
 proof –
 have *gap-bound*:
 wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
 checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap
 adversary-initial-state $\leq P + Q$
 by (rule
 checked-staged-security-with-actual-alpha-prefix-same-run-gap-bound-from-data-pre-and-new
 [*OF data-pre-bound witness-new-bound*])
 show ?thesis
 by (rule
 checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-same-run-gap
 [*OF wf controlled gap-bound data-pre-bound transcript-new-bound*])
 qed

theorem *stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-from-same-*

fixes G P *empty-query-error* :: *prob*
and C :: *nat* $\Rightarrow$ *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
trace-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *composition-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
composition-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \Rightarrow$
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
adversary-initial-state $\leq C$ i
and *gap-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state
A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap
adversary-initial-state $\leq G$
and *data-pre-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$
and *current-empty-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ empty-query-error
shows *checked-staged-adversary-acceptance-probability A $\leq$*
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
($\sum i < \text{rounds}. C$ i) +
trace-fri-error + composition-fri-error +
($\sum i < \text{rounds}. C$ i) +
(composition-error-bound +

staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) +
 (*hash-collision-budget-value 0*
 (*staged-alpha-search-queries budgets 0*) + *G* +
 (*P* + *staged-concrete-transcript-target-error-bound*) +
 (*P* + *staged-concrete-transcript-target-error-bound*)) +
staged-phase-relation-error size
 (*staged-alpha-search-queries budgets 0*) +
trace-fri-error + *empty-query-error*)
proof –
have *trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-partial-candidates)
adversary-initial-state ≤ *trace-fri-error*
by (rule
trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction
 [*OF trace-fri-reduction*])
have *comp-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-partial-candidates)
adversary-initial-state ≤ *composition-fri-error*
by (rule
composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction
 [*OF composition-fri-reduction*])
have *empty-trace-fri-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤ *trace-fri-error*
by (rule
trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
 [*OF trace-fri-reduction*])
have *transcript-new-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-transcript-new-hit
adversary-initial-state ≤ *staged-concrete-transcript-target-error-bound*
by (rule *checked-staged-security-with-data-state-transcript-new-hit-bound*
 [*OF wf controlled*])
have *drift-bound*:
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state ≤
hash-collision-budget-value 0
 (*staged-alpha-search-queries budgets 0*) + *G* +


```

(P + staged-concrete-transcript-target-error-bound) +
(P + staged-concrete-transcript-target-error-bound)
by (rule
  checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-same-run-gap
  [OF wf controlled gap-bound data-pre-bound transcript-new-bound])
have base:
  checked-staged-adversary-acceptance-probability A ≤
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
   staged-challenge-query-budget + verifier-hash-query-budget) +
  (∑ i<rounds. C i) +
  trace-fri-error + composition-fri-error +
  (∑ i<rounds. C i) +
  (composition-error-bound +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
   hash-collision-budget-value 0
   (staged-alpha-search-queries budgets 0) +
   staged-phase-relation-error size
   (staged-alpha-search-queries budgets 0) +
   (hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) + G +
    (P + staged-concrete-transcript-target-error-bound) +
    (P + staged-concrete-transcript-target-error-bound))) +
  trace-fri-error + empty-query-error)
by (rule
  checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix-ro
  [OF false-statement wf controlled trace-fri-bound comp-fri-bound
   round-bound drift-bound empty-trace-fri-bound current-empty-bound])
then show ?thesis
  by (simp add: algebra-simps)
qed

```

```

lemma powers-scaled-domain-bound-imp-query-sample-space:
  assumes bounded:
    ∧j. j ∈ set (powers-scaled idx) ⇒ j < clength * scale
  shows idx ∈ query-sample-space
proof -
  let ?m = Max (set [0..<powers])
  have m-in: ?m ∈ set [0..<powers]
  using powers-pos by (intro Max-in) simp-all
  have m-eq: ?m = powers - 1
  using powers-pos by (intro Max-eqI) auto
  have idx-m-bound: idx + ?m * scale < clength * scale
  by (rule bounded) (use m-in in ⟨auto simp: powers-scaled-def⟩)
  have m-le: ?m * scale ≤ clength * scale
proof -
  have ?m < clength
  using m-eq powers-le-clength powers-pos by simp

```

```

    then show ?thesis
      by (simp add: mult-le-mono)
qed
have idx < clength * scale - ?m * scale
  using idx-m-bound m-le by linarith
then show ?thesis
  unfolding query-sample-space-def query-sample-space-size-def by simp
qed

```

lemma *partial-authenticated-trace-openings-query-index-in-sample-space:*

```

  assumes indices:
    map opening-index openings = powers-scaled idx
  and auth:
    partial-authenticated-table fr (scale * clength) openings s
  shows idx ∈ query-sample-space
proof (rule powers-scaled-domain-bound-imp-query-sample-space)
  fix j
  assume j-in: j ∈ set (powers-scaled idx)
  have j ∈ opening-index ‘ set openings
    by (metis indices j-in set-map)
  then obtain opn where opn-in: opn ∈ set openings
    and j-eq: j = opening-index opn
    by blast
  have opening-index opn < opening-length opn
    using auth opn-in unfolding partial-authenticated-table-def
      authenticated-opening-in-def by blast
  moreover have opening-length opn = scale * clength
    using auth opn-in unfolding partial-authenticated-table-def by blast
  ultimately show j < clength * scale
    using j-eq by (simp add: mult.commute)
qed

```

lemma *accepted-partial-trace-openings-leave-domain-gap-if-sparse-queries:*

```

  assumes partial:
    accepted-with-partial-trace-openings s out fr query-idxs trace-openings
  and sparse: rounds * powers < scale * clength
  shows partial-trace-openings-leave-domain-gap trace-openings
proof –
  have rounds-len: length trace-openings = rounds
    by (rule accepted-with-partial-trace-openings-shapes(2)[OF partial])
  have concat-sparse:
    length (List.concat trace-openings) < scale * clength
    using accepted-with-partial-trace-openings-concat-length[OF partial]
      sparse by simp
  obtain j where j-bound: j < scale * clength
    and unopened:
       $\bigwedge i \text{ opn. } i < \text{rounds} \implies$ 
       $\text{opn} \in \text{set } (\text{trace-openings } ! i) \implies$ 
       $\text{opening-index opn} \neq j$ 

```

```

    using sparse-openings-obtain-unopened-index[OF rounds-len concat-sparse]
    by blast
show ?thesis
  unfolding partial-trace-openings-leave-domain-gap-def
  using j-bound unopened by blast
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap-if-sparse-queries:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and sparse: rounds * powers < scale * clength
  and support:
    out ∈
      set-dist
        (execute
          (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
            A)
          adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
    bad-sets out
  shows
    checked-staged-security-with-actual-alpha-prefix-not-prefix-collision out ∨
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate out ∨
    checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output out ∨
    checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap out
proof (cases out)
case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound-def
    by simp
next
case (Some packed)
  then obtain prefix prefix-state data attacker-state result final-state
    where out-eq:
      out =
        Some (((((prefix, prefix-state), data), attacker-state), result),
          final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from hit obtain fr f-fri-roots f-final as where not-prefix:
    alpha-header-partial-candidate-not-prefix-bound prefix-state ?s fr
    f-fri-roots f-final
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound-def

```

```

    Let-def
  by (auto split: option.splits prod.splits)
from checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-obtain-witness-final-ext
  [OF wf controlled support[unfolded out-eq] not-prefix]
obtain trace-table witness-final-state trace-openings where witness:
  alpha-header-supported-partial-trace-table-candidate-witness ?s fr
  f-fri-roots f-final trace-table witness-final-state trace-openings
and table-not-prefix:
  trace-table  $\notin$  alpha-prefix-trace-table-candidates prefix-state fr
and prefix-ext: prefix-state  $\leq$  witness-final-state
by blast
from witness obtain result' query-idxs as' dg composition-fri-roots final rest
where partial:
  accepted-with-partial-trace-openings ?s
  (Some (result', witness-final-state)) fr query-idxs trace-openings
unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
by blast
have gap:
  partial-trace-openings-leave-domain-gap trace-openings
by (rule accepted-partial-trace-openings-leave-domain-gap-if-sparse-queries
    [OF partial sparse])
have
  checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap
  out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap-def
  Let-def
using witness table-not-prefix prefix-ext gap
by (simp split: prod.splits, blast)
then show ?thesis by simp
qed

lemma alpha-header-witness-prefix-candidate-values-agree-on-openings:
assumes prefix-candidate:
  prefix-trace-table  $\in$ 
  alpha-prefix-trace-table-candidates prefix-state fr
and witness:
  alpha-header-supported-partial-trace-table-candidate-witness s fr
  f-fri-roots f-final trace-table final-state trace-openings
and pullback:
  partial-authenticated-table fr (scale * clength)
  (trace-openings ! i) prefix-state
and clean-prefix:  $\neg$  hash-map-output-collision prefix-state
and i-bound:  $i < \text{rounds}$ 
shows
  map (! trace-table) (map opening-index (trace-openings ! i)) =
  map (! prefix-trace-table) (map opening-index (trace-openings ! i))
proof –
have prefix-len:

```

```

length prefix-trace-table = clength * scale
using prefix-candidate
unfolding alpha-prefix-trace-table-candidates-def by simp
have prefix-bind:
merkle-root-binds-table fr prefix-trace-table prefix-state
using prefix-candidate
unfolding alpha-prefix-trace-table-candidates-def by simp
from witness obtain result query-idxs as dg composition-fri-roots final rest
where trace-cand:
partial-trace-table-candidate trace-table trace-openings
unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
by blast
have trace-agrees:
table-agrees-with-authenticated-openings trace-table
(scale * clength) (trace-openings ! i)
using trace-cand i-bound
unfolding partial-trace-table-candidate-def by blast
have point-eq:
 $\bigwedge \text{opn. } \text{opn} \in \text{set } (\text{trace-openings} ! i) \implies$ 
trace-table ! opening-index opn =
prefix-trace-table ! opening-index opn
proof -
fix opn
assume opn-in: opn  $\in$  set (trace-openings ! i)
have trace-value:
trace-table ! opening-index opn = opening-value opn
using trace-agrees opn-in
unfolding table-agrees-with-authenticated-openings-def by blast
have root: opening-root opn = fr
using pullback opn-in unfolding partial-authenticated-table-def
by blast
have len: opening-length opn = length prefix-trace-table
using pullback opn-in prefix-len
unfolding partial-authenticated-table-def by simp
have auth: authenticated-opening-in prefix-state opn
using pullback opn-in unfolding partial-authenticated-table-def
by blast
obtain n where len-pow: clength * scale =  $2^n$ 
using eval-domain-length-power by blast
have table-pow: length prefix-trace-table =  $2^n$ 
using prefix-len len-pow by (simp add: mult.commute)
have prefix-value:
prefix-trace-table ! opening-index opn = opening-value opn
using
merkle-bound-table-agrees-with-authenticated-opening-if-clean
[OF prefix-bind table-pow auth root len clean-prefix]
by simp
show
trace-table ! opening-index opn =

```

```

    prefix-trace-table ! opening-index opn
  using trace-value prefix-value by simp
qed
show ?thesis
  by (rule nth-equalityI)
    (use point-eq in auto)
qed

lemma alpha-header-witness-prefix-candidate-trace-table-agreement-index:
  assumes prefix-candidate:
    prefix-trace-table ∈
      alpha-prefix-trace-table-candidates prefix-state fr
  and witness:
    alpha-header-supported-partial-trace-table-candidate-witness s fr
    f-fri-roots f-final trace-table final-state trace-openings
  and pullback:
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! i) prefix-state
  and clean-prefix: ¬ hash-map-output-collision prefix-state
  and i-bound: i < rounds
  and idx-sample: idx ∈ query-sample-space
  and idx-openings:
    map opening-index (trace-openings ! i) = powers-scaled idx
  shows idx ∈ trace-table-agreement-indices trace-table prefix-trace-table
proof -
  have agree:
    map ((! trace-table) (map opening-index (trace-openings ! i))) =
    map ((! prefix-trace-table) (map opening-index (trace-openings ! i)))
  by (rule alpha-header-witness-prefix-candidate-values-agree-on-openings
    [OF prefix-candidate witness pullback clean-prefix i-bound])
  show ?thesis
    unfolding trace-table-agreement-indices-def
    using idx-sample agree idx-openings by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-imp-sampled-agreement-on-support:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    out ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
          A)
        adversary-initial-state)
  and hit:
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
    bad-sets out
  shows

```

```

checked-staged-security-with-actual-alpha-prefix-not-prefix-collision out ∨
checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate out ∨
checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output out ∨
checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state data attacker-state result final-state
    where out-eq:
      out =
        Some (((((prefix, prefix-state), data), attacker-state), result),
              final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  from hit obtain fr f-fri-roots f-final as where not-prefix-bound:
    alpha-header-partial-candidate-not-prefix-bound prefix-state ?s fr
    f-fri-roots f-final
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound-def
    Let-def
  by (auto split: option.splits prod.splits)
  from checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-obtain-witness-final-ext
    [OF wf controlled support[unfolded out-eq] not-prefix-bound]
  obtain trace-table witness-final-state trace-openings where witness:
    alpha-header-supported-partial-trace-table-candidate-witness ?s fr
    f-fri-roots f-final trace-table witness-final-state trace-openings
  and table-not-prefix:
    trace-table ∉ alpha-prefix-trace-table-candidates prefix-state fr
  and prefix-ext: prefix-state ≤ witness-final-state
  by blast
  show ?thesis
  proof (cases hash-map-output-collision prefix-state)
    case True
    then show ?thesis
      unfolding out-eq
        checked-staged-security-with-actual-alpha-prefix-not-prefix-collision-def
      by simp
  next
    case clean-prefix: False
    show ?thesis
    proof (cases alpha-prefix-trace-table-candidates prefix-state fr = {})

```

```

case True
then have
  checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
    out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate-def
    Let-def
  using witness table-not-prefix prefix-ext by (simp, blast)
then show ?thesis by simp
next
case nonempty: False
then obtain prefix-trace-table where prefix-candidate:
  prefix-trace-table  $\in$ 
    alpha-prefix-trace-table-candidates prefix-state fr
  by blast
have pullback-or-new:
  ( $\forall i < \text{rounds}.$ 
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! i prefix-state)  $\vee$ 
    ( $\exists i < \text{rounds}.$   $\exists \text{opn} \in \text{set } (\text{trace-openings ! } i).$ 
      hash-map-new-output-hit
      (merkle-path-target-roots witness-final-state fr
        (scale * clength) (opening-index opn)
        (opening-value opn) (opening-path opn))
      prefix-state witness-final-state)
    by (rule alpha-header-witness-openings-pullback-or-new-output-hit
      [OF prefix-ext witness])
  from pullback-or-new show ?thesis
proof
  assume pullback-all:
     $\forall i < \text{rounds}.$ 
      partial-authenticated-table fr (scale * clength)
      (trace-openings ! i prefix-state)
  from witness obtain witness-result query-idxs as dg
    composition-fri-roots final rest where partial:
    accepted-with-partial-trace-openings ?s
    (Some (witness-result, witness-final-state)) fr query-idxs
    trace-openings
  unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
  by blast
have i-bound: 0 < rounds
  by (rule rounds-positive)
have idx-openings:
  map opening-index (trace-openings ! 0) =
    powers-scaled (query-idxs ! 0)
  by (rule accepted-with-partial-trace-openings-shapes(3)
    [OF partial i-bound])
have pullback0:
  partial-authenticated-table fr (scale * clength)

```



```

    (trace-openings ! 0) prefix-state
  using pullback-all i-bound by blast
have idx-sample: query-idxs ! 0 ∈ query-sample-space
  by (rule partial-authenticated-trace-openings-query-index-in-sample-space
    [OF idx-openings pullback0])
have agreement:
  query-idxs ! 0 ∈
    trace-table-agreement-indices trace-table prefix-trace-table
  by (rule
    alpha-header-witness-prefix-candidate-trace-table-agreement-index
    [OF prefix-candidate witness pullback0 clean-prefix i-bound
      idx-sample idx-openings])
have
  checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement
    out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement-def
    Let-def
  using witness table-not-prefix prefix-candidate prefix-ext partial
    i-bound agreement
  by (simp split: prod.splits, blast)
then show ?thesis by simp
next
assume new:
  ∃ i < rounds. ∃ opn ∈ set (trace-openings ! i).
    hash-map-new-output-hit
      (merkle-path-target-roots witness-final-state fr
        (scale * clength) (opening-index opn)
        (opening-value opn) (opening-path opn))
      prefix-state witness-final-state
have
  checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
    out
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output-def
    Let-def
  using witness table-not-prefix prefix-ext new
  by (simp split: prod.splits, blast)
then show ?thesis by simp
qed
qed
qed
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-sampled-agreement-union-bound:
  fixes C S N P :: prob
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and collision-bound:

```

```

wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
   A)
  checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
  adversary-initial-state  $\leq C$ 
and sampled-agreement-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
     A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement
    adversary-initial-state  $\leq S$ 
and no-prefix-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
     A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
    adversary-initial-state  $\leq N$ 
and path-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
     A)
    checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
    adversary-initial-state  $\leq P$ 
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
     A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
     bad-sets)
    adversary-initial-state  $\leq C + S + N + P$ 
proof –
  let ?M =
    checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
  let ?B =
    checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
    bad-sets
  let ?C = checked-staged-security-with-actual-alpha-prefix-not-prefix-collision
  let ?S =
    checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement
  let ?N =
    checked-staged-security-with-actual-alpha-prefix-not-prefix-no-prefix-candidate
  let ?P = checked-staged-security-with-actual-alpha-prefix-not-prefix-path-output
  have wp-event ?M ?B adversary-initial-state  $\leq$ 
    wp-event ?M ( $\lambda out. ?C out \vee ?S out \vee ?N out \vee ?P out$ )
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out  $\in$  set-dist (execute ?M adversary-initial-state)

```

```

    and hit: ?B out
  from checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-imp-sampled-agreement-on-support
    [OF wf controlled support hit]
  show ?C out  $\vee$  ?S out  $\vee$  ?N out  $\vee$  ?P out
  by blast
qed
also have ...  $\leq$ 
  wp-event ?M ?C adversary-initial-state +
  wp-event ?M ?S adversary-initial-state +
  wp-event ?M ?N adversary-initial-state +
  wp-event ?M ?P adversary-initial-state
  by (rule wp-event-union-bound4)
also have ...  $\leq C + S + N + P$ 
  by (intro add-mono collision-bound sampled-agreement-bound
    no-prefix-bound path-bound)
finally show ?thesis .
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement-imp-witness-transcript-hit:
  assumes
    checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement
    out
  shows
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
    out
proof (cases out)
case None
  then show ?thesis
  using assms
  unfolding
    checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement-def
  by simp
next
case (Some packed)
  then obtain prefix prefix-state data attacker-state result final-state
  where out-eq:
    out =
      Some (((((prefix, prefix-state), data), attacker-state), result),
        final-state)
  by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from assms obtain fr f-fri-roots f-final trace-table prefix-trace-table
    witness-final-state trace-openings witness-result query-idxs i where
    witness:
      alpha-header-supported-partial-trace-table-candidate-witness ?s fr
      f-fri-roots f-final trace-table witness-final-state trace-openings
  unfolding out-eq

```

```

    checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement-def
    Let-def
    by (auto split: prod.splits)
from witness obtain result' query-idxs' as dg composition-fri-roots final
    rest where outcome:
    Some (result', witness-final-state) ∈
    set-dist (execute verify-monad ?s)
    unfolding alpha-header-supported-partial-trace-table-candidate-witness-def
    by blast
have tr-eq: PTranscript ?s = staged-proof-transcript data
    by simp
have transcript-hit:
    hash-map-output-values ?s ∩ set (staged-proof-transcript data)
    ≠ {} ∨
    hash-map-new-output-hit (set (staged-proof-transcript data))
    ?s witness-final-state
    by (rule verify-monad-trace-root-target-preexisting-or-new
    [OF outcome tr-eq])
show ?thesis
    unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-def
    Let-def
    using outcome transcript-hit
    by (auto split: prod.splits)
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement-le-witness-transcript-hit:
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement
  adversary-initial-state ≤
wp-event
  (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit
  adversary-initial-state
by (rule wp-event-mono)
  (rule
    checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement-imp-witness-transcript-hit)

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement-bound-from-pre-and-new:
assumes pre-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
      A)
    checked-staged-security-with-actual-alpha-prefix-witness-transcript-pre-hit
    adversary-initial-state ≤ P
and new-bound:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state

```

A)
checked-staged-security-with-actual-alpha-prefix-witness-transcript-new-hit
adversary-initial-state $\leq Q$

shows

wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement
adversary-initial-state $\leq P + Q$

by (rule *order-trans*
 [OF
checked-staged-security-with-actual-alpha-prefix-not-prefix-sampled-agreement-le-witness-transcript-hit
checked-staged-security-with-actual-alpha-prefix-witness-transcript-hit-bound-from-pre-and-new
 [OF *pre-bound new-bound*]])

lemma *checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-imp-sampled-transcript-hit-on-support*:
assumes *support*:
out $\in$
set-dist
 (execute
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state*
 A)
adversary-initial-state)
and *hit*:
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
bad-sets out

shows
checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
out

proof (cases *out*)
case *None*
then show *?thesis*
using *hit*
unfolding
checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound-def
by *simp*

next
case (Some *packed*)
then obtain *prefix prefix-state data attacker-state result final-state*
where *out-eq*:
out =
 Some (((((*prefix*, *prefix-state*), *data*), *attacker-state*), *result*),
final-state)
by (cases *packed*, auto split: *prod.splits*)
let *?s* =
verifier-state-from-adversary attacker-state
 (staged-proof-transcript *data*)
have *data-support*:
 Some (((*data*, *attacker-state*), *result*), *final-state*) $\in$

```

    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
  by (rule
    checked-staged-security-with-actual-alpha-prefix-support-imp-data-state-support
    [OF support[unfolded out-eq]])
  have verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  using checked-staged-security-experiment-with-data-state-outcomeE
    [OF data-support]
  by blast
  have tr-eq: PTranscript ?s = staged-proof-transcript data
  by simp
  have split:
    hash-map-output-values ?s ∩ set (staged-proof-transcript data)
      ≠ {} ∨
    hash-map-new-output-hit (set (staged-proof-transcript data))
      ?s final-state
  by (rule verify-monad-trace-root-target-preexisting-or-new
    [OF verifier tr-eq])
  show ?thesis
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit-def
    staged-security-with-data-state-transcript-pre-hit-def
    staged-security-with-data-state-transcript-new-hit-def
  using split by simp
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-le-sampled-transcript-hit:
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    (checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
      bad-sets)
  adversary-initial-state ≤
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
    checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit
    adversary-initial-state
  by (rule wp-event-mono-on-support)
  (rule
    checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-imp-sampled-transcript-hit-on-support)

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-from-sampled-transcript:
  assumes pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-transcript-pre-hit
      adversary-initial-state ≤ P
  and new-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

staged-security-with-data-state-transcript-new-hit
adversary-initial-state $\leq R$
shows
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
bad-sets)
adversary-initial-state $\leq P + R$
by (rule *order-trans*
 [*OF*
checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-le-sampled-transcript-hit
checked-staged-security-with-actual-alpha-prefix-sampled-transcript-hit-bound-from-data-state
 [*OF pre-bound new-bound*]])

lemma *checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-sampled-t*
fixes *P R :: prob*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *data-pre-bound:*
wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$
and *transcript-new-bound:*
wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-transcript-new-hit
adversary-initial-state $\leq R$
shows
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq P + R$

proof –
have *not-prefix-bound:*
wp-event
 (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
 (*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*
composition-trace-bad-alpha-space)
adversary-initial-state $\leq P + R$
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-from-sampled-transcript
 [*OF data-pre-bound transcript-new-bound*])
show *?thesis*
by (rule *order-trans*
 [*OF*
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-le-not-prefix-bound
not-prefix-bound])
 (use *wf controlled* **in** *simp-all*)
qed

theorem *stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-from-sampling*
fixes P *empty-query-error* :: *prob*
and C :: *nat* $\Rightarrow$ *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
trace-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *composition-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
composition-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \Rightarrow$
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
adversary-initial-state $\leq C i$
and *data-pre-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$
and *current-empty-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ empty-query-error
shows *checked-staged-adversary-acceptance-probability A $\leq$*
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
($\sum i < \text{rounds}. C i$) +
trace-fri-error + composition-fri-error +
($\sum i < \text{rounds}. C i$) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size

$(\text{staged-alpha-search-queries budgets } 0) +$
 $(P + \text{staged-concrete-transcript-target-error-bound}) +$
 $\text{trace-fri-error} + \text{empty-query-error})$

proof –

have *trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *trace-fri-error*

by (*rule*
trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction
 $[OF \text{ trace-fri-reduction}]$)

have *comp-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-partial-candidates)
adversary-initial-state $\leq$ *composition-fri-error*

by (*rule*
composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction
 $[OF \text{ composition-fri-reduction}]$)

have *empty-trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *trace-fri-error*

by (*rule*
trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction
 $[OF \text{ trace-fri-reduction}]$)

have *transcript-new-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-transcript-new-hit
adversary-initial-state $\leq$ *staged-concrete-transcript-target-error-bound*

by (*rule* *checked-staged-security-with-data-state-transcript-new-hit-bound*
 $[OF \text{ wf controlled}]$)

have *drift-bound*:

wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$
 $P + \text{staged-concrete-transcript-target-error-bound}$

by (*rule*
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-sampled-trans
 $[OF \text{ wf controlled data-pre-bound transcript-new-bound}]$)

show *?thesis*

by (*rule*
checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix-ro
 $[OF \text{ false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\text{round-bound drift-bound empty-trace-fri-bound current-empty-bound}]$)

qed

theorem *stark-soundness-from-aligned-partial-components-current-query-prefix-and-current-empty-query-from-sample*

fixes P *empty-query-error* :: *prob*

and C :: *nat* $\Rightarrow$ *prob*

assumes *false-statement*: $\neg$ *exists-valid-trace*

and *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *trace-fri-reduction*:

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

trace-fri-reachable-partial-candidate-reduction-assumption

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data))

and *composition-fri-reduction*:

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

composition-fri-reachable-partial-candidate-reduction-assumption

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data))

and *round-bound*:

$\bigwedge i. i < \text{rounds} \Rightarrow$

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A i)

(checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)

adversary-initial-state $\leq C i$

and *data-pre-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-transcript-pre-hit

adversary-initial-state $\leq P$

and *current-empty-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-current-empty-query-opening-hit-at 0)

adversary-initial-state $\leq$ empty-query-error

shows *checked-staged-adversary-acceptance-probability A $\leq$*

hash-collision-budget-value 0

(staged-attacker-query-budget budgets +

staged-challenge-query-budget + verifier-hash-query-budget) +

($\sum i < \text{rounds}. C i$) +

trace-fri-error + composition-fri-error +

(composition-error-bound +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +

hash-collision-budget-value 0

(staged-alpha-search-queries budgets 0) +

staged-phase-relation-error size

$(\text{staged-alpha-search-queries budgets } 0) +$
 $(P + \text{staged-concrete-transcript-target-error-bound}) +$
 $\text{trace-fri-error} + \text{empty-query-error})$

proof –

have *trace-fri-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$

by (rule
 $\text{trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction}$
 $[\text{OF trace-fri-reduction}])$

have *comp-fri-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}$

by (rule
 $\text{composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction}$
 $[\text{OF composition-fri-reduction}])$

have *empty-trace-fri-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$

by (rule
 $\text{trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction}$
 $[\text{OF trace-fri-reduction}])$

have *transcript-new-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-new-hit}$
 $\text{adversary-initial-state} \leq \text{staged-concrete-transcript-target-error-bound}$

by (rule $\text{checked-staged-security-with-data-state-transcript-new-hit-bound}$
 $[\text{OF wf controlled}])$

have *drift-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift}$
 $\text{adversary-initial-state} \leq$
 $P + \text{staged-concrete-transcript-target-error-bound}$

by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-sampled-trans}$
 $[\text{OF wf controlled data-pre-bound transcript-new-bound}])$

show *?thesis*

by (rule
 $\text{checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-current-query-prefix-ro}$
 $[\text{OF false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\text{round-bound drift-bound empty-trace-fri-bound current-empty-bound}])$

qed

theorem *stark-soundness-from-aligned-partial-components-structured-query-and-current-empty-query-from-sampled-t*
fixes P *empty-query-error* :: *prob*
and C :: *nat* $\Rightarrow$ *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \Rightarrow$
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *composition-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \Rightarrow$
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *round-bound*:
 $\bigwedge i. i < \text{rounds} \Rightarrow$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at } i)$
 $\text{adversary-initial-state} \leq C \ i$
and *data-pre-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-pre-hit}$
 $\text{adversary-initial-state} \leq P$
and *current-empty-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state} \leq \text{empty-query-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\sum i < \text{rounds}. C \ i) +$
 $\text{trace-fri-error} + \text{composition-fri-error} + 1 +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$

$(\text{staged-alpha-search-queries budgets } 0) +$
 $(P + \text{staged-concrete-transcript-target-error-bound}) +$
 $\text{trace-fri-error} + \text{empty-query-error})$

proof –

have *trace-fri-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$

by (rule
 $\text{trace-fri-bad-with-partial-candidates-bound-from-reachable-reduction}$
 $[\text{OF trace-fri-reduction}])$

have *comp-fri-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-partial-candidates})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}$

by (rule
 $\text{composition-fri-bad-with-partial-candidates-bound-from-reachable-reduction}$
 $[\text{OF composition-fri-reduction}])$

have *empty-trace-fri-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$

by (rule
 $\text{trace-fri-bad-with-empty-header-candidates-bound-from-reachable-reduction}$
 $[\text{OF trace-fri-reduction}])$

have *transcript-new-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-new-hit}$
 $\text{adversary-initial-state} \leq \text{staged-concrete-transcript-target-error-bound}$

by (rule $\text{checked-staged-security-with-data-state-transcript-new-hit-bound}$
 $[\text{OF wf controlled}])$

have *drift-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state } A)$
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift}$
 $\text{adversary-initial-state} \leq$
 $P + \text{staged-concrete-transcript-target-error-bound}$

by (rule
 $\text{checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-sampled-trans}$
 $[\text{OF wf controlled data-pre-bound transcript-new-bound}])$

have *header-query-bound*:

wp-event
 $(\text{checked-staged-security-experiment-with-actual-alpha-prefix-data-state}$
 $A)$
 $(\text{checked-staged-security-with-actual-alpha-prefix-header-authenticated-candidate-opening-query-hit-from-prefix}$
 $A)$

```

    adversary-initial-state  $\leq$  (1::prob)
  by (rule wp-event-le-1)
show ?thesis
  by (rule
    checked-staged-soundness-from-aligned-transcript-aligned-randomization-components-and-structured-query-from-
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
      round-bound header-query-bound drift-bound empty-trace-fri-bound
      current-empty-bound])
qed

theorem stark-soundness-from-transcript-indexed-partial-opening-route-from-sampled-transcript:
  fixes P empty-query-error :: prob
  and C :: nat  $\Rightarrow$  prob
  assumes false-statement:  $\neg$  exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-reduction:
     $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      trace-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
  and composition-fri-reduction:
     $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      composition-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
  and round-bound:
     $\bigwedge i. i < \text{rounds} \implies$ 
      wp-event
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
        adversary-initial-state  $\leq$  C i
  and data-pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-transcript-pre-hit
      adversary-initial-state  $\leq$  P
  and current-empty-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
      adversary-initial-state  $\leq$  empty-query-error
  shows checked-staged-adversary-acceptance-probability A  $\leq$ 
    hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +

```

$$\begin{aligned}
& \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) + \\
& (\sum i < \text{rounds}. C\ i) + \\
& \text{trace-fri-error} + \text{composition-fri-error} + \\
& (\sum i < \text{rounds}. C\ i) + \\
& (\text{composition-error-bound} + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \quad \text{hash-collision-budget-value } 0 \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& ((\text{composition-error-bound} + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \quad \text{hash-collision-budget-value } 0 \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \quad (P + \text{staged-concrete-transcript-target-error-bound})) + \\
& (\text{trace-fri-error} + \\
& \quad \text{low-degree-trace-pair-agreement-error budgets})) + \\
& \text{trace-fri-error} + \text{empty-query-error})
\end{aligned}$$

proof –

have *sampld-transcript-new-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

staged-security-with-data-state-transcript-new-hit

adversary-initial-state ≤ *staged-concrete-transcript-target-error-bound*

by (*rule*

checked-staged-security-with-data-state-transcript-new-hit-bound

[*OF wf controlled*])

have *not-prefix-bound*:

wp-event

(*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)

(*checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound*

composition-trace-bad-alpha-space)

adversary-initial-state ≤

P + staged-concrete-transcript-target-error-bound

by (*rule*

checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-from-sampled-transcript

[*OF data-pre-bound sampld-transcript-new-bound*])

show *?thesis*

by (*rule*

stark-soundness-from-transcript-indexed-partial-opening-route-from-not-prefix

[*OF false-statement wf controlled trace-fri-reduction*

composition-fri-reduction round-bound not-prefix-bound

current-empty-bound])

qed

lemma *checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-sampled-agreement-imp-prefix-pair-hit*:

assumes

checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-sampled-agreement

```

    A out
  shows
     $\exists \text{pair} \in \text{low-degree-trace-pair-space}.$ 
     $\exists i \in \{..<\text{rounds}\}.$ 
    checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-from-prefix
    A (fst pair) (snd pair) i out
  proof (cases out)
  case None
  then show ?thesis
    using assms
    unfolding
      checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-sampled-agreement-def
    by simp
  next
  case (Some packed)
  then obtain alpha-prefix alpha-prefix-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((alpha-prefix, alpha-prefix-state), data), attacker-state),
        result), final-state)
    by (cases packed, auto split: prod.splits)
  from assms obtain fr f-fri-roots f-final trace-table prefix-trace-table
    witness-final-state trace-openings witness-result query-idxs as i
    prefix prefix-state' raw raw-state where
    trace-low: trace-table-low-degree trace-table
    and prefix-low: trace-table-low-degree prefix-trace-table
    and distinct: trace-table  $\neq$  prefix-trace-table
    and i-bound:  $i < \text{rounds}$ 
    and builder:
      Some (data, attacker-state)  $\in$ 
        set-dist
          (execute (checked-staged-transcript-program A)
            adversary-initial-state)
    and prefix-receive:
      Some (((prefix, prefix-state'), raw), raw-state)  $\in$ 
        set-dist
          (execute (checked-staged-query-prefix-receive-with-state A i)
            adversary-initial-state)
    and raw-agree:
      index (to-nat raw)  $\in$ 
        trace-table-agreement-indices trace-table prefix-trace-table
  unfolding out-eq
    checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-sampled-agreement-def
    Let-def
  by (auto split: prod.splits)
  have pair-in:
    (trace-table, prefix-trace-table)  $\in$  low-degree-trace-pair-space
  unfolding low-degree-trace-pair-space-def
  using trace-low prefix-low distinct by simp

```



```

have prefix-hit:
  checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-from-prefix
  A trace-table prefix-trace-table i out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-from-prefix-def
  checked-staged-query-prefix-dynamic-index-hit-def
apply (simp add: builder)
using prefix-receive raw-agree by blast
show ?thesis
by (intro bexI[of - (trace-table, prefix-trace-table)]
  bexI[of - i])
  (use pair-in i-bound prefix-hit in simp-all)
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement-imp-pair-hit:
assumes
  checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement
  out
shows
   $\exists \text{pair} \in \text{low-degree-trace-pair-space}.$ 
   $\exists i \in \{..<\text{rounds}\}.$ 
  checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
  (fst pair) (snd pair) i out
proof (cases out)
case None
then show ?thesis
using assms
unfolding
  checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement-def
by simp
next
case (Some packed)
then obtain alpha-prefix prefix-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((alpha-prefix, prefix-state), data), attacker-state),
      result), final-state)
by (cases packed, auto split: prod.splits)
from assms obtain fr f-fri-roots f-final trace-table prefix-trace-table
  witness-final-state trace-openings as query-idxs i where
  trace-low: trace-table-low-degree trace-table
and prefix-low: trace-table-low-degree prefix-trace-table
and distinct: trace-table  $\neq$  prefix-trace-table
and shape:
  accepted-transcript-shape
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (Some (result, final-state)) as query-idxs
and i-bound: i < rounds

```

```

and idx-agree:
  query-idxs ! i ∈
    trace-table-agreement-indices trace-table prefix-trace-table
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement-def
    Let-def
by auto
have pair-in:
  (trace-table, prefix-trace-table) ∈ low-degree-trace-pair-space
unfolding low-degree-trace-pair-space-def
using trace-low prefix-low distinct by simp
have pair-hit:
  checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
    trace-table prefix-trace-table i out
unfolding out-eq
  checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at-def
  staged-security-with-data-state-trace-pair-agreement-hit-at-def
using shape i-bound idx-agree by auto
show ?thesis
  by (intro bexI[of - (trace-table, prefix-trace-table)])
    bexI[of - i])
    (use pair-in i-bound pair-hit in simp-all)
qed

lemma checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event
    (checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
  checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement
    adversary-initial-state ≤
    low-degree-trace-pair-agreement-error budgets
proof -
let ?M =
  checked-staged-security-experiment-with-actual-alpha-prefix-data-state A
let ?pair-hit =
   $\lambda(pair, i) out.$ 
    checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at
      (fst pair) (snd pair) i out
let ?A = low-degree-trace-pair-space × {..<rounds}
have event-le:
  wp-event ?M
  checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement
    adversary-initial-state ≤
  wp-event ?M ( $\lambda out. \exists x \in ?A. ?pair-hit\ x\ out$ )
    adversary-initial-state
by (rule wp-event-mono)
  (auto dest:

```

checked-staged-security-with-actual-alpha-prefix-not-prefix-low-degree-actual-sampled-agreement-imp-pair-hit)
have *finite-A*: *finite* ?*A*
by *simp*
have *union-bound*:
wp-event ?*M* ($\lambda out. \exists x \in ?A. ?pair\text{-}hit\ x\ out$)
adversary-initial-state $\leq$
 $(\sum x \in ?A.$
 case *x* of (*pair*, *i*) $\Rightarrow$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets* *i* + 1) +
 nnreal clength / *nnreal (card query-sample-space)*)
proof (*rule wp-event-finite-UN-bound*[*OF finite-A*])
fix *x*
assume *x-in*: *x* $\in$?*A*
then obtain *trace-table prefix-trace-table i* **where** *x-eq*:
 x = ((*trace-table*, *prefix-trace-table*), *i*)
and *pair-in*:
 (*trace-table*, *prefix-trace-table*) $\in$ *low-degree-trace-pair-space*
and *i-bound*: *i* < *rounds*
by (*cases x, auto*)
have *trace-low*: *trace-table-low-degree trace-table*
 using *pair-in unfolding low-degree-trace-pair-space-def* **by** *auto*
have *prefix-low*: *trace-table-low-degree prefix-trace-table*
 using *pair-in unfolding low-degree-trace-pair-space-def* **by** *auto*
have *distinct*: *trace-table* $\neq$ *prefix-trace-table*
 using *pair-in unfolding low-degree-trace-pair-space-def* **by** *auto*
have *bound*:
 wp-event ?*M*
 (*checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at*
 trace-table prefix-trace-table i)
 adversary-initial-state $\leq$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets* *i* + 1) +
 nnreal clength / *nnreal (card query-sample-space)*
 by (*rule*
 checked-staged-security-with-actual-alpha-prefix-trace-pair-agreement-hit-at-bound
 [*OF wf controlled i-bound trace-low prefix-low distinct*])
show *wp-event* ?*M* (?*pair-hit* *x*) *adversary-initial-state*
 $\leq$ (*case* *x* of (*pair*, *i*) $\Rightarrow$
 staged-phase-relation-error size
 (*staged-query-search-queries budgets* *i* + 1) +
 nnreal clength / *nnreal (card query-sample-space)*)
 unfolding *x-eq*
 using *bound* **by** *simp*
qed
have *sum-eq*:
 $(\sum x \in ?A.$
 case *x* of (*pair*, *i*) $\Rightarrow$
 staged-phase-relation-error size

```

      (staged-query-search-queries budgets  $i + 1$ ) +
      nnreal clength / nnreal (card query-sample-space)) =
    low-degree-trace-pair-agreement-error budgets
  unfolding low-degree-trace-pair-agreement-error-def
    low-degree-trace-pair-agreement-round-error-def
  by (simp add: sum.cartesian-product case-prod-unfold)
show ?thesis
proof (rule order-trans[OF event-le])
  show wp-event ?M ( $\lambda out. \exists x \in ?A. ?pair\text{-}hit\ x\ out$ )
    adversary-initial-state  $\leq$ 
    low-degree-trace-pair-agreement-error budgets
  using union-bound sum-eq by simp
qed
qed

theorem stark-soundness-from-transcript-indexed-partial-opening-route-from-sampled-transcript-not-prefix:
  fixes P empty-query-error :: prob
  and C :: nat  $\Rightarrow$  prob
  assumes false-statement:  $\neg$  exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-reduction:
     $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      trace-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
  and composition-fri-reduction:
     $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      composition-fri-reachable-partial-candidate-reduction-assumption
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
  and round-bound:
     $\bigwedge i. i < rounds \implies$ 
      wp-event
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i)
        adversary-initial-state  $\leq C\ i$ 
  and data-pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      staged-security-with-data-state-transcript-pre-hit
      adversary-initial-state  $\leq P$ 
  and current-empty-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ *empty-query-error*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
 $(\sum i < \text{rounds}. C i)$ +
trace-fri-error + composition-fri-error +
 $(\sum i < \text{rounds}. C i)$ +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
((composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(P + staged-concrete-transcript-target-error-bound)) +
(trace-fri-error +
low-degree-trace-pair-agreement-error budgets)) +
trace-fri-error + empty-query-error)
proof –
have *transcript-new-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-transcript-new-hit
adversary-initial-state $\leq$
staged-concrete-transcript-target-error-bound
by (rule
checked-staged-security-with-data-state-transcript-new-hit-bound
[OF wf controlled])
have *not-prefix-bound*:
wp-event
(checked-staged-security-experiment-with-actual-alpha-prefix-data-state A)
(checked-staged-security-with-actual-alpha-prefix-partial-header-candidate-not-prefix-bound
composition-trace-bad-alpha-space)
adversary-initial-state $\leq$
P + staged-concrete-transcript-target-error-bound
by (rule
checked-staged-security-with-actual-alpha-prefix-not-prefix-bound-from-sampled-transcript
[OF data-pre-bound transcript-new-bound])
show *?thesis*
by (rule
stark-soundness-from-transcript-indexed-partial-opening-route-from-not-prefix
[OF false-statement wf controlled trace-fri-reduction
composition-fri-reduction round-bound not-prefix-bound

current-empty-bound])

qed

theorem *stark-soundness-from-transcript-indexed-partial-opening-route-from-residual-sampled-transcript-not-prefix*:

fixes P *empty-query-error* :: *prob*
and R :: *nat* $\Rightarrow$ *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
trace-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *composition-fri-reduction*:
 $\bigwedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
composition-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
and *residual-bound*:
 $\bigwedge i. i < \text{rounds} \Rightarrow$
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A i)
(checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
i)
adversary-initial-state $\leq R i$
and *data-pre-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq P$
and *current-empty-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-current-empty-query-opening-hit-at 0)
adversary-initial-state $\leq$ empty-query-error
shows *checked-staged-adversary-acceptance-probability A $\leq$*
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
 $(\sum i < \text{rounds}.$
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound + R i +
(P + staged-concrete-transcript-target-error-bound) +

$(P + \text{staged-concrete-transcript-target-error-bound})) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\sum_{i < \text{rounds.}}$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-query-search-queries budgets } i + 1) +$
 $\quad \text{query-error-bound} + R \ i +$
 $\quad (P + \text{staged-concrete-transcript-target-error-bound}) +$
 $\quad (P + \text{staged-concrete-transcript-target-error-bound})) +$
 $(\text{composition-error-bound} +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $((\text{composition-error-bound} +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\quad (P + \text{staged-concrete-transcript-target-error-bound})) +$
 $\quad (\text{trace-fri-error} +$
 $\quad \quad \text{low-degree-trace-pair-agreement-error budgets})) +$
 $\quad \text{trace-fri-error} + \text{empty-query-error})$
proof –
have *transcript-new-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-new-hit}$
 $\text{adversary-initial-state} \leq$
 $\text{staged-concrete-transcript-target-error-bound}$
by (rule
 $\text{checked-staged-security-with-data-state-transcript-new-hit-bound}$
 $[\text{OF wf controlled}])$
have *trace-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at}$
 $\quad i)$
 $\text{adversary-initial-state} \leq$
 $P + \text{staged-concrete-transcript-target-error-bound}$
by (rule
 $\text{checked-staged-security-with-query-prefix-current-trace-path-output-hit-at-bound-from-transcript}$
 $[\text{OF - data-pre-bound transcript-new-bound}])$
have *composition-path-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-composition-path-output-hit-at}$

i)
adversary-initial-state $\leq$
P + *staged-concrete-transcript-target-error-bound*
by (rule
checked-staged-security-with-query-prefix-current-composition-path-output-hit-at-bound-from-transcript
[*OF* - *data-pre-bound transcript-new-bound*])
have *round-bound*:
 $\bigwedge i. i < \text{rounds} \implies$
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state A i*)
(*checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at i*)
adversary-initial-state $\leq$
staged-phase-relation-error size
(*staged-query-search-queries budgets i + 1*) +
query-error-bound + *R i* +
(*P* + *staged-concrete-transcript-target-error-bound*) +
(*P* + *staged-concrete-transcript-target-error-bound*)
by (rule
checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-conceptual-residual-and
[*OF wf controlled - residual-bound trace-path-bound*
composition-path-bound])
show ?thesis
by (rule
stark-soundness-from-transcript-indexed-partial-opening-route-from-sampled-transcript-not-prefix
[*OF false-statement wf controlled trace-fri-reduction*
composition-fri-reduction round-bound data-pre-bound
current-empty-bound])

qed

theorem *stark-soundness-from-transcript-indexed-partial-opening-route-from-residual-split-sampled-transcript-not-pr*
fixes *P empty-query-error* :: prob
and *LT LC Alpha* :: nat $\Rightarrow$ prob
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
Some (*data, attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
trace-fri-reachable-partial-candidate-reduction-assumption
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))
and *composition-fri-reduction*:
 $\bigwedge \text{data attacker-state.}$
Some (*data, attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
composition-fri-reachable-partial-candidate-reduction-assumption

(verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
and trace-low-degree-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-conceptual-trace-low-degree-bad-at
 i)
 adversary-initial-state $\leq LT\ i$
and composition-low-degree-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-conceptual-composition-low-degree-bad-at
 i)
 adversary-initial-state $\leq LC\ i$
and alpha-bound:
 $\bigwedge i. i < \text{rounds} \implies$
 wp-event
 (checked-staged-security-experiment-with-query-prefix-data-state A i)
 (checked-staged-security-with-query-prefix-conceptual-alpha-bad-set-hit-at
 i)
 adversary-initial-state $\leq Alpha\ i$
and data-pre-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 staged-security-with-data-state-transcript-pre-hit
 adversary-initial-state $\leq P$
and current-empty-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
 adversary-initial-state $\leq \text{empty-query-error}$
shows checked-staged-adversary-acceptance-probability A $\leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 ($\sum i < \text{rounds}.$
 staged-phase-relation-error size
 (staged-query-search-queries budgets i + 1) +
 query-error-bound +
 hash-collision-budget-value 0
 (staged-query-search-queries budgets i + 1) +
 (LT i + LC i + Alpha i) +
 (P + staged-concrete-transcript-target-error-bound) +
 (P + staged-concrete-transcript-target-error-bound)) +
 trace-fri-error + composition-fri-error +
 ($\sum i < \text{rounds}.$
 staged-phase-relation-error size
 (staged-query-search-queries budgets i + 1) +
 query-error-bound +

```

hash-collision-budget-value 0
  (staged-query-search-queries budgets  $i + 1$ ) +
  ( $LT\ i + LC\ i + Alpha\ i$ ) +
  ( $P + staged-concrete-transcript-target-error-bound$ ) +
  ( $P + staged-concrete-transcript-target-error-bound$ )) +
(composition-error-bound +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  staged-phase-relation-error size
  (staged-alpha-search-queries budgets 0) +
  ((composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
  (staged-alpha-search-queries budgets 0) +
  ( $P + staged-concrete-transcript-target-error-bound$ )) +
  (trace-fri-error +
    low-degree-trace-pair-agreement-error budgets)) +
  trace-fri-error + empty-query-error)
proof -
have residual-bound:
 $\bigwedge i. i < rounds \implies$ 
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state  $A\ i$ )
    (checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
       $i$ )
    adversary-initial-state  $\leq$ 
hash-collision-budget-value 0
  (staged-query-search-queries budgets  $i + 1$ ) +
  ( $LT\ i + LC\ i + Alpha\ i$ )
proof -
fix  $i$ 
assume  $i$ -bound:  $i < rounds$ 
have algebraic-bound:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state  $A\ i$ )
    (checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at
       $i$ )
    adversary-initial-state  $\leq LT\ i + LC\ i + Alpha\ i$ 
by (rule
  checked-staged-security-with-query-prefix-conceptual-context-algebraic-residual-hit-at-bound-from-split
    [OF false-statement trace-low-degree-bound[OF  $i$ -bound]
      composition-low-degree-bound[OF  $i$ -bound]
      alpha-bound[OF  $i$ -bound]])
show
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state  $A\ i$ )

```

(checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at
i)
adversary-initial-state $\leq$
hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
(LT i + LC i + Alpha i)
by (rule
checked-staged-security-with-query-prefix-conceptual-context-residual-hit-at-bound-from-split
[OF wf controlled i-bound algebraic-bound])
qed
have bound:
checked-staged-adversary-acceptance-probability A $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
($\sum i < \text{rounds}$.
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
(hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
(LT i + LC i + Alpha i)) +
(P + staged-concrete-transcript-target-error-bound) +
(P + staged-concrete-transcript-target-error-bound)) +
trace-fri-error + composition-fri-error +
($\sum i < \text{rounds}$.
staged-phase-relation-error size
(staged-query-search-queries budgets i + 1) +
query-error-bound +
(hash-collision-budget-value 0
(staged-query-search-queries budgets i + 1) +
(LT i + LC i + Alpha i)) +
(P + staged-concrete-transcript-target-error-bound) +
(P + staged-concrete-transcript-target-error-bound)) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
((composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
(P + staged-concrete-transcript-target-error-bound)) +
(trace-fri-error +
low-degree-trace-pair-agreement-error budgets)) +

```

    trace-fri-error + empty-query-error)
  by (rule
    stark-soundness-from-transcript-indexed-partial-opening-route-from-residual-sampled-transcript-not-prefix
    [OF false-statement wf controlled trace-fri-reduction
      composition-fri-reduction residual-bound data-pre-bound
      current-empty-bound])
  show ?thesis
  using bound by (simp add: algebra-simps)
qed

theorem stark-soundness-from-transcript-indexed-partial-opening-route-from-same-run-partial-opening-gap-and-tran
fixes  $G$   $P$  empty-query-error :: prob
and  $C :: \text{nat} \Rightarrow \text{prob}$ 
assumes false-statement:  $\neg \text{exists-valid-trace}$ 
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and trace-fri-reduction:
   $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program  $A$ )
      adversary-initial-state)  $\implies$ 
    trace-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
and composition-fri-reduction:
   $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program  $A$ )
      adversary-initial-state)  $\implies$ 
    composition-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
and round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state  $A$   $i$ )
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at  $i$ )
      adversary-initial-state  $\leq C$   $i$ 
and gap-bound:
    wp-event
      (checked-staged-security-experiment-with-actual-alpha-prefix-data-state
         $A$ )
      checked-staged-security-with-actual-alpha-prefix-not-prefix-same-run-partial-opening-gap
      adversary-initial-state  $\leq G$ 
and data-pre-bound:
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
      staged-security-with-data-state-transcript-pre-hit
      adversary-initial-state  $\leq P$ 
and current-empty-bound:

```

$wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}current\text{-}empty\text{-}query\text{-}opening\text{-}hit\text{-}at \ 0)$
 $adversary\text{-}initial\text{-}state \leq empty\text{-}query\text{-}error$
shows $checked\text{-}staged\text{-}adversary\text{-}acceptance\text{-}probability \ A \leq$
 $hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $(staged\text{-}attacker\text{-}query\text{-}budget \ budgets +$
 $staged\text{-}challenge\text{-}query\text{-}budget + verifier\text{-}hash\text{-}query\text{-}budget) +$
 $(\sum i < rounds. \ C \ i) +$
 $trace\text{-}fri\text{-}error + composition\text{-}fri\text{-}error +$
 $(\sum i < rounds. \ C \ i) +$
 $(composition\text{-}error\text{-}bound +$
 $staged\text{-}phase\text{-}relation\text{-}error \ size$
 $(staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) +$
 $hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $(staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) +$
 $staged\text{-}phase\text{-}relation\text{-}error \ size$
 $(staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) +$
 $((composition\text{-}error\text{-}bound +$
 $staged\text{-}phase\text{-}relation\text{-}error \ size$
 $(staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) +$
 $hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $(staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) +$
 $(hash\text{-}collision\text{-}budget\text{-}value \ 0$
 $(staged\text{-}alpha\text{-}search\text{-}queries \ budgets \ 0) + G +$
 $(P + staged\text{-}concrete\text{-}transcript\text{-}target\text{-}error\text{-}bound) +$
 $(P + staged\text{-}concrete\text{-}transcript\text{-}target\text{-}error\text{-}bound)))) +$
 $(trace\text{-}fri\text{-}error +$
 $low\text{-}degree\text{-}trace\text{-}pair\text{-}agreement\text{-}error \ budgets)) +$
 $trace\text{-}fri\text{-}error + empty\text{-}query\text{-}error)$
proof –
have $uncovered\text{-}bound$:
 $wp\text{-}event$
 $(checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}data\text{-}state$
 $A)$
 $checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}not\text{-}prefix\text{-}uncovered\text{-}index$
 $adversary\text{-}initial\text{-}state \leq G$
by (rule $order\text{-}trans$
 $[OF \ checked\text{-}staged\text{-}security\text{-}with\text{-}actual\text{-}alpha\text{-}prefix\text{-}not\text{-}prefix\text{-}uncovered\text{-}index\text{-}le\text{-}same\text{-}run\text{-}partial\text{-}opening\text{-}gap$
 $gap\text{-}bound]$)
show $?thesis$
by (rule
 $stark\text{-}soundness\text{-}from\text{-}transcript\text{-}indexed\text{-}partial\text{-}opening\text{-}route\text{-}from\text{-}uncovered\text{-}and\text{-}transcript$
 $[OF \ false\text{-}statement \ wf \ controlled \ trace\text{-}fri\text{-}reduction$
 $composition\text{-}fri\text{-}reduction \ round\text{-}bound \ uncovered\text{-}bound \ data\text{-}pre\text{-}bound$
 $current\text{-}empty\text{-}bound]$)
qed
end

end

```
theory Soundness-Public-Route-Single-Query
imports
  Soundness-FRI-Verifier-Tied
  Soundness-Not-Prefix-Partial-Opening
begin
```

Narrow downstream wrappers for the public route that use the transcript-indexed single-current-query charge. This theory depends directly on the partial-opening route needed by the final endpoint.

```
context soundness
begin
```

```
theorem stark-soundness-from-transcript-indexed-partial-opening-route-from-sampled-transcript-relevant-drift-only-s
fixes  $P$  trace-fri-error' composition-fri-error'
  empty-trace-fri-error' empty-query-error :: prob
and  $C :: \text{nat} \Rightarrow \text{prob}$ 
assumes false-statement:  $\neg \text{exists-valid-trace}$ 
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and round-bound:
   $\bigwedge i. i < \text{rounds} \implies$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state  $A$   $i$ )
      (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at  $i$ )
      adversary-initial-state  $\leq C$   $i$ 
and data-pre-bound:
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
    staged-security-with-data-state-transcript-pre-hit
    adversary-initial-state  $\leq P$ 
and current-empty-bound:
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
    (staged-security-with-data-state-current-empty-query-opening-hit-at 0)
    adversary-initial-state  $\leq \text{empty-query-error}$ 
and trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state  $\leq \text{trace-fri-error}'$ 
and comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-verifier-tied-partial-candidate)
    adversary-initial-state  $\leq \text{composition-fri-error}'$ 
and empty-trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
    (staged-security-with-data-state-verifier-event
```

$\text{trace-fri-bad-with-empty-composition-header-candidates}$
 $\text{adversary-initial-state} \leq \text{empty-trace-fri-error}'$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(P + \text{staged-concrete-transcript-target-error-bound})) +$
 $\text{trace-fri-error}' + \text{composition-fri-error}' +$
 $(\sum_{i < \text{rounds}} C i) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(P + \text{staged-concrete-transcript-target-error-bound}) +$
 $\text{empty-trace-fri-error}' + \text{empty-query-error})$
proof –
have $\text{current-query-bound}$:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-current-query-partial-opening-hit}$
 $\text{adversary-initial-state} \leq (\sum_{i < \text{rounds}} C i)$
by (rule
 $\text{checked-staged-security-with-data-state-current-query-partial-opening-hit-bound-from-query-prefix-rounds}$
 $[\text{OF round-bound wf controlled}])$
have query-bound :
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{query-bad-with-aligned-transcript-partial-candidates})$
 $\text{adversary-initial-state} \leq (\sum_{i < \text{rounds}} C i)$
by (rule order-trans [
 $\text{OF checked-staged-security-with-data-state-aligned-transcript-query-bad-bound-from-current-partial-opening-hit}$
 $\text{current-query-bound}]$)
 $(\text{use wf controlled in simp-all})$
have $\text{transcript-new-bound}$:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-new-hit}$
 $\text{adversary-initial-state} \leq \text{staged-concrete-transcript-target-error-bound}$
by (rule
 $\text{checked-staged-security-with-data-state-transcript-new-hit-bound}$
 $[\text{OF wf controlled}])$

have *drift-bound*:
wp-event (*checked-staged-security-experiment-with-actual-alpha-prefix-data-state A*)
checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift
adversary-initial-state $\leq$
 $P + \text{staged-concrete-transcript-target-error-bound}$
by (*rule* *checked-staged-security-with-actual-alpha-prefix-relevant-partial-header-candidate-drift-bound-from-sampled-trans*
 $[OF\ wf\ controlled\ data-pre-bound\ transcript-new-bound]$)
have *comp-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
composition-bad-with-partial-candidates)
adversary-initial-state $\leq$
 $\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(P + \text{staged-concrete-transcript-target-error-bound})$
by (*rule* *checked-staged-security-with-data-state-partial-candidate-composition-bad-bound-from-relevant-drift-and-budgets*
 $[OF\ false-statement\ wf\ controlled\ drift-bound]$)
have *merkle-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad)
adversary-initial-state $\leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets } +$
 $\text{staged-challenge-query-budget } + \text{verifier-hash-query-budget})$
by (*rule* *checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled*
 $[OF\ wf\ controlled]$)
have *empty-query-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
query-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq \text{empty-query-error}$
by (*rule* *order-trans*
 $OF\ \text{checked-staged-security-with-data-state-empty-query-bad-le-current-empty-opening-hit-at-zero}$
 $OF\ wf\ controlled]$ *current-empty-bound*)
have *empty-header-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
soundness-bad-event-partial-candidate-empty-header)
adversary-initial-state $\leq$


```

    composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    (P + staged-concrete-transcript-target-error-bound) +
    empty-trace-fri-error' + empty-query-error
  by (rule
    checked-staged-security-with-data-state-empty-header-bad-event-bound-from-relevant-drift-trace-fri-and-query
    [OF false-statement wf controlled drift-bound empty-trace-fri-bound
      empty-query-bound])
  show ?thesis
  by (rule
    checked-staged-soundness-from-verifier-tied-aligned-transcript-partial-candidate-components-and-empty-header-b
    [OF false-statement merkle-bound comp-bound trace-fri-bound
      comp-fri-bound query-bound empty-header-bound])
qed

```

end

These single-query-charge wrappers are downstream plumbing for the public same-run route. The visible theorem remains in *Soundness-Public-Route-Same-Run*.

hide-fact

soundness.stark-soundness-from-transcript-indexed-partial-opening-route-from-sampled-transcript-relevant-drift-only

end

theory *Soundness-FRI-Header-Tied-Reduction*

imports *Soundness-FRI-Verifier-Tied*

begin

Header-tied trace FRI reductions.

The active public route uses trace candidates tied to the verifier header root. This layer mirrors the verifier-tied sampled/missing decomposition without widening back to arbitrary authenticated roots.

context *soundness*

begin

lemma *trace-fri-bad-with-header-tied-partial-candidate-None[simp]*:

¬ *trace-fri-bad-with-header-tied-partial-candidate s None*

unfolding *trace-fri-bad-with-header-tied-partial-candidate-def*
accepted-fri-opening-transcript-def

by *simp*

definition *trace-fri-bad-with-header-tied-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-bad-with-header-tied-sampled-layer-chain } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table}$
 doms layers.
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-sampled-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers doms layers})$

definition *trace-fri-header-tied-missing-sampled-layer-chain*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-missing-sampled-layer-chain } s \text{ out} \longleftrightarrow$
 $\text{trace-fri-bad-with-header-tied-partial-candidate } s \text{ out} \wedge$
 $\neg \text{trace-fri-bad-with-header-tied-sampled-layer-chain } s \text{ out}$

definition *trace-fri-header-tied-sampled-layer-assignment-obstruction*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-sampled-layer-assignment-obstruction } s \text{ out}$
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-sampled-layer-assignment-obstruction}$
 $\text{trace-table-low-degree trace-table (clength - 1) trace-roots}$

trace-bs trace-final fri-query-idxs trace-round-layers)

definition *trace-fri-header-tied-sampled-assignment-conflict*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-sampled-assignment-conflict s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{ accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final dg composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{ verifier-header-transcript s fr trace-roots trace-final as dg}$
 $\text{ composition-roots composition-final rest} \wedge$
 $\text{ accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{ trace-table-low-degree trace-table} \wedge$
 $\text{ generic-fri-sampled-assignment-conflict trace-table trace-roots}$
 $\text{ trace-bs trace-final fri-query-idxs trace-round-layers})$

lemma *trace-fri-header-tied-sampled-assignment-conflict-imp-sampled:*

assumes

trace-fri-header-tied-sampled-assignment-conflict s out

shows *trace-fri-sampled-assignment-conflict s out*

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table}$

where *fri-openings:*

$\text{ accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final dg composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers}$

and *partial:*

$\text{ accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 trace-openings

and *cand:* $\text{ partial-trace-table-candidate trace-table trace-openings}$

and *not-low:* $\neg \text{ trace-table-low-degree trace-table}$

and *conflict:*

$\text{ generic-fri-sampled-assignment-conflict trace-table trace-roots}$
 $\text{ trace-bs trace-final fri-query-idxs trace-round-layers}$

unfolding *trace-fri-header-tied-sampled-assignment-conflict-def*
by *blast*

have *evidence:*

$\text{ trace-fri-partial-candidate-opening-evidence s out trace-roots}$
 $\text{ trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$

```

    composition-round-layers fr fri-query-idxs trace-openings
    trace-table
  unfolding trace-fri-partial-candidate-opening-evidence-def
    trace-fri-partial-candidate-evidence-def
  by (intro conjI fri-openings partial cand not-low)
show ?thesis
  unfolding trace-fri-sampled-assignment-conflict-def
  by (intro exI conjI)
    (rule evidence, rule conflict)
qed

lemma wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-sampled:
  assumes sampled-bound:
    wp-event verify-monad (trace-fri-sampled-assignment-conflict s) s ≤ C
  shows
    wp-event verify-monad
      (trace-fri-header-tied-sampled-assignment-conflict s) s ≤ C
proof -
  have wp-event verify-monad
    (trace-fri-header-tied-sampled-assignment-conflict s) s ≤
      wp-event verify-monad (trace-fri-sampled-assignment-conflict s) s
  by (rule wp-event-mono)
    (rule trace-fri-header-tied-sampled-assignment-conflict-imp-sampled)
  then show ?thesis
    by (rule order-trans[OF - sampled-bound])
qed

lemma trace-fri-header-tied-missing-sampled-imp-assignment-obstruction:
  assumes missing:
    trace-fri-header-tied-missing-sampled-layer-chain s out
  shows
    trace-fri-header-tied-sampled-layer-assignment-obstruction s out
proof -
  have bad:
    trace-fri-bad-with-header-tied-partial-candidate s out
  and not-sampled:
    ¬ trace-fri-bad-with-header-tied-sampled-layer-chain s out
  using missing
  unfolding trace-fri-header-tied-missing-sampled-layer-chain-def
  by blast+
  from bad obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as dg

```

```

    composition-roots composition-final rest
and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
and cand: partial-trace-table-candidate trace-table trace-openings
and not-low:  $\neg$  trace-table-low-degree trace-table
unfolding trace-fri-bad-with-header-tied-partial-candidate-def
by blast
have partial-generic:
    generic-fri-partial-evidence trace-table-low-degree trace-table
    (clength - 1) trace-roots trace-bs trace-final fri-query-idxs
    trace-round-layers
by (rule accepted-fri-opening-transcript-trace-generic-partial-evidence
    [OF fri-openings])
have no-chain:
     $\bigwedge$  doms layers. \neg
    generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
proof
    fix doms layers
    assume chain:
    generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
    have trace-fri-bad-with-header-tied-sampled-layer-chain s out
    unfolding trace-fri-bad-with-header-tied-sampled-layer-chain-def
    by (intro exI conjI)
    (rule fri-openings, rule header, rule partial, rule cand,
    rule not-low, rule chain)
    then show False
    using not-sampled by contradiction
qed
have obstruction:
    generic-fri-sampled-layer-assignment-obstruction
    trace-table-low-degree trace-table (clength - 1) trace-roots
    trace-bs trace-final fri-query-idxs trace-round-layers
    unfolding generic-fri-sampled-layer-assignment-obstruction-def
    using partial-generic no-chain by blast
show ?thesis
    unfolding trace-fri-header-tied-sampled-layer-assignment-obstruction-def
    by (intro exI conjI)
    (rule fri-openings, rule header, rule partial, rule cand,
    rule not-low, rule obstruction)
qed

lemma trace-fri-header-tied-assignment-obstruction-imp-conflict-or-zero:
assumes obstruction:
    trace-fri-header-tied-sampled-layer-assignment-obstruction s out

```

shows

trace-fri-header-tied-sampled-assignment-conflict s out $\vee$
trace-fri-header-tied-zero-round-final-obstruction s out

proof –

obtain *trace-roots trace-bs' trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table

where *fri-openings*:

accepted-fri-opening-transcript s out trace-roots trace-bs'
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers

and *header*:

verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest

and *partial*:

accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings

and *cand*: *partial-trace-table-candidate trace-table trace-openings*

and *not-low*: $\neg$ *trace-table-low-degree trace-table*

and *generic*:

generic-fri-sampled-layer-assignment-obstruction
trace-table-low-degree trace-table (clength – 1) trace-roots
trace-bs' trace-final fri-query-idxs trace-round-layers

using *obstruction*

unfolding *trace-fri-header-tied-sampled-layer-assignment-obstruction-def*

by *blast*

obtain *N* **where** *eval-power*: $\text{clength} * \text{scale} = 2^N$

using *eval-domain-length-power* **by** *blast*

have *trace-rounds-le*: $\text{length } \text{trace-bs}' \leq N$

proof –

have *len*: $\text{length } \text{trace-bs}' = \text{ceil-log } \text{clength}$
using *accepted-fri-opening-transcript-shapes(1,2)[OF fri-openings]*
by *simp*

have $\text{clength} \leq \text{length} * \text{scale}$

using *scale-pos* **by** *simp*

also have $\dots = 2^N$

by (rule *eval-power*)

finally have $\text{ceil-log } \text{clength} \leq N$

by (rule *ceil-log-le-power*)

then show *?thesis*

using *len* **by** *simp*

qed

have *query-bounds'*:

$\bigwedge \text{round-idx. round-idx} < \text{length } \text{fri-query-idxs} \implies$
 $\text{fri-query-idxs} ! \text{round-idx} < \text{clength} * \text{scale}$

proof –

fix *round-idx*

assume *round-bound*: $\text{round-idx} < \text{length } \text{fri-query-idxs}$

have *fri-query-idxs* ! $\text{round-idx} \in \text{set } \text{fri-query-idxs}$

```

    using round-bound by simp
  then show fri-query-idxs ! round-idx < clength * scale
    by (rule accepted-fri-opening-transcript-query-idx-bound
        [OF fri-openings])
qed
have split:
  generic-fri-sampled-assignment-conflict trace-table trace-roots
    trace-bs' trace-final fri-query-idxs trace-round-layers  $\vee$ 
    length trace-bs' = 0  $\wedge$ 
     $\neg$  fri-final-constant-consistent trace-table trace-final
  by (rule
      generic-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final
        [OF generic query-bounds' eval-power trace-rounds-le])
then show ?thesis
proof
  assume conflict:
    generic-fri-sampled-assignment-conflict trace-table trace-roots
      trace-bs' trace-final fri-query-idxs trace-round-layers
  then show ?thesis
    unfolding trace-fri-header-tied-sampled-assignment-conflict-def
    by (intro disjI1 exI conjI)
      (rule fri-openings, rule header, rule partial, rule cand,
        rule not-low, rule conflict)
next
  assume zero:
    length trace-bs' = 0  $\wedge$ 
     $\neg$  fri-final-constant-consistent trace-table trace-final
  then show ?thesis
    unfolding trace-fri-header-tied-zero-round-final-obstruction-def
    by (intro disjI2 exI conjI)
      (rule fri-openings, rule header, rule partial, rule cand,
        rule not-low,
        simp-all)
qed
qed

lemma wp-trace-fri-header-tied-missing-sampled-bound-from-assignment-obstruction:
  assumes assignment-bound:
    wp-event verify-monad
      (trace-fri-header-tied-sampled-layer-assignment-obstruction s) s
     $\leq M$ 
  shows
    wp-event verify-monad
      (trace-fri-header-tied-missing-sampled-layer-chain s) s  $\leq M$ 
proof -
  have wp-event verify-monad
    (trace-fri-header-tied-missing-sampled-layer-chain s) s  $\leq$ 
    wp-event verify-monad
      (trace-fri-header-tied-sampled-layer-assignment-obstruction s) s

```

by (rule wp-event-mono)
 (rule trace-fri-header-tied-missing-sampled-imp-assignment-obstruction)
 then show ?thesis
 by (rule order-trans[OF - assignment-bound])
 qed

lemma wp-trace-fri-header-tied-bound-from-sampled-and-missing:
assumes sampled-bound:
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) $s \leq R$
and missing-bound:
 wp-event verify-monad
 (trace-fri-header-tied-missing-sampled-layer-chain s) $s \leq M$
shows
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) $s \leq R + M$
proof –
have wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) $s \leq$
 wp-event verify-monad
 ($\lambda out. \text{trace-fri-bad-with-header-tied-sampled-layer-chain } s \text{ out} \vee$
 $\text{trace-fri-header-tied-missing-sampled-layer-chain } s \text{ out}$) s
unfolding trace-fri-header-tied-missing-sampled-layer-chain-def
 by (rule wp-event-mono) blast
also have ... $\leq$
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) $s +$
 wp-event verify-monad
 (trace-fri-header-tied-missing-sampled-layer-chain s) s
 by (rule wp-event-union-bound)
also have ... $\leq R + M$
 by (rule add-mono[OF sampled-bound missing-bound])
finally show ?thesis .
 qed

lemma wp-trace-fri-header-tied-assignment-obstruction-bound-from-conflict-and-zero:
assumes conflict-bound:
 wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict s) $s \leq C$
and zero-bound:
 wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction s) $s \leq Z$
shows
 wp-event verify-monad
 (trace-fri-header-tied-sampled-layer-assignment-obstruction s) s
 $\leq C + Z$
proof –
have wp-event verify-monad
 (trace-fri-header-tied-sampled-layer-assignment-obstruction s) s

$\leq$ *wp-event verify-monad*
 $(\lambda out. \text{trace-fri-header-tied-sampled-assignment-conflict } s \text{ out} \vee$
 $\text{trace-fri-header-tied-zero-round-final-obstruction } s \text{ out}) \ s$
by (*rule wp-event-mono*)
 $(\text{rule trace-fri-header-tied-assignment-obstruction-imp-conflict-or-zero})$
also have ... $\leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict } s) \ s +$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s$
by (*rule wp-event-union-bound*)
also have ... $\leq C + Z$
by (*rule add-mono*[*OF conflict-bound zero-bound*])
finally show ?thesis .
qed

lemma *wp-trace-fri-header-tied-bound-from-sampled-conflict-and-zero*:
assumes *sampled-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-sampled-layer-chain } s) \ s \leq R$
and *conflict-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict } s) \ s \leq C$
and *zero-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s \leq Z$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \ s$
 $\leq R + (C + Z)$
by (*rule wp-trace-fri-header-tied-bound-from-sampled-and-missing*
 $[\text{OF sampled-bound}$
 $\text{wp-trace-fri-header-tied-missing-sampled-bound-from-assignment-obstruction}$
 $[\text{OF wp-trace-fri-header-tied-assignment-obstruction-bound-from-conflict-and-zero}$
 $[\text{OF conflict-bound zero-bound}]]])$

lemma *trace-fri-bad-with-header-tied-partial-candidates-staged-bound*:
assumes *verifier-bound*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq C$
shows

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state $\leq C$
by (*rule checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
 (*simp-all add: verifier-bound*)

lemma *wp-trace-fri-header-tied-zero-round-bound-from-checked-and-merkle:*

assumes *checked-bound:*

$wp\text{-}event$ *verify-monad*
 (*trace-fri-zero-round-checked-final-obstruction* s) $s \leq Z$

and *merkle-bound:*

$wp\text{-}event$ *verify-monad* (*partial-merkle-inconsistency-bad* s) $s \leq M$

shows

$wp\text{-}event$ *verify-monad*
 (*trace-fri-header-tied-zero-round-final-obstruction* s) $s \leq Z + M$

proof –

have $wp\text{-}event$ *verify-monad*

(*trace-fri-header-tied-zero-round-final-obstruction* s) $s \leq$

$wp\text{-}event$ *verify-monad*

($\lambda out. trace\text{-}fri\text{-}zero\text{-}round\text{-}checked\text{-}final\text{-}obstruction\ s\ out \vee$
partial-merkle-inconsistency-bad $s\ out$) s

by (*rule wp-event-mono*)

(*rule trace-fri-header-tied-zero-round-final-obstruction-imp-checked-or-merkle*)

also have $\dots \leq$

$wp\text{-}event$ *verify-monad*

(*trace-fri-zero-round-checked-final-obstruction* s) $s +$

$wp\text{-}event$ *verify-monad* (*partial-merkle-inconsistency-bad* s) s

by (*rule wp-event-union-bound*)

also have $\dots \leq Z + M$

by (*rule add-mono[OF checked-bound merkle-bound]*)

finally show *?thesis* .

qed

end

end

theory *Soundness-FRI-Layer-Merkle*

imports *Soundness-FRI-Header-Tied-Reduction*

begin

Authenticated-opening facts for sampled FRI layer chunks.

The sampled-assignment conflict predicates are stated over transcript chunks. This layer connects those chunks back to the Merkle checks performed by the verifier. It is intentionally local: it extracts only sampled openings from the verifier execution and does not reconstruct complete FRI tables.

context *soundness*

begin

definition *fri-layer-chunk-authenticated*

$:: 'f \Rightarrow \text{nat} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

fri-layer-chunk-authenticated *rt len idx chunk s* $\longleftrightarrow$
 $(\exists xp \text{ xp-path } xn \text{ xn-path.}$
 $\text{fri-layer-opening-chunk } \text{len } xp \text{ xp-path } xn \text{ xn-path } \text{chunk} \wedge$
 $\text{authenticated-opening-in } s$
 $(\parallel \text{opening-root} = \text{rt},$
 $\text{opening-length} = \text{len},$
 $\text{opening-index} = \text{idx},$
 $\text{opening-value} = xp,$
 $\text{opening-path} = \text{xp-path} \parallel) \wedge$
 $\text{authenticated-opening-in } s$
 $(\parallel \text{opening-root} = \text{rt},$
 $\text{opening-length} = \text{len},$
 $\text{opening-index} = \text{fri-sibling-index } \text{len } \text{idx},$
 $\text{opening-value} = xn,$
 $\text{opening-path} = \text{xn-path} \parallel))$

definition *fri-mfold-chunks-authenticated*

$:: ('f \times 'f) \text{ list} \Rightarrow \text{nat} \Rightarrow \text{nat} \Rightarrow 'f \text{ list list} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

fri-mfold-chunks-authenticated *bfs len idx chunks s* $\longleftrightarrow$
 $\text{length } \text{chunks} = \text{length } \text{bfs} \wedge$
 $(\forall j < \text{length } \text{bfs.}$
 $\text{fri-layer-chunk-authenticated } (\text{snd } (\text{bfs } ! j))$
 $(\text{fri-layer-lengths } (\text{length } \text{bfs}) \text{ len } ! j)$
 $(\text{fri-layer-indices } (\text{length } \text{bfs}) \text{ idx } \text{len } ! j)$
 $(\text{chunks } ! j) \text{ s})$

lemma *fri-layers-transcript-nth-length:*

assumes *layers:* *fri-layers-transcript* *n len layer-chunks chunk*
and *j-bound:* *j < n*

shows

$\text{length } (\text{layer-chunks } ! j) =$
 $2 + 2 * \text{floor-log } (\text{fri-layer-lengths } n \text{ len } ! j)$

proof –

from *fri-layers-transcript-nth-opening*[*OF layers j-bound*]

obtain *xp xp-path xn xn-path* **where**

layer-open:

$\text{fri-layer-opening-chunk } (\text{fri-layer-lengths } n \text{ len } ! j)$
 $xp \text{ xp-path } xn \text{ xn-path } (\text{layer-chunks } ! j)$

by *blast*

show *?thesis*

using *fri-layer-opening-chunk-length*[*OF layer-open*] .

qed

lemma *fri-layers-transcript-unique*:

assumes *left*: *fri-layers-transcript* *n* *len* *layer-chunks* *chunk*
and *right*: *fri-layers-transcript* *n* *len* *layer-chunks'* *chunk*
shows *layer-chunks* = *layer-chunks'*

proof –

have *len-left*: *length* *layer-chunks* = *n*
using *left* **unfolding** *fri-layers-transcript-def* **by** *simp*
have *len-right*: *length* *layer-chunks'* = *n*
using *right* **unfolding** *fri-layers-transcript-def* **by** *simp*
have *concat-eq*:
List.concat *layer-chunks* @ [] = *List.concat* *layer-chunks'*
using *left* *right* **unfolding** *fri-layers-transcript-def* **by** *simp*
have *same-lens*:
 $\bigwedge j. j < \text{length } \text{layer-chunks} \implies$
 $\text{length } (\text{layer-chunks} ! j) = \text{length } (\text{layer-chunks}' ! j)$
using *fri-layers-transcript-nth-length* [*OF left*]
fri-layers-transcript-nth-length [*OF right*]
len-left *len-right*
by *simp*
have *layer-chunks* = *layer-chunks'* $\wedge$ ([] :: 'f list) = []
by (*rule* *concat-append-eq-concat-same-chunk-lengths*
[*OF* - *same-lens concat-eq*])
(*use* *len-left* *len-right* **in** *simp*)
then show *?thesis*
by *simp*

qed

lemma *query-round-fri-layer-transcripts-unique*:

assumes *left*:
query-round-fri-layer-transcripts *idx* *trace-roots* *composition-roots*
chunk *trace-layers* *composition-layers*
and *right*:
query-round-fri-layer-transcripts *idx* *trace-roots* *composition-roots*
chunk *trace-layers'* *composition-layers'*
shows *trace-layers* = *trace-layers'*
and *composition-layers* = *composition-layers'*

proof –

from *left* **obtain** *query-chunk* *trace-fri-chunk* *composition-fri-chunk*
leaves *paths* **where**
chunk-left:
chunk = *query-chunk* @ *trace-fri-chunk* @ *composition-fri-chunk*
and *query-left*:
query-decommitment-transcript *idx* *leaves* *paths* *query-chunk*
and *trace-left*:
fri-layers-transcript (*length* *trace-roots*) (*length* * *scale*)
trace-layers *trace-fri-chunk*
and *composition-left*:

```

    fri-layers-transcript (length composition-roots) (clength * scale)
    composition-layers composition-fri-chunk
  unfolding query-round-fri-layer-transcripts-def by blast
from right obtain query-chunk' trace-fri-chunk' composition-fri-chunk'
  leaves' paths' where
chunk-right:
  chunk = query-chunk' @ trace-fri-chunk' @ composition-fri-chunk'
and query-right:
  query-decommitment-transcript idx leaves' paths' query-chunk'
and trace-right:
  fri-layers-transcript (length trace-roots) (clength * scale)
  trace-layers' trace-fri-chunk'
and composition-right:
  fri-layers-transcript (length composition-roots) (clength * scale)
  composition-layers' composition-fri-chunk'
  unfolding query-round-fri-layer-transcripts-def by blast
have len-query:
  length query-chunk = length query-chunk'
  using query-decommitment-transcript-length[OF query-left]
  query-decommitment-transcript-length[OF query-right]
  by simp
have len-trace:
  length trace-fri-chunk = length trace-fri-chunk'
  using fri-layers-transcript-length[OF trace-left]
  fri-layers-transcript-length[OF trace-right]
  by simp
have trace-chunk-eq: trace-fri-chunk = trace-fri-chunk'
proof -
  have left-drop:
    drop (length query-chunk) chunk =
      trace-fri-chunk @ composition-fri-chunk
    using chunk-left by simp
  have right-drop:
    drop (length query-chunk) chunk =
      trace-fri-chunk' @ composition-fri-chunk'
    using chunk-right len-query by simp
  have left-take:
    trace-fri-chunk =
      take (length trace-fri-chunk) (drop (length query-chunk) chunk)
    using left-drop by simp
  have right-take:
    trace-fri-chunk' =
      take (length trace-fri-chunk') (drop (length query-chunk) chunk)
    using right-drop by simp
  show ?thesis
    using left-take right-take len-trace by simp
qed
have composition-chunk-eq:
  composition-fri-chunk = composition-fri-chunk'

```

```

proof –
  have left-drop:
    drop (length query-chunk + length trace-fri-chunk) chunk =
      composition-fri-chunk
    using chunk-left by simp
  have right-drop:
    drop (length query-chunk + length trace-fri-chunk) chunk =
      composition-fri-chunk'
    using chunk-right len-query len-trace by simp
  show ?thesis
    using left-drop right-drop by simp
qed
have trace-right':
  fri-layers-transcript (length trace-roots) (clength * scale)
    trace-layers' trace-fri-chunk
  using trace-right trace-chunk-eq by simp
have composition-right':
  fri-layers-transcript (length composition-roots) (clength * scale)
    composition-layers' composition-fri-chunk
  using composition-right composition-chunk-eq by simp
show trace-layers = trace-layers'
  by (rule fri-layers-transcript-unique[OF trace-left trace-right'])
show composition-layers = composition-layers'
  by (rule fri-layers-transcript-unique
    [OF composition-left composition-right'])
qed

lemma fri-layer-chunk-authenticatedI:
assumes fri-layer-opening-chunk len xp xp-path xn xn-path chunk
and authenticated-opening-in s
  (opening-root = rt,
   opening-length = len,
   opening-index = idx,
   opening-value = xp,
   opening-path = xp-path)
and authenticated-opening-in s
  (opening-root = rt,
   opening-length = len,
   opening-index = fri-sibling-index len idx,
   opening-value = xn,
   opening-path = xn-path)
shows fri-layer-chunk-authenticated rt len idx chunk s
using assms unfolding fri-layer-chunk-authenticated-def by blast

lemma fri-layer-chunk-authenticatedE:
assumes fri-layer-chunk-authenticated rt len idx chunk s
obtains xp xp-path xn xn-path where
  fri-layer-opening-chunk len xp xp-path xn xn-path chunk
  authenticated-opening-in s

```

```

  (opening-root = rt,
   opening-length = len,
   opening-index = idx,
   opening-value = xp,
   opening-path = xp-path)
authenticated-opening-in s
  (opening-root = rt,
   opening-length = len,
   opening-index = fri-sibling-index len idx,
   opening-value = xn,
   opening-path = xn-path)
using assms unfolding fri-layer-chunk-authenticated-def by blast

```

lemma *fri-layer-chunk-authenticated-matching-values:*

assumes *auth: fri-layer-chunk-authenticated rt len idx chunk s*

and *chunk:*

fri-layer-opening-chunk len yp yp-path yn yn-path chunk

obtains *xp-path xn-path* **where**

fri-layer-opening-chunk len yp xp-path yn xn-path chunk

authenticated-opening-in s

```

  (opening-root = rt,
   opening-length = len,
   opening-index = idx,
   opening-value = yp,
   opening-path = xp-path)

```

authenticated-opening-in s

```

  (opening-root = rt,
   opening-length = len,
   opening-index = fri-sibling-index len idx,
   opening-value = yn,
   opening-path = xn-path)

```

proof –

from *auth* **obtain** *xp xp-path xn xn-path* **where**

chunk': fri-layer-opening-chunk len xp xp-path xn xn-path chunk

and *xp-auth:*

authenticated-opening-in s

```

  (opening-root = rt,
   opening-length = len,
   opening-index = idx,
   opening-value = xp,
   opening-path = xp-path)

```

and *xn-auth:*

authenticated-opening-in s

```

  (opening-root = rt,
   opening-length = len,
   opening-index = fri-sibling-index len idx,
   opening-value = xn,
   opening-path = xn-path)

```

by (*rule fri-layer-chunk-authenticatedE*)

```

have xp-eq: xp = yp
  by (rule fri-layer-opening-chunk-values-unique(1)[OF chunk' chunk])
have xn-eq: xn = yn
  by (rule fri-layer-opening-chunk-values-unique(2)[OF chunk' chunk])
have chunk-y:
  fri-layer-opening-chunk len yp xp-path yn xn-path chunk
  using chunk' xp-eq xn-eq by simp
show ?thesis
  by (rule that[OF chunk-y])
    (use xp-auth xn-auth xp-eq xn-eq in simp-all)
qed

```

```

lemma authenticated-openings-values-eq-if-no-partial-merkle:
  assumes no-bad:
     $\neg$  partial-merkle-inconsistency-bad s
    (Some (result, final-state))
  and first: authenticated-opening-in final-state opn
  and second: authenticated-opening-in final-state opn'
  and same-root: opening-root opn = opening-root opn'
  and same-length: opening-length opn = opening-length opn'
  and same-index: opening-index opn = opening-index opn'
  shows opening-value opn = opening-value opn'
proof (rule ccontr)
  assume neg: opening-value opn  $\neq$  opening-value opn'
  have partial-merkle-inconsistency-bad s
    (Some (result, final-state))
  unfolding partial-merkle-inconsistency-bad-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state]
    exI[of - opn] exI[of - opn'])
    (use first second same-root same-length same-index neg in simp-all)
  then show False using no-bad by contradiction
qed

```

```

lemma fri-layer-authenticated-base-values-eq-if-no-partial-merkle:
  assumes no-bad:
     $\neg$  partial-merkle-inconsistency-bad s
    (Some (result, final-state))
  and auth:
    fri-layer-chunk-authenticated rt len idx chunk final-state
  and auth':
    fri-layer-chunk-authenticated rt len idx chunk' final-state
  and chunk:
    fri-layer-opening-chunk len xp xp-path xn xn-path chunk
  and chunk':
    fri-layer-opening-chunk len yp yp-path yn yn-path chunk'
  shows xp = yp
proof (rule ccontr)
  assume neg: xp  $\neq$  yp
  from fri-layer-chunk-authenticated-matching-values[OF auth chunk]

```



```

obtain xp-path' xn-path' where
  xp-auth:
    authenticated-opening-in final-state
    ( $\langle$ opening-root = rt,
      opening-length = len,
      opening-index = idx,
      opening-value = xp,
      opening-path = xp-path' $\rangle$ )
  by blast
from fri-layer-chunk-authenticated-matching-values[OF auth' chunk']
obtain yp-path' yn-path' where
  yp-auth:
    authenticated-opening-in final-state
    ( $\langle$ opening-root = rt,
      opening-length = len,
      opening-index = idx,
      opening-value = yp,
      opening-path = yp-path' $\rangle$ )
  by blast
have partial-merkle-inconsistency-bad s
  (Some (result, final-state))
unfolding partial-merkle-inconsistency-bad-def accepted-def
by (intro conjI exI[of - result] exI[of - final-state]
    exI[of - ( $\langle$ opening-root = rt,
      opening-length = len,
      opening-index = idx,
      opening-value = xp,
      opening-path = xp-path' $\rangle$ )]
    exI[of - ( $\langle$ opening-root = rt,
      opening-length = len,
      opening-index = idx,
      opening-value = yp,
      opening-path = yp-path' $\rangle$ )]])
  (use xp-auth yp-auth neq in simp-all)
then show False using no-bad by contradiction
qed

lemma fri-layer-authenticated-sibling-values-eq-if-no-partial-merkle:
assumes no-bad:
   $\neg$  partial-merkle-inconsistency-bad s
  (Some (result, final-state))
and auth:
  fri-layer-chunk-authenticated rt len idx chunk final-state
and auth':
  fri-layer-chunk-authenticated rt len idx chunk' final-state
and chunk:
  fri-layer-opening-chunk len xp xp-path xn xn-path chunk
and chunk':
  fri-layer-opening-chunk len yp yp-path yn yn-path chunk'

```

```

shows  $xn = yn$ 
proof (rule ccontr)
  assume  $neg: xn \neq yn$ 
  from fri-layer-chunk-authenticated-matching-values[OF auth chunk]
  obtain  $xp\text{-}path'$   $xn\text{-}path'$  where
    xn-auth:
      authenticated-opening-in final-state
      ( $\langle opening\text{-}root = rt,$ 
         $opening\text{-}length = len,$ 
         $opening\text{-}index = fri\text{-}sibling\text{-}index\ len\ idx,$ 
         $opening\text{-}value = xn,$ 
         $opening\text{-}path = xn\text{-}path' \rangle$ )
    by blast
  from fri-layer-chunk-authenticated-matching-values[OF auth' chunk']
  obtain  $yp\text{-}path'$   $yn\text{-}path'$  where
    yn-auth:
      authenticated-opening-in final-state
      ( $\langle opening\text{-}root = rt,$ 
         $opening\text{-}length = len,$ 
         $opening\text{-}index = fri\text{-}sibling\text{-}index\ len\ idx,$ 
         $opening\text{-}value = yn,$ 
         $opening\text{-}path = yn\text{-}path' \rangle$ )
    by blast
  have partial-merkle-inconsistency-bad s
    (Some (result, final-state))
  unfolding partial-merkle-inconsistency-bad-def accepted-def
  by (intro conjI exI[of - result] exI[of - final-state]
     $exI[of - \langle opening\text{-}root = rt,$ 
       $opening\text{-}length = len,$ 
       $opening\text{-}index = fri\text{-}sibling\text{-}index\ len\ idx,$ 
       $opening\text{-}value = xn,$ 
       $opening\text{-}path = xn\text{-}path' \rangle]$ 
     $exI[of - \langle opening\text{-}root = rt,$ 
       $opening\text{-}length = len,$ 
       $opening\text{-}index = fri\text{-}sibling\text{-}index\ len\ idx,$ 
       $opening\text{-}value = yn,$ 
       $opening\text{-}path = yn\text{-}path' \rangle]$ )
    (use xn-auth yn-auth neg in simp-all)
  then show False using no-bad by contradiction
qed

lemma fri-layer-authenticated-base-sibling-values-eq-if-no-partial-merkle:
  assumes no-bad:
     $\neg$  partial-merkle-inconsistency-bad s
    (Some (result, final-state))
  and auth:
    fri-layer-chunk-authenticated rt len idx chunk final-state
  and auth':
    fri-layer-chunk-authenticated rt len idx' chunk' final-state

```

```

and chunk:
  fri-layer-opening-chunk len xp xp-path xn xn-path chunk
and chunk':
  fri-layer-opening-chunk len yp yp-path yn yn-path chunk'
and same-index: idx = fri-sibling-index len idx'
shows xp = yn
proof –
from fri-layer-chunk-authenticated-matching-values[OF auth chunk]
obtain xp-path' xn-path' where
  xp-auth:
    authenticated-opening-in final-state
    (opening-root = rt,
     opening-length = len,
     opening-index = idx,
     opening-value = xp,
     opening-path = xp-path')
  by blast
from fri-layer-chunk-authenticated-matching-values[OF auth' chunk']
obtain yp-path' yn-path' where
  yn-auth:
    authenticated-opening-in final-state
    (opening-root = rt,
     opening-length = len,
     opening-index = fri-sibling-index len idx',
     opening-value = yn,
     opening-path = yn-path')
  by blast
show ?thesis
using authenticated-openings-values-eq-if-no-partial-merkle
  [OF no-bad xp-auth yn-auth] same-index
by simp
qed

lemma fri-layer-authenticated-sibling-base-values-eq-if-no-partial-merkle:
assumes no-bad:
   $\neg$  partial-merkle-inconsistency-bad s
  (Some (result, final-state))
and auth:
  fri-layer-chunk-authenticated rt len idx chunk final-state
and auth':
  fri-layer-chunk-authenticated rt len idx' chunk' final-state
and chunk:
  fri-layer-opening-chunk len xp xp-path xn xn-path chunk
and chunk':
  fri-layer-opening-chunk len yp yp-path yn yn-path chunk'
and same-index: fri-sibling-index len idx = idx'
shows xn = yp
proof –
from fri-layer-chunk-authenticated-matching-values[OF auth chunk]

```

```

obtain xp-path' xn-path' where
  xn-auth:
    authenticated-opening-in final-state
    ( $\parallel$  opening-root = rt,
      opening-length = len,
      opening-index = fri-sibling-index len idx,
      opening-value = xn,
      opening-path = xn-path')
  by blast
from fri-layer-chunk-authenticated-matching-values[OF auth' chunk']
obtain yp-path' yn-path' where
  yp-auth:
    authenticated-opening-in final-state
    ( $\parallel$  opening-root = rt,
      opening-length = len,
      opening-index = idx',
      opening-value = yp,
      opening-path = yp-path')
  by blast
show ?thesis
  using authenticated-openings-values-eq-if-no-partial-merkle
    [OF no-bad xn-auth yp-auth] same-index
  by simp
qed

lemma fri-layer-authenticated-sibling-sibling-values-eq-if-no-partial-merkle:
  assumes no-bad:
     $\neg$  partial-merkle-inconsistency-bad s
    (Some (result, final-state))
  and auth:
    fri-layer-chunk-authenticated rt len idx chunk final-state
  and auth':
    fri-layer-chunk-authenticated rt len idx' chunk' final-state
  and chunk:
    fri-layer-opening-chunk len xp xp-path xn xn-path chunk
  and chunk':
    fri-layer-opening-chunk len yp yp-path yn yn-path chunk'
  and same-index:
    fri-sibling-index len idx = fri-sibling-index len idx'
  shows xn = yn
proof –
from fri-layer-chunk-authenticated-matching-values[OF auth chunk]
obtain xp-path' xn-path' where
  xn-auth:
    authenticated-opening-in final-state
    ( $\parallel$  opening-root = rt,
      opening-length = len,
      opening-index = fri-sibling-index len idx,
      opening-value = xn,

```

```

    opening-path = xn-path')
  by blast
from fri-layer-chunk-authenticated-matching-values[OF auth' chunk']
obtain yp-path' yn-path' where
  yn-auth:
    authenticated-opening-in final-state
    (opening-root = rt,
     opening-length = len,
     opening-index = fri-sibling-index len idx',
     opening-value = yn,
     opening-path = yn-path')
  by blast
show ?thesis
  using authenticated-openings-values-eq-if-no-partial-merkle
    [OF no-bad xn-auth yn-auth] same-index
  by simp
qed

```

```

lemma fri-layer-chunk-authenticated-mono:
  assumes auth: fri-layer-chunk-authenticated rt len idx chunk s
    and ext:  $s \leq t$ 
  shows fri-layer-chunk-authenticated rt len idx chunk t
  using auth
  unfolding fri-layer-chunk-authenticated-def
  by (meson authenticated-opening-in-mono ext)

```

```

lemma fri-mfold-chunks-authenticated-mono:
  assumes auth: fri-mfold-chunks-authenticated bfs len idx chunks s
    and ext:  $s \leq t$ 
  shows fri-mfold-chunks-authenticated bfs len idx chunks t
  using auth
  unfolding fri-mfold-chunks-authenticated-def
  by (meson fri-layer-chunk-authenticated-mono ext)

```

```

lemma fri-layer-opening-step-authenticated-chunk:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes len-pos:  $0 < len$ 
    and idx-bound:  $idx < len$ 
    and outcome:
      Some (out, t) ∈
        set-dist
          (execute (fri-layer-opening-step (b, rt) (idx, x, len, pw)) s)
  obtains xp xp-path xn xn-path where
    fri-layer-opening-chunk len xp xp-path xn xn-path
      ([xp] @ xp-path @ [xn] @ xn-path)
    authenticated-opening-in t
      (opening-root = rt,
       opening-length = len,
       opening-index = idx,

```

```

    opening-value = xp,
    opening-path = xp-path)
authenticated-opening-in t
  (opening-root = rt,
   opening-length = len,
   opening-index = fri-sibling-index len idx,
   opening-value = xn,
   opening-path = xn-path)
proof –
from outcome[unfolded fri-layer-opening-step-unfold-finish]
obtain xp s1 xp-path s2 xn s3 xn-path s4 where
  read-xp: Some (xp, s1) ∈ set-dist (execute read s)
and read-xp-path:
  Some (xp-path, s2) ∈
    set-dist (execute (ntimes read (floor-log len)) s1)
and read-xn: Some (xn, s3) ∈ set-dist (execute read s2)
and read-xn-path:
  Some (xn-path, s4) ∈
    set-dist (execute (ntimes read (floor-log len)) s3)
and finish:
  Some (out, t) ∈
    set-dist
      (execute
        (fri-layer-opening-finish b rt idx x len pw
         xp xp-path xn xn-path) s4)
by (auto elim!: set-dist-bindE)
have xp-path-length: length xp-path = floor-log len
  using ntimes-read-any-outcome[OF read-xp-path] by simp
have xn-path-length: length xn-path = floor-log len
  using ntimes-read-any-outcome[OF read-xn-path] by simp
have sibling-bound: (idx + len div 2) mod len < len
  using len-pos by simp
from fri-layer-opening-finish-authenticated-openings
  [OF idx-bound sibling-bound xp-path-length xn-path-length finish]
have xp-auth:
  authenticated-opening-in t
    (opening-root = rt,
     opening-length = len,
     opening-index = idx,
     opening-value = xp,
     opening-path = xp-path)
and xn-auth:
  authenticated-opening-in t
    (opening-root = rt,
     opening-length = len,
     opening-index = fri-sibling-index len idx,
     opening-value = xn,
     opening-path = xn-path)
unfolding fri-sibling-index-def

```

```

    by blast+
  have chunk:
    fri-layer-opening-chunk len xp xp-path xn xn-path
      ([xp] @ xp-path @ [xn] @ xn-path)
    using xp-path-length xn-path-length
    unfolding fri-layer-opening-chunk-def by simp
  show ?thesis
    by (rule that[OF chunk xp-auth xn-auth])
qed

lemma fri-layer-opening-step-authenticated-chunk-pred:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes len-pos: 0 < len
  and idx-bound: idx < len
  and outcome:
    Some (out, t) ∈
      set-dist
        (execute (fri-layer-opening-step (b, rt) (idx, x, len, pw)) s)
  obtains chunk where fri-layer-chunk-authenticated rt len idx chunk t
proof -
  from fri-layer-opening-step-authenticated-chunk
    [OF len-pos idx-bound outcome]
  obtain xp xp-path xn xn-path where
    chunk:
      fri-layer-opening-chunk len xp xp-path xn xn-path
        ([xp] @ xp-path @ [xn] @ xn-path)
  and xp-auth:
    authenticated-opening-in t
      (|opening-root = rt,
        opening-length = len,
        opening-index = idx,
        opening-value = xp,
        opening-path = xp-path)
  and xn-auth:
    authenticated-opening-in t
      (|opening-root = rt,
        opening-length = len,
        opening-index = fri-sibling-index len idx,
        opening-value = xn,
        opening-path = xn-path)
  by blast
  have fri-layer-chunk-authenticated rt len idx
    ([xp] @ xp-path @ [xn] @ xn-path) t
    by (rule fri-layer-chunk-authenticatedI[OF chunk xp-auth xn-auth])
  then show ?thesis
    by (rule that)
qed

lemma fri-layer-opening-step-outcome-authenticated:

```

fixes $s\ t :: ('f, 'a) \text{ protocol-channel-scheme}$
assumes $\text{len-pos}: 0 < \text{len}$
and $\text{idx-bound}: \text{idx} < \text{len}$
and outcome :
 $\text{Some } (\text{out}, t) \in$
 set-dist
 $(\text{execute } (\text{fri-layer-opening-step } (b, rt) (\text{idx}, x, \text{len}, pw))\ s)$
obtains $xp\ xp\text{-path}\ xn\ xn\text{-path}\ x'\ \text{chunk}$ **where**
 $\text{out} = (\text{idx} \bmod (\text{len} \div 2), x', \text{len} \div 2, pw + pw)$
 $xp = x$
 $x' = \text{fri-fold-value } b\ xp\ xn$
 $(\text{fri-fold-denominator } ((h \wedge \text{idx}) * \text{shift})\ pw)$
 $\text{chunk} = [xp] @ xp\text{-path} @ [xn] @ xn\text{-path}$
 $\text{fri-layer-opening-chunk } \text{len}\ xp\ xp\text{-path}\ xn\ xn\text{-path}\ \text{chunk}$
 $P\text{Transcript } s = \text{chunk} @ P\text{Transcript } t$
 $P\text{State } t = \text{foldl concat } (P\text{State } s)\ \text{chunk}$
 $s \leq t$
 $P\text{QueryCounter } t = P\text{QueryCounter } s$
 $\text{authenticated-opening-in } t$
 $(\text{opening-root} = rt,$
 $\text{opening-length} = \text{len},$
 $\text{opening-index} = \text{idx},$
 $\text{opening-value} = xp,$
 $\text{opening-path} = xp\text{-path})$
 $\text{authenticated-opening-in } t$
 $(\text{opening-root} = rt,$
 $\text{opening-length} = \text{len},$
 $\text{opening-index} = \text{fri-sibling-index } \text{len}\ \text{idx},$
 $\text{opening-value} = xn,$
 $\text{opening-path} = xn\text{-path})$
proof –
from outcome **obtain** $xp\ s1\ xp\text{-path}\ s2\ xn\ s3\ xn\text{-path}\ s4\ s5\ ap1\ s6\ s7$
 $ap2\ s8\ s9$ **where**
 $\text{read-xp}: \text{Some } (xp, s1) \in \text{set-dist } (\text{execute read } s)$
and read-xp-path :
 $\text{Some } (xp\text{-path}, s2) \in$
 $\text{set-dist } (\text{execute } (\text{ntimes read } (\text{floor-log } \text{len}))\ s1)$
and $\text{read-xn}: \text{Some } (xn, s3) \in \text{set-dist } (\text{execute read } s2)$
and read-xn-path :
 $\text{Some } (xn\text{-path}, s4) \in$
 $\text{set-dist } (\text{execute } (\text{ntimes read } (\text{floor-log } \text{len}))\ s3)$
and $\text{assert-xp}: \text{Some } ((), s5) \in \text{set-dist } (\text{execute } (\text{assert } (xp = x))\ s4)$
and check1 :
 $\text{Some } (ap1, s6) \in$
 $\text{set-dist } (\text{execute } (\text{check-authentication-path } \text{len}\ \text{idx}\ xp\ xp\text{-path})\ s5)$
and assert-ap1 :
 $\text{Some } ((), s7) \in \text{set-dist } (\text{execute } (\text{assert } (ap1 = rt))\ s6)$
and check2 :
 $\text{Some } (ap2, s8) \in$


```

    set-dist (execute
      (check-authentication-path len ((idx + len div 2) mod len) xn
        xn-path) s7)
  and assert-ap2:
    Some ((), s9) ∈ set-dist (execute (assert (ap2 = rt)) s8)
  and ret:
    Some (out, t) ∈
      set-dist (execute
        (return
          (idx mod (len div 2),
            (xp + xn) div 2 +
              b * ((xp - xn) div (2 * ((h ^ idx) * shift) ^ pw)),
            len div 2, pw + pw)) s9)
  unfolding fri-layer-opening-step-def
  by (auto simp: Let-def elim!: set-dist-bindE)
from read-outcome[OF read-xp] obtain rest1 where
  tr-s: PTranscript s = xp # rest1
  and st-s1: PState s1 = concat (PState s) xp
  and tr-s1: PTranscript s1 = rest1
  and ext-s-s1: s ≤ s1
  and query-count-s1: PQueryCounter s1 = PQueryCounter s
  by blast
have xp-path-res:
  length xp-path = floor-log len ∧
  PTranscript s1 = xp-path @ PTranscript s2 ∧
  PState s2 = foldl concat (PState s1) xp-path ∧
  s1 ≤ s2 ∧
  PQueryCounter s2 = PQueryCounter s1
  using ntimes-read-any-outcome[OF read-xp-path] by simp
from read-outcome[OF read-xn] obtain rest3 where
  tr-s2: PTranscript s2 = xn # rest3
  and st-s3: PState s3 = concat (PState s2) xn
  and tr-s3: PTranscript s3 = rest3
  and ext-s2-s3: s2 ≤ s3
  and query-count-s3: PQueryCounter s3 = PQueryCounter s2
  by blast
have xn-path-res:
  length xn-path = floor-log len ∧
  PTranscript s3 = xn-path @ PTranscript s4 ∧
  PState s4 = foldl concat (PState s3) xn-path ∧
  s3 ≤ s4 ∧
  PQueryCounter s4 = PQueryCounter s3
  using ntimes-read-any-outcome[OF read-xn-path] by simp
have xp-eq: xp = x
  using assert-xp unfolding assert-def
  by (cases xp = x) (auto simp: throw-no-outcome)
have s5-eq: s5 = s4
  using assert-xp xp-eq unfolding assert-def by simp
have s5-s6: s5 ≤ s6

```

```

    by (rule check-authentication-path-hash-extends[OF check1])
  have st-s6: PState s6 = PState s5
    using check-authentication-path-preserves-channel(1)[OF check1] .
  have tr-s6: PTranscript s6 = PTranscript s5
    using check-authentication-path-preserves-channel(2)[OF check1] .
  have query-count-s6: PQueryCounter s6 = PQueryCounter s5
    using check-authentication-path-preserves-channel(6)[OF check1] .
  have ap1-eq: ap1 = rt
    using assert-ap1 unfolding assert-def
    by (cases ap1 = rt) (auto simp: throw-no-outcome)
  have s7-eq: s7 = s6
    using assert-ap1 ap1-eq unfolding assert-def by simp
  have s7-s8: s7 ≤ s8
    by (rule check-authentication-path-hash-extends[OF check2])
  have st-s8: PState s8 = PState s7
    using check-authentication-path-preserves-channel(1)[OF check2] .
  have tr-s8: PTranscript s8 = PTranscript s7
    using check-authentication-path-preserves-channel(2)[OF check2] .
  have query-count-s8: PQueryCounter s8 = PQueryCounter s7
    using check-authentication-path-preserves-channel(6)[OF check2] .
  have ap2-eq: ap2 = rt
    using assert-ap2 unfolding assert-def
    by (cases ap2 = rt) (auto simp: throw-no-outcome)
  have s9-eq: s9 = s8
    using assert-ap2 ap2-eq unfolding assert-def by simp
  let ?x' =
    (xp + xn) div 2 +
    b * ((xp - xn) div (2 * ((h ^ idx) * shift) ^ pw))
  let ?chunk = [xp] @ xp-path @ [xn] @ xn-path
  have out-t:
    out = (idx mod (len div 2), ?x', len div 2, pw + pw) ∧ t = s9
    using ret by simp
  have x'-fold:
    ?x' = fri-fold-value b xp xn
      (fri-fold-denominator ((h ^ idx) * shift) pw)
    unfolding fri-fold-value-def fri-fold-denominator-def by simp
  have chunk-shape:
    fri-layer-opening-chunk len xp xp-path xn xn-path ?chunk
    using xp-path-res xn-path-res unfolding fri-layer-opening-chunk-def by simp
  have tr-t: PTranscript s = ?chunk @ PTranscript t
    using tr-s tr-s1 xp-path-res tr-s2 tr-s3 xn-path-res s5-eq tr-s6
    s7-eq tr-s8 s9-eq out-t
    by simp
  have st-t: PState t = foldl concat (PState s) ?chunk
    using st-s1 xp-path-res st-s3 xn-path-res s5-eq st-s6 s7-eq st-s8
    s9-eq out-t
    by simp
  have s-t: s ≤ t
    using ext-s-s1 xp-path-res ext-s2-s3 xn-path-res s5-s6 s7-s8 out-t

```

```

unfolding s5-eq s7-eq s9-eq
by (meson hash-ext-trans)
have query-count-t: PQueryCounter t = PQueryCounter s
using out-t query-count-s1
  xp-path-res query-count-s3 xn-path-res query-count-s6 query-count-s8
  s5-eq s7-eq s9-eq
by simp
have sibling-bound: (idx + len div 2) mod len < len
  using len-pos by simp
have xp-path-length: length xp-path = floor-log len
  using xp-path-res by simp
have xn-path-length: length xn-path = floor-log len
  using xn-path-res by simp
have s6-t: s6 ≤ t
  using s7-s8 out-t unfolding s7-eq s9-eq by simp
have xp-bound:
  merkle-path-bound rt len idx xp xp-path t
proof –
  have merkle-path-bound ap1 len idx xp xp-path s6
    by (rule check-authentication-path-outcome-bound[OF check1])
  then have merkle-path-bound rt len idx xp xp-path s6
    unfolding ap1-eq .
  then show ?thesis
    by (rule merkle-path-bound-mono[OF - s6-t])
qed
have xn-bound:
  merkle-path-bound rt len (fri-sibling-index len idx) xn xn-path t
proof –
  have merkle-path-bound ap2 len ((idx + len div 2) mod len)
    xn xn-path s8
    by (rule check-authentication-path-outcome-bound[OF check2])
  then show ?thesis
    using ap2-eq out-t unfolding fri-sibling-index-def s9-eq by simp
qed
have xp-auth:
  authenticated-opening-in t
  (|opening-root = rt, opening-length = len,
   opening-index = idx, opening-value = xp,
   opening-path = xp-path|)
  unfolding authenticated-opening-in-def
  using idx-bound xp-path-length xp-bound by simp
have xn-auth:
  authenticated-opening-in t
  (|opening-root = rt, opening-length = len,
   opening-index = fri-sibling-index len idx,
   opening-value = xn, opening-path = xn-path|)
  unfolding authenticated-opening-in-def
  using sibling-bound xn-path-length xn-bound
  unfolding fri-sibling-index-def by simp

```

```

show ?thesis
proof (rule that)
  show out = (idx mod (len div 2), ?x', len div 2, pw + pw)
    using out-t by simp
  show xp = x
    by (rule xp-eq)
  show ?x' =
    fri-fold-value b xp xn
    (fri-fold-denominator ((h ^ idx) * shift) pw)
    by (rule x'-fold)
  show ?chunk = [xp] @ xp-path @ [xn] @ xn-path
    by simp
  show fri-layer-opening-chunk len xp xp-path xn xn-path ?chunk
    by (rule chunk-shape)
  show PTranscript s = ?chunk @ PTranscript t
    by (rule tr-t)
  show PState t = foldl concat (PState s) ?chunk
    by (rule st-t)
  show s ≤ t
    by (rule s-t)
  show PQueryCounter t = PQueryCounter s
    by (rule query-count-t)
  show authenticated-opening-in t
    (opening-root = rt, opening-length = len,
     opening-index = idx, opening-value = xp,
     opening-path = xp-path)
    by (rule xp-auth)
  show authenticated-opening-in t
    (opening-root = rt, opening-length = len,
     opening-index = fri-sibling-index len idx,
     opening-value = xn, opening-path = xn-path)
    by (rule xn-auth)
qed
qed

lemma mfold-fri-layer-opening-steps-hash-extends:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (out, t) ∈
      set-dist
        (execute (mfold (idx, x, len, pw)
          (map fri-layer-opening-step bfs)) s)
  shows s ≤ t
proof –
  from mfold-fri-layer-openings-outcome[OF outcome]
  obtain layer-chunks chunk where
    s ≤ t
    by blast
  then show ?thesis .

```

qed

lemma *mfold-fri-layer-opening-steps-authenticated-opening-mono*:
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes *auth*: *authenticated-opening-in s opening*
and *outcome*:
 $\text{Some } (out, t) \in$
 set-dist
 $(\text{execute } (\text{mfold } (idx, x, len, pw)$
 $(\text{map } \text{fri-layer-opening-step } bfs))\ s)$
shows *authenticated-opening-in t opening*
by (*rule authenticated-opening-in-mono*
 $[OF\ \text{auth}\ \text{mfold-fri-layer-opening-steps-hash-extends}[OF\ \text{outcome}]])$

lemma *mfold-fri-layer-opening-head-chunk-authenticated*:
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes *len-pos*: $0 < len$
and *idx-bound*: $idx < len$
and *outcome*:
 $\text{Some } (out, t) \in$
 set-dist
 $(\text{execute } (\text{mfold } (idx, x, len, pw)$
 $(\text{map } \text{fri-layer-opening-step } ((b, rt) \# bfs)))\ s)$
obtains *chunk* **where** *fri-layer-chunk-authenticated rt len idx chunk t*
proof –
from *outcome* **obtain** *out1 s1* **where**
head:
 $\text{Some } (out1, s1) \in$
 set-dist
 $(\text{execute } (\text{fri-layer-opening-step } (b, rt) (idx, x, len, pw))\ s)$
and *tail*:
 $\text{Some } (out, t) \in$
 $\text{set-dist } (\text{execute } (\text{mfold } out1\ (\text{map } \text{fri-layer-opening-step } bfs))\ s1)$
by (*auto elim!*: *set-dist-bindE*)
from *fri-layer-opening-step-authenticated-chunk-pred*
 $[OF\ \text{len-pos}\ \text{idx-bound}\ \text{head}]$
obtain *chunk* **where** *auth-s1*:
fri-layer-chunk-authenticated rt len idx chunk s1
by *blast*
obtain *idx1 x1 len1 pw1* **where** *out1-eq*:
 $out1 = (idx1, x1, len1, pw1)$
by (*cases out1*)
have *s1-t*: $s1 \leq t$
by (*rule mfold-fri-layer-opening-steps-hash-extends*
 $[OF\ \text{tail}[\text{unfolded } out1\text{-eq}]])$
have *fri-layer-chunk-authenticated rt len idx chunk t*
by (*rule fri-layer-chunk-authenticated-mono* $[OF\ \text{auth-s1}\ s1\text{-}t])$
then show *?thesis*

```

    by (rule that)
qed

lemma mfold-fri-layer-opening-selected-chunk-authenticated-at-exists:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (out, t) ∈
      set-dist
        (execute
          (mfold (idx, x, len, pw)
            (map fri-layer-opening-step bfs)) s)
  and j-bound: j < length bfs
  and layer-len-pos:
    0 < fri-layer-lengths (length bfs) len ! j
  and layer-idx-bound:
    fri-layer-indices (length bfs) idx len ! j <
      fri-layer-lengths (length bfs) len ! j
  shows ∃ chunk.
    fri-layer-chunk-authenticated (snd (bfs ! j))
      (fri-layer-lengths (length bfs) len ! j)
      (fri-layer-indices (length bfs) idx len ! j)
      chunk t
  using outcome j-bound layer-len-pos layer-idx-bound
proof (induction bfs arbitrary: idx x len pw s out t j)
  case Nil
  then show ?case by simp
next
  case (Cons bf bfs)
  obtain b rt where bf-eq: bf = (b, rt)
  by (cases bf)
  from Cons.prem1[unfolded bf-eq]
  obtain idx1 x1 len1 pw1 s1 where
    head:
      Some ((idx1, x1, len1, pw1), s1) ∈
        set-dist
          (execute (fri-layer-opening-step (b, rt) (idx, x, len, pw)) s)
  and tail:
    Some (out, t) ∈
      set-dist
        (execute
          (mfold (idx1, x1, len1, pw1)
            (map fri-layer-opening-step bfs)) s1)
  by (auto elim!: set-dist-bindE)
  show ?case
proof (cases j)
  case 0
  have len-pos: 0 < len
  using Cons.prem3 0 by simp
  have idx-bound: idx < len

```

```

    using Cons.premis(4) 0 by simp
  from fri-layer-opening-step-authenticated-chunk-pred
    [OF len-pos idx-bound head]
  obtain chunk where auth-s1:
    fri-layer-chunk-authenticated rt len idx chunk s1
  by blast
  have s1-t: s1 ≤ t
  by (rule mfold-fri-layer-opening-steps-hash-extends
    [OF tail])
  have auth-t:
    fri-layer-chunk-authenticated rt len idx chunk t
  by (rule fri-layer-chunk-authenticated-mono[OF auth-s1 s1-t])
  show ?thesis
  using auth-t 0 bf-eq by auto
next
case (Suc k)
from fri-layer-opening-step-outcome[OF head]
obtain xp xp-path xn xn-path x' where out1-eq:
  (idx1, x1, len1, pw1) =
  (idx mod (len div 2), x', len div 2, pw + pw)
  by blast
have k-bound: k < length bfs
  using Cons.premis(2) Suc by simp
have tail-len-pos:
  0 < fri-layer-lengths (length bfs) (len div 2) ! k
  using Cons.premis(3) Suc by simp
have tail-idx-bound:
  fri-layer-indices (length bfs) (idx mod (len div 2))
  (len div 2) ! k <
  fri-layer-lengths (length bfs) (len div 2) ! k
  using Cons.premis(4) Suc by simp
show ?thesis
  using Cons.IH[OF tail[unfolded out1-eq] k-bound
    tail-len-pos tail-idx-bound] Suc
  by auto
qed
qed

lemma mfold-fri-layer-opening-selected-chunk-authenticated-at:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (out, t) ∈
    set-dist
    (execute
      (mfold (idx, x, len, pw)
        (map fri-layer-opening-step bfs)) s)
  and j-bound: j < length bfs
  and layer-len-pos:
    0 < fri-layer-lengths (length bfs) len ! j

```

```

and layer-idx-bound:
  fri-layer-indices (length bfs) idx len ! j <
    fri-layer-lengths (length bfs) len ! j
obtains chunk where
  fri-layer-chunk-authenticated (snd (bfs ! j))
    (fri-layer-lengths (length bfs) len ! j)
    (fri-layer-indices (length bfs) idx len ! j)
    chunk t
using mfold-fri-layer-opening-selected-chunk-authenticated-at-exists
  [OF outcome j-bound layer-len-pos layer-idx-bound]
by blast

lemma receive-query-commits-selected-chunk-authenticated-at:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (out, t) ∈
    set-dist
    (execute
      (mfold (idx, x, len, pw) (receive-query-commits bfs)) s)
and j-bound: j < length bfs
and layer-len-pos:
  0 < fri-layer-lengths (length bfs) len ! j
and layer-idx-bound:
  fri-layer-indices (length bfs) idx len ! j <
    fri-layer-lengths (length bfs) len ! j
obtains chunk where
  fri-layer-chunk-authenticated (snd (bfs ! j))
    (fri-layer-lengths (length bfs) len ! j)
    (fri-layer-indices (length bfs) idx len ! j)
    chunk t
proof –
  have map-eq: receive-query-commits bfs = map fri-layer-opening-step bfs
  unfolding receive-query-commits-def fri-layer-opening-step-def
  by (simp add: fun-eq-iff split: prod.splits)
  show ?thesis
  by (rule mfold-fri-layer-opening-selected-chunk-authenticated-at
    [OF outcome[unfolded map-eq] j-bound layer-len-pos
      layer-idx-bound that])
qed

lemma receive-query-commits-mfold-hash-extends:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (out, t) ∈
    set-dist
    (execute (mfold (idx, x, len, pw)
      (receive-query-commits bfs)) s)
shows s ≤ t
proof –

```


have *map-eq*: *receive-query-commits bfs* = *map fri-layer-opening-step bfs*
unfolding *receive-query-commits-def fri-layer-opening-step-def*
by (*simp add: fun-eq-iff split: prod.splits*)
show *?thesis*
by (*rule mfold-fri-layer-opening-steps-hash-extends*
[OF outcome[unfolded map-eq]])
qed

lemma *receive-query-commits-mfold-authenticated-opening-mono*:
fixes *s t :: ('f, 'a) protocol-channel-scheme*
assumes *auth: authenticated-opening-in s opening*
and *outcome:*
Some (out, t) ∈
set-dist
(execute (mfold (idx, x, len, pw)
(receive-query-commits bfs)) s)
shows *authenticated-opening-in t opening*
by (*rule authenticated-opening-in-mono*
[OF auth receive-query-commits-mfold-hash-extends[OF outcome]])

lemma *receive-query-commits-mfold-head-chunk-authenticated*:
fixes *s t :: ('f, 'a) protocol-channel-scheme*
assumes *len-pos: 0 < len*
and *idx-bound: idx < len*
and *outcome:*
Some (out, t) ∈
set-dist
(execute
(mfold (idx, x, len, pw)
(receive-query-commits ((b, rt) # bfs))) s)
obtains *chunk where fri-layer-chunk-authenticated rt len idx chunk t*
proof –
have *map-eq*:
receive-query-commits ((b, rt) # bfs) =
map fri-layer-opening-step ((b, rt) # bfs)
unfolding *receive-query-commits-def fri-layer-opening-step-def*
by (*simp add: fun-eq-iff split: prod.splits*)
from *mfold-fri-layer-opening-head-chunk-authenticated*
[OF len-pos idx-bound outcome[unfolded map-eq]]
obtain *chunk where fri-layer-chunk-authenticated rt len idx chunk t*
by *blast*
then show *?thesis*
by (*rule that*)
qed

lemma *verifier-query-round-after-index-trace-fri-head-chunk-authenticated*:
fixes *s t :: ('f, 'a) protocol-channel-scheme*
assumes *f-fl-eq: f-fl = (b, rt) # f-fl-tail*
and *outcome:*

```

    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s)
obtains idx chunk where
  idx = index (to-nat raw)
  fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
proof –
  let ?idx = index (to-nat raw)
from outcome obtain fv s1 f-out s2 s3 c-out s4 where
  query-decommit:
    Some (fv, s1) ∈
      set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
and trace-fri:
  Some (f-out, s2) ∈
    set-dist (execute
      (mfold (?idx, hd fv, clength * scale, 1)
        (receive-query-commits f-fl)) s1)
and assert-trace:
  Some (( ), s3) ∈
    set-dist
      (execute
        (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
  Some (c-out, s4) ∈
    set-dist (execute
      (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
        clength * scale, 1)
        (receive-query-commits fl)) s3)
and assert-comp:
  Some (( ), t) ∈
    set-dist
      (execute
        (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
  unfolding verifier-query-round-after-index-program-def
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have len-pos: 0 < clength * scale
  using eval-domain-nontrivial by linarith
have idx-bound: ?idx < clength * scale
  by (rule index-less-domain)
from receive-query-commits-mfold-head-chunk-authenticated
  [OF len-pos idx-bound trace-fri[unfolded f-fl-eq]]
obtain chunk where auth-s2:
  fri-layer-chunk-authenticated rt (clength * scale) ?idx chunk s2
  by blast
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def
  by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)

```

```

    (auto simp: throw-no-outcome)
  have s3-s4:  $s3 \leq s4$ 
  by (rule receive-query-commits-mfold-hash-extends[OF comp-fri])
  have t-eq:  $t = s4$ 
  using assert-comp unfolding assert-def
  by (cases case c-out of  $(-, x, -, -) \Rightarrow x = final$ )
    (auto simp: throw-no-outcome)
  have s2-t:  $s2 \leq t$ 
  using s3-s4 unfolding s3-eq t-eq by simp
  have auth-t:
    fri-layer-chunk-authenticated rt (clength * scale) ?idx chunk t
  by (rule fri-layer-chunk-authenticated-mono[OF auth-s2 s2-t])
  show ?thesis
  by (rule that[OF refl auth-t])
qed

lemma verifier-query-round-after-index-trace-fri-selected-chunk-authenticated:
  fixes  $s\ t :: ('f, 'a) \text{protocol-channel-scheme}$ 
  assumes outcome:
    Some  $(((), t) \in$ 
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s)
    and j-bound:  $j < \text{length } f\text{-fl}$ 
    and layer-len-pos:
       $0 < \text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) ! j$ 
    and layer-idx-bound:
      fri-layer-indices  $(\text{length } f\text{-fl}) (\text{index } (\text{to-nat raw}))$ 
       $(\text{clength} * \text{scale}) ! j <$ 
      fri-layer-lengths  $(\text{length } f\text{-fl}) (\text{clength} * \text{scale}) ! j$ 
  obtains chunk where
    fri-layer-chunk-authenticated  $(\text{snd } (f\text{-fl} ! j))$ 
     $(\text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) ! j)$ 
     $(\text{fri-layer-indices } (\text{length } f\text{-fl}) (\text{index } (\text{to-nat raw}))$ 
     $(\text{clength} * \text{scale}) ! j)$ 
    chunk t
  proof -
    let ?idx =  $\text{index } (\text{to-nat raw})$ 
    from outcome obtain fv s1 f-out s2 s3 c-out s4 where
      trace-fri:
        Some  $(f\text{-out}, s2) \in$ 
          set-dist (execute
            (mfold (?idx, hd fv, clength * scale, 1)
              (receive-query-commits f-fl)) s1)
    and assert-trace:
      Some  $(((), s3) \in$ 
        set-dist
          (execute

```

```

      (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
  Some (c-out, s4) ∈
    set-dist (execute
      (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
        clength * scale, 1)
      (receive-query-commits fl)) s3)
and assert-comp:
  Some (( ), t) ∈
    set-dist
      (execute
        (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
unfolding verifier-query-round-after-index-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
from receive-query-commits-selected-chunk-authenticated-at
  [OF trace-fri j-bound layer-len-pos layer-idx-bound]
obtain chunk where auth-s2:
  fri-layer-chunk-authenticated (snd (f-fl ! j))
  (fri-layer-lengths (length f-fl) (clength * scale) ! j)
  (fri-layer-indices (length f-fl) ?idx (clength * scale) ! j)
  chunk s2
by blast
have s3-eq: s3 = s2
using assert-trace unfolding assert-def
by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
  (auto simp: throw-no-outcome)
have s3-s4: s3 ≤ s4
by (rule receive-query-commits-mfold-hash-extends[OF comp-fri])
have t-eq: t = s4
using assert-comp unfolding assert-def
by (cases case c-out of (-, x, -, -) ⇒ x = final)
  (auto simp: throw-no-outcome)
have s2-t: s2 ≤ t
using s3-s4 unfolding s3-eq t-eq by simp
have auth-t:
  fri-layer-chunk-authenticated (snd (f-fl ! j))
  (fri-layer-lengths (length f-fl) (clength * scale) ! j)
  (fri-layer-indices (length f-fl) ?idx (clength * scale) ! j)
  chunk t
by (rule fri-layer-chunk-authenticated-mono[OF auth-s2 s2-t])
then show ?thesis
by (rule that)
qed

```

```

lemma verifier-query-round-after-index-composition-fri-head-chunk-authenticated:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-eq: fl = (b, rt) # fl-tail
and outcome:
  Some (( ), t) ∈

```

```

    set-dist
    (execute
      (verifier-query-round-after-index-program
        fr f-fl f-final as fl final raw) s)
obtains idx chunk where
  idx = index (to-nat raw)
  fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
proof -
let ?idx = index (to-nat raw)
from outcome obtain fv s1 f-out s2 s3 c-out s4 where
  query-decommit:
    Some (fv, s1) ∈
      set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
and trace-fri:
    Some (f-out, s2) ∈
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
and assert-trace:
    Some ((), s3) ∈
      set-dist
        (execute
          (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
    Some (c-out, s4) ∈
      set-dist (execute
        (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
          clength * scale, 1)
          (receive-query-commits fl)) s3)
and assert-comp:
    Some ((), t) ∈
      set-dist
        (execute
          (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
unfolding verifier-query-round-after-index-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have len-pos: 0 < clength * scale
using eval-domain-nontrivial by linarith
have idx-bound: ?idx < clength * scale
by (rule index-less-domain)
from receive-query-commits-mfold-head-chunk-authenticated
  [OF len-pos idx-bound comp-fri[unfolded fl-eq]]
obtain chunk where auth-s4:
  fri-layer-chunk-authenticated rt (clength * scale) ?idx chunk s4
by blast
have t-eq: t = s4
using assert-comp unfolding assert-def
by (cases case c-out of (-, x, -, -) ⇒ x = final)
  (auto simp: throw-no-outcome)

```

```

have auth-t:
  fri-layer-chunk-authenticated rt (clength * scale) ?idx chunk t
  using auth-s4 unfolding t-eq .
show ?thesis
  by (rule that[OF refl auth-t])
qed

lemma verifier-query-round-after-index-composition-fri-selected-chunk-authenticated:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s)
and j-bound: j < length fl
and layer-len-pos:
  0 < fri-layer-lengths (length fl) (clength * scale) ! j
and layer-idx-bound:
  fri-layer-indices (length fl) (index (to-nat raw))
    (clength * scale) ! j <
    fri-layer-lengths (length fl) (clength * scale) ! j
obtains chunk where
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j)
    chunk t
proof —
let ?idx = index (to-nat raw)
from outcome obtain fv s1 f-out s2 s3 c-out s4 where
  comp-fri:
    Some (c-out, s4) ∈
      set-dist (execute
        (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
          clength * scale, 1)
          (receive-query-commits fl) s3)
and assert-comp:
  Some ((), t) ∈
    set-dist
      (execute
        (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
unfolding verifier-query-round-after-index-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
from receive-query-commits-selected-chunk-authenticated-at
  [OF comp-fri j-bound layer-len-pos layer-idx-bound]
obtain chunk where auth-s4:
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)

```

```

    (fri-layer-indices (length fl) ?idx (clength * scale) ! j)
    chunk s4
  by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -)  $\Rightarrow$  x = final)
    (auto simp: throw-no-outcome)
have auth-t:
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) ?idx (clength * scale) ! j)
    chunk t
  using auth-s4 unfolding t-eq .
then show ?thesis
  by (rule that)
qed

lemma verifier-query-round-program-trace-fri-head-chunk-authenticated:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes f-fl-eq: f-fl = (b, rt) # f-fl-tail
  and outcome:
    Some ((), t)  $\in$ 
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  obtains raw idx chunk where
    idx = index (to-nat raw)
    fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
proof -
  from outcome[unfolded verifier-query-round-program-alt-def]
  obtain raw s0 where
    raw:
      Some (raw, s0)  $\in$ 
        set-dist (execute receive-query-index-challenge s)
  and tail:
    Some ((), t)  $\in$ 
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s0)
  by (auto elim!: set-dist-bindE)
  from verifier-query-round-after-index-trace-fri-head-chunk-authenticated
    [OF f-fl-eq tail]
  obtain idx chunk where
    idx-eq: idx = index (to-nat raw)
    and auth: fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
  by blast
show ?thesis
  by (rule that[OF idx-eq auth])

```

qed

lemma *verifier-query-round-program-trace-fri-selected-chunk-authenticated:*
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes *outcome:*
Some $((\),\ t) \in$
set-dist
(execute
(verifier-query-round-program fr f-fl f-final as fl final) s)
and *j-bound:* $j < \text{length } f\text{-fl}$
and *raw-layer:*
 $\bigwedge_{\text{raw.}}$
 $0 < \text{fri-layer-lengths } (\text{length } f\text{-fl})\ (\text{clength} * \text{scale})\ !\ j \wedge$
 $\text{fri-layer-indices } (\text{length } f\text{-fl})\ (\text{index } (\text{to-nat } \text{raw}))$
 $(\text{clength} * \text{scale})\ !\ j <$
 $\text{fri-layer-lengths } (\text{length } f\text{-fl})\ (\text{clength} * \text{scale})\ !\ j$
obtains *raw chunk where*
fri-layer-chunk-authenticated $(\text{snd } (f\text{-fl}\ !\ j))$
(fri-layer-lengths $(\text{length } f\text{-fl})\ (\text{clength} * \text{scale})\ !\ j)$
(fri-layer-indices $(\text{length } f\text{-fl})\ (\text{index } (\text{to-nat } \text{raw}))$
 $(\text{clength} * \text{scale})\ !\ j)$
chunk t
proof –
from *outcome[unfolded verifier-query-round-program-alt-def]*
obtain *raw s0 where*
raw:
Some $(\text{raw},\ s0) \in$
set-dist $(\text{execute } \text{receive-query-index-challenge } s)$
and *tail:*
Some $((\),\ t) \in$
set-dist
(execute
(verifier-query-round-after-index-program
fr f-fl f-final as fl final raw) s0)
by $(\text{auto elim!}:\ \text{set-dist-bindE})$
have *layer-len-pos:*
 $0 < \text{fri-layer-lengths } (\text{length } f\text{-fl})\ (\text{clength} * \text{scale})\ !\ j$
using *raw-layer[of raw]* **by** *simp*
have *layer-idx-bound:*
 $\text{fri-layer-indices } (\text{length } f\text{-fl})\ (\text{index } (\text{to-nat } \text{raw}))$
 $(\text{clength} * \text{scale})\ !\ j <$
 $\text{fri-layer-lengths } (\text{length } f\text{-fl})\ (\text{clength} * \text{scale})\ !\ j$
using *raw-layer[of raw]* **by** *simp*
from *verifier-query-round-after-index-trace-fri-selected-chunk-authenticated*
 $[\text{OF } \text{tail } j\text{-bound } \text{layer-len-pos } \text{layer-idx-bound}]$
obtain *chunk where auth:*
fri-layer-chunk-authenticated $(\text{snd } (f\text{-fl}\ !\ j))$
(fri-layer-lengths $(\text{length } f\text{-fl})\ (\text{clength} * \text{scale})\ !\ j)$
(fri-layer-indices $(\text{length } f\text{-fl})\ (\text{index } (\text{to-nat } \text{raw}))$


```

      (clength * scale) ! j)
    chunk t
  by blast
show ?thesis
  by (rule that[OF auth])
qed

```

lemma *verifier-query-round-program-trace-fri-selected-chunk-authenticated-with-lookup:*

```

fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (((), t) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
and j-bound: j < length f-fl
and raw-layer:
  ∧ raw.
    0 < fri-layer-lengths (length f-fl) (clength * scale) ! j ∧
    fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! j <
    fri-layer-lengths (length f-fl) (clength * scale) ! j
obtains raw chunk where
  fri-layer-chunk-authenticated (snd (f-fl ! j))
  (fri-layer-lengths (length f-fl) (clength * scale) ! j)
  (fri-layer-indices (length f-fl) (index (to-nat raw))
    (clength * scale) ! j)
  chunk t
  fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw

```

proof –

```

from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where
  raw:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s)
and tail:
  Some (((), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
  by (auto elim!: set-dist-bindE)
have layer-len-pos:
  0 < fri-layer-lengths (length f-fl) (clength * scale) ! j
using raw-layer[of raw] by simp
have layer-idx-bound:
  fri-layer-indices (length f-fl) (index (to-nat raw))
    (clength * scale) ! j <
  fri-layer-lengths (length f-fl) (clength * scale) ! j

```

```

using raw-layer[of raw] by simp
from verifier-query-round-after-index-trace-fri-selected-chunk-authenticated
  [OF tail j-bound layer-len-pos layer-idx-bound]
obtain chunk where auth:
  fri-layer-chunk-authenticated (snd (f-fl ! j))
  (fri-layer-lengths (length f-fl) (clength * scale) ! j)
  (fri-layer-indices (length f-fl) (index (to-nat raw))
    (clength * scale) ! j)
  chunk t
by blast
have lookup-s0:
  fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
using receive-query-index-challenge-outcome[OF raw] by simp
have s0-t:  $s0 \leq t$ 
using verifier-query-round-after-index-program-outcome[OF tail] by simp
have lookup-t:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
by (rule hash-extension-lookup[OF lookup-s0 s0-t])
show ?thesis
by (rule that[OF auth lookup-t])
qed

lemma verifier-query-round-program-composition-fri-head-chunk-authenticated:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes fl-eg: fl = (b, rt) # fl-tail
and outcome:
  Some (( ), t) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
obtains raw idx chunk where
  idx = index (to-nat raw)
  fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where
  raw:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s)
and tail:
  Some (( ), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
by (auto elim!: set-dist-bindE)
from verifier-query-round-after-index-composition-fri-head-chunk-authenticated

```

```

    [OF fl-eq tail]
obtain idx chunk where
  idx-eq: idx = index (to-nat raw)
  and auth: fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
  by blast
show ?thesis
  by (rule that[OF idx-eq auth])
qed

lemma verifier-query-round-program-composition-fri-selected-chunk-authenticated:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (( ), t) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
and j-bound: j < length fl
and raw-layer:
   $\bigwedge$ raw.
    0 < fri-layer-lengths (length fl) (clength * scale) ! j ∧
    fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j <
    fri-layer-lengths (length fl) (clength * scale) ! j
obtains raw chunk where
  fri-layer-chunk-authenticated (snd (fl ! j))
  (fri-layer-lengths (length fl) (clength * scale) ! j)
  (fri-layer-indices (length fl) (index (to-nat raw))
    (clength * scale) ! j)
  chunk t
proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where
  raw:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s)
and tail:
  Some (( ), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
  by (auto elim!: set-dist-bindE)
have layer-len-pos:
  0 < fri-layer-lengths (length fl) (clength * scale) ! j
  using raw-layer[of raw] by simp
have layer-idx-bound:
  fri-layer-indices (length fl) (index (to-nat raw))
    (clength * scale) ! j <
  fri-layer-lengths (length fl) (clength * scale) ! j

```

```

    using raw-layer[of raw] by simp
  from
    verifier-query-round-after-index-composition-fri-selected-chunk-authenticated
    [OF tail j-bound layer-len-pos layer-idx-bound]
  obtain chunk where auth:
    fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (index (to-nat raw))
     (clength * scale) ! j)
    chunk t
  by blast
  show ?thesis
  by (rule that[OF auth])
qed

lemma verifier-query-round-program-composition-fri-selected-chunk-authenticated-with-lookup:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and j-bound: j < length fl
  and raw-layer:
     $\bigwedge$  raw.
      0 < fri-layer-lengths (length fl) (clength * scale) ! j ∧
      fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! j <
      fri-layer-lengths (length fl) (clength * scale) ! j
  obtains raw chunk where
    fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (index (to-nat raw))
     (clength * scale) ! j)
    chunk t
    fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof -
  from outcome[unfolded verifier-query-round-program-alt-def]
  obtain raw s0 where
    raw:
      Some (raw, s0) ∈
        set-dist (execute receive-query-index-challenge s)
  and tail:
    Some (( ), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s0)

```

```

    by (auto elim!: set-dist-bindE)
  have layer-len-pos:
    0 < fri-layer-lengths (length fl) (clength * scale) ! j
    using raw-layer[of raw] by simp
  have layer-idx-bound:
    fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j <
      fri-layer-lengths (length fl) (clength * scale) ! j
    using raw-layer[of raw] by simp
  from
    verifier-query-round-after-index-composition-fri-selected-chunk-authenticated
      [OF tail j-bound layer-len-pos layer-idx-bound]
  obtain chunk where auth:
    fri-layer-chunk-authenticated (snd (fl ! j))
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! j)
    chunk t
  by blast
  have lookup-s0:
    fmlookup (HashMap s0)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
    using receive-query-index-challenge-outcome[OF raw] by simp
  have s0-t: s0 ≤ t
    using verifier-query-round-after-index-program-outcome[OF tail] by simp
  have lookup-t:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
    by (rule hash-extension-lookup[OF lookup-s0 s0-t])
  show ?thesis
    by (rule that[OF auth lookup-t])
qed

lemma ntimes-verifier-query-rounds-selected-trace-fri-head-chunk-authenticated-at:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes f-fl-eq: f-fl = (b, rt) # f-fl-tail
  and outcome:
    Some (results, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s)
  and i-bound: i < n
  obtains raw idx chunk where
    idx = index (to-nat raw)
    fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
proof -
  let ?round = verifier-query-round-program fr f-fl f-final as fl final

```

```

from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
obtain prefix x suffix s-i s-suc where
  round:
    Some (x, s-suc) ∈ set-dist (execute ?round s-i)
  and suffix:
    Some (suffix, t) ∈
      set-dist (execute (ntimes ?round (n - Suc i)) s-suc)
  by blast
have x-eq: x = ()
  by (cases x) simp
from verifier-query-round-program-trace-fri-head-chunk-authenticated
  [OF f-fl-eq round[unfolded x-eq]]
obtain raw idx chunk where
  idx-eq: idx = index (to-nat raw)
  and auth-suc:
    fri-layer-chunk-authenticated rt (clength * scale) idx chunk s-suc
  by blast
have s-suc-t: s-suc ≤ t
  using ntimes-verifier-query-rounds-outcome[OF suffix] by blast
have auth-t:
  fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
  by (rule fri-layer-chunk-authenticated-mono[OF auth-suc s-suc-t])
show ?thesis
  by (rule that[OF idx-eq auth-t])
qed

lemma ntimes-verifier-query-rounds-selected-trace-fri-chunk-authenticated-at:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (results, t) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final) n)
        s)
and i-bound: i < n
and j-bound: j < length f-fl
and raw-layer:
   $\bigwedge^{\text{raw.}}$ 
     $0 < \text{fri-layer-lengths } (\text{length } f\text{-fl}) \text{ (clength * scale) ! } j \wedge$ 
     $\text{fri-layer-indices } (\text{length } f\text{-fl}) \text{ (index (to-nat raw))}$ 
     $(\text{clength * scale) ! } j <$ 
     $\text{fri-layer-lengths } (\text{length } f\text{-fl}) \text{ (clength * scale) ! } j$ 
obtains raw chunk where
  fri-layer-chunk-authenticated (snd (f-fl ! j))
  (fri-layer-lengths (length f-fl) (clength * scale) ! j)
  (fri-layer-indices (length f-fl) (index (to-nat raw)))
  (clength * scale) ! j)
  chunk t

```

proof –

let $?round = \text{verifier-query-round-program } fr \ f-fl \ f-final \text{ as } fl \ final$
from $\text{query-opening-ntimes-outcome-decomp-at}[OF \ outcome \ i-bound]$
obtain $\text{prefix } x \ \text{suffix } s-i \ s-suc$ **where**
 $round:$
 $\text{Some } (x, s-suc) \in \text{set-dist } (\text{execute } ?round \ s-i)$
and $\text{suffix}:$
 $\text{Some } (suffix, t) \in$
 $\text{set-dist } (\text{execute } (ntimes \ ?round \ (n - Suc \ i)) \ s-suc)$
by blast
have $x-eq: x = ()$
by $(\text{cases } x) \ \text{simp}$
from $\text{verifier-query-round-program-trace-fri-selected-chunk-authenticated}$
 $[OF \ round[unfolded \ x-eq] \ j-bound \ raw-layer]$
obtain raw chunk **where** $\text{auth-suc}:$
 $\text{fri-layer-chunk-authenticated } (snd \ (f-fl \ ! \ j))$
 $(\text{fri-layer-lengths } (length \ f-fl) \ (clength * scale) \ ! \ j)$
 $(\text{fri-layer-indices } (length \ f-fl) \ (index \ (to-nat \ raw))$
 $(clength * scale) \ ! \ j)$
 $\text{chunk } s-suc$
by blast
have $s-suc-t: s-suc \leq t$
using $\text{ntimes-verifier-query-rounds-outcome}[OF \ suffix]$ **by** blast
have $\text{auth-t}:$
 $\text{fri-layer-chunk-authenticated } (snd \ (f-fl \ ! \ j))$
 $(\text{fri-layer-lengths } (length \ f-fl) \ (clength * scale) \ ! \ j)$
 $(\text{fri-layer-indices } (length \ f-fl) \ (index \ (to-nat \ raw))$
 $(clength * scale) \ ! \ j)$
 $\text{chunk } t$
by $(\text{rule } \text{fri-layer-chunk-authenticated-mono}[OF \ \text{auth-suc } s-suc-t])$
show $?thesis$
by $(\text{rule } \text{that}[OF \ \text{auth-t}])$
qed

lemma $\text{ntimes-verifier-query-rounds-selected-trace-fri-chunk-authenticated-with-prefix-at}:$

fixes $s \ t :: ('f, 'a) \ \text{protocol-channel-scheme}$

assumes $\text{outcome}:$

$\text{Some } (results, t) \in$
 set-dist
 $(\text{execute}$
 $(ntimes$
 $(\text{verifier-query-round-program } fr \ f-fl \ f-final \text{ as } fl \ final) \ n)$
 $s)$

and $i-bound: i < n$

and $j-bound: j < length \ f-fl$

and $\text{raw-layer}:$

$\bigwedge_{\text{raw.}}$
 $0 < \text{fri-layer-lengths } (length \ f-fl) \ (clength * scale) \ ! \ j \wedge$
 $\text{fri-layer-indices } (length \ f-fl) \ (index \ (to-nat \ raw))$

```

      (clength * scale) ! j <
      fri-layer-lengths (length f-fl) (clength * scale) ! j
obtain raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
      suffix-chunks layer-chunk where
      idx = index (to-nat raw)
      length prefix-query-idxs = i
      length prefix-chunks = i
      length suffix-query-idxs = n - Suc i
      length suffix-chunks = n - Suc i
      verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
       $\wedge k. k < i \implies$ 
        verifier-query-round-chunk (prefix-query-idxs ! k)
          (map snd f-fl) (map snd fl) (prefix-chunks ! k)
       $\wedge k. k < n - \text{Suc } i \implies$ 
        verifier-query-round-chunk (suffix-query-idxs ! k)
          (map snd f-fl) (map snd fl) (suffix-chunks ! k)
      PTranscript s =
        List.concat (prefix-chunks @ chunk # suffix-chunks) @ PTranscript t
      fri-layer-chunk-authenticated (snd (f-fl ! j))
        (fri-layer-lengths (length f-fl) (clength * scale) ! j)
        (fri-layer-indices (length f-fl) idx (clength * scale) ! j)
        layer-chunk t
      fmlookup (HashMap t)
        (QueryIndexChallenge (PQueryCounter s + i)
          (state-after-query-chunks (PState s) prefix-chunks i)) =
        Some raw
proof -
let ?round = verifier-query-round-program fr f-fl f-final as fl final
from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
obtain prefix x suffix s-i s-suc where
  prefix:
    Some (prefix, s-i)  $\in$ 
      set-dist (execute (ntimes ?round i) s)
and round:
    Some (x, s-suc)  $\in$  set-dist (execute ?round s-i)
and suffix:
    Some (suffix, t)  $\in$ 
      set-dist (execute (ntimes ?round (n - Suc i)) s-suc)
by blast
have x-eq: x = ()
by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
fix prefix-query-idxs prefix-chunks
assume len-prefix-idxs: length prefix-query-idxs = i
and len-prefix-chunks: length prefix-chunks = i
and transcript-prefix:
  PTranscript s = List.concat prefix-chunks @ PTranscript s-i

```



```

and state-i:
  PState s-i =
    state-after-query-chunks (PState s) prefix-chunks i
and counter-i: PQueryCounter s-i = PQueryCounter s + i
and prefix-ext: s ≤ s-i
and prefix-rounds:
   $\bigwedge k. k < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
show ?thesis
proof (rule verifier-query-round-program-outcome-with-authenticated-trace-openings
  [OF round[unfolded x-eq]])
fix raw idx chunk openings
assume idx-eq: idx = index (to-nat raw)
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
    chunk
and transcript-round:
  PTranscript s-i = chunk @ PTranscript s-suc
and state-suc:
  PState s-suc = foldl concat (PState s-i) chunk
and round-ext: s-i ≤ s-suc
and counter-suc: PQueryCounter s-suc = Suc (PQueryCounter s-i)
and lookup-suc:
  fmlookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
      Some raw
and len-openings:
  length openings = length (powers-scaled idx)
and trace-indices:
  map opening-index openings = powers-scaled idx
and trace-table-suc:
  partial-authenticated-table fr (scale * clength) openings
    s-suc
show ?thesis
proof (rule
  verifier-query-round-program-trace-fri-selected-chunk-authenticated-with-lookup
  [OF round[unfolded x-eq] j-bound raw-layer])
fix raw' layer-chunk
assume auth-suc:
  fri-layer-chunk-authenticated (snd (f-fl ! j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! j)
    (fri-layer-indices (length f-fl) (index (to-nat raw')))
    (clength * scale ! j)
    layer-chunk s-suc
and lookup-suc':
  fmlookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
      Some raw'

```

```

have raw'-eq: raw' = raw
  using lookup-suc lookup-suc' by simp
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
  fix suffix-query-idxs suffix-chunks
  assume len-suffix-idxs: length suffix-query-idxs = n - Suc i
  and len-suffix-chunks:
    length suffix-chunks = n - Suc i
  and transcript-suffix:
    PTranscript s-suc =
      List.concat suffix-chunks @ PTranscript t
  and suffix-state:
    PState t =
      state-after-query-chunks (PState s-suc) suffix-chunks
      (n - Suc i)
  and suffix-counter:
    PQueryCounter t = PQueryCounter s-suc + (n - Suc i)
  and suffix-ext: s-suc ≤ t
  and suffix-rounds:
     $\bigwedge k. k < n - \text{Suc } i \implies$ 
      verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)
  have transcript-all:
    PTranscript s =
      List.concat (prefix-chunks @ chunk # suffix-chunks) @
      PTranscript t
  using transcript-prefix transcript-round transcript-suffix
  by simp
  have auth-t-row:
    fri-layer-chunk-authenticated (snd (f-fl ! j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! j)
    (fri-layer-indices (length f-fl) (index (to-nat raw)))
    (clength * scale) ! j)
    layer-chunk t
  by (rule fri-layer-chunk-authenticated-mono
    [OF auth-suc[unfolded raw'-eq] suffix-ext])
  have auth-t:
    fri-layer-chunk-authenticated (snd (f-fl ! j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! j)
    (fri-layer-indices (length f-fl) idx (clength * scale) ! j)
    layer-chunk t
  using auth-t-row idx-eq by simp
  have lookup-t:
    fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks (PState s) prefix-chunks i)) =
      Some raw
  using hash-extension-lookup[OF lookup-suc suffix-ext]

```

```

    unfolding counter-i state-i .
  show ?thesis
  by (rule that[OF idx-eq len-prefix-idxs len-prefix-chunks
    len-suffix-idxs len-suffix-chunks chunk-shape
    prefix-rounds suffix-rounds transcript-all auth-t
    lookup-t])
qed
qed
qed
qed
qed

lemma ntimes-verifier-query-rounds-selected-trace-fri-chunk-aligned-with-replay-at:
  fixes query-state final-state :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (results, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            rounds)
          query-state)
  and i-bound: i < rounds
  and j-bound: j < length f-fl
  and raw-layer:
     $\bigwedge \text{raw.}$ 
    0 < fri-layer-lengths (length f-fl) (clength * scale) ! j  $\wedge$ 
    fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! j <
    fri-layer-lengths (length f-fl) (clength * scale) ! j
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda \text{raw. index (to-nat raw)}$ ) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
  and query-chunk:
     $\bigwedge i. i < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
  and replay-lookup:
     $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
    Some (raw-idxs ! i)
  obtains layer-chunk where

```

```

    fri-layer-chunk-authenticated (snd (f-fl ! j))
      (fri-layer-lengths (length f-fl) (clength * scale) ! j)
      (fri-layer-indices (length f-fl) (query-idxs ! i)
        (clength * scale) ! j)
      layer-chunk final-state
  proof (rule
    ntimes-verifier-query-rounds-selected-trace-fri-chunk-authenticated-with-prefix-at
      [OF outcome i-bound j-bound raw-layer])
  fix raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
    suffix-chunks layer-chunk
  assume selected-idx: idx = index (to-nat raw)
  and len-prefix-idxs: length prefix-query-idxs = i
  and len-prefix-chunks: length prefix-chunks = i
  and len-suffix-idxs: length suffix-query-idxs = rounds - Suc i
  and len-suffix-chunks: length suffix-chunks = rounds - Suc i
  and chunk-shape:
    verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
  and prefix-rounds:
     $\bigwedge k. k < i \implies$ 
      verifier-query-round-chunk (prefix-query-idxs ! k)
        (map snd f-fl) (map snd fl) (prefix-chunks ! k)
  and suffix-rounds:
     $\bigwedge k. k < rounds - Suc i \implies$ 
      verifier-query-round-chunk (suffix-query-idxs ! k)
        (map snd f-fl) (map snd fl) (suffix-chunks ! k)
  and selected-transcript:
    PTranscript query-state =
      List.concat (prefix-chunks @ chunk # suffix-chunks) @
        PTranscript final-state
  and auth:
    fri-layer-chunk-authenticated (snd (f-fl ! j))
      (fri-layer-lengths (length f-fl) (clength * scale) ! j)
      (fri-layer-indices (length f-fl) idx (clength * scale) ! j)
      layer-chunk final-state
  and selected-lookup:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) prefix-chunks i)) =
      Some raw
  let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
  have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks i-bound by simp
  have selected-chunk-lens:
     $\bigwedge k. k < rounds \implies$ 
      length (?selected-chunks ! k) =
        verifier-query-round-transcript-length (query-idxs ! k)
          (map snd f-fl) (map snd fl)
  proof -
  fix k

```

```

assume k-bound:  $k < \text{rounds}$ 
show
   $\text{length } (?selected\text{-chunks } ! k) =$ 
     $\text{verifier-query-round-transcript-length } (query\text{-idxs } ! k)$ 
     $(\text{map } snd\ f\text{-fl}) (\text{map } snd\ fl)$ 
proof (cases  $k < i$ )
  case True
  have nth-eq:  $?selected\text{-chunks } ! k = \text{prefix-chunks } ! k$ 
    using True len-prefix-chunks by (simp add: nth-append)
  have
     $\text{length } (\text{prefix-chunks } ! k) =$ 
       $\text{verifier-query-round-transcript-length}$ 
       $(\text{prefix-query-idxs } ! k) (\text{map } snd\ f\text{-fl}) (\text{map } snd\ fl)$ 
    by (rule verifier-query-round-chunk-length[OF prefix-rounds[OF True]])
  then show ?thesis
    using nth-eq
    by (simp add: verifier-query-round-transcript-length-index-irrelevant)
next
  case False
  show ?thesis
  proof (cases  $k = i$ )
    case True
    have nth-eq:  $?selected\text{-chunks } ! k = \text{chunk}$ 
      using True len-prefix-chunks by (simp add: nth-append)
    have
       $\text{length } \text{chunk} =$ 
         $\text{verifier-query-round-transcript-length } idx$ 
         $(\text{map } snd\ f\text{-fl}) (\text{map } snd\ fl)$ 
      by (rule verifier-query-round-chunk-length[OF chunk-shape])
    then show ?thesis
      using nth-eq
      by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
    next
    case False
    let  $?k = k - \text{Suc } i$ 
    have ge:  $\text{Suc } i \leq k$ 
      using  $\langle \neg k < i \rangle$  False by linarith
    have k-bound':  $?k < \text{rounds} - \text{Suc } i$ 
      using ge k-bound by linarith
    have nth-eq:  $?selected\text{-chunks } ! k = \text{suffix-chunks } ! ?k$ 
      using ge len-prefix-chunks by (simp add: nth-append)
    have
       $\text{length } (\text{suffix-chunks } ! ?k) =$ 
         $\text{verifier-query-round-transcript-length}$ 
         $(\text{suffix-query-idxs } ! ?k) (\text{map } snd\ f\text{-fl}) (\text{map } snd\ fl)$ 
      by (rule verifier-query-round-chunk-length
        [OF suffix-rounds[OF k-bound']])
    then show ?thesis

```

```

    using nth-eq
    by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
  qed
  qed
  qed
  have query-chunk-lens:
     $\bigwedge k. k < \text{rounds} \implies$ 
    length (query-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
    using query-chunk verifier-query-round-chunk-length by blast
  have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
    using selected-transcript transcript-query by simp
  have chunks-eq: ?selected-chunks = query-chunks
  proof -
    have same-lens:
       $\bigwedge k. k < \text{length } ?\text{selected-chunks} \implies$ 
      length (?selected-chunks ! k) = length (query-chunks ! k)
      using selected-chunk-lens query-chunk-lens len-selected-chunks by simp
    have concat-eq-empty:
      List.concat ?selected-chunks @ [] = List.concat query-chunks
      using concat-eq by simp
    have len-eq: length ?selected-chunks = length query-chunks
      using len-selected-chunks len-query-chunks by simp
    have ?selected-chunks = query-chunks  $\wedge$  ([] :: 'f list) = []
      by (rule concat-append-eq-concat-same-chunk-lengths
          [OF len-eq same-lens concat-eq-empty])
    then show ?thesis
      by simp
  qed
  have prefix-state-eq:
    state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
  proof -
    have
      state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
      by (rule state-after-query-chunks-append-prefix[OF len-prefix-chunks])
    then show ?thesis
      using chunks-eq by simp
  qed
  have selected-lookup-global:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw
    using selected-lookup prefix-state-eq by simp
  have raw-eq: raw = raw-idxs ! i

```

```

    using selected-lookup-global replay-lookup[OF i-bound] by simp
  have idx-eq:  $idx = query-idxs ! i$ 
    using selected-idx raw-eq query-idxs-eq len-raw i-bound by simp
  show ?thesis
    by (rule that[of layer-chunk]) (use auth idx-eq in simp)
qed

lemma ntimes-verifier-query-rounds-selected-composition-fri-head-chunk-authenticated-at:
  fixes  $s\ t :: ('f, 'a) protocol-channel-scheme$ 
  assumes fl-eq:  $fl = (b, rt) \# fl\text{-tail}$ 
  and outcome:
    Some (results, t)  $\in$ 
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s)
  and i-bound:  $i < n$ 
  obtains raw idx chunk where
     $idx = index\ (to\text{-nat}\ raw)$ 
    fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
proof -
  let ?round = verifier-query-round-program fr f-fl f-final as fl final
  from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
  obtain prefix x suffix s-i s-suc where
    round:
      Some (x, s-suc)  $\in$  set-dist (execute ?round s-i)
  and suffix:
    Some (suffix, t)  $\in$ 
      set-dist (execute (ntimes ?round (n - Suc i)) s-suc)
  by blast
  have x-eq:  $x = ()$ 
  by (cases x) simp
  from verifier-query-round-program-composition-fri-head-chunk-authenticated
    [OF fl-eq round[unfolded x-eq]]
  obtain raw idx chunk where
    idx-eq:  $idx = index\ (to\text{-nat}\ raw)$ 
  and auth-suc:
    fri-layer-chunk-authenticated rt (clength * scale) idx chunk s-suc
  by blast
  have s-suc-t:  $s-suc \leq t$ 
  using ntimes-verifier-query-rounds-outcome[OF suffix] by blast
  have auth-t:
    fri-layer-chunk-authenticated rt (clength * scale) idx chunk t
  by (rule fri-layer-chunk-authenticated-mono[OF auth-suc s-suc-t])
  show ?thesis
    by (rule that[OF idx-eq auth-t])
qed

```

lemma *ntimes-verifier-query-rounds-selected-composition-fri-chunk-authenticated-at:*

fixes $s\ t :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *outcome:*

Some (results, t) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final) n)
s)

and *i-bound: i < n*

and *j-bound: j < length fl*

and *raw-layer:*

$\bigwedge_{\text{raw.}}$
 $0 < \text{fri-layer-lengths } (\text{length } fl) (\text{clength} * \text{scale}) ! j \wedge$
 $\text{fri-layer-indices } (\text{length } fl) (\text{index } (\text{to-nat } \text{raw}))$
 $(\text{clength} * \text{scale}) ! j <$
 $\text{fri-layer-lengths } (\text{length } fl) (\text{clength} * \text{scale}) ! j$

obtains *raw chunk where*

fri-layer-chunk-authenticated (snd (fl ! j))
*(fri-layer-lengths (length fl) (clength * scale) ! j)*
(fri-layer-indices (length fl) (index (to-nat raw)))
*(clength * scale) ! j)*
chunk t

proof –

let *?round = verifier-query-round-program fr f-fl f-final as fl final*

from *query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]*

obtain *prefix x suffix s-i s-suc where*

round:
Some (x, s-suc) ∈ set-dist (execute ?round s-i)
and *suffix:*
Some (suffix, t) ∈
set-dist (execute (ntimes ?round (n – Suc i)) s-suc)

by *blast*

have *x-eq: x = ()*

by *(cases x) simp*

from

verifier-query-round-program-composition-fri-selected-chunk-authenticated
[OF round[unfolded x-eq] j-bound raw-layer]

obtain *raw chunk where auth-suc:*

fri-layer-chunk-authenticated (snd (fl ! j))
*(fri-layer-lengths (length fl) (clength * scale) ! j)*
(fri-layer-indices (length fl) (index (to-nat raw)))
*(clength * scale) ! j)*
chunk s-suc

by *blast*

have *s-suc-t: s-suc ≤ t*

using *ntimes-verifier-query-rounds-outcome[OF suffix] by blast*

have *auth-t:*

fri-layer-chunk-authenticated (snd (fl ! j))


```

    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j)
    chunk t
  by (rule fri-layer-chunk-authenticated-mono[OF auth-suc s-suc-t])
show ?thesis
  by (rule that[OF auth-t])
qed

lemma ntimes-verifier-query-rounds-selected-composition-fri-chunk-authenticated-with-prefix-at:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (results, t) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final) n)
        s)
and i-bound: i < n
and j-bound: j < length fl
and raw-layer:
  ∧ raw.
    0 < fri-layer-lengths (length fl) (clength * scale) ! j ∧
    fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j <
    fri-layer-lengths (length fl) (clength * scale) ! j
obtains raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
  suffix-chunks layer-chunk where
  idx = index (to-nat raw)
  length prefix-query-idxs = i
  length prefix-chunks = i
  length suffix-query-idxs = n - Suc i
  length suffix-chunks = n - Suc i
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
  ∧ k. k < i ⇒
    verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
  ∧ k. k < n - Suc i ⇒
    verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)
  PTranscript s =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @ PTranscript t
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) idx (clength * scale) ! j)
    layer-chunk t
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + i)
      (state-after-query-chunks (PState s) prefix-chunks i)) =

```

Some raw

proof –

let ?round = verifier-query-round-program fr f-fl f-final as fl final

from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]

obtain prefix x suffix s-i s-suc **where**

prefix:

Some (prefix, s-i) ∈

set-dist (execute (ntimes ?round i) s)

and round:

Some (x, s-suc) ∈ set-dist (execute ?round s-i)

and suffix:

Some (suffix, t) ∈

set-dist (execute (ntimes ?round (n – Suc i)) s-suc)

by blast

have x-eq: x = ()

by (cases x) simp

show ?thesis

proof (rule query-opening-ntimes-prefix-transcript-state-alignment
[OF prefix])

fix prefix-query-idxs prefix-chunks

assume len-prefix-idxs: length prefix-query-idxs = i

and len-prefix-chunks: length prefix-chunks = i

and transcript-prefix:

PTranscript s = List.concat prefix-chunks @ PTranscript s-i

and state-i:

PState s-i =

state-after-query-chunks (PState s) prefix-chunks i

and counter-i: PQueryCounter s-i = PQueryCounter s + i

and prefix-ext: s ≤ s-i

and prefix-rounds:

$\bigwedge k. k < i \implies$

verifier-query-round-chunk (prefix-query-idxs ! k)

(map snd f-fl) (map snd fl) (prefix-chunks ! k)

show ?thesis

proof (rule verifier-query-round-program-outcome-with-authenticated-trace-openings
[OF round[unfolded x-eq]])

fix raw idx chunk openings

assume idx-eq: idx = index (to-nat raw)

and chunk-shape:

verifier-query-round-chunk idx (map snd f-fl) (map snd fl)

chunk

and transcript-round:

PTranscript s-i = chunk @ PTranscript s-suc

and state-suc:

PState s-suc = foldl concat (PState s-i) chunk

and round-ext: s-i ≤ s-suc

and counter-suc: PQueryCounter s-suc = Suc (PQueryCounter s-i)

and lookup-suc:

fmlookup (HashMap s-suc)

```

    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
      Some raw
  and len-openings:
    length openings = length (powers-scaled idx)
  and trace-indices:
    map opening-index openings = powers-scaled idx
  and trace-table-suc:
    partial-authenticated-table fr (scale * clength) openings
      s-suc
show ?thesis
proof (rule
  verifier-query-round-program-composition-fri-selected-chunk-authenticated-with-lookup
  [OF round[unfolded x-eq] j-bound raw-layer])
fix raw' layer-chunk
assume auth-suc:
  fri-layer-chunk-authenticated (snd (fl ! j))
  (fri-layer-lengths (length fl) (clength * scale) ! j)
  (fri-layer-indices (length fl) (index (to-nat raw'))
    (clength * scale) ! j)
  layer-chunk s-suc
and lookup-suc':
  fmlookup (HashMap s-suc)
  (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw'
have raw'-eq: raw' = raw
  using lookup-suc lookup-suc' by simp
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-idxs: length suffix-query-idxs = n - Suc i
  and len-suffix-chunks:
    length suffix-chunks = n - Suc i
  and transcript-suffix:
    PTranscript s-suc =
      List.concat suffix-chunks @ PTranscript t
  and suffix-state:
    PState t =
      state-after-query-chunks (PState s-suc) suffix-chunks
      (n - Suc i)
  and suffix-counter:
    PQueryCounter t = PQueryCounter s-suc + (n - Suc i)
  and suffix-ext: s-suc ≤ t
  and suffix-rounds:
     $\bigwedge k. k < n - \text{Suc } i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)
have transcript-all:
  PTranscript s =

```

```

      List.concat (prefix-chunks @ chunk # suffix-chunks) @
      PTranscript t
    using transcript-prefix transcript-round transcript-suffix
    by simp
  have auth-t-raw:
    fri-layer-chunk-authenticated (snd (fl ! j))
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! j)
      layer-chunk t
    by (rule fri-layer-chunk-authenticated-mono
      [OF auth-suc[unfolded raw'-eq] suffix-ext])
  have auth-t:
    fri-layer-chunk-authenticated (snd (fl ! j))
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) idx (clength * scale) ! j)
      layer-chunk t
    using auth-t-raw idx-eq by simp
  have lookup-t:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) prefix-chunks i)) =
      Some raw
    using hash-extension-lookup[OF lookup-suc suffix-ext]
    unfolding counter-i state-i .
  show ?thesis
    by (rule that[OF idx-eq len-prefix-idxs len-prefix-chunks
      len-suffix-idxs len-suffix-chunks chunk-shape
      prefix-rounds suffix-rounds transcript-all auth-t
      lookup-t])
  qed
  qed
  qed
  qed
  qed

```

lemma *ntimes-verifier-query-rounds-selected-composition-fri-chunk-aligned-with-replay-at:*
fixes *query-state final-state* :: ('f, 'a) *protocol-channel-scheme*
assumes *outcome:*
 Some (results, final-state) ∈
 set-dist
 (*execute*
 (*ntimes*
 (*verifier-query-round-program* fr f-fl f-final as fl final)
 rounds)
 query-state)
and *i-bound: i < rounds*
and *j-bound: j < length fl*
and *raw-layer:*

```

 $\wedge$ raw.
  0 < fri-layer-lengths (length fl) (clength * scale) ! j  $\wedge$ 
  fri-layer-indices (length fl) (index (to-nat raw))
    (clength * scale) ! j <
  fri-layer-lengths (length fl) (clength * scale) ! j
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and query-chunk:
   $\wedge$ i. i < rounds  $\implies$ 
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and replay-lookup:
   $\wedge$ i. i < rounds  $\implies$ 
    fmllookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
obtains layer-chunk where
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (query-idxs ! i)
      (clength * scale) ! j)
  layer-chunk final-state
proof (rule
  ntimes-verifier-query-rounds-selected-composition-fri-chunk-authenticated-with-prefix-at
    [OF outcome i-bound j-bound raw-layer])
fix raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
  suffix-chunks layer-chunk
assume selected-idx: idx = index (to-nat raw)
and len-prefix-idxs: length prefix-query-idxs = i
and len-prefix-chunks: length prefix-chunks = i
and len-suffix-idxs: length suffix-query-idxs = rounds - Suc i
and len-suffix-chunks: length suffix-chunks = rounds - Suc i
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
and prefix-rounds:
   $\wedge$ k. k < i  $\implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
and suffix-rounds:
   $\wedge$ k. k < rounds - Suc i  $\implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)

```

```

and selected-transcript:
  PTranscript query-state =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @
      PTranscript final-state
and auth:
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) idx (clength * scale) ! j)
    layer-chunk final-state
and selected-lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) prefix-chunks i)) =
    Some raw
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks i-bound by simp
have selected-chunk-lens:
   $\wedge k. k < rounds \implies$ 
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
proof –
  fix k
  assume k-bound: k < rounds
  show
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  proof (cases k < i)
    case True
    have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
      using True len-prefix-chunks by (simp add: nth-append)
    have
      length (prefix-chunks ! k) =
        verifier-query-round-transcript-length
          (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
      by (rule verifier-query-round-chunk-length[OF prefix-rounds[OF True]])
    then show ?thesis
      using nth-eq
      by (simp add: verifier-query-round-transcript-length-index-irrelevant)
  next
  case False
  show ?thesis
  proof (cases k = i)
    case True
    have nth-eq: ?selected-chunks ! k = chunk
      using True len-prefix-chunks by (simp add: nth-append)
    have

```

```

    length chunk =
      verifier-query-round-transcript-length idx
      (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length[OF chunk-shape])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
next
case False
let ?k = k - Suc i
have ge: Suc i ≤ k
  using <¬ k < i> False by linarith
have k-bound': ?k < rounds - Suc i
  using ge k-bound by linarith
have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k
  using ge len-prefix-chunks by (simp add: nth-append)
have
  length (suffix-chunks ! ?k) =
    verifier-query-round-transcript-length
    (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
  by (rule verifier-query-round-chunk-length
    [OF suffix-rounds[OF k-bound']])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-lens:
  ∧k. k < rounds ⇒
    length (query-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
  using query-chunk verifier-query-round-chunk-length by blast
have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
  using selected-transcript transcript-query by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
    ∧k. k < length ?selected-chunks ⇒
      length (?selected-chunks ! k) = length (query-chunks ! k)
    using selected-chunk-lens query-chunk-lens len-selected-chunks by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp

```

```

have ?selected-chunks = query-chunks  $\wedge$  ( $\square :: 'f \text{ list}$ ) =  $\square$ 
  by (rule concat-append-eq-concat-same-chunk-lengths
    [OF len-eq same-len concat-eq-empty])
then show ?thesis
  by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof –
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix[OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have selected-lookup-global:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw
    using selected-lookup prefix-state-eq by simp
have raw-eq: raw = raw-idxs ! i
    using selected-lookup-global replay-lookup[OF i-bound] by simp
have idx-eq: idx = query-idxs ! i
    using selected-idx raw-eq query-idxs-eq len-raw i-bound by simp
show ?thesis
  by (rule that[of layer-chunk]) (use auth idx-eq in simp)
qed

lemma accepted-fri-opening-transcript-trace-selected-chunk-authenticated-at:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
and out-eq: out = Some (result, final-state)
and round-bound: i < rounds
and layer-bound: j < length trace-roots
and raw-layer:
   $\bigwedge_{\text{raw.}}$ 
  0 < fri-layer-lengths (length trace-roots) (clength * scale) ! j  $\wedge$ 
  fri-layer-indices (length trace-roots) (index (to-nat raw))
    (clength * scale) ! j <
  fri-layer-lengths (length trace-roots) (clength * scale) ! j
obtains layer-chunk where
  fri-layer-chunk-authenticated (trace-roots ! j)
  (fri-layer-lengths (length trace-roots) (clength * scale) ! j)
  (fri-layer-indices (length trace-roots) (query-idxs ! i))

```



```

      (clength * scale) ! j)
    layer-chunk final-state
  proof -
  from fri-openings obtain result' final-state' fr f-fl as fl query-state
    raw-idxs query-chunks where
    out-def: out = Some (result', final-state')
  and trace-roots-eq: trace-roots = map snd f-fl
  and composition-roots-eq: composition-roots = map snd fl
  and outcome:
    Some (result', final-state') ∈
      set-dist (execute
        (ntimes
          (verifier-query-round-program fr f-fl trace-final as fl
            composition-final)
          rounds)
        query-state)
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state'
  and query-chunk:
    ∧ i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
      trace-roots composition-roots (query-chunks ! i)
  and replay-lookup:
    ∧ i. i < rounds ⇒
      fmlookup (HashMap final-state')
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
        Some (raw-idxs ! i)
  unfolding accepted-fri-opening-transcript-def by blast
  have result-eq: result' = result
  using out-def out-eq by simp
  have final-state-eq: final-state' = final-state
  using out-def out-eq by simp
  have outcome':
    Some (result, final-state) ∈
      set-dist (execute
        (ntimes
          (verifier-query-round-program fr f-fl trace-final as fl
            composition-final)
          rounds)
        query-state)
  using outcome result-eq final-state-eq by simp
  have j-bound-f: j < length f-fl
  using layer-bound trace-roots-eq by simp

```

```

have raw-layer-f:
   $\bigwedge$ raw.
    0 < fri-layer-lengths (length f-fl) (clength * scale) ! j  $\wedge$ 
    fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! j <
    fri-layer-lengths (length f-fl) (clength * scale) ! j
  using raw-layer trace-roots-eq by simp
have query-chunk-f:
   $\bigwedge$ i. i < rounds  $\implies$ 
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
  using query-chunk trace-roots-eq composition-roots-eq by simp
from ntimes-verifier-query-rounds-selected-trace-fri-chunk-aligned-with-replay-at
  [OF outcome' round-bound j-bound-f raw-layer-f len-raw query-idxs-eq
    len-query-chunks transcript-query[unfolded final-state-eq]
    query-chunk-f replay-lookup[unfolded final-state-eq]]
obtain layer-chunk where auth:
  fri-layer-chunk-authenticated (snd (f-fl ! j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! j)
    (fri-layer-indices (length f-fl) (query-idxs ! i)
      (clength * scale) ! j)
  layer-chunk final-state
by blast
have root-eq: trace-roots ! j = snd (f-fl ! j)
  using trace-roots-eq layer-bound by simp
show ?thesis
  by (rule that[of layer-chunk])
    (use auth trace-roots-eq root-eq in simp)
qed

```

lemma accepted-fri-opening-transcript-composition-selected-chunk-authenticated-at:

```

assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
and out-eq: out = Some (result, final-state)
and round-bound: i < rounds
and layer-bound: j < length composition-roots
and raw-layer:
   $\bigwedge$ raw.
    0 < fri-layer-lengths (length composition-roots) (clength * scale) ! j  $\wedge$ 
    fri-layer-indices (length composition-roots) (index (to-nat raw))
      (clength * scale) ! j <
    fri-layer-lengths (length composition-roots) (clength * scale) ! j
obtains layer-chunk where
  fri-layer-chunk-authenticated (composition-roots ! j)
    (fri-layer-lengths (length composition-roots) (clength * scale) ! j)
    (fri-layer-indices (length composition-roots) (query-idxs ! i)
      (clength * scale) ! j)

```

layer-chunk final-state

proof –

from *fri-openings* **obtain** *result' final-state'* fr *f-fl* as *fl* *query-state*
raw-idxs query-chunks **where**
out-def: *out* = *Some (result', final-state')*
and *trace-roots-eq*: *trace-roots* = *map snd f-fl*
and *composition-roots-eq*: *composition-roots* = *map snd fl*
and *outcome*:
Some (result', final-state') ∈
set-dist (execute
(ntimes
(verifier-query-round-program fr f-fl trace-final as fl
composition-final)
rounds)
query-state)
and *len-raw*: *length raw-idxs* = *rounds*
and *query-idxs-eq*:
query-idxs = *map (λraw. index (to-nat raw)) raw-idxs*
and *len-query-chunks*: *length query-chunks* = *rounds*
and *transcript-query*:
PTranscript query-state =
List.concat query-chunks @ PTranscript final-state'
and *query-chunk*:
 $\bigwedge i. i < \text{rounds} \implies$
verifier-query-round-chunk (query-idxs ! i)
trace-roots composition-roots (query-chunks ! i)
and *replay-lookup*:
 $\bigwedge i. i < \text{rounds} \implies$
fmlookup (HashMap final-state')
(QueryIndexChallenge (PQueryCounter query-state + i)
(state-after-query-chunks (PState query-state) query-chunks i)) =
Some (raw-idxs ! i)
unfolding *accepted-fri-opening-transcript-def* **by** *blast*
have *result-eq*: *result' = result*
using *out-def out-eq* **by** *simp*
have *final-state-eq*: *final-state' = final-state*
using *out-def out-eq* **by** *simp*
have *outcome'*:
Some (result, final-state) ∈
set-dist (execute
(ntimes
(verifier-query-round-program fr f-fl trace-final as fl
composition-final)
rounds)
query-state)
using *outcome result-eq final-state-eq* **by** *simp*
have *j-bound-f*: *j < length fl*
using *layer-bound composition-roots-eq* **by** *simp*
have *raw-layer-f*:

```

 $\wedge$ raw.
  0 < fri-layer-lengths (length fl) (clength * scale) ! j  $\wedge$ 
  fri-layer-indices (length fl) (index (to-nat raw))
    (clength * scale) ! j <
  fri-layer-lengths (length fl) (clength * scale) ! j
  using raw-layer composition-roots-eq by simp
have query-chunk-f:
 $\wedge$ i. i < rounds  $\implies$ 
  verifier-query-round-chunk (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
  using query-chunk trace-roots-eq composition-roots-eq by simp
from ntimes-verifier-query-rounds-selected-composition-fri-chunk-aligned-with-replay-at
  [OF outcome' round-bound j-bound-f raw-layer-f len-raw query-idxs-eq
    len-query-chunks transcript-query[unfolded final-state-eq]
    query-chunk-f replay-lookup[unfolded final-state-eq]]
obtain layer-chunk where auth:
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (query-idxs ! i)
      (clength * scale) ! j)
  layer-chunk final-state
by blast
have root-eq: composition-roots ! j = snd (fl ! j)
  using composition-roots-eq layer-bound by simp
show ?thesis
  by (rule that[of layer-chunk])
    (use auth composition-roots-eq root-eq in simp)
qed

end

end

```

```

theory Soundness-FRI-Layer-Authenticated
  imports Soundness-FRI-Layer-Merkle
begin

```

Authenticated FRI layer evidence paired with transcript layer evidence.

The exact equality between the layer chunk recorded in *accepted-fri-opening-transcript* and the layer chunk authenticated by the verifier is still intentionally not asserted here. Instead this theory packages the checked facts that are currently available without a broad replay induction: the transcript gives a local FRI step, and the verifier execution gives an authenticated FRI chunk at the same root, layer length, and replayed query index.

```

context soundness
begin

```

definition *generic-fri-transcript-step-with-authenticated-chunk*
 $:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $\text{nat} \Rightarrow \text{nat} \Rightarrow \text{bool}$

where

generic-fri-transcript-step-with-authenticated-chunk roots challenges
 $\text{query-idxs round-layers final-state round-idx layer-idx} \longleftrightarrow$
fri-round-layer-evidence roots challenges query-idxs round-idx
 $\text{layer-idx round-layers} \wedge$
 $(\exists \text{ auth-chunk.}$
 $\text{fri-layer-chunk-authenticated (roots ! layer-idx)}$
 $\text{(fri-evidence-layer-len roots layer-idx)}$
 $\text{(fri-evidence-layer-idx roots query-idxs round-idx layer-idx)}$
 $\text{auth-chunk final-state})$

lemma *generic-fri-transcript-step-with-authenticated-chunkI:*

assumes *step:*

fri-round-layer-evidence roots challenges query-idxs round-idx
 $\text{layer-idx round-layers}$

and *auth:*

fri-layer-chunk-authenticated (roots ! layer-idx)
 $\text{(fri-evidence-layer-len roots layer-idx)}$
 $\text{(fri-evidence-layer-idx roots query-idxs round-idx layer-idx)}$
 $\text{auth-chunk final-state}$

shows

generic-fri-transcript-step-with-authenticated-chunk roots challenges
 $\text{query-idxs round-layers final-state round-idx layer-idx}$

using *assms*

unfolding *generic-fri-transcript-step-with-authenticated-chunk-def*

by *blast*

lemma *generic-fri-transcript-step-with-authenticated-chunkE:*

assumes

generic-fri-transcript-step-with-authenticated-chunk roots challenges
 $\text{query-idxs round-layers final-state round-idx layer-idx}$

obtains *auth-chunk where*

fri-round-layer-evidence roots challenges query-idxs round-idx
 $\text{layer-idx round-layers}$

fri-layer-chunk-authenticated (roots ! layer-idx)
 $\text{(fri-evidence-layer-len roots layer-idx)}$
 $\text{(fri-evidence-layer-idx roots query-idxs round-idx layer-idx)}$
 $\text{auth-chunk final-state}$

using *assms*

unfolding *generic-fri-transcript-step-with-authenticated-chunk-def*

by *blast*

definition *generic-fri-authenticated-same-layer-opening-conflict*

$:: 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

generic-fri-authenticated-same-layer-opening-conflict roots query-idxs
 final-state $\longleftrightarrow$
 $(\exists \text{ round-idx round-idx' layer-idx opn opn'.$
 $\text{round-idx} < \text{length query-idxs} \wedge$
 $\text{round-idx'} < \text{length query-idxs} \wedge$
 $\text{layer-idx} < \text{length roots} \wedge$
 $\text{authenticated-opening-in final-state opn} \wedge$
 $\text{authenticated-opening-in final-state opn'} \wedge$
 $\text{opening-root opn} = \text{roots} ! \text{layer-idx} \wedge$
 $\text{opening-root opn'} = \text{roots} ! \text{layer-idx} \wedge$
 $\text{opening-length opn} = \text{fri-evidence-layer-len roots layer-idx} \wedge$
 $\text{opening-length opn'} = \text{fri-evidence-layer-len roots layer-idx} \wedge$
 $(\text{opening-index opn} =$
 $\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} \vee$
 $\text{opening-index opn} =$
 $\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx})) \wedge$
 $(\text{opening-index opn'} =$
 $\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx} \vee$
 $\text{opening-index opn'} =$
 $\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx})) \wedge$
 $\text{opening-index opn} = \text{opening-index opn'} \wedge$
 $\text{opening-value opn} \neq \text{opening-value opn'})$

lemma generic-fri-authenticated-same-layer-opening-conflictI:

assumes round-bound: round-idx < length query-idxs
and round-bound': round-idx' < length query-idxs
and layer-bound: layer-idx < length roots
and auth: authenticated-opening-in final-state opn
 $\text{authenticated-opening-in final-state opn'}$
and root: opening-root opn = roots ! layer-idx
 $\text{opening-root opn'} = \text{roots} ! \text{layer-idx}$
and len:
 $\text{opening-length opn} = \text{fri-evidence-layer-len roots layer-idx}$
 $\text{opening-length opn'} = \text{fri-evidence-layer-len roots layer-idx}$
and idx:
 $\text{opening-index opn} =$
 $\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} \vee$
 $\text{opening-index opn} =$
 $\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx})$
 $\text{opening-index opn'} =$
 $\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx} \vee$
 $\text{opening-index opn'} =$
 $\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx})$
and same-index: opening-index opn = opening-index opn'

and *neq*: opening-value *opn* $\neq$ opening-value *opn'*
shows
generic-fri-authenticated-same-layer-opening-conflict roots query-idxs final-state
unfolding *generic-fri-authenticated-same-layer-opening-conflict-def*
by (*intro exI*[*of* - *round-idx*] *exI*[*of* - *round-idx*']
exI[*of* - *layer-idx*] *exI*[*of* - *opn*] *exI*[*of* - *opn'*] *conjI*)
(use assms in auto)

lemma *generic-fri-authenticated-same-layer-opening-conflict-imp-partial-merkle*:
assumes
generic-fri-authenticated-same-layer-opening-conflict roots query-idxs final-state
shows *partial-merkle-inconsistency-bad s* (*Some (result, final-state)*)
proof –
from *assms* **obtain** *round-idx round-idx' layer-idx opn opn'* **where**
auth: authenticated-opening-in final-state opn
authenticated-opening-in final-state opn'
and *roots*:
opening-root opn = roots ! layer-idx
opening-root opn' = roots ! layer-idx
and *lengths*:
opening-length opn = fri-evidence-layer-len roots layer-idx
opening-length opn' = fri-evidence-layer-len roots layer-idx
and *same-index*: *opening-index opn = opening-index opn'*
and *neq*: opening-value *opn* $\neq$ opening-value *opn'*
unfolding *generic-fri-authenticated-same-layer-opening-conflict-def*
by *blast*
have *same-root*: *opening-root opn = opening-root opn'*
using *roots* **by** *simp*
have *same-length*: *opening-length opn = opening-length opn'*
using *lengths* **by** *simp*
show *?thesis*
unfolding *partial-merkle-inconsistency-bad-def accepted-def*
by (*intro conjI exI*[*of* - *result*] *exI*[*of* - *final-state*]
exI[*of* - *opn*] *exI*[*of* - *opn'*])
(use auth same-root same-length same-index neq in simp-all)
qed

lemma *generic-fri-authenticated-same-layer-opening-conflictI-base-base*:
assumes *round-bound*: *round-idx* < *length query-idxs*
and *round-bound'*: *round-idx'* < *length query-idxs*
and *layer-bound*: *layer-idx* < *length roots*
and *auth*:
fri-layer-chunk-authenticated (roots ! layer-idx)
(fri-evidence-layer-len roots layer-idx)
(fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
chunk final-state
and *auth'*:

```

    fri-layer-chunk-authenticated (roots ! layer-idx)
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
      chunk' final-state
  and chunk:
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
      xp xp-path xn xn-path chunk
  and chunk':
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
      yp yp-path yn yn-path chunk'
  and same-index:
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
    fri-evidence-layer-idx roots query-idxs round-idx' layer-idx
  and neq: xp ≠ yp
shows
  generic-fri-authenticated-same-layer-opening-conflict roots query-idxs
    final-state
proof -
  from fri-layer-chunk-authenticated-matching-values[OF auth chunk]
  obtain xp-path' xn-path' where auth-xp:
    authenticated-opening-in final-state
    (|opening-root = roots ! layer-idx,
     opening-length = fri-evidence-layer-len roots layer-idx,
     opening-index = fri-evidence-layer-idx roots query-idxs round-idx
       layer-idx,
     opening-value = xp,
     opening-path = xp-path'|)
  by blast
  from fri-layer-chunk-authenticated-matching-values[OF auth' chunk']
  obtain yp-path' yn-path' where auth-yp:
    authenticated-opening-in final-state
    (|opening-root = roots ! layer-idx,
     opening-length = fri-evidence-layer-len roots layer-idx,
     opening-index = fri-evidence-layer-idx roots query-idxs round-idx'
       layer-idx,
     opening-value = yp,
     opening-path = yp-path'|)
  by blast
  let ?opn =
    (|opening-root = roots ! layer-idx,
     opening-length = fri-evidence-layer-len roots layer-idx,
     opening-index = fri-evidence-layer-idx roots query-idxs round-idx
       layer-idx,
     opening-value = xp,
     opening-path = xp-path'|)
  let ?opn' =
    (|opening-root = roots ! layer-idx,
     opening-length = fri-evidence-layer-len roots layer-idx,
     opening-index = fri-evidence-layer-idx roots query-idxs round-idx'

```



```

    layer-idx,
    opening-value = yp,
    opening-path = yp-path')
show ?thesis
by (rule generic-fri-authenticated-same-layer-opening-conflictI
    [OF round-bound round-bound' layer-bound auth-xp auth-yp])
    (use same-index neq in simp-all)
qed

lemma generic-fri-authenticated-same-layer-opening-conflictI-base-sibling:
assumes round-bound: round-idx < length query-idxs
and round-bound': round-idx' < length query-idxs
and layer-bound: layer-idx < length roots
and auth:
    fri-layer-chunk-authenticated (roots ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    chunk final-state
and auth':
    fri-layer-chunk-authenticated (roots ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
    chunk' final-state
and chunk:
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
    xp xp-path xn xn-path chunk
and chunk':
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
    yp yp-path yn yn-path chunk'
and same-index:
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
and neq: xp ≠ yn
shows
    generic-fri-authenticated-same-layer-opening-conflict roots query-idxs
    final-state
proof -
from fri-layer-chunk-authenticated-matching-values[OF auth chunk]
obtain xp-path' xn-path' where auth-xp:
    authenticated-opening-in final-state
    (|opening-root = roots ! layer-idx,
    opening-length = fri-evidence-layer-len roots layer-idx,
    opening-index = fri-evidence-layer-idx roots query-idxs round-idx
    layer-idx,
    opening-value = xp,
    opening-path = xp-path'|)
by blast
from fri-layer-chunk-authenticated-matching-values[OF auth' chunk']

```

```

obtain yp-path' yn-path' where auth-yn:
  authenticated-opening-in final-state
  (opening-root = roots ! layer-idx,
   opening-length = fri-evidence-layer-len roots layer-idx,
   opening-index =
     fri-sibling-index (fri-evidence-layer-len roots layer-idx)
     (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx),
   opening-value = yn,
   opening-path = yn-path')
by blast
show ?thesis
by (rule generic-fri-authenticated-same-layer-opening-conflictI
  [OF round-bound round-bound' layer-bound auth-xp auth-yn])
  (use same-index neq in simp-all)
qed

lemma generic-fri-authenticated-same-layer-opening-conflictI-sibling-base:
assumes round-bound: round-idx < length query-idxs
and round-bound': round-idx' < length query-idxs
and layer-bound: layer-idx < length roots
and auth:
  fri-layer-chunk-authenticated (roots ! layer-idx)
  (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
  chunk final-state
and auth':
  fri-layer-chunk-authenticated (roots ! layer-idx)
  (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
  chunk' final-state
and chunk:
  fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
  xp xp-path xn xn-path chunk
and chunk':
  fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
  yp yp-path yn yn-path chunk'
and same-index:
  fri-sibling-index (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
  fri-evidence-layer-idx roots query-idxs round-idx' layer-idx
and neq: xn ≠ yp
shows
  generic-fri-authenticated-same-layer-opening-conflict roots query-idxs
  final-state
proof –
from fri-layer-chunk-authenticated-matching-values[OF auth chunk]
obtain xp-path' xn-path' where auth-xn:
  authenticated-opening-in final-state
  (opening-root = roots ! layer-idx,

```

```

    opening-length = fri-evidence-layer-len roots layer-idx,
    opening-index =
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx layer-idx),
    opening-value = xn,
    opening-path = xn-path')
  by blast
from fri-layer-chunk-authenticated-matching-values[OF auth' chunk']
obtain yp-path' yn-path' where auth-yp:
  authenticated-opening-in final-state
  (|opening-root = roots ! layer-idx,
   opening-length = fri-evidence-layer-len roots layer-idx,
   opening-index = fri-evidence-layer-idx roots query-idxs round-idx'
    layer-idx,
   opening-value = yp,
   opening-path = yp-path')
  by blast
show ?thesis
  by (rule generic-fri-authenticated-same-layer-opening-conflictI
      [OF round-bound round-bound' layer-bound auth-xn auth-yp])
    (use same-index neq in simp-all)
qed

lemma generic-fri-authenticated-same-layer-opening-conflictI-sibling-sibling:
  assumes round-bound: round-idx < length query-idxs
  and round-bound': round-idx' < length query-idxs
  and layer-bound: layer-idx < length roots
  and auth:
    fri-layer-chunk-authenticated (roots ! layer-idx)
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      chunk final-state
  and auth':
    fri-layer-chunk-authenticated (roots ! layer-idx)
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
      chunk' final-state
  and chunk:
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
      xp xp-path xn xn-path chunk
  and chunk':
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
      yp yp-path yn yn-path chunk'
  and same-index:
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
  and neq: xn ≠ yn

```

shows
generic-fri-authenticated-same-layer-opening-conflict roots query-idxs
final-state

proof –
from *fri-layer-chunk-authenticated-matching-values*[*OF auth chunk*]
obtain *xp-path' xn-path'* **where** *auth-xn*:
authenticated-opening-in final-state
 (|*opening-root* = *roots ! layer-idx*,
opening-length = *fri-evidence-layer-len roots layer-idx*,
opening-index =
 fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*),
opening-value = *xn*,
opening-path = *xn-path'*)
by *blast*
from *fri-layer-chunk-authenticated-matching-values*[*OF auth' chunk'*]
obtain *yp-path' yn-path'* **where** *auth-yn*:
authenticated-opening-in final-state
 (|*opening-root* = *roots ! layer-idx*,
opening-length = *fri-evidence-layer-len roots layer-idx*,
opening-index =
 fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (*fri-evidence-layer-idx roots query-idxs round-idx' layer-idx*),
opening-value = *yn*,
opening-path = *yn-path'*)
by *blast*
show *?thesis*
by (*rule generic-fri-authenticated-same-layer-opening-conflictI*
 [*OF round-bound round-bound' layer-bound auth-xn auth-yn*])
 (*use same-index neq in simp-all*)

qed

definition *generic-fri-recorded-layer-chunk-authenticated*
 :: '*f list* ⇒ *nat list* ⇒ '*f list list list* ⇒
 (*f, 'a*) *protocol-channel-scheme* ⇒ *nat* ⇒ *nat* ⇒ *bool*

where
generic-fri-recorded-layer-chunk-authenticated roots query-idxs
round-layers final-state round-idx layer-idx ⇔
fri-layer-chunk-authenticated (roots ! layer-idx)
 (*fri-evidence-layer-len roots layer-idx*)
 (*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*)
 (*round-layers ! round-idx ! layer-idx*) *final-state*

definition *generic-fri-authenticated-slot-recorded-chunk-gap*
 :: '*f list* ⇒ *nat list* ⇒ '*f list list list* ⇒
 (*f, 'a*) *protocol-channel-scheme* ⇒ *bool*

where
generic-fri-authenticated-slot-recorded-chunk-gap roots query-idxs
round-layers final-state ⇔

$(\exists \text{ round-idx layer-idx auth-chunk.}$
 $\text{round-idx} < \text{length query-idxs} \wedge$
 $\text{layer-idx} < \text{length roots} \wedge$
 $\text{fri-layer-chunk-authenticated (roots ! layer-idx)}$
 $(\text{fri-evidence-layer-len roots layer-idx})$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx})$
 $\text{auth-chunk final-state} \wedge$
 $\neg \text{generic-fri-recorded-layer-chunk-authenticated roots query-idxs}$
 $\text{round-layers final-state round-idx layer-idx})$

lemma *generic-fri-transcript-step-auth-missing-imp-authenticated-slot-recorded-chunk-gap:*

assumes *step:*

generic-fri-transcript-step-with-authenticated-chunk roots challenges
query-idxs round-layers final-state round-idx layer-idx

and *missing:*

$\neg \text{generic-fri-recorded-layer-chunk-authenticated roots query-idxs}$
round-layers final-state round-idx layer-idx

shows

generic-fri-authenticated-slot-recorded-chunk-gap roots query-idxs
round-layers final-state

proof —

from *step* **obtain** *auth-chunk* **where** *evidence:*

fri-round-layer-evidence roots challenges query-idxs round-idx layer-idx
round-layers

and *auth:*

fri-layer-chunk-authenticated (roots ! layer-idx)
 $(\text{fri-evidence-layer-len roots layer-idx})$
 $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx})$
auth-chunk final-state

by (rule *generic-fri-transcript-step-with-authenticated-chunkE*)

from *evidence* **have** *round-bound: round-idx < length query-idxs*

and *layer-bound: layer-idx < length roots*

by (auto elim: *fri-round-layer-evidenceE*)

show *?thesis*

unfolding *generic-fri-authenticated-slot-recorded-chunk-gap-def*

by (intro *exI conjI*)

(rule *round-bound*, rule *layer-bound*, rule *auth*, rule *missing*)

qed

definition *generic-fri-sampled-same-layer-recorded-auth-gap*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow$

$'f \text{ list list list} \Rightarrow ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

generic-fri-sampled-same-layer-recorded-auth-gap roots challenges
query-idxs round-layers final-state $\longleftrightarrow$

$(\exists \text{ round-idx round-idx'} \text{ layer-idx.}$

$\text{round-idx} < \text{length query-idxs} \wedge$

$\text{round-idx'} < \text{length query-idxs} \wedge$

$\text{layer-idx} < \text{length challenges} \wedge$

generic-fri-sampled-same-layer-opening-conflict roots challenges
query-idxs round-layers $\wedge$
 $(\neg \text{generic-fri-recorded-layer-chunk-authenticated roots query-idxs}$
round-layers final-state round-idx layer-idx $\vee$
 $\neg \text{generic-fri-recorded-layer-chunk-authenticated roots query-idxs}$
round-layers final-state round-idx' layer-idx'))

definition *generic-fri-sampled-same-layer-authenticated-recorded-auth-gap*
 $:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

generic-fri-sampled-same-layer-authenticated-recorded-auth-gap roots
challenges query-idxs round-layers final-state $\longleftrightarrow$
 $(\exists \text{round-idx round-idx' layer-idx.}$
 $\text{round-idx} < \text{length query-idxs} \wedge$
 $\text{round-idx'} < \text{length query-idxs} \wedge$
 $\text{layer-idx} < \text{length challenges} \wedge$
generic-fri-sampled-same-layer-opening-conflict roots challenges
query-idxs round-layers $\wedge$
 $((\text{generic-fri-transcript-step-with-authenticated-chunk roots challenges}$
 $\text{query-idxs round-layers final-state round-idx layer-idx} \wedge$
 $\neg \text{generic-fri-recorded-layer-chunk-authenticated roots query-idxs}$
 $\text{round-layers final-state round-idx layer-idx}) \vee$
 $(\text{generic-fri-transcript-step-with-authenticated-chunk roots challenges}$
 $\text{query-idxs round-layers final-state round-idx' layer-idx} \wedge$
 $\neg \text{generic-fri-recorded-layer-chunk-authenticated roots query-idxs}$
 $\text{round-layers final-state round-idx' layer-idx})))$

lemma *generic-fri-authenticated-recorded-auth-gap-imp-slot-recorded-chunk-gap:*

assumes

generic-fri-sampled-same-layer-authenticated-recorded-auth-gap roots
challenges query-idxs round-layers final-state

shows

generic-fri-authenticated-slot-recorded-chunk-gap roots query-idxs
round-layers final-state

proof —

from *assms* **obtain** *round-idx round-idx' layer-idx* **where**

gap:

$(\text{generic-fri-transcript-step-with-authenticated-chunk roots challenges}$
 $\text{query-idxs round-layers final-state round-idx layer-idx} \wedge$
 $\neg \text{generic-fri-recorded-layer-chunk-authenticated roots query-idxs}$
 $\text{round-layers final-state round-idx layer-idx}) \vee$
 $(\text{generic-fri-transcript-step-with-authenticated-chunk roots challenges}$
 $\text{query-idxs round-layers final-state round-idx' layer-idx} \wedge$
 $\neg \text{generic-fri-recorded-layer-chunk-authenticated roots query-idxs}$
 $\text{round-layers final-state round-idx' layer-idx})$

unfolding *generic-fri-sampled-same-layer-authenticated-recorded-auth-gap-def*

by *blast*

then show *?thesis*

proof

assume *left*:

generic-fri-transcript-step-with-authenticated-chunk roots challenges
query-idxs round-layers final-state round-idx layer-idx $\wedge$
 $\neg$ *generic-fri-recorded-layer-chunk-authenticated roots query-idxs*
round-layers final-state round-idx layer-idx

show *?thesis*

by (rule

generic-fri-transcript-step-auth-missing-imp-authenticated-slot-recorded-chunk-gap
[of *roots challenges query-idxs round-layers final-state round-idx*
layer-idx])

(use *left* in *simp-all*)

next

assume *right*:

generic-fri-transcript-step-with-authenticated-chunk roots challenges
query-idxs round-layers final-state round-idx' layer-idx $\wedge$
 $\neg$ *generic-fri-recorded-layer-chunk-authenticated roots query-idxs*
round-layers final-state round-idx' layer-idx

show *?thesis*

by (rule

generic-fri-transcript-step-auth-missing-imp-authenticated-slot-recorded-chunk-gap
[of *roots challenges query-idxs round-layers final-state round-idx'*
layer-idx])

(use *right* in *simp-all*)

qed

qed

lemma *generic-fri-recorded-auth-gap-imp-authenticated-recorded-auth-gap*:

assumes *gap*:

generic-fri-sampled-same-layer-recorded-auth-gap roots challenges
query-idxs round-layers final-state

and *auth-step*:

$\bigwedge$ *round-idx layer-idx*.

round-idx < length *query-idxs* $\implies$

layer-idx < length *challenges* $\implies$

generic-fri-transcript-step-with-authenticated-chunk roots
challenges query-idxs round-layers final-state round-idx layer-idx

shows

generic-fri-sampled-same-layer-authenticated-recorded-auth-gap roots
challenges query-idxs round-layers final-state

proof –

from *gap* **obtain** *round-idx round-idx' layer-idx* **where**

round-bound: *round-idx* < length *query-idxs*

and *round-bound'*: *round-idx'* < length *query-idxs*

and *layer-bound*: *layer-idx* < length *challenges*

and *conflict*:

generic-fri-sampled-same-layer-opening-conflict roots challenges
query-idxs round-layers

and *missing*:

```

    ¬ generic-fri-recorded-layer-chunk-authenticated roots query-idxs
      round-layers final-state round-idx layer-idx ∨
    ¬ generic-fri-recorded-layer-chunk-authenticated roots query-idxs
      round-layers final-state round-idx' layer-idx
  unfolding generic-fri-sampled-same-layer-recorded-auth-gap-def
  by blast
show ?thesis
proof (cases
  ¬ generic-fri-recorded-layer-chunk-authenticated roots query-idxs
    round-layers final-state round-idx layer-idx)
case True
have step:
  generic-fri-transcript-step-with-authenticated-chunk roots challenges
    query-idxs round-layers final-state round-idx layer-idx
  by (rule auth-step[OF round-bound layer-bound])
show ?thesis
  unfolding
    generic-fri-sampled-same-layer-authenticated-recorded-auth-gap-def
  by (intro exI conjI disjI1)
    (rule round-bound, rule round-bound', rule layer-bound,
      rule conflict, rule step, rule True)
next
case False
then have missing':
  ¬ generic-fri-recorded-layer-chunk-authenticated roots query-idxs
    round-layers final-state round-idx' layer-idx
  using missing by simp
have step:
  generic-fri-transcript-step-with-authenticated-chunk roots challenges
    query-idxs round-layers final-state round-idx' layer-idx
  by (rule auth-step[OF round-bound' layer-bound])
show ?thesis
  unfolding
    generic-fri-sampled-same-layer-authenticated-recorded-auth-gap-def
  by (intro exI conjI disjI2)
    (rule round-bound, rule round-bound', rule layer-bound,
      rule conflict, rule step, rule missing')
qed
qed

definition generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
  :: 'f list ⇒ 'f list ⇒ nat list ⇒
    'f list list list ⇒ ('f, 'a) protocol-channel-scheme ⇒ bool
where
  generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    roots challenges query-idxs round-layers final-state ⟷
  (∃ round-idx round-idx' layer-idx
    xp xp-path xn xn-path yp yp-path yn yn-path.
    round-idx < length query-idxs ∧

```


$round_idx' < length\ query_idxs \wedge$
 $layer_idx < length\ challenges \wedge$
 $fri_layer_step_evidence$
 $(roots\ !\ layer_idx)$
 $(challenges\ !\ layer_idx)$
 $(fri_evidence_layer_len\ roots\ layer_idx)$
 $(fri_evidence_layer_idx\ roots\ query_idxs\ round_idx\ layer_idx)$
 $(2\ ^\wedge\ layer_idx)$
 $(fri_sibling_index\ (fri_evidence_layer_len\ roots\ layer_idx)$
 $\quad (fri_evidence_layer_idx\ roots\ query_idxs\ round_idx\ layer_idx))$
 $xp\ xp_path\ xn\ xn_path$
 $(fri_evidence_next_idx\ roots\ query_idxs\ round_idx\ layer_idx)$
 $(fri_evidence_next_value\ roots\ challenges\ query_idxs\ round_idx$
 $\quad layer_idx\ xp\ xn)$
 $(round_layers\ !\ round_idx\ !\ layer_idx) \wedge$
 $fri_layer_step_evidence$
 $(roots\ !\ layer_idx)$
 $(challenges\ !\ layer_idx)$
 $(fri_evidence_layer_len\ roots\ layer_idx)$
 $(fri_evidence_layer_idx\ roots\ query_idxs\ round_idx'\ layer_idx)$
 $(2\ ^\wedge\ layer_idx)$
 $(fri_sibling_index\ (fri_evidence_layer_len\ roots\ layer_idx)$
 $\quad (fri_evidence_layer_idx\ roots\ query_idxs\ round_idx'\ layer_idx))$
 $yp\ yp_path\ yn\ yn_path$
 $(fri_evidence_next_idx\ roots\ query_idxs\ round_idx'\ layer_idx)$
 $(fri_evidence_next_value\ roots\ challenges\ query_idxs\ round_idx'$
 $\quad layer_idx\ yp\ yn)$
 $(round_layers\ !\ round_idx'\ !\ layer_idx) \wedge$
 $fri_layer_chunk_authenticated\ (roots\ !\ layer_idx)$
 $(fri_evidence_layer_len\ roots\ layer_idx)$
 $(fri_evidence_layer_idx\ roots\ query_idxs\ round_idx\ layer_idx)$
 $(round_layers\ !\ round_idx\ !\ layer_idx)\ final_state \wedge$
 $fri_layer_chunk_authenticated\ (roots\ !\ layer_idx)$
 $(fri_evidence_layer_len\ roots\ layer_idx)$
 $(fri_evidence_layer_idx\ roots\ query_idxs\ round_idx'\ layer_idx)$
 $(round_layers\ !\ round_idx'\ !\ layer_idx)\ final_state \wedge$
 $((fri_evidence_layer_idx\ roots\ query_idxs\ round_idx\ layer_idx =$
 $\quad fri_evidence_layer_idx\ roots\ query_idxs\ round_idx'\ layer_idx \wedge$
 $\quad xp \neq yp) \vee$
 $(fri_evidence_layer_idx\ roots\ query_idxs\ round_idx\ layer_idx =$
 $\quad fri_sibling_index\ (fri_evidence_layer_len\ roots\ layer_idx)$
 $\quad (fri_evidence_layer_idx\ roots\ query_idxs\ round_idx'\ layer_idx) \wedge$
 $\quad xp \neq yn) \vee$
 $(fri_sibling_index\ (fri_evidence_layer_len\ roots\ layer_idx)$
 $\quad (fri_evidence_layer_idx\ roots\ query_idxs\ round_idx\ layer_idx) =$
 $\quad fri_evidence_layer_idx\ roots\ query_idxs\ round_idx'\ layer_idx \wedge$
 $\quad xn \neq yp) \vee$
 $(fri_sibling_index\ (fri_evidence_layer_len\ roots\ layer_idx)$
 $\quad (fri_evidence_layer_idx\ roots\ query_idxs\ round_idx\ layer_idx) =$

$$\begin{aligned} & \text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)} \\ & \text{(fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) } \wedge \\ & \text{xn} \neq \text{yn})) \end{aligned}$$

lemma *generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-authenticated:*

assumes *conflict:*

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks

roots challenges query-idxs round-layers final-state

and *len-challenges:* $\text{length challenges} = \text{length roots}$

shows

*generic-fri-authenticated-same-layer-opening-conflict roots query-idxs
final-state*

proof –

from *conflict* **obtain** *round-idx round-idx' layer-idx*

xp xp-path xn xn-path yp yp-path yn yn-path **where**

round-bound: $\text{round-idx} < \text{length query-idxs}$

and *round-bound':* $\text{round-idx}' < \text{length query-idxs}$

and *layer-bound-ch:* $\text{layer-idx} < \text{length challenges}$

and *step:*

fri-layer-step-evidence

(roots ! layer-idx)

(challenges ! layer-idx)

(fri-evidence-layer-len roots layer-idx)

(fri-evidence-layer-idx roots query-idxs round-idx layer-idx)

(2 ^ layer-idx)

(fri-sibling-index (fri-evidence-layer-len roots layer-idx)

(fri-evidence-layer-idx roots query-idxs round-idx layer-idx))

xp xp-path xn xn-path

(fri-evidence-next-idx roots query-idxs round-idx layer-idx)

*(fri-evidence-next-value roots challenges query-idxs round-idx
layer-idx xp xn)*

(round-layers ! round-idx ! layer-idx)

and *step':*

fri-layer-step-evidence

(roots ! layer-idx)

(challenges ! layer-idx)

(fri-evidence-layer-len roots layer-idx)

(fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)

(2 ^ layer-idx)

(fri-sibling-index (fri-evidence-layer-len roots layer-idx)

(fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))

yp yp-path yn yn-path

(fri-evidence-next-idx roots query-idxs round-idx' layer-idx)

*(fri-evidence-next-value roots challenges query-idxs round-idx'
layer-idx yp yn)*

(round-layers ! round-idx' ! layer-idx)

and *auth:*

fri-layer-chunk-authenticated (roots ! layer-idx)

(fri-evidence-layer-len roots layer-idx)

```

    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (round-layers ! round-idx ! layer-idx) final-state
and auth':
    fri-layer-chunk-authenticated (roots ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
    (round-layers ! round-idx' ! layer-idx) final-state
and cases:
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
      fri-evidence-layer-idx roots query-idxs round-idx' layer-idx  $\wedge$ 
       $xp \neq yp$ )  $\vee$ 
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)  $\wedge$ 
       $xp \neq yn$ )  $\vee$ 
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
      fri-evidence-layer-idx roots query-idxs round-idx' layer-idx  $\wedge$ 
       $xn \neq yp$ )  $\vee$ 
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)  $\wedge$ 
       $xn \neq yn$ )
unfolding
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-def
by blast
have layer-bound: layer-idx < length roots
    using layer-bound-ch len-challenges by simp
have chunk:
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
    xp xp-path xn xn-path (round-layers ! round-idx ! layer-idx)
    using fri-layer-step-evidenceD(4)[OF step] .
have chunk':
    fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
    yp yp-path yn yn-path (round-layers ! round-idx' ! layer-idx)
    using fri-layer-step-evidenceD(4)[OF step'] .
from cases show ?thesis
proof
    assume c:
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
      fri-evidence-layer-idx roots query-idxs round-idx' layer-idx  $\wedge$ 
       $xp \neq yp$ 
    show ?thesis
    by (rule generic-fri-authenticated-same-layer-opening-conflictI-base-base
      [OF round-bound round-bound' layer-bound auth auth' chunk chunk'])
      (use c in simp-all)
next
    assume rest:

```

```

(fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
  fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
  xp ≠ yn) ∨
(fri-sibling-index (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
  fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ∧
  xn ≠ yp) ∨
(fri-sibling-index (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
  fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
  xn ≠ yn)
then show ?thesis
proof
  assume c:
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
      xp ≠ yn
  show ?thesis
  by (rule generic-fri-authenticated-same-layer-opening-conflictI-base-sibling
    [OF round-bound round-bound' layer-bound auth auth' chunk chunk'])
    (use c in simp-all)
next
  assume rest':
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
      fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ∧
      xn ≠ yp) ∨
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
      xn ≠ yn)
  then show ?thesis
  proof
    assume c:
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
      fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ∧
      xn ≠ yp
    show ?thesis
    by (rule generic-fri-authenticated-same-layer-opening-conflictI-sibling-base
      [OF round-bound round-bound' layer-bound auth auth' chunk chunk'])
      (use c in simp-all)
  next
    assume c:
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)

```

```

      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
    xn ≠ yn
  show ?thesis
  by (rule generic-fri-authenticated-same-layer-opening-conflictI-sibling-sibling
      [OF round-bound round-bound' layer-bound auth auth' chunk chunk'])
      (use c in simp-all)
qed
qed
qed
qed

lemma generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-sampled:
  assumes
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    roots challenges query-idxs round-layers final-state
  shows
    generic-fri-sampled-same-layer-opening-conflict roots challenges
    query-idxs round-layers
  using assms
  unfolding
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-def
    generic-fri-sampled-same-layer-opening-conflict-def
  by blast

lemma generic-fri-sampled-same-layer-imp-authenticated-or-recorded-auth-gap:
  assumes conflict:
    generic-fri-sampled-same-layer-opening-conflict roots challenges
    query-idxs round-layers
  shows
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    roots challenges query-idxs round-layers final-state ∨
    generic-fri-sampled-same-layer-recorded-auth-gap roots challenges
    query-idxs round-layers final-state
proof —
  from conflict obtain round-idx round-idx' layer-idx
    xp xp-path xn xn-path yp yp-path yn yn-path where
    round-bound: round-idx < length query-idxs
    and round-bound': round-idx' < length query-idxs
    and layer-bound: layer-idx < length challenges
    and step:
      fri-layer-step-evidence
      (roots ! layer-idx)
      (challenges ! layer-idx)
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      (2 ^ layer-idx)
      (fri-sibling-index (fri-evidence-layer-len roots layer-idx)

```

```

    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
  xp xp-path xn xn-path
  (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  (fri-evidence-next-value roots challenges query-idxs round-idx
    layer-idx xp xn)
  (round-layers ! round-idx ! layer-idx)
and step':
  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx))
    yp yp-path yn yn-path
    (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx'
      layer-idx yp yn)
    (round-layers ! round-idx' ! layer-idx)
and cases:
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
    fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ∧
    xp ≠ yp) ∨
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
    xp ≠ yn) ∨
  (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
    fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ∧
    xn ≠ yp) ∨
  (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
    xn ≠ yn)
unfolding generic-fri-sampled-same-layer-opening-conflict-def
by blast
show ?thesis
proof (cases
  generic-fri-recorded-layer-chunk-authenticated roots query-idxs
    round-layers final-state round-idx layer-idx ∧
  generic-fri-recorded-layer-chunk-authenticated roots query-idxs
    round-layers final-state round-idx' layer-idx)
case True
have auth:
  fri-layer-chunk-authenticated (roots ! layer-idx)
  (fri-evidence-layer-len roots layer-idx)

```

```

    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (round-layers ! round-idx ! layer-idx) final-state
  using True
  unfolding generic-fri-recorded-layer-chunk-authenticated-def
  by simp
  have auth':
    fri-layer-chunk-authenticated (roots ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
    (round-layers ! round-idx' ! layer-idx) final-state
  using True
  unfolding generic-fri-recorded-layer-chunk-authenticated-def
  by simp
  have
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    roots challenges query-idxs round-layers final-state
  unfolding
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-def
  by (intro exI conjI)
    (rule round-bound, rule round-bound', rule layer-bound, rule step,
     rule step', rule auth, rule auth', rule cases)
  then show ?thesis by simp
next
case False
have
  generic-fri-sampled-same-layer-recorded-auth-gap roots challenges
  query-idxs round-layers final-state
  unfolding generic-fri-sampled-same-layer-recorded-auth-gap-def
  by (intro exI conjI)
    (rule round-bound, rule round-bound', rule layer-bound,
     rule conflict, use False in simp)
  then show ?thesis by simp
qed
qed

lemma generic-fri-sampled-same-layer-auth-gap-imp-recorded-auth-gap:
  assumes conflict:
    generic-fri-sampled-same-layer-opening-conflict roots challenges
    query-idxs round-layers
  and no-auth:
    ¬ generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    roots challenges query-idxs round-layers final-state
  shows
    generic-fri-sampled-same-layer-recorded-auth-gap roots challenges
    query-idxs round-layers final-state
  using generic-fri-sampled-same-layer-imp-authenticated-or-recorded-auth-gap
  [OF conflict, of final-state] no-auth
  by blast

```

lemma *generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-partial-merkle:*

assumes *conflict:*

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks

roots challenges query-idxs round-layers final-state

and *len-challenges:* $\text{length } \text{challenges} = \text{length } \text{roots}$

shows *partial-merkle-inconsistency-bad s* (*Some (result, final-state)*)

proof –

have *auth-conflict:*

generic-fri-authenticated-same-layer-opening-conflict roots query-idxs

final-state

by (*rule*

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-authenticated

[*OF conflict len-challenges*])

show *?thesis*

by (*rule generic-fri-authenticated-same-layer-opening-conflict-imp-partial-merkle*

[*OF auth-conflict*])

qed

definition *generic-fri-sampled-assignment-conflict-with-authenticated-same-layer*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$

$\text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow$

$('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$

where

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer

candidate-table roots challenges final-value query-idxs round-layers

final-state $\longleftrightarrow$

generic-fri-sampled-base-opening-conflict candidate-table roots

challenges query-idxs round-layers $\vee$

generic-fri-sampled-next-value-conflict roots challenges query-idxs

round-layers $\vee$

generic-fri-sampled-successor-opening-conflict roots challenges

query-idxs round-layers $\vee$

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks

roots challenges query-idxs round-layers final-state $\vee$

generic-fri-sampled-final-value-conflict roots challenges final-value

query-idxs round-layers

definition *generic-fri-sampled-assignment-conflict-without-same-layer*

$:: 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$

$\text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow \text{bool}$

where

generic-fri-sampled-assignment-conflict-without-same-layer

candidate-table roots challenges final-value query-idxs round-layers

$\longleftrightarrow$

generic-fri-sampled-base-opening-conflict candidate-table roots

challenges query-idxs round-layers $\vee$

generic-fri-sampled-next-value-conflict roots challenges query-idxs

round-layers $\vee$

generic-fri-sampled-successor-opening-conflict roots challenges

query-idxs round-layers $\vee$
generic-fri-sampled-final-value-conflict roots challenges final-value
query-idxs round-layers

lemma *generic-fri-sampled-assignment-conflict-without-same-layer-split:*

assumes

generic-fri-sampled-assignment-conflict-without-same-layer
candidate-table roots challenges final-value query-idxs round-layers

shows

generic-fri-sampled-base-opening-conflict candidate-table roots
challenges query-idxs round-layers $\vee$
generic-fri-sampled-next-value-conflict roots challenges query-idxs
round-layers $\vee$
generic-fri-sampled-successor-opening-conflict roots challenges
query-idxs round-layers $\vee$
generic-fri-sampled-final-value-conflict roots challenges final-value
query-idxs round-layers

using *assms*

unfolding *generic-fri-sampled-assignment-conflict-without-same-layer-def*
by *blast*

lemma *generic-fri-sampled-assignment-conflict-without-same-layerI-base:*

assumes

generic-fri-sampled-base-opening-conflict candidate-table roots
challenges query-idxs round-layers

shows

generic-fri-sampled-assignment-conflict-without-same-layer
candidate-table roots challenges final-value query-idxs round-layers

using *assms*

unfolding *generic-fri-sampled-assignment-conflict-without-same-layer-def*
by *blast*

lemma *generic-fri-sampled-assignment-conflict-without-same-layerI-next:*

assumes

generic-fri-sampled-next-value-conflict roots challenges query-idxs
round-layers

shows

generic-fri-sampled-assignment-conflict-without-same-layer
candidate-table roots challenges final-value query-idxs round-layers

using *assms*

unfolding *generic-fri-sampled-assignment-conflict-without-same-layer-def*
by *blast*

lemma *generic-fri-sampled-assignment-conflict-without-same-layerI-successor:*

assumes

generic-fri-sampled-successor-opening-conflict roots challenges
query-idxs round-layers

shows

generic-fri-sampled-assignment-conflict-without-same-layer

candidate-table roots challenges final-value query-idxs round-layers
using *assms*
unfolding *generic-fri-sampled-assignment-conflict-without-same-layer-def*
by *blast*

lemma *generic-fri-sampled-assignment-conflict-without-same-layerI-final:*
assumes
generic-fri-sampled-final-value-conflict roots challenges final-value
query-idxs round-layers
shows
generic-fri-sampled-assignment-conflict-without-same-layer
candidate-table roots challenges final-value query-idxs round-layers
using *assms*
unfolding *generic-fri-sampled-assignment-conflict-without-same-layer-def*
by *blast*

lemma *generic-fri-sampled-assignment-conflict-without-same-layer-iff:*
generic-fri-sampled-assignment-conflict-without-same-layer
candidate-table roots challenges final-value query-idxs round-layers
 $\longleftrightarrow$
generic-fri-sampled-base-opening-conflict candidate-table roots
challenges query-idxs round-layers $\vee$
generic-fri-sampled-next-value-conflict roots challenges query-idxs
round-layers $\vee$
generic-fri-sampled-successor-opening-conflict roots challenges
query-idxs round-layers $\vee$
generic-fri-sampled-final-value-conflict roots challenges final-value
query-idxs round-layers
unfolding *generic-fri-sampled-assignment-conflict-without-same-layer-def*
by *simp*

lemma *generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-split:*
assumes
generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
candidate-table roots challenges final-value query-idxs round-layers
final-state
shows
generic-fri-sampled-assignment-conflict-without-same-layer
candidate-table roots challenges final-value query-idxs round-layers $\vee$
generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
roots challenges query-idxs round-layers final-state
using *assms*
unfolding
generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-def
generic-fri-sampled-assignment-conflict-without-same-layer-def
by *blast*

lemma *generic-fri-sampled-assignment-conflict-with-authenticated-same-layerI-without:*
assumes

```

    generic-fri-sampled-assignment-conflict-without-same-layer
    candidate-table roots challenges final-value query-idxs round-layers
shows
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
    candidate-table roots challenges final-value query-idxs round-layers
    final-state
using assms
unfolding
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-def
    generic-fri-sampled-assignment-conflict-without-same-layer-def
by blast

lemma generic-fri-sampled-assignment-conflict-with-authenticated-same-layerI-same:
assumes
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    roots challenges query-idxs round-layers final-state
shows
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
    candidate-table roots challenges final-value query-idxs round-layers
    final-state
using assms
unfolding generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-def
by blast

lemma generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-iff:
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
    candidate-table roots challenges final-value query-idxs round-layers
    final-state  $\longleftrightarrow$ 
    generic-fri-sampled-assignment-conflict-without-same-layer
    candidate-table roots challenges final-value query-idxs round-layers  $\vee$ 
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    roots challenges query-idxs round-layers final-state
proof
assume
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
    candidate-table roots challenges final-value query-idxs round-layers
    final-state
then show
    generic-fri-sampled-assignment-conflict-without-same-layer
    candidate-table roots challenges final-value query-idxs round-layers  $\vee$ 
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    roots challenges query-idxs round-layers final-state
by (rule generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-split)
next
assume
    generic-fri-sampled-assignment-conflict-without-same-layer
    candidate-table roots challenges final-value query-idxs round-layers  $\vee$ 
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    roots challenges query-idxs round-layers final-state

```

then show

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
candidate-table roots challenges final-value query-idxs round-layers
final-state

by (*elim disjE*)

(*rule generic-fri-sampled-assignment-conflict-with-authenticated-same-layerI-without,*
assumption,
rule generic-fri-sampled-assignment-conflict-with-authenticated-same-layerI-same,
assumption)

qed

lemma *generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-imp-sampled:*

assumes

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
candidate-table roots challenges final-value query-idxs round-layers
final-state

shows

generic-fri-sampled-assignment-conflict candidate-table roots challenges
final-value query-idxs round-layers

using *assms*

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-sampled

unfolding

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-def
generic-fri-sampled-assignment-conflict-def

by *blast*

lemma *generic-fri-sampled-assignment-conflict-imp-authenticated-or-same-layer-auth-gap:*

assumes

generic-fri-sampled-assignment-conflict candidate-table roots
challenges final-value query-idxs round-layers

shows

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
candidate-table roots challenges final-value query-idxs round-layers
final-state $\vee$

(*generic-fri-sampled-same-layer-opening-conflict roots challenges*
query-idxs round-layers $\wedge$

$\neg$ *generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks*
roots challenges query-idxs round-layers final-state)

using *assms*

unfolding *generic-fri-sampled-assignment-conflict-def*

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-def
generic-fri-sampled-assignment-conflict-without-same-layer-def

by *blast*

lemma *generic-fri-authenticated-same-layer-branch-imp-partial-merkle:*

assumes

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
roots challenges query-idxs round-layers final-state

and *length challenges = length roots*

shows *partial-merkle-inconsistency-bad s* (Some (result, final-state))
by (rule
generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-partial-merkle
[OF assms])

lemma *generic-fri-sampled-same-layer-opening-conflict-imp-authenticated-if-recorded:*

assumes *conflict:*

generic-fri-sampled-same-layer-opening-conflict roots challenges
query-idxs round-layers

and *len-challenges:* *length challenges = length roots*

and *recorded:*

$\bigwedge$ *round-idx layer-idx.*

round-idx < length query-idxs $\implies$

layer-idx < length roots $\implies$

generic-fri-recorded-layer-chunk-authenticated roots query-idxs

round-layers final-state round-idx layer-idx

shows

generic-fri-authenticated-same-layer-opening-conflict roots query-idxs
final-state

proof –

from *conflict* **obtain** *round-idx round-idx' layer-idx*

xp xp-path xn xn-path yp yp-path yn yn-path **where**

round-bound: *round-idx < length query-idxs*

and *round-bound':* *round-idx' < length query-idxs*

and *layer-bound-ch:* *layer-idx < length challenges*

and *step:*

fri-layer-step-evidence

(*roots ! layer-idx*)

(*challenges ! layer-idx*)

(*fri-evidence-layer-len roots layer-idx*)

(*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*)

($2 \wedge$ *layer-idx*)

(*fri-sibling-index (fri-evidence-layer-len roots layer-idx)*

(*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*))

xp xp-path xn xn-path

(*fri-evidence-next-idx roots query-idxs round-idx layer-idx*)

(*fri-evidence-next-value roots challenges query-idxs round-idx*
layer-idx xp xn)

(*round-layers ! round-idx ! layer-idx*)

and *step':*

fri-layer-step-evidence

(*roots ! layer-idx*)

(*challenges ! layer-idx*)

(*fri-evidence-layer-len roots layer-idx*)

(*fri-evidence-layer-idx roots query-idxs round-idx' layer-idx*)

($2 \wedge$ *layer-idx*)

(*fri-sibling-index (fri-evidence-layer-len roots layer-idx)*

(*fri-evidence-layer-idx roots query-idxs round-idx' layer-idx*))

yp yp-path yn yn-path

```

    (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx'
      layer-idx yp yn)
    (round-layers ! round-idx' ! layer-idx)
  and cases:
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
      fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ∧
      xp ≠ yp) ∨
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx =
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
      xp ≠ yn) ∨
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
      fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ∧
      xn ≠ yp) ∨
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
      fri-sibling-index (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
      xn ≠ yn)
  unfolding generic-fri-sampled-same-layer-opening-conflict-def
    fri-evidence-layer-len-def fri-evidence-layer-idx-def
    fri-evidence-next-idx-def fri-evidence-next-value-def
  by blast
have layer-bound: layer-idx < length roots
  using layer-bound-ch len-challenges by simp
have chunk:
  fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
    xp xp-path xn xn-path (round-layers ! round-idx ! layer-idx)
  using fri-layer-step-evidenceD(4)[OF step] .
have chunk':
  fri-layer-opening-chunk (fri-evidence-layer-len roots layer-idx)
    yp yp-path yn yn-path (round-layers ! round-idx' ! layer-idx)
  using fri-layer-step-evidenceD(4)[OF step] .
have auth:
  fri-layer-chunk-authenticated (roots ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (round-layers ! round-idx ! layer-idx) final-state
  using recorded[OF round-bound layer-bound]
  unfolding generic-fri-recorded-layer-chunk-authenticated-def .
have auth':
  fri-layer-chunk-authenticated (roots ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx)
    (round-layers ! round-idx' ! layer-idx) final-state
  using recorded[OF round-bound' layer-bound]
  unfolding generic-fri-recorded-layer-chunk-authenticated-def .

```

```

from cases show ?thesis
proof
  assume c:
     $\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} =$ 
     $\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx} \wedge$ 
     $xp \neq yp$ 
  show ?thesis
  by (rule generic-fri-authenticated-same-layer-opening-conflictI-base-base
    [OF round-bound round-bound' layer-bound auth auth' chunk chunk'])
    (use c in simp-all)
next
  assume rest:
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} =$ 
     $\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx}) \wedge$ 
     $xp \neq yn) \vee$ 
     $(\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx}) =$ 
     $\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx} \wedge$ 
     $xn \neq yp) \vee$ 
     $(\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx}) =$ 
     $\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx}) \wedge$ 
     $xn \neq yn)$ 
  then show ?thesis
proof
  assume c:
     $\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx} =$ 
     $\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx}) \wedge$ 
     $xp \neq yn$ 
  show ?thesis
  by (rule generic-fri-authenticated-same-layer-opening-conflictI-base-sibling
    [OF round-bound round-bound' layer-bound auth auth' chunk chunk'])
    (use c in simp-all)
next
  assume rest':
     $(\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx}) =$ 
     $\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx} \wedge$ 
     $xn \neq yp) \vee$ 
     $(\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx layer-idx}) =$ 
     $\text{fri-sibling-index (fri-evidence-layer-len roots layer-idx)}$ 
     $(\text{fri-evidence-layer-idx roots query-idxs round-idx' layer-idx}) \wedge$ 
     $xn \neq yn)$ 
  then show ?thesis
proof

```

```

assume c:
  fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
    fri-evidence-layer-idx roots query-idxs round-idx' layer-idx ∧
    xn ≠ yp
show ?thesis
  by (rule generic-fri-authenticated-same-layer-opening-conflictI-sibling-base
    [OF round-bound round-bound' layer-bound auth auth' chunk chunk'])
    (use c in simp-all)
next
assume c:
  fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx) =
    fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx' layer-idx) ∧
    xn ≠ yn
show ?thesis
  by (rule generic-fri-authenticated-same-layer-opening-conflictI-sibling-sibling
    [OF round-bound round-bound' layer-bound auth auth' chunk chunk'])
    (use c in simp-all)
qed
qed
qed
qed

lemma accepted-fri-opening-transcript-trace-step-with-authenticated-chunk-at:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
and out-eq: out = Some (result, final-state)
and round-bound: i < rounds
and layer-bound: j < length trace-roots
and raw-layer:
  ∧ raw.
    0 < fri-layer-lengths (length trace-roots) (clength * scale) ! j ∧
    fri-layer-indices (length trace-roots) (index (to-nat raw))
      (clength * scale) ! j <
    fri-layer-lengths (length trace-roots) (clength * scale) ! j
shows
  generic-fri-transcript-step-with-authenticated-chunk trace-roots
  trace-bs query-idxs trace-round-layers final-state i j
proof —
have step:
  fri-round-layer-evidence trace-roots trace-bs query-idxs i j
  trace-round-layers
by (rule accepted-fri-opening-transcript-trace-round-layer-evidence
  [OF fri-openings round-bound layer-bound])
from accepted-fri-opening-transcript-trace-selected-chunk-authenticated-at

```



```

    [OF fri-openings out-eq round-bound layer-bound raw-layer]
obtain auth-chunk where auth:
  fri-layer-chunk-authenticated (trace-roots ! j)
  (fri-layer-lengths (length trace-roots) (clength * scale) ! j)
  (fri-layer-indices (length trace-roots) (query-idxs ! i)
    (clength * scale) ! j)
  auth-chunk final-state
by blast
have auth-compact:
  fri-layer-chunk-authenticated (trace-roots ! j)
  (fri-evidence-layer-len trace-roots j)
  (fri-evidence-layer-idx trace-roots query-idxs i j)
  auth-chunk final-state
using auth
unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
by simp
show ?thesis
by (rule generic-fri-transcript-step-with-authenticated-chunkI
  [OF step auth-compact])
qed

lemma accepted-fri-opening-transcript-composition-step-with-authenticated-chunk-at:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
and out-eq: out = Some (result, final-state)
and round-bound: i < rounds
and layer-bound: j < length composition-roots
and raw-layer:
   $\bigwedge$ raw.
    0 < fri-layer-lengths (length composition-roots) (clength * scale) ! j  $\wedge$ 
    fri-layer-indices (length composition-roots) (index (to-nat raw))
      (clength * scale) ! j <
    fri-layer-lengths (length composition-roots) (clength * scale) ! j
shows
  generic-fri-transcript-step-with-authenticated-chunk composition-roots
  composition-bs query-idxs composition-round-layers final-state i j
proof –
have step:
  fri-round-layer-evidence composition-roots composition-bs query-idxs i j
  composition-round-layers
by (rule accepted-fri-opening-transcript-composition-round-layer-evidence
  [OF fri-openings round-bound layer-bound])
from
  accepted-fri-opening-transcript-composition-selected-chunk-authenticated-at
  [OF fri-openings out-eq round-bound layer-bound raw-layer]
obtain auth-chunk where auth:
  fri-layer-chunk-authenticated (composition-roots ! j)

```

```

    (fri-layer-lengths (length composition-roots) (clength * scale) ! j)
    (fri-layer-indices (length composition-roots) (query-idxs ! i)
      (clength * scale) ! j)
    auth-chunk final-state
  by blast
have auth-compact:
  fri-layer-chunk-authenticated (composition-roots ! j)
  (fri-evidence-layer-len composition-roots j)
  (fri-evidence-layer-idx composition-roots query-idxs i j)
  auth-chunk final-state
using auth
unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
by simp
show ?thesis
  by (rule generic-fri-transcript-step-with-authenticated-chunkI
    [OF step auth-compact])
qed

```

end

end

```

theory Soundness-FRI-Layer-Replay
imports Soundness-FRI-Layer-Authenticated
begin

```

Narrow replay facts for selected FRI layer openings.

The first lemma is a generic selected-position decomposition for *mfold*. It avoids adding another proof by induction over the full FRI verifier layer in the already-large Merkle theory.

```

context soundness
begin

```

```

lemma fri-layer-lengths-prefix-nth:
  assumes  $j < n$ 
  shows  $\text{fri-layer-lengths } n \text{ len } ! j = \text{fri-layer-lengths } (\text{Suc } j) \text{ len } ! j$ 
  using assms
proof (induction  $j$  arbitrary:  $n \text{ len}$ )
  case 0
  then show ?case
    by (cases  $n$ ) simp-all
next
  case (Suc  $j$ )
  then obtain  $n'$  where  $n\text{-eq}: n = \text{Suc } n'$ 
    by (cases  $n$ ) auto
  have  $j\text{-bound}: j < n'$ 
    using Suc.prem unfolding  $n\text{-eq}$  by simp
  show ?case

```

```

    using Suc.IH[OF j-bound, of len div 2]
    unfolding n-eq by simp
qed

lemma fri-layer-indices-prefix-nth:
  assumes j < n
  shows
    fri-layer-indices n idx len ! j =
      fri-layer-indices (Suc j) idx len ! j
  using assms
proof (induction j arbitrary: n idx len)
  case 0
  then show ?case
    by (cases n) simp-all
next
  case (Suc j)
  then obtain n' where n-eq: n = Suc n'
    by (cases n) auto
  have j-bound: j < n'
    using Suc.premis unfolding n-eq by simp
  show ?case
    using Suc.IH[OF j-bound, of idx mod (len div 2) len div 2]
    unfolding n-eq by simp
qed

lemma fri-layer-lengths-prefix-same:
  assumes k-bound: k < j
  shows fri-layer-lengths (Suc j) len ! k = fri-layer-lengths j len ! k
  using fri-layer-lengths-prefix-nth[OF less-trans[OF k-bound lessI], of len]
    fri-layer-lengths-prefix-nth[OF k-bound, of len]
  by simp

lemma fri-layers-transcript-snoc:
  assumes prefix:
    fri-layers-transcript j len prefix-chunks prefix-chunk
  and last:
    fri-layer-opening-chunk (fri-layer-lengths (Suc j) len ! j)
    xp xp-path xn xn-path last-chunk
  shows
    fri-layers-transcript (Suc j) len (prefix-chunks @ [last-chunk])
    (prefix-chunk @ last-chunk)
  using prefix last
proof (induction j arbitrary: len prefix-chunks prefix-chunk)
  case 0
  then have chunks-eq: prefix-chunks = []
    and prefix-chunk-eq: prefix-chunk = []
    unfolding fri-layers-transcript-def by simp-all
  have last-fact:
    fri-layer-opening-chunk len xp xp-path xn xn-path last-chunk

```

```

    using 0.premis(2) by simp
have last-open:
  ∃ yp yp-path yn yn-path.
    fri-layer-opening-chunk len yp yp-path yn yn-path last-chunk
  apply (rule exI[where x=yp])
  apply (rule exI[where x=yp-path])
  apply (rule exI[where x=yn])
  apply (rule exI[where x=yn-path])
  apply (rule last-fact)
done
show ?case
  unfolding chunks-eq prefix-chunk-eq fri-layers-transcript-def
  using last-open by simp
next
case (Suc j)
obtain first rest where chunks-eq: prefix-chunks = first # rest
  using Suc.premis(1) unfolding fri-layers-transcript-def
  by (cases prefix-chunks) auto
have first-open:
  ∃ yp yp-path yn yn-path.
    fri-layer-opening-chunk len yp yp-path yn yn-path first
  using Suc.premis(1) unfolding fri-layers-transcript-def chunks-eq
  by auto
have rest-layers:
  fri-layers-transcript j (len div 2) rest (List.concat rest)
  using Suc.premis(1) unfolding fri-layers-transcript-def chunks-eq
  by auto
have last-tail:
  fri-layer-opening-chunk
    (fri-layer-lengths (Suc j) (len div 2) ! j)
    xp xp-path xn xn-path last-chunk
  using Suc.premis(2) by simp
have tail:
  fri-layers-transcript (Suc j) (len div 2) (rest @ [last-chunk])
    (List.concat rest @ last-chunk)
  by (rule Suc.IH[OF rest-layers last-tail])
have prefix-chunk-eq: prefix-chunk = first @ List.concat rest
  using Suc.premis(1) unfolding fri-layers-transcript-def chunks-eq by simp
show ?case
  unfolding chunks-eq prefix-chunk-eq fri-layers-transcript-def
  using first-open tail unfolding fri-layers-transcript-def by auto
qed

lemma fri-layers-transcript-last-chunk-eq:
  assumes layers:
    fri-layers-transcript (Suc j) len layer-chunks chunk
  and prefix:
    fri-layers-transcript j len prefix-chunks prefix-chunk
  and last:

```

```

    fri-layer-opening-chunk (fri-layer-lengths (Suc j) len ! j)
      xp xp-path xn xn-path last-chunk
    and chunk-eq: chunk = prefix-chunk @ last-chunk
  shows layer-chunks ! j = last-chunk
proof -
  have combined:
    fri-layers-transcript (Suc j) len (prefix-chunks @ [last-chunk])
      (prefix-chunk @ last-chunk)
  by (rule fri-layers-transcript-snoc[OF prefix last])
  have layers-eq: layer-chunks = prefix-chunks @ [last-chunk]
  by (rule fri-layers-transcript-unique
      [OF layers combined[unfolded chunk-eq[symmetric]]])
  have len-prefix: length prefix-chunks = j
  using prefix unfolding fri-layers-transcript-def by simp
  show ?thesis
  using layers-eq len-prefix by (simp add: nth-append)
qed

```

lemma *generic-fri-round-forced-next-value-from-step-evidence:*

assumes *step:*

```

  fri-layer-step-evidence
    (roots ! layer-idx)
    (challenges ! layer-idx)
    (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
  xp xp-path xn xn-path
  (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  v
  (round-layers ! round-idx ! layer-idx)

```

shows

```

  generic-fri-round-forced-next-value roots challenges query-idxs
    round-layers round-idx layer-idx v

```

using *step*

unfolding *generic-fri-round-forced-next-value-def* **by** *blast*

lemma *mfold-outcome-decomp-at:*

assumes *outcome:*

```

  Some (out, t) ∈ set-dist (execute (mfold a ms) s)
  and j-bound: j < length ms

```

shows

```

  ∃ a-j s-j out-j s-suc.
    Some (a-j, s-j) ∈ set-dist (execute (mfold a (take j ms)) s) ∧
    Some (out-j, s-suc) ∈ set-dist (execute ((ms ! j) a-j) s-j) ∧
    Some (out, t) ∈
      set-dist (execute (mfold out-j (drop (Suc j) ms)) s-suc)

```

using *outcome j-bound*

```

proof (induction j arbitrary: a ms s)
  case 0
  then obtain m ms' where ms-eq: ms = m # ms'
    by (cases ms) auto
  from 0.premis(1)[unfolded ms-eq]
  obtain out-j s-suc where
    head: Some (out-j, s-suc) ∈ set-dist (execute (m a) s)
    and tail:
      Some (out, t) ∈ set-dist (execute (mfold out-j ms') s-suc)
    by (auto elim!: set-dist-bindE)
  have prefix: Some (a, s) ∈ set-dist (execute (mfold a (take 0 ms)) s)
    by simp
  have selected-orig:
    Some (out-j, s-suc) ∈ set-dist (execute ((ms ! 0) a) s)
    using head unfolding ms-eq by simp
  have suffix-orig:
    Some (out, t) ∈
      set-dist (execute (mfold out-j (drop (Suc 0) ms)) s-suc)
    using tail unfolding ms-eq by simp
  show ?case
    by (intro exI[of - a] exI[of - s] exI[of - out-j]
      exI[of - s-suc] conjI)
      (use prefix selected-orig suffix-orig in simp-all)
  next
  case (Suc j)
  then obtain m ms' where ms-eq: ms = m # ms'
    by (cases ms) auto
  from Suc.premis(1)[unfolded ms-eq]
  obtain out1 s1 where
    head: Some (out1, s1) ∈ set-dist (execute (m a) s)
    and tail:
      Some (out, t) ∈ set-dist (execute (mfold out1 ms') s1)
    by (auto elim!: set-dist-bindE)
  have j-bound': j < length ms'
    using Suc.premis(2) unfolding ms-eq by simp
  from Suc.IH[OF tail j-bound']
  obtain a-j s-j out-j s-suc where
    prefix-tail:
      Some (a-j, s-j) ∈
        set-dist (execute (mfold out1 (take j ms')) s1)
    and selected:
      Some (out-j, s-suc) ∈
        set-dist (execute ((ms' ! j) a-j) s-j)
    and suffix:
      Some (out, t) ∈
        set-dist (execute (mfold out-j (drop (Suc j) ms')) s-suc)
    by blast
  have prefix-unfolded:
    Some (a-j, s-j) ∈

```

```

    set-dist (execute ((m a) >>= (λx. mfold x (take j ms'))) s)
  by (rule-tac x=out1 and t=s1 in set-dist-bindI)
    (use head prefix-tail in simp-all)
have prefix-mfold:
  Some (a-j, s-j) ∈
    set-dist (execute (mfold a (m # take j ms'))) s)
  using prefix-unfolded by simp
have prefix:
  Some (a-j, s-j) ∈
    set-dist (execute (mfold a (take (Suc j) ms))) s)
  using prefix-mfold unfolding ms-eq by simp
have selected-orig:
  Some (out-j, s-suc) ∈
    set-dist (execute ((ms ! Suc j) a-j) s-j)
  using selected unfolding ms-eq by simp
have suffix-orig:
  Some (out, t) ∈
    set-dist (execute (mfold out-j (drop (Suc (Suc j)) ms))) s-suc)
  using suffix unfolding ms-eq by simp
show ?case
  by (intro exI[of - a-j] exI[of - s-j] exI[of - out-j]
      exI[of - s-suc] conjI)
    (use prefix selected-orig suffix-orig in simp-all)
qed

```

lemma *mfold-fri-layer-openings-decomp-at:*
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes *outcome:*
 Some (out, t) ∈
 set-dist
 (execute
 (mfold (idx, x, len, pw) (map fri-layer-opening-step bfs)) s)
and *j-bound:* $j < \text{length bfs}$
obtains $\text{idx-j } x\text{-j } \text{len-j } \text{pw-j } \text{out-j } s\text{-j } s\text{-suc}$ **where**
 Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
 set-dist
 (execute
 (mfold (idx, x, len, pw)
 (take j (map fri-layer-opening-step bfs)))) s)
 Some (out-j, s-suc) ∈
 set-dist
 (execute
 (fri-layer-opening-step (bfs ! j)
 (idx-j, x-j, len-j, pw-j)) s-j)
 Some (out, t) ∈
 set-dist
 (execute
 (mfold out-j (drop (Suc j) (map fri-layer-opening-step bfs)))
 s-suc)

proof –
have *mapped-bound*: $j < \text{length } (\text{map } \text{fri-layer-opening-step } \text{bfs})$
using *j-bound* **by** *simp*
from *mfold-outcome-decomp-at*[*OF outcome mapped-bound*]
obtain *a-j s-j out-j s-suc* **where**
prefix:
Some (*a-j*, *s-j*) $\in$
set-dist
(execute
(mfold (idx, x, len, pw)
(take j (map fri-layer-opening-step bfs)))) s)
and *selected*:
Some (*out-j*, *s-suc*) $\in$
set-dist
(execute
((map fri-layer-opening-step bfs ! j) a-j) s-j)
and *suffix*:
Some (*out*, *t*) $\in$
set-dist
(execute
(mfold out-j (drop (Suc j) (map fri-layer-opening-step bfs))))
s-suc)
by *blast*
obtain *idx-j x-j len-j pw-j* **where** *a-j-eq*:
a-j = (idx-j, x-j, len-j, pw-j)
by (*cases a-j*)
have *selected'*:
Some (*out-j*, *s-suc*) $\in$
set-dist
(execute
(fri-layer-opening-step (bfs ! j)
(idx-j, x-j, len-j, pw-j))) s-j)
using *selected j-bound unfolding a-j-eq* **by** *simp*
show *?thesis*
by (*rule that*[*OF prefix[unfolded a-j-eq] selected' suffix*])
qed

lemma *mfold-fri-layer-openings-prefix-idx-len*:
fixes *s s-j* :: (*'f*, *'a*) *protocol-channel-scheme*
assumes *prefix*:
Some ((*idx-j*, *x-j*, *len-j*, *pw-j*), *s-j*) $\in$
set-dist
(execute
(mfold (idx, x, len, pw)
(take j (map fri-layer-opening-step bfs)))) s)
and *j-bound*: $j \leq \text{length } \text{bfs}$
shows
idx-j = fri-layer-indices (Suc j) idx len ! j $\wedge$
len-j = fri-layer-lengths (Suc j) len ! j


```

    using prefix j-bound
  proof (induction j arbitrary: idx x len pw bfs s idx-j x-j len-j pw-j s-j)
    case 0
    then show ?case
      by simp
  next
    case (Suc j)
    then obtain bf bfs' where bfs-eq: bfs = bf # bfs'
      by (cases bfs) auto
    obtain b rt where bf-eq: bf = (b, rt)
      by (cases bf)
    from Suc.premis(1)[unfolded bfs-eq]
    obtain out1 s1 where
      head:
        Some (out1, s1) ∈
          set-dist
            (execute (fri-layer-opening-step bf (idx, x, len, pw)) s)
      and tail:
        Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
          set-dist
            (execute
              (mfold out1 (take j (map fri-layer-opening-step bfs')) s1)
            )
      by (auto elim!: set-dist-bindE)
    from fri-layer-opening-step-outcome[OF head[unfolded bf-eq]]
    obtain x' xp xp-path xn xn-path where out1-eq:
      out1 = (idx mod (len div 2), x', len div 2, pw + pw)
      by blast
    have j-bound': j ≤ length bfs'
      using Suc.premis(2) unfolding bfs-eq by simp
    have tail-shape:
      idx-j =
        fri-layer-indices (Suc j) (idx mod (len div 2)) (len div 2) ! j ∧
      len-j = fri-layer-lengths (Suc j) (len div 2) ! j
      by (rule Suc.IH[OF tail[unfolded out1-eq] j-bound'])
    show ?case
      using tail-shape by simp
  qed

```

```

lemma mfold-fri-layer-openings-prefix-pow:
  fixes s s-j :: ('f, 'a) protocol-channel-scheme
  assumes prefix:
    Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
      set-dist
        (execute
          (mfold (idx, x, len, pw)
            (take j (map fri-layer-opening-step bfs))) s)
  and j-bound: j ≤ length bfs
  shows pw-j = pw * 2 ^ j
  using prefix j-bound

```

```

proof (induction j arbitrary: idx x len pw bfs s idx-j x-j len-j pw-j s-j)
  case 0
  then show ?case
    by simp
next
  case (Suc j)
  then obtain bf bfs' where bfs-eq: bfs = bf # bfs'
    by (cases bfs) auto
  obtain b rt where bf-eq: bf = (b, rt)
    by (cases bf)
  from Suc.premis(1)[unfolded bfs-eq]
  obtain out1 s1 where
    head:
      Some (out1, s1) ∈
        set-dist
          (execute (fri-layer-opening-step bf (idx, x, len, pw)) s)
    and tail:
      Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
        set-dist
          (execute
            (mfold out1 (take j (map fri-layer-opening-step bfs'))) s1)
    by (auto elim!: set-dist-bindE)
  from fri-layer-opening-step-outcome[OF head[unfolded bf-eq]]
  obtain x' xp xp-path xn xn-path where out1-eq:
    out1 = (idx mod (len div 2), x', len div 2, pw + pw)
  by blast
  have j-bound': j ≤ length bfs'
    using Suc.premis(2) unfolding bfs-eq by simp
  have tail-pow: pw-j = (pw + pw) * 2 ^ j
    by (rule Suc.IH[OF tail[unfolded out1-eq] j-bound'])
  then show ?case
    by simp
qed

```

```

lemma mfold-fri-layer-openings-prefix-pow-full:
  fixes s s-j :: ('f, 'a) protocol-channel-scheme
  assumes prefix:
    Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
      set-dist
        (execute
          (mfold (idx, x, len, pw)
            (take j (map fri-layer-opening-step bfs)))) s)
  and j-bound: j < length bfs
  shows pw-j = pw * 2 ^ j
  by (rule mfold-fri-layer-openings-prefix-pow[OF prefix])
    (use j-bound in simp)

```

```

lemma mfold-fri-layer-openings-prefix-idx-len-full:
  fixes s s-j :: ('f, 'a) protocol-channel-scheme

```

```

assumes prefix:
  Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
    set-dist
      (execute
        (mfold (idx, x, len, pw)
          (take j (map fri-layer-opening-step bfs)))) s)
and j-bound: j < length bfs
shows
  idx-j = fri-layer-indices (length bfs) idx len ! j ∧
  len-j = fri-layer-lengths (length bfs) len ! j
proof –
  have shape:
    idx-j = fri-layer-indices (Suc j) idx len ! j ∧
    len-j = fri-layer-lengths (Suc j) len ! j
  by (rule mfold-fri-layer-openings-prefix-idx-len[OF prefix])
    (use j-bound in simp)
  show ?thesis
  using shape fri-layer-indices-prefix-nth[OF j-bound, of idx len]
    fri-layer-lengths-prefix-nth[OF j-bound, of len]
  by simp
qed

```

lemma *fri-layer-opening-selected-step-authenticated-full*:

fixes *s s-j s-suc* :: ('f, 'a) protocol-channel-scheme

assumes *prefix*:

```

  Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
    set-dist
      (execute
        (mfold (idx, x, len, pw)
          (take j (map fri-layer-opening-step bfs)))) s)

```

and *selected*:

```

  Some (out-j, s-suc) ∈
    set-dist
      (execute
        (fri-layer-opening-step (bfs ! j)
          (idx-j, x-j, len-j, pw-j)) s-j)

```

and *j-bound*: *j* < length bfs

and *layer-len-pos*:

```

  0 < fri-layer-lengths (length bfs) len ! j

```

and *layer-idx-bound*:

```

  fri-layer-indices (length bfs) idx len ! j <
  fri-layer-lengths (length bfs) len ! j

```

obtains *chunk* **where**

```

  fri-layer-chunk-authenticated (snd (bfs ! j))
  (fri-layer-lengths (length bfs) len ! j)
  (fri-layer-indices (length bfs) idx len ! j)
  chunk s-suc

```

proof –

have *shape*:

```

    idx-j = fri-layer-indices (length bfs) idx len ! j ∧
    len-j = fri-layer-lengths (length bfs) len ! j
  by (rule mfold-fri-layer-openings-prefix-idx-len-full
      [OF prefix j-bound])
  obtain b rt where bf-eq: bfs ! j = (b, rt)
  by (cases bfs ! j)
  from fri-layer-opening-step-authenticated-chunk-pred
      [OF - - selected[unfolded bf-eq]]
  obtain chunk where auth:
    fri-layer-chunk-authenticated rt len-j idx-j chunk s-suc
  using layer-len-pos layer-idx-bound shape
  by blast
  show ?thesis
  by (rule that[of chunk])
      (use auth shape bf-eq in simp)
qed

lemma receive-query-commits-decomp-at:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (out, t) ∈
      set-dist
        (execute
          (mfold (idx, x, len, pw) (receive-query-commits bfs)) s)
  and j-bound: j < length bfs
  obtains idx-j x-j len-j pw-j out-j s-j s-suc where
    Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
      set-dist
        (execute
          (mfold (idx, x, len, pw)
            (take j (map fri-layer-opening-step bfs)))) s)
    Some (out-j, s-suc) ∈
      set-dist
        (execute
          (fri-layer-opening-step (bfs ! j)
            (idx-j, x-j, len-j, pw-j)) s-j)
    Some (out, t) ∈
      set-dist
        (execute
          (mfold out-j (drop (Suc j) (map fri-layer-opening-step bfs)))
            s-suc)
  proof -
  have map-eq: receive-query-commits bfs = map fri-layer-opening-step bfs
  unfolding receive-query-commits-def fri-layer-opening-step-def
  by (simp add: fun-eq-iff split: prod.splits)
  show ?thesis
  by (rule mfold-fri-layer-openings-decomp-at
      [OF outcome[unfolded map-eq] j-bound that])
qed

```

lemma *receive-query-commits-prefix-idx-len*:
fixes $s \ s\text{-}j :: ('f, 'a) \text{protocol-channel-scheme}$
assumes *prefix*:
Some $((idx\text{-}j, x\text{-}j, len\text{-}j, pw\text{-}j), s\text{-}j) \in$
set-dist
(execute
(mfold (idx, x, len, pw)
(take j (map fri-layer-opening-step bfs)))) s)
and *j-bound*: $j \leq \text{length } bfs$
shows
 $idx\text{-}j = \text{fri-layer-indices } (Suc\ j) \ idx \ len \ ! \ j \wedge$
 $len\text{-}j = \text{fri-layer-lengths } (Suc\ j) \ len \ ! \ j$
by (rule *mfold-fri-layer-openings-prefix-idx-len*[*OF prefix j-bound*])

lemma *receive-query-commits-prefix-pow*:
fixes $s \ s\text{-}j :: ('f, 'a) \text{protocol-channel-scheme}$
assumes *prefix*:
Some $((idx\text{-}j, x\text{-}j, len\text{-}j, pw\text{-}j), s\text{-}j) \in$
set-dist
(execute
(mfold (idx, x, len, pw)
(take j (map fri-layer-opening-step bfs)))) s)
and *j-bound*: $j \leq \text{length } bfs$
shows $pw\text{-}j = pw * 2^j$
by (rule *mfold-fri-layer-openings-prefix-pow*[*OF prefix j-bound*])

lemma *receive-query-commits-prefix-pow-full*:
fixes $s \ s\text{-}j :: ('f, 'a) \text{protocol-channel-scheme}$
assumes *prefix*:
Some $((idx\text{-}j, x\text{-}j, len\text{-}j, pw\text{-}j), s\text{-}j) \in$
set-dist
(execute
(mfold (idx, x, len, pw)
(take j (map fri-layer-opening-step bfs)))) s)
and *j-bound*: $j < \text{length } bfs$
shows $pw\text{-}j = pw * 2^j$
by (rule *mfold-fri-layer-openings-prefix-pow-full*[*OF prefix j-bound*])

lemma *receive-query-commits-prefix-idx-len-full*:
fixes $s \ s\text{-}j :: ('f, 'a) \text{protocol-channel-scheme}$
assumes *prefix*:
Some $((idx\text{-}j, x\text{-}j, len\text{-}j, pw\text{-}j), s\text{-}j) \in$
set-dist
(execute
(mfold (idx, x, len, pw)
(take j (map fri-layer-opening-step bfs)))) s)
and *j-bound*: $j < \text{length } bfs$
shows

$idx-j = \text{fri-layer-indices } (\text{length } bfs) \text{ } idx \text{ } len ! j \wedge$
 $len-j = \text{fri-layer-lengths } (\text{length } bfs) \text{ } len ! j$
by (rule mfold-fri-layer-openings-prefix-idx-len-full[OF prefix j-bound])

lemma *fri-layer-opening-step-outcome-evidence:*

fixes $s \ t :: ('f, 'a) \text{ protocol-channel-scheme}$

assumes *outcome:*

$\text{Some } ((\text{next-idx}, \text{next-value}, \text{next-len}, \text{next-pow}), t) \in$
 $\text{set-dist } (\text{execute } (\text{fri-layer-opening-step } (b, rt) \text{ } (idx, x, len, pw)) \text{ } s)$

obtains $xp \text{ } xp\text{-path} \text{ } xn \text{ } xn\text{-path}$ **where**

$xp = x$
 $\text{next-idx} = idx \text{ mod } (len \text{ div } 2)$
 $\text{next-len} = len \text{ div } 2$
 $\text{next-pow} = pw + pw$
 $\text{fri-layer-step-evidence } rt \text{ } b \text{ } len \text{ } idx \text{ } pw$
 $(\text{fri-sibling-index } len \text{ } idx) \text{ } xp \text{ } xp\text{-path} \text{ } xn \text{ } xn\text{-path}$
 $\text{next-idx next-value } ([xp] @ xp\text{-path} @ [xn] @ xn\text{-path})$

proof –

from *fri-layer-opening-step-outcome*[OF *outcome*]

obtain $xp \text{ } xp\text{-path} \text{ } xn \text{ } xn\text{-path} \text{ } x'$ **where** *out-eq:*

$(\text{next-idx}, \text{next-value}, \text{next-len}, \text{next-pow}) =$
 $(idx \text{ mod } (len \text{ div } 2), x', len \text{ div } 2, pw + pw)$

and *xp-eq:* $xp = x$

and *x'-eq:*

$x' = \text{fri-fold-value } b \text{ } xp \text{ } xn$
 $(\text{fri-fold-denominator } ((h \wedge idx) * \text{shift}) \text{ } pw)$

and *chunk:*

$\text{fri-layer-opening-chunk } len \text{ } xp \text{ } xp\text{-path} \text{ } xn \text{ } xn\text{-path}$
 $([xp] @ xp\text{-path} @ [xn] @ xn\text{-path})$

by *blast*

have *next-idx-eq:* $\text{next-idx} = idx \text{ mod } (len \text{ div } 2)$

using *out-eq* **by** *simp*

have *next-value-eq:* $\text{next-value} = x'$

using *out-eq* **by** *simp*

have *next-len-eq:* $\text{next-len} = len \text{ div } 2$

using *out-eq* **by** *simp*

have *next-pow-eq:* $\text{next-pow} = pw + pw$

using *out-eq* **by** *simp*

have *evidence:*

$\text{fri-layer-step-evidence } rt \text{ } b \text{ } len \text{ } idx \text{ } pw$
 $(\text{fri-sibling-index } len \text{ } idx) \text{ } xp \text{ } xp\text{-path} \text{ } xn \text{ } xn\text{-path}$
 $\text{next-idx next-value } ([xp] @ xp\text{-path} @ [xn] @ xn\text{-path})$

by (rule *fri-layer-step-evidenceI*)

(use *next-idx-eq next-value-eq x'-eq chunk* **in** *simp-all*)

show *?thesis*

by (rule *that*[OF *xp-eq next-idx-eq next-len-eq next-pow-eq evidence*])

qed

lemma *receive-query-commits-last-step-evidence:*

```

fixes  $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$ 
assumes  $\text{outcome:}$ 
   $\text{Some } ((\text{out-idx}, \text{out-value}, \text{out-len}, \text{out-pow}), t) \in$ 
     $\text{set-dist}$ 
     $(\text{execute}$ 
       $(\text{mfold } (\text{idx}, x, \text{len}, \text{pw}) (\text{receive-query-commits bfs}))\ s)$ 
    and  $\text{nonempty: } 0 < \text{length bfs}$ 
obtains  $j\ \text{idx-j}\ x\text{-j}\ \text{len-j}\ \text{pw-j}\ b\ \text{rt}\ \text{xp}\ \text{xp-path}\ \text{xn}\ \text{xn-path}\ s\text{-j}$ 
   $s\text{-suc}$  where
     $j = \text{length bfs} - 1$ 
     $j < \text{length bfs}$ 
     $\text{bfs} ! j = (b, \text{rt})$ 
     $\text{Some } ((\text{idx-j}, x\text{-j}, \text{len-j}, \text{pw-j}), s\text{-j}) \in$ 
       $\text{set-dist}$ 
       $(\text{execute}$ 
         $(\text{mfold } (\text{idx}, x, \text{len}, \text{pw})$ 
           $(\text{take } j\ (\text{map } \text{fri-layer-opening-step bfs})))\ s)$ 
       $\text{Some } ((\text{out-idx}, \text{out-value}, \text{out-len}, \text{out-pow}), s\text{-suc}) \in$ 
         $\text{set-dist}$ 
         $(\text{execute}$ 
           $(\text{fri-layer-opening-step } (b, \text{rt})$ 
             $(\text{idx-j}, x\text{-j}, \text{len-j}, \text{pw-j}))\ s\text{-j})$ 
         $t = s\text{-suc}$ 
         $\text{idx-j} = \text{fri-layer-indices } (\text{length bfs})\ \text{idx}\ \text{len} ! j$ 
         $\text{len-j} = \text{fri-layer-lengths } (\text{length bfs})\ \text{len} ! j$ 
         $\text{pw-j} = \text{pw} * 2^j$ 
         $\text{fri-layer-step-evidence } \text{rt}\ b\ \text{len-j}\ \text{idx-j}\ \text{pw-j}$ 
         $(\text{fri-sibling-index } \text{len-j}\ \text{idx-j})\ \text{xp}\ \text{xp-path}\ \text{xn}\ \text{xn-path}$ 
         $\text{out-idx}\ \text{out-value}\ ([\text{xp}]\ @\ \text{xp-path}\ @\ [\text{xn}]\ @\ \text{xn-path})$ 
proof  $-$ 
let  $?j = \text{length bfs} - 1$ 
have  $j\text{-bound: } ?j < \text{length bfs}$ 
using  $\text{nonempty by simp}$ 
show  $?thesis$ 
proof  $(\text{rule } \text{receive-query-commits-decomp-at}[\text{OF outcome } j\text{-bound}])$ 
fix  $\text{idx-j}\ x\text{-j}\ \text{len-j}\ \text{pw-j}\ \text{out-j}\ s\text{-j}\ s\text{-suc}$ 
assume  $\text{prefix:}$ 
   $\text{Some } ((\text{idx-j}, x\text{-j}, \text{len-j}, \text{pw-j}), s\text{-j}) \in$ 
     $\text{set-dist}$ 
     $(\text{execute}$ 
       $(\text{mfold } (\text{idx}, x, \text{len}, \text{pw})$ 
         $(\text{take } ?j\ (\text{map } \text{fri-layer-opening-step bfs})))\ s)$ 
    and  $\text{selected:}$ 
     $\text{Some } (\text{out-j}, s\text{-suc}) \in$ 
       $\text{set-dist}$ 
       $(\text{execute}$ 
         $(\text{fri-layer-opening-step } (\text{bfs} ! ?j)$ 
           $(\text{idx-j}, x\text{-j}, \text{len-j}, \text{pw-j}))\ s\text{-j})$ 
      and  $\text{suffix:}$ 

```

```

    Some ((out-idx, out-value, out-len, out-pow), t) ∈
      set-dist
        (execute
          (mfold out-j
            (drop (Suc ?j) (map fri-layer-opening-step bfs)))) s-suc)
  have drop-empty:
    drop (Suc ?j) (map fri-layer-opening-step bfs) = []
  using nonempty by simp
  have out-j-eq: out-j = (out-idx, out-value, out-len, out-pow)
  and t-eq: t = s-suc
  using suffix unfolding drop-empty by simp-all
  obtain b rt where bf-eq: bfs ! ?j = (b, rt)
  by (cases bfs ! ?j)
  have shape:
    idx-j = fri-layer-indices (length bfs) idx len ! ?j ∧
    len-j = fri-layer-lengths (length bfs) len ! ?j
  by (rule receive-query-commits-prefix-idx-len-full
    [OF prefix j-bound])
  have pow: pw-j = pw * 2 ^ ?j
  by (rule receive-query-commits-prefix-pow-full[OF prefix j-bound])
  show ?thesis
  proof (rule fri-layer-opening-step-outcome-evidence
    [OF selected[unfolded bf-eq out-j-eq]])
    fix xp xp-path xn xn-path
    assume
      xp = x-j
      out-idx = idx-j mod (len-j div 2)
      out-len = len-j div 2
      out-pow = pw-j + pw-j
    and step:
      fri-layer-step-evidence rt b len-j idx-j pw-j
      (fri-sibling-index len-j idx-j) xp xp-path xn xn-path
      out-idx out-value ([xp] @ xp-path @ [xn] @ xn-path)
    show ?thesis
    by (rule that[OF refl j-bound bf-eq prefix
      selected[unfolded bf-eq out-j-eq] t-eq conjunct1[OF shape]
      conjunct2[OF shape] pow step])
  qed
qed
qed

```

lemma *verifier-query-round-after-index-trace-fri-last-step-evidence:*
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes *outcome:*
Some $((\),\ t) \in$
set-dist
(execute
(verifier-query-round-after-index-program
fr f-fl f-final as fl final raw) s)

and *nonempty*: $0 < \text{length } f\text{-fl}$
and *raw-layer*:
 $\wedge \text{raw}'.$
 $0 < \text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) !$
 $(\text{length } f\text{-fl} - 1) \wedge$
 $\text{fri-layer-indices } (\text{length } f\text{-fl}) (\text{index } (\text{to-nat } \text{raw}'))$
 $(\text{clength} * \text{scale}) ! (\text{length } f\text{-fl} - 1) <$
 $\text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) !$
 $(\text{length } f\text{-fl} - 1)$
obtains *actual-chunk* *axp* *axp-path* *axn* *axn-path* **where**
fri-layer-chunk-authenticated
 $(\text{snd } (f\text{-fl} ! (\text{length } f\text{-fl} - 1)))$
 $(\text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) !$
 $(\text{length } f\text{-fl} - 1))$
 $(\text{fri-layer-indices } (\text{length } f\text{-fl}) (\text{index } (\text{to-nat } \text{raw})))$
 $(\text{clength} * \text{scale}) ! (\text{length } f\text{-fl} - 1))$
actual-chunk *t*
fri-layer-step-evidence
 $(\text{snd } (f\text{-fl} ! (\text{length } f\text{-fl} - 1)))$
 $(\text{fst } (f\text{-fl} ! (\text{length } f\text{-fl} - 1)))$
 $(\text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) !$
 $(\text{length } f\text{-fl} - 1))$
 $(\text{fri-layer-indices } (\text{length } f\text{-fl}) (\text{index } (\text{to-nat } \text{raw})))$
 $(\text{clength} * \text{scale}) ! (\text{length } f\text{-fl} - 1))$
 $(2 \wedge (\text{length } f\text{-fl} - 1))$
fri-sibling-index
 $(\text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) !$
 $(\text{length } f\text{-fl} - 1))$
 $(\text{fri-layer-indices } (\text{length } f\text{-fl}) (\text{index } (\text{to-nat } \text{raw})))$
 $(\text{clength} * \text{scale}) ! (\text{length } f\text{-fl} - 1)))$
axp *axp-path* *axn* *axn-path*
 $((\text{fri-layer-indices } (\text{length } f\text{-fl}) (\text{index } (\text{to-nat } \text{raw})))$
 $(\text{clength} * \text{scale}) ! (\text{length } f\text{-fl} - 1)) \bmod$
 $((\text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) !$
 $(\text{length } f\text{-fl} - 1)) \text{ div } 2))$
f-final *actual-chunk*
proof –
show *?thesis*
proof (*rule verifier-query-round-after-index-final-checks*[*OF* *outcome*])
fix *idx* *fv* *s1* *f-i* *f-len* *f-pow* *s2* *c-i* *c-len* *c-pow* *s4*
assume *idx-eq*: *idx* = *index* (*to-nat* *raw*)
and *trace-fri*:
Some ((*f-i*, *f-final*, *f-len*, *f-pow*), *s2*) ∈
set-dist
(execute
(mfold (*idx*, *hd* *fv*, *clength* * *scale*, 1)
(receive-query-commits *f-fl*)) *s1*)
and *comp-fri*:
Some ((*c-i*, *final*, *c-len*, *c-pow*), *s4*) ∈

```

    set-dist
      (execute
        (mfold (idx, cp-eval as fv (h ^ idx * shift),
          clength * scale, 1)
          (receive-query-commits fl)) s2)
    and t-eq: t = s4
  let ?j = length f-fl - 1
  let ?len = fri-layer-lengths (length f-fl) (clength * scale) ! ?j
  let ?idx =
    fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! ?j
  have len-pos: 0 < ?len
    using raw-layer[of raw] by simp
  have idx-bound: ?idx < ?len
    using raw-layer[of raw] by simp
  show ?thesis
  proof (rule receive-query-commits-last-step-evidence
    [OF trace-fri[unfolded idx-eq] nonempty])
    fix j idx-j x-j len-j pw-j b rt xp xp-path xn xn-path s-j s-suc
    assume j-eq: j = ?j
    and bf-eq: f-fl ! j = (b, rt)
    and selected:
      Some ((f-i, f-final, f-len, f-pow), s-suc) ∈
        set-dist
          (execute
            (fri-layer-opening-step (b, rt)
              (idx-j, x-j, len-j, pw-j)) s-j)
    and s2-eq: s2 = s-suc
    and idx-j-eq:
      idx-j =
        fri-layer-indices (length f-fl) (index (to-nat raw))
          (clength * scale) ! j
    and len-j-eq:
      len-j =
        fri-layer-lengths (length f-fl) (clength * scale) ! j
    and pw-j-eq: pw-j = 1 * 2 ^ j
  have len-pos-j: 0 < len-j
    using len-pos len-j-eq j-eq by simp
  have idx-bound-j: idx-j < len-j
    using idx-bound idx-j-eq len-j-eq j-eq by simp
  show ?thesis
  proof (rule fri-layer-opening-step-outcome-authenticated
    [OF len-pos-j idx-bound-j selected])
    fix axp axp-path axn axn-path next-value actual-chunk
    assume selected-out:
      (f-i, f-final, f-len, f-pow) =
        (idx-j mod (len-j div 2), next-value, len-j div 2,
          pw-j + pw-j)
    and next-value-eq:

```

```

    next-value =
      fri-fold-value b axp axn
      (fri-fold-denominator ((h ^ idx-j) * shift) pw-j)
  and chunk:
    fri-layer-opening-chunk len-j axp axp-path axn axn-path
    actual-chunk
  and axp-auth:
    authenticated-opening-in s-suc
    (opening-root = rt, opening-length = len-j,
     opening-index = idx-j, opening-value = axp,
     opening-path = axp-path)
  and axn-auth:
    authenticated-opening-in s-suc
    (opening-root = rt, opening-length = len-j,
     opening-index = fri-sibling-index len-j idx-j,
     opening-value = axn, opening-path = axn-path)
  have f-final-eq: f-final = next-value
  using selected-out by simp
  have auth-s2:
    fri-layer-chunk-authenticated rt len-j idx-j actual-chunk s2
  unfolding s2-eq
  by (rule fri-layer-chunk-authenticatedI[OF chunk axp-auth axn-auth])
  have s2-t: s2 ≤ t
  using receive-query-commits-mfold-hash-extends[OF comp-fri] t-eq
  by simp
  have auth-t:
    fri-layer-chunk-authenticated rt len-j idx-j actual-chunk t
  by (rule fri-layer-chunk-authenticated-mono[OF auth-s2 s2-t])
  have step:
    fri-layer-step-evidence rt b len-j idx-j pw-j
    (fri-sibling-index len-j idx-j) axp axp-path axn axn-path
    (idx-j mod (len-j div 2)) f-final actual-chunk
  by (rule fri-layer-step-evidenceI)
  (use next-value-eq f-final-eq chunk in simp-all)
  have step':
    fri-layer-step-evidence
    (snd (f-fl ! ?j)) (fst (f-fl ! ?j)) ?len ?idx (2 ^ ?j)
    (fri-sibling-index ?len ?idx) axp axp-path axn axn-path
    (?idx mod (?len div 2)) f-final actual-chunk
  using step bf-eq idx-j-eq len-j-eq pw-j-eq j-eq by simp
  show ?thesis
  by (rule that[OF - step'])
  (use auth-t bf-eq idx-j-eq len-j-eq j-eq in simp)
qed
qed
qed
qed

```

lemma *verifier-query-round-after-index-composition-fri-last-step-evidence:*

```

fixes  $s\ t :: ('f, 'a) \text{ protocol-channel-scheme}$ 
assumes  $\text{outcome}$ :
   $\text{Some } ((), t) \in$ 
     $\text{set-dist}$ 
     $(\text{execute}$ 
       $(\text{verifier-query-round-after-index-program}$ 
         $\text{fr } f\text{-fl } f\text{-final as fl final raw})\ s)$ 
and  $\text{nonempty}$ :  $0 < \text{length fl}$ 
and  $\text{raw-layer}$ :
   $\bigwedge \text{raw'}$ .
     $0 < \text{fri-layer-lengths } (\text{length fl})\ (\text{clength} * \text{scale}) !$ 
     $(\text{length fl} - 1) \wedge$ 
     $\text{fri-layer-indices } (\text{length fl})\ (\text{index } (\text{to-nat raw}'))$ 
     $(\text{clength} * \text{scale}) ! (\text{length fl} - 1) <$ 
     $\text{fri-layer-lengths } (\text{length fl})\ (\text{clength} * \text{scale}) !$ 
     $(\text{length fl} - 1)$ 
obtains  $\text{actual-chunk axp axp-path axn axn-path}$  where
   $\text{fri-layer-chunk-authenticated}$ 
     $(\text{snd } (\text{fl} ! (\text{length fl} - 1)))$ 
     $(\text{fri-layer-lengths } (\text{length fl})\ (\text{clength} * \text{scale}) !$ 
     $(\text{length fl} - 1))$ 
     $(\text{fri-layer-indices } (\text{length fl})\ (\text{index } (\text{to-nat raw})))$ 
     $(\text{clength} * \text{scale}) ! (\text{length fl} - 1))$ 
     $\text{actual-chunk } t$ 
   $\text{fri-layer-step-evidence}$ 
     $(\text{snd } (\text{fl} ! (\text{length fl} - 1)))$ 
     $(\text{fst } (\text{fl} ! (\text{length fl} - 1)))$ 
     $(\text{fri-layer-lengths } (\text{length fl})\ (\text{clength} * \text{scale}) !$ 
     $(\text{length fl} - 1))$ 
     $(\text{fri-layer-indices } (\text{length fl})\ (\text{index } (\text{to-nat raw})))$ 
     $(\text{clength} * \text{scale}) ! (\text{length fl} - 1))$ 
     $(2 \wedge (\text{length fl} - 1))$ 
     $(\text{fri-sibling-index}$ 
       $(\text{fri-layer-lengths } (\text{length fl})\ (\text{clength} * \text{scale}) !$ 
       $(\text{length fl} - 1))$ 
       $(\text{fri-layer-indices } (\text{length fl})\ (\text{index } (\text{to-nat raw})))$ 
       $(\text{clength} * \text{scale}) ! (\text{length fl} - 1)))$ 
     $\text{axp axp-path axn axn-path}$ 
     $((\text{fri-layer-indices } (\text{length fl})\ (\text{index } (\text{to-nat raw})))$ 
     $(\text{clength} * \text{scale}) ! (\text{length fl} - 1)) \text{ mod}$ 
     $((\text{fri-layer-lengths } (\text{length fl})\ (\text{clength} * \text{scale}) !$ 
     $(\text{length fl} - 1)) \text{ div } 2))$ 
     $\text{final actual-chunk}$ 
proof –
  show  $?thesis$ 
proof  $(\text{rule verifier-query-round-after-index-final-checks}[OF\ \text{outcome}])$ 
  fix  $\text{idx fv s1 f-i f-len f-pow s2 c-i c-len c-pow s4}$ 
  assume  $\text{idx-eq}$ :  $\text{idx} = \text{index } (\text{to-nat raw})$ 
  and  $\text{comp-fri}$ :

```

```

Some ((c-i, final, c-len, c-pow), s4) ∈
  set-dist
  (execute
    (mfold (idx, cp-eval as fv (h ^ idx * shift),
      clength * scale, 1)
      (receive-query-commits fl)) s2)
and t-eq: t = s4
let ?j = length fl - 1
let ?len = fri-layer-lengths (length fl) (clength * scale) ! ?j
let ?idx =
  fri-layer-indices (length fl) (index (to-nat raw))
  (clength * scale) ! ?j
have len-pos: 0 < ?len
  using raw-layer[of raw] by simp
have idx-bound: ?idx < ?len
  using raw-layer[of raw] by simp
show ?thesis
proof (rule receive-query-commits-last-step-evidence
  [OF comp-fri[unfolded idx-eq] nonempty])
fix j idx-j x-j len-j pw-j b rt xp xp-path xn xn-path s-j s-suc
assume j-eq: j = ?j
  and bf-eq: fl ! j = (b, rt)
  and selected:
    Some ((c-i, final, c-len, c-pow), s-suc) ∈
      set-dist
      (execute
        (fri-layer-opening-step (b, rt)
          (idx-j, x-j, len-j, pw-j)) s-j)
  and s4-eq: s4 = s-suc
  and idx-j-eq:
    idx-j =
      fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j
  and len-j-eq:
    len-j =
      fri-layer-lengths (length fl) (clength * scale) ! j
  and pw-j-eq: pw-j = 1 * 2 ^ j
have len-pos-j: 0 < len-j
  using len-pos len-j-eq j-eq by simp
have idx-bound-j: idx-j < len-j
  using idx-bound idx-j-eq len-j-eq j-eq by simp
show ?thesis
proof (rule fri-layer-opening-step-outcome-authenticated
  [OF len-pos-j idx-bound-j selected])
fix axp axp-path axn axn-path next-value actual-chunk
assume selected-out:
  (c-i, final, c-len, c-pow) =
    (idx-j mod (len-j div 2), next-value, len-j div 2,
      pw-j + pw-j)

```

```

and next-value-eq:
  next-value =
    fri-fold-value b axp axn
      (fri-fold-denominator ((h  $\wedge$  idx-j) * shift) pw-j)
and chunk:
  fri-layer-opening-chunk len-j axp axp-path axn axn-path
    actual-chunk
and axp-auth:
  authenticated-opening-in s-suc
    (opening-root = rt, opening-length = len-j,
     opening-index = idx-j, opening-value = axp,
     opening-path = axp-path)
and axn-auth:
  authenticated-opening-in s-suc
    (opening-root = rt, opening-length = len-j,
     opening-index = fri-sibling-index len-j idx-j,
     opening-value = axn, opening-path = axn-path)
have final-eq: final = next-value
  using selected-out by simp
have auth-t:
  fri-layer-chunk-authenticated rt len-j idx-j actual-chunk t
  unfolding t-eq s4-eq
  by (rule fri-layer-chunk-authenticatedI[OF chunk axp-auth axn-auth])
have step:
  fri-layer-step-evidence rt b len-j idx-j pw-j
    (fri-sibling-index len-j idx-j) axp axp-path axn axn-path
    (idx-j mod (len-j div 2)) final actual-chunk
  by (rule fri-layer-step-evidenceI)
    (use next-value-eq final-eq chunk in simp-all)
have step':
  fri-layer-step-evidence
    (snd (fl ! ?j)) (fst (fl ! ?j)) ?len ?idx (2  $\wedge$  ?j)
    (fri-sibling-index ?len ?idx) axp axp-path axn axn-path
    (?idx mod (?len div 2)) final actual-chunk
  using step bf-eq idx-j-eq len-j-eq pw-j-eq j-eq by simp
show ?thesis
  by (rule that[OF - step'])
    (use auth-t bf-eq idx-j-eq len-j-eq j-eq in simp)
qed
qed
qed
qed

lemma ntimes-verifier-query-rounds-selected-trace-fri-last-step-with-prefix-at:
  fixes query-state final-state :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (results, final-state)  $\in$ 
      set-dist
      (execute

```

```

      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
      query-state)
and i-bound: i < rounds
and nonempty: 0 < length f-fl
and raw-layer:
   $\bigwedge$ raw.
    0 < fri-layer-lengths (length f-fl) (clength * scale) !
    (length f-fl - 1)  $\wedge$ 
    fri-layer-indices (length f-fl) (index (to-nat raw))
    (clength * scale) ! (length f-fl - 1) <
    fri-layer-lengths (length f-fl) (clength * scale) !
    (length f-fl - 1)
obtains raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
  suffix-chunks actual-chunk axp axp-path axn axn-path where
  idx = index (to-nat raw)
  length prefix-query-idxs = i
  length prefix-chunks = i
  length suffix-query-idxs = rounds - Suc i
  length suffix-chunks = rounds - Suc i
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
   $\bigwedge$ k. k < i  $\implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! k)
    (map snd f-fl) (map snd fl) (prefix-chunks ! k)
   $\bigwedge$ k. k < rounds - Suc i  $\implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)
    (map snd f-fl) (map snd fl) (suffix-chunks ! k)
  PTranscript query-state =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @
    PTranscript final-state
  fri-layer-chunk-authenticated
    (snd (f-fl ! (length f-fl - 1)))
    (fri-layer-lengths (length f-fl) (clength * scale) !
      (length f-fl - 1))
    (fri-layer-indices (length f-fl) idx (clength * scale) !
      (length f-fl - 1))
    actual-chunk final-state
  fri-layer-step-evidence
    (snd (f-fl ! (length f-fl - 1)))
    (fst (f-fl ! (length f-fl - 1)))
    (fri-layer-lengths (length f-fl) (clength * scale) !
      (length f-fl - 1))
    (fri-layer-indices (length f-fl) idx (clength * scale) !
      (length f-fl - 1))
    (2  $\wedge$  (length f-fl - 1))
    (fri-sibling-index
      (fri-layer-lengths (length f-fl) (clength * scale) !
        (length f-fl - 1))

```

```

    (fri-layer-indices (length f-fl) idx (clength * scale) !
      (length f-fl - 1)))
  axp axp-path axn axn-path
  ((fri-layer-indices (length f-fl) idx (clength * scale) !
    (length f-fl - 1)) mod
    ((fri-layer-lengths (length f-fl) (clength * scale) !
      (length f-fl - 1)) div 2))
  f-final actual-chunk
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) prefix-chunks i)) =
    Some raw
proof -
  let ?round = verifier-query-round-program fr f-fl f-final as fl final
  from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
  obtain prefix x suffix s-i s-suc where
    prefix:
      Some (prefix, s-i) ∈
        set-dist (execute (ntimes ?round i) query-state)
  and round:
      Some (x, s-suc) ∈ set-dist (execute ?round s-i)
  and suffix:
      Some (suffix, final-state) ∈
        set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
  by blast
  have x-eq: x = ()
  by (cases x) simp
  show ?thesis
  proof (rule query-opening-ntimes-prefix-transcript-state-alignment
    [OF prefix])
    fix prefix-query-idxs prefix-chunks
    assume len-prefix-idxs: length prefix-query-idxs = i
    and len-prefix-chunks: length prefix-chunks = i
    and transcript-prefix:
      PTranscript query-state =
        List.concat prefix-chunks @ PTranscript s-i
    and state-i:
      PState s-i =
        state-after-query-chunks (PState query-state) prefix-chunks i
    and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
    and prefix-rounds:
       $\bigwedge k. k < i \implies$ 
        verifier-query-round-chunk (prefix-query-idxs ! k)
          (map snd f-fl) (map snd fl) (prefix-chunks ! k)
    from round[unfolded x-eq verifier-query-round-program-alt-def]
    obtain raw s0 where
      raw-out:
        Some (raw, s0) ∈
          set-dist (execute receive-query-index-challenge s-i)

```



```

and after:
  Some ((), s-suc) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program fr f-fl f-final as
          fl final raw) s0)
by (auto elim!: set-dist-bindE)
have lookup-s0:
  fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
  Some raw
using receive-query-index-challenge-outcome[OF raw-out] by simp
show ?thesis
proof (rule verifier-query-round-program-outcome[OF round[unfolded x-eq]])
  fix raw' idx chunk
  assume idx-eq: idx = index (to-nat raw')
  and chunk-shape:
    verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
    chunk
  and transcript-round:
    PTranscript s-i = chunk @ PTranscript s-suc
  and lookup-suc:
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw'
  have s0-suc: s0 ≤ s-suc
    using verifier-query-round-after-index-program-outcome[OF after]
    by simp
  have lookup-suc-raw:
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
    by (rule hash-extension-lookup[OF lookup-s0 s0-suc])
  have raw-eq': raw = raw'
    using lookup-suc lookup-suc-raw by simp
  from verifier-query-round-after-index-trace-fri-last-step-evidence
    [OF after nonempty raw-layer]
  obtain actual-chunk axp axp-path arn arn-path where
    auth-suc:
      fri-layer-chunk-authenticated
        (snd (f-fl ! (length f-fl - 1)))
        (fri-layer-lengths (length f-fl) (clength * scale) !
          (length f-fl - 1))
        (fri-layer-indices (length f-fl) (index (to-nat raw))
          (clength * scale) ! (length f-fl - 1))
      actual-chunk s-suc
  and step-raw:
    fri-layer-step-evidence
      (snd (f-fl ! (length f-fl - 1)))

```

```

(fst (f-fl ! (length f-fl - 1)))
(fri-layer-lengths (length f-fl) (clength * scale) !
  (length f-fl - 1))
(fri-layer-indices (length f-fl) (index (to-nat raw))
  (clength * scale) ! (length f-fl - 1))
(2 ^ (length f-fl - 1))
(fri-sibling-index
  (fri-layer-lengths (length f-fl) (clength * scale) !
    (length f-fl - 1))
  (fri-layer-indices (length f-fl) (index (to-nat raw))
    (clength * scale) ! (length f-fl - 1)))
axp axp-path axn axn-path
((fri-layer-indices (length f-fl) (index (to-nat raw))
  (clength * scale) ! (length f-fl - 1)) mod
  ((fri-layer-lengths (length f-fl) (clength * scale) !
    (length f-fl - 1)) div 2))
f-final actual-chunk
by blast
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-idxs:
  length suffix-query-idxs = rounds - Suc i
and len-suffix-chunks:
  length suffix-chunks = rounds - Suc i
and transcript-suffix:
  PTranscript s-suc =
    List.concat suffix-chunks @ PTranscript final-state
and suffix-ext: s-suc ≤ final-state
and suffix-rounds:
  ∧k. k < rounds - Suc i ⇒
    verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)
have transcript-all:
  PTranscript query-state =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @
      PTranscript final-state
using transcript-prefix transcript-round transcript-suffix by simp
have auth-final-raw:
  fri-layer-chunk-authenticated
    (snd (f-fl ! (length f-fl - 1)))
    (fri-layer-lengths (length f-fl) (clength * scale) !
      (length f-fl - 1))
    (fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) !
      (length f-fl - 1))
    actual-chunk final-state
by (rule fri-layer-chunk-authenticated-mono[OF auth-suc suffix-ext])

```

```

have auth-final:
  fri-layer-chunk-authenticated
    (snd (f-fl ! (length f-fl - 1)))
    (fri-layer-lengths (length f-fl) (clength * scale) !
      (length f-fl - 1))
    (fri-layer-indices (length f-fl) idx (clength * scale) !
      (length f-fl - 1))
  actual-chunk final-state
using auth-final-raw raw-eq' idx-eq by simp
have step-idx:
  fri-layer-step-evidence
    (snd (f-fl ! (length f-fl - 1)))
    (fst (f-fl ! (length f-fl - 1)))
    (fri-layer-lengths (length f-fl) (clength * scale) !
      (length f-fl - 1))
    (fri-layer-indices (length f-fl) idx (clength * scale) !
      (length f-fl - 1))
    (2 ^ (length f-fl - 1))
    (fri-sibling-index
      (fri-layer-lengths (length f-fl) (clength * scale) !
        (length f-fl - 1))
      (fri-layer-indices (length f-fl) idx (clength * scale) !
        (length f-fl - 1)))
    axp axp-path axn axn-path
    ((fri-layer-indices (length f-fl) idx (clength * scale) !
      (length f-fl - 1)) mod
      ((fri-layer-lengths (length f-fl) (clength * scale) !
        (length f-fl - 1)) div 2))
    f-final actual-chunk
using step-raw raw-eq' idx-eq by simp
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state)
        prefix-chunks i)) =
    Some raw
proof -
have lookup-suc':
  fmlookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state)
        prefix-chunks i)) =
    Some raw
using lookup-suc-raw state-i counter-i by simp
show ?thesis
by (rule hash-extension-lookup[OF lookup-suc' suffix-ext])
qed
show ?thesis
by (rule that[OF - len-prefix-idxs len-prefix-chunks

```

```

    len-suffix-idxs len-suffix-chunks chunk-shape prefix-rounds
    suffix-rounds transcript-all auth-final step-idx
    lookup-final])
  (use idx-eq raw-eq' in simp)
qed
qed
qed
qed

lemma ntimes-verifier-query-rounds-selected-composition-fri-last-step-with-prefix-at:
fixes query-state final-state :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (results, final-state) ∈
    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
      query-state)
and i-bound: i < rounds
and nonempty: 0 < length fl
and raw-layer:
   $\bigwedge_{\text{raw.}}$ 
    0 < fri-layer-lengths (length fl) (clength * scale) !
    (length fl - 1)  $\wedge$ 
    fri-layer-indices (length fl) (index (to-nat raw))
    (clength * scale) ! (length fl - 1) <
    fri-layer-lengths (length fl) (clength * scale) !
    (length fl - 1)
obtains raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
  suffix-chunks actual-chunk axp axp-path axn axn-path where
  idx = index (to-nat raw)
  length prefix-query-idxs = i
  length prefix-chunks = i
  length suffix-query-idxs = rounds - Suc i
  length suffix-chunks = rounds - Suc i
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
   $\bigwedge k. k < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! k)
    (map snd f-fl) (map snd fl) (prefix-chunks ! k)
   $\bigwedge k. k < \text{rounds} - \text{Suc } i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)
    (map snd f-fl) (map snd fl) (suffix-chunks ! k)
  PTranscript query-state =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @
    PTranscript final-state
  fri-layer-chunk-authenticated
    (snd (fl ! (length fl - 1)))
    (fri-layer-lengths (length fl) (clength * scale) !

```

```

    (length fl - 1))
  (fri-layer-indices (length fl) idx (clength * scale) !
    (length fl - 1))
  actual-chunk final-state
fri-layer-step-evidence
  (snd (fl ! (length fl - 1)))
  (fst (fl ! (length fl - 1)))
  (fri-layer-lengths (length fl) (clength * scale) !
    (length fl - 1))
  (fri-layer-indices (length fl) idx (clength * scale) !
    (length fl - 1))
  (2 ^ (length fl - 1))
  (fri-sibling-index
    (fri-layer-lengths (length fl) (clength * scale) !
      (length fl - 1))
    (fri-layer-indices (length fl) idx (clength * scale) !
      (length fl - 1)))
  arp arp-path arn arn-path
  ((fri-layer-indices (length fl) idx (clength * scale) !
    (length fl - 1)) mod
    ((fri-layer-lengths (length fl) (clength * scale) !
      (length fl - 1)) div 2))
  final actual-chunk
fmlookup (HashMap final-state)
  (QueryIndexChallenge (PQueryCounter query-state + i)
    (state-after-query-chunks (PState query-state) prefix-chunks i)) =
  Some raw
proof -
let ?round = verifier-query-round-program fr f-fl f-final as fl final
from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
obtain prefix x suffix s-i s-suc where
  prefix:
    Some (prefix, s-i) ∈
      set-dist (execute (ntimes ?round i) query-state)
and round:
    Some (x, s-suc) ∈ set-dist (execute ?round s-i)
and suffix:
    Some (suffix, final-state) ∈
      set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
by blast
have x-eq: x = ()
by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
  fix prefix-query-idxs prefix-chunks
  assume len-prefix-idxs: length prefix-query-idxs = i
  and len-prefix-chunks: length prefix-chunks = i
  and transcript-prefix:

```

```

    PTranscript query-state =
      List.concat prefix-chunks @ PTranscript s-i
  and state-i:
    PState s-i =
      state-after-query-chunks (PState query-state) prefix-chunks i
  and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
  and prefix-rounds:
     $\bigwedge k. k < i \implies$ 
      verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
  from round[unfolded x-eq verifier-query-round-program-alt-def]
  obtain raw s0 where
    raw-out:
      Some (raw, s0)  $\in$ 
        set-dist (execute receive-query-index-challenge s-i)
  and after:
    Some ((), s-suc)  $\in$ 
      set-dist
      (execute
        (verifier-query-round-after-index-program fr f-fl f-final as
          fl final raw) s0)
  by (auto elim!: set-dist-bindE)
  have lookup-s0:
    fmlookup (HashMap s0)
      (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
      Some raw
  using receive-query-index-challenge-outcome[OF raw-out] by simp
  show ?thesis
  proof (rule verifier-query-round-program-outcome[OF round[unfolded x-eq]])
    fix raw' idx chunk
    assume idx-eq: idx = index (to-nat raw')
    and chunk-shape:
      verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
      chunk
    and transcript-round:
      PTranscript s-i = chunk @ PTranscript s-suc
    and lookup-suc:
      fmlookup (HashMap s-suc)
        (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
        Some raw'
    have s0-suc: s0  $\leq$  s-suc
    using verifier-query-round-after-index-program-outcome[OF after]
    by simp
    have lookup-suc-raw:
      fmlookup (HashMap s-suc)
        (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
        Some raw
    by (rule hash-extension-lookup[OF lookup-s0 s0-suc])
    have raw-eq': raw = raw'
  end

```

```

using lookup-suc lookup-suc-raw by simp
from verifier-query-round-after-index-composition-fri-last-step-evidence
  [OF after_nonempty raw-layer]
obtain actual-chunk axp axp-path axn axn-path where
  auth-suc:
    fri-layer-chunk-authenticated
      (snd (fl ! (length fl - 1)))
      (fri-layer-lengths (length fl) (clength * scale) !
        (length fl - 1))
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! (length fl - 1))
    actual-chunk s-suc
and step-raw:
  fri-layer-step-evidence
    (snd (fl ! (length fl - 1)))
    (fst (fl ! (length fl - 1)))
    (fri-layer-lengths (length fl) (clength * scale) !
      (length fl - 1))
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! (length fl - 1))
    (2 ^ (length fl - 1))
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) !
        (length fl - 1))
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! (length fl - 1)))
    axp axp-path axn axn-path
    ((fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! (length fl - 1)) mod
      ((fri-layer-lengths (length fl) (clength * scale) !
        (length fl - 1)) div 2))
    final actual-chunk
by blast
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-idxs:
  length suffix-query-idxs = rounds - Suc i
and len-suffix-chunks:
  length suffix-chunks = rounds - Suc i
and transcript-suffix:
  PTranscript s-suc =
    List.concat suffix-chunks @ PTranscript final-state
and suffix-ext: s-suc ≤ final-state
and suffix-rounds:
   $\bigwedge k. k < rounds - Suc i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)

```

```

have transcript-all:
  PTranscript query-state =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @
      PTranscript final-state
  using transcript-prefix transcript-round transcript-suffix by simp
have auth-final-raw:
  fri-layer-chunk-authenticated
    (snd (fl ! (length fl - 1)))
    (fri-layer-lengths (length fl) (clength * scale) !
      (length fl - 1))
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) !
      (length fl - 1))
    actual-chunk final-state
  by (rule fri-layer-chunk-authenticated-mono[OF auth-suc suffix-ext])
have auth-final:
  fri-layer-chunk-authenticated
    (snd (fl ! (length fl - 1)))
    (fri-layer-lengths (length fl) (clength * scale) !
      (length fl - 1))
    (fri-layer-indices (length fl) idx (clength * scale) !
      (length fl - 1))
    actual-chunk final-state
  using auth-final-raw raw-eq' idx-eq by simp
have step-idx:
  fri-layer-step-evidence
    (snd (fl ! (length fl - 1)))
    (fst (fl ! (length fl - 1)))
    (fri-layer-lengths (length fl) (clength * scale) !
      (length fl - 1))
    (fri-layer-indices (length fl) idx (clength * scale) !
      (length fl - 1))
    (2 ^ (length fl - 1))
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) !
        (length fl - 1))
      (fri-layer-indices (length fl) idx (clength * scale) !
        (length fl - 1)))
    arp arp-path arn arn-path
    ((fri-layer-indices (length fl) idx (clength * scale) !
      (length fl - 1)) mod
      ((fri-layer-lengths (length fl) (clength * scale) !
        (length fl - 1)) div 2))
    final actual-chunk
  using step-raw raw-eq' idx-eq by simp
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state)

```



```

    prefix-chunks i)) =
  Some raw
proof -
  have lookup-suc':
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
    Some raw
  using lookup-suc-raw state-i counter-i by simp
  show ?thesis
    by (rule hash-extension-lookup[OF lookup-suc' suffix-ext])
qed
show ?thesis
  by (rule that[OF - len-prefix-idxs len-prefix-chunks
    len-suffix-idxs len-suffix-chunks chunk-shape prefix-rounds
    suffix-rounds transcript-all auth-final step-idx
    lookup-final])
    (use idx-eq raw-eq' in simp)
qed
qed
qed
qed

```

lemma generic-fri-forced-value-eq-from-authenticated-step:

assumes forced:

generic-fri-round-forced-next-value roots challenges query-idxs
 round-layers round-idx layer-idx v

and recorded-auth:

generic-fri-recorded-layer-chunk-authenticated roots query-idxs
 round-layers final-state round-idx layer-idx

and actual-auth:

fri-layer-chunk-authenticated (roots ! layer-idx)
 (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
 actual-chunk final-state

and actual-step:

fri-layer-step-evidence
 (roots ! layer-idx)
 (challenges ! layer-idx)
 (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
 (2 ^ layer-idx)
 (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
 axp axp-path axn axn-path
 (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
 actual-value actual-chunk

and no-merkle:

$\neg$ partial-merkle-inconsistency-bad s (Some (result, final-state))
shows $v = \text{actual-value}$
proof –
from forced obtain $\text{rxp rxp-path rxn rxn-path}$ **where** *recorded-step*:
fri-layer-step-evidence
 (roots ! layer-idx)
 (challenges ! layer-idx)
 (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
 ($2 \wedge$ layer-idx)
 (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
 $\text{rxp rxp-path rxn rxn-path}$
 (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
 v
 (round-layers ! round-idx ! layer-idx)
unfolding generic-fri-round-forced-next-value-def **by** blast
have *recorded-chunk*:
fri-layer-chunk-authenticated (roots ! layer-idx)
 (fri-evidence-layer-len roots layer-idx)
 (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
 (round-layers ! round-idx ! layer-idx) *final-state*
using *recorded-auth*
unfolding generic-fri-recorded-layer-chunk-authenticated-def .
have $\text{rxp-eq: rxp} = \text{axp}$
by (rule fri-layer-authenticated-base-values-eq-if-no-partial-merkle
 [OF no-merkle recorded-chunk actual-auth])
 (use fri-layer-step-evidenceD(4)[OF recorded-step]
 fri-layer-step-evidenceD(4)[OF actual-step] **in** simp-all)
have $\text{rxn-eq: rxn} = \text{axn}$
by (rule fri-layer-authenticated-sibling-values-eq-if-no-partial-merkle
 [OF no-merkle recorded-chunk actual-auth])
 (use fri-layer-step-evidenceD(4)[OF recorded-step]
 fri-layer-step-evidenceD(4)[OF actual-step] **in** simp-all)
show ?thesis
using fri-layer-step-evidenceD(3)[OF recorded-step]
 fri-layer-step-evidenceD(3)[OF actual-step]
 rxp-eq rxn-eq
by simp
qed

lemma generic-fri-final-conflict-false-from-authenticated-step:
assumes forced:
 generic-fri-round-forced-next-value roots challenges query-idxs
 round-layers round-idx layer-idx v
and recorded-auth:
 generic-fri-recorded-layer-chunk-authenticated roots query-idxs
 round-layers final-state round-idx layer-idx
and actual-auth:

```

fri-layer-chunk-authenticated (roots ! layer-idx)
  (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
  actual-chunk final-state
and actual-step:
fri-layer-step-evidence
  (roots ! layer-idx)
  (challenges ! layer-idx)
  (fri-evidence-layer-len roots layer-idx)
  (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
  ( $2 \wedge \text{layer-idx}$ )
  (fri-sibling-index (fri-evidence-layer-len roots layer-idx)
    (fri-evidence-layer-idx roots query-idxs round-idx layer-idx))
  axp axp-path axn axn-path
  (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
  final-value actual-chunk
and no-merkle:
   $\neg$  partial-merkle-inconsistency-bad s (Some (result, final-state))
and conflict: v  $\neq$  final-value
shows False
using generic-fri-forced-value-eq-from-authenticated-step
  [OF forced recorded-auth actual-auth actual-step no-merkle]
  conflict
by simp

lemma generic-fri-selected-final-step-missing-false-from-authenticated-step:
  fixes root challenge final-value :: 'f'
  and final-state :: ('f, 'a) protocol-channel-scheme
assumes actual-auth:
  fri-layer-chunk-authenticated root len idx actual-chunk final-state
and actual-step:
  fri-layer-step-evidence root challenge len idx pw sibling-idx
  axp axp-path axn axn-path next-idx final-value actual-chunk
and missing:
   $\wedge$  actual-chunk axp axp-path axn axn-path.
   $\neg$  (fri-layer-chunk-authenticated root len idx actual-chunk final-state  $\wedge$ 
    fri-layer-step-evidence root challenge len idx pw sibling-idx
    axp axp-path axn axn-path next-idx final-value actual-chunk)
shows False
using actual-auth actual-step missing by blast

end

end

theory Soundness-FRI-Authenticated-Route
  imports Soundness-FRI-Layer-Replay
begin

```

Route-level FRI events that keep the existing public reduction shape but refine the sampled same-layer branch to carry authenticated Merkle evidence.

context *soundness*

begin

lemma *fri-first-layer-raw-bound*:

assumes *roots-nonempty*: $0 < \text{length roots}$

shows

$0 < \text{fri-layer-lengths } (\text{length roots}) \ (\text{clength} * \text{scale}) ! 0 \wedge$
 $\text{fri-layer-indices } (\text{length roots}) \ (\text{index } (\text{to-nat raw}))$
 $(\text{clength} * \text{scale}) ! 0 <$
 $\text{fri-layer-lengths } (\text{length roots}) \ (\text{clength} * \text{scale}) ! 0$

proof –

have *idx-bound*: $\text{index } (\text{to-nat raw}) < \text{clength} * \text{scale}$

proof –

have $\text{index } (\text{to-nat raw}) < \text{query-sample-space-size}$

by (*rule index-less-query-sample-space*)

also have $\dots \leq \text{clength} * \text{scale}$

by (*rule query-sample-space-size-le-domain*)

finally show *?thesis* .

qed

show *?thesis*

using *roots-nonempty idx-bound clength-pos scale-pos*

by (*cases roots*) *simp-all*

qed

lemma *fri-layer-raw-bound*:

assumes *layer-bound*: $j < \text{length roots}$

and *eval-power*: $\text{clength} * \text{scale} = 2^N$

and *roots-rounds*: $\text{length roots} \leq N$

shows

$0 < \text{fri-layer-lengths } (\text{length roots}) \ (\text{clength} * \text{scale}) ! j \wedge$
 $\text{fri-layer-indices } (\text{length roots}) \ (\text{index } (\text{to-nat raw}))$
 $(\text{clength} * \text{scale}) ! j <$
 $\text{fri-layer-lengths } (\text{length roots}) \ (\text{clength} * \text{scale}) ! j$

proof –

have *idx-bound*: $\text{index } (\text{to-nat raw}) < \text{clength} * \text{scale}$

proof –

have $\text{index } (\text{to-nat raw}) < \text{query-sample-space-size}$

by (*rule index-less-query-sample-space*)

also have $\dots \leq \text{clength} * \text{scale}$

by (*rule query-sample-space-size-le-domain*)

finally show *?thesis* .

qed

have *idx-layer*:

$\text{fri-layer-indices } (\text{length roots}) \ (\text{index } (\text{to-nat raw}))$

$(\text{clength} * \text{scale}) ! j <$

$\text{fri-layer-lengths } (\text{length roots}) \ (\text{clength} * \text{scale}) ! j$

by (*rule fri-layer-indices-nth-bound*)

```

      [OF idx-bound layer-bound eval-power roots-rounds])
have len-pos:
  0 < fri-layer-lengths (length roots) (clength * scale) ! j
  using idx-layer by linarith
show ?thesis
  using len-pos idx-layer by simp
qed

definition trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
    s out  $\longleftrightarrow$ 
    ( $\exists$  trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final fri-query-idxs trace-round-layers
      composition-round-layers fr as rest trace-openings trace-table
      result final-state.
      accepted-fri-opening-transcript s out trace-roots trace-bs
        trace-final dg composition-roots composition-bs composition-final
        fri-query-idxs trace-round-layers composition-round-layers  $\wedge$ 
      out = Some (result, final-state)  $\wedge$ 
      verifier-header-transcript s fr trace-roots trace-final as dg
        composition-roots composition-final rest  $\wedge$ 
      accepted-with-partial-trace-openings s out fr fri-query-idxs
        trace-openings  $\wedge$ 
      partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
       $\neg$  trace-table-low-degree trace-table  $\wedge$ 
      generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
        trace-table trace-roots trace-bs trace-final fri-query-idxs
        trace-round-layers final-state)

definition trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
    s out  $\longleftrightarrow$ 
    ( $\exists$  trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final fri-query-idxs trace-round-layers
      composition-round-layers fr as rest trace-openings trace-table.
      accepted-fri-opening-transcript s out trace-roots trace-bs
        trace-final dg composition-roots composition-bs composition-final
        fri-query-idxs trace-round-layers composition-round-layers  $\wedge$ 
      verifier-header-transcript s fr trace-roots trace-final as dg
        composition-roots composition-final rest  $\wedge$ 
      accepted-with-partial-trace-openings s out fr fri-query-idxs
        trace-openings  $\wedge$ 
      partial-trace-table-candidate trace-table trace-openings  $\wedge$ 

```

$\neg$ trace-table-low-degree trace-table $\wedge$
 generic-fri-sampled-assignment-conflict-without-same-layer
 trace-table trace-roots trace-bs trace-final fri-query-idxs
 trace-round-layers)

definition trace-fri-header-tied-sampled-base-opening-conflict

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-sampled-base-opening-conflict s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{ accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final dg composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{ verifier-header-transcript s fr trace-roots trace-final as dg}$
 $\text{ composition-roots composition-final rest} \wedge$
 $\text{ accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{ trace-table-low-degree trace-table} \wedge$
 $\text{ generic-fri-sampled-base-opening-conflict trace-table trace-roots}$
 $\text{ trace-bs fri-query-idxs trace-round-layers})$

definition trace-fri-header-tied-sampled-next-value-conflict

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-sampled-next-value-conflict s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{ accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final dg composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{ verifier-header-transcript s fr trace-roots trace-final as dg}$
 $\text{ composition-roots composition-final rest} \wedge$
 $\text{ accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{ trace-table-low-degree trace-table} \wedge$
 $\text{ generic-fri-sampled-next-value-conflict trace-roots trace-bs}$
 $\text{ fri-query-idxs trace-round-layers})$

definition trace-fri-header-tied-sampled-successor-opening-conflict

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-sampled-successor-opening-conflict } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\quad \text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\quad \text{composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\quad \text{trace-final dg composition-roots composition-bs composition-final}$
 $\quad \text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as dg}$
 $\quad \text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\quad \text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-sampled-successor-opening-conflict trace-roots trace-bs}$
 $\quad \text{fri-query-idxs trace-round-layers})$

definition *trace-fri-header-tied-sampled-final-value-conflict*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-sampled-final-value-conflict } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\quad \text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\quad \text{composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\quad \text{trace-final dg composition-roots composition-bs composition-final}$
 $\quad \text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as dg}$
 $\quad \text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\quad \text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-sampled-final-value-conflict trace-roots trace-bs}$
 $\quad \text{trace-final fri-query-idxs trace-round-layers})$

lemma *trace-fri-header-tied-without-same-imp-branch:*

assumes

$\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer}$
 $s \text{ out}$

shows

$\text{trace-fri-header-tied-sampled-base-opening-conflict } s \text{ out} \vee$
 $\text{trace-fri-header-tied-sampled-next-value-conflict } s \text{ out} \vee$
 $\text{trace-fri-header-tied-sampled-successor-opening-conflict } s \text{ out} \vee$
 $\text{trace-fri-header-tied-sampled-final-value-conflict } s \text{ out}$

proof —

from *assms* **obtain** $\text{trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table}$

where *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *header*:
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
and *partial*:
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *cand*: *partial-trace-table-candidate trace-table trace-openings*
and *not-low*: $\neg$ *trace-table-low-degree trace-table*
and *without*:
generic-fri-sampled-assignment-conflict-without-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers
unfolding
trace-fri-header-tied-sampled-assignment-conflict-without-same-layer-def
by *blast*
have *split*:
generic-fri-sampled-base-opening-conflict trace-table trace-roots
trace-bs fri-query-idxs trace-round-layers $\vee$
generic-fri-sampled-next-value-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers $\vee$
generic-fri-sampled-successor-opening-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers $\vee$
generic-fri-sampled-final-value-conflict trace-roots trace-bs
trace-final fri-query-idxs trace-round-layers
by (rule *generic-fri-sampled-assignment-conflict-without-same-layer-split*
[*OF without*])
then show *?thesis*
proof
assume *base*:
generic-fri-sampled-base-opening-conflict trace-table trace-roots
trace-bs fri-query-idxs trace-round-layers
then show *?thesis*
unfolding *trace-fri-header-tied-sampled-base-opening-conflict-def*
by (*intro disjI1 exI conjI*)
(rule *fri-openings*, rule *header*, rule *partial*, rule *cand*,
rule *not-low*, rule *base*)
next
assume *rest-split*:
generic-fri-sampled-next-value-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers $\vee$
generic-fri-sampled-successor-opening-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers $\vee$
generic-fri-sampled-final-value-conflict trace-roots trace-bs
trace-final fri-query-idxs trace-round-layers
then show *?thesis*

proof
assume *next-conflict*:
generic-fri-sampled-next-value-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers
then show *?thesis*
unfolding *trace-fri-header-tied-sampled-next-value-conflict-def*
by (*intro disjI2 disjI1 exI conjI*)
(*rule fri-openings, rule header, rule partial, rule cand,*
rule not-low, rule next-conflict)
next
assume *rest-split'*:
generic-fri-sampled-successor-opening-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers $\vee$
generic-fri-sampled-final-value-conflict trace-roots trace-bs
trace-final fri-query-idxs trace-round-layers
then show *?thesis*
proof
assume *successor*:
generic-fri-sampled-successor-opening-conflict trace-roots
trace-bs fri-query-idxs trace-round-layers
then show *?thesis*
unfolding
trace-fri-header-tied-sampled-successor-opening-conflict-def
by (*intro disjI2 disjI2 disjI1 exI conjI*)
(*rule fri-openings, rule header, rule partial, rule cand,*
rule not-low, rule successor)
next
assume *final-conflict*:
generic-fri-sampled-final-value-conflict trace-roots trace-bs
trace-final fri-query-idxs trace-round-layers
then show *?thesis*
unfolding *trace-fri-header-tied-sampled-final-value-conflict-def*
by (*intro disjI2 disjI2 disjI2 exI conjI*)
(*rule fri-openings, rule header, rule partial, rule cand,*
rule not-low, rule final-conflict)
qed
qed
qed
qed

lemma *wp-trace-fri-header-tied-without-same-bound-from-branches*:
assumes *base-bound*:
wp-event verify-monad
(*trace-fri-header-tied-sampled-base-opening-conflict s*) $s \leq B$
and *next-bound*:
wp-event verify-monad
(*trace-fri-header-tied-sampled-next-value-conflict s*) $s \leq N$
and *successor-bound*:
wp-event verify-monad

(trace-fri-header-tied-sampled-successor-opening-conflict s) $s \leq S$
and final-bound:
 wp-event verify-monad
 (trace-fri-header-tied-sampled-final-value-conflict s) $s \leq F$
shows
 wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict-without-same-layer s)
 $s \leq B + N + S + F$
proof –
have wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict-without-same-layer s)
 $s \leq$
 wp-event verify-monad
 ($\lambda out.$
 trace-fri-header-tied-sampled-base-opening-conflict s out $\vee$
 trace-fri-header-tied-sampled-next-value-conflict s out $\vee$
 trace-fri-header-tied-sampled-successor-opening-conflict s out $\vee$
 trace-fri-header-tied-sampled-final-value-conflict s out) s
by (rule wp-event-mono)
 (rule trace-fri-header-tied-without-same-imp-branch)
also have ... $\leq$
 wp-event verify-monad
 (trace-fri-header-tied-sampled-base-opening-conflict s) s +
 wp-event verify-monad
 (trace-fri-header-tied-sampled-next-value-conflict s) s +
 wp-event verify-monad
 (trace-fri-header-tied-sampled-successor-opening-conflict s) s +
 wp-event verify-monad
 (trace-fri-header-tied-sampled-final-value-conflict s) s
by (rule wp-event-union-bound₄)
also have ... $\leq B + N + S + F$
by (intro add-mono base-bound next-bound successor-bound final-bound)
finally show ?thesis .
qed

lemma trace-fri-header-tied-without-sameI-base:
assumes trace-fri-header-tied-sampled-base-opening-conflict s out
shows
 trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
 s out
using assms
unfolding trace-fri-header-tied-sampled-base-opening-conflict-def
 trace-fri-header-tied-sampled-assignment-conflict-without-same-layer-def
by (blast intro:
 generic-fri-sampled-assignment-conflict-without-same-layerI-base)

lemma trace-fri-header-tied-without-sameI-next:
assumes trace-fri-header-tied-sampled-next-value-conflict s out
shows

```

    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
      s out
using assms
unfolding trace-fri-header-tied-sampled-next-value-conflict-def
    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer-def
by (blast intro:
    generic-fri-sampled-assignment-conflict-without-same-layerI-next)

lemma trace-fri-header-tied-without-sameI-successor:
assumes trace-fri-header-tied-sampled-successor-opening-conflict s out
shows
    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
      s out
using assms
unfolding trace-fri-header-tied-sampled-successor-opening-conflict-def
    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer-def
by (blast intro:
    generic-fri-sampled-assignment-conflict-without-same-layerI-successor)

lemma trace-fri-header-tied-without-sameI-final:
assumes trace-fri-header-tied-sampled-final-value-conflict s out
shows
    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
      s out
using assms
unfolding trace-fri-header-tied-sampled-final-value-conflict-def
    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer-def
by (blast intro:
    generic-fri-sampled-assignment-conflict-without-same-layerI-final)

lemma trace-fri-header-tied-without-same-iff-branches:
    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer s out
     $\longleftrightarrow$ 
    trace-fri-header-tied-sampled-base-opening-conflict s out  $\vee$ 
    trace-fri-header-tied-sampled-next-value-conflict s out  $\vee$ 
    trace-fri-header-tied-sampled-successor-opening-conflict s out  $\vee$ 
    trace-fri-header-tied-sampled-final-value-conflict s out
using trace-fri-header-tied-without-same-imp-branch
    trace-fri-header-tied-without-sameI-base
    trace-fri-header-tied-without-sameI-next
    trace-fri-header-tied-without-sameI-successor
    trace-fri-header-tied-without-sameI-final
by blast

lemma trace-fri-header-tied-base-conflict-obtains-authenticated-first-chunk:
assumes base:
    trace-fri-header-tied-sampled-base-opening-conflict
      s (Some (result, final-state))
obtains trace-roots trace-bs trace-final dg composition-roots

```

composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr as rest trace-openings trace-table
 round-idx xp xp-path xn xn-path auth-chunk

where

accepted-fri-opening-transcript s (Some (result, final-state))
 trace-roots trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers
 verifier-header-transcript s fr trace-roots trace-final as dg
 composition-roots composition-final rest
 accepted-with-partial-trace-openings s
 (Some (result, final-state)) fr fri-query-idxs trace-openings
 partial-trace-table-candidate trace-table trace-openings
 $\neg$ trace-table-low-degree trace-table
 round-idx < length fri-query-idxs
 0 < length trace-bs
 fri-layer-step-evidence
 (trace-roots ! 0)
 (trace-bs ! 0)
 (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
 1
 (fri-sibling-index (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
 xp xp-path xn xn-path
 (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
 (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
 round-idx 0 xp xn)
 (trace-round-layers ! round-idx ! 0)
 $\neg$ fri-opening-matches-table
 (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
 trace-table xp xn
 fri-layer-chunk-authenticated (trace-roots ! 0)
 (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
 auth-chunk final-state

proof —

from base obtain trace-roots trace-bs trace-final dg composition-roots
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr as rest trace-openings trace-table

where fri-openings:

accepted-fri-opening-transcript s (Some (result, final-state))
 trace-roots trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers

and header:

verifier-header-transcript s fr trace-roots trace-final as dg
 composition-roots composition-final rest

```

and partial:
  accepted-with-partial-trace-openings s
  (Some (result, final-state)) fr fri-query-idxs trace-openings
and cand: partial-trace-table-candidate trace-table trace-openings
and not-low:  $\neg$  trace-table-low-degree trace-table
and generic:
  generic-fri-sampled-base-opening-conflict trace-table trace-roots
  trace-bs fri-query-idxs trace-round-layers
unfolding trace-fri-header-tied-sampled-base-opening-conflict-def
by blast
from generic obtain round-idx xp xp-path xn xn-path where
  trace-bs-nonempty:  $0 < \text{length } \text{trace-bs}$ 
and round-bound: round-idx  $< \text{length } \text{fri-query-idxs}$ 
and step:
  fri-layer-step-evidence
  (trace-roots ! 0)
  (trace-bs ! 0)
  (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
  1
  (fri-sibling-index (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
  xp xp-path xn xn-path
  (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
  (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
    round-idx 0 xp xn)
  (trace-round-layers ! round-idx ! 0)
and no-match:
   $\neg$  fri-opening-matches-table
  (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
  trace-table xp xn
unfolding generic-fri-sampled-base-opening-conflict-def
by blast
have roots-nonempty:  $0 < \text{length } \text{trace-roots}$ 
using accepted-fri-opening-transcript-shapes(2)[OF fri-openings]
  trace-bs-nonempty by simp
have round-bound-rounds: round-idx  $< \text{rounds}$ 
using round-bound accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
by simp
obtain auth-chunk where auth:
  fri-layer-chunk-authenticated (trace-roots ! 0)
  (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
  auth-chunk final-state
proof –
have layer-bound:  $0 < \text{length } \text{trace-roots}$ 
by (rule roots-nonempty)
have raw-layer:

```

```

 $\wedge$ raw.
  0 < fri-layer-lengths (length trace-roots) (clength * scale) ! 0  $\wedge$ 
    fri-layer-indices (length trace-roots) (index (to-nat raw))
      (clength * scale) ! 0 <
    fri-layer-lengths (length trace-roots) (clength * scale) ! 0
  by (rule fri-first-layer-raw-bound[OF roots-nonempty])
from accepted-fri-opening-transcript-trace-step-with-authenticated-chunk-at
  [OF fri-openings refl round-bound-rounds layer-bound raw-layer]
obtain auth-chunk where
  generic-fri-transcript-step-with-authenticated-chunk trace-roots
    trace-bs fri-query-idxs trace-round-layers final-state round-idx 0
  by blast
then obtain auth-chunk where
  fri-layer-chunk-authenticated (trace-roots ! 0)
    (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    auth-chunk final-state
  by (rule generic-fri-transcript-step-with-authenticated-chunkE)
then show ?thesis
  by (rule that)
qed
show ?thesis
  by (rule that[OF fri-openings header partial cand not-low round-bound
    trace-bs-nonempty step no-match auth])
qed

lemma trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer-imp-header-tied:
  assumes
    trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
      s out
  shows trace-fri-header-tied-sampled-assignment-conflict s out
proof -
  from assms obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
    result final-state
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
      trace-final dg composition-roots composition-bs composition-final
      fri-query-idxs trace-round-layers composition-round-layers
  and out-eq: out = Some (result, final-state)
  and header:
    verifier-header-transcript s fr trace-roots trace-final as dg
      composition-roots composition-final rest
  and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
      trace-openings
  and cand: partial-trace-table-candidate trace-table trace-openings
  and not-low:  $\neg$  trace-table-low-degree trace-table

```

and *conflict*:
generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers final-state
unfolding
trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
by *blast*
have *sampled*:
generic-fri-sampled-assignment-conflict trace-table trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers
by (*rule*
generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-imp-sampled
[OF conflict])
show *?thesis*
unfolding *trace-fri-header-tied-sampled-assignment-conflict-def*
by (*intro exI conjI*)
(rule fri-openings, rule header, rule partial, rule cand,
rule not-low, rule sampled)
qed

lemma *wp-trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer-bound-from-header-tied*:
assumes *header-bound*:
wp-event verify-monad
(trace-fri-header-tied-sampled-assignment-conflict s) s ≤ C
shows
wp-event verify-monad
(trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer s)
s ≤ C
proof –
have *wp-event verify-monad*
(trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer s)
s ≤
wp-event verify-monad
(trace-fri-header-tied-sampled-assignment-conflict s) s
by (*rule wp-event-mono*)
(rule
trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer-imp-header-tied)
then show *?thesis*
by (*rule order-trans[OF - header-bound]*)
qed

definition *trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks
s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$

composition-round-layers fr as rest trace-openings trace-table
result final-state.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers $\wedge$
out = Some (result, final-state) $\wedge$
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ *trace-table-low-degree trace-table $\wedge$*
generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
trace-roots trace-bs fri-query-idxs trace-round-layers final-state)

lemma *trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks-imp-partial-merkle:*

assumes

trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks
s out

shows *partial-merkle-inconsistency-bad s out*

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table result
final-state

where *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers

and *out-eq: out = Some (result, final-state)*

and *conflict:*

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
trace-roots trace-bs fri-query-idxs trace-round-layers final-state

unfolding

trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks-def

by *blast*

have *len: length trace-bs = length trace-roots*

using *accepted-fri-opening-transcript-shapes(2)[OF fri-openings]* .

have *partial-merkle-inconsistency-bad s (Some (result, final-state))*

by *(rule*

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-partial-merkle
[OF conflict len])

then show *?thesis*

by *(simp add: out-eq)*

qed

lemma *wp-trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks-bound-from-partial-merkle:*

assumes *merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s $\leq$ M

shows
wp-event verify-monad
(trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks s)
 $s \leq M$
proof –
have *wp-event verify-monad*
(trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks s)
 $s \leq \text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \text{ } s$
by (rule *wp-event-mono*)
(rule
trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks-imp-partial-merkle)
then show *?thesis*
by (rule *order-trans[OF - merkle-bound]*)
qed

lemma *trace-fri-header-tied-authenticated-conflict-imp-without-same-or-same-layer:*
assumes
trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
 $s \text{ out}$
shows
trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
 $s \text{ out} \vee$
trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks s out
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table result
final-state
where *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *out-eq:* $\text{out} = \text{Some } (\text{result}, \text{final-state})$
and *header:*
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *cand:* *partial-trace-table-candidate trace-table trace-openings*
and *not-low:* $\neg \text{trace-table-low-degree trace-table}$
and *conflict:*
generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers final-state
unfolding
trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
by *blast*
have *split:*

```

generic-fri-sampled-assignment-conflict-without-same-layer
  trace-table trace-roots trace-bs trace-final fri-query-idxs
  trace-round-layers  $\vee$ 
generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
  trace-roots trace-bs fri-query-idxs trace-round-layers final-state
by (rule
  generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-split
  [OF conflict])
then show ?thesis
proof
  assume without:
    generic-fri-sampled-assignment-conflict-without-same-layer
    trace-table trace-roots trace-bs trace-final fri-query-idxs
    trace-round-layers
  then show ?thesis
  unfolding
    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer-def
  by (intro disjI1 exI conjI)
    (rule fri-openings, rule header, rule partial, rule cand,
     rule not-low, rule without)
next
  assume same:
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    trace-roots trace-bs fri-query-idxs trace-round-layers final-state
  then show ?thesis
  unfolding
    trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks-def
  by (intro disjI2 exI conjI)
    (rule fri-openings, rule out-eq, rule header, rule partial,
     rule cand, rule not-low, rule same)
qed
qed

lemma trace-fri-header-tied-authenticated-conflictI-without-same:
  assumes
    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
    s out
  shows
    trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
    s out
proof -
  from assms obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and header:

```

```

    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
and cand: partial-trace-table-candidate trace-table trace-openings
and not-low:  $\neg$  trace-table-low-degree trace-table
and without:
    generic-fri-sampled-assignment-conflict-without-same-layer
    trace-table trace-roots trace-bs trace-final fri-query-idxs
    trace-round-layers
unfolding
    trace-fri-header-tied-sampled-assignment-conflict-without-same-layer-def
by blast
from fri-openings obtain result final-state where out-eq:
    out = Some (result, final-state)
    unfolding accepted-fri-opening-transcript-def by blast
have conflict:
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
    trace-table trace-roots trace-bs trace-final fri-query-idxs
    trace-round-layers final-state
by (rule
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layerI-without
    [OF without])
show ?thesis
    unfolding
    trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
by (intro exI conjI)
    (rule fri-openings, rule out-eq, rule header, rule partial, rule cand,
    rule not-low, rule conflict)
qed

lemma trace-fri-header-tied-authenticated-conflictI-same-layer:
assumes
    trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks s out
shows
    trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
    s out
proof –
from assms obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table result
    final-state
where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
and out-eq: out = Some (result, final-state)
and header:

```

verifier-header-transcript s fr trace-roots trace-final as dg
 composition-roots composition-final rest
and partial:
 accepted-with-partial-trace-openings s out fr fri-query-idxs
 trace-openings
and cand: partial-trace-table-candidate trace-table trace-openings
and not-low: $\neg$ trace-table-low-degree trace-table
and same:
 generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
 trace-roots trace-bs fri-query-idxs trace-round-layers final-state
unfolding
 trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks-def
by blast
have conflict:
 generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
 trace-table trace-roots trace-bs trace-final fri-query-idxs
 trace-round-layers final-state
by (rule
 generic-fri-sampled-assignment-conflict-with-authenticated-same-layerI-same
 [OF same])
show ?thesis
unfolding
 trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
by (intro exI conjI)
 (rule fri-openings, rule out-eq, rule header, rule partial, rule cand,
 rule not-low, rule conflict)
qed

lemma trace-fri-header-tied-authenticated-conflict-iff:
 trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
 s out $\longleftrightarrow$
 trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
 s out $\vee$
 trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks s out
using trace-fri-header-tied-authenticated-conflict-imp-without-same-or-same-layer
 trace-fri-header-tied-authenticated-conflictI-without-same
 trace-fri-header-tied-authenticated-conflictI-same-layer
by blast

lemma wp-trace-fri-header-tied-authenticated-conflict-bound-from-without-same-and-merkle:
assumes without-bound:
 wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict-without-same-layer s)
 $s \leq W$
and merkle-bound:
 wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq M$
shows
 wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer s)

$s \leq W + M$
proof –
have *wp-event verify-monad*
(trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer s)
 $s \leq$
wp-event verify-monad
 $(\lambda out.$
trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
 $s \text{ out} \vee$
trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks
 $s \text{ out}) s$
by *(rule wp-event-mono)*
(rule
trace-fri-header-tied-authenticated-conflict-imp-without-same-or-same-layer)
also have $\dots \leq$
wp-event verify-monad
(trace-fri-header-tied-sampled-assignment-conflict-without-same-layer s)
 $s +$
wp-event verify-monad
(trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks s)
 s
by *(rule wp-event-union-bound)*
also have $\dots \leq W + M$
by *(rule add-mono[OF without-bound*
wp-trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks-bound-from-partial-merkle
[OF merkle-bound]])
finally show *?thesis* .
qed

definition *trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
 $s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr trace-openings trace-table result}$
 final-state.
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-sampled-assignment-conflict-with-authenticated-same-layer}$
 $\text{trace-table trace-roots trace-bs trace-final fri-query-idxs}$

trace-round-layers final-state)

lemma *trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-imp-verifier-tied:*

assumes

trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
s out

shows *trace-fri-verifier-tied-sampled-assignment-conflict s out*

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr trace-openings trace-table result
final-state

where *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers

and *out-eq:* *out = Some (result, final-state)*

and *partial:*

accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings

and *cand:* *partial-trace-table-candidate trace-table trace-openings*

and *not-low:* $\neg$ *trace-table-low-degree trace-table*

and *conflict:*

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers final-state

unfolding

trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-def

by *blast*

have *sampled:*

generic-fri-sampled-assignment-conflict trace-table trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers

by (*rule*

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-imp-sampled
[OF conflict])

show *?thesis*

unfolding *trace-fri-verifier-tied-sampled-assignment-conflict-def*

by (*intro exI conjI*)

(*rule fri-openings, rule partial, rule cand, rule not-low,*
rule sampled)

qed

lemma *wp-trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-bound-from-verifier-tied:*

assumes *verifier-bound:*

wp-event verify-monad
(trace-fri-verifier-tied-sampled-assignment-conflict s) s ≤ C

shows

wp-event verify-monad
(trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s)

$s \leq C$
proof –
have *wp-event verify-monad*
 $(\text{trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer } s)$
 $s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-verifier-tied-sampled-assignment-conflict } s) \ s$
by (rule *wp-event-mono*)
(rule
 $\text{trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-imp-verifier-tied}$)
then show ?thesis
by (rule *order-trans*[*OF* - *verifier-bound*])
qed

definition *trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks}$
 $s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr trace-openings trace-table result}$
 final-state.
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks}$
 $\text{trace-roots trace-bs fri-query-idxs trace-round-layers final-state})$

lemma *trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-imp-partial-merkle:*

assumes

$\text{trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks}$
 $s \text{ out}$

shows *partial-merkle-inconsistency-bad* $s \text{ out}$

proof –

from *assms* **obtain** $\text{trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr trace-openings trace-table result}$
 final-state

where *fri-openings:*

$\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers}$

and *out-eq*: *out* = *Some* (*result*, *final-state*)
and *conflict*:
generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
trace-roots trace-bs fri-query-idxs trace-round-layers final-state
unfolding
trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-def
by *blast*
have *len*: *length trace-bs* = *length trace-roots*
using *accepted-fri-opening-transcript-shapes*(2)[*OF fri-openings*] .
have *partial-merkle-inconsistency-bad s* (*Some* (*result*, *final-state*))
by (*rule*
generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-partial-merkle
[*OF conflict len*])
then show *?thesis*
by (*simp add*: *out-eq*)
qed

lemma *wp-trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-bound-from-partial-merkle*:
assumes *merkle-bound*:
wp-event verify-monad (*partial-merkle-inconsistency-bad s*) $s \leq M$
shows
wp-event verify-monad
(*trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks s*)
 $s \leq M$
proof –
have *wp-event verify-monad*
(*trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks s*)
 $s \leq wp\text{-}event\ verify\text{-}monad\ (partial\text{-}merkle\text{-}inconsistency\text{-}bad\ s)\ s$
by (*rule wp-event-mono*)
(*rule*
trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-imp-partial-merkle)
then show *?thesis*
by (*rule order-trans*[*OF - merkle-bound*])
qed

definition *composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer*
 $:: ('f, 'a)\ protocol\text{-}channel\text{-}scheme \Rightarrow$
 $(unit\ list \times ('f, 'a)\ protocol\text{-}channel\text{-}scheme)\ option \Rightarrow bool$
where
composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
 $s\ out \longleftrightarrow$
 $(\exists\ trace\text{-}roots\ trace\text{-}bs\ trace\text{-}final\ fri\text{-}dg\ composition\text{-}roots$
 $\ composition\text{-}bs\ composition\text{-}final\ fri\text{-}query\text{-}idxs\ trace\text{-}round\text{-}layers$
 $\ composition\text{-}round\text{-}layers\ fr\ as\ trace\text{-}openings\ composition\text{-}openings$
 $\ trace\text{-}table\ composition\text{-}table\ result\ final\text{-}state.$
 $accepted\text{-}fri\text{-}opening\text{-}transcript\ s\ out\ trace\text{-}roots\ trace\text{-}bs$
 $\ trace\text{-}final\ fri\text{-}dg\ composition\text{-}roots\ composition\text{-}bs$
 $\ composition\text{-}final\ fri\text{-}query\text{-}idxs\ trace\text{-}round\text{-}layers$
 $\ composition\text{-}round\text{-}layers \wedge$

$out = Some (result, final-state) \wedge$
 $accepted-with-partial-initial-openings-aligned\ s\ out\ fr\ trace-roots$
 $\quad trace-final\ as\ fri-dg\ composition-roots\ composition-final\ fri-query-idxs$
 $\quad trace-openings\ composition-openings \wedge$
 $partial-trace-table-candidate\ trace-table\ trace-openings \wedge$
 $partial-composition-table-candidate\ composition-table$
 $\quad composition-openings \wedge$
 $trace-table-low-degree\ trace-table \wedge$
 $\neg composition-table-low-degree\ maxDegree\ composition-table \wedge$
 $generic-fri-sampled-assignment-conflict-with-authenticated-same-layer$
 $\quad composition-table\ composition-roots\ composition-bs$
 $\quad composition-final\ fri-query-idxs\ composition-round-layers$
 $\quad final-state)$

definition *composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer*

$:: (f, 'a)\ protocol-channel-scheme \Rightarrow$
 $(unit\ list \times (f, 'a)\ protocol-channel-scheme)\ option \Rightarrow bool$

where

$composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer$
 $\quad s\ out \longleftrightarrow$
 $(\exists\ trace-roots\ trace-bs\ trace-final\ fri-dg\ composition-roots$
 $\quad composition-bs\ composition-final\ fri-query-idxs\ trace-round-layers$
 $\quad composition-round-layers\ fr\ as\ trace-openings\ composition-openings$
 $\quad trace-table\ composition-table.$
 $accepted-fri-opening-transcript\ s\ out\ trace-roots\ trace-bs$
 $\quad trace-final\ fri-dg\ composition-roots\ composition-bs$
 $\quad composition-final\ fri-query-idxs\ trace-round-layers$
 $\quad composition-round-layers \wedge$
 $accepted-with-partial-initial-openings-aligned\ s\ out\ fr\ trace-roots$
 $\quad trace-final\ as\ fri-dg\ composition-roots\ composition-final\ fri-query-idxs$
 $\quad trace-openings\ composition-openings \wedge$
 $partial-trace-table-candidate\ trace-table\ trace-openings \wedge$
 $partial-composition-table-candidate\ composition-table$
 $\quad composition-openings \wedge$
 $trace-table-low-degree\ trace-table \wedge$
 $\neg composition-table-low-degree\ maxDegree\ composition-table \wedge$
 $generic-fri-sampled-assignment-conflict-without-same-layer$
 $\quad composition-table\ composition-roots\ composition-bs$
 $\quad composition-final\ fri-query-idxs\ composition-round-layers)$

definition *composition-fri-verifier-tied-sampled-base-opening-conflict*

$:: (f, 'a)\ protocol-channel-scheme \Rightarrow$
 $(unit\ list \times (f, 'a)\ protocol-channel-scheme)\ option \Rightarrow bool$

where

$composition-fri-verifier-tied-sampled-base-opening-conflict\ s\ out$
 $\longleftrightarrow$
 $(\exists\ trace-roots\ trace-bs\ trace-final\ fri-dg\ composition-roots$
 $\quad composition-bs\ composition-final\ fri-query-idxs\ trace-round-layers$
 $\quad composition-round-layers\ fr\ as\ trace-openings\ composition-openings$

trace-table composition-table.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
generic-fri-sampled-base-opening-conflict composition-table
composition-roots composition-bs fri-query-idxs
composition-round-layers)

definition *composition-fri-verifier-tied-sampled-next-value-conflict*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-sampled-next-value-conflict s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table.}$
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
generic-fri-sampled-next-value-conflict composition-roots
composition-bs fri-query-idxs composition-round-layers)

definition *composition-fri-verifier-tied-sampled-successor-opening-conflict*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-sampled-successor-opening-conflict s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$

composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
generic-fri-sampled-successor-opening-conflict composition-roots
composition-bs fri-query-idxs composition-round-layers)

definition *composition-fri-verifier-tied-sampled-final-value-conflict*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-sampled-final-value-conflict s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table.}$
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
generic-fri-sampled-final-value-conflict composition-roots
composition-bs composition-final fri-query-idxs
composition-round-layers)

lemma *composition-fri-verifier-tied-without-same-imp-branch:*

assumes

composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s out

shows

composition-fri-verifier-tied-sampled-base-opening-conflict s out $\vee$
composition-fri-verifier-tied-sampled-next-value-conflict s out $\vee$
composition-fri-verifier-tied-sampled-successor-opening-conflict
s out $\vee$
composition-fri-verifier-tied-sampled-final-value-conflict s out

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as trace-openings
composition-openings trace-table composition-table

where *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

and *aligned*:
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings

and *trace-cand*:
partial-trace-table-candidate trace-table trace-openings

and *comp-cand*:
partial-composition-table-candidate composition-table
composition-openings

and *trace-low*: *trace-table-low-degree trace-table*

and *comp-not-low*:
 $\neg$ *composition-table-low-degree maxDegree composition-table*

and *without*:
generic-fri-sampled-assignment-conflict-without-same-layer
composition-table composition-roots composition-bs composition-final
fri-query-idxs composition-round-layers

unfolding
composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def

by *blast*

have *split*:
generic-fri-sampled-base-opening-conflict composition-table
composition-roots composition-bs fri-query-idxs
composition-round-layers $\vee$
generic-fri-sampled-next-value-conflict composition-roots
composition-bs fri-query-idxs composition-round-layers $\vee$
generic-fri-sampled-successor-opening-conflict composition-roots
composition-bs fri-query-idxs composition-round-layers $\vee$
generic-fri-sampled-final-value-conflict composition-roots
composition-bs composition-final fri-query-idxs
composition-round-layers

by (rule *generic-fri-sampled-assignment-conflict-without-same-layer-split*
 $[OF\ without]$)

then show *?thesis*

proof

```

assume base:
  generic-fri-sampled-base-opening-conflict composition-table
  composition-roots composition-bs fri-query-idxs
  composition-round-layers
then show ?thesis
  unfolding composition-fri-verifier-tied-sampled-base-opening-conflict-def
  by (intro disjI1 exI conjI)
    (rule fri-openings, rule aligned, rule trace-cand, rule comp-cand,
     rule trace-low, rule comp-not-low, rule base)
next
assume rest-split:
  generic-fri-sampled-next-value-conflict composition-roots
  composition-bs fri-query-idxs composition-round-layers  $\vee$ 
  generic-fri-sampled-successor-opening-conflict composition-roots
  composition-bs fri-query-idxs composition-round-layers  $\vee$ 
  generic-fri-sampled-final-value-conflict composition-roots
  composition-bs composition-final fri-query-idxs
  composition-round-layers
then show ?thesis
proof
  assume next-conflict:
    generic-fri-sampled-next-value-conflict composition-roots
    composition-bs fri-query-idxs composition-round-layers
  then show ?thesis
    unfolding
      composition-fri-verifier-tied-sampled-next-value-conflict-def
    by (intro disjI2 disjI1 exI conjI)
      (rule fri-openings, rule aligned, rule trace-cand, rule comp-cand,
       rule trace-low, rule comp-not-low, rule next-conflict)
next
assume rest-split':
  generic-fri-sampled-successor-opening-conflict composition-roots
  composition-bs fri-query-idxs composition-round-layers  $\vee$ 
  generic-fri-sampled-final-value-conflict composition-roots
  composition-bs composition-final fri-query-idxs
  composition-round-layers
then show ?thesis
proof
  assume successor:
    generic-fri-sampled-successor-opening-conflict composition-roots
    composition-bs fri-query-idxs composition-round-layers
  then show ?thesis
    unfolding
      composition-fri-verifier-tied-sampled-successor-opening-conflict-def
    by (intro disjI2 disjI2 disjI1 exI conjI)
      (rule fri-openings, rule aligned, rule trace-cand,
       rule comp-cand, rule trace-low, rule comp-not-low,
       rule successor)
next

```

```

assume final-conflict:
  generic-fri-sampled-final-value-conflict composition-roots
  composition-bs composition-final fri-query-idxs
  composition-round-layers
then show ?thesis
  unfolding
    composition-fri-verifier-tied-sampled-final-value-conflict-def
  by (intro disjI2 disjI2 disjI2 exI conjI)
    (rule fri-openings, rule aligned, rule trace-cand,
     rule comp-cand, rule trace-low, rule comp-not-low,
     rule final-conflict)
qed
qed
qed
qed

lemma wp-composition-fri-verifier-tied-without-same-bound-from-branches:
assumes base-bound:
  wp-event verify-monad
  (composition-fri-verifier-tied-sampled-base-opening-conflict s)  $s \leq B$ 
and next-bound:
  wp-event verify-monad
  (composition-fri-verifier-tied-sampled-next-value-conflict s)  $s \leq N$ 
and successor-bound:
  wp-event verify-monad
  (composition-fri-verifier-tied-sampled-successor-opening-conflict s)
   $s \leq S$ 
and final-bound:
  wp-event verify-monad
  (composition-fri-verifier-tied-sampled-final-value-conflict s)  $s \leq F$ 
shows
  wp-event verify-monad
  (composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s)
   $s \leq B + N + S + F$ 
proof –
have wp-event verify-monad
  (composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s)
   $s \leq$ 
  wp-event verify-monad
  ( $\lambda out.$ 
    composition-fri-verifier-tied-sampled-base-opening-conflict
     $s out \vee$ 
    composition-fri-verifier-tied-sampled-next-value-conflict
     $s out \vee$ 
    composition-fri-verifier-tied-sampled-successor-opening-conflict
     $s out \vee$ 
    composition-fri-verifier-tied-sampled-final-value-conflict
     $s out$ )  $s$ 
by (rule wp-event-mono)

```

```

    (rule composition-fri-verifier-tied-without-same-imp-branch)
  also have ...  $\leq$ 
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-base-opening-conflict s) s +
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-next-value-conflict s) s +
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-successor-opening-conflict s)
      s +
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-final-value-conflict s) s
  by (rule wp-event-union-bound4)
  also have ...  $\leq B + N + S + F$ 
  by (intro add-mono base-bound next-bound successor-bound final-bound)
  finally show ?thesis .
qed

lemma composition-fri-verifier-tied-without-sameI-base:
  assumes composition-fri-verifier-tied-sampled-base-opening-conflict s out
  shows
    composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
      s out
  using assms
  unfolding
    composition-fri-verifier-tied-sampled-base-opening-conflict-def
    composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def
  by (blast intro:
    generic-fri-sampled-assignment-conflict-without-same-layerI-base)

lemma composition-fri-verifier-tied-without-sameI-next:
  assumes composition-fri-verifier-tied-sampled-next-value-conflict s out
  shows
    composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
      s out
  using assms
  unfolding
    composition-fri-verifier-tied-sampled-next-value-conflict-def
    composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def
  by (blast intro:
    generic-fri-sampled-assignment-conflict-without-same-layerI-next)

lemma composition-fri-verifier-tied-without-sameI-successor:
  assumes
    composition-fri-verifier-tied-sampled-successor-opening-conflict s out
  shows
    composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
      s out
  using assms
  unfolding

```

composition-fri-verifier-tied-sampled-successor-opening-conflict-def
composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def
by (blast intro:
generic-fri-sampled-assignment-conflict-without-same-layerI-successor)

lemma *composition-fri-verifier-tied-without-sameI-final*:
assumes *composition-fri-verifier-tied-sampled-final-value-conflict s out*
shows
composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s out
using *assms*
unfolding
composition-fri-verifier-tied-sampled-final-value-conflict-def
composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def
by (blast intro:
generic-fri-sampled-assignment-conflict-without-same-layerI-final)

lemma *composition-fri-verifier-tied-without-same-iff-branches*:
composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s out
 $\longleftrightarrow$
composition-fri-verifier-tied-sampled-base-opening-conflict s out $\vee$
composition-fri-verifier-tied-sampled-next-value-conflict s out $\vee$
composition-fri-verifier-tied-sampled-successor-opening-conflict s out $\vee$
composition-fri-verifier-tied-sampled-final-value-conflict s out
using *composition-fri-verifier-tied-without-same-imp-branch*
composition-fri-verifier-tied-without-sameI-base
composition-fri-verifier-tied-without-sameI-next
composition-fri-verifier-tied-without-sameI-successor
composition-fri-verifier-tied-without-sameI-final
by *blast*

lemma *composition-fri-verifier-tied-base-conflict-obtains-authenticated-first-chunk*:
assumes *base*:
composition-fri-verifier-tied-sampled-base-opening-conflict
s (Some (result, final-state))
obtains *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table round-idx xp xp-path xn xn-path
auth-chunk
where
accepted-fri-opening-transcript s (Some (result, final-state))
trace-roots trace-bs trace-final fri-dg composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers
accepted-with-partial-initial-openings-aligned s
(Some (result, final-state)) fr trace-roots trace-final as fri-dg
composition-roots composition-final fri-query-idxs trace-openings


```

    composition-openings
partial-trace-table-candidate trace-table trace-openings
partial-composition-table-candidate composition-table
    composition-openings
trace-table-low-degree trace-table
¬ composition-table-low-degree maxDegree composition-table
round-idx < length fri-query-idxs
0 < length composition-bs
fri-layer-step-evidence
  (composition-roots ! 0)
  (composition-bs ! 0)
  (fri-evidence-layer-len composition-roots 0)
  (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
  1
  (fri-sibling-index (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0))
  xp xp-path xn xn-path
  (fri-evidence-next-idx composition-roots fri-query-idxs round-idx 0)
  (fri-evidence-next-value composition-roots composition-bs
    fri-query-idxs round-idx 0 xp xn)
  (composition-round-layers ! round-idx ! 0)
¬ fri-opening-matches-table
  (fri-evidence-layer-len composition-roots 0)
  (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
  composition-table xp xn
fri-layer-chunk-authenticated (composition-roots ! 0)
  (fri-evidence-layer-len composition-roots 0)
  (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
  auth-chunk final-state
proof –
from base obtain trace-roots trace-bs trace-final fri-dg
  composition-roots composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers fr as trace-openings
  composition-openings trace-table composition-table
where fri-openings:
  accepted-fri-opening-transcript s (Some (result, final-state))
  trace-roots trace-bs trace-final fri-dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers
and aligned:
  accepted-with-partial-initial-openings-aligned s
  (Some (result, final-state)) fr trace-roots trace-final as fri-dg
  composition-roots composition-final fri-query-idxs trace-openings
  composition-openings
and trace-cand:
  partial-trace-table-candidate trace-table trace-openings
and comp-cand:
  partial-composition-table-candidate composition-table
  composition-openings

```

```

and trace-low: trace-table-low-degree trace-table
and comp-not-low:
   $\neg$  composition-table-low-degree maxDegree composition-table
and generic:
  generic-fri-sampled-base-opening-conflict composition-table
    composition-roots composition-bs fri-query-idxs
    composition-round-layers
unfolding composition-fri-verifier-tied-sampled-base-opening-conflict-def
by blast
from generic obtain round-idx xp xp-path xn xn-path where
  composition-bs-nonempty:  $0 < \text{length } \text{composition-bs}$ 
and round-bound: round-idx < length fri-query-idxs
and step:
  fri-layer-step-evidence
    (composition-roots ! 0)
    (composition-bs ! 0)
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
    1
    (fri-sibling-index (fri-evidence-layer-len composition-roots 0)
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0))
    xp xp-path xn xn-path
    (fri-evidence-next-idx composition-roots fri-query-idxs round-idx 0)
    (fri-evidence-next-value composition-roots composition-bs
      fri-query-idxs round-idx 0 xp xn)
    (composition-round-layers ! round-idx ! 0)
and no-match:
   $\neg$  fri-opening-matches-table
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
    composition-table xp xn
unfolding generic-fri-sampled-base-opening-conflict-def
by blast
have roots-nonempty:  $0 < \text{length } \text{composition-roots}$ 
using accepted-fri-opening-transcript-shapes(4)[OF fri-openings]
  composition-bs-nonempty by simp
have round-bound-rounds: round-idx < rounds
using round-bound accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
by simp
obtain auth-chunk where auth:
  fri-layer-chunk-authenticated (composition-roots ! 0)
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
  auth-chunk final-state
proof –
have layer-bound:  $0 < \text{length } \text{composition-roots}$ 
by (rule roots-nonempty)
have raw-layer:
   $\bigwedge \text{raw.}$ 

```

```

    0 < fri-layer-lengths (length composition-roots) (clength * scale) ! 0 ∧
    fri-layer-indices (length composition-roots) (index (to-nat raw))
    (clength * scale) ! 0 <
    fri-layer-lengths (length composition-roots) (clength * scale) ! 0
  by (rule fri-first-layer-raw-bound[OF roots-nonempty])
from accepted-fri-opening-transcript-composition-step-with-authenticated-chunk-at
  [OF fri-openings refl round-bound-rounds layer-bound raw-layer]
obtain auth-chunk where
  generic-fri-transcript-step-with-authenticated-chunk composition-roots
  composition-bs fri-query-idxs composition-round-layers final-state
  round-idx 0
  by blast
then obtain auth-chunk where
  fri-layer-chunk-authenticated (composition-roots ! 0)
  (fri-evidence-layer-len composition-roots 0)
  (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
  auth-chunk final-state
  by (rule generic-fri-transcript-step-with-authenticated-chunkE)
then show ?thesis
  by (rule that)
qed
show ?thesis
  by (rule that[OF fri-openings aligned trace-cand comp-cand trace-low
    comp-not-low round-bound composition-bs-nonempty step no-match auth])
qed

```

definition *composition-fri-verifier-tied-authenticated-base-opening-conflict*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-authenticated-base-opening-conflict s out
 $\iff$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table round-idx xp xp-path xn xn-path}$
 $\text{ result final-state.}$
 $out = \text{Some (result, final-state)} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{ trace-final fri-dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers} \wedge$
 $\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr trace-roots}$
 $\text{ trace-final as fri-dg composition-roots composition-final fri-query-idxs}$
 $\text{ trace-openings composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{ composition-openings} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$

$\neg$ composition-table-low-degree maxDegree composition-table $\wedge$
round-idx < length fri-query-idxs $\wedge$
0 < length composition-bs $\wedge$
fri-layer-step-evidence
(composition-roots ! 0)
(composition-bs ! 0)
(fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
1
(fri-sibling-index (fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0))
xp xp-path xn xn-path
(fri-evidence-next-idx composition-roots fri-query-idxs round-idx 0)
(fri-evidence-next-value composition-roots composition-bs
fri-query-idxs round-idx 0 xp xn)
(composition-round-layers ! round-idx ! 0) $\wedge$
 $\neg$ fri-opening-matches-table
(fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
composition-table xp xn $\wedge$
fri-layer-chunk-authenticated (composition-roots ! 0)
(fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
(composition-round-layers ! round-idx ! 0) final-state)

definition composition-fri-verifier-tied-base-chunk-auth-gap

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-base-chunk-auth-gap s out $\longleftrightarrow$
composition-fri-verifier-tied-sampled-base-opening-conflict s out $\wedge$
 $\neg$ composition-fri-verifier-tied-authenticated-base-opening-conflict
s out

lemma composition-fri-verifier-tied-sampled-base-imp-authenticated-or-auth-gap:

assumes

composition-fri-verifier-tied-sampled-base-opening-conflict s out

shows

composition-fri-verifier-tied-authenticated-base-opening-conflict s out $\vee$
composition-fri-verifier-tied-base-chunk-auth-gap s out

using assms

unfolding composition-fri-verifier-tied-base-chunk-auth-gap-def

by blast

lemma wp-composition-fri-verifier-tied-sampled-base-bound-from-authenticated-and-auth-gap:

fixes A G :: prob

assumes authenticated-bound:

wp-event verify-monad

(composition-fri-verifier-tied-authenticated-base-opening-conflict s)

$s \leq A$
and *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-base-chunk-auth-gap } s) \ s \leq G$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-base-opening-conflict } s) \ s$
 $\leq A + G$
proof –
have $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-base-opening-conflict } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\lambda \text{out.}$
 $\text{composition-fri-verifier-tied-authenticated-base-opening-conflict}$
 $s \ \text{out} \ \vee$
 $\text{composition-fri-verifier-tied-base-chunk-auth-gap } s \ \text{out}) \ s$
by $(\text{rule wp-event-mono})$
 $(\text{rule composition-fri-verifier-tied-sampled-base-imp-authenticated-or-auth-gap})$
also have $\dots \leq$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-authenticated-base-opening-conflict } s) \ s +$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-base-chunk-auth-gap } s) \ s$
by $(\text{rule wp-event-union-bound})$
also have $\dots \leq A + G$
by $(\text{intro add-mono authenticated-bound gap-bound})$
finally show $?thesis$.
qed

lemma *composition-fri-verifier-tied-authenticated-base-conflict-imp-sampled-base*:
assumes

$\text{composition-fri-verifier-tied-authenticated-base-opening-conflict } s \ \text{out}$

shows $\text{composition-fri-verifier-tied-sampled-base-opening-conflict } s \ \text{out}$

proof –

from *assms* **obtain** $\text{trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as trace-openings composition-openings}$
 $\text{trace-table composition-table round-idx xp xp-path xn xn-path}$
 $\text{result final-state}$

where $\text{out-eq: out} = \text{Some (result, final-state)}$

and *fri-openings*:

$\text{accepted-fri-opening-transcript } s \ \text{out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers}$

and *aligned*:

$\text{accepted-with-partial-initial-openings-aligned } s \ \text{out fr trace-roots}$
 $\text{trace-final as fri-dg composition-roots composition-final fri-query-idxs}$
 $\text{trace-openings composition-openings}$

```

and trace-cand:
  partial-trace-table-candidate trace-table trace-openings
and comp-cand:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-not-low:
  ¬ composition-table-low-degree maxDegree composition-table
and round-bound: round-idx < length fri-query-idxs
and composition-bs-nonempty: 0 < length composition-bs
and step:
  fri-layer-step-evidence
    (composition-roots ! 0)
    (composition-bs ! 0)
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
    1
    (fri-sibling-index (fri-evidence-layer-len composition-roots 0)
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0))
    xp xp-path xn xn-path
    (fri-evidence-next-idx composition-roots fri-query-idxs round-idx 0)
    (fri-evidence-next-value composition-roots composition-bs
      fri-query-idxs round-idx 0 xp xn)
    (composition-round-layers ! round-idx ! 0)
and no-match:
  ¬ fri-opening-matches-table
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
    composition-table xp xn
and auth:
  fri-layer-chunk-authenticated (composition-roots ! 0)
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
    (composition-round-layers ! round-idx ! 0) final-state
unfolding
  composition-fri-verifier-tied-authenticated-base-opening-conflict-def
by auto
have base:
  generic-fri-sampled-base-opening-conflict composition-table
    composition-roots composition-bs fri-query-idxs composition-round-layers
unfolding generic-fri-sampled-base-opening-conflict-def
by (intro exI[of - round-idx] exI[of - xp] exI[of - xp-path]
  exI[of - xn] exI[of - xn-path] conjI)
  (use round-bound composition-bs-nonempty step no-match in auto)
show ?thesis
unfolding composition-fri-verifier-tied-sampled-base-opening-conflict-def
by (intro exI conjI)
  (use out-eq fri-openings aligned trace-cand comp-cand trace-low
    comp-not-low base in auto)

```

qed

lemma *composition-fri-verifier-tied-authenticated-base-conflict-imp-partial-merkle:*

assumes *base:*

composition-fri-verifier-tied-authenticated-base-opening-conflict
s (Some (result, final-state))

shows

partial-merkle-inconsistency-bad s (Some (result, final-state))

proof –

show *?thesis*

proof (*rule ccontr*)

assume *no-bad:*

$\neg$ *partial-merkle-inconsistency-bad s*
(Some (result, final-state))

obtain *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table round-idx xp xp-path xn xn-path

where *aligned:*

accepted-with-partial-initial-openings-aligned s
(Some (result, final-state)) fr trace-roots trace-final as fri-dg
composition-roots composition-final fri-query-idxs trace-openings
composition-openings

and *comp-cand:*

partial-composition-table-candidate composition-table
composition-openings

and *round-bound:* *round-idx < length fri-query-idxs*

and *step:*

fri-layer-step-evidence
(composition-roots ! 0)
(composition-bs ! 0)
(fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
1
(fri-sibling-index (fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0))
xp xp-path xn xn-path
(fri-evidence-next-idx composition-roots fri-query-idxs round-idx 0)
(fri-evidence-next-value composition-roots composition-bs
fri-query-idxs round-idx 0 xp xn)
(composition-round-layers ! round-idx ! 0)

and *no-match:*

$\neg$ *fri-opening-matches-table*
(fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
composition-table xp xn

and *auth:*

fri-layer-chunk-authenticated (composition-roots ! 0)
(fri-evidence-layer-len composition-roots 0)

```

    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
    (composition-round-layers ! round-idx ! 0) final-state
using base
unfolding
    composition-fri-verifier-tied-authenticated-base-opening-conflict-def
by blast
have initial:
    accepted-with-partial-initial-openings s
    (Some (result, final-state)) fr trace-roots trace-final as fri-dg
    composition-roots composition-final fri-query-idxs trace-openings
    fri-query-idxs composition-openings
by (rule accepted-with-partial-initial-openings-aligned-imp-unaligned
    [OF aligned])
have comp-partial:
    accepted-with-partial-composition-openings s
    (Some (result, final-state)) (hd composition-roots) fri-query-idxs
    composition-openings
by (rule accepted-with-partial-initial-openings-shapes(4)
    [OF initial])
have roots-nonempty: composition-roots  $\neq$  []
by (rule accepted-with-partial-initial-openings-shapes(1)[OF initial])
have round-bound-rounds: round-idx < rounds
using round-bound
    accepted-with-partial-composition-openings-shapes(1)[OF comp-partial]
by simp
let ?idx =
    fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0
have idx-eq:
    ?idx = fri-query-idxs ! round-idx
unfolding fri-evidence-layer-idx-def
using roots-nonempty by (cases composition-roots) simp-all
have len-eq:
    fri-evidence-layer-len composition-roots 0 = scale * clength
unfolding fri-evidence-layer-len-def
using roots-nonempty by (cases composition-roots) simp-all
have root-eq: composition-roots ! 0 = hd composition-roots
using roots-nonempty by (cases composition-roots) simp-all
have idxs:
    map opening-index (composition-openings ! round-idx) =
    [?idx, fri-sibling-index (scale * clength) ?idx]
using accepted-with-partial-composition-openings-shapes(3)
    [OF comp-partial round-bound-rounds]
unfolding idx-eq by simp
have comp-value:
    composition-table ! ?idx =
    opening-value ((composition-openings ! round-idx) ! 0)
by (rule partial-composition-table-candidate-query-value
    [OF comp-cand round-bound-rounds idxs])
have len-openings: length (composition-openings ! round-idx) = 2

```



```

proof –
  have length (map opening-index (composition-openings ! round-idx)) =
    length [?idx, fri-sibling-index (scale * clength) ?idx]
    using arg-cong[OF idxs, of length] by simp
  then show ?thesis by simp
qed
have opn0-in:
  (composition-openings ! round-idx) ! 0
  ∈ set (composition-openings ! round-idx)
  using len-openings by (cases composition-openings ! round-idx) auto
have opn1-in:
  (composition-openings ! round-idx) ! 1
  ∈ set (composition-openings ! round-idx)
  using len-openings
  by (cases composition-openings ! round-idx; cases tl (composition-openings ! round-idx))
    auto
have comp-table:
  partial-authenticated-table (hd composition-roots) (scale * clength)
    (composition-openings ! round-idx) final-state
  using comp-partial round-bound-rounds
  unfolding accepted-with-partial-composition-openings-def by blast
have opn0-auth:
  authenticated-opening-in final-state
    ((composition-openings ! round-idx) ! 0)
  using comp-table opn0-in
  unfolding partial-authenticated-table-def by blast
have opn0-root:
  opening-root ((composition-openings ! round-idx) ! 0) =
    composition-roots ! 0
  using comp-table opn0-in root-eq
  unfolding partial-authenticated-table-def by simp
have opn0-len:
  opening-length ((composition-openings ! round-idx) ! 0) =
    fri-evidence-layer-len composition-roots 0
  using comp-table opn0-in len-eq
  unfolding partial-authenticated-table-def by simp
have opn0-idx:
  opening-index ((composition-openings ! round-idx) ! 0) = ?idx
  using idxs len-openings by (cases composition-openings ! round-idx) auto
have opn0-value: opening-value ((composition-openings ! round-idx) ! 0) = xp
proof –
  from step have chunk:
    fri-layer-opening-chunk
      (fri-evidence-layer-len composition-roots 0)
      xp xp-path xn xn-path
      (composition-round-layers ! round-idx ! 0)
    unfolding fri-layer-step-evidence-def by blast
  from fri-layer-chunk-authenticated-matching-values[OF auth chunk]
  obtain xp-path' xn-path' where xp-auth:

```

```

    authenticated-opening-in final-state
    (|opening-root = composition-roots ! 0,
     opening-length = fri-evidence-layer-len composition-roots 0,
     opening-index =
       fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0,
     opening-value = xp,
     opening-path = xp-path'|)
  by (metis)
have opening-value ((composition-openings ! round-idx) ! 0) =
  opening-value
  (|opening-root = composition-roots ! 0,
   opening-length = fri-evidence-layer-len composition-roots 0,
   opening-index =
     fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0,
   opening-value = xp,
   opening-path = xp-path'|)
  by (rule authenticated-openings-values-eq-if-no-partial-merkle
      [OF no-bad opn0-auth xp-auth])
    (use opn0-root opn0-len opn0-idx in simp-all)
  then show ?thesis by simp
qed
have comp-match-base: composition-table ! ?idx = xp
  using comp-value opn0-value by simp
have comp-sibling:
  composition-table ! fri-sibling-index (scale * clength) ?idx =
    opening-value ((composition-openings ! round-idx) ! 1)
proof -
  have vals:
    map (!) composition-table
      [?idx, fri-sibling-index (scale * clength) ?idx] =
    map opening-value (composition-openings ! round-idx)
  by (rule partial-composition-table-candidate-query-values
      [OF comp-cand round-bound-rounds idxs])
  have lhs:
    map (!) composition-table
      [?idx, fri-sibling-index (scale * clength) ?idx] ! 1 =
    composition-table ! fri-sibling-index (scale * clength) ?idx
  by simp
  have rhs:
    map opening-value (composition-openings ! round-idx) ! 1 =
    opening-value ((composition-openings ! round-idx) ! 1)
  using len-openings by simp
  show ?thesis
    using arg-cong[OF vals, of  $\lambda x s. xs ! 1$ ] lhs rhs by simp
qed
have opn1-auth:
  authenticated-opening-in final-state
  ((composition-openings ! round-idx) ! 1)
  using comp-table opn1-in

```

```

unfolding partial-authenticated-table-def by blast
have opn1-root:
  opening-root ((composition-openings ! round-idx) ! 1) =
    composition-roots ! 0
using comp-table opn1-in root-eq
unfolding partial-authenticated-table-def by simp
have opn1-len:
  opening-length ((composition-openings ! round-idx) ! 1) =
    fri-evidence-layer-len composition-roots 0
using comp-table opn1-in len-eq
unfolding partial-authenticated-table-def by simp
have opn1-idx:
  opening-index ((composition-openings ! round-idx) ! 1) =
    fri-sibling-index (fri-evidence-layer-len composition-roots 0) ?idx
using idxs len-openings len-eq
by (cases composition-openings ! round-idx; cases tl (composition-openings ! round-idx))
  auto
have opn1-value: opening-value ((composition-openings ! round-idx) ! 1) = xn
proof –
  from step have chunk:
    fri-layer-opening-chunk
    (fri-evidence-layer-len composition-roots 0)
    xp xp-path xn xn-path
    (composition-round-layers ! round-idx ! 0)
    unfolding fri-layer-step-evidence-def by blast
  from fri-layer-chunk-authenticated-matching-values[OF auth chunk]
  obtain xp-path' xn-path' where xn-auth:
    authenticated-opening-in final-state
    (|opening-root = composition-roots ! 0,
      opening-length = fri-evidence-layer-len composition-roots 0,
      opening-index =
        fri-sibling-index (fri-evidence-layer-len composition-roots 0)
        (fri-evidence-layer-idx composition-roots fri-query-idxs
          round-idx 0),
      opening-value = xn,
      opening-path = xn-path')
  by (metis)
  have opening-value ((composition-openings ! round-idx) ! 1) =
    opening-value
    (|opening-root = composition-roots ! 0,
      opening-length = fri-evidence-layer-len composition-roots 0,
      opening-index =
        fri-sibling-index (fri-evidence-layer-len composition-roots 0)
        (fri-evidence-layer-idx composition-roots fri-query-idxs
          round-idx 0),
      opening-value = xn,
      opening-path = xn-path')
  by (rule authenticated-openings-values-eq-if-no-partial-merkle
    [OF no-bad opn1-auth xn-auth])

```

```

      (use opn1-root opn1-len opn1-idx in simp-all)
    then show ?thesis by simp
  qed
  have comp-match-sibling:
    composition-table !
    fri-sibling-index (fri-evidence-layer-len composition-roots 0) ?idx =
    xn
  using comp-sibling opn1-value len-eq by simp
  have idx-bound: ?idx < fri-evidence-layer-len composition-roots 0
  proof -
    from auth obtain xp0 xp0-path xn0 xn0-path where
      authenticated-opening-in final-state
      (|opening-root = composition-roots ! 0,
        opening-length = fri-evidence-layer-len composition-roots 0,
        opening-index = ?idx,
        opening-value = xp0,
        opening-path = xp0-path|)
    by (rule fri-layer-chunk-authenticatedE)
  then show ?thesis
    unfolding authenticated-opening-in-def by simp
  qed
  have table-len:
    fri-evidence-layer-len composition-roots 0 ≤
    length composition-table
  using partial-composition-table-candidateD(1)
    [OF comp-cand round-bound-rounds opn0-in]
  unfolding len-eq by simp
  have match:
    fri-opening-matches-table
    (fri-evidence-layer-len composition-roots 0)
    ?idx composition-table xp xn
  unfolding fri-opening-matches-table-def
  using idx-bound table-len comp-match-base comp-match-sibling
  by simp
  then show False
    using no-match by contradiction
  qed
qed

lemma wp-composition-fri-verifier-tied-authenticated-base-conflict-bound-by-partial-merkle:
  wp-event verify-monad
    (composition-fri-verifier-tied-authenticated-base-opening-conflict s) s
  ≤ wp-event verify-monad (partial-merkle-inconsistency-bad s) s
proof (rule wp-event-mono)
  fix out
  assume conflict:
    composition-fri-verifier-tied-authenticated-base-opening-conflict s out
  from conflict obtain result final-state where out-eq:
    out = Some (result, final-state)

```

```

    unfolding composition-fri-verifier-tied-authenticated-base-opening-conflict-def
    by blast
show partial-merkle-inconsistency-bad s out
    unfolding out-eq
    by (rule composition-fri-verifier-tied-authenticated-base-conflict-imp-partial-merkle)
      (use conflict out-eq in simp)
qed

lemma wp-composition-fri-verifier-tied-sampled-base-bound-from-partial-merkle-and-auth-gap:
  fixes G :: prob
  assumes gap-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-base-chunk-auth-gap s) s ≤ G
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-base-opening-conflict s) s
      ≤ wp-event verify-monad (partial-merkle-inconsistency-bad s) s + G
proof (rule wp-composition-fri-verifier-tied-sampled-base-bound-from-authenticated-and-auth-gap)
  show wp-event verify-monad
    (composition-fri-verifier-tied-authenticated-base-opening-conflict s) s
    ≤ wp-event verify-monad (partial-merkle-inconsistency-bad s) s
  by (rule wp-composition-fri-verifier-tied-authenticated-base-conflict-bound-by-partial-merkle)
  show wp-event verify-monad
    (composition-fri-verifier-tied-base-chunk-auth-gap s) s ≤ G
  by (rule gap-bound)
qed

lemma wp-composition-fri-verifier-tied-without-same-bound-from-partial-merkle-auth-gap-and-branches:
  fixes G N S F :: prob
  assumes gap-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-base-chunk-auth-gap s) s ≤ G
  and next-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-next-value-conflict s) s ≤ N
  and successor-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-successor-opening-conflict s)
      s ≤ S
  and final-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-final-value-conflict s) s ≤ F
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s)
      s ≤ (wp-event verify-monad (partial-merkle-inconsistency-bad s) s + G) + N + S + F
  by (rule wp-composition-fri-verifier-tied-without-same-bound-from-branches)
    (rule wp-composition-fri-verifier-tied-sampled-base-bound-from-partial-merkle-and-auth-gap
      [OF gap-bound],

```

rule next-bound,
rule successor-bound,
rule final-bound)

lemma *composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-imp-verifier-tied:*

assumes

composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
s out

shows *composition-fri-verifier-tied-sampled-assignment-conflict s out*

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*

composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as trace-openings
composition-openings trace-table composition-table result final-state

where *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

and *out-eq:* *out = Some (result, final-state)*

and *aligned:*

accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings

and *trace-cand:*

partial-trace-table-candidate trace-table trace-openings

and *comp-cand:*

partial-composition-table-candidate composition-table
composition-openings

and *trace-low:* *trace-table-low-degree trace-table*

and *comp-not-low:*

$\neg$ *composition-table-low-degree maxDegree composition-table*

and *conflict:*

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
composition-table composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers
final-state

unfolding

composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-def

by *blast*

have *sampled:*

generic-fri-sampled-assignment-conflict composition-table
composition-roots composition-bs composition-final fri-query-idxs
composition-round-layers

by *(rule*

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-imp-sampled
[OF conflict])

show *?thesis*

unfolding *composition-fri-verifier-tied-sampled-assignment-conflict-def*

by (*intro exI conjI*)
 (rule *fri-openings*, rule *aligned*, rule *trace-cand*, rule *comp-cand*,
 rule *trace-low*, rule *comp-not-low*, rule *sampling*)
qed

lemma *wp-composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-bound-from-verifier*
assumes *verifier-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict *s*) $s \leq C$
shows
wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer *s*)
 $s \leq C$
proof –
have *wp-event verify-monad*
 (composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer *s*)
 $s \leq$
wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict *s*) *s*
by (rule *wp-event-mono*)
 (rule
 composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-imp-verifier-tied)
then show ?thesis
by (rule *order-trans[OF - verifier-bound]*)
qed

definition *composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 (unit list $\times$ (*f*, 'a) *protocol-channel-scheme*) *option* $\Rightarrow$ bool

where

composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
 $s \text{ out} \longleftrightarrow$
 ($\exists$ *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table result final-state.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
out = Some (result, final-state) $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
composition-roots composition-bs fri-query-idxs
composition-round-layers final-state)

lemma *composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-imp-partial-merkle:*
assumes

composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
s out

shows *partial-merkle-inconsistency-bad s out*

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as trace-openings
composition-openings trace-table composition-table result final-state

where *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

and *out-eq: out = Some (result, final-state)*

and *conflict:*

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
composition-roots composition-bs fri-query-idxs
composition-round-layers final-state

unfolding

composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-def

by *blast*

have *len: length composition-bs = length composition-roots*

using *accepted-fri-opening-transcript-shapes(4)[OF fri-openings]* .

have *partial-merkle-inconsistency-bad s (Some (result, final-state))*

by *(rule*

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-partial-merkle
[OF conflict len])

then show *?thesis*

by *(simp add: out-eq)*

qed

lemma *wp-composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-bound-from-partial-merkle:*
assumes *merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M

shows

wp-event verify-monad
(composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks s)
s ≤ M

proof —

have *wp-event verify-monad*

(composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks s)
s ≤ wp-event verify-monad (partial-merkle-inconsistency-bad s) s

by *(rule wp-event-mono)*


```

    (rule
      composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-imp-partial-merkle)
  then show ?thesis
  by (rule order-trans[OF - merkle-bound])
qed

```

lemma *composition-fri-verifier-tied-authenticated-conflict-imp-without-same-or-same-layer:*

assumes

composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
s out

shows

composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s out $\vee$
composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
s out

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*

composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as trace-openings
composition-openings trace-table composition-table result final-state

where *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

and *out-eq:* *out* = *Some (result, final-state)*

and *aligned:*

accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings

and *trace-cand:*

partial-trace-table-candidate trace-table trace-openings

and *comp-cand:*

partial-composition-table-candidate composition-table
composition-openings

and *trace-low:* *trace-table-low-degree trace-table*

and *comp-not-low:*

$\neg$ *composition-table-low-degree maxDegree composition-table*

and *conflict:*

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
composition-table composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers
final-state

unfolding

composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-def

by *blast*

have *split:*

generic-fri-sampled-assignment-conflict-without-same-layer
composition-table composition-roots composition-bs composition-final

```

    fri-query-idxs composition-round-layers ∨
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    composition-roots composition-bs fri-query-idxs composition-round-layers
    final-state
  by (rule
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-split
    [OF conflict])
  then show ?thesis
proof
  assume without:
    generic-fri-sampled-assignment-conflict-without-same-layer
    composition-table composition-roots composition-bs composition-final
    fri-query-idxs composition-round-layers
  then show ?thesis
  unfolding
    composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def
  by (intro disjI1 exI conjI)
    (rule fri-openings, rule aligned, rule trace-cand, rule comp-cand,
     rule trace-low, rule comp-not-low, rule without)
next
  assume same:
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    composition-roots composition-bs fri-query-idxs
    composition-round-layers final-state
  then show ?thesis
  unfolding
    composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-def
  by (intro disjI2 exI conjI)
    (rule fri-openings, rule out-eq, rule aligned, rule trace-cand,
     rule comp-cand, rule trace-low, rule comp-not-low, rule same)
qed
qed

lemma composition-fri-verifier-tied-authenticated-conflictI-without-same:
  assumes
    composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
    s out
  shows
    composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
    s out
proof -
  from assms obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr as trace-openings
    composition-openings trace-table composition-table
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers

```

```

    composition-round-layers
and aligned:
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final fri-query-idxs
    trace-openings composition-openings
and trace-cand:
    partial-trace-table-candidate trace-table trace-openings
and comp-cand:
    partial-composition-table-candidate composition-table
    composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-not-low:
    ¬ composition-table-low-degree maxDegree composition-table
and without:
    generic-fri-sampled-assignment-conflict-without-same-layer
    composition-table composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
unfolding
    composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def
by blast
from fri-openings obtain result final-state where out-eq:
    out = Some (result, final-state)
    unfolding accepted-fri-opening-transcript-def by blast
have conflict:
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
    composition-table composition-roots composition-bs composition-final
    fri-query-idxs composition-round-layers final-state
by (rule
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layerI-without
    [OF without])
show ?thesis
    unfolding
        composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
    by (intro exI conjI)
        (rule fri-openings, rule out-eq, rule aligned, rule trace-cand,
         rule comp-cand, rule trace-low, rule comp-not-low, rule conflict)
qed

lemma composition-fri-verifier-tied-authenticated-conflictI-same-layer:
    assumes
        composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
        s out
    shows
        composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
        s out
proof -
    from assms obtain trace-roots trace-bs trace-final fri-dg
        composition-roots composition-bs composition-final fri-query-idxs
        trace-round-layers composition-round-layers fr as trace-openings

```

```

    composition-openings trace-table composition-table result final-state
where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
and out-eq: out = Some (result, final-state)
and aligned:
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final fri-query-idxs
    trace-openings composition-openings
and trace-cand:
    partial-trace-table-candidate trace-table trace-openings
and comp-cand:
    partial-composition-table-candidate composition-table
    composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-not-low:
    ¬ composition-table-low-degree maxDegree composition-table
and same:
    generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    composition-roots composition-bs fri-query-idxs
    composition-round-layers final-state
unfolding
    composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-def
by blast
have conflict:
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
    composition-table composition-roots composition-bs composition-final
    fri-query-idxs composition-round-layers final-state
by (rule
    generic-fri-sampled-assignment-conflict-with-authenticated-same-layerI-same
    [OF same])
show ?thesis
unfolding
    composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
by (intro exI conjI)
    (rule fri-openings, rule out-eq, rule aligned, rule trace-cand,
    rule comp-cand, rule trace-low, rule comp-not-low, rule conflict)
qed

lemma composition-fri-verifier-tied-authenticated-conflict-iff:
    composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
    s out  $\longleftrightarrow$ 
    composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
    s out  $\vee$ 
    composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
    s out
using composition-fri-verifier-tied-authenticated-conflict-imp-without-same-or-same-layer

```

composition-fri-verifier-tied-authenticated-conflictI-without-same
composition-fri-verifier-tied-authenticated-conflictI-same-layer
by *blast*

lemma *wp-composition-fri-verifier-tied-authenticated-conflict-bound-from-without-same-and-merkle:*

assumes *without-bound:*

wp-event verify-monad

(*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s*)

$s \leq W$

and *merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M

shows

wp-event verify-monad

(*composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s*)

$s \leq W + M$

proof –

have *wp-event verify-monad*

(*composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s*)

$s \leq$

wp-event verify-monad

($\lambda out.$

composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer

$s out \vee$

composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks

$s out) s$

by (*rule wp-event-mono*)

(*rule*

composition-fri-verifier-tied-authenticated-conflict-imp-without-same-or-same-layer)

also have $\dots \leq$

wp-event verify-monad

(*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s*)

$s +$

wp-event verify-monad

(*composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks s*)

s

by (*rule wp-event-union-bound*)

also have $\dots \leq W + M$

by (*rule add-mono[OF without-bound*

wp-composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-bound-from-partial-merkle

[*OF merkle-bound*]])

finally show *?thesis* .

qed

end

end

theory *Soundness-FRI-Base-Recorded-Chunk*

```

imports Soundness-FRI-Layer-Merkle
begin

```

4.9 FRI Replay and Active Bounds

Base-layer recorded chunk replay. This is the small case needed for the existing first-layer composition residual: the head chunk read by *receive-query-commits* is the first chunk in the extracted layer transcript, hence the recorded first layer is authenticated.

```

context soundness
begin

```

lemma *mfold-fri-layer-opening-head-recorded-chunk-authenticated:*

assumes *outcome:*

```

  Some (out, t) ∈
    set-dist
    (execute
      (mfold (idx, x, len, pw)
        (map fri-layer-opening-step ((b, rt) # bfs))) s)

```

and *layers:*

```

  fri-layers-transcript (length ((b, rt) # bfs)) len layer-chunks
  total-chunk

```

and *transcript:* *PTranscript s = total-chunk @ PTranscript t*

and *len-pos:* *0 < len*

and *idx-bound:* *idx < len*

shows

```

  fri-layer-chunk-authenticated rt len idx (layer-chunks ! 0) t

```

proof —

from *outcome* **obtain** *out1 s1* **where**

head:

```

  Some (out1, s1) ∈
    set-dist
    (execute (fri-layer-opening-step (b, rt) (idx, x, len, pw)) s)

```

and *tail:*

```

  Some (out, t) ∈
    set-dist (execute (mfold out1 (map fri-layer-opening-step bfs)) s1)

```

by (*auto elim!:* *set-dist-bindE*)

obtain *xp xp-path xn xn-path x' head-chunk* **where**

out1-eq:

```

  out1 = (idx mod (len div 2), x', len div 2, pw + pw)

```

and *head-open:*

```

  fri-layer-opening-chunk len xp xp-path xn xn-path head-chunk

```

and *tr-head:* *PTranscript s = head-chunk @ PTranscript s1*

and *auth-xp:*

```

  authenticated-opening-in s1
  (| opening-root = rt,
    opening-length = len,
    opening-index = idx,
    opening-value = xp,
    opening-path = xp-path)

```

```

and auth-xn:
  authenticated-opening-in s1
  ( $\parallel$  opening-root = rt,
   opening-length = len,
   opening-index = fri-sibling-index len idx,
   opening-value = xn,
   opening-path = xn-path)
by (rule fri-layer-opening-step-outcome-authenticated
  [OF len-pos idx-bound head])
from mfold-fri-layer-openings-outcome[OF tail[unfolded out1-eq]]
obtain tail-layers tail-chunk where
  tail-layers:
    fri-layers-transcript (length bfs) (len div 2) tail-layers
    tail-chunk
  and tr-tail: PTranscript s1 = tail-chunk @ PTranscript t
  and ext-tail: s1 ≤ t
  by auto
let ?layers = head-chunk # tail-layers
let ?chunk = head-chunk @ tail-chunk
have exec-layers:
  fri-layers-transcript (length ((b, rt) # bfs)) len ?layers ?chunk
  using head-open tail-layers
  unfolding fri-layers-transcript-def by auto
have exec-transcript: PTranscript s = ?chunk @ PTranscript t
  using tr-head tr-tail by simp
have chunk-eq: ?chunk = total-chunk
  using exec-transcript transcript by simp
have layers-eq: ?layers = layer-chunks
  by (rule fri-layers-transcript-unique[OF exec-layers])
  (use layers chunk-eq in simp)
have head-auth-s1:
  fri-layer-chunk-authenticated rt len idx head-chunk s1
  by (rule fri-layer-chunk-authenticatedI[OF head-open auth-xp auth-xn])
have head-auth-t:
  fri-layer-chunk-authenticated rt len idx head-chunk t
  by (rule fri-layer-chunk-authenticated-mono[OF head-auth-s1 ext-tail])
show ?thesis
  using head-auth-t layers-eq[symmetric] by simp
qed

```

lemma *receive-query-commits-head-recorded-chunk-authenticated*:

assumes *outcome*:

Some (out, t) ∈

set-dist

(*execute*

(*mfold (idx, x, len, pw)*

(*receive-query-commits ((b, rt) # bfs)*)) *s*)

and *layers*:

fri-layers-transcript (length ((b, rt) # bfs)) len layer-chunks

```

    total-chunk
  and transcript: PTranscript s = total-chunk @ PTranscript t
  and len-pos: 0 < len
  and idx-bound: idx < len
shows
  fri-layer-chunk-authenticated rt len idx (layer-chunks ! 0) t
proof -
  have map-eq:
    receive-query-commits ((b, rt) # bfs) =
      map fri-layer-opening-step ((b, rt) # bfs)
  unfolding receive-query-commits-def fri-layer-opening-step-def
  by (simp add: fun-eq-iff split: prod.splits)
  show ?thesis
  by (rule mfold-fri-layer-opening-head-recorded-chunk-authenticated
      [OF outcome[unfolded map-eq] layers transcript len-pos idx-bound])
qed

lemma verifier-query-round-after-index-trace-base-recorded-chunk-authenticated:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes f-fl-eq: f-fl = (b, rt) # f-fl-tail
  and outcome:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s)
  and recorded:
    query-round-fri-layer-transcripts (index (to-nat raw))
      (map snd f-fl) (map snd fl) round-chunk trace-layers
      composition-layers
  and transcript: PTranscript s = round-chunk @ PTranscript t
shows
  fri-layer-chunk-authenticated rt (clength * scale)
    (index (to-nat raw)) (trace-layers ! 0) t
proof -
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
    query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
  and trace-fri:
    Some (f-out, s2) ∈
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
  and assert-trace:
    Some ((), s3) ∈
      set-dist
        (execute

```



```

      (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
  Some (c-out, s4) ∈
    set-dist (execute
      (mfold
        (?idx, cp-eval as fv (h ^ ?idx * shift), clength * scale, 1)
        (receive-query-commits fl)) s3)
and assert-comp:
  Some (( ), t) ∈
    set-dist
      (execute
        (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
unfolding verifier-query-round-after-index-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
from check-decommit-on-query-outcome[OF query-decommit]
obtain query-paths query-chunk where
  query-chunk-shape:
    query-decommitment-transcript ?idx fv query-paths query-chunk
and tr-s: PTranscript s = query-chunk @ PTranscript s1
by blast
from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where
  trace-layers-actual:
    fri-layers-transcript (length f-fl) (clength * scale)
    trace-layer-chunks trace-fri-chunk
and tr-s1: PTranscript s1 = trace-fri-chunk @ PTranscript s2
and ext-s1-s2: s1 ≤ s2
by blast
have s3-eq: s3 = s2
using assert-trace unfolding assert-def
by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
  (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  comp-layers-actual:
    fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
and ext-s3-s4: s3 ≤ s4
by blast
have t-eq: t = s4
using assert-comp unfolding assert-def
by (cases case c-out of (-, x, -, -) ⇒ x = final)
  (auto simp: throw-no-outcome)
let ?actual-chunk = query-chunk @ trace-fri-chunk @ comp-fri-chunk
have actual-recorded:
  query-round-fri-layer-transcripts ?idx (map snd f-fl) (map snd fl)
    ?actual-chunk trace-layer-chunks comp-layer-chunks
unfolding query-round-fri-layer-transcripts-def

```

```

  by (intro exI[of - query-chunk] exI[of - trace-fri-chunk]
      exI[of - comp-fri-chunk] exI[of - fv] exI[of - query-paths] conjI)
    (use query-chunk-shape trace-layers-actual comp-layers-actual in
      ⟨simp-all add: mult.commute⟩)
have actual-transcript:
  PTranscript s = ?actual-chunk @ PTranscript t
  using tr-s tr-s1 tr-s3 s3-eq t-eq by simp
have actual-chunk-eq: ?actual-chunk = round-chunk
  using actual-transcript transcript by simp
have trace-layer-chunks-eq:
  trace-layer-chunks = trace-layers
  by (rule query-round-fri-layer-transcripts-unique(1)
      [OF actual-recorded[unfolded actual-chunk-eq] recorded])
have len-pos: 0 < clength * scale
  by (rule eval-domain-size-pos)
have idx-bound: ?idx < clength * scale
  by (rule index-less-domain)
have auth-s2:
  fri-layer-chunk-authenticated rt (clength * scale) ?idx
    (trace-layer-chunks ! 0) s2
  by (rule receive-query-commits-head-recorded-chunk-authenticated
      [OF trace-fri[unfolded f-fl-eq]
        trace-layers-actual[unfolded f-fl-eq] tr-s1 len-pos idx-bound])
have s2-t: s2 ≤ t
  using ext-s3-s4 s3-eq t-eq by simp
have auth-t:
  fri-layer-chunk-authenticated rt (clength * scale) ?idx
    (trace-layer-chunks ! 0) t
  by (rule fri-layer-chunk-authenticated-mono[OF auth-s2 s2-t])
show ?thesis
  using auth-t trace-layer-chunks-eq by simp
qed

lemma verifier-query-round-after-index-composition-base-recorded-chunk-authenticated:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes fl-eq: fl = (b, rt) # fl-tail
  and outcome:
    Some (((), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s)
  and recorded:
    query-round-fri-layer-transcripts (index (to-nat raw))
      (map snd f-fl) (map snd fl) round-chunk trace-layers
      composition-layers
  and transcript: PTranscript s = round-chunk @ PTranscript t
  shows
    fri-layer-chunk-authenticated rt (clength * scale)

```

```

    (index (to-nat raw)) (composition-layers ! 0) t
proof -
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
    query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
  and trace-fri:
    Some (f-out, s2) ∈
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
  and assert-trace:
    Some ((), s3) ∈
      set-dist
        (execute
          (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
  and comp-fri:
    Some (c-out, s4) ∈
      set-dist (execute
        (mfold
          (?idx, cp-eval as fv (h ^ ?idx * shift), clength * scale, 1)
          (receive-query-commits fl)) s3)
  and assert-comp:
    Some ((), t) ∈
      set-dist
        (execute
          (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
  unfolding verifier-query-round-after-index-program-def
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
from check-decommit-on-query-outcome[OF query-decommit]
obtain query-paths query-chunk where
  query-chunk-shape:
    query-decommitment-transcript ?idx fv query-paths query-chunk
  and tr-s: PTranscript s = query-chunk @ PTranscript s1
  by blast
from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where
  trace-layers-actual:
    fri-layers-transcript (length f-fl) (clength * scale)
    trace-layer-chunks trace-fri-chunk
  and tr-s1: PTranscript s1 = trace-fri-chunk @ PTranscript s2
  by blast
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def
  by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
    (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where

```

```

comp-layers-actual:
  fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
  and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
  by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -)  $\Rightarrow$  x = final)
    (auto simp: throw-no-outcome)
let ?actual-chunk = query-chunk @ trace-fri-chunk @ comp-fri-chunk
have actual-recorded:
  query-round-fri-layer-transcripts ?idx (map snd f-fl) (map snd fl)
    ?actual-chunk trace-layer-chunks comp-layer-chunks
  unfolding query-round-fri-layer-transcripts-def
  by (intro exI[of - query-chunk] exI[of - trace-fri-chunk]
    exI[of - comp-fri-chunk] exI[of - fv] exI[of - query-paths] conjI)
    (use query-chunk-shape trace-layers-actual comp-layers-actual in
       $\langle \text{simp-all add: mult.commute} \rangle$ )
have actual-transcript:
  PTranscript s = ?actual-chunk @ PTranscript t
  using tr-s tr-s1 tr-s3 s3-eq t-eq by simp
have actual-chunk-eq: ?actual-chunk = round-chunk
  using actual-transcript transcript by simp
have comp-layer-chunks-eq:
  comp-layer-chunks = composition-layers
  by (rule query-round-fri-layer-transcripts-unique(2)
    [OF actual-recorded[unfolded actual-chunk-eq] recorded])
have len-pos:  $0 < \text{length} * \text{scale}$ 
  by (rule eval-domain-size-pos)
have idx-bound:  $?idx < \text{length} * \text{scale}$ 
  by (rule index-less-domain)
have auth-s4:
  fri-layer-chunk-authenticated rt (clength * scale) ?idx
    (comp-layer-chunks ! 0) s4
  by (rule receive-query-commits-head-recorded-chunk-authenticated
    [OF comp-fri[unfolded fl-eq] comp-layers-actual[unfolded fl-eq]
      tr-s3 len-pos idx-bound])
show ?thesis
  using auth-s4 comp-layer-chunks-eq t-eq by simp
qed

lemma accepted-fri-opening-transcript-trace-base-recorded-chunk-authenticated-at:
  assumes fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
      dg composition-roots composition-bs composition-final query-idxs
      trace-round-layers composition-round-layers
  and out-eq: out = Some (result, final-state)
  and round-bound:  $i < \text{rounds}$ 
  and trace-nonempty:  $0 < \text{length trace-roots}$ 

```

shows

fri-layer-chunk-authenticated (*trace-roots* ! 0) (*clength* * *scale*)
 (*query-idxs* ! *i*) (*trace-round-layers* ! *i* ! 0) *final-state*

proof –

from *fri-openings* **obtain** *result'* *final-state'* *fr f-fl as fl query-state*

raw-idxs query-chunks **where**

out-def: *out* = *Some* (*result'*, *final-state'*)

and *trace-roots-eq*: *trace-roots* = *map snd f-fl*

and *composition-roots-eq*: *composition-roots* = *map snd fl*

and *outcome*:

Some (*result'*, *final-state'*) ∈

set-dist (*execute*

(*ntimes*

(*verifier-query-round-program fr f-fl trace-final as fl*
composition-final)

rounds)

query-state)

and *len-raw*: *length raw-idxs* = *rounds*

and *query-idxs-eq*:

query-idxs = *map* (*λraw. index (to-nat raw)*) *raw-idxs*

and *len-query-chunks*: *length query-chunks* = *rounds*

and *transcript-query*:

PTranscript query-state =

List.concat query-chunks @ PTranscript final-state'

and *query-chunk*:

Λi. i < rounds ⇒

verifier-query-round-chunk (*query-idxs* ! *i*)

(*map snd f-fl*) (*map snd fl*) (*query-chunks* ! *i*)

and *replay-lookup*:

Λi. i < rounds ⇒

fmlookup (*HashMap final-state'*)

(*QueryIndexChallenge* (*PQueryCounter query-state* + *i*)

(*state-after-query-chunks* (*PState query-state*) *query-chunks i*)) =

Some (*raw-idxs* ! *i*)

and *recorded*:

Λi. i < rounds ⇒

query-round-fri-layer-transcripts (*query-idxs* ! *i*)

trace-roots composition-roots (*query-chunks* ! *i*)

(*trace-round-layers* ! *i*) (*composition-round-layers* ! *i*)

unfolding *accepted-fri-opening-transcript-def* **by** *blast*

have *result-eq*: *result'* = *result*

using *out-def out-eq* **by** *simp*

have *final-state-eq*: *final-state'* = *final-state*

using *out-def out-eq* **by** *simp*

have *outcome'*:

Some (*result*, *final-state*) ∈

set-dist (*execute*

(*ntimes*

(*verifier-query-round-program fr f-fl trace-final as fl*

```

      composition-final)
    rounds)
  query-state)
  using outcome result-eq final-state-eq by simp
have f-fl-nonempty: f-fl ≠ []
  using trace-nonempty trace-roots-eq by auto
then obtain bf f-fl-tail where f-fl-cons: f-fl = bf # f-fl-tail
  by (cases f-fl) auto
obtain b rt where bf-eq: bf = (b, rt)
  by (cases bf)
have f-fl-eq: f-fl = (b, rt) # f-fl-tail
  using f-fl-cons bf-eq by simp
have rt-eq: rt = trace-roots ! 0
  using f-fl-cons bf-eq trace-roots-eq by simp
let ?round =
  verifier-query-round-program fr f-fl trace-final as fl composition-final
from query-opening-ntimes-outcome-decomp-at[OF outcome' round-bound]
obtain prefix x suffix s-i s-suc where
  prefix:
    Some (prefix, s-i) ∈
      set-dist (execute (ntimes ?round i) query-state)
  and round:
    Some (x, s-suc) ∈ set-dist (execute ?round s-i)
  and suffix:
    Some (suffix, final-state) ∈
      set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
  by blast
have x-eq: x = ()
  by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
  fix prefix-query-idxs prefix-chunks
  assume len-prefix-idxs: length prefix-query-idxs = i
  and len-prefix-chunks: length prefix-chunks = i
  and transcript-prefix:
    PTranscript query-state =
      List.concat prefix-chunks @ PTranscript s-i
  and state-i:
    PState s-i =
      state-after-query-chunks (PState query-state) prefix-chunks i
  and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
  and prefix-ext: query-state ≤ s-i
  and prefix-rounds:
     $\bigwedge k. k < i \implies$ 
      verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
  from round[unfolded x-eq verifier-query-round-program-alt-def]
  obtain raw s0 where

```

```

raw-out:
  Some (raw, s0) ∈
    set-dist (execute receive-query-index-challenge s-i)
and after:
  Some ((), s-suc) ∈
    set-dist (execute
      (verifier-query-round-after-index-program fr f-fl trace-final as
        fl composition-final raw) s0)
  by (auto elim!: set-dist-bindE)
have tr-s0: PTranscript s0 = PTranscript s-i
  using receive-query-index-challenge-outcome[OF raw-out] by simp
show ?thesis
proof (rule verifier-query-round-program-outcome[OF round[unfolded x-eq]])
  fix raw' idx chunk
  assume idx-eq: idx = index (to-nat raw')
  and chunk-shape:
    verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
    chunk
  and transcript-round:
    PTranscript s-i = chunk @ PTranscript s-suc
  and state-suc:
    PState s-suc = foldl concat (PState s-i) chunk
  and round-ext: s-i ≤ s-suc
  and counter-suc:
    PQueryCounter s-suc = Suc (PQueryCounter s-i)
  and lookup-suc:
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
      Some raw'
  have raw-eq': raw = raw'
  proof -
    have lookup-s0:
      fmlookup (HashMap s0)
        (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
        Some raw
      using receive-query-index-challenge-outcome[OF raw-out] by simp
    have s0-suc: s0 ≤ s-suc
      using verifier-query-round-after-index-program-outcome[OF after]
      by simp
    have lookup-suc-raw:
      fmlookup (HashMap s-suc)
        (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
        Some raw
      using hash-extension-lookup[OF lookup-s0 s0-suc] .
    show ?thesis
      using lookup-suc lookup-suc-raw by simp
  qed
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment

```

```

[OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-idxs:
  length suffix-query-idxs = rounds - Suc i
and len-suffix-chunks:
  length suffix-chunks = rounds - Suc i
and transcript-suffix:
  PTranscript s-suc =
    List.concat suffix-chunks @ PTranscript final-state
and suffix-state:
  PState final-state =
    state-after-query-chunks (PState s-suc) suffix-chunks
    (rounds - Suc i)
and suffix-counter:
  PQueryCounter final-state =
    PQueryCounter s-suc + (rounds - Suc i)
and suffix-ext: s-suc ≤ final-state
and suffix-rounds:
   $\bigwedge k. k < \text{rounds} - \text{Suc } i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)
    (map snd f-fl) (map snd fl) (suffix-chunks ! k)
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks round-bound by simp
have selected-chunk-lens:
   $\bigwedge k. k < \text{rounds} \implies$ 
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
proof -
  fix k
  assume k-bound: k < rounds
  show
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
proof (cases k < i)
  case True
  have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length (prefix-chunks ! k) =
      verifier-query-round-transcript-length
        (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
      [OF prefix-rounds[OF True]])
  then show ?thesis
    using nth-eq
    by (simp add:

```



```

      verifier-query-round-transcript-length-index-irrelevant)
next
  case False
  show ?thesis
  proof (cases k = i)
    case True
    have nth-eq: ?selected-chunks ! k = chunk
      using True len-prefix-chunks by (simp add: nth-append)
    have
      length chunk =
        verifier-query-round-transcript-length idx
          (map snd f-fl) (map snd fl)
      by (rule verifier-query-round-chunk-length[OF chunk-shape])
    then show ?thesis
      using nth-eq
      by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
  next
  case False
  let ?k = k - Suc i
  have ge: Suc i ≤ k
    using ⟨¬ k < i⟩ False by linarith
  have k-bound!: ?k < rounds - Suc i
    using ge k-bound by linarith
  have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k
    using ge len-prefix-chunks by (simp add: nth-append)
  have
    length (suffix-chunks ! ?k) =
      verifier-query-round-transcript-length
        (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
      [OF suffix-rounds[OF k-bound']])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
  qed
qed
qed
have query-chunk-lens:
   $\bigwedge k. k < rounds \implies$ 
  length (query-chunks ! k) =
    verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
  using query-chunk verifier-query-round-chunk-length by blast
have selected-transcript:
  PTranscript query-state =
    List.concat ?selected-chunks @ PTranscript final-state
  using transcript-prefix transcript-round transcript-suffix by simp

```

```

have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
  using selected-transcript transcript-query final-state-eq by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
     $\bigwedge k. k < \text{length } ?\text{selected-chunks} \implies$ 
       $\text{length } (?selected-chunks ! k) = \text{length } (query-chunks ! k)$ 
    using selected-chunk-lens query-chunk-lens len-selected-chunks
    by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks  $\wedge$  ( [] :: 'f list) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
      [OF len-eq same-lens concat-eq-empty])
  then show ?thesis
    by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix
      [OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw'
proof -
  have lookup-suc':
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
      Some raw'
    using lookup-suc state-i counter-i by simp
  have lookup-final-prefix:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
      Some raw'
    using lookup-suc state-i counter-i by simp

```

```

    prefix-chunks i)) =
    Some raw'
    using hash-extension-lookup[OF lookup-suc' suffix-ext] .
  show ?thesis
    using lookup-final-prefix prefix-state-eq by simp
qed
have raw-idx-eq: raw' = raw-idxs ! i
  using lookup-final replay-lookup[OF round-bound] final-state-eq
  by simp
have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-idx-eq query-idxs-eq len-raw round-bound by simp
have selected-i: ?selected-chunks ! i = chunk
  using len-prefix-chunks by (simp add: nth-append)
have chunk-i-eq: chunk = query-chunks ! i
  using chunks-eq selected-i by simp
have recorded-i:
  query-round-fri-layer-transcripts (index (to-nat raw))
    (map snd f-fl) (map snd fl) chunk (trace-round-layers ! i)
    (composition-round-layers ! i)
  using recorded[OF round-bound] idx-eq idx-eq-query raw-eq'
    chunk-i-eq trace-roots-eq composition-roots-eq
  by simp
have tr-s0-suc:
  PTranscript s0 = chunk @ PTranscript s-suc
  using tr-s0 transcript-round by simp
have auth-suc:
  fri-layer-chunk-authenticated rt (clength * scale)
    (index (to-nat raw)) (trace-round-layers ! i ! 0) s-suc
  by (rule
    verifier-query-round-after-index-trace-base-recorded-chunk-authenticated
    [OF f-fl-eq after recorded-i tr-s0-suc])
have auth-final:
  fri-layer-chunk-authenticated rt (clength * scale)
    (index (to-nat raw)) (trace-round-layers ! i ! 0)
    final-state
  by (rule fri-layer-chunk-authenticated-mono[OF auth-suc suffix-ext])
have raw-idx-query: index (to-nat raw) = query-idxs ! i
  using idx-eq idx-eq-query raw-eq' by simp
show ?thesis
  using auth-final rt-eq raw-idx-query by simp
qed
qed
qed
qed

```

lemma *accepted-fri-opening-transcript-composition-base-recorded-chunk-authenticated-at:*
assumes *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs

```

    trace-round-layers composition-round-layers
  and out-eq: out = Some (result, final-state)
  and round-bound: i < rounds
  and composition-nonempty: 0 < length composition-roots
shows
  fri-layer-chunk-authenticated (composition-roots ! 0) (clength * scale)
    (query-idxs ! i) (composition-round-layers ! i ! 0) final-state
proof -
  from fri-openings obtain result' final-state' fr f-fl as fl query-state
    raw-idxs query-chunks where
    out-def: out = Some (result', final-state')
  and trace-roots-eq: trace-roots = map snd f-fl
  and composition-roots-eq: composition-roots = map snd fl
  and outcome:
    Some (result', final-state') ∈
      set-dist (execute
        (ntimes
          (verifier-query-round-program fr f-fl trace-final as fl
            composition-final)
          rounds)
        query-state)
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state'
  and query-chunk:
    ∧ i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
        (map snd f-fl) (map snd fl) (query-chunks ! i)
  and replay-lookup:
    ∧ i. i < rounds ⇒
      fmlookup (HashMap final-state')
        (QueryIndexChallenge (PQueryCounter query-state + i)
          (state-after-query-chunks (PState query-state) query-chunks i)) =
        Some (raw-idxs ! i)
  and recorded:
    ∧ i. i < rounds ⇒
      query-round-fri-layer-transcripts (query-idxs ! i)
        trace-roots composition-roots (query-chunks ! i)
        (trace-round-layers ! i) (composition-round-layers ! i)
  unfolding accepted-fri-opening-transcript-def by blast
  have result-eq: result' = result
  using out-def out-eq by simp
  have final-state-eq: final-state' = final-state
  using out-def out-eq by simp
  have outcome':

```

```

Some (result, final-state) ∈
  set-dist (execute
    (ntimes
      (verifier-query-round-program fr f-fl trace-final as fl
        composition-final)
      rounds)
    query-state)
using outcome result-eq final-state-eq by simp
have fl-nonempty: fl ≠ []
  using composition-nonempty composition-roots-eq by auto
then obtain bf fl-tail where fl-cons: fl = bf # fl-tail
  by (cases fl) auto
obtain b rt where bf-eq: bf = (b, rt)
  by (cases bf)
have fl-eq: fl = (b, rt) # fl-tail
  using fl-cons bf-eq by simp
have rt-eq: rt = composition-roots ! 0
  using fl-cons bf-eq composition-roots-eq by simp
let ?round =
  verifier-query-round-program fr f-fl trace-final as fl composition-final
from query-opening-ntimes-outcome-decomp-at[OF outcome' round-bound]
obtain prefix x suffix s-i s-suc where
  prefix:
    Some (prefix, s-i) ∈
      set-dist (execute (ntimes ?round i) query-state)
  and round:
    Some (x, s-suc) ∈ set-dist (execute ?round s-i)
  and suffix:
    Some (suffix, final-state) ∈
      set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
  by blast
have x-eq: x = ()
  by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
  fix prefix-query-idxs prefix-chunks
  assume len-prefix-idxs: length prefix-query-idxs = i
  and len-prefix-chunks: length prefix-chunks = i
  and transcript-prefix:
    PTranscript query-state =
      List.concat prefix-chunks @ PTranscript s-i
  and state-i:
    PState s-i =
      state-after-query-chunks (PState query-state) prefix-chunks i
  and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
  and prefix-ext: query-state ≤ s-i
  and prefix-rounds:
     $\bigwedge k. k < i \implies$ 

```

```

    verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
from round[unfolded x-eq verifier-query-round-program-alt-def]
obtain raw s0 where
  raw-out:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s-i)
and after:
    Some ((), s-suc) ∈
      set-dist (execute
        (verifier-query-round-after-index-program fr f-fl trace-final as
          fl composition-final raw) s0)
by (auto elim!: set-dist-bindE)
have tr-s0: PTranscript s0 = PTranscript s-i
  using receive-query-index-challenge-outcome[OF raw-out] by simp
show ?thesis
proof (rule verifier-query-round-program-outcome[OF round[unfolded x-eq]])
  fix raw' idx chunk
  assume idx-eq: idx = index (to-nat raw')
  and chunk-shape:
    verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
      chunk
  and transcript-round:
    PTranscript s-i = chunk @ PTranscript s-suc
  and state-suc:
    PState s-suc = foldl concat (PState s-i) chunk
  and round-ext: s-i ≤ s-suc
  and counter-suc:
    PQueryCounter s-suc = Suc (PQueryCounter s-i)
  and lookup-suc:
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
      Some raw'
  have raw-eq': raw = raw'
  proof -
    have lookup-s0:
      fmlookup (HashMap s0)
        (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
        Some raw
      using receive-query-index-challenge-outcome[OF raw-out] by simp
    have s0-suc: s0 ≤ s-suc
      using verifier-query-round-after-index-program-outcome[OF after]
      by simp
    have lookup-suc-raw:
      fmlookup (HashMap s-suc)
        (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
        Some raw
      using hash-extension-lookup[OF lookup-s0 s0-suc] .
  show ?thesis

```

```

    using lookup-suc lookup-suc-raw by simp
qed
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
  fix suffix-query-idxs suffix-chunks
  assume len-suffix-idxs:
    length suffix-query-idxs = rounds - Suc i
  and len-suffix-chunks:
    length suffix-chunks = rounds - Suc i
  and transcript-suffix:
    PTranscript s-suc =
      List.concat suffix-chunks @ PTranscript final-state
  and suffix-state:
    PState final-state =
      state-after-query-chunks (PState s-suc) suffix-chunks
      (rounds - Suc i)
  and suffix-counter:
    PQueryCounter final-state =
      PQueryCounter s-suc + (rounds - Suc i)
  and suffix-ext: s-suc ≤ final-state
  and suffix-rounds:
     $\bigwedge k. k < \text{rounds} - \text{Suc } i \implies$ 
      verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)
  let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
  have len-selected-chunks: length ?selected-chunks = rounds
    using len-prefix-chunks len-suffix-chunks round-bound by simp
  have selected-chunk-lens:
     $\bigwedge k. k < \text{rounds} \implies$ 
      length (?selected-chunks ! k) =
        verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  proof -
    fix k
    assume k-bound: k < rounds
    show
      length (?selected-chunks ! k) =
        verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
    proof (cases k < i)
      case True
      have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
        using True len-prefix-chunks by (simp add: nth-append)
      have
        length (prefix-chunks ! k) =
          verifier-query-round-transcript-length
            (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
        by (rule verifier-query-round-chunk-length

```

```

      [OF prefix-rounds[OF True]])
    then show ?thesis
      using nth-eq
      by (simp add:
          verifier-query-round-transcript-length-index-irrelevant)
  next
    case False
    show ?thesis
    proof (cases k = i)
      case True
      have nth-eq: ?selected-chunks ! k = chunk
        using True len-prefix-chunks by (simp add: nth-append)
      have
        length chunk =
          verifier-query-round-transcript-length idx
            (map snd f-fl) (map snd fl)
        by (rule verifier-query-round-chunk-length[OF chunk-shape])
      then show ?thesis
        using nth-eq
        by (simp add:
            verifier-query-round-transcript-length-index-irrelevant)
    next
      case False
      let ?k = k - Suc i
      have ge: Suc i ≤ k
        using <¬ k < i> False by linarith
      have k-bound!: ?k < rounds - Suc i
        using ge k-bound by linarith
      have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k
        using ge len-prefix-chunks by (simp add: nth-append)
      have
        length (suffix-chunks ! ?k) =
          verifier-query-round-transcript-length
            (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
        by (rule verifier-query-round-chunk-length
            [OF suffix-rounds[OF k-bound']])
      then show ?thesis
        using nth-eq
        by (simp add:
            verifier-query-round-transcript-length-index-irrelevant)
    qed
  qed
qed
have query-chunk-lens:
   $\bigwedge k. k < rounds \implies$ 
  length (query-chunks ! k) =
    verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
  using query-chunk verifier-query-round-chunk-length by blast

```



```

have selected-transcript:
  PTranscript query-state =
    List.concat ?selected-chunks @ PTranscript final-state
  using transcript-prefix transcript-round transcript-suffix by simp
have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
  using selected-transcript transcript-query final-state-eq by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
     $\bigwedge k. k < \text{length } ?\text{selected-chunks} \implies$ 
       $\text{length } (?selected-chunks ! k) = \text{length } (query-chunks ! k)$ 
    using selected-chunk-lens query-chunk-lens len-selected-chunks
    by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks  $\wedge$  ( [] :: 'f list) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
      [OF len-eq same-lens concat-eq-empty])
  then show ?thesis
    by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix
      [OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw'
proof -
  have lookup-suc':
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
      Some raw'
    using lookup-suc state-i counter-i by simp

```

```

have lookup-final-prefix:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state)
        prefix-chunks i)) =
    Some raw'
  using hash-extension-lookup[OF lookup-suc' suffix-ext] .
show ?thesis
  using lookup-final-prefix prefix-state-eq by simp
qed
have raw-idx-eq: raw' = raw-idxs ! i
  using lookup-final replay-lookup[OF round-bound] final-state-eq
  by simp
have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-idx-eq query-idxs-eq len-row round-bound by simp
have selected-i: ?selected-chunks ! i = chunk
  using len-prefix-chunks by (simp add: nth-append)
have chunk-i-eq: chunk = query-chunks ! i
  using chunks-eq selected-i by simp
have recorded-i:
  query-round-fri-layer-transcripts (index (to-nat raw))
    (map snd f-fl) (map snd fl) chunk (trace-round-layers ! i)
    (composition-round-layers ! i)
  using recorded[OF round-bound] idx-eq idx-eq-query raw-eq'
    chunk-i-eq trace-roots-eq composition-roots-eq
  by simp
have tr-s0-suc:
  PTranscript s0 = chunk @ PTranscript s-suc
  using tr-s0 transcript-round by simp
have auth-suc:
  fri-layer-chunk-authenticated rt (clength * scale)
    (index (to-nat raw)) (composition-round-layers ! i ! 0) s-suc
  by (rule
    verifier-query-round-after-index-composition-base-recorded-chunk-authenticated
      [OF fl-eq after recorded-i tr-s0-suc])
have auth-final:
  fri-layer-chunk-authenticated rt (clength * scale)
    (index (to-nat raw)) (composition-round-layers ! i ! 0)
    final-state
  by (rule fri-layer-chunk-authenticated-mono[OF auth-suc suffix-ext])
have raw-idx-query: index (to-nat raw) = query-idxs ! i
  using idx-eq idx-eq-query raw-eq' by simp
show ?thesis
  using auth-final rt-eq raw-idx-query by simp
qed
qed
qed
qed

```

end

end

theory *Soundness-FRI-Raw-Layer-Bounds*
 imports *Soundness-FRI-Authenticated-Route*
begin

Small reusable bounds for FRI layer indices derived from accepted verifier transcripts.
These keep later replay proofs from repeatedly rebuilding the same *fri-layer-raw-bound*
arithmetic.

context *soundness*
begin

lemma *accepted-fri-opening-transcript-trace-raw-layer-bound*:

assumes *fri-openings*:

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
 dg composition-roots composition-bs composition-final query-idxs
 trace-round-layers composition-round-layers

and *layer-bound*: $j < \text{length } \text{trace-roots}$

shows

$\bigwedge \text{raw.}$
 $0 < \text{fri-layer-lengths } (\text{length } \text{trace-roots}) (\text{clength} * \text{scale}) ! j \wedge$
 $\text{fri-layer-indices } (\text{length } \text{trace-roots}) (\text{index } (\text{to-nat } \text{raw}))$
 $(\text{clength} * \text{scale}) ! j <$
 $\text{fri-layer-lengths } (\text{length } \text{trace-roots}) (\text{clength} * \text{scale}) ! j$

proof –

obtain N **where** *eval-power*: $\text{clength} * \text{scale} = 2^N$

using *eval-domain-length-power* **by** *blast*

have *trace-roots-le*: $\text{length } \text{trace-roots} \leq N$

proof –

have *len*: $\text{length } \text{trace-roots} = \text{ceil-log } \text{clength}$

using *accepted-fri-opening-transcript-shapes(1)[OF fri-openings]* .

have $\text{clength} \leq \text{clength} * \text{scale}$

using *scale-pos* **by** *simp*

also have $\dots = 2^N$

by (*rule eval-power*)

finally have $\text{ceil-log } \text{clength} \leq N$

by (*rule ceil-log-le-power*)

then show *?thesis*

using *len* **by** *simp*

qed

show $\bigwedge \text{raw.}$

$0 < \text{fri-layer-lengths } (\text{length } \text{trace-roots}) (\text{clength} * \text{scale}) ! j \wedge$

$\text{fri-layer-indices } (\text{length } \text{trace-roots}) (\text{index } (\text{to-nat } \text{raw}))$

$(\text{clength} * \text{scale}) ! j <$

$\text{fri-layer-lengths } (\text{length } \text{trace-roots}) (\text{clength} * \text{scale}) ! j$

by (*rule fri-layer-raw-bound[OF layer-bound eval-power trace-roots-le]*)

qed

lemma *accepted-fri-opening-transcript-composition-raw-layer-bound:*

assumes *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers

and *degree-bound: to-nat dg ≤ maxDegree*

and *layer-bound: j < length composition-roots*

shows

$\bigwedge_{\text{raw.}}$

$0 < \text{fri-layer-lengths } (\text{length } \text{composition-roots}) (\text{clength} * \text{scale}) ! j \wedge$
 $\text{fri-layer-indices } (\text{length } \text{composition-roots}) (\text{index } (\text{to-nat raw}))$
 $(\text{clength} * \text{scale}) ! j <$
 $\text{fri-layer-lengths } (\text{length } \text{composition-roots}) (\text{clength} * \text{scale}) ! j$

proof –

obtain *N* **where** *eval-power: clength * scale = 2 ^ N*

using *eval-domain-length-power* **by** *blast*

have *composition-roots-le: length composition-roots ≤ N*

proof –

have *len: length composition-roots = ceil-log (to-nat dg + 1)*

using *accepted-fri-opening-transcript-shapes(3)[OF fri-openings]* .

have *to-nat dg + 1 ≤ clength * scale*

using *degree-bound maxDegree-less-eval-domain* **by** *linarith*

also have $\dots = 2 ^ N$

by *(rule eval-power)*

finally have *ceil-log (to-nat dg + 1) ≤ N*

by *(rule ceil-log-le-power)*

then show *?thesis*

using *len* **by** *simp*

qed

show $\bigwedge_{\text{raw.}}$

$0 < \text{fri-layer-lengths } (\text{length } \text{composition-roots})$
 $(\text{clength} * \text{scale}) ! j \wedge$
 $\text{fri-layer-indices } (\text{length } \text{composition-roots}) (\text{index } (\text{to-nat raw}))$
 $(\text{clength} * \text{scale}) ! j <$
 $\text{fri-layer-lengths } (\text{length } \text{composition-roots}) (\text{clength} * \text{scale}) ! j$

by *(rule fri-layer-raw-bound*

[OF layer-bound eval-power composition-roots-le])

qed

lemma *accepted-fri-opening-transcript-trace-step-with-authenticated-chunk-at-raw:*

assumes *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers

and *out-eq: out = Some (result, final-state)*

and *round-bound: i < rounds*

and *layer-bound: j < length trace-roots*

shows

*generic-fri-transcript-step-with-authenticated-chunk trace-roots
trace-bs query-idxs trace-round-layers final-state i j*

by (rule *accepted-fri-opening-transcript-trace-step-with-authenticated-chunk-at*
[*OF fri-openings out-eq round-bound layer-bound*])
(rule *accepted-fri-opening-transcript-trace-raw-layer-bound*
[*OF fri-openings layer-bound*])

lemma *accepted-fri-opening-transcript-composition-step-with-authenticated-chunk-at-raw:*

assumes *fri-openings:*

*accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers*

and *out-eq:* *out = Some (result, final-state)*

and *degree-bound:* *to-nat dg ≤ maxDegree*

and *round-bound:* *i < rounds*

and *layer-bound:* *j < length composition-roots*

shows

*generic-fri-transcript-step-with-authenticated-chunk composition-roots
composition-bs query-idxs composition-round-layers final-state i j*

by (rule
accepted-fri-opening-transcript-composition-step-with-authenticated-chunk-at
[*OF fri-openings out-eq round-bound layer-bound*])
(rule *accepted-fri-opening-transcript-composition-raw-layer-bound*
[*OF fri-openings degree-bound layer-bound*])

end

end

theory *Soundness-FRI-Final-Step-Replay*

imports *Soundness-FRI-Raw-Layer-Bounds*

begin

Selected final-step replay for trace FRI openings.

This layer lifts the monadic replay fact for the last trace FRI fold step through the accepted FRI transcript. It keeps the result local to sampled verifier evidence and does not reconstruct complete FRI tables.

context *soundness*

begin

lemma *accepted-fri-opening-transcript-trace-final-step-evidence-at:*

assumes *fri-openings:*

*accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers*

and *out-eq:* *out = Some (result, final-state)*

and *round-bound:* *i < rounds*

and *nonempty*: $0 < \text{length trace-roots}$
and *raw-layer*:
 $\wedge_{\text{raw.}}$
 $0 < \text{fri-layer-lengths } (\text{length trace-roots}) (\text{clength} * \text{scale}) !$
 $(\text{length trace-roots} - 1) \wedge$
 $\text{fri-layer-indices } (\text{length trace-roots}) (\text{index } (\text{to-nat raw}))$
 $(\text{clength} * \text{scale}) ! (\text{length trace-roots} - 1) <$
 $\text{fri-layer-lengths } (\text{length trace-roots}) (\text{clength} * \text{scale}) !$
 $(\text{length trace-roots} - 1)$
obtains *actual-chunk* *axp* *axp-path* *axn* *axn-path* **where**
fri-layer-chunk-authenticated
 $(\text{trace-roots} ! (\text{length trace-roots} - 1))$
 $(\text{fri-evidence-layer-len trace-roots } (\text{length trace-roots} - 1))$
 $(\text{fri-evidence-layer-idx trace-roots query-idxs } i$
 $(\text{length trace-roots} - 1))$
actual-chunk final-state
fri-layer-step-evidence
 $(\text{trace-roots} ! (\text{length trace-roots} - 1))$
 $(\text{trace-bs} ! (\text{length trace-roots} - 1))$
 $(\text{fri-evidence-layer-len trace-roots } (\text{length trace-roots} - 1))$
 $(\text{fri-evidence-layer-idx trace-roots query-idxs } i$
 $(\text{length trace-roots} - 1))$
 $(2 \wedge (\text{length trace-roots} - 1))$
 $(\text{fri-sibling-index}$
 $(\text{fri-evidence-layer-len trace-roots } (\text{length trace-roots} - 1))$
 $(\text{fri-evidence-layer-idx trace-roots query-idxs } i$
 $(\text{length trace-roots} - 1)))$
axp *axp-path* *axn* *axn-path*
 $(\text{fri-evidence-next-idx trace-roots query-idxs } i$
 $(\text{length trace-roots} - 1))$
trace-final actual-chunk
proof –
from *fri-openings* **obtain** *result'* *final-state'* *fr f-fl as fl query-state*
raw-idxs query-chunks **where**
out-def: *out* = *Some (result', final-state')*
and *trace-roots-eq*: *trace-roots* = *map snd f-fl*
and *trace-bs-eq*: *trace-bs* = *map fst f-fl*
and *composition-roots-eq*: *composition-roots* = *map snd fl*
and *query-out*:
 $\text{Some } (\text{result}', \text{final-state}') \in$
 set-dist
 $(\text{execute}$
 $(\text{ntimes}$
 $(\text{verifier-query-round-program fr f-fl trace-final as fl}$
 $\text{composition-final})$
 $\text{rounds})$
 $\text{query-state})$
and *len-raw*: *length raw-idxs* = *rounds*
and *query-idxs-eq*:

```

    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state'
and query-chunk:
    ∧i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
      trace-roots composition-roots (query-chunks ! i)
and replay-lookup:
    ∧i. i < rounds ⇒
      fmllookup (HashMap final-state')
        (QueryIndexChallenge (PQueryCounter query-state + i)
          (state-after-query-chunks (PState query-state) query-chunks i)) =
        Some (raw-idxs ! i)
unfolding accepted-fri-opening-transcript-def by blast
have result-eq: result' = result
using out-def out-eq by simp
have final-state-eq: final-state' = final-state
using out-def out-eq by simp
have query-out':
  Some (result, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl trace-final as fl
            composition-final)
            rounds)
          query-state)
using query-out result-eq final-state-eq by simp
have f-fl-nonempty: 0 < length f-fl
using nonempty trace-roots-eq by simp
have raw-layer-f:
  ∧raw.
    0 < fri-layer-lengths (length f-fl) (clength * scale) !
    (length f-fl - 1) ∧
    fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! (length f-fl - 1) <
    fri-layer-lengths (length f-fl) (clength * scale) !
    (length f-fl - 1)
using raw-layer trace-roots-eq by simp
from ntimes-verifier-query-rounds-selected-trace-fri-last-step-with-prefix-at
  [OF query-out' round-bound f-fl-nonempty raw-layer-f]
obtain raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
  suffix-chunks actual-chunk axp axp-path axn axn-path where
  idx-eq: idx = index (to-nat raw)
and len-prefix-idxs: length prefix-query-idxs = i
and len-prefix-chunks: length prefix-chunks = i
and len-suffix-idxs: length suffix-query-idxs = rounds - Suc i

```

```

and len-suffix-chunks: length suffix-chunks = rounds - Suc i
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl) chunk
and prefix-rounds:
   $\bigwedge k. k < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
and suffix-rounds:
   $\bigwedge k. k < \text{rounds} - \text{Suc } i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)
and selected-transcript:
  PTranscript query-state =
    List.concat (prefix-chunks @ chunk # suffix-chunks) @
      PTranscript final-state
and auth:
  fri-layer-chunk-authenticated
    (snd (f-fl ! (length f-fl - 1)))
    (fri-layer-lengths (length f-fl) (clength * scale) !
      (length f-fl - 1))
    (fri-layer-indices (length f-fl) idx (clength * scale) !
      (length f-fl - 1))
    actual-chunk final-state
and step:
  fri-layer-step-evidence
    (snd (f-fl ! (length f-fl - 1)))
    (fst (f-fl ! (length f-fl - 1)))
    (fri-layer-lengths (length f-fl) (clength * scale) !
      (length f-fl - 1))
    (fri-layer-indices (length f-fl) idx (clength * scale) !
      (length f-fl - 1))
    ( $2 \wedge (\text{length } f\text{-fl} - 1)$ )
    (fri-sibling-index
      (fri-layer-lengths (length f-fl) (clength * scale) !
        (length f-fl - 1))
      (fri-layer-indices (length f-fl) idx (clength * scale) !
        (length f-fl - 1)))
    axp axp-path axn axn-path
    ((fri-layer-indices (length f-fl) idx (clength * scale) !
      (length f-fl - 1)) mod
      ((fri-layer-lengths (length f-fl) (clength * scale) !
        (length f-fl - 1)) div 2))
    trace-final actual-chunk
and selected-lookup:
  fmllookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) prefix-chunks i)) =
    Some raw
by (rule

```



```

    ntimes-verifier-query-rounds-selected-trace-fri-last-step-with-prefix-at
    [OF query-out' round-bound f-fl-nonempty raw-layer-f])
  (rule that, assumption+)
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks round-bound by simp
have selected-chunk-lens:
   $\wedge k. k < rounds \implies$ 
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
proof -
  fix k
  assume k-bound: k < rounds
  show
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
  proof (cases k < i)
    case True
    have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
      using True len-prefix-chunks by (simp add: nth-append)
    have
      length (prefix-chunks ! k) =
        verifier-query-round-transcript-length
          (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
      by (rule verifier-query-round-chunk-length[OF prefix-rounds[OF True]])
    then show ?thesis
      using nth-eq
      by (simp add: verifier-query-round-transcript-length-index-irrelevant)
  next
    case False
    show ?thesis
    proof (cases k = i)
      case True
      have nth-eq: ?selected-chunks ! k = chunk
        using True len-prefix-chunks by (simp add: nth-append)
      have
        length chunk =
          verifier-query-round-transcript-length idx
            (map snd f-fl) (map snd fl)
        by (rule verifier-query-round-chunk-length[OF chunk-shape])
      then show ?thesis
        using nth-eq
        by (simp add: verifier-query-round-transcript-length-index-irrelevant)
    next
      case False
      let ?k = k - Suc i
      have ge: Suc i  $\leq$  k

```

```

    using  $\neg k < i$  False by linarith
  have  $k\text{-bound}'$ :  $?k < \text{rounds} - \text{Suc } i$ 
    using  $ge\ k\text{-bound}$  by linarith
  have  $nth\text{-eq}$ :  $?selected\text{-chunks} ! k = \text{suffix}\text{-chunks} ! ?k$ 
    using  $ge\ len\text{-prefix}\text{-chunks}$  by (simp add:  $nth\text{-append}$ )
  have
    length (suffix-chunks !  $?k$ ) =
      verifier-query-round-transcript-length
        (suffix-query-idxs !  $?k$ ) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
        [OF suffix-rounds [OF  $k\text{-bound}'$ ]])
  then show  $?thesis$ 
    using  $nth\text{-eq}$ 
    by (simp add: verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-lens:
 $\bigwedge k. k < \text{rounds} \implies$ 
  length (query-chunks !  $k$ ) =
    verifier-query-round-transcript-length (query-idxs !  $k$ )
      (map snd f-fl) (map snd fl)
proof -
  fix  $k$ 
  assume  $k\text{-bound}$ :  $k < \text{rounds}$ 
  have
    verifier-query-round-chunk (query-idxs !  $k$ )
      (map snd f-fl) (map snd fl) (query-chunks !  $k$ )
    using query-chunk [OF  $k\text{-bound}$ ] trace-roots-eq composition-roots-eq
    by simp
  then show
    length (query-chunks !  $k$ ) =
      verifier-query-round-transcript-length (query-idxs !  $k$ )
        (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length)
qed
have concat-eq:  $List.concat\ ?selected\text{-chunks} = List.concat\ query\text{-chunks}$ 
  using selected-transcript transcript-query final-state-eq by simp
have chunks-eq:  $?selected\text{-chunks} = query\text{-chunks}$ 
proof -
  have same-lens:
 $\bigwedge k. k < length\ ?selected\text{-chunks} \implies$ 
    length ( $?selected\text{-chunks} ! k$ ) = length (query-chunks !  $k$ )
    using selected-chunk-lens query-chunk-lens len-selected-chunks by simp
  have concat-eq-empty:
 $List.concat\ ?selected\text{-chunks} @ [] = List.concat\ query\text{-chunks}$ 
    using concat-eq by simp
  have len-eq:  $length\ ?selected\text{-chunks} = length\ query\text{-chunks}$ 
    using len-selected-chunks len-query-chunks by simp

```

```

have ?selected-chunks = query-chunks  $\wedge$  ( $[] :: 'f \text{ list}$ ) = []
  by (rule concat-append-eq-concat-same-chunk-lengths
    [OF len-eq same-len concat-eq-empty])
then show ?thesis by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof –
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix[OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have selected-lookup-global:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw
  using selected-lookup prefix-state-eq by simp
have raw-eq: raw = raw-idxs ! i
  using selected-lookup-global replay-lookup[OF round-bound]
    final-state-eq by simp
have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-eq query-idxs-eq len-raw round-bound by simp
have auth':
  fri-layer-chunk-authenticated
    (trace-roots ! (length trace-roots – 1))
    (fri-evidence-layer-len trace-roots (length trace-roots – 1))
    (fri-evidence-layer-idx trace-roots query-idxs i
      (length trace-roots – 1))
    actual-chunk final-state
  using auth idx-eq-query trace-roots-eq f-fl-nonempty
  unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
  by simp
have step':
  fri-layer-step-evidence
    (trace-roots ! (length trace-roots – 1))
    (trace-bs ! (length trace-roots – 1))
    (fri-evidence-layer-len trace-roots (length trace-roots – 1))
    (fri-evidence-layer-idx trace-roots query-idxs i
      (length trace-roots – 1))
    (2  $\wedge$  (length trace-roots – 1))
    (fri-sibling-index
      (fri-evidence-layer-len trace-roots (length trace-roots – 1))
      (fri-evidence-layer-idx trace-roots query-idxs i
        (length trace-roots – 1)))

```

```

    arp arp-path arn arn-path
    (fri-evidence-next-idx trace-roots query-idxs i
     (length trace-roots - 1))
    trace-final actual-chunk
using step idx-eq-query trace-roots-eq trace-bs-eq f-fl-nonempty
unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
    fri-evidence-next-idx-def
by simp
show ?thesis
    by (rule that[OF auth' step'])
qed

```

lemma *accepted-fri-opening-transcript-composition-final-step-evidence-at:*

assumes *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers

and *out-eq:* *out = Some (result, final-state)*

and *round-bound:* *i < rounds*

and *nonempty:* *0 < length composition-roots*

and *raw-layer:*

$\bigwedge_{\text{raw.}}$

*0 < fri-layer-lengths (length composition-roots) (clength * scale) !*
(length composition-roots - 1) $\wedge$
fri-layer-indices (length composition-roots) (index (to-nat raw))
*(clength * scale) ! (length composition-roots - 1) <*
*fri-layer-lengths (length composition-roots) (clength * scale) !*
(length composition-roots - 1)

obtains *actual-chunk arp arp-path arn arn-path* **where**

fri-layer-chunk-authenticated

(composition-roots ! (length composition-roots - 1))
(fri-evidence-layer-len composition-roots (length composition-roots - 1))
(fri-evidence-layer-idx composition-roots query-idxs i
(length composition-roots - 1))

actual-chunk final-state

fri-layer-step-evidence

(composition-roots ! (length composition-roots - 1))
(composition-bs ! (length composition-roots - 1))
(fri-evidence-layer-len composition-roots (length composition-roots - 1))
(fri-evidence-layer-idx composition-roots query-idxs i
(length composition-roots - 1))
(2 $\wedge$ (length composition-roots - 1))
(fri-sibling-index
(fri-evidence-layer-len composition-roots (length composition-roots - 1))
(fri-evidence-layer-idx composition-roots query-idxs i
(length composition-roots - 1)))

arp arp-path arn arn-path

(fri-evidence-next-idx composition-roots query-idxs i
(length composition-roots - 1))

composition-final actual-chunk

proof –

from *fri-openings* **obtain** *result'* *final-state'* *fr f-fl as fl query-state*
raw-idxs query-chunks **where**
out-def: *out* = *Some (result', final-state')*
and *trace-roots-eq*: *trace-roots* = *map snd f-fl*
and *composition-bs-eq*: *composition-bs* = *map fst fl*
and *composition-roots-eq*: *composition-roots* = *map snd fl*
and *query-out*:
Some (result', final-state') ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl trace-final as fl
composition-final)
rounds)
query-state)
and *len-raw*: *length raw-idxs* = *rounds*
and *query-idxs-eq*:
query-idxs = *map (λraw. index (to-nat raw)) raw-idxs*
and *len-query-chunks*: *length query-chunks* = *rounds*
and *transcript-query*:
PTranscript query-state =
List.concat query-chunks @ PTranscript final-state'
and *query-chunk*:
 $\bigwedge i. i < \text{rounds} \implies$
verifier-query-round-chunk (query-idxs ! i)
trace-roots composition-roots (query-chunks ! i)
and *replay-lookup*:
 $\bigwedge i. i < \text{rounds} \implies$
fmllookup (HashMap final-state')
(QueryIndexChallenge (PQueryCounter query-state + i)
(state-after-query-chunks (PState query-state) query-chunks i)) =
Some (raw-idxs ! i)
unfolding *accepted-fri-opening-transcript-def* **by** *blast*
have *result-eq*: *result'* = *result*
using *out-def out-eq* **by** *simp*
have *final-state-eq*: *final-state'* = *final-state*
using *out-def out-eq* **by** *simp*
have *query-out'*:
Some (result, final-state) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl trace-final as fl
composition-final)
rounds)
query-state)
using *query-out result-eq final-state-eq* **by** *simp*

```

have fl-nonempty:  $0 < \text{length } fl$ 
  using nonempty composition-roots-eq by simp
have raw-layer-fl:
   $\wedge$ raw.
     $0 < \text{fri-layer-lengths } (\text{length } fl) (\text{clength} * \text{scale}) !$ 
     $(\text{length } fl - 1) \wedge$ 
     $\text{fri-layer-indices } (\text{length } fl) (\text{index } (\text{to-nat } raw))$ 
     $(\text{clength} * \text{scale}) ! (\text{length } fl - 1) <$ 
     $\text{fri-layer-lengths } (\text{length } fl) (\text{clength} * \text{scale}) !$ 
     $(\text{length } fl - 1)$ 
  using raw-layer composition-roots-eq by simp
from ntimes-verifier-query-rounds-selected-composition-fri-last-step-with-prefix-at
  [OF query-out' round-bound fl-nonempty raw-layer-fl]
obtain raw idx prefix-query-idxs prefix-chunks chunk suffix-query-idxs
  suffix-chunks actual-chunk axp axp-path axn axn-path where
idx-eq:  $idx = \text{index } (\text{to-nat } raw)$ 
and len-prefix-idxs:  $\text{length } \text{prefix-query-idxs} = i$ 
and len-prefix-chunks:  $\text{length } \text{prefix-chunks} = i$ 
and len-suffix-idxs:  $\text{length } \text{suffix-query-idxs} = \text{rounds} - \text{Suc } i$ 
and len-suffix-chunks:  $\text{length } \text{suffix-chunks} = \text{rounds} - \text{Suc } i$ 
and chunk-shape:
  verifier-query-round-chunk idx ( $\text{map } \text{snd } f\text{-}fl$ ) ( $\text{map } \text{snd } fl$ ) chunk
and prefix-rounds:
   $\wedge k. k < i \implies$ 
    verifier-query-round-chunk ( $\text{prefix-query-idxs} ! k$ )
    ( $\text{map } \text{snd } f\text{-}fl$ ) ( $\text{map } \text{snd } fl$ ) ( $\text{prefix-chunks} ! k$ )
and suffix-rounds:
   $\wedge k. k < \text{rounds} - \text{Suc } i \implies$ 
    verifier-query-round-chunk ( $\text{suffix-query-idxs} ! k$ )
    ( $\text{map } \text{snd } f\text{-}fl$ ) ( $\text{map } \text{snd } fl$ ) ( $\text{suffix-chunks} ! k$ )
and selected-transcript:
  PTranscript query-state =
    List.concat ( $\text{prefix-chunks} @ \text{chunk} \# \text{suffix-chunks}$ ) @
    PTranscript final-state
and auth:
  fri-layer-chunk-authenticated
    ( $\text{snd } (fl ! (\text{length } fl - 1)))$ 
    ( $\text{fri-layer-lengths } (\text{length } fl) (\text{clength} * \text{scale}) !$ 
    ( $\text{length } fl - 1$ ))
    ( $\text{fri-layer-indices } (\text{length } fl) \text{idx} (\text{clength} * \text{scale}) !$ 
    ( $\text{length } fl - 1$ ))
    actual-chunk final-state
and step:
  fri-layer-step-evidence
    ( $\text{snd } (fl ! (\text{length } fl - 1)))$ 
    ( $\text{fst } (fl ! (\text{length } fl - 1)))$ 
    ( $\text{fri-layer-lengths } (\text{length } fl) (\text{clength} * \text{scale}) !$ 
    ( $\text{length } fl - 1$ ))
    ( $\text{fri-layer-indices } (\text{length } fl) \text{idx} (\text{clength} * \text{scale}) !$ 

```

```

    (length fl - 1))
  (2 ^ (length fl - 1))
  (fri-sibling-index
    (fri-layer-lengths (length fl) (clength * scale) !
      (length fl - 1))
    (fri-layer-indices (length fl) idx (clength * scale) !
      (length fl - 1)))
  arp arp-path arn arn-path
  ((fri-layer-indices (length fl) idx (clength * scale) !
    (length fl - 1)) mod
    ((fri-layer-lengths (length fl) (clength * scale) !
      (length fl - 1)) div 2))
  composition-final actual-chunk
and selected-lookup:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) prefix-chunks i)) =
    Some raw
by (rule
  ntimes-verifier-query-rounds-selected-composition-fri-last-step-with-prefix-at
    [OF query-out' round-bound fl-nonempty raw-layer-fl])
  (rule that, assumption+)
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks round-bound by simp
have selected-chunk-lens:
   $\bigwedge k. k < rounds \implies$ 
  length (?selected-chunks ! k) =
    verifier-query-round-transcript-length (query-idxs ! k)
    (map snd f-fl) (map snd fl)
proof -
  fix k
  assume k-bound: k < rounds
  show
  length (?selected-chunks ! k) =
    verifier-query-round-transcript-length (query-idxs ! k)
    (map snd f-fl) (map snd fl)
proof (cases k < i)
  case True
  have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
    using True len-prefix-chunks by (simp add: nth-append)
  have
  length (prefix-chunks ! k) =
    verifier-query-round-transcript-length
      (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length[OF prefix-rounds[OF True]])
  then show ?thesis
    using nth-eq
    by (simp add: verifier-query-round-transcript-length-index-irrelevant)

```

```

next
  case False
  show ?thesis
  proof (cases k = i)
    case True
    have nth-eq: ?selected-chunks ! k = chunk
      using True len-prefix-chunks by (simp add: nth-append)
    have
      length chunk =
        verifier-query-round-transcript-length idx
          (map snd f-fl) (map snd fl)
      by (rule verifier-query-round-chunk-length[OF chunk-shape])
    then show ?thesis
      using nth-eq
      by (simp add: verifier-query-round-transcript-length-index-irrelevant)
  next
  case False
  let ?k = k - Suc i
  have ge: Suc i ≤ k
    using <¬ k < i> False by linarith
  have k-bound': ?k < rounds - Suc i
    using ge k-bound by linarith
  have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k
    using ge len-prefix-chunks by (simp add: nth-append)
  have
    length (suffix-chunks ! ?k) =
      verifier-query-round-transcript-length
        (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
        [OF suffix-rounds[OF k-bound']])
  then show ?thesis
    using nth-eq
    by (simp add: verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-lens:
  ∧k. k < rounds ⟹
    length (query-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
proof -
  fix k
  assume k-bound: k < rounds
  have
    verifier-query-round-chunk (query-idxs ! k)
      (map snd f-fl) (map snd fl) (query-chunks ! k)
  using query-chunk[OF k-bound] trace-roots-eq composition-roots-eq
  by simp

```



```

then show
  length (query-chunks ! k) =
    verifier-query-round-transcript-length (query-idxs ! k)
    (map snd f-fl) (map snd fl)
  by (rule verifier-query-round-chunk-length)
qed
have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
  using selected-transcript transcript-query final-state-eq by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
     $\bigwedge k. k < \text{length } ?\text{selected-chunks} \implies$ 
    length (?selected-chunks ! k) = length (query-chunks ! k)
    using selected-chunk-lens query-chunk-lens len-selected-chunks by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks  $\wedge$  ([] :: 'f list) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
      [OF len-eq same-lens concat-eq-empty])
  then show ?thesis by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix[OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have selected-lookup-global:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw
  using selected-lookup prefix-state-eq by simp
have raw-eq: raw = raw-idxs ! i
  using selected-lookup-global replay-lookup[OF round-bound]
    final-state-eq by simp
have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-eq query-idxs-eq len-raw round-bound by simp
have auth':
  fri-layer-chunk-authenticated
    (composition-roots ! (length composition-roots - 1))

```

```

    (fri-evidence-layer-len composition-roots (length composition-roots - 1))
    (fri-evidence-layer-idx composition-roots query-idxs i
      (length composition-roots - 1))
    actual-chunk final-state
  using auth idx-eq-query composition-roots-eq fl-nonempty
  unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
  by simp
  have step':
    fri-layer-step-evidence
      (composition-roots ! (length composition-roots - 1))
      (composition-bs ! (length composition-roots - 1))
      (fri-evidence-layer-len composition-roots (length composition-roots - 1))
      (fri-evidence-layer-idx composition-roots query-idxs i
        (length composition-roots - 1))
      (2 ^ (length composition-roots - 1))
      (fri-sibling-index
        (fri-evidence-layer-len composition-roots (length composition-roots - 1))
        (fri-evidence-layer-idx composition-roots query-idxs i
          (length composition-roots - 1)))
      arp arp-path arn arn-path
      (fri-evidence-next-idx composition-roots query-idxs i
        (length composition-roots - 1))
      composition-final actual-chunk
  using step idx-eq-query composition-roots-eq composition-bs-eq fl-nonempty
  unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
    fri-evidence-next-idx-def
  by simp
  show ?thesis
  by (rule that[OF auth' step'])
qed

end

end

```

```

theory Soundness-FRI-Recorded-Chunk-Replay
imports Soundness-FRI-Raw-Layer-Bounds
begin

```

Selected recorded FRI layer replay.

The authenticated-route layer already proves that the verifier authenticates some chunk at the replayed root/index of a selected FRI layer. The residual proofs need the stronger fact that the specific chunk recorded in the extracted verifier transcript is authenticated. This theory proves that narrow replay fact for one selected layer, avoiding the earlier broad all-layer replay induction.

```

context soundness
begin

```

```

lemma fri-layer-lengths-nth-pos-imp-initial-pos:
  assumes j-bound:  $j < n$ 
    and pos:  $0 < \text{fri-layer-lengths } n \text{ len} \ ! \ j$ 
  shows  $0 < \text{len}$ 
  using assms
proof (induction j arbitrary: n len)
  case 0
  then show ?case
    by (cases n) simp-all
next
  case (Suc j)
  then obtain n' where n-eq:  $n = \text{Suc } n'$ 
    by (cases n) auto
  have j-bound':  $j < n'$ 
    using Suc.prem1 unfolding n-eq by simp
  have pos':  $0 < \text{fri-layer-lengths } n' (\text{len div } 2) \ ! \ j$ 
    using Suc.prem2 unfolding n-eq by simp
  have  $0 < \text{len div } 2$ 
    by (rule Suc.IH[OF j-bound' pos'])
  then show ?case
    by simp
qed

lemma mfold-fri-layer-opening-selected-recorded-chunk-authenticated-at:
  fixes s t :: (f, a) protocol-channel-scheme
  assumes outcome:
    Some (out, t)  $\in$ 
      set-dist
        (execute
          (mfold (idx, x, len, pw)
            (map fri-layer-opening-step bfs)) s)
  and layers:
    fri-layers-transcript (length bfs) len layer-chunks total-chunk
  and transcript:
    PTranscript s = total-chunk @ PTranscript t
  and initial-idx-bound:  $\text{idx} < \text{len}$ 
  and all-layer-bounds:
     $\bigwedge k. k < \text{length } \text{bfs} \implies$ 
       $0 < \text{fri-layer-lengths } (\text{length } \text{bfs}) \text{ len} \ ! \ k \wedge$ 
       $\text{fri-layer-indices } (\text{length } \text{bfs}) \ \text{idx} \ \text{len} \ ! \ k <$ 
       $\text{fri-layer-lengths } (\text{length } \text{bfs}) \ \text{len} \ ! \ k$ 
  and j-bound:  $j < \text{length } \text{bfs}$ 
shows
  fri-layer-chunk-authenticated (snd (bfs ! j))
    (fri-layer-lengths (length bfs) len ! j)
    (fri-layer-indices (length bfs) idx len ! j)
    (layer-chunks ! j) t
using outcome layers transcript initial-idx-bound all-layer-bounds j-bound

```

```

proof (induction bfs arbitrary: idx x len pw s out t j layer-chunks total-chunk)
  case Nil
  then show ?case by simp
next
  case (Cons bf bfs)
  obtain b rt where bf-eq: bf = (b, rt)
  by (cases bf)
  from Cons.premis(1)[unfolded bf-eq]
  obtain out1 s1 where
    head:
      Some (out1, s1) ∈
        set-dist
          (execute (fri-layer-opening-step (b, rt) (idx, x, len, pw)) s)
    and tail:
      Some (out, t) ∈
        set-dist
          (execute (mfold out1 (map fri-layer-opening-step bfs)) s1)
  by (auto elim!: set-dist-bindE)
  obtain first rest where chunks-eq: layer-chunks = first # rest
  using Cons.premis(6) Cons.premis(2)
  unfolding fri-layers-transcript-def by (cases layer-chunks) auto
  have total-eq: total-chunk = first @ List.concat rest
  using Cons.premis(2) unfolding fri-layers-transcript-def chunks-eq
  by simp
  have first-open:
    ∃ xp xp-path xn xn-path.
      fri-layer-opening-chunk len xp xp-path xn xn-path first
  using Cons.premis(2) unfolding fri-layers-transcript-def chunks-eq
  by auto
  have rest-layers:
    fri-layers-transcript (length bfs) (len div 2) rest
    (List.concat rest)
  using Cons.premis(2) unfolding fri-layers-transcript-def chunks-eq
  by auto
  obtain x' xp xp-path xn xn-path head-chunk where
    out1-eq:
      out1 = (idx mod (len div 2), x', len div 2, pw + pw)
  and head-chunk-def: head-chunk = [xp] @ xp-path @ [xn] @ xn-path
  and head-open:
    fri-layer-opening-chunk len xp xp-path xn xn-path head-chunk
  and tr-head: PTranscript s = head-chunk @ PTranscript s1
  and xp-auth:
    authenticated-opening-in s1
    (|opening-root = rt,
      opening-length = len,
      opening-index = idx,
      opening-value = xp,
      opening-path = xp-path|)
  and xn-auth:

```

```

    authenticated-opening-in s1
    (opening-root = rt,
     opening-length = len,
     opening-index = fri-sibling-index len idx,
     opening-value = xn,
     opening-path = xn-path)
  proof -
    have len-pos: 0 < len
      using Cons.premis(5)[of 0] by simp
    have idx-bound: idx < len
      by (rule Cons.premis(4))
    show ?thesis
  proof (rule fri-layer-opening-step-outcome-authenticated
    [OF len-pos idx-bound head])
    fix xp' xp-path' xn' xn-path' x'' chunk
    assume out-eq:
      out1 = (idx mod (len div 2), x'', len div 2, pw + pw)
    and chunk-eq: chunk = [xp'] @ xp-path' @ [xn'] @ xn-path'
    and chunk-open:
      fri-layer-opening-chunk len xp' xp-path' xn' xn-path' chunk
    and tr: PTranscript s = chunk @ PTranscript s1
    and ext: s ≤ s1
    and auth-xp:
      authenticated-opening-in s1
      (opening-root = rt, opening-length = len,
       opening-index = idx, opening-value = xp',
       opening-path = xp-path')
    and auth-xn:
      authenticated-opening-in s1
      (opening-root = rt, opening-length = len,
       opening-index = fri-sibling-index len idx,
       opening-value = xn', opening-path = xn-path')
    show ?thesis
      by (rule that[OF out-eq chunk-eq chunk-open tr auth-xp auth-xn])
  qed
qed
from mfold-fri-layer-openings-outcome[OF tail[unfolded out1-eq]]
obtain tail-layers tail-chunk where
  tail-transcript:
    PTranscript s1 = tail-chunk @ PTranscript t
  and tail-ext: s1 ≤ t
  by blast
have head-len: length head-chunk = length first
proof -
  obtain yp yp-path yn yn-path where first-open':
    fri-layer-opening-chunk len yp yp-path yn yn-path first
  using first-open by blast
show ?thesis
  using fri-layer-opening-chunk-length[OF head-open]

```

```

    fri-layer-opening-chunk-length[OF first-open']
  by simp
qed
have chunks-total-eq:
  head-chunk @ tail-chunk = first @ List.concat rest
  using tr-head tail-transcript Cons.premis(3) total-eq by simp
have head-eq: head-chunk = first
  using arg-cong[OF chunks-total-eq, of take (length head-chunk)]
  head-len
  by simp
have tail-transcript-rest:
  PTranscript s1 = List.concat rest @ PTranscript t
  using chunks-total-eq head-eq tail-transcript by simp
show ?case
proof (cases j)
  case 0
  have auth-s1:
    fri-layer-chunk-authenticated rt len idx head-chunk s1
    by (rule fri-layer-chunk-authenticatedI[OF head-open xp-auth xn-auth])
  have auth-t:
    fri-layer-chunk-authenticated rt len idx head-chunk t
    by (rule fri-layer-chunk-authenticated-mono[OF auth-s1 tail-ext])
  show ?thesis
    using auth-t head-eq 0 chunks-eq bf-eq by simp
next
  case (Suc k)
  have k-bound: k < length bfs
    using Cons.premis(6) Suc by simp
  have tail-all-layer-bounds:
     $\bigwedge m. m < \text{length bfs} \implies$ 
    0 < fri-layer-lengths (length bfs) (len div 2) ! m  $\wedge$ 
    fri-layer-indices (length bfs) (idx mod (len div 2))
    (len div 2) ! m <
    fri-layer-lengths (length bfs) (len div 2) ! m
    using Cons.premis(5)[of Suc m for m] by simp
  have tail-initial-idx-bound: idx mod (len div 2) < len div 2
    using tail-all-layer-bounds[of 0] k-bound by (cases bfs) simp-all
  have tail-auth:
    fri-layer-chunk-authenticated (snd (bfs ! k))
    (fri-layer-lengths (length bfs) (len div 2) ! k)
    (fri-layer-indices (length bfs) (idx mod (len div 2))
    (len div 2) ! k)
    (rest ! k) t
    by (rule Cons.IH[OF tail[unfolded out1-eq] rest-layers
    tail-transcript-rest tail-initial-idx-bound tail-all-layer-bounds
    k-bound])
  show ?thesis
    using tail-auth Suc chunks-eq by simp
qed

```

qed

lemma *receive-query-commits-selected-recorded-chunk-authenticated-at:*
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes *outcome:*
 $\text{Some } (out, t) \in$
 set-dist
 $(\text{execute}$
 $(\text{mfold } (idx, x, len, pw) (\text{receive-query-commits } bfs))\ s)$
and *layers:*
 $\text{fri-layers-transcript } (length\ bfs)\ len\ \text{layer-chunks}\ \text{total-chunk}$
and *transcript:*
 $P\text{Transcript } s = \text{total-chunk} @ P\text{Transcript } t$
and *initial-idx-bound:* $idx < len$
and *all-layer-bounds:*
 $\bigwedge k. k < length\ bfs \implies$
 $0 < \text{fri-layer-lengths } (length\ bfs)\ len ! k \wedge$
 $\text{fri-layer-indices } (length\ bfs)\ idx\ len ! k <$
 $\text{fri-layer-lengths } (length\ bfs)\ len ! k$
and *j-bound:* $j < length\ bfs$
shows
 $\text{fri-layer-chunk-authenticated } (snd\ (bfs ! j))$
 $(\text{fri-layer-lengths } (length\ bfs)\ len ! j)$
 $(\text{fri-layer-indices } (length\ bfs)\ idx\ len ! j)$
 $(\text{layer-chunks } ! j)\ t$
proof –
have *map-eq:* $\text{receive-query-commits } bfs = \text{map } \text{fri-layer-opening-step } bfs$
unfolding *receive-query-commits-def fri-layer-opening-step-def*
by $(\text{simp add: fun-eq-iff split: prod.splits})$
show *?thesis*
by $(\text{rule mfold-fri-layer-opening-selected-recorded-chunk-authenticated-at}$
 $[\text{OF outcome[unfolded map-eq] layers transcript initial-idx-bound}$
 $\text{all-layer-bounds j-bound}])$

qed

lemma *verifier-query-round-after-index-trace-recorded-layer-chunk-authenticated:*
fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$
assumes *outcome:*
 $\text{Some } ((), t) \in$
 set-dist
 $(\text{execute}$
 $(\text{verifier-query-round-after-index-program}$
 $\text{fr } f\text{-fl } f\text{-final as fl final raw})\ s)$
and *recorded:*
 $\text{query-round-fri-layer-transcripts } (index\ (to\text{-nat } raw))$
 $(\text{map } snd\ f\text{-fl})\ (\text{map } snd\ fl)\ \text{round-chunk}\ \text{trace-layers}$
 $\text{composition-layers}$
and *transcript:* $P\text{Transcript } s = \text{round-chunk} @ P\text{Transcript } t$
and *j-bound:* $j < length\ f\text{-fl}$

and *all-layer-bounds*:
 $\bigwedge k. k < \text{length } f\text{-fl} \implies$
 $0 < \text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) ! k \wedge$
 $\text{fri-layer-indices } (\text{length } f\text{-fl}) (\text{index } (\text{to-nat } \text{raw}))$
 $(\text{clength} * \text{scale}) ! k <$
 $\text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) ! k$
shows
 $\text{fri-layer-chunk-authenticated } (\text{snd } (f\text{-fl} ! j))$
 $(\text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) ! j)$
 $(\text{fri-layer-indices } (\text{length } f\text{-fl}) (\text{index } (\text{to-nat } \text{raw})))$
 $(\text{clength} * \text{scale}) ! j)$
 $(\text{trace-layers} ! j) \text{ } t$
proof –
let $?idx = \text{index } (\text{to-nat } \text{raw})$
from *outcome* **obtain** $fv \text{ } s1 \text{ } f\text{-out } s2 \text{ } s3 \text{ } c\text{-out } s4$ **where**
query-decommit:
 $\text{Some } (fv, s1) \in$
 $\text{set-dist } (\text{execute } (\text{mmap } (\text{check-decommit-on-query } fr \text{ } ?idx)) \text{ } s)$
and *trace-fri*:
 $\text{Some } (f\text{-out}, s2) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{mfold } (?idx, \text{hd } fv, \text{clength} * \text{scale}, 1)$
 $(\text{receive-query-commits } f\text{-fl})) \text{ } s1)$
and *assert-trace*:
 $\text{Some } ((), s3) \in$
 set-dist
 $(\text{execute}$
 $(\text{assert } (\text{case } f\text{-out of } (-, f\text{-x}, -, -) \Rightarrow f\text{-x} = f\text{-final})) \text{ } s2)$
and *comp-fri*:
 $\text{Some } (c\text{-out}, s4) \in$
 $\text{set-dist } (\text{execute}$
 $(\text{mfold}$
 $(?idx, \text{cp-eval as } fv \text{ } (h \wedge ?idx * \text{shift}), \text{clength} * \text{scale}, 1)$
 $(\text{receive-query-commits } fl)) \text{ } s3)$
and *assert-comp*:
 $\text{Some } ((), t) \in$
 set-dist
 $(\text{execute}$
 $(\text{assert } (\text{case } c\text{-out of } (-, x, -, -) \Rightarrow x = \text{final})) \text{ } s4)$
unfolding *verifier-query-round-after-index-program-def*
by (*auto simp: Let-def split: prod.splits elim!: set-dist-bindE*)
from *check-decommit-on-query-outcome*[*OF query-decommit*]
obtain *query-paths query-chunk* **where**
query-chunk-shape:
 $\text{query-decommitment-transcript } ?idx \text{ } fv \text{ } \text{query-paths } \text{query-chunk}$
and *tr-s*: $P\text{Transcript } s = \text{query-chunk} @ P\text{Transcript } s1$
by *blast*
from *receive-query-commits-layers-outcome*[*OF trace-fri*]
obtain *trace-layer-chunks trace-fri-chunk* **where**


```

trace-layers-actual:
  fri-layers-transcript (length f-fl) (clength * scale)
  trace-layer-chunks trace-fri-chunk
and tr-s1: PTranscript s1 = trace-fri-chunk @ PTranscript s2
and ext-s1-s2: s1 ≤ s2
by blast
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def
  by (cases case f-out of (-, f-x, -, -)  $\Rightarrow$  f-x = f-final)
    (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  comp-layers-actual:
    fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
and ext-s3-s4: s3 ≤ s4
by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -)  $\Rightarrow$  x = final)
    (auto simp: throw-no-outcome)
let ?actual-chunk = query-chunk @ trace-fri-chunk @ comp-fri-chunk
have actual-recorded:
  query-round-fri-layer-transcripts ?idx (map snd f-fl) (map snd fl)
  ?actual-chunk trace-layer-chunks comp-layer-chunks
unfolding query-round-fri-layer-transcripts-def
by (intro exI[of - query-chunk] exI[of - trace-fri-chunk]
  exI[of - comp-fri-chunk] exI[of - fv] exI[of - query-paths] conjI)
  (use query-chunk-shape trace-layers-actual comp-layers-actual in
  ⟨simp-all add: mult.commute⟩)
have actual-transcript:
  PTranscript s = ?actual-chunk @ PTranscript t
  using tr-s tr-s1 tr-s3 s3-eq t-eq by simp
have actual-chunk-eq: ?actual-chunk = round-chunk
  using actual-transcript transcript by simp
have trace-layer-chunks-eq:
  trace-layer-chunks = trace-layers
  by (rule query-round-fri-layer-transcripts-unique(1)
  [OF actual-recorded[unfolded actual-chunk-eq] recorded])
have idx-bound: ?idx < clength * scale
  by (rule index-less-domain)
have auth-s2:
  fri-layer-chunk-authenticated (snd (f-fl ! j))
  (fri-layer-lengths (length f-fl) (clength * scale) ! j)
  (fri-layer-indices (length f-fl) ?idx (clength * scale) ! j)
  (trace-layer-chunks ! j) s2
by (rule receive-query-commits-selected-recorded-chunk-authenticated-at
  [OF trace-fri trace-layers-actual tr-s1 idx-bound

```

```

      all-layer-bounds j-bound])
have s2-t: s2 ≤ t
  using ext-s3-s4 s3-eq t-eq by simp
have auth-t:
  fri-layer-chunk-authenticated (snd (f-fl ! j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! j)
    (fri-layer-indices (length f-fl) ?idx (clength * scale) ! j)
    (trace-layer-chunks ! j) t
  by (rule fri-layer-chunk-authenticated-mono[OF auth-s2 s2-t])
show ?thesis
  using auth-t trace-layer-chunks-eq by simp
qed

lemma verifier-query-round-after-index-composition-recorded-layer-chunk-authenticated:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s)
  and recorded:
    query-round-fri-layer-transcripts (index (to-nat raw))
      (map snd f-fl) (map snd fl) round-chunk trace-layers
      composition-layers
  and transcript: PTranscript s = round-chunk @ PTranscript t
  and j-bound: j < length fl
  and all-layer-bounds:
    ∧k. k < length fl ⇒
      0 < fri-layer-lengths (length fl) (clength * scale) ! k ∧
      fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! k <
      fri-layer-lengths (length fl) (clength * scale) ! k
  shows
    fri-layer-chunk-authenticated (snd (fl ! j))
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! j)
      (composition-layers ! j) t
proof -
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
    query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
  and trace-fri:
    Some (f-out, s2) ∈
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)

```

```

    (receive-query-commits f-fl)) s1)
and assert-trace:
  Some (( ), s3) ∈
    set-dist
      (execute
        (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
  Some (c-out, s4) ∈
    set-dist (execute
      (mfold
        (?idx, cp-eval as fv (h ^ ?idx * shift), clength * scale, 1)
        (receive-query-commits fl)) s3)
and assert-comp:
  Some (( ), t) ∈
    set-dist
      (execute
        (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
unfolding verifier-query-round-after-index-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
from check-decommit-on-query-outcome[OF query-decommit]
obtain query-paths query-chunk where
  query-chunk-shape:
    query-decommitment-transcript ?idx fv query-paths query-chunk
and tr-s: PTranscript s = query-chunk @ PTranscript s1
by blast
from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where
  trace-layers-actual:
    fri-layers-transcript (length f-fl) (clength * scale)
    trace-layer-chunks trace-fri-chunk
and tr-s1: PTranscript s1 = trace-fri-chunk @ PTranscript s2
by blast
have s3-eq: s3 = s2
using assert-trace unfolding assert-def
by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
  (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  comp-layers-actual:
    fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
by blast
have t-eq: t = s4
using assert-comp unfolding assert-def
by (cases case c-out of (-, x, -, -) ⇒ x = final)
  (auto simp: throw-no-outcome)
let ?actual-chunk = query-chunk @ trace-fri-chunk @ comp-fri-chunk
have actual-recorded:

```

```

query-round-fri-layer-transcripts ?idx (map snd f-fl) (map snd fl)
  ?actual-chunk trace-layer-chunks comp-layer-chunks
unfolding query-round-fri-layer-transcripts-def
by (intro exI[of - query-chunk] exI[of - trace-fri-chunk]
    exI[of - comp-fri-chunk] exI[of - fv] exI[of - query-paths] conjI)
  (use query-chunk-shape trace-layers-actual comp-layers-actual in
    ⟨simp-all add: mult.commute⟩)
have actual-transcript:
  PTranscript s = ?actual-chunk @ PTranscript t
  using tr-s tr-s1 tr-s3 s3-eq t-eq by simp
have actual-chunk-eq: ?actual-chunk = round-chunk
  using actual-transcript transcript by simp
have comp-layer-chunks-eq:
  comp-layer-chunks = composition-layers
  by (rule query-round-fri-layer-transcripts-unique(2)
    [OF actual-recorded[unfolded actual-chunk-eq] recorded])
have idx-bound: ?idx < clength * scale
  by (rule index-less-domain)
have auth-s4:
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) ?idx (clength * scale) ! j)
    (comp-layer-chunks ! j) s4
  by (rule receive-query-commits-selected-recorded-chunk-authenticated-at
    [OF comp-fri comp-layers-actual tr-s3 idx-bound
      all-layer-bounds j-bound])
show ?thesis
  using auth-s4 comp-layer-chunks-eq t-eq by simp
qed

lemma accepted-fri-opening-transcript-trace-recorded-layer-chunk-authenticated-at:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
and out-eq: out = Some (result, final-state)
and round-bound: i < rounds
and layer-bound: j < length trace-roots
and all-layer-bounds:
   $\bigwedge \text{raw } k. k < \text{length trace-roots} \implies$ 
   $0 < \text{fri-layer-lengths } (\text{length trace-roots}) (\text{clength} * \text{scale}) ! k \wedge$ 
   $\text{fri-layer-indices } (\text{length trace-roots}) (\text{index } (\text{to-nat raw}))$ 
   $(\text{clength} * \text{scale}) ! k <$ 
   $\text{fri-layer-lengths } (\text{length trace-roots}) (\text{clength} * \text{scale}) ! k$ 
shows
  generic-fri-recorded-layer-chunk-authenticated trace-roots query-idxs
  trace-round-layers final-state i j
proof –
from fri-openings obtain result' final-state' fr f-fl as fl query-state

```

```

    raw-idxs query-chunks where
  out-def: out = Some (result', final-state')
and trace-roots-eq: trace-roots = map snd f-fl
and trace-bs-eq: trace-bs = map fst f-fl
and composition-roots-eq: composition-roots = map snd fl
and outcome:
  Some (result', final-state') ∈
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr f-fl trace-final as fl
          composition-final)
        rounds)
      query-state)
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state'
and query-chunk:
  ∧i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
and replay-lookup:
  ∧i. i < rounds ⇒
    fmlookup (HashMap final-state')
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
and recorded:
  ∧i. i < rounds ⇒
    query-round-fri-layer-transcripts (query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
    (trace-round-layers ! i) (composition-round-layers ! i)
unfolding accepted-fri-opening-transcript-def by blast
have result-eq: result' = result
using out-def out-eq by simp
have final-state-eq: final-state' = final-state
using out-def out-eq by simp
have outcome':
  Some (result, final-state) ∈
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr f-fl trace-final as fl
          composition-final)
        rounds)
      query-state)
using outcome result-eq final-state-eq by simp

```

```

let ?round =
  verifier-query-round-program fr f-fl trace-final as fl composition-final
from query-opening-ntimes-outcome-decomp-at[OF outcome' round-bound]
obtain prefix x suffix s-i s-suc where
  prefix:
    Some (prefix, s-i) ∈
      set-dist (execute (ntimes ?round i) query-state)
and round:
    Some (x, s-suc) ∈ set-dist (execute ?round s-i)
and suffix:
    Some (suffix, final-state) ∈
      set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
by blast
have x-eq: x = ()
  by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
  fix prefix-query-idxs prefix-chunks
  assume len-prefix-idxs: length prefix-query-idxs = i
  and len-prefix-chunks: length prefix-chunks = i
  and transcript-prefix:
    PTranscript query-state =
      List.concat prefix-chunks @ PTranscript s-i
  and state-i:
    PState s-i =
      state-after-query-chunks (PState query-state) prefix-chunks i
  and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
  and prefix-ext: query-state ≤ s-i
  and prefix-rounds:
     $\bigwedge k. k < i \implies$ 
      verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
  from round[unfolded x-eq verifier-query-round-program-alt-def]
  obtain raw s0 where
    raw-out:
      Some (raw, s0) ∈
        set-dist (execute receive-query-index-challenge s-i)
  and after:
    Some ((), s-suc) ∈
      set-dist (execute
        (verifier-query-round-after-index-program fr f-fl trace-final as
          fl composition-final raw) s0)
  by (auto elim!: set-dist-bindE)
  have tr-s0: PTranscript s0 = PTranscript s-i
  using receive-query-index-challenge-outcome[OF raw-out] by simp
  show ?thesis
  proof (rule verifier-query-round-program-outcome[OF round[unfolded x-eq]])
    fix raw' idx chunk

```

```

assume idx-eq: idx = index (to-nat raw')
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
  chunk
and transcript-round:
  PTranscript s-i = chunk @ PTranscript s-suc
and state-suc:
  PState s-suc = foldl concat (PState s-i) chunk
and round-ext: s-i ≤ s-suc
and counter-suc:
  PQueryCounter s-suc = Suc (PQueryCounter s-i)
and lookup-suc:
  fmlookup (HashMap s-suc)
  (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
  Some raw'
have raw-eq': raw = raw'
proof –
  have lookup-s0:
    fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
    using receive-query-index-challenge-outcome[OF raw-out] by simp
  have s0-suc: s0 ≤ s-suc
    using verifier-query-round-after-index-program-outcome[OF after]
    by simp
  have lookup-suc-raw:
    fmlookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
    using hash-extension-lookup[OF lookup-s0 s0-suc] .
  show ?thesis
    using lookup-suc lookup-suc-raw by simp
qed
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
  fix suffix-query-idxs suffix-chunks
  assume len-suffix-idxs:
    length suffix-query-idxs = rounds – Suc i
  and len-suffix-chunks:
    length suffix-chunks = rounds – Suc i
  and transcript-suffix:
    PTranscript s-suc =
    List.concat suffix-chunks @ PTranscript final-state
  and suffix-state:
    PState final-state =
    state-after-query-chunks (PState s-suc) suffix-chunks
    (rounds – Suc i)
  and suffix-counter:

```

```

    PQueryCounter final-state =
      PQueryCounter s-suc + (rounds - Suc i)
  and suffix-ext: s-suc ≤ final-state
  and suffix-rounds:
     $\bigwedge k. k < rounds - Suc\ i \implies$ 
      verifier-query-round-chunk (suffix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (suffix-chunks ! k)
  let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
  have len-selected-chunks: length ?selected-chunks = rounds
    using len-prefix-chunks len-suffix-chunks round-bound by simp
  have selected-chunk-lens:
     $\bigwedge k. k < rounds \implies$ 
      length (?selected-chunks ! k) =
        verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  proof -
    fix k
    assume k-bound: k < rounds
    show
      length (?selected-chunks ! k) =
        verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  proof (cases k < i)
    case True
      have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
        using True len-prefix-chunks by (simp add: nth-append)
      have
        length (prefix-chunks ! k) =
          verifier-query-round-transcript-length
            (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
        by (rule verifier-query-round-chunk-length
            [OF prefix-rounds[OF True]])
      then show ?thesis
        using nth-eq
        by (simp add:
            verifier-query-round-transcript-length-index-irrelevant)
    next
      case False
      show ?thesis
      proof (cases k = i)
        case True
          have nth-eq: ?selected-chunks ! k = chunk
            using True len-prefix-chunks by (simp add: nth-append)
          have
            length chunk =
              verifier-query-round-transcript-length idx
                (map snd f-fl) (map snd fl)
            by (rule verifier-query-round-chunk-length[OF chunk-shape])
          then show ?thesis

```



```

    using nth-eq
    by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
next
case False
let ?k = k - Suc i
have ge: Suc i ≤ k
    using ⟨¬ k < i⟩ False by linarith
have k-bound': ?k < rounds - Suc i
    using ge k-bound by linarith
have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k
    using ge len-prefix-chunks by (simp add: nth-append)
have
    length (suffix-chunks ! ?k) =
        verifier-query-round-transcript-length
            (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
        [OF suffix-rounds[OF k-bound']])
then show ?thesis
    using nth-eq
    by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-f:
    ∧k. k < rounds ⇒
        verifier-query-round-chunk (query-idxs ! k)
            (map snd f-fl) (map snd fl) (query-chunks ! k)
    using query-chunk trace-roots-eq composition-roots-eq by simp
have query-chunk-lens:
    ∧k. k < rounds ⇒
        length (query-chunks ! k) =
            verifier-query-round-transcript-length (query-idxs ! k)
                (map snd f-fl) (map snd fl)
    using query-chunk-f verifier-query-round-chunk-length by blast
have selected-transcript:
    PTranscript query-state =
        List.concat ?selected-chunks @ PTranscript final-state
    using transcript-prefix transcript-round transcript-suffix by simp
have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
    using selected-transcript transcript-query final-state-eq by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
    have same-lens:
        ∧k. k < length ?selected-chunks ⇒
            length (?selected-chunks ! k) = length (query-chunks ! k)
        using selected-chunk-lens query-chunk-lens len-selected-chunks
        by simp

```

```

have concat-eq-empty:
  List.concat ?selected-chunks @ [] = List.concat query-chunks
  using concat-eq by simp
have len-eq: length ?selected-chunks = length query-chunks
  using len-selected-chunks len-query-chunks by simp
have ?selected-chunks = query-chunks  $\wedge$  ( [] :: 'f list) = []
  by (rule concat-append-eq-concat-same-chunk-lengths
    [OF len-eq same-lens concat-eq-empty])
then show ?thesis
  by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix
      [OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw'
proof -
  have lookup-suc':
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
      Some raw'
    using lookup-suc state-i counter-i by simp
  have lookup-final-prefix:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
      Some raw'
    using hash-extension-lookup [OF lookup-suc' suffix-ext] .
  show ?thesis
    using lookup-final-prefix prefix-state-eq by simp
qed
have raw-idx-eq: raw' = raw-idxs ! i
  using lookup-final replay-lookup [OF round-bound] final-state-eq
  by simp

```

```

have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-idx-eq query-idxs-eq len-raw round-bound by simp
have selected-i: ?selected-chunks ! i = chunk
  using len-prefix-chunks by (simp add: nth-append)
have chunk-i-eq: chunk = query-chunks ! i
  using chunks-eq selected-i by simp
have recorded-i:
  query-round-fri-layer-transcripts (index (to-nat raw))
    (map snd f-fl) (map snd fl) chunk (trace-round-layers ! i)
    (composition-round-layers ! i)
  using recorded[OF round-bound] idx-eq idx-eq-query raw-eq'
    chunk-i-eq trace-roots-eq composition-roots-eq
  by simp
have tr-s0-suc:
  PTranscript s0 = chunk @ PTranscript s-suc
  using tr-s0 transcript-round by simp
have j-bound-f: j < length f-fl
  using layer-bound trace-roots-eq by simp
have all-layer-bounds-f:
   $\bigwedge k. k < \text{length } f\text{-fl} \implies$ 
     $0 < \text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) ! k \wedge$ 
     $\text{fri-layer-indices } (\text{length } f\text{-fl}) (\text{index } (\text{to-nat raw}))$ 
     $(\text{clength} * \text{scale}) ! k <$ 
     $\text{fri-layer-lengths } (\text{length } f\text{-fl}) (\text{clength} * \text{scale}) ! k$ 
  using all-layer-bounds trace-roots-eq by simp
have auth-suc:
  fri-layer-chunk-authenticated (snd (f-fl ! j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! j)
    (fri-layer-indices (length f-fl) (index (to-nat raw)))
    (clength * scale) ! j)
    (trace-round-layers ! i ! j) s-suc
  by (rule
    verifier-query-round-after-index-trace-recorded-layer-chunk-authenticated
    [OF after recorded-i tr-s0-suc j-bound-f
      all-layer-bounds-f])
have auth-final:
  fri-layer-chunk-authenticated (snd (f-fl ! j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! j)
    (fri-layer-indices (length f-fl) (index (to-nat raw)))
    (clength * scale) ! j)
    (trace-round-layers ! i ! j) final-state
  by (rule fri-layer-chunk-authenticated-mono[OF auth-suc suffix-ext])
have raw-idx-query: index (to-nat raw) = query-idxs ! i
  using idx-eq idx-eq-query raw-eq' by simp
have auth:
  fri-layer-chunk-authenticated (trace-roots ! j)
    (fri-evidence-layer-len trace-roots j)
    (fri-evidence-layer-idx trace-roots query-idxs i j)
    (trace-round-layers ! i ! j) final-state

```

```

    using auth-final trace-roots-eq raw-idx-query layer-bound
    unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
    by simp
  show ?thesis
    unfolding generic-fri-recorded-layer-chunk-authenticated-def
    by (rule auth)
qed
qed
qed
qed

lemma accepted-fri-opening-transcript-composition-recorded-layer-chunk-authenticated-at:
  assumes fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers
  and out-eq: out = Some (result, final-state)
  and round-bound: i < rounds
  and layer-bound: j < length composition-roots
  and all-layer-bounds:
     $\bigwedge \text{raw } k. k < \text{length } \text{composition-roots} \implies$ 
     $0 < \text{fri-layer-lengths } (\text{length } \text{composition-roots}) \text{ (clength * scale) ! } k \wedge$ 
     $\text{fri-layer-indices } (\text{length } \text{composition-roots}) \text{ (index (to-nat raw))}$ 
     $\text{(clength * scale) ! } k <$ 
     $\text{fri-layer-lengths } (\text{length } \text{composition-roots}) \text{ (clength * scale) ! } k$ 
  shows
    generic-fri-recorded-layer-chunk-authenticated composition-roots
    query-idxs composition-round-layers final-state i j
proof -
  from fri-openings obtain result' final-state' fr f-fl as fl query-state
    raw-idxs query-chunks where
    out-def: out = Some (result', final-state')
  and trace-roots-eq: trace-roots = map snd f-fl
  and composition-roots-eq: composition-roots = map snd fl
  and composition-bs-eq: composition-bs = map fst fl
  and outcome:
    Some (result', final-state')  $\in$ 
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr f-fl trace-final as fl
          composition-final)
        rounds)
      query-state)
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda \text{raw}. \text{index (to-nat raw)}$ ) raw-idxs
  and len-query-chunks: length query-chunks = rounds
  and transcript-query:
    PTranscript query-state =

```

```

    List.concat query-chunks @ PTranscript final-state'
and query-chunk:
   $\wedge i. i < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
and replay-lookup:
   $\wedge i. i < \text{rounds} \implies$ 
    fmllookup (HashMap final-state')
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some (raw-idxs ! i)
and recorded:
   $\wedge i. i < \text{rounds} \implies$ 
    query-round-fri-layer-transcripts (query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
    (trace-round-layers ! i) (composition-round-layers ! i)
unfolding accepted-fri-opening-transcript-def by blast
have result-eq: result' = result
  using out-def out-eq by simp
have final-state-eq: final-state' = final-state
  using out-def out-eq by simp
have outcome':
  Some (result, final-state)  $\in$ 
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr f-fl trace-final as fl
          composition-final)
        rounds)
      query-state)
  using outcome result-eq final-state-eq by simp
let ?round =
  verifier-query-round-program fr f-fl trace-final as fl composition-final
from query-opening-ntimes-outcome-decomp-at[OF outcome' round-bound]
obtain prefix x suffix s-i s-suc where
  prefix:
    Some (prefix, s-i)  $\in$ 
    set-dist (execute (ntimes ?round i) query-state)
and round:
  Some (x, s-suc)  $\in$  set-dist (execute ?round s-i)
and suffix:
  Some (suffix, final-state)  $\in$ 
    set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
by blast
have x-eq: x = ()
  by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
  fix prefix-query-idxs prefix-chunks

```

```

assume len-prefix-idxs: length prefix-query-idxs = i
and len-prefix-chunks: length prefix-chunks = i
and transcript-prefix:
  PTranscript query-state =
    List.concat prefix-chunks @ PTranscript s-i
and state-i:
  PState s-i =
    state-after-query-chunks (PState query-state) prefix-chunks i
and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
and prefix-ext: query-state ≤ s-i
and prefix-rounds:
   $\bigwedge k. k < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
from round[unfolded x-eq verifier-query-round-program-alt-def]
obtain raw s0 where
  raw-out:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s-i)
and after:
  Some ((), s-suc) ∈
    set-dist (execute
      (verifier-query-round-after-index-program fr f-fl trace-final as
        fl composition-final raw) s0)
  by (auto elim!: set-dist-bindE)
have tr-s0: PTranscript s0 = PTranscript s-i
  using receive-query-index-challenge-outcome[OF raw-out] by simp
show ?thesis
proof (rule verifier-query-round-program-outcome[OF round[unfolded x-eq]])
  fix raw' idx chunk
  assume idx-eq: idx = index (to-nat raw')
  and chunk-shape:
    verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
      chunk
  and transcript-round:
    PTranscript s-i = chunk @ PTranscript s-suc
  and state-suc:
    PState s-suc = foldl concat (PState s-i) chunk
  and round-ext: s-i ≤ s-suc
  and counter-suc:
    PQueryCounter s-suc = Suc (PQueryCounter s-i)
  and lookup-suc:
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
        Some raw'
  have raw-eq': raw = raw'
  proof –
    have lookup-s0:
      fmlookup (HashMap s0)

```

```

    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
  using receive-query-index-challenge-outcome[OF raw-out] by simp
have s0-suc:  $s0 \leq s\text{-suc}$ 
  using verifier-query-round-after-index-program-outcome[OF after]
  by simp
have lookup-suc-raw:
  fmllookup (HashMap s-suc)
  (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
  Some raw
  using hash-extension-lookup[OF lookup-s0 s0-suc] .
show ?thesis
  using lookup-suc lookup-suc-raw by simp
qed
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-idxs:
  length suffix-query-idxs = rounds - Suc i
and len-suffix-chunks:
  length suffix-chunks = rounds - Suc i
and transcript-suffix:
  PTranscript s-suc =
  List.concat suffix-chunks @ PTranscript final-state
and suffix-state:
  PState final-state =
  state-after-query-chunks (PState s-suc) suffix-chunks
  (rounds - Suc i)
and suffix-counter:
  PQueryCounter final-state =
  PQueryCounter s-suc + (rounds - Suc i)
and suffix-ext:  $s\text{-suc} \leq \text{final-state}$ 
and suffix-rounds:
   $\bigwedge k. k < \text{rounds} - \text{Suc } i \implies$ 
  verifier-query-round-chunk (suffix-query-idxs ! k)
  (map snd f-fl) (map snd fl) (suffix-chunks ! k)
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks round-bound by simp
have selected-chunk-lens:
   $\bigwedge k. k < \text{rounds} \implies$ 
  length (?selected-chunks ! k) =
  verifier-query-round-transcript-length (query-idxs ! k)
  (map snd f-fl) (map snd fl)
proof -
fix k
assume k-bound:  $k < \text{rounds}$ 
show

```

```

length (?selected-chunks ! k) =
  verifier-query-round-transcript-length (query-idxs ! k)
    (map snd f-fl) (map snd fl)
proof (cases k < i)
  case True
  have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length (prefix-chunks ! k) =
      verifier-query-round-transcript-length
        (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
      [OF prefix-rounds[OF True]])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
next
  case False
  show ?thesis
  proof (cases k = i)
    case True
    have nth-eq: ?selected-chunks ! k = chunk
      using True len-prefix-chunks by (simp add: nth-append)
    have
      length chunk =
        verifier-query-round-transcript-length idx
          (map snd f-fl) (map snd fl)
      by (rule verifier-query-round-chunk-length[OF chunk-shape])
    then show ?thesis
      using nth-eq
      by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
    next
    case False
    let ?k = k - Suc i
    have ge: Suc i ≤ k
      using ⟨¬ k < i⟩ False by linarith
    have k-bound': ?k < rounds - Suc i
      using ge k-bound by linarith
    have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k
      using ge len-prefix-chunks by (simp add: nth-append)
    have
      length (suffix-chunks ! ?k) =
        verifier-query-round-transcript-length
          (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
      by (rule verifier-query-round-chunk-length
        [OF suffix-rounds[OF k-bound']])
    then show ?thesis

```



```

    using nth-eq
    by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
  qed
qed
qed
have query-chunk-f:
   $\bigwedge k. k < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! k)
    (map snd f-fl) (map snd fl) (query-chunks ! k)
  using query-chunk trace-roots-eq composition-roots-eq by simp
have query-chunk-lens:
   $\bigwedge k. k < \text{rounds} \implies$ 
    length (query-chunks ! k) =
    verifier-query-round-transcript-length (query-idxs ! k)
    (map snd f-fl) (map snd fl)
  using query-chunk-f verifier-query-round-chunk-length by blast
have selected-transcript:
  PTranscript query-state =
    List.concat ?selected-chunks @ PTranscript final-state
  using transcript-prefix transcript-round transcript-suffix by simp
have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
  using selected-transcript transcript-query final-state-eq by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
     $\bigwedge k. k < \text{length } ?\text{selected-chunks} \implies$ 
      length (?selected-chunks ! k) = length (query-chunks ! k)
    using selected-chunk-lens query-chunk-lens len-selected-chunks
    by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks  $\wedge$  ([] :: 'f list) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
        [OF len-eq same-lens concat-eq-empty])
  then show ?thesis
    by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
  state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
    state-after-query-chunks (PState query-state) prefix-chunks i
  by (rule state-after-query-chunks-append-prefix

```

```

      [OF len-prefix-chunks])
    then show ?thesis
      using chunks-eq by simp
qed
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw'
proof -
  have lookup-suc':
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
      Some raw'
    using lookup-suc state-i counter-i by simp
  have lookup-final-prefix:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
      Some raw'
    using hash-extension-lookup[OF lookup-suc' suffix-ext] .
  show ?thesis
    using lookup-final-prefix prefix-state-eq by simp
qed
have raw-idx-eq: raw' = raw-idxs ! i
  using lookup-final replay-lookup[OF round-bound] final-state-eq
  by simp
have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-idx-eq query-idxs-eq len-raw round-bound by simp
have selected-i: ?selected-chunks ! i = chunk
  using len-prefix-chunks by (simp add: nth-append)
have chunk-i-eq: chunk = query-chunks ! i
  using chunks-eq selected-i by simp
have recorded-i:
  query-round-fri-layer-transcripts (index (to-nat raw))
    (map snd f-fl) (map snd fl) chunk (trace-round-layers ! i)
    (composition-round-layers ! i)
  using recorded[OF round-bound] idx-eq idx-eq-query raw-eq'
    chunk-i-eq trace-roots-eq composition-roots-eq
  by simp
have tr-s0-suc:
  PTranscript s0 = chunk @ PTranscript s-suc
  using tr-s0 transcript-round by simp
have j-bound-fl: j < length fl
  using layer-bound composition-roots-eq by simp
have all-layer-bounds-fl:

```

```

 $\bigwedge k. k < \text{length } fl \implies$ 
  0 < fri-layer-lengths (length fl) (clength * scale) ! k  $\wedge$ 
  fri-layer-indices (length fl) (index (to-nat raw))
    (clength * scale) ! k <
  fri-layer-lengths (length fl) (clength * scale) ! k
using all-layer-bounds composition-roots-eq by simp
have auth-suc:
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j)
    (composition-round-layers ! i ! j) s-suc
by (rule
  verifier-query-round-after-index-composition-recorded-layer-chunk-authenticated
    [OF after recorded-i tr-s0-suc j-bound-fl
      all-layer-bounds-fl])
have auth-final:
  fri-layer-chunk-authenticated (snd (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j)
    (composition-round-layers ! i ! j) final-state
by (rule fri-layer-chunk-authenticated-mono[OF auth-suc suffix-ext])
have raw-idx-query: index (to-nat raw) = query-idxs ! i
using idx-eq idx-eq-query raw-eq' by simp
have auth:
  fri-layer-chunk-authenticated (composition-roots ! j)
    (fri-evidence-layer-len composition-roots j)
    (fri-evidence-layer-idx composition-roots query-idxs i j)
    (composition-round-layers ! i ! j) final-state
using auth-final composition-roots-eq raw-idx-query layer-bound
unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
by simp
show ?thesis
unfolding generic-fri-recorded-layer-chunk-authenticated-def
by (rule auth)
qed
qed
qed
qed

end

end

```

```

theory Soundness-FRI-Composition-Residuals
imports
  Soundness-FRI-Authenticated-Route

```

Soundness-FRI-Base-Recorded-Chunk
Soundness-FRI-Final-Step-Replay
Soundness-FRI-Raw-Layer-Bounds
Soundness-FRI-Recorded-Chunk-Replay

begin

Narrow residual refinements for the verifier-tied composition FRI route. This layer keeps the larger authenticated route theory stable while naming the exact remaining base-opening gap: the recorded first-layer composition FRI chunk is not known to be authenticated.

context *soundness*

begin

definition *composition-fri-reachable-verifier-tie-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-reachable-verifier-tie-gap } s \text{ out} \longleftrightarrow$
 $\text{composition-fri-bad-with-reachable-partial-candidate } s \text{ out} \wedge$
 $\neg \text{composition-fri-bad-with-verifier-tied-partial-candidate } s \text{ out}$

definition *composition-fri-verifier-tied-candidate-transfer-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-candidate-transfer-gap } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as trace-openings composition-openings}$
 $\text{trace-table composition-table.}$
 $\text{composition-fri-bad-with-reachable-partial-candidate } s \text{ out} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr trace-roots}$
 $\text{trace-final as fri-dg composition-roots composition-final}$
 $\text{fri-query-idxs trace-openings composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\neg (\text{trace-table-low-degree trace-table} \wedge$
 $\neg \text{composition-table-low-degree maxDegree composition-table}))$

definition *composition-fri-verifier-tied-candidate-available-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-candidate-available-gap } s \text{ out} \longleftrightarrow$

$(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as trace-openings composition-openings}$
 $\text{trace-table composition-table.}$
 $\text{composition-fri-reachable-verifier-tie-gap s out} \wedge$
 $\text{accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\text{accepted-with-partial-initial-openings-aligned s out fr trace-roots}$
 $\text{trace-final as fri-dg composition-roots composition-final}$
 $\text{fri-query-idxs trace-openings composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings})$

lemma *composition-fri-aligned-candidate-bad-contradicts-verifier-tie-gap:*

assumes *gap:* *composition-fri-reachable-verifier-tie-gap s out*

and *fri-openings:*

$\text{accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers}$

and *aligned:*

$\text{accepted-with-partial-initial-openings-aligned s out fr trace-roots}$
 $\text{trace-final as fri-dg composition-roots composition-final}$
 $\text{fri-query-idxs trace-openings composition-openings}$

and *trace-candidate:*

$\text{partial-trace-table-candidate trace-table trace-openings}$

and *composition-candidate:*

$\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings}$

and *trace-low:* *trace-table-low-degree trace-table*

and *composition-not-low:*

$\neg \text{composition-table-low-degree maxDegree composition-table}$

shows *False*

proof —

have *composition-fri-bad-with-verifier-tied-partial-candidate s out*

unfolding *composition-fri-bad-with-verifier-tied-partial-candidate-def*

by (*intro exI conjI*)

$(\text{rule fri-openings, rule aligned, rule trace-candidate,}$
 $\text{rule composition-candidate, rule trace-low, rule composition-not-low})$

then show *False*

using *gap unfolding composition-fri-reachable-verifier-tie-gap-def* **by** *simp*

qed

lemma *composition-fri-aligned-candidate-transfer-imp-transfer-gap:*

assumes *gap:* *composition-fri-reachable-verifier-tie-gap s out*

and *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *aligned:*
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *composition-candidate:*
partial-composition-table-candidate composition-table
composition-openings
and *transfer:*
 $\neg$ (*trace-table-low-degree trace-table* $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table*)
shows *composition-fri-verifier-tied-candidate-transfer-gap s out*
proof –
have *raw: composition-fri-bad-with-reachable-partial-candidate s out*
using *gap unfolding composition-fri-reachable-verifier-tie-gap-def*
by *simp*
show *?thesis*
unfolding *composition-fri-verifier-tied-candidate-transfer-gap-def*
by (*intro exI conjI*)
(rule raw, rule fri-openings, rule aligned, rule trace-candidate,
rule composition-candidate, rule transfer)
qed

lemma *composition-fri-verifier-tied-candidate-available-gap-imp-transfer-gap:*
assumes *composition-fri-verifier-tied-candidate-available-gap s out*
shows *composition-fri-verifier-tied-candidate-transfer-gap s out*
proof –
obtain *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table
where *gap: composition-fri-reachable-verifier-tie-gap s out*
and *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *aligned:*
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *composition-candidate:*

```

    partial-composition-table-candidate composition-table
    composition-openings
using assms
unfolding composition-fri-verifier-tied-candidate-available-gap-def
by blast
show ?thesis
proof (cases
    trace-table-low-degree trace-table  $\wedge$ 
     $\neg$  composition-table-low-degree maxDegree composition-table)
case True
have False
    by (rule composition-fri-aligned-candidate-bad-contradicts-verifier-tie-gap
        [OF gap fri-openings aligned trace-candidate composition-candidate
         True[THEN conjunct1] True[THEN conjunct2]])
    then show ?thesis by simp
next
case False
show ?thesis
    by (rule composition-fri-aligned-candidate-transfer-imp-transfer-gap
        [OF gap fri-openings aligned trace-candidate composition-candidate
         False])
qed
qed

lemma wp-composition-fri-verifier-tied-candidate-available-gap-bound-from-transfer-gap:
assumes transfer-bound:
    wp-event verify-mono
    (composition-fri-verifier-tied-candidate-transfer-gap s)  $s \leq T$ 
shows
    wp-event verify-mono
    (composition-fri-verifier-tied-candidate-available-gap s)  $s \leq T$ 
by (rule order-trans[OF - transfer-bound])
    (rule wp-event-mono,
     rule composition-fri-verifier-tied-candidate-available-gap-imp-transfer-gap)

lemma composition-fri-verifier-tied-reduction-staged-bound:
assumes reductions:
     $\wedge$  data attacker-state.
    Some (data, attacker-state) \in
        set-dist (execute (checked-staged-transcript-program A)
         adversary-initial-state) \implies
        composition-fri-verifier-tied-reduction
        (verifier-state-from-adversary attacker-state
         (staged-proof-transcript data))
shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-fri-bad-with-verifier-tied-partial-candidate)
    adversary-initial-state \leq composition-fri-error

```

by (rule composition-fri-bad-with-verifier-tied-partial-candidates-staged-bound)
 (rule composition-fri-verifier-tied-reductionD[OF reductions])

lemma composition-fri-verifier-tied-reduction-from-reachable-reduction:

assumes composition-fri-reachable-partial-candidate-reduction-assumption s

shows composition-fri-verifier-tied-reduction s

proof –

have reachable:

wp-event verify-monad

(composition-fri-bad-with-reachable-partial-candidate s) $s \leq$

composition-fri-error

using assms

unfolding

composition-fri-reachable-partial-candidate-reduction-assumption-def

by simp

show ?thesis

unfolding composition-fri-verifier-tied-reduction-def

by (rule order-trans[OF - reachable])

(rule wp-event-mono,

rule composition-fri-bad-with-verifier-tied-imp-reachable-partial-candidate)

qed

lemma composition-fri-reachable-imp-verifier-tied-or-gap:

assumes composition-fri-bad-with-reachable-partial-candidate s out

shows

composition-fri-bad-with-verifier-tied-partial-candidate s out $\vee$

composition-fri-reachable-verifier-tie-gap s out

using assms

unfolding composition-fri-reachable-verifier-tie-gap-def

by blast

lemma wp-composition-fri-reachable-bound-from-verifier-tied-and-gap:

fixes $V G :: \text{prob}$

assumes verifier-bound:

wp-event verify-monad

(composition-fri-bad-with-verifier-tied-partial-candidate s) $s \leq V$

and gap-bound:

wp-event verify-monad

(composition-fri-reachable-verifier-tie-gap s) $s \leq G$

shows

wp-event verify-monad

(composition-fri-bad-with-reachable-partial-candidate s) $s \leq V + G$

proof –

have wp-event verify-monad

(composition-fri-bad-with-reachable-partial-candidate s) $s \leq$

wp-event verify-monad

(λ out.

composition-fri-bad-with-verifier-tied-partial-candidate s out $\vee$

composition-fri-reachable-verifier-tie-gap s out) s


```

    by (rule wp-event-mono)
      (rule composition-fri-reachable-imp-verifier-tied-or-gap)
  also have ... ≤
    wp-event verify-monad
      (composition-fri-bad-with-verifier-tied-partial-candidate s) s +
    wp-event verify-monad
      (composition-fri-reachable-verifier-tie-gap s) s
  by (rule wp-event-union-bound)
  also have ... ≤ V + G
  by (rule add-mono[OF verifier-bound gap-bound])
  finally show ?thesis .
qed

lemma composition-fri-reachable-partial-candidate-reduction-from-verifier-tied-and-gap:
  fixes V G :: prob
  assumes verifier-bound:
    wp-event verify-monad
      (composition-fri-bad-with-verifier-tied-partial-candidate s) s ≤ V
  and gap-bound:
    wp-event verify-monad
      (composition-fri-reachable-verifier-tie-gap s) s ≤ G
  and total-bound: V + G ≤ composition-fri-error
  shows composition-fri-reachable-partial-candidate-reduction-assumption s
proof -
  have old-bound:
    wp-event verify-monad
      (composition-fri-bad-with-reachable-partial-candidate s) s ≤ V + G
  by (rule wp-composition-fri-reachable-bound-from-verifier-tied-and-gap
      [OF verifier-bound gap-bound])
  show ?thesis
  unfolding
    composition-fri-reachable-partial-candidate-reduction-assumption-def
  by (rule order-trans[OF old-bound total-bound])
qed

definition composition-fri-verifier-tied-recorded-base-chunk-auth-missing
  :: ('f, 'a) protocol-channel-scheme ⇒
    (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool
where
  composition-fri-verifier-tied-recorded-base-chunk-auth-missing s out
  ⇔
  (∃ trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings composition-openings
    trace-table composition-table round-idx xp xp-path xn xn-path
    result final-state.
    out = Some (result, final-state) ∧
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs

```

composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
round-idx < length fri-query-idxs $\wedge$
0 < length composition-bs $\wedge$
fri-layer-step-evidence
(composition-roots ! 0)
(composition-bs ! 0)
(fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
1
(fri-sibling-index (fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0))
xp xp-path xn xn-path
(fri-evidence-next-idx composition-roots fri-query-idxs round-idx 0)
(fri-evidence-next-value composition-roots composition-bs
fri-query-idxs round-idx 0 xp xn)
(composition-round-layers ! round-idx ! 0) $\wedge$
 $\neg$ *fri-opening-matches-table*
(fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
composition-table xp xn $\wedge$
 $\neg$ *fri-layer-chunk-authenticated (composition-roots ! 0)*
(fri-evidence-layer-len composition-roots 0)
(fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
(composition-round-layers ! round-idx ! 0) final-state)

lemma *composition-fri-verifier-tied-recorded-base-chunk-auth-missing-false:*

assumes

composition-fri-verifier-tied-recorded-base-chunk-auth-missing s out

shows *False*

proof —

from *assms obtain* *trace-roots trace-bs trace-final fri-dg*

composition-roots composition-bs composition-final fri-query-idxs

trace-round-layers composition-round-layers fr as trace-openings

composition-openings trace-table composition-table round-idx xp xp-path

xn xn-path result final-state

where *out-eq: out = Some (result, final-state)*

and *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final fri-dg composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

```

    composition-round-layers
  and round-bound: round-idx < length fri-query-idxs
  and composition-bs-nonempty: 0 < length composition-bs
  and missing:
    ¬ fri-layer-chunk-authenticated (composition-roots ! 0)
      (fri-evidence-layer-len composition-roots 0)
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
      (composition-round-layers ! round-idx ! 0) final-state
  unfolding
    composition-fri-verifier-tied-recorded-base-chunk-auth-missing-def
  by blast
  have len-composition-bs:
    length composition-bs = length composition-roots
    using accepted-fri-opening-transcript-shapes(4)[OF fri-openings] .
  have composition-nonempty: 0 < length composition-roots
    using composition-bs-nonempty len-composition-bs by simp
  have len-query: length fri-query-idxs = rounds
    using accepted-fri-opening-transcript-shapes(7)[OF fri-openings] .
  have round-bound-rounds: round-idx < rounds
    using round-bound len-query by simp
  have auth:
    fri-layer-chunk-authenticated (composition-roots ! 0) (clength * scale)
      (fri-query-idxs ! round-idx)
      (composition-round-layers ! round-idx ! 0) final-state
  by (rule
    accepted-fri-opening-transcript-composition-base-recorded-chunk-authenticated-at
      [OF fri-openings out-eq round-bound-rounds composition-nonempty])
  have len0:
    fri-evidence-layer-len composition-roots 0 = clength * scale
    using composition-nonempty
    unfolding fri-evidence-layer-len-def
    by (cases composition-roots) simp-all
  have idx0:
    fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0 =
      fri-query-idxs ! round-idx
    using composition-nonempty
    unfolding fri-evidence-layer-idx-def
    by (cases composition-roots) simp-all
  show False
    using missing auth unfolding len0 idx0 by contradiction
qed

lemma wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero:
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
      s ≤ 0
  proof -
    have wp-event verify-monad

```

```

      (composition-fri-verifier-tied-recorded-base-chunk-auth-missing s) s
    ≤ wp-event verify-monad (λ-. False) s
  by (rule wp-event-mono)
    (rule composition-fri-verifier-tied-recorded-base-chunk-auth-missing-false)
  also have ... = 0
    unfolding wp-event-def wp-def dist-expect-def by simp
  finally show ?thesis .
qed

lemma composition-fri-verifier-tied-base-chunk-auth-gap-imp-recorded-missing:
  assumes composition-fri-verifier-tied-base-chunk-auth-gap s out
  shows
    composition-fri-verifier-tied-recorded-base-chunk-auth-missing s out
proof -
  from assms have base:
    composition-fri-verifier-tied-sampled-base-opening-conflict s out
  and no-auth:
    ¬ composition-fri-verifier-tied-authenticated-base-opening-conflict
      s out
  unfolding composition-fri-verifier-tied-base-chunk-auth-gap-def
  by simp-all
  from base obtain trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings composition-openings
    trace-table composition-table
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
  and aligned:
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final fri-query-idxs
    trace-openings composition-openings
  and trace-cand:
    partial-trace-table-candidate trace-table trace-openings
  and comp-cand:
    partial-composition-table-candidate composition-table
    composition-openings
  and trace-low: trace-table-low-degree trace-table
  and comp-not-low:
    ¬ composition-table-low-degree maxDegree composition-table
  and generic:
    generic-fri-sampled-base-opening-conflict composition-table
    composition-roots composition-bs fri-query-idxs
    composition-round-layers
  unfolding composition-fri-verifier-tied-sampled-base-opening-conflict-def
  by blast
  from fri-openings obtain result final-state where out-eg:

```

```

out = Some (result, final-state)
unfolding accepted-fri-opening-transcript-def by blast
from generic obtain round-idx xp xp-path xn xn-path where
  round-bound: round-idx < length fri-query-idxs
and composition-bs-nonempty: 0 < length composition-bs
and step:
  fri-layer-step-evidence
    (composition-roots ! 0)
    (composition-bs ! 0)
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
  1
  (fri-sibling-index (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0))
  xp xp-path xn xn-path
  (fri-evidence-next-idx composition-roots fri-query-idxs round-idx 0)
  (fri-evidence-next-value composition-roots composition-bs
    fri-query-idxs round-idx 0 xp xn)
  (composition-round-layers ! round-idx ! 0)
and no-match:
  ¬ fri-opening-matches-table
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
    composition-table xp xn
unfolding generic-fri-sampled-base-opening-conflict-def
by blast
have missing:
  ¬ fri-layer-chunk-authenticated (composition-roots ! 0)
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
    (composition-round-layers ! round-idx ! 0) final-state
proof
assume auth:
  fri-layer-chunk-authenticated (composition-roots ! 0)
    (fri-evidence-layer-len composition-roots 0)
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
    (composition-round-layers ! round-idx ! 0) final-state
have
  composition-fri-verifier-tied-authenticated-base-opening-conflict
    s out
unfolding composition-fri-verifier-tied-authenticated-base-opening-conflict-def
by (intro exI conjI)
  (rule out-eq, rule fri-openings, rule aligned, rule trace-cand,
    rule comp-cand, rule trace-low, rule comp-not-low, rule round-bound,
    rule composition-bs-nonempty, rule step, rule no-match, rule auth)
then show False
using no-auth by contradiction
qed
show ?thesis

```

unfolding *composition-fri-verifier-tied-recorded-base-chunk-auth-missing-def*
by (intro exI conjI)
(rule out-eq, rule fri-openings, rule aligned, rule trace-cand,
rule comp-cand, rule trace-low, rule comp-not-low, rule round-bound,
rule composition-bs-nonempty, rule step, rule no-match, rule missing)
qed

lemma *wp-composition-fri-verifier-tied-base-chunk-auth-gap-bound-from-recorded-missing*:
fixes $M :: \text{prob}$
assumes *missing-bound*:
wp-event verify-monad
(*composition-fri-verifier-tied-recorded-base-chunk-auth-missing* s)
 $s \leq M$
shows
wp-event verify-monad
(*composition-fri-verifier-tied-base-chunk-auth-gap* s) $s \leq M$
proof –
have wp-event verify-monad
(*composition-fri-verifier-tied-base-chunk-auth-gap* s) $s \leq$
wp-event verify-monad
(*composition-fri-verifier-tied-recorded-base-chunk-auth-missing* s) s
by (rule wp-event-mono)
(rule *composition-fri-verifier-tied-base-chunk-auth-gap-imp-recorded-missing*)
then show ?thesis
by (rule order-trans[*OF* - *missing-bound*])
qed

lemma *wp-composition-fri-verifier-tied-sampled-base-bound-from-partial-merkle-and-recorded-missing*:
fixes $M :: \text{prob}$
assumes *missing-bound*:
wp-event verify-monad
(*composition-fri-verifier-tied-recorded-base-chunk-auth-missing* s)
 $s \leq M$
shows
wp-event verify-monad
(*composition-fri-verifier-tied-sampled-base-opening-conflict* s) s
 $\leq$ wp-event verify-monad (*partial-merkle-inconsistency-bad* s) $s + M$
by (rule wp-composition-fri-verifier-tied-sampled-base-bound-from-partial-merkle-and-auth-gap)
(rule
wp-composition-fri-verifier-tied-base-chunk-auth-gap-bound-from-recorded-missing
[*OF* *missing-bound*])

lemma *wp-composition-fri-verifier-tied-without-same-bound-from-partial-merkle-recorded-missing-and-branches*:
fixes $M N S F :: \text{prob}$
assumes *missing-bound*:
wp-event verify-monad
(*composition-fri-verifier-tied-recorded-base-chunk-auth-missing* s)
 $s \leq M$
and *next-bound*:

wp-event verify-monad
 (composition-fri-verifier-tied-sampled-next-value-conflict s) $s \leq N$
and *successor-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-sampled-successor-opening-conflict s)
 $s \leq S$
and *final-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-sampled-final-value-conflict s) $s \leq F$
shows
wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s)
 $s \leq (\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) s + M) + N + S + F$
by (rule wp-composition-fri-verifier-tied-without-same-bound-from-branches)
 (rule wp-composition-fri-verifier-tied-sampled-base-bound-from-partial-merkle-and-recorded-missing
 [OF missing-bound],
 rule next-bound,
 rule successor-bound,
 rule final-bound)

lemma wp-composition-fri-verifier-tied-authenticated-conflict-bound-from-recorded-missing-and-branches:

fixes $P M N S F :: \text{prob}$

assumes *partial-merkle-bound*:

wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq P$

and *missing-bound*:

wp-event verify-monad

(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)

$s \leq M$

and *next-bound*:

wp-event verify-monad

(composition-fri-verifier-tied-sampled-next-value-conflict s) $s \leq N$

and *successor-bound*:

wp-event verify-monad

(composition-fri-verifier-tied-sampled-successor-opening-conflict s)

$s \leq S$

and *final-bound*:

wp-event verify-monad

(composition-fri-verifier-tied-sampled-final-value-conflict s) $s \leq F$

shows

wp-event verify-monad

(composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s)

$s \leq (P + M) + N + S + F + P$

proof –

have *without-bound*:

wp-event verify-monad

(composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s)

$s \leq (P + M) + N + S + F$

proof –

have *wp-event verify-monad*

(*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer* s)
 $s \leq$
 (*wp-event verify-mono* (*partial-merkle-inconsistency-bad* s) $s + M$)
 $+ N + S + F$
by (*rule*
wp-composition-fri-verifier-tied-without-same-bound-from-partial-merkle-recorded-missing-and-branches
 [*OF missing-bound next-bound successor-bound final-bound*])
also have $\dots \leq (P + M) + N + S + F$
by (*intro add-mono partial-merkle-bound order-refl*)
finally show *?thesis* .
qed
show *?thesis*
by (*rule*
wp-composition-fri-verifier-tied-authenticated-conflict-bound-from-without-same-and-merkle
 [*OF without-bound partial-merkle-bound*])
qed

definition *composition-fri-verifier-tied-next-value-replay-checked-conflict*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-next-value-replay-checked-conflict s *out*
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table round-idx round-idx' layer-idx } v \ v'.$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{ trace-final fri-dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers} \wedge$
 $\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr trace-roots}$
 $\text{ trace-final as fri-dg composition-roots composition-final fri-query-idxs}$
 $\text{ trace-openings composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{ composition-openings} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\neg \text{ composition-table-low-degree maxDegree composition-table} \wedge$
 $\text{round-idx} < \text{length fri-query-idxs} \wedge$
 $\text{round-idx}' < \text{length fri-query-idxs} \wedge$
 $\text{layer-idx} < \text{length composition-bs} \wedge$
 $\text{fri-evidence-next-idx composition-roots fri-query-idxs round-idx}$
 $\text{ layer-idx} =$
 $\text{fri-evidence-next-idx composition-roots fri-query-idxs round-idx}'$
 $\text{ layer-idx} \wedge$
 $\text{generic-fri-round-forced-next-value composition-roots composition-bs}$
 $\text{ fri-query-idxs composition-round-layers round-idx layer-idx } v \wedge$
 $\text{generic-fri-round-forced-next-value composition-roots composition-bs}$

$fri_query_idxs \text{ composition-round-layers round-idx' layer-idx } v' \wedge$
 $v \neq v' \wedge$
 $v = v')$

definition *composition-fri-verifier-tied-next-value-replay-gap*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-next-value-replay-gap } s \text{ out} \longleftrightarrow$
 $\text{composition-fri-verifier-tied-sampled-next-value-conflict } s \text{ out} \wedge$
 $\neg \text{composition-fri-verifier-tied-next-value-replay-checked-conflict}$
 $s \text{ out}$

lemma *composition-fri-verifier-tied-next-value-replay-checked-conflict-false:*

$\neg \text{composition-fri-verifier-tied-next-value-replay-checked-conflict } s \text{ out}$

unfolding *composition-fri-verifier-tied-next-value-replay-checked-conflict-def*
by *blast*

lemma *composition-fri-verifier-tied-sampled-next-value-imp-replay-gap:*

assumes *composition-fri-verifier-tied-sampled-next-value-conflict* $s \text{ out}$

shows *composition-fri-verifier-tied-next-value-replay-gap* $s \text{ out}$

using *assms composition-fri-verifier-tied-next-value-replay-checked-conflict-false*

unfolding *composition-fri-verifier-tied-next-value-replay-gap-def*
by *simp*

lemma *composition-fri-verifier-tied-sampled-next-value-conflictE:*

assumes *composition-fri-verifier-tied-sampled-next-value-conflict* $s \text{ out}$

obtains *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table round-idx round-idx' layer-idx v v'

where

$\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers}$
 $\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr trace-roots}$
 $\text{trace-final as fri-dg composition-roots composition-final fri-query-idxs}$
 $\text{trace-openings composition-openings}$
 $\text{partial-trace-table-candidate trace-table trace-openings}$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings}$
 $\text{trace-table-low-degree trace-table}$
 $\neg \text{composition-table-low-degree maxDegree composition-table}$
 $\text{round-idx} < \text{length fri-query-idxs}$
 $\text{round-idx'} < \text{length fri-query-idxs}$
 $\text{layer-idx} < \text{length composition-bs}$
 $\text{fri-evidence-next-idx composition-roots fri-query-idxs round-idx}$
 $\text{layer-idx} =$
 $\text{fri-evidence-next-idx composition-roots fri-query-idxs round-idx'}$

layer-idx
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx layer-idx v
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx' layer-idx v'
 $v \neq v'$
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as trace-openings
composition-openings trace-table composition-table
where *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *aligned*:
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
and *trace-cand*:
partial-trace-table-candidate trace-table trace-openings
and *comp-cand*:
partial-composition-table-candidate composition-table
composition-openings
and *trace-low*: *trace-table-low-degree trace-table*
and *comp-not-low*:
 $\neg$ *composition-table-low-degree maxDegree composition-table*
and *conflict*:
generic-fri-sampled-next-value-conflict composition-roots
composition-bs fri-query-idxs composition-round-layers
unfolding *composition-fri-verifier-tied-sampled-next-value-conflict-def*
by *blast*
from *generic-fri-sampled-next-value-conflictE[OF conflict]*
obtain *round-idx round-idx' layer-idx v v'*
where *round-bound*: *round-idx < length fri-query-idxs*
and *round-bound'*: *round-idx' < length fri-query-idxs*
and *layer-bound*: *layer-idx < length composition-bs*
and *same-idx*:
fri-evidence-next-idx composition-roots fri-query-idxs round-idx
layer-idx =
fri-evidence-next-idx composition-roots fri-query-idxs round-idx'
layer-idx
and *forced*:
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx layer-idx v
and *forced'*:
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx' layer-idx v'

```

    and neg:  $v \neq v'$ 
  by blast
show ?thesis
  by (rule that[OF fri-openings aligned trace-cand comp-cand trace-low
    comp-not-low round-bound round-bound' layer-bound same-idx forced
    forced' neg])
qed

lemma wp-composition-fri-verifier-tied-sampled-next-value-bound-from-replay-gap:
  fixes  $G :: \text{prob}$ 
  assumes gap-bound:
    wp-event verify-monad
    (composition-fri-verifier-tied-next-value-replay-gap  $s$ )  $s \leq G$ 
  shows
    wp-event verify-monad
    (composition-fri-verifier-tied-sampled-next-value-conflict  $s$ )  $s \leq G$ 
proof –
  have wp-event verify-monad
    (composition-fri-verifier-tied-sampled-next-value-conflict  $s$ )  $s \leq$ 
    wp-event verify-monad
    (composition-fri-verifier-tied-next-value-replay-gap  $s$ )  $s$ 
  by (rule wp-event-mono)
    (rule composition-fri-verifier-tied-sampled-next-value-imp-replay-gap)
  then show ?thesis
    by (rule order-trans[OF - gap-bound])
qed

definition composition-fri-verifier-tied-successor-replay-checked-conflict
  :: ( $f, 'a$ ) protocol-channel-scheme  $\Rightarrow$ 
    ( $\text{unit list} \times (\mathbf{f}, 'a) \text{ protocol-channel-scheme}$ ) option  $\Rightarrow \text{bool}$ 
where
  composition-fri-verifier-tied-successor-replay-checked-conflict  $s$  out
 $\iff$ 
  ( $\exists$  trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings composition-openings
    trace-table composition-table source-round target-round layer-idx
     $v$   $xp$   $xp\text{-path}$   $xn$   $xn\text{-path}$ .
    accepted-fri-opening-transcript  $s$  out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers  $\wedge$ 
    accepted-with-partial-initial-openings-aligned  $s$  out fr trace-roots
    trace-final as fri-dg composition-roots composition-final fri-query-idxs
    trace-openings composition-openings  $\wedge$ 
    partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
    partial-composition-table-candidate composition-table
    composition-openings  $\wedge$ 
    trace-table-low-degree trace-table  $\wedge$ 

```

$\neg$ composition-table-low-degree maxDegree composition-table $\wedge$
source-round < length fri-query-idxs $\wedge$
target-round < length fri-query-idxs $\wedge$
Suc layer-idx < length composition-bs $\wedge$
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers source-round layer-idx v $\wedge$
fri-layer-step-evidence
(composition-roots ! Suc layer-idx)
(composition-bs ! Suc layer-idx)
(fri-evidence-layer-len composition-roots (Suc layer-idx))
(fri-evidence-layer-idx composition-roots fri-query-idxs target-round
(Suc layer-idx))
(2 ^ Suc layer-idx)
(fri-sibling-index
(fri-evidence-layer-len composition-roots (Suc layer-idx))
(fri-evidence-layer-idx composition-roots fri-query-idxs
target-round (Suc layer-idx)))
xp xp-path xn xn-path
(fri-evidence-next-idx composition-roots fri-query-idxs target-round
(Suc layer-idx))
(fri-evidence-next-value composition-roots composition-bs
fri-query-idxs target-round (Suc layer-idx) xp xn)
(composition-round-layers ! target-round ! Suc layer-idx) $\wedge$
((fri-evidence-next-idx composition-roots fri-query-idxs source-round
layer-idx =
fri-evidence-layer-idx composition-roots fri-query-idxs target-round
(Suc layer-idx) $\wedge$
v $\neq$ xp $\wedge$ v = xp) $\vee$
(fri-evidence-next-idx composition-roots fri-query-idxs source-round
layer-idx =
fri-sibling-index
(fri-evidence-layer-len composition-roots (Suc layer-idx))
(fri-evidence-layer-idx composition-roots fri-query-idxs
target-round (Suc layer-idx)) $\wedge$
v $\neq$ xn $\wedge$ v = xn)))

definition composition-fri-verifier-tied-successor-replay-gap

$::$ ('f, 'a) protocol-channel-scheme $\Rightarrow$
(unit list $\times$ ('f, 'a) protocol-channel-scheme) option $\Rightarrow$ bool

where

composition-fri-verifier-tied-successor-replay-gap s out $\longleftrightarrow$
composition-fri-verifier-tied-sampled-successor-opening-conflict s out $\wedge$
 $\neg$ composition-fri-verifier-tied-successor-replay-checked-conflict
s out

lemma composition-fri-verifier-tied-successor-replay-checked-conflict-false:

$\neg$ composition-fri-verifier-tied-successor-replay-checked-conflict s out

unfolding

composition-fri-verifier-tied-successor-replay-checked-conflict-def

by blast

lemma *composition-fri-verifier-tied-sampled-successor-imp-replay-gap:*

assumes

composition-fri-verifier-tied-sampled-successor-opening-conflict s out

shows *composition-fri-verifier-tied-successor-replay-gap s out*

using *assms*

composition-fri-verifier-tied-successor-replay-checked-conflict-false

unfolding *composition-fri-verifier-tied-successor-replay-gap-def*

by *simp*

lemma *composition-fri-verifier-tied-sampled-successor-opening-conflictE:*

assumes

composition-fri-verifier-tied-sampled-successor-opening-conflict s out

obtains *trace-roots trace-bs trace-final fri-dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as trace-openings composition-openings

trace-table composition-table source-round target-round layer-idx v

xp xp-path xn xn-path

where

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final fri-dg composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers

accepted-with-partial-initial-openings-aligned s out fr trace-roots

trace-final as fri-dg composition-roots composition-final fri-query-idxs

trace-openings composition-openings

partial-trace-table-candidate trace-table trace-openings

partial-composition-table-candidate composition-table

composition-openings

trace-table-low-degree trace-table

$\neg$ *composition-table-low-degree maxDegree composition-table*

source-round < length fri-query-idxs

target-round < length fri-query-idxs

Suc layer-idx < length composition-bs

generic-fri-round-forced-next-value composition-roots composition-bs

fri-query-idxs composition-round-layers source-round layer-idx v

fri-layer-step-evidence

(composition-roots ! Suc layer-idx)

(composition-bs ! Suc layer-idx)

(fri-evidence-layer-len composition-roots (Suc layer-idx))

(fri-evidence-layer-idx composition-roots fri-query-idxs target-round

(Suc layer-idx))

(2 $\wedge$ Suc layer-idx)

(fri-sibling-index

(fri-evidence-layer-len composition-roots (Suc layer-idx))

(fri-evidence-layer-idx composition-roots fri-query-idxs

target-round (Suc layer-idx)))

xp xp-path xn xn-path

(fri-evidence-next-idx composition-roots fri-query-idxs target-round

```

    (Suc layer-idx))
  (fri-evidence-next-value composition-roots composition-bs
    fri-query-idxs target-round (Suc layer-idx) xp xn)
  (composition-round-layers ! target-round ! Suc layer-idx)
  (fri-evidence-next-idx composition-roots fri-query-idxs source-round
    layer-idx =
    fri-evidence-layer-idx composition-roots fri-query-idxs target-round
    (Suc layer-idx) ∧
    v ≠ xp) ∨
  (fri-evidence-next-idx composition-roots fri-query-idxs source-round
    layer-idx =
    fri-sibling-index
    (fri-evidence-layer-len composition-roots (Suc layer-idx))
    (fri-evidence-layer-idx composition-roots fri-query-idxs
      target-round (Suc layer-idx)) ∧
    v ≠ xn)
proof -
from assms obtain trace-roots trace-bs trace-final fri-dg
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as trace-openings
composition-openings trace-table composition-table
where fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final fri-dg composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers
and aligned:
  accepted-with-partial-initial-openings-aligned s out fr trace-roots
  trace-final as fri-dg composition-roots composition-final
  fri-query-idxs trace-openings composition-openings
and trace-cand:
  partial-trace-table-candidate trace-table trace-openings
and comp-cand:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-not-low:
  ¬ composition-table-low-degree maxDegree composition-table
and conflict:
  generic-fri-sampled-successor-opening-conflict composition-roots
  composition-bs fri-query-idxs composition-round-layers
unfolding
  composition-fri-verifier-tied-sampled-successor-opening-conflict-def
by blast
from generic-fri-sampled-successor-opening-conflictE[OF conflict]
obtain source-round target-round layer-idx v xp xp-path xn xn-path
where source-bound: source-round < length fri-query-idxs
and target-bound: target-round < length fri-query-idxs
and layer-bound: Suc layer-idx < length composition-bs

```

and forced:
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers source-round layer-idx v
and step:
fri-layer-step-evidence
(composition-roots ! Suc layer-idx)
(composition-bs ! Suc layer-idx)
(fri-evidence-layer-len composition-roots (Suc layer-idx))
(fri-evidence-layer-idx composition-roots fri-query-idxs
target-round (Suc layer-idx))
(2 ^ Suc layer-idx)
(fri-sibling-index
(fri-evidence-layer-len composition-roots (Suc layer-idx))
(fri-evidence-layer-idx composition-roots fri-query-idxs
target-round (Suc layer-idx)))
xp xp-path xn xn-path
(fri-evidence-next-idx composition-roots fri-query-idxs
target-round (Suc layer-idx))
(fri-evidence-next-value composition-roots composition-bs
fri-query-idxs target-round (Suc layer-idx) xp xn)
(composition-round-layers ! target-round ! Suc layer-idx)
and cases:
(fri-evidence-next-idx composition-roots fri-query-idxs source-round
layer-idx =
fri-evidence-layer-idx composition-roots fri-query-idxs target-round
(Suc layer-idx) ∧
v ≠ xp) ∨
(fri-evidence-next-idx composition-roots fri-query-idxs source-round
layer-idx =
fri-sibling-index
(fri-evidence-layer-len composition-roots (Suc layer-idx))
(fri-evidence-layer-idx composition-roots fri-query-idxs
target-round (Suc layer-idx)) ∧
v ≠ xn)
by blast
show ?thesis
by (rule that[*OF fri-openings aligned trace-cand comp-cand trace-low*
comp-not-low source-bound target-bound layer-bound forced step
cases])
qed

lemma *wp-composition-fri-verifier-tied-sampled-successor-bound-from-replay-gap:*
fixes *G :: prob*
assumes *gap-bound:*
wp-event verify-monad
(composition-fri-verifier-tied-successor-replay-gap s) s ≤ G
shows
wp-event verify-monad
(composition-fri-verifier-tied-sampled-successor-opening-conflict s)

$s \leq G$
proof –
have *wp-event verify-monad*
 $(\text{composition-fri-verifier-tied-sampled-successor-opening-conflict } s)$
 $s \leq$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap } s) \ s$
by $(\text{rule wp-event-mono})$
 $(\text{rule composition-fri-verifier-tied-sampled-successor-imp-replay-gap})$
then show *?thesis*
by $(\text{rule order-trans}[OF - \text{gap-bound}])$
qed

definition *composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap}$
 $s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as trace-openings composition-openings}$
 $\text{trace-table composition-table result final-state.}$
 $\text{out} = \text{Some}(\text{result}, \text{final-state}) \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr trace-roots}$
 $\text{trace-final as fri-dg composition-roots composition-final}$
 $\text{fri-query-idxs trace-openings composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\neg \text{composition-table-low-degree maxDegree composition-table} \wedge$
 $\text{generic-fri-authenticated-slot-recorded-chunk-gap composition-roots}$
 $\text{fri-query-idxs composition-round-layers final-state})$

definition *composition-fri-verifier-tied-final-value-replay-checked-conflict*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-final-value-replay-checked-conflict } s \text{ out}$
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as trace-openings composition-openings}$
 $\text{trace-table composition-table round-idx } v.$

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
 $0 < \text{length } \text{composition-bs}$ $\wedge$
 $\text{round-idx} < \text{length } \text{fri-query-idxs}$ $\wedge$
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx
 $(\text{length } \text{composition-bs} - 1) \vee \wedge$
 $v \neq \text{composition-final}$ $\wedge$
 $v = \text{composition-final}$

definition *composition-fri-verifier-tied-final-value-replay-gap*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-final-value-replay-gap s out $\longleftrightarrow$
composition-fri-verifier-tied-sampled-final-value-conflict s out $\wedge$
 $\neg$ *composition-fri-verifier-tied-final-value-replay-checked-conflict*
s out

lemma *composition-fri-verifier-tied-sampled-final-value-conflictE*:

assumes

composition-fri-verifier-tied-sampled-final-value-conflict s out

obtains *trace-roots trace-bs trace-final fri-dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as trace-openings composition-openings

trace-table composition-table round-idx v

where

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final fri-dg composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers

accepted-with-partial-initial-openings-aligned s out fr trace-roots

trace-final as fri-dg composition-roots composition-final fri-query-idxs

trace-openings composition-openings

partial-trace-table-candidate trace-table trace-openings

partial-composition-table-candidate composition-table

composition-openings

trace-table-low-degree trace-table

$\neg$ *composition-table-low-degree maxDegree composition-table*

$0 < \text{length } \text{composition-bs}$

round-idx < *length fri-query-idxs*
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx
(length composition-bs - 1) v
v ≠ *composition-final*
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as trace-openings
composition-openings trace-table composition-table **where**
fri-openings:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *aligned:*
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *composition-candidate:*
partial-composition-table-candidate composition-table
composition-openings
and *trace-low:* *trace-table-low-degree trace-table*
and *composition-nonlow:*
 ¬ *composition-table-low-degree maxDegree composition-table*
and *final-conflict:*
generic-fri-sampled-final-value-conflict composition-roots
composition-bs composition-final fri-query-idxs
composition-round-layers
unfolding *composition-fri-verifier-tied-sampled-final-value-conflict-def*
by *blast*
from *generic-fri-sampled-final-value-conflictE[OF final-conflict]*
obtain *round-idx v* **where**
nonempty: *0 < length composition-bs*
and *round-bound:* *round-idx < length fri-query-idxs*
and *forced:*
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx
(length composition-bs - 1) v
and *neq:* *v* ≠ *composition-final*
by *blast*
show *?thesis*
by (*rule that[OF fri-openings aligned trace-candidate*
composition-candidate trace-low composition-nonlow nonempty
round-bound forced neq])
qed

lemma *composition-fri-verifier-tied-final-value-replay-checked-conflict-false*:
 $\neg$ *composition-fri-verifier-tied-final-value-replay-checked-conflict* *s* *out*
unfolding *composition-fri-verifier-tied-final-value-replay-checked-conflict-def*
by *blast*

lemma *composition-fri-verifier-tied-sampled-final-value-imp-replay-gap*:
assumes *composition-fri-verifier-tied-sampled-final-value-conflict* *s* *out*
shows *composition-fri-verifier-tied-final-value-replay-gap* *s* *out*
using *assms composition-fri-verifier-tied-final-value-replay-checked-conflict-false*
unfolding *composition-fri-verifier-tied-final-value-replay-gap-def*
by *simp*

lemma *wp-composition-fri-verifier-tied-sampled-final-value-bound-from-replay-gap*:
fixes *G* :: *prob*
assumes *gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-final-value-replay-gap s) s $\leq$ *G*
shows
wp-event verify-monad
(composition-fri-verifier-tied-sampled-final-value-conflict s) s $\leq$ *G*
proof –
have *wp-event verify-monad*
(composition-fri-verifier-tied-sampled-final-value-conflict s) s $\leq$
wp-event verify-monad
(composition-fri-verifier-tied-final-value-replay-gap s) s
by (*rule wp-event-mono*)
(rule composition-fri-verifier-tied-sampled-final-value-imp-replay-gap)
then show *?thesis*
by (*rule order-trans[OF - gap-bound]*)
qed

definition *composition-fri-verifier-tied-final-value-authenticated-replay-gap*
:: (*f*, *a*) *protocol-channel-scheme* $\Rightarrow$
(*unit list* $\times$ (*f*, *a*) *protocol-channel-scheme*) *option* $\Rightarrow$ *bool*
where
composition-fri-verifier-tied-final-value-authenticated-replay-gap s out
 $\iff$
 $(\exists$ *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table result final-state round-idx v.
out = Some (result, final-state) $\wedge$
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings $\wedge$

partial-trace-table-candidate *trace-table* *trace-openings* $\wedge$
partial-composition-table-candidate *composition-table*
composition-openings $\wedge$
trace-table-low-degree *trace-table* $\wedge$
 $\neg$ *composition-table-low-degree* *maxDegree* *composition-table* $\wedge$
 $0 < \text{length } \text{composition-bs}$ $\wedge$
 $\text{round-idx} < \text{length } \text{fri-query-idxs}$ $\wedge$
generic-fri-round-forced-next-value *composition-roots* *composition-bs*
fri-query-idxs *composition-round-layers* *round-idx*
 $(\text{length } \text{composition-bs} - 1) \vee \wedge$
 $v \neq \text{composition-final}$ $\wedge$
 $\neg$ *partial-merkle-inconsistency-bad* *s* *out* $\wedge$
generic-fri-recorded-layer-chunk-authenticated *composition-roots*
fri-query-idxs *composition-round-layers* *final-state* *round-idx*
 $(\text{length } \text{composition-bs} - 1))$

definition *composition-fri-verifier-tied-final-value-selected-step-missing-gap*

$:: ('f, 'a) \text{protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{protocol-channel-scheme}) \text{option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-final-value-selected-step-missing-gap *s* *out*
 $\longleftrightarrow$
composition-fri-verifier-tied-final-value-authenticated-replay-gap *s* *out* $\wedge$
 $(\exists \text{trace-roots } \text{trace-bs } \text{trace-final } \text{fri-dg } \text{composition-roots}$
 $\text{composition-bs } \text{composition-final } \text{fri-query-idxs } \text{trace-round-layers}$
 $\text{composition-round-layers } \text{result } \text{final-state } \text{round-idx } v.$
 $\text{out} = \text{Some } (\text{result}, \text{final-state}) \wedge$
accepted-fri-opening-transcript *s* *out* *trace-roots* *trace-bs*
trace-final *fri-dg* *composition-roots* *composition-bs*
composition-final *fri-query-idxs* *trace-round-layers*
composition-round-layers $\wedge$
 $0 < \text{length } \text{composition-bs}$ $\wedge$
 $\text{round-idx} < \text{length } \text{fri-query-idxs}$ $\wedge$
generic-fri-round-forced-next-value *composition-roots* *composition-bs*
fri-query-idxs *composition-round-layers* *round-idx*
 $(\text{length } \text{composition-bs} - 1) \vee \wedge$
 $v \neq \text{composition-final}$ $\wedge$
 $\neg$ *partial-merkle-inconsistency-bad* *s* *out* $\wedge$
generic-fri-recorded-layer-chunk-authenticated *composition-roots*
fri-query-idxs *composition-round-layers* *final-state* *round-idx*
 $(\text{length } \text{composition-bs} - 1) \wedge$
 $(\forall \text{actual-chunk } \text{axp } \text{axp-path } \text{axn } \text{axn-path}.$
 $\neg (\text{fri-layer-chunk-authenticated}$
 $(\text{composition-roots } ! (\text{length } \text{composition-bs} - 1))$
 $(\text{fri-evidence-layer-len } \text{composition-roots } (\text{length } \text{composition-bs} - 1))$
 $(\text{fri-evidence-layer-idx } \text{composition-roots } \text{fri-query-idxs } \text{round-idx}$
 $(\text{length } \text{composition-bs} - 1))$
 $\text{actual-chunk } \text{final-state} \wedge$
 $\text{fri-layer-step-evidence}$

```

    (composition-roots ! (length composition-bs - 1))
    (composition-bs ! (length composition-bs - 1))
    (fri-evidence-layer-len composition-roots (length composition-bs - 1))
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
      (length composition-bs - 1))
    (2 ^ (length composition-bs - 1))
    (fri-sibling-index
      (fri-evidence-layer-len composition-roots (length composition-bs - 1))
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
        (length composition-bs - 1)))
    arp arp-path arn arn-path
    (fri-evidence-next-idx composition-roots fri-query-idxs round-idx
      (length composition-bs - 1))
    composition-final actual-chunk)))

```

lemma *composition-fri-verifier-tied-final-value-selected-step-missing-gap-false-on-support:*
assumes *support:* $out \in \text{set-dist } (\text{execute verify-monad } s)$

and *gap:*

composition-fri-verifier-tied-final-value-selected-step-missing-gap
s out

shows *False*

proof —

from *gap* **obtain** *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers result final-state round-idx v

where *out-eq:* $out = \text{Some } (result, final-state)$

and *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

and *nonempty-bs:* $0 < \text{length } composition-bs$

and *round-bound:* $\text{round-idx} < \text{length } \text{fri-query-idxs}$

and *missing:*

$\bigwedge \text{actual-chunk } arp \text{ arp-path } arn \text{ arn-path.}$

$\neg (\text{fri-layer-chunk-authenticated}$

$(\text{composition-roots ! } (\text{length } \text{composition-bs} - 1))$

$(\text{fri-evidence-layer-len } \text{composition-roots } (\text{length } \text{composition-bs} - 1))$

$(\text{fri-evidence-layer-idx } \text{composition-roots } \text{fri-query-idxs } \text{round-idx}$

$(\text{length } \text{composition-bs} - 1))$

$\text{actual-chunk } \text{final-state} \wedge$

$\text{fri-layer-step-evidence}$

$(\text{composition-roots ! } (\text{length } \text{composition-bs} - 1))$

$(\text{composition-bs ! } (\text{length } \text{composition-bs} - 1))$

$(\text{fri-evidence-layer-len } \text{composition-roots } (\text{length } \text{composition-bs} - 1))$

$(\text{fri-evidence-layer-idx } \text{composition-roots } \text{fri-query-idxs } \text{round-idx}$

$(\text{length } \text{composition-bs} - 1))$

$(2 \wedge (\text{length } \text{composition-bs} - 1))$

fri-sibling-index

```

    (fri-evidence-layer-len composition-roots (length composition-bs - 1))
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
      (length composition-bs - 1)))
  arp arp-path arn arn-path
  (fri-evidence-next-idx composition-roots fri-query-idxs round-idx
    (length composition-bs - 1))
  composition-final actual-chunk)
unfolding
  composition-fri-verifier-tied-final-value-selected-step-missing-gap-def
by blast
have outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
using support out-eq by simp
have challenges:
  accepted-fri-challenges s (Some (result, final-state))
  trace-bs fri-dg composition-bs
using accepted-fri-opening-transcript-challenges[OF fri-openings]
  out-eq by simp
have degree-bound: to-nat fri-dg ≤ maxDegree
by (rule verify-monad-accepted-fri-challenges-degree-bound
  [OF outcome challenges])
have len-bs-roots: length composition-bs = length composition-roots
using accepted-fri-opening-transcript-shapes(4)[OF fri-openings] .
have nonempty-roots: 0 < length composition-roots
using nonempty-bs len-bs-roots by simp
have round-bound-rounds: round-idx < rounds
using round-bound
  accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
by simp
have raw-layer:
  ∧ raw.
  0 < fri-layer-lengths (length composition-roots) (clength * scale) !
  (length composition-roots - 1) ∧
  fri-layer-indices (length composition-roots) (index (to-nat raw))
  (clength * scale) ! (length composition-roots - 1) <
  fri-layer-lengths (length composition-roots) (clength * scale) !
  (length composition-roots - 1)
by (rule accepted-fri-opening-transcript-composition-raw-layer-bound
  [OF fri-openings degree-bound])
  (use nonempty-roots in simp)
from accepted-fri-opening-transcript-composition-final-step-evidence-at
  [OF fri-openings out-eq round-bound-rounds nonempty-roots raw-layer]
obtain actual-chunk arp arp-path arn arn-path where
  auth:
  fri-layer-chunk-authenticated
  (composition-roots ! (length composition-roots - 1))
  (fri-evidence-layer-len composition-roots (length composition-roots - 1))
  (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
    (length composition-roots - 1))

```

```

    actual-chunk final-state
  and step:
    fri-layer-step-evidence
      (composition-roots ! (length composition-roots - 1))
      (composition-bs ! (length composition-roots - 1))
      (fri-evidence-layer-len composition-roots (length composition-roots - 1))
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
        (length composition-roots - 1))
      (2 ^ (length composition-roots - 1))
      (fri-sibling-index
        (fri-evidence-layer-len composition-roots (length composition-roots - 1))
        (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
          (length composition-roots - 1)))
      arp arp-path arn arn-path
      (fri-evidence-next-idx composition-roots fri-query-idxs round-idx
        (length composition-roots - 1))
      composition-final actual-chunk
    by blast
  show False
  using missing[of actual-chunk arp arp-path arn arn-path]
    auth step len-bs-roots
  by simp
qed

lemma wp-composition-fri-verifier-tied-final-value-selected-step-missing-gap-zero:
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-final-value-selected-step-missing-gap s)
      s ≤ (0::prob)
  proof -
    have
      wp-event verify-monad
        (composition-fri-verifier-tied-final-value-selected-step-missing-gap s)
        s ≤ wp-event verify-monad (λ-. False) s
      by (rule wp-event-mono-on-support)
      (rule
        composition-fri-verifier-tied-final-value-selected-step-missing-gap-false-on-support)
    also have ... = 0
      unfolding wp-event-def wp-def dist-expect-def by simp
    finally show ?thesis by simp
  qed

lemma composition-fri-verifier-tied-final-value-authenticated-replay-gap-imp-selected-step-missing:
  assumes
    composition-fri-verifier-tied-final-value-authenticated-replay-gap s out
  shows
    composition-fri-verifier-tied-final-value-selected-step-missing-gap s out
  proof -
    from assms obtain trace-roots trace-bs trace-final fri-dg composition-roots

```

```

    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings composition-openings
    trace-table composition-table result final-state round-idx v
where out-eq: out = Some (result, final-state)
and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
and nonempty: 0 < length composition-bs
and round-bound: round-idx < length fri-query-idxs
and forced:
    generic-fri-round-forced-next-value composition-roots composition-bs
    fri-query-idxs composition-round-layers round-idx
    (length composition-bs - 1) v
and neg: v ≠ composition-final
and no-merkle: ¬ partial-merkle-inconsistency-bad s out
and recorded-auth:
    generic-fri-recorded-layer-chunk-authenticated composition-roots
    fri-query-idxs composition-round-layers final-state round-idx
    (length composition-bs - 1)
unfolding
    composition-fri-verifier-tied-final-value-authenticated-replay-gap-def
by blast
have no-step:
    ∧ actual-chunk axp axp-path axn axn-path.
    ¬ (fri-layer-chunk-authenticated
      (composition-roots ! (length composition-bs - 1))
      (fri-evidence-layer-len composition-roots (length composition-bs - 1))
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
        (length composition-bs - 1))
      actual-chunk final-state ∧
      fri-layer-step-evidence
      (composition-roots ! (length composition-bs - 1))
      (composition-bs ! (length composition-bs - 1))
      (fri-evidence-layer-len composition-roots (length composition-bs - 1))
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
        (length composition-bs - 1))
      (2 ^ (length composition-bs - 1))
      (fri-sibling-index
        (fri-evidence-layer-len composition-roots (length composition-bs - 1))
        (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
          (length composition-bs - 1)))
      axp axp-path axn axn-path
      (fri-evidence-next-idx composition-roots fri-query-idxs round-idx
        (length composition-bs - 1))
      composition-final actual-chunk)
proof
    fix actual-chunk axp axp-path axn axn-path

```



```

assume step-auth:
  fri-layer-chunk-authenticated
    (composition-roots ! (length composition-bs - 1))
    (fri-evidence-layer-len composition-roots (length composition-bs - 1))
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
      (length composition-bs - 1))
    actual-chunk final-state ∧
  fri-layer-step-evidence
    (composition-roots ! (length composition-bs - 1))
    (composition-bs ! (length composition-bs - 1))
    (fri-evidence-layer-len composition-roots (length composition-bs - 1))
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
      (length composition-bs - 1))
    (2 ^ (length composition-bs - 1))
    (fri-sibling-index
      (fri-evidence-layer-len composition-roots (length composition-bs - 1))
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
        (length composition-bs - 1)))
    axp axp-path axn axn-path
    (fri-evidence-next-idx composition-roots fri-query-idxs round-idx
      (length composition-bs - 1))
    composition-final actual-chunk
have False
  by (rule generic-fri-final-conflict-false-from-authenticated-step
    [OF forced recorded-auth conjunct1[OF step-auth]
      conjunct2[OF step-auth] no-merkle[unfolded out-eq] neg])
then show False .
qed
have witness:
  ∃ trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers result final-state round-idx v.
  out = Some (result, final-state) ∧
  accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers ∧
  0 < length composition-bs ∧
  round-idx < length fri-query-idxs ∧
  generic-fri-round-forced-next-value composition-roots composition-bs
    fri-query-idxs composition-round-layers round-idx
    (length composition-bs - 1) v ∧
  v ≠ composition-final ∧
  ¬ partial-merkle-inconsistency-bad s out ∧
  generic-fri-recorded-layer-chunk-authenticated composition-roots
    fri-query-idxs composition-round-layers final-state round-idx
    (length composition-bs - 1) ∧
  (∀ actual-chunk axp axp-path axn axn-path.
    ¬ (fri-layer-chunk-authenticated

```

```

    (composition-roots ! (length composition-bs - 1))
    (fri-evidence-layer-len composition-roots (length composition-bs - 1))
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
      (length composition-bs - 1))
    actual-chunk final-state ∧
  fri-layer-step-evidence
    (composition-roots ! (length composition-bs - 1))
    (composition-bs ! (length composition-bs - 1))
    (fri-evidence-layer-len composition-roots (length composition-bs - 1))
    (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
      (length composition-bs - 1))
    (2 ^ (length composition-bs - 1))
    (fri-sibling-index
      (fri-evidence-layer-len composition-roots (length composition-bs - 1))
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
        (length composition-bs - 1)))
    arp arp-path arn arn-path
    (fri-evidence-next-idx composition-roots fri-query-idxs round-idx
      (length composition-bs - 1))
    composition-final actual-chunk))
using out-eq fri-openings nonempty round-bound forced neq no-merkle
  recorded-auth no-step
by blast
show ?thesis
unfolding
  composition-fri-verifier-tied-final-value-selected-step-missing-gap-def
using assms witness by blast
qed

lemma wp-composition-fri-verifier-tied-final-value-authenticated-replay-gap-bound-from-selected-step-missing:
  fixes G :: prob
  assumes gap-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-final-value-selected-step-missing-gap s)
    s ≤ G
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-final-value-authenticated-replay-gap s)
    s ≤ G
proof -
  have wp-event verify-monad
    (composition-fri-verifier-tied-final-value-authenticated-replay-gap s)
    s ≤
  wp-event verify-monad
    (composition-fri-verifier-tied-final-value-selected-step-missing-gap s)
  s
  by (rule wp-event-mono)
  (rule
    composition-fri-verifier-tied-final-value-authenticated-replay-gap-imp-selected-step-missing)

```

then show ?thesis
by (rule order-trans[*OF* - gap-bound])
qed

lemma *composition-fri-verifier-tied-final-value-replay-gap-imp-merkle-or-slot-or-authenticated-on-support*:
assumes *support*: *out* ∈ *set-dist* (execute *verify-monad* *s*)
and *gap*: *composition-fri-verifier-tied-final-value-replay-gap* *s out*
shows
partial-merkle-inconsistency-bad *s out* ∨
composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
s out ∨
composition-fri-verifier-tied-final-value-authenticated-replay-gap *s out*

proof –
from *gap* **have** *sampled*:
composition-fri-verifier-tied-sampled-final-value-conflict *s out*
unfolding *composition-fri-verifier-tied-final-value-replay-gap-def*
by *simp*
from *composition-fri-verifier-tied-sampled-final-value-conflictE*[*OF* *sampled*]
obtain *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table round-idx v
where *fri-openings*:
accepted-fri-opening-transcript *s out trace-roots trace-bs*
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *aligned*:
accepted-with-partial-initial-openings-aligned *s out fr trace-roots*
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
and *trace-cand*: *partial-trace-table-candidate trace-table trace-openings*
and *comp-cand*:
partial-composition-table-candidate composition-table
composition-openings
and *trace-low*: *trace-table-low-degree trace-table*
and *comp-not-low*:
 $\neg$ *composition-table-low-degree maxDegree composition-table*
and *nonempty*: $0 < \text{length } \text{composition-bs}$
and *round-bound*: *round-idx* < *length fri-query-idxs*
and *forced*:
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx
 $(\text{length } \text{composition-bs} - 1) v$
and *neq*: $v \neq \text{composition-final}$
by *blast*
from *fri-openings* **obtain** *result final-state* **where** *out-eq*:
out = *Some (result, final-state)*
unfolding *accepted-fri-opening-transcript-def* **by** *blast*

```

show ?thesis
proof (cases partial-merkle-inconsistency-bad s out)
  case True
  then show ?thesis by simp
next
case no-merkle: False
let ?j = length composition-bs - 1
have len-composition-bs: length composition-bs = length composition-roots
  using accepted-fri-opening-transcript-shapes(4)[OF fri-openings] .
have layer-bound: ?j < length composition-roots
  using nonempty len-composition-bs by simp
show ?thesis
proof (cases
  generic-fri-recorded-layer-chunk-authenticated composition-roots
  fri-query-idxs composition-round-layers final-state round-idx ?j)
  case True
  have authenticated-gap:
    composition-fri-verifier-tied-final-value-authenticated-replay-gap
    s out
  unfolding
    composition-fri-verifier-tied-final-value-authenticated-replay-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule aligned, rule trace-cand,
     rule comp-cand, rule trace-low, rule comp-not-low, rule nonempty,
     rule round-bound, rule forced, rule neq,
     use no-merkle True in simp-all)
  then show ?thesis by simp
next
case missing-recorded: False
have outcome:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using support out-eq by simp
have challenges:
  accepted-fri-challenges s (Some (result, final-state))
  trace-bs fri-dg composition-bs
  using accepted-fri-opening-transcript-challenges[OF fri-openings]
  out-eq by simp
have degree-bound: to-nat fri-dg ≤ maxDegree
  by (rule verify-monad-accepted-fri-challenges-degree-bound
    [OF outcome challenges])
have round-bound-rounds: round-idx < rounds
  using round-bound
  accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
  by simp
have raw-layer:
  ∧ raw.
  0 < fri-layer-lengths (length composition-roots)
  (clength * scale) ! ?j ∧
  fri-layer-indices (length composition-roots) (index (to-nat raw))

```

```

      (clength * scale) ! ?j <
      fri-layer-lengths (length composition-roots)
      (clength * scale) ! ?j
    by (rule accepted-fri-opening-transcript-composition-raw-layer-bound
        [OF fri-openings degree-bound layer-bound])
  from accepted-fri-opening-transcript-composition-selected-chunk-authenticated-at
    [OF fri-openings out-eq round-bound-rounds layer-bound raw-layer]
  obtain auth-chunk where auth:
    fri-layer-chunk-authenticated (composition-roots ! ?j)
    (fri-layer-lengths (length composition-roots) (clength * scale) ! ?j)
    (fri-layer-indices (length composition-roots)
      (fri-query-idxs ! round-idx) (clength * scale) ! ?j)
    auth-chunk final-state
  by blast
  have step:
    fri-round-layer-evidence composition-roots composition-bs
    fri-query-idxs round-idx ?j composition-round-layers
  by (rule accepted-fri-opening-transcript-composition-round-layer-evidence
      [OF fri-openings round-bound-rounds layer-bound])
  have generic-step:
    generic-fri-transcript-step-with-authenticated-chunk
    composition-roots composition-bs fri-query-idxs
    composition-round-layers final-state round-idx ?j
  by (rule generic-fri-transcript-step-with-authenticated-chunkI
      [OF step])
    (use auth in
      ‹simp add: fri-evidence-layer-len-def
        fri-evidence-layer-idx-def›)
  have slot-gap:
    generic-fri-authenticated-slot-recorded-chunk-gap
    composition-roots fri-query-idxs composition-round-layers
    final-state
  by (rule
    generic-fri-transcript-step-auth-missing-imp-authenticated-slot-recorded-chunk-gap
    [OF generic-step missing-recorded])
  have route-slot-gap:
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
    s out
  unfolding
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule aligned, rule trace-cand,
      rule comp-cand, rule trace-low, rule comp-not-low, rule slot-gap)
  then show ?thesis by simp
qed
qed
qed

```

lemma *wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-merkle-slot-and-authenticated:*

fixes $M A R :: \text{prob}$

assumes *merkle-bound*:

$\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq M$

and *slot-bound*:

$\text{wp-event verify-monad}$

$(\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap } s)$

$s \leq A$

and *replay-bound*:

$\text{wp-event verify-monad}$

$(\text{composition-fri-verifier-tied-final-value-authenticated-replay-gap } s)$

$s \leq R$

shows

$\text{wp-event verify-monad}$

$(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \ s$

$\leq M + A + R$

proof –

let $?P = \text{partial-merkle-inconsistency-bad } s$

let $?Q = \text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap } s$

let $?R =$

$\text{composition-fri-verifier-tied-final-value-authenticated-replay-gap } s$

have *event-le*:

$\text{wp-event verify-monad}$

$(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \ s \leq$

$\text{wp-event verify-monad } (\lambda \text{out. } ?P \ \text{out} \vee ?Q \ \text{out} \vee ?R \ \text{out}) \ s$

by (*rule wp-event-mono-on-support*)

(*rule*

$\text{composition-fri-verifier-tied-final-value-replay-gap-imp-merkle-or-slot-or-authenticated-on-support}$)

have *union-le*:

$\text{wp-event verify-monad } (\lambda \text{out. } ?P \ \text{out} \vee ?Q \ \text{out} \vee ?R \ \text{out}) \ s \leq$

$\text{wp-event verify-monad } ?P \ s +$

$\text{wp-event verify-monad } ?Q \ s +$

$\text{wp-event verify-monad } ?R \ s$

proof –

have $\text{wp-event verify-monad } (\lambda \text{out. } ?P \ \text{out} \vee ?Q \ \text{out} \vee ?R \ \text{out}) \ s \leq$

$\text{wp-event verify-monad } ?P \ s +$

$\text{wp-event verify-monad } (\lambda \text{out. } ?Q \ \text{out} \vee ?R \ \text{out}) \ s$

by (*rule wp-event-union-bound*)

also have $\dots \leq$

$\text{wp-event verify-monad } ?P \ s +$

$(\text{wp-event verify-monad } ?Q \ s + \text{wp-event verify-monad } ?R \ s)$

by (*intro add-mono order.refl wp-event-union-bound*)

finally show *?thesis*

by (*simp add: algebra-simps*)

qed

have *bounds*:

$\text{wp-event verify-monad } ?P \ s +$

$\text{wp-event verify-monad } ?Q \ s +$

$\text{wp-event verify-monad } ?R \ s \leq M + A + R$

using *merkle-bound slot-bound replay-bound*

```

    by (intro add-mono)
  show ?thesis
    by (rule order-trans[OF event-le order-trans[OF union-le bounds]])
qed

lemma wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step:
  fixes M A R :: prob
  assumes merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M
  and slot-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s)
      s ≤ A
  and selected-step-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-final-value-selected-step-missing-gap s)
      s ≤ R
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-final-value-replay-gap s) s
      ≤ M + A + R
proof -
  have authenticated-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-final-value-authenticated-replay-gap s)
      s ≤ R
  by (rule
    wp-composition-fri-verifier-tied-final-value-authenticated-replay-gap-bound-from-selected-step-missing
    [OF selected-step-bound])
  show ?thesis
    by (rule
      wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-merkle-slot-and-authenticated
      (rule merkle-bound, rule slot-bound, rule authenticated-bound))
qed

```

```

lemma wp-composition-fri-verifier-tied-authenticated-conflict-bound-from-recorded-missing-next-successor-and-final-g
  fixes P M N S G :: prob
  assumes partial-merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P
  and missing-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
      s ≤ M
  and next-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-next-value-conflict s) s ≤ N
  and successor-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-successor-opening-conflict s)

```

$s \leq S$
and *final-gap-bound*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq G$
shows
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}sampled\text{-}assignment\text{-}conflict\text{-}with\text{-}authenticated\text{-}same\text{-}layer\ s)$
 $s \leq (P + M) + N + S + G + P$
by (rule
 $wp\text{-}composition\text{-}fri\text{-}verifier\text{-}tied\text{-}authenticated\text{-}conflict\text{-}bound\text{-}from\text{-}recorded\text{-}missing\text{-}and\text{-}branches)$
(rule *partial-merkle-bound*,
rule *missing-bound*,
rule *next-bound*,
rule *successor-bound*,
rule *wp-composition-fri-verifier-tied-sampled-final-value-bound-from-replay-gap*
[*OF final-gap-bound*])

lemma *wp-composition-fri-verifier-tied-authenticated-conflict-bound-from-all-replay-gaps*:

fixes $P\ M\ N\ S\ G :: prob$

assumes *partial-merkle-bound*:

$wp\text{-}event\ verify\text{-}monad\ (partial\text{-}merkle\text{-}inconsistency\text{-}bad\ s)\ s \leq P$

and *missing-bound*:

$wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}recorded\text{-}base\text{-}chunk\text{-}auth\text{-}missing\ s)$
 $s \leq M$

and *next-gap-bound*:

$wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}next\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq N$

and *successor-gap-bound*:

$wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}successor\text{-}replay\text{-}gap\ s)\ s \leq S$

and *final-gap-bound*:

$wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq G$

shows

$wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}sampled\text{-}assignment\text{-}conflict\text{-}with\text{-}authenticated\text{-}same\text{-}layer\ s)$
 $s \leq (P + M) + N + S + G + P$

by (rule

$wp\text{-}composition\text{-}fri\text{-}verifier\text{-}tied\text{-}authenticated\text{-}conflict\text{-}bound\text{-}from\text{-}recorded\text{-}missing\text{-}next\text{-}successor\text{-}and\text{-}final\text{-}gap)$
(rule *partial-merkle-bound*,
rule *missing-bound*,
rule *wp-composition-fri-verifier-tied-sampled-next-value-bound-from-replay-gap*
[*OF next-gap-bound*],
rule *wp-composition-fri-verifier-tied-sampled-successor-bound-from-replay-gap*
[*OF successor-gap-bound*],
rule *final-gap-bound*)

definition *composition-fri-verifier-tied-assignment-auth-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-assignment-auth-gap } s \text{ out} \longleftrightarrow$
 $\text{composition-fri-verifier-tied-sampled-assignment-conflict } s \text{ out} \wedge$
 $\neg \text{composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer}$
 $s \text{ out}$

definition *composition-fri-verifier-tied-same-layer-auth-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-same-layer-auth-gap } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as trace-openings composition-openings}$
 $\text{trace-table composition-table result final-state.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr trace-roots}$
 $\text{trace-final as fri-dg composition-roots composition-final}$
 $\text{fri-query-idxs trace-openings composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\neg \text{composition-table-low-degree maxDegree composition-table} \wedge$
 $\text{generic-fri-sampled-same-layer-opening-conflict composition-roots}$
 $\text{composition-bs fri-query-idxs composition-round-layers} \wedge$
 $\neg \text{generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks}$
 $\text{composition-roots composition-bs fri-query-idxs composition-round-layers}$
 $\text{final-state})$

definition *composition-fri-verifier-tied-same-layer-recorded-auth-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-same-layer-recorded-auth-gap } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as trace-openings composition-openings}$
 $\text{trace-table composition-table result final-state.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$

composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned *s out fr trace-roots*
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table* $\wedge$
generic-fri-sampled-same-layer-recorded-auth-gap composition-roots
composition-bs fri-query-idxs composition-round-layers final-state)

definition *composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap
s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table result final-state.}$
 $\text{ out} = \text{Some (result, final-state)} \wedge$
 $\text{ accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final fri-dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers} \wedge$
 $\text{ accepted-with-partial-initial-openings-aligned s out fr trace-roots}$
 $\text{ trace-final as fri-dg composition-roots composition-final}$
 $\text{ fri-query-idxs trace-openings composition-openings} \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{ partial-composition-table-candidate composition-table}$
 $\text{ composition-openings} \wedge$
 $\text{ trace-table-low-degree trace-table} \wedge$
 $\neg \text{ composition-table-low-degree maxDegree composition-table} \wedge$
 $\text{ generic-fri-sampled-same-layer-authenticated-recorded-auth-gap}$
 $\text{ composition-roots composition-bs fri-query-idxs composition-round-layers}$
 $\text{ final-state})$

lemma *composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-false-on-support:*

assumes *support: out* $\in$ *set-dist (execute verify-monad s)*

and event:

composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
s out

shows *False*

proof —

from event obtain *trace-roots trace-bs trace-final fri-dg*
 $\text{ composition-roots composition-bs composition-final fri-query-idxs}$
 $\text{ trace-round-layers composition-round-layers fr as trace-openings}$

```

    composition-openings trace-table composition-table result final-state
where out-eq: out = Some (result, final-state)
and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
and gap:
    generic-fri-authenticated-slot-recorded-chunk-gap composition-roots
    fri-query-idxs composition-round-layers final-state
unfolding
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-def
by blast
from gap obtain round-idx layer-idx auth-chunk where
    round-bound: round-idx < length fri-query-idxs
and layer-bound: layer-idx < length composition-roots
and missing:
    ¬ generic-fri-recorded-layer-chunk-authenticated composition-roots
    fri-query-idxs composition-round-layers final-state round-idx
    layer-idx
unfolding generic-fri-authenticated-slot-recorded-chunk-gap-def
by blast
have support-some:
    Some (result, final-state) ∈ set-dist (execute verify-monad s)
using support out-eq by simp
have challenges:
    accepted-fri-challenges s (Some (result, final-state))
    trace-bs fri-dg composition-bs
using accepted-fri-opening-transcript-challenges[OF fri-openings]
    out-eq by simp
have degree-bound: to-nat fri-dg ≤ maxDegree
by (rule verify-monad-accepted-fri-challenges-degree-bound
    [OF support-some challenges])
have round-bound-rounds: round-idx < rounds
using round-bound accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
by simp
have all-layer-bounds:
    ∧ raw k. k < length composition-roots ⇒
    0 < fri-layer-lengths (length composition-roots) (clength * scale) ! k ∧
    fri-layer-indices (length composition-roots) (index (to-nat raw))
    (clength * scale) ! k <
    fri-layer-lengths (length composition-roots) (clength * scale) ! k
by (rule accepted-fri-opening-transcript-composition-raw-layer-bound
    [OF fri-openings degree-bound])
have recorded:
    generic-fri-recorded-layer-chunk-authenticated composition-roots
    fri-query-idxs composition-round-layers final-state round-idx layer-idx
by (rule
    accepted-fri-opening-transcript-composition-recorded-layer-chunk-authenticated-at

```

```

      [OF fri-openings out-eq round-bound-rounds layer-bound
        all-layer-bounds])
  show False
    using missing recorded by simp
qed

lemma wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero:
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s)
      s ≤ (0::prob)
  proof -
    have
      wp-event verify-monad
        (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s)
        s ≤ wp-event verify-monad (λ-. False) s
    by (rule wp-event-mono-on-support)
      (rule
        composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-false-on-support)
    also have ... = 0
      unfolding wp-event-def wp-def dist-expect-def by simp
    finally show ?thesis by simp
  qed

lemma composition-fri-verifier-tied-recorded-base-chunk-auth-missing-imp-slot-recorded-chunk-gap:
  assumes
    composition-fri-verifier-tied-recorded-base-chunk-auth-missing s out
  shows
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s out
  proof -
    from assms obtain trace-roots trace-bs trace-final fri-dg
      composition-roots composition-bs composition-final fri-query-idxs
      trace-round-layers composition-round-layers fr as trace-openings
      composition-openings trace-table composition-table round-idx xp xp-path
      xn xn-path result final-state
    where out-eq: out = Some (result, final-state)
    and fri-openings:
      accepted-fri-opening-transcript s out trace-roots trace-bs
      trace-final fri-dg composition-roots composition-bs
      composition-final fri-query-idxs trace-round-layers
      composition-round-layers
    and aligned:
      accepted-with-partial-initial-openings-aligned s out fr trace-roots
      trace-final as fri-dg composition-roots composition-final
      fri-query-idxs trace-openings composition-openings
    and trace-cand:
      partial-trace-table-candidate trace-table trace-openings
    and comp-cand:
      partial-composition-table-candidate composition-table

```

```

    composition-openings
  and trace-low: trace-table-low-degree trace-table
  and comp-not-low:
    ¬ composition-table-low-degree maxDegree composition-table
  and round-bound: round-idx < length fri-query-idxs
  and composition-bs-nonempty: 0 < length composition-bs
  and missing:
    ¬ fri-layer-chunk-authenticated (composition-roots ! 0)
      (fri-evidence-layer-len composition-roots 0)
      (fri-evidence-layer-idx composition-roots fri-query-idxs round-idx 0)
      (composition-round-layers ! round-idx ! 0) final-state
  unfolding
    composition-fri-verifier-tied-recorded-base-chunk-auth-missing-def
  by blast
  have len-composition-bs:
    length composition-bs = length composition-roots
    using accepted-fri-opening-transcript-shapes(4)[OF fri-openings] .
  have layer-bound: 0 < length composition-roots
    using composition-bs-nonempty len-composition-bs by simp
  have len-query: length fri-query-idxs = rounds
    using accepted-fri-opening-transcript-shapes(7)[OF fri-openings] .
  have round-bound-rounds: round-idx < rounds
    using round-bound len-query by simp
  have raw-layer:
    ∧ raw.
    0 < fri-layer-lengths (length composition-roots) (clength * scale) ! 0 ∧
    fri-layer-indices (length composition-roots) (index (to-nat raw))
      (clength * scale) ! 0 <
    fri-layer-lengths (length composition-roots) (clength * scale) ! 0
    by (rule fri-first-layer-raw-bound[OF layer-bound])
  have auth-step:
    generic-fri-transcript-step-with-authenticated-chunk composition-roots
      composition-bs fri-query-idxs composition-round-layers final-state
      round-idx 0
    by (rule
      accepted-fri-opening-transcript-composition-step-with-authenticated-chunk-at
      [OF fri-openings out-eq round-bound-rounds layer-bound raw-layer])
  have recorded-missing:
    ¬ generic-fri-recorded-layer-chunk-authenticated composition-roots
      fri-query-idxs composition-round-layers final-state round-idx 0
    using missing
  unfolding generic-fri-recorded-layer-chunk-authenticated-def
  by simp
  have slot-gap:
    generic-fri-authenticated-slot-recorded-chunk-gap composition-roots
      fri-query-idxs composition-round-layers final-state
    by (rule
      generic-fri-transcript-step-auth-missing-imp-authenticated-slot-recorded-chunk-gap
      [OF auth-step recorded-missing])

```

```

show ?thesis
  unfolding
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule aligned, rule trace-cand,
     rule comp-cand, rule trace-low, rule comp-not-low, rule slot-gap)
qed

lemma wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-bound-from-slot-recorded-chunk-gap:
  fixes A :: prob
  assumes slot-gap-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s)
    s ≤ A
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
    s ≤ A
proof –
  have wp-event verify-monad
    (composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
    s ≤
    wp-event verify-monad
      (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s)
    s
  by (rule wp-event-mono)
    (rule
      composition-fri-verifier-tied-recorded-base-chunk-auth-missing-imp-slot-recorded-chunk-gap)
  then show ?thesis
    by (rule order-trans[OF slot-gap-bound])
qed

lemma composition-fri-verifier-tied-assignment-imp-authenticated-or-auth-gap:
  assumes composition-fri-verifier-tied-sampled-assignment-conflict s out
  shows
    composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
      s out ∨
    composition-fri-verifier-tied-assignment-auth-gap s out
  using assms
  unfolding composition-fri-verifier-tied-assignment-auth-gap-def
  by blast

lemma composition-fri-verifier-tied-assignment-auth-gap-imp-same-layer-auth-gap:
  assumes composition-fri-verifier-tied-assignment-auth-gap s out
  shows composition-fri-verifier-tied-same-layer-auth-gap s out
proof –
  from assms have conflict:
    composition-fri-verifier-tied-sampled-assignment-conflict s out
  and no-auth:

```

```

   $\neg$  composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
    s out
  unfolding composition-fri-verifier-tied-assignment-auth-gap-def
  by simp-all
from conflict obtain trace-roots trace-bs trace-final fri-dg
  composition-roots composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers fr as trace-openings
  composition-openings trace-table composition-table
where fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final fri-dg composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers
and aligned:
  accepted-with-partial-initial-openings-aligned s out fr trace-roots
  trace-final as fri-dg composition-roots composition-final
  fri-query-idxs trace-openings composition-openings
and trace-cand:
  partial-trace-table-candidate trace-table trace-openings
and comp-cand:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-not-low:
   $\neg$  composition-table-low-degree maxDegree composition-table
and sampled:
  generic-fri-sampled-assignment-conflict composition-table
  composition-roots composition-bs composition-final fri-query-idxs
  composition-round-layers
unfolding
  composition-fri-verifier-tied-sampled-assignment-conflict-def
by blast
from fri-openings obtain result final-state where out-eg:
  out = Some (result, final-state)
unfolding accepted-fri-opening-transcript-def by blast
have split:
  generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
  composition-table composition-roots composition-bs composition-final
  fri-query-idxs composition-round-layers final-state  $\vee$ 
  (generic-fri-sampled-same-layer-opening-conflict composition-roots
   composition-bs fri-query-idxs composition-round-layers  $\wedge$ 
    $\neg$  generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
   composition-roots composition-bs fri-query-idxs composition-round-layers
   final-state)
by (rule
  generic-fri-sampled-assignment-conflict-imp-authenticated-or-same-layer-auth-gap
  [OF sampled])
then show ?thesis
proof

```

```

assume auth:
  generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
    composition-table composition-roots composition-bs composition-final
    fri-query-idxs composition-round-layers final-state
have
  composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
    s out
unfolding
  composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
by (intro exI conjI)
  (rule fri-openings, rule out-eq, rule aligned, rule trace-cand,
    rule comp-cand, rule trace-low, rule comp-not-low, rule auth)
then show ?thesis
  using no-auth by contradiction
next
assume gap:
  generic-fri-sampled-same-layer-opening-conflict composition-roots
    composition-bs fri-query-idxs composition-round-layers  $\wedge$ 
   $\neg$  generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    composition-roots composition-bs fri-query-idxs composition-round-layers
    final-state
show ?thesis
unfolding composition-fri-verifier-tied-same-layer-auth-gap-def
by (intro exI conjI)
  (rule out-eq, rule fri-openings, rule aligned, rule trace-cand,
    rule comp-cand, rule trace-low, rule comp-not-low, use gap in simp-all)
qed
qed

lemma composition-fri-verifier-tied-same-layer-auth-gap-imp-recorded-auth-gap:
  assumes composition-fri-verifier-tied-same-layer-auth-gap s out
  shows composition-fri-verifier-tied-same-layer-recorded-auth-gap s out
proof –
  from assms obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr as trace-openings
    composition-openings trace-table composition-table result final-state
  where out-eq: out = Some (result, final-state)
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
  and aligned:
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final
    fri-query-idxs trace-openings composition-openings
  and trace-cand:
    partial-trace-table-candidate trace-table trace-openings

```


and *comp-cand*:
partial-composition-table-candidate composition-table
composition-openings
and *trace-low*: *trace-table-low-degree trace-table*
and *comp-not-low*:
 $\neg$ *composition-table-low-degree maxDegree composition-table*
and *same*:
generic-fri-sampled-same-layer-opening-conflict composition-roots
composition-bs fri-query-idxs composition-round-layers
and *no-auth*:
 $\neg$ *generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks*
composition-roots composition-bs fri-query-idxs composition-round-layers
final-state
unfolding *composition-fri-verifier-tied-same-layer-auth-gap-def*
by *blast*
have *recorded-gap*:
generic-fri-sampled-same-layer-recorded-auth-gap composition-roots
composition-bs fri-query-idxs composition-round-layers final-state
by (rule *generic-fri-sampled-same-layer-auth-gap-imp-recorded-auth-gap*
[*OF same no-auth*])
show *?thesis*
unfolding *composition-fri-verifier-tied-same-layer-recorded-auth-gap-def*
by (*intro exI conjI*)
(rule *out-eq*, rule *fri-openings*, rule *aligned*, rule *trace-cand*,
rule *comp-cand*, rule *trace-low*, rule *comp-not-low*, rule *recorded-gap*)
qed

lemma *composition-fri-verifier-tied-same-layer-recorded-auth-gap-imp-authenticated-recorded-auth-gap-on-support*:
assumes *support*: *out* $\in$ *set-dist* (*execute verify-monad s*)
and *gap*: *composition-fri-verifier-tied-same-layer-recorded-auth-gap s out*
shows
composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap
s out
proof –
from *gap* **obtain** *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table result final-state
where *out-eq*: *out* = *Some (result, final-state)*
and *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *aligned*:
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
and *trace-cand*:

```

    partial-trace-table-candidate trace-table trace-openings
and comp-cand:
    partial-composition-table-candidate composition-table
    composition-openings
and trace-low: trace-table-low-degree trace-table
and comp-not-low:
     $\neg$  composition-table-low-degree maxDegree composition-table
and recorded-gap:
    generic-fri-sampled-same-layer-recorded-auth-gap composition-roots
    composition-bs fri-query-idxs composition-round-layers final-state
unfolding composition-fri-verifier-tied-same-layer-recorded-auth-gap-def
by blast
have outcome:
    Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
    using support out-eq by simp
have challenges:
    accepted-fri-challenges s (Some (result, final-state))
    trace-bs fri-dg composition-bs
    using accepted-fri-opening-transcript-challenges[OF fri-openings]
    out-eq by simp
have degree-bound: to-nat fri-dg  $\leq$  maxDegree
    by (rule verify-monad-accepted-fri-challenges-degree-bound
    [OF outcome challenges])
have len-composition-bs:
    length composition-bs = length composition-roots
    using accepted-fri-opening-transcript-shapes(4)[OF fri-openings] .
have len-query: length fri-query-idxs = rounds
    using accepted-fri-opening-transcript-shapes(7)[OF fri-openings] .
have auth-step:
     $\bigwedge$  round-idx layer-idx.
    round-idx < length fri-query-idxs  $\implies$ 
    layer-idx < length composition-bs  $\implies$ 
    generic-fri-transcript-step-with-authenticated-chunk composition-roots
    composition-bs fri-query-idxs composition-round-layers final-state
    round-idx layer-idx
proof –
    fix round-idx layer-idx
    assume round-bound: round-idx < length fri-query-idxs
    and layer-bound-bs: layer-idx < length composition-bs
    have round-bound-rounds: round-idx < rounds
    using round-bound len-query by simp
    have layer-bound-roots: layer-idx < length composition-roots
    using layer-bound-bs len-composition-bs by simp
    show
    generic-fri-transcript-step-with-authenticated-chunk
    composition-roots composition-bs fri-query-idxs
    composition-round-layers final-state round-idx layer-idx
    by (rule
    accepted-fri-opening-transcript-composition-step-with-authenticated-chunk-at-raw

```

[*OF fri-openings out-eq degree-bound round-bound-rounds*
layer-bound-roots])

qed

have *authenticated-recorded-gap*:
generic-fri-sampled-same-layer-authenticated-recorded-auth-gap
composition-roots composition-bs fri-query-idxs composition-round-layers
final-state
by (*rule*
generic-fri-recorded-auth-gap-imp-authenticated-recorded-auth-gap
[*OF recorded-gap auth-step*])

show *?thesis*
unfolding
composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap-def
by (*intro exI conjI*)
(*rule out-eq, rule fri-openings, rule aligned, rule trace-cand,*
rule comp-cand, rule trace-low, rule comp-not-low,
rule authenticated-recorded-gap)

qed

lemma *composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap-imp-slot-recorded-chunk-gap*:
assumes
composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap
s out
shows
composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
s out

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table result final-state

where *out-eq*: *out = Some (result, final-state)*

and *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

and *aligned*:
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings

and *trace-cand*:
partial-trace-table-candidate trace-table trace-openings

and *comp-cand*:
partial-composition-table-candidate composition-table
composition-openings

and *trace-low*: *trace-table-low-degree trace-table*

and *comp-not-low*:
 $\neg$ *composition-table-low-degree maxDegree composition-table*

```

and gap:
  generic-fri-sampled-same-layer-authenticated-recorded-auth-gap
    composition-roots composition-bs fri-query-idxs composition-round-layers
    final-state
unfolding
  composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap-def
by blast
have slot-gap:
  generic-fri-authenticated-slot-recorded-chunk-gap composition-roots
    fri-query-idxs composition-round-layers final-state
by (rule
  generic-fri-authenticated-recorded-auth-gap-imp-slot-recorded-chunk-gap
  [OF gap])
show ?thesis
unfolding
  composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-def
by (intro exI conjI)
  (rule out-eq, rule fri-openings, rule aligned, rule trace-cand,
  rule comp-cand, rule trace-low, rule comp-not-low, rule slot-gap)
qed

lemma wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-authenticated-and-auth-gap:
fixes C G :: prob
assumes authenticated-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s)
  s ≤ C
and auth-gap-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-assignment-auth-gap s) s ≤ G
shows
  wp-event verify-monad
    (composition-fri-verifier-tied-sampled-assignment-conflict s) s
    ≤ C + G
proof –
have wp-event verify-monad
  (composition-fri-verifier-tied-sampled-assignment-conflict s) s ≤
  wp-event verify-monad
    (λout.
      composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
        s out ∨
        composition-fri-verifier-tied-assignment-auth-gap s out) s
by (rule wp-event-mono)
  (rule
    composition-fri-verifier-tied-assignment-imp-authenticated-or-auth-gap)
also have ... ≤
  wp-event verify-monad
    (composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s)
  s +

```

```

    wp-event verify-monad
      (composition-fri-verifier-tied-assignment-auth-gap s) s
  by (rule wp-event-union-bound)
also have ... ≤ C + G
  by (intro add-mono authenticated-bound auth-gap-bound)
finally show ?thesis .
qed

lemma wp-composition-fri-verifier-tied-assignment-auth-gap-bound-from-same-layer-auth-gap:
  fixes G :: prob
  assumes gap-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-same-layer-auth-gap s) s ≤ G
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-assignment-auth-gap s) s ≤ G
proof -
  have wp-event verify-monad
    (composition-fri-verifier-tied-assignment-auth-gap s) s ≤
      wp-event verify-monad
        (composition-fri-verifier-tied-same-layer-auth-gap s) s
  by (rule wp-event-mono)
  (rule
    composition-fri-verifier-tied-assignment-auth-gap-imp-same-layer-auth-gap)
  then show ?thesis
  by (rule order-trans[OF - gap-bound])
qed

lemma wp-composition-fri-verifier-tied-same-layer-auth-gap-bound-from-recorded-auth-gap:
  fixes G :: prob
  assumes gap-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-same-layer-recorded-auth-gap s) s
      ≤ G
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-same-layer-auth-gap s) s ≤ G
proof -
  have wp-event verify-monad
    (composition-fri-verifier-tied-same-layer-auth-gap s) s ≤
      wp-event verify-monad
        (composition-fri-verifier-tied-same-layer-recorded-auth-gap s) s
  by (rule wp-event-mono)
  (rule
    composition-fri-verifier-tied-same-layer-auth-gap-imp-recorded-auth-gap)
  then show ?thesis
  by (rule order-trans[OF - gap-bound])
qed

```

lemma *wp-composition-fri-verifier-tied-same-layer-recorded-auth-gap-bound-from-authenticated-recorded-auth-gap*:
fixes $G :: \text{prob}$
assumes *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap}$
 $s) s \leq G$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-same-layer-recorded-auth-gap } s) s$
 $\leq G$
proof –
have $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-same-layer-recorded-auth-gap } s) s \leq$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap}$
 $s) s$
by (rule *wp-event-mono-on-support*)
 (rule
 $\text{composition-fri-verifier-tied-same-layer-recorded-auth-gap-imp-authenticated-recorded-auth-gap-on-support}$)
then show ?thesis
by (rule *order-trans*[*OF* - *gap-bound*])
qed

lemma *wp-composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap-bound-from-slot-recorded-chunk-gap*:
fixes $G :: \text{prob}$
assumes *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap } s)$
 $s \leq G$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap}$
 $s) s \leq G$
proof –
have $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap}$
 $s) s \leq$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap } s)$
 s
by (rule *wp-event-mono*)
 (rule
 $\text{composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap-imp-slot-recorded-chunk-gap}$)
then show ?thesis
by (rule *order-trans*[*OF* - *gap-bound*])
qed

lemma *wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-auth-gap-and-residuals*:
fixes $A P M N S G :: \text{prob}$

assumes *auth-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-assignment-auth-gap s) s ≤ A
and *partial-merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P
and *missing-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
s ≤ M
and *next-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-next-value-replay-gap s) s ≤ N
and *successor-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-successor-replay-gap s) s ≤ S
and *final-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-final-value-replay-gap s) s ≤ G
shows
wp-event verify-monad
(composition-fri-verifier-tied-sampled-assignment-conflict s) s
 $\leq ((P + M) + N + S + G + P) + A$
by (*rule*
wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-authenticated-and-auth-gap)
(rule wp-composition-fri-verifier-tied-authenticated-conflict-bound-from-all-replay-gaps
[OF partial-merkle-bound missing-bound next-gap-bound
successor-gap-bound final-gap-bound],
rule auth-gap-bound)

lemma *wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-same-layer-auth-gap-and-residuals*:

fixes *A P M N S G :: prob*
assumes *same-layer-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-same-layer-auth-gap s) s ≤ A
and *partial-merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P
and *missing-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
s ≤ M
and *next-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-next-value-replay-gap s) s ≤ N
and *successor-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-successor-replay-gap s) s ≤ S
and *final-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-final-value-replay-gap s) s ≤ G

shows

wp-event verify-monad
 (*composition-fri-verifier-tied-sampled-assignment-conflict s*) *s*
 $\leq ((P + M) + N + S + G + P) + A$

by (*rule*

wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-auth-gap-and-residuals)
 (*rule*
wp-composition-fri-verifier-tied-assignment-auth-gap-bound-from-same-layer-auth-gap
 [*OF same-layer-gap-bound*],
rule partial-merkle-bound,
rule missing-bound,
rule next-gap-bound,
rule successor-gap-bound,
rule final-gap-bound)

lemma *wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-recorded-auth-gap-and-residuals*:

fixes *A P M N S G :: prob*

assumes *recorded-gap-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-same-layer-recorded-auth-gap s*) *s*
 $\leq A$

and *partial-merkle-bound*:

wp-event verify-monad (*partial-merkle-inconsistency-bad s*) *s* $\leq P$

and *missing-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-recorded-base-chunk-auth-missing s*)
s $\leq M$

and *next-gap-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-next-value-replay-gap s*) *s* $\leq N$

and *successor-gap-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-successor-replay-gap s*) *s* $\leq S$

and *final-gap-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-final-value-replay-gap s*) *s* $\leq G$

shows

wp-event verify-monad
 (*composition-fri-verifier-tied-sampled-assignment-conflict s*) *s*
 $\leq ((P + M) + N + S + G + P) + A$

by (*rule*

wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-same-layer-auth-gap-and-residuals)
 (*rule*
wp-composition-fri-verifier-tied-same-layer-auth-gap-bound-from-recorded-auth-gap
 [*OF recorded-gap-bound*],
rule partial-merkle-bound,
rule missing-bound,
rule next-gap-bound,
rule successor-gap-bound,

rule final-gap-bound)

lemma *wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-authenticated-recorded-auth-gap-and*

fixes $A\ P\ M\ N\ S\ G :: \text{prob}$

assumes *gap-bound:*

wp-event verify-monad

(composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap
s) s ≤ A

and *partial-merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P

and *missing-bound:*

wp-event verify-monad

(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
s ≤ M

and *next-gap-bound:*

wp-event verify-monad

(composition-fri-verifier-tied-next-value-replay-gap s) s ≤ N

and *successor-gap-bound:*

wp-event verify-monad

(composition-fri-verifier-tied-successor-replay-gap s) s ≤ S

and *final-gap-bound:*

wp-event verify-monad

(composition-fri-verifier-tied-final-value-replay-gap s) s ≤ G

shows

wp-event verify-monad

(composition-fri-verifier-tied-sampled-assignment-conflict s) s
≤ ((P + M) + N + S + G + P) + A

by *(rule*

wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-recorded-auth-gap-and-residuals)

(rule

wp-composition-fri-verifier-tied-same-layer-recorded-auth-gap-bound-from-authenticated-recorded-auth-gap

[OF gap-bound],

rule partial-merkle-bound,

rule missing-bound,

rule next-gap-bound,

rule successor-gap-bound,

rule final-gap-bound)

lemma *wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-slot-recorded-chunk-gap-and-residual*

fixes $A\ P\ M\ N\ S\ G :: \text{prob}$

assumes *gap-bound:*

wp-event verify-monad

(composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s)
s ≤ A

and *partial-merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P

and *missing-bound:*

wp-event verify-monad

(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)

$s \leq M$
and *next-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap } s) \ s \leq N$
and *successor-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap } s) \ s \leq S$
and *final-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \ s \leq G$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-assignment-conflict } s) \ s$
 $\leq ((P + M) + N + S + G + P) + A$
by (rule
 $\text{wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-authenticated-recorded-auth-gap-and-residuals}$
(rule
 $\text{wp-composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap-bound-from-slot-recorded-chunk-gap}$
 $[\text{OF gap-bound}]$,
rule *partial-merkle-bound*,
rule *missing-bound*,
rule *next-gap-bound*,
rule *successor-gap-bound*,
rule *final-gap-bound*)

lemma *wp-composition-fri-verifier-tied-bound-from-sampled-auth-gap-and-residuals*:

fixes $R \ A \ P \ M \ N \ S \ G :: \text{prob}$

assumes *sampled-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s \leq R$

and *auth-gap-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-assignment-auth-gap } s) \ s \leq A$

and *partial-merkle-bound*:

$\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq P$

and *missing-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-recorded-base-chunk-auth-missing } s)$
 $s \leq M$

and *next-gap-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap } s) \ s \leq N$

and *successor-gap-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap } s) \ s \leq S$

and *final-gap-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \ s \leq G$

shows

wp-event verify-monad
 (*composition-fri-bad-with-verifier-tied-partial-candidate s*) *s*
 $\leq R + (((P + M) + N + S + G + P) + A)$
proof –
have *conflict-bound*:
wp-event verify-monad
 (*composition-fri-verifier-tied-sampled-assignment-conflict s*) *s*
 $\leq ((P + M) + N + S + G + P) + A$
by (rule
wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-auth-gap-and-residuals)
 (rule *auth-gap-bound*,
 rule *partial-merkle-bound*,
 rule *missing-bound*,
 rule *next-gap-bound*,
 rule *successor-gap-bound*,
 rule *final-gap-bound*)
show ?thesis
by (rule
wp-composition-fri-verifier-tied-bound-from-sampled-conflict
 [OF *sampled-bound conflict-bound*])
qed

lemma *wp-composition-fri-verifier-tied-bound-from-sampled-same-layer-auth-gap-and-residuals*:

fixes *R A P M N S G* :: prob

assumes *sampled-bound*:

wp-event verify-monad
 (*composition-fri-bad-with-verifier-tied-sampled-layer-chain s*) *s* $\leq R$

and *same-layer-gap-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-same-layer-auth-gap s*) *s* $\leq A$

and *partial-merkle-bound*:

wp-event verify-monad (*partial-merkle-inconsistency-bad s*) *s* $\leq P$

and *missing-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-recorded-base-chunk-auth-missing s*)
s $\leq M$

and *next-gap-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-next-value-replay-gap s*) *s* $\leq N$

and *successor-gap-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-successor-replay-gap s*) *s* $\leq S$

and *final-gap-bound*:

wp-event verify-monad
 (*composition-fri-verifier-tied-final-value-replay-gap s*) *s* $\leq G$

shows

wp-event verify-monad
 (*composition-fri-bad-with-verifier-tied-partial-candidate s*) *s*
 $\leq R + (((P + M) + N + S + G + P) + A)$

by (rule
 wp-composition-fri-verifier-tied-bound-from-sampled-auth-gap-and-residuals)
 (rule sampled-bound,
 rule
 wp-composition-fri-verifier-tied-assignment-auth-gap-bound-from-same-layer-auth-gap
 [OF same-layer-gap-bound],
 rule partial-merkle-bound,
 rule missing-bound,
 rule next-gap-bound,
 rule successor-gap-bound,
 rule final-gap-bound)

lemma wp-composition-fri-verifier-tied-bound-from-sampled-recorded-auth-gap-and-residuals:

fixes $R A P M N S G :: \text{prob}$

assumes sampled-bound:

wp-event verify-monad

(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) $s \leq R$

and recorded-gap-bound:

wp-event verify-monad

(composition-fri-verifier-tied-same-layer-recorded-auth-gap s) s
 $\leq A$

and partial-merkle-bound:

wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq P$

and missing-bound:

wp-event verify-monad

(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
 $s \leq M$

and next-gap-bound:

wp-event verify-monad

(composition-fri-verifier-tied-next-value-replay-gap s) $s \leq N$

and successor-gap-bound:

wp-event verify-monad

(composition-fri-verifier-tied-successor-replay-gap s) $s \leq S$

and final-gap-bound:

wp-event verify-monad

(composition-fri-verifier-tied-final-value-replay-gap s) $s \leq G$

shows

wp-event verify-monad

(composition-fri-bad-with-verifier-tied-partial-candidate s) s
 $\leq R + (((P + M) + N + S + G + P) + A)$

by (rule

wp-composition-fri-verifier-tied-bound-from-sampled-same-layer-auth-gap-and-residuals)

(rule sampled-bound,

rule

wp-composition-fri-verifier-tied-same-layer-auth-gap-bound-from-recorded-auth-gap

[OF recorded-gap-bound],

rule partial-merkle-bound,

rule missing-bound,

rule next-gap-bound,

rule successor-gap-bound,
rule final-gap-bound)

lemma *wp-composition-fri-verifier-tied-bound-from-sampled-authenticated-recorded-auth-gap-and-residuals:*
fixes $R\ A\ P\ M\ N\ S\ G :: \text{prob}$
assumes *sampled-bound:*
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R
and *gap-bound:*
wp-event verify-monad
(composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap
s) s ≤ A
and *partial-merkle-bound:*
wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P
and *missing-bound:*
wp-event verify-monad
(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
s ≤ M
and *next-gap-bound:*
wp-event verify-monad
(composition-fri-verifier-tied-next-value-replay-gap s) s ≤ N
and *successor-gap-bound:*
wp-event verify-monad
(composition-fri-verifier-tied-successor-replay-gap s) s ≤ S
and *final-gap-bound:*
wp-event verify-monad
(composition-fri-verifier-tied-final-value-replay-gap s) s ≤ G
shows
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-partial-candidate s) s
 $\leq R + (((P + M) + N + S + G + P) + A)$
by (*rule*
wp-composition-fri-verifier-tied-bound-from-sampled-recorded-auth-gap-and-residuals)
(*rule sampled-bound,*
rule
wp-composition-fri-verifier-tied-same-layer-recorded-auth-gap-bound-from-authenticated-recorded-auth-gap
[OF gap-bound],
rule partial-merkle-bound,
rule missing-bound,
rule next-gap-bound,
rule successor-gap-bound,
rule final-gap-bound)

lemma *wp-composition-fri-verifier-tied-bound-from-sampled-slot-recorded-chunk-gap-and-residuals:*
fixes $R\ A\ P\ M\ N\ S\ G :: \text{prob}$
assumes *sampled-bound:*
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R
and *gap-bound:*

wp-event verify-monad
 (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
 s) $s \leq A$
and *partial-merkle-bound*:
 wp-event verify-monad (partial-merkle-inconsistency-bad *s*) $s \leq P$
and *missing-bound*:
 wp-event verify-monad
 (composition-fri-verifier-tied-recorded-base-chunk-auth-missing *s*)
 s $\leq M$
and *next-gap-bound*:
 wp-event verify-monad
 (composition-fri-verifier-tied-next-value-replay-gap *s*) $s \leq N$
and *successor-gap-bound*:
 wp-event verify-monad
 (composition-fri-verifier-tied-successor-replay-gap *s*) $s \leq S$
and *final-gap-bound*:
 wp-event verify-monad
 (composition-fri-verifier-tied-final-value-replay-gap *s*) $s \leq G$
shows
 wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-partial-candidate *s*) *s*
 $\leq R + (((P + M) + N + S + G + P) + A)$
by (rule
 wp-composition-fri-verifier-tied-bound-from-sampled-authenticated-recorded-auth-gap-and-residuals)
 (rule *sampled-bound*,
 rule
 wp-composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap-bound-from-slot-recorded-chunk-gap
 [*OF gap-bound*],
 rule *partial-merkle-bound*,
 rule *missing-bound*,
 rule *next-gap-bound*,
 rule *successor-gap-bound*,
 rule *final-gap-bound*)

lemma *composition-fri-reachable-partial-candidate-reduction-from-verifier-tied-residual-bounds*:

fixes *R A P M N S G T* :: *prob*

assumes *sampled-bound*:

wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-sampled-layer-chain *s*) $s \leq R$
and *slot-gap-bound*:
 wp-event verify-monad
 (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
 s) $s \leq A$

and *partial-merkle-bound*:

wp-event verify-monad (partial-merkle-inconsistency-bad *s*) $s \leq P$

and *missing-bound*:

wp-event verify-monad
 (composition-fri-verifier-tied-recorded-base-chunk-auth-missing *s*)
 s $\leq M$

and *next-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap } s) \ s \leq N$
and *successor-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap } s) \ s \leq S$
and *final-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \ s \leq G$
and *verifier-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-reachable-verifier-tie-gap } s) \ s \leq T$
and *total-bound*:
 $R + (((P + M) + N + S + G + P) + A) + T$
 $\leq \text{composition-fri-error}$
shows *composition-fri-reachable-partial-candidate-reduction-assumption* s
proof –
have *verifier-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-partial-candidate } s) \ s$
 $\leq R + (((P + M) + N + S + G + P) + A)$
by (rule
 $\text{wp-composition-fri-verifier-tied-bound-from-sampled-slot-recorded-chunk-gap-and-residuals}$
 $[\text{OF sampled-bound slot-gap-bound partial-merkle-bound missing-bound}$
 $\text{next-gap-bound successor-gap-bound final-gap-bound}])$
show *?thesis*
by (rule
 $\text{composition-fri-reachable-partial-candidate-reduction-from-verifier-tied-and-gap}$
 $[\text{OF verifier-bound verifier-gap-bound total-bound}])$
qed

lemma *composition-fri-reachable-partial-candidate-reduction-from-verifier-tied-residual-bounds-with-recorded-missing*

fixes $R \ A \ P \ N \ S \ G \ T :: \text{prob}$

assumes *sampled-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s \leq R$

and *slot-gap-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap}$
 $s) \ s \leq A$

and *partial-merkle-bound*:

$\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq P$

and *next-gap-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap } s) \ s \leq N$

and *successor-gap-bound*:

$\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap } s) \ s \leq S$

and *final-gap-bound*:

wp-event verify-monad
 (composition-fri-verifier-tied-final-value-replay-gap s) $s \leq G$
and *verifier-gap-bound*:
wp-event verify-monad
 (composition-fri-reachable-verifier-tie-gap s) $s \leq T$
and *total-bound*:
 $R + (((P + 0) + N + S + G + P) + A) + T$
 $\leq$ *composition-fri-error*
shows *composition-fri-reachable-partial-candidate-reduction-assumption s*
by (rule *composition-fri-reachable-partial-candidate-reduction-from-verifier-tied-residual-bounds*
 [where $M = 0$])
 (rule *sampled-bound*,
 rule *slot-gap-bound*,
 rule *partial-merkle-bound*,
 rule *wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero*,
 rule *next-gap-bound*,
 rule *successor-gap-bound*,
 rule *final-gap-bound*,
 rule *verifier-gap-bound*,
 rule *total-bound*)

lemma *composition-fri-verifier-tied-reduction-from-residual-bounds*:
fixes $R\ A\ P\ M\ N\ S\ G :: \text{prob}$
assumes *sampled-bound*:
wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-sampled-layer-chain s) $s \leq R$
and *slot-gap-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
 s) $s \leq A$
and *partial-merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq P$
and *missing-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
 $s \leq M$
and *next-gap-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-next-value-replay-gap s) $s \leq N$
and *successor-gap-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-successor-replay-gap s) $s \leq S$
and *final-gap-bound*:
wp-event verify-monad
 (composition-fri-verifier-tied-final-value-replay-gap s) $s \leq G$
and *total-bound*:
 $R + (((P + M) + N + S + G + P) + A)$
 $\leq$ *composition-fri-error*
shows *composition-fri-verifier-tied-reduction s*

unfolding *composition-fri-verifier-tied-reduction-def*
by (rule *order-trans*[
OF wp-composition-fri-verifier-tied-bound-from-sampled-slot-recorded-chunk-gap-and-residuals
[OF sampled-bound slot-gap-bound partial-merkle-bound missing-bound
next-gap-bound successor-gap-bound final-gap-bound] total-bound])

lemma *composition-fri-verifier-tied-reduction-from-residual-bounds-without-recorded-missing*:
fixes $R\ A\ P\ N\ S\ G :: \text{prob}$
assumes *sampled-bound*:
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R
and *slot-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
s) s ≤ A
and *partial-merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P
and *next-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-next-value-replay-gap s) s ≤ N
and *successor-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-successor-replay-gap s) s ≤ S
and *final-gap-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-final-value-replay-gap s) s ≤ G
and *total-bound*:
 $R + (((P + A) + N + S + G + P) + A)$
 $\leq \text{composition-fri-error}$
shows *composition-fri-verifier-tied-reduction s*
proof –
have *missing-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
 $s \leq A$
by (rule
wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-bound-from-slot-recorded-chunk-gap
[OF slot-gap-bound])
show *?thesis*
by (rule *composition-fri-verifier-tied-reduction-from-residual-bounds*
[OF sampled-bound slot-gap-bound partial-merkle-bound missing-bound
next-gap-bound successor-gap-bound final-gap-bound total-bound])
qed

lemma *composition-fri-verifier-tied-reduction-from-residual-bounds-with-recorded-missing-zero*:
fixes $R\ A\ P\ N\ S\ G :: \text{prob}$
assumes *sampled-bound*:
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R

```

and slot-gap-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
     s)  $s \leq A$ 
and partial-merkle-bound:
  wp-event verify-monad (partial-merkle-inconsistency-bad s)  $s \leq P$ 
and next-gap-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-next-value-replay-gap s)  $s \leq N$ 
and successor-gap-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-successor-replay-gap s)  $s \leq S$ 
and final-gap-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-final-value-replay-gap s)  $s \leq G$ 
and total-bound:
   $R + (((P + 0) + N + S + G + P) + A)$ 
     $\leq \text{composition-fri-error}$ 
shows composition-fri-verifier-tied-reduction s
by (rule composition-fri-verifier-tied-reduction-from-residual-bounds
  [where  $M = 0$ , OF sampled-bound slot-gap-bound partial-merkle-bound
   wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero
   next-gap-bound successor-gap-bound final-gap-bound total-bound])

end

end

```

```

theory Soundness-FRI-Value-Aligned-Events
  imports Soundness-FRI-Composition-Residuals
begin

```

Value-aligned FRI branch events.

These events are narrower than the raw sampled base-opening conflicts: they keep the trace query opening that supplies the first trace FRI input value. The intended use is to avoid exporting broad verifier-query-round decomposition lemmas, which are expensive for Isabelle to finalize.

```

context soundness
begin

```

```

lemma receive-query-commits-head-layer-outcome:
  fixes  $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$ 
  assumes bfs-eq:  $bfs = (b, rt) \# bfs\text{-tail}$ 
  and outcome:
     $\text{Some } (out, t) \in$ 
      set-dist
      (execute
       (mfold (idx, x, len, pw) (receive-query-commits bfs)) s)

```

obtains *layer-chunks chunk xp xp-path xn xn-path* **where**
fri-layers-transcript (length bfs) len layer-chunks chunk
PTranscript s = chunk @ PTranscript t
PState t = foldl concat (PState s) chunk
s ≤ t
PQueryCounter t = PQueryCounter s
layer-chunks ≠ []
xp = x
fri-layer-opening-chunk len xp xp-path xn xn-path (hd layer-chunks)
proof –
have *steps*:
receive-query-commits bfs =
fri-layer-opening-step (b, rt) #
receive-query-commits bfs-tail
unfolding *bfs-eq receive-query-commits-def fri-layer-opening-step-def*
by *simp*
from *outcome[unfolded steps]*
obtain *out1 s1* **where** *head*:
Some (out1, s1) ∈
set-dist
(execute
(fri-layer-opening-step (b, rt) (idx, x, len, pw)) s)
and *tail*:
Some (out, t) ∈
set-dist (execute
(mfold out1 (receive-query-commits bfs-tail)) s1)
by *(auto elim!: set-dist-bindE)*
from *fri-layer-opening-step-outcome[OF head]*
obtain *xp xp-path xn xn-path x'* **where** *out1-eq*:
out1 = (idx mod (len div 2), x', len div 2, pw + pw)
and *xp-eq*: *xp = x*
and *head-chunk*:
fri-layer-opening-chunk len xp xp-path xn xn-path
([xp] @ xp-path @ [xn] @ xn-path)
and *transcript-head*:
PTranscript s =
([xp] @ xp-path @ [xn] @ xn-path) @ PTranscript s1
and *state-head*:
PState s1 =
foldl concat (PState s) ([xp] @ xp-path @ [xn] @ xn-path)
and *ext-head*: *s ≤ s1*
and *counter-head*: *PQueryCounter s1 = PQueryCounter s*
using *fri-layer-opening-step-outcome[OF head]*
by *blast*
from *receive-query-commits-layers-outcome[OF tail[unfolded out1-eq]]*
obtain *tail-layers tail-chunk* **where** *tail-layers*:
fri-layers-transcript (length bfs-tail) (len div 2)
tail-layers tail-chunk
and *transcript-tail*:

```

    PTranscript s1 = tail-chunk @ PTranscript t
  and state-tail:
    PState t = foldl concat (PState s1) tail-chunk
  and ext-tail: s1 ≤ t
  and counter-tail: PQueryCounter t = PQueryCounter s1
  by blast
let ?head-chunk = [xp] @ xp-path @ [xn] @ xn-path
let ?layers = ?head-chunk # tail-layers
let ?chunk = ?head-chunk @ tail-chunk
have layers:
  fri-layers-transcript (length bfs) len ?layers ?chunk
  using tail-layers head-chunk unfolding bfs-eq fri-layers-transcript-def
  by auto
have transcript:
  PTranscript s = ?chunk @ PTranscript t
  using transcript-head transcript-tail by simp
have state:
  PState t = foldl concat (PState s) ?chunk
  using state-head state-tail by simp
have ext: s ≤ t
  by (rule hash-ext-trans[OF ext-head ext-tail])
have counter: PQueryCounter t = PQueryCounter s
  using counter-head counter-tail by simp
show ?thesis
  by (rule that[OF layers transcript state ext counter - xp-eq])
    (use head-chunk in simp-all)
qed

```

lemma verifier-query-round-after-index-trace-fri-head-value-aligned:

```

fixes s t :: ('f, 'a) protocol-channel-scheme
assumes f-fl-eq: f-fl = (b, rt) # f-fl-tail
and outcome:
  Some (( ), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s)
obtains idx openings xp xp-path xn xn-path where
  idx = index (to-nat raw)
  map opening-index openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) openings t
  openings ≠ []
  opening-value (hd openings) = xp
  fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
    ([xp] @ xp-path @ [xn] @ xn-path)

```

proof –

```

let ?idx = index (to-nat raw)
from outcome obtain fv s1 f-out s2 s3 c-out s4 where
  query-decommit:

```

```

    Some (fv, s1) ∈
      set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
and trace-fri:
    Some (f-out, s2) ∈
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
and assert-trace:
    Some ((), s3) ∈
      set-dist
        (execute
          (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
    Some (c-out, s4) ∈
      set-dist (execute
        (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
          clength * scale, 1)
          (receive-query-commits fl)) s3)
and assert-comp:
    Some ((), t) ∈
      set-dist
        (execute
          (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
unfolding verifier-query-round-after-index-program-def
by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have idx-in: ?idx ∈ query-sample-space
using index-less-query-sample-space
unfolding query-sample-space-def by simp
from check-decommit-on-query-authenticated-openings
  [OF idx-in query-decommit]
obtain openings where len-openings:
  length openings = length (powers-scaled ?idx)
and values-openings: map opening-value openings = fv
and idx-openings: map opening-index openings = powers-scaled ?idx
and table-s1:
  partial-authenticated-table fr (scale * clength) openings s1
by blast
from receive-query-commits-decomp-at
  [OF trace-fri[unfolded f-fl-eq], of 0]
obtain idx0 x0 len0 pw0 out1 s1' s-suc where
  head:
    Some (out1, s1') ∈
      set-dist
        (execute (fri-layer-opening-step (b, rt)
          (?idx, hd fv, clength * scale, 1)) s1)
by simp blast
from fri-layer-opening-step-outcome[OF head]
obtain xp xp-path xn xn-path x' where
  xp-eq: xp = hd fv

```

```

and chunk:
  fri-layer-opening-chunk (clength * scale) xp xp-path xn xn-path
    ([xp] @ xp-path @ [xn] @ xn-path)
by blast
have s1-t: s1 ≤ t
proof –
  obtain idx1 x1 len1 pw1 where f-out-eq:
    f-out = (idx1, x1, len1, pw1)
    by (cases f-out)
  have s1-s2:
    s1 ≤ s2
    by (rule receive-query-commits-mfold-hash-extends
      [OF trace-fri[unfolded f-out-eq]])
  have s3-eq: s3 = s2
    using assert-trace unfolding assert-def f-out-eq
    by (cases x1 = f-final) (auto simp: throw-no-outcome)
  have s3-s4:
    s3 ≤ s4
    by (rule receive-query-commits-mfold-hash-extends[OF comp-fri])
  have t-eq: t = s4
    using assert-comp unfolding assert-def
    by (cases case c-out of (-, x, -, -) ⇒ x = final)
      (auto simp: throw-no-outcome)
  have s2-s4: s2 ≤ s4
    using s3-s4 unfolding s3-eq .
  show ?thesis
    using hash-ext-trans[OF s1-s2 s2-s4] t-eq by simp
qed
have table-t:
  partial-authenticated-table fr (scale * clength) openings t
    by (rule partial-authenticated-table-mono[OF table-s1 s1-t])
have openings-nonempty: openings ≠ []
  using len-openings powers-pos unfolding powers-scaled-def by auto
have hd-value: opening-value (hd openings) = hd fv
  using values-openings openings-nonempty
  by (cases openings; cases fv) simp-all
show ?thesis
  by (rule that[OF refl idx-openings table-t openings-nonempty])
    (use hd-value xp-eq chunk in ⟨simp-all add: mult.commute⟩)
qed

lemma verifier-query-round-after-index-trace-fri-head-layer-value-aligned:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes f-fl-eq: f-fl = (b, rt) # f-fl-tail
  and outcome:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program

```

```

      fr f-fl f-final as fl final raw) s)
obtains idx openings query-chunk trace-layer-chunks composition-layer-chunks
  xp xp-path xn xn-path where
  idx = index (to-nat raw)
  query-round-fri-layer-transcripts idx (map snd f-fl) (map snd fl)
  query-chunk trace-layer-chunks composition-layer-chunks
  PTranscript s = query-chunk @ PTranscript t
  PState t = foldl concat (PState s) query-chunk
  s ≤ t
  PQueryCounter t = PQueryCounter s
  map opening-index openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) openings t
  openings ≠ []
  opening-value (hd openings) = xp
  fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
    (trace-layer-chunks ! 0)
proof –
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
    query-decommit:
      Some (fv, s1) ∈
        set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
    and trace-fri:
      Some (f-out, s2) ∈
        set-dist (execute
          (mfold (?idx, hd fv, clength * scale, 1)
            (receive-query-commits f-fl)) s1)
    and assert-trace:
      Some ((), s3) ∈
        set-dist
          (execute
            (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
    and comp-fri:
      Some (c-out, s4) ∈
        set-dist (execute
          (mfold (?idx, cp-eval as fv (h ^ ?idx * shift),
            clength * scale, 1)
            (receive-query-commits fl)) s3)
    and assert-comp:
      Some ((), t) ∈
        set-dist
          (execute
            (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
  unfolding verifier-query-round-after-index-program-def
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
have idx-in: ?idx ∈ query-sample-space
  using index-less-query-sample-space
  unfolding query-sample-space-def by simp
from check-decommit-on-query-authenticated-openings

```

```

    [OF idx-in query-decommit]
obtain openings where values-openings:
    map opening-value openings = fv
    and idx-openings: map opening-index openings = powers-scaled ?idx
    and table-s1:
    partial-authenticated-table fr (scale * clength) openings s1
    by blast
from check-decommit-on-query-outcome[OF query-decommit]
obtain query-paths decommit-chunk where query-chunk-shape:
    query-decommitment-transcript ?idx fv query-paths decommit-chunk
    and tr-s-s1:
    PTranscript s = decommit-chunk @ PTranscript s1
    and st-s1:
    PState s1 = foldl concat (PState s) decommit-chunk
    and ext-s-s1: s ≤ s1
    and query-counter-s1: PQueryCounter s1 = PQueryCounter s
    by blast
from receive-query-commits-head-layer-outcome[OF f-fl-eq trace-fri]
obtain trace-layer-chunks trace-fri-chunk xp xp-path xn xn-path where
    trace-layers:
    fri-layers-transcript (length f-fl) (clength * scale)
    trace-layer-chunks trace-fri-chunk
    and tr-s1-s2:
    PTranscript s1 = trace-fri-chunk @ PTranscript s2
    and st-s2:
    PState s2 = foldl concat (PState s1) trace-fri-chunk
    and ext-s1-s2: s1 ≤ s2
    and query-counter-s2: PQueryCounter s2 = PQueryCounter s1
    and trace-layers-nonempty: trace-layer-chunks ≠ []
    and xp-eq: xp = hd fv
    and head-chunk:
    fri-layer-opening-chunk (clength * scale) xp xp-path xn xn-path
    (hd trace-layer-chunks)
    by blast
have s3-eq: s3 = s2
    using assert-trace unfolding assert-def
    by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
    (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain composition-layer-chunks composition-fri-chunk where
    composition-layers:
    fri-layers-transcript (length fl) (clength * scale)
    composition-layer-chunks composition-fri-chunk
    and tr-s3-s4:
    PTranscript s3 = composition-fri-chunk @ PTranscript s4
    and st-s4:
    PState s4 = foldl concat (PState s3) composition-fri-chunk
    and ext-s3-s4: s3 ≤ s4
    and query-counter-s4: PQueryCounter s4 = PQueryCounter s3

```



```

  by blast
have t-eq:  $t = s_4$ 
  using assert-comp unfolding assert-def
  by (cases case c-out of  $(-, x, -, -) \Rightarrow x = \text{final}$ )
    (auto simp: throw-no-outcome)
let ?query-chunk =
  decommit-chunk @ trace-fri-chunk @ composition-fri-chunk
have round-layers:
  query-round-fri-layer-transcripts ?idx (map snd f-fl) (map snd fl)
  ?query-chunk trace-layer-chunks composition-layer-chunks
unfolding query-round-fri-layer-transcripts-def
by (intro exI[of - decommit-chunk] exI[of - trace-fri-chunk]
    exI[of - composition-fri-chunk] exI[of - fv]
    exI[of - query-paths] conjI)
  (use query-chunk-shape trace-layers composition-layers in
     $\langle \text{simp-all add: mult.commute} \rangle$ )
have tr-t:  $P\text{Transcript } s = ?\text{query-chunk} @ P\text{Transcript } t$ 
  using tr-s-s1 tr-s1-s2 tr-s3-s4 s3-eq t-eq by simp
have st-t:  $P\text{State } t = \text{foldl concat } (P\text{State } s) ?\text{query-chunk}$ 
  using st-s1 st-s2 st-s4 s3-eq t-eq by simp
have ext-s-t:  $s \leq t$ 
  using ext-s-s1 ext-s1-s2 ext-s3-s4 t-eq unfolding s3-eq
  by (meson hash-ext-trans)
have query-counter-t:  $P\text{QueryCounter } t = P\text{QueryCounter } s$ 
  using query-counter-s1 query-counter-s2 query-counter-s4
  unfolding s3-eq t-eq by simp
have table-t:
  partial-authenticated-table fr (scale * clength) openings t
  by (rule partial-authenticated-table-mono[OF table-s1])
    (use ext-s1-s2 ext-s3-s4 t-eq s3-eq in
       $\langle \text{meson hash-ext-trans} \rangle$ )
have openings-nonempty:  $\text{openings} \neq []$ 
  using idx-openings powers-pos unfolding powers-scaled-def by auto
have hd-value:  $\text{opening-value } (\text{hd openings}) = \text{hd } fv$ 
  using values-openings openings-nonempty
  by (cases openings; cases fv) simp-all
have head-chunk-nth:
  fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
  (trace-layer-chunks ! 0)
  using head-chunk trace-layers-nonempty
  by (cases trace-layer-chunks) (simp-all add: mult.commute)
show ?thesis
  by (rule that[OF refl round-layers tr-t st-t ext-s-t query-counter-t
    idx-openings table-t openings-nonempty - head-chunk-nth])
    (use hd-value xp-eq in simp)
qed

lemma verifier-query-round-program-trace-fri-head-value-aligned-with-lookup:
  fixes  $s \ t :: ('f, 'a) \text{protocol-channel-scheme}$ 

```

```

assumes f-fl-eq: f-fl = (b, rt) # f-fl-tail
and outcome:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
obtains raw idx openings xp xp-path xn xn-path where
  idx = index (to-nat raw)
  map opening-index openings = powers-scaled idx
  partial-authenticated-table fr (scale * clength) openings t
  openings ≠ []
  opening-value (hd openings) = xp
  fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
    ([xp] @ xp-path @ [xn] @ xn-path)
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where raw:
  Some (raw, s0) ∈
    set-dist (execute receive-query-index-challenge s)
and tail:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
by (auto elim!: set-dist-bindE)
obtain idx openings xp xp-path xn xn-path where
  idx-eq: idx = index (to-nat raw)
and idxs: map opening-index openings = powers-scaled idx
and auth-table:
  partial-authenticated-table fr (scale * clength) openings t
and nonempty: openings ≠ []
and opening-value-eq: opening-value (hd openings) = xp
and chunk:
  fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
    ([xp] @ xp-path @ [xn] @ xn-path)
using verifier-query-round-after-index-trace-fri-head-value-aligned
  [OF f-fl-eq tail]
by metis
have lookup-s0:
  fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
using receive-query-index-challenge-outcome[OF raw] by simp
have s0-t: s0 ≤ t
using verifier-query-round-after-index-program-outcome[OF tail]
by simp
have lookup-t:

```

```

    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  by (rule hash-extension-lookup[OF lookup-s0 s0-t])
show ?thesis
  by (rule that[OF idx-eq idxs auth-table nonempty opening-value-eq chunk lookup-t])
qed

lemma ntimes-verifier-query-rounds-selected-trace-fri-head-value-aligned-with-replay-at:
fixes query-state final-state :: ('f, 'a) protocol-channel-scheme
assumes f-fl-eq: f-fl = (b, rt) # f-fl-tail
and outcome:
  Some (results, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
and i-bound: i < rounds
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and query-chunk:
  ∧ i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and replay-lookup:
  ∧ i. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
obtains openings xp xp-path xn xn-path where
  map opening-index openings = powers-scaled (query-idxs ! i)
  partial-authenticated-table fr (scale * clength) openings final-state
  openings ≠ []
  opening-value (hd openings) = xp
  fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
    ([xp] @ xp-path @ [xn] @ xn-path)
proof –
  let ?round = verifier-query-round-program fr f-fl f-final as fl final
from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
obtain prefix x suffix s-i s-suc where
  prefix:

```

```

    Some (prefix, s-i) ∈
      set-dist (execute (ntimes ?round i) query-state)
and round:
    Some (x, s-suc) ∈ set-dist (execute ?round s-i)
and suffix:
    Some (suffix, final-state) ∈
      set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
by blast
have x-eq: x = ()
by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
fix prefix-query-idxs prefix-chunks
assume len-prefix-idxs: length prefix-query-idxs = i
and len-prefix-chunks: length prefix-chunks = i
and transcript-prefix:
  PTranscript query-state =
    List.concat prefix-chunks @ PTranscript s-i
and state-i:
  PState s-i =
    state-after-query-chunks (PState query-state) prefix-chunks i
and counter-i:
  PQueryCounter s-i = PQueryCounter query-state + i
and prefix-ext: query-state ≤ s-i
and prefix-rounds:
   $\bigwedge k. k < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
show ?thesis
proof (rule verifier-query-round-program-outcome-with-authenticated-trace-openings
  [OF round[unfolded x-eq]])
fix raw idx chunk round-openings
assume idx-eq: idx = index (to-nat raw)
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
    chunk
and transcript-round:
  PTranscript s-i = chunk @ PTranscript s-suc
and state-suc:
  PState s-suc = foldl concat (PState s-i) chunk
and round-ext: s-i ≤ s-suc
and counter-suc:
  PQueryCounter s-suc = Suc (PQueryCounter s-i)
and lookup-suc:
  fmlookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
and len-round-openings:

```

```

    length round-openings = length (powers-scaled idx)
and trace-indices:
    map opening-index round-openings = powers-scaled idx
and trace-table-suc:
    partial-authenticated-table fr (scale * clength)
    round-openings s-suc
show ?thesis
proof (rule verifier-query-round-program-trace-fri-head-value-aligned-with-lookup
    [OF f-ft-eq round[unfolded x-eq]])
fix raw' idx' openings xp xp-path xn xn-path
assume idx'-eq: idx' = index (to-nat raw')
and opening-indices:
    map opening-index openings = powers-scaled idx'
and auth-table-suc:
    partial-authenticated-table fr (scale * clength) openings
    s-suc
and openings-nonempty: openings ≠ []
and opening-value-eq: opening-value (hd openings) = xp
and head-chunk:
    fri-layer-opening-chunk (scale * clength) xp xp-path xn
    xn-path ([xp] @ xp-path @ [xn] @ xn-path)
and lookup-suc':
    fmlookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw'
have raw'-eq: raw' = raw
using lookup-suc lookup-suc' by simp
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
    [OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-idxs:
    length suffix-query-idxs = rounds - Suc i
and len-suffix-chunks:
    length suffix-chunks = rounds - Suc i
and transcript-suffix:
    PTranscript s-suc =
    List.concat suffix-chunks @ PTranscript final-state
and suffix-state:
    PState final-state =
    state-after-query-chunks (PState s-suc) suffix-chunks
    (rounds - Suc i)
and suffix-counter:
    PQueryCounter final-state =
    PQueryCounter s-suc + (rounds - Suc i)
and suffix-ext: s-suc ≤ final-state
and suffix-rounds:
     $\bigwedge k. k < rounds - Suc i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)

```

```

      (map snd f-fl) (map snd fl) (suffix-chunks ! k)
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks i-bound by simp
have selected-chunk-lens:
   $\bigwedge k. k < rounds \implies$ 
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
proof -
  fix k
  assume k-bound: k < rounds
  show
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
proof (cases k < i)
  case True
  have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length (prefix-chunks ! k) =
      verifier-query-round-transcript-length
        (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
      [OF prefix-rounds[OF True]])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
next
  case False
  show ?thesis
  proof (cases k = i)
  case True
  have nth-eq: ?selected-chunks ! k = chunk
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length chunk =
      verifier-query-round-transcript-length idx
      (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length[OF chunk-shape])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
next
  case False
  let ?k = k - Suc i

```

```

have ge:  $Suc\ i \leq k$ 
  using  $\langle \neg k < i \rangle$  False by linarith
have k-bound':  $?k < rounds - Suc\ i$ 
  using ge k-bound by linarith
have nth-eq:  $?selected-chunks ! k = suffix-chunks ! ?k$ 
  using ge len-prefix-chunks by (simp add: nth-append)
have
  length (suffix-chunks ! ?k) =
    verifier-query-round-transcript-length
      (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
  by (rule verifier-query-round-chunk-length
      [OF suffix-rounds [OF k-bound']])
then show ?thesis
  using nth-eq
  by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-lens:
   $\bigwedge k. k < rounds \implies$ 
    length (query-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  using query-chunk verifier-query-round-chunk-length by blast
have transcript-all:
  PTranscript query-state =
    List.concat ?selected-chunks @ PTranscript final-state
  using transcript-prefix transcript-round transcript-suffix by simp
have concat-eq:
  List.concat ?selected-chunks = List.concat query-chunks
  using transcript-all transcript-query by simp
have chunks-eq:  $?selected-chunks = query-chunks$ 
proof -
  have same-lens:
     $\bigwedge k. k < length\ ?selected-chunks \implies$ 
      length (?selected-chunks ! k) = length (query-chunks ! k)
    using selected-chunk-lens query-chunk-lens len-selected-chunks
    by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks  $\wedge$  ( $[] :: 'f\ list$ ) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
        [OF len-eq same-lens concat-eq-empty])
  then show ?thesis
    by simp

```

```

qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state)
      ?selected-chunks i =
        state-after-query-chunks (PState query-state)
          prefix-chunks i
    by (rule state-after-query-chunks-append-prefix
      [OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks
        (PState query-state) query-chunks i)) =
    Some raw
  using hash-extension-lookup[OF lookup-suc suffix-ext]
  unfolding counter-i state-i prefix-state-eq .
have raw-eq: raw = raw-idxs ! i
  using lookup-final replay-lookup[OF i-bound] by simp
have idx'-eq-query: idx' = query-idxs ! i
  using idx'-eq raw'-eq raw-eq query-idxs-eq len-raw i-bound
  by simp
have auth-table-final:
  partial-authenticated-table fr (scale * clength) openings
    final-state
  by (rule partial-authenticated-table-mono
    [OF auth-table-suc suffix-ext])
show ?thesis
  by (rule that[OF - auth-table-final openings-nonempty
    opening-value-eq head-chunk])
    (use opening-indices idx'-eq-query in simp)
qed
qed
qed
qed
qed

definition trace-fri-header-tied-value-aligned-base-opening-conflict
  :: (f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  (f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  trace-fri-header-tied-value-aligned-base-opening-conflict s out
     $\longleftrightarrow$ 

```


$(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table}$
 $\text{round-idx xp xp-path xn xn-path result final-state.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{accepted-fri-opening-transcript s out trace-roots trace-bs trace-final}$
 $\text{dg composition-roots composition-bs composition-final fri-query-idxs}$
 $\text{trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript s fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{round-idx} < \text{length fri-query-idxs} \wedge$
 $0 < \text{length trace-bs} \wedge$
 $\text{trace-openings} ! \text{round-idx} \neq [] \wedge$
 $\text{opening-index (hd (trace-openings ! round-idx))} =$
 $\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0 \wedge$
 $\text{opening-value (hd (trace-openings ! round-idx))} = \text{xp} \wedge$
 $\text{fri-layer-step-evidence}$
 $(\text{trace-roots} ! 0)$
 $(\text{trace-bs} ! 0)$
 $(\text{fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0)$
 1
 $(\text{fri-sibling-index (fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0))$
 $\text{xp xp-path xn xn-path}$
 $(\text{fri-evidence-next-idx trace-roots fri-query-idxs round-idx } 0)$
 $(\text{fri-evidence-next-value trace-roots trace-bs fri-query-idxs}$
 $\text{round-idx } 0 \text{ xp xn})$
 $(\text{trace-round-layers} ! \text{round-idx} ! 0) \wedge$
 $\neg \text{fri-opening-matches-table}$
 $(\text{fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0)$
 $\text{trace-table xp xn})$

definition *trace-fri-header-tied-value-aligned-sibling-mismatch*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-value-aligned-sibling-mismatch s out

$\longleftrightarrow$

$(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table}$
 $\text{round-idx xp xp-path xn xn-path result final-state.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
 dg composition-roots composition-bs composition-final fri-query-idxs
 trace-round-layers composition-round-layers $\wedge$
 verifier-header-transcript s fr trace-roots trace-final as dg
 composition-roots composition-final rest $\wedge$
 accepted-with-partial-trace-openings s out fr fri-query-idxs
 trace-openings $\wedge$
 partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ trace-table-low-degree trace-table $\wedge$
 round-idx < length fri-query-idxs $\wedge$
 0 < length trace-bs $\wedge$
 trace-openings ! round-idx $\neq []$ $\wedge$
 opening-index (hd (trace-openings ! round-idx)) =
 fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 $\wedge$
 opening-value (hd (trace-openings ! round-idx)) = x_p $\wedge$
 fri-layer-step-evidence
 (trace-roots ! 0)
 (trace-bs ! 0)
 (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
 1
 (fri-sibling-index (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
 x_p x_p -path x_n x_n -path
 (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
 (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
 round-idx 0 x_p x_n)
 (trace-round-layers ! round-idx ! 0) $\wedge$
 trace-table !
 fri-sibling-index (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
 $\neq x_n$)

lemma trace-fri-header-tied-value-aligned-base-imp-sampled-base:

assumes

trace-fri-header-tied-value-aligned-base-opening-conflict s out

shows trace-fri-header-tied-sampled-base-opening-conflict s out

proof —

from **assms** **obtain** trace-roots trace-bs trace-final dg composition-roots

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as rest trace-openings trace-table

round-idx x_p x_p -path x_n x_n -path result final-state

where out-eq: out = Some (result, final-state)

and fri-openings:

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final

dg composition-roots composition-bs composition-final fri-query-idxs

trace-round-layers composition-round-layers

and header:

verifier-header-transcript s fr trace-roots trace-final as dg

```

    composition-roots composition-final rest
and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
and cand: partial-trace-table-candidate trace-table trace-openings
and not-low:  $\neg$  trace-table-low-degree trace-table
and round-bound: round-idx < length fri-query-idxs
and trace-bs-nonempty: 0 < length trace-bs
and openings-nonempty: trace-openings ! round-idx  $\neq$  []
and hd-idx:
    opening-index (hd (trace-openings ! round-idx)) =
        fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0
and hd-value: opening-value (hd (trace-openings ! round-idx)) = xp
and step:
    fri-layer-step-evidence
    (trace-roots ! 0)
    (trace-bs ! 0)
    (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    1
    (fri-sibling-index (fri-evidence-layer-len trace-roots 0)
        (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
    xp xp-path xn xn-path
    (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
    (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
        round-idx 0 xp xn)
    (trace-round-layers ! round-idx ! 0)
and no-match:
     $\neg$  fri-opening-matches-table
    (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    trace-table xp xn
unfolding
    trace-fri-header-tied-value-aligned-base-opening-conflict-def
by auto
have base:
    generic-fri-sampled-base-opening-conflict trace-table trace-roots
    trace-bs fri-query-idxs trace-round-layers
unfolding generic-fri-sampled-base-opening-conflict-def
by (intro exI[of - round-idx] exI[of - xp] exI[of - xp-path]
    exI[of - xn] exI[of - xn-path] conjI)
    (use round-bound trace-bs-nonempty step no-match in auto)
show ?thesis
unfolding trace-fri-header-tied-sampled-base-opening-conflict-def
by (intro exI conjI)
    (use out-eq fri-openings header partial cand not-low base in auto)
qed

```

lemma trace-fri-header-tied-value-aligned-base-imp-sibling-mismatch:

assumes

trace-fri-header-tied-value-aligned-base-opening-conflict s out

shows *trace-fri-header-tied-value-aligned-sibling-mismatch s out*

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as rest trace-openings trace-table

round-idx xp xp-path xn xn-path result final-state

where *out-eq*: *out* = *Some (result, final-state)*

and *fri-openings*:

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final

dg composition-roots composition-bs composition-final fri-query-idxs

trace-round-layers composition-round-layers

and *header*:

verifier-header-transcript s fr trace-roots trace-final as dg

composition-roots composition-final rest

and *partial*:

accepted-with-partial-trace-openings s out fr fri-query-idxs

trace-openings

and *cand*: *partial-trace-table-candidate trace-table trace-openings*

and *not-low*: $\neg$ *trace-table-low-degree trace-table*

and *round-bound*: *round-idx* < *length fri-query-idxs*

and *trace-bs-nonempty*: *0* < *length trace-bs*

and *openings-nonempty*: *trace-openings* ! *round-idx* $\neq$ []

and *hd-idx*:

opening-index (hd (trace-openings ! round-idx)) =

fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0

and *hd-value*: *opening-value (hd (trace-openings ! round-idx)) = xp*

and *step*:

fri-layer-step-evidence

(trace-roots ! 0)

(trace-bs ! 0)

(fri-evidence-layer-len trace-roots 0)

(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)

1

(fri-sibling-index (fri-evidence-layer-len trace-roots 0)

(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))

xp xp-path xn xn-path

(fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)

(fri-evidence-next-value trace-roots trace-bs fri-query-idxs

round-idx 0 xp xn)

(trace-round-layers ! round-idx ! 0)

and *no-match*:

$\neg$ *fri-opening-matches-table*

(fri-evidence-layer-len trace-roots 0)

(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)

trace-table xp xn

unfolding

trace-fri-header-tied-value-aligned-base-opening-conflict-def

```

  by auto
let ?idx =
  fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0
let ?len = fri-evidence-layer-len trace-roots 0
let ?opn = hd (trace-openings ! round-idx)
have round-bound-rounds: round-idx < rounds
  using round-bound accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
  by simp
have opn-in: ?opn ∈ set (trace-openings ! round-idx)
  using openings-nonempty by simp
have table-len: length trace-table = scale * clength
  by (rule partial-trace-table-candidateD(1)
    [OF cand round-bound-rounds opn-in])
have idx-bound-eval: ?idx < scale * clength
  using partial-trace-table-candidateD(2)
    [OF cand round-bound-rounds opn-in] hd-idx
  by simp
have table-left: trace-table ! ?idx = xp
  using partial-trace-table-candidateD(3)
    [OF cand round-bound-rounds opn-in] hd-idx hd-value
  by simp
have roots-nonempty: 0 < length trace-roots
  using accepted-fri-opening-transcript-shapes(2)[OF fri-openings]
    trace-bs-nonempty
  by simp
have len-eq: ?len = scale * clength
  unfolding fri-evidence-layer-len-def
  using fri-layer-lengths-nth-div[of 0 length trace-roots
    clength * scale] roots-nonempty
  by (simp add: mult.commute)
have sibling-mismatch:
  trace-table ! fri-sibling-index ?len ?idx ≠ xn
proof
  assume sibling-eq: trace-table ! fri-sibling-index ?len ?idx = xn
  have fri-opening-matches-table ?len ?idx trace-table xp xn
    unfolding fri-opening-matches-table-def
    using idx-bound-eval table-len table-left sibling-eq len-eq
    by (simp add: mult.commute)
  then show False
    using no-match by contradiction
qed
show ?thesis
  unfolding trace-fri-header-tied-value-aligned-sibling-mismatch-def
  by (intro exI conjI)
  (use out-eq fri-openings header partial cand not-low round-bound
    trace-bs-nonempty openings-nonempty hd-idx hd-value step
    sibling-mismatch in auto)
qed

```

lemma *wp-trace-fri-header-tied-value-aligned-base-bound-from-sibling-mismatch:*

assumes *mismatch-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-value-aligned-sibling-mismatch } s) \ s \leq M$

shows

wp-event verify-monad

$(\text{trace-fri-header-tied-value-aligned-base-opening-conflict } s) \ s \leq M$

proof –

have *wp-event verify-monad*

$(\text{trace-fri-header-tied-value-aligned-base-opening-conflict } s) \ s \leq$

wp-event verify-monad

$(\text{trace-fri-header-tied-value-aligned-sibling-mismatch } s) \ s$

by *(rule wp-event-mono)*

(rule trace-fri-header-tied-value-aligned-base-imp-sibling-mismatch)

then show *?thesis*

by *(rule order-trans[OF - mismatch-bound])*

qed

definition *trace-fri-header-tied-sibling-mismatch-structural-gap*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-sibling-mismatch-structural-gap s out

$\longleftrightarrow$

trace-fri-header-tied-value-aligned-sibling-mismatch s out $\wedge$

$\neg \text{partial-merkle-inconsistency-bad } s \text{ out}$

lemma *trace-fri-header-tied-value-aligned-sibling-mismatch-imp-partial-merkle-or-structural-gap:*

assumes *mismatch:*

trace-fri-header-tied-value-aligned-sibling-mismatch s out

shows

partial-merkle-inconsistency-bad s out $\vee$

trace-fri-header-tied-sibling-mismatch-structural-gap s out

proof –

show *?thesis*

using *mismatch*

unfolding *trace-fri-header-tied-sibling-mismatch-structural-gap-def*

by *blast*

qed

lemma *wp-trace-fri-header-tied-value-aligned-sibling-mismatch-bound-from-partial-merkle-and-structural-gap:*

fixes *P G :: prob*

assumes *partial-merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s $\leq P$

and *gap-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s \leq G$

shows

wp-event verify-monad

$(\text{trace-fri-header-tied-value-aligned-sibling-mismatch } s) \ s \leq P + G$
proof –
have $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-value-aligned-sibling-mismatch } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\lambda \text{out. partial-merkle-inconsistency-bad } s \ \text{out} \ \vee$
 $\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s \ \text{out}) \ s$
by $(\text{rule wp-event-mono})$
 $(\text{rule}$
 $\text{trace-fri-header-tied-value-aligned-sibling-mismatch-imp-partial-merkle-or-structural-gap})$
also have $\dots \leq$
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s +$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s$
by $(\text{rule wp-event-union-bound})$
also have $\dots \leq P + G$
by $(\text{intro add-mono partial-merkle-bound gap-bound})$
finally show $?thesis$.
qed

definition $\text{trace-fri-header-tied-base-value-alignment-gap}$
 $:: (\text{'f}, \text{'a}) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-header-tied-base-value-alignment-gap } s \ \text{out} \longleftrightarrow$
 $\text{trace-fri-header-tied-sampled-base-opening-conflict } s \ \text{out} \wedge$
 $\neg \text{trace-fri-header-tied-value-aligned-base-opening-conflict } s \ \text{out}$

lemma $\text{trace-fri-header-tied-sampled-base-imp-value-aligned-or-gap}$:
assumes $\text{trace-fri-header-tied-sampled-base-opening-conflict } s \ \text{out}$
shows
 $\text{trace-fri-header-tied-value-aligned-base-opening-conflict } s \ \text{out} \vee$
 $\text{trace-fri-header-tied-base-value-alignment-gap } s \ \text{out}$
using assms
unfolding $\text{trace-fri-header-tied-base-value-alignment-gap-def}$
by blast

lemma $\text{wp-trace-fri-header-tied-sampled-base-bound-from-value-aligned-and-gap}$:
fixes $V \ G :: \text{prob}$
assumes $\text{value-aligned-bound}$:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-value-aligned-base-opening-conflict } s) \ s \leq V$
and gap-bound :
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-value-alignment-gap } s) \ s \leq G$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-base-opening-conflict } s) \ s \leq V + G$
proof –

```

have wp-event verify-monad
  (trace-fri-header-tied-sampled-base-opening-conflict s) s ≤
  wp-event verify-monad
    (λout.
      trace-fri-header-tied-value-aligned-base-opening-conflict s out ∨
      trace-fri-header-tied-base-value-alignment-gap s out) s
by (rule wp-event-mono)
  (rule trace-fri-header-tied-sampled-base-imp-value-aligned-or-gap)
also have ... ≤
  wp-event verify-monad
    (trace-fri-header-tied-value-aligned-base-opening-conflict s) s +
  wp-event verify-monad
    (trace-fri-header-tied-base-value-alignment-gap s) s
by (rule wp-event-union-bound)
also have ... ≤ V + G
by (intro add-mono value-aligned-bound gap-bound)
finally show ?thesis .
qed

lemma wp-trace-fri-header-tied-sampled-base-bound-from-sibling-mismatch-and-gap:
fixes M G :: prob
assumes mismatch-bound:
  wp-event verify-monad
    (trace-fri-header-tied-value-aligned-sibling-mismatch s) s ≤ M
and gap-bound:
  wp-event verify-monad
    (trace-fri-header-tied-base-value-alignment-gap s) s ≤ G
shows
  wp-event verify-monad
    (trace-fri-header-tied-sampled-base-opening-conflict s) s ≤ M + G
by (rule
  wp-trace-fri-header-tied-sampled-base-bound-from-value-aligned-and-gap)
  (rule wp-trace-fri-header-tied-value-aligned-base-bound-from-sibling-mismatch
    [OF mismatch-bound], rule gap-bound)

end

end

theory Soundness-FRI-Trace-Residuals
imports
  Soundness-FRI-Value-Aligned-Events
  Soundness-FRI-Final-Step-Replay
  Soundness-FRI-Recorded-Chunk-Replay
begin

```

Narrow residual refinements for the header-tied trace FRI route. These predicates name the remaining replay gaps for non-base FRI branches without changing the protocol or

exporting broad selected-step replay lemmas.

context *soundness*

begin

definition *trace-fri-reachable-header-tie-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-reachable-header-tie-gap } s \text{ out} \longleftrightarrow$
 $\text{trace-fri-bad-with-reachable-partial-candidate } s \text{ out} \wedge$
 $\neg \text{trace-fri-bad-with-header-tied-partial-candidate } s \text{ out}$

definition *trace-fri-header-tied-candidate-transfer-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-candidate-transfer-gap } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{trace-fri-bad-with-reachable-partial-candidate } s \text{ out} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as fri-dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{trace-table-low-degree trace-table})$

definition *trace-fri-header-tied-candidate-available-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-candidate-available-gap } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{trace-fri-reachable-header-tie-gap } s \text{ out} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as fri-dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$

```

    partial-trace-table-candidate trace-table trace-openings)

lemma trace-fri-zero-round-final-obstruction-imp-header-tied-or-gap:
  assumes trace-fri-zero-round-final-obstruction s out
  shows
    trace-fri-bad-with-header-tied-partial-candidate s out  $\vee$ 
    trace-fri-reachable-header-tie-gap s out
proof –
  have reachable:
    trace-fri-bad-with-reachable-partial-candidate s out
proof –
  from assms obtain trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr candidate-query-idxs
    trace-openings trace-table
  where evidence:
    trace-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
  unfolding trace-fri-zero-round-final-obstruction-def by blast
  have partial:
    trace-fri-partial-candidate-evidence s out fr candidate-query-idxs
    trace-openings trace-table
  using trace-fri-partial-candidate-opening-evidenceD(2)[OF evidence] .
  show ?thesis
  unfolding trace-fri-bad-with-reachable-partial-candidate-def
  by (intro exI conjI)
    (rule trace-fri-partial-candidate-evidenceD(1)[OF partial],
     rule trace-fri-partial-candidate-evidenceD(2)[OF partial],
     rule trace-fri-partial-candidate-evidenceD(3)[OF partial])
qed
show ?thesis
proof (cases trace-fri-bad-with-header-tied-partial-candidate s out)
  case True
  then show ?thesis by simp
next
  case False
  then have trace-fri-reachable-header-tie-gap s out
    unfolding trace-fri-reachable-header-tie-gap-def
    using reachable by simp
  then show ?thesis by simp
qed
qed

lemma wp-trace-fri-zero-round-final-obstruction-bound-from-header-tied-and-gap:
  fixes H G :: prob
  assumes header-bound:

```

wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) $s \leq H$
and gap-bound:
wp-event verify-monad
 (trace-fri-reachable-header-tie-gap s) $s \leq G$
shows
wp-event verify-monad
 (trace-fri-zero-round-final-obstruction s) $s \leq H + G$
proof –
have *wp-event verify-monad*
 (trace-fri-zero-round-final-obstruction s) $s \leq$
wp-event verify-monad
 ($\lambda out.$
 trace-fri-bad-with-header-tied-partial-candidate s out $\vee$
 trace-fri-reachable-header-tie-gap s out) s
by (rule *wp-event-mono*)
 (rule trace-fri-zero-round-final-obstruction-imp-header-tied-or-gap)
also have ... $\leq$
wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) $s +$
wp-event verify-monad (trace-fri-reachable-header-tie-gap s) s
by (rule *wp-event-union-bound*)
also have ... $\leq H + G$
by (rule *add-mono*[*OF* header-bound gap-bound])
finally show ?thesis .
qed

definition trace-fri-header-tied-reduction
 $:: (f, 'a)$ protocol-channel-scheme $\Rightarrow bool$
where

trace-fri-header-tied-reduction $s \longleftrightarrow$
wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) $s \leq$
 trace-fri-error

lemma trace-fri-header-tied-reductionD:
assumes trace-fri-header-tied-reduction s
shows
wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) $s \leq$
 trace-fri-error
using *assms unfolding* trace-fri-header-tied-reduction-def **by** *simp*

lemma trace-fri-header-tied-reduction-staged-bound:
assumes reductions:
 $\bigwedge data$ attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

```

    trace-fri-header-tied-reduction
      (verifier-state-from-adversary attacker-state
       (staged-proof-transcript data))
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state ≤ trace-fri-error
  by (rule trace-fri-bad-with-header-tied-partial-candidates-staged-bound)
     (rule trace-fri-header-tied-reductionD[OF reductions])

lemma trace-fri-header-tied-reduction-from-reachable-reduction:
  assumes trace-fri-reachable-partial-candidate-reduction-assumption s
  shows trace-fri-header-tied-reduction s
proof –
  have reachable:
    wp-event verify-monad
      (trace-fri-bad-with-reachable-partial-candidate s) s ≤
      trace-fri-error
  using assms
  unfolding trace-fri-reachable-partial-candidate-reduction-assumption-def
  by simp
  have verifier-tied:
    wp-event verify-monad
      (trace-fri-bad-with-verifier-tied-partial-candidate s) s ≤
      trace-fri-error
  by (rule order-trans[OF - reachable])
     (rule wp-event-mono,
      rule trace-fri-bad-with-verifier-tied-imp-reachable-partial-candidate)
  show ?thesis
  unfolding trace-fri-header-tied-reduction-def
  by (rule order-trans[OF - verifier-tied])
     (rule wp-event-mono,
      rule trace-fri-bad-with-header-tied-imp-verifier-tied)
qed

lemma trace-fri-header-candidate-non-low-contradicts-header-tie-gap:
  assumes gap: trace-fri-reachable-header-tie-gap s out
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings

```

and *candidate*:
 partial-trace-table-candidate trace-table trace-openings
and *not-low*: $\neg$ *trace-table-low-degree trace-table*
shows *False*
proof –
 have *trace-fri-bad-with-header-tied-partial-candidate s out*
 unfolding *trace-fri-bad-with-header-tied-partial-candidate-def*
 by (*intro exI conjI*)
 (*rule fri-openings, rule header, rule partial, rule candidate,*
 rule not-low)
 then show *False*
 using *gap unfolding trace-fri-reachable-header-tie-gap-def* **by** *simp*
qed

lemma *trace-fri-header-candidate-low-imp-transfer-gap*:
assumes *gap: trace-fri-reachable-header-tie-gap s out*
and *fri-openings*:
 accepted-fri-opening-transcript s out trace-roots trace-bs
 trace-final fri-dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers
and *header*:
 verifier-header-transcript s fr trace-roots trace-final as fri-dg
 composition-roots composition-final rest
and *partial*:
 accepted-with-partial-trace-openings s out fr fri-query-idxs
 trace-openings
and *candidate*:
 partial-trace-table-candidate trace-table trace-openings
and *low*: *trace-table-low-degree trace-table*
shows *trace-fri-header-tied-candidate-transfer-gap s out*
proof –
 have *raw: trace-fri-bad-with-reachable-partial-candidate s out*
 using *gap unfolding trace-fri-reachable-header-tie-gap-def* **by** *simp*
 show *?thesis*
 unfolding *trace-fri-header-tied-candidate-transfer-gap-def*
 by (*intro exI conjI*)
 (*rule raw, rule fri-openings, rule header, rule partial,*
 rule candidate, rule low)
qed

lemma *trace-fri-header-tied-candidate-available-gap-imp-transfer-gap*:
assumes *trace-fri-header-tied-candidate-available-gap s out*
shows *trace-fri-header-tied-candidate-transfer-gap s out*
proof –
 obtain *trace-roots trace-bs trace-final fri-dg composition-roots*
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr as rest trace-openings trace-table
 where *gap: trace-fri-reachable-header-tie-gap s out*

and *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *header*:
verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest
and *partial*:
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *candidate*:
partial-trace-table-candidate trace-table trace-openings
using *assms*
unfolding *trace-fri-header-tied-candidate-available-gap-def* **by** *blast*
show *?thesis*
proof (*cases trace-table-low-degree trace-table*)
case *True*
show *?thesis*
by (*rule trace-fri-header-candidate-low-imp-transfer-gap*
[OF gap fri-openings header partial candidate True])
next
case *False*
have *False*
by (*rule trace-fri-header-candidate-non-low-contradicts-header-tie-gap*
[OF gap fri-openings header partial candidate False])
then show *?thesis* **by** *simp*
qed
qed

lemma *wp-trace-fri-header-tied-candidate-available-gap-bound-from-transfer-gap*:
assumes *transfer-bound*:
wp-event verify-monad
(trace-fri-header-tied-candidate-transfer-gap s) s ≤ T
shows
wp-event verify-monad
(trace-fri-header-tied-candidate-available-gap s) s ≤ T
by (*rule order-trans[OF - transfer-bound]*)
(rule wp-event-mono,
rule trace-fri-header-tied-candidate-available-gap-imp-transfer-gap)

lemma *trace-fri-reachable-imp-header-tied-or-gap*:
assumes *trace-fri-bad-with-reachable-partial-candidate s out*
shows
trace-fri-bad-with-header-tied-partial-candidate s out ∨
trace-fri-reachable-header-tie-gap s out
using *assms*
unfolding *trace-fri-reachable-header-tie-gap-def*
by *blast*

lemma *wp-trace-fri-reachable-bound-from-header-tied-and-gap:*
fixes $H\ G :: \text{prob}$
assumes *header-bound:*
 wp-event verify-monad
 $(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \ s \leq H$
and *gap-bound:*
 wp-event verify-monad
 $(\text{trace-fri-reachable-header-tie-gap } s) \ s \leq G$
shows
 wp-event verify-monad
 $(\text{trace-fri-bad-with-reachable-partial-candidate } s) \ s \leq H + G$
proof –
 have *wp-event verify-monad*
 $(\text{trace-fri-bad-with-reachable-partial-candidate } s) \ s \leq$
 wp-event verify-monad
 $(\lambda \text{out.}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate } s \ \text{out} \vee$
 $\text{trace-fri-reachable-header-tie-gap } s \ \text{out}) \ s$
 by (rule *wp-event-mono*)
 (rule *trace-fri-reachable-imp-header-tied-or-gap*)
 also have $\dots \leq$
 wp-event verify-monad
 $(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \ s +$
 wp-event verify-monad
 $(\text{trace-fri-reachable-header-tie-gap } s) \ s$
 by (rule *wp-event-union-bound*)
 also have $\dots \leq H + G$
 by (rule *add-mono*[*OF header-bound gap-bound*])
 finally show *?thesis* .
qed

lemma *trace-fri-reachable-partial-candidate-reduction-from-header-tied-and-gap:*
fixes $H\ G :: \text{prob}$
assumes *header-bound:*
 wp-event verify-monad
 $(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \ s \leq H$
and *gap-bound:*
 wp-event verify-monad
 $(\text{trace-fri-reachable-header-tie-gap } s) \ s \leq G$
and *total-bound:* $H + G \leq \text{trace-fri-error}$
shows *trace-fri-reachable-partial-candidate-reduction-assumption s*
proof –
 have *old-bound:*
 wp-event verify-monad
 $(\text{trace-fri-bad-with-reachable-partial-candidate } s) \ s \leq H + G$
 by (rule *wp-trace-fri-reachable-bound-from-header-tied-and-gap*
 [*OF header-bound gap-bound*])
 show *?thesis*

unfolding *trace-fri-reachable-partial-candidate-reduction-assumption-def*
by (rule order-trans[*OF old-bound total-bound*])
qed

definition *trace-fri-header-tied-recorded-base-chunk-auth-missing*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-recorded-base-chunk-auth-missing s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table}$
 $\text{ round-idx xp xp-path xn xn-path result final-state.}$
 $\text{out} = \text{Some}(\text{result}, \text{final-state}) \wedge$
 $\text{accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final dg composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript s fr trace-roots trace-final as dg}$
 $\text{ composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{round-idx} < \text{length fri-query-idxs} \wedge$
 $0 < \text{length trace-bs} \wedge$
 $\text{fri-layer-step-evidence}$
 $(\text{trace-roots} ! 0)$
 $(\text{trace-bs} ! 0)$
 $(\text{fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0)$
 1
 $(\text{fri-sibling-index} (\text{fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0))$
 $\text{xp xp-path xn xn-path}$
 $(\text{fri-evidence-next-idx trace-roots fri-query-idxs round-idx } 0)$
 $(\text{fri-evidence-next-value trace-roots trace-bs fri-query-idxs}$
 $\text{ round-idx } 0 \text{ xp xn})$
 $(\text{trace-round-layers} ! \text{round-idx} ! 0) \wedge$
 $\neg \text{fri-opening-matches-table}$
 $(\text{fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0)$
 $\text{trace-table xp xn} \wedge$
 $\neg \text{fri-layer-chunk-authenticated} (\text{trace-roots} ! 0)$
 $(\text{fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0)$
 $(\text{trace-round-layers} ! \text{round-idx} ! 0) \text{ final-state})$

lemma *trace-fri-header-tied-recorded-base-chunk-auth-missing-false:*
assumes


```

    trace-fri-header-tied-recorded-base-chunk-auth-missing s out
  shows False
proof -
  from assms obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
    round-idx xp xp-path xn xn-path result final-state
  where out-eq: out = Some (result, final-state)
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and round-bound: round-idx < length fri-query-idxs
  and trace-bs-nonempty: 0 < length trace-bs
  and missing:
    ¬ fri-layer-chunk-authenticated (trace-roots ! 0)
    (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    (trace-round-layers ! round-idx ! 0) final-state
  unfolding
    trace-fri-header-tied-recorded-base-chunk-auth-missing-def
  by blast
  have len-trace-bs:
    length trace-bs = length trace-roots
  using accepted-fri-opening-transcript-shapes(2)[OF fri-openings] .
  have trace-nonempty: 0 < length trace-roots
  using trace-bs-nonempty len-trace-bs by simp
  have len-query: length fri-query-idxs = rounds
  using accepted-fri-opening-transcript-shapes(7)[OF fri-openings] .
  have round-bound-rounds: round-idx < rounds
  using round-bound len-query by simp
  have auth:
    fri-layer-chunk-authenticated (trace-roots ! 0) (clength * scale)
    (fri-query-idxs ! round-idx)
    (trace-round-layers ! round-idx ! 0) final-state
  by (rule
    accepted-fri-opening-transcript-trace-base-recorded-chunk-authenticated-at
    [OF fri-openings out-eq round-bound-rounds trace-nonempty])
  have len0:
    fri-evidence-layer-len trace-roots 0 = clength * scale
  using trace-nonempty
  unfolding fri-evidence-layer-len-def
  by (cases trace-roots) simp-all
  have idx0:
    fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 =
    fri-query-idxs ! round-idx
  using trace-nonempty
  unfolding fri-evidence-layer-idx-def
  by (cases trace-roots) simp-all

```

show *False*
 using *missing auth unfolding len0 idx0* by contradiction
 qed

lemma *wp-trace-fri-header-tied-recorded-base-chunk-auth-missing-zero*:
 shows

wp-event verify-monad
 (*trace-fri-header-tied-recorded-base-chunk-auth-missing s*) *s* ≤ 0

proof –

have *wp-event verify-monad*
 (*trace-fri-header-tied-recorded-base-chunk-auth-missing s*) *s*
 $\leq$ *wp-event verify-monad* ($\lambda\cdot$. *False*) *s*
by (*rule wp-event-mono*)
 (*rule trace-fri-header-tied-recorded-base-chunk-auth-missing-false*)
also have ... = 0
unfolding *wp-event-def wp-def dist-expect-def* **by** *simp*
finally show *?thesis* .

qed

definition *trace-fri-header-tied-next-value-replay-checked-conflict*

:: (*f*, *a*) *protocol-channel-scheme* $\Rightarrow$
 (*unit list* $\times$ (*f*, *a*) *protocol-channel-scheme*) *option* $\Rightarrow$ *bool*

where

trace-fri-header-tied-next-value-replay-checked-conflict s out
 $\longleftrightarrow$
 ($\exists$ *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr trace-openings trace-table
round-idx round-idx' layer-idx v v'.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
fr = *hd trace-roots* $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ *trace-table-low-degree trace-table* $\wedge$
round-idx < *length fri-query-idxs* $\wedge$
round-idx' < *length fri-query-idxs* $\wedge$
layer-idx < *length trace-bs* $\wedge$
fri-evidence-next-idx trace-roots fri-query-idxs round-idx layer-idx =
fri-evidence-next-idx trace-roots fri-query-idxs round-idx'
layer-idx $\wedge$
generic-fri-round-forced-next-value trace-roots trace-bs
fri-query-idxs trace-round-layers round-idx layer-idx v $\wedge$
generic-fri-round-forced-next-value trace-roots trace-bs
fri-query-idxs trace-round-layers round-idx' layer-idx v' $\wedge$
v $\neq$ *v'* $\wedge$

$$v = v')$$

definition *trace-fri-header-tied-next-value-replay-gap*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-next-value-replay-gap } s \text{ out} \longleftrightarrow$
 $\text{trace-fri-header-tied-sampled-next-value-conflict } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-next-value-replay-checked-conflict } s \text{ out}$

lemma *trace-fri-header-tied-next-value-replay-checked-conflict-false:*

$\neg \text{trace-fri-header-tied-next-value-replay-checked-conflict } s \text{ out}$

unfolding *trace-fri-header-tied-next-value-replay-checked-conflict-def*
by *blast*

lemma *trace-fri-header-tied-sampled-next-value-imp-replay-gap:*

assumes *trace-fri-header-tied-sampled-next-value-conflict s out*

shows *trace-fri-header-tied-next-value-replay-gap s out*

using *assms trace-fri-header-tied-next-value-replay-checked-conflict-false*

unfolding *trace-fri-header-tied-next-value-replay-gap-def*

by *simp*

lemma *wp-trace-fri-header-tied-sampled-next-value-bound-from-replay-gap:*

fixes $G :: \text{prob}$

assumes *gap-bound:*

$\text{wp-event verify-monad}$

$(\text{trace-fri-header-tied-next-value-replay-gap } s) \leq G$

shows

$\text{wp-event verify-monad}$

$(\text{trace-fri-header-tied-sampled-next-value-conflict } s) \leq G$

proof –

have *wp-event verify-monad*

$(\text{trace-fri-header-tied-sampled-next-value-conflict } s) \leq$

$\text{wp-event verify-monad}$

$(\text{trace-fri-header-tied-next-value-replay-gap } s) \leq$

by $(\text{rule wp-event-mono})$

$(\text{rule trace-fri-header-tied-sampled-next-value-imp-replay-gap})$

then show *?thesis*

by $(\text{rule order-trans}[OF - \text{gap-bound}])$

qed

definition *trace-fri-header-tied-successor-replay-checked-conflict*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-successor-replay-checked-conflict } s \text{ out}$

$\longleftrightarrow$

$(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$

$\text{composition-bs composition-final fri-query-idxs trace-round-layers}$

composition-round-layers *fr* *trace-openings* *trace-table*
source-round *target-round* *layer-idx* *v* *xp* *xp-path* *xn* *xn-path*.
accepted-fri-opening-transcript *s* *out* *trace-roots* *trace-bs*
trace-final *fri-dg* *composition-roots* *composition-bs*
composition-final *fri-query-idxs* *trace-round-layers*
composition-round-layers $\wedge$
fr = *hd* *trace-roots* $\wedge$
accepted-with-partial-trace-openings *s* *out* *fr* *fri-query-idxs*
trace-openings $\wedge$
partial-trace-table-candidate *trace-table* *trace-openings* $\wedge$
 $\neg$ *trace-table-low-degree* *trace-table* $\wedge$
source-round < *length* *fri-query-idxs* $\wedge$
target-round < *length* *fri-query-idxs* $\wedge$
Suc *layer-idx* < *length* *trace-bs* $\wedge$
generic-fri-round-forced-next-value *trace-roots* *trace-bs*
fri-query-idxs *trace-round-layers* *source-round* *layer-idx* *v* $\wedge$
fri-layer-step-evidence
(*trace-roots* ! *Suc* *layer-idx*)
(*trace-bs* ! *Suc* *layer-idx*)
(*fri-evidence-layer-len* *trace-roots* (*Suc* *layer-idx*))
(*fri-evidence-layer-idx* *trace-roots* *fri-query-idxs* *target-round*
(*Suc* *layer-idx*))
($2 \wedge$ *Suc* *layer-idx*)
(*fri-sibling-index*
(*fri-evidence-layer-len* *trace-roots* (*Suc* *layer-idx*))
(*fri-evidence-layer-idx* *trace-roots* *fri-query-idxs*
target-round (*Suc* *layer-idx*)))
xp *xp-path* *xn* *xn-path*
(*fri-evidence-next-idx* *trace-roots* *fri-query-idxs* *target-round*
(*Suc* *layer-idx*))
(*fri-evidence-next-value* *trace-roots* *trace-bs* *fri-query-idxs*
target-round (*Suc* *layer-idx*) *xp* *xn*)
(*trace-round-layers* ! *target-round* ! *Suc* *layer-idx*) $\wedge$
((*fri-evidence-next-idx* *trace-roots* *fri-query-idxs* *source-round*
layer-idx =
fri-evidence-layer-idx *trace-roots* *fri-query-idxs* *target-round*
(*Suc* *layer-idx*) $\wedge$
v $\neq$ *xp* $\wedge$ *v* = *xp*) $\vee$
(*fri-evidence-next-idx* *trace-roots* *fri-query-idxs* *source-round*
layer-idx =
fri-sibling-index
(*fri-evidence-layer-len* *trace-roots* (*Suc* *layer-idx*))
(*fri-evidence-layer-idx* *trace-roots* *fri-query-idxs* *target-round*
(*Suc* *layer-idx*)) $\wedge$
v $\neq$ *xn* $\wedge$ *v* = *xn*)))

definition *trace-fri-header-tied-successor-replay-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-successor-replay-gap } s \text{ out} \longleftrightarrow$
 $\text{trace-fri-header-tied-sampled-successor-opening-conflict } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-successor-replay-checked-conflict } s \text{ out}$

lemma *trace-fri-header-tied-successor-replay-checked-conflict-false:*

$\neg \text{trace-fri-header-tied-successor-replay-checked-conflict } s \text{ out}$
unfolding *trace-fri-header-tied-successor-replay-checked-conflict-def*
by *blast*

lemma *trace-fri-header-tied-sampled-successor-imp-replay-gap:*

assumes *trace-fri-header-tied-sampled-successor-opening-conflict* *s out*
shows *trace-fri-header-tied-successor-replay-gap* *s out*
using *assms trace-fri-header-tied-successor-replay-checked-conflict-false*
unfolding *trace-fri-header-tied-successor-replay-gap-def*
by *simp*

lemma *wp-trace-fri-header-tied-sampled-successor-bound-from-replay-gap:*

fixes *G :: prob*
assumes *gap-bound:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s \leq G$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-successor-opening-conflict } s) \ s \leq G$

proof –

have *wp-event verify-monad*
 $(\text{trace-fri-header-tied-sampled-successor-opening-conflict } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s$
by (*rule wp-event-mono*)
 $(\text{rule trace-fri-header-tied-sampled-successor-imp-replay-gap})$
then show *?thesis*
by (*rule order-trans[OF - gap-bound]*)

qed

definition *trace-fri-header-tied-final-value-replay-checked-conflict*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-final-value-replay-checked-conflict } s \text{ out}$
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr trace-openings trace-table round-idx } v.$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$

$fr = hd\ trace\ roots \wedge$
 $accepted\ with\ partial\ trace\ openings\ s\ out\ fr\ fri\ query\ idxs$
 $trace\ openings \wedge$
 $partial\ trace\ table\ candidate\ trace\ table\ trace\ openings \wedge$
 $\neg trace\ table\ low\ degree\ trace\ table \wedge$
 $0 < length\ trace\ bs \wedge$
 $round\ idx < length\ fri\ query\ idxs \wedge$
 $generic\ fri\ round\ forced\ next\ value\ trace\ roots\ trace\ bs$
 $fri\ query\ idxs\ trace\ round\ layers\ round\ idx\ (length\ trace\ bs - 1)$
 $v \wedge$
 $v \neq trace\ final \wedge$
 $v = trace\ final$

definition *trace-fri-header-tied-final-value-replay-gap*

$:: (f, 'a)\ protocol\ channel\ scheme \Rightarrow$
 $(unit\ list \times (f, 'a)\ protocol\ channel\ scheme)\ option \Rightarrow bool$

where

$trace\ fri\ header\ tied\ final\ value\ replay\ gap\ s\ out \longleftrightarrow$
 $trace\ fri\ header\ tied\ sampled\ final\ value\ conflict\ s\ out \wedge$
 $\neg trace\ fri\ header\ tied\ final\ value\ replay\ checked\ conflict\ s\ out$

lemma *trace-fri-header-tied-sampled-final-value-conflictE:*

assumes *trace-fri-header-tied-sampled-final-value-conflict s out*

obtains *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table
round-idx v

where

$accepted\ fri\ opening\ transcript\ s\ out\ trace\ roots\ trace\ bs$
 $trace\ final\ fri\ dg\ composition\ roots\ composition\ bs\ composition\ final$
 $fri\ query\ idxs\ trace\ round\ layers\ composition\ round\ layers$
 $verifier\ header\ transcript\ s\ fr\ trace\ roots\ trace\ final\ as\ fri\ dg$
 $composition\ roots\ composition\ final\ rest$
 $accepted\ with\ partial\ trace\ openings\ s\ out\ fr\ fri\ query\ idxs$
 $trace\ openings$
 $partial\ trace\ table\ candidate\ trace\ table\ trace\ openings$
 $\neg trace\ table\ low\ degree\ trace\ table$
 $0 < length\ trace\ bs$
 $round\ idx < length\ fri\ query\ idxs$
 $generic\ fri\ round\ forced\ next\ value\ trace\ roots\ trace\ bs$
 $fri\ query\ idxs\ trace\ round\ layers\ round\ idx\ (length\ trace\ bs - 1)\ v$
 $v \neq trace\ final$

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as rest trace-openings
trace-table **where**
fri-openings:
 $accepted\ fri\ opening\ transcript\ s\ out\ trace\ roots\ trace\ bs$

trace-final fri-dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *header:*
verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *candidate:*
partial-trace-table-candidate trace-table trace-openings
and *nonlow:* $\neg$ *trace-table-low-degree trace-table*
and *final-conflict:*
generic-fri-sampled-final-value-conflict trace-roots trace-bs
trace-final fri-query-idxs trace-round-layers
unfolding *trace-fri-header-tied-sampled-final-value-conflict-def*
by *blast*
from *generic-fri-sampled-final-value-conflictE[OF final-conflict]*
obtain *round-idx v* **where**
nonempty: $0 < \text{length } \text{trace-bs}$
and *round-bound:* *round-idx < length fri-query-idxs*
and *forced:*
generic-fri-round-forced-next-value trace-roots trace-bs
fri-query-idxs trace-round-layers round-idx (length trace-bs - 1) v
and *neq:* $v \neq \text{trace-final}$
by *blast*
show *?thesis*
by (*rule that[OF fri-openings header partial candidate nonlow nonempty*
round-bound forced neq])
qed

lemma *trace-fri-header-tied-final-value-replay-checked-conflict-false:*
 $\neg$ *trace-fri-header-tied-final-value-replay-checked-conflict s out*
unfolding *trace-fri-header-tied-final-value-replay-checked-conflict-def*
by *blast*

lemma *trace-fri-header-tied-sampled-final-value-imp-replay-gap:*
assumes *trace-fri-header-tied-sampled-final-value-conflict s out*
shows *trace-fri-header-tied-final-value-replay-gap s out*
using *assms trace-fri-header-tied-final-value-replay-checked-conflict-false*
unfolding *trace-fri-header-tied-final-value-replay-gap-def*
by *simp*

lemma *wp-trace-fri-header-tied-sampled-final-value-bound-from-replay-gap:*
fixes $G :: \text{prob}$
assumes *gap-bound:*
wp-event verify-monad
(trace-fri-header-tied-final-value-replay-gap s) s $\leq$ G
shows
wp-event verify-monad

$(\text{trace-fri-header-tied-sampled-final-value-conflict } s) \ s \leq G$
proof –
have $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-final-value-conflict } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-replay-gap } s) \ s$
by $(\text{rule wp-event-mono})$
 $(\text{rule trace-fri-header-tied-sampled-final-value-imp-replay-gap})$
then show $?thesis$
by $(\text{rule order-trans}[OF - \text{gap-bound}])$
qed

lemma $\text{wp-trace-fri-header-tied-without-same-bound-from-replay-gaps}$:
fixes $B \ N \ S \ F :: \text{prob}$
assumes base-bound :
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-base-opening-conflict } s) \ s \leq B$
and next-gap-bound :
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq N$
and $\text{successor-gap-bound}$:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s \leq S$
and final-gap-bound :
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-replay-gap } s) \ s \leq F$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer } s)$
 $s \leq B + N + S + F$
by $(\text{rule wp-trace-fri-header-tied-without-same-bound-from-branches})$
 $(\text{rule base-bound},$
 $\text{rule wp-trace-fri-header-tied-sampled-next-value-bound-from-replay-gap}$
 $[\text{OF next-gap-bound}],$
 $\text{rule wp-trace-fri-header-tied-sampled-successor-bound-from-replay-gap}$
 $[\text{OF successor-gap-bound}],$
 $\text{rule wp-trace-fri-header-tied-sampled-final-value-bound-from-replay-gap}$
 $[\text{OF final-gap-bound}])$

lemma $\text{wp-trace-fri-header-tied-without-same-bound-from-value-alignment-and-replay-gaps}$:
fixes $M \ G \ N \ S \ F :: \text{prob}$
assumes mismatch-bound :
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-value-aligned-sibling-mismatch } s) \ s \leq M$
and $\text{alignment-gap-bound}$:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-value-alignment-gap } s) \ s \leq G$
and next-gap-bound :
 $\text{wp-event verify-monad}$

(trace-fri-header-tied-next-value-replay-gap s) $s \leq N$
and successor-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-successor-replay-gap s) $s \leq S$
and final-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-final-value-replay-gap s) $s \leq F$
shows
 wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict-without-same-layer s)
 $s \leq (M + G) + N + S + F$
by (rule wp-trace-fri-header-tied-without-same-bound-from-replay-gaps)
 (rule
 wp-trace-fri-header-tied-sampled-base-bound-from-sibling-mismatch-and-gap
 [OF mismatch-bound alignment-gap-bound],
 rule next-gap-bound,
 rule successor-gap-bound,
 rule final-gap-bound)

lemma wp-trace-fri-header-tied-without-same-bound-from-partial-merkle-structural-and-replay-gaps:

fixes $P H G N S F :: \text{prob}$

assumes partial-merkle-bound:

wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq P$

and structural-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-sibling-mismatch-structural-gap s) $s \leq H$

and alignment-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-base-value-alignment-gap s) $s \leq G$

and next-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-next-value-replay-gap s) $s \leq N$

and successor-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-successor-replay-gap s) $s \leq S$

and final-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-final-value-replay-gap s) $s \leq F$

shows

wp-event verify-monad

(trace-fri-header-tied-sampled-assignment-conflict-without-same-layer s)

$s \leq ((P + H) + G) + N + S + F$

by (rule

wp-trace-fri-header-tied-without-same-bound-from-value-alignment-and-replay-gaps)

(rule

wp-trace-fri-header-tied-value-aligned-sibling-mismatch-bound-from-partial-merkle-and-structural-gap

[OF partial-merkle-bound structural-gap-bound],

rule alignment-gap-bound,

rule next-gap-bound,

rule successor-gap-bound,
rule final-gap-bound)

lemma *wp-trace-fri-header-tied-authenticated-conflict-bound-from-partial-merkle-structural-and-replay-gaps:*

fixes $P\ H\ G\ N\ S\ F :: \text{prob}$

assumes *partial-merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P

and *structural-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-sibling-mismatch-structural-gap s) s ≤ H

and *alignment-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-base-value-alignment-gap s) s ≤ G

and *next-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-next-value-replay-gap s) s ≤ N

and *successor-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-successor-replay-gap s) s ≤ S

and *final-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-final-value-replay-gap s) s ≤ F

shows

wp-event verify-monad

(trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer s)

$s \leq (((P + H) + G) + N + S + F) + P$

proof –

have *without-bound:*

wp-event verify-monad

(trace-fri-header-tied-sampled-assignment-conflict-without-same-layer s)

$s \leq ((P + H) + G) + N + S + F$

by (rule

wp-trace-fri-header-tied-without-same-bound-from-partial-merkle-structural-and-replay-gaps)

(rule *partial-merkle-bound,*

rule *structural-gap-bound,*

rule *alignment-gap-bound,*

rule *next-gap-bound,*

rule *successor-gap-bound,*

rule *final-gap-bound)*

show *?thesis*

by (rule

wp-trace-fri-header-tied-authenticated-conflict-bound-from-without-same-and-merkle)

(rule *without-bound,* rule *partial-merkle-bound)*

qed

definition *trace-fri-header-tied-assignment-auth-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$

$(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-assignment-auth-gap } s \text{ out} \longleftrightarrow$
 $\text{trace-fri-header-tied-sampled-assignment-conflict } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer}$
 $s \text{ out}$

definition *trace-fri-header-tied-same-layer-auth-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-same-layer-auth-gap } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table}$
 $\text{result final-state.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-sampled-same-layer-opening-conflict trace-roots trace-bs}$
 $\text{fri-query-idxs trace-round-layers} \wedge$
 $\neg \text{generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks}$
 $\text{trace-roots trace-bs fri-query-idxs trace-round-layers final-state})$

definition *trace-fri-header-tied-same-layer-recorded-auth-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-same-layer-recorded-auth-gap } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table}$
 $\text{result final-state.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-sampled-same-layer-recorded-auth-gap trace-roots trace-bs}$

fri-query-idxs trace-round-layers final-state)

definition *trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table}$
 $\text{ result final-state.}$
 $\text{ out} = \text{Some} (\text{result}, \text{final-state}) \wedge$
 $\text{accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final dg composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript s fr trace-roots trace-final as dg}$
 $\text{ composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-sampled-same-layer-authenticated-recorded-auth-gap}$
 $\text{ trace-roots trace-bs fri-query-idxs trace-round-layers final-state})$

definition *trace-fri-header-tied-authenticated-slot-recorded-chunk-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table}$
 $\text{ result final-state.}$
 $\text{ out} = \text{Some} (\text{result}, \text{final-state}) \wedge$
 $\text{accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final dg composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript s fr trace-roots trace-final as dg}$
 $\text{ composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{generic-fri-authenticated-slot-recorded-chunk-gap trace-roots}$
 $\text{ fri-query-idxs trace-round-layers final-state})$

definition *trace-fri-header-tied-final-value-authenticated-replay-gap*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  trace-fri-header-tied-final-value-authenticated-replay-gap s out
 $\longleftrightarrow$ 
  ( $\exists$  trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
    result final-state round-idx v.
    out = Some (result, final-state)  $\wedge$ 
    accepted-fri-opening-transcript s out trace-roots trace-bs
      trace-final dg composition-roots composition-bs composition-final
      fri-query-idxs trace-round-layers composition-round-layers  $\wedge$ 
    verifier-header-transcript s fr trace-roots trace-final as dg
      composition-roots composition-final rest  $\wedge$ 
    accepted-with-partial-trace-openings s out fr fri-query-idxs
      trace-openings  $\wedge$ 
    partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
 $\neg$  trace-table-low-degree trace-table  $\wedge$ 
    0 < length trace-bs  $\wedge$ 
    round-idx < length fri-query-idxs  $\wedge$ 
    generic-fri-round-forced-next-value trace-roots trace-bs
      fri-query-idxs trace-round-layers round-idx (length trace-bs - 1)
      v  $\wedge$ 
    v  $\neq$  trace-final  $\wedge$ 
 $\neg$  partial-merkle-inconsistency-bad s out  $\wedge$ 
    generic-fri-recorded-layer-chunk-authenticated trace-roots
      fri-query-idxs trace-round-layers final-state round-idx
      (length trace-bs - 1))

lemma trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-false:
assumes
  trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s out
shows False
proof -
from assms obtain trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr as rest trace-openings trace-table
  result final-state
where out-eg: out = Some (result, final-state)
and fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
and gap:
  generic-fri-authenticated-slot-recorded-chunk-gap trace-roots
    fri-query-idxs trace-round-layers final-state
unfolding trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-def
by blast

```

```

from gap obtain round-idx layer-idx auth-chunk where
  round-bound: round-idx < length fri-query-idxs
  and layer-bound: layer-idx < length trace-roots
  and missing:
    ¬ generic-fri-recorded-layer-chunk-authenticated trace-roots
      fri-query-idxs trace-round-layers final-state round-idx layer-idx
  unfolding generic-fri-authenticated-slot-recorded-chunk-gap-def
  by blast
have round-bound-rounds: round-idx < rounds
  using round-bound accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
  by simp
have all-layer-bounds:
   $\bigwedge \text{raw } k. k < \text{length trace-roots} \implies$ 
    0 < fri-layer-lengths (length trace-roots) (clength * scale) ! k  $\wedge$ 
    fri-layer-indices (length trace-roots) (index (to-nat raw))
      (clength * scale) ! k <
    fri-layer-lengths (length trace-roots) (clength * scale) ! k
  by (rule accepted-fri-opening-transcript-trace-raw-layer-bound
    [OF fri-openings])
have recorded:
  generic-fri-recorded-layer-chunk-authenticated trace-roots
    fri-query-idxs trace-round-layers final-state round-idx layer-idx
  by (rule
    accepted-fri-opening-transcript-trace-recorded-layer-chunk-authenticated-at
    [OF fri-openings out-eq round-bound-rounds layer-bound
      all-layer-bounds])
show False
  using missing recorded by simp
qed

lemma wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero:
  shows
    wp-event verify-monad
      (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s) s
       $\leq (0::\text{prob})$ 
proof –
  have
    wp-event verify-monad
      (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s) s
       $\leq$  wp-event verify-monad ( $\lambda\cdot$ . False) s
  by (rule wp-event-mono)
    (rule trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-false)
  also have ... = 0
  unfolding wp-event-def wp-def dist-expect-def by simp
  finally show ?thesis by simp
qed

definition trace-fri-header-tied-final-value-selected-step-missing-gap
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 

```

$(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-header-tied-final-value-selected-step-missing-gap } s \text{ out}$
 $\longleftrightarrow$
 $\text{trace-fri-header-tied-final-value-authenticated-replay-gap } s \text{ out} \wedge$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\quad \text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\quad \text{composition-round-layers result final-state round-idx } v.$
 $\text{out} = \text{Some } (\text{result}, \text{final-state}) \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\quad \text{trace-final dg composition-roots composition-bs composition-final}$
 $\quad \text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $0 < \text{length trace-bs} \wedge$
 $\text{round-idx} < \text{length fri-query-idxs} \wedge$
 $\text{generic-fri-round-forced-next-value trace-roots trace-bs}$
 $\quad \text{fri-query-idxs trace-round-layers round-idx } (\text{length trace-bs} - 1)$
 $\quad v \wedge$
 $v \neq \text{trace-final} \wedge$
 $\neg \text{partial-merkle-inconsistency-bad } s \text{ out} \wedge$
 $\text{generic-fri-recorded-layer-chunk-authenticated trace-roots}$
 $\quad \text{fri-query-idxs trace-round-layers final-state round-idx}$
 $\quad (\text{length trace-bs} - 1) \wedge$
 $(\forall \text{ actual-chunk axp axp-path axn axn-path.}$
 $\quad \neg (\text{fri-layer-chunk-authenticated}$
 $\quad \quad (\text{trace-roots ! } (\text{length trace-bs} - 1))$
 $\quad \quad (\text{fri-evidence-layer-len trace-roots } (\text{length trace-bs} - 1))$
 $\quad \quad (\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx}$
 $\quad \quad \quad (\text{length trace-bs} - 1))$
 $\quad \quad \text{actual-chunk final-state} \wedge$
 $\quad \text{fri-layer-step-evidence}$
 $\quad \quad (\text{trace-roots ! } (\text{length trace-bs} - 1))$
 $\quad \quad (\text{trace-bs ! } (\text{length trace-bs} - 1))$
 $\quad \quad (\text{fri-evidence-layer-len trace-roots } (\text{length trace-bs} - 1))$
 $\quad \quad (\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx}$
 $\quad \quad \quad (\text{length trace-bs} - 1))$
 $\quad \quad (2 \wedge (\text{length trace-bs} - 1))$
 $\quad \quad (\text{fri-sibling-index}$
 $\quad \quad \quad (\text{fri-evidence-layer-len trace-roots } (\text{length trace-bs} - 1))$
 $\quad \quad \quad (\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx}$
 $\quad \quad \quad \quad (\text{length trace-bs} - 1)))$
 $\quad \text{axp axp-path axn axn-path}$
 $\quad (\text{fri-evidence-next-idx trace-roots fri-query-idxs round-idx}$
 $\quad \quad (\text{length trace-bs} - 1))$
 $\quad \text{trace-final actual-chunk})))$

lemma $\text{trace-fri-header-tied-final-value-selected-step-missing-gap-false:}$
assumes gap:
 $\text{trace-fri-header-tied-final-value-selected-step-missing-gap } s \text{ out}$
shows False

proof –

from *gap* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers result final-state round-idx v
where *out-eq*: *out* = *Some (result, final-state)*
and *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *nonempty-bs*: $0 < \text{length } \text{trace-bs}$
and *round-bound*: $\text{round-idx} < \text{length } \text{fri-query-idxs}$
and *missing*:
 $\bigwedge \text{actual-chunk } \text{axp } \text{axp-path } \text{axn } \text{axn-path}.$
 $\neg (\text{fri-layer-chunk-authenticated}$
 $(\text{trace-roots } ! (\text{length } \text{trace-bs} - 1))$
 $(\text{fri-evidence-layer-len } \text{trace-roots } (\text{length } \text{trace-bs} - 1))$
 $(\text{fri-evidence-layer-idx } \text{trace-roots } \text{fri-query-idxs } \text{round-idx}$
 $(\text{length } \text{trace-bs} - 1))$
 $\text{actual-chunk } \text{final-state} \wedge$
 $\text{fri-layer-step-evidence}$
 $(\text{trace-roots } ! (\text{length } \text{trace-bs} - 1))$
 $(\text{trace-bs } ! (\text{length } \text{trace-bs} - 1))$
 $(\text{fri-evidence-layer-len } \text{trace-roots } (\text{length } \text{trace-bs} - 1))$
 $(\text{fri-evidence-layer-idx } \text{trace-roots } \text{fri-query-idxs } \text{round-idx}$
 $(\text{length } \text{trace-bs} - 1))$
 $(2 \wedge (\text{length } \text{trace-bs} - 1))$
 $(\text{fri-sibling-index}$
 $(\text{fri-evidence-layer-len } \text{trace-roots } (\text{length } \text{trace-bs} - 1))$
 $(\text{fri-evidence-layer-idx } \text{trace-roots } \text{fri-query-idxs } \text{round-idx}$
 $(\text{length } \text{trace-bs} - 1)))$
 $\text{axp } \text{axp-path } \text{axn } \text{axn-path}$
 $(\text{fri-evidence-next-idx } \text{trace-roots } \text{fri-query-idxs } \text{round-idx}$
 $(\text{length } \text{trace-bs} - 1))$
 $\text{trace-final } \text{actual-chunk})$
unfolding *trace-fri-header-tied-final-value-selected-step-missing-gap-def*
by *blast*
have *len-bs-roots*: $\text{length } \text{trace-bs} = \text{length } \text{trace-roots}$
using *accepted-fri-opening-transcript-shapes(2)[OF fri-openings]* .
have *nonempty-roots*: $0 < \text{length } \text{trace-roots}$
using *nonempty-bs len-bs-roots* **by** *simp*
have *round-bound-rounds*: $\text{round-idx} < \text{rounds}$
using *round-bound*
accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
by *simp*
have *raw-layer*:
 $\bigwedge_{\text{raw}}.$
 $0 < \text{fri-layer-lengths } (\text{length } \text{trace-roots}) (\text{clength } * \text{scale}) !$
 $(\text{length } \text{trace-roots} - 1) \wedge$
 $\text{fri-layer-indices } (\text{length } \text{trace-roots}) (\text{index } (\text{to-nat } \text{raw}))$


```

    (clength * scale) ! (length trace-roots - 1) <
    fri-layer-lengths (length trace-roots) (clength * scale) !
    (length trace-roots - 1)
  by (rule accepted-fri-opening-transcript-trace-raw-layer-bound
      [OF fri-openings])
    (use nonempty-roots in simp)
  from accepted-fri-opening-transcript-trace-final-step-evidence-at
    [OF fri-openings out-eq round-bound-rounds nonempty-roots raw-layer]
  obtain actual-chunk arp arp-path arn arn-path where
  auth:
    fri-layer-chunk-authenticated
    (trace-roots ! (length trace-roots - 1))
    (fri-evidence-layer-len trace-roots (length trace-roots - 1))
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
      (length trace-roots - 1))
    actual-chunk final-state
  and step:
    fri-layer-step-evidence
    (trace-roots ! (length trace-roots - 1))
    (trace-bs ! (length trace-roots - 1))
    (fri-evidence-layer-len trace-roots (length trace-roots - 1))
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
      (length trace-roots - 1))
    (2 ^ (length trace-roots - 1))
    (fri-sibling-index
      (fri-evidence-layer-len trace-roots (length trace-roots - 1))
      (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
        (length trace-roots - 1)))
    arp arp-path arn arn-path
    (fri-evidence-next-idx trace-roots fri-query-idxs round-idx
      (length trace-roots - 1))
    trace-final actual-chunk
  by blast
  show False
  using missing[of actual-chunk arp arp-path arn arn-path]
    auth step len-bs-roots
  by simp
qed

```

lemma *wp-trace-fri-header-tied-final-value-selected-step-missing-gap-zero:*

shows

wp-event verify-monad
 (trace-fri-header-tied-final-value-selected-step-missing-gap s) s
 ≤ (0::prob)

proof –

have

wp-event verify-monad
 (trace-fri-header-tied-final-value-selected-step-missing-gap s) s ≤
wp-event verify-monad (λ-. False) s

```

    by (rule wp-event-mono)
      (rule trace-fri-header-tied-final-value-selected-step-missing-gap-false)
  also have ... = 0
    unfolding wp-event-def wp-def dist-expect-def by simp
  finally show ?thesis by simp
qed

lemma trace-fri-header-tied-final-value-authenticated-replay-gap-imp-selected-step-missing:
  assumes
    trace-fri-header-tied-final-value-authenticated-replay-gap s out
  shows
    trace-fri-header-tied-final-value-selected-step-missing-gap s out
proof -
  from assms obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
    result final-state round-idx v
  where out-eq: out = Some (result, final-state)
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and nonempty: 0 < length trace-bs
  and round-bound: round-idx < length fri-query-idxs
  and forced:
    generic-fri-round-forced-next-value trace-roots trace-bs
    fri-query-idxs trace-round-layers round-idx (length trace-bs - 1) v
  and neq: v ≠ trace-final
  and no-merkle: ¬ partial-merkle-inconsistency-bad s out
  and recorded-auth:
    generic-fri-recorded-layer-chunk-authenticated trace-roots
    fri-query-idxs trace-round-layers final-state round-idx
    (length trace-bs - 1)
  unfolding trace-fri-header-tied-final-value-authenticated-replay-gap-def
  by blast
have no-step:
  ∧ actual-chunk axp axp-path axn axn-path.
  ¬ (fri-layer-chunk-authenticated
    (trace-roots ! (length trace-bs - 1))
    (fri-evidence-layer-len trace-roots (length trace-bs - 1))
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
      (length trace-bs - 1))
    actual-chunk final-state ∧
    fri-layer-step-evidence
    (trace-roots ! (length trace-bs - 1))
    (trace-bs ! (length trace-bs - 1))
    (fri-evidence-layer-len trace-roots (length trace-bs - 1))
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
      (length trace-bs - 1)))

```

```

    (2 ^ (length trace-bs - 1))
    (fri-sibling-index
      (fri-evidence-layer-len trace-roots (length trace-bs - 1))
      (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
        (length trace-bs - 1)))
    arp arp-path arn arn-path
    (fri-evidence-next-idx trace-roots fri-query-idxs round-idx
      (length trace-bs - 1))
    trace-final actual-chunk)
proof
  fix actual-chunk arp arp-path arn arn-path
  assume step-auth:
    fri-layer-chunk-authenticated
      (trace-roots ! (length trace-bs - 1))
      (fri-evidence-layer-len trace-roots (length trace-bs - 1))
      (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
        (length trace-bs - 1))
    actual-chunk final-state ∧
    fri-layer-step-evidence
      (trace-roots ! (length trace-bs - 1))
      (trace-bs ! (length trace-bs - 1))
      (fri-evidence-layer-len trace-roots (length trace-bs - 1))
      (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
        (length trace-bs - 1))
      (2 ^ (length trace-bs - 1))
      (fri-sibling-index
        (fri-evidence-layer-len trace-roots (length trace-bs - 1))
        (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
          (length trace-bs - 1)))
      arp arp-path arn arn-path
      (fri-evidence-next-idx trace-roots fri-query-idxs round-idx
        (length trace-bs - 1))
      trace-final actual-chunk
  have False
    by (rule generic-fri-final-conflict-false-from-authenticated-step
      [OF forced-recorded-auth conjunct1[OF step-auth]
        conjunct2[OF step-auth] no-merkle[unfolded out-eq] neq])
  then show False .
qed
have witness:
  ∃ trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers result final-state round-idx v.
  out = Some (result, final-state) ∧
  accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers ∧
  0 < length trace-bs ∧
  round-idx < length fri-query-idxs ∧

```

```

generic-fri-round-forced-next-value trace-roots trace-bs
  fri-query-idxs trace-round-layers round-idx (length trace-bs - 1)
  v ∧
  v ≠ trace-final ∧
  ¬ partial-merkle-inconsistency-bad s out ∧
generic-fri-recorded-layer-chunk-authenticated trace-roots
  fri-query-idxs trace-round-layers final-state round-idx
  (length trace-bs - 1) ∧
(∀ actual-chunk axp axp-path axn axn-path.
  ¬ (fri-layer-chunk-authenticated
    (trace-roots ! (length trace-bs - 1))
    (fri-evidence-layer-len trace-roots (length trace-bs - 1))
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
      (length trace-bs - 1))
    actual-chunk final-state ∧
    fri-layer-step-evidence
      (trace-roots ! (length trace-bs - 1))
      (trace-bs ! (length trace-bs - 1))
      (fri-evidence-layer-len trace-roots (length trace-bs - 1))
      (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
        (length trace-bs - 1))
      (2 ^ (length trace-bs - 1))
      (fri-sibling-index
        (fri-evidence-layer-len trace-roots (length trace-bs - 1))
        (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx
          (length trace-bs - 1)))
      axp axp-path axn axn-path
      (fri-evidence-next-idx trace-roots fri-query-idxs round-idx
        (length trace-bs - 1))
      trace-final actual-chunk))
using out-eq fri-openings nonempty round-bound forced neq no-merkle
  recorded-auth no-step
by blast
show ?thesis
unfolding trace-fri-header-tied-final-value-selected-step-missing-gap-def
using assms witness by blast
qed

```

```

lemma wp-trace-fri-header-tied-final-value-authenticated-replay-gap-bound-from-selected-step-missing:
  fixes G :: prob
  assumes gap-bound:
    wp-event verify-monad
      (trace-fri-header-tied-final-value-selected-step-missing-gap s) s
    ≤ G
  shows
    wp-event verify-monad
      (trace-fri-header-tied-final-value-authenticated-replay-gap s) s
    ≤ G
proof -

```

```

have wp-event verify-monad
  (trace-fri-header-tied-final-value-authenticated-replay-gap s) s ≤
  wp-event verify-monad
  (trace-fri-header-tied-final-value-selected-step-missing-gap s) s
by (rule wp-event-mono)
  (rule
    trace-fri-header-tied-final-value-authenticated-replay-gap-imp-selected-step-missing)
then show ?thesis
  by (rule order-trans[OF - gap-bound])
qed

lemma trace-fri-header-tied-final-value-replay-gap-imp-merkle-or-slot-or-authenticated:
  assumes trace-fri-header-tied-final-value-replay-gap s out
  shows
    partial-merkle-inconsistency-bad s out ∨
    trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s out ∨
    trace-fri-header-tied-final-value-authenticated-replay-gap s out
proof –
  from assms have sampled:
    trace-fri-header-tied-sampled-final-value-conflict s out
  unfolding trace-fri-header-tied-final-value-replay-gap-def by simp
  from trace-fri-header-tied-sampled-final-value-conflictE[OF sampled]
  obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
    round-idx v
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  and cand: partial-trace-table-candidate trace-table trace-openings
  and not-low: ¬ trace-table-low-degree trace-table
  and nonempty: 0 < length trace-bs
  and round-bound: round-idx < length fri-query-idxs
  and forced:
    generic-fri-round-forced-next-value trace-roots trace-bs
    fri-query-idxs trace-round-layers round-idx (length trace-bs - 1) v
  and neq: v ≠ trace-final
  by blast
  from fri-openings obtain result final-state where out-eq:
    out = Some (result, final-state)
  unfolding accepted-fri-opening-transcript-def by blast
  show ?thesis

```

```

proof (cases partial-merkle-inconsistency-bad s out)
  case True
  then show ?thesis by simp
next
  case no-merkle: False
  let ?j = length trace-bs - 1
  have len-trace-bs: length trace-bs = length trace-roots
    using accepted-fri-opening-transcript-shapes(2)[OF fri-openings] .
  have layer-bound: ?j < length trace-roots
    using nonempty len-trace-bs by simp
  show ?thesis
proof (cases generic-fri-recorded-layer-chunk-authenticated trace-roots
  fri-query-idxs trace-round-layers final-state round-idx ?j)
  case True
  have authenticated-gap:
    trace-fri-header-tied-final-value-authenticated-replay-gap s out
  unfolding
    trace-fri-header-tied-final-value-authenticated-replay-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule header, rule partial,
     rule cand, rule not-low, rule nonempty, rule round-bound,
     rule forced, rule neg, use no-merkle True in simp-all)
  then show ?thesis by simp
next
  case missing-recorded: False
  have round-bound-rounds: round-idx < rounds
    using round-bound
      accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
    by simp
  have raw-layer:
     $\bigwedge_{\text{raw.}}$ 
    0 < fri-layer-lengths (length trace-roots) (clength * scale) !
    ?j  $\wedge$ 
    fri-layer-indices (length trace-roots) (index (to-nat raw))
    (clength * scale) ! ?j <
    fri-layer-lengths (length trace-roots) (clength * scale) ! ?j
  by (rule accepted-fri-opening-transcript-trace-raw-layer-bound
    [OF fri-openings layer-bound])
  from accepted-fri-opening-transcript-trace-selected-chunk-authenticated-at
    [OF fri-openings out-eq round-bound-rounds layer-bound raw-layer]
  obtain auth-chunk where auth:
    fri-layer-chunk-authenticated (trace-roots ! ?j)
    (fri-layer-lengths (length trace-roots) (clength * scale) ! ?j)
    (fri-layer-indices (length trace-roots) (fri-query-idxs ! round-idx)
    (clength * scale) ! ?j)
    auth-chunk final-state
  by blast
  have step:
    fri-round-layer-evidence trace-roots trace-bs fri-query-idxs

```

```

    round-idx ?j trace-round-layers
  by (rule accepted-fri-opening-transcript-trace-round-layer-evidence
      [OF fri-openings round-bound-rounds layer-bound])
have generic-step:
  generic-fri-transcript-step-with-authenticated-chunk trace-roots
  trace-bs fri-query-idxs trace-round-layers final-state
  round-idx ?j
  by (rule generic-fri-transcript-step-with-authenticated-chunkI
      [OF step])
    (use auth in
      ⟨simp add: fri-evidence-layer-len-def
        fri-evidence-layer-idx-def⟩)
have slot-gap:
  generic-fri-authenticated-slot-recorded-chunk-gap trace-roots
  fri-query-idxs trace-round-layers final-state
  by (rule
      generic-fri-transcript-step-auth-missing-imp-authenticated-slot-recorded-chunk-gap
      [OF generic-step missing-recorded])
have route-slot-gap:
  trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s out
  unfolding trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule header, rule partial,
      rule cand, rule not-low, rule slot-gap)
then show ?thesis by simp
qed
qed
qed

lemma wp-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-authenticated:
  fixes M A R :: prob
  assumes merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M
  and slot-bound:
    wp-event verify-monad
      (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s) s ≤ A
  and replay-bound:
    wp-event verify-monad
      (trace-fri-header-tied-final-value-authenticated-replay-gap s) s ≤ R
  shows
    wp-event verify-monad
      (trace-fri-header-tied-final-value-replay-gap s) s ≤ M + A + R
proof -
  let ?P = partial-merkle-inconsistency-bad s
  let ?Q = trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s
  let ?R =
    trace-fri-header-tied-final-value-authenticated-replay-gap s
  have event-le:
    wp-event verify-monad

```

```

    (trace-fri-header-tied-final-value-replay-gap s) s ≤
    wp-event verify-monad (λout. ?P out ∨ ?Q out ∨ ?R out) s
  by (rule wp-event-mono)
    (rule
      trace-fri-header-tied-final-value-replay-gap-imp-merkle-or-slot-or-authenticated)
  have union-le:
    wp-event verify-monad (λout. ?P out ∨ ?Q out ∨ ?R out) s ≤
    wp-event verify-monad ?P s +
    wp-event verify-monad ?Q s +
    wp-event verify-monad ?R s
  proof -
    have wp-event verify-monad (λout. ?P out ∨ ?Q out ∨ ?R out) s ≤
      wp-event verify-monad ?P s +
      wp-event verify-monad (λout. ?Q out ∨ ?R out) s
    by (rule wp-event-union-bound)
    also have ... ≤
      wp-event verify-monad ?P s +
      (wp-event verify-monad ?Q s + wp-event verify-monad ?R s)
    by (intro add-mono order.refl wp-event-union-bound)
    finally show ?thesis
    by (simp add: algebra-simps)
  qed
  have bounds:
    wp-event verify-monad ?P s +
    wp-event verify-monad ?Q s +
    wp-event verify-monad ?R s ≤ M + A + R
  using merkle-bound slot-bound replay-bound
  by (intro add-mono)
  show ?thesis
  by (rule order-trans[OF event-le order-trans[OF union-le bounds]])
qed

lemma wp-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step:
  fixes M A R :: prob
  assumes merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M
  and slot-bound:
    wp-event verify-monad
      (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s) s ≤ A
  and selected-step-bound:
    wp-event verify-monad
      (trace-fri-header-tied-final-value-selected-step-missing-gap s) s
      ≤ R
  shows
    wp-event verify-monad
      (trace-fri-header-tied-final-value-replay-gap s) s ≤ M + A + R
  proof -
    have authenticated-bound:
      wp-event verify-monad

```


$(\text{trace-fri-header-tied-final-value-authenticated-replay-gap } s) \ s$
 $\leq R$
by (rule
 $\text{wp-trace-fri-header-tied-final-value-authenticated-replay-gap-bound-from-selected-step-missing}$
 $[OF \ \text{selected-step-bound}]$)
show ?thesis
by (rule
 $\text{wp-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-authenticated}$
 $(\text{rule merkle-bound, rule slot-bound, rule authenticated-bound})$)
qed

definition *trace-fri-header-tied-base-head-value-mismatch-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-base-head-value-mismatch-gap } s \ \text{out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table}$
 $\text{round-idx xp xp-path xn xn-path result final-state.}$
 $\text{out} = \text{Some (result, final-state)} \wedge$
 $\text{accepted-fri-opening-transcript } s \ \text{out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \ \text{fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \ \text{out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{trace-table-low-degree trace-table} \wedge$
 $\text{round-idx} < \text{length fri-query-idxs} \wedge$
 $0 < \text{length trace-bs} \wedge$
 $\text{fri-layer-step-evidence}$
 $(\text{trace-roots} ! 0)$
 $(\text{trace-bs} ! 0)$
 $(\text{fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0)$
 1
 $(\text{fri-sibling-index (fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0))$
 $\text{xp xp-path xn xn-path}$
 $(\text{fri-evidence-next-idx trace-roots fri-query-idxs round-idx } 0)$
 $(\text{fri-evidence-next-value trace-roots trace-bs fri-query-idxs}$
 $\text{round-idx } 0 \ \text{xp xn})$
 $(\text{trace-round-layers} ! \text{round-idx} ! 0) \wedge$
 $\neg \text{fri-opening-matches-table}$
 $(\text{fri-evidence-layer-len trace-roots } 0)$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs round-idx } 0)$
 $\text{trace-table xp xn} \wedge$

$\neg$ (trace-openings ! round-idx $\neq$ [] $\wedge$
 opening-index (hd (trace-openings ! round-idx)) =
 fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 $\wedge$
 opening-value (hd (trace-openings ! round-idx)) = xp))

lemma trace-fri-header-tied-base-value-alignment-gap-imp-head-value-mismatch-gap:

assumes trace-fri-header-tied-base-value-alignment-gap s out

shows trace-fri-header-tied-base-head-value-mismatch-gap s out

proof –

from assms **have** sampled:

trace-fri-header-tied-sampled-base-opening-conflict s out

and no-aligned:

$\neg$ trace-fri-header-tied-value-aligned-base-opening-conflict s out

unfolding trace-fri-header-tied-base-value-alignment-gap-def **by** simp-all

from sampled **obtain** trace-roots trace-bs trace-final dg composition-roots

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as rest trace-openings trace-table

where fri-openings:

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final dg composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers

and header:

verifier-header-transcript s fr trace-roots trace-final as dg

composition-roots composition-final rest

and partial:

accepted-with-partial-trace-openings s out fr fri-query-idxs

trace-openings

and cand: partial-trace-table-candidate trace-table trace-openings

and not-low: $\neg$ trace-table-low-degree trace-table

and generic:

generic-fri-sampled-base-opening-conflict trace-table trace-roots

trace-bs fri-query-idxs trace-round-layers

unfolding trace-fri-header-tied-sampled-base-opening-conflict-def

by blast

from fri-openings **obtain** result final-state **where** out-eq:

out = Some (result, final-state)

unfolding accepted-fri-opening-transcript-def **by** blast

from generic **obtain** round-idx xp xp-path xn xn-path **where**

trace-bs-nonempty: 0 < length trace-bs

and round-bound: round-idx < length fri-query-idxs

and step:

fri-layer-step-evidence

(trace-roots ! 0)

(trace-bs ! 0)

(fri-evidence-layer-len trace-roots 0)

(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)

1

(fri-sibling-index (fri-evidence-layer-len trace-roots 0)

(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))

```

    xp xp-path xn xn-path
    (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
    (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
      round-idx 0 xp xn)
    (trace-round-layers ! round-idx ! 0)
  and no-match:
    ¬ fri-opening-matches-table
      (fri-evidence-layer-len trace-roots 0)
      (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
      trace-table xp xn
  unfolding generic-fri-sampled-base-opening-conflict-def
  by blast
have no-head:
  ¬ (trace-openings ! round-idx ≠ [] ∧
    opening-index (hd (trace-openings ! round-idx)) =
      fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 ∧
    opening-value (hd (trace-openings ! round-idx)) = xp)
proof
  assume head:
    trace-openings ! round-idx ≠ [] ∧
    opening-index (hd (trace-openings ! round-idx)) =
      fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 ∧
    opening-value (hd (trace-openings ! round-idx)) = xp
  have aligned:
    trace-fri-header-tied-value-aligned-base-opening-conflict s out
  unfolding trace-fri-header-tied-value-aligned-base-opening-conflict-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
      rule not-low, rule round-bound, rule trace-bs-nonempty,
      use head step no-match in simp-all)
  then show False
    using no-aligned by contradiction
qed
show ?thesis
  unfolding trace-fri-header-tied-base-head-value-mismatch-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
      rule not-low, rule round-bound, rule trace-bs-nonempty, rule step,
      rule no-match, rule no-head)
qed

lemma wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap:
  fixes G :: prob
  assumes mismatch-bound:
    wp-event verify-monad
      (trace-fri-header-tied-base-head-value-mismatch-gap s) s ≤ G
  shows
    wp-event verify-monad
      (trace-fri-header-tied-base-value-alignment-gap s) s ≤ G

```

proof –
have *wp-event verify-monad*
 (trace-fri-header-tied-base-value-alignment-gap *s*) *s* ≤
wp-event verify-monad
 (trace-fri-header-tied-base-head-value-mismatch-gap *s*) *s*
by (rule *wp-event-mono*)
 (rule
trace-fri-header-tied-base-value-alignment-gap-imp-head-value-mismatch-gap)
then show ?thesis
by (rule *order-trans*[*OF* - *mismatch-bound*])
qed

definition *trace-fri-header-tied-base-head-value-neq-gap*
 :: ('f, 'a) protocol-channel-scheme ⇒
 (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool

where
trace-fri-header-tied-base-head-value-neq-gap s out ⇔
 (∃ *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table
round-idx xp xp-path xn xn-path result final-state.
out = Some (*result, final-state*) ∧
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers ∧
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest ∧
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings ∧
partial-trace-table-candidate trace-table trace-openings ∧
 ¬ *trace-table-low-degree trace-table* ∧
round-idx < *length fri-query-idxs* ∧
 0 < *length trace-bs* ∧
trace-openings ! *round-idx* ≠ [] ∧
opening-index (*hd (trace-openings ! round-idx)*) =
fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 ∧
opening-value (*hd (trace-openings ! round-idx)*) ≠ *xp* ∧
fri-layer-step-evidence
 (*trace-roots* ! 0)
 (*trace-bs* ! 0)
 (*fri-evidence-layer-len trace-roots 0*)
 (*fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0*)
 1
 (*fri-sibling-index (fri-evidence-layer-len trace-roots 0)*
 (*fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0*))
xp xp-path xn xn-path
 (*fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0*)
 (*fri-evidence-next-value trace-roots trace-bs fri-query-idxs*
round-idx 0 xp xn)

```

    (trace-round-layers ! round-idx ! 0) ∧
  ¬ fri-opening-matches-table
  (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
  trace-table xp xn)

```

lemma *trace-fri-header-tied-base-head-value-mismatch-gap-imp-neq-gap:*

assumes *trace-fri-header-tied-base-head-value-mismatch-gap s out*

shows *trace-fri-header-tied-base-head-value-neq-gap s out*

proof —

from *assms show ?thesis*

unfolding *trace-fri-header-tied-base-head-value-mismatch-gap-def*

proof (*elim exE conjE*)

fix *trace-roots trace-bs trace-final dg composition-roots composition-bs*

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as rest trace-openings trace-table

round-idx xp xp-path xn xn-path result final-state

assume *out-eq: out = Some (result, final-state)*

assume *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final dg composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers

assume *header:*

verifier-header-transcript s fr trace-roots trace-final as dg

composition-roots composition-final rest

assume *partial:*

accepted-with-partial-trace-openings s out fr fri-query-idxs

trace-openings

assume *cand: partial-trace-table-candidate trace-table trace-openings*

assume *not-low: ¬ trace-table-low-degree trace-table*

assume *round-bound: round-idx < length fri-query-idxs*

assume *trace-bs-nonempty: 0 < length trace-bs*

assume *step:*

fri-layer-step-evidence

(trace-roots ! 0)

(trace-bs ! 0)

(fri-evidence-layer-len trace-roots 0)

(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)

1

(fri-sibling-index (fri-evidence-layer-len trace-roots 0)

(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))

xp xp-path xn xn-path

(fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)

(fri-evidence-next-value trace-roots trace-bs fri-query-idxs

round-idx 0 xp xn)

(trace-round-layers ! round-idx ! 0)

assume *no-match:*

¬ fri-opening-matches-table

(fri-evidence-layer-len trace-roots 0)

```

    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    trace-table xp xn
assume no-head:
  ¬ (trace-openings ! round-idx ≠ [] ∧
    opening-index (hd (trace-openings ! round-idx)) =
      fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 ∧
    opening-value (hd (trace-openings ! round-idx)) = xp)
have round-bound-rounds: round-idx < rounds
  using round-bound
    accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
  by simp
have roots-nonempty: 0 < length trace-roots
  using accepted-fri-opening-transcript-shapes(2)[OF fri-openings]
    trace-bs-nonempty by simp
have idxs:
  map opening-index (trace-openings ! round-idx) =
    powers-scaled (fri-query-idxs ! round-idx)
  by (rule accepted-with-partial-trace-openings-shapes(3)
    [OF partial round-bound-rounds])
have openings-nonempty: trace-openings ! round-idx ≠ []
  using idxs powers-pos unfolding powers-scaled-def by auto
have hd-idx:
  opening-index (hd (trace-openings ! round-idx)) =
    fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0
proof –
  have opening-index (hd (trace-openings ! round-idx)) =
    fri-query-idxs ! round-idx
  proof –
    have opening-index (hd (trace-openings ! round-idx)) =
      hd (map opening-index (trace-openings ! round-idx))
    using openings-nonempty by (cases trace-openings ! round-idx) auto
    also have ... = hd (powers-scaled (fri-query-idxs ! round-idx))
    using idxs by simp
    also have ... = fri-query-idxs ! round-idx
    by (rule powers-scaled-hd)
    finally show ?thesis .
qed
moreover have
  fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 =
    fri-query-idxs ! round-idx
  using roots-nonempty
  unfolding fri-evidence-layer-idx-def
  by (cases trace-roots) simp-all
  ultimately show ?thesis by simp
qed
have hd-neq:
  opening-value (hd (trace-openings ! round-idx)) ≠ xp
  using no-head openings-nonempty hd-idx by blast
show trace-fri-header-tied-base-head-value-neq-gap s out

```

```

unfolding trace-fri-header-tied-base-head-value-neq-gap-def
by (intro exI conjI)
  (rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
   rule not-low, rule round-bound, rule trace-bs-nonempty,
   rule openings-nonempty, rule hd-idx, rule hd-neq, rule step,
   rule no-match)
qed
qed

lemma wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap:
fixes  $G :: prob$ 
assumes neq-bound:
  wp-event verify-monad
  (trace-fri-header-tied-base-head-value-neq-gap  $s$ )  $s \leq G$ 
shows
  wp-event verify-monad
  (trace-fri-header-tied-base-head-value-mismatch-gap  $s$ )  $s \leq G$ 
proof -
have wp-event verify-monad
  (trace-fri-header-tied-base-head-value-mismatch-gap  $s$ )  $s \leq$ 
  wp-event verify-monad
  (trace-fri-header-tied-base-head-value-neq-gap  $s$ )  $s$ 
by (rule wp-event-mono)
  (rule trace-fri-header-tied-base-head-value-mismatch-gap-imp-neq-gap)
then show ?thesis
by (rule order-trans[OF - neq-bound])
qed

lemma trace-fri-header-tied-assignment-imp-authenticated-or-auth-gap:
assumes trace-fri-header-tied-sampled-assignment-conflict  $s$  out
shows
  trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
   $s$  out  $\vee$ 
  trace-fri-header-tied-assignment-auth-gap  $s$  out
using assms
unfolding trace-fri-header-tied-assignment-auth-gap-def
by blast

lemma trace-fri-header-tied-assignment-auth-gap-imp-same-layer-auth-gap:
assumes trace-fri-header-tied-assignment-auth-gap  $s$  out
shows trace-fri-header-tied-same-layer-auth-gap  $s$  out
proof -
from assms have conflict:
  trace-fri-header-tied-sampled-assignment-conflict  $s$  out
and no-auth:
 $\neg$  trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
 $s$  out
unfolding trace-fri-header-tied-assignment-auth-gap-def
by simp-all

```

from *conflict* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table
where *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *header*:
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
and *partial*:
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *cand*: *partial-trace-table-candidate trace-table trace-openings*
and *not-low*: $\neg$ *trace-table-low-degree trace-table*
and *sampled*:
generic-fri-sampled-assignment-conflict trace-table trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers
unfolding *trace-fri-header-tied-sampled-assignment-conflict-def*
by *blast*
from *fri-openings* **obtain** *result final-state* **where** *out-eq*:
out = Some (result, final-state)
unfolding *accepted-fri-opening-transcript-def* **by** *blast*
have *split*:
generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers final-state $\vee$
(generic-fri-sampled-same-layer-opening-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers $\wedge$
 $\neg$ *generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks*
trace-roots trace-bs fri-query-idxs trace-round-layers final-state)
by (rule
generic-fri-sampled-assignment-conflict-imp-authenticated-or-same-layer-auth-gap
[OF *sampled*])
then show *?thesis*
proof
assume *auth*:
generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers final-state
have
trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
s out
unfolding
trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
by (intro *exI conjI*)
(*rule fri-openings, rule out-eq, rule header, rule partial, rule cand,*
rule not-low, rule auth)
then show *?thesis*


```

    using no-auth by contradiction
next
assume gap:
  generic-fri-sampled-same-layer-opening-conflict trace-roots trace-bs
  fri-query-idxs trace-round-layers  $\wedge$ 
   $\neg$  generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
  trace-roots trace-bs fri-query-idxs trace-round-layers final-state
show ?thesis
  unfolding trace-fri-header-tied-same-layer-auth-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
     rule not-low, use gap in simp-all)
qed
qed

lemma trace-fri-header-tied-same-layer-auth-gap-imp-recorded-auth-gap:
  assumes trace-fri-header-tied-same-layer-auth-gap s out
  shows trace-fri-header-tied-same-layer-recorded-auth-gap s out
proof -
  from assms obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
    result final-state
  where out-eq: out = Some (result, final-state)
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  and cand: partial-trace-table-candidate trace-table trace-openings
  and not-low:  $\neg$  trace-table-low-degree trace-table
  and same:
    generic-fri-sampled-same-layer-opening-conflict trace-roots trace-bs
    fri-query-idxs trace-round-layers
  and no-auth:
     $\neg$  generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
    trace-roots trace-bs fri-query-idxs trace-round-layers final-state
  unfolding trace-fri-header-tied-same-layer-auth-gap-def
  by blast
have recorded-gap:
  generic-fri-sampled-same-layer-recorded-auth-gap trace-roots trace-bs
  fri-query-idxs trace-round-layers final-state
by (rule generic-fri-sampled-same-layer-auth-gap-imp-recorded-auth-gap
  [OF same no-auth])

```

show ?thesis

unfolding trace-fri-header-tied-same-layer-recorded-auth-gap-def

by (intro exI conjI)

(rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
rule not-low, rule recorded-gap)

qed

lemma trace-fri-header-tied-same-layer-recorded-auth-gap-imp-authenticated-recorded-auth-gap:

assumes trace-fri-header-tied-same-layer-recorded-auth-gap s out

shows

trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap s out

proof –

from assms **obtain** trace-roots trace-bs trace-final dg composition-roots

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as rest trace-openings trace-table

result final-state

where out-eq: out = Some (result, final-state)

and fri-openings:

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final dg composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers

and header:

verifier-header-transcript s fr trace-roots trace-final as dg

composition-roots composition-final rest

and partial:

accepted-with-partial-trace-openings s out fr fri-query-idxs

trace-openings

and cand: partial-trace-table-candidate trace-table trace-openings

and not-low: $\neg$ trace-table-low-degree trace-table

and recorded-gap:

generic-fri-sampled-same-layer-recorded-auth-gap trace-roots

trace-bs fri-query-idxs trace-round-layers final-state

unfolding trace-fri-header-tied-same-layer-recorded-auth-gap-def

by blast

have len-trace-bs: length trace-bs = length trace-roots

using accepted-fri-opening-transcript-shapes(2)[OF fri-openings] .

have len-query: length fri-query-idxs = rounds

using accepted-fri-opening-transcript-shapes(7)[OF fri-openings] .

have auth-step:

$\bigwedge$ round-idx layer-idx.

round-idx < length fri-query-idxs $\implies$

layer-idx < length trace-bs $\implies$

generic-fri-transcript-step-with-authenticated-chunk trace-roots

trace-bs fri-query-idxs trace-round-layers final-state round-idx

layer-idx

proof –

fix round-idx layer-idx

assume round-bound: round-idx < length fri-query-idxs

and layer-bound-bs: layer-idx < length trace-bs

```

have round-bound-rounds: round-idx < rounds
  using round-bound len-query by simp
have layer-bound-roots: layer-idx < length trace-roots
  using layer-bound-bs len-trace-bs by simp
show
  generic-fri-transcript-step-with-authenticated-chunk trace-roots
    trace-bs fri-query-idxs trace-round-layers final-state round-idx
    layer-idx
  by (rule
    accepted-fri-opening-transcript-trace-step-with-authenticated-chunk-at-raw
    [OF fri-openings out-eq round-bound-rounds layer-bound-roots])
qed
have authenticated-recorded-gap:
  generic-fri-sampled-same-layer-authenticated-recorded-auth-gap
    trace-roots trace-bs fri-query-idxs trace-round-layers final-state
  by (rule
    generic-fri-recorded-auth-gap-imp-authenticated-recorded-auth-gap
    [OF recorded-gap auth-step])
show ?thesis
  unfolding
    trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
    rule not-low, rule authenticated-recorded-gap)
qed

lemma trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap-imp-slot-recorded-chunk-gap:
  assumes
    trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap s out
  shows trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s out
proof —
  from assms obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
    result final-state
  where out-eq: out = Some (result, final-state)
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  and cand: partial-trace-table-candidate trace-table trace-openings
  and not-low:  $\neg$  trace-table-low-degree trace-table
  and gap:

```

```

    generic-fri-sampled-same-layer-authenticated-recorded-auth-gap
    trace-roots trace-bs fri-query-idxs trace-round-layers final-state
unfolding
    trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap-def
by blast
have slot-gap:
    generic-fri-authenticated-slot-recorded-chunk-gap trace-roots
    fri-query-idxs trace-round-layers final-state
by (rule
    generic-fri-authenticated-recorded-auth-gap-imp-slot-recorded-chunk-gap
    [OF gap])
show ?thesis
unfolding trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-def
by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
    rule not-low, rule slot-gap)
qed

lemma wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-authenticated-and-auth-gap:
fixes C G :: prob
assumes authenticated-bound:
    wp-event verify-monad
    (trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer s)
    s ≤ C
and auth-gap-bound:
    wp-event verify-monad
    (trace-fri-header-tied-assignment-auth-gap s) s ≤ G
shows
    wp-event verify-monad
    (trace-fri-header-tied-sampled-assignment-conflict s) s ≤ C + G
proof –
have wp-event verify-monad
    (trace-fri-header-tied-sampled-assignment-conflict s) s ≤
    wp-event verify-monad
    (λout.
    trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
    s out ∨
    trace-fri-header-tied-assignment-auth-gap s out) s
by (rule wp-event-mono)
    (rule trace-fri-header-tied-assignment-imp-authenticated-or-auth-gap)
also have ... ≤
    wp-event verify-monad
    (trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer s)
    s +
    wp-event verify-monad
    (trace-fri-header-tied-assignment-auth-gap s) s
by (rule wp-event-union-bound)
also have ... ≤ C + G
by (intro add-mono authenticated-bound auth-gap-bound)

```

finally show ?thesis .
qed

lemma *wp-trace-fri-header-tied-assignment-auth-gap-bound-from-same-layer-auth-gap*:
fixes $G :: \text{prob}$
assumes *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-auth-gap } s) \ s \leq G$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-assignment-auth-gap } s) \ s \leq G$
proof –
 have $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-assignment-auth-gap } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-auth-gap } s) \ s$
 by (rule *wp-event-mono*)
 (rule *trace-fri-header-tied-assignment-auth-gap-imp-same-layer-auth-gap*)
 then show ?thesis
 by (rule *order-trans[OF - gap-bound]*)
 qed

lemma *wp-trace-fri-header-tied-same-layer-auth-gap-bound-from-recorded-auth-gap*:
fixes $G :: \text{prob}$
assumes *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-recorded-auth-gap } s) \ s \leq G$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-auth-gap } s) \ s \leq G$
proof –
 have $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-auth-gap } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-recorded-auth-gap } s) \ s$
 by (rule *wp-event-mono*)
 (rule *trace-fri-header-tied-same-layer-auth-gap-imp-recorded-auth-gap*)
 then show ?thesis
 by (rule *order-trans[OF - gap-bound]*)
 qed

lemma *wp-trace-fri-header-tied-same-layer-recorded-auth-gap-bound-from-authenticated-recorded-auth-gap*:
fixes $G :: \text{prob}$
assumes *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap } s)$
 $s \leq G$
shows
 $\text{wp-event verify-monad}$

$(\text{trace-fri-header-tied-same-layer-recorded-auth-gap } s) \ s \leq G$
proof –
have $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-recorded-auth-gap } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap } s)$
 s
by $(\text{rule wp-event-mono})$
 $(\text{rule}$
 $\text{trace-fri-header-tied-same-layer-recorded-auth-gap-imp-authenticated-recorded-auth-gap})$
then show $?thesis$
by $(\text{rule order-trans}[OF - \text{gap-bound}])$
qed

lemma $\text{wp-trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap-bound-from-slot-recorded-chunk-gap}$:
fixes $G :: \text{prob}$
assumes gap-bound :
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-authenticated-slot-recorded-chunk-gap } s) \ s$
 $\leq G$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap } s)$
 $s \leq G$
proof –
have $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap } s)$
 $s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-authenticated-slot-recorded-chunk-gap } s) \ s$
by $(\text{rule wp-event-mono})$
 $(\text{rule}$
 $\text{trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap-imp-slot-recorded-chunk-gap})$
then show $?thesis$
by $(\text{rule order-trans}[OF - \text{gap-bound}])$
qed

lemma $\text{wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-auth-gap-and-residuals}$:
fixes $A \ P \ H \ G \ N \ S \ F :: \text{prob}$
assumes auth-gap-bound :
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-assignment-auth-gap } s) \ s \leq A$
and $\text{partial-merkle-bound}$:
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq P$
and $\text{structural-gap-bound}$:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s \leq H$
and $\text{alignment-gap-bound}$:
 $\text{wp-event verify-monad}$

(trace-fri-header-tied-base-value-alignment-gap s) $s \leq G$
and next-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-next-value-replay-gap s) $s \leq N$
and successor-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-successor-replay-gap s) $s \leq S$
and final-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-final-value-replay-gap s) $s \leq F$
shows
 wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict s) s
 $\leq (((((P + H) + G) + N + S + F) + P) + A$
by (rule
 wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-authenticated-and-auth-gap)
 (rule
 wp-trace-fri-header-tied-authenticated-conflict-bound-from-partial-merkle-structural-and-replay-gaps
 [OF partial-merkle-bound structural-gap-bound alignment-gap-bound
 next-gap-bound successor-gap-bound final-gap-bound],
 rule auth-gap-bound)

lemma wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-same-layer-auth-gap-and-residuals:

fixes $A P H G N S F :: \text{prob}$

assumes same-layer-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-same-layer-auth-gap s) $s \leq A$

and partial-merkle-bound:

wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq P$

and structural-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-sibling-mismatch-structural-gap s) $s \leq H$

and alignment-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-base-value-alignment-gap s) $s \leq G$

and next-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-next-value-replay-gap s) $s \leq N$

and successor-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-successor-replay-gap s) $s \leq S$

and final-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-final-value-replay-gap s) $s \leq F$

shows

wp-event verify-monad

(trace-fri-header-tied-sampled-assignment-conflict s) s

$\leq (((((P + H) + G) + N + S + F) + P) + A$

by (rule

$wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}sampled\text{-}assignment\text{-}conflict\text{-}bound\text{-}from\text{-}auth\text{-}gap\text{-}and\text{-}residuals)$
 (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}assignment\text{-}auth\text{-}gap\text{-}bound\text{-}from\text{-}same\text{-}layer\text{-}auth\text{-}gap$
 $[OF\ same\text{-}layer\text{-}gap\text{-}bound]$,
 rule $partial\text{-}merkle\text{-}bound$,
 rule $structural\text{-}gap\text{-}bound$,
 rule $alignment\text{-}gap\text{-}bound$,
 rule $next\text{-}gap\text{-}bound$,
 rule $successor\text{-}gap\text{-}bound$,
 rule $final\text{-}gap\text{-}bound$)

lemma $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}sampled\text{-}assignment\text{-}conflict\text{-}bound\text{-}from\text{-}recorded\text{-}auth\text{-}gap\text{-}and\text{-}residuals$:

fixes $A\ P\ H\ G\ N\ S\ F :: prob$

assumes $recorded\text{-}gap\text{-}bound$:

$wp\text{-}event\ verify\text{-}monad$

$(trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}recorded\text{-}auth\text{-}gap\ s)\ s \leq A$

and $partial\text{-}merkle\text{-}bound$:

$wp\text{-}event\ verify\text{-}monad\ (partial\text{-}merkle\text{-}inconsistency\text{-}bad\ s)\ s \leq P$

and $structural\text{-}gap\text{-}bound$:

$wp\text{-}event\ verify\text{-}monad$

$(trace\text{-}fri\text{-}header\text{-}tied\text{-}sibling\text{-}mismatch\text{-}structural\text{-}gap\ s)\ s \leq H$

and $alignment\text{-}gap\text{-}bound$:

$wp\text{-}event\ verify\text{-}monad$

$(trace\text{-}fri\text{-}header\text{-}tied\text{-}base\text{-}value\text{-}alignment\text{-}gap\ s)\ s \leq G$

and $next\text{-}gap\text{-}bound$:

$wp\text{-}event\ verify\text{-}monad$

$(trace\text{-}fri\text{-}header\text{-}tied\text{-}next\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq N$

and $successor\text{-}gap\text{-}bound$:

$wp\text{-}event\ verify\text{-}monad$

$(trace\text{-}fri\text{-}header\text{-}tied\text{-}successor\text{-}replay\text{-}gap\ s)\ s \leq S$

and $final\text{-}gap\text{-}bound$:

$wp\text{-}event\ verify\text{-}monad$

$(trace\text{-}fri\text{-}header\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq F$

shows

$wp\text{-}event\ verify\text{-}monad$

$(trace\text{-}fri\text{-}header\text{-}tied\text{-}sampled\text{-}assignment\text{-}conflict\ s)\ s$

$\leq (((((P + H) + G) + N + S + F) + P) + A$

by (rule

$wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}sampled\text{-}assignment\text{-}conflict\text{-}bound\text{-}from\text{-}same\text{-}layer\text{-}auth\text{-}gap\text{-}and\text{-}residuals)$

(rule

$wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}auth\text{-}gap\text{-}bound\text{-}from\text{-}recorded\text{-}auth\text{-}gap$

$[OF\ recorded\text{-}gap\text{-}bound]$,
 rule $partial\text{-}merkle\text{-}bound$,
 rule $structural\text{-}gap\text{-}bound$,
 rule $alignment\text{-}gap\text{-}bound$,
 rule $next\text{-}gap\text{-}bound$,
 rule $successor\text{-}gap\text{-}bound$,
 rule $final\text{-}gap\text{-}bound$)

lemma *wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-authenticated-recorded-auth-gap-and-residuals*
fixes $A\ P\ H\ G\ N\ S\ F :: \text{prob}$
assumes *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap } s)$
 $s \leq A$
and *partial-merkle-bound*:
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq P$
and *structural-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s \leq H$
and *alignment-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-value-alignment-gap } s) \ s \leq G$
and *next-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq N$
and *successor-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s \leq S$
and *final-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-replay-gap } s) \ s \leq F$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict } s) \ s$
 $\leq (((((P + H) + G) + N + S + F) + P) + A)$
by (*rule*
 $\text{wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-recorded-auth-gap-and-residuals}$)
(*rule*
 $\text{wp-trace-fri-header-tied-same-layer-recorded-auth-gap-bound-from-authenticated-recorded-auth-gap}$
 $[\text{OF gap-bound}]$,
 $\text{rule partial-merkle-bound}$,
 $\text{rule structural-gap-bound}$,
 $\text{rule alignment-gap-bound}$,
 $\text{rule next-gap-bound}$,
 $\text{rule successor-gap-bound}$,
 $\text{rule final-gap-bound}$)

lemma *wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-slot-recorded-chunk-gap-and-residuals*:
fixes $A\ P\ H\ G\ N\ S\ F :: \text{prob}$
assumes *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-authenticated-slot-recorded-chunk-gap } s) \ s$
 $\leq A$
and *partial-merkle-bound*:
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq P$
and *structural-gap-bound*:
 $\text{wp-event verify-monad}$

(trace-fri-header-tied-sibling-mismatch-structural-gap s) $s \leq H$
and alignment-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-base-value-alignment-gap s) $s \leq G$
and next-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-next-value-replay-gap s) $s \leq N$
and successor-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-successor-replay-gap s) $s \leq S$
and final-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-final-value-replay-gap s) $s \leq F$
shows
 wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict s) s
 $\leq (((((P + H) + G) + N + S + F) + P) + A$
by (rule
 wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-authenticated-recorded-auth-gap-and-residuals)
 (rule
 wp-trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap-bound-from-slot-recorded-chunk-gap
 [OF gap-bound],
 rule partial-merkle-bound,
 rule structural-gap-bound,
 rule alignment-gap-bound,
 rule next-gap-bound,
 rule successor-gap-bound,
 rule final-gap-bound)
lemma wp-trace-fri-header-tied-bound-from-sampled-zero-auth-gap-and-residuals:
fixes $R\ Z\ A\ P\ H\ G\ N\ S\ F :: \text{prob}$
assumes sampled-bound:
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) $s \leq R$
and zero-bound:
 wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction s) $s \leq Z$
and auth-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-assignment-auth-gap s) $s \leq A$
and partial-merkle-bound:
 wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq P$
and structural-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-sibling-mismatch-structural-gap s) $s \leq H$
and alignment-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-base-value-alignment-gap s) $s \leq G$
and next-gap-bound:

$wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq N$
and $\text{successor-gap-bound}$:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s \leq S$
and final-gap-bound :
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-replay-gap } s) \ s \leq F$
shows
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \ s$
 $\leq R + (((((P + H) + G) + N + S + F) + P) + A + Z)$
proof –
have conflict-bound :
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict } s) \ s$
 $\leq (((((P + H) + G) + N + S + F) + P) + A$
by (rule
 $wp\text{-trace-fri-header-tied-sampled-assignment-conflict-bound-from-auth-gap-and-residuals})$
(rule auth-gap-bound ,
rule $\text{partial-merkle-bound}$,
rule $\text{structural-gap-bound}$,
rule $\text{alignment-gap-bound}$,
rule next-gap-bound ,
rule $\text{successor-gap-bound}$,
rule final-gap-bound)
have $wp\text{-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \ s$
 $\leq R + (((((P + H) + G) + N + S + F) + P) + A + Z)$
by (rule order-trans [
 $OF \text{ } wp\text{-trace-fri-header-tied-bound-from-sampled-conflict-and-zero}$
 $[OF \text{ } \text{sampled-bound } \text{conflict-bound } \text{zero-bound}]]])$
 $(\text{simp add: algebra-simps})$
then show $?thesis$.
qed

lemma $wp\text{-trace-fri-header-tied-bound-from-sampled-zero-same-layer-auth-gap-and-residuals}$:
fixes $R \ Z \ A \ P \ H \ G \ N \ S \ F :: \text{prob}$
assumes sampled-bound :
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-sampled-layer-chain } s) \ s \leq R$
and zero-bound :
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s \leq Z$
and $\text{same-layer-gap-bound}$:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-auth-gap } s) \ s \leq A$
and $\text{partial-merkle-bound}$:
 $wp\text{-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq P$

and *structural-gap-bound*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s \leq H$
and *alignment-gap-bound*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-base-value-alignment-gap } s) \ s \leq G$
and *next-gap-bound*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq N$
and *successor-gap-bound*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s \leq S$
and *final-gap-bound*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-replay-gap } s) \ s \leq F$
shows
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \ s$
 $\leq R + (((((P + H) + G) + N + S + F) + P) + A + Z)$
by (rule
 $wp\text{-trace-fri-header-tied-bound-from-sampled-zero-auth-gap-and-residuals})$
(rule *sampled-bound*,
rule *zero-bound*,
rule
 $wp\text{-trace-fri-header-tied-assignment-auth-gap-bound-from-same-layer-auth-gap}$
 $[OF \text{ same-layer-gap-bound}]$,
rule *partial-merkle-bound*,
rule *structural-gap-bound*,
rule *alignment-gap-bound*,
rule *next-gap-bound*,
rule *successor-gap-bound*,
rule *final-gap-bound*)

lemma *wp-trace-fri-header-tied-bound-from-sampled-zero-recorded-auth-gap-and-residuals*:

fixes $R \ Z \ A \ P \ H \ G \ N \ S \ F :: \text{prob}$

assumes *sampled-bound*:

$wp\text{-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-sampled-layer-chain } s) \ s \leq R$

and *zero-bound*:

$wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s \leq Z$

and *recorded-gap-bound*:

$wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-recorded-auth-gap } s) \ s \leq A$

and *partial-merkle-bound*:

$wp\text{-event verify-monad} (\text{partial-merkle-inconsistency-bad } s) \ s \leq P$

and *structural-gap-bound*:

$wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s \leq H$

and *alignment-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-base-value-alignment-gap s*) $s \leq G$
and *next-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-next-value-replay-gap s*) $s \leq N$
and *successor-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-successor-replay-gap s*) $s \leq S$
and *final-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-final-value-replay-gap s*) $s \leq F$
shows
 wp-event verify-monad
 (*trace-fri-bad-with-header-tied-partial-candidate s*) s
 $\leq R + (((((P + H) + G) + N + S + F) + P) + A + Z)$
by (*rule*
 wp-trace-fri-header-tied-bound-from-sampled-zero-same-layer-auth-gap-and-residuals)
 (*rule sampled-bound*,
 rule zero-bound,
 rule
 wp-trace-fri-header-tied-same-layer-auth-gap-bound-from-recorded-auth-gap
 [*OF recorded-gap-bound*],
 rule partial-merkle-bound,
 rule structural-gap-bound,
 rule alignment-gap-bound,
 rule next-gap-bound,
 rule successor-gap-bound,
 rule final-gap-bound)

lemma *wp-trace-fri-header-tied-bound-from-sampled-zero-authenticated-recorded-auth-gap-and-residuals*:

fixes $R\ Z\ A\ P\ H\ G\ N\ S\ F :: \text{prob}$

assumes *sampled-bound*:

wp-event verify-monad
 (*trace-fri-bad-with-header-tied-sampled-layer-chain s*) $s \leq R$

and *zero-bound*:

wp-event verify-monad
 (*trace-fri-header-tied-zero-round-final-obstruction s*) $s \leq Z$

and *gap-bound*:

wp-event verify-monad
 (*trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap s*)
 $s \leq A$

and *partial-merkle-bound*:

wp-event verify-monad (*partial-merkle-inconsistency-bad s*) $s \leq P$

and *structural-gap-bound*:

wp-event verify-monad
 (*trace-fri-header-tied-sibling-mismatch-structural-gap s*) $s \leq H$

and *alignment-gap-bound*:

wp-event verify-monad

(trace-fri-header-tied-base-value-alignment-gap s) $s \leq G$
and next-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-next-value-replay-gap s) $s \leq N$
and successor-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-successor-replay-gap s) $s \leq S$
and final-gap-bound:
 wp-event verify-monad
 (trace-fri-header-tied-final-value-replay-gap s) $s \leq F$
shows
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) s
 $\leq R + (((((P + H) + G) + N + S + F) + P) + A + Z)$
by (rule
 wp-trace-fri-header-tied-bound-from-sampled-zero-recorded-auth-gap-and-residuals)
 (rule sampled-bound,
 rule zero-bound,
 rule
 wp-trace-fri-header-tied-same-layer-recorded-auth-gap-bound-from-authenticated-recorded-auth-gap
 [OF gap-bound],
 rule partial-merkle-bound,
 rule structural-gap-bound,
 rule alignment-gap-bound,
 rule next-gap-bound,
 rule successor-gap-bound,
 rule final-gap-bound)

lemma wp-trace-fri-header-tied-bound-from-sampled-zero-slot-recorded-chunk-gap-and-residuals:

fixes $R\ Z\ A\ P\ H\ G\ N\ S\ F :: \text{prob}$

assumes sampled-bound:

wp-event verify-monad

(trace-fri-bad-with-header-tied-sampled-layer-chain s) $s \leq R$

and zero-bound:

wp-event verify-monad

(trace-fri-header-tied-zero-round-final-obstruction s) $s \leq Z$

and gap-bound:

wp-event verify-monad

(trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s)

$s \leq A$

and partial-merkle-bound:

wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq P$

and structural-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-sibling-mismatch-structural-gap s) $s \leq H$

and alignment-gap-bound:

wp-event verify-monad

(trace-fri-header-tied-base-value-alignment-gap s) $s \leq G$

and next-gap-bound:

wp-event verify-monad
 (trace-fri-header-tied-next-value-replay-gap s) $s \leq N$
and *successor-gap-bound*:
wp-event verify-monad
 (trace-fri-header-tied-successor-replay-gap s) $s \leq S$
and *final-gap-bound*:
wp-event verify-monad
 (trace-fri-header-tied-final-value-replay-gap s) $s \leq F$
shows
wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) s
 $\leq R + (((((P + H) + G) + N + S + F) + P) + A + Z)$
by (rule
wp-trace-fri-header-tied-bound-from-sampled-zero-authenticated-recorded-auth-gap-and-residuals)
 (rule *sampled-bound*,
 rule *zero-bound*,
 rule
wp-trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap-bound-from-slot-recorded-chunk-gap
 [OF *gap-bound*],
 rule *partial-merkle-bound*,
 rule *structural-gap-bound*,
 rule *alignment-gap-bound*,
 rule *next-gap-bound*,
 rule *successor-gap-bound*,
 rule *final-gap-bound*)

lemma *wp-trace-fri-header-tied-bound-from-sampled-zero-slot-and-authenticated-final-residuals*:

fixes $R\ Z\ A\ P\ H\ G\ N\ S\ F :: \text{prob}$

assumes *sampled-bound*:

wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) $s \leq R$

and *zero-bound*:

wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction s) $s \leq Z$

and *slot-gap-bound*:

wp-event verify-monad
 (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s)
 $s \leq A$

and *partial-merkle-bound*:

wp-event verify-monad (*partial-merkle-inconsistency-bad* s) $s \leq P$

and *structural-gap-bound*:

wp-event verify-monad
 (trace-fri-header-tied-sibling-mismatch-structural-gap s) $s \leq H$

and *alignment-gap-bound*:

wp-event verify-monad
 (trace-fri-header-tied-base-value-alignment-gap s) $s \leq G$

and *next-gap-bound*:

wp-event verify-monad
 (trace-fri-header-tied-next-value-replay-gap s) $s \leq N$

and *successor-gap-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-header-tied-successor-replay-gap } s) \ s \leq S$
and *authenticated-final-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-header-tied-final-value-authenticated-replay-gap } s) \ s$
 $\leq F$
shows
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-bad-with-header-tied-partial-candidate } s) \ s$
 $\leq R + (((((P + H) + G) + N + S + (P + A + F)) + P) + A + Z)$
proof –
have *final-gap-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-header-tied-final-value-replay-gap } s) \ s \leq P + A + F$
by (rule
 $wp\text{-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-authenticated}$
(rule *partial-merkle-bound*, rule *slot-gap-bound*,
rule *authenticated-final-bound*)
have *old-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-bad-with-header-tied-partial-candidate } s) \ s$
 $\leq R + (((((P + H) + G) + N + S + (P + A + F)) + P) + A + Z)$
by (rule
 $wp\text{-trace-fri-header-tied-bound-from-sampled-zero-slot-recorded-chunk-gap-and-residuals}$
(rule *sampled-bound*,
rule *zero-bound*,
rule *slot-gap-bound*,
rule *partial-merkle-bound*,
rule *structural-gap-bound*,
rule *alignment-gap-bound*,
rule *next-gap-bound*,
rule *successor-gap-bound*,
rule *final-gap-bound*)
then show *?thesis* .
qed

lemma *wp-trace-fri-header-tied-bound-from-sampled-zero-slot-and-selected-step-final-residuals*:
fixes $R \ Z \ A \ P \ H \ G \ N \ S \ F :: prob$
assumes *sampled-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-bad-with-header-tied-sampled-layer-chain } s) \ s \leq R$
and *zero-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-header-tied-zero-round-final-obstruction } s) \ s \leq Z$
and *slot-gap-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-header-tied-authenticated-slot-recorded-chunk-gap } s)$
 $s \leq A$

and *partial-merkle-bound*:
wp-event verify-monad (*partial-merkle-inconsistency-bad* s) $s \leq P$
and *structural-gap-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-sibling-mismatch-structural-gap* s) $s \leq H$
and *alignment-gap-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-base-value-alignment-gap* s) $s \leq G$
and *next-gap-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-next-value-replay-gap* s) $s \leq N$
and *successor-gap-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-successor-replay-gap* s) $s \leq S$
and *selected-step-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-final-value-selected-step-missing-gap* s) s
 $\leq F$
shows
wp-event verify-monad
 (*trace-fri-bad-with-header-tied-partial-candidate* s) s
 $\leq R + (((((P + H) + G) + N + S + (P + A + F)) + P) + A + Z)$
proof –
have *final-gap-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-final-value-replay-gap* s) $s \leq P + A + F$
by (rule
wp-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step)
 (rule *partial-merkle-bound*, rule *slot-gap-bound*,
 rule *selected-step-bound*)
show *?thesis*
by (rule
wp-trace-fri-header-tied-bound-from-sampled-zero-slot-recorded-chunk-gap-and-residuals)
 (rule *sampled-bound*,
 rule *zero-bound*,
 rule *slot-gap-bound*,
 rule *partial-merkle-bound*,
 rule *structural-gap-bound*,
 rule *alignment-gap-bound*,
 rule *next-gap-bound*,
 rule *successor-gap-bound*,
 rule *final-gap-bound*)
qed

lemma *trace-fri-reachable-partial-candidate-reduction-from-header-tied-residual-bounds*:

fixes $R\ Z\ A\ P\ H\ G\ N\ S\ F\ T :: \text{prob}$

assumes *sampled-bound*:

wp-event verify-monad

(*trace-fri-bad-with-header-tied-sampled-layer-chain* s) $s \leq R$

and *zero-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-zero-round-final-obstruction s*) $s \leq Z$
and *slot-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s*)
 $s \leq A$
and *partial-merkle-bound*:
 wp-event verify-monad (*partial-merkle-inconsistency-bad s*) $s \leq P$
and *structural-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-sibling-mismatch-structural-gap s*) $s \leq H$
and *alignment-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-base-value-alignment-gap s*) $s \leq G$
and *next-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-next-value-replay-gap s*) $s \leq N$
and *successor-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-successor-replay-gap s*) $s \leq S$
and *final-gap-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-final-value-replay-gap s*) $s \leq F$
and *header-gap-bound*:
 wp-event verify-monad
 (*trace-fri-reachable-header-tie-gap s*) $s \leq T$
and *total-bound*:
 $R + (((((P + H) + G) + N + S + F) + P) + A + Z) + T$
 $\leq \text{trace-fri-error}$
shows *trace-fri-reachable-partial-candidate-reduction-assumption s*
proof –
have *header-bound*:
 wp-event verify-monad
 (*trace-fri-bad-with-header-tied-partial-candidate s*) s
 $\leq R + (((((P + H) + G) + N + S + F) + P) + A + Z)$
by (rule
 wp-trace-fri-header-tied-bound-from-sampled-zero-slot-recorded-chunk-gap-and-residuals
 [*OF sampled-bound zero-bound slot-gap-bound partial-merkle-bound*
 structural-gap-bound alignment-gap-bound next-gap-bound
 successor-gap-bound final-gap-bound])
show *?thesis*
by (rule
 trace-fri-reachable-partial-candidate-reduction-from-header-tied-and-gap
 [*OF header-bound header-gap-bound total-bound*])
qed

lemma *trace-fri-header-tied-reduction-from-residual-bounds*:
fixes $R\ Z\ A\ P\ H\ G\ N\ S\ F :: \text{prob}$

```

assumes sampled-bound:
  wp-event verify-monad
    (trace-fri-bad-with-header-tied-sampled-layer-chain s)  $s \leq R$ 
and zero-bound:
  wp-event verify-monad
    (trace-fri-header-tied-zero-round-final-obstruction s)  $s \leq Z$ 
and slot-gap-bound:
  wp-event verify-monad
    (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s)
       $s \leq A$ 
and partial-merkle-bound:
  wp-event verify-monad (partial-merkle-inconsistency-bad s)  $s \leq P$ 
and structural-gap-bound:
  wp-event verify-monad
    (trace-fri-header-tied-sibling-mismatch-structural-gap s)  $s \leq H$ 
and alignment-gap-bound:
  wp-event verify-monad
    (trace-fri-header-tied-base-value-alignment-gap s)  $s \leq G$ 
and next-gap-bound:
  wp-event verify-monad
    (trace-fri-header-tied-next-value-replay-gap s)  $s \leq N$ 
and successor-gap-bound:
  wp-event verify-monad
    (trace-fri-header-tied-successor-replay-gap s)  $s \leq S$ 
and final-gap-bound:
  wp-event verify-monad
    (trace-fri-header-tied-final-value-replay-gap s)  $s \leq F$ 
and total-bound:
   $R + (((((P + H) + G) + N + S + F) + P) + A + Z)$ 
     $\leq \text{trace-fri-error}$ 
shows trace-fri-header-tied-reduction s
unfolding trace-fri-header-tied-reduction-def
by (rule order-trans[
  OF wp-trace-fri-header-tied-bound-from-sampled-zero-slot-recorded-chunk-gap-and-residuals
    [OF sampled-bound zero-bound slot-gap-bound partial-merkle-bound
      structural-gap-bound alignment-gap-bound next-gap-bound
      successor-gap-bound final-gap-bound] total-bound])

end

end

```

```

theory Soundness-FRI-Query-Openings
imports
  Soundness-FRI
  Soundness-Query-Transcript-Openings
begin

```

Bridge from the FRI opening transcript evidence to the authenticated trace openings

for the exact verifier-derived query-index list.

context *soundness*

begin

lemma *accepted-fri-opening-transcript-trace-openings-for-query-indices:*

assumes *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
query-idxs trace-round-layers composition-round-layers

and *header:*

verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest

and *out-eq:* *out = Some (result, final-state)*

obtains *trace-openings* **where**

accepted-with-partial-trace-openings s out fr query-idxs trace-openings

proof *—*

let *?P =*

λi openings. map opening-index openings =
powers-scaled (query-idxs ! i) ∧
*partial-authenticated-table fr (scale * clength) openings final-state*

have *len-query-idxs:* *length query-idxs = rounds*

by *(rule accepted-fri-opening-transcript-shapes(7)[OF fri-openings])*

from *fri-openings* **obtain** *result' final-state' fr' f-fl header-as fl*

query-state raw-idxs query-chunks

where *out-def:* *out = Some (result', final-state')*

and *trace-roots-eq:* *trace-roots = map snd f-fl*

and *composition-roots-eq:* *composition-roots = map snd fl*

and *header':*

verifier-header-transcript s fr' trace-roots trace-final header-as dg
composition-roots composition-final (PTranscript query-state)

and *query-out:*

Some (result', final-state') ∈

set-dist

(execute

(ntimes

(verifier-query-round-program fr' f-fl trace-final header-as fl
composition-final)

rounds)

query-state)

and *len-raw:* *length raw-idxs = rounds*

and *query-idxs-eq:*

query-idxs = map (λraw. index (to-nat raw)) raw-idxs

and *len-query-chunks:* *length query-chunks = rounds*

and *transcript-query:*

PTranscript query-state =

List.concat query-chunks @ PTranscript final-state'

and *query-chunk:*

∧ i. i < rounds ⇒

verifier-query-round-chunk (query-idxs ! i)

```

      trace-roots composition-roots (query-chunks ! i)
and replay-lookup:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state')
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
unfolding accepted-fri-opening-transcript-def
by blast
have final-state-eq: final-state' = final-state
  using out-def out-eq by simp
have fr-eq: fr' = fr
  using verifier-header-transcript-unique[OF header' header] by simp
have ex-round:  $\bigwedge i. i < \text{rounds} \implies \exists \text{openings}. ?P\ i\ \text{openings}$ 
proof -
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  show  $\exists \text{openings}. ?P\ i\ \text{openings}$ 
  proof (rule
    ntimes-verifier-query-rounds-selected-trace-openings-aligned-with-replay-at
    [OF query-out i-bound len-raw query-idxs-eq len-query-chunks])
  show PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state'
    by (rule transcript-query)
  show  $\bigwedge j. j < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! j)
      (map snd f-fl) (map snd fl) (query-chunks ! j)
    using query-chunk trace-roots-eq composition-roots-eq by simp
  show  $\bigwedge j. j < \text{rounds} \implies$ 
    fmlookup (HashMap final-state')
      (QueryIndexChallenge (PQueryCounter query-state + j)
        (state-after-query-chunks
          (PState query-state) query-chunks j)) =
      Some (raw-idxs ! j)
    using replay-lookup by simp
  next
  fix openings
  assume
    map opening-index openings = powers-scaled (query-idxs ! i)
    partial-authenticated-table fr' (scale * clength) openings
    final-state'
  then show ?thesis
    using fr-eq final-state-eq by blast
qed
qed
let ?w =  $\lambda i. \text{SOME openings}. ?P\ i\ \text{openings}$ 
let ?trace-openings = map ?w [0.. $\text{rounds}$ ]
have len-trace: length ?trace-openings = rounds

```

```

    by simp
  have props:  $\bigwedge i. i < \text{rounds} \implies ?P\ i\ (?trace\text{-openings}\ !\ i)$ 
  proof -
    fix i
    assume i-bound:  $i < \text{rounds}$ 
    have  $?P\ i\ (?w\ i)$ 
      by (rule someI-ex[OF ex-round[OF i-bound]])
    then show  $?P\ i\ (?trace\text{-openings}\ !\ i)$ 
      using i-bound by simp
  qed
  have partial:
    accepted-with-partial-trace-openings s out fr query-idxs
      ?trace-openings
    unfolding accepted-with-partial-trace-openings-def accepted-def
    by (intro conjI exI[of - result] exI[of - final-state])
      (use out-eq len-query-idxs len-trace props in auto)
  show ?thesis
    by (rule that[OF partial])
  qed
end
end
end

```

theory *Soundness-FRI-Trace-Head-Value*

imports

Soundness-FRI-Trace-Residuals

Soundness-FRI-Query-Openings

begin

Local replay facts for the trace base head-value residual. This theory keeps the exact selected trace-layer replay needed by the residual proof out of the already large trace residual layer.

context *soundness*

begin

lemma *verifier-query-round-program-trace-fri-head-layer-value-aligned-with-lookup:*

fixes $s\ t :: ('f, 'a)\ \text{protocol-channel-scheme}$

assumes $f\text{-fl}\text{-eq}: f\text{-fl} = (b, rt) \# f\text{-fl}\text{-tail}$

and *outcome:*

$\text{Some}\ (\(), t) \in$

set-dist

(execute

(verifier-query-round-program fr f-fl f-final as fl final) s)

obtains *raw idx openings query-chunk trace-layer-chunks*

composition-layer-chunks xp xp-path xn xn-path **where**

$idx = \text{index}\ (\text{to-nat}\ \text{raw})$

query-round-fri-layer-transcripts idx (map snd f-fl) (map snd fl)

```

    query-chunk trace-layer-chunks composition-layer-chunks
PTranscript s = query-chunk @ PTranscript t
PState t = foldl concat (PState s) query-chunk
s ≤ t
map opening-index openings = powers-scaled idx
partial-authenticated-table fr (scale * clength) openings t
openings ≠ []
opening-value (hd openings) = xp
fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
  (trace-layer-chunks ! 0)
fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof –
from outcome[unfolded verifier-query-round-program-alt-def]
obtain raw s0 where raw:
  Some (raw, s0) ∈
    set-dist (execute receive-query-index-challenge s)
and tail:
  Some ((), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s0)
by (auto elim!: set-dist-bindE)
obtain idx openings query-chunk trace-layer-chunks composition-layer-chunks
  xp xp-path xn xn-path
where idx-eq: idx = index (to-nat raw)
and layers:
  query-round-fri-layer-transcripts idx (map snd f-fl) (map snd fl)
  query-chunk trace-layer-chunks composition-layer-chunks
and tr-t: PTranscript s0 = query-chunk @ PTranscript t
and st-t: PState t = foldl concat (PState s0) query-chunk
and ext-s0-t: s0 ≤ t
and idxs: map opening-index openings = powers-scaled idx
and auth-table:
  partial-authenticated-table fr (scale * clength) openings t
and nonempty: openings ≠ []
and opening-value-eq: opening-value (hd openings) = xp
and head-chunk:
  fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
    (trace-layer-chunks ! 0)
using verifier-query-round-after-index-trace-fri-head-layer-value-aligned
  [OF f-fl-eq tail]
by metis
have lookup-s0:
  fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
using receive-query-index-challenge-outcome[OF raw] by simp
have lookup-t:

```

```

  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  by (rule hash-extension-lookup[OF lookup-s0 ext-s0-t])
  have tr-s0: PTranscript s = PTranscript s0
    using receive-query-index-challenge-outcome[OF raw] by simp
  have st-s0: PState s0 = PState s
    using receive-query-index-challenge-outcome[OF raw] by simp
  have ext-s-s0: s ≤ s0
    using receive-query-index-challenge-outcome[OF raw] by simp
  have ext-s-t: s ≤ t
    by (rule hash-ext-trans[OF ext-s-s0 ext-s0-t])
  have tr-goal: PTranscript s = query-chunk @ PTranscript t
    using tr-t tr-s0 by simp
  have st-goal: PState t = foldl concat (PState s) query-chunk
    using st-t st-s0 by simp
  show ?thesis
    by (rule that[OF idx-eq layers tr-goal st-goal ext-s-t idxs auth-table
      nonempty opening-value-eq head-chunk lookup-t])
qed

```

lemma *ntimes-verifier-query-rounds-selected-trace-fri-head-layer-value-aligned-with-replay-at:*

```

fixes query-state final-state :: ('f, 'a) protocol-channel-scheme
assumes f-fl-eq: f-fl = (b, rt) # f-fl-tail
and outcome:
  Some (results, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
and i-bound: i < rounds
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and query-chunk:
  ∧ i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and query-layers:
  query-round-fri-layer-transcripts (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
    trace-layer-chunks composition-layer-chunks
and replay-lookup:
  ∧ i. i < rounds ⇒

```



```

    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
    Some (raw-idxs ! i)
obtains openings xp xp-path xn xn-path where
  map opening-index openings = powers-scaled (query-idxs ! i)
  partial-authenticated-table fr (scale * clength) openings final-state
  openings ≠ []
  opening-value (hd openings) = xp
  fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
    (trace-layer-chunks ! 0)
proof –
  let ?round = verifier-query-round-program fr f-fl f-final as fl final
  from query-opening-ntimes-outcome-decomp-at[OF outcome i-bound]
  obtain prefix x suffix s-i s-suc where
    prefix:
      Some (prefix, s-i) ∈
        set-dist (execute (ntimes ?round i) query-state)
    and round:
      Some (x, s-suc) ∈ set-dist (execute ?round s-i)
    and suffix:
      Some (suffix, final-state) ∈
        set-dist (execute (ntimes ?round (rounds – Suc i)) s-suc)
    by blast
  have x-eq: x = ()
    by (cases x) simp
  show ?thesis
  proof (rule query-opening-ntimes-prefix-transcript-state-alignment
    [OF prefix])
    fix prefix-query-idxs prefix-chunks
    assume len-prefix-idxs: length prefix-query-idxs = i
    and len-prefix-chunks: length prefix-chunks = i
    and transcript-prefix:
      PTranscript query-state =
        List.concat prefix-chunks @ PTranscript s-i
    and state-i:
      PState s-i =
        state-after-query-chunks (PState query-state) prefix-chunks i
    and counter-i:
      PQueryCounter s-i = PQueryCounter query-state + i
    and prefix-ext: query-state ≤ s-i
    and prefix-rounds:
       $\bigwedge k. k < i \implies$ 
        verifier-query-round-chunk (prefix-query-idxs ! k)
          (map snd f-fl) (map snd fl) (prefix-chunks ! k)
    show ?thesis
  proof (rule
    verifier-query-round-program-trace-fri-head-layer-value-aligned-with-lookup

```

```

[OF f-fl-eq round[unfolded x-eq]]
fix raw idx chunk actual-trace-layers actual-composition-layers
  openings xp xp-path xn xn-path
assume idx-eq: idx = index (to-nat raw)
and actual-layers:
  query-round-fri-layer-transcripts idx (map snd f-fl)
  (map snd fl) chunk actual-trace-layers actual-composition-layers
and transcript-round: PTranscript s-i = chunk @ PTranscript s-suc
and state-suc: PState s-suc = foldl concat (PState s-i) chunk
and round-ext: s-i ≤ s-suc
and opening-indices: map opening-index openings = powers-scaled idx
and auth-table-suc:
  partial-authenticated-table fr (scale * clength) openings s-suc
and openings-nonempty: openings ≠ []
and opening-value-eq: opening-value (hd openings) = xp
and head-chunk:
  fri-layer-opening-chunk (scale * clength) xp xp-path xn
  xn-path (actual-trace-layers ! 0)
and lookup-suc:
  fmlookup (HashMap s-suc)
  (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
  Some raw
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
fix suffix-query-idxs suffix-chunks
assume len-suffix-idxs:
  length suffix-query-idxs = rounds - Suc i
and len-suffix-chunks:
  length suffix-chunks = rounds - Suc i
and transcript-suffix:
  PTranscript s-suc =
  List.concat suffix-chunks @ PTranscript final-state
and suffix-state:
  PState final-state =
  state-after-query-chunks (PState s-suc) suffix-chunks
  (rounds - Suc i)
and suffix-counter:
  PQueryCounter final-state =
  PQueryCounter s-suc + (rounds - Suc i)
and suffix-ext: s-suc ≤ final-state
and suffix-rounds:
  ∧k. k < rounds - Suc i ⇒
  verifier-query-round-chunk (suffix-query-idxs ! k)
  (map snd f-fl) (map snd fl) (suffix-chunks ! k)
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks i-bound by simp
have selected-chunk-lens:

```

```

 $\bigwedge k. k < \text{rounds} \implies$ 
  length (?selected-chunks ! k) =
    verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
proof -
  fix k
  assume k-bound: k < rounds
  show
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  proof (cases k < i)
    case True
    have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
      using True len-prefix-chunks by (simp add: nth-append)
    have
      length (prefix-chunks ! k) =
        verifier-query-round-transcript-length
          (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
      by (rule verifier-query-round-chunk-length
        [OF prefix-rounds[OF True]])
    then show ?thesis
      using nth-eq
      by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
  next
  case False
  show ?thesis
  proof (cases k = i)
    case True
    have nth-eq: ?selected-chunks ! k = chunk
      using True len-prefix-chunks by (simp add: nth-append)
    have chunk-shape:
      verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
        chunk
    unfolding verifier-query-round-chunk-def
    using actual-layers
    unfolding query-round-fri-layer-transcripts-def by blast
    have
      length chunk =
        verifier-query-round-transcript-length idx
          (map snd f-fl) (map snd fl)
      by (rule verifier-query-round-chunk-length[OF chunk-shape])
    then show ?thesis
      using nth-eq
      by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
  next
  case False

```

```

let ?k = k - Suc i
have ge: Suc i ≤ k
  using ⟨¬ k < i⟩ False by linarith
have k-bound': ?k < rounds - Suc i
  using ge k-bound by linarith
have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k
  using ge len-prefix-chunks by (simp add: nth-append)
have
  length (suffix-chunks ! ?k) =
    verifier-query-round-transcript-length
      (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
  by (rule verifier-query-round-chunk-length
      [OF suffix-rounds[OF k-bound']])
then show ?thesis
  using nth-eq
  by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-lens:
  ∧k. k < rounds ⇒
    length (query-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  using query-chunk verifier-query-round-chunk-length by blast
have transcript-all:
  PTranscript query-state =
    List.concat ?selected-chunks @ PTranscript final-state
  using transcript-prefix transcript-round transcript-suffix by simp
have concat-eq:
  List.concat ?selected-chunks = List.concat query-chunks
  using transcript-all transcript-query by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
    ∧k. k < length ?selected-chunks ⇒
      length (?selected-chunks ! k) = length (query-chunks ! k)
    using selected-chunk-lens query-chunk-lens len-selected-chunks
    by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks ∧ ([] :: 'f list) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
        [OF len-eq same-lens concat-eq-empty])
  then show ?thesis

```

```

    by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state)
      ?selected-chunks i =
    state-after-query-chunks (PState query-state)
      prefix-chunks i
  by (rule state-after-query-chunks-append-prefix
    [OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks
        (PState query-state) query-chunks i)) =
    Some raw
  using hash-extension-lookup[OF lookup-suc suffix-ext]
  unfolding counter-i state-i prefix-state-eq .
have raw-eq: raw = raw-idxs ! i
  using lookup-final replay-lookup[OF i-bound] by simp
have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-eq query-idxs-eq len-raw i-bound by simp
have auth-table-final:
  partial-authenticated-table fr (scale * clength) openings
    final-state
  by (rule partial-authenticated-table-mono
    [OF auth-table-suc suffix-ext])
have selected-nth: ?selected-chunks ! i = chunk
  using len-prefix-chunks by (simp add: nth-append)
have actual-layers-query:
  query-round-fri-layer-transcripts (query-idxs ! i)
    (map snd f-fl) (map snd fl) (query-chunks ! i)
    actual-trace-layers actual-composition-layers
  using actual-layers idx-eq-query chunks-eq selected-nth by simp
have actual-trace-layers-eq:
  actual-trace-layers = trace-layer-chunks
  by (rule query-round-fri-layer-transcripts-unique(1)
    [OF actual-layers-query query-layers])
show ?thesis
  by (rule that[OF - auth-table-final openings-nonempty
    opening-value-eq])
  (use opening-indices idx-eq-query head-chunk
    actual-trace-layers-eq in simp-all)

```

qed
 qed
 qed
 qed

lemma *accepted-fri-opening-transcript-trace-head-layer-value-aligned-at:*
assumes *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
query-idxs trace-round-layers composition-round-layers
and *header:*
verifier-header-transcript s fr trace-roots trace-final header-as dg
composition-roots composition-final header-rest
and *out-eq:* *out = Some (result, final-state)*
and *round-bound:* *round-idx < length query-idxs*
and *trace-bs-nonempty:* *0 < length trace-bs*
obtains *openings xp xp-path xn xn-path* **where**
map opening-index openings = powers-scaled (query-idxs ! round-idx)
*partial-authenticated-table fr (scale * clength) openings final-state*
openings ≠ []
opening-value (hd openings) = xp
*fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path*
(trace-round-layers ! round-idx ! 0)
proof –
from *fri-openings* **obtain** *result' final-state' fr' f-fl as fl query-state*
raw-idxs query-chunks
where *out-def:* *out = Some (result', final-state')*
and *trace-roots-eq:* *trace-roots = map snd f-fl*
and *trace-bs-eq:* *trace-bs = map fst f-fl*
and *composition-roots-eq:* *composition-roots = map snd fl*
and *header':*
verifier-header-transcript s fr' trace-roots trace-final as dg
composition-roots composition-final (PTranscript query-state)
and *query-out:*
Some (result', final-state') ∈
set-dist (execute
(ntimes
(verifier-query-round-program fr' f-fl trace-final as fl
composition-final)
rounds)
query-state)
and *len-raw:* *length raw-idxs = rounds*
and *query-idxs-eq:* *query-idxs = map (λraw. index (to-nat raw)) raw-idxs*
and *len-query-chunks:* *length query-chunks = rounds*
and *transcript-query:*
PTranscript query-state =
List.concat query-chunks @ PTranscript final-state'
and *query-chunk:*
 $\bigwedge i. i < rounds \implies$

```

    verifier-query-round-chunk (query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
and query-layers:
   $\bigwedge i. i < \text{rounds} \implies$ 
    query-round-fri-layer-transcripts (query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
    (trace-round-layers ! i) (composition-round-layers ! i)
and replay-lookup:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmllookup (HashMap final-state')
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks
        (PState query-state) query-chunks i)) =
    Some (raw-idxs ! i)
unfolding accepted-fri-opening-transcript-def by blast
have fr'-eq: fr' = fr
  using verifier-header-transcript-unique[OF header header'] by simp
have result'-eq: result' = result
  using out-def out-eq by simp
have final-state'-eq: final-state' = final-state
  using out-def out-eq by simp
have query-out':
  Some (result, final-state)  $\in$ 
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr' f-fl trace-final as fl
          composition-final)
        rounds)
      query-state)
  using query-out result'-eq final-state'-eq by simp
have round-bound-rounds: round-idx < rounds
  using round-bound
    accepted-fri-opening-transcript-shapes( $\gamma$ )[OF fri-openings]
  by simp
obtain b rt f-fl-tail where f-fl-eq: f-fl = (b, rt) # f-fl-tail
proof (cases f-fl)
case Nil
then show ?thesis
  using trace-bs-nonempty trace-bs-eq by simp
next
case (Cons head tail)
then obtain b rt where head-eq: head = (b, rt)
  by (cases head)
then show ?thesis
  using Cons that by simp
qed
have query-layer:
  query-round-fri-layer-transcripts (query-idxs ! round-idx)
  (map snd f-fl) (map snd fl) (query-chunks ! round-idx)

```

```

    (trace-round-layers ! round-idx)
    (composition-round-layers ! round-idx)
using query-layers[OF round-bound-rounds]
    trace-roots-eq composition-roots-eq by simp
have ex:
  ∃ openings xp xp-path xn xn-path.
    map opening-index openings = powers-scaled (query-idxs ! round-idx) ∧
    partial-authenticated-table fr' (scale * clength) openings final-state ∧
    openings ≠ [] ∧
    opening-value (hd openings) = xp ∧
    fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
    (trace-round-layers ! round-idx ! 0)
proof (rule
  ntimes-verifier-query-rounds-selected-trace-fri-head-layer-value-aligned-with-replay-at
  [OF f-fl-eq query-out' round-bound-rounds len-raw query-idxs-eq
    len-query-chunks])
show PTranscript query-state =
  List.concat query-chunks @ PTranscript final-state
using transcript-query final-state'-eq by simp
show ∧i. i < rounds ⇒
  verifier-query-round-chunk (query-idxs ! i)
  (map snd f-fl) (map snd fl) (query-chunks ! i)
using query-chunk trace-roots-eq composition-roots-eq by simp
show query-round-fri-layer-transcripts (query-idxs ! round-idx)
  (map snd f-fl) (map snd fl) (query-chunks ! round-idx)
  (trace-round-layers ! round-idx)
  (composition-round-layers ! round-idx)
by (rule query-layer)
show ∧i. i < rounds ⇒
  fmlookup (HashMap final-state)
  (QueryIndexChallenge (PQueryCounter query-state + i)
    (state-after-query-chunks
      (PState query-state) query-chunks i)) =
  Some (raw-idxs ! i)
using replay-lookup final-state'-eq by simp
next
fix openings xp xp-path xn xn-path
assume map opening-index openings =
  powers-scaled (query-idxs ! round-idx)
and partial-authenticated-table fr' (scale * clength) openings
  final-state
and openings ≠ []
and opening-value (hd openings) = xp
and fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
  (trace-round-layers ! round-idx ! 0)
then show ?thesis
by blast
qed
from ex show ?thesis

```


by (metis fr'-eq that)
qed

lemma *trace-fri-header-tied-base-head-value-neq-gap-imp-partial-merkle:*
assumes *trace-fri-header-tied-base-head-value-neq-gap s out*
shows *partial-merkle-inconsistency-bad s out*

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table
round-idx xp xp-path xn xn-path result final-state

where *out-eq*: *out* = *Some (result, final-state)*

and *fri-openings*:

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers

and *header*:

verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest

and *partial*:

accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings

and *candidate*: *partial-trace-table-candidate trace-table trace-openings*

and *nonlow*: $\neg$ *trace-table-low-degree trace-table*

and *round-bound*: *round-idx* < *length fri-query-idxs*

and *trace-bs-nonempty*: *0* < *length trace-bs*

and *openings-nonempty*: *trace-openings* ! *round-idx* $\neq$ []

and *hd-idx*:

opening-index (hd (trace-openings ! round-idx)) =
fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0

and *hd-neq*: *opening-value (hd (trace-openings ! round-idx))* $\neq$ *xp*

and *step*:

fri-layer-step-evidence
(trace-roots ! 0)
(trace-bs ! 0)
(fri-evidence-layer-len trace-roots 0)
(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
1
(fri-sibling-index (fri-evidence-layer-len trace-roots 0)
(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
xp xp-path xn xn-path
(fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
(fri-evidence-next-value trace-roots trace-bs fri-query-idxs
round-idx 0 xp xn)
(trace-round-layers ! round-idx ! 0)

and *no-match*:

$\neg$ *fri-opening-matches-table (fri-evidence-layer-len trace-roots 0)*
(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
trace-table xp xn

```

unfolding trace-fri-header-tied-base-head-value-neq-gap-def
by metis
have round-bound-rounds: round-idx < rounds
using round-bound
  accepted-fri-opening-transcript-shapes(7)[OF fri-openings]
by simp
from accepted-fri-opening-transcript-trace-head-layer-value-aligned-at
  [OF fri-openings header out-eq round-bound trace-bs-nonempty]
obtain openings yp yp-path yn yn-path where actual-idxs:
  map opening-index openings = powers-scaled (fri-query-idxs ! round-idx)
and actual-auth:
  partial-authenticated-table fr (scale * clength) openings final-state
and actual-nonempty: openings ≠ []
and actual-value: opening-value (hd openings) = yp
and actual-chunk:
  fri-layer-opening-chunk (scale * clength) yp yp-path yn yn-path
  (trace-round-layers ! round-idx ! 0)
by blast
have roots-nonempty: 0 < length trace-roots
using accepted-fri-opening-transcript-shapes(2)[OF fri-openings]
  trace-bs-nonempty by simp
have len-eq:
  fri-evidence-layer-len trace-roots 0 = scale * clength
unfolding fri-evidence-layer-len-def
using roots-nonempty
by (cases trace-roots) (simp-all add: mult.commute)
have step-chunk:
  fri-layer-opening-chunk (scale * clength) xp xp-path xn xn-path
  (trace-round-layers ! round-idx ! 0)
using fri-layer-step-evidenceD(4)[OF step]
unfolding len-eq .
have yp-eq: yp = xp
by (rule fri-layer-opening-chunk-values-unique(1)
  [OF actual-chunk step-chunk])
have actual-head-idx:
  opening-index (hd openings) =
  fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0
proof –
have opening-index (hd openings) = fri-query-idxs ! round-idx
proof –
have opening-index (hd openings) =
  hd (map opening-index openings)
using actual-nonempty by (cases openings) auto
also have ... = hd (powers-scaled (fri-query-idxs ! round-idx))
using actual-idxs by simp
also have ... = fri-query-idxs ! round-idx
by (rule powers-scaled-hd)
finally show ?thesis .
qed

```

```

moreover have
  fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 =
    fri-query-idxs ! round-idx
using roots-nonempty
unfolding fri-evidence-layer-idx-def
by (cases trace-roots) simp-all
ultimately show ?thesis by simp
qed
let ?opn = hd (trace-openings ! round-idx)
let ?opn' = hd openings
have trace-table:
  partial-authenticated-table fr (scale * clength)
    (trace-openings ! round-idx) final-state
using partial round-bound-rounds out-eq
unfolding accepted-with-partial-trace-openings-def by blast
have opn-auth: authenticated-opening-in final-state ?opn
using trace-table openings-nonempty
unfolding partial-authenticated-table-def by simp
have opn'-auth: authenticated-opening-in final-state ?opn'
using actual-auth actual-nonempty
unfolding partial-authenticated-table-def by simp
have same-root: opening-root ?opn = opening-root ?opn'
using trace-table actual-auth openings-nonempty actual-nonempty
unfolding partial-authenticated-table-def by auto
have same-length: opening-length ?opn = opening-length ?opn'
using trace-table actual-auth openings-nonempty actual-nonempty
unfolding partial-authenticated-table-def by auto
have same-index: opening-index ?opn = opening-index ?opn'
using hd-idx actual-head-idx by simp
have diff-value: opening-value ?opn  $\neq$  opening-value ?opn'
using hd-neq actual-value yp-eq by simp
have acc: accepted out
unfolding accepted-def out-eq by simp
show ?thesis
unfolding partial-merkle-inconsistency-bad-def
by (intro conjI exI)
  (rule acc, rule out-eq, rule opn-auth, rule opn'-auth,
   rule same-root, rule same-length, rule same-index, rule diff-value)
qed

lemma wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle:
assumes merkle-bound:
  wp-event verify-monad (partial-merkle-inconsistency-bad s)  $s \leq M$ 
shows
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-neq-gap s)  $s \leq M$ 
proof -
have wp-event verify-monad
  (trace-fri-header-tied-base-head-value-neq-gap s)  $s \leq$ 

```

```

      wp-event verify-monad (partial-merkle-inconsistency-bad s) s
    by (rule wp-event-mono)
      (rule trace-fri-header-tied-base-head-value-neq-gap-imp-partial-merkle)
  then show ?thesis
    by (rule order-trans[OF - merkle-bound])
qed

end

end

```

```

theory Soundness-FRI-Trace-Candidate-Gap
imports Soundness-FRI-Trace-Head-Value
begin

```

Candidate-availability split for the reachable/header-tied trace FRI gap. This layer depends on the exact FRI-query opening bridge but keeps the exported split separate from the base head-value residual theory.

```

context soundness
begin

```

```

lemma trace-fri-reachable-header-tie-gap-imp-merkle-or-candidate-available-on-support:

```

```

  assumes support: out ∈ set-dist (execute verify-monad s)

```

```

  and gap: trace-fri-reachable-header-tie-gap s out

```

```

  shows

```

```

    partial-merkle-inconsistency-bad s out ∨

```

```

    trace-fri-header-tied-candidate-available-gap s out

```

```

proof (cases partial-merkle-inconsistency-bad s out)

```

```

  case True

```

```

    then show ?thesis by simp

```

```

next

```

```

  case no-merkle: False

```

```

  from gap have raw: trace-fri-bad-with-reachable-partial-candidate s out

```

```

    unfolding trace-fri-reachable-header-tie-gap-def by simp

```

```

  from trace-fri-bad-with-reachable-partial-candidateE[OF raw]

```

```

  obtain fr0 query-idxs0 trace-openings0 trace-table0 where partial0:

```

```

    accepted-with-partial-trace-openings s out fr0 query-idxs0

```

```

    trace-openings0

```

```

  using raw unfolding trace-fri-bad-with-reachable-partial-candidate-def

```

```

  by blast

```

```

  have acc: accepted out

```

```

    by (rule accepted-with-partial-trace-openings-imp-accepted[OF partial0])

```

```

  from acc obtain result final-state where out-eq:

```

```

    out = Some (result, final-state)

```

```

    unfolding accepted-def by (cases out) auto

```

```

  from verify-monad-trace-fri-bad-extracts-opening-evidence

```

```

    [OF support[unfolded out-eq] raw[unfolded out-eq]]

```

```

  obtain trace-roots trace-bs trace-final dg composition-roots

```

```

    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr1 candidate-query-idxs trace-openings1
    trace-table1
where evidence:
    trace-fri-partial-candidate-opening-evidence s
    (Some (result, final-state)) trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr1 candidate-query-idxs
    trace-openings1 trace-table1
by (rule verify-monad-trace-fri-bad-extracts-opening-evidence
    [OF support[unfolded out-eq] raw[unfolded out-eq]])
have fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
using trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence]
unfolding out-eq .
from accepted-fri-opening-transcript-headerE[OF fri-openings]
obtain fr as rest where header:
    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
by (rule accepted-fri-opening-transcript-headerE[OF fri-openings])
from accepted-fri-opening-transcript-trace-openings-for-query-indices
    [OF fri-openings header out-eq]
obtain trace-openings where partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
by (rule accepted-fri-opening-transcript-trace-openings-for-query-indices
    [OF fri-openings header out-eq])
obtain trace-table where candidate:
    partial-trace-table-candidate trace-table trace-openings
using accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
    [OF partial[unfolded out-eq] no-merkle[unfolded out-eq]]
by blast
show ?thesis
unfolding trace-fri-header-tied-candidate-available-gap-def
by (intro disjI2 exI conjI)
    (rule gap, rule fri-openings, rule header, rule partial,
    rule candidate)
qed

lemma wp-trace-fri-reachable-header-tie-gap-bound-from-merkle-and-candidate-transfer:
assumes merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s)  $s \leq M$ 
and transfer-bound:
    wp-event verify-monad
    (trace-fri-header-tied-candidate-transfer-gap s)  $s \leq T$ 
shows
    wp-event verify-monad

```

$(\text{trace-fri-reachable-header-tie-gap } s) \ s \leq M + T$
proof –
have *split-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-reachable-header-tie-gap } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\lambda \text{out. partial-merkle-inconsistency-bad } s \ \text{out} \ \vee$
 $\text{trace-fri-header-tied-candidate-available-gap } s \ \text{out}) \ s$
by (rule *wp-event-mono-on-support*)
(rule
 $\text{trace-fri-reachable-header-tie-gap-imp-merkle-or-candidate-available-on-support}$)
also have $\dots \leq$
 $\text{wp-event verify-monad (partial-merkle-inconsistency-bad } s) \ s +$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-candidate-available-gap } s) \ s$
by (rule *wp-event-union-bound*)
also have $\dots \leq M + T$
by (rule *add-mono[OF merkle-bound]*)
(rule
 $\text{wp-trace-fri-header-tied-candidate-available-gap-bound-from-transfer-gap}$
 $[\text{OF transfer-bound}]$)
finally show *?thesis* .
qed

lemma *trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-candidate-transfer*:

assumes *header-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \ s \leq Ht$
and *merkle-bound*:
 $\text{wp-event verify-monad (partial-merkle-inconsistency-bad } s) \ s \leq Mt$
and *transfer-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-candidate-transfer-gap } s) \ s \leq Tt$
and *total-bound*: $Ht + (Mt + Tt) \leq \text{trace-fri-error}$
shows *trace-fri-reachable-partial-candidate-reduction-assumption* s

proof –
have *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-reachable-header-tie-gap } s) \ s \leq Mt + Tt$
by (rule
 $\text{wp-trace-fri-reachable-header-tie-gap-bound-from-merkle-and-candidate-transfer}$
 $[\text{OF merkle-bound transfer-bound}]$)
have *reachable-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-reachable-partial-candidate } s) \ s \leq$
 $Ht + (Mt + Tt)$
by (rule *wp-trace-fri-reachable-bound-from-header-tied-and-gap*
 $[\text{OF header-bound gap-bound}]$)
show *?thesis*

```

    unfolding trace-fri-reachable-partial-candidate-reduction-assumption-def
    by (rule order-trans[OF reachable-bound total-bound])
qed

end

end

```

```

theory Soundness-FRI-Composition-Support
  imports Soundness-FRI-Composition-Residuals
begin

```

Small support/extraction facts for the composition FRI reachable gap. These are factored out to avoid large direct proofs that stall theory finalization.

```

context soundness
begin

```

```

definition composition-fri-reachable-verifier-tie-empty-header-gap
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where

```

```

  composition-fri-reachable-verifier-tie-empty-header-gap s out
 $\longleftrightarrow$ 
  ( $\exists$  trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers.
    composition-fri-reachable-verifier-tie-gap s out  $\wedge$ 
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers  $\wedge$ 
    composition-roots = [])

```

```

lemma composition-fri-bad-with-reachable-partial-candidate-imp-accepted:
  assumes composition-fri-bad-with-reachable-partial-candidate s out
  shows accepted out

```

```

proof (rule composition-fri-bad-with-reachable-partial-candidateE[OF assms])

```

```

  fix fr f-roots f-final as dg composition-roots final
    trace-query-idxs trace-openings composition-query-idxs
    composition-openings trace-table composition-table

```

```

  assume evidence:

```

```

    composition-fri-partial-candidate-evidence s out fr f-roots
    f-final as dg composition-roots final trace-query-idxs
    trace-openings composition-query-idxs composition-openings
    trace-table composition-table

```

```

  have partial:

```

```

    accepted-with-partial-initial-openings s out fr f-roots f-final
    as dg composition-roots final trace-query-idxs trace-openings

```

composition-query-idxs composition-openings
by (rule *composition-fri-partial-candidate-evidenceD(1)[OF evidence]*)
then show *accepted out*
by (rule *accepted-with-partial-initial-openings-imp-accepted*)
qed

lemma *composition-fri-reachable-verifier-tie-gap-imp-reachable-bad:*
assumes *composition-fri-reachable-verifier-tie-gap s out*
shows *composition-fri-bad-with-reachable-partial-candidate s out*
using *assms unfolding composition-fri-reachable-verifier-tie-gap-def* **by** *simp*

lemma *composition-fri-reachable-verifier-tie-gap-imp-accepted:*
assumes *composition-fri-reachable-verifier-tie-gap s out*
shows *accepted out*
by (rule *composition-fri-bad-with-reachable-partial-candidate-imp-accepted*)
(rule *composition-fri-reachable-verifier-tie-gap-imp-reachable-bad[OF assms]*)

lemma *composition-fri-reachable-verifier-tie-empty-header-gapI:*
assumes *gap: composition-fri-reachable-verifier-tie-gap s out*
and *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *empty: composition-roots = []*
shows *composition-fri-reachable-verifier-tie-empty-header-gap s out*
unfolding *composition-fri-reachable-verifier-tie-empty-header-gap-def*
by (*intro exI conjI*)
(*rule gap, rule fri-openings, rule empty*)

lemma *composition-fri-reachable-verifier-tie-empty-header-gap-false:*
 $\neg$ *composition-fri-reachable-verifier-tie-empty-header-gap s out*
proof
assume *empty-gap:*
composition-fri-reachable-verifier-tie-empty-header-gap s out
then obtain *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers
where *gap: composition-fri-reachable-verifier-tie-gap s out*
and *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *empty: composition-roots = []*
unfolding *composition-fri-reachable-verifier-tie-empty-header-gap-def*
by *blast*
obtain *fr as rest where verifier-header:*
verifier-header-transcript s fr trace-roots trace-final as fri-dg


```

    composition-roots composition-final rest
  by (rule accepted-fri-opening-transcript-headerE[OF fri-openings])
have raw: composition-fri-bad-with-reachable-partial-candidate s out
  using gap unfolding composition-fri-reachable-verifier-tie-gap-def
  by simp
from composition-fri-bad-with-reachable-partial-candidateE[OF raw]
obtain fr' f-fri-roots f-final as' dg' composition-fri-roots final
  trace-query-idxs trace-openings composition-query-idxs
  composition-openings trace-table composition-table
where evidence:
  composition-fri-partial-candidate-evidence s out fr' f-fri-roots
  f-final as' dg' composition-fri-roots final trace-query-idxs
  trace-openings composition-query-idxs composition-openings
  trace-table composition-table
by blast
have partial:
  accepted-with-partial-initial-openings s out fr' f-fri-roots
  f-final as' dg' composition-fri-roots final trace-query-idxs
  trace-openings composition-query-idxs composition-openings
  by (rule composition-fri-partial-candidate-evidenceD(1)[OF evidence])
have nonempty: composition-fri-roots  $\neq$  []
  by (rule accepted-with-partial-initial-openings-shapes(1)[OF partial])
obtain rest' where candidate-header:
  verifier-header-transcript s fr' f-fri-roots f-final as' dg'
  composition-fri-roots final rest'
  using accepted-with-partial-initial-openings-shapes(2)[OF partial]
  by blast
have composition-fri-roots = composition-roots
  using verifier-header-transcript-unique
  [OF verifier-header candidate-header]
  by simp
then show False
  using empty nonempty by simp
qed

```

lemma wp-composition-fri-reachable-verifier-tie-empty-header-gap-zero:
 wp-event verify-monad
 (composition-fri-reachable-verifier-tie-empty-header-gap s) s = 0
unfolding wp-event-def wp-def dist-expect-def
by (simp add:
 composition-fri-reachable-verifier-tie-empty-header-gap-false)

lemma composition-fri-reachable-verifier-tie-gap-with-fri-openings-empty-or-nonempty:
assumes gap: composition-fri-reachable-verifier-tie-gap s out
and fri-openings:
 accepted-fri-opening-transcript s out trace-roots trace-bs
 trace-final fri-dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers

shows
composition-fri-reachable-verifier-tie-empty-header-gap s out $\vee$
composition-roots $\neq []$
proof (*cases composition-roots* = [])
case *True*
have *composition-fri-reachable-verifier-tie-empty-header-gap s out*
by (*rule composition-fri-reachable-verifier-tie-empty-header-gapI*
[OF gap fri-openings True])
then show ?thesis **by** *simp*
next
case *False*
then show ?thesis **by** *simp*
qed

lemma *composition-fri-verifier-tied-candidate-available-gapI*:
assumes *gap*: *composition-fri-reachable-verifier-tie-gap s out*
and *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *aligned*:
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
and *trace-candidate*:
partial-trace-table-candidate trace-table trace-openings
and *composition-candidate*:
partial-composition-table-candidate composition-table
composition-openings
shows *composition-fri-verifier-tied-candidate-available-gap s out*
unfolding *composition-fri-verifier-tied-candidate-available-gap-def*
by (*intro exI conjI*)
(rule gap, rule fri-openings, rule aligned, rule trace-candidate,
rule composition-candidate)

lemma *composition-fri-reachable-verifier-tie-gap-with-aligned-openings-imp-merkle-or-candidate-available*:
assumes *out-eq*: *out = Some (result, final-state)*
and *gap*: *composition-fri-reachable-verifier-tie-gap s out*
and *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *aligned*:
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
shows

partial-merkle-inconsistency-bad s out $\vee$
composition-fri-verifier-tied-candidate-available-gap s out
proof (*cases partial-merkle-inconsistency-bad s out*)
 case *True*
 then show ?thesis **by** *simp*
next
 case *no-merkle: False*
 obtain *trace-table composition-table* **where** *trace-candidate:*
 partial-trace-table-candidate trace-table trace-openings
 and *composition-candidate:*
 partial-composition-table-candidate composition-table
 composition-openings
 using
 accepted-with-partial-initial-openings-aligned-candidates-if-no-partial-merkle-bad
 [*OF aligned[unfolded out-eq] no-merkle[unfolded out-eq]*]
 by *blast*
 have *available:*
 composition-fri-verifier-tied-candidate-available-gap s out
 by (*rule composition-fri-verifier-tied-candidate-available-gapI*
 [*OF gap fri-openings aligned trace-candidate composition-candidate*])
 then show ?thesis **by** *simp*
qed

lemma *composition-fri-reachable-verifier-tie-gap-with-fri-openings-headerE:*

assumes

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

obtains *fr as rest* **where**

verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest

by (*rule accepted-fri-opening-transcript-headerE*[*OF assms*])

lemma *composition-fri-reachable-verifier-tie-gap-with-fri-openings-header-and-branchE:*

assumes *gap: composition-fri-reachable-verifier-tie-gap s out*

and *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

obtains *fr as rest* **where**

verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest
composition-fri-reachable-verifier-tie-empty-header-gap s out $\vee$
composition-roots $\neq$ []

proof –

obtain *fr as rest* **where** *header:*

verifier-header-transcript s fr trace-roots trace-final as fri-dg

```

    composition-roots composition-final rest
  by (rule composition-fri-reachable-verifier-tie-gap-with-fri-openings-headerE
    [OF fri-openings])
  have branch:
    composition-fri-reachable-verifier-tie-empty-header-gap s out  $\vee$ 
    composition-roots  $\neq []$ 
  by (rule
    composition-fri-reachable-verifier-tie-gap-with-fri-openings-empty-or-nonempty
    [OF gap fri-openings])
  show ?thesis
    by (rule that[OF header branch])
qed

lemma composition-fri-reachable-verifier-tie-gap-nonempty-imp-merkle-or-candidate-available:
  assumes out-eq: out = Some (result, final-state)
  and gap: composition-fri-reachable-verifier-tie-gap s out
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
  and nonempty: composition-roots  $\neq []$ 
  shows
    partial-merkle-inconsistency-bad s out  $\vee$ 
    composition-fri-verifier-tied-candidate-available-gap s out
proof -
  have split:
    partial-merkle-inconsistency-bad s (Some (result, final-state))  $\vee$ 
    ( $\exists$  fr as rest trace-openings composition-openings trace-table
    composition-table.
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest  $\wedge$ 
    accepted-with-partial-initial-openings-aligned s
    (Some (result, final-state)) fr trace-roots trace-final as fri-dg
    composition-roots composition-final fri-query-idxs trace-openings
    composition-openings  $\wedge$ 
    partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
    partial-composition-table-candidate composition-table
    composition-openings)
  by (rule
    accepted-fri-opening-transcript-obtains-aligned-partial-candidates-or-partial-merkle
    [OF fri-openings[unfolded out-eq] nonempty])
  show ?thesis
proof (rule disjE[OF split])
  assume partial-merkle-inconsistency-bad s (Some (result, final-state))
  then show ?thesis
    by (simp add: out-eq)
next
  assume

```

```

     $\exists$  fr as rest trace-openings composition-openings trace-table
      composition-table.
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
      composition-roots composition-final rest  $\wedge$ 
    accepted-with-partial-initial-openings-aligned s
      (Some (result, final-state)) fr trace-roots trace-final as fri-dg
      composition-roots composition-final fri-query-idxs trace-openings
      composition-openings  $\wedge$ 
    partial-trace-table-candidate trace-table trace-openings  $\wedge$ 
    partial-composition-table-candidate composition-table
      composition-openings
  then obtain fr as rest trace-openings composition-openings trace-table
    composition-table where aligned:
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
      trace-final as fri-dg composition-roots composition-final
      fri-query-idxs trace-openings composition-openings
  and trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and composition-candidate:
    partial-composition-table-candidate composition-table
      composition-openings
  by (auto simp: out-eq)
  have available:
    composition-fri-verifier-tied-candidate-available-gap s out
  by (rule composition-fri-verifier-tied-candidate-available-gapI
    [OF gap fri-openings aligned trace-candidate composition-candidate])
  then show ?thesis by simp
qed
qed

lemma composition-fri-reachable-verifier-tie-gap-imp-empty-or-merkle-or-candidate-available-on-support:
  assumes support: out  $\in$  set-dist (execute verify-monad s)
  and gap: composition-fri-reachable-verifier-tie-gap s out
  shows
    composition-fri-reachable-verifier-tie-empty-header-gap s out  $\vee$ 
    partial-merkle-inconsistency-bad s out  $\vee$ 
    composition-fri-verifier-tied-candidate-available-gap s out
proof –
  have raw: composition-fri-bad-with-reachable-partial-candidate s out
  using gap
  unfolding composition-fri-reachable-verifier-tie-gap-def by simp
  have acc: accepted out
  by (rule composition-fri-bad-with-reachable-partial-candidate-imp-accepted
    [OF raw])
  obtain result final-state where out-eq: out = Some (result, final-state)
  using acc unfolding accepted-def by (cases out) auto
  from verify-monad-accepted-fri-opening-transcript[OF support[unfolded out-eq]]
  obtain trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers

```

```

    composition-round-layers
where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
    unfolding out-eq by blast
show ?thesis
proof (cases composition-roots = [])
    case True
    have composition-fri-reachable-verifier-tie-empty-header-gap s out
    by (rule composition-fri-reachable-verifier-tie-empty-header-gapI
        [OF gap fri-openings True])
    then show ?thesis by simp
next
    case False
    have
    partial-merkle-inconsistency-bad s out  $\vee$ 
    composition-fri-verifier-tied-candidate-available-gap s out
    by (rule
        composition-fri-reachable-verifier-tie-gap-nonempty-imp-merkle-or-candidate-available
        [OF out-eq gap fri-openings False])
    then show ?thesis by simp
qed
qed

lemma wp-composition-fri-reachable-verifier-tie-gap-bound-from-empty-merkle-and-candidate-transfer:
assumes empty-bound:
    wp-event verify-monad
    (composition-fri-reachable-verifier-tie-empty-header-gap s)  $s \leq E$ 
and merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s)  $s \leq M$ 
and transfer-bound:
    wp-event verify-monad
    (composition-fri-verifier-tied-candidate-transfer-gap s)  $s \leq T$ 
shows
    wp-event verify-monad
    (composition-fri-reachable-verifier-tie-gap s)  $s \leq E + (M + T)$ 
proof –
    have split-bound:
    wp-event verify-monad
    (composition-fri-reachable-verifier-tie-gap s)  $s \leq$ 
    wp-event verify-monad
    ( $\lambda$ out.
        composition-fri-reachable-verifier-tie-empty-header-gap s out  $\vee$ 
        partial-merkle-inconsistency-bad s out  $\vee$ 
        composition-fri-verifier-tied-candidate-available-gap s out) s
    by (rule wp-event-mono-on-support)
    (rule

```

composition-fri-reachable-verifier-tie-gap-imp-empty-or-merkle-or-candidate-available-on-support)
also have ... $\leq$
wp-event verify-monad
(composition-fri-reachable-verifier-tie-empty-header-gap s) s +
wp-event verify-monad
(λout.
partial-merkle-inconsistency-bad s out ∨
composition-fri-verifier-tied-candidate-available-gap s out) s
by (rule *wp-event-union-bound*)
also have ... $\leq$
wp-event verify-monad
(composition-fri-reachable-verifier-tie-empty-header-gap s) s +
(wp-event verify-monad (partial-merkle-inconsistency-bad s) s +
wp-event verify-monad
(composition-fri-verifier-tied-candidate-available-gap s) s)
by (rule *add-mono[OF order-refl wp-event-union-bound]*)
also have ... $\leq E + (M + T)$
by (intro *add-mono empty-bound merkle-bound*
wp-composition-fri-verifier-tied-candidate-available-gap-bound-from-transfer-gap
transfer-bound)
finally show ?thesis .
qed

lemma *composition-fri-reachable-partial-candidate-reduction-from-split-residual-bounds:*

fixes *R A P M N S G E Q T :: prob*

assumes *sampled-bound:*

wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R

and *slot-gap-bound:*

wp-event verify-monad
(composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
s) s ≤ A

and *verifier-partial-merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P

and *missing-bound:*

wp-event verify-monad
(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
s ≤ M

and *next-gap-bound:*

wp-event verify-monad
(composition-fri-verifier-tied-next-value-replay-gap s) s ≤ N

and *successor-gap-bound:*

wp-event verify-monad
(composition-fri-verifier-tied-successor-replay-gap s) s ≤ S

and *final-gap-bound:*

wp-event verify-monad
(composition-fri-verifier-tied-final-value-replay-gap s) s ≤ G

and *empty-bound:*

wp-event verify-monad

(composition-fri-reachable-verifier-tie-empty-header-gap s) $s \leq E$
and gap-partial-merkle-bound:
 wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq Q$
and candidate-transfer-bound:
 wp-event verify-monad
 (composition-fri-verifier-tied-candidate-transfer-gap s) $s \leq T$
and total-bound:
 $R + (((P + M) + N + S + G + P) + A) + (E + (Q + T))$
 $\leq$ composition-fri-error
shows composition-fri-reachable-partial-candidate-reduction-assumption s
proof –
have verifier-bound:
 wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-partial-candidate s) s
 $\leq R + (((P + M) + N + S + G + P) + A)$
by (rule
 wp-composition-fri-verifier-tied-bound-from-sampled-slot-recorded-chunk-gap-and-residuals
 [OF sampled-bound slot-gap-bound verifier-partial-merkle-bound
 missing-bound next-gap-bound successor-gap-bound final-gap-bound])
have gap-bound:
 wp-event verify-monad
 (composition-fri-reachable-verifier-tie-gap s) $s \leq$
 $E + (Q + T)$
by (rule
 wp-composition-fri-reachable-verifier-tie-gap-bound-from-empty-merkle-and-candidate-transfer
 [OF empty-bound gap-partial-merkle-bound candidate-transfer-bound])
show ?thesis
by (rule
 composition-fri-reachable-partial-candidate-reduction-from-verifier-tied-and-gap
 [OF verifier-bound gap-bound total-bound])
qed

lemma composition-fri-reachable-partial-candidate-reduction-from-split-residual-bounds-with-recorded-missing-zero:
fixes $R A P N S G E Q T :: \text{prob}$
assumes sampled-bound:
 wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-sampled-layer-chain s) $s \leq R$
and slot-gap-bound:
 wp-event verify-monad
 (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
 s) $s \leq A$
and verifier-partial-merkle-bound:
 wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq P$
and next-gap-bound:
 wp-event verify-monad
 (composition-fri-verifier-tied-next-value-replay-gap s) $s \leq N$
and successor-gap-bound:
 wp-event verify-monad
 (composition-fri-verifier-tied-successor-replay-gap s) $s \leq S$


```

and final-gap-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-final-value-replay-gap s)  $s \leq G$ 
and empty-bound:
  wp-event verify-monad
    (composition-fri-reachable-verifier-tie-empty-header-gap s)  $s \leq E$ 
and gap-partial-merkle-bound:
  wp-event verify-monad (partial-merkle-inconsistency-bad s)  $s \leq Q$ 
and candidate-transfer-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-candidate-transfer-gap s)  $s \leq T$ 
and total-bound:
   $R + (((P + 0) + N + S + G + P) + A) + (E + (Q + T))$ 
     $\leq \text{composition-fri-error}$ 
shows composition-fri-reachable-partial-candidate-reduction-assumption s
by (rule
  composition-fri-reachable-partial-candidate-reduction-from-split-residual-bounds
  [where  $M = 0$ ])
  (rule sampled-bound,
   rule slot-gap-bound,
   rule verifier-partial-merkle-bound,
   rule wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero,
   rule next-gap-bound,
   rule successor-gap-bound,
   rule final-gap-bound,
   rule empty-bound,
   rule gap-partial-merkle-bound,
   rule candidate-transfer-bound,
   rule total-bound)

end

end

```

```

theory Soundness-FRI-Candidate-Transfer-Support
imports
  Soundness-FRI-Trace-Candidate-Gap
  Soundness-FRI-Composition-Support
begin

```

Small destructors for the remaining FRI candidate-transfer branches. These branches are intentionally not closed by complete-table uniqueness: sampled partial openings can admit multiple completions. The lemmas here expose the exact low/non-low candidate shape that later proofs must charge to a partial-opening inconsistency or a narrower candidate-indexed event.

```

context soundness
begin

```

definition *trace-fri-header-tied-low-candidate-available*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-low-candidate-available s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final fri-dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers} \wedge$
 $\text{verifier-header-transcript s fr trace-roots trace-final as fri-dg}$
 $\text{ composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{trace-table-low-degree trace-table})$

definition *trace-fri-header-tied-low-candidate-gap*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-low-candidate-gap s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{trace-fri-reachable-header-tie-gap s out} \wedge$
 $\text{accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{ trace-final fri-dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers} \wedge$
 $\text{verifier-header-transcript s fr trace-roots trace-final as fri-dg}$
 $\text{ composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings s out fr fri-query-idxs}$
 $\text{ trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{trace-table-low-degree trace-table})$

definition *composition-fri-verifier-tied-misaligned-candidate-classification*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-misaligned-candidate-classification s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings composition-openings}$
 $\text{ trace-table composition-table.})$

$\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\text{accepted-with-partial-initial-openings-aligned } s \text{ out fr trace-roots}$
 $\text{trace-final as fri-dg composition-roots composition-final}$
 $\text{fri-query-idxs trace-openings composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\neg (\text{trace-table-low-degree trace-table} \wedge$
 $\neg \text{composition-table-low-degree maxDegree composition-table}))$

definition *composition-fri-verifier-tied-transcript-consistent-misaligned-classification*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-transcript-consistent-misaligned-classification}$
 $s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings}$
 $\text{composition-openings trace-table composition-table.}$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as fri-dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-initial-openings-aligned-transcript-consistent } s$
 $\text{out fr trace-roots trace-final as fri-dg composition-roots}$
 $\text{composition-final fri-query-idxs trace-openings composition-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{partial-composition-table-candidate composition-table}$
 $\text{composition-openings} \wedge$
 $\neg (\text{trace-table-low-degree trace-table} \wedge$
 $\neg \text{composition-table-low-degree maxDegree composition-table}))$

definition *composition-fri-verifier-tied-transcript-consistent-classification-cover*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-verifier-tied-transcript-consistent-classification-cover}$
 $s \text{ out} \longleftrightarrow$
 $\text{composition-bad-with-partial-candidates } s \text{ out} \vee$
 $\text{trace-fri-bad-with-header-tied-partial-candidate } s \text{ out} \vee$
 $\text{query-bad-with-aligned-transcript-partial-candidates } s \text{ out}$

lemma *trace-fri-header-tied-candidate-transfer-gapD:*

assumes *trace-fri-header-tied-candidate-transfer-gap s out*
shows
trace-fri-bad-with-reachable-partial-candidate s out
trace-fri-header-tied-low-candidate-available s out
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as rest trace-openings
trace-table **where**
bad: trace-fri-bad-with-reachable-partial-candidate s out
and *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *header:*
verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *candidate:*
partial-trace-table-candidate trace-table trace-openings
and *low: trace-table-low-degree trace-table*
unfolding *trace-fri-header-tied-candidate-transfer-gap-def*
by *blast*
show *trace-fri-bad-with-reachable-partial-candidate s out*
by *(rule bad)*
show *trace-fri-header-tied-low-candidate-available s out*
unfolding *trace-fri-header-tied-low-candidate-available-def*
by *(intro exI conjI)*
(rule fri-openings, rule header, rule partial, rule candidate,
rule low)
qed

lemma *trace-fri-header-tied-candidate-transfer-gap-imp-accepted:*
assumes *trace-fri-header-tied-candidate-transfer-gap s out*
shows *accepted out*
proof –
have *bad: trace-fri-bad-with-reachable-partial-candidate s out*
by *(rule trace-fri-header-tied-candidate-transfer-gapD(1)[OF assms])*
from *trace-fri-bad-with-reachable-partial-candidateE[OF bad]*
obtain *fr query-idxs trace-openings trace-table* **where** *evidence:*
trace-fri-partial-candidate-evidence s out fr query-idxs
trace-openings trace-table
by *blast*
have *partial:*
accepted-with-partial-trace-openings s out fr query-idxs
trace-openings

by (rule trace-fri-partial-candidate-evidenceD(1)[OF evidence])
 show ?thesis
 by (rule accepted-with-partial-trace-openings-imp-accepted[OF partial])
 qed

lemma wp-trace-fri-header-tied-candidate-transfer-gap-bound-from-low-candidate:
 assumes low-bound:
 wp-event verify-monad
 (trace-fri-header-tied-low-candidate-available s) $s \leq T$
 shows
 wp-event verify-monad
 (trace-fri-header-tied-candidate-transfer-gap s) $s \leq T$
proof –
 have wp-event verify-monad
 (trace-fri-header-tied-candidate-transfer-gap s) $s \leq$
 wp-event verify-monad
 (trace-fri-header-tied-low-candidate-available s) s
 by (rule wp-event-mono)
 (rule trace-fri-header-tied-candidate-transfer-gapD(2))
 then show ?thesis
 by (rule order-trans[OF low-bound])
 qed

lemma trace-fri-header-tied-low-candidate-gap-imp-low-candidate-available:
 assumes trace-fri-header-tied-low-candidate-gap s out
 shows trace-fri-header-tied-low-candidate-available s out
 using assms
 unfolding trace-fri-header-tied-low-candidate-gap-def
 trace-fri-header-tied-low-candidate-available-def
 by blast

lemma trace-fri-header-tied-low-candidate-gap-imp-transfer-gap:
 assumes trace-fri-header-tied-low-candidate-gap s out
 shows trace-fri-header-tied-candidate-transfer-gap s out
proof –
 from assms obtain trace-roots trace-bs trace-final fri-dg
 composition-roots composition-bs composition-final fri-query-idxs
 trace-round-layers composition-round-layers fr as rest trace-openings
 trace-table **where**
 gap: trace-fri-reachable-header-tie-gap s out
 and fri-openings:
 accepted-fri-opening-transcript s out trace-roots trace-bs
 trace-final fri-dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers
 and header:
 verifier-header-transcript s fr trace-roots trace-final as fri-dg
 composition-roots composition-final rest
 and partial:

```

    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  and candidate:
    partial-trace-table-candidate trace-table trace-openings
  and low: trace-table-low-degree trace-table
  unfolding trace-fri-header-tied-low-candidate-gap-def by blast
show ?thesis
  by (rule trace-fri-header-candidate-low-imp-transfer-gap
      [OF gap fri-openings header partial candidate low])
qed

lemma trace-fri-header-tied-candidate-available-gap-imp-low-candidate-gap:
  assumes trace-fri-header-tied-candidate-available-gap s out
  shows trace-fri-header-tied-low-candidate-gap s out
proof -
  from assms obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr as rest trace-openings
    trace-table where
  gap: trace-fri-reachable-header-tie-gap s out
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  and candidate:
    partial-trace-table-candidate trace-table trace-openings
  unfolding trace-fri-header-tied-candidate-available-gap-def by blast
show ?thesis
proof (cases trace-table-low-degree trace-table)
  case True
  show ?thesis
    unfolding trace-fri-header-tied-low-candidate-gap-def
    by (intro exI conjI)
      (rule gap, rule fri-openings, rule header, rule partial,
       rule candidate, rule True)
  next
  case False
  have False
    by (rule trace-fri-header-candidate-non-low-contradicts-header-tie-gap
        [OF gap fri-openings header partial candidate False])
  then show ?thesis by simp
qed

```

qed

lemma *wp-trace-fri-header-tied-candidate-available-gap-bound-from-low-candidate-gap:*

assumes *low-gap-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-gap } s) \ s \leq T$

shows

wp-event verify-monad

$(\text{trace-fri-header-tied-candidate-available-gap } s) \ s \leq T$

proof –

have *wp-event verify-monad*

$(\text{trace-fri-header-tied-candidate-available-gap } s) \ s \leq$

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-gap } s) \ s$

by (rule *wp-event-mono*)

(rule *trace-fri-header-tied-candidate-available-gap-imp-low-candidate-gap*)

then show *?thesis*

by (rule *order-trans*[*OF* - *low-gap-bound*])

qed

lemma *wp-trace-fri-reachable-header-tie-gap-bound-from-merkle-and-low-candidate-gap:*

assumes *merkle-bound:*

wp-event verify-monad $(\text{partial-merkle-inconsistency-bad } s) \ s \leq M$

and *low-gap-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-gap } s) \ s \leq T$

shows

wp-event verify-monad

$(\text{trace-fri-reachable-header-tie-gap } s) \ s \leq M + T$

proof –

have *split-bound:*

wp-event verify-monad

$(\text{trace-fri-reachable-header-tie-gap } s) \ s \leq$

wp-event verify-monad

$(\lambda \text{out. } \text{partial-merkle-inconsistency-bad } s \ \text{out} \ \vee$

$\text{trace-fri-header-tied-candidate-available-gap } s \ \text{out}) \ s$

by (rule *wp-event-mono-on-support*)

(rule

trace-fri-reachable-header-tie-gap-imp-merkle-or-candidate-available-on-support)

also have $\dots \leq$

wp-event verify-monad $(\text{partial-merkle-inconsistency-bad } s) \ s +$

wp-event verify-monad

$(\text{trace-fri-header-tied-candidate-available-gap } s) \ s$

by (rule *wp-event-union-bound*)

also have $\dots \leq M + T$

by (rule *add-mono*[*OF* *merkle-bound*])

(rule

wp-trace-fri-header-tied-candidate-available-gap-bound-from-low-candidate-gap

[*OF* *low-gap-bound*])

finally show ?thesis .
qed

lemma *trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-low-candidate-gap:*

assumes *header-bound:*

wp-event verify-monad

(trace-fri-bad-with-header-tied-partial-candidate s) s ≤ Ht

and *merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ Mt

and *low-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-low-candidate-gap s) s ≤ Tt

and *total-bound: Ht + (Mt + Tt) ≤ trace-fri-error*

shows *trace-fri-reachable-partial-candidate-reduction-assumption s*

proof –

have *gap-bound:*

wp-event verify-monad

(trace-fri-reachable-header-tie-gap s) s ≤ Mt + Tt

by (rule

wp-trace-fri-reachable-header-tie-gap-bound-from-merkle-and-low-candidate-gap
[OF *merkle-bound low-gap-bound*])

have *reachable-bound:*

wp-event verify-monad

(trace-fri-bad-with-reachable-partial-candidate s) s ≤
Ht + (Mt + Tt)

by (rule *wp-trace-fri-reachable-bound-from-header-tied-and-gap*

[OF *header-bound gap-bound*])

show ?thesis

unfolding *trace-fri-reachable-partial-candidate-reduction-assumption-def*

by (rule *order-trans*[OF *reachable-bound total-bound*])

qed

lemma *composition-fri-verifier-tied-candidate-transfer-gapD:*

assumes *composition-fri-verifier-tied-candidate-transfer-gap s out*

shows

composition-fri-bad-with-reachable-partial-candidate s out

composition-fri-verifier-tied-misaligned-candidate-classification s out

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*

composition-roots composition-bs composition-final fri-query-idxs

trace-round-layers composition-round-layers fr as trace-openings

composition-openings trace-table composition-table **where**

bad: composition-fri-bad-with-reachable-partial-candidate s out

and *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final fri-dg composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers

and *aligned:*


```

    accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final
    fri-query-idxs trace-openings composition-openings
  and trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and composition-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  and classified:
    ¬ (trace-table-low-degree trace-table ∧
      ¬ composition-table-low-degree maxDegree composition-table)
  unfolding composition-fri-verifier-tied-candidate-transfer-gap-def
  by blast
show composition-fri-bad-with-reachable-partial-candidate s out
  by (rule bad)
show
  composition-fri-verifier-tied-misaligned-candidate-classification s out
  unfolding
    composition-fri-verifier-tied-misaligned-candidate-classification-def
  by (intro exI conjI)
    (rule fri-openings, rule aligned, rule trace-candidate,
     rule composition-candidate, rule classified)
qed

lemma composition-fri-verifier-tied-candidate-transfer-gap-imp-accepted:
  assumes composition-fri-verifier-tied-candidate-transfer-gap s out
  shows accepted out
  by (rule composition-fri-bad-with-reachable-partial-candidate-imp-accepted)
    (rule composition-fri-verifier-tied-candidate-transfer-gapD(1)[OF assms])

lemma wp-composition-fri-verifier-tied-candidate-transfer-gap-bound-from-misaligned-classification:
  assumes classification-bound:
    wp-event verify-monad
    (composition-fri-verifier-tied-misaligned-candidate-classification s)
    s ≤ T
  shows
    wp-event verify-monad
    (composition-fri-verifier-tied-candidate-transfer-gap s) s ≤ T
proof -
  have wp-event verify-monad
    (composition-fri-verifier-tied-candidate-transfer-gap s) s ≤
    wp-event verify-monad
    (composition-fri-verifier-tied-misaligned-candidate-classification s)
    s
  by (rule wp-event-mono)
    (rule composition-fri-verifier-tied-candidate-transfer-gapD(2))
  then show ?thesis
  by (rule order-trans[OF - classification-bound])
qed

```

lemma *composition-fri-verifier-tied-transcript-consistent-misaligned-classification-imp-misaligned-classification:*
assumes
composition-fri-verifier-tied-transcript-consistent-misaligned-classification
s out
shows *composition-fri-verifier-tied-misaligned-candidate-classification*
s out
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as rest trace-openings
composition-openings trace-table composition-table **where**
fri-openings:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *partial:*
accepted-with-partial-initial-openings-aligned-transcript-consistent s
out fr trace-roots trace-final as fri-dg composition-roots
composition-final fri-query-idxs trace-openings composition-openings
and *trace-candidate:*
partial-trace-table-candidate trace-table trace-openings
and *composition-candidate:*
partial-composition-table-candidate composition-table
composition-openings
and *classified:*
 $\neg$ (*trace-table-low-degree trace-table* $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table*)
unfolding
composition-fri-verifier-tied-transcript-consistent-misaligned-classification-def
by *blast*
have *aligned:*
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final fri-query-idxs
trace-openings composition-openings
using *partial*
by (*blast dest:*
accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
accepted-with-partial-initial-openings-aligned-consistent-imp-aligned)
show *?thesis*
unfolding
composition-fri-verifier-tied-misaligned-candidate-classification-def
by (*intro exI conjI*)
(rule fri-openings, rule aligned, rule trace-candidate,
rule composition-candidate, rule classified)
qed

lemma *composition-fri-verifier-tied-transcript-consistent-misaligned-classification-imp-aligned-bad-event:*

assumes *false-statement*: $\neg$ *exists-valid-trace*
and *classification*:
composition-fri-verifier-tied-transcript-consistent-misaligned-classification
s out
shows
soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate
s out
proof –
from *classification* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as rest trace-openings
composition-openings trace-table composition-table **where**
fri-openings:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *header*:
verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest
and *partial*:
accepted-with-partial-initial-openings-aligned-transcript-consistent s
out fr trace-roots trace-final as fri-dg composition-roots
composition-final fri-query-idxs trace-openings composition-openings
and *trace-candidate*:
partial-trace-table-candidate trace-table trace-openings
and *composition-candidate*:
partial-composition-table-candidate composition-table
composition-openings
unfolding
composition-fri-verifier-tied-transcript-consistent-misaligned-classification-def
by *blast*
show *?thesis*
by (*rule*
accepted-verifier-tied-aligned-transcript-partial-candidate-partition
[*OF false-statement fri-openings partial header trace-candidate*
composition-candidate])
qed

lemma *wp-composition-fri-verifier-tied-transcript-consistent-misaligned-classification-bound-from-aligned-bad-event*:
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *aligned-bound*:
wp-event verify-monad
(*soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate*
s) $s \leq T$
shows
wp-event verify-monad
(*composition-fri-verifier-tied-transcript-consistent-misaligned-classification*
s) $s \leq T$

```

proof –
  have wp-event verify-monad
    (composition-fri-verifier-tied-transcript-consistent-misaligned-classification
     s) s ≤
  wp-event verify-monad
    (soundness-bad-event-verifier-tied-aligned-transcript-partial-candidate
     s) s
  by (rule wp-event-mono)
    (rule
     composition-fri-verifier-tied-transcript-consistent-misaligned-classification-imp-aligned-bad-event
     [OF false-statement])
  then show ?thesis
    by (rule order-trans[OF - aligned-bound])
qed

lemma composition-fri-verifier-tied-transcript-consistent-misaligned-classification-imp-cover:
  assumes false-statement:  $\neg$  exists-valid-trace
  and classification:
    composition-fri-verifier-tied-transcript-consistent-misaligned-classification
    s out
  shows
    composition-fri-verifier-tied-transcript-consistent-classification-cover
    s out
proof –
  from classification obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr as rest trace-openings
    composition-openings trace-table composition-table where
    fri-openings:
      accepted-fri-opening-transcript s out trace-roots trace-bs
      trace-final fri-dg composition-roots composition-bs
      composition-final fri-query-idxs trace-round-layers
      composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-initial-openings-aligned-transcript-consistent s
    out fr trace-roots trace-final as fri-dg composition-roots
    composition-final fri-query-idxs trace-openings composition-openings
  and trace-candidate:
    partial-trace-table-candidate trace-table trace-openings
  and composition-candidate:
    partial-composition-table-candidate composition-table
    composition-openings
  and classified:
     $\neg$  (trace-table-low-degree trace-table  $\wedge$ 
       $\neg$  composition-table-low-degree maxDegree composition-table)
  unfolding

```

```

    composition-fri-verifier-tied-transcript-consistent-misaligned-classification-def
  by blast
show ?thesis
proof (cases trace-table-low-degree trace-table)
  case False
  have aligned:
    accepted-with-partial-initial-openings-aligned-consistent s out fr
      trace-roots trace-final as fri-dg composition-roots
      composition-final fri-query-idxs trace-openings composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
  have aligned0:
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
      trace-final as fri-dg composition-roots composition-final
      fri-query-idxs trace-openings composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-consistent-imp-aligned
    [OF aligned])
  have trace-partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
      trace-openings
  by (rule accepted-with-partial-initial-openings-aligned-shapes(3)
    [OF aligned0])
  have trace-fri-bad-with-header-tied-partial-candidate s out
    unfolding trace-fri-bad-with-header-tied-partial-candidate-def
  by (intro exI conjI)
    (rule fri-openings, rule header, rule trace-partial,
    rule trace-candidate, rule False)
  then show ?thesis
    unfolding
      composition-fri-verifier-tied-transcript-consistent-classification-cover-def
    by blast
next
case True
note trace-low = True
have composition-low:
  composition-table-low-degree maxDegree composition-table
  using classified trace-low by blast
show ?thesis
proof (cases all-queries-consistent trace-table composition-table as)
  case False
  have query-bad-with-aligned-transcript-partial-candidates s out
    unfolding query-bad-with-aligned-transcript-partial-candidates-def
  using partial trace-candidate composition-candidate trace-low
    composition-low False
  by blast
  then show ?thesis
    unfolding

```

```

    composition-fri-verifier-tied-transcript-consistent-classification-cover-def
  by blast
next
case True
note all-queries = True
from trace-low obtain f where
  deg-f: degree f < clength
  and trace-table-eq: trace-table = map (poly f) eval-domain
  unfolding trace-table-low-degree-def by blast
have violated: violated-constraints f ≠ {}
  by (rule false-statement-violated-constraints
    [OF false-statement deg-f])
have aligned:
  accepted-with-partial-initial-openings-aligned-consistent s out fr
  trace-roots trace-final as fri-dg composition-roots
  composition-final fri-query-idxs trace-openings
  composition-openings
  by (rule
    accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
    [OF partial])
have composition-bad-with-partial-candidates s out
  unfolding composition-bad-with-partial-candidates-def
  using accepted-with-partial-initial-openings-aligned-consistent-imp-unaligned
    [OF aligned]
  trace-candidate composition-candidate deg-f trace-table-eq
  violated composition-low all-queries
  by blast
then show ?thesis
  unfolding
    composition-fri-verifier-tied-transcript-consistent-classification-cover-def
  by blast
qed
qed
qed

```

lemma *wp-composition-fri-verifier-tied-transcript-consistent-classification-cover-union-bound:*
assumes *comp-bound:*
 wp-event verify-monad
 (composition-bad-with-partial-candidates s) s ≤ C
and *trace-bound:*
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) s ≤ Ft
and *query-bound:*
 wp-event verify-monad
 (query-bad-with-aligned-transcript-partial-candidates s) s ≤ Q
shows
 wp-event verify-monad
 (composition-fri-verifier-tied-transcript-consistent-classification-cover
 s) s ≤ C + Ft + Q

proof –
have *wp-event verify-monad*
 (*composition-fri-verifier-tied-transcript-consistent-classification-cover*
s) $s \leq$
wp-event verify-monad
 (*composition-bad-with-partial-candidates s*) $s +$
wp-event verify-monad
 (*trace-fri-bad-with-header-tied-partial-candidate s*) $s +$
wp-event verify-monad
 (*query-bad-with-aligned-transcript-partial-candidates s*) s
proof –
have *wp-event verify-monad*
 (*composition-fri-verifier-tied-transcript-consistent-classification-cover*
s) $s \leq$
wp-event verify-monad
 (*composition-bad-with-partial-candidates s*) $s +$
wp-event verify-monad
 ($\lambda out.$
trace-fri-bad-with-header-tied-partial-candidate s out $\vee$
query-bad-with-aligned-transcript-partial-candidates s out) s
unfolding
composition-fri-verifier-tied-transcript-consistent-classification-cover-def
by (*rule wp-event-union-bound*)
also have $\dots \leq$
wp-event verify-monad
 (*composition-bad-with-partial-candidates s*) $s +$
wp-event verify-monad
 (*trace-fri-bad-with-header-tied-partial-candidate s*) $s +$
wp-event verify-monad
 (*query-bad-with-aligned-transcript-partial-candidates s*) s)
by (*rule add-mono[OF order-refl wp-event-union-bound]*)
finally show *?thesis* **by** (*simp add: algebra-simps*)
qed
also have $\dots \leq C + Ft + Q$
by (*intro add-mono comp-bound trace-bound query-bound*)
finally show *?thesis* .
qed

lemma *wp-composition-fri-verifier-tied-transcript-consistent-misaligned-classification-bound-from-cover-components:*
assumes *false-statement: $\neg$ exists-valid-trace*
and *comp-bound:*
wp-event verify-monad
 (*composition-bad-with-partial-candidates s*) $s \leq C$
and *trace-bound:*
wp-event verify-monad
 (*trace-fri-bad-with-header-tied-partial-candidate s*) $s \leq Ft$
and *query-bound:*
wp-event verify-monad
 (*query-bad-with-aligned-transcript-partial-candidates s*) $s \leq Q$

shows
wp-event verify-monad
(composition-fri-verifier-tied-transcript-consistent-misaligned-classification
s) s ≤ C + Ft + Q
proof –
have *wp-event verify-monad*
(composition-fri-verifier-tied-transcript-consistent-misaligned-classification
s) s ≤
wp-event verify-monad
(composition-fri-verifier-tied-transcript-consistent-classification-cover
s) s
by (rule *wp-event-mono*)
(rule
composition-fri-verifier-tied-transcript-consistent-misaligned-classification-imp-cover
[*OF false-statement*])
also have ... ≤ *C + Ft + Q*
by (rule
wp-composition-fri-verifier-tied-transcript-consistent-classification-cover-union-bound
[*OF comp-bound trace-bound query-bound*])
finally show ?thesis .
qed

lemma *composition-fri-verifier-tied-candidate-available-gap-imp-empty-or-merkle-or-transcript-consistent-classification*
assumes *available:*

composition-fri-verifier-tied-candidate-available-gap s out

shows

composition-fri-reachable-verifier-tie-empty-header-gap s out ∨

partial-merkle-inconsistency-bad s out ∨

composition-fri-verifier-tied-transcript-consistent-misaligned-classification
s out

proof –

from *available* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as trace-openings
composition-openings trace-table composition-table **where**
gap: composition-fri-reachable-verifier-tie-gap s out

and *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

and *aligned:*

accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *composition-candidate:*

partial-composition-table-candidate composition-table


```

    composition-openings
  unfolding composition-fri-verifier-tied-candidate-available-gap-def
  by blast
from fri-openings obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-fri-opening-transcript-def by blast
show ?thesis
proof (cases composition-roots = [])
  case True
  have composition-fri-reachable-verifier-tie-empty-header-gap s out
    by (rule composition-fri-reachable-verifier-tie-empty-header-gapI
      [OF gap fri-openings True])
  then show ?thesis by simp
next
case comp-nonempty: False
show ?thesis
proof (cases partial-merkle-inconsistency-bad s out)
  case True
  then show ?thesis by simp
next
case no-merkle: False
from accepted-fri-opening-transcript-obtains-aligned-partial-openings
  [OF fri-openings[unfolded out-eq] comp-nonempty]
obtain fr' as' rest' trace-openings' composition-openings'
where header':
  verifier-header-transcript s fr' trace-roots trace-final as' fri-dg
  composition-roots composition-final rest'
and partial':
  accepted-with-partial-initial-openings-aligned-transcript-consistent s
  (Some (result, final-state)) fr' trace-roots trace-final as'
  fri-dg composition-roots composition-final fri-query-idxs
  trace-openings' composition-openings'
by blast
have aligned':
  accepted-with-partial-initial-openings-aligned s out fr'
  trace-roots trace-final as' fri-dg composition-roots
  composition-final fri-query-idxs trace-openings'
  composition-openings'
using partial'[folded out-eq]
by (blast dest:
  accepted-with-partial-initial-openings-aligned-transcript-consistent-imp-aligned
  accepted-with-partial-initial-openings-aligned-consistent-imp-aligned)
obtain trace-table' composition-table'
where trace-candidate':
  partial-trace-table-candidate trace-table' trace-openings'
and composition-candidate':
  partial-composition-table-candidate composition-table'
  composition-openings'
using

```

```

    accepted-with-partial-initial-openings-aligned-candidates-if-no-partial-merkle-bad
    [OF aligned'[unfolded out-eq] no-merkle[unfolded out-eq]]
  by blast
have classified:
  ¬ (trace-table-low-degree trace-table' ∧
    ¬ composition-table-low-degree maxDegree composition-table')
proof
  assume bad-class:
    trace-table-low-degree trace-table' ∧
    ¬ composition-table-low-degree maxDegree composition-table'
  have composition-fri-bad-with-verifier-tied-partial-candidate s out
  unfolding composition-fri-bad-with-verifier-tied-partial-candidate-def
  by (intro exI conjI)
    (rule fri-openings, rule aligned', rule trace-candidate',
     rule composition-candidate', rule bad-class[THEN conjunct1],
     rule bad-class[THEN conjunct2])
  then show False
  using gap
  unfolding composition-fri-reachable-verifier-tie-gap-def by simp
qed
have
  composition-fri-verifier-tied-transcript-consistent-misaligned-classification
  s out
  unfolding
    composition-fri-verifier-tied-transcript-consistent-misaligned-classification-def
  by (intro exI conjI)
    (rule fri-openings, rule header'[folded out-eq],
     rule partial'[folded out-eq], rule trace-candidate',
     rule composition-candidate', rule classified)
  then show ?thesis by simp
qed
qed
qed
lemma wp-composition-fri-verifier-tied-candidate-available-gap-bound-from-empty-merkle-and-transcript-consistent-c
  assumes empty-bound:
    wp-event verify-monad
      (composition-fri-reachable-verifier-tie-empty-header-gap s) s ≤ E
  and merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M
  and classification-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-transcript-consistent-misaligned-classification
       s) s ≤ T
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-candidate-available-gap s) s ≤
      E + (M + T)
proof -

```

have *wp-event verify-monad*
 (*composition-fri-verifier-tied-candidate-available-gap s*) $s \leq$
wp-event verify-monad
 ($\lambda out.$
 composition-fri-reachable-verifier-tie-empty-header-gap s out $\vee$
 partial-merkle-inconsistency-bad s out $\vee$
 composition-fri-verifier-tied-transcript-consistent-misaligned-classification
 s out) s
by (*rule wp-event-mono*)
 (*rule*
 composition-fri-verifier-tied-candidate-available-gap-imp-empty-or-merkle-or-transcript-consistent-classification)
also have ... $\leq$
wp-event verify-monad
 (*composition-fri-reachable-verifier-tie-empty-header-gap s*) $s +$
wp-event verify-monad
 ($\lambda out.$
 partial-merkle-inconsistency-bad s out $\vee$
 composition-fri-verifier-tied-transcript-consistent-misaligned-classification
 s out) s
by (*rule wp-event-union-bound*)
also have ... $\leq$
wp-event verify-monad
 (*composition-fri-reachable-verifier-tie-empty-header-gap s*) $s +$
 (*wp-event verify-monad* (*partial-merkle-inconsistency-bad s*)) $s +$
wp-event verify-monad
 (*composition-fri-verifier-tied-transcript-consistent-misaligned-classification*
 s) s
by (*rule add-mono[OF order-refl wp-event-union-bound]*)
also have ... $\leq E + (M + T)$
by (*intro add-mono empty-bound merkle-bound classification-bound*)
finally show ?thesis .
qed

lemma *wp-composition-fri-reachable-verifier-tie-gap-bound-from-empty-merkle-and-candidate-available:*
assumes *empty-bound:*
 wp-event verify-monad
 (*composition-fri-reachable-verifier-tie-empty-header-gap s*) $s \leq E$
and *merkle-bound:*
 wp-event verify-monad (*partial-merkle-inconsistency-bad s*) $s \leq M$
and *available-bound:*
 wp-event verify-monad
 (*composition-fri-verifier-tied-candidate-available-gap s*) $s \leq T$
shows
 wp-event verify-monad
 (*composition-fri-reachable-verifier-tie-gap s*) $s \leq E + (M + T)$
proof –
have *split-bound:*
 wp-event verify-monad
 (*composition-fri-reachable-verifier-tie-gap s*) $s \leq$

```

wp-event verify-monad
  (λout.
    composition-fri-reachable-verifier-tie-empty-header-gap s out ∨
    partial-merkle-inconsistency-bad s out ∨
    composition-fri-verifier-tied-candidate-available-gap s out) s
by (rule wp-event-mono-on-support)
  (rule
    composition-fri-reachable-verifier-tie-gap-imp-empty-or-merkle-or-candidate-available-on-support)
also have ... ≤
  wp-event verify-monad
    (composition-fri-reachable-verifier-tie-empty-header-gap s) s +
  wp-event verify-monad
    (λout.
      partial-merkle-inconsistency-bad s out ∨
      composition-fri-verifier-tied-candidate-available-gap s out) s
by (rule wp-event-union-bound)
also have ... ≤
  wp-event verify-monad
    (composition-fri-reachable-verifier-tie-empty-header-gap s) s +
  (wp-event verify-monad (partial-merkle-inconsistency-bad s) s) +
  wp-event verify-monad
    (composition-fri-verifier-tied-candidate-available-gap s) s)
by (rule add-mono[OF order_refl wp-event-union-bound])
also have ... ≤ E + (M + T)
by (intro add-mono empty-bound merkle-bound available-bound)
finally show ?thesis .
qed

lemma composition-fri-reachable-partial-candidate-reduction-from-split-residual-bounds-with-candidate-available:
fixes R A P M N S G E Q T :: prob
assumes sampled-bound:
  wp-event verify-monad
    (composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R
and slot-gap-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
      s) s ≤ A
and verifier-partial-merkle-bound:
  wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P
and missing-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-recorded-base-chunk-auth-missing s)
      s ≤ M
and next-gap-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-next-value-replay-gap s) s ≤ N
and successor-gap-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-successor-replay-gap s) s ≤ S

```

and *final-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \ s \leq G$
and *empty-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-reachable-verifier-tie-empty-header-gap } s) \ s \leq E$
and *gap-partial-merkle-bound*:
 $\text{wp-event verify-monad}$ $(\text{partial-merkle-inconsistency-bad } s) \ s \leq Q$
and *candidate-available-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-candidate-available-gap } s) \ s \leq T$
and *total-bound*:
 $R + (((P + M) + N + S + G + P) + A) + (E + (Q + T))$
 $\leq \text{composition-fri-error}$
shows *composition-fri-reachable-partial-candidate-reduction-assumption* s
proof –
have *verifier-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-partial-candidate } s) \ s$
 $\leq R + (((P + M) + N + S + G + P) + A)$
by (rule
 $\text{wp-composition-fri-verifier-tied-bound-from-sampled-slot-recorded-chunk-gap-and-residuals}$
 $[\text{OF sampled-bound slot-gap-bound verifier-partial-merkle-bound}$
 $\text{missing-bound next-gap-bound successor-gap-bound final-gap-bound}])$
have *gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-reachable-verifier-tie-gap } s) \ s \leq$
 $E + (Q + T)$
by (rule
 $\text{wp-composition-fri-reachable-verifier-tie-gap-bound-from-empty-merkle-and-candidate-available}$
 $[\text{OF empty-bound gap-partial-merkle-bound candidate-available-bound}])$
show *?thesis*
by (rule
 $\text{composition-fri-reachable-partial-candidate-reduction-from-verifier-tied-and-gap}$
 $[\text{OF verifier-bound gap-bound total-bound}])$
qed

lemma *composition-fri-reachable-partial-candidate-reduction-from-split-residual-bounds-with-candidate-available-and-*
fixes $R \ A \ P \ N \ S \ G \ E \ Q \ T :: \text{prob}$
assumes *sampled-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s \leq R$
and *slot-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap}$
 $s) \ s \leq A$
and *verifier-partial-merkle-bound*:
 $\text{wp-event verify-monad}$ $(\text{partial-merkle-inconsistency-bad } s) \ s \leq P$
and *next-gap-bound*:

```

    wp-event verify-monad
      (composition-fri-verifier-tied-next-value-replay-gap s) s ≤ N
  and successor-gap-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-successor-replay-gap s) s ≤ S
  and final-gap-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-final-value-replay-gap s) s ≤ G
  and empty-bound:
    wp-event verify-monad
      (composition-fri-reachable-verifier-tie-empty-header-gap s) s ≤ E
  and gap-partial-merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ Q
  and candidate-available-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-candidate-available-gap s) s ≤ T
  and total-bound:
    R + (((P + 0) + N + S + G + P) + A) + (E + (Q + T))
      ≤ composition-fri-error
shows composition-fri-reachable-partial-candidate-reduction-assumption s
by (rule
  composition-fri-reachable-partial-candidate-reduction-from-split-residual-bounds-with-candidate-available
  [where M = 0])
(rule sampled-bound,
  rule slot-gap-bound,
  rule verifier-partial-merkle-bound,
  rule wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero,
  rule next-gap-bound,
  rule successor-gap-bound,
  rule final-gap-bound,
  rule empty-bound,
  rule gap-partial-merkle-bound,
  rule candidate-available-bound,
  rule total-bound)

end

end

theory Soundness-FRI-Empty-Header-Reduction
  imports Soundness-FRI-Trace-Residuals
begin

Named reduction interface for the empty-composition-header trace FRI branch. This
keeps the large reachable-FRI theory stable while giving the final route the same active-
bound shape for all three FRI components.

context soundness
begin

```

definition *trace-fri-empty-header-reduction*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{bool}$
where
 $\text{trace-fri-empty-header-reduction } s \longleftrightarrow$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-empty-composition-header-candidates } s) \ s \leq$
 trace-fri-error

lemma *trace-fri-empty-header-reductionD*:
assumes *trace-fri-empty-header-reduction s*
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-empty-composition-header-candidates } s) \ s \leq$
 trace-fri-error
using *assms unfolding trace-fri-empty-header-reduction-def by simp*

lemma *trace-fri-empty-header-reduction-staged-bound*:
assumes *reductions*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})} \implies$
 $\text{trace-fri-empty-header-reduction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
shows
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
by (*rule checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
(simp-all add: trace-fri-bad-with-empty-composition-header-candidates-not-None
trace-fri-empty-header-reductionD reductions)

lemma *trace-fri-empty-header-reduction-from-reachable-reduction*:
assumes *trace-fri-reachable-partial-candidate-reduction-assumption s*
shows *trace-fri-empty-header-reduction s*
proof –
have *reachable*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-reachable-partial-candidate } s) \ s \leq$
 trace-fri-error
using *assms*
unfolding *trace-fri-reachable-partial-candidate-reduction-assumption-def*
by *simp*
show *?thesis*
unfolding *trace-fri-empty-header-reduction-def*
by (*rule order-trans[OF - reachable]*)

```

      (rule wp-event-mono,
       rule
         trace-fri-bad-with-empty-composition-header-candidates-imp-reachable-partial-candidate)
    qed

  end

end

```

```

theory Soundness-FRI-Active-Reductions
imports
  Soundness-FRI-Candidate-Transfer-Support
  Soundness-FRI-Empty-Header-Reduction
begin

```

Staged packaging for the active FRI reductions used by the public route. This layer does not add assumptions; it names the exact verifier-local reductions consumed by the public-route bounds.

Internal symbolic FRI soundness interface for the public staged theorem.

The trace side is intentionally stated in terms of the verifier-consumed header-tied branch and the empty-composition-header branch. The composition side is stated in terms of the verifier-tied branch consumed by the active public route.

```

locale soundness-fri = soundness +
assumes trace-fri-active-reductions:
   $\bigwedge A$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      trace-fri-header-tied-reduction
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))  $\wedge$ 
      trace-fri-empty-header-reduction
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
assumes composition-fri-active-reduction:
   $\bigwedge A$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
      composition-fri-verifier-tied-reduction
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))

```

```

context soundness
begin

```

```

lemma active-composition-candidate-transfer-bound-from-classification-bound:

```


assumes *classification-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (execute (checked-staged-transcript-program *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(composition-fri-verifier-tied-misaligned-candidate-classification
(verifier-state-from-adversary *attacker-state*
(staged-proof-transcript *data*)))
(verifier-state-from-adversary *attacker-state*
(staged-proof-transcript *data*)) $\leq$
T data attacker-state

and *support*:

Some (*data*, *attacker-state*) $\in$
set-dist (execute (checked-staged-transcript-program *A*)
adversary-initial-state)

shows

wp-event verify-monad
(composition-fri-verifier-tied-candidate-transfer-gap
(verifier-state-from-adversary *attacker-state*
(staged-proof-transcript *data*)))
(verifier-state-from-adversary *attacker-state*
(staged-proof-transcript *data*)) $\leq$
T data attacker-state

by (rule

wp-composition-fri-verifier-tied-candidate-transfer-gap-bound-from-misaligned-classification)
(rule *classification-bound*[OF *support*])

lemma *active-composition-candidate-available-bound-from-transcript-consistent-classification-bound*:

assumes *empty-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (execute (checked-staged-transcript-program *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(composition-fri-reachable-verifier-tie-empty-header-gap
(verifier-state-from-adversary *attacker-state*
(staged-proof-transcript *data*)))
(verifier-state-from-adversary *attacker-state*
(staged-proof-transcript *data*)) $\leq$
E data attacker-state

and *merkle-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (execute (checked-staged-transcript-program *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary *attacker-state*

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $M \text{ data attacker-state}$
and *classification-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-transcript-consistent-misaligned-classification}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $T \text{ data attacker-state}$
and *support*:
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)}$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-candidate-available-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $E \text{ data attacker-state} + (M \text{ data attacker-state} +$
 $T \text{ data attacker-state})$
by (rule
 $\text{wp-composition-fri-verifier-tied-candidate-available-gap-bound-from-empty-merkle-and-transcript-consistent-classification-bound}$
 $(\text{rule empty-bound[OF support], rule merkle-bound[OF support],}$
 $\text{rule classification-bound[OF support]})$

lemma *active-composition-transcript-consistent-classification-bound-from-components*:

assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *comp-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-bad-with-partial-candidates}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $C \text{ data attacker-state}$
and *trace-fri-bound*:

$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$
 Ft data attacker-state
and query-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (query-bad-with-aligned-transcript-partial-candidates
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$
 Q data attacker-state
and support:
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state)
shows
 wp-event verify-monad
 (composition-fri-verifier-tied-transcript-consistent-misaligned-classification
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$
 C data attacker-state + Ft data attacker-state +
 Q data attacker-state
by (rule
 wp-composition-fri-verifier-tied-transcript-consistent-misaligned-classification-bound-from-cover-components
 [OF false-statement])
 (rule comp-bound[OF support], rule trace-fri-bound[OF support],
 rule query-bound[OF support])

lemma active-trace-fri-reachable-reduction-from-header-merkle-and-low-candidate-gap:

fixes Ht Mt Tt ::

($\text{'f staged-proof-data} \Rightarrow \text{'f protocol-channel} \Rightarrow \text{prob}$)

assumes trace-header-bound:

$\wedge$ data attacker-state.

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

$\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Ht data attacker-state}$
and $\text{trace-merkle-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state}) \implies}$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Mt data attacker-state}$
and $\text{trace-low-gap-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state}) \implies}$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-low-candidate-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Tt data attacker-state}$
and total-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state}) \implies}$
 $\text{Ht data attacker-state} +$
 $(\text{Mt data attacker-state} + \text{Tt data attacker-state}) \leq$
 trace-fri-error
and support:
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state})$
shows
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
by $(\text{rule}$

$\text{trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-low-candidate-gap})$
 $(\text{rule } \text{trace-header-bound}[OF \text{ support}],$
 $\text{rule } \text{trace-merkle-bound}[OF \text{ support}],$
 $\text{rule } \text{trace-low-gap-bound}[OF \text{ support}],$
 $\text{rule } \text{total-bound}[OF \text{ support}])$

definition *active-fri-reductions-for* **where**

$\text{active-fri-reductions-for } A \longleftrightarrow$
 $(\forall \text{ data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \longrightarrow$
 $\text{trace-fri-header-tied-reduction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \wedge$
 $\text{composition-fri-verifier-tied-reduction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \wedge$
 $\text{trace-fri-empty-header-reduction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$

definition *trace-fri-header-tied-residual-obligations*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $\text{prob} \Rightarrow \text{prob} \Rightarrow \text{prob} \Rightarrow \text{prob} \Rightarrow \text{prob} \Rightarrow \text{prob} \Rightarrow$
 $\text{prob} \Rightarrow \text{prob} \Rightarrow \text{prob} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-residual-obligations } s \text{ } r\text{-}b \text{ } z\text{-}b \text{ } a\text{-}b \text{ } p\text{-}b \text{ } h\text{-}b \text{ } g\text{-}b \text{ } n\text{-}b \text{ } s\text{-}b \text{ } f\text{-}b$
 $\longleftrightarrow$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-sampled-layer-chain } s) \text{ } s \leq r\text{-}b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \text{ } s \leq z\text{-}b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-authenticated-slot-recorded-chunk-gap } s) \text{ } s$
 $\leq a\text{-}b \wedge$
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \text{ } s \leq p\text{-}b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \text{ } s \leq h\text{-}b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-head-value-neq-gap } s) \text{ } s \leq g\text{-}b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap } s) \text{ } s \leq n\text{-}b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \text{ } s \leq s\text{-}b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-selected-step-missing-gap } s) \text{ } s$
 $\leq f\text{-}b \wedge$
 $r\text{-}b + (((((p\text{-}b + h\text{-}b) + g\text{-}b) + n\text{-}b + s\text{-}b +$

$$\begin{aligned}
& (p-b + a-b + f-b)) + p-b) + a-b + z-b) \\
& \leq \text{trace-fri-error}
\end{aligned}$$

definition *composition-fri-verifier-tied-residual-obligations*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $\text{prob} \Rightarrow \text{prob} \Rightarrow \text{prob} \Rightarrow \text{prob} \Rightarrow \text{prob} \Rightarrow$
 $\text{prob} \Rightarrow \text{bool}$

where

composition-fri-verifier-tied-residual-obligations s r - b a - b p - b n - b s - b g - b
 $\longleftrightarrow$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s$
 $\leq r-b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap } s)$
 $s \leq a-b \wedge$
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq p-b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap } s) \ s \leq n-b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap } s) \ s \leq s-b \wedge$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \ s \leq g-b \wedge$
 $r-b + (((p-b + a-b) + n-b + s-b + g-b + p-b) + a-b)$
 $\leq \text{composition-fri-error}$

lemma *trace-fri-header-tied-residual-obligations-from-bounds-without-head:*

assumes *sampled:*

$\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-sampled-layer-chain } s) \ s \leq r-b$

and *zero:*

$\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s \leq z-b$

and *slot:*

$\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-authenticated-slot-recorded-chunk-gap } s) \ s$
 $\leq a-b$

and *merkle:*

$\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq p-b$

and *structural:*

$\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s \leq h-b$

and *next-bound:*

$\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq n-b$

and *successor:*

$\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s \leq s-b$

and *final-bound:*

wp-event verify-monad
 (trace-fri-header-tied-final-value-selected-step-missing-gap s) s
 $\leq f-b$
and total:
 $r-b + (((((p-b + h-b) + p-b) + n-b + s-b +$
 $(p-b + a-b + f-b)) + p-b) + a-b + z-b)$
 $\leq \text{trace-fri-error}$
shows
 trace-fri-header-tied-residual-obligations s
 $r-b \ z-b \ a-b \ p-b \ h-b \ p-b \ n-b \ s-b \ f-b$
proof –
have head:
wp-event verify-monad
 (trace-fri-header-tied-base-head-value-neq-gap s) $s \leq p-b$
by (rule
 wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle
 [OF merkle])
show ?thesis
using sampled zero slot merkle structural head next-bound successor
 final-bound total
unfolding trace-fri-header-tied-residual-obligations-def **by** blast
qed

lemma trace-fri-header-tied-residual-obligations-from-bounds-without-head-and-selected-zero:
assumes sampled:
wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) $s \leq r-b$
and zero:
wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction s) $s \leq z-b$
and slot:
wp-event verify-monad
 (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s) s
 $\leq a-b$
and merkle:
wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq p-b$
and structural:
wp-event verify-monad
 (trace-fri-header-tied-sibling-mismatch-structural-gap s) $s \leq h-b$
and next-bound:
wp-event verify-monad
 (trace-fri-header-tied-next-value-replay-gap s) $s \leq n-b$
and successor:
wp-event verify-monad
 (trace-fri-header-tied-successor-replay-gap s) $s \leq s-b$
and total:
 $r-b + (((((p-b + h-b) + p-b) + n-b + s-b +$
 $(p-b + a-b + 0)) + p-b) + a-b + z-b)$
 $\leq \text{trace-fri-error}$

shows

trace-fri-header-tied-residual-obligations s

r-b z-b a-b p-b h-b p-b n-b s-b 0

by (rule *trace-fri-header-tied-residual-obligations-from-bounds-without-head*)

(rule *sampled*, rule *zero*, rule *slot*, rule *merkle*, rule *structural*,

rule *next-bound*, rule *successor*,

rule *wp-trace-fri-header-tied-final-value-selected-step-missing-gap-zero*,

rule *total*)

lemma *trace-fri-header-tied-residual-obligations-from-bounds-without-head-slot-and-selected-zero*:

assumes *sampled*:

wp-event verify-monad

(*trace-fri-bad-with-header-tied-sampled-layer-chain s*) $s \leq r-b$

and *zero*:

wp-event verify-monad

(*trace-fri-header-tied-zero-round-final-obstruction s*) $s \leq z-b$

and *merkle*:

wp-event verify-monad (*partial-merkle-inconsistency-bad s*) $s \leq p-b$

and *structural*:

wp-event verify-monad

(*trace-fri-header-tied-sibling-mismatch-structural-gap s*) $s \leq h-b$

and *next-bound*:

wp-event verify-monad

(*trace-fri-header-tied-next-value-replay-gap s*) $s \leq n-b$

and *successor*:

wp-event verify-monad

(*trace-fri-header-tied-successor-replay-gap s*) $s \leq s-b$

and *total*:

$r-b + (((((p-b + h-b) + p-b) + n-b + s-b +$

$(p-b + 0 + 0)) + p-b) + 0 + z-b)$

$\leq \text{trace-fri-error}$

shows

trace-fri-header-tied-residual-obligations s

r-b z-b 0 p-b h-b p-b n-b s-b 0

by (rule

trace-fri-header-tied-residual-obligations-from-bounds-without-head-and-selected-zero)

(rule *sampled*, rule *zero*,

rule *wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero*,

rule *merkle*, rule *structural*, rule *next-bound*, rule *successor*, rule *total*)

lemma *composition-fri-verifier-tied-residual-obligations-from-bounds*:

assumes *sampled*:

wp-event verify-monad

(*composition-fri-bad-with-verifier-tied-sampled-layer-chain s*) s

$\leq r-b$

and *slot*:

wp-event verify-monad

(*composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s*)

$s \leq a-b$

and merkle:
 $wp\text{-}event\ verify\text{-}monad\ (partial\text{-}merkle\text{-}inconsistency\text{-}bad\ s)\ s \leq p\text{-}b$
and next-bound:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}next\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq n\text{-}b$
and successor:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}successor\text{-}replay\text{-}gap\ s)\ s \leq s\text{-}b$
and final-bound:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq g\text{-}b$
and total:
 $r\text{-}b + (((p\text{-}b + a\text{-}b) + n\text{-}b + s\text{-}b + g\text{-}b + p\text{-}b) + a\text{-}b)$
 $\leq composition\text{-}fri\text{-}error$
shows
 $composition\text{-}fri\text{-}verifier\text{-}tied\text{-}residual\text{-}obligations\ s$
 $r\text{-}b\ a\text{-}b\ p\text{-}b\ n\text{-}b\ s\text{-}b\ g\text{-}b$
using *sampled slot merkle next-bound successor final-bound total*
unfolding *composition-fri-verifier-tied-residual-obligations-def* **by** *blast*

lemma *composition-fri-verifier-tied-residual-obligations-from-bounds-slot-zero:*

assumes *sampled:*
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}bad\text{-}with\text{-}verifier\text{-}tied\text{-}sampled\text{-}layer\text{-}chain\ s)\ s$
 $\leq r\text{-}b$
and merkle:
 $wp\text{-}event\ verify\text{-}monad\ (partial\text{-}merkle\text{-}inconsistency\text{-}bad\ s)\ s \leq p\text{-}b$
and next-bound:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}next\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq n\text{-}b$
and successor:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}successor\text{-}replay\text{-}gap\ s)\ s \leq s\text{-}b$
and final-bound:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq g\text{-}b$
and total:
 $r\text{-}b + (((p\text{-}b + 0) + n\text{-}b + s\text{-}b + g\text{-}b + p\text{-}b) + 0)$
 $\leq composition\text{-}fri\text{-}error$
shows
 $composition\text{-}fri\text{-}verifier\text{-}tied\text{-}residual\text{-}obligations\ s$
 $r\text{-}b\ 0\ p\text{-}b\ n\text{-}b\ s\text{-}b\ g\text{-}b$
by (*rule composition-fri-verifier-tied-residual-obligations-from-bounds*)
(*rule sampled,*
rule wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero,
rule merkle, rule next-bound, rule successor, rule final-bound,
rule total)

lemma *active-composition-fri-reachable-reduction-from-split-bounds-without-empty-with-candidate-available:*

fixes
Rc Ac Pc Nc Sc Gc Qc Tc ::
('f staged-proof-data $\Rightarrow$
'f protocol-channel $\Rightarrow$ prob)
assumes *comp-sampled-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Rc data attacker-state
and *comp-slot-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Ac data attacker-state
and *comp-merkle-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Pc data attacker-state
and *comp-next-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(composition-fri-verifier-tied-next-value-replay-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Nc \text{ data attacker-state}$
and *comp-successor-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Sc \text{ data attacker-state}$
and *comp-final-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Gc \text{ data attacker-state}$
and *comp-gap-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Qc \text{ data attacker-state}$
and *comp-candidate-available-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-candidate-available-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data}) \leq$
 $Tc \text{ data attacker-state}$
and *comp-total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $Rc \text{ data attacker-state} +$
 $((Pc \text{ data attacker-state} + 0) +$
 $Nc \text{ data attacker-state} + Sc \text{ data attacker-state} +$
 $Gc \text{ data attacker-state} + Pc \text{ data attacker-state}) +$
 $Ac \text{ data attacker-state}) +$
 $(Qc \text{ data attacker-state} + Tc \text{ data attacker-state})$
 $\leq \text{composition-fri-error}$
assumes *support*:
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)}$
shows
 $\text{composition-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
proof –
let $?s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
show $?thesis$
by (rule
 $\text{composition-fri-reachable-partial-candidate-reduction-from-split-residual-bounds-with-candidate-available-and-rec}$
 $[\textbf{where } E = 0])$
 $(\text{rule comp-sampled-bound}[OF \text{ support}],$
 $\text{rule comp-slot-bound}[OF \text{ support}],$
 $\text{rule comp-merkle-bound}[OF \text{ support}],$
 $\text{rule comp-next-bound}[OF \text{ support}],$
 $\text{rule comp-successor-bound}[OF \text{ support}],$
 $\text{rule comp-final-bound}[OF \text{ support}],$
 $\text{simp add: wp-composition-fri-reachable-verifier-tie-empty-header-gap-zero,}$
 $\text{rule comp-gap-merkle-bound}[OF \text{ support}],$
 $\text{rule comp-candidate-available-bound}[OF \text{ support}],$
 $\text{insert comp-total-bound}[OF \text{ support}], \text{ simp})$
qed

definition *active-fri-residual-obligations-for* **where**
 $\text{active-fri-residual-obligations-for } A \text{ Rt Zt At Pt Ht Gt Nt St Ft}$
 $Rc \text{ Ac Pc Nc Sc Gc} \longleftrightarrow$
 $(\forall \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \longrightarrow$

(let $s =$
 verifier-state-from-adversary attacker-state
 (*staged-proof-transcript data*)
 in
 trace-fri-header-tied-residual-obligations s
 (*Rt data attacker-state*) (*Zt data attacker-state*)
 (*At data attacker-state*) (*Pt data attacker-state*)
 (*Ht data attacker-state*) (*Gt data attacker-state*)
 (*Nt data attacker-state*) (*St data attacker-state*)
 (*Ft data attacker-state*) $\wedge$
 composition-fri-verifier-tied-residual-obligations s
 (*Rc data attacker-state*) (*Ac data attacker-state*)
 (*Pc data attacker-state*)
 (*Nc data attacker-state*) (*Sc data attacker-state*)
 (*Gc data attacker-state*) $\wedge$
 trace-fri-empty-header-reduction s))

definition *active-trace-fri-residual-obligations-for* **where**

active-trace-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft

$\longleftrightarrow$

($\forall$ *data attacker-state*.

 Some (*data, attacker-state*) $\in$

set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\longrightarrow$

trace-fri-header-tied-residual-obligations

 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*))

 (*Rt data attacker-state*) (*Zt data attacker-state*)

 (*At data attacker-state*) (*Pt data attacker-state*)

 (*Ht data attacker-state*) (*Gt data attacker-state*)

 (*Nt data attacker-state*) (*St data attacker-state*)

 (*Ft data attacker-state*))

definition *active-composition-fri-residual-obligations-for* **where**

active-composition-fri-residual-obligations-for A Rc Ac Pc Nc Sc Gc

$\longleftrightarrow$

($\forall$ *data attacker-state*.

 Some (*data, attacker-state*) $\in$

set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\longrightarrow$

composition-fri-verifier-tied-residual-obligations

 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*))

 (*Rc data attacker-state*) (*Ac data attacker-state*)

 (*Pc data attacker-state*)

 (*Nc data attacker-state*) (*Sc data attacker-state*)

 (*Gc data attacker-state*))

definition *active-empty-trace-fri-reduction-for* **where**

active-empty-trace-fri-reduction-for A $\longleftrightarrow$
 $(\forall \text{ data attacker-state.}$
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) →
 trace-fri-empty-header-reduction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))

lemma *active-fri-residual-obligations-from-components:*

assumes

active-trace-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft
active-composition-fri-residual-obligations-for A Rc Ac Pc Nc Sc Gc
active-empty-trace-fri-reduction-for A

shows *active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft*
Rc Ac Pc Nc Sc Gc

using *assms*

unfolding *active-fri-residual-obligations-for-def*

active-trace-fri-residual-obligations-for-def
active-composition-fri-residual-obligations-for-def
active-empty-trace-fri-reduction-for-def

by (*simp add: Let-def*)

lemma *active-fri-residual-obligations-trace-component:*

assumes *active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft*
Rc Ac Pc Nc Sc Gc

shows *active-trace-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft*

using *assms*

unfolding *active-fri-residual-obligations-for-def*

active-trace-fri-residual-obligations-for-def

by (*simp add: Let-def*)

lemma *active-fri-residual-obligations-composition-component:*

assumes *active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft*
Rc Ac Pc Nc Sc Gc

shows *active-composition-fri-residual-obligations-for A Rc Ac Pc Nc Sc Gc*

using *assms*

unfolding *active-fri-residual-obligations-for-def*

active-composition-fri-residual-obligations-for-def

by (*simp add: Let-def*)

lemma *active-fri-residual-obligations-empty-trace-component:*

assumes *active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft*
Rc Ac Pc Nc Sc Gc

shows *active-empty-trace-fri-reduction-for A*

using *assms*

unfolding *active-fri-residual-obligations-for-def*

active-empty-trace-fri-reduction-for-def

by (*simp add: Let-def*)

lemma *active-trace-fri-residual-obligations-from-bounds-without-head:*

fixes

Rt Zt At Pt Ht Nt St Ft ::
('f staged-proof-data $\Rightarrow$
'f protocol-channel $\Rightarrow$ prob)

assumes *trace-sampled-bound:*

$\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-bad-with-header-tied-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Rt data attacker-state

and *trace-zero-bound:*

$\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-header-tied-zero-round-final-obstruction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Zt data attacker-state

and *trace-slot-bound:*

$\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-header-tied-authenticated-slot-recorded-chunk-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
At data attacker-state

and *trace-merkle-bound:*

$\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(partial-merkle-inconsistency-bad

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Pt \text{ data attacker-state}$
and *trace-structural-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ht \text{ data attacker-state}$
and *trace-next-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Nt \text{ data attacker-state}$
and *trace-successor-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $St \text{ data attacker-state}$
and *trace-final-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-selected-step-missing-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$


```

      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤
    Ft data attacker-state
and trace-total-bound:
 $\wedge$  data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    Rt data attacker-state +
      (((((Pt data attacker-state + Ht data attacker-state) +
        Pt data attacker-state) +
        Nt data attacker-state + St data attacker-state +
        (Pt data attacker-state + At data attacker-state +
        Ft data attacker-state)) +
        Pt data attacker-state) +
        At data attacker-state + Zt data attacker-state)
    ≤ trace-fri-error
shows active-trace-fri-residual-obligations-for A Rt Zt At Pt Ht Pt
  Nt St Ft
unfolding active-trace-fri-residual-obligations-for-def
proof (intro allI impI)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show trace-fri-header-tied-residual-obligations ?s
  (Rt data attacker-state) (Zt data attacker-state)
  (At data attacker-state) (Pt data attacker-state)
  (Ht data attacker-state) (Pt data attacker-state)
  (Nt data attacker-state) (St data attacker-state)
  (Ft data attacker-state)
by (rule trace-fri-header-tied-residual-obligations-from-bounds-without-head)
  (rule trace-sampled-bound[OF support],
    rule trace-zero-bound[OF support],
    rule trace-slot-bound[OF support],
    rule trace-merkle-bound[OF support],
    rule trace-structural-bound[OF support],
    rule trace-next-bound[OF support],
    rule trace-successor-bound[OF support],
    rule trace-final-bound[OF support],
    rule trace-total-bound[OF support])
qed

lemma active-trace-fri-residual-obligations-from-bounds-without-head-and-selected-zero:

```

fixes
Rt Zt At Pt Ht Nt St ::
('f staged-proof-data $\Rightarrow$
'f protocol-channel $\Rightarrow$ prob)
assumes *trace-sampled-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-bad-with-header-tied-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Rt data attacker-state
and *trace-zero-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-header-tied-zero-round-final-obstruction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Zt data attacker-state
and *trace-slot-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-header-tied-authenticated-slot-recorded-chunk-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
At data attacker-state
and *trace-merkle-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Pt \text{ data attacker-state}$
and *trace-structural-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ht \text{ data attacker-state}$
and *trace-next-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Nt \text{ data attacker-state}$
and *trace-successor-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $St \text{ data attacker-state}$
and *trace-total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $Rt \text{ data attacker-state} +$
 $(((((Pt \text{ data attacker-state} + Ht \text{ data attacker-state}) +$
 $Pt \text{ data attacker-state}) +$
 $Nt \text{ data attacker-state} + St \text{ data attacker-state} +$
 $(Pt \text{ data attacker-state} + At \text{ data attacker-state} + 0))) +$

```

    Pt data attacker-state) +
    At data attacker-state + Zt data attacker-state)
  ≤ trace-fri-error
shows active-trace-fri-residual-obligations-for A Rt Zt At Pt Ht Pt
  Nt St (λ- -. (0::prob))
unfolding active-trace-fri-residual-obligations-for-def
proof (intro allI impI)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
show trace-fri-header-tied-residual-obligations ?s
  (Rt data attacker-state) (Zt data attacker-state)
  (At data attacker-state) (Pt data attacker-state)
  (Ht data attacker-state) (Pt data attacker-state)
  (Nt data attacker-state) (St data attacker-state) 0
by (rule
  trace-fri-header-tied-residual-obligations-from-bounds-without-head-and-selected-zero)
  (rule trace-sampled-bound[OF support],
  rule trace-zero-bound[OF support],
  rule trace-slot-bound[OF support],
  rule trace-merkle-bound[OF support],
  rule trace-structural-bound[OF support],
  rule trace-next-bound[OF support],
  rule trace-successor-bound[OF support],
  rule trace-total-bound[OF support])
qed

lemma active-trace-fri-residual-obligations-from-bounds-without-head-slot-and-selected-zero:
fixes
  Rt Zt Pt Ht Nt St ::
  ('f staged-proof-data ⇒
  'f protocol-channel ⇒ prob)
assumes trace-sampled-bound:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  wp-event verify-monad
  (trace-fri-bad-with-header-tied-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  Rt data attacker-state

```

and *trace-zero-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$

wp-event verify-monad
(*trace-fri-header-tied-zero-round-final-obstruction*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$

Zt data attacker-state

and *trace-merkle-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$

wp-event verify-monad
(*partial-merkle-inconsistency-bad*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$

Pt data attacker-state

and *trace-structural-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$

wp-event verify-monad
(*trace-fri-header-tied-sibling-mismatch-structural-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$

Ht data attacker-state

and *trace-next-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$

wp-event verify-monad
(*trace-fri-header-tied-next-value-replay-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$

Nt data attacker-state

and *trace-successor-bound*:

```

 $\wedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (trace-fri-header-tied-successor-replay-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq$ 
  St data attacker-state
and trace-total-bound:
 $\wedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
  Rt data attacker-state +
    (((((Pt data attacker-state + Ht data attacker-state) +
      Pt data attacker-state) +
      Nt data attacker-state + St data attacker-state +
      (Pt data attacker-state + 0 + 0)) +
      Pt data attacker-state) +
      0 + Zt data attacker-state)
     $\leq$  trace-fri-error
shows
  active-trace-fri-residual-obligations-for A Rt Zt
    ( $\lambda$ - -. (0::prob)) Pt Ht Pt Nt St ( $\lambda$ - -. (0::prob))
unfolding active-trace-fri-residual-obligations-for-def
proof (intro allI impI)
fix data attacker-state
assume support:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show trace-fri-header-tied-residual-obligations ?s
  (Rt data attacker-state) (Zt data attacker-state) 0
  (Pt data attacker-state) (Ht data attacker-state)
  (Pt data attacker-state) (Nt data attacker-state)
  (St data attacker-state) 0
by (rule
  trace-fri-header-tied-residual-obligations-from-bounds-without-head-slot-and-selected-zero)
  (rule trace-sampled-bound[OF support],
  rule trace-zero-bound[OF support],
  rule trace-merkle-bound[OF support],
  rule trace-structural-bound[OF support],
  rule trace-next-bound[OF support],

```

rule trace-successor-bound[*OF support*],
 rule trace-total-bound[*OF support*])

qed

lemma *active-composition-fri-residual-obligations-from-bounds:*

fixes

Rc Ac Pc Nc Sc Gc ::

('f staged-proof-data $\Rightarrow$

'f protocol-channel $\Rightarrow$ prob)

assumes *comp-sampled-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

wp-event verify-monad

(composition-fri-bad-with-verifier-tied-sampled-layer-chain

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)) $\leq$

Rc data attacker-state

and *comp-slot-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

wp-event verify-monad

(composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)) $\leq$

Ac data attacker-state

and *comp-merkle-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

wp-event verify-monad

(partial-merkle-inconsistency-bad

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)) $\leq$

Pc data attacker-state

and *comp-next-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

```

    adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (composition-fri-verifier-tied-next-value-replay-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq$ 
Nc data attacker-state
and comp-successor-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (composition-fri-verifier-tied-successor-replay-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq$ 
Sc data attacker-state
and comp-final-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (composition-fri-verifier-tied-final-value-replay-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq$ 
Gc data attacker-state
and comp-total-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
Rc data attacker-state +
  (((Pc data attacker-state + Ac data attacker-state) +
    Nc data attacker-state + Sc data attacker-state +
    Gc data attacker-state + Pc data attacker-state) +
    Ac data attacker-state)
 $\leq$  composition-fri-error
shows active-composition-fri-residual-obligations-for A Rc Ac Pc Nc Sc Gc
unfolding active-composition-fri-residual-obligations-for-def
proof (intro allI impI)
fix data attacker-state
assume support:
  Some (data, attacker-state)  $\in$ 

```



```

    set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  show composition-fri-verifier-tied-residual-obligations ?s
    (Rc data attacker-state) (Ac data attacker-state)
    (Pc data attacker-state)
    (Nc data attacker-state) (Sc data attacker-state)
    (Gc data attacker-state)
  by (rule composition-fri-verifier-tied-residual-obligations-from-bounds)
    (rule comp-sampled-bound[OF support],
    rule comp-slot-bound[OF support],
    rule comp-merkle-bound[OF support],
    rule comp-next-bound[OF support],
    rule comp-successor-bound[OF support],
    rule comp-final-bound[OF support],
    rule comp-total-bound[OF support])
qed

lemma active-composition-fri-residual-obligations-from-bounds-slot-zero:
  fixes
    Rc Pc Nc Sc Gc ::
      ('f staged-proof-data  $\Rightarrow$ 
       'f protocol-channel  $\Rightarrow$  prob)
  assumes comp-sampled-bound:
     $\bigwedge$  data attacker-state.
      Some (data, attacker-state)  $\in$ 
        set-dist (execute (checked-staged-transcript-program A)
            adversary-initial-state)  $\implies$ 
        wp-event verify-monad
          (composition-fri-bad-with-verifier-tied-sampled-layer-chain
            (verifier-state-from-adversary attacker-state
              (staged-proof-transcript data)))
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))  $\leq$ 
        Rc data attacker-state
  and comp-merkle-bound:
     $\bigwedge$  data attacker-state.
      Some (data, attacker-state)  $\in$ 
        set-dist (execute (checked-staged-transcript-program A)
            adversary-initial-state)  $\implies$ 
        wp-event verify-monad
          (partial-merkle-inconsistency-bad
            (verifier-state-from-adversary attacker-state
              (staged-proof-transcript data)))
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))  $\leq$ 
        Pc data attacker-state

```

and *comp-next-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$

wp-event verify-monad
(*composition-fri-verifier-tied-next-value-replay-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$

Nc data attacker-state

and *comp-successor-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$

wp-event verify-monad
(*composition-fri-verifier-tied-successor-replay-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$

Sc data attacker-state

and *comp-final-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$

wp-event verify-monad
(*composition-fri-verifier-tied-final-value-replay-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$

Gc data attacker-state

and *comp-total-bound*:

$\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$

Rc data attacker-state +
(((*Pc data attacker-state* + 0) +
Nc data attacker-state + *Sc data attacker-state* +
Gc data attacker-state + *Pc data attacker-state*) +
0)

$\leq$ *composition-fri-error*

shows

active-composition-fri-residual-obligations-for A Rc

```

    (λ- -. (0::prob)) Pc Nc Sc Gc
unfolding active-composition-fri-residual-obligations-for-def
proof (intro allI impI)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show composition-fri-verifier-tied-residual-obligations ?s
  (Rc data attacker-state) 0 (Pc data attacker-state)
  (Nc data attacker-state) (Sc data attacker-state)
  (Gc data attacker-state)
by (rule composition-fri-verifier-tied-residual-obligations-from-bounds-slot-zero)
  (rule comp-sampled-bound[OF support],
    rule comp-merkle-bound[OF support],
    rule comp-next-bound[OF support],
    rule comp-successor-bound[OF support],
    rule comp-final-bound[OF support],
    rule comp-total-bound[OF support])
qed

```

```

lemma active-fri-reductions-forD:
assumes active-fri-reductions-for A
and Some (data, attacker-state) ∈
  set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state)
shows
  trace-fri-header-tied-reduction
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  composition-fri-verifier-tied-reduction
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  trace-fri-empty-header-reduction
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
using assms unfolding active-fri-reductions-for-def by blast+

```

```

lemma trace-fri-header-tied-reduction-from-residual-obligations:
assumes
  trace-fri-header-tied-residual-obligations s r-b z-b a-b p-b h-b g-b n-b s-b f-b
shows trace-fri-header-tied-reduction s
proof -
have sampled:
  wp-event verify-monad
    (trace-fri-bad-with-header-tied-sampled-layer-chain s) s ≤ r-b

```

```

using assms unfolding trace-fri-header-tied-residual-obligations-def
by blast
have zero:
  wp-event verify-monad
    (trace-fri-header-tied-zero-round-final-obstruction s)  $s \leq z-b$ 
  using assms unfolding trace-fri-header-tied-residual-obligations-def
  by blast
have slot:
  wp-event verify-monad
    (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s)  $s \leq a-b$ 
  using assms unfolding trace-fri-header-tied-residual-obligations-def
  by blast
have merkle:
  wp-event verify-monad (partial-merkle-inconsistency-bad s)  $s \leq p-b$ 
  using assms unfolding trace-fri-header-tied-residual-obligations-def
  by blast
have structural:
  wp-event verify-monad
    (trace-fri-header-tied-sibling-mismatch-structural-gap s)  $s \leq h-b$ 
  using assms unfolding trace-fri-header-tied-residual-obligations-def
  by blast
have head-neq:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-neq-gap s)  $s \leq g-b$ 
  using assms unfolding trace-fri-header-tied-residual-obligations-def
  by blast
have head-mismatch:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-mismatch-gap s)  $s \leq g-b$ 
  by (rule
    wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
    [OF head-neq])
have alignment:
  wp-event verify-monad
    (trace-fri-header-tied-base-value-alignment-gap s)  $s \leq g-b$ 
  by (rule
    wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap
    [OF head-mismatch])
have next-bound:
  wp-event verify-monad
    (trace-fri-header-tied-next-value-replay-gap s)  $s \leq n-b$ 
  using assms unfolding trace-fri-header-tied-residual-obligations-def
  by blast
have successor:
  wp-event verify-monad
    (trace-fri-header-tied-successor-replay-gap s)  $s \leq s-b$ 
  using assms unfolding trace-fri-header-tied-residual-obligations-def
  by blast

```

have *selected-final*:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-header-tied-final-value-selected-step-missing-gap } s) \ s$
 $\leq f\text{-}b$
using *assms* **unfolding** *trace-fri-header-tied-residual-obligations-def*
by *blast*
have *final-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-header-tied-final-value-replay-gap } s) \ s \leq$
 $p\text{-}b + a\text{-}b + f\text{-}b$
by (*rule*
 $wp\text{-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step}$)
 $(rule \text{ merkle}, rule \text{ slot}, rule \text{ selected-final})$)
have *total*:
 $r\text{-}b + (((((p\text{-}b + h\text{-}b) + g\text{-}b) + n\text{-}b + s\text{-}b +$
 $(p\text{-}b + a\text{-}b + f\text{-}b)) + p\text{-}b) + a\text{-}b + z\text{-}b)$
 $\leq trace\text{-fri-error}$
using *assms* **unfolding** *trace-fri-header-tied-residual-obligations-def*
by *blast*
show *?thesis*
by (*rule* *trace-fri-header-tied-reduction-from-residual-bounds*
 $[OF \text{ sampled zero slot merkle structural alignment next-bound successor final-bound}$
 $total])$)
qed

lemma *composition-fri-verifier-tied-reduction-from-residual-obligations*:
assumes
 $composition\text{-fri-verifier-tied-residual-obligations } s \ r\text{-}b \ a\text{-}b \ p\text{-}b \ n\text{-}b \ s\text{-}b \ g\text{-}b$
shows *composition-fri-verifier-tied-reduction* s
proof –
have *sampled*:
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-bad-with-verifier-tied-sampled-layer-chain } s) \ s$
 $\leq r\text{-}b$
using *assms*
unfolding *composition-fri-verifier-tied-residual-obligations-def*
by *blast*
have *slot*:
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-verifier-tied-authenticated-slot-recorded-chunk-gap } s)$
 $s \leq a\text{-}b$
using *assms*
unfolding *composition-fri-verifier-tied-residual-obligations-def*
by *blast*
have *merkle*:
 $wp\text{-event } verify\text{-monad } (partial\text{-merkle-inconsistency-bad } s) \ s \leq p\text{-}b$
using *assms*
unfolding *composition-fri-verifier-tied-residual-obligations-def*
by *blast*

```

have next-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-next-value-replay-gap s)  $s \leq n-b$ 
  using assms
  unfolding composition-fri-verifier-tied-residual-obligations-def
  by blast
have successor:
  wp-event verify-monad
    (composition-fri-verifier-tied-successor-replay-gap s)  $s \leq s-b$ 
  using assms
  unfolding composition-fri-verifier-tied-residual-obligations-def
  by blast
have final-bound:
  wp-event verify-monad
    (composition-fri-verifier-tied-final-value-replay-gap s)  $s \leq g-b$ 
  using assms
  unfolding composition-fri-verifier-tied-residual-obligations-def
  by blast
have total:
   $r-b + (((p-b + a-b) + n-b + s-b + g-b + p-b) + a-b)$ 
     $\leq$  composition-fri-error
  using assms
  unfolding composition-fri-verifier-tied-residual-obligations-def
  by blast
have total-zero:
   $r-b + (((p-b + 0) + n-b + s-b + g-b + p-b) + a-b)$ 
     $\leq$  composition-fri-error
  by (rule order-trans[OF - total]) simp
show ?thesis
  by (rule
    composition-fri-verifier-tied-reduction-from-residual-bounds-with-recorded-missing-zero
    [OF sampled slot merkle next-bound successor final-bound total-zero])
qed

lemma active-fri-reductions-from-residual-obligations:
  assumes active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft
    Rc Ac Pc Nc Sc Gc
  shows active-fri-reductions-for A
  unfolding active-fri-reductions-for-def
proof (intro allI impI conjI)
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have trace-obs:

```

```

trace-fri-header-tied-residual-obligations ?s
  (Rt data attacker-state) (Zt data attacker-state)
  (At data attacker-state) (Pt data attacker-state)
  (Ht data attacker-state) (Gt data attacker-state)
  (Nt data attacker-state) (St data attacker-state)
  (Ft data attacker-state)
using assms support
unfolding active-fri-residual-obligations-for-def by (simp add: Let-def)
have composition-obs:
  composition-fri-verifier-tied-residual-obligations ?s
  (Rc data attacker-state) (Ac data attacker-state)
  (Pc data attacker-state)
  (Nc data attacker-state) (Sc data attacker-state)
  (Gc data attacker-state)
using assms support
unfolding active-fri-residual-obligations-for-def by (simp add: Let-def)
have empty-obs:
  trace-fri-empty-header-reduction ?s
using assms support
unfolding active-fri-residual-obligations-for-def by (simp add: Let-def)
show trace-fri-header-tied-reduction ?s
by (rule trace-fri-header-tied-reduction-from-residual-obligations
  [OF trace-obs])
show composition-fri-verifier-tied-reduction ?s
by (rule
  composition-fri-verifier-tied-reduction-from-residual-obligations
  [OF composition-obs])
show trace-fri-empty-header-reduction ?s
by (rule empty-obs)
qed

lemma active-fri-reductions-from-reachable:
assumes trace-reachable:
   $\bigwedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    trace-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
and composition-reachable:
   $\bigwedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    composition-fri-reachable-partial-candidate-reduction-assumption
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
shows active-fri-reductions-for A

```

```

unfolding active-fri-reductions-for-def
proof (intro allI impI conjI)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show trace-fri-header-tied-reduction
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
by (rule trace-fri-header-tied-reduction-from-reachable-reduction)
  (rule trace-reachable[OF support])
show composition-fri-verifier-tied-reduction
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
by (rule composition-fri-verifier-tied-reduction-from-reachable-reduction)
  (rule composition-reachable[OF support])
show trace-fri-empty-header-reduction
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
by (rule trace-fri-empty-header-reduction-from-reachable-reduction)
  (rule trace-reachable[OF support])
qed

```

lemma *active-fri-residual-obligations-from-bounds*:

```

fixes
  Rt Zt At Pt Ht Gt Nt St Ft ::
    ('f staged-proof-data ⇒
      'f protocol-channel ⇒ prob)
and Rc Ac Pc Nc Sc Gc ::
  ('f staged-proof-data ⇒
    'f protocol-channel ⇒ prob)
assumes trace-sampled-bound:
   $\bigwedge$  data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      wp-event verify-monad
        (trace-fri-bad-with-header-tied-sampled-layer-chain
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))  $\leq$ 
      Rt data attacker-state
and trace-zero-bound:
   $\bigwedge$  data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒

```


wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤
 Zt data attacker-state
and trace-slot-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
wp-event verify-monad
 (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤
 At data attacker-state
and trace-merkle-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤
 Pt data attacker-state
and trace-structural-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
wp-event verify-monad
 (trace-fri-header-tied-sibling-mismatch-structural-gap
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤
 Ht data attacker-state
and trace-head-neq-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
wp-event verify-monad

$(\text{trace-fri-header-tied-base-head-value-neq-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Gt \text{ data attacker-state}$
and trace-next-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Nt \text{ data attacker-state}$
and $\text{trace-successor-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $St \text{ data attacker-state}$
and $\text{trace-final-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-selected-step-missing-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ft \text{ data attacker-state}$
and $\text{trace-total-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $Rt \text{ data attacker-state} +$
 $(((((Pt \text{ data attacker-state} + Ht \text{ data attacker-state}) +$

$Gt \text{ data attacker-state}) +$
 $Nt \text{ data attacker-state} + St \text{ data attacker-state} +$
 $(Pt \text{ data attacker-state} + At \text{ data attacker-state} +$
 $Ft \text{ data attacker-state})) +$
 $Pt \text{ data attacker-state}) +$
 $At \text{ data attacker-state} + Zt \text{ data attacker-state})$
 $\leq \text{trace-fri-error}$
and comp-sampled-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Rc \text{ data attacker-state}$
and comp-slot-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ac \text{ data attacker-state}$
and comp-merkle-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Pc \text{ data attacker-state}$
and comp-next-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$

$(\text{composition-fri-verifier-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Nc \text{ data attacker-state}$
and *comp-successor-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Sc \text{ data attacker-state}$
and *comp-final-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Gc \text{ data attacker-state}$
and *comp-total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $Rc \text{ data attacker-state} +$
 $((Pc \text{ data attacker-state} + Ac \text{ data attacker-state}) +$
 $Nc \text{ data attacker-state} + Sc \text{ data attacker-state} +$
 $Gc \text{ data attacker-state} + Pc \text{ data attacker-state}) +$
 $Ac \text{ data attacker-state})$
 $\leq \text{composition-fri-error}$
and *empty-header-reduction*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-empty-header-reduction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$

```

shows active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft
  Rc Ac Pc Nc Sc Gc
unfolding active-fri-residual-obligations-for-def
proof (intro allI impI)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show (let s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  in
    trace-fri-header-tied-residual-obligations s
      (Rt data attacker-state) (Zt data attacker-state)
      (At data attacker-state) (Pt data attacker-state)
      (Ht data attacker-state) (Gt data attacker-state)
      (Nt data attacker-state) (St data attacker-state)
      (Ft data attacker-state)  $\wedge$ 
    composition-fri-verifier-tied-residual-obligations s
      (Rc data attacker-state) (Ac data attacker-state)
      (Pc data attacker-state)
      (Nc data attacker-state) (Sc data attacker-state)
      (Gc data attacker-state)  $\wedge$ 
    trace-fri-empty-header-reduction s)
using trace-sampled-bound[OF support] trace-zero-bound[OF support]
  trace-slot-bound[OF support] trace-merkle-bound[OF support]
  trace-structural-bound[OF support] trace-head-neq-bound[OF support]
  trace-next-bound[OF support] trace-successor-bound[OF support]
  trace-final-bound[OF support] trace-total-bound[OF support]
  comp-sampled-bound[OF support] comp-slot-bound[OF support]
  comp-merkle-bound[OF support] comp-next-bound[OF support]
  comp-successor-bound[OF support]
  comp-final-bound[OF support] comp-total-bound[OF support]
  empty-header-reduction[OF support]
unfolding trace-fri-header-tied-residual-obligations-def
  composition-fri-verifier-tied-residual-obligations-def
by (simp add: Let-def)
qed

```

```

lemma active-fri-residual-obligations-from-bounds-without-head:
fixes
  Rt Zt At Pt Ht Nt St Ft ::
    ('f staged-proof-data ⇒
      'f protocol-channel ⇒ prob)
and Rc Ac Pc Nc Sc Gc ::
    ('f staged-proof-data ⇒
      'f protocol-channel ⇒ prob)
assumes trace-sampled-bound:

```

$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$
 Rt data attacker-state
and trace-zero-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$
 Zt data attacker-state
and trace-slot-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$
 At data attacker-state
and trace-merkle-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$
 Pt data attacker-state
and trace-structural-bound:
 $\wedge$ data attacker-state.

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-sibling-mismatch-structural-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Ht data attacker-state
and *trace-next-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-next-value-replay-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Nt data attacker-state
and *trace-successor-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-successor-replay-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
St data attacker-state
and *trace-final-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-final-value-selected-step-missing-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Ft data attacker-state
and *trace-total-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈

$set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $Rt \ data \ attacker-state +$
 $(((((Pt \ data \ attacker-state + Ht \ data \ attacker-state) +$
 $Pt \ data \ attacker-state) +$
 $Nt \ data \ attacker-state + St \ data \ attacker-state +$
 $(Pt \ data \ attacker-state + At \ data \ attacker-state +$
 $Ft \ data \ attacker-state))) +$
 $Pt \ data \ attacker-state) +$
 $At \ data \ attacker-state + Zt \ data \ attacker-state)$
 $\leq trace-fri-error$
and *comp-sampled-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-bad-with-verifier-tied-sampled-layer-chain$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq$
 $Rc \ data \ attacker-state$
and *comp-slot-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq$
 $Ac \ data \ attacker-state$
and *comp-merkle-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(partial-merkle-inconsistency-bad$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq$
 $Pc \ data \ attacker-state$
and *comp-next-bound*:
 $\wedge data \ attacker-state.$

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(composition-fri-verifier-tied-next-value-replay-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Nc data attacker-state
and comp-successor-bound:
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(composition-fri-verifier-tied-successor-replay-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Sc data attacker-state
and comp-final-bound:
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(composition-fri-verifier-tied-final-value-replay-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Gc data attacker-state
and comp-total-bound:
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
Rc data attacker-state +
((Pc data attacker-state + Ac data attacker-state) +
Nc data attacker-state + Sc data attacker-state +
Gc data attacker-state + Pc data attacker-state) +
Ac data attacker-state)
 $\leq$ *composition-fri-error*
and empty-header-reduction:
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)

```

    adversary-initial-state)  $\implies$ 
    trace-fri-empty-header-reduction
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))
shows active-fri-residual-obligations-for A Rt Zt At Pt Ht Pt Nt St Ft
    Rc Ac Pc Nc Sc Gc
proof (rule active-fri-residual-obligations-from-bounds)
fix data attacker-state
assume support:
    Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
               adversary-initial-state)
show wp-event verify-monad
    (trace-fri-bad-with-header-tied-sampled-layer-chain
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))  $\leq$ 
    Rt data attacker-state
by (rule trace-sampled-bound[OF support])
show wp-event verify-monad
    (trace-fri-header-tied-zero-round-final-obstruction
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))  $\leq$ 
    Zt data attacker-state
by (rule trace-zero-bound[OF support])
show wp-event verify-monad
    (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))  $\leq$ 
    At data attacker-state
by (rule trace-slot-bound[OF support])
show wp-event verify-monad
    (partial-merkle-inconsistency-bad
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))  $\leq$ 
    Pt data attacker-state
by (rule trace-merkle-bound[OF support])
show wp-event verify-monad
    (trace-fri-header-tied-sibling-mismatch-structural-gap
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state

```

```

    (staged-proof-transcript data)) ≤
  Ht data attacker-state
  by (rule trace-structural-bound[OF support])
show wp-event verify-monad
  (trace-fri-header-tied-base-head-value-neq-gap
   (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data)) ≤
  Pt data attacker-state
  by (rule
      wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle)
  (rule trace-merkle-bound[OF support])
show wp-event verify-monad
  (trace-fri-header-tied-next-value-replay-gap
   (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data)) ≤
  Nt data attacker-state
  by (rule trace-next-bound[OF support])
show wp-event verify-monad
  (trace-fri-header-tied-successor-replay-gap
   (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data)) ≤
  St data attacker-state
  by (rule trace-successor-bound[OF support])
show wp-event verify-monad
  (trace-fri-header-tied-final-value-selected-step-missing-gap
   (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data)) ≤
  Ft data attacker-state
  by (rule trace-final-bound[OF support])
show Rt data attacker-state +
  (((((Pt data attacker-state + Ht data attacker-state) +
      Pt data attacker-state) +
      Nt data attacker-state + St data attacker-state +
      (Pt data attacker-state + At data attacker-state +
      Ft data attacker-state)) +
      Pt data attacker-state) +
      At data attacker-state + Zt data attacker-state)
  ≤ trace-fri-error
  by (rule trace-total-bound[OF support])
show wp-event verify-monad
  (composition-fri-bad-with-verifier-tied-sampled-layer-chain

```

```

    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  Rc data attacker-state
  by (rule comp-sampled-bound[OF support])
show wp-event verify-monad
  (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  Ac data attacker-state
  by (rule comp-slot-bound[OF support])
show wp-event verify-monad
  (partial-merkle-inconsistency-bad
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  Pc data attacker-state
  by (rule comp-merkle-bound[OF support])
show wp-event verify-monad
  (composition-fri-verifier-tied-next-value-replay-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  Nc data attacker-state
  by (rule comp-next-bound[OF support])
show wp-event verify-monad
  (composition-fri-verifier-tied-successor-replay-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  Sc data attacker-state
  by (rule comp-successor-bound[OF support])
show wp-event verify-monad
  (composition-fri-verifier-tied-final-value-replay-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  Gc data attacker-state
  by (rule comp-final-bound[OF support])
show Rc data attacker-state +
  (((Pc data attacker-state + Ac data attacker-state) +
    Nc data attacker-state + Sc data attacker-state +

```

```

      Gc data attacker-state + Pc data attacker-state) +
      Ac data attacker-state)
    ≤ composition-fri-error
  by (rule comp-total-bound[OF support])
show trace-fri-empty-header-reduction
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))
  by (rule empty-header-reduction[OF support])
qed

```

lemma *active-fri-residual-obligations-from-bounds-without-head-and-selected-zero:*

```

fixes
  Rt Zt At Pt Ht Nt St ::
    ('f staged-proof-data ⇒
     'f protocol-channel ⇒ prob)
and Rc Ac Pc Nc Sc Gc ::
    ('f staged-proof-data ⇒
     'f protocol-channel ⇒ prob)
assumes trace-sampled-bound:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                      adversary-initial-state) ⇒
      wp-event verify-monad
        (trace-fri-bad-with-header-tied-sampled-layer-chain
         (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
        (verifier-state-from-adversary attacker-state
         (staged-proof-transcript data)) ≤
      Rt data attacker-state
and trace-zero-bound:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                      adversary-initial-state) ⇒
      wp-event verify-monad
        (trace-fri-header-tied-zero-round-final-obstruction
         (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
        (verifier-state-from-adversary attacker-state
         (staged-proof-transcript data)) ≤
      Zt data attacker-state
and trace-slot-bound:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                      adversary-initial-state) ⇒
      wp-event verify-monad
        (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap

```

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{At data attacker-state}$
and $\text{trace-merkle-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Pt data attacker-state}$
and $\text{trace-structural-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Ht data attacker-state}$
and trace-next-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Nt data attacker-state}$
and $\text{trace-successor-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{St data attacker-state}$
and *trace-total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{Rt data attacker-state} +$
 $(((((\text{Pt data attacker-state} + \text{Ht data attacker-state}) +$
 $\text{Pt data attacker-state}) +$
 $\text{Nt data attacker-state} + \text{St data attacker-state} +$
 $(\text{Pt data attacker-state} + \text{At data attacker-state} + 0)) +$
 $\text{Pt data attacker-state}) +$
 $\text{At data attacker-state} + \text{Zt data attacker-state})$
 $\leq \text{trace-fri-error}$
and *comp-sampled-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Rc data attacker-state}$
and *comp-slot-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Ac data attacker-state}$
and *comp-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $P_c \text{ data attacker-state}$
and *comp-next-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $N_c \text{ data attacker-state}$
and *comp-successor-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $S_c \text{ data attacker-state}$
and *comp-final-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $G_c \text{ data attacker-state}$
and *comp-total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $R_c \text{ data attacker-state} +$
 $((P_c \text{ data attacker-state} + A_c \text{ data attacker-state}) +$
 $N_c \text{ data attacker-state} + S_c \text{ data attacker-state} +$
 $G_c \text{ data attacker-state} + P_c \text{ data attacker-state}) +$


```

    Ac data attacker-state)
  ≤ composition-fri-error
and empty-header-reduction:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      trace-fri-empty-header-reduction
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
shows active-fri-residual-obligations-for A Rt Zt At Pt Ht Pt Nt St
  (λ- -. (0::prob)) Rc Ac Pc Nc Sc Gc
proof (rule active-fri-residual-obligations-from-components)
show active-trace-fri-residual-obligations-for A Rt Zt At Pt Ht Pt
  Nt St (λ- -. (0::prob))
by (rule
  active-trace-fri-residual-obligations-from-bounds-without-head-and-selected-zero)
  (erule trace-sampled-bound, erule trace-zero-bound,
  erule trace-slot-bound, erule trace-merkle-bound,
  erule trace-structural-bound, erule trace-next-bound,
  erule trace-successor-bound, erule trace-total-bound)
show active-composition-fri-residual-obligations-for A Rc Ac Pc Nc Sc Gc
by (rule active-composition-fri-residual-obligations-from-bounds)
  (erule comp-sampled-bound, erule comp-slot-bound,
  erule comp-merkle-bound, erule comp-next-bound,
  erule comp-successor-bound, erule comp-final-bound,
  erule comp-total-bound)
show active-empty-trace-fri-reduction-for A
  unfolding active-empty-trace-fri-reduction-for-def
  by (intro allI impI, erule empty-header-reduction)
qed

```

lemma active-fri-residual-obligations-from-bounds-without-head-slot-and-selected-zero:
fixes

```

  Rt Zt Pt Ht Nt St ::
    ('f staged-proof-data ⇒
      'f protocol-channel ⇒ prob)
and Rc Pc Nc Sc Gc ::
    ('f staged-proof-data ⇒
      'f protocol-channel ⇒ prob)
assumes trace-sampled-bound:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      wp-event verify-monad
      (trace-fri-bad-with-header-tied-sampled-layer-chain
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))

```

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Rt \text{ data attacker-state}$
and *trace-zero-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Zt \text{ data attacker-state}$
and *trace-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Pt \text{ data attacker-state}$
and *trace-structural-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ht \text{ data attacker-state}$
and *trace-next-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data}) \leq$
 $Nt \text{ data attacker-state}$
and *trace-successor-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $St \text{ data attacker-state}$
and *trace-total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $Rt \text{ data attacker-state} +$
 $(((((Pt \text{ data attacker-state} + Ht \text{ data attacker-state}) +$
 $Pt \text{ data attacker-state}) +$
 $Nt \text{ data attacker-state} + St \text{ data attacker-state} +$
 $(Pt \text{ data attacker-state} + 0 + 0)) +$
 $Pt \text{ data attacker-state}) +$
 $0 + Zt \text{ data attacker-state})$
 $\leq \text{trace-fri-error}$
and *comp-sampled-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Rc \text{ data attacker-state}$
and *comp-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data})) \leq$
 $Pc \text{ data attacker-state}$
and *comp-next-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Nc \text{ data attacker-state}$
and *comp-successor-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Sc \text{ data attacker-state}$
and *comp-final-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Gc \text{ data attacker-state}$
and *comp-total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $Rc \text{ data attacker-state} +$
 $((Pc \text{ data attacker-state} + 0) +$
 $Nc \text{ data attacker-state} + Sc \text{ data attacker-state} +$
 $Gc \text{ data attacker-state} + Pc \text{ data attacker-state}) +$
 $0)$
 $\leq \text{composition-fri-error}$

and *empty-header-reduction*:
 $\wedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
trace-fri-empty-header-reduction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
shows
active-fri-residual-obligations-for A Rt Zt (λ - -. (0::prob))
Pt Ht Pt Nt St (λ - -. (0::prob)) Rc (λ - -. (0::prob))
Pc Nc Sc Gc
proof (rule *active-fri-residual-obligations-from-components*)
show *active-trace-fri-residual-obligations-for A Rt Zt*
(λ - -. (0::prob)) Pt Ht Pt Nt St (λ - -. (0::prob))
by (rule
active-trace-fri-residual-obligations-from-bounds-without-head-slot-and-selected-zero)
(erule trace-sampled-bound, erule trace-zero-bound,
erule trace-merkle-bound, erule trace-structural-bound,
erule trace-next-bound, erule trace-successor-bound,
erule trace-total-bound)
show *active-composition-fri-residual-obligations-for A Rc*
(λ - -. (0::prob)) Pc Nc Sc Gc
by (rule *active-composition-fri-residual-obligations-from-bounds-slot-zero*)
(erule comp-sampled-bound, erule comp-merkle-bound,
erule comp-next-bound, erule comp-successor-bound,
erule comp-final-bound, erule comp-total-bound)
show *active-empty-trace-fri-reduction-for A*
unfolding *active-empty-trace-fri-reduction-for-def*
by (intro *allI impI*, erule *empty-header-reduction*)
qed

lemmas *active-fri-reductions-from-residual-bounds =*
active-fri-reductions-from-residual-obligations[
OF active-fri-residual-obligations-from-bounds]

lemmas *active-fri-reductions-from-residual-bounds-without-head =*
active-fri-reductions-from-residual-obligations[
OF active-fri-residual-obligations-from-bounds-without-head]

lemmas *active-fri-reductions-from-residual-bounds-without-head-and-selected-zero =*
active-fri-reductions-from-residual-obligations[
OF active-fri-residual-obligations-from-bounds-without-head-and-selected-zero]

lemmas *active-fri-reductions-from-residual-bounds-without-head-slot-and-selected-zero =*
active-fri-reductions-from-residual-obligations[
OF active-fri-residual-obligations-from-bounds-without-head-slot-and-selected-zero]

lemma *active-fri-reductions-staged-trace-bound*:

assumes *active: active-fri-reductions-for A*
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state ≤ trace-fri-error
by (rule *trace-fri-header-tied-reduction-staged-bound*)
(rule *active-fri-reductions-forD(1)[OF active]*)

lemma *active-fri-reductions-staged-composition-bound:*
assumes *active: active-fri-reductions-for A*
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state ≤ composition-fri-error
by (rule *composition-fri-verifier-tied-reduction-staged-bound*)
(rule *active-fri-reductions-forD(2)[OF active]*)

lemma *active-fri-reductions-staged-empty-header-bound:*
assumes *active: active-fri-reductions-for A*
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤ trace-fri-error
by (rule *trace-fri-empty-header-reduction-staged-bound*)
(rule *active-fri-reductions-forD(3)[OF active]*)

lemma *active-trace-fri-residual-obligations-staged-trace-bound:*
assumes
active-trace-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state ≤ trace-fri-error
proof (rule *trace-fri-header-tied-reduction-staged-bound*)
fix *data attacker-state*
assume *support:*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)
let *?s =*
verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
have *residual:*
trace-fri-header-tied-residual-obligations ?s
(Rt data attacker-state) (Zt data attacker-state)

```

    (At data attacker-state) (Pt data attacker-state)
    (Ht data attacker-state) (Gt data attacker-state)
    (Nt data attacker-state) (St data attacker-state)
    (Ft data attacker-state)
  using assms support
  unfolding active-trace-fri-residual-obligations-for-def by simp
show trace-fri-header-tied-reduction ?s
  by (rule trace-fri-header-tied-reduction-from-residual-obligations
      [OF residual])
qed

lemma active-composition-fri-residual-obligations-staged-composition-bound:
  assumes
    active-composition-fri-residual-obligations-for A Rc Ac Pc Nc Sc Gc
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-verifier-tied-partial-candidate)
    adversary-initial-state ≤ composition-fri-error
proof (rule composition-fri-verifier-tied-reduction-staged-bound)
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have residual:
    composition-fri-verifier-tied-residual-obligations ?s
    (Rc data attacker-state) (Ac data attacker-state)
    (Pc data attacker-state)
    (Nc data attacker-state) (Sc data attacker-state)
    (Gc data attacker-state)
  using assms support
  unfolding active-composition-fri-residual-obligations-for-def by simp
show composition-fri-verifier-tied-reduction ?s
  by (rule composition-fri-verifier-tied-reduction-from-residual-obligations
      [OF residual])
qed

lemma active-empty-trace-fri-reduction-staged-empty-header-bound:
  assumes active-empty-trace-fri-reduction-for A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state ≤ trace-fri-error
  by (rule trace-fri-empty-header-reduction-staged-bound)

```

```

  (use assms in
    ⟨unfold active-empty-trace-fri-reduction-for-def, blast⟩)

lemma active-fri-residual-obligations-staged-trace-bound:
  assumes active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft
    Rc Ac Pc Nc Sc Gc
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state ≤ trace-fri-error
  by (rule active-fri-reductions-staged-trace-bound)
    (rule active-fri-reductions-from-residual-obligations[OF assms])

lemma active-fri-residual-obligations-staged-composition-bound:
  assumes active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft
    Rc Ac Pc Nc Sc Gc
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-verifier-tied-partial-candidate)
    adversary-initial-state ≤ composition-fri-error
  by (rule active-fri-reductions-staged-composition-bound)
    (rule active-fri-reductions-from-residual-obligations[OF assms])

lemma active-fri-residual-obligations-staged-empty-header-bound:
  assumes active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft
    Rc Ac Pc Nc Sc Gc
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state ≤ trace-fri-error
  by (rule active-fri-reductions-staged-empty-header-bound)
    (rule active-fri-reductions-from-residual-obligations[OF assms])

end

context soundness-fri
begin

lemma active-fri-reductions-for-from-locale:
  active-fri-reductions-for A
  unfolding active-fri-reductions-for-def
proof (intro allI impI conjI)
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)

```



```

      adversary-initial-state)
show trace-fri-header-tied-reduction
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))
  using trace-fri-active-reductions[OF support] by blast
next
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                    adversary-initial-state)
  show composition-fri-verifier-tied-reduction
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))
  by (rule composition-fri-active-reduction[OF support])
next
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                    adversary-initial-state)
  show trace-fri-empty-header-reduction
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))
  using trace-fri-active-reductions[OF support] by blast
qed

end

end

```

```

theory Soundness-FRI-Obligation-Normalization
  imports Soundness-FRI-Active-Reductions
begin

```

Normal form for the three FRI obligations that remain internal to locale *soundness-fri*. This theory does not change the protocol or the public soundness route. It only records the exact verifier-local events behind the internal FRI assumptions and exposes the generic partial FRI evidence already available from accepted verifier transcripts.

```

context soundness
begin

```

```

lemma trace-fri-header-tied-reduction-event:
  trace-fri-header-tied-reduction s  $\longleftrightarrow$ 
  wp-event verify-monad
    (λout. ∃ trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final fri-query-idxs trace-round-layers
      composition-round-layers fr as rest trace-openings trace-table.

```

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers $\wedge$
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ *trace-table-low-degree trace-table*) $s \leq$
trace-fri-error
unfolding *trace-fri-header-tied-reduction-def*
trace-fri-bad-with-header-tied-partial-candidate-def
by *simp*

lemma *trace-fri-empty-header-reduction-event:*
trace-fri-empty-header-reduction s $\longleftrightarrow$
wp-event verify-monad
 $(\lambda out. \exists fr f\text{-}fri\text{-}roots f\text{-}final as dg final trace\text{-}query\text{-}idxs$
trace-openings trace-table composition-table.
accepted-with-empty-composition-header-candidates s out fr
f-fri-roots f-final as dg final trace-query-idxs trace-openings
trace-table composition-table $\wedge$
 $\neg$ *trace-table-low-degree trace-table*) $s \leq$
trace-fri-error
unfolding *trace-fri-empty-header-reduction-def*
trace-fri-bad-with-empty-composition-header-candidates-def
by *simp*

lemma *composition-fri-verifier-tied-reduction-event:*
composition-fri-verifier-tied-reduction s $\longleftrightarrow$
wp-event verify-monad
 $(\lambda out. \exists trace\text{-}roots trace\text{-}bs trace\text{-}final fri\text{-}dg composition\text{-}roots$
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers $\wedge$
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
partial-composition-table-candidate composition-table
composition-openings $\wedge$
trace-table-low-degree trace-table $\wedge$
 $\neg$ *composition-table-low-degree maxDegree composition-table*) $s \leq$
composition-fri-error
unfolding *composition-fri-verifier-tied-reduction-def*

composition-fri-bad-with-verifier-tied-partial-candidate-def
by *simp*

lemma *trace-fri-header-tied-badE*:
assumes *trace-fri-bad-with-header-tied-partial-candidate s out*
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table **where**
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
partial-trace-table-candidate trace-table trace-openings
 $\neg$ *trace-table-low-degree trace-table*
using *assms*
unfolding *trace-fri-bad-with-header-tied-partial-candidate-def*
by *blast*

lemma *trace-fri-empty-header-badE*:
assumes *trace-fri-bad-with-empty-composition-header-candidates s out*
obtains *fr f-fri-roots f-final as dg final trace-query-idxs*
trace-openings trace-table composition-table **where**
accepted-with-empty-composition-header-candidates s out fr
f-fri-roots f-final as dg final trace-query-idxs trace-openings
trace-table composition-table
 $\neg$ *trace-table-low-degree trace-table*
using *assms*
unfolding *trace-fri-bad-with-empty-composition-header-candidates-def*
by *blast*

lemma *trace-fri-empty-header-bad-extracts-generic-partial-evidence*:
assumes *outcome*:
Some (result, final-state) $\in$ set-dist (execute verify-monad s)
and *bad*:
trace-fri-bad-with-empty-composition-header-candidates s
(Some (result, final-state))
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table **where**
trace-fri-partial-candidate-opening-evidence s
(Some (result, final-state)) trace-roots trace-bs trace-final dg
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr candidate-query-idxs
trace-openings trace-table
generic-fri-partial-evidence trace-table-low-degree trace-table

$(clength - 1)$ *trace-roots trace-bs trace-final fri-query-idxs*
trace-round-layers
proof –
have *reachable*:
trace-fri-bad-with-reachable-partial-candidate s
(Some (result, final-state))
by (rule
trace-fri-bad-with-empty-composition-header-candidates-imp-reachable-partial-candidate
[OF bad])
from *verify-monad-trace-fri-bad-extracts-opening-evidence*
[OF outcome reachable]
obtain *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
where *evidence*:
trace-fri-partial-candidate-opening-evidence s
(Some (result, final-state)) trace-roots trace-bs trace-final dg
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr candidate-query-idxs
trace-openings trace-table
by *blast*
have *generic*:
generic-fri-partial-evidence trace-table-low-degree trace-table
(clength - 1) trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers
by (rule *trace-fri-partial-candidate-opening-evidence-generic-partial*
[OF evidence])
show *?thesis*
by (rule *that[OF evidence generic]*)
qed

lemma *trace-fri-header-tied-sampled-layer-chain-imp-sampled-layer-chain*:
assumes *trace-fri-bad-with-header-tied-sampled-layer-chain s out*
shows *trace-fri-bad-with-sampled-layer-chain s out*
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table doms
layers
where *fri*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *partial*:
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *candidate*:
partial-trace-table-candidate trace-table trace-openings

and *not-low*: $\neg$ *trace-table-low-degree* *trace-table*
and *chain*:
generic-fri-sampled-layer-chain-evidence *trace-table-low-degree*
trace-table (*clength* - 1) *trace-roots* *trace-bs* *trace-final*
fri-query-idxs *trace-round-layers* *doms* *layers*
unfolding *trace-fri-bad-with-header-tied-sampled-layer-chain-def*
by *blast*
have *evidence*:
trace-fri-partial-candidate-opening-evidence *s* *out* *trace-roots*
trace-bs *trace-final* *dg* *composition-roots* *composition-bs*
composition-final *fri-query-idxs* *trace-round-layers*
composition-round-layers *fr* *fri-query-idxs* *trace-openings*
trace-table
unfolding *trace-fri-partial-candidate-opening-evidence-def*
trace-fri-partial-candidate-evidence-def
by (*intro conjI* *fri* *partial* *candidate* *not-low*)
show *?thesis*
unfolding *trace-fri-bad-with-sampled-layer-chain-def*
by (*intro exI conjI*)
(*rule evidence*, *rule chain*)
qed

lemma *wp-trace-fri-header-tied-sampled-layer-chain-bound-from-sampled*:
assumes *sampled-bound*:
wp-event *verify-monad* (*trace-fri-bad-with-sampled-layer-chain* *s*) $s \leq R$
shows
wp-event *verify-monad*
(*trace-fri-bad-with-header-tied-sampled-layer-chain* *s*) $s \leq R$
proof -
have *wp-event* *verify-monad*
(*trace-fri-bad-with-header-tied-sampled-layer-chain* *s*) $s \leq$
wp-event *verify-monad* (*trace-fri-bad-with-sampled-layer-chain* *s*) *s*
by (*rule wp-event-mono*)
(*rule trace-fri-header-tied-sampled-layer-chain-imp-sampled-layer-chain*)
then show *?thesis*
by (*rule order-trans*[*OF* - *sampled-bound*])
qed

lemma *wp-trace-fri-header-tied-sampled-layer-chain-bound-from-full-cover-and-missing*:
assumes *full-bound*:
wp-event *verify-monad*
(*trace-fri-bad-with-full-cover-layer-chain* *s* *bad*) $s \leq R$
and *missing-bound*:
wp-event *verify-monad*
(*trace-fri-sampled-missing-full-cover* *s* *bad*) $s \leq M$
shows
wp-event *verify-monad*
(*trace-fri-bad-with-header-tied-sampled-layer-chain* *s*) $s \leq R + M$
by (*rule wp-trace-fri-header-tied-sampled-layer-chain-bound-from-sampled*)

(rule *wp-trace-fri-sampled-bound-from-full-cover-and-missing*
 [OF *full-bound missing-bound*])

lemma *composition-fri-verifier-tied-sampled-layer-chain-imp-sampled-layer-chain:*

assumes *composition-fri-bad-with-verifier-tied-sampled-layer-chain s out*

shows *composition-fri-bad-with-sampled-layer-chain s out*

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*

composition-roots composition-bs composition-final fri-query-idxs

trace-round-layers composition-round-layers fr as trace-openings

composition-openings trace-table composition-table doms layers

where *fri:*

accepted-fri-opening-transcript s out trace-roots trace-bs

trace-final fri-dg composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers

and *aligned:*

accepted-with-partial-initial-openings-aligned s out fr trace-roots

trace-final as fri-dg composition-roots composition-final

fri-query-idxs trace-openings composition-openings

and *trace-candidate:*

partial-trace-table-candidate trace-table trace-openings

and *composition-candidate:*

partial-composition-table-candidate composition-table

composition-openings

and *trace-low: trace-table-low-degree trace-table*

and *composition-not-low:*

$\neg$ *composition-table-low-degree maxDegree composition-table*

and *chain:*

generic-fri-sampled-layer-chain-evidence

(composition-table-low-degree (to-nat fri-dg)) composition-table

(to-nat fri-dg) composition-roots composition-bs composition-final

fri-query-idxs composition-round-layers doms layers

unfolding *composition-fri-bad-with-verifier-tied-sampled-layer-chain-def*

by *blast*

have *unaligned:*

accepted-with-partial-initial-openings s out fr trace-roots

trace-final as fri-dg composition-roots composition-final

fri-query-idxs trace-openings fri-query-idxs composition-openings

by (rule *accepted-with-partial-initial-openings-aligned-imp-unaligned*

[OF *aligned*])

have *evidence:*

composition-fri-partial-candidate-opening-evidence s out trace-roots

trace-bs trace-final fri-dg composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr trace-roots trace-final as fri-dg

composition-roots composition-final fri-query-idxs trace-openings

fri-query-idxs composition-openings trace-table composition-table

unfolding *composition-fri-partial-candidate-opening-evidence-def*

composition-fri-partial-candidate-evidence-def
by (*intro conjI fri unaligned trace-candidate composition-candidate*
trace-low composition-not-low)
show ?thesis
unfolding *composition-fri-bad-with-sampled-layer-chain-def*
by (*intro exI conjI*)
(rule evidence, rule chain)
qed

lemma *wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-sampled:*
assumes *sampled-bound:*
wp-event verify-monad (composition-fri-bad-with-sampled-layer-chain s) s
 $\leq R$
shows
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s
 $\leq R$
proof –
have *wp-event verify-monad*
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s $\leq$
wp-event verify-monad (composition-fri-bad-with-sampled-layer-chain s) s
by (*rule wp-event-mono*)
(rule
composition-fri-verifier-tied-sampled-layer-chain-imp-sampled-layer-chain)
then show ?thesis
by (*rule order-trans[OF - sampled-bound]*)
qed

lemma *wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-full-cover-and-missing:*
assumes *full-bound:*
wp-event verify-monad
(composition-fri-bad-with-full-cover-layer-chain s bad) s $\leq R$
and *missing-bound:*
wp-event verify-monad
(composition-fri-sampled-missing-full-cover s bad) s $\leq M$
shows
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s
 $\leq R + M$
by (*rule*
wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-sampled)
(rule wp-composition-fri-sampled-bound-from-full-cover-and-missing
[OF full-bound missing-bound])

lemma *trace-fri-header-tied-residual-obligations-from-full-cover-missing-without-head-slot-and-selected-zero:*
assumes *full-bound:*
wp-event verify-monad
(trace-fri-bad-with-full-cover-layer-chain s bad) s $\leq R$
and *missing-bound:*

wp-event verify-monad
 (trace-fri-sampled-missing-full-cover s bad) $s \leq M$
and zero:
wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction s) $s \leq z-b$
and merkle:
wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq p-b$
and structural:
wp-event verify-monad
 (trace-fri-header-tied-sibling-mismatch-structural-gap s) $s \leq h-b$
and next-bound:
wp-event verify-monad
 (trace-fri-header-tied-next-value-replay-gap s) $s \leq n-b$
and successor:
wp-event verify-monad
 (trace-fri-header-tied-successor-replay-gap s) $s \leq s-b$
and total:
 $(R + M) + (((((p-b + h-b) + p-b) + n-b + s-b +$
 $(p-b + 0 + 0)) + p-b) + 0 + z-b)$
 $\leq \text{trace-fri-error}$
shows
trace-fri-header-tied-residual-obligations s
 $(R + M) \ z-b \ 0 \ p-b \ h-b \ p-b \ n-b \ s-b \ 0$
by (rule
trace-fri-header-tied-residual-obligations-from-bounds-without-head-slot-and-selected-zero)
 (rule
wp-trace-fri-header-tied-sampled-layer-chain-bound-from-full-cover-and-missing
 $[OF \ \text{full-bound} \ \text{missing-bound}]$,
rule zero, *rule merkle*, *rule structural*, *rule next-bound*,
rule successor, *rule total*)

lemma *composition-fri-verifier-tied-residual-obligations-from-full-cover-missing-slot-zero:*

assumes *full-bound:*
wp-event verify-monad
 (composition-fri-bad-with-full-cover-layer-chain s bad) $s \leq R$
and *missing-bound:*
wp-event verify-monad
 (composition-fri-sampled-missing-full-cover s bad) $s \leq M$
and merkle:
wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq p-b$
and next-bound:
wp-event verify-monad
 (composition-fri-verifier-tied-next-value-replay-gap s) $s \leq n-b$
and successor:
wp-event verify-monad
 (composition-fri-verifier-tied-successor-replay-gap s) $s \leq s-b$
and final-bound:
wp-event verify-monad
 (composition-fri-verifier-tied-final-value-replay-gap s) $s \leq g-b$

and total:
 $(R + M) + ((p-b + 0) + n-b + s-b + g-b + p-b) + 0)$
 $\leq \text{composition-fri-error}$
shows
composition-fri-verifier-tied-residual-obligations s
 $(R + M) \ 0 \ p-b \ n-b \ s-b \ g-b$
by (rule *composition-fri-verifier-tied-residual-obligations-from-bounds-slot-zero*)
(rule
wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-full-cover-and-missing
[*OF full-bound missing-bound*],
rule merkle, rule next-bound, rule successor, rule final-bound,
rule total)

lemma *composition-fri-verifier-tied-badE:*
assumes *composition-fri-bad-with-verifier-tied-partial-candidate s out*
obtains *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table **where**
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
partial-trace-table-candidate trace-table trace-openings
partial-composition-table-candidate composition-table
composition-openings
trace-table-low-degree trace-table
 $\neg$ *composition-table-low-degree maxDegree composition-table*
using *assms*
unfolding *composition-fri-bad-with-verifier-tied-partial-candidate-def*
by *blast*

lemma *trace-fri-header-tied-bad-generic-partial-evidence:*
assumes *trace-fri-bad-with-header-tied-partial-candidate s out*
obtains *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table **where**
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
partial-trace-table-candidate trace-table trace-openings
 $\neg$ *trace-table-low-degree trace-table*

generic-fri-partial-evidence trace-table-low-degree trace-table
(clength - 1) trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers
proof –
from *trace-fri-header-tied-badE[OF assms]* **obtain** *trace-roots trace-bs*
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr as rest
trace-openings trace-table
where *fri*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *header*:
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
and *partial*:
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *candidate*:
partial-trace-table-candidate trace-table trace-openings
and *not-low*: $\neg$ *trace-table-low-degree trace-table*
by *blast*
have *generic*:
generic-fri-partial-evidence trace-table-low-degree trace-table
(clength - 1) trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers
by (rule *accepted-fri-opening-transcript-trace-generic-partial-evidence*
 $[OF\ fri]$)
show *?thesis*
by (rule *that[OF fri header partial candidate not-low generic]*)
qed

lemma *composition-fri-verifier-tied-bad-generic-partial-evidence*:
assumes *composition-fri-bad-with-verifier-tied-partial-candidate s out*
obtains *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table **where**
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
partial-trace-table-candidate trace-table trace-openings
partial-composition-table-candidate composition-table
composition-openings
trace-table-low-degree trace-table

```

    ¬ composition-table-low-degree maxDegree composition-table
generic-fri-partial-evidence
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) composition-roots composition-bs composition-final
  fri-query-idxs composition-round-layers
proof –
from composition-fri-verifier-tied-badE[OF assms] obtain trace-roots
  trace-bs trace-final fri-dg composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr as trace-openings composition-openings
  trace-table composition-table
where fri:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final fri-dg composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers
and aligned:
  accepted-with-partial-initial-openings-aligned s out fr trace-roots
  trace-final as fri-dg composition-roots composition-final
  fri-query-idxs trace-openings composition-openings
and trace-candidate:
  partial-trace-table-candidate trace-table trace-openings
and composition-candidate:
  partial-composition-table-candidate composition-table
  composition-openings
and trace-low: trace-table-low-degree trace-table
and composition-not-low:
  ¬ composition-table-low-degree maxDegree composition-table
by blast
have generic:
  generic-fri-partial-evidence
  (composition-table-low-degree (to-nat fri-dg)) composition-table
  (to-nat fri-dg) composition-roots composition-bs composition-final
  fri-query-idxs composition-round-layers
by (rule
  accepted-fri-opening-transcript-composition-generic-partial-evidence
  [OF fri])
show ?thesis
by (rule that[OF fri aligned trace-candidate composition-candidate
  trace-low composition-not-low generic])
qed

end

end

```

```

theory Soundness-FRI-Prequery-Core
imports

```

begin

Verifier-local FRI challenge prequery accounting.

The concrete FRI challenge-list bounds require freshness of future Trace/Composition FRI challenge keys. In the staged adversary experiment the verifier receives the adversary's final oracle map, so freshness is not automatic. This layer defines the exact verifier-header key paths and proves deterministic fresh/prequeried splits. It does not add assumptions or change the oracle model.

context *soundness*

begin

definition *trace-fri-challenge-key-at*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow$
 $\text{nat} \Rightarrow 'f \text{ protocol-hash-input}$

where

trace-fri-challenge-key-at *s* *fr* *roots* *i* =
 $\text{TraceFriChallenge } (P\text{TraceFriCounter } s + i)$
 $(\text{foldl concat } (\text{concat } (P\text{State } s) \text{ fr}) (\text{take } (\text{Suc } i) \text{ roots}))$

definition *composition-fri-challenge-key-at*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow$
 $'f \text{ protocol-hash-input}$

where

composition-fri-challenge-key-at *s* *fr* *trace-roots* *trace-final* *as* *dg*
 $\text{composition-roots } i =$
 $\text{CompositionFriChallenge } (P\text{CompositionFriCounter } s + i)$
 $(\text{foldl concat}$
 $(\text{concat}$
 $(\text{foldl concat}$
 $(\text{concat}$
 $(\text{foldl concat } (\text{concat } (P\text{State } s) \text{ fr}) \text{ trace-roots})$
 $\text{trace-final})$
 $\text{as})$
 $\text{dg})$
 $(\text{take } (\text{Suc } i) \text{ composition-roots}))$

definition *trace-fri-challenge-path-fresh*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow$
 $\text{nat} \Rightarrow \text{bool}$

where

trace-fri-challenge-path-fresh *s* *fr* *roots* *n* $\longleftrightarrow$
 $(\forall i < n.$
 $i < \text{length roots} \longrightarrow$
 $\text{fmlookup } (\text{HashMap } s) (\text{trace-fri-challenge-key-at } s \text{ fr roots } i) =$
 $\text{None})$

definition *trace-fri-challenge-path-prequeried*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow$
 $\text{nat} \Rightarrow \text{bool}$

where

$\text{trace-fri-challenge-path-prequeried } s \text{ fr roots } n \longleftrightarrow$
 $(\exists i < n.$
 $i < \text{length roots} \wedge$
 $\text{fmlookup } (\text{HashMap } s) (\text{trace-fri-challenge-key-at } s \text{ fr roots } i)$
 $\neq \text{None})$

definition *composition-fri-challenge-path-fresh*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow \text{bool}$

where

$\text{composition-fri-challenge-path-fresh } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots } n \longleftrightarrow$
 $(\forall i < n.$
 $i < \text{length composition-roots} \longrightarrow$
 $\text{fmlookup } (\text{HashMap } s)$
 $(\text{composition-fri-challenge-key-at } s \text{ fr trace-roots trace-final}$
 $\text{as dg composition-roots } i) = \text{None})$

definition *composition-fri-challenge-path-prequeried*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow \text{bool}$

where

$\text{composition-fri-challenge-path-prequeried } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots } n \longleftrightarrow$
 $(\exists i < n.$
 $i < \text{length composition-roots} \wedge$
 $\text{fmlookup } (\text{HashMap } s)$
 $(\text{composition-fri-challenge-key-at } s \text{ fr trace-roots trace-final}$
 $\text{as dg composition-roots } i) \neq \text{None})$

definition *trace-fri-challenge-list-fresh-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list set} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-challenge-list-fresh-hit } s \text{ B out} \longleftrightarrow$
 $(\exists \text{ trace-bs dg comp-bs fr trace-roots trace-final as}$
 $\text{composition-roots final rest.}$
 $\text{accepted-fri-challenges } s \text{ out trace-bs dg comp-bs} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots final rest} \wedge$
 $\text{trace-bs} \in B \wedge$
 $\text{trace-fri-challenge-path-fresh } s \text{ fr trace-roots}$
 $(\text{length trace-roots}))$

definition *trace-fri-challenge-list-prequery-hit*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list set} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-challenge-list-prequery-hit } s \ B \ \text{out} \longleftrightarrow$
 $(\exists \text{ trace-bs } dg \ \text{comp-bs } fr \ \text{trace-roots } \text{trace-final} \ \text{as}$
 $\quad \text{composition-roots } \text{final } \text{rest}.$
 $\text{accepted-fri-challenges } s \ \text{out } \text{trace-bs } dg \ \text{comp-bs} \wedge$
 $\text{verifier-header-transcript } s \ \text{fr } \text{trace-roots } \text{trace-final} \ \text{as } dg$
 $\quad \text{composition-roots } \text{final } \text{rest} \wedge$
 $\text{trace-bs} \in B \wedge$
 $\text{trace-fri-challenge-path-prequeried } s \ \text{fr } \text{trace-roots}$
 $\quad (\text{length } \text{trace-roots}))$

definition *composition-fri-challenge-list-fresh-hit*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow ('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-challenge-list-fresh-hit } s \ B \ \text{out} \longleftrightarrow$
 $(\exists \text{ trace-bs } dg \ \text{comp-bs } fr \ \text{trace-roots } \text{trace-final} \ \text{as}$
 $\quad \text{composition-roots } \text{final } \text{rest}.$
 $\text{accepted-fri-challenges } s \ \text{out } \text{trace-bs } dg \ \text{comp-bs} \wedge$
 $\text{verifier-header-transcript } s \ \text{fr } \text{trace-roots } \text{trace-final} \ \text{as } dg$
 $\quad \text{composition-roots } \text{final } \text{rest} \wedge$
 $\text{comp-bs} \in B \ dg \wedge$
 $\text{composition-fri-challenge-path-fresh } s \ \text{fr } \text{trace-roots } \text{trace-final}$
 $\quad \text{as } dg \ \text{composition-roots } (\text{length } \text{composition-roots}))$

definition *composition-fri-challenge-list-prequery-hit*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow ('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-challenge-list-prequery-hit } s \ B \ \text{out} \longleftrightarrow$
 $(\exists \text{ trace-bs } dg \ \text{comp-bs } fr \ \text{trace-roots } \text{trace-final} \ \text{as}$
 $\quad \text{composition-roots } \text{final } \text{rest}.$
 $\text{accepted-fri-challenges } s \ \text{out } \text{trace-bs } dg \ \text{comp-bs} \wedge$
 $\text{verifier-header-transcript } s \ \text{fr } \text{trace-roots } \text{trace-final} \ \text{as } dg$
 $\quad \text{composition-roots } \text{final } \text{rest} \wedge$
 $\text{comp-bs} \in B \ dg \wedge$
 $\text{composition-fri-challenge-path-prequeried } s \ \text{fr } \text{trace-roots}$
 $\quad \text{trace-final } \text{as } dg \ \text{composition-roots } (\text{length } \text{composition-roots}))$

definition *trace-fri-challenge-key-relation*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list set} \Rightarrow \text{nat} \Rightarrow$
 $'f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}$

where

$\text{trace-fri-challenge-key-relation } s \ B \ n \ x \ y \longleftrightarrow$
 $(\exists bs \in B. \exists \text{fr roots } i.$
 $\quad i < n \wedge$

$i < \text{length } bs \wedge$
 $i < \text{length } roots \wedge$
 $x = \text{trace-fri-challenge-key-at } s \text{ fr } roots \ i \wedge$
 $y = bs \ ! \ i)$

definition *composition-fri-challenge-key-relation*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow ('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $\text{nat} \Rightarrow 'f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}$

where

$\text{composition-fri-challenge-key-relation } s \ B \ n \ x \ y \longleftrightarrow$
 $(\exists dg. \exists bs \in B \ dg. \exists fr \ \text{trace-roots } \text{trace-final } as$
 $\quad \text{composition-roots } i.$
 $i < n \wedge$
 $i < \text{length } bs \wedge$
 $i < \text{length } \text{composition-roots} \wedge$
 $x = \text{composition-fri-challenge-key-at } s \text{ fr } \text{trace-roots } \text{trace-final}$
 $\quad as \ dg \ \text{composition-roots } i \wedge$
 $y = bs \ ! \ i)$

definition *checked-staged-trace-fri-prequery-hit*

$:: 'f \text{ list set} \Rightarrow$
 $('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

$\text{checked-staged-trace-fri-prequery-hit } B \ out \longleftrightarrow$
 $(\text{case out of}$
 $\quad \text{None} \Rightarrow \text{False}$
 $\mid \text{Some } (data, \text{attacker-state}) \Rightarrow$
 $\quad \text{let } s = \text{verifier-state-from-adversary } \text{attacker-state}$
 $\quad (\text{staged-proof-transcript } data)$
 $\quad \text{in } \text{staged-trace-fri-challenges } data \in B \wedge$
 $\quad \text{trace-fri-challenge-path-prequeried } s$
 $\quad (\text{staged-trace-root } data)$
 $\quad (\text{staged-trace-fri-roots } data)$
 $\quad (\text{length } (\text{staged-trace-fri-roots } data))))$

definition *checked-staged-composition-fri-prequery-hit*

$:: ('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $('f \text{ staged-proof-data} \times 'f \text{ protocol-channel}) \text{ option} \Rightarrow \text{bool}$

where

$\text{checked-staged-composition-fri-prequery-hit } B \ out \longleftrightarrow$
 $(\text{case out of}$
 $\quad \text{None} \Rightarrow \text{False}$
 $\mid \text{Some } (data, \text{attacker-state}) \Rightarrow$
 $\quad \text{let } s = \text{verifier-state-from-adversary } \text{attacker-state}$
 $\quad (\text{staged-proof-transcript } data)$
 $\quad \text{in } \text{staged-composition-fri-challenges } data \in$
 $\quad B \ (\text{staged-degree } data) \wedge$
 $\quad \text{composition-fri-challenge-path-prequeried } s$
 $\quad (\text{staged-trace-root } data))$

(staged-trace-fri-roots data)
 (staged-trace-final data)
 (staged-alphas data)
 (staged-degree data)
 (staged-composition-fri-roots data)
 (length (staged-composition-fri-roots data)))

definition *checked-staged-trace-fri-prequery-relation*

:: 'f staged-adversary $\Rightarrow$ 'f list set $\Rightarrow$
 'f protocol-hash-input $\Rightarrow$ 'f $\Rightarrow$ bool

where

checked-staged-trace-fri-prequery-relation $A\ B\ x\ y \longleftrightarrow$
 ($\exists$ data attacker-state i .
 Some (data, attacker-state) $\in$
 set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\wedge$
 staged-trace-fri-challenges data $\in B \wedge$
 $i < \text{ceil-log } \text{clength} \wedge$
 $i < \text{length } (\text{staged-trace-fri-challenges data}) \wedge$
 $i < \text{length } (\text{staged-trace-fri-roots data}) \wedge$
 $x = \text{trace-fri-challenge-key-at}$
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data) (staged-trace-fri-roots data) $i \wedge$
 $y = \text{staged-trace-fri-challenges data } !\ i$)

definition *checked-staged-composition-fri-prequery-relation*

:: 'f staged-adversary $\Rightarrow$ ('f $\Rightarrow$ 'f list set) $\Rightarrow$
 'f protocol-hash-input $\Rightarrow$ 'f $\Rightarrow$ bool

where

checked-staged-composition-fri-prequery-relation $A\ B\ x\ y \longleftrightarrow$
 ($\exists$ data attacker-state i .
 Some (data, attacker-state) $\in$
 set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\wedge$
 staged-composition-fri-challenges data $\in B$ (staged-degree data) $\wedge$
 $i < \text{ceil-log } (\text{maxDegree} + 1) \wedge$
 $i < \text{length } (\text{staged-composition-fri-challenges data}) \wedge$
 $i < \text{length } (\text{staged-composition-fri-roots data}) \wedge$
 $x = \text{composition-fri-challenge-key-at}$
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
 (staged-trace-root data) (staged-trace-fri-roots data)
 (staged-trace-final data) (staged-alphas data)
 (staged-degree data) (staged-composition-fri-roots data) $i \wedge$
 $y = \text{staged-composition-fri-challenges data } !\ i$)

definition *checked-staged-trace-fri-verifier-prequery-relation*
 $:: 'f \text{ staged-adversary} \Rightarrow 'f \text{ list set} \Rightarrow$
 $'f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}$
where
 $\text{checked-staged-trace-fri-verifier-prequery-relation } A \ B \ x \ y \longleftrightarrow$
 $(\exists \text{ data attacker-state } bs \text{ fr roots } i.$
 $\text{Some } (data, \text{attacker-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \wedge$
 $bs \in B \wedge$
 $i < \text{ceil-log } \text{clength} \wedge$
 $i < \text{length } bs \wedge$
 $i < \text{length } roots \wedge$
 $x = \text{trace-fri-challenge-key-at}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $\text{fr roots } i \wedge$
 $y = bs ! i)$

definition *checked-staged-composition-fri-verifier-prequery-relation*
 $:: 'f \text{ staged-adversary} \Rightarrow ('f \Rightarrow 'f \text{ list set}) \Rightarrow$
 $'f \text{ protocol-hash-input} \Rightarrow 'f \Rightarrow \text{bool}$
where
 $\text{checked-staged-composition-fri-verifier-prequery-relation } A \ B \ x \ y$
 $\longleftrightarrow$
 $(\exists \text{ data attacker-state } dg \ bs \text{ fr trace-roots trace-final as}$
 $\text{composition-roots } i.$
 $\text{Some } (data, \text{attacker-state}) \in$
 set-dist
 $(\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \wedge$
 $bs \in B \ dg \wedge$
 $i < \text{ceil-log } (\text{maxDegree} + 1) \wedge$
 $i < \text{length } bs \wedge$
 $i < \text{length } \text{composition-roots} \wedge$
 $x = \text{composition-fri-challenge-key-at}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $\text{fr trace-roots trace-final as } dg \text{ composition-roots } i \wedge$
 $y = bs ! i)$

lemma *trace-fri-challenge-key-relation-fiber-card-bound:*
assumes *finite-B:* *finite B*
shows
 $\text{card } \{y. \text{trace-fri-challenge-key-relation } s \ B \ n \ x \ y\} \leq \text{card } B * n$
proof –
let *?pairs* = $B \times \{..<n\}$
let *?values* = $(\lambda(bs, i). bs ! i) \text{ ` } ?pairs$

```

have subset:
  {y. trace-fri-challenge-key-relation s B n x y}  $\subseteq$  ?values
  unfolding trace-fri-challenge-key-relation-def
  by force
have finite-values: finite ?values
  using finite-B by simp
have card {y. trace-fri-challenge-key-relation s B n x y}  $\leq$ 
  card ?values
  by (rule card-mono[OF finite-values subset])
also have ...  $\leq$  card ?pairs
  by (rule card-image-le) (use finite-B in simp)
also have ... = card B * n
  using finite-B by simp
finally show ?thesis .
qed

lemma composition-fri-challenge-key-relation-fiber-card-bound:
  assumes finite-B:  $\bigwedge dg. \text{finite } (B \ dg)$ 
  shows
    card {y. composition-fri-challenge-key-relation s B n x y}  $\leq$ 
    ( $\sum dg \in (UNIV :: 'f \text{ set}). \text{card } (B \ dg) * n$ )
proof -
  let ?pairs = ( $\text{SIGMA } dg:(UNIV :: 'f \text{ set}). B \ dg \times \{..<n\}$ )
  let ?values = ( $\lambda(dg, bs, i). bs \ ! \ i$ ) ' ?pairs
  have subset:
    {y. composition-fri-challenge-key-relation s B n x y}  $\subseteq$  ?values
    unfolding composition-fri-challenge-key-relation-def
    by force
  have finite-values: finite ?values
  using finite-B by simp
  have card {y. composition-fri-challenge-key-relation s B n x y}  $\leq$ 
    card ?values
    by (rule card-mono[OF finite-values subset])
  also have ...  $\leq$  card ?pairs
    by (rule card-image-le) (use finite-B in simp)
  also have ... =
    ( $\sum dg \in (UNIV :: 'f \text{ set}). \text{card } (B \ dg \times \{..<n\})$ )
    using finite-B by (simp add: card-SigmaI)
  also have ... = ( $\sum dg \in (UNIV :: 'f \text{ set}). \text{card } (B \ dg) * n$ )
    using finite-B by simp
  finally show ?thesis .
qed

lemma hash-relation-program-checked-staged-transcript-trace-fri-prequery:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B: finite B
  shows
    hash-relation-program (trace-fri-challenge-key-relation s B n)

```

```

    (card  $B * n$ )
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    (checked-staged-transcript-program A)
  by (rule hash-relation-program-checked-staged-transcript-program
      [OF wf controlled])
    (rule trace-fri-challenge-key-relation-fiber-card-bound[OF finite-B])

lemma checked-staged-transcript-trace-fri-prequery-relation-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and finite-B: finite B
  shows
    wp-event (checked-staged-transcript-program A)
      (hash-relation-hit-event (trace-fri-challenge-key-relation s B n)
        adversary-initial-state)
    adversary-initial-state  $\leq$ 
    hash-relation-budget-value (card  $B * n$ )
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
proof -
  have program:
    hash-relation-program (trace-fri-challenge-key-relation s B n)
      (card  $B * n$ )
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (checked-staged-transcript-program A)
  by (rule hash-relation-program-checked-staged-transcript-trace-fri-prequery
      [OF wf controlled finite-B])
  show ?thesis
    using program
    unfolding hash-relation-program-def hash-relation-budget-def by blast
qed

lemma hash-relation-program-checked-staged-transcript-composition-fri-prequery:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and finite-B:  $\bigwedge dg. \text{finite } (B \ dg)$ 
  shows
    hash-relation-program (composition-fri-challenge-key-relation s B n)
      ( $\sum dg \in (UNIV :: 'f \text{ set}). \text{card } (B \ dg) * n$ )
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (checked-staged-transcript-program A)
  by (rule hash-relation-program-checked-staged-transcript-program
      [OF wf controlled])
    (rule composition-fri-challenge-key-relation-fiber-card-bound
      [OF finite-B])

lemma checked-staged-transcript-composition-fri-prequery-relation-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and finite-B:  $\bigwedge dg. \text{finite } (B \ dg)$ 

```

```

shows
  wp-event (checked-staged-transcript-program A)
    (hash-relation-hit-event
      (composition-fri-challenge-key-relation s B n)
      adversary-initial-state)
  adversary-initial-state ≤
  hash-relation-budget-value
    (∑ dg ∈ (UNIV :: 'f set). card (B dg) * n)
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
proof –
  have program:
    hash-relation-program (composition-fri-challenge-key-relation s B n)
      (∑ dg ∈ (UNIV :: 'f set). card (B dg) * n)
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (checked-staged-transcript-program A)
  by (rule
    hash-relation-program-checked-staged-transcript-composition-fri-prequery
      [OF wf controlled finite-B])
  show ?thesis
    using program
    unfolding hash-relation-program-def hash-relation-budget-def by blast
qed

end

context soundness
begin

lemma checked-staged-trace-fri-prequery-relation-fiber-card-bound:
  assumes finite-B: finite B
  shows
    card {y. checked-staged-trace-fri-prequery-relation A B x y} ≤
      card B * ceil-log clength
proof –
  let ?pairs = B × {.. $\text{ceil-log clength}$ }
  let ?values = (λ(bs, i). bs ! i) ‘ ?pairs
  have subset:
    {y. checked-staged-trace-fri-prequery-relation A B x y} ⊆
      ?values
  unfolding checked-staged-trace-fri-prequery-relation-def
  by force
  have finite-values: finite ?values
  using finite-B by simp
  have card {y. checked-staged-trace-fri-prequery-relation A B x y} ≤
    card ?values
  by (rule card-mono[OF finite-values subset])
  also have ... ≤ card ?pairs
  by (rule card-image-le) (use finite-B in simp)
  also have ... = card B * ceil-log clength

```

```

    using finite-B by simp
    finally show ?thesis .
qed

lemma checked-staged-composition-fri-prequery-relation-fiber-card-bound:
  assumes finite-B:  $\bigwedge dg. \text{finite } (B \ dg)$ 
  shows
    card  $\{y. \text{checked-staged-composition-fri-prequery-relation } A \ B \ x \ y\}$ 
       $\leq (\sum dg \in (UNIV :: 'f \ set). \text{card } (B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$ 
  proof -
    let ?pairs =
      (SIGMA dg:(UNIV :: 'f set). B dg  $\times$   $\{..<\text{ceil-log } (\text{maxDegree} + 1)\}$ )
    let ?values =  $(\lambda(dg, bs, i). bs \ ! \ i) \ ` \ ?pairs$ 
    have subset:
       $\{y. \text{checked-staged-composition-fri-prequery-relation } A \ B \ x \ y\}$ 
         $\subseteq ?values$ 
    unfolding checked-staged-composition-fri-prequery-relation-def
    by force
    have finite-values: finite ?values
      using finite-B by simp
    have card  $\{y. \text{checked-staged-composition-fri-prequery-relation } A \ B \ x \ y\}$ 
       $\leq \text{card } ?values$ 
      by (rule card-mono[OF finite-values subset])
    also have ...  $\leq \text{card } ?pairs$ 
      by (rule card-image-le) (use finite-B in simp)
    also have ... =
       $(\sum dg \in (UNIV :: 'f \ set). \text{card } (B \ dg \times \{..<\text{ceil-log } (\text{maxDegree} + 1)\}))$ 
      using finite-B by (simp add: card-SigmaI)
    also have ... =
       $(\sum dg \in (UNIV :: 'f \ set). \text{card } (B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$ 
      using finite-B by simp
    finally show ?thesis .
  qed

```

```

lemma checked-staged-trace-fri-verifier-prequery-relation-fiber-card-bound:
  assumes finite-B: finite B
  shows
    card  $\{y. \text{checked-staged-trace-fri-verifier-prequery-relation } A \ B \ x \ y\}$ 
       $\leq \text{card } B * \text{ceil-log clength}$ 
  proof -
    let ?pairs =  $B \times \{..<\text{ceil-log clength}\}$ 
    let ?values =  $(\lambda(bs, i). bs \ ! \ i) \ ` \ ?pairs$ 
    have subset:
       $\{y. \text{checked-staged-trace-fri-verifier-prequery-relation } A \ B \ x \ y\}$ 
         $\subseteq ?values$ 
    unfolding checked-staged-trace-fri-verifier-prequery-relation-def

```

by auto
 have card {y. checked-staged-trace-fri-verifier-prequery-relation A B x y}
 ≤ card ?values
 by (rule card-mono[OF finite-imageI[OF finite-cartesian-product
 [OF finite-B finite-lessThan]] subset])
 also have ... ≤ card ?pairs
 by (rule card-image-le) (use finite-B in simp)
 also have ... = card B * ceil-log clength
 using finite-B by simp
 finally show ?thesis .
 qed

lemma

checked-staged-composition-fri-verifier-prequery-relation-fiber-card-bound:

assumes finite-B: $\bigwedge dg. \text{finite } (B \ dg)$

shows

card
 {y. checked-staged-composition-fri-verifier-prequery-relation A B x y}
 ≤ $(\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))

proof –

let ?pairs =
 SIGMA dg:(UNIV :: 'f set). B dg × {.. $\text{ceil-log } (\text{maxDegree} + 1)$ }
 let ?values = $(\lambda(dg, bs, i). bs \ ! \ i) \ ` \ ?pairs$
 have subset:
 {y. checked-staged-composition-fri-verifier-prequery-relation A B x y}
 ⊆ ?values
 unfolding checked-staged-composition-fri-verifier-prequery-relation-def
 by force
 have finite-pairs: finite ?pairs
 using finite-B by simp
 have card
 {y. checked-staged-composition-fri-verifier-prequery-relation A B x y}
 ≤ card ?values
 by (rule card-mono[OF finite-imageI[OF finite-pairs] subset])
 also have ... ≤ card ?pairs
 by (rule card-image-le) (rule finite-pairs)
 also have ... =
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 using finite-B by (simp add: card-SigmaI)
 finally show ?thesis .

qed

lemma *checked-staged-transcript-trace-fri-verifier-prequery-relation-bound:*

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and finite-B: finite B

shows

wp-event (*checked-staged-transcript-program* *A*)
 (*hash-relation-hit-event*
 (*checked-staged-trace-fri-verifier-prequery-relation* *A* *B*)
adversary-initial-state)
adversary-initial-state $\leq$
hash-relation-budget-value (*card* *B* * *ceil-log* *clength*)
 (*staged-attacker-query-budget* *budgets* + *staged-challenge-query-budget*)
proof –
have *program*:
hash-relation-program
 (*checked-staged-trace-fri-verifier-prequery-relation* *A* *B*)
 (*card* *B* * *ceil-log* *clength*)
 (*staged-attacker-query-budget* *budgets* + *staged-challenge-query-budget*)
 (*checked-staged-transcript-program* *A*)
by (*rule* *hash-relation-program-checked-staged-transcript-program*
 [*OF wf controlled*])
 (*rule* *checked-staged-trace-fri-verifier-prequery-relation-fiber-card-bound*
 [*OF finite-B*])
then show *?thesis*
unfolding *hash-relation-program-def* *hash-relation-budget-def* **by** *blast*
qed

lemma

checked-staged-transcript-composition-fri-verifier-prequery-relation-bound:

assumes *wf*: *staged-budget-wellformed* *budgets*

and *controlled*: *staged-adversary-controlled* *budgets* *A*

and *finite-B*: $\bigwedge dg. \text{finite } (B \ dg)$

shows

wp-event (*checked-staged-transcript-program* *A*)
 (*hash-relation-hit-event*
 (*checked-staged-composition-fri-verifier-prequery-relation* *A* *B*)
adversary-initial-state)
adversary-initial-state $\leq$
hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
card (*B* *dg*) * *ceil-log* (*maxDegree* + 1))
 (*staged-attacker-query-budget* *budgets* + *staged-challenge-query-budget*)

proof –

have *program*:

hash-relation-program
 (*checked-staged-composition-fri-verifier-prequery-relation* *A* *B*)
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
card (*B* *dg*) * *ceil-log* (*maxDegree* + 1))
 (*staged-attacker-query-budget* *budgets* + *staged-challenge-query-budget*)
 (*checked-staged-transcript-program* *A*)
by (*rule* *hash-relation-program-checked-staged-transcript-program*
 [*OF wf controlled*])
 (*rule*
checked-staged-composition-fri-verifier-prequery-relation-fiber-card-bound

```

    [OF finite-B])
  then show ?thesis
    unfolding hash-relation-program-def hash-relation-budget-def by blast
qed

end

context soundness
begin

lemma checked-staged-trace-fri-verifier-prequery-hit-imp-relation-hit:
  assumes builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:
    Some (result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
  and hit:
    trace-fri-challenge-list-prequery-hit
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
    B (Some (result, final-state))
  shows
    hash-relation-hit
      (checked-staged-trace-fri-verifier-prequery-relation A B)
      adversary-initial-state attacker-state
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  from hit obtain trace-bs dg comp-bs fr roots trace-final as
    composition-roots final rest where
    challenges:
      accepted-fri-challenges ?s (Some (result, final-state))
      trace-bs dg comp-bs
  and header:
    verifier-header-transcript ?s fr roots trace-final as dg
    composition-roots final rest
  and trace-bs-B: trace-bs ∈ B
  and prequeried:
    trace-fri-challenge-path-prequeried ?s fr roots (length roots)
  unfolding trace-fri-challenge-list-prequery-hit-def by blast
  from prequeried obtain i y where
    i-len: i < length roots

```



```

and lookup-s:
  fmlookup (HashMap ?s) (trace-fri-challenge-key-at ?s fr roots i) =
    Some y
unfolding trace-fri-challenge-path-prequeried-def
by (auto split: option.splits)
have lookup-info:
  i < length trace-bs ∧ y = trace-bs ! i
by (rule verify-monad-trace-fri-prequery-lookup
    [OF verifier challenges header i-len])
    (use lookup-s in ⟨simp add: trace-fri-challenge-key-at-def⟩)
have i-round: i < ceil-log clength
proof –
  have trace-bs ∈ fri-challenge-space (ceil-log clength)
    by (rule accepted-fri-challenges-trace-space[OF challenges])
  then have length trace-bs = ceil-log clength
    unfolding fri-challenge-space-def by simp
  then show ?thesis
    using lookup-info by simp
qed
have relation:
  checked-staged-trace-fri-verifier-prequery-relation A B
    (trace-fri-challenge-key-at ?s fr roots i) y
unfolding checked-staged-trace-fri-verifier-prequery-relation-def
apply (rule exI[where x = data])
apply (rule exI[where x = attacker-state])
apply (rule exI[where x = trace-bs])
apply (rule exI[where x = fr])
apply (rule exI[where x = roots])
apply (rule exI[where x = i])
using builder trace-bs-B i-round i-len lookup-info by simp
have initial-absent:
  fmlookup (HashMap adversary-initial-state)
    (trace-fri-challenge-key-at ?s fr roots i) = None
unfolding adversary-initial-state-def by simp
have attacker-lookup:
  fmlookup (HashMap attacker-state)
    (trace-fri-challenge-key-at ?s fr roots i) = Some y
using lookup-s unfolding verifier-state-from-adversary-def by simp
show ?thesis
  unfolding hash-relation-hit-def
  using initial-absent attacker-lookup relation by blast
qed

lemma checked-staged-composition-fri-verifier-prequery-hit-imp-relation-hit:
assumes builder:
  Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)

```

```

and verifier:
  Some (result, final-state) ∈
    set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
and hit:
  composition-fri-challenge-list-prequery-hit
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    B (Some (result, final-state))
shows
  hash-relation-hit
    (checked-staged-composition-fri-verifier-prequery-relation A B)
    adversary-initial-state attacker-state
proof –
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
from hit obtain trace-bs dg comp-bs fr trace-roots trace-final as
  composition-roots final rest where
    challenges:
      accepted-fri-challenges ?s (Some (result, final-state))
      trace-bs dg comp-bs
and header:
  verifier-header-transcript ?s fr trace-roots trace-final as dg
  composition-roots final rest
and comp-bs-B: comp-bs ∈ B dg
and prequeried:
  composition-fri-challenge-path-prequeried ?s fr trace-roots
  trace-final as dg composition-roots (length composition-roots)
unfolding composition-fri-challenge-list-prequery-hit-def by blast
from prequeried obtain i y where
  i-len: i < length composition-roots
and lookup-s:
  fmlookup (HashMap ?s)
  (composition-fri-challenge-key-at ?s fr trace-roots trace-final as
    dg composition-roots i) =
    Some y
unfolding composition-fri-challenge-path-prequeried-def
by (auto split: option.splits)
have lookup-info:
  i < length comp-bs ∧ y = comp-bs ! i
by (rule verify-monad-composition-fri-prequery-lookup
    [OF verifier challenges header i-len])
  (use lookup-s in
    ⟨simp add: composition-fri-challenge-key-at-def⟩)
have i-round: i < ceil-log (maxDegree + 1)
proof –

```

```

have comp-bs  $\in$  fri-challenge-space (ceil-log (to-nat dg + 1))
  by (rule accepted-fri-challenges-composition-space[OF challenges])
then have len-comp: length comp-bs = ceil-log (to-nat dg + 1)
  unfolding fri-challenge-space-def by simp
have dg-bound: to-nat dg  $\leq$  maxDegree
  by (rule verify-monad-accepted-fri-challenges-degree-bound
    [OF verifier challenges])
have ceil-log (to-nat dg + 1)  $\leq$  ceil-log (maxDegree + 1)
  by (rule ceil-log-to-nat-degree-bound[OF dg-bound])
then show ?thesis
  using lookup-info len-comp by simp
qed

have relation:
  checked-staged-composition-fri-verifier-prequery-relation A B
    (composition-fri-challenge-key-at ?s fr trace-roots trace-final as dg
      composition-roots i) y
  unfolding checked-staged-composition-fri-verifier-prequery-relation-def
  apply (rule exI[where x = data])
  apply (rule exI[where x = attacker-state])
  apply (rule exI[where x = dg])
  apply (rule exI[where x = comp-bs])
  apply (rule exI[where x = fr])
  apply (rule exI[where x = trace-roots])
  apply (rule exI[where x = trace-final])
  apply (rule exI[where x = as])
  apply (rule exI[where x = composition-roots])
  apply (rule exI[where x = i])
  using builder comp-bs-B i-round i-len lookup-info by simp
have initial-absent:
  fmlookup (HashMap adversary-initial-state)
    (composition-fri-challenge-key-at ?s fr trace-roots trace-final as dg
      composition-roots i) = None
  unfolding adversary-initial-state-def by simp
have attacker-lookup:
  fmlookup (HashMap attacker-state)
    (composition-fri-challenge-key-at ?s fr trace-roots trace-final as dg
      composition-roots i) = Some y
  using lookup-s unfolding verifier-state-from-adversary-def by simp
show ?thesis
  unfolding hash-relation-hit-def
  using initial-absent attacker-lookup relation by blast
qed

lemma checked-staged-security-trace-fri-verifier-prequery-hit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B: finite B
  shows
  wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (staged-security-with-data-state-verifier-event
     (λs. trace-fri-challenge-list-prequery-hit s B))
    adversary-initial-state ≤
    hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  unfolding checked-staged-security-experiment-with-data-state-def
proof (rule wp-event-bind-bound-by-head-event)
  show wp-event (checked-staged-transcript-program A)
    (hash-relation-hit-event
     (checked-staged-trace-fri-verifier-prequery-relation A B)
     adversary-initial-state)
    adversary-initial-state
    ≤ hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule checked-staged-transcript-trace-fri-verifier-prequery-relation-bound
      [OF wf controlled finite-B])
next
  show staged-security-with-data-state-verifier-event
    (λs. trace-fri-challenge-list-prequery-hit s B) None ⇒
    hash-relation-hit-event
    (checked-staged-trace-fri-verifier-prequery-relation A B)
    adversary-initial-state None
  unfolding staged-security-with-data-state-verifier-event-def by simp
next
  fix data attacker-state out
  assume builder:
    Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
     adversary-initial-state)
  and cont:
    out ∈
    set-dist
    (execute
     (get >=>
      (λs. put
       (verifier-state-from-adversary s
        (staged-proof-transcript data)) >=>
        (λ-. verify-monad >=>
         (λresult. return ((data, s), result))))))
     attacker-state)
  and hit:
    staged-security-with-data-state-verifier-event
    (λs. trace-fri-challenge-list-prequery-hit s B) out
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from cont hit obtain result final-state where
    verifier:

```

Some (result, final-state) $\in$ set-dist (execute verify-monad ?s)
and prequery:
 trace-fri-challenge-list-prequery-hit ?s B
 (Some (result, final-state))
unfolding staged-security-with-data-state-verifier-event-def
by (auto elim!: set-dist-bindE split: option.splits prod.splits)
have relation-hit:
 hash-relation-hit
 (checked-staged-trace-fri-verifier-prequery-relation A B)
 adversary-initial-state attacker-state
by (rule checked-staged-trace-fri-verifier-prequery-hit-imp-relation-hit
 [OF builder verifier prequery])
then show
 hash-relation-hit-event
 (checked-staged-trace-fri-verifier-prequery-relation A B)
 adversary-initial-state (Some (data, attacker-state))
unfolding hash-relation-hit-event-def **by** simp
qed

lemma checked-staged-security-composition-fri-verifier-prequery-hit-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: $\bigwedge dg. \text{finite } (B \ dg)$
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s. \text{composition-fri-challenge-list-prequery-hit } s \ B$))
 adversary-initial-state $\leq$
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets + staged-challenge-query-budget)
unfolding checked-staged-security-experiment-with-data-state-def
proof (rule wp-event-bind-bound-by-head-event)
show wp-event (checked-staged-transcript-program A)
 (hash-relation-hit-event
 (checked-staged-composition-fri-verifier-prequery-relation A B)
 adversary-initial-state)
 adversary-initial-state
 $\leq$ hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets + staged-challenge-query-budget)
by (rule
 checked-staged-transcript-composition-fri-verifier-prequery-relation-bound
 [OF wf controlled finite-B])
next
show staged-security-with-data-state-verifier-event
 ($\lambda s. \text{composition-fri-challenge-list-prequery-hit } s \ B$) None $\implies$

```

    hash-relation-hit-event
      (checked-staged-composition-fri-verifier-prequery-relation A B)
      adversary-initial-state None
    unfolding staged-security-with-data-state-verifier-event-def by simp
next
fix data attacker-state out
assume builder:
  Some (data, attacker-state) ∈
    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
and cont:
  out ∈
    set-dist
      (execute
        (get >>=
          (λs. put
            (verifier-state-from-adversary s
              (staged-proof-transcript data)) >>=
            (λ-. verify-monad >>=
              (λresult. return ((data, s), result))))))
          attacker-state)
and hit:
  staged-security-with-data-state-verifier-event
    (λs. composition-fri-challenge-list-prequery-hit s B) out
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from cont hit obtain result final-state where
  verifier:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  and prequery:
    composition-fri-challenge-list-prequery-hit ?s B
    (Some (result, final-state))
  unfolding staged-security-with-data-state-verifier-event-def
  by (auto elim!: set-dist-bindE split: option.splits prod.splits)
have relation-hit:
  hash-relation-hit
    (checked-staged-composition-fri-verifier-prequery-relation A B)
    adversary-initial-state attacker-state
  by (rule
    checked-staged-composition-fri-verifier-prequery-hit-imp-relation-hit
      [OF builder verifier prequery])
then show
  hash-relation-hit-event
    (checked-staged-composition-fri-verifier-prequery-relation A B)
    adversary-initial-state (Some (data, attacker-state))
  unfolding hash-relation-hit-event-def by simp
qed

```

lemma *checked-staged-trace-fri-prequery-hit-imp-fixed-relation-hit:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *outcome:*
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
and *hit:*
checked-staged-trace-fri-prequery-hit B
(Some (data, attacker-state))
shows
hash-relation-hit
(checked-staged-trace-fri-prequery-relation A B)
adversary-initial-state attacker-state
proof –
let *?s =*
verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
have *member: staged-trace-fri-challenges data ∈ B*
using *hit unfolding checked-staged-trace-fri-prequery-hit-def* **by** *simp*
have *prequeried:*
trace-fri-challenge-path-prequeried ?s
(staged-trace-root data) (staged-trace-fri-roots data)
(length (staged-trace-fri-roots data))
using *hit unfolding checked-staged-trace-fri-prequery-hit-def* **by** *simp*
from *prequeried* **obtain** *i* **where**
i-len: i < length (staged-trace-fri-roots data)
and *lookup-s:*
fmlookup (HashMap ?s)
(trace-fri-challenge-key-at ?s (staged-trace-root data)
(staged-trace-fri-roots data) i) ≠ None
unfolding *trace-fri-challenge-path-prequeried-def* **by** *blast*
have *shape:*
length (staged-trace-fri-roots data) = ceil-log clength ∧
length (staged-trace-fri-challenges data) = ceil-log clength ∧
length (staged-alphas data) = length spec ∧
length (staged-composition-fri-roots data) =
ceil-log (to-nat (staged-degree data) + 1) ∧
length (staged-composition-fri-challenges data) =
ceil-log (to-nat (staged-degree data) + 1) ∧
ceil-log (to-nat (staged-degree data) + 1) ≤
ceil-log (maxDegree + 1) ∧
length (staged-query-chunks data) = rounds
by *(rule checked-staged-transcript-program-outcome-shape*
[OF wf controlled outcome])
have *unchecked-out:*
Some (data, attacker-state) ∈

```

    set-dist (execute (staged-transcript-program A)
      adversary-initial-state)
  by (rule checked-staged-transcript-program-outcome-imp-staged-transcript-program
    [OF outcome])
have staged-lookup:
  fmlookup (HashMap attacker-state)
    (TraceFriChallenge i
      (foldl concat (staged-trace-fri-start-hash data)
        (take (Suc i) (staged-trace-fri-roots data)))) =
    Some (staged-trace-fri-challenges data ! i)
  by (rule staged-transcript-program-outcome-trace-challenge-lookup
    [OF wf controlled unchecked-out])
    (use i-len shape in simp)
have key-eq:
  trace-fri-challenge-key-at ?s (staged-trace-root data)
    (staged-trace-fri-roots data) i =
    TraceFriChallenge i
      (foldl concat (staged-trace-fri-start-hash data)
        (take (Suc i) (staged-trace-fri-roots data)))
  unfolding trace-fri-challenge-key-at-def staged-trace-fri-start-hash-def
  by simp
have relation:
  checked-staged-trace-fri-prequery-relation A B
    (trace-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) i)
    (staged-trace-fri-challenges data ! i)
  unfolding checked-staged-trace-fri-prequery-relation-def
  by (intro exI[of - data] exI[of - attacker-state] exI[of - i])
    (use outcome member i-len shape in simp-all)
have initial-absent:
  fmlookup (HashMap adversary-initial-state)
    (trace-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) i) = None
  unfolding adversary-initial-state-def by simp
have staged-lookup-key:
  fmlookup (HashMap attacker-state)
    (trace-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) i) =
    Some (staged-trace-fri-challenges data ! i)
  using staged-lookup key-eq by simp
show ?thesis
  unfolding hash-relation-hit-def
  using initial-absent staged-lookup-key relation by blast
qed

```

```

lemma checked-staged-composition-fri-prequery-hit-imp-fixed-relation-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:

```



```

    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and hit:
    checked-staged-composition-fri-prequery-hit B
    (Some (data, attacker-state))
  shows
    hash-relation-hit
      (checked-staged-composition-fri-prequery-relation A B)
      adversary-initial-state attacker-state
  proof -
    let ?s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have member:
    staged-composition-fri-challenges data ∈ B (staged-degree data)
  using hit unfolding checked-staged-composition-fri-prequery-hit-def
  by simp
  have prequeried:
    composition-fri-challenge-path-prequeried ?s
      (staged-trace-root data) (staged-trace-fri-roots data)
      (staged-trace-final data) (staged-alphas data) (staged-degree data)
      (staged-composition-fri-roots data)
      (length (staged-composition-fri-roots data))
  using hit unfolding checked-staged-composition-fri-prequery-hit-def
  by simp
  from prequeried obtain i where
    i-len: i < length (staged-composition-fri-roots data)
  and lookup-s:
    fmllookup (HashMap ?s)
      (composition-fri-challenge-key-at ?s (staged-trace-root data)
        (staged-trace-fri-roots data) (staged-trace-final data)
        (staged-alphas data) (staged-degree data)
        (staged-composition-fri-roots data) i) ≠ None
  unfolding composition-fri-challenge-path-prequeried-def by blast
  have shape:
    length (staged-trace-fri-roots data) = ceil-log clength ∧
    length (staged-trace-fri-challenges data) = ceil-log clength ∧
    length (staged-alphas data) = length spec ∧
    length (staged-composition-fri-roots data) =
      ceil-log (to-nat (staged-degree data) + 1) ∧
    length (staged-composition-fri-challenges data) =
      ceil-log (to-nat (staged-degree data) + 1) ∧
    ceil-log (to-nat (staged-degree data) + 1) ≤
      ceil-log (maxDegree + 1) ∧
    length (staged-query-chunks data) = rounds
  by (rule checked-staged-transcript-program-outcome-shape
    [OF wf controlled outcome])

```

```

have unchecked-out:
  Some (data, attacker-state) ∈
    set-dist (execute (staged-transcript-program A)
      adversary-initial-state)
by (rule checked-staged-transcript-program-outcome-imp-staged-transcript-program
  [OF outcome])
have staged-lookup:
  fmlookup (HashMap attacker-state)
    (CompositionFriChallenge i
      (foldl concat (staged-composition-fri-start-hash data)
        (take (Suc i) (staged-composition-fri-roots data)))) =
    Some (staged-composition-fri-challenges data ! i)
by (rule staged-transcript-program-outcome-composition-challenge-lookup
  [OF wf controlled unchecked-out])
  (use i-len shape in simp)
have key-eq:
  composition-fri-challenge-key-at ?s (staged-trace-root data)
    (staged-trace-fri-roots data) (staged-trace-final data)
    (staged-alphas data) (staged-degree data)
    (staged-composition-fri-roots data) i =
    CompositionFriChallenge i
      (foldl concat (staged-composition-fri-start-hash data)
        (take (Suc i) (staged-composition-fri-roots data)))
unfolding composition-fri-challenge-key-at-def
  staged-composition-fri-start-hash-def staged-trace-fri-start-hash-def
by simp
have relation:
  checked-staged-composition-fri-prequery-relation A B
    (composition-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) (staged-trace-final data)
      (staged-alphas data) (staged-degree data)
      (staged-composition-fri-roots data) i)
    (staged-composition-fri-challenges data ! i)
unfolding checked-staged-composition-fri-prequery-relation-def
by (intro exI[of - data] exI[of - attacker-state] exI[of - i])
  (use outcome member i-len shape in simp-all)
have initial-absent:
  fmlookup (HashMap adversary-initial-state)
    (composition-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) (staged-trace-final data)
      (staged-alphas data) (staged-degree data)
      (staged-composition-fri-roots data) i) = None
unfolding adversary-initial-state-def by simp
have staged-lookup-key:
  fmlookup (HashMap attacker-state)
    (composition-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) (staged-trace-final data)
      (staged-alphas data) (staged-degree data)
      (staged-composition-fri-roots data) i) =

```

```

    Some (staged-composition-fri-challenges data ! i)
  using staged-lookup key-eq by simp
show ?thesis
  unfolding hash-relation-hit-def
  using initial-absent staged-lookup-key relation by blast
qed

lemma checked-staged-transcript-trace-fri-prequery-hit-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and finite-B: finite B
  shows
    wp-event (checked-staged-transcript-program A)
      (checked-staged-trace-fri-prequery-hit B)
      adversary-initial-state ≤
      hash-relation-budget-value (card B * ceil-log clength)
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
proof -
  have prehit-le-relation:
    wp-event (checked-staged-transcript-program A)
      (checked-staged-trace-fri-prequery-hit B)
      adversary-initial-state ≤
    wp-event (checked-staged-transcript-program A)
      (hash-relation-hit-event
        (checked-staged-trace-fri-prequery-relation A B)
        adversary-initial-state)
      adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support:
      out ∈ set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
    and hit: checked-staged-trace-fri-prequery-hit B out
    show
      hash-relation-hit-event
        (checked-staged-trace-fri-prequery-relation A B)
        adversary-initial-state out
  proof (cases out)
    case None
    then show ?thesis
      using hit unfolding checked-staged-trace-fri-prequery-hit-def by simp
  next
    case (Some result)
    then obtain data attacker-state where out-eq:
      out = Some (data, attacker-state)
    by (cases result) simp
    have relation-hit:
      hash-relation-hit

```

```

      (checked-staged-trace-fri-prequery-relation A B)
      adversary-initial-state attacker-state
    by (rule checked-staged-trace-fri-prequery-hit-imp-fixed-relation-hit
        [OF wf controlled - -])
        (use support hit out-eq in simp-all)
    then show ?thesis
      unfolding out-eq hash-relation-hit-event-def by simp
qed
qed
also have ... ≤
  hash-relation-budget-value (card B * ceil-log clength)
  (staged-attacker-query-budget budgets + staged-challenge-query-budget)
proof -
  have program:
    hash-relation-program
    (checked-staged-trace-fri-prequery-relation A B)
    (card B * ceil-log clength)
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    (checked-staged-transcript-program A)
  by (rule hash-relation-program-checked-staged-transcript-program
      [OF wf controlled])
      (rule checked-staged-trace-fri-prequery-relation-fiber-card-bound
          [OF finite-B])
  show ?thesis
    using program
    unfolding hash-relation-program-def hash-relation-budget-def by blast
qed
finally show ?thesis .
qed

```

lemma *checked-staged-transcript-composition-fri-prequery-hit-bound:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: $\bigwedge dg. \text{finite } (B \ dg)$*

shows

wp-event (checked-staged-transcript-program A)
(checked-staged-composition-fri-prequery-hit B)
adversary-initial-state ≤
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$
(staged-attacker-query-budget budgets + staged-challenge-query-budget)

proof -

have *prehit-le-relation:*

wp-event (checked-staged-transcript-program A)
(checked-staged-composition-fri-prequery-hit B)
adversary-initial-state ≤
wp-event (checked-staged-transcript-program A)
(hash-relation-hit-event

```

    (checked-staged-composition-fri-prequery-relation A B)
    adversary-initial-state)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈ set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  and hit: checked-staged-composition-fri-prequery-hit B out
show
  hash-relation-hit-event
  (checked-staged-composition-fri-prequery-relation A B)
  adversary-initial-state out
proof (cases out)
  case None
  then show ?thesis
    using hit unfolding checked-staged-composition-fri-prequery-hit-def
    by simp
  next
  case (Some result)
  then obtain data attacker-state where out-eq:
    out = Some (data, attacker-state)
  by (cases result) simp
  have relation-hit:
    hash-relation-hit
    (checked-staged-composition-fri-prequery-relation A B)
    adversary-initial-state attacker-state
  by (rule
    checked-staged-composition-fri-prequery-hit-imp-fixed-relation-hit
    [OF wf controlled - -])
    (use support hit out-eq in simp-all)
  then show ?thesis
    unfolding out-eq hash-relation-hit-event-def by simp
  qed
qed
also have ... ≤
  hash-relation-budget-value
  (∑ dg ∈ (UNIV :: 'f set).
    card (B dg) * ceil-log (maxDegree + 1))
  (staged-attacker-query-budget budgets + staged-challenge-query-budget)
proof –
  have program:
    hash-relation-program
    (checked-staged-composition-fri-prequery-relation A B)
    (∑ dg ∈ (UNIV :: 'f set).
      card (B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    (checked-staged-transcript-program A)

```

```

    by (rule hash-relation-program-checked-staged-transcript-program
        [OF wf controlled])
      (rule
        checked-staged-composition-fri-prequery-relation-fiber-card-bound
        [OF finite-B])
  show ?thesis
    using program
    unfolding hash-relation-program-def hash-relation-budget-def by blast
qed
finally show ?thesis .
qed

end

context soundness
begin

lemma checked-staged-trace-fri-prequery-hit-imp-relation-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and hit:
    checked-staged-trace-fri-prequery-hit B
    (Some (data, attacker-state))
  shows
    hash-relation-hit
      (trace-fri-challenge-key-relation
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
        B (length (staged-trace-fri-roots data)))
      adversary-initial-state attacker-state
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have member: staged-trace-fri-challenges data ∈ B
  using hit unfolding checked-staged-trace-fri-prequery-hit-def by simp
  have prequeried:
    trace-fri-challenge-path-prequeried ?s
    (staged-trace-root data) (staged-trace-fri-roots data)
    (length (staged-trace-fri-roots data))
  using hit unfolding checked-staged-trace-fri-prequery-hit-def by simp
  from prequeried obtain i where
    i-len: i < length (staged-trace-fri-roots data)
  and lookup-s:

```

```

    fmlookup (HashMap ?s)
      (trace-fri-challenge-key-at ?s (staged-trace-root data)
        (staged-trace-fri-roots data) i) ≠ None
  unfolding trace-fri-challenge-path-prequeried-def by blast
have shape:
  length (staged-trace-fri-roots data) = ceil-log clength ∧
  length (staged-trace-fri-challenges data) = ceil-log clength ∧
  length (staged-alphas data) = length spec ∧
  length (staged-composition-fri-roots data) =
    ceil-log (to-nat (staged-degree data) + 1) ∧
  length (staged-composition-fri-challenges data) =
    ceil-log (to-nat (staged-degree data) + 1) ∧
  ceil-log (to-nat (staged-degree data) + 1) ≤
    ceil-log (maxDegree + 1) ∧
  length (staged-query-chunks data) = rounds
by (rule checked-staged-transcript-program-outcome-shape
  [OF wf controlled outcome])
have unchecked-out:
  Some (data, attacker-state) ∈
    set-dist (execute (staged-transcript-program A)
      adversary-initial-state)
by (rule checked-staged-transcript-program-outcome-imp-staged-transcript-program
  [OF outcome])
have staged-lookup:
  fmlookup (HashMap attacker-state)
    (TraceFriChallenge i
      (foldl concat (staged-trace-fri-start-hash data)
        (take (Suc i) (staged-trace-fri-roots data)))) =
    Some (staged-trace-fri-challenges data ! i)
by (rule staged-transcript-program-outcome-trace-challenge-lookup
  [OF wf controlled unchecked-out])
  (use i-len shape in simp)
have key-eq:
  trace-fri-challenge-key-at ?s (staged-trace-root data)
    (staged-trace-fri-roots data) i =
    TraceFriChallenge i
      (foldl concat (staged-trace-fri-start-hash data)
        (take (Suc i) (staged-trace-fri-roots data)))
  unfolding trace-fri-challenge-key-at-def staged-trace-fri-start-hash-def
  by simp
have relation:
  trace-fri-challenge-key-relation ?s B
    (length (staged-trace-fri-roots data))
    (trace-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) i)
    (staged-trace-fri-challenges data ! i)
proof -
  have i-challenges: i < length (staged-trace-fri-challenges data)
  using i-len shape by simp

```

```

show ?thesis
  unfolding trace-fri-challenge-key-relation-def
  by (intro bexI[of - staged-trace-fri-challenges data]
    exI[of - staged-trace-root data]
    exI[of - staged-trace-fri-roots data]
    exI[of - i])
    (use member i-len i-challenges in simp-all)
qed
have initial-absent:
  fmlookup (HashMap adversary-initial-state)
    (trace-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) i) = None
  unfolding adversary-initial-state-def by simp
have staged-lookup-key:
  fmlookup (HashMap attacker-state)
    (trace-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) i) =
    Some (staged-trace-fri-challenges data ! i)
  using staged-lookup key-eq by simp
show ?thesis
  unfolding hash-relation-hit-def
  using initial-absent staged-lookup-key relation by blast
qed

lemma checked-staged-composition-fri-prequery-hit-imp-relation-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and outcome:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and hit:
    checked-staged-composition-fri-prequery-hit B
      (Some (data, attacker-state))
  shows
    hash-relation-hit
      (composition-fri-challenge-key-relation
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      B (length (staged-composition-fri-roots data)))
      adversary-initial-state attacker-state
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have member:
    staged-composition-fri-challenges data ∈ B (staged-degree data)
  using hit unfolding checked-staged-composition-fri-prequery-hit-def

```



```

by simp
have prequeried:
  composition-fri-challenge-path-prequeried ?s
  (staged-trace-root data) (staged-trace-fri-roots data)
  (staged-trace-final data) (staged-alphas data) (staged-degree data)
  (staged-composition-fri-roots data)
  (length (staged-composition-fri-roots data))
using hit unfolding checked-staged-composition-fri-prequery-hit-def
by simp
from prequeried obtain i where
  i-len: i < length (staged-composition-fri-roots data)
and lookup-s:
  fmlookup (HashMap ?s)
  (composition-fri-challenge-key-at ?s (staged-trace-root data)
    (staged-trace-fri-roots data) (staged-trace-final data)
    (staged-alphas data) (staged-degree data)
    (staged-composition-fri-roots data) i) ≠ None
unfolding composition-fri-challenge-path-prequeried-def by blast
have shape:
  length (staged-trace-fri-roots data) = ceil-log clength ∧
  length (staged-trace-fri-challenges data) = ceil-log clength ∧
  length (staged-alphas data) = length spec ∧
  length (staged-composition-fri-roots data) =
    ceil-log (to-nat (staged-degree data) + 1) ∧
  length (staged-composition-fri-challenges data) =
    ceil-log (to-nat (staged-degree data) + 1) ∧
  ceil-log (to-nat (staged-degree data) + 1) ≤
    ceil-log (maxDegree + 1) ∧
  length (staged-query-chunks data) = rounds
by (rule checked-staged-transcript-program-outcome-shape
  [OF wf controlled outcome])
have unchecked-out:
  Some (data, attacker-state) ∈
    set-dist (execute (staged-transcript-program A)
      adversary-initial-state)
by (rule checked-staged-transcript-program-outcome-imp-staged-transcript-program
  [OF outcome])
have staged-lookup:
  fmlookup (HashMap attacker-state)
  (CompositionFriChallenge i
    (foldl concat (staged-composition-fri-start-hash data)
      (take (Suc i) (staged-composition-fri-roots data)))) =
    Some (staged-composition-fri-challenges data ! i)
by (rule staged-transcript-program-outcome-composition-challenge-lookup
  [OF wf controlled unchecked-out])
  (use i-len shape in simp)
have key-eq:
  composition-fri-challenge-key-at ?s (staged-trace-root data)
  (staged-trace-fri-roots data) (staged-trace-final data)

```

```

    (staged-alphas data) (staged-degree data)
    (staged-composition-fri-roots data) i =
    CompositionFriChallenge i
    (foldl concat (staged-composition-fri-start-hash data)
      (take (Suc i) (staged-composition-fri-roots data)))
  unfolding composition-fri-challenge-key-at-def
    staged-composition-fri-start-hash-def staged-trace-fri-start-hash-def
  by simp
have relation:
  composition-fri-challenge-key-relation ?s B
  (length (staged-composition-fri-roots data))
  (composition-fri-challenge-key-at ?s (staged-trace-root data)
    (staged-trace-fri-roots data) (staged-trace-final data)
    (staged-alphas data) (staged-degree data)
    (staged-composition-fri-roots data) i)
  (staged-composition-fri-challenges data ! i)
proof -
  have i-challenges: i < length (staged-composition-fri-challenges data)
    using i-len shape by simp
  show ?thesis
    unfolding composition-fri-challenge-key-relation-def
    by (intro exI[of - staged-degree data]
      bexI[of - staged-composition-fri-challenges data]
      exI[of - staged-trace-root data]
      exI[of - staged-trace-fri-roots data]
      exI[of - staged-trace-final data]
      exI[of - staged-alphas data]
      exI[of - staged-composition-fri-roots data]
      exI[of - i])
      (use member i-len i-challenges in simp-all)
qed
have initial-absent:
  fmlookup (HashMap adversary-initial-state)
    (composition-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) (staged-trace-final data)
      (staged-alphas data) (staged-degree data)
      (staged-composition-fri-roots data) i) = None
  unfolding adversary-initial-state-def by simp
have staged-lookup-key:
  fmlookup (HashMap attacker-state)
    (composition-fri-challenge-key-at ?s (staged-trace-root data)
      (staged-trace-fri-roots data) (staged-trace-final data)
      (staged-alphas data) (staged-degree data)
      (staged-composition-fri-roots data) i) =
    Some (staged-composition-fri-challenges data ! i)
  using staged-lookup key-eq by simp
show ?thesis
  unfolding hash-relation-hit-def
  using initial-absent staged-lookup-key relation by blast

```

qed

end

end

theory *Soundness-FRI-Prequery-Challenge-Bounds*
 imports *Soundness-FRI-Prequery-Core*
begin

Staged and verifier-local FRI challenge-list accounting built on the prequery core definitions and relation bounds.

context *soundness*
begin

definition *trace-fri-root-path-fresh-from*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow \text{bool}$
where
 $\text{trace-fri-root-path-fresh-from } s \text{ roots } n \longleftrightarrow$
 $(\forall i < n.$
 $i < \text{length roots} \longrightarrow$
 $\text{fmlookup } (\text{HashMap } s)$
 $(\text{TraceFriChallenge } (P\text{TraceFriCounter } s + i)$
 $(\text{foldl concat } (P\text{State } s) (\text{take } (\text{Suc } i) \text{ roots}))) = \text{None})$

definition *composition-fri-root-path-fresh-from*
 $:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow \text{bool}$
where
 $\text{composition-fri-root-path-fresh-from } s \text{ roots } n \longleftrightarrow$
 $(\forall i < n.$
 $i < \text{length roots} \longrightarrow$
 $\text{fmlookup } (\text{HashMap } s)$
 $(\text{CompositionFriChallenge } (P\text{CompositionFriCounter } s + i)$
 $(\text{foldl concat } (P\text{State } s) (\text{take } (\text{Suc } i) \text{ roots}))) = \text{None})$

lemma *trace-fri-challenge-path-fresh-after-read-iff:*
 assumes *read-fr:* $\text{Some } (fr, s1) \in \text{set-dist } (\text{execute read } s)$
 shows

$\text{trace-fri-challenge-path-fresh } s \text{ fr roots } n \longleftrightarrow$
 $\text{trace-fri-root-path-fresh-from } s1 \text{ roots } n$

proof –

have *state:* $P\text{State } s1 = \text{concat } (P\text{State } s) \text{ fr}$
 using *read-outcome*[*OF read-fr*] **by** *blast*
 have *counter:* $P\text{TraceFriCounter } s1 = P\text{TraceFriCounter } s$
 using *read-outcome*[*OF read-fr*] **by** *blast*
 have *hash:* $\text{HashMap } s1 = \text{HashMap } s$
 by (*rule read-preserves-hash-map*[*OF read-fr*])
 show *?thesis*

```

unfolding trace-fri-challenge-path-fresh-def
  trace-fri-root-path-fresh-from-def
by (simp add: trace-fri-challenge-key-at-def state counter hash)
qed

lemma receive-trace-fri-commits-preserves-trace-root-path-fresh-tail:
  assumes path:
    trace-fri-root-path-fresh-from s (r # roots) (Suc n)
  and outcome:
    Some ((b, r), t) ∈ set-dist (execute receive-trace-fri-commits s)
  shows trace-fri-root-path-fresh-from t roots n
proof -
  from outcome obtain s-read where
    read-r: Some (r, s-read) ∈ set-dist (execute read s)
  and rand-b:
    Some (b, t) ∈ set-dist (execute receive-trace-fri-challenge s-read)
  unfolding receive-trace-fri-commits-def receive-fri-commits-with-def
  by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-r] have read-state:
    PState s-read = concat (PState s) r
  by blast
  from read-outcome[OF read-r] have read-counter:
    PTraceFriCounter s-read = PTraceFriCounter s
  by blast
  have read-hash: HashMap s-read = HashMap s
  by (rule read-preserves-hash-map[OF read-r])
  have counter-t:
    PTraceFriCounter t = Suc (PTraceFriCounter s)
  using receive-trace-fri-challenge-counter-outcome[OF rand-b]
  read-counter by simp
  have state-t: PState t = concat (PState s) r
  using receive-trace-fri-challenge-outcome[OF rand-b] read-state by simp
  show ?thesis
  unfolding trace-fri-root-path-fresh-from-def
  proof (intro allI impI)
  fix i
  assume i-lt: i < n
  and i-len: i < length roots
  let ?key =
    TraceFriChallenge (PTraceFriCounter t + i)
    (foldl concat (PState t) (take (Suc i) roots))
  have key-eq:
    ?key =
    TraceFriChallenge (PTraceFriCounter s + Suc i)
    (foldl concat (PState s) (take (Suc (Suc i)) (r # roots)))
  using counter-t state-t by simp
  have neg:
    ?key ≠ TraceFriChallenge (PTraceFriCounter s-read) (PState s-read)
  using counter-t read-counter by simp

```

```

have fmlookup (HashMap t) ?key =
  fmlookup (HashMap s-read) ?key
by (rule receive-trace-fri-challenge-preserves-other-lookup
  [OF rand-b neq])
also have ... = fmlookup (HashMap s) ?key
  using read-hash by simp
also have ... = None
proof -
  have Suc i < Suc n
    using i-lt by simp
  moreover have Suc i < length (r # roots)
    using i-len by simp
  ultimately show ?thesis
    using path unfolding trace-fri-root-path-fresh-from-def key-eq
    by blast
qed
finally show fmlookup (HashMap t) ?key = None .
qed
qed

lemma wp-receive-trace-fri-commits-actual-fresh-set:
shows
  wp-event receive-trace-fri-commits
  (λout. case out of
    None ⇒ False
  | Some ((b, r), -) ⇒
    b ∈ B ∧
    fmlookup (HashMap s)
      (TraceFriChallenge (PTraceFriCounter s)
        (concat (PState s) r)) = None) s
  ≤ nnreal (card B) / nnreal size
  unfolding receive-trace-fri-commits-def receive-fri-commits-with-def
proof (rule wp-event-bind-bound-by-cont)
  show ¬ (case None of None ⇒ False
  | Some ((b, r), -) ⇒
    b ∈ B ∧
    fmlookup (HashMap s)
      (TraceFriChallenge (PTraceFriCounter s)
        (concat (PState s) r)) = None)
  by simp
next
  fix r s-read
  assume read-r: Some (r, s-read) ∈ set-dist (execute read s)
  have read-state: PState s-read = concat (PState s) r
    using read-outcome[OF read-r] by blast
  have read-counter: PTraceFriCounter s-read = PTraceFriCounter s
    using read-outcome[OF read-r] by blast
  have read-hash: HashMap s-read = HashMap s
    by (rule read-preserves-hash-map[OF read-r])

```

```

let ?fresh =
  fmlookup (HashMap s-read)
    (TraceFriChallenge (PTraceFriCounter s-read) (PState s-read)) = None
let ?Q =
  λout :: (('f × 'f) × ('f, 'a) protocol-channel-scheme) option.
    case out of
      None ⇒ False
    | Some ((b, r'), -) ⇒
      b ∈ B ∧
      fmlookup (HashMap s)
        (TraceFriChallenge (PTraceFriCounter s)
          (concat (PState s) r')) = None
show wp-event
  (receive-trace-fri-challenge ≫= (λb. return (b, r))) ?Q
  s-read
  ≤ nnreal (card B) / nnreal size
proof (cases ?fresh)
case True
have fresh-key:
  fmlookup (HashMap s)
    (TraceFriChallenge (PTraceFriCounter s) (concat (PState s) r)) =
    None
using True by (simp add: read-state read-counter read-hash)
have cont-eq:
  (λout. case out of
    None ⇒ (0 :: prob)
    | Some (b, u) ⇒
      wp (return (b, r)) (λout. if ?Q out then 1 else 0) u) =
  (λout :: (('f × ('f, 'a) protocol-channel-scheme) option.
    if (case out of None ⇒ False | Some (b, -) ⇒ b ∈ B)
    then 1 else 0)
  by (rule ext)
  (auto simp: wpsimps fresh-key split: option.splits prod.splits)
have wp-event
  (receive-trace-fri-challenge ≫= (λb. return (b, r))) ?Q
  s-read =
  wp-event receive-trace-fri-challenge
  (λout. case out of None ⇒ False | Some (b, -) ⇒ b ∈ B)
  s-read
  unfolding wp-event-def by (simp add: wpsimps cont-eq)
also have ... = nnreal (card B) / nnreal size
  by (rule wp-receive-trace-fri-challenge-fresh-set[OF True])
finally show ?thesis by simp
next
case False
have not-fresh-s:
  fmlookup (HashMap s)
    (TraceFriChallenge (PTraceFriCounter s) (concat (PState s) r))
    ≠ None

```

```

    using False by (simp add: read-state read-counter read-hash)
  have wp-event
    (receive-trace-fri-challenge  $\gg$  ( $\lambda b$ . return (b, r))) ?Q
    s-read
     $\leq$  wp-event
    (receive-trace-fri-challenge  $\gg$  ( $\lambda b$ . return (b, r)))
    ( $\lambda$ - :: (('f  $\times$  'f)  $\times$ 
      ('f, 'a) protocol-channel-scheme) option. False) s-read
  by (rule wp-event-mono-on-support)
    (use not-fresh-s in
       $\langle$ auto simp: wpsimps elim!: set-dist-bindE
        split: option.splits prod.splits $\rangle$ )
  also have ... = 0
    unfolding wp-event-def wp-def dist-expect-def by simp
  finally have zero-bound:
    wp-event
    (receive-trace-fri-challenge  $\gg$  ( $\lambda b$ . return (b, r))) ?Q
    s-read  $\leq$  0 .
  also have ...  $\leq$  nnreal (card B) / nnreal size
    by simp
  finally show ?thesis .
qed
qed

lemma wp-ntimes-receive-trace-fri-commits-exact-challenges-actual-fresh-bound:
  assumes len: length bs = n
  shows
    wp-event (ntimes receive-trace-fri-commits n)
    ( $\lambda$ out. case out of
      None  $\Rightarrow$  False
      | Some (brs, -)  $\Rightarrow$ 
        map fst brs = bs  $\wedge$ 
        trace-fri-root-path-fresh-from s (map snd brs) n) s
     $\leq$  (1 / nnreal size)  $^n$ 
  using len
proof (induction n arbitrary: s bs)
  case 0
  then have bs-empty: bs = []
    by simp
  show ?case
    unfolding bs-empty by (simp add: wp-event-def wpsimps
      trace-fri-root-path-fresh-from-def)
next
  case (Suc n)
  obtain b bs' where bs-eq: bs = b # bs'
    using Suc.prem by (cases bs) auto
  have len-bs': length bs' = n
    using Suc.prem bs-eq by simp
  let ?tail =

```

```

(ntimes receive-trace-fri-commits n ::
  (('f × 'f) list, ('f, 'a) protocol-channel-scheme) state-monad)
let ?Q =
  λout :: (((('f × 'f) list) ×
    ('f, 'a) protocol-channel-scheme) option.
    case out of
      None ⇒ False
    | Some (brs, -) ⇒
      map fst brs = bs ∧
      trace-fri-root-path-fresh-from s (map snd brs) (Suc n)
let ?Head =
  λout :: (((('f × 'f) × ('f, 'a) protocol-channel-scheme) option.
    case out of
      None ⇒ False
    | Some ((b0, r0), -) ⇒
      b0 ∈ {b} ∧
      fmlookup (HashMap s)
        (TraceFriChallenge (PTraceFriCounter s)
          (concat (PState s) r0)) = None
have head-bound:
  wp-event receive-trace-fri-commits ?Head s ≤ 1 / nnreal size
proof -
  have wp-event receive-trace-fri-commits ?Head s
    ≤ nnreal (card ({b} :: 'f set)) / nnreal size
  by (rule wp-receive-trace-fri-commits-actual-fresh-set)
  then show ?thesis by simp
qed
have bind-bound:
  wp-event
    (receive-trace-fri-commits ≫=
      (λbr :: 'f × 'f.
        ?tail ≫= (λbrs :: ('f × 'f) list. return (br # brs))))
  ?Q s
  ≤ wp-event receive-trace-fri-commits ?Head s *
    (1 / nnreal size) ^ n
proof (rule wp-event-bind-bound-by-head-and-cont[where P = ?Head])
  show ¬ ?Q None
  by simp
next
  fix br t
  assume not-head: ¬ ?Head (Some (br, t))
  obtain b0 r0 where br-eq: br = (b0, r0)
  by (cases br)
  have cont-false:
    wp-event (?tail ≫= (λbrs :: ('f × 'f) list. return (br # brs)))
    ?Q t
    ≤ wp-event (?tail ≫= (λbrs :: ('f × 'f) list. return (br # brs)))
      (λ- :: (((('f × 'f) list) ×
        ('f, 'a) protocol-channel-scheme) option. False) t

```



```

  by (rule wp-event-mono-on-support)
    (use not-head bs-eq br-eq in
      ‹auto simp: wpsimps trace-fri-root-path-fresh-from-def
        elim!: set-dist-bindE split: option.splits prod.splits›)
also have ... = 0
  unfolding wp-event-def wp-def dist-expect-def by simp
finally have le-zero:
  wp-event (?tail  $\gg$  ( $\lambda$ brs :: ('f  $\times$  'f) list. return (br # brs)))
  ?Q t  $\leq$  0 .
show
  wp-event (?tail  $\gg$  ( $\lambda$ brs :: ('f  $\times$  'f) list. return (br # brs)))
  ?Q t = 0
  by (rule antisym[OF le-zero]) simp
next
fix br t
assume head:
  Some (br, t)  $\in$  set-dist (execute receive-trace-fri-commits s)
  and head-hit: ?Head (Some (br, t))
obtain b0 r0 where br-eq: br = (b0, r0)
  by (cases br)
have b0-eq: b0 = b
  using head-hit unfolding br-eq by simp
let ?TailQ =
   $\lambda$ out :: (((('f  $\times$  'f) list)  $\times$ 
    ('f, 'a) protocol-channel-scheme) option.
    case out of
      None  $\Rightarrow$  False
    | Some (brs, -)  $\Rightarrow$ 
      map fst brs = bs'  $\wedge$ 
      trace-fri-root-path-fresh-from t (map snd brs) n
have cont-le:
  wp-event (?tail  $\gg$  ( $\lambda$ brs :: ('f  $\times$  'f) list. return (br # brs)))
  ?Q t  $\leq$  wp-event ?tail ?TailQ t
proof (rule wp-event-bind-bound-by-head-event)
  show wp-event ?tail ?TailQ t  $\leq$  wp-event ?tail ?TailQ t
    by simp
next
  show ?Q None  $\Longrightarrow$  ?TailQ None
    by simp
next
fix brs u out
assume tail-out:
  Some (brs, u)  $\in$  set-dist (execute ?tail t)
  and ret-out:
    out  $\in$  set-dist (execute (return (br # brs)) u)
  and hit: ?Q out
have out-eq: out = Some (br # brs, u)
  using ret-out
  unfolding set-dist-def return.rep-eq dist-return-def

```

```

    dist-delta-dist delta-map-def by simp
  have map-eq: map fst brs = bs'
    using hit out-eq b0-eq bs-eq br-eq by simp
  have path-full:
    trace-fri-root-path-fresh-from s (map snd (br # brs)) (Suc n)
    using hit out-eq by simp
  have path-full':
    trace-fri-root-path-fresh-from s (r0 # map snd brs) (Suc n)
    using path-full unfolding br-eq by simp
  have path-tail:
    trace-fri-root-path-fresh-from t (map snd brs) n
    by (rule receive-trace-fri-commits-preserves-trace-root-path-fresh-tail
        [OF path-full' head[unfolded br-eq]])
  show ?TailQ (Some (brs, u))
    using map-eq path-tail by simp
qed
also have ... ≤ (1 / nnreal size) ^ n
  by (rule Suc.IH[OF len-bs])
finally show
  wp-event (?tail ≫ (λbrs :: ('f × 'f) list. return (br # brs)))
    ?Q t ≤ (1 / nnreal size) ^ n .
qed
have wp-event (ntimes receive-trace-fri-commits (Suc n)) ?Q s
  ≤ wp-event receive-trace-fri-commits ?Head s *
    (1 / nnreal size) ^ n
  using bind-bound by simp
also have ... ≤ (1 / nnreal size) * (1 / nnreal size) ^ n
  by (rule mult-right-mono[OF head-bound]) simp
also have ... = (1 / nnreal size) ^ Suc n
  by simp
finally show ?case .
qed

lemma wp-ntimes-receive-trace-fri-commits-finite-set-actual-fresh-bound:
  assumes finite-B: finite B
  and lengths: ⋀bs. bs ∈ B ⇒ length bs = n
  shows
    wp-event (ntimes receive-trace-fri-commits n)
      (λout. case out of
        None ⇒ False
        | Some (brs, -) ⇒
          map fst brs ∈ B ∧
          trace-fri-root-path-fresh-from s (map snd brs) n) s
      ≤ nnreal (card B) / (nnreal size) ^ n
proof -
  let ?P =
    λbs out. case out of
      None ⇒ False
      | Some (brs, -) ⇒

```

```

      map fst brs = bs ∧
      trace-fri-root-path-fresh-from s (map snd brs) n
have event-mono:
  wp-event (ntimes receive-trace-fri-commits n)
    (λout. case out of
      None ⇒ False
      | Some (brs, -) ⇒
        map fst brs ∈ B ∧
        trace-fri-root-path-fresh-from s (map snd brs) n) s
    ≤ wp-event (ntimes receive-trace-fri-commits n)
      (λout. ∃ bs ∈ B. ?P bs out) s
  by (rule wp-event-mono) (auto split: option.splits prod.splits)
also have ... ≤ (∑ bs ∈ B. (1 / nnreal size) ^ n)
  by (rule wp-event-finite-UN-bound[OF finite-B])
    (rule wp-ntimes-receive-trace-fri-commits-exact-challenges-actual-fresh-bound
      [OF lengths])
also have ... = nnreal (card B) * ((1 / nnreal size) ^ n)
  using finite-B by simp
also have ... = nnreal (card B) / (nnreal size) ^ n
  using size-card by transfer (simp add: power-divide)
finally show ?thesis .
qed

```

lemma wp-ntimes-receive-trace-fri-commits-challenge-space-actual-fresh-bound:

assumes subset: $B \subseteq \text{fri-challenge-space } n$

shows

```

  wp-event (ntimes receive-trace-fri-commits n)
    (λout. case out of
      None ⇒ False
      | Some (brs, -) ⇒
        map fst brs ∈ B ∧
        trace-fri-root-path-fresh-from s (map snd brs) n) s
    ≤ nnreal (card B) / nnreal (CARD('f) ^ n)

```

proof –

have finite-B: finite B

by (rule finite-subset[OF subset finite-fri-challenge-space])

have lengths: $\bigwedge bs. bs \in B \implies \text{length } bs = n$

using subset **unfolding** fri-challenge-space-def **by** auto

have wp-event (ntimes receive-trace-fri-commits n)

```

  (λout. case out of
    None ⇒ False
    | Some (brs, -) ⇒
      map fst brs ∈ B ∧
      trace-fri-root-path-fresh-from s (map snd brs) n) s
    ≤ nnreal (card B) / (nnreal size) ^ n

```

by (rule wp-ntimes-receive-trace-fri-commits-finite-set-actual-fresh-bound
 [OF finite-B lengths])

also have ... = nnreal (card B) / nnreal (CARD('f) ^ n)

using size-card **by** simp

```

finally show ?thesis .
qed

lemma wp-verifier-trace-fri-prefix-challenge-list-actual-fresh-bound:
  assumes subset:  $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$ 
  shows
    wp-event verifier-trace-fri-prefix
      ( $\lambda \text{out. case out of}$ 
         $\text{None} \Rightarrow \text{False}$ 
         $| \text{Some } ((\text{fr}, \text{f-fl}), -) \Rightarrow$ 
           $\text{map fst f-fl} \in B \wedge$ 
           $\text{trace-fri-challenge-path-fresh } s \text{ fr } (\text{map snd f-fl})$ 
           $(\text{length } (\text{map snd f-fl})) \text{ } s$ 
           $\leq \text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD('f)} \wedge \text{ceil-log } \text{clength})$ )
  proof -
    let ?Q =
       $\lambda \text{out. case out of}$ 
         $\text{None} \Rightarrow \text{False}$ 
         $| \text{Some } ((\text{fr}, \text{f-fl}), -) \Rightarrow$ 
           $\text{map fst f-fl} \in B \wedge$ 
           $\text{trace-fri-challenge-path-fresh } s \text{ fr } (\text{map snd f-fl})$ 
           $(\text{length } (\text{map snd f-fl}))$ 
    let ?C =  $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD('f)} \wedge \text{ceil-log } \text{clength})$ 
    show ?thesis
      unfolding verifier-trace-fri-prefix-def
    proof (rule wp-event-bind-bound-by-cont)
      show  $\neg ?Q \text{ None}$ 
      by simp
    next
      fix fr s1
      assume read-fr:  $\text{Some } (\text{fr}, s1) \in \text{set-dist } (\text{execute read } s)$ 
      let ?Head =
         $\lambda \text{out. case out of}$ 
           $\text{None} \Rightarrow \text{False}$ 
           $| \text{Some } (\text{f-fl}, -) \Rightarrow$ 
             $\text{map fst f-fl} \in B \wedge$ 
             $\text{trace-fri-root-path-fresh-from } s1 \text{ } (\text{map snd f-fl})$ 
             $(\text{ceil-log } \text{clength})$ 
      have trace-bound:
        wp-event ( $\text{ntimes receive-trace-fri-commits } (\text{ceil-log } \text{clength})$ )
          ?Head s1  $\leq ?C$ 
      by (rule wp-ntimes-receive-trace-fri-commits-challenge-space-actual-fresh-bound
        [OF subset])
      show wp-event
        ( $\text{ntimes receive-trace-fri-commits } (\text{ceil-log } \text{clength}) \gg=$ 
           $(\lambda \text{f-fl. return } (\text{fr}, \text{f-fl}))$ )
        ?Q s1  $\leq ?C$ 
      proof (rule wp-event-bind-bound-by-head-event[OF trace-bound])
        show ?Q None  $\implies ?Head \text{ None}$ 

```

```

    by simp
next
fix f-fl t out
assume trace-out:
  Some (f-fl, t) ∈
    set-dist
      (execute (ntimes receive-trace-fri-commits (ceil-log clength))
        s1)
and cont:
  out ∈ set-dist (execute (return (fr, f-fl)) t)
and hit: ?Q out
have out-eq: out = Some ((fr, f-fl), t)
using cont
unfolding set-dist-def return.rep-eq dist-return-def
  dist-delta-dist delta-map-def by simp
have len: length f-fl = ceil-log clength
using ntimes-receive-trace-fri-commits-outcome[OF trace-out] by simp
have path:
  trace-fri-root-path-fresh-from s1 (map snd f-fl)
    (ceil-log clength)
using hit out-eq len
  trace-fri-challenge-path-fresh-after-read-iff[OF read-fr]
by simp
show ?Head (Some (f-fl, t))
using hit out-eq path by simp
qed
qed
qed

lemma wp-verify-monad-trace-fri-challenge-list-fresh-hit-actual-bound:
  assumes subset:  $B \subseteq \text{fri-challenge-space } (\text{ceil-log clength})$ 
  shows
    wp-event verify-monad (trace-fri-challenge-list-fresh-hit s B)  $s \leq$ 
      nnreal (card B) / nnreal ( $\text{CARD}(f) \wedge \text{ceil-log clength}$ )
proof -
  let ?Head =
    λout. case out of
      None ⇒ False
    | Some ((fr, f-fl), -) ⇒
      map fst f-fl ∈ B ∧
      trace-fri-challenge-path-fresh s fr (map snd f-fl)
      (length (map snd f-fl))
  have prefix-bound:
    wp-event verifier-trace-fri-prefix ?Head  $s \leq$ 
      nnreal (card B) / nnreal ( $\text{CARD}(f) \wedge \text{ceil-log clength}$ )
  by (rule wp-verifier-trace-fri-prefix-challenge-list-actual-fresh-bound
    [OF subset])
show ?thesis
  unfolding verify-monad-trace-fri-decomposition

```

```

proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
  show trace-fri-challenge-list-fresh-hit s B None  $\implies$ 
    ?Head None
    unfolding trace-fri-challenge-list-fresh-hit-def
      accepted-fri-challenges-def by blast
next
  fix header prefix-state out
  assume prefix:
    Some (header, prefix-state)  $\in$ 
      set-dist (execute verifier-trace-fri-prefix s)
  and suffix:
    out  $\in$ 
      set-dist (execute (verifier-after-trace-fri header) prefix-state)
  and hit: trace-fri-challenge-list-fresh-hit s B out
  obtain fr f-fl where header-eq: header = (fr, f-fl)
  by (cases header) auto
  from hit obtain trace-bs dg comp-bs fr' trace-roots' trace-final' as'
    composition-roots' final' rest' where
    challenges-hit: accepted-fri-challenges s out trace-bs dg comp-bs
  and header-hit:
    verifier-header-transcript s fr' trace-roots' trace-final' as' dg
    composition-roots' final' rest'
  and trace-bs-B: trace-bs  $\in$  B
  and fresh-hit:
    trace-fri-challenge-path-fresh s fr' trace-roots'
    (length trace-roots')
  unfolding trace-fri-challenge-list-fresh-hit-def by blast
  from challenges-hit obtain result final-state where out-eq:
    out = Some (result, final-state)
  unfolding accepted-fri-challenges-def by blast
  have prefix':
    Some ((fr, f-fl), prefix-state)  $\in$ 
      set-dist (execute verifier-trace-fri-prefix s)
  using prefix header-eq by simp
  have suffix':
    Some (result, final-state)  $\in$ 
      set-dist
        (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
  using suffix header-eq out-eq by simp
  from verifier-after-trace-fri-accepted-fri-challenges[OF prefix' suffix']
  obtain dg' comp-bs' where
    challenges-prefix:
      accepted-fri-challenges s (Some (result, final-state))
      (map fst f-fl) dg' comp-bs'
  by blast
  have trace-bs-eq: trace-bs = map fst f-fl
  using accepted-fri-challenges-unique
    [OF challenges-prefix challenges-hit[unfolded out-eq]]
  by simp

```

```

from challenges-prefix obtain result0 final-state0 fr0 f-fl0
  trace-final0 as0 fl0 final0 query-state0 where body0:
  Some (result, final-state) = Some (result0, final-state0) ∧
  verifier-header-transcript s fr0 (map snd f-fl0) trace-final0 as0
  dg' (map snd fl0) final0 (PTranscript query-state0) ∧
  PState query-state0 =
    verifier-header-state s fr0 (map snd f-fl0) trace-final0 as0
    dg' (map snd fl0) final0 ∧
  Some (result0, final-state0) ∈
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr0 f-fl0 trace-final0 as0
          fl0 final0)
        rounds)
      query-state0) ∧
  PQueryCounter query-state0 = PQueryCounter s ∧
  map fst f-fl = map fst f-fl0 ∧
  comp-bs' = map fst fl0 ∧
  (∀ i < length f-fl0.
    fmlookup (HashMap query-state0)
      (TraceFriChallenge (PTraceFriCounter s + i)
        (foldl concat (concat (PState s) fr0)
          (take (Suc i) (map snd f-fl0)))) =
    Some (fst (f-fl0 ! i))) ∧
  (∀ i < length fl0.
    fmlookup (HashMap query-state0)
      (CompositionFriChallenge (PCompositionFriCounter s + i)
        (foldl concat
          (concat
            (foldl concat
              (concat
                (foldl concat (concat (PState s) fr0)
                  (map snd f-fl0))
                trace-final0)
              as0)
            dg')
          (take (Suc i) (map snd fl0)))) =
    Some (fst (fl0 ! i)))
  unfolding accepted-fri-challenges-def by blast
have header-prefix0:
  verifier-header-transcript s fr0 (map snd f-fl0) trace-final0 as0
  dg' (map snd fl0) final0 (PTranscript query-state0)
using body0 by simp
have trace-bs0: map fst f-fl = map fst f-fl0
using body0 by simp
have prefix-res:
  PTranscript s = [fr] @ map snd f-fl @ PTranscript prefix-state
using verifier-trace-fri-prefix-outcome[OF prefix'] by simp
have len-f-fl: length f-fl = ceil-log clength

```

```

using verifier-trace-fri-prefix-outcome[OF prefix'] by simp
have fr0-eq: fr0 = fr
using prefix-res header-prefix0
unfolding verifier-header-transcript-def verifier-header-messages-def
by simp
have len-f-fl0: length f-fl0 = ceil-log clength
using header-prefix0
unfolding verifier-header-transcript-def by simp
have roots0-eq: map snd f-fl0 = map snd f-fl
proof -
  let ?rest0 =
    trace-final0 # as0 @ dg' # map snd fl0 @ final0 #
    PTranscript query-state0
  have prefix0:
    PTranscript s = [fr] @ map snd f-fl0 @ ?rest0
  using header-prefix0 fr0-eq
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
  have eq:
    [fr] @ map snd f-fl @ PTranscript prefix-state =
    [fr] @ map snd f-fl0 @ ?rest0
  using prefix-res prefix0 by simp
  have take-eq:
    take (Suc (length (map snd f-fl)))
      ([fr] @ map snd f-fl @ PTranscript prefix-state) =
    take (Suc (length (map snd f-fl)))
      ([fr] @ map snd f-fl0 @ ?rest0)
  using eq by simp
  then show ?thesis
  using len-f-fl len-f-fl0 by simp
qed
have header-prefix:
  verifier-header-transcript s fr (map snd f-fl) trace-final0 as0
  dg' (map snd fl0) final0 (PTranscript query-state0)
using header-prefix0 fr0-eq roots0-eq by simp
have roots-eq:
  fr' = fr  $\wedge$  trace-roots' = map snd f-fl
using verifier-header-transcript-unique[OF header-prefix header-hit]
by simp
then have fresh-prefix:
  trace-fri-challenge-path-fresh s fr (map snd f-fl)
  (length (map snd f-fl))
using fresh-hit by simp
show ?Head (Some (header, prefix-state))
using trace-bs-B trace-bs-eq fresh-prefix header-eq by simp
qed
qed

```

lemma receive-composition-fri-commits-preserves-composition-root-path-fresh-tail:


```

assumes path:
  composition-fri-root-path-fresh-from s (r # roots) (Suc n)
and outcome:
  Some ((b, r), t) ∈ set-dist (execute receive-composition-fri-commits s)
shows composition-fri-root-path-fresh-from t roots n
proof -
from outcome obtain s-read where
  read-r: Some (r, s-read) ∈ set-dist (execute read s)
and rand-b:
  Some (b, t) ∈
    set-dist (execute receive-composition-fri-challenge s-read)
unfolding receive-composition-fri-commits-def receive-fri-commits-with-def
by (auto elim!: set-dist-bindE)
from read-outcome[OF read-r] have read-state:
  PState s-read = concat (PState s) r
by blast
from read-outcome[OF read-r] have read-counter:
  PCompositionFriCounter s-read = PCompositionFriCounter s
by blast
have read-hash: HashMap s-read = HashMap s
by (rule read-preserves-hash-map[OF read-r])
have counter-t:
  PCompositionFriCounter t = Suc (PCompositionFriCounter s)
using receive-composition-fri-challenge-counter-outcome[OF rand-b]
  read-counter by simp
have state-t: PState t = concat (PState s) r
using receive-composition-fri-challenge-outcome[OF rand-b] read-state
by simp
show ?thesis
unfolding composition-fri-root-path-fresh-from-def
proof (intro allI impI)
  fix i
  assume i-lt: i < n
  and i-len: i < length roots
  let ?key =
    CompositionFriChallenge (PCompositionFriCounter t + i)
    (foldl concat (PState t) (take (Suc i) roots))
  have key-eq:
    ?key =
      CompositionFriChallenge (PCompositionFriCounter s + Suc i)
      (foldl concat (PState s) (take (Suc (Suc i)) (r # roots)))
  using counter-t state-t by simp
  have neg:
    ?key ≠
      CompositionFriChallenge (PCompositionFriCounter s-read)
      (PState s-read)
  using counter-t read-counter by simp
  have fmlookup (HashMap t) ?key =
    fmlookup (HashMap s-read) ?key

```

```

    by (rule receive-composition-fri-challenge-preserves-other-lookup
        [OF rand-b neq])
  also have ... = fmlookup (HashMap s) ?key
    using read-hash by simp
  also have ... = None
  proof -
    have Suc i < Suc n
      using i-lt by simp
    moreover have Suc i < length (r # roots)
      using i-len by simp
    ultimately show ?thesis
      using path unfolding composition-fri-root-path-fresh-from-def key-eq
      by blast
  qed
  finally show fmlookup (HashMap t) ?key = None .
  qed
qed

lemma wp-receive-composition-fri-commits-actual-fresh-set:
  shows
    wp-event receive-composition-fri-commits
      (λout. case out of
        None ⇒ False
        | Some ((b, r), -) ⇒
          b ∈ B ∧
          fmlookup (HashMap s)
            (CompositionFriChallenge (PCompositionFriCounter s)
              (concat (PState s) r)) = None) s
      ≤ nnreal (card B) / nnreal size
  unfolding receive-composition-fri-commits-def receive-fri-commits-with-def
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ (case None of None ⇒ False
      | Some ((b, r), -) ⇒
        b ∈ B ∧
        fmlookup (HashMap s)
          (CompositionFriChallenge (PCompositionFriCounter s)
            (concat (PState s) r)) = None)
      by simp
  next
    fix r s-read
    assume read-r: Some (r, s-read) ∈ set-dist (execute read s)
    have read-state: PState s-read = concat (PState s) r
      using read-outcome[OF read-r] by blast
    have read-counter:
      PCompositionFriCounter s-read = PCompositionFriCounter s
      using read-outcome[OF read-r] by blast
    have read-hash: HashMap s-read = HashMap s
      by (rule read-preserves-hash-map[OF read-r])
    let ?fresh =

```

```

    fmlookup (HashMap s-read)
      (CompositionFriChallenge (PCompositionFriCounter s-read)
        (PState s-read)) = None
  let ?Q =
    λout :: (('f × 'f) × ('f, 'a) protocol-channel-scheme) option.
      case out of
        None ⇒ False
      | Some ((b, r'), -) ⇒
          b ∈ B ∧
            fmlookup (HashMap s)
              (CompositionFriChallenge (PCompositionFriCounter s)
                (concat (PState s) r')) = None
  show wp-event
    (receive-composition-fri-challenge ≫= (λb. return (b, r))) ?Q
    s-read
    ≤ nnreal (card B) / nnreal size
  proof (cases ?fresh)
  case True
  have fresh-key:
    fmlookup (HashMap s)
      (CompositionFriChallenge (PCompositionFriCounter s)
        (concat (PState s) r)) = None
  using True by (simp add: read-state read-counter read-hash)
  have cont-eq:
    (λout. case out of
      None ⇒ (0 :: prob)
      | Some (b, u) ⇒
          wp (return (b, r)) (λout. if ?Q out then 1 else 0) u) =
      (λout :: ('f × ('f, 'a) protocol-channel-scheme) option.
        if (case out of None ⇒ False | Some (b, -) ⇒ b ∈ B)
          then 1 else 0)
  by (rule ext)
  (auto simp: wpsimps fresh-key split: option.splits prod.splits)
  have wp-event
    (receive-composition-fri-challenge ≫= (λb. return (b, r))) ?Q
    s-read =
    wp-event receive-composition-fri-challenge
      (λout. case out of None ⇒ False | Some (b, -) ⇒ b ∈ B)
    s-read
  unfolding wp-event-def by (simp add: wpsimps cont-eq)
  also have ... = nnreal (card B) / nnreal size
  by (rule wp-receive-composition-fri-challenge-fresh-set[OF True])
  finally show ?thesis by simp
next
case False
have not-fresh-s:
  fmlookup (HashMap s)
    (CompositionFriChallenge (PCompositionFriCounter s)
      (concat (PState s) r)) ≠ None

```

```

    using False by (simp add: read-state read-counter read-hash)
  have wp-event
    (receive-composition-fri-challenge  $\gg$  ( $\lambda b$ . return (b, r))) ?Q
    s-read
     $\leq$  wp-event
    (receive-composition-fri-challenge  $\gg$  ( $\lambda b$ . return (b, r)))
    ( $\lambda$ - :: (('f  $\times$  'f)  $\times$ 
      ('f, 'a) protocol-channel-scheme) option. False) s-read
  by (rule wp-event-mono-on-support)
    (use not-fresh-s in
       $\langle$ auto simp: wpsimps elim!: set-dist-bindE
        split: option.splits prod.splits $\rangle$ )
  also have ... = 0
    unfolding wp-event-def wp-def dist-expect-def by simp
  finally have zero-bound:
    wp-event
    (receive-composition-fri-challenge  $\gg$  ( $\lambda b$ . return (b, r))) ?Q
    s-read  $\leq 0$  .
  also have ...  $\leq$  nnreal (card B) / nnreal size
    by simp
  finally show ?thesis .
qed
qed

lemma wp-ntimes-receive-composition-fri-commits-exact-challenges-actual-fresh-bound:
  assumes len: length bs = n
  shows
    wp-event (ntimes receive-composition-fri-commits n)
    ( $\lambda$ out. case out of
      None  $\Rightarrow$  False
      | Some (brs, -)  $\Rightarrow$ 
        map fst brs = bs  $\wedge$ 
        composition-fri-root-path-fresh-from s (map snd brs) n) s
     $\leq (1 / \text{nnreal size})^n$ 
  using len
proof (induction n arbitrary: s bs)
  case 0
  then have bs-empty: bs = []
    by simp
  show ?case
    unfolding bs-empty by (simp add: wp-event-def wpsimps
      composition-fri-root-path-fresh-from-def)
next
  case (Suc n)
  obtain b bs' where bs-eq: bs = b # bs'
    using Suc.prem by (cases bs) auto
  have len-bs': length bs' = n
    using Suc.prem bs-eq by simp
  let ?tail =

```

```

  (ntimes receive-composition-fri-commits n ::
    (('f × 'f) list, ('f, 'a) protocol-channel-scheme) state-monad)
let ?Q =
  λout :: (((('f × 'f) list) ×
    ('f, 'a) protocol-channel-scheme) option.
  case out of
    None ⇒ False
  | Some (brs, -) ⇒
    map fst brs = bs ∧
    composition-fri-root-path-fresh-from s (map snd brs) (Suc n)
let ?Head =
  λout :: (((('f × 'f) × ('f, 'a) protocol-channel-scheme) option.
  case out of
    None ⇒ False
  | Some ((b0, r0), -) ⇒
    b0 ∈ {b} ∧
    fmlookup (HashMap s)
      (CompositionFriChallenge (PCompositionFriCounter s)
        (concat (PState s) r0)) = None
have head-bound:
  wp-event receive-composition-fri-commits ?Head s ≤ 1 / nnreal size
proof -
  have wp-event receive-composition-fri-commits ?Head s
    ≤ nnreal (card ({b} :: 'f set)) / nnreal size
  by (rule wp-receive-composition-fri-commits-actual-fresh-set)
  then show ?thesis by simp
qed
have bind-bound:
  wp-event
    (receive-composition-fri-commits ≫=
      (λbr :: 'f × 'f.
        ?tail ≫= (λbrs :: ('f × 'f) list. return (br # brs))))
  ?Q s
  ≤ wp-event receive-composition-fri-commits ?Head s *
    (1 / nnreal size) ^ n
proof (rule wp-event-bind-bound-by-head-and-cont[where P = ?Head])
  show ¬ ?Q None
  by simp
next
fix br t
assume not-head: ¬ ?Head (Some (br, t))
obtain b0 r0 where br-eq: br = (b0, r0)
  by (cases br)
have cont-false:
  wp-event (?tail ≫= (λbrs :: ('f × 'f) list. return (br # brs)))
  ?Q t
  ≤ wp-event (?tail ≫= (λbrs :: ('f × 'f) list. return (br # brs)))
  (λ- :: (((('f × 'f) list) ×
    ('f, 'a) protocol-channel-scheme) option. False) t

```

```

  by (rule wp-event-mono-on-support)
    (use not-head bs-eq br-eq in
      ‹auto simp: wpsimps composition-fri-root-path-fresh-from-def
        elim!: set-dist-bindE split: option.splits prod.splits›)
also have ... = 0
  unfolding wp-event-def wp-def dist-expect-def by simp
finally have le-zero:
  wp-event (?tail  $\gg$  ( $\lambda$ brs :: ('f  $\times$  'f) list. return (br # brs)))
  ?Q t  $\leq$  0 .
show
  wp-event (?tail  $\gg$  ( $\lambda$ brs :: ('f  $\times$  'f) list. return (br # brs)))
  ?Q t = 0
  by (rule antisym[OF le-zero]) simp
next
fix br t
assume head:
  Some (br, t)  $\in$ 
    set-dist (execute receive-composition-fri-commits s)
  and head-hit: ?Head (Some (br, t))
obtain b0 r0 where br-eq: br = (b0, r0)
  by (cases br)
have b0-eq: b0 = b
  using head-hit unfolding br-eq by simp
let ?TailQ =
   $\lambda$ out :: (((('f  $\times$  'f) list)  $\times$ 
    ('f, 'a) protocol-channel-scheme) option.
    case out of
      None  $\Rightarrow$  False
    | Some (brs, -)  $\Rightarrow$ 
      map fst brs = bs'  $\wedge$ 
      composition-fri-root-path-fresh-from t (map snd brs) n
have cont-le:
  wp-event (?tail  $\gg$  ( $\lambda$ brs :: ('f  $\times$  'f) list. return (br # brs)))
  ?Q t  $\leq$  wp-event ?tail ?TailQ t
proof (rule wp-event-bind-bound-by-head-event)
  show wp-event ?tail ?TailQ t  $\leq$  wp-event ?tail ?TailQ t
    by simp
next
  show ?Q None  $\Longrightarrow$  ?TailQ None
    by simp
next
fix brs u out
assume tail-out:
  Some (brs, u)  $\in$  set-dist (execute ?tail t)
  and ret-out:
    out  $\in$  set-dist (execute (return (br # brs)) u)
  and hit: ?Q out
have out-eq: out = Some (br # brs, u)
  using ret-out

```

```

unfolding set-dist-def return.rep-eq dist-return-def
  dist-delta-dist delta-map-def by simp
have map-eq: map fst brs = bs'
  using hit out-eq b0-eq bs-eq br-eq by simp
have path-full:
  composition-fri-root-path-fresh-from s (map snd (br # brs))
    (Suc n)
  using hit out-eq by simp
have path-full':
  composition-fri-root-path-fresh-from s (r0 # map snd brs)
    (Suc n)
  using path-full unfolding br-eq by simp
have path-tail:
  composition-fri-root-path-fresh-from t (map snd brs) n
  by (rule
    receive-composition-fri-commits-preserves-composition-root-path-fresh-tail
    [OF path-full' head[unfolded br-eq]])
show ?TailQ (Some (brs, u))
  using map-eq path-tail by simp
qed
also have ... ≤ (1 / nnreal size) ^ n
  by (rule Suc.IH[OF len-bs])
finally show
  wp-event (?tail ≫ (λbrs :: ('f × 'f) list. return (br # brs)))
    ?Q t ≤ (1 / nnreal size) ^ n .
qed
have wp-event (ntimes receive-composition-fri-commits (Suc n)) ?Q s
  ≤ wp-event receive-composition-fri-commits ?Head s *
    (1 / nnreal size) ^ n
  using bind-bound by simp
also have ... ≤ (1 / nnreal size) * (1 / nnreal size) ^ n
  by (rule mult-right-mono[OF head-bound]) simp
also have ... = (1 / nnreal size) ^ Suc n
  by simp
finally show ?case .
qed

lemma wp-ntimes-receive-composition-fri-commits-finite-set-actual-fresh-bound:
assumes finite-B: finite B
and lengths: ⋀bs. bs ∈ B ⇒ length bs = n
shows
  wp-event (ntimes receive-composition-fri-commits n)
    (λout. case out of
      None ⇒ False
    | Some (brs, -) ⇒
      map fst brs ∈ B ∧
      composition-fri-root-path-fresh-from s (map snd brs) n) s
    ≤ nnreal (card B) / (nnreal size) ^ n
proof -

```

```

let ?P =
  λbs out. case out of
    None ⇒ False
  | Some (brs, -) ⇒
    map fst brs = bs ∧
    composition-fri-root-path-fresh-from s (map snd brs) n
have event-mono:
  wp-event (ntimes receive-composition-fri-commits n)
    (λout. case out of
      None ⇒ False
    | Some (brs, -) ⇒
      map fst brs ∈ B ∧
      composition-fri-root-path-fresh-from s (map snd brs) n) s
  ≤ wp-event (ntimes receive-composition-fri-commits n)
    (λout. ∃ bs ∈ B. ?P bs out) s
  by (rule wp-event-mono) (auto split: option.splits prod.splits)
also have ... ≤ (∑ bs ∈ B. (1 / nnreal size) ^ n)
  by (rule wp-event-finite-UN-bound[OF finite-B])
    (rule wp-ntimes-receive-composition-fri-commits-exact-challenges-actual-fresh-bound
      [OF lengths])
also have ... = nnreal (card B) * ((1 / nnreal size) ^ n)
  using finite-B by simp
also have ... = nnreal (card B) / (nnreal size) ^ n
  using size-card by transfer (simp add: power-divide)
finally show ?thesis .
qed

```

lemma wp-ntimes-receive-composition-fri-commits-challenge-space-actual-fresh-bound:

assumes subset: $B \subseteq \text{fri-challenge-space } n$

shows

```

wp-event (ntimes receive-composition-fri-commits n)
  (λout. case out of
    None ⇒ False
  | Some (brs, -) ⇒
    map fst brs ∈ B ∧
    composition-fri-root-path-fresh-from s (map snd brs) n) s
  ≤ nnreal (card B) / nnreal (CARD('f) ^ n)

```

proof –

```

have finite-B: finite B
  by (rule finite-subset[OF subset finite-fri-challenge-space])
have lengths:  $\bigwedge bs. bs \in B \implies \text{length } bs = n$ 
  using subset unfolding fri-challenge-space-def by auto
have wp-event (ntimes receive-composition-fri-commits n)
  (λout. case out of
    None ⇒ False
  | Some (brs, -) ⇒
    map fst brs ∈ B ∧
    composition-fri-root-path-fresh-from s (map snd brs) n) s
  ≤ nnreal (card B) / (nnreal size) ^ n

```


by (rule
 $wp_ntimes_receive_composition_fri_commits_finite_set_actual_fresh_bound$
 $[OF\ finite_B\ lengths]$)
 also have ... = $nnreal\ (card\ B) / nnreal\ (CARD('f) \wedge n)$
 using size-card by simp
 finally show ?thesis .
 qed

lemma *composition-fri-challenge-path-fresh-before-composition-iff*:
assumes state:
 $PState\ s_comp =$
 concat
 (foldl concat
 (concat
 (foldl concat (concat (PState s) fr) trace-roots)
 trace-final)
 as)
 dg
 and counter: $PCompositionFriCounter\ s_comp = PCompositionFriCounter\ s$
 and hash: $HashMap\ s_comp = HashMap\ s$
shows
 $composition_fri_challenge_path_fresh\ s\ fr\ trace_roots\ trace_final\ as\ dg$
 $composition_roots\ n \longleftrightarrow$
 $composition_fri_root_path_fresh_from\ s_comp\ composition_roots\ n$
unfolding *composition-fri-challenge-path-fresh-def*
 $composition_fri_root_path_fresh_from_def$
by (simp add: *composition-fri-challenge-key-at-def* state counter hash)

lemma *composition-fri-challenge-path-fresh-from-preservedI*:
assumes state:
 $PState\ s_comp =$
 concat
 (foldl concat
 (concat
 (foldl concat (concat (PState s) fr) trace-roots)
 trace-final)
 as)
 dg
 and counter: $PCompositionFriCounter\ s_comp = PCompositionFriCounter\ s$
and lookup:
 $\bigwedge i\ x. fmlookup\ (HashMap\ s_comp)\ (CompositionFriChallenge\ i\ x) =$
 $fmlookup\ (HashMap\ s)\ (CompositionFriChallenge\ i\ x)$
and fresh:
 $composition_fri_root_path_fresh_from\ s_comp\ composition_roots\ n$
shows
 $composition_fri_challenge_path_fresh\ s\ fr\ trace_roots\ trace_final\ as\ dg$
 $composition_roots\ n$
unfolding *composition-fri-challenge-path-fresh-def*
 $composition_fri_challenge_key_at_def$

```

proof (intro allI impI)
  fix i
  assume i-lt:  $i < n$ 
  and i-len:  $i < \text{length } \text{composition-roots}$ 
  let ?key =
    CompositionFriChallenge (PCompositionFriCounter s + i)
    (foldl concat
      (concat
        (foldl concat
          (concat (foldl concat (concat (PState s) fr) trace-roots)
            trace-final)
          as)
        dg)
      (take (Suc i) composition-roots))
  have ?key =
    CompositionFriChallenge (PCompositionFriCounter s-comp + i)
    (foldl concat (PState s-comp) (take (Suc i) composition-roots))
  using state counter by simp
  moreover have
    fmlookup (HashMap s-comp)
      (CompositionFriChallenge (PCompositionFriCounter s-comp + i)
        (foldl concat (PState s-comp) (take (Suc i) composition-roots))) =
      None
  using fresh i-lt i-len
  unfolding composition-fri-root-path-fresh-from-def by blast
  ultimately show fmlookup (HashMap s) ?key = None
  using lookup[of PCompositionFriCounter s-comp + i
    foldl concat (PState s-comp) (take (Suc i) composition-roots)]
  by simp
qed

lemma receive-trace-fri-commits-preserves-composition-fri-lookup:
  assumes outcome:
    Some (br, t)  $\in \text{set-dist } (\text{execute } \text{receive-trace-fri-commits } s)$ 
  shows
    fmlookup (HashMap t) (CompositionFriChallenge i x) =
    fmlookup (HashMap s) (CompositionFriChallenge i x)
proof –
  obtain b r where br-eq:  $br = (b, r)$ 
  by (cases br)
  from outcome[unfolded br-eq] obtain s-read where
    read-r: Some (r, s-read)  $\in \text{set-dist } (\text{execute } \text{read } s)$ 
  and rand-b:
    Some (b, t)  $\in \text{set-dist } (\text{execute } \text{receive-trace-fri-challenge } s\text{-read})$ 
  unfolding receive-trace-fri-commits-def receive-fri-commits-with-def
  by (auto elim!: set-dist-bindE)
  have read-hash:  $\text{HashMap } s\text{-read} = \text{HashMap } s$ 
  by (rule read-preserves-hash-map[OF read-r])
  have neq:

```

$CompositionFriChallenge\ i\ x \neq$
 $TraceFriChallenge\ (PTraceFriCounter\ s-read)\ (PState\ s-read)$
by *simp*
have $fmlookup\ (HashMap\ t)\ (CompositionFriChallenge\ i\ x) =$
 $fmlookup\ (HashMap\ s-read)\ (CompositionFriChallenge\ i\ x)$
by (rule *receive-trace-fri-challenge-preserves-other-lookup*
 $[OF\ rand-b\ neq]$)
also have $\dots = fmlookup\ (HashMap\ s)\ (CompositionFriChallenge\ i\ x)$
using *read-hash* **by** *simp*
finally show *?thesis* .
qed

lemma *ntimes-receive-trace-fri-commits-preserves-composition-fri-lookup*:
assumes *outcome*:
 $Some\ (brs,\ t) \in$
 $set-dist\ (execute\ (ntimes\ receive-trace-fri-commits\ n)\ s)$
shows
 $fmlookup\ (HashMap\ t)\ (CompositionFriChallenge\ i\ x) =$
 $fmlookup\ (HashMap\ s)\ (CompositionFriChallenge\ i\ x)$
using *outcome*
proof (induction *n* arbitrary: *s brs t*)
case 0
then show *?case*
by *simp*
next
case (*Suc n*)
from *Suc.prem*s **obtain** *br brs' s1* **where**
 $head: Some\ (br,\ s1) \in set-dist\ (execute\ receive-trace-fri-commits\ s)$
and *tail*:
 $Some\ (brs',\ t) \in$
 $set-dist\ (execute\ (ntimes\ receive-trace-fri-commits\ n)\ s1)$
by (auto elim!: *set-dist-bindE*)
have $fmlookup\ (HashMap\ t)\ (CompositionFriChallenge\ i\ x) =$
 $fmlookup\ (HashMap\ s1)\ (CompositionFriChallenge\ i\ x)$
by (rule *Suc.IH* [*OF tail*])
also have $\dots = fmlookup\ (HashMap\ s)\ (CompositionFriChallenge\ i\ x)$
by (rule *receive-trace-fri-commits-preserves-composition-fri-lookup*
 $[OF\ head]$)
finally show *?case* .
qed

lemma *alpha-round-preserves-composition-fri-lookup*:
assumes *outcome*: $Some\ (a,\ t) \in set-dist\ (execute\ alpha-round\ s)$
shows
 $fmlookup\ (HashMap\ t)\ (CompositionFriChallenge\ i\ x) =$
 $fmlookup\ (HashMap\ s)\ (CompositionFriChallenge\ i\ x)$
proof –
from *outcome* **obtain** *a0 s0 a1 s1 s2* **where**
 $rand: Some\ (a0,\ s0) \in set-dist\ (execute\ receive-alpha-challenge\ s)$

and *read-a*: $\text{Some } (a1, s1) \in \text{set-dist } (\text{execute read } s0)$
and *assert-a*: $\text{Some } ((), s2) \in \text{set-dist } (\text{execute } (\text{assert } (a0 = a1)) s1)$
and *ret-a*: $\text{Some } (a, t) \in \text{set-dist } (\text{execute } (\text{return } a1) s2)$
unfolding *alpha-round-def*
by (*auto simp*: *Let-def elim!*: *set-dist-bindE*)
have *neg*:
 CompositionFriChallenge i x $\neq$
 AlphaChallenge (PAlphaCounter s) (PState s)
by *simp*
have *fmlookup* (*HashMap s0*) (*CompositionFriChallenge i x*) =
 fmlookup (*HashMap s*) (*CompositionFriChallenge i x*)
by (*rule receive-alpha-challenge-preserves-other-lookup*[*OF rand neg*])
moreover **have** *HashMap s1* = *HashMap s0*
by (*rule read-preserves-hash-map*[*OF read-a*])
moreover **have** *s2* = *s1*
using *assert-a* **unfolding** *assert-def*
by (*cases a0 = a1*) (*auto simp*: *throw-no-outcome*)
moreover **have** *t* = *s2*
using *ret-a* **by** *simp*
ultimately **show** *?thesis*
by *simp*
qed

lemma *mmap-alpha-round-preserves-composition-fri-lookup*:

assumes *outcome*:
 $\text{Some } (as, t) \in$
 $\text{set-dist } (\text{execute } (\text{mmap } (\text{replicate } n \text{ alpha-round})) s)$
shows
 fmlookup (*HashMap t*) (*CompositionFriChallenge i x*) =
 fmlookup (*HashMap s*) (*CompositionFriChallenge i x*)
using *outcome*
proof (*induction n arbitrary: s as t*)
case 0
then **show** *?case*
by *simp*
next
case (*Suc n*)
from *Suc.prem*s **obtain** *a as' s1* **where**
 head: $\text{Some } (a, s1) \in \text{set-dist } (\text{execute alpha-round } s)$
and *tail*:
 $\text{Some } (as', t) \in$
 $\text{set-dist } (\text{execute } (\text{mmap } (\text{replicate } n \text{ alpha-round})) s1)$
by (*auto elim!*: *set-dist-bindE*)
have *fmlookup* (*HashMap t*) (*CompositionFriChallenge i x*) =
 fmlookup (*HashMap s1*) (*CompositionFriChallenge i x*)
by (*rule Suc.IH*[*OF tail*])
also **have** ... = *fmlookup* (*HashMap s*) (*CompositionFriChallenge i x*)
by (*rule alpha-round-preserves-composition-fri-lookup*[*OF head*])
finally **show** *?case* .

qed

lemma *wp-verifier-composition-fri-prefix-challenge-list-actual-fresh-bound:*

fixes $C :: \text{prob}$

assumes *subset:*

$\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$

and *bound:*

$\bigwedge dg. \text{nnreal } (\text{card } (B \ dg)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$

shows

wp-event verifier-composition-fri-prefix

$(\lambda \text{out. case out of}$

$\text{None} \Rightarrow \text{False}$

$| \text{Some } ((fr, f-fl, f-final, as, dg, fl), -) \Rightarrow$

$\text{map fst fl} \in B \ dg \wedge$

$\text{composition-fri-challenge-path-fresh } s \ fr \ (\text{map snd f-fl})$

$f-final \ as \ dg \ (\text{map snd fl}) \ (\text{length } (\text{map snd fl})) \ s$

$\leq C$

proof $-$

let $?Q =$

$\lambda \text{out. case out of}$

$\text{None} \Rightarrow \text{False}$

$| \text{Some } ((fr, f-fl, f-final, as, dg, fl), -) \Rightarrow$

$\text{map fst fl} \in B \ dg \wedge$

$\text{composition-fri-challenge-path-fresh } s \ fr \ (\text{map snd f-fl})$

$f-final \ as \ dg \ (\text{map snd fl}) \ (\text{length } (\text{map snd fl}))$

show $?thesis$

unfolding *verifier-composition-fri-prefix-def*

proof $(\text{rule wp-event-bind-bound-by-cont})$

show $\neg ?Q \ \text{None}$

by *simp*

next

fix $fr \ s1$

assume *read-fr:* $\text{Some } (fr, s1) \in \text{set-dist } (\text{execute read } s)$

from *read-outcome*[*OF* *read-fr*] **have** *st-s1:*

$PState \ s1 = \text{concat } (PState \ s) \ fr$

by *blast*

from *read-outcome*[*OF* *read-fr*] **have** *comp-count-s1:*

$PCompositionFriCounter \ s1 = PCompositionFriCounter \ s$

by *blast*

have *hash-s1:* $\text{HashMap } s1 = \text{HashMap } s$

by $(\text{rule read-preserves-hash-map}[OF \ \text{read-fr}])$

show *wp-event*

$(\text{ntimes receive-trace-fri-commits } (\text{ceil-log } \text{clength}) \gg=$

$(\lambda f-fl. \text{read } \gg=$

$(\lambda f-final.$

$\text{mmap } (\text{replicate } (\text{length spec}) \ \text{alpha-round}) \gg=$

$(\lambda as. \text{read } \gg=$

$(\lambda dg.$

```

      assert (to-nat dg ≤ maxDegree) >>=
        (λ-. ntimes receive-composition-fri-commits
          (ceil-log (to-nat dg + 1)) >>=
            (λfl.
              return (fr, f-fl, f-final, as, dg, fl))))))
    ?Q s1 ≤ C
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ ?Q None
      by simp
  next
    fix f-fl s2
    assume trace-fri:
      Some (f-fl, s2) ∈
        set-dist
        (execute (ntimes receive-trace-fri-commits (ceil-log clength)) s1)
    have trace-res:
      PState s2 = foldl concat (PState s1) (map snd f-fl) ∧
      PCompositionFriCounter s2 = PCompositionFriCounter s1
    using ntimes-receive-trace-fri-commits-outcome[OF trace-fri]
    by simp
    have lookup-s2:
      ∧ i x. fmlookup (HashMap s2) (CompositionFriChallenge i x) =
        fmlookup (HashMap s) (CompositionFriChallenge i x)
    proof -
      fix i x
      have fmlookup (HashMap s2) (CompositionFriChallenge i x) =
        fmlookup (HashMap s1) (CompositionFriChallenge i x)
      by (rule ntimes-receive-trace-fri-commits-preserves-composition-fri-lookup
        [OF trace-fri])
      also have ... = fmlookup (HashMap s) (CompositionFriChallenge i x)
      using hash-s1 by simp
      finally show
        fmlookup (HashMap s2) (CompositionFriChallenge i x) =
          fmlookup (HashMap s) (CompositionFriChallenge i x) .
    qed
    show wp-event
      (read >>=
        (λf-final.
          mmap (replicate (length spec) alpha-round) >>=
            (λas. read >>=
              (λdg.
                assert (to-nat dg ≤ maxDegree) >>=
                  (λ-. ntimes receive-composition-fri-commits
                    (ceil-log (to-nat dg + 1)) >>=
                      (λfl.
                        return (fr, f-fl, f-final, as, dg, fl))))))
                ?Q s2 ≤ C
      proof (rule wp-event-bind-bound-by-cont)
        show ¬ ?Q None

```

```

    by simp
next
fix f-final s3
assume read-final: Some (f-final, s3) ∈ set-dist (execute read s2)
from read-outcome[OF read-final] have st-s3:
  PState s3 = concat (PState s2) f-final
  by blast
from read-outcome[OF read-final] have comp-count-s3:
  PCompositionFriCounter s3 = PCompositionFriCounter s2
  by blast
have hash-s3: HashMap s3 = HashMap s2
  by (rule read-preserves-hash-map[OF read-final])
have lookup-s3:
   $\bigwedge i x. \text{fmlookup } (\text{HashMap } s3) (\text{CompositionFriChallenge } i x) =$ 
     $\text{fmlookup } (\text{HashMap } s) (\text{CompositionFriChallenge } i x)$ 
  using hash-s3 lookup-s2 by simp
show wp-event
  (mmap (replicate (length spec) alpha-round)  $\gg$ 
    ( $\lambda as. \text{read} \gg$ 
      ( $\lambda dg.$ 
        assert (to-nat dg ≤ maxDegree)  $\gg$ 
        ( $\lambda -. \text{ntimes receive-composition-fri-commits}$ 
          (ceil-log (to-nat dg + 1))  $\gg$ 
          ( $\lambda fl.$ 
            return (fr, f-fl, f-final, as, dg, fl)))))))
  ?Q s3 ≤ C
proof (rule wp-event-bind-bound-by-cont)
  show  $\neg ?Q \text{ None}$ 
    by simp
next
fix as s4
assume alpha:
  Some (as, s4) ∈
    set-dist
      (execute (mmap (replicate (length spec) alpha-round)) s3)
have alpha-res:
  PState s4 = foldl concat (PState s3) as  $\wedge$ 
  PCompositionFriCounter s4 = PCompositionFriCounter s3
  using mmap-alpha-round-outcome[OF alpha] by simp
have lookup-s4:
   $\bigwedge i x. \text{fmlookup } (\text{HashMap } s4) (\text{CompositionFriChallenge } i x) =$ 
     $\text{fmlookup } (\text{HashMap } s) (\text{CompositionFriChallenge } i x)$ 
proof -
  fix i x
  have fmlookup (HashMap s4) (CompositionFriChallenge i x) =
    fmlookup (HashMap s3) (CompositionFriChallenge i x)
    by (rule mmap-alpha-round-preserves-composition-fri-lookup
      [OF alpha])
  also have ... =

```

```

      fmlookup (HashMap s) (CompositionFriChallenge i x)
    by (rule lookup-s3)
  finally show
    fmlookup (HashMap s4) (CompositionFriChallenge i x) =
      fmlookup (HashMap s) (CompositionFriChallenge i x) .
qed
show wp-event
  (read  $\gg$ 
    ( $\lambda dg.$ 
      assert (to-nat dg  $\leq$  maxDegree)  $\gg$ 
      ( $\lambda-. ntimes receive-composition-fri-commits$ 
        (ceil-log (to-nat dg + 1))  $\gg$ 
        ( $\lambda fl.$ 
          return (fr, f-fl, f-final, as, dg, fl))))))
  ?Q s4  $\leq$  C
proof (rule wp-event-bind-bound-by-cont)
  show  $\neg$  ?Q None
  by simp
next
  fix dg s5
  assume read-dg: Some (dg, s5)  $\in$  set-dist (execute read s4)
  from read-outcome[OF read-dg] have st-s5:
    PState s5 = concat (PState s4) dg
  by blast
  from read-outcome[OF read-dg] have comp-count-s5:
    PCompositionFriCounter s5 = PCompositionFriCounter s4
  by blast
  have hash-s5: HashMap s5 = HashMap s4
  by (rule read-preserves-hash-map[OF read-dg])
  have lookup-s5:
     $\bigwedge i x. fmlookup (HashMap s5) (CompositionFriChallenge i x) =$ 
      fmlookup (HashMap s) (CompositionFriChallenge i x)
  using hash-s5 lookup-s4 by simp
  show wp-event
    (assert (to-nat dg  $\leq$  maxDegree)  $\gg$ 
      ( $\lambda-. ntimes receive-composition-fri-commits$ 
        (ceil-log (to-nat dg + 1))  $\gg$ 
        ( $\lambda fl.$ 
          return (fr, f-fl, f-final, as, dg, fl))))))
    ?Q s5  $\leq$  C
proof (rule wp-event-bind-bound-by-cont)
  show  $\neg$  ?Q None
  by simp
next
  fix u s6
  assume assert-out:
    Some (u, s6)  $\in$ 
      set-dist (execute (assert (to-nat dg  $\leq$  maxDegree)) s5)
  have s6-eq: s6 = s5

```



```

using assert-out unfolding assert-def
by (cases to-nat dg ≤ maxDegree)
    (auto simp: throw-no-outcome)
have state-s6:
  PState s6 =
    concat
      (foldl concat
        (concat
          (foldl concat (concat (PState s) fr)
            (map snd f-fl))
          f-final)
        as)
      dg
using st-s1 trace-res st-s3 alpha-res st-s5 s6-eq
by simp
have counter-s6:
  PCompositionFriCounter s6 = PCompositionFriCounter s
using comp-count-s1 trace-res comp-count-s3 alpha-res
    comp-count-s5 s6-eq
by simp
have lookup-s6:
   $\bigwedge i x. \text{fmlookup } (\text{HashMap } s6) \text{ (CompositionFriChallenge } i \text{ } x) =$ 
     $\text{fmlookup } (\text{HashMap } s) \text{ (CompositionFriChallenge } i \text{ } x)$ 
using lookup-s5 s6-eq by simp
let ?Head =
   $\lambda out. \text{case out of}$ 
     $\text{None} \Rightarrow \text{False}$ 
     $| \text{Some } (fl, -) \Rightarrow$ 
       $\text{map fst } fl \in B \text{ } dg \wedge$ 
       $\text{composition-fri-root-path-fresh-from } s6 \text{ (map snd } fl)$ 
       $(\text{ceil-log } (\text{to-nat } dg + 1))$ 
have comp-bound:
  wp-event
    (ntimes receive-composition-fri-commits
      (ceil-log (to-nat dg + 1)))
  ?Head s6 ≤
     $\text{nnreal } (\text{card } (B \text{ } dg)) /$ 
     $\text{nnreal } (\text{CARD('f)} \wedge \text{ceil-log } (\text{to-nat } dg + 1))$ 
by (rule
    wp-ntimes-receive-composition-fri-commits-challenge-space-actual-fresh-bound
    [OF subset])
have comp-bound-C:
  wp-event
    (ntimes receive-composition-fri-commits
      (ceil-log (to-nat dg + 1)))
  ?Head s6 ≤ C
by (rule order-trans[OF comp-bound bound])
show wp-event

```

```

      (ntimes receive-composition-fri-commits
       (ceil-log (to-nat dg + 1)) >>=
       (λfl.
        return (fr, f-fl, f-final, as, dg, fl)))
    ?Q s6 ≤ C
proof (rule wp-event-bind-bound-by-head-event[OF comp-bound-C])
  show ?Q None ⇒ ?Head None
  by simp
next
  fix fl t out
  assume comp-out:
    Some (fl, t) ∈
      set-dist
        (execute
         (ntimes receive-composition-fri-commits
          (ceil-log (to-nat dg + 1)))) s6)
  and cont:
    out ∈
      set-dist
        (execute
         (return (fr, f-fl, f-final, as, dg, fl)) t)
  and hit: ?Q out
have out-eq:
  out = Some ((fr, f-fl, f-final, as, dg, fl), t)
  using cont
unfolding set-dist-def return.rep-eq dist-return-def
  dist-delta-dist delta-map-def
by simp
have len-fl: length fl = ceil-log (to-nat dg + 1)
  using ntimes-receive-composition-fri-commits-outcome
  [OF comp-out] by simp
have path-fresh-len:
  composition-fri-challenge-path-fresh s fr
    (map snd f-fl) f-final as dg (map snd fl)
    (length fl)
  using hit out-eq by simp
have root-fresh-head:
  composition-fri-root-path-fresh-from s6 (map snd fl)
    (ceil-log (to-nat dg + 1))
  using path-fresh-len len-fl state-s6 counter-s6 lookup-s6
unfolding composition-fri-challenge-path-fresh-def
  composition-fri-root-path-fresh-from-def
  composition-fri-challenge-key-at-def
by simp
have root-fresh:
  composition-fri-root-path-fresh-from s6 (map snd fl)
    (ceil-log (Suc (to-nat dg)))
  using root-fresh-head by simp
have path-fresh:

```

```

      composition-fri-challenge-path-fresh s fr (map snd f-fl)
      f-final as dg (map snd fl) (length (map snd fl))
    using composition-fri-challenge-path-fresh-from-preservedI
      [OF state-s6 counter-s6 lookup-s6 root-fresh]
      len-fl
    by simp
  show ?Head (Some (fl, t))
    using hit out-eq root-fresh-head by simp
qed
qed
qed
qed
qed
qed
qed
qed
lemma wp-verify-monad-composition-fri-challenge-list-fresh-hit-actual-bound:
  fixes C :: prob
  assumes subset:
     $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$ 
  and bound:
     $\bigwedge dg. \text{nnreal } (\text{card } (B \ dg)) / \text{nnreal } (\text{CARD('f)} \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$ 
  shows
    wp-event verify-monad (composition-fri-challenge-list-fresh-hit s B) s
       $\leq C$ 
proof -
  let ?Head =
     $\lambda out. \text{case out of}$ 
      None  $\Rightarrow$  False
    | Some ((fr, f-fl, f-final, as, dg, fl), -)  $\Rightarrow$ 
      map fst fl  $\in B \ dg \wedge$ 
      composition-fri-challenge-path-fresh s fr (map snd f-fl)
      f-final as dg (map snd fl) (length (map snd fl))
  have prefix-bound:
    wp-event verifier-composition-fri-prefix ?Head s  $\leq C$ 
  by (rule
    wp-verifier-composition-fri-prefix-challenge-list-actual-fresh-bound
    [OF subset bound])
  show ?thesis
    unfolding verify-monad-composition-fri-decomposition
  proof (rule wp-event-bind-bound-by-head-event[OF prefix-bound])
    show composition-fri-challenge-list-fresh-hit s B None  $\Rightarrow$ 
      ?Head None
    unfolding composition-fri-challenge-list-fresh-hit-def
      accepted-fri-challenges-def by blast
  next
    fix header prefix-state out

```

assume *prefix*:
Some (header, prefix-state) ∈
set-dist (execute verifier-composition-fri-prefix s)
and *suffix*:
out ∈
set-dist (execute
(verifier-after-composition-fri header) prefix-state)
and *hit*: *composition-fri-challenge-list-fresh-hit s B out*
obtain *fr f-fl f-final as dg fl* **where** *header-eq*:
header = (fr, f-fl, f-final, as, dg, fl)
by *(cases header) auto*
from *hit* **obtain** *trace-bs dg' comp-bs fr' trace-roots' trace-final'*
as' composition-roots' final' rest' where
challenges-hit: accepted-fri-challenges s out trace-bs dg' comp-bs
and *header-hit*:
verifier-header-transcript s fr' trace-roots' trace-final' as' dg'
composition-roots' final' rest'
and *comp-bs-B*: *comp-bs ∈ B dg'*
and *fresh-hit*:
composition-fri-challenge-path-fresh s fr' trace-roots'
trace-final' as' dg' composition-roots'
(length composition-roots')
unfolding *composition-fri-challenge-list-fresh-hit-def* **by** *blast*
from *challenges-hit* **obtain** *result final-state* **where** *out-eq*:
out = Some (result, final-state)
unfolding *accepted-fri-challenges-def* **by** *blast*
have *prefix'*:
Some ((fr, f-fl, f-final, as, dg, fl), prefix-state) ∈
set-dist (execute verifier-composition-fri-prefix s)
using *prefix header-eq* **by** *simp*
have *suffix'*:
Some (result, final-state) ∈
set-dist
(execute
(verifier-after-composition-fri
(fr, f-fl, f-final, as, dg, fl))
prefix-state)
using *suffix header-eq out-eq* **by** *simp*
from *suffix'* **obtain** *final query-state* **where**
read-final:
Some (final, query-state) ∈ set-dist (execute read prefix-state)
and *query-out*:
Some (result, final-state) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final)
rounds)
query-state)

```

unfolding verifier-after-composition-fri-def
by (auto elim!: set-dist-bindE)
have prefix-res:
  length f-fl = ceil-log clength  $\wedge$ 
  length as = length spec  $\wedge$ 
  length fl = ceil-log (to-nat dg + 1)  $\wedge$ 
  PTranscript s =
    [fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
    map snd fl @ PTranscript prefix-state
using verifier-composition-fri-prefix-outcome[OF prefix'] by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-prefix: PTranscript prefix-state = final # rest-query
and tr-query: PTranscript query-state = rest-query
by blast
have header-prefix:
  verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
using prefix-res tr-prefix tr-query
unfolding verifier-header-transcript-def verifier-header-messages-def
by simp
have challenges-prefix:
  accepted-fri-challenges s (Some (result, final-state))
    (map fst f-fl) dg (map fst fl)
by (rule verifier-after-composition-fri-accepted-fri-challenges
    [OF prefix' suffix'])
have eqs:
  trace-bs = map fst f-fl  $\wedge$  dg' = dg  $\wedge$  comp-bs = map fst fl
using accepted-fri-challenges-unique
  [OF challenges-prefix challenges-hit[unfolded out-eq]]
by simp
have header-eqs:
  fr' = fr  $\wedge$ 
  trace-roots' = map snd f-fl  $\wedge$ 
  trace-final' = f-final  $\wedge$ 
  as' = as  $\wedge$ 
  composition-roots' = map snd fl
using verifier-header-transcript-unique[OF header-prefix header-hit]
  eqs
by simp
then have fresh-prefix:
  composition-fri-challenge-path-fresh s fr (map snd f-fl)
    f-final as dg (map snd fl) (length (map snd fl))
using fresh-hit eqs by simp
show ?Head (Some (header, prefix-state))
using comp-bs-B eqs fresh-prefix header-eq by simp
qed
qed

```

lemma trace-fri-challenge-list-fresh-hit-imp-set-hit:

assumes *trace-fri-challenge-list-fresh-hit s B out*
shows *trace-fri-challenge-list-set-hit s B out*
using *assms*
unfolding *trace-fri-challenge-list-fresh-hit-def*
trace-fri-challenge-list-set-hit-def
by *blast*

lemma *composition-fri-challenge-list-fresh-hit-imp-set-hit:*
assumes *composition-fri-challenge-list-fresh-hit s B out*
shows *composition-fri-challenge-list-set-hit s B out*
using *assms*
unfolding *composition-fri-challenge-list-fresh-hit-def*
composition-fri-challenge-list-set-hit-def
by *blast*

lemma *wp-trace-fri-challenge-list-fresh-hit-bound-from-future-fresh:*
assumes *future: trace-fri-future-fresh s*
and subset: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
shows
 $\text{wp-event verify-monad } (\text{trace-fri-challenge-list-fresh-hit } s \ B) \ s \leq$
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
proof –
have $\text{wp-event verify-monad } (\text{trace-fri-challenge-list-fresh-hit } s \ B) \ s$
 $\leq \text{wp-event verify-monad } (\text{trace-fri-challenge-list-set-hit } s \ B) \ s$
by (*rule wp-event-mono*)
(rule trace-fri-challenge-list-fresh-hit-imp-set-hit)
also have $\dots \leq$
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
by (*rule wp-verify-monad-trace-fri-challenge-list-set-bound*
[OF future subset])
finally show *?thesis* .
qed

lemma *wp-composition-fri-challenge-list-fresh-hit-bound-from-future-fresh:*
assumes *future: composition-fri-future-fresh s*
and subset:
 $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and bound:
 $\bigwedge dg. \text{nnreal } (\text{card } (B \ dg)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$
shows
 $\text{wp-event verify-monad } (\text{composition-fri-challenge-list-fresh-hit } s \ B) \ s$
 $\leq C$
proof –
have $\text{wp-event verify-monad } (\text{composition-fri-challenge-list-fresh-hit } s \ B) \ s$
 $\leq \text{wp-event verify-monad } (\text{composition-fri-challenge-list-set-hit } s \ B) \ s$
by (*rule wp-event-mono*)
(rule composition-fri-challenge-list-fresh-hit-imp-set-hit)
also have $\dots \leq C$

```

    by (rule up-verify-monad-composition-fri-challenge-list-set-bound
        [OF future subset bound])
  finally show ?thesis .
qed

lemma accepted-fri-challenges-headerE:
  assumes accepted-fri-challenges s out trace-bs dg comp-bs
  obtains fr f-fl trace-final alphas fl final rest where
    verifier-header-transcript s fr (map snd f-fl) trace-final alphas dg
      (map snd fl) final rest
    trace-bs = map fst f-fl
    comp-bs = map fst fl
proof -
  from assms obtain result final-state fr f-fl trace-final alphas fl final
    query-state where body:
    out = Some (result, final-state) ∧
    verifier-header-transcript s fr (map snd f-fl) trace-final alphas dg
      (map snd fl) final (PTranscript query-state) ∧
    PState query-state =
      verifier-header-state s fr (map snd f-fl) trace-final alphas dg
        (map snd fl) final ∧
    Some (result, final-state) ∈
      set-dist (execute
        (ntimes (verifier-query-round-program fr f-fl trace-final alphas fl final)
          rounds)
        query-state) ∧
    PQueryCounter query-state = PQueryCounter s ∧
    trace-bs = map fst f-fl ∧
    comp-bs = map fst fl ∧
    (∀ i < length f-fl.
      fmllookup (HashMap query-state)
        (TraceFriChallenge (PTraceFriCounter s + i)
          (foldl concat (concat (PState s) fr)
            (take (Suc i) (map snd f-fl))))) =
      Some (fst (f-fl ! i))) ∧
    (∀ i < length fl.
      fmllookup (HashMap query-state)
        (CompositionFriChallenge (PCompositionFriCounter s + i)
          (foldl concat
            (concat
              (foldl concat
                (concat
                  (foldl concat (concat (PState s) fr) (map snd f-fl))
                    trace-final)
                alphas)
              dg)
            (take (Suc i) (map snd fl))))) =
      Some (fst (fl ! i)))
  unfolding accepted-fri-challenges-def by blast

```

```

have header:
  verifier-header-transcript s fr (map snd f-fl) trace-final alphas dg
  (map snd fl) final (PTranscript query-state)
using body by simp
have trace-eq: trace-bs = map fst f-fl
using body by simp
have comp-eq: comp-bs = map fst fl
using body by simp
show thesis
  by (rule that[of fr f-fl trace-final alphas fl final
    PTranscript query-state,
    OF header trace-eq comp-eq])
qed

lemma trace-fri-challenge-path-fresh-or-prequeried:
  trace-fri-challenge-path-fresh s fr roots n  $\vee$ 
  trace-fri-challenge-path-prequeried s fr roots n
proof (cases  $\exists i < n. i < \text{length roots} \wedge$ 
  fmlookup (HashMap s) (trace-fri-challenge-key-at s fr roots i)  $\neq$ 
  None)
  case True
  then have trace-fri-challenge-path-prequeried s fr roots n
    unfolding trace-fri-challenge-path-prequeried-def .
  then show ?thesis by simp
next
  case False
  then have trace-fri-challenge-path-fresh s fr roots n
    unfolding trace-fri-challenge-path-fresh-def
    by auto
  then show ?thesis by simp
qed

lemma composition-fri-challenge-path-fresh-or-prequeried:
  composition-fri-challenge-path-fresh s fr trace-roots trace-final as dg
  composition-roots n  $\vee$ 
  composition-fri-challenge-path-prequeried s fr trace-roots trace-final as
  dg composition-roots n
proof (cases  $\exists i < n. i < \text{length composition-roots} \wedge$ 
  fmlookup (HashMap s)
  (composition-fri-challenge-key-at s fr trace-roots trace-final as dg
  composition-roots i)  $\neq$  None)
  case True
  then have composition-fri-challenge-path-prequeried s fr trace-roots
    trace-final as dg composition-roots n
    unfolding composition-fri-challenge-path-prequeried-def .
  then show ?thesis by simp
next
  case False
  then have composition-fri-challenge-path-fresh s fr trace-roots

```



```

    trace-final as dg composition-roots n
  unfolding composition-fri-challenge-path-fresh-def
  by auto
  then show ?thesis by simp
qed

lemma trace-fri-challenge-list-set-hit-imp-fresh-or-prequery:
  assumes trace-fri-challenge-list-set-hit s B out
  shows
    trace-fri-challenge-list-fresh-hit s B out ∨
    trace-fri-challenge-list-prequery-hit s B out
proof -
  from assms obtain trace-bs dg comp-bs where hit:
    accepted-fri-challenges s out trace-bs dg comp-bs
    trace-bs ∈ B
  unfolding trace-fri-challenge-list-set-hit-def by auto
  from hit(1) obtain fr f-fl trace-final alphas fl final rest where
    header:
      verifier-header-transcript s fr (map snd f-fl) trace-final alphas dg
      (map snd fl) final rest
    trace-bs = map fst f-fl
    comp-bs = map fst fl
  by (rule accepted-fri-challenges-headerE)
  have split: trace-fri-challenge-path-fresh s fr (map snd f-fl)
    (length (map snd f-fl)) ∨
    trace-fri-challenge-path-prequeried s fr (map snd f-fl)
    (length (map snd f-fl))
  by (rule trace-fri-challenge-path-fresh-or-prequeried)
  then show ?thesis
proof
  assume fresh: trace-fri-challenge-path-fresh s fr (map snd f-fl)
    (length (map snd f-fl))
  have trace-fri-challenge-list-fresh-hit s B out
  unfolding trace-fri-challenge-list-fresh-hit-def
  by (intro exI conjI; (rule hit(1) | rule header | rule hit(2) |
    rule fresh))
  then show ?thesis by simp
next
  assume prequeried: trace-fri-challenge-path-prequeried s fr
    (map snd f-fl) (length (map snd f-fl))
  have trace-fri-challenge-list-prequery-hit s B out
  unfolding trace-fri-challenge-list-prequery-hit-def
  by (intro exI conjI; (rule hit(1) | rule header | rule hit(2) |
    rule prequeried))
  then show ?thesis by simp
qed
qed

```

lemma composition-fri-challenge-list-set-hit-imp-fresh-or-prequery:

assumes *composition-fri-challenge-list-set-hit s B out*
shows
composition-fri-challenge-list-fresh-hit s B out $\vee$
composition-fri-challenge-list-prequery-hit s B out
proof –
from *assms* **obtain** *trace-bs dg comp-bs* **where** *hit*:
accepted-fri-challenges s out trace-bs dg comp-bs
comp-bs $\in$ *B dg*
unfolding *composition-fri-challenge-list-set-hit-def* **by** *auto*
from *hit(1)* **obtain** *fr f-fl trace-final alphas fl final rest* **where**
header:
verifier-header-transcript s fr (map snd f-fl) trace-final alphas dg
(map snd fl) final rest
trace-bs = map fst f-fl
comp-bs = map fst fl
by (*rule accepted-fri-challenges-headerE*)
have *split*: *composition-fri-challenge-path-fresh s fr (map snd f-fl)*
trace-final alphas dg (map snd fl) (length (map snd fl)) $\vee$
composition-fri-challenge-path-prequeried s fr (map snd f-fl)
trace-final alphas dg (map snd fl) (length (map snd fl))
by (*rule composition-fri-challenge-path-fresh-or-prequeried*)
then show *?thesis*
proof
assume *fresh*: *composition-fri-challenge-path-fresh s fr (map snd f-fl)*
trace-final alphas dg (map snd fl) (length (map snd fl))
have *composition-fri-challenge-list-fresh-hit s B out*
unfolding *composition-fri-challenge-list-fresh-hit-def*
by (*intro exI conjI; (rule hit(1) | rule header | rule hit(2) |*
rule fresh))
then show *?thesis* **by** *simp*
next
assume *prequeried*: *composition-fri-challenge-path-prequeried s fr*
(map snd f-fl) trace-final alphas dg (map snd fl) (length (map snd fl))
have *composition-fri-challenge-list-prequery-hit s B out*
unfolding *composition-fri-challenge-list-prequery-hit-def*
by (*intro exI conjI; (rule hit(1) | rule header | rule hit(2) |*
rule prequeried))
then show *?thesis* **by** *simp*
qed
qed

lemma *wp-trace-fri-challenge-list-set-hit-bound-from-fresh-and-prequery*:
assumes *fresh-bound*:
wp-event verify-monad (trace-fri-challenge-list-fresh-hit s B) s $\leq$ *F*
and *prequery-bound*:
wp-event verify-monad (trace-fri-challenge-list-prequery-hit s B) s
 $\leq$ *P*
shows *wp-event verify-monad (trace-fri-challenge-list-set-hit s B) s*
 $\leq$ *F + P*

proof –
have $wp\text{-event}\ \text{verify-monad}\ (\text{trace-fri-challenge-list-set-hit}\ s\ B)\ s \leq$
 $wp\text{-event}\ \text{verify-monad}$
 $(\lambda out. \text{trace-fri-challenge-list-fresh-hit}\ s\ B\ out \vee$
 $\text{trace-fri-challenge-list-prequery-hit}\ s\ B\ out)\ s$
by (rule $wp\text{-event-mono}$)
(rule $\text{trace-fri-challenge-list-set-hit-imp-fresh-or-prequery}$)
also have $\dots \leq$
 $wp\text{-event}\ \text{verify-monad}\ (\text{trace-fri-challenge-list-fresh-hit}\ s\ B)\ s +$
 $wp\text{-event}\ \text{verify-monad}\ (\text{trace-fri-challenge-list-prequery-hit}\ s\ B)\ s$
by (rule $wp\text{-event-union-bound}$)
also have $\dots \leq F + P$
by (rule $\text{add-mono}[OF\ \text{fresh-bound}\ \text{prequery-bound}]$)
finally show $?thesis$.
qed

lemma $wp\text{-composition-fri-challenge-list-set-hit-bound-from-fresh-and-prequery}$:
assumes fresh-bound :
 $wp\text{-event}\ \text{verify-monad}\ (\text{composition-fri-challenge-list-fresh-hit}\ s\ B)\ s$
 $\leq F$
and prequery-bound :
 $wp\text{-event}\ \text{verify-monad}\ (\text{composition-fri-challenge-list-prequery-hit}\ s\ B)\ s$
 $\leq P$
shows $wp\text{-event}\ \text{verify-monad}\ (\text{composition-fri-challenge-list-set-hit}\ s\ B)\ s$
 $\leq F + P$

proof –
have $wp\text{-event}\ \text{verify-monad}\ (\text{composition-fri-challenge-list-set-hit}\ s\ B)\ s$
 $\leq wp\text{-event}\ \text{verify-monad}$
 $(\lambda out. \text{composition-fri-challenge-list-fresh-hit}\ s\ B\ out \vee$
 $\text{composition-fri-challenge-list-prequery-hit}\ s\ B\ out)\ s$
by (rule $wp\text{-event-mono}$)
(rule $\text{composition-fri-challenge-list-set-hit-imp-fresh-or-prequery}$)
also have $\dots \leq$
 $wp\text{-event}\ \text{verify-monad}\ (\text{composition-fri-challenge-list-fresh-hit}\ s\ B)\ s +$
 $wp\text{-event}\ \text{verify-monad}\ (\text{composition-fri-challenge-list-prequery-hit}\ s\ B)\ s$
by (rule $wp\text{-event-union-bound}$)
also have $\dots \leq F + P$
by (rule $\text{add-mono}[OF\ \text{fresh-bound}\ \text{prequery-bound}]$)
finally show $?thesis$.
qed

end

context soundness
begin

lemma $\text{checked-staged-security-trace-fri-challenge-list-set-hit-bound-from-fresh-and-prequery}$:
assumes fresh-bound :
 $wp\text{-event}\ (\text{checked-staged-security-experiment-with-data-state}\ A)$

(staged-security-with-data-state-verifier-event
 (λs. trace-fri-challenge-list-fresh-hit s B))
 adversary-initial-state ≤ F
and prequery-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 (λs. trace-fri-challenge-list-prequery-hit s B))
 adversary-initial-state ≤ P
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 (λs. trace-fri-challenge-list-set-hit s B))
 adversary-initial-state ≤ F + P
proof –
have wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 (λs. trace-fri-challenge-list-set-hit s B))
 adversary-initial-state ≤
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 (λs out.
 trace-fri-challenge-list-fresh-hit s B out ∨
 trace-fri-challenge-list-prequery-hit s B out))
 adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume
 out ∈
 set-dist
 (execute (checked-staged-security-experiment-with-data-state A)
 adversary-initial-state)
and hit:
 staged-security-with-data-state-verifier-event
 (λs. trace-fri-challenge-list-set-hit s B) out
then show
 staged-security-with-data-state-verifier-event
 (λs out.
 trace-fri-challenge-list-fresh-hit s B out ∨
 trace-fri-challenge-list-prequery-hit s B out) out
unfolding staged-security-with-data-state-verifier-event-def
by (auto dest: trace-fri-challenge-list-set-hit-imp-fresh-or-prequery
 split: option.splits prod.splits)
qed
also have ... ≤
 wp-event (checked-staged-security-experiment-with-data-state A)
 (λout.
 staged-security-with-data-state-verifier-event
 (λs. trace-fri-challenge-list-fresh-hit s B) out ∨
 staged-security-with-data-state-verifier-event

$(\lambda s. \text{trace-fri-challenge-list-prequery-hit } s \ B) \ \text{out})$
 $\text{adversary-initial-state}$
by (rule $\text{wp-event-mono-on-support}$)
 $(\text{auto simp: staged-security-with-data-state-verifier-event-def}$
 $\text{split: option.splits prod.splits})$
also have $\dots \leq$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B))$
 $\text{adversary-initial-state} +$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-challenge-list-prequery-hit } s \ B))$
 $\text{adversary-initial-state}$
by (rule $\text{wp-event-union-bound}$)
also have $\dots \leq F + P$
by (rule $\text{add-mono}[OF \ \text{fresh-bound prequery-bound}]$)
finally show $?thesis$.
qed

lemma $\text{checked-staged-security-trace-fri-challenge-list-set-hit-bound}$:

assumes $\text{wf: staged-budget-wellformed budgets}$

and $\text{controlled: staged-adversary-controlled budgets } A$

and $\text{finite-B: finite } B$

and fresh-bound :

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$

$(\text{staged-security-with-data-state-verifier-event}$

$(\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B))$

$\text{adversary-initial-state} \leq F$

shows

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$

$(\text{staged-security-with-data-state-verifier-event}$

$(\lambda s. \text{trace-fri-challenge-list-set-hit } s \ B))$

$\text{adversary-initial-state} \leq$

$F +$

$\text{hash-relation-budget-value } (\text{card } B * \text{ceil-log clength})$

$(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$

by (rule

$\text{checked-staged-security-trace-fri-challenge-list-set-hit-bound-from-fresh-and-prequery})$

(rule fresh-bound ,

rule $\text{checked-staged-security-trace-fri-verifier-prequery-hit-bound}$

$[OF \ \text{wf controlled finite-B}]$)

lemma $\text{checked-staged-security-trace-fri-challenge-list-fresh-hit-actual-bound}$:

assumes $\text{subset: } B \subseteq \text{fri-challenge-space } (\text{ceil-log clength})$

shows

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$

$(\text{staged-security-with-data-state-verifier-event}$

$(\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B))$

$\text{adversary-initial-state} \leq$
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show $\bigwedge s. \neg \text{trace-fri-challenge-list-fresh-hit } s \ B \ \text{None}$
unfolding trace-fri-challenge-list-fresh-hit-def
accepted-fri-challenges-def **by** blast
next
fix data attacker-state
assume Some (data, attacker-state) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
show wp-event verify-monad
(trace-fri-challenge-list-fresh-hit
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) B)
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
 $\leq \text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
by (rule wp-verify-monad-trace-fri-challenge-list-fresh-hit-actual-bound
[OF subset])
qed

lemma checked-staged-security-trace-fri-bad-challenge-list-fresh-hit-bound-from-round-bounds:
assumes local-bound:
 $\bigwedge i \text{ prefix.}$
 $i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \implies$
 $\text{length prefix} = i \implies$
 $\text{card } (\text{bad } i \text{ prefix}) \leq \text{bd } i$
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
($\lambda s. \text{trace-fri-challenge-list-fresh-hit } s$
(generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound (clength - 1)) bad)))
adversary-initial-state $\leq$
nnreal
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound } (\text{clength} - 1)\}.$
 $\text{CARD}('f) \wedge i * \text{bd } i *$
 $\text{CARD}('f) \wedge$
 $(\text{fri-round-count-for-degree-bound } (\text{clength} - 1) - \text{Suc } i)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{fri-round-count-for-degree-bound } (\text{clength} - 1))$
proof -
let ?n = fri-round-count-for-degree-bound (clength - 1)
let ?B = generic-fri-bad-challenge-lists ?n bad
let ?K =
 $(\sum i \in \{..<?n\}.$
 $\text{CARD}('f) \wedge i * \text{bd } i * \text{CARD}('f) \wedge (?n - \text{Suc } i))$
have subset: ?B $\subseteq$ fri-challenge-space (ceil-log clength)

```

using generic-fri-bad-challenge-lists-subset[of ?n bad]
by (simp add: fri-round-count-for-degree-bound-def clength-pos)
have base:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s.$  trace-fri-challenge-list-fresh-hit s ?B))
  adversary-initial-state  $\leq$ 
  nnreal (card ?B) / nnreal (CARD('f)  $\wedge$  ceil-log clength)
by (rule checked-staged-security-trace-fri-challenge-list-fresh-hit-actual-bound
  [OF subset])
have card-bound: card ?B  $\leq$  ?K
by (rule generic-fri-bad-challenge-lists-card-bound-from-round-bounds)
  (rule local-bound)
have ratio-bound:
  nnreal (card ?B) / nnreal (CARD('f)  $\wedge$  ceil-log clength)
     $\leq$  nnreal ?K / nnreal (CARD('f)  $\wedge$  ?n)
proof -
  have nnreal (card ?B) / nnreal (CARD('f)  $\wedge$  ceil-log clength)
     $\leq$  nnreal ?K / nnreal (CARD('f)  $\wedge$  ceil-log clength)
    using card-bound by (rule nnreal-nat-divide-right-mono)
    also have ... = nnreal ?K / nnreal (CARD('f)  $\wedge$  ?n)
    by (simp add: fri-round-count-for-degree-bound-def clength-pos)
    finally show ?thesis .
qed
show ?thesis
by (rule order-trans[OF base ratio-bound])
qed

lemma checked-staged-security-composition-fri-challenge-list-fresh-hit-actual-bound:
fixes C :: prob
assumes subset:
   $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$ 
and bound:
   $\bigwedge dg. \text{nnreal } (\text{card } (B \ dg)) / \text{nnreal } (\text{CARD('f) } \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s.$  composition-fri-challenge-list-fresh-hit s B))
  adversary-initial-state  $\leq C$ 
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show  $\bigwedge s. \neg \text{composition-fri-challenge-list-fresh-hit } s \ B \ \text{None}$ 
unfolding composition-fri-challenge-list-fresh-hit-def
  accepted-fri-challenges-def by blast
next
fix data attacker-state
assume Some (data, attacker-state)  $\in$ 
  set-dist
  (execute (checked-staged-transcript-program A)

```

```

    adversary-initial-state)
  show wp-event verify-monad
    (composition-fri-challenge-list-fresh-hit
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) B)
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    ≤ C
  by (rule wp-verify-monad-composition-fri-challenge-list-fresh-hit-actual-bound
    [OF subset bound])
qed

lemma checked-staged-security-composition-fri-bad-challenge-list-fresh-hit-bound-from-round-bounds:
  fixes C :: prob
  assumes local-bound:
     $\bigwedge dg \ i \text{ prefix.}$ 
     $i < \text{fri-round-count-for-degree-bound } (to\text{-nat } dg) \implies$ 
     $\text{length prefix} = i \implies$ 
     $\text{card } (\text{bad } dg \ i \text{ prefix}) \leq \text{bd } dg \ i$ 
  and fraction-bound:
     $\bigwedge dg.$ 
     $\text{nnreal}$ 
     $(\sum i \in \{..<\text{fri-round-count-for-degree-bound } (to\text{-nat } dg)\}.$ 
     $\text{CARD('f)} \wedge^i * \text{bd } dg \ i *$ 
     $\text{CARD('f)} \wedge$ 
     $(\text{fri-round-count-for-degree-bound } (to\text{-nat } dg) - \text{Suc } i)) /$ 
     $\text{nnreal } (\text{CARD('f)} \wedge \text{fri-round-count-for-degree-bound } (to\text{-nat } dg)) \leq C$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        ( $\lambda s.$  composition-fri-challenge-list-fresh-hit s
          ( $\lambda dg.$  generic-fri-bad-challenge-lists
            ( $\text{fri-round-count-for-degree-bound } (to\text{-nat } dg)$ ) (bad dg))))
      adversary-initial-state ≤ C
proof -
  let ?B =
     $\lambda dg.$  generic-fri-bad-challenge-lists
      ( $\text{fri-round-count-for-degree-bound } (to\text{-nat } dg)$ ) (bad dg)
  have subset:
     $\bigwedge dg.$  ?B dg  $\subseteq$  fri-challenge-space (ceil-log (to-nat dg + 1))
  using generic-fri-bad-challenge-lists-subset
  by (simp add: fri-round-count-for-degree-bound-def)
  have ratio-bound:
     $\bigwedge dg.$  nnreal (card (?B dg)) /
    nnreal (CARD('f)  $\wedge^{\text{ceil-log } (to\text{-nat } dg + 1)}$ ) ≤ C
proof -
  fix dg
  let ?n = fri-round-count-for-degree-bound (to-nat dg)
  let ?K =

```



```

  ( $\sum i \in \{..<?n\}$ .
     $CARD('f) \wedge i * bd \ dg \ i * CARD('f) \wedge (?n - Suc \ i))$ 
have card-bound:  $card \ ( ?B \ dg) \leq ?K$ 
  by (rule generic-fri-bad-challenge-lists-card-bound-from-round-bounds)
    (rule local-bound)
have nnreal ( $card \ ( ?B \ dg) /$ 
   $nnreal \ (CARD('f) \wedge ceil\text{-}log \ (to\text{-}nat \ dg + 1))$ 
   $\leq nnreal \ ?K / nnreal \ (CARD('f) \wedge ceil\text{-}log \ (to\text{-}nat \ dg + 1))$ )
using card-bound by (rule nnreal-nat-divide-right-mono)
also have ... =  $nnreal \ ?K / nnreal \ (CARD('f) \wedge ?n)$ 
  by (simp add: fri-round-count-for-degree-bound-def)
also have ...  $\leq C$ 
  by (rule fraction-bound)
finally show
   $nnreal \ (card \ ( ?B \ dg) /$ 
   $nnreal \ (CARD('f) \wedge ceil\text{-}log \ (to\text{-}nat \ dg + 1)) \leq C \ .$ 
qed
show ?thesis
  by (rule
    checked-staged-security-composition-fri-challenge-list-fresh-hit-actual-bound
    [OF subset ratio-bound])
qed

lemma checked-staged-security-composition-fri-challenge-list-set-hit-bound-from-fresh-and-prequery:
assumes fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s. composition\text{-}fri\text{-}challenge\text{-}list\text{-}fresh\text{-}hit \ s \ B$ ))
  adversary-initial-state  $\leq F$ 
and prequery-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s. composition\text{-}fri\text{-}challenge\text{-}list\text{-}prequery\text{-}hit \ s \ B$ ))
  adversary-initial-state  $\leq P$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s. composition\text{-}fri\text{-}challenge\text{-}list\text{-}set\text{-}hit \ s \ B$ ))
  adversary-initial-state  $\leq F + P$ 
proof –
have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s. composition\text{-}fri\text{-}challenge\text{-}list\text{-}set\text{-}hit \ s \ B$ ))
  adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s \ out.$ 
       $composition\text{-}fri\text{-}challenge\text{-}list\text{-}fresh\text{-}hit \ s \ B \ out \ \vee$ 
       $composition\text{-}fri\text{-}challenge\text{-}list\text{-}prequery\text{-}hit \ s \ B \ out$ ))

```

```

    adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume
    out ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  and hit:
    staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-set-hit s B) out
  then show
    staged-security-with-data-state-verifier-event
      (λs out.
        composition-fri-challenge-list-fresh-hit s B out ∨
        composition-fri-challenge-list-prequery-hit s B out) out
  unfolding staged-security-with-data-state-verifier-event-def
  by (auto dest:
    composition-fri-challenge-list-set-hit-imp-fresh-or-prequery
    split: option.splits prod.splits)
qed
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (λout.
      staged-security-with-data-state-verifier-event
        (λs. composition-fri-challenge-list-fresh-hit s B) out ∨
        staged-security-with-data-state-verifier-event
          (λs. composition-fri-challenge-list-prequery-hit s B) out)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      split: option.splits prod.splits)
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-fresh-hit s B))
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-prequery-hit s B))
    adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤ F + P
  by (rule add-mono[OF fresh-bound prequery-bound])
finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-challenge-list-set-hit-bound:
  assumes wf: staged-budget-wellformed budgets

```

```

and controlled: staged-adversary-controlled budgets A
and finite-B:  $\bigwedge dg. \text{finite } (B \ dg)$ 
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B$ ))
    adversary-initial-state  $\leq F$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s. \text{composition-fri-challenge-list-set-hit } s \ B$ ))
    adversary-initial-state  $\leq$ 
       $F +$ 
      hash-relation-budget-value
        ( $\sum dg \in (UNIV :: 'f \ \text{set}).$ 
           $\text{card } (B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$ 
        (staged-attacker-query-budget budgets + staged-challenge-query-budget)
by (rule
  checked-staged-security-composition-fri-challenge-list-set-hit-bound-from-fresh-and-prequery)
(rule fresh-bound,
  rule checked-staged-security-composition-fri-verifier-prequery-hit-bound
    [OF wf controlled finite-B])

```

end

end

theory *Soundness-FRI-Prequery-Trace-Sampled-Bounds*
imports *Soundness-FRI-Prequery-Challenge-Bounds*
begin

Staged bounds for trace FRI sampled-layer events. The lemmas combine the fresh-challenge case with prequery accounting from the shared random oracle.

context *soundness*
begin

lemma *checked-staged-security-trace-fri-sampled-layer-chain-bound-from-fresh-and-prequery*:
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *finite-B*: *finite B*
and *cover*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
trace-fri-sampled-layer-chain-cover
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) *bad*

and *envelope*:
 $\bigwedge \text{data attacker-state trace-table.}$
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
trace-fri-multiround-bad-sets bad trace-table $\subseteq B$
and *fresh-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
 $(\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B)$)
adversary-initial-state $\leq F$
shows
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-sampled-layer-chain)
adversary-initial-state $\leq$
 $F +$
hash-relation-budget-value (*card* *B* * *ceil-log clength*)
(*staged-attacker-query-budget budgets* + *staged-challenge-query-budget*)
proof –
have *wp-event* (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-sampled-layer-chain)
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
 $(\lambda s. \text{trace-fri-challenge-list-set-hit } s \ B)$)
adversary-initial-state
proof (*rule wp-event-mono-on-support*)
fix *out*
assume *support*:
out $\in$
set-dist
(*execute* (*checked-staged-security-experiment-with-data-state* *A*)
adversary-initial-state)
and *hit*:
staged-security-with-data-state-verifier-event
trace-fri-bad-with-sampled-layer-chain out
from *hit* **obtain** *data attacker-state result final-state* **where**
out-eq: *out* = *Some* (((*data*, *attacker-state*), *result*), *final-state*)
unfolding *staged-security-with-data-state-verifier-event-def*
by (*cases out*) (*auto split*: *option.splits prod.splits*)
have *builder*:
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state)
using *checked-staged-security-experiment-with-data-state-outcomeE*
 $[OF \ \text{support}[\text{unfolded out-eq}]]$
by *blast*

```

let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have sampled:
  trace-fri-bad-with-sampled-layer-chain ?s
  (Some (result, final-state))
using hit unfolding staged-security-with-data-state-verifier-event-def
  out-eq by simp
have list-hit:
  trace-fri-challenge-list-set-hit ?s B (Some (result, final-state))
by (rule trace-fri-bad-with-sampled-layer-chain-imp-list-hit
  [OF sampled cover [OF builder]])
  (rule envelope [OF builder])
show
  staged-security-with-data-state-verifier-event
  ( $\lambda s.$  trace-fri-challenge-list-set-hit s B) out
unfolding staged-security-with-data-state-verifier-event-def out-eq
using list-hit by simp
qed
also have ...  $\leq$ 
  F +
  hash-relation-budget-value (card B * ceil-log clength)
  (staged-attacker-query-budget budgets + staged-challenge-query-budget)
by (rule checked-staged-security-trace-fri-challenge-list-set-hit-bound
  [OF wf controlled finite-B fresh-bound])
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-fresh-and-prequery:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: finite B
and cover:
 $\wedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
  set-dist (execute (checked-staged-transcript-program A)
  adversary-initial-state)  $\implies$ 
  trace-fri-sampled-layer-chain-cover
  (verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)) bad
and envelope:
 $\wedge$  data attacker-state trace-table.
  Some (data, attacker-state)  $\in$ 
  set-dist (execute (checked-staged-transcript-program A)
  adversary-initial-state)  $\implies$ 
  trace-fri-multi-round-bad-sets bad trace-table  $\subseteq B$ 
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event

```

```

      (λs. trace-fri-challenge-list-fresh-hit s B))
    adversary-initial-state ≤ F
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-header-tied-sampled-layer-chain)
    adversary-initial-state ≤
      F +
      hash-relation-budget-value (card B * ceil-log clength)
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  proof -
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-sampled-layer-chain)
    adversary-initial-state ≤
      wp-event (checked-staged-security-experiment-with-data-state A)
        (staged-security-with-data-state-verifier-event
          trace-fri-bad-with-sampled-layer-chain)
        adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro: trace-fri-header-tied-sampled-layer-chain-imp-sampled-layer-chain
        split: option.splits prod.splits)
  also have ... ≤
    F +
    hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule
    checked-staged-security-trace-fri-sampled-layer-chain-bound-from-fresh-and-prequery
      [OF wf controlled finite-B cover envelope fresh-bound])
  finally show ?thesis .
qed

end

end

theory Soundness-FRI-Prequery-Composition-Sampled-Bounds
  imports Soundness-FRI-Prequery-Trace-Sampled-Bounds
begin

Staged bounds for composition FRI sampled-layer events. This theory mirrors the trace-
side prequery accounting with the composition degree parameter.

context soundness
begin

lemma checked-staged-security-composition-fri-sampled-layer-chain-bound-from-fresh-and-prequery:
  assumes wf: staged-budget-wellformed budgets

```

and *controlled*: *staged-adversary-controlled budgets A*
and *finite-B*: $\bigwedge dg. \text{finite } (B \text{ dg})$
and *cover*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{composition-fri-sampled-layer-chain-cover}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad}$
and *envelope*:
 $\bigwedge \text{data attacker-state dg composition-table.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{composition-fri-multiround-bad-sets bad dg composition-table}$
 $\subseteq B \text{ dg}$
and *fresh-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \text{ B}))$
 $\text{adversary-initial-state} \leq F$
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-sampled-layer-chain})$
 $\text{adversary-initial-state} \leq$
 $F +$
 $\text{hash-relation-budget-value}$
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$
proof –
have $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-sampled-layer-chain})$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-challenge-list-set-hit } s \text{ B}))$
 $\text{adversary-initial-state}$
proof (rule *wp-event-mono-on-support*)
fix *out*
assume *support*:
 $out \in$
 set-dist
 $(\text{execute } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{adversary-initial-state})$
and *hit*:

```

    staged-security-with-data-state-verifier-event
    composition-fri-bad-with-sampled-layer-chain out
from hit obtain data attacker-state result final-state where
    out-eq: out = Some (((data, attacker-state), result), final-state)
    unfolding staged-security-with-data-state-verifier-event-def
    by (cases out) (auto split: option.splits prod.splits)
have builder:
    Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state)
    using checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
    by blast
let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have sampled:
    composition-fri-bad-with-sampled-layer-chain ?s
    (Some (result, final-state))
    using hit unfolding staged-security-with-data-state-verifier-event-def
    out-eq by simp
have list-hit:
    composition-fri-challenge-list-set-hit ?s B
    (Some (result, final-state))
    by (rule composition-fri-bad-with-sampled-layer-chain-imp-list-hit
    [OF sampled cover[OF builder]])
    (rule envelope[OF builder])
show
    staged-security-with-data-state-verifier-event
    (λs. composition-fri-challenge-list-set-hit s B) out
    unfolding staged-security-with-data-state-verifier-event-def out-eq
    using list-hit by simp
qed
also have ... ≤
    F +
    hash-relation-budget-value
    (∑ dg ∈ (UNIV :: 'f set).
    card (B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
    by (rule
    checked-staged-security-composition-fri-challenge-list-set-hit-bound
    [OF wf controlled finite-B fresh-bound])
finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-fresh-and-prequery:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: ∧ dg. finite (B dg)

```


and *cover*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
composition-fri-sampled-layer-chain-cover
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) *bad*
and *envelope*:
 $\wedge$ *data attacker-state dg composition-table*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
composition-fri-multiround-bad-sets *bad dg composition-table*
 $\subseteq B$ *dg*
and *fresh-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
($\lambda s.$ *composition-fri-challenge-list-fresh-hit* *s B*))
adversary-initial-state $\leq F$
shows
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-verifier-tied-sampled-layer-chain)
adversary-initial-state $\leq$
 $F +$
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \text{ } dg) * \text{ceil-log } (\text{maxDegree} + 1))$
(*staged-attacker-query-budget* *budgets* + *staged-challenge-query-budget*)
proof –
have *wp-event* (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-verifier-tied-sampled-layer-chain)
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-sampled-layer-chain)
adversary-initial-state
by (*rule wp-event-mono-on-support*)
(*auto simp: staged-security-with-data-state-verifier-event-def*
intro:
composition-fri-verifier-tied-sampled-layer-chain-imp-sampled-layer-chain
split: option.splits prod.splits)
also have ... $\leq$
 $F +$
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \text{ } dg) * \text{ceil-log } (\text{maxDegree} + 1))$

```

    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule
    checked-staged-security-composition-fri-sampled-layer-chain-bound-from-fresh-and-prequery
    [OF wf controlled finite-B cover envelope fresh-bound])
  finally show ?thesis .
qed

end

end

```

```

theory Soundness-FRI-Staged-Active-Bounds
imports Soundness-FRI-Prequery-Composition-Sampled-Bounds
begin

```

Staged integration lemmas for the active FRI route.

The prequery accounting layer proves whole checked-security-experiment bounds for sampled FRI layer-chain events. The active route still reasons about partial-candidate FRI events. This layer bridges those levels by splitting a partial-candidate FRI bad event into the sampled layer-chain case and the remaining missing-sampled-chain residual.

```

context soundness
begin

```

```

lemma trace-fri-header-tied-partial-candidate-imp-sampled-conflict-or-zero:
  assumes bad: trace-fri-bad-with-header-tied-partial-candidate s out
  shows
    trace-fri-bad-with-header-tied-sampled-layer-chain s out ∨
    trace-fri-header-tied-sampled-assignment-conflict s out ∨
    trace-fri-header-tied-zero-round-final-obstruction s out
proof (cases trace-fri-bad-with-header-tied-sampled-layer-chain s out)
  case True
    then show ?thesis by blast
  next
    case False
    have missing: trace-fri-header-tied-missing-sampled-layer-chain s out
    unfolding trace-fri-header-tied-missing-sampled-layer-chain-def
    using bad False by blast
    have obstruction:
      trace-fri-header-tied-sampled-layer-assignment-obstruction s out
    by (rule trace-fri-header-tied-missing-sampled-imp-assignment-obstruction
      [OF missing])
    then show ?thesis
      using trace-fri-header-tied-assignment-obstruction-imp-conflict-or-zero
      by blast
qed

```

```

lemma checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-and-missing:
  assumes sampled-bound:

```

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-sampled-layer-chain)
 $adversary\text{-}initial\text{-}state \leq R$
and *missing-bound*:
 $\wedge data$ *attacker-state*.
 Some ($data$, *attacker-state*) $\in$
 $set\text{-}dist$ ($execute$ (*checked-staged-transcript-program* A)
 $adversary\text{-}initial\text{-}state$) $\implies$
 $wp\text{-}event$ *verify-monad*
 (*trace-fri-header-tied-missing-sampled-layer-chain*
 (*verifier-state-from-adversary* *attacker-state*
 (*staged-proof-transcript* $data$)))
 (*verifier-state-from-adversary* *attacker-state*
 (*staged-proof-transcript* $data$)) $\leq M$
shows
 $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)
 $adversary\text{-}initial\text{-}state \leq R + M$
proof –
have *missing-staged*:
 $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-header-tied-missing-sampled-layer-chain)
 $adversary\text{-}initial\text{-}state \leq M$
by (*rule* *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
 (*auto simp*: *trace-fri-header-tied-missing-sampled-layer-chain-def*
missing-bound)
have $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)
 $adversary\text{-}initial\text{-}state \leq$
 $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 ($\lambda out.$
staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-sampled-layer-chain $out \vee$
staged-security-with-data-state-verifier-event
trace-fri-header-tied-missing-sampled-layer-chain out)
 $adversary\text{-}initial\text{-}state$
unfolding *trace-fri-header-tied-missing-sampled-layer-chain-def*
by (*rule* *wp-event-mono-on-support*)
 (*auto simp*: *staged-security-with-data-state-verifier-event-def*
split: *option.splits* *prod.splits*)
also have $\dots \leq$
 $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-sampled-layer-chain)
 $adversary\text{-}initial\text{-}state +$

```

    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-missing-sampled-layer-chain)
      adversary-initial-state
  by (rule wp-event-union-bound)
also have ...  $\leq R + M$ 
  by (rule add-mono[OF sampled-bound missing-staged])
finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-conflict-and-staged-zero:*

assumes *sampled-bound:*

```

    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-header-tied-sampled-layer-chain)
    adversary-initial-state  $\leq R$ 

```

and *conflict-bound:*

```

    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-sampled-assignment-conflict)
    adversary-initial-state  $\leq C$ 

```

and *zero-bound:*

```

    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-zero-round-final-obstruction)
    adversary-initial-state  $\leq Z$ 

```

shows

```

    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state  $\leq R + (C + Z)$ 

```

proof –

let *?M* = *checked-staged-security-experiment-with-data-state A*

let *?sampled* =

```

    staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-sampled-layer-chain

```

let *?conflict* =

```

    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-sampled-assignment-conflict

```

let *?zero* =

```

    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-zero-round-final-obstruction

```

have *split:*

```

    wp-event ?M
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state  $\leq$ 
    wp-event ?M ( $\lambda out. ?sampled\ out \vee ?conflict\ out \vee ?zero\ out$ )
    adversary-initial-state

```

```

unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono-on-support)
  (auto dest:
    trace-fri-header-tied-partial-candidate-imp-sampled-conflict-or-zero
    split: option.splits prod.splits)
also have ...  $\leq$ 
  wp-event ?M ?sampled adversary-initial-state +
  wp-event ?M ( $\lambda$ out. ?conflict out  $\vee$  ?zero out)
  adversary-initial-state
by (rule wp-event-union-bound)
also have ...  $\leq$ 
  wp-event ?M ?sampled adversary-initial-state +
  (wp-event ?M ?conflict adversary-initial-state +
  wp-event ?M ?zero adversary-initial-state)
by (intro add-mono order-refl wp-event-union-bound)
also have ...  $\leq R + (C + Z)$ 
by (intro add-mono sampled-bound conflict-bound zero-bound)
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-and-missing:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: finite B
and cover:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    trace-fri-sampled-layer-chain-cover
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad
and envelope:
 $\wedge$ data attacker-state trace-table.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    trace-fri-multiround-bad-sets bad trace-table  $\subseteq B$ 
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda$ s. trace-fri-challenge-list-fresh-hit s B))
    adversary-initial-state  $\leq F$ 
and missing-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    wp-event verify-monad

```

$$\begin{aligned}
& (\text{trace-fri-header-tied-missing-sampled-layer-chain} \\
& \quad (\text{verifier-state-from-adversary attacker-state} \\
& \quad \quad (\text{staged-proof-transcript data}))) \\
& (\text{verifier-state-from-adversary attacker-state} \\
& \quad (\text{staged-proof-transcript data})) \leq M
\end{aligned}$$

shows

$$\begin{aligned}
& \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\
& \quad (\text{staged-security-with-data-state-verifier-event} \\
& \quad \quad \text{trace-fri-bad-with-header-tied-partial-candidate}) \\
& \text{adversary-initial-state} \leq \\
& (F + \\
& \quad \text{hash-relation-budget-value } (\text{card } B * \text{ceil-log clength}) \\
& \quad (\text{staged-attacker-query-budget budgets} + \\
& \quad \quad \text{staged-challenge-query-budget})) + M
\end{aligned}$$

proof –

have *sampled*:

$$\begin{aligned}
& \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\
& \quad (\text{staged-security-with-data-state-verifier-event} \\
& \quad \quad \text{trace-fri-bad-with-header-tied-sampled-layer-chain}) \\
& \text{adversary-initial-state} \leq \\
& F + \\
& \quad \text{hash-relation-budget-value } (\text{card } B * \text{ceil-log clength}) \\
& \quad (\text{staged-attacker-query-budget budgets} + \\
& \quad \quad \text{staged-challenge-query-budget})
\end{aligned}$$

by (rule

$$\begin{aligned}
& \text{checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-fresh-and-prequery} \\
& \quad [\text{OF wf controlled finite-B cover envelope fresh-bound}]
\end{aligned}$$

show *?thesis*

by (rule

$$\begin{aligned}
& \text{checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-and-missing} \\
& \quad [\text{OF sampled missing-bound}]
\end{aligned}$$

qed

lemma *checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-and-assignment-obs*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *finite-B: finite B*

and *cover:*

$$\bigwedge \text{data attacker-state.}$$

$$\begin{aligned}
& \text{Some } (\text{data, attacker-state}) \in \\
& \quad \text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \\
& \quad \quad \text{adversary-initial-state}) \implies \\
& \text{trace-fri-sampled-layer-chain-cover} \\
& \quad (\text{verifier-state-from-adversary attacker-state} \\
& \quad \quad (\text{staged-proof-transcript data})) \text{ bad}
\end{aligned}$$

and *envelope:*

$$\bigwedge \text{data attacker-state trace-table.}$$

$$\begin{aligned}
& \text{Some } (\text{data, attacker-state}) \in \\
& \quad \text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)
\end{aligned}$$

$\text{adversary-initial-state}) \implies$
 $\text{trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
and *fresh-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B))$
 $\text{adversary-initial-state} \leq F$
and *assignment-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-layer-assignment-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq M$
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq$
 $(F +$
 $\text{hash-relation-budget-value } (\text{card } B * \text{ceil-log clength})$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) + M$
proof (rule
 $\text{checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-and-missing}$
 $[OF \text{ wf controlled finite-B cover envelope fresh-bound}])$
fix *data attacker-state*
assume *support*:
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$
show $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-missing-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq M$
by (rule $\text{wp-trace-fri-header-tied-missing-sampled-bound-from-assignment-obstruction}$)
(rule $\text{assignment-bound}[OF \text{ support}]$)
qed

lemma $\text{checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-conflict-and-staged-}$
assumes *wf*: $\text{staged-budget-wellformed budgets}$
and *controlled*: $\text{staged-adversary-controlled budgets } A$
and *finite-B*: $\text{finite } B$

and *cover*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
trace-fri-sampled-layer-chain-cover
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) *bad*
and *envelope*:
 $\wedge$ *data attacker-state trace-table*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
trace-fri-multiround-bad-sets bad trace-table $\subseteq B$
and *fresh-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
($\lambda s.$ *trace-fri-challenge-list-fresh-hit s B*))
adversary-initial-state $\leq F$
and *conflict-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-header-tied-sampled-assignment-conflict*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq C$
and *zero-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-header-tied-zero-round-final-obstruction)
adversary-initial-state $\leq Z$
shows
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state $\leq$
(*F* +
hash-relation-budget-value (*card B* * *ceil-log clength*)
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)) + (*C* + *Z*)
proof (*rule*
checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-conflict-and-staged-zero)
show *wp-event* (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-sampled-layer-chain)

$adversary-initial-state \leq$
 $F +$
 $hash-relation-budget-value (card\ B * ceil-log\ clength)$
 $(staged-attacker-query-budget\ budgets +$
 $staged-challenge-query-budget)$
by (rule
 $checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-fresh-and-prequery$
 $[OF\ wf\ controlled\ finite-B\ cover\ envelope\ fresh-bound])$
show $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $trace-fri-header-tied-sampled-assignment-conflict)$
 $adversary-initial-state \leq C$
by (rule $checked-staged-security-with-data-state-verifier-event-bound-from-cont$)
 $(simp-all\ add:\ conflict-bound$
 $trace-fri-header-tied-sampled-assignment-conflict-def$
 $accepted-fri-opening-transcript-def)$
show $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $trace-fri-header-tied-zero-round-final-obstruction)$
 $adversary-initial-state \leq Z$
by (rule $zero-bound$)
qed

lemma $checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-conflict-and-zero:$
assumes $wf:\ staged-budget-wellformed\ budgets$
and $controlled:\ staged-adversary-controlled\ budgets\ A$
and $finite-B:\ finite\ B$
and $cover:$
 $\bigwedge data\ attacker-state.$
 $Some\ (data,\ attacker-state) \in$
 $set-dist\ (execute\ (checked-staged-transcript-program\ A)$
 $adversary-initial-state) \implies$
 $trace-fri-sampled-layer-chain-cover$
 $(verifier-state-from-adversary\ attacker-state$
 $(staged-proof-transcript\ data))\ bad$
and $envelope:$
 $\bigwedge data\ attacker-state\ trace-table.$
 $Some\ (data,\ attacker-state) \in$
 $set-dist\ (execute\ (checked-staged-transcript-program\ A)$
 $adversary-initial-state) \implies$
 $trace-fri-multi-round-bad-sets\ bad\ trace-table \subseteq B$
and $fresh-bound:$
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $(\lambda s.\ trace-fri-challenge-list-fresh-hit\ s\ B))$
 $adversary-initial-state \leq F$
and $conflict-bound:$
 $\bigwedge data\ attacker-state.$
 $Some\ (data,\ attacker-state) \in$

$\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq C$
and zero-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq Z$
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq$
 $(F +$
 $\text{hash-relation-budget-value } (\text{card } B * \text{ceil-log clength})$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) + (C + Z)$
proof (rule
 $\text{checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-and-assignment-obstruction}$
 $[\text{OF wf controlled finite-}B \text{ cover envelope fresh-bound}])$
fix $\text{data attacker-state}$
assume support:
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state})$
show $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-layer-assignment-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq C + Z$
by (rule
 $\text{wp-trace-fri-header-tied-assignment-obstruction-bound-from-conflict-and-zero})$
 $(\text{rule conflict-bound}[\text{OF support}], \text{rule zero-bound}[\text{OF support}])$
qed

lemma $\text{checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-and-residual-gaps:}$
assumes $\text{wf: staged-budget-wellformed budgets}$

and *controlled*: staged-adversary-controlled budgets A
and *finite-B*: finite B
and *cover*:
 $\wedge$ *data attacker-state*.
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{trace-fri-sampled-layer-chain-cover}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad}$
and *envelope*:
 $\wedge$ *data attacker-state trace-table*.
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
and *fresh-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B))$
 $\text{adversary-initial-state} \leq \text{Fresh}$
and *slot-bound*:
 $\wedge$ *data attacker-state*.
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-authenticated-slot-recorded-chunk-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Slot}$
and *merkle-bound*:
 $\wedge$ *data attacker-state*.
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Merkle}$
and *structural-bound*:
 $\wedge$ *data attacker-state*.
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$

$(\text{trace-fri-header-tied-sibling-mismatch-structural-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Structural}$
and *alignment-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-value-alignment-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Align}$
and *next-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Next}$
and *successor-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Successor}$
and *final-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Final}$

and *zero-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-header-tied-zero-round-final-obstruction*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$ *Zero*

shows
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state $\leq$
(*Fresh* +
hash-relation-budget-value (*card* *B* * *ceil-log* *clength*)
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)) +
((((*Merkle* + *Structural*) + *Align*) + *Next* + *Successor* + *Final*) +
Merkle) + *Slot* + *Zero*)

proof (*rule*
checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-conflict-and-zero
[*OF* *wf* *controlled* *finite-B* *cover* *envelope* *fresh-bound*])
fix *data attacker-state*
assume *support*:
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state)
let *?s* =
verifier-state-from-adversary attacker-state
(*staged-proof-transcript data*)
show *wp-event verify-monad*
(*trace-fri-header-tied-sampled-assignment-conflict ?s*) *?s*
 $\leq$ (((((*Merkle* + *Structural*) + *Align*) + *Next* + *Successor* + *Final*) +
Merkle) + *Slot*)
by (*rule*
wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-slot-recorded-chunk-gap-and-residuals)
(*rule slot-bound*[*OF* *support*],
rule merkle-bound[*OF* *support*],
rule structural-bound[*OF* *support*],
rule alignment-bound[*OF* *support*],
rule next-bound[*OF* *support*],
rule successor-bound[*OF* *support*],
rule final-bound[*OF* *support*])
show *wp-event verify-monad*
(*trace-fri-header-tied-zero-round-final-obstruction ?s*) *?s* $\leq$ *Zero*
by (*rule zero-bound*[*OF* *support*])

qed

lemma *checked-staged-security-trace-fri-empty-header-bound-from-sampled-and-missing:*

assumes *sampled-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-sampled-layer-chain)
adversary-initial-state ≤ R

and *missing-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒

wp-event verify-monad
(trace-fri-reachable-missing-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ M

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤ R + M

proof —

have *missing-staged:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-reachable-missing-sampled-layer-chain)
adversary-initial-state ≤ M

by (*rule checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
(*simp add: trace-fri-reachable-missing-sampled-layer-chain-def*
trace-fri-bad-with-reachable-partial-candidate-def
accepted-with-partial-trace-openings-def,
rule missing-bound)

have *wp-event (checked-staged-security-experiment-with-data-state A)*

(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤

wp-event (checked-staged-security-experiment-with-data-state A)
(λout.

staged-security-with-data-state-verifier-event
trace-fri-bad-with-sampled-layer-chain out ∨
staged-security-with-data-state-verifier-event
trace-fri-reachable-missing-sampled-layer-chain out)
adversary-initial-state

by (*rule wp-event-mono-on-support*)

(*auto simp: staged-security-with-data-state-verifier-event-def*
dest:

$\text{trace-fri-bad-with-empty-composition-header-candidates-imp-reachable-partial-candidate}$
 $\text{trace-fri-reachable-imp-sampled-or-missing-layer-chain}$
 $\text{split: option.splits prod.splits}$
also have $\dots \leq$
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-sampled-layer-chain})$
 $\text{adversary-initial-state} +$
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-reachable-missing-sampled-layer-chain})$
 $\text{adversary-initial-state}$
by $(\text{rule wp-event-union-bound})$
also have $\dots \leq R + M$
by $(\text{rule add-mono}[OF \text{ sampled-bound missing-staged}])$
finally show $?thesis$.
qed

lemma *checked-staged-security-trace-fri-empty-header-bound-from-fresh-prequery-and-missing:*

assumes $\text{wf: staged-budget-wellformed budgets}$
and $\text{controlled: staged-adversary-controlled budgets } A$
and $\text{finite-B: finite } B$
and cover:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})} \implies$
 $\text{trace-fri-sampled-layer-chain-cover}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad}$
and envelope:
 $\bigwedge \text{data attacker-state trace-table.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})} \implies$
 $\text{trace-fri-multi-round-bad-sets bad trace-table} \subseteq B$
and fresh-bound:
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B))$
 $\text{adversary-initial-state} \leq F$
and missing-bound:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-reachable-missing-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$

$$\begin{aligned} & (staged\text{-}proof\text{-}transcript\ data))) \\ & (verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state \\ & (staged\text{-}proof\text{-}transcript\ data)) \leq M \end{aligned}$$

shows

$$\begin{aligned} & wp\text{-}event\ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A) \\ & (staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event \\ & \quad trace\text{-}fri\text{-}bad\text{-}with\text{-}empty\text{-}composition\text{-}header\text{-}candidates) \\ & adversary\text{-}initial\text{-}state \leq \\ & (F + \\ & \quad hash\text{-}relation\text{-}budget\text{-}value\ (card\ B * ceil\text{-}log\ clength) \\ & \quad (staged\text{-}attacker\text{-}query\text{-}budget\ budgets + \\ & \quad \quad staged\text{-}challenge\text{-}query\text{-}budget)) + M \end{aligned}$$

proof –

have *sampled*:

$$\begin{aligned} & wp\text{-}event\ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A) \\ & (staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event \\ & \quad trace\text{-}fri\text{-}bad\text{-}with\text{-}sampled\text{-}layer\text{-}chain) \\ & adversary\text{-}initial\text{-}state \leq \\ & F + \\ & \quad hash\text{-}relation\text{-}budget\text{-}value\ (card\ B * ceil\text{-}log\ clength) \\ & \quad (staged\text{-}attacker\text{-}query\text{-}budget\ budgets + \\ & \quad \quad staged\text{-}challenge\text{-}query\text{-}budget) \end{aligned}$$

by (rule
checked-staged-security-trace-fri-sampled-layer-chain-bound-from-fresh-and-prequery
[OF wf controlled finite-B cover envelope fresh-bound])

show *?thesis*

by (rule
checked-staged-security-trace-fri-empty-header-bound-from-sampled-and-missing
[OF sampled missing-bound])

qed

lemma *checked-staged-security-trace-fri-empty-header-bound-from-fresh-prequery-and-assignment-obstruction*:

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *finite-B: finite B*

and *cover*:

$\bigwedge data\ attacker\text{-}state.$

Some $(data, attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (checked\text{-}staged\text{-}transcript\text{-}program\ A)$
 $adversary\text{-}initial\text{-}state) \implies$
 $trace\text{-}fri\text{-}sampled\text{-}layer\text{-}chain\text{-}cover$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data))\ bad$

and *envelope*:

$\bigwedge data\ attacker\text{-}state\ trace\text{-}table.$

Some $(data, attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (checked\text{-}staged\text{-}transcript\text{-}program\ A)$
 $adversary\text{-}initial\text{-}state) \implies$
 $trace\text{-}fri\text{-}multi\text{-}round\text{-}bad\text{-}sets\ bad\ trace\text{-}table \subseteq B$

and *fresh-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B$))
adversary-initial-state $\leq F$
and *assignment-bound*:
 $\wedge \text{data attacker-state.}$
 Some (*data*, *attacker-state*) $\in$
 set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
wp-event verify-monad
 (*trace-fri-sampled-layer-assignment-obstruction*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\leq M$
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$
 ($F +$
hash-relation-budget-value (*card B* * *ceil-log clength*)
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)) + M
proof (*rule*
checked-staged-security-trace-fri-empty-header-bound-from-fresh-prequery-and-missing
 [*OF wf controlled finite-B cover envelope fresh-bound*])
fix *data attacker-state*
assume *support*:
 Some (*data*, *attacker-state*) $\in$
 set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state)
show *wp-event verify-monad*
 (*trace-fri-reachable-missing-sampled-layer-chain*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\leq M$
by (*rule wp-trace-fri-reachable-missing-sampled-bound-from-assignment-obstruction*)
 (*rule assignment-bound*[*OF support*])
qed

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-sampled-and-missing*:
assumes *sampled-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-verifier-tied-sampled-layer-chain)
adversary-initial-state $\leq R$

and *missing-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-missing-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq M$
shows
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq R + M$
proof –
have *missing-staged*:
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-verifier-tied-missing-sampled-layer-chain})$
 $\text{adversary-initial-state} \leq M$
by (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
 $(\text{auto simp: composition-fri-verifier-tied-missing-sampled-layer-chain-def}$
 $\text{missing-bound})$
have $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\lambda \text{out.}$
 $\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain out } \vee$
 $\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-verifier-tied-missing-sampled-layer-chain out})$
 $\text{adversary-initial-state}$
unfolding *composition-fri-verifier-tied-missing-sampled-layer-chain-def*
by (rule *wp-event-mono-on-support*)
 $(\text{auto simp: staged-security-with-data-state-verifier-event-def}$
 $\text{split: option.splits prod.splits})$
also have $\dots \leq$
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain})$
 $\text{adversary-initial-state} +$
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-verifier-tied-missing-sampled-layer-chain})$
 $\text{adversary-initial-state}$

```

    by (rule wp-event-union-bound)
  also have ...  $\leq R + M$ 
    by (rule add-mono[OF sampled-bound missing-staged])
  finally show ?thesis .
qed

```

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-and-missing*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B:  $\bigwedge dg. \text{finite } (B \ dg)$ 
and cover:
 $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    composition-fri-sampled-layer-chain-cover
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) bad
and envelope:
 $\bigwedge \text{data attacker-state dg composition-table.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    composition-fri-multiround-bad-sets bad dg composition-table
       $\subseteq B \ dg$ 
and fresh-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B$ ))
  adversary-initial-state  $\leq F$ 
and missing-bound:
 $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (composition-fri-verifier-tied-missing-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq M$ 
shows
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-fri-bad-with-verifier-tied-partial-candidate)
  adversary-initial-state  $\leq$ 
  (F +
    hash-relation-budget-value
      ( $\sum dg \in (UNIV :: 'f \ \text{set}).$ 

```

$\text{card } (B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) + M$

proof –

have *sampled*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-verifier-tied-sampled-layer-chain)
adversary-initial-state $\leq$
 $F +$
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})$

by (*rule*
checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-fresh-and-prequery
 $[OF \text{ wf controlled finite-B cover envelope fresh-bound}]$)

show *?thesis*

by (*rule*
checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-sampled-and-missing
 $[OF \text{ sampled missing-bound}]$)

qed

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-and-assignment*

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *finite-B*: $\bigwedge dg. \text{finite } (B \text{ dg})$

and *cover*:

$\bigwedge \text{data attacker-state.}$
Some (*data, attacker-state*) $\in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
composition-fri-sampled-layer-chain-cover
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad}$

and *envelope*:

$\bigwedge \text{data attacker-state dg composition-table.}$
Some (*data, attacker-state*) $\in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{composition-fri-multiround-bad-sets bad dg composition-table}$
 $\subseteq B \text{ dg}$

and *fresh-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
 $(\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \text{ B}))$
adversary-initial-state $\leq F$

and *assignment-bound*:

$\wedge \text{data attacker-state.}$
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-verifier-tied-sampled-layer-assignment-obstruction*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq M$

shows

wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state $\leq$
(*F* +
hash-relation-budget-value
($\sum dg \in (\text{UNIV} :: 'f \text{ set}).$
card (*B dg*) * *ceil-log* (*maxDegree* + 1))
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)) + *M*

proof (*rule*
checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-and-missing
[*OF wf controlled finite-B cover envelope fresh-bound*])

fix *data attacker-state*

assume *support*:

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state)

show *wp-event verify-monad*
(*composition-fri-verifier-tied-missing-sampled-layer-chain*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq M$

by (*rule*
wp-composition-fri-verifier-tied-missing-sampled-bound-from-assignment-obstruction)
(*rule assignment-bound*[*OF support*])

qed

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-conflict-and-*

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *finite-B*: $\wedge dg. \text{finite } (B \ dg)$

and *cover*:

$\wedge \text{data attacker-state.}$
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$

composition-fri-sampled-layer-chain-cover
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad
and envelope:
 $\wedge$ data attacker-state dg composition-table.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B$ dg
and fresh-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s.$ composition-fri-challenge-list-fresh-hit s B))
 adversary-initial-state $\leq F$
and conflict-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq C$
and zero-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq Z$
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state $\leq$
 (F +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) + (C + Z)
proof (rule

checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-and-assignment-ob
 [OF wf controlled finite-B cover envelope fresh-bound])
fix data attacker-state
assume support:
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state)
show wp-event verify-monad
 (composition-fri-verifier-tied-sampled-layer-assignment-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ C + Z
by (rule
 wp-composition-fri-verifier-tied-assignment-obstruction-bound-from-conflict-and-zero)
 (rule conflict-bound[OF support], rule zero-bound[OF support])
qed

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-and-residual*
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: $\bigwedge dg. \text{finite } (B \ dg)$
and cover:
 $\bigwedge$ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 composition-fri-sampled-layer-chain-cover
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad
and envelope:
 $\bigwedge$ data attacker-state dg composition-table.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B \ dg$
and fresh-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B$))
 adversary-initial-state ≤ Fresh
and slot-bound:
 $\bigwedge$ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Slot}$
and merkle-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Merkle}$
and missing-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-recorded-base-chunk-auth-missing}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Missing}$
and next-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Next}$
and successor-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Successor}$
and final-bound:

$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-final-value-replay-gap
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Final
and zero-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Zero
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state $\leq$
 (Fresh +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (((Merkle + Missing) + Next + Successor + Final + Merkle) +
 Slot + Zero)
proof (rule
 checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-conflict-and-zero
 [OF wf controlled finite-B cover envelope fresh-bound])
fix data attacker-state
assume support:
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state)
let ?s =
 verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)
show wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict ?s) ?s
 $\leq$ ((Merkle + Missing) + Next + Successor + Final + Merkle) + Slot
by (rule

```

    wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-slot-recorded-chunk-gap-and-residuals)
  (rule slot-bound[OF support],
   rule merkle-bound[OF support],
   rule missing-bound[OF support],
   rule next-bound[OF support],
   rule successor-bound[OF support],
   rule final-bound[OF support])
  show wp-event verify-monad
    (composition-fri-verifier-tied-zero-round-final-obstruction ?s) ?s ≤ Zero
  by (rule zero-bound[OF support])
qed

end

end

```

```

theory Soundness-FRI-Trace-Candidate-Blocker
imports Soundness-FRI-Candidate-Transfer-Support
begin

```

Diagnostic normalization facts for the remaining trace candidate-transfer branch. These lemmas do not close the branch; they identify the exact obstacle: the broad reachable witness may use a different authenticated root, query-index list, or opening set than the verifier-consumed/header-tied FRI witness.

```

context soundness
begin

```

```

definition trace-fri-header-tied-low-candidate-untied-reachable-witness-gap
  :: ('f, 'a) protocol-channel-scheme ⇒
    (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool

```

```

where

```

```

  trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
  ⇔
  (∃ trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
    fr0 query-idxs0 trace-openings0 trace-table0.
    trace-fri-reachable-header-tie-gap s out ∧
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers ∧
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest ∧
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings ∧
    partial-trace-table-candidate trace-table trace-openings ∧
    trace-table-low-degree trace-table ∧

```

accepted-with-partial-trace-openings *s* out *fr0* *query-idxs0*
trace-openings0 $\wedge$
partial-trace-table-candidate *trace-table0* *trace-openings0* $\wedge$
 $\neg$ *trace-table-low-degree* *trace-table0* $\wedge$
(*fr0* $\neq$ *fr* $\vee$ *query-idxs0* $\neq$ *fri-query-idxs* $\vee$
trace-openings0 $\neq$ *trace-openings*)

lemma *trace-fri-header-tied-low-candidate-gap-imp-untied-reachable-witness*:

assumes *trace-fri-header-tied-low-candidate-gap* *s* out

shows

trace-fri-header-tied-low-candidate-untied-reachable-witness-gap *s* out

proof –

from *assms* **obtain** *trace-roots* *trace-bs* *trace-final* *fri-dg*

composition-roots *composition-bs* *composition-final* *fri-query-idxs*

trace-round-layers *composition-round-layers* *fr* as *rest* *trace-openings*

trace-table

where *gap*: *trace-fri-reachable-header-tie-gap* *s* out

and *fri-openings*:

accepted-fri-opening-transcript *s* out *trace-roots* *trace-bs*

trace-final *fri-dg* *composition-roots* *composition-bs*

composition-final *fri-query-idxs* *trace-round-layers*

composition-round-layers

and *header*:

verifier-header-transcript *s* *fr* *trace-roots* *trace-final* as *fri-dg*

composition-roots *composition-final* *rest*

and *partial*:

accepted-with-partial-trace-openings *s* out *fr* *fri-query-idxs*

trace-openings

and *candidate*:

partial-trace-table-candidate *trace-table* *trace-openings*

and *low*: *trace-table-low-degree* *trace-table*

unfolding *trace-fri-header-tied-low-candidate-gap-def* **by** *blast*

have *raw*: *trace-fri-bad-with-reachable-partial-candidate* *s* out

using *gap* **unfolding** *trace-fri-reachable-header-tie-gap-def* **by** *simp*

from *trace-fri-bad-with-reachable-partial-candidateE*[*OF raw*]

obtain *fr0* *query-idxs0* *trace-openings0* *trace-table0* **where** *evidence0*:

trace-fri-partial-candidate-evidence *s* out *fr0* *query-idxs0*

trace-openings0 *trace-table0*

by *blast*

have *partial0*:

accepted-with-partial-trace-openings *s* out *fr0* *query-idxs0*

trace-openings0

by (rule *trace-fri-partial-candidate-evidenceD*(1)[*OF evidence0*])

have *candidate0*:

partial-trace-table-candidate *trace-table0* *trace-openings0*

by (rule *trace-fri-partial-candidate-evidenceD*(2)[*OF evidence0*])

have *not-low0*: $\neg$ *trace-table-low-degree* *trace-table0*

by (rule *trace-fri-partial-candidate-evidenceD*(3)[*OF evidence0*])

have *untied*:

```

fr0 ≠ fr ∨ query-idxs0 ≠ fri-query-idxs ∨
  trace-openings0 ≠ trace-openings
proof (rule ccontr)
  assume ¬ (fr0 ≠ fr ∨ query-idxs0 ≠ fri-query-idxs ∨
    trace-openings0 ≠ trace-openings)
  then have eqs:
    fr0 = fr
    query-idxs0 = fri-query-idxs
    trace-openings0 = trace-openings
  by auto
  have trace-fri-bad-with-header-tied-partial-candidate s out
  unfolding trace-fri-bad-with-header-tied-partial-candidate-def
  by (intro exI conjI)
    (rule fri-openings, rule header,
      use partial0 candidate0 not-low0 eqs in auto)
  then show False
    using gap unfolding trace-fri-reachable-header-tie-gap-def by simp
qed
show ?thesis
  unfolding
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap-def
  by (intro exI conjI)
    (rule gap, rule fri-openings, rule header, rule partial,
      rule candidate, rule low, rule partial0, rule candidate0,
      rule not-low0, rule untied)
qed

lemma wp-trace-fri-header-tied-low-candidate-gap-bound-from-untied-witness:
  assumes
    wp-event verify-monad
    (trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)
    s ≤ T
  shows
    wp-event verify-monad
    (trace-fri-header-tied-low-candidate-gap s) s ≤ T
  by (rule order-trans[OF - assms])
    (rule wp-event-mono,
      rule trace-fri-header-tied-low-candidate-gap-imp-untied-reachable-witness)

lemma trace-fri-reachable-header-tie-gap-imp-merkle-or-untied-on-support:
  assumes support: out ∈ set-dist (execute verify-monad s)
  and gap: trace-fri-reachable-header-tie-gap s out
  shows
    partial-merkle-inconsistency-bad s out ∨
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
proof –
  have split:
    partial-merkle-inconsistency-bad s out ∨
    trace-fri-header-tied-candidate-available-gap s out

```

by (rule
 trace-fri-reachable-header-tie-gap-imp-merkle-or-candidate-available-on-support
 [*OF support gap*])
 then show ?thesis
 proof
 assume *partial-merkle-inconsistency-bad s out*
 then show ?thesis by simp
 next
 assume available: *trace-fri-header-tied-candidate-available-gap s out*
 have low-gap: *trace-fri-header-tied-low-candidate-gap s out*
 by (rule *trace-fri-header-tied-candidate-available-gap-imp-low-candidate-gap*
 [*OF available*])
 have
 trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
 by (rule
 trace-fri-header-tied-low-candidate-gap-imp-untied-reachable-witness
 [*OF low-gap*])
 then show ?thesis by simp
 qed
 qed

lemma *wp-trace-fri-reachable-header-tie-gap-bound-from-merkle-and-untied:*
assumes *merkle-bound:*
 wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M
and *untied-bound:*
 wp-event verify-monad
 (*trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s*)
 s ≤ U
shows
 wp-event verify-monad
 (*trace-fri-reachable-header-tie-gap s*) *s ≤ M + U*
proof –
have *split-bound:*
 wp-event verify-monad
 (*trace-fri-reachable-header-tie-gap s*) *s ≤*
 wp-event verify-monad
 ($\lambda out. \text{partial-merkle-inconsistency-bad } s \text{ out} \vee$
 trace-fri-header-tied-low-candidate-untied-reachable-witness-gap
 s out) *s*
 by (rule *wp-event-mono-on-support*)
 (rule *trace-fri-reachable-header-tie-gap-imp-merkle-or-untied-on-support*)
also have ... $\leq$
 wp-event verify-monad (partial-merkle-inconsistency-bad s) s +
 wp-event verify-monad
 (*trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s*)
 s
 by (rule *wp-event-union-bound*)
also have ... $\leq M + U$
 by (rule *add-mono*[*OF merkle-bound untied-bound*])

finally show ?thesis .
qed

lemma *trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied:*

fixes $H\ M\ U :: \text{prob}$

assumes *header-bound:*

wp-event verify-monad

(trace-fri-bad-with-header-tied-partial-candidate s) s ≤ H

and *merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M

and *untied-bound:*

wp-event verify-monad

(trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)

s ≤ U

and *total-bound: $H + (M + U) \leq \text{trace-fri-error}$*

shows *trace-fri-reachable-partial-candidate-reduction-assumption s*

proof –

have *gap-bound:*

wp-event verify-monad

(trace-fri-reachable-header-tie-gap s) s ≤ M + U

by (rule

wp-trace-fri-reachable-header-tie-gap-bound-from-merkle-and-untied

[OF merkle-bound untied-bound])

have *reachable-bound:*

wp-event verify-monad

(trace-fri-bad-with-reachable-partial-candidate s) s ≤ H + (M + U)

by (rule *wp-trace-fri-reachable-bound-from-header-tied-and-gap*

[OF header-bound gap-bound])

show ?thesis

unfolding *trace-fri-reachable-partial-candidate-reduction-assumption-def*

by (rule *order-trans[OF reachable-bound total-bound]*)

qed

end

end

theory *Soundness-FRI-Trace-Untied-Cases*

imports *Soundness-FRI-Trace-Candidate-Blocker*

begin

Case split for the remaining trace FRI untied-witness residual.

The broad residual says that the reachable low-degree witness consumed by the verifier and the separate non-low-degree reachable witness disagree in at least one of: authenticated root, query-index list, or sampled openings. This theory exposes these three causes as separate events so subsequent proof layers can bound them independently.

context *soundness*

begin

definition *trace-fri-header-tied-low-candidate-untied-reachable-witness-case*

$:: ('f \Rightarrow 'f \Rightarrow \text{nat list} \Rightarrow \text{nat list} \Rightarrow$
 $\quad 'f \text{ authenticated-opening list list} \Rightarrow$
 $\quad 'f \text{ authenticated-opening list list} \Rightarrow \text{bool}) \Rightarrow$
 $(('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $\quad (\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-low-candidate-untied-reachable-witness-case P s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\quad \text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\quad \text{composition-round-layers fr as rest trace-openings trace-table}$
 $\quad \text{fr0 query-idxs0 trace-openings0 trace-table0.}$
 $\text{trace-fri-reachable-header-tie-gap } s \text{ out} \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\quad \text{trace-final fri-dg composition-roots composition-bs}$
 $\quad \text{composition-final fri-query-idxs trace-round-layers}$
 $\quad \text{composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as fri-dg}$
 $\quad \text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\quad \text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{trace-table-low-degree trace-table} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr0 query-idxs0}$
 $\quad \text{trace-openings0} \wedge$
 $\text{partial-trace-table-candidate trace-table0 trace-openings0} \wedge$
 $\neg \text{trace-table-low-degree trace-table0} \wedge$
 $P \text{ fr0 fr query-idxs0 fri-query-idxs trace-openings0 trace-openings})$

lemma *trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI*:

assumes

trace-fri-reachable-header-tie-gap s out
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\quad \text{trace-final fri-dg composition-roots composition-bs}$
 $\quad \text{composition-final fri-query-idxs trace-round-layers}$
 $\quad \text{composition-round-layers}$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as fri-dg}$
 $\quad \text{composition-roots composition-final rest}$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\quad \text{trace-openings}$
 $\text{partial-trace-table-candidate trace-table trace-openings}$
 $\text{trace-table-low-degree trace-table}$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr0 query-idxs0}$
 $\quad \text{trace-openings0}$
 $\text{partial-trace-table-candidate trace-table0 trace-openings0}$
 $\neg \text{trace-table-low-degree trace-table0}$
 $P \text{ fr0 fr query-idxs0 fri-query-idxs trace-openings0 trace-openings}$

shows

trace-fri-header-tied-low-candidate-untied-reachable-witness-case
P s out

unfolding

trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
by (*intro exI conjI*)
(*rule assms*)+

definition *trace-fri-header-tied-low-candidate-untied-root-gap*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-low-candidate-untied-root-gap s out
 $\longleftrightarrow$
trace-fri-header-tied-low-candidate-untied-reachable-witness-case
 $(\lambda fr0 \text{ fr} \dots \text{fr0} \neq \text{fr}) \text{ s out}$

definition *trace-fri-header-tied-low-candidate-untied-query-gap*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-low-candidate-untied-query-gap s out
 $\longleftrightarrow$
trace-fri-header-tied-low-candidate-untied-reachable-witness-case
 $(\lambda - \text{ query-idxs0 fri-query-idxs} \dots$
 $\text{ query-idxs0} \neq \text{ fri-query-idxs}) \text{ s out}$

definition *trace-fri-header-tied-low-candidate-untied-openings-gap*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-low-candidate-untied-openings-gap s out
 $\longleftrightarrow$
trace-fri-header-tied-low-candidate-untied-reachable-witness-case
 $(\lambda - \dots \text{ trace-openings0 trace-openings}.$
 $\text{ trace-openings0} \neq \text{ trace-openings}) \text{ s out}$

definition *trace-fri-header-tied-low-candidate-untied-normalized-openings-gap*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-low-candidate-untied-normalized-openings-gap s out
 $\longleftrightarrow$
trace-fri-header-tied-low-candidate-untied-reachable-witness-case
 $(\lambda fr0 \text{ fr query-idxs0 fri-query-idxs trace-openings0 trace-openings}.$
 $\text{ fr0} = \text{ fr} \wedge \text{ query-idxs0} = \text{ fri-query-idxs} \wedge$
 $\text{ trace-openings0} \neq \text{ trace-openings}) \text{ s out}$

definition *trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap*


```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap s out
   $\longleftrightarrow$ 
  trace-fri-header-tied-low-candidate-untied-reachable-witness-case
    ( $\lambda$ fr0 fr query-idxs0 fri-query-idxs trace-openings0 trace-openings.
      (fr0  $\neq$  fr  $\vee$  query-idxs0  $\neq$  fri-query-idxs)  $\wedge$ 
      trace-openings0  $\neq$  trace-openings) s out

lemma trace-fri-header-tied-low-candidate-untied-reachable-witness-gap-imp-cases:
assumes
  trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
shows
  trace-fri-header-tied-low-candidate-untied-root-gap s out  $\vee$ 
  trace-fri-header-tied-low-candidate-untied-query-gap s out  $\vee$ 
  trace-fri-header-tied-low-candidate-untied-openings-gap s out
proof -
from assms obtain trace-roots trace-bs trace-final fri-dg
  composition-roots composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers fr as rest trace-openings
  trace-table fr0 query-idxs0 trace-openings0 trace-table0 where
  gap: trace-fri-reachable-header-tie-gap s out
and fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final fri-dg composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers
and header:
  verifier-header-transcript s fr trace-roots trace-final as fri-dg
  composition-roots composition-final rest
and partial:
  accepted-with-partial-trace-openings s out fr fri-query-idxs
  trace-openings
and candidate:
  partial-trace-table-candidate trace-table trace-openings
and low: trace-table-low-degree trace-table
and partial0:
  accepted-with-partial-trace-openings s out fr0 query-idxs0
  trace-openings0
and candidate0:
  partial-trace-table-candidate trace-table0 trace-openings0
and not-low0:  $\neg$  trace-table-low-degree trace-table0
and mismatch:
  fr0  $\neq$  fr  $\vee$  query-idxs0  $\neq$  fri-query-idxs  $\vee$ 
  trace-openings0  $\neq$  trace-openings
unfolding
  trace-fri-header-tied-low-candidate-untied-reachable-witness-gap-def
by blast

```

```

from mismatch show ?thesis
proof
  assume root-mismatch:  $fr0 \neq fr$ 
  have trace-fri-header-tied-low-candidate-untied-root-gap s out
    unfolding trace-fri-header-tied-low-candidate-untied-root-gap-def
    by (rule
      trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI)
      (rule gap, rule fri-openings, rule header, rule partial,
        rule candidate, rule low, rule partial0, rule candidate0,
        rule not-low0, rule root-mismatch)
  then show ?thesis by simp
next
  assume tail:
     $query-idxs0 \neq fri-query-idxs \vee$ 
     $trace-openings0 \neq trace-openings$ 
  then show ?thesis
  proof
    assume query-mismatch:  $query-idxs0 \neq fri-query-idxs$ 
    have trace-fri-header-tied-low-candidate-untied-query-gap s out
      unfolding trace-fri-header-tied-low-candidate-untied-query-gap-def
      by (rule
        trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI)
        (rule gap, rule fri-openings, rule header, rule partial,
          rule candidate, rule low, rule partial0, rule candidate0,
          rule not-low0, rule query-mismatch)
    then show ?thesis by simp
  next
    assume openings-mismatch:
       $trace-openings0 \neq trace-openings$ 
    have trace-fri-header-tied-low-candidate-untied-openings-gap s out
      unfolding trace-fri-header-tied-low-candidate-untied-openings-gap-def
      by (rule
        trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI)
        (rule gap, rule fri-openings, rule header, rule partial,
          rule candidate, rule low, rule partial0, rule candidate0,
          rule not-low0, rule openings-mismatch)
    then show ?thesis by simp
  qed
qed
qed

lemma trace-fri-header-tied-low-candidate-untied-root-gap-imp-witness-gap:
assumes trace-fri-header-tied-low-candidate-untied-root-gap s out
shows
  trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
using assms
unfolding
  trace-fri-header-tied-low-candidate-untied-reachable-witness-gap-def
  trace-fri-header-tied-low-candidate-untied-root-gap-def

```

trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
by *blast*

lemma *trace-fri-header-tied-low-candidate-untied-query-gap-imp-witness-gap:*
assumes *trace-fri-header-tied-low-candidate-untied-query-gap s out*
shows
trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
using *assms*
unfolding
trace-fri-header-tied-low-candidate-untied-reachable-witness-gap-def
trace-fri-header-tied-low-candidate-untied-query-gap-def
trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
by *blast*

lemma *trace-fri-header-tied-low-candidate-untied-openings-gap-imp-witness-gap:*
assumes *trace-fri-header-tied-low-candidate-untied-openings-gap s out*
shows
trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
using *assms*
unfolding
trace-fri-header-tied-low-candidate-untied-reachable-witness-gap-def
trace-fri-header-tied-low-candidate-untied-openings-gap-def
trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
by *blast*

lemma *merkle-path-bound-unique-path-if-clean:*
assumes *clean: $\neg$ hash-map-output-collision s*
and *first: merkle-path-bound rt len idx v path s*
and *second: merkle-path-bound rt len idx v path' s*
and *same-length: length path = length path'*
shows *path = path'*
using *first second same-length*
proof (*induction path arbitrary: path' rt len idx*)
case *Nil*
then show *?case* **by** *simp*
next
case (*Cons sibling path*)
from *Cons.prem1(3)* **obtain** *sibling' path''* **where**
path': path' = sibling' # path''
and *lengths: length path = length path''*
by (*cases path'*) *auto*
show *?case*
proof (*cases idx < len div 2*)
case *True*
obtain *child* **where**
child:
merkle-path-bound child (len div 2) idx v path s
and *first-lookup:*
fmlookup (HashMap s) (MerkleNode child sibling) = Some rt

```

    using Cons.premis(1) True by auto
obtain child' where
  child':
    merkle-path-bound child' (len div 2) idx v path'' s
  and second-lookup:
    fmllookup (HashMap s) (MerkleNode child' sibling') = Some rt
  using Cons.premis(2) True path' by auto
have node-eq:
  MerkleNode child sibling = MerkleNode child' sibling'
proof (rule ccontr)
  assume neg:
    MerkleNode child sibling ≠ MerkleNode child' sibling'
  have hash-map-output-collision s
  by (rule hash-map-output-collisionI
    [OF neg first-lookup second-lookup])
  then show False using clean by contradiction
qed
have child-eq: child = child' using node-eq by simp
have sibling-eq: sibling = sibling' using node-eq by simp
have path-eq: path = path''
  by (rule Cons.IH[OF child - lengths])
  (use child' child-eq in simp)
show ?thesis
  using path' sibling-eq path-eq by simp
next
case False
obtain child where
  child:
    merkle-path-bound child (len div 2) (idx - len div 2)
    v path s
  and first-lookup:
    fmllookup (HashMap s) (MerkleNode sibling child) = Some rt
  using Cons.premis(1) False by auto
obtain child' where
  child':
    merkle-path-bound child' (len div 2) (idx - len div 2)
    v path'' s
  and second-lookup:
    fmllookup (HashMap s) (MerkleNode sibling' child') = Some rt
  using Cons.premis(2) False path' by auto
have node-eq:
  MerkleNode sibling child = MerkleNode sibling' child'
proof (rule ccontr)
  assume neg:
    MerkleNode sibling child ≠ MerkleNode sibling' child'
  have hash-map-output-collision s
  by (rule hash-map-output-collisionI
    [OF neg first-lookup second-lookup])
  then show False using clean by contradiction

```

```

qed
have child-eq: child = child' using node-eq by simp
have sibling-eq: sibling = sibling' using node-eq by simp
have path-eq: path = path''
  by (rule Cons.IH[OF child - lengths])
  (use child' child-eq in simp)
show ?thesis
  using path' sibling-eq path-eq by simp
qed
qed

lemma authenticated-opening-eq-if-clean:
  assumes clean:  $\neg$  hash-map-output-collision s
  and first: authenticated-opening-in s opening
  and second: authenticated-opening-in s opening'
  and same-root: opening-root opening = opening-root opening'
  and same-length: opening-length opening = opening-length opening'
  and same-index: opening-index opening = opening-index opening'
  and same-value: opening-value opening = opening-value opening'
  shows opening = opening'
proof -
  have path-lengths:
    length (opening-path opening) = length (opening-path opening')
  using first second same-length
  unfolding authenticated-opening-in-def by simp
  have first-bound:
    merkle-path-bound
      (opening-root opening)
      (opening-length opening)
      (opening-index opening)
      (opening-value opening)
      (opening-path opening)
    s
  using first unfolding authenticated-opening-in-def by blast
  have second-bound:
    merkle-path-bound
      (opening-root opening)
      (opening-length opening)
      (opening-index opening)
      (opening-value opening)
      (opening-path opening')
    s
  using second same-root same-length same-index same-value
  unfolding authenticated-opening-in-def by simp
  have paths:
    opening-path opening = opening-path opening'
  by (rule merkle-path-bound-unique-path-if-clean
    [OF clean first-bound second-bound path-lengths])
show ?thesis

```

```

    using same-root same-length same-index same-value paths
    by (cases opening; cases opening'; simp)
qed

lemma accepted-with-partial-trace-openings-same-root-query-openings-eq-if-clean:
  assumes partial:
    accepted-with-partial-trace-openings s (Some (result, final-state)) fr
      query-idxs trace-openings
  and partial0:
    accepted-with-partial-trace-openings s (Some (result, final-state)) fr
      query-idxs trace-openings0
  and clean:  $\neg$  hash-map-output-collision final-state
  shows trace-openings0 = trace-openings
proof -
  have len-openings: length trace-openings = rounds
    using partial unfolding accepted-with-partial-trace-openings-def by blast
  have len-openings0: length trace-openings0 = rounds
    using partial0 unfolding accepted-with-partial-trace-openings-def by blast
  have outer-len: length trace-openings0 = length trace-openings
    using len-openings len-openings0 by simp
  have outer-nth:
     $\bigwedge i. i < \text{length trace-openings0} \implies$ 
      trace-openings0 ! i = trace-openings ! i
  proof -
    fix i
    assume i-bound0: i < length trace-openings0
    then have i-bound: i < rounds
      using len-openings0 by simp
    have idxs:
      map opening-index (trace-openings ! i) =
        powers-scaled (query-idxs ! i)
      using partial i-bound
      unfolding accepted-with-partial-trace-openings-def by blast
    have idxs0:
      map opening-index (trace-openings0 ! i) =
        powers-scaled (query-idxs ! i)
      using partial0 i-bound
      unfolding accepted-with-partial-trace-openings-def by blast
    have table:
      partial-authenticated-table fr (scale * clength)
        (trace-openings ! i) final-state
      using partial i-bound
      unfolding accepted-with-partial-trace-openings-def by blast
    have table0:
      partial-authenticated-table fr (scale * clength)
        (trace-openings0 ! i) final-state
      using partial0 i-bound
      unfolding accepted-with-partial-trace-openings-def by blast
    have len-inner:

```

```

length (trace-openings0 ! i) = length (trace-openings ! i)
using idxs idxs0 by (metis length-map)
have inner-nth:
 $\bigwedge j. j < \text{length} (\text{trace-openings0} ! i) \implies$ 
  nth (nth trace-openings0 i) j = nth (nth trace-openings i) j
proof -
  fix j
  assume j-bound0: j < length (trace-openings0 ! i)
  have j-bound: j < length (trace-openings ! i)
    using j-bound0 len-inner by simp
  define opn0 where opn0 = nth (nth trace-openings0 i) j
  define opn where opn = nth (nth trace-openings i) j
  have opn0-in: opn0  $\in$  set (trace-openings0 ! i)
    unfolding opn0-def using j-bound0 by simp
  have opn-in: opn  $\in$  set (trace-openings ! i)
    unfolding opn-def using j-bound by simp
  have auth0: authenticated-opening-in final-state opn0
    using table0 opn0-in unfolding partial-authenticated-table-def
    by blast
  have auth: authenticated-opening-in final-state opn
    using table opn-in unfolding partial-authenticated-table-def
    by blast
  have root0: opening-root opn0 = fr
    using table0 opn0-in unfolding partial-authenticated-table-def
    by blast
  have root: opening-root opn = fr
    using table opn-in unfolding partial-authenticated-table-def
    by blast
  have length0: opening-length opn0 = scale * clength
    using table0 opn0-in unfolding partial-authenticated-table-def
    by blast
  have length: opening-length opn = scale * clength
    using table opn-in unfolding partial-authenticated-table-def
    by blast
  have same-index: opening-index opn0 = opening-index opn
  proof -
    have map opening-index (trace-openings0 ! i) ! j =
      map opening-index (trace-openings ! i) ! j
      using idxs idxs0 by simp
    then show ?thesis
      unfolding opn0-def opn-def using j-bound0 j-bound by simp
  qed
  have same-value: opening-value opn0 = opening-value opn
  proof (rule ccontr)
    assume neq: opening-value opn0  $\neq$  opening-value opn
    have hash-map-output-collision final-state
      by (rule inconsistent-authenticated-openings-imp-hash-collision
        [OF auth0 auth])
    (use root0 root length0 length same-index neq in simp-all)
  
```

```

    then show False using clean by contradiction
  qed
  have opn-eq: opn0 = opn
    by (rule authenticated-opening-eq-if-clean[OF clean auth0 auth])
      (use root0 root length0 length same-index same-value in simp-all)
  show nth (nth trace-openings0 i) j = nth (nth trace-openings i) j
    using opn-eq unfolding opn0-def opn-def by simp
  qed
  show trace-openings0 ! i = trace-openings ! i
    using len-inner inner-nth by (metis nth-equalityI)
  qed
  show ?thesis
    using outer-len outer-nth by (metis nth-equalityI)
  qed

lemma trace-fri-header-tied-low-candidate-untied-normalized-openings-gap-imp-hash-collision:
  assumes
    trace-fri-header-tied-low-candidate-untied-normalized-openings-gap s out
  shows hash-map-output-collision-bad s out
proof -
  from assms obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr as rest trace-openings
    trace-table fr0 query-idxs0 trace-openings0 trace-table0 where
  partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  and partial0:
    accepted-with-partial-trace-openings s out fr0 query-idxs0
    trace-openings0
  and root-eq: fr0 = fr
  and query-eq: query-idxs0 = fri-query-idxs
  and openings-neq: trace-openings0 ≠ trace-openings
  unfolding
    trace-fri-header-tied-low-candidate-untied-normalized-openings-gap-def
    trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
  by blast
  from partial obtain result final-state where out:
    out = Some (result, final-state)
  unfolding accepted-with-partial-trace-openings-def by blast
  show ?thesis
proof (cases hash-map-output-collision final-state)
  case True
  then show ?thesis
    unfolding hash-map-output-collision-bad-def accepted-def
    using out by simp
next
  case clean: False
  have partial-some:

```



```

    accepted-with-partial-trace-openings s (Some (result, final-state))
    fr fri-query-idxs trace-openings
    using partial out by simp
  have partial0-some:
    accepted-with-partial-trace-openings s (Some (result, final-state))
    fr fri-query-idxs trace-openings0
    using partial0 out root-eq query-eq by simp
  have trace-openings0 = trace-openings
    by (rule
        accepted-with-partial-trace-openings-same-root-query-openings-eq-if-clean
        [OF partial-some partial0-some clean])
  then show ?thesis
    using openings-neq by contradiction
qed
qed

lemma trace-fri-header-tied-low-candidate-untied-normalized-openings-gap-mono-hash-collision:
  wp-event verify-monad
    (trace-fri-header-tied-low-candidate-untied-normalized-openings-gap s) s ≤
  wp-event verify-monad (hash-map-output-collision-bad s) s
by (rule wp-event-mono)
  (rule
    trace-fri-header-tied-low-candidate-untied-normalized-openings-gap-imp-hash-collision)

lemma trace-fri-header-tied-low-candidate-untied-openings-gap-imp-normalized-or-nonnormalized:
  assumes trace-fri-header-tied-low-candidate-untied-openings-gap s out
  shows
    trace-fri-header-tied-low-candidate-untied-normalized-openings-gap s out ∨
    trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap s out
proof –
  from assms obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr as rest trace-openings
    trace-table fr0 query-idxs0 trace-openings0 trace-table0 where
    gap: trace-fri-reachable-header-tie-gap s out
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  and candidate:
    partial-trace-table-candidate trace-table trace-openings
  and low: trace-table-low-degree trace-table

```

```

and partial0:
  accepted-with-partial-trace-openings s out fr0 query-idxs0
    trace-openings0
and candidate0:
  partial-trace-table-candidate trace-table0 trace-openings0
and not-low0:  $\neg$  trace-table-low-degree trace-table0
and openings-neq: trace-openings0  $\neq$  trace-openings
unfolding trace-fri-header-tied-low-candidate-untied-openings-gap-def
  trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
by blast
show ?thesis
proof (cases fr0 = fr  $\wedge$  query-idxs0 = fri-query-idxs)
  case True
  have trace-fri-header-tied-low-candidate-untied-normalized-openings-gap
    s out
  unfolding
    trace-fri-header-tied-low-candidate-untied-normalized-openings-gap-def
  by (rule
    trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI
    [OF gap fri-openings header partial candidate low partial0 candidate0
      not-low0])
    (use True openings-neq in simp)
  then show ?thesis by simp
next
  case False
  then have diff: fr0  $\neq$  fr  $\vee$  query-idxs0  $\neq$  fri-query-idxs
    by blast
  have trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap
    s out
  unfolding
    trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-def
  by (rule
    trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI
    [OF gap fri-openings header partial candidate low partial0 candidate0
      not-low0])
    (use diff openings-neq in simp)
  then show ?thesis by simp
qed
qed

lemma wp-trace-fri-header-tied-low-candidate-untied-openings-gap-bound-from-hash-and-nonnormalized:
fixes hash-err nonnormalized-err :: prob
assumes hash-bound:
  wp-event verify-monad (hash-map-output-collision-bad s) s  $\leq$ 
    hash-err
and nonnormalized-bound:
  wp-event verify-monad
    (trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap s)
    s  $\leq$  nonnormalized-err

```

shows
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-low-candidate-untied-openings-gap } s) \ s \leq$
 $\text{hash-err} + \text{nonnormalized-err}$

proof –
have $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-low-candidate-untied-openings-gap } s) \ s \leq$
 $wp\text{-event verify-monad}$
 $(\lambda out.$
 $\text{trace-fri-header-tied-low-candidate-untied-normalized-openings-gap}$
 $s \ out \ \vee$
 $\text{trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap}$
 $s \ out) \ s$
by $(\text{rule } wp\text{-event-mono})$
 $(\text{rule}$
 $\text{trace-fri-header-tied-low-candidate-untied-openings-gap-imp-normalized-or-nonnormalized})$
also have $\dots \leq$
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-low-candidate-untied-normalized-openings-gap } s)$
 $s +$
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap}$
 $s) \ s$
by $(\text{rule } wp\text{-event-union-bound})$
also have $\dots \leq \text{hash-err} + \text{nonnormalized-err}$
by $(\text{rule } add\text{-mono})$
 $(\text{rule } order\text{-trans}[$
 $OF \text{ trace-fri-header-tied-low-candidate-untied-normalized-openings-gap-mono-hash-collision}$
 $\text{hash-bound}],$
 $\text{rule nonnormalized-bound})$
finally show $?thesis$.
qed

lemma $\text{trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-imp-root-or-query}$:
assumes
 $\text{trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap}$
 $s \ out$
shows
 $\text{trace-fri-header-tied-low-candidate-untied-root-gap } s \ out \ \vee$
 $\text{trace-fri-header-tied-low-candidate-untied-query-gap } s \ out$
proof –
from $assms$ **obtain** $\text{trace-roots } \text{trace-bs } \text{trace-final } \text{fri-dg}$
 $\text{composition-roots } \text{composition-bs } \text{composition-final } \text{fri-query-idxs}$
 $\text{trace-round-layers } \text{composition-round-layers } \text{fr } \text{as } \text{rest } \text{trace-openings}$
 $\text{trace-table } \text{fr0 } \text{query-idxs0 } \text{trace-openings0 } \text{trace-table0}$ **where**
 $\text{gap: trace-fri-reachable-header-tie-gap } s \ out$
and fri-openings :
 $\text{accepted-fri-opening-transcript } s \ out \ \text{trace-roots } \text{trace-bs}$
 $\text{trace-final } \text{fri-dg } \text{composition-roots } \text{composition-bs}$

```

    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
and header:
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest
and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
and candidate:
    partial-trace-table-candidate trace-table trace-openings
and low: trace-table-low-degree trace-table
and partial0:
    accepted-with-partial-trace-openings s out fr0 query-idxs0
    trace-openings0
and candidate0:
    partial-trace-table-candidate trace-table0 trace-openings0
and not-low0:  $\neg$  trace-table-low-degree trace-table0
and mismatch:  $fr0 \neq fr \vee query-idxs0 \neq fri-query-idxs$ 
unfolding
    trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-def
    trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
by blast
from mismatch show ?thesis
proof
    assume root-mismatch:  $fr0 \neq fr$ 
    have trace-fri-header-tied-low-candidate-untied-root-gap s out
    unfolding trace-fri-header-tied-low-candidate-untied-root-gap-def
    by (rule
        trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI)
    (rule gap, rule fri-openings, rule header, rule partial,
    rule candidate, rule low, rule partial0, rule candidate0,
    rule not-low0, rule root-mismatch)
    then show ?thesis by simp
next
    assume query-mismatch:  $query-idxs0 \neq fri-query-idxs$ 
    have trace-fri-header-tied-low-candidate-untied-query-gap s out
    unfolding trace-fri-header-tied-low-candidate-untied-query-gap-def
    by (rule
        trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI)
    (rule gap, rule fri-openings, rule header, rule partial,
    rule candidate, rule low, rule partial0, rule candidate0,
    rule not-low0, rule query-mismatch)
    then show ?thesis by simp
qed
qed

lemma wp-trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-bound-from-root-query:
    fixes root-err query-err :: prob
    assumes root-bound:

```

wp-event verify-monad
 (trace-fri-header-tied-low-candidate-untied-root-gap s) $s \leq$
root-err
and *query-bound*:
wp-event verify-monad
 (trace-fri-header-tied-low-candidate-untied-query-gap s) $s \leq$
query-err
shows
wp-event verify-monad
 (trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap s)
 $s \leq$ *root-err* + *query-err*
proof –
have *wp-event verify-monad*
 (trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap s)
 $s \leq$
wp-event verify-monad
 ($\lambda out.$
 trace-fri-header-tied-low-candidate-untied-root-gap s *out* $\vee$
 trace-fri-header-tied-low-candidate-untied-query-gap s *out*) s
by (rule *wp-event-mono*)
 (rule
 trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-imp-root-or-query)
also have ... $\leq$
wp-event verify-monad
 (trace-fri-header-tied-low-candidate-untied-root-gap s) s +
wp-event verify-monad
 (trace-fri-header-tied-low-candidate-untied-query-gap s) s
by (rule *wp-event-union-bound*)
also have ... $\leq$ *root-err* + *query-err*
by (rule *add-mono*[*OF* *root-bound* *query-bound*])
finally show ?thesis .
qed

lemma *trace-fri-header-tied-low-candidate-untied-root-gap-imp-query-or-openings*:

assumes *trace-fri-header-tied-low-candidate-untied-root-gap* s *out*

shows

trace-fri-header-tied-low-candidate-untied-query-gap s *out* $\vee$
trace-fri-header-tied-low-candidate-untied-openings-gap s *out*

proof –

from *assms* **obtain** *trace-roots* *trace-bs* *trace-final* *fri-dg*
composition-roots *composition-bs* *composition-final* *fri-query-idxs*
trace-round-layers *composition-round-layers* *fr* *as* *rest* *trace-openings*
trace-table *fr0* *query-idxs0* *trace-openings0* *trace-table0* **where**
gap: *trace-fri-reachable-header-tie-gap* s *out*
and *fri-openings*:
accepted-fri-opening-transcript s *out* *trace-roots* *trace-bs*
trace-final *fri-dg* *composition-roots* *composition-bs*
composition-final *fri-query-idxs* *trace-round-layers*
composition-round-layers

```

and header:
  verifier-header-transcript s fr trace-roots trace-final as fri-dg
  composition-roots composition-final rest
and partial:
  accepted-with-partial-trace-openings s out fr fri-query-idxs
  trace-openings
and candidate:
  partial-trace-table-candidate trace-table trace-openings
and low: trace-table-low-degree trace-table
and partial0:
  accepted-with-partial-trace-openings s out fr0 query-idxs0
  trace-openings0
and candidate0:
  partial-trace-table-candidate trace-table0 trace-openings0
and not-low0:  $\neg$  trace-table-low-degree trace-table0
and root-mismatch: fr0  $\neq$  fr
unfolding trace-fri-header-tied-low-candidate-untied-root-gap-def
  trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
by blast
show ?thesis
proof (cases query-idxs0  $\neq$  fri-query-idxs)
  case True
  have trace-fri-header-tied-low-candidate-untied-query-gap s out
  unfolding trace-fri-header-tied-low-candidate-untied-query-gap-def
  by (rule
    trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI)
    (rule gap, rule fri-openings, rule header, rule partial,
     rule candidate, rule low, rule partial0, rule candidate0,
     rule not-low0, rule True)
  then show ?thesis by simp
next
  case query-same: False
  show ?thesis
  proof (cases trace-openings0  $\neq$  trace-openings)
    case True
    have trace-fri-header-tied-low-candidate-untied-openings-gap s out
    unfolding trace-fri-header-tied-low-candidate-untied-openings-gap-def
    by (rule
      trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI)
      (rule gap, rule fri-openings, rule header, rule partial,
       rule candidate, rule low, rule partial0, rule candidate0,
       rule not-low0, rule True)
    then show ?thesis by simp
  next
  case openings-same: False
  have same-query: query-idxs0 = fri-query-idxs
  using query-same by simp
  have same-openings: trace-openings0 = trace-openings
  using openings-same by simp

```

```

have len-openings: length trace-openings = rounds
  using partial unfolding accepted-with-partial-trace-openings-def
  by blast
have idxs0:
  map opening-index (trace-openings ! 0) =
    powers-scaled (fri-query-idxs ! 0)
  using partial rounds-positive
  unfolding accepted-with-partial-trace-openings-def by blast
have nonempty: trace-openings ! 0 ≠ []
proof -
  have powers-scaled (fri-query-idxs ! 0) ≠ []
  unfolding powers-scaled-def using powers-pos by simp
  then show ?thesis
  using idxs0 by auto
qed
then obtain opn where opn-in: opn ∈ set (trace-openings ! 0)
  by (cases trace-openings ! 0) auto
have table:
  partial-authenticated-table fr (scale * clength)
  (trace-openings ! 0) (snd (the out))
proof -
  from partial obtain result final-state where
    out: out = Some (result, final-state)
  and table':
    partial-authenticated-table fr (scale * clength)
    (trace-openings ! 0) final-state
  using rounds-positive
  unfolding accepted-with-partial-trace-openings-def by blast
  then show ?thesis by simp
qed
have table0:
  partial-authenticated-table fr0 (scale * clength)
  (trace-openings ! 0) (snd (the out))
proof -
  from partial0 obtain result final-state where
    out: out = Some (result, final-state)
  and table':
    partial-authenticated-table fr0 (scale * clength)
    (trace-openings0 ! 0) final-state
  using rounds-positive
  unfolding accepted-with-partial-trace-openings-def by blast
  then show ?thesis
  using same-openings by simp
qed
have opening-root opn = fr
  using table opn-in unfolding partial-authenticated-table-def by blast
moreover have opening-root opn = fr0
  using table0 opn-in unfolding partial-authenticated-table-def by blast
ultimately have fr0 = fr by simp

```

```

    then show ?thesis
      using root-mismatch by contradiction
  qed
qed
qed

lemma trace-fri-header-tied-low-candidate-untied-gap-imp-query-or-openings:
  assumes
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
  shows
    trace-fri-header-tied-low-candidate-untied-query-gap s out ∨
    trace-fri-header-tied-low-candidate-untied-openings-gap s out
proof -
  have cases:
    trace-fri-header-tied-low-candidate-untied-root-gap s out ∨
    trace-fri-header-tied-low-candidate-untied-query-gap s out ∨
    trace-fri-header-tied-low-candidate-untied-openings-gap s out
  by (rule
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap-imp-cases
    [OF assms])
  then show ?thesis
proof
  assume trace-fri-header-tied-low-candidate-untied-root-gap s out
  then show ?thesis
    by (rule
      trace-fri-header-tied-low-candidate-untied-root-gap-imp-query-or-openings)
next
  assume
    trace-fri-header-tied-low-candidate-untied-query-gap s out ∨
    trace-fri-header-tied-low-candidate-untied-openings-gap s out
  then show ?thesis .
qed
qed

lemma trace-untied-powers-scaled-eq-imp-eq:
  assumes powers-scaled idx = powers-scaled idx'
  shows idx = idx'
proof -
  have left: powers-scaled idx ! 0 = idx
    unfolding powers-scaled-def using powers-pos by simp
  have right: powers-scaled idx' ! 0 = idx'
    unfolding powers-scaled-def using powers-pos by simp
  have powers-scaled idx ! 0 = powers-scaled idx' ! 0
    using assms by simp
  then show ?thesis
    using left right by simp
qed

lemma trace-fri-header-tied-low-candidate-untied-query-gap-imp-openings:

```


assumes *trace-fri-header-tied-low-candidate-untied-query-gap s out*
shows *trace-fri-header-tied-low-candidate-untied-openings-gap s out*
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as rest trace-openings
trace-table fr0 query-idxs0 trace-openings0 trace-table0 **where**
gap: trace-fri-reachable-header-tie-gap s out
and *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *header:*
verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *candidate:*
partial-trace-table-candidate trace-table trace-openings
and *low: trace-table-low-degree trace-table*
and *partial0:*
accepted-with-partial-trace-openings s out fr0 query-idxs0
trace-openings0
and *candidate0:*
partial-trace-table-candidate trace-table0 trace-openings0
and *not-low0: $\neg$ trace-table-low-degree trace-table0*
and *query-mismatch: query-idxs0 $\neq$ fri-query-idxs*
unfolding *trace-fri-header-tied-low-candidate-untied-query-gap-def*
trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
by *blast*
show *?thesis*
proof (*cases trace-openings0 $\neq$ trace-openings*)
case *True*
show *?thesis*
unfolding *trace-fri-header-tied-low-candidate-untied-openings-gap-def*
by (*rule*
trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI)
(rule gap, rule fri-openings, rule header, rule partial,
rule candidate, rule low, rule partial0, rule candidate0,
rule not-low0, rule True)
next
case *openings-same: False*
have *same-openings: trace-openings0 = trace-openings*
using *openings-same* **by** *simp*
have *len-query: length fri-query-idxs = rounds*
using *partial* **unfolding** *accepted-with-partial-trace-openings-def*
by *blast*

```

have len-query0: length query-idxs0 = rounds
  using partial0 unfolding accepted-with-partial-trace-openings-def
  by blast
have nth-round-eq:
   $\bigwedge i. i < rounds \implies query-idxs0 ! i = fri-query-idxs ! i$ 
proof –
  fix i
  assume i-bound: i < rounds
  have idxs:
    map opening-index (trace-openings ! i) =
      powers-scaled (fri-query-idxs ! i)
  using partial i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
  have idxs0:
    map opening-index (trace-openings0 ! i) =
      powers-scaled (query-idxs0 ! i)
  using partial0 i-bound
  unfolding accepted-with-partial-trace-openings-def by blast
  have powers-scaled (query-idxs0 ! i) =
    powers-scaled (fri-query-idxs ! i)
  using idxs idxs0 same-openings by simp
  then show query-idxs0 ! i = fri-query-idxs ! i
    by (rule trace-untied-powers-scaled-eq-imp-eq)
qed
have len-eq: length query-idxs0 = length fri-query-idxs
  using len-query len-query0 by simp
have query-idxs0 = fri-query-idxs
proof (rule nth-equalityI)
  show length query-idxs0 = length fri-query-idxs
    by (rule len-eq)
next
  fix i
  assume i < length query-idxs0
  then have i < rounds
    using len-query0 by simp
  then show query-idxs0 ! i = fri-query-idxs ! i
    by (rule nth-round-eq)
qed
then show ?thesis
  using query-mismatch by contradiction
qed
qed

lemma trace-fri-header-tied-low-candidate-untied-gap-imp-openings:
  assumes
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
  shows
    trace-fri-header-tied-low-candidate-untied-openings-gap s out
proof –

```

```

have split:
  trace-fri-header-tied-low-candidate-untied-query-gap s out ∨
  trace-fri-header-tied-low-candidate-untied-openings-gap s out
by (rule
  trace-fri-header-tied-low-candidate-untied-gap-imp-query-or-openings
  [OF assms])
then show ?thesis
proof
  assume trace-fri-header-tied-low-candidate-untied-query-gap s out
  then show ?thesis
    by (rule
      trace-fri-header-tied-low-candidate-untied-query-gap-imp-openings)
  next
    assume trace-fri-header-tied-low-candidate-untied-openings-gap s out
    then show ?thesis .
qed
qed

lemma trace-fri-header-tied-low-candidate-untied-gap-imp-hash-root-or-query:
assumes
  trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
shows
  hash-map-output-collision-bad s out ∨
  trace-fri-header-tied-low-candidate-untied-root-gap s out ∨
  trace-fri-header-tied-low-candidate-untied-query-gap s out
proof —
have openings:
  trace-fri-header-tied-low-candidate-untied-openings-gap s out
by (rule trace-fri-header-tied-low-candidate-untied-gap-imp-openings
  [OF assms])
have normalized-or:
  trace-fri-header-tied-low-candidate-untied-normalized-openings-gap
  s out ∨
  trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap
  s out
by (rule
  trace-fri-header-tied-low-candidate-untied-openings-gap-imp-normalized-or-nonnormalized
  [OF openings])
then show ?thesis
proof
  assume
    trace-fri-header-tied-low-candidate-untied-normalized-openings-gap
    s out
  then have hash-map-output-collision-bad s out
    by (rule
      trace-fri-header-tied-low-candidate-untied-normalized-openings-gap-imp-hash-collision)
  then show ?thesis by simp
next
  assume

```

```

    trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap
    s out
  then show ?thesis
  using
    trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-imp-root-or-query
  by blast
qed
qed

lemma wp-trace-fri-header-tied-low-candidate-untied-gap-bound-from-cases:
  fixes root-err query-err opening-err :: prob
  assumes root-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-root-gap s) s ≤
      root-err
  and query-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-query-gap s) s ≤
      query-err
  and openings-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-openings-gap s) s ≤
      opening-err
  shows
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)
      s ≤ root-err + (query-err + opening-err)
proof -
  have split-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)
      s ≤
    wp-event verify-monad
      (λout.
        trace-fri-header-tied-low-candidate-untied-root-gap s out ∨
        trace-fri-header-tied-low-candidate-untied-query-gap s out ∨
        trace-fri-header-tied-low-candidate-untied-openings-gap s out)
      s
  by (rule wp-event-mono)
  (rule
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap-imp-cases)
  also have ... ≤
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-root-gap s) s +
    wp-event verify-monad
      (λout.
        trace-fri-header-tied-low-candidate-untied-query-gap s out ∨
        trace-fri-header-tied-low-candidate-untied-openings-gap s out)
      s

```

```

    by (rule wp-event-union-bound)
  also have ... ≤
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-root-gap s) s +
    (wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-query-gap s) s +
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-openings-gap s) s)
  by (intro add-mono order-reft wp-event-union-bound)
  also have ... ≤ root-err + (query-err + opening-err)
  by (intro add-mono root-bound query-bound openings-bound)
  finally show ?thesis .
qed

lemma wp-trace-fri-header-tied-low-candidate-untied-gap-bound-from-query-and-openings:
  fixes query-err opening-err :: prob
  assumes query-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-query-gap s) s ≤
      query-err
  and openings-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-openings-gap s) s ≤
      opening-err
  shows
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)
      s ≤ query-err + opening-err
proof -
  have split-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)
      s ≤
    wp-event verify-monad
      (λout.
        trace-fri-header-tied-low-candidate-untied-query-gap s out ∨
        trace-fri-header-tied-low-candidate-untied-openings-gap s out)
      s
  by (rule wp-event-mono)
  (rule trace-fri-header-tied-low-candidate-untied-gap-imp-query-or-openings)
  also have ... ≤
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-query-gap s) s +
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-openings-gap s) s
  by (rule wp-event-union-bound)
  also have ... ≤ query-err + opening-err
  by (rule add-mono[OF query-bound openings-bound])
  finally show ?thesis .

```

qed

lemma *wp-trace-fri-header-tied-low-candidate-untied-gap-bound-from-openings:*

fixes *opening-err* :: *prob*

assumes *openings-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-untied-openings-gap } s) \leq$

opening-err

shows

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-untied-reachable-witness-gap } s)$

$s \leq \text{opening-err}$

by (*rule order-trans[OF - openings-bound]*)

(*rule wp-event-mono,*

rule trace-fri-header-tied-low-candidate-untied-gap-imp-openings)

lemma *trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied-cases:*

fixes *header-err merkle-err root-err query-err opening-err* :: *prob*

assumes *header-bound:*

wp-event verify-monad

$(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \leq$

header-err

and *merkle-bound:*

wp-event verify-monad (*partial-merkle-inconsistency-bad* *s*) $s \leq$

merkle-err

and *root-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-untied-root-gap } s) \leq$

root-err

and *query-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-untied-query-gap } s) \leq$

query-err

and *openings-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-untied-openings-gap } s) \leq$

opening-err

and *total-bound:*

$\text{header-err} + (\text{merkle-err} + (\text{root-err} + (\text{query-err} + \text{opening-err})))$

$\leq \text{trace-fri-error}$

shows *trace-fri-reachable-partial-candidate-reduction-assumption* *s*

proof –

have *untied-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-untied-reachable-witness-gap } s)$

$s \leq \text{root-err} + (\text{query-err} + \text{opening-err})$

by (*rule*

wp-trace-fri-header-tied-low-candidate-untied-gap-bound-from-cases

[*OF root-bound query-bound openings-bound*])

```

have reduction:
  trace-fri-reachable-partial-candidate-reduction-assumption s
by (rule
    trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied
    [OF header-bound merkle-bound untied-bound])
    (rule total-bound)
show ?thesis
  by (rule reduction)
qed

lemma trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied-openings:
fixes header-err merkle-err opening-err :: prob
assumes header-bound:
  wp-event verify-monad
    (trace-fri-bad-with-header-tied-partial-candidate s) s ≤
    header-err
and merkle-bound:
  wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤
  merkle-err
and openings-bound:
  wp-event verify-monad
    (trace-fri-header-tied-low-candidate-untied-openings-gap s) s ≤
    opening-err
and total-bound:
  header-err + (merkle-err + opening-err) ≤ trace-fri-error
shows trace-fri-reachable-partial-candidate-reduction-assumption s
proof –
have untied-bound:
  wp-event verify-monad
    (trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)
    s ≤ opening-err
by (rule
    wp-trace-fri-header-tied-low-candidate-untied-gap-bound-from-openings
    [OF openings-bound])
have reduction:
  trace-fri-reachable-partial-candidate-reduction-assumption s
by (rule
    trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied
    [OF header-bound merkle-bound untied-bound])
    (rule total-bound)
show ?thesis
  by (rule reduction)
qed

lemma trace-fri-reachable-partial-candidate-reduction-from-header-merkle-hash-and-nonnormalized-openings:
fixes header-err merkle-err hash-err nonnormalized-err :: prob
assumes header-bound:
  wp-event verify-monad
    (trace-fri-bad-with-header-tied-partial-candidate s) s ≤

```

header-err
and merkle-bound:
 wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤
 merkle-err
and hash-bound:
 wp-event verify-monad (hash-map-output-collision-bad s) s ≤
 hash-err
and nonnormalized-bound:
 wp-event verify-monad
 (trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap s)
 s ≤ nonnormalized-err
and total-bound:
 header-err + (merkle-err + (hash-err + nonnormalized-err)) ≤
 trace-fri-error
shows trace-fri-reachable-partial-candidate-reduction-assumption s
proof –
have openings-bound:
 wp-event verify-monad
 (trace-fri-header-tied-low-candidate-untied-openings-gap s) s ≤
 hash-err + nonnormalized-err
by (rule
 wp-trace-fri-header-tied-low-candidate-untied-openings-gap-bound-from-hash-and-nonnormalized
 [OF hash-bound nonnormalized-bound])
show ?thesis
by (rule
 trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied-openings
 [OF header-bound merkle-bound openings-bound])
 (rule total-bound)
qed

lemma trace-fri-reachable-partial-candidate-reduction-from-header-merkle-hash-root-query:

fixes header-err merkle-err hash-err root-err query-err :: prob

assumes header-bound:

wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) s ≤
 header-err

and merkle-bound:

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤
 merkle-err

and hash-bound:

wp-event verify-monad (hash-map-output-collision-bad s) s ≤
 hash-err

and root-bound:

wp-event verify-monad
 (trace-fri-header-tied-low-candidate-untied-root-gap s) s ≤
 root-err

and query-bound:

wp-event verify-monad
 (trace-fri-header-tied-low-candidate-untied-query-gap s) s ≤

query-err
and *total-bound*:
 $\text{header-err} + (\text{merkle-err} + (\text{hash-err} + (\text{root-err} + \text{query-err}))) \leq \text{trace-fri-error}$
shows *trace-fri-reachable-partial-candidate-reduction-assumption s*
proof –
have *nonnormalized-bound*:
wp-event verify-monad
 $(\text{trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap } s)$
 $s \leq \text{root-err} + \text{query-err}$
by (rule
wp-trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-bound-from-root-query
 $[OF \text{ root-bound } \text{query-bound}]$)
show *?thesis*
by (rule
trace-fri-reachable-partial-candidate-reduction-from-header-merkle-hash-and-nonnormalized-openings
 $[OF \text{ header-bound } \text{merkle-bound } \text{hash-bound } \text{nonnormalized-bound}]$)
 $(\text{rule } \text{total-bound})$
qed

lemma *wp-trace-fri-header-tied-low-candidate-untied-gap-bound-from-hash-root-query*:
fixes *hash-err root-err query-err :: prob*
assumes *hash-bound*:
wp-event verify-monad $(\text{hash-map-output-collision-bad } s) \ s \leq \text{hash-err}$
and *root-bound*:
wp-event verify-monad
 $(\text{trace-fri-header-tied-low-candidate-untied-root-gap } s) \ s \leq \text{root-err}$
and *query-bound*:
wp-event verify-monad
 $(\text{trace-fri-header-tied-low-candidate-untied-query-gap } s) \ s \leq \text{query-err}$
shows
wp-event verify-monad
 $(\text{trace-fri-header-tied-low-candidate-untied-reachable-witness-gap } s)$
 $s \leq \text{hash-err} + (\text{root-err} + \text{query-err})$
proof –
have *nonnormalized-bound*:
wp-event verify-monad
 $(\text{trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap } s)$
 $s \leq \text{root-err} + \text{query-err}$
by (rule
wp-trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-bound-from-root-query
 $[OF \text{ root-bound } \text{query-bound}]$)
have *openings-bound*:
wp-event verify-monad
 $(\text{trace-fri-header-tied-low-candidate-untied-openings-gap } s) \ s \leq \text{hash-err} + (\text{root-err} + \text{query-err})$

```

    by (rule
      wp-trace-fri-header-tied-low-candidate-untied-openings-gap-bound-from-hash-and-nonnormalized
      [OF hash-bound nonnormalized-bound])
  show ?thesis
    by (rule
      wp-trace-fri-header-tied-low-candidate-untied-gap-bound-from-openings
      [OF openings-bound])
qed

lemma trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied-query-openings:
  fixes header-err merkle-err query-err opening-err :: prob
  assumes header-bound:
    wp-event verify-monad
      (trace-fri-bad-with-header-tied-partial-candidate s) s ≤
      header-err
  and merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤
    merkle-err
  and query-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-query-gap s) s ≤
      query-err
  and openings-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-openings-gap s) s ≤
      opening-err
  and total-bound:
    header-err + (merkle-err + (query-err + opening-err))
    ≤ trace-fri-error
  shows trace-fri-reachable-partial-candidate-reduction-assumption s
proof -
  have untied-bound:
    wp-event verify-monad
      (trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)
      s ≤ query-err + opening-err
  by (rule
    wp-trace-fri-header-tied-low-candidate-untied-gap-bound-from-query-and-openings
    [OF query-bound openings-bound])
  have reduction:
    trace-fri-reachable-partial-candidate-reduction-assumption s
  by (rule
    trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied
    [OF header-bound merkle-bound untied-bound])
    (rule total-bound)
  show ?thesis
    by (rule reduction)
qed

end

```

end

theory *Soundness-FRI-Trace-Untied-Active*

imports

Soundness-FRI-Active-Reductions

Soundness-FRI-Trace-Untied-Cases

begin

Active-level packaging for the trace FRI untied-witness blocker. This theory is intentionally downstream of the large active-reduction layer so that the new diagnostic event can be used without growing that file further.

context *soundness*

begin

lemma *active-trace-fri-reachable-reduction-from-header-merkle-and-untied:*

fixes *Ht Mt Ut ::*

('f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ prob)

assumes *trace-header-bound:*

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

wp-event verify-monad

(trace-fri-bad-with-header-tied-partial-candidate

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)) $\leq$

Ht data attacker-state

and *trace-merkle-bound:*

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

wp-event verify-monad

(partial-merkle-inconsistency-bad

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)) $\leq$

Mt data attacker-state

and *trace-untied-bound:*

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

wp-event verify-monad

$(\text{trace-fri-header-tied-low-candidate-untied-reachable-witness-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $U_t \text{ data attacker-state}$
and *total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $H_t \text{ data attacker-state} +$
 $(M_t \text{ data attacker-state} + U_t \text{ data attacker-state}) \leq$
 trace-fri-error
and *support*:
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state})}$
shows
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
by (*rule*
 $\text{trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied)}$
 $(\text{rule trace-header-bound[OF support]},$
 $\text{rule trace-merkle-bound[OF support]},$
 $\text{rule trace-untied-bound[OF support]},$
 $\text{rule total-bound[OF support]})$

lemma *active-trace-fri-reachable-reduction-from-header-merkle-and-untied-cases*:

fixes $H_t M_t R_t Q_t O_t ::$

$(\text{'f staged-proof-data} \Rightarrow \text{'f protocol-channel} \Rightarrow \text{prob})$

assumes *trace-header-bound*:

$\wedge \text{data attacker-state.}$

$\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$

$\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$

$H_t \text{ data attacker-state}$

and *trace-merkle-bound*:

$\wedge \text{data attacker-state.}$

$\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$

$wp\text{-}event\ verify\text{-}monad$
 $(partial\text{-}merkle\text{-}inconsistency\text{-}bad$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data)))$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data)) \leq$
 $Mt\ data\ attacker\text{-}state$
and $trace\text{-}root\text{-}bound$:
 $\wedge data\ attacker\text{-}state.$
 $Some\ (data,\ attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (checked\text{-}staged\text{-}transcript\text{-}program\ A)$
 $adversary\text{-}initial\text{-}state) \implies$
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}low\text{-}candidate\text{-}untied\text{-}root\text{-}gap$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data)))$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data)) \leq$
 $Rt\ data\ attacker\text{-}state$
and $trace\text{-}query\text{-}bound$:
 $\wedge data\ attacker\text{-}state.$
 $Some\ (data,\ attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (checked\text{-}staged\text{-}transcript\text{-}program\ A)$
 $adversary\text{-}initial\text{-}state) \implies$
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}low\text{-}candidate\text{-}untied\text{-}query\text{-}gap$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data)))$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data)) \leq$
 $Qt\ data\ attacker\text{-}state$
and $trace\text{-}openings\text{-}bound$:
 $\wedge data\ attacker\text{-}state.$
 $Some\ (data,\ attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (checked\text{-}staged\text{-}transcript\text{-}program\ A)$
 $adversary\text{-}initial\text{-}state) \implies$
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}low\text{-}candidate\text{-}untied\text{-}openings\text{-}gap$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data)))$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data)) \leq$
 $Ot\ data\ attacker\text{-}state$
and $total\text{-}bound$:
 $\wedge data\ attacker\text{-}state.$
 $Some\ (data,\ attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (checked\text{-}staged\text{-}transcript\text{-}program\ A)$
 $adversary\text{-}initial\text{-}state) \implies$
 $Ht\ data\ attacker\text{-}state +$

$(Mt \text{ data attacker-state} +$
 $(Rt \text{ data attacker-state} +$
 $(Qt \text{ data attacker-state} + Ot \text{ data attacker-state}))) \leq$
 trace-fri-error
and *support*:
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$
shows
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
by (*rule*
 $\text{trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied-cases})$
 $(\text{rule trace-header-bound}[OF \text{ support}],$
 $\text{rule trace-merkle-bound}[OF \text{ support}],$
 $\text{rule trace-root-bound}[OF \text{ support}],$
 $\text{rule trace-query-bound}[OF \text{ support}],$
 $\text{rule trace-openings-bound}[OF \text{ support}],$
 $\text{rule total-bound}[OF \text{ support}])$

lemma *active-trace-fri-reachable-reduction-from-header-merkle-and-untied-query-openings:*
fixes $Ht \ Mt \ Qt \ Ot ::$
 $(\text{'f staged-proof-data} \Rightarrow \text{'f protocol-channel} \Rightarrow \text{prob})$
assumes *trace-header-bound:*
 $\bigwedge data \text{ attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ht \text{ data attacker-state}$
and *trace-merkle-bound:*
 $\bigwedge data \text{ attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Mt \text{ data attacker-state}$
and *trace-query-bound:*

$\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-header-tied-low-candidate-untied-query-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
Qt data attacker-state
and *trace-openings-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-header-tied-low-candidate-untied-openings-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
Ot data attacker-state
and *total-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
Ht data attacker-state +
(*Mt data attacker-state* +
(*Qt data attacker-state* + *Ot data attacker-state*)) $\leq$
trace-fri-error
and *support*:
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state)
shows
trace-fri-reachable-partial-candidate-reduction-assumption
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*))
by (*rule*
trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied-query-openings)
(*rule trace-header-bound*[*OF support*],
rule trace-merkle-bound[*OF support*],
rule trace-query-bound[*OF support*],
rule trace-openings-bound[*OF support*],
rule total-bound[*OF support*])

lemma *active-trace-fri-reachable-reduction-from-header-merkle-and-untied-openings*:

```

fixes Ht Mt Ot ::
  ('f staged-proof-data  $\Rightarrow$  'f protocol-channel  $\Rightarrow$  prob)
assumes trace-header-bound:
   $\wedge$ data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\Rightarrow$ 
      wp-event verify-monad
        (trace-fri-bad-with-header-tied-partial-candidate
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))  $\leq$ 
      Ht data attacker-state
and trace-merkle-bound:
   $\wedge$ data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\Rightarrow$ 
      wp-event verify-monad
        (partial-merkle-inconsistency-bad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))  $\leq$ 
      Mt data attacker-state
and trace-openings-bound:
   $\wedge$ data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\Rightarrow$ 
      wp-event verify-monad
        (trace-fri-header-tied-low-candidate-untied-openings-gap
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))  $\leq$ 
      Ot data attacker-state
and total-bound:
   $\wedge$ data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\Rightarrow$ 
      Ht data attacker-state +
        (Mt data attacker-state + Ot data attacker-state)  $\leq$ 
      trace-fri-error
and support:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)

```


adversary-initial-state)
shows
trace-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
by (*rule*
trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied-openings)
(rule trace-header-bound[OF support],
rule trace-merkle-bound[OF support],
rule trace-openings-bound[OF support],
rule total-bound[OF support])

lemma *active-trace-fri-reachable-reduction-from-header-merkle-hash-root-query:*

fixes *Ht Mt Kt Rt Qt ::*

('f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ prob)

assumes *trace-header-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$

wp-event verify-monad
(trace-fri-bad-with-header-tied-partial-candidate
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$

Ht data attacker-state

and *trace-merkle-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$

wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$

Mt data attacker-state

and *trace-hash-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$

wp-event verify-monad
(hash-map-output-collision-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state

$(\text{staged-proof-transcript data}) \leq$
 $Kt \text{ data attacker-state}$
and *trace-root-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-low-candidate-untied-root-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Rt \text{ data attacker-state}$
and *trace-query-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-low-candidate-untied-query-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Qt \text{ data attacker-state}$
and *total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $Ht \text{ data attacker-state} +$
 $(Mt \text{ data attacker-state} +$
 $(Kt \text{ data attacker-state} +$
 $(Rt \text{ data attacker-state} + Qt \text{ data attacker-state}))) \leq$
 trace-fri-error
and *support*:
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)}$
shows
 $\text{trace-fri-reachable-partial-candidate-reduction-assumption}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
by (rule
 $\text{trace-fri-reachable-partial-candidate-reduction-from-header-merkle-hash-root-query})$
(rule *trace-header-bound*[*OF support*],
rule *trace-merkle-bound*[*OF support*],
rule *trace-hash-bound*[*OF support*],

```

    rule trace-root-bound[OF support],
    rule trace-query-bound[OF support],
    rule total-bound[OF support])

```

end

end

theory *Soundness-FRI-Replay-Gap-Assignment*

imports

Soundness-FRI-Trace-Untied-Active

begin

Small adapters from the replay-gap residual predicates to the existing sampled-assignment conflict branches. The replay gaps are represented as sampled next, successor, or final conflicts, so they are charged to the same FRI sampled-assignment layer rather than treated as independent probabilistic obligations.

context *soundness*

begin

lemma *trace-fri-header-tied-next-value-replay-gap-imp-without-same:*

assumes *trace-fri-header-tied-next-value-replay-gap s out*

shows

trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
s out

using *assms*

unfolding *trace-fri-header-tied-next-value-replay-gap-def*

by (*blast intro: trace-fri-header-tied-without-sameI-next*)

lemma *trace-fri-header-tied-successor-replay-gap-imp-without-same:*

assumes *trace-fri-header-tied-successor-replay-gap s out*

shows

trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
s out

using *assms*

unfolding *trace-fri-header-tied-successor-replay-gap-def*

by (*blast intro: trace-fri-header-tied-without-sameI-successor*)

lemma *trace-fri-header-tied-final-value-replay-gap-imp-without-same:*

assumes *trace-fri-header-tied-final-value-replay-gap s out*

shows

trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
s out

using *assms*

unfolding *trace-fri-header-tied-final-value-replay-gap-def*

by (*blast intro: trace-fri-header-tied-without-sameI-final*)

lemma *trace-fri-header-tied-without-same-imp-sampled-assignment-conflict:*

assumes
trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
s out
shows *trace-fri-header-tied-sampled-assignment-conflict s out*
using *assms*
unfolding
trace-fri-header-tied-sampled-assignment-conflict-without-same-layer-def
trace-fri-header-tied-sampled-assignment-conflict-def
generic-fri-sampled-assignment-conflict-without-same-layer-def
generic-fri-sampled-assignment-conflict-def
by *blast*

lemma *wp-trace-fri-header-tied-next-value-replay-gap-bound-from-without-same:*
fixes *C :: prob*
assumes *conflict-bound:*
wp-event verify-monad
(trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
s) s ≤ C
shows
wp-event verify-monad
(trace-fri-header-tied-next-value-replay-gap s) s ≤ C
by (*rule order-trans[OF - conflict-bound]*)
(rule wp-event-mono,
rule trace-fri-header-tied-next-value-replay-gap-imp-without-same)

lemma *wp-trace-fri-header-tied-successor-replay-gap-bound-from-without-same:*
fixes *C :: prob*
assumes *conflict-bound:*
wp-event verify-monad
(trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
s) s ≤ C
shows
wp-event verify-monad
(trace-fri-header-tied-successor-replay-gap s) s ≤ C
by (*rule order-trans[OF - conflict-bound]*)
(rule wp-event-mono,
rule trace-fri-header-tied-successor-replay-gap-imp-without-same)

lemma *wp-trace-fri-header-tied-final-value-replay-gap-bound-from-without-same:*
fixes *C :: prob*
assumes *conflict-bound:*
wp-event verify-monad
(trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
s) s ≤ C
shows
wp-event verify-monad
(trace-fri-header-tied-final-value-replay-gap s) s ≤ C
by (*rule order-trans[OF - conflict-bound]*)
(rule wp-event-mono,

rule trace-fri-header-tied-final-value-replay-gap-imp-without-same)

lemma *wp-trace-fri-header-tied-without-same-bound-from-sampled-assignment-conflict:*

fixes *C :: prob*

assumes *conflict-bound:*

wp-event verify-monad

(trace-fri-header-tied-sampled-assignment-conflict s) s ≤ C

shows

wp-event verify-monad

*(trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
s) s ≤ C*

by *(rule order-trans[OF - conflict-bound])*

(rule wp-event-mono,

rule trace-fri-header-tied-without-same-imp-sampled-assignment-conflict)

lemma *composition-fri-verifier-tied-next-value-replay-gap-imp-without-same:*

assumes *composition-fri-verifier-tied-next-value-replay-gap s out*

shows

*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s out*

using *assms*

unfolding *composition-fri-verifier-tied-next-value-replay-gap-def*

by *(blast intro: composition-fri-verifier-tied-without-sameI-next)*

lemma *composition-fri-verifier-tied-successor-replay-gap-imp-without-same:*

assumes *composition-fri-verifier-tied-successor-replay-gap s out*

shows

*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s out*

using *assms*

unfolding *composition-fri-verifier-tied-successor-replay-gap-def*

by *(blast intro: composition-fri-verifier-tied-without-sameI-successor)*

lemma *composition-fri-verifier-tied-final-value-replay-gap-imp-without-same:*

assumes *composition-fri-verifier-tied-final-value-replay-gap s out*

shows

*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s out*

using *assms*

unfolding *composition-fri-verifier-tied-final-value-replay-gap-def*

by *(blast intro: composition-fri-verifier-tied-without-sameI-final)*

lemma *composition-fri-verifier-tied-without-same-imp-sampled-assignment-conflict:*

assumes

*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s out*

shows *composition-fri-verifier-tied-sampled-assignment-conflict s out*

using *assms*

unfolding

composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def
composition-fri-verifier-tied-sampled-assignment-conflict-def
generic-fri-sampled-assignment-conflict-without-same-layer-def
generic-fri-sampled-assignment-conflict-def
 by blast

lemma *wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-without-same:*
fixes $C :: \text{prob}$
assumes *conflict-bound:*
 wp-event verify-monad
 (*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer*
 s) $s \leq C$
shows
 wp-event verify-monad
 (*composition-fri-verifier-tied-next-value-replay-gap s*) $s \leq C$
by (rule order-trans[*OF* - *conflict-bound*])
 (rule wp-event-mono,
 rule *composition-fri-verifier-tied-next-value-replay-gap-imp-without-same*)

lemma *wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-without-same:*
fixes $C :: \text{prob}$
assumes *conflict-bound:*
 wp-event verify-monad
 (*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer*
 s) $s \leq C$
shows
 wp-event verify-monad
 (*composition-fri-verifier-tied-successor-replay-gap s*) $s \leq C$
by (rule order-trans[*OF* - *conflict-bound*])
 (rule wp-event-mono,
 rule *composition-fri-verifier-tied-successor-replay-gap-imp-without-same*)

lemma *wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-without-same:*
fixes $C :: \text{prob}$
assumes *conflict-bound:*
 wp-event verify-monad
 (*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer*
 s) $s \leq C$
shows
 wp-event verify-monad
 (*composition-fri-verifier-tied-final-value-replay-gap s*) $s \leq C$
by (rule order-trans[*OF* - *conflict-bound*])
 (rule wp-event-mono,
 rule *composition-fri-verifier-tied-final-value-replay-gap-imp-without-same*)

lemma *wp-composition-fri-verifier-tied-without-same-bound-from-sampled-assignment-conflict:*
fixes $C :: \text{prob}$
assumes *conflict-bound:*
 wp-event verify-monad

(*composition-fri-verifier-tied-sampled-assignment-conflict* s) $s \leq C$
shows
wp-event verify-monad
 (*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer*
 s) $s \leq C$
by (*rule order-trans*[*OF - conflict-bound*])
 (*rule wp-event-mono*,
rule composition-fri-verifier-tied-without-same-imp-sampled-assignment-conflict)

lemma *trace-fri-header-tied-residual-obligations-from-without-same-bounds*:

fixes $C :: \text{prob}$

assumes *sampled*:

wp-event verify-monad

(*trace-fri-bad-with-header-tied-sampled-layer-chain* s) $s \leq r\text{-}b$

and *zero*:

wp-event verify-monad

(*trace-fri-header-tied-zero-round-final-obstruction* s) $s \leq z\text{-}b$

and *merkle*:

wp-event verify-monad (*partial-merkle-inconsistency-bad* s) $s \leq p\text{-}b$

and *structural*:

wp-event verify-monad

(*trace-fri-header-tied-sibling-mismatch-structural-gap* s) $s \leq h\text{-}b$

and *without-same*:

wp-event verify-monad

(*trace-fri-header-tied-sampled-assignment-conflict-without-same-layer*
 s) $s \leq C$

and *total*:

$r\text{-}b + (((((p\text{-}b + h\text{-}b) + p\text{-}b) + C + C +$
 $(p\text{-}b + 0 + 0)) + p\text{-}b) + 0 + z\text{-}b)$
 $\leq \text{trace-fri-error}$

shows

trace-fri-header-tied-residual-obligations s

$r\text{-}b\ z\text{-}b\ 0\ p\text{-}b\ h\text{-}b\ p\text{-}b\ C\ C\ 0$

by (*rule*

trace-fri-header-tied-residual-obligations-from-bounds-without-head-slot-and-selected-zero)

(*rule sampled*, *rule zero*, *rule merkle*, *rule structural*,

rule wp-trace-fri-header-tied-next-value-replay-gap-bound-from-without-same
 $[OF\ without\ same]$,

rule wp-trace-fri-header-tied-successor-replay-gap-bound-from-without-same
 $[OF\ without\ same]$,

rule total)

lemma *composition-fri-verifier-tied-residual-obligations-from-without-same-bounds*:

fixes $C :: \text{prob}$

assumes *sampled*:

wp-event verify-monad

(*composition-fri-bad-with-verifier-tied-sampled-layer-chain* s) s
 $\leq r\text{-}b$

and *merkle*:

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ p-b
and *without-same:*
wp-event verify-monad
(composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s) s ≤ C
and *total:*
r-b + (((p-b + 0) + C + C + C + p-b) + 0)
≤ composition-fri-error
shows
composition-fri-verifier-tied-residual-obligations s r-b 0 p-b C C C
by (*rule composition-fri-verifier-tied-residual-obligations-from-bounds-slot-zero*)
(*rule sampled, rule merkle,*
rule wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-without-same
[OF without-same],
rule wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-without-same
[OF without-same],
rule wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-without-same
[OF without-same],
rule total)

lemma *active-trace-fri-residual-obligations-from-without-same-bounds:*

fixes

Rt Zt Pt Ht Ct ::
('f staged-proof-data ⇒
'f protocol-channel ⇒ prob)

assumes *trace-sampled-bound:*

$\bigwedge \text{data attacker-state.}$
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-bad-with-header-tied-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Rt data attacker-state

and *trace-zero-bound:*

$\bigwedge \text{data attacker-state.}$
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-zero-round-final-obstruction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Zt data attacker-state

and *trace-merkle-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*partial-merkle-inconsistency-bad*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
Pt data attacker-state
and *trace-structural-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-header-tied-sibling-mismatch-structural-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
Ht data attacker-state
and *trace-without-same-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-header-tied-sampled-assignment-conflict-without-same-layer*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
Ct data attacker-state
and *trace-total-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
Rt data attacker-state +
((((*Pt data attacker-state* + *Ht data attacker-state*) +
Pt data attacker-state) +
Ct data attacker-state + *Ct data attacker-state* +
(*Pt data attacker-state* + 0 + 0)) +
Pt data attacker-state) +
0 + *Zt data attacker-state*)
 $\leq$ *trace-fri-error*

```

shows active-trace-fri-residual-obligations-for  $A$   $Rt$   $Zt$ 
   $(\lambda - . (0::prob))$   $Pt$   $Ht$   $Pt$   $Ct$   $Ct$   $(\lambda - . (0::prob))$ 
unfolding active-trace-fri-residual-obligations-for-def
proof (intro allI impI)
fix data attacker-state
assume support:
   $Some (data, attacker-state) \in$ 
     $set-dist (execute (checked-staged-transcript-program A)$ 
       $adversary-initial-state)$ 
let  $?s =$ 
   $verifier-state-from-adversary attacker-state$ 
   $(staged-proof-transcript data)$ 
show trace-fri-header-tied-residual-obligations  $?s$ 
   $(Rt data attacker-state) (Zt data attacker-state) 0$ 
   $(Pt data attacker-state) (Ht data attacker-state)$ 
   $(Pt data attacker-state) (Ct data attacker-state)$ 
   $(Ct data attacker-state) 0$ 
by (rule trace-fri-header-tied-residual-obligations-from-without-same-bounds)
  (rule trace-sampled-bound[ $OF support$ ],
    rule trace-zero-bound[ $OF support$ ],
    rule trace-merkle-bound[ $OF support$ ],
    rule trace-structural-bound[ $OF support$ ],
    rule trace-without-same-bound[ $OF support$ ],
    rule trace-total-bound[ $OF support$ ])
qed

```

lemma *active-composition-fri-residual-obligations-from-without-same-bounds*:

```

fixes
   $Rc Pc Cc ::$ 
     $(f staged-proof-data \Rightarrow$ 
       $f protocol-channel \Rightarrow prob)$ 
assumes comp-sampled-bound:
   $\bigwedge data attacker-state.$ 
     $Some (data, attacker-state) \in$ 
       $set-dist (execute (checked-staged-transcript-program A)$ 
         $adversary-initial-state) \implies$ 
     $wp-event verify-monad$ 
       $(composition-fri-bad-with-verifier-tied-sampled-layer-chain$ 
         $(verifier-state-from-adversary attacker-state$ 
           $(staged-proof-transcript data)))$ 
       $(verifier-state-from-adversary attacker-state$ 
         $(staged-proof-transcript data)) \leq$ 
     $Rc data attacker-state$ 
and comp-merkle-bound:
   $\bigwedge data attacker-state.$ 
     $Some (data, attacker-state) \in$ 
       $set-dist (execute (checked-staged-transcript-program A)$ 
         $adversary-initial-state) \implies$ 
     $wp-event verify-monad$ 

```

```

    (partial-merkle-inconsistency-bad
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤
  Pc data attacker-state
and comp-without-same-bound:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
    wp-event verify-monad
      (composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) ≤
    Cc data attacker-state
and comp-total-bound:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
    Rc data attacker-state +
      (((Pc data attacker-state + 0) +
        Cc data attacker-state + Cc data attacker-state +
        Cc data attacker-state + Pc data attacker-state) +
        0)
    ≤ composition-fri-error
shows active-composition-fri-residual-obligations-for A Rc
  (λ- -. (0::prob)) Pc Cc Cc Cc
unfolding active-composition-fri-residual-obligations-for-def
proof (intro allI impI)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show composition-fri-verifier-tied-residual-obligations ?s
  (Rc data attacker-state) 0 (Pc data attacker-state)
  (Cc data attacker-state) (Cc data attacker-state)
  (Cc data attacker-state)
by (rule composition-fri-verifier-tied-residual-obligations-from-without-same-bounds)
  (rule comp-sampled-bound[OF support],
    rule comp-merkle-bound[OF support],
    rule comp-without-same-bound[OF support],

```

$\text{rule comp-total-bound}[OF \text{ support}]$
qed

lemma *trace-fri-reachable-partial-candidate-reduction-from-header-without-same-and-untied:*
fixes $R \ Z \ P \ H \ C \ Q \ U :: \text{prob}$
assumes *sampled-bound:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-sampled-layer-chain } s) \ s \leq R$
and *zero-bound:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s \leq Z$
and *header-partial-merkle-bound:*
 $\text{wp-event verify-monad}$ $(\text{partial-merkle-inconsistency-bad } s) \ s \leq P$
and *structural-gap-bound:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s \leq H$
and *without-same-bound:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer } s) \ s \leq C$
and *gap-partial-merkle-bound:*
 $\text{wp-event verify-monad}$ $(\text{partial-merkle-inconsistency-bad } s) \ s \leq Q$
and *untied-bound:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-low-candidate-untied-reachable-witness-gap } s) \ s \leq U$
and *total-bound:*
 $R + (((((P + H) + P) + C + C + C) + P) + 0 + Z) + (Q + U) \leq \text{trace-fri-error}$
shows *trace-fri-reachable-partial-candidate-reduction-assumption* s
proof –
have *head-neq:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-head-value-neq-gap } s) \ s \leq P$
by (rule
 $\text{wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle}$
 $[OF \text{ header-partial-merkle-bound}]$)
have *head-mismatch:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-head-value-mismatch-gap } s) \ s \leq P$
by (rule
 $\text{wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap}$
 $[OF \text{ head-neq}]$)
have *alignment:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-value-alignment-gap } s) \ s \leq P$
by (rule
 $\text{wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap}$
 $[OF \text{ head-mismatch}]$)

have *next-bound*:
 wp-event verify-monad
 (trace-fri-header-tied-next-value-replay-gap s) $s \leq C$
by (rule
 wp-trace-fri-header-tied-next-value-replay-gap-bound-from-without-same
 [OF without-same-bound])
have *successor-bound*:
 wp-event verify-monad
 (trace-fri-header-tied-successor-replay-gap s) $s \leq C$
by (rule
 wp-trace-fri-header-tied-successor-replay-gap-bound-from-without-same
 [OF without-same-bound])
have *final-bound*:
 wp-event verify-monad
 (trace-fri-header-tied-final-value-replay-gap s) $s \leq C$
by (rule
 wp-trace-fri-header-tied-final-value-replay-gap-bound-from-without-same
 [OF without-same-bound])
have *header-bound*:
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) $s \leq$
 $R + (((((P + H) + P) + C + C + C) + P) + 0 + Z)$
by (rule
 wp-trace-fri-header-tied-bound-from-sampled-zero-slot-recorded-chunk-gap-and-residuals)
 (rule sampled-bound, rule zero-bound,
 rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero,
 rule header-partial-merkle-bound, rule structural-gap-bound,
 rule alignment, rule next-bound, rule successor-bound,
 rule final-bound)
show ?thesis
by (rule
 trace-fri-reachable-partial-candidate-reduction-from-header-merkle-and-untied)
 (rule header-bound, rule gap-partial-merkle-bound,
 rule untied-bound, rule total-bound)

qed

lemma trace-fri-reachable-partial-candidate-reduction-from-header-without-same-hash-root-query:

fixes $R\ Z\ P\ H\ C\ Q\ K\ Rt\ Qt :: \text{prob}$

assumes sampled-bound:

wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) $s \leq R$

and zero-bound:

wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction s) $s \leq Z$

and header-partial-merkle-bound:

wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq P$

and structural-gap-bound:

wp-event verify-monad
 (trace-fri-header-tied-sibling-mismatch-structural-gap s) $s \leq H$

and *without-same-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-sampled-assignment-conflict-without-same-layer*
 s) $s \leq C$
and *gap-partial-merkle-bound*:
 wp-event verify-monad (*partial-merkle-inconsistency-bad s*) $s \leq Q$
and *hash-bound*:
 wp-event verify-monad (*hash-map-output-collision-bad s*) $s \leq K$
and *root-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-low-candidate-untied-root-gap s*) $s \leq Rt$
and *query-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-low-candidate-untied-query-gap s*) $s \leq Qt$
and *total-bound*:
 $R + (((((P + H) + P) + C + C + C) + P) + 0 + Z) +$
 $(Q + (K + (Rt + Qt))) \leq \text{trace-fri-error}$
shows *trace-fri-reachable-partial-candidate-reduction-assumption s*
proof –
have *head-neq*:
 wp-event verify-monad
 (*trace-fri-header-tied-base-head-value-neq-gap s*) $s \leq P$
by (rule
 wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle
 [*OF header-partial-merkle-bound*])
have *head-mismatch*:
 wp-event verify-monad
 (*trace-fri-header-tied-base-head-value-mismatch-gap s*) $s \leq P$
by (rule
 wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
 [*OF head-neq*])
have *alignment*:
 wp-event verify-monad
 (*trace-fri-header-tied-base-value-alignment-gap s*) $s \leq P$
by (rule
 wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap
 [*OF head-mismatch*])
have *next-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-next-value-replay-gap s*) $s \leq C$
by (rule
 wp-trace-fri-header-tied-next-value-replay-gap-bound-from-without-same
 [*OF without-same-bound*])
have *successor-bound*:
 wp-event verify-monad
 (*trace-fri-header-tied-successor-replay-gap s*) $s \leq C$
by (rule
 wp-trace-fri-header-tied-successor-replay-gap-bound-from-without-same
 [*OF without-same-bound*])

have *final-bound*:
 wp-event verify-monad
 (trace-fri-header-tied-final-value-replay-gap s) $s \leq C$
by (rule
 wp-trace-fri-header-tied-final-value-replay-gap-bound-from-without-same
 [OF without-same-bound])
have *header-bound*:
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-partial-candidate s) $s \leq$
 $R + (((((P + H) + P) + C + C + C) + P) + 0 + Z)$
by (rule
 wp-trace-fri-header-tied-bound-from-sampled-zero-slot-recorded-chunk-gap-and-residuals)
 (rule sampled-bound, rule zero-bound,
 rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero,
 rule header-partial-merkle-bound, rule structural-gap-bound,
 rule alignment, rule next-bound, rule successor-bound,
 rule final-bound)
show ?thesis
by (rule
 trace-fri-reachable-partial-candidate-reduction-from-header-merkle-hash-root-query)
 (rule header-bound, rule gap-partial-merkle-bound, rule hash-bound,
 rule root-bound, rule query-bound, rule total-bound)
qed

lemma *active-trace-fri-reachable-reduction-from-header-without-same-and-untied*:

fixes $Rt\ Zt\ Pt\ Ht\ Ct\ Qt\ Ut ::$

($'f$ staged-proof-data $\Rightarrow$ $'f$ protocol-channel $\Rightarrow$ prob)

assumes *trace-sampled-bound*:

$\wedge$ data attacker-state.

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$

Rt data attacker-state

and *trace-zero-bound*:

$\wedge$ data attacker-state.

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state

$(\text{staged-proof-transcript data}) \leq$
 $Zt \text{ data attacker-state}$
and *trace-header-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Pt \text{ data attacker-state}$
and *trace-structural-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ht \text{ data attacker-state}$
and *trace-without-same-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ct \text{ data attacker-state}$
and *trace-gap-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$

Qt data attacker-state
and *trace-untied-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(trace-fri-header-tied-low-candidate-untied-reachable-witness-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Ut data attacker-state
and *trace-total-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
Rt data attacker-state +
(((((Pt data attacker-state + Ht data attacker-state) +
Pt data attacker-state) +
Ct data attacker-state + Ct data attacker-state +
Ct data attacker-state) +
Pt data attacker-state) +
0 + Zt data attacker-state) +
(Qt data attacker-state + Ut data attacker-state)
 $\leq$ *trace-fri-error*
and *support:*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)
shows
trace-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
by (*rule*
trace-fri-reachable-partial-candidate-reduction-from-header-without-same-and-untied)
(rule trace-sampled-bound[OF support],
rule trace-zero-bound[OF support],
rule trace-header-merkle-bound[OF support],
rule trace-structural-bound[OF support],
rule trace-without-same-bound[OF support],
rule trace-gap-merkle-bound[OF support],
rule trace-untied-bound[OF support],
rule trace-total-bound[OF support])

lemma *active-trace-fri-reachable-reduction-from-header-without-same-hash-root-query:*
fixes *Rt Zt Pt Ht Ct Qt Kt Ut Vt ::*
('f staged-proof-data $\implies$ 'f protocol-channel $\implies$ prob)

assumes *trace-sampled-bound*:

$\wedge$ *data attacker-state*.

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$

wp-event verify-monad

(trace-fri-bad-with-header-tied-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$

Rt data attacker-state

and *trace-zero-bound*:

$\wedge$ *data attacker-state*.

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$

wp-event verify-monad

(trace-fri-header-tied-zero-round-final-obstruction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$

Zt data attacker-state

and *trace-header-merkle-bound*:

$\wedge$ *data attacker-state*.

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$

wp-event verify-monad

(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$

Pt data attacker-state

and *trace-structural-bound*:

$\wedge$ *data attacker-state*.

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$

wp-event verify-monad

(trace-fri-header-tied-sibling-mismatch-structural-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$

Ht data attacker-state

and *trace-without-same-bound*:

$\wedge$ *data attacker-state*.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-sampled-assignment-conflict-without-same-layer
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Ct data attacker-state
and *trace-gap-merkle-bound:*
 $\wedge$ *data attacker-state*.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Qt data attacker-state
and *trace-hash-bound:*
 $\wedge$ *data attacker-state*.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(hash-map-output-collision-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Kt data attacker-state
and *trace-root-bound:*
 $\wedge$ *data attacker-state*.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-low-candidate-untied-root-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Ut data attacker-state
and *trace-query-bound:*
 $\wedge$ *data attacker-state*.

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-low-candidate-untied-query-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
Vt data attacker-state
and *trace-total-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
Rt data attacker-state +
(((((Pt data attacker-state + Ht data attacker-state) +
Pt data attacker-state) +
Ct data attacker-state + Ct data attacker-state +
Ct data attacker-state) +
Pt data attacker-state) +
0 + Zt data attacker-state) +
(Qt data attacker-state +
(Kt data attacker-state +
(Ut data attacker-state + Vt data attacker-state))))
≤ trace-fri-error
and *support:*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)
shows
trace-fri-reachable-partial-candidate-reduction-assumption
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))
by *(rule*
trace-fri-reachable-partial-candidate-reduction-from-header-without-same-hash-root-query)
(rule trace-sampled-bound[OF support],
rule trace-zero-bound[OF support],
rule trace-header-merkle-bound[OF support],
rule trace-structural-bound[OF support],
rule trace-without-same-bound[OF support],
rule trace-gap-merkle-bound[OF support],
rule trace-hash-bound[OF support],
rule trace-root-bound[OF support],
rule trace-query-bound[OF support],
rule trace-total-bound[OF support])

lemma *composition-fri-reachable-partial-candidate-reduction-from-split-without-same-bounds:*
fixes *R P C E Q T :: prob*

assumes *sampled-bound*:
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R
and *verifier-partial-merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P
and *without-same-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s) s ≤ C
and *empty-bound*:
wp-event verify-monad
(composition-fri-reachable-verifier-tie-empty-header-gap s) s ≤ E
and *gap-partial-merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ Q
and *candidate-available-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-candidate-available-gap s) s ≤ T
and *total-bound*:
 $R + (((P + 0) + C + C + C + P) + 0) + (E + (Q + T))$
 $\leq \text{composition-fri-error}$
shows *composition-fri-reachable-partial-candidate-reduction-assumption s*
by (rule
composition-fri-reachable-partial-candidate-reduction-from-split-residual-bounds-with-candidate-available-and-recon
(rule *sampled-bound*,
rule *wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero*,
rule *verifier-partial-merkle-bound*,
rule *wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-without-same*
[*OF without-same-bound*],
rule *wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-without-same*
[*OF without-same-bound*],
rule *wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-without-same*
[*OF without-same-bound*],
rule *empty-bound*,
rule *gap-partial-merkle-bound*,
rule *candidate-available-bound*,
rule *total-bound*)

lemma *composition-fri-reachable-partial-candidate-reduction-from-split-sampled-assignment-bounds*:

fixes $R\ P\ C\ E\ Q\ T :: \text{prob}$

assumes *sampled-bound*:

wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain s) s ≤ R

and *verifier-partial-merkle-bound*:

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P

and *sampled-assignment-bound*:

wp-event verify-monad
(composition-fri-verifier-tied-sampled-assignment-conflict s) s
 $\leq C$

and *empty-bound*:

wp-event verify-monad
 (*composition-fri-reachable-verifier-tie-empty-header-gap s*) $s \leq E$
and *gap-partial-merkle-bound*:
wp-event verify-monad (*partial-merkle-inconsistency-bad s*) $s \leq Q$
and *candidate-available-bound*:
wp-event verify-monad
 (*composition-fri-verifier-tied-candidate-available-gap s*) $s \leq T$
and *total-bound*:
 $R + (((P + 0) + C + C + C + P) + 0) + (E + (Q + T))$
 $\leq \text{composition-fri-error}$
shows *composition-fri-reachable-partial-candidate-reduction-assumption s*
by (*rule*
composition-fri-reachable-partial-candidate-reduction-from-split-without-same-bounds)
 (*rule sampled-bound*,
rule verifier-partial-merkle-bound,
rule wp-composition-fri-verifier-tied-without-same-bound-from-sampled-assignment-conflict
 [*OF sampled-assignment-bound*],
rule empty-bound,
rule gap-partial-merkle-bound,
rule candidate-available-bound,
rule total-bound)

lemma *active-composition-fri-reachable-reduction-from-split-without-same-bounds*:

fixes *Rc Pc Cc Ec Qc Tc* ::

(*'f staged-proof-data* $\Rightarrow$ *'f protocol-channel* $\Rightarrow$ *prob*)

assumes *comp-sampled-bound*:

$\wedge$ *data attacker-state*.

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
 (*composition-fri-bad-with-verifier-tied-sampled-layer-chain*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\leq$
Rc data attacker-state

and *comp-merkle-bound*:

$\wedge$ *data attacker-state*.

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
 (*partial-merkle-inconsistency-bad*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\leq$
Pc data attacker-state

and *comp-without-same-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
Cc data attacker-state
and *comp-empty-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-reachable-verifier-tie-empty-header-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
Ec data attacker-state
and *comp-gap-merkle-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*partial-merkle-inconsistency-bad*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
Qc data attacker-state
and *comp-candidate-available-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-verifier-tied-candidate-available-gap*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
Tc data attacker-state
and *comp-total-bound*:

$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 Rc data attacker-state +
 (((Pc data attacker-state + 0) +
 Cc data attacker-state + Cc data attacker-state +
 Cc data attacker-state + Pc data attacker-state) +
 0) +
 (Ec data attacker-state +
 (Qc data attacker-state + Tc data attacker-state))
 $\leq$ composition-fri-error
and support:
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state)
shows
 composition-fri-reachable-partial-candidate-reduction-assumption
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
by (rule
 composition-fri-reachable-partial-candidate-reduction-from-split-without-same-bounds)
 (rule comp-sampled-bound[OF support],
 rule comp-merkle-bound[OF support],
 rule comp-without-same-bound[OF support],
 rule comp-empty-bound[OF support],
 rule comp-gap-merkle-bound[OF support],
 rule comp-candidate-available-bound[OF support],
 rule comp-total-bound[OF support])

lemma active-fri-reductions-from-without-same-and-untied-bounds:

fixes
 Rt Zt Pt Ht Ct Qt Ut Rc Pc Cc Ec Qc Tc ::
 ('f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ prob)
assumes trace-sampled-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$
 Rt data attacker-state
and trace-zero-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$

$\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Zt data attacker-state}$
and $\text{trace-header-merkle-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Pt data attacker-state}$
and $\text{trace-structural-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Ht data attacker-state}$
and $\text{trace-without-same-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{Ct data attacker-state}$
and $\text{trace-gap-merkle-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$

adversary-initial-state) $\implies$
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Qt data attacker-state
and *trace-untied-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(trace-fri-header-tied-low-candidate-untied-reachable-witness-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Ut data attacker-state
and *trace-total-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
Rt data attacker-state +
(((((Pt data attacker-state + Ht data attacker-state) +
Pt data attacker-state) +
Ct data attacker-state + Ct data attacker-state +
Ct data attacker-state) +
Pt data attacker-state) +
0 + Zt data attacker-state) +
(Qt data attacker-state + Ut data attacker-state)
 $\leq$ *trace-fri-error*
and *comp-sampled-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Rc data attacker-state
and *comp-merkle-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$

$\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $P_c \text{ data attacker-state}$
and $\text{comp-without-same-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $C_c \text{ data attacker-state}$
and comp-empty-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-reachable-verifier-tie-empty-header-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $E_c \text{ data attacker-state}$
and $\text{comp-gap-merkle-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Q_c \text{ data attacker-state}$
and $\text{comp-candidate-available-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$

```

    adversary-initial-state)  $\impl$ 
wp-event verify-monad
  (composition-fri-verifier-tied-candidate-available-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq$ 
Tc data attacker-state
and comp-total-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\impl$ 
Rc data attacker-state +
  (((Pc data attacker-state + 0) +
    Cc data attacker-state + Cc data attacker-state +
    Cc data attacker-state + Pc data attacker-state) +
    0) +
  (Ec data attacker-state +
    (Qc data attacker-state + Tc data attacker-state))
 $\leq$  composition-fri-error
shows active-fri-reductions-for A
proof (rule active-fri-reductions-from-reachable)
fix data attacker-state
assume support:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
have total:
  Rt data attacker-state +
  (Pt data attacker-state + Ht data attacker-state +
    Pt data attacker-state +
    Ct data attacker-state +
    Ct data attacker-state +
    Ct data attacker-state +
    Pt data attacker-state +
    0 +
    Zt data attacker-state) +
  (Qt data attacker-state + Ut data attacker-state)
 $\leq$  trace-fri-error
using trace-total-bound[OF support] by (simp add: add.assoc)
show trace-fri-reachable-partial-candidate-reduction-assumption
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
using trace-sampled-bound[OF support]
  trace-zero-bound[OF support]
  trace-header-merkle-bound[OF support]
  trace-structural-bound[OF support]
  trace-without-same-bound[OF support]

```

```

    trace-gap-merkle-bound[OF support]
    trace-untied-bound[OF support]
    total
  by (rule
    trace-fri-reachable-partial-candidate-reduction-from-header-without-same-and-untied
    [where R = Rt data attacker-state
      and Z = Zt data attacker-state
      and P = Pt data attacker-state
      and H = Ht data attacker-state
      and C = Ct data attacker-state
      and Q = Qt data attacker-state
      and U = Ut data attacker-state])
next
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  have total:
    Rc data attacker-state +
      (Pc data attacker-state + 0 + Cc data attacker-state +
        Cc data attacker-state +
        Cc data attacker-state +
        Pc data attacker-state +
        0) +
      (Ec data attacker-state +
        (Qc data attacker-state + Tc data attacker-state))
    ≤ composition-fri-error
  using comp-total-bound[OF support] by (simp add: add.assoc)
  show composition-fri-reachable-partial-candidate-reduction-assumption
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  using comp-sampled-bound[OF support]
    comp-merkle-bound[OF support]
    comp-without-same-bound[OF support]
    comp-empty-bound[OF support]
    comp-gap-merkle-bound[OF support]
    comp-candidate-available-bound[OF support]
    total
  by (rule
    composition-fri-reachable-partial-candidate-reduction-from-split-without-same-bounds
    [where R = Rc data attacker-state
      and P = Pc data attacker-state
      and C = Cc data attacker-state
      and E = Ec data attacker-state
      and Q = Qc data attacker-state
      and T = Tc data attacker-state])
qed

```

lemma *active-fri-reductions-from-without-same-hash-root-query-bounds:*

fixes

Rt Zt Pt Ht Ct Qt Kt Ut Vt Rc Pc Cc Ec Qc Tc ::
('f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ prob)

assumes *trace-sampled-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-bad-with-header-tied-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$

Rt data attacker-state

and *trace-zero-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-header-tied-zero-round-final-obstruction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$

Zt data attacker-state

and *trace-header-merkle-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$

Pt data attacker-state

and *trace-structural-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-header-tied-sibling-mismatch-structural-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ht \text{ data attacker-state}$
and $\text{trace-without-same-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Ct \text{ data attacker-state}$
and $\text{trace-gap-merkle-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Qt \text{ data attacker-state}$
and trace-hash-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{hash-map-output-collision-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Kt \text{ data attacker-state}$
and trace-root-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-low-candidate-untied-root-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data}) \leq$
Ut data attacker-state
and *trace-query-bound:*
 $\wedge \text{data attacker-state.}$
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(trace-fri-header-tied-low-candidate-untied-query-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Vt data attacker-state
and *trace-total-bound:*
 $\wedge \text{data attacker-state.}$
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
Rt data attacker-state +
(((((Pt data attacker-state + Ht data attacker-state) +
Pt data attacker-state) +
Ct data attacker-state + Ct data attacker-state +
Ct data attacker-state) +
Pt data attacker-state) +
0 + Zt data attacker-state) +
(Qt data attacker-state +
(Kt data attacker-state +
(Ut data attacker-state + Vt data attacker-state))))
 $\leq \text{trace-fri-error}$
and *comp-sampled-bound:*
 $\wedge \text{data attacker-state.}$
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
Rc data attacker-state
and *comp-merkle-bound:*
 $\wedge \text{data attacker-state.}$
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(partial-merkle-inconsistency-bad

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $P_c \text{ data attacker-state}$
and *comp-without-same-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $C_c \text{ data attacker-state}$
and *comp-empty-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-reachable-verifier-tie-empty-header-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $E_c \text{ data attacker-state}$
and *comp-gap-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $Q_c \text{ data attacker-state}$
and *comp-candidate-available-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-candidate-available-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$

```

    (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  Tc data attacker-state
and comp-total-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
  Rc data attacker-state +
    (((Pc data attacker-state + 0) +
      Cc data attacker-state + Cc data attacker-state +
      Cc data attacker-state + Pc data attacker-state) +
      0) +
    (Ec data attacker-state +
      (Qc data attacker-state + Tc data attacker-state))
  ≤ composition-fri-error
shows active-fri-reductions-for A
apply (rule active-fri-reductions-from-without-same-and-untied-bounds
  [where Rt = Rt and Zt = Zt and Pt = Pt and Ht = Ht and Ct = Ct
    and Qt = Qt
    and Ut =
       $\lambda$ data attacker-state.
        Kt data attacker-state +
          (Ut data attacker-state + Vt data attacker-state)
    and Rc = Rc and Pc = Pc and Cc = Cc and Ec = Ec and Qc = Qc
    and Tc = Tc])
    apply (rule trace-sampled-bound, assumption)
    apply (rule trace-zero-bound, assumption)
    apply (rule trace-header-merkle-bound, assumption)
    apply (rule trace-structural-bound, assumption)
    apply (rule trace-without-same-bound, assumption)
    apply (rule trace-gap-merkle-bound, assumption)
    apply (rule
      wp-trace-fri-header-tied-low-candidate-untied-gap-bound-from-hash-root-query)
    apply (rule trace-hash-bound, assumption)
    apply (rule trace-root-bound, assumption)
    apply (rule trace-query-bound, assumption)
    apply (rule trace-total-bound, assumption)
    apply (rule comp-sampled-bound, assumption)
    apply (rule comp-merkle-bound, assumption)
    apply (rule comp-without-same-bound, assumption)
    apply (rule comp-empty-bound, assumption)
    apply (rule comp-gap-merkle-bound, assumption)
    apply (rule comp-candidate-available-bound, assumption)
    apply (rule comp-total-bound, assumption)
  done

end

```

end

theory *Soundness-FRI-Composition-Replay-Gaps*

imports

Soundness-FRI-Composition-Support

Soundness-FRI-Generic-Low-Degree

begin

Narrow downstream layer for the remaining composition verifier-tied replay gaps. This avoids growing the composition residuals theory, which is already above the advisory size threshold.

context *soundness*

begin

lemma *fri-same-mod-half-index-cases*:

fixes *len i j :: nat*

assumes *even-len: 2 dvd len*

and *i-bound: i < len*

and *j-bound: j < len*

and *same-mod: i mod (len div 2) = j mod (len div 2)*

shows

i = j $\vee$

i = fri-sibling-index len j $\vee$

fri-sibling-index len i = j $\vee$

fri-sibling-index len i = fri-sibling-index len j

proof –

let *?half = len div 2*

have *len-eq: len = 2 * ?half*

using *even-len* **by** *simp*

have *half-pos: 0 < ?half*

using *i-bound len-eq* **by** *linarith*

consider

(ff) i < ?half j < ?half

| *(fs) i < ?half $\neg$ j < ?half*

| *(sf) $\neg$ i < ?half j < ?half*

| *(ss) $\neg$ i < ?half $\neg$ j < ?half*

by *blast*

then show *?thesis*

proof *cases*

case *ff*

then have *i = j*

using *same-mod half-pos* **by** *simp*

then show *?thesis* **by** *simp*

next

case *fs*

have *j-eq: j = fri-sibling-index len (j mod ?half)*

by (*rule fri-sibling-index-mod-half-second-half*)

```

    [OF even-len j-bound fs(2)])
  have i-eq: j mod ?half = i
    using same-mod fs(1) by simp
  have fri-sibling-index len i = j
    using j-eq i-eq by simp
  then show ?thesis by simp
next
case sf
have i-eq: i = fri-sibling-index len (i mod ?half)
  by (rule fri-sibling-index-mod-half-second-half
    [OF even-len i-bound sf(1)])
have j-eq: i mod ?half = j
  using same-mod sf(2) by simp
have i = fri-sibling-index len j
  using i-eq j-eq by simp
then show ?thesis by simp
next
case ss
have i-eq: i = fri-sibling-index len (i mod ?half)
  by (rule fri-sibling-index-mod-half-second-half
    [OF even-len i-bound ss(1)])
have j-eq: j = fri-sibling-index len (j mod ?half)
  by (rule fri-sibling-index-mod-half-second-half
    [OF even-len j-bound ss(2)])
have i = j
  using i-eq j-eq same-mod by simp
then show ?thesis by simp
qed
qed

lemma fri-same-mod-half-index-cases3:
  fixes len i j :: nat
  assumes even-len: 2 dvd len
    and i-bound: i < len
    and j-bound: j < len
    and same-mod: i mod (len div 2) = j mod (len div 2)
  shows
    i = j ∨
    i = fri-sibling-index len j ∨
    fri-sibling-index len i = j
proof -
  let ?half = len div 2
  have len-eq: len = 2 * ?half
    using even-len by simp
  have half-pos: 0 < ?half
    using i-bound len-eq by linarith
  consider
    (ff) i < ?half j < ?half
  | (fs) i < ?half ¬ j < ?half

```

```

| (sf)  $\neg i < ?half\ j < ?half$ 
| (ss)  $\neg i < ?half \neg j < ?half$ 
  by blast
then show ?thesis
proof cases
  case ff
  then have  $i = j$ 
    using same-mod half-pos by simp
  then show ?thesis by simp
next
  case fs
  have j-eq:  $j = \text{fri-sibling-index len } (j \bmod ?half)$ 
    by (rule fri-sibling-index-mod-half-second-half
        [OF even-len j-bound fs(2)])
  have i-eq:  $j \bmod ?half = i$ 
    using same-mod fs(1) by simp
  have fri-sibling-index len  $i = j$ 
    using j-eq i-eq by simp
  then show ?thesis by simp
next
  case sf
  have i-eq:  $i = \text{fri-sibling-index len } (i \bmod ?half)$ 
    by (rule fri-sibling-index-mod-half-second-half
        [OF even-len i-bound sf(1)])
  have j-eq:  $i \bmod ?half = j$ 
    using same-mod sf(2) by simp
  have  $i = \text{fri-sibling-index len } j$ 
    using i-eq j-eq by simp
  then show ?thesis by simp
next
  case ss
  have i-eq:  $i = \text{fri-sibling-index len } (i \bmod ?half)$ 
    by (rule fri-sibling-index-mod-half-second-half
        [OF even-len i-bound ss(1)])
  have j-eq:  $j = \text{fri-sibling-index len } (j \bmod ?half)$ 
    by (rule fri-sibling-index-mod-half-second-half
        [OF even-len j-bound ss(2)])
  have  $i = j$ 
    using i-eq j-eq same-mod by simp
  then show ?thesis by simp
qed
qed

lemma fri-sibling-index-involution:
  fixes len idx :: nat
  assumes even-len: 2 dvd len
  and idx-bound: idx < len
  shows fri-sibling-index len (fri-sibling-index len idx) = idx
proof -

```

```

let ?half = len div 2
have len-eq: len = 2 * ?half
  using even-len by simp
have half-pos: 0 < ?half
  using idx-bound len-eq by linarith
show ?thesis
proof (cases idx < ?half)
  case True
  then show ?thesis
    by (rule fri-sibling-index-involution-first-half[OF even-len])
next
  case False
  define k where k = idx - ?half
  have idx-eq: idx = ?half + k
    using False unfolding k-def by simp
  have k-bound: k < ?half
    using idx-bound len-eq idx-eq by linarith
  have sib-eq: fri-sibling-index len idx = k
  proof -
    have sum-eq: (?half + k + ?half) = k + len
      using len-eq by linarith
    have k-len: k < len
      using k-bound len-eq by linarith
    have (?half + k + ?half) mod len = (k + len) mod len
      by (simp only: sum-eq)
    also have ... = k
      using k-len by simp
    finally have mod-eq: (?half + k + ?half) mod len = k .
    then show ?thesis
      unfolding fri-sibling-index-def idx-eq
      using mod-eq by simp
  qed
  have fri-sibling-index len (fri-sibling-index len idx) =
    fri-sibling-index len k
    using sib-eq by simp
  also have ... = idx
    using fri-sibling-index-first-half[OF even-len k-bound]
    unfolding idx-eq by simp
  finally show ?thesis .
qed
qed

```

```

lemma fri-sibling-fold-denominator-verifier:
  assumes len-pos: 0 < len
  and even-len: 2 dvd len
  and round: len * pw = clength * scale
  shows
    fri-fold-denominator ((h ^ fri-sibling-index len i) * shift) pw =
      - fri-fold-denominator ((h ^ i) * shift) pw

```

using *fri-sibling-domain-round*[*OF len-pos even-len round, of i*]
unfolding *fri-sibling-index-def fri-fold-denominator-def*
by *simp*

lemma *fri-fold-value-sibling-verifier*:

assumes *len-pos*: $0 < \text{len}$
and *even-len*: $2 \text{ dvd } \text{len}$
and *round*: $\text{len} * \text{pw} = \text{clength} * \text{scale}$
shows
fri-fold-value *b xn xp*
*(fri-fold-denominator ((h ^ fri-sibling-index len i) * shift) pw) =*
fri-fold-value b xp xn
*(fri-fold-denominator ((h ^ i) * shift) pw)*

proof –

have *denom*:
*fri-fold-denominator ((h ^ fri-sibling-index len i) * shift) pw =*
– *fri-fold-denominator ((h ^ i) * shift) pw*
by (*rule fri-sibling-fold-denominator-verifier*
[*OF len-pos even-len round*])
show *?thesis*
unfolding *denom*
by (*rule fri-fold-value-swap-neg-denominator*)

qed

lemma *generic-fri-forced-next-value-eq-if-no-same-layer-conflict*:

assumes *no-same*:
 $\neg$ *generic-fri-sampled-same-layer-opening-conflict roots challenges*
query-idxs round-layers
and *round-bound*: $\text{round-idx} < \text{length } \text{query-idxs}$
and *round-bound'*: $\text{round-idx}' < \text{length } \text{query-idxs}$
and *layer-bound*: $\text{layer-idx} < \text{length } \text{challenges}$
and *even-len*: $2 \text{ dvd } \text{fri-evidence-layer-len roots layer-idx}$
and *raw-bound*:
fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
fri-evidence-layer-len roots layer-idx
and *raw-bound'*:
fri-evidence-layer-idx roots query-idxs round-idx' layer-idx <
fri-evidence-layer-len roots layer-idx
and *round-product*:
*fri-evidence-layer-len roots layer-idx * 2 ^ layer-idx =*
*clength * scale*
and *same-next*:
fri-evidence-next-idx roots query-idxs round-idx layer-idx =
fri-evidence-next-idx roots query-idxs round-idx' layer-idx
and *forced*:
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers round-idx layer-idx v
and *forced'*:
generic-fri-round-forced-next-value roots challenges query-idxs

```

      round-layers round-idx' layer-idx v'
shows v = v'
proof -
  let ?len = fri-evidence-layer-len roots layer-idx
  let ?raw = fri-evidence-layer-idx roots query-idxs round-idx layer-idx
  let ?raw' =
    fri-evidence-layer-idx roots query-idxs round-idx' layer-idx
  let ?ch = challenges ! layer-idx
  let ?pw = 2 ^ layer-idx
  have len-pos: 0 < ?len
  using raw-bound by simp
  from forced obtain xp xp-path xn xn-path where step:
    fri-layer-step-evidence
      (roots ! layer-idx)
      ?ch
      ?len
      ?raw
      ?pw
      (fri-sibling-index ?len ?raw)
      xp xp-path xn xn-path
      (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
      v
      (round-layers ! round-idx ! layer-idx)
  unfolding generic-fri-round-forced-next-value-def by blast
  have v-eq:
    v = fri-evidence-next-value roots challenges query-idxs round-idx
      layer-idx xp xn
  using fri-layer-step-evidenceD(3)[OF step]
  unfolding fri-evidence-next-value-def by simp
  have step-norm:
    fri-layer-step-evidence
      (roots ! layer-idx)
      ?ch
      ?len
      ?raw
      ?pw
      (fri-sibling-index ?len ?raw)
      xp xp-path xn xn-path
      (fri-evidence-next-idx roots query-idxs round-idx layer-idx)
      (fri-evidence-next-value roots challenges query-idxs round-idx
        layer-idx xp xn)
      (round-layers ! round-idx ! layer-idx)
  using step v-eq by simp
  from forced' obtain yp yp-path yn yn-path where step':
    fri-layer-step-evidence
      (roots ! layer-idx)
      ?ch
      ?len
      ?raw'

```



```

    ?pw
    (fri-sibling-index ?len ?raw')
    yp yp-path yn yn-path
    (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
    v'
    (round-layers ! round-idx' ! layer-idx)
  unfolding generic-fri-round-forced-next-value-def by blast
have v'-eq:
  v' = fri-evidence-next-value roots challenges query-idxs round-idx'
    layer-idx yp yn
  using fri-layer-step-evidenceD(3)[OF step']
  unfolding fri-evidence-next-value-def by simp
have step'-norm:
  fri-layer-step-evidence
    (roots ! layer-idx)
    ?ch
    ?len
    ?raw'
    ?pw
    (fri-sibling-index ?len ?raw')
    yp yp-path yn yn-path
    (fri-evidence-next-idx roots query-idxs round-idx' layer-idx)
    (fri-evidence-next-value roots challenges query-idxs round-idx'
      layer-idx yp yn)
    (round-layers ! round-idx' ! layer-idx)
  using step' v'-eq by simp
have same-mod: ?raw mod (?len div 2) = ?raw' mod (?len div 2)
  using same-next unfolding fri-evidence-next-idx-def by simp
have idx-cases:
  ?raw = ?raw' ∨
  ?raw = fri-sibling-index ?len ?raw' ∨
  fri-sibling-index ?len ?raw = ?raw'
  by (rule fri-same-mod-half-index-cases3
    [OF even-len raw-bound raw-bound' same-mod])
from idx-cases show ?thesis
proof
  assume raw-eq: ?raw = ?raw'
  have xp-eq: xp = yp
    by (rule generic-fri-no-same-layer-opening-conflict-left-left
      [OF no-same round-bound round-bound' layer-bound step-norm
        step'-norm])
    (rule raw-eq)
  have xn-eq: xn = yn
    by (rule generic-fri-no-same-layer-opening-conflict-right-right
      [OF no-same round-bound round-bound' layer-bound step-norm
        step'-norm])
    (use raw-eq in simp)
  show ?thesis
    using v-eq v'-eq raw-eq xp-eq xn-eq

```

```

    unfolding fri-evidence-next-value-def
  by simp
next
assume raw-or-sib:
  ?raw = fri-sibling-index ?len ?raw'  $\vee$ 
  fri-sibling-index ?len ?raw = ?raw'
then show ?thesis
proof
  assume raw-sib: ?raw = fri-sibling-index ?len ?raw'
  have sib-row: fri-sibling-index ?len ?raw = ?raw'
    using raw-sib
    fri-sibling-index-involution[OF even-len raw-bound]
  by simp
  have xp-eq: xp = yn
    by (rule generic-fri-no-same-layer-opening-conflict-left-right
      [OF no-same round-bound round-bound' layer-bound step-norm
        step'-norm])
    (rule raw-sib)
  have xn-eq: xn = yp
    by (rule generic-fri-no-same-layer-opening-conflict-right-left
      [OF no-same round-bound round-bound' layer-bound step-norm
        step'-norm])
    (rule sib-row)
  have fold-eq:
    fri-fold-value ?ch xp xn
    (fri-fold-denominator ((h  $\wedge$  ?raw) * shift) ?pw) =
    fri-fold-value ?ch yp yn
    (fri-fold-denominator ((h  $\wedge$  ?raw') * shift) ?pw)
  using fri-fold-value-sibling-verifier
  [OF len-pos even-len round-product,
    where i=?raw' and b=?ch and xn=yn and xp=yp]
    raw-sib xp-eq xn-eq
  by simp
show ?thesis
  using v-eq v'-eq fold-eq
  unfolding fri-evidence-next-value-def
  by simp
next
assume sib-row: fri-sibling-index ?len ?raw = ?raw'
have raw-sib: ?raw = fri-sibling-index ?len ?raw'
  using sib-row fri-sibling-index-involution[OF even-len raw-bound]
  by simp
have xn-eq: xn = yp
  by (rule generic-fri-no-same-layer-opening-conflict-right-left
    [OF no-same round-bound round-bound' layer-bound step-norm
      step'-norm])
    (rule sib-row)
have xp-eq: xp = yn
  by (rule generic-fri-no-same-layer-opening-conflict-left-right

```

```

    [OF no-same round-bound round-bound' layer-bound step-norm
      step'-norm])
  (rule raw-sib)
have fold-eq:
  fri-fold-value ?ch yp yn
    (fri-fold-denominator ((h ^ ?raw') * shift) ?pw) =
  fri-fold-value ?ch xp xn
    (fri-fold-denominator ((h ^ ?raw) * shift) ?pw)
using fri-fold-value-sibling-verifier
  [OF len-pos even-len round-product,
    where i=?raw and b=?ch and xn=xn and xp=xp]
  sib-raw xp-eq xn-eq
by simp
show ?thesis
using v-eq v'-eq fold-eq
unfolding fri-evidence-next-value-def
by simp
qed
qed
qed

lemma accepted-fri-opening-transcript-composition-layer-arithmetic:
assumes fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
and degree-bound: to-nat dg ≤ maxDegree
and layer-bound: layer-idx < length composition-bs
shows 2 dvd fri-evidence-layer-len composition-roots layer-idx
and fri-evidence-layer-len composition-roots layer-idx * 2 ^ layer-idx =
  clength * scale
proof -
obtain N where eval-power: clength * scale = 2 ^ N
using eval-domain-length-power by blast
have bs-len: length composition-bs = length composition-roots
using accepted-fri-opening-transcript-shapes(4)[OF fri-openings] .
have roots-bound: layer-idx < length composition-roots
using layer-bound bs-len by simp
have roots-le: length composition-roots ≤ N
proof -
have len: length composition-roots = ceil-log (to-nat dg + 1)
using accepted-fri-opening-transcript-shapes(3)[OF fri-openings] .
have to-nat dg + 1 ≤ clength * scale
using degree-bound maxDegree-less-eval-domain by linarith
also have ... = 2 ^ N
by (rule eval-power)
finally have ceil-log (to-nat dg + 1) ≤ N
by (rule ceil-log-le-power)
then show ?thesis

```

```

    using len by simp
qed
have suc-le: Suc layer-idx ≤ N
  using roots-bound roots-le by simp
have dvd-eval-suc: (2::nat) ^ Suc layer-idx dvd clength * scale
  using le-imp-power-dvd[OF suc-le] eval-power by simp
show 2 dvd fri-evidence-layer-len composition-roots layer-idx
  unfolding fri-evidence-layer-len-def
  by (rule fri-layer-lengths-even-if-dvd[OF roots-bound dvd-eval-suc])
have le-N: layer-idx ≤ N
  using roots-bound roots-le by simp
have dvd-eval: (2::nat) ^ layer-idx dvd clength * scale
  using le-imp-power-dvd[OF le-N] eval-power by simp
show fri-evidence-layer-len composition-roots layer-idx *
  2 ^ layer-idx = clength * scale
  unfolding fri-evidence-layer-len-def
  by (rule fri-layer-lengths-round-product-if-dvd
    [OF roots-bound dvd-eval])
qed

lemma accepted-fri-opening-transcript-composition-evidence-layer-idx-bound:
  assumes fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers
  and degree-bound: to-nat dg ≤ maxDegree
  and round-bound: round-idx < length query-idxs
  and layer-bound: layer-idx < length composition-roots
  shows
    fri-evidence-layer-idx composition-roots query-idxs round-idx layer-idx <
    fri-evidence-layer-len composition-roots layer-idx
proof -
  obtain raw-idxs query-chunks query-state where
    len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  using fri-openings
  by (auto simp: accepted-fri-opening-transcript-def)
  have len-query: length query-idxs = rounds
    using accepted-fri-opening-transcript-shapes(7)[OF fri-openings] .
  have round-raw-bound: round-idx < length raw-idxs
    using round-bound len-query len-raw by simp
  have query-idx-eq:
    query-idxs ! round-idx = index (to-nat (raw-idxs ! round-idx))
    using query-idxs-eq round-raw-bound by simp
  have raw-layer:
    0 < fri-layer-lengths (length composition-roots)
    (clength * scale) ! layer-idx ∧
    fri-layer-indices (length composition-roots)

```

```

      (index (to-nat (raw-idxs ! round-idx))) (clength * scale) !
      layer-idx <
    fri-layer-lengths (length composition-roots)
      (clength * scale) ! layer-idx
  by (rule accepted-fri-opening-transcript-composition-raw-layer-bound
      [OF fri-openings degree-bound layer-bound])
show ?thesis
  using raw-layer query-idx-eq
  unfolding fri-evidence-layer-idx-def fri-evidence-layer-len-def
  by simp
qed

```

lemma *generic-fri-forced-next-value-eq-from-recorded-step*:

assumes *forced*:

generic-fri-round-forced-next-value roots challenges query-idxs
round-layers round-idx layer-idx v

and *step*:

fri-layer-step-evidence
 (*roots ! layer-idx*)
 (*challenges ! layer-idx*)
 (*fri-evidence-layer-len roots layer-idx*)
 (*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*)
 ($2 \wedge \text{layer-idx}$)
 (*fri-sibling-index (fri-evidence-layer-len roots layer-idx)*
 (*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*))
xp xp-path xn xn-path
 (*fri-evidence-next-idx roots query-idxs round-idx layer-idx*)
v'
 (*round-layers ! round-idx ! layer-idx*)

shows $v = v'$

proof —

from *forced* **obtain** *yp yp-path yn yn-path* **where** *forced-step*:

fri-layer-step-evidence
 (*roots ! layer-idx*)
 (*challenges ! layer-idx*)
 (*fri-evidence-layer-len roots layer-idx*)
 (*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*)
 ($2 \wedge \text{layer-idx}$)
 (*fri-sibling-index (fri-evidence-layer-len roots layer-idx)*
 (*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*))
yp yp-path yn yn-path
 (*fri-evidence-next-idx roots query-idxs round-idx layer-idx*)
v
 (*round-layers ! round-idx ! layer-idx*)

unfolding *generic-fri-round-forced-next-value-def* **by** *blast*

have *yp-eq*: $yp = xp$

by (rule *fri-layer-opening-chunk-values-unique*(1))
 (rule *fri-layer-step-evidenceD*(4)[OF *forced-step*],
 rule *fri-layer-step-evidenceD*(4)[OF *step*])

```

have yn-eq: yn = xn
  by (rule fri-layer-opening-chunk-values-unique(2))
    (rule fri-layer-step-evidenceD(4)[OF forced-step],
      rule fri-layer-step-evidenceD(4)[OF step])
show ?thesis
  using fri-layer-step-evidenceD(3)[OF forced-step]
    fri-layer-step-evidenceD(3)[OF step]
    yp-eq yn-eq
  by simp
qed

lemma fri-layer-indices-tail-nth:
  fri-layer-indices (Suc j) (q mod (len div 2)) (len div 2) ! j =
  fri-layer-indices (Suc j) q len ! j mod
    (fri-layer-lengths (Suc j) len ! j div 2)
  by (induction j arbitrary: q len) simp-all

lemma fri-layer-indices-Suc-nth:
  assumes layer-bound: Suc j < n
  shows
    fri-layer-indices n q len ! Suc j =
    fri-layer-indices n q len ! j mod
      (fri-layer-lengths n len ! j div 2)
proof -
  have idx-Suc:
    fri-layer-indices n q len ! Suc j =
    fri-layer-indices (Suc (Suc j)) q len ! Suc j
  using fri-layer-indices-prefix-nth[OF layer-bound, of q len] by simp
  have idx-j:
    fri-layer-indices n q len ! j =
    fri-layer-indices (Suc j) q len ! j
  using fri-layer-indices-prefix-nth[of j n q len] layer-bound by simp
  have len-j:
    fri-layer-lengths n len ! j =
    fri-layer-lengths (Suc j) len ! j
  using fri-layer-lengths-prefix-nth[of j n len] layer-bound by simp
  show ?thesis
    unfolding idx-Suc idx-j len-j
    using fri-layer-indices-tail-nth[of j q len]
    by simp
qed

lemma fri-evidence-layer-idx-Suc:
  assumes layer-bound: Suc layer-idx < length roots
  shows
    fri-evidence-layer-idx roots query-idxs round-idx (Suc layer-idx) =
    fri-evidence-next-idx roots query-idxs round-idx layer-idx
  using fri-layer-indices-Suc-nth[OF layer-bound,
    of query-idxs ! round-idx clength * scale]

```

unfolding *fri-evidence-layer-idx-def* *fri-evidence-next-idx-def*
fri-evidence-layer-len-def
by *simp*

lemma *mfold-fri-layer-openings-successor-prefix-step*:

fixes *s s-suc* :: (*'f*, *'a*) *protocol-channel-scheme*

assumes *prefix-suc*:

Some ((idx-suc, x-suc, len-suc, pw-suc), s-suc) ∈
set-dist

(execute
(mfold (idx, x, len, pw)
(take (Suc j) (map fri-layer-opening-step bfs))) s)

and *j-bound*: *j < length bfs*

obtains *idx-j x-j len-j pw-j b rt xp xp-path xn xn-path s-j* **where**
bfs ! j = (b, rt)

Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
set-dist

(execute
(mfold (idx, x, len, pw)
(take j (map fri-layer-opening-step bfs))) s)

Some ((idx-suc, x-suc, len-suc, pw-suc), s-suc) ∈
set-dist

(execute
(fri-layer-opening-step (b, rt) (idx-j, x-j, len-j, pw-j))
s-j)

fri-layer-step-evidence rt b len-j idx-j pw-j
(fri-sibling-index len-j idx-j) xp xp-path xn xn-path
idx-suc x-suc ([xp] @ xp-path @ [xn] @ xn-path)

proof –

let *?bfs = take (Suc j) bfs*

have *j-bound-take*: *j < length ?bfs*

using *j-bound* **by** *simp*

have *take-map*:

take (Suc j) (map fri-layer-opening-step bfs) =
map fri-layer-opening-step ?bfs

by (*simp add: take-map*)

show *?thesis*

proof (*rule mfold-fri-layer-openings-decomp-at*
 $[OF\ prefix-suc[unfolded\ take-map]\ j-bound-take]$)

fix *idx-j x-j len-j pw-j out-j s-j s-j-suc*

assume *prefix*:

Some ((idx-j, x-j, len-j, pw-j), s-j) ∈
set-dist

(execute
(mfold (idx, x, len, pw)
(take j (map fri-layer-opening-step ?bfs))) s)

and *selected*:

Some (out-j, s-j-suc) ∈
set-dist

```

      (execute
        (fri-layer-opening-step (?bfs ! j)
          (idx-j, x-j, len-j, pw-j)) s-j)
    and suffix:
    Some ((idx-suc, x-suc, len-suc, pw-suc), s-suc) ∈
      set-dist
      (execute
        (mfold out-j
          (drop (Suc j) (map fri-layer-opening-step ?bfs))) s-j-suc)
  have drop-empty:
    drop (Suc j) (map fri-layer-opening-step ?bfs) = []
  by simp
  have out-j-eq: out-j = (idx-suc, x-suc, len-suc, pw-suc)
  and s-suc-eq: s-suc = s-j-suc
  using suffix unfolding drop-empty by simp-all
  have take-prefix-list: take j ?bfs = take j bfs
  by (simp add: take-take)
  have take-prefix:
    take j (map fri-layer-opening-step ?bfs) =
      take j (map fri-layer-opening-step bfs)
  proof (rule nth-equalityI)
    show length (take j (map fri-layer-opening-step ?bfs)) =
      length (take j (map fri-layer-opening-step bfs))
    by simp
  next
  fix i
  assume i-bound: i < length (take j (map fri-layer-opening-step ?bfs))
  then have i < j
  by simp
  then show take j (map fri-layer-opening-step ?bfs) ! i =
    take j (map fri-layer-opening-step bfs) ! i
  using j-bound by simp
qed
  have bfs-j: ?bfs ! j = bfs ! j
  using j-bound by simp
  obtain b rt where bf-eq: bfs ! j = (b, rt)
  by (cases bfs ! j)
  show ?thesis
  proof (rule fri-layer-opening-step-outcome-evidence
    [OF selected[unfolded bfs-j bf-eq out-j-eq]])
    fix xp xp-path xn xn-path
    assume xp-eq: xp = x-j
    and next-idx-eq: idx-suc = idx-j mod (len-j div 2)
    and next-len-eq: len-suc = len-j div 2
    and next-pow-eq: pw-suc = pw-j + pw-j
    and step:
      fri-layer-step-evidence rt b len-j idx-j pw-j
      (fri-sibling-index len-j idx-j) xp xp-path xn xn-path
      idx-suc x-suc ([xp] @ xp-path @ [xn] @ xn-path)
  end

```



```

have selected-suc:
  Some ((idx-suc, x-suc, len-suc, pw-suc), s-suc) ∈
    set-dist
      (execute
        (fri-layer-opening-step (b, rt)
          (idx-j, x-j, len-j, pw-j)) s-j)
    using selected[unfolded bfs-j bf-eq out-j-eq] s-suc-eq by simp
show ?thesis
  by (rule that[OF bf-eq prefix[unfolded take-prefix]
    selected-suc step])
qed
qed
qed

lemma mfold-fri-layer-opening-selected-recorded-step-evidence-at:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (out, t) ∈
    set-dist
      (execute
        (mfold (idx, x, len, pw)
          (map fri-layer-opening-step bfs)) s)
and layers:
  fri-layers-transcript (length bfs) len layer-chunks total-chunk
and transcript:
  PTranscript s = total-chunk @ PTranscript t
and initial-idx-bound: idx < len
and all-layer-bounds:
   $\bigwedge k. k < \text{length } bfs \implies$ 
     $0 < \text{fri-layer-lengths } (\text{length } bfs) \text{ len } ! k \wedge$ 
     $\text{fri-layer-indices } (\text{length } bfs) \text{ idx len } ! k <$ 
     $\text{fri-layer-lengths } (\text{length } bfs) \text{ len } ! k$ 
and j-bound: j < length bfs
shows
   $\exists xp \text{ } xp\text{-path } xn \text{ } xn\text{-path } next\text{-value.}$ 
  fri-layer-step-evidence (snd (bfs ! j)) (fst (bfs ! j))
  (fri-layer-lengths (length bfs) len ! j)
  (fri-layer-indices (length bfs) idx len ! j)
  (pw * 2 ^ j)
  (fri-sibling-index (fri-layer-lengths (length bfs) len ! j)
    (fri-layer-indices (length bfs) idx len ! j))
  xp xp-path xn xn-path
  (fri-layer-indices (length bfs) idx len ! j mod
    (fri-layer-lengths (length bfs) len ! j div 2))
  next-value
  (layer-chunks ! j)
proof —
show ?thesis
using outcome layers transcript initial-idx-bound all-layer-bounds

```

```

  j-bound
proof (induction bfs arbitrary: idx x len pw s out t j layer-chunks total-chunk)
  case Nil
  then show ?case by simp
next
  case (Cons bf bfs)
  obtain b rt where bf-eq: bf = (b, rt)
  by (cases bf)
  from Cons.premis(1)[unfolded bf-eq]
  obtain out1 s1 where
    head:
      Some (out1, s1) ∈
        set-dist
          (execute (fri-layer-opening-step (b, rt)
            (idx, x, len, pw)) s)
    and tail:
      Some (out, t) ∈
        set-dist
          (execute (mfold out1 (map fri-layer-opening-step bfs)) s1)
  by (auto elim!: set-dist-bindE)
  obtain first rest where chunks-eq: layer-chunks = first # rest
  using Cons.premis(6) Cons.premis(2)
  unfolding fri-layers-transcript-def by (cases layer-chunks) auto
  have total-eq: total-chunk = first @ List.concat rest
  using Cons.premis(2) unfolding fri-layers-transcript-def chunks-eq
  by simp
  have first-open:
    ∃ yp yp-path yn yn-path.
      fri-layer-opening-chunk len yp yp-path yn yn-path first
  using Cons.premis(2) unfolding fri-layers-transcript-def chunks-eq
  by auto
  have rest-layers:
    fri-layers-transcript (length bfs) (len div 2) rest
    (List.concat rest)
  using Cons.premis(2) unfolding fri-layers-transcript-def chunks-eq
  by auto
  obtain x1 xp xp-path xn xn-path head-chunk where out1-eq:
    out1 = (idx mod (len div 2), x1, len div 2, pw + pw)
  and head-chunk-def: head-chunk = [xp] @ xp-path @ [xn] @ xn-path
  and head-open:
    fri-layer-opening-chunk len xp xp-path xn xn-path head-chunk
  and head-transcript:
    PTranscript s = head-chunk @ PTranscript s1
  and head-step:
    fri-layer-step-evidence rt b len idx pw
    (fri-sibling-index len idx) xp xp-path xn xn-path
    (idx mod (len div 2)) x1 head-chunk
proof –
  have len-pos: 0 < len

```

```

    using Cons.premis(5)[of 0] by simp
have idx-bound:  $idx < len$ 
  by (rule Cons.premis(4))
show ?thesis
proof (rule fri-layer-opening-step-outcome-authenticated
  [OF len-pos idx-bound head])
  fix  $xp'$   $xp\text{-}path'$   $xn'$   $xn\text{-}path'$   $x'$   $chunk$ 
  assume out-eq:
     $out1 = (idx \bmod (len \div 2), x', len \div 2, pw + pw)$ 
  and xp-eq:  $xp' = x$ 
  and x'-eq:
     $x' = \text{fri-fold-value } b \text{ } xp' \text{ } xn'$ 
     $(\text{fri-fold-denominator } ((h \wedge idx) * \text{shift}) \text{ } pw)$ 
  and chunk-eq:  $chunk = [xp'] @ xp\text{-}path' @ [xn'] @ xn\text{-}path'$ 
  and chunk-open:
     $\text{fri-layer-opening-chunk } len \text{ } xp' \text{ } xp\text{-}path' \text{ } xn' \text{ } xn\text{-}path'$ 
     $chunk$ 
  and tr:  $PTranscript \text{ } s = chunk @ PTranscript \text{ } s1$ 
  assume
     $s \leq s1$ 
     $PState \text{ } s1 = \text{foldl concat } (PState \text{ } s) \text{ } chunk$ 
     $PQueryCounter \text{ } s1 = PQueryCounter \text{ } s$ 
     $\text{authenticated-opening-in } s1$ 
     $(\text{opening-root} = rt, \text{opening-length} = len,$ 
     $\text{opening-index} = idx, \text{opening-value} = xp',$ 
     $\text{opening-path} = xp\text{-}path')$ 
     $\text{authenticated-opening-in } s1$ 
     $(\text{opening-root} = rt, \text{opening-length} = len,$ 
     $\text{opening-index} = \text{fri-sibling-index } len \text{ } idx,$ 
     $\text{opening-value} = xn', \text{opening-path} = xn\text{-}path')$ 
  have step:
     $\text{fri-layer-step-evidence } rt \text{ } b \text{ } len \text{ } idx \text{ } pw$ 
     $(\text{fri-sibling-index } len \text{ } idx) \text{ } xp' \text{ } xp\text{-}path' \text{ } xn' \text{ } xn\text{-}path'$ 
     $(idx \bmod (len \div 2)) \text{ } x' \text{ } chunk$ 
  by (rule fri-layer-step-evidenceI)
  (use x'-eq chunk-open in simp-all)
show ?thesis
  by (rule that[OF out-eq chunk-eq chunk-open tr step])
qed
qed
from mfold-fri-layer-openings-outcome[OF tail[unfolding out1-eq]]
obtain tail-layers tail-chunk where
  tail-transcript:
     $PTranscript \text{ } s1 = tail\text{-}chunk @ PTranscript \text{ } t$ 
  by blast
have head-len:
   $length \text{ } head\text{-}chunk = length \text{ } first$ 
proof -
  obtain  $yp$   $yp\text{-}path$   $yn$   $yn\text{-}path$  where first-open':

```

```

    fri-layer-opening-chunk len yp yp-path yn yn-path first
    using first-open by blast
show ?thesis
    using fri-layer-opening-chunk-length[OF head-open]
    fri-layer-opening-chunk-length[OF first-open]
    by simp
qed
have chunks-total-eq:
    head-chunk @ tail-chunk = first @ List.concat rest
    using head-transcript tail-transcript Cons.premis(3) total-eq by simp
have head-eq: head-chunk = first
    using arg-cong[OF chunks-total-eq,
    of take (length head-chunk)]
    head-len
    by simp
have tail-chunk-eq: tail-chunk = List.concat rest
    using chunks-total-eq head-eq by simp
have tail-transcript-rest:
    PTranscript s1 = List.concat rest @ PTranscript t
    using tail-transcript tail-chunk-eq by simp
show ?case
proof (cases j)
case 0
have step:
    fri-layer-step-evidence (snd ((bf # bfs) ! j))
    (fst ((bf # bfs) ! j))
    (fri-layer-lengths (length (bf # bfs)) len ! j)
    (fri-layer-indices (length (bf # bfs)) idx len ! j)
    (pw * 2 ^ j)
    (fri-sibling-index
    (fri-layer-lengths (length (bf # bfs)) len ! j)
    (fri-layer-indices (length (bf # bfs)) idx len ! j))
    xp xp-path xn xn-path
    (fri-layer-indices (length (bf # bfs)) idx len ! j mod
    (fri-layer-lengths (length (bf # bfs)) len ! j div 2))
    x1 (layer-chunks ! j)
    using head-step 0 bf-eq head-eq chunks-eq by simp
show ?thesis
    using step by blast
next
case (Suc k)
have k-bound: k < length bfs
    using Cons.premis(6) Suc by simp
have tail-all-layer-bounds:
     $\bigwedge m. m < \text{length } \text{bfs} \implies$ 
     $0 < \text{fri-layer-lengths } (\text{length } \text{bfs}) \ (\text{len div } 2) \ ! \ m \wedge$ 
     $\text{fri-layer-indices } (\text{length } \text{bfs}) \ (\text{idx mod } (\text{len div } 2))$ 
     $(\text{len div } 2) \ ! \ m <$ 
     $\text{fri-layer-lengths } (\text{length } \text{bfs}) \ (\text{len div } 2) \ ! \ m$ 

```

```

    using Cons.premis(5)[of Suc m for m] by simp
  have tail-initial-idx-bound:
    idx mod (len div 2) < len div 2
    using tail-all-layer-bounds[of 0] k-bound by (cases bfs) simp-all
  from Cons.IH[OF tail[unfolded out1-eq] rest-layers
    tail-transcript-rest tail-initial-idx-bound tail-all-layer-bounds
    k-bound]
  obtain txp txp-path txn txn-path tnext where tail-step:
    fri-layer-step-evidence (snd (bfs ! k)) (fst (bfs ! k))
    (fri-layer-lengths (length bfs) (len div 2) ! k)
    (fri-layer-indices (length bfs) (idx mod (len div 2))
      (len div 2) ! k)
    ((pw + pw) * 2 ^ k)
    (fri-sibling-index
      (fri-layer-lengths (length bfs) (len div 2) ! k)
      (fri-layer-indices (length bfs) (idx mod (len div 2))
        (len div 2) ! k))
    txp txp-path txn txn-path
    (fri-layer-indices (length bfs) (idx mod (len div 2))
      (len div 2) ! k mod
      (fri-layer-lengths (length bfs) (len div 2) ! k div 2))
    tnext (rest ! k)
  by blast
  have pow-eq: (pw + pw) * 2 ^ k = pw * 2 ^ Suc k
  by (simp add: algebra-simps)
  have step:
    fri-layer-step-evidence (snd ((bf # bfs) ! j))
    (fst ((bf # bfs) ! j))
    (fri-layer-lengths (length (bf # bfs)) len ! j)
    (fri-layer-indices (length (bf # bfs)) idx len ! j)
    (pw * 2 ^ j)
    (fri-sibling-index
      (fri-layer-lengths (length (bf # bfs)) len ! j)
      (fri-layer-indices (length (bf # bfs)) idx len ! j))
    txp txp-path txn txn-path
    (fri-layer-indices (length (bf # bfs)) idx len ! j mod
      (fri-layer-lengths (length (bf # bfs)) len ! j div 2))
    tnext (layer-chunks ! j)
  using tail-step Suc chunks-eq pow-eq by simp
  show ?thesis
  using step by blast
qed
qed
qed

```

```

lemma mfold-fri-layer-opening-first-recorded-step-evidence:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some (out, t) ∈

```

```

    set-dist
      (execute
        (mfold (idx, x, len, pw)
          (map fri-layer-opening-step bfs)) s)
  and layers:
    fri-layers-transcript (length bfs) len layer-chunks total-chunk
  and transcript:
    PTranscript s = total-chunk @ PTranscript t
  and nonempty: 0 < length bfs
shows
  ∃ xp-path xn xn-path next-value.
    fri-layer-step-evidence (snd (bfs ! 0)) (fst (bfs ! 0))
    (fri-layer-lengths (length bfs) len ! 0)
    (fri-layer-indices (length bfs) idx len ! 0)
    pw
    (fri-sibling-index (fri-layer-lengths (length bfs) len ! 0)
      (fri-layer-indices (length bfs) idx len ! 0))
    x xp-path xn xn-path
    (fri-layer-indices (length bfs) idx len ! 0 mod
      (fri-layer-lengths (length bfs) len ! 0 div 2))
    next-value
    (layer-chunks ! 0)
proof -
  obtain bf bfs' where bfs-eq: bfs = bf # bfs'
  using nonempty by (cases bfs) auto
  obtain b rt where bf-eq: bf = (b, rt)
  by (cases bf)
  from outcome[unfolded bfs-eq bf-eq]
  obtain out1 s1 where
    head:
      Some (out1, s1) ∈
        set-dist
          (execute (fri-layer-opening-step (b, rt)
            (idx, x, len, pw)) s)
  and tail:
    Some (out, t) ∈
      set-dist
        (execute (mfold out1 (map fri-layer-opening-step bfs')) s1)
  by (auto elim!: set-dist-bindE)
  obtain first rest where chunks-eq: layer-chunks = first # rest
  using layers nonempty unfolding bfs-eq fri-layers-transcript-def
  by (cases layer-chunks) auto
  have total-eq: total-chunk = first @ List.concat rest
  using layers unfolding bfs-eq fri-layers-transcript-def chunks-eq
  by simp
  have first-open:
    ∃ yp yp-path yn yn-path.
      fri-layer-opening-chunk len yp yp-path yn yn-path first
  using layers unfolding bfs-eq fri-layers-transcript-def chunks-eq

```

```

by auto
obtain  $x1\ xp\ xp\text{-}path\ xn\ xn\text{-}path\ head\text{-}chunk$  where  $out1\text{-}eq$ :
   $out1 = (idx\ mod\ (len\ div\ 2),\ x1,\ len\ div\ 2,\ pw + pw)$ 
and  $head\text{-}chunk\text{-}def$ :  $head\text{-}chunk = [xp] @ xp\text{-}path @ [xn] @ xn\text{-}path$ 
and  $xp\text{-}eq$ :  $xp = x$ 
and  $head\text{-}open$ :
   $fri\text{-}layer\text{-}opening\text{-}chunk\ len\ xp\ xp\text{-}path\ xn\ xn\text{-}path\ head\text{-}chunk$ 
and  $head\text{-}transcript$ :
   $PTranscript\ s = head\text{-}chunk @ PTranscript\ s1$ 
and  $head\text{-}step$ :
   $fri\text{-}layer\text{-}step\text{-}evidence\ rt\ b\ len\ idx\ pw$ 
   $(fri\text{-}sibling\text{-}index\ len\ idx)\ xp\ xp\text{-}path\ xn\ xn\text{-}path$ 
   $(idx\ mod\ (len\ div\ 2))\ x1\ head\text{-}chunk$ 
proof –
from  $fri\text{-}layer\text{-}opening\text{-}step\text{-}outcome[OF\ head]$ 
obtain  $xp'\ xp'\text{-}path'\ xn'\ xn'\text{-}path'\ x'$  where  $out1\text{-}eq'$ :
   $out1 = (idx\ mod\ (len\ div\ 2),\ x',\ len\ div\ 2,\ pw + pw)$ 
and  $xp'\text{-}eq$ :  $xp' = x$ 
and  $x'\text{-}eq$ :
   $x' = fri\text{-}fold\text{-}value\ b\ xp'\ xn'$ 
   $(fri\text{-}fold\text{-}denominator\ ((h \wedge idx) * shift)\ pw)$ 
and  $chunk\text{-}open$ :
   $fri\text{-}layer\text{-}opening\text{-}chunk\ len\ xp'\ xp'\text{-}path'\ xn'\ xn'\text{-}path'$ 
   $([xp'] @ xp'\text{-}path' @ [xn'] @ xn'\text{-}path')$ 
and  $tr$ :
   $PTranscript\ s =$ 
   $([xp'] @ xp'\text{-}path' @ [xn'] @ xn'\text{-}path') @ PTranscript\ s1$ 
by blast
have  $step$ :
   $fri\text{-}layer\text{-}step\text{-}evidence\ rt\ b\ len\ idx\ pw$ 
   $(fri\text{-}sibling\text{-}index\ len\ idx)\ xp'\ xp'\text{-}path'\ xn'\ xn'\text{-}path'$ 
   $(idx\ mod\ (len\ div\ 2))\ x'$ 
   $([xp'] @ xp'\text{-}path' @ [xn'] @ xn'\text{-}path')$ 
by (rule  $fri\text{-}layer\text{-}step\text{-}evidenceI$ )
  (use  $x'\text{-}eq\ chunk\text{-}open$  in  $simp\text{-}all$ )
show  $?thesis$ 
by (rule  $that[OF\ out1\text{-}eq'\ refl\ xp'\text{-}eq\ chunk\text{-}open\ tr\ step]$ )
qed
from  $mfold\text{-}fri\text{-}layer\text{-}openings\text{-}outcome[OF\ tail[unfolded\ out1\text{-}eq]]$ 
obtain  $tail\text{-}layers\ tail\text{-}chunk$  where
   $tail\text{-}transcript$ :
   $PTranscript\ s1 = tail\text{-}chunk @ PTranscript\ t$ 
by blast
have  $head\text{-}len$ :
   $length\ head\text{-}chunk = length\ first$ 
proof –
obtain  $yp\ yp\text{-}path\ yn\ yn\text{-}path$  where  $first\text{-}open'$ :
   $fri\text{-}layer\text{-}opening\text{-}chunk\ len\ yp\ yp\text{-}path\ yn\ yn\text{-}path\ first$ 
using  $first\text{-}open$  by blast

```

```

show ?thesis
  using fri-layer-opening-chunk-length[OF head-open]
    fri-layer-opening-chunk-length[OF first-open]
  by simp
qed
have chunks-total-eq:
  head-chunk @ tail-chunk = first @ List.concat rest
  using head-transcript tail-transcript transcript total-eq by simp
have head-eq: head-chunk = first
  using arg-cong[OF chunks-total-eq,
    of take (length head-chunk)]
    head-len
  by simp
have step:
  fri-layer-step-evidence (snd (bfs ! 0)) (fst (bfs ! 0))
    (fri-layer-lengths (length bfs) len ! 0)
    (fri-layer-indices (length bfs) idx len ! 0)
  pw
    (fri-sibling-index (fri-layer-lengths (length bfs) len ! 0)
      (fri-layer-indices (length bfs) idx len ! 0))
  x xp-path xn xn-path
    (fri-layer-indices (length bfs) idx len ! 0 mod
      (fri-layer-lengths (length bfs) len ! 0 div 2))
  x1
    (layer-chunks ! 0)
  using head-step bfs-eq bf-eq chunks-eq head-eq xp-eq by simp
show ?thesis
  using step by blast
qed

lemma mfold-fri-layer-opening-successor-recorded-step-from-previous:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (out, t) ∈
    set-dist
      (execute
        (mfold (idx, x, len, pw)
          (map fri-layer-opening-step bfs)) s)
and layers:
  fri-layers-transcript (length bfs) len layer-chunks total-chunk
and transcript:
  PTranscript s = total-chunk @ PTranscript t
and initial-idx-bound: idx < len
and all-layer-bounds:
   $\bigwedge k. k < \text{length } \text{bfs} \implies$ 
    0 < fri-layer-lengths (length bfs) len ! k  $\wedge$ 
    fri-layer-indices (length bfs) idx len ! k <
      fri-layer-lengths (length bfs) len ! k
and j-bound: Suc j < length bfs

```



```

and previous-step:
  fri-layer-step-evidence (snd (bfs ! j)) (fst (bfs ! j))
    (fri-layer-lengths (length bfs) len ! j)
    (fri-layer-indices (length bfs) idx len ! j)
    (pw * 2 ^ j)
    (fri-sibling-index
      (fri-layer-lengths (length bfs) len ! j)
      (fri-layer-indices (length bfs) idx len ! j))
    xp xp-path xn xn-path
    (fri-layer-indices (length bfs) idx len ! j mod
      (fri-layer-lengths (length bfs) len ! j div 2))
    v
    (layer-chunks ! j)
shows
  ∃ xp-path' xn' xn-path' next-value.
    fri-layer-step-evidence (snd (bfs ! Suc j)) (fst (bfs ! Suc j))
      (fri-layer-lengths (length bfs) len ! Suc j)
      (fri-layer-indices (length bfs) idx len ! Suc j)
      (pw * 2 ^ Suc j)
      (fri-sibling-index
        (fri-layer-lengths (length bfs) len ! Suc j)
        (fri-layer-indices (length bfs) idx len ! Suc j))
      v xp-path' xn' xn-path'
      (fri-layer-indices (length bfs) idx len ! Suc j mod
        (fri-layer-lengths (length bfs) len ! Suc j div 2))
      next-value
      (layer-chunks ! Suc j)
using outcome layers transcript initial-idx-bound all-layer-bounds j-bound
previous-step
proof (induction bfs arbitrary: idx x len pw s out t j layer-chunks total-chunk
  xp xp-path xn xn-path v)
case Nil
then show ?case by simp
next
case (Cons bf bfs)
obtain b rt where bf-eq: bf = (b, rt)
by (cases bf)
from Cons.prem1[unfolded bf-eq]
obtain out1 s1 where
  head:
    Some (out1, s1) ∈
      set-dist
        (execute (fri-layer-opening-step (b, rt)
          (idx, x, len, pw)) s)
and tail:
    Some (out, t) ∈
      set-dist
        (execute (mfold out1 (map fri-layer-opening-step bfs)) s1)
by (auto elim!: set-dist-bindE)

```

```

obtain first rest where chunks-eq: layer-chunks = first # rest
  using Cons.premis(2) Cons.premis(6)
  unfolding fri-layers-transcript-def by (cases layer-chunks) auto
have total-eq: total-chunk = first @ List.concat rest
  using Cons.premis(2) unfolding fri-layers-transcript-def chunks-eq
  by simp
have first-open:
   $\exists$  yp yp-path yn yn-path.
    fri-layer-opening-chunk len yp yp-path yn yn-path first
  using Cons.premis(2) unfolding fri-layers-transcript-def chunks-eq
  by auto
have rest-layers:
  fri-layers-transcript (length bfs) (len div 2) rest
  (List.concat rest)
  using Cons.premis(2) unfolding fri-layers-transcript-def chunks-eq
  by auto
obtain x1 hxp hxp-path hxn hxn-path head-chunk where out1-eq:
  out1 = (idx mod (len div 2), x1, len div 2, pw + pw)
and hxp-eq: hxp = x
and head-open:
  fri-layer-opening-chunk len hxp hxp-path hxn hxn-path head-chunk
and head-transcript:
  PTranscript s = head-chunk @ PTranscript s1
and head-step:
  fri-layer-step-evidence rt b len idx pw
  (fri-sibling-index len idx) hxp hxp-path hxn hxn-path
  (idx mod (len div 2)) x1 head-chunk
proof –
from fri-layer-opening-step-outcome[OF head]
obtain xp' xp-path' xn' xn-path' x' where out1-eq':
  out1 = (idx mod (len div 2), x', len div 2, pw + pw)
and xp'-eq: xp' = x
and x'-eq:
  x' = fri-fold-value b xp' xn'
  (fri-fold-denominator ((h ^ idx) * shift) pw)
and chunk-open:
  fri-layer-opening-chunk len xp' xp-path' xn' xn-path'
  ([xp'] @ xp-path' @ [xn'] @ xn-path')
and tr:
  PTranscript s =
  ([xp'] @ xp-path' @ [xn'] @ xn-path') @ PTranscript s1
by blast
have step:
  fri-layer-step-evidence rt b len idx pw
  (fri-sibling-index len idx) xp' xp-path' xn' xn-path'
  (idx mod (len div 2)) x'
  ([xp'] @ xp-path' @ [xn'] @ xn-path')
by (rule fri-layer-step-evidenceI)
  (use x'-eq chunk-open in simp-all)

```

```

show ?thesis
  by (rule that[OF out1-eq' xp'-eq chunk-open tr step])
qed
from mfold-fri-layer-openings-outcome[OF tail[unfolded out1-eq]]
obtain tail-layers tail-chunk where
  tail-transcript:
    PTranscript s1 = tail-chunk @ PTranscript t
  by blast
have head-len:
  length head-chunk = length first
proof –
  obtain yp yp-path yn yn-path where first-open':
    fri-layer-opening-chunk len yp yp-path yn yn-path first
  using first-open by blast
show ?thesis
  using fri-layer-opening-chunk-length[OF head-open]
    fri-layer-opening-chunk-length[OF first-open']
  by simp
qed
have chunks-total-eq:
  head-chunk @ tail-chunk = first @ List.concat rest
  using head-transcript tail-transcript Cons.prem3 total-eq by simp
have head-eq: head-chunk = first
  using arg-cong[OF chunks-total-eq,
    of take (length head-chunk)]
    head-len
  by simp
have tail-chunk-eq: tail-chunk = List.concat rest
  using chunks-total-eq head-eq by simp
have tail-transcript-rest:
  PTranscript s1 = List.concat rest @ PTranscript t
  using tail-transcript tail-chunk-eq by simp
show ?case
proof (cases j)
  case 0
  have prev-head:
    fri-layer-step-evidence rt b len idx pw
    (fri-sibling-index len idx) xp xp-path xn xn-path
    (idx mod (len div 2)) v first
  using Cons.prem7 0 bf-eq chunks-eq by simp
have v-eq-x1: v = x1
proof –
  have xp-eq: xp = hxp
  by (rule fri-layer-opening-chunk-values-unique(1))
    (rule fri-layer-step-evidenceD(4)[OF prev-head],
    rule fri-layer-step-evidenceD(4)[OF head-step[unfolded head-eq]])
  have xn-eq: xn = hxn
  by (rule fri-layer-opening-chunk-values-unique(2))
    (rule fri-layer-step-evidenceD(4)[OF prev-head],

```

```

    rule fri-layer-step-evidenceD(4)[OF head-step[unfolded head-eq]]
  show ?thesis
    using fri-layer-step-evidenceD(3)[OF prev-head]
      fri-layer-step-evidenceD(3)[OF head-step[unfolded head-eq]]
      xp-eq xn-eq
    by simp
  qed
  have bfs-nonempty: 0 < length bfs
    using Cons.premis(6) 0 by simp
  from mfold-fri-layer-opening-first-recorded-step-evidence
    [OF tail[unfolded out1-eq] rest-layers tail-transcript-rest
      bfs-nonempty]
  obtain sxp-path sxn sxn-path snext where succ-step-tail:
    fri-layer-step-evidence (snd (bfs ! 0)) (fst (bfs ! 0))
      (fri-layer-lengths (length bfs) (len div 2) ! 0)
      (fri-layer-indices (length bfs) (idx mod (len div 2))
        (len div 2) ! 0)
      (pw + pw)
      (fri-sibling-index
        (fri-layer-lengths (length bfs) (len div 2) ! 0)
        (fri-layer-indices (length bfs) (idx mod (len div 2))
          (len div 2) ! 0))
      x1 sxp-path sxn sxn-path
      (fri-layer-indices (length bfs) (idx mod (len div 2))
        (len div 2) ! 0 mod
        (fri-layer-lengths (length bfs) (len div 2) ! 0 div 2))
      snext
      (rest ! 0)
  by blast
  have pow-eq: pw + pw = pw * 2 ^ Suc 0
    by (simp add: algebra-simps)
  have succ-step:
    fri-layer-step-evidence (snd ((bf # bfs) ! Suc j))
      (fst ((bf # bfs) ! Suc j))
      (fri-layer-lengths (length (bf # bfs)) len ! Suc j)
      (fri-layer-indices (length (bf # bfs)) idx len ! Suc j)
      (pw * 2 ^ Suc j)
      (fri-sibling-index
        (fri-layer-lengths (length (bf # bfs)) len ! Suc j)
        (fri-layer-indices (length (bf # bfs)) idx len ! Suc j))
      v sxp-path sxn sxn-path
      (fri-layer-indices (length (bf # bfs)) idx len ! Suc j mod
        (fri-layer-lengths (length (bf # bfs)) len ! Suc j div 2))
      snext
      (layer-chunks ! Suc j)
    using succ-step-tail 0 chunks-eq v-eq-x1 pow-eq by simp
  show ?thesis
    using succ-step by blast
next

```

```

case (Suc k)
have k-bound: Suc k < length bfs
  using Cons.premis(6) Suc by simp
have tail-all-layer-bounds:
   $\wedge m. m < \text{length } bfs \implies$ 
    0 < fri-layer-lengths (length bfs) (len div 2) ! m  $\wedge$ 
    fri-layer-indices (length bfs) (idx mod (len div 2))
      (len div 2) ! m <
    fri-layer-lengths (length bfs) (len div 2) ! m
  using Cons.premis(5)[of Suc m for m] by simp
have tail-initial-idx-bound:
  idx mod (len div 2) < len div 2
  using tail-all-layer-bounds[of 0] k-bound by (cases bfs) simp-all
have prev-tail:
  fri-layer-step-evidence (snd (bfs ! k)) (fst (bfs ! k))
    (fri-layer-lengths (length bfs) (len div 2) ! k)
    (fri-layer-indices (length bfs) (idx mod (len div 2))
      (len div 2) ! k)
    ((pw + pw) * 2 ^ k)
    (fri-sibling-index
      (fri-layer-lengths (length bfs) (len div 2) ! k)
      (fri-layer-indices (length bfs) (idx mod (len div 2))
        (len div 2) ! k))
    xp xp-path xn xn-path
    (fri-layer-indices (length bfs) (idx mod (len div 2))
      (len div 2) ! k mod
      (fri-layer-lengths (length bfs) (len div 2) ! k div 2))
    v (rest ! k)
proof -
  have pow-eq: (pw + pw) * 2 ^ k = pw * (2 * 2 ^ k)
    by (simp add: algebra-simps)
  have raw:
    fri-layer-step-evidence (snd (bfs ! k)) (fst (bfs ! k))
      (fri-layer-lengths (length bfs) (len div 2) ! k)
      (fri-layer-indices (length bfs) (idx mod (len div 2))
        (len div 2) ! k)
      (pw * (2 * 2 ^ k))
      (fri-sibling-index
        (fri-layer-lengths (length bfs) (len div 2) ! k)
        (fri-layer-indices (length bfs) (idx mod (len div 2))
          (len div 2) ! k))
      xp xp-path xn xn-path
      (fri-layer-indices (length bfs) (idx mod (len div 2))
        (len div 2) ! k mod
        (fri-layer-lengths (length bfs) (len div 2) ! k div 2))
      v (rest ! k)
    using Cons.premis(7) Suc chunks-eq pow-eq by simp
show ?thesis
  unfolding pow-eq by (rule raw)

```

```

qed
from Cons.IH[OF tail[unfolded out1-eq] rest-layers
  tail-transcript-rest tail-initial-idx-bound tail-all-layer-bounds
  k-bound prev-tail]
obtain sxp-path sn sn-path snext where succ-tail:
  fri-layer-step-evidence (snd (bfs ! Suc k))
    (fst (bfs ! Suc k))
    (fri-layer-lengths (length bfs) (len div 2) ! Suc k)
    (fri-layer-indices (length bfs) (idx mod (len div 2))
      (len div 2) ! Suc k)
    ((pw + pw) * 2 ^ Suc k)
    (fri-sibling-index
      (fri-layer-lengths (length bfs) (len div 2) ! Suc k)
      (fri-layer-indices (length bfs) (idx mod (len div 2))
        (len div 2) ! Suc k))
  v sxp-path sn sn-path
    (fri-layer-indices (length bfs) (idx mod (len div 2))
      (len div 2) ! Suc k mod
      (fri-layer-lengths (length bfs) (len div 2) ! Suc k div 2))
    snext (rest ! Suc k)
by blast
have pow-eq: (pw + pw) * (2 * 2 ^ k) = pw * (4 * 2 ^ k)
by (simp add: algebra-simps)
have succ-tail-raw:
  fri-layer-step-evidence (snd (bfs ! Suc k))
    (fst (bfs ! Suc k))
    (fri-layer-lengths (length bfs) (len div 2) ! Suc k)
    (fri-layer-indices (length bfs) (idx mod (len div 2))
      (len div 2) ! Suc k)
    ((pw + pw) * (2 * 2 ^ k))
    (fri-sibling-index
      (fri-layer-lengths (length bfs) (len div 2) ! Suc k)
      (fri-layer-indices (length bfs) (idx mod (len div 2))
        (len div 2) ! Suc k))
  v sxp-path sn sn-path
    (fri-layer-indices (length bfs) (idx mod (len div 2))
      (len div 2) ! Suc k mod
      (fri-layer-lengths (length bfs) (len div 2) ! Suc k div 2))
    snext (rest ! Suc k)
using succ-tail by simp
have succ-step:
  fri-layer-step-evidence (snd ((bf # bfs) ! Suc j))
    (fst ((bf # bfs) ! Suc j))
    (fri-layer-lengths (length (bf # bfs)) len ! Suc j)
    (fri-layer-indices (length (bf # bfs)) idx len ! Suc j)
    (pw * 2 ^ Suc j)
    (fri-sibling-index
      (fri-layer-lengths (length (bf # bfs)) len ! Suc j)
      (fri-layer-indices (length (bf # bfs)) idx len ! Suc j))

```

```

    v xsp-path xsn xsn-path
    (fri-layer-indices (length (bf # bfs)) idx len ! Suc j mod
      (fri-layer-lengths (length (bf # bfs)) len ! Suc j div 2))
    snext
    (layer-chunks ! Suc j)
  using succ-tail-raw Suc chunks-eq pow-eq
  by (simp add: algebra-simps)
show ?thesis
  using succ-step by blast
qed
qed

lemma verifier-query-round-after-index-composition-recorded-layer-step-evidence:
  fixes s t :: ('f, 'a) protocol-channel-scheme
  assumes outcome:
    Some ((), t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program
            fr f-fl f-final as fl final raw) s)
  and recorded:
    query-round-fri-layer-transcripts (index (to-nat raw))
      (map snd f-fl) (map snd fl) round-chunk trace-layers
      composition-layers
  and transcript: PTranscript s = round-chunk @ PTranscript t
  and j-bound: j < length fl
  and all-layer-bounds:
     $\bigwedge k. k < \text{length } fl \implies$ 
    0 < fri-layer-lengths (length fl) (clength * scale) ! k  $\wedge$ 
    fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! k <
    fri-layer-lengths (length fl) (clength * scale) ! k
  shows
     $\exists xp \ xp\text{-path } xn \ xn\text{-path next-value.}$ 
    fri-layer-step-evidence (snd (fl ! j)) (fst (fl ! j))
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! j)
      (2 ^ j)
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! j))
    xp xp-path xn xn-path
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j mod
      (fri-layer-lengths (length fl) (clength * scale) ! j div 2))
    next-value
    (composition-layers ! j)

```

```

proof –
  let  $?idx = index\ (to\text{-}nat\ raw)$ 
  from outcome obtain  $fv\ s1\ f\text{-}out\ s2\ s3\ c\text{-}out\ s4$  where
    query-decommit:
       $Some\ (fv,\ s1) \in$ 
         $set\text{-}dist\ (execute\ (mmap\ (check\text{-}decommit\text{-}on\text{-}query\ fr\ ?idx))\ s)$ 
    and trace-fri:
       $Some\ (f\text{-}out,\ s2) \in$ 
         $set\text{-}dist\ (execute$ 
           $(mfold\ (?idx,\ hd\ fv,\ clength * scale,\ 1)$ 
             $(receive\text{-}query\text{-}commits\ f\text{-}fl))\ s1)$ 
    and assert-trace:
       $Some\ ((),\ s3) \in$ 
         $set\text{-}dist$ 
           $(execute$ 
             $(assert\ (case\ f\text{-}out\ of\ (-,\ f\text{-}x,\ -, -) \Rightarrow f\text{-}x = f\text{-}final))\ s2)$ 
    and comp-fri:
       $Some\ (c\text{-}out,\ s4) \in$ 
         $set\text{-}dist\ (execute$ 
           $(mfold$ 
             $(?idx,\ cp\text{-}eval\ as\ fv\ (h \wedge ?idx * shift),\ clength * scale,\ 1)$ 
             $(receive\text{-}query\text{-}commits\ fl))\ s3)$ 
    and assert-comp:
       $Some\ ((),\ t) \in$ 
         $set\text{-}dist$ 
           $(execute$ 
             $(assert\ (case\ c\text{-}out\ of\ (-,\ x,\ -, -) \Rightarrow x = final))\ s4)$ 
    unfolding verifier-query-round-after-index-program-def
    by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
  from check-decommit-on-query-outcome[OF query-decommit]
  obtain query-paths query-chunk where
    query-chunk-shape:
       $query\text{-}decommitment\text{-}transcript\ ?idx\ fv\ query\text{-}paths\ query\text{-}chunk$ 
    and tr-s:  $PTranscript\ s = query\text{-}chunk\ @\ PTranscript\ s1$ 
    by blast
  from receive-query-commits-layers-outcome[OF trace-fri]
  obtain trace-layer-chunks trace-fri-chunk where
    trace-layers-actual:
       $fri\text{-}layers\text{-}transcript\ (length\ f\text{-}fl)\ (clength * scale)$ 
       $trace\text{-}layer\text{-}chunks\ trace\text{-}fri\text{-}chunk$ 
    and tr-s1:  $PTranscript\ s1 = trace\text{-}fri\text{-}chunk\ @\ PTranscript\ s2$ 
    by blast
  have s3-eq:  $s3 = s2$ 
    using assert-trace unfolding assert-def
    by (cases case f-out of (-, f-x, -, -)  $\Rightarrow$  f-x = f-final)
      (auto simp: throw-no-outcome)
  from receive-query-commits-layers-outcome[OF comp-fri]
  obtain comp-layer-chunks comp-fri-chunk where
    comp-layers-actual:

```



```

    fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
  and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
  by blast
  have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -)  $\Rightarrow$  x = final)
    (auto simp: throw-no-outcome)
  let ?actual-chunk = query-chunk @ trace-fri-chunk @ comp-fri-chunk
  have actual-recorded:
    query-round-fri-layer-transcripts ?idx (map snd f-fl) (map snd fl)
    ?actual-chunk trace-layer-chunks comp-layer-chunks
  unfolding query-round-fri-layer-transcripts-def
  by (intro exI[of - query-chunk] exI[of - trace-fri-chunk]
    exI[of - comp-fri-chunk] exI[of - fv] exI[of - query-paths] conjI)
    (use query-chunk-shape trace-layers-actual comp-layers-actual in
       $\langle$ simp-all add: mult.commute $\rangle$ )
  have actual-transcript:
    PTranscript s = ?actual-chunk @ PTranscript t
  using tr-s tr-s1 tr-s3 s3-eq t-eq by simp
  have actual-chunk-eq: ?actual-chunk = round-chunk
  using actual-transcript transcript by simp
  have comp-layer-chunks-eq:
    comp-layer-chunks = composition-layers
  by (rule query-round-fri-layer-transcripts-unique(2)
    [OF actual-recorded[unfolded actual-chunk-eq] recorded])
  have idx-bound: ?idx < clength * scale
  by (rule index-less-domain)
  have map-eq:
    receive-query-commits fl = map fri-layer-opening-step fl
  unfolding receive-query-commits-def fri-layer-opening-step-def
  by (simp add: fun-eq-iff split: prod.splits)
  from mfold-fri-layer-opening-selected-recorded-step-evidence-at
    [OF comp-fri[unfolded map-eq] comp-layers-actual tr-s3 idx-bound
      all-layer-bounds j-bound]
  obtain xp xp-path xn xn-path next-value where step-actual:
    fri-layer-step-evidence (snd (fl ! j)) (fst (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) ?idx (clength * scale) ! j)
    (1 * 2 ^ j)
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) ?idx (clength * scale) ! j))
    xp xp-path xn xn-path
    (fri-layer-indices (length fl) ?idx (clength * scale) ! j mod
      (fri-layer-lengths (length fl) (clength * scale) ! j div 2))
    next-value
    (comp-layer-chunks ! j)
  by blast

```

```

have step:
  fri-layer-step-evidence (snd (fl ! j)) (fst (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) ?idx (clength * scale) ! j)
    (2 ^ j)
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) ?idx (clength * scale) ! j))
  xp xp-path xn xn-path
    (fri-layer-indices (length fl) ?idx (clength * scale) ! j mod
      (fri-layer-lengths (length fl) (clength * scale) ! j div 2))
  next-value
    (composition-layers ! j)
using step-actual comp-layer-chunks-eq by simp
show ?thesis
using step by blast
qed

lemma verifier-query-round-after-index-composition-successor-recorded-step-from-previous:
fixes s t :: ('f, 'a) protocol-channel-scheme
assumes outcome:
  Some (( ), t) ∈
    set-dist
      (execute
        (verifier-query-round-after-index-program
          fr f-fl f-final as fl final raw) s)
and recorded:
  query-round-fri-layer-transcripts (index (to-nat raw))
    (map snd f-fl) (map snd fl) round-chunk trace-layers
    composition-layers
and transcript: PTranscript s = round-chunk @ PTranscript t
and j-bound: Suc j < length fl
and all-layer-bounds:
   $\bigwedge k. k < \text{length } fl \implies$ 
    0 < fri-layer-lengths (length fl) (clength * scale) ! k  $\wedge$ 
    fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! k <
    fri-layer-lengths (length fl) (clength * scale) ! k
and previous-step:
  fri-layer-step-evidence (snd (fl ! j)) (fst (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j)
    (2 ^ j)
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! j))
  xp xp-path xn xn-path

```

```

(fri-layer-indices (length fl) (index (to-nat raw))
 (clength * scale) ! j mod
 (fri-layer-lengths (length fl) (clength * scale) ! j div 2))
v
(composition-layers ! j)
shows
∃ xp-path' xn' xn-path' next-value.
fri-layer-step-evidence (snd (fl ! Suc j)) (fst (fl ! Suc j))
(fri-layer-lengths (length fl) (clength * scale) ! Suc j)
(fri-layer-indices (length fl) (index (to-nat raw))
 (clength * scale) ! Suc j)
(2 ^ Suc j)
(fri-sibling-index
 (fri-layer-lengths (length fl) (clength * scale) ! Suc j)
 (fri-layer-indices (length fl) (index (to-nat raw))
 (clength * scale) ! Suc j))
v xp-path' xn' xn-path'
(fri-layer-indices (length fl) (index (to-nat raw))
 (clength * scale) ! Suc j mod
 (fri-layer-lengths (length fl) (clength * scale) ! Suc j div 2))
next-value
(composition-layers ! Suc j)
proof -
let ?idx = index (to-nat raw)
from outcome obtain fv s1 f-out s2 s3 c-out s4 where
query-decommit:
Some (fv, s1) ∈
set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
and trace-fri:
Some (f-out, s2) ∈
set-dist (execute
(mfold (?idx, hd fv, clength * scale, 1)
(receive-query-commits f-fl)) s1)
and assert-trace:
Some ((), s3) ∈
set-dist
(execute
(assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
and comp-fri:
Some (c-out, s4) ∈
set-dist (execute
(mfold
(?idx, cp-eval as fv (h ^ ?idx * shift), clength * scale, 1)
(receive-query-commits fl)) s3)
and assert-comp:
Some ((), t) ∈
set-dist
(execute
(assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)

```

```

unfolding verifier-query-round-after-index-program-def
  by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
from check-decommit-on-query-outcome[OF query-decommit]
obtain query-paths query-chunk where
  query-chunk-shape:
    query-decommitment-transcript ?idx fv query-paths query-chunk
  and tr-s: PTranscript s = query-chunk @ PTranscript s1
  by blast
from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where
  trace-layers-actual:
    fri-layers-transcript (length f-fl) (clength * scale)
    trace-layer-chunks trace-fri-chunk
  and tr-s1: PTranscript s1 = trace-fri-chunk @ PTranscript s2
  by blast
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def
  by (cases case f-out of (-, f-x, -, -)  $\Rightarrow$  f-x = f-final)
    (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  comp-layers-actual:
    fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
  and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
  by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -)  $\Rightarrow$  x = final)
    (auto simp: throw-no-outcome)
let ?actual-chunk = query-chunk @ trace-fri-chunk @ comp-fri-chunk
have actual-recorded:
  query-round-fri-layer-transcripts ?idx (map snd f-fl) (map snd fl)
    ?actual-chunk trace-layer-chunks comp-layer-chunks
  unfolding query-round-fri-layer-transcripts-def
  by (intro exI[of - query-chunk] exI[of - trace-fri-chunk]
    exI[of - comp-fri-chunk] exI[of - fv] exI[of - query-paths] conjI)
    (use query-chunk-shape trace-layers-actual comp-layers-actual in
      ⟨simp-all add: mult commute⟩)
have actual-transcript:
  PTranscript s = ?actual-chunk @ PTranscript t
  using tr-s tr-s1 tr-s3 s3-eq t-eq by simp
have actual-chunk-eq: ?actual-chunk = round-chunk
  using actual-transcript transcript by simp
have comp-layer-chunks-eq:
  comp-layer-chunks = composition-layers
  by (rule query-round-fri-layer-transcripts-unique(2)
    [OF actual-recorded[unfolded actual-chunk-eq] recorded])
have idx-bound: ?idx < clength * scale

```

```

by (rule index-less-domain)
have map-eq:
  receive-query-commits fl = map fri-layer-opening-step fl
  unfolding receive-query-commits-def fri-layer-opening-step-def
  by (simp add: fun-eq-iff split: prod.splits)
have previous-step-actual:
  fri-layer-step-evidence (snd (fl ! j)) (fst (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) ?idx (clength * scale) ! j)
    (1 * 2 ^ j)
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) ?idx (clength * scale) ! j))
  xp xp-path xn xn-path
    (fri-layer-indices (length fl) ?idx (clength * scale) ! j mod
      (fri-layer-lengths (length fl) (clength * scale) ! j div 2))
  v
    (comp-layer-chunks ! j)
  using previous-step comp-layer-chunks-eq by simp
from mfold-fri-layer-opening-successor-recorded-step-from-previous
  [OF comp-fri[unfolded map-eq] comp-layers-actual tr-s3 idx-bound
    all-layer-bounds j-bound previous-step-actual]
obtain sxp-path sxn sxn-path snext where succ-actual:
  fri-layer-step-evidence (snd (fl ! Suc j)) (fst (fl ! Suc j))
    (fri-layer-lengths (length fl) (clength * scale) ! Suc j)
    (fri-layer-indices (length fl) ?idx (clength * scale) ! Suc j)
    (1 * 2 ^ Suc j)
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) ! Suc j)
      (fri-layer-indices (length fl) ?idx (clength * scale) ! Suc j))
  v sxp-path sxn sxn-path
    (fri-layer-indices (length fl) ?idx (clength * scale) ! Suc j mod
      (fri-layer-lengths (length fl) (clength * scale) ! Suc j div 2))
  snext
    (comp-layer-chunks ! Suc j)
  by blast
have succ:
  fri-layer-step-evidence (snd (fl ! Suc j)) (fst (fl ! Suc j))
    (fri-layer-lengths (length fl) (clength * scale) ! Suc j)
    (fri-layer-indices (length fl) ?idx (clength * scale) ! Suc j)
    (2 ^ Suc j)
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) ! Suc j)
      (fri-layer-indices (length fl) ?idx (clength * scale) ! Suc j))
  v sxp-path sxn sxn-path
    (fri-layer-indices (length fl) ?idx (clength * scale) ! Suc j mod
      (fri-layer-lengths (length fl) (clength * scale) ! Suc j div 2))
  snext
    (composition-layers ! Suc j)

```

```

    using succ-actual comp-layer-chunks-eq by simp
  show ?thesis
    using succ by blast
qed

lemma accepted-fri-opening-transcript-composition-successor-recorded-step-from-previous-at:
  assumes fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers
  and out-eq: out = Some (result, final-state)
  and round-bound: i < rounds
  and layer-bound: Suc j < length composition-roots
  and all-layer-bounds:
     $\bigwedge \text{raw } k. k < \text{length } \text{composition-roots} \implies$ 
    0 < fri-layer-lengths (length composition-roots)
    (clength * scale) ! k  $\wedge$ 
    fri-layer-indices (length composition-roots) (index (to-nat raw))
    (clength * scale) ! k <
    fri-layer-lengths (length composition-roots) (clength * scale) ! k
  and previous-step:
    fri-layer-step-evidence (composition-roots ! j)
    (composition-bs ! j)
    (fri-evidence-layer-len composition-roots j)
    (fri-evidence-layer-idx composition-roots query-idxs i j)
    (2 ^ j)
    (fri-sibling-index (fri-evidence-layer-len composition-roots j)
      (fri-evidence-layer-idx composition-roots query-idxs i j))
    xp xp-path xn xn-path
    (fri-evidence-next-idx composition-roots query-idxs i j)
    v
    (composition-round-layers ! i ! j)
  shows
     $\exists \text{xp-path}' \text{xn}' \text{xn-path}' \text{next-value}.$ 
    fri-layer-step-evidence (composition-roots ! Suc j)
    (composition-bs ! Suc j)
    (fri-evidence-layer-len composition-roots (Suc j))
    (fri-evidence-layer-idx composition-roots query-idxs i (Suc j))
    (2 ^ Suc j)
    (fri-sibling-index
      (fri-evidence-layer-len composition-roots (Suc j))
      (fri-evidence-layer-idx composition-roots query-idxs i
        (Suc j)))
    v xp-path' xn' xn-path'
    (fri-evidence-next-idx composition-roots query-idxs i (Suc j))
    next-value
    (composition-round-layers ! i ! Suc j)
  proof -
    from fri-openings obtain result' final-state' fr f-fl as fl query-state

```

```

    raw-idxs query-chunks where
out-def: out = Some (result', final-state')
and trace-roots-eq: trace-roots = map snd f-fl
and composition-roots-eq: composition-roots = map snd fl
and composition-bs-eq: composition-bs = map fst fl
and outcome:
  Some (result', final-state') ∈
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr f-fl trace-final as fl
          composition-final)
        rounds)
      query-state)
and len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-query-chunks: length query-chunks = rounds
and transcript-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state'
and query-chunk:
  ∧ i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
and replay-lookup:
  ∧ i. i < rounds ⇒
    fmlookup (HashMap final-state')
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
and recorded:
  ∧ i. i < rounds ⇒
    query-round-fri-layer-transcripts (query-idxs ! i)
    trace-roots composition-roots (query-chunks ! i)
    (trace-round-layers ! i) (composition-round-layers ! i)
unfolding accepted-fri-opening-transcript-def by blast
have result-eq: result' = result
using out-def out-eq by simp
have final-state-eq: final-state' = final-state
using out-def out-eq by simp
have outcome':
  Some (result, final-state) ∈
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr f-fl trace-final as fl
          composition-final)
        rounds)
      query-state)
using outcome result-eq final-state-eq by simp

```

```

let ?round =
  verifier-query-round-program fr f-fl trace-final as fl composition-final
from query-opening-ntimes-outcome-decomp-at[OF outcome' round-bound]
obtain prefix x suffix s-i s-suc where
  prefix:
    Some (prefix, s-i) ∈
      set-dist (execute (ntimes ?round i) query-state)
and round:
    Some (x, s-suc) ∈ set-dist (execute ?round s-i)
and suffix:
    Some (suffix, final-state) ∈
      set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
by blast
have x-eq: x = ()
by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
  fix prefix-query-idxs prefix-chunks
  assume len-prefix-idxs: length prefix-query-idxs = i
  and len-prefix-chunks: length prefix-chunks = i
  and transcript-prefix:
    PTranscript query-state =
      List.concat prefix-chunks @ PTranscript s-i
  and state-i:
    PState s-i =
      state-after-query-chunks (PState query-state) prefix-chunks i
  and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
  and prefix-ext: query-state ≤ s-i
  and prefix-rounds:
     $\bigwedge k. k < i \implies$ 
      verifier-query-round-chunk (prefix-query-idxs ! k)
      (map snd f-fl) (map snd fl) (prefix-chunks ! k)
  from round[unfolded x-eq verifier-query-round-program-alt-def]
  obtain raw s0 where
    raw-out:
      Some (raw, s0) ∈
        set-dist (execute receive-query-index-challenge s-i)
  and after:
    Some ((), s-suc) ∈
      set-dist (execute
        (verifier-query-round-after-index-program fr f-fl trace-final as
          fl composition-final raw) s0)
  by (auto elim!: set-dist-bindE)
  have tr-s0: PTranscript s0 = PTranscript s-i
  using receive-query-index-challenge-outcome[OF raw-out] by simp
  show ?thesis
  proof (rule verifier-query-round-program-outcome[OF round[unfolded x-eq]])
    fix raw' idx chunk

```



```

assume idx-eq: idx = index (to-nat raw')
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
  chunk
and transcript-round:
  PTranscript s-i = chunk @ PTranscript s-suc
and state-suc:
  PState s-suc = foldl concat (PState s-i) chunk
and round-ext: s-i ≤ s-suc
and counter-suc:
  PQueryCounter s-suc = Suc (PQueryCounter s-i)
and lookup-suc:
  fmlookup (HashMap s-suc)
  (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
  Some raw'
have raw-eq': raw = raw'
proof –
  have lookup-s0:
    fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
    using receive-query-index-challenge-outcome[OF raw-out] by simp
  have s0-suc: s0 ≤ s-suc
    using verifier-query-round-after-index-program-outcome[OF after]
    by simp
  have lookup-suc-raw:
    fmlookup (HashMap s-suc)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
    using hash-extension-lookup[OF lookup-s0 s0-suc] .
  show ?thesis
    using lookup-suc lookup-suc-raw by simp
qed
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
  fix suffix-query-idxs suffix-chunks
  assume len-suffix-idxs:
    length suffix-query-idxs = rounds – Suc i
  and len-suffix-chunks:
    length suffix-chunks = rounds – Suc i
  and transcript-suffix:
    PTranscript s-suc =
    List.concat suffix-chunks @ PTranscript final-state
  and suffix-state:
    PState final-state =
    state-after-query-chunks (PState s-suc) suffix-chunks
    (rounds – Suc i)
  and suffix-counter:

```

```

PQueryCounter final-state =
  PQueryCounter s-suc + (rounds - Suc i)
and suffix-ext: s-suc ≤ final-state
and suffix-rounds:
  ∧ k. k < rounds - Suc i ⇒
    verifier-query-round-chunk (suffix-query-idxs ! k)
    (map snd f-fl) (map snd fl) (suffix-chunks ! k)
let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks round-bound by simp
have selected-chunk-lens:
  ∧ k. k < rounds ⇒
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
proof -
  fix k
  assume k-bound: k < rounds
  show
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
  proof (cases k < i)
  case True
  have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length (prefix-chunks ! k) =
      verifier-query-round-transcript-length
        (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
      [OF prefix-rounds[OF True]])
  then show ?thesis
    using nth-eq
    by (simp add:
      verifier-query-round-transcript-length-index-irrelevant)
  next
  case False
  show ?thesis
  proof (cases k = i)
  case True
  have nth-eq: ?selected-chunks ! k = chunk
    using True len-prefix-chunks by (simp add: nth-append)
  have
    length chunk =
      verifier-query-round-transcript-length idx
      (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length[OF chunk-shape])
  then show ?thesis

```

```

    using nth-eq
    by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
next
case False
let ?k = k - Suc i
have ge: Suc i ≤ k
    using ⟨¬ k < i⟩ False by linarith
have k-bound': ?k < rounds - Suc i
    using ge k-bound by linarith
have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k
    using ge len-prefix-chunks by (simp add: nth-append)
have
    length (suffix-chunks ! ?k) =
        verifier-query-round-transcript-length
            (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
        [OF suffix-rounds[OF k-bound']])
then show ?thesis
    using nth-eq
    by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-f:
    ∧k. k < rounds ⇒
        verifier-query-round-chunk (query-idxs ! k)
            (map snd f-fl) (map snd fl) (query-chunks ! k)
    using query-chunk trace-roots-eq composition-roots-eq by simp
have query-chunk-lens:
    ∧k. k < rounds ⇒
        length (query-chunks ! k) =
            verifier-query-round-transcript-length (query-idxs ! k)
                (map snd f-fl) (map snd fl)
    using query-chunk-f verifier-query-round-chunk-length by blast
have selected-transcript:
    PTranscript query-state =
        List.concat ?selected-chunks @ PTranscript final-state
    using transcript-prefix transcript-round transcript-suffix by simp
have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
    using selected-transcript transcript-query final-state-eq by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
    have same-lens:
        ∧k. k < length ?selected-chunks ⇒
            length (?selected-chunks ! k) = length (query-chunks ! k)
        using selected-chunk-lens query-chunk-lens len-selected-chunks
        by simp

```

```

have concat-eq-empty:
  List.concat ?selected-chunks @ [] = List.concat query-chunks
  using concat-eq by simp
have len-eq: length ?selected-chunks = length query-chunks
  using len-selected-chunks len-query-chunks by simp
have ?selected-chunks = query-chunks  $\wedge$  ( [] :: 'f list) = []
  by (rule concat-append-eq-concat-same-chunk-lengths
    [OF len-eq same-lens concat-eq-empty])
then show ?thesis
  by simp
qed
have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
  state-after-query-chunks (PState query-state) query-chunks i
proof -
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
    state-after-query-chunks (PState query-state) prefix-chunks i
  by (rule state-after-query-chunks-append-prefix
    [OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
  Some raw'
proof -
  have lookup-suc':
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
    Some raw'
  using lookup-suc state-i counter-i by simp
have lookup-final-prefix:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state)
        prefix-chunks i)) =
  Some raw'
  using hash-extension-lookup [OF lookup-suc' suffix-ext] .
show ?thesis
  using lookup-final-prefix prefix-state-eq by simp
qed
have raw-idx-eq: raw' = raw-idxs ! i
  using lookup-final replay-lookup [OF round-bound] final-state-eq
  by simp

```

```

have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-idx-eq query-idxs-eq len-row round-bound by simp
have selected-i: ?selected-chunks ! i = chunk
  using len-prefix-chunks by (simp add: nth-append)
have chunk-i-eq: chunk = query-chunks ! i
  using chunks-eq selected-i by simp
have recorded-i:
  query-round-fri-layer-transcripts (index (to-nat raw))
    (map snd f-fl) (map snd fl) chunk (trace-round-layers ! i)
    (composition-round-layers ! i)
  using recorded[OF round-bound] idx-eq idx-eq-query raw-eq'
    chunk-i-eq trace-roots-eq composition-roots-eq
  by simp
have tr-s0-suc:
  PTranscript s0 = chunk @ PTranscript s-suc
  using tr-s0 transcript-round by simp
have suc-j-bound-fl: Suc j < length fl
  using layer-bound composition-roots-eq by simp
have all-layer-bounds-fl:
   $\bigwedge k. k < \text{length } fl \implies$ 
     $0 < \text{fri-layer-lengths } (\text{length } fl) (\text{clength} * \text{scale}) ! k \wedge$ 
     $\text{fri-layer-indices } (\text{length } fl) (\text{index } (\text{to-nat } raw))$ 
     $(\text{clength} * \text{scale}) ! k <$ 
     $\text{fri-layer-lengths } (\text{length } fl) (\text{clength} * \text{scale}) ! k$ 
  using all-layer-bounds composition-roots-eq by simp
have raw-idx-query: index (to-nat raw) = query-idxs ! i
  using idx-eq idx-eq-query raw-eq' by simp
have previous-step-fl:
  fri-layer-step-evidence (snd (fl ! j)) (fst (fl ! j))
    (fri-layer-lengths (length fl) (clength * scale) ! j)
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j)
    ( $2^j$ )
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) ! j)
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! j))
    xp xp-path xn xn-path
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! j mod
      (fri-layer-lengths (length fl) (clength * scale) ! j div 2))
    v
    (composition-round-layers ! i ! j)
  using previous-step composition-roots-eq composition-bs-eq
    raw-idx-query layer-bound
  unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
    fri-evidence-next-idx-def
  by simp
from verifier-query-round-after-index-composition-successor-recorded-step-from-previous

```

```

    [OF after recorded-i tr-s0-suc suc-j-bound-fl
      all-layer-bounds-fl previous-step-fl]
obtain sxp-path sxn sxn-path snext where succ-fl:
  fri-layer-step-evidence (snd (fl ! Suc j))
    (fst (fl ! Suc j))
    (fri-layer-lengths (length fl) (clength * scale) ! Suc j)
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! Suc j)
    (2 ^ Suc j)
    (fri-sibling-index
      (fri-layer-lengths (length fl) (clength * scale) ! Suc j)
      (fri-layer-indices (length fl) (index (to-nat raw))
        (clength * scale) ! Suc j))
  v sxp-path sxn sxn-path
    (fri-layer-indices (length fl) (index (to-nat raw))
      (clength * scale) ! Suc j mod
      (fri-layer-lengths (length fl) (clength * scale) ! Suc j
        div 2))
  snext
    (composition-round-layers ! i ! Suc j)
by blast
have succ:
  fri-layer-step-evidence (composition-roots ! Suc j)
    (composition-bs ! Suc j)
    (fri-evidence-layer-len composition-roots (Suc j))
    (fri-evidence-layer-idx composition-roots query-idxs i
      (Suc j))
    (2 ^ Suc j)
    (fri-sibling-index
      (fri-evidence-layer-len composition-roots (Suc j))
      (fri-evidence-layer-idx composition-roots query-idxs i
        (Suc j)))
  v sxp-path sxn sxn-path
    (fri-evidence-next-idx composition-roots query-idxs i (Suc j))
  snext
    (composition-round-layers ! i ! Suc j)
using succ-fl composition-roots-eq composition-bs-eq
  raw-idx-query layer-bound
unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
  fri-evidence-next-idx-def
by simp
show ?thesis
using succ by blast
qed
qed
qed
qed

```

lemma generic-fri-sampled-successor-conflict-imp-same-layer-with-source-steps:

assumes *successor*:
generic-fri-sampled-successor-opening-conflict roots challenges
query-idxs round-layers
and *len-challenges*: *length challenges = length roots*
and *source-step*:
 $\bigwedge$ *source-round layer-idx v*.
source-round < length query-idxs $\implies$
Suc layer-idx < length challenges $\implies$
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers source-round layer-idx v $\implies$
 $\exists$ *xp-path xn xn-path*.
fri-layer-step-evidence
(roots ! Suc layer-idx)
(challenges ! Suc layer-idx)
(fri-evidence-layer-len roots (Suc layer-idx))
(fri-evidence-layer-idx roots query-idxs source-round
(Suc layer-idx))
(2 $\wedge$ Suc layer-idx)
(fri-sibling-index
(fri-evidence-layer-len roots (Suc layer-idx))
(fri-evidence-layer-idx roots query-idxs source-round
(Suc layer-idx)))
v xp-path xn xn-path
(fri-evidence-next-idx roots query-idxs source-round
(Suc layer-idx))
(fri-evidence-next-value roots challenges query-idxs
source-round (Suc layer-idx) v xn)
(round-layers ! source-round ! Suc layer-idx)
shows
generic-fri-sampled-same-layer-opening-conflict roots challenges
query-idxs round-layers
proof –
from *generic-fri-sampled-successor-opening-conflictE[OF successor]*
obtain *source-round target-round layer-idx v xp xp-path xn xn-path*
where *source-bound*: *source-round < length query-idxs*
and *target-bound*: *target-round < length query-idxs*
and *layer-bound*: *Suc layer-idx < length challenges*
and *forced*:
generic-fri-round-forced-next-value roots challenges query-idxs
round-layers source-round layer-idx v
and *target-step*:
fri-layer-step-evidence
(roots ! Suc layer-idx)
(challenges ! Suc layer-idx)
(fri-evidence-layer-len roots (Suc layer-idx))
(fri-evidence-layer-idx roots query-idxs target-round
(Suc layer-idx))
(2 $\wedge$ Suc layer-idx)
(fri-sibling-index

```

    (fri-evidence-layer-len roots (Suc layer-idx))
    (fri-evidence-layer-idx roots query-idxs target-round
      (Suc layer-idx)))
  xp xp-path xn xn-path
  (fri-evidence-next-idx roots query-idxs target-round
    (Suc layer-idx))
  (fri-evidence-next-value roots challenges query-idxs target-round
    (Suc layer-idx) xp xn)
  (round-layers ! target-round ! Suc layer-idx)
and conflict:
  (fri-evidence-next-idx roots query-idxs source-round layer-idx =
    fri-evidence-layer-idx roots query-idxs target-round
      (Suc layer-idx)  $\wedge$ 
     $v \neq xp$ )  $\vee$ 
  (fri-evidence-next-idx roots query-idxs source-round layer-idx =
    fri-sibling-index
      (fri-evidence-layer-len roots (Suc layer-idx))
      (fri-evidence-layer-idx roots query-idxs target-round
        (Suc layer-idx))  $\wedge$ 
     $v \neq xn$ )
by blast
from source-step[OF source-bound layer-bound forced]
obtain vp-path vn vn-path where source-suc-step:
  fri-layer-step-evidence
    (roots ! Suc layer-idx)
    (challenges ! Suc layer-idx)
    (fri-evidence-layer-len roots (Suc layer-idx))
    (fri-evidence-layer-idx roots query-idxs source-round
      (Suc layer-idx))
    ( $2 \wedge$  Suc layer-idx)
    (fri-sibling-index
      (fri-evidence-layer-len roots (Suc layer-idx))
      (fri-evidence-layer-idx roots query-idxs source-round
        (Suc layer-idx)))
  v vp-path vn vn-path
  (fri-evidence-next-idx roots query-idxs source-round
    (Suc layer-idx))
  (fri-evidence-next-value roots challenges query-idxs source-round
    (Suc layer-idx) v vn)
  (round-layers ! source-round ! Suc layer-idx)
by blast
have source-idx-suc:
  fri-evidence-layer-idx roots query-idxs source-round (Suc layer-idx) =
    fri-evidence-next-idx roots query-idxs source-round layer-idx
by (rule fri-evidence-layer-idx-Suc)
  (use layer-bound len-challenges in simp)
have layer-suc-bound: Suc layer-idx < length challenges
  using layer-bound .
have same-conflict:

```



```

(fri-evidence-layer-idx roots query-idxs source-round (Suc layer-idx) =
  fri-evidence-layer-idx roots query-idxs target-round
  (Suc layer-idx) ∧
  v ≠ xp) ∨
(fri-evidence-layer-idx roots query-idxs source-round (Suc layer-idx) =
  fri-sibling-index
  (fri-evidence-layer-len roots (Suc layer-idx))
  (fri-evidence-layer-idx roots query-idxs target-round
  (Suc layer-idx)) ∧
  v ≠ xn)
proof -
from conflict show ?thesis
proof
  assume c:
    fri-evidence-next-idx roots query-idxs source-round layer-idx =
    fri-evidence-layer-idx roots query-idxs target-round
    (Suc layer-idx) ∧
    v ≠ xp
  then show ?thesis
    using source-idx-suc by simp
next
  assume c:
    fri-evidence-next-idx roots query-idxs source-round layer-idx =
    fri-sibling-index
    (fri-evidence-layer-len roots (Suc layer-idx))
    (fri-evidence-layer-idx roots query-idxs target-round
    (Suc layer-idx)) ∧
    v ≠ xn
  then show ?thesis
    using source-idx-suc by simp
qed
qed
show ?thesis
  unfolding generic-fri-sampled-same-layer-opening-conflict-def
  apply (rule exI[where x=source-round])
  apply (rule exI[where x=target-round])
  apply (rule exI[where x=Suc layer-idx])
  apply (rule exI[where x=v])
  apply (rule exI[where x=vp-path])
  apply (rule exI[where x=vn])
  apply (rule exI[where x=vn-path])
  apply (rule exI[where x=xp])
  apply (rule exI[where x=xp-path])
  apply (rule exI[where x=xn])
  apply (rule exI[where x=xn-path])
  apply (intro conjI)
  using source-bound target-bound layer-suc-bound source-suc-step
    target-step same-conflict
  by blast+

```

qed

lemma *generic-fri-sampled-same-layer-conflict-imp-merkle-or-slot-gap:*

assumes *conflict:*

generic-fri-sampled-same-layer-opening-conflict roots challenges

query-idxs round-layers

and *len-challenges:* *length challenges = length roots*

and *auth-step:*

$\bigwedge \text{round-idx layer-idx.}$

$\text{round-idx} < \text{length query-idxs} \implies$

$\text{layer-idx} < \text{length challenges} \implies$

generic-fri-transcript-step-with-authenticated-chunk roots

challenges query-idxs round-layers final-state round-idx layer-idx

shows

partial-merkle-inconsistency-bad s (Some (result, final-state)) $\vee$

generic-fri-authenticated-slot-recorded-chunk-gap roots query-idxs

round-layers final-state

proof –

from *generic-fri-sampled-same-layer-imp-authenticated-or-recorded-auth-gap*

[*OF conflict, of final-state*]

consider

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks

roots challenges query-idxs round-layers final-state

| *generic-fri-sampled-same-layer-recorded-auth-gap roots challenges*

query-idxs round-layers final-state

by *blast*

then show *?thesis*

proof *cases*

case 1

then have *partial-merkle-inconsistency-bad s (Some (result, final-state))*

by (*rule*

generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks-imp-partial-merkle)

(*rule len-challenges*)

then show *?thesis* **by** *simp*

next

case 2

then have *auth-gap:*

generic-fri-sampled-same-layer-authenticated-recorded-auth-gap roots

challenges query-idxs round-layers final-state

by (*rule generic-fri-recorded-auth-gap-imp-authenticated-recorded-auth-gap*)

(*rule auth-step*)

have *generic-fri-authenticated-slot-recorded-chunk-gap roots query-idxs*

round-layers final-state

by (*rule generic-fri-authenticated-recorded-auth-gap-imp-slot-recorded-chunk-gap*

[*OF auth-gap*])

then show *?thesis* **by** *simp*

qed

qed

lemma *composition-fri-verifier-tied-next-value-replay-gap-imp-merkle-or-slot-on-support:*
assumes *support: out ∈ set-dist (execute verify-monad s)*
and *gap:*
composition-fri-verifier-tied-next-value-replay-gap s out
shows
partial-merkle-inconsistency-bad s out ∨
composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
s out
proof –
from *gap* **have** *sampled:*
composition-fri-verifier-tied-sampled-next-value-conflict s out
unfolding *composition-fri-verifier-tied-next-value-replay-gap-def*
by *simp*
from *composition-fri-verifier-tied-sampled-next-value-conflictE[OF sampled]*
obtain *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings composition-openings
trace-table composition-table round-idx round-idx' layer-idx v v'
where *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *aligned:*
accepted-with-partial-initial-openings-aligned s out fr trace-roots
trace-final as fri-dg composition-roots composition-final
fri-query-idxs trace-openings composition-openings
and *trace-cand: partial-trace-table-candidate trace-table trace-openings*
and *comp-cand:*
partial-composition-table-candidate composition-table
composition-openings
and *trace-low: trace-table-low-degree trace-table*
and *comp-not-low:*
 $\neg$ *composition-table-low-degree maxDegree composition-table*
and *round-bound: round-idx < length fri-query-idxs*
and *round-bound': round-idx' < length fri-query-idxs*
and *layer-bound: layer-idx < length composition-bs*
and *same-idx:*
fri-evidence-next-idx composition-roots fri-query-idxs round-idx
layer-idx =
fri-evidence-next-idx composition-roots fri-query-idxs round-idx'
layer-idx
and *forced:*
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx layer-idx v
and *forced':*
generic-fri-round-forced-next-value composition-roots composition-bs
fri-query-idxs composition-round-layers round-idx' layer-idx v'
and *neq: v ≠ v'*
using *sampled*

```

  by (elim composition-fri-verifier-tied-sampled-next-value-conflictE)
from fri-openings obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-fri-opening-transcript-def by blast
have support-some:
  Some (result, final-state) ∈ set-dist (execute verify-monad s)
  using support out-eq by simp
have challenges:
  accepted-fri-challenges s (Some (result, final-state))
  trace-bs fri-dg composition-bs
  using accepted-fri-opening-transcript-challenges[OF fri-openings]
  out-eq by simp
have degree-bound: to-nat fri-dg ≤ maxDegree
  by (rule verify-monad-accepted-fri-challenges-degree-bound
    [OF support-some challenges])
have len-bs-roots: length composition-bs = length composition-roots
  using accepted-fri-opening-transcript-shapes(4)[OF fri-openings] .
have layer-bound-roots: layer-idx < length composition-roots
  using layer-bound len-bs-roots by simp
have len-query: length fri-query-idxs = rounds
  using accepted-fri-opening-transcript-shapes(7)[OF fri-openings] .
have round-bound-rounds: round-idx < rounds
  using round-bound len-query by simp
have round-bound-rounds': round-idx' < rounds
  using round-bound' len-query by simp
have raw-bound:
  fri-evidence-layer-idx composition-roots fri-query-idxs round-idx
  layer-idx <
  fri-evidence-layer-len composition-roots layer-idx
  by (rule
    accepted-fri-opening-transcript-composition-evidence-layer-idx-bound
    [OF fri-openings degree-bound round-bound layer-bound-roots])
have raw-bound':
  fri-evidence-layer-idx composition-roots fri-query-idxs round-idx'
  layer-idx <
  fri-evidence-layer-len composition-roots layer-idx
  by (rule
    accepted-fri-opening-transcript-composition-evidence-layer-idx-bound
    [OF fri-openings degree-bound round-bound' layer-bound-roots])
have even-len:
  2 dvd fri-evidence-layer-len composition-roots layer-idx
  by (rule accepted-fri-opening-transcript-composition-layer-arithmetic(1)
    [OF fri-openings degree-bound layer-bound])
have round-product:
  fri-evidence-layer-len composition-roots layer-idx * 2 ^ layer-idx =
  clength * scale
  by (rule accepted-fri-opening-transcript-composition-layer-arithmetic(2)
    [OF fri-openings degree-bound layer-bound])
have auth-step:

```

$\wedge i j.$
 $i < \text{length } \text{fri-query-idxs} \implies$
 $j < \text{length } \text{composition-bs} \implies$
 $\text{generic-fri-transcript-step-with-authenticated-chunk } \text{composition-roots}$
 $\text{composition-bs } \text{fri-query-idxs } \text{composition-round-layers } \text{final-state } i j$

proof –

fix $i j$
assume $i\text{-bound}: i < \text{length } \text{fri-query-idxs}$
and $j\text{-bound}: j < \text{length } \text{composition-bs}$
have $i\text{-rounds}: i < \text{rounds}$
using $i\text{-bound } \text{len-query}$ **by** simp
have $j\text{-roots}: j < \text{length } \text{composition-roots}$
using $j\text{-bound } \text{len-bs-roots}$ **by** simp
show
 $\text{generic-fri-transcript-step-with-authenticated-chunk}$
 $\text{composition-roots } \text{composition-bs } \text{fri-query-idxs}$
 $\text{composition-round-layers } \text{final-state } i j$
by (rule
 $\text{accepted-fri-opening-transcript-composition-step-with-authenticated-chunk-at-raw}$
 $[\text{OF } \text{fri-openings } \text{out-eq } \text{degree-bound } i\text{-rounds } j\text{-roots}])$

qed

show $?thesis$

proof (cases
 $\text{generic-fri-sampled-same-layer-opening-conflict } \text{composition-roots}$
 $\text{composition-bs } \text{fri-query-idxs } \text{composition-round-layers}$)

case True

have conflict-route :

$\text{partial-merkle-inconsistency-bad } s (\text{Some } (\text{result}, \text{final-state})) \vee$
 $\text{generic-fri-authenticated-slot-recorded-chunk-gap } \text{composition-roots}$
 $\text{fri-query-idxs } \text{composition-round-layers } \text{final-state}$
by (rule $\text{generic-fri-sampled-same-layer-conflict-imp-merkle-or-slot-gap}$
 $[\text{OF } \text{True } \text{len-bs-roots } \text{auth-step}])$

then show $?thesis$

proof

assume $\text{partial-merkle-inconsistency-bad } s (\text{Some } (\text{result}, \text{final-state}))$

then show $?thesis$

using out-eq **by** simp

next

assume slot-gap :

$\text{generic-fri-authenticated-slot-recorded-chunk-gap } \text{composition-roots}$
 $\text{fri-query-idxs } \text{composition-round-layers } \text{final-state}$
have route-slot-gap :
 $\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap}$
 $s \text{ out}$

unfolding
 $\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-def}$

by (intro $\text{exI } \text{conjI}$)
(rule out-eq , rule fri-openings , rule aligned , rule trace-cand ,
rule comp-cand , rule trace-low , rule comp-not-low , rule slot-gap)

```

    then show ?thesis by simp
qed
next
case False
then have no-same:
  ¬ generic-fri-sampled-same-layer-opening-conflict
    composition-roots composition-bs fri-query-idxs
    composition-round-layers
  by simp
have v = v'
  by (rule generic-fri-forced-next-value-eq-if-no-same-layer-conflict
    [OF no-same round-bound round-bound' layer-bound even-len
      raw-bound raw-bound' round-product same-idx forced forced'])
then show ?thesis
  using neq by simp
qed
qed

lemma composition-fri-verifier-tied-successor-replay-gap-imp-merkle-or-slot-on-support:
  assumes support: out ∈ set-dist (execute verify-monad s)
  and gap:
    composition-fri-verifier-tied-successor-replay-gap s out
  shows
    partial-merkle-inconsistency-bad s out ∨
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
      s out
proof -
  from gap have sampled:
    composition-fri-verifier-tied-sampled-successor-opening-conflict s out
  unfolding composition-fri-verifier-tied-successor-replay-gap-def
  by simp
obtain trace-roots trace-bs trace-final fri-dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr as trace-openings composition-openings
  trace-table composition-table source-round target-round layer-idx v
  xp xp-path xn xn-path
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and aligned:
    accepted-with-partial-initial-openings-aligned s out fr trace-roots
    trace-final as fri-dg composition-roots composition-final
    fri-query-idxs trace-openings composition-openings
  and trace-cand: partial-trace-table-candidate trace-table trace-openings
  and comp-cand:
    partial-composition-table-candidate composition-table
    composition-openings
  and trace-low: trace-table-low-degree trace-table

```

```

and comp-not-low:
   $\neg$  composition-table-low-degree maxDegree composition-table
and source-bound: source-round < length fri-query-idxs
and target-bound: target-round < length fri-query-idxs
and layer-bound: Suc layer-idx < length composition-bs
and forced:
  generic-fri-round-forced-next-value composition-roots composition-bs
  fri-query-idxs composition-round-layers source-round layer-idx v
and target-step:
  fri-layer-step-evidence
    (composition-roots ! Suc layer-idx)
    (composition-bs ! Suc layer-idx)
    (fri-evidence-layer-len composition-roots (Suc layer-idx))
    (fri-evidence-layer-idx composition-roots fri-query-idxs
      target-round (Suc layer-idx))
    ( $2 \wedge$  Suc layer-idx)
    (fri-sibling-index
      (fri-evidence-layer-len composition-roots (Suc layer-idx))
      (fri-evidence-layer-idx composition-roots fri-query-idxs
        target-round (Suc layer-idx)))
    xp xp-path xn xn-path
    (fri-evidence-next-idx composition-roots fri-query-idxs
      target-round (Suc layer-idx))
    (fri-evidence-next-value composition-roots composition-bs
      fri-query-idxs target-round (Suc layer-idx) xp xn)
    (composition-round-layers ! target-round ! Suc layer-idx)
and conflict:
  (fri-evidence-next-idx composition-roots fri-query-idxs source-round
    layer-idx =
    fri-evidence-layer-idx composition-roots fri-query-idxs target-round
      (Suc layer-idx)  $\wedge$ 
    v  $\neq$  xp)  $\vee$ 
  (fri-evidence-next-idx composition-roots fri-query-idxs source-round
    layer-idx =
    fri-sibling-index
      (fri-evidence-layer-len composition-roots (Suc layer-idx))
      (fri-evidence-layer-idx composition-roots fri-query-idxs
        target-round (Suc layer-idx))  $\wedge$ 
    v  $\neq$  xn)
using sampled
by (elim composition-fri-verifier-tied-sampled-successor-opening-conflictE)
from fri-openings obtain result final-state where out-eq:
  out = Some (result, final-state)
unfolding accepted-fri-opening-transcript-def by blast
have support-some:
  Some (result, final-state)  $\in$  set-dist (execute verify-monad s)
using support out-eq by simp
have challenges:
  accepted-fri-challenges s (Some (result, final-state))

```

```

    trace-bs fri-dg composition-bs
using accepted-fri-opening-transcript-challenges[OF fri-openings]
    out-eq by simp
have degree-bound: to-nat fri-dg ≤ maxDegree
    by (rule verify-monad-accepted-fri-challenges-degree-bound
        [OF support-some challenges])
have len-bs-roots: length composition-bs = length composition-roots
    using accepted-fri-opening-transcript-shapes(4)[OF fri-openings] .
have len-query: length fri-query-idxs = rounds
    using accepted-fri-opening-transcript-shapes(7)[OF fri-openings] .
have sampled-generic:
    generic-fri-sampled-successor-opening-conflict composition-roots
        composition-bs fri-query-idxs composition-round-layers
unfolding generic-fri-sampled-successor-opening-conflict-def
by (intro exI conjI)
    (rule source-bound, rule target-bound, rule layer-bound,
     rule forced, rule target-step, rule conflict)
have all-layer-bounds:
    ∧ raw k. k < length composition-roots ⇒
        0 < fri-layer-lengths (length composition-roots)
        (clength * scale) ! k ∧
        fri-layer-indices (length composition-roots) (index (to-nat raw))
        (clength * scale) ! k <
        fri-layer-lengths (length composition-roots) (clength * scale) ! k
    by (rule accepted-fri-opening-transcript-composition-raw-layer-bound
        [OF fri-openings degree-bound])
have source-step:
    ∧ source-round layer-idx v.
        source-round < length fri-query-idxs ⇒
        Suc layer-idx < length composition-bs ⇒
        generic-fri-round-forced-next-value composition-roots composition-bs
            fri-query-idxs composition-round-layers source-round layer-idx v ⇒
        ∃ xp-path xn xn-path.
            fri-layer-step-evidence
            (composition-roots ! Suc layer-idx)
            (composition-bs ! Suc layer-idx)
            (fri-evidence-layer-len composition-roots (Suc layer-idx))
            (fri-evidence-layer-idx composition-roots fri-query-idxs
                source-round (Suc layer-idx))
            (2 ^ Suc layer-idx)
            (fri-sibling-index
                (fri-evidence-layer-len composition-roots (Suc layer-idx))
                (fri-evidence-layer-idx composition-roots fri-query-idxs
                    source-round (Suc layer-idx)))
            v xp-path xn xn-path
            (fri-evidence-next-idx composition-roots fri-query-idxs
                source-round (Suc layer-idx))
            (fri-evidence-next-value composition-roots composition-bs
                fri-query-idxs source-round (Suc layer-idx) v xn)

```



```

      (composition-round-layers ! source-round ! Suc layer-idx)
proof –
  fix source-round layer-idx v
  assume source-round-bound: source-round < length fri-query-idxs
    and layer-bound': Suc layer-idx < length composition-bs
    and forced':
      generic-fri-round-forced-next-value composition-roots composition-bs
        fri-query-idxs composition-round-layers source-round layer-idx v
  have source-round-rounds: source-round < rounds
    using source-round-bound len-query by simp
  have layer-bound-roots: Suc layer-idx < length composition-roots
    using layer-bound' len-bs-roots by simp
  from forced' obtain fxp fxp-path fxn fxn-path where previous-step:
    fri-layer-step-evidence (composition-roots ! layer-idx)
      (composition-bs ! layer-idx)
      (fri-evidence-layer-len composition-roots layer-idx)
      (fri-evidence-layer-idx composition-roots fri-query-idxs
        source-round layer-idx)
      (2 ^ layer-idx)
      (fri-sibling-index
        (fri-evidence-layer-len composition-roots layer-idx)
        (fri-evidence-layer-idx composition-roots fri-query-idxs
          source-round layer-idx))
    fxp fxp-path fxn fxn-path
    (fri-evidence-next-idx composition-roots fri-query-idxs
      source-round layer-idx)
    v
    (composition-round-layers ! source-round ! layer-idx)
  unfolding generic-fri-round-forced-next-value-def by blast
  from accepted-fri-opening-transcript-composition-successor-recorded-step-from-previous-at
    [OF fri-openings out-eq source-round-rounds layer-bound-roots
      all-layer-bounds previous-step]
  obtain sxp-path sxn sxn-path snext where succ:
    fri-layer-step-evidence (composition-roots ! Suc layer-idx)
      (composition-bs ! Suc layer-idx)
      (fri-evidence-layer-len composition-roots (Suc layer-idx))
      (fri-evidence-layer-idx composition-roots fri-query-idxs
        source-round (Suc layer-idx))
      (2 ^ Suc layer-idx)
      (fri-sibling-index
        (fri-evidence-layer-len composition-roots (Suc layer-idx))
        (fri-evidence-layer-idx composition-roots fri-query-idxs
          source-round (Suc layer-idx)))
    v sxp-path sxn sxn-path
    (fri-evidence-next-idx composition-roots fri-query-idxs
      source-round (Suc layer-idx))
    snext
    (composition-round-layers ! source-round ! Suc layer-idx)
  by blast

```

```

have snext-eq:
  snext =
    fri-evidence-next-value composition-roots composition-bs
      fri-query-idxs source-round (Suc layer-idx) v sn
  using fri-layer-step-evidenceD( $\beta$ )[OF succ]
  unfolding fri-evidence-next-value-def by simp
show
   $\exists$  xp-path xn xn-path.
    fri-layer-step-evidence
      (composition-roots ! Suc layer-idx)
      (composition-bs ! Suc layer-idx)
      (fri-evidence-layer-len composition-roots (Suc layer-idx))
      (fri-evidence-layer-idx composition-roots fri-query-idxs
        source-round (Suc layer-idx))
      ( $2 \wedge$  Suc layer-idx)
      (fri-sibling-index
        (fri-evidence-layer-len composition-roots (Suc layer-idx))
        (fri-evidence-layer-idx composition-roots fri-query-idxs
          source-round (Suc layer-idx)))
      v xp-path xn xn-path
      (fri-evidence-next-idx composition-roots fri-query-idxs
        source-round (Suc layer-idx))
      (fri-evidence-next-value composition-roots composition-bs
        fri-query-idxs source-round (Suc layer-idx) v xn)
      (composition-round-layers ! source-round ! Suc layer-idx)
    using succ snext-eq by blast
qed
have same-layer:
  generic-fri-sampled-same-layer-opening-conflict composition-roots
    composition-bs fri-query-idxs composition-round-layers
  by (rule generic-fri-sampled-successor-conflict-imp-same-layer-with-source-steps
    [OF sampled-generic len-bs-roots source-step])
have auth-step:
   $\bigwedge^i j.$ 
     $i < \text{length } \textit{fri-query-idxs} \implies$ 
     $j < \text{length } \textit{composition-bs} \implies$ 
    generic-fri-transcript-step-with-authenticated-chunk composition-roots
      composition-bs fri-query-idxs composition-round-layers final-state i j
proof –
  fix i j
  assume i-bound:  $i < \text{length } \textit{fri-query-idxs}$ 
  and j-bound:  $j < \text{length } \textit{composition-bs}$ 
  have i-rounds:  $i < \text{rounds}$ 
  using i-bound len-query by simp
  have j-roots:  $j < \text{length } \textit{composition-roots}$ 
  using j-bound len-bs-roots by simp
  show
    generic-fri-transcript-step-with-authenticated-chunk
      composition-roots composition-bs fri-query-idxs

```

```

    composition-round-layers final-state i j
  by (rule
    accepted-fri-opening-transcript-composition-step-with-authenticated-chunk-at-raw
    [OF fri-openings out-eq degree-bound i-rounds j-roots])
qed
have route:
  partial-merkle-inconsistency-bad s (Some (result, final-state)) ∨
  generic-fri-authenticated-slot-recorded-chunk-gap composition-roots
  fri-query-idxs composition-round-layers final-state
  by (rule generic-fri-sampled-same-layer-conflict-imp-merkle-or-slot-gap
    [OF same-layer len-bs-roots auth-step])
then show ?thesis
proof
  assume partial-merkle-inconsistency-bad s (Some (result, final-state))
  then show ?thesis
    using out-eq by simp
next
  assume slot-gap:
    generic-fri-authenticated-slot-recorded-chunk-gap composition-roots
    fri-query-idxs composition-round-layers final-state
  have route-slot-gap:
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
    s out
  unfolding
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule aligned, rule trace-cand,
    rule comp-cand, rule trace-low, rule comp-not-low, rule slot-gap)
  then show ?thesis by simp
qed
qed

lemma wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-merkle-and-slot:
  fixes M A :: prob
  assumes merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M
  and slot-bound:
    wp-event verify-monad
    (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s)
    s ≤ A
  shows
    wp-event verify-monad
    (composition-fri-verifier-tied-successor-replay-gap s) s
    ≤ M + A
proof -
  let ?P = partial-merkle-inconsistency-bad s
  let ?Q = composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s
  have event-le:
    wp-event verify-monad

```

```

    (composition-fri-verifier-tied-successor-replay-gap s) s ≤
    wp-event verify-monad (λout. ?P out ∨ ?Q out) s
  by (rule wp-event-mono-on-support)
    (rule
      composition-fri-verifier-tied-successor-replay-gap-imp-merkle-or-slot-on-support)
  also have ... ≤ wp-event verify-monad ?P s +
    wp-event verify-monad ?Q s
  by (rule wp-event-union-bound)
  also have ... ≤ M + A
  by (intro add-mono merkle-bound slot-bound)
  finally show ?thesis .
qed

lemma wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-merkle-and-slot:
  fixes M A :: prob
  assumes merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M
  and slot-bound:
    wp-event verify-monad
      (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s)
      s ≤ A
  shows
    wp-event verify-monad
      (composition-fri-verifier-tied-next-value-replay-gap s) s
      ≤ M + A
proof -
  let ?P = partial-merkle-inconsistency-bad s
  let ?Q = composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s
  have event-le:
    wp-event verify-monad
      (composition-fri-verifier-tied-next-value-replay-gap s) s ≤
    wp-event verify-monad (λout. ?P out ∨ ?Q out) s
  by (rule wp-event-mono-on-support)
    (rule
      composition-fri-verifier-tied-next-value-replay-gap-imp-merkle-or-slot-on-support)
  also have ... ≤ wp-event verify-monad ?P s +
    wp-event verify-monad ?Q s
  by (rule wp-event-union-bound)
  also have ... ≤ M + A
  by (intro add-mono merkle-bound slot-bound)
  finally show ?thesis .
qed

end

end

theory Soundness-FRI-Trace-Replay-Gaps

```

```

imports
  Soundness-FRI-Trace-Residuals
  Soundness-FRI-Composition-Replay-Gaps
begin

```

Narrow downstream layer for trace header-tied replay gaps. This mirrors the composition replay-gap layer without growing the already-large trace residual theory.

```

context soundness
begin

```

```

lemma accepted-fri-opening-transcript-trace-layer-arithmetic:
  assumes fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers
  and layer-bound: layer-idx < length trace-bs
  shows 2 dvd fri-evidence-layer-len trace-roots layer-idx
  and fri-evidence-layer-len trace-roots layer-idx * 2 ^ layer-idx =
    clength * scale
proof –
  obtain N where eval-power: clength * scale = 2 ^ N
  using eval-domain-length-power by blast
  have bs-len: length trace-bs = length trace-roots
  using accepted-fri-opening-transcript-shapes(2)[OF fri-openings] .
  have roots-bound: layer-idx < length trace-roots
  using layer-bound bs-len by simp
  have roots-le: length trace-roots ≤ N
  proof –
    have len: length trace-roots = ceil-log clength
    using accepted-fri-opening-transcript-shapes(1)[OF fri-openings] .
    have clength ≤ clength * scale
    using scale-pos by simp
    also have ... = 2 ^ N
    by (rule eval-power)
    finally have ceil-log clength ≤ N
    by (rule ceil-log-le-power)
    then show ?thesis
    using len by simp
  qed
  have suc-le: Suc layer-idx ≤ N
  using roots-bound roots-le by simp
  have dvd-eval-suc: (2::nat) ^ Suc layer-idx dvd clength * scale
  using le-imp-power-dvd[OF suc-le] eval-power by simp
  show 2 dvd fri-evidence-layer-len trace-roots layer-idx
  unfolding fri-evidence-layer-len-def
  by (rule fri-layer-lengths-even-if-dvd[OF roots-bound dvd-eval-suc])
  have le-N: layer-idx ≤ N
  using roots-bound roots-le by simp
  have dvd-eval: (2::nat) ^ layer-idx dvd clength * scale

```

```

    using le-imp-power-dvd[OF le-N] eval-power by simp
show fri-evidence-layer-len trace-roots layer-idx *
  2 ^ layer-idx = clength * scale
unfolding fri-evidence-layer-len-def
by (rule fri-layer-lengths-round-product-if-dvd
  [OF roots-bound dvd-eval])
qed

lemma accepted-fri-opening-transcript-trace-evidence-layer-idx-bound:
  assumes fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers
  and round-bound: round-idx < length query-idxs
  and layer-bound: layer-idx < length trace-roots
  shows
    fri-evidence-layer-idx trace-roots query-idxs round-idx layer-idx <
    fri-evidence-layer-len trace-roots layer-idx
proof -
  obtain N where eval-power: clength * scale = 2 ^ N
  using eval-domain-length-power by blast
  have roots-le: length trace-roots ≤ N
  proof -
    have len: length trace-roots = ceil-log clength
    using accepted-fri-opening-transcript-shapes(1)[OF fri-openings] .
    have clength ≤ clength * scale
    using scale-pos by simp
    also have ... = 2 ^ N
    by (rule eval-power)
    finally have ceil-log clength ≤ N
    by (rule ceil-log-le-power)
    then show ?thesis
    using len by simp
  qed
  have query-bound: query-idxs ! round-idx < clength * scale
  by (rule accepted-fri-opening-transcript-query-idx-bound
    [OF fri-openings])
    (use round-bound in simp)
  have idx-bound:
    fri-layer-indices (length trace-roots) (query-idxs ! round-idx)
    (clength * scale) ! layer-idx <
    fri-layer-lengths (length trace-roots) (clength * scale) ! layer-idx
  by (rule fri-layer-indices-nth-bound
    [OF query-bound layer-bound eval-power roots-le])
  then show ?thesis
  unfolding fri-evidence-layer-idx-def fri-evidence-layer-len-def by simp
qed

lemma trace-fri-header-tied-next-value-replay-gap-imp-merkle-or-slot-on-support:

```

assumes *support*: $out \in \text{set-dist } (\text{execute verify-monad } s)$
and *gap*: *trace-fri-header-tied-next-value-replay-gap* *s* *out*
shows
partial-merkle-inconsistency-bad *s* *out* $\vee$
trace-fri-header-tied-authenticated-slot-recorded-chunk-gap *s* *out*
proof –
from *gap* **have** *sampled*:
trace-fri-header-tied-sampled-next-value-conflict *s* *out*
unfolding *trace-fri-header-tied-next-value-replay-gap-def* **by** *simp*
from *sampled* **obtain** *trace-roots* *trace-bs* *trace-final* *fri-dg*
composition-roots *composition-bs* *composition-final* *fri-query-idxs*
trace-round-layers *composition-round-layers* *fr* *as* *rest* *trace-openings*
trace-table
where *fri-openings*:
accepted-fri-opening-transcript *s* *out* *trace-roots* *trace-bs* *trace-final*
fri-dg *composition-roots* *composition-bs* *composition-final*
fri-query-idxs *trace-round-layers* *composition-round-layers*
and *header*:
verifier-header-transcript *s* *fr* *trace-roots* *trace-final* *as* *fri-dg*
composition-roots *composition-final* *rest*
and *partial*:
accepted-with-partial-trace-openings *s* *out* *fr* *fri-query-idxs*
trace-openings
and *cand*: *partial-trace-table-candidate* *trace-table* *trace-openings*
and *not-low*: $\neg$ *trace-table-low-degree* *trace-table*
and *generic*:
generic-fri-sampled-next-value-conflict *trace-roots* *trace-bs*
fri-query-idxs *trace-round-layers*
unfolding *trace-fri-header-tied-sampled-next-value-conflict-def* **by** *blast*
from *generic-fri-sampled-next-value-conflictE*[*OF* *generic*]
obtain *round-idx* *round-idx'* *layer-idx* *v* *v'*
where *round-bound*: *round-idx* < *length* *fri-query-idxs*
and *round-bound'*: *round-idx'* < *length* *fri-query-idxs*
and *layer-bound*: *layer-idx* < *length* *trace-bs*
and *same-idx*:
fri-evidence-next-idx *trace-roots* *fri-query-idxs* *round-idx*
layer-idx =
fri-evidence-next-idx *trace-roots* *fri-query-idxs* *round-idx'*
layer-idx
and *forced*:
generic-fri-round-forced-next-value *trace-roots* *trace-bs*
fri-query-idxs *trace-round-layers* *round-idx* *layer-idx* *v*
and *forced'*:
generic-fri-round-forced-next-value *trace-roots* *trace-bs*
fri-query-idxs *trace-round-layers* *round-idx'* *layer-idx* *v'*
and *neq*: $v \neq v'$
by *blast*
from *fri-openings* **obtain** *result* *final-state* **where** *out-eq*:
out = *Some* (*result*, *final-state*)

```

unfolding accepted-fri-opening-transcript-def by blast
have len-bs-roots: length trace-bs = length trace-roots
  using accepted-fri-opening-transcript-shapes(2)[OF fri-openings] .
have layer-bound-roots: layer-idx < length trace-roots
  using layer-bound len-bs-roots by simp
have len-query: length fri-query-idxs = rounds
  using accepted-fri-opening-transcript-shapes(7)[OF fri-openings] .
have raw-bound:
  fri-evidence-layer-idx trace-roots fri-query-idxs round-idx layer-idx <
  fri-evidence-layer-len trace-roots layer-idx
  by (rule accepted-fri-opening-transcript-trace-evidence-layer-idx-bound
    [OF fri-openings round-bound layer-bound-roots])
have raw-bound':
  fri-evidence-layer-idx trace-roots fri-query-idxs round-idx' layer-idx <
  fri-evidence-layer-len trace-roots layer-idx
  by (rule accepted-fri-opening-transcript-trace-evidence-layer-idx-bound
    [OF fri-openings round-bound' layer-bound-roots])
have even-len: 2 dvd fri-evidence-layer-len trace-roots layer-idx
  by (rule accepted-fri-opening-transcript-trace-layer-arithmetic(1)
    [OF fri-openings layer-bound])
have round-product:
  fri-evidence-layer-len trace-roots layer-idx * 2 ^ layer-idx =
  clength * scale
  by (rule accepted-fri-opening-transcript-trace-layer-arithmetic(2)
    [OF fri-openings layer-bound])
have auth-step:
   $\bigwedge i j.$ 
   $i < \text{length } \text{fri-query-idxs} \implies$ 
   $j < \text{length } \text{trace-bs} \implies$ 
  generic-fri-transcript-step-with-authenticated-chunk trace-roots
  trace-bs fri-query-idxs trace-round-layers final-state i j
proof –
  fix i j
  assume i-bound: i < length fri-query-idxs
  and j-bound: j < length trace-bs
  have i-rounds: i < rounds
  using i-bound len-query by simp
  have j-roots: j < length trace-roots
  using j-bound len-bs-roots by simp
  show
  generic-fri-transcript-step-with-authenticated-chunk trace-roots
  trace-bs fri-query-idxs trace-round-layers final-state i j
  by (rule
    accepted-fri-opening-transcript-trace-step-with-authenticated-chunk-at-raw
    [OF fri-openings out-eq i-rounds j-roots])
qed
show ?thesis
proof (cases
  generic-fri-sampled-same-layer-opening-conflict trace-roots trace-bs

```



```

    fri-query-idxs trace-round-layers)
  case True
  have route:
    partial-merkle-inconsistency-bad s (Some (result, final-state)) ∨
    generic-fri-authenticated-slot-recorded-chunk-gap trace-roots
    fri-query-idxs trace-round-layers final-state
  by (rule generic-fri-sampled-same-layer-conflict-imp-merkle-or-slot-gap
    [OF True len-bs-roots auth-step])
  then show ?thesis
  proof
    assume partial-merkle-inconsistency-bad s (Some (result, final-state))
    then show ?thesis
      using out-eq by simp
  next
    assume slot-gap:
      generic-fri-authenticated-slot-recorded-chunk-gap trace-roots
      fri-query-idxs trace-round-layers final-state
    have route-slot-gap:
      trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s out
    unfolding trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-def
    by (intro exI conjI)
      (rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
        rule not-low, rule slot-gap)
    then show ?thesis by simp
  qed
next
case False
then have no-same:
  ¬ generic-fri-sampled-same-layer-opening-conflict trace-roots
  trace-bs fri-query-idxs trace-round-layers
  by simp
have v = v'
  by (rule generic-fri-forced-next-value-eq-if-no-same-layer-conflict
    [OF no-same round-bound round-bound' layer-bound even-len
      raw-bound raw-bound' round-product same-idx forced forced'])
then show ?thesis
  using neq by simp
qed
qed

lemma wp-trace-fri-header-tied-next-value-replay-gap-bound-from-merkle-and-slot:
  fixes M A :: prob
  assumes merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M
  and slot-bound:
    wp-event verify-monad
      (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s) s ≤ A
  shows
    wp-event verify-monad

```

$(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq M + A$
proof –
let $?P = \text{partial-merkle-inconsistency-bad } s$
let $?Q = \text{trace-fri-header-tied-authenticated-slot-recorded-chunk-gap } s$
have *event-le*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq$
 $\text{wp-event verify-monad } (\lambda \text{out. } ?P \text{ out} \vee ?Q \text{ out}) \ s$
by (rule *wp-event-mono-on-support*)
(rule
 $\text{trace-fri-header-tied-next-value-replay-gap-imp-merkle-or-slot-on-support}$)
also have $\dots \leq \text{wp-event verify-monad } ?P \ s +$
 $\text{wp-event verify-monad } ?Q \ s$
by (rule *wp-event-union-bound*)
also have $\dots \leq M + A$
by (intro *add-mono merkle-bound slot-bound*)
finally show *?thesis* .
qed

lemma *verifier-query-round-after-index-trace-successor-recorded-step-from-previous*:
fixes $s \ t :: ('f, 'a) \text{protocol-channel-scheme}$
assumes *outcome*:
 $\text{Some } ((\), t) \in$
 set-dist
 $(\text{execute}$
 $(\text{verifier-query-round-after-index-program}$
 $\text{fr } f\text{-fl } f\text{-final as fl final raw}) \ s)$
and *recorded*:
 $\text{query-round-fri-layer-transcripts } (\text{index } (to\text{-nat raw}))$
 $(\text{map snd } f\text{-fl}) \ (\text{map snd fl}) \ \text{round-chunk trace-layers}$
 $\text{composition-layers}$
and *transcript*: $P\text{Transcript } s = \text{round-chunk } @ \ P\text{Transcript } t$
and *j-bound*: $\text{Suc } j < \text{length } f\text{-fl}$
and *all-layer-bounds*:
 $\bigwedge k. k < \text{length } f\text{-fl} \implies$
 $0 < \text{fri-layer-lengths } (\text{length } f\text{-fl}) \ (\text{clength } * \text{scale}) ! k \wedge$
 $\text{fri-layer-indices } (\text{length } f\text{-fl}) \ (\text{index } (to\text{-nat raw}))$
 $(\text{clength } * \text{scale}) ! k <$
 $\text{fri-layer-lengths } (\text{length } f\text{-fl}) \ (\text{clength } * \text{scale}) ! k$
and *previous-step*:
 $\text{fri-layer-step-evidence } (\text{snd } (f\text{-fl} ! j)) \ (\text{fst } (f\text{-fl} ! j))$
 $(\text{fri-layer-lengths } (\text{length } f\text{-fl}) \ (\text{clength } * \text{scale}) ! j)$
 $(\text{fri-layer-indices } (\text{length } f\text{-fl}) \ (\text{index } (to\text{-nat raw}))$
 $(\text{clength } * \text{scale}) ! j)$
 $(2 \wedge j)$
 $(\text{fri-sibling-index}$
 $(\text{fri-layer-lengths } (\text{length } f\text{-fl}) \ (\text{clength } * \text{scale}) ! j)$
 $(\text{fri-layer-indices } (\text{length } f\text{-fl}) \ (\text{index } (to\text{-nat raw}))$
 $(\text{clength } * \text{scale}) ! j))$

```

  xp xp-path xn xn-path
  (fri-layer-indices (length f-fl) (index (to-nat raw))
    (clength * scale) ! j mod
    (fri-layer-lengths (length f-fl) (clength * scale) ! j div 2))
  v
  (trace-layers ! j)
shows
  ∃ xp-path' xn' xn-path' next-value.
  fri-layer-step-evidence (snd (f-fl ! Suc j)) (fst (f-fl ! Suc j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j)
    (fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! Suc j)
    (2 ^ Suc j)
  (fri-sibling-index
    (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j)
    (fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! Suc j))
  v xp-path' xn' xn-path'
  (fri-layer-indices (length f-fl) (index (to-nat raw))
    (clength * scale) ! Suc j mod
    (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j div 2))
  next-value
  (trace-layers ! Suc j)
proof -
  let ?idx = index (to-nat raw)
  from outcome obtain fv s1 f-out s2 s3 c-out s4 where
  query-decommit:
    Some (fv, s1) ∈
      set-dist (execute (mmap (check-decommit-on-query fr ?idx)) s)
  and trace-fri:
    Some (f-out, s2) ∈
      set-dist (execute
        (mfold (?idx, hd fv, clength * scale, 1)
          (receive-query-commits f-fl)) s1)
  and assert-trace:
    Some ((), s3) ∈
      set-dist
        (execute
          (assert (case f-out of (-, f-x, -, -) ⇒ f-x = f-final)) s2)
  and comp-fri:
    Some (c-out, s4) ∈
      set-dist (execute
        (mfold
          (?idx, cp-eval as fv (h ^ ?idx * shift), clength * scale, 1)
          (receive-query-commits fl)) s3)
  and assert-comp:
    Some ((), t) ∈
      set-dist
        (execute

```

```

      (assert (case c-out of (-, x, -, -) ⇒ x = final)) s4)
    unfolding verifier-query-round-after-index-program-def
    by (auto simp: Let-def split: prod.splits elim!: set-dist-bindE)
from check-decommit-on-query-outcome[OF query-decommit]
obtain query-paths query-chunk where
  query-chunk-shape:
    query-decommitment-transcript ?idx fv query-paths query-chunk
  and tr-s: PTranscript s = query-chunk @ PTranscript s1
  by blast
from receive-query-commits-layers-outcome[OF trace-fri]
obtain trace-layer-chunks trace-fri-chunk where
  trace-layers-actual:
    fri-layers-transcript (length f-fl) (clength * scale)
    trace-layer-chunks trace-fri-chunk
  and tr-s1: PTranscript s1 = trace-fri-chunk @ PTranscript s2
  by blast
have s3-eq: s3 = s2
  using assert-trace unfolding assert-def
  by (cases case f-out of (-, f-x, -, -) ⇒ f-x = f-final)
    (auto simp: throw-no-outcome)
from receive-query-commits-layers-outcome[OF comp-fri]
obtain comp-layer-chunks comp-fri-chunk where
  comp-layers-actual:
    fri-layers-transcript (length fl) (clength * scale)
    comp-layer-chunks comp-fri-chunk
  and tr-s3: PTranscript s3 = comp-fri-chunk @ PTranscript s4
  by blast
have t-eq: t = s4
  using assert-comp unfolding assert-def
  by (cases case c-out of (-, x, -, -) ⇒ x = final)
    (auto simp: throw-no-outcome)
let ?actual-chunk = query-chunk @ trace-fri-chunk @ comp-fri-chunk
have actual-recorded:
  query-round-fri-layer-transcripts ?idx (map snd f-fl) (map snd fl)
    ?actual-chunk trace-layer-chunks comp-layer-chunks
  unfolding query-round-fri-layer-transcripts-def
  by (intro exI[of - query-chunk] exI[of - trace-fri-chunk]
    exI[of - comp-fri-chunk] exI[of - fv] exI[of - query-paths] conjI)
    (use query-chunk-shape trace-layers-actual comp-layers-actual in
      ⟨simp-all add: mult.commute⟩)
have actual-transcript:
  PTranscript s = ?actual-chunk @ PTranscript t
  using tr-s tr-s1 tr-s3 s3-eq t-eq by simp
have actual-chunk-eq: ?actual-chunk = round-chunk
  using actual-transcript transcript by simp
have trace-layer-chunks-eq:
  trace-layer-chunks = trace-layers
  by (rule query-round-fri-layer-transcripts-unique(1)
    [OF actual-recorded[unfolded actual-chunk-eq] recorded])

```

```

have idx-bound: ?idx < clength * scale
  by (rule index-less-domain)
have map-eq:
  receive-query-commits f-fl = map fri-layer-opening-step f-fl
  unfolding receive-query-commits-def fri-layer-opening-step-def
  by (simp add: fun-eq-iff split: prod.splits)
have previous-step-actual:
  fri-layer-step-evidence (snd (f-fl ! j)) (fst (f-fl ! j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! j)
    (fri-layer-indices (length f-fl) ?idx (clength * scale) ! j)
    (1 * 2 ^ j)
    (fri-sibling-index
      (fri-layer-lengths (length f-fl) (clength * scale) ! j)
      (fri-layer-indices (length f-fl) ?idx (clength * scale) ! j))
  xp xp-path xn xn-path
    (fri-layer-indices (length f-fl) ?idx (clength * scale) ! j mod
      (fri-layer-lengths (length f-fl) (clength * scale) ! j div 2))
  v
    (trace-layer-chunks ! j)
  using previous-step trace-layer-chunks-eq by simp
from mfold-fri-layer-opening-successor-recorded-step-from-previous
  [OF trace-fri[unfolded map-eq] trace-layers-actual tr-s1 idx-bound
    all-layer-bounds j-bound previous-step-actual]
obtain sxp-path sxn sxn-path snext where succ-actual:
  fri-layer-step-evidence (snd (f-fl ! Suc j)) (fst (f-fl ! Suc j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j)
    (fri-layer-indices (length f-fl) ?idx (clength * scale) ! Suc j)
    (1 * 2 ^ Suc j)
    (fri-sibling-index
      (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j)
      (fri-layer-indices (length f-fl) ?idx (clength * scale) ! Suc j))
  v sxp-path sxn sxn-path
    (fri-layer-indices (length f-fl) ?idx (clength * scale) ! Suc j mod
      (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j div 2))
  snext
    (trace-layer-chunks ! Suc j)
  by blast
have succ:
  fri-layer-step-evidence (snd (f-fl ! Suc j)) (fst (f-fl ! Suc j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j)
    (fri-layer-indices (length f-fl) ?idx (clength * scale) ! Suc j)
    (2 ^ Suc j)
    (fri-sibling-index
      (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j)
      (fri-layer-indices (length f-fl) ?idx (clength * scale) ! Suc j))
  v sxp-path sxn sxn-path
    (fri-layer-indices (length f-fl) ?idx (clength * scale) ! Suc j mod
      (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j div 2))
  snext

```

```

    (trace-layers ! Suc j)
  using succ-actual trace-layer-chunks-eq by simp
show ?thesis
  using succ by blast
qed

lemma accepted-fri-opening-transcript-trace-successor-recorded-step-from-previous-at:
  assumes fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers
  and out-eq: out = Some (result, final-state)
  and round-bound: i < rounds
  and layer-bound: Suc j < length trace-roots
  and all-layer-bounds:
     $\bigwedge \text{raw } k. k < \text{length trace-roots} \implies$ 
    0 < fri-layer-lengths (length trace-roots)
    (clength * scale) ! k  $\wedge$ 
    fri-layer-indices (length trace-roots) (index (to-nat raw))
    (clength * scale) ! k <
    fri-layer-lengths (length trace-roots) (clength * scale) ! k
  and previous-step:
    fri-layer-step-evidence (trace-roots ! j)
    (trace-bs ! j)
    (fri-evidence-layer-len trace-roots j)
    (fri-evidence-layer-idx trace-roots query-idxs i j)
    (2 ^ j)
    (fri-sibling-index (fri-evidence-layer-len trace-roots j)
      (fri-evidence-layer-idx trace-roots query-idxs i j))
    xp xp-path xn xn-path
    (fri-evidence-next-idx trace-roots query-idxs i j)
    v
    (trace-round-layers ! i ! j)
  shows
     $\exists \text{xp-path}' \text{xn}' \text{xn-path}' \text{next-value}.$ 
    fri-layer-step-evidence (trace-roots ! Suc j)
    (trace-bs ! Suc j)
    (fri-evidence-layer-len trace-roots (Suc j))
    (fri-evidence-layer-idx trace-roots query-idxs i (Suc j))
    (2 ^ Suc j)
    (fri-sibling-index
      (fri-evidence-layer-len trace-roots (Suc j))
      (fri-evidence-layer-idx trace-roots query-idxs i
        (Suc j)))
    v xp-path' xn' xn-path'
    (fri-evidence-next-idx trace-roots query-idxs i (Suc j))
    next-value
    (trace-round-layers ! i ! Suc j)
proof -

```

```

from fri-openings obtain result' final-state' fr f-fl as fl query-state
  raw-idxs query-chunks where
    out-def: out = Some (result', final-state')
    and trace-roots-eq: trace-roots = map snd f-fl
    and trace-bs-eq: trace-bs = map fst f-fl
    and composition-roots-eq: composition-roots = map snd fl
    and outcome:
      Some (result', final-state') ∈
        set-dist (execute
          (ntimes
            (verifier-query-round-program fr f-fl trace-final as fl
              composition-final)
            rounds)
          query-state)
    and len-raw: length raw-idxs = rounds
    and query-idxs-eq:
      query-idxs = map (λraw. index (to-nat raw)) raw-idxs
    and len-query-chunks: length query-chunks = rounds
    and transcript-query:
      PTranscript query-state =
        List.concat query-chunks @ PTranscript final-state'
    and query-chunk:
       $\bigwedge i. i < rounds \implies$ 
        verifier-query-round-chunk (query-idxs ! i)
        trace-roots composition-roots (query-chunks ! i)
    and replay-lookup:
       $\bigwedge i. i < rounds \implies$ 
        fmllookup (HashMap final-state')
        (QueryIndexChallenge (PQueryCounter query-state + i)
          (state-after-query-chunks (PState query-state) query-chunks i)) =
          Some (raw-idxs ! i)
    and recorded:
       $\bigwedge i. i < rounds \implies$ 
        query-round-fri-layer-transcripts (query-idxs ! i)
        trace-roots composition-roots (query-chunks ! i)
        (trace-round-layers ! i) (composition-round-layers ! i)
    unfolding accepted-fri-opening-transcript-def by blast
have result-eq: result' = result
  using out-def out-eq by simp
have final-state-eq: final-state' = final-state
  using out-def out-eq by simp
have outcome':
  Some (result, final-state) ∈
    set-dist (execute
      (ntimes
        (verifier-query-round-program fr f-fl trace-final as fl
          composition-final)
        rounds)
      query-state)

```

```

using outcome result-eq final-state-eq by simp
let ?round =
  verifier-query-round-program fr f-fl trace-final as fl composition-final
from query-opening-ntimes-outcome-decomp-at[OF outcome' round-bound]
obtain prefix x suffix s-i s-suc where
  prefix:
    Some (prefix, s-i) ∈
      set-dist (execute (ntimes ?round i) query-state)
and round:
    Some (x, s-suc) ∈ set-dist (execute ?round s-i)
and suffix:
    Some (suffix, final-state) ∈
      set-dist (execute (ntimes ?round (rounds - Suc i)) s-suc)
by blast
have x-eq: x = ()
by (cases x) simp
show ?thesis
proof (rule query-opening-ntimes-prefix-transcript-state-alignment
  [OF prefix])
fix prefix-query-idxs prefix-chunks
assume len-prefix-idxs: length prefix-query-idxs = i
and len-prefix-chunks: length prefix-chunks = i
and transcript-prefix:
  PTranscript query-state =
    List.concat prefix-chunks @ PTranscript s-i
and state-i:
  PState s-i =
    state-after-query-chunks (PState query-state) prefix-chunks i
and counter-i: PQueryCounter s-i = PQueryCounter query-state + i
and prefix-ext: query-state ≤ s-i
and prefix-rounds:
   $\bigwedge k. k < i \implies$ 
    verifier-query-round-chunk (prefix-query-idxs ! k)
    (map snd f-fl) (map snd fl) (prefix-chunks ! k)
from round[unfolded x-eq verifier-query-round-program-alt-def]
obtain raw s0 where
  raw-out:
    Some (raw, s0) ∈
      set-dist (execute receive-query-index-challenge s-i)
and after:
    Some ((), s-suc) ∈
      set-dist (execute
        (verifier-query-round-after-index-program fr f-fl trace-final as
          fl composition-final raw) s0)
by (auto elim!: set-dist-bindE)
have tr-s0: PTranscript s0 = PTranscript s-i
using receive-query-index-challenge-outcome[OF raw-out] by simp
show ?thesis
proof (rule verifier-query-round-program-outcome[OF round[unfolded x-eq]])

```



```

fix raw' idx chunk
assume idx-eq: idx = index (to-nat raw')
and chunk-shape:
  verifier-query-round-chunk idx (map snd f-fl) (map snd fl)
  chunk
and transcript-round:
  PTranscript s-i = chunk @ PTranscript s-suc
and lookup-suc:
  fmlookup (HashMap s-suc)
  (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
  Some raw'
have raw-eq': raw = raw'
proof –
  have lookup-s0:
    fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
    Some raw
  using receive-query-index-challenge-outcome[OF raw-out] by simp
have s0-suc: s0 ≤ s-suc
  using verifier-query-round-after-index-program-outcome[OF after]
  by simp
have lookup-suc-raw:
  fmlookup (HashMap s-suc)
  (QueryIndexChallenge (PQueryCounter s-i) (PState s-i)) =
  Some raw
  using hash-extension-lookup[OF lookup-s0 s0-suc] .
show ?thesis
  using lookup-suc lookup-suc-raw by simp
qed
show ?thesis
proof (rule query-opening-ntimes-suffix-transcript-state-alignment
  [OF suffix])
  fix suffix-query-idxs suffix-chunks
  assume len-suffix-idxs:
    length suffix-query-idxs = rounds – Suc i
  and len-suffix-chunks:
    length suffix-chunks = rounds – Suc i
  and transcript-suffix:
    PTranscript s-suc =
    List.concat suffix-chunks @ PTranscript final-state
  and suffix-ext: s-suc ≤ final-state
  and suffix-rounds:
     $\bigwedge k. k < \text{rounds} - \text{Suc } i \implies$ 
    verifier-query-round-chunk (suffix-query-idxs ! k)
    (map snd f-fl) (map snd fl) (suffix-chunks ! k)
  let ?selected-chunks = prefix-chunks @ chunk # suffix-chunks
  have len-selected-chunks: length ?selected-chunks = rounds
  using len-prefix-chunks len-suffix-chunks round-bound by simp
  have selected-chunk-lens:

```

```

 $\bigwedge k. k < \text{rounds} \implies$ 
  length (?selected-chunks ! k) =
    verifier-query-round-transcript-length (query-idxs ! k)
      (map snd f-fl) (map snd fl)
proof -
  fix k
  assume k-bound: k < rounds
  show
    length (?selected-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  proof (cases k < i)
    case True
      have nth-eq: ?selected-chunks ! k = prefix-chunks ! k
        using True len-prefix-chunks by (simp add: nth-append)
      have
        length (prefix-chunks ! k) =
          verifier-query-round-transcript-length
            (prefix-query-idxs ! k) (map snd f-fl) (map snd fl)
        by (rule verifier-query-round-chunk-length
            [OF prefix-rounds[OF True]])
      then show ?thesis
        using nth-eq
        by (simp add:
            verifier-query-round-transcript-length-index-irrelevant)
    next
      case False
      show ?thesis
      proof (cases k = i)
        case True
          have nth-eq: ?selected-chunks ! k = chunk
            using True len-prefix-chunks by (simp add: nth-append)
          have
            length chunk =
              verifier-query-round-transcript-length idx
                (map snd f-fl) (map snd fl)
            by (rule verifier-query-round-chunk-length[OF chunk-shape])
          then show ?thesis
            using nth-eq
            by (simp add:
                verifier-query-round-transcript-length-index-irrelevant)
        next
          case False
          let ?k = k - Suc i
          have ge: Suc i ≤ k
            using ⟨¬ k < i⟩ False by linarith
          have k-bound': ?k < rounds - Suc i
            using ge k-bound by linarith
          have nth-eq: ?selected-chunks ! k = suffix-chunks ! ?k

```

```

    using ge len-prefix-chunks by (simp add: nth-append)
  have
    length (suffix-chunks ! ?k) =
      verifier-query-round-transcript-length
        (suffix-query-idxs ! ?k) (map snd f-fl) (map snd fl)
    by (rule verifier-query-round-chunk-length
        [OF suffix-rounds[OF k-bound']])
  then show ?thesis
    using nth-eq
    by (simp add:
        verifier-query-round-transcript-length-index-irrelevant)
qed
qed
qed
have query-chunk-f:
   $\bigwedge k. k < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! k)
      (map snd f-fl) (map snd fl) (query-chunks ! k)
  using query-chunk trace-roots-eq composition-roots-eq by simp
have query-chunk-lens:
   $\bigwedge k. k < \text{rounds} \implies$ 
    length (query-chunks ! k) =
      verifier-query-round-transcript-length (query-idxs ! k)
        (map snd f-fl) (map snd fl)
  using query-chunk-f verifier-query-round-chunk-length by blast
have selected-transcript:
  PTranscript query-state =
    List.concat ?selected-chunks @ PTranscript final-state
  using transcript-prefix transcript-round transcript-suffix by simp
have concat-eq: List.concat ?selected-chunks = List.concat query-chunks
  using selected-transcript transcript-query final-state-eq by simp
have chunks-eq: ?selected-chunks = query-chunks
proof -
  have same-lens:
     $\bigwedge k. k < \text{length } ?\text{selected-chunks} \implies$ 
      length (?selected-chunks ! k) = length (query-chunks ! k)
    using selected-chunk-lens query-chunk-lens len-selected-chunks
    by simp
  have concat-eq-empty:
    List.concat ?selected-chunks @ [] = List.concat query-chunks
    using concat-eq by simp
  have len-eq: length ?selected-chunks = length query-chunks
    using len-selected-chunks len-query-chunks by simp
  have ?selected-chunks = query-chunks  $\wedge$  ([] :: 'f list) = []
    by (rule concat-append-eq-concat-same-chunk-lengths
        [OF len-eq same-lens concat-eq-empty])
  then show ?thesis
    by simp
qed

```

```

have prefix-state-eq:
  state-after-query-chunks (PState query-state) prefix-chunks i =
    state-after-query-chunks (PState query-state) query-chunks i
proof —
  have
    state-after-query-chunks (PState query-state) ?selected-chunks i =
      state-after-query-chunks (PState query-state) prefix-chunks i
    by (rule state-after-query-chunks-append-prefix
      [OF len-prefix-chunks])
  then show ?thesis
    using chunks-eq by simp
qed
have lookup-final:
  fmlookup (HashMap final-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    Some raw'
proof —
  have lookup-suc':
    fmlookup (HashMap s-suc)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
      Some raw'
    using lookup-suc state-i counter-i by simp
  have lookup-final-prefix:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state)
          prefix-chunks i)) =
      Some raw'
    using hash-extension-lookup [OF lookup-suc' suffix-ext] .
  show ?thesis
    using lookup-final-prefix prefix-state-eq by simp
qed
have raw-idx-eq: raw' = raw-idxs ! i
  using lookup-final replay-lookup [OF round-bound] final-state-eq
  by simp
have idx-eq-query: idx = query-idxs ! i
  using idx-eq raw-idx-eq query-idxs-eq len-raw round-bound by simp
have selected-i: ?selected-chunks ! i = chunk
  using len-prefix-chunks by (simp add: nth-append)
have chunk-i-eq: chunk = query-chunks ! i
  using chunks-eq selected-i by simp
have recorded-i:
  query-round-fri-layer-transcripts (index (to-nat raw))
    (map snd f-fl) (map snd fl) chunk (trace-round-layers ! i)
    (composition-round-layers ! i)
  using recorded [OF round-bound] idx-eq idx-eq-query raw-eq'

```

```

    chunk-i-eq trace-roots-eq composition-roots-eq
  by simp
have tr-s0-suc:
  PTranscript s0 = chunk @ PTranscript s-suc
  using tr-s0 transcript-round by simp
have suc-j-bound-fl: Suc j < length f-fl
  using layer-bound trace-roots-eq by simp
have all-layer-bounds-fl:
   $\bigwedge k. k < \text{length } f\text{-fl} \implies$ 
    0 < fri-layer-lengths (length f-fl) (clength * scale) ! k  $\wedge$ 
    fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! k <
    fri-layer-lengths (length f-fl) (clength * scale) ! k
  using all-layer-bounds trace-roots-eq by simp
have raw-idx-query: index (to-nat raw) = query-idxs ! i
  using idx-eq idx-eq-query raw-eq' by simp
have previous-step-fl:
  fri-layer-step-evidence (snd (f-fl ! j)) (fst (f-fl ! j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! j)
    (fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! j)
    (2 ^ j)
    (fri-sibling-index
      (fri-layer-lengths (length f-fl) (clength * scale) ! j)
      (fri-layer-indices (length f-fl) (index (to-nat raw))
        (clength * scale) ! j))
    xp xp-path xn xn-path
    (fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! j mod
      (fri-layer-lengths (length f-fl) (clength * scale) ! j div 2))
    v
    (trace-round-layers ! i ! j)
  using previous-step trace-roots-eq trace-bs-eq raw-idx-query
    layer-bound
  unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
    fri-evidence-next-idx-def
  by simp
from verifier-query-round-after-index-trace-successor-recorded-step-from-previous
  [OF after recorded-i tr-s0-suc suc-j-bound-fl
    all-layer-bounds-fl previous-step-fl]
obtain sxp-path sxn sxn-path snext where succ-fl:
  fri-layer-step-evidence (snd (f-fl ! Suc j))
    (fst (f-fl ! Suc j))
    (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j)
    (fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! Suc j)
    (2 ^ Suc j)
    (fri-sibling-index
      (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j)

```

```

      (fri-layer-indices (length f-fl) (index (to-nat raw))
        (clength * scale) ! Suc j))
    v sxp-path sxn sxn-path
    (fri-layer-indices (length f-fl) (index (to-nat raw))
      (clength * scale) ! Suc j mod
      (fri-layer-lengths (length f-fl) (clength * scale) ! Suc j
        div 2))
    snext
    (trace-round-layers ! i ! Suc j)
  by blast
have succ:
  fri-layer-step-evidence (trace-roots ! Suc j)
  (trace-bs ! Suc j)
  (fri-evidence-layer-len trace-roots (Suc j))
  (fri-evidence-layer-idx trace-roots query-idxs i (Suc j))
  (2 ^ Suc j)
  (fri-sibling-index
    (fri-evidence-layer-len trace-roots (Suc j))
    (fri-evidence-layer-idx trace-roots query-idxs i
      (Suc j)))
  v sxp-path sxn sxn-path
  (fri-evidence-next-idx trace-roots query-idxs i (Suc j))
  snext
  (trace-round-layers ! i ! Suc j)
using succ-fl trace-roots-eq trace-bs-eq raw-idx-query layer-bound
unfolding fri-evidence-layer-len-def fri-evidence-layer-idx-def
  fri-evidence-next-idx-def
by simp
show ?thesis
using succ by blast
qed
qed
qed
qed

```

lemma *trace-fri-header-tied-successor-replay-gap-imp-merkle-or-slot-on-support:*

assumes *support: out ∈ set-dist (execute verify-monad s)*

and *gap: trace-fri-header-tied-successor-replay-gap s out*

shows

partial-merkle-inconsistency-bad s out ∨

trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s out

proof –

from *gap* **have** *sampled:*

trace-fri-header-tied-sampled-successor-opening-conflict s out

unfolding *trace-fri-header-tied-successor-replay-gap-def* **by** *simp*

from *sampled* **obtain** *trace-roots trace-bs trace-final fri-dg*

composition-roots composition-bs composition-final fri-query-idxs

trace-round-layers composition-round-layers fr as rest trace-openings

trace-table

where *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *header*:
verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest
and *partial*:
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *cand*: *partial-trace-table-candidate trace-table trace-openings*
and *not-low*: $\neg$ *trace-table-low-degree trace-table*
and *generic*:
generic-fri-sampled-successor-opening-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers
unfolding *trace-fri-header-tied-sampled-successor-opening-conflict-def*
by *blast*
from *fri-openings* **obtain** *result final-state* **where** *out-eq*:
out = Some (result, final-state)
unfolding *accepted-fri-opening-transcript-def* **by** *blast*
have *len-bs-roots*: *length trace-bs = length trace-roots*
using *accepted-fri-opening-transcript-shapes(2)[OF fri-openings]* .
have *len-query*: *length fri-query-idxs = rounds*
using *accepted-fri-opening-transcript-shapes(7)[OF fri-openings]* .
have *all-layer-bounds*:
 $\bigwedge \text{raw } k. k < \text{length trace-roots} \implies$
 $0 < \text{fri-layer-lengths } (\text{length trace-roots})$
 $(\text{clength} * \text{scale}) ! k \wedge$
 $\text{fri-layer-indices } (\text{length trace-roots}) (\text{index } (\text{to-nat raw}))$
 $(\text{clength} * \text{scale}) ! k <$
 $\text{fri-layer-lengths } (\text{length trace-roots}) (\text{clength} * \text{scale}) ! k$
by (*rule accepted-fri-opening-transcript-trace-raw-layer-bound*
[OF fri-openings])
have *source-step*:
 $\bigwedge \text{source-round layer-idx } v.$
 $\text{source-round} < \text{length fri-query-idxs} \implies$
 $\text{Suc layer-idx} < \text{length trace-bs} \implies$
 $\text{generic-fri-round-forced-next-value trace-roots trace-bs}$
 $\text{fri-query-idxs trace-round-layers source-round layer-idx } v \implies$
 $\exists \text{xp-path } \text{xn } \text{xn-path}.$
fri-layer-step-evidence
 $(\text{trace-roots} ! \text{Suc layer-idx})$
 $(\text{trace-bs} ! \text{Suc layer-idx})$
 $(\text{fri-evidence-layer-len trace-roots } (\text{Suc layer-idx}))$
 $(\text{fri-evidence-layer-idx trace-roots fri-query-idxs}$
 $\text{source-round } (\text{Suc layer-idx}))$
 $(2 \wedge \text{Suc layer-idx})$
 $(\text{fri-sibling-index}$
 $(\text{fri-evidence-layer-len trace-roots } (\text{Suc layer-idx}))$

```

    (fri-evidence-layer-idx trace-roots fri-query-idxs
     source-round (Suc layer-idx)))
  v xp-path xn xn-path
  (fri-evidence-next-idx trace-roots fri-query-idxs
   source-round (Suc layer-idx))
  (fri-evidence-next-value trace-roots trace-bs
   fri-query-idxs source-round (Suc layer-idx) v xn)
  (trace-round-layers ! source-round ! Suc layer-idx)
proof -
  fix source-round layer-idx v
  assume source-round-bound: source-round < length fri-query-idxs
  and layer-bound': Suc layer-idx < length trace-bs
  and forced:
    generic-fri-round-forced-next-value trace-roots trace-bs
    fri-query-idxs trace-round-layers source-round layer-idx v
  have source-round-rounds: source-round < rounds
  using source-round-bound len-query by simp
  have layer-bound-roots: Suc layer-idx < length trace-roots
  using layer-bound' len-bs-roots by simp
  from forced obtain fxp fxp-path fxn fxn-path where previous-step:
    fri-layer-step-evidence (trace-roots ! layer-idx)
    (trace-bs ! layer-idx)
    (fri-evidence-layer-len trace-roots layer-idx)
    (fri-evidence-layer-idx trace-roots fri-query-idxs
     source-round layer-idx)
    (2 ^ layer-idx)
    (fri-sibling-index
     (fri-evidence-layer-len trace-roots layer-idx)
     (fri-evidence-layer-idx trace-roots fri-query-idxs
      source-round layer-idx))
    fxp fxp-path fxn fxn-path
    (fri-evidence-next-idx trace-roots fri-query-idxs
     source-round layer-idx)
    v
    (trace-round-layers ! source-round ! layer-idx)
  unfolding generic-fri-round-forced-next-value-def by blast
  from accepted-fri-opening-transcript-trace-successor-recorded-step-from-previous-at
    [OF fri-openings out-eq source-round-rounds layer-bound-roots
     all-layer-bounds previous-step]
  obtain sxp-path sxn sxn-path snext where succ:
    fri-layer-step-evidence (trace-roots ! Suc layer-idx)
    (trace-bs ! Suc layer-idx)
    (fri-evidence-layer-len trace-roots (Suc layer-idx))
    (fri-evidence-layer-idx trace-roots fri-query-idxs
     source-round (Suc layer-idx))
    (2 ^ Suc layer-idx)
    (fri-sibling-index
     (fri-evidence-layer-len trace-roots (Suc layer-idx))
     (fri-evidence-layer-idx trace-roots fri-query-idxs
      source-round (Suc layer-idx)))

```



```

      source-round (Suc layer-idx)))
    v xsp-path sxn sxn-path
    (fri-evidence-next-idx trace-roots fri-query-idxs
      source-round (Suc layer-idx))
    snext
    (trace-round-layers ! source-round ! Suc layer-idx)
  by blast
have snext-eq:
  snext =
    fri-evidence-next-value trace-roots trace-bs
    fri-query-idxs source-round (Suc layer-idx) v sxn
  using fri-layer-step-evidenceD(3)[OF succ]
  unfolding fri-evidence-next-value-def by simp
show
  ∃ xp-path xn xn-path.
    fri-layer-step-evidence
      (trace-roots ! Suc layer-idx)
      (trace-bs ! Suc layer-idx)
      (fri-evidence-layer-len trace-roots (Suc layer-idx))
      (fri-evidence-layer-idx trace-roots fri-query-idxs
        source-round (Suc layer-idx))
      (2 ^ Suc layer-idx)
      (fri-sibling-index
        (fri-evidence-layer-len trace-roots (Suc layer-idx))
        (fri-evidence-layer-idx trace-roots fri-query-idxs
          source-round (Suc layer-idx)))
      v xp-path xn xn-path
      (fri-evidence-next-idx trace-roots fri-query-idxs
        source-round (Suc layer-idx))
      (fri-evidence-next-value trace-roots trace-bs
        fri-query-idxs source-round (Suc layer-idx) v xn)
      (trace-round-layers ! source-round ! Suc layer-idx)
    using succ snext-eq by blast
qed
have same-layer:
  generic-fri-sampled-same-layer-opening-conflict trace-roots trace-bs
  fri-query-idxs trace-round-layers
  by (rule generic-fri-sampled-successor-conflict-imp-same-layer-with-source-steps
    [OF generic len-bs-roots source-step])
have auth-step:
  ∧i j.
    i < length fri-query-idxs ⇒
    j < length trace-bs ⇒
    generic-fri-transcript-step-with-authenticated-chunk trace-roots
    trace-bs fri-query-idxs trace-round-layers final-state i j
proof -
  fix i j
  assume i-bound: i < length fri-query-idxs
  and j-bound: j < length trace-bs

```

```

have i-rounds:  $i < \text{rounds}$ 
  using i-bound len-query by simp
have j-roots:  $j < \text{length trace-roots}$ 
  using j-bound len-bs-roots by simp
show
  generic-fri-transcript-step-with-authenticated-chunk trace-roots
    trace-bs fri-query-idxs trace-round-layers final-state i j
  by (rule
    accepted-fri-opening-transcript-trace-step-with-authenticated-chunk-at-raw
    [OF fri-openings out-eq i-rounds j-roots])
qed
have route:
  partial-merkle-inconsistency-bad s (Some (result, final-state))  $\vee$ 
  generic-fri-authenticated-slot-recorded-chunk-gap trace-roots
    fri-query-idxs trace-round-layers final-state
  by (rule generic-fri-sampled-same-layer-conflict-imp-merkle-or-slot-gap
    [OF same-layer len-bs-roots auth-step])
then show ?thesis
proof
  assume partial-merkle-inconsistency-bad s (Some (result, final-state))
  then show ?thesis
    using out-eq by simp
next
  assume slot-gap:
    generic-fri-authenticated-slot-recorded-chunk-gap trace-roots
      fri-query-idxs trace-round-layers final-state
  have route-slot-gap:
    trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s out
  unfolding trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-def
  by (intro exI conjI)
    (rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
      rule not-low, rule slot-gap)
  then show ?thesis by simp
qed
qed

lemma wp-trace-fri-header-tied-successor-replay-gap-bound-from-merkle-and-slot:
fixes  $M A :: \text{prob}$ 
assumes merkle-bound:
  wp-event verify-monad (partial-merkle-inconsistency-bad s)  $s \leq M$ 
and slot-bound:
  wp-event verify-monad
    (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap s)  $s \leq A$ 
shows
  wp-event verify-monad
    (trace-fri-header-tied-successor-replay-gap s)  $s \leq M + A$ 
proof –
  let  $?P = \text{partial-merkle-inconsistency-bad } s$ 
  let  $?Q = \text{trace-fri-header-tied-authenticated-slot-recorded-chunk-gap } s$ 

```

```

have event-le:
  wp-event verify-monad
    (trace-fri-header-tied-successor-replay-gap s) s ≤
    wp-event verify-monad ( $\lambda out. ?P\ out \vee ?Q\ out$ ) s
  by (rule wp-event-mono-on-support)
    (rule
      trace-fri-header-tied-successor-replay-gap-imp-merkle-or-slot-on-support)
also have ... ≤ wp-event verify-monad ?P s +
  wp-event verify-monad ?Q s
  by (rule wp-event-union-bound)
also have ... ≤ M + A
  by (intro add-mono merkle-bound slot-bound)
finally show ?thesis .
qed

end

end

```

```

theory Soundness-FRI-Trace-Sibling-Evidence
  imports Soundness-FRI-Value-Aligned-Events
begin

```

Proof-only refinement of the trace FRI sibling mismatch branch.

The existing trace candidate predicate records agreement with the trace-query openings only. The first trace FRI sibling value is read from the FRI layer transcript, so the sibling branch cannot be closed from the current candidate predicate alone. This theory isolates the exact missing evidence: a trace candidate must also agree with the recorded first-layer FRI sibling values.

```

context soundness
begin

```

definition *trace-table-agrees-with-recorded-first-fri-siblings*

```

:: 'f list ⇒ 'f list ⇒ 'f list ⇒ nat list ⇒
  'f list list list ⇒ bool

```

where

```

trace-table-agrees-with-recorded-first-fri-siblings trace-table
  trace-roots trace-bs query-idxs trace-round-layers  $\longleftrightarrow$ 
  ( $\forall$  round-idx xp xp-path xn xn-path.
    round-idx < length query-idxs  $\longrightarrow$ 
    0 < length trace-roots  $\longrightarrow$ 
    0 < length trace-bs  $\longrightarrow$ 
    fri-layer-step-evidence
      (trace-roots ! 0)
      (trace-bs ! 0)
      (fri-evidence-layer-len trace-roots 0)
      (fri-evidence-layer-idx trace-roots query-idxs round-idx 0))

```

1
 $(\text{fri-sibling-index } (\text{fri-evidence-layer-len } \text{trace-roots } 0)$
 $\quad (\text{fri-evidence-layer-idx } \text{trace-roots } \text{query-idxs } \text{round-idx } 0))$
 $\text{xp } \text{xp-path } \text{xn } \text{xn-path}$
 $(\text{fri-evidence-next-idx } \text{trace-roots } \text{query-idxs } \text{round-idx } 0)$
 $(\text{fri-evidence-next-value } \text{trace-roots } \text{trace-bs } \text{query-idxs}$
 $\quad \text{round-idx } 0 \text{ xp } \text{xn})$
 $(\text{trace-round-layers } ! \text{round-idx } ! 0) \longrightarrow$
 $\text{trace-table } !$
 $\text{fri-sibling-index } (\text{fri-evidence-layer-len } \text{trace-roots } 0)$
 $\quad (\text{fri-evidence-layer-idx } \text{trace-roots } \text{query-idxs } \text{round-idx } 0) =$
 $\text{xn})$

definition *partial-trace-table-candidate-with-recorded-first-fri-siblings*

$:: 'f \text{ list} \Rightarrow 'f \text{ authenticated-opening list list} \Rightarrow$
 $\quad 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow \text{nat list} \Rightarrow 'f \text{ list list list} \Rightarrow$
 bool

where

$\text{partial-trace-table-candidate-with-recorded-first-fri-siblings}$
 $\quad \text{trace-table } \text{trace-openings } \text{trace-roots } \text{trace-bs } \text{query-idxs}$
 $\quad \text{trace-round-layers} \longleftarrow \longrightarrow$
 $\text{partial-trace-table-candidate } \text{trace-table } \text{trace-openings} \wedge$
 $\text{trace-table-agrees-with-recorded-first-fri-siblings } \text{trace-table}$
 $\quad \text{trace-roots } \text{trace-bs } \text{query-idxs } \text{trace-round-layers}$

lemma *partial-trace-table-candidate-with-recorded-first-fri-siblingsD:*

assumes

$\text{partial-trace-table-candidate-with-recorded-first-fri-siblings}$
 $\quad \text{trace-table } \text{trace-openings } \text{trace-roots } \text{trace-bs } \text{query-idxs}$
 $\quad \text{trace-round-layers}$

shows $\text{partial-trace-table-candidate } \text{trace-table } \text{trace-openings}$

and $\text{trace-table-agrees-with-recorded-first-fri-siblings } \text{trace-table}$
 $\quad \text{trace-roots } \text{trace-bs } \text{query-idxs } \text{trace-round-layers}$

using *assms*

unfolding *partial-trace-table-candidate-with-recorded-first-fri-siblings-def*

by *blast+*

definition *trace-fri-header-tied-recorded-sibling-candidate-mismatch*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $\quad (\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{trace-fri-header-tied-recorded-sibling-candidate-mismatch } s \text{ out}$
 $\longleftarrow \longrightarrow$
 $(\exists \text{ trace-roots } \text{trace-bs } \text{trace-final } \text{dg } \text{composition-roots}$
 $\quad \text{composition-bs } \text{composition-final } \text{fri-query-idxs } \text{trace-round-layers}$
 $\quad \text{composition-round-layers } \text{fr } \text{as } \text{rest } \text{trace-openings } \text{trace-table}$
 $\quad \text{round-idx } \text{xp } \text{xp-path } \text{xn } \text{xn-path } \text{result } \text{final-state.}$
 $\text{out} = \text{Some } (\text{result}, \text{final-state}) \wedge$
 $\text{accepted-fri-opening-transcript } s \text{ out } \text{trace-roots } \text{trace-bs } \text{trace-final}$

dg composition-roots composition-bs composition-final fri-query-idxs
 trace-round-layers composition-round-layers $\wedge$
 verifier-header-transcript s fr trace-roots trace-final as dg
 composition-roots composition-final rest $\wedge$
 accepted-with-partial-trace-openings s out fr fri-query-idxs
 trace-openings $\wedge$
 partial-trace-table-candidate trace-table trace-openings $\wedge$
 trace-table-agrees-with-recorded-first-fri-siblings trace-table
 trace-roots trace-bs fri-query-idxs trace-round-layers $\wedge$
 $\neg$ trace-table-low-degree trace-table $\wedge$
 round-idx < length fri-query-idxs $\wedge$
 $0 < \text{length trace-bs}$ $\wedge$
 trace-openings ! round-idx $\neq []$ $\wedge$
 opening-index (hd (trace-openings ! round-idx)) =
 fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0 $\wedge$
 opening-value (hd (trace-openings ! round-idx)) = xp $\wedge$
 fri-layer-step-evidence
 (trace-roots ! 0)
 (trace-bs ! 0)
 (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
 1
 (fri-sibling-index (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
 xp xp -path xn xn -path
 (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
 (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
 round-idx 0 xp xn)
 (trace-round-layers ! round-idx ! 0) $\wedge$
 trace-table !
 fri-sibling-index (fri-evidence-layer-len trace-roots 0)
 (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
 $\neq xn$)

definition trace-fri-header-tied-recorded-sibling-candidate-missing

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-recorded-sibling-candidate-missing s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final } dg \text{ composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as rest trace-openings trace-table}$
 $\text{ round-idx } xp \text{ } xp\text{-path } xn \text{ } xn\text{-path result final-state.}$
 $\text{ out} = \text{Some (result, final-state)} \wedge$
 $\text{ accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs trace-final}$
 $dg \text{ composition-roots composition-bs composition-final fri-query-idxs}$
 $\text{ trace-round-layers composition-round-layers} \wedge$
 $\text{ verifier-header-transcript } s \text{ fr trace-roots trace-final as } dg$

composition-roots composition-final rest $\wedge$
accepted-with-partial-trace-openings *s out fr fri-query-idxs*
trace-openings $\wedge$
partial-trace-table-candidate *trace-table trace-openings* $\wedge$
 $\neg$ *trace-table-agrees-with-recorded-first-fri-siblings* *trace-table*
trace-roots trace-bs fri-query-idxs trace-round-layers $\wedge$
 $\neg$ *trace-table-low-degree* *trace-table* $\wedge$
round-idx < length fri-query-idxs $\wedge$
0 < length trace-bs $\wedge$
trace-openings ! round-idx $\neq []$ $\wedge$
opening-index (*hd (trace-openings ! round-idx)*) =
fri-evidence-layer-idx *trace-roots fri-query-idxs round-idx 0* $\wedge$
opening-value (*hd (trace-openings ! round-idx)*) = *xp* $\wedge$
fri-layer-step-evidence
(*trace-roots ! 0*)
(*trace-bs ! 0*)
(*fri-evidence-layer-len* *trace-roots 0*)
(*fri-evidence-layer-idx* *trace-roots fri-query-idxs round-idx 0*)
1
(*fri-sibling-index* (*fri-evidence-layer-len* *trace-roots 0*)
(*fri-evidence-layer-idx* *trace-roots fri-query-idxs round-idx 0*))
xp xp-path xn xn-path
(*fri-evidence-next-idx* *trace-roots fri-query-idxs round-idx 0*)
(*fri-evidence-next-value* *trace-roots trace-bs fri-query-idxs*
round-idx 0 xp xn)
(*trace-round-layers ! round-idx ! 0*) $\wedge$
trace-table !
fri-sibling-index (*fri-evidence-layer-len* *trace-roots 0*)
(*fri-evidence-layer-idx* *trace-roots fri-query-idxs round-idx 0*)
 $\neq$ *xn*)

lemma *generic-trace-sampled-layer-chain-evidence-recorded-first-sibling-agreement:*

assumes *chain:*

generic-fri-sampled-layer-chain-evidence *trace-table-low-degree*
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers

shows

trace-table-agrees-with-recorded-first-fri-siblings *trace-table*
trace-roots trace-bs fri-query-idxs trace-round-layers

proof (*unfold trace-table-agrees-with-recorded-first-fri-siblings-def,*
intro allI impI)

fix *round-idx xp xp-path xn xn-path*

assume *round-bound: round-idx < length fri-query-idxs*

and *roots-nonempty: 0 < length trace-roots*

and *challenges-nonempty: 0 < length trace-bs*

and *step:*

fri-layer-step-evidence
(*trace-roots ! 0*)
(*trace-bs ! 0*)

```

(fri-evidence-layer-len trace-roots 0)
(fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
1
(fri-sibling-index (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
xp xp-path xn xn-path
(fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
(fri-evidence-next-value trace-roots trace-bs fri-query-idxs
  round-idx 0 xp xn)
(trace-round-layers ! round-idx ! 0)
have layer0: layers ! 0 = trace-table
  by (rule generic-fri-sampled-layer-chain-evidenceD(5)[OF chain])
have sampled:
  fri-sampled-table-fold
    (trace-bs ! 0)
    (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    (2 ^ 0)
    (doms ! 0)
    (layers ! 0)
    (trace-round-layers ! round-idx ! 0)
    (layers ! Suc 0 !
      fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
  by (rule generic-fri-sampled-layer-chain-evidence-sample(2)
    [OF chain round-bound challenges-nonempty])
from sampled obtain yp yp-path yn yn-path
  where chunk:
    fri-layer-opening-chunk (fri-evidence-layer-len trace-roots 0)
    yp yp-path yn yn-path (trace-round-layers ! round-idx ! 0)
  and match:
    fri-opening-matches-table (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    (layers ! 0) yp yn
  by (elim fri-sampled-table-foldE)
have step-chunk:
  fri-layer-opening-chunk (fri-evidence-layer-len trace-roots 0)
  xp xp-path xn xn-path (trace-round-layers ! round-idx ! 0)
  by (rule fri-layer-step-evidenceD(4)[OF step])
have yn-eq: yn = xn
  using fri-layer-opening-chunk-values-unique(2)[OF chunk step-chunk]
  by simp
have (layers ! 0) !
  fri-sibling-index (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0) =
  yn
  by (rule fri-opening-matches-tableD(4)[OF match])
then show
  trace-table !
  fri-sibling-index (fri-evidence-layer-len trace-roots 0)

```

$(\text{fri-evidence-layer-idx } \text{trace-roots } \text{fri-query-idxs } \text{round-idx } 0) =$
 xn
using layer0 yn-eq **by** simp
qed

lemma *generic-trace-sampled-layer-chain-evidence-augments-partial-trace-candidate:*
assumes $\text{cand: partial-trace-table-candidate trace-table trace-openings}$
and chain:
 $\text{generic-fri-sampled-layer-chain-evidence trace-table-low-degree}$
 $\text{trace-table (clength - 1) trace-roots trace-bs trace-final}$
 $\text{query-idxs trace-round-layers doms layers}$
shows
 $\text{partial-trace-table-candidate-with-recorded-first-fri-siblings}$
 $\text{trace-table trace-openings trace-roots trace-bs query-idxs}$
 $\text{trace-round-layers}$
unfolding $\text{partial-trace-table-candidate-with-recorded-first-fri-siblings-def}$
by $(\text{intro conjI cand}$
 $\text{generic-trace-sampled-layer-chain-evidence-recorded-first-sibling-agreement}$
 $[\text{OF chain}])$

definition *trace-fri-bad-with-header-tied-augmented-partial-candidate*
 $:: (\text{'f}, \text{'a}) \text{protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{protocol-channel-scheme}) \text{option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-bad-with-header-tied-augmented-partial-candidate } s \text{ out}$
 $\longleftrightarrow$
 $(\exists \text{trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr as rest trace-openings trace-table.}$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{verifier-header-transcript } s \text{ fr trace-roots trace-final as dg}$
 $\text{composition-roots composition-final rest} \wedge$
 $\text{accepted-with-partial-trace-openings } s \text{ out fr fri-query-idxs}$
 $\text{trace-openings} \wedge$
 $\text{partial-trace-table-candidate-with-recorded-first-fri-siblings}$
 $\text{trace-table trace-openings trace-roots trace-bs fri-query-idxs}$
 $\text{trace-round-layers} \wedge$
 $\neg \text{trace-table-low-degree trace-table})$

definition *trace-fri-bad-with-header-tied-augmented-sampled-layer-chain*
 $:: (\text{'f}, \text{'a}) \text{protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (\text{'f}, \text{'a}) \text{protocol-channel-scheme}) \text{option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-bad-with-header-tied-augmented-sampled-layer-chain } s \text{ out}$
 $\longleftrightarrow$
 $(\exists \text{trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$

composition-round-layers fr as rest trace-openings trace-table
doms layers.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers $\wedge$
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate-with-recorded-first-fri-siblings
trace-table trace-openings trace-roots trace-bs fri-query-idxs
trace-round-layers $\wedge$
 $\neg$ *trace-table-low-degree trace-table $\wedge$*
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers)

definition *trace-fri-header-tied-augmented-missing-sampled-layer-chain*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-augmented-missing-sampled-layer-chain s out
 $\longleftrightarrow$
trace-fri-bad-with-header-tied-augmented-partial-candidate s out $\wedge$
 $\neg$ *trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s out*

definition *trace-fri-header-tied-unaugmented-partial-candidate*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-unaugmented-partial-candidate s out
 $\longleftrightarrow$
trace-fri-bad-with-header-tied-partial-candidate s out $\wedge$
 $\neg$ *trace-fri-bad-with-header-tied-augmented-partial-candidate s out*

lemma *trace-fri-bad-with-header-tied-augmented-partial-candidate-imp-plain:*

assumes

trace-fri-bad-with-header-tied-augmented-partial-candidate s out

shows *trace-fri-bad-with-header-tied-partial-candidate s out*

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table

where *fri:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers

and *header:*

verifier-header-transcript s fr trace-roots trace-final as dg

composition-roots composition-final rest
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *cand:*
partial-trace-table-candidate-with-recorded-first-fri-siblings
trace-table trace-openings trace-roots trace-bs fri-query-idxs
trace-round-layers
and *nonlow:* $\neg$ *trace-table-low-degree trace-table*
unfolding *trace-fri-bad-with-header-tied-augmented-partial-candidate-def*
by *blast*
have *plain:* *partial-trace-table-candidate trace-table trace-openings*
by (*rule*
partial-trace-table-candidate-with-recorded-first-fri-siblingsD(1)
[OF cand])
show *?thesis*
unfolding *trace-fri-bad-with-header-tied-partial-candidate-def*
by (*intro exI conjI*)
(rule fri, rule header, rule partial, rule plain, rule nonlow)
qed

lemma *trace-fri-bad-with-header-tied-partial-candidate-imp-augmented-or-unaugmented:*
assumes *trace-fri-bad-with-header-tied-partial-candidate s out*
shows
trace-fri-bad-with-header-tied-augmented-partial-candidate s out $\vee$
trace-fri-header-tied-unaugmented-partial-candidate s out
using *assms*
unfolding *trace-fri-header-tied-unaugmented-partial-candidate-def*
by *blast*

lemma *wp-trace-fri-header-tied-partial-candidate-bound-from-augmented-and-unaugmented:*
assumes *augmented-bound:*
wp-event verify-monad
(trace-fri-bad-with-header-tied-augmented-partial-candidate s) s
 $\leq A$
and *unaugmented-bound:*
wp-event verify-monad
(trace-fri-header-tied-unaugmented-partial-candidate s) s $\leq U$
shows
wp-event verify-monad
(trace-fri-bad-with-header-tied-partial-candidate s) s $\leq A + U$
proof –
have *wp-event verify-monad*
(trace-fri-bad-with-header-tied-partial-candidate s) s $\leq$
wp-event verify-monad
($\lambda out.$
trace-fri-bad-with-header-tied-augmented-partial-candidate s out $\vee$
trace-fri-header-tied-unaugmented-partial-candidate s out) s
by (*rule wp-event-mono*)

(rule
 trace-fri-bad-with-header-tied-partial-candidate-imp-augmented-or-unaugmented)
 also have ... $\leq$
 wp-event verify-monad
 (*trace-fri-bad-with-header-tied-augmented-partial-candidate* s) s +
 wp-event verify-monad
 (*trace-fri-header-tied-unaugmented-partial-candidate* s) s
 by (rule wp-event-union-bound)
 also have ... $\leq A + U$
 by (rule add-mono[*OF augmented-bound unaugmented-bound*])
 finally show ?thesis .
 qed

lemma *trace-fri-bad-with-header-tied-augmented-sampled-layer-chain-imp-plain:*

assumes

trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s out

shows *trace-fri-bad-with-header-tied-sampled-layer-chain* s out

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table doms
layers

where *fri*:

accepted-fri-opening-transcript s out *trace-roots trace-bs*
 trace-final dg composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers

and *header*:

verifier-header-transcript s fr *trace-roots trace-final as dg*
 composition-roots composition-final rest

and *partial*:

accepted-with-partial-trace-openings s out fr *fri-query-idxs*
 trace-openings

and *cand*:

partial-trace-table-candidate-with-recorded-first-fri-siblings
 trace-table trace-openings trace-roots trace-bs fri-query-idxs
 trace-round-layers

and *nonlow*: $\neg$ *trace-table-low-degree trace-table*

and *chain*:

generic-fri-sampled-layer-chain-evidence trace-table-low-degree
 trace-table (clength - 1) trace-roots trace-bs trace-final
 fri-query-idxs trace-round-layers doms layers

unfolding

trace-fri-bad-with-header-tied-augmented-sampled-layer-chain-def

by *blast*

have *plain*: *partial-trace-table-candidate trace-table trace-openings*

by (rule

partial-trace-table-candidate-with-recorded-first-fri-siblingsD(1)
 [*OF cand*])

show ?thesis

unfolding *trace-fri-bad-with-header-tied-sampled-layer-chain-def*
by (*intro exI conjI*)
 (*rule fri, rule header, rule partial, rule plain, rule nonlow,*
rule chain)
qed

lemma *trace-fri-bad-with-header-tied-sampled-layer-chain-imp-augmented:*
assumes *trace-fri-bad-with-header-tied-sampled-layer-chain s out*
shows *trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s out*
proof –
from *assms obtain trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table doms
layers
where *fri:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *header:*
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *cand:* *partial-trace-table-candidate trace-table trace-openings*
and *nonlow:* $\neg$ *trace-table-low-degree trace-table*
and *chain:*
generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength – 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers
unfolding *trace-fri-bad-with-header-tied-sampled-layer-chain-def*
by *blast*
have *augmented:*
partial-trace-table-candidate-with-recorded-first-fri-siblings
trace-table trace-openings trace-roots trace-bs fri-query-idxs
trace-round-layers
by (*rule*
generic-trace-sampled-layer-chain-evidence-augments-partial-trace-candidate
[OF cand chain])
show *?thesis*
unfolding
trace-fri-bad-with-header-tied-augmented-sampled-layer-chain-def
by (*intro exI conjI*)
 (*rule fri, rule header, rule partial, rule augmented, rule nonlow,*
rule chain)
qed

lemma *wp-trace-fri-header-tied-augmented-sampled-layer-chain-bound-from-plain:*
assumes *sampled-bound:*

wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) $s \leq R$
shows
wp-event verify-monad
 (trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s) s
 $\leq R$
proof –
have *wp-event verify-monad*
 (trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s) s
 $\leq$ *wp-event verify-monad*
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) s
by (rule *wp-event-mono*)
 (rule
 trace-fri-bad-with-header-tied-augmented-sampled-layer-chain-imp-plain)
then show ?thesis
by (rule *order-trans*[*OF* - *sampled-bound*])
qed

lemma *wp-trace-fri-header-tied-sampled-layer-chain-bound-from-augmented:*

assumes *sampled-bound:*

wp-event verify-monad
 (trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s) s
 $\leq R$

shows

wp-event verify-monad
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) $s \leq R$

proof –

have *wp-event verify-monad*
 (trace-fri-bad-with-header-tied-sampled-layer-chain s) s
 $\leq$ *wp-event verify-monad*
 (trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s) s
by (rule *wp-event-mono*)
 (rule *trace-fri-bad-with-header-tied-sampled-layer-chain-imp-augmented*)
then show ?thesis
by (rule *order-trans*[*OF* - *sampled-bound*])
qed

lemma *trace-fri-header-tied-augmented-missing-sampled-imp-assignment-obstruction:*

assumes *missing:*

trace-fri-header-tied-augmented-missing-sampled-layer-chain s out

shows

trace-fri-header-tied-sampled-layer-assignment-obstruction s out

proof –

have *bad:*

trace-fri-bad-with-header-tied-augmented-partial-candidate s out

and *not-sampled:*

$\neg$ *trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s out*

using *missing*

unfolding *trace-fri-header-tied-augmented-missing-sampled-layer-chain-def*

```

by blast+
from bad obtain trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr as rest trace-openings trace-table
where fri:
  accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
and header:
  verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
and partial:
  accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
and augmented:
  partial-trace-table-candidate-with-recorded-first-fri-siblings
    trace-table trace-openings trace-roots trace-bs fri-query-idxs
    trace-round-layers
and nonlow:  $\neg$  trace-table-low-degree trace-table
unfolding trace-fri-bad-with-header-tied-augmented-partial-candidate-def
by blast
have cand: partial-trace-table-candidate trace-table trace-openings
by (rule
  partial-trace-table-candidate-with-recorded-first-fri-siblingsD(1)
  [OF augmented])
have partial-generic:
  generic-fri-partial-evidence trace-table-low-degree trace-table
    (clength - 1) trace-roots trace-bs trace-final fri-query-idxs
    trace-round-layers
by (rule accepted-fri-opening-transcript-trace-generic-partial-evidence
  [OF fri])
have no-chain:
 $\bigwedge$  doms layers.  $\neg$ 
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
proof
fix doms layers
assume chain:
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
have trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s out
unfolding
  trace-fri-bad-with-header-tied-augmented-sampled-layer-chain-def
by (intro exI conjI)
  (rule fri, rule header, rule partial, rule augmented, rule nonlow,
    rule chain)
then show False

```

```

    using not-sampled by contradiction
qed
have obstruction:
  generic-fri-sampled-layer-assignment-obstruction
  trace-table-low-degree trace-table (clength - 1) trace-roots
  trace-bs trace-final fri-query-idxs trace-round-layers
  unfolding generic-fri-sampled-layer-assignment-obstruction-def
  using partial-generic no-chain by blast
show ?thesis
  unfolding trace-fri-header-tied-sampled-layer-assignment-obstruction-def
  by (intro exI conjI)
    (rule fri, rule header, rule partial, rule cand, rule nonlow,
     rule obstruction)
qed

lemma wp-trace-fri-header-tied-augmented-missing-sampled-bound-from-assignment-obstruction:
  assumes obstruction-bound:
    wp-event verify-monad
      (trace-fri-header-tied-sampled-layer-assignment-obstruction s) s
      ≤ M
  shows
    wp-event verify-monad
      (trace-fri-header-tied-augmented-missing-sampled-layer-chain s) s
      ≤ M
  proof -
    have wp-event verify-monad
      (trace-fri-header-tied-augmented-missing-sampled-layer-chain s) s
      ≤ wp-event verify-monad
        (trace-fri-header-tied-sampled-layer-assignment-obstruction s) s
    by (rule wp-event-mono)
      (rule
        trace-fri-header-tied-augmented-missing-sampled-imp-assignment-obstruction)
    then show ?thesis
      by (rule order-trans[OF - obstruction-bound])
  qed

lemma wp-trace-fri-header-tied-augmented-partial-candidate-bound-from-sampled-and-missing:
  assumes sampled-bound:
    wp-event verify-monad
      (trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s) s
      ≤ R
  and missing-bound:
    wp-event verify-monad
      (trace-fri-header-tied-augmented-missing-sampled-layer-chain s) s
      ≤ M
  shows
    wp-event verify-monad
      (trace-fri-bad-with-header-tied-augmented-partial-candidate s) s
      ≤ R + M

```

proof –

have *wp-event verify-monad*
 (*trace-fri-bad-with-header-tied-augmented-partial-candidate s*) *s*
 $\leq$ *wp-event verify-monad*
 ($\lambda out.$
 trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s out $\vee$
 trace-fri-header-tied-augmented-missing-sampled-layer-chain s out) *s*
unfolding *trace-fri-header-tied-augmented-missing-sampled-layer-chain-def*
by (*rule wp-event-mono*) *blast*
also have ... $\leq$
 wp-event verify-monad
 (*trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s*) *s* +
 wp-event verify-monad
 (*trace-fri-header-tied-augmented-missing-sampled-layer-chain s*) *s*
by (*rule wp-event-union-bound*)
also have ... $\leq R + M$
by (*rule add-mono*[*OF sampled-bound missing-bound*])
finally show *?thesis* .
qed

lemma *wp-trace-fri-header-tied-augmented-partial-candidate-bound-from-sampled-conflict-and-zero*:

assumes *sampled-bound*:

wp-event verify-monad
 (*trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s*) *s*
 $\leq R$

and *conflict-bound*:

wp-event verify-monad
 (*trace-fri-header-tied-sampled-assignment-conflict s*) *s* $\leq C$

and *zero-bound*:

wp-event verify-monad
 (*trace-fri-header-tied-zero-round-final-obstruction s*) *s* $\leq Z$

shows

wp-event verify-monad
 (*trace-fri-bad-with-header-tied-augmented-partial-candidate s*) *s*
 $\leq R + (C + Z)$

proof –

have *obstruction-bound*:

wp-event verify-monad
 (*trace-fri-header-tied-sampled-layer-assignment-obstruction s*) *s*
 $\leq C + Z$
by (*rule*
 wp-trace-fri-header-tied-assignment-obstruction-bound-from-conflict-and-zero
 [*OF conflict-bound zero-bound*])

have *missing-bound*:

wp-event verify-monad
 (*trace-fri-header-tied-augmented-missing-sampled-layer-chain s*) *s*
 $\leq C + Z$
by (*rule*
 wp-trace-fri-header-tied-augmented-missing-sampled-bound-from-assignment-obstruction


```

    [OF obstruction-bound])
show ?thesis
  by (rule
    wp-trace-fri-header-tied-augmented-partial-candidate-bound-from-sampled-and-missing
    [OF sampled-bound missing-bound])
qed

lemma trace-fri-header-tied-augmented-partial-candidate-imp-sampled-conflict-or-zero:
  assumes bad:
    trace-fri-bad-with-header-tied-augmented-partial-candidate s out
  shows
    trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s out  $\vee$ 
    trace-fri-header-tied-sampled-assignment-conflict s out  $\vee$ 
    trace-fri-header-tied-zero-round-final-obstruction s out
proof (cases
  trace-fri-bad-with-header-tied-augmented-sampled-layer-chain s out)
case True
  then show ?thesis by blast
next
case False
have missing:
  trace-fri-header-tied-augmented-missing-sampled-layer-chain s out
  unfolding trace-fri-header-tied-augmented-missing-sampled-layer-chain-def
  using bad False by blast
have obstruction:
  trace-fri-header-tied-sampled-layer-assignment-obstruction s out
  by (rule
    trace-fri-header-tied-augmented-missing-sampled-imp-assignment-obstruction
    [OF missing])
  then show ?thesis
    using trace-fri-header-tied-assignment-obstruction-imp-conflict-or-zero
    by blast
qed

lemma trace-fri-header-tied-recorded-sibling-candidate-mismatch-false:
   $\neg$  trace-fri-header-tied-recorded-sibling-candidate-mismatch s out
proof
  assume bad:
    trace-fri-header-tied-recorded-sibling-candidate-mismatch s out
  from bad obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as rest trace-openings trace-table
    round-idx xp xp-path xn xn-path result final-state where
    fri-openings:
      accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
      dg composition-roots composition-bs composition-final fri-query-idxs
      trace-round-layers composition-round-layers
  and agrees:
    trace-table-agrees-with-recorded-first-fri-siblings trace-table

```

```

    trace-roots trace-bs fri-query-idxs trace-round-layers
and round-bound: round-idx < length fri-query-idxs
and bs-nonempty: 0 < length trace-bs
and step:
  fri-layer-step-evidence
    (trace-roots ! 0)
    (trace-bs ! 0)
    (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    1
    (fri-sibling-index (fri-evidence-layer-len trace-roots 0)
      (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
    xp xp-path xn xn-path
    (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
    (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
      round-idx 0 xp xn)
    (trace-round-layers ! round-idx ! 0)
and neq:
  trace-table !
  fri-sibling-index (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    ≠ xn
unfolding trace-fri-header-tied-recorded-sibling-candidate-mismatch-def
by blast
have roots-nonempty: 0 < length trace-roots
using accepted-fri-opening-transcript-shapes(2)[OF fri-openings]
  bs-nonempty by simp
have eq:
  trace-table !
  fri-sibling-index (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0) =
  xn
using agrees round-bound roots-nonempty bs-nonempty step
unfolding trace-table-agrees-with-recorded-first-fri-siblings-def
by blast
show False
using eq neq by simp
qed

```

```

lemma trace-fri-header-tied-value-aligned-sibling-mismatch-imp-recorded-sibling-missing:
assumes mismatch:
  trace-fri-header-tied-value-aligned-sibling-mismatch s out
shows
  trace-fri-header-tied-recorded-sibling-candidate-missing s out
proof —
from mismatch obtain trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr as rest trace-openings trace-table
  round-idx xp xp-path xn xn-path result final-state where

```

```

out-eq: out = Some (result, final-state)
and fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers
and header:
  verifier-header-transcript s fr trace-roots trace-final as dg
  composition-roots composition-final rest
and partial:
  accepted-with-partial-trace-openings s out fr fri-query-idxs
  trace-openings
and cand: partial-trace-table-candidate trace-table trace-openings
and nonlow:  $\neg$  trace-table-low-degree trace-table
and round-bound: round-idx < length fri-query-idxs
and bs-nonempty: 0 < length trace-bs
and openings-nonempty: trace-openings ! round-idx  $\neq$  []
and hd-idx:
  opening-index (hd (trace-openings ! round-idx)) =
    fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0
and hd-value:
  opening-value (hd (trace-openings ! round-idx)) = xp
and step:
  fri-layer-step-evidence
  (trace-roots ! 0)
  (trace-bs ! 0)
  (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
  1
  (fri-sibling-index (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
  xp xp-path xn xn-path
  (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
  (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
    round-idx 0 xp xn)
  (trace-round-layers ! round-idx ! 0)
and neg:
  trace-table !
  fri-sibling-index (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
   $\neq$  xn
unfolding trace-fri-header-tied-value-aligned-sibling-mismatch-def
by auto
have roots-nonempty: 0 < length trace-roots
using accepted-fri-opening-transcript-shapes(2)[OF fri-openings]
bs-nonempty by simp
have missing:
 $\neg$  trace-table-agrees-with-recorded-first-fri-siblings trace-table
  trace-roots trace-bs fri-query-idxs trace-round-layers
unfolding trace-table-agrees-with-recorded-first-fri-siblings-def

```

using *round-bound roots-nonempty bs-nonempty step neq* **by** *blast*
show *?thesis*
unfolding *trace-fri-header-tied-recorded-sibling-candidate-missing-def*
by (*intro exI conjI*)
(rule out-eq, rule fri-openings, rule header, rule partial, rule cand,
rule missing, rule nonlow, rule round-bound, rule bs-nonempty,
rule openings-nonempty, rule hd-idx, rule hd-value, rule step, rule neq)
qed

lemma *trace-fri-header-tied-sibling-mismatch-structural-gap-imp-recorded-sibling-missing:*
assumes *gap: trace-fri-header-tied-sibling-mismatch-structural-gap s out*
shows *trace-fri-header-tied-recorded-sibling-candidate-missing s out*
using *gap*
unfolding *trace-fri-header-tied-sibling-mismatch-structural-gap-def*
by (*blast intro:*
trace-fri-header-tied-value-aligned-sibling-mismatch-imp-recorded-sibling-missing)

lemma *trace-fri-header-tied-unaugmented-partial-candidate-imp-missing-or-base-gap:*
assumes *unaugmented:*
trace-fri-header-tied-unaugmented-partial-candidate s out
shows
trace-fri-header-tied-recorded-sibling-candidate-missing s out $\vee$
trace-fri-header-tied-base-value-alignment-gap s out

proof –
from *unaugmented* **have** *plain:*
trace-fri-bad-with-header-tied-partial-candidate s out
and *not-augmented:*
 $\neg$ *trace-fri-bad-with-header-tied-augmented-partial-candidate s out*
unfolding *trace-fri-header-tied-unaugmented-partial-candidate-def*
by *simp-all*
from *plain* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as rest trace-openings trace-table
where *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *header:*
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *cand: partial-trace-table-candidate trace-table trace-openings*
and *not-low: $\neg$ trace-table-low-degree trace-table*
unfolding *trace-fri-bad-with-header-tied-partial-candidate-def*
by *blast*
have *not-agrees:*
 $\neg$ *trace-table-agrees-with-recorded-first-fri-siblings trace-table*

```

    trace-roots trace-bs fri-query-idxs trace-round-layers
proof
  assume agrees:
    trace-table-agrees-with-recorded-first-fri-siblings trace-table
    trace-roots trace-bs fri-query-idxs trace-round-layers
  have augmented-candidate:
    partial-trace-table-candidate-with-recorded-first-fri-siblings
    trace-table trace-openings trace-roots trace-bs fri-query-idxs
    trace-round-layers
  unfolding
    partial-trace-table-candidate-with-recorded-first-fri-siblings-def
  by (intro conjI cand agrees)
  have trace-fri-bad-with-header-tied-augmented-partial-candidate s out
  unfolding trace-fri-bad-with-header-tied-augmented-partial-candidate-def
  by (intro exI conjI)
    (rule fri-openings, rule header, rule partial,
     rule augmented-candidate, rule not-low)
  then show False
    using not-augmented by contradiction
qed
from not-agrees obtain round-idx xp xp-path xn xn-path where
  round-bound: round-idx < length fri-query-idxs
  and roots-nonempty: 0 < length trace-roots
  and trace-bs-nonempty: 0 < length trace-bs
  and step:
    fri-layer-step-evidence
    (trace-roots ! 0)
    (trace-bs ! 0)
    (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    1
    (fri-sibling-index (fri-evidence-layer-len trace-roots 0)
     (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
    xp xp-path xn xn-path
    (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
    (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
     round-idx 0 xp xn)
    (trace-round-layers ! round-idx ! 0)
  and sibling-neq:
    trace-table !
    fri-sibling-index (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
    ≠ xn
  unfolding trace-table-agrees-with-recorded-first-fri-siblings-def
  by blast
have no-match:
  ¬ fri-opening-matches-table (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
  trace-table xp xn

```

```

proof
  assume match:
    fri-opening-matches-table (fri-evidence-layer-len trace-roots 0)
      (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
      trace-table xp xn
  have trace-table !
    fri-sibling-index (fri-evidence-layer-len trace-roots 0)
      (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0) =
      xn
    by (rule fri-opening-matches-tableD(4)[OF match])
  then show False
    using sibling-neq by contradiction
qed
have generic-base:
  generic-fri-sampled-base-opening-conflict trace-table trace-roots
    trace-bs fri-query-idxs trace-round-layers
  unfolding generic-fri-sampled-base-opening-conflict-def
  by (intro exI conjI)
    (rule trace-bs-nonempty, rule round-bound, rule step, rule no-match)
have sampled-base:
  trace-fri-header-tied-sampled-base-opening-conflict s out
  unfolding trace-fri-header-tied-sampled-base-opening-conflict-def
  by (intro exI conjI)
    (rule fri-openings, rule header, rule partial, rule cand, rule not-low,
      rule generic-base)
show ?thesis
proof (cases
  trace-fri-header-tied-value-aligned-base-opening-conflict s out)
  case True
  have trace-fri-header-tied-value-aligned-sibling-mismatch s out
    by (rule trace-fri-header-tied-value-aligned-base-imp-sibling-mismatch
      [OF True])
  then show ?thesis
    by (blast intro:
      trace-fri-header-tied-value-aligned-sibling-mismatch-imp-recorded-sibling-missing)
  next
  case False
  then have trace-fri-header-tied-base-value-alignment-gap s out
    using sampled-base
    unfolding trace-fri-header-tied-base-value-alignment-gap-def
    by simp
  then show ?thesis by simp
qed
qed

lemma wp-trace-fri-header-tied-unaugmented-partial-candidate-bound-from-recorded-missing-and-base-gap:
  fixes M G :: prob
  assumes missing-bound:
    wp-event verify-monad

```

(*trace-fri-header-tied-recorded-sibling-candidate-missing* s) $s \leq M$
and *gap-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-base-value-alignment-gap* s) $s \leq G$
shows
wp-event verify-monad
 (*trace-fri-header-tied-unaugmented-partial-candidate* s) $s \leq M + G$
proof –
have *wp-event verify-monad*
 (*trace-fri-header-tied-unaugmented-partial-candidate* s) $s \leq$
wp-event verify-monad
 ($\lambda out.$
 trace-fri-header-tied-recorded-sibling-candidate-missing s $out \vee$
 trace-fri-header-tied-base-value-alignment-gap s out) s
by (*rule wp-event-mono*)
 (*rule*
 trace-fri-header-tied-unaugmented-partial-candidate-imp-missing-or-base-gap)
also have $\dots \leq$
wp-event verify-monad
 (*trace-fri-header-tied-recorded-sibling-candidate-missing* s) $s +$
wp-event verify-monad
 (*trace-fri-header-tied-base-value-alignment-gap* s) s
by (*rule wp-event-union-bound*)
also have $\dots \leq M + G$
by (*rule add-mono*[*OF missing-bound gap-bound*])
finally show *?thesis* .
qed

lemma *wp-trace-fri-header-tied-sibling-structural-gap-bound-from-recorded-sibling-missing*:
assumes *missing-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-recorded-sibling-candidate-missing* s) $s \leq M$
shows
wp-event verify-monad
 (*trace-fri-header-tied-sibling-mismatch-structural-gap* s) $s \leq M$
proof –
have *wp-event verify-monad*
 (*trace-fri-header-tied-sibling-mismatch-structural-gap* s) $s \leq$
wp-event verify-monad
 (*trace-fri-header-tied-recorded-sibling-candidate-missing* s) s
by (*rule wp-event-mono*)
 (*rule*
 trace-fri-header-tied-sibling-mismatch-structural-gap-imp-recorded-sibling-missing)
then show *?thesis*
by (*rule order-trans*[*OF - missing-bound*])
qed

lemma *trace-fri-header-tied-sibling-structural-gap-without-recorded-missing-false*:
assumes *gap*: *trace-fri-header-tied-sibling-mismatch-structural-gap* s out

and *not-missing*:
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}$
shows *False*
using *trace-fri-header-tied-sibling-mismatch-structural-gap-imp-recorded-sibling-missing*
 $[OF \text{ gap}] \text{ not-missing}$
by *contradiction*

lemma *wp-trace-fri-header-tied-sibling-structural-gap-without-recorded-missing-zero*:
 $\text{wp-event verify-monad}$
 $(\lambda \text{out. trace-fri-header-tied-sibling-mismatch-structural-gap } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \text{ } s$
 ≤ 0
proof –
have *wp-event verify-monad*
 $(\lambda \text{out. trace-fri-header-tied-sibling-mismatch-structural-gap } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \text{ } s$
 $\leq \text{wp-event verify-monad } (\lambda \text{. False}) \text{ } s$
by (*rule wp-event-mono*)
 $(\text{auto dest:}$
 $\text{trace-fri-header-tied-sibling-structural-gap-without-recorded-missing-false})$
also have $\dots = 0$
unfolding *wp-event-def wp-def dist-expect-def* **by** *simp*
finally show *?thesis* .
qed

lemma *trace-fri-header-tied-value-aligned-sibling-mismatch-without-recorded-missing-imp-partial-merkle*:
assumes *mismatch*:
 $\text{trace-fri-header-tied-value-aligned-sibling-mismatch } s \text{ out}$
and *not-missing*:
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}$
shows *partial-merkle-inconsistency-bad* $s \text{ out}$
proof –
have *split*:
 $\text{partial-merkle-inconsistency-bad } s \text{ out} \vee$
 $\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s \text{ out}$
by (*rule*
 $\text{trace-fri-header-tied-value-aligned-sibling-mismatch-imp-partial-merkle-or-structural-gap}$
 $[OF \text{ mismatch}]$)
then show *?thesis*
using *not-missing*
 $\text{trace-fri-header-tied-sibling-mismatch-structural-gap-imp-recorded-sibling-missing}$
by *blast*
qed

lemma *wp-trace-fri-header-tied-value-aligned-sibling-mismatch-without-recorded-missing-bound-from-partial-merkle*:
assumes *merkle-bound*:
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \text{ } s \leq P$
shows
 $\text{wp-event verify-monad}$

$(\lambda out. \text{trace-fri-header-tied-value-aligned-sibling-mismatch } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
 $\leq P$
proof –
have *wp-event verify-monad*
 $(\lambda out. \text{trace-fri-header-tied-value-aligned-sibling-mismatch } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
 $\leq \text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s$
by (*rule wp-event-mono*)
 $(\text{auto dest:}$
 $\text{trace-fri-header-tied-value-aligned-sibling-mismatch-without-recorded-missing-imp-partial-merkle})$
then show *?thesis*
by (*rule order-trans[OF - merkle-bound]*)
qed

lemma *wp-trace-fri-header-tied-value-aligned-base-without-recorded-missing-bound-from-partial-merkle:*
assumes *merkle-bound:*

$\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq P$

shows

$\text{wp-event verify-monad}$
 $(\lambda out.$
 $\text{trace-fri-header-tied-value-aligned-base-opening-conflict } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
 $\leq P$

proof –

have *wp-event verify-monad*
 $(\lambda out.$
 $\text{trace-fri-header-tied-value-aligned-base-opening-conflict } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
 $\leq \text{wp-event verify-monad}$
 $(\lambda out. \text{trace-fri-header-tied-value-aligned-sibling-mismatch } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
by (*rule wp-event-mono*)
 $(\text{auto intro: trace-fri-header-tied-value-aligned-base-imp-sibling-mismatch})$
also have $\dots \leq P$
by (*rule*
 $\text{wp-trace-fri-header-tied-value-aligned-sibling-mismatch-without-recorded-missing-bound-from-partial-merkle}$
 $[\text{OF merkle-bound}])$
finally show *?thesis* .
qed

lemma *wp-trace-fri-header-tied-sampled-base-without-recorded-missing-bound-from-partial-merkle-and-gap:*

fixes $P \ G :: \text{prob}$

assumes *merkle-bound:*

$\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq P$

and *gap-bound:*

$\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-value-alignment-gap } s) \ s \leq G$

shows

wp-event verify-monad
 (λout.
 trace-fri-header-tied-sampled-base-opening-conflict s out ∧
 ¬ *trace-fri-header-tied-recorded-sibling-candidate-missing s out*) *s*
 ≤ *P* + *G*
proof –
have *wp-event verify-monad*
 (λout.
 trace-fri-header-tied-sampled-base-opening-conflict s out ∧
 ¬ *trace-fri-header-tied-recorded-sibling-candidate-missing s out*) *s*
 ≤ *wp-event verify-monad*
 (λout.
 (*trace-fri-header-tied-value-aligned-base-opening-conflict s out* ∧
 ¬ *trace-fri-header-tied-recorded-sibling-candidate-missing s out*) ∨
 trace-fri-header-tied-base-value-alignment-gap s out) *s*
by (rule *wp-event-mono*)
 (auto dest:
 trace-fri-header-tied-sampled-base-imp-value-aligned-or-gap)
also have ... ≤
wp-event verify-monad
 (λout.
 trace-fri-header-tied-value-aligned-base-opening-conflict s out ∧
 ¬ *trace-fri-header-tied-recorded-sibling-candidate-missing s out*) *s* +
wp-event verify-monad
 (*trace-fri-header-tied-base-value-alignment-gap s*) *s*
by (rule *wp-event-union-bound*)
also have ... ≤ *P* + *G*
by (intro *add-mono gap-bound*
 wp-trace-fri-header-tied-value-aligned-base-without-recorded-missing-bound-from-partial-merkle
 [*OF merkle-bound*])
finally show ?thesis .
qed

lemma *trace-fri-header-tied-recorded-sibling-candidate-missingE*:
assumes *missing*:
 trace-fri-header-tied-recorded-sibling-candidate-missing s out
obtains *result final-state trace-roots trace-bs trace-final dg*
 composition-roots composition-bs composition-final fri-query-idxs
 trace-round-layers composition-round-layers fr as rest trace-openings
 trace-table round-idx xp xp-path xn xn-path
where
 out = *Some (result, final-state)*
 accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
 dg composition-roots composition-bs composition-final fri-query-idxs
 trace-round-layers composition-round-layers
 verifier-header-transcript s fr trace-roots trace-final as dg
 composition-roots composition-final rest
 accepted-with-partial-trace-openings s out fr fri-query-idxs
 trace-openings

```

partial-trace-table-candidate trace-table trace-openings
¬ trace-table-agrees-with-recorded-first-fri-siblings trace-table
  trace-roots trace-bs fri-query-idxs trace-round-layers
¬ trace-table-low-degree trace-table
round-idx < length fri-query-idxs
0 < length trace-bs
trace-openings ! round-idx ≠ []
opening-index (hd (trace-openings ! round-idx)) =
  fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0
opening-value (hd (trace-openings ! round-idx)) = xp
fri-layer-step-evidence
  (trace-roots ! 0)
  (trace-bs ! 0)
  (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
  1
  (fri-sibling-index (fri-evidence-layer-len trace-roots 0)
    (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0))
  xp xp-path xn xn-path
  (fri-evidence-next-idx trace-roots fri-query-idxs round-idx 0)
  (fri-evidence-next-value trace-roots trace-bs fri-query-idxs
    round-idx 0 xp xn)
  (trace-round-layers ! round-idx ! 0)
trace-table !
  fri-sibling-index (fri-evidence-layer-len trace-roots 0)
  (fri-evidence-layer-idx trace-roots fri-query-idxs round-idx 0)
  ≠ xn
using missing
unfolding trace-fri-header-tied-recorded-sibling-candidate-missing-def
apply (elim exE conjE)
apply (rule that)
apply assumption+
done

```

lemma trace-fri-header-tied-recorded-sibling-candidate-missing-no-same-sampled-chain:

```

assumes missing:
  trace-fri-header-tied-recorded-sibling-candidate-missing s out
obtains result final-state trace-roots trace-bs trace-final dg
  composition-roots composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers fr as rest trace-openings
  trace-table round-idx xp xp-path xn xn-path
where
  out = Some (result, final-state)
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    dg composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers
  verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
  accepted-with-partial-trace-openings s out fr fri-query-idxs

```

```

    trace-openings
  partial-trace-table-candidate trace-table trace-openings
  ¬ trace-table-low-degree trace-table
  ∧ doms layers.
  ¬ generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
proof -
  from missing show thesis
proof (rule trace-fri-header-tied-recorded-sibling-candidate-missingE)
  fix result final-state trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr as rest trace-openings
    trace-table round-idx xp xp-path xn xn-path
  assume out-eq: out = Some (result, final-state)
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  and candidate:
    partial-trace-table-candidate trace-table trace-openings
  and no-agree:
    ¬ trace-table-agrees-with-recorded-first-fri-siblings trace-table
    trace-roots trace-bs fri-query-idxs trace-round-layers
  and nonlow: ¬ trace-table-low-degree trace-table
  have no-chain:
    ∧ doms layers.
    ¬ generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
proof
  fix doms layers
  assume chain:
    generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers doms layers
  have trace-table-agrees-with-recorded-first-fri-siblings trace-table
    trace-roots trace-bs fri-query-idxs trace-round-layers
  by (rule
    generic-trace-sampled-layer-chain-evidence-recorded-first-sibling-agreement
    [OF chain])
  then show False
  using no-agree by contradiction

```

qed
show *thesis*
 by (rule that[*OF out-eq fri-openings header partial candidate nonlow no-chain*])
qed
qed

lemma *trace-fri-header-tied-recorded-sibling-candidate-missing-imp-assignment-obstruction:*
assumes *missing:*
trace-fri-header-tied-recorded-sibling-candidate-missing s out
shows
trace-fri-header-tied-sampled-layer-assignment-obstruction s out
proof (rule
trace-fri-header-tied-recorded-sibling-candidate-missing-no-same-sampled-chain
[*OF missing*])
fix *result final-state trace-roots trace-bs trace-final dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as rest trace-openings
trace-table round-idx xp xp-path xn xn-path
assume *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers
and *header:*
verifier-header-transcript s fr trace-roots trace-final as dg
composition-roots composition-final rest
and *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
and *cand:* *partial-trace-table-candidate trace-table trace-openings*
and *not-low:* $\neg$ *trace-table-low-degree trace-table*
and *no-chain:*
 $\bigwedge$ *doms layers.*
 $\neg$ *generic-fri-sampled-layer-chain-evidence trace-table-low-degree*
trace-table (clength - 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers doms layers
have *partial-generic:*
generic-fri-partial-evidence trace-table-low-degree trace-table
(clength - 1) trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers
by (rule *accepted-fri-opening-transcript-trace-generic-partial-evidence*
[*OF fri-openings*])
have *obstruction:*
generic-fri-sampled-layer-assignment-obstruction
trace-table-low-degree trace-table (clength - 1) trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers
unfolding *generic-fri-sampled-layer-assignment-obstruction-def*
using *partial-generic no-chain* **by** *blast*
show *trace-fri-header-tied-sampled-layer-assignment-obstruction s out*

unfolding *trace-fri-header-tied-sampled-layer-assignment-obstruction-def*
by (*intro exI conjI*)
 (rule *fri-openings*, rule *header*, rule *partial*, rule *cand*,
 rule *not-low*, rule *obstruction*)
qed

lemma *trace-fri-header-tied-recorded-sibling-candidate-missing-imp-conflict-or-zero*:
assumes *missing*:
trace-fri-header-tied-recorded-sibling-candidate-missing s out
shows
trace-fri-header-tied-sampled-assignment-conflict s out $\vee$
trace-fri-header-tied-zero-round-final-obstruction s out
by (rule *trace-fri-header-tied-assignment-obstruction-imp-conflict-or-zero*)
 (rule
trace-fri-header-tied-recorded-sibling-candidate-missing-imp-assignment-obstruction
 [*OF missing*])

lemma *wp-trace-fri-header-tied-recorded-sibling-candidate-missing-bound-from-conflict-and-zero*:
assumes *conflict-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-sampled-assignment-conflict s*) $s \leq C$
and *zero-bound*:
wp-event verify-monad
 (*trace-fri-header-tied-zero-round-final-obstruction s*) $s \leq Z$
shows
wp-event verify-monad
 (*trace-fri-header-tied-recorded-sibling-candidate-missing s*) s
 $\leq C + Z$
proof –
have *wp-event verify-monad*
 (*trace-fri-header-tied-recorded-sibling-candidate-missing s*) $s \leq$
wp-event verify-monad
 ($\lambda out. trace-fri-header-tied-sampled-assignment-conflict s out \vee$
trace-fri-header-tied-zero-round-final-obstruction s out) s
by (rule *wp-event-mono*)
 (rule
trace-fri-header-tied-recorded-sibling-candidate-missing-imp-conflict-or-zero)
also have $\dots \leq$
wp-event verify-monad
 (*trace-fri-header-tied-sampled-assignment-conflict s*) $s +$
wp-event verify-monad
 (*trace-fri-header-tied-zero-round-final-obstruction s*) s
by (rule *wp-event-union-bound*)
also have $\dots \leq C + Z$
by (rule *add-mono*[*OF conflict-bound zero-bound*])
finally show *?thesis* .
qed

end

end

theory *Soundness-FRI-Staged-Replay-Bounds*

imports

Soundness-FRI-Staged-Active-Bounds
Soundness-FRI-Replay-Gap-Assignment
Soundness-FRI-Composition-Replay-Gaps
Soundness-FRI-Trace-Replay-Gaps
Soundness-FRI-Trace-Sibling-Evidence

begin

Staged active FRI bounds that additionally use the replay-gap assignment adapters. This theory is deliberately downstream of both the staged prequery-accounting layer and the replay-gap layer.

context *soundness*

begin

lemma *wp-trace-fri-header-tied-without-same-without-recorded-missing-bound-from-partial-merkle-and-replay-gaps:*

fixes $P\ G\ N\ S\ F :: \text{prob}$

assumes *merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s $\leq P$

and *alignment-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-base-value-alignment-gap s) s $\leq G$

and *next-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-next-value-replay-gap s) s $\leq N$

and *successor-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-successor-replay-gap s) s $\leq S$

and *final-gap-bound:*

wp-event verify-monad

(trace-fri-header-tied-final-value-replay-gap s) s $\leq F$

shows

wp-event verify-monad

(λ out.

trace-fri-header-tied-sampled-assignment-conflict-without-same-layer

s out $\wedge$

$\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out})\ s$

$\leq (P + G) + N + S + F$

proof $-$

have *base-bound:*

wp-event verify-monad

(λ out.

trace-fri-header-tied-sampled-base-opening-conflict s out $\wedge$

$\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out})\ s$

$\leq P + G$

by (rule
wp-trace-fri-header-tied-sampled-base-without-recorded-missing-bound-from-partial-merkle-and-gap
 $[OF \text{ merkle-bound alignment-gap-bound}]$)
have *wp-event verify-monad*
 $(\lambda out.$
 $\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer}$
 $s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) s$
 $\leq \text{wp-event verify-monad}$
 $(\lambda out.$
 $(\text{trace-fri-header-tied-sampled-base-opening-conflict } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \vee$
 $\text{trace-fri-header-tied-sampled-next-value-conflict } s \text{ out} \vee$
 $\text{trace-fri-header-tied-sampled-successor-opening-conflict } s \text{ out} \vee$
 $\text{trace-fri-header-tied-sampled-final-value-conflict } s \text{ out}) s$
by (rule *wp-event-mono*)
 $(\text{auto dest: trace-fri-header-tied-without-same-imp-branch})$
also have $\dots \leq$
 $\text{wp-event verify-monad}$
 $(\lambda out.$
 $\text{trace-fri-header-tied-sampled-base-opening-conflict } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) s +$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-next-value-conflict } s) s +$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-successor-opening-conflict } s) s +$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-final-value-conflict } s) s$
by (rule *wp-event-union-bound4*)
also have $\dots \leq (P + G) + N + S + F$
by (intro *add-mono base-bound*
 $\text{wp-trace-fri-header-tied-sampled-next-value-bound-from-replay-gap}$
 $[OF \text{ next-gap-bound}]$
 $\text{wp-trace-fri-header-tied-sampled-successor-bound-from-replay-gap}$
 $[OF \text{ successor-gap-bound}]$
 $\text{wp-trace-fri-header-tied-sampled-final-value-bound-from-replay-gap}$
 $[OF \text{ final-gap-bound}])$
finally show *?thesis* .
qed

lemma *wp-trace-fri-header-tied-authenticated-conflict-without-recorded-missing-bound-from-partial-merkle-and-replay*
fixes $P \ G \ N \ S \ F :: \text{prob}$
assumes *merkle-bound*:
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) s \leq P$
and *alignment-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-base-value-alignment-gap } s) s \leq G$
and *next-gap-bound*:
 $\text{wp-event verify-monad}$

$(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq N$
and *successor-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s \leq S$
and *final-gap-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-final-value-replay-gap } s) \ s \leq F$
shows
 $\text{wp-event verify-monad}$
 $(\lambda \text{out.}$
 $\text{trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer}$
 $\text{ } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
 $\leq ((P + G) + N + S + F) + P$
proof –
have *without-bound*:
 $\text{wp-event verify-monad}$
 $(\lambda \text{out.}$
 $\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer}$
 $\text{ } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
 $\leq (P + G) + N + S + F$
by (rule
 $\text{wp-trace-fri-header-tied-without-same-without-recorded-missing-bound-from-partial-merkle-and-replay-gaps}$
 $[\text{OF merkle-bound alignment-gap-bound next-gap-bound successor-gap-bound}$
 $\text{final-gap-bound}])$
have $\text{wp-event verify-monad}$
 $(\lambda \text{out.}$
 $\text{trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer}$
 $\text{ } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
 $\leq \text{wp-event verify-monad}$
 $(\lambda \text{out.}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer}$
 $\text{ } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \vee$
 $\text{trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks}$
 $\text{ } s \text{ out}) \ s$
by (rule *wp-event-mono*)
 $(\text{auto dest:}$
 $\text{trace-fri-header-tied-authenticated-conflict-imp-without-same-or-same-layer})$
also have $\dots \leq$
 $\text{wp-event verify-monad}$
 $(\lambda \text{out.}$
 $\text{trace-fri-header-tied-sampled-assignment-conflict-without-same-layer}$
 $\text{ } s \text{ out} \wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s +$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks } s)$

s
by (*rule wp-event-union-bound*)
also have $\dots \leq ((P + G) + N + S + F) + P$
by (*intro add-mono without-bound*
wp-trace-fri-header-tied-same-layer-conflict-with-authenticated-chunks-bound-from-partial-merkle
[OF merkle-bound])
finally show *?thesis* .
qed

lemma *wp-trace-fri-header-tied-sampled-assignment-conflict-without-recorded-missing-bound-from-auth-gap-and-replay-gap*
fixes $A\ P\ G\ N\ S\ F :: \text{prob}$
assumes *auth-gap-bound*:
wp-event verify-monad
(trace-fri-header-tied-assignment-auth-gap s) s ≤ A
and *merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ P
and *alignment-gap-bound*:
wp-event verify-monad
(trace-fri-header-tied-base-value-alignment-gap s) s ≤ G
and *next-gap-bound*:
wp-event verify-monad
(trace-fri-header-tied-next-value-replay-gap s) s ≤ N
and *successor-gap-bound*:
wp-event verify-monad
(trace-fri-header-tied-successor-replay-gap s) s ≤ S
and *final-gap-bound*:
wp-event verify-monad
(trace-fri-header-tied-final-value-replay-gap s) s ≤ F
shows
wp-event verify-monad
(λout.
trace-fri-header-tied-sampled-assignment-conflict s out ∧
¬ trace-fri-header-tied-recorded-sibling-candidate-missing s out) s
 $\leq (((P + G) + N + S + F) + P) + A$
proof –
have *auth-bound*:
wp-event verify-monad
(λout.
trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer
s out ∧
¬ trace-fri-header-tied-recorded-sibling-candidate-missing s out) s
 $\leq ((P + G) + N + S + F) + P$
by (*rule*
wp-trace-fri-header-tied-authenticated-conflict-without-recorded-missing-bound-from-partial-merkle-and-replay-gap
[OF merkle-bound alignment-gap-bound next-gap-bound successor-gap-bound
final-gap-bound])
have *wp-event verify-monad*
(λout.
trace-fri-header-tied-sampled-assignment-conflict s out ∧

$\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
 $\leq \text{wp-event verify-monad}$
 $(\lambda \text{out.}$
 $\quad (\text{trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer}$
 $\quad \quad s \text{ out} \wedge$
 $\quad \quad \neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \vee$
 $\quad \quad \text{trace-fri-header-tied-assignment-auth-gap } s \text{ out}) \ s$
by (rule wp-event-mono)
 $(\text{auto dest: trace-fri-header-tied-assignment-imp-authenticated-or-auth-gap})$
also have ... $\leq$
 $\text{wp-event verify-monad}$
 $(\lambda \text{out.}$
 $\quad \text{trace-fri-header-tied-sampled-assignment-conflict-with-authenticated-same-layer}$
 $\quad \quad s \text{ out} \wedge$
 $\quad \quad \neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s +$
 $\text{wp-event verify-monad}$
 $\quad (\text{trace-fri-header-tied-assignment-auth-gap } s) \ s$
by (rule wp-event-union-bound)
also have ... $\leq (((P + G) + N + S + F) + P) + A$
by (intro add-mono auth-bound auth-gap-bound)
finally show ?thesis .
qed

lemma wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-recorded-missing-and-complement:

fixes Missing Complement :: prob

assumes missing-bound:

$\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-recorded-sibling-candidate-missing } s) \ s$
 $\leq \text{Missing}$

and complement-bound:

$\text{wp-event verify-monad}$
 $(\lambda \text{out.}$
 $\quad \text{trace-fri-header-tied-sampled-assignment-conflict } s \text{ out} \wedge$
 $\quad \neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out}) \ s$
 $\leq \text{Complement}$

shows

$\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict } s) \ s$
 $\leq \text{Missing} + \text{Complement}$

proof –

have wp-event verify-monad

$(\text{trace-fri-header-tied-sampled-assignment-conflict } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\lambda \text{out.}$

$\quad \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out} \vee$
 $\quad (\text{trace-fri-header-tied-sampled-assignment-conflict } s \text{ out} \wedge$
 $\quad \quad \neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } s \text{ out})) \ s$

by (rule wp-event-mono) blast

also have ... $\leq$

```

wp-event verify-monad
  (trace-fri-header-tied-recorded-sibling-candidate-missing s) s +
wp-event verify-monad
  (λout.
    trace-fri-header-tied-sampled-assignment-conflict s out ∧
    ¬ trace-fri-header-tied-recorded-sibling-candidate-missing s out) s
by (rule wp-event-union-bound)
also have ... ≤ Missing + Complement
  by (intro add-mono missing-bound complement-bound)
finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-header-tied-augmented-sampled-layer-chain-bound-from-fresh-and-prequery:*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: finite B
and cover:
  ∧data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      trace-fri-sampled-layer-chain-cover
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) bad
and envelope:
  ∧data attacker-state trace-table.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      trace-fri-multiround-bad-sets bad trace-table ⊆ B
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-challenge-list-fresh-hit s B))
  adversary-initial-state ≤ Fresh
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-augmented-sampled-layer-chain)
  adversary-initial-state ≤
    Fresh +
    hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)

```

proof —

```

have sampled:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-sampled-layer-chain)

```

```

    adversary-initial-state ≤
    Fresh +
    hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)
  by (rule
    checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-fresh-and-prequery
    [OF wf controlled finite-B cover envelope fresh-bound])
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-augmented-sampled-layer-chain)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-sampled-layer-chain)
    adversary-initial-state
  unfolding staged-security-with-data-state-verifier-event-def
  by (rule wp-event-mono)
    (auto dest:
    trace-fri-bad-with-header-tied-augmented-sampled-layer-chain-imp-plain
    split: option.splits prod.splits)
  then show ?thesis
    by (rule order-trans[OF - sampled])
qed

lemma checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-augmented-and-unaugmented:
  assumes augmented-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-augmented-partial-candidate)
    adversary-initial-state ≤ Augmented
  and unaugmented-bound:
    ∧ data attacker-state.
    Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state) ⇒
    wp-event verify-monad
    (trace-fri-header-tied-unaugmented-partial-candidate
    (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ Unaugmented
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state ≤ Augmented + Unaugmented
proof -
  have unaugmented-staged:

```

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-header-tied-unaugmented-partial-candidate)
adversary-initial-state ≤ Unaugmented
by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  (auto simp: trace-fri-header-tied-unaugmented-partial-candidate-def
   unaugmented-bound)
have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (λout.
   staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-augmented-partial-candidate out ∨
   staged-security-with-data-state-verifier-event
    trace-fri-header-tied-unaugmented-partial-candidate out)
adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono-on-support)
  (auto dest:
   trace-fri-bad-with-header-tied-partial-candidate-imp-augmented-or-unaugmented
   split: option.splits prod.splits)
also have ... ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-bad-with-header-tied-augmented-partial-candidate)
adversary-initial-state +
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-header-tied-unaugmented-partial-candidate)
adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤ Augmented + Unaugmented
by (rule add-mono[OF augmented-bound unaugmented-staged])
finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-header-tied-unaugmented-partial-candidate-bound-from-recorded-missing-and*

assumes merkle-bound:

$\bigwedge \text{data attacker-state.}$

Some (data, attacker-state) ∈

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\implies$

wp-event verify-monad

(partial-merkle-inconsistency-bad

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state

```

    (staged-proof-transcript data)) ≤ Merkle
and missing-bound:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
    wp-event verify-monad
      (trace-fri-header-tied-recorded-sibling-candidate-missing
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) ≤ Missing
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-unaugmented-partial-candidate)
    adversary-initial-state ≤ Missing + Merkle
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have head-neq:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-neq-gap ?s) ?s ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle)
  (rule merkle-bound[OF support])
have head-mismatch:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-mismatch-gap ?s) ?s ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
    [OF head-neq])
have alignment:
  wp-event verify-monad
    (trace-fri-header-tied-base-value-alignment-gap ?s) ?s ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap
    [OF head-mismatch])
show
  wp-event verify-monad
    (trace-fri-header-tied-unaugmented-partial-candidate ?s) ?s
    ≤ Missing + Merkle
by (rule

```

```

      wp-trace-fri-header-tied-unaugmented-partial-candidate-bound-from-recorded-missing-and-base-gap)
      (rule missing-bound[OF support], rule alignment)
next
  fix s
  show  $\neg$  trace-fri-header-tied-unaugmented-partial-candidate s None
  unfolding trace-fri-header-tied-unaugmented-partial-candidate-def
    trace-fri-bad-with-header-tied-partial-candidate-def
    accepted-fri-opening-transcript-def
  by simp
qed

lemma trace-fri-header-tied-reduction-from-augmented-and-recorded-missing-bounds:
  fixes Augmented Missing Merkle :: prob
  assumes augmented-bound:
    wp-event verify-monad
      (trace-fri-bad-with-header-tied-augmented-partial-candidate s) s
       $\leq$  Augmented
  and missing-bound:
    wp-event verify-monad
      (trace-fri-header-tied-recorded-sibling-candidate-missing s) s
       $\leq$  Missing
  and merkle-bound:
    wp-event verify-monad (partial-merkle-inconsistency-bad s) s
       $\leq$  Merkle
  and total-bound:
    Augmented + (Missing + Merkle)  $\leq$  trace-fri-error
  shows trace-fri-header-tied-reduction s
proof -
  have head-neq:
    wp-event verify-monad
      (trace-fri-header-tied-base-head-value-neq-gap s) s  $\leq$  Merkle
  by (rule
    wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle)
    (rule merkle-bound)
  have head-mismatch:
    wp-event verify-monad
      (trace-fri-header-tied-base-head-value-mismatch-gap s) s
       $\leq$  Merkle
  by (rule
    wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
    [OF head-neq])
  have alignment:
    wp-event verify-monad
      (trace-fri-header-tied-base-value-alignment-gap s) s  $\leq$  Merkle
  by (rule
    wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap
    [OF head-mismatch])
  have unaugmented:
    wp-event verify-monad

```


$(\text{trace-fri-header-tied-unaugmented-partial-candidate } s) \ s$
 $\leq \text{Missing} + \text{Merkle}$
by (rule
 $\text{wp-trace-fri-header-tied-unaugmented-partial-candidate-bound-from-recorded-missing-and-base-gap}$
(rule *missing-bound*, rule *alignment*)
have *partial*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-partial-candidate } s) \ s$
 $\leq \text{Augmented} + (\text{Missing} + \text{Merkle})$
by (rule
 $\text{wp-trace-fri-header-tied-partial-candidate-bound-from-augmented-and-unaugmented}$
(rule *augmented-bound*, rule *unaugmented*)
show *?thesis*
unfolding *trace-fri-header-tied-reduction-def*
by (rule *order-trans[OF partial total-bound]*)
qed

lemma *active-trace-fri-header-tied-reductions-from-augmented-and-recorded-missing-bounds*:

fixes *Augmented Missing Merkle* ::

$'f \text{ staged-proof-data} \Rightarrow 'f \text{ protocol-channel} \Rightarrow \text{prob}$

assumes *augmented-bound*:

$\wedge \text{data attacker-state.}$

$\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-augmented-partial-candidate}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Augmented data attacker-state}$

and *missing-bound*:

$\wedge \text{data attacker-state.}$

$\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-recorded-sibling-candidate-missing}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Missing data attacker-state}$

and *merkle-bound*:

$\wedge \text{data attacker-state.}$

$\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Merkle data attacker-state}$
and *total-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{Augmented data attacker-state} +$
 $(\text{Missing data attacker-state} + \text{Merkle data attacker-state})$
 $\leq \text{trace-fri-error}$
and *support*:
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state})}$
shows
 $\text{trace-fri-header-tied-reduction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
by (*rule*
 $\text{trace-fri-header-tied-reduction-from-augmented-and-recorded-missing-bounds})$
 $(\text{rule augmented-bound[OF support]}, \text{rule missing-bound[OF support]},$
 $\text{rule merkle-bound[OF support]}, \text{rule total-bound[OF support]})$

lemma *checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-without-same-residu*

assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *finite-B*: *finite B*
and *cover*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-sampled-layer-chain-cover}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad}$
and *envelope*:
 $\wedge \text{data attacker-state trace-table.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-multi-round-bad-sets bad trace-table} \subseteq B$
and *fresh-bound*:
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-challenge-list-fresh-hit s B}))$
 $\text{adversary-initial-state} \leq \text{Fresh}$
and *merkle-bound*:

$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Merkle
and structural-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-header-tied-sibling-mismatch-structural-gap
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Structural
and zero-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Zero
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-partial-candidate)
 adversary-initial-state $\leq$
 (Fresh +
 hash-relation-budget-value (card B * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (((((Merkle + Structural) + Merkle) + (Merkle + 0) + (Merkle + 0) +
 (Merkle + 0 + 0)) + Merkle) + 0 + Zero)
proof (rule
 checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-and-residual-gaps
 [OF wf controlled finite-B cover envelope fresh-bound])
fix data attacker-state
assume support:
 Some (data, attacker-state) $\in$

```

    set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
    (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap ?s) ?s
    ≤ 0
    by (rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)
show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
    ≤ Merkle
    by (rule merkle-bound[OF support])
show wp-event verify-monad
    (trace-fri-header-tied-sibling-mismatch-structural-gap ?s) ?s
    ≤ Structural
    by (rule structural-bound[OF support])
have head-neq:
    wp-event verify-monad
    (trace-fri-header-tied-base-head-value-neq-gap ?s) ?s ≤ Merkle
    by (rule
        wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle)
    (rule merkle-bound[OF support])
have head-mismatch:
    wp-event verify-monad
    (trace-fri-header-tied-base-head-value-mismatch-gap ?s) ?s
    ≤ Merkle
    by (rule
        wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
        [OF head-neq])
show wp-event verify-monad
    (trace-fri-header-tied-base-value-alignment-gap ?s) ?s ≤ Merkle
    by (rule
        wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap
        [OF head-mismatch])
show wp-event verify-monad
    (trace-fri-header-tied-next-value-replay-gap ?s) ?s ≤ Merkle + 0
    by (rule
        wp-trace-fri-header-tied-next-value-replay-gap-bound-from-merkle-and-slot)
    (rule merkle-bound[OF support],
        rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)
show wp-event verify-monad
    (trace-fri-header-tied-successor-replay-gap ?s) ?s ≤ Merkle + 0
    by (rule
        wp-trace-fri-header-tied-successor-replay-gap-bound-from-merkle-and-slot)
    (rule merkle-bound[OF support],
        rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)
show wp-event verify-monad
    (trace-fri-header-tied-final-value-replay-gap ?s) ?s
    ≤ Merkle + 0 + 0

```

```

by (rule
  wp-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step)
(rule merkle-bound[OF support],
  rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero,
  rule wp-trace-fri-header-tied-final-value-selected-step-missing-gap-zero)
show wp-event verify-monad
  (trace-fri-header-tied-zero-round-final-obstruction ?s) ?s ≤ Zero
by (rule zero-bound[OF support])
qed

lemma checked-staged-security-trace-fri-header-tied-zero-round-bound-from-staged-checked-and-merkle:
  assumes checked-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-zero-round-checked-final-obstruction)
    adversary-initial-state ≤ CheckedZero
  and merkle-bound:
    ∧data attacker-state.
      Some (data, attacker-state) ∈
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state) ⇒
      wp-event verify-monad
        (partial-merkle-inconsistency-bad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)) ≤ Merkle
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-zero-round-final-obstruction)
      adversary-initial-state ≤ CheckedZero + Merkle
proof -
  have merkle-staged:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        partial-merkle-inconsistency-bad)
    adversary-initial-state ≤ Merkle
  by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
    (auto simp: partial-merkle-inconsistency-bad-def merkle-bound)
  have split:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-zero-round-final-obstruction)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (λout.
        staged-security-with-data-state-verifier-event
          trace-fri-zero-round-checked-final-obstruction out ∨

```

```

    staged-security-with-data-state-verifier-event
    partial-merkle-inconsistency-bad out)
  adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono-on-support)
  (auto dest:
    trace-fri-header-tied-zero-round-final-obstruction-imp-checked-or-merkle
    split: option.splits prod.splits)
also have ...  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-zero-round-checked-final-obstruction)
  adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    partial-merkle-inconsistency-bad)
  adversary-initial-state
by (rule wp-event-union-bound)
also have ...  $\leq$  CheckedZero + Merkle
by (intro add-mono checked-bound merkle-staged)
finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-staged-checked-zero-*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: finite B*
and *cover:*
 $\bigwedge$ *data attacker-state.*
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 trace-fri-sampled-layer-chain-cover
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad
and *envelope:*
 $\bigwedge$ *data attacker-state trace-table.*
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 trace-fri-multiround-bad-sets bad trace-table $\subseteq B$
and *fresh-bound:*
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s.$ trace-fri-challenge-list-fresh-hit s B))
 adversary-initial-state $\leq$ Fresh
and *merkle-bound:*
 $\bigwedge$ *data attacker-state.*
 Some (data, attacker-state) $\in$

$\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Merkle}$
and structural-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Structural}$
and $\text{checked-zero-bound:}$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-zero-round-checked-final-obstruction})$
 $\text{adversary-initial-state} \leq \text{CheckedZero}$
shows
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq$
 $(\text{Fresh} +$
 $\text{hash-relation-budget-value } (\text{card } B * \text{ceil-log clength})$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(((((\text{Merkle} + \text{Structural}) + \text{Merkle}) + (\text{Merkle} + 0) + (\text{Merkle} + 0) +$
 $(\text{Merkle} + 0 + 0)) + \text{Merkle}) + 0 + (\text{CheckedZero} + \text{Merkle}))$
proof –
have sampled:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-sampled-layer-chain})$
 $\text{adversary-initial-state} \leq$
 $\text{Fresh} +$
 $\text{hash-relation-budget-value } (\text{card } B * \text{ceil-log clength})$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})$
by $(\text{rule}$
 $\text{checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-fresh-and-prequery}$
 $[\text{OF wf controlled finite-B cover envelope fresh-bound}])$
have conflict:

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-header-tied-sampled-assignment-conflict)
  adversary-initial-state ≤
  (((((Merkle + Structural) + Merkle) + (Merkle + 0) +
    (Merkle + 0) + (Merkle + 0 + 0)) + Merkle) + 0)
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have head-neq:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-neq-gap ?s) ?s ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle)
  (rule merkle-bound[OF support])
have head-mismatch:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-mismatch-gap ?s) ?s
    ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
  [OF head-neq])
have alignment:
  wp-event verify-monad
    (trace-fri-header-tied-base-value-alignment-gap ?s) ?s ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap
  [OF head-mismatch])
have next-bound:
  wp-event verify-monad
    (trace-fri-header-tied-next-value-replay-gap ?s) ?s ≤ Merkle + 0
by (rule
  wp-trace-fri-header-tied-next-value-replay-gap-bound-from-merkle-and-slot)
  (rule merkle-bound[OF support],
  rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)
have successor:
  wp-event verify-monad
    (trace-fri-header-tied-successor-replay-gap ?s) ?s ≤ Merkle + 0
by (rule
  wp-trace-fri-header-tied-successor-replay-gap-bound-from-merkle-and-slot)
  (rule merkle-bound[OF support],
  rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)
have final:

```



```

wp-event verify-monad
  (trace-fri-header-tied-final-value-replay-gap ?s) ?s
  ≤ Merkle + 0 + 0
by (rule
  wp-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step)
(rule merkle-bound[OF support],
  rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero,
  rule wp-trace-fri-header-tied-final-value-selected-step-missing-gap-zero)
show wp-event verify-monad
  (trace-fri-header-tied-sampled-assignment-conflict ?s) ?s
  ≤ (((((Merkle + Structural) + Merkle) + (Merkle + 0) +
    (Merkle + 0) + (Merkle + 0 + 0)) + Merkle) + 0)
by (rule
  wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-slot-recorded-chunk-gap-and-residuals)
(rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero,
  rule merkle-bound[OF support],
  rule structural-bound[OF support],
  rule alignment, rule next-bound, rule successor, rule final)
next
fix s
show ¬ trace-fri-header-tied-sampled-assignment-conflict s None
  unfolding trace-fri-header-tied-sampled-assignment-conflict-def
    accepted-fri-opening-transcript-def
  by simp
qed
have zero:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-zero-round-final-obstruction)
    adversary-initial-state ≤ CheckedZero + Merkle
  by (rule
    checked-staged-security-trace-fri-header-tied-zero-round-bound-from-staged-checked-and-merkle)
  (rule checked-zero-bound, rule merkle-bound)
show ?thesis
  by (rule
    checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-conflict-and-staged-zero
      [OF sampled conflict zero])
qed

lemma checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-staged-checked-zero-
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B: finite B
  and cover:
    ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
    trace-fri-sampled-layer-chain-cover

```

(verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad
and envelope:
 $\wedge$ data attacker-state trace-table.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 trace-fri-multiround-bad-sets bad trace-table $\subseteq B$
and fresh-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s.$ trace-fri-challenge-list-fresh-hit s B))
 adversary-initial-state $\leq$ Fresh
and merkle-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Merkle
and missing-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-header-tied-recorded-sibling-candidate-missing
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Missing
and checked-zero-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-zero-round-checked-final-obstruction)
 adversary-initial-state $\leq$ CheckedZero
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-partial-candidate)
 adversary-initial-state $\leq$
 (Fresh +
 hash-relation-budget-value (card B * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +

$(((((\text{Merkle} + \text{Missing}) + \text{Merkle}) + (\text{Merkle} + 0) + (\text{Merkle} + 0) + (\text{Merkle} + 0 + 0)) + \text{Merkle}) + 0 + (\text{CheckedZero} + \text{Merkle}))$
proof (rule
checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-staged-checked-zero-and-w
[OF wf controlled finite-B cover envelope fresh-bound merkle-bound - checked-zero-bound])
fix data attacker-state
assume support:
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)
let ?s =
verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
show wp-event verify-monad
(trace-fri-header-tied-sibling-mismatch-structural-gap ?s) ?s
≤ Missing
by (rule
wp-trace-fri-header-tied-sibling-structural-gap-bound-from-recorded-sibling-missing)
(rule missing-bound[OF support])
qed

lemma *checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-staged-checked-zero-*
assumes wf: *staged-budget-wellformed budgets*
and controlled: *staged-adversary-controlled budgets A*
and finite-B: *finite B*
and cover:
 $\bigwedge$ data attacker-state.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
trace-fri-sampled-layer-chain-cover
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) bad
and envelope:
 $\bigwedge$ data attacker-state trace-table.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
trace-fri-multi-round-bad-sets bad trace-table ⊆ B
and fresh-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
(λs. trace-fri-challenge-list-fresh-hit s B))
adversary-initial-state ≤ Fresh
and merkle-bound:
 $\bigwedge$ data attacker-state.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒

wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Merkle
and missing-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
wp-event verify-monad
 (trace-fri-header-tied-recorded-sibling-candidate-missing
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Missing
and checked-zero-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-zero-round-checked-final-obstruction)
 adversary-initial-state $\leq$ CheckedZero
shows
wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-partial-candidate)
 adversary-initial-state $\leq$
 (Fresh +
 hash-relation-budget-value (card B * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 ((Missing +
 (((Merkle + Merkle) + (Merkle + 0) + (Merkle + 0) +
 (Merkle + 0 + 0)) + Merkle) + 0)) +
 (CheckedZero + Merkle))
proof –
have sampled:
wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-sampled-layer-chain)
 adversary-initial-state $\leq$
 Fresh +
 hash-relation-budget-value (card B * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)
by (rule
 checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-fresh-and-prequery
 [OF wf controlled finite-B cover envelope fresh-bound])
have conflict:

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-header-tied-sampled-assignment-conflict)
adversary-initial-state ≤
Missing +
  (((Merkle + Merkle) + (Merkle + 0) + (Merkle + 0) +
    (Merkle + 0 + 0)) + Merkle) + 0)
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have head-neq:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-neq-gap ?s) ?s ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle)
  (rule merkle-bound[OF support])
have head-mismatch:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-mismatch-gap ?s) ?s
    ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
  [OF head-neq])
have alignment:
  wp-event verify-monad
    (trace-fri-header-tied-base-value-alignment-gap ?s) ?s ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap
  [OF head-mismatch])
have next-bound:
  wp-event verify-monad
    (trace-fri-header-tied-next-value-replay-gap ?s) ?s ≤ Merkle + 0
by (rule
  wp-trace-fri-header-tied-next-value-replay-gap-bound-from-merkle-and-slot)
  (rule merkle-bound[OF support],
  rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)
have successor:
  wp-event verify-monad
    (trace-fri-header-tied-successor-replay-gap ?s) ?s ≤ Merkle + 0
by (rule
  wp-trace-fri-header-tied-successor-replay-gap-bound-from-merkle-and-slot)
  (rule merkle-bound[OF support],
  rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)

```

have *final*:
 wp-event verify-monad
 (*trace-fri-header-tied-final-value-replay-gap* ?*s*) ?*s*
 $\leq \text{Merkle} + 0 + 0$
by (*rule*
 wp-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step)
 (*rule merkle-bound*[*OF support*],
 rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero,
 rule wp-trace-fri-header-tied-final-value-selected-step-missing-gap-zero)
have *slot*:
 wp-event verify-monad
 (*trace-fri-header-tied-authenticated-slot-recorded-chunk-gap* ?*s*) ?*s*
 ≤ 0
by (*rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero*)
have *authenticated-recorded-auth*:
 wp-event verify-monad
 (*trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap* ?*s*)
 ?*s* ≤ 0
by (*rule*
 wp-trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap-bound-from-slot-recorded-chunk-gap
 [*OF slot*])
have *recorded-auth*:
 wp-event verify-monad
 (*trace-fri-header-tied-same-layer-recorded-auth-gap* ?*s*) ?*s* ≤ 0
by (*rule*
 wp-trace-fri-header-tied-same-layer-recorded-auth-gap-bound-from-authenticated-recorded-auth-gap
 [*OF authenticated-recorded-auth*])
have *same-auth*:
 wp-event verify-monad
 (*trace-fri-header-tied-same-layer-auth-gap* ?*s*) ?*s* ≤ 0
by (*rule*
 wp-trace-fri-header-tied-same-layer-auth-gap-bound-from-recorded-auth-gap
 [*OF recorded-auth*])
have *auth-gap*:
 wp-event verify-monad
 (*trace-fri-header-tied-assignment-auth-gap* ?*s*) ?*s* ≤ 0
by (*rule*
 wp-trace-fri-header-tied-assignment-auth-gap-bound-from-same-layer-auth-gap
 [*OF same-auth*])
have *complement*:
 wp-event verify-monad
 ($\lambda out.$
 trace-fri-header-tied-sampled-assignment-conflict ?*s* *out* $\wedge$
 $\neg \text{trace-fri-header-tied-recorded-sibling-candidate-missing } ?s \text{ out}$)
 ?*s* $\leq$
 ($((((\text{Merkle} + \text{Merkle}) + (\text{Merkle} + 0) + (\text{Merkle} + 0) +$
 $(\text{Merkle} + 0 + 0)) + \text{Merkle}) + 0)$)
by (*rule*
 wp-trace-fri-header-tied-sampled-assignment-conflict-without-recorded-missing-bound-from-auth-gap-and-replay-

(rule *auth-gap*, rule *merkle-bound*[*OF support*], rule *alignment*,
 rule *next-bound*, rule *successor*, rule *final*)
show *wp-event verify-monad*
 (trace-fri-header-tied-sampled-assignment-conflict ?s) ?s
 $\leq$ *Missing* +
 (((*Merkle* + *Merkle*) + (*Merkle* + 0) + (*Merkle* + 0) +
 (*Merkle* + 0 + 0)) + *Merkle*) + 0)
by (rule
wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-recorded-missing-and-complement)
 (rule *missing-bound*[*OF support*], rule *complement*)
next
fix *s*
show $\neg$ *trace-fri-header-tied-sampled-assignment-conflict s None*
unfolding *trace-fri-header-tied-sampled-assignment-conflict-def*
accepted-fri-opening-transcript-def
by *simp*
qed
have *zero*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-header-tied-zero-round-final-obstruction)
adversary-initial-state $\leq$ *CheckedZero* + *Merkle*
by (rule
checked-staged-security-trace-fri-header-tied-zero-round-bound-from-staged-checked-and-merkle)
 (rule *checked-zero-bound*, rule *merkle-bound*)
show ?thesis
by (rule
checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-conflict-and-staged-zero
 [*OF sampled conflict zero*])
qed

lemma *checked-staged-security-trace-fri-header-tied-augmented-partial-candidate-bound-from-fresh-prequery-staged-ch*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: finite B*
and *cover*:
 $\bigwedge$ *data attacker-state.*
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
trace-fri-sampled-layer-chain-cover
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) *bad*
and *envelope*:
 $\bigwedge$ *data attacker-state trace-table.*
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
trace-fri-multi-round-bad-sets bad trace-table $\subseteq$ *B*

and *fresh-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B$))
adversary-initial-state $\leq$ *Fresh*
and *merkle-bound*:
 $\wedge$ *data attacker-state*.
 Some (*data*, *attacker-state*) $\in$
 set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
wp-event verify-monad
 (*partial-merkle-inconsistency-bad*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\leq$ *Merkle*
and *missing-bound*:
 $\wedge$ *data attacker-state*.
 Some (*data*, *attacker-state*) $\in$
 set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
wp-event verify-monad
 (*trace-fri-header-tied-recorded-sibling-candidate-missing*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\leq$ *Missing*
and *checked-zero-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-zero-round-checked-final-obstruction)
adversary-initial-state $\leq$ *CheckedZero*
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-augmented-partial-candidate)
adversary-initial-state $\leq$
 (*Fresh* +
hash-relation-budget-value (*card B* * *ceil-log clength*)
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)) +
 ((*Missing* +
 (((*Merkle* + *Merkle*) + (*Merkle* + 0) + (*Merkle* + 0) +
 (*Merkle* + 0 + 0)) + *Merkle*) + 0)) +
 (*CheckedZero* + *Merkle*))
proof –
have *sampled*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*


```

    trace-fri-bad-with-header-tied-augmented-sampled-layer-chain)
  adversary-initial-state ≤
  Fresh +
  hash-relation-budget-value (card B * ceil-log clength)
  (staged-attacker-query-budget budgets +
   staged-challenge-query-budget)
by (rule
  checked-staged-security-trace-fri-header-tied-augmented-sampled-layer-chain-bound-from-fresh-and-prequery
  [OF wf controlled finite-B cover envelope fresh-bound])
have conflict:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-header-tied-sampled-assignment-conflict)
  adversary-initial-state ≤
  Missing +
  (((Merkle + Merkle) + (Merkle + 0) + (Merkle + 0) +
   (Merkle + 0 + 0)) + Merkle) + 0)
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
  set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have head-neq:
  wp-event verify-monad
  (trace-fri-header-tied-base-head-value-neq-gap ?s) ?s ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle)
(rule merkle-bound[OF support])
have head-mismatch:
  wp-event verify-monad
  (trace-fri-header-tied-base-head-value-mismatch-gap ?s) ?s
  ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
  [OF head-neq])
have alignment:
  wp-event verify-monad
  (trace-fri-header-tied-base-value-alignment-gap ?s) ?s ≤ Merkle
by (rule
  wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap
  [OF head-mismatch])
have next-bound:
  wp-event verify-monad
  (trace-fri-header-tied-next-value-replay-gap ?s) ?s ≤ Merkle + 0
by (rule

```

```

    wp-trace-fri-header-tied-next-value-replay-gap-bound-from-merkle-and-slot)
  (rule merkle-bound[OF support],
    rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)
have successor:
  wp-event verify-monad
    (trace-fri-header-tied-successor-replay-gap ?s) ?s ≤ Merkle + 0
  by (rule
    wp-trace-fri-header-tied-successor-replay-gap-bound-from-merkle-and-slot)
  (rule merkle-bound[OF support],
    rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)
have final:
  wp-event verify-monad
    (trace-fri-header-tied-final-value-replay-gap ?s) ?s
    ≤ Merkle + 0 + 0
  by (rule
    wp-trace-fri-header-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step)
  (rule merkle-bound[OF support],
    rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero,
    rule wp-trace-fri-header-tied-final-value-selected-step-missing-gap-zero)
have slot:
  wp-event verify-monad
    (trace-fri-header-tied-authenticated-slot-recorded-chunk-gap ?s) ?s
    ≤ 0
  by (rule wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero)
have authenticated-recorded-auth:
  wp-event verify-monad
    (trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap ?s)
    ?s ≤ 0
  by (rule
    wp-trace-fri-header-tied-same-layer-authenticated-recorded-auth-gap-bound-from-slot-recorded-chunk-gap
    [OF slot])
have recorded-auth:
  wp-event verify-monad
    (trace-fri-header-tied-same-layer-recorded-auth-gap ?s) ?s ≤ 0
  by (rule
    wp-trace-fri-header-tied-same-layer-recorded-auth-gap-bound-from-authenticated-recorded-auth-gap
    [OF authenticated-recorded-auth])
have same-auth:
  wp-event verify-monad
    (trace-fri-header-tied-same-layer-auth-gap ?s) ?s ≤ 0
  by (rule
    wp-trace-fri-header-tied-same-layer-auth-gap-bound-from-recorded-auth-gap
    [OF recorded-auth])
have auth-gap:
  wp-event verify-monad
    (trace-fri-header-tied-assignment-auth-gap ?s) ?s ≤ 0
  by (rule
    wp-trace-fri-header-tied-assignment-auth-gap-bound-from-same-layer-auth-gap
    [OF same-auth])

```

have *complement*:
 $wp\text{-event } verify\text{-monad}$
 $(\lambda out.$
 $\quad trace\text{-fri-header-tied-sampled-assignment-conflict } ?s \text{ out} \wedge$
 $\quad \neg trace\text{-fri-header-tied-recorded-sibling-candidate-missing } ?s \text{ out})$
 $?s \leq$
 $\quad (((Merkle + Merkle) + (Merkle + 0) + (Merkle + 0) +$
 $\quad (Merkle + 0 + 0)) + Merkle) + 0)$
by (*rule*
 $wp\text{-trace-fri-header-tied-sampled-assignment-conflict-without-recorded-missing-bound-from-auth-gap-and-replay-}$
 $(rule \text{ auth-gap}, rule \text{ merkle-bound}[OF \text{ support}], rule \text{ alignment},$
 $rule \text{ next-bound}, rule \text{ successor}, rule \text{ final})$
show $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-header-tied-sampled-assignment-conflict } ?s) ?s$
 $\leq Missing +$
 $\quad (((Merkle + Merkle) + (Merkle + 0) + (Merkle + 0) +$
 $\quad (Merkle + 0 + 0)) + Merkle) + 0)$
by (*rule*
 $wp\text{-trace-fri-header-tied-sampled-assignment-conflict-bound-from-recorded-missing-and-complement})$
 $(rule \text{ missing-bound}[OF \text{ support}], rule \text{ complement})$
next
fix s
show $\neg trace\text{-fri-header-tied-sampled-assignment-conflict } s \text{ None}$
unfolding $trace\text{-fri-header-tied-sampled-assignment-conflict-def}$
 $accepted\text{-fri-opening-transcript-def}$
by *simp*
qed
have *zero*:
 $wp\text{-event } (checked\text{-staged-security-experiment-with-data-state } A)$
 $(staged\text{-security-with-data-state-verifier-event}$
 $\quad trace\text{-fri-header-tied-zero-round-final-obstruction})$
 $adversary\text{-initial-state} \leq CheckedZero + Merkle$
by (*rule*
 $checked\text{-staged-security-trace-fri-header-tied-zero-round-bound-from-staged-checked-and-merkle})$
 $(rule \text{ checked-zero-bound}, rule \text{ merkle-bound})$
have $wp\text{-event } (checked\text{-staged-security-experiment-with-data-state } A)$
 $(staged\text{-security-with-data-state-verifier-event}$
 $\quad trace\text{-fri-bad-with-header-tied-augmented-partial-candidate})$
 $adversary\text{-initial-state} \leq$
 $wp\text{-event } (checked\text{-staged-security-experiment-with-data-state } A)$
 $(\lambda out.$
 $\quad staged\text{-security-with-data-state-verifier-event}$
 $\quad trace\text{-fri-bad-with-header-tied-augmented-sampled-layer-chain } out \vee$
 $\quad staged\text{-security-with-data-state-verifier-event}$
 $\quad trace\text{-fri-header-tied-sampled-assignment-conflict } out \vee$
 $\quad staged\text{-security-with-data-state-verifier-event}$
 $\quad trace\text{-fri-header-tied-zero-round-final-obstruction } out)$
 $adversary\text{-initial-state}$
unfolding $staged\text{-security-with-data-state-verifier-event-def}$

```

by (rule wp-event-mono-on-support)
  (auto dest:
    trace-fri-header-tied-augmented-partial-candidate-imp-sampled-conflict-or-zero
    split: option.splits prod.splits)
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-augmented-sampled-layer-chain)
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (λout.
      staged-security-with-data-state-verifier-event
        trace-fri-header-tied-sampled-assignment-conflict out ∨
      staged-security-with-data-state-verifier-event
        trace-fri-header-tied-zero-round-final-obstruction out)
    adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-augmented-sampled-layer-chain)
    adversary-initial-state +
  (wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-sampled-assignment-conflict)
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-zero-round-final-obstruction)
    adversary-initial-state)
by (intro add-mono order-refl wp-event-union-bound)
also have ... ≤
  (Fresh +
    hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)) +
  ((Missing +
    (((Merkle + Merkle) + (Merkle + 0) + (Merkle + 0) +
      (Merkle + 0 + 0)) + Merkle) + 0)) +
  (CheckedZero + Merkle))
by (intro add-mono sampled conflict zero)
finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-augmented-route-and-unaugmented*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: finite B*
and *cover:*

$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 trace-fri-sampled-layer-chain-cover
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad
and envelope:
 $\wedge$ data attacker-state trace-table.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 trace-fri-multiround-bad-sets bad trace-table $\subseteq B$
and fresh-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 (λs . trace-fri-challenge-list-fresh-hit s B))
 adversary-initial-state $\leq$ Fresh
and merkle-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Merkle
and missing-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-header-tied-recorded-sibling-candidate-missing
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Missing
and checked-zero-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-zero-round-checked-final-obstruction)
 adversary-initial-state $\leq$ CheckedZero
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-partial-candidate)

adversary-initial-state $\leq$
 ((*Fresh* +
 hash-relation-budget-value (*card B* * *ceil-log clength*)
 (*staged-attacker-query-budget budgets* +
 staged-challenge-query-budget)) +
 ((*Missing* +
 (((*Merkle* + *Merkle*) + (*Merkle* + 0) + (*Merkle* + 0) +
 (*Merkle* + 0 + 0)) + *Merkle*) + 0)) +
 (*CheckedZero* + *Merkle*))) +
 (*Missing* + *Merkle*)
proof –
have *augmented*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-header-tied-augmented-partial-candidate)
adversary-initial-state $\leq$
 (*Fresh* +
 hash-relation-budget-value (*card B* * *ceil-log clength*)
 (*staged-attacker-query-budget budgets* +
 staged-challenge-query-budget)) +
 ((*Missing* +
 (((*Merkle* + *Merkle*) + (*Merkle* + 0) + (*Merkle* + 0) +
 (*Merkle* + 0 + 0)) + *Merkle*) + 0)) +
 (*CheckedZero* + *Merkle*)))
by (rule
 checked-staged-security-trace-fri-header-tied-augmented-partial-candidate-bound-from-fresh-prequery-staged-check
 [*OF wf controlled finite-B cover envelope fresh-bound merkle-bound*
 missing-bound checked-zero-bound])
have *unaugmented*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-header-tied-unaugmented-partial-candidate)
adversary-initial-state $\leq$ *Missing* + *Merkle*
by (rule
 checked-staged-security-trace-fri-header-tied-unaugmented-partial-candidate-bound-from-recorded-missing-and-me
 (erule *merkle-bound*, erule *missing-bound*)
have *wp-event* (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (λ out.
 staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-augmented-partial-candidate out $\vee$
 staged-security-with-data-state-verifier-event
 trace-fri-header-tied-unaugmented-partial-candidate out)
 adversary-initial-state
unfolding *staged-security-with-data-state-verifier-event-def*
by (rule *wp-event-mono-on-support*)

```

(auto dest:
  trace-fri-bad-with-header-tied-partial-candidate-imp-augmented-or-unaugmented
  split: option.splits prod.splits)
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-augmented-partial-candidate)
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-unaugmented-partial-candidate)
    adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤
  ((Fresh +
    hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)) +
  ((Missing +
    (((Merkle + Merkle) + (Merkle + 0) + (Merkle + 0) +
      (Merkle + 0 + 0)) + Merkle) + 0)) +
  (CheckedZero + Merkle))) +
  (Missing + Merkle)
by (rule add-mono[OF augmented unaugmented])
finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-checked-zero-and-wi*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: finite B*
and *cover:*

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 trace-fri-sampled-layer-chain-cover
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad

and *envelope:*

$\bigwedge$ *data attacker-state trace-table.*

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 trace-fri-multiround-bad-sets bad trace-table $\subseteq B$

and *fresh-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s.$ trace-fri-challenge-list-fresh-hit s B))

$adversary-initial-state \leq Fresh$
and merkle-bound:
 $\wedge data\ attacker-state.$
 $Some\ (data,\ attacker-state) \in$
 $set-dist\ (execute\ (checked-staged-transcript-program\ A)$
 $adversary-initial-state) \implies$
 $wp-event\ verify-monad$
 $(partial-merkle-inconsistency-bad$
 $(verifier-state-from-adversary\ attacker-state$
 $(staged-proof-transcript\ data)))$
 $(verifier-state-from-adversary\ attacker-state$
 $(staged-proof-transcript\ data)) \leq Merkle$
and structural-bound:
 $\wedge data\ attacker-state.$
 $Some\ (data,\ attacker-state) \in$
 $set-dist\ (execute\ (checked-staged-transcript-program\ A)$
 $adversary-initial-state) \implies$
 $wp-event\ verify-monad$
 $(trace-fri-header-tied-sibling-mismatch-structural-gap$
 $(verifier-state-from-adversary\ attacker-state$
 $(staged-proof-transcript\ data)))$
 $(verifier-state-from-adversary\ attacker-state$
 $(staged-proof-transcript\ data)) \leq Structural$
and checked-zero-bound:
 $\wedge data\ attacker-state.$
 $Some\ (data,\ attacker-state) \in$
 $set-dist\ (execute\ (checked-staged-transcript-program\ A)$
 $adversary-initial-state) \implies$
 $wp-event\ verify-monad$
 $(trace-fri-zero-round-checked-final-obstruction$
 $(verifier-state-from-adversary\ attacker-state$
 $(staged-proof-transcript\ data)))$
 $(verifier-state-from-adversary\ attacker-state$
 $(staged-proof-transcript\ data)) \leq CheckedZero$
shows
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-verifier-event$
 $trace-fri-bad-with-header-tied-partial-candidate)$
 $adversary-initial-state \leq$
 $(Fresh +$
 $hash-relation-budget-value\ (card\ B * ceil-log\ clength)$
 $(staged-attacker-query-budget\ budgets +$
 $staged-challenge-query-budget)) +$
 $(((((Merkle + Structural) + Merkle) + (Merkle + 0) + (Merkle + 0) +$
 $(Merkle + 0 + 0)) + Merkle) + 0 + (CheckedZero + Merkle))$
proof (rule
 $checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-fresh-prequery-without-same-residuals$
 $[OF\ wf\ controlled\ finite-B\ cover\ envelope\ fresh-bound\ merkle-bound$
 $structural-bound])$

fix *data attacker-state*
assume *support*:
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)
show *wp-event verify-monad*
(trace-fri-header-tied-zero-round-final-obstruction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ CheckedZero + Merkle
by (*rule wp-trace-fri-header-tied-zero-round-bound-from-checked-and-merkle*)
(rule checked-zero-bound[OF support],
rule merkle-bound[OF support])
qed

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-without-sam*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: $\bigwedge dg. \text{finite } (B \text{ dg})$*
and *cover*:
 $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
composition-fri-sampled-layer-chain-cover
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) bad
and *envelope*:
 $\bigwedge \text{data attacker-state dg composition-table.}$
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B \text{ dg}$
and *fresh-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \text{ B}$))
adversary-initial-state ≤ Fresh
and *merkle-bound*:
 $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

```

    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data)) ≤ Merkle
and missing-bound:
 $\wedge$ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                      adversary-initial-state)  $\implies$ 
wp-event verify-monad
    (composition-fri-verifier-tied-recorded-base-chunk-auth-missing
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data)) ≤ Missing
and without-same-bound:
 $\wedge$ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                      adversary-initial-state)  $\implies$ 
wp-event verify-monad
    (composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data)) ≤ WithoutSame
shows
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state ≤
(Fresh +
 hash-relation-budget-value
  ( $\sum dg \in (UNIV :: 'f \text{ set}).$ 
   card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
  staged-challenge-query-budget)) +
(((Merkle + Missing) + (Merkle + 0) + (Merkle + 0) + WithoutSame +
 Merkle) + 0 + 0)
proof (rule
  checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-and-residual-gaps
  [OF wf controlled finite-B cover envelope fresh-bound])
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
                  adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad

```

```

    (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap ?s)
    ?s ≤ 0
  by (rule
    wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
  show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
    ≤ Merkle
  by (rule merkle-bound[OF support])
  show wp-event verify-monad
    (composition-fri-verifier-tied-recorded-base-chunk-auth-missing ?s) ?s
    ≤ Missing
  by (rule missing-bound[OF support])
  show wp-event verify-monad
    (composition-fri-verifier-tied-next-value-replay-gap ?s) ?s
    ≤ Merkle + 0
  by (rule
    wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-merkle-and-slot)
    (rule merkle-bound[OF support],
    rule
    wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
  show wp-event verify-monad
    (composition-fri-verifier-tied-successor-replay-gap ?s) ?s
    ≤ Merkle + 0
  by (rule
    wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-merkle-and-slot)
    (rule merkle-bound[OF support],
    rule
    wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
  show wp-event verify-monad
    (composition-fri-verifier-tied-final-value-replay-gap ?s) ?s
    ≤ WithoutSame
  by (rule
    wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-without-same)
    (rule without-same-bound[OF support])
  show wp-event verify-monad
    (composition-fri-verifier-tied-zero-round-final-obstruction ?s) ?s
    ≤ 0
  by (subst wp-composition-fri-verifier-tied-zero-round-final-obstruction-zero)
    simp
qed

```

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-without-miss*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: $\bigwedge dg. \text{finite } (B \ dg)$*
and *cover:*
 $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$

composition-fri-sampled-layer-chain-cover
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad
and envelope:
 $\wedge$ data attacker-state dg composition-table.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B$ dg
and fresh-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s.$ composition-fri-challenge-list-fresh-hit s B))
 adversary-initial-state $\leq$ Fresh
and merkle-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Merkle
and without-same-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ WithoutSame
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state $\leq$
 (Fresh +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (((Merkle + 0) + (Merkle + 0) + (Merkle + 0) + WithoutSame +

$\text{Merkle}) + 0 + 0)$
proof (rule
checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-without-same-resi
[**where** $\text{Missing} = 0$])
show *staged-budget-wellformed budgets*
by (rule *wf*)
show *staged-adversary-controlled budgets A*
by (rule *controlled*)
show $\bigwedge dg. \text{finite } (B \ dg)$
by (rule *finite-B*)
show $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state)
 $\in \text{set-dist}$
(execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
composition-fri-sampled-layer-chain-cover
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) bad
by (rule *cover*)
show $\bigwedge \text{data attacker-state dg composition-table.}$
Some (data, attacker-state)
 $\in \text{set-dist}$
(execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B \ dg$
by (rule *envelope*)
show *wp-event (checked-staged-security-experiment-with-data-state A)*
(staged-security-with-data-state-verifier-event
($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B$))
adversary-initial-state $\leq \text{Fresh}$
by (rule *fresh-bound*)
show $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state)
 $\in \text{set-dist}$
(execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq \text{Merkle}$
by (rule *merkle-bound*)
show $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state)
 $\in \text{set-dist}$
(execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$

wp-event verify-monad
 (composition-fri-verifier-tied-recorded-base-chunk-auth-missing
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ 0
by (subst wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero)
simp
show $\bigwedge$ data attacker-state.
 Some (data, attacker-state)
 $\in$ set-dist
 (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ WithoutSame
by (rule without-same-bound)
qed

lemma checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-zero-slot-mis

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and finite-B: $\bigwedge dg. \text{finite } (B \ dg)$

and cover:

$\bigwedge$ data attacker-state.

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

composition-fri-sampled-layer-chain-cover
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad

and envelope:

$\bigwedge$ data attacker-state dg composition-table.

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B \ dg$

and fresh-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B$))
 adversary-initial-state $\leq$ Fresh

and merkle-bound:

$\bigwedge$ data attacker-state.

Some (data, attacker-state) $\in$

$set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(partial-merkle-inconsistency-bad$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq Merkle$
and *next-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-verifier-tied-next-value-replay-gap$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq Next$
and *successor-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-verifier-tied-successor-replay-gap$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq Successor$
and *final-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-verifier-tied-final-value-replay-gap$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq Final$
shows
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $composition-fri-bad-with-verifier-tied-partial-candidate)$
 $adversary-initial-state \leq$
 $(Fresh +$
 $hash-relation-budget-value$
 $(\sum dg \in (UNIV :: 'f \ set).$

```

      card (B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget)) +
    (((Merkle + 0) + Next + Successor + Final + Merkle) + 0 + 0)
proof (rule
  checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-and-residual-gaps
  [where Slot = 0 and Missing = 0 and Zero = 0])
show staged-budget-wellformed budgets
  by (rule wf)
show staged-adversary-controlled budgets A
  by (rule controlled)
show  $\bigwedge dg.$  finite (B dg)
  by (rule finite-B)
show  $\bigwedge$  data attacker-state.
  Some (data, attacker-state)
   $\in$  set-dist
    (execute (checked-staged-transcript-program A)
     adversary-initial-state)  $\implies$ 
    composition-fri-sampled-layer-chain-cover
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data)) bad
  by (rule cover)
show  $\bigwedge$  data attacker-state dg composition-table.
  Some (data, attacker-state)
   $\in$  set-dist
    (execute (checked-staged-transcript-program A)
     adversary-initial-state)  $\implies$ 
    composition-fri-multiround-bad-sets bad dg composition-table
     $\subseteq$  B dg
  by (rule envelope)
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   ( $\lambda s.$  composition-fri-challenge-list-fresh-hit s B))
  adversary-initial-state  $\leq$  Fresh
  by (rule fresh-bound)
show  $\bigwedge$  data attacker-state.
  Some (data, attacker-state)
   $\in$  set-dist
    (execute (checked-staged-transcript-program A)
     adversary-initial-state)  $\implies$ 
  wp-event verify-monad
    (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))  $\leq$  0
  by (rule
    wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
show  $\bigwedge$  data attacker-state.

```


Some (data, attacker-state)
 $\in \text{set-dist}$
 $(\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Merkle}$
by (rule merkle-bound)
show $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state)
 $\in \text{set-dist}$
 $(\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-recorded-base-chunk-auth-missing}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq 0$
by (subst wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero)
 simp
show $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state)
 $\in \text{set-dist}$
 $(\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Next}$
by (rule next-bound)
show $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state)
 $\in \text{set-dist}$
 $(\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Successor}$
by (rule successor-bound)
show $\bigwedge \text{data attacker-state.}$

Some (data, attacker-state)
 $\in \text{set-dist}$
 $(\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Final}$
by (rule final-bound)
show $\bigwedge \text{data attacker-state.}$
Some (data, attacker-state)
 $\in \text{set-dist}$
 $(\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq 0$
by (subst wp-composition-fri-verifier-tied-zero-round-final-obstruction-zero)
simp
qed

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-zero-slot-mis*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *finite-B: $\bigwedge dg. \text{finite } (B \ dg)$*

and *cover:*

$\bigwedge \text{data attacker-state.}$

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$

composition-fri-sampled-layer-chain-cover
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) bad

and *envelope:*

$\bigwedge \text{data attacker-state dg composition-table.}$

Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B \ dg$

and *fresh-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B)$)

$\text{adversary-initial-state} \leq \text{Fresh}$
and merkle-bound:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{Merkle}$
shows
 $\text{wp-event (checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq$
 $(\text{Fresh} +$
 $\text{hash-relation-budget-value}$
 $(\sum dg \in (\text{UNIV} :: 'f \text{ set}).$
 $\text{card } (B \text{ } dg) * \text{ceil-log (maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $((\text{Merkle} + 0) + (\text{Merkle} + 0) + (\text{Merkle} + 0) + (\text{Merkle} + 0 + 0) +$
 $\text{Merkle}) + 0 + 0)$
proof (rule
 $\text{checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-zero-slot-missing-}$
show staged-budget-wellformed budgets
by (rule wf)
show staged-adversary-controlled budgets A
by (rule controlled)
show $\bigwedge dg. \text{finite } (B \text{ } dg)$
by (rule finite-B)
show $\bigwedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)}$
 $\in \text{set-dist}$
 $(\text{execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{composition-fri-sampled-layer-chain-cover}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad}$
by (rule cover)
show $\bigwedge \text{data attacker-state } dg \text{ composition-table.}$
 $\text{Some (data, attacker-state)}$
 $\in \text{set-dist}$
 $(\text{execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{composition-fri-multiround-bad-sets bad } dg \text{ composition-table}$
 $\subseteq B \text{ } dg$

```

    by (rule envelope)
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s.$  composition-fri-challenge-list-fresh-hit s B))
    adversary-initial-state  $\leq$  Fresh
  by (rule fresh-bound)
  show  $\bigwedge$ data attacker-state.
    Some (data, attacker-state)
     $\in$  set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
    wp-event verify-monad
      (partial-merkle-inconsistency-bad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))  $\leq$  Merkle
  by (rule merkle-bound)
  show  $\bigwedge$ data attacker-state.
    Some (data, attacker-state)
     $\in$  set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
    wp-event verify-monad
      (composition-fri-verifier-tied-next-value-replay-gap
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))  $\leq$  Merkle + 0
  proof -
    fix data attacker-state
    assume support:
      Some (data, attacker-state)
       $\in$  set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
    let ?s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    show wp-event verify-monad
      (composition-fri-verifier-tied-next-value-replay-gap ?s) ?s
       $\leq$  Merkle + 0
    by (rule
      wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-merkle-and-slot)
      (rule merkle-bound[OF support],
        rule
          wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
  qed
  show  $\bigwedge$ data attacker-state.

```

```

    Some (data, attacker-state)
  ∈ set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
wp-event verify-monad
  (composition-fri-verifier-tied-successor-replay-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ Merkle + 0
proof -
  fix data attacker-state
  assume support:
    Some (data, attacker-state)
  ∈ set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  show wp-event verify-monad
    (composition-fri-verifier-tied-successor-replay-gap ?s) ?s
    ≤ Merkle + 0
  by (rule
    wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-merkle-and-slot)
  (rule merkle-bound[OF support],
    rule
      wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
qed
show ∧data attacker-state.
  Some (data, attacker-state)
  ∈ set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
wp-event verify-monad
  (composition-fri-verifier-tied-final-value-replay-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ Merkle + 0 + 0
proof -
  fix data attacker-state
  assume support:
    Some (data, attacker-state)
  ∈ set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state

```

```

    (staged-proof-transcript data)
  show wp-event verify-monad
    (composition-fri-verifier-tied-final-value-replay-gap ?s) ?s
    ≤ Merkle + 0 + 0
  by (rule
    wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step)
    (rule merkle-bound[OF support],
    rule
      wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero,
    rule
      wp-composition-fri-verifier-tied-final-value-selected-step-missing-gap-zero)
qed
qed

```

lemma *active-composition-fri-verifier-tied-reductions-from-sampled-and-merkle-bounds:*

fixes *Sampled Merkle ::*

'f staged-proof-data ⇒ 'f protocol-channel ⇒ prob

assumes *sampled-bound:*

$\bigwedge \text{data attacker-state.}$

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (composition-fri-bad-with-verifier-tied-sampled-layer-chain
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ Sampled data attacker-state

and *merkle-bound:*

$\bigwedge \text{data attacker-state.}$

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ Merkle data attacker-state

and *total-bound:*

$\bigwedge \text{data attacker-state.}$

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 Sampled data attacker-state +
 (((Merkle data attacker-state + 0) +
 (Merkle data attacker-state + 0) +
 (Merkle data attacker-state + 0) +
 (Merkle data attacker-state + 0 + 0) +

$\text{Merkle data attacker-state}) + 0)$
 $\leq \text{composition-fri-error}$
and support:
Some (*data*, *attacker-state*) $\in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$
shows
 $\text{composition-fri-verifier-tied-reduction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
proof –
let $?s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
have slot:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap}$
 $?s) ?s \leq 0$
by (*rule*
 $\text{wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero})$
have next-gap:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap } ?s) ?s$
 $\leq \text{Merkle data attacker-state} + 0$
by (*rule*
 $\text{wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-merkle-and-slot})$
 $(\text{rule merkle-bound}[OF \text{ support}], \text{rule slot})$
have successor:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap } ?s) ?s$
 $\leq \text{Merkle data attacker-state} + 0$
by (*rule*
 $\text{wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-merkle-and-slot})$
 $(\text{rule merkle-bound}[OF \text{ support}], \text{rule slot})$
have final:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap } ?s) ?s$
 $\leq \text{Merkle data attacker-state} + 0 + 0$
by (*rule*
 $\text{wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step})$
 $(\text{rule merkle-bound}[OF \text{ support}], \text{rule slot},$
 rule
 $\text{wp-composition-fri-verifier-tied-final-value-selected-step-missing-gap-zero})$
show $?thesis$
by (*rule*
 $\text{composition-fri-verifier-tied-reduction-from-residual-bounds-with-recorded-missing-zero}$
 $[OF \text{ sampled-bound}[OF \text{ support}] \text{ slot merkle-bound}[OF \text{ support}]$
 $\text{next-gap successor final total-bound}[OF \text{ support}]])$
qed

end

end

theory *Soundness-FRI-Full-Cover-Active-Bounds*

imports

Soundness-FRI-Obligation-Normalization

Soundness-FRI-Full-Cover-Domains

begin

Compact active-route bridges for the full-cover FRI split.

The imported layers already prove:

- a full-cover challenge-list/envelope bound;
- a missing-full-cover domain split;
- active trace/composition residual adapters.

This theory only packages those ingredients in the shape consumed by the same-run public soundness route. It does not add protocol assumptions or change the FRI model.

context *soundness*

begin

lemma *wp-trace-fri-header-tied-sampled-layer-chain-bound-from-envelope-domain-split:*

assumes *future: trace-fri-future-fresh s*

and *envelope:*

$\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$

and *subset: $B \subseteq \text{fri-challenge-space } (\text{ceil-log clength})$*

and *challenge-bound:*

$\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log clength}) \leq R$

and *proximity:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$

and *index:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$

and *domain-length:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-domain-length-failure } s) \ s \leq DL$

and *sampled-domain:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-sampled-domain-failure } s) \ s \leq SD$

shows

wp-event verify-monad

$(\text{trace-fri-bad-with-header-tied-sampled-layer-chain } s) \ s$

$\leq R + (P + I + (I + DL) + (I + SD))$

proof *(rule*

$wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}sampled\text{-}layer\text{-}chain\text{-}bound\text{-}from\text{-}full\text{-}cover\text{-}and\text{-}missing)$
show $wp\text{-}event\text{-}verify\text{-}monad$
 $(trace\text{-}fri\text{-}bad\text{-}with\text{-}full\text{-}cover\text{-}layer\text{-}chain\text{-}s\text{-}bad)\ s \leq R$
by (rule $wp\text{-}trace\text{-}fri\text{-}bad\text{-}with\text{-}full\text{-}cover\text{-}layer\text{-}chain\text{-}bound\text{-}from\text{-}envelope$
 $[OF\ future\ envelope\ subset\ challenge\text{-}bound])$)
show $wp\text{-}event\text{-}verify\text{-}monad$
 $(trace\text{-}fri\text{-}sampled\text{-}missing\text{-}full\text{-}cover\text{-}s\text{-}bad)\ s$
 $\leq P + I + (I + DL) + (I + SD)$
by (rule $wp\text{-}trace\text{-}fri\text{-}sampled\text{-}missing\text{-}full\text{-}cover\text{-}bound\text{-}from\text{-}domain\text{-}split$
 $[OF\ proximity\ index\ domain\text{-}length\ sampled\text{-}domain])$)
qed

lemma $wp\text{-}composition\text{-}fri\text{-}verifier\text{-}tied\text{-}sampled\text{-}layer\text{-}chain\text{-}bound\text{-}from\text{-}envelope\text{-}domain\text{-}split$:
assumes $future$: $composition\text{-}fri\text{-}future\text{-}fresh\ s$
and $envelope$:
 $\bigwedge dg\ composition\text{-}table.$
 $composition\text{-}fri\text{-}multi\text{-}round\text{-}bad\text{-}sets\ bad\ dg\ composition\text{-}table$
 $\subseteq B\ dg$
and $subset$:
 $\bigwedge dg.\ B\ dg \subseteq fri\text{-}challenge\text{-}space\ (ceil\text{-}log\ (to\text{-}nat\ dg + 1))$
and $challenge\text{-}bound$:
 $\bigwedge dg.\ nnreal\ (card\ (B\ dg))\ /\$
 $nnreal\ (CARD('f) \wedge ceil\text{-}log\ (to\text{-}nat\ dg + 1)) \leq R$
and $proximity$:
 $wp\text{-}event\text{-}verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}proximity\text{-}failure\ s\ bad)\ s$
 $\leq P$
and $index$:
 $wp\text{-}event\text{-}verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}index\text{-}failure\ s)\ s \leq I$
and $domain\text{-}length$:
 $wp\text{-}event\text{-}verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}domain\text{-}length\text{-}failure\ s)\ s$
 $\leq DL$
and $sampled\text{-}domain$:
 $wp\text{-}event\text{-}verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}sampled\text{-}domain\text{-}failure\ s)\ s$
 $\leq SD$
shows
 $wp\text{-}event\text{-}verify\text{-}monad$
 $(composition\text{-}fri\text{-}bad\text{-}with\text{-}verifier\text{-}tied\text{-}sampled\text{-}layer\text{-}chain\ s)\ s$
 $\leq R + (P + I + (I + DL) + (I + SD))$
proof (rule
 $wp\text{-}composition\text{-}fri\text{-}verifier\text{-}tied\text{-}sampled\text{-}layer\text{-}chain\text{-}bound\text{-}from\text{-}full\text{-}cover\text{-}and\text{-}missing)$
show $wp\text{-}event\text{-}verify\text{-}monad$
 $(composition\text{-}fri\text{-}bad\text{-}with\text{-}full\text{-}cover\text{-}layer\text{-}chain\ s\ bad)\ s \leq R$
by (rule
 $wp\text{-}composition\text{-}fri\text{-}bad\text{-}with\text{-}full\text{-}cover\text{-}layer\text{-}chain\text{-}bound\text{-}from\text{-}envelope$
 $[OF\ future\ envelope\ subset\ challenge\text{-}bound])$)

show *wp-event verify-monad*
 (*composition-fri-sampled-missing-full-cover s bad*) *s*
 $\leq P + I + (I + DL) + (I + SD)$
by (rule
wp-composition-fri-sampled-missing-full-cover-bound-from-domain-split
 [OF *proximity index domain-length sampled-domain*])
qed

lemma *wp-trace-fri-header-tied-sampled-layer-chain-bound-from-envelope-witness-domain:*
assumes *future: trace-fri-future-fresh s*
and *envelope:*
 $\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
and *subset: $B \subseteq \text{fri-challenge-space (ceil-log clength)}$*
and *challenge-bound:*
 $\text{nnreal (card } B) / \text{nnreal (CARD('f) } ^{\text{ceil-log clength}}) \leq R$
and *proximity:*
wp-event verify-monad
 (*trace-fri-sampled-full-cover-proximity-failure s bad*) *s* $\leq P$
and *index:*
wp-event verify-monad
 (*trace-fri-sampled-full-cover-index-failure s*) *s* $\leq I$
and *witness-length:*
wp-event verify-monad
 (*trace-fri-sampled-full-cover-witness-length-failure s*) *s* $\leq W$
and *domain:*
wp-event verify-monad
 (*trace-fri-sampled-full-cover-domain-failure s*) *s* $\leq D$
shows
wp-event verify-monad
 (*trace-fri-bad-with-header-tied-sampled-layer-chain s*) *s*
 $\leq R + (P + I + W + D)$
proof (rule
wp-trace-fri-header-tied-sampled-layer-chain-bound-from-full-cover-and-missing)
show *wp-event verify-monad*
 (*trace-fri-bad-with-full-cover-layer-chain s bad*) *s* $\leq R$
by (rule *wp-trace-fri-bad-with-full-cover-layer-chain-bound-from-envelope*
 [OF *future envelope subset challenge-bound*])
show *wp-event verify-monad*
 (*trace-fri-sampled-missing-full-cover s bad*) *s* $\leq P + I + W + D$
by (rule
wp-trace-fri-sampled-missing-full-cover-bound-from-witness-length-subcases
 [OF *proximity index witness-length domain*])
qed

lemma *wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-envelope-witness-domain:*
assumes *future: composition-fri-future-fresh s*
and *envelope:*
 $\bigwedge dg \text{ composition-table. }$
composition-fri-multiround-bad-sets bad dg composition-table

$\subseteq B \text{ dg}$
and subset:
 $\bigwedge \text{dg. } B \text{ dg} \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat dg} + 1))$
and challenge-bound:
 $\bigwedge \text{dg. } \text{nnreal } (\text{card } (B \text{ dg})) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat dg} + 1)) \leq R$
and proximity:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \text{ } s$
 $\leq P$
and index:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-index-failure } s) \text{ } s \leq I$
and witness-length:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-witness-length-failure } s) \text{ } s$
 $\leq W$
and domain:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-domain-failure } s) \text{ } s \leq D$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain } s) \text{ } s$
 $\leq R + (P + I + W + D)$
proof (rule
 $\text{wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-full-cover-and-missing})$
show $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad}) \text{ } s \leq R$
by (rule
 $\text{wp-composition-fri-bad-with-full-cover-layer-chain-bound-from-envelope}$
 $[\text{OF future envelope subset challenge-bound}])$
show $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-missing-full-cover } s \text{ bad}) \text{ } s$
 $\leq P + I + W + D$
by (rule
 $\text{wp-composition-fri-sampled-missing-full-cover-bound-from-witness-length-subcases}$
 $[\text{OF proximity index witness-length domain}])$
qed

lemma *trace-fri-header-tied-residual-obligations-from-envelope-domain-split-without-head-slot-and-selected-zero:*

assumes *future: trace-fri-future-fresh s*
and envelope:
 $\bigwedge \text{trace-table. } \text{trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
and subset: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and challenge-bound:
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength}) \leq R$
and proximity:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \text{ } s \leq P$

and *index*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$
and *domain-length*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-domain-length-failure } s) \ s \leq DL$
and *sampled-domain*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-sampled-domain-failure } s) \ s \leq SD$
and *zero*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s \leq z\text{-}b$
and *merkle*:
 $wp\text{-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq p\text{-}b$
and *structural*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s$
 $\leq h\text{-}b$
and *next-bound*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq n\text{-}b$
and *successor*:
 $wp\text{-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s \leq s\text{-}b$
and *total*:
 $(R + (P + I + (I + DL) + (I + SD))) +$
 $(((((p\text{-}b + h\text{-}b) + p\text{-}b) + n\text{-}b + s\text{-}b +$
 $(p\text{-}b + 0 + 0)) + p\text{-}b) + 0 + z\text{-}b)$
 $\leq \text{trace-fri-error}$

shows

$\text{trace-fri-header-tied-residual-obligations } s$
 $(R + (P + I + (I + DL) + (I + SD))) \ z\text{-}b \ 0 \ p\text{-}b \ h\text{-}b \ p\text{-}b$
 $n\text{-}b \ s\text{-}b \ 0$

by (rule

$\text{trace-fri-header-tied-residual-obligations-from-bounds-without-head-slot-and-selected-zero})$
(rule
 $wp\text{-trace-fri-header-tied-sampled-layer-chain-bound-from-envelope-domain-split}$
 $[OF \text{ future envelope subset challenge-bound proximity index}$
 $\text{domain-length sampled-domain}]$,
rule zero, rule merkle, rule structural, rule next-bound,
rule successor, rule total)

lemma *composition-fri-verifier-tied-residual-obligations-from-envelope-domain-split-slot-zero:*

assumes *future: composition-fri-future-fresh* s

and *envelope*:

$\bigwedge dg \text{ composition-table.}$

$\text{composition-fri-multiround-bad-sets bad } dg \text{ composition-table}$

$\subseteq B \ dg$

and *subset*:

$\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (to\text{-nat } dg + 1))$
and *challenge-bound*:
 $\bigwedge dg. \text{nnreal } (\text{card } (B \ dg)) /$
 $\text{nnreal } (CARD('f) \wedge \text{ceil-log } (to\text{-nat } dg + 1)) \leq R$
and *proximity*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s$
 $\leq P$
and *index*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-index-failure } s) \ s \leq I$
and *domain-length*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-domain-length-failure } s) \ s$
 $\leq DL$
and *sampled-domain*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-sampled-domain-failure } s) \ s$
 $\leq SD$
and *merkle*:
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq p\text{-b}$
and *next-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap } s) \ s \leq n\text{-b}$
and *successor*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap } s) \ s \leq s\text{-b}$
and *final-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \ s \leq g\text{-b}$
and *total*:
 $(R + (P + I + (I + DL) + (I + SD))) +$
 $((p\text{-b} + 0) + n\text{-b} + s\text{-b} + g\text{-b} + p\text{-b}) + 0)$
 $\leq \text{composition-fri-error}$

shows

$\text{composition-fri-verifier-tied-residual-obligations } s$
 $(R + (P + I + (I + DL) + (I + SD))) \ 0 \ p\text{-b} \ n\text{-b} \ s\text{-b} \ g\text{-b}$
by $(\text{rule composition-fri-verifier-tied-residual-obligations-from-bounds-slot-zero})$
 $(\text{rule}$
 $\text{wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-envelope-domain-split}$
 $[\text{OF future envelope subset challenge-bound proximity index}$
 $\text{domain-length sampled-domain}],$
 $\text{rule merkle, rule next-bound, rule successor, rule final-bound,}$
 $\text{rule total})$

lemma *trace-fri-header-tied-residual-obligations-from-envelope-witness-domain-without-head-slot-and-selected-zero*:

assumes *future*: $\text{trace-fri-future-fresh } s$

and *envelope*:

$\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$

and *subset*: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and *challenge-bound*:
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log } \text{clength}) \leq R$
and *proximity*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$
and *index*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$
and *witness-length*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-witness-length-failure } s) \ s \leq W$
and *domain*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-domain-failure } s) \ s \leq D$
and *zero*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s \leq z-b$
and *merkle*:
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \ s \leq p-b$
and *structural*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s$
 $\leq h-b$
and *next-bound*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq n-b$
and *successor*:
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-successor-replay-gap } s) \ s \leq s-b$
and *total*:
 $(R + (P + I + W + D)) +$
 $(((((p-b + h-b) + p-b) + n-b + s-b +$
 $(p-b + 0 + 0)) + p-b) + 0 + z-b)$
 $\leq \text{trace-fri-error}$
shows
 $\text{trace-fri-header-tied-residual-obligations } s$
 $(R + (P + I + W + D)) \ z-b \ 0 \ p-b \ h-b \ p-b \ n-b \ s-b \ 0$
by (rule
 $\text{trace-fri-header-tied-residual-obligations-from-bounds-without-head-slot-and-selected-zero})$
(rule
 $\text{wp-trace-fri-header-tied-sampled-layer-chain-bound-from-envelope-witness-domain}$
 $[\text{OF future envelope subset challenge-bound proximity index}$
 $\text{witness-length domain}]$,
rule zero, rule merkle, rule structural, rule next-bound,
rule successor, rule total)

lemma *composition-fri-verifier-tied-residual-obligations-from-envelope-witness-domain-slot-zero*:
assumes *future*: *composition-fri-future-fresh* s

and *envelope*:
 $\bigwedge dg \text{ composition-table.}$
 $\text{composition-fri-multiround-bad-sets bad } dg \text{ composition-table}$
 $\subseteq B \text{ } dg$
and *subset*:
 $\bigwedge dg. B \text{ } dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (to\text{-nat } dg + 1))$
and *challenge-bound*:
 $\bigwedge dg. \text{nnreal } (\text{card } (B \text{ } dg)) /$
 $\text{nnreal } (CARD('f) \wedge \text{ceil-log } (to\text{-nat } dg + 1)) \leq R$
and *proximity*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \text{ } s$
 $\leq P$
and *index*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-index-failure } s) \text{ } s \leq I$
and *witness-length*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-witness-length-failure } s) \text{ } s$
 $\leq W$
and *domain*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-domain-failure } s) \text{ } s \leq D$
and *merkle*:
 $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } s) \text{ } s \leq p\text{-b}$
and *next-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-next-value-replay-gap } s) \text{ } s \leq n\text{-b}$
and *successor*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-successor-replay-gap } s) \text{ } s \leq s\text{-b}$
and *final-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \text{ } s \leq g\text{-b}$
and *total*:
 $(R + (P + I + W + D)) +$
 $((p\text{-b} + 0) + n\text{-b} + s\text{-b} + g\text{-b} + p\text{-b}) + 0)$
 $\leq \text{composition-fri-error}$
shows
 $\text{composition-fri-verifier-tied-residual-obligations } s$
 $(R + (P + I + W + D)) \text{ } 0 \text{ } p\text{-b } n\text{-b } s\text{-b } g\text{-b}$
by (rule $\text{composition-fri-verifier-tied-residual-obligations-from-bounds-slot-zero}$)
(rule
 $\text{wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-envelope-witness-domain}$
 $[OF \text{ future envelope subset challenge-bound proximity index}$
 $\text{witness-length domain}]$,
rule merkle, rule next-bound, rule successor, rule final-bound,
rule total)

end

end

theory *Soundness-FRI-Full-Cover-Staged-Bounds*

imports

Soundness-FRI-Prequery-Composition-Sampled-Bounds

Soundness-FRI-Full-Cover-Active-Bounds

begin

Staged bounds for the full-cover FRI split.

Unlike the verifier-local full-cover envelope lemmas, these results work in the staged adversary experiment where future FRI keys may have been queried by the attacker. The full-cover part is therefore routed through the existing staged fresh/prequery accounting for FRI challenge-list hits.

context *soundness*

begin

lemma *trace-fri-bad-with-sampled-layer-chain-not-None:*

$\neg$ *trace-fri-bad-with-sampled-layer-chain* *s* *None*

unfolding *trace-fri-bad-with-sampled-layer-chain-def*

trace-fri-partial-candidate-opening-evidence-def

accepted-fri-opening-transcript-def

by *blast*

lemma *composition-fri-bad-with-sampled-layer-chain-not-None:*

$\neg$ *composition-fri-bad-with-sampled-layer-chain* *s* *None*

unfolding *composition-fri-bad-with-sampled-layer-chain-def*

composition-fri-partial-candidate-opening-evidence-def

accepted-fri-opening-transcript-def

by *blast*

lemma *trace-fri-sampled-missing-full-cover-not-None:*

$\neg$ *trace-fri-sampled-missing-full-cover* *s* *bad* *None*

unfolding *trace-fri-sampled-missing-full-cover-def*

using *trace-fri-bad-with-sampled-layer-chain-not-None* **by** *blast*

lemma *composition-fri-sampled-missing-full-cover-not-None:*

$\neg$ *composition-fri-sampled-missing-full-cover* *s* *bad* *None*

unfolding *composition-fri-sampled-missing-full-cover-def*

using *composition-fri-bad-with-sampled-layer-chain-not-None* **by** *blast*

lemma *checked-staged-security-trace-fri-full-cover-layer-chain-bound-from-fresh-and-prequery:*

assumes *wf: staged-budget-wellformed* *budgets*

and *controlled: staged-adversary-controlled* *budgets* *A*

and *finite-B: finite* *B*

and *envelope:*

$\wedge$ *data* *attacker-state* *trace-table*.

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
trace-fri-multiround-bad-sets bad trace-table ⊆ B
and *fresh-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
(λs. trace-fri-challenge-list-fresh-hit s B))
adversary-initial-state ≤ F
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
(λs. trace-fri-bad-with-full-cover-layer-chain s bad))
adversary-initial-state
≤ F +
*hash-relation-budget-value (card B * ceil-log clength)*
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)
proof –
have *wp-event (checked-staged-security-experiment-with-data-state A)*
(staged-security-with-data-state-verifier-event
(λs. trace-fri-bad-with-full-cover-layer-chain s bad))
adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
(λs. trace-fri-challenge-list-set-hit s B))
adversary-initial-state
proof (rule *wp-event-mono-on-support*)
fix *out*
assume *support:*
out ∈
set-dist
(execute (checked-staged-security-experiment-with-data-state A)
adversary-initial-state)
and *hit:*
staged-security-with-data-state-verifier-event
(λs. trace-fri-bad-with-full-cover-layer-chain s bad) out
from *hit* **obtain** *data attacker-state result final-state* **where**
out-eq: out = Some (((data, attacker-state), result), final-state)
unfolding *staged-security-with-data-state-verifier-event-def*
by (cases *out*) (auto split: *option.splits prod.splits*)
have *builder:*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)
using *checked-staged-security-experiment-with-data-state-outcomeE*
[OF support[unfolded out-eq]]
by *blast*
let *?s =*

```

    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have full:
    trace-fri-bad-with-full-cover-layer-chain ?s bad
      (Some (result, final-state))
  using hit unfolding staged-security-with-data-state-verifier-event-def
    out-eq by simp
  have list-hit:
    trace-fri-challenge-list-set-hit ?s B
      (Some (result, final-state))
  by (rule trace-fri-bad-with-full-cover-layer-chain-imp-list-hit
      [OF full])
    (rule envelope[OF builder])
  show
    staged-security-with-data-state-verifier-event
      ( $\lambda s.$  trace-fri-challenge-list-set-hit s B) out
  unfolding staged-security-with-data-state-verifier-event-def out-eq
  using list-hit by simp
qed
also have ...  $\leq$ 
  F +
    hash-relation-budget-value (card B * ceil-log clength)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  by (rule checked-staged-security-trace-fri-challenge-list-set-hit-bound
      [OF wf controlled finite-B fresh-bound])
  finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-full-cover-layer-chain-bound-from-fresh-and-prequery:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B:  $\bigwedge dg.$  finite (B dg)
  and envelope:
     $\bigwedge data$  attacker-state dg composition-table.
      Some (data, attacker-state)  $\in$ 
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state)  $\implies$ 
        composition-fri-multiround-bad-sets bad dg composition-table
           $\subseteq$  B dg
  and fresh-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        ( $\lambda s.$  composition-fri-challenge-list-fresh-hit s B))
    adversary-initial-state  $\leq$  F
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        ( $\lambda s.$  composition-fri-bad-with-full-cover-layer-chain s bad))

```

```

    adversary-initial-state
  ≤ F +
    hash-relation-budget-value
    (∑ dg ∈ (UNIV :: 'f set).
      card (B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
proof –
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-bad-with-full-cover-layer-chain s bad))
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. composition-fri-challenge-list-set-hit s B))
      adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
      set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  and hit:
    staged-security-with-data-state-verifier-event
      (λs. composition-fri-bad-with-full-cover-layer-chain s bad) out
  from hit obtain data attacker-state result final-state where
    out-eq: out = Some (((data, attacker-state), result), final-state)
  unfolding staged-security-with-data-state-verifier-event-def
  by (cases out) (auto split: option.splits prod.splits)
  have builder:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  using checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
  by blast
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have full:
    composition-fri-bad-with-full-cover-layer-chain ?s bad
      (Some (result, final-state))
  using hit unfolding staged-security-with-data-state-verifier-event-def
    out-eq by simp
  have list-hit:
    composition-fri-challenge-list-set-hit ?s B
      (Some (result, final-state))
  by (rule composition-fri-bad-with-full-cover-layer-chain-imp-list-hit)

```

```

    [OF full])
  (rule envelope[OF builder])
show
  staged-security-with-data-state-verifier-event
  (λs. composition-fri-challenge-list-set-hit s B) out
unfolding staged-security-with-data-state-verifier-event-def out-eq
using list-hit by simp
qed
also have ... ≤
  F +
  hash-relation-budget-value
  (∑ dg ∈ (UNIV :: 'f set).
    card (B dg) * ceil-log (maxDegree + 1))
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)
by (rule
  checked-staged-security-composition-fri-challenge-list-set-hit-bound
  [OF wf controlled finite-B fresh-bound])
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-sampled-missing-full-cover-bound-from-domain-split:
assumes proximity:
  ∧data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  wp-event verify-monad
    (trace-fri-sampled-full-cover-proximity-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) bad)
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ P
and index:
  ∧data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  wp-event verify-monad
    (trace-fri-sampled-full-cover-index-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ I
and domain-length:
  ∧data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒

```

```

wp-event verify-monad
  (trace-fri-sampled-full-cover-domain-length-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ DL
and sampled-domain:
 $\wedge$ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (trace-fri-sampled-full-cover-sampled-domain-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ SD
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s$ . trace-fri-sampled-missing-full-cover s bad))
    adversary-initial-state ≤ P + I + (I + DL) + (I + SD)
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\wedge s$ .  $\neg$  trace-fri-sampled-missing-full-cover s bad None
  by (rule trace-fri-sampled-missing-full-cover-not-None)
next
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
  (trace-fri-sampled-missing-full-cover ?s bad) ?s
  ≤ P + I + (I + DL) + (I + SD)
by (rule wp-trace-fri-sampled-missing-full-cover-bound-from-domain-split)
  (rule proximity[OF support], rule index[OF support],
    rule domain-length[OF support], rule sampled-domain[OF support])
qed

lemma checked-staged-security-trace-fri-sampled-layer-chain-bound-from-full-cover-fresh-and-domain-split:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B: finite B
  and envelope:
 $\wedge$ data attacker-state trace-table.
    Some (data, attacker-state) ∈

```

$\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
and *fresh-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B))$
 $\text{adversary-initial-state} \leq F$
and *proximity*:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-proximity-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad})$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq P$
and *index*:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-index-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq I$
and *domain-length*:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-domain-length-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq DL$
and *sampled-domain*:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-sampled-domain-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$

$$\begin{aligned}
& (staged-proof-transcript\ data))) \\
& (verifier-state-from-adversary\ attacker-state \\
& (staged-proof-transcript\ data)) \leq SD
\end{aligned}$$

shows

$$\begin{aligned}
& wp-event\ (checked-staged-security-experiment-with-data-state\ A) \\
& (staged-security-with-data-state-verifier-event \\
& \quad trace-fri-bad-with-sampled-layer-chain) \\
& adversary-initial-state \\
& \leq \\
& (F + \\
& \quad hash-relation-budget-value\ (card\ B * ceil-log\ clength) \\
& \quad (staged-attacker-query-budget\ budgets + \\
& \quad \quad staged-challenge-query-budget)) + \\
& (P + I + (I + DL) + (I + SD))
\end{aligned}$$

proof –

have *full-bound*:

$$\begin{aligned}
& wp-event\ (checked-staged-security-experiment-with-data-state\ A) \\
& (staged-security-with-data-state-verifier-event \\
& \quad (\lambda s. trace-fri-bad-with-full-cover-layer-chain\ s\ bad)) \\
& adversary-initial-state \\
& \leq \\
& F + \\
& \quad hash-relation-budget-value\ (card\ B * ceil-log\ clength) \\
& \quad (staged-attacker-query-budget\ budgets + \\
& \quad \quad staged-challenge-query-budget)
\end{aligned}$$

by (rule

$$\begin{aligned}
& checked-staged-security-trace-fri-full-cover-layer-chain-bound-from-fresh-and-prequery \\
& [OF\ wf\ controlled\ finite-B\ envelope\ fresh-bound])
\end{aligned}$$

have *missing-bound*:

$$\begin{aligned}
& wp-event\ (checked-staged-security-experiment-with-data-state\ A) \\
& (staged-security-with-data-state-verifier-event \\
& \quad (\lambda s. trace-fri-sampled-missing-full-cover\ s\ bad)) \\
& adversary-initial-state \leq P + I + (I + DL) + (I + SD)
\end{aligned}$$

by (rule

$$\begin{aligned}
& checked-staged-security-trace-fri-sampled-missing-full-cover-bound-from-domain-split \\
& [OF\ proximity\ index\ domain-length\ sampled-domain])
\end{aligned}$$

have *wp-event* (checked-staged-security-experiment-with-data-state *A*)

$$\begin{aligned}
& (staged-security-with-data-state-verifier-event \\
& \quad trace-fri-bad-with-sampled-layer-chain) \\
& adversary-initial-state \leq \\
& wp-event\ (checked-staged-security-experiment-with-data-state\ A) \\
& (\lambda out. \\
& \quad staged-security-with-data-state-verifier-event \\
& \quad \quad (\lambda s. trace-fri-bad-with-full-cover-layer-chain\ s\ bad)\ out \vee \\
& \quad staged-security-with-data-state-verifier-event \\
& \quad \quad (\lambda s. trace-fri-sampled-missing-full-cover\ s\ bad)\ out) \\
& adversary-initial-state
\end{aligned}$$

by (rule *wp-event-mono-on-support*)

(auto simp: staged-security-with-data-state-verifier-event-def

```

    dest: trace-fri-sampled-imp-full-cover-or-missing
    split: option.splits prod.splits)
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-bad-with-full-cover-layer-chain s bad))
  adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-sampled-missing-full-cover s bad))
  adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤
  (F +
    hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)) +
  (P + I + (I + DL) + (I + SD))
by (rule add-mono[OF full-bound missing-bound])
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-canonical-sampled-layer-chain-bound-from-full-cover-fresh-and-domain-split:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B: finite B
  and envelope:
    ∧data attacker-state trace-table.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
      trace-fri-multiround-bad-sets bad trace-table ⊆ B
  and fresh-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. trace-fri-challenge-list-fresh-hit s B))
    adversary-initial-state ≤ F
  and proximity:
    ∧data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
    wp-event verify-monad
      (trace-fri-sampled-full-cover-proximity-failure
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)) bad)
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) ≤ P
  and index:

```


$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-sampled-full-cover-index-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq I$
and domain-length:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-sampled-full-cover-domain-length-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq DL$
and sampled-domain:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-sampled-full-cover-sampled-domain-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq SD$
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-canonical-sampled-layer-chain)
 adversary-initial-state
 $\leq$
 (F +
 hash-relation-budget-value (card B * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (P + I + (I + DL) + (I + SD))
proof –
have wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-canonical-sampled-layer-chain)
 adversary-initial-state $\leq$
 wp-event (checked-staged-security-experiment-with-data-state A)

```

    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-sampled-layer-chain)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
     intro: trace-fri-bad-with-canonical-sampled-layer-chain-imp-sampled
     split: option.splits prod.splits)
  also have ... ≤
    (F +
     hash-relation-budget-value (card B * ceil-log clength)
     (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)) +
    (P + I + (I + DL) + (I + SD))
  by (rule
    checked-staged-security-trace-fri-sampled-layer-chain-bound-from-full-cover-fresh-and-domain-split
    [OF wf_controlled_finite-B envelope_fresh-bound_proximity_index
     domain-length sampled-domain])
  finally show ?thesis .
qed

```

lemma *checked-staged-security-composition-fri-sampled-missing-full-cover-bound-from-domain-split:*

assumes *proximity:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad

(composition-fri-sampled-full-cover-proximity-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad)
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ P

and *index:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad

(composition-fri-sampled-full-cover-index-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ I

and *domain-length:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad

```

    (composition-fri-sampled-full-cover-domain-length-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ DL
  and sampled-domain:
  ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
    wp-event verify-monad
      (composition-fri-sampled-full-cover-sampled-domain-failure
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) ≤ SD
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. composition-fri-sampled-missing-full-cover s bad))
      adversary-initial-state ≤ P + I + (I + DL) + (I + SD)
  proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
    show ∧s. ¬ composition-fri-sampled-missing-full-cover s bad None
      by (rule composition-fri-sampled-missing-full-cover-not-None)
  next
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  show wp-event verify-monad
    (composition-fri-sampled-missing-full-cover ?s bad) ?s
    ≤ P + I + (I + DL) + (I + SD)
  by (rule
    wp-composition-fri-sampled-missing-full-cover-bound-from-domain-split)
    (rule proximity[OF support], rule index[OF support],
      rule domain-length[OF support], rule sampled-domain[OF support])
qed

lemma checked-staged-security-composition-fri-sampled-layer-chain-bound-from-full-cover-fresh-and-domain-split:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B: ∧dg. finite (B dg)
  and envelope:
    ∧ data attacker-state dg composition-table.
      Some (data, attacker-state) ∈

```

$\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \\ \text{adversary-initial-state}) \implies \\ \text{composition-fri-multiround-bad-sets bad dg composition-table} \\ \subseteq B \text{ dg}$
and *fresh-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\ (\text{staged-security-with-data-state-verifier-event} \\ (\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B)) \\ \text{adversary-initial-state} \leq F$
and *proximity*:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (\text{data, attacker-state}) \in \\ \text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \\ \text{adversary-initial-state}) \implies \\ \text{wp-event verify-monad} \\ (\text{composition-fri-sampled-full-cover-proximity-failure} \\ (\text{verifier-state-from-adversary attacker-state} \\ (\text{staged-proof-transcript data})) \text{ bad}) \\ (\text{verifier-state-from-adversary attacker-state} \\ (\text{staged-proof-transcript data})) \leq P$
and *index*:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (\text{data, attacker-state}) \in \\ \text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \\ \text{adversary-initial-state}) \implies \\ \text{wp-event verify-monad} \\ (\text{composition-fri-sampled-full-cover-index-failure} \\ (\text{verifier-state-from-adversary attacker-state} \\ (\text{staged-proof-transcript data}))) \\ (\text{verifier-state-from-adversary attacker-state} \\ (\text{staged-proof-transcript data})) \leq I$
and *domain-length*:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (\text{data, attacker-state}) \in \\ \text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \\ \text{adversary-initial-state}) \implies \\ \text{wp-event verify-monad} \\ (\text{composition-fri-sampled-full-cover-domain-length-failure} \\ (\text{verifier-state-from-adversary attacker-state} \\ (\text{staged-proof-transcript data}))) \\ (\text{verifier-state-from-adversary attacker-state} \\ (\text{staged-proof-transcript data})) \leq DL$
and *sampled-domain*:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (\text{data, attacker-state}) \in \\ \text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \\ \text{adversary-initial-state}) \implies \\ \text{wp-event verify-monad} \\ (\text{composition-fri-sampled-full-cover-sampled-domain-failure}$

$$\begin{aligned}
& (\text{verifier-state-from-adversary attacker-state} \\
& \quad (\text{staged-proof-transcript data})) \\
& (\text{verifier-state-from-adversary attacker-state} \\
& \quad (\text{staged-proof-transcript data})) \leq SD
\end{aligned}$$

shows

$$\begin{aligned}
& \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\
& \quad (\text{staged-security-with-data-state-verifier-event} \\
& \quad \quad \text{composition-fri-bad-with-sampled-layer-chain}) \\
& \text{adversary-initial-state} \\
& \leq \\
& (F + \\
& \quad \text{hash-relation-budget-value} \\
& \quad (\sum dg \in (\text{UNIV} :: 'f \text{ set}). \\
& \quad \quad \text{card } (B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1)) \\
& \quad (\text{staged-attacker-query-budget budgets} + \\
& \quad \quad \text{staged-challenge-query-budget})) + \\
& (P + I + (I + DL) + (I + SD))
\end{aligned}$$

proof –

have *full-bound*:

$$\begin{aligned}
& \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\
& \quad (\text{staged-security-with-data-state-verifier-event} \\
& \quad \quad (\lambda s. \text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad})) \\
& \text{adversary-initial-state} \\
& \leq \\
& F + \\
& \quad \text{hash-relation-budget-value} \\
& \quad (\sum dg \in (\text{UNIV} :: 'f \text{ set}). \\
& \quad \quad \text{card } (B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1)) \\
& \quad (\text{staged-attacker-query-budget budgets} + \\
& \quad \quad \text{staged-challenge-query-budget})
\end{aligned}$$

by (rule

$$\begin{aligned}
& \text{checked-staged-security-composition-fri-full-cover-layer-chain-bound-from-fresh-and-prequery} \\
& \quad [\text{OF wf controlled finite-B envelope fresh-bound}]
\end{aligned}$$

have *missing-bound*:

$$\begin{aligned}
& \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\
& \quad (\text{staged-security-with-data-state-verifier-event} \\
& \quad \quad (\lambda s. \text{composition-fri-sampled-missing-full-cover } s \text{ bad})) \\
& \text{adversary-initial-state} \leq P + I + (I + DL) + (I + SD)
\end{aligned}$$

by (rule

$$\begin{aligned}
& \text{checked-staged-security-composition-fri-sampled-missing-full-cover-bound-from-domain-split} \\
& \quad [\text{OF proximity index domain-length sampled-domain}]
\end{aligned}$$

have *wp-event* (checked-staged-security-experiment-with-data-state A)

$$\begin{aligned}
& \quad (\text{staged-security-with-data-state-verifier-event} \\
& \quad \quad \text{composition-fri-bad-with-sampled-layer-chain}) \\
& \text{adversary-initial-state} \leq \\
& \text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) \\
& \quad (\lambda out. \\
& \quad \quad \text{staged-security-with-data-state-verifier-event} \\
& \quad \quad (\lambda s. \text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad}))
\end{aligned}$$

```

      out  $\vee$ 
      staged-security-with-data-state-verifier-event
      ( $\lambda s.$  composition-fri-sampled-missing-full-cover  $s$  bad) out)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
  (auto simp: staged-security-with-data-state-verifier-event-def
    dest: composition-fri-sampled-imp-full-cover-or-missing
    split: option.splits prod.splits)
also have ...  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
  (staged-security-with-data-state-verifier-event
    ( $\lambda s.$  composition-fri-bad-with-full-cover-layer-chain  $s$  bad))
  adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
  (staged-security-with-data-state-verifier-event
    ( $\lambda s.$  composition-fri-sampled-missing-full-cover  $s$  bad))
  adversary-initial-state
  by (rule wp-event-union-bound)
also have ...  $\leq$ 
  ( $F$  +
    hash-relation-budget-value
    ( $\sum dg \in (UNIV :: 'f \text{ set}).$ 
      card ( $B \ dg$ ) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)) +
  ( $P$  +  $I$  + ( $I$  +  $DL$ ) + ( $I$  +  $SD$ ))
  by (rule add-mono[OF full-bound missing-bound])
finally show ?thesis .
qed

```

lemma *checked-staged-security-composition-fri-canonical-sampled-layer-chain-bound-from-full-cover-fresh-and-domain*

assumes *wf*: staged-budget-wellformed budgets

and *controlled*: staged-adversary-controlled budgets A

and *finite-B*: $\bigwedge dg. \text{finite } (B \ dg)$

and *envelope*:

$\bigwedge \text{data attacker-state } dg \text{ composition-table.}$

Some ($\text{data}, \text{attacker-state}$) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\implies$

composition-fri-multiround-bad-sets bad dg composition-table

$\subseteq B \ dg$

and *fresh-bound*:

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

($\lambda s.$ composition-fri-challenge-list-fresh-hit $s \ B$))

adversary-initial-state $\leq F$

and *proximity*:

$\bigwedge \text{data attacker-state.}$

Some ($\text{data}, \text{attacker-state}$) $\in$

$set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-sampled-full-cover-proximity-failure$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \ bad)$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq P$
and $index:$
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-sampled-full-cover-index-failure$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq I$
and $domain-length:$
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-sampled-full-cover-domain-length-failure$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq DL$
and $sampling-domain:$
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-sampled-full-cover-sampled-domain-failure$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq SD$
shows
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $composition-fri-bad-with-canonical-sampled-layer-chain)$
 $adversary-initial-state$
 $\leq$
 $(F +$
 $hash-relation-budget-value$

$(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\quad \text{card } (B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\quad \text{staged-challenge-query-budget})) +$
 $(P + I + (I + DL) + (I + SD))$

proof –

have *wp-event (checked-staged-security-experiment-with-data-state A)*
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-canonical-sampled-layer-chain)
adversary-initial-state $\leq$
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-sampled-layer-chain)
adversary-initial-state

by (rule *wp-event-mono-on-support*)
(auto simp: staged-security-with-data-state-verifier-event-def
intro: composition-fri-bad-with-canonical-sampled-layer-chain-imp-sampled
split: option.splits prod.splits)

also have ... $\leq$
 $(F +$
 $\quad \text{hash-relation-budget-value}$
 $\quad (\sum dg \in (UNIV :: 'f \text{ set}).$
 $\quad \quad \text{card } (B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1))$
 $\quad (\text{staged-attacker-query-budget budgets} +$
 $\quad \quad \text{staged-challenge-query-budget})) +$
 $(P + I + (I + DL) + (I + SD))$

by (rule
checked-staged-security-composition-fri-sampled-layer-chain-bound-from-full-cover-fresh-and-domain-split
[OF wf controlled finite-B envelope fresh-bound proximity index
domain-length sampled-domain])

finally show *?thesis* .

qed

lemma *checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-full-cover-fresh-and-domain-split*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: finite B*
and *envelope:*
 $\bigwedge \text{data attacker-state trace-table.}$
 $\text{Some } (\text{data}, \text{attacker-state}) \in$
 $\quad \text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\quad \quad \text{adversary-initial-state}) \implies$
 $\text{trace-fri-multi-round-bad-sets bad trace-table} \subseteq B$
and *fresh-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
 $(\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \text{ } B))$
adversary-initial-state $\leq F$
and *proximity:*

$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-sampled-full-cover-proximity-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad)
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq P$
and index:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-sampled-full-cover-index-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq I$
and domain-length:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-sampled-full-cover-domain-length-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq DL$
and sampled-domain:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-sampled-full-cover-sampled-domain-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq SD$
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-sampled-layer-chain)
 adversary-initial-state
 $\leq$

$(F +$
 $\text{hash-relation-budget-value } (\text{card } B * \text{ceil-log } \text{clength})$
 $(\text{staged-attacker-query-budget } \text{budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(P + I + (I + DL) + (I + SD))$

proof –

have *full-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-bad-with-full-cover-layer-chain } s \text{ bad}))$
 $\text{adversary-initial-state}$
 $\leq$
 $F +$
 $\text{hash-relation-budget-value } (\text{card } B * \text{ceil-log } \text{clength})$
 $(\text{staged-attacker-query-budget } \text{budgets} +$
 $\text{staged-challenge-query-budget})$

by (rule
 $\text{checked-staged-security-trace-fri-full-cover-layer-chain-bound-from-fresh-and-prequery}$
 $[\text{OF } \text{wf controlled finite-}B \text{ envelope fresh-bound}])$

have *missing-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-sampled-missing-full-cover } s \text{ bad}))$
 $\text{adversary-initial-state} \leq P + I + (I + DL) + (I + SD)$

by (rule
 $\text{checked-staged-security-trace-fri-sampled-missing-full-cover-bound-from-domain-split}$
 $[\text{OF } \text{proximity index domain-length sampled-domain}])$

have $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-sampled-layer-chain})$
 $\text{adversary-initial-state} \leq$

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\lambda \text{out.}$
 $\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-bad-with-full-cover-layer-chain } s \text{ bad}) \text{ out} \vee$
 $\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-sampled-missing-full-cover } s \text{ bad}) \text{ out})$
 $\text{adversary-initial-state}$

by (rule *wp-event-mono-on-support*)
 $(\text{auto simp: staged-security-with-data-state-verifier-event-def}$
 $\text{dest!: trace-fri-header-tied-sampled-layer-chain-imp-sampled-layer-chain}$
 $\text{dest: trace-fri-sampled-imp-full-cover-or-missing}$
 $\text{split: option.splits prod.splits})$

also have $\dots \leq$

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-bad-with-full-cover-layer-chain } s \text{ bad}))$
 $\text{adversary-initial-state} +$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$

(staged-security-with-data-state-verifier-event
 (λs. trace-fri-sampled-missing-full-cover s bad))
 adversary-initial-state
 by (rule wp-event-union-bound)
 also have ... ≤
 (F +
 hash-relation-budget-value (card B * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (P + I + (I + DL) + (I + SD))
 by (rule add-mono[OF full-bound missing-bound])
 finally show ?thesis .
 qed

lemma *checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-full-cover-fresh-and-dom*

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: $\bigwedge dg. \text{finite } (B \ dg)$
and envelope:
 $\bigwedge \text{data attacker-state } dg \text{ composition-table.}$
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B \ dg$
and fresh-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 (λs. composition-fri-challenge-list-fresh-hit s B))
 adversary-initial-state ≤ F
and proximity:
 $\bigwedge \text{data attacker-state.}$
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-proximity-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad)
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ P
and index:
 $\bigwedge \text{data attacker-state.}$
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-index-failure
 (verifier-state-from-adversary attacker-state

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq I$
and *domain-length*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-domain-length-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq DL$
and *sampling-domain*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-sampled-domain-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq SD$
shows
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain})$
 $\text{adversary-initial-state}$
 $\leq$
 $(F +$
 $\text{hash-relation-budget-value}$
 $(\sum dg \in (\text{UNIV} :: 'f \text{ set}).$
 $\text{card (B dg) * ceil-log (maxDegree + 1))}$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(P + I + (I + DL) + (I + SD))$
proof –
have *full-bound*:
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-bad-with-full-cover-layer-chain s bad}))$
 $\text{adversary-initial-state}$
 $\leq$
 $F +$
 $\text{hash-relation-budget-value}$
 $(\sum dg \in (\text{UNIV} :: 'f \text{ set}).$
 $\text{card (B dg) * ceil-log (maxDegree + 1))}$

$(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$
by (rule
checked-staged-security-composition-fri-full-cover-layer-chain-bound-from-fresh-and-prequery
 $[OF \text{ wf controlled finite-}B \text{ envelope fresh-bound}]$)
have *missing-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-sampled-missing-full-cover } s \text{ bad}))$
 $\text{adversary-initial-state} \leq P + I + (I + DL) + (I + SD)$
by (rule
checked-staged-security-composition-fri-sampled-missing-full-cover-bound-from-domain-split
 $[OF \text{ proximity index domain-length sampled-domain}]$)
have $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain})$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\lambda \text{out.}$
 $\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad})$
 $\text{out} \vee$
 $\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-sampled-missing-full-cover } s \text{ bad})$
 $\text{out})$
 $\text{adversary-initial-state}$
by (rule *wp-event-mono-on-support*)
 $(\text{auto simp: staged-security-with-data-state-verifier-event-def}$
 $\text{dest!: composition-fri-verifier-tied-sampled-layer-chain-imp-sampled-layer-chain}$
 $\text{dest: composition-fri-sampled-imp-full-cover-or-missing}$
 $\text{split: option.splits prod.splits})$
also have $\dots \leq$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad}))$
 $\text{adversary-initial-state} +$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-sampled-missing-full-cover } s \text{ bad}))$
 $\text{adversary-initial-state}$
by (rule *wp-event-union-bound*)
also have $\dots \leq$
 $(F +$
 $\text{hash-relation-budget-value}$
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(P + I + (I + DL) + (I + SD))$

by (rule add-mono[OF full-bound missing-bound])
 finally show ?thesis .
 qed

lemma *checked-staged-security-trace-fri-sampled-missing-full-cover-bound-from-witness-domain:*

assumes *proximity:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad
 (trace-fri-sampled-full-cover-proximity-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad)
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq P$

and *index:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad
 (trace-fri-sampled-full-cover-index-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq I$

and *witness-length:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad
 (trace-fri-sampled-full-cover-witness-length-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq W$

and *domain:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad
 (trace-fri-sampled-full-cover-domain-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq D$

```

shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s.$  trace-fri-sampled-missing-full-cover s bad))
  adversary-initial-state  $\leq P + I + W + D$ 
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\bigwedge s. \neg \text{trace-fri-sampled-missing-full-cover } s \text{ bad None}$ 
    by (rule trace-fri-sampled-missing-full-cover-not-None)
next
  fix data attacker-state
  assume support:
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  show wp-event verify-monad
    (trace-fri-sampled-missing-full-cover ?s bad) ?s  $\leq$ 
      P + I + W + D
  by (rule
    wp-trace-fri-sampled-missing-full-cover-bound-from-witness-length-subcases)
    (rule proximity[OF support], rule index[OF support],
      rule witness-length[OF support], rule domain[OF support])
qed

lemma checked-staged-security-composition-fri-sampled-missing-full-cover-bound-from-witness-domain:
  assumes proximity:
     $\bigwedge$ data attacker-state.
      Some (data, attacker-state)  $\in$ 
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state)  $\implies$ 
        wp-event verify-monad
          (composition-fri-sampled-full-cover-proximity-failure
            (verifier-state-from-adversary attacker-state
              (staged-proof-transcript data)) bad)
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))  $\leq P$ 
  and index:
     $\bigwedge$ data attacker-state.
      Some (data, attacker-state)  $\in$ 
        set-dist (execute (checked-staged-transcript-program A)
          adversary-initial-state)  $\implies$ 
        wp-event verify-monad
          (composition-fri-sampled-full-cover-index-failure
            (verifier-state-from-adversary attacker-state
              (staged-proof-transcript data)))
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))  $\leq I$ 

```

and *witness-length*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-sampled-full-cover-witness-length-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq W$
and *domain*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-sampled-full-cover-domain-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq D$
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
($\lambda s.$ *composition-fri-sampled-missing-full-cover s bad*))
adversary-initial-state $\leq P + I + W + D$
proof (*rule checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
show $\wedge s. \neg$ *composition-fri-sampled-missing-full-cover s bad None*
by (*rule composition-fri-sampled-missing-full-cover-not-None*)
next
fix *data attacker-state*
assume *support*:
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state)
let *?s* =
verifier-state-from-adversary attacker-state
(*staged-proof-transcript data*)
show *wp-event verify-monad*
(*composition-fri-sampled-missing-full-cover ?s bad*) *?s* $\leq$
P + I + W + D
by (*rule*
wp-composition-fri-sampled-missing-full-cover-bound-from-witness-length-subcases)
(*rule proximity*[*OF support*], *rule index*[*OF support*],
rule witness-length[*OF support*], *rule domain*[*OF support*])
qed

lemma *checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-full-cover-fresh-and-witness-dom*

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *finite-B*: finite *B*
and *envelope*:
 $\wedge$ *data* *attacker-state* *trace-table*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
trace-fri-multiround-bad-sets *bad* *trace-table* $\subseteq B$
and *fresh-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
($\lambda s.$ *trace-fri-challenge-list-fresh-hit* *s* *B*))
adversary-initial-state $\leq F$
and *proximity*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-sampled-full-cover-proximity-failure*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) *bad*)
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq P$
and *index*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-sampled-full-cover-index-failure*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)))
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq I$
and *witness-length*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-sampled-full-cover-witness-length-failure*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)))
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq W$
and *domain*:
 $\wedge$ *data* *attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-sampled-full-cover-domain-failure*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)))
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq D$

shows

wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-sampled-layer-chain)
adversary-initial-state
 $\leq$
(*F* +
hash-relation-budget-value (*card* *B* * *ceil-log* *clength*)
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget)) +
(*P* + *I* + *W* + *D*)

proof –

have *full-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
($\lambda s.$ *trace-fri-bad-with-full-cover-layer-chain* *s* *bad*))
adversary-initial-state
 $\leq$
F +
hash-relation-budget-value (*card* *B* * *ceil-log* *clength*)
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget)

by (*rule*
checked-staged-security-trace-fri-full-cover-layer-chain-bound-from-fresh-and-prequery
[*OF* *wf* *controlled* *finite-B* *envelope* *fresh-bound*])

have *missing-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
($\lambda s.$ *trace-fri-sampled-missing-full-cover* *s* *bad*))
adversary-initial-state $\leq P + I + W + D$

by (*rule*
checked-staged-security-trace-fri-sampled-missing-full-cover-bound-from-witness-domain
[*OF* *proximity* *index* *witness-length* *domain*])

have *wp-event* (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-sampled-layer-chain)
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
($\lambda out.$
staged-security-with-data-state-verifier-event

```

      (λs. trace-fri-bad-with-full-cover-layer-chain s bad) out ∨
      staged-security-with-data-state-verifier-event
      (λs. trace-fri-sampled-missing-full-cover s bad) out)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      dest!: trace-fri-header-tied-sampled-layer-chain-imp-sampled-layer-chain
      dest: trace-fri-sampled-imp-full-cover-or-missing
      split: option.splits prod.splits)
  also have ... ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. trace-fri-bad-with-full-cover-layer-chain s bad))
    adversary-initial-state +
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. trace-fri-sampled-missing-full-cover s bad))
    adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤
    (F +
      hash-relation-budget-value (card B * ceil-log clength)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) +
    (P + I + W + D)
  by (rule add-mono[OF full-bound missing-bound])
  finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-sampled-layer-chain-bound-from-full-cover-fresh-and-witness-domain:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: finite B
and envelope:
  ∧ data attacker-state trace-table.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
    trace-fri-multi-round-bad-sets bad trace-table ⊆ B
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-challenge-list-fresh-hit s B))
    adversary-initial-state ≤ F
and proximity:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒

```

$wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-full-cover-proximity-failure}$
 $(verifier\text{-state-from-adversary } attacker\text{-state}$
 $(staged\text{-proof-transcript } data)) \text{ bad})$
 $(verifier\text{-state-from-adversary } attacker\text{-state}$
 $(staged\text{-proof-transcript } data)) \leq P$
and $index$:
 $\wedge data \text{ attacker-state.}$
 $Some (data, attacker\text{-state}) \in$
 $set\text{-dist } (execute (checked\text{-staged-transcript-program } A)$
 $adversary\text{-initial-state}) \implies$
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-full-cover-index-failure}$
 $(verifier\text{-state-from-adversary } attacker\text{-state}$
 $(staged\text{-proof-transcript } data)))$
 $(verifier\text{-state-from-adversary } attacker\text{-state}$
 $(staged\text{-proof-transcript } data)) \leq I$
and $witness\text{-length}$:
 $\wedge data \text{ attacker-state.}$
 $Some (data, attacker\text{-state}) \in$
 $set\text{-dist } (execute (checked\text{-staged-transcript-program } A)$
 $adversary\text{-initial-state}) \implies$
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-full-cover-witness-length-failure}$
 $(verifier\text{-state-from-adversary } attacker\text{-state}$
 $(staged\text{-proof-transcript } data)))$
 $(verifier\text{-state-from-adversary } attacker\text{-state}$
 $(staged\text{-proof-transcript } data)) \leq W$
and $domain$:
 $\wedge data \text{ attacker-state.}$
 $Some (data, attacker\text{-state}) \in$
 $set\text{-dist } (execute (checked\text{-staged-transcript-program } A)$
 $adversary\text{-initial-state}) \implies$
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-full-cover-domain-failure}$
 $(verifier\text{-state-from-adversary } attacker\text{-state}$
 $(staged\text{-proof-transcript } data)))$
 $(verifier\text{-state-from-adversary } attacker\text{-state}$
 $(staged\text{-proof-transcript } data)) \leq D$
shows
 $wp\text{-event } (checked\text{-staged-security-experiment-with-data-state } A)$
 $(staged\text{-security-with-data-state-verifier-event}$
 $trace\text{-fri-bad-with-sampled-layer-chain})$
 $adversary\text{-initial-state}$
 $\leq$
 $(F +$
 $hash\text{-relation-budget-value } (card \text{ } B * \text{ceil-log } clength)$
 $(staged\text{-attacker-query-budget } budgets +$
 $staged\text{-challenge-query-budget})) +$

$(P + I + W + D)$
proof –
have *full-bound*:
 $wp\text{-event}$ (*checked-staged-security-experiment-with-data-state* A)
 $(staged\text{-security-with-data-state-verifier-event}$
 $(\lambda s. trace\text{-fri-bad-with-full-cover-layer-chain } s \text{ bad}))$
 $adversary\text{-initial-state}$
 $\leq$
 $F +$
 $hash\text{-relation-budget-value}$ ($card\ B * ceil\text{-log } clength$)
 $(staged\text{-attacker-query-budget } budgets +$
 $staged\text{-challenge-query-budget})$
by (*rule*
 $checked\text{-staged-security-trace-fri-full-cover-layer-chain-bound-from-fresh-and-prequery}$
 $[OF\ wf\ controlled\ finite\text{-}B\ envelope\ fresh\text{-}bound])$)
have *missing-bound*:
 $wp\text{-event}$ (*checked-staged-security-experiment-with-data-state* A)
 $(staged\text{-security-with-data-state-verifier-event}$
 $(\lambda s. trace\text{-fri-sampled-missing-full-cover } s \text{ bad}))$
 $adversary\text{-initial-state} \leq P + I + W + D$
by (*rule*
 $checked\text{-staged-security-trace-fri-sampled-missing-full-cover-bound-from-witness-domain}$
 $[OF\ proximity\ index\ witness\text{-}length\ domain])$)
have $wp\text{-event}$ (*checked-staged-security-experiment-with-data-state* A)
 $(staged\text{-security-with-data-state-verifier-event}$
 $trace\text{-fri-bad-with-sampled-layer-chain})$
 $adversary\text{-initial-state} \leq$
 $wp\text{-event}$ (*checked-staged-security-experiment-with-data-state* A)
 $(\lambda out.$
 $staged\text{-security-with-data-state-verifier-event}$
 $(\lambda s. trace\text{-fri-bad-with-full-cover-layer-chain } s \text{ bad})\ out \vee$
 $staged\text{-security-with-data-state-verifier-event}$
 $(\lambda s. trace\text{-fri-sampled-missing-full-cover } s \text{ bad})\ out)$
 $adversary\text{-initial-state}$
by (*rule* $wp\text{-event-mono-on-support}$)
 $(auto\ simp: staged\text{-security-with-data-state-verifier-event-def}$
 $dest: trace\text{-fri-sampled-imp-full-cover-or-missing}$
 $split: option.splits\ prod.splits)$
also have $\dots \leq$
 $wp\text{-event}$ (*checked-staged-security-experiment-with-data-state* A)
 $(staged\text{-security-with-data-state-verifier-event}$
 $(\lambda s. trace\text{-fri-bad-with-full-cover-layer-chain } s \text{ bad}))$
 $adversary\text{-initial-state} +$
 $wp\text{-event}$ (*checked-staged-security-experiment-with-data-state* A)
 $(staged\text{-security-with-data-state-verifier-event}$
 $(\lambda s. trace\text{-fri-sampled-missing-full-cover } s \text{ bad}))$
 $adversary\text{-initial-state}$
by (*rule* $wp\text{-event-union-bound}$)
also have $\dots \leq$

```

(F +
  hash-relation-budget-value (card B * ceil-log clength)
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)) +
(P + I + W + D)
by (rule add-mono[OF full-bound missing-bound])
finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-full-cover-fresh-and-witn
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B:  $\bigwedge dg. \text{finite } (B \ dg)$ 
and envelope:
 $\bigwedge \text{data attacker-state } dg \text{ composition-table.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    composition-fri-multiround-bad-sets bad dg composition-table
       $\subseteq B \ dg$ 
and fresh-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B$ ))
  adversary-initial-state  $\leq F$ 
and proximity:
 $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (composition-fri-sampled-full-cover-proximity-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad)
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq P$ 
and index:
 $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (composition-fri-sampled-full-cover-index-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq I$ 
and witness-length:
 $\bigwedge \text{data attacker-state.}$ 

```

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-witness-length-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq W$
and domain:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-domain-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq D$
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-sampled-layer-chain)
 adversary-initial-state
 $\leq$
 (F +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (P + I + W + D)
proof –
have full-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s.$ composition-fri-bad-with-full-cover-layer-chain s bad))
 adversary-initial-state
 $\leq$
 F +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)
by (rule
 checked-staged-security-composition-fri-full-cover-layer-chain-bound-from-fresh-and-prequery
 [OF wf controlled finite-B envelope fresh-bound])

```

have missing-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s. \text{composition-fri-sampled-missing-full-cover } s \text{ bad}$ ))
    adversary-initial-state  $\leq P + I + W + D$ 
  by (rule
    checked-staged-security-composition-fri-sampled-missing-full-cover-bound-from-witness-domain
      [OF proximity index witness-length domain])
have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-fri-bad-with-verifier-tied-sampled-layer-chain)
  adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    ( $\lambda out.$ 
      staged-security-with-data-state-verifier-event
        ( $\lambda s. \text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad}$ )
        out  $\vee$ 
      staged-security-with-data-state-verifier-event
        ( $\lambda s. \text{composition-fri-sampled-missing-full-cover } s \text{ bad}$ )
        out)
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      dest!: composition-fri-verifier-tied-sampled-layer-chain-imp-sampled-layer-chain
      dest: composition-fri-sampled-imp-full-cover-or-missing
      split: option.splits prod.splits)
also have ...  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s. \text{composition-fri-bad-with-full-cover-layer-chain } s \text{ bad}$ ))
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s. \text{composition-fri-sampled-missing-full-cover } s \text{ bad}$ ))
    adversary-initial-state
  by (rule wp-event-union-bound)
also have ...  $\leq$ 
  (F +
    hash-relation-budget-value
      ( $\sum dg \in (UNIV :: 'f \text{ set}).$ 
        card (B dg) * ceil-log (maxDegree + 1))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) +
    (P + I + W + D)
  by (rule add-mono[OF full-bound missing-bound])
finally show ?thesis .
qed

end

```


end

theory *Soundness-FRI-Derived-Active-Reductions*

imports

Soundness-FRI-Staged-Replay-Bounds

Soundness-FRI-Full-Cover-Staged-Bounds

begin

Derived active-FRI packaging.

This layer keeps the near-threshold staged replay theory stable. It packages the already proved trace and composition verifier-state reductions into the active-FRI interface consumed by the public route. It does not add protocol assumptions or change the FRI model.

context *soundness*

begin

lemma *trace-fri-header-tied-reduction-from-envelope-witness-domain-without-head-slot-and-selected-zero:*

assumes *future: trace-fri-future-fresh s*

and *envelope:*

$\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$

and *subset: $B \subseteq \text{fri-challenge-space } (\text{ceil-log clength})$*

and *challenge-bound:*

$\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log clength}) \leq R$

and *proximity:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$

and *index:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$

and *witness-length:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-witness-length-failure } s) \ s \leq W$

and *domain:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-domain-failure } s) \ s \leq D$

and *zero:*

wp-event verify-monad

$(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s \leq z-b$

and *merkle:*

wp-event verify-monad $(\text{partial-merkle-inconsistency-bad } s) \ s \leq p-b$

and *structural:*

wp-event verify-monad

$(\text{trace-fri-header-tied-sibling-mismatch-structural-gap } s) \ s \leq h-b$

and *next-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-next-value-replay-gap } s) \ s \leq n-b$

and *successor*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}successor\text{-}replay\text{-}gap\ s)\ s \leq s\text{-}b$
and *total*:
 $(R + (P + I + W + D)) +$
 $(((((p\text{-}b + h\text{-}b) + p\text{-}b) + n\text{-}b + s\text{-}b +$
 $(p\text{-}b + 0 + 0)) + p\text{-}b) + 0 + z\text{-}b)$
 $\leq trace\text{-}fri\text{-}error$
shows *trace-fri-header-tied-reduction s*
by (*rule trace-fri-header-tied-reduction-from-residual-obligations*)
(rule
trace-fri-header-tied-residual-obligations-from-envelope-witness-domain-without-head-slot-and-selected-zero
[OF future envelope subset challenge-bound proximity index
witness-length domain zero merkle structural next-bound successor
total])

lemma *composition-fri-verifier-tied-reduction-from-envelope-witness-domain-slot-zero:*

assumes *future: composition-fri-future-fresh s*
and *envelope*:
 $\bigwedge dg\ composition\text{-}table.$
 $composition\text{-}fri\text{-}multiround\text{-}bad\text{-}sets\ bad\ dg\ composition\text{-}table$
 $\subseteq B\ dg$
and *subset*:
 $\bigwedge dg.\ B\ dg \subseteq fri\text{-}challenge\text{-}space\ (ceil\text{-}log\ (to\text{-}nat\ dg + 1))$
and *challenge-bound*:
 $\bigwedge dg.\ nnreal\ (card\ (B\ dg))\ /\$
 $nnreal\ (CARD('f) \wedge^{ceil\text{-}log\ (to\text{-}nat\ dg + 1)}) \leq R$
and *proximity*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}proximity\text{-}failure\ s\ bad)\ s$
 $\leq P$
and *index*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}index\text{-}failure\ s)\ s \leq I$
and *witness-length*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}witness\text{-}length\text{-}failure\ s)\ s$
 $\leq W$
and *domain*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}domain\text{-}failure\ s)\ s \leq D$
and *merkle*:
 $wp\text{-}event\ verify\text{-}monad\ (partial\text{-}merkle\text{-}inconsistency\text{-}bad\ s)\ s \leq p\text{-}b$
and *next-bound*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}next\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq n\text{-}b$
and *successor*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}successor\text{-}replay\text{-}gap\ s)\ s \leq s\text{-}b$

and *final-bound*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}verifier\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\ s)\ s \leq g\text{-}b$
and *total*:
 $(R + (P + I + W + D)) +$
 $((p\text{-}b + 0) + n\text{-}b + s\text{-}b + g\text{-}b + p\text{-}b) + 0)$
 $\leq composition\text{-}fri\text{-}error$
shows *composition-fri-verifier-tied-reduction s*
by (*rule composition-fri-verifier-tied-reduction-from-residual-obligations*)
(rule
composition-fri-verifier-tied-residual-obligations-from-envelope-witness-domain-slot-zero
[OF future envelope subset challenge-bound proximity index
witness-length domain merkle next-bound successor final-bound total])

lemma *composition-fri-verifier-tied-reduction-from-envelope-witness-domain-merkle*:
assumes *future: composition-fri-future-fresh s*
and *envelope*:
 $\bigwedge dg\ composition\text{-}table.$
 $composition\text{-}fri\text{-}multiround\text{-}bad\text{-}sets\ bad\ dg\ composition\text{-}table$
 $\subseteq B\ dg$
and *subset*:
 $\bigwedge dg.\ B\ dg \subseteq fri\text{-}challenge\text{-}space\ (ceil\text{-}log\ (to\text{-}nat\ dg + 1))$
and *challenge-bound*:
 $\bigwedge dg.\ nnreal\ (card\ (B\ dg))\ /\$
 $nnreal\ (CARD('f) \wedge^{ceil\text{-}log\ (to\text{-}nat\ dg + 1)}) \leq R$
and *proximity*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}proximity\text{-}failure\ s\ bad)\ s$
 $\leq P$
and *index*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}index\text{-}failure\ s)\ s \leq I$
and *witness-length*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}witness\text{-}length\text{-}failure\ s)\ s$
 $\leq W$
and *domain*:
 $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}domain\text{-}failure\ s)\ s \leq D$
and *merkle*:
 $wp\text{-}event\ verify\text{-}monad\ (partial\text{-}merkle\text{-}inconsistency\text{-}bad\ s)\ s \leq M$
and *total*:
 $(R + (P + I + W + D)) +$
 $((M + 0) + (M + 0) + (M + 0) + (M + 0 + 0) + M) + 0)$
 $\leq composition\text{-}fri\text{-}error$
shows *composition-fri-verifier-tied-reduction s*
proof –
have *slot*:
 $wp\text{-}event\ verify\text{-}monad$

(composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s)
 $s \leq 0$
by (rule
 wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
have next-bound:
 wp-event verify-monad
 (composition-fri-verifier-tied-next-value-replay-gap s) $s \leq M + 0$
by (rule
 wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-merkle-and-slot)
 (rule merkle, rule slot)
have successor:
 wp-event verify-monad
 (composition-fri-verifier-tied-successor-replay-gap s) $s \leq M + 0$
by (rule
 wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-merkle-and-slot)
 (rule merkle, rule slot)
have final-bound:
 wp-event verify-monad
 (composition-fri-verifier-tied-final-value-replay-gap s) s
 $\leq M + 0 + 0$
by (rule
 wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step)
 (rule merkle, rule slot,
 rule
 wp-composition-fri-verifier-tied-final-value-selected-step-missing-gap-zero)
show ?thesis
by (rule
 composition-fri-verifier-tied-reduction-from-envelope-witness-domain-slot-zero
 [OF future envelope subset challenge-bound proximity index
 witness-length domain merkle next-bound successor final-bound total])
qed

lemma checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-full-cover-domain-conflict-and-zero

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and finite-B: finite B

and envelope:

$\bigwedge$ data attacker-state trace-table.

Some $(data, attacker-state) \in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\implies$

trace-fri-multi-round-bad-sets bad trace-table $\subseteq B$

and fresh-bound:

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

($\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B$))

adversary-initial-state $\leq F$

and proximity:

$\bigwedge$ data attacker-state.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-sampled-full-cover-proximity-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) *bad*)
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq P$

and *index*:

$\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-sampled-full-cover-index-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq I$

and *domain-length*:

$\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-sampled-full-cover-domain-length-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq DL$

and *sampled-domain*:

$\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-sampled-full-cover-sampled-domain-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq SD$

and *conflict-bound*:

$\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-header-tied-sampled-assignment-conflict*

$$\begin{aligned} & (\text{verifier-state-from-adversary attacker-state} \\ & \quad (\text{staged-proof-transcript data}))) \\ & (\text{verifier-state-from-adversary attacker-state} \\ & \quad (\text{staged-proof-transcript data})) \leq C \end{aligned}$$

and zero-bound:

$\wedge$ data attacker-state.

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq Z$

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-partial-candidate)
 adversary-initial-state $\leq$
 ((F +
 hash-relation-budget-value (card B * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (P + I + (I + DL) + (I + SD))) + (C + Z)

proof (rule
 checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-conflict-and-staged-zero)

show wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-sampled-layer-chain)
 adversary-initial-state
 $\leq$
 (F +
 hash-relation-budget-value (card B * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (P + I + (I + DL) + (I + SD))

by (rule
 checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-full-cover-fresh-and-domain-split
 [OF wf controlled finite-B envelope fresh-bound proximity index
 domain-length sampled-domain])

show wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-header-tied-sampled-assignment-conflict)
 adversary-initial-state $\leq C$

by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
 (auto simp: trace-fri-header-tied-sampled-assignment-conflict-def
 accepted-fri-opening-transcript-def conflict-bound)

show wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event
 trace-fri-header-tied-zero-round-final-obstruction)
 adversary-initial-state $\leq Z$
by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
 (auto simp: trace-fri-header-tied-zero-round-final-obstruction-def
 accepted-fri-opening-transcript-def zero-bound)
qed

lemma checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-full-cover-witness-domain-conflict-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: finite B
and envelope:
 $\wedge$ data attacker-state trace-table.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 trace-fri-multiround-bad-sets bad trace-table $\subseteq B$
and fresh-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 (λs . trace-fri-challenge-list-fresh-hit s B))
 adversary-initial-state $\leq F$
and proximity:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-sampled-full-cover-proximity-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad)
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq P$
and index:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-sampled-full-cover-index-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq I$
and witness-length:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)

$\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-witness-length-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq W$
and domain:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state}) \implies}$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-domain-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq D$
and conflict-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state}) \implies}$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq C$
and zero-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state}) \implies}$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq Z$
shows
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq$
 $((F +$
 $\text{hash-relation-budget-value (card B * ceil-log clength)}$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$

$(P + I + W + D)) + (C + Z)$
proof (rule
checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-conflict-and-staged-zero)
show *wp-event* (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-sampled-layer-chain)
adversary-initial-state
 $\leq$
 $(F +$
hash-relation-budget-value (*card B * ceil-log clength*)
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)) +
 $(P + I + W + D)$
by (rule
checked-staged-security-trace-fri-header-tied-sampled-layer-chain-bound-from-full-cover-fresh-and-witness-domain
[OF wf controlled finite-B envelope fresh-bound proximity index
witness-length domain])
show *wp-event* (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-sampled-assignment-conflict)
adversary-initial-state $\leq C$
by (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
(auto simp: trace-fri-header-tied-sampled-assignment-conflict-def
accepted-fri-opening-transcript-def conflict-bound)
show *wp-event* (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-zero-round-final-obstruction)
adversary-initial-state $\leq Z$
by (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
(auto simp: trace-fri-header-tied-zero-round-final-obstruction-def
accepted-fri-opening-transcript-def zero-bound)
qed

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-full-cover-domain-conflict-a*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: $\bigwedge dg. \text{finite } (B \text{ dg})$*
and *envelope:*
 $\bigwedge \text{data attacker-state dg composition-table.}$
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B \text{ dg}$
and *fresh-bound:*
wp-event (*checked-staged-security-experiment-with-data-state A*)
(staged-security-with-data-state-verifier-event
($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \text{ B})$)
adversary-initial-state $\leq F$

and proximity:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-proximity-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad)
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq P$

and index:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-index-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq I$

and domain-length:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-domain-length-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq DL$

and sampled-domain:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-sampled-domain-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq SD$

and conflict-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$

wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq C$
and zero-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
wp-event verify-monad
 (composition-fri-verifier-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq Z$
shows
wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state $\leq$
 ((F +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (P + I + (I + DL) + (I + SD))) + (C + Z)
proof (rule
 checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-sampled-and-missing)
show *wp-event* (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-sampled-layer-chain)
 adversary-initial-state $\leq$
 (F +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (P + I + (I + DL) + (I + SD))
by (rule
 checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-full-cover-fresh-and-domain
 [OF wf controlled finite-B envelope fresh-bound proximity index
 domain-length sampled-domain])
fix data attacker-state
assume support:
 Some (data, attacker-state) $\in$

$\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state})$
show $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-missing-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $\leq C + Z$
by (rule
 $\text{wp-composition-fri-verifier-tied-missing-sampled-bound-from-assignment-obstruction})$
(rule
 $\text{wp-composition-fri-verifier-tied-assignment-obstruction-bound-from-conflict-and-zero}$
 $[OF \text{ conflict-bound}[OF \text{ support}] \text{ zero-bound}[OF \text{ support}]])$
qed

lemma $\text{checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-full-cover-witness-domain-c}$

assumes $\text{wf: staged-budget-wellformed budgets}$
and $\text{controlled: staged-adversary-controlled budgets } A$
and $\text{finite-B: } \bigwedge dg. \text{finite } (B \text{ dg})$
and envelope:
 $\bigwedge \text{data attacker-state dg composition-table.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{composition-fri-multiround-bad-sets bad dg composition-table}$
 $\subseteq B \text{ dg}$
and fresh-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \text{ B}))$
 $\text{adversary-initial-state} \leq F$
and proximity:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-proximity-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ bad})$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq P$
and index:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$

(composition-fri-sampled-full-cover-index-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq I$
and witness-length:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-witness-length-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq W$
and domain:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-full-cover-domain-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq D$
and conflict-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq C$
and zero-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq Z$

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state $\leq$
 $((F +$
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget } \text{budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(P + I + W + D)) + (C + Z)$

proof (rule

checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-sampled-and-missing)

show *wp-event (checked-staged-security-experiment-with-data-state A)*

(staged-security-with-data-state-verifier-event
composition-fri-bad-with-verifier-tied-sampled-layer-chain)

adversary-initial-state $\leq$

$(F +$
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget } \text{budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(P + I + W + D)$

by (rule

checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-full-cover-fresh-and-witness

[OF wf controlled finite-B envelope fresh-bound proximity index
witness-length domain]]

fix *data attacker-state*

assume *support:*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)

show *wp-event verify-monad*

(composition-fri-verifier-tied-missing-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state
(staged-proof-transcript data))

$\leq C + Z$

by (rule

wp-composition-fri-verifier-tied-missing-sampled-bound-from-assignment-obstruction)

(rule

wp-composition-fri-verifier-tied-assignment-obstruction-bound-from-conflict-and-zero
 $[OF \text{ conflict-bound}[OF \text{ support}] \text{ zero-bound}[OF \text{ support}]]$)

qed

lemma *checked-staged-security-trace-fri-empty-header-bound-from-full-cover-domain-conflict-and-zero:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *finite-B: finite B*
and *envelope:*
 $\wedge$ *data attacker-state trace-table.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
trace-fri-multiround-bad-sets bad trace-table $\subseteq B$
and *fresh-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
($\lambda s.$ trace-fri-challenge-list-fresh-hit s B))
adversary-initial-state $\leq F$
and *proximity:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(trace-fri-sampled-full-cover-proximity-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) bad)
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq P$
and *index:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(trace-fri-sampled-full-cover-index-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq I$
and *domain-length:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(trace-fri-sampled-full-cover-domain-length-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq DL$
and *sampled-domain:*
 $\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-sampled-full-cover-sampled-domain-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ SD
and *conflict-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-sampled-assignment-conflict
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ C
and *zero-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-zero-round-final-obstruction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ Z
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state ≤
((F +
*hash-relation-budget-value (card B * ceil-log clength)*
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)) +
(P + I + (I + DL) + (I + SD))) + (C + Z)
proof *(rule checked-staged-security-trace-fri-empty-header-bound-from-sampled-and-missing)*
show *wp-event (checked-staged-security-experiment-with-data-state A)*
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-sampled-layer-chain)
adversary-initial-state
 $\leq$
(F +
*hash-relation-budget-value (card B * ceil-log clength)*


```

      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget)) +
      (P + I + (I + DL) + (I + SD))
  by (rule
      checked-staged-security-trace-fri-sampled-layer-chain-bound-from-full-cover-fresh-and-domain-split
      [OF wf controlled finite-B envelope fresh-bound proximity index
       domain-length sampled-domain])
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                  adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have obstruction:
    wp-event verify-monad
      (trace-fri-sampled-layer-assignment-obstruction ?s) ?s ≤ C + Z
  proof -
    have wp-event verify-monad
      (trace-fri-sampled-layer-assignment-obstruction ?s) ?s ≤
        wp-event verify-monad
          (λout. trace-fri-sampled-assignment-conflict ?s out ∨
              trace-fri-zero-round-final-obstruction ?s out) ?s
    by (rule wp-event-mono)
      (rule
        trace-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final)
    also have ... ≤
      wp-event verify-monad (trace-fri-sampled-assignment-conflict ?s) ?s +
      wp-event verify-monad (trace-fri-zero-round-final-obstruction ?s) ?s
    by (rule wp-event-union-bound)
    also have ... ≤ C + Z
    by (rule add-mono[OF conflict-bound[OF support]
                     zero-bound[OF support]])
    finally show ?thesis .
  qed
  show wp-event verify-monad
    (trace-fri-reachable-missing-sampled-layer-chain
     ?s) ?s ≤ C + Z
  by (rule wp-trace-fri-reachable-missing-sampled-bound-from-assignment-obstruction)
    (rule obstruction)
  qed

lemma checked-staged-security-trace-fri-empty-header-bound-from-full-cover-witness-domain-conflict-and-zero:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B: finite B
  and envelope:
    ∧ data attacker-state trace-table.

```

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
trace-fri-multiround-bad-sets *bad* *trace-table* $\subseteq B$
and *fresh-bound*:
wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(*staged-security-with-data-state-verifier-event*
($\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B$))
adversary-initial-state $\leq F$
and *proximity*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-sampled-full-cover-proximity-failure*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) *bad*)
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq P$
and *index*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-sampled-full-cover-index-failure*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)))
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq I$
and *witness-length*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-sampled-full-cover-witness-length-failure*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)))
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq W$
and *domain*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-sampled-full-cover-domain-failure*

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq D$
and *conflict-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-assignment-conflict}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq C$
and *zero-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq Z$
shows
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq$
 $((F +$
 $\text{hash-relation-budget-value (card B * ceil-log clength)}$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(P + I + W + D)) + (C + Z)$
proof (rule *checked-staged-security-trace-fri-empty-header-bound-from-sampled-and-missing*)
show $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-sampled-layer-chain})$
 $\text{adversary-initial-state}$
 $\leq$
 $(F +$
 $\text{hash-relation-budget-value (card B * ceil-log clength)}$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(P + I + W + D)$
by (rule
checked-staged-security-trace-fri-sampled-layer-chain-bound-from-full-cover-fresh-and-witness-domain

```

    [OF wf controlled finite-B envelope fresh-bound proximity index
      witness-length domain])
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have obstruction:
  wp-event verify-monad
    (trace-fri-sampled-layer-assignment-obstruction ?s) ?s ≤ C + Z
proof –
  have wp-event verify-monad
    (trace-fri-sampled-layer-assignment-obstruction ?s) ?s ≤
  wp-event verify-monad
    (λout. trace-fri-sampled-assignment-conflict ?s out ∨
      trace-fri-zero-round-final-obstruction ?s out) ?s
  by (rule wp-event-mono)
    (rule
      trace-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final)
  also have ... ≤
    wp-event verify-monad (trace-fri-sampled-assignment-conflict ?s) ?s +
    wp-event verify-monad (trace-fri-zero-round-final-obstruction ?s) ?s
  by (rule wp-event-union-bound)
  also have ... ≤ C + Z
  by (rule add-mono[OF conflict-bound[OF support]
    zero-bound[OF support]])
  finally show ?thesis .
qed
show wp-event verify-monad
  (trace-fri-reachable-missing-sampled-layer-chain
    ?s) ?s ≤ C + Z
  by (rule wp-trace-fri-reachable-missing-sampled-bound-from-assignment-obstruction)
    (rule obstruction)
qed

lemma active-fri-reductions-for-from-component-reductions:
assumes trace-header:
  ∧data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  trace-fri-header-tied-reduction
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
and composition:
  ∧data attacker-state.

```

Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 composition-fri-verifier-tied-reduction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
and empty-trace:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 trace-fri-empty-header-reduction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data))
shows active-fri-reductions-for A
unfolding active-fri-reductions-for-def
by (intro allI impI conjI)
 (erule trace-header, erule composition, erule empty-trace)

lemma trace-fri-empty-header-reduction-from-sampled-and-missing-bounds:
fixes R M :: prob
assumes sampled-bound:
 wp-event verify-monad
 (trace-fri-bad-with-sampled-layer-chain s) $s \leq R$
and missing-bound:
 wp-event verify-monad
 (trace-fri-reachable-missing-sampled-layer-chain s) $s \leq M$
and total-bound:
 $R + M \leq \text{trace-fri-error}$
shows trace-fri-empty-header-reduction s
proof –
have reachable:
 wp-event verify-monad
 (trace-fri-bad-with-reachable-partial-candidate s) $s \leq R + M$
by (rule wp-trace-fri-reachable-bound-from-sampled-and-missing
 [OF sampled-bound missing-bound])
have empty:
 wp-event verify-monad
 (trace-fri-bad-with-empty-composition-header-candidates s) $s \leq$
 $R + M$
by (rule order-trans[OF - reachable])
 (rule wp-event-mono,
 rule
 trace-fri-bad-with-empty-composition-header-candidates-imp-reachable-partial-candidate)
show ?thesis
unfolding trace-fri-empty-header-reduction-def
by (rule order-trans[OF empty total-bound])
qed

lemma *trace-fri-empty-header-reduction-from-sampled-and-assignment-obstruction-bounds:*

fixes $R\ M :: \text{prob}$

assumes *sampled-bound:*

wp-event verify-monad

$(\text{trace-fri-bad-with-sampled-layer-chain } s) \ s \leq R$

and *obstruction-bound:*

wp-event verify-monad

$(\text{trace-fri-sampled-layer-assignment-obstruction } s) \ s \leq M$

and *total-bound:*

$R + M \leq \text{trace-fri-error}$

shows *trace-fri-empty-header-reduction s*

proof (rule *trace-fri-empty-header-reduction-from-sampled-and-missing-bounds*)

show *wp-event verify-monad*

$(\text{trace-fri-bad-with-sampled-layer-chain } s) \ s \leq R$

by (rule *sampled-bound*)

show *wp-event verify-monad*

$(\text{trace-fri-reachable-missing-sampled-layer-chain } s) \ s \leq M$

by (rule

wp-trace-fri-reachable-missing-sampled-bound-from-assignment-obstruction

[*OF obstruction-bound*])

show $R + M \leq \text{trace-fri-error}$

by (rule *total-bound*)

qed

lemma *wp-trace-fri-sampled-layer-assignment-obstruction-bound-from-conflict-and-zero:*

assumes *conflict-bound:*

wp-event verify-monad

$(\text{trace-fri-sampled-assignment-conflict } s) \ s \leq C$

and *zero-bound:*

wp-event verify-monad

$(\text{trace-fri-zero-round-final-obstruction } s) \ s \leq Z$

shows

wp-event verify-monad

$(\text{trace-fri-sampled-layer-assignment-obstruction } s) \ s \leq C + Z$

proof –

have *wp-event verify-monad*

$(\text{trace-fri-sampled-layer-assignment-obstruction } s) \ s \leq$

wp-event verify-monad

$(\lambda \text{out. } \text{trace-fri-sampled-assignment-conflict } s \ \text{out} \ \vee$

$\text{trace-fri-zero-round-final-obstruction } s \ \text{out}) \ s$

by (rule *wp-event-mono*)

(rule *trace-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final*)

also have $\dots \leq$

wp-event verify-monad $(\text{trace-fri-sampled-assignment-conflict } s) \ s +$

wp-event verify-monad $(\text{trace-fri-zero-round-final-obstruction } s) \ s$

by (rule *wp-event-union-bound*)

also have $\dots \leq C + Z$

by (rule *add-mono*[*OF conflict-bound zero-bound*])

finally show *?thesis* .

qed

lemma *trace-fri-empty-header-reduction-from-sampled-conflict-and-zero-bounds:*

fixes $R\ C\ Z :: \text{prob}$

assumes *sampled-bound:*

wp-event verify-monad

$(\text{trace-fri-bad-with-sampled-layer-chain } s) \ s \leq R$

and *conflict-bound:*

wp-event verify-monad

$(\text{trace-fri-sampled-assignment-conflict } s) \ s \leq C$

and *zero-bound:*

wp-event verify-monad

$(\text{trace-fri-zero-round-final-obstruction } s) \ s \leq Z$

and *total-bound:*

$R + (C + Z) \leq \text{trace-fri-error}$

shows *trace-fri-empty-header-reduction s*

proof (rule

trace-fri-empty-header-reduction-from-sampled-and-assignment-obstruction-bounds)

show *wp-event verify-monad*

$(\text{trace-fri-bad-with-sampled-layer-chain } s) \ s \leq R$

by (rule *sampled-bound*)

show *wp-event verify-monad*

$(\text{trace-fri-sampled-layer-assignment-obstruction } s) \ s \leq C + Z$

by (rule

wp-trace-fri-sampled-layer-assignment-obstruction-bound-from-conflict-and-zero
[OF conflict-bound zero-bound])

show $R + (C + Z) \leq \text{trace-fri-error}$

by (rule *total-bound*)

qed

lemma *trace-fri-header-tied-reduction-from-augmented-conflict-zero-merkle-bounds:*

fixes $A\ C\ Z\ M :: \text{prob}$

assumes *augmented-bound:*

wp-event verify-monad

$(\text{trace-fri-bad-with-header-tied-augmented-partial-candidate } s) \ s$
 $\leq A$

and *conflict-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-sampled-assignment-conflict } s) \ s \leq C$

and *zero-bound:*

wp-event verify-monad

$(\text{trace-fri-header-tied-zero-round-final-obstruction } s) \ s \leq Z$

and *merkle-bound:*

wp-event verify-monad $(\text{partial-merkle-inconsistency-bad } s) \ s \leq M$

and *total-bound:*

$A + ((C + Z) + M) \leq \text{trace-fri-error}$

shows *trace-fri-header-tied-reduction s*

proof –

have *missing-bound:*

```

wp-event verify-monad
  (trace-fri-header-tied-recorded-sibling-candidate-missing s) s
  ≤ C + Z
by (rule
  wp-trace-fri-header-tied-recorded-sibling-candidate-missing-bound-from-conflict-and-zero)
  (rule conflict-bound, rule zero-bound)
show ?thesis
by (rule
  trace-fri-header-tied-reduction-from-augmented-and-recorded-missing-bounds
  [OF augmented-bound missing-bound merkle-bound total-bound])
qed

```

lemma *active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty:*

fixes *TraceAug TraceMissing TraceMerkle CompSampled CompMerkle ::*

'f staged-proof-data ⇒ 'f protocol-channel ⇒ prob

assumes *trace-augmented-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-bad-with-header-tied-augmented-partial-candidate
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ TraceAug data attacker-state

and *trace-missing-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-recorded-sibling-candidate-missing
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
TraceMissing data attacker-state

and *trace-merkle-bound:*

$\wedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ TraceMerkle data attacker-state

and *trace-total*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
 $\text{TraceAug } \text{data attacker-state} +$
 $(\text{TraceMissing } \text{data attacker-state} + \text{TraceMerkle } \text{data attacker-state})$
 $\leq \text{trace-fri-error}$

and *composition-sampled-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{CompSampled data attacker-state}$

and *composition-merkle-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{CompMerkle data attacker-state}$

and *composition-total*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
 $\text{CompSampled data attacker-state} +$
 $((\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0 + 0) +$
 $\text{CompMerkle data attacker-state}) + 0)$
 $\leq \text{composition-fri-error}$

and *empty-trace*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
 $\text{trace-fri-empty-header-reduction}$

```

      (verifier-state-from-adversary attacker-state
       (staged-proof-transcript data))
shows active-fri-reductions-for A
proof (rule active-fri-reductions-for-from-component-reductions)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
                  adversary-initial-state)
show trace-fri-header-tied-reduction
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))
proof –
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have aug:
  wp-event verify-monad
    (trace-fri-bad-with-header-tied-augmented-partial-candidate ?s)
    ?s ≤ TraceAug data attacker-state
by (rule trace-augmented-bound[OF support])
have missing:
  wp-event verify-monad
    (trace-fri-header-tied-recorded-sibling-candidate-missing ?s)
    ?s ≤ TraceMissing data attacker-state
by (rule trace-missing-bound[OF support])
have merkle:
  wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
    ≤ TraceMerkle data attacker-state
by (rule trace-merkle-bound[OF support])
have total:
  TraceAug data attacker-state +
    (TraceMissing data attacker-state + TraceMerkle data attacker-state)
    ≤ trace-fri-error
by (rule trace-total[OF support])
show ?thesis
by (rule
      trace-fri-header-tied-reduction-from-augmented-and-recorded-missing-bounds
      [OF aug missing merkle total])
qed
show composition-fri-verifier-tied-reduction
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))
by (rule active-composition-fri-verifier-tied-reductions-from-sampled-and-merkle-bounds
      [OF composition-sampled-bound composition-merkle-bound
       composition-total support])
show trace-fri-empty-header-reduction
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))

```

by (rule empty-trace[OF support])
qed

lemma *active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty-sampled:*
fixes *TraceAug TraceMissing TraceMerkle CompSampled CompMerkle EmptySampled EmptyMissing ::*

'f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ prob

assumes *trace-augmented-bound:*

$\wedge$ *data attacker-state.*

*Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-bad-with-header-tied-augmented-partial-candidate
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$ TraceAug data attacker-state*

and *trace-missing-bound:*

$\wedge$ *data attacker-state.*

*Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-header-tied-recorded-sibling-candidate-missing
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
TraceMissing data attacker-state*

and *trace-merkle-bound:*

$\wedge$ *data attacker-state.*

*Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$ TraceMerkle data attacker-state*

and *trace-total:*

$\wedge$ *data attacker-state.*

*Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
TraceAug data attacker-state +
(TraceMissing data attacker-state + TraceMerkle data attacker-state)
 $\leq$ trace-fri-error*

and *composition-sampled-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-bad-with-verifier-tied-sampled-layer-chain*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
CompSampled data attacker-state
and *composition-merkle-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*partial-merkle-inconsistency-bad*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$ *CompMerkle data attacker-state*
and *composition-total*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
CompSampled data attacker-state +
(((*CompMerkle data attacker-state* + 0) +
(*CompMerkle data attacker-state* + 0) +
(*CompMerkle data attacker-state* + 0) +
(*CompMerkle data attacker-state* + 0 + 0) +
CompMerkle data attacker-state) + 0)
 $\leq$ *composition-fri-error*
and *empty-sampled-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-bad-with-sampled-layer-chain*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
EmptySampled data attacker-state
and *empty-missing-bound*:
 $\wedge$ *data attacker-state*.

$\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-reachable-missing-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{EmptyMissing data attacker-state}$
and empty-total:
 $\wedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{EmptySampled data attacker-state} + \text{EmptyMissing data attacker-state}$
 $\leq \text{trace-fri-error}$
shows $\text{active-fri-reductions-for } A$
proof $(\text{rule active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty})$
fix $\text{data attacker-state}$
assume support:
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state})$
let $?s =$
 $\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})$
show $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-augmented-partial-candidate } ?s) ?s$
 $\leq \text{TraceAug data attacker-state}$
by $(\text{rule trace-augmented-bound}[OF \text{ support}])$
show $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-recorded-sibling-candidate-missing } ?s) ?s$
 $\leq \text{TraceMissing data attacker-state}$
by $(\text{rule trace-missing-bound}[OF \text{ support}])$
show $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } ?s) ?s$
 $\leq \text{TraceMerkle data attacker-state}$
by $(\text{rule trace-merkle-bound}[OF \text{ support}])$
show $\text{TraceAug data attacker-state} +$
 $(\text{TraceMissing data attacker-state} + \text{TraceMerkle data attacker-state})$
 $\leq \text{trace-fri-error}$
by $(\text{rule trace-total}[OF \text{ support}])$
show $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain } ?s) ?s$
 $\leq \text{CompSampled data attacker-state}$
by $(\text{rule composition-sampled-bound}[OF \text{ support}])$
show $\text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } ?s) ?s$
 $\leq \text{CompMerkle data attacker-state}$
by $(\text{rule composition-merkle-bound}[OF \text{ support}])$

```

show CompSampled data attacker-state +
  (((CompMerkle data attacker-state + 0) +
    (CompMerkle data attacker-state + 0) +
    (CompMerkle data attacker-state + 0) +
    (CompMerkle data attacker-state + 0 + 0) +
    CompMerkle data attacker-state) + 0)
  ≤ composition-fri-error
by (rule composition-total[OF support])
show trace-fri-empty-header-reduction ?s
by (rule trace-fri-empty-header-reduction-from-sampled-and-missing-bounds
  [OF empty-sampled-bound[OF support]
   empty-missing-bound[OF support] empty-total[OF support]])
qed

lemma active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty-assignment:
  fixes TraceAug TraceMissing TraceMerkle CompSampled CompMerkle EmptySampled Empty-
Obstruction ::
  'f staged-proof-data ⇒ 'f protocol-channel ⇒ prob
assumes trace-augmented-bound:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
    wp-event verify-monad
      (trace-fri-bad-with-header-tied-augmented-partial-candidate
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) ≤ TraceAug data attacker-state
and trace-missing-bound:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
    wp-event verify-monad
      (trace-fri-header-tied-recorded-sibling-candidate-missing
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) ≤
      TraceMissing data attacker-state
and trace-merkle-bound:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
    wp-event verify-monad
      (partial-merkle-inconsistency-bad
        (verifier-state-from-adversary attacker-state

```

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceMerkle data attacker-state}$
and *trace-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{TraceAug data attacker-state} +$
 $(\text{TraceMissing data attacker-state} + \text{TraceMerkle data attacker-state})$
 $\leq \text{trace-fri-error}$
and *composition-sampled-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{CompSampled data attacker-state}$
and *composition-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{CompMerkle data attacker-state}$
and *composition-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{CompSampled data attacker-state} +$
 $((\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0 + 0) +$
 $\text{CompMerkle data attacker-state}) + 0)$
 $\leq \text{composition-fri-error}$
and *empty-sampled-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$

```

    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (trace-fri-bad-with-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq$ 
  EmptySampled data attacker-state
and empty-obstruction-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (trace-fri-sampled-layer-assignment-obstruction
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq$ 
  EmptyObstruction data attacker-state
and empty-total:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    EmptySampled data attacker-state + EmptyObstruction data attacker-state
     $\leq$  trace-fri-error
shows active-fri-reductions-for A
proof (rule active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty)
fix data attacker-state
assume support:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
  (trace-fri-bad-with-header-tied-augmented-partial-candidate ?s) ?s
   $\leq$  TraceAug data attacker-state
by (rule trace-augmented-bound[OF support])
show wp-event verify-monad
  (trace-fri-header-tied-recorded-sibling-candidate-missing ?s) ?s
   $\leq$  TraceMissing data attacker-state
by (rule trace-missing-bound[OF support])
show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
   $\leq$  TraceMerkle data attacker-state
by (rule trace-merkle-bound[OF support])

```



```

show TraceAug data attacker-state +
  (TraceMissing data attacker-state + TraceMerkle data attacker-state)
  ≤ trace-fri-error
by (rule trace-total[OF support])
show wp-event verify-monad
  (composition-fri-bad-with-verifier-tied-sampled-layer-chain ?s) ?s
  ≤ CompSampled data attacker-state
by (rule composition-sampled-bound[OF support])
show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
  ≤ CompMerkle data attacker-state
by (rule composition-merkle-bound[OF support])
show CompSampled data attacker-state +
  (((CompMerkle data attacker-state + 0) +
    (CompMerkle data attacker-state + 0) +
    (CompMerkle data attacker-state + 0) +
    (CompMerkle data attacker-state + 0 + 0) +
    CompMerkle data attacker-state) + 0)
  ≤ composition-fri-error
by (rule composition-total[OF support])
show trace-fri-empty-header-reduction ?s
by (rule
  trace-fri-empty-header-reduction-from-sampled-and-assignment-obstruction-bounds
  [OF empty-sampled-bound[OF support]
  empty-obstruction-bound[OF support] empty-total[OF support]])

```

qed

lemma active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty-conflict-zero:

fixes TraceAug TraceMissing TraceMerkle CompSampled CompMerkle EmptySampled Empty-
Conflict EmptyZero ::

'f staged-proof-data ⇒ 'f protocol-channel ⇒ prob

assumes trace-augmented-bound:

∧ data attacker-state.

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-bad-with-header-tied-augmented-partial-candidate
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ TraceAug data attacker-state

and trace-missing-bound:

∧ data attacker-state.

Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-header-tied-recorded-sibling-candidate-missing
 (verifier-state-from-adversary attacker-state

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{TraceMissing data attacker-state}$
and *trace-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceMerkle data attacker-state}$
and *trace-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{TraceAug data attacker-state} +$
 $(\text{TraceMissing data attacker-state} + \text{TraceMerkle data attacker-state})$
 $\leq \text{trace-fri-error}$
and *composition-sampled-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{CompSampled data attacker-state}$
and *composition-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{CompMerkle data attacker-state}$
and *composition-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$

$set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $CompSampled \ data \ attacker-state +$
 $((CompMerkle \ data \ attacker-state + 0) +$
 $(CompMerkle \ data \ attacker-state + 0) +$
 $(CompMerkle \ data \ attacker-state + 0) +$
 $(CompMerkle \ data \ attacker-state + 0 + 0) +$
 $CompMerkle \ data \ attacker-state) + 0)$
 $\leq composition-fri-error$
and *empty-sampled-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(trace-fri-bad-with-sampled-layer-chain$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq$
 $EmptySampled \ data \ attacker-state$
and *empty-conflict-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(trace-fri-sampled-assignment-conflict$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq$
 $EmptyConflict \ data \ attacker-state$
and *empty-zero-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(trace-fri-zero-round-final-obstruction$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq$
 $EmptyZero \ data \ attacker-state$
and *empty-total*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A)$

```

    adversary-initial-state)  $\implies$ 
    EmptySampled data attacker-state +
    (EmptyConflict data attacker-state + EmptyZero data attacker-state)
     $\leq$  trace-fri-error
shows active-fri-reductions-for A
proof (rule active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty-assignment
    [where EmptyObstruction =
         $\lambda$ data attacker-state.
        EmptyConflict data attacker-state + EmptyZero data attacker-state])
fix data attacker-state
assume support:
    Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
    (trace-fri-bad-with-header-tied-augmented-partial-candidate ?s) ?s
     $\leq$  TraceAug data attacker-state
    by (rule trace-augmented-bound[OF support])
show wp-event verify-monad
    (trace-fri-header-tied-recorded-sibling-candidate-missing ?s) ?s
     $\leq$  TraceMissing data attacker-state
    by (rule trace-missing-bound[OF support])
show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
     $\leq$  TraceMerkle data attacker-state
    by (rule trace-merkle-bound[OF support])
show TraceAug data attacker-state +
    (TraceMissing data attacker-state + TraceMerkle data attacker-state)
     $\leq$  trace-fri-error
    by (rule trace-total[OF support])
show wp-event verify-monad
    (composition-fri-bad-with-verifier-tied-sampled-layer-chain ?s) ?s
     $\leq$  CompSampled data attacker-state
    by (rule composition-sampled-bound[OF support])
show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
     $\leq$  CompMerkle data attacker-state
    by (rule composition-merkle-bound[OF support])
show CompSampled data attacker-state +
    (((CompMerkle data attacker-state + 0) +
    (CompMerkle data attacker-state + 0) +
    (CompMerkle data attacker-state + 0) +
    (CompMerkle data attacker-state + 0 + 0) +
    CompMerkle data attacker-state) + 0)
     $\leq$  composition-fri-error
    by (rule composition-total[OF support])
show wp-event verify-monad
    (trace-fri-bad-with-sampled-layer-chain ?s) ?s

```

$\leq \text{EmptySampled data attacker-state}$
by (rule empty-sampled-bound[OF support])
show wp-event verify-monad
 $(\text{trace-fri-sampled-layer-assignment-obstruction } ?s) ?s$
 $\leq \text{EmptyConflict data attacker-state} + \text{EmptyZero data attacker-state}$
by (rule
 $\text{wp-trace-fri-sampled-layer-assignment-obstruction-bound-from-conflict-and-zero}$
(rule empty-conflict-bound[OF support],
rule empty-zero-bound[OF support])
show EmptySampled data attacker-state +
 $(\text{EmptyConflict data attacker-state} + \text{EmptyZero data attacker-state})$
 $\leq \text{trace-fri-error}$
by (rule empty-total[OF support])
qed

lemma active-fri-reductions-for-from-trace-augmented-conflict-zero-composition-sampled-and-empty-conflict-zero:

fixes TraceAug TraceConflict TraceZero TraceMerkle CompSampled CompMerkle

EmptySampled EmptyConflict EmptyZero ::

'f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ prob

assumes trace-augmented-bound:

$\wedge \text{data attacker-state.}$

$\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \Rightarrow$
wp-event verify-monad
 $(\text{trace-fri-bad-with-header-tied-augmented-partial-candidate}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceAug data attacker-state}$

and trace-conflict-bound:

$\wedge \text{data attacker-state.}$

$\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \Rightarrow$
wp-event verify-monad
 $(\text{trace-fri-header-tied-sampled-assignment-conflict}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceConflict data attacker-state}$

and trace-zero-bound:

$\wedge \text{data attacker-state.}$

$\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \Rightarrow$
wp-event verify-monad
 $(\text{trace-fri-header-tied-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceZero data attacker-state}$
and *trace-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceMerkle data attacker-state}$
and *trace-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{TraceAug data attacker-state} +$
 $((\text{TraceConflict data attacker-state} + \text{TraceZero data attacker-state}) +$
 $\text{TraceMerkle data attacker-state})$
 $\leq \text{trace-fri-error}$
and *composition-sampled-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{CompSampled data attacker-state}$
and *composition-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{CompMerkle data attacker-state}$
and *composition-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$

$set-dist \ (execute \ (checked-staged-transcript-program \ A) \ \text{adversary-initial-state}) \implies$
 $CompSampled \ data \ \text{attacker-state} +$
 $((CompMerkle \ data \ \text{attacker-state} + 0) +$
 $(CompMerkle \ data \ \text{attacker-state} + 0) +$
 $(CompMerkle \ data \ \text{attacker-state} + 0) +$
 $(CompMerkle \ data \ \text{attacker-state} + 0 + 0) +$
 $CompMerkle \ data \ \text{attacker-state}) + 0)$
 $\leq composition-fri-error$
and *empty-sampled-bound*:
 $\wedge data \ \text{attacker-state}.$
 $Some \ (data, \ \text{attacker-state}) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ \text{adversary-initial-state}) \implies$
 $wp-event \ \text{verify-monad}$
 $(trace-fri-bad-with-sampled-layer-chain$
 $(verifier-state-from-adversary \ \text{attacker-state}$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ \text{attacker-state}$
 $(staged-proof-transcript \ data)) \leq$
 $EmptySampled \ data \ \text{attacker-state}$
and *empty-conflict-bound*:
 $\wedge data \ \text{attacker-state}.$
 $Some \ (data, \ \text{attacker-state}) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ \text{adversary-initial-state}) \implies$
 $wp-event \ \text{verify-monad}$
 $(trace-fri-sampled-assignment-conflict$
 $(verifier-state-from-adversary \ \text{attacker-state}$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ \text{attacker-state}$
 $(staged-proof-transcript \ data)) \leq$
 $EmptyConflict \ data \ \text{attacker-state}$
and *empty-zero-bound*:
 $\wedge data \ \text{attacker-state}.$
 $Some \ (data, \ \text{attacker-state}) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ \text{adversary-initial-state}) \implies$
 $wp-event \ \text{verify-monad}$
 $(trace-fri-zero-round-final-obstruction$
 $(verifier-state-from-adversary \ \text{attacker-state}$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ \text{attacker-state}$
 $(staged-proof-transcript \ data)) \leq$
 $EmptyZero \ data \ \text{attacker-state}$
and *empty-total*:
 $\wedge data \ \text{attacker-state}.$
 $Some \ (data, \ \text{attacker-state}) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A)$

```

    adversary-initial-state)  $\implies$ 
    EmptySampled data attacker-state +
    (EmptyConflict data attacker-state + EmptyZero data attacker-state)
     $\leq$  trace-fri-error
shows active-fri-reductions-for A
proof (rule active-fri-reductions-for-from-component-reductions)
fix data attacker-state
assume support:
    Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state)
let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show trace-fri-header-tied-reduction ?s
by (rule
    trace-fri-header-tied-reduction-from-augmented-conflict-zero-merkle-bounds)
    (rule trace-augmented-bound[OF support],
    rule trace-conflict-bound[OF support],
    rule trace-zero-bound[OF support],
    rule trace-merkle-bound[OF support],
    rule trace-total[OF support])
show composition-fri-verifier-tied-reduction ?s
by (rule active-composition-fri-verifier-tied-reductions-from-sampled-and-merkle-bounds
    [OF composition-sampled-bound composition-merkle-bound
    composition-total support])
show trace-fri-empty-header-reduction ?s
by (rule trace-fri-empty-header-reduction-from-sampled-conflict-and-zero-bounds)
    (rule empty-sampled-bound[OF support],
    rule empty-conflict-bound[OF support],
    rule empty-zero-bound[OF support],
    rule empty-total[OF support])
qed

lemma active-fri-reductions-for-from-trace-augmented-composition-canonical-and-empty-conflict-zero:
fixes TraceAug TraceMissing TraceMerkle CompCanonical CompMerkle EmptyCanonical EmptyConflict EmptyZero ::
    'f staged-proof-data  $\Rightarrow$  'f protocol-channel  $\Rightarrow$  prob
assumes trace-augmented-bound:
     $\bigwedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state)  $\implies$ 
    wp-event verify-monad
    (trace-fri-bad-with-header-tied-augmented-partial-candidate
    (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq$  TraceAug data attacker-state

```


and *trace-missing-bound*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-header-tied-recorded-sibling-candidate-missing*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)))
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq$
TraceMissing *data* *attacker-state*
and *trace-merkle-bound*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*partial-merkle-inconsistency-bad*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)))
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq$ *TraceMerkle* *data* *attacker-state*
and *trace-total*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
TraceAug *data* *attacker-state* +
(*TraceMissing* *data* *attacker-state* + *TraceMerkle* *data* *attacker-state*)
 $\leq$ *trace-fri-error*
and *composition-canonical-bound*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*composition-fri-bad-with-canonical-sampled-layer-chain*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)))
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq$
CompCanonical *data* *attacker-state*
and *composition-merkle-bound*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*

(partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ CompMerkle data attacker-state
and composition-total:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 CompCanonical data attacker-state +
 (((CompMerkle data attacker-state + 0) +
 (CompMerkle data attacker-state + 0) +
 (CompMerkle data attacker-state + 0) +
 (CompMerkle data attacker-state + 0 + 0) +
 CompMerkle data attacker-state) + 0)
 ≤ composition-fri-error
and empty-canonical-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-bad-with-canonical-sampled-layer-chain
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤
 EmptyCanonical data attacker-state
and empty-conflict-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-sampled-assignment-conflict
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤
 EmptyConflict data attacker-state
and empty-zero-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state

```

      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤
    EmptyZero data attacker-state
and empty-total:
 $\wedge$ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    EmptyCanonical data attacker-state +
      (EmptyConflict data attacker-state + EmptyZero data attacker-state)
    ≤ trace-fri-error
shows active-fri-reductions-for A
proof (rule
  active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty-conflict-zero)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
  (trace-fri-bad-with-header-tied-augmented-partial-candidate ?s) ?s
  ≤ TraceAug data attacker-state
by (rule trace-augmented-bound[OF support])
show wp-event verify-monad
  (trace-fri-header-tied-recorded-sibling-candidate-missing ?s) ?s
  ≤ TraceMissing data attacker-state
by (rule trace-missing-bound[OF support])
show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
  ≤ TraceMerkle data attacker-state
by (rule trace-merkle-bound[OF support])
show TraceAug data attacker-state +
  (TraceMissing data attacker-state + TraceMerkle data attacker-state)
  ≤ trace-fri-error
by (rule trace-total[OF support])
show wp-event verify-monad
  (composition-fri-bad-with-verifier-tied-sampled-layer-chain ?s) ?s
  ≤ CompCanonical data attacker-state
by (rule
  wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-sampled)
  (rule wp-composition-fri-sampled-bound-from-canonical,
  rule composition-canonical-bound[OF support])
show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
  ≤ CompMerkle data attacker-state
by (rule composition-merkle-bound[OF support])
show CompCanonical data attacker-state +

```

```

      (((CompMerkle data attacker-state + 0) +
        (CompMerkle data attacker-state + 0) +
        (CompMerkle data attacker-state + 0) +
        (CompMerkle data attacker-state + 0 + 0) +
        CompMerkle data attacker-state) + 0)
    ≤ composition-fri-error
  by (rule composition-total[OF support])
show wp-event verify-monad
  (trace-fri-bad-with-sampled-layer-chain ?s) ?s
  ≤ EmptyCanonical data attacker-state
  by (rule wp-trace-fri-sampled-bound-from-canonical)
    (rule empty-canonical-bound[OF support])
show wp-event verify-monad
  (trace-fri-sampled-assignment-conflict ?s) ?s
  ≤ EmptyConflict data attacker-state
  by (rule empty-conflict-bound[OF support])
show wp-event verify-monad
  (trace-fri-zero-round-final-obstruction ?s) ?s
  ≤ EmptyZero data attacker-state
  by (rule empty-zero-bound[OF support])
show EmptyCanonical data attacker-state +
  (EmptyConflict data attacker-state + EmptyZero data attacker-state)
  ≤ trace-fri-error
  by (rule empty-total[OF support])
qed

end

end

```

```

theory Soundness-FRI-Sampled-Interface
imports
  Soundness-FRI-Derived-Active-Reductions
begin

```

4.10 Concrete FRI Query Bounds and Public Endpoint

Sampled-query FRI interface.

This theory names the FRI bad-candidate event that matches the verifier's sampled evidence: authenticated partial openings, sampled fold checks on the canonical domains, final consistency, and a non-low-degree candidate. It is deliberately separate from the full-cover compatibility route, where sampled openings are temporarily strengthened to full fold-index coverage.

```

context soundness
begin

```

```

definition fri-query-index-list-space :: nat list set

```

where

fri-query-index-list-space =
 $\{ \text{query-idxs}. \text{length } \text{query-idxs} = \text{rounds} \wedge$
 $\text{set } \text{query-idxs} \subseteq \text{query-sample-space} \}$

definition *trace-fri-query-index-list-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{nat list set} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-query-index-list-set-hit *s Q out* $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers.}$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $\text{fri-query-idxs} \in Q)$

definition *composition-fri-query-index-list-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{nat list set} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-query-index-list-set-hit *s Q out* $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers.}$
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $\text{fri-query-idxs} \in Q)$

definition *query-index-list-entries* $:: \text{nat list set} \Rightarrow \text{nat set}$

where

query-index-list-entries *Q* = $(\bigcup \text{query-idxs} \in Q. \text{set } \text{query-idxs})$

lemma *query-index-list-entries-subset-query-sample-space:*

assumes $Q \subseteq \text{fri-query-index-list-space}$

shows $\text{query-index-list-entries } Q \subseteq \text{query-sample-space}$

using *assms*

unfolding *query-index-list-entries-def fri-query-index-list-space-def*

by *auto*

lemma *finite-query-index-list-entries:*

assumes $Q \subseteq \text{fri-query-index-list-space}$

shows *finite* $(\text{query-index-list-entries } Q)$

using *query-index-list-entries-subset-query-sample-space[OF assms]*

by $(\text{rule finite-subset}) (\text{rule finite-query-sample-space})$

lemma *query-index-list-entries-card-le*:
assumes *finite-Q*: *finite Q*
and *subset*: $Q \subseteq \text{fri-query-index-list-space}$
shows $\text{card } (\text{query-index-list-entries } Q) \leq \text{card } Q * \text{rounds}$
proof –
have
 $\text{card } (\text{query-index-list-entries } Q) \leq (\sum \text{query-idxs} \in Q.$
 $\text{card } (\text{set query-idxs}))$
unfolding *query-index-list-entries-def*
by (*rule card-UN-le*) (*rule finite-Q*)
also have $\dots \leq (\sum \text{query-idxs} \in Q. \text{rounds})$
proof (*rule sum-mono*)
fix *query-idxs*
assume $\text{query-idxs} \in Q$
then have $\text{length query-idxs} = \text{rounds}$
using *subset* **unfolding** *fri-query-index-list-space-def* **by** *auto*
then show $\text{card } (\text{set query-idxs}) \leq \text{rounds}$
using *card-length[of query-idxs]* **by** *simp*
qed
also have $\dots = \text{card } Q * \text{rounds}$
using *finite-Q* **by** *simp*
finally show *?thesis* .
qed

definition *generic-fri-sampled-query-candidate-evidence*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow$
 $'f \text{ list} \Rightarrow \text{nat} \Rightarrow$
 $'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat list} \Rightarrow$
 $'f \text{ list list list} \Rightarrow 'f \text{ list list} \Rightarrow \text{bool}$

where

generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers $\longleftrightarrow$
generic-fri-canonical-sampled-layer-chain-evidence low-degree
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers $\wedge$
candidate-bad candidate-table

lemma *generic-fri-sampled-query-candidate-evidenceI*:

assumes

generic-fri-canonical-sampled-layer-chain-evidence low-degree
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers
candidate-bad candidate-table

shows

generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers

using *assms*

unfolding *generic-fri-sampled-query-candidate-evidence-def* **by** *blast*

lemma *generic-fri-sampled-query-candidate-evidenceD:*

assumes

*generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers*

shows

*generic-fri-canonical-sampled-layer-chain-evidence low-degree
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers
candidate-bad candidate-table*

using *assms*

unfolding *generic-fri-sampled-query-candidate-evidence-def* **by** *blast+*

definition *generic-fri-sampled-query-bad-pair-set*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow$
 $'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list list list} \Rightarrow 'f \text{ list list} \Rightarrow$
 $(\text{nat list} \times 'f \text{ list}) \text{ set}$

where

*generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
candidate-table degree-bound roots final-value round-layers layers =*
 $\{(query-idxs, challenges).$
*generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers}*

lemma *generic-fri-sampled-query-bad-pair-setI:*

assumes

*generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers*

shows

$(query-idxs, challenges) \in$
*generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
candidate-table degree-bound roots final-value round-layers layers*

using *assms* **unfolding** *generic-fri-sampled-query-bad-pair-set-def* **by** *simp*

lemma *generic-fri-sampled-query-bad-pair-setD:*

assumes

$(query-idxs, challenges) \in$
*generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
candidate-table degree-bound roots final-value round-layers layers*

shows

*generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers*

using *assms* **unfolding** *generic-fri-sampled-query-bad-pair-set-def* **by** *simp*

lemma *generic-fri-sampled-query-bad-pair-set-canonical-evidence*:
assumes *pair*:
 (*query-idxs*, *challenges*) $\in$
generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
candidate-table degree-bound roots final-value round-layers layers
shows
generic-fri-canonical-sampled-layer-chain-evidence low-degree
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers
proof –
have *evidence*:
generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers
by (rule *generic-fri-sampled-query-bad-pair-setD*[*OF pair*])
show ?thesis
by (rule *generic-fri-sampled-query-candidate-evidenceD*(1)[*OF evidence*])
qed

lemma *generic-fri-sampled-query-bad-pair-set-sampled-evidence*:
assumes *pair*:
 (*query-idxs*, *challenges*) $\in$
generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
candidate-table degree-bound roots final-value round-layers layers
shows
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges)) layers
proof –
have *canonical*:
generic-fri-canonical-sampled-layer-chain-evidence low-degree
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers
by (rule *generic-fri-sampled-query-bad-pair-set-canonical-evidence*
 [*OF pair*])
show ?thesis
by (rule *generic-fri-canonical-sampled-layer-chain-evidenceD*
 [*OF canonical*])
qed

lemma *generic-fri-sampled-query-bad-pair-set-partial-evidence*:
assumes *pair*:
 (*query-idxs*, *challenges*) $\in$
generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
candidate-table degree-bound roots final-value round-layers layers
shows
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers

proof –
have *sampled*:
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges)) layers
by (rule *generic-fri-sampled-query-bad-pair-set-sampled-evidence*
[*OF pair*])
show *?thesis*
by (rule *generic-fri-sampled-layer-chain-evidenceD(1)*[*OF sampled*])
qed

lemma *generic-fri-sampled-query-bad-pair-set-challenge-length*:
assumes *pair*:
(*query-idxs, challenges*) ∈
generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
candidate-table degree-bound roots final-value round-layers layers
shows *length challenges = fri-round-count-for-degree-bound degree-bound*
proof –
have *evidence*:
generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers
by (rule *generic-fri-sampled-query-bad-pair-setD*[*OF pair*])
have *canonical*:
generic-fri-canonical-sampled-layer-chain-evidence low-degree
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers
by (rule *generic-fri-sampled-query-candidate-evidenceD(1)*[*OF evidence*])
have *sampled*:
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges)) layers
by (rule *generic-fri-canonical-sampled-layer-chain-evidenceD*
[*OF canonical*])
have *partial*:
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers
by (rule *generic-fri-sampled-layer-chain-evidenceD(1)*[*OF sampled*])
show *?thesis*
using *generic-fri-partial-evidence-shapes(1,2)*[*OF partial*] **by** *simp*
qed

lemma *generic-fri-sampled-query-bad-pair-set-round-layers-length*:
assumes *pair*:
(*query-idxs, challenges*) ∈
generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
candidate-table degree-bound roots final-value round-layers layers
shows *length round-layers = length query-idxs*
proof –

have *partial*:
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers
by (rule *generic-fri-sampled-query-bad-pair-set-partial-evidence*
[*OF pair*])
have *fri-all-round-layer-evidence roots challenges query-idxs round-layers*
by (rule *generic-fri-partial-evidence-shapes*(3)[*OF partial*])
then show *?thesis*
unfolding *fri-all-round-layer-evidence-def* **by** *simp*
qed

lemma *generic-fri-sampled-query-bad-pair-set-sample*:
assumes *pair*:
(*query-idxs, challenges*) $\in$
generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
candidate-table degree-bound roots final-value round-layers layers
and *round-bound*: *round-idx* < *length query-idxs*
and *layer-bound*: *layer-idx* < *length challenges*
shows
fri-evidence-next-idx roots query-idxs round-idx layer-idx <
length (layers ! Suc layer-idx)
fri-sampled-table-fold
(*challenges ! layer-idx*)
(*fri-evidence-layer-len roots layer-idx*)
(*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*)
($2 \wedge \text{layer-idx}$)
(*fri-canonical-domains (length challenges) ! layer-idx*)
(*layers ! layer-idx*)
(*round-layers ! round-idx ! layer-idx*)
(*layers ! Suc layer-idx !*
fri-evidence-next-idx roots query-idxs round-idx layer-idx)

proof –
have *sampled*:
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(*fri-canonical-domains (length challenges) layers*)
by (rule *generic-fri-sampled-query-bad-pair-set-sampled-evidence*
[*OF pair*])
show *fri-evidence-next-idx roots query-idxs round-idx layer-idx* <
length (layers ! Suc layer-idx)
by (rule *generic-fri-sampled-layer-chain-evidence-sample*(1)
[*OF sampled round-bound layer-bound*])
show *fri-sampled-table-fold*
(*challenges ! layer-idx*)
(*fri-evidence-layer-len roots layer-idx*)
(*fri-evidence-layer-idx roots query-idxs round-idx layer-idx*)
($2 \wedge \text{layer-idx}$)
(*fri-canonical-domains (length challenges) ! layer-idx*)
(*layers ! layer-idx*)

(round-layers ! round-idx ! layer-idx)
 (layers ! Suc layer-idx !
 fri-evidence-next-idx roots query-idxs round-idx layer-idx)
 by (rule generic-fri-sampled-layer-chain-evidence-sample(2)
 [OF sampled round-bound layer-bound])
 qed

lemma generic-fri-sampled-query-bad-pair-set-candidate-bad:
 assumes pair:
 (query-idxs, challenges) ∈
 generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
 candidate-table degree-bound roots final-value round-layers layers
 shows candidate-bad candidate-table
proof –
 have evidence:
 generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
 candidate-table degree-bound roots challenges final-value query-idxs
 round-layers layers
 by (rule generic-fri-sampled-query-bad-pair-setD[OF pair])
 show ?thesis
 by (rule generic-fri-sampled-query-candidate-evidenceD(2)[OF evidence])
 qed

lemma generic-fri-sampled-query-bad-pair-set-challenge-space:
 assumes
 (query-idxs, challenges) ∈
 generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
 candidate-table degree-bound roots final-value round-layers layers
 shows challenges ∈
 fri-challenge-space (fri-round-count-for-degree-bound degree-bound)
 using generic-fri-sampled-query-bad-pair-set-challenge-length[OF assms]
 unfolding fri-challenge-space-def by simp

lemma generic-fri-sampled-query-bad-pair-set-subset-challenge-space:
 generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
 candidate-table degree-bound roots final-value round-layers layers
 ⊆ (UNIV :: nat list set) ×
 fri-challenge-space (fri-round-count-for-degree-bound degree-bound)
 by (auto intro: generic-fri-sampled-query-bad-pair-set-challenge-space)

definition fri-query-challenge-pair-query-fiber
 :: (nat list × 'f list) set ⇒ 'f list ⇒ nat list set
where
 fri-query-challenge-pair-query-fiber P challenges =
 {query-idxs. (query-idxs, challenges) ∈ P}

lemma finite-fri-query-index-list-space:
 finite fri-query-index-list-space
proof –

```

have subset:
  fri-query-index-list-space  $\subseteq$ 
    {query-idxs. set query-idxs  $\subseteq$  query-sample-space  $\wedge$ 
      length query-idxs = rounds}
  unfolding fri-query-index-list-space-def by auto
have finite-super:
  finite
    {query-idxs. set query-idxs  $\subseteq$  query-sample-space  $\wedge$ 
      length query-idxs = rounds}
  by (rule finite-lists-length-eq[OF finite-query-sample-space])
show ?thesis
  by (rule finite-subset[OF subset finite-super])
qed

lemma fri-query-challenge-pair-query-fiber-iff[simp]:
  query-idxs  $\in$  fri-query-challenge-pair-query-fiber  $P$  challenges  $\longleftrightarrow$ 
    (query-idxs, challenges)  $\in P$ 
  unfolding fri-query-challenge-pair-query-fiber-def by simp

lemma fri-query-challenge-pair-query-projection-restricted:
  fst ‘ (P  $\cap$  (fri-query-index-list-space  $\times$  UNIV))
     $\subseteq$  fri-query-index-list-space
proof
  fix query-idxs
  assume query-idxs  $\in$ 
    fst ‘ (P  $\cap$  (fri-query-index-list-space  $\times$  UNIV))
  then show query-idxs  $\in$  fri-query-index-list-space
    by auto
qed

lemma finite-fri-query-challenge-pair-query-fiber-restricted:
  finite
    (fri-query-challenge-pair-query-fiber
      (P  $\cap$  (fri-query-index-list-space  $\times$  UNIV)) challenges)
proof –
  have
    fri-query-challenge-pair-query-fiber
      (P  $\cap$  (fri-query-index-list-space  $\times$  UNIV)) challenges
       $\subseteq$  fri-query-index-list-space
  unfolding fri-query-challenge-pair-query-fiber-def by blast
  then show ?thesis
    by (rule finite-subset) (rule finite-fri-query-index-list-space)
qed

definition trace-fri-query-challenge-pair-set-hit
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$  (nat list  $\times$  'f list) set  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  trace-fri-query-challenge-pair-set-hit s P out  $\longleftrightarrow$ 

```

$(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers.}$
 $\text{accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{trace-final dg composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers} \wedge$
 $(\text{fri-query-idxs, trace-bs}) \in P)$

definition *composition-fri-query-challenge-pair-set-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow (\text{nat list} \times 'f \text{ list}) \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

$\text{composition-fri-query-challenge-pair-set-hit s P out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers.}$
 $\text{accepted-fri-opening-transcript s out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers} \wedge$
 $(\text{fri-query-idxs, composition-bs}) \in P \text{ fri-dg})$

lemma *accepted-fri-opening-transcript-query-index-list-space:*

assumes

$\text{accepted-fri-opening-transcript s out trace-roots trace-bs trace-final}$
 $\text{dg composition-roots composition-bs composition-final fri-query-idxs}$
 $\text{trace-round-layers composition-round-layers}$

shows $\text{fri-query-idxs} \in \text{fri-query-index-list-space}$

proof –

have $\text{len: length fri-query-idxs} = \text{rounds}$

using $\text{accepted-fri-opening-transcript-shapes[OF assms]}$ **by** *blast*

have $\text{subset: set fri-query-idxs} \subseteq \text{query-sample-space}$

proof

fix idx

assume $\text{idx} \in \text{set fri-query-idxs}$

then show $\text{idx} \in \text{query-sample-space}$

by $(\text{rule accepted-fri-opening-transcript-query-idx-in-sample-space}$
 $[\text{OF assms}])$

qed

show *?thesis*

unfolding $\text{fri-query-index-list-space-def}$

using len subset **by** *simp*

qed

lemma *trace-fri-query-challenge-pair-set-hit-restrict-to-query-space:*

assumes $\text{trace-fri-query-challenge-pair-set-hit s P out}$

shows

$\text{trace-fri-query-challenge-pair-set-hit s}$

$(P \cap (\text{fri-query-index-list-space} \times \text{UNIV})) \text{ out}$
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers **where**
transcript:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *pair*: $(\text{fri-query-idxs}, \text{trace-bs}) \in P$
unfolding *trace-fri-query-challenge-pair-set-hit-def* **by** *blast*
have *query-space*: $\text{fri-query-idxs} \in \text{fri-query-index-list-space}$
by $(\text{rule } \text{accepted-fri-opening-transcript-query-index-list-space}$
 $[\text{OF } \text{transcript}])$
show *?thesis*
unfolding *trace-fri-query-challenge-pair-set-hit-def*
using *transcript pair query-space* **by** *blast*
qed

lemma *composition-fri-query-challenge-pair-set-hit-restrict-to-query-space:*
assumes *composition-fri-query-challenge-pair-set-hit s P out*
shows
composition-fri-query-challenge-pair-set-hit s
 $(\lambda dg. P \ dg \cap (\text{fri-query-index-list-space} \times \text{UNIV})) \text{ out}$
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers **where**
transcript:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *pair*: $(\text{fri-query-idxs}, \text{composition-bs}) \in P \text{ fri-dg}$
unfolding *composition-fri-query-challenge-pair-set-hit-def* **by** *blast*
have *query-space*: $\text{fri-query-idxs} \in \text{fri-query-index-list-space}$
by $(\text{rule } \text{accepted-fri-opening-transcript-query-index-list-space}$
 $[\text{OF } \text{transcript}])$
show *?thesis*
unfolding *composition-fri-query-challenge-pair-set-hit-def*
using *transcript pair query-space* **by** *blast*
qed

lemma *trace-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit:*
assumes *hit: trace-fri-query-challenge-pair-set-hit s P out*
and *projection: fst ‘ P ⊆ Q*
shows *trace-fri-query-index-list-set-hit s Q out*
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers
where *openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
and *pair*: (*fri-query-idxs*, *trace-bs*) $\in P$
unfolding *trace-fri-query-challenge-pair-set-hit-def* **by** *blast*
have *fri-query-idxs* $\in Q$
using *pair projection* **by** *force*
then show *?thesis*
unfolding *trace-fri-query-index-list-set-hit-def*
using *openings* **by** *blast*
qed

lemma *composition-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit*:
assumes *hit*: *composition-fri-query-challenge-pair-set-hit s P out*
and *projection*: $\bigwedge dg. \text{fst } dg \subseteq Q$
shows *composition-fri-query-index-list-set-hit s Q out*
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers
where *openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
and *pair*: (*fri-query-idxs*, *composition-bs*) $\in P$ *fri-dg*
unfolding *composition-fri-query-challenge-pair-set-hit-def* **by** *blast*
have *fri-query-idxs* $\in Q$
using *pair projection[of fri-dg]* **by** *force*
then show *?thesis*
unfolding *composition-fri-query-index-list-set-hit-def*
using *openings* **by** *blast*
qed

lemma *trace-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit*:
assumes *hit*: *trace-fri-query-challenge-pair-set-hit s P out*
and *projection*: $\text{snd } P \subseteq B$
shows *trace-fri-challenge-list-set-hit s B out*
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers
where *openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers

and *pair*: (*fri-query-idxs*, *trace-bs*) ∈ *P*
unfolding *trace-fri-query-challenge-pair-set-hit-def* **by** *blast*
have *challenges*:
 accepted-fri-challenges *s* *out* *trace-bs* *dg* *composition-bs*
 by (*rule* *accepted-fri-opening-transcript-challenges*[*OF* *openings*])
have *trace-bs* ∈ *B*
 using *pair* *projection* **by** *force*
then show *?thesis*
 unfolding *trace-fri-challenge-list-set-hit-def*
 using *challenges* **by** *blast*
qed

lemma *composition-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit*:
assumes *hit*: *composition-fri-query-challenge-pair-set-hit* *s* *P* *out*
and *projection*: $\bigwedge dg. \text{snd } 'P \text{ } dg \subseteq B \text{ } dg$
shows *composition-fri-challenge-list-set-hit* *s* *B* *out*
proof –
from *hit* **obtain** *trace-roots* *trace-bs* *trace-final* *fri-dg* *composition-roots*
 composition-bs *composition-final* *fri-query-idxs* *trace-round-layers*
 composition-round-layers
where *openings*:
 accepted-fri-opening-transcript *s* *out* *trace-roots* *trace-bs*
 trace-final *fri-dg* *composition-roots* *composition-bs*
 composition-final *fri-query-idxs* *trace-round-layers*
 composition-round-layers
and *pair*: (*fri-query-idxs*, *composition-bs*) ∈ *P* *fri-dg*
unfolding *composition-fri-query-challenge-pair-set-hit-def* **by** *blast*
have *challenges*:
 accepted-fri-challenges *s* *out* *trace-bs* *fri-dg* *composition-bs*
 by (*rule* *accepted-fri-opening-transcript-challenges*[*OF* *openings*])
have *composition-bs* ∈ *B* *fri-dg*
 using *pair* *projection*[*of* *fri-dg*] **by** *force*
then show *?thesis*
 unfolding *composition-fri-challenge-list-set-hit-def*
 using *challenges* **by** *blast*
qed

definition *trace-fri-sampled-query-pair-set-hit*
 $:: ('f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow 'f \text{ list list list} \Rightarrow$
 $'f \text{ list list} \Rightarrow (\text{nat list} \times 'f \text{ list}) \text{ set}) \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-sampled-query-pair-set-hit } R \text{ } s \text{ } out \iff$
 $(\exists \text{ trace-roots } \text{trace-bs } \text{trace-final } dg \text{ } \text{composition-roots}$
 $\text{composition-bs } \text{composition-final } \text{fri-query-idxs } \text{trace-round-layers}$
 $\text{composition-round-layers } fr \text{ } \text{candidate-query-idxs } \text{trace-openings}$
 $\text{trace-table } \text{layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ } out \text{ } \text{trace-roots}$

trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table $\wedge$
(fri-query-idxs, trace-bs) $\in$
R trace-table trace-roots trace-final trace-round-layers layers)

definition *composition-fri-sampled-query-pair-set-hit*

$:: ('f \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list} \Rightarrow 'f \Rightarrow$
 $'f \text{ list list list} \Rightarrow 'f \text{ list list} \Rightarrow$
 $(\text{nat list} \times 'f \text{ list}) \text{ set}) \Rightarrow$
 $('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

composition-fri-sampled-query-pair-set-hit R s out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{opening-composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table layers.}$
 $\text{composition-fri-partial-candidate-opening-evidence s out trace-roots}$
 $\text{trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{composition-query-idxs composition-openings trace-table}$
 $\text{composition-table} \wedge$
 $(\text{fri-query-idxs, composition-bs}) \in$
 $R \text{ fri-dg composition-table opening-composition-roots composition-final}$
 $\text{composition-round-layers layers})$

lemma *trace-fri-sampled-query-pair-set-hit-imp-challenge-list-set-hit:*

assumes *hit: trace-fri-sampled-query-pair-set-hit R s out*

and projection:

$\bigwedge \text{trace-table roots final round-layers layers.}$
 snd '

$R \text{ trace-table roots final round-layers layers} \subseteq B$

shows *trace-fri-challenge-list-set-hit s B out*

proof –

from *hit* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table layers

where *evidence:*

trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings

trace-table
and pair:
(fri-query-idxs, trace-bs) ∈
R trace-table trace-roots trace-final trace-round-layers layers
unfolding *trace-fri-sampled-query-pair-set-hit-def* **by** *blast*
have *challenges:*
accepted-fri-challenges s out trace-bs dg composition-bs
by *(rule accepted-fri-opening-transcript-challenges)*
(rule trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence])
have *trace-bs ∈ B*
using *pair projection[of trace-table trace-roots trace-final*
trace-round-layers layers] **by** *force*
then show *?thesis*
unfolding *trace-fri-challenge-list-set-hit-def*
using *challenges* **by** *blast*
qed

lemma *composition-fri-sampled-query-pair-set-hit-imp-challenge-list-set-hit:*
assumes *hit: composition-fri-sampled-query-pair-set-hit R s out*
and *projection:*
 $\bigwedge$ *fri-dg composition-table roots final round-layers layers.*
snd ‘
R fri-dg composition-table roots final round-layers layers
 $\subseteq$ *B fri-dg*
shows *composition-fri-challenge-list-set-hit s B out*
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table layers
where *evidence:*
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and pair:
(fri-query-idxs, composition-bs) ∈
R fri-dg composition-table opening-composition-roots
composition-final composition-round-layers layers
unfolding *composition-fri-sampled-query-pair-set-hit-def* **by** *auto*
have *challenges:*
accepted-fri-challenges s out trace-bs fri-dg composition-bs
by *(rule accepted-fri-opening-transcript-challenges)*
(rule composition-fri-partial-candidate-opening-evidenceD(1))

[*OF evidence*])
have *composition-bs* $\in B$ *fri-dg*
using *pair projection*[*of fri-dg composition-table*
opening-composition-roots composition-final composition-round-layers
layers] **by** *force*
then show *?thesis*
unfolding *composition-fri-challenge-list-set-hit-def*
using *challenges* **by** *blast*
qed

lemma *trace-fri-sampled-query-pair-set-hit-imp-query-index-list-set-hit:*

assumes *hit: trace-fri-sampled-query-pair-set-hit R s out*

and *projection:*

$\bigwedge \text{trace-table roots final round-layers layers.}$
fst ‘

$R \text{ trace-table roots final round-layers layers} \subseteq Q$

shows *trace-fri-query-index-list-set-hit s Q out*

proof –

from *hit* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table layers

where *evidence:*

trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table

and *pair:*

$(\text{fri-query-idxs}, \text{trace-bs}) \in$

$R \text{ trace-table trace-roots trace-final trace-round-layers layers}$

unfolding *trace-fri-sampled-query-pair-set-hit-def* **by** *blast*

have *openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers

by $(\text{rule } \text{trace-fri-partial-candidate-opening-evidenceD}(1)[\text{OF evidence}])$

have *fri-query-idxs* $\in Q$

using *pair projection*[*of trace-table trace-roots trace-final*
trace-round-layers layers] **by** *force*

then show *?thesis*

unfolding *trace-fri-query-index-list-set-hit-def*

using *openings* **by** *blast*

qed

lemma *composition-fri-sampled-query-pair-set-hit-imp-query-index-list-set-hit:*

assumes *hit: composition-fri-sampled-query-pair-set-hit R s out*

and *projection:*

$\bigwedge \text{fri-dg composition-table roots final round-layers layers.}$

$\text{fst } \langle$
 $R \text{ fri-dg composition-table roots final round-layers layers}$
 $\subseteq Q$
shows *composition-fri-query-index-list-set-hit* $s \ Q \text{ out}$
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as declared-dg composition-fri-roots final-value
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table layers
where *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as declared-dg
composition-fri-roots final-value trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *pair*:
 $(\text{fri-query-idxs}, \text{composition-bs}) \in$
 $R \text{ fri-dg composition-table opening-composition-roots}$
 $\text{composition-final composition-round-layers layers}$
unfolding *composition-fri-sampled-query-pair-set-hit-def* **by** *auto*
have *openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
by $(\text{rule } \text{composition-fri-partial-candidate-opening-evidenceD}(1)$
 $[\text{OF } \text{evidence}])$
have $\text{fri-query-idxs} \in Q$
using *pair projection*[*of fri-dg composition-table*
opening-composition-roots composition-final composition-round-layers
layers] **by** *force*
then show *?thesis*
unfolding *composition-fri-query-index-list-set-hit-def*
using *openings* **by** *blast*
qed

lemma *trace-fri-sampled-query-pair-set-hit-imp-query-challenge-pair-set-hit*:
assumes *hit*: *trace-fri-sampled-query-pair-set-hit* $R \ s \text{ out}$
and *projection*:
 $\bigwedge \text{trace-table roots final round-layers layers.}$
 $R \text{ trace-table roots final round-layers layers} \subseteq P$
shows *trace-fri-query-challenge-pair-set-hit* $s \ P \text{ out}$
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr candidate-query-idxs trace-openings
trace-table layers
where *evidence*:
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *pair*:
(fri-query-idxs, trace-bs) ∈
R trace-table trace-roots trace-final trace-round-layers layers
unfolding *trace-fri-sampled-query-pair-set-hit-def* **by** *blast*
have *openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
by *(rule trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence])*
have *(fri-query-idxs, trace-bs) ∈ P*
using *pair projection[of trace-table trace-roots trace-final*
trace-round-layers layers] **by** *blast*
then show *?thesis*
unfolding *trace-fri-query-challenge-pair-set-hit-def*
using *openings* **by** *blast*
qed

lemma *composition-fri-sampled-query-pair-set-hit-imp-query-challenge-pair-set-hit*:
assumes *hit: composition-fri-sampled-query-pair-set-hit R s out*
and *projection*:
 $\bigwedge \text{fri-dg composition-table roots final round-layers layers.}$
 $R \text{ fri-dg composition-table roots final round-layers layers}$
 $\subseteq P \text{ fri-dg}$
shows *composition-fri-query-challenge-pair-set-hit s P out*
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as declared-dg composition-fri-roots final-value
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table layers
where *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as declared-dg
composition-fri-roots final-value trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *pair*:
(fri-query-idxs, composition-bs) ∈

R fri-dg composition-table opening-composition-roots
composition-final composition-round-layers layers
unfolding *composition-fri-sampled-query-pair-set-hit-def* **by** *auto*
have *openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
by (*rule composition-fri-partial-candidate-opening-evidenceD(1)*
[OF evidence])
have (*fri-query-idxs, composition-bs*) $\in P$ *fri-dg*
using *pair projection[of fri-dg composition-table*
opening-composition-roots composition-final composition-round-layers
layers] **by** *blast*
then show *?thesis*
unfolding *composition-fri-query-challenge-pair-set-hit-def*
using *openings* **by** *blast*
qed

lemma *trace-fri-sampled-query-pair-set-hit-imp-query-index-list-space-hit*:

assumes *hit: trace-fri-sampled-query-pair-set-hit R s out*

shows

trace-fri-query-index-list-set-hit s fri-query-index-list-space out

proof —

from *hit* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table layers

where *evidence*:

trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table

unfolding *trace-fri-sampled-query-pair-set-hit-def* **by** *blast*

have *openings*:

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
by (*rule trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence]*)

have *fri-query-idxs* $\in$ *fri-query-index-list-space*

by (*rule accepted-fri-opening-transcript-query-index-list-space*
[OF openings])

then show *?thesis*

unfolding *trace-fri-query-index-list-set-hit-def*

using *openings* **by** *blast*

qed

lemma *composition-fri-sampled-query-pair-set-hit-imp-query-index-list-space-hit*:

assumes *hit: composition-fri-sampled-query-pair-set-hit R s out*
shows
composition-fri-query-index-list-set-hit s fri-query-index-list-space out
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as declared-dg composition-fri-roots final-value
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table layers
where *evidence:*
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as declared-dg
composition-fri-roots final-value trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
unfolding *composition-fri-sampled-query-pair-set-hit-def* **by** *auto*
have *openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
by (*rule composition-fri-partial-candidate-opening-evidenceD(1)*
[OF evidence])
have *fri-query-idxs ∈ fri-query-index-list-space*
by (*rule accepted-fri-opening-transcript-query-index-list-space*
[OF openings])
then show *?thesis*
unfolding *composition-fri-query-index-list-set-hit-def*
using *openings* **by** *blast*
qed

lemma *trace-fri-sampled-query-pair-set-hit-query-indices:*
assumes *hit: trace-fri-sampled-query-pair-set-hit R s out*
obtains *query-idxs challenges candidate-table roots final round-layers layers*
where
length query-idxs = rounds
 $\forall idx \in \text{set } query-idxs. idx \in query-sample-space$
 $\forall idx \in \text{set } query-idxs. idx < clength * scale$
 $(query-idxs, challenges) \in$
R candidate-table roots final round-layers layers
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
candidate-table candidate-layers
where *evidence:*

trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
candidate-table
and *pair*:
(fri-query-idxs, trace-bs) ∈
R candidate-table trace-roots trace-final trace-round-layers
candidate-layers
unfolding *trace-fri-sampled-query-pair-set-hit-def* **by** *blast*
have *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
by *(rule trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence])*
have *len: length fri-query-idxs = rounds*
using *accepted-fri-opening-transcript-shapes[OF fri-openings]* **by** *blast*
have *sample-space: ∀ idx ∈ set fri-query-idxs. idx ∈ query-sample-space*
proof
fix *idx*
assume *idx ∈ set fri-query-idxs*
then show *idx ∈ query-sample-space*
by *(rule accepted-fri-opening-transcript-query-idx-in-sample-space*
[OF fri-openings])
qed
have *bounds: ∀ idx ∈ set fri-query-idxs. idx < clength * scale*
using *fri-openings* **unfolding** *accepted-fri-opening-transcript-def*
by *blast*
show *?thesis*
by *(rule that[OF len sample-space bounds pair])*
qed

lemma *composition-fri-sampled-query-pair-set-hit-query-indices*:
assumes *hit: composition-fri-sampled-query-pair-set-hit R s out*
obtains *query-idxs challenges dg candidate-table roots final round-layers*
layers
where
length query-idxs = rounds
∀ idx ∈ set query-idxs. idx ∈ query-sample-space
*∀ idx ∈ set query-idxs. idx < clength * scale*
(query-idxs, challenges) ∈
R dg candidate-table roots final round-layers layers
proof –
from *hit* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as declared-dg composition-fri-roots final-value
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table candidate-table candidate-layers

where *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as declared-dg
composition-fri-roots final-value trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
candidate-table
and *pair*:
(fri-query-idxs, composition-bs) ∈
R fri-dg candidate-table opening-composition-roots
composition-final composition-round-layers candidate-layers
unfolding *composition-fri-sampled-query-pair-set-hit-def* **by** *auto*
have *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers
by *(rule composition-fri-partial-candidate-opening-evidenceD(1)*
[OF evidence])
have *len: length fri-query-idxs = rounds*
using *accepted-fri-opening-transcript-shapes[OF fri-openings]* **by** *blast*
have *sample-space: ∀ idx ∈ set fri-query-idxs. idx ∈ query-sample-space*
proof
fix *idx*
assume *idx ∈ set fri-query-idxs*
then show *idx ∈ query-sample-space*
by *(rule accepted-fri-opening-transcript-query-idx-in-sample-space*
[OF fri-openings])
qed
have *bounds: ∀ idx ∈ set fri-query-idxs. idx < clength * scale*
using *fri-openings* **unfolding** *accepted-fri-opening-transcript-def*
by *blast*
show *?thesis*
by *(rule that[OF len sample-space bounds pair])*
qed

lemma *wp-trace-fri-sampled-query-pair-set-hit-bound-from-projection*:
assumes *future: trace-fri-future-fresh s*
and *projection*:
 $\bigwedge \text{trace-table roots final round-layers layers.}$
 snd '
 $R \text{ trace-table roots final round-layers layers} \subseteq B$
and *subset: $B \subseteq \text{fri-challenge-space } (\text{ceil-log clength})$*
and *bound*:
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD('f)} \wedge \text{ceil-log clength}) \leq C$
shows
wp-event verify-monad
 $(\text{trace-fri-sampled-query-pair-set-hit } R \text{ s}) \text{ s} \leq C$

proof –
have *wp-event verify-monad*
 (*trace-fri-sampled-query-pair-set-hit R s*) *s* ≤
wp-event verify-monad (trace-fri-challenge-list-set-hit s B) s
by (*rule wp-event-mono*)
 (*rule trace-fri-sampled-query-pair-set-hit-imp-challenge-list-set-hit*
 [*OF - projection*])
also have ... ≤ *nnreal (card B) /*
nnreal (CARD('f) ^ ceil-log clength)
by (*rule wp-verify-monad-trace-fri-challenge-list-set-bound*
 [*OF future subset*])
also have ... ≤ *C*
by (*rule bound*)
finally show ?thesis .
qed

lemma *wp-composition-fri-sampled-query-pair-set-hit-bound-from-projection:*
fixes *C :: prob*
assumes *future: composition-fri-future-fresh s*
and *projection:*
 $\bigwedge \text{fri-dg } \text{composition-table roots final round-layers layers.}$
snd ' P
 $R \text{ fri-dg composition-table roots final round-layers layers}$
 $\subseteq B \text{ fri-dg}$
and *subset:*
 $\bigwedge \text{dg. } B \text{ dg} \subseteq \text{fri-challenge-space (ceil-log (to-nat dg + 1))}$
and *bound:*
 $\bigwedge \text{dg. } \text{nnreal (card (B dg)) /}$
 $\text{nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))} \leq C$
shows
wp-event verify-monad
 (*composition-fri-sampled-query-pair-set-hit R s*) *s* ≤ *C*
proof –
have *wp-event verify-monad*
 (*composition-fri-sampled-query-pair-set-hit R s*) *s* ≤
wp-event verify-monad (composition-fri-challenge-list-set-hit s B) s
by (*rule wp-event-mono*)
 (*rule*
composition-fri-sampled-query-pair-set-hit-imp-challenge-list-set-hit
 [*OF - projection*])
also have ... ≤ *C*
by (*rule wp-verify-monad-composition-fri-challenge-list-set-bound*
 [*OF future subset bound*])
finally show ?thesis .
qed

lemma *wp-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-projection:*
assumes *future: trace-fri-future-fresh s*
and *projection: snd ' P* $\subseteq B$

and *subset*: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and *bound*:
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength}) \leq C$
shows
 $\text{wp-event verify-monad } (\text{trace-fri-query-challenge-pair-set-hit } s \ P) \ s \leq C$
proof –
have *wp-event verify-monad*
 $(\text{trace-fri-query-challenge-pair-set-hit } s \ P) \ s \leq$
 $\text{wp-event verify-monad } (\text{trace-fri-challenge-list-set-hit } s \ B) \ s$
by (*rule wp-event-mono*)
 $(\text{rule trace-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit}$
 $[\text{OF - projection}])$
also have $\dots \leq \text{nnreal } (\text{card } B) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})$
by (*rule wp-verify-monad-trace-fri-challenge-list-set-bound*
 $[\text{OF future subset}])$
also have $\dots \leq C$
by (*rule bound*)
finally show *?thesis* .
qed

lemma *wp-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-projection*:
fixes $C :: \text{prob}$
assumes *future: composition-fri-future-fresh s*
and *projection*: $\bigwedge dg. \text{snd } 'P \ dg \subseteq B \ dg$
and *subset*:
 $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and *bound*:
 $\bigwedge dg. \text{nnreal } (\text{card } (B \ dg)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$
shows
 $\text{wp-event verify-monad } (\text{composition-fri-query-challenge-pair-set-hit } s \ P) \ s \leq C$
proof –
have *wp-event verify-monad*
 $(\text{composition-fri-query-challenge-pair-set-hit } s \ P) \ s \leq$
 $\text{wp-event verify-monad } (\text{composition-fri-challenge-list-set-hit } s \ B) \ s$
by (*rule wp-event-mono*)
 $(\text{rule}$
 $\text{composition-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit}$
 $[\text{OF - projection}])$
also have $\dots \leq C$
by (*rule wp-verify-monad-composition-fri-challenge-list-set-bound*
 $[\text{OF future subset bound}])$
finally show *?thesis* .
qed

definition *trace-fri-sampled-query-bad-candidate*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
 $\text{trace-fri-sampled-query-bad-candidate } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-query-candidate-evidence trace-table-low-degree}$
 $(\text{Not} \circ \text{trace-table-low-degree}) \text{ trace-table (clength} - 1)$
 $\text{trace-roots trace-bs trace-final fri-query-idxs trace-round-layers}$
 $\text{layers})$

definition *composition-fri-sampled-query-bad-candidate*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where

$\text{composition-fri-sampled-query-bad-candidate } s \text{ out} \longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{opening-composition-roots composition-bs composition-final}$
 $\text{fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{trace-query-idxs trace-openings composition-query-idxs}$
 $\text{composition-openings trace-table composition-table layers.}$
 $\text{composition-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{composition-query-idxs composition-openings trace-table}$
 $\text{composition-table} \wedge$
 $\text{generic-fri-sampled-query-candidate-evidence}$
 $(\text{composition-table-low-degree (to-nat fri-dg)})$
 $(\text{Not} \circ \text{composition-table-low-degree maxDegree}) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs composition-round-layers layers})$

definition *generic-fri-sampled-query-bad-pair-union*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow \text{nat} \Rightarrow$
 $(\text{nat list} \times 'f \text{ list}) \text{ set}$

where

$\text{generic-fri-sampled-query-bad-pair-union low-degree candidate-bad}$
 $\text{degree-bound} =$
 $\{ \text{pair. } \exists \text{ candidate-table roots final round-layers layers.}$

pair ∈
generic-fri-sampled-query-bad-pair-set *low-degree* *candidate-bad*
candidate-table *degree-bound*
roots *final* *round-layers* *layers*}

definition *generic-fri-sampled-query-projection-fraction-bound*

:: ('f list ⇒ bool) ⇒ ('f list ⇒ bool) ⇒ nat ⇒
prob ⇒ bool

where

generic-fri-sampled-query-projection-fraction-bound *low-degree*
candidate-bad *degree-bound* *C* $\longleftrightarrow$
nnreal
(card (snd ‘
(*generic-fri-sampled-query-bad-pair-union* *low-degree* *candidate-bad*
degree-bound $\cap$ (*fri-query-index-list-space* $\times$ *UNIV*)))) /
nnreal (*CARD*('f) $\wedge$ *fri-round-count-for-degree-bound* *degree-bound*)
 $\leq C$

lemma *generic-fri-sampled-query-projection-fraction-boundD*:

assumes

generic-fri-sampled-query-projection-fraction-bound *low-degree*
candidate-bad *degree-bound* *C*

shows

nnreal
(card (snd ‘
(*generic-fri-sampled-query-bad-pair-union* *low-degree* *candidate-bad*
degree-bound $\cap$ (*fri-query-index-list-space* $\times$ *UNIV*)))) /
nnreal (*CARD*('f) $\wedge$ *fri-round-count-for-degree-bound* *degree-bound*)
 $\leq C$

using *assms*

unfolding *generic-fri-sampled-query-projection-fraction-bound-def* .

lemma *generic-fri-sampled-query-projection-fraction-bound-mono*:

assumes

generic-fri-sampled-query-projection-fraction-bound *low-degree*
candidate-bad *degree-bound* *C*

and $C \leq D$

shows

generic-fri-sampled-query-projection-fraction-bound *low-degree*
candidate-bad *degree-bound* *D*

using *assms*

unfolding *generic-fri-sampled-query-projection-fraction-bound-def*

by (*rule* *order-trans*)

lemma *generic-fri-sampled-query-projection-fraction-bound-from-card*:

assumes *card-bound*:

card (snd ‘
(*generic-fri-sampled-query-bad-pair-union* *low-degree* *candidate-bad*
degree-bound $\cap$ (*fri-query-index-list-space* $\times$ *UNIV*))) $\leq K$

and *fraction-bound*:

$$\text{nnreal } K / \text{nnreal } (\text{CARD}('f) \wedge \text{fri-round-count-for-degree-bound degree-bound}) \leq C$$
shows
generic-fri-sampled-query-projection-fraction-bound low-degree candidate-bad degree-bound C
proof –
have *nnreal*

$$(\text{card } (\text{snd } ' (\text{generic-fri-sampled-query-bad-pair-union low-degree candidate-bad degree-bound} \cap (\text{fri-query-index-list-space} \times \text{UNIV})))) / \text{nnreal } (\text{CARD}('f) \wedge \text{fri-round-count-for-degree-bound degree-bound}) \leq \text{nnreal } K / \text{nnreal } (\text{CARD}('f) \wedge \text{fri-round-count-for-degree-bound degree-bound})$$
using *card-bound* **by** (*rule nnreal-nat-divide-right-mono*)
also have $\dots \leq C$
by (*rule fraction-bound*)
finally show *?thesis*
unfolding *generic-fri-sampled-query-projection-fraction-bound-def* .
qed

lemma *generic-fri-sampled-query-projection-fraction-bound-from-finite-cover*:
assumes *cover*:

$$\text{snd } ' (\text{generic-fri-sampled-query-bad-pair-union low-degree candidate-bad degree-bound} \cap (\text{fri-query-index-list-space} \times \text{UNIV})) \subseteq B$$
and *finite-B: finite B*
and *fraction-bound*:

$$\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{fri-round-count-for-degree-bound degree-bound}) \leq C$$
shows
generic-fri-sampled-query-projection-fraction-bound low-degree candidate-bad degree-bound C
proof (*rule generic-fri-sampled-query-projection-fraction-bound-from-card*)
show *card* ($\text{snd } ' (\text{generic-fri-sampled-query-bad-pair-union low-degree candidate-bad degree-bound} \cap (\text{fri-query-index-list-space} \times \text{UNIV}))$)

$$\leq \text{card } B$$
by (*rule card-mono[OF finite-B cover]*)
show *nnreal* ($\text{card } B$) /

$$\text{nnreal } (\text{CARD}('f) \wedge \text{fri-round-count-for-degree-bound degree-bound}) \leq C$$
by (*rule fraction-bound*)
qed

definition *trace-fri-sampled-query-bad-pair-union*

```

:: (nat list × 'f list) set
where
  trace-fri-sampled-query-bad-pair-union =
    generic-fri-sampled-query-bad-pair-union trace-table-low-degree
      (Not ∘ trace-table-low-degree) (clength - 1)

definition composition-fri-sampled-query-bad-pair-union
  :: 'f ⇒ (nat list × 'f list) set
where
  composition-fri-sampled-query-bad-pair-union fri-dg =
    generic-fri-sampled-query-bad-pair-union
      (composition-table-low-degree (to-nat fri-dg))
      (Not ∘ composition-table-low-degree maxDegree) (to-nat fri-dg)

lemma generic-fri-sampled-query-bad-pair-unionI:
  assumes
    (query-idxs, challenges) ∈
      generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
        candidate-table degree-bound roots final round-layers layers
  shows
    (query-idxs, challenges) ∈
      generic-fri-sampled-query-bad-pair-union low-degree candidate-bad
        degree-bound
  using assms
  unfolding generic-fri-sampled-query-bad-pair-union-def by blast

lemma generic-fri-sampled-query-bad-pair-union-subset-challenge-space:
  generic-fri-sampled-query-bad-pair-union low-degree candidate-bad
    degree-bound
  ⊆ (UNIV :: nat list set) ×
    fri-challenge-space (fri-round-count-for-degree-bound degree-bound)
  unfolding generic-fri-sampled-query-bad-pair-union-def
  by (auto intro: generic-fri-sampled-query-bad-pair-set-challenge-space)

lemma generic-fri-sampled-query-bad-pair-union-challenge-projection:
  snd ‘
    generic-fri-sampled-query-bad-pair-union low-degree candidate-bad
      degree-bound
  ⊆ fri-challenge-space
    (fri-round-count-for-degree-bound degree-bound)
  using generic-fri-sampled-query-bad-pair-union-subset-challenge-space
    [of low-degree candidate-bad degree-bound]
  by force

Fallback sanity bound: the restricted projection is always a subset of the full challenge
space. This lemma is intentionally not the main FRI low-degree theorem; the useful
symbolic bounds must come from a sharper sampled-query FRI argument.

lemma generic-fri-sampled-query-projection-fraction-bound-trivial:
  generic-fri-sampled-query-projection-fraction-bound low-degree

```

candidate-bad degree-bound 1
proof –
let $?S =$
snd ‘
(generic-fri-sampled-query-bad-pair-union low-degree candidate-bad
degree-bound $\cap$ (fri-query-index-list-space $\times$ UNIV))
let $?N = \text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound degree-bound}$
have *subset:*
 $?S \subseteq \text{fri-challenge-space}$
(fri-round-count-for-degree-bound degree-bound)
using *generic-fri-sampled-query-bad-pair-union-challenge-projection*
[of low-degree candidate-bad degree-bound]
by *blast*
have *card-le: card ?S $\leq$?N*
using *card-mono[OF finite-fri-challenge-space subset]*
by *(simp add: card-fri-challenge-space)*
have $?N \neq 0$
by *simp*
then have *nnreal ?N / nnreal ?N = (1::prob)*
by *(rule nnreal-nat-divide-self)*
moreover have *nnreal (card ?S) / nnreal ?N $\leq$ nnreal ?N / nnreal ?N*
using *card-le by (rule nnreal-nat-divide-right-mono)*
ultimately show *?thesis*
unfolding *generic-fri-sampled-query-projection-fraction-bound-def*
by *simp*
qed

lemma *trace-fri-sampled-query-bad-pair-union-challenge-projection:*
snd ‘ trace-fri-sampled-query-bad-pair-union $\subseteq$
fri-challenge-space (ceil-log clength)
unfolding *trace-fri-sampled-query-bad-pair-union-def*
using *generic-fri-sampled-query-bad-pair-union-challenge-projection*
[of trace-table-low-degree Not $\circ$ trace-table-low-degree
clength $- 1$]
by *(simp add: fri-round-count-for-degree-bound-def clength-pos)*

lemma *composition-fri-sampled-query-bad-pair-union-challenge-projection:*
snd ‘ composition-fri-sampled-query-bad-pair-union fri-dg $\subseteq$
fri-challenge-space (ceil-log (to-nat fri-dg + 1))
unfolding *composition-fri-sampled-query-bad-pair-union-def*
using *generic-fri-sampled-query-bad-pair-union-challenge-projection*
[of composition-table-low-degree (to-nat fri-dg)
Not $\circ$ composition-table-low-degree maxDegree to-nat fri-dg]
by *(simp add: fri-round-count-for-degree-bound-def)*

lemma *trace-fri-restricted-sampled-query-bad-pair-union-challenge-projection:*
snd ‘
(trace-fri-sampled-query-bad-pair-union $\cap$
(fri-query-index-list-space $\times$ UNIV))

$\subseteq$ *fri-challenge-space* (*ceil-log* *clength*)
using *trace-fri-sampled-query-bad-pair-union-challenge-projection* **by** *blast*

lemma *composition-fri-restricted-sampled-query-bad-pair-union-challenge-projection*:

snd ‘
 (*composition-fri-sampled-query-bad-pair-union* *fri-dg* $\cap$
 (*fri-query-index-list-space* $\times$ *UNIV*))
 $\subseteq$ *fri-challenge-space* (*ceil-log* (*to-nat* *fri-dg* + 1))
using *composition-fri-sampled-query-bad-pair-union-challenge-projection*
 [*of* *fri-dg*]
by *blast*

lemma *trace-fri-sampled-query-bad-candidate-iff-pair-set-hit*:

trace-fri-sampled-query-bad-candidate *s out* $\longleftrightarrow$
trace-fri-sampled-query-pair-set-hit
 (λ *trace-table* *roots* *final* *round-layers* *layers*.
generic-fri-sampled-query-bad-pair-set *trace-table-low-degree*
 (*Not* $\circ$ *trace-table-low-degree*) *trace-table* (*clength* - 1)
roots *final* *round-layers* *layers*)
s out
unfolding *trace-fri-sampled-query-bad-candidate-def*
trace-fri-sampled-query-pair-set-hit-def
generic-fri-sampled-query-bad-pair-set-def
by *blast*

lemma *composition-fri-sampled-query-bad-candidate-iff-pair-set-hit*:

composition-fri-sampled-query-bad-candidate *s out* $\longleftrightarrow$
composition-fri-sampled-query-pair-set-hit
 (λ *fri-dg* *composition-table* *roots* *final* *round-layers* *layers*.
generic-fri-sampled-query-bad-pair-set
 (*composition-table-low-degree* (*to-nat* *fri-dg*))
 (*Not* $\circ$ *composition-table-low-degree* *maxDegree*)
composition-table (*to-nat* *fri-dg*) *roots* *final* *round-layers* *layers*)
s out
unfolding *composition-fri-sampled-query-bad-candidate-def*
composition-fri-sampled-query-pair-set-hit-def
generic-fri-sampled-query-bad-pair-set-def
by *auto*

lemma *trace-fri-sampled-query-bad-candidate-imp-query-challenge-pair-union-hit*:

assumes *trace-fri-sampled-query-bad-candidate* *s out*
shows

trace-fri-query-challenge-pair-set-hit *s*
trace-fri-sampled-query-bad-pair-union *out*

proof –

have *hit*:
trace-fri-sampled-query-pair-set-hit
 (λ *trace-table* *roots* *final* *round-layers* *layers*.
generic-fri-sampled-query-bad-pair-set *trace-table-low-degree*

```

      (Not ◦ trace-table-low-degree) trace-table (clength - 1)
      roots final round-layers layers)
  s out
using assms trace-fri-sampled-query-bad-candidate-iff-pair-set-hit
by blast
show ?thesis
proof (rule trace-fri-sampled-query-pair-set-hit-imp-query-challenge-pair-set-hit
  [OF hit])
fix trace-table roots final round-layers layers
show
  generic-fri-sampled-query-bad-pair-set trace-table-low-degree
    (Not ◦ trace-table-low-degree) trace-table (clength - 1)
    roots final round-layers layers
  ⊆ trace-fri-sampled-query-bad-pair-union
unfolding trace-fri-sampled-query-bad-pair-union-def
by (auto intro: generic-fri-sampled-query-bad-pair-unionI)
qed
qed

lemma composition-fri-sampled-query-bad-candidate-imp-query-challenge-pair-union-hit:
assumes composition-fri-sampled-query-bad-candidate s out
shows
  composition-fri-query-challenge-pair-set-hit s
  composition-fri-sampled-query-bad-pair-union out
proof -
have hit:
  composition-fri-sampled-query-pair-set-hit
    (λfri-dg composition-table roots final round-layers layers.
      generic-fri-sampled-query-bad-pair-set
        (composition-table-low-degree (to-nat fri-dg))
        (Not ◦ composition-table-low-degree maxDegree)
        composition-table (to-nat fri-dg) roots final round-layers
        layers)
  s out
using assms composition-fri-sampled-query-bad-candidate-iff-pair-set-hit
by blast
show ?thesis
proof (rule
  composition-fri-sampled-query-pair-set-hit-imp-query-challenge-pair-set-hit
  [OF hit])
fix fri-dg composition-table roots final round-layers layers
show
  generic-fri-sampled-query-bad-pair-set
    (composition-table-low-degree (to-nat fri-dg))
    (Not ◦ composition-table-low-degree maxDegree)
    composition-table (to-nat fri-dg) roots final round-layers layers
  ⊆ composition-fri-sampled-query-bad-pair-union fri-dg
unfolding composition-fri-sampled-query-bad-pair-union-def
by (auto intro: generic-fri-sampled-query-bad-pair-unionI)

```

qed
qed

lemma *trace-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit:*
assumes *trace-fri-sampled-query-bad-candidate s out*
shows
trace-fri-query-challenge-pair-set-hit s
(trace-fri-sampled-query-bad-pair-union $\cap$
(fri-query-index-list-space $\times$ UNIV)) out
by (rule *trace-fri-query-challenge-pair-set-hit-restrict-to-query-space*)
(rule
trace-fri-sampled-query-bad-candidate-imp-query-challenge-pair-union-hit
[OF *assms*])

lemma *composition-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit:*
assumes *composition-fri-sampled-query-bad-candidate s out*
shows
composition-fri-query-challenge-pair-set-hit s
(*ldg. composition-fri-sampled-query-bad-pair-union dg $\cap$*
(fri-query-index-list-space $\times$ UNIV)) out
by (rule *composition-fri-query-challenge-pair-set-hit-restrict-to-query-space*)
(rule
composition-fri-sampled-query-bad-candidate-imp-query-challenge-pair-union-hit
[OF *assms*])

lemma *wp-trace-fri-sampled-query-bad-candidate-iff-pair-set-hit:*
wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s =
wp-event verify-monad
(trace-fri-sampled-query-pair-set-hit
(λ trace-table roots final round-layers layers.
generic-fri-sampled-query-bad-pair-set trace-table-low-degree
(Not $\circ$ trace-table-low-degree) trace-table (clength - 1)
roots final round-layers layers)
s) s

proof (rule *antisym*)
show *wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s*
 $\leq$ *wp-event verify-monad*
(trace-fri-sampled-query-pair-set-hit
(λ trace-table roots final round-layers layers.
generic-fri-sampled-query-bad-pair-set trace-table-low-degree
(Not $\circ$ trace-table-low-degree) trace-table (clength - 1)
roots final round-layers layers)
s) s
by (rule *wp-event-mono*)
(use *trace-fri-sampled-query-bad-candidate-iff-pair-set-hit* **in** *blast*)
show *wp-event verify-monad*
(trace-fri-sampled-query-pair-set-hit
(λ trace-table roots final round-layers layers.
generic-fri-sampled-query-bad-pair-set trace-table-low-degree

(Not $\circ$ trace-table-low-degree) trace-table (clength - 1)
 roots final round-layers layers)
 s) s
 $\leq$ wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s
by (rule wp-event-mono)
 (use trace-fri-sampled-query-bad-candidate-iff-pair-set-hit **in** blast)
qed

lemma wp-composition-fri-sampled-query-bad-candidate-iff-pair-set-hit:

wp-event verify-monad
 (composition-fri-sampled-query-bad-candidate s) s =
 wp-event verify-monad
 (composition-fri-sampled-query-pair-set-hit
 (λ fri-dg composition-table roots final round-layers layers.
 generic-fri-sampled-query-bad-pair-set
 (composition-table-low-degree (to-nat fri-dg))
 (Not $\circ$ composition-table-low-degree maxDegree)
 composition-table (to-nat fri-dg) roots final round-layers
 layers)
 s) s
proof (rule antisym)
show wp-event verify-monad
 (composition-fri-sampled-query-bad-candidate s) s
 $\leq$ wp-event verify-monad
 (composition-fri-sampled-query-pair-set-hit
 (λ fri-dg composition-table roots final round-layers layers.
 generic-fri-sampled-query-bad-pair-set
 (composition-table-low-degree (to-nat fri-dg))
 (Not $\circ$ composition-table-low-degree maxDegree)
 composition-table (to-nat fri-dg) roots final round-layers
 layers)
 s) s
by (rule wp-event-mono)
 (use composition-fri-sampled-query-bad-candidate-iff-pair-set-hit
in blast)
show wp-event verify-monad
 (composition-fri-sampled-query-pair-set-hit
 (λ fri-dg composition-table roots final round-layers layers.
 generic-fri-sampled-query-bad-pair-set
 (composition-table-low-degree (to-nat fri-dg))
 (Not $\circ$ composition-table-low-degree maxDegree)
 composition-table (to-nat fri-dg) roots final round-layers
 layers)
 s) s
 $\leq$ wp-event verify-monad
 (composition-fri-sampled-query-bad-candidate s) s
by (rule wp-event-mono)
 (use composition-fri-sampled-query-bad-candidate-iff-pair-set-hit
in blast)

qed

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-by-query-challenge-union-hit:*

wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s ≤

wp-event verify-monad

(trace-fri-query-challenge-pair-set-hit s
trace-fri-sampled-query-bad-pair-union) s

by *(rule wp-event-mono)*

(rule

trace-fri-sampled-query-bad-candidate-imp-query-challenge-pair-union-hit)

lemma *wp-composition-fri-sampled-query-bad-candidate-bound-by-query-challenge-union-hit:*

wp-event verify-monad (composition-fri-sampled-query-bad-candidate s) s ≤

wp-event verify-monad

(composition-fri-query-challenge-pair-set-hit s
composition-fri-sampled-query-bad-pair-union) s

by *(rule wp-event-mono)*

(rule

composition-fri-sampled-query-bad-candidate-imp-query-challenge-pair-union-hit)

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-by-restricted-query-challenge-union-hit:*

wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s ≤

wp-event verify-monad

(trace-fri-query-challenge-pair-set-hit s
(trace-fri-sampled-query-bad-pair-union ∩
(fri-query-index-list-space × UNIV))) s

by *(rule wp-event-mono)*

(rule

trace-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit)

lemma *wp-composition-fri-sampled-query-bad-candidate-bound-by-restricted-query-challenge-union-hit:*

wp-event verify-monad (composition-fri-sampled-query-bad-candidate s) s ≤

wp-event verify-monad

(composition-fri-query-challenge-pair-set-hit s
(λdg. composition-fri-sampled-query-bad-pair-union dg ∩
(fri-query-index-list-space × UNIV))) s

by *(rule wp-event-mono)*

(rule

composition-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit)

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-from-bad-pair-union-projection:*

assumes *future: trace-fri-future-fresh s*

and *bound:*

nnreal (card (snd ‘ trace-fri-sampled-query-bad-pair-union)) /
nnreal (CARD('f) ^ ceil-log clength) ≤ C

shows

wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s ≤ C

proof —

have *wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s*

$\leq$ *wp-event verify-monad*
 (*trace-fri-query-challenge-pair-set-hit s*
trace-fri-sampled-query-bad-pair-union) *s*
by (rule
wp-trace-fri-sampled-query-bad-candidate-bound-by-query-challenge-union-hit)
also have ... $\leq C$
by (rule *wp-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-projection*
 [*OF future subset-refl*
trace-fri-sampled-query-bad-pair-union-challenge-projection
bound])
finally show ?thesis .
qed

lemma *wp-composition-fri-sampled-query-bad-candidate-bound-from-bad-pair-union-projection*:
fixes *C* :: *prob*
assumes *future: composition-fri-future-fresh s*
and *bound*:
 $\bigwedge dg. \text{nnreal}$
 $(\text{card } (\text{snd } ' \text{ composition-fri-sampled-query-bad-pair-union } dg)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C$
shows
wp-event verify-monad (*composition-fri-sampled-query-bad-candidate s*) *s*
 $\leq C$
proof –
have *wp-event verify-monad*
 (*composition-fri-sampled-query-bad-candidate s*) *s* $\leq$
wp-event verify-monad
 (*composition-fri-query-challenge-pair-set-hit s*
composition-fri-sampled-query-bad-pair-union) *s*
by (rule
wp-composition-fri-sampled-query-bad-candidate-bound-by-query-challenge-union-hit)
also have ... $\leq C$
by (rule
wp-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-projection
 [*OF future subset-refl*
composition-fri-sampled-query-bad-pair-union-challenge-projection
bound])
finally show ?thesis .
qed

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-from-restricted-bad-pair-union-projection*:
assumes *future: trace-fri-future-fresh s*
and *bound*:
 nnreal
 $(\text{card } (\text{snd } ' ($
 $(\text{trace-fri-sampled-query-bad-pair-union } \cap$
 $(\text{fri-query-index-list-space } \times \text{ UNIV})))) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength}) \leq C$
shows

$wp\text{-event verify-monad } (trace\text{-fri-sampled-query-bad-candidate } s) \ s \leq C$
proof –
have $wp\text{-event verify-monad } (trace\text{-fri-sampled-query-bad-candidate } s) \ s$
 $\leq wp\text{-event verify-monad}$
 $(trace\text{-fri-query-challenge-pair-set-hit } s$
 $(trace\text{-fri-sampled-query-bad-pair-union } \cap$
 $(fri\text{-query-index-list-space } \times UNIV))) \ s$
by (rule
 $wp\text{-trace-fri-sampled-query-bad-candidate-bound-by-restricted-query-challenge-union-hit})$
also have $\dots \leq C$
by (rule $wp\text{-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-projection}$
 $[OF \text{ future subset-refl}$
 $trace\text{-fri-restricted-sampled-query-bad-pair-union-challenge-projection}$
 $bound])$
finally show ?thesis .
qed

lemma $wp\text{-composition-fri-sampled-query-bad-candidate-bound-from-restricted-bad-pair-union-projection}$:
fixes $C :: prob$
assumes $future: composition\text{-fri-future-fresh } s$
and $bound$:
 $\bigwedge dg. nnreal$
 $(card \ (snd \ ‘$
 $(composition\text{-fri-sampled-query-bad-pair-union } dg \cap$
 $(fri\text{-query-index-list-space } \times UNIV)))) \ /$
 $nnreal \ (CARD('f) \wedge \text{ceil-log } (to\text{-nat } dg + 1)) \leq C$
shows
 $wp\text{-event verify-monad } (composition\text{-fri-sampled-query-bad-candidate } s) \ s$
 $\leq C$
proof –
have $wp\text{-event verify-monad}$
 $(composition\text{-fri-sampled-query-bad-candidate } s) \ s \leq$
 $wp\text{-event verify-monad}$
 $(composition\text{-fri-query-challenge-pair-set-hit } s$
 $(\lambda dg. composition\text{-fri-sampled-query-bad-pair-union } dg \cap$
 $(fri\text{-query-index-list-space } \times UNIV))) \ s$
by (rule
 $wp\text{-composition-fri-sampled-query-bad-candidate-bound-by-restricted-query-challenge-union-hit})$
also have $\dots \leq C$
by (rule
 $wp\text{-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-projection}$
 $[OF \text{ future subset-refl}$
 $composition\text{-fri-restricted-sampled-query-bad-pair-union-challenge-projection}$
 $bound])$
finally show ?thesis .
qed

lemma $trace\text{-fri-sampled-query-bad-pair-union-generic-projection-fraction-bound}$:
assumes

generic-fri-sampled-query-projection-fraction-bound
trace-table-low-degree (*Not* $\circ$ *trace-table-low-degree*) (*clength* - 1) *C*
shows
nnreal
 (*card* (*snd* ‘
 (*trace-fri-sampled-query-bad-pair-union* $\cap$
 (*fri-query-index-list-space* $\times$ *UNIV*)))) /
nnreal (*CARD*(*f*) $\wedge$ *ceil-log* *clength*) $\leq$ *C*
using *assms*
unfolding *generic-fri-sampled-query-projection-fraction-bound-def*
trace-fri-sampled-query-bad-pair-union-def
by (*simp add: fri-round-count-for-degree-bound-def clength-pos*)

lemma *composition-fri-sampled-query-bad-pair-union-generic-projection-fraction-bound:*

assumes
generic-fri-sampled-query-projection-fraction-bound
 (*composition-table-low-degree* (*to-nat dg*))
 (*Not* $\circ$ *composition-table-low-degree maxDegree*) (*to-nat dg*) *C*
shows
nnreal
 (*card* (*snd* ‘
 (*composition-fri-sampled-query-bad-pair-union dg* $\cap$
 (*fri-query-index-list-space* $\times$ *UNIV*)))) /
nnreal (*CARD*(*f*) $\wedge$ *ceil-log* (*to-nat dg* + 1)) $\leq$ *C*
using *assms*
unfolding *generic-fri-sampled-query-projection-fraction-bound-def*
composition-fri-sampled-query-bad-pair-union-def
by (*simp add: fri-round-count-for-degree-bound-def*)

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction:*

assumes *future: trace-fri-future-fresh s*
and *generic-bound:*
generic-fri-sampled-query-projection-fraction-bound
trace-table-low-degree (*Not* $\circ$ *trace-table-low-degree*)
 (*clength* - 1) *C*
shows
wp-event verify-monad (*trace-fri-sampled-query-bad-candidate s*) *s* $\leq$ *C*
by (*rule*
wp-trace-fri-sampled-query-bad-candidate-bound-from-restricted-bad-pair-union-projection
 [*OF future*])
 (*rule*
trace-fri-sampled-query-bad-pair-union-generic-projection-fraction-bound
 [*OF generic-bound*])

lemma *wp-composition-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction:*

fixes *C :: prob*
assumes *future: composition-fri-future-fresh s*
and *generic-bound:*
 $\bigwedge dg.$ *generic-fri-sampled-query-projection-fraction-bound*

$(\text{composition-table-low-degree } (\text{to-nat } dg))$
 $(\text{Not } \circ \text{ composition-table-low-degree } \text{maxDegree}) (\text{to-nat } dg) \ C$
shows
 $\text{wp-event verify-monad } (\text{composition-fri-sampled-query-bad-candidate } s) \ s$
 $\leq C$
by (rule
 $\text{wp-composition-fri-sampled-query-bad-candidate-bound-from-restricted-bad-pair-union-projection}$
 $[OF \ \text{future}]$)
(rule
 $\text{composition-fri-sampled-query-bad-pair-union-generic-projection-fraction-bound}$
 $[OF \ \text{generic-bound}]$)

lemma $\text{trace-fri-sampled-query-bad-candidate-imp-query-index-list-space-hit}$:
assumes $\text{trace-fri-sampled-query-bad-candidate } s \ \text{out}$
shows
 $\text{trace-fri-query-index-list-set-hit } s \ \text{fri-query-index-list-space } \text{out}$
using $\text{assms trace-fri-sampled-query-bad-candidate-iff-pair-set-hit}$
 $\text{trace-fri-sampled-query-pair-set-hit-imp-query-index-list-space-hit}$
by blast

lemma $\text{composition-fri-sampled-query-bad-candidate-imp-query-index-list-space-hit}$:
assumes $\text{composition-fri-sampled-query-bad-candidate } s \ \text{out}$
shows
 $\text{composition-fri-query-index-list-set-hit } s \ \text{fri-query-index-list-space } \text{out}$
using $\text{assms composition-fri-sampled-query-bad-candidate-iff-pair-set-hit}$
 $\text{composition-fri-sampled-query-pair-set-hit-imp-query-index-list-space-hit}$
by blast

lemma $\text{wp-trace-fri-sampled-query-bad-candidate-bound-by-query-index-space-hit}$:
 $\text{wp-event verify-monad } (\text{trace-fri-sampled-query-bad-candidate } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-query-index-list-set-hit } s \ \text{fri-query-index-list-space}) \ s$
by (rule wp-event-mono)
(rule $\text{trace-fri-sampled-query-bad-candidate-imp-query-index-list-space-hit}$)

lemma $\text{wp-composition-fri-sampled-query-bad-candidate-bound-by-query-index-space-hit}$:
 $\text{wp-event verify-monad } (\text{composition-fri-sampled-query-bad-candidate } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-query-index-list-set-hit } s \ \text{fri-query-index-list-space}) \ s$
by (rule wp-event-mono)
(rule
 $\text{composition-fri-sampled-query-bad-candidate-imp-query-index-list-space-hit}$)

lemma $\text{trace-fri-sampled-query-bad-candidate-iff-canonical}$:
 $\text{trace-fri-sampled-query-bad-candidate } s \ \text{out} \longleftrightarrow$
 $\text{trace-fri-bad-with-canonical-sampled-layer-chain } s \ \text{out}$
proof
assume $\text{trace-fri-sampled-query-bad-candidate } s \ \text{out}$
then show $\text{trace-fri-bad-with-canonical-sampled-layer-chain } s \ \text{out}$

unfolding *trace-fri-sampled-query-bad-candidate-def*
trace-fri-bad-with-canonical-sampled-layer-chain-def
generic-fri-sampled-query-candidate-evidence-def
by *blast*
next
assume *canonical: trace-fri-bad-with-canonical-sampled-layer-chain s out*
then obtain *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table layers
where *evidence:*
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *sampled:*
generic-fri-canonical-sampled-layer-chain-evidence
trace-table-low-degree trace-table (clength - 1) trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers layers
unfolding *trace-fri-bad-with-canonical-sampled-layer-chain-def* **by** *blast*
have *not-low: (Not ◦ trace-table-low-degree) trace-table*
using *trace-fri-partial-candidate-opening-evidenceD(2)[OF evidence]*
by *(simp add: o-def trace-fri-partial-candidate-evidenceD(3))*
have *bad:*
generic-fri-sampled-query-candidate-evidence trace-table-low-degree
(Not ◦ trace-table-low-degree) trace-table (clength - 1)
trace-roots trace-bs trace-final fri-query-idxs trace-round-layers
layers
using *sampled not-low*
unfolding *generic-fri-sampled-query-candidate-evidence-def* **by** *blast*
show *trace-fri-sampled-query-bad-candidate s out*
unfolding *trace-fri-sampled-query-bad-candidate-def*
by *(intro exI conjI) (rule evidence, rule bad)*
qed

lemma *composition-fri-partial-candidate-opening-evidence-candidate-bad:*
assumes
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
shows *(Not ◦ composition-table-low-degree maxDegree) composition-table*
using *composition-fri-partial-candidate-opening-evidenceD(2)[OF assms]*
by *(simp add: o-def composition-fri-partial-candidate-evidenceD(5))*

lemma *composition-fri-sampled-query-bad-candidate-iff-canonical:*
composition-fri-sampled-query-bad-candidate s out $\longleftrightarrow$
composition-fri-bad-with-canonical-sampled-layer-chain s out

proof (rule *iffI*)
assume *composition-fri-sampled-query-bad-candidate s out*
then show *composition-fri-bad-with-canonical-sampled-layer-chain s out*
unfolding *composition-fri-sampled-query-bad-candidate-def*
composition-fri-bad-with-canonical-sampled-layer-chain-def
generic-fri-sampled-query-candidate-evidence-def
apply (elim *exE conjE*)
apply (intro *exI conjI*)
apply *assumption*
apply *assumption*
done

next
assume *composition-fri-bad-with-canonical-sampled-layer-chain s out*
then show *composition-fri-sampled-query-bad-candidate s out*
unfolding *composition-fri-sampled-query-bad-candidate-def*
composition-fri-bad-with-canonical-sampled-layer-chain-def
generic-fri-sampled-query-candidate-evidence-def
using *composition-fri-partial-candidate-opening-evidence-candidate-bad*
apply (elim *exE conjE*)
apply (intro *exI conjI*)
apply *assumption*
apply *assumption*
apply (rule *composition-fri-partial-candidate-opening-evidence-candidate-bad*)
apply *assumption*
done

qed

lemma *wp-trace-fri-sampled-query-bad-candidate-iff-canonical:*
wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s =
wp-event verify-monad (trace-fri-bad-with-canonical-sampled-layer-chain s)
s

proof (rule *antisym*)
show *wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s*
 $\leq$ *wp-event verify-monad*
(trace-fri-bad-with-canonical-sampled-layer-chain s) s
by (rule *wp-event-mono*)
(use trace-fri-sampled-query-bad-candidate-iff-canonical in blast)
show *wp-event verify-monad*
(trace-fri-bad-with-canonical-sampled-layer-chain s) s
 $\leq$ *wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s*
by (rule *wp-event-mono*)
(use trace-fri-sampled-query-bad-candidate-iff-canonical in blast)

qed

lemma *wp-composition-fri-sampled-query-bad-candidate-iff-canonical:*
wp-event verify-monad

```

      (composition-fri-sampled-query-bad-candidate s) s =
wp-event verify-monad
      (composition-fri-bad-with-canonical-sampled-layer-chain s) s
proof (rule antisym)
show wp-event verify-monad
      (composition-fri-sampled-query-bad-candidate s) s
≤ wp-event verify-monad
      (composition-fri-bad-with-canonical-sampled-layer-chain s) s
by (rule wp-event-mono)
      (use composition-fri-sampled-query-bad-candidate-iff-canonical in blast)
show wp-event verify-monad
      (composition-fri-bad-with-canonical-sampled-layer-chain s) s
≤ wp-event verify-monad
      (composition-fri-sampled-query-bad-candidate s) s
by (rule wp-event-mono)
      (use composition-fri-sampled-query-bad-candidate-iff-canonical in blast)
qed

```

The following bounds expose the current internal sampled-FRI interface: a sampled layer-chain cover plus a bad-set envelope implies a challenge-list hit, which is then bounded by the existing Fiat-Shamir challenge-list calculus. The cover and envelope parameters are internal proof-layer interfaces; they are not public protocol assumptions.

```

lemma trace-fri-sampled-query-bad-candidate-imp-list-hit:
assumes sampled: trace-fri-sampled-query-bad-candidate s out
and cover: trace-fri-sampled-layer-chain-cover s bad
and envelope:
   $\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$ 
shows trace-fri-challenge-list-set-hit s B out
proof –
have canonical: trace-fri-bad-with-canonical-sampled-layer-chain s out
using sampled trace-fri-sampled-query-bad-candidate-iff-canonical
by blast
then have sampled-chain: trace-fri-bad-with-sampled-layer-chain s out
by (rule trace-fri-bad-with-canonical-sampled-layer-chain-imp-sampled)
show ?thesis
by (rule trace-fri-bad-with-sampled-layer-chain-imp-list-hit
  [OF sampled-chain cover envelope])
qed

```

```

lemma composition-fri-sampled-query-bad-candidate-imp-list-hit:
assumes sampled: composition-fri-sampled-query-bad-candidate s out
and cover: composition-fri-sampled-layer-chain-cover s bad
and envelope:
   $\bigwedge dg \text{ composition-table. } \text{composition-fri-multiround-bad-sets bad dg composition-table} \subseteq B dg$ 
shows composition-fri-challenge-list-set-hit s B out
proof –
have canonical:

```

composition-fri-bad-with-canonical-sampled-layer-chain s out
using *sampled composition-fri-sampled-query-bad-candidate-iff-canonical*
by *blast*
then have *sampled-chain:*
composition-fri-bad-with-sampled-layer-chain s out
by (rule *composition-fri-bad-with-canonical-sampled-layer-chain-imp-sampled*)
show *?thesis*
by (rule *composition-fri-bad-with-sampled-layer-chain-imp-list-hit*
[OF sampled-chain cover envelope])
qed

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-from-envelope:*
assumes *future: trace-fri-future-fresh s*
and *cover: trace-fri-sampled-layer-chain-cover s bad*
and *envelope:*
 $\bigwedge \text{trace-table. trace-fri-multiround-bad-sets bad trace-table} \subseteq B$
and *subset: $B \subseteq \text{fri-challenge-space (ceil-log clength)}$*
and *bound:*
 $\text{nnreal (card B) / nnreal (CARD('f) } ^{\wedge} \text{ceil-log clength)} \leq C$
shows
 $\text{wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s} \leq C$
proof –
have $\text{wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s}$
 $\leq \text{wp-event verify-monad (trace-fri-challenge-list-set-hit s B) s}$
by (rule *wp-event-mono*)
(rule trace-fri-sampled-query-bad-candidate-imp-list-hit
[OF - cover envelope])
also have $\dots \leq \text{nnreal (card B) /}$
 $\text{nnreal (CARD('f) } ^{\wedge} \text{ceil-log clength)}$
by (rule *wp-verify-monad-trace-fri-challenge-list-set-bound*
[OF future subset])
also have $\dots \leq C$
by (rule *bound*)
finally show *?thesis .*
qed

lemma *wp-composition-fri-sampled-query-bad-candidate-bound-from-envelope:*
assumes *future: composition-fri-future-fresh s*
and *cover: composition-fri-sampled-layer-chain-cover s bad*
and *envelope:*
 $\bigwedge dg \text{ composition-table.}$
 $\text{composition-fri-multiround-bad-sets bad dg composition-table}$
 $\subseteq B dg$
and *subset:*
 $\bigwedge dg. B dg \subseteq \text{fri-challenge-space (ceil-log (to-nat dg + 1))}$
and *bound:*
 $\bigwedge dg. \text{nnreal (card (B dg)) /}$
 $\text{nnreal (CARD('f) } ^{\wedge} \text{ceil-log (to-nat dg + 1))} \leq C$
shows

$wp\text{-}event\ verify\text{-}monad\ (composition\text{-}fri\text{-}sampled\text{-}query\text{-}bad\text{-}candidate\ s)\ s$
 $\leq C$
proof –
have $wp\text{-}event\ verify\text{-}monad$
 $(composition\text{-}fri\text{-}sampled\text{-}query\text{-}bad\text{-}candidate\ s)\ s \leq$
 $wp\text{-}event\ verify\text{-}monad\ (composition\text{-}fri\text{-}challenge\text{-}list\text{-}set\text{-}hit\ s\ B)\ s$
by $(rule\ wp\text{-}event\text{-}mono)$
 $(rule\ composition\text{-}fri\text{-}sampled\text{-}query\text{-}bad\text{-}candidate\text{-}imp\text{-}list\text{-}hit$
 $[OF\text{-}cover\ envelope])$
also have $\dots \leq C$
by $(rule\ wp\text{-}verify\text{-}monad\text{-}composition\text{-}fri\text{-}challenge\text{-}list\text{-}set\text{-}bound$
 $[OF\ future\ subset\ bound])$
finally show $?thesis$.
qed

lemma *checked-staged-security-trace-fri-sampled-query-bad-candidate-bound:*

assumes wf : *staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *finite-B: finite B*

and *cover:*

$\wedge data\ attacker\text{-}state.$

$Some\ (data,\ attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (checked\text{-}staged\text{-}transcript\text{-}program\ A)$
 $adversary\text{-}initial\text{-}state) \implies$
 $trace\text{-}fri\text{-}sampled\text{-}layer\text{-}chain\text{-}cover$
 $(verifier\text{-}state\text{-}from\text{-}adversary\ attacker\text{-}state$
 $(staged\text{-}proof\text{-}transcript\ data))\ bad$

and *envelope:*

$\wedge data\ attacker\text{-}state\ trace\text{-}table.$

$Some\ (data,\ attacker\text{-}state) \in$
 $set\text{-}dist\ (execute\ (checked\text{-}staged\text{-}transcript\text{-}program\ A)$
 $adversary\text{-}initial\text{-}state) \implies$
 $trace\text{-}fri\text{-}multi\text{-}round\text{-}bad\text{-}sets\ bad\ trace\text{-}table \subseteq B$

and *fresh-bound:*

$wp\text{-}event\ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $(\lambda s.\ trace\text{-}fri\text{-}challenge\text{-}list\text{-}fresh\text{-}hit\ s\ B))$
 $adversary\text{-}initial\text{-}state \leq F$

shows

$wp\text{-}event\ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $trace\text{-}fri\text{-}sampled\text{-}query\text{-}bad\text{-}candidate)$
 $adversary\text{-}initial\text{-}state \leq$
 $F +$
 $hash\text{-}relation\text{-}budget\text{-}value\ (card\ B * ceil\text{-}log\ clength)$
 $(staged\text{-}attacker\text{-}query\text{-}budget\ budgets + staged\text{-}challenge\text{-}query\text{-}budget)$

proof –

have $wp\text{-}event\ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state\ A)$
 $(staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$

```

    trace-fri-sampled-query-bad-candidate)
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-challenge-list-set-hit s B))
  adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  and hit:
    staged-security-with-data-state-verifier-event
    trace-fri-sampled-query-bad-candidate out
from hit obtain data attacker-state result final-state where
  out-eq: out = Some (((data, attacker-state), result), final-state)
  unfolding staged-security-with-data-state-verifier-event-def
  by (cases out) (auto split: option.splits prod.splits)
have builder:
  Some (data, attacker-state) ∈
  set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state)
  using checked-staged-security-experiment-with-data-state-outcomeE
  [OF support[unfolded out-eq]]
  by blast
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
have sampled:
  trace-fri-sampled-query-bad-candidate ?s
  (Some (result, final-state))
  using hit unfolding staged-security-with-data-state-verifier-event-def
  out-eq by simp
have list-hit:
  trace-fri-challenge-list-set-hit ?s B (Some (result, final-state))
  by (rule trace-fri-sampled-query-bad-candidate-imp-list-hit
    [OF sampled cover[OF builder]])
  (rule envelope[OF builder])
show
  staged-security-with-data-state-verifier-event
  (λs. trace-fri-challenge-list-set-hit s B) out
  unfolding staged-security-with-data-state-verifier-event-def out-eq
  using list-hit by simp
qed
also have ... ≤
  F +
  hash-relation-budget-value (card B * ceil-log clength)

```

(staged-attacker-query-budget budgets + staged-challenge-query-budget)
 by (rule checked-staged-security-trace-fri-challenge-list-set-hit-bound
 [OF wf controlled finite-B fresh-bound])
 finally show ?thesis .
 qed

lemma checked-staged-security-composition-fri-sampled-query-bad-candidate-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: $\bigwedge dg. \text{finite } (B \ dg)$
and cover:
 $\bigwedge \text{data attacker-state.}$
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 composition-fri-sampled-layer-chain-cover
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) bad
and envelope:
 $\bigwedge \text{data attacker-state dg composition-table.}$
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 composition-fri-multiround-bad-sets bad dg composition-table
 $\subseteq B \ dg$
and fresh-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B$))
 adversary-initial-state $\leq F$
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-sampled-query-bad-candidate)
 adversary-initial-state $\leq$
 F +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets + staged-challenge-query-budget)

proof –
have wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-sampled-query-bad-candidate)
 adversary-initial-state $\leq$
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s. \text{composition-fri-challenge-list-set-hit } s \ B$))
 adversary-initial-state


```

proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  and hit:
    staged-security-with-data-state-verifier-event
      composition-fri-sampled-query-bad-candidate out
  from hit obtain data attacker-state result final-state where
    out-eq: out = Some (((data, attacker-state), result), final-state)
  unfolding staged-security-with-data-state-verifier-event-def
  by (cases out) (auto split: option.splits prod.splits)
  have builder:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  using checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
  by blast
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have sampled:
    composition-fri-sampled-query-bad-candidate ?s
      (Some (result, final-state))
  using hit unfolding staged-security-with-data-state-verifier-event-def
    out-eq by simp
  have list-hit:
    composition-fri-challenge-list-set-hit ?s B
      (Some (result, final-state))
  by (rule composition-fri-sampled-query-bad-candidate-imp-list-hit
    [OF sampled cover[OF builder]])
    (rule envelope[OF builder])
  show
    staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-set-hit s B) out
  unfolding staged-security-with-data-state-verifier-event-def out-eq
  using list-hit by simp
qed
also have ... ≤
  F +
  hash-relation-budget-value
    (∑ dg ∈ (UNIV :: 'f set).
      card (B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule checked-staged-security-composition-fri-challenge-list-set-hit-bound
    [OF wf controlled finite-B fresh-bound])

```

finally show ?thesis .
qed

lemma *active-fri-reductions-for-from-trace-augmented-composition-sampled-query-and-empty-conflict-zero:*

fixes *TraceAug TraceMissing TraceMerkle CompSampled CompMerkle EmptySampled Empty-Conflict EmptyZero ::*

'f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ prob

assumes *trace-augmented-bound:*

$\wedge$ *data attacker-state.*

*Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-bad-with-header-tied-augmented-partial-candidate
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$ TraceAug data attacker-state*

and *trace-missing-bound:*

$\wedge$ *data attacker-state.*

*Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(trace-fri-header-tied-recorded-sibling-candidate-missing
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$
TraceMissing data attacker-state*

and *trace-merkle-bound:*

$\wedge$ *data attacker-state.*

*Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$ TraceMerkle data attacker-state*

and *trace-total:*

$\wedge$ *data attacker-state.*

*Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
TraceAug data attacker-state +
(TraceMissing data attacker-state + TraceMerkle data attacker-state)
 $\leq$ trace-fri-error*

and *composition-sampled-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-sampled-query-bad-candidate*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
CompSampled data attacker-state
and *composition-merkle-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*partial-merkle-inconsistency-bad*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$ *CompMerkle data attacker-state*
and *composition-total*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
CompSampled data attacker-state +
(((*CompMerkle data attacker-state* + 0) +
(*CompMerkle data attacker-state* + 0) +
(*CompMerkle data attacker-state* + 0) +
(*CompMerkle data attacker-state* + 0 + 0) +
CompMerkle data attacker-state) + 0)
 $\leq$ *composition-fri-error*
and *empty-sampled-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-sampled-query-bad-candidate*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
EmptySampled data attacker-state
and *empty-conflict-bound*:
 $\wedge$ *data attacker-state*.

```

Some (data, attacker-state) ∈
  set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state) ⇒
wp-event verify-monad
  (trace-fri-sampled-assignment-conflict
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  EmptyConflict data attacker-state
and empty-zero-bound:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
wp-event verify-monad
  (trace-fri-zero-round-final-obstruction
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤
  EmptyZero data attacker-state
and empty-total:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  EmptySampled data attacker-state +
    (EmptyConflict data attacker-state + EmptyZero data attacker-state)
  ≤ trace-fri-error
shows active-fri-reductions-for A
proof (rule
  active-fri-reductions-for-from-trace-augmented-composition-canonical-and-empty-conflict-zero
  [OF trace-augmented-bound trace-missing-bound trace-merkle-bound
    trace-total - composition-merkle-bound composition-total -
    empty-conflict-bound empty-zero-bound empty-total])
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (composition-fri-bad-with-canonical-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  ≤ CompSampled data attacker-state
using composition-sampled-bound[OF support]

```

```

    by (simp add: wp-composition-fri-sampled-query-bad-candidate-iff-canonical)
next
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (trace-fri-bad-with-canonical-sampled-layer-chain
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  ≤ EmptySampled data attacker-state
using empty-sampled-bound[OF support]
by (simp add: wp-trace-fri-sampled-query-bad-candidate-iff-canonical)
qed

```

definition *generic-fri-sampled-query-finite-cover*
 $:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow \text{nat} \Rightarrow$
 $'f \text{ list set} \Rightarrow \text{bool}$

where

generic-fri-sampled-query-finite-cover low-degree candidate-bad
 $\text{degree-bound } B \longleftrightarrow$
 $\text{finite } B \wedge$
 snd '
 $(\text{generic-fri-sampled-query-bad-pair-union low-degree candidate-bad}$
 $\text{degree-bound} \cap (\text{fri-query-index-list-space} \times \text{UNIV})) \subseteq B$

lemma *generic-fri-sampled-query-finite-coverD:*

assumes

generic-fri-sampled-query-finite-cover low-degree candidate-bad
 $\text{degree-bound } B$

shows

$\text{finite } B$
 snd '
 $(\text{generic-fri-sampled-query-bad-pair-union low-degree candidate-bad}$
 $\text{degree-bound} \cap (\text{fri-query-index-list-space} \times \text{UNIV})) \subseteq B$

using *assms*

unfolding *generic-fri-sampled-query-finite-cover-def* **by** *simp-all*

lemma *generic-fri-sampled-query-finite-cover-from-candidate-challenge-cover:*

assumes *finite-B: finite B*

and *cover:*

$\bigwedge \text{candidate-table roots final-value round-layers layers query-idxs}$
 challenges.

$\text{query-idxs} \in \text{fri-query-index-list-space} \implies$

generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
 $\text{candidate-table degree-bound roots challenges final-value query-idxs}$

$\text{round-layers layers} \implies$
 $\text{challenges} \in B$
shows
 $\text{generic-fri-sampled-query-finite-cover low-degree candidate-bad}$
 $\text{degree-bound } B$
proof –
have *subset*:
 snd '
 $(\text{generic-fri-sampled-query-bad-pair-union low-degree candidate-bad}$
 $\text{degree-bound} \cap (\text{fri-query-index-list-space} \times \text{UNIV})) \subseteq B$
proof
fix *challenges*
assume $\text{challenges} \in \text{snd '}$
 $(\text{generic-fri-sampled-query-bad-pair-union low-degree candidate-bad}$
 $\text{degree-bound} \cap (\text{fri-query-index-list-space} \times \text{UNIV}))$
then obtain pair where pair-mem:
 $\text{pair} \in \text{generic-fri-sampled-query-bad-pair-union low-degree}$
 $\text{candidate-bad degree-bound} \cap (\text{fri-query-index-list-space} \times \text{UNIV})$
and *challenges-eq*: $\text{challenges} = \text{snd pair}$
by *blast*
obtain query-idxs challenges' where pair-eq: $\text{pair} = (\text{query-idxs}, \text{challenges'})$
by (*cases pair*)
from *pair-mem pair-eq challenges-eq*
obtain *candidate-table roots final-value round-layers layers*
where *query-space*: $\text{query-idxs} \in \text{fri-query-index-list-space}$
and *pair*:
 $(\text{query-idxs}, \text{challenges}) \in$
 $\text{generic-fri-sampled-query-bad-pair-set low-degree candidate-bad}$
 $\text{candidate-table degree-bound roots final-value round-layers layers}$
unfolding *generic-fri-sampled-query-bad-pair-union-def* **by** *auto*
have *evidence*:
 $\text{generic-fri-sampled-query-candidate-evidence low-degree candidate-bad}$
 $\text{candidate-table degree-bound roots challenges final-value query-idxs}$
 $\text{round-layers layers}$
by (*rule generic-fri-sampled-query-bad-pair-setD[OF pair]*)
show $\text{challenges} \in B$
by (*rule cover[OF query-space evidence]*)
qed
show *?thesis*
unfolding *generic-fri-sampled-query-finite-cover-def*
using *finite-B subset* **by** *simp*
qed

lemma *generic-fri-sampled-query-projection-fraction-bound-from-cover*:
assumes *cover*:
 $\text{generic-fri-sampled-query-finite-cover low-degree candidate-bad}$
 $\text{degree-bound } B$
and *fraction-bound*:
 $\text{nnreal } (\text{card } B) /$

$$\text{nnreal } (\text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound degree-bound}) \leq C$$

shows

generic-fri-sampled-query-projection-fraction-bound low-degree
candidate-bad degree-bound C

by (rule *generic-fri-sampled-query-projection-fraction-bound-from-finite-cover*)
(rule *generic-fri-sampled-query-finite-coverD(2)[OF cover]*,
rule *generic-fri-sampled-query-finite-coverD(1)[OF cover]*,
rule *fraction-bound*)

lemma *trace-fri-sampled-query-bad-candidate-bound-from-generic-cover:*

assumes *future: trace-fri-future-fresh s*

and *cover:*

generic-fri-sampled-query-finite-cover trace-table-low-degree
(Not o trace-table-low-degree) (clength - 1) B

and *fraction-bound:*

$$\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log clength}) \leq C$$

shows

wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s ≤ C

proof (rule
wp-trace-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction
[OF future])

show

generic-fri-sampled-query-projection-fraction-bound
trace-table-low-degree (Not o trace-table-low-degree)
(clength - 1) C

by (rule *generic-fri-sampled-query-projection-fraction-bound-from-cover*
[OF cover])
(use *fraction-bound* **in**
 $\langle \text{simp add: fri-round-count-for-degree-bound-def clength-pos} \rangle$)

qed

lemma *composition-fri-sampled-query-bad-candidate-bound-from-generic-cover:*

fixes *C :: prob*

assumes *future: composition-fri-future-fresh s*

and *cover:*

$\bigwedge dg. \text{generic-fri-sampled-query-finite-cover}$
(composition-table-low-degree (to-nat dg))
(Not o composition-table-low-degree maxDegree) (to-nat dg) (B dg)

and *fraction-bound:*

$\bigwedge dg. \text{nnreal } (\text{card } (B dg)) /$
 $\text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log } (\text{to-nat dg} + 1)) \leq C$

shows

wp-event verify-monad (composition-fri-sampled-query-bad-candidate s) s
 $\leq C$

proof (rule
wp-composition-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction
[OF future])

fix *dg*

```

show
  generic-fri-sampled-query-projection-fraction-bound
    (composition-table-low-degree (to-nat dg))
    (Not  $\circ$  composition-table-low-degree maxDegree) (to-nat dg) C
by (rule generic-fri-sampled-query-projection-fraction-bound-from-cover
      [OF cover[of dg]])
    (use fraction-bound[of dg] in
       $\langle$ simp add: fri-round-count-for-degree-bound-def $\rangle$ )
qed

end

end

theory Soundness-FRI-Query-Aware-Interface
imports Soundness-FRI-Sampled-Interface
begin

Query-aware sampled FRI interface.

This theory names the joint query/challenge boundary used by the sampled FRI low-
degree reduction. The definitions are generic and are instantiated by the trace and
composition FRI routes.

context soundness
begin

definition generic-fri-sampled-query-restricted-bad-pairs
  :: ('f list  $\Rightarrow$  bool)  $\Rightarrow$  ('f list  $\Rightarrow$  bool)  $\Rightarrow$  nat  $\Rightarrow$ 
    (nat list  $\times$  'f list) set
where
  generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad
    degree-bound =
  generic-fri-sampled-query-bad-pair-union low-degree candidate-bad
    degree-bound  $\cap$  (fri-query-index-list-space  $\times$  UNIV)

definition generic-fri-sampled-query-pair-fraction-bound
  :: ('f list  $\Rightarrow$  bool)  $\Rightarrow$  ('f list  $\Rightarrow$  bool)  $\Rightarrow$  nat  $\Rightarrow$ 
    prob  $\Rightarrow$  bool
where
  generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad
    degree-bound C  $\longleftrightarrow$ 
    nnreal
    (card
      (generic-fri-sampled-query-restricted-bad-pairs low-degree
        candidate-bad degree-bound)) /
    nnreal
    (card fri-query-index-list-space *
      CARD('f)  $\wedge$  fri-round-count-for-degree-bound degree-bound)
   $\leq$  C

```


definition *generic-fri-sampled-query-challenge-projection*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow \text{nat} \Rightarrow$
 $\quad 'f \text{ list set}$

where

generic-fri-sampled-query-challenge-projection low-degree candidate-bad
degree-bound =
snd ‘
generic-fri-sampled-query-restricted-bad-pairs low-degree
candidate-bad degree-bound

definition *generic-fri-sampled-query-query-fiber*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow \text{nat} \Rightarrow$
 $\quad 'f \text{ list} \Rightarrow \text{nat list set}$

where

generic-fri-sampled-query-query-fiber low-degree candidate-bad
degree-bound challenges =
fri-query-challenge-pair-query-fiber
(generic-fri-sampled-query-restricted-bad-pairs low-degree
candidate-bad degree-bound) challenges

definition *query-rounds-index-list-hit*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{nat list} \Rightarrow \text{nat} \Rightarrow$
 $\quad (\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

query-rounds-index-list-hit s query-idxs n out $\longleftrightarrow$
 $(\exists \text{ results } t \text{ raw-idxs.}$
 $\quad \text{out} = \text{Some } (\text{results}, t) \wedge$
 $\quad \text{length raw-idxs} = n \wedge$
 $\quad \text{length query-idxs} = n \wedge$
 $\quad \text{query-idxs} = \text{map } (\lambda \text{ raw. index } (\text{to-nat raw})) \text{ raw-idxs} \wedge$
 $\quad (\forall i < n. \exists x.$
 $\quad \quad \text{fmlookup } (\text{HashMap } t)$
 $\quad \quad (\text{QueryIndexChallenge } (\text{PQueryCounter } s + i) x) =$
 $\quad \quad \text{Some } (\text{raw-idxs } ! i)))$

lemma *card-fri-query-index-list-space:*

card fri-query-index-list-space = *card query-sample-space* $\wedge$ *rounds*
unfolding *fri-query-index-list-space-def*
using *card-lists-length-eq[OF finite-query-sample-space, of rounds]*
by (*simp add: conj-commute*)

lemma *query-rounds-index-list-hit-None[simp]:*

$\neg \text{query-rounds-index-list-hit } s \text{ query-idxs } n \text{ None}$
unfolding *query-rounds-index-list-hit-def* **by** *simp*

lemma *query-rounds-index-list-hit-tail:*

assumes *head:*
 $\text{Some } ((), s1) \in$

```

    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s1)
  and hit:
    query-rounds-index-list-hit s query-idxs (Suc n)
      (Some (u # results, t))
  shows
    query-rounds-index-list-hit s1 (tl query-idxs) n
      (Some (results, t))
  proof -
  from hit obtain raw-idxs where
    len-raw: length raw-idxs = Suc n
  and len-query: length query-idxs = Suc n
  and query-eq: query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and lookups:
    ∧ i. i < Suc n ⇒ ∃ x.
      fmllookup (HashMap t)
        (QueryIndexChallenge (PQueryCounter s + i) x) =
        Some (raw-idxs ! i)
  unfolding query-rounds-index-list-hit-def by auto
  have counter-s1: PQueryCounter s1 = Suc (PQueryCounter s)
  using verifier-query-round-program-outcome[OF head] by blast
  let ?raw-tail = tl raw-idxs
  have len-tail-raw: length ?raw-tail = n
  using len-raw by simp
  have len-tail-query: length (tl query-idxs) = n
  using len-query by simp
  have tail-eq:
    tl query-idxs = map (λraw. index (to-nat raw)) ?raw-tail
  using query-eq by (cases raw-idxs; cases query-idxs; simp)
  have tail-lookups:
    ∀ i < n. ∃ x.
      fmllookup (HashMap t)
        (QueryIndexChallenge (PQueryCounter s1 + i) x) =
        Some (?raw-tail ! i)
  proof (intro allI impI)
  fix i
  assume i-bound: i < n
  then obtain x where lookup:
    fmllookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + Suc i) x) =
      Some (raw-idxs ! Suc i)

```

```

    using lookups[of Suc i] by auto
  have counter-eq: PQueryCounter s1 + i = PQueryCounter s + Suc i
    using counter-s1 by presburger
  have tail-raw: ?raw-tail ! i = raw-idxs ! Suc i
    using i-bound len-raw by (cases raw-idxs) auto
  show  $\exists x$ .
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s1 + i) x) =
        Some (?raw-tail ! i)
    by (rule exI[where x=x])
      (use lookup counter-eq tail-raw in simp)
qed
show ?thesis
  unfolding query-rounds-index-list-hit-def
  by (intro exI[of - results] exI[of - t] exI[of - ?raw-tail] conjI)
    (use len-tail-raw len-tail-query tail-eq tail-lookups in simp-all)
qed

lemma query-rounds-index-list-hit-head-index:
  assumes future: query-future-fresh s
  and head:
    Some ((), s1)  $\in$ 
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results, t)  $\in$ 
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s1)
  and hit:
    query-rounds-index-list-hit s query-idxs (Suc n)
      (Some (u # results, t))
  shows
    query-round-any-index-set-hit s {query-idxs ! 0} (Some ((), s1))
proof -
  from head[unfolded verifier-query-round-program-alt-def]
  obtain raw s0 where
    recv: Some (raw, s0)  $\in$  set-dist (execute receive-query-index-challenge s)
  and after:
    Some ((), s1)  $\in$ 
      set-dist
        (execute
          (verifier-query-round-after-index-program fr f-fl f-final as fl
            final raw) s0)
  by (auto elim!: set-dist-bindE)
  from hit obtain raw-idxs x where

```

```

len-raw: length raw-idxs = Suc n
and len-query: length query-idxs = Suc n
and query-eq: query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and lookup-t:
  fmlookup (HashMap t)
  (QueryIndexChallenge (PQueryCounter s) x) = Some (raw-idxs ! 0)
unfolding query-rounds-index-list-hit-def by auto
have counter-s1: PQueryCounter s1 = Suc (PQueryCounter s)
  using verifier-query-round-program-outcome[OF head] by blast
have past: PQueryCounter s < PQueryCounter s1
  using counter-s1 by simp
have lookup-s1:
  fmlookup (HashMap s1) (QueryIndexChallenge (PQueryCounter s) x) =
    Some (raw-idxs ! 0)
  using ntimes-verifier-query-round-program-preserves-past-query-lookup
    [OF past tail, of x] lookup-t
  by simp
have lookup-s0-x:
  fmlookup (HashMap s0) (QueryIndexChallenge (PQueryCounter s) x) =
    Some (raw-idxs ! 0)
  using verifier-query-round-after-index-program-preserves-query-lookup
    [OF after, of PQueryCounter s x] lookup-s1
  by simp
have lookup-s0-current:
  fmlookup (HashMap s0)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  using receive-query-index-challenge-outcome[OF recv] by blast
have key-eq:
  QueryIndexChallenge (PQueryCounter s) x =
    QueryIndexChallenge (PQueryCounter s) (PState s)
proof (rule ccontr)
  assume neg:
    QueryIndexChallenge (PQueryCounter s) x  $\neq$ 
      QueryIndexChallenge (PQueryCounter s) (PState s)
  have lookup-old:
    fmlookup (HashMap s0) (QueryIndexChallenge (PQueryCounter s) x) =
      fmlookup (HashMap s) (QueryIndexChallenge (PQueryCounter s) x)
    by (rule receive-query-index-challenge-preserves-other-lookup
      [OF recv neg])
  have fmlookup (HashMap s) (QueryIndexChallenge (PQueryCounter s) x) =
    None
    using future unfolding query-future-fresh-def by simp
  then show False
    using lookup-s0-x lookup-old by simp
qed
have raw-eq: raw-idxs ! 0 = raw
  using lookup-s0-x lookup-s0-current key-eq by simp
have index (to-nat raw) = query-idxs ! 0
  using query-eq raw-eq len-raw by simp

```

```

then show ?thesis
  unfolding query-round-any-index-set-hit-def
  using lookup-s1 raw-eq query-eq len-row by auto
qed

lemma wp-ntimes-verifier-query-round-program-index-list-bound:
  assumes future: query-future-fresh s
  and raw-bound: query-index-raw-preimage-bound
  and len-query: length query-idxs = n
  and subset: set query-idxs  $\subseteq$  query-sample-space
  shows
    wp-event
      (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n)
      (query-rounds-index-list-hit s query-idxs n) s  $\leq$ 
      (1 / nnreal (card query-sample-space))  $^n$ 
  using future len-query subset
proof (induction n arbitrary: s query-idxs)
  case 0
  show ?case
    using wp-event-le-1
    [of ntimes (verifier-query-round-program fr f-fl f-final as fl final) 0
      query-rounds-index-list-hit s query-idxs 0 s]
  by simp
next
  case (Suc n)
  let ?prog = verifier-query-round-program fr f-fl f-final as fl final
  let ?p = 1 / nnreal (card query-sample-space)
  let ?Q = query-rounds-index-list-hit s query-idxs (Suc n)
  let ?Head = query-round-any-index-set-hit s {query-idxs ! 0}
  have query0-in: query-idxs ! 0  $\in$  query-sample-space
    using Suc.premis(2,3) by (auto intro: nth-mem)
  have singleton-subset: {query-idxs ! 0}  $\subseteq$  query-sample-space
    using query0-in by simp
  have head-bound:
    wp-event ?prog ?Head s  $\leq$  ?p
  proof -
    have base:
      wp-event ?prog ?Head s  $\leq$ 
      nnreal (card {query-idxs ! 0}) /
      nnreal (card query-sample-space)
      by (rule wp-verifier-query-round-program-any-index-set-bound
        [OF Suc.premis(1) raw-bound singleton-subset])
    then show ?thesis
      by simp
  qed
  show ?case
    unfolding ntimes.simps
  proof (rule order-trans)
    show

```

```

wp-event
  (?prog  $\gg$  ( $\lambda u. \text{ntimes } ?prog \ n \gg (\lambda xs. \text{return } (u \# \ xs))$ ))
  ?Q s
 $\leq$  wp-event ?prog ?Head s * (?p  $\wedge$  n)
proof (rule wp-event-bind-bound-by-head-and-cont)
show  $\neg$  ?Q None
  by simp
next
fix u s1
assume head:
  Some (u, s1)  $\in$  set-dist (execute ?prog s)
  and not-head:  $\neg$  ?Head (Some (u, s1))
have u-eq: u = ()
  by (cases u) simp
show
  wp-event
  (ntimes ?prog n  $\gg$  ( $\lambda xs. \text{return } (u \# \ xs)$ ))
  ?Q s1 = 0
proof (rule ccontr)
assume nonzero:
  wp-event
  (ntimes ?prog n  $\gg$  ( $\lambda xs. \text{return } (u \# \ xs)$ ))
  ?Q s1  $\neq$  0
have positive:
  0 <
  wp-event
  (ntimes ?prog n  $\gg$  ( $\lambda xs. \text{return } (u \# \ xs)$ ))
  ?Q s1
  using nonzero by simp
from wp-event-pos-imp-exists-support[OF positive]
obtain out where out-support:
  out  $\in$ 
  set-dist
  (execute
  (ntimes ?prog n  $\gg$  ( $\lambda xs. \text{return } (u \# \ xs)$ )) s1)
  and hit: ?Q out
  by blast
from hit obtain ys t0 raw-idxs where out-some: out = Some (ys, t0)
  unfolding query-rounds-index-list-hit-def by blast
have  $\exists$  results t.
  Some (results, t)  $\in$  set-dist (execute (ntimes ?prog n) s1)  $\wedge$ 
  out = Some (u # results, t)
proof (rule set-dist-bindE[OF out-support[unfolded out-some]])
fix results t
assume tail:
  Some (results, t)  $\in$  set-dist (execute (ntimes ?prog n) s1)
  and ret:
  Some (ys, t0)  $\in$  set-dist (execute (return (u # results)) t)
have out-eq: out = Some (u # results, t)

```

```

    using ret out-some
    unfolding set-dist-def return.rep-eq dist-return-def
      dist-delta-dist delta-map-def
    by simp
    show ?thesis
      using tail out-eq by blast
qed
then obtain results t where
  tail:
    Some (results, t) ∈ set-dist (execute (ntimes ?prog n) s1)
  and out-eq: out = Some (u # results, t)
  by blast
have head-hit: ?Head (Some (u, s1))
  unfolding u-eq
  by (rule query-rounds-index-list-hit-head-index
    [OF Suc.prem1(1) head[unfolded u-eq] tail])
  (use hit out-eq u-eq in simp)
then show False
  using not-head by simp
qed
next
fix u s1
assume head:
  Some (u, s1) ∈ set-dist (execute ?prog s)
  and head-hit: ?Head (Some (u, s1))
have u-eq: u = ()
  by (cases u) simp
have future-s1: query-future-fresh s1
  by (rule verifier-query-round-program-preserves-query-future-fresh
    [OF Suc.prem1(1) head[unfolded u-eq]])
have tail-len: length (tl query-idxs) = n
  using Suc.prem2(2) by simp
have tail-subset: set (tl query-idxs) ⊆ query-sample-space
  using Suc.prem3(3) by (cases query-idxs) auto
have tail-bound:
  wp-event (ntimes ?prog n)
    (query-rounds-index-list-hit s1 (tl query-idxs) n) s1 ≤
    ?p ^ n
  by (rule Suc.IH[OF future-s1 tail-len tail-subset])
show
  wp-event
    (ntimes ?prog n ≫ (λxs. return (u # xs)))
    ?Q s1 ≤ ?p ^ n
proof (rule order-trans[OF - tail-bound])
  show
    wp-event
      (ntimes ?prog n ≫ (λxs. return (u # xs)))
      ?Q s1
    ≤ wp-event (ntimes ?prog n)

```

```

      (query-rounds-index-list-hit s1 (tl query-idxs) n) s1
proof (rule wp-event-bind-bound-by-head-event)
  show
    wp-event (ntimes ?prog n)
      (query-rounds-index-list-hit s1 (tl query-idxs) n) s1
    ≤ wp-event (ntimes ?prog n)
      (query-rounds-index-list-hit s1 (tl query-idxs) n) s1
    by simp
  next
  show
    ?Q None ⇒
      query-rounds-index-list-hit s1 (tl query-idxs) n None
    by simp
  next
  fix results t out
  assume tail:
    Some (results, t) ∈ set-dist (execute (ntimes ?prog n) s1)
  and ret:
    out ∈ set-dist (execute (return (u # results)) t)
  and hit: ?Q out
  have out-eq: out = Some (u # results, t)
  using ret
  unfolding set-dist-def return.rep-eq dist-return-def
    dist-delta-dist delta-map-def
  by simp
  show query-rounds-index-list-hit s1 (tl query-idxs) n
    (Some (results, t))
  by (rule query-rounds-index-list-hit-tail
    [OF head[unfolded u-eq] tail])
    (use hit out-eq u-eq in simp)
  qed
qed
qed
next
  show wp-event ?prog ?Head s * ?p ^ n ≤ ?p ^ Suc n
  using head-bound by (simp add: mult-right-mono)
qed
qed

lemma wp-ntimes-verifier-query-round-program-index-list-set-bound:
  assumes future: query-future-fresh s
  and raw-bound: query-index-raw-preimage-bound
  and finite-Q: finite Q
  and lengths:  $\bigwedge \text{query-idxs}. \text{query-idxs} \in Q \implies \text{length } \text{query-idxs} = n$ 
  and subsets:
     $\bigwedge \text{query-idxs}. \text{query-idxs} \in Q \implies$ 
      set query-idxs  $\subseteq$  query-sample-space
  shows
    wp-event

```



```

    (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n)
    (λout. ∃ query-idxs ∈ Q.
      query-rounds-index-list-hit s query-idxs n out) s ≤
    nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ n
proof -
  have
    wp-event
    (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n)
    (λout. ∃ query-idxs ∈ Q.
      query-rounds-index-list-hit s query-idxs n out) s ≤
    (∑ query-idxs ∈ Q.
      (1 / nnreal (card query-sample-space)) ^ n)
  by (rule wp-event-finite-UN-bound[OF finite-Q])
    (rule wp-ntimes-verifier-query-round-program-index-list-bound
      [OF future raw-bound lengths subsets])
  also have ... =
    nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ n
  using finite-Q by simp
  finally show ?thesis .
qed

```

lemma wp-ntimes-verifier-query-round-program-fri-index-list-set-bound:

assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and subset: $Q \subseteq \text{fri-query-index-list-space}$

shows

```

wp-event
  (ntimes
    (verifier-query-round-program fr f-fl f-final as fl final) rounds)
  (λout. ∃ query-idxs ∈ Q.
    query-rounds-index-list-hit s query-idxs rounds out) s ≤
  nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds

```

proof -

have finite-Q: finite Q
using subset **by** (rule finite-subset) (rule finite-fri-query-index-list-space)
show ?thesis

proof (rule wp-ntimes-verifier-query-round-program-index-list-set-bound
 [OF future raw-bound finite-Q])

fix query-idxs

assume query-idxs ∈ Q

then show length query-idxs = rounds

using subset **unfolding** fri-query-index-list-space-def **by** auto

next

fix query-idxs

assume query-idxs ∈ Q

then show set query-idxs ⊆ query-sample-space

using subset unfolding fri-query-index-list-space-def by auto
 qed
 qed

lemma generic-fri-sampled-query-query-fiber-entries-subset:
 query-index-list-entries
 (generic-fri-sampled-query-query-fiber low-degree candidate-bad
 degree-bound challenges)
 $\subseteq$ query-sample-space
unfolding query-index-list-entries-def
 generic-fri-sampled-query-query-fiber-def
 generic-fri-sampled-query-restricted-bad-pairs-def
 fri-query-challenge-pair-query-fiber-def
 fri-query-index-list-space-def
by auto

lemma finite-generic-fri-sampled-query-query-fiber-entries:
 finite
 (query-index-list-entries
 (generic-fri-sampled-query-query-fiber low-degree candidate-bad
 degree-bound challenges))
using generic-fri-sampled-query-query-fiber-entries-subset
by (rule finite-subset) simp

lemma generic-fri-sampled-query-restricted-bad-pairs-subset:
 generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad
 degree-bound
 $\subseteq$ fri-query-index-list-space $\times$
 fri-challenge-space
 (fri-round-count-for-degree-bound degree-bound)
unfolding generic-fri-sampled-query-restricted-bad-pairs-def
using generic-fri-sampled-query-bad-pair-union-subset-challenge-space
 [of low-degree candidate-bad degree-bound]
by blast

lemma finite-generic-fri-sampled-query-restricted-bad-pairs:
 finite
 (generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad
 degree-bound)
proof (rule finite-subset)
show
 generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad
 degree-bound
 $\subseteq$ fri-query-index-list-space $\times$
 fri-challenge-space
 (fri-round-count-for-degree-bound degree-bound)
by (rule generic-fri-sampled-query-restricted-bad-pairs-subset)
show
 finite

$(\text{fri-query-index-list-space} \times \text{fri-challenge-space} \text{ (fri-round-count-for-degree-bound degree-bound)})$
by (*simp add: finite-fri-query-index-list-space finite-fri-challenge-space*)
qed

lemma *generic-fri-sampled-query-pair-fraction-boundD*:
assumes
generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad degree-bound C
shows
 $\text{nnreal} \text{ (card (generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad degree-bound)) /}$
 $\text{nnreal} \text{ (card fri-query-index-list-space * CARD('f) ^ fri-round-count-for-degree-bound degree-bound)}$
 $\leq C$
using *assms*
unfolding *generic-fri-sampled-query-pair-fraction-bound-def* .

lemma *generic-fri-sampled-query-challenge-projection-subset*:
generic-fri-sampled-query-challenge-projection low-degree candidate-bad degree-bound
 $\subseteq \text{fri-challenge-space} \text{ (fri-round-count-for-degree-bound degree-bound)}$
unfolding *generic-fri-sampled-query-challenge-projection-def*
using *generic-fri-sampled-query-restricted-bad-pairs-subset [of low-degree candidate-bad degree-bound]*
by *force*

lemma *generic-fri-sampled-query-query-fiber-subset*:
generic-fri-sampled-query-query-fiber low-degree candidate-bad degree-bound challenges
 $\subseteq \text{fri-query-index-list-space}$
unfolding *generic-fri-sampled-query-query-fiber-def generic-fri-sampled-query-restricted-bad-pairs-def fri-query-challenge-pair-query-fiber-def*
by *auto*

lemma *finite-generic-fri-sampled-query-query-fiber*:
finite (generic-fri-sampled-query-query-fiber low-degree candidate-bad degree-bound challenges)
using *generic-fri-sampled-query-query-fiber-subset [of low-degree candidate-bad degree-bound challenges]*
by (*rule finite-subset*) (*rule finite-fri-query-index-list-space*)

lemma *generic-fri-sampled-query-query-fiber-entries-card-le:*
 card
 $(\text{query-index-list-entries}$
 $(\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad}$
 $\text{degree-bound challenges}))$
 $\leq$
 card
 $(\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad}$
 $\text{degree-bound challenges}) * \text{rounds}$
by $(\text{rule query-index-list-entries-card-le})$
 $(\text{rule finite-generic-fri-sampled-query-query-fiber},$
 $\text{rule generic-fri-sampled-query-query-fiber-subset})$

lemma *generic-fri-sampled-query-restricted-bad-pairs-as-fiber-Sigma:*
 $\text{generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad}$
 $\text{degree-bound} =$
 $(\lambda(\text{challenges}, \text{query-idxs}). (\text{query-idxs}, \text{challenges}))$ ‘
 $(\text{SIGMA challenges}$
 $\text{fri-challenge-space (fri-round-count-for-degree-bound degree-bound).}$
 $\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad}$
 $\text{degree-bound challenges})$
unfolding $\text{generic-fri-sampled-query-query-fiber-def}$
 $\text{fri-query-challenge-pair-query-fiber-def}$
using $\text{generic-fri-sampled-query-restricted-bad-pairs-subset}$
by force

lemma *generic-fri-sampled-query-restricted-bad-pairs-card-eq-fiber-sum:*
 card
 $(\text{generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad}$
 $\text{degree-bound}) =$
 $(\sum \text{challenges} \in$
 $\text{fri-challenge-space (fri-round-count-for-degree-bound degree-bound).}$
 card
 $(\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad}$
 $\text{degree-bound challenges}))$

proof –
let $?S =$
 SIGMA challenges
 $\text{fri-challenge-space (fri-round-count-for-degree-bound degree-bound).}$
 $\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad}$
 $\text{degree-bound challenges}$
let $?swap = \lambda(\text{challenges}, \text{query-idxs}). (\text{query-idxs}, \text{challenges})$
have $\text{finite-S: finite } ?S$
by $(\text{rule finite-SigmaI})$
 $(\text{rule finite-fri-challenge-space},$
 $\text{rule finite-generic-fri-sampled-query-query-fiber})$
have $\text{inj: inj-on } ?swap ?S$
unfolding inj-on-def **by auto**

```

have finite-challenge-space:
  finite
    (fri-challenge-space
      (fri-round-count-for-degree-bound degree-bound))
  by (rule finite-fri-challenge-space)
have finite-fibers:
   $\bigwedge$ challenges. challenges  $\in$ 
    fri-challenge-space (fri-round-count-for-degree-bound degree-bound)
     $\implies$ 
    finite
      (generic-fri-sampled-query-query-fiber low-degree candidate-bad
        degree-bound challenges)
  by (rule finite-generic-fri-sampled-query-query-fiber)
have card-S:
  card ?S =
    ( $\sum$  challenges  $\in$ 
      fri-challenge-space (fri-round-count-for-degree-bound degree-bound).
      card
        (generic-fri-sampled-query-query-fiber low-degree candidate-bad
          degree-bound challenges))
  by (rule card-SigmaI)
    (use finite-challenge-space finite-fibers in auto)
have card
  (generic-fri-sampled-query-restricted-bad-pairs low-degree
    candidate-bad degree-bound) =
  card (?swap ‘ ?S)
  unfolding generic-fri-sampled-query-restricted-bad-pairs-as-fiber-Sigma
  by simp
also have ... = card ?S
  by (rule card-image[OF inj])
also have ... =
  ( $\sum$  challenges  $\in$ 
    fri-challenge-space (fri-round-count-for-degree-bound degree-bound).
    card
      (generic-fri-sampled-query-query-fiber low-degree candidate-bad
        degree-bound challenges))
  by (rule card-S)
finally show ?thesis .
qed

```

```

lemma generic-fri-sampled-query-pair-fraction-bound-from-fiber-sum:
assumes fraction-bound:
  nnreal
    ( $\sum$  challenges  $\in$ 
      fri-challenge-space (fri-round-count-for-degree-bound degree-bound).
      card
        (generic-fri-sampled-query-query-fiber low-degree candidate-bad
          degree-bound challenges)) /
  nnreal

```

$$\begin{aligned}
& (\text{card } \text{fri-query-index-list-space} * \\
& \quad \text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound } \text{degree-bound}) \\
& \leq C
\end{aligned}$$

shows

generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad degree-bound C

proof –

have *card-eq*:

card

$$\begin{aligned}
& (\text{generic-fri-sampled-query-restricted-bad-pairs low-degree} \\
& \quad \text{candidate-bad degree-bound}) = \\
& (\sum \text{challenges} \in \\
& \quad \text{fri-challenge-space } (\text{fri-round-count-for-degree-bound } \text{degree-bound}). \\
& \quad \text{card} \\
& \quad (\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad} \\
& \quad \quad \text{degree-bound challenges}))
\end{aligned}$$

by (rule *generic-fri-sampled-query-restricted-bad-pairs-card-eq-fiber-sum*)

show *?thesis*

unfolding *generic-fri-sampled-query-pair-fraction-bound-def*

using *fraction-bound card-eq* **by** *simp*

qed

lemma *generic-fri-sampled-query-pair-fraction-bound-iff-fiber-sum*:

generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad degree-bound C

$\longleftrightarrow$

nnreal

$$\begin{aligned}
& (\sum \text{challenges} \in \\
& \quad \text{fri-challenge-space } (\text{fri-round-count-for-degree-bound } \text{degree-bound}). \\
& \quad \text{card} \\
& \quad (\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad} \\
& \quad \quad \text{degree-bound challenges})) / \\
& \text{nnreal} \\
& (\text{card } \text{fri-query-index-list-space} * \\
& \quad \text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound } \text{degree-bound}) \\
& \leq C
\end{aligned}$$

unfolding *generic-fri-sampled-query-pair-fraction-bound-def*

by (*simp add*:

generic-fri-sampled-query-restricted-bad-pairs-card-eq-fiber-sum)

lemma *trace-fri-sampled-query-restricted-bad-pairs-card-eq-fiber-sum*:

card

$$\begin{aligned}
& (\text{trace-fri-sampled-query-bad-pair-union} \cap \\
& \quad (\text{fri-query-index-list-space} \times \text{UNIV})) = \\
& (\sum \text{challenges} \in \\
& \quad \text{fri-challenge-space } (\text{fri-round-count-for-degree-bound } (\text{clength} - 1))). \\
& \text{card} \\
& (\text{generic-fri-sampled-query-query-fiber trace-table-low-degree} \\
& \quad (\text{Not} \circ \text{trace-table-low-degree}) (\text{clength} - 1) \text{ challenges}))
\end{aligned}$$

proof –
have *generic*:
 card
 (generic-fri-sampled-query-restricted-bad-pairs trace-table-low-degree
 (Not ◦ trace-table-low-degree) (clength – 1)) =
 (∑ challenges ∈
 fri-challenge-space (fri-round-count-for-degree-bound (clength – 1))).
 card
 (generic-fri-sampled-query-query-fiber trace-table-low-degree
 (Not ◦ trace-table-low-degree) (clength – 1) challenges))
by (rule generic-fri-sampled-query-restricted-bad-pairs-card-eq-fiber-sum)
then show ?thesis
unfolding trace-fri-sampled-query-bad-pair-union-def
 generic-fri-sampled-query-restricted-bad-pairs-def
by simp
qed

lemma composition-fri-sampled-query-restricted-bad-pairs-card-eq-fiber-sum:
 card
 (composition-fri-sampled-query-bad-pair-union dg ∩
 (fri-query-index-list-space × UNIV)) =
 (∑ challenges ∈
 fri-challenge-space (fri-round-count-for-degree-bound (to-nat dg))).
 card
 (generic-fri-sampled-query-query-fiber
 (composition-table-low-degree (to-nat dg))
 (Not ◦ composition-table-low-degree maxDegree) (to-nat dg)
 challenges))

proof –
have *generic*:
 card
 (generic-fri-sampled-query-restricted-bad-pairs
 (composition-table-low-degree (to-nat dg))
 (Not ◦ composition-table-low-degree maxDegree) (to-nat dg)) =
 (∑ challenges ∈
 fri-challenge-space (fri-round-count-for-degree-bound (to-nat dg))).
 card
 (generic-fri-sampled-query-query-fiber
 (composition-table-low-degree (to-nat dg))
 (Not ◦ composition-table-low-degree maxDegree) (to-nat dg)
 challenges))
by (rule generic-fri-sampled-query-restricted-bad-pairs-card-eq-fiber-sum)
then show ?thesis
unfolding composition-fri-sampled-query-bad-pair-union-def
 generic-fri-sampled-query-restricted-bad-pairs-def
by simp
qed

lemma trace-fri-sampled-query-pair-fraction-bound-iff-fiber-sum:

generic-fri-sampled-query-pair-fraction-bound trace-table-low-degree
 $(\text{Not} \circ \text{trace-table-low-degree}) (\text{clength} - 1) C$
 $\longleftrightarrow$
nnreal
 $(\sum \text{challenges} \in$
 $\text{fri-challenge-space } (\text{fri-round-count-for-degree-bound } (\text{clength} - 1)).$
 card
 $(\text{generic-fri-sampled-query-query-fiber trace-table-low-degree}$
 $(\text{Not} \circ \text{trace-table-low-degree}) (\text{clength} - 1) \text{ challenges})) /$
nnreal
 $(\text{card fri-query-index-list-space} *$
 $\text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound } (\text{clength} - 1))$
 $\leq C$
by (*rule generic-fri-sampled-query-pair-fraction-bound-iff-fiber-sum*)

lemma *composition-fri-sampled-query-pair-fraction-bound-iff-fiber-sum:*

generic-fri-sampled-query-pair-fraction-bound
 $(\text{composition-table-low-degree } (\text{to-nat } dg))$
 $(\text{Not} \circ \text{composition-table-low-degree } \text{maxDegree}) (\text{to-nat } dg) C$
 $\longleftrightarrow$
nnreal
 $(\sum \text{challenges} \in$
 $\text{fri-challenge-space } (\text{fri-round-count-for-degree-bound } (\text{to-nat } dg)).$
 card
 $(\text{generic-fri-sampled-query-query-fiber}$
 $(\text{composition-table-low-degree } (\text{to-nat } dg))$
 $(\text{Not} \circ \text{composition-table-low-degree } \text{maxDegree}) (\text{to-nat } dg)$
 $\text{challenges})) /$
nnreal
 $(\text{card fri-query-index-list-space} *$
 $\text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound } (\text{to-nat } dg))$
 $\leq C$
by (*rule generic-fri-sampled-query-pair-fraction-bound-iff-fiber-sum*)

lemma *trace-fri-query-index-list-set-hit-imp-query-rounds-any-index-set-hit:*

assumes *hit:* *trace-fri-query-index-list-set-hit* *s* *Q* *out*

shows

query-rounds-any-index-set-hit *s* (*query-index-list-entries* *Q*) *rounds*
out

proof –

from *hit* **obtain** *trace-roots* *trace-bs* *trace-final* *dg* *composition-roots*
composition-bs *composition-final* *query-idxs* *trace-round-layers*
composition-round-layers **where**

openings:

accepted-fri-opening-transcript *s* *out* *trace-roots* *trace-bs*
trace-final *dg* *composition-roots* *composition-bs* *composition-final*
query-idxs *trace-round-layers* *composition-round-layers*

and *query-in:* *query-idxs* $\in Q$

unfolding *trace-fri-query-index-list-set-hit-def* **by** *blast*


```

from openings obtain result final-state query-start-state rest raw-idxs
  query-chunks trailing where
    out-eq: out = Some (result, final-state)
    and len-raw: length raw-idxs = rounds
    and query-idxs-eq: query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
    and lookup:
       $\bigwedge i. i < \text{rounds} \implies$ 
        fmllookup (HashMap final-state)
          (QueryIndexChallenge (PQueryCounter s + i)
            (state-after-query-chunks query-start-state query-chunks i)) =
            Some (raw-idxs ! i)
    unfolding accepted-fri-opening-transcript-def
      verifier-query-indices-derived-def
    by blast
have zero-bound: 0 < rounds
  by (rule rounds-positive)
have query-len: length query-idxs = rounds
  using query-idxs-eq len-raw by simp
have idx-in-entries:
  query-idxs ! 0  $\in$  query-index-list-entries Q
  using query-in zero-bound query-len
  unfolding query-index-list-entries-def
  by (auto intro: nth-mem)
have idx-eq: query-idxs ! 0 = index (to-nat (raw-idxs ! 0))
  using query-idxs-eq len-raw zero-bound by simp
show ?thesis
  unfolding query-rounds-any-index-set-hit-def
  apply (rule exI[where x=result])
  apply (rule exI[where x=final-state])
  apply (rule exI[where x=0])
  apply (rule exI[where x=state-after-query-chunks query-start-state
    query-chunks 0])
  apply (rule exI[where x=raw-idxs ! 0])
  using out-eq zero-bound lookup[OF zero-bound] idx-in-entries idx-eq
  by simp
qed

```

lemma composition-fri-query-index-list-set-hit-imp-query-rounds-any-index-set-hit:

assumes hit: composition-fri-query-index-list-set-hit s Q out

shows

query-rounds-any-index-set-hit s (query-index-list-entries Q) rounds
out

proof –

from hit **obtain** trace-roots trace-bs trace-final dg composition-roots
composition-bs composition-final query-idxs trace-round-layers
composition-round-layers **where**

openings:

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final

```

    query-idxs trace-round-layers composition-round-layers
  and query-in: query-idxs ∈ Q
  unfolding composition-fri-query-index-list-set-hit-def by blast
from openings obtain result final-state query-start-state rest raw-idxs
  query-chunks trailing where
  out-eq: out = Some (result, final-state)
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq: query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and lookup:
    ∧ i. i < rounds ⇒
      fmllookup (HashMap final-state)
        (QueryIndexChallenge (PQueryCounter s + i)
          (state-after-query-chunks query-start-state query-chunks i)) =
          Some (raw-idxs ! i)
  unfolding accepted-fri-opening-transcript-def
  verifier-query-indices-derived-def
  by blast
have zero-bound: 0 < rounds
  by (rule rounds-positive)
have query-len: length query-idxs = rounds
  using query-idxs-eq len-raw by simp
have idx-in-entries:
  query-idxs ! 0 ∈ query-index-list-entries Q
  using query-in zero-bound query-len
  unfolding query-index-list-entries-def
  by (auto intro: nth-mem)
have idx-eq: query-idxs ! 0 = index (to-nat (raw-idxs ! 0))
  using query-idxs-eq len-raw zero-bound by simp
show ?thesis
  unfolding query-rounds-any-index-set-hit-def
  apply (rule exI[where x=result])
  apply (rule exI[where x=final-state])
  apply (rule exI[where x=0])
  apply (rule exI[where x=state-after-query-chunks query-start-state
    query-chunks 0])
  apply (rule exI[where x=raw-idxs ! 0])
  using out-eq zero-bound lookup[OF zero-bound] idx-in-entries idx-eq
  by simp
qed

```

lemma *wp-trace-fri-query-index-list-set-hit-bound-from-entries:*
assumes *future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *subset: Q ⊆ fri-query-index-list-space*
shows

$$\text{wp-event verify-monad (trace-fri-query-index-list-set-hit s Q) } s \leq$$

$$\text{nnreal rounds} * \text{nnreal (card (query-index-list-entries Q))} /$$

$$\text{nnreal (card query-sample-space)}$$

proof –
have *event-bound*:
 wp-event verify-monad
 (query-rounds-any-index-set-hit *s* (query-index-list-entries *Q*) rounds)
 $s \leq$
 nnreal rounds *
 (nnreal (card (query-index-list-entries *Q*)) /
 nnreal (card query-sample-space))
by (rule wp-verify-monad-query-rounds-any-index-set-bound
 [OF future raw-bound
 query-index-list-entries-subset-query-sample-space[OF subset]])
show ?thesis
by (rule order-trans[OF - event-bound])
 (rule wp-event-mono,
 rule trace-fri-query-index-list-set-hit-imp-query-rounds-any-index-set-hit)
qed

lemma *wp-composition-fri-query-index-list-set-hit-bound-from-entries*:
assumes future: query-future-fresh *s*
and raw-bound: query-index-raw-preimage-bound
and subset: $Q \subseteq \text{fri-query-index-list-space}$
shows
 wp-event verify-monad (composition-fri-query-index-list-set-hit *s* *Q*) $s \leq$
 nnreal rounds *
 (nnreal (card (query-index-list-entries *Q*)) /
 nnreal (card query-sample-space))

proof –
have *event-bound*:
 wp-event verify-monad
 (query-rounds-any-index-set-hit *s* (query-index-list-entries *Q*) rounds)
 $s \leq$
 nnreal rounds *
 (nnreal (card (query-index-list-entries *Q*)) /
 nnreal (card query-sample-space))
by (rule wp-verify-monad-query-rounds-any-index-set-bound
 [OF future raw-bound
 query-index-list-entries-subset-query-sample-space[OF subset]])
show ?thesis
by (rule order-trans[OF - event-bound])
 (rule wp-event-mono,
 rule composition-fri-query-index-list-set-hit-imp-query-rounds-any-index-set-hit)
qed

lemma *generic-fri-sampled-query-pair-fraction-bound-mono*:
assumes
 generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad
 degree-bound *C*
and $C \leq D$
shows

```

generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad
degree-bound D
using assms
unfolding generic-fri-sampled-query-pair-fraction-bound-def
by (rule order-trans)

lemma generic-fri-sampled-query-pair-fraction-bound-from-card:
assumes card-bound:
  card
  (generic-fri-sampled-query-restricted-bad-pairs low-degree
   candidate-bad degree-bound)  $\leq K$ 
and fraction-bound:
  nnreal  $K$  /
  nnreal
  (card fri-query-index-list-space *
   CARD('f) ^ fri-round-count-for-degree-bound degree-bound)
   $\leq C$ 
shows
  generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad
  degree-bound C
proof –
have nnreal
  (card
   (generic-fri-sampled-query-restricted-bad-pairs low-degree
    candidate-bad degree-bound)) /
  nnreal
  (card fri-query-index-list-space *
   CARD('f) ^ fri-round-count-for-degree-bound degree-bound)
   $\leq$ 
  nnreal  $K$  /
  nnreal
  (card fri-query-index-list-space *
   CARD('f) ^ fri-round-count-for-degree-bound degree-bound)
using card-bound by (rule nnreal-nat-divide-right-mono)
also have ...  $\leq C$ 
by (rule fraction-bound)
finally show ?thesis
unfolding generic-fri-sampled-query-pair-fraction-bound-def .
qed

```

```

lemma generic-fri-sampled-query-restricted-bad-pairs-subset-Sigma:
assumes projection:
  generic-fri-sampled-query-challenge-projection low-degree candidate-bad
  degree-bound  $\subseteq B$ 
shows
  generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad
  degree-bound
   $\subseteq$ 
  ( $\lambda(challenges, query-idxs). (query-idxs, challenges)$ ) ‘

```

(SIGMA challenges:B.
generic-fri-sampled-query-query-fiber low-degree candidate-bad
degree-bound challenges)

proof

fix *pair*

assume *pair-mem*:

pair $\in$
generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad
degree-bound

obtain *query-idxs challenges* **where** *pair-eq*: *pair* = (*query-idxs*, *challenges*)
by (*cases pair*)

have *challenges-in*: *challenges* $\in B$
using *projection pair-mem pair-eq*
unfolding *generic-fri-sampled-query-challenge-projection-def*
by *force*

have *query-in*:

query-idxs $\in$
generic-fri-sampled-query-query-fiber low-degree candidate-bad
degree-bound challenges

using *pair-mem pair-eq*
unfolding *generic-fri-sampled-query-query-fiber-def*
fri-query-challenge-pair-query-fiber-def
by *simp*

have (*challenges*, *query-idxs*) $\in$
(SIGMA challenges:B.
generic-fri-sampled-query-query-fiber low-degree candidate-bad
degree-bound challenges)

using *challenges-in query-in* **by** *simp*

then show

pair $\in$
 $(\lambda(\text{challenges}, \text{query-idxs}). (\text{query-idxs}, \text{challenges}))$ ‘
(SIGMA challenges:B.
generic-fri-sampled-query-query-fiber low-degree candidate-bad
degree-bound challenges)

using *pair-eq* **by** *force*

qed

lemma *generic-fri-sampled-query-restricted-bad-pairs-card-le-from-fibers*:

assumes *finite-B*: *finite B*

and *projection*:

generic-fri-sampled-query-challenge-projection low-degree
candidate-bad degree-bound $\subseteq B$

and *fiber-bound*:

$\bigwedge \text{challenges. challenges} \in B \implies$
card
 $(\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad}$
 $\text{degree-bound challenges}) \leq K$

shows

card

```

    (generic-fri-sampled-query-restricted-bad-pairs low-degree
     candidate-bad degree-bound)
  ≤ card B * K
proof —
  have subset:
    generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad
    degree-bound
  ⊆
  (λ(challenges, query-idxs). (query-idxs, challenges)) ‘
  (SIGMA challenges:B.
   generic-fri-sampled-query-query-fiber low-degree candidate-bad
   degree-bound challenges)
  by (rule generic-fri-sampled-query-restricted-bad-pairs-subset-Sigma
    [OF projection])
  have finite-fibers:
    ∧challenges. challenges ∈ B ⇒
    finite
    (generic-fri-sampled-query-query-fiber low-degree candidate-bad
     degree-bound challenges)
  by (rule finite-generic-fri-sampled-query-query-fiber)
  have card-sigma:
    card
    (SIGMA challenges:B.
     generic-fri-sampled-query-query-fiber low-degree candidate-bad
     degree-bound challenges)
    ≤ card B * K
  using finite-B finite-fibers fiber-bound
  proof (induction rule: finite-induct)
  case empty
  then show ?case by simp
  next
  case (insert challenges B)
  have fiber-le:
    card
    (generic-fri-sampled-query-query-fiber low-degree candidate-bad
     degree-bound challenges) ≤ K
  by (rule insert.prem(2)) simp
  have tail-le:
    card
    (SIGMA challenges:B.
     generic-fri-sampled-query-query-fiber low-degree candidate-bad
     degree-bound challenges)
    ≤ card B * K
  by (rule insert.IH) (use insert.prem in auto)
  have card
    (SIGMA challenges:insert challenges B.
     generic-fri-sampled-query-query-fiber low-degree candidate-bad
     degree-bound challenges)
  =

```

```

card
  (generic-fri-sampled-query-query-fiber low-degree candidate-bad
   degree-bound challenges) +
card
  (SIGMA challenges:B.
   generic-fri-sampled-query-query-fiber low-degree candidate-bad
   degree-bound challenges)
  using insert.hyps insert.premis by simp
also have ... ≤ K + card B * K
  using fiber-le tail-le by simp
also have ... = card (insert challenges B) * K
  using insert.hyps by simp
finally show ?case .
qed
have finite-sigma:
  finite
    (SIGMA challenges:B.
     generic-fri-sampled-query-query-fiber low-degree candidate-bad
     degree-bound challenges)
  by (rule finite-SigmaI[OF finite-B finite-fibers])
then have finite-swap:
  finite
    ((λ(challenges, query-idxs). (query-idxs, challenges)) ‘
     (SIGMA challenges:B.
      generic-fri-sampled-query-query-fiber low-degree candidate-bad
      degree-bound challenges))
  by simp
have card
  (generic-fri-sampled-query-restricted-bad-pairs low-degree candidate-bad
   degree-bound)
≤
card
  ((λ(challenges, query-idxs). (query-idxs, challenges)) ‘
   (SIGMA challenges:B.
    generic-fri-sampled-query-query-fiber low-degree candidate-bad
    degree-bound challenges))
  by (rule card-mono[OF finite-swap subset])
also have ... ≤
card
  (SIGMA challenges:B.
   generic-fri-sampled-query-query-fiber low-degree candidate-bad
   degree-bound challenges)
  by (rule card-image-le[OF finite-sigma])
also have ... ≤ card B * K
  by (rule card-sigma)
finally show ?thesis .
qed

```

lemma generic-fri-sampled-query-pair-fraction-bound-from-projection-and-fibers:

```

assumes finite-B: finite B
and projection:
  generic-fri-sampled-query-challenge-projection low-degree
  candidate-bad degree-bound  $\subseteq B$ 
and fiber-bound:
   $\bigwedge \text{challenges. challenges} \in B \implies$ 
    card
    (generic-fri-sampled-query-query-fiber low-degree candidate-bad
     degree-bound challenges)  $\leq K$ 
and fraction-bound:
  nnreal (card B * K) /
  nnreal
  (card fri-query-index-list-space *
   CARD('f) ^ fri-round-count-for-degree-bound degree-bound)
   $\leq C$ 
shows
  generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad
  degree-bound C
proof (rule generic-fri-sampled-query-pair-fraction-bound-from-card)
show
  card
  (generic-fri-sampled-query-restricted-bad-pairs low-degree
   candidate-bad degree-bound)
   $\leq \text{card } B * K$ 
by (rule generic-fri-sampled-query-restricted-bad-pairs-card-le-from-fibers
     [OF finite-B projection fiber-bound])
show
  nnreal (card B * K) /
  nnreal
  (card fri-query-index-list-space *
   CARD('f) ^ fri-round-count-for-degree-bound degree-bound)
   $\leq C$ 
by (rule fraction-bound)
qed

end

end

```

```

theory Soundness-FRI-Query-Exact-Verifier
imports Soundness-FRI-Query-Aware-Interface
begin

```

Exact verifier-level consumption of sampled query-index lists.

The query-aware interface defines the strict query-list events and proves exact bounds for the raw verifier query-round sequence. This narrow layer lifts those bounds through the verifier decomposition.

```

context soundness

```


begin

lemma *query-rounds-index-list-hit-counter-eq:*

assumes *counter:* $PQueryCounter\ t = PQueryCounter\ s$
and *hit:* *query-rounds-index-list-hit* $s\ query\ idxs\ n\ out$
shows *query-rounds-index-list-hit* $t\ query\ idxs\ n\ out$
using *assms* **unfolding** *query-rounds-index-list-hit-def* **by** *auto*

lemma *wp-verifier-after-composition-fri-query-index-list-set-bound:*

assumes *future:* *query-future-fresh* t
and *counter:* $PQueryCounter\ t = PQueryCounter\ s$
and *raw-bound:* *query-index-raw-preimage-bound*
and *subset:* $Q \subseteq fri\ query\ index\ list\ space$
shows
wp-event (*verifier-after-composition-fri* *header*)
 $(\lambda out. \exists query\ idxs \in Q.$
 $\quad query\ rounds\ index\ list\ hit\ s\ query\ idxs\ rounds\ out) \ t \leq$
 $nnreal\ (card\ Q) \ast$
 $(1 / nnreal\ (card\ query\ sample\ space)) \wedge rounds$

proof –

obtain *fr f-fl f-final as dg fl* **where** *header-eq:*

header = (*fr, f-fl, f-final, as, dg, fl*)

by (*cases header*) *auto*

have *after-eq:*

verifier-after-composition-fri header =
 $(read \gg=$
 $\quad (\lambda final.$
 $\quad \quad ntimes$
 $\quad \quad (verifier\ query\ round\ program\ fr\ f\text{-}fl\ f\text{-}final\ as\ fl\ final)$
 $\quad \quad rounds))$

unfolding *header-eq* *verifier-after-composition-fri-def* **by** *simp*

show *?thesis*

unfolding *after-eq*

proof (*rule wp-event-bind-bound-by-cont*)

show $\neg (\exists query\ idxs \in Q.$
 $\quad query\ rounds\ index\ list\ hit\ s\ query\ idxs\ rounds\ None)$
by *simp*

next

fix *final query-state*

assume *read-final:*

Some (final, query-state) \in set-dist (execute read t)

have *future-query:* *query-future-fresh* *query-state*

by (*rule read-preserves-query-future-fresh[OF future read-final]*)

from *read-outcome[OF read-final]* **obtain** *rest* **where**

counter-query-t: $PQueryCounter\ query\ state = PQueryCounter\ t$

by *blast*

have *counter-query:* $PQueryCounter\ query\ state = PQueryCounter\ s$

using *counter-query-t counter* **by** *simp*

have *event-mono:*

```

wp-event
  (ntimes
    (verifier-query-round-program fr f-fl f-final as fl final)
    rounds)
  (λout. ∃ query-idxs ∈ Q.
    query-rounds-index-list-hit s query-idxs rounds out)
  query-state ≤
wp-event
  (ntimes
    (verifier-query-round-program fr f-fl f-final as fl final)
    rounds)
  (λout. ∃ query-idxs ∈ Q.
    query-rounds-index-list-hit query-state query-idxs rounds out)
  query-state
proof (rule wp-event-mono)
fix out
assume ∃ query-idxs ∈ Q.
  query-rounds-index-list-hit s query-idxs rounds out
then obtain query-idxs where
  query-in: query-idxs ∈ Q
  and hit: query-rounds-index-list-hit s query-idxs rounds out
  by blast
have query-rounds-index-list-hit query-state query-idxs rounds out
  by (rule query-rounds-index-list-hit-counter-eq[OF counter-query hit])
then show ∃ query-idxs ∈ Q.
  query-rounds-index-list-hit query-state query-idxs rounds out
  using query-in by blast
qed
also have ... ≤
  nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds
by (rule wp-ntimes-verifier-query-round-program-fri-index-list-set-bound
  [OF future-query raw-bound subset])
finally show
  wp-event
    (ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds)
    (λout. ∃ query-idxs ∈ Q.
      query-rounds-index-list-hit s query-idxs rounds out)
    query-state ≤
  nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds .
qed
qed

lemma wp-verify-monad-query-index-list-set-bound:
assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound

```

```

and subset:  $Q \subseteq \text{fri-query-index-list-space}$ 
shows
  wp-event verify-monad
    ( $\lambda \text{out. } \exists \text{ query-idxs} \in Q.$ 
      query-rounds-index-list-hit  $s$  query-idxs rounds out)  $s \leq$ 
      nnreal (card  $Q$ ) *
      ( $1 / \text{nnreal (card query-sample-space)}$ )  $\wedge$  rounds
  unfolding verify-monad-composition-fri-decomposition
proof (rule wp-event-bind-bound-by-cont)
  show  $\neg (\exists \text{ query-idxs} \in Q.$ 
    query-rounds-index-list-hit  $s$  query-idxs rounds None)
  by simp
next
fix header  $t$ 
assume prefix:
  Some (header,  $t$ )  $\in$ 
    set-dist (execute verifier-composition-fri-prefix  $s$ )
obtain fr f-fl f-final as dg fl where header-eq:
  header = (fr, f-fl, f-final, as, dg, fl)
by (cases header) auto
have future-t: query-future-fresh  $t$ 
by (rule verifier-composition-fri-prefix-preserves-query-future-fresh
  [OF future prefix])
have counter-t: PQueryCounter  $t = \text{PQueryCounter } s$ 
using verifier-composition-fri-prefix-outcome[OF prefix[unfolded header-eq]]
by simp
show
  wp-event (verifier-after-composition-fri header)
    ( $\lambda \text{out. } \exists \text{ query-idxs} \in Q.$ 
      query-rounds-index-list-hit  $s$  query-idxs rounds out)  $t \leq$ 
      nnreal (card  $Q$ ) *
      ( $1 / \text{nnreal (card query-sample-space)}$ )  $\wedge$  rounds
  by (rule wp-verifier-after-composition-fri-query-index-list-set-bound
  [OF future-t counter-t raw-bound subset])
qed

end

end

```

```

theory Soundness-FRI-Query-Exact-Bounds
imports Soundness-FRI-Query-Exact-Verifier
begin

```

Exact sampled-query bounds for the FRI query-list events.

This layer connects the verifier-level query-index-list consumption theorem to the trace/composition sampled FRI events. It stays separate from the raw verifier decomposition to keep theory finalization predictable.

context *soundness*
begin

lemma *wp-event-finite-union-bound-fri-exact*:
fixes $B :: 'i \Rightarrow \text{prob}$
assumes *finite*: $\text{finite } I$
and *bounds*: $\bigwedge i. i \in I \implies \text{wp-event } m (E \ i) \ s \leq B \ i$
shows
 $\text{wp-event } m (\lambda \text{out}. \exists i \in I. E \ i \ \text{out}) \ s \leq (\sum i \in I. B \ i)$
using *finite bounds*
proof (*induction I*)
case *empty*
show *?case*
unfolding *wp-event-def wp-def dist-expect-def* **by** *simp*
next
case (*insert i I*)
have *split*:
 $\text{wp-event } m (\lambda \text{out}. \exists j \in \text{insert } i \ I. E \ j \ \text{out}) \ s \leq$
 $\text{wp-event } m (E \ i) \ s +$
 $\text{wp-event } m (\lambda \text{out}. \exists j \in I. E \ j \ \text{out}) \ s$
proof –
have $\text{wp-event } m (\lambda \text{out}. \exists j \in \text{insert } i \ I. E \ j \ \text{out}) \ s \leq$
 $\text{wp-event } m (\lambda \text{out}. E \ i \ \text{out} \vee (\exists j \in I. E \ j \ \text{out})) \ s$
by (*rule wp-event-mono*) *auto*
also have $\dots \leq$
 $\text{wp-event } m (E \ i) \ s +$
 $\text{wp-event } m (\lambda \text{out}. \exists j \in I. E \ j \ \text{out}) \ s$
by (*rule wp-event-union-bound*)
finally show *?thesis* .
qed
have *tail*:
 $\text{wp-event } m (\lambda \text{out}. \exists j \in I. E \ j \ \text{out}) \ s \leq (\sum j \in I. B \ j)$
by (*rule insert.IH*) (*use insert.prem*s **in** *blast*)
have $\text{wp-event } m (E \ i) \ s +$
 $\text{wp-event } m (\lambda \text{out}. \exists j \in I. E \ j \ \text{out}) \ s \leq$
 $B \ i + (\sum j \in I. B \ j)$
by (*intro add-mono tail insert.prem*s) *simp*
also have $\dots = (\sum j \in \text{insert } i \ I. B \ j)$
using *insert.hyps* **by** *simp*
finally show *?case*
by (*rule order-trans[OF split]*)
qed

lemma *weighted-query-fiber-sum-eq*:
fixes $I :: 'i \text{ set}$ **and** $f :: 'i \Rightarrow \text{nat}$
assumes *fin*: $\text{finite } I$
and *A-pos*: $0 < A$
and *B-pos*: $0 < B$
shows

$(\sum x \in I.$
 $(1 / \text{nnreal } A) * ((1 / \text{nnreal } B) ^ \wedge \text{rounds} * \text{nnreal } (f x))) =$
 $\text{nnreal } (\sum x \in I. f x) / (\text{nnreal } A * \text{nnreal } B ^ \wedge \text{rounds})$

proof –

have *sum-nnreal*:

$(\sum x \in I. \text{nnreal } (f x)) = \text{nnreal } (\sum x \in I. f x)$

using *fin by induction simp-all*

have

$(\sum x \in I.$
 $(1 / \text{nnreal } A) * ((1 / \text{nnreal } B) ^ \wedge \text{rounds} * \text{nnreal } (f x))) =$
 $(\sum x \in I. \text{nnreal } (f x) / (\text{nnreal } A * \text{nnreal } B ^ \wedge \text{rounds}))$

using *A-pos B-pos*

by (*intro sum.cong refl*)

(*transfer, simp add: field-simps power-divide*)

also have ... =

$(\sum x \in I. \text{nnreal } (f x)) / (\text{nnreal } A * \text{nnreal } B ^ \wedge \text{rounds})$

by (*rule sum-divide-nnreal[OF fin]*)

also have ... =

$\text{nnreal } (\sum x \in I. f x) / (\text{nnreal } A * \text{nnreal } B ^ \wedge \text{rounds})$

by (*simp add: sum-nnreal*)

finally show *?thesis* .

qed

lemma *accepted-fri-opening-transcript-imp-query-rounds-index-list-hit*:

assumes *openings*:

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers

shows *query-rounds-index-list-hit s query-idxs rounds out*

proof –

from *openings* **obtain** *result final-state query-start-state rest raw-idxs*
query-chunks trailing **where**
out-eq: out = Some (result, final-state)
and *len-raw: length raw-idxs = rounds*
and *query-idxs-eq: query-idxs = map (λraw. index (to-nat raw)) raw-idxs*
and *lookup:*
 $\bigwedge i. i < \text{rounds} \implies$
 $\text{fmlookup } (\text{HashMap } \text{final-state})$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } s + i)$
 $(\text{state-after-query-chunks } \text{query-start-state } \text{query-chunks } i)) =$
 $\text{Some } (\text{raw-idxs } ! i)$

unfolding *accepted-fri-opening-transcript-def*
verifier-query-indices-derived-def

by *blast*

show *?thesis*

unfolding *query-rounds-index-list-hit-def*

apply (*rule exI[where x=result]*)

apply (*rule exI[where x=final-state]*)

apply (*rule exI[where x=raw-idxs]*)

```

    using out-eq len-raw query-idxs-eq lookup
  by auto
qed

lemma trace-fri-query-index-list-set-hit-imp-query-rounds-index-list-hit:
  assumes hit: trace-fri-query-index-list-set-hit s Q out
  shows
     $\exists$  query-idxs  $\in$  Q.
      query-rounds-index-list-hit s query-idxs rounds out
  proof -
    from hit obtain trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final query-idxs trace-round-layers
      composition-round-layers where
      openings:
        accepted-fri-opening-transcript s out trace-roots trace-bs
        trace-final dg composition-roots composition-bs composition-final
        query-idxs trace-round-layers composition-round-layers
    and query-in: query-idxs  $\in$  Q
    unfolding trace-fri-query-index-list-set-hit-def by blast
    have query-rounds-index-list-hit s query-idxs rounds out
      by (rule accepted-fri-opening-transcript-imp-query-rounds-index-list-hit
        [OF openings])
    then show ?thesis
      using query-in by blast
  qed

lemma composition-fri-query-index-list-set-hit-imp-query-rounds-index-list-hit:
  assumes hit: composition-fri-query-index-list-set-hit s Q out
  shows
     $\exists$  query-idxs  $\in$  Q.
      query-rounds-index-list-hit s query-idxs rounds out
  proof -
    from hit obtain trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final query-idxs trace-round-layers
      composition-round-layers where
      openings:
        accepted-fri-opening-transcript s out trace-roots trace-bs
        trace-final dg composition-roots composition-bs composition-final
        query-idxs trace-round-layers composition-round-layers
    and query-in: query-idxs  $\in$  Q
    unfolding composition-fri-query-index-list-set-hit-def by blast
    have query-rounds-index-list-hit s query-idxs rounds out
      by (rule accepted-fri-opening-transcript-imp-query-rounds-index-list-hit
        [OF openings])
    then show ?thesis
      using query-in by blast
  qed

lemma wp-trace-fri-query-index-list-set-hit-exact-bound:

```

assumes *future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *subset: $Q \subseteq \text{fri-query-index-list-space}$*
shows
 $\text{wp-event verify-monad } (\text{trace-fri-query-index-list-set-hit } s \ Q) \ s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
proof –
have *event-bound:*
 $\text{wp-event verify-monad}$
 $(\lambda \text{out. } \exists \text{query-idxs} \in Q.$
 $\text{query-rounds-index-list-hit } s \ \text{query-idxs} \ \text{rounds out}) \ s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
by (*rule wp-verify-monad-query-index-list-set-bound*
 $[\text{OF future raw-bound subset}]$)
show *?thesis*
by (*rule order-trans*[*OF - event-bound*])
 $(\text{rule wp-event-mono},$
 $\text{rule trace-fri-query-index-list-set-hit-imp-query-rounds-index-list-hit})$
qed

lemma *wp-composition-fri-query-index-list-set-hit-exact-bound:*
assumes *future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *subset: $Q \subseteq \text{fri-query-index-list-space}$*
shows
 $\text{wp-event verify-monad } (\text{composition-fri-query-index-list-set-hit } s \ Q) \ s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
proof –
have *event-bound:*
 $\text{wp-event verify-monad}$
 $(\lambda \text{out. } \exists \text{query-idxs} \in Q.$
 $\text{query-rounds-index-list-hit } s \ \text{query-idxs} \ \text{rounds out}) \ s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
by (*rule wp-verify-monad-query-index-list-set-bound*
 $[\text{OF future raw-bound subset}]$)
show *?thesis*
by (*rule order-trans*[*OF - event-bound*])
 $(\text{rule wp-event-mono},$
 $\text{rule composition-fri-query-index-list-set-hit-imp-query-rounds-index-list-hit})$
qed

lemma *wp-trace-fri-query-challenge-pair-set-hit-bound-from-query-projection:*
assumes *future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *projection: fst ‘ $P \subseteq Q$*

and *subset*: $Q \subseteq \text{fri-query-index-list-space}$
shows
 $\text{wp-event verify-monad } (\text{trace-fri-query-challenge-pair-set-hit } s \ P) \ s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
proof –
have *event-bound*:
 $\text{wp-event verify-monad } (\text{trace-fri-query-index-list-set-hit } s \ Q) \ s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
by (rule *wp-trace-fri-query-index-list-set-hit-exact-bound*
 $[\text{OF future raw-bound subset}]$)
show *?thesis*
by (rule *order-trans*[*OF - event-bound*])
(rule *wp-event-mono*,
rule *trace-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit*
 $[\text{OF - projection}]$)
qed

lemma
wp-composition-fri-query-challenge-pair-set-hit-bound-from-query-projection:
assumes *future*: *query-future-fresh* *s*
and *raw-bound*: *query-index-raw-preimage-bound*
and *projection*: $\bigwedge dg. \text{fst } 'P \ dg \subseteq Q$
and *subset*: $Q \subseteq \text{fri-query-index-list-space}$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-query-challenge-pair-set-hit } s \ P) \ s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
proof –
have *event-bound*:
 $\text{wp-event verify-monad } (\text{composition-fri-query-index-list-set-hit } s \ Q) \ s$
 $\leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
by (rule *wp-composition-fri-query-index-list-set-hit-exact-bound*
 $[\text{OF future raw-bound subset}]$)
show *?thesis*
by (rule *order-trans*[*OF - event-bound*])
(rule *wp-event-mono*,
rule *composition-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit*
 $[\text{OF - projection}]$)
qed

lemma *wp-trace-fri-sampled-query-pair-set-hit-bound-from-query-projection*:
assumes *future*: *query-future-fresh* *s*
and *raw-bound*: *query-index-raw-preimage-bound*
and *projection*:

$\bigwedge \text{trace-table roots final round-layers layers.}$
 $\text{fst } \text{' } R \text{ trace-table roots final round-layers layers } \subseteq Q$
and *subset*: $Q \subseteq \text{fri-query-index-list-space}$
shows
 $\text{wp-event verify-monad } (\text{trace-fri-sampled-query-pair-set-hit } R \text{ } s) \text{ } s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card query-sample-space})) \wedge \text{rounds}$
proof –
have *event-bound*:
 $\text{wp-event verify-monad } (\text{trace-fri-query-index-list-set-hit } s \text{ } Q) \text{ } s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card query-sample-space})) \wedge \text{rounds}$
by (*rule wp-trace-fri-query-index-list-set-hit-exact-bound*
 $[\text{OF future raw-bound subset}]$)
show *?thesis*
by (*rule order-trans*[*OF - event-bound*])
 $(\text{rule wp-event-mono,}$
 $\text{rule trace-fri-sampled-query-pair-set-hit-imp-query-index-list-set-hit}$
 $[\text{OF - projection}])$
qed

lemma *wp-composition-fri-sampled-query-pair-set-hit-bound-from-query-projection*:
assumes *future*: *query-future-fresh* s
and *raw-bound*: *query-index-raw-preimage-bound*
and *projection*:
 $\bigwedge \text{dg composition-table roots final round-layers layers.}$
 $\text{fst } \text{' } R \text{ dg composition-table roots final round-layers layers } \subseteq Q$
and *subset*: $Q \subseteq \text{fri-query-index-list-space}$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-query-pair-set-hit } R \text{ } s) \text{ } s \leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card query-sample-space})) \wedge \text{rounds}$
proof –
have *event-bound*:
 $\text{wp-event verify-monad } (\text{composition-fri-query-index-list-set-hit } s \text{ } Q) \text{ } s$
 $\leq$
 $\text{nnreal } (\text{card } Q) *$
 $(1 / \text{nnreal } (\text{card query-sample-space})) \wedge \text{rounds}$
by (*rule wp-composition-fri-query-index-list-set-hit-exact-bound*
 $[\text{OF future raw-bound subset}]$)
show *?thesis*
by (*rule order-trans*[*OF - event-bound*])
 $(\text{rule wp-event-mono,}$
 $\text{rule composition-fri-sampled-query-pair-set-hit-imp-query-index-list-set-hit}$
 $[\text{OF - projection}])$
qed

lemma *wp-verifier-after-trace-fri-query-index-list-set-bound*:

```

assumes future: query-future-fresh t
and counter: PQueryCounter t = PQueryCounter s
and raw-bound: query-index-raw-preimage-bound
and subset:  $Q \subseteq \text{fri-query-index-list-space}$ 
shows
  wp-event (verifier-after-trace-fri header)
    ( $\lambda \text{out. } \exists \text{ query-idxs} \in Q.$ 
      query-rounds-index-list-hit s query-idxs rounds out)  $t \leq$ 
      nnreal (card Q) *
      ( $1 / \text{nnreal} (\text{card } \text{query-sample-space})$ )  $\wedge$  rounds
proof -
obtain fr f-fl where header-eq: header = (fr, f-fl)
by (cases header) auto
let ?E =
   $\lambda \text{out. } \exists \text{ query-idxs} \in Q.$ 
    query-rounds-index-list-hit s query-idxs rounds out
let ?C =
  nnreal (card Q) *
  ( $1 / \text{nnreal} (\text{card } \text{query-sample-space})$ )  $\wedge$  rounds
have after-eq:
  verifier-after-trace-fri header =
    (read  $\gg=$ 
      ( $\lambda f\text{-final.}$ 
        mmap (replicate (length spec) alpha-round)  $\gg=$ 
        ( $\lambda as. \text{read } \gg=$ 
          ( $\lambda dg. \text{assert } (\text{to-nat } dg \leq \text{maxDegree}) \gg=$ 
            ( $\lambda -. \text{ntimes } \text{receive-composition-fri-commits}$ 
              (ceil-log (to-nat dg + 1))  $\gg=$ 
                ( $\lambda fl. \text{read } \gg=$ 
                  ( $\lambda \text{final.}$ 
                    ntimes
                      (verifier-query-round-program fr f-fl f-final as fl
                        final)
                      rounds))))))
    unfolding header-eq verifier-after-trace-fri-def by simp
show ?thesis
unfolding after-eq
proof (rule wp-event-bind-bound-by-cont)
show  $\neg ?E$  None
by simp
next
fix f-final s1
assume read-f-final: Some (f-final, s1)  $\in$  set-dist (execute read t)
have future-s1: query-future-fresh s1
by (rule read-preserves-query-future-fresh[OF future read-f-final])
from read-outcome[OF read-f-final] obtain rest1 where
  counter-s1-t: PQueryCounter s1 = PQueryCounter t
by blast
have counter-s1: PQueryCounter s1 = PQueryCounter s

```

```

using counter-s1-t counter by simp
show
  wp-event
    (mmap (replicate (length spec) alpha-round)  $\gg=$ 
      ( $\lambda$ as. read  $\gg=$ 
        ( $\lambda$ dg. assert (to-nat dg  $\leq$  maxDegree)  $\gg=$ 
          ( $\lambda$ -. ntimes receive-composition-fri-commits
            (ceil-log (to-nat dg + 1))  $\gg=$ 
              ( $\lambda$ fl. read  $\gg=$ 
                ( $\lambda$ final.
                  ntimes
                    (verifier-query-round-program fr f-fl f-final as fl
                      final)
                    rounds)))))) ?E s1  $\leq$  ?C
proof (rule wp-event-bind-bound-by-cont)
  show  $\neg$  ?E None
    by simp
next
  fix as s2
  assume alpha-out:
    Some (as, s2)  $\in$ 
      set-dist
        (execute (mmap (replicate (length spec) alpha-round)) s1)
  have future-s2: query-future-fresh s2
    by (rule mmap-alpha-round-preserves-query-future-fresh
      [OF future-s1 alpha-out])
  have counter-s2-s1: PQueryCounter s2 = PQueryCounter s1
    using mmap-alpha-round-outcome[OF alpha-out] by simp
  have counter-s2: PQueryCounter s2 = PQueryCounter s
    using counter-s2-s1 counter-s1 by simp
  show
    wp-event
      (read  $\gg=$ 
        ( $\lambda$ dg. assert (to-nat dg  $\leq$  maxDegree)  $\gg=$ 
          ( $\lambda$ -. ntimes receive-composition-fri-commits
            (ceil-log (to-nat dg + 1))  $\gg=$ 
              ( $\lambda$ fl. read  $\gg=$ 
                ( $\lambda$ final.
                  ntimes
                    (verifier-query-round-program fr f-fl f-final as fl
                      final)
                    rounds)))))) ?E s2  $\leq$  ?C
  proof (rule wp-event-bind-bound-by-cont)
    show  $\neg$  ?E None
      by simp
  next
    fix dg s3
    assume read-dg: Some (dg, s3)  $\in$  set-dist (execute read s2)
    have future-s3: query-future-fresh s3

```

```

    by (rule read-preserves-query-future-fresh[OF future-s2 read-dg])
  from read-outcome[OF read-dg] obtain rest3 where
    counter-s3-s2: PQueryCounter s3 = PQueryCounter s2
  by blast
  have counter-s3: PQueryCounter s3 = PQueryCounter s
    using counter-s3-s2 counter-s2 by simp
  show
    wp-event
    (assert (to-nat dg ≤ maxDegree) ≥=
      (λ-. ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1)) ≥=
        (λfl. read ≥=
          (λfinal.
            ntimes
              (verifier-query-round-program fr f-fl f-final as fl
                final)
            rounds)))) ?E s3 ≤ ?C
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ ?E None
      by simp
  next
    fix unit s4
    assume assert-out:
      Some (unit, s4) ∈
        set-dist (execute (assert (to-nat dg ≤ maxDegree)) s3)
    have s4-eq: s4 = s3
      using assert-out unfolding assert-def
      by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
    have future-s4: query-future-fresh s4
      using future-s3 s4-eq by simp
    have counter-s4: PQueryCounter s4 = PQueryCounter s
      using counter-s3 s4-eq by simp
    show
      wp-event
      (ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1)) ≥=
        (λfl. read ≥=
          (λfinal.
            ntimes
              (verifier-query-round-program fr f-fl f-final as fl
                final)
            rounds)))) ?E s4 ≤ ?C
  proof (rule wp-event-bind-bound-by-cont)
    show ¬ ?E None
      by simp
  next
    fix fl s5
    assume comp-fri:
      Some (fl, s5) ∈

```

```

    set-dist
    (execute
      (ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1))) s4)
  have future-s5: query-future-fresh s5
  by (rule ntimes-receive-composition-fri-commits-preserves-query-future-fresh
    [OF comp-fri future-s4])
  have counter-s5-s4: PQueryCounter s5 = PQueryCounter s4
  using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
  by simp
  have counter-s5: PQueryCounter s5 = PQueryCounter s
  using counter-s5-s4 counter-s4 by simp
show
  wp-event
  (read  $\gg$ 
    ( $\lambda$ final.
      ntimes
        (verifier-query-round-program fr f-fl f-final as fl
          final)
        rounds)) ?E s5  $\leq$  ?C
proof (rule wp-event-bind-bound-by-cont)
  show  $\neg$  ?E None
  by simp
next
  fix final query-state
  assume read-final:
    Some (final, query-state)  $\in$  set-dist (execute read s5)
  have future-query: query-future-fresh query-state
  by (rule read-preserves-query-future-fresh
    [OF future-s5 read-final])
  from read-outcome[OF read-final] obtain rest where
    counter-query-s5:
      PQueryCounter query-state = PQueryCounter s5
  by blast
  have counter-query:
    PQueryCounter query-state = PQueryCounter s
  using counter-query-s5 counter-s5 by simp
  have event-mono:
    wp-event
    (ntimes
      (verifier-query-round-program fr f-fl f-final as fl
        final)
      rounds)
    ?E query-state  $\leq$ 
    wp-event
    (ntimes
      (verifier-query-round-program fr f-fl f-final as fl
        final)
      rounds)

```

```

    ( $\lambda out. \exists query\_idxs \in Q.$ 
       $query\_rounds\_index\_list\_hit\ query\_state\ query\_idxs$ 
       $rounds\ out$ )
     $query\_state$ 
  proof (rule wp-event-mono)
    fix  $out$ 
    assume  $\exists query\_idxs \in Q.$ 
       $query\_rounds\_index\_list\_hit\ s\ query\_idxs\ rounds\ out$ 
    then obtain  $query\_idxs$  where
       $query\_in: query\_idxs \in Q$ 
    and  $hit:$ 
       $query\_rounds\_index\_list\_hit\ s\ query\_idxs\ rounds\ out$ 
    by  $blast$ 
  have
     $query\_rounds\_index\_list\_hit\ query\_state\ query\_idxs\ rounds$ 
     $out$ 
    by (rule query-rounds-index-list-hit-counter-eq
      [OF counter-query hit])
  then show  $\exists query\_idxs \in Q.$ 
     $query\_rounds\_index\_list\_hit\ query\_state\ query\_idxs\ rounds$ 
     $out$ 
    using  $query\_in$  by  $blast$ 
  qed
also have  $\dots \leq ?C$ 
  by (rule wp-ntimes-verifier-query-round-program-fri-index-list-set-bound
    [OF future-query raw-bound subset])
finally show
   $wp\_event$ 
  ( $ntimes$ 
    ( $verifier\_query\_round\_program\ fr\ f\_fl\ f\_final\ as\ fl$ 
       $final$ )
     $rounds$ )
   $?E\ query\_state \leq ?C$  .
  qed
qed
qed
qed
qed
qed
qed

```

lemma *verifier-trace-fri-prefix-preserves-query-future-fresh-exact:*
assumes $future: query_future_fresh\ s$
and $prefix:$
 $Some\ (header, t) \in set_dist\ (execute\ verifier_trace_fri_prefix\ s)$
shows $query_future_fresh\ t$
proof –
obtain $fr\ f_fl$ **where** $header_eq: header = (fr, f_fl)$
by (*cases header*) $auto$

```

from prefix[unfolded verifier-trace-fri-prefix-def header-eq]
obtain s1 where
  read-fr: Some (fr, s1) ∈ set-dist (execute read s)
  and trace-fri:
    Some (f-fl, t) ∈
      set-dist
        (execute (ntimes receive-trace-fri-commits (ceil-log clength))
          s1)
  by (auto elim!: set-dist-bindE)
have future-s1: query-future-fresh s1
  by (rule read-preserves-query-future-fresh[OF future read-fr])
show ?thesis
  by (rule ntimes-receive-trace-fri-commits-preserves-query-future-fresh
    [OF trace-fri future-s1])
qed

lemma wp-trace-fri-query-challenge-pair-set-hit-fixed-challenge-bound:
assumes trace-future: trace-fri-future-fresh s
and query-future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and challenges-space:
  challenges ∈ fri-challenge-space (ceil-log clength)
and projection:
  fst ‘ (P ∩ (UNIV × {challenges})) ⊆ Q
and subset: Q ⊆ fri-query-index-list-space
shows
  wp-event verify-monad
    (λout. trace-fri-challenge-list-set-hit s {challenges} out ∧
      trace-fri-query-challenge-pair-set-hit s P out) s ≤
    (1 / nnreal (CARD('f) ^ ceil-log clength)) *
    (nnreal (card Q) *
      (1 / nnreal (card query-sample-space)) ^ rounds)
proof –
  let ?Hit =
    λout. trace-fri-challenge-list-set-hit s {challenges} out ∧
      trace-fri-query-challenge-pair-set-hit s P out
  let ?Head =
    λout. case out of
      None ⇒ False
    | Some ((-, f-fl), -) ⇒ map fst f-fl ∈ {challenges}
  let ?D =
    nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds
  have bind-bound:
    wp-event (verifier-trace-fri-prefix ≫ verifier-after-trace-fri)
      ?Hit s ≤ wp-event verifier-trace-fri-prefix ?Head s * ?D
  proof (rule wp-event-bind-bound-by-head-and-cont)
  show ¬ ?Hit None
    unfolding trace-fri-challenge-list-set-hit-def

```

```

    trace-fri-query-challenge-pair-set-hit-def
    accepted-fri-opening-transcript-def
  by blast
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
and not-head:
  ¬ (case Some (header, prefix-state) of None ⇒ False
    | Some ((-, f-fl), -) ⇒ map fst f-fl ∈ {challenges})
obtain fr f-fl where header-eq: header = (fr, f-fl)
  by (cases header) auto
have neq: map fst f-fl ≠ challenges
  using not-head unfolding header-eq by simp
have zero-le:
  wp-event (verifier-after-trace-fri header)
    (λout. trace-fri-challenge-list-set-hit s {challenges} out ∧
      trace-fri-query-challenge-pair-set-hit s P out) prefix-state
    ≤ 0
proof (rule order-trans)
show
  wp-event (verifier-after-trace-fri header)
    (λout. trace-fri-challenge-list-set-hit s {challenges} out ∧
      trace-fri-query-challenge-pair-set-hit s P out) prefix-state
    ≤
  wp-event (verifier-after-trace-fri header)
    (λ-. False) prefix-state
proof (rule wp-event-mono-on-support)
fix out
assume suffix:
  out ∈
    set-dist (execute (verifier-after-trace-fri header) prefix-state)
and hit:
  trace-fri-challenge-list-set-hit s {challenges} out ∧
  trace-fri-query-challenge-pair-set-hit s P out
from hit obtain trace-bs fri-dg composition-bs where
  challenges-hit:
    accepted-fri-challenges s out trace-bs fri-dg composition-bs
and trace-in: trace-bs ∈ {challenges}
unfolding trace-fri-challenge-list-set-hit-def
by blast
from challenges-hit obtain result final-state where out-eq:
  out = Some (result, final-state)
unfolding accepted-fri-challenges-def by blast
have prefix':
  Some ((fr, f-fl), prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
  using prefix header-eq by simp

```



```

have suffix':
  Some (result, final-state) ∈
    set-dist
      (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
  using suffix header-eq out-eq by simp
from verifier-after-trace-fri-accepted-fri-challenges[OF prefix' suffix']
obtain dg' comp-bs' where challenges-prefix:
  accepted-fri-challenges s (Some (result, final-state))
    (map fst f-fl) dg' comp-bs'
  by blast
have trace-eq: trace-bs = map fst f-fl
  using accepted-fri-challenges-unique
    [OF challenges-prefix challenges-hit[unfolded out-eq]]
  by simp
show False
  using neq trace-eq trace-in by simp
qed
next
show
  wp-event (verifier-after-trace-fri header)
    ( $\lambda\cdot. \text{False}$ ) prefix-state ≤ 0
  unfolding wp-event-def wp-def dist-expect-def by simp
qed
show
  wp-event (verifier-after-trace-fri header)
    ( $\lambda\text{out. trace-fri-challenge-list-set-hit } s \{ \text{challenges} \} \text{ out} \wedge$ 
       $\text{trace-fri-query-challenge-pair-set-hit } s \text{ } P \text{ out}$ ) prefix-state = 0
  by (rule antisym[OF zero-le]) simp
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
and head:
  (case Some (header, prefix-state) of None ⇒ False
    | Some ((-, f-fl), -) ⇒ map fst f-fl ∈ {challenges})
obtain fr f-fl where header-eq: header = (fr, f-fl)
  by (cases header) auto
have f-fl-eq: map fst f-fl = challenges
  using head unfolding header-eq by simp
have future-prefix: query-future-fresh prefix-state
  by (rule verifier-trace-fri-prefix-preserves-query-future-fresh-exact
    [OF query-future prefix])
have counter-prefix: PQueryCounter prefix-state = PQueryCounter s
  using verifier-trace-fri-prefix-outcome[OF prefix[unfolded header-eq]]
  by simp
have event-bound:
  wp-event (verifier-after-trace-fri header)
    ( $\lambda\text{out. } \exists \text{query-idxs} \in Q.$ 

```

```

    query-rounds-index-list-hit s query-idxs rounds out)
  prefix-state ≤ ?D
by (rule wp-verifier-after-trace-fri-query-index-list-set-bound
    [OF future-prefix counter-prefix raw-bound subset])
have hit-to-query:
  wp-event (verifier-after-trace-fri header)
    (λout. trace-fri-challenge-list-set-hit s {challenges} out ∧
      trace-fri-query-challenge-pair-set-hit s P out) prefix-state
    ≤
  wp-event (verifier-after-trace-fri header)
    (λout. ∃ query-idxs ∈ Q.
      query-rounds-index-list-hit s query-idxs rounds out) prefix-state
proof (rule wp-event-mono-on-support)
fix out
assume suffix:
  out ∈
    set-dist (execute (verifier-after-trace-fri header) prefix-state)
and hit:
  trace-fri-challenge-list-set-hit s {challenges} out ∧
  trace-fri-query-challenge-pair-set-hit s P out
from hit obtain trace-roots trace-bs trace-final fri-dg
  composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers where
  openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final query-idxs trace-round-layers
    composition-round-layers
  and pair: (query-idxs, trace-bs) ∈ P
  unfolding trace-fri-query-challenge-pair-set-hit-def by blast
from openings obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-fri-opening-transcript-def by blast
have prefix':
  Some ((fr, f-fl), prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
  using prefix header-eq by simp
have suffix':
  Some (result, final-state) ∈
    set-dist
      (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
  using suffix header-eq out-eq by simp
from verifier-after-trace-fri-accepted-fri-challenges[OF prefix' suffix']
obtain dg' comp-bs' where challenges-prefix:
  accepted-fri-challenges s (Some (result, final-state))
  (map fst f-fl) dg' comp-bs'
by blast
have challenges-openings:
  accepted-fri-challenges s out trace-bs fri-dg composition-bs

```

```

using openings
unfolding accepted-fri-opening-transcript-def
apply (elim exE conjE)
apply assumption
done
have trace-eq: trace-bs = map fst f-fl
  using accepted-fri-challenges-unique
  [OF challenges-prefix challenges-openings[unfolded out-eq]]
  by simp
have query-in: query-idxs ∈ Q
proof –
  have query-idxs ∈ fst ‘ (P ∩ (UNIV × {challenges}))
    using pair trace-eq f-fl-eq by force
  then show ?thesis
    using projection by blast
qed
have query-hit:
  query-rounds-index-list-hit s query-idxs rounds out
  by (rule accepted-fri-opening-transcript-imp-query-rounds-index-list-hit
    [OF openings])
show ∃ query-idxs ∈ Q.
  query-rounds-index-list-hit s query-idxs rounds out
  using query-in query-hit by blast
qed
show
  wp-event (verifier-after-trace-fri header)
  (λout. trace-fri-challenge-list-set-hit s {challenges} out ∧
    trace-fri-query-challenge-pair-set-hit s P out) prefix-state
  ≤ ?D
  by (rule order-trans[OF hit-to-query event-bound])
qed
have head-bound:
  wp-event verifier-trace-fri-prefix ?Head s ≤
    1 / nnreal (CARD('f) ^ ceil-log clength)
proof –
  have wp-event verifier-trace-fri-prefix ?Head s ≤
    nnreal (card {challenges}) / nnreal (CARD('f) ^ ceil-log clength)
  by (rule wp-verifier-trace-fri-prefix-challenge-space-set-bound
    [where B = {challenges}, OF trace-future])
    (use challenges-space in auto)
  then show ?thesis by simp
qed
show ?thesis
  unfolding verify-monad-trace-fri-decomposition
  apply (rule order-trans)
  apply (rule bind-bound)
  apply (rule mult-right-mono[OF head-bound])
  apply simp
done

```

qed

lemma *wp-trace-fri-query-challenge-pair-set-hit-bound-from-query-fiber-sum:*

assumes *trace-future: trace-fri-future-fresh s*
and *query-future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *projection:*
 $\bigwedge \text{challenges.}$
 $\text{fst } ' (P \cap (UNIV \times \{\text{challenges}\})) \subseteq$
 $Q \text{ challenges}$
and *subset: $\bigwedge \text{challenges. } Q \text{ challenges} \subseteq \text{fri-query-index-list-space}$*
shows

wp-event verify-monad
 $(\text{trace-fri-query-challenge-pair-set-hit } s \ P) \ s \leq$
 $(\sum \text{challenges} \in \text{fri-challenge-space } (\text{ceil-log } \text{clength}).$
 $(1 / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})) *$
 $(\text{nnreal } (\text{card } (Q \text{ challenges})) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}))$

proof –

let *?I = fri-challenge-space (ceil-log clength)*
let *?E =*
 $\lambda \text{challenges out.}$
 $\text{trace-fri-challenge-list-set-hit } s \ \{\text{challenges}\} \ \text{out} \wedge$
 $\text{trace-fri-query-challenge-pair-set-hit } s \ P \ \text{out}$
let *?B =*
 $\lambda \text{challenges.}$
 $(1 / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength})) *$
 $(\text{nnreal } (\text{card } (Q \text{ challenges})) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}))$

have *split:*

wp-event verify-monad
 $(\text{trace-fri-query-challenge-pair-set-hit } s \ P) \ s \leq$
wp-event verify-monad
 $(\lambda \text{out. } \exists \text{challenges} \in ?I. ?E \text{ challenges out}) \ s$

proof (rule *wp-event-mono*)

fix *out*
assume *hit: trace-fri-query-challenge-pair-set-hit s P out*
from *hit* **obtain** *trace-roots trace-bs trace-final fri-dg*
 $\text{composition-roots composition-bs composition-final query-idxs}$
 $\text{trace-round-layers composition-round-layers}$ **where**
openings:
 $\text{accepted-fri-opening-transcript } s \ \text{out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final query-idxs trace-round-layers}$
 $\text{composition-round-layers}$
unfolding *trace-fri-query-challenge-pair-set-hit-def* **by** *blast*
have *challenges:*
 $\text{accepted-fri-challenges } s \ \text{out trace-bs fri-dg composition-bs}$
using *openings*

```

unfolding accepted-fri-opening-transcript-def
apply (elim exE conjE)
apply assumption
done
have trace-space: trace-bs ∈ ?I
  by (rule accepted-fri-challenges-trace-space[OF challenges])
have challenge-hit:
  trace-fri-challenge-list-set-hit s {trace-bs} out
  unfolding trace-fri-challenge-list-set-hit-def
  using challenges by blast
show ∃ challenges ∈ ?I. ?E challenges out
  using trace-space challenge-hit hit by blast
qed
also have ... ≤ (∑ challenges ∈ ?I. ?B challenges)
proof (rule wp-event-finite-union-bound-fri-exact)
  show finite ?I
    by (rule finite-fri-challenge-space)
next
  fix challenges
  assume challenges: challenges ∈ ?I
  show
    wp-event verify-monad (?E challenges) s ≤ ?B challenges
    by (rule wp-trace-fri-query-challenge-pair-set-hit-fixed-challenge-bound
      [OF trace-future query-future raw-bound challenges
        projection subset])
qed
finally show ?thesis .
qed

lemma wp-trace-fri-query-challenge-pair-set-hit-bound-from-query-fibers:
fixes C :: prob
assumes future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and projection:
   $\bigwedge \text{challenges.}$ 
   $\text{fst } '(P \cap (UNIV \times \{\text{challenges}\})) \subseteq$ 
   $Q \text{ challenges}$ 
and subset:  $\bigwedge \text{challenges. } Q \text{ challenges} \subseteq \text{fri-query-index-list-space}$ 
and bound:
   $\bigwedge \text{challenges.}$ 
   $\text{nnreal } (\text{card } (Q \text{ challenges})) *$ 
   $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds} \leq C$ 
shows
  wp-event verify-monad
   $(\text{trace-fri-query-challenge-pair-set-hit } s \ P) \ s \leq C$ 
unfolding verify-monad-trace-fri-decomposition
proof (rule wp-event-bind-bound-by-cont)
show  $\neg \text{trace-fri-query-challenge-pair-set-hit } s \ P \ \text{None}$ 
unfolding trace-fri-query-challenge-pair-set-hit-def

```

```

    accepted-fri-opening-transcript-def
  by blast
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
obtain fr f-fl where header-eq: header = (fr, f-fl)
  by (cases header) auto
have future-prefix: query-future-fresh prefix-state
  by (rule verifier-trace-fri-prefix-preserves-query-future-fresh-exact
    [OF future prefix])
have counter-prefix: PQueryCounter prefix-state = PQueryCounter s
  using verifier-trace-fri-prefix-outcome[OF prefix[unfolded header-eq]]
  by simp
let ?Q = Q (map fst f-fl)
have event-bound:
  wp-event (verifier-after-trace-fri header)
    (λout. ∃ query-idxs ∈ ?Q.
      query-rounds-index-list-hit s query-idxs rounds out)
    prefix-state ≤
    nnreal (card ?Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds
  by (rule wp-verifier-after-trace-fri-query-index-list-set-bound
    [OF future-prefix counter-prefix raw-bound subset])
have pair-to-query:
  wp-event (verifier-after-trace-fri header)
    (trace-fri-query-challenge-pair-set-hit s P) prefix-state ≤
  wp-event (verifier-after-trace-fri header)
    (λout. ∃ query-idxs ∈ ?Q.
      query-rounds-index-list-hit s query-idxs rounds out) prefix-state
proof (rule wp-event-mono-on-support)
fix out
assume suffix:
  out ∈
    set-dist (execute (verifier-after-trace-fri header) prefix-state)
  and hit: trace-fri-query-challenge-pair-set-hit s P out
from hit obtain trace-roots trace-bs trace-final fri-dg
  composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers where
  openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final query-idxs trace-round-layers
    composition-round-layers
  and pair: (query-idxs, trace-bs) ∈ P
  unfolding trace-fri-query-challenge-pair-set-hit-def by blast
from openings obtain result final-state where out-eq:
  out = Some (result, final-state)

```

```

unfolding accepted-fri-opening-transcript-def by blast
have prefix':
  Some ((fr, f-fl), prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
using prefix header-eq by simp
have suffix':
  Some (result, final-state) ∈
    set-dist
      (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
using suffix header-eq out-eq by simp
from verifier-after-trace-fri-accepted-fri-challenges[OF prefix' suffix']
obtain dg' comp-bs' where challenges-prefix:
  accepted-fri-challenges s (Some (result, final-state))
  (map fst f-fl) dg' comp-bs'
by blast
have challenges-openings:
  accepted-fri-challenges s out trace-bs fri-dg composition-bs
using openings
unfolding accepted-fri-opening-transcript-def
apply (elim exE conjE)
apply assumption
done
have trace-eq: trace-bs = map fst f-fl
using accepted-fri-challenges-unique
  [OF challenges-prefix challenges-openings[unfolded out-eq]]
by simp
have query-in: query-idxs ∈ ?Q
proof –
  have query-idxs ∈
    fst ‘ (P ∩ (UNIV × {map fst f-fl}))
  using pair trace-eq by force
then show ?thesis
  using projection[of map fst f-fl] by blast
qed
have query-hit:
  query-rounds-index-list-hit s query-idxs rounds out
by (rule accepted-fri-opening-transcript-imp-query-rounds-index-list-hit
  [OF openings])
show  $\exists$  query-idxs ∈ ?Q.
  query-rounds-index-list-hit s query-idxs rounds out
using query-in query-hit by blast
qed
show
  wp-event (verifier-after-trace-fri header)
  (trace-fri-query-challenge-pair-set-hit s P) prefix-state ≤ C
by (rule order-trans[OF pair-to-query])
  (rule order-trans[OF event-bound bound])
qed

```

lemma *wp-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-fibers:*
fixes *C-challenge C-query :: prob*
assumes *trace-future: trace-fri-future-fresh s*
and *query-future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *challenge-subset: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$*
and *challenge-bound:*
 $\text{nnreal } (\text{card } B) / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } \text{clength}) \leq$
 C-challenge
and *projection:*
 $\bigwedge \text{challenges.}$
 $\text{fst } ' ((P \cap (\text{UNIV} \times (- B))) \cap$
 $(\text{UNIV} \times \{\text{challenges}\})) \subseteq Q \text{ challenges}$
and *subset: $\bigwedge \text{challenges. } Q \text{ challenges} \subseteq \text{fri-query-index-list-space}$*
and *query-bound:*
 $\bigwedge \text{challenges.}$
 $\text{nnreal } (\text{card } (Q \text{ challenges})) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds} \leq \text{C-query}$
shows
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-query-challenge-pair-set-hit } s P) s \leq$
 $\text{C-challenge} + \text{C-query}$
proof –
let *?Outside = $P \cap (\text{UNIV} \times (- B))$*
have *split:*
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-query-challenge-pair-set-hit } s P) s \leq$
 $\text{wp-event verify-monad}$
 $(\lambda \text{out. trace-fri-challenge-list-set-hit } s B \text{ out} \vee$
 $\text{trace-fri-query-challenge-pair-set-hit } s ?\text{Outside out}) s$
proof (rule *wp-event-mono*)
fix *out*
assume *hit: trace-fri-query-challenge-pair-set-hit s P out*
from *hit obtain* *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers **where**
openings:
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final query-idxs trace-round-layers}$
 $\text{composition-round-layers}$
and *pair: $(\text{query-idxs}, \text{trace-bs}) \in P$*
unfolding *trace-fri-query-challenge-pair-set-hit-def* **by** *blast*
show
 $\text{trace-fri-challenge-list-set-hit } s B \text{ out} \vee$
 $\text{trace-fri-query-challenge-pair-set-hit } s ?\text{Outside out}$
proof (cases *trace-bs* $\in B$)
case *True*
have *challenges:*


```

    accepted-fri-challenges s out trace-bs fri-dg composition-bs
  using openings
  unfolding accepted-fri-opening-transcript-def
  apply (elim exE conjE)
  apply assumption
  done
then have trace-fri-challenge-list-set-hit s B out
  using True unfolding trace-fri-challenge-list-set-hit-def by blast
then show ?thesis by simp
next
case False
have (query-idxs, trace-bs) ∈ ?Outside
  using pair False by simp
then have trace-fri-query-challenge-pair-set-hit s ?Outside out
  unfolding trace-fri-query-challenge-pair-set-hit-def
  using openings by blast
then show ?thesis by simp
qed
qed
also have ... ≤
  wp-event verify-monad (trace-fri-challenge-list-set-hit s B) s +
  wp-event verify-monad
    (trace-fri-query-challenge-pair-set-hit s ?Outside) s
  by (rule wp-event-union-bound)
also have ... ≤ C-challenge + C-query
proof (rule add-mono)
  show wp-event verify-monad (trace-fri-challenge-list-set-hit s B) s
    ≤ C-challenge
  by (rule order-trans[
    OF wp-verify-monad-trace-fri-challenge-list-set-bound
    [OF trace-future challenge-subset] challenge-bound])
  show
    wp-event verify-monad
      (trace-fri-query-challenge-pair-set-hit s ?Outside) s ≤
      C-query
  by (rule wp-trace-fri-query-challenge-pair-set-hit-bound-from-query-fibers
    [OF query-future raw-bound projection subset query-bound])
qed
finally show ?thesis .
qed

lemma
  wp-composition-fri-query-challenge-pair-set-hit-bound-from-query-fibers:
  fixes C :: prob
  assumes future: query-future-fresh s
  and raw-bound: query-index-raw-preimage-bound
  and projection:
     $\bigwedge dg \text{ challenges. } fst \text{ ' } (P \ dg \ \cap \ (UNIV \times \ \{challenges\})) \subseteq$ 

```

Q dg challenges
and subset: $\bigwedge$ dg challenges. Q dg challenges $\subseteq$ fri-query-index-list-space
and bound:
 $\bigwedge$ dg challenges.
 $\text{nnreal}(\text{card}(Q \text{ dg challenges})) * (1 / \text{nnreal}(\text{card query-sample-space})) ^ \wedge \text{rounds} \leq C$
shows
 $\text{wp-event verify-monad} (\text{composition-fri-query-challenge-pair-set-hit } s P) s \leq C$
unfolding verify-monad-composition-fri-decomposition
proof (rule wp-event-bind-bound-by-cont)
show $\neg \text{composition-fri-query-challenge-pair-set-hit } s P \text{ None}$
unfolding composition-fri-query-challenge-pair-set-hit-def
accepted-fri-opening-transcript-def
by blast
next
fix header prefix-state
assume prefix:
Some (header, prefix-state) $\in$
set-dist (execute verifier-composition-fri-prefix s)
obtain fr f-fl f-final as dg fl **where** header-eq:
header = (fr, f-fl, f-final, as, dg, fl)
by (cases header) auto
have future-prefix: query-future-fresh prefix-state
by (rule verifier-composition-fri-prefix-preserves-query-future-fresh [OF future prefix])
have counter-prefix: PQueryCounter prefix-state = PQueryCounter s
using verifier-composition-fri-prefix-outcome[OF prefix[unfolded header-eq]]
by simp
let ?Q = Q dg (map fst fl)
have event-bound:
wp-event (verifier-after-composition-fri header)
(λ out. $\exists$ query-idxs $\in$?Q.
query-rounds-index-list-hit s query-idxs rounds out)
prefix-state $\leq$
 $\text{nnreal}(\text{card } ?Q) * (1 / \text{nnreal}(\text{card query-sample-space})) ^ \wedge \text{rounds}$
by (rule wp-verifier-after-composition-fri-query-index-list-set-bound [OF future-prefix counter-prefix raw-bound subset])
have pair-to-query:
wp-event (verifier-after-composition-fri header)
(composition-fri-query-challenge-pair-set-hit s P) prefix-state $\leq$
wp-event (verifier-after-composition-fri header)
(λ out. $\exists$ query-idxs $\in$?Q.
query-rounds-index-list-hit s query-idxs rounds out) prefix-state
proof (rule wp-event-mono-on-support)
fix out
assume suffix:
out $\in$

```

    set-dist (execute (verifier-after-composition-fri header)
      prefix-state)
  and hit: composition-fri-query-challenge-pair-set-hit s P out
from hit obtain trace-roots trace-bs trace-final fri-dg
  composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers where
openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final fri-dg composition-roots composition-bs
  composition-final query-idxs trace-round-layers
  composition-round-layers
  and pair: (query-idxs, composition-bs) ∈ P fri-dg
  unfolding composition-fri-query-challenge-pair-set-hit-def by blast
from openings obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-fri-opening-transcript-def by blast
have prefix':
  Some ((fr, f-fl, f-final, as, dg, fl), prefix-state) ∈
    set-dist (execute verifier-composition-fri-prefix s)
  using prefix header-eq by simp
have suffix':
  Some (result, final-state) ∈
    set-dist
      (execute
        (verifier-after-composition-fri
          (fr, f-fl, f-final, as, dg, fl))
        prefix-state)
  using suffix header-eq out-eq by simp
have challenges-prefix:
  accepted-fri-challenges s (Some (result, final-state))
  (map fst f-fl) dg (map fst fl)
  by (rule verifier-after-composition-fri-accepted-fri-challenges
    [OF prefix' suffix'])
have challenges-openings:
  accepted-fri-challenges s out trace-bs fri-dg composition-bs
  using openings
  unfolding accepted-fri-opening-transcript-def
  apply (elim exE conjE)
  apply assumption
  done
have eqs: fri-dg = dg ∧ composition-bs = map fst fl
  using accepted-fri-challenges-unique
  [OF challenges-prefix challenges-openings[unfolded out-eq]]
  by simp
have query-in: query-idxs ∈ ?Q
proof -
  have query-idxs ∈
    fst ' (P dg ∩ (UNIV × {map fst fl}))
  using pair eqs by force

```

```

    then show ?thesis
      using projection[of dg map fst fl] by blast
  qed
  have query-hit:
    query-rounds-index-list-hit s query-idxs rounds out
    by (rule accepted-fri-opening-transcript-imp-query-rounds-index-list-hit
        [OF openings])
  show  $\exists$  query-idxs  $\in$  ?Q.
    query-rounds-index-list-hit s query-idxs rounds out
    using query-in query-hit by blast
  qed
  show
    wp-event (verifier-after-composition-fri header)
      (composition-fri-query-challenge-pair-set-hit s P) prefix-state  $\leq$  C
  by (rule order-trans[OF pair-to-query])
    (rule order-trans[OF event-bound bound])
  qed

lemma wp-composition-fri-query-challenge-pair-set-hit-fixed-challenge-bound:
assumes composition-future: composition-fri-future-fresh s
and query-future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and challenges-space:
  challenges  $\in$  fri-challenge-space (ceil-log (to-nat dg + 1))
and projection:
  fst ‘ (P dg  $\cap$  (UNIV  $\times$  {challenges}))  $\subseteq$  Q
and subset: Q  $\subseteq$  fri-query-index-list-space
shows
  wp-event verify-monad
    ( $\lambda$ out.
      composition-fri-challenge-list-set-hit s
        ( $\lambda$ dg'. if dg' = dg then {challenges} else {}) out  $\wedge$ 
        composition-fri-query-challenge-pair-set-hit s P out) s  $\leq$ 
      (1 / nnreal (CARD('f)  $\wedge$  ceil-log (to-nat dg + 1))) *
      (nnreal (card Q) *
        (1 / nnreal (card query-sample-space))  $\wedge$  rounds)
proof –
  let ?B =  $\lambda$ dg'. if dg' = dg then {challenges} else {}
  let ?Hit =
     $\lambda$ out. composition-fri-challenge-list-set-hit s ?B out  $\wedge$ 
      composition-fri-query-challenge-pair-set-hit s P out
  let ?Head =
     $\lambda$ out. case out of
      None  $\Rightarrow$  False
    | Some ((-, -, -, dg', fl), -)  $\Rightarrow$ 
      dg' = dg  $\wedge$  map fst fl = challenges
  let ?D =
    nnreal (card Q) *
    (1 / nnreal (card query-sample-space))  $\wedge$  rounds

```

```

have bind-bound:
  wp-event
    (verifier-composition-fri-prefix  $\gg$  verifier-after-composition-fri)
    ?Hit  $s \leq$  wp-event verifier-composition-fri-prefix ?Head  $s * ?D$ 
proof (rule wp-event-bind-bound-by-head-and-cont)
show  $\neg$  ?Hit None
  unfolding composition-fri-challenge-list-set-hit-def
    composition-fri-query-challenge-pair-set-hit-def
    accepted-fri-opening-transcript-def
  by blast
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state)  $\in$ 
    set-dist (execute verifier-composition-fri-prefix  $s$ )
and not-head:
   $\neg$  (case Some (header, prefix-state) of None  $\Rightarrow$  False
    | Some ((-, -, -, -, dg', fl), -)  $\Rightarrow$ 
      dg' = dg  $\wedge$  map fst fl = challenges)
obtain fr f-fl f-final as dg' fl where header-eq:
  header = (fr, f-fl, f-final, as, dg', fl)
by (cases header) auto
have not-fixed: dg'  $\neq$  dg  $\vee$  map fst fl  $\neq$  challenges
using not-head unfolding header-eq by simp
have zero-le:
  wp-event (verifier-after-composition-fri header)
    ( $\lambda$ out.
      composition-fri-challenge-list-set-hit  $s$ 
      ( $\lambda$ dg'. if dg' = dg then {challenges} else {}) out  $\wedge$ 
      composition-fri-query-challenge-pair-set-hit  $s$  P out) prefix-state
     $\leq 0$ 
proof (rule order-trans)
show
  wp-event (verifier-after-composition-fri header)
    ( $\lambda$ out.
      composition-fri-challenge-list-set-hit  $s$ 
      ( $\lambda$ dg'. if dg' = dg then {challenges} else {}) out  $\wedge$ 
      composition-fri-query-challenge-pair-set-hit  $s$  P out) prefix-state
     $\leq$ 
  wp-event (verifier-after-composition-fri header)
    ( $\lambda$ -. False) prefix-state
proof (rule wp-event-mono-on-support)
fix out
assume suffix:
  out  $\in$ 
    set-dist
      (execute (verifier-after-composition-fri header) prefix-state)
and hit:
  composition-fri-challenge-list-set-hit  $s$ 

```

```

    ( $\lambda dg'.$  if  $dg' = dg$  then  $\{\text{challenges}\}$  else  $\{\}$ ) out  $\wedge$ 
    composition-fri-query-challenge-pair-set-hit  $s$   $P$  out
from hit obtain trace-bs fri-dg composition-bs where
    challenges-hit:
        accepted-fri-challenges  $s$  out trace-bs fri-dg composition-bs
    and comp-in:
        composition-bs  $\in$ 
        (if fri-dg = dg then  $\{\text{challenges}\}$  else  $\{\}$ )
    unfolding composition-fri-challenge-list-set-hit-def
    by blast
from challenges-hit obtain result final-state where out-eq:
    out = Some (result, final-state)
    unfolding accepted-fri-challenges-def by blast
have prefix':
    Some ((fr, f-fl, f-final, as, dg', fl), prefix-state)  $\in$ 
    set-dist (execute verifier-composition-fri-prefix  $s$ )
    using prefix header-eq by simp
have suffix':
    Some (result, final-state)  $\in$ 
    set-dist
    (execute
    (verifier-after-composition-fri
    (fr, f-fl, f-final, as, dg', fl))
    prefix-state)
    using suffix header-eq out-eq by simp
have challenges-prefix:
    accepted-fri-challenges  $s$  (Some (result, final-state))
    (map fst f-fl) dg' (map fst fl)
    by (rule verifier-after-composition-fri-accepted-fri-challenges
    [OF prefix' suffix'])
have eqs: fri-dg = dg'  $\wedge$  composition-bs = map fst fl
    using accepted-fri-challenges-unique
    [OF challenges-prefix challenges-hit[unfolding out-eq]]
    by simp
have fixed: fri-dg = dg  $\wedge$  composition-bs = challenges
    using comp-in by (cases fri-dg = dg) auto
show False
    using not-fixed eqs fixed by simp
qed
next
show
    wp-event (verifier-after-composition-fri header)
    ( $\lambda.$  False) prefix-state  $\leq 0$ 
    unfolding wp-event-def wp-def dist-expect-def by simp
qed
show
    wp-event (verifier-after-composition-fri header)
    ( $\lambda out.$ 
    composition-fri-challenge-list-set-hit  $s$ 

```

```

      ( $\lambda dg'. \text{ if } dg' = dg \text{ then } \{\text{challenges}\} \text{ else } \{\}\rangle \text{ out} \wedge$ 
       $\text{composition-fri-query-challenge-pair-set-hit } s \ P \text{ out}) \text{ prefix-state}$ 
      = 0
    by (rule antisym[OF zero-le]) simp
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state)  $\in$ 
  set-dist (execute verifier-composition-fri-prefix s)
and head:
  (case Some (header, prefix-state) of None  $\Rightarrow$  False
   | Some ((-, -, -, -, dg', fl), -)  $\Rightarrow$ 
     dg' = dg  $\wedge$  map fst fl = challenges)
obtain fr f-fl f-final as dg' fl where header-eq:
  header = (fr, f-fl, f-final, as, dg', fl)
by (cases header) auto
have dg-eq: dg' = dg and fl-eq: map fst fl = challenges
  using head unfolding header-eq by simp-all
have future-prefix: query-future-fresh prefix-state
  by (rule verifier-composition-fri-prefix-preserves-query-future-fresh
      [OF query-future prefix])
have counter-prefix: PQueryCounter prefix-state = PQueryCounter s
  using verifier-composition-fri-prefix-outcome[OF prefix[unfolded header-eq]]
  by simp
have event-bound:
  wp-event (verifier-after-composition-fri header)
  ( $\lambda \text{out}. \exists \text{query-idxs} \in Q.$ 
    query-rounds-index-list-hit s query-idxs rounds out)
  prefix-state  $\leq$  ?D
  by (rule wp-verifier-after-composition-fri-query-index-list-set-bound
      [OF future-prefix counter-prefix raw-bound subset])
have hit-to-query:
  wp-event (verifier-after-composition-fri header)
  ( $\lambda \text{out}.$ 
    composition-fri-challenge-list-set-hit s
    ( $\lambda dg'. \text{ if } dg' = dg \text{ then } \{\text{challenges}\} \text{ else } \{\}\rangle \text{ out} \wedge$ 
    composition-fri-query-challenge-pair-set-hit s P out) prefix-state
     $\leq$ 
    wp-event (verifier-after-composition-fri header)
    ( $\lambda \text{out}. \exists \text{query-idxs} \in Q.$ 
      query-rounds-index-list-hit s query-idxs rounds out) prefix-state
proof (rule wp-event-mono-on-support)
fix out
assume suffix:
  out  $\in$ 
  set-dist (execute (verifier-after-composition-fri header)
    prefix-state)
and hit:
  composition-fri-challenge-list-set-hit s

```

$(\lambda dg'. \text{ if } dg' = dg \text{ then } \{\text{challenges}\} \text{ else } \{\}) \text{ out } \wedge$
 $\text{composition-fri-query-challenge-pair-set-hit } s \text{ } P \text{ out}$
from *hit* **obtain** *trace-roots trace-bs trace-final fri-dg*
 $\text{composition-roots composition-bs composition-final query-idxs}$
 $\text{trace-round-layers composition-round-layers}$ **where**
openings:
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final query-idxs trace-round-layers}$
 $\text{composition-round-layers}$
and *pair*: $(\text{query-idxs}, \text{composition-bs}) \in P \text{ fri-dg}$
unfolding *composition-fri-query-challenge-pair-set-hit-def* **by** *blast*
from *openings* **obtain** *result final-state* **where** *out-eq*:
 $\text{out} = \text{Some } (\text{result}, \text{final-state})$
unfolding *accepted-fri-opening-transcript-def* **by** *blast*
have *prefix'*:
 $\text{Some } ((\text{fr}, \text{f-fl}, \text{f-final}, \text{as}, \text{dg}', \text{fl}), \text{prefix-state}) \in$
 $\text{set-dist } (\text{execute verifier-composition-fri-prefix } s)$
using *prefix header-eq* **by** *simp*
have *suffix'*:
 $\text{Some } (\text{result}, \text{final-state}) \in$
 set-dist
 $(\text{execute}$
 $(\text{verifier-after-composition-fri}$
 $(\text{fr}, \text{f-fl}, \text{f-final}, \text{as}, \text{dg}', \text{fl}))$
 $\text{prefix-state})$
using *suffix header-eq out-eq* **by** *simp*
have *challenges-prefix*:
 $\text{accepted-fri-challenges } s \text{ } (\text{Some } (\text{result}, \text{final-state}))$
 $(\text{map fst f-fl}) \text{ dg}' (\text{map fst fl})$
by $(\text{rule verifier-after-composition-fri-accepted-fri-challenges}$
 $[\text{OF prefix' suffix'}])$
have *challenges-openings*:
 $\text{accepted-fri-challenges } s \text{ out trace-bs fri-dg composition-bs}$
using *openings*
unfolding *accepted-fri-opening-transcript-def*
apply (elim exE conjE)
apply *assumption*
done
have *eqs*: $\text{fri-dg} = \text{dg}' \wedge \text{composition-bs} = \text{map fst fl}$
using *accepted-fri-challenges-unique*
 $[\text{OF challenges-prefix challenges-openings}[\text{unfolded out-eq}]]$
by *simp*
have *query-in*: $\text{query-idxs} \in Q$
proof –
have $\text{query-idxs} \in \text{fst } ' (P \text{ dg} \cap (UNIV \times \{\text{challenges}\}))$
using *pair eqs dg-eq fl-eq* **by** *force*
then show *?thesis*
using *projection* **by** *blast*


```

qed
have query-hit:
  query-rounds-index-list-hit s query-idxs rounds out
  by (rule accepted-fri-opening-transcript-imp-query-rounds-index-list-hit
      [OF openings])
show  $\exists$  query-idxs  $\in Q$ .
  query-rounds-index-list-hit s query-idxs rounds out
  using query-in query-hit by blast
qed
show
  wp-event (verifier-after-composition-fri header)
    ( $\lambda$ out.
      composition-fri-challenge-list-set-hit s
        ( $\lambda$ dg'. if dg' = dg then {challenges} else {}) out  $\wedge$ 
        composition-fri-query-challenge-pair-set-hit s P out) prefix-state
     $\leq$  ?D
  by (rule order-trans[OF hit-to-query event-bound])
qed
have head-bound:
  wp-event verifier-composition-fri-prefix ?Head s  $\leq$ 
    1 / nnreal (CARD('f)  $\wedge$  ceil-log (to-nat dg + 1))
proof (rule order-trans)
  let ?SetHead =
     $\lambda$ out. case out of
      None  $\Rightarrow$  False
    | Some ((-, -, -, dg', fl), -)  $\Rightarrow$  map fst fl  $\in$  ?B dg'
  show wp-event verifier-composition-fri-prefix ?Head s  $\leq$ 
    wp-event verifier-composition-fri-prefix ?SetHead s
  by (rule wp-event-mono-on-support)
  (auto split: option.splits prod.splits)
  show wp-event verifier-composition-fri-prefix ?SetHead s  $\leq$ 
    1 / nnreal (CARD('f)  $\wedge$  ceil-log (to-nat dg + 1))
  by (rule wp-verifier-composition-fri-prefix-challenge-space-set-bound
      [OF composition-future])
  (use challenges-space in auto)
qed
show ?thesis
  unfolding verify-monad-composition-fri-decomposition
  apply (rule order-trans)
  apply (rule bind-bound)
  apply (rule mult-right-mono[OF head-bound])
  apply simp
done
qed

lemma wp-composition-fri-query-challenge-pair-set-hit-bound-from-query-fiber-sum:
  assumes composition-future: composition-fri-future-fresh s
  and query-future: query-future-fresh s
  and raw-bound: query-index-raw-preimage-bound

```

and *projection*:
 $\bigwedge dg \text{ challenges.}$
 $\text{fst } ' (P \text{ dg} \cap (UNIV \times \{\text{challenges}\})) \subseteq$
 $Q \text{ dg challenges}$
and *subset*:
 $\bigwedge dg \text{ challenges. } Q \text{ dg challenges} \subseteq \text{fri-query-index-list-space}$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-query-challenge-pair-set-hit } s \text{ } P) \text{ } s \leq$
 $(\sum dg \in UNIV.$
 $\sum \text{challenges} \in \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1)).$
 $(1 / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1))) *$
 $(\text{nnreal } (\text{card } (Q \text{ dg challenges})) *$
 $(1 / \text{nnreal } (\text{card query-sample-space})) \wedge \text{rounds}))$
proof –
let $?I =$
 $SIGMA \text{ dg:} UNIV. \text{ fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
let $?E =$
 $\lambda p \text{ out.}$
 $\text{composition-fri-challenge-list-set-hit } s$
 $(\lambda dg'. \text{ if } dg' = \text{fst } p \text{ then } \{\text{snd } p\} \text{ else } \{\}) \text{ out} \wedge$
 $\text{composition-fri-query-challenge-pair-set-hit } s \text{ } P \text{ out}$
let $?B =$
 $\lambda p.$
 $(1 / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } (\text{fst } p) + 1))) *$
 $(\text{nnreal } (\text{card } (Q (\text{fst } p) (\text{snd } p)))) *$
 $(1 / \text{nnreal } (\text{card query-sample-space})) \wedge \text{rounds})$
have *split*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-query-challenge-pair-set-hit } s \text{ } P) \text{ } s \leq$
 $\text{wp-event verify-monad}$
 $(\lambda out. \exists p \in ?I. ?E \text{ } p \text{ out}) \text{ } s$
proof (rule *wp-event-mono*)
fix out
assume $hit: \text{composition-fri-query-challenge-pair-set-hit } s \text{ } P \text{ out}$
from hit **obtain** $\text{trace-roots trace-bs trace-final fri-dg}$
 $\text{composition-roots composition-bs composition-final query-idxs}$
 $\text{trace-round-layers composition-round-layers}$ **where**
 openings:
 $\text{accepted-fri-opening-transcript } s \text{ out trace-roots trace-bs}$
 $\text{trace-final fri-dg composition-roots composition-bs}$
 $\text{composition-final query-idxs trace-round-layers}$
 $\text{composition-round-layers}$
unfolding $\text{composition-fri-query-challenge-pair-set-hit-def}$ **by** blast
have *challenges*:
 $\text{accepted-fri-challenges } s \text{ out trace-bs fri-dg composition-bs}$
using openings
unfolding $\text{accepted-fri-opening-transcript-def}$
apply (elim exE conjE)

```

    apply assumption
  done
have comp-space:
  composition-bs ∈ fri-challenge-space (ceil-log (to-nat fri-dg + 1))
  by (rule accepted-fri-challenges-composition-space[OF challenges])
have challenge-hit:
  composition-fri-challenge-list-set-hit s
  (λdg'. if dg' = fri-dg then {composition-bs} else {}) out
  unfolding composition-fri-challenge-list-set-hit-def
  using challenges by auto
have (fri-dg, composition-bs) ∈ ?I
  using comp-space by simp
then show ∃ p ∈ ?I. ?E p out
proof (intro bexI[where x = (fri-dg, composition-bs)])
  show ?E (fri-dg, composition-bs) out
    using challenge-hit hit by (simp only: fst-conv snd-conv)
  show (fri-dg, composition-bs) ∈ ?I
    using comp-space by simp
qed
qed
also have ... ≤ (∑ p ∈ ?I. ?B p)
proof (rule wp-event-finite-union-bound-fri-exact)
  show finite ?I
    by (simp add: finite-fri-challenge-space)
next
fix p
assume p-in: p ∈ ?I
obtain dg challenges where p-eq: p = (dg, challenges)
  by (cases p)
have challenges:
  challenges ∈ fri-challenge-space (ceil-log (to-nat dg + 1))
  using p-in p-eq by simp
have fixed-bound:
  wp-event verify-monad
  (λout. composition-fri-challenge-list-set-hit s
    (λdg'. if dg' = dg then {challenges} else {}) out ∧
    composition-fri-query-challenge-pair-set-hit s P out) s
  ≤ 1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) *
    (nnreal (card (Q dg challenges)) *
    (1 / nnreal (card query-sample-space)) ^ rounds)
  by (rule
    wp-composition-fri-query-challenge-pair-set-hit-fixed-challenge-bound
    [OF composition-future query-future raw-bound challenges
    projection subset])
show wp-event verify-monad (?E p) s ≤ ?B p
  unfolding p-eq using fixed-bound by (simp only: fst-conv snd-conv)
qed
also have ... =
  (∑ dg ∈ UNIV.

```

$\sum \text{challenges} \in \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1)).$
 $(1 / \text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log } (\text{to-nat } dg + 1))) *$
 $(\text{nnreal } (\text{card } (Q \text{ } dg \text{ } \text{challenges}))) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}))$
by (*subst sum.Sigma*) (*simp-all add: finite-fri-challenge-space case-prod-beta*)
finally show ?thesis .
qed

lemma *wp-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-fibers:*
fixes *C-challenge C-query :: prob*
assumes *composition-future: composition-fri-future-fresh s*
and *query-future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *challenge-subset:*
 $\bigwedge dg. B \text{ } dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and *challenge-bound:*
 $\bigwedge dg. \text{nnreal } (\text{card } (B \text{ } dg)) /$
 $\text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C\text{-challenge}$
and *projection:*
 $\bigwedge dg \text{ } \text{challenges.}$
 $\text{fst } ' ((P \text{ } dg \cap (UNIV \times (- B \text{ } dg))) \cap$
 $(UNIV \times \{\text{challenges}\})) \subseteq Q \text{ } dg \text{ } \text{challenges}$
and *subset:*
 $\bigwedge dg \text{ } \text{challenges. } Q \text{ } dg \text{ } \text{challenges} \subseteq \text{fri-query-index-list-space}$
and *query-bound:*
 $\bigwedge dg \text{ } \text{challenges.}$
 $\text{nnreal } (\text{card } (Q \text{ } dg \text{ } \text{challenges})) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds} \leq C\text{-query}$
shows
wp-event verify-monad
 $(\text{composition-fri-query-challenge-pair-set-hit } s \text{ } P) \text{ } s \leq$
 $C\text{-challenge} + C\text{-query}$
proof –
let ?Outside = $\lambda dg. P \text{ } dg \cap (UNIV \times (- B \text{ } dg))$
have *split:*
wp-event verify-monad
 $(\text{composition-fri-query-challenge-pair-set-hit } s \text{ } P) \text{ } s \leq$
wp-event verify-monad
 $(\lambda out. \text{composition-fri-challenge-list-set-hit } s \text{ } B \text{ } out \vee$
 $\text{composition-fri-query-challenge-pair-set-hit } s \text{ } ?\text{Outside } out) \text{ } s$
proof (*rule wp-event-mono*)
fix *out*
assume *hit: composition-fri-query-challenge-pair-set-hit s P out*
from *hit obtain* *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers **where**
openings:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs

```

    composition-final query-idxs trace-round-layers
    composition-round-layers
  and pair: (query-idxs, composition-bs) ∈ P fri-dg
  unfolding composition-fri-query-challenge-pair-set-hit-def by blast
show
  composition-fri-challenge-list-set-hit s B out ∨
  composition-fri-query-challenge-pair-set-hit s ?Outside out
proof (cases composition-bs ∈ B fri-dg)
  case True
  have challenges:
    accepted-fri-challenges s out trace-bs fri-dg composition-bs
  using openings
  unfolding accepted-fri-opening-transcript-def
  apply (elim exE conjE)
  apply assumption
  done
  then have composition-fri-challenge-list-set-hit s B out
  using True unfolding composition-fri-challenge-list-set-hit-def
  by blast
  then show ?thesis by simp
next
  case False
  have (query-idxs, composition-bs) ∈ ?Outside fri-dg
  using pair False by simp
  then have
    composition-fri-query-challenge-pair-set-hit s ?Outside out
  unfolding composition-fri-query-challenge-pair-set-hit-def
  using openings by blast
  then show ?thesis by simp
qed
qed
also have ... ≤
  wp-event verify-monad (composition-fri-challenge-list-set-hit s B) s +
  wp-event verify-monad
  (composition-fri-query-challenge-pair-set-hit s ?Outside) s
  by (rule wp-event-union-bound)
also have ... ≤ C-challenge + C-query
proof (rule add-mono)
  show
    wp-event verify-monad (composition-fri-challenge-list-set-hit s B) s
    ≤ C-challenge
  by (rule wp-verify-monad-composition-fri-challenge-list-set-bound
    [OF composition-future challenge-subset challenge-bound])
  show
    wp-event verify-monad
    (composition-fri-query-challenge-pair-set-hit s ?Outside) s ≤
    C-query
  by (rule
    wp-composition-fri-query-challenge-pair-set-hit-bound-from-query-fibers

```

$[OF \text{ query-future raw-bound projection subset query-bound}]$
qed
finally show *?thesis* .
qed

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-fibers*:
fixes *C-challenge C-query :: prob*
assumes *trace-future: trace-fri-future-fresh s*
and *query-future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *challenge-subset: $B \subseteq \text{fri-challenge-space}$ (ceil-log clength)*
and *challenge-bound:*
 $\text{nnreal (card } B) / \text{nnreal (CARD('f) } ^{\text{ceil-log clength}} \leq$
 $C\text{-challenge}$
and *projection:*
 $\bigwedge \text{challenges.}$
 fst '
 $((\text{trace-fri-sampled-query-bad-pair-union} \cap$
 $(\text{fri-query-index-list-space} \times \text{UNIV})) \cap$
 $(\text{UNIV} \times (- B))) \cap$
 $(\text{UNIV} \times \{\text{challenges}\})) \subseteq Q \text{ challenges}$
and *subset: $\bigwedge \text{challenges. } Q \text{ challenges} \subseteq \text{fri-query-index-list-space}$*
and *query-bound:*
 $\bigwedge \text{challenges.}$
 $\text{nnreal (card (} Q \text{ challenges))} *$
 $(1 / \text{nnreal (card query-sample-space)}) ^{\text{rounds}} \leq C\text{-query}$
shows
 $\text{wp-event verify-monad (trace-fri-sampled-query-bad-candidate } s) s \leq$
 $C\text{-challenge} + C\text{-query}$
proof –
let *?P =*
 $\text{trace-fri-sampled-query-bad-pair-union} \cap$
 $(\text{fri-query-index-list-space} \times \text{UNIV})$
have *wp-event verify-monad (trace-fri-sampled-query-bad-candidate } s) s*
 $\leq \text{wp-event verify-monad}$
 $(\text{trace-fri-query-challenge-pair-set-hit } s \text{ ?P}) s$
by (rule
 $\text{wp-trace-fri-sampled-query-bad-candidate-bound-by-restricted-query-challenge-union-hit})$
also have $\dots \leq C\text{-challenge} + C\text{-query}$
by (rule
 $\text{wp-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-fibers}$
 $[OF \text{ trace-future query-future raw-bound challenge-subset}$
 $\text{challenge-bound projection subset query-bound}]$
finally show *?thesis* .
qed

lemma *wp-composition-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-fibers*:
fixes *C-challenge C-query :: prob*
assumes *composition-future: composition-fri-future-fresh s*

and *query-future*: *query-future-fresh s*
and *raw-bound*: *query-index-raw-preimage-bound*
and *challenge-subset*:
 $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and *challenge-bound*:
 $\bigwedge dg. \text{nnreal } (\text{card } (B \ dg)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1)) \leq C\text{-challenge}$
and *projection*:
 $\bigwedge dg \text{ challenges.}$
 fst '
 $(((\text{composition-fri-sampled-query-bad-pair-union } dg \cap$
 $(\text{fri-query-index-list-space} \times \text{UNIV})) \cap$
 $(\text{UNIV} \times (- \ B \ dg))) \cap$
 $(\text{UNIV} \times \{\text{challenges}\})) \subseteq Q \ dg \text{ challenges}$
and *subset*:
 $\bigwedge dg \text{ challenges. } Q \ dg \text{ challenges} \subseteq \text{fri-query-index-list-space}$
and *query-bound*:
 $\bigwedge dg \text{ challenges.}$
 $\text{nnreal } (\text{card } (Q \ dg \text{ challenges})) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds} \leq C\text{-query}$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-query-bad-candidate } s) \ s \leq$
 $C\text{-challenge} + C\text{-query}$
proof –
let $?P =$
 $\bigwedge dg. \text{composition-fri-sampled-query-bad-pair-union } dg \cap$
 $(\text{fri-query-index-list-space} \times \text{UNIV})$
have $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-query-bad-candidate } s) \ s \leq$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-query-challenge-pair-set-hit } s \ ?P) \ s$
by (rule
 $\text{wp-composition-fri-sampled-query-bad-candidate-bound-by-restricted-query-challenge-union-hit})$
also have $\dots \leq C\text{-challenge} + C\text{-query}$
by (rule
 $\text{wp-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-fibers}$
 $[\text{OF composition-future query-future raw-bound challenge-subset}$
 $\text{challenge-bound projection subset query-bound}])$
finally show $?thesis$.
qed

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-fibers*:
fixes $C\text{-challenge } C\text{-query} :: \text{prob}$
assumes *trace-future*: *trace-fri-future-fresh s*
and *query-future*: *query-future-fresh s*
and *raw-bound*: *query-index-raw-preimage-bound*
and *challenge-subset*: $B \subseteq \text{fri-challenge-space } (\text{ceil-log } \text{clength})$
and *challenge-bound*:

```

nnreal (card B) / nnreal (CARD('f) ^ ceil-log clength) ≤
  C-challenge
and query-bound:
  ∧ challenges.
  nnreal
    (card
      (generic-fri-sampled-query-query-fiber trace-table-low-degree
        (Not ◦ trace-table-low-degree) (clength - 1) challenges)) *
      (1 / nnreal (card query-sample-space)) ^ rounds ≤ C-query
shows
  wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s ≤
    C-challenge + C-query
proof (rule
  wp-trace-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-fibers
  [OF trace-future query-future raw-bound challenge-subset challenge-bound])
fix challenges
show
  fst ‘
    (((trace-fri-sampled-query-bad-pair-union ∩
      (fri-query-index-list-space × UNIV)) ∩
      (UNIV × (- B))) ∩
      (UNIV × {challenges}))
    ⊆
    generic-fri-sampled-query-query-fiber trace-table-low-degree
      (Not ◦ trace-table-low-degree) (clength - 1) challenges
unfolding trace-fri-sampled-query-bad-pair-union-def
  generic-fri-sampled-query-query-fiber-def
  generic-fri-sampled-query-restricted-bad-pairs-def
  fri-query-challenge-pair-query-fiber-def
by auto
next
fix challenges
show
  generic-fri-sampled-query-query-fiber trace-table-low-degree
    (Not ◦ trace-table-low-degree) (clength - 1) challenges
    ⊆ fri-query-index-list-space
by (rule generic-fri-sampled-query-query-fiber-subset)
next
fix challenges
show
  nnreal
    (card
      (generic-fri-sampled-query-query-fiber trace-table-low-degree
        (Not ◦ trace-table-low-degree) (clength - 1) challenges)) *
      (1 / nnreal (card query-sample-space)) ^ rounds ≤ C-query
by (rule query-bound)
qed

```

lemma wp-trace-fri-sampled-query-bad-candidate-bound-from-pair-fraction:

assumes *trace-future*: *trace-fri-future-fresh s*
and *query-future*: *query-future-fresh s*
and *raw-bound*: *query-index-raw-preimage-bound*
and *fraction*:
generic-fri-sampled-query-pair-fraction-bound trace-table-low-degree
(Not ◦ trace-table-low-degree) (clength - 1) C
shows
wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s ≤ C
proof –
let *?P* =
trace-fri-sampled-query-bad-pair-union ∩
(fri-query-index-list-space × UNIV)
let *?Q* =
λchallenges.
generic-fri-sampled-query-query-fiber trace-table-low-degree
(Not ◦ trace-table-low-degree) (clength - 1) challenges
let *?I* = *fri-challenge-space (ceil-log clength)*
have *sampled-to-pair*:
wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s
≤ wp-event verify-monad
(trace-fri-query-challenge-pair-set-hit s ?P) s
by (rule
wp-trace-fri-sampled-query-bad-candidate-bound-by-restricted-query-challenge-union-hit)
also have ... ≤
(∑ challenges ∈ ?I.
*(1 / nnreal (CARD('f) ^ ceil-log clength))) **
*(nnreal (card (?Q challenges)) **
(1 / nnreal (card query-sample-space)) ^ rounds))
proof (rule
wp-trace-fri-query-challenge-pair-set-hit-bound-from-query-fiber-sum
[OF trace-future query-future raw-bound])
fix *challenges*
show *fst ' (?P ∩ (UNIV × {challenges})) ⊆*
?Q challenges
unfolding *trace-fri-sampled-query-bad-pair-union-def*
generic-fri-sampled-query-query-fiber-def
generic-fri-sampled-query-restricted-bad-pairs-def
fri-query-challenge-pair-query-fiber-def
by *auto*
next
fix *challenges*
show *?Q challenges ⊆ fri-query-index-list-space*
by (rule *generic-fri-sampled-query-query-fiber-subset*)
qed
also have ... ≤ *C*
proof –
have *round-eq*:
fri-round-count-for-degree-bound (clength - Suc 0) = ceil-log clength
unfolding *fri-round-count-for-degree-bound-def*

```

    by (simp add: clength-pos)
  have fiber-fraction:
    nnreal
      (∑ challenges ∈ ?I. card (?Q challenges)) /
      (nnreal (CARD('f) ^ ceil-log clength) *
        nnreal query-sample-space-size ^ rounds)
    ≤ C
  using fraction
  unfolding trace-fri-sampled-query-pair-fraction-bound-iff-fiber-sum
  by (simp add: round-eq card-fri-query-index-list-space
    query-sample-space-size-pos algebra-simps)
  have prob-eq:
    (∑ challenges ∈ ?I.
      (1 / nnreal (CARD('f) ^ ceil-log clength)) *
      (nnreal (card (?Q challenges)) *
        (1 / nnreal (card query-sample-space)) ^ rounds)) =
    nnreal
      (∑ challenges ∈ ?I. card (?Q challenges)) /
      (nnreal (CARD('f) ^ ceil-log clength) *
        nnreal query-sample-space-size ^ rounds)
  using weighted-query-fiber-sum-eq
  [where I = ?I
    and f = λchallenges. card (?Q challenges)
    and A = CARD('f) ^ ceil-log clength
    and B = query-sample-space-size]
  by (simp add: finite-fri-challenge-space query-sample-space-size-pos
    card-query-sample-space algebra-simps)
  show ?thesis
    using fiber-fraction unfolding prob-eq .
qed
finally show ?thesis .
qed

lemma wp-composition-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-fibers:
  fixes C-challenge C-query :: prob
  assumes composition-future: composition-fri-future-fresh s
    and query-future: query-future-fresh s
    and raw-bound: query-index-raw-preimage-bound
    and challenge-subset:
      ∧ dg. B dg ⊆ fri-challenge-space (ceil-log (to-nat dg + 1))
    and challenge-bound:
      ∧ dg. nnreal (card (B dg)) /
        nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) ≤ C-challenge
    and query-bound:
      ∧ dg challenges.
        nnreal
          (card
            (generic-fri-sampled-query-query-fiber
              (composition-table-low-degree (to-nat dg)))

```

```

      (Not  $\circ$  composition-table-low-degree maxDegree) (to-nat dg)
      challenges)) *
      (1 / nnreal (card query-sample-space))  $\wedge$  rounds  $\leq$  C-query
shows
  wp-event verify-monad
    (composition-fri-sampled-query-bad-candidate s) s  $\leq$ 
    C-challenge + C-query
proof (rule
  wp-composition-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-fibers
  [OF composition-future query-future raw-bound challenge-subset
    challenge-bound])
fix dg challenges
show
  fst ‘
    (((composition-fri-sampled-query-bad-pair-union dg  $\cap$ 
      (fri-query-index-list-space  $\times$  UNIV))  $\cap$ 
      (UNIV  $\times$  ( $-$  B dg)))  $\cap$ 
      (UNIV  $\times$  {challenges}))
     $\subseteq$ 
    generic-fri-sampled-query-query-fiber
      (composition-table-low-degree (to-nat dg))
      (Not  $\circ$  composition-table-low-degree maxDegree) (to-nat dg)
      challenges
unfolding composition-fri-sampled-query-bad-pair-union-def
  generic-fri-sampled-query-query-fiber-def
  generic-fri-sampled-query-restricted-bad-pairs-def
  fri-query-challenge-pair-query-fiber-def
by auto
next
fix dg challenges
show
  generic-fri-sampled-query-query-fiber
    (composition-table-low-degree (to-nat dg))
    (Not  $\circ$  composition-table-low-degree maxDegree) (to-nat dg)
    challenges
     $\subseteq$  fri-query-index-list-space
by (rule generic-fri-sampled-query-query-fiber-subset)
next
fix dg challenges
show
  nnreal
    (card
      (generic-fri-sampled-query-query-fiber
        (composition-table-low-degree (to-nat dg))
        (Not  $\circ$  composition-table-low-degree maxDegree) (to-nat dg)
        challenges)) *
      (1 / nnreal (card query-sample-space))  $\wedge$  rounds  $\leq$  C-query
by (rule query-bound)
qed

```

lemma *wp-composition-fri-sampled-query-bad-candidate-bound-from-pair-fraction-sum:*
fixes $C :: 'f \Rightarrow \text{prob}$
assumes *composition-future: composition-fri-future-fresh s*
and *query-future: query-future-fresh s*
and *raw-bound: query-index-raw-preimage-bound*
and *fraction:*
 $\wedge dg. \text{generic-fri-sampled-query-pair-fraction-bound}$
 $(\text{composition-table-low-degree } (to\text{-nat } dg))$
 $(\text{Not } \circ \text{composition-table-low-degree } \text{maxDegree}) (to\text{-nat } dg) (C \ dg)$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-query-bad-candidate } s) \ s \leq$
 $(\sum dg \in \text{UNIV}. C \ dg)$
proof –
let $?P =$
 $\lambda dg. \text{composition-fri-sampled-query-bad-pair-union } dg \cap$
 $(\text{fri-query-index-list-space} \times \text{UNIV})$
let $?Q =$
 $\lambda dg \text{ challenges.}$
 $\text{generic-fri-sampled-query-query-fiber}$
 $(\text{composition-table-low-degree } (to\text{-nat } dg))$
 $(\text{Not } \circ \text{composition-table-low-degree } \text{maxDegree}) (to\text{-nat } dg)$
 challenges
have *sampled-to-pair:*
 $\text{wp-event verify-monad } (\text{composition-fri-sampled-query-bad-candidate } s) \ s$
 $\leq \text{wp-event verify-monad}$
 $(\text{composition-fri-query-challenge-pair-set-hit } s \ ?P) \ s$
by (rule
 $\text{wp-composition-fri-sampled-query-bad-candidate-bound-by-restricted-query-challenge-union-hit})$
also have $\dots \leq$
 $(\sum dg \in \text{UNIV}.$
 $\sum \text{challenges} \in \text{fri-challenge-space } (\text{ceil-log } (to\text{-nat } dg + 1)).$
 $(1 / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (to\text{-nat } dg + 1))) *$
 $(\text{nnreal } (\text{card } (?Q \ dg \ \text{challenges}))) *$
 $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}))$
proof (rule
 $\text{wp-composition-fri-query-challenge-pair-set-hit-bound-from-query-fiber-sum}$
 $[\text{OF composition-future query-future raw-bound}])$
fix $dg \ \text{challenges}$
show $\text{fst } ' (?P \ dg \cap (\text{UNIV} \times \{\text{challenges}\})) \subseteq$
 $?Q \ dg \ \text{challenges}$
unfolding *composition-fri-sampled-query-bad-pair-union-def*
generic-fri-sampled-query-query-fiber-def
generic-fri-sampled-query-restricted-bad-pairs-def
fri-query-challenge-pair-query-fiber-def
by *auto*
next
fix $dg \ \text{challenges}$

```

show ?Q dg challenges  $\subseteq$  fri-query-index-list-space
  by (rule generic-fri-sampled-query-query-fiber-subset)
qed
also have ...  $\leq$  ( $\sum$  dg  $\in$  UNIV. C dg)
proof (rule sum-mono)
  fix dg :: 'f
  assume dg  $\in$  UNIV
  let ?I = fri-challenge-space (ceil-log (to-nat dg + 1))
  have round-eq:
    fri-round-count-for-degree-bound (to-nat dg) =
      ceil-log (to-nat dg + 1)
  unfolding fri-round-count-for-degree-bound-def by simp
  have fiber-fraction:
    nnreal
      ( $\sum$  challenges  $\in$  ?I. card (?Q dg challenges)) /
      (nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) *
        nnreal query-sample-space-size ^ rounds)
     $\leq$  C dg
  using fraction[of dg]
  unfolding composition-fri-sampled-query-pair-fraction-bound-iff-fiber-sum
  by (simp add: round-eq card-fri-query-index-list-space
    query-sample-space-size-pos algebra-simps)
  have prob-eq:
    ( $\sum$  challenges  $\in$  ?I.
      (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
      (nnreal (card (?Q dg challenges)) *
        (1 / nnreal (card query-sample-space)) ^ rounds)) =
    nnreal
      ( $\sum$  challenges  $\in$  ?I. card (?Q dg challenges)) /
      (nnreal (CARD('f) ^ ceil-log (to-nat dg + 1)) *
        nnreal query-sample-space-size ^ rounds)
  using weighted-query-fiber-sum-eq
  [where I = ?I
    and f =  $\lambda$ challenges. card (?Q dg challenges)
    and A = CARD('f) ^ ceil-log (to-nat dg + 1)
    and B = query-sample-space-size]
  by (simp add: finite-fri-challenge-space query-sample-space-size-pos
    card-query-sample-space algebra-simps)
  show
    ( $\sum$  challenges  $\in$  ?I.
      (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
      (nnreal (card (?Q dg challenges)) *
        (1 / nnreal (card query-sample-space)) ^ rounds))
     $\leq$  C dg
  using fiber-fraction unfolding prob-eq .
qed
finally show ?thesis .
qed

```

end

end

theory *Soundness-FRI-Query-Index-Staged-Bounds*
 imports *Soundness-FRI-Sampled-Interface*
begin

Staged query-index accounting for FRI query-list events.

The verifier-local exact query-list bounds still assume query freshness of the transferred verifier state. This layer charges accepted FRI query-list hits to the checked staged transcript query-index target event, using only the query indices actually derived by the verifier.

lemma *pair-second-set-cover-split*:

assumes $(x, y) \in P$

shows $y \in B \vee (x, y) \in P \cap (UNIV \times (- B))$

proof (*cases* $y \in B$)

case *True*

then show *?thesis* ..

next

case *False*

then have $(x, y) \in P \cap (UNIV \times (- B))$

using *assms* **by** *simp*

then show *?thesis* ..

qed

context *soundness*

begin

lemma *accepted-fri-opening-transcript-query-headerE*:

assumes

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final

dg composition-roots composition-bs composition-final query-idxs

trace-round-layers composition-round-layers

obtains *fr as query-state* **where**

verifier-header-transcript s fr trace-roots trace-final as dg

composition-roots composition-final (PTranscript query-state)

PState query-state =

verifier-header-state s fr trace-roots trace-final as dg

composition-roots composition-final

verifier-query-indices-derived s out

 (*verifier-header-state s fr trace-roots trace-final as dg*

composition-roots composition-final)

 (*PTranscript query-state*) *trace-roots composition-roots query-idxs*

proof –

show *?thesis*

using *assms*

unfolding *accepted-fri-opening-transcript-def*

```

apply (elim exE conjE)
apply (rule that)
by (auto intro: Channel-Core.protocol-channel.select-convs(1))
qed

lemma accepted-fri-opening-transcript-imp-staged-query-index-set-hit:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and builder:
  Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
and openings:
  accepted-fri-opening-transcript
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  (Some (result, final-state)) trace-roots trace-bs trace-final dg
  composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers
and query-in: query-idxs ∈ Q
shows staged-security-with-data-query-index-set-hit
  (query-index-list-entries Q) (Some ((data, result), final-state))
proof –
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from accepted-fri-opening-transcript-query-headerE[OF openings]
obtain fr as query-state where
  header:
    verifier-header-transcript ?s fr trace-roots trace-final as dg
    composition-roots composition-final (PTranscript query-state)
and query-state-hash:
  PState query-state =
    verifier-header-state ?s fr trace-roots trace-final as dg
    composition-roots composition-final
and derived:
  verifier-query-indices-derived ?s (Some (result, final-state))
  (verifier-header-state ?s fr trace-roots trace-final as dg
    composition-roots composition-final)
  (PTranscript query-state) trace-roots composition-roots query-idxs
by blast
have staged-header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)
  (staged-degree data)

```

```

    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled builder])
  have header-eq:
    fr = staged-trace-root data ∧
    trace-roots = staged-trace-fri-roots data ∧
    trace-final = staged-trace-final data ∧
    as = staged-alphas data ∧
    dg = staged-degree data ∧
    composition-roots = staged-composition-fri-roots data ∧
    composition-final = staged-composition-final data ∧
    PTranscript query-state = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header staged-header]
  by simp
  have query-start-eq:
    verifier-header-state ?s fr trace-roots trace-final as dg
    composition-roots composition-final =
    staged-query-start-hash data
  using query-state-hash header-eq
  unfolding staged-query-start-hash-def
    staged-composition-fri-start-hash-def
    staged-trace-fri-start-hash-def
    verifier-header-state-def verifier-header-messages-def
    verifier-state-from-adversary-def
  by simp
  have query-counter: PQueryCounter ?s = 0
  unfolding verifier-state-from-adversary-def by simp
  from verifier-query-indices-derivedE[OF derived]
  obtain result' final-state' raw-idxs query-chunks trailing where
    out-derived: Some (result, final-state) = Some (result', final-state')
  and len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and len-chunks: length query-chunks = rounds
  and concat-eq:
    List.concat query-chunks @ trailing = PTranscript query-state
  and round-chunks:
    ∧ i. i < rounds ⇒
      verifier-query-round-chunk (query-idxs ! i)
      trace-roots composition-roots (query-chunks ! i)
  and lookups':
    ∧ i. i < rounds ⇒
      fmlookup (HashMap final-state')
      (QueryIndexChallenge (PQueryCounter ?s + i)
        (state-after-query-chunks
          (verifier-header-state ?s fr trace-roots trace-final as dg
            composition-roots composition-final)))

```



```

      query-chunks i)) =
    Some (raw-idxs ! i)
  by metis
have final-state'-eq: final-state' = final-state
  using out-derived by simp
have lookups:
   $\wedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter ?s + i)
        (state-after-query-chunks
          (verifier-header-state ?s fr trace-roots trace-final as dg
            composition-roots composition-final)
          query-chunks i)) =
      Some (raw-idxs ! i)
  using lookups' final-state'-eq by simp
have len-query: length query-idxs = rounds
  using len-raw query-idxs-eq by simp
have parser-chunk-len:
   $\wedge i. i < \text{rounds} \implies$ 
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
proof -
  fix i
  assume i-bound: i < rounds
  show
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
  using verifier-query-round-chunk-length[OF round-chunks[OF i-bound]]
    header-eq
  by simp
qed
have concat-staged:
  List.concat query-chunks @ trailing =
    List.concat (staged-query-chunks data)
  using concat-eq header-eq by simp
have chunks-eq: query-chunks = staged-query-chunks data
  using selected-query-chunks-eq-staged
    [OF builder len-query len-chunks concat-staged parser-chunk-len]
  by simp
from rounds-positive obtain i where i-bound: i < rounds
  by auto
have idx-in: query-idxs ! i  $\in$  query-index-list-entries Q
  using query-in i-bound len-query unfolding query-index-list-entries-def
  by (auto intro: nth-mem)
have lookup:

```

```

fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
  Some (raw-idxs ! i)
using lookups[OF i-bound] query-counter query-start-eq chunks-eq
by simp
have raw-hit:
  index (to-nat (raw-idxs ! i)) ∈ query-index-list-entries Q
  using idx-in query-idxs-eq i-bound len-raw by simp
show ?thesis
  unfolding staged-security-with-data-query-index-set-hit-def
  using i-bound lookup raw-hit by auto
qed

lemma checked-staged-security-trace-fri-query-index-list-set-hit-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-index-list-set-hit s Q))
  adversary-initial-state ≤
  staged-phase-target-error
    (query-index-raw-preimage (query-index-list-entries Q))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
proof –
let ?B = query-index-list-entries Q
let ?projected =
  λout. case out of None ⇒
    staged-security-with-data-query-index-set-hit ?B None
  | Some (((data, -), result), t) ⇒
    staged-security-with-data-query-index-set-hit ?B
      (Some ((data, result), t))
have event-le:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-index-list-set-hit s Q))
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?projected adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume out-support:
  out ∈
  set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)

```

```

assume hit:
  staged-security-with-data-state-verifier-event
  ( $\lambda s. \text{trace-fri-query-index-list-set-hit } s \ Q$ ) out
show ?projected out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding staged-security-with-data-state-verifier-event-def
      staged-security-with-data-query-index-set-hit-def
    by simp
next
  case (Some pair)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases pair) (auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have builder:
    Some (data, attacker-state) ∈
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
    using out-support
    unfolding out-eq checked-staged-security-experiment-with-data-state-def
    by (auto elim!: set-dist-bindE)
  from hit out-eq have trace-hit:
    trace-fri-query-index-list-set-hit ?s Q
    (Some (result, final-state))
    unfolding staged-security-with-data-state-verifier-event-def by simp
  from trace-hit obtain trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers where
    fri:
      accepted-fri-opening-transcript ?s (Some (result, final-state))
      trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final query-idxs trace-round-layers
      composition-round-layers
    and query-in: query-idxs ∈ Q
    unfolding trace-fri-query-index-list-set-hit-def by blast
  have staged-hit:
    staged-security-with-data-query-index-set-hit
    (query-index-list-entries Q)
    (Some ((data, result), final-state))
    by (rule
      accepted-fri-opening-transcript-imp-staged-query-index-set-hit
      [OF wf controlled builder fri query-in])
  show ?thesis

```

```

    unfolding out-eq using staged-hit by simp
  qed
qed
also have
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?projected adversary-initial-state =
  wp-event (checked-staged-security-experiment-with-data A)
    (staged-security-with-data-query-index-set-hit ?B)
    adversary-initial-state
  using checked-staged-security-experiment-with-data-event-from-data-state
    [of A staged-security-with-data-query-index-set-hit ?B]
  by simp
also have ... ≤
  staged-phase-target-error
    (query-index-raw-preimage (query-index-list-entries Q))
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule checked-staged-security-with-data-query-index-set-hit-bound
    [OF wf controlled])
  finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-query-index-list-set-hit-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. composition-fri-query-index-list-set-hit s Q))
      adversary-initial-state ≤
      staged-phase-target-error
        (query-index-raw-preimage (query-index-list-entries Q))
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget)
  proof -
    let ?B = query-index-list-entries Q
    let ?projected =
      λout. case out of None ⇒
        staged-security-with-data-query-index-set-hit ?B None
      | Some (((data, -), result), t) ⇒
        staged-security-with-data-query-index-set-hit ?B
          (Some ((data, result), t))
    have event-le:
      wp-event (checked-staged-security-experiment-with-data-state A)
        (staged-security-with-data-state-verifier-event
          (λs. composition-fri-query-index-list-set-hit s Q))
        adversary-initial-state ≤
      wp-event (checked-staged-security-experiment-with-data-state A)
        ?projected adversary-initial-state
    proof (rule wp-event-mono-on-support)

```

```

fix out
assume out-support:
  out ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
assume hit:
  staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-index-list-set-hit s Q) out
show ?projected out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding staged-security-with-data-state-verifier-event-def
      staged-security-with-data-query-index-set-hit-def
    by simp
next
  case (Some pair)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases pair) (auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
    using out-support
    unfolding out-eq checked-staged-security-experiment-with-data-state-def
    by (auto elim!: set-dist-bindE)
  from hit out-eq have composition-hit:
    composition-fri-query-index-list-set-hit ?s Q
    (Some (result, final-state))
    unfolding staged-security-with-data-state-verifier-event-def by simp
  from composition-hit obtain trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers where
    fri:
      accepted-fri-opening-transcript ?s (Some (result, final-state))
      trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final query-idxs trace-round-layers
      composition-round-layers
    and query-in: query-idxs ∈ Q
    unfolding composition-fri-query-index-list-set-hit-def by blast
  have staged-hit:
    staged-security-with-data-query-index-set-hit

```

```

      (query-index-list-entries Q)
      (Some ((data, result), final-state))
    apply (rule
      accepted-fri-opening-transcript-imp-staged-query-index-set-hit
      [OF wf controlled builder fri query-in])
  .
  show ?thesis
    unfolding out-eq using staged-hit by simp
qed
qed
also have
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?projected adversary-initial-state =
  wp-event (checked-staged-security-experiment-with-data A)
    (staged-security-with-data-query-index-set-hit ?B)
    adversary-initial-state
  using checked-staged-security-experiment-with-data-event-from-data-state
    [of A staged-security-with-data-query-index-set-hit ?B]
  by simp
also have ... ≤
  staged-phase-target-error
    (query-index-raw-preimage (query-index-list-entries Q))
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule checked-staged-security-with-data-query-index-set-hit-bound
    [OF wf controlled])
  finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-query-projection:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and projection: fst ' P ⊆ Q
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. trace-fri-query-challenge-pair-set-hit s P))
    adversary-initial-state ≤
    staged-phase-target-error
      (query-index-raw-preimage (query-index-list-entries Q))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  proof (rule order-trans)
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-challenge-pair-set-hit s P))
    adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-index-list-set-hit s Q))

```

```

    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro:
        trace-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit
        [OF - projection]
      split: option.splits prod.splits)
next
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. trace-fri-query-index-list-set-hit s Q))
  adversary-initial-state
  ≤ staged-phase-target-error
    (query-index-raw-preimage (query-index-list-entries Q))
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  apply (rule checked-staged-security-trace-fri-query-index-list-set-hit-bound
    [OF wf controlled]) .
qed

```

lemma *checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-query-projection:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *projection: $\bigwedge dg. \text{fst } P \text{ dg} \subseteq Q$*

shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-challenge-pair-set-hit s P))
  adversary-initial-state ≤
  staged-phase-target-error
    (query-index-raw-preimage (query-index-list-entries Q))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)

```

proof (rule order-trans)

```

show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-challenge-pair-set-hit s P))
  adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-index-list-set-hit s Q))
  adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro:
        composition-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit
        [OF - projection]
      split: option.splits prod.splits)

```

next

```

show wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (staged-security-with-data-state-verifier-event
      ( $\lambda s$ . composition-fri-query-index-list-set-hit s Q))
    adversary-initial-state
   $\leq$  staged-phase-target-error
    (query-index-raw-preimage (query-index-list-entries Q))
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule
    checked-staged-security-composition-fri-query-index-list-set-hit-bound
    [OF wf controlled])
qed

lemma checked-staged-security-trace-fri-sampled-query-pair-set-hit-bound-from-query-projection:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and projection:
     $\bigwedge$  trace-table roots final round-layers layers.
    fst ' R trace-table roots final round-layers layers  $\subseteq$  Q
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s$ . trace-fri-sampled-query-pair-set-hit R s))
    adversary-initial-state  $\leq$ 
    staged-phase-target-error
      (query-index-raw-preimage (query-index-list-entries Q))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  proof (rule order-trans)
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s$ . trace-fri-sampled-query-pair-set-hit R s))
    adversary-initial-state  $\leq$ 
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        ( $\lambda s$ . trace-fri-query-index-list-set-hit s Q))
      adversary-initial-state
  by (rule wp-event-mono-on-support)
  (auto simp: staged-security-with-data-state-verifier-event-def
    intro:
      trace-fri-sampled-query-pair-set-hit-imp-query-index-list-set-hit
      [OF - projection]
      split: option.splits prod.splits)
  next
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s$ . trace-fri-query-index-list-set-hit s Q))
    adversary-initial-state
   $\leq$  staged-phase-target-error
    (query-index-raw-preimage (query-index-list-entries Q))
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)

```


by (rule checked-staged-security-trace-fri-query-index-list-set-hit-bound
[OF wf controlled])
qed

lemma *checked-staged-security-composition-fri-sampled-query-pair-set-hit-bound-from-query-projection:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *projection:*
 $\bigwedge \text{fri-dg composition-table roots final round-layers layers.}$
 $\text{fst } R \text{ fri-dg composition-table roots final round-layers layers}$
 $\subseteq Q$
shows
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-sampled-query-pair-set-hit } R \ s))$
 $\text{adversary-initial-state} \leq$
 $\text{staged-phase-target-error}$
 $(\text{query-index-raw-preimage (query-index-list-entries } Q))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})$
proof (rule order-trans)
show $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-sampled-query-pair-set-hit } R \ s))$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-query-index-list-set-hit } s \ Q))$
 $\text{adversary-initial-state}$
by (rule wp-event-mono-on-support)
(auto simp: staged-security-with-data-state-verifier-event-def
intro:
composition-fri-sampled-query-pair-set-hit-imp-query-index-list-set-hit
[OF - projection]
split: option.splits prod.splits)
next
show $\text{wp-event (checked-staged-security-experiment-with-data-state A)}$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-query-index-list-set-hit } s \ Q))$
 $\text{adversary-initial-state}$
 $\leq \text{staged-phase-target-error}$
 $(\text{query-index-raw-preimage (query-index-list-entries } Q))$
 $(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$
by (rule
checked-staged-security-composition-fri-query-index-list-set-hit-bound
[OF wf controlled])
qed

lemma *trace-fri-query-challenge-pair-set-hit-split-by-pair-cover:*

```

assumes hit: trace-fri-query-challenge-pair-set-hit s P out
shows
  trace-fri-query-challenge-pair-set-hit s
    ( $P \cap (UNIV \times B)$ ) out  $\vee$ 
    trace-fri-query-challenge-pair-set-hit s
      ( $P \cap (UNIV \times (- B))$ ) out
proof –
  show ?thesis
  using hit
  unfolding trace-fri-query-challenge-pair-set-hit-def
  apply (elim exE conjE)
  apply (subgoal-tac
    (fri-query-idxs, trace-bs)  $\in P \cap (UNIV \times B) \vee$ 
    (fri-query-idxs, trace-bs)  $\in P \cap (UNIV \times (- B))$ )
  apply (erule disjE)
  apply (rule disjI1)
  apply (intro exI conjI)
  apply assumption
  apply assumption
  apply (rule disjI2)
  apply (intro exI conjI)
  apply assumption
  apply assumption
  by simp
qed

lemma composition-fri-query-challenge-pair-set-hit-split-by-pair-cover:
assumes hit: composition-fri-query-challenge-pair-set-hit s P out
shows
  composition-fri-query-challenge-pair-set-hit s
    ( $\lambda dg. P \ dg \cap (UNIV \times B \ dg)$ ) out  $\vee$ 
    composition-fri-query-challenge-pair-set-hit s
      ( $\lambda dg. P \ dg \cap (UNIV \times (- B \ dg))$ ) out
proof –
  show ?thesis
  using hit
  unfolding composition-fri-query-challenge-pair-set-hit-def
  apply (elim exE conjE)
  apply (subgoal-tac
    (fri-query-idxs, composition-bs)  $\in P \ fri-dg \cap (UNIV \times B \ fri-dg) \vee$ 
    (fri-query-idxs, composition-bs)  $\in$ 
       $P \ fri-dg \cap (UNIV \times (- B \ fri-dg))$ )
  apply (erule disjE)
  apply (rule disjI1)
  apply (intro exI conjI)
  apply assumption
  apply assumption
  apply (rule disjI2)
  apply (intro exI conjI)

```

apply assumption
 apply assumption
 by auto
 qed

lemma *checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-projection*

assumes *wf*: staged-budget-wellformed budgets

and *controlled*: staged-adversary-controlled budgets *A*

and *finite-B*: finite *B*

and *fresh-bound*:

wp-event (checked-staged-security-experiment-with-data-state *A*)

(staged-security-with-data-state-verifier-event

($\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B$))

adversary-initial-state $\leq F$

and *projection*:

fst ' ($P \cap (UNIV \times (- B))$) $\subseteq Q$

shows

wp-event (checked-staged-security-experiment-with-data-state *A*)

(staged-security-with-data-state-verifier-event

($\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ P$))

adversary-initial-state $\leq$

F +

hash-relation-budget-value (card *B* * ceil-log clength)

(staged-attacker-query-budget budgets + staged-challenge-query-budget) +

staged-phase-target-error

(query-index-row-preimage (query-index-list-entries *Q*))

(staged-attacker-query-budget budgets +

staged-challenge-query-budget)

proof –

let *?Inside* = $P \cap (UNIV \times B)$

let *?Outside* = $P \cap (UNIV \times (- B))$

let *?challenge-event* =

staged-security-with-data-state-verifier-event

($\lambda s. \text{trace-fri-challenge-list-set-hit } s \ B$)

let *?inside-event* =

staged-security-with-data-state-verifier-event

($\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ ?\text{Inside}$)

let *?outside-event* =

staged-security-with-data-state-verifier-event

($\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ ?\text{Outside}$)

let *?hbudget* =

hash-relation-budget-value (card *B* * ceil-log clength)

(staged-attacker-query-budget budgets + staged-challenge-query-budget)

let *?qbudget* =

staged-phase-target-error

(query-index-row-preimage (query-index-list-entries *Q*))

(staged-attacker-query-budget budgets + staged-challenge-query-budget)

have *split*:

wp-event (checked-staged-security-experiment-with-data-state *A*)

```

  (staged-security-with-data-state-verifier-event
   (λs. trace-fri-query-challenge-pair-set-hit s P))
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
  (λout.
   ?inside-event out ∨ ?outside-event out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume hit:
    staged-security-with-data-state-verifier-event
    (λs. trace-fri-query-challenge-pair-set-hit s P) out
  then show
    ?inside-event out ∨ ?outside-event out
  unfolding staged-security-with-data-state-verifier-event-def
  by (auto
    dest: trace-fri-query-challenge-pair-set-hit-split-by-pair-cover
    split: option.splits prod.splits)
qed
have union-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (λout. ?inside-event out ∨ ?outside-event out)
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
  ?inside-event adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
  ?outside-event adversary-initial-state
  by (rule wp-event-union-bound)
have split-union-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   (λs. trace-fri-query-challenge-pair-set-hit s P))
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
  ?inside-event adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
  ?outside-event adversary-initial-state
  apply (rule order-trans)
  apply (rule split)
  apply (rule union-bound)
done
have sum-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  ?inside-event adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
  ?outside-event adversary-initial-state ≤ (F + ?hbudget) + ?qbudget
proof –
  have challenge-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    ?challenge-event adversary-initial-state  $\leq F + ?hbudget$ 
  by (rule checked-staged-security-trace-fri-challenge-list-set-hit-bound
    [OF wf controlled finite-B fresh-bound])
have inside-projection: snd ‘ ?Inside  $\subseteq B$ 
  by force
have inside-to-challenge:
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?inside-event adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?challenge-event adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro:
        trace-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit
        [OF - inside-projection]
        split: option.splits prod.splits)
have inside-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?inside-event adversary-initial-state  $\leq F + ?hbudget$ 
  apply (rule order-trans)
  apply (rule inside-to-challenge)
  apply (rule challenge-bound)
  done
have outside-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?outside-event adversary-initial-state  $\leq ?qbudget$ 
  by (rule
    checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-query-projection
    [OF wf controlled projection])
show ?thesis
  by (rule add-mono[OF inside-bound outside-bound])
qed
show ?thesis
  using order-trans[OF split-union-bound sum-bound]
  by (simp add: add.assoc)
qed

```

lemma *checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-pr*
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: $\bigwedge dg. \text{finite } (B \ dg)$
and fresh-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 ($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B$))
 adversary-initial-state $\leq F$
and projection:
 $\bigwedge dg. \text{fst } ‘ (P \ dg \cap (UNIV \times (- \ B \ dg))) \subseteq Q$
shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s$ . composition-fri-query-challenge-pair-set-hit s P))
adversary-initial-state  $\leq$ 
  F +
  hash-relation-budget-value
    ( $\sum dg \in (UNIV :: 'f \text{ set})$ .
      card (B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget) +
  staged-phase-target-error
    (query-index-raw-preimage (query-index-list-entries Q))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
proof -
  let ?Inside =  $\lambda dg$ . P dg  $\cap$  (UNIV  $\times$  B dg)
  let ?Outside =  $\lambda dg$ . P dg  $\cap$  (UNIV  $\times$  (- B dg))
  let ?challenge-event =
    staged-security-with-data-state-verifier-event
      ( $\lambda s$ . composition-fri-challenge-list-set-hit s B)
  let ?inside-event =
    staged-security-with-data-state-verifier-event
      ( $\lambda s$ . composition-fri-query-challenge-pair-set-hit s ?Inside)
  let ?outside-event =
    staged-security-with-data-state-verifier-event
      ( $\lambda s$ . composition-fri-query-challenge-pair-set-hit s ?Outside)
  let ?hbudget =
    hash-relation-budget-value
      ( $\sum dg \in (UNIV :: 'f \text{ set})$ .
        card (B dg) * ceil-log (maxDegree + 1))
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  let ?qbudget =
    staged-phase-target-error
      (query-index-raw-preimage (query-index-list-entries Q))
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  have split:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        ( $\lambda s$ . composition-fri-query-challenge-pair-set-hit s P))
      adversary-initial-state  $\leq$ 
      wp-event (checked-staged-security-experiment-with-data-state A)
        ( $\lambda out$ .
          ?inside-event out  $\vee$  ?outside-event out)
        adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume hit:
      staged-security-with-data-state-verifier-event
        ( $\lambda s$ . composition-fri-query-challenge-pair-set-hit s P) out

```

```

then show
  ?inside-event out  $\vee$  ?outside-event out
unfolding staged-security-with-data-state-verifier-event-def
by (auto
  dest: composition-fri-query-challenge-pair-set-hit-split-by-pair-cover
  split: option.splits prod.splits)
qed
have union-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    ( $\lambda$ out. ?inside-event out  $\vee$  ?outside-event out)
    adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?inside-event adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?outside-event adversary-initial-state
by (rule wp-event-union-bound)
have split-union-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda$ s. composition-fri-query-challenge-pair-set-hit s P))
    adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?inside-event adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?outside-event adversary-initial-state
apply (rule order-trans)
apply (rule split)
apply (rule union-bound)
done
have sum-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?inside-event adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?outside-event adversary-initial-state  $\leq$  (F + ?hbudget) + ?qbudget
proof –
have challenge-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?challenge-event adversary-initial-state  $\leq$  F + ?hbudget
by (rule
  checked-staged-security-composition-fri-challenge-list-set-hit-bound
  [OF wf controlled finite-B fresh-bound])
have inside-projection:  $\bigwedge dg. \text{snd } dg \subseteq B \text{ } dg$ 
by force
have inside-to-challenge:
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?inside-event adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?challenge-event adversary-initial-state
by (rule wp-event-mono-on-support)

```

```

(auto simp: staged-security-with-data-state-verifier-event-def
intro:
  composition-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit
  [OF - inside-projection]
split: option.splits prod.splits)
have inside-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  ?inside-event adversary-initial-state  $\leq F + ?hbudget$ 
  apply (rule order-trans)
  apply (rule inside-to-challenge)
  apply (rule challenge-bound)
  done
have outside-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  ?outside-event adversary-initial-state  $\leq ?qbudget$ 
  by (rule
    checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-query-projection
    [OF wf controlled projection])
show ?thesis
  by (rule add-mono[OF inside-bound outside-bound])
qed
show ?thesis
  using order-trans[OF split-union-bound sum-bound]
  by (simp add: add.assoc)
qed

lemma checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-projection
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: finite B
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B$ ))
  adversary-initial-state  $\leq F$ 
and projection:
  fst ‘
    ((trace-fri-sampled-query-bad-pair-union  $\cap$ 
      (fri-query-index-list-space  $\times UNIV$ ))  $\cap$ 
      (UNIV  $\times (- B)$ ))  $\subseteq Q$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-sampled-query-bad-candidate)
  adversary-initial-state  $\leq$ 
  F +
  hash-relation-budget-value (card B * ceil-log clength)
  (staged-attacker-query-budget budgets + staged-challenge-query-budget) +
  staged-phase-target-error

```



```

      (query-index-row-preimage (query-index-list-entries Q))
      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget)
proof –
  let ?P =
    trace-fri-sampled-query-bad-pair-union  $\cap$ 
    (fri-query-index-list-space  $\times$  UNIV)
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-sampled-query-bad-candidate)
    adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     ( $\lambda s$ . trace-fri-query-challenge-pair-set-hit s ?P))
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
     intro:
      trace-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit
      split: option.splits prod.splits)
  also have ...  $\leq$ 
    F +
    hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget) +
    staged-phase-target-error
    (query-index-row-preimage (query-index-list-entries Q))
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget)
  by (rule
    checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-projection
    [OF wf controlled finite-B fresh-bound projection])
  finally show ?thesis .
qed

```

```

lemma checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-p
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B:  $\bigwedge dg$ . finite (B dg)
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     ( $\lambda s$ . composition-fri-challenge-list-fresh-hit s B))
    adversary-initial-state  $\leq$  F
and projection:
   $\bigwedge dg$ . fst ‘
    ((composition-fri-sampled-query-bad-pair-union dg  $\cap$ 
     (fri-query-index-list-space  $\times$  UNIV))  $\cap$ 
     (UNIV  $\times$  ( $-$  B dg)))  $\subseteq$  Q

```

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-sampled-query-bad-candidate)
adversary-initial-state $\leq$
F +
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \text{ } dg) * \text{ceil-log } (\text{maxDegree} + 1))$
(staged-attacker-query-budget budgets +
staged-challenge-query-budget) +
staged-phase-target-error
(query-index-row-preimage (query-index-list-entries Q))
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)

proof –

let *?P* =

$\lambda dg. \text{composition-fri-sampled-query-bad-pair-union } dg \cap$
 $(\text{fri-query-index-list-space} \times UNIV)$

have *wp-event (checked-staged-security-experiment-with-data-state A)*

(staged-security-with-data-state-verifier-event
composition-fri-sampled-query-bad-candidate)

adversary-initial-state $\leq$

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event
 $(\lambda s. \text{composition-fri-query-challenge-pair-set-hit } s \text{ } ?P))$

adversary-initial-state

by *(rule wp-event-mono-on-support)*

(auto simp: staged-security-with-data-state-verifier-event-def
intro:

composition-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit

split: option.splits prod.splits)

also have ... $\leq$

F +

hash-relation-budget-value

$(\sum dg \in (UNIV :: 'f \text{ set}).$

$\text{card } (B \text{ } dg) * \text{ceil-log } (\text{maxDegree} + 1))$

(staged-attacker-query-budget budgets +

staged-challenge-query-budget) +

staged-phase-target-error

(query-index-row-preimage (query-index-list-entries Q))

(staged-attacker-query-budget budgets +

staged-challenge-query-budget)

by *(rule*

checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-projection

[OF wf controlled finite-B fresh-bound projection])

finally show *?thesis* .

qed

lemma *checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-generic-cover*:

assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *cover*:
generic-fri-sampled-query-finite-cover trace-table-low-degree
(Not $\circ$ trace-table-low-degree) (clength - 1) B
and *fresh-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
(λs . trace-fri-challenge-list-fresh-hit s B))
adversary-initial-state $\leq$ F

shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-sampled-query-bad-candidate)
adversary-initial-state $\leq$
F +
*hash-relation-budget-value (card B * ceil-log clength)*
(staged-attacker-query-budget budgets + staged-challenge-query-budget)

proof –
have *finite-B*: *finite B*
by (rule *generic-fri-sampled-query-finite-coverD(1)[OF cover]*)
have *projection*:
snd ‘
(trace-fri-sampled-query-bad-pair-union $\cap$
(fri-query-index-list-space $\times$ UNIV)) $\subseteq$ B
using *generic-fri-sampled-query-finite-coverD(2)[OF cover]*
unfolding *trace-fri-sampled-query-bad-pair-union-def* **by** *simp*
have *wp-event (checked-staged-security-experiment-with-data-state A)*
(staged-security-with-data-state-verifier-event
trace-fri-sampled-query-bad-candidate)
adversary-initial-state $\leq$
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
(λs . trace-fri-challenge-list-set-hit s B))
adversary-initial-state
by (rule *wp-event-mono-on-support*)
(auto simp: staged-security-with-data-state-verifier-event-def
intro:
trace-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit
[OF trace-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit
projection]
split: option.splits prod.splits)
also have ... $\leq$
F +
*hash-relation-budget-value (card B * ceil-log clength)*
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)
by (rule *checked-staged-security-trace-fri-challenge-list-set-hit-bound*

$[OF \text{ wf controlled finite-}B \text{ fresh-bound}]$
finally show ?thesis .
qed

lemma *checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-generic-cover*:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and cover:
 $\bigwedge dg. \text{generic-fri-sampled-query-finite-cover}$
 $(\text{composition-table-low-degree } (to\text{-nat } dg))$
 $(Not \circ \text{composition-table-low-degree } maxDegree) (to\text{-nat } dg) (B \ dg)$
and fresh-bound:
 $wp\text{-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B))$
 $\text{adversary-initial-state} \leq F$
shows
 $wp\text{-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-sampled-query-bad-candidate})$
 $\text{adversary-initial-state} \leq$
 $F +$
 $\text{hash-relation-budget-value}$
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \ dg) * \text{ceil-log } (maxDegree + 1))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})$
proof –
have finite-B: $\bigwedge dg. \text{finite } (B \ dg)$
by (rule generic-fri-sampled-query-finite-coverD(1)[OF cover])
have projection:
 $\bigwedge dg. \text{snd '}$
 $(\text{composition-fri-sampled-query-bad-pair-union } dg \cap$
 $(\text{fri-query-index-list-space} \times UNIV)) \subseteq B \ dg$
proof –
fix dg
show snd '
 $(\text{composition-fri-sampled-query-bad-pair-union } dg \cap$
 $(\text{fri-query-index-list-space} \times UNIV)) \subseteq B \ dg$
using generic-fri-sampled-query-finite-coverD(2)[OF cover[of dg]]
unfolding composition-fri-sampled-query-bad-pair-union-def **by** simp
qed
have wp-event (checked-staged-security-experiment-with-data-state A)
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-sampled-query-bad-candidate})$
 $\text{adversary-initial-state} \leq$
 $wp\text{-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-challenge-list-set-hit } s \ B))$

```

    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro:
        composition-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit
        [OF composition-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit
          projection]
        split: option.splits prod.splits)
  also have ... ≤
    F +
    hash-relation-budget-value
    (∑ dg ∈ (UNIV :: 'f set).
      card (B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  by (rule checked-staged-security-composition-fri-challenge-list-set-hit-bound
    [OF wf controlled finite-B fresh-bound])
  finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-sampled-layer-chain-bound-from-sampled-query:*

assumes *sampled-query:*

```

  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-sampled-query-bad-candidate)
  adversary-initial-state ≤ R

```

shows

```

  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-sampled-layer-chain)
  adversary-initial-state ≤ R

```

proof —

have *missing:*

```

  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-sampled-missing-canonical-sampled-layer-chain)
  adversary-initial-state ≤ 0

```

proof (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)

show $\bigwedge s. \neg \text{trace-fri-sampled-missing-canonical-sampled-layer-chain } s \text{ None}$

unfolding *trace-fri-sampled-missing-canonical-sampled-layer-chain-def*

```

  trace-fri-bad-with-sampled-layer-chain-def
  accepted-fri-opening-transcript-def
  trace-fri-partial-candidate-opening-evidence-def
  trace-fri-partial-candidate-evidence-def

```

by *blast*

next

fix *data attacker-state*

assume

Some (data, attacker-state) ∈

```

    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  show wp-event verify-monad
    (trace-fri-sampled-missing-canonical-sampled-layer-chain
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ 0
  by (rule wp-trace-fri-missing-canonical-sampled-zero)
qed
have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-sampled-layer-chain)
  adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (λout.
    staged-security-with-data-state-verifier-event
      trace-fri-sampled-query-bad-candidate out ∨
    staged-security-with-data-state-verifier-event
      trace-fri-sampled-missing-canonical-sampled-layer-chain out)
  adversary-initial-state
by (rule wp-event-mono-on-support)
  (auto simp: staged-security-with-data-state-verifier-event-def
    trace-fri-sampled-query-bad-candidate-iff-canonical
    dest: trace-fri-sampled-layer-chain-canonical-or-missing
    split: option.splits prod.splits)
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-query-bad-candidate)
  adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-missing-canonical-sampled-layer-chain)
  adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤ R + 0
  by (rule add-mono[OF sampled-query missing])
finally show ?thesis
  by simp
qed

lemma checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-sampled-query:
  assumes sampled-query:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-sampled-query-bad-candidate)
    adversary-initial-state ≤ R
  shows

```

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   composition-fri-bad-with-verifier-tied-sampled-layer-chain)
  adversary-initial-state  $\leq R$ 
proof –
  have missing:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       composition-fri-sampled-missing-canonical-sampled-layer-chain)
      adversary-initial-state  $\leq 0$ 
  proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show
     $\bigwedge s. \neg \text{composition-fri-sampled-missing-canonical-sampled-layer-chain}$ 
     $s \text{ None}$ 
  unfolding composition-fri-sampled-missing-canonical-sampled-layer-chain-def
    composition-fri-bad-with-sampled-layer-chain-def
    accepted-fri-opening-transcript-def
    composition-fri-partial-candidate-opening-evidence-def
    composition-fri-partial-candidate-evidence-def
  by blast
next
  fix data attacker-state
  assume
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  show wp-event verify-monad
    (composition-fri-sampled-missing-canonical-sampled-layer-chain
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))  $\leq 0$ 
  by (rule wp-composition-fri-missing-canonical-sampled-zero)
qed
have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   composition-fri-bad-with-verifier-tied-sampled-layer-chain)
  adversary-initial-state  $\leq$ 
wp-event (checked-staged-security-experiment-with-data-state A)
  ( $\lambda out.$ 
    staged-security-with-data-state-verifier-event
      composition-fri-sampled-query-bad-candidate out  $\vee$ 
    staged-security-with-data-state-verifier-event
      composition-fri-sampled-missing-canonical-sampled-layer-chain out)
  adversary-initial-state
by (rule wp-event-mono-on-support)
  (auto simp: staged-security-with-data-state-verifier-event-def
    composition-fri-sampled-query-bad-candidate-iff-canonical
    dest!: composition-fri-verifier-tied-sampled-layer-chain-imp-sampled-layer-chain)

```

```

    dest: composition-fri-sampled-layer-chain-canonical-or-missing
    split: option.splits prod.splits)
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-sampled-query-bad-candidate)
  adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-sampled-missing-canonical-sampled-layer-chain)
  adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤ R + 0
by (rule add-mono[OF sampled-query missing])
finally show ?thesis
by simp
qed

lemma checked-staged-security-trace-fri-empty-header-bound-from-sampled-query-conflict-and-zero:
  assumes sampled-query:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-sampled-query-bad-candidate)
    adversary-initial-state ≤ R
  and conflict-bound:
    ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
    wp-event verify-monad
      (trace-fri-sampled-assignment-conflict
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) ≤ C
  and zero-bound:
    ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state) ⇒
    wp-event verify-monad
      (trace-fri-zero-round-final-obstruction
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) ≤ Z
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event

```


trace-fri-bad-with-empty-composition-header-candidates
adversary-initial-state $\leq R + (C + Z)$
proof (rule *checked-staged-security-trace-fri-empty-header-bound-from-sampled-and-missing*)
show *wp-event* (*checked-staged-security-experiment-with-data-state* *A*)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-sampled-layer-chain)
adversary-initial-state $\leq R$
by (rule
checked-staged-security-trace-fri-sampled-layer-chain-bound-from-sampled-query
[OF sampled-query])
next
fix *data attacker-state*
assume *support*:
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)
let *?s =*
verifier-state-from-adversary attacker-state
(staged-proof-transcript data)
have *obstruction*:
wp-event verify-monad
(trace-fri-sampled-layer-assignment-obstruction ?s) ?s ≤ C + Z
by (rule
wp-trace-fri-sampled-layer-assignment-obstruction-bound-from-conflict-and-zero)
(rule conflict-bound[OF support], rule zero-bound[OF support])
show *wp-event verify-monad*
(trace-fri-reachable-missing-sampled-layer-chain ?s) ?s ≤ C + Z
by (rule *wp-trace-fri-reachable-missing-sampled-bound-from-assignment-obstruction*)
(rule obstruction)
qed

lemma *wp-composition-fri-verifier-tied-missing-sampled-layer-chain-bound-from-residual-gaps*:

assumes *slot-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s) s
 $\leq \text{Slot}$
and *merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) s
 $\leq \text{Merkle}$
and *missing-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-recorded-base-chunk-auth-missing s) s
 $\leq \text{Missing}$
and *next-bound*:
wp-event verify-monad
(composition-fri-verifier-tied-next-value-replay-gap s) s
 $\leq \text{Next}$
and *successor-bound*:
wp-event verify-monad

$(\text{composition-fri-verifier-tied-successor-replay-gap } s) \ s$
 $\leq \text{Successor}$
and *final-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-final-value-replay-gap } s) \ s$
 $\leq \text{Final}$
and *zero-bound*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-zero-round-final-obstruction } s) \ s$
 $\leq \text{Zero}$
shows
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-missing-sampled-layer-chain } s) \ s$
 $\leq$
 $((\text{Merkle} + \text{Missing}) + \text{Next} + \text{Successor} + \text{Final} + \text{Merkle}) +$
 $\text{Slot} + \text{Zero})$
proof –
have *conflict*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-assignment-conflict } s) \ s$
 $\leq ((\text{Merkle} + \text{Missing}) + \text{Next} + \text{Successor} + \text{Final} + \text{Merkle}) + \text{Slot}$
by (rule
 $\text{wp-composition-fri-verifier-tied-sampled-assignment-conflict-bound-from-slot-recorded-chunk-gap-and-residuals}$
 $[\text{OF slot-bound merkle-bound missing-bound next-bound successor-bound}$
 $\text{final-bound}])$
have *obstruction*:
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-layer-assignment-obstruction } s) \ s$
 $\leq$
 $((\text{Merkle} + \text{Missing}) + \text{Next} + \text{Successor} + \text{Final} + \text{Merkle}) +$
 $\text{Slot} + \text{Zero})$
by (rule
 $\text{wp-composition-fri-verifier-tied-assignment-obstruction-bound-from-conflict-and-zero})$
(rule *conflict*, rule *zero-bound*)
show *?thesis*
by (rule
 $\text{wp-composition-fri-verifier-tied-missing-sampled-bound-from-assignment-obstruction})$
(rule *obstruction*)
qed
lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-sampled-query-and-residual*
assumes *sampled-query*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-sampled-query-bad-candidate})$
 $\text{adversary-initial-state} \leq R$
and *slot-bound*:
 $\bigwedge \text{data attacker-state.}$
 $\text{Some } (\text{data}, \text{attacker-state}) \in$

$set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq Slot$
and *merkle-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(partial-merkle-inconsistency-bad$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq Merkle$
and *missing-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-verifier-tied-recorded-base-chunk-auth-missing$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq Missing$
and *next-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-verifier-tied-next-value-replay-gap$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq Next$
and *successor-bound*:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(composition-fri-verifier-tied-successor-replay-gap$
 $(verifier-state-from-adversary \ attacker-state$

(staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Successor
and final-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-final-value-replay-gap
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Final
and zero-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ Zero
shows
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state $\leq$
 R + (((Merkle + Missing) + Next + Successor + Final + Merkle) +
 Slot + Zero)
proof –
have sampled:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-sampled-layer-chain)
 adversary-initial-state $\leq$ R
by (rule
 checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-sampled-query
 [OF sampled-query])
have missing:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-verifier-tied-missing-sampled-layer-chain
 (verifier-state-from-adversary attacker-state

```

      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤
    (((Merkle + Missing) + Next + Successor + Final + Merkle) +
      Slot + Zero)
  proof -
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  show wp-event verify-monad
    (composition-fri-verifier-tied-missing-sampled-layer-chain ?s) ?s
    ≤
    (((Merkle + Missing) + Next + Successor + Final + Merkle) +
      Slot + Zero)
  by (rule
    wp-composition-fri-verifier-tied-missing-sampled-layer-chain-bound-from-residual-gaps)
  (rule slot-bound[OF support],
    rule merkle-bound[OF support],
    rule missing-bound[OF support],
    rule next-bound[OF support],
    rule successor-bound[OF support],
    rule final-bound[OF support],
    rule zero-bound[OF support])
  qed
  show ?thesis
  by (rule
    checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-sampled-and-missing
      [OF sampled missing])
  qed
end

end

end

```

```

theory Soundness-Public-Route-Active-Base
imports
  Soundness-Public-Route-Single-Query
  Soundness-FRI-Query-Exact-Bounds
  Soundness-FRI-Query-Index-Staged-Bounds
begin

```

Base active-FRI public-route endpoint.

This theory is deliberately upstream of the downstream FRI route adapters. It exposes

the common route theorem from staged active FRI event bounds, without importing the final same-run endpoint theory. This avoids an import cycle when later route layers are consumed by the final public endpoint.

context *soundness*
begin

definition *active-route-transcript-pre-error-for*
where

active-route-transcript-pre-error-for $A =$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $\text{staged-security-with-data-state-transcript-pre-hit}$
 $\text{adversary-initial-state}$

definition *active-route-current-empty-query-error-for*
where

active-route-current-empty-query-error-for $A =$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-current-empty-query-opening-hit-at } 0)$
 $\text{adversary-initial-state}$

definition *active-route-current-prefix-authenticated-error-for*
where

active-route-current-prefix-authenticated-error-for $A \ i =$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at } i)$
 $\text{adversary-initial-state}$

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-bounds:*

fixes *trace-fri-error'* *composition-fri-error'*

empty-trace-fri-error' :: prob

assumes *false-statement: $\neg \text{exists-valid-trace}$*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *trace-fri-bound:*

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}'$

and *comp-fri-bound:*

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}'$

and *empty-trace-fri-bound:*

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$

$adversary-initial-state \leq empty-trace-fri-error'$
shows $checked-staged-adversary-acceptance-probability\ A \leq$
 $hash-collision-budget-value\ 0$
 $(staged-attacker-query-budget\ budgets +$
 $staged-challenge-query-budget + verifier-hash-query-budget) +$
 $(composition-error-bound +$
 $staged-phase-relation-error\ size$
 $(staged-alpha-search-queries\ budgets\ 0) +$
 $hash-collision-budget-value\ 0$
 $(staged-alpha-search-queries\ budgets\ 0) +$
 $staged-phase-relation-error\ size$
 $(staged-alpha-search-queries\ budgets\ 0) +$
 $(active-route-transcript-pre-error-for\ A +$
 $staged-concrete-transcript-target-error-bound)) +$
 $(\sum i < rounds.$
 $active-route-current-prefix-authenticated-error-for\ A\ i +$
 $(active-route-transcript-pre-error-for\ A +$
 $staged-concrete-transcript-target-error-bound) +$
 $(active-route-transcript-pre-error-for\ A +$
 $staged-concrete-transcript-target-error-bound)) +$
 $trace-fri-error' + composition-fri-error' +$
 $(composition-error-bound +$
 $staged-phase-relation-error\ size$
 $(staged-alpha-search-queries\ budgets\ 0) +$
 $hash-collision-budget-value\ 0$
 $(staged-alpha-search-queries\ budgets\ 0) +$
 $staged-phase-relation-error\ size$
 $(staged-alpha-search-queries\ budgets\ 0) +$
 $(active-route-transcript-pre-error-for\ A +$
 $staged-concrete-transcript-target-error-bound) +$
 $empty-trace-fri-error' + active-route-current-empty-query-error-for\ A)$
proof –
have *transcript-new-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $staged-security-with-data-state-transcript-new-hit$
 $adversary-initial-state \leq staged-concrete-transcript-target-error-bound$
by (rule *checked-staged-security-with-data-state-transcript-new-hit-bound*
 $[OF\ wf\ controlled]$)
have *data-pre-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $staged-security-with-data-state-transcript-pre-hit$
 $adversary-initial-state \leq active-route-transcript-pre-error-for\ A$
by (simp add: *active-route-transcript-pre-error-for-def*)
have *current-empty-bound*:
 $wp-event\ (checked-staged-security-experiment-with-data-state\ A)$
 $(staged-security-with-data-state-current-empty-query-opening-hit-at\ 0)$
 $adversary-initial-state \leq active-route-current-empty-query-error-for\ A$
by (simp add: *active-route-current-empty-query-error-for-def*)
have *prefix-bound*:

```

 $\wedge i. i < \text{rounds} \implies$ 
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
      i)
    adversary-initial-state  $\leq$ 
      active-route-current-prefix-authenticated-error-for A i
  by (simp add: active-route-current-prefix-authenticated-error-for-def)
have trace-path-bound:
 $\wedge i. i < \text{rounds} \implies$ 
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-trace-path-output-hit-at
      i)
    adversary-initial-state  $\leq$ 
      active-route-transcript-pre-error-for A +
      staged-concrete-transcript-target-error-bound
  by (rule
    checked-staged-security-with-query-prefix-current-trace-path-output-hit-at-bound-from-transcript
      [OF - data-pre-bound transcript-new-bound])
have composition-path-bound:
 $\wedge i. i < \text{rounds} \implies$ 
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-composition-path-output-hit-at
      i)
    adversary-initial-state  $\leq$ 
      active-route-transcript-pre-error-for A +
      staged-concrete-transcript-target-error-bound
  by (rule
    checked-staged-security-with-query-prefix-current-composition-path-output-hit-at-bound-from-transcript
      [OF - data-pre-bound transcript-new-bound])
define C :: nat  $\Rightarrow$  prob where
  C =
    ( $\lambda i. \text{active-route-current-prefix-authenticated-error-for A } i +$ 
      (active-route-transcript-pre-error-for A +
        staged-concrete-transcript-target-error-bound) +
      (active-route-transcript-pre-error-for A +
        staged-concrete-transcript-target-error-bound))
have round-bound:
 $\wedge i. i < \text{rounds} \implies$ 
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
      i)
    adversary-initial-state  $\leq$  C i
proof -
  fix i
  assume i-bound: i < rounds

```



```

show
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A i)
    (checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at
      i)
    adversary-initial-state  $\leq C$  i
  unfolding C-def
  by (rule
    checked-staged-security-with-query-prefix-current-query-partial-opening-hit-at-bound-from-prefix-and-structured-
      [OF wf controlled i-bound prefix-bound[OF i-bound]
        trace-path-bound[OF i-bound] composition-path-bound[OF i-bound]])
  qed
have route:
  checked-staged-adversary-acceptance-probability A  $\leq$ 
    hash-collision-budget-value 0
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget + verifier-hash-query-budget) +
    (composition-error-bound +
      staged-phase-relation-error size
        (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
        (staged-alpha-search-queries budgets 0) +
      staged-phase-relation-error size
        (staged-alpha-search-queries budgets 0) +
      (active-route-transcript-pre-error-for A +
        staged-concrete-transcript-target-error-bound)) +
    trace-fri-error' + composition-fri-error' +
    ( $\sum i < \text{rounds. } C$  i) +
    (composition-error-bound +
      staged-phase-relation-error size
        (staged-alpha-search-queries budgets 0) +
      hash-collision-budget-value 0
        (staged-alpha-search-queries budgets 0) +
      staged-phase-relation-error size
        (staged-alpha-search-queries budgets 0) +
      (active-route-transcript-pre-error-for A +
        staged-concrete-transcript-target-error-bound) +
      empty-trace-fri-error' +
      active-route-current-empty-query-error-for A)
  by (rule
    stark-soundness-from-transcript-indexed-partial-opening-route-from-sampled-transcript-relevant-drift-only-single-
      [OF false-statement wf controlled round-bound data-pre-bound
        current-empty-bound trace-fri-bound comp-fri-bound
        empty-trace-fri-bound])
  then show ?thesis
  unfolding C-def by (simp add: algebra-simps)
qed

end

```

end

theory *Soundness-FRI-Sampled-Interface-Conflict-Zero*
imports *Soundness-FRI-Sampled-Interface*
begin

Conflict/zero active-FRI wrapper for the sampled-query interface.

This narrow layer keeps the sampled-interface theory from growing further while exposing the cleaner trace route that does not carry a separate recorded-sibling-missing component.

context *soundness*
begin

lemma *active-fri-reductions-for-from-trace-augmented-conflict-zero-composition-sampled-query-and-empty-conflict-zero*

fixes *TraceAug TraceConflict TraceZero TraceMerkle CompSampled CompMerkle*

EmptySampled EmptyConflict EmptyZero ::

'f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ prob

assumes *trace-augmented-bound:*

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\implies$

wp-event verify-monad

(trace-fri-bad-with-header-tied-augmented-partial-candidate

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)) $\leq$ TraceAug data attacker-state

and *trace-conflict-bound:*

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\implies$

wp-event verify-monad

(trace-fri-header-tied-sampled-assignment-conflict

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)) $\leq$ TraceConflict data attacker-state

and *trace-zero-bound:*

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\implies$

wp-event verify-monad

(trace-fri-header-tied-zero-round-final-obstruction

(verifier-state-from-adversary attacker-state

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceZero data attacker-state}$
and *trace-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceMerkle data attacker-state}$
and *trace-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{TraceAug data attacker-state} +$
 $((\text{TraceConflict data attacker-state} + \text{TraceZero data attacker-state}) +$
 $\text{TraceMerkle data attacker-state})$
 $\leq \text{trace-fri-error}$
and *composition-sampled-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-query-bad-candidate}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{CompSampled data attacker-state}$
and *composition-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{CompMerkle data attacker-state}$
and *composition-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$

$set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $CompSampled \ data \ attacker-state +$
 $((CompMerkle \ data \ attacker-state + 0) +$
 $(CompMerkle \ data \ attacker-state + 0) +$
 $(CompMerkle \ data \ attacker-state + 0) +$
 $(CompMerkle \ data \ attacker-state + 0 + 0) +$
 $CompMerkle \ data \ attacker-state) + 0)$
 $\leq composition-fri-error$
and empty-sampled-bound:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(trace-fri-sampled-query-bad-candidate$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq$
 $EmptySampled \ data \ attacker-state$
and empty-conflict-bound:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(trace-fri-sampled-assignment-conflict$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq$
 $EmptyConflict \ data \ attacker-state$
and empty-zero-bound:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A) \ adversary-initial-state) \implies$
 $wp-event \ verify-monad$
 $(trace-fri-zero-round-final-obstruction$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)))$
 $(verifier-state-from-adversary \ attacker-state$
 $(staged-proof-transcript \ data)) \leq$
 $EmptyZero \ data \ attacker-state$
and empty-total:
 $\wedge data \ attacker-state.$
 $Some \ (data, \ attacker-state) \in$
 $set-dist \ (execute \ (checked-staged-transcript-program \ A)$

```

    adversary-initial-state)  $\implies$ 
    EmptySampled data attacker-state +
    (EmptyConflict data attacker-state + EmptyZero data attacker-state)
     $\leq$  trace-fri-error
  shows active-fri-reductions-for A
proof (rule
  active-fri-reductions-for-from-trace-augmented-conflict-zero-composition-sampled-and-empty-conflict-zero)
fix data attacker-state
assume support:
  Some (data, attacker-state)  $\in$ 
  set-dist (execute (checked-staged-transcript-program A)
    adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
show wp-event verify-monad
  (trace-fri-bad-with-header-tied-augmented-partial-candidate ?s) ?s
   $\leq$  TraceAug data attacker-state
  by (rule trace-augmented-bound[OF support])
show wp-event verify-monad
  (trace-fri-header-tied-sampled-assignment-conflict ?s) ?s
   $\leq$  TraceConflict data attacker-state
  by (rule trace-conflict-bound[OF support])
show wp-event verify-monad
  (trace-fri-header-tied-zero-round-final-obstruction ?s) ?s
   $\leq$  TraceZero data attacker-state
  by (rule trace-zero-bound[OF support])
show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
   $\leq$  TraceMerkle data attacker-state
  by (rule trace-merkle-bound[OF support])
show TraceAug data attacker-state +
  ((TraceConflict data attacker-state + TraceZero data attacker-state) +
  TraceMerkle data attacker-state)
   $\leq$  trace-fri-error
  by (rule trace-total[OF support])
show wp-event verify-monad
  (composition-fri-bad-with-verifier-tied-sampled-layer-chain ?s) ?s
   $\leq$  CompSampled data attacker-state
  by (rule wp-composition-fri-verifier-tied-sampled-layer-chain-bound-from-sampled)
  (rule wp-composition-fri-sampled-bound-from-canonical,
    insert composition-sampled-bound[OF support],
    simp add: wp-composition-fri-sampled-query-bad-candidate-iff-canonical)
show wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
   $\leq$  CompMerkle data attacker-state
  by (rule composition-merkle-bound[OF support])
show CompSampled data attacker-state +
  (((CompMerkle data attacker-state + 0) +
  (CompMerkle data attacker-state + 0) +
  (CompMerkle data attacker-state + 0) +

```

```

      (CompMerkle data attacker-state + 0 + 0) +
      CompMerkle data attacker-state) + 0)
    ≤ composition-fri-error
  by (rule composition-total[OF support])
show wp-event verify-monad
  (trace-fri-bad-with-sampled-layer-chain ?s) ?s
  ≤ EmptySampled data attacker-state
  by (rule wp-trace-fri-sampled-bound-from-canonical)
    (insert empty-sampled-bound[OF support],
      simp add: wp-trace-fri-sampled-query-bad-candidate-iff-canonical)
show wp-event verify-monad
  (trace-fri-sampled-assignment-conflict ?s) ?s
  ≤ EmptyConflict data attacker-state
  by (rule empty-conflict-bound[OF support])
show wp-event verify-monad
  (trace-fri-zero-round-final-obstruction ?s) ?s
  ≤ EmptyZero data attacker-state
  by (rule empty-zero-bound[OF support])
show EmptySampled data attacker-state +
  (EmptyConflict data attacker-state + EmptyZero data attacker-state)
  ≤ trace-fri-error
  by (rule empty-total[OF support])
qed

end

end

```

```

theory Soundness-Public-Route-Same-Run
imports
  Soundness-Public-Route-Active-Base
  Soundness-FRI-Sampled-Interface-Conflict-Zero
begin

```

Public-route infrastructure for the staged soundness theorem. This layer assembles the checked current-empty and transcript-empty reductions and states them in the plain locale *soundness*. FRI losses are expressed as concrete staged verifier-event probabilities that are bounded in later FRI layers.

```

context soundness
begin

```

```

lemma current-query-route-bound-from-active-fri-bounds-internal:
fixes trace-fri-error' composition-fri-error'
  empty-trace-fri-error' :: prob
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fri-bound:

```

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state $\leq$ *trace-fri-error'*
and *comp-fri-bound*:
 $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state $\leq$ *composition-fri-error'*
and *empty-trace-fri-bound*:
 $wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *empty-trace-fri-error'*
shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
 ($1 + \text{staged-concrete-transcript-target-error-bound}$)) +
 ($\sum_{i < \text{rounds.}}$
 $1 +$
 ($1 + \text{staged-concrete-transcript-target-error-bound}$) +
 ($1 + \text{staged-concrete-transcript-target-error-bound}$)) +
trace-fri-error' + *composition-fri-error'* +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
 ($1 + \text{staged-concrete-transcript-target-error-bound}$) +
empty-trace-fri-error' + 1)
proof –
let *?exact* =
hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0

(staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds}$.
 active-route-current-prefix-authenticated-error-for A i +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 trace-fri-error' + composition-fri-error' +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 empty-trace-fri-error' + active-route-current-empty-query-error-for A)
have route:
 checked-staged-adversary-acceptance-probability A $\leq$?exact
by (rule
 stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-bounds
 [OF false-statement wf controlled trace-fri-bound comp-fri-bound
 empty-trace-fri-bound])
have exact-le-route:
 ?exact $\leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds}$.
 1 +
 (1 + staged-concrete-transcript-target-error-bound) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 trace-fri-error' + composition-fri-error' +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +

hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
(1 + staged-concrete-transcript-target-error-bound) +
empty-trace-fri-error' + 1)
unfolding *active-route-transcript-pre-error-for-def*
active-route-current-prefix-authenticated-error-for-def
active-route-current-empty-query-error-for-def
by (*intro add-mono sum-mono order-refl wp-event-le-1*)
show *?thesis*
by (*rule order-trans[OF route exact-le-route]*)
qed

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-staged-active-fri-bounds:*

fixes *trace-bad ::*
'f list $\Rightarrow$ nat $\Rightarrow$ 'f list $\Rightarrow$ 'f set
and *composition-bad ::*
'f $\Rightarrow$ 'f list $\Rightarrow$ nat $\Rightarrow$ 'f list $\Rightarrow$ 'f set
and *trace-B :: 'f list set*
and *composition-B :: 'f $\Rightarrow$ 'f list set*
and *trace-fresh trace-merkle trace-missing empty-fresh*
empty-assignment composition-fresh composition-merkle :: prob
and *trace-fri-error' composition-fri-error'*
empty-trace-fri-error' :: prob
assumes *false-statement: $\neg$ exists-valid-trace*
and *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *trace-finite: finite trace-B*
and *trace-cover:*
 $\bigwedge$ *data attacker-state.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
trace-fri-sampled-layer-chain-cover
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) trace-bad
and *trace-envelope:*
 $\bigwedge$ *data attacker-state trace-table.*
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$
trace-fri-multi-round-bad-sets trace-bad trace-table $\subseteq$ trace-B
and *trace-fresh-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
($\lambda s.$ trace-fri-challenge-list-fresh-hit s trace-B))
adversary-initial-state $\leq$ trace-fresh
and *trace-merkle-bound:*

$\wedge$ *data attacker-state*.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ trace-merkle
and *trace-missing-bound*:
 $\wedge$ *data attacker-state*.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-recorded-sibling-candidate-missing
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ trace-missing
and *trace-total*:
(trace-fresh +
*hash-relation-budget-value (card trace-B * ceil-log clength)*
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)) +
((trace-missing +
(((((trace-merkle + trace-merkle) + (trace-merkle + 0) +
(trace-merkle + 0) + (trace-merkle + 0 + 0)) +
trace-merkle) + 0))) +
(1 + trace-merkle)) +
(trace-missing + trace-merkle)
≤ trace-fri-error'
and *empty-fresh-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
(λs . trace-fri-challenge-list-fresh-hit s trace-B))
adversary-initial-state ≤ empty-fresh
and *empty-assignment-bound*:
 $\wedge$ *data attacker-state*.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-sampled-layer-assignment-obstruction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ empty-assignment

and *empty-total*:
 (*empty-fresh* +
 hash-relation-budget-value (*card trace-B* * *ceil-log clength*)
 (*staged-attacker-query-budget budgets* +
 staged-challenge-query-budget)) + *empty-assignment*
 ≤ *empty-trace-fri-error'*
and *composition-finite*: $\bigwedge dg. \text{finite } (composition-B \ dg)$
and *composition-cover*:
 $\bigwedge data \text{ attacker-state.}$
 Some (*data*, *attacker-state*) ∈
 set-dist (*execute* (*checked-staged-transcript-program A*)
 adversary-initial-state) ⇒
 composition-fri-sampled-layer-chain-cover
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) *composition-bad*
and *composition-envelope*:
 $\bigwedge data \text{ attacker-state } dg \text{ composition-table.}$
 Some (*data*, *attacker-state*) ∈
 set-dist (*execute* (*checked-staged-transcript-program A*)
 adversary-initial-state) ⇒
 composition-fri-multiround-bad-sets composition-bad dg composition-table
 ⊆ *composition-B dg*
and *composition-fresh-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \text{ composition-B}$))
 adversary-initial-state ≤ *composition-fresh*
and *composition-merkle-bound*:
 $\bigwedge data \text{ attacker-state.}$
 Some (*data*, *attacker-state*) ∈
 set-dist (*execute* (*checked-staged-transcript-program A*)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (*partial-merkle-inconsistency-bad*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) ≤ *composition-merkle*
and *composition-total*:
 (*composition-fresh* +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \text{ set}).$
 card (*composition-B dg*) * *ceil-log (maxDegree + 1)*)
 (*staged-attacker-query-budget budgets* +
 staged-challenge-query-budget)) +
 (((*composition-merkle* + 0) +
 (*composition-merkle* + 0) + (*composition-merkle* + 0) +
 (*composition-merkle* + 0 + 0) + *composition-merkle*) + 0 + 0)
 ≤ *composition-fri-error'*

shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
 (1 + *staged-concrete-transcript-target-error-bound*)) +
 ($\sum_{i < \text{rounds.}}$
 1 +
 (1 + *staged-concrete-transcript-target-error-bound*) +
 (1 + *staged-concrete-transcript-target-error-bound*)) +
trace-fri-error' + *composition-fri-error'* +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
 (1 + *staged-concrete-transcript-target-error-bound*) +
empty-trace-fri-error' + 1)

proof –

have *trace-fri-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state $\leq$ *trace-fri-error'*

proof (rule *order-trans*)

show *wp-event* (*checked-staged-security-experiment-with-data-state* A)

(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state $\leq$

((*trace-fresh* +
hash-relation-budget-value (*card* *trace-B* * *ceil-log* *clength*)
 (*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget)) +
 ((*trace-missing* +
 ((((*trace-merkle* + *trace-merkle*) + (*trace-merkle* + 0) +
 (*trace-merkle* + 0) + (*trace-merkle* + 0 + 0)) +
trace-merkle) + 0)) +
 (1 + *trace-merkle*))) +
 (*trace-missing* + *trace-merkle*)

by (rule

checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-augmented-route-and-unaugmented-sp

```

    [OF wf controlled trace-finite trace-cover trace-envelope
      trace-fresh-bound trace-merkle-bound trace-missing-bound
      wp-event-le-1])
  show ... ≤ trace-fri-error'
  by (rule trace-total)
qed
have empty-trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state ≤ empty-trace-fri-error'
proof (rule order-trans)
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state ≤
    (empty-fresh +
      hash-relation-budget-value (card trace-B * ceil-log clength)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) + empty-assignment
  by (rule
    checked-staged-security-trace-fri-empty-header-bound-from-fresh-prequery-and-assignment-obstruction
    [OF wf controlled trace-finite trace-cover trace-envelope
      empty-fresh-bound empty-assignment-bound])
  show ... ≤ empty-trace-fri-error'
  by (rule empty-total)
qed
have comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-verifier-tied-partial-candidate)
  adversary-initial-state ≤ composition-fri-error'
proof (rule order-trans)
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-verifier-tied-partial-candidate)
  adversary-initial-state ≤
    (composition-fresh +
      hash-relation-budget-value
        (∑ dg ∈ (UNIV :: 'f set).
          card (composition-B dg) * ceil-log (maxDegree + 1))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) +
    (((composition-merkle + 0) +
      (composition-merkle + 0) + (composition-merkle + 0) +
      (composition-merkle + 0 + 0) + composition-merkle) + 0 + 0)
  by (rule
    checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-fresh-prequery-zero-slot-missin
    [OF wf controlled composition-finite composition-cover

```

```

    composition-envelope composition-fresh-bound composition-merkle-bound]])
  show ... ≤ composition-fri-error'
  by (rule composition-total)
qed
show ?thesis
  by (rule
    current-query-route-bound-from-active-fri-bounds-internal
    [OF false-statement wf controlled trace-fri-bound comp-fri-bound
      empty-trace-fri-bound])
qed

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-residual-obliga
assumes false-statement: ¬ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and residuals:
  active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft
  Rc Ac Pc Nc Sc Gc
shows checked-staged-adversary-acceptance-probability A ≤
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget) +
  (composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    (1 + staged-concrete-transcript-target-error-bound)) +
  (∑ i<rounds.
    1 +
    (1 + staged-concrete-transcript-target-error-bound) +
    (1 + staged-concrete-transcript-target-error-bound)) +
  trace-fri-error + composition-fri-error +
  (composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    (1 + staged-concrete-transcript-target-error-bound) +
    trace-fri-error + 1)
proof –
have trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-partial-candidate)

```

$\text{adversary-initial-state} \leq \text{trace-fri-error}$
by (rule *active-fri-residual-obligations-staged-trace-bound*[*OF residuals*])
have *comp-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}$
by (rule
 $\text{active-fri-residual-obligations-staged-composition-bound}$ [*OF residuals*])
have *empty-trace-fri-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
by (rule
 $\text{active-fri-residual-obligations-staged-empty-header-bound}$ [*OF residuals*])
show *?thesis*
by (rule
 $\text{current-query-route-bound-from-active-fri-bounds-internal}$
 $[OF \text{ false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\text{empty-trace-fri-bound}]$)
qed

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-residual-compo*

assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-residuals*:
 $\text{active-trace-fri-residual-obligations-for } A \text{ Rt Zt At Pt Ht Gt Nt St Ft}$
and *composition-residuals*:
 $\text{active-composition-fri-residual-obligations-for } A \text{ Rc Ac Pc Nc Sc Gc}$
and *empty-trace*: *active-empty-trace-fri-reduction-for A*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound})) +$
 $(\sum_{i < \text{rounds.}} 1 +$
 $(1 + \text{staged-concrete-transcript-target-error-bound}) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound})) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$

(composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound) +
 trace-fri-error + 1)
proof –
have residuals:
 active-fri-residual-obligations-for A Rt Zt At Pt Ht Gt Nt St Ft
 Rc Ac Pc Nc Sc Gc
by (rule active-fri-residual-obligations-from-components
 [OF trace-residuals composition-residuals empty-trace])
show ?thesis
by (rule
 stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-residual-obligations
 [OF false-statement wf controlled residuals])
qed

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-reductions:
assumes false-statement: $\neg$ exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and active-fri: active-fri-reductions-for A
shows checked-staged-adversary-acceptance-probability A $\leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds.}$
 1 +
 (1 + staged-concrete-transcript-target-error-bound) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 trace-fri-error + composition-fri-error +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size

$(\text{staged-alpha-search-queries budgets } 0) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound}) +$
 $\text{trace-fri-error} + 1)$
proof –
have *trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
by (rule *active-fri-reductions-staged-trace-bound*[*OF active-fri*])
have *comp-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq \text{composition-fri-error}$
by (rule *active-fri-reductions-staged-composition-bound*[*OF active-fri*])
have *empty-trace-fri-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{trace-fri-error}$
by (rule *active-fri-reductions-staged-empty-header-bound*[*OF active-fri*])
show *?thesis*
by (rule
 $\text{current-query-route-bound-from-active-fri-bounds-internal}$
 $[\text{OF false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\text{empty-trace-fri-bound}]$)

qed

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-derived-active-fri-composition*
fixes *TraceAug TraceMissing TraceMerkle CompSampled CompMerkle EmptySampled EmptyMissing* ::

'f staged-proof-data $\Rightarrow$ 'f protocol-channel $\Rightarrow$ prob

assumes *false-statement: $\neg \text{exists-valid-trace}$*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *trace-augmented-bound:*

$\bigwedge \text{data attacker-state.}$

Some (data, attacker-state) $\in$

set-dist (execute (checked-staged-transcript-program A)

adversary-initial-state) $\Rightarrow$

wp-event verify-monad

(trace-fri-bad-with-header-tied-augmented-partial-candidate

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)) $\leq$ TraceAug data attacker-state

and *trace-missing-bound:*

$\bigwedge \text{data attacker-state.}$

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-recorded-sibling-candidate-missing
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
TraceMissing data attacker-state
and *trace-merkle-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ TraceMerkle data attacker-state
and *trace-total:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
TraceAug data attacker-state +
(TraceMissing data attacker-state + TraceMerkle data attacker-state)
 $\leq$ *trace-fri-error*
and *composition-sampled-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(composition-fri-bad-with-verifier-tied-sampled-layer-chain
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤
CompSampled data attacker-state
and *composition-merkle-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(partial-merkle-inconsistency-bad
(verifier-state-from-adversary attacker-state

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{CompMerkle data attacker-state}$
and *composition-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{CompSampled data attacker-state} +$
 $((\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0 + 0) +$
 $\text{CompMerkle data attacker-state}) + 0)$
 $\leq \text{composition-fri-error}$
and *empty-sampled-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{EmptySampled data attacker-state}$
and *empty-missing-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-reachable-missing-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{EmptyMissing data attacker-state}$
and *empty-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{EmptySampled data attacker-state} + \text{EmptyMissing data attacker-state}$
 $\leq \text{trace-fri-error}$
shows *checked-staged-adversary-acceptance-probability A* $\leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$

$$\begin{aligned}
& \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) + \\
& (\text{composition-error-bound} + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \text{hash-collision-budget-value } 0 \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& (1 + \text{staged-concrete-transcript-target-error-bound})) + \\
& (\sum_{i < \text{rounds.}} \\
& \quad 1 + \\
& \quad (1 + \text{staged-concrete-transcript-target-error-bound}) + \\
& \quad (1 + \text{staged-concrete-transcript-target-error-bound})) + \\
& \text{trace-fri-error} + \text{composition-fri-error} + \\
& (\text{composition-error-bound} + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \text{hash-collision-budget-value } 0 \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets } 0) + \\
& (1 + \text{staged-concrete-transcript-target-error-bound}) + \\
& \text{trace-fri-error} + 1)
\end{aligned}$$

proof –
have *active-fri*: *active-fri-reductions-for A*
by (rule
active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty-sampled
[*OF trace-augmented-bound trace-missing-bound trace-merkle-bound*
trace-total composition-sampled-bound composition-merkle-bound
composition-total empty-sampled-bound empty-missing-bound
empty-total])
show ?thesis
by (rule
stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-reductions
[*OF false-statement wf controlled active-fri*])
qed

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-derived-active-fri-conflict*
fixes *TraceAug TraceMissing TraceMerkle CompSampled CompMerkle EmptySampled Empty-*
Conflict EmptyZero ::
'*f staged-proof-data* $\Rightarrow$ '*f protocol-channel* $\Rightarrow$ *prob*
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-augmented-bound*:
 $\bigwedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\Rightarrow$

$\text{wp-event verify-monad}$
 $(\text{trace-fri-bad-with-header-tied-augmented-partial-candidate}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceAug data attacker-state}$
and $\text{trace-missing-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-recorded-sibling-candidate-missing}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{TraceMissing data attacker-state}$
and $\text{trace-merkle-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceMerkle data attacker-state}$
and trace-total:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{TraceAug data attacker-state} +$
 $(\text{TraceMissing data attacker-state} + \text{TraceMerkle data attacker-state})$
 $\leq \text{trace-fri-error}$
and $\text{composition-sampled-bound:}$
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-verifier-tied-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{CompSampled data attacker-state}$

and *composition-merkle-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*partial-merkle-inconsistency-bad*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$ *CompMerkle data attacker-state*
and *composition-total*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
CompSampled data attacker-state +
(((*CompMerkle data attacker-state* + 0) +
(*CompMerkle data attacker-state* + 0) +
(*CompMerkle data attacker-state* + 0) +
(*CompMerkle data attacker-state* + 0 + 0) +
CompMerkle data attacker-state) + 0)
 $\leq$ *composition-fri-error*
and *empty-sampled-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-bad-with-sampled-layer-chain*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
EmptySampled data attacker-state
and *empty-conflict-bound*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*trace-fri-sampled-assignment-conflict*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq$
EmptyConflict data attacker-state
and *empty-zero-bound*:
 $\wedge$ *data attacker-state*.

Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
wp-event *verify-monad*
(*trace-fri-zero-round-final-obstruction*
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)))
(*verifier-state-from-adversary* *attacker-state*
(*staged-proof-transcript* *data*)) $\leq$
EmptyZero *data* *attacker-state*
and *empty-total*:
 $\wedge$ *data* *attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program* *A*)
adversary-initial-state) $\implies$
EmptySampled *data* *attacker-state* +
(*EmptyConflict* *data* *attacker-state* + *EmptyZero* *data* *attacker-state*)
 $\leq$ *trace-fri-error*
shows *checked-staged-adversary-acceptance-probability* *A* $\leq$
hash-collision-budget-value 0
(*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
(*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
(*staged-alpha-search-queries* *budgets* 0) +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
(1 + *staged-concrete-transcript-target-error-bound*)) +
($\sum$ *i* < *rounds*.
1 +
(1 + *staged-concrete-transcript-target-error-bound*) +
(1 + *staged-concrete-transcript-target-error-bound*)) +
trace-fri-error + *composition-fri-error* +
(*composition-error-bound* +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
(*staged-alpha-search-queries* *budgets* 0) +
staged-phase-relation-error *size*
(*staged-alpha-search-queries* *budgets* 0) +
(1 + *staged-concrete-transcript-target-error-bound*) +
trace-fri-error + 1)
proof –
have *active-fri*: *active-fri-reductions-for* *A*
by (*rule*
active-fri-reductions-for-from-trace-augmented-composition-sampled-and-empty-conflict-zero
[*OF* *trace-augmented-bound* *trace-missing-bound* *trace-merkle-bound*

```

    trace-total composition-sampled-bound composition-merkle-bound
    composition-total empty-sampled-bound empty-conflict-bound
    empty-zero-bound empty-total])
show ?thesis
by (rule
  stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-reductions
  [OF false-statement wf controlled active-fri])
qed

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-derived-active-fri-canonical
fixes TraceAug TraceMissing TraceMerkle CompCanonical CompMerkle EmptyCanonical EmptyConflict EmptyZero ::
  'f staged-proof-data  $\Rightarrow$  'f protocol-channel  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-augmented-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\Rightarrow$ 
    wp-event verify-monad
      (trace-fri-bad-with-header-tied-augmented-partial-candidate
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))  $\leq$  TraceAug data attacker-state
and trace-missing-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\Rightarrow$ 
    wp-event verify-monad
      (trace-fri-header-tied-recorded-sibling-candidate-missing
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))  $\leq$ 
      TraceMissing data attacker-state
and trace-merkle-bound:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\Rightarrow$ 
    wp-event verify-monad
      (partial-merkle-inconsistency-bad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state

```


$(\text{staged-proof-transcript data})) \leq \text{TraceMerkle data attacker-state}$
and *trace-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{TraceAug data attacker-state} +$
 $(\text{TraceMissing data attacker-state} + \text{TraceMerkle data attacker-state})$
 $\leq \text{trace-fri-error}$
and *composition-canonical-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-bad-with-canonical-sampled-layer-chain}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{CompCanonical data attacker-state}$
and *composition-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{CompMerkle data attacker-state}$
and *composition-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{CompCanonical data attacker-state} +$
 $((\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0 + 0) +$
 $\text{CompMerkle data attacker-state}) + 0)$
 $\leq \text{composition-fri-error}$
and *empty-canonical-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$

$$\begin{aligned} & \text{wp-event verify-monad} \\ & \quad (\text{trace-fri-bad-with-canonical-sampled-layer-chain} \\ & \quad \quad (\text{verifier-state-from-adversary attacker-state} \\ & \quad \quad \quad (\text{staged-proof-transcript data}))) \\ & \quad (\text{verifier-state-from-adversary attacker-state} \\ & \quad \quad (\text{staged-proof-transcript data})) \leq \\ & \quad \text{EmptyCanonical data attacker-state} \\ \text{and empty-conflict-bound:} \\ \wedge \text{data attacker-state.} \\ & \quad \text{Some (data, attacker-state)} \in \\ & \quad \quad \text{set-dist (execute (checked-staged-transcript-program A)} \\ & \quad \quad \quad \text{adversary-initial-state)} \implies \\ & \quad \text{wp-event verify-monad} \\ & \quad \quad (\text{trace-fri-sampled-assignment-conflict} \\ & \quad \quad \quad (\text{verifier-state-from-adversary attacker-state} \\ & \quad \quad \quad \quad (\text{staged-proof-transcript data}))) \\ & \quad \quad (\text{verifier-state-from-adversary attacker-state} \\ & \quad \quad \quad (\text{staged-proof-transcript data})) \leq \\ & \quad \quad \text{EmptyConflict data attacker-state} \\ \text{and empty-zero-bound:} \\ \wedge \text{data attacker-state.} \\ & \quad \text{Some (data, attacker-state)} \in \\ & \quad \quad \text{set-dist (execute (checked-staged-transcript-program A)} \\ & \quad \quad \quad \text{adversary-initial-state)} \implies \\ & \quad \text{wp-event verify-monad} \\ & \quad \quad (\text{trace-fri-zero-round-final-obstruction} \\ & \quad \quad \quad (\text{verifier-state-from-adversary attacker-state} \\ & \quad \quad \quad \quad (\text{staged-proof-transcript data}))) \\ & \quad \quad (\text{verifier-state-from-adversary attacker-state} \\ & \quad \quad \quad (\text{staged-proof-transcript data})) \leq \\ & \quad \quad \text{EmptyZero data attacker-state} \\ \text{and empty-total:} \\ \wedge \text{data attacker-state.} \\ & \quad \text{Some (data, attacker-state)} \in \\ & \quad \quad \text{set-dist (execute (checked-staged-transcript-program A)} \\ & \quad \quad \quad \text{adversary-initial-state)} \implies \\ & \quad \text{EmptyCanonical data attacker-state} + \\ & \quad \quad (\text{EmptyConflict data attacker-state} + \text{EmptyZero data attacker-state}) \\ & \quad \leq \text{trace-fri-error} \\ \text{shows checked-staged-adversary-acceptance-probability } A & \leq \\ \text{hash-collision-budget-value } 0 & \\ & \quad (\text{staged-attacker-query-budget budgets} + \\ & \quad \quad \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) + \\ & \quad (\text{composition-error-bound} + \\ & \quad \quad \text{staged-phase-relation-error size} \\ & \quad \quad (\text{staged-alpha-search-queries budgets } 0)) + \\ \text{hash-collision-budget-value } 0 & \\ & \quad (\text{staged-alpha-search-queries budgets } 0) + \\ & \quad \text{staged-phase-relation-error size} \end{aligned}$$

(staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 (Σ i < rounds.
 1 +
 (1 + staged-concrete-transcript-target-error-bound) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 trace-fri-error + composition-fri-error +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound) +
 trace-fri-error + 1)

proof –

have active-fri: active-fri-reductions-for A

by (rule
 active-fri-reductions-for-from-trace-augmented-composition-canonical-and-empty-conflict-zero
 [OF trace-augmented-bound trace-missing-bound trace-merkle-bound
 trace-total composition-canonical-bound composition-merkle-bound
 composition-total empty-canonical-bound empty-conflict-bound
 empty-zero-bound empty-total])

show ?thesis

by (rule
 stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-reductions
 [OF false-statement wf controlled active-fri])

qed

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-sampled-query-fri-compo

fixes TraceAug TraceConflict TraceZero TraceMerkle CompSampled CompMerkle EmptySam-
 pled EmptyConflict EmptyZero ::

'f staged-proof-data ⇒ 'f protocol-channel ⇒ prob

assumes false-statement: ¬ exists-valid-trace

and wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and trace-augmented-bound:

∧ data attacker-state.

Some (data, attacker-state) ∈

set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒

wp-event verify-monad

(trace-fri-bad-with-header-tied-augmented-partial-candidate
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))

(verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ TraceAug data attacker-state

and trace-conflict-bound:

$\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ TraceConflict data attacker-state
and trace-zero-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ TraceZero data attacker-state
and trace-merkle-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ TraceMerkle data attacker-state
and trace-total:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 TraceAug data attacker-state +
 ((TraceConflict data attacker-state + TraceZero data attacker-state) +
 TraceMerkle data attacker-state)
 $\leq$ trace-fri-error
and composition-sampled-bound:
 $\wedge$ data attacker-state.
 Some (data, attacker-state) $\in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (composition-fri-sampled-query-bad-candidate
 (verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤
 CompSampled data attacker-state
and composition-merkle-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (partial-merkle-inconsistency-bad
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ CompMerkle data attacker-state
and composition-total:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 CompSampled data attacker-state +
 (((CompMerkle data attacker-state + 0) +
 (CompMerkle data attacker-state + 0) +
 (CompMerkle data attacker-state + 0) +
 (CompMerkle data attacker-state + 0 + 0) +
 CompMerkle data attacker-state) + 0)
 ≤ composition-fri-error
and empty-sampled-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-sampled-query-bad-candidate
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤
 EmptySampled data attacker-state
and empty-conflict-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-sampled-assignment-conflict
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state

$(\text{staged-proof-transcript data}) \leq$
 $\text{EmptyConflict data attacker-state}$
and empty-zero-bound:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{EmptyZero data attacker-state}$
and empty-total:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program } A)$
 $\text{adversary-initial-state)} \implies$
 $\text{EmptySampled data attacker-state} +$
 $(\text{EmptyConflict data attacker-state} + \text{EmptyZero data attacker-state})$
 $\leq \text{trace-fri-error}$
shows $\text{checked-staged-adversary-acceptance-probability } A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound})) +$
 $(\sum_{i < \text{rounds.}}$
 $1 +$
 $(1 + \text{staged-concrete-transcript-target-error-bound}) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound})) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound}) +$
 $\text{trace-fri-error} + 1)$
proof –

```

have active-fri: active-fri-reductions-for A
by (rule
  active-fri-reductions-for-from-trace-augmented-conflict-zero-composition-sampled-query-and-empty-conflict-zero
  [OF trace-augmented-bound trace-conflict-bound trace-zero-bound
    trace-merkle-bound trace-total composition-sampled-bound composition-merkle-bound
    composition-total empty-sampled-bound empty-conflict-bound
    empty-zero-bound empty-total])
show ?thesis
by (rule
  stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-reductions
  [OF false-statement wf controlled active-fri])
qed

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-sampled-query-projection
fixes TraceAug TraceConflict TraceZero TraceMerkle CompSampled CompMerkle EmptySam-
pled EmptyConflict EmptyZero ::
  'f staged-proof-data  $\Rightarrow$  'f protocol-channel  $\Rightarrow$  prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-augmented-bound:
 $\wedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\Rightarrow$ 
wp-event verify-monad
    (trace-fri-bad-with-header-tied-augmented-partial-candidate
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))  $\leq$  TraceAug data attacker-state
and trace-conflict-bound:
 $\wedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\Rightarrow$ 
wp-event verify-monad
    (trace-fri-header-tied-sampled-assignment-conflict
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))  $\leq$  TraceConflict data attacker-state
and trace-zero-bound:
 $\wedge$  data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\Rightarrow$ 
wp-event verify-monad
    (trace-fri-header-tied-zero-round-final-obstruction

```

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceZero data attacker-state}$
and *trace-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{TraceMerkle data attacker-state}$
and *trace-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{TraceAug data attacker-state} +$
 $((\text{TraceConflict data attacker-state} + \text{TraceZero data attacker-state}) +$
 $\text{TraceMerkle data attacker-state})$
 $\leq \text{trace-fri-error}$
and *composition-sampled-future*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{composition-fri-future-fresh}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *composition-sampled-generic*:
 $\wedge \text{data attacker-state dg.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{generic-fri-sampled-query-projection-fraction-bound}$
 $(\text{composition-table-low-degree (to-nat dg)})$
 $(\text{Not } \circ \text{ composition-table-low-degree maxDegree) (to-nat dg)}$
 $(\text{CompSampled data attacker-state})$
and *composition-merkle-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{partial-merkle-inconsistency-bad}$
 $(\text{verifier-state-from-adversary attacker-state}$

$(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{CompMerkle data attacker-state}$
and *composition-total*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{CompSampled data attacker-state} +$
 $((\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0) +$
 $(\text{CompMerkle data attacker-state} + 0 + 0) +$
 $\text{CompMerkle data attacker-state}) + 0)$
 $\leq \text{composition-fri-error}$
and *empty-sampled-future*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{trace-fri-future-fresh}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
and *empty-sampled-generic*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{generic-fri-sampled-query-projection-fraction-bound}$
 $\text{trace-table-low-degree (Not } \circ \text{ trace-table-low-degree)}$
 $(\text{clength} - 1) (\text{EmptySampled data attacker-state})$
and *empty-conflict-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-assignment-conflict}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq$
 $\text{EmptyConflict data attacker-state}$
and *empty-zero-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$

$$\begin{aligned}
& (\text{trace-fri-zero-round-final-obstruction} \\
& \quad (\text{verifier-state-from-adversary attacker-state} \\
& \quad \quad (\text{staged-proof-transcript data}))) \\
& (\text{verifier-state-from-adversary attacker-state} \\
& \quad (\text{staged-proof-transcript data})) \leq \\
& \text{EmptyZero data attacker-state} \\
\text{and empty-total:} \\
& \wedge \text{data attacker-state.} \\
& \text{Some (data, attacker-state)} \in \\
& \quad \text{set-dist (execute (checked-staged-transcript-program A) } \\
& \quad \quad \text{adversary-initial-state)} \implies \\
& \text{EmptySampled data attacker-state} + \\
& \quad (\text{EmptyConflict data attacker-state} + \text{EmptyZero data attacker-state}) \\
& \leq \text{trace-fri-error} \\
\text{shows checked-staged-adversary-acceptance-probability A} \leq \\
& \text{hash-collision-budget-value 0} \\
& \quad (\text{staged-attacker-query-budget budgets} + \\
& \quad \quad \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) + \\
& (\text{composition-error-bound} + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets 0}) + \\
& \text{hash-collision-budget-value 0} \\
& \quad (\text{staged-alpha-search-queries budgets 0}) + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets 0}) + \\
& (1 + \text{staged-concrete-transcript-target-error-bound})) + \\
& (\sum i < \text{rounds.} \\
& \quad 1 + \\
& \quad (1 + \text{staged-concrete-transcript-target-error-bound}) + \\
& \quad (1 + \text{staged-concrete-transcript-target-error-bound})) + \\
& \text{trace-fri-error} + \text{composition-fri-error} + \\
& (\text{composition-error-bound} + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets 0}) + \\
& \text{hash-collision-budget-value 0} \\
& \quad (\text{staged-alpha-search-queries budgets 0}) + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets 0}) + \\
& (1 + \text{staged-concrete-transcript-target-error-bound}) + \\
& \text{trace-fri-error} + 1) \\
\text{proof (rule} \\
& \text{stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-sampled-query-fri-components} \\
& \quad [\text{OF false-statement wf controlled trace-augmented-bound} \\
& \quad \quad \text{trace-conflict-bound trace-zero-bound trace-merkle-bound trace-total -} \\
& \quad \quad \text{composition-merkle-bound composition-total - empty-conflict-bound} \\
& \quad \quad \text{empty-zero-bound empty-total}]) \\
\text{fix data attacker-state} \\
\text{assume support:} \\
& \text{Some (data, attacker-state)} \in
\end{aligned}$$

```

    set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  show
    wp-event verify-monad (composition-fri-sampled-query-bad-candidate ?s)
    ?s ≤ CompSampled data attacker-state
  by (rule
    wp-composition-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction
    [OF composition-sampled-future[OF support]])
    (rule composition-sampled-generic[OF support])
next
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  show
    wp-event verify-monad (trace-fri-sampled-query-bad-candidate ?s)
    ?s ≤ EmptySampled data attacker-state
  by (rule
    wp-trace-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction
    [OF empty-sampled-future[OF support]])
    (rule empty-sampled-generic[OF support])
qed

```

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-trace-active-and-staged-s

```

  fixes composition-B :: 'f ⇒ 'f list set
  and composition-Q empty-Q :: nat list set
  and CompSampled CompMerkle EmptySampled EmptyConflict EmptyZero
    composition-fresh empty-fresh :: prob
  assumes false-statement: ¬ exists-valid-trace
  and wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-fri-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state ≤ trace-fri-error
  and composition-finite: ∧dg. finite (composition-B dg)
  and composition-fresh-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-fresh-hit s composition-B))
    adversary-initial-state ≤ composition-fresh

```

and *composition-projection*:
 $\bigwedge dg. \text{fst } ((\text{composition-fri-sampled-query-bad-pair-union } dg \cap (\text{fri-query-index-list-space} \times \text{UNIV})) \cap (\text{UNIV} \times (- \text{composition-B } dg))) \subseteq \text{composition-Q}$
and *composition-sampled-total*:
 $\text{composition-fresh} + \text{hash-relation-budget-value} (\sum dg \in (\text{UNIV} :: 'f \text{ set}). \text{card } (\text{composition-B } dg) * \text{ceil-log } (\text{maxDegree} + 1)) (\text{staged-attacker-query-budget } \text{budgets} + \text{staged-challenge-query-budget}) + \text{staged-phase-target-error} (\text{query-index-raw-preimage} (\text{query-index-list-entries } \text{composition-Q})) (\text{staged-attacker-query-budget } \text{budgets} + \text{staged-challenge-query-budget}) \leq \text{CompSampled}$
and *composition-merkle-bound*:
 $\bigwedge \text{data attacker-state. Some } (\text{data}, \text{attacker-state}) \in \text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A) \text{ adversary-initial-state}) \implies \text{wp-event verify-monad } (\text{partial-merkle-inconsistency-bad } (\text{verifier-state-from-adversary } \text{attacker-state} (\text{staged-proof-transcript } \text{data}))) (\text{verifier-state-from-adversary } \text{attacker-state} (\text{staged-proof-transcript } \text{data})) \leq \text{CompMerkle}$
and *composition-total*:
 $\text{CompSampled} + (((\text{CompMerkle} + 0) + (\text{CompMerkle} + 0) + (\text{CompMerkle} + 0) + (\text{CompMerkle} + 0 + 0) + \text{CompMerkle}) + 0 + 0) \leq \text{composition-fri-error}$
and *empty-finite*: *finite empty-B*
and *empty-fresh-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A) (\text{staged-security-with-data-state-verifier-event } (\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \text{ empty-B})) \text{ adversary-initial-state} \leq \text{empty-fresh}$
and *empty-projection*:
 $\text{fst } ((\text{trace-fri-sampled-query-bad-pair-union} \cap (\text{fri-query-index-list-space} \times \text{UNIV})) \cap (\text{UNIV} \times (- \text{empty-B}))) \subseteq \text{empty-Q}$
and *empty-sampled-total*:
 $\text{empty-fresh} + \text{hash-relation-budget-value } (\text{card } \text{empty-B} * \text{ceil-log } \text{clength}) (\text{staged-attacker-query-budget } \text{budgets} +$

$$\begin{aligned}
& \text{staged-challenge-query-budget}) + \\
& \text{staged-phase-target-error} \\
& (\text{query-index-raw-preimage } (\text{query-index-list-entries empty-Q})) \\
& (\text{staged-attacker-query-budget budgets} + \\
& \quad \text{staged-challenge-query-budget}) \\
& \leq \text{EmptySampled} \\
\text{and empty-conflict-bound:} \\
& \wedge \text{data attacker-state.} \\
& \text{Some (data, attacker-state)} \in \\
& \quad \text{set-dist (execute (checked-staged-transcript-program A) } \\
& \quad \quad \text{adversary-initial-state)} \implies \\
& \text{wp-event verify-monad} \\
& \quad (\text{trace-fri-sampled-assignment-conflict} \\
& \quad \quad (\text{verifier-state-from-adversary attacker-state} \\
& \quad \quad \quad (\text{staged-proof-transcript data}))) \\
& \quad (\text{verifier-state-from-adversary attacker-state} \\
& \quad \quad (\text{staged-proof-transcript data})) \leq \text{EmptyConflict} \\
\text{and empty-zero-bound:} \\
& \wedge \text{data attacker-state.} \\
& \text{Some (data, attacker-state)} \in \\
& \quad \text{set-dist (execute (checked-staged-transcript-program A) } \\
& \quad \quad \text{adversary-initial-state)} \implies \\
& \text{wp-event verify-monad} \\
& \quad (\text{trace-fri-zero-round-final-obstruction} \\
& \quad \quad (\text{verifier-state-from-adversary attacker-state} \\
& \quad \quad \quad (\text{staged-proof-transcript data}))) \\
& \quad (\text{verifier-state-from-adversary attacker-state} \\
& \quad \quad (\text{staged-proof-transcript data})) \leq \text{EmptyZero} \\
\text{and empty-total:} \\
& \text{EmptySampled} + (\text{EmptyConflict} + \text{EmptyZero}) \leq \text{trace-fri-error} \\
\text{shows checked-staged-adversary-acceptance-probability A} \leq \\
& \text{hash-collision-budget-value 0} \\
& (\text{staged-attacker-query-budget budgets} + \\
& \quad \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) + \\
& (\text{composition-error-bound} + \\
& \quad \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets 0}) + \\
& \text{hash-collision-budget-value 0} \\
& \quad (\text{staged-alpha-search-queries budgets 0}) + \\
& \text{staged-phase-relation-error size} \\
& \quad (\text{staged-alpha-search-queries budgets 0}) + \\
& (1 + \text{staged-concrete-transcript-target-error-bound})) + \\
& (\sum_{i < \text{rounds.}} \\
& \quad 1 + \\
& \quad (1 + \text{staged-concrete-transcript-target-error-bound}) + \\
& \quad (1 + \text{staged-concrete-transcript-target-error-bound})) + \\
& \text{trace-fri-error} + \text{composition-fri-error} + \\
& (\text{composition-error-bound} + \\
& \quad \text{staged-phase-relation-error size}
\end{aligned}$$

(staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound) +
 trace-fri-error + 1)
proof –
have composition-sampled:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-sampled-query-bad-candidate)
 adversary-initial-state ≤ CompSampled
by (rule order-trans
 [OF checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-
 [OF wf controlled composition-finite composition-fresh-bound
 composition-projection]
 composition-sampled-total])
have comp-fri-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state ≤ composition-fri-error
proof –
have wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state ≤
 CompSampled +
 (((CompMerkle + 0) + (CompMerkle + 0) + (CompMerkle + 0) +
 (CompMerkle + 0 + 0) + CompMerkle) + 0 + 0)
apply (rule
 checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-sampled-query-and-residual-g
 [OF composition-sampled])
apply (rule wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
apply (rule composition-merkle-bound)
apply assumption
apply (rule wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero)
apply (rule wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-merkle-and-slot)
apply (rule composition-merkle-bound)
apply assumption
apply (rule wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
apply (rule wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-merkle-and-slot)
apply (rule composition-merkle-bound)
apply assumption
apply (rule wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
apply (rule wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step)
apply (rule composition-merkle-bound)
apply assumption

```

    apply (rule wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
    apply (rule wp-composition-fri-verifier-tied-final-value-selected-step-missing-gap-zero)
    apply (subst wp-composition-fri-verifier-tied-zero-round-final-obstruction-zero)
    apply simp
  done
then show ?thesis
  by (rule order-trans[OF - composition-total])
qed
have empty-sampled:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-query-bad-candidate)
  adversary-initial-state ≤ EmptySampled
by (rule order-trans
  [OF checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-project
    [OF wf controlled empty-finite empty-fresh-bound empty-projection]
    empty-sampled-total])
have empty-trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state ≤ trace-fri-error
proof -
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state ≤ EmptySampled + (EmptyConflict + EmptyZero)
  by (rule
    checked-staged-security-trace-fri-empty-header-bound-from-sampled-query-conflict-and-zero
      [OF empty-sampled empty-conflict-bound empty-zero-bound])
  then show ?thesis
    by (rule order-trans[OF - empty-total])
qed
show ?thesis
  by (rule
    current-query-route-bound-from-active-fri-bounds-internal
      [OF false-statement wf controlled trace-fri-bound comp-fri-bound
        empty-trace-fri-bound])
qed

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-full-cover-domain-confli
fixes trace-bad ::
  'f list ⇒ nat ⇒ 'f list ⇒ 'f set
and composition-bad ::
  'f ⇒ 'f list ⇒ nat ⇒ 'f list ⇒ 'f set
and trace-B :: 'f list set
and composition-B :: 'f ⇒ 'f list set
and trace-fresh trace-proximity trace-index trace-domain-length
  trace-sampled-domain trace-conflict trace-zero

```

composition-fresh composition-proximity composition-index
composition-domain-length composition-sampled-domain
composition-conflict composition-zero
empty-conflict empty-zero :: prob
assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *trace-finite*: *finite trace-B*
and *trace-envelope*:
 $\wedge$ *data attacker-state trace-table*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
trace-fri-multiround-bad-sets trace-bad trace-table $\subseteq$ trace-B
and *trace-fresh-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
*(λs . *trace-fri-challenge-list-fresh-hit s trace-B*))*
adversary-initial-state $\leq$ trace-fresh
and *trace-proximity-bound*:
 $\wedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(trace-fri-sampled-full-cover-proximity-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) trace-bad)
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$ trace-proximity
and *trace-index-bound*:
 $\wedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(trace-fri-sampled-full-cover-index-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) $\leq$ trace-index
and *trace-domain-length-bound*:
 $\wedge$ *data attacker-state*.
Some (data, attacker-state) $\in$
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$
wp-event verify-monad
(trace-fri-sampled-full-cover-domain-length-failure
(verifier-state-from-adversary attacker-state

(staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ trace-domain-length
and trace-sampled-domain-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-sampled-full-cover-sampled-domain-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ trace-sampled-domain
and trace-conflict-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-header-tied-sampled-assignment-conflict
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ trace-conflict
and trace-zero-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-header-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ trace-zero
and trace-total:
 ((trace-fresh +
 hash-relation-budget-value (card trace-B * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (trace-proximity + trace-index +
 (trace-index + trace-domain-length) +
 (trace-index + trace-sampled-domain))) +
 (trace-conflict + trace-zero)
 ≤ trace-fri-error
and composition-finite: ∧ dg. finite (composition-B dg)
and composition-envelope:
 ∧ data attacker-state dg composition-table.

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
composition-fri-multiround-bad-sets composition-bad dg composition-table
⊆ composition-B dg
and *composition-fresh-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
(λs. composition-fri-challenge-list-fresh-hit s composition-B))
adversary-initial-state ≤ composition-fresh
and *composition-proximity-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(composition-fri-sampled-full-cover-proximity-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) composition-bad)
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ composition-proximity
and *composition-index-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(composition-fri-sampled-full-cover-index-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ composition-index
and *composition-domain-length-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(composition-fri-sampled-full-cover-domain-length-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ composition-domain-length
and *composition-sampled-domain-bound:*
 $\wedge$ *data attacker-state.*
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad

(composition-fri-sampled-full-cover-sampled-domain-failure
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ composition-sampled-domain
and composition-conflict-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (composition-fri-verifier-tied-sampled-assignment-conflict
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ composition-conflict
and composition-zero-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (composition-fri-verifier-tied-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) ≤ composition-zero
and composition-total:
 ((composition-fresh +
 hash-relation-budget-value
 (∑ dg ∈ (UNIV :: 'f set).
 card (composition-B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (composition-proximity + composition-index +
 (composition-index + composition-domain-length) +
 (composition-index + composition-sampled-domain))) +
 (composition-conflict + composition-zero)
 ≤ composition-fri-error
and empty-conflict-bound:
 ∧ data attacker-state.
 Some (data, attacker-state) ∈
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) ⇒
 wp-event verify-monad
 (trace-fri-sampled-assignment-conflict
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state

$(\text{staged-proof-transcript data})) \leq \text{empty-conflict}$
and empty-zero-bound :
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{empty-zero}$
and empty-total :
 $((\text{trace-fresh} +$
 $\text{hash-relation-budget-value (card trace-B * ceil-log clength)}$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(\text{trace-proximity} + \text{trace-index} +$
 $(\text{trace-index} + \text{trace-domain-length}) +$
 $(\text{trace-index} + \text{trace-sampled-domain}))) +$
 $(\text{empty-conflict} + \text{empty-zero})$
 $\leq \text{trace-fri-error}$
shows $\text{checked-staged-adversary-acceptance-probability A} \leq$
 $\text{hash-collision-budget-value 0}$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets 0}) +$
 $\text{hash-collision-budget-value 0}$
 $(\text{staged-alpha-search-queries budgets 0}) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets 0}) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound})) +$
 $(\sum i < \text{rounds.}$
 $1 +$
 $(1 + \text{staged-concrete-transcript-target-error-bound}) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound})) +$
 $\text{trace-fri-error} + \text{composition-fri-error} +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets 0}) +$
 $\text{hash-collision-budget-value 0}$
 $(\text{staged-alpha-search-queries budgets 0}) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets 0}) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound}) +$
 $\text{trace-fri-error} + 1)$
proof –

```

have trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-partial-candidate)
  adversary-initial-state  $\leq$  trace-fri-error
proof –
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-partial-candidate)
  adversary-initial-state  $\leq$ 
    ((trace-fresh +
      hash-relation-budget-value (card trace-B * ceil-log clength)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) +
      (trace-proximity + trace-index +
        (trace-index + trace-domain-length) +
        (trace-index + trace-sampled-domain))) +
      (trace-conflict + trace-zero)
  by (rule
    checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-full-cover-domain-conflict-and-zero
    [where bad = trace-bad];
    (rule wf | rule controlled | rule trace-finite |
      rule trace-envelope | rule trace-fresh-bound |
      rule trace-proximity-bound | rule trace-index-bound |
      rule trace-domain-length-bound | rule trace-sampled-domain-bound |
      rule trace-conflict-bound | rule trace-zero-bound))
  then show ?thesis
  by (rule order-trans[OF - trace-total])
qed
have comp-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-verifier-tied-partial-candidate)
  adversary-initial-state  $\leq$  composition-fri-error
proof –
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-verifier-tied-partial-candidate)
  adversary-initial-state  $\leq$ 
    ((composition-fresh +
      hash-relation-budget-value
        ( $\sum dg \in (UNIV :: 'f \text{ set}).$ 
          card (composition-B dg) * ceil-log (maxDegree + 1))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) +
      (composition-proximity + composition-index +
        (composition-index + composition-domain-length) +
        (composition-index + composition-sampled-domain))) +
      (composition-conflict + composition-zero)

```

```

by (rule
  checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-full-cover-domain-conflict-and-
    [where bad = composition-bad];
  (rule wf | rule controlled | rule composition-finite |
    rule composition-envelope | rule composition-fresh-bound |
    rule composition-proximity-bound | rule composition-index-bound |
    rule composition-domain-length-bound |
    rule composition-sampled-domain-bound |
    rule composition-conflict-bound | rule composition-zero-bound))
then show ?thesis
  by (rule order-trans[OF - composition-total])
qed
have empty-trace-fri-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state  $\leq$  trace-fri-error
proof -
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state  $\leq$ 
    ((trace-fresh +
      hash-relation-budget-value (card trace-B * ceil-log clength)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) +
      (trace-proximity + trace-index +
        (trace-index + trace-domain-length) +
        (trace-index + trace-sampled-domain))) +
      (empty-conflict + empty-zero)
  by (rule
    checked-staged-security-trace-fri-empty-header-bound-from-full-cover-domain-conflict-and-zero
      [where bad = trace-bad];
    (rule wf | rule controlled | rule trace-finite |
      rule trace-envelope | rule trace-fresh-bound |
      rule trace-proximity-bound | rule trace-index-bound |
      rule trace-domain-length-bound | rule trace-sampled-domain-bound |
      rule empty-conflict-bound | rule empty-zero-bound))
  then show ?thesis
    by (rule order-trans[OF - empty-total])
qed
show ?thesis
  by (rule
    current-query-route-bound-from-active-fri-bounds-internal
      [OF false-statement wf controlled trace-fri-bound comp-fri-bound
        empty-trace-fri-bound])
qed

```

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-full-cover-domain-conflict-and-zero

```

fixes trace-bad ::
  'f list  $\Rightarrow$  nat  $\Rightarrow$  'f list  $\Rightarrow$  'f set
and composition-bad ::
  'f  $\Rightarrow$  'f list  $\Rightarrow$  nat  $\Rightarrow$  'f list  $\Rightarrow$  'f set
and trace-fresh trace-proximity trace-index trace-domain-length
  trace-sampled-domain trace-conflict trace-zero
  composition-fresh composition-proximity composition-index
  composition-domain-length composition-sampled-domain
  composition-conflict composition-zero
  empty-conflict empty-zero :: prob
assumes false-statement:  $\neg$  exists-valid-trace
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and trace-fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s.$ 
        trace-fri-challenge-list-fresh-hit s
        (fri-challenge-space (ceil-log clength))))
    adversary-initial-state  $\leq$  trace-fresh
and trace-proximity-bound:
   $\wedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\Rightarrow$ 
      wp-event verify-monad
        (trace-fri-sampled-full-cover-proximity-failure
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))) trace-bad)
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))  $\leq$  trace-proximity
and trace-index-bound:
   $\wedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\Rightarrow$ 
      wp-event verify-monad
        (trace-fri-sampled-full-cover-index-failure
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data))))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))  $\leq$  trace-index
and trace-domain-length-bound:
   $\wedge$  data attacker-state.
    Some (data, attacker-state)  $\in$ 
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\Rightarrow$ 
      wp-event verify-monad
        (trace-fri-sampled-full-cover-domain-length-failure

```

$(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{trace-domain-length}$
and $\text{trace-sampled-domain-bound}$:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-full-cover-sampled-domain-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{trace-sampled-domain}$
and $\text{trace-conflict-bound}$:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-sampled-assignment-conflict}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{trace-conflict}$
and trace-zero-bound :
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-header-tied-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data}))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{trace-zero}$
and trace-total :
 $((\text{trace-fresh} +$
 $\text{hash-relation-budget-value}$
 $(\text{card (fri-challenge-space (ceil-log clength))} * \text{ceil-log clength})$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(\text{trace-proximity} + \text{trace-index} +$
 $(\text{trace-index} + \text{trace-domain-length}) +$
 $(\text{trace-index} + \text{trace-sampled-domain}))) +$
 $(\text{trace-conflict} + \text{trace-zero})$
 $\leq \text{trace-fri-error}$
and $\text{composition-fresh-bound}$:

$wp\text{-}event$ (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s.$
 composition-fri-challenge-list-fresh-hit s
 ($\lambda dg.$ *fri-challenge-space* (*ceil-log* (*to-nat* $dg + 1$))))))
adversary-initial-state $\leq$ *composition-fresh*
and *composition-proximity-bound*:
 $\wedge$ *data attacker-state*.
 Some (*data*, *attacker-state*) $\in$
 set-dist (*execute* (*checked-staged-transcript-program* A)
 adversary-initial-state) $\implies$
 $wp\text{-}event$ *verify-monad*
 (*composition-fri-sampled-full-cover-proximity-failure*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) *composition-bad*)
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\leq$ *composition-proximity*
and *composition-index-bound*:
 $\wedge$ *data attacker-state*.
 Some (*data*, *attacker-state*) $\in$
 set-dist (*execute* (*checked-staged-transcript-program* A)
 adversary-initial-state) $\implies$
 $wp\text{-}event$ *verify-monad*
 (*composition-fri-sampled-full-cover-index-failure*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\leq$ *composition-index*
and *composition-domain-length-bound*:
 $\wedge$ *data attacker-state*.
 Some (*data*, *attacker-state*) $\in$
 set-dist (*execute* (*checked-staged-transcript-program* A)
 adversary-initial-state) $\implies$
 $wp\text{-}event$ *verify-monad*
 (*composition-fri-sampled-full-cover-domain-length-failure*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)) $\leq$ *composition-domain-length*
and *composition-sampled-domain-bound*:
 $\wedge$ *data attacker-state*.
 Some (*data*, *attacker-state*) $\in$
 set-dist (*execute* (*checked-staged-transcript-program* A)
 adversary-initial-state) $\implies$
 $wp\text{-}event$ *verify-monad*
 (*composition-fri-sampled-full-cover-sampled-domain-failure*
 (*verifier-state-from-adversary attacker-state*
 (*staged-proof-transcript data*)))
 (*verifier-state-from-adversary attacker-state*

$(\text{staged-proof-transcript data})) \leq \text{composition-sampled-domain}$
and *composition-conflict-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-sampled-assignment-conflict}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{composition-conflict}$
and *composition-zero-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-verifier-tied-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{composition-zero}$
and *composition-total*:
 $((\text{composition-fresh} +$
 $\text{hash-relation-budget-value}$
 $(\sum dg \in (\text{UNIV} :: 'f \text{ set}).$
 $\text{card (fri-challenge-space (ceil-log (to-nat dg + 1)))} *$
 $\text{ceil-log (maxDegree + 1))}$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(\text{composition-proximity} + \text{composition-index} +$
 $(\text{composition-index} + \text{composition-domain-length}) +$
 $(\text{composition-index} + \text{composition-sampled-domain}))) +$
 $(\text{composition-conflict} + \text{composition-zero})$
 $\leq \text{composition-fri-error}$
and *empty-conflict-bound*:
 $\wedge \text{data attacker-state.}$
 $\text{Some (data, attacker-state)} \in$
 $\text{set-dist (execute (checked-staged-transcript-program A)}$
 $\text{adversary-initial-state)} \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-assignment-conflict}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{empty-conflict}$
and *empty-zero-bound*:
 $\wedge \text{data attacker-state.}$

Some $(data, attacker-state) \in$
 set-dist (execute (checked-staged-transcript-program A)
 adversary-initial-state) $\implies$
 wp-event verify-monad
 (trace-fri-zero-round-final-obstruction
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)) $\leq$ empty-zero
and empty-total:
 ((trace-fresh +
 hash-relation-budget-value
 (card (fri-challenge-space (ceil-log clength)) * ceil-log clength)
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget)) +
 (trace-proximity + trace-index +
 (trace-index + trace-domain-length) +
 (trace-index + trace-sampled-domain))) +
 (empty-conflict + empty-zero)
 $\leq$ trace-fri-error
shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds.}$
 1 +
 (1 + staged-concrete-transcript-target-error-bound) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 trace-fri-error + composition-fri-error +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound) +
 trace-fri-error + 1)
proof (rule
 stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-full-cover-domain-conflict-zero-c
 [where trace-B = fri-challenge-space (ceil-log clength)

```

and composition-B =
   $\lambda dg. \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$ 
and trace-bad = trace-bad
and composition-bad = composition-bad])
show  $\neg \text{exists-valid-trace}$ 
  by (rule false-statement)
show staged-budget-wellformed budgets
  by (rule wf)
show staged-adversary-controlled budgets A
  by (rule controlled)
show finite (fri-challenge-space (ceil-log clength))
  by simp
show  $\bigwedge \text{data attacker-state trace-table.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
    trace-fri-multiround-bad-sets trace-bad trace-table
     $\subseteq$  fri-challenge-space (ceil-log clength)
  by (rule trace-fri-multiround-bad-sets-subset)
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s.$ 
      trace-fri-challenge-list-fresh-hit s
      (fri-challenge-space (ceil-log clength))))
  adversary-initial-state  $\leq$  trace-fresh
  by (rule trace-fresh-bound)
show  $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
  wp-event verify-monad
    (trace-fri-sampled-full-cover-proximity-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) trace-bad)
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))  $\leq$  trace-proximity
  by (rule trace-proximity-bound)
show  $\bigwedge \text{data attacker-state.}$ 
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
  wp-event verify-monad
    (trace-fri-sampled-full-cover-index-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))  $\leq$  trace-index
  by (rule trace-index-bound)
show  $\bigwedge \text{data attacker-state.}$ 

```

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-sampled-full-cover-domain-length-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ trace-domain-length
by (rule trace-domain-length-bound)
show $\bigwedge$ data attacker-state.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-sampled-full-cover-sampled-domain-failure
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ trace-sampled-domain
by (rule trace-sampled-domain-bound)
show $\bigwedge$ data attacker-state.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-sampled-assignment-conflict
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ trace-conflict
by (rule trace-conflict-bound)
show $\bigwedge$ data attacker-state.
Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒
wp-event verify-monad
(trace-fri-header-tied-zero-round-final-obstruction
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)) ≤ trace-zero
by (rule trace-zero-bound)
show ((trace-fresh +
hash-relation-budget-value
(card (fri-challenge-space (ceil-log clength)) * ceil-log clength)
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)) +
(trace-proximity + trace-index +

$(\text{trace-index} + \text{trace-domain-length}) +$
 $(\text{trace-index} + \text{trace-sampled-domain})) +$
 $(\text{trace-conflict} + \text{trace-zero})$
 $\leq \text{trace-fri-error}$
by (rule trace-total)
show $\bigwedge dg. \text{finite} (\text{fri-challenge-space} (\text{ceil-log} (\text{to-nat } dg + 1)))$
by simp
show $\bigwedge \text{data attacker-state } dg \text{ composition-table.}$
 $\text{Some} (\text{data}, \text{attacker-state}) \in$
 $\text{set-dist} (\text{execute} (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{composition-fri-multiround-bad-sets composition-bad } dg$
 composition-table
 $\subseteq \text{fri-challenge-space} (\text{ceil-log} (\text{to-nat } dg + 1))$
by (rule composition-fri-multiround-bad-sets-subset)
show $\text{wp-event} (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s.$
 $\text{composition-fri-challenge-list-fresh-hit } s$
 $(\lambda dg. \text{fri-challenge-space} (\text{ceil-log} (\text{to-nat } dg + 1))))))$
 $\text{adversary-initial-state} \leq \text{composition-fresh}$
by (rule composition-fresh-bound)
show $\bigwedge \text{data attacker-state.}$
 $\text{Some} (\text{data}, \text{attacker-state}) \in$
 $\text{set-dist} (\text{execute} (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-proximity-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \text{ composition-bad})$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{composition-proximity}$
by (rule composition-proximity-bound)
show $\bigwedge \text{data attacker-state.}$
 $\text{Some} (\text{data}, \text{attacker-state}) \in$
 $\text{set-dist} (\text{execute} (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{composition-fri-sampled-full-cover-index-failure}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{composition-index}$
by (rule composition-index-bound)
show $\bigwedge \text{data attacker-state.}$
 $\text{Some} (\text{data}, \text{attacker-state}) \in$
 $\text{set-dist} (\text{execute} (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$

```

    (composition-fri-sampled-full-cover-domain-length-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ composition-domain-length
  by (rule composition-domain-length-bound)
show ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  wp-event verify-monad
    (composition-fri-sampled-full-cover-sampled-domain-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ composition-sampled-domain
  by (rule composition-sampled-domain-bound)
show ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  wp-event verify-monad
    (composition-fri-verifier-tied-sampled-assignment-conflict
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ composition-conflict
  by (rule composition-conflict-bound)
show ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  wp-event verify-monad
    (composition-fri-verifier-tied-zero-round-final-obstruction
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ composition-zero
  by (rule composition-zero-bound)
show ((composition-fresh +
  hash-relation-budget-value
    (∑ dg ∈ (UNIV :: 'f set).
      card (fri-challenge-space (ceil-log (to-nat dg + 1))) *
        ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)) +
  (composition-proximity + composition-index +
    (composition-index + composition-domain-length) +
    (composition-index + composition-sampled-domain))) +

```

$(\text{composition-conflict} + \text{composition-zero})$
 $\leq \text{composition-fri-error}$
by (rule *composition-total*)
show $\bigwedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-sampled-assignment-conflict}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{empty-conflict}$
by (rule *empty-conflict-bound*)
show $\bigwedge \text{data attacker-state.}$
 $\text{Some } (data, \text{attacker-state}) \in$
 $\text{set-dist } (\text{execute } (\text{checked-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$
 $\text{wp-event verify-monad}$
 $(\text{trace-fri-zero-round-final-obstruction}$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})))$
 $(\text{verifier-state-from-adversary attacker-state}$
 $(\text{staged-proof-transcript data})) \leq \text{empty-zero}$
by (rule *empty-zero-bound*)
show $((\text{trace-fresh} +$
 $\text{hash-relation-budget-value}$
 $(\text{card } (\text{fri-challenge-space } (\text{ceil-log clength})) * \text{ceil-log clength})$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})) +$
 $(\text{trace-proximity} + \text{trace-index} +$
 $(\text{trace-index} + \text{trace-domain-length}) +$
 $(\text{trace-index} + \text{trace-sampled-domain}))) +$
 $(\text{empty-conflict} + \text{empty-zero})$
 $\leq \text{trace-fri-error}$
by (rule *empty-total*)
qed

definition *concrete-trace-fri-header-error-for*

where

$\text{concrete-trace-fri-header-error-for } A =$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$
 $\text{adversary-initial-state}$

definition *concrete-composition-fri-error-for*

where

$\text{concrete-composition-fri-error-for } A =$

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state

definition *concrete-trace-fri-empty-error-for*

where

concrete-trace-fri-empty-error-for A =
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route:*

assumes *false-statement: $\neg$ exists-valid-trace*

and *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows *checked-staged-adversary-acceptance-probability A $\leq$*

hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
(1 + staged-concrete-transcript-target-error-bound)) +
($\sum_{i < \text{rounds.}}$
1 +
(1 + staged-concrete-transcript-target-error-bound) +
(1 + staged-concrete-transcript-target-error-bound)) +
concrete-trace-fri-header-error-for A +
concrete-composition-fri-error-for A +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
(1 + staged-concrete-transcript-target-error-bound) +
concrete-trace-fri-empty-error-for A + 1)

proof –

have *trace-fri-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-partial-candidate)

$\text{adversary-initial-state} \leq \text{concrete-trace-fri-header-error-for } A$
by (simp add: concrete-trace-fri-header-error-for-def)
have comp-fri-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{composition-fri-bad-with-verifier-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq \text{concrete-composition-fri-error-for } A$
by (simp add: concrete-composition-fri-error-for-def)
have empty-trace-fri-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state} \leq \text{concrete-trace-fri-empty-error-for } A$
by (simp add: concrete-trace-fri-empty-error-for-def)
show ?thesis
by (rule
 $\text{current-query-route-bound-from-active-fri-bounds-internal}$
 $[OF \text{ false-statement wf controlled trace-fri-bound comp-fri-bound}$
 $\text{empty-trace-fri-bound}]$)
qed

theorem stark-soundness-same-run-endpoint:
assumes false-statement: $\neg \text{exists-valid-trace}$
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows checked-staged-adversary-acceptance-probability $A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound})) +$
 $(\sum_{i < \text{rounds.}} 1 +$
 $(1 + \text{staged-concrete-transcript-target-error-bound}) +$
 $(1 + \text{staged-concrete-transcript-target-error-bound})) +$
 $\text{concrete-trace-fri-header-error-for } A +$
 $\text{concrete-composition-fri-error-for } A +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$

```

    (staged-alpha-search-queries budgets 0) +
    (1 + staged-concrete-transcript-target-error-bound) +
    concrete-trace-fri-empty-error-for A + 1)
  by (rule
    stark-soundness-from-current-public-prefix-current-query-relevant-drift-route
    [OF false-statement wf controlled])

end

hide-fact
  soundness.stark-soundness-same-run-endpoint
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-staged-active-fri-bound
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-bounds
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-reductions
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-derived-active-fri-comp
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-derived-active-fri-conf
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-derived-active-fri-cano
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-sampled-query-fri-comp
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-sampled-query-projecti
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-trace-active-and-staged
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-full-cover-domain-conf
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-full-cover-domain-conf
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-residual-obli
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-residual-com

end

theory Soundness-Public-Route-Concrete-FRI-Endpoint
  imports Soundness-Public-Route-Same-Run
begin

Public-route endpoint with concrete FRI losses expanded one step below the top-level
bad-event probabilities.

This theory is retained as infrastructure for the current sharpened endpoint: downstream
theories reuse the reachable verifier-event envelope definitions and one-step concrete FRI
bounds. The active public theorem is exported by Soundness_FRI_Trace_Restricted_Query_Endpoint.th

context soundness
begin

definition reachable-verifier-event-bound-for
where
  reachable-verifier-event-bound-for A E =
    Max (insert 0
      ((λout. case out of
        None ⇒ 0
        | Some y ⇒
          let s =

```

```

      verifier-state-from-adversary (snd y)
      (staged-proof-transcript (fst y))
    in wp-event verify-monad (E s) s) ‘
  set-dist
  (execute (checked-staged-transcript-program A)
    adversary-initial-state)))

```

lemma *reachable-verifier-event-bound-for-ge:*

assumes *support:*

```

  Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)
    adversary-initial-state)

```

shows

```

  wp-event verify-monad
  (E
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  ≤ reachable-verifier-event-bound-for A E

```

proof –

```

  let ?F =
    (λout. case out of
      None ⇒ 0
    | Some y ⇒
      let s =
        verifier-state-from-adversary (snd y)
        (staged-proof-transcript (fst y))
      in wp-event verify-monad (E s) s)
  let ?S =
    insert 0
    (?F ‘
      set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state))

```

have *finite-S: finite ?S*

by *simp*

have *mem: ?F (Some (data, attacker-state)) ∈ ?S*

using *support* **by** *blast*

have *le: ?F (Some (data, attacker-state)) ≤ Max ?S*

by (rule *Max-ge[OF finite-S mem]*)

have *F-eq:*

```

  ?F (Some (data, attacker-state)) =
  wp-event verify-monad
  (E
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state

```

```

      (staged-proof-transcript data))
    by (simp add: Let-def)
  show ?thesis
    unfolding reachable-verifier-event-bound-for-def
    using le F-eq by simp
qed

```

definition *trace-fri-full-challenge-bad*

where

trace-fri-full-challenge-bad - - - = *UNIV*

definition *composition-fri-full-challenge-bad*

where

composition-fri-full-challenge-bad - - - = *UNIV*

definition *concrete-trace-fri-full-cover-domain-error-for*

where

concrete-trace-fri-full-cover-domain-error-for budgets *A* =
 ((*wp-event* (*checked-staged-security-experiment-with-data-state* *A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s.$
 trace-fri-challenge-list-fresh-hit *s*
 (*fri-challenge-space* (*ceil-log* *clength*))))
 adversary-initial-state +
 hash-relation-budget-value
 (*card* (*fri-challenge-space* (*ceil-log* *clength*))) * *ceil-log* *clength*)
 (*staged-attacker-query-budget* budgets +
 staged-challenge-query-budget)) +
 (*reachable-verifier-event-bound-for* *A*
 ($\lambda s.$
 trace-fri-sampled-full-cover-proximity-failure *s*
 trace-fri-full-challenge-bad) +
 reachable-verifier-event-bound-for *A*
 trace-fri-sampled-full-cover-index-failure +
 (*reachable-verifier-event-bound-for* *A*
 trace-fri-sampled-full-cover-index-failure +
 reachable-verifier-event-bound-for *A*
 trace-fri-sampled-full-cover-domain-length-failure) +
 (*reachable-verifier-event-bound-for* *A*
 trace-fri-sampled-full-cover-index-failure +
 reachable-verifier-event-bound-for *A*
 trace-fri-sampled-full-cover-sampled-domain-failure))))))

definition *concrete-trace-fri-header-error-for'*

where

concrete-trace-fri-header-error-for' budgets *A* =
concrete-trace-fri-full-cover-domain-error-for budgets *A* +
 (*reachable-verifier-event-bound-for* *A*
 trace-fri-header-tied-sampled-assignment-conflict +

reachable-verifier-event-bound-for A
trace-fri-header-tied-zero-round-final-obstruction)

definition *concrete-trace-fri-empty-error-for'*

where

concrete-trace-fri-empty-error-for' budgets $A =$
concrete-trace-fri-full-cover-domain-error-for budgets $A +$
(reachable-verifier-event-bound-for A
trace-fri-sampled-assignment-conflict +
reachable-verifier-event-bound-for A
trace-fri-zero-round-final-obstruction)

definition *concrete-composition-fri-full-cover-domain-error-for*

where

concrete-composition-fri-full-cover-domain-error-for budgets $A =$
((wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
(λs.
composition-fri-challenge-list-fresh-hit s
(λdg. fri-challenge-space (ceil-log (to-nat dg + 1))))))
adversary-initial-state +
hash-relation-budget-value
(∑ dg ∈ (UNIV :: 'f set).
*card (fri-challenge-space (ceil-log (to-nat dg + 1))) **
ceil-log (maxDegree + 1))
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)) +
(reachable-verifier-event-bound-for A
(λs.
composition-fri-sampled-full-cover-proximity-failure s
composition-fri-full-challenge-bad) +
reachable-verifier-event-bound-for A
composition-fri-sampled-full-cover-index-failure +
(reachable-verifier-event-bound-for A
composition-fri-sampled-full-cover-index-failure +
reachable-verifier-event-bound-for A
composition-fri-sampled-full-cover-domain-length-failure) +
(reachable-verifier-event-bound-for A
composition-fri-sampled-full-cover-index-failure +
reachable-verifier-event-bound-for A
composition-fri-sampled-full-cover-sampled-domain-failure))))

definition *concrete-composition-fri-error-for'*

where

concrete-composition-fri-error-for' budgets $A =$
concrete-composition-fri-full-cover-domain-error-for budgets $A +$
(reachable-verifier-event-bound-for A
composition-fri-verifier-tied-sampled-assignment-conflict +
reachable-verifier-event-bound-for A

composition-fri-verifier-tied-zero-round-final-obstruction)

lemma *concrete-trace-fri-header-error-for'-bound:*

assumes *wf: staged-budget-wellformed budgets*

and controlled: *staged-adversary-controlled budgets A*

shows

wp-event (checked-staged-security-experiment-with-data-state A)

*(staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-partial-candidate)*

adversary-initial-state ≤ concrete-trace-fri-header-error-for' budgets A

unfolding *concrete-trace-fri-header-error-for'-def*

concrete-trace-fri-full-cover-domain-error-for-def

by *(rule*

checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-full-cover-domain-conflict-and-zero

[where bad = trace-fri-full-challenge-bad

and B = fri-challenge-space (ceil-log clength),

OF wf controlled])

(auto intro!: reachable-verifier-event-bound-for-ge

dest: trace-fri-multiround-bad-sets-subset[THEN subsetD])

lemma *concrete-trace-fri-empty-error-for'-bound:*

assumes *wf: staged-budget-wellformed budgets*

and controlled: *staged-adversary-controlled budgets A*

shows

wp-event (checked-staged-security-experiment-with-data-state A)

*(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)*

adversary-initial-state ≤ concrete-trace-fri-empty-error-for' budgets A

unfolding *concrete-trace-fri-empty-error-for'-def*

concrete-trace-fri-full-cover-domain-error-for-def

by *(rule*

checked-staged-security-trace-fri-empty-header-bound-from-full-cover-domain-conflict-and-zero

[where bad = trace-fri-full-challenge-bad

and B = fri-challenge-space (ceil-log clength),

OF wf controlled])

(auto intro!: reachable-verifier-event-bound-for-ge

dest: trace-fri-multiround-bad-sets-subset[THEN subsetD])

lemma *concrete-composition-fri-error-for'-bound:*

assumes *wf: staged-budget-wellformed budgets*

and controlled: *staged-adversary-controlled budgets A*

shows

wp-event (checked-staged-security-experiment-with-data-state A)

*(staged-security-with-data-state-verifier-event
composition-fri-bad-with-verifier-tied-partial-candidate)*

adversary-initial-state ≤ concrete-composition-fri-error-for' budgets A

unfolding *concrete-composition-fri-error-for'-def*

concrete-composition-fri-full-cover-domain-error-for-def

by *(rule*

checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-full-cover-domain-conflict-and-z
 [where bad = composition-fri-full-challenge-bad
 and B = $\lambda dg. \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$,
 OF wf controlled])
 (auto intro!: reachable-verifier-event-bound-for-ge
 dest: composition-fri-multiround-bad-sets-subset[THEN subsetD])

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route:

assumes false-statement: $\neg \text{exists-valid-trace}$
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows checked-staged-adversary-acceptance-probability A $\leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds.}$
 1 +
 (1 + staged-concrete-transcript-target-error-bound) +
 (1 + staged-concrete-transcript-target-error-bound)) +
 concrete-trace-fri-header-error-for' budgets A +
 concrete-composition-fri-error-for' budgets A +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (1 + staged-concrete-transcript-target-error-bound) +
 concrete-trace-fri-empty-error-for' budgets A + 1)
by (rule
 current-query-route-bound-from-active-fri-bounds-internal
 [OF false-statement wf controlled
 concrete-trace-fri-header-error-for'-bound[OF wf controlled]
 concrete-composition-fri-error-for'-bound[OF wf controlled]
 concrete-trace-fri-empty-error-for'-bound[OF wf controlled]])

theorem stark-soundness-concrete-fri-endpoint:

assumes false-statement: $\neg \text{exists-valid-trace}$
and wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A

shows *checked-staged-adversary-acceptance-probability* $A \leq$
hash-collision-budget-value 0
 (*staged-attacker-query-budget* *budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
 (1 + *staged-concrete-transcript-target-error-bound*)) +
 ($\sum_{i < \text{rounds}}$
 1 +
 (1 + *staged-concrete-transcript-target-error-bound*) +
 (1 + *staged-concrete-transcript-target-error-bound*)) +
concrete-trace-fri-header-error-for' *budgets* A +
concrete-composition-fri-error-for' *budgets* A +
 (*composition-error-bound* +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
hash-collision-budget-value 0
 (*staged-alpha-search-queries* *budgets* 0) +
staged-phase-relation-error *size*
 (*staged-alpha-search-queries* *budgets* 0) +
 (1 + *staged-concrete-transcript-target-error-bound*) +
concrete-trace-fri-empty-error-for' *budgets* A + 1)
by (*rule*
stark-soundness-from-current-public-prefix-current-query-relevant-drift-route
 [*OF false-statement* *wf controlled*])

end

hide-fact

soundness.stark-soundness-concrete-fri-endpoint
soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route

end

theory *Soundness-FRI-Query-Fiber-Staged-Bounds*

imports

Soundness-FRI-Query-Index-Staged-Bounds
Soundness-FRI-Query-Aware-Interface

begin

Staged sampled-query FRI bounds from generic query fibers.

The checked staged query-index accounting currently consumes a fixed query target.
 This layer derives such a target from the union of the generic per-challenge query fibers

outside a chosen challenge-list cover. It keeps the query-aware FRI boundary explicit without changing the protocol or the random-oracle model.

context *soundness*

begin

definition *generic-fri-sampled-query-outside-query-cover*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow \text{nat} \Rightarrow$
 $'f \text{ list set} \Rightarrow \text{nat list set}$

where

generic-fri-sampled-query-outside-query-cover low-degree candidate-bad
degree-bound B =
 $(\bigcup \text{challenges} \in$
 $\text{fri-challenge-space } (\text{fri-round-count-for-degree-bound degree-bound}) - B.$
generic-fri-sampled-query-query-fiber low-degree candidate-bad
degree-bound challenges)

definition *generic-fri-sampled-query-missing-full-cover-fiber*

$:: ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow ('f \text{ list} \Rightarrow \text{bool}) \Rightarrow \text{nat} \Rightarrow$
 $'f \text{ list} \Rightarrow \text{nat list set}$

where

generic-fri-sampled-query-missing-full-cover-fiber low-degree candidate-bad
degree-bound challenges =
 $\{\text{query-idxs} \in \text{fri-query-index-list-space}.$
 $\exists \text{candidate-table roots final-value round-layers layers}.$
 $\text{generic-fri-sampled-query-candidate-evidence low-degree candidate-bad}$
 $\text{candidate-table degree-bound roots challenges final-value query-idxs}$
 $\text{round-layers layers} \wedge$
 $\neg \text{fri-sampled-layers-cover-fold-indices roots query-idxs}\}$

lemma *generic-fri-sampled-query-missing-full-cover-fiber-subset:*

generic-fri-sampled-query-missing-full-cover-fiber low-degree candidate-bad
degree-bound challenges $\subseteq$ fri-query-index-list-space

unfolding *generic-fri-sampled-query-missing-full-cover-fiber-def by auto*

lemma *finite-generic-fri-sampled-query-missing-full-cover-fiber:*

finite
 $(\text{generic-fri-sampled-query-missing-full-cover-fiber low-degree}$
 $\text{candidate-bad degree-bound challenges})$

using *generic-fri-sampled-query-missing-full-cover-fiber-subset*

by $(\text{rule finite-subset}) (\text{rule finite-fri-query-index-list-space})$

lemma *generic-fri-sampled-query-query-fiber-subset-missing-full-cover:*

assumes *full-cover-imp:*

$\bigwedge \text{candidate-table roots final-value round-layers layers query-idxs}.$
 $\text{query-idxs} \in \text{fri-query-index-list-space} \implies$
 $\text{generic-fri-sampled-query-candidate-evidence low-degree candidate-bad}$
 $\text{candidate-table degree-bound roots challenges final-value query-idxs}$
 $\text{round-layers layers} \implies$
 $\text{fri-sampled-layers-cover-fold-indices roots query-idxs} \implies$

challenges $\in B$
and *challenges-notin*: *challenges* $\notin B$
shows
generic-fri-sampled-query-query-fiber low-degree candidate-bad
degree-bound challenges $\subseteq$
generic-fri-sampled-query-missing-full-cover-fiber low-degree
candidate-bad degree-bound challenges
proof
fix *query-idxs*
assume *query-in*:
query-idxs $\in$
generic-fri-sampled-query-query-fiber low-degree candidate-bad
degree-bound challenges
then have *query-space*: *query-idxs* $\in$ *fri-query-index-list-space*
using *generic-fri-sampled-query-query-fiber-subset* **by** *blast*
from *query-in* **obtain** *candidate-table roots final-value round-layers layers*
where *evidence*:
generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers
unfolding *generic-fri-sampled-query-query-fiber-def*
generic-fri-sampled-query-restricted-bad-pairs-def
generic-fri-sampled-query-bad-pair-union-def
generic-fri-sampled-query-bad-pair-set-def
fri-query-challenge-pair-query-fiber-def
by *auto*
have *not-cover*: $\neg$ *fri-sampled-layers-cover-fold-indices roots query-idxs*
proof
assume *cover*: *fri-sampled-layers-cover-fold-indices roots query-idxs*
have *challenges* $\in B$
by (*rule full-cover-imp*[*OF query-space evidence cover*])
then show *False*
using *challenges-notin* **by** *simp*
qed
show *query-idxs* $\in$
generic-fri-sampled-query-missing-full-cover-fiber low-degree
candidate-bad degree-bound challenges
unfolding *generic-fri-sampled-query-missing-full-cover-fiber-def*
by (*intro CollectI conjI exI*)
(*rule query-space, rule evidence, rule not-cover*)
qed
lemma *generic-fri-sampled-query-query-fiber-entries-card-le-missing-full-cover*:
assumes *full-cover-imp*:
 $\bigwedge$ *candidate-table roots final-value round-layers layers query-idxs*.
query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers $\implies$

$fri\text{-}sampled\text{-}layers\text{-}cover\text{-}fold\text{-}indices\ roots\ query\text{-}idxs \implies$
 $challenges \in B$
and $challenges\text{-}notin: challenges \notin B$
shows
 $card$
 $(query\text{-}index\text{-}list\text{-}entries$
 $(generic\text{-}fri\text{-}sampled\text{-}query\text{-}query\text{-}fiber\ low\text{-}degree\ candidate\text{-}bad$
 $degree\text{-}bound\ challenges))$
 $\leq$
 $card$
 $(query\text{-}index\text{-}list\text{-}entries$
 $(generic\text{-}fri\text{-}sampled\text{-}query\text{-}missing\text{-}full\text{-}cover\text{-}fiber\ low\text{-}degree$
 $candidate\text{-}bad\ degree\text{-}bound\ challenges))$
proof –
have $subset:$
 $generic\text{-}fri\text{-}sampled\text{-}query\text{-}query\text{-}fiber\ low\text{-}degree\ candidate\text{-}bad$
 $degree\text{-}bound\ challenges \subseteq$
 $generic\text{-}fri\text{-}sampled\text{-}query\text{-}missing\text{-}full\text{-}cover\text{-}fiber\ low\text{-}degree$
 $candidate\text{-}bad\ degree\text{-}bound\ challenges$
by $(rule\ generic\text{-}fri\text{-}sampled\text{-}query\text{-}query\text{-}fiber\text{-}subset\text{-}missing\text{-}full\text{-}cover$
 $[OF\ full\text{-}cover\text{-}imp\ challenges\text{-}notin])$
have $entries\text{-}subset:$
 $query\text{-}index\text{-}list\text{-}entries$
 $(generic\text{-}fri\text{-}sampled\text{-}query\text{-}query\text{-}fiber\ low\text{-}degree\ candidate\text{-}bad$
 $degree\text{-}bound\ challenges)$
 $\subseteq$
 $query\text{-}index\text{-}list\text{-}entries$
 $(generic\text{-}fri\text{-}sampled\text{-}query\text{-}missing\text{-}full\text{-}cover\text{-}fiber\ low\text{-}degree$
 $candidate\text{-}bad\ degree\text{-}bound\ challenges)$
using $subset\ unfolding\ query\text{-}index\text{-}list\text{-}entries\text{-}def\ by\ blast$
show $?thesis$
by $(rule\ card\text{-}mono[OF - entries\text{-}subset])$
 $(rule\ finite\text{-}subset[OF - finite\text{-}query\text{-}sample\text{-}space],$
 $unfold\ query\text{-}index\text{-}list\text{-}entries\text{-}def$
 $generic\text{-}fri\text{-}sampled\text{-}query\text{-}missing\text{-}full\text{-}cover\text{-}fiber\text{-}def$
 $fri\text{-}query\text{-}index\text{-}list\text{-}space\text{-}def, auto)$
qed

lemma $generic\text{-}fri\text{-}sampled\text{-}query\text{-}outside\text{-}query\text{-}cover\text{-}subset:$
 $generic\text{-}fri\text{-}sampled\text{-}query\text{-}outside\text{-}query\text{-}cover\ low\text{-}degree\ candidate\text{-}bad$
 $degree\text{-}bound\ B \subseteq fri\text{-}query\text{-}index\text{-}list\text{-}space$
unfolding $generic\text{-}fri\text{-}sampled\text{-}query\text{-}outside\text{-}query\text{-}cover\text{-}def$
by $(auto\ intro: generic\text{-}fri\text{-}sampled\text{-}query\text{-}query\text{-}fiber\text{-}subset[THEN\ subsetD])$

lemma $finite\text{-}generic\text{-}fri\text{-}sampled\text{-}query\text{-}outside\text{-}query\text{-}cover:$
 $finite$
 $(generic\text{-}fri\text{-}sampled\text{-}query\text{-}outside\text{-}query\text{-}cover\ low\text{-}degree\ candidate\text{-}bad$
 $degree\text{-}bound\ B)$
using $generic\text{-}fri\text{-}sampled\text{-}query\text{-}outside\text{-}query\text{-}cover\text{-}subset$

by (rule finite-subset) (rule finite-fri-query-index-list-space)

lemma generic-fri-sampled-query-outside-query-cover-card-le-from-fibers:

$$\begin{aligned} & \text{card} \\ & (\text{generic-fri-sampled-query-outside-query-cover low-degree candidate-bad} \\ & \quad \text{degree-bound } B) \\ & \leq \\ & (\sum_{\text{card}} \text{challenges} \in \\ & \quad \text{fri-challenge-space (fri-round-count-for-degree-bound degree-bound)} - B. \\ & \quad (\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad} \\ & \quad \text{degree-bound challenges})) \end{aligned}$$

unfolding generic-fri-sampled-query-outside-query-cover-def

by (rule card-UN-le) simp

lemma generic-fri-sampled-query-outside-query-cover-entries-card-le-from-fibers:

$$\begin{aligned} & \text{card} \\ & (\text{query-index-list-entries} \\ & \quad (\text{generic-fri-sampled-query-outside-query-cover low-degree candidate-bad} \\ & \quad \text{degree-bound } B)) \\ & \leq \\ & (\sum_{\text{card}} \text{challenges} \in \\ & \quad \text{fri-challenge-space (fri-round-count-for-degree-bound degree-bound)} - B. \\ & \quad (\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad} \\ & \quad \text{degree-bound challenges})) * \text{rounds} \end{aligned}$$

proof –

let ?Q =

generic-fri-sampled-query-outside-query-cover low-degree candidate-bad
degree-bound B

let ?K =

$$\begin{aligned} & (\sum_{\text{card}} \text{challenges} \in \\ & \quad \text{fri-challenge-space (fri-round-count-for-degree-bound degree-bound)} - B. \\ & \quad (\text{generic-fri-sampled-query-query-fiber low-degree candidate-bad} \\ & \quad \text{degree-bound challenges})) \end{aligned}$$

have entries-le: card (query-index-list-entries ?Q) ≤ card ?Q * rounds

by (rule query-index-list-entries-card-le)

(rule finite-generic-fri-sampled-query-outside-query-cover,
rule generic-fri-sampled-query-outside-query-cover-subset)

have cover-le: card ?Q ≤ ?K

by (rule generic-fri-sampled-query-outside-query-cover-card-le-from-fibers)

have card ?Q * rounds ≤ ?K * rounds

by (rule mult-right-mono[OF cover-le]) simp

then show ?thesis

using entries-le by linarith

qed

lemma trace-fri-sampled-query-outside-projection-subset:

$\text{fst } \langle$
 $((\text{trace-fri-sampled-query-bad-pair-union} \cap$
 $(\text{fri-query-index-list-space} \times \text{UNIV})) \cap$
 $(\text{UNIV} \times (- B)))$
 $\subseteq$
 $\text{generic-fri-sampled-query-outside-query-cover trace-table-low-degree}$
 $(\text{Not} \circ \text{trace-table-low-degree}) (\text{clength} - 1) B$
unfolding $\text{trace-fri-sampled-query-bad-pair-union-def}$
 $\text{generic-fri-sampled-query-outside-query-cover-def}$
 $\text{generic-fri-sampled-query-query-fiber-def}$
 $\text{generic-fri-sampled-query-restricted-bad-pairs-def}$
 $\text{fri-query-challenge-pair-query-fiber-def}$
using $\text{generic-fri-sampled-query-bad-pair-union-subset-challenge-space}$
 $[\text{of trace-table-low-degree Not} \circ \text{trace-table-low-degree clength} - 1]$
by force

definition $\text{composition-fri-sampled-query-outside-query-cover}$
 $:: ('f \Rightarrow 'f \text{ list set}) \Rightarrow \text{nat list set}$

where

$\text{composition-fri-sampled-query-outside-query-cover } B =$
 $(\bigcup dg \in (\text{UNIV} :: 'f \text{ set}).$
 $\text{generic-fri-sampled-query-outside-query-cover}$
 $(\text{composition-table-low-degree } (\text{to-nat } dg))$
 $(\text{Not} \circ \text{composition-table-low-degree maxDegree})$
 $(\text{to-nat } dg) (B dg))$

lemma $\text{composition-fri-sampled-query-outside-query-cover-subset}:$

$\text{composition-fri-sampled-query-outside-query-cover } B$
 $\subseteq \text{fri-query-index-list-space}$

unfolding $\text{composition-fri-sampled-query-outside-query-cover-def}$

by $(\text{auto intro}:$

$\text{generic-fri-sampled-query-outside-query-cover-subset}[\text{THEN subsetD}])$

lemma $\text{finite-composition-fri-sampled-query-outside-query-cover}:$

$\text{finite } (\text{composition-fri-sampled-query-outside-query-cover } B)$

using $\text{composition-fri-sampled-query-outside-query-cover-subset}$

by $(\text{rule finite-subset}) (\text{rule finite-fri-query-index-list-space})$

lemma $\text{composition-fri-sampled-query-outside-query-cover-card-le-from-fibers}:$

$\text{card } (\text{composition-fri-sampled-query-outside-query-cover } B)$

$\leq$

$(\sum dg \in (\text{UNIV} :: 'f \text{ set}).$

$\sum \text{challenges} \in$

$\text{fri-challenge-space } (\text{fri-round-count-for-degree-bound } (\text{to-nat } dg)) -$

$B dg.$

card

$(\text{generic-fri-sampled-query-query-fiber}$

$(\text{composition-table-low-degree } (\text{to-nat } dg))$

$(\text{Not} \circ \text{composition-table-low-degree maxDegree})$

(to-nat dg) challenges))

proof –

let ?Q = composition-fri-sampled-query-outside-query-cover B

let ?G = λdg.

generic-fri-sampled-query-outside-query-cover

(composition-table-low-degree (to-nat dg))

(Not ∘ composition-table-low-degree maxDegree)

(to-nat dg) (B dg)

let ?K = λdg.

(∑ challenges ∈

fri-challenge-space (fri-round-count-for-degree-bound (to-nat dg)) –

B dg.

card

(generic-fri-sampled-query-query-fiber

(composition-table-low-degree (to-nat dg))

(Not ∘ composition-table-low-degree maxDegree)

(to-nat dg) challenges))

have card ?Q ≤ (∑ dg ∈ (UNIV :: 'f set). card (?G dg))

unfolding composition-fri-sampled-query-outside-query-cover-def

by (rule card-UN-le) simp

also have ... ≤ (∑ dg ∈ (UNIV :: 'f set). ?K dg)

by (rule sum-mono)

(rule generic-fri-sampled-query-outside-query-cover-card-le-from-fibers)

finally show ?thesis .

qed

lemma composition-fri-sampled-query-outside-query-cover-entries-card-le-from-fibers:

card

(query-index-list-entries

(composition-fri-sampled-query-outside-query-cover B))

≤

(∑ dg ∈ (UNIV :: 'f set).

∑ challenges ∈

fri-challenge-space (fri-round-count-for-degree-bound (to-nat dg)) –

B dg.

card

(generic-fri-sampled-query-query-fiber

(composition-table-low-degree (to-nat dg))

(Not ∘ composition-table-low-degree maxDegree)

(to-nat dg) challenges)) * rounds

proof –

let ?Q = composition-fri-sampled-query-outside-query-cover B

let ?K =

(∑ dg ∈ (UNIV :: 'f set).

∑ challenges ∈

fri-challenge-space (fri-round-count-for-degree-bound (to-nat dg)) –

B dg.

card

(generic-fri-sampled-query-query-fiber

```

      (composition-table-low-degree (to-nat dg))
      (Not  $\circ$  composition-table-low-degree maxDegree)
      (to-nat dg) challenges))
have entries-le: card (query-index-list-entries ?Q)  $\leq$  card ?Q * rounds
by (rule query-index-list-entries-card-le)
      (rule finite-composition-fri-sampled-query-outside-query-cover,
       rule composition-fri-sampled-query-outside-query-cover-subset)
have cover-le: card ?Q  $\leq$  ?K
by (rule composition-fri-sampled-query-outside-query-cover-card-le-from-fibers)
have card ?Q * rounds  $\leq$  ?K * rounds
by (rule mult-right-mono[OF cover-le]) simp
then show ?thesis
using entries-le by linarith
qed

```

lemma composition-fri-sampled-query-outside-projection-subset:

```

fst ‘
  ((composition-fri-sampled-query-bad-pair-union dg  $\cap$ 
    (fri-query-index-list-space  $\times$  UNIV))  $\cap$ 
   (UNIV  $\times$  ( $-$  B dg)))
 $\subseteq$  composition-fri-sampled-query-outside-query-cover B
unfolding composition-fri-sampled-query-bad-pair-union-def
  composition-fri-sampled-query-outside-query-cover-def
  generic-fri-sampled-query-outside-query-cover-def
  generic-fri-sampled-query-query-fiber-def
  generic-fri-sampled-query-restricted-bad-pairs-def
  fri-query-challenge-pair-query-fiber-def
using generic-fri-sampled-query-bad-pair-union-subset-challenge-space
  [of composition-table-low-degree (to-nat dg)
    Not  $\circ$  composition-table-low-degree maxDegree to-nat dg]
by force

```

lemma checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-outsides

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: finite B
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s$ . trace-fri-challenge-list-fresh-hit s B))
  adversary-initial-state  $\leq$  F
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-sampled-query-bad-candidate)
  adversary-initial-state  $\leq$ 
  F +
  hash-relation-budget-value (card B * ceil-log clength)
  (staged-attacker-query-budget budgets + staged-challenge-query-budget) +

```


staged-phase-target-error
 (*query-index-raw-preimage*
 (*query-index-list-entries*
 (*generic-fri-sampled-query-outside-query-cover*
trace-table-low-degree (*Not* $\circ$ *trace-table-low-degree*)
 (*clength* $-$ *1*) *B*)))
 (*staged-attacker-query-budget* *budgets* $+$
staged-challenge-query-budget)
by (*rule*
checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-projection
 [*OF* *wf* *controlled* *finite-B* *fresh-bound*
trace-fri-sampled-query-outside-projection-subset])

lemma *checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-outsid*

assumes *wf*: *staged-budget-wellformed* *budgets*

and *controlled*: *staged-adversary-controlled* *budgets* *A*

and *finite-B*: *finite* *B*

and *fresh-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* *A*)

(*staged-security-with-data-state-verifier-event*

($\lambda s.$ *trace-fri-challenge-list-fresh-hit* *s* *B*))

adversary-initial-state $\leq$ *F*

and *query-fraction*:

nnreal

(*card*

(*query-index-list-entries*

(*generic-fri-sampled-query-outside-query-cover*

trace-table-low-degree (*Not* $\circ$ *trace-table-low-degree*)

(*clength* $-$ *1*) *B*))) /

nnreal (*card* *query-sample-space*) $\leq$ *Q*

shows

wp-event (*checked-staged-security-experiment-with-data-state* *A*)

(*staged-security-with-data-state-verifier-event*

trace-fri-sampled-query-bad-candidate)

adversary-initial-state $\leq$

F $+$

hash-relation-budget-value (*card* *B* $*$ *ceil-log* *clength*)

(*staged-attacker-query-budget* *budgets* $+$ *staged-challenge-query-budget*) $+$

nnreal (*staged-attacker-query-budget* *budgets* $+$

staged-challenge-query-budget) $*$ *Q*

proof $-$

let *?Qset* $=$

generic-fri-sampled-query-outside-query-cover *trace-table-low-degree*

(*Not* $\circ$ *trace-table-low-degree*) (*clength* $-$ *1*) *B*

have *base*:

wp-event (*checked-staged-security-experiment-with-data-state* *A*)

(*staged-security-with-data-state-verifier-event*

trace-fri-sampled-query-bad-candidate)

adversary-initial-state $\leq$

```

    F +
    hash-relation-budget-value (card B * ceil-log clength)
    (staged-attacker-query-budget budgets + staged-challenge-query-budget) +
    staged-phase-target-error
    (query-index-row-preimage (query-index-list-entries ?Qset))
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget)
  by (rule
    checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-outside-fil
    [OF wf controlled finite-B fresh-bound])
  have subset:
    query-index-list-entries ?Qset  $\subseteq$  query-sample-space
  by (rule query-index-list-entries-subset-query-sample-space)
    (rule generic-fri-sampled-query-outside-query-cover-subset)
  have target-bound:
    staged-phase-target-error
    (query-index-row-preimage (query-index-list-entries ?Qset))
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget)
     $\leq$ 
    nnreal (staged-attacker-query-budget budgets +
     staged-challenge-query-budget) * Q
  by (rule staged-phase-query-index-target-error-query-bound
    [OF query-index-row-preimage-bound-from-sampler-wellformed subset
     query-fraction])
  show ?thesis
  by (rule order-trans[OF base])
    (use target-bound in simp)
qed

lemma checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-outside-fil
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: finite B
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   ( $\lambda s.$  trace-fri-challenge-list-fresh-hit s B))
  adversary-initial-state  $\leq$  F
and query-fraction:
  nnreal
  (( $\sum$  challenges  $\in$ 
    fri-challenge-space (fri-round-count-for-degree-bound (clength - 1)) -
    B.
    card
    (generic-fri-sampled-query-query-fiber trace-table-low-degree
     (Not  $\circ$  trace-table-low-degree) (clength - 1) challenges)) *
   rounds) /
  nnreal (card query-sample-space)  $\leq$  Q

```

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-sampled-query-bad-candidate)
adversary-initial-state $\leq$
F +
*hash-relation-budget-value (card B * ceil-log clength)*
(staged-attacker-query-budget budgets + staged-challenge-query-budget) +
nnreal (staged-attacker-query-budget budgets +
*staged-challenge-query-budget) * Q*

proof (rule

checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-outside-fiber
[OF wf controlled finite-B fresh-bound])

let *?Entries* =

query-index-list-entries
(generic-fri-sampled-query-outside-query-cover trace-table-low-degree
(Not $\circ$ trace-table-low-degree) (clength - 1) B)

let *?K* =

$(\sum \text{challenges} \in$
fri-challenge-space (fri-round-count-for-degree-bound (clength - 1)) -
B.
card
(generic-fri-sampled-query-query-fiber trace-table-low-degree
*(Not $\circ$ trace-table-low-degree) (clength - 1) challenges)) **
rounds

have *card-le: card ?Entries* $\leq$ *?K*

by (rule

generic-fri-sampled-query-outside-query-cover-entries-card-le-from-fibers)

have *nnreal (card ?Entries) / nnreal (card query-sample-space)*

$\leq$ *nnreal ?K / nnreal (card query-sample-space)*

using *card-le* **by** (rule *nnreal-nat-divide-right-mono*)

also have ... $\leq$ *Q*

by (rule *query-fraction*)

finally show

nnreal (card ?Entries) / nnreal (card query-sample-space) $\leq$ Q .

qed

lemma *checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *finite-B: $\bigwedge dg. \text{finite } (B dg)$*

and *fresh-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

($\lambda s. \text{composition-fri-challenge-list-fresh-hit } s B$))

adversary-initial-state $\leq$ F

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

composition-fri-sampled-query-bad-candidate)
adversary-initial-state $\leq$
 $F +$
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \text{ } dg) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget}) +$
staged-phase-target-error
 $(\text{query-index-raw-preimage}$
 $(\text{query-index-list-entries}$
 $(\text{composition-fri-sampled-query-outside-query-cover } B)))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})$
by (rule
checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-challenge-cover-and-query-projection
 $[OF \text{ wf controlled finite-}B \text{ fresh-bound}$
 $\text{composition-fri-sampled-query-outside-projection-subset}]$)

lemma *checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-query-projection*

assumes *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *finite-B*: $\bigwedge dg. \text{finite } (B \text{ } dg)$

and *fresh-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

$(\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \text{ } B))$

adversary-initial-state $\leq F$

and *query-fraction*:

nnreal

(*card*

(*query-index-list-entries*

$(\text{composition-fri-sampled-query-outside-query-cover } B))) /$

nnreal (*card query-sample-space*) $\leq Q$

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*

composition-fri-sampled-query-bad-candidate)

adversary-initial-state $\leq$

$F +$

hash-relation-budget-value

$(\sum dg \in (UNIV :: 'f \text{ set}).$

$\text{card } (B \text{ } dg) * \text{ceil-log } (\text{maxDegree} + 1))$

$(\text{staged-attacker-query-budget budgets} +$

$\text{staged-challenge-query-budget}) +$

nnreal ($(\text{staged-attacker-query-budget budgets} +$

$\text{staged-challenge-query-budget}) * Q$

proof –

let $?Qset = \text{composition-fri-sampled-query-outside-query-cover } B$

have *base*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-sampled-query-bad-candidate)
adversary-initial-state $\leq$
F +
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \text{ } dg) * \text{ceil-log } (\text{maxDegree} + 1))$
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget) +
staged-phase-target-error
(*query-index-row-preimage* (*query-index-list-entries ?Qset*))
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)
by (*rule*
checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-ou
[*OF wf controlled finite-B fresh-bound*])
have *subset*:
query-index-list-entries ?Qset $\subseteq$ *query-sample-space*
by (*rule query-index-list-entries-subset-query-sample-space*)
(*rule composition-fri-sampled-query-outside-query-cover-subset*)
have *target-bound*:
staged-phase-target-error
(*query-index-row-preimage* (*query-index-list-entries ?Qset*))
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)
 $\leq$
nnreal (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget) * *Q*
by (*rule staged-phase-query-index-target-error-query-bound*
[*OF query-index-row-preimage-bound-from-sampler-wellformed subset*
query-fraction])
show *?thesis*
by (*rule order-trans*[*OF base*])
(*use target-bound in simp*)
qed

lemma *checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and*
assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *finite-B*: $\bigwedge dg. \text{finite } (B \text{ } dg)$
and *fresh-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
 $(\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \text{ } B)$)
adversary-initial-state $\leq F$
and *query-fraction*:
nnreal

```

(( $\sum dg \in (UNIV :: 'f \text{ set}).$ 
 $\sum challenges \in$ 
  fri-challenge-space (fri-round-count-for-degree-bound (to-nat dg)) –
    B dg.
  card
    (generic-fri-sampled-query-query-fiber
      (composition-table-low-degree (to-nat dg))
      (Not  $\circ$  composition-table-low-degree maxDegree)
      (to-nat dg) challenges)) * rounds) /
  nnreal (card query-sample-space)  $\leq Q$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-sampled-query-bad-candidate)
  adversary-initial-state  $\leq$ 
    F +
    hash-relation-budget-value
      ( $\sum dg \in (UNIV :: 'f \text{ set}).$ 
        card (B dg) * ceil-log (maxDegree + 1))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget) +
    nnreal (staged-attacker-query-budget budgets +
      staged-challenge-query-budget) * Q
proof (rule
  checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-outside
  [OF wf controlled finite-B fresh-bound])
let ?Entries =
  query-index-list-entries
    (composition-fri-sampled-query-outside-query-cover B)
let ?K =
  ( $\sum dg \in (UNIV :: 'f \text{ set}).$ 
     $\sum challenges \in$ 
      fri-challenge-space (fri-round-count-for-degree-bound (to-nat dg)) –
        B dg.
    card
      (generic-fri-sampled-query-query-fiber
        (composition-table-low-degree (to-nat dg))
        (Not  $\circ$  composition-table-low-degree maxDegree)
        (to-nat dg) challenges)) * rounds
have card-le: card ?Entries  $\leq$  ?K
by (rule
  composition-fri-sampled-query-outside-query-cover-entries-card-le-from-fibers)
have nnreal (card ?Entries) / nnreal (card query-sample-space)
   $\leq$  nnreal ?K / nnreal (card query-sample-space)
using card-le by (rule nnreal-nat-divide-right-mono)
also have ...  $\leq Q$ 
by (rule query-fraction)
finally show
  nnreal (card ?Entries) / nnreal (card query-sample-space)  $\leq Q$  .

```

qed

end

end

theory *Soundness-FRI-Query-List-Prequery*
 imports
 Stark-Core.Merkle-Tree
 Soundness-Core-Events
 Soundness-FRI-Query-Exact-Bounds
 Soundness-FRI-Query-Index-Staged-Bounds
begin

Query-list prequery accounting for FRI. The lemmas in this theory relate verifier-generated query index lists to random-oracle budget events used in the staged soundness proof.

context *soundness*
begin

lemma *set-dist-return-valueD*:
 assumes $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (\text{return } x) \ s)$
 shows $y = x$
 using *assms*
 unfolding *set-dist-def return.rep-eq dist-return-def*
 by (*simp add: dist-delta-dist delta-map-def*)

lemma *set-dist-return-stateD*:
 assumes $\text{Some } (y, t) \in \text{set-dist } (\text{execute } (\text{return } x) \ s)$
 shows $t = s$
 using *assms*
 unfolding *set-dist-def return.rep-eq dist-return-def*
 by (*simp add: dist-delta-dist delta-map-def*)

lemma *set-dist-bind-return-consE*:
 assumes *support*:
 $\text{out} \in \text{set-dist } (\text{execute } (m \gg (\lambda xs. \text{return } (u \# xs))) \ s)$
 and *out-eq*: $\text{out} = \text{Some } (ys, t0)$
 obtains *results t where*
 $\text{Some } (\text{results}, t) \in \text{set-dist } (\text{execute } m \ s)$
 $ys = u \# \text{results}$
 $t0 = t$
 $\text{out} = \text{Some } (u \# \text{results}, t)$
proof –
 from *support[unfolded out-eq]*
 obtain *results t where tail*:
 $\text{Some } (\text{results}, t) \in \text{set-dist } (\text{execute } m \ s)$
 and *ret*:
 $\text{Some } (ys, t0) \in \text{set-dist } (\text{execute } (\text{return } (u \# \text{results})) \ t)$

```

    by (auto elim!: set-dist-bindE)
  have ys-eq: ys = u # results
    by (rule set-dist-return-valueD[OF ret])
  have t0-eq: t0 = t
    by (rule set-dist-return-stateD[OF ret])
  have out-eq': out = Some (u # results, t)
    using out-eq ys-eq t0-eq by simp
  show ?thesis
    by (rule that[OF tail ys-eq t0-eq out-eq'])
qed

lemma set-dist-bind-return-cons-SomeE:
  assumes support:
    Some (ys, t) ∈ set-dist (execute (m ≫ (λxs. return (u # xs))) s)
  obtains xs where
    Some (xs, t) ∈ set-dist (execute m s)
    ys = u # xs
proof -
  from support obtain xs t0 where tail:
    Some (xs, t0) ∈ set-dist (execute m s)
  and ys-eq: ys = u # xs
  and t-eq: t = t0
  by (elim set-dist-bind-return-consE) simp
  have tail-t: Some (xs, t) ∈ set-dist (execute m s)
    by (subst t-eq, rule tail)
  show ?thesis
    by (rule that[OF tail-t ys-eq])
qed

lemma wp-event-nonzero-imp-exists-support:
  assumes wp-event m P s ≠ 0
  obtains out where out ∈ set-dist (execute m s) and P out
  by (rule wp-event-pos-imp-exists-support)
  (rule zero-less-iff-neq-zero[THEN iffD2], rule assms)

lemma wp-event-false [simp]:
  wp-event m (λ-. False) s = 0
  unfolding wp-event-def wp-def dist-expect-def
  by simp

lemma wp-event-bind-return-cons-mono:
  fixes m :: ('a list, 's) state-monad
  assumes none-imp: Q None ⇒ R None
  and some-imp:
    ∧xs t. Some (xs, t) ∈ set-dist (execute m s) ⇒
      Q (Some (u # xs, t)) ⇒ R (Some (xs, t))
  shows
    wp-event (m ≫ (λxs. return (u # xs))) Q s ≤
    wp-event m R s

```



```

proof (rule wp-event-bind-bound-by-head-event)
  show wp-event m R s ≤ wp-event m R s
    by (rule order.refl)
next
  show Q None ⇒ R None
    by (rule none-imp)
next
  fix xs t out
  assume support: Some (xs, t) ∈ set-dist (execute m s)
    and ret: out ∈ set-dist (execute (return (u # xs)) t)
    and q: Q out
  have out-eq: out = Some (u # xs, t)
    using ret
  unfolding set-dist-def return.rep-eq dist-return-def
    dist-delta-dist delta-map-def
  by simp
  have q-some: Q (Some (u # xs, t))
    using q by (simp only: out-eq)
  show R (Some (xs, t))
    by (rule some-imp[OF support q-some])
qed

lemma eq-Suc-imp-less-nat:
  assumes m = Suc n
  shows n < m
  by (subst assms, rule lessI)

lemma list-length-SucE:
  assumes length xs = Suc n
  obtains x xs' where xs = x # xs'
  using assms by (cases xs) simp-all

lemma state-after-query-chunks-cons-shift:
  assumes s1 = state-after-query-chunks s (chunk # chunks) 1
  shows
    state-after-query-chunks s1 chunks n =
      state-after-query-chunks s (chunk # chunks) (Suc n)
  using assms unfolding state-after-query-chunks-def by simp

lemma state-after-query-chunks-cons-shift-index:
  assumes s1 = state-after-query-chunks s (chunk # chunks) 1
  shows
    state-after-query-chunks s1 chunks i =
      state-after-query-chunks s (chunk # chunks) (Suc i)
  using assms unfolding state-after-query-chunks-def by simp

lemma state-after-query-chunks-one-cons:
  state-after-query-chunks s (chunk # chunks) 1 = foldl concat s chunk
  unfolding state-after-query-chunks-def by simp

```

end

Query-list prequery accounting for sampled FRI.

The exact verifier-local sampled-query bounds are list-valued: they bound the probability of drawing a whole query-index list from a target set. In the staged adversary experiment the verifier reuses the adversary's oracle map, so an exact staged theorem needs a fresh/prequeried split for the whole query-index list. This layer provides the conservative relation-budget side of that split without changing the oracle model.

context *soundness*

begin

lemma *verifier-query-round-program-alt-outcomeE*:

assumes *outcome*:

Some $((\cdot), t) \in$

set-dist

(execute

(verifier-query-round-program fr f-fl f-final as fl final) s)

obtains *raw s0* **where**

Some $(\text{raw}, s0) \in \text{set-dist } (\text{execute receive-query-index-challenge } s)$

Some $((\cdot), t) \in$

set-dist

(execute

*(verifier-query-round-after-index-program fr f-fl f-final as fl
final raw) s0)*

using *outcome[unfolded verifier-query-round-program-alt-def]*

by *(auto elim!: set-dist-bindE)*

lemma *verifier-query-round-parts-counter*:

assumes *recv*:

Some $(\text{raw}, s0) \in \text{set-dist } (\text{execute receive-query-index-challenge } s)$

and *after*:

Some $((\cdot), t) \in$

set-dist

(execute

*(verifier-query-round-after-index-program fr f-fl f-final as fl
final raw) s0)*

shows *PQueryCounter t = Suc (PQueryCounter s)*

proof –

have *counter-s0*: *PQueryCounter s0 = Suc (PQueryCounter s)*

using *receive-query-index-challenge-counter-outcome[OF recv]* **by** *simp*

have *counter-t-s0*: *PQueryCounter t = PQueryCounter s0*

by *(rule conjunct2[OF verifier-query-round-after-index-program-outcome
[OF after]])*

show *?thesis*

proof *(rule HOL.trans)*

show *PQueryCounter t = PQueryCounter s0*

by *(rule counter-t-s0)*

```

show PQueryCounter s0 = Suc (PQueryCounter s)
  by (rule counter-s0)
qed
qed

definition query-index-raw-list-preimage :: nat list set  $\Rightarrow$  'f list set
where
  query-index-raw-list-preimage Q =
    {raws. length raws = rounds  $\wedge$ 
      map ( $\lambda$ raw. index (to-nat raw)) raws  $\in$  Q}

definition query-index-raw-list-position-values
  :: nat list set  $\Rightarrow$  nat  $\Rightarrow$  'f set
where
  query-index-raw-list-position-values Q i =
    {raw.  $\exists$  raws  $\in$  query-index-raw-list-preimage Q.
      i < rounds  $\wedge$  raw = raws ! i}

definition query-index-raw-list-relation-fiber-bound
  :: nat list set  $\Rightarrow$  nat
where
  query-index-raw-list-relation-fiber-bound Q =
    ( $\sum$  i < rounds. card (query-index-raw-list-position-values Q i))

definition staged-transcript-query-index-list-set-hit
  :: nat list set  $\Rightarrow$ 
    ('f staged-proof-data  $\times$  'f protocol-channel) option  $\Rightarrow$  bool
where
  staged-transcript-query-index-list-set-hit Q out  $\longleftrightarrow$ 
    (case out of
      None  $\Rightarrow$  False
    | Some (data, attacker-state)  $\Rightarrow$ 
      ( $\exists$  raw-idxs query-idxs.
        length raw-idxs = rounds  $\wedge$ 
        query-idxs  $\in$  Q  $\wedge$ 
        query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs  $\wedge$ 
        ( $\forall$  i < rounds.
          fmlookup (HashMap attacker-state)
            (QueryIndexChallenge i
              (state-after-query-chunks
                (staged-query-start-hash data)
                (staged-query-chunks data) i)) =
            Some (raw-idxs ! i))))

definition staged-security-with-data-query-index-list-set-hit
  :: nat list set  $\Rightarrow$ 
    (((('f staged-proof-data  $\times$  'f protocol-channel)  $\times$ 
      unit list)  $\times$  'f protocol-channel) option  $\Rightarrow$  bool
where

```

staged-security-with-data-query-index-list-set-hit Q $out \longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((data, -), result), final-state) $\Rightarrow$
 ($\exists$ raw-idxs query-idxs.
 length raw-idxs = rounds $\wedge$
 query-idxs $\in Q$ $\wedge$
 query-idxs = map (λ raw. index (to-nat raw)) raw-idxs $\wedge$
 ($\forall i < rounds$.
 fmlookup (HashMap final-state)
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data)
 (staged-query-chunks data) i)) =
 Some (raw-idxs ! i))))

definition *staged-security-with-data-state-query-index-list-path-fresh*

:: nat list set $\Rightarrow$
 ((('f staged-proof-data $\times$ 'f protocol-channel) $\times$
 unit list) $\times$ 'f protocol-channel) option $\Rightarrow$ bool

where

staged-security-with-data-state-query-index-list-path-fresh Q $out \longleftrightarrow$
 (case out of
 None $\Rightarrow$ False
 | Some (((data, attacker-state), result), final-state) $\Rightarrow$
 ($\exists$ raw-idxs query-idxs.
 length raw-idxs = rounds $\wedge$
 query-idxs $\in Q$ $\wedge$
 query-idxs = map (λ raw. index (to-nat raw)) raw-idxs $\wedge$
 ($\forall i < rounds$.
 fmlookup (HashMap final-state)
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data)
 (staged-query-chunks data) i)) =
 Some (raw-idxs ! i) $\wedge$
 fmlookup
 (HashMap
 (verifier-state-from-adversary attacker-state
 (staged-proof-transcript data)))
 (QueryIndexChallenge i
 (state-after-query-chunks
 (staged-query-start-hash data)
 (staged-query-chunks data) i)) =
 None))))

definition *checked-staged-query-index-list-relation*

:: 'f staged-adversary $\Rightarrow$ nat list set $\Rightarrow$
 'f protocol-hash-input $\Rightarrow$ 'f $\Rightarrow$ bool

where

$checked_staged_query_index_list_relation\ A\ Q\ x\ y \longleftrightarrow$
 $(\exists\ data\ attacker_state\ raw_idxs\ i.$
 $\quad Some\ (data,\ attacker_state) \in$
 $\quad\quad set_dist$
 $\quad\quad (execute\ (checked_staged_transcript_program\ A)$
 $\quad\quad\quad adversary_initial_state) \wedge$
 $\quad raw_idxs \in query_index_raw_list_preimage\ Q \wedge$
 $\quad i < rounds \wedge$
 $\quad x = QueryIndexChallenge\ i$
 $\quad\quad (state_after_query_chunks$
 $\quad\quad\quad (staged_query_start_hash\ data)$
 $\quad\quad\quad (staged_query_chunks\ data)\ i) \wedge$
 $\quad y = raw_idxs\ !\ i)$

definition *staged-security-with-data-state-query-index-list-prequery-hit*

$:: 'f\ staged_adversary \Rightarrow nat\ list\ set \Rightarrow$
 $((('f\ staged_proof_data \times 'f\ protocol_channel) \times$
 $\quad unit\ list) \times 'f\ protocol_channel)\ option \Rightarrow bool$

where

$staged_security_with_data_state_query_index_list_prequery_hit\ A\ Q\ out$
 $\longleftrightarrow$
 $(case\ out\ of$
 $\quad None \Rightarrow False$
 $\quad | Some\ (((data,\ attacker_state),\ result),\ final_state) \Rightarrow$
 $\quad\quad hash_relation_hit$
 $\quad\quad (checked_staged_query_index_list_relation\ A\ Q)$
 $\quad\quad\quad adversary_initial_state\ attacker_state)$

definition *query-index-list-path-fresh*

$:: ('f,\ 'a)\ protocol_channel_scheme \Rightarrow nat \Rightarrow 'f\ list\ list \Rightarrow$
 $\quad 'f\ list \Rightarrow bool$

where

$query_index_list_path_fresh\ s\ n\ chunks\ raws \longleftrightarrow$
 $\quad length\ raws = n \wedge$
 $\quad length\ chunks = n \wedge$
 $\quad (\forall\ i < n.$
 $\quad\quad fmlookup\ (HashMap\ s)$
 $\quad\quad (QueryIndexChallenge\ (PQueryCounter\ s + i)$
 $\quad\quad\quad (state_after_query_chunks\ (PState\ s)\ chunks\ i)) = None)$

lemma *query-index-list-path-fresh-head-fresh:*

assumes

$query_index_list_path_fresh\ s\ (Suc\ n)\ chunks\ (raw\ \# \ raws)$

shows

$fmlookup\ (HashMap\ s)$
 $\quad (QueryIndexChallenge\ (PQueryCounter\ s)\ (PState\ s)) = None$

using *assms* **unfolding** *query-index-list-path-fresh-def*

state-after-query-chunks-def

by auto

lemma *query-index-list-path-fresh-lookupD*:

assumes *fresh*: *query-index-list-path-fresh s n chunks raws*
and *i-bound*: $i < n$

shows

fmlookup (HashMap s)
 (*QueryIndexChallenge (PQueryCounter s + i)*
 (*state-after-query-chunks (PState s) chunks i*)) = *None*

proof –

have *all*:

$\forall j < n.$

fmlookup (HashMap s)
 (*QueryIndexChallenge (PQueryCounter s + j)*
 (*state-after-query-chunks (PState s) chunks j*)) = *None*

using *fresh* **unfolding** *query-index-list-path-fresh-def*

by (*elim conjE*)

show *?thesis*

by (*rule all[rule-format, OF i-bound]*)

qed

definition *query-rounds-raw-list-path-hit*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow$
 ($\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}$) $\text{option} \Rightarrow \text{bool}$

where

query-rounds-raw-list-path-hit s raws n out $\longleftrightarrow$
 (*case out of*
 None $\Rightarrow$ *False*
 | *Some (results, t)* $\Rightarrow$
 ($\exists$ *chunks*.
 length raws = *n* $\wedge$
 length chunks = *n* $\wedge$
 PState t = *state-after-query-chunks (PState s) chunks n* $\wedge$
 query-index-list-path-fresh s n chunks raws $\wedge$
 ($\forall i < n.$
 fmlookup (HashMap t)
 (*QueryIndexChallenge (PQueryCounter s + i)*
 (*state-after-query-chunks (PState s) chunks i*)) =
 Some (raws ! i))))

lemma *query-rounds-raw-list-path-hitI*:

assumes *len-raws*: *length raws* = *n*

and *len-chunks*: *length chunks* = *n*

and *state*:

PState t = *state-after-query-chunks (PState s) chunks n*

and *fresh*: *query-index-list-path-fresh s n chunks raws*

and *lookup*:

$\bigwedge i. i < n \implies$

fmlookup (HashMap t)

```

      (QueryIndexChallenge (PQueryCounter s + i)
       (state-after-query-chunks (PState s) chunks i)) =
    Some (raws ! i)
shows query-rounds-raw-list-path-hit s raws n (Some (results, t))
proof -
have rhs-eq:
  query-rounds-raw-list-path-hit s raws n (Some (results, t)) =
    (∃ chunks.
     length raws = n ∧
     length chunks = n ∧
     PState t = state-after-query-chunks (PState s) chunks n ∧
     query-index-list-path-fresh s n chunks raws ∧
     (∀ i < n.
      fmlookup (HashMap t)
        (QueryIndexChallenge (PQueryCounter s + i)
          (state-after-query-chunks (PState s) chunks i)) =
        Some (raws ! i)))
unfolding query-rounds-raw-list-path-hit-def by simp
show ?thesis
by (subst rhs-eq, rule exI[of - chunks],
     intro conjI allI impI,
     rule len-raws, rule len-chunks, rule state,
     rule fresh, rule lookup)
qed

```

```

lemma query-rounds-raw-list-path-hit-cons-SomeE:
assumes
  query-rounds-raw-list-path-hit s (raw # raws) (Suc n)
  (Some (ys, t))
obtains chunks where
  length raws = n
  length chunks = Suc n
  PState t = state-after-query-chunks (PState s) chunks (Suc n)
  query-index-list-path-fresh s (Suc n) chunks (raw # raws)
  ∧ i. i < Suc n ⟹
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) chunks i)) =
    Some ((raw # raws) ! i)
proof -
have hit-eq:
  query-rounds-raw-list-path-hit s (raw # raws) (Suc n)
  (Some (ys, t)) =
    (∃ chunks.
     length (raw # raws) = Suc n ∧
     length chunks = Suc n ∧
     PState t = state-after-query-chunks (PState s) chunks (Suc n) ∧
     query-index-list-path-fresh s (Suc n) chunks (raw # raws) ∧
     (∀ i < Suc n.

```

```

      fmlookup (HashMap t)
        (QueryIndexChallenge (PQueryCounter s + i)
          (state-after-query-chunks (PState s) chunks i)) =
      Some ((raw # rows) ! i)))
  unfolding query-rounds-row-list-path-hit-def by simp
have ex-chunks:
   $\exists$  chunks.
    length (raw # rows) = Suc n  $\wedge$ 
    length chunks = Suc n  $\wedge$ 
    PState t = state-after-query-chunks (PState s) chunks (Suc n)  $\wedge$ 
    query-index-list-path-fresh s (Suc n) chunks (raw # rows)  $\wedge$ 
    ( $\forall i < \text{Suc } n.$ 
      fmlookup (HashMap t)
        (QueryIndexChallenge (PQueryCounter s + i)
          (state-after-query-chunks (PState s) chunks i)) =
      Some ((raw # rows) ! i))
  using assms by (simp only: hit-eq)
then obtain chunks where len-rows-full: length (raw # rows) = Suc n
and len-chunks: length chunks = Suc n
and state: PState t = state-after-query-chunks (PState s) chunks (Suc n)
and fresh: query-index-list-path-fresh s (Suc n) chunks (raw # rows)
and lookup-all:
   $\forall i < \text{Suc } n.$ 
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) chunks i)) =
    Some ((raw # rows) ! i)
  by blast
have len-rows: length rows = n
  using len-rows-full by simp
have lookup:
   $\bigwedge i. i < \text{Suc } n \implies$ 
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) chunks i)) =
    Some ((raw # rows) ! i)
  using lookup-all by blast
show ?thesis
  by (rule that[OF len-rows len-chunks state fresh lookup])
qed

```

definition query-round-any-row-set-hit

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$  'f set  $\Rightarrow$ 
  (unit  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  query-round-any-row-set-hit s B out  $\longleftrightarrow$ 
    (case out of
      None  $\Rightarrow$  False
    | Some (-, t)  $\Rightarrow$ 

```


$(\exists \text{raw} \in B.$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) =$
 $\text{Some raw}))$

lemma *query-round-any-row-set-hit-None*:
 $\neg \text{query-round-any-row-set-hit } s \ B \ \text{None}$
unfolding *query-round-any-row-set-hit-def* **by** *simp*

lemma *query-round-any-row-set-hit-NoneE*:
assumes *query-round-any-row-set-hit* $s \ B \ \text{None}$
shows P
using *assms query-round-any-row-set-hit-None* **by** *contradiction*

lemma *query-round-any-row-set-hitI*:
assumes $\text{raw} \in B$
and $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) = \text{Some raw}$
shows *query-round-any-row-set-hit* $s \ B \ (\text{Some } (u, t))$
proof –
have *hit-eq*:
 $\text{query-round-any-row-set-hit } s \ B \ (\text{Some } (u, t)) =$
 $(\exists \text{raw} \in B.$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) = \text{Some raw})$
unfolding *query-round-any-row-set-hit-def* **by** *simp*
show *?thesis*
by (*subst hit-eq*, *rule bexI[of - raw]*, *rule assms(2)*, *rule assms(1)*)
qed

lemma *query-round-any-row-set-hit-SomeE*:
assumes *query-round-any-row-set-hit* $s \ B \ (\text{Some } (u, t))$
obtains *raw* **where**
 $\text{raw} \in B$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) = \text{Some raw}$
proof –
have *hit-eq*:
 $\text{query-round-any-row-set-hit } s \ B \ (\text{Some } (u, t)) =$
 $(\exists \text{raw} \in B.$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) = \text{Some raw})$
unfolding *query-round-any-row-set-hit-def* **by** *simp*
have $\exists \text{raw} \in B.$
 $\text{fmlookup } (\text{HashMap } t)$
 $(\text{QueryIndexChallenge } (P\text{QueryCounter } s) (P\text{State } s)) = \text{Some raw}$
using *assms* **by** (*simp only: hit-eq*)
then show *?thesis*
by (*blast intro: that*)

qed

lemma *receive-query-index-challenge-current-lookupD*:

assumes *Some (raw, s0) ∈ set-dist (execute receive-query-index-challenge s)*

shows

fmlookup (HashMap s0)

(QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw

proof –

have *recv-info*:

s ≤ s0 ∧

PState s0 = PState s ∧

PTranscript s0 = PTranscript s ∧

fmlookup (HashMap s0)

(QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw

by *(rule receive-query-index-challenge-outcome[OF assms])*

show *?thesis*

by *(rule conjunct2[OF conjunct2[OF conjunct2[OF recv-info]])*

qed

lemma *verifier-query-round-after-index-program-preserves-current-lookupD*:

assumes *after*:

Some (u, t) ∈

set-dist

(execute

(verifier-query-round-after-index-program fr f-fl f-final as fl

final raw) s0)

and *lookup*:

fmlookup (HashMap s0)

(QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw

shows

fmlookup (HashMap t)

(QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw

proof –

let *?key = QueryIndexChallenge (PQueryCounter s) (PState s)*

have *u-eq: u = ()*

by *(cases u) simp*

have *preserve*:

fmlookup (HashMap t) ?key = fmlookup (HashMap s0) ?key

by *(rule verifier-query-round-after-index-program-preserves-query-lookup
[OF after[unfolded u-eq]])*

show *?thesis*

proof *(rule HOL.trans)*

show *fmlookup (HashMap t) ?key = fmlookup (HashMap s0) ?key*

by *(rule preserve)*

show *fmlookup (HashMap s0) ?key = Some raw*

by *(rule lookup)*

qed

qed

lemma *wp-receive-query-index-challenge-row-set-le*:
assumes *fresh*:
fmlookup (*HashMap s*)
 (*QueryIndexChallenge* (*PQueryCounter s*) (*PState s*)) = *None*
shows
wp-event receive-query-index-challenge
 ($\lambda out. \text{case out of None} \Rightarrow \text{False} \mid \text{Some (raw, -)} \Rightarrow \text{raw} \in B$)
 $s \leq \text{nnreal (card B)} / \text{nnreal size}$
proof –
have *eq*:
wp-event receive-query-index-challenge
 ($\lambda out. \text{case out of None} \Rightarrow \text{False} \mid \text{Some (raw, -)} \Rightarrow \text{raw} \in B$)
 $s = \text{nnreal (card B)} / \text{nnreal size}$
by (*rule wp-receive-query-index-challenge-fresh-set* [*OF fresh*])
show *?thesis*
by (*subst eq, rule order-refl*)
qed

lemma *finite-query-index-row-list-preimage[simp]*:
finite (*query-index-row-list-preimage Q*)
proof –
have *query-index-row-list-preimage* $Q \subseteq \{xs. \text{length } xs = \text{rounds}\}$
unfolding *query-index-row-list-preimage-def* **by** *auto*
then show *?thesis*
apply (*rule finite-subset*) **by** (*simp add: finite-list-length*)
qed

lemma *finite-query-index-row-list-position-values[simp]*:
finite (*query-index-row-list-position-values Q i*)
proof –
have *query-index-row-list-position-values* $Q\ i \subseteq \text{UNIV}$
by *simp*
then show *?thesis*
by (*rule finite-subset*) *simp*
qed

lemma *query-index-list-path-fresh-tail-lookup*:
assumes *fresh*:
query-index-list-path-fresh $s\ (\text{Suc } n)\ (\text{chunk } \# \text{ chunks})\ (\text{raw } \# \text{ raws})$
and head:
 $\text{Some } ((), s1) \in$
set-dist
 (*execute*
 (*verifier-query-round-program* *fr f-fl f-final as fl final*) s)
and head-chunk:
 $PState\ s1 = \text{foldl concat } (PState\ s)\ \text{chunk}$
and head-counter:
 $PQueryCounter\ s1 = \text{Suc } (PQueryCounter\ s)$
and i-bound: $i < n$

```

shows
  fmlookup (HashMap s1)
    (QueryIndexChallenge (PQueryCounter s1 + i)
      (state-after-query-chunks (PState s1) chunks i)) = None
proof -
  from verifier-query-round-program-alt-outcomeE[OF head]
  obtain raw-idx s0 where recv:
    Some (raw-idx, s0) ∈ set-dist (execute receive-query-index-challenge s)
  and after:
    Some ((), s1) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program fr f-fl f-final as fl
            final raw-idx) s0)
  by blast
  let ?tail-key =
    QueryIndexChallenge (PQueryCounter s1 + i)
      (state-after-query-chunks (PState s1) chunks i)
  let ?full-key =
    QueryIndexChallenge (PQueryCounter s + Suc i)
      (state-after-query-chunks (PState s) (chunk # chunks) (Suc i))
  have key-eq: ?tail-key = ?full-key
    using head-chunk head-counter
    unfolding state-after-query-chunks-def
    by simp
  have suc-i-bound: Suc i < Suc n
    using i-bound by simp
  have full-none:
    fmlookup (HashMap s) ?full-key = None
    by (rule query-index-list-path-fresh-lookupD[OF fresh suc-i-bound])
  have neq:
    ?full-key ≠ QueryIndexChallenge (PQueryCounter s) (PState s)
    by simp
  have lookup-s0:
    fmlookup (HashMap s0) ?full-key = None
  proof -
    have preserve:
      fmlookup (HashMap s0) ?full-key =
        fmlookup (HashMap s) ?full-key
      by (rule receive-query-index-challenge-preserves-other-lookup
        [OF recv neq])
    show ?thesis
  proof (rule HOL.trans)
    show fmlookup (HashMap s0) ?full-key =
      fmlookup (HashMap s) ?full-key
      by (rule preserve)
    show fmlookup (HashMap s) ?full-key = None
      by (rule full-none)
  qed
qed

```

```

qed
have lookup-s1:
  fmlookup (HashMap s1) ?full-key = None
proof -
  have preserve:
    fmlookup (HashMap s1) ?full-key =
      fmlookup (HashMap s0) ?full-key
  by (rule verifier-query-round-after-index-program-preserves-query-lookup
      [OF after])
  show ?thesis
proof (rule HOL.trans)
  show fmlookup (HashMap s1) ?full-key =
    fmlookup (HashMap s0) ?full-key
  by (rule preserve)
  show fmlookup (HashMap s0) ?full-key = None
  by (rule lookup-s0)
qed
qed
show ?thesis
  unfolding key-eq by (rule lookup-s1)
qed

lemma query-index-list-path-fresh-tail:
  assumes fresh:
    query-index-list-path-fresh s (Suc n) (chunk # chunks) (raw # raws)
  and head:
    Some (( ), s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and head-chunk:
    PState s1 = foldl concat (PState s) chunk
  and head-counter:
    PQueryCounter s1 = Suc (PQueryCounter s)
  shows query-index-list-path-fresh s1 n chunks raws
proof -
  have len-raws: length raws = n
  using fresh unfolding query-index-list-path-fresh-def by simp
  have len-chunks: length chunks = n
  using fresh unfolding query-index-list-path-fresh-def by simp
  have lookup-none:
     $\bigwedge i. i < n \implies$ 
    fmlookup (HashMap s1)
      (QueryIndexChallenge (PQueryCounter s1 + i)
        (state-after-query-chunks (PState s1) chunks i)) = None
  by (rule query-index-list-path-fresh-tail-lookup
      [OF fresh head head-chunk head-counter])
  show ?thesis
  unfolding query-index-list-path-fresh-def

```

```

    using len-rows len-chunks lookup-none by simp
qed

lemma query-rounds-first-tail-key-fresh-after-head:
  assumes fresh:
    query-index-list-path-fresh s (Suc n) chunks (raw # rows)
  and head:
    Some ((), s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and one-bound: 1 < Suc n
  and head-counter: PQueryCounter s1 = Suc (PQueryCounter s)
  shows
    fmlookup (HashMap s1)
      (QueryIndexChallenge (PQueryCounter s1)
        (state-after-query-chunks (PState s) chunks 1)) = None
  proof -
    let ?y = state-after-query-chunks (PState s) chunks 1
    let ?full-key = QueryIndexChallenge (PQueryCounter s + 1) ?y
    have full-none-s: fmlookup (HashMap s) ?full-key = None
      by (rule query-index-list-path-fresh-lookupD[OF fresh one-bound])
    from verifier-query-round-program-alt-outcomeE[OF head]
    obtain r s0 where recv:
      Some (r, s0) ∈ set-dist (execute receive-query-index-challenge s)
    and after:
      Some ((), s1) ∈
        set-dist
          (execute
            (verifier-query-round-after-index-program fr f-fl f-final as fl
              final r) s0)
    by blast
    have neq:
      ?full-key ≠ QueryIndexChallenge (PQueryCounter s) (PState s)
    by simp
    have s0-none:
      fmlookup (HashMap s0) ?full-key = None
    proof -
      have preserve:
        fmlookup (HashMap s0) ?full-key =
          fmlookup (HashMap s) ?full-key
      by (rule receive-query-index-challenge-preserves-other-lookup
        [OF recv neq])
    show ?thesis
    proof (rule HOL.trans)
      show fmlookup (HashMap s0) ?full-key =
        fmlookup (HashMap s) ?full-key
      by (rule preserve)
      show fmlookup (HashMap s) ?full-key = None
    end
  end

```

```

      by (rule full-none-s)
    qed
  qed
  have s1-none: fmlookup (HashMap s1) ?full-key = None
  proof -
    have preserve:
      fmlookup (HashMap s1) ?full-key =
        fmlookup (HashMap s0) ?full-key
    by (rule verifier-query-round-after-index-program-preserves-query-lookup
        [OF after])
    show ?thesis
  proof (rule HOL.trans)
    show fmlookup (HashMap s1) ?full-key =
      fmlookup (HashMap s0) ?full-key
    by (rule preserve)
    show fmlookup (HashMap s0) ?full-key = None
    by (rule s0-none)
  qed
  qed
  show ?thesis
  using head-counter s1-none by simp
  qed

lemma query-rounds-raw-list-path-hit-tail-head-state:
  assumes head:
    Some (((), s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s1)
  and len-chunks-suc: length chunks = Suc n
  and state-t:
    PState t = state-after-query-chunks (PState s) chunks (Suc n)
  and fresh:
    query-index-list-path-fresh s (Suc n) chunks (raw # raws)
  and lookups:
     $\bigwedge i. i < \text{Suc } n \implies$ 
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) chunks i)) =
      Some ((raw # raws) ! i)
  and chunks-eq: chunks = chunk # chunks-tail
  and head-counter: PQueryCounter s1 = Suc (PQueryCounter s)

```

```

  shows PState s1 = state-after-query-chunks (PState s) chunks 1
proof (cases n)
  case 0
  then have results = [] t = s1
    using tail by simp-all
  then show ?thesis
    using state-t 0 unfolding chunks-eq state-after-query-chunks-def by simp
next
  case (Suc m)
  from tail[unfolded Suc ntimes.simps] obtain u' s2 rest t0 where tail-head:
    Some (u', s2) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s1)
  and tail-rest0:
    Some (rest, t0) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) m)
          s2)
  and ret:
    Some (results, t) ∈ set-dist (execute (return (u' # rest)) t0)
  by (auto elim!: set-dist-bindE)
  have results-eq: results = u' # rest
    by (rule set-dist-return-valueD[OF ret])
  have t-eq: t = t0
    by (rule set-dist-return-stateD[OF ret])
  have tail-rest:
    Some (rest, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) m)
          s2)
  by (subst t-eq, rule tail-rest0)
  have u'-eq: u' = ()
    by (cases u') simp
  let ?y = state-after-query-chunks (PState s) chunks 1
  let ?key = QueryIndexChallenge (PQueryCounter s1) ?y
  have full-none-s1:
    fmlookup (HashMap s1) ?key = None
  by (rule query-rounds-first-tail-key-fresh-after-head
    [OF fresh head - head-counter])
    (use Suc in simp)
  have lookup-t:
    fmlookup (HashMap t) ?key = Some (raws ! 0)
  using lookups[of 1] Suc head-counter
  unfolding chunks-eq state-after-query-chunks-def by simp

```



```

have key-eq:
  ?key = QueryIndexChallenge (PQueryCounter s1) (PState s1)
proof (rule ccontr)
  assume neq:
    ?key ≠ QueryIndexChallenge (PQueryCounter s1) (PState s1)
  have tail-head-unit:
    Some ((), s2) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s1)
    using tail-head u'-eq by simp
  from verifier-query-round-program-alt-outcomeE[OF tail-head-unit]
  obtain r s0 where recv:
    Some (r, s0) ∈ set-dist (execute receive-query-index-challenge s1)
  and after:
    Some ((), s2) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program fr f-fl f-final as fl
            final r) s0)
    by blast
  have s0-none:
    fmlookup (HashMap s0) ?key = None
    using receive-query-index-challenge-preserves-other-lookup[OF recv neq]
    full-none-s1
    by simp
  have s2-none: fmlookup (HashMap s2) ?key = None
  proof –
    have preserve:
      fmlookup (HashMap s2) ?key = fmlookup (HashMap s0) ?key
      by (rule verifier-query-round-after-index-program-preserves-query-lookup
        [OF after])
    show ?thesis
    proof (rule HOL.trans)
      show fmlookup (HashMap s2) ?key = fmlookup (HashMap s0) ?key
      by (rule preserve)
      show fmlookup (HashMap s0) ?key = None
      by (rule s0-none)
    qed
  qed
  from verifier-query-round-program-outcome[OF tail-head[unfolded u'-eq]]
  obtain r2::'f and idx2::nat and chunk2::'f list where
    counter-s2: PQueryCounter s2 = Suc (PQueryCounter s1)
    by blast
  have lookup-pres:
    fmlookup (HashMap t) ?key = fmlookup (HashMap s2) ?key
    by (rule ntimes-verifier-query-round-program-preserves-past-query-lookup
      [OF - tail-rest])
    (use counter-s2 in simp)

```

```

    show False
    using lookup-t s2-none lookup-pres by simp
qed
then show ?thesis
  by simp
qed

end

context soundness
begin

lemma query-rounds-raw-list-path-hit-tail:
  assumes head:
    Some ((), s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and tail:
    Some (results, t) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final) n)
          s1)
  and hit:
    query-rounds-raw-list-path-hit s (raw # rows) (Suc n)
      (Some (u # results, t))
  shows query-rounds-raw-list-path-hit s1 rows n (Some (results, t))
proof -
  from query-rounds-raw-list-path-hit-cons-SomeE[OF hit] obtain chunks where
    len-rows: length rows = n
  and len-chunks-suc: length chunks = Suc n
  and state-t:
    PState t = state-after-query-chunks (PState s) chunks (Suc n)
  and fresh:
    query-index-list-path-fresh s (Suc n) chunks (raw # rows)
  and lookups:
     $\bigwedge i. i < \text{Suc } n \implies$ 
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) chunks i)) =
    Some ((raw # rows) ! i)
  by blast
  obtain chunk chunks-tail where chunks-eq: chunks = chunk # chunks-tail
  using len-chunks-suc by (cases chunks) auto
  have len-chunks-tail: length chunks-tail = n
  using len-chunks-suc unfolding chunks-eq by simp
  from verifier-query-round-program-outcome[OF head]

```

```

obtain raw0::'f and idx::nat and chunk0:: 'f list where
  head-counter: PQueryCounter s1 = Suc (PQueryCounter s)
by blast
have head-state:
  PState s1 = state-after-query-chunks (PState s) chunks 1
by (rule query-rounds-raw-list-path-hit-tail-head-state
    [OF head tail len-chunks-suc state-t fresh lookups chunks-eq
      head-counter])
have fresh-tail:
  query-index-list-path-fresh s1 n chunks-tail raws
proof -
  have fresh-chunks:
    query-index-list-path-fresh s (Suc n) (chunk # chunks-tail)
      (raw # raws)
    by (subst chunks-eq[symmetric], rule fresh)
  have head-chunk: PState s1 = foldl concat (PState s) chunk
proof -
  have head-state-cons:
    PState s1 =
      state-after-query-chunks (PState s) (chunk # chunks-tail) 1
    by (subst chunks-eq[symmetric], rule head-state)
  show ?thesis
    by (subst state-after-query-chunks-one-cons[symmetric],
      rule head-state-cons)
  qed
show ?thesis
  by (rule query-index-list-path-fresh-tail
    [OF fresh-chunks head head-chunk head-counter])
qed
have state-tail:
  PState t =
    state-after-query-chunks (PState s1) chunks-tail n
proof -
  have shift:
    state-after-query-chunks (PState s1) chunks-tail n =
      state-after-query-chunks (PState s) (chunk # chunks-tail) (Suc n)
    by (rule state-after-query-chunks-cons-shift)
      (use head-state chunks-eq in simp)
  show ?thesis
    by (subst shift, subst chunks-eq[symmetric], rule state-t)
  qed
have lookup-tail:
   $\bigwedge i. i < n \implies$ 
    fmllookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s1 + i)
        (state-after-query-chunks (PState s1) chunks-tail i)) =
      Some (raws ! i)
proof -
  fix i

```

```

assume i-bound:  $i < n$ 
have key-eq:
  QueryIndexChallenge (PQueryCounter  $s1 + i$ )
    (state-after-query-chunks (PState  $s1$ ) chunks-tail  $i$ ) =
  QueryIndexChallenge (PQueryCounter  $s + \text{Suc } i$ )
    (state-after-query-chunks (PState  $s$ ) chunks (Suc  $i$ ))
proof –
  have st-shift:
    state-after-query-chunks (PState  $s1$ ) chunks-tail  $i$  =
    state-after-query-chunks (PState  $s$ ) (chunk # chunks-tail) (Suc  $i$ )
  by (rule state-after-query-chunks-cons-shift-index)
    (use head-state chunks-eq in simp)
  show ?thesis
    using head-counter st-shift unfolding chunks-eq by simp
qed
show fmlookup (HashMap  $t$ )
  (QueryIndexChallenge (PQueryCounter  $s1 + i$ )
    (state-after-query-chunks (PState  $s1$ ) chunks-tail  $i$ )) =
  Some (rows !  $i$ )
  using lookups[of Suc  $i$ ] i-bound key-eq by simp
qed
show ?thesis
  by (rule query-rounds-raw-list-path-hitI
    [OF len-rows len-chunks-tail state-tail fresh-tail lookup-tail])
qed

end

context soundness
begin

lemma query-rounds-raw-list-path-hit-head:
  assumes head:
    Some  $((\ ), s1) \in$ 
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final)  $s$ )
  and tail:
    Some (results,  $t$ )  $\in$ 
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)  $n$ )
           $s1$ )
  and hit:
    query-rounds-raw-list-path-hit  $s$  (raw # rows) (Suc  $n$ )
      (Some ( $u$  # results,  $t$ ))
  shows query-round-any-raw-set-hit  $s$  {raw} (Some  $((\ ), s1)$ )
proof –

```

```

have lookup-t:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof -
  from query-rounds-raw-list-path-hit-cons-SomeE[OF hit]
  obtain chunks where lookup:
     $\bigwedge i. i < \text{Suc } n \implies$ 
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s + i)
        (state-after-query-chunks (PState s) chunks i)) =
      Some ((raw # rows) ! i)
  by blast
have lookup0:
  fmlookup (HashMap t)
    (QueryIndexChallenge (PQueryCounter s + 0)
      (state-after-query-chunks (PState s) chunks 0)) =
    Some ((raw # rows) ! 0)
  using lookup[of 0] by simp
show ?thesis
  using lookup0 unfolding state-after-query-chunks-def by simp
qed
from verifier-query-round-program-alt-outcomeE[OF head]
obtain raw0 s0 where
  recv:
    Some (raw0, s0)  $\in$  set-dist (execute receive-query-index-challenge s)
  and after:
    Some ((), s1)  $\in$ 
      set-dist
        (execute
          (verifier-query-round-after-index-program fr f-fl f-final as fl
            final raw0) s0)
  by blast
have counter-s1: PQueryCounter s1 = Suc (PQueryCounter s)
  by (rule verifier-query-round-parts-counter[OF recv after])
have past: PQueryCounter s < PQueryCounter s1
  by (rule eq-Suc-imp-less-nat[OF counter-s1])
have lookup-s1:
  fmlookup (HashMap s1)
    (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
proof -
  let ?key = QueryIndexChallenge (PQueryCounter s) (PState s)
  have preserve:
    fmlookup (HashMap t) ?key = fmlookup (HashMap s1) ?key
  by (rule ntimes-verifier-query-round-program-preserves-past-query-lookup
    [OF past tail])
  show ?thesis
  proof (rule HOL.trans)
    show fmlookup (HashMap s1) ?key = fmlookup (HashMap t) ?key
    by (rule HOL.sym[OF preserve])
  qed

```

```

    show fmlookup (HashMap t) ?key = Some raw
    by (rule lookup-t)
  qed
qed
show ?thesis
  by (rule query-round-any-raw-set-hitI[OF singletonI lookup-s1])
qed

lemma verifier-query-round-program-raw-set-hit-imp-raw-in-Some:
  assumes recv:
    Some (raw, s0) ∈ set-dist (execute receive-query-index-challenge s)
  and after:
    Some (u, t) ∈
      set-dist
        (execute
          (verifier-query-round-after-index-program fr f-fl f-final as fl
            final raw) s0)
  and hit: query-round-any-raw-set-hit s B (Some (u, t))
  shows raw ∈ B
proof -
  from query-round-any-raw-set-hit-SomeE[OF hit]
  obtain raw' where
    raw'-in: raw' ∈ B
  and lookup-t:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw'
  by blast
  have lookup-s0:
    fmlookup (HashMap s0)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  by (rule receive-query-index-challenge-current-lookupD[OF recv])
  have lookup-t-current:
    fmlookup (HashMap t)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = Some raw
  by (rule verifier-query-round-after-index-program-preserves-current-lookupD
    [OF after lookup-s0])
  have some-eq: Some raw = Some raw'
  using lookup-t-current lookup-t by simp
  have raw-eq: raw = raw'
  using some-eq by simp
  show ?thesis
  by (subst raw-eq, rule raw'-in)
qed

```

```

lemma verifier-query-round-program-raw-set-hit-imp-raw-in:
  assumes recv:
    Some (raw, s0) ∈ set-dist (execute receive-query-index-challenge s)
  and after:
    out ∈

```

```

    set-dist
    (execute
      (verifier-query-round-after-index-program fr f-fl f-final as fl
        final raw) s0)
  and hit: query-round-any-row-set-hit s B out
shows raw ∈ B
proof -
  have rhs:
    (case out of
      None ⇒ False
    | Some (-, t) ⇒
      (∃ raw' ∈ B.
        fmlookup (HashMap t)
          (QueryIndexChallenge (PQueryCounter s) (PState s)) =
            Some raw'))
  using hit by (simp only: query-round-any-row-set-hit-def)
obtain u t where out-eq: out = Some (u, t)
proof (cases out)
  case None
  have False
  using rhs None by simp
  then show ?thesis
  by (rule FalseE)
next
  case (Some pair)
  then obtain a b where out = Some (a, b)
  by (cases pair) auto
  then show ?thesis
  by (rule that)
qed
have after-some:
  Some (u, t) ∈
  set-dist
  (execute
    (verifier-query-round-after-index-program fr f-fl f-final as fl
      final raw) s0)
  by (subst out-eq[symmetric]) (rule after)
have hit-some: query-round-any-row-set-hit s B (Some (u, t))
  by (subst out-eq[symmetric]) (rule hit)
show ?thesis
  by (rule verifier-query-round-program-row-set-hit-imp-row-in-Some
    [OF recv after-some hit-some])
qed

lemma wp-verifier-query-round-program-any-row-set-bound:
  assumes fresh:
    fmlookup (HashMap s)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = None
  shows

```

```

    wp-event (verifier-query-round-program fr f-fl f-final as fl final)
      (query-round-any-raw-set-hit s B) s ≤
      nnreal (card B) / nnreal size
  unfolding verifier-query-round-program-alt-def
proof (rule wp-event-bind-bound-by-head-event)
  show wp-event receive-query-index-challenge
    (λout. case out of None ⇒ False | Some (raw, -) ⇒ raw ∈ B)
    s ≤ nnreal (card B) / nnreal size
  by (rule wp-receive-query-index-challenge-raw-set-le[OF fresh])
next
  show query-round-any-raw-set-hit s B None ⇒
    (case None of None ⇒ False | Some (raw, -) ⇒ raw ∈ B)
  by (rule query-round-any-raw-set-hit-NoneE)
next
  fix raw s0 out
  assume recv:
    Some (raw, s0) ∈ set-dist (execute receive-query-index-challenge s)
  and out:
    out ∈
      set-dist
      (execute
        (verifier-query-round-after-index-program fr f-fl f-final as fl
          final raw) s0)
  and hit: query-round-any-raw-set-hit s B out
  show (case Some (raw, s0) of None ⇒ False | Some (raw, -) ⇒ raw ∈ B)
proof -
  have raw-in: raw ∈ B
  by (rule verifier-query-round-program-raw-set-hit-imp-raw-in
    [where s=s and B=B and raw=raw and s0=s0 and out=out
      and fr=fr and f-fl=f-fl and f-final=f-final and as=as
      and fl=fl and final=final, OF recv out hit])
  then show ?thesis
  by simp
qed
qed

lemma query-rounds-raw-list-path-hit-SomeE:
  assumes query-rounds-raw-list-path-hit s raws n out
  obtains results t where out = Some (results, t)
proof -
  show ?thesis
proof (cases out)
  case None
  have none-eq: query-rounds-raw-list-path-hit s raws n None = False
  unfolding query-rounds-raw-list-path-hit-def by simp
  have hit-none: query-rounds-raw-list-path-hit s raws n None
  by (subst None[symmetric], rule assms)
  then show ?thesis
  using hit-none by (simp only: none-eq)

```



```

next
  case (Some pair)
  then obtain results t where out = Some (results, t)
    by (cases pair) auto
  then show ?thesis
    by (rule that)
qed
qed

lemma query-rounds-raw-list-path-hit-some-fresh-currentD:
  assumes hit:
    query-rounds-raw-list-path-hit s (raw # rows-tail) (Suc n)
    (Some (ys, t))
  shows
    fmlookup (HashMap s)
      (QueryIndexChallenge (PQueryCounter s) (PState s)) = None
proof -
  have some-eq:
    query-rounds-raw-list-path-hit s (raw # rows-tail) (Suc n)
    (Some (ys, t)) =
    (∃ chunks.
      length (raw # rows-tail) = Suc n ∧
      length chunks = Suc n ∧
      PState t = state-after-query-chunks (PState s) chunks (Suc n) ∧
      query-index-list-path-fresh s (Suc n) chunks (raw # rows-tail) ∧
      (∀ i < Suc n.
        fmlookup (HashMap t)
          (QueryIndexChallenge (PQueryCounter s + i)
            (state-after-query-chunks (PState s) chunks i)) =
          Some ((raw # rows-tail) ! i)))
    unfolding query-rounds-raw-list-path-hit-def by simp
  have ex-chunks:
    ∃ chunks.
      length (raw # rows-tail) = Suc n ∧
      length chunks = Suc n ∧
      PState t = state-after-query-chunks (PState s) chunks (Suc n) ∧
      query-index-list-path-fresh s (Suc n) chunks (raw # rows-tail) ∧
      (∀ i < Suc n.
        fmlookup (HashMap t)
          (QueryIndexChallenge (PQueryCounter s + i)
            (state-after-query-chunks (PState s) chunks i)) =
          Some ((raw # rows-tail) ! i))
    using hit by (simp only: some-eq)
  then obtain chunks where fresh-path:
    query-index-list-path-fresh s (Suc n) chunks (raw # rows-tail)
    by (elim exE conjE)
  show ?thesis
    by (rule query-index-list-path-fresh-head-fresh[OF fresh-path])
qed

```

lemma *query-rounds-raw-list-path-hit-some-head-lookupD*:
assumes *hit*:
 $\text{query-rounds-raw-list-path-hit } s \text{ (raw \# rows) (Suc } n\text{)}$
 $(\text{Some } (ys, t))$
shows
 $\text{fmlookup (HashMap } t\text{)}$
 $(\text{QueryIndexChallenge (PQueryCounter } s\text{) (PState } s\text{)}) = \text{Some raw}$
proof –
have *some-eq*:
 $\text{query-rounds-raw-list-path-hit } s \text{ (raw \# rows) (Suc } n\text{)}$
 $(\text{Some } (ys, t)) =$
 $(\exists \text{ chunks.}$
 $\text{length (raw \# rows) = Suc } n \wedge$
 $\text{length chunks = Suc } n \wedge$
 $\text{PState } t = \text{state-after-query-chunks (PState } s\text{) chunks (Suc } n\text{) } \wedge$
 $\text{query-index-list-path-fresh } s \text{ (Suc } n\text{) chunks (raw \# rows) } \wedge$
 $(\forall i < \text{Suc } n.$
 $\text{fmlookup (HashMap } t\text{)}$
 $(\text{QueryIndexChallenge (PQueryCounter } s + i\text{)}$
 $(\text{state-after-query-chunks (PState } s\text{) chunks } i\text{)}) =$
 $\text{Some } ((\text{raw \# rows}) ! i)))$
unfolding *query-rounds-raw-list-path-hit-def* **by** *simp*
have *ex-chunks*:
 $\exists \text{ chunks.}$
 $\text{length (raw \# rows) = Suc } n \wedge$
 $\text{length chunks = Suc } n \wedge$
 $\text{PState } t = \text{state-after-query-chunks (PState } s\text{) chunks (Suc } n\text{) } \wedge$
 $\text{query-index-list-path-fresh } s \text{ (Suc } n\text{) chunks (raw \# rows) } \wedge$
 $(\forall i < \text{Suc } n.$
 $\text{fmlookup (HashMap } t\text{)}$
 $(\text{QueryIndexChallenge (PQueryCounter } s + i\text{)}$
 $(\text{state-after-query-chunks (PState } s\text{) chunks } i\text{)}) =$
 $\text{Some } ((\text{raw \# rows}) ! i))$
using *hit* **by** (*simp only: some-eq*)
then obtain *chunks* **where** *lookup-all*:
 $\forall i < \text{Suc } n.$
 $\text{fmlookup (HashMap } t\text{)}$
 $(\text{QueryIndexChallenge (PQueryCounter } s + i\text{)}$
 $(\text{state-after-query-chunks (PState } s\text{) chunks } i\text{)}) =$
 $\text{Some } ((\text{raw \# rows}) ! i)$
by (*elim exE conjE*)
have *lookup0*:
 $\text{fmlookup (HashMap } t\text{)}$
 $(\text{QueryIndexChallenge (PQueryCounter } s + 0\text{)}$
 $(\text{state-after-query-chunks (PState } s\text{) chunks } 0\text{)}) =$
 $\text{Some } ((\text{raw \# rows}) ! 0)$
proof –
have $0 < \text{Suc } n$

```

    by simp
  from lookup-all this show ?thesis
    by blast
qed
show ?thesis
  using lookup0 unfolding state-after-query-chunks-def by simp
qed

lemma query-rounds-raw-list-path-hit-bind-headD:
  assumes head:
    Some (u, s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  and support:
    out ∈
      set-dist
        (execute
          (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n
            >>= (λxs. return (u # xs))) s1)
  and hit:
    query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n) out
  shows query-round-any-raw-set-hit s {raw} (Some (u, s1))
proof -
  let ?tail =
    ntimes (verifier-query-round-program fr f-fl f-final as fl final) n
  from hit obtain ys t0 where out-some: out = Some (ys, t0)
  by (rule query-rounds-raw-list-path-hit-SomeE)
  from support out-some obtain results t where tail:
    Some (results, t) ∈ set-dist (execute ?tail s1)
  and out-eq: out = Some (u # results, t)
  by (elim set-dist-bind-return-consE)
  have u-eq: u = ()
  by (cases u) simp
  have head-unit:
    Some ((), s1) ∈
      set-dist
        (execute
          (verifier-query-round-program fr f-fl f-final as fl final) s)
  using head u-eq by simp
  have hit-head-out:
    query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n)
      (Some (u # results, t))
  by (subst out-eq[symmetric], rule hit)
  have query-round-any-raw-set-hit s {raw} (Some ((), s1))
  by (rule query-rounds-raw-list-path-hit-head
    [OF head-unit tail hit-head-out])
  then show ?thesis
  using u-eq by simp

```

qed

lemma *query-rounds-raw-list-path-hit-cont-zero-if-not-head*:

assumes *head*:

Some (*u*, *s1*) ∈
set-dist
 (*execute*
 (*verifier-query-round-program* *fr* *f-fl* *f-final* *as* *fl* *final*) *s*)

and *not-head*:

¬ (*fmlookup* (*HashMap* *s*)
 (*QueryIndexChallenge* (*PQueryCounter* *s*) (*PState* *s*)) = *None* ∧
query-round-any-raw-set-hit *s* {*raw*} (*Some* (*u*, *s1*)))

shows

wp-event
 (*ntimes* (*verifier-query-round-program* *fr* *f-fl* *f-final* *as* *fl* *final*) *n*
 ≫ (λ*xs*. *return* (*u* # *xs*)))
 (*query-rounds-raw-list-path-hit* *s* (*raw* # *raws-tail*) (*Suc* *n*))
s1 = 0

proof (*rule* *ccontr*)

let *?tail* =

ntimes (*verifier-query-round-program* *fr* *f-fl* *f-final* *as* *fl* *final*) *n*

assume *nonzero*:

wp-event (*?tail* ≫ (λ*xs*. *return* (*u* # *xs*)))
 (*query-rounds-raw-list-path-hit* *s* (*raw* # *raws-tail*) (*Suc* *n*))
s1 ≠ 0

from *wp-event-nonzero-imp-exists-support*[*OF nonzero*]

obtain *out* **where** *out-support*:

out ∈ *set-dist* (*execute* (*?tail* ≫ (λ*xs*. *return* (*u* # *xs*))) *s1*)

and *hit*:

query-rounds-raw-list-path-hit *s* (*raw* # *raws-tail*) (*Suc* *n*) *out*

by *blast*

from *hit* **obtain** *ys* *t0* **where** *out-some*: *out* = *Some* (*ys*, *t0*)

by (*rule* *query-rounds-raw-list-path-hit-SomeE*)

have *hit-some*:

query-rounds-raw-list-path-hit *s* (*raw* # *raws-tail*) (*Suc* *n*)
 (*Some* (*ys*, *t0*))

by (*subst* *out-some*[*symmetric*], *rule* *hit*)

have *fresh-current*:

fmlookup (*HashMap* *s*)
 (*QueryIndexChallenge* (*PQueryCounter* *s*) (*PState* *s*)) = *None*
by (*rule* *query-rounds-raw-list-path-hit-some-fresh-currentD*[*OF hit-some*])

have *raw-head*:

query-round-any-raw-set-hit *s* {*raw*} (*Some* (*u*, *s1*))
by (*rule* *query-rounds-raw-list-path-hit-bind-headD*[*OF head out-support hit*])

show *False*

using *not-head* *fresh-current* *raw-head* **by** *blast*

qed

lemma *query-rounds-raw-list-path-hit-cont-le-tail*:

```

assumes head:
  Some (u, s1) ∈
    set-dist
      (execute
        (verifier-query-round-program fr f-fl f-final as fl final) s)
shows
  wp-event
    (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n
      ≫ (λxs. return (u # xs)))
    (query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n))
    s1
  ≤ wp-event
    (ntimes (verifier-query-round-program fr f-fl f-final as fl final) n)
    (query-rounds-raw-list-path-hit s1 raws-tail n) s1
proof -
  let ?tail =
    ntimes (verifier-query-round-program fr f-fl f-final as fl final) n
have u-eq: u = ()
  by (cases u) simp
show ?thesis
proof (rule wp-event-bind-bound-by-head-event)
  show wp-event ?tail
    (query-rounds-raw-list-path-hit s1 raws-tail n) s1
  ≤ wp-event ?tail
    (query-rounds-raw-list-path-hit s1 raws-tail n) s1
  by simp
next
show
  query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n) None
    ⇒ query-rounds-raw-list-path-hit s1 raws-tail n None
  unfolding query-rounds-raw-list-path-hit-def by simp
next
fix results t out
assume tail:
  Some (results, t) ∈ set-dist (execute ?tail s1)
  and ret: out ∈ set-dist (execute (return (u # results)) t)
  and hit:
    query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n) out
from ret obtain out-eq:
  out = Some (u # results, t)
unfolding set-dist-def return.rep-eq dist-return-def
  dist-delta-dist delta-map-def
by simp
show query-rounds-raw-list-path-hit s1 raws-tail n
  (Some (results, t))
by (rule query-rounds-raw-list-path-hit-tail
  [OF head[unfolded u-eq] tail])
  (use hit out-eq u-eq in simp)
qed

```

qed

lemma *wp-ntimes-verifier-query-round-program-raw-list-path-hit-bound:*
assumes *len-rows: length rows = n*
shows
 wp-event
 (*ntimes (verifier-query-round-program fr f-fl f-final as fl final) n*)
 (*query-rounds-raw-list-path-hit s rows n*) *s* ≤
 (*1 / nnreal size*) ^ *n*
using *len-rows*
proof (*induction n arbitrary: s rows*)
case 0
then show ?*case*
 unfolding *query-rounds-raw-list-path-hit-def*
 query-index-list-path-fresh-def state-after-query-chunks-def
 by (*simp add: wp-event-def wpsimps*)
next
case (*Suc n*)
obtain *raw rows-tail where rows-eq: rows = raw # rows-tail*
 by (*rule list-length-SucE[OF Suc.premis]*)
have *len-tail: length rows-tail = n*
 using *Suc.premis unfolding rows-eq by simp*
let ?*prog* = *verifier-query-round-program fr f-fl f-final as fl final*
let ?*tail* =
 ntimes ?prog n
let ?*Q* = *query-rounds-raw-list-path-hit s rows (Suc n)*
let ?*fresh* =
 fmlookup (HashMap s)
 (*QueryIndexChallenge (PQueryCounter s) (PState s)*) = *None*
let ?*Head* =
 λout. ?fresh ∧ query-round-any-raw-set-hit s {raw} out
have *head-bound:*
 wp-event ?prog ?Head s ≤ 1 / nnreal size
proof (*cases ?fresh*)
case *True*
have *mono:*
 wp-event ?prog ?Head s ≤
 wp-event ?prog (query-round-any-raw-set-hit s {raw}) s
 by (*rule wp-event-mono simp*)
also have ... ≤ *nnreal (card {raw}) / nnreal size*
 by (*rule wp-verifier-query-round-program-any-raw-set-bound[OF True]*)
also have ... = *1 / nnreal size*
 by *simp*
finally show ?*thesis* .
next
case *False*
have *impossible: ?Head = (λ-. False)*
 using *False by auto*
have *zero: wp-event ?prog ?Head s = 0*

```

    by (simp only: impossible wp-event-false)
  then show ?thesis
    by simp
qed
show ?case
  unfolding ntimes.simps raws-eq
proof (rule order-trans)
  have none-Q:
     $\neg$  query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n) None
  by (simp add: query-rounds-raw-list-path-hit-def)
show
  wp-event
    ( $?prog \gg (\lambda u. ?tail \gg (\lambda xs. \text{return } (u \# xs)))$ )
    (query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n)) s
   $\leq$  wp-event ?prog ?Head s * ((1 / nnreal size) ^ n)
proof (rule wp-event-bind-bound-by-head-and-cont
  [where m = ?prog
    and k =  $\lambda u. ?tail \gg (\lambda xs. \text{return } (u \# xs))$ 
    and Q = query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n)
    and P = ?Head
    and D = (1 / nnreal size) ^ n,
    OF none-Q])
  fix u s1
  assume head:
    Some (u, s1)  $\in$  set-dist (execute ?prog s)
  and not-head:  $\neg$  ?Head (Some (u, s1))
show
  wp-event ( $?tail \gg (\lambda xs. \text{return } (u \# xs))$ )
    (query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n))
    s1 = 0
  by (rule query-rounds-raw-list-path-hit-cont-zero-if-not-head
    [OF head not-head])
next
  fix u s1
  assume head:
    Some (u, s1)  $\in$  set-dist (execute ?prog s)
  and head-hit: ?Head (Some (u, s1))
  have tail-bound:
    wp-event ?tail
      (query-rounds-raw-list-path-hit s1 raws-tail n) s1  $\leq$ 
      (1 / nnreal size) ^ n
  by (rule Suc.IH[OF len-tail])
show
  wp-event ( $?tail \gg (\lambda xs. \text{return } (u \# xs))$ )
    (query-rounds-raw-list-path-hit s (raw # raws-tail) (Suc n))
    s1  $\leq$  (1 / nnreal size) ^ n
proof (rule order-trans[OF tail-bound])
  show
    wp-event ( $?tail \gg (\lambda xs. \text{return } (u \# xs))$ )

```

```

      (query-rounds-raw-list-path-hit s (raw # rows-tail) (Suc n))
    s1
  ≤ wp-event ?tail
    (query-rounds-raw-list-path-hit s1 rows-tail n) s1
proof (rule wp-event-bind-return-cons-mono
  [where m = ?tail
    and Q =
      query-rounds-raw-list-path-hit s (raw # rows-tail) (Suc n)
    and R = query-rounds-raw-list-path-hit s1 rows-tail n
    and u = u])
show
  query-rounds-raw-list-path-hit s (raw # rows-tail) (Suc n)
    None  $\implies$ 
  query-rounds-raw-list-path-hit s1 rows-tail n None
unfolding query-rounds-raw-list-path-hit-def by simp
next
fix xs t
assume tail:
  Some (xs, t)  $\in$  set-dist (execute ?tail s1)
and hit:
  query-rounds-raw-list-path-hit s (raw # rows-tail) (Suc n)
    (Some (u # xs, t))
have u-eq: u = ()
by (cases u) simp
show query-rounds-raw-list-path-hit s1 rows-tail n
  (Some (xs, t))
by (rule query-rounds-raw-list-path-hit-tail
  [OF head[unfolded u-eq] tail])
  (use hit u-eq in simp)
qed
qed
qed
next
show wp-event ?prog ?Head s * (1 / nnreal size) ^ n
  ≤ (1 / nnreal size) ^ Suc n
using head-bound by (simp add: mult-right-mono)
qed
qed

lemma wp-ntimes-verifier-query-round-program-raw-list-preimage-path-hit-bound:
shows
  wp-event
    (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
      rounds)
  (λout.  $\exists$  rows  $\in$  query-index-raw-list-preimage Q.
    query-rounds-raw-list-path-hit s rows rounds out)
  s ≤
  nnreal (card (query-index-raw-list-preimage Q)) *
  (1 / nnreal size) ^ rounds

```


proof –
have *finite-rows*: *finite* (*query-index-row-list-preimage* *Q*)
 by (*rule finite-query-index-row-list-preimage*)
have
wp-event
 (*ntimes* (*verifier-query-round-program* *fr f-fl f-final as fl final*)
rounds)
 ($\lambda out. \exists rows \in query-index-row-list-preimage\ Q.$
query-rounds-row-list-path-hit *s rows rounds out*)
 $s \leq$
 ($\sum rows \in query-index-row-list-preimage\ Q.$
 ($1 / nreal\ size$) $\wedge rounds$)
by (*rule wp-event-finite-UN-bound*[*OF finite-rows*])
 (*rule wp-ntimes-verifier-query-round-program-row-list-path-hit-bound*,
simp add: query-index-row-list-preimage-def)
also have ... =
nreal (*card* (*query-index-row-list-preimage* *Q*)) *
 ($1 / nreal\ size$) $\wedge rounds$
using *finite-rows* **by** *simp*
finally show *?thesis* .
qed

lemma *checked-staged-query-index-list-relation-fiber-card-bound*:
card {*y. checked-staged-query-index-list-relation* *A Q x y*}
 $\leq query-index-row-list-relation-fiber-bound\ Q$

proof –
let *?values* =
 ($\bigcup i < rounds. query-index-row-list-position-values\ Q\ i$)
have *subset*:
 {*y. checked-staged-query-index-list-relation* *A Q x y*} $\subseteq ?values$
unfolding *checked-staged-query-index-list-relation-def*
query-index-row-list-position-values-def
by *blast*
have *finite-values*: *finite* *?values*
by *simp*
have *card* {*y. checked-staged-query-index-list-relation* *A Q x y*}
 $\leq card\ ?values$
by (*rule card-mono*[*OF finite-values subset*])
also have ... $\leq$
 ($\sum i < rounds. card\ (query-index-row-list-position-values\ Q\ i)$)
apply (*rule card-UN-le*) **by** *simp*
also have ... = *query-index-row-list-relation-fiber-bound* *Q*
unfolding *query-index-row-list-relation-fiber-bound-def* **by** *simp*
finally show *?thesis* .
qed

lemma *hash-relation-program-checked-staged-transcript-query-index-list*:
assumes *wf*: *staged-budget-wellformed* *budgets*
and *controlled*: *staged-adversary-controlled* *budgets* *A*

shows

hash-relation-program
 (*checked-staged-query-index-list-relation* *A* *Q*)
 (*query-index-raw-list-relation-fiber-bound* *Q*)
 (*staged-attacker-query-budget* *budgets* + *staged-challenge-query-budget*)
 (*checked-staged-transcript-program* *A*)
by (*rule hash-relation-program-checked-staged-transcript-program*
 [*OF wf controlled*])
 (*rule checked-staged-query-index-list-relation-fiber-card-bound*)

lemma *staged-transcript-query-index-list-set-hit-imp-relation-hit*:

assumes *support*:

Some (*data*, *attacker-state*) ∈
 set-dist
 (*execute* (*checked-staged-transcript-program* *A*)
 adversary-initial-state)

and *hit*:

staged-transcript-query-index-list-set-hit *Q*
 (*Some* (*data*, *attacker-state*))

shows

hash-relation-hit
 (*checked-staged-query-index-list-relation* *A* *Q*)
 adversary-initial-state *attacker-state*

proof –

from *hit* **obtain** *raw-idxs* *query-idxs* **where**

len-raw: *length raw-idxs* = *rounds*

and *query-in*: *query-idxs* ∈ *Q*

and *query-idxs-eq*:

query-idxs = *map* (*λraw. index (to-nat raw)*) *raw-idxs*

and *lookup*:

$\bigwedge i. i < \text{rounds} \implies$
 fmlookup (*HashMap attacker-state*)
 (*QueryIndexChallenge* *i*
 (*state-after-query-chunks*
 (*staged-query-start-hash data*)
 (*staged-query-chunks data*) *i*)) =
 Some (*raw-idxs* ! *i*)

unfolding *staged-transcript-query-index-list-set-hit-def* **by** *auto*

have *raw-in*: *raw-idxs* ∈ *query-index-raw-list-preimage* *Q*

unfolding *query-index-raw-list-preimage-def*

using *len-raw query-in query-idxs-eq* **by** *blast*

let *?x* =

QueryIndexChallenge 0
 (*state-after-query-chunks*
 (*staged-query-start-hash data*) (*staged-query-chunks data*) 0)

have *zero-bound*: 0 < *rounds*

by (*rule rounds-positive*)

have *initial-none*: *fmlookup* (*HashMap adversary-initial-state*) *?x* = *None*

by (*simp add: adversary-initial-state-def*)

```

have final-lookup:
  fmlookup (HashMap attacker-state) ?x = Some (raw-idxs ! 0)
  using lookup[OF zero-bound] by simp
have rel:
  checked-staged-query-index-list-relation A Q ?x (raw-idxs ! 0)
  unfolding checked-staged-query-index-list-relation-def
  by (intro exI[of - data] exI[of - attacker-state]
    exI[of - raw-idxs] exI[of - 0] conjI)
    (use support raw-in zero-bound in simp-all)
show ?thesis
  unfolding hash-relation-hit-def
  by (intro exI[of - ?x] exI[of - raw-idxs ! 0] conjI)
    (use initial-none final-lookup rel in simp-all)
qed

lemma staged-transcript-query-index-list-set-hit-imp-relation-hit-event:
assumes support:
  out ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
and hit: staged-transcript-query-index-list-set-hit Q out
shows
  hash-relation-hit-event
    (checked-staged-query-index-list-relation A Q)
    adversary-initial-state out
proof (cases out)
case None
then show ?thesis
  using hit unfolding staged-transcript-query-index-list-set-hit-def
    hash-relation-hit-event-def
  by simp
next
case (Some result)
then obtain data attacker-state where out-eq:
  out = Some (data, attacker-state)
  by (cases result) simp
have support-some:
  Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  using support by (simp only: out-eq)
have hit-some:
  staged-transcript-query-index-list-set-hit Q
    (Some (data, attacker-state))
  using hit by (simp only: out-eq)
have rel-hit:
  hash-relation-hit

```

```

    (checked-staged-query-index-list-relation A Q)
    adversary-initial-state attacker-state
  by (rule staged-transcript-query-index-list-set-hit-imp-relation-hit
      [OF support-some hit-some])
  show ?thesis
    unfolding out-eq hash-relation-hit-event-def
    using rel-hit by simp
qed

lemma staged-transcript-query-index-list-set-hit-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-transcript-program A)
      (staged-transcript-query-index-list-set-hit Q)
      adversary-initial-state ≤
      hash-relation-budget-value
        (query-index-row-list-relation-fiber-bound Q)
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget)
  proof -
    have event-mono:
      wp-event (checked-staged-transcript-program A)
        (staged-transcript-query-index-list-set-hit Q)
        adversary-initial-state ≤
      wp-event (checked-staged-transcript-program A)
        (hash-relation-hit-event
          (checked-staged-query-index-list-relation A Q)
          adversary-initial-state)
        adversary-initial-state
    proof (rule wp-event-mono-on-support)
      fix out
      assume support:
        out ∈
        set-dist
          (execute (checked-staged-transcript-program A)
            adversary-initial-state)
      and hit: staged-transcript-query-index-list-set-hit Q out
      show hash-relation-hit-event
        (checked-staged-query-index-list-relation A Q)
        adversary-initial-state out
      by (rule staged-transcript-query-index-list-set-hit-imp-relation-hit-event
          [OF support hit])
    qed
  qed

lemma relation-bound:
  wp-event (checked-staged-transcript-program A)
    (hash-relation-hit-event
      (checked-staged-query-index-list-relation A Q)
      adversary-initial-state)

```

```

    adversary-initial-state ≤
    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound Q)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  proof —
    have program:
      hash-relation-program
        (checked-staged-query-index-list-relation A Q)
        (query-index-raw-list-relation-fiber-bound Q)
        (staged-attacker-query-budget budgets + staged-challenge-query-budget)
        (checked-staged-transcript-program A)
    by (rule hash-relation-program-checked-staged-transcript-query-index-list
      [OF wf controlled])
    then show ?thesis
      unfolding hash-relation-program-def hash-relation-budget-def by blast
  qed
  show ?thesis
    by (rule order-trans[OF event-mono relation-bound])
  qed

lemma checked-staged-security-with-data-query-index-list-set-bound-from-transcript:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and transcript-bound:
    wp-event (checked-staged-transcript-program A)
      (staged-transcript-query-index-list-set-hit Q)
      adversary-initial-state ≤ C
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-query-index-list-set-hit Q)
      adversary-initial-state ≤ C
  unfolding checked-staged-security-experiment-with-data-state-def
proof (rule wp-event-bind-bound-by-head-event[OF transcript-bound])
  show staged-security-with-data-query-index-list-set-hit Q None ⇒
    staged-transcript-query-index-list-set-hit Q None
  unfolding staged-security-with-data-query-index-list-set-hit-def
    staged-transcript-query-index-list-set-hit-def
  by simp
next
  fix data attacker-state out
  assume builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and cont:
    out ∈
      set-dist

```

```

    (execute
      (get >>=
        (λs. put
          (verifier-state-from-adversary s
            (staged-proof-transcript data)) >>=
            (λ-.
              verify-monad >>=
                (λresult. return ((data, s), result))))))
      attacker-state)
  and hit: staged-security-with-data-query-index-list-set-hit Q out
from checked-staged-transcript-program-outcome-query-lookups
  [OF wf controlled builder]
obtain attacker-raw-idxs attacker-query-idxs where
  len-attacker-raw: length attacker-raw-idxs = rounds
and attacker-query-idxs-eq:
  attacker-query-idxs =
    map (λraw. index (to-nat raw)) attacker-raw-idxs
and lookup-attacker:
  ∀ j < rounds.
    fmllookup (HashMap attacker-state)
      (QueryIndexChallenge j
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) j)) =
      Some (attacker-raw-idxs ! j)
by blast
from hit obtain data' attacker-state' result final-state raw-idxs query-idxs where
  out-eq: out = Some (((data', attacker-state'), result), final-state))
and len-raw: length raw-idxs = rounds
and query-in: query-idxs ∈ Q
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and lookup-final:
  ∧ i. i < rounds ⇒
    fmllookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data')
          (staged-query-chunks data') i)) =
      Some (raw-idxs ! i)
unfolding staged-security-with-data-query-index-list-set-hit-def
by (auto split: option.splits prod.splits)
from cont[unfolded out-eq] obtain verifier-result where
  verifier:
    Some (verifier-result, final-state) ∈
      set-dist
      (execute verify-monad
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))

```

```

and data'-eq: data' = data
and attacker-state'-eq: attacker-state' = attacker-state
by (auto elim!: set-dist-bindE)
have ext:
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data) ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
have raw-eq:
  ∧i. i < rounds ⇒ attacker-raw-idxs ! i = raw-idxs ! i
proof -
  fix i
  assume i-bound: i < rounds
  have lookup-initial:
    fmlookup
      (HashMap
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
        Some (attacker-raw-idxs ! i)
  using lookup-attacker i-bound by simp
  have lookup-final-from-initial:
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
        Some (attacker-raw-idxs ! i)
  by (rule hash-extension-lookup[OF lookup-initial ext])
  show attacker-raw-idxs ! i = raw-idxs ! i
  using lookup-final-from-initial lookup-final[OF i-bound] data'-eq
  by simp
qed
have query-idxs-attacker:
  query-idxs = map (λraw. index (to-nat raw)) attacker-raw-idxs
proof (rule nth-equalityI)
  show length query-idxs =
    length (map (λraw. index (to-nat raw)) attacker-raw-idxs)
  using len-raw len-attacker-raw query-idxs-eq by simp
next
  fix i
  assume i-bound:
    i < length query-idxs
  then have i-rounds: i < rounds
  using len-raw query-idxs-eq by simp
  then show query-idxs ! i =
    map (λraw. index (to-nat raw)) attacker-raw-idxs ! i

```

```

    using query-idxs-eq raw-eq[OF i-rounds] len-row apply simp
    using len-attacker-row by auto
qed
show staged-transcript-query-index-list-set-hit Q
  (Some (data, attacker-state))
unfolding staged-transcript-query-index-list-set-hit-def
using len-attacker-row query-in query-idxs-attacker lookup-attacker
by auto
qed

lemma checked-staged-security-with-data-query-index-list-set-hit-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-query-index-list-set-hit Q)
    adversary-initial-state ≤
    hash-relation-budget-value
      (query-index-row-list-relation-fiber-bound Q)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  by (rule checked-staged-security-with-data-query-index-list-set-bound-from-transcript
    [OF wf controlled staged-transcript-query-index-list-set-hit-bound
      [OF wf controlled]])

lemma checked-staged-security-with-data-state-query-index-list-fresh-or-prequeried:
  assumes support:
    Some (((data, attacker-state), result), final-state) ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  and hit:
    staged-security-with-data-query-index-list-set-hit Q
    (Some (((data, attacker-state), result), final-state))
  shows
    staged-security-with-data-state-query-index-list-path-fresh Q
    (Some (((data, attacker-state), result), final-state)) ∨
    hash-relation-hit
      (checked-staged-query-index-list-relation A Q)
      adversary-initial-state attacker-state
proof —
  from checked-staged-security-experiment-with-data-state-outcomeE[OF support]
  obtain builder verifier where
    builder:
      Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and verifier:

```



```

    Some (result, final-state) ∈
      set-dist
        (execute verify-monad
          (verifier-state-from-adversary attacker-state
            (staged-proof-transcript data)))
  by blast
from hit obtain raw-idxs query-idxs where
  len-raw: length raw-idxs = rounds
  and query-in: query-idxs ∈ Q
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and lookup-final:
    ∧ i. i < rounds ⇒
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
  unfolding staged-security-with-data-query-index-list-set-hit-def
  by auto
let ?vstate =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
have raw-in: raw-idxs ∈ query-index-raw-list-preimage Q
  unfolding query-index-raw-list-preimage-def
  using len-raw query-in query-idxs-eq by blast
have ext: ?vstate ≤ final-state
  by (rule verify-monad-hash-extends[OF verifier])
show ?thesis
proof (cases
  ∀ i < rounds.
    fmllookup (HashMap ?vstate)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      None)
  case True
  then have
    staged-security-with-data-state-query-index-list-path-fresh Q
      (Some (((data, attacker-state), result), final-state))
  unfolding staged-security-with-data-state-query-index-list-path-fresh-def
  using len-raw query-in query-idxs-eq lookup-final
  by (auto intro: exI[of - raw-idxs] exI[of - query-idxs])
  then show ?thesis by simp
next
  case False
  then obtain i old where

```

```

i-bound:  $i < \text{rounds}$ 
and lookup-initial:
  fmlookup (HashMap ?vstate)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
      Some old
by (metis option.exhaust)
let ?key =
  QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data) (staged-query-chunks data) i)
have lookup-final-old:
  fmlookup (HashMap final-state) ?key = Some old
by (rule hash-extension-lookup[OF lookup-initial ext])
have old-eq:  $\text{old} = \text{raw-idxs} ! i$ 
  using lookup-final-old lookup-final[OF i-bound] by simp
have lookup-attacker:
  fmlookup (HashMap attacker-state) ?key = Some (raw-idxs ! i)
using lookup-initial old-eq
unfolding verifier-state-from-adversary-def by simp
have relation:
  checked-staged-query-index-list-relation A Q ?key ( $\text{raw-idxs} ! i$ )
unfolding checked-staged-query-index-list-relation-def
by (intro exI[of - data] exI[of - attacker-state]
  exI[of - raw-idxs] exI[of - i] conjI)
  (use builder raw-in i-bound in simp-all)
have initial-none:
  fmlookup (HashMap adversary-initial-state) ?key = None
by (simp add: adversary-initial-state-def)
have
  hash-relation-hit
    (checked-staged-query-index-list-relation A Q)
    adversary-initial-state attacker-state
unfolding hash-relation-hit-def
by (intro exI[of - ?key] exI[of - raw-idxs ! i] conjI)
  (use initial-none lookup-attacker relation in simp-all)
then show ?thesis by simp
qed
qed

lemma checked-staged-security-with-data-state-query-index-list-prequery-hit-bound:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-query-index-list-prequery-hit A Q)
    adversary-initial-state  $\leq$ 

```

```

    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound Q)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  unfolding checked-staged-security-experiment-with-data-state-def
proof (rule wp-event-bind-bound-by-head-event)
  show wp-event (checked-staged-transcript-program A)
    (hash-relation-hit-event
      (checked-staged-query-index-list-relation A Q)
      adversary-initial-state)
  adversary-initial-state
  ≤ hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound Q)
    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
proof -
  have program:
    hash-relation-program
      (checked-staged-query-index-list-relation A Q)
      (query-index-raw-list-relation-fiber-bound Q)
      (staged-attacker-query-budget budgets + staged-challenge-query-budget)
      (checked-staged-transcript-program A)
  by (rule hash-relation-program-checked-staged-transcript-query-index-list
    [OF wf controlled])
  then show ?thesis
    unfolding hash-relation-program-def hash-relation-budget-def by blast
qed
next
  show
    staged-security-with-data-state-query-index-list-prequery-hit A Q None
    ⇒
    hash-relation-hit-event
      (checked-staged-query-index-list-relation A Q)
      adversary-initial-state None
  unfolding staged-security-with-data-state-query-index-list-prequery-hit-def
    hash-relation-hit-event-def
  by simp
next
  fix data attacker-state out
  assume cont:
    out ∈
      set-dist
        (execute
          (get ≥=
            (λs. put
              (verifier-state-from-adversary s
                (staged-proof-transcript data)) ≥=
              (λ-. verify-monad ≥=
                (λresult. return ((data, s), result))))))
          attacker-state)

```

```

assume prequery:
  staged-security-with-data-state-query-index-list-prequery-hit A Q out
show
  hash-relation-hit-event
    (checked-staged-query-index-list-relation A Q)
    adversary-initial-state (Some (data, attacker-state))
proof (cases out)
  case None
  then show ?thesis
    using prequery
    unfolding staged-security-with-data-state-query-index-list-prequery-hit-def
    by simp
next
  case (Some packed)
  from cont Some have cont-some:
    Some packed  $\in$ 
      set-dist
      (execute
        (get  $\gg$ 
          ( $\lambda s.$  put
            (verifier-state-from-adversary s
              (staged-proof-transcript data))  $\gg$ 
            ( $\lambda -. \text{verify-monad } \gg$ 
              ( $\lambda \text{result. return } ((\text{data}, s), \text{result})$ ))))
          attacker-state)
        by simp
      )
  obtain packed-value packed-state where packed-pair:
    packed = (packed-value, packed-state)
    by (cases packed)
  from cont-some[unfolded packed-pair] obtain captured get-state where get-out:
    Some (captured, get-state)  $\in$  set-dist (execute get attacker-state)
  and after-get0:
    Some (packed-value, packed-state)  $\in$ 
      set-dist
      (execute
        (put
          (verifier-state-from-adversary captured
            (staged-proof-transcript data))  $\gg$ 
          ( $\lambda -. \text{verify-monad } \gg$ 
            ( $\lambda \text{result. return } ((\text{data}, \text{captured}), \text{result})$ ))))
          get-state)
        by (rule set-dist-bindE) (rule that)
      )
  have captured-eq: captured = attacker-state
  and get-state-eq: get-state = attacker-state
  using get-out
  unfolding set-dist-def get.rep-eq dist-get-def
  by (simp-all add: dist-delta-dist delta-map-def)
  have after-get:
    Some packed  $\in$ 

```

```

    set-dist
    (execute
      (put
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))  $\gg=$ 
        ( $\lambda$ -. verify-monad  $\gg=$ 
          ( $\lambda$ result. return ((data, attacker-state), result))))
      attacker-state)
  using after-get0 captured-eq get-state-eq packed-pair by simp
from after-get[unfolded packed-pair] obtain unit put-state where put-out:
  Some (unit, put-state)  $\in$ 
  set-dist
  (execute
    (put
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
      attacker-state)
  and after-put0:
    Some (packed-value, packed-state)  $\in$ 
    set-dist
    (execute
      (verify-monad  $\gg=$ 
        ( $\lambda$ result. return ((data, attacker-state), result)))
      put-state)
  by (rule set-dist-bindE) (rule that)
have put-state-eq:
  put-state =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
using put-out
unfolding set-dist-def put.rep-eq dist-put-def
by (simp add: dist-delta-dist delta-map-def)
have after-put:
  Some packed  $\in$ 
  set-dist
  (execute
    (verify-monad  $\gg=$ 
      ( $\lambda$ result. return ((data, attacker-state), result)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  using after-put0 put-state-eq packed-pair by simp
from after-put[unfolded packed-pair] obtain result final-state where verify-out:
  Some (result, final-state)  $\in$ 
  set-dist
  (execute verify-monad
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  and ret-out:
    Some (packed-value, packed-state)  $\in$ 

```

```

      set-dist (execute (return ((data, attacker-state), result)) final-state)
    by (rule set-dist-bindE) (rule that)
  have packed-eq: packed = (((data, attacker-state), result), final-state)
  proof -
    have value-eq: packed-value = ((data, attacker-state), result)
      by (rule set-dist-return-valueD[OF ret-out])
    have state-eq: packed-state = final-state
      by (rule set-dist-return-stateD[OF ret-out])
    show ?thesis
      using packed-pair value-eq state-eq by simp
  qed
  have out-eq:
    out = Some (((data, attacker-state), result), final-state)
    using Some packed-eq by simp
  from prequery have hit:
    hash-relation-hit
      (checked-staged-query-index-list-relation A Q)
      adversary-initial-state attacker-state
  unfolding out-eq
    staged-security-with-data-state-query-index-list-prequery-hit-def
  by simp
  then show ?thesis
    using hit unfolding hash-relation-hit-event-def by simp
  qed
  qed

lemma checked-staged-security-with-data-state-query-index-list-hit-bound-from-path-fresh:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and path-fresh-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-query-index-list-path-fresh Q)
      adversary-initial-state  $\leq$  C
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-query-index-list-set-hit Q)
      adversary-initial-state  $\leq$ 
      C +
      hash-relation-budget-value
        (query-index-raw-list-relation-fiber-bound Q)
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget)
  proof -
    have event-mono:
      wp-event (checked-staged-security-experiment-with-data-state A)
        (staged-security-with-data-state-query-index-list-set-hit Q)
        adversary-initial-state  $\leq$ 
      wp-event (checked-staged-security-experiment-with-data-state A)
        ( $\lambda$ out.

```

```

    staged-security-with-data-state-query-index-list-path-fresh Q out ∨
    staged-security-with-data-state-query-index-list-prequery-hit A Q out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
  and hit: staged-security-with-data-query-index-list-set-hit Q out
show
  staged-security-with-data-state-query-index-list-path-fresh Q out ∨
  staged-security-with-data-state-query-index-list-prequery-hit A Q out
proof (cases out)
  case None
  then show ?thesis
    using hit unfolding staged-security-with-data-query-index-list-set-hit-def
    by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
  by (cases packed) (auto split: prod.splits)
  have split:
    staged-security-with-data-state-query-index-list-path-fresh Q
    (Some (((data, attacker-state), result), final-state)) ∨
    hash-relation-hit
    (checked-staged-query-index-list-relation A Q)
    adversary-initial-state attacker-state
  apply (rule
    checked-staged-security-with-data-state-query-index-list-fresh-or-prequeried)
  using support hit unfolding out-eq by simp-all
  then show ?thesis
    unfolding out-eq
    staged-security-with-data-state-query-index-list-prequery-hit-def
    by simp
  qed
qed
have union-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (λout.
    staged-security-with-data-state-query-index-list-path-fresh Q out ∨
    staged-security-with-data-state-query-index-list-prequery-hit A Q out)
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-query-index-list-path-fresh Q)
  adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)

```

```

      (staged-security-with-data-state-query-index-list-prequery-hit A Q)
      adversary-initial-state
    by (rule wp-event-union-bound)
  have prequery-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-query-index-list-prequery-hit A Q)
      adversary-initial-state ≤
      hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound Q)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  by (rule
    checked-staged-security-with-data-state-query-index-list-prequery-hit-bound
    [OF wf controlled])
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-query-index-list-set-hit Q)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-query-index-list-path-fresh Q)
      adversary-initial-state +
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-query-index-list-prequery-hit A Q)
      adversary-initial-state
  by (rule order-trans[OF event-mono union-bound])
  also have ... ≤
    C +
    hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound Q)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  by (rule add-mono[OF path-fresh-bound prequery-bound])
  finally show ?thesis .
qed

```

lemma *accepted-fri-opening-transcript-imp-accepted-transcript-shape:*
assumes *fri:*
 accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
 dg composition-roots composition-bs composition-final query-idxs
 trace-round-layers composition-round-layers
shows $\exists$ *as*. *accepted-transcript-shape s out as query-idxs*
using *fri*
unfolding *accepted-fri-opening-transcript-def*
 accepted-transcript-shape-def
by *blast*

lemma *checked-staged-security-trace-fri-query-index-list-set-hit-bound-list:*
assumes *wf: staged-budget-wellformed budgets*
 and *controlled: staged-adversary-controlled budgets A*
shows


```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. trace-fri-query-index-list-set-hit s Q))
adversary-initial-state ≤
hash-relation-budget-value
  (query-index-raw-list-relation-fiber-bound Q)
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)
proof (rule order-trans)
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-index-list-set-hit s Q))
    adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-query-index-list-set-hit Q)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
      set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  and hit:
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-index-list-set-hit s Q) out
  show staged-security-with-data-query-index-list-set-hit Q out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding staged-security-with-data-state-verifier-event-def
      staged-security-with-data-query-index-list-set-hit-def
    by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed) (auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from hit out-eq have trace-hit:
    trace-fri-query-index-list-set-hit ?s Q
    (Some (result, final-state))
    unfolding staged-security-with-data-state-verifier-event-def by simp
  from trace-hit obtain trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers where

```

```

fri:
  accepted-fri-opening-transcript ?s (Some (result, final-state))
  trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final query-idxs trace-round-layers
  composition-round-layers
  and query-in: query-idxs ∈ Q
  unfolding trace-fri-query-index-list-set-hit-def by blast
from accepted-fri-opening-transcript-imp-accepted-transcript-shape[OF fri]
obtain as where shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
  query-idxs
  by blast
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support[unfolded out-eq] shape]
obtain raw-idxs where
  len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and lookup:
    ∧ i. i < rounds ⇒
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
    by blast
show ?thesis
  unfolding out-eq staged-security-with-data-query-index-list-set-hit-def
  using len-raw query-in query-idxs-eq lookup by auto
qed
qed
next
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-query-index-list-set-hit Q)
  adversary-initial-state
  ≤ hash-relation-budget-value
  (query-index-raw-list-relation-fiber-bound Q)
  (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule checked-staged-security-with-data-query-index-list-set-hit-bound
    [OF wf controlled])
qed

lemma checked-staged-security-composition-fri-query-index-list-set-hit-bound-list:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event

```

```

    (λs. composition-fri-query-index-list-set-hit s Q))
  adversary-initial-state ≤
  hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound Q)
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget)
proof (rule order-trans)
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     (λs. composition-fri-query-index-list-set-hit s Q))
    adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-query-index-list-set-hit Q)
    adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
               adversary-initial-state)
  and hit:
    staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-index-list-set-hit s Q) out
  show staged-security-with-data-query-index-list-set-hit Q out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding staged-security-with-data-state-verifier-event-def
      staged-security-with-data-query-index-list-set-hit-def
    by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
  by (cases packed) (auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from hit out-eq have composition-hit:
    composition-fri-query-index-list-set-hit ?s Q
    (Some (result, final-state))
  unfolding staged-security-with-data-state-verifier-event-def by simp
  from composition-hit obtain trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers where
  fri:
    accepted-fri-opening-transcript ?s (Some (result, final-state))

```

```

    trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final query-idxs trace-round-layers
    composition-round-layers
  and query-in: query-idxs ∈ Q
  unfolding composition-fri-query-index-list-set-hit-def by blast
from accepted-fri-opening-transcript-imp-accepted-transcript-shape[OF fri]
obtain as where shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as
  query-idxs
  by blast
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support[unfolded out-eq] shape]
obtain raw-idxs where
  len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  and lookup:
    ∧i. i < rounds ⇒
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
    by blast
show ?thesis
  unfolding out-eq staged-security-with-data-query-index-list-set-hit-def
  using len-raw query-in query-idxs-eq lookup by auto
qed
qed
next
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-query-index-list-set-hit Q)
  adversary-initial-state
  ≤ hash-relation-budget-value
  (query-index-raw-list-relation-fiber-bound Q)
  (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule checked-staged-security-with-data-query-index-list-set-hit-bound
    [OF wf controlled])
qed

lemma checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-list:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and projection: fst ‘ P ⊆ Q
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. trace-fri-query-challenge-pair-set-hit s P))

```

$adversary-initial-state \leq$
 $hash-relation-budget-value$
 $(query-index-raw-list-relation-fiber-bound \ Q)$
 $(staged-attacker-query-budget \ budgets +$
 $staged-challenge-query-budget)$
proof (rule order-trans)
show $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $(\lambda s. trace-fri-query-challenge-pair-set-hit \ s \ P))$
 $adversary-initial-state \leq$
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $(\lambda s. trace-fri-query-index-list-set-hit \ s \ Q))$
 $adversary-initial-state$
by (rule wp-event-mono-on-support)
 $(auto \ simp: staged-security-with-data-state-verifier-event-def$
 $intro:$
 $trace-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit$
 $[OF \ - \ projection]$
 $split: option.splits \ prod.splits)$
next
show $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $(\lambda s. trace-fri-query-index-list-set-hit \ s \ Q))$
 $adversary-initial-state$
 $\leq hash-relation-budget-value$
 $(query-index-raw-list-relation-fiber-bound \ Q)$
 $(staged-attacker-query-budget \ budgets + staged-challenge-query-budget)$
by (rule checked-staged-security-trace-fri-query-index-list-set-hit-bound-list
 $[OF \ wf \ controlled])$
qed

lemma *checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-list:*
assumes $wf: staged-budget-wellformed \ budgets$
and $controlled: staged-adversary-controlled \ budgets \ A$
and $projection: \bigwedge dg. fst \ 'P \ dg \subseteq Q$
shows
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $(\lambda s. composition-fri-query-challenge-pair-set-hit \ s \ P))$
 $adversary-initial-state \leq$
 $hash-relation-budget-value$
 $(query-index-raw-list-relation-fiber-bound \ Q)$
 $(staged-attacker-query-budget \ budgets +$
 $staged-challenge-query-budget)$
proof (rule order-trans)
show $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $(\lambda s. composition-fri-query-challenge-pair-set-hit \ s \ P))$

$\text{adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-query-index-list-set-hit } s \ Q))$
 $\text{adversary-initial-state}$
by (rule $\text{wp-event-mono-on-support}$)
 $(\text{auto simp: staged-security-with-data-state-verifier-event-def}$
 intro:
 $\text{composition-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit}$
 $[OF - \text{projection}]$
 $\text{split: option.splits prod.splits})$

next

show $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{composition-fri-query-index-list-set-hit } s \ Q))$
 $\text{adversary-initial-state}$
 $\leq \text{hash-relation-budget-value}$
 $(\text{query-index-raw-list-relation-fiber-bound } Q)$
 $(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$
by (rule
 $\text{checked-staged-security-composition-fri-query-index-list-set-hit-bound-list}$
 $[OF \text{ wf controlled}])$

qed

lemma $\text{checked-staged-security-trace-fri-query-index-list-set-hit-bound-from-path-fresh:}$

assumes $\text{wf: staged-budget-wellformed budgets}$
and $\text{controlled: staged-adversary-controlled budgets } A$
and path-fresh-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-query-index-list-path-fresh } Q)$
 $\text{adversary-initial-state} \leq C$

shows

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-query-index-list-set-hit } s \ Q))$
 $\text{adversary-initial-state} \leq$
 $C +$
 $\text{hash-relation-budget-value}$
 $(\text{query-index-raw-list-relation-fiber-bound } Q)$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget})$

proof (rule order-trans)

show $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-query-index-list-set-hit } s \ Q))$
 $\text{adversary-initial-state} \leq$
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-query-index-list-set-hit } Q)$
 $\text{adversary-initial-state}$

```

proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  and hit:
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-index-list-set-hit s Q) out
  show staged-security-with-data-query-index-list-set-hit Q out
  proof (cases out)
    case None
    then show ?thesis
      using hit
      unfolding staged-security-with-data-state-verifier-event-def
        staged-security-with-data-query-index-list-set-hit-def
      by simp
  next
    case (Some packed)
    then obtain data attacker-state result final-state where out-eq:
      out = Some (((data, attacker-state), result), final-state)
      by (cases packed) (auto split: prod.splits)
    let ?s =
      verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)
    from hit out-eq have trace-hit:
      trace-fri-query-index-list-set-hit ?s Q
      (Some (result, final-state))
    unfolding staged-security-with-data-state-verifier-event-def by simp
    from trace-hit obtain trace-roots trace-bs trace-final dg
      composition-roots composition-bs composition-final query-idxs
      trace-round-layers composition-round-layers where
        fri:
          accepted-fri-opening-transcript ?s (Some (result, final-state))
          trace-roots trace-bs trace-final dg composition-roots
          composition-bs composition-final query-idxs trace-round-layers
          composition-round-layers
    and query-in: query-idxs ∈ Q
    unfolding trace-fri-query-index-list-set-hit-def by blast
    from accepted-fri-opening-transcript-imp-accepted-transcript-shape[OF fri]
    obtain as where shape:
      accepted-transcript-shape ?s (Some (result, final-state)) as
        query-idxs
    by blast
    from checked-staged-security-with-data-state-accepted-shape-query-keys
      [OF wf controlled support[unfolded out-eq] shape]
    obtain raw-idxs where
      len-raw: length raw-idxs = rounds

```

```

and query-idxs-eq:
  query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
and lookup:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
  by blast
show ?thesis
  unfolding out-eq staged-security-with-data-query-index-list-set-hit-def
  using len-raw query-in query-idxs-eq lookup by auto
qed
qed
next
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-query-index-list-set-hit Q)
  adversary-initial-state
   $\leq C +$ 
    hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound Q)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  by (rule
    checked-staged-security-with-data-state-query-index-list-hit-bound-from-path-fresh
    [OF wf controlled path-fresh-bound])
qed

lemma checked-staged-security-composition-fri-query-index-list-set-hit-bound-from-path-fresh:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and path-fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-query-index-list-path-fresh Q)
  adversary-initial-state  $\leq C$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s$ . composition-fri-query-index-list-set-hit s Q))
  adversary-initial-state  $\leq$ 
    C +
    hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound Q)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
proof (rule order-trans)
show wp-event (checked-staged-security-experiment-with-data-state A)

```



```

    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-index-list-set-hit s Q))
    adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-query-index-list-set-hit Q)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈
    set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
and hit:
  staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-index-list-set-hit s Q) out
show staged-security-with-data-query-index-list-set-hit Q out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding staged-security-with-data-state-verifier-event-def
    staged-security-with-data-query-index-list-set-hit-def
  by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed) (auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from hit out-eq have composition-hit:
  composition-fri-query-index-list-set-hit ?s Q
  (Some (result, final-state))
  unfolding staged-security-with-data-state-verifier-event-def by simp
from composition-hit obtain trace-roots trace-bs trace-final dg
  composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers where
fri:
  accepted-fri-opening-transcript ?s (Some (result, final-state))
  trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final query-idxs trace-round-layers
  composition-round-layers
and query-in: query-idxs ∈ Q
  unfolding composition-fri-query-index-list-set-hit-def by blast
from accepted-fri-opening-transcript-imp-accepted-transcript-shape[OF fri]
obtain as where shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as

```

```

    query-idxs
  by blast
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support[unfolded out-eq] shape]
obtain raw-idxs where
  len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
  and lookup:
     $\bigwedge i. i < rounds \implies$ 
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
  by blast
show ?thesis
  unfolding out-eq staged-security-with-data-query-index-list-set-hit-def
  using len-raw query-in query-idxs-eq lookup by auto
qed
qed
next
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-query-index-list-set-hit Q)
  adversary-initial-state
   $\leq C +$ 
  hash-relation-budget-value
  (query-index-raw-list-relation-fiber-bound Q)
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)
  by (rule
    checked-staged-security-with-data-state-query-index-list-hit-bound-from-path-fresh
    [OF wf controlled path-fresh-bound])
qed

lemma checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-path-fresh:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and projection: fst '  $P \subseteq Q$ 
and path-fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-query-index-list-path-fresh Q)
    adversary-initial-state  $\leq C$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ P$ ))
    adversary-initial-state  $\leq$ 

```

```

    C +
    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound Q)
      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget)
  proof (rule order-trans)
    show wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       (λs. trace-fri-query-challenge-pair-set-hit s P))
      adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       (λs. trace-fri-query-index-list-set-hit s Q))
      adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
     intro:
       trace-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit
       [OF - projection]
     split: option.splits prod.splits)
  next
    show wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       (λs. trace-fri-query-index-list-set-hit s Q))
      adversary-initial-state
      ≤ C +
      hash-relation-budget-value
        (query-index-raw-list-relation-fiber-bound Q)
        (staged-attacker-query-budget budgets +
         staged-challenge-query-budget)
    by (rule
        checked-staged-security-trace-fri-query-index-list-set-hit-bound-from-path-fresh
        [OF wf controlled path-fresh-bound])
  qed

lemma checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-path-fresh:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and projection:  $\bigwedge dg. \text{fst } dg \subseteq P \implies dg \subseteq Q$ 
  and path-fresh-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-query-index-list-path-fresh Q)
      adversary-initial-state ≤ C
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       (λs. composition-fri-query-challenge-pair-set-hit s P))
      adversary-initial-state ≤
    C +

```

```

    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound Q)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
proof (rule order-trans)
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-challenge-pair-set-hit s P))
    adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-index-list-set-hit s Q))
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro:
        composition-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit
        [OF - projection]
        split: option.splits prod.splits)
next
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-index-list-set-hit s Q))
    adversary-initial-state
    ≤ C +
      hash-relation-budget-value
        (query-index-raw-list-relation-fiber-bound Q)
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget)
  by (rule
    checked-staged-security-composition-fri-query-index-list-set-hit-bound-from-path-fresh
    [OF wf controlled path-fresh-bound])
qed

end

end

```

```

theory Soundness-FRI-Query-List-Exact-Product
  imports Soundness-FRI-Query-List-Prequery
begin

```

Exact product accounting for sampled FRI query-index lists.

The preceding layer proves the product theorem in raw field space. This layer converts raw-list preimages to query-index-list targets using the current exact-uniformity assumptions for the modulo sampler.

```

context soundness
begin

```

definition *verifier-after-composition-fri-raw-list-path-hit*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow \text{nat list set} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
verifier-after-composition-fri-raw-list-path-hit $s \ Q \ \text{out} \longleftrightarrow$
 $(\exists \text{ final query-state rows.}$
 $\text{Some (final, query-state)} \in \text{set-dist (execute read s)} \wedge$
 $\text{rows} \in \text{query-index-raw-list-preimage } Q \wedge$
 $\text{query-rounds-raw-list-path-hit query-state rows rounds out})$

lemma *verifier-after-composition-fri-raw-list-path-hitI*:
assumes *read-final*:
 $\text{Some (final, query-state)} \in \text{set-dist (execute read s)}$
and *raw-in*: $\text{rows} \in \text{query-index-raw-list-preimage } Q$
and *len-rows*: $\text{length rows} = \text{rounds}$
and *len-chunks*: $\text{length chunks} = \text{rounds}$
and *state*:
 $P\text{State } t = \text{state-after-query-chunks (PState query-state) chunks rounds}$
and *fresh*:
 $\text{query-index-list-path-fresh query-state rounds chunks rows}$
and *lookup*:
 $\bigwedge i. i < \text{rounds} \implies$
 $\text{fmlookup (HashMap } t)$
 $(\text{QueryIndexChallenge (PQueryCounter query-state} + i)$
 $(\text{state-after-query-chunks (PState query-state) chunks } i)) =$
 $\text{Some (rows ! } i)$
and *out-eq*: $\text{out} = \text{Some (results, } t)$
shows *verifier-after-composition-fri-raw-list-path-hit* $s \ Q \ \text{out}$
proof –
have *path*: $\text{query-rounds-raw-list-path-hit query-state rows rounds out}$
unfolding *out-eq*
by (*rule query-rounds-raw-list-path-hitI*
 $[\text{OF len-rows len-chunks state fresh lookup}]$)
show *?thesis*
unfolding *verifier-after-composition-fri-raw-list-path-hit-def*
by (*intro* $\text{exI[of - final] exI[of - query-state] exI[of - rows]}$
 $\text{conjI read-final raw-in path})$
qed

lemma *staged-security-with-data-state-query-index-list-path-fresh-SomeE*:
assumes *hit*:
 $\text{staged-security-with-data-state-query-index-list-path-fresh } Q$
 $(\text{Some } (((\text{data, attacker-state}), \text{result}), \text{final-state}))$
obtains *raw-idxs query-idxs* **where**
 $\text{length raw-idxs} = \text{rounds}$
 $\text{query-idxs} \in Q$
 $\text{query-idxs} = \text{map } (\lambda \text{ raw. index (to-nat raw)}) \ \text{raw-idxs}$
 $\bigwedge i. i < \text{rounds} \implies$

```

fmlookup (HashMap final-state)
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
Some (raw-idxs ! i)
 $\wedge i. i < \text{rounds} \implies$ 
fmlookup
  (HashMap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
None
proof -
have hit-ex:
 $\exists \text{raw-idxs query-idxs.}$ 
length raw-idxs = rounds  $\wedge$ 
query-idxs  $\in Q$   $\wedge$ 
query-idxs = map ( $\lambda \text{raw. index (to-nat raw)}$ ) raw-idxs  $\wedge$ 
( $\forall i < \text{rounds.}$ 
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
Some (raw-idxs ! i)  $\wedge$ 
  fmlookup
    (HashMap
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
None)
using hit
unfolding staged-security-with-data-state-query-index-list-path-fresh-def
by simp
from hit-ex obtain raw-idxs query-idxs where len:
length raw-idxs = rounds
and in-Q: query-idxs  $\in Q$ 
and map-eq:
query-idxs = map ( $\lambda \text{raw. index (to-nat raw)}$ ) raw-idxs
and lookups:
 $\forall i < \text{rounds.}$ 
  fmlookup (HashMap final-state)

```

```

      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
Some (raw-idxs ! i) ∧
fmlookup
  (HashMap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))))
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
None
  by blast
show ?thesis
proof (rule that[OF len in-Q map-eq])
  fix i
  assume i < rounds
  then show
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
Some (raw-idxs ! i)
    using lookups by blast
next
  fix i
  assume i < rounds
  then show
    fmlookup
      (HashMap
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))))
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
None
    using lookups by blast
qed
qed

lemma staged-security-with-data-state-query-index-list-path-fresh-None[simp]:
  ¬ staged-security-with-data-state-query-index-list-path-fresh Q None
  unfolding staged-security-with-data-state-query-index-list-path-fresh-def
  by simp

```

lemma *staged-security-with-data-state-query-index-list-path-fresh-caseD*:

assumes *hit*:

staged-security-with-data-state-query-index-list-path-fresh *Q*
 (case out of
 None $\Rightarrow$ *None*
 | *Some* (*result*, *final-state*) $\Rightarrow$
 Some (((*data*, *attacker-state*), *result*), *final-state*))

shows

$\exists$ *result final-state raw-idxs query-idxs*.
out = *Some* (*result*, *final-state*) $\wedge$
length *raw-idxs* = *rounds* $\wedge$
query-idxs $\in$ *Q* $\wedge$
query-idxs = *map* (λ *raw*. *index* (*to-nat* *raw*)) *raw-idxs* $\wedge$
 $(\forall i < \text{rounds}.$
 fmlookup (*HashMap* *final-state*)
 (*QueryIndexChallenge* *i*
 (*state-after-query-chunks*
 (*staged-query-start-hash* *data*)
 (*staged-query-chunks* *data*) *i*)) =
 Some (*raw-idxs* ! *i*) $\wedge$
 $(\forall i < \text{rounds}.$
 fmlookup
 (*HashMap*
 (*verifier-state-from-adversary* *attacker-state*
 (*staged-proof-transcript* *data*))))
 (*QueryIndexChallenge* *i*
 (*state-after-query-chunks*
 (*staged-query-start-hash* *data*)
 (*staged-query-chunks* *data*) *i*)) =
 None)

proof (*cases out*)

case *None*

then show *?thesis*

using *hit* **by** *simp*

next

case (*Some pair*)

then obtain *result final-state* **where** *out-eq*:

out = *Some* (*result*, *final-state*)

by (*cases pair*) *simp*

have *hit-ex*:

$\exists$ *raw-idxs query-idxs*.
length *raw-idxs* = *rounds* $\wedge$
query-idxs $\in$ *Q* $\wedge$
query-idxs = *map* (λ *raw*. *index* (*to-nat* *raw*)) *raw-idxs* $\wedge$
 $(\forall i < \text{rounds}.$
 fmlookup (*HashMap* *final-state*)
 (*QueryIndexChallenge* *i*
 (*state-after-query-chunks*
 (*staged-query-start-hash* *data*)


```

      (staged-query-chunks data) i)) =
Some (raw-idxs ! i) ∧
fmlookup
  (HashMap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (QueryIndexChallenge i
    (state-after-query-chunks
      (staged-query-start-hash data)
      (staged-query-chunks data) i)) =
None)
using hit out-eq
unfolding staged-security-with-data-state-query-index-list-path-fresh-def
by simp
from hit-ex obtain raw-idxs query-idxs where
  length raw-idxs = rounds
  query-idxs ∈ Q
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
  ∀ i < rounds.
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i) ∧
    fmlookup
      (HashMap
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      None
by blast
then show ?thesis
using out-eq by blast
qed

lemma staged-security-with-data-state-query-index-list-path-fresh-caseE:
assumes hit:
  staged-security-with-data-state-query-index-list-path-fresh Q
  (case out of
    None ⇒ None
  | Some (result, final-state) ⇒
    Some (((data, attacker-state), result), final-state))
obtains result final-state raw-idxs query-idxs where
  out = Some (result, final-state)
  length raw-idxs = rounds

```

```

query-idxs ∈ Q
query-idxs = map (λraw. index (to-nat raw)) raw-idxs
∧ i. i < rounds ⇒
  fmlookup (HashMap final-state)
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
∧ i. i < rounds ⇒
  fmlookup
    (HashMap
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (QueryIndexChallenge i
      (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    None
proof (cases out)
  case None
  then have False
    using hit by simp
  then show ?thesis by simp
next
  case (Some pair)
  then obtain result final-state where out-eq:
    out = Some (result, final-state)
  by (cases pair) simp
  have hit-some:
    staged-security-with-data-state-query-index-list-path-fresh Q
      (Some (((data, attacker-state), result), final-state))
  using hit out-eq by simp
  from staged-security-with-data-state-query-index-list-path-fresh-SomeE
    [OF hit-some]
  obtain raw-idxs query-idxs where
    len: length raw-idxs = rounds
    and in-Q: query-idxs ∈ Q
    and map-eq: query-idxs = map (λraw. index (to-nat raw)) raw-idxs
    and lookup:
      ∧ i. i < rounds ⇒
        fmlookup (HashMap final-state)
          (QueryIndexChallenge i
            (state-after-query-chunks
              (staged-query-start-hash data)
              (staged-query-chunks data) i)) =
          Some (raw-idxs ! i)
    and fresh:
      ∧ i. i < rounds ⇒

```

```

    fmllookup
      (HashMap
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data)))
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
    None
  by blast
show ?thesis
  by (rule that[OF out-eq len in-Q map-eq lookup fresh])
qed

lemma verifier-header-transcript-from-prefix-read:
  assumes len-trace: length f-roots = ceil-log clength
  and len-alpha: length as = length spec
  and len-comp: length comp-roots = ceil-log (to-nat dg + 1)
  and transcript-prefix:
    PTranscript s =
      [fr] @ f-roots @ [f-final] @ as @ [dg] @
      comp-roots @ PTranscript prefix-state
  and read-prefix: PTranscript prefix-state = final # rest
  and query-rest: PTranscript query-state = rest
  shows
    verifier-header-transcript s fr f-roots f-final as dg
    comp-roots final (PTranscript query-state)
proof -
  have transcript:
    PTranscript s =
      verifier-header-messages fr f-roots f-final as dg comp-roots final @
      PTranscript query-state
  using transcript-prefix read-prefix query-rest
  unfolding verifier-header-messages-def
  by simp
show ?thesis
  unfolding verifier-header-transcript-def
  using transcript len-trace len-alpha len-comp by simp
qed

```

```

lemma verifier-composition-fri-prefix-outcome-basicD:
  assumes outcome:
    Some ((fr, f-fl, f-final, as, dg, fl), t) ∈
      set-dist (execute verifier-composition-fri-prefix s)
  shows
    length f-fl = ceil-log clength ∧
    length as = length spec ∧
    length fl = ceil-log (to-nat dg + 1) ∧
    PTranscript s =

```



```

      (take (Suc i) (map snd fl)))) =
    Some (fst (fl ! i)))
  by (rule verifier-composition-fri-prefix-outcome[OF outcome])
  then show ?thesis by simp
qed

lemma alpha-round-preserves-query-lookup:
  assumes outcome: Some (a, t) ∈ set-dist (execute alpha-round s)
  shows
    fmlookup (HashMap t) (QueryIndexChallenge i x) =
      fmlookup (HashMap s) (QueryIndexChallenge i x)
  proof -
    from outcome obtain a0 s0 a1 s1 s2 where
      rand: Some (a0, s0) ∈ set-dist (execute receive-alpha-challenge s)
      and read-a: Some (a1, s1) ∈ set-dist (execute read s0)
      and assert-a: Some ((), s2) ∈ set-dist (execute (assert (a0 = a1)) s1)
      and ret-a: Some (a, t) ∈ set-dist (execute (return a1) s2)
      unfolding alpha-round-def
      by (auto simp: Let-def elim!: set-dist-bindE)
    have neg:
      QueryIndexChallenge i x ≠
        AlphaChallenge (PAlphaCounter s) (PState s)
      by simp
    have rand-lookup:
      fmlookup (HashMap s0) (QueryIndexChallenge i x) =
        fmlookup (HashMap s) (QueryIndexChallenge i x)
      by (rule receive-alpha-challenge-preserves-other-lookup[OF rand neg])
    have read-lookup:
      HashMap s1 = HashMap s0
      by (rule read-preserves-hash-map[OF read-a])
    have s2-eq: s2 = s1
      using assert-a unfolding assert-def
      by (cases a0 = a1) (auto simp: throw-no-outcome)
    have t-eq: t = s2
      using ret-a by simp
    show ?thesis
      using rand-lookup read-lookup unfolding t-eq s2-eq by simp
  qed

qed

lemma receive-trace-fri-commits-preserves-query-lookup:
  assumes outcome:
    Some ((b, r), t) ∈ set-dist (execute receive-trace-fri-commits s)
  shows
    fmlookup (HashMap t) (QueryIndexChallenge i x) =
      fmlookup (HashMap s) (QueryIndexChallenge i x)
  proof -
    from outcome obtain s-read where
      read-r: Some (r, s-read) ∈ set-dist (execute read s)
      and rand-b:

```

Some $(b, t) \in \text{set-dist (execute receive-trace-fri-challenge s-read)}$
unfolding receive-trace-fri-commits-def receive-fri-commits-with-def
 by (auto elim!: set-dist-bindE)
have read-lookup:
 HashMap s-read = HashMap s
 by (rule read-preserves-hash-map[OF read-r])
have neq:
 QueryIndexChallenge i x $\neq$
 TraceFriChallenge (PTraceFriCounter s-read) (PState s-read)
 by simp
have rand-lookup:
 fmlookup (HashMap t) (QueryIndexChallenge i x) =
 fmlookup (HashMap s-read) (QueryIndexChallenge i x)
 by (rule receive-trace-fri-challenge-preserves-other-lookup
 [OF rand-b neq])
show ?thesis
 using rand-lookup read-lookup by simp
qed

lemma receive-composition-fri-commits-preserves-query-lookup:
assumes outcome:
 Some $((b, r), t) \in$
 set-dist (execute receive-composition-fri-commits s)
shows
 fmlookup (HashMap t) (QueryIndexChallenge i x) =
 fmlookup (HashMap s) (QueryIndexChallenge i x)
proof –
from outcome **obtain** s-read **where**
 read-r: Some $(r, s-read) \in \text{set-dist (execute read s)}$
and rand-b:
 Some $(b, t) \in$
 set-dist (execute receive-composition-fri-challenge s-read)
unfolding receive-composition-fri-commits-def receive-fri-commits-with-def
 by (auto elim!: set-dist-bindE)
have read-lookup:
 HashMap s-read = HashMap s
 by (rule read-preserves-hash-map[OF read-r])
have neq:
 QueryIndexChallenge i x $\neq$
 CompositionFriChallenge
 (PCompositionFriCounter s-read) (PState s-read)
 by simp
have rand-lookup:
 fmlookup (HashMap t) (QueryIndexChallenge i x) =
 fmlookup (HashMap s-read) (QueryIndexChallenge i x)
 by (rule receive-composition-fri-challenge-preserves-other-lookup
 [OF rand-b neq])
show ?thesis
 using rand-lookup read-lookup by simp

qed

lemma *verifier-composition-fri-prefix-preserves-query-lookup*:
assumes *outcome*:
Some ((fr, f-fl, f-final, as, dg, fl), t) ∈
set-dist (execute verifier-composition-fri-prefix s)
shows
fmlookup (HashMap t) (QueryIndexChallenge i x) =
fmlookup (HashMap s) (QueryIndexChallenge i x)
proof –
let *?key = QueryIndexChallenge i x*
from *outcome* **obtain** *s1 s2 s3 s4 s5 s6 s7* **where**
read-fr: Some (fr, s1) ∈ set-dist (execute read s)
and *trace-fri*:
Some (f-fl, s2) ∈
set-dist (execute
(ntimes receive-trace-fri-commits (ceil-log clength)) s1)
and *read-f-final: Some (f-final, s3) ∈ set-dist (execute read s2)*
and *alpha-out*:
Some (as, s4) ∈
set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)
and *read-dg: Some (dg, s5) ∈ set-dist (execute read s4)*
and *degree-assert*:
Some ((), s6) ∈
set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
and *comp-fri*:
Some (fl, s7) ∈
set-dist (execute
(ntimes receive-composition-fri-commits
(ceil-log (to-nat dg + 1))) s6)
and *ret*:
Some ((fr, f-fl, f-final, as, dg, fl), t) ∈
set-dist (execute (return (fr, f-fl, f-final, as, dg, fl)) s7)
unfolding *verifier-composition-fri-prefix-def*
by (*auto elim!: set-dist-bindE*)
have *lookup-s1*:
fmlookup (HashMap s1) ?key = fmlookup (HashMap s) ?key
using *read-preserves-hash-map[OF read-fr]* **by** *simp*
have *lookup-s2*:
fmlookup (HashMap s2) ?key = fmlookup (HashMap s1) ?key
proof (*rule ntimes-preserves-lookup[OF - trace-fri]*)
fix *y u v*
assume *out: Some (y, v) ∈ set-dist (execute receive-trace-fri-commits u)*
obtain *b r* **where** *y-eq: y = (b, r)*
by (*cases y*) *simp*
show *fmlookup (HashMap v) ?key = fmlookup (HashMap u) ?key*
by (*rule receive-trace-fri-commits-preserves-query-lookup*
[OF out[unfolded y-eq]])
qed

```

have lookup-s3:
  fmlookup (HashMap s3) ?key = fmlookup (HashMap s2) ?key
  using read-preserves-hash-map[OF read-f-final] by simp
have lookup-s4:
  fmlookup (HashMap s4) ?key = fmlookup (HashMap s3) ?key
proof (rule mmap-preserves-lookup[OF - alpha-out])
  fix m y u v
  assume m-in:  $m \in \text{set } (\text{replicate } (\text{length spec}) \text{ alpha-round})$ 
  and out:  $\text{Some } (y, v) \in \text{set-dist } (\text{execute } m \ u)$ 
  then have m-eq:  $m = \text{alpha-round}$ 
  by simp
  show fmlookup (HashMap v) ?key = fmlookup (HashMap u) ?key
  using alpha-round-preserves-query-lookup[OF out[unfolded m-eq]] .
qed
have lookup-s5:
  fmlookup (HashMap s5) ?key = fmlookup (HashMap s4) ?key
  using read-preserves-hash-map[OF read-dg] by simp
have s6-eq:  $s6 = s5$ 
  using degree-assert unfolding assert-def
  by (cases to-nat dg  $\leq \text{maxDegree}$ ) (auto simp: throw-no-outcome)
have lookup-s7:
  fmlookup (HashMap s7) ?key = fmlookup (HashMap s6) ?key
proof (rule ntimes-preserves-lookup[OF - comp-fri])
  fix y u v
  assume out:  $\text{Some } (y, v) \in \text{set-dist } (\text{execute receive-composition-fri-commits } u)$ 
  obtain b r where y-eq:  $y = (b, r)$ 
  by (cases y) simp
  show fmlookup (HashMap v) ?key = fmlookup (HashMap u) ?key
  by (rule receive-composition-fri-commits-preserves-query-lookup
    [OF out[unfolded y-eq]])
qed
have t-eq:  $t = s7$ 
  using ret by simp
show ?thesis
  using lookup-s1 lookup-s2 lookup-s3 lookup-s4 lookup-s5 lookup-s7
  unfolding t-eq s6-eq by simp
qed

lemma query-index-raw-singleton-card-uniform:
  assumes b-in:  $b \in \text{query-sample-space}$ 
  shows card (query-index-raw-preimage {b}) =
    size div query-sample-space-size
proof -
  have subset:  $\{b\} \subseteq \text{query-sample-space}$ 
  using b-in by simp
  have card (query-index-raw-preimage {b}) =
    (size div query-sample-space-size) * card {b}
  unfolding card-query-index-raw-preimage-eq-nat-preimage

```



```

    apply (rule card-query-index-nat-preimage-uniform-range)
    apply (rule to-nat-range)
    using query-sample-space-size-dvd subset
      unfolding query-sample-space-size-def by simp-all
  then show ?thesis
    by simp
qed

lemma query-index-row-list-fixed-preimage-card-bound:
  assumes len: length query-idxs = rounds
    and subset: set query-idxs  $\subseteq$  query-sample-space
  shows card {rows.
    length rows = rounds  $\wedge$ 
    map ( $\lambda$ raw. index (to-nat raw)) rows = query-idxs}
     $\leq$  (size div query-sample-space-size)  $\wedge$  rounds
  using len subset
proof (induction rounds arbitrary: query-idxs)
  case 0
  then have {rows.
    length rows = 0  $\wedge$ 
    map ( $\lambda$ raw. index (to-nat raw)) rows = query-idxs} = {}
  by auto
  then show ?case by simp
next
  case (Suc n)
  obtain q qs where query-eq: query-idxs = q # qs
    using Suc.prem1 by (cases query-idxs) auto
  have len-qs: length qs = n
    using Suc.prem1 unfolding query-eq by simp
  have q-in: q  $\in$  query-sample-space
    using Suc.prem2 unfolding query-eq by simp
  have qs-subset: set qs  $\subseteq$  query-sample-space
    using Suc.prem2 unfolding query-eq by simp
  let ?A = {rows.
    length rows = Suc n  $\wedge$ 
    map ( $\lambda$ raw. index (to-nat raw)) rows = q # qs}
  let ?B = query-index-row-preimage {q}
  let ?C = {rows.
    length rows = n  $\wedge$ 
    map ( $\lambda$ raw. index (to-nat raw)) rows = qs}
  have A-eq: ?A = ( $\lambda$ (raw, rows). raw # rows) ' (?B  $\times$  ?C)
  proof
    show ?A  $\subseteq$  ( $\lambda$ (raw, rows). raw # rows) ' (?B  $\times$  ?C)
  proof
    fix rows
    assume rows-in: rows  $\in$  ?A
    then obtain raw tail where rows-eq: rows = raw # tail
      by (cases rows) auto
    have raw-in: raw  $\in$  ?B

```

```

    using raws-in unfolding raws-eq query-index-row-preimage-def by simp
  have tail-in: tail ∈ ?C
    using raws-in unfolding raws-eq by simp
  show raws ∈ (λ(raw, raws). raw # raws) ‘ (?B × ?C)
    unfolding raws-eq by (intro image-eqI[of - - (raw, tail)])
    (use raw-in tail-in in simp-all)
qed
next
  show (λ(raw, raws). raw # raws) ‘ (?B × ?C) ⊆ ?A
    unfolding query-index-row-preimage-def by auto
qed
have inj: inj-on (λ(raw, raws). raw # raws) (?B × ?C)
  unfolding inj-on-def by auto
have finite-B: finite ?B
  by simp
have finite-C: finite ?C
proof -
  have ?C ⊆ {xs. length xs = n}
    by auto
  then show ?thesis
    by (auto intro: List.finite-list-length)
qed
have card ?A = card ((λ(raw, raws). raw # raws) ‘ (?B × ?C))
  unfolding A-eq by simp
also have ... = card (?B × ?C)
  by (rule card-image[OF inj])
also have ... = card ?B * card ?C
  using finite-B finite-C by simp
also have ... ≤
  (size div query-sample-space-size) *
  (size div query-sample-space-size) ^ n
proof (rule mult-mono)
  show card ?B ≤ size div query-sample-space-size
    using query-index-row-singleton-card-uniform[OF q-in] by simp
  show card ?C ≤ (size div query-sample-space-size) ^ n
    by (rule Suc.IH[OF len-qs qs-subset])
  show 0 ≤ size div query-sample-space-size by simp
  show 0 ≤ card ?C by simp
qed
also have ... = (size div query-sample-space-size) ^ Suc n
  by simp
finally show ?case
  unfolding query-eq .
qed

```

```

lemma query-index-row-list-preimage-card-bound:
  assumes subset: Q ⊆ fri-query-index-list-space
  shows card (query-index-row-list-preimage Q)
    ≤ card Q * (size div query-sample-space-size) ^ rounds

```

```

proof –
  let ?fiber =  $\lambda \text{query-idxs}. \{\text{rows}.$ 
     $\text{length rows} = \text{rounds} \wedge$ 
     $\text{map } (\lambda \text{raw}. \text{index } (\text{to-nat } \text{raw})) \text{ rows} = \text{query-idxs}\}$ 
  have finite-Q: finite Q
    using subset by (rule finite-subset) (rule finite-fri-query-index-list-space)
  have preimage-eq:
     $\text{query-index-raw-list-preimage } Q = (\bigcup \text{query-idxs} \in Q. ?\text{fiber } \text{query-idxs})$ 
    unfolding query-index-raw-list-preimage-def by auto
  have finite-fiber:  $\bigwedge \text{query-idxs}. \text{finite } (?\text{fiber } \text{query-idxs})$ 
proof –
  fix query-idxs
  have ?fiber query-idxs  $\subseteq \{xs. \text{length } xs = \text{rounds}\}$ 
    by auto
  then show finite (?fiber query-idxs)
    by (auto intro: List.finite-list-length)
qed
have card (query-index-raw-list-preimage Q) =
  card  $(\bigcup \text{query-idxs} \in Q. ?\text{fiber } \text{query-idxs})$ 
  unfolding preimage-eq ..
also have ...  $\leq (\sum \text{query-idxs} \in Q. \text{card } (?\text{fiber } \text{query-idxs}))$ 
  using finite-Q by (auto intro: card-UN-le)
also have ...  $\leq$ 
   $(\sum \text{query-idxs} \in Q.$ 
     $(\text{size div query-sample-space-size}) \wedge \text{rounds})$ 
proof (rule sum-mono)
  fix query-idxs
  assume query-in: query-idxs  $\in Q$ 
  have len: length query-idxs = rounds
    using query-in subset unfolding fri-query-index-list-space-def by auto
  have entries: set query-idxs  $\subseteq \text{query-sample-space}$ 
    using query-in subset unfolding fri-query-index-list-space-def by auto
  show card (?fiber query-idxs)
     $\leq (\text{size div query-sample-space-size}) \wedge \text{rounds}$ 
    by (rule query-index-raw-list-fixed-preimage-card-bound[OF len entries])
qed
also have ... = card Q *  $(\text{size div query-sample-space-size}) \wedge \text{rounds}$ 
  using finite-Q by simp
  finally show ?thesis .
qed

lemma query-index-raw-list-preimage-probability-bound:
  assumes subset:  $Q \subseteq \text{fri-query-index-list-space}$ 
  shows
     $\text{nnreal } (\text{card } (\text{query-index-raw-list-preimage } Q)) *$ 
     $(1 / \text{nnreal size}) \wedge \text{rounds}$ 
     $\leq \text{nnreal } (\text{card } Q) *$ 
     $(1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$ 
proof –

```

```

let ?d = size div query-sample-space-size
have card-le:
  card (query-index-raw-list-preimage Q)
    ≤ card Q * ?d ^ rounds
  by (rule query-index-raw-list-preimage-card-bound[OF subset])
have dvd: query-sample-space-size dvd size
  using query-sample-space-size-dvd
  unfolding query-sample-space-size-def by simp
have size-eq: size = ?d * query-sample-space-size
  using dvd query-sample-space-size-pos by (simp add: dvd-eq-mod-eq-0)
have size-pos: 0 < size
  using size-card by simp
have d-pos: 0 < ?d
proof (cases ?d)
  case 0
  then show ?thesis
    using size-eq size-pos by simp
next
  case (Suc n)
  then show ?thesis by simp
qed
have le:
  nnreal (card (query-index-raw-list-preimage Q)) *
    (1 / nnreal size) ^ rounds
  ≤ nnreal (card Q * ?d ^ rounds) *
    (1 / nnreal size) ^ rounds
  using card-le apply (simp add: mult-right-mono)
  by (metis mult-right-mono of-nat-le-iff of-nat-mult of-nat-power zero-least)
have prod-eq:
  nnreal (card Q * ?d ^ rounds) *
    (1 / nnreal size) ^ rounds =
  nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds
proof -
  have card-space-eq:
    card query-sample-space = query-sample-space-size
  by (rule card-query-sample-space)
  have base-eq:
    nnreal ?d * (1 / nnreal size) =
    1 / nnreal (card query-sample-space)
  proof (rule nn2real-eq-iff[THEN iffD1])
    have real-base:
      real ?d / real size = 1 / real query-sample-space-size
    proof -
      have real-size-eq:
        real size = real ?d * real query-sample-space-size
      using arg-cong[OF size-eq, of real]
      by (simp add: of-nat-mult)
      have real ?d / real size =

```

```

      real ?d / (real ?d * real query-sample-space-size)
    using real-size-eq by simp
  also have ... = 1 / real query-sample-space-size
    using d-pos query-sample-space-size-pos
    by (simp add: field-simps)
  finally show ?thesis .
qed
show nn2real (nnreal ?d * (1 / nnreal size)) =
  nn2real (1 / nnreal (card query-sample-space))
proof -
  have lhs-eq:
    nn2real (nnreal ?d * (1 / nnreal size)) =
      real ?d / real size
  proof -
    have nn2real (nnreal ?d * (1 / nnreal size)) =
      real ?d * (1 / real size)
    by (simp only: nn2real-mult nn2real-nnreal
      nn2real-divide nn2real-1)
    also have ... = real ?d / real size
    by (simp add: divide-inverse)
    finally show ?thesis .
  qed
  have rhs-eq:
    nn2real (1 / nnreal (card query-sample-space)) =
      1 / real query-sample-space-size
  unfolding card-space-eq
  by (simp only: nn2real-divide nn2real-1 nn2real-nnreal)
  show ?thesis
    using real-base lhs-eq rhs-eq by simp
qed
qed
have
  nnreal (card Q * ?d ^ rounds) * (1 / nnreal size) ^ rounds =
  nnreal (card Q) * (nnreal ?d * (1 / nnreal size)) ^ rounds
  by (simp add: of-nat-mult of-nat-power power-mult-distrib
    ac-simps)
  also have ... =
    nnreal (card Q) *
      (1 / nnreal (card query-sample-space)) ^ rounds
    using base-eq by simp
  finally show ?thesis .
qed
show ?thesis
  using le prod-eq by simp
qed

lemma wp-ntimes-verifier-query-round-program-query-list-path-product-bound:
  assumes subset:  $Q \subseteq \text{fri-query-index-list-space}$ 
  shows

```

```

wp-event
  (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
    rounds)
  (λout. ∃ rows ∈ query-index-raw-list-preimage Q.
    query-rounds-raw-list-path-hit s rows rounds out)
  s ≤
  nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds
proof –
  have raw-bound:
    wp-event
      (ntimes (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
      (λout. ∃ rows ∈ query-index-raw-list-preimage Q.
        query-rounds-raw-list-path-hit s rows rounds out)
      s ≤
      nnreal (card (query-index-raw-list-preimage Q)) *
      (1 / nnreal size) ^ rounds
    by (rule wp-ntimes-verifier-query-round-program-raw-list-preimage-path-hit-bound)
  also have ... ≤
    nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds
  by (rule query-index-raw-list-preimage-probability-bound[OF subset])
  finally show ?thesis .
qed

```

```

lemma read-outcome-unique-state:
  assumes out1: Some (x, t) ∈ set-dist (execute read s)
  and out2: Some (y, u) ∈ set-dist (execute read s)
  shows x = y ∧ t = u
proof –
  from read-outcome[OF out1] obtain rest where
    tr: PTranscript s = x # rest
  by blast
  have out1-eq:
    t = s(PState := concat (PState s) x, PTranscript := rest)
  using read-nonempty-outcome[OF out1 tr] by simp
  have out2-eq:
    y = x ∧
    u = s(PState := concat (PState s) x, PTranscript := rest)
  using read-nonempty-outcome[OF out2 tr] by simp
  then show ?thesis
  using out1-eq by simp
qed

```

```

lemma wp-verifier-after-composition-fri-raw-list-path-product-bound:
  assumes subset: Q ⊆ fri-query-index-list-space
  shows
    wp-event (verifier-after-composition-fri header)

```

```

    (verifier-after-composition-fri-raw-list-path-hit s Q) s ≤
    nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds
proof -
obtain fr f-fl f-final as dg fl where header-eq:
  header = (fr, f-fl, f-final, as, dg, fl)
by (cases header) auto
have after-eq:
  verifier-after-composition-fri header =
    (read >>=
      (λfinal.
        ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
            rounds))
unfolding header-eq verifier-after-composition-fri-def by simp
show ?thesis
unfolding after-eq
proof (rule wp-event-bind-bound-by-cont)
show ¬ verifier-after-composition-fri-raw-list-path-hit s Q None
unfolding verifier-after-composition-fri-raw-list-path-hit-def
  query-rounds-raw-list-path-hit-def
by simp
next
fix final query-state
assume read-final:
  Some (final, query-state) ∈ set-dist (execute read s)
have mono:
  wp-event
    (ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
    (verifier-after-composition-fri-raw-list-path-hit s Q)
    query-state ≤
  wp-event
    (ntimes
      (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
    (λout. ∃ raws ∈ query-index-raw-list-preimage Q.
      query-rounds-raw-list-path-hit query-state raws rounds out)
    query-state
proof (rule wp-event-mono-on-support)
fix out
assume hit:
  verifier-after-composition-fri-raw-list-path-hit s Q out
from hit obtain final' query-state' raws where
  read':
    Some (final', query-state') ∈ set-dist (execute read s)
and raw-in: raws ∈ query-index-raw-list-preimage Q
and path:

```

```

    query-rounds-raw-list-path-hit query-state' raws rounds out
  unfolding verifier-after-composition-fri-raw-list-path-hit-def
  by blast
  have query-state' = query-state
    using read-outcome-unique-state[OF read' read-final] by simp
  then show  $\exists$  raws  $\in$  query-index-raw-list-preimage  $Q$ .
    query-rounds-raw-list-path-hit query-state raws rounds out
    using raw-in path by blast
qed
also have ...  $\leq$ 
  nnreal (card  $Q$ ) *
  (1 / nnreal (card query-sample-space))  $\wedge$  rounds
  by (rule
    wp-ntimes-verifier-query-round-program-query-list-path-product-bound
    [OF subset])
finally show
  wp-event
  (ntimes
    (verifier-query-round-program fr f-fl f-final as fl final)
    rounds)
  (verifier-after-composition-fri-raw-list-path-hit s  $Q$ )
  query-state
 $\leq$  nnreal (card  $Q$ ) *
  (1 / nnreal (card query-sample-space))  $\wedge$  rounds .
qed
qed

lemma verifier-after-composition-fri-query-index-list-path-fresh-imp-raw-hit-after-prefix:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and builder:
  Some (data, attacker-state)  $\in$ 
  set-dist
  (execute (checked-staged-transcript-program  $A$ )
    adversary-initial-state)
and prefix:
  Some (header, prefix-state)  $\in$ 
  set-dist
  (execute verifier-composition-fri-prefix
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
and suffix:
  out  $\in$  set-dist
  (execute (verifier-after-composition-fri header) prefix-state)
and hit:
  staged-security-with-data-state-query-index-list-path-fresh  $Q$ 
  (case out of
    None  $\Rightarrow$  None
    | Some (result, final-state)  $\Rightarrow$ 

```



```

      Some (((data, attacker-state), result), final-state))
shows verifier-after-composition-fri-raw-list-path-hit prefix-state Q out
proof (cases out)
  case None
  then show ?thesis
    using hit by simp
next
  case (Some result-state)
  then obtain result final-state where
    out-eq: out = Some (result, final-state)
    by (cases result-state) simp
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have hit-some:
    staged-security-with-data-state-query-index-list-path-fresh Q
    (Some (((data, attacker-state), result), final-state))
    using hit out-eq by simp
  from staged-security-with-data-state-query-index-list-path-fresh-SomeE
    [OF hit-some]
  obtain raw-idxs-hit query-idxs-hit where
    len-hit: length raw-idxs-hit = rounds
    and query-hit-in: query-idxs-hit ∈ Q
    and query-hit-eq:
      query-idxs-hit = map (λraw. index (to-nat raw)) raw-idxs-hit
    and lookup-hit:
      ∧ i. i < rounds ⇒
        fmllookup (HashMap final-state)
          (QueryIndexChallenge i
            (state-after-query-chunks
              (staged-query-start-hash data)
              (staged-query-chunks data) i)) =
          Some (raw-idxs-hit ! i)
    and fresh-hit:
      ∧ i. i < rounds ⇒
        fmllookup (HashMap ?s)
          (QueryIndexChallenge i
            (state-after-query-chunks
              (staged-query-start-hash data)
              (staged-query-chunks data) i)) =
          None
    by blast
  obtain fr f-fl f-final as dg fl where header-eq:
    header = (fr, f-fl, f-final, as, dg, fl)
    by (cases header) auto
  from suffix[unfolded out-eq header-eq verifier-after-composition-fri-def]
  obtain final query-state where
    read-final:
      Some (final, query-state) ∈ set-dist (execute read prefix-state)

```

```

and query-out:
  Some (result, final-state) ∈
    set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
by (auto elim!: set-dist-bindE)
from ntimes-verifier-query-rounds-outcome[OF query-out]
obtain raw-idxs query-idxs query-chunks where
  len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-chunks: length query-chunks = rounds
and tr-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and state-final:
  PState final-state =
    state-after-query-chunks (PState query-state) query-chunks rounds
and round-chunks:
  ∧i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and lookups:
  ∧i. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
by blast
from read-outcome[OF read-final] have query-counter:
  PQueryCounter query-state = PQueryCounter prefix-state
by blast
have prefix-basic:
  length f-fl = ceil-log clength ∧
  length as = length spec ∧
  length fl = ceil-log (to-nat dg + 1) ∧
  PTranscript ?s =
    [fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
    map snd fl @ PTranscript prefix-state ∧
  PState prefix-state =
    foldl concat
      (concat
        (foldl concat
          (concat
            (foldl concat (concat (PState ?s) fr) (map snd f-fl))

```

```

      f-final)
    as)
  dg)
  (map snd fl) ∧
  PQueryCounter prefix-state = PQueryCounter ?s
by (rule verifier-composition-fri-prefix-outcome-basicD
  [OF prefix[unfolded header-eq]])
have prefix-counter: PQueryCounter prefix-state = PQueryCounter ?s
  using prefix-basic by simp
have local-counter: PQueryCounter query-state = 0
  using query-counter prefix-counter
  unfolding verifier-state-from-adversary-def by simp
have prefix-res:
  length f-fl = ceil-log clength ∧
  length as = length spec ∧
  length fl = ceil-log (to-nat dg + 1) ∧
  PTranscript ?s =
    [fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
    map snd fl @ PTranscript prefix-state ∧
  PState prefix-state =
    foldl concat
      (concat
        (foldl concat
          (concat
            (foldl concat (concat (PState ?s) fr) (map snd f-fl))
            f-final)
          as)
        dg)
      (map snd fl)
  using prefix-basic by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-prefix: PTranscript prefix-state = final # rest-query
  and state-query:
    PState query-state = concat (PState prefix-state) final
  and tr-read-query: PTranscript query-state = rest-query
  by blast
have lookup-query:
  ∧i x. fmllookup (HashMap query-state)
    (QueryIndexChallenge i x) =
    fmllookup (HashMap ?s) (QueryIndexChallenge i x)
proof -
  have query-state-eq:
    query-state =
      prefix-state(PState := concat (PState prefix-state) final,
        PTranscript := rest-query)
  using read-nonempty-outcome[OF read-final tr-prefix] by simp
have hash-prefix:
  ∧i x. fmllookup (HashMap prefix-state)
    (QueryIndexChallenge i x) =

```

```

    fmlookup (HashMap ?s) (QueryIndexChallenge i x)
  by (rule verifier-composition-fri-prefix-preserves-query-lookup
      [OF prefix[unfolded header-eq]])
show  $\bigwedge i x.$  fmlookup (HashMap query-state)
    (QueryIndexChallenge i x) =
    fmlookup (HashMap ?s) (QueryIndexChallenge i x)
  using query-state-eq hash-prefix by simp
qed
have len-trace-actual: length (map snd f-fl) = ceil-log clength
  using prefix-res by simp
have len-alpha-actual: length as = length spec
  using prefix-res by simp
have len-comp-actual:
  length (map snd fl) = ceil-log (to-nat dg + 1)
  using prefix-res by simp
have transcript-prefix-actual:
  PTranscript ?s =
    [fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
    map snd fl @ PTranscript prefix-state
  using prefix-res by simp
have header-actual:
  verifier-header-transcript ?s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
  by (rule verifier-header-transcript-from-prefix-read
      [OF len-trace-actual len-alpha-actual len-comp-actual
        transcript-prefix-actual tr-prefix tr-read-query])
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled builder])
have header-unique:
  fr = staged-trace-root data  $\wedge$ 
  map snd f-fl = staged-trace-fri-roots data  $\wedge$ 
  f-final = staged-trace-final data  $\wedge$ 
  as = staged-alphas data  $\wedge$ 
  dg = staged-degree data  $\wedge$ 
  map snd fl = staged-composition-fri-roots data  $\wedge$ 
  final = staged-composition-final data  $\wedge$ 
  PTranscript query-state = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header-actual staged-header]
  by simp

```

```

have prefix-state-query:
  PState query-state = staged-query-start-hash data
  using state-query prefix-res header-unique
  unfolding staged-query-start-hash-def
    staged-composition-fri-start-hash-def
    staged-trace-fri-start-hash-def
  by simp
have chunks-eq: query-chunks = staged-query-chunks data
proof -
  from checked-staged-transcript-program-query-chunks-match-verifier-lengths
    [OF builder]
  obtain staged-query-idxs where
    staged-match:
      staged-query-chunks-match-verifier-lengths data staged-query-idxs
  by blast
have query-idxs-len: length query-idxs = rounds
  using query-idxs-eq len-raw by simp
have staged-match-query-idxs:
  staged-query-chunks-match-verifier-lengths data query-idxs
  by (rule staged-query-chunks-match-verifier-lengths-transfer
    [OF staged-match query-idxs-len])
have parser-chunk-len:
   $\bigwedge i. i < \text{rounds} \implies$ 
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
proof -
  fix i
  assume i-bound: i < rounds
  show
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
  using verifier-query-round-chunk-length
    [OF round-chunks [OF i-bound]] header-unique
  by simp
qed
have concat-staged:
  List.concat query-chunks @ PTranscript final-state =
    List.concat (staged-query-chunks data)
  using tr-query header-unique by simp
then show ?thesis
  using query-chunks-eq-staged-if-matching-lengths
    [OF len-chunks concat-staged parser-chunk-len staged-match-query-idxs]
  by simp
qed
have raw-eq:

```

```

 $\wedge i. i < \text{rounds} \implies \text{raw-idxs} ! i = \text{raw-idxs-hit} ! i$ 
using lookups lookup-hit local-counter prefix-state-query chunks-eq
by simp
have raw-idxs-eq:  $\text{raw-idxs} = \text{raw-idxs-hit}$ 
by (rule nth-equalityI)
  (use len-raw len-hit raw-eq in simp-all)
have raw-in:  $\text{raw-idxs} \in \text{query-index-raw-list-preimage } Q$ 
unfolding query-index-raw-list-preimage-def
using len-raw raw-idxs-eq query-hit-in query-hit-eq by blast
have local-fresh:
  query-index-list-path-fresh query-state rounds query-chunks raw-idxs
unfolding query-index-list-path-fresh-def
using len-raw len-chunks fresh-hit local-counter prefix-state-query
  chunks-eq lookup-query
by simp
show ?thesis
by (rule verifier-after-composition-fri-raw-list-path-hitI
  [where final=final and query-state=query-state and raws=raw-idxs
  and chunks=query-chunks and t=final-state and results=result,
  OF read-final raw-in len-raw len-chunks state-final local-fresh
  lookups out-eq])
qed

lemma verifier-after-composition-fri-query-index-list-path-fresh-product-bound-after-prefix:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and subset:  $Q \subseteq \text{fri-query-index-list-space}$ 
and builder:
  Some (data, attacker-state) \in
    set-dist
    (execute (checked-staged-transcript-program A)
    adversary-initial-state)
and prefix:
  Some (header, prefix-state) \in
    set-dist
    (execute verifier-composition-fri-prefix
    (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))))
shows
  wp-event (verifier-after-composition-fri header)
  ( $\lambda \text{out.}$ 
    staged-security-with-data-state-query-index-list-path-fresh Q
    (case out of
      None \implies None
      | Some (result, final-state) \implies
        Some (((data, attacker-state), result), final-state))))
  prefix-state
   $\leq \text{nnreal (card } Q) *$ 
   $(1 / \text{nnreal (card query-sample-space)}) \wedge \text{rounds}$ 

```

```

proof –
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  let ?P =
    λout.
      staged-security-with-data-state-query-index-list-path-fresh Q
      (case out of
        None ⇒ None
        | Some (result, final-state) ⇒
          Some (((data, attacker-state), result), final-state))
  have event-mono:
    wp-event (verifier-after-composition-fri header)
      ?P prefix-state
    ≤
    wp-event (verifier-after-composition-fri header)
      (verifier-after-composition-fri-raw-list-path-hit prefix-state Q)
      prefix-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume suffix:
      out ∈
        set-dist
        (execute (verifier-after-composition-fri header) prefix-state)
    and hit:
      staged-security-with-data-state-query-index-list-path-fresh Q
      (case out of
        None ⇒ None
        | Some (result, final-state) ⇒
          Some (((data, attacker-state), result), final-state))
    show verifier-after-composition-fri-raw-list-path-hit prefix-state Q out
    by (rule
      verifier-after-composition-fri-query-index-list-path-fresh-imp-raw-hit-after-prefix
      [OF wf controlled builder prefix suffix hit])
  qed
  also have ... ≤
    nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds
  by (rule wp-verifier-after-composition-fri-raw-list-path-product-bound
    [OF subset])
  finally show ?thesis .
qed

lemma verify-monad-query-index-list-path-fresh-product-bound-after-builder:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and subset: Q ⊆ fri-query-index-list-space
  and builder:
    Some (data, attacker-state) ∈

```

```

    set-dist
      (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  shows
    wp-event verify-monad
      (λout.
        staged-security-with-data-state-query-index-list-path-fresh Q
          (case out of
            None ⇒ None
            | Some (result, final-state) ⇒
              Some (((data, attacker-state), result), final-state)))
        (verifier-state-from-adversary attacker-state
          (staged-proof-transcript data))
      ≤ nnreal (card Q) *
        (1 / nnreal (card query-sample-space)) ^ rounds
  proof -
    let ?s =
      verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)
    note verify-decomp = verify-monad-composition-fri-decomposition
    show ?thesis
      unfolding verify-decomp
    proof (rule wp-event-bind-bound-by-cont)
      show ¬ staged-security-with-data-state-query-index-list-path-fresh Q
        (case None of None ⇒ None
          | Some (result, final-state) ⇒
            Some (((data, attacker-state), result), final-state))
        by simp
    next
      fix header prefix-state
      assume prefix:
        Some (header, prefix-state) ∈
          set-dist (execute verifier-composition-fri-prefix ?s)
      show
        wp-event (verifier-after-composition-fri header)
          (λout.
            staged-security-with-data-state-query-index-list-path-fresh Q
              (case out of
                None ⇒ None
                | Some (result, final-state) ⇒
                  Some (((data, attacker-state), result), final-state)))
            prefix-state
          ≤ nnreal (card Q) *
            (1 / nnreal (card query-sample-space)) ^ rounds
        by (rule
          verifier-after-composition-fri-query-index-list-path-fresh-product-bound-after-prefix
            [OF wf controlled subset builder prefix])
    qed
  qed

```


lemma *checked-staged-security-with-data-state-query-index-list-path-fresh-product-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *subset: $Q \subseteq \text{fri-query-index-list-space}$*

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-query-index-list-path-fresh Q)
adversary-initial-state $\leq$
*nnreal (card Q) **
(1 / nnreal (card query-sample-space)) $\wedge$ rounds

proof –

let *?P = staged-security-with-data-state-query-index-list-path-fresh Q*

have *none: $\neg ?P$ None*

by *simp*

have *cont-bound:*

$\wedge$ data attacker-state.

Some (data, attacker-state) $\in$
set-dist

(execute (checked-staged-transcript-program A)
adversary-initial-state) $\implies$

wp-event verify-monad

(λ out.

?P

(case out of

None $\implies$ None

| Some (result, final-state) $\implies$

Some (((data, attacker-state), result), final-state)))

(verifier-state-from-adversary attacker-state

(staged-proof-transcript data))

*$\leq$ nnreal (card Q) **

(1 / nnreal (card query-sample-space)) $\wedge$ rounds

by *(rule verify-monad-query-index-list-path-fresh-product-bound-after-builder*

[OF wf controlled subset])

show *?thesis*

by *(rule checked-staged-security-with-data-state-bound-from-data-cont*

[OF none cont-bound])

qed

lemma *checked-staged-security-with-data-state-query-index-list-hit-exact-product-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *subset: $Q \subseteq \text{fri-query-index-list-space}$*

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-query-index-list-set-hit Q)
adversary-initial-state $\leq$
*nnreal (card Q) **
(1 / nnreal (card query-sample-space)) $\wedge$ rounds +

hash-relation-budget-value
 (*query-index-raw-list-relation-fiber-bound* Q)
 (*staged-attacker-query-budget* budgets +
 staged-challenge-query-budget)
by (*rule*
 checked-staged-security-with-data-state-query-index-list-hit-bound-from-path-fresh
 [*OF* *wf* *controlled*
 checked-staged-security-with-data-state-query-index-list-path-fresh-product-bound
 [*OF* *wf* *controlled* *subset*]])

lemma *checked-staged-security-trace-fri-query-index-list-set-hit-exact-product-bound:*

assumes *wf*: *staged-budget-wellformed* budgets
and *controlled*: *staged-adversary-controlled* budgets A
and *subset*: $Q \subseteq \text{fri-query-index-list-space}$

shows

wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{trace-fri-query-index-list-set-hit } s \ Q$))
adversary-initial-state $\leq$
nnreal (*card* Q) *
 ($1 / \text{nnreal} (\text{card } \text{query-sample-space})$) $\wedge$ *rounds* +
hash-relation-budget-value
 (*query-index-raw-list-relation-fiber-bound* Q)
 (*staged-attacker-query-budget* budgets +
 staged-challenge-query-budget)
by (*rule*
 checked-staged-security-trace-fri-query-index-list-set-hit-bound-from-path-fresh
 [*OF* *wf* *controlled*
 checked-staged-security-with-data-state-query-index-list-path-fresh-product-bound
 [*OF* *wf* *controlled* *subset*]])

lemma *checked-staged-security-composition-fri-query-index-list-set-hit-exact-product-bound:*

assumes *wf*: *staged-budget-wellformed* budgets
and *controlled*: *staged-adversary-controlled* budgets A
and *subset*: $Q \subseteq \text{fri-query-index-list-space}$

shows

wp-event (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{composition-fri-query-index-list-set-hit } s \ Q$))
adversary-initial-state $\leq$
nnreal (*card* Q) *
 ($1 / \text{nnreal} (\text{card } \text{query-sample-space})$) $\wedge$ *rounds* +
hash-relation-budget-value
 (*query-index-raw-list-relation-fiber-bound* Q)
 (*staged-attacker-query-budget* budgets +
 staged-challenge-query-budget)
by (*rule*
 checked-staged-security-composition-fri-query-index-list-set-hit-bound-from-path-fresh
 [*OF* *wf* *controlled*

checked-staged-security-with-data-state-query-index-list-path-fresh-product-bound
 $[OF \text{ wf controlled subset}]])$

lemma *checked-staged-security-trace-fri-query-challenge-pair-set-hit-exact-product-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *projection: fst ' P $\subseteq$ Q*

and *subset: Q $\subseteq$ fri-query-index-list-space*

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

($\lambda s.$ trace-fri-query-challenge-pair-set-hit s P))

adversary-initial-state $\leq$

*nnreal (card Q) **

(1 / nnreal (card query-sample-space)) $\wedge$ rounds +

hash-relation-budget-value

(query-index-raw-list-relation-fiber-bound Q)

(staged-attacker-query-budget budgets +

staged-challenge-query-budget)

by *(rule*

checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-path-fresh

[OF wf controlled projection

checked-staged-security-with-data-state-query-index-list-path-fresh-product-bound

[OF wf controlled subset]])

lemma *checked-staged-security-composition-fri-query-challenge-pair-set-hit-exact-product-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *projection: $\bigwedge dg.$ fst ' P dg $\subseteq$ Q*

and *subset: Q $\subseteq$ fri-query-index-list-space*

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

($\lambda s.$ composition-fri-query-challenge-pair-set-hit s P))

adversary-initial-state $\leq$

*nnreal (card Q) **

(1 / nnreal (card query-sample-space)) $\wedge$ rounds +

hash-relation-budget-value

(query-index-raw-list-relation-fiber-bound Q)

(staged-attacker-query-budget budgets +

staged-challenge-query-budget)

by *(rule*

checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-path-fresh

[OF wf controlled projection

checked-staged-security-with-data-state-query-index-list-path-fresh-product-bound

[OF wf controlled subset]])

end

end

theory *Soundness-FRI-Query-Fiber-Exact-Staged-Bounds*

imports

Soundness-FRI-Query-Fiber-Staged-Bounds

Soundness-FRI-Query-List-Exact-Product

begin

Exact query-list staged bounds for the outside-query-cover branch of sampled FRI query events.

This layer keeps the challenge-cover split and uses exact product query-list accounting for the outside branch.

context *soundness*

begin

lemma *checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-list-exact*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *finite-B: finite B*

and *fresh-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

($\lambda s. \text{trace-fri-challenge-list-fresh-hit } s \ B$))

adversary-initial-state $\leq F$

and *projection:*

fst ‘ $(P \cap (UNIV \times (- B))) \subseteq Q$

and *subset: $Q \subseteq \text{fri-query-index-list-space}$*

shows

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

($\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ P$))

adversary-initial-state $\leq$

F +

*hash-relation-budget-value (card B * ceil-log clength)*

(staged-attacker-query-budget budgets + staged-challenge-query-budget) +

*(nnreal (card Q) **

(1 / nnreal (card query-sample-space)) ^ rounds +

hash-relation-budget-value

(query-index-raw-list-relation-fiber-bound Q)

(staged-attacker-query-budget budgets +

staged-challenge-query-budget))

proof –

let *?Outside = $P \cap (UNIV \times (- B))$*

have *split:*

wp-event (checked-staged-security-experiment-with-data-state A)

(staged-security-with-data-state-verifier-event

($\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ P$))

adversary-initial-state $\leq$

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (λout.
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-challenge-list-set-hit s B) out ∨
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-challenge-pair-set-hit s ?Outside) out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume hit:
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-challenge-pair-set-hit s P) out
  then show
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-challenge-list-set-hit s B) out ∨
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-challenge-pair-set-hit s ?Outside) out
  proof (cases out)
  case None
  then show ?thesis
    using hit unfolding staged-security-with-data-state-verifier-event-def
    by simp
  next
  case (Some full)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
  by (cases full) (auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have pair-hit:
    trace-fri-query-challenge-pair-set-hit ?s P
    (Some (result, final-state))
  using hit out-eq
  unfolding staged-security-with-data-state-verifier-event-def by simp
  from pair-hit obtain trace-roots trace-bs trace-final dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers
  where openings:
    accepted-fri-opening-transcript ?s (Some (result, final-state))
    trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers
  and pair: (fri-query-idxs, trace-bs) ∈ P
  unfolding trace-fri-query-challenge-pair-set-hit-def by blast
  show ?thesis
  proof (cases trace-bs ∈ B)
  case True
  have challenges:

```

```

    accepted-fri-challenges ?s (Some (result, final-state))
    trace-bs dg composition-bs
  by (rule accepted-fri-opening-transcript-challenges[OF openings])
have trace-fri-challenge-list-set-hit ?s B
  (Some (result, final-state))
  unfolding trace-fri-challenge-list-set-hit-def
  using challenges True by blast
then show ?thesis
  using out-eq unfolding staged-security-with-data-state-verifier-event-def
  by simp
next
case False
have outside-pair: (fri-query-idxs, trace-bs) ∈ ?Outside
  using pair False by simp
have trace-fri-query-challenge-pair-set-hit ?s ?Outside
  (Some (result, final-state))
  unfolding trace-fri-query-challenge-pair-set-hit-def
  using openings outside-pair by blast
then show ?thesis
  using out-eq unfolding staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
qed
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-challenge-list-set-hit s B))
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-challenge-pair-set-hit s ?Outside))
    adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤
  (F +
    hash-relation-budget-value (card B * ceil-log clength)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)) +
  (nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds +
    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound Q)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget))
proof (rule add-mono)
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. trace-fri-challenge-list-set-hit s B))

```

```

    adversary-initial-state
  ≤ F +
    hash-relation-budget-value (card B * ceil-log clength)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  by (rule checked-staged-security-trace-fri-challenge-list-set-hit-bound
    [OF wf controlled finite-B fresh-bound])
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. trace-fri-query-challenge-pair-set-hit s ?Outside))
  adversary-initial-state
  ≤ nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds +
    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound Q)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
  by (rule
    checked-staged-security-trace-fri-query-challenge-pair-set-hit-exact-product-bound
    [OF wf controlled projection subset])
qed
finally show ?thesis
  by (simp add: add.assoc)
qed

```

lemma *checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-lis*

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: $\bigwedge dg. \text{finite } (B \ dg)$
and fresh-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 (λs. composition-fri-challenge-list-fresh-hit s B))
 adversary-initial-state ≤ F

and projection:

$\bigwedge dg. \text{fst } ' (P \ dg \cap (UNIV \times (- \ B \ dg))) \subseteq Q$

and subset: $Q \subseteq \text{fri-query-index-list-space}$

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 (λs. composition-fri-query-challenge-pair-set-hit s P))
 adversary-initial-state ≤
 F +
 hash-relation-budget-value
 ($\sum dg \in (UNIV :: 'f \ \text{set}).$
 card (B dg) * ceil-log (maxDegree + 1))
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget) +
 (nnreal (card Q) *

```

    (1 / nnreal (card query-sample-space)) ^ rounds +
    hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound Q)
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget))
proof –
  let ?Outside = λdg. P dg ∩ (UNIV × (– B dg))
  have split:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       (λs. composition-fri-query-challenge-pair-set-hit s P))
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (λout.
        staged-security-with-data-state-verifier-event
          (λs. composition-fri-challenge-list-set-hit s B) out ∨
        staged-security-with-data-state-verifier-event
          (λs. composition-fri-query-challenge-pair-set-hit s ?Outside) out)
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume hit:
    staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-challenge-pair-set-hit s P) out
  then show
    staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-set-hit s B) out ∨
    staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-challenge-pair-set-hit s ?Outside) out
  proof (cases out)
  case None
  then show ?thesis
    using hit unfolding staged-security-with-data-state-verifier-event-def
    by simp
  next
  case (Some full)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases full) (auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  have pair-hit:
    composition-fri-query-challenge-pair-set-hit ?s P
      (Some (result, final-state))
  using hit out-eq
  unfolding staged-security-with-data-state-verifier-event-def by simp
  from pair-hit obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final fri-query-idxs

```



```

    trace-round-layers composition-round-layers
  where openings:
    accepted-fri-opening-transcript ?s (Some (result, final-state))
    trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers
  and pair: (fri-query-idxs, composition-bs) ∈ P fri-dg
  unfolding composition-fri-query-challenge-pair-set-hit-def by blast
show ?thesis
proof (cases composition-bs ∈ B fri-dg)
  case True
  have challenges:
    accepted-fri-challenges ?s (Some (result, final-state))
    trace-bs fri-dg composition-bs
  by (rule accepted-fri-opening-transcript-challenges[OF openings])
  have composition-fri-challenge-list-set-hit ?s B
    (Some (result, final-state))
  unfolding composition-fri-challenge-list-set-hit-def
  using challenges True by blast
  then show ?thesis
    using out-eq unfolding staged-security-with-data-state-verifier-event-def
    by simp
next
  case False
  have outside-pair:
    (fri-query-idxs, composition-bs) ∈ ?Outside fri-dg
  using pair False by simp
  have composition-fri-query-challenge-pair-set-hit ?s ?Outside
    (Some (result, final-state))
  unfolding composition-fri-query-challenge-pair-set-hit-def
  using openings outside-pair by blast
  then show ?thesis
    using out-eq unfolding staged-security-with-data-state-verifier-event-def
    by simp
qed
qed
qed
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-set-hit s B))
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-challenge-pair-set-hit s ?Outside))
    adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤
  (F +

```

```

hash-relation-budget-value
  (∑ dg ∈ (UNIV :: 'f set).
    card (B dg) * ceil-log (maxDegree + 1))
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)) +
(nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds +
hash-relation-budget-value
  (query-index-raw-list-relation-fiber-bound Q)
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget))
proof (rule add-mono)
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. composition-fri-challenge-list-set-hit s B))
  adversary-initial-state
≤ F +
  hash-relation-budget-value
    (∑ dg ∈ (UNIV :: 'f set).
      card (B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
by (rule
  checked-staged-security-composition-fri-challenge-list-set-hit-bound
  [OF wf controlled finite-B fresh-bound])
show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-challenge-pair-set-hit s ?Outside))
  adversary-initial-state
≤ nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds +
  hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound Q)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
by (rule
  checked-staged-security-composition-fri-query-challenge-pair-set-hit-exact-product-bound
  [OF wf controlled projection subset])
qed
finally show ?thesis
by (simp add: add.assoc)
qed

lemma checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-outsid
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B: finite B
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (staged-security-with-data-state-verifier-event
      ( $\lambda s.$  trace-fri-challenge-list-fresh-hit  $s$   $B$ ))
    adversary-initial-state  $\leq F$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
      (staged-security-with-data-state-verifier-event
        trace-fri-sampled-query-bad-candidate)
    adversary-initial-state  $\leq$ 
       $F +$ 
      hash-relation-budget-value (card  $B * \text{ceil-log } \text{clength}$ )
      (staged-attacker-query-budget budgets + staged-challenge-query-budget) +
      (nnreal
        (card
          (generic-fri-sampled-query-outside-query-cover
            trace-table-low-degree (Not  $\circ$  trace-table-low-degree)
            ( $\text{clength} - 1$ )  $B$ )) *
          (1 / nnreal (card query-sample-space))  $\wedge$  rounds +
          hash-relation-budget-value
            (query-index-row-list-relation-fiber-bound
              (generic-fri-sampled-query-outside-query-cover
                trace-table-low-degree (Not  $\circ$  trace-table-low-degree)
                ( $\text{clength} - 1$ )  $B$ ))
              (staged-attacker-query-budget budgets +
                staged-challenge-query-budget)))
  proof -
  let ?P =
    trace-fri-sampled-query-bad-pair-union  $\cap$ 
    (fri-query-index-list-space  $\times$  UNIV)
  let ?Q =
    generic-fri-sampled-query-outside-query-cover trace-table-low-degree
    (Not  $\circ$  trace-table-low-degree) ( $\text{clength} - 1$ )  $B$ 
  have sampled-to-pair:
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
      (staged-security-with-data-state-verifier-event
        trace-fri-sampled-query-bad-candidate)
    adversary-initial-state  $\leq$ 
    wp-event (checked-staged-security-experiment-with-data-state  $A$ )
      (staged-security-with-data-state-verifier-event
        ( $\lambda s.$  trace-fri-query-challenge-pair-set-hit  $s$  ?P))
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro:
        trace-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit
        split: option.splits prod.splits)
  also have ...  $\leq$ 
     $F +$ 
    hash-relation-budget-value (card  $B * \text{ceil-log } \text{clength}$ )
    (staged-attacker-query-budget budgets + staged-challenge-query-budget) +

```

```

    (nnreal (card ?Q) *
      (1 / nnreal (card query-sample-space)) ^ rounds +
      hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound ?Q)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget))
  by (rule
    checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-list-exact
    [OF wf controlled finite-B fresh-bound
      trace-fri-sampled-query-outside-projection-subset
      generic-fri-sampled-query-outside-query-cover-subset])
  finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and finite-B:  $\bigwedge dg. \text{finite } (B \ dg)$ 
and fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s. \text{composition-fri-challenge-list-fresh-hit } s \ B$ ))
  adversary-initial-state  $\leq F$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-fri-sampled-query-bad-candidate)
  adversary-initial-state  $\leq$ 
    F +
    hash-relation-budget-value
    ( $\sum dg \in (UNIV :: 'f \text{ set}).$ 
      card (B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget) +
    (nnreal (card (composition-fri-sampled-query-outside-query-cover B)) *
      (1 / nnreal (card query-sample-space)) ^ rounds +
      hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound
        (composition-fri-sampled-query-outside-query-cover B))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget))
proof -
let ?P =
   $\lambda dg. \text{composition-fri-sampled-query-bad-pair-union } dg \cap$ 
  (fri-query-index-list-space  $\times$  UNIV)
let ?Q = composition-fri-sampled-query-outside-query-cover B
have sampled-to-pair:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event

```

```

      composition-fri-sampled-query-bad-candidate)
    adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-challenge-pair-set-hit s ?P))
  adversary-initial-state
by (rule wp-event-mono-on-support)
  (auto simp: staged-security-with-data-state-verifier-event-def
    intro:
      composition-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit
      split: option.splits prod.splits)
also have ... ≤
  F +
  hash-relation-budget-value
  (∑ dg ∈ (UNIV :: 'f set).
    card (B dg) * ceil-log (maxDegree + 1))
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget) +
  (nnreal (card ?Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds +
    hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound ?Q)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget))
by (rule
  checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-cover-and-query-list-ex
  [OF wf controlled finite-B fresh-bound
    composition-fri-sampled-query-outside-projection-subset
    composition-fri-sampled-query-outside-query-cover-subset])
finally show ?thesis .
qed

end

end

```

```

theory Soundness-FRI-Query-Challenge-Fresh-Split
imports Soundness-FRI-Query-Fiber-Exact-Staged-Bounds
begin

```

Staged split for FRI query/challenge pair events.

The exact pair-fraction route must keep the query/challenge pair event until the challenge-list probability has been accounted for. This layer separates only the challenge prequery side event, leaving the fresh branch as the original pair hit conjoined with path freshness of the relevant FRI challenge keys.

```

context soundness
begin

```

lemma *verifier-after-trace-fri-concrete-header-transcript:*
fixes $s :: ('f, 'a) \text{ protocol-channel-scheme}$
assumes *prefix:*
Some ((fr, f-fl), prefix-state) ∈
set-dist (execute verifier-trace-fri-prefix s)
and *suffix:*
Some (result, final-state) ∈
set-dist
(execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
obtains *f-final as dg fl final query-state* **where**
verifier-header-transcript s fr (map snd f-fl) f-final as dg
(map snd fl) final (PTranscript query-state)
Some (result, final-state) ∈
set-dist
(execute
(ntimes
(verifier-query-round-program fr f-fl f-final as fl final)
rounds)
query-state)
proof –
have *prefix-res:*
PTranscript s = [fr] @ map snd f-fl @ PTranscript prefix-state ∧
PState prefix-state =
foldl concat (concat (PState s) fr) (map snd f-fl)
using *verifier-trace-fri-prefix-outcome[OF prefix] by simp*
have *len-f-fl: length f-fl = ceil-log clength*
using *verifier-trace-fri-prefix-outcome[OF prefix] by simp*
from *suffix* **obtain** *f-final s3 as s4 dg s5 s6 fl s7 final query-state* **where**
read-trace-final:
Some (f-final, s3) ∈ set-dist (execute read prefix-state)
and *alpha-out:*
Some (as, s4) ∈
set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)
and *read-dg: Some (dg, s5) ∈ set-dist (execute read s4)*
and *degree-assert:*
Some ((), s6) ∈
set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
and *comp-fri:*
Some (fl, s7) ∈
set-dist (execute
(ntimes receive-composition-fri-commits
(ceil-log (to-nat dg + 1))) s6)
and *read-final:*
Some (final, query-state) ∈ set-dist (execute read s7)
and *query-out:*
Some (result, final-state) ∈
set-dist (execute
(ntimes (verifier-query-round-program fr f-fl f-final as fl final)
rounds)

```

      query-state)
    unfolding verifier-after-trace-fri-def
    by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-trace-final] obtain rest3 where
    tr-prefix: PTranscript prefix-state = f-final # rest3
    and tr-s3: PTranscript s3 = rest3
    by blast
  have alpha-res:
    length as = length spec ∧
    PTranscript s3 = as @ PTranscript s4
    using mmap-alpha-round-outcome[OF alpha-out] by simp
  from read-outcome[OF read-dg] obtain rest5 where
    tr-s4: PTranscript s4 = dg # rest5
    and tr-s5: PTranscript s5 = rest5
    by blast
  have s6-eq: s6 = s5
    using degree-assert unfolding assert-def
    by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
  have comp-res:
    length fl = ceil-log (to-nat dg + 1) ∧
    PTranscript s6 = map snd fl @ PTranscript s7
    using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
    by simp
  from read-outcome[OF read-final] obtain rest-query where
    tr-s7: PTranscript s7 = final # rest-query
    and tr-query: PTranscript query-state = rest-query
    by blast
  have header:
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
    using prefix-res len-f-fl tr-prefix tr-s3 alpha-res tr-s4 tr-s5 s6-eq comp-res
    tr-s7 tr-query
    unfolding verifier-header-transcript-def verifier-header-messages-def
    by simp
  show thesis
    by (rule that[OF header query-out])
qed

```

lemma *verifier-after-trace-fri-header-agrees-with-accepted-challenges:*

```

assumes prefix:
  Some ((fr, f-fl), prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
and suffix:
  Some (result, final-state) ∈
    set-dist
      (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
and accepted:
  accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
  comp-bs

```

```

shows trace-bs = map fst f-fl
proof –
from verifier-after-trace-fri-accepted-fri-challenges[OF prefix suffix]
obtain dg' comp-bs' where replay:
  accepted-fri-challenges s (Some (result, final-state))
  (map fst f-fl) dg' comp-bs'
by blast
show ?thesis
using accepted-fri-challenges-unique[OF replay accepted] by simp
qed

lemma verifier-after-trace-fri-header-roots-agree-with-fresh-hit:
assumes prefix:
  Some ((fr, f-fl), prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
and suffix:
  Some (result, final-state) ∈
    set-dist
      (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
and fresh:
  trace-fri-challenge-list-fresh-hit s B
  (Some (result, final-state))
shows trace-fri-challenge-path-fresh s fr (map snd f-fl)
  (length (map snd f-fl))
proof –
from fresh obtain trace-bs dg comp-bs fr' trace-roots' trace-final' as'
  composition-roots' final' rest' where
  accepted:
    accepted-fri-challenges s (Some (result, final-state)) trace-bs dg
    comp-bs
and header-hit:
  verifier-header-transcript s fr' trace-roots' trace-final' as' dg
  composition-roots' final' rest'
and fresh-path:
  trace-fri-challenge-path-fresh s fr' trace-roots'
  (length trace-roots')
unfolding trace-fri-challenge-list-fresh-hit-def by blast
from verifier-after-trace-fri-accepted-fri-challenges[OF prefix suffix]
obtain dg0 comp-bs0 where replay:
  accepted-fri-challenges s (Some (result, final-state))
  (map fst f-fl) dg0 comp-bs0
by blast
have dg-eq: dg = dg0
using accepted-fri-challenges-unique[OF accepted replay] by simp
from verifier-after-trace-fri-concrete-header-transcript[OF prefix suffix]
obtain f-final0::'f and as0:: 'f list and dg1::'f and fl0:: ('f × 'f) list and final0:: 'f and
  query-state0:: ('f, 'a) protocol-channel-scheme where
  header-concrete:
    verifier-header-transcript s fr (map snd f-fl) f-final0 as0 dg1

```



```

    (map snd fl0) final0 (PTranscript query-state0)
  by metis
  from accepted-fri-challenges-headerE[OF replay]
  obtain fr0::'f and f-fl0::('f × 'f) list and f-final0'::'f and as0':: 'f list and fl0':: ('f × 'f)
  list and final0'::'f and rest0':: 'f list where
    header-replay:
      verifier-header-transcript s fr0 (map snd f-fl0) f-final0' as0'
      dg0 (map snd fl0') final0' rest0'
  by metis
  have replay-unique:
    fr0 = fr ∧ map snd f-fl0 = map snd f-fl ∧ dg0 = dg1
  using verifier-header-transcript-unique[OF header-replay header-concrete]
  by simp
  from accepted-fri-challenges-headerE[OF accepted]
  obtain fr1::'f and f-fl1::('f × 'f) list and f-final1::'f and as1:: 'f list and fl1:: ('f × 'f) list
  and final1::'f and rest1 where
    header-accepted:
      verifier-header-transcript s fr1 (map snd f-fl1) f-final1 as1 dg
      (map snd fl1) final1 rest1
  by metis
  have accepted-unique-header:
    fr1 = fr0 ∧ map snd f-fl1 = map snd f-fl0
  using verifier-header-transcript-unique[OF header-accepted header-replay]
  dg-eq
  by simp
  have hit-unique:
    fr' = fr1 ∧ trace-roots' = map snd f-fl1
  using verifier-header-transcript-unique[OF header-hit header-accepted]
  by simp
  show ?thesis
  using fresh-path hit-unique accepted-unique-header replay-unique by simp
qed

```

lemma *verifier-after-composition-fri-concrete-header-transcript:*

```

  fixes s :: ('f, 'a) protocol-channel-scheme
  assumes prefix:
    Some ((fr, f-fl, f-final, as, dg, fl), prefix-state) ∈
      set-dist (execute verifier-composition-fri-prefix s)
  and suffix:
    Some (result, final-state) ∈
      set-dist
        (execute
          (verifier-after-composition-fri
            (fr, f-fl, f-final, as, dg, fl))
          prefix-state)
  obtains final query-state where
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
      (map snd fl) final (PTranscript query-state)
    Some (result, final-state) ∈

```

```

    set-dist
    (execute
      (ntimes
        (verifier-query-round-program fr f-fl f-final as fl final)
        rounds)
      query-state)
  proof -
  have prefix-res:
    PTranscript s =
      [fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
      map snd fl @ PTranscript prefix-state
  using verifier-composition-fri-prefix-outcome[OF prefix] by simp
  from suffix obtain final query-state where
  read-final:
    Some (final, query-state) ∈
      set-dist (execute read prefix-state)
  and query-out:
    Some (result, final-state) ∈
      set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
  unfolding verifier-after-composition-fri-def
  by (auto elim!: set-dist-bindE)
  from read-outcome[OF read-final] obtain rest-query where
  tr-prefix: PTranscript prefix-state = final # rest-query
  and tr-query: PTranscript query-state = rest-query
  by blast
  have len-f-fl: length f-fl = ceil-log clength
  using verifier-composition-fri-prefix-outcome[OF prefix] by simp
  have len-as: length as = length spec
  using verifier-composition-fri-prefix-outcome[OF prefix] by simp
  have len-fl: length fl = ceil-log (to-nat dg + 1)
  using verifier-composition-fri-prefix-outcome[OF prefix] by simp
  have header:
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
      (map snd fl) final (PTranscript query-state)
  using prefix-res len-f-fl len-as len-fl tr-prefix tr-query
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
  show thesis
  by (rule that[OF header query-out])
qed

```

lemma *verifier-after-composition-fri-header-agrees-with-accepted-challenges:*
assumes *prefix:*
Some ((fr, f-fl, f-final, as, dg, fl), prefix-state) ∈

```

    set-dist (execute verifier-composition-fri-prefix s)
and suffix:
    Some (result, final-state) ∈
    set-dist
    (execute
    (verifier-after-composition-fri
    (fr, f-fl, f-final, as, dg, fl))
    prefix-state)
and accepted:
    accepted-fri-challenges s (Some (result, final-state)) trace-bs dg'
    comp-bs
shows dg' = dg ∧ comp-bs = map fst fl
proof –
have replay:
    accepted-fri-challenges s (Some (result, final-state))
    (map fst f-fl) dg (map fst fl)
by (rule verifier-after-composition-fri-accepted-fri-challenges
    [OF prefix suffix])
show ?thesis
    using accepted-fri-challenges-unique[OF replay accepted] by simp
qed

lemma verifier-after-composition-fri-header-roots-agree-with-fresh-hit:
assumes prefix:
    Some ((fr, f-fl, f-final, as, dg, fl), prefix-state) ∈
    set-dist (execute verifier-composition-fri-prefix s)
and suffix:
    Some (result, final-state) ∈
    set-dist
    (execute
    (verifier-after-composition-fri
    (fr, f-fl, f-final, as, dg, fl))
    prefix-state)
and fresh:
    composition-fri-challenge-list-fresh-hit s B
    (Some (result, final-state))
shows composition-fri-challenge-path-fresh s fr (map snd f-fl) f-final
    as dg (map snd fl) (length (map snd fl))
proof –
from fresh obtain trace-bs dg' comp-bs fr' trace-roots' trace-final' as'
    composition-roots' final' rest' where
    accepted:
    accepted-fri-challenges s (Some (result, final-state)) trace-bs dg'
    comp-bs
and header-hit:
    verifier-header-transcript s fr' trace-roots' trace-final' as' dg'
    composition-roots' final' rest'
and fresh-path:
    composition-fri-challenge-path-fresh s fr' trace-roots' trace-final'

```

```

    as' dg' composition-roots' (length composition-roots')
  unfolding composition-fri-challenge-list-fresh-hit-def by blast
from verifier-after-composition-fri-concrete-header-transcript[OF prefix suffix]
obtain final0:: 'f and query-state0:: ('f, 'a) protocol-channel-scheme where
  header-concrete:
    verifier-header-transcript s fr (map snd f-fl) f-final as dg
    (map snd fl) final0 (PTranscript query-state0)
  by blast
have fr' = fr ∧
  trace-roots' = map snd f-fl ∧
  trace-final' = f-final ∧
  as' = as ∧
  dg' = dg ∧
  composition-roots' = map snd fl
using verifier-header-transcript-unique[OF header-hit header-concrete]
by simp
then show ?thesis
  using fresh-path by simp
qed

```

definition staged-security-trace-fri-pair-challenge-fresh

```

:: (nat list × 'f list) set ⇒ 'f list set ⇒
  (((('f staged-proof-data × 'f protocol-channel) ×
    unit list) × 'f protocol-channel) option ⇒ bool

```

where

```

  staged-security-trace-fri-pair-challenge-fresh P B out ⟷
  staged-security-with-data-state-verifier-event
    (λs. trace-fri-query-challenge-pair-set-hit s P) out ∧
  staged-security-with-data-state-verifier-event
    (λs. trace-fri-challenge-list-fresh-hit s B) out

```

definition staged-security-composition-fri-pair-challenge-fresh

```

:: ('f ⇒ (nat list × 'f list) set) ⇒
  ('f ⇒ 'f list set) ⇒
  (((('f staged-proof-data × 'f protocol-channel) ×
    unit list) × 'f protocol-channel) option ⇒ bool

```

where

```

  staged-security-composition-fri-pair-challenge-fresh P B out ⟷
  staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-challenge-pair-set-hit s P) out ∧
  staged-security-with-data-state-verifier-event
    (λs. composition-fri-challenge-list-fresh-hit s B) out

```

lemma checked-staged-security-trace-fri-query-challenge-pair-set-hit-split-challenge-fresh:

assumes projection: $\text{snd} \, 'P \subseteq B$

shows

```

  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-query-challenge-pair-set-hit s P))

```

```

    adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-trace-fri-pair-challenge-fresh P B)
    adversary-initial-state +
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
  (λs. trace-fri-challenge-list-prequery-hit s B))
    adversary-initial-state
proof –
have event-mono:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
  (λs. trace-fri-query-challenge-pair-set-hit s P))
    adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
(λout.
  staged-security-trace-fri-pair-challenge-fresh P B out ∨
  staged-security-with-data-state-verifier-event
    (λs. trace-fri-challenge-list-prequery-hit s B) out)
    adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume hit:
  staged-security-with-data-state-verifier-event
    (λs. trace-fri-query-challenge-pair-set-hit s P) out
show
  staged-security-trace-fri-pair-challenge-fresh P B out ∨
  staged-security-with-data-state-verifier-event
    (λs. trace-fri-challenge-list-prequery-hit s B) out
proof (cases out)
case None
then show ?thesis
  using hit unfolding staged-security-with-data-state-verifier-event-def
  by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed) (auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from hit out-eq have pair-hit:
  trace-fri-query-challenge-pair-set-hit ?s P
  (Some (result, final-state))
  unfolding staged-security-with-data-state-verifier-event-def by simp
have challenge-hit:
  trace-fri-challenge-list-set-hit ?s B
  (Some (result, final-state))

```

```

    by (rule trace-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit
        [OF pair-hit projection])
  from trace-fri-challenge-list-set-hit-imp-fresh-or-prequery
    [OF challenge-hit]
  show ?thesis
proof
  assume fresh:
    trace-fri-challenge-list-fresh-hit ?s B
    (Some (result, final-state))
  then have staged-security-trace-fri-pair-challenge-fresh P B out
    using hit out-eq unfolding
      staged-security-trace-fri-pair-challenge-fresh-def
      staged-security-with-data-state-verifier-event-def
    by simp
  then show ?thesis by simp
next
  assume prequery:
    trace-fri-challenge-list-prequery-hit ?s B
    (Some (result, final-state))
  then have
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-challenge-list-prequery-hit s B) out
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
  then show ?thesis by simp
qed
qed
qed
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-trace-fri-pair-challenge-fresh P B)
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-challenge-list-prequery-hit s B))
    adversary-initial-state
  by (rule wp-event-union-bound)
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-fresh:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B: finite B
  and projection: snd ' P ⊆ B
  and fresh-branch-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-trace-fri-pair-challenge-fresh P B)
      adversary-initial-state ≤ F

```

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ P$))
adversary-initial-state $\leq$
 $F +$
hash-relation-budget-value (*card B* * *ceil-log clength*)
 (*staged-attacker-query-budget budgets* + *staged-challenge-query-budget*)

proof –

have *split*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ P$))
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-trace-fri-pair-challenge-fresh P B*)
adversary-initial-state +
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{trace-fri-challenge-list-prequery-hit } s \ B$))
adversary-initial-state
by (*rule*
checked-staged-security-trace-fri-query-challenge-pair-set-hit-split-challenge-fresh
 [*OF projection*])

have *prequery-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{trace-fri-challenge-list-prequery-hit } s \ B$))
adversary-initial-state $\leq$
hash-relation-budget-value (*card B* * *ceil-log clength*)
 (*staged-attacker-query-budget budgets* + *staged-challenge-query-budget*)
by (*rule checked-staged-security-trace-fri-verifier-prequery-hit-bound*
 [*OF wf controlled finite-B*])

have *wp-event* (*checked-staged-security-experiment-with-data-state A*)

(*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ P$))
adversary-initial-state $\leq$
 $F +$
hash-relation-budget-value (*card B* * *ceil-log clength*)
 (*staged-attacker-query-budget budgets* + *staged-challenge-query-budget*)

by (*rule order-trans*[*OF split*])
 (*rule add-mono*[*OF fresh-branch-bound prequery-bound*])

then show *?thesis* .

qed

lemma *checked-staged-security-composition-fri-query-challenge-pair-set-hit-split-challenge-fresh*:

assumes *projection*: $\bigwedge dg. \text{snd } P \ dg \subseteq B \ dg$

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)

```

    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-challenge-pair-set-hit s P))
    adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-composition-fri-pair-challenge-fresh P B)
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-prequery-hit s B))
    adversary-initial-state
proof –
have event-mono:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-challenge-pair-set-hit s P))
    adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (λout.
      staged-security-composition-fri-pair-challenge-fresh P B out ∨
      staged-security-with-data-state-verifier-event
        (λs. composition-fri-challenge-list-prequery-hit s B) out)
    adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume hit:
  staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-challenge-pair-set-hit s P) out
show
  staged-security-composition-fri-pair-challenge-fresh P B out ∨
  staged-security-with-data-state-verifier-event
    (λs. composition-fri-challenge-list-prequery-hit s B) out
proof (cases out)
case None
then show ?thesis
  using hit unfolding staged-security-with-data-state-verifier-event-def
  by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
by (cases packed) (auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from hit out-eq have pair-hit:
  composition-fri-query-challenge-pair-set-hit ?s P
  (Some (result, final-state))
unfolding staged-security-with-data-state-verifier-event-def by simp
have challenge-hit:

```



```

    composition-fri-challenge-list-set-hit ?s B
    (Some (result, final-state))
  by (rule
    composition-fri-query-challenge-pair-set-hit-imp-challenge-list-set-hit
    [OF pair-hit projection])
  from composition-fri-challenge-list-set-hit-imp-fresh-or-prequery
    [OF challenge-hit]
  show ?thesis
  proof
    assume fresh:
      composition-fri-challenge-list-fresh-hit ?s B
      (Some (result, final-state))
    then have
      staged-security-composition-fri-pair-challenge-fresh P B out
    using hit out-eq unfolding
      staged-security-composition-fri-pair-challenge-fresh-def
      staged-security-with-data-state-verifier-event-def
    by simp
    then show ?thesis by simp
  next
    assume prequery:
      composition-fri-challenge-list-prequery-hit ?s B
      (Some (result, final-state))
    then have
      staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-prequery-hit s B) out
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
    then show ?thesis by simp
  qed
  qed
  qed
  also have ... ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-composition-fri-pair-challenge-fresh P B)
    adversary-initial-state +
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-prequery-hit s B))
    adversary-initial-state
  by (rule wp-event-union-bound)
  finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-fresh:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and finite-B:  $\bigwedge dg. \text{finite } (B \ dg)$ 
  and projection:  $\bigwedge dg. \text{snd } ' P \ dg \subseteq B \ dg$ 

```

and *fresh-branch-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-composition-fri-pair-challenge-fresh P B*)
adversary-initial-state $\leq F$

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{composition-fri-query-challenge-pair-set-hit } s \ P$))
adversary-initial-state $\leq$
 $F +$
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$

proof –

have *split*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{composition-fri-query-challenge-pair-set-hit } s \ P$))
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-composition-fri-pair-challenge-fresh P B*)
adversary-initial-state $+$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{composition-fri-challenge-list-prequery-hit } s \ B$))
adversary-initial-state

by (*rule*
checked-staged-security-composition-fri-query-challenge-pair-set-hit-split-challenge-fresh
 $[OF \ \text{projection}]$)

have *prequery-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{composition-fri-challenge-list-prequery-hit } s \ B$))
adversary-initial-state $\leq$
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$

by (*rule* *checked-staged-security-composition-fri-verifier-prequery-hit-bound*
 $[OF \ wf \ \text{controlled } \text{finite-}B]$)

have *wp-event* (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{composition-fri-query-challenge-pair-set-hit } s \ P$))
adversary-initial-state $\leq$
 $F +$
hash-relation-budget-value
 $(\sum dg \in (UNIV :: 'f \text{ set}).$
 $\text{card } (B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$

```

    (staged-attacker-query-budget budgets + staged-challenge-query-budget)
  by (rule order-trans[OF split])
    (rule add-mono[OF fresh-branch-bound prequery-bound])
  then show ?thesis .
qed

end

end

```

```

theory Soundness-FRI-Query-Challenge-Actual-Product
imports Soundness-FRI-Query-Challenge-Fresh-Split
begin

```

Verifier-local pair-product bounds with actual FRI challenge-path freshness.

This replaces the global trace-FRI future-fresh premise in the fixed challenge product bound by the concrete freshness evidence carried by the trace FRI challenge-list freshness event.

```

context soundness
begin

```

lemma *verifier-trace-fri-prefix-preserves-query-lookup*:

assumes *outcome*:

Some ((fr, f-fl), t) ∈
set-dist (execute verifier-trace-fri-prefix s)

shows

fmlookup (HashMap t) (QueryIndexChallenge i x) =
fmlookup (HashMap s) (QueryIndexChallenge i x)

proof –

let *?key = QueryIndexChallenge i x*

from *outcome* **obtain** *s1 s2* **where**

read-fr: Some (fr, s1) ∈ set-dist (execute read s)

and *trace-fri*:

Some (f-fl, s2) ∈
set-dist (execute
(ntimes receive-trace-fri-commits (ceil-log clength)) s1)

and *ret*:

Some ((fr, f-fl), t) ∈ set-dist (execute (return (fr, f-fl)) s2)

unfolding *verifier-trace-fri-prefix-def*

by (*auto elim!: set-dist-bindE*)

have *lookup-s1*:

fmlookup (HashMap s1) ?key = fmlookup (HashMap s) ?key

using *read-preserves-hash-map[OF read-fr]* **by** *simp*

have *lookup-s2*:

fmlookup (HashMap s2) ?key = fmlookup (HashMap s1) ?key

proof (*rule ntimes-preserves-lookup[OF - trace-fri]*)

fix *y u v*

assume *out: Some (y, v) ∈ set-dist (execute receive-trace-fri-commits u)*

```

obtain  $b\ r$  where  $y\text{-eq}$ :  $y = (b, r)$ 
  by (cases  $y$ ) simp
show  $\text{fmlookup } (\text{HashMap } v) \text{ ?key} = \text{fmlookup } (\text{HashMap } u) \text{ ?key}$ 
  by (rule receive-trace-fri-commits-preserves-query-lookup
    [OF out[unfolded y-eq]])
qed
have  $t\text{-eq}$ :  $t = s2$ 
  using ret by simp
show ?thesis
  using lookup-s1 lookup-s2 unfolding t-eq by simp
qed

lemma wp-trace-fri-query-challenge-pair-set-hit-fixed-challenge-actual-fresh-bound:
assumes query-future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and challenges-space:
   $\text{challenges} \in \text{fri-challenge-space } (\text{ceil-log } \text{clength})$ 
and projection:
   $\text{fst } ' ( \text{Pairs} \cap ( \text{UNIV} \times \{ \text{challenges} \} ) ) \subseteq \text{Queries}$ 
and subset:  $\text{Queries} \subseteq \text{fri-query-index-list-space}$ 
shows
  wp-event verify-monad
    ( $\lambda \text{out. trace-fri-challenge-list-fresh-hit } s \{ \text{challenges} \} \text{ out} \wedge$ 
       $\text{trace-fri-query-challenge-pair-set-hit } s \text{ Pairs out} ) s \leq$ 
    ( $1 / \text{nnreal } (\text{CARD } 'f) ^{\text{ceil-log } \text{clength}} ) *$ 
    ( $\text{nnreal } (\text{card } \text{Queries}) *$ 
      ( $1 / \text{nnreal } (\text{card } \text{query-sample-space}) ) ^{\text{rounds}} )$ 
proof –
  let ?Hit =
     $\lambda \text{out. trace-fri-challenge-list-fresh-hit } s \{ \text{challenges} \} \text{ out} \wedge$ 
     $\text{trace-fri-query-challenge-pair-set-hit } s \text{ Pairs out}$ 
  let ?Head =
     $\lambda \text{out. case out of}$ 
     $\text{None} \Rightarrow \text{False}$ 
    |  $\text{Some } ((\text{fr}, \text{f-fl}), -) \Rightarrow$ 
       $\text{map fst f-fl} = \text{challenges} \wedge$ 
       $\text{trace-fri-challenge-path-fresh } s \text{ fr } (\text{map snd f-fl})$ 
      ( $\text{length } (\text{map snd f-fl})$ )
  let ?D =
     $\text{nnreal } (\text{card } \text{Queries}) *$ 
    ( $1 / \text{nnreal } (\text{card } \text{query-sample-space}) ) ^{\text{rounds}}$ 
have bind-bound:
  wp-event verify-monad ?Hit s  $\leq$ 
  wp-event verifier-trace-fri-prefix ?Head s  $*$  ?D
unfolding verify-monad-trace-fri-decomposition
proof (rule wp-event-bind-bound-by-head-and-cont)
show  $\neg ?Hit \text{ None}$ 
unfolding trace-fri-challenge-list-fresh-hit-def
  trace-fri-query-challenge-pair-set-hit-def

```

```

    accepted-fri-challenges-def
    accepted-fri-opening-transcript-def
  by blast
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix s)
  and not-head:  $\neg$  ?Head (Some (header, prefix-state))
obtain fr f-fl where header-eq: header = (fr, f-fl)
  by (cases header) auto
have not-head':
  map fst f-fl  $\neq$  challenges  $\vee$ 
     $\neg$  trace-fri-challenge-path-fresh s fr (map snd f-fl)
    (length (map snd f-fl))
  using not-head unfolding header-eq by simp
have zero-le:
  wp-event (verifier-after-trace-fri header) ?Hit prefix-state  $\leq$  0
proof (rule order-trans)
  show wp-event (verifier-after-trace-fri header) ?Hit prefix-state  $\leq$ 
    wp-event (verifier-after-trace-fri header)
    ( $\lambda$ -. False) prefix-state
proof (rule wp-event-mono-on-support)
  fix out
  assume suffix:
    out ∈
      set-dist (execute (verifier-after-trace-fri header) prefix-state)
  and hit: ?Hit out
  from hit obtain result :: unit list and final-state where out-eq:
    out = Some (result, final-state)
  unfolding trace-fri-challenge-list-fresh-hit-def
    accepted-fri-challenges-def
  by blast
  have prefix':
    Some ((fr, f-fl), prefix-state) ∈
      set-dist (execute verifier-trace-fri-prefix s)
  using prefix header-eq by simp
  have suffix':
    Some (result, final-state) ∈
      set-dist
        (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
  using suffix header-eq out-eq by simp
  from hit obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers where
    openings:
      accepted-fri-opening-transcript s out trace-roots trace-bs
      trace-final fri-dg composition-roots composition-bs
      composition-final query-idxs trace-round-layers

```

```

    composition-round-layers
  unfolding trace-fri-query-challenge-pair-set-hit-def by blast
have pair-challenges:
  accepted-fri-challenges s out trace-bs fri-dg composition-bs
  by (rule accepted-fri-opening-transcript-challenges[OF openings])
from hit obtain trace-bs' dg' comp-bs' where
  fresh-challenges:
    accepted-fri-challenges s out trace-bs' dg' comp-bs'
  and trace-bs'-in: trace-bs' ∈ {challenges}
  unfolding trace-fri-challenge-list-fresh-hit-def by blast
have trace-bs-eq: trace-bs = trace-bs'
  using accepted-fri-challenges-unique
  [OF pair-challenges[unfolded out-eq]
    fresh-challenges[unfolded out-eq]]
  by simp
have prefix-trace-eq: trace-bs = map fst f-fl
  by (rule verifier-after-trace-fri-header-agrees-with-accepted-challenges
    [OF prefix' suffix' pair-challenges[unfolded out-eq]])
have fresh-hit:
  trace-fri-challenge-list-fresh-hit s {challenges}
  (Some (result, final-state))
  using hit out-eq by simp
have concrete-fresh:
  trace-fri-challenge-path-fresh s fr (map snd f-fl)
  (length (map snd f-fl))
  by (rule verifier-after-trace-fri-header-roots-agree-with-fresh-hit
    [OF prefix' suffix' fresh-hit])
have head-holds:
  map fst f-fl = challenges ∧
  trace-fri-challenge-path-fresh s fr (map snd f-fl)
  (length (map snd f-fl))
  using prefix-trace-eq trace-bs-eq trace-bs'-in concrete-fresh by simp
show False
  using not-head' head-holds by blast
qed
next
show wp-event (verifier-after-trace-fri header)
  (λ-. False) prefix-state ≤ 0
  unfolding wp-event-def wp-def dist-expect-def by simp
qed
show wp-event (verifier-after-trace-fri header) ?Hit prefix-state = 0
  by (rule antisym[OF zero-le]) simp
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state) ∈
  set-dist (execute verifier-trace-fri-prefix s)
  and head: ?Head (Some (header, prefix-state))
obtain fr f-fl where header-eq: header = (fr, f-fl)

```

```

by (cases header) auto
have f-fl-eq: map fst f-fl = challenges
  using head unfolding header-eq by simp
have future-prefix: query-future-fresh prefix-state
  by (rule verifier-trace-fri-prefix-preserves-query-future-fresh-exact
    [OF query-future prefix])
have counter-prefix: PQueryCounter prefix-state = PQueryCounter s
  using verifier-trace-fri-prefix-outcome[OF prefix[unfolded header-eq]]
  by simp
have event-bound:
  wp-event (verifier-after-trace-fri header)
    (λout. ∃ query-idxs ∈ Queries.
      query-rounds-index-list-hit s query-idxs rounds out)
    prefix-state ≤ ?D
  by (rule wp-verifier-after-trace-fri-query-index-list-set-bound
    [OF future-prefix counter-prefix raw-bound subset])
have hit-to-query:
  wp-event (verifier-after-trace-fri header) ?Hit prefix-state ≤
    wp-event (verifier-after-trace-fri header)
      (λout. ∃ query-idxs ∈ Queries.
        query-rounds-index-list-hit s query-idxs rounds out) prefix-state
proof (rule wp-event-mono-on-support)
  fix out
  assume suffix:
    out ∈
      set-dist (execute (verifier-after-trace-fri header) prefix-state)
  and hit: ?Hit out
  from hit obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers where
    openings:
      accepted-fri-opening-transcript s out trace-roots trace-bs
      trace-final fri-dg composition-roots composition-bs
      composition-final query-idxs trace-round-layers
      composition-round-layers
  and pair: (query-idxs, trace-bs) ∈ Pairs
  unfolding trace-fri-query-challenge-pair-set-hit-def by blast
  from openings obtain result :: unit list and final-state where out-eq:
    out = Some (result, final-state)
  unfolding accepted-fri-opening-transcript-def by blast
  have prefix':
    Some ((fr, f-fl), prefix-state) ∈
      set-dist (execute verifier-trace-fri-prefix s)
  using prefix header-eq by simp
  have suffix':
    Some (result, final-state) ∈
      set-dist
        (execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)
  using suffix header-eq out-eq by simp

```

```

have challenges-openings:
  accepted-fri-challenges s out trace-bs fri-dg composition-bs
  by (rule accepted-fri-opening-transcript-challenges[OF openings])
have trace-eq: trace-bs = map fst f-fl
  by (rule verifier-after-trace-fri-header-agrees-with-accepted-challenges
    [OF prefix' suffix' challenges-openings[unfolded out-eq]])
have query-in: query-idxs ∈ Queries
proof –
  have query-idxs ∈ fst ‘ (Pairs ∩ (UNIV × {challenges}))
    using pair trace-eq f-fl-eq by force
  then show ?thesis
    using projection by blast
qed
have query-hit:
  query-rounds-index-list-hit s query-idxs rounds out
  by (rule accepted-fri-opening-transcript-imp-query-rounds-index-list-hit
    [OF openings])
show ∃ query-idxs ∈ Queries.
  query-rounds-index-list-hit s query-idxs rounds out
  using query-in query-hit by blast
qed
show wp-event (verifier-after-trace-fri header) ?Hit prefix-state ≤ ?D
  by (rule order-trans[OF hit-to-query event-bound])
qed
have head-bound:
  wp-event verifier-trace-fri-prefix ?Head s ≤
    1 / nnreal (CARD('f) ^ ceil-log clength)
proof –
  let ?HeadSet =
    λout. case out of
      None ⇒ False
    | Some ((fr, f-fl), -) ⇒
      map fst f-fl ∈ {challenges} ∧
      trace-fri-challenge-path-fresh s fr (map snd f-fl)
      (length (map snd f-fl))
  have subset: {challenges} ⊆ fri-challenge-space (ceil-log clength)
    using challenges-space by simp
  have mono:
    wp-event verifier-trace-fri-prefix ?Head s ≤
      wp-event verifier-trace-fri-prefix ?HeadSet s
    by (rule wp-event-mono) (auto split: option.splits prod.splits)
  have wp-event verifier-trace-fri-prefix ?Head s ≤
    nnreal (card {challenges}) /
    nnreal (CARD('f) ^ ceil-log clength)
    by (rule order-trans[OF mono])
    (rule wp-verifier-trace-fri-prefix-challenge-list-actual-fresh-bound
      [OF subset])
  then show ?thesis
    by simp

```


qed
have *wp-event verify-monad* ?Hit $s \leq$
 wp-event verifier-trace-fri-prefix ?Head $s * ?D$
by (*rule bind-bound*)
also have ... $\leq$
 $(1 / \text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log clength})) * ?D$
by (*rule mult-right-mono*[*OF head-bound*]) *simp*
finally show ?thesis .
qed

lemma *wp-trace-fri-query-challenge-pair-set-hit-challenge-fresh-fiber-sum*:
assumes *query-future: query-future-fresh* s
and *raw-bound: query-index-raw-preimage-bound*
and *challenge-subset: $B \subseteq \text{fri-challenge-space} (\text{ceil-log clength})$*
and *projection:*
 $\bigwedge \text{challenges. challenges} \in B \implies$
 $\text{fst } ' (P \cap (\text{UNIV} \times \{\text{challenges}\})) \subseteq Q \text{ challenges}$
and *subset:*
 $\bigwedge \text{challenges. challenges} \in B \implies$
 $Q \text{ challenges} \subseteq \text{fri-query-index-list-space}$
shows
 wp-event verify-monad
 $(\lambda \text{out. trace-fri-challenge-list-fresh-hit } s \text{ } B \text{ out} \wedge$
 $\text{trace-fri-query-challenge-pair-set-hit } s \text{ } P \text{ out}) \text{ } s \leq$
 $(\sum \text{challenges} \in B.$
 $(1 / \text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log clength})) *$
 $(\text{nnreal} (\text{card } (Q \text{ challenges})) *$
 $(1 / \text{nnreal} (\text{card query-sample-space})) \wedge \text{rounds}))$
proof –
let ?E =
 $\lambda \text{challenges out.}$
 $\text{trace-fri-challenge-list-fresh-hit } s \text{ } \{\text{challenges}\} \text{ out} \wedge$
 $\text{trace-fri-query-challenge-pair-set-hit } s \text{ } P \text{ out}$
let ?C =
 $\lambda \text{challenges.}$
 $(1 / \text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log clength})) *$
 $(\text{nnreal} (\text{card } (Q \text{ challenges})) *$
 $(1 / \text{nnreal} (\text{card query-sample-space})) \wedge \text{rounds})$
have *split:*
 wp-event verify-monad
 $(\lambda \text{out. trace-fri-challenge-list-fresh-hit } s \text{ } B \text{ out} \wedge$
 $\text{trace-fri-query-challenge-pair-set-hit } s \text{ } P \text{ out}) \text{ } s \leq$
 wp-event verify-monad
 $(\lambda \text{out. } \exists \text{challenges} \in B. ?E \text{ challenges out}) \text{ } s$
proof (*rule wp-event-mono*)
fix out
assume hit:
 $\text{trace-fri-challenge-list-fresh-hit } s \text{ } B \text{ out} \wedge$

```

    trace-fri-query-challenge-pair-set-hit s P out
then obtain trace-bs :: 'f list and dg :: 'f and comp-bs :: 'f list
    and fr :: 'f and trace-roots :: 'f list and trace-final :: 'f
    and as :: 'f list and composition-roots :: 'f list
    and final :: 'f and rest :: 'f list where
    accepted:
        accepted-fri-challenges s out trace-bs dg comp-bs
and header:
        verifier-header-transcript s fr trace-roots trace-final as dg
        composition-roots final rest
and trace-bs-in: trace-bs ∈ B
and fresh-path:
        trace-fri-challenge-path-fresh s fr trace-roots
        (length trace-roots)
    unfolding trace-fri-challenge-list-fresh-hit-def by blast
have singleton-fresh:
    trace-fri-challenge-list-fresh-hit s {trace-bs} out
    unfolding trace-fri-challenge-list-fresh-hit-def
proof (intro exI conjI)
    show accepted-fri-challenges s out trace-bs dg comp-bs
        by (rule accepted)
    show verifier-header-transcript s fr trace-roots trace-final as dg
        composition-roots final rest
        by (rule header)
    show trace-bs ∈ {trace-bs}
        by simp
    show trace-fri-challenge-path-fresh s fr trace-roots
        (length trace-roots)
        by (rule fresh-path)
qed

    show ∃ challenges ∈ B. ?E challenges out
        using singleton-fresh hit trace-bs-in
        by (intro bexI[of - trace-bs]) simp-all
qed
also have ... ≤ (∑ challenges ∈ B. ?C challenges)
proof (rule wp-event-finite-union-bound-fri-exact)
    show finite B
        using challenge-subset by (rule finite-subset)
        (rule finite-fri-challenge-space)
next
    fix challenges
    assume challenges-in: challenges ∈ B
    have challenges-space:
        challenges ∈ fri-challenge-space (ceil-log clength)
        using challenge-subset challenges-in by blast
    show wp-event verify-monad (?E challenges) s ≤ ?C challenges
        by (rule
            wp-trace-fri-query-challenge-pair-set-hit-fixed-challenge-actual-fresh-bound

```

```

    [OF query-future raw-bound challenges-space
      projection[OF challenges-in] subset[OF challenges-in]])
qed
finally show ?thesis .
qed

lemma wp-composition-fri-query-challenge-pair-set-hit-fixed-challenge-actual-fresh-bound:
  assumes query-future: query-future-fresh s
    and raw-bound: query-index-raw-preimage-bound
    and challenges-space:
      challenges ∈ fri-challenge-space (ceil-log (to-nat dg + 1))
    and projection:
      fst ‘ (Pairs dg ∩ (UNIV × {challenges})) ⊆ Queries
    and subset: Queries ⊆ fri-query-index-list-space
  shows
    wp-event verify-monad
      (λout.
        composition-fri-challenge-list-fresh-hit s
          (λdg'. if dg' = dg then {challenges} else {}) out ∧
          composition-fri-query-challenge-pair-set-hit s Pairs out) s ≤
      (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
      (nnreal (card Queries) *
        (1 / nnreal (card query-sample-space)) ^ rounds)
  proof -
    let ?B = λdg'. if dg' = dg then {challenges} else {}
    let ?Hit =
      λout. composition-fri-challenge-list-fresh-hit s ?B out ∧
        composition-fri-query-challenge-pair-set-hit s Pairs out
    let ?Head =
      λout. case out of
        None ⇒ False
      | Some ((fr, f-fl, f-final, as, dg', fl), -) ⇒
        dg' = dg ∧ map fst fl = challenges ∧
        composition-fri-challenge-path-fresh s fr (map snd f-fl)
        f-final as dg' (map snd fl) (length (map snd fl))
    let ?D =
      nnreal (card Queries) *
      (1 / nnreal (card query-sample-space)) ^ rounds
    have bind-bound:
      wp-event verify-monad ?Hit s ≤
        wp-event verifier-composition-fri-prefix ?Head s * ?D
    unfolding verify-monad-composition-fri-decomposition
  proof (rule wp-event-bind-bound-by-head-and-cont)
    show ¬ ?Hit None
    unfolding composition-fri-challenge-list-fresh-hit-def
      composition-fri-query-challenge-pair-set-hit-def
      accepted-fri-challenges-def
      accepted-fri-opening-transcript-def
    by blast
  end

```

```

next
  fix header prefix-state
  assume prefix:
    Some (header, prefix-state) ∈
      set-dist (execute verifier-composition-fri-prefix s)
  and not-head:  $\neg$  ?Head (Some (header, prefix-state))
  obtain fr f-fl f-final as dg' fl where header-eq:
    header = (fr, f-fl, f-final, as, dg', fl)
  by (cases header) auto
  have not-head':
    dg'  $\neq$  dg  $\vee$  map fst fl  $\neq$  challenges  $\vee$ 
       $\neg$  composition-fri-challenge-path-fresh s fr (map snd f-fl)
      f-final as dg' (map snd fl) (length (map snd fl))
  using not-head unfolding header-eq by simp
  have zero-le:
    wp-event (verifier-after-composition-fri header) ?Hit prefix-state  $\leq$  0
  proof (rule order-trans)
  show wp-event (verifier-after-composition-fri header) ?Hit prefix-state  $\leq$ 
    wp-event (verifier-after-composition-fri header)
      ( $\lambda$ -. False) prefix-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume suffix:
    out ∈
      set-dist
        (execute (verifier-after-composition-fri header) prefix-state)
  and hit: ?Hit out
  from hit obtain result :: unit list and final-state where out-eq:
    out = Some (result, final-state)
  unfolding composition-fri-challenge-list-fresh-hit-def
    accepted-fri-challenges-def
  by blast
  have prefix':
    Some ((fr, f-fl, f-final, as, dg', fl), prefix-state) ∈
      set-dist (execute verifier-composition-fri-prefix s)
  using prefix header-eq by simp
  have suffix':
    Some (result, final-state) ∈
      set-dist
        (execute
          (verifier-after-composition-fri
            (fr, f-fl, f-final, as, dg', fl))
          prefix-state)
  using suffix header-eq out-eq by simp
  from hit obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers where
  openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs

```

```

    trace-final fri-dg composition-roots composition-bs
    composition-final query-idxs trace-round-layers
    composition-round-layers
  unfolding composition-fri-query-challenge-pair-set-hit-def by blast
have pair-challenges:
  accepted-fri-challenges s out trace-bs fri-dg composition-bs
  by (rule accepted-fri-opening-transcript-challenges[OF openings])
from hit obtain trace-bs' dg'' comp-bs' where
  fresh-challenges:
    accepted-fri-challenges s out trace-bs' dg'' comp-bs'
  and comp-bs'-in:
    comp-bs' ∈ (if dg'' = dg then {challenges} else {})
  unfolding composition-fri-challenge-list-fresh-hit-def by blast
have challenge-eqs:
  fri-dg = dg'' ∧ composition-bs = comp-bs'
  using accepted-fri-challenges-unique
  [OF pair-challenges[unfolded out-eq]
   fresh-challenges[unfolded out-eq]]
  by simp
have prefix-eqs:
  fri-dg = dg' ∧ composition-bs = map fst fl
  by (rule verifier-after-composition-fri-header-agrees-with-accepted-challenges
      [OF prefix' suffix' pair-challenges[unfolded out-eq]])
have fresh-hit:
  composition-fri-challenge-list-fresh-hit s ?B
  (Some (result, final-state))
  using hit out-eq by simp
have concrete-fresh:
  composition-fri-challenge-path-fresh s fr (map snd f-fl)
  f-final as dg' (map snd fl) (length (map snd fl))
  by (rule verifier-after-composition-fri-header-roots-agree-with-fresh-hit
      [OF prefix' suffix' fresh-hit])
have dg-eq': dg'' = dg
  using comp-bs'-in by (cases dg'' = dg) auto
have comp-bs'-eq: comp-bs' = map fst fl
  using challenge-eqs prefix-eqs by simp
have comp-bs'-challenges: comp-bs' = challenges
  using comp-bs'-in dg-eq' by simp
have dg'-eq: dg' = dg
  using challenge-eqs prefix-eqs dg-eq' by simp
have fl-challenges: map fst fl = challenges
  using comp-bs'-eq comp-bs'-challenges by simp
have head-holds:
  dg' = dg ∧ map fst fl = challenges ∧
  composition-fri-challenge-path-fresh s fr (map snd f-fl)
  f-final as dg' (map snd fl) (length (map snd fl))
  using dg'-eq fl-challenges concrete-fresh by simp
show False
  using not-head' head-holds by blast

```

```

qed
next
  show wp-event (verifier-after-composition-fri header)
    ( $\lambda$ -. False) prefix-state  $\leq 0$ 
    unfolding wp-event-def wp-def dist-expect-def by simp
qed
show wp-event (verifier-after-composition-fri header) ?Hit prefix-state = 0
  by (rule antisym[OF zero-le]) simp
next
  fix header prefix-state
  assume prefix:
    Some (header, prefix-state)  $\in$ 
      set-dist (execute verifier-composition-fri-prefix s)
    and head: ?Head (Some (header, prefix-state))
  obtain fr f-fl f-final as dg' fl where header-eq:
    header = (fr, f-fl, f-final, as, dg', fl)
  by (cases header) auto
  have dg-eq: dg' = dg and fl-eq: map fst fl = challenges
    using head unfolding header-eq by simp-all
  have future-prefix: query-future-fresh prefix-state
    by (rule verifier-composition-fri-prefix-preserves-query-future-fresh
      [OF query-future prefix])
  have counter-prefix: PQueryCounter prefix-state = PQueryCounter s
    using verifier-composition-fri-prefix-outcome[OF prefix[unfolded header-eq]]
    by simp
  have event-bound:
    wp-event (verifier-after-composition-fri header)
      ( $\lambda$ out.  $\exists$  query-idxs  $\in$  Queries.
        query-rounds-index-list-hit s query-idxs rounds out)
      prefix-state  $\leq$  ?D
    by (rule wp-verifier-after-composition-fri-query-index-list-set-bound
      [OF future-prefix counter-prefix raw-bound subset])
  have hit-to-query:
    wp-event (verifier-after-composition-fri header) ?Hit prefix-state  $\leq$ 
      wp-event (verifier-after-composition-fri header)
        ( $\lambda$ out.  $\exists$  query-idxs  $\in$  Queries.
          query-rounds-index-list-hit s query-idxs rounds out) prefix-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume suffix:
      out  $\in$ 
        set-dist
          (execute (verifier-after-composition-fri header) prefix-state)
    and hit: ?Hit out
    from hit obtain trace-roots trace-bs trace-final fri-dg
      composition-roots composition-bs composition-final query-idxs
      trace-round-layers composition-round-layers where
      openings:
        accepted-fri-opening-transcript s out trace-roots trace-bs

```

```

    trace-final fri-dg composition-roots composition-bs
    composition-final query-idxs trace-round-layers
    composition-round-layers
  and pair: (query-idxs, composition-bs) ∈ Pairs fri-dg
  unfolding composition-fri-query-challenge-pair-set-hit-def by blast
from openings obtain result :: unit list and final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-fri-opening-transcript-def by blast
have prefix':
  Some ((fr, f-fl, f-final, as, dg', fl), prefix-state) ∈
    set-dist (execute verifier-composition-fri-prefix s)
  using prefix header-eq by simp
have suffix':
  Some (result, final-state) ∈
    set-dist
      (execute
        (verifier-after-composition-fri
          (fr, f-fl, f-final, as, dg', fl))
        prefix-state)
  using suffix header-eq out-eq by simp
have challenges-openings:
  accepted-fri-challenges s out trace-bs fri-dg composition-bs
  by (rule accepted-fri-opening-transcript-challenges[OF openings])
have eqs: fri-dg = dg' ∧ composition-bs = map fst fl
  by (rule verifier-after-composition-fri-header-agrees-with-accepted-challenges
    [OF prefix' suffix' challenges-openings[unfolded out-eq]])
have query-in: query-idxs ∈ Queries
proof -
  have query-idxs ∈ fst ' (Pairs dg ∩ (UNIV × {challenges}))
    using pair eqs dg-eq fl-eq by force
  then show ?thesis
    using projection by blast
qed
have query-hit:
  query-rounds-index-list-hit s query-idxs rounds out
  by (rule accepted-fri-opening-transcript-imp-query-rounds-index-list-hit
    [OF openings])
show ∃ query-idxs ∈ Queries.
  query-rounds-index-list-hit s query-idxs rounds out
  using query-in query-hit by blast
qed
show wp-event (verifier-after-composition-fri header) ?Hit prefix-state ≤
  ?D
  by (rule order-trans[OF hit-to-query event-bound])
qed
have head-bound:
  wp-event verifier-composition-fri-prefix ?Head s ≤
    1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
proof -

```

```

let ?B =  $\lambda dg'. \text{ if } dg' = dg \text{ then } \{\text{challenges}\} \text{ else } \{\}$ 
let ?HeadSet =
   $\lambda out. \text{ case out of }$ 
    None  $\Rightarrow$  False
    | Some ((fr, f-fl, f-final, as, dg', fl), -)  $\Rightarrow$ 
      map fst fl  $\in$  ?B dg'  $\wedge$ 
      composition-fri-challenge-path-fresh s fr (map snd f-fl)
      f-final as dg' (map snd fl) (length (map snd fl))
have subset:
   $\bigwedge dg'. ?B dg' \subseteq$ 
    fri-challenge-space (ceil-log (to-nat dg' + 1))
  using challenges-space by auto
have bound:
   $\bigwedge dg'. \text{ nreal } (\text{card } (?B dg')) /$ 
    nreal (CARD('f)  $\wedge$  ceil-log (to-nat dg' + 1))  $\leq$ 
    1 / nreal (CARD('f)  $\wedge$  ceil-log (to-nat dg + 1))
  using challenges-space by auto
have mono:
  wp-event verifier-composition-fri-prefix ?Head s  $\leq$ 
  wp-event verifier-composition-fri-prefix ?HeadSet s
  by (rule wp-event-mono) (auto split: option.splits prod.splits)
show ?thesis
  by (rule order-trans[OF mono])
    (rule wp-verifier-composition-fri-prefix-challenge-list-actual-fresh-bound
      [OF subset bound])
qed
have wp-event verify-mono ?Hit s  $\leq$ 
  wp-event verifier-composition-fri-prefix ?Head s * ?D
  by (rule bind-bound)
also have ...  $\leq$ 
  (1 / nreal (CARD('f)  $\wedge$  ceil-log (to-nat dg + 1))) * ?D
  by (rule mult-right-mono[OF head-bound]) simp
finally show ?thesis .
qed

lemma wp-composition-fri-query-challenge-pair-set-hit-challenge-fresh-fiber-sum:
assumes query-future: query-future-fresh s
and raw-bound: query-index-raw-preimage-bound
and challenge-subset:
   $\bigwedge dg. B dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$ 
and projection:
   $\bigwedge dg \text{ challenges. challenges } \in B dg \Rightarrow$ 
    fst ' (P dg  $\cap$  (UNIV  $\times$  {challenges}))  $\subseteq$ 
    Q dg challenges
and subset:
   $\bigwedge dg \text{ challenges. challenges } \in B dg \Rightarrow$ 
    Q dg challenges  $\subseteq \text{fri-query-index-list-space}$ 
shows

```



```

wp-event verify-monad
  (λout. composition-fri-challenge-list-fresh-hit s B out ∧
    composition-fri-query-challenge-pair-set-hit s P out) s ≤
  (Σ dg ∈ UNIV.
    Σ challenges ∈ B dg.
      (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
      (nnreal (card (Q dg challenges)) *
        (1 / nnreal (card query-sample-space)) ^ rounds))
proof –
  let ?I = SIGMA dg:UNIV. B dg
  let ?E =
    λp out.
      composition-fri-challenge-list-fresh-hit s
        (λdg'. if dg' = fst p then {snd p} else {}) out ∧
      composition-fri-query-challenge-pair-set-hit s P out
  let ?C =
    λp.
      (1 / nnreal (CARD('f) ^ ceil-log (to-nat (fst p) + 1))) *
      (nnreal (card (Q (fst p) (snd p))) *
        (1 / nnreal (card query-sample-space)) ^ rounds)
  have split:
    wp-event verify-monad
      (λout. composition-fri-challenge-list-fresh-hit s B out ∧
        composition-fri-query-challenge-pair-set-hit s P out) s ≤
    wp-event verify-monad
      (λout. ∃ p ∈ ?I. ?E p out) s
  proof (rule wp-event-mono)
  fix out
  assume hit:
    composition-fri-challenge-list-fresh-hit s B out ∧
    composition-fri-query-challenge-pair-set-hit s P out
  then obtain trace-bs dg comp-bs fr trace-roots trace-final as
    composition-roots final rest where
    accepted:
      accepted-fri-challenges s out trace-bs dg comp-bs
  and comp-bs-in: comp-bs ∈ B dg
  and header:
    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots final rest
  and fresh-path:
    composition-fri-challenge-path-fresh s fr trace-roots trace-final
    as dg composition-roots (length composition-roots)
  unfolding composition-fri-challenge-list-fresh-hit-def by blast
  have singleton-fresh:
    composition-fri-challenge-list-fresh-hit s
      (λdg'. if dg' = dg then {comp-bs} else {}) out
  unfolding composition-fri-challenge-list-fresh-hit-def
  proof (intro exI conjI)
  show accepted-fri-challenges s out trace-bs dg comp-bs

```

```

    by (rule accepted)
  show verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots final rest
    by (rule header)
  show comp-bs ∈ (if dg = dg then {comp-bs} else {})
    by simp
  show composition-fri-challenge-path-fresh s fr trace-roots trace-final
    as dg composition-roots (length composition-roots)
    by (rule fresh-path)
qed
have p-in: (dg, comp-bs) ∈ ?I
  using comp-bs-in by simp
have singleton-fresh':
  composition-fri-challenge-list-fresh-hit s
  (λdg'. if dg' = fst (dg, comp-bs) then {snd (dg, comp-bs)}
    else {}) out
  using singleton-fresh by (simp only: fst-conv snd-conv)
have e-hit: ?E (dg, comp-bs) out
  using singleton-fresh' hit by simp
show ∃ p ∈ ?I. ?E p out
  using p-in e-hit by blast
qed
also have ... ≤ (∑ p ∈ ?I. ?C p)
proof (rule wp-event-finite-union-bound-fri-exact)
  show finite ?I
    apply (rule finite-SigmaI)
    using challenge-subset finite-fri-challenge-space
    apply (auto intro: finite-subset)
    by (meson finite-fri-challenge-space rev-finite-subset)
next
fix p
assume p-in: p ∈ ?I
obtain dg challenges where p-eq: p = (dg, challenges)
  by (cases p) auto
have challenges-in: challenges ∈ B dg
  using p-in p-eq by simp
have challenges-space:
  challenges ∈ fri-challenge-space (ceil-log (to-nat dg + 1))
  using challenge-subset challenges-in by blast
show wp-event verify-monad (?E p) s ≤ ?C p
  unfolding p-eq
  apply (rule wp-composition-fri-query-challenge-pair-set-hit-fixed-challenge-actual-fresh-bound)
  using query-future raw-bound challenges-space
    projection[OF challenges-in] subset[OF challenges-in]
  by auto
qed
finally have bound:
  wp-event verify-monad
  (λout. composition-fri-challenge-list-fresh-hit s B out ∧

```

$\text{composition-fri-query-challenge-pair-set-hit } s \ P \ \text{out}) \ s \leq$
 $(\sum p \in ?I. ?C \ p) \ .$
have *finite-B*: $\bigwedge dg. \text{finite } (B \ dg)$
by (*meson challenge-subset finite-fri-challenge-space finite-subset*)
have *sigma-eq*:
 $(\sum p \in ?I. ?C \ p) =$
 $(\sum dg \in UNIV.$
 $\sum challenges \in B \ dg.$
 $(1 / \text{nnreal } (CARD('f) \wedge \text{ceil-log } (to-nat \ dg + 1))) *$
 $(\text{nnreal } (card \ (Q \ dg \ challenges)) *$
 $(1 / \text{nnreal } (card \ \text{query-sample-space})) \wedge \text{rounds}))$
by (*subst sum.Sigma*) (*simp-all add: finite-B case-prod-beta*)
show *?thesis*
using *bound sigma-eq by simp*
qed

definition *staged-security-trace-fri-pair-challenge-query-path-fresh*

$:: (\text{nat list} \times 'f \ \text{list}) \ \text{set} \Rightarrow 'f \ \text{list set} \Rightarrow$
 $\text{nat list set} \Rightarrow$
 $((('f \ \text{staged-proof-data} \times 'f \ \text{protocol-channel}) \times$
 $\text{unit list}) \times 'f \ \text{protocol-channel}) \ \text{option} \Rightarrow \text{bool}$

where

$\text{staged-security-trace-fri-pair-challenge-query-path-fresh } P \ B \ Q \ \text{out}$
 $\longleftrightarrow$
 $\text{staged-security-trace-fri-pair-challenge-fresh } P \ B \ \text{out} \wedge$
 $\text{staged-security-with-data-state-query-index-list-path-fresh } Q \ \text{out}$

definition *staged-security-composition-fri-pair-challenge-query-path-fresh*

$:: ('f \Rightarrow (\text{nat list} \times 'f \ \text{list}) \ \text{set}) \Rightarrow$
 $('f \Rightarrow 'f \ \text{list set}) \Rightarrow \text{nat list set} \Rightarrow$
 $((('f \ \text{staged-proof-data} \times 'f \ \text{protocol-channel}) \times$
 $\text{unit list}) \times 'f \ \text{protocol-channel}) \ \text{option} \Rightarrow \text{bool}$

where

$\text{staged-security-composition-fri-pair-challenge-query-path-fresh } P \ B \ Q \ \text{out}$
 $\longleftrightarrow$
 $\text{staged-security-composition-fri-pair-challenge-fresh } P \ B \ \text{out} \wedge$
 $\text{staged-security-with-data-state-query-index-list-path-fresh } Q \ \text{out}$

definition *verifier-after-trace-fri-raw-list-path-hit*

$:: 'f \times ('f \times 'f) \ \text{list} \Rightarrow$
 $('f, 'a) \ \text{protocol-channel-scheme} \Rightarrow \text{nat list set} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \ \text{protocol-channel-scheme}) \ \text{option} \Rightarrow \text{bool}$

where

$\text{verifier-after-trace-fri-raw-list-path-hit header } s \ Q \ \text{out} \longleftrightarrow$
 $(\exists \text{fr f-fl f-final s3 as s4 dg s5 s6 fl s7 final query-state raws.}$
 $\text{header} = (\text{fr, f-fl}) \wedge$
 $\text{Some } (f\text{-final, s3}) \in \text{set-dist } (\text{execute read } s) \wedge$
 $\text{Some } (\text{as, s4}) \in$
 $\text{set-dist } (\text{execute } (\text{mmap } (\text{replicate } (\text{length spec}) \ \text{alpha-round})) \ s3) \wedge$

Some ($dg, s5$) $\in$ *set-dist* (*execute read* $s4$) $\wedge$
Some ($()$, $s6$) $\in$
set-dist (*execute* (*assert* (*to-nat* $dg \leq \text{maxDegree}$)) $s5$) $\wedge$
Some ($fl, s7$) $\in$
set-dist (*execute*
(ntimes receive-composition-fri-commits
(ceil-log (to-nat dg + 1))) $s6$) $\wedge$
Some (*final, query-state*) $\in$ *set-dist* (*execute read* $s7$) $\wedge$
 $raws \in \text{query-index-raw-list-preimage } Q \wedge$
query-rounds-raw-list-path-hit query-state raws rounds out)

lemma *verifier-after-trace-fri-staged-query-path-fresh-imp-raw-list-path-hit:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *builder:*

Some (*data, attacker-state*) $\in$
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)

and *prefix:*

Some ($(fr, f-fl)$, *prefix-state*) $\in$
set-dist
(execute verifier-trace-fri-prefix
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))

and *suffix:*

out $\in$
set-dist
(execute (verifier-after-trace-fri (fr, f-fl)) prefix-state)

and *hit:*

staged-security-with-data-state-query-index-list-path-fresh Q
(case out of
None $\Rightarrow$ None
| Some (result, final-state) $\Rightarrow$
Some (((data, attacker-state), result), final-state))

shows *verifier-after-trace-fri-raw-list-path-hit (fr, f-fl) prefix-state Q out*

proof (*cases out*)

case *None*

then show *?thesis*

using *hit*

unfolding *staged-security-with-data-state-query-index-list-path-fresh-def*

by *simp*

next

case (*Some pair*)

then obtain *result :: unit list and final-state where out-eq:*

out = Some (result, final-state)

by (*cases pair*) *simp*

let *?s =*

verifier-state-from-adversary attacker-state

```

    (staged-proof-transcript data)
from suffix[unfolded out-eq verifier-after-trace-fri-def]
obtain f-final s3 as s4 dg s5 s6 fl s7 final query-state where
  read-trace-final:
    Some (f-final, s3) ∈ set-dist (execute read prefix-state)
and alpha-out:
    Some (as, s4) ∈
      set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)
and read-dg: Some (dg, s5) ∈ set-dist (execute read s4)
and degree-assert:
    Some ((), s6) ∈
      set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
and comp-fri:
    Some (fl, s7) ∈
      set-dist (execute
        (ntimes receive-composition-fri-commits
          (ceil-log (to-nat dg + 1))) s6)
and read-final:
    Some (final, query-state) ∈ set-dist (execute read s7)
and query-out:
    Some (result, final-state) ∈
      set-dist
        (execute
          (ntimes
            (verifier-query-round-program fr f-fl f-final as fl final)
            rounds)
          query-state)
by (auto elim!: set-dist-bindE)
from ntimes-verifier-query-rounds-outcome[OF query-out]
obtain raw-idxs query-idxs query-chunks where
  len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs
and len-chunks: length query-chunks = rounds
and tr-query:
  PTranscript query-state =
    List.concat query-chunks @ PTranscript final-state
and state-final:
  PState final-state =
    state-after-query-chunks (PState query-state) query-chunks rounds
and round-chunks:
  ∧ i. i < rounds ⇒
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and lookups:
  ∧ i. i < rounds ⇒
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks

```

```

      (PState query-state) query-chunks i)) =
    Some (raw-idxs ! i)
  by blast
have trace-prefix-res:
  length f-fl = ceil-log clength ∧
  PTranscript ?s = [fr] @ map snd f-fl @ PTranscript prefix-state ∧
  PState prefix-state =
    foldl concat (concat (PState ?s) fr) (map snd f-fl) ∧
  PQueryCounter prefix-state = PQueryCounter ?s
  using verifier-trace-fri-prefix-outcome[OF prefix] by simp
from read-outcome[OF read-trace-final] obtain rest3 where
  tr-prefix: PTranscript prefix-state = f-final # rest3
  and tr-s3: PTranscript s3 = rest3
  and state-s3: PState s3 = concat (PState prefix-state) f-final
  and counter-s3: PQueryCounter s3 = PQueryCounter prefix-state
  by blast
have alpha-res:
  length as = length spec ∧
  PTranscript s3 = as @ PTranscript s4 ∧
  PState s4 = foldl concat (PState s3) as ∧
  PQueryCounter s4 = PQueryCounter s3
  using mmap-alpha-round-outcome[OF alpha-out] by simp
from read-outcome[OF read-dg] obtain rest5 where
  tr-s4: PTranscript s4 = dg # rest5
  and tr-s5: PTranscript s5 = rest5
  and state-s5: PState s5 = concat (PState s4) dg
  and counter-s5: PQueryCounter s5 = PQueryCounter s4
  by blast
have s6-eq: s6 = s5
  using degree-assert unfolding assert-def
  by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
have comp-res:
  length fl = ceil-log (to-nat dg + 1) ∧
  PTranscript s6 = map snd fl @ PTranscript s7 ∧
  PState s7 = foldl concat (PState s6) (map snd fl) ∧
  PQueryCounter s7 = PQueryCounter s6
  using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
  by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-s7: PTranscript s7 = final # rest-query
  and tr-read-query: PTranscript query-state = rest-query
  and state-query: PState query-state = concat (PState s7) final
  and counter-query-s7: PQueryCounter query-state = PQueryCounter s7
  by blast
have local-counter: PQueryCounter query-state = 0
  using trace-prefix-res counter-s3 alpha-res counter-s5 s6-eq comp-res
  counter-query-s7
  unfolding verifier-state-from-adversary-def by simp
have prefix-lookup:

```

```

 $\wedge i x. \text{fmlookup } (\text{HashMap } \text{prefix-state})$ 
  ( $\text{QueryIndexChallenge } i x$ ) =
   $\text{fmlookup } (\text{HashMap } ?s) (\text{QueryIndexChallenge } i x)$ 
  by (rule verifier-trace-fri-prefix-preserves-query-lookup[OF prefix])
have s3-lookup:
   $\text{HashMap } s3 = \text{HashMap } \text{prefix-state}$ 
  by (rule read-preserves-hash-map[OF read-trace-final])
have s4-lookup:
 $\wedge i x. \text{fmlookup } (\text{HashMap } s4) (\text{QueryIndexChallenge } i x) =$ 
   $\text{fmlookup } (\text{HashMap } s3) (\text{QueryIndexChallenge } i x)$ 
proof -
  fix i x
  show  $\text{fmlookup } (\text{HashMap } s4) (\text{QueryIndexChallenge } i x) =$ 
     $\text{fmlookup } (\text{HashMap } s3) (\text{QueryIndexChallenge } i x)$ 
  proof (rule mmap-preserves-lookup[OF - alpha-out])
    fix m y u v
    assume m-in:  $m \in \text{set } (\text{replicate } (\text{length spec}) \text{alpha-round})$ 
    and out:  $\text{Some } (y, v) \in \text{set-dist } (\text{execute } m u)$ 
    then have m-eq:  $m = \text{alpha-round}$ 
    by simp
    show  $\text{fmlookup } (\text{HashMap } v) (\text{QueryIndexChallenge } i x) =$ 
       $\text{fmlookup } (\text{HashMap } u) (\text{QueryIndexChallenge } i x)$ 
    using alpha-round-preserves-query-lookup[OF out[unfolded m-eq]] .
  qed
qed
have s5-lookup:
   $\text{HashMap } s5 = \text{HashMap } s4$ 
  by (rule read-preserves-hash-map[OF read-dg])
have s7-lookup:
 $\wedge i x. \text{fmlookup } (\text{HashMap } s7) (\text{QueryIndexChallenge } i x) =$ 
   $\text{fmlookup } (\text{HashMap } s6) (\text{QueryIndexChallenge } i x)$ 
proof -
  fix i x
  show  $\text{fmlookup } (\text{HashMap } s7) (\text{QueryIndexChallenge } i x) =$ 
     $\text{fmlookup } (\text{HashMap } s6) (\text{QueryIndexChallenge } i x)$ 
  proof (rule ntimes-preserves-lookup[OF - comp-fri])
    fix y u v
    assume out:
       $\text{Some } (y, v) \in$ 
       $\text{set-dist } (\text{execute } \text{receive-composition-fri-commits } u)$ 
    obtain b r where y-eq:  $y = (b, r)$ 
    by (cases y) simp
    show  $\text{fmlookup } (\text{HashMap } v) (\text{QueryIndexChallenge } i x) =$ 
       $\text{fmlookup } (\text{HashMap } u) (\text{QueryIndexChallenge } i x)$ 
    by (rule receive-composition-fri-commits-preserves-query-lookup
      [OF out[unfolded y-eq]])
  qed
qed
have query-lookup:

```

```

    HashMap query-state = HashMap s7
  by (rule read-preserves-hash-map[OF read-final])
have header-actual:
  verifier-header-transcript ?s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
  using trace-prefix-res tr-prefix tr-s3 alpha-res tr-s4 tr-s5 s6-eq
    comp-res tr-s7 tr-read-query
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
have header-unique:
  fr = staged-trace-root data ∧
  map snd f-fl = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  map snd fl = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  PTranscript query-state = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header-actual staged-header]
  by simp
have prefix-state-query:
  PState query-state = staged-query-start-hash data
  using trace-prefix-res state-s3 alpha-res state-s5 s6-eq comp-res
    state-query header-unique
  unfolding staged-query-start-hash-def
    staged-composition-fri-start-hash-def
    staged-trace-fri-start-hash-def
  by simp
have chunks-eq: query-chunks = staged-query-chunks data
proof -
  from checked-staged-transcript-program-query-chunks-match-verifier-lengths
    [OF builder]
  obtain staged-query-idxs where
    staged-match:
      staged-query-chunks-match-verifier-lengths data staged-query-idxs
  by blast
have query-idxs-len: length query-idxs = rounds

```



```

    using query-idxs-eq len-raw by simp
  have staged-match-query-idxs:
    staged-query-chunks-match-verifier-lengths data query-idxs
  by (rule staged-query-chunks-match-verifier-lengths-transfer
      [OF staged-match query-idxs-len])
  have parser-chunk-len:
     $\bigwedge i. i < \text{rounds} \implies$ 
      length (query-chunks ! i) =
        verifier-query-round-transcript-length (query-idxs ! i)
          (staged-trace-fri-roots data)
          (staged-composition-fri-roots data)
  proof -
    fix i
    assume i-bound: i < rounds
    show
      length (query-chunks ! i) =
        verifier-query-round-transcript-length (query-idxs ! i)
          (staged-trace-fri-roots data)
          (staged-composition-fri-roots data)
    using verifier-query-round-chunk-length[OF round-chunks[OF i-bound]]
      header-unique
    by simp
  qed
  have concat-staged:
    List.concat query-chunks @ PTranscript final-state =
      List.concat (staged-query-chunks data)
  using tr-query header-unique by simp
  then show ?thesis
    using query-chunks-eq-staged-if-matching-lengths
      [OF len-chunks concat-staged parser-chunk-len
        staged-match-query-idxs]
    by simp
  qed
  from hit out-eq obtain raw-idxs-hit query-idxs-hit where
    query-hit-in: query-idxs-hit  $\in$  Q
  and len-hit: length raw-idxs-hit = rounds
  and query-hit-eq:
    query-idxs-hit =
      map ( $\lambda$ raw. index (to-nat raw)) raw-idxs-hit
  and lookup-hit:
     $\bigwedge i. i < \text{rounds} \implies$ 
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs-hit ! i)
  and fresh-hit:
     $\bigwedge i. i < \text{rounds} \implies$ 

```

```

    fmlookup (HashMap ?s)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
    None
  unfolding staged-security-with-data-state-query-index-list-path-fresh-def
  by auto
have raw-eq:
   $\bigwedge i. i < \text{rounds} \implies \text{raw-idxs} ! i = \text{raw-idxs-hit} ! i$ 
  using lookups lookup-hit local-counter prefix-state-query chunks-eq
  by simp
have raw-idxs-eq:  $\text{raw-idxs} = \text{raw-idxs-hit}$ 
  by (rule nth-equalityI)
  (use len-raw len-hit raw-eq in simp-all)
have raw-in:  $\text{raw-idxs} \in \text{query-index-raw-list-preimage } Q$ 
  unfolding query-index-raw-list-preimage-def
  using len-raw raw-idxs-eq query-hit-in query-hit-eq by blast
have query-lookup-start:
   $\bigwedge i x. \text{fmlookup} (\text{HashMap } \text{query-state}) (\text{QueryIndexChallenge } i x) =$ 
   $\text{fmlookup} (\text{HashMap } ?s) (\text{QueryIndexChallenge } i x)$ 
  using prefix-lookup s3-lookup s4-lookup s5-lookup s6-eq
  s7-lookup query-lookup by simp
have local-fresh:
  query-index-list-path-fresh query-state rounds query-chunks raw-idxs
  unfolding query-index-list-path-fresh-def
proof (intro conjI allI impI)
  show  $\text{length } \text{raw-idxs} = \text{rounds}$ 
    using len-raw .
next
  show  $\text{length } \text{query-chunks} = \text{rounds}$ 
    using len-chunks .
next
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  let ?x =
    state-after-query-chunks (staged-query-start-hash data)
    (staged-query-chunks data) i
  have fmlookup (HashMap query-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    fmlookup (HashMap ?s) (QueryIndexChallenge i ?x)
    using local-counter prefix-state-query chunks-eq
    query-lookup-start[of i ?x] by simp
  also have ... = None
    by (rule fresh-hit[OF i-bound])
  finally show
    fmlookup (HashMap query-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)

```

```

      (state-after-query-chunks (PState query-state) query-chunks i)) =
      None .
qed
have path:
  query-rounds-raw-list-path-hit query-state raw-idxs rounds out
  unfolding query-rounds-raw-list-path-hit-def out-eq
  using len-raw len-chunks state-final local-fresh lookups
  by auto
have witness:
  (fr, f-fl) = (fr, f-fl) ∧
  Some (f-final, s3) ∈ set-dist (execute read prefix-state) ∧
  Some (as, s4) ∈
    set-dist (execute (mmap (replicate (length spec) alpha-round)) s3) ∧
  Some (dg, s5) ∈ set-dist (execute read s4) ∧
  Some ((), s6) ∈
    set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5) ∧
  Some (fl, s7) ∈
    set-dist (execute
      (ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1))) s6) ∧
  Some (final, query-state) ∈ set-dist (execute read s7) ∧
  raw-idxs ∈ query-index-raw-list-preimage Q ∧
  query-rounds-raw-list-path-hit query-state raw-idxs rounds out
  using read-trace-final alpha-out read-dg degree-assert comp-fri
  read-final raw-in path by simp
show ?thesis
  unfolding verifier-after-trace-fri-raw-list-path-hit-def
  using witness by blast
qed

lemma verifier-after-composition-fri-staged-query-path-fresh-imp-raw-list-path-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  and prefix:
    Some (header, prefix-state) ∈
      set-dist
        (execute
          verifier-composition-fri-prefix
            (verifier-state-from-adversary attacker-state
              (staged-proof-transcript data)))
  and suffix:
    out ∈
      set-dist
        (execute (verifier-after-composition-fri header) prefix-state)

```

```

and hit:
  staged-security-with-data-state-query-index-list-path-fresh Q
  (case out of
    None  $\Rightarrow$  None
    | Some (result, final-state)  $\Rightarrow$ 
      Some (((data, attacker-state), result), final-state))
shows verifier-after-composition-fri-raw-list-path-hit prefix-state Q out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding staged-security-with-data-state-query-index-list-path-fresh-def
  by simp
next
case (Some pair)
then obtain result :: unit list and final-state where out-eq:
  out = Some (result, final-state)
  by (cases pair) simp
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
obtain fr f-fl f-final as dg fl where header-eq:
  header = (fr, f-fl, f-final, as, dg, fl)
  by (cases header) auto
from suffix[unfolded out-eq header-eq verifier-after-composition-fri-def]
obtain final query-state where
  read-final:
    Some (final, query-state)  $\in$  set-dist (execute read prefix-state)
  and query-out:
    Some (result, final-state)  $\in$ 
      set-dist
      (execute
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final)
          rounds)
        query-state)
  by (auto elim!: set-dist-bindE)
from ntimes-verifier-query-rounds-outcome[OF query-out]
obtain raw-idxs query-idxs query-chunks where
  len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
  and len-chunks: length query-chunks = rounds
  and tr-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
  and state-final:
    PState final-state =
      state-after-query-chunks (PState query-state) query-chunks rounds

```

```

and round-chunks:
   $\bigwedge i. i < \text{rounds} \implies$ 
    verifier-query-round-chunk (query-idxs ! i)
      (map snd f-fl) (map snd fl) (query-chunks ! i)
and lookups:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks
          (PState query-state) query-chunks i)) =
      Some (raw-idxs ! i)
by blast
from read-outcome[OF read-final] have query-counter:
  PQueryCounter query-state = PQueryCounter prefix-state
by blast
have prefix-counter:
  PQueryCounter prefix-state = PQueryCounter ?s
using verifier-composition-fri-prefix-outcome
  [OF prefix[unfolded header-eq]]
by simp
have local-counter: PQueryCounter query-state = 0
using query-counter prefix-counter
unfolding verifier-state-from-adversary-def by simp
have prefix-lookup:
   $\bigwedge i x. \text{fmlookup} (\text{HashMap prefix-state})$ 
    (QueryIndexChallenge i x) =
    fmlookup (HashMap ?s) (QueryIndexChallenge i x)
by (rule verifier-composition-fri-prefix-preserves-query-lookup
  [OF prefix[unfolded header-eq]])
have read-lookup: HashMap query-state = HashMap prefix-state
by (rule read-preserves-hash-map[OF read-final])
have prefix-res:
  length f-fl = ceil-log clength  $\wedge$ 
  length as = length spec  $\wedge$ 
  length fl = ceil-log (to-nat dg + 1)  $\wedge$ 
  PTranscript ?s =
    [fr] @ map snd f-fl @ [f-final] @ as @ [dg] @
    map snd fl @ PTranscript prefix-state  $\wedge$ 
  PState prefix-state =
    foldl concat
      (concat
        (foldl concat
          (concat
            (foldl concat (concat (PState ?s) fr) (map snd f-fl))
              f-final)
            as)
          dg)
        (map snd fl)
      )
using verifier-composition-fri-prefix-outcome

```

```

    [OF prefix[unfolded header-eq]]
  by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-prefix: PTranscript prefix-state = final # rest-query
  and state-query: PState query-state = concat (PState prefix-state) final
  and tr-read-query: PTranscript query-state = rest-query
  by blast
have header-actual:
  verifier-header-transcript ?s fr (map snd f-fl) f-final as dg
    (map snd fl) final (PTranscript query-state)
  unfolding verifier-header-transcript-def verifier-header-messages-def
  using prefix-res tr-prefix tr-read-query by simp
have staged-header:
  verifier-header-transcript ?s
    (staged-trace-root data)
    (staged-trace-fri-roots data)
    (staged-trace-final data)
    (staged-alphas data)
    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
    [OF wf controlled builder])
have header-unique:
  fr = staged-trace-root data ∧
  map snd f-fl = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  map snd fl = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  PTranscript query-state = List.concat (staged-query-chunks data)
  using verifier-header-transcript-unique[OF header-actual staged-header]
  by simp
have prefix-state-query:
  PState query-state = staged-query-start-hash data
  using state-query prefix-res header-unique
  unfolding staged-query-start-hash-def
    staged-composition-fri-start-hash-def
    staged-trace-fri-start-hash-def
  by simp
have chunks-eq: query-chunks = staged-query-chunks data
proof –
  from checked-staged-transcript-program-query-chunks-match-verifier-lengths
    [OF builder]
  obtain staged-query-idxs where
    staged-match:
      staged-query-chunks-match-verifier-lengths data staged-query-idxs

```

```

    by blast
  have query-idxs-len: length query-idxs = rounds
    using query-idxs-eq len-raw by simp
  have staged-match-query-idxs:
    staged-query-chunks-match-verifier-lengths data query-idxs
    by (rule staged-query-chunks-match-verifier-lengths-transfer
        [OF staged-match query-idxs-len])
  have parser-chunk-len:
     $\bigwedge i. i < rounds \implies$ 
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
        (staged-trace-fri-roots data)
        (staged-composition-fri-roots data)
  proof -
    fix i
    assume i-bound: i < rounds
    show
      length (query-chunks ! i) =
        verifier-query-round-transcript-length (query-idxs ! i)
          (staged-trace-fri-roots data)
          (staged-composition-fri-roots data)
      using verifier-query-round-chunk-length[OF round-chunks[OF i-bound]]
        header-unique
      by simp
  qed
  have concat-staged:
    List.concat query-chunks @ PTranscript final-state =
      List.concat (staged-query-chunks data)
    using tr-query header-unique by simp
  then show ?thesis
    using query-chunks-eq-staged-if-matching-lengths
      [OF len-chunks concat-staged parser-chunk-len
        staged-match-query-idxs]
    by simp
  qed
  from hit out-eq obtain raw-idxs-hit query-idxs-hit where
    query-hit-in: query-idxs-hit  $\in$  Q
    and len-hit: length raw-idxs-hit = rounds
    and query-hit-eq:
      query-idxs-hit =
        map ( $\lambda$ raw. index (to-nat raw)) raw-idxs-hit
    and lookup-hit:
       $\bigwedge i. i < rounds \implies$ 
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs-hit ! i)

```

```

and fresh-hit:
   $\wedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap ?s)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
        None
unfolding staged-security-with-data-state-query-index-list-path-fresh-def
by auto
have raw-eq:
   $\wedge i. i < \text{rounds} \implies \text{raw-idxs} ! i = \text{raw-idxs-hit} ! i$ 
using lookups lookup-hit local-counter prefix-state-query chunks-eq
by simp
have raw-idxs-eq: raw-idxs = raw-idxs-hit
by (rule nth-equalityI)
  (use len-raw len-hit raw-eq in simp-all)
have raw-in: raw-idxs  $\in$  query-index-raw-list-preimage Q
unfolding query-index-raw-list-preimage-def
using len-raw raw-idxs-eq query-hit-in query-hit-eq by blast
have query-lookup-start:
   $\wedge i x. \text{fmlookup} (\text{HashMap } \text{query-state}) (\text{QueryIndexChallenge } i x) =$ 
    fmlookup (HashMap ?s) (QueryIndexChallenge i x)
using prefix-lookup read-lookup by simp
have local-fresh:
  query-index-list-path-fresh query-state rounds query-chunks raw-idxs
unfolding query-index-list-path-fresh-def
proof (intro conjI allI impI)
  show length raw-idxs = rounds
    using len-raw .
next
  show length query-chunks = rounds
    using len-chunks .
next
fix i
assume i-bound: i < rounds
let ?x =
  state-after-query-chunks (staged-query-start-hash data)
    (staged-query-chunks data) i
have fmlookup (HashMap query-state)
  (QueryIndexChallenge (PQueryCounter query-state + i)
    (state-after-query-chunks (PState query-state) query-chunks i)) =
  fmlookup (HashMap ?s) (QueryIndexChallenge i ?x)
using local-counter prefix-state-query chunks-eq
  query-lookup-start[of i ?x] by simp
also have ... = None
by (rule fresh-hit[OF i-bound])
finally show
  fmlookup (HashMap query-state)

```



```

      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
      None .
qed
have path:
  query-rounds-raw-list-path-hit query-state raw-idxs rounds out
  unfolding query-rounds-raw-list-path-hit-def out-eq
  using len-raw len-chunks state-final local-fresh lookups
  by auto
show ?thesis
  unfolding verifier-after-composition-fri-raw-list-path-hit-def
  by (intro exI[of - final] exI[of - query-state]
      exI[of - raw-idxs] conjI)
    (use read-final raw-in path in simp-all)
qed

lemma staged-security-trace-fri-pair-challenge-fresh-imp-query-index-list-hit:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and support:
    out ∈
    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
  and projection: fst ' P ⊆ Q
  and hit: staged-security-trace-fri-pair-challenge-fresh P B out
  shows staged-security-with-data-query-index-list-set-hit Q out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding staged-security-trace-fri-pair-challenge-fresh-def
    staged-security-with-data-state-verifier-event-def
    staged-security-with-data-query-index-list-set-hit-def
  by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed) (auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from hit out-eq have pair-hit:
  trace-fri-query-challenge-pair-set-hit ?s P
  (Some (result, final-state))
unfolding staged-security-trace-fri-pair-challenge-fresh-def
  staged-security-with-data-state-verifier-event-def
by simp

```

```

have trace-hit:
  trace-fri-query-index-list-set-hit ?s Q
  (Some (result, final-state))
by (rule trace-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit
  [OF pair-hit projection])
from trace-hit obtain trace-roots trace-bs trace-final dg
  composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers where
  fri:
    accepted-fri-opening-transcript ?s (Some (result, final-state))
    trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final query-idxs trace-round-layers
    composition-round-layers
  and query-in: query-idxs  $\in$  Q
  unfolding trace-fri-query-index-list-set-hit-def by blast
from accepted-fri-opening-transcript-imp-accepted-transcript-shape[OF fri]
obtain as where shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as query-idxs
by blast
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support[unfolded out-eq] shape]
obtain raw-idxs where
  len-raw: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
  and lookup:
     $\bigwedge i. i < rounds \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i)
  by blast
show ?thesis
  unfolding out-eq staged-security-with-data-query-index-list-set-hit-def
  using len-raw query-in query-idxs-eq lookup by auto
qed

lemma staged-security-composition-fri-pair-challenge-fresh-imp-query-index-list-hit:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and support:
  out  $\in$ 
  set-dist
  (execute (checked-staged-security-experiment-with-data-state A)
    adversary-initial-state)
and projection:  $\bigwedge dg. fst \text{ ` } P \ dg \subseteq Q$ 
and hit: staged-security-composition-fri-pair-challenge-fresh P B out

```

```

  shows staged-security-with-data-query-index-list-set-hit Q out
proof (cases out)
case None
then show ?thesis
  using hit
  unfolding staged-security-composition-fri-pair-challenge-fresh-def
    staged-security-with-data-state-verifier-event-def
    staged-security-with-data-query-index-list-set-hit-def
  by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed) (auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from hit out-eq have pair-hit:
  composition-fri-query-challenge-pair-set-hit ?s P
  (Some (result, final-state))
unfolding staged-security-composition-fri-pair-challenge-fresh-def
  staged-security-with-data-state-verifier-event-def
by simp
have composition-hit:
  composition-fri-query-index-list-set-hit ?s Q
  (Some (result, final-state))
by (rule
  composition-fri-query-challenge-pair-set-hit-imp-query-index-list-set-hit
  [OF pair-hit projection])
from composition-hit obtain trace-roots trace-bs trace-final dg
  composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers where
fri:
  accepted-fri-opening-transcript ?s (Some (result, final-state))
  trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final query-idxs trace-round-layers
  composition-round-layers
and query-in: query-idxs ∈ Q
unfolding composition-fri-query-index-list-set-hit-def by blast
from accepted-fri-opening-transcript-imp-accepted-transcript-shape[OF fri]
obtain as where shape:
  accepted-transcript-shape ?s (Some (result, final-state)) as query-idxs
by blast
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support[unfolded out-eq] shape]
obtain raw-idxs where
  len-raw: length raw-idxs = rounds
and query-idxs-eq:
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs

```

```

and lookup:
   $\bigwedge i. i < \text{rounds} \implies$ 
    fmllookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
  by blast
show ?thesis
  unfolding out-eq staged-security-with-data-query-index-list-set-hit-def
  using len-raw query-in query-idxs-eq lookup by auto
qed

lemma checked-staged-security-trace-fri-pair-challenge-fresh-bound-from-query-path-fresh:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and projection: fst ‘  $P \subseteq Q$ 
  and path-branch-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-trace-fri-pair-challenge-query-path-fresh P B Q)
    adversary-initial-state  $\leq C$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-trace-fri-pair-challenge-fresh P B)
    adversary-initial-state  $\leq$ 
      C +
      hash-relation-budget-value
        (query-index-raw-list-relation-fiber-bound Q)
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget)

proof –
  have event-mono:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-trace-fri-pair-challenge-fresh P B)
    adversary-initial-state  $\leq$ 
    wp-event (checked-staged-security-experiment-with-data-state A)
      ( $\lambda out.$ 
        staged-security-trace-fri-pair-challenge-query-path-fresh P B Q out  $\vee$ 
        staged-security-with-data-state-query-index-list-prequery-hit A Q out)
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out  $\in$ 
      set-dist
        (execute (checked-staged-security-experiment-with-data-state A)
          adversary-initial-state)
  and hit: staged-security-trace-fri-pair-challenge-fresh P B out

```

```

have query-hit:
  staged-security-with-data-query-index-list-set-hit Q out
by (rule
    staged-security-trace-fri-pair-challenge-fresh-imp-query-index-list-hit
    [OF wf controlled support projection hit])
show
  staged-security-trace-fri-pair-challenge-query-path-fresh P B Q out  $\vee$ 
  staged-security-with-data-state-query-index-list-prequery-hit A Q out
proof (cases out)
  case None
  then show ?thesis
    using query-hit
    unfolding staged-security-with-data-query-index-list-set-hit-def
    by simp
next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed) (auto split: prod.splits)
  have split:
    staged-security-with-data-state-query-index-list-path-fresh Q
    (Some (((data, attacker-state), result), final-state))  $\vee$ 
    hash-relation-hit
    (checked-staged-query-index-list-relation A Q)
    adversary-initial-state attacker-state
  apply (rule
    checked-staged-security-with-data-state-query-index-list-fresh-or-prequeried)
  using support query-hit unfolding out-eq by simp-all
  then show ?thesis
  proof
    assume fresh:
      staged-security-with-data-state-query-index-list-path-fresh Q
      (Some (((data, attacker-state), result), final-state))
    have pair-fresh:
      staged-security-trace-fri-pair-challenge-fresh P B
      (Some (((data, attacker-state), result), final-state))
      using hit out-eq by simp
    then have
      staged-security-trace-fri-pair-challenge-query-path-fresh P B Q
      out
      unfolding out-eq
      staged-security-trace-fri-pair-challenge-query-path-fresh-def
      using fresh pair-fresh by simp
    then show ?thesis ..
  next
  assume prequery:
    hash-relation-hit
    (checked-staged-query-index-list-relation A Q)
    adversary-initial-state attacker-state

```

```

then have
  staged-security-with-data-state-query-index-list-prequery-hit A Q
  out
unfolding out-eq
  staged-security-with-data-state-query-index-list-prequery-hit-def
by simp
then show ?thesis ..
qed
qed
qed
have union-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (λout.
    staged-security-trace-fri-pair-challenge-query-path-fresh P B Q out ∨
    staged-security-with-data-state-query-index-list-prequery-hit A Q out)
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-trace-fri-pair-challenge-query-path-fresh P B Q)
  adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-query-index-list-prequery-hit A Q)
  adversary-initial-state
by (rule wp-event-union-bound)
have prequery-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-query-index-list-prequery-hit A Q)
  adversary-initial-state ≤
  hash-relation-budget-value
  (query-index-raw-list-relation-fiber-bound Q)
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)
by (rule
  checked-staged-security-with-data-state-query-index-list-prequery-hit-bound
  [OF wf controlled])
have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-trace-fri-pair-challenge-fresh P B)
  adversary-initial-state ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-trace-fri-pair-challenge-query-path-fresh P B Q)
  adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-query-index-list-prequery-hit A Q)
  adversary-initial-state
by (rule order-trans[OF event-mono union-bound])
also have ... ≤
  C +
  hash-relation-budget-value
  (query-index-raw-list-relation-fiber-bound Q)
  (staged-attacker-query-budget budgets +

```

staged-challenge-query-budget)
 by (rule add-mono[OF path-branch-bound prequery-bound])
 finally show ?thesis .
 qed

lemma *checked-staged-security-composition-fri-pair-challenge-fresh-bound-from-query-path-fresh:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *projection: $\bigwedge dg. \text{fst } dg \subseteq P \implies dg \subseteq Q$*
and *path-branch-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-composition-fri-pair-challenge-query-path-fresh P B Q)
adversary-initial-state $\leq C$
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-composition-fri-pair-challenge-fresh P B)
adversary-initial-state $\leq$
C +
hash-relation-budget-value
(query-index-row-list-relation-fiber-bound Q)
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)

proof –
have *event-mono:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-composition-fri-pair-challenge-fresh P B)
adversary-initial-state $\leq$
wp-event (checked-staged-security-experiment-with-data-state A)
($\lambda out.$
staged-security-composition-fri-pair-challenge-query-path-fresh P B Q out $\vee$
staged-security-with-data-state-query-index-list-prequery-hit A Q out)
adversary-initial-state

proof (rule *wp-event-mono-on-support*)
fix *out*
assume *support:*
out $\in$
set-dist
(execute (checked-staged-security-experiment-with-data-state A)
adversary-initial-state)
and *hit: staged-security-composition-fri-pair-challenge-fresh P B out*
have *query-hit:*
staged-security-with-data-query-index-list-set-hit Q out
by (rule
staged-security-composition-fri-pair-challenge-fresh-imp-query-index-list-hit
[OF wf controlled support projection hit])
show
staged-security-composition-fri-pair-challenge-query-path-fresh P B Q out $\vee$
staged-security-with-data-state-query-index-list-prequery-hit A Q out
proof (cases *out*)

```

case None
then show ?thesis
  using query-hit
  unfolding staged-security-with-data-query-index-list-set-hit-def
  by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases packed) (auto split: prod.splits)
have split:
  staged-security-with-data-state-query-index-list-path-fresh Q
    (Some (((data, attacker-state), result), final-state))  $\vee$ 
  hash-relation-hit
    (checked-staged-query-index-list-relation A Q)
    adversary-initial-state attacker-state
apply (rule
  checked-staged-security-with-data-state-query-index-list-fresh-or-prequeried)
using support query-hit unfolding out-eq by simp-all
then show ?thesis
proof
  assume fresh:
    staged-security-with-data-state-query-index-list-path-fresh Q
      (Some (((data, attacker-state), result), final-state))
  have pair-fresh:
    staged-security-composition-fri-pair-challenge-fresh P B
      (Some (((data, attacker-state), result), final-state))
    using hit out-eq by simp
  then have
    staged-security-composition-fri-pair-challenge-query-path-fresh
      P B Q out
    unfolding out-eq
      staged-security-composition-fri-pair-challenge-query-path-fresh-def
    using fresh pair-fresh by simp
  then show ?thesis ..
next
assume prequery:
  hash-relation-hit
    (checked-staged-query-index-list-relation A Q)
    adversary-initial-state attacker-state
then have
  staged-security-with-data-state-query-index-list-prequery-hit A Q
    out
  unfolding out-eq
    staged-security-with-data-state-query-index-list-prequery-hit-def
  by simp
  then show ?thesis ..
qed
qed

```


qed

have *union-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 ($\lambda out.$
staged-security-composition-fri-pair-challenge-query-path-fresh P B Q out $\vee$
staged-security-with-data-state-query-index-list-prequery-hit A Q out)
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-composition-fri-pair-challenge-query-path-fresh P B Q*)
adversary-initial-state $+$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-query-index-list-prequery-hit A Q*)
adversary-initial-state
by (*rule wp-event-union-bound*)

have *prequery-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-query-index-list-prequery-hit A Q*)
adversary-initial-state $\leq$
hash-relation-budget-value
 (*query-index-row-list-relation-fiber-bound Q*)
 (*staged-attacker-query-budget budgets* $+$
staged-challenge-query-budget)
by (*rule*
checked-staged-security-with-data-state-query-index-list-prequery-hit-bound
 [*OF wf controlled*])

have *wp-event* (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-composition-fri-pair-challenge-fresh P B*)
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-composition-fri-pair-challenge-query-path-fresh P B Q*)
adversary-initial-state $+$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-query-index-list-prequery-hit A Q*)
adversary-initial-state
by (*rule order-trans*[*OF event-mono union-bound*])

also have ... $\leq$
C $+$
hash-relation-budget-value
 (*query-index-row-list-relation-fiber-bound Q*)
 (*staged-attacker-query-budget budgets* $+$
staged-challenge-query-budget)
by (*rule add-mono*[*OF path-branch-bound prequery-bound*])

finally show *?thesis* **by** *simp*

qed

lemma *checked-staged-security-composition-fri-pair-challenge-query-path-fresh-fixed-bound*:

assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *challenges-space*:

```

    challenges ∈ fri-challenge-space (ceil-log (to-nat dg + 1))
  and projection:
    fst ‘ (P dg ∩ (UNIV × {challenges})) ⊆ Q
  and subset: Q ⊆ fri-query-index-list-space
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-composition-fri-pair-challenge-query-path-fresh P
      (λdg'. if dg' = dg then {challenges} else {}) Q)
  adversary-initial-state ≤
    (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
    (nnreal (card Q) *
      (1 / nnreal (card query-sample-space)) ^ rounds)
proof (rule checked-staged-security-with-data-state-bound-from-data-cont)
  let ?B = λdg'. if dg' = dg then {challenges} else {}
  let ?D =
    nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds
  show ¬ staged-security-composition-fri-pair-challenge-query-path-fresh
    P ?B Q None
  unfolding
    staged-security-composition-fri-pair-challenge-query-path-fresh-def
    staged-security-composition-fri-pair-challenge-fresh-def
    staged-security-with-data-state-verifier-event-def
  by simp
next
  fix data attacker-state
  assume builder:
    Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  let ?B = λdg'. if dg' = dg then {challenges} else {}
  let ?Event =
    λout.
      staged-security-composition-fri-pair-challenge-query-path-fresh P ?B Q
      (case out of
        None ⇒ None
        | Some (result, final-state) ⇒
          Some (((data, attacker-state), result), final-state))
  let ?Head =
    λout. case out of
      None ⇒ False
      | Some ((fr, f-fl, f-final, as, dg', fl), -) ⇒
        dg' = dg ∧ map fst fl = challenges ∧
        composition-fri-challenge-path-fresh ?s fr (map snd f-fl)
        f-final as dg' (map snd fl) (length (map snd fl))

```

```

show wp-event verify-monad ?Event ?s ≤
  (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
  (nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds)
unfolding verify-monad-composition-fri-decomposition
proof (rule order-trans)
show wp-event
  (verifier-composition-fri-prefix >>=
    verifier-after-composition-fri)
  ?Event ?s
≤ wp-event verifier-composition-fri-prefix ?Head ?s *
  (nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds)
proof (rule wp-event-bind-bound-by-head-and-cont)
show ¬ ?Event None
unfolding
  staged-security-composition-fri-pair-challenge-query-path-fresh-def
  staged-security-composition-fri-pair-challenge-fresh-def
  staged-security-with-data-state-verifier-event-def
by simp
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state) ∈
    set-dist (execute verifier-composition-fri-prefix ?s)
and not-head: ¬ ?Head (Some (header, prefix-state))
obtain fr f-fl f-final as dg' fl where header-eq:
  header = (fr, f-fl, f-final, as, dg', fl)
by (cases header) auto
have not-head':
  dg' ≠ dg ∨ map fst fl ≠ challenges ∨
    ¬ composition-fri-challenge-path-fresh ?s fr (map snd f-fl)
    f-final as dg' (map snd fl) (length (map snd fl))
using not-head unfolding header-eq by simp
show wp-event (verifier-after-composition-fri header) ?Event
  prefix-state = 0
proof (rule antisym)
show wp-event (verifier-after-composition-fri header) ?Event
  prefix-state ≤ 0
proof (rule order-trans)
show wp-event (verifier-after-composition-fri header) ?Event
  prefix-state ≤
    wp-event (verifier-after-composition-fri header)
    (λ-. False) prefix-state
proof (rule wp-event-mono-on-support)
fix out
assume suffix:
  out ∈
    set-dist

```

```

      (execute (verifier-after-composition-fri header)
        prefix-state)
    and hit: ?Event out
show False
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding
      staged-security-composition-fri-pair-challenge-query-path-fresh-def
      staged-security-composition-fri-pair-challenge-fresh-def
      staged-security-with-data-state-verifier-event-def
    by simp
next
case (Some pair)
then obtain result :: unit list and final-state where out-eq:
  out = Some (result, final-state)
  by (cases pair) simp
from hit out-eq have fresh:
  composition-fri-challenge-list-fresh-hit ?s ?B
  (Some (result, final-state))
and pair-hit:
  composition-fri-query-challenge-pair-set-hit ?s P
  (Some (result, final-state))
unfolding
  staged-security-composition-fri-pair-challenge-query-path-fresh-def
  staged-security-composition-fri-pair-challenge-fresh-def
  staged-security-with-data-state-verifier-event-def
by simp-all
have prefix':
  Some ((fr, f-fl, f-final, as, dg', fl), prefix-state) ∈
  set-dist (execute verifier-composition-fri-prefix ?s)
using prefix header-eq by simp
have suffix':
  Some (result, final-state) ∈
  set-dist
  (execute
    (verifier-after-composition-fri
      (fr, f-fl, f-final, as, dg', fl))
    prefix-state)
using suffix header-eq out-eq by simp
from pair-hit obtain trace-roots trace-bs trace-final fri-dg
  composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers where
openings:
  accepted-fri-opening-transcript ?s
  (Some (result, final-state)) trace-roots trace-bs
  trace-final fri-dg composition-roots composition-bs
  composition-final query-idxs trace-round-layers

```

```

      composition-round-layers
unfolding composition-fri-query-challenge-pair-set-hit-def
by blast
have pair-challenges:
  accepted-fri-challenges ?s (Some (result, final-state))
  trace-bs fri-dg composition-bs
by (rule accepted-fri-opening-transcript-challenges
    [OF openings])
from fresh obtain trace-bs' dg'' comp-bs' where
  fresh-challenges:
    accepted-fri-challenges ?s (Some (result, final-state))
    trace-bs' dg'' comp-bs'
and comp-bs'-in:
  comp-bs' ∈
    (if dg'' = dg then {challenges} else {})
unfolding composition-fri-challenge-list-fresh-hit-def
by blast
have challenge-eqs:
  fri-dg = dg'' ∧ composition-bs = comp-bs'
using accepted-fri-challenges-unique
  [OF pair-challenges fresh-challenges]
by simp
have prefix-eqs:
  fri-dg = dg' ∧ composition-bs = map fst fl
by (rule
    verifier-after-composition-fri-header-agrees-with-accepted-challenges
    [OF prefix' suffix' pair-challenges])
have concrete-fresh:
  composition-fri-challenge-path-fresh ?s fr (map snd f-fl)
  f-final as dg' (map snd fl) (length (map snd fl))
by (rule
    verifier-after-composition-fri-header-roots-agree-with-fresh-hit
    [OF prefix' suffix' fresh])
have dg''-eq: dg'' = dg
using comp-bs'-in by (cases dg'' = dg) auto
have dg'-eq: dg' = dg
using challenge-eqs prefix-eqs dg''-eq by simp
have fl-challenges: map fst fl = challenges
using comp-bs'-in prefix-eqs challenge-eqs dg''-eq by simp
have head-holds:
  dg' = dg ∧ map fst fl = challenges ∧
    composition-fri-challenge-path-fresh ?s fr (map snd f-fl)
    f-final as dg' (map snd fl) (length (map snd fl))
using dg'-eq fl-challenges concrete-fresh by simp
show ?thesis
using not-head' head-holds by blast
qed
qed
next

```

```

    show wp-event (verifier-after-composition-fri header)
      ( $\lambda\cdot$ . False) prefix-state  $\leq 0$ 
    unfolding wp-event-def wp-def dist-expect-def by simp
  qed
next
  show 0  $\leq$ 
    wp-event (verifier-after-composition-fri header) ?Event
      prefix-state
    by simp
  qed
next
  fix header prefix-state
  assume prefix:
    Some (header, prefix-state)  $\in$ 
      set-dist (execute verifier-composition-fri-prefix ?s)
  and head: ?Head (Some (header, prefix-state))
  have event-mono:
    wp-event (verifier-after-composition-fri header) ?Event
      prefix-state  $\leq$ 
    wp-event (verifier-after-composition-fri header)
      (verifier-after-composition-fri-raw-list-path-hit prefix-state Q)
      prefix-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume suffix:
      out  $\in$ 
        set-dist
          (execute (verifier-after-composition-fri header) prefix-state)
    and hit: ?Event out
    from hit have path-hit:
      staged-security-with-data-state-query-index-list-path-fresh Q
      (case out of
        None  $\Rightarrow$  None
      | Some (result, final-state)  $\Rightarrow$ 
        Some (((data, attacker-state), result), final-state))
    unfolding
      staged-security-composition-fri-pair-challenge-query-path-fresh-def
    by simp
  show
    verifier-after-composition-fri-raw-list-path-hit prefix-state Q
      out
    by (rule
      verifier-after-composition-fri-staged-query-path-fresh-imp-raw-list-path-hit
      [OF wf controlled builder prefix suffix path-hit])
  qed
  also have ...  $\leq$ 
    nnreal (card Q) *
      (1 / nnreal (card query-sample-space))  $\wedge$  rounds
  by (rule wp-verifier-after-composition-fri-raw-list-path-product-bound)

```

```

      [OF subset])
finally show
  wp-event (verifier-after-composition-fri header) ?Event
    prefix-state
  ≤ nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds .
qed
next
show wp-event verifier-composition-fri-prefix ?Head ?s *
  (nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds)
  ≤
  (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
  (nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds)
proof (rule mult-right-mono)
show wp-event verifier-composition-fri-prefix ?Head ?s ≤
  1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
proof -
  let ?B = λdg'. if dg' = dg then {challenges} else {}
  let ?HeadSet =
    λout. case out of
      None ⇒ False
    | Some ((fr, f-fl, f-final, as, dg', fl), -) ⇒
      map fst fl ∈ ?B dg' ∧
      composition-fri-challenge-path-fresh ?s fr (map snd f-fl)
      f-final as dg' (map snd fl) (length (map snd fl))
  have mono:
    wp-event verifier-composition-fri-prefix ?Head ?s ≤
    wp-event verifier-composition-fri-prefix ?HeadSet ?s
  by (rule wp-event-mono) (auto split: option.splits prod.splits)
  have subset:
    ∧dg'. ?B dg' ⊆
      fri-challenge-space (ceil-log (to-nat dg' + 1))
  using challenges-space by auto
  have bound:
    ∧dg'. nnreal (card (?B dg')) /
      nnreal (CARD('f) ^ ceil-log (to-nat dg' + 1)) ≤
      1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
  using challenges-space by auto
  have set-bound:
    wp-event verifier-composition-fri-prefix ?HeadSet ?s ≤
      1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))
  by (rule
    wp-verifier-composition-fri-prefix-challenge-list-actual-fresh-bound
    [OF subset bound])
  show ?thesis
  by (rule order-trans[OF mono set-bound])
qed

```

```

    show  $0 \leq \text{nnreal } (\text{card } Q) * (1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$ 
    by simp
qed
qed
qed

lemma checked-staged-security-composition-fri-pair-challenge-query-path-fresh-fiber-sum:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and challenge-subset:
     $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$ 
  and projection:
     $\bigwedge dg \text{ challenges. challenges} \in B \ dg \implies \text{fst } ' (P \ dg \cap (UNIV \times \{\text{challenges}\})) \subseteq Q \ dg \text{ challenges}$ 
  and subset:
     $\bigwedge dg \text{ challenges. challenges} \in B \ dg \implies Q \ dg \text{ challenges} \subseteq \text{fri-query-index-list-space}$ 
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-composition-fri-pair-challenge-query-path-fresh P B
         $(\bigcup dg \in UNIV. \bigcup \text{challenges} \in B \ dg. Q \ dg \text{ challenges})$ 
        adversary-initial-state  $\leq$ 
         $(\sum dg \in UNIV. \sum \text{challenges} \in B \ dg. (1 / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1))) * (\text{nnreal } (\text{card } (Q \ dg \text{ challenges})) * (1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}))$ 
      )
  proof -
    let ?Q =  $(\bigcup dg \in UNIV. \bigcup \text{challenges} \in B \ dg. Q \ dg \text{ challenges})$ 
    let ?I =  $SIGMA \ dg:UNIV. B \ dg$ 
    let ?E =
       $\lambda p. \text{staged-security-composition-fri-pair-challenge-query-path-fresh } P$ 
       $(\lambda dg'. \text{if } dg' = \text{fst } p \text{ then } \{\text{snd } p\} \text{ else } \{\})$ 
       $(Q \ (\text{fst } p) \ (\text{snd } p))$ 
    let ?C =
       $\lambda p. (1 / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } (\text{fst } p) + 1))) * (\text{nnreal } (\text{card } (Q \ (\text{fst } p) \ (\text{snd } p)))) * (1 / \text{nnreal } (\text{card } \text{query-sample-space})) \wedge \text{rounds}$ 
    have split:
      wp-event (checked-staged-security-experiment-with-data-state A)
        (staged-security-composition-fri-pair-challenge-query-path-fresh P B ?Q)
      adversary-initial-state  $\leq$ 
      wp-event (checked-staged-security-experiment-with-data-state A)

```



```

    ( $\lambda out. \exists p \in ?I. ?E\ p\ out$ )
    adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
    out  $\in$ 
    set-dist
    (execute (checked-staged-security-experiment-with-data-state A)
      adversary-initial-state)
and hit:
    staged-security-composition-fri-pair-challenge-query-path-fresh P B ?Q
    out
show  $\exists p \in ?I. ?E\ p\ out$ 
proof (cases out)
case None
then show ?thesis
    using hit
    unfolding
      staged-security-composition-fri-pair-challenge-query-path-fresh-def
      staged-security-composition-fri-pair-challenge-fresh-def
      staged-security-with-data-state-verifier-event-def
    by simp
next
case (Some packed)
then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed) (auto split: prod.splits)
let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
from hit out-eq have pair-hit:
    composition-fri-query-challenge-pair-set-hit ?s P
    (Some (result, final-state))
and fresh:
    composition-fri-challenge-list-fresh-hit ?s B
    (Some (result, final-state))
and path:
    staged-security-with-data-state-query-index-list-path-fresh ?Q out
unfolding
    staged-security-composition-fri-pair-challenge-query-path-fresh-def
    staged-security-composition-fri-pair-challenge-fresh-def
    staged-security-with-data-state-verifier-event-def
by simp-all
from fresh obtain trace-bs dg comp-bs fr trace-roots trace-final as
    composition-roots final rest where
    accepted:
    accepted-fri-challenges ?s (Some (result, final-state))
    trace-bs dg comp-bs
and comp-bs-in: comp-bs  $\in B\ dg$ 

```

```

and header:
  verifier-header-transcript ?s fr trace-roots trace-final as dg
  composition-roots final rest
and fresh-path:
  composition-fri-challenge-path-fresh ?s fr trace-roots
  trace-final as dg composition-roots (length composition-roots)
  unfolding composition-fri-challenge-list-fresh-hit-def by blast
from pair-hit obtain trace-roots' trace-bs' trace-final' fri-dg
  composition-roots' composition-bs' composition-final' query-idxs
  trace-round-layers composition-round-layers where
  openings:
    accepted-fri-opening-transcript ?s
    (Some (result, final-state)) trace-roots' trace-bs'
    trace-final' fri-dg composition-roots' composition-bs'
    composition-final' query-idxs trace-round-layers
    composition-round-layers
  and pair: (query-idxs, composition-bs') ∈ P fri-dg
  unfolding composition-fri-query-challenge-pair-set-hit-def by blast
have pair-challenges:
  accepted-fri-challenges ?s (Some (result, final-state))
  trace-bs' fri-dg composition-bs'
  by (rule accepted-fri-opening-transcript-challenges[OF openings])
have eqs: fri-dg = dg ∧ composition-bs' = comp-bs
  using accepted-fri-challenges-unique[OF pair-challenges accepted]
  by simp
have query-in: query-idxs ∈ Q dg comp-bs
proof –
  have query-idxs ∈
    fst ‘ (P dg ∩ (UNIV × {comp-bs}))
  using pair eqs by force
  then show ?thesis
  using projection[OF comp-bs-in] by blast
qed
from path out-eq obtain raw-idxs query-idxs-path where
  len-raw: length raw-idxs = rounds
  and query-path-in: query-idxs-path ∈ ?Q
  and query-path-eq:
    query-idxs-path =
      map (λraw. index (to-nat raw)) raw-idxs
  and lookup:
    ∧i. i < rounds ⇒
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)
  and fresh-lookup:
    ∧i. i < rounds ⇒

```

```

    fmlookup (HashMap ?s)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
    None
  unfolding staged-security-with-data-state-query-index-list-path-fresh-def
  by auto
from accepted-fri-opening-transcript-imp-accepted-transcript-shape[OF openings]
obtain as-shape where shape:
  accepted-transcript-shape ?s (Some (result, final-state))
  as-shape query-idxs
  by blast
from checked-staged-security-with-data-state-accepted-shape-query-keys
  [OF wf controlled support[unfolded out-eq] shape]
obtain raw-idxs-pair where
  len-raw-pair: length raw-idxs-pair = rounds
  and query-idxs-eq:
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs-pair
  and lookup-pair:
    ∧ i. i < rounds ⇒
      fmlookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs-pair ! i)
  by blast
have raw-eq:
  ∧ i. i < rounds ⇒ raw-idxs ! i = raw-idxs-pair ! i
  using lookup lookup-pair by simp
have raw-idxs-eq: raw-idxs = raw-idxs-pair
  by (rule nth-equalityI)
  (use len-raw len-raw-pair raw-eq in simp-all)
have path-witness:
  length raw-idxs = rounds ∧
  query-idxs ∈ Q dg comp-bs ∧
  query-idxs = map (λraw. index (to-nat raw)) raw-idxs ∧
  (∀ i < rounds.
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
    Some (raw-idxs ! i) ∧
    fmlookup (HashMap ?s)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)

```

```

      (staged-query-chunks data) i)) =
    None)
using len-row raw-idxs-eq query-in query-idxs-eq lookup
  fresh-lookup by auto
have path-body:
   $\exists$  raw-idxs query-idxs.
    length raw-idxs = rounds  $\wedge$ 
    query-idxs  $\in Q$  dg comp-bs  $\wedge$ 
    query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs  $\wedge$ 
    ( $\forall i < \text{rounds}$ .
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        Some (raw-idxs ! i)  $\wedge$ 
      fmllookup (HashMap ?s)
        (QueryIndexChallenge i
          (state-after-query-chunks
            (staged-query-start-hash data)
            (staged-query-chunks data) i)) =
        None)
    using path-witness by blast
have path-single:
  staged-security-with-data-state-query-index-list-path-fresh
    (Q dg comp-bs) out
unfolding out-eq
  staged-security-with-data-state-query-index-list-path-fresh-def
using path-body by simp
have fresh-single:
  composition-fri-challenge-list-fresh-hit ?s
    ( $\lambda$ dg'. if dg' = dg then {comp-bs} else {})
    (Some (result, final-state))
proof –
  have witness:
    accepted-fri-challenges ?s (Some (result, final-state))
      trace-bs dg comp-bs  $\wedge$ 
      verifier-header-transcript ?s fr trace-roots trace-final as dg
      composition-roots final rest  $\wedge$ 
      comp-bs  $\in$  (if dg = dg then {comp-bs} else {})  $\wedge$ 
      composition-fri-challenge-path-fresh ?s fr trace-roots
        trace-final as dg composition-roots (length composition-roots)
    using accepted header fresh-path by simp
  show ?thesis
    unfolding composition-fri-challenge-list-fresh-hit-def
    using witness by blast
qed
have singleton:
  ?E (dg, comp-bs) out

```

```

proof –
  have fresh-single-pair:
    composition-fri-challenge-list-fresh-hit ?s
    ( $\lambda dg'. \text{ if } dg' = dg \text{ then } \{\text{snd } (dg, \text{comp-bs})\} \text{ else } \{\}$ )
    (Some (result, final-state))
  proof –
    have
      ( $\lambda dg'. \text{ if } dg' = dg \text{ then } \{\text{comp-bs}\} \text{ else } \{\}$ ) =
      ( $\lambda dg'. \text{ if } dg' = dg \text{ then } \{\text{snd } (dg, \text{comp-bs})\} \text{ else } \{\}$ )
      by (rule ext) (simp add: prod.sel)
    then show ?thesis
      using fresh-single by simp
  qed
show ?thesis
unfolding out-eq
  staged-security-composition-fri-pair-challenge-query-path-fresh-def
  staged-security-composition-fri-pair-challenge-fresh-def
  staged-security-with-data-state-verifier-event-def
  using pair-hit fresh-single-pair path-single out-eq by simp
qed
show ?thesis
  by (intro beI[of - (dg, comp-bs)])
  (use comp-bs-in singleton in simp-all)
qed
qed
also have ...  $\leq (\sum p \in ?I. ?C p)$ 
proof (rule wp-event-finite-union-bound-fri-exact)
  show finite ?I
  by (rule finite-SigmaI)
  (use challenge-subset finite-fri-challenge-space in
     $\langle \text{auto intro: finite-subset}[OF \text{ challenge-subset}] \rangle$ )
next
fix p
assume p-in:  $p \in ?I$ 
obtain dg challenges where p-eq:  $p = (dg, \text{challenges})$ 
  by (cases p) auto
have challenges-in:  $\text{challenges} \in B \text{ dg}$ 
  using p-in p-eq by simp
have challenges-space:
   $\text{challenges} \in \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$ 
  using challenge-subset[of dg] challenges-in by blast
have fixed-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-composition-fri-pair-challenge-query-path-fresh P
    ( $\lambda dg'. \text{ if } dg' = dg \text{ then } \{\text{challenges}\} \text{ else } \{\}$ )
    (Q dg challenges))
  adversary-initial-state  $\leq$ 
  ( $1 / \text{nnreal } (\text{CARD}('f) \wedge \text{ceil-log } (\text{to-nat } dg + 1))) *$ 
    ( $\text{nnreal } (\text{card } (Q \text{ dg challenges})) *$ 

```

```

      (1 / nnreal (card query-sample-space)) ^ rounds)
  by (rule
    checked-staged-security-composition-fri-pair-challenge-query-path-fresh-fixed-bound
    [where dg = dg and challenges = challenges
      and Q = Q dg challenges and P = P,
      OF wf controlled challenges-space
      projection[OF challenges-in] subset[OF challenges-in]])
  have event-eq:
    ?E p =
      staged-security-composition-fri-pair-challenge-query-path-fresh P
      (λdg'. if dg' = dg then {challenges} else {})
      (Q dg challenges)
  unfolding p-eq
    staged-security-composition-fri-pair-challenge-query-path-fresh-def
    staged-security-composition-fri-pair-challenge-fresh-def
    composition-fri-challenge-list-fresh-hit-def
  by (rule ext) (simp add: prod.sel)
  have cost-eq:
    ?C p =
      (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
      (nnreal (card (Q dg challenges)) *
      (1 / nnreal (card query-sample-space)) ^ rounds)
  unfolding p-eq by simp
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (?E p) adversary-initial-state ≤ ?C p
  using fixed-bound unfolding event-eq cost-eq .
qed
finally have bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-composition-fri-pair-challenge-query-path-fresh P
      B ?Q)
  adversary-initial-state ≤ (∑ p ∈ ?I. ?C p) .
  have finite-B: ∧ dg. finite (B dg)
  by (meson challenge-subset finite-fri-challenge-space finite-subset)
  have sigma-eq:
    (∑ p ∈ ?I. ?C p) =
    (∑ dg ∈ UNIV.
      ∑ challenges ∈ B dg.
      (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
      (nnreal (card (Q dg challenges)) *
      (1 / nnreal (card query-sample-space)) ^ rounds))
  by (subst sum.Sigma) (simp-all add: finite-B case-prod-beta)
  show ?thesis
  using bound sigma-eq by simp
qed

end

end

```

```

theory Soundness-FRI-Query-Challenge-Trace-Staged-Product
  imports Soundness-FRI-Query-Challenge-Actual-Product
begin

```

Trace-side staged product accounting for the challenge-fresh/query-path-fresh FRI pair branch.

This theory is intentionally narrow. The preceding layer contains the verifier-local actual-fresh product bounds and the raw-list path bridges. Here we package the trace bridge into staged fixed-challenge and finite-cover product bounds.

```

context soundness
begin

```

lemma *verifier-after-trace-fri-query-index-list-path-fresh-imp-actual-raw-path:*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and builder:
  Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
and prefix:
  Some ((fr, f-fl), prefix-state) ∈
    set-dist
    (execute verifier-trace-fri-prefix
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))))
and read-trace-final:
  Some (f-final, s3) ∈ set-dist (execute read prefix-state)
and alpha-out:
  Some (as, s4) ∈
    set-dist (execute (mmap (replicate (length spec) alpha-round)) s3)
and read-dg: Some (dg, s5) ∈ set-dist (execute read s4)
and degree-assert:
  Some ((), s6) ∈
    set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
and comp-fri:
  Some (fl, s7) ∈
    set-dist (execute
      (ntimes receive-composition-fri-commits
        (ceil-log (to-nat dg + 1)))) s6)
and read-final:
  Some (final, query-state) ∈ set-dist (execute read s7)
and query-out:
  Some (result, final-state) ∈
    set-dist
    (execute
      (ntimes

```

```

      (verifier-query-round-program fr f-fl f-final as fl final)
      rounds)
    query-state)
  and hit:
    staged-security-with-data-state-query-index-list-path-fresh Q
      (Some (((data, attacker-state), result), final-state))
  shows  $\exists$  raws  $\in$  query-index-row-list-preimage Q.
    query-rounds-row-list-path-hit query-state raws rounds
      (Some (result, final-state))
proof -
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  from ntimes-verifier-query-rounds-outcome[OF query-out]
  obtain raw-idxs query-idxs query-chunks where
    len-row: length raw-idxs = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda$ raw. index (to-nat raw)) raw-idxs
  and len-chunks: length query-chunks = rounds
  and tr-query:
    PTranscript query-state =
      List.concat query-chunks @ PTranscript final-state
  and state-final:
    PState final-state =
      state-after-query-chunks (PState query-state) query-chunks rounds
  and round-chunks:
     $\bigwedge i. i < \text{rounds} \implies$ 
      verifier-query-round-chunk (query-idxs ! i)
        (map snd f-fl) (map snd fl) (query-chunks ! i)
  and lookups:
     $\bigwedge i. i < \text{rounds} \implies$ 
      fmlookup (HashMap final-state)
        (QueryIndexChallenge (PQueryCounter query-state + i)
          (state-after-query-chunks
            (PState query-state) query-chunks i)) =
        Some (raw-idxs ! i)
  by blast
  have trace-prefix-res:
    length f-fl = ceil-log clength  $\wedge$ 
      PTranscript ?s = [fr] @ map snd f-fl @ PTranscript prefix-state  $\wedge$ 
      PState prefix-state =
        foldl concat (concat (PState ?s) fr) (map snd f-fl)  $\wedge$ 
        PQueryCounter prefix-state = PQueryCounter ?s
  using verifier-trace-fri-prefix-outcome[OF prefix] by simp
  from read-outcome[OF read-trace-final] obtain rest3 where
    tr-prefix: PTranscript prefix-state = f-final # rest3
  and tr-s3: PTranscript s3 = rest3
  and state-s3: PState s3 = concat (PState prefix-state) f-final
  and counter-s3: PQueryCounter s3 = PQueryCounter prefix-state

```



```

  by blast
have alpha-res:
  length as = length spec ∧
  PTranscript s3 = as @ PTranscript s4 ∧
  PState s4 = foldl concat (PState s3) as ∧
  PQueryCounter s4 = PQueryCounter s3
  using mmap-alpha-round-outcome[OF alpha-out] by simp
from read-outcome[OF read-dg] obtain rest5 where
  tr-s4: PTranscript s4 = dg # rest5
  and tr-s5: PTranscript s5 = rest5
  and state-s5: PState s5 = concat (PState s4) dg
  and counter-s5: PQueryCounter s5 = PQueryCounter s4
  by blast
have s6-eq: s6 = s5
  using degree-assert unfolding assert-def
  by (cases to-nat dg ≤ maxDegree) (auto simp: throw-no-outcome)
have comp-res:
  length fl = ceil-log (to-nat dg + 1) ∧
  PTranscript s6 = map snd fl @ PTranscript s7 ∧
  PState s7 = foldl concat (PState s6) (map snd fl) ∧
  PQueryCounter s7 = PQueryCounter s6
  using ntimes-receive-composition-fri-commits-outcome[OF comp-fri]
  by simp
from read-outcome[OF read-final] obtain rest-query where
  tr-s7: PTranscript s7 = final # rest-query
  and tr-read-query: PTranscript query-state = rest-query
  and state-query: PState query-state = concat (PState s7) final
  and counter-query-s7: PQueryCounter query-state = PQueryCounter s7
  by blast
have local-counter: PQueryCounter query-state = 0
  using trace-prefix-res counter-s3 alpha-res counter-s5 s6-eq comp-res
  counter-query-s7
  unfolding verifier-state-from-adversary-def by simp
have prefix-lookup:
  ∧ i x. fmlookup (HashMap prefix-state)
    (QueryIndexChallenge i x) =
    fmlookup (HashMap ?s) (QueryIndexChallenge i x)
  by (rule verifier-trace-fri-prefix-preserves-query-lookup[OF prefix])
have s3-lookup:
  HashMap s3 = HashMap prefix-state
  by (rule read-preserves-hash-map[OF read-trace-final])
have s4-lookup:
  ∧ i x. fmlookup (HashMap s4) (QueryIndexChallenge i x) =
    fmlookup (HashMap s3) (QueryIndexChallenge i x)
proof -
  fix i x
  show fmlookup (HashMap s4) (QueryIndexChallenge i x) =
    fmlookup (HashMap s3) (QueryIndexChallenge i x)
  proof (rule mmap-preserves-lookup[OF - alpha-out])

```

```

fix m y u v
assume m-in: m ∈ set (replicate (length spec) alpha-round)
  and out: Some (y, v) ∈ set-dist (execute m u)
then have m-eq: m = alpha-round
  by simp
show fmlookup (HashMap v) (QueryIndexChallenge i x) =
  fmlookup (HashMap u) (QueryIndexChallenge i x)
  using alpha-round-preserves-query-lookup[OF out[unfolded m-eq]] .
qed
qed
have s5-lookup:
  HashMap s5 = HashMap s4
  by (rule read-preserves-hash-map[OF read-dg])
have s7-lookup:
  ∧i x. fmlookup (HashMap s7) (QueryIndexChallenge i x) =
  fmlookup (HashMap s6) (QueryIndexChallenge i x)
proof –
  fix i x
  show fmlookup (HashMap s7) (QueryIndexChallenge i x) =
  fmlookup (HashMap s6) (QueryIndexChallenge i x)
  proof (rule ntimes-preserves-lookup[OF - comp-fri])
    fix y u v
    assume out:
      Some (y, v) ∈
        set-dist (execute receive-composition-fri-commits u)
    obtain b r where y-eq: y = (b, r)
    by (cases y) simp
    show fmlookup (HashMap v) (QueryIndexChallenge i x) =
    fmlookup (HashMap u) (QueryIndexChallenge i x)
    by (rule receive-composition-fri-commits-preserves-query-lookup
      [OF out[unfolded y-eq]])
  qed
qed
have query-lookup:
  HashMap query-state = HashMap s7
  by (rule read-preserves-hash-map[OF read-final])
have header-actual:
  verifier-header-transcript ?s fr (map snd f-fl) f-final as dg
  (map snd fl) final (PTranscript query-state)
  using trace-prefix-res tr-prefix tr-s3 alpha-res tr-s4 tr-s5 s6-eq
  comp-res tr-s7 tr-read-query
  unfolding verifier-header-transcript-def verifier-header-messages-def
  by simp
have staged-header:
  verifier-header-transcript ?s
  (staged-trace-root data)
  (staged-trace-fri-roots data)
  (staged-trace-final data)
  (staged-alphas data)

```

```

    (staged-degree data)
    (staged-composition-fri-roots data)
    (staged-composition-final data)
    (List.concat (staged-query-chunks data))
  by (rule checked-staged-transcript-program-outcome-header-transcript
      [OF wf controlled builder])
have header-unique:
  fr = staged-trace-root data ∧
  map snd f-fl = staged-trace-fri-roots data ∧
  f-final = staged-trace-final data ∧
  as = staged-alphas data ∧
  dg = staged-degree data ∧
  map snd fl = staged-composition-fri-roots data ∧
  final = staged-composition-final data ∧
  PTranscript query-state = List.concat (staged-query-chunks data)
using verifier-header-transcript-unique[OF header-actual staged-header]
by simp
have prefix-state-query:
  PState query-state = staged-query-start-hash data
using trace-prefix-res state-s3 alpha-res state-s5 s6-eq comp-res
  state-query header-unique
unfolding staged-query-start-hash-def
  staged-composition-fri-start-hash-def
  staged-trace-fri-start-hash-def
by simp
have chunks-eq: query-chunks = staged-query-chunks data
proof -
  from checked-staged-transcript-program-query-chunks-match-verifier-lengths
    [OF builder]
  obtain staged-query-idxs where
    staged-match:
      staged-query-chunks-match-verifier-lengths data staged-query-idxs
  by blast
have query-idxs-len: length query-idxs = rounds
  using query-idxs-eq len-raw by simp
have staged-match-query-idxs:
  staged-query-chunks-match-verifier-lengths data query-idxs
  by (rule staged-query-chunks-match-verifier-lengths-transfer
      [OF staged-match query-idxs-len])
have parser-chunk-len:
  ∧i. i < rounds ⇒
    length (query-chunks ! i) =
      verifier-query-round-transcript-length (query-idxs ! i)
      (staged-trace-fri-roots data)
      (staged-composition-fri-roots data)
proof -
  fix i
  assume i-bound: i < rounds
  show

```

```

length (query-chunks ! i) =
  verifier-query-round-transcript-length (query-idxs ! i)
    (staged-trace-fri-roots data)
    (staged-composition-fri-roots data)
using verifier-query-round-chunk-length[OF round-chunks[OF i-bound]]
  header-unique
by simp
qed
have concat-staged:
  List.concat query-chunks @ PTranscript final-state =
    List.concat (staged-query-chunks data)
using tr-query header-unique by simp
then show ?thesis
using query-chunks-eq-staged-if-matching-lengths
  [OF len-chunks concat-staged parser-chunk-len
    staged-match-query-idxs]
by simp
qed
from hit obtain raw-idxs-hit query-idxs-hit where
  query-hit-in: query-idxs-hit ∈ Q
and len-hit: length raw-idxs-hit = rounds
and query-hit-eq:
  query-idxs-hit =
    map (λraw. index (to-nat raw)) raw-idxs-hit
and lookup-hit:
  ∧i. i < rounds ⇒
    fmllookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs-hit ! i)
and fresh-hit:
  ∧i. i < rounds ⇒
    fmllookup (HashMap ?s)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      None
unfolding staged-security-with-data-state-query-index-list-path-fresh-def
by auto
have raw-eq:
  ∧i. i < rounds ⇒ raw-idxs ! i = raw-idxs-hit ! i
using lookups lookup-hit local-counter prefix-state-query chunks-eq
by simp
have raw-idxs-eq: raw-idxs = raw-idxs-hit
by (rule nth-equalityI)
  (use len-raw len-hit raw-eq in simp-all)

```

```

have raw-in: raw-idxs ∈ query-index-raw-list-preimage Q
  unfolding query-index-raw-list-preimage-def
  using len-raw raw-idxs-eq query-hit-in query-hit-eq by blast
have query-lookup-start:
   $\bigwedge i x. \text{fmlookup } (\text{HashMap } \text{query-state}) (\text{QueryIndexChallenge } i x) =$ 
     $\text{fmlookup } (\text{HashMap } ?s) (\text{QueryIndexChallenge } i x)$ 
  using prefix-lookup s3-lookup s4-lookup s5-lookup s6-eq
    s7-lookup query-lookup by simp
have local-fresh:
  query-index-list-path-fresh query-state rounds query-chunks raw-idxs
  unfolding query-index-list-path-fresh-def
proof (intro conjI allI impI)
  show length raw-idxs = rounds
    using len-raw .
next
  show length query-chunks = rounds
    using len-chunks .
next
  fix i
  assume i-bound:  $i < \text{rounds}$ 
  let ?x =
    state-after-query-chunks (staged-query-start-hash data)
      (staged-query-chunks data) i
  have fmlookup (HashMap query-state)
    (QueryIndexChallenge (PQueryCounter query-state + i)
      (state-after-query-chunks (PState query-state) query-chunks i)) =
    fmlookup (HashMap ?s) (QueryIndexChallenge i ?x)
  using local-counter prefix-state-query chunks-eq
    query-lookup-start[of i ?x] by simp
  also have ... = None
    by (rule fresh-hit[OF i-bound])
  finally show
    fmlookup (HashMap query-state)
      (QueryIndexChallenge (PQueryCounter query-state + i)
        (state-after-query-chunks (PState query-state) query-chunks i)) =
      None .
qed
have path:
  query-rounds-raw-list-path-hit query-state raw-idxs rounds
    (Some (result, final-state))
  unfolding query-rounds-raw-list-path-hit-def
  using len-raw len-chunks state-final local-fresh lookups
  by auto
show ?thesis
  using raw-in path by blast
qed

lemma verifier-after-trace-fri-query-index-list-path-fresh-product-bound-after-prefix:
  assumes wf: staged-budget-wellformed budgets

```

and *controlled*: staged-adversary-controlled budgets A
and *subset*: $Q \subseteq \text{fri-query-index-list-space}$
and *builder*:
 Some (data , attacker-state) $\in$
 set-dist
 (*execute* (*checked-staged-transcript-program* A)
 adversary-initial-state)
and *prefix*:
 Some (header , prefix-state) $\in$
 set-dist
 (*execute* *verifier-trace-fri-prefix*
 (*verifier-state-from-adversary* attacker-state
 (*staged-proof-transcript* data))))
shows
 wp-event (*verifier-after-trace-fri* header)
 ($\lambda \text{out.}$
 staged-security-with-data-state-query-index-list-path-fresh Q
 (*case out of*
 $\text{None} \Rightarrow \text{None}$
 | $\text{Some} (\text{result}, \text{final-state}) \Rightarrow$
 Some (((data , attacker-state), result), final-state)))
 prefix-state
 $\leq \text{nnreal} (\text{card } Q) *$
 $(1 / \text{nnreal} (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
proof –
 obtain fr f-fl **where** *header-eq*: $\text{header} = (\text{fr}, \text{f-fl})$
 by (*cases* header) *auto*
 let $?P =$
 $\lambda \text{out.}$
 staged-security-with-data-state-query-index-list-path-fresh Q
 (*case out of*
 $\text{None} \Rightarrow \text{None}$
 | $\text{Some} (\text{result}, \text{final-state}) \Rightarrow$
 Some (((data , attacker-state), result), final-state)))
 let $?C =$
 $\text{nnreal} (\text{card } Q) *$
 $(1 / \text{nnreal} (\text{card } \text{query-sample-space})) \wedge \text{rounds}$
 have *prefix'*:
 Some ((fr , f-fl), prefix-state) $\in$
 set-dist
 (*execute* *verifier-trace-fri-prefix*
 (*verifier-state-from-adversary* attacker-state
 (*staged-proof-transcript* data))))
 using *prefix* *header-eq* **by** *simp*
 have *after-eq*:
 verifier-after-trace-fri $\text{header} =$
 (*read* $\gg=$
 ($\lambda \text{f-final.}$
 mmap (*replicate* (*length spec*) $\alpha\text{-round}$) $\gg=$


```

      (ceil-log (to-nat dg + 1)) >>=
      (λfl. read >>=
      (λfinal.
      ntimes
      (verifier-query-round-program fr f-fl f-final as fl
      final)
      rounds)))) ?P s4 ≤ ?C
proof (rule wp-event-bind-bound-by-cont)
show ¬ ?P None
  unfolding staged-security-with-data-state-query-index-list-path-fresh-def
  by simp
next
fix dg s5
assume read-dg:
  Some (dg, s5) ∈ set-dist (execute read s4)
show
  wp-event
  (assert (to-nat dg ≤ maxDegree) >>=
  (λ-. ntimes receive-composition-fri-commits
  (ceil-log (to-nat dg + 1)) >>=
  (λfl. read >>=
  (λfinal.
  ntimes
  (verifier-query-round-program fr f-fl f-final as fl
  final)
  rounds)))) ?P s5 ≤ ?C
proof (rule wp-event-bind-bound-by-cont)
show ¬ ?P None
  unfolding staged-security-with-data-state-query-index-list-path-fresh-def
  by simp
next
fix unit s6
assume degree-assert:
  Some (unit, s6) ∈
  set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
show
  wp-event
  (ntimes receive-composition-fri-commits
  (ceil-log (to-nat dg + 1)) >>=
  (λfl. read >>=
  (λfinal.
  ntimes
  (verifier-query-round-program fr f-fl f-final as fl
  final)
  rounds)))) ?P s6 ≤ ?C
proof (rule wp-event-bind-bound-by-cont)
show ¬ ?P None
  unfolding staged-security-with-data-state-query-index-list-path-fresh-def
  by simp

```



```

next
  fix fl s7
  assume comp-fri:
    Some (fl, s7) ∈
      set-dist
      (execute
        (ntimes receive-composition-fri-commits
          (ceil-log (to-nat dg + 1)))) s6)
  show
    wp-event
      (read  $\gg$ 
        ( $\lambda$ final.
          ntimes
            (verifier-query-round-program fr f-fl f-final as fl
              final)
            rounds)) ?P s7 ≤ ?C
  proof (rule wp-event-bind-bound-by-cont)
  show  $\neg ?P \text{ None}$ 
    unfolding staged-security-with-data-state-query-index-list-path-fresh-def
    by simp
  next
    fix final query-state
    assume read-final:
      Some (final, query-state) ∈ set-dist (execute read s7)
    have degree-assert-unit:
      Some ((), s6) ∈
        set-dist (execute (assert (to-nat dg ≤ maxDegree)) s5)
    using degree-assert by (cases unit) simp
    have mono:
      wp-event
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final
            rounds)
          ?P query-state ≤
      wp-event
        (ntimes
          (verifier-query-round-program fr f-fl f-final as fl final
            rounds)
          ( $\lambda$ out. ∃ raws ∈ query-index-raw-list-preimage Q.
            query-rounds-raw-list-path-hit query-state raws rounds out)
          query-state
      proof (rule wp-event-mono-on-support)
      fix out
      assume query-out:
        out ∈
          set-dist
          (execute
            (ntimes
              (verifier-query-round-program fr f-fl f-final as fl

```

```

      final)
      rounds)
      query-state)
    and hit: ?P out
  show  $\exists \text{raws} \in \text{query-index-raw-list-preimage } Q.$ 
      query-rounds-raw-list-path-hit query-state raws rounds out
  proof (cases out)
    case None
    then show ?thesis
      using hit
      unfolding staged-security-with-data-state-query-index-list-path-fresh-def
      by simp
  next
  case (Some pair)
  then obtain result final-state where out-eq:
    out = Some (result, final-state)
  by (cases pair) simp
  have query-out':
    Some (result, final-state)  $\in$ 
      set-dist
      (execute
      (ntimes
      (verifier-query-round-program fr f-fl f-final as fl
      final)
      rounds)
      query-state)
  using query-out out-eq by simp
  have hit':
    staged-security-with-data-state-query-index-list-path-fresh Q
    (Some (((data, attacker-state), result), final-state))
  using hit out-eq by simp
  show ?thesis
    using
      verifier-after-trace-fri-query-index-list-path-fresh-imp-actual-raw-path
      [OF wf controlled builder prefix' read-trace-final
      alpha-out read-dg degree-assert-unit comp-fri read-final
      query-out' hit']
    unfolding out-eq .
  qed
  qed
  also have ...  $\leq$  ?C
  by (rule
      wp-ntimes-verifier-query-round-program-query-list-path-product-bound
      [OF subset])
  finally show
    wp-event
    (ntimes
    (verifier-query-round-program fr f-fl f-final as fl final)
    rounds)

```

```

      ?P query-state ≤ ?C .
    qed
  qed
  qed
  qed
  qed
  qed
  qed
  lemma checked-staged-security-trace-fri-pair-challenge-query-path-fresh-fixed-bound:
    assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and challenges-space:
      challenges ∈ fri-challenge-space (ceil-log clength)
    and projection:
      fst ‘ (P ∩ (UNIV × {challenges})) ⊆ Q
    and subset: Q ⊆ fri-query-index-list-space
    shows
      wp-event (checked-staged-security-experiment-with-data-state A)
        (staged-security-trace-fri-pair-challenge-query-path-fresh P
          {challenges} Q)
      adversary-initial-state ≤
        (1 / nnreal (CARD('f) ^ ceil-log clength)) *
        (nnreal (card Q) *
          (1 / nnreal (card query-sample-space)) ^ rounds)
  proof (rule checked-staged-security-with-data-state-bound-from-data-cont)
    let ?D =
      nnreal (card Q) *
      (1 / nnreal (card query-sample-space)) ^ rounds
    show ¬ staged-security-trace-fri-pair-challenge-query-path-fresh
      P {challenges} Q None
    unfolding staged-security-trace-fri-pair-challenge-query-path-fresh-def
      staged-security-trace-fri-pair-challenge-fresh-def
      staged-security-with-data-state-verifier-event-def
    by simp
  next
    fix data attacker-state
    assume builder:
      Some (data, attacker-state) ∈
      set-dist
        (execute (checked-staged-transcript-program A)
          adversary-initial-state)
    let ?s =
      verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
    let ?Event =
      λout.
        staged-security-trace-fri-pair-challenge-query-path-fresh P
          {challenges} Q

```

```

    (case out of
      None  $\Rightarrow$  None
    | Some (result, final-state)  $\Rightarrow$ 
      Some (((data, attacker-state), result), final-state))
let ?Head =
   $\lambda$ out. case out of
    None  $\Rightarrow$  False
  | Some ((fr, f-fl), -)  $\Rightarrow$ 
    map fst f-fl = challenges  $\wedge$ 
    trace-fri-challenge-path-fresh ?s fr (map snd f-fl)
    (length (map snd f-fl))
show wp-event verify-monad ?Event ?s  $\leq$ 
  (1 / nnreal (CARD('f)  $\wedge$  ceil-log clength)) *
  (nnreal (card Q) *
    (1 / nnreal (card query-sample-space))  $\wedge$  rounds)
unfolding verify-monad-trace-fri-decomposition
proof (rule order-trans)
show wp-event
  (verifier-trace-fri-prefix  $\gg=$  verifier-after-trace-fri)
  ?Event ?s
 $\leq$  wp-event verifier-trace-fri-prefix ?Head ?s *
  (nnreal (card Q) *
    (1 / nnreal (card query-sample-space))  $\wedge$  rounds)
proof (rule wp-event-bind-bound-by-head-and-cont)
show  $\neg$  ?Event None
  unfolding staged-security-trace-fri-pair-challenge-query-path-fresh-def
    staged-security-trace-fri-pair-challenge-fresh-def
    staged-security-with-data-state-verifier-event-def
  by simp
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state)  $\in$ 
    set-dist (execute verifier-trace-fri-prefix ?s)
  and not-head:  $\neg$  ?Head (Some (header, prefix-state))
obtain fr f-fl where header-eq: header = (fr, f-fl)
  by (cases header) auto
have not-head':
  map fst f-fl  $\neq$  challenges  $\vee$ 
     $\neg$  trace-fri-challenge-path-fresh ?s fr (map snd f-fl)
    (length (map snd f-fl))
  using not-head unfolding header-eq by simp
show wp-event (verifier-after-trace-fri header) ?Event prefix-state =
  0
proof (rule antisym)
show wp-event (verifier-after-trace-fri header) ?Event
  prefix-state  $\leq$  0
proof (rule order-trans)
  show wp-event (verifier-after-trace-fri header) ?Event

```

```

    prefix-state ≤
    wp-event (verifier-after-trace-fri header)
    (λ-. False) prefix-state
proof (rule wp-event-mono-on-support)
fix out
assume suffix:
    out ∈
    set-dist
    (execute (verifier-after-trace-fri header) prefix-state)
and hit: ?Event out
show False
proof (cases out)
case None
then show ?thesis
using hit
unfolding
    staged-security-trace-fri-pair-challenge-query-path-fresh-def
    staged-security-trace-fri-pair-challenge-fresh-def
    staged-security-with-data-state-verifier-event-def
by simp
next
case (Some pair)
then obtain result final-state where out-eq:
    out = Some (result, final-state)
by (cases pair) simp
from hit out-eq have fresh:
    trace-fri-challenge-list-fresh-hit ?s {challenges}
    (Some (result, final-state))
and pair-hit:
    trace-fri-query-challenge-pair-set-hit ?s P
    (Some (result, final-state))
unfolding
    staged-security-trace-fri-pair-challenge-query-path-fresh-def
    staged-security-trace-fri-pair-challenge-fresh-def
    staged-security-with-data-state-verifier-event-def
by simp-all
have prefix':
    Some ((fr, f-fl), prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix ?s)
using prefix header-eq by simp
have suffix':
    Some (result, final-state) ∈
    set-dist
    (execute (verifier-after-trace-fri (fr, f-fl))
    prefix-state)
using suffix header-eq out-eq by simp
from pair-hit obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final query-idxs
    trace-round-layers composition-round-layers where

```

```

    openings:
      accepted-fri-opening-transcript ?s
      (Some (result, final-state)) trace-roots trace-bs
      trace-final fri-dg composition-roots composition-bs
      composition-final query-idxs trace-round-layers
      composition-round-layers
    unfolding trace-fri-query-challenge-pair-set-hit-def by blast
  have pair-challenges:
    accepted-fri-challenges ?s (Some (result, final-state))
    trace-bs fri-dg composition-bs
  by (rule accepted-fri-opening-transcript-challenges
    [OF openings])
  from fresh obtain trace-bs' dg' comp-bs' where
    fresh-challenges:
      accepted-fri-challenges ?s (Some (result, final-state))
      trace-bs' dg' comp-bs'
  and trace-bs'-in: trace-bs'  $\in$  {challenges}
  unfolding trace-fri-challenge-list-fresh-hit-def by blast
  have trace-bs-eq: trace-bs = trace-bs'
  using accepted-fri-challenges-unique
    [OF pair-challenges fresh-challenges]
  by simp
  have prefix-trace-eq: trace-bs = map fst f-fl
  by (rule
    verifier-after-trace-fri-header-agrees-with-accepted-challenges
    [OF prefix' suffix' pair-challenges])
  have concrete-fresh:
    trace-fri-challenge-path-fresh ?s fr (map snd f-fl)
    (length (map snd f-fl))
  by (rule verifier-after-trace-fri-header-roots-agree-with-fresh-hit
    [OF prefix' suffix' fresh])
  have head-holds:
    map fst f-fl = challenges  $\wedge$ 
    trace-fri-challenge-path-fresh ?s fr (map snd f-fl)
    (length (map snd f-fl))
  using prefix-trace-eq trace-bs-eq trace-bs'-in
    concrete-fresh by simp
  show ?thesis
  using not-head' head-holds by blast
qed
qed
next
  show wp-event (verifier-after-trace-fri header)
    ( $\lambda$ -. False) prefix-state  $\leq 0$ 
  unfolding wp-event-def wp-def dist-expect-def by simp
qed
next
  show  $0 \leq$  wp-event (verifier-after-trace-fri header) ?Event
    prefix-state

```

```

    by simp
qed
next
fix header prefix-state
assume prefix:
  Some (header, prefix-state) ∈
    set-dist (execute verifier-trace-fri-prefix ?s)
  and head: ?Head (Some (header, prefix-state))
obtain fr f-fl where header-eq: header = (fr, f-fl)
  by (cases header) auto
have event-mono:
  wp-event (verifier-after-trace-fri header) ?Event prefix-state ≤
  wp-event (verifier-after-trace-fri header)
  (λout.
    staged-security-with-data-state-query-index-list-path-fresh Q
    (case out of
      None ⇒ None
      | Some (result, final-state) ⇒
        Some (((data, attacker-state), result), final-state)))
    prefix-state
  by (rule wp-event-mono-on-support)
  (auto simp: staged-security-trace-fri-pair-challenge-query-path-fresh-def)
also have ... ≤
  nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds
  by (rule
    verifier-after-trace-fri-query-index-list-path-fresh-product-bound-after-prefix
    [OF wf controlled subset builder prefix])
finally show wp-event (verifier-after-trace-fri header) ?Event
  prefix-state ≤
  nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds .
qed
next
show wp-event verifier-trace-fri-prefix ?Head ?s *
  (nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds)
  ≤
  (1 / nnreal (CARD('f) ^ ceil-log clength)) *
  (nnreal (card Q) *
  (1 / nnreal (card query-sample-space)) ^ rounds)
proof (rule mult-right-mono)
show wp-event verifier-trace-fri-prefix ?Head ?s ≤
  1 / nnreal (CARD('f) ^ ceil-log clength)
proof -
  let ?HeadSet =
  λout. case out of
    None ⇒ False
    | Some ((fr, f-fl), -) ⇒

```

```

      map fst f-fl ∈ {challenges} ∧
      trace-fri-challenge-path-fresh ?s fr (map snd f-fl)
      (length (map snd f-fl))
    have mono:
      wp-event verifier-trace-fri-prefix ?Head ?s ≤
      wp-event verifier-trace-fri-prefix ?HeadSet ?s
    by (rule wp-event-mono) (auto split: option.splits prod.splits)
    have set-bound:
      wp-event verifier-trace-fri-prefix ?HeadSet ?s ≤
      nnreal (card {challenges}) /
      nnreal (CARD('f) ^ ceil-log clength)
    by (rule
      wp-verifier-trace-fri-prefix-challenge-list-actual-fresh-bound
      [where B = {challenges}])
      (use challenges-space in auto)
    show ?thesis
      using order-trans[OF mono set-bound] by simp
  qed
  show 0 ≤ nnreal (card Q) *
    (1 / nnreal (card query-sample-space)) ^ rounds
  by simp
  qed
  qed
  qed

```

lemma *checked-staged-security-trace-fri-pair-challenge-query-path-fresh-fiber-sum:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *challenge-subset: B ⊆ fri-challenge-space (ceil-log clength)*

and *projection:*

$\bigwedge \text{challenges. challenges} \in B \implies$

$\text{fst } '(P \cap (UNIV \times \{\text{challenges}\})) \subseteq Q \text{ challenges}$

and *subset:*

$\bigwedge \text{challenges. challenges} \in B \implies$

$Q \text{ challenges} \subseteq \text{fri-query-index-list-space}$

shows

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$

$(\text{staged-security-trace-fri-pair-challenge-query-path-fresh } P B$

$(\bigcup \text{challenges} \in B. Q \text{ challenges}))$

$\text{adversary-initial-state} \leq$

$(\sum \text{challenges} \in B.$

$(1 / \text{nnreal } (\text{CARD}('f) ^ \text{ceil-log clength})) *$

$(\text{nnreal } (\text{card } (Q \text{ challenges})) *$

$(1 / \text{nnreal } (\text{card query-sample-space})) ^ \text{rounds}))$

proof –

let $?Q = (\bigcup \text{challenges} \in B. Q \text{ challenges})$

let $?E =$

$\lambda \text{challenges.}$


```

    staged-security-trace-fri-pair-challenge-query-path-fresh P
      {challenges} (Q challenges)
  let ?C =
    λchallenges.
      (1 / nnreal (CARD('f) ^ ceil-log clength)) *
      (nnreal (card (Q challenges)) *
        (1 / nnreal (card query-sample-space)) ^ rounds)
  have split:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-trace-fri-pair-challenge-query-path-fresh P B ?Q)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (λout. ∃ challenges ∈ B. ?E challenges out)
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈
      set-dist
      (execute (checked-staged-security-experiment-with-data-state A)
        adversary-initial-state)
  and hit:
    staged-security-trace-fri-pair-challenge-query-path-fresh P B ?Q out
  show ∃ challenges ∈ B. ?E challenges out
  proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding staged-security-trace-fri-pair-challenge-query-path-fresh-def
      staged-security-trace-fri-pair-challenge-fresh-def
      staged-security-with-data-state-verifier-event-def
    by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed) (auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  from hit out-eq have pair-hit:
    trace-fri-query-challenge-pair-set-hit ?s P
      (Some (result, final-state))
  and fresh:
    trace-fri-challenge-list-fresh-hit ?s B
      (Some (result, final-state))
  and path:
    staged-security-with-data-state-query-index-list-path-fresh ?Q out
  unfolding staged-security-trace-fri-pair-challenge-query-path-fresh-def

```

```

    staged-security-trace-fri-pair-challenge-fresh-def
    staged-security-with-data-state-verifier-event-def
  by simp-all
from fresh obtain trace-bs dg comp-bs fr trace-roots trace-final as
  composition-roots final rest where
  accepted:
    accepted-fri-challenges ?s (Some (result, final-state))
    trace-bs dg comp-bs
  and trace-bs-in: trace-bs ∈ B
  and header:
    verifier-header-transcript ?s fr trace-roots trace-final as dg
    composition-roots final rest
  and fresh-path:
    trace-fri-challenge-path-fresh ?s fr trace-roots
    (length trace-roots)
  unfolding trace-fri-challenge-list-fresh-hit-def by blast
from pair-hit obtain trace-roots' trace-bs' trace-final' fri-dg
  composition-roots' composition-bs' composition-final' query-idxs
  trace-round-layers composition-round-layers where
  openings:
    accepted-fri-opening-transcript ?s
    (Some (result, final-state)) trace-roots' trace-bs'
    trace-final' fri-dg composition-roots' composition-bs'
    composition-final' query-idxs trace-round-layers
    composition-round-layers
  and pair: (query-idxs, trace-bs') ∈ P
  unfolding trace-fri-query-challenge-pair-set-hit-def by blast
have pair-challenges:
  accepted-fri-challenges ?s (Some (result, final-state))
  trace-bs' fri-dg composition-bs'
  by (rule accepted-fri-opening-transcript-challenges[OF openings])
have trace-eq: trace-bs' = trace-bs
  using accepted-fri-challenges-unique[OF pair-challenges accepted]
  by simp
have query-in: query-idxs ∈ Q trace-bs
proof -
  have query-idxs ∈ fst ' (P ∩ (UNIV × {trace-bs}))
  using pair trace-eq by force
  then show ?thesis
  using projection[OF trace-bs-in] by blast
qed
from path out-eq obtain raw-idxs query-idxs-path where
  len-raw: length raw-idxs = rounds
  and query-path-in: query-idxs-path ∈ ?Q
  and query-path-eq:
    query-idxs-path =
      map (λraw. index (to-nat raw)) raw-idxs
  and lookup:
    ∧i. i < rounds ⇒

```

```

    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
    Some (raw-idxs ! i)
  and fresh-lookup:
     $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap ?s)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
    None
  unfolding staged-security-with-data-state-query-index-list-path-fresh-def
  by auto
  from accepted-fri-opening-transcript-imp-accepted-transcript-shape[OF openings]
  obtain as-shape where shape:
    accepted-transcript-shape ?s (Some (result, final-state))
    as-shape query-idxs
  by blast
  from checked-staged-security-with-data-state-accepted-shape-query-keys
    [OF wf controlled support[unfolded out-eq] shape]
  obtain raw-idxs-pair where
    len-raw-pair: length raw-idxs-pair = rounds
  and query-idxs-eq:
    query-idxs = map ( $\lambda \text{raw}. \text{index } (\text{to-nat } \text{raw})$ ) raw-idxs-pair
  and lookup-pair:
     $\bigwedge i. i < \text{rounds} \implies$ 
    fmlookup (HashMap final-state)
      (QueryIndexChallenge i
        (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
    Some (raw-idxs-pair ! i)
  by blast
  have raw-eq:
     $\bigwedge i. i < \text{rounds} \implies \text{raw-idxs } ! i = \text{raw-idxs-pair } ! i$ 
  using lookup lookup-pair by simp
  have raw-idxs-eq: raw-idxs = raw-idxs-pair
  by (rule nth-equalityI)
    (use len-raw len-raw-pair raw-eq in simp-all)
  have path-witness:
    length raw-idxs = rounds  $\wedge$ 
    query-idxs  $\in Q \text{ trace-bs } \wedge$ 
    query-idxs = map ( $\lambda \text{raw}. \text{index } (\text{to-nat } \text{raw})$ ) raw-idxs  $\wedge$ 
    ( $\forall i < \text{rounds}. \text{fmlookup } (\text{HashMap } \text{final-state})$ 
      (QueryIndexChallenge i

```

```

      (state-after-query-chunks
       (staged-query-start-hash data)
       (staged-query-chunks data) i)) =
    Some (raw-idxs ! i) ∧
    fmllookup (HashMap ?s)
      (QueryIndexChallenge i
       (state-after-query-chunks
        (staged-query-start-hash data)
        (staged-query-chunks data) i)) =
    None)
  using len-raw raw-idxs-eq query-in query-idxs-eq lookup
    fresh-lookup by auto
have path-body:
  ∃ raw-idxs query-idxs.
    length raw-idxs = rounds ∧
    query-idxs ∈ Q trace-bs ∧
    query-idxs = map (λraw. index (to-nat raw)) raw-idxs ∧
    (∀ i < rounds.
      fmllookup (HashMap final-state)
        (QueryIndexChallenge i
         (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      Some (raw-idxs ! i) ∧
      fmllookup (HashMap ?s)
        (QueryIndexChallenge i
         (state-after-query-chunks
          (staged-query-start-hash data)
          (staged-query-chunks data) i)) =
      None)
    using path-witness by blast
have path-single:
  staged-security-with-data-state-query-index-list-path-fresh
    (Q trace-bs) out
  unfolding out-eq
    staged-security-with-data-state-query-index-list-path-fresh-def
  using path-body by simp
have fresh-single:
  trace-fri-challenge-list-fresh-hit ?s {trace-bs}
    (Some (result, final-state))
proof -
  have witness:
    accepted-fri-challenges ?s (Some (result, final-state))
      trace-bs dg comp-bs ∧
      verifier-header-transcript ?s fr trace-roots trace-final as dg
      composition-roots final rest ∧
      trace-bs ∈ {trace-bs} ∧
      trace-fri-challenge-path-fresh ?s fr trace-roots
      (length trace-roots)

```

```

    using accepted header fresh-path by simp
  show ?thesis
    unfolding trace-fri-challenge-list-fresh-hit-def
    using witness by blast
qed
have singleton:
  ?E trace-bs out
  unfolding out-eq
    staged-security-trace-fri-pair-challenge-query-path-fresh-def
    staged-security-trace-fri-pair-challenge-fresh-def
    staged-security-with-data-state-verifier-event-def
  using pair-hit fresh-single path-single out-eq by simp
show ?thesis
  using singleton trace-bs-in by blast
qed
qed
also have ... ≤ (∑ challenges ∈ B. ?C challenges)
proof (rule wp-event-finite-union-bound-fri-exact)
  show finite B
    using challenge-subset by (rule finite-subset)
    (rule finite-fri-challenge-space)
next
  fix challenges
  assume challenges-in: challenges ∈ B
  have challenges-space:
    challenges ∈ fri-challenge-space (ceil-log clength)
    using challenge-subset challenges-in by blast
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (?E challenges) adversary-initial-state ≤ ?C challenges
    by (rule
      checked-staged-security-trace-fri-pair-challenge-query-path-fresh-fixed-bound
      [OF wf controlled challenges-space
        projection[OF challenges-in] subset[OF challenges-in]])
qed
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-query-challenge-pair-set-hit-challenge-weighted-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and challenge-subset: B ⊆ fri-challenge-space (ceil-log clength)
  and challenge-projection: snd ‘ P ⊆ B
  and query-projection:
    ∧ challenges.
      challenges ∈ B ⇒
      fst ‘ (P ∩ (UNIV × {challenges})) ⊆ Q challenges
  and subset:
    ∧ challenges. challenges ∈ B ⇒
      Q challenges ⊆ fri-query-index-list-space

```

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s. \text{trace-fri-query-challenge-pair-set-hit } s \ P$))
adversary-initial-state $\leq$
 ($\sum \text{challenges} \in B.$
 ($1 / \text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log clength})) *$
 ($\text{nnreal} (\text{card } (Q \text{ challenges})) *$
 ($1 / \text{nnreal} (\text{card query-sample-space})) \wedge \text{rounds}$)) +
hash-relation-budget-value
 (*query-index-raw-list-relation-fiber-bound*
 ($\bigcup \text{challenges} \in B. Q \text{ challenges}$))
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget) +
hash-relation-budget-value ($\text{card } B * \text{ceil-log clength}$)
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)

proof –

let $?Q = (\bigcup \text{challenges} \in B. Q \text{ challenges})$
let $?Path =$
 ($\sum \text{challenges} \in B.$
 ($1 / \text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log clength})) *$
 ($\text{nnreal} (\text{card } (Q \text{ challenges})) *$
 ($1 / \text{nnreal} (\text{card query-sample-space})) \wedge \text{rounds}$))
let $?QueryPre =$
hash-relation-budget-value
 (*query-index-raw-list-relation-fiber-bound* $?Q$)
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)
let $?ChallengePre =$
hash-relation-budget-value ($\text{card } B * \text{ceil-log clength}$)
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget)
have *finite-B*: *finite B*
using *challenge-subset* **by** (*rule finite-subset*)
 (*rule finite-fri-challenge-space*)
have *fst-projection*: *fst* ‘ $P \subseteq ?Q$
proof
fix q
assume $q\text{-in}: q \in \text{fst} \text{ ‘ } P$
then obtain c **where** $\text{pair}: (q, c) \in P$
by *auto*
have $c\text{-in}: c \in B$
using *challenge-projection pair* **by** *force*
have $q \in \text{fst} \text{ ‘ } (P \cap (UNIV \times \{c\}))$
using pair **by** *force*
then have $q \in Q \ c$
using *query-projection*[*OF c-in*] **by** *blast*
then show $q \in ?Q$

using *c-in* by *blast*
qed
have *path-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-trace-fri-pair-challenge-query-path-fresh P B ?Q*)
 adversary-initial-state $\leq$ *?Path*
 by (rule
 checked-staged-security-trace-fri-pair-challenge-query-path-fresh-fiber-sum
 [*OF wf controlled challenge-subset query-projection subset*])
have *fresh-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-trace-fri-pair-challenge-fresh P B*)
 adversary-initial-state $\leq$ *?Path* + *?QueryPre*
 by (rule
 checked-staged-security-trace-fri-pair-challenge-fresh-bound-from-query-path-fresh
 [*OF wf controlled fst-projection path-bound*])
have *pair-bound*:
 wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s.$ *trace-fri-query-challenge-pair-set-hit s P*))
 adversary-initial-state $\leq$
 (*?Path* + *?QueryPre*) + *?ChallengePre*
 by (rule
 checked-staged-security-trace-fri-query-challenge-pair-set-hit-bound-from-challenge-fresh
 [*OF wf controlled finite-B challenge-projection fresh-bound*])
then show *?thesis*
 by (*simp add: add.assoc*)
qed

lemma *checked-staged-security-composition-fri-query-challenge-pair-set-hit-challenge-weighted-bound*:

assumes *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
and *challenge-subset*:
 $\bigwedge dg. B \ dg \subseteq \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))$
and *challenge-projection*: $\bigwedge dg. \text{snd } ' P \ dg \subseteq B \ dg$
and *query-projection*:
 $\bigwedge dg \text{ challenges. } challenges \in B \ dg \implies$
 $\text{fst } ' (P \ dg \cap (UNIV \times \{challenges\})) \subseteq$
 $Q \ dg \text{ challenges}$
and *subset*:
 $\bigwedge dg \text{ challenges. } challenges \in B \ dg \implies$
 $Q \ dg \text{ challenges} \subseteq \text{fri-query-index-list-space}$

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s.$ *composition-fri-query-challenge-pair-set-hit s P*))
adversary-initial-state $\leq$
 ($\sum dg \in UNIV.$

```

     $\sum dg \in B$ 
     $(1 / \text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log} (\text{to-nat } dg + 1))) *$ 
     $(\text{nnreal} (\text{card } (Q \text{ } dg \text{ challenges})) *$ 
     $(1 / \text{nnreal} (\text{card } \text{query-sample-space})) \wedge \text{rounds})) +$ 
    hash-relation-budget-value
     $(\text{query-index-raw-list-relation-fiber-bound}$ 
     $(\bigcup dg \in UNIV. \bigcup dg \in B dg. Q \text{ } dg \text{ challenges}))$ 
     $(\text{staged-attacker-query-budget budgets} +$ 
     $\text{staged-challenge-query-budget}) +$ 
    hash-relation-budget-value
     $(\sum dg \in (UNIV :: 'f \text{ set}).$ 
     $\text{card } (B \text{ } dg) * \text{ceil-log} (\text{maxDegree} + 1))$ 
     $(\text{staged-attacker-query-budget budgets} +$ 
     $\text{staged-challenge-query-budget})$ 
proof –
    let ?Q =
     $(\bigcup dg \in UNIV. \bigcup dg \in B dg. Q \text{ } dg \text{ challenges})$ 
    let ?Path =
     $(\sum dg \in UNIV.$ 
     $\sum dg \in B dg.$ 
     $(1 / \text{nnreal} (\text{CARD}('f) \wedge \text{ceil-log} (\text{to-nat } dg + 1))) *$ 
     $(\text{nnreal} (\text{card } (Q \text{ } dg \text{ challenges})) *$ 
     $(1 / \text{nnreal} (\text{card } \text{query-sample-space})) \wedge \text{rounds}))$ 
    let ?QueryPre =
    hash-relation-budget-value
     $(\text{query-index-raw-list-relation-fiber-bound } ?Q)$ 
     $(\text{staged-attacker-query-budget budgets} +$ 
     $\text{staged-challenge-query-budget})$ 
    let ?ChallengePre =
    hash-relation-budget-value
     $(\sum dg \in (UNIV :: 'f \text{ set}).$ 
     $\text{card } (B \text{ } dg) * \text{ceil-log} (\text{maxDegree} + 1))$ 
     $(\text{staged-attacker-query-budget budgets} +$ 
     $\text{staged-challenge-query-budget})$ 
    have finite-B:  $\bigwedge dg. \text{finite } (B \text{ } dg)$ 
    by (rule finite-subset[OF challenge-subset])
    (rule finite-fri-challenge-space)
    have fst-projection:  $\bigwedge dg. \text{fst } ' P \text{ } dg \subseteq ?Q$ 
proof
    fix dg q
    assume q-in:  $q \in \text{fst } ' P \text{ } dg$ 
    then obtain c where pair:  $(q, c) \in P \text{ } dg$ 
    by auto
    have c-in:  $c \in B \text{ } dg$ 
    using challenge-projection[of dg] pair by force
    have q  $\in \text{fst } ' (P \text{ } dg \cap (UNIV \times \{c\}))$ 
    using pair by force
    then have q  $\in Q \text{ } dg \text{ } c$ 

```



```

    using query-projection[OF c-in] by blast
  then show  $q \in ?Q$ 
    using c-in by blast
qed
have path-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-composition-fri-pair-challenge-query-path-fresh P
      B ?Q)
  adversary-initial-state  $\leq$  ?Path
by (rule
  checked-staged-security-composition-fri-pair-challenge-query-path-fresh-fiber-sum
  [OF wf controlled challenge-subset query-projection subset])
have fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-composition-fri-pair-challenge-fresh P B)
  adversary-initial-state  $\leq$  ?Path + ?QueryPre
by (rule
  checked-staged-security-composition-fri-pair-challenge-fresh-bound-from-query-path-fresh
  [OF wf controlled fst-projection path-bound])
have pair-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s.$  composition-fri-query-challenge-pair-set-hit s P))
  adversary-initial-state  $\leq$ 
    (?Path + ?QueryPre) + ?ChallengePre
by (rule
  checked-staged-security-composition-fri-query-challenge-pair-set-hit-bound-from-challenge-fresh
  [OF wf controlled finite-B challenge-projection fresh-bound])
then show ?thesis
  by (simp add: add.assoc)
qed

```

lemma *checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-pair-fraction-challenge-weighted:*

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

and fraction:

generic-fri-sampled-query-pair-fraction-bound trace-table-low-degree
 (Not $\circ$ trace-table-low-degree) (clength - 1) C

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-sampled-query-bad-candidate)
 adversary-initial-state $\leq$
 C +
 hash-relation-budget-value
 (query-index-raw-list-relation-fiber-bound
 ($\bigcup$ challenges $\in$ fri-challenge-space (ceil-log clength).
 generic-fri-sampled-query-query-fiber trace-table-low-degree
 (Not $\circ$ trace-table-low-degree) (clength - 1) challenges))

```

      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget) +
    hash-relation-budget-value
    (card (fri-challenge-space (ceil-log clength)) * ceil-log clength)
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget)
proof -
  let ?P =
    trace-fri-sampled-query-bad-pair-union  $\cap$ 
    (fri-query-index-list-space  $\times$  UNIV)
  let ?B = fri-challenge-space (ceil-log clength)
  let ?Q =
     $\lambda$ challenges.
      generic-fri-sampled-query-query-fiber trace-table-low-degree
      (Not  $\circ$  trace-table-low-degree) (clength - 1) challenges
  let ?Path =
    ( $\sum$  challenges  $\in$  ?B.
     (1 / nnreal (CARD('f)  $\wedge$  ceil-log clength)) *
     (nnreal (card (?Q challenges)) *
      (1 / nnreal (card query-sample-space))  $\wedge$  rounds))
  let ?QueryPre =
    hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound
     ( $\bigcup$  challenges  $\in$  ?B. ?Q challenges))
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget)
  let ?ChallengePre =
    hash-relation-budget-value
    (card ?B * ceil-log clength)
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget)
  have sampled-to-pair:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-sampled-query-bad-candidate)
    adversary-initial-state  $\leq$ 
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     ( $\lambda s$ . trace-fri-query-challenge-pair-set-hit s ?P))
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
     intro:
      trace-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit
      split: option.splits prod.splits)
  also have ...  $\leq$  ?Path + ?QueryPre + ?ChallengePre
  proof -
    have challenge-subset:
      ?B  $\subseteq$  fri-challenge-space (ceil-log clength)

```

```

  by simp
have challenge-projection: snd ‘ ?P ⊆ ?B
  by (rule trace-fri-restricted-sampled-query-bad-pair-union-challenge-projection)
have query-projection:
  ∧ challenges. challenges ∈ ?B ⇒
  fst ‘ (?P ∩ (UNIV × {challenges})) ⊆ ?Q challenges
  by (auto
    intro: generic-fri-sampled-query-query-fiber-subset
    simp: generic-fri-sampled-query-query-fiber-def
    generic-fri-sampled-query-restricted-bad-pairs-def
    fri-query-challenge-pair-query-fiber-def
    trace-fri-sampled-query-bad-pair-union-def)
have subset-Q:
  ∧ challenges. challenges ∈ ?B ⇒
  ?Q challenges ⊆ fri-query-index-list-space
  by (rule generic-fri-sampled-query-query-fiber-subset)
show ?thesis
  by (rule
    checked-staged-security-trace-fri-query-challenge-pair-set-hit-challenge-weighted-bound
    [OF wf controlled challenge-subset challenge-projection
      query-projection subset-Q])
qed
also have ... ≤ C + ?QueryPre + ?ChallengePre
proof -
  have round-eq:
    fri-round-count-for-degree-bound (clength - Suc 0) = ceil-log clength
  unfolding fri-round-count-for-degree-bound-def
  by (simp add: clength-pos)
have fiber-fraction:
  nnreal
    (∑ challenges ∈ ?B. card (?Q challenges)) /
    (nnreal (CARD('f) ^ ceil-log clength) *
      nnreal query-sample-space-size ^ rounds)
  ≤ C
  using fraction
  unfolding trace-fri-sampled-query-pair-fraction-bound-iff-fiber-sum
  by (simp add: round-eq card-fri-query-index-list-space
    query-sample-space-size-pos algebra-simps)
have prob-eq:
  ?Path =
  nnreal
    (∑ challenges ∈ ?B. card (?Q challenges)) /
    (nnreal (CARD('f) ^ ceil-log clength) *
      nnreal query-sample-space-size ^ rounds)
  using weighted-query-fiber-sum-eq
  [where I = ?B
    and f = λchallenges. card (?Q challenges)
    and A = CARD('f) ^ ceil-log clength
    and B = query-sample-space-size]

```

```

    by (simp add: finite-fri-challenge-space query-sample-space-size-pos
        card-query-sample-space algebra-simps)
  have ?Path ≤ C
    using fiber-fraction unfolding prob-eq .
  then show ?thesis
    by (simp add: add-mono)
qed
finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-pair-fraction-challenge-weig
  fixes C :: 'f ⇒ prob
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and fraction:
    ∧ dg. generic-fri-sampled-query-pair-fraction-bound
      (composition-table-low-degree (to-nat dg))
      (Not ∘ composition-table-low-degree maxDegree) (to-nat dg) (C dg)
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-sampled-query-bad-candidate)
    adversary-initial-state ≤
      (∑ dg ∈ UNIV. C dg) +
      hash-relation-budget-value
        (query-index-raw-list-relation-fiber-bound
          (⋃ dg ∈ UNIV.
            ⋃ challenges ∈ fri-challenge-space (ceil-log (to-nat dg + 1)).
              generic-fri-sampled-query-query-fiber
                (composition-table-low-degree (to-nat dg))
                (Not ∘ composition-table-low-degree maxDegree)
                (to-nat dg) challenges))
          (staged-attacker-query-budget budgets +
            staged-challenge-query-budget) +
          hash-relation-budget-value
            (∑ dg ∈ (UNIV :: 'f set).
              card (fri-challenge-space (ceil-log (to-nat dg + 1))) *
              ceil-log (maxDegree + 1))
            (staged-attacker-query-budget budgets +
              staged-challenge-query-budget))
  proof -
    let ?P =
      λ dg. composition-fri-sampled-query-bad-pair-union dg ∩
        (fri-query-index-list-space × UNIV)
    let ?B =
      λ dg. fri-challenge-space (ceil-log (to-nat dg + 1))
    let ?Q =
      λ dg challenges.
        generic-fri-sampled-query-query-fiber

```

```

    (composition-table-low-degree (to-nat dg))
    (Not o composition-table-low-degree maxDegree)
    (to-nat dg) challenges
  let ?Path =
    (∑ dg ∈ UNIV.
      ∑ challenges ∈ ?B dg.
        (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
        (nnreal (card (?Q dg challenges)) *
          (1 / nnreal (card query-sample-space)) ^ rounds))
  let ?QueryPre =
    hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound
      (∪ dg ∈ UNIV. ∪ challenges ∈ ?B dg. ?Q dg challenges))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  let ?ChallengePre =
    hash-relation-budget-value
    (∑ dg ∈ (UNIV :: 'f set). card (?B dg) * ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  have sampled-to-pair:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-sampled-query-bad-candidate)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-query-challenge-pair-set-hit s ?P))
    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro:
        composition-fri-sampled-query-bad-candidate-imp-restricted-query-challenge-pair-union-hit
        split: option.splits prod.splits)
  also have ... ≤ ?Path + ?QueryPre + ?ChallengePre
  proof -
    have challenge-subset:
      ∧ dg. ?B dg ⊆ fri-challenge-space (ceil-log (to-nat dg + 1))
    by simp
    have challenge-projection: ∧ dg. snd ' ?P dg ⊆ ?B dg
    by (rule
      composition-fri-restricted-sampled-query-bad-pair-union-challenge-projection)
    have query-projection:
      ∧ dg challenges. challenges ∈ ?B dg ⇒
      fst ' (?P dg ∩ (UNIV × {challenges})) ⊆
      ?Q dg challenges
    by (auto
      intro: generic-fri-sampled-query-query-fiber-subset
      simp: generic-fri-sampled-query-query-fiber-def)

```

```

    generic-fri-sampled-query-restricted-bad-pairs-def
    fri-query-challenge-pair-query-fiber-def
    composition-fri-sampled-query-bad-pair-union-def)
have subset-Q:
   $\bigwedge dg \text{ challenges. } challenges \in ?B \ dg \implies$ 
     $?Q \ dg \ challenges \subseteq \text{fri-query-index-list-space}$ 
  by (rule generic-fri-sampled-query-query-fiber-subset)
show ?thesis
  by (rule
    checked-staged-security-composition-fri-query-challenge-pair-set-hit-challenge-weighted-bound
    [OF wf controlled challenge-subset challenge-projection
     query-projection subset-Q])
qed
also have ...  $\leq (\sum dg \in UNIV. \ C \ dg) + ?QueryPre + ?ChallengePre$ 
proof –
  have path-le:
     $?Path \leq (\sum dg \in UNIV. \ C \ dg)$ 
  proof (rule sum-mono)
    fix dg :: 'f
    assume dg  $\in UNIV$ 
    let ?I = ?B dg
    have round-eq:
      fri-round-count-for-degree-bound (to-nat dg) =
        ceil-log (to-nat dg + 1)
    unfolding fri-round-count-for-degree-bound-def by simp
  have fiber-fraction:
    nnreal
      (
         $(\sum challenges \in ?I. \ \text{card } (?Q \ dg \ challenges)) /$ 
         $(\text{nnreal } (CARD('f) \wedge \text{ceil-log } (to-nat \ dg + 1))) *$ 
         $\text{nnreal query-sample-space-size} \wedge \text{rounds})$ 
         $\leq C \ dg$ 
      )
    using fraction[of dg]
    unfolding composition-fri-sampled-query-pair-fraction-bound-iff-fiber-sum
    by (simp add: round-eq card-fri-query-index-list-space
      query-sample-space-size-pos algebra-simps)
  have prob-eq:
    (
       $(\sum challenges \in ?I. \$ 
         $(1 / \text{nnreal } (CARD('f) \wedge \text{ceil-log } (to-nat \ dg + 1))) *$ 
         $(\text{nnreal } (\text{card } (?Q \ dg \ challenges)) *$ 
         $(1 / \text{nnreal } (\text{card query-sample-space})) \wedge \text{rounds})) =$ 
      )
    nnreal
      (
         $(\sum challenges \in ?I. \ \text{card } (?Q \ dg \ challenges)) /$ 
         $(\text{nnreal } (CARD('f) \wedge \text{ceil-log } (to-nat \ dg + 1))) *$ 
         $\text{nnreal query-sample-space-size} \wedge \text{rounds})$ 
      )
    using weighted-query-fiber-sum-eq
    [where I = ?I
     and f =  $\lambda challenges. \ \text{card } (?Q \ dg \ challenges)$ 
     and A =  $CARD('f) \wedge \text{ceil-log } (to-nat \ dg + 1)$ 
     and B = query-sample-space-size]

```

```

    by (simp add: finite-fri-challenge-space query-sample-space-size-pos
        card-query-sample-space algebra-simps)
  show
    ( $\sum challenges \in ?I.$ 
      ( $1 / nnreal (CARD('f) \wedge \text{ceil-log } (to\text{-nat } dg + 1))) *$ 
      ( $nnreal (card (?Q \ dg \ challenges)) *$ 
      ( $1 / nnreal (card \ query\text{-sample}\text{-space})) \wedge \text{rounds}))$ 
       $\leq C \ dg$ 
    using fiber-fraction unfolding prob-eq .
  qed
  then show ?thesis
    by (simp add: add-mono)
  qed
  finally show ?thesis .
qed

end

end

```

```

theory Soundness-FRI-Sampled-Challenge-Cover
imports
  Soundness-FRI-Query-Challenge-Trace-Staged-Product
  Soundness-FRI-Internal-Bounds
begin

```

Generic sampled-query FRI challenge-cover reductions.

This layer is deliberately independent of trace/composition naming. The intended later FRI low-degree proof has to show that every sampled bad candidate produces a challenge list in an algebraic multiround bad set. The lemmas here turn exactly that fact, plus local round cardinality bounds, into the finite-cover/fraction interface consumed by the staged route.

```

lemma nnreal-nat-mult-divide-cancel-left:
  fixes  $a \ b \ c :: nat$ 
  assumes  $0 < b$ 
  shows  $nnreal (a * b) / nnreal (b * c) = nnreal a / nnreal c$ 
proof -
  have  $b\text{-ne}: (real \ b) \neq 0$ 
  using  $assms$  by  $simp$ 
  have  $nn2real (nnreal (a * b) / nnreal (b * c)) =$ 
     $real (a * b) / real (b * c)$ 
  by ( $simp \ only: nn2real\text{-divide } nn2real\text{-nnreal}$ )
  also have  $\dots = (real \ a * real \ b) / (real \ b * real \ c)$ 
  by ( $simp \ add: of\text{-nat}\text{-mult}$ )
  also have  $\dots = real \ a / real \ c$ 
  using  $b\text{-ne}$  by ( $simp \ add: field\text{-simps}$ )
  also have  $\dots = nn2real (nnreal \ a / nnreal \ c)$ 

```

```

    by (simp only: nn2real-divide nn2real-nnreal)
  finally have
    nn2real (nnreal (a * b) / nnreal (b * c)) =
      nn2real (nnreal a / nnreal c) .
  then show ?thesis
    by (simp only: nn2real-eq-iff)
qed

context soundness
begin

definition trace-fri-sampled-query-no-full-cover-failure-candidate
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    ('f list  $\Rightarrow$  nat  $\Rightarrow$  'f list  $\Rightarrow$  'f set)  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  trace-fri-sampled-query-no-full-cover-failure-candidate s bad out
     $\longleftrightarrow$ 
    ( $\exists$  trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final fri-query-idxs trace-round-layers
      composition-round-layers fr candidate-query-idxs trace-openings
      trace-table layers.
      trace-fri-partial-candidate-opening-evidence s out trace-roots
      trace-bs trace-final dg composition-roots composition-bs
      composition-final fri-query-idxs trace-round-layers
      composition-round-layers fr candidate-query-idxs trace-openings
      trace-table  $\wedge$ 
      generic-fri-sampled-query-candidate-evidence trace-table-low-degree
      (Not  $\circ$  trace-table-low-degree) trace-table (clength - 1)
      trace-roots trace-bs trace-final fri-query-idxs trace-round-layers
      layers  $\wedge$ 
       $\neg$  generic-fri-sampled-full-cover-failure trace-table-low-degree
      trace-table (clength - 1) trace-roots trace-bs trace-final
      fri-query-idxs trace-round-layers
      (fri-canonical-domains (length trace-bs)) layers
      (bad trace-table))

definition composition-fri-sampled-query-no-full-cover-failure-candidate
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    ('f  $\Rightarrow$  'f list  $\Rightarrow$  nat  $\Rightarrow$  'f list  $\Rightarrow$  'f set)  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  composition-fri-sampled-query-no-full-cover-failure-candidate s bad out
     $\longleftrightarrow$ 
    ( $\exists$  trace-roots trace-bs trace-final fri-dg
      opening-composition-roots composition-bs composition-final
      fri-query-idxs trace-round-layers composition-round-layers fr
      f-fri-roots f-final as dg composition-fri-roots final
      trace-query-idxs trace-openings composition-query-idxs

```


composition-openings trace-table composition-table layers.
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table $\wedge$
generic-fri-sampled-query-candidate-evidence
(composition-table-low-degree (to-nat fri-dg))
(Not $\circ$ composition-table-low-degree maxDegree) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers layers $\wedge$
 $\neg$ *generic-fri-sampled-full-cover-failure*
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers
(fri-canonical-domains (length composition-bs)) layers
(bad fri-dg composition-table))

definition *trace-fri-sampled-query-no-full-cover-failure-pairs*

$:: ('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{nat list} \times 'f \text{ list}) \text{ set}$

where

trace-fri-sampled-query-no-full-cover-failure-pairs bad =
 $(\{(query\text{-}idxs, challenges).\}$
 $\exists$ *candidate-table roots final-value round-layers layers.*
generic-fri-sampled-query-candidate-evidence trace-table-low-degree
(Not $\circ$ trace-table-low-degree) candidate-table (clength - 1)
roots challenges final-value query-idxs round-layers layers $\wedge$
 $\neg$ *generic-fri-sampled-full-cover-failure trace-table-low-degree*
candidate-table (clength - 1) roots challenges final-value
query-idxs round-layers (fri-canonical-domains (length challenges))
layers (bad candidate-table))
 $\cap (\text{fri-query-index-list-space} \times \text{UNIV}))$

definition *trace-fri-sampled-query-no-full-cover-failure-query-fiber*

$:: ('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $'f \text{ list} \Rightarrow \text{nat list set}$

where

trace-fri-sampled-query-no-full-cover-failure-query-fiber bad
challenges =
fri-query-challenge-pair-query-fiber
(trace-fri-sampled-query-no-full-cover-failure-pairs bad)
challenges

definition *composition-fri-sampled-query-no-full-cover-failure-pairs*

$:: ('f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $'f \Rightarrow (\text{nat list} \times 'f \text{ list}) \text{ set}$

where

composition-fri-sampled-query-no-full-cover-failure-pairs bad dg =
 (*{(query-idxs, challenges).*
 $\exists$ *candidate-table roots final-value round-layers layers.*
 generic-fri-sampled-query-candidate-evidence
 (*composition-table-low-degree (to-nat dg)*)
 (*Not* $\circ$ *composition-table-low-degree maxDegree*)
 candidate-table (to-nat dg) roots challenges final-value
 query-idxs round-layers layers $\wedge$
 $\neg$ *generic-fri-sampled-full-cover-failure*
 (*composition-table-low-degree (to-nat dg)*) *candidate-table*
 (*to-nat dg*) *roots challenges final-value query-idxs round-layers*
 (*fri-canonical-domains (length challenges)*) *layers*
 (*bad dg candidate-table*)}
 $\cap$ (*fri-query-index-list-space* $\times$ *UNIV*))

definition *composition-fri-sampled-query-no-full-cover-failure-query-fiber*

$:: ('f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $'f \Rightarrow 'f \text{ list} \Rightarrow \text{nat list set}$

where

composition-fri-sampled-query-no-full-cover-failure-query-fiber bad dg
challenges =
fri-query-challenge-pair-query-fiber
 (*composition-fri-sampled-query-no-full-cover-failure-pairs bad dg*)
challenges

lemma *trace-fri-sampled-query-no-full-cover-failure-pairs-subset:*

trace-fri-sampled-query-no-full-cover-failure-pairs bad
 $\subseteq$ *trace-fri-sampled-query-bad-pair-union* $\cap$
 (*fri-query-index-list-space* $\times$ *UNIV*)

proof

fix *pair*

assume *pair-in:*

pair $\in$ *trace-fri-sampled-query-no-full-cover-failure-pairs bad*

obtain *query-idxs challenges* **where** *pair-eq:* *pair = (query-idxs, challenges)*

by (*cases pair*)

from *pair-in pair-eq* **obtain** *candidate-table roots final-value round-layers*

layers **where** *query-space:* *query-idxs* $\in$ *fri-query-index-list-space*

and *evidence:*

generic-fri-sampled-query-candidate-evidence trace-table-low-degree

(*Not* $\circ$ *trace-table-low-degree*) *candidate-table (clength - 1)*

roots challenges final-value query-idxs round-layers layers

unfolding *trace-fri-sampled-query-no-full-cover-failure-pairs-def*

by *blast*

have *pair-set:*

(*query-idxs, challenges*) $\in$

generic-fri-sampled-query-bad-pair-set trace-table-low-degree

(*Not* $\circ$ *trace-table-low-degree*) *candidate-table (clength - 1)*

roots final-value round-layers layers

by (rule generic-fri-sampled-query-bad-pair-setI[OF evidence])
 then have union:
 (query-idxs, challenges) ∈ trace-fri-sampled-query-bad-pair-union
 unfolding trace-fri-sampled-query-bad-pair-union-def
 by (rule generic-fri-sampled-query-bad-pair-unionI)
 show
 pair ∈ trace-fri-sampled-query-bad-pair-union ∩
 (fri-query-index-list-space × UNIV)
 using pair-eq query-space union by simp
 qed

lemma trace-fri-sampled-query-no-full-cover-failure-pairs-challenge-projection:
 snd ‘ trace-fri-sampled-query-no-full-cover-failure-pairs bad
 ⊆ fri-challenge-space (ceil-log clength)
using trace-fri-sampled-query-no-full-cover-failure-pairs-subset
 trace-fri-restricted-sampled-query-bad-pair-union-challenge-projection
by blast

lemma trace-fri-sampled-query-no-full-cover-failure-query-fiber-subset:
 trace-fri-sampled-query-no-full-cover-failure-query-fiber bad challenges
 ⊆ fri-query-index-list-space
unfolding trace-fri-sampled-query-no-full-cover-failure-query-fiber-def
using trace-fri-sampled-query-no-full-cover-failure-pairs-subset
by (auto simp: fri-query-challenge-pair-query-fiber-def)

lemma composition-fri-sampled-query-no-full-cover-failure-pairs-subset:
 composition-fri-sampled-query-no-full-cover-failure-pairs bad dg
 ⊆ composition-fri-sampled-query-bad-pair-union dg ∩
 (fri-query-index-list-space × UNIV)

proof
 fix pair
 assume pair-in:
 pair ∈ composition-fri-sampled-query-no-full-cover-failure-pairs
 bad dg
obtain query-idxs challenges **where** pair-eq: pair = (query-idxs, challenges)
by (cases pair)
from pair-in pair-eq **obtain** candidate-table roots final-value round-layers
 layers **where** query-space: query-idxs ∈ fri-query-index-list-space
and evidence:
 generic-fri-sampled-query-candidate-evidence
 (composition-table-low-degree (to-nat dg))
 (Not ∘ composition-table-low-degree maxDegree)
 candidate-table (to-nat dg) roots challenges final-value
 query-idxs round-layers layers
unfolding composition-fri-sampled-query-no-full-cover-failure-pairs-def
by blast
have pair-set:
 (query-idxs, challenges) ∈
 generic-fri-sampled-query-bad-pair-set

(composition-table-low-degree (to-nat dg))
 (Not $\circ$ composition-table-low-degree maxDegree)
 candidate-table (to-nat dg) roots final-value round-layers layers
by (rule generic-fri-sampled-query-bad-pair-setI [OF evidence])
then have union:
 (query-idxs, challenges) $\in$
 composition-fri-sampled-query-bad-pair-union dg
unfolding composition-fri-sampled-query-bad-pair-union-def
by (rule generic-fri-sampled-query-bad-pair-unionI)
show
 pair $\in$ composition-fri-sampled-query-bad-pair-union dg $\cap$
 (fri-query-index-list-space $\times$ UNIV)
using pair-eq query-space union **by** simp
qed

lemma composition-fri-sampled-query-no-full-cover-failure-pairs-challenge-projection:
 snd ‘ composition-fri-sampled-query-no-full-cover-failure-pairs bad dg
 $\subseteq$ fri-challenge-space (ceil-log (to-nat dg + 1))
using composition-fri-sampled-query-no-full-cover-failure-pairs-subset
 composition-fri-restricted-sampled-query-bad-pair-union-challenge-projection
by blast

lemma composition-fri-sampled-query-no-full-cover-failure-query-fiber-subset:
 composition-fri-sampled-query-no-full-cover-failure-query-fiber bad dg
 challenges
 $\subseteq$ fri-query-index-list-space
unfolding composition-fri-sampled-query-no-full-cover-failure-query-fiber-def
using composition-fri-sampled-query-no-full-cover-failure-pairs-subset
by (auto simp: fri-query-challenge-pair-query-fiber-def)

lemma trace-fri-sampled-query-no-full-cover-failure-candidate-imp-pair-hit:
assumes
 trace-fri-sampled-query-no-full-cover-failure-candidate s bad out
shows
 trace-fri-query-challenge-pair-set-hit s
 (trace-fri-sampled-query-no-full-cover-failure-pairs bad) out
proof –
from assms **obtain** trace-roots trace-bs trace-final dg composition-roots
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table layers **where** evidence:
 trace-fri-partial-candidate-opening-evidence s out trace-roots
 trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr candidate-query-idxs trace-openings
 trace-table
and candidate:
 generic-fri-sampled-query-candidate-evidence trace-table-low-degree
 (Not $\circ$ trace-table-low-degree) trace-table (clength – 1)

```

    trace-roots trace-bs trace-final fri-query-idxs trace-round-layers
    layers
and no-failure:
  ¬ generic-fri-sampled-full-cover-failure trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers
    (fri-canonical-domains (length trace-bs)) layers
    (bad trace-table)
unfolding trace-fri-sampled-query-no-full-cover-failure-candidate-def
by fastforce
have transcript:
  accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers
using evidence
unfolding trace-fri-partial-candidate-opening-evidence-def by blast
have query-space:
  fri-query-idxs ∈ fri-query-index-list-space
by (rule accepted-fri-opening-transcript-query-index-list-space
    [OF transcript])
have pair:
  (fri-query-idxs, trace-bs) ∈
    trace-fri-sampled-query-no-full-cover-failure-pairs bad
unfolding trace-fri-sampled-query-no-full-cover-failure-pairs-def
using candidate no-failure query-space by blast
show ?thesis
unfolding trace-fri-query-challenge-pair-set-hit-def
using transcript pair by blast
qed

lemma composition-fri-sampled-query-no-full-cover-failure-candidate-imp-pair-hit:
assumes
  composition-fri-sampled-query-no-full-cover-failure-candidate s bad out
shows
  composition-fri-query-challenge-pair-set-hit s
    (composition-fri-sampled-query-no-full-cover-failure-pairs bad) out
proof —
from assms obtain trace-roots trace-bs trace-final fri-dg
  opening-composition-roots composition-bs composition-final
  fri-query-idxs trace-round-layers composition-round-layers fr
  f-fri-roots f-final as dg composition-fri-roots final
  trace-query-idxs trace-openings composition-query-idxs
  composition-openings trace-table composition-table layers where
  evidence:
  composition-fri-partial-candidate-opening-evidence s out trace-roots
  trace-bs trace-final fri-dg opening-composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers fr f-fri-roots f-final as dg
  composition-fri-roots final trace-query-idxs trace-openings

```

```

    composition-query-idxs composition-openings trace-table
    composition-table
and candidate:
    generic-fri-sampled-query-candidate-evidence
    (composition-table-low-degree (to-nat fri-dg))
    (Not  $\circ$  composition-table-low-degree maxDegree) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers layers
and no-failure:
     $\neg$  generic-fri-sampled-full-cover-failure
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
    (fri-canonical-domains (length composition-bs)) layers
    (bad fri-dg composition-table)
unfolding composition-fri-sampled-query-no-full-cover-failure-candidate-def
by fastforce
have transcript:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    fri-dg opening-composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
using evidence
unfolding composition-fri-partial-candidate-opening-evidence-def by blast
have query-space:
    fri-query-idxs  $\in$  fri-query-index-list-space
by (rule accepted-fri-opening-transcript-query-index-list-space
    [OF transcript])
have pair:
    (fri-query-idxs, composition-bs)  $\in$ 
    composition-fri-sampled-query-no-full-cover-failure-pairs bad fri-dg
unfolding composition-fri-sampled-query-no-full-cover-failure-pairs-def
using candidate no-failure query-space by blast
show ?thesis
unfolding composition-fri-query-challenge-pair-set-hit-def
using transcript pair by blast
qed

lemma trace-fri-sampled-query-no-full-cover-failure-candidate-not-None[simp]:
     $\neg$  trace-fri-sampled-query-no-full-cover-failure-candidate s bad None
unfolding trace-fri-sampled-query-no-full-cover-failure-candidate-def
    trace-fri-partial-candidate-opening-evidence-def
    accepted-fri-opening-transcript-def
by simp

lemma composition-fri-sampled-query-no-full-cover-failure-candidate-not-None[simp]:
     $\neg$  composition-fri-sampled-query-no-full-cover-failure-candidate s bad
    None
unfolding composition-fri-sampled-query-no-full-cover-failure-candidate-def
    composition-fri-partial-candidate-opening-evidence-def

```

accepted-fri-opening-transcript-def
by *simp*

lemma *trace-fri-sampled-query-no-full-cover-failure-candidate-imp-bad-candidate:*
assumes
trace-fri-sampled-query-no-full-cover-failure-candidate s bad out
shows *trace-fri-sampled-query-bad-candidate s out*
using *assms*
unfolding *trace-fri-sampled-query-no-full-cover-failure-candidate-def*
trace-fri-sampled-query-bad-candidate-def
by *force*

lemma *composition-fri-sampled-query-no-full-cover-failure-candidate-imp-bad-candidate:*
assumes
composition-fri-sampled-query-no-full-cover-failure-candidate s bad out
shows *composition-fri-sampled-query-bad-candidate s out*
using *assms*
unfolding
composition-fri-sampled-query-no-full-cover-failure-candidate-def
composition-fri-sampled-query-bad-candidate-def
by *force*

lemma *trace-fri-sampled-query-bad-candidate-imp-no-full-cover-or-failure:*
assumes *sampled: trace-fri-sampled-query-bad-candidate s out*
shows
trace-fri-sampled-query-no-full-cover-failure-candidate s bad out $\vee$
trace-fri-sampled-full-cover-failure s bad out

proof —

from *sampled* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table layers

where *evidence:*

trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table

and *candidate:*

generic-fri-sampled-query-candidate-evidence trace-table-low-degree
(Not $\circ$ trace-table-low-degree) trace-table (clength — 1)
trace-roots trace-bs trace-final fri-query-idxs trace-round-layers
layers

unfolding *trace-fri-sampled-query-bad-candidate-def* **by** *blast*

have *canonical:*

generic-fri-canonical-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength — 1) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers layers

by *(rule generic-fri-sampled-query-candidate-evidenceD(1))*

```

    [OF candidate])
  have chain:
    generic-fri-sampled-layer-chain-evidence trace-table-low-degree
      trace-table (clength - 1) trace-roots trace-bs trace-final
      fri-query-idxs trace-round-layers
      (fri-canonical-domains (length trace-bs)) layers
  proof -
    have sampled-chain:
      generic-fri-sampled-layer-chain-evidence trace-table-low-degree
        trace-table (clength - 1) trace-roots trace-bs trace-final
        fri-query-idxs trace-round-layers
        (fri-canonical-domains (length trace-bs)) layers
    by (rule generic-fri-canonical-sampled-layer-chain-evidenceD
        [OF canonical])
    then show ?thesis .
  qed
  show ?thesis
  proof (cases
    generic-fri-sampled-full-cover-failure trace-table-low-degree
      trace-table (clength - 1) trace-roots trace-bs trace-final
      fri-query-idxs trace-round-layers
      (fri-canonical-domains (length trace-bs)) layers
      (bad trace-table))
    case False
    then have
      trace-fri-sampled-query-no-full-cover-failure-candidate s bad out
    unfolding trace-fri-sampled-query-no-full-cover-failure-candidate-def
    using evidence candidate by blast
    then show ?thesis by simp
  next
    case True
    then have trace-fri-sampled-full-cover-failure s bad out
    unfolding trace-fri-sampled-full-cover-failure-def
    using evidence chain by blast
    then show ?thesis by simp
  qed
qed

lemma composition-fri-sampled-query-bad-candidate-imp-no-full-cover-or-failure:
  assumes sampled: composition-fri-sampled-query-bad-candidate s out
  shows
    composition-fri-sampled-query-no-full-cover-failure-candidate s bad out  $\vee$ 
    composition-fri-sampled-full-cover-failure s bad out
  proof -
    from sampled obtain trace-roots trace-bs trace-final fri-dg
      opening-composition-roots composition-bs composition-final
      fri-query-idxs trace-round-layers composition-round-layers fr
      f-fri-roots f-final as dg composition-fri-roots final
      trace-query-idxs trace-openings composition-query-idxs

```



```

    composition-openings trace-table composition-table layers
where evidence:
    composition-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final fri-dg opening-composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr f-fri-roots f-final as dg
    composition-fri-roots final trace-query-idxs trace-openings
    composition-query-idxs composition-openings trace-table
    composition-table
and candidate:
    generic-fri-sampled-query-candidate-evidence
    (composition-table-low-degree (to-nat fri-dg))
    (Not o composition-table-low-degree maxDegree) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers layers
unfolding composition-fri-sampled-query-bad-candidate-def by meson
have canonical:
    generic-fri-canonical-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers layers
by (rule generic-fri-sampled-query-candidate-evidenceD(1)
    [OF candidate])
have chain:
    generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
    (fri-canonical-domains (length composition-bs)) layers
proof –
have sampled-chain:
    generic-fri-sampled-layer-chain-evidence
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
    (fri-canonical-domains (length composition-bs)) layers
by (rule generic-fri-canonical-sampled-layer-chain-evidenceD
    [OF canonical])
then show ?thesis .
qed
show ?thesis
proof (cases
    generic-fri-sampled-full-cover-failure
    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
    (fri-canonical-domains (length composition-bs)) layers
    (bad fri-dg composition-table))
case False

```

then have
composition-fri-sampled-query-no-full-cover-failure-candidate s bad out
unfolding
composition-fri-sampled-query-no-full-cover-failure-candidate-def
using *evidence candidate* **by** *meson*
then show *?thesis* **by** *simp*
next
case *True*
then have *composition-fri-sampled-full-cover-failure s bad out*
unfolding *composition-fri-sampled-full-cover-failure-def*
using *evidence chain* **by** *meson*
then show *?thesis* **by** *simp*
qed
qed

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-failure:*
assumes *no-failure-bound:*
wp-event verify-monad
(trace-fri-sampled-query-no-full-cover-failure-candidate s bad) s
 $\leq N$
and *failure-bound:*
wp-event verify-monad
(trace-fri-sampled-full-cover-failure s bad) s $\leq F$
shows
wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s
 $\leq N + F$
proof –
have *wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s*
 $\leq$ *wp-event verify-monad*
(λ out.
trace-fri-sampled-query-no-full-cover-failure-candidate s bad out $\vee$
trace-fri-sampled-full-cover-failure s bad out) s
by *(rule wp-event-mono)*
(rule trace-fri-sampled-query-bad-candidate-imp-no-full-cover-or-failure)
also have $\dots \leq$
wp-event verify-monad
(trace-fri-sampled-query-no-full-cover-failure-candidate s bad) s +
wp-event verify-monad
(trace-fri-sampled-full-cover-failure s bad) s
by *(rule wp-event-union-bound)*
also have $\dots \leq N + F$
by *(rule add-mono[OF no-failure-bound failure-bound])*
finally show *?thesis* .
qed

lemma *wp-composition-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-failure:*
assumes *no-failure-bound:*
wp-event verify-monad
(composition-fri-sampled-query-no-full-cover-failure-candidate s bad) s

$\leq N$
and *failure-bound*:
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-sampled-full-cover-failure } s \text{ bad}) \ s \leq F$
shows
 $wp\text{-event } verify\text{-monad } (composition\text{-fri-sampled-query-bad-candidate } s) \ s$
 $\leq N + F$
proof –
have $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-sampled-query-bad-candidate } s) \ s$
 $\leq wp\text{-event } verify\text{-monad}$
 $(\lambda out.$
 $\quad composition\text{-fri-sampled-query-no-full-cover-failure-candidate } s \text{ bad}$
 $\quad out \vee$
 $\quad composition\text{-fri-sampled-full-cover-failure } s \text{ bad } out) \ s$
by (rule *wp-event-mono*)
(rule
 $composition\text{-fri-sampled-query-bad-candidate-imp-no-full-cover-or-failure}$)
also have $\dots \leq$
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-sampled-query-no-full-cover-failure-candidate } s \text{ bad})$
 $s +$
 $wp\text{-event } verify\text{-monad}$
 $(composition\text{-fri-sampled-full-cover-failure } s \text{ bad}) \ s$
by (rule *wp-event-union-bound*)
also have $\dots \leq N + F$
by (rule *add-mono[OF no-failure-bound failure-bound]*)
finally show *?thesis* .
qed

lemma *checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-failure*:

assumes *no-failure-bound*:

$\wedge data \text{ attacker-state.}$

Some $(data, \text{attacker-state}) \in$
 $set\text{-dist } (execute \ (checked\text{-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$

$wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-query-no-full-cover-failure-candidate}$
 $\quad (verifier\text{-state-from-adversary } \text{attacker-state}$
 $\quad \quad (staged\text{-proof-transcript } data)) \text{ bad})$
 $\quad (verifier\text{-state-from-adversary } \text{attacker-state}$
 $\quad \quad (staged\text{-proof-transcript } data)) \leq N$

and *failure-bound*:

$\wedge data \text{ attacker-state.}$

Some $(data, \text{attacker-state}) \in$
 $set\text{-dist } (execute \ (checked\text{-staged-transcript-program } A)$
 $\text{adversary-initial-state}) \implies$

$wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-full-cover-failure}$

```

      (verifier-state-from-adversary attacker-state
       (staged-proof-transcript data)) bad)
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data)) ≤ F
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       trace-fri-sampled-query-bad-candidate)
    adversary-initial-state ≤ N + F
  proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
    show  $\bigwedge s. \neg \text{trace-fri-sampled-query-bad-candidate } s \text{ None}$ 
    unfolding trace-fri-sampled-query-bad-candidate-def
      trace-fri-partial-candidate-opening-evidence-def
      accepted-fri-opening-transcript-def
    by simp
  next
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  show wp-event verify-monad (trace-fri-sampled-query-bad-candidate ?s) ?s
    ≤ N + F
  by (rule
      wp-trace-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-failure)
    (rule no-failure-bound[OF support], rule failure-bound[OF support])
qed

lemma checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-failure:
  assumes no-failure-bound:
     $\bigwedge \text{data attacker-state.}$ 
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
    wp-event verify-monad
      (composition-fri-sampled-query-no-full-cover-failure-candidate
       (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) bad)
      (verifier-state-from-adversary attacker-state
       (staged-proof-transcript data)) ≤ N
  and failure-bound:
     $\bigwedge \text{data attacker-state.}$ 
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)  $\implies$ 
    wp-event verify-monad

```

```

    (composition-fri-sampled-full-cover-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) bad)
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ F
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-sampled-query-bad-candidate)
    adversary-initial-state ≤ N + F
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\bigwedge s. \neg \text{composition-fri-sampled-query-bad-candidate } s \text{ None}$ 
  unfolding composition-fri-sampled-query-bad-candidate-def
    composition-fri-partial-candidate-opening-evidence-def
    accepted-fri-opening-transcript-def
  by simp
next
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  show wp-event verify-monad
    (composition-fri-sampled-query-bad-candidate ?s) ?s ≤ N + F
  by (rule
    wp-composition-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-failure)
    (rule no-failure-bound[OF support], rule failure-bound[OF support])
qed

lemma checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-staged-no-full-cover-or-failure:
assumes no-failure-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s. \text{trace-fri-sampled-query-no-full-cover-failure-candidate } s \text{ bad}$ ))
  adversary-initial-state ≤ N
and failure-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s. \text{trace-fri-sampled-full-cover-failure } s \text{ bad}$ ))
  adversary-initial-state ≤ F
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-query-bad-candidate)
  adversary-initial-state ≤ N + F
proof –

```

have *wp-event* (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-sampled-query-bad-candidate)
adversary-initial-state $\leq$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 (λs *out.*
trace-fri-sampled-query-no-full-cover-failure-candidate s bad
out $\vee$
trace-fri-sampled-full-cover-failure s bad out))
adversary-initial-state
by (*rule wp-event-mono*)
 (*auto simp: staged-security-with-data-state-verifier-event-def*
dest: trace-fri-sampled-query-bad-candidate-imp-no-full-cover-or-failure
split: option.splits prod.splits)
also have ... $\leq$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 ($\lambda out.$
staged-security-with-data-state-verifier-event
 ($\lambda s.$ *trace-fri-sampled-query-no-full-cover-failure-candidate s*
bad) *out* $\vee$
staged-security-with-data-state-verifier-event
 ($\lambda s.$ *trace-fri-sampled-full-cover-failure s bad*) *out*)
adversary-initial-state
by (*rule wp-event-mono-on-support*)
 (*auto simp: staged-security-with-data-state-verifier-event-def*
split: option.splits prod.splits)
also have ... $\leq$
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s.$ *trace-fri-sampled-query-no-full-cover-failure-candidate s*
bad))
adversary-initial-state +
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s.$ *trace-fri-sampled-full-cover-failure s bad*))
adversary-initial-state
by (*rule wp-event-union-bound*)
also have ... $\leq N + F$
by (*rule add-mono[OF no-failure-bound failure-bound]*)
finally show *?thesis* .
qed

lemma *checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-staged-no-full-cover-or-failure*
assumes *no-failure-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s.$
composition-fri-sampled-query-no-full-cover-failure-candidate s

$bad))$
 $adversary-initial-state \leq N$
and *failure-bound*:
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $(\lambda s. composition-fri-sampled-full-cover-failure \ s \ bad))$
 $adversary-initial-state \leq F$
shows
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $composition-fri-sampled-query-bad-candidate)$
 $adversary-initial-state \leq N + F$
proof –
have $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $composition-fri-sampled-query-bad-candidate)$
 $adversary-initial-state \leq$
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $(\lambda s \ out.$
 $composition-fri-sampled-query-no-full-cover-failure-candidate \ s$
 $bad \ out \ \vee$
 $composition-fri-sampled-full-cover-failure \ s \ bad \ out))$
 $adversary-initial-state$
by (rule *wp-event-mono*)
 $(auto \ simp: staged-security-with-data-state-verifier-event-def$
 $dest:$
 $composition-fri-sampled-query-bad-candidate-imp-no-full-cover-or-failure$
 $split: option.splits \ prod.splits)$
also have $\dots \leq$
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(\lambda out.$
 $staged-security-with-data-state-verifier-event$
 $(\lambda s.$
 $composition-fri-sampled-query-no-full-cover-failure-candidate \ s$
 $bad) \ out \ \vee$
 $staged-security-with-data-state-verifier-event$
 $(\lambda s. composition-fri-sampled-full-cover-failure \ s \ bad) \ out)$
 $adversary-initial-state$
by (rule *wp-event-mono-on-support*)
 $(auto \ simp: staged-security-with-data-state-verifier-event-def$
 $split: option.splits \ prod.splits)$
also have $\dots \leq$
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $(\lambda s.$
 $composition-fri-sampled-query-no-full-cover-failure-candidate \ s$
 $bad))$
 $adversary-initial-state +$

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s$ . composition-fri-sampled-full-cover-failure s bad))
  adversary-initial-state
by (rule wp-event-union-bound)
also have ...  $\leq N + F$ 
  by (rule add-mono[OF no-failure-bound failure-bound])
finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-sampled-query-no-full-cover-failure-candidate-bound-from-pair-fraction-chall*

```

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and fraction:
  generic-fri-sampled-query-pair-fraction-bound trace-table-low-degree
    (Not  $\circ$  trace-table-low-degree) (clength - 1) C

```

shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s$ . trace-fri-sampled-query-no-full-cover-failure-candidate s bad))
  adversary-initial-state  $\leq$ 
    C +
    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound
        ( $\bigcup$  challenges  $\in$  fri-challenge-space (ceil-log clength).
          generic-fri-sampled-query-query-fiber trace-table-low-degree
            (Not  $\circ$  trace-table-low-degree) (clength - 1) challenges))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget) +
    hash-relation-budget-value
      (card (fri-challenge-space (ceil-log clength)) * ceil-log clength)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)

```

proof –

```

have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s$ . trace-fri-sampled-query-no-full-cover-failure-candidate s bad))
  adversary-initial-state
 $\leq$  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-sampled-query-bad-candidate)
  adversary-initial-state
by (rule wp-event-mono)
  (auto simp: staged-security-with-data-state-verifier-event-def
    intro:
      trace-fri-sampled-query-no-full-cover-failure-candidate-imp-bad-candidate
      split: option.splits prod.splits)
also have ...  $\leq$ 
  C +

```



```

hash-relation-budget-value
  (query-index-raw-list-relation-fiber-bound
    ( $\bigcup$  challenges  $\in$  fri-challenge-space (ceil-log clength).
      generic-fri-sampled-query-query-fiber trace-table-low-degree
      (Not  $\circ$  trace-table-low-degree) (clength - 1) challenges))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget) +
hash-relation-budget-value
  (card (fri-challenge-space (ceil-log clength)) * ceil-log clength)
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)
by (rule
  checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-pair-fraction-challenge-weighted
  [OF wf controlled fraction])
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-sampled-query-no-full-cover-failure-candidate-bound-from-restricted-pairs:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        ( $\lambda s$ . trace-fri-sampled-query-no-full-cover-failure-candidate s bad))
    adversary-initial-state  $\leq$ 
    ( $\sum$  challenges  $\in$  fri-challenge-space (ceil-log clength).
      (1 / nnreal (CARD('f)  $\wedge$  ceil-log clength)) *
      (nnreal
        (card
          (trace-fri-sampled-query-no-full-cover-failure-query-fiber
            bad challenges)) *
          (1 / nnreal (card query-sample-space))  $\wedge$  rounds)) +
    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound
        ( $\bigcup$  challenges  $\in$  fri-challenge-space (ceil-log clength).
          trace-fri-sampled-query-no-full-cover-failure-query-fiber
          bad challenges))
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget) +
    hash-relation-budget-value
      (card (fri-challenge-space (ceil-log clength)) * ceil-log clength)
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget)
proof -
  let ?P = trace-fri-sampled-query-no-full-cover-failure-pairs bad
  let ?B = fri-challenge-space (ceil-log clength)
  let ?Q =
    trace-fri-sampled-query-no-full-cover-failure-query-fiber bad
  have event-le:

```

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s.$  trace-fri-sampled-query-no-full-cover-failure-candidate s bad))
adversary-initial-state  $\leq$ 
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s.$  trace-fri-query-challenge-pair-set-hit s ?P))
adversary-initial-state
by (rule wp-event-mono-on-support)
  (auto simp: staged-security-with-data-state-verifier-event-def
    intro:
      trace-fri-sampled-query-no-full-cover-failure-candidate-imp-pair-hit
      split: option.splits prod.splits)
have pair-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    ( $\lambda s.$  trace-fri-query-challenge-pair-set-hit s ?P))
adversary-initial-state  $\leq$ 
( $\sum$  challenges  $\in$  ?B.
  (1 / nnreal (CARD('f) ^ ceil-log clength)) *
  (nnreal (card (?Q challenges)) *
    (1 / nnreal (card query-sample-space)) ^ rounds)) +
hash-relation-budget-value
  (query-index-row-list-relation-fiber-bound
    ( $\bigcup$  challenges  $\in$  ?B. ?Q challenges))
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget) +
hash-relation-budget-value (card ?B * ceil-log clength)
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)
proof (rule
  checked-staged-security-trace-fri-query-challenge-pair-set-hit-challenge-weighted-bound
  [OF wf controlled])
show ?B  $\subseteq$  fri-challenge-space (ceil-log clength)
  by simp
next
show snd ' ?P  $\subseteq$  ?B
  by (rule
    trace-fri-sampled-query-no-full-cover-failure-pairs-challenge-projection)
next
fix challenges
assume challenges  $\in$  ?B
show fst ' (?P  $\cap$  (UNIV  $\times$  {challenges}))  $\subseteq$ 
  ?Q challenges
  unfolding trace-fri-sampled-query-no-full-cover-failure-query-fiber-def
    fri-query-challenge-pair-query-fiber-def
  by force
next
fix challenges

```

```

assume challenges ∈ ?B
show ?Q challenges ⊆ fri-query-index-list-space
  by (rule
    trace-fri-sampled-query-no-full-cover-failure-query-fiber-subset)
qed
show ?thesis
  by (rule order-trans[OF event-le pair-bound])
qed

```

lemma checked-staged-security-composition-fri-sampled-query-no-full-cover-failure-candidate-bound-from-restricted-p

assumes wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs.
      composition-fri-sampled-query-no-full-cover-failure-candidate s
      bad))
adversary-initial-state ≤
  (∑ dg ∈ UNIV.
    ∑ challenges ∈ fri-challenge-space (ceil-log (to-nat dg + 1)).
      (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
      (nnreal
        (card
          (composition-fri-sampled-query-no-full-cover-failure-query-fiber
            bad dg challenges)) *
          (1 / nnreal (card query-sample-space)) ^ rounds)) +
    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound
        (∪ dg ∈ UNIV.
          ∪ challenges ∈ fri-challenge-space (ceil-log (to-nat dg + 1)).
            composition-fri-sampled-query-no-full-cover-failure-query-fiber
              bad dg challenges))
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget) +
      hash-relation-budget-value
        (∑ dg ∈ (UNIV :: 'f set).
          card (fri-challenge-space (ceil-log (to-nat dg + 1))) *
          ceil-log (maxDegree + 1))
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget)

```

proof –

let ?P =

composition-fri-sampled-query-no-full-cover-failure-pairs bad

let ?B =

λdg. fri-challenge-space (ceil-log (to-nat dg + 1))

let ?Q =

composition-fri-sampled-query-no-full-cover-failure-query-fiber bad

have event-le:

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs.
      composition-fri-sampled-query-no-full-cover-failure-candidate s
      bad))
adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-challenge-pair-set-hit s ?P))
adversary-initial-state
by (rule wp-event-mono-on-support)
  (auto simp: staged-security-with-data-state-verifier-event-def
    intro:
      composition-fri-sampled-query-no-full-cover-failure-candidate-imp-pair-hit
      split: option.splits prod.splits)
have pair-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. composition-fri-query-challenge-pair-set-hit s ?P))
adversary-initial-state ≤
  (∑ dg ∈ UNIV.
    ∑ challenges ∈ ?B dg.
      (1 / nnreal (CARD('f) ^ ceil-log (to-nat dg + 1))) *
      (nnreal (card (?Q dg challenges)) *
        (1 / nnreal (card query-sample-space)) ^ rounds)) +
  hash-relation-budget-value
  (query-index-row-list-relation-fiber-bound
    (∪ dg ∈ UNIV. ∪ challenges ∈ ?B dg.
      ?Q dg challenges))
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget) +
  hash-relation-budget-value
  (∑ dg ∈ (UNIV :: 'f set).
    card (?B dg) * ceil-log (maxDegree + 1))
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget)
proof (rule
  checked-staged-security-composition-fri-query-challenge-pair-set-hit-challenge-weighted-bound
  [OF wf controlled])
fix dg
show ?B dg ⊆ fri-challenge-space (ceil-log (to-nat dg + 1))
  by simp
next
fix dg
show snd ' ?P dg ⊆ ?B dg
  by (rule
    composition-fri-sampled-query-no-full-cover-failure-pairs-challenge-projection)
next
fix dg challenges

```

```

assume challenges ∈ ?B dg
show fst ‘ (?P dg ∩ (UNIV × {challenges})) ⊆
  ?Q dg challenges
unfolding
  composition-fri-sampled-query-no-full-cover-failure-query-fiber-def
  fri-query-challenge-pair-query-fiber-def
by force
next
fix dg challenges
assume challenges ∈ ?B dg
show ?Q dg challenges ⊆ fri-query-index-list-space
by (rule
  composition-fri-sampled-query-no-full-cover-failure-query-fiber-subset)
qed
show ?thesis
by (rule order-trans[OF event-le pair-bound])
qed

lemma checked-staged-security-composition-fri-sampled-query-no-full-cover-failure-candidate-bound-from-pair-fraction
fixes C :: 'f ⇒ prob
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and fraction:
  ∧ dg. generic-fri-sampled-query-pair-fraction-bound
    (composition-table-low-degree (to-nat dg))
    (Not ∘ composition-table-low-degree maxDegree) (to-nat dg) (C dg)
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs.
        composition-fri-sampled-query-no-full-cover-failure-candidate s
        bad))
  adversary-initial-state ≤
    (∑ dg ∈ UNIV. C dg) +
    hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound
        (∪ dg ∈ UNIV.
          ∪ challenges ∈ fri-challenge-space (ceil-log (to-nat dg + 1)).
            generic-fri-sampled-query-query-fiber
              (composition-table-low-degree (to-nat dg))
              (Not ∘ composition-table-low-degree maxDegree)
              (to-nat dg challenges))
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget) +
        hash-relation-budget-value
          (∑ dg ∈ (UNIV :: 'f set).
            card (fri-challenge-space (ceil-log (to-nat dg + 1))) *
            ceil-log (maxDegree + 1))
          (staged-attacker-query-budget budgets +

```

```

      staged-challenge-query-budget)
proof –
  have wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s.$ 
        composition-fri-sampled-query-no-full-cover-failure-candidate s
        bad))
    adversary-initial-state
   $\leq$  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-sampled-query-bad-candidate)
    adversary-initial-state
  by (rule wp-event-mono)
    (auto simp: staged-security-with-data-state-verifier-event-def
      intro:
        composition-fri-sampled-query-no-full-cover-failure-candidate-imp-bad-candidate
        split: option.splits prod.splits)
  also have ...  $\leq$ 
    ( $\sum dg \in UNIV. C\ dg$ ) +
    hash-relation-budget-value
    (query-index-row-list-relation-fiber-bound
      ( $\bigcup dg \in UNIV.$ 
         $\bigcup challenges \in fri\text{-}challenge\text{-}space\ (ceil\text{-}log\ (to\text{-}nat\ dg + 1)).$ 
        generic-fri-sampled-query-query-fiber
          (composition-table-low-degree (to-nat dg))
          (Not  $\circ$  composition-table-low-degree maxDegree)
          (to-nat dg) challenges))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget) +
    hash-relation-budget-value
    ( $\sum dg \in (UNIV :: 'f\ set).$ 
      card (fri-challenge-space (ceil-log (to-nat dg + 1))) *
      ceil-log (maxDegree + 1))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  by (rule
    checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-pair-fraction-challenge-weighted
    [OF wf controlled fraction])
  finally show ?thesis .
qed

lemma finite-generic-fri-bad-challenge-lists:
  finite (generic-fri-bad-challenge-lists n bad)
proof (rule finite-subset)
  show generic-fri-bad-challenge-lists n bad  $\subseteq$  fri-challenge-space n
    by (rule generic-fri-bad-challenge-lists-subset)
  show finite (fri-challenge-space n)
    by (rule finite-fri-challenge-space)
qed

```

lemma *generic-fri-sampled-query-finite-cover-from-bad-challenge-lists:*
assumes *cover:*
 $\bigwedge$ *candidate-table roots final-value round-layers layers query-idxs*
challenges.
query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers $\implies$
challenges $\in$
generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound degree-bound) *bad*
shows
generic-fri-sampled-query-finite-cover low-degree candidate-bad
degree-bound
(generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound degree-bound) *bad)*
proof *(rule generic-fri-sampled-query-finite-cover-from-candidate-challenge-cover)*
show *finite*
(generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound degree-bound) *bad)*
by *(rule finite-generic-fri-bad-challenge-lists)*
fix *candidate-table roots final-value round-layers layers query-idxs*
challenges
assume *query-space: query-idxs* $\in$ *fri-query-index-list-space*
and *evidence:*
generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers
show *challenges* $\in$
generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound degree-bound) *bad*
by *(rule cover[OF query-space evidence])*
qed

lemma *generic-fri-sampled-query-projection-fraction-bound-from-bad-challenge-round-bounds:*
assumes *cover:*
 $\bigwedge$ *candidate-table roots final-value round-layers layers query-idxs*
challenges.
query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers $\implies$
challenges $\in$
generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound degree-bound) *bad*
and *local-bound:*
 $\bigwedge i$ *prefix.*
i $<$ *fri-round-count-for-degree-bound degree-bound* $\implies$

$length\ prefix = i \implies$
 $card\ (bad\ i\ prefix) \leq bd\ i$
and *fraction-bound*:
 $nnreal$
 $(\sum i \in \{..<fri-round-count-for-degree-bound\ degree-bound\}).$
 $CARD('f) \wedge i * bd\ i *$
 $CARD('f) \wedge$
 $(fri-round-count-for-degree-bound\ degree-bound - Suc\ i)) /$
 $nnreal\ (CARD('f) \wedge fri-round-count-for-degree-bound\ degree-bound)$
 $\leq C$
shows
 $generic-fri-sampled-query-projection-fraction-bound\ low-degree$
 $candidate-bad\ degree-bound\ C$
proof –
let $?n = fri-round-count-for-degree-bound\ degree-bound$
let $?B = generic-fri-bad-challenge-lists\ ?n\ bad$
let $?K =$
 $(\sum i \in \{..<?n\}.$
 $CARD('f) \wedge i * bd\ i * CARD('f) \wedge (?n - Suc\ i))$
have *cover-B*:
 $generic-fri-sampled-query-finite-cover\ low-degree\ candidate-bad$
 $degree-bound\ ?B$
by (rule *generic-fri-sampled-query-finite-cover-from-bad-challenge-lists*
[OF *cover*])
have *card-bound*: $card\ ?B \leq ?K$
by (rule *generic-fri-bad-challenge-lists-card-bound-from-round-bounds*)
(rule *local-bound*)
have *ratio-bound*:
 $nnreal\ (card\ ?B) / nnreal\ (CARD('f) \wedge ?n) \leq C$
proof –
have $nnreal\ (card\ ?B) / nnreal\ (CARD('f) \wedge ?n)$
 $\leq nnreal\ ?K / nnreal\ (CARD('f) \wedge ?n)$
using *card-bound* **by** (rule *nnreal-nat-divide-right-mono*)
also have $\dots \leq C$
by (rule *fraction-bound*)
finally show *?thesis* .
qed
show *?thesis*
by (rule *generic-fri-sampled-query-projection-fraction-bound-from-cover*
[OF *cover-B ratio-bound*])
qed
lemma *generic-fri-sampled-query-projection-fraction-bound-from-one-step-disagreement-subset*:
assumes *cover*:
 $\bigwedge candidate-table\ roots\ final-value\ round-layers\ layers\ query-idxs$
 $challenges.$
 $query-idxs \in fri-query-index-list-space \implies$
 $generic-fri-sampled-query-candidate-evidence\ low-degree\ candidate-bad$
 $candidate-table\ degree-bound\ roots\ challenges\ final-value\ query-idxs$

round-layers layers $\implies$
challenges $\in$
generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound degree-bound) bad
and *local-subset*:
 $\bigwedge i$ *prefix*.
 $i < \text{fri-round-count-for-degree-bound degree-bound} \implies$
 $\text{length prefix} = i \implies$
 $\text{bad } i \text{ prefix} \subseteq$
fri-one-step-disagreement-challenges
(len i) (pw i) (layer i prefix) (claimed i prefix) (domains i prefix)
and *fraction-bound*:
 nnreal
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound degree-bound}\}.$
 $\text{CARD}(f) \wedge i * (\text{len } i \text{ div } 2) *$
 $\text{CARD}(f) \wedge$
 $(\text{fri-round-count-for-degree-bound degree-bound} - \text{Suc } i)) /$
 $\text{nnreal } (\text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound degree-bound})$
 $\leq C$
shows
generic-fri-sampled-query-projection-fraction-bound low-degree
candidate-bad degree-bound C
proof (rule
generic-fri-sampled-query-projection-fraction-bound-from-bad-challenge-round-bounds
[OF cover - fraction-bound])
fix $i :: \text{nat}$ **and** $\text{prefix} :: 'f \text{ list}$
assume $i: i < \text{fri-round-count-for-degree-bound degree-bound}$
and $\text{prefix}: \text{length prefix} = i$
have $\text{card } (\text{bad } i \text{ prefix}) \leq$
 card
(fri-one-step-disagreement-challenges
(len i) (pw i) (layer i prefix) (claimed i prefix)
(domains i prefix))
by (rule *card-mono*) (*simp-all add: local-subset[OF i prefix]*)
also have $\dots \leq \text{len } i \text{ div } 2$
by (rule *fri-one-step-disagreement-challenges-card-bound-local*)
finally show $\text{card } (\text{bad } i \text{ prefix}) \leq \text{len } i \text{ div } 2$.
qed

lemma *generic-fri-sampled-query-projection-fraction-bound-from-one-step-disagreement-subset-exact*:
assumes *cover*:
 $\bigwedge \text{candidate-table roots final-value round-layers layers query-idxs}$
 $\text{challenges}.$
 $\text{query-idxs} \in \text{fri-query-index-list-space} \implies$
generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers $\implies$
challenges $\in$
generic-fri-bad-challenge-lists

```

    (fri-round-count-for-degree-bound degree-bound) bad
and local-subset:
   $\bigwedge i \text{ prefix.}$ 
     $i < \text{fri-round-count-for-degree-bound degree-bound} \implies$ 
     $\text{length prefix} = i \implies$ 
     $\text{bad } i \text{ prefix} \subseteq$ 
     $\text{fri-one-step-disagreement-challenges}$ 
     $(\text{len } i) (\text{pw } i) (\text{layer } i \text{ prefix}) (\text{claimed } i \text{ prefix}) (\text{domains } i \text{ prefix})$ 
shows
  generic-fri-sampled-query-projection-fraction-bound low-degree
  candidate-bad degree-bound
  (fri-one-step-disagreement-round-fraction
    (fri-round-count-for-degree-bound degree-bound) len)
proof (rule
  generic-fri-sampled-query-projection-fraction-bound-from-one-step-disagreement-subset)
fix candidate-table roots final-value round-layers layers query-idxs challenges
assume query-idxs: query-idxs  $\in$  fri-query-index-list-space
and evidence:
  generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
  candidate-table degree-bound roots challenges final-value query-idxs
  round-layers layers
show challenges  $\in$ 
  generic-fri-bad-challenge-lists
  (fri-round-count-for-degree-bound degree-bound) bad
by (rule cover[OF query-idxs evidence])
next
fix i :: nat and prefix :: 'f list
assume i:  $i < \text{fri-round-count-for-degree-bound degree-bound}$ 
and prefix:  $\text{length prefix} = i$ 
show  $\text{bad } i \text{ prefix} \subseteq$ 
  fri-one-step-disagreement-challenges
   $(\text{len } i) (\text{pw } i) (\text{layer } i \text{ prefix}) (\text{claimed } i \text{ prefix}) (\text{domains } i \text{ prefix})$ 
by (rule local-subset[OF i prefix])
next
show nnreal
   $(\sum i \in \{..<\text{fri-round-count-for-degree-bound degree-bound}\}.$ 
     $\text{CARD('f)} ^ i * (\text{len } i \text{ div } 2) *$ 
     $\text{CARD('f)} ^$ 
     $(\text{fri-round-count-for-degree-bound degree-bound} - \text{Suc } i)) /$ 
  nnreal (CARD('f) ^ fri-round-count-for-degree-bound degree-bound)
   $\leq \text{fri-one-step-disagreement-round-fraction}$ 
   $(\text{fri-round-count-for-degree-bound degree-bound}) \text{ len}$ 
by (rule fri-one-step-disagreement-round-fraction-refl-bound)
qed

lemma generic-fri-sampled-query-pair-fraction-bound-from-bad-challenge-round-bounds:
assumes cover:
   $\bigwedge \text{candidate-table roots final-value round-layers layers query-idxs}$ 
  challenges.

```

query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence *low-degree* *candidate-bad*
candidate-table *degree-bound* *roots* *challenges* *final-value* *query-idxs*
round-layers *layers* $\implies$
challenges $\in$
generic-fri-bad-challenge-lists
(*fri-round-count-for-degree-bound* *degree-bound*) *bad*
and *local-bound*:
 $\bigwedge i$ *prefix*.
i < *fri-round-count-for-degree-bound* *degree-bound* $\implies$
length prefix = *i* $\implies$
card (*bad i prefix*) $\leq$ *bd i*
and *fraction-bound*:
nnreal
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound degree-bound}\}.$
 $\text{CARD}(f) \wedge^i * \text{bd } i *$
 $\text{CARD}(f) \wedge$
 $(\text{fri-round-count-for-degree-bound degree-bound} - \text{Suc } i)) /$
nnreal ($\text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound degree-bound}$)
 $\leq C$
shows
generic-fri-sampled-query-pair-fraction-bound *low-degree* *candidate-bad*
degree-bound *C*
proof –
let *?n* = *fri-round-count-for-degree-bound* *degree-bound*
let *?B* = *generic-fri-bad-challenge-lists* *?n* *bad*
let *?K* =
 $(\sum i \in \{..<?n\}.$
 $\text{CARD}(f) \wedge^i * \text{bd } i * \text{CARD}(f) \wedge (?n - \text{Suc } i))$
have *projection*:
generic-fri-sampled-query-challenge-projection *low-degree*
candidate-bad *degree-bound* $\subseteq$ *?B*
proof
fix *challenges*
assume *challenges-in*:
challenges $\in$
generic-fri-sampled-query-challenge-projection *low-degree*
candidate-bad *degree-bound*
then obtain *query-idxs* **where** *pair*:
(*query-idxs*, *challenges*) $\in$
generic-fri-sampled-query-restricted-bad-pairs *low-degree*
candidate-bad *degree-bound*
unfolding *generic-fri-sampled-query-challenge-projection-def* **by** *force*
then have *query-space*: *query-idxs* $\in$ *fri-query-index-list-space*
unfolding *generic-fri-sampled-query-restricted-bad-pairs-def* **by** *simp*
have *pair-union*:
(*query-idxs*, *challenges*) $\in$
generic-fri-sampled-query-bad-pair-union *low-degree* *candidate-bad*
degree-bound

```

using pair
unfolding generic-fri-sampled-query-restricted-bad-pairs-def by simp
then obtain candidate-table roots final-value round-layers layers where
  pair-set:
    (query-idxs, challenges) ∈
      generic-fri-sampled-query-bad-pair-set low-degree candidate-bad
      candidate-table degree-bound roots final-value round-layers layers
unfolding generic-fri-sampled-query-bad-pair-union-def by auto
have evidence:
  generic-fri-sampled-query-candidate-evidence low-degree
  candidate-bad candidate-table degree-bound roots challenges
  final-value query-idxs round-layers layers
by (rule generic-fri-sampled-query-bad-pair-setD[OF pair-set])
show challenges ∈ ?B
by (rule cover[OF query-space evidence])
qed
have card-bound: card ?B ≤ ?K
by (rule generic-fri-bad-challenge-lists-card-bound-from-round-bounds)
  (rule local-bound)
have ratio-bound:
  nnreal (card ?B * card fri-query-index-list-space) /
    nnreal (card fri-query-index-list-space * CARD('f) ^ ?n)
  ≤ C
proof –
have card-q-pos: 0 < card fri-query-index-list-space
by (simp add: card-fri-query-index-list-space
  query-sample-space-size-pos)
then have card-q-ne: card fri-query-index-list-space ≠ 0
by simp
have nnreal (card ?B * card fri-query-index-list-space) /
  nnreal (card fri-query-index-list-space * CARD('f) ^ ?n) =
  nnreal (card ?B) / nnreal (CARD('f) ^ ?n)
by (rule nnreal-nat-mult-divide-cancel-left[OF card-q-pos])
also have ... ≤ nnreal ?K / nnreal (CARD('f) ^ ?n)
using card-bound by (rule nnreal-nat-divide-right-mono)
also have ... ≤ C
by (rule fraction-bound)
finally show ?thesis .
qed
show ?thesis
by (rule generic-fri-sampled-query-pair-fraction-bound-from-projection-and-fibers
  [OF finite-generic-fri-bad-challenge-lists projection - ratio-bound])
  (meson card-mono finite-fri-query-index-list-space
  generic-fri-sampled-query-query-fiber-subset)
qed

lemma generic-fri-sampled-query-pair-fraction-bound-from-one-step-disagreement-subset:
assumes cover:
  ∧ candidate-table roots final-value round-layers layers query-idxs

```

challenges.
query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence *low-degree candidate-bad*
candidate-table *degree-bound* *roots* *challenges* *final-value* *query-idxs*
round-layers *layers* $\implies$
challenges $\in$
generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound degree-bound) *bad*
and *local-subset*:
 $\bigwedge i$ *prefix*.
i < *fri-round-count-for-degree-bound degree-bound* $\implies$
length prefix = *i* $\implies$
bad i prefix $\subseteq$
fri-one-step-disagreement-challenges
(len i) *(pw i)* *(layer i prefix)* *(claimed i prefix)* *(domains i prefix)*
and *fraction-bound*:
nnreal
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound degree-bound}\}.$
 $\text{CARD}(f) \wedge^i * (\text{len } i \text{ div } 2) *$
 $\text{CARD}(f) \wedge$
 $(\text{fri-round-count-for-degree-bound degree-bound} - \text{Suc } i)) /$
 $\text{nnreal } (\text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound degree-bound})$
 $\leq C$
shows
generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad
degree-bound C
proof (*rule*
generic-fri-sampled-query-pair-fraction-bound-from-bad-challenge-round-bounds
 $[OF \text{ cover } - \text{fraction-bound}]$)
fix *i* :: *nat* **and** *prefix* :: *'f list*
assume *i*: *i* < *fri-round-count-for-degree-bound degree-bound*
and *prefix*: *length prefix* = *i*
have *card (bad i prefix)* $\leq$
card
(fri-one-step-disagreement-challenges
(len i) *(pw i)* *(layer i prefix)* *(claimed i prefix)*
(domains i prefix))
by (*rule card-mono*) (*simp-all add: local-subset[OF i prefix]*)
also have ... $\leq \text{len } i \text{ div } 2$
by (*rule fri-one-step-disagreement-challenges-card-bound-local*)
finally show *card (bad i prefix)* $\leq \text{len } i \text{ div } 2$.
qed

lemma *generic-fri-sampled-query-pair-fraction-bound-from-one-step-disagreement-subset-exact*:
assumes *cover*:
 $\bigwedge$ *candidate-table* *roots* *final-value* *round-layers* *layers* *query-idxs*
challenges.
query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence *low-degree candidate-bad*

```

    candidate-table degree-bound roots challenges final-value query-idxs
    round-layers layers  $\implies$ 
    challenges  $\in$ 
      generic-fri-bad-challenge-lists
      (fri-round-count-for-degree-bound degree-bound) bad
  and local-subset:
     $\wedge i$  prefix.
       $i < \text{fri-round-count-for-degree-bound degree-bound} \implies$ 
      length prefix =  $i \implies$ 
      bad  $i$  prefix  $\subseteq$ 
        fri-one-step-disagreement-challenges
        (len  $i$ ) (pw  $i$ ) (layer  $i$  prefix) (claimed  $i$  prefix) (domains  $i$  prefix)
  shows
    generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad
    degree-bound
    (fri-one-step-disagreement-round-fraction
      (fri-round-count-for-degree-bound degree-bound) len)
  proof (rule
    generic-fri-sampled-query-pair-fraction-bound-from-one-step-disagreement-subset)
  fix candidate-table roots final-value round-layers layers query-idxs challenges
  assume query-idxs: query-idxs  $\in$  fri-query-index-list-space
  and evidence:
    generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
    candidate-table degree-bound roots challenges final-value query-idxs
    round-layers layers
  show challenges  $\in$ 
    generic-fri-bad-challenge-lists
    (fri-round-count-for-degree-bound degree-bound) bad
  by (rule cover[OF query-idxs evidence])
next
  fix  $i :: \text{nat}$  and prefix  $:: 'f \text{ list}$ 
  assume  $i$ :  $i < \text{fri-round-count-for-degree-bound degree-bound}$ 
  and prefix: length prefix =  $i$ 
  show bad  $i$  prefix  $\subseteq$ 
    fri-one-step-disagreement-challenges
    (len  $i$ ) (pw  $i$ ) (layer  $i$  prefix) (claimed  $i$  prefix) (domains  $i$  prefix)
  by (rule local-subset[OF  $i$  prefix])
next
  show nnreal
    ( $\sum i \in \{..<\text{fri-round-count-for-degree-bound degree-bound}\}.$ 
      CARD('f)  $\wedge i * (\text{len } i \text{ div } 2) * \text{CARD('f)}$ 
       $\wedge (\text{fri-round-count-for-degree-bound degree-bound} - \text{Suc } i)) /$ 
    nnreal (CARD('f)  $\wedge \text{fri-round-count-for-degree-bound degree-bound}$ )
     $\leq \text{fri-one-step-disagreement-round-fraction}$ 
    (fri-round-count-for-degree-bound degree-bound) len
  by (rule fri-one-step-disagreement-round-fraction-refl-bound)
qed

```

lemma *trace-fri-sampled-query-pair-fraction-bound-from-bad-challenge-round-bounds:*

assumes *cover:*

$\bigwedge$ *candidate-table roots final-value round-layers layers query-idxs*
challenges.
query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence trace-table-low-degree
(Not $\circ$ *trace-table-low-degree)* *candidate-table (clength - 1)*
roots challenges final-value query-idxs round-layers layers $\implies$
challenges $\in$
generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound (clength - 1)) bad

and *local-bound:*

$\bigwedge i$ *prefix.*
i $<$ *fri-round-count-for-degree-bound (clength - 1)* $\implies$
length prefix = i $\implies$
card (bad i prefix) $\leq$ *bd i*

and *fraction-bound:*

nnreal
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound (clength - 1)}\}.$
 $\text{CARD('f)} \wedge i * \text{bd } i *$
 $\text{CARD('f)} \wedge$
 $(\text{fri-round-count-for-degree-bound (clength - 1)} - \text{Suc } i)) /$
 $\text{nnreal (CARD('f)} \wedge \text{fri-round-count-for-degree-bound (clength - 1))}$
 $\leq C$

shows

generic-fri-sampled-query-pair-fraction-bound trace-table-low-degree
(Not $\circ$ *trace-table-low-degree)* *(clength - 1) C*

by (*rule*

generic-fri-sampled-query-pair-fraction-bound-from-bad-challenge-round-bounds
 $[OF \text{ cover local-bound fraction-bound}]$)

lemma *generic-fri-sampled-query-pair-fraction-bound-from-full-cover-round-bounds:*

assumes *full-cover:*

$\bigwedge$ *candidate-table roots final-value round-layers layers query-idxs*
challenges.
query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers $\implies$
 $\exists$ *doms. generic-fri-sampled-layer-chain-full-cover low-degree*
candidate-table degree-bound roots challenges final-value query-idxs
round-layers doms layers bad

and *local-bound:*

$\bigwedge i$ *prefix.*
i $<$ *fri-round-count-for-degree-bound degree-bound* $\implies$
length prefix = i $\implies$
card (bad i prefix) $\leq$ *bd i*

and *fraction-bound:*

nnreal

$$\begin{aligned}
& (\sum i \in \{..<\text{fri-round-count-for-degree-bound degree-bound}\}. \\
& \quad \text{CARD}(f) \wedge i * \text{bd } i * \\
& \quad \text{CARD}(f) \wedge \\
& \quad (\text{fri-round-count-for-degree-bound degree-bound} - \text{Suc } i)) / \\
& \quad \text{nnreal } (\text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound degree-bound}) \\
& \leq C
\end{aligned}$$

shows

generic-fri-sampled-query-pair-fraction-bound low-degree candidate-bad degree-bound C

proof (rule *generic-fri-sampled-query-pair-fraction-bound-from-bad-challenge-round-bounds [OF - local-bound fraction-bound]*)

fix *candidate-table roots final-value round-layers layers query-idxs challenges*

assume *query-space: query-idxs ∈ fri-query-index-list-space*

and *evidence:*

generic-fri-sampled-query-candidate-evidence low-degree candidate-bad candidate-table degree-bound roots challenges final-value query-idxs round-layers layers

obtain *doms where cover:*

generic-fri-sampled-layer-chain-full-cover low-degree candidate-table degree-bound roots challenges final-value query-idxs round-layers doms layers bad

using *full-cover[OF query-space evidence]* **by** *blast*

have *bad:*

challenges ∈ generic-fri-bad-challenge-lists (length challenges) bad

by (rule *generic-fri-sampled-layer-chain-full-cover-bad-challenge-list [OF cover]*)

have *canonical:*

generic-fri-canonical-sampled-layer-chain-evidence low-degree candidate-table degree-bound roots challenges final-value query-idxs round-layers layers

by (rule *generic-fri-sampled-query-candidate-evidenceD(1)[OF evidence]*)

have *sampled:*

generic-fri-sampled-layer-chain-evidence low-degree candidate-table degree-bound roots challenges final-value query-idxs round-layers (fri-canonical-domains (length challenges)) layers

by (rule *generic-fri-canonical-sampled-layer-chain-evidenceD [OF canonical]*)

have *partial:*

generic-fri-partial-evidence low-degree candidate-table degree-bound roots challenges final-value query-idxs round-layers

by (rule *generic-fri-sampled-layer-chain-evidenceD(1)[OF sampled]*)

have *len:*

length challenges = fri-round-count-for-degree-bound degree-bound

using *generic-fri-partial-evidence-shapes(1,2)[OF partial]* **by** *simp*

show *challenges ∈ generic-fri-bad-challenge-lists (fri-round-count-for-degree-bound degree-bound) bad*

using bad by (simp add: len)
qed

lemma generic-fri-sampled-query-candidate-evidence-full-cover-if-no-failure:

assumes evidence:

generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers

and no-failure:

¬ generic-fri-sampled-full-cover-failure low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges)) layers bad

shows

∃ doms. generic-fri-sampled-layer-chain-full-cover low-degree
candidate-table degree-bound roots challenges final-value query-idxs
round-layers doms layers bad

proof —

have canonical:

generic-fri-canonical-sampled-layer-chain-evidence low-degree
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers

by (rule generic-fri-sampled-query-candidate-evidenceD(1)[OF evidence])

have sampled:

generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges)) layers

by (rule generic-fri-canonical-sampled-layer-chain-evidenceD
[OF canonical])

show ?thesis

using sampled no-failure

unfolding generic-fri-sampled-layer-chain-full-cover-def

generic-fri-sampled-full-cover-failure-def

by blast

qed

lemma generic-fri-sampled-query-pair-fraction-bound-from-no-full-cover-failure-round-bounds:

assumes no-failure:

∧ candidate-table roots final-value round-layers layers query-idxs
challenges.

query-idxs ∈ fri-query-index-list-space ⇒

generic-fri-sampled-query-candidate-evidence low-degree candidate-bad
candidate-table degree-bound roots challenges final-value query-idxs
round-layers layers ⇒

¬ generic-fri-sampled-full-cover-failure low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges)) layers bad

and local-bound:

∧ i prefix.

i < fri-round-count-for-degree-bound degree-bound ⇒

$length\ prefix = i \implies$
 $card\ (bad\ i\ prefix) \leq bd\ i$
and *fraction-bound*:
 $nnreal$
 $(\sum i \in \{..<fri-round-count-for-degree-bound\ degree-bound\}).$
 $CARD('f) \wedge i * bd\ i *$
 $CARD('f) \wedge$
 $(fri-round-count-for-degree-bound\ degree-bound - Suc\ i)) /$
 $nnreal\ (CARD('f) \wedge fri-round-count-for-degree-bound\ degree-bound)$
 $\leq C$
shows
 $generic-fri-sampled-query-pair-fraction-bound\ low-degree\ candidate-bad$
 $degree-bound\ C$
proof (rule
 $generic-fri-sampled-query-pair-fraction-bound-from-full-cover-round-bounds$
 $[OF - local-bound\ fraction-bound])$
fix $candidate-table\ roots\ final-value\ round-layers\ layers\ query-idxs$
 $challenges$
assume $query-space: query-idxs \in fri-query-index-list-space$
and *evidence*:
 $generic-fri-sampled-query-candidate-evidence\ low-degree$
 $candidate-bad\ candidate-table\ degree-bound\ roots\ challenges$
 $final-value\ query-idxs\ round-layers\ layers$
show $\exists\ doms.\ generic-fri-sampled-layer-chain-full-cover\ low-degree$
 $candidate-table\ degree-bound\ roots\ challenges\ final-value\ query-idxs$
 $round-layers\ doms\ layers\ bad$
by (rule $generic-fri-sampled-query-candidate-evidence-full-cover-if-no-failure$
 $[OF\ evidence\ no-failure[OF\ query-space\ evidence]])$
qed

lemma *trace-fri-sampled-query-pair-fraction-bound-from-full-cover-round-bounds*:
assumes *full-cover*:
 $\bigwedge candidate-table\ roots\ final-value\ round-layers\ layers\ query-idxs$
 $challenges.$
 $query-idxs \in fri-query-index-list-space \implies$
 $generic-fri-sampled-query-candidate-evidence\ trace-table-low-degree$
 $(Not \circ trace-table-low-degree)\ candidate-table\ (clength - 1)$
 $roots\ challenges\ final-value\ query-idxs\ round-layers\ layers \implies$
 $\exists\ doms.\ generic-fri-sampled-layer-chain-full-cover$
 $trace-table-low-degree\ candidate-table\ (clength - 1)\ roots$
 $challenges\ final-value\ query-idxs\ round-layers\ doms\ layers\ bad$
and *local-bound*:
 $\bigwedge i\ prefix.$
 $i < fri-round-count-for-degree-bound\ (clength - 1) \implies$
 $length\ prefix = i \implies$
 $card\ (bad\ i\ prefix) \leq bd\ i$
and *fraction-bound*:
 $nnreal$
 $(\sum i \in \{..<fri-round-count-for-degree-bound\ (clength - 1)\}).$

$$\begin{aligned}
& \text{CARD}('f) \wedge i * \text{bd } i * \\
& \text{CARD}('f) \wedge \\
& (\text{fri-round-count-for-degree-bound } (\text{clength} - 1) - \text{Suc } i)) / \\
& \text{nnreal } (\text{CARD}('f) \wedge \text{fri-round-count-for-degree-bound } (\text{clength} - 1)) \\
& \leq C
\end{aligned}$$

shows

generic-fri-sampled-query-pair-fraction-bound trace-table-low-degree
(Not ◦ trace-table-low-degree) (clength - 1) C

by (rule

generic-fri-sampled-query-pair-fraction-bound-from-full-cover-round-bounds
[OF full-cover local-bound fraction-bound])

lemma *trace-fri-sampled-query-pair-fraction-bound-from-no-full-cover-failure-round-bounds:*

assumes *no-failure:*

$\bigwedge$ *candidate-table roots final-value round-layers layers query-idxs*
challenges.

query-idxs ∈ fri-query-index-list-space ⇒
generic-fri-sampled-query-candidate-evidence trace-table-low-degree

(Not ◦ trace-table-low-degree) candidate-table (clength - 1)
roots challenges final-value query-idxs round-layers layers ⇒

$\neg$ *generic-fri-sampled-full-cover-failure trace-table-low-degree*
candidate-table (clength - 1) roots challenges final-value
query-idxs round-layers (fri-canonical-domains (length challenges))
layers bad

and *local-bound:*

$\bigwedge i$ *prefix.*
i < fri-round-count-for-degree-bound (clength - 1) ⇒
length prefix = i ⇒
card (bad i prefix) ≤ bd i

and *fraction-bound:*

nnreal
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound } (\text{clength} - 1)\}).$
 $\text{CARD}('f) \wedge i * \text{bd } i *$
 $\text{CARD}('f) \wedge$
 $(\text{fri-round-count-for-degree-bound } (\text{clength} - 1) - \text{Suc } i)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{fri-round-count-for-degree-bound } (\text{clength} - 1))$
 $\leq C$

shows

generic-fri-sampled-query-pair-fraction-bound trace-table-low-degree
(Not ◦ trace-table-low-degree) (clength - 1) C

by (rule

generic-fri-sampled-query-pair-fraction-bound-from-no-full-cover-failure-round-bounds
[OF no-failure local-bound fraction-bound])

lemma *composition-fri-sampled-query-pair-fraction-bound-from-bad-challenge-round-bounds:*

fixes *dg :: 'f*

assumes *cover:*

$\bigwedge$ *candidate-table roots final-value round-layers layers query-idxs*
challenges.

```

query-idxs ∈ fri-query-index-list-space ⇒
generic-fri-sampled-query-candidate-evidence
  (composition-table-low-degree (to-nat dg))
  (Not ∘ composition-table-low-degree maxDegree) candidate-table
  (to-nat dg) roots challenges final-value query-idxs round-layers
  layers ⇒
challenges ∈
  generic-fri-bad-challenge-lists
  (fri-round-count-for-degree-bound (to-nat dg)) (bad dg)
and local-bound:
  ∧dg i prefix.
  i < fri-round-count-for-degree-bound (to-nat dg) ⇒
  length prefix = i ⇒
  card (bad dg i prefix) ≤ bd dg i
and fraction-bound:
  ∧dg.
  nnreal
  (∑ i ∈ {..fri-round-count-for-degree-bound (to-nat dg)}.
    CARD('f) ^ i * bd dg i *
    CARD('f) ^
      (fri-round-count-for-degree-bound (to-nat dg) - Suc i) /
    nnreal (CARD('f) ^ fri-round-count-for-degree-bound (to-nat dg))
  ≤ C dg
shows
generic-fri-sampled-query-pair-fraction-bound
  (composition-table-low-degree (to-nat dg))
  (Not ∘ composition-table-low-degree maxDegree) (to-nat dg) (C dg)
proof (rule
  generic-fri-sampled-query-pair-fraction-bound-from-bad-challenge-round-bounds)
fix candidate-table roots final-value round-layers layers query-idxs
challenges
assume query-space: query-idxs ∈ fri-query-index-list-space
and evidence:
  generic-fri-sampled-query-candidate-evidence
  (composition-table-low-degree (to-nat dg))
  (Not ∘ composition-table-low-degree maxDegree) candidate-table
  (to-nat dg) roots challenges final-value query-idxs round-layers
  layers
show challenges ∈
  generic-fri-bad-challenge-lists
  (fri-round-count-for-degree-bound (to-nat dg)) (bad dg)
by (rule cover[OF query-space evidence])
next
fix i and prefix :: 'f list
assume i: i < fri-round-count-for-degree-bound (to-nat dg)
and len: length prefix = i
show card (bad dg i prefix) ≤ bd dg i
by (rule local-bound[OF i len])
next

```

show *nnreal*
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound } (to\text{-nat } dg)\}.$
 $CARD('f) \wedge i * bd\ dg\ i *$
 $CARD('f) \wedge$
 $(\text{fri-round-count-for-degree-bound } (to\text{-nat } dg) - Suc\ i)) /$
 $nnreal\ (CARD('f) \wedge \text{fri-round-count-for-degree-bound } (to\text{-nat } dg))$
 $\leq C\ dg$
by (*rule fraction-bound*)
qed

lemma *composition-fri-sampled-query-pair-fraction-bound-from-full-cover-round-bounds:*
fixes *dg :: 'f*
assumes *full-cover:*
 $\bigwedge \text{candidate-table roots final-value round-layers layers query-idxs}$
 challenges.
 $\text{query-idxs} \in \text{fri-query-index-list-space} \implies$
 $\text{generic-fri-sampled-query-candidate-evidence}$
 $(\text{composition-table-low-degree } (to\text{-nat } dg))$
 $(Not \circ \text{composition-table-low-degree } \text{maxDegree})\ \text{candidate-table}$
 $(to\text{-nat } dg)\ \text{roots challenges final-value query-idxs round-layers}$
 $\text{layers} \implies$
 $\exists \text{doms. generic-fri-sampled-layer-chain-full-cover}$
 $(\text{composition-table-low-degree } (to\text{-nat } dg))\ \text{candidate-table}$
 $(to\text{-nat } dg)\ \text{roots challenges final-value query-idxs round-layers doms}$
 $\text{layers } (bad\ dg)$
and *local-bound:*
 $\bigwedge dg\ i\ \text{prefix.}$
 $i < \text{fri-round-count-for-degree-bound } (to\text{-nat } dg) \implies$
 $\text{length prefix} = i \implies$
 $\text{card } (bad\ dg\ i\ \text{prefix}) \leq bd\ dg\ i$
and *fraction-bound:*
 $\bigwedge dg.$
 $nnreal$
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound } (to\text{-nat } dg)\}.$
 $CARD('f) \wedge i * bd\ dg\ i *$
 $CARD('f) \wedge$
 $(\text{fri-round-count-for-degree-bound } (to\text{-nat } dg) - Suc\ i)) /$
 $nnreal\ (CARD('f) \wedge \text{fri-round-count-for-degree-bound } (to\text{-nat } dg))$
 $\leq C\ dg$
shows
 $\text{generic-fri-sampled-query-pair-fraction-bound}$
 $(\text{composition-table-low-degree } (to\text{-nat } dg))$
 $(Not \circ \text{composition-table-low-degree } \text{maxDegree})\ (to\text{-nat } dg)\ (C\ dg)$
proof (*rule*
 $\text{generic-fri-sampled-query-pair-fraction-bound-from-full-cover-round-bounds})$
fix *candidate-table roots final-value round-layers layers query-idxs*
 challenges
assume *query-space:* $\text{query-idxs} \in \text{fri-query-index-list-space}$
and *evidence:*

```

    generic-fri-sampled-query-candidate-evidence
      (composition-table-low-degree (to-nat dg))
      (Not o composition-table-low-degree maxDegree) candidate-table
      (to-nat dg) roots challenges final-value query-idxs round-layers
      layers
  show  $\exists$  doms. generic-fri-sampled-layer-chain-full-cover
    (composition-table-low-degree (to-nat dg)) candidate-table
    (to-nat dg) roots challenges final-value query-idxs round-layers doms
    layers (bad dg)
  by (rule full-cover[OF query-space evidence])
next
  fix i and prefix :: 'f list
  assume i:  $i < \text{fri-round-count-for-degree-bound (to-nat dg)}$ 
  and len: length prefix = i
  show card (bad dg i prefix)  $\leq$  bd dg i
  by (rule local-bound[OF i len])
next
  show nreal
    ( $\sum i \in \{..<\text{fri-round-count-for-degree-bound (to-nat dg)}\}$ .
      CARD('f)  $\wedge$  i * bd dg i *
      CARD('f)  $\wedge$ 
      (fri-round-count-for-degree-bound (to-nat dg) - Suc i)) /
    nreal (CARD('f)  $\wedge$  fri-round-count-for-degree-bound (to-nat dg))
     $\leq$  C dg
  by (rule fraction-bound)
qed

```

lemma composition-fri-sampled-query-pair-fraction-bound-from-no-full-cover-failure-round-bounds:

```

  fixes dg :: 'f
  assumes no-failure:
     $\bigwedge$  candidate-table roots final-value round-layers layers query-idxs
      challenges.
      query-idxs  $\in$  fri-query-index-list-space  $\implies$ 
      generic-fri-sampled-query-candidate-evidence
        (composition-table-low-degree (to-nat dg))
        (Not o composition-table-low-degree maxDegree) candidate-table
        (to-nat dg) roots challenges final-value query-idxs round-layers
        layers  $\implies$ 
       $\neg$  generic-fri-sampled-full-cover-failure
        (composition-table-low-degree (to-nat dg)) candidate-table
        (to-nat dg) roots challenges final-value query-idxs round-layers
        (fri-canonical-domains (length challenges)) layers (bad dg)
  and local-bound:
     $\bigwedge$  dg i prefix.
       $i < \text{fri-round-count-for-degree-bound (to-nat dg)} \implies$ 
      length prefix = i  $\implies$ 
      card (bad dg i prefix)  $\leq$  bd dg i
  and fraction-bound:
     $\bigwedge$  dg.

```

$$\begin{aligned} & \text{nnreal} \\ & (\sum i \in \{..<\text{fri-round-count-for-degree-bound } (to\text{-nat } dg)\}. \\ & \quad \text{CARD}(f) \wedge i * \text{bd } dg \ i * \\ & \quad \text{CARD}(f) \wedge \\ & \quad (\text{fri-round-count-for-degree-bound } (to\text{-nat } dg) - \text{Suc } i)) / \\ & \text{nnreal } (\text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound } (to\text{-nat } dg)) \\ & \leq C \ dg \end{aligned}$$

shows

$$\begin{aligned} & \text{generic-fri-sampled-query-pair-fraction-bound} \\ & (\text{composition-table-low-degree } (to\text{-nat } dg)) \\ & (\text{Not } \circ \text{ composition-table-low-degree } \text{maxDegree}) (to\text{-nat } dg) (C \ dg) \end{aligned}$$

proof (rule

$$\begin{aligned} & \text{generic-fri-sampled-query-pair-fraction-bound-from-no-full-cover-failure-round-bounds}) \\ & \text{fix candidate-table roots final-value round-layers layers query-idxs} \\ & \text{challenges} \\ & \text{assume query-space: query-idxs } \in \text{fri-query-index-list-space} \\ & \text{and evidence:} \\ & \text{generic-fri-sampled-query-candidate-evidence} \\ & (\text{composition-table-low-degree } (to\text{-nat } dg)) \\ & (\text{Not } \circ \text{ composition-table-low-degree } \text{maxDegree}) \text{ candidate-table} \\ & (to\text{-nat } dg) \text{ roots challenges final-value query-idxs round-layers} \\ & \text{layers} \end{aligned}$$

show $\neg$ generic-fri-sampled-full-cover-failure

$$\begin{aligned} & (\text{composition-table-low-degree } (to\text{-nat } dg)) \text{ candidate-table} \\ & (to\text{-nat } dg) \text{ roots challenges final-value query-idxs round-layers} \\ & (\text{fri-canonical-domains } (\text{length challenges})) \text{ layers } (bad \ dg) \\ & \text{by (rule no-failure[OF query-space evidence])} \end{aligned}$$

next

fix i and prefix :: 'f list

assume $i < \text{fri-round-count-for-degree-bound } (to\text{-nat } dg)$

and len: length prefix = i

show card (bad dg i prefix) $\leq \text{bd } dg \ i$

by (rule local-bound[OF i len])

next

show nnreal

$$\begin{aligned} & (\sum i \in \{..<\text{fri-round-count-for-degree-bound } (to\text{-nat } dg)\}. \\ & \quad \text{CARD}(f) \wedge i * \text{bd } dg \ i * \\ & \quad \text{CARD}(f) \wedge \\ & \quad (\text{fri-round-count-for-degree-bound } (to\text{-nat } dg) - \text{Suc } i)) / \\ & \text{nnreal } (\text{CARD}(f) \wedge \text{fri-round-count-for-degree-bound } (to\text{-nat } dg)) \\ & \leq C \ dg \\ & \text{by (rule fraction-bound)} \end{aligned}$$

qed

lemma wp-trace-fri-sampled-query-bad-candidate-bound-from-bad-challenge-round-bounds:

assumes future: trace-fri-future-fresh s

and cover:

$$\bigwedge \text{candidate-table roots final-value round-layers layers query-idxs}$$

$$\text{challenges.}$$

query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence trace-table-low-degree
(Not $\circ$ *trace-table-low-degree)* *candidate-table (clength - 1)*
roots challenges final-value query-idxs round-layers layers $\implies$
challenges $\in$
generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound (clength - 1)) bad
and *local-bound*:
 $\bigwedge i$ *prefix*.
i $<$ *fri-round-count-for-degree-bound (clength - 1)* $\implies$
length prefix = i $\implies$
card (bad i prefix) $\leq$ *bd i*
and *fraction-bound*:
nnreal
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound (clength - 1)}\}.$
 $\text{CARD('f)} \wedge i * \text{bd i} *$
 $\text{CARD('f)} \wedge$
 $(\text{fri-round-count-for-degree-bound (clength - 1)} - \text{Suc } i)) /$
 $\text{nnreal (CARD('f)} \wedge \text{fri-round-count-for-degree-bound (clength - 1))}$
 $\leq C$
shows
wp-event verify-monad (trace-fri-sampled-query-bad-candidate s) s
 $\leq C$
proof (*rule*
wp-trace-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction
[*OF future*])
show
generic-fri-sampled-query-projection-fraction-bound
trace-table-low-degree (Not $\circ$ *trace-table-low-degree)*
(clength - 1) C
by (*rule*
generic-fri-sampled-query-projection-fraction-bound-from-bad-challenge-round-bounds
[*OF cover local-bound fraction-bound*])
qed

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-from-one-step-disagreement-subset*:
assumes *future: trace-fri-future-fresh s*
and *cover*:
 $\bigwedge$ *candidate-table roots final-value round-layers layers query-idxs*
challenges.
query-idxs $\in$ *fri-query-index-list-space* $\implies$
generic-fri-sampled-query-candidate-evidence trace-table-low-degree
(Not $\circ$ *trace-table-low-degree)* *candidate-table (clength - 1)*
roots challenges final-value query-idxs round-layers layers $\implies$
challenges $\in$
generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound (clength - 1)) bad
and *local-subset*:
 $\bigwedge i$ *prefix*.

$i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \implies$
 $\text{length prefix} = i \implies$
 $\text{bad } i \text{ prefix} \subseteq$
 $\text{fri-one-step-disagreement-challenges}$
 $(\text{len } i) (\text{pw } i) (\text{layer } i \text{ prefix}) (\text{claimed } i \text{ prefix}) (\text{domains } i \text{ prefix})$
and fraction-bound :
 nnreal
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound } (\text{clength} - 1)\}.$
 $\text{CARD('f)} \wedge i * (\text{len } i \text{ div } 2) *$
 $\text{CARD('f)} \wedge$
 $(\text{fri-round-count-for-degree-bound } (\text{clength} - 1) - \text{Suc } i)) /$
 $\text{nnreal } (\text{CARD('f)} \wedge \text{fri-round-count-for-degree-bound } (\text{clength} - 1))$
 $\leq C$
shows
 $\text{wp-event verify-monad } (\text{trace-fri-sampled-query-bad-candidate } s) s$
 $\leq C$
proof (rule
 $\text{wp-trace-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction}$
 $[\text{OF future}]$)
show
 $\text{generic-fri-sampled-query-projection-fraction-bound}$
 $\text{trace-table-low-degree } (\text{Not } \circ \text{ trace-table-low-degree})$
 $(\text{clength} - 1) C$
by (rule
 $\text{generic-fri-sampled-query-projection-fraction-bound-from-one-step-disagreement-subset}$
 $[\text{OF cover local-subset fraction-bound}]$)
qed

lemma $\text{wp-trace-fri-sampled-query-bad-candidate-bound-from-one-step-disagreement-subset-exact}$:
assumes $\text{future: trace-fri-future-fresh } s$
and cover :
 $\bigwedge \text{candidate-table roots final-value round-layers layers query-idxs}$
 challenges.
 $\text{query-idxs} \in \text{fri-query-index-list-space} \implies$
 $\text{generic-fri-sampled-query-candidate-evidence trace-table-low-degree}$
 $(\text{Not } \circ \text{ trace-table-low-degree}) \text{ candidate-table } (\text{clength} - 1)$
 $\text{roots challenges final-value query-idxs round-layers layers} \implies$
 $\text{challenges} \in$
 $\text{generic-fri-bad-challenge-lists}$
 $(\text{fri-round-count-for-degree-bound } (\text{clength} - 1)) \text{ bad}$
and local-subset :
 $\bigwedge i \text{ prefix.}$
 $i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \implies$
 $\text{length prefix} = i \implies$
 $\text{bad } i \text{ prefix} \subseteq$
 $\text{fri-one-step-disagreement-challenges}$
 $(\text{len } i) (\text{pw } i) (\text{layer } i \text{ prefix}) (\text{claimed } i \text{ prefix}) (\text{domains } i \text{ prefix})$
shows
 $\text{wp-event verify-monad } (\text{trace-fri-sampled-query-bad-candidate } s) s$

```

    ≤
    fri-one-step-disagreement-round-fraction
    (fri-round-count-for-degree-bound (clength - 1)) len
proof (rule wp-trace-fri-sampled-query-bad-candidate-bound-from-one-step-disagreement-subset)
  show trace-fri-future-fresh s
  by (rule future)
next
  fix candidate-table roots final-value round-layers layers query-idxs challenges
  assume query-idxs: query-idxs ∈ fri-query-index-list-space
  and evidence:
    generic-fri-sampled-query-candidate-evidence trace-table-low-degree
    (Not ∘ trace-table-low-degree) candidate-table (clength - 1)
    roots challenges final-value query-idxs round-layers layers
  show challenges ∈
    generic-fri-bad-challenge-lists
    (fri-round-count-for-degree-bound (clength - 1)) bad
  by (rule cover[OF query-idxs evidence])
next
  fix i :: nat and prefix :: 'f list
  assume i: i < fri-round-count-for-degree-bound (clength - 1)
  and prefix: length prefix = i
  show bad i prefix ⊆
    fri-one-step-disagreement-challenges
    (len i) (pw i) (layer i prefix) (claimed i prefix) (domains i prefix)
  by (rule local-subset[OF i prefix])
next
  show nnreal
    (∑ i ∈ {.. $\text{fri-round-count-for-degree-bound (clength - 1)}$ }.
      CARD('f) ^ i * (len i div 2) *
      CARD('f) ^
        (fri-round-count-for-degree-bound (clength - 1) - Suc i)) /
    nnreal (CARD('f) ^ fri-round-count-for-degree-bound (clength - 1))
    ≤ fri-one-step-disagreement-round-fraction
      (fri-round-count-for-degree-bound (clength - 1)) len
  by (rule fri-one-step-disagreement-round-fraction-refl-bound)
qed

```

lemma wp-trace-fri-sampled-query-bad-candidate-bound-from-canonical-one-step-disagreement-subset:

```

assumes future: trace-fri-future-fresh s
and cover:
  ∧ candidate-table roots final-value round-layers layers query-idxs
    challenges.
  query-idxs ∈ fri-query-index-list-space ⇒
  generic-fri-sampled-query-candidate-evidence trace-table-low-degree
    (Not ∘ trace-table-low-degree) candidate-table (clength - 1)
    roots challenges final-value query-idxs round-layers layers ⇒
  challenges ∈
    generic-fri-bad-challenge-lists
    (fri-round-count-for-degree-bound (clength - 1)) bad

```

and *local-subset*:
 $\bigwedge i \text{ prefix.}$
 $i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \implies$
 $\text{length prefix} = i \implies$
 $\text{bad } i \text{ prefix} \subseteq$
 $\text{fri-one-step-disagreement-challenges}$
 $(\text{trace-fri-canonical-layer-len } i) (pw \ i)$
 $(\text{layer } i \text{ prefix}) (\text{claimed } i \text{ prefix}) (\text{domains } i \text{ prefix})$
shows
 $\text{wp-event verify-monad } (\text{trace-fri-sampled-query-bad-candidate } s) \ s$
 $\leq \text{concrete-trace-fri-bad-challenge-fraction}$
using
 $\text{wp-trace-fri-sampled-query-bad-candidate-bound-from-one-step-disagreement-subset-exact}$
 $[\text{OF future cover local-subset}]$
by (*simp add: concrete-trace-fri-bad-challenge-fraction-def*)

lemma *wp-composition-fri-sampled-query-bad-candidate-bound-from-bad-challenge-round-bounds*:
assumes *future: composition-fri-future-fresh s*
and *cover*:
 $\bigwedge dg \text{ candidate-table roots final-value round-layers layers query-idxs}$
 challenges.
 $\text{query-idxs} \in \text{fri-query-index-list-space} \implies$
 $\text{generic-fri-sampled-query-candidate-evidence}$
 $(\text{composition-table-low-degree } (\text{to-nat } dg))$
 $(\text{Not } \circ \text{composition-table-low-degree } \text{maxDegree}) \text{ candidate-table}$
 $(\text{to-nat } dg) \text{ roots challenges final-value query-idxs round-layers}$
 $\text{layers} \implies$
 $\text{challenges} \in$
 $\text{generic-fri-bad-challenge-lists}$
 $(\text{fri-round-count-for-degree-bound } (\text{to-nat } dg)) (\text{bad } dg)$
and *local-bound*:
 $\bigwedge dg \ i \text{ prefix.}$
 $i < \text{fri-round-count-for-degree-bound } (\text{to-nat } dg) \implies$
 $\text{length prefix} = i \implies$
 $\text{card } (\text{bad } dg \ i \text{ prefix}) \leq \text{bd } dg \ i$
and *fraction-bound*:
 $\bigwedge dg.$
 nnreal
 $(\sum i \in \{..<\text{fri-round-count-for-degree-bound } (\text{to-nat } dg)\}).$
 $\text{CARD}('f) \wedge i * \text{bd } dg \ i *$
 $\text{CARD}('f) \wedge$
 $(\text{fri-round-count-for-degree-bound } (\text{to-nat } dg) - \text{Suc } i)) /$
 $\text{nnreal } (\text{CARD}('f) \wedge \text{fri-round-count-for-degree-bound } (\text{to-nat } dg))$
 $\leq C$
shows
 $\text{wp-event verify-monad } (\text{composition-fri-sampled-query-bad-candidate } s) \ s$
 $\leq C$
proof (*rule*
 $\text{wp-composition-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction}$

```

[OF future])
fix dg :: 'f
show
  generic-fri-sampled-query-projection-fraction-bound
    (composition-table-low-degree (to-nat dg))
    (Not o composition-table-low-degree maxDegree)
    (to-nat dg) C
proof (rule
  generic-fri-sampled-query-projection-fraction-bound-from-bad-challenge-round-bounds)
fix candidate-table roots final-value round-layers layers query-idxs
  challenges
assume query-space: query-idxs ∈ fri-query-index-list-space
and evidence:
  generic-fri-sampled-query-candidate-evidence
    (composition-table-low-degree (to-nat dg))
    (Not o composition-table-low-degree maxDegree) candidate-table
    (to-nat dg) roots challenges final-value query-idxs round-layers
    layers
show challenges ∈
  generic-fri-bad-challenge-lists
    (fri-round-count-for-degree-bound (to-nat dg)) (bad dg)
by (rule cover[OF query-space evidence])
next
fix i and prefix :: 'f list
assume i: i < fri-round-count-for-degree-bound (to-nat dg)
and len: length prefix = i
show card (bad dg i prefix) ≤ bd dg i
by (rule local-bound[OF i len])
next
show nnreal
  (∑ i ∈ {.. $\text{fri-round-count-for-degree-bound (to-nat dg)}$ }.
    CARD('f) ^ i * bd dg i *
    CARD('f) ^
      (fri-round-count-for-degree-bound (to-nat dg) - Suc i)) /
  nnreal (CARD('f) ^ fri-round-count-for-degree-bound (to-nat dg))
  ≤ C
by (rule fraction-bound)
qed
qed

lemma wp-composition-fri-sampled-query-bad-candidate-bound-from-one-step-disagreement-subset:
  assumes future: composition-fri-future-fresh s
  and cover:
     $\bigwedge dg$  candidate-table roots final-value round-layers layers query-idxs
      challenges.
    query-idxs ∈ fri-query-index-list-space  $\implies$ 
    generic-fri-sampled-query-candidate-evidence
      (composition-table-low-degree (to-nat dg))
      (Not o composition-table-low-degree maxDegree) candidate-table

```

```

    (to-nat dg) roots challenges final-value query-idxs round-layers
    layers  $\implies$ 
    challenges  $\in$ 
      generic-fri-bad-challenge-lists
      (fri-round-count-for-degree-bound (to-nat dg)) (bad dg)
  and local-subset:
     $\bigwedge dg$  i prefix.
      i < fri-round-count-for-degree-bound (to-nat dg)  $\implies$ 
      length prefix = i  $\implies$ 
      bad dg i prefix  $\subseteq$ 
        fri-one-step-disagreement-challenges
        (len dg i) (pw dg i) (layer dg i prefix)
        (claimed dg i prefix) (domains dg i prefix)
  and fraction-bound:
     $\bigwedge dg.$ 
      nreal
       $(\sum i \in \{..<fri-round-count-for-degree-bound (to-nat dg)\}.$ 
        CARD('f)  $\wedge$  i * (len dg i div 2) *
        CARD('f)  $\wedge$ 
        (fri-round-count-for-degree-bound (to-nat dg) - Suc i)) /
      nreal (CARD('f)  $\wedge$  fri-round-count-for-degree-bound (to-nat dg))
       $\leq C$ 
  shows
    wp-event verify-monad (composition-fri-sampled-query-bad-candidate s) s
     $\leq C$ 
  proof (rule
    wp-composition-fri-sampled-query-bad-candidate-bound-from-generic-projection-fraction
    [OF future])
  fix dg :: 'f
  show
    generic-fri-sampled-query-projection-fraction-bound
      (composition-table-low-degree (to-nat dg))
      (Not  $\circ$  composition-table-low-degree maxDegree)
      (to-nat dg) C
  proof (rule
    generic-fri-sampled-query-projection-fraction-bound-from-one-step-disagreement-subset)
  fix candidate-table roots final-value round-layers layers query-idxs
  challenges
  assume query-space: query-idxs  $\in$  fri-query-index-list-space
  and evidence:
    generic-fri-sampled-query-candidate-evidence
      (composition-table-low-degree (to-nat dg))
      (Not  $\circ$  composition-table-low-degree maxDegree) candidate-table
      (to-nat dg) roots challenges final-value query-idxs round-layers
      layers
  show challenges  $\in$ 
    generic-fri-bad-challenge-lists
      (fri-round-count-for-degree-bound (to-nat dg)) (bad dg)
  by (rule cover[OF query-space evidence])

```

```

next
  fix i and prefix :: 'f list
  assume i: i < fri-round-count-for-degree-bound (to-nat dg)
    and prefix: length prefix = i
  show bad dg i prefix  $\subseteq$ 
    fri-one-step-disagreement-challenges
      (len dg i) (pw dg i) (layer dg i prefix)
      (claimed dg i prefix) (domains dg i prefix)
    by (rule local-subset[OF i prefix])
next
show nnreal
  ( $\sum i \in \{..<\text{fri-round-count-for-degree-bound (to-nat dg)}\}$ ).
  CARD('f)  $\wedge$  i * (len dg i div 2) *
  CARD('f)  $\wedge$ 
    (fri-round-count-for-degree-bound (to-nat dg) - Suc i) /
  nnreal (CARD('f)  $\wedge$  fri-round-count-for-degree-bound (to-nat dg))
 $\leq C$ 
  by (rule fraction-bound)
qed
qed

```

lemma wp-composition-fri-sampled-query-bad-candidate-bound-from-one-step-disagreement-subset-exact-bound:

```

assumes future: composition-fri-future-fresh s
and cover:
 $\bigwedge dg$  candidate-table roots final-value round-layers layers query-idxs
  challenges.
  query-idxs  $\in$  fri-query-index-list-space  $\implies$ 
  generic-fri-sampled-query-candidate-evidence
    (composition-table-low-degree (to-nat dg))
    (Not  $\circ$  composition-table-low-degree maxDegree) candidate-table
    (to-nat dg) roots challenges final-value query-idxs round-layers
    layers  $\implies$ 
  challenges  $\in$ 
    generic-fri-bad-challenge-lists
      (fri-round-count-for-degree-bound (to-nat dg)) (bad dg)
and local-subset:
 $\bigwedge dg$  i prefix.
  i < fri-round-count-for-degree-bound (to-nat dg)  $\implies$ 
  length prefix = i  $\implies$ 
  bad dg i prefix  $\subseteq$ 
    fri-one-step-disagreement-challenges
      (len dg i) (pw dg i) (layer dg i prefix)
      (claimed dg i prefix) (domains dg i prefix)
and fraction-bound:
 $\bigwedge dg$ .
  fri-one-step-disagreement-round-fraction
    (fri-round-count-for-degree-bound (to-nat dg)) (len dg)
 $\leq C$ 
shows

```

```

wp-event verify-monad (composition-fri-sampled-query-bad-candidate s) s
  ≤ C
proof (rule
  wp-composition-fri-sampled-query-bad-candidate-bound-from-one-step-disagreement-subset
  [OF future cover local-subset])
fix dg
show nreal
  (∑ i ∈ {..fri-round-count-for-degree-bound (to-nat dg)}.
    CARD('f) ^ i * (len dg i div 2) *
    CARD('f) ^
      (fri-round-count-for-degree-bound (to-nat dg) - Suc i)) /
  nreal (CARD('f) ^ fri-round-count-for-degree-bound (to-nat dg))
  ≤ C
using fraction-bound[of dg]
by (simp add: fri-one-step-disagreement-round-fraction-def
  fri-bad-challenge-round-fraction-def
  fri-one-step-disagreement-round-mass-def
  fri-bad-challenge-round-mass-def)
qed

lemma wp-composition-fri-sampled-query-bad-candidate-bound-from-canonical-one-step-disagreement-subset:
assumes future: composition-fri-future-fresh s
and cover:
  ∧ dg candidate-table roots final-value round-layers layers query-idxs
    challenges.
  query-idxs ∈ fri-query-index-list-space ⇒
  generic-fri-sampled-query-candidate-evidence
    (composition-table-low-degree (to-nat dg))
    (Not ∘ composition-table-low-degree maxDegree) candidate-table
    (to-nat dg) roots challenges final-value query-idxs round-layers
    layers ⇒
  challenges ∈
    generic-fri-bad-challenge-lists
    (fri-round-count-for-degree-bound (to-nat dg)) (bad dg)
and local-subset:
  ∧ dg i prefix.
  i < fri-round-count-for-degree-bound (to-nat dg) ⇒
  length prefix = i ⇒
  bad dg i prefix ⊆
    fri-one-step-disagreement-challenges
    (composition-fri-canonical-layer-len dg i) (pw dg i)
    (layer dg i prefix) (claimed dg i prefix) (domains dg i prefix)
shows
  wp-event verify-monad (composition-fri-sampled-query-bad-candidate s) s
    ≤ concrete-composition-fri-bad-challenge-fraction
proof (rule
  wp-composition-fri-sampled-query-bad-candidate-bound-from-one-step-disagreement-subset-exact-bound)
show composition-fri-future-fresh s
by (rule future)

```

```

next
  fix dg candidate-table roots final-value round-layers layers query-idxs
    challenges
  assume query-idxs: query-idxs ∈ fri-query-index-list-space
  and evidence:
    generic-fri-sampled-query-candidate-evidence
      (composition-table-low-degree (to-nat dg))
      (Not ∘ composition-table-low-degree maxDegree) candidate-table
      (to-nat dg) roots challenges final-value query-idxs round-layers
      layers
  show challenges ∈
    generic-fri-bad-challenge-lists
      (fri-round-count-for-degree-bound (to-nat dg)) (bad dg)
  by (rule cover[OF query-idxs evidence])
next
  fix dg and i :: nat and prefix :: 'f list
  assume i: i < fri-round-count-for-degree-bound (to-nat dg)
  and prefix: length prefix = i
  show bad dg i prefix ⊆
    fri-one-step-disagreement-challenges
      (composition-fri-canonical-layer-len dg i) (pw dg i)
      (layer dg i prefix) (claimed dg i prefix) (domains dg i prefix)
  by (rule local-subset[OF i prefix])
next
  fix dg
  show fri-one-step-disagreement-round-fraction
    (fri-round-count-for-degree-bound (to-nat dg))
    (composition-fri-canonical-layer-len dg)
    ≤ concrete-composition-fri-bad-challenge-fraction
  by (rule concrete-composition-fri-bad-challenge-fraction-ge)
qed

end

end

```

theory *Soundness-FRI-Zero-Round-Query-List-Bounds*

imports

Soundness-FRI-Zero-Round-Target

Soundness-FRI-Query-List-Exact-Product

begin

Exact staged query-list bounds for fixed zero-round FRI targets.

The target table and final value must already be fixed before query sampling. This theory deliberately does not close the remaining candidate/verifier alignment gap; it only packages the exact-product query accounting for the fixed targets isolated in *Soundness-FRI-Zero-Round-Target*.

context *soundness*

begin

definition *fri-zero-round-first-query-index-lists*

$:: 'f \text{ list} \Rightarrow 'f \Rightarrow \text{nat list set}$

where

fri-zero-round-first-query-index-lists table final =

$\{ \text{query-idxs} \in \text{fri-query-index-list-space}.$

$\text{query-idxs} ! 0 \in \text{fri-final-value-agreement-set table final} \}$

lemma *fri-zero-round-first-query-index-lists-subset*:

fri-zero-round-first-query-index-lists table final

$\subseteq \text{fri-query-index-list-space}$

unfolding *fri-zero-round-first-query-index-lists-def* **by** *blast*

lemma *finite-fri-zero-round-first-query-index-lists[simp]*:

finite (fri-zero-round-first-query-index-lists table final)

by (*rule finite-subset*

$[OF \text{ fri-zero-round-first-query-index-lists-subset}]$

$(\text{rule finite-fri-query-index-list-space})$

lemma *fri-zero-round-first-query-index-listsD*:

assumes *query-idxs* $\in$

fri-zero-round-first-query-index-lists table final

shows *query-idxs* $\in \text{fri-query-index-list-space}$

and *query-idxs* ! 0 $\in \text{fri-final-value-agreement-set table final}$

using *assms*

unfolding *fri-zero-round-first-query-index-lists-def* **by** *blast+*

lemma *fri-zero-round-first-query-index-lists-card-le*:

card (fri-zero-round-first-query-index-lists table final) $\leq$

card (fri-final-value-agreement-set table final) *

card query-sample-space $\wedge (\text{rounds} - 1)$

proof –

let *?T* = *fri-zero-round-first-query-index-lists table final*

let *?Tail* =

$\{xs. \text{set } xs \subseteq \text{query-sample-space} \wedge$

$\text{length } xs = \text{rounds} - 1\}$

let *?F* = $\lambda \text{query-idxs}. (\text{query-idxs} ! 0, \text{tl } \text{query-idxs})$

have *image-subset*:

?F ‘ *?T* $\subseteq$

fri-final-value-agreement-set table final $\times$ *?Tail*

proof

fix *y*

assume *y* $\in$ *?F* ‘ *?T*

then obtain *query-idxs* **where** *query-idxs-in*: *query-idxs* $\in$ *?T*

and *y-def*: *y* = *?F query-idxs*

by *blast*

have *query-space*: *query-idxs* $\in \text{fri-query-index-list-space}$

and *first-in*:

```

    query-idxs ! 0 ∈ fri-final-value-agreement-set table final
  using fri-zero-round-first-query-index-listsD[OF query-idxs-in]
  by blast+
have len: length query-idxs = rounds
  and set-subset: set query-idxs ⊆ query-sample-space
  using query-space unfolding fri-query-index-list-space-def by auto
have tail-len: length (tl query-idxs) = rounds - 1
  using len rounds-positive by (cases query-idxs) simp-all
have tail-set: set (tl query-idxs) ⊆ set query-idxs
  by (cases query-idxs) auto
have tail-subset: set (tl query-idxs) ⊆ query-sample-space
  using tail-set set-subset by blast
show y ∈ fri-final-value-agreement-set table final × ?Tail
  using y-def first-in tail-len tail-subset by simp
qed
have inj: inj-on ?F ?T
proof (rule inj-onI)
  fix xs ys
  assume xs-in: xs ∈ ?T
    and ys-in: ys ∈ ?T
    and eq: ?F xs = ?F ys
  have xs-len: length xs = rounds
    and ys-len: length ys = rounds
    using fri-zero-round-first-query-index-listsD(1)[OF xs-in]
      fri-zero-round-first-query-index-listsD(1)[OF ys-in]
    unfolding fri-query-index-list-space-def by auto
  have xs-ne: xs ≠ []
    and ys-ne: ys ≠ []
    using xs-len ys-len rounds-positive by auto
  have head-eq: xs ! 0 = ys ! 0
    using eq by simp
  have tail-eq: tl xs = tl ys
    using eq by simp
  from xs-ne obtain x xs' where xs-def: xs = x # xs'
    by (cases xs) auto
  from ys-ne obtain y ys' where ys-def: ys = y # ys'
    by (cases ys) auto
  show xs = ys
    using head-eq tail-eq unfolding xs-def ys-def by simp
qed
have card ?T = card (?F ' ?T)
  using inj by (rule card-image[symmetric])
also have ... ≤
  card (fri-final-value-agreement-set table final × ?Tail)
proof (rule card-mono)
  show finite (fri-final-value-agreement-set table final × ?Tail)
    by (intro finite-cartesian-product finite-fri-final-value-agreement-set
      finite-lists-length-eq[OF finite-query-sample-space])
  show ?F ' ?T ⊆ fri-final-value-agreement-set table final × ?Tail

```

by (rule image-subset)
 qed
 also have ... =
 card (fri-final-value-agreement-set table final) * card ?Tail
 by (simp add: card-cartesian-product)
 also have card ?Tail = card query-sample-space $^{\wedge}(\text{rounds} - 1)$
 by (rule card-lists-length-eq[OF finite-query-sample-space])
 finally show ?thesis .
 qed

lemma fri-zero-round-first-query-index-lists-card-bound-from-small:

assumes small:

 card (fri-final-value-agreement-set table final) <
 query-sample-space-size

shows card (fri-zero-round-first-query-index-lists table final) $\leq$
 (query-sample-space-size - 1) *
 query-sample-space-size $^{\wedge}(\text{rounds} - 1)$

proof -

have agreement-le:

 card (fri-final-value-agreement-set table final) $\leq$
 query-sample-space-size - 1

using small **by** linarith

have card (fri-zero-round-first-query-index-lists table final) $\leq$
 card (fri-final-value-agreement-set table final) *
 card query-sample-space $^{\wedge}(\text{rounds} - 1)$

by (rule fri-zero-round-first-query-index-lists-card-le)

also have ... $\leq$
 (query-sample-space-size - 1) *
 query-sample-space-size $^{\wedge}(\text{rounds} - 1)$

using agreement-le **by** simp

finally show ?thesis .

qed

definition trace-fri-zero-round-query-list-target-hit

 :: ('f, 'a) protocol-channel-scheme $\Rightarrow$
 (unit list $\times$ ('f, 'a) protocol-channel-scheme) option $\Rightarrow$ bool

where

 trace-fri-zero-round-query-list-target-hit s out $\longleftrightarrow$
 ($\exists$ trace-table trace-final.
 trace-fri-query-index-list-set-hit s
 (fri-zero-round-first-query-index-lists trace-table trace-final)
 out $\wedge$
 card (fri-final-value-agreement-set trace-table trace-final) <
 query-sample-space-size)

lemma trace-fri-zero-round-query-list-target-hitE:

assumes trace-fri-zero-round-query-list-target-hit s out

obtains trace-table trace-final **where**

 trace-fri-query-index-list-set-hit s

```

    (fri-zero-round-first-query-index-lists trace-table trace-final)
  out
  card (fri-final-value-agreement-set trace-table trace-final) <
    query-sample-space-size
using assms
unfolding trace-fri-zero-round-query-list-target-hit-def by blast

lemma trace-fri-zero-round-checked-first-query-target-hit-imp-query-list-target:
  assumes trace-fri-zero-round-checked-first-query-target-hit s out
  shows trace-fri-zero-round-query-list-target-hit s out
proof –
  from assms obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr trace-openings trace-table
  where fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
  and first-hit:
    fri-query-idxs ! 0 ∈
    fri-final-value-agreement-set trace-table trace-final
  and card-lt:
    card (fri-final-value-agreement-set trace-table trace-final) <
      query-sample-space-size
  unfolding trace-fri-zero-round-checked-first-query-target-hit-def
  by blast
  have query-space:
    fri-query-idxs ∈ fri-query-index-list-space
  by (rule accepted-fri-opening-transcript-query-index-list-space
    [OF fri-openings])
  have query-target:
    fri-query-idxs ∈
    fri-zero-round-first-query-index-lists trace-table trace-final
  using query-space first-hit
  unfolding fri-zero-round-first-query-index-lists-def by simp
  have hit:
    trace-fri-query-index-list-set-hit s
    (fri-zero-round-first-query-index-lists trace-table trace-final) out
  unfolding trace-fri-query-index-list-set-hit-def
  using fri-openings query-target by blast
  show ?thesis
  unfolding trace-fri-zero-round-query-list-target-hit-def
  using hit card-lt by blast
qed

lemma trace-fri-zero-round-candidate-first-query-target-hit-imp-query-list-target-or-alignment-gap:
  assumes trace-fri-zero-round-candidate-first-query-target-hit s out
  shows
    trace-fri-zero-round-query-list-target-hit s out ∨

```

```

    trace-fri-zero-round-candidate-first-query-alignment-gap s out
proof (rule trace-fri-zero-round-candidate-first-query-target-hitE[OF assms])
fix trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
assume evidence:
    trace-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
and rounds-empty: length trace-bs = 0
and not-final:
    ¬ fri-final-constant-consistent trace-table trace-final
and candidate-first-hit:
    candidate-query-idxs ! 0 ∈
    fri-final-value-agreement-set trace-table trace-final
and card-lt:
    card (fri-final-value-agreement-set trace-table trace-final) <
    query-sample-space-size
have fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
by (rule trace-fri-partial-candidate-opening-evidenceD(1)
    [OF evidence])
show ?thesis
proof (cases fri-query-idxs ! 0 ∈
    fri-final-value-agreement-set trace-table trace-final)
case True
have query-space:
    fri-query-idxs ∈ fri-query-index-list-space
by (rule accepted-fri-opening-transcript-query-index-list-space
    [OF fri-openings])
have query-target:
    fri-query-idxs ∈
    fri-zero-round-first-query-index-lists trace-table trace-final
using query-space True
unfolding fri-zero-round-first-query-index-lists-def by simp
have hit:
    trace-fri-query-index-list-set-hit s
    (fri-zero-round-first-query-index-lists trace-table trace-final)
    out
unfolding trace-fri-query-index-list-set-hit-def
using fri-openings query-target by blast
have trace-fri-zero-round-query-list-target-hit s out
unfolding trace-fri-zero-round-query-list-target-hit-def
using hit card-lt by blast

```

```

    then show ?thesis by blast
next
case False
have gap:
  trace-fri-zero-round-candidate-first-query-alignment-gap s out
  unfolding trace-fri-zero-round-candidate-first-query-alignment-gap-def
  by (intro exI conjI)
    (rule evidence, rule rounds-empty, rule not-final,
     rule candidate-first-hit, rule False, rule card-lt)
then show ?thesis by blast
qed
qed

lemma trace-fri-zero-round-final-obstruction-imp-query-list-target-or-candidate-gaps:
  assumes trace-fri-zero-round-final-obstruction s out
  shows
    trace-fri-zero-round-query-list-target-hit s out  $\vee$ 
    trace-fri-zero-round-candidate-first-query-value-gap s out  $\vee$ 
    trace-fri-zero-round-candidate-first-query-alignment-gap s out
proof (rule trace-fri-zero-round-final-obstruction-agreement-set-card-lt
  [OF assms])
  fix trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
  assume evidence:
    trace-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr candidate-query-idxs trace-openings
    trace-table
  and rounds-empty: length trace-bs = 0
  and not-final:
     $\neg$  fri-final-constant-consistent trace-table trace-final
  and card-lt:
    card (fri-final-value-agreement-set trace-table trace-final) <
      query-sample-space-size
  show ?thesis
proof (cases candidate-query-idxs ! 0  $\in$ 
  fri-final-value-agreement-set trace-table trace-final)
case True
have candidate-hit:
  trace-fri-zero-round-candidate-first-query-target-hit s out
  unfolding trace-fri-zero-round-candidate-first-query-target-hit-def
  by (intro exI conjI)
    (rule evidence, rule rounds-empty, rule not-final, rule True,
     rule card-lt)
then show ?thesis
  using

```

```

      trace-fri-zero-round-candidate-first-query-target-hit-imp-query-list-target-or-alignment-gap
    by blast
  next
  case False
  have value-gap:
    trace-fri-zero-round-candidate-first-query-value-gap s out
  unfolding trace-fri-zero-round-candidate-first-query-value-gap-def
  by (intro exI conjI)
    (rule evidence, rule rounds-empty, rule not-final, rule False,
     rule card-lt)
  then show ?thesis by blast
qed
qed

lemma wp-trace-fri-zero-round-final-obstruction-bound-from-query-list-target-and-candidate-gaps:
  assumes target-bound:
    wp-event verify-monad
      (trace-fri-zero-round-query-list-target-hit s) s ≤ Q
  and value-bound:
    wp-event verify-monad
      (trace-fri-zero-round-candidate-first-query-value-gap s) s ≤ V
  and alignment-bound:
    wp-event verify-monad
      (trace-fri-zero-round-candidate-first-query-alignment-gap s) s ≤ A
  shows
    wp-event verify-monad
      (trace-fri-zero-round-final-obstruction s) s ≤ Q + V + A
proof -
  have event-le:
    wp-event verify-monad
      (trace-fri-zero-round-final-obstruction s) s ≤
    wp-event verify-monad
      (λout.
        trace-fri-zero-round-query-list-target-hit s out ∨
        trace-fri-zero-round-candidate-first-query-value-gap s out ∨
        trace-fri-zero-round-candidate-first-query-alignment-gap s out) s
  by (rule wp-event-mono)
    (rule trace-fri-zero-round-final-obstruction-imp-query-list-target-or-candidate-gaps)
  also have ... ≤
    wp-event verify-monad
      (trace-fri-zero-round-query-list-target-hit s) s +
    wp-event verify-monad
      (λout.
        trace-fri-zero-round-candidate-first-query-value-gap s out ∨
        trace-fri-zero-round-candidate-first-query-alignment-gap s out) s
  by (rule wp-event-union-bound)
  also have ... ≤
    wp-event verify-monad
      (trace-fri-zero-round-query-list-target-hit s) s +

```

```

    (wp-event verify-monad
      (trace-fri-zero-round-candidate-first-query-value-gap s) s +
      wp-event verify-monad
        (trace-fri-zero-round-candidate-first-query-alignment-gap s) s)
  by (intro add-mono order.refl wp-event-union-bound)
also have ... ≤ Q + (V + A)
  by (intro add-mono target-bound value-bound alignment-bound)
also have ... = Q + V + A
  by (simp add: add.assoc)
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-zero-round-query-list-target-exact-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs.
          trace-fri-query-index-list-set-hit s
            (fri-zero-round-first-query-index-lists table final)))
    adversary-initial-state ≤
      nnreal
        (card (fri-zero-round-first-query-index-lists table final)) *
        (1 / nnreal (card query-sample-space)) ^ rounds +
      hash-relation-budget-value
        (query-index-row-list-relation-fiber-bound
          (fri-zero-round-first-query-index-lists table final))
        (staged-attacker-query-budget budgets +
          staged-challenge-query-budget)
  by (rule checked-staged-security-trace-fri-query-index-list-set-hit-exact-product-bound
    [OF wf controlled fri-zero-round-first-query-index-lists-subset])

lemma checked-staged-security-trace-fri-zero-round-query-list-target-small-bound:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
    and small:
      card (fri-final-value-agreement-set table final) <
        query-sample-space-size
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs.
          trace-fri-query-index-list-set-hit s
            (fri-zero-round-first-query-index-lists table final)))
    adversary-initial-state ≤
      nnreal
        ((query-sample-space-size - 1) *
          query-sample-space-size ^ (rounds - 1)) *

```



```

      (1 / nnreal (card query-sample-space)) ^ rounds +
      hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound
       (fri-zero-round-first-query-index-lists table final))
      (staged-attacker-query-budget budgets +
       staged-challenge-query-budget)
proof -
have exact:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     (λs.
      trace-fri-query-index-list-set-hit s
      (fri-zero-round-first-query-index-lists table final)))
  adversary-initial-state ≤
  nnreal
    (card (fri-zero-round-first-query-index-lists table final)) *
    (1 / nnreal (card query-sample-space)) ^ rounds +
    hash-relation-budget-value
    (query-index-raw-list-relation-fiber-bound
     (fri-zero-round-first-query-index-lists table final))
    (staged-attacker-query-budget budgets +
     staged-challenge-query-budget)
by (rule
      checked-staged-security-trace-fri-zero-round-query-list-target-exact-bound
      [OF wf controlled])
have card-le:
  card (fri-zero-round-first-query-index-lists table final) ≤
    (query-sample-space-size - 1) *
    query-sample-space-size ^ (rounds - 1)
by (rule fri-zero-round-first-query-index-lists-card-bound-from-small
      [OF small])
have product-le:
  nnreal
    (card (fri-zero-round-first-query-index-lists table final)) *
    (1 / nnreal (card query-sample-space)) ^ rounds ≤
  nnreal
    ((query-sample-space-size - 1) *
     query-sample-space-size ^ (rounds - 1)) *
    (1 / nnreal (card query-sample-space)) ^ rounds
proof -
have card-nn:
  nnreal (card (fri-zero-round-first-query-index-lists table final)) ≤
  nnreal
    ((query-sample-space-size - 1) *
     query-sample-space-size ^ (rounds - 1))
using of-nat-mono[OF card-le, where 'a = nnreal] by simp
show ?thesis
by (rule mult-right-mono[OF card-nn]) simp
qed

```

```

show ?thesis
  by (rule order-trans[OF exact add-mono[OF product-le order-refl]])
qed

end

end

```

```

theory Soundness-FRI-Zero-Round-Agreement-Set-Bounds
  imports Soundness-FRI-Zero-Round-Query-List-Bounds
begin

```

Global agreement-set bound for the zero-round trace FRI target.

The verifier-local event *trace-fri-zero-round-query-list-target-hit* is existential in a candidate table and final value. The query-list target depends on these witnesses only through the agreement set inside *query-sample-space*. This layer replaces the arbitrary-list existential by a finite union over agreement sets, preserving the exact query-list product accounting for each fixed set.

```

context soundness
begin

```

```

definition fri-zero-round-first-query-index-lists-for-set
  :: nat set  $\Rightarrow$  nat list set

```

where

```

  fri-zero-round-first-query-index-lists-for-set S =
    {query-idxs  $\in$  fri-query-index-list-space. query-idxs ! 0  $\in$  S}

```

```

definition fri-zero-round-small-agreement-sets :: nat set set

```

where

```

  fri-zero-round-small-agreement-sets =
    {S. S  $\subseteq$  query-sample-space  $\wedge$  card S < query-sample-space-size}

```

```

definition trace-fri-zero-round-agreement-set-target-error-for

```

where

```

  trace-fri-zero-round-agreement-set-target-error-for budgets =
    ( $\sum$  S  $\in$  fri-zero-round-small-agreement-sets.
      nnreal (card (fri-zero-round-first-query-index-lists-for-set S)) *
      (1 / nnreal (card query-sample-space)) ^ rounds +
      hash-relation-budget-value
      (query-index-raw-list-relation-fiber-bound
        (fri-zero-round-first-query-index-lists-for-set S))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget))

```

```

lemma finite-fri-zero-round-small-agreement-sets[simp]:

```

```

  finite fri-zero-round-small-agreement-sets

```

proof –

have *fri-zero-round-small-agreement-sets* $\subseteq$ *Pow query-sample-space*
unfolding *fri-zero-round-small-agreement-sets-def* **by** *blast*
then show *?thesis*
by (rule *finite-subset*) *simp*
qed

lemma *fri-zero-round-first-query-index-lists-for-set-subset*:
fri-zero-round-first-query-index-lists-for-set S
 $\subseteq$ *fri-query-index-list-space*
unfolding *fri-zero-round-first-query-index-lists-for-set-def* **by** *blast*

lemma *trace-fri-zero-round-table-target-eq-agreement-set-target*:
fri-zero-round-first-query-index-lists table final =
fri-zero-round-first-query-index-lists-for-set
(fri-final-value-agreement-set table final)
unfolding *fri-zero-round-first-query-index-lists-def*
fri-zero-round-first-query-index-lists-for-set-def
by *simp*

lemma *checked-staged-security-trace-fri-zero-round-agreement-set-target-bound*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
and *S-in: S $\in$ fri-zero-round-small-agreement-sets*
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
($\lambda s.$
trace-fri-query-index-list-set-hit s
(fri-zero-round-first-query-index-lists-for-set S)))
adversary-initial-state $\leq$
*nnreal (card (fri-zero-round-first-query-index-lists-for-set S)) **
(1 / nnreal (card query-sample-space)) ^ rounds +
hash-relation-budget-value
(query-index-raw-list-relation-fiber-bound
(fri-zero-round-first-query-index-lists-for-set S))
(staged-attacker-query-budget budgets +
staged-challenge-query-budget)
by (rule *checked-staged-security-trace-fri-query-index-list-set-hit-exact-product-bound*
 $[OF\ wf\ controlled\ fri-zero-round-first-query-index-lists-for-set-subset]$)

lemma *checked-staged-security-trace-fri-zero-round-query-list-target-bound-from-agreement-sets*:
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-zero-round-query-list-target-hit)
adversary-initial-state $\leq$
trace-fri-zero-round-agreement-set-target-error-for budgets

```

proof –
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?A = fri-zero-round-small-agreement-sets
  let ?P =
     $\lambda S.$  staged-security-with-data-state-verifier-event
      ( $\lambda s.$ 
        trace-fri-query-index-list-set-hit s
        (fri-zero-round-first-query-index-lists-for-set S))
  have event-le:
    wp-event ?M
      (staged-security-with-data-state-verifier-event
        trace-fri-zero-round-query-list-target-hit)
    adversary-initial-state  $\leq$ 
    wp-event ?M ( $\lambda out. \exists S \in ?A. ?P S out$ )
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume
    out  $\in$  set-dist (execute ?M adversary-initial-state)
  and hit:
    staged-security-with-data-state-verifier-event
      trace-fri-zero-round-query-list-target-hit out
  show  $\exists S \in ?A. ?P S out$ 
  proof (cases out)
  case None
  then show ?thesis
    using hit unfolding staged-security-with-data-state-verifier-event-def
      trace-fri-zero-round-query-list-target-hit-def
    by simp
  next
  case (Some out-data)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
  by (cases out-data, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have local-hit:
    trace-fri-zero-round-query-list-target-hit ?s
    (Some (result, final-state))
  using hit unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
  then obtain trace-table trace-final where query-hit:
    trace-fri-query-index-list-set-hit ?s
    (fri-zero-round-first-query-index-lists trace-table trace-final)
    (Some (result, final-state))
  and small:
    card (fri-final-value-agreement-set trace-table trace-final)
    < query-sample-space-size

```

```

    by (rule trace-fri-zero-round-query-list-target-hitE)
  let ?S = fri-final-value-agreement-set trace-table trace-final
  have S-in: ?S ∈ ?A
    unfolding fri-zero-round-small-agreement-sets-def
    using fri-final-value-agreement-set-subset small by simp
  have query-hit':
    trace-fri-query-index-list-set-hit ?s
      (fri-zero-round-first-query-index-lists-for-set ?S)
      (Some (result, final-state))
    using query-hit
    by (simp add:
        trace-fri-zero-round-table-target-eq-agreement-set-target)
  have ?P ?S out
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    using query-hit' by simp
  show ?thesis
    using S-in ⟨?P ?S out⟩ by blast
qed
qed
have union-bound:
  wp-event ?M (λout. ∃ S ∈ ?A. ?P S out)
  adversary-initial-state ≤
  (∑ S ∈ ?A.
    nnreal (card (fri-zero-round-first-query-index-lists-for-set S)) *
      (1 / nnreal (card query-sample-space)) ^ rounds +
    hash-relation-budget-value
      (query-index-row-list-relation-fiber-bound
        (fri-zero-round-first-query-index-lists-for-set S))
      (staged-attacker-query-budget budgets +
        staged-challenge-query-budget))
proof (rule wp-event-finite-UN-bound)
  show finite ?A
    by simp
next
fix S
assume S-in: S ∈ ?A
show wp-event ?M (?P S) adversary-initial-state ≤
  nnreal (card (fri-zero-round-first-query-index-lists-for-set S)) *
    (1 / nnreal (card query-sample-space)) ^ rounds +
  hash-relation-budget-value
    (query-index-row-list-relation-fiber-bound
      (fri-zero-round-first-query-index-lists-for-set S))
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  by (rule checked-staged-security-trace-fri-zero-round-agreement-set-target-bound
      [OF wf controlled S-in])
qed
show ?thesis
  by (rule order-trans[OF event-le])

```

```

      (use union-bound in
        ⟨simp add: trace-fri-zero-round-agreement-set-target-error-for-def⟩)
qed
end
end

```

```

theory Soundness-FRI-Trace-Empty-Verifier-Tied
imports
  Soundness-FRI-Trace-Untied-Cases
  Soundness-FRI-Zero-Round-Query-List-Bounds
begin

```

Verifier-tied normalization for the empty-composition-header trace FRI branch.

The broad empty-header branch can carry a non-low candidate built from existential partial openings. This layer separates that branch into the verifier-checked zero-round obstruction, Merkle inconsistency, or the existing candidate-transfer residual. It deliberately avoids complete-table reconstruction from sampled openings.

```

context soundness
begin

```

lemma *trace-zero-round-candidate-low-degree-if-final-consistent:*

```

assumes rounds-empty: length trace-bs = 0
and fri-openings:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final dg composition-roots composition-bs composition-final
  fri-query-idxs trace-round-layers composition-round-layers
and candidate:
  partial-trace-table-candidate trace-table trace-openings
and final:
  fri-final-constant-consistent trace-table trace-final
shows trace-table-low-degree trace-table

```

proof –

```

have len-trace-bs: length trace-bs = ceil-log clength
using accepted-fri-opening-transcript-shapes(1,2)[OF fri-openings]
by simp
have ceil-zero: ceil-log clength = 0
using rounds-empty len-trace-bs by simp
have clength-le-one: clength ≤ 1
using ceil-zero unfolding ceil-log-def
by (cases clength ≤ 1) simp-all
have clength-eq: clength = 1
using clength-le-one clength-pos by simp
have len-table: length trace-table = length eval-domain
using candidate eval-domain-length
unfolding partial-trace-table-candidate-def
by simp

```

```

have table-eq:
  trace-table = map (poly [:trace-final:]) eval-domain
proof (rule nth-equalityI)
  show length trace-table =
    length (map (poly [:trace-final:]) eval-domain)
    using len-table by simp
next
  fix i
  assume i-bound:  $i < \text{length trace-table}$ 
  have trace-table ! i = trace-final
    by (rule fri-final-constant-consistent-nth[OF final i-bound])
  then show trace-table ! i =
    map (poly [:trace-final:]) eval-domain ! i
    using i-bound len-table by simp
qed
show ?thesis
  unfolding trace-table-low-degree-def
  by (intro exI[of - [:trace-final:]] conjI)
    (use clength-eq table-eq in simp-all)
qed

lemma trace-fri-zero-round-final-obstruction-imp-checked-merkle-transfer-or-header:
  assumes support:  $\text{out} \in \text{set-dist} (\text{execute verify-monad } s)$ 
  and obstruction: trace-fri-zero-round-final-obstruction s out
  shows
    trace-fri-zero-round-checked-final-obstruction s out  $\vee$ 
    partial-merkle-inconsistency-bad s out  $\vee$ 
    trace-fri-header-tied-candidate-transfer-gap s out  $\vee$ 
    trace-fri-bad-with-header-tied-partial-candidate s out
proof -
  from trace-fri-zero-round-final-obstructionE[OF obstruction]
  obtain trace-roots trace-bs trace-final dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr0 candidate-query-idxs trace-openings0
    trace-table0 where evidence:
    trace-fri-partial-candidate-opening-evidence s out trace-roots
    trace-bs trace-final dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr0 candidate-query-idxs trace-openings0
    trace-table0
  and rounds-empty:  $\text{length trace-bs} = 0$ 
  and not-final0:
     $\neg \text{fri-final-constant-consistent trace-table0 trace-final}$ 
  by blast
  have fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
    using trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence] .

```

```

from fri-openings obtain result final-state where out-eq:
  out = Some (result, final-state)
  unfolding accepted-fri-opening-transcript-def by blast
from accepted-fri-opening-transcript-headerE[OF fri-openings]
obtain fr as rest where header:
  verifier-header-transcript s fr trace-roots trace-final as dg
  composition-roots composition-final rest
by blast
show ?thesis
proof (cases partial-merkle-inconsistency-bad s out)
  case True
  then show ?thesis by simp
next
  case no-merkle: False
  from accepted-fri-opening-transcript-trace-openings-for-query-indices
    [OF fri-openings header out-eq]
  obtain trace-openings where partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  by blast
  obtain trace-table where candidate:
    partial-trace-table-candidate trace-table trace-openings
  using accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
    [OF partial[unfolded out-eq] no-merkle[unfolded out-eq]]
  by blast
  show ?thesis
  proof (cases trace-table-low-degree trace-table)
    case True
    then show ?thesis
    proof (cases trace-fri-bad-with-header-tied-partial-candidate s out)
      case True
      then show ?thesis by simp
    next
    case no-header: False
    have reachable:
      trace-fri-bad-with-reachable-partial-candidate s out
    using trace-fri-zero-round-final-obstruction-imp-header-tied-or-gap
      [OF obstruction] no-header
    unfolding trace-fri-reachable-header-tie-gap-def by simp
    have gap: trace-fri-reachable-header-tie-gap s out
    unfolding trace-fri-reachable-header-tie-gap-def
    using reachable no-header by simp
    have transfer:
      trace-fri-header-tied-candidate-transfer-gap s out
    by (rule trace-fri-header-candidate-low-imp-transfer-gap
      [OF gap fri-openings header partial candidate True])
    then show ?thesis by simp
  qed
next

```



```

case not-low: False
have not-final:
   $\neg$  fri-final-constant-consistent trace-table trace-final
proof
  assume final:
    fri-final-constant-consistent trace-table trace-final
  have low:
    trace-table-low-degree trace-table
  by (rule
    trace-zero-round-candidate-low-degree-if-final-consistent
    [OF rounds-empty fri-openings candidate final])
  then show False
    using not-low by contradiction
qed
have header-zero:
  trace-fri-header-tied-zero-round-final-obstruction s out
  unfolding trace-fri-header-tied-zero-round-final-obstruction-def
  by (intro exI conjI)
    (rule fri-openings, rule header, rule partial, rule candidate,
    rule not-low, rule rounds-empty, rule not-final)
  then show ?thesis
    using
      trace-fri-header-tied-zero-round-final-obstruction-imp-checked-or-merkle
    by blast
qed
qed
qed

lemma wp-trace-fri-zero-round-checked-final-obstruction-bound-from-query-list-target:
  assumes target-bound:
    wp-event verify-monad
    (trace-fri-zero-round-query-list-target-hit s)  $s \leq Q$ 
  shows
    wp-event verify-monad
    (trace-fri-zero-round-checked-final-obstruction s)  $s \leq Q$ 
proof –
  have wp-event verify-monad
    (trace-fri-zero-round-checked-final-obstruction s)  $s \leq$ 
    wp-event verify-monad
    (trace-fri-zero-round-checked-first-query-target-hit s)  $s$ 
  by (rule wp-event-mono)
    (rule trace-fri-zero-round-checked-final-obstruction-imp-first-query-target-hit)
  also have ...  $\leq$ 
    wp-event verify-monad
    (trace-fri-zero-round-query-list-target-hit s)  $s$ 
  by (rule wp-event-mono)
    (rule trace-fri-zero-round-checked-first-query-target-hit-imp-query-list-target)
  also have ...  $\leq Q$ 
  by (rule target-bound)

```

finally show ?thesis .
qed

lemma *wp-trace-fri-zero-round-final-obstruction-bound-from-checked-merkle-transfer-header:*

fixes $H\ M\ Q\ T :: \text{prob}$

assumes *checked-bound:*

wp-event verify-monad

(trace-fri-zero-round-checked-final-obstruction s) s ≤ Q

and *merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad s) s ≤ M

and *transfer-bound:*

wp-event verify-monad

(trace-fri-header-tied-candidate-transfer-gap s) s ≤ T

and *header-bound:*

wp-event verify-monad

(trace-fri-bad-with-header-tied-partial-candidate s) s ≤ H

shows

wp-event verify-monad

(trace-fri-zero-round-final-obstruction s) s ≤ Q + M + T + H

proof –

have *wp-event verify-monad*

(trace-fri-zero-round-final-obstruction s) s ≤

wp-event verify-monad

(λout.

trace-fri-zero-round-checked-final-obstruction s out ∨

partial-merkle-inconsistency-bad s out ∨

trace-fri-header-tied-candidate-transfer-gap s out ∨

trace-fri-bad-with-header-tied-partial-candidate s out) s

by *(rule wp-event-mono-on-support)*

(rule trace-fri-zero-round-final-obstruction-imp-checked-merkle-transfer-or-header)

also have $\dots \leq$

wp-event verify-monad

(trace-fri-zero-round-checked-final-obstruction s) s +

wp-event verify-monad (partial-merkle-inconsistency-bad s) s +

wp-event verify-monad

(trace-fri-header-tied-candidate-transfer-gap s) s +

wp-event verify-monad

(trace-fri-bad-with-header-tied-partial-candidate s) s

by *(rule wp-event-union-bound4)*

also have $\dots \leq Q + M + T + H$

by *(intro add-mono checked-bound merkle-bound transfer-bound header-bound)*

finally show ?thesis .

qed

lemma *trace-fri-header-tied-candidate-transfer-gap-imp-header-or-untied:*

assumes *transfer: trace-fri-header-tied-candidate-transfer-gap s out*

shows

trace-fri-bad-with-header-tied-partial-candidate s out ∨

```

    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
proof (cases trace-fri-bad-with-header-tied-partial-candidate s out)
  case True
  then show ?thesis by simp
next
  case no-header: False
  from transfer obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr as rest trace-openings
    trace-table where
  raw: trace-fri-bad-with-reachable-partial-candidate s out
  and fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs
    trace-final fri-dg composition-roots composition-bs
    composition-final fri-query-idxs trace-round-layers
    composition-round-layers
  and header:
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest
  and partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
  and candidate:
    partial-trace-table-candidate trace-table trace-openings
  and low: trace-table-low-degree trace-table
  unfolding trace-fri-header-tied-candidate-transfer-gap-def by blast
  have gap: trace-fri-reachable-header-tie-gap s out
  unfolding trace-fri-reachable-header-tie-gap-def
  using raw no-header by simp
  have low-gap: trace-fri-header-tied-low-candidate-gap s out
  unfolding trace-fri-header-tied-low-candidate-gap-def
  by (intro exI conjI)
    (rule gap, rule fri-openings, rule header, rule partial, rule candidate,
     rule low)
  have untied:
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
  by (rule trace-fri-header-tied-low-candidate-gap-imp-untied-reachable-witness
    [OF low-gap])
  then show ?thesis by simp
qed

```

```

lemma wp-trace-fri-header-tied-candidate-transfer-gap-bound-from-header-and-untied:
  fixes H U :: prob
  assumes header-bound:
    wp-event verify-monad
    (trace-fri-bad-with-header-tied-partial-candidate s) s ≤ H
  and untied-bound:
    wp-event verify-monad
    (trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)

```

```

       $s \leq U$ 
shows
  wp-event verify-monad
    (trace-fri-header-tied-candidate-transfer-gap s)  $s \leq H + U$ 
proof –
  have wp-event verify-monad
    (trace-fri-header-tied-candidate-transfer-gap s)  $s \leq$ 
wp-event verify-monad
    ( $\lambda out.$ 
      trace-fri-bad-with-header-tied-partial-candidate s out  $\vee$ 
      trace-fri-header-tied-low-candidate-untied-reachable-witness-gap
         $s out$ )  $s$ 
  by (rule wp-event-mono)
    (rule trace-fri-header-tied-candidate-transfer-gap-imp-header-or-untied)
also have ...  $\leq$ 
  wp-event verify-monad
    (trace-fri-bad-with-header-tied-partial-candidate s)  $s +$ 
wp-event verify-monad
    (trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s)
     $s$ 
  by (rule wp-event-union-bound)
also have ...  $\leq H + U$ 
  by (rule add-mono[OF header-bound untied-bound])
finally show ?thesis .
qed

end

end

```

```

theory Soundness-Public-Route-Concrete-FRI-Sharpened
imports
  Soundness-Public-Route-Concrete-FRI-Endpoint
  Soundness-FRI-Query-Fiber-Staged-Bounds
  Soundness-FRI-Prequery-Challenge-Bounds
  Soundness-FRI-Sampled-Challenge-Cover
  Soundness-FRI-Staged-Replay-Bounds
  Soundness-FRI-Zero-Round-Agreement-Set-Bounds
  Soundness-FRI-Trace-Empty-Verifier-Tied
begin

```

First-stage API for sharpening the concrete FRI endpoint.

The public endpoint already removed the FRI soundness assumptions by using explicit staged verifier-event error definitions. This theory factors those definitions into named challenge, full-cover/domain, conflict, and zero-round components. At this stage the components are definitionally equal to the endpoint terms; later proof layers can replace individual components by sharper product/fiber or residual sums without touching the public theorem interface.

context *soundness*
begin

definition *trace-fri-challenge-prequery-error-for*
where

trace-fri-challenge-prequery-error-for budgets $A =$
 (*wp-event* (*checked-staged-security-experiment-with-data-state* A)
 (*staged-security-with-data-state-verifier-event*
 ($\lambda s.$
 trace-fri-challenge-list-fresh-hit s
 (*fri-challenge-space* (*ceil-log* *clength*))))
adversary-initial-state +
hash-relation-budget-value
 (*card* (*fri-challenge-space* (*ceil-log* *clength*))) * *ceil-log* *clength*)
 (*staged-attacker-query-budget* budgets +
 staged-challenge-query-budget))

definition *trace-fri-full-cover-domain-residual-error-for*
where

trace-fri-full-cover-domain-residual-error-for $A =$
 (*reachable-verifier-event-bound-for* A
 ($\lambda s.$
 trace-fri-sampled-full-cover-proximity-failure s
 trace-fri-full-challenge-bad) +
reachable-verifier-event-bound-for A
trace-fri-sampled-full-cover-index-failure +
 (*reachable-verifier-event-bound-for* A
 trace-fri-sampled-full-cover-index-failure +
 reachable-verifier-event-bound-for A
 trace-fri-sampled-full-cover-domain-length-failure) +
 (*reachable-verifier-event-bound-for* A
 trace-fri-sampled-full-cover-index-failure +
 reachable-verifier-event-bound-for A
 trace-fri-sampled-full-cover-sampled-domain-failure))

definition *sharpened-trace-fri-full-cover-domain-error-for*
where

sharpened-trace-fri-full-cover-domain-error-for budgets $A =$
trace-fri-challenge-prequery-error-for budgets A +
trace-fri-full-cover-domain-residual-error-for A

definition *trace-fri-header-conflict-zero-error-for*
where

trace-fri-header-conflict-zero-error-for $A =$
 (*reachable-verifier-event-bound-for* A
 trace-fri-header-tied-sampled-assignment-conflict +
reachable-verifier-event-bound-for A
 trace-fri-header-tied-zero-round-final-obstruction)

definition *trace-fri-header-replay-conflict-error-for*

where

trace-fri-header-replay-conflict-error-for $A =$
(let $M = \text{reachable-verifier-event-bound-for } A$
 $\text{partial-merkle-inconsistency-bad};$
 $H = \text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-recorded-sibling-candidate-missing}$
in $H + (((M + M) + (M + 0) + (M + 0) + (M + 0 + 0)) + M) + 0))$

definition *trace-fri-controlled-merkle-error-for*

where

trace-fri-controlled-merkle-error-for $\text{budgets} =$
 $\text{hash-collision-budget-value } 0$
($\text{staged-attacker-query-budget } \text{budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}$)

definition *trace-fri-header-replay-conflict-zero-error-for*

where

trace-fri-header-replay-conflict-zero-error-for $\text{budgets } A =$
 $\text{trace-fri-header-replay-conflict-error-for } A +$
($\text{trace-fri-zero-round-agreement-set-target-error-for } \text{budgets} +$
 $\text{trace-fri-controlled-merkle-error-for } \text{budgets}$)

definition *sharpened-trace-fri-header-error-for*

where

sharpened-trace-fri-header-error-for $\text{budgets } A =$
 $\text{sharpened-trace-fri-full-cover-domain-error-for } \text{budgets } A +$
 $\text{trace-fri-header-conflict-zero-error-for } A$

definition *trace-fri-empty-conflict-zero-error-for*

where

trace-fri-empty-conflict-zero-error-for $A =$
($\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-sampled-assignment-conflict} +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-zero-round-final-obstruction}$)

definition *trace-fri-empty-small-conflict-zero-error-for*

where

trace-fri-empty-small-conflict-zero-error-for $\text{budgets } A =$
($\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-sampled-assignment-conflict} +$
 $\text{trace-fri-zero-round-agreement-set-target-error-for } \text{budgets} +$
 $\text{trace-fri-controlled-merkle-error-for } \text{budgets} +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-bad-with-header-tied-partial-candidate} +$
($\text{trace-fri-controlled-merkle-error-for } \text{budgets} +$
 ($\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-low-candidate-untied-root-gap} +$

$\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-low-candidate-untied-query-gap})) +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$

definition *sharpened-trace-fri-empty-error-for*

where

$\text{sharpened-trace-fri-empty-error-for budgets } A =$
 $\text{sharpened-trace-fri-full-cover-domain-error-for budgets } A +$
 $\text{trace-fri-empty-conflict-zero-error-for } A$

definition *composition-fri-challenge-prequery-error-for*

where

$\text{composition-fri-challenge-prequery-error-for budgets } A =$
 $(\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s.$
 $\text{composition-fri-challenge-list-fresh-hit } s$
 $(\lambda dg. \text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))))))$
 $\text{adversary-initial-state} +$
 $\text{hash-relation-budget-value}$
 $(\sum dg \in (\text{UNIV} :: 'f \text{ set}).$
 $\text{card } (\text{fri-challenge-space } (\text{ceil-log } (\text{to-nat } dg + 1))) *$
 $\text{ceil-log } (\text{maxDegree} + 1))$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget}))$

definition *composition-fri-full-cover-domain-residual-error-for*

where

$\text{composition-fri-full-cover-domain-residual-error-for } A =$
 $(\text{reachable-verifier-event-bound-for } A$
 $(\lambda s.$
 $\text{composition-fri-sampled-full-cover-proximity-failure } s$
 $\text{composition-fri-full-challenge-bad}) +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{composition-fri-sampled-full-cover-index-failure} +$
 $(\text{reachable-verifier-event-bound-for } A$
 $\text{composition-fri-sampled-full-cover-index-failure} +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{composition-fri-sampled-full-cover-domain-length-failure}) +$
 $(\text{reachable-verifier-event-bound-for } A$
 $\text{composition-fri-sampled-full-cover-index-failure} +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{composition-fri-sampled-full-cover-sampled-domain-failure}))$

definition *sharpened-composition-fri-full-cover-domain-error-for*

where

$\text{sharpened-composition-fri-full-cover-domain-error-for budgets } A =$
 $\text{composition-fri-challenge-prequery-error-for budgets } A +$

composition-fri-full-cover-domain-residual-error-for A

definition *composition-fri-conflict-zero-error-for*

where

composition-fri-conflict-zero-error-for A =
(reachable-verifier-event-bound-for A
composition-fri-verifier-tied-sampled-assignment-conflict +
reachable-verifier-event-bound-for A
composition-fri-verifier-tied-zero-round-final-obstruction)

definition *sharpened-composition-fri-error-for*

where

sharpened-composition-fri-error-for budgets A =
sharpened-composition-fri-full-cover-domain-error-for budgets A +
composition-fri-conflict-zero-error-for A

definition *composition-one-step-query-fraction-for*

:: ('f ⇒ nat ⇒ 'f list ⇒ 'f set) ⇒ prob

where

composition-one-step-query-fraction-for comp-bad =
nnreal
((∑ dg ∈ (UNIV :: 'f set).
∑ challenges ∈
fri-challenge-space (fri-round-count-for-degree-bound (to-nat dg)) -
generic-fri-bad-challenge-lists
(fri-round-count-for-degree-bound (to-nat dg))
(comp-bad dg).
card
(generic-fri-sampled-query-query-fiber
(composition-table-low-degree (to-nat dg))
(Not ∘ composition-table-low-degree maxDegree)
*(to-nat dg) challenges)) * rounds) /*
nnreal (card query-sample-space)

definition *composition-one-step-challenge-hash-count :: nat*

where

composition-one-step-challenge-hash-count =
(∑ dg ∈ (UNIV :: 'f set).
fri-one-step-disagreement-round-mass
(fri-round-count-for-degree-bound (to-nat dg))
*(composition-fri-canonical-layer-len dg) **
ceil-log (maxDegree + 1))

definition *composition-one-step-sampled-error-for*

where

composition-one-step-sampled-error-for budgets comp-bad =
concrete-composition-fri-bad-challenge-fraction +
hash-relation-budget-value
composition-one-step-challenge-hash-count

$$\begin{aligned} & (\text{staged-attacker-query-budget budgets} + \\ & \quad \text{staged-challenge-query-budget}) + \\ \text{nnreal } & (\text{staged-attacker-query-budget budgets} + \\ & \quad \text{staged-challenge-query-budget}) * \\ & \text{composition-one-step-query-fraction-for comp-bad} \end{aligned}$$

definition *composition-one-step-reachable-merkle-error-for*

where

$$\begin{aligned} \text{composition-one-step-reachable-merkle-error-for } A = \\ \text{reachable-verifier-event-bound-for } A \text{ partial-merkle-inconsistency-bad} \end{aligned}$$

definition *composition-one-step-controlled-merkle-error-for*

where

$$\begin{aligned} \text{composition-one-step-controlled-merkle-error-for budgets} = \\ \text{hash-collision-budget-value } 0 \\ (\text{staged-attacker-query-budget budgets} + \\ \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) \end{aligned}$$

definition *composition-one-step-verifier-tied-error-for*

where

$$\begin{aligned} \text{composition-one-step-verifier-tied-error-for budgets } A \text{ comp-bad} = \\ \text{composition-one-step-sampled-error-for budgets comp-bad} + \\ (((\text{composition-one-step-reachable-merkle-error-for } A + 0) + \\ (\text{composition-one-step-reachable-merkle-error-for } A + 0) + \\ (\text{composition-one-step-reachable-merkle-error-for } A + 0) + \\ (\text{composition-one-step-reachable-merkle-error-for } A + 0 + 0) + \\ \text{composition-one-step-reachable-merkle-error-for } A) + 0 + 0) \end{aligned}$$

definition *composition-one-step-verifier-tied-controlled-error-for*

where

$$\begin{aligned} \text{composition-one-step-verifier-tied-controlled-error-for budgets comp-bad} = \\ \text{composition-one-step-sampled-error-for budgets comp-bad} + \\ (((\text{composition-one-step-controlled-merkle-error-for budgets} + 0) + \\ (\text{composition-one-step-controlled-merkle-error-for budgets} + 0) + \\ (\text{composition-one-step-controlled-merkle-error-for budgets} + 0) + \\ (\text{composition-one-step-controlled-merkle-error-for budgets} + 0 + 0) + \\ \text{composition-one-step-controlled-merkle-error-for budgets}) + 0 + 0) \end{aligned}$$

definition *trace-one-step-query-fraction-for*

$:: (\text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow \text{prob}$

where

$$\begin{aligned} \text{trace-one-step-query-fraction-for empty-bad} = \\ \text{nnreal} \\ ((\sum \text{challenges} \in \\ \text{fri-challenge-space } (\text{fri-round-count-for-degree-bound } (\text{clength} - 1)) - \\ \text{generic-fri-bad-challenge-lists} \\ (\text{fri-round-count-for-degree-bound } (\text{clength} - 1)) \text{ empty-bad}. \\ \text{card} \\ (\text{generic-fri-sampled-query-query-fiber trace-table-low-degree} \end{aligned}$$

$$\frac{(Not \circ trace-table-low-degree) (clength - 1) challenges)}{rounds} /$$

$$nnreal (card \ query-sample-space)$$

definition *trace-one-step-challenge-hash-count* :: nat

where

$$trace-one-step-challenge-hash-count =$$

$$fri-one-step-disagreement-round-mass$$

$$(fri-round-count-for-degree-bound (clength - 1))$$

$$trace-fri-canonical-layer-len *$$

$$ceil-log \ clength$$

definition *trace-one-step-disagreement-bad-for*

$$:: (nat \Rightarrow nat) \Rightarrow$$

$$(nat \Rightarrow 'f \ list \Rightarrow 'f \ list) \Rightarrow$$

$$(nat \Rightarrow 'f \ list \Rightarrow 'f \ list) \Rightarrow$$

$$(nat \Rightarrow 'f \ list \Rightarrow 'f \ list) \Rightarrow$$

$$nat \Rightarrow 'f \ list \Rightarrow 'f \ set$$

where

$$trace-one-step-disagreement-bad-for \ pw \ layer \ claimed \ doms \ i \ prefix =$$

$$fri-one-step-disagreement-challenges$$

$$(trace-fri-canonical-layer-len \ i) (pw \ i)$$

$$(layer \ i \ prefix) (claimed \ i \ prefix) (doms \ i \ prefix)$$

definition *composition-one-step-disagreement-bad-for*

$$:: ('f \Rightarrow nat \Rightarrow nat) \Rightarrow$$

$$('f \Rightarrow nat \Rightarrow 'f \ list \Rightarrow 'f \ list) \Rightarrow$$

$$('f \Rightarrow nat \Rightarrow 'f \ list \Rightarrow 'f \ list) \Rightarrow$$

$$('f \Rightarrow nat \Rightarrow 'f \ list \Rightarrow 'f \ list) \Rightarrow$$

$$'f \Rightarrow nat \Rightarrow 'f \ list \Rightarrow 'f \ set$$

where

$$composition-one-step-disagreement-bad-for \ pw \ layer \ claimed \ doms \ dg \ i \ prefix =$$

$$fri-one-step-disagreement-challenges$$

$$(composition-fri-canonical-layer-len \ dg \ i) (pw \ dg \ i)$$

$$(layer \ dg \ i \ prefix) (claimed \ dg \ i \ prefix) (doms \ dg \ i \ prefix)$$

definition *trace-one-step-sampled-error-for*

where

$$trace-one-step-sampled-error-for \ budgets \ empty-bad =$$

$$concrete-trace-fri-bad-challenge-fraction +$$

$$hash-relation-budget-value$$

$$trace-one-step-challenge-hash-count$$

$$(staged-attacker-query-budget \ budgets +$$

$$staged-challenge-query-budget) +$$

$$nnreal (staged-attacker-query-budget \ budgets +$$

$$staged-challenge-query-budget) *$$

$$trace-one-step-query-fraction-for \ empty-bad$$

definition *trace-empty-one-step-error-for*

where

trace-empty-one-step-error-for budgets *A* *empty-bad* =
trace-one-step-sampled-error-for budgets *empty-bad* +
trace-fri-empty-small-conflict-zero-error-for budgets *A*

definition *trace-header-one-step-error-for*

where

trace-header-one-step-error-for budgets *A* *trace-bad* =
trace-one-step-sampled-error-for budgets *trace-bad* +
trace-fri-header-replay-conflict-zero-error-for budgets *A*

definition *trace-one-step-split-pw* :: nat $\Rightarrow$ nat

where

trace-one-step-split-pw - = 0

definition *trace-one-step-split-layer*

:: nat $\Rightarrow$ 'f list $\Rightarrow$ 'f list

where

trace-one-step-split-layer *i* - =
replicate (*trace-fri-canonical-layer-len* *i*) 0

definition *trace-one-step-split-claimed*

:: nat $\Rightarrow$ 'f list $\Rightarrow$ 'f list

where

trace-one-step-split-claimed *i* - =
replicate (*trace-fri-canonical-layer-len* *i*) 1

definition *trace-one-step-split-domain*

:: nat $\Rightarrow$ 'f list $\Rightarrow$ 'f list

where

trace-one-step-split-domain *i* - =
replicate (*trace-fri-canonical-layer-len* *i*) 1

definition *trace-one-step-split-bad*

:: nat $\Rightarrow$ 'f list $\Rightarrow$ 'f set

where

trace-one-step-split-bad =
trace-one-step-disagreement-bad-for
trace-one-step-split-pw *trace-one-step-split-layer*
trace-one-step-split-claimed *trace-one-step-split-domain*

definition *composition-one-step-split-pw*

:: 'f $\Rightarrow$ nat $\Rightarrow$ nat

where

composition-one-step-split-pw - - = 0

definition *composition-one-step-split-layer*

:: 'f $\Rightarrow$ nat $\Rightarrow$ 'f list $\Rightarrow$ 'f list

where

composition-one-step-split-layer dg i - =
replicate (composition-fri-canonical-layer-len dg i) 0

definition *composition-one-step-split-claimed*
 $:: 'f \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list}$

where

composition-one-step-split-claimed dg i - =
replicate (composition-fri-canonical-layer-len dg i) 1

definition *composition-one-step-split-domain*
 $:: 'f \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ list}$

where

composition-one-step-split-domain dg i - =
replicate (composition-fri-canonical-layer-len dg i) 1

definition *composition-one-step-split-bad*
 $:: 'f \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}$

where

composition-one-step-split-bad =
composition-one-step-disagreement-bad-for
composition-one-step-split-pw composition-one-step-split-layer
composition-one-step-split-claimed composition-one-step-split-domain

definition *concrete-trace-header-one-step-error-for*

where

concrete-trace-header-one-step-error-for budgets A =
trace-header-one-step-error-for budgets A
trace-one-step-split-bad

definition *concrete-composition-one-step-error-for*

where

concrete-composition-one-step-error-for budgets A =
composition-one-step-verifier-tied-controlled-error-for budgets
composition-one-step-split-bad

definition *concrete-trace-empty-one-step-error-for*

where

concrete-trace-empty-one-step-error-for budgets A =
trace-empty-one-step-error-for budgets A
trace-one-step-split-bad

lemma *sharpened-trace-fri-full-cover-domain-error-for-eq:*

sharpened-trace-fri-full-cover-domain-error-for budgets A =
concrete-trace-fri-full-cover-domain-error-for budgets A

by (*simp add:*

sharpened-trace-fri-full-cover-domain-error-for-def
trace-fri-challenge-prequery-error-for-def
trace-fri-full-cover-domain-residual-error-for-def
concrete-trace-fri-full-cover-domain-error-for-def)

lemma *sharpened-trace-fri-header-error-for-eq:*
sharpened-trace-fri-header-error-for budgets $A =$
concrete-trace-fri-header-error-for' budgets A

by (*simp add:*
sharpened-trace-fri-header-error-for-def
sharpened-trace-fri-full-cover-domain-error-for-eq
trace-fri-header-conflict-zero-error-for-def
concrete-trace-fri-header-error-for'-def)

lemma *sharpened-trace-fri-empty-error-for-eq:*
sharpened-trace-fri-empty-error-for budgets $A =$
concrete-trace-fri-empty-error-for' budgets A

by (*simp add:*
sharpened-trace-fri-empty-error-for-def
sharpened-trace-fri-full-cover-domain-error-for-eq
trace-fri-empty-conflict-zero-error-for-def
concrete-trace-fri-empty-error-for'-def)

lemma *sharpened-composition-fri-full-cover-domain-error-for-eq:*
sharpened-composition-fri-full-cover-domain-error-for budgets $A =$
concrete-composition-fri-full-cover-domain-error-for budgets A

by (*simp add:*
sharpened-composition-fri-full-cover-domain-error-for-def
composition-fri-challenge-prequery-error-for-def
composition-fri-full-cover-domain-residual-error-for-def
concrete-composition-fri-full-cover-domain-error-for-def)

lemma *sharpened-composition-fri-error-for-eq:*
sharpened-composition-fri-error-for budgets $A =$
concrete-composition-fri-error-for' budgets A

by (*simp add:*
sharpened-composition-fri-error-for-def
sharpened-composition-fri-full-cover-domain-error-for-eq
composition-fri-conflict-zero-error-for-def
concrete-composition-fri-error-for'-def)

lemma *sharpened-trace-fri-header-error-for-bound:*

assumes *wf: staged-budget-wellformed* budgets

and *controlled: staged-adversary-controlled* budgets A

shows

wp-event (*checked-staged-security-experiment-with-data-state* A)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-header-tied-partial-candidate)

adversary-initial-state $\leq$ *sharpened-trace-fri-header-error-for* budgets A

using *concrete-trace-fri-header-error-for'-bound*[*OF wf controlled*]

by (*simp add: sharpened-trace-fri-header-error-for-eq*)

lemma *sharpened-trace-fri-empty-error-for-bound:*

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state $\leq$ *sharpened-trace-fri-empty-error-for budgets A*
using *concrete-trace-fri-empty-error-for'-bound*[*OF wf controlled*]
by (*simp add: sharpened-trace-fri-empty-error-for-eq*)

lemma *sharpened-composition-fri-error-for-bound*:
assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state $\leq$ *sharpened-composition-fri-error-for budgets A*
using *concrete-composition-fri-error-for'-bound*[*OF wf controlled*]
by (*simp add: sharpened-composition-fri-error-for-eq*)

lemma *composition-one-step-sampled-error-for-bound*:
fixes *comp-bad* :: '*f* $\Rightarrow$ nat $\Rightarrow$ '*f* list $\Rightarrow$ '*f* set
assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets *A*
and *comp-subset*:
 $\bigwedge dg\ i\ prefix.$
 $i < \text{fri-round-count-for-degree-bound } (to\text{-nat } dg) \implies$
 $\text{length } prefix = i \implies$
 $\text{comp-bad } dg\ i\ prefix \subseteq$
fri-one-step-disagreement-challenges
(*composition-fri-canonical-layer-len dg i*) (*comp-pw dg i*)
(*comp-layer dg i prefix*) (*comp-claimed dg i prefix*)
(*comp-dom dg i prefix*)

shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
composition-fri-sampled-query-bad-candidate)
adversary-initial-state
 $\leq$ *composition-one-step-sampled-error-for budgets comp-bad*

proof –
let *?B* =
 $\lambda dg.$ *generic-fri-bad-challenge-lists*
(*fri-round-count-for-degree-bound (to-nat dg)*) (*comp-bad dg*)
let *?Q* = *composition-one-step-query-fraction-for comp-bad*
have *round-bound*:
 $\bigwedge dg\ i\ prefix.$
 $i < \text{fri-round-count-for-degree-bound } (to\text{-nat } dg) \implies$
 $\text{length } prefix = i \implies$

```

    card (comp-bad dg i prefix)
      ≤ composition-fri-canonical-layer-len dg i div 2
proof –
  fix dg i and prefix :: 'f list
  assume i: i < fri-round-count-for-degree-bound (to-nat dg)
    and prefix-len: length prefix = i
  have card (comp-bad dg i prefix) ≤
    card
      (fri-one-step-disagreement-challenges
        (composition-fri-canonical-layer-len dg i) (comp-pw dg i)
        (comp-layer dg i prefix) (comp-claimed dg i prefix)
        (comp-dom dg i prefix))
    by (rule card-mono) (simp-all add: comp-subset[OF i prefix-len])
  also have ... ≤ composition-fri-canonical-layer-len dg i div 2
    by (rule fri-one-step-disagreement-challenges-card-bound-local)
  finally show card (comp-bad dg i prefix)
    ≤ composition-fri-canonical-layer-len dg i div 2 .
qed
have fraction-bound:
   $\bigwedge dg. \text{nnreal}$ 
     $(\sum i \in \{.. < \text{fri-round-count-for-degree-bound (to-nat dg)}\}. \text{CARD('f)} ^ i * \text{composition-fri-canonical-layer-len dg i div 2}) * \text{CARD('f)} ^ (\text{fri-round-count-for-degree-bound (to-nat dg)} - \text{Suc } i)) / \text{nnreal (CARD('f)} ^ \text{fri-round-count-for-degree-bound (to-nat dg)})$ 
    ≤ concrete-composition-fri-bad-challenge-fraction
proof –
  fix dg
  show nnreal
     $(\sum i \in \{.. < \text{fri-round-count-for-degree-bound (to-nat dg)}\}. \text{CARD('f)} ^ i * \text{composition-fri-canonical-layer-len dg i div 2}) * \text{CARD('f)} ^ (\text{fri-round-count-for-degree-bound (to-nat dg)} - \text{Suc } i)) / \text{nnreal (CARD('f)} ^ \text{fri-round-count-for-degree-bound (to-nat dg)})$ 
    ≤ concrete-composition-fri-bad-challenge-fraction
  using concrete-composition-fri-bad-challenge-fraction-ge[of dg]
  by (simp add: fri-one-step-disagreement-round-fraction-def
    fri-bad-challenge-round-fraction-def
    fri-one-step-disagreement-round-mass-def
    fri-bad-challenge-round-mass-def)
qed
have fresh-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-challenge-list-fresh-hit s ?B))
  adversary-initial-state
    ≤ concrete-composition-fri-bad-challenge-fraction

```

by (rule
checked-staged-security-composition-fri-bad-challenge-list-fresh-hit-bound-from-round-bounds
 [OF round-bound fraction-bound])
have fiber-fraction:
nnreal
 (($\sum_{dg \in UNIV} \sum_{challenges \in$
fri-challenge-space (*fri-round-count-for-degree-bound* (*to-nat dg*)) –
 ?B *dg*.
card
 (*generic-fri-sampled-query-query-fiber*
 (*composition-table-low-degree* (*to-nat dg*))
 (*Not* $\circ$ *composition-table-low-degree* *maxDegree*)
 (*to-nat dg* *challenges*)) *
rounds) /
nnreal (*card query-sample-space*) $\leq$?Q
by (*simp add: composition-one-step-query-fraction-for-def*)
have sampled:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
composition-fri-sampled-query-bad-candidate)
adversary-initial-state
 $\leq$
concrete-composition-fri-bad-challenge-fraction +
hash-relation-budget-value
 ($\sum_{dg \in UNIV} \text{card } (?B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1)$)
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget) +
nnreal
 (*staged-attacker-query-budget budgets* +
staged-challenge-query-budget) * ?Q
by (rule
checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-ou
 [OF *wf controlled finite-generic-fri-bad-challenge-lists*
fresh-bound fiber-fraction])
have card-le:
 ($\sum_{dg \in UNIV} \text{card } (?B \text{ dg}) * \text{ceil-log } (\text{maxDegree} + 1)$)
 $\leq$ *composition-one-step-challenge-hash-count*
proof (*unfold composition-one-step-challenge-hash-count-def, rule sum-mono*)
fix *dg* :: 'f
have raw:
card (?B *dg*) $\leq$
 ($\sum_{i \in \{..<\text{fri-round-count-for-degree-bound } (\text{to-nat dg})\}} \text{CARD}('f) \wedge i * \text{composition-fri-canonical-layer-len } dg \text{ } i \text{ div } 2$) *
 $\text{CARD}('f) \wedge (\text{fri-round-count-for-degree-bound } (\text{to-nat dg}) - \text{Suc } i)$)
proof (rule *generic-fri-bad-challenge-lists-card-bound-from-round-bounds*)
fix *i* :: nat **and** *prefix* :: 'f list


```

assume  $i: i < \text{fri-round-count-for-degree-bound } (\text{to-nat } dg)$ 
and  $\text{prefix-len}: \text{length prefix} = i$ 
show  $\text{card } (\text{comp-bad } dg \ i \ \text{prefix})$ 
 $\leq \text{composition-fri-canonical-layer-len } dg \ i \ \text{div } 2$ 
by  $(\text{rule round-bound}[OF \ i \ \text{prefix-len}])$ 
qed
have  $\text{card } (?B \ dg) \leq$ 
 $\text{fri-one-step-disagreement-round-mass}$ 
 $(\text{fri-round-count-for-degree-bound } (\text{to-nat } dg))$ 
 $(\text{composition-fri-canonical-layer-len } dg)$ 
using  $\text{raw}$ 
by  $(\text{simp add: fri-one-step-disagreement-round-mass-def}$ 
 $\text{fri-bad-challenge-round-mass-def})$ 
then show  $\text{card } (?B \ dg) * \text{ceil-log } (\text{maxDegree} + 1)$ 
 $\leq \text{fri-one-step-disagreement-round-mass}$ 
 $(\text{fri-round-count-for-degree-bound } (\text{to-nat } dg))$ 
 $(\text{composition-fri-canonical-layer-len } dg) *$ 
 $\text{ceil-log } (\text{maxDegree} + 1)$ 
by  $(\text{rule mult-right-mono}) \text{ simp}$ 
qed
have  $\text{budget-le:}$ 
 $\text{hash-relation-budget-value}$ 
 $(\sum dg \in UNIV. \text{card } (?B \ dg) * \text{ceil-log } (\text{maxDegree} + 1))$ 
 $(\text{staged-attacker-query-budget budgets} +$ 
 $\text{staged-challenge-query-budget})$ 
 $\leq$ 
 $\text{hash-relation-budget-value}$ 
 $\text{composition-one-step-challenge-hash-count}$ 
 $(\text{staged-attacker-query-budget budgets} +$ 
 $\text{staged-challenge-query-budget})$ 
by  $(\text{rule hash-relation-budget-value-mono-left}[OF \ \text{card-le}])$ 
show  $?thesis$ 
by  $(\text{rule order-trans}[OF \ \text{sampled}])$ 
 $(\text{use budget-le in}$ 
 $\langle \text{simp add: composition-one-step-sampled-error-for-def} \rangle)$ 
qed

lemma  $\text{trace-one-step-sampled-error-for-bound:}$ 
fixes  $\text{empty-bad} :: \text{nat} \Rightarrow 'f \ \text{list} \Rightarrow 'f \ \text{set}$ 
assumes  $\text{wf: staged-budget-wellformed budgets}$ 
and  $\text{controlled: staged-adversary-controlled budgets } A$ 
and  $\text{empty-subset:}$ 
 $\bigwedge i \ \text{prefix.}$ 
 $i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \implies$ 
 $\text{length prefix} = i \implies$ 
 $\text{empty-bad } i \ \text{prefix} \subseteq$ 
 $\text{fri-one-step-disagreement-challenges}$ 
 $(\text{trace-fri-canonical-layer-len } i) \ (\text{empty-pw } i)$ 
 $(\text{empty-layer } i \ \text{prefix}) \ (\text{empty-claimed } i \ \text{prefix})$ 

```

```

      (empty-dom i prefix)
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-query-bad-candidate)
    adversary-initial-state
  ≤ trace-one-step-sampled-error-for budgets empty-bad
proof -
  let ?B =
    generic-fri-bad-challenge-lists
    (fri-round-count-for-degree-bound (clength - 1)) empty-bad
  let ?Q = trace-one-step-query-fraction-for empty-bad
  have round-bound:
    ∧ i prefix.
      i < fri-round-count-for-degree-bound (clength - 1) ⇒
        length prefix = i ⇒
          card (empty-bad i prefix) ≤ trace-fri-canonical-layer-len i div 2
  proof -
    fix i :: nat and prefix :: 'f list
    assume i: i < fri-round-count-for-degree-bound (clength - 1)
      and prefix-len: length prefix = i
    have card (empty-bad i prefix) ≤
      card
        (fri-one-step-disagreement-challenges
          (trace-fri-canonical-layer-len i) (empty-pw i)
          (empty-layer i prefix) (empty-claimed i prefix)
          (empty-dom i prefix))
      by (rule card-mono) (simp-all add: empty-subset[OF i prefix-len])
    also have ... ≤ trace-fri-canonical-layer-len i div 2
      by (rule fri-one-step-disagreement-challenges-card-bound-local)
    finally show card (empty-bad i prefix)
      ≤ trace-fri-canonical-layer-len i div 2 .
  qed
  have fresh-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. trace-fri-challenge-list-fresh-hit s ?B))
      adversary-initial-state
    ≤ concrete-trace-fri-bad-challenge-fraction
  proof -
    have base:
      wp-event (checked-staged-security-experiment-with-data-state A)
        (staged-security-with-data-state-verifier-event
          (λs. trace-fri-challenge-list-fresh-hit s ?B))
        adversary-initial-state
      ≤ nnreal
        (∑ i ∈ {..

```

```

      (fri-round-count-for-degree-bound (clength - 1) - Suc i)) /
    nnreal (CARD('f) ^ fri-round-count-for-degree-bound (clength - 1))
  by (rule
      checked-staged-security-trace-fri-bad-challenge-list-fresh-hit-bound-from-round-bounds
      [OF round-bound])
  show ?thesis
  by (rule order-trans[OF base])
    (simp add: concrete-trace-fri-bad-challenge-fraction-def
      fri-one-step-disagreement-round-fraction-def
      fri-bad-challenge-round-fraction-def
      fri-one-step-disagreement-round-mass-def
      fri-bad-challenge-round-mass-def)
qed
have fiber-fraction:
  nnreal
    (( $\sum$  challenges $\in$ 
      fri-challenge-space (fri-round-count-for-degree-bound (clength - 1)) -
      ?B.
      card
        (generic-fri-sampled-query-query-fiber trace-table-low-degree
          (Not  $\circ$  trace-table-low-degree) (clength - 1) challenges)) *
      rounds) /
    nnreal (card query-sample-space)  $\leq$  ?Q
  by (simp add: trace-one-step-query-fraction-for-def)
have sampled:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-query-bad-candidate)
  adversary-initial-state
   $\leq$ 
  concrete-trace-fri-bad-challenge-fraction +
  hash-relation-budget-value (card ?B * ceil-log clength)
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget) +
  nnreal
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget) * ?Q
  by (rule
      checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-generic-challenge-cover-and-outside-fil
      [OF wf controlled finite-generic-fri-bad-challenge-lists
        fresh-bound fiber-fraction])
have card-le:
  card ?B * ceil-log clength  $\leq$  trace-one-step-challenge-hash-count
proof -
  have raw:
    card ?B  $\leq$ 
    ( $\sum i \in \{..< \text{fri-round-count-for-degree-bound (clength - 1)}\}$ .
      CARD('f) ^ i * (trace-fri-canonical-layer-len i div 2) *
      CARD('f) ^

```

```

      (fri-round-count-for-degree-bound (clength - 1) - Suc i))
proof (rule generic-fri-bad-challenge-lists-card-bound-from-round-bounds)
  fix i :: nat and prefix :: 'f list
  assume i: i < fri-round-count-for-degree-bound (clength - 1)
    and prefix-len: length prefix = i
  show card (empty-bad i prefix)
    ≤ trace-fri-canonical-layer-len i div 2
    by (rule round-bound[OF i prefix-len])
qed
have card ?B ≤
  fri-one-step-disagreement-round-mass
    (fri-round-count-for-degree-bound (clength - 1))
    trace-fri-canonical-layer-len
  using raw
  by (simp add: fri-one-step-disagreement-round-mass-def
    fri-bad-challenge-round-mass-def)
then show ?thesis
  unfolding trace-one-step-challenge-hash-count-def
  by (rule mult-right-mono) simp
qed
have budget-le:
  hash-relation-budget-value (card ?B * ceil-log clength)
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
    ≤
  hash-relation-budget-value trace-one-step-challenge-hash-count
    (staged-attacker-query-budget budgets +
      staged-challenge-query-budget)
  by (rule hash-relation-budget-value-mono-left[OF card-le])
show ?thesis
  by (rule order-trans[OF sampled])
    (use budget-le in ⟨simp add: trace-one-step-sampled-error-for-def⟩)
qed

lemma checked-staged-security-trace-fri-empty-header-bound-from-staged-sampled-conflict-and-zero:
assumes sampled-query:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-query-bad-candidate)
  adversary-initial-state ≤ R
and conflict-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-assignment-conflict)
  adversary-initial-state ≤ C
and zero-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-zero-round-final-obstruction)

```

$adversary-initial-state \leq Z$
shows
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $\quad trace-fri-bad-with-empty-composition-header-candidates)$
 $adversary-initial-state \leq R + (C + Z)$
proof –
let $?M = checked-staged-security-experiment-with-data-state \ A$
let $?sampled =$
 $\quad staged-security-with-data-state-verifier-event$
 $\quad \quad trace-fri-bad-with-sampled-layer-chain$
let $?missing =$
 $\quad staged-security-with-data-state-verifier-event$
 $\quad \quad trace-fri-reachable-missing-sampled-layer-chain$
let $?conflict =$
 $\quad staged-security-with-data-state-verifier-event$
 $\quad \quad trace-fri-sampled-assignment-conflict$
let $?zero =$
 $\quad staged-security-with-data-state-verifier-event$
 $\quad \quad trace-fri-zero-round-final-obstruction$
have $sampled:$
 $wp-event \ ?M \ ?sampled \ adversary-initial-state \leq R$
by (rule
 $\quad checked-staged-security-trace-fri-sampled-layer-chain-bound-from-sampled-query$
 $\quad [OF \ sampled-query])$
have $missing-split:$
 $wp-event \ ?M \ ?missing \ adversary-initial-state \leq$
 $\quad wp-event \ ?M \ (\lambda out. \ ?conflict \ out \vee \ ?zero \ out)$
 $\quad \quad adversary-initial-state$
proof (rule $wp-event-mono-on-support$)
fix out
assume $support:$
 $\quad out \in set-dist \ (execute \ ?M \ adversary-initial-state)$
and $missing: \ ?missing \ out$
show $?conflict \ out \vee \ ?zero \ out$
proof (cases out)
case $None$
then show $?thesis$
using $missing \ unfolding \ staged-security-with-data-state-verifier-event-def$
by $simp$
next
case $(Some \ out-data)$
then obtain $data \ attacker-state \ result \ final-state$ **where** $out-eq:$
 $\quad out = Some \ (((data, \ attacker-state), \ result), \ final-state)$
by (cases $out-data$, auto split: $prod.splits$)
let $?s =$
 $\quad verifier-state-from-adversary \ attacker-state$
 $\quad \quad (staged-proof-transcript \ data)$
from $checked-staged-security-experiment-with-data-state-outcomeE$

```

    [OF support[unfolded out-eq]]
have verify-support:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
by blast
have missing-local:
  trace-fri-reachable-missing-sampled-layer-chain ?s
  (Some (result, final-state))
using missing unfolding out-eq
  staged-security-with-data-state-verifier-event-def
by simp
have obstruction:
  trace-fri-sampled-layer-assignment-obstruction ?s
  (Some (result, final-state))
by (rule
    trace-fri-reachable-missing-sampled-imp-assignment-obstruction-on-support
    [OF verify-support missing-local])
then have
  trace-fri-sampled-assignment-conflict ?s
  (Some (result, final-state)) ∨
  trace-fri-zero-round-final-obstruction ?s
  (Some (result, final-state))
by (rule
  trace-fri-sampled-layer-assignment-obstruction-imp-conflict-or-zero-round-final)
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
have missing-bound:
  wp-event ?M ?missing adversary-initial-state ≤ C + Z
proof –
  have wp-event ?M ?missing adversary-initial-state ≤
    wp-event ?M (λout. ?conflict out ∨ ?zero out)
    adversary-initial-state
  by (rule missing-split)
  also have ... ≤
    wp-event ?M ?conflict adversary-initial-state +
    wp-event ?M ?zero adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤ C + Z
  by (rule add-mono[OF conflict-bound zero-bound])
  finally show ?thesis .
qed
have split:
  wp-event ?M
  (staged-security-with-data-state-verifier-event
   trace-fri-bad-with-empty-composition-header-candidates)
  adversary-initial-state ≤
  wp-event ?M (λout. ?sampled out ∨ ?missing out)

```

```

    adversary-initial-state
  by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      dest:
        trace-fri-bad-with-empty-composition-header-candidates-imp-reachable-partial-candidate
        trace-fri-reachable-imp-sampled-or-missing-layer-chain
      split: option.splits prod.splits)
  also have ... ≤
    wp-event ?M ?sampled adversary-initial-state +
    wp-event ?M ?missing adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤ R + (C + Z)
  by (rule add-mono[OF sampled missing-bound])
  finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-header-tied-candidate-transfer-gap-bound-from-staged-header-and-untied:
  fixes H U :: prob
  assumes header-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state ≤ H
  and untied-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-low-candidate-untied-reachable-witness-gap)
    adversary-initial-state ≤ U
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-candidate-transfer-gap)
    adversary-initial-state ≤ H + U
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?header =
    staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-partial-candidate
  let ?untied =
    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap
  have wp-event ?M
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-candidate-transfer-gap)
    adversary-initial-state ≤
    wp-event ?M (λout. ?header out ∨ ?untied out)
    adversary-initial-state
  unfolding staged-security-with-data-state-verifier-event-def
  by (rule wp-event-mono-on-support)

```

```

(auto dest:
  trace-fri-header-tied-candidate-transfer-gap-imp-header-or-untied
  split: option.splits prod.splits)
also have ... ≤
  wp-event ?M ?header adversary-initial-state +
  wp-event ?M ?untied adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤ H + U
by (rule add-mono[OF header-bound untied-bound])
finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-staged-checked-merkle-transfer-head*

fixes $H\ M\ Q\ T :: \text{prob}$

assumes *checked-bound*:

```

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
  trace-fri-zero-round-checked-final-obstruction)
adversary-initial-state ≤ Q

```

and *merkle-bound*:

```

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
  partial-merkle-inconsistency-bad)
adversary-initial-state ≤ M

```

and *transfer-bound*:

```

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
  trace-fri-header-tied-candidate-transfer-gap)
adversary-initial-state ≤ T

```

and *header-bound*:

```

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
  trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state ≤ H

```

shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
  trace-fri-zero-round-final-obstruction)
adversary-initial-state ≤ Q + M + T + H

```

proof –

let $?M = \text{checked-staged-security-experiment-with-data-state } A$

let $?checked =$

```

  staged-security-with-data-state-verifier-event
  trace-fri-zero-round-checked-final-obstruction

```

let $?merkle =$

```

  staged-security-with-data-state-verifier-event
  partial-merkle-inconsistency-bad

```

let $?transfer =$

```

  staged-security-with-data-state-verifier-event

```



```

    trace-fri-header-tied-candidate-transfer-gap
  let ?header =
    staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-partial-candidate
  have wp-event ?M
    (staged-security-with-data-state-verifier-event
     trace-fri-zero-round-final-obstruction)
  adversary-initial-state ≤
  wp-event ?M
    (λout.
     ?checked out ∨ ?merkle out ∨ ?transfer out ∨ ?header out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈ set-dist (execute ?M adversary-initial-state)
  and obstruction:
    staged-security-with-data-state-verifier-event
    trace-fri-zero-round-final-obstruction out
  show ?checked out ∨ ?merkle out ∨ ?transfer out ∨ ?header out
proof (cases out)
  case None
  then show ?thesis
    using obstruction
    unfolding staged-security-with-data-state-verifier-event-def
    by simp
next
  case (Some out-data)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases out-data, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
  have verify-support:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
    by blast
  have obstruction-local:
    trace-fri-zero-round-final-obstruction ?s
    (Some (result, final-state))
    using obstruction
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
  have
    trace-fri-zero-round-checked-final-obstruction ?s
    (Some (result, final-state)) ∨
    partial-merkle-inconsistency-bad ?s

```

```

    (Some (result, final-state)) ∨
    trace-fri-header-tied-candidate-transfer-gap ?s
    (Some (result, final-state)) ∨
    trace-fri-bad-with-header-tied-partial-candidate ?s
    (Some (result, final-state))
  by (rule
    trace-fri-zero-round-final-obstruction-imp-checked-merkle-transfer-or-header
    [OF verify-support obstruction-local])
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
qed
qed
also have ... ≤
  wp-event ?M ?checked adversary-initial-state +
  wp-event ?M ?merkle adversary-initial-state +
  wp-event ?M ?transfer adversary-initial-state +
  wp-event ?M ?header adversary-initial-state
  by (rule wp-event-union-bound4)
also have ... ≤ Q + M + T + H
  by (intro add-mono checked-bound merkle-bound transfer-bound
    header-bound)
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-zero-round-checked-final-obstruction-bound-from-agreement-sets:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-zero-round-checked-final-obstruction)
    adversary-initial-state ≤
    trace-fri-zero-round-agreement-set-target-error-for budgets
proof -
  have event-le:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-zero-round-checked-final-obstruction)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-zero-round-query-list-target-hit)
    adversary-initial-state
  unfolding staged-security-with-data-state-verifier-event-def
  by (rule wp-event-mono-on-support)
  (auto
    intro:
      trace-fri-zero-round-checked-first-query-target-hit-imp-query-list-target

```

```

    dest:
      trace-fri-zero-round-checked-final-obstruction-imp-first-query-target-hit
    split: option.splits prod.splits)
show ?thesis
by (rule order-trans[OF event-le])
  (rule checked-staged-security-trace-fri-zero-round-query-list-target-bound-from-agreement-sets
    [OF wf controlled])
qed

lemma checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-agreement-sets:
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-zero-round-final-obstruction)
    adversary-initial-state ≤
    trace-fri-zero-round-agreement-set-target-error-for budgets +
    trace-fri-controlled-merkle-error-for budgets +
    reachable-verifier-event-bound-for A
      trace-fri-bad-with-header-tied-partial-candidate +
    (trace-fri-controlled-merkle-error-for budgets +
      (reachable-verifier-event-bound-for A
        trace-fri-header-tied-low-candidate-untied-root-gap +
        reachable-verifier-event-bound-for A
          trace-fri-header-tied-low-candidate-untied-query-gap)) +
    reachable-verifier-event-bound-for A
      trace-fri-bad-with-header-tied-partial-candidate
proof -
  have checked:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-zero-round-checked-final-obstruction)
    adversary-initial-state ≤
    trace-fri-zero-round-agreement-set-target-error-for budgets
  by (rule checked-staged-security-trace-fri-zero-round-checked-final-obstruction-bound-from-agreement-sets
    [OF wf controlled])
  have merkle:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        partial-merkle-inconsistency-bad)
    adversary-initial-state ≤
    trace-fri-controlled-merkle-error-for budgets
  unfolding trace-fri-controlled-merkle-error-for-def
  by (rule
    checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
    [OF wf controlled])
  have header:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state ≤
    reachable-verifier-event-bound-for A
    trace-fri-bad-with-header-tied-partial-candidate
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show  $\bigwedge s. \neg \text{trace-fri-bad-with-header-tied-partial-candidate } s \text{ None}$ 
unfolding trace-fri-bad-with-header-tied-partial-candidate-def
    accepted-fri-opening-transcript-def
by simp
next
fix data attacker-state
assume support: Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
     adversary-initial-state)
show wp-event verify-monad
    (trace-fri-bad-with-header-tied-partial-candidate
     (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))
    ≤ reachable-verifier-event-bound-for A
    trace-fri-bad-with-header-tied-partial-candidate
by (rule reachable-verifier-event-bound-for-ge[OF support])
qed
have hash:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     hash-map-output-collision-bad)
    adversary-initial-state ≤
    trace-fri-controlled-merkle-error-for budgets
unfolding trace-fri-controlled-merkle-error-for-def
by (rule
    checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
    [OF wf controlled])
have root:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-header-tied-low-candidate-untied-root-gap)
    adversary-initial-state ≤
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-root-gap
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show  $\bigwedge s. \neg \text{trace-fri-header-tied-low-candidate-untied-root-gap } s \text{ None}$ 
unfolding trace-fri-header-tied-low-candidate-untied-root-gap-def
    trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
    accepted-fri-opening-transcript-def
by simp

```

```

next
  fix data attacker-state
  assume support: Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  show wp-event verify-monad
    (trace-fri-header-tied-low-candidate-untied-root-gap
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    ≤ reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-untied-root-gap
  by (rule reachable-verifier-event-bound-for-ge[OF support])
qed
have query:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-low-candidate-untied-query-gap)
  adversary-initial-state ≤
  reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-query-gap
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\bigwedge s. \neg \text{trace-fri-header-tied-low-candidate-untied-query-gap } s \text{ None}$ 
  unfolding trace-fri-header-tied-low-candidate-untied-query-gap-def
    trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
    accepted-fri-opening-transcript-def
  by simp
next
  fix data attacker-state
  assume support: Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  show wp-event verify-monad
    (trace-fri-header-tied-low-candidate-untied-query-gap
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    ≤ reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-untied-query-gap
  by (rule reachable-verifier-event-bound-for-ge[OF support])
qed
have untied:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-low-candidate-untied-reachable-witness-gap)

```

```

adversary-initial-state ≤
trace-fri-controlled-merkle-error-for budgets +
(reachable-verifier-event-bound-for A
  trace-fri-header-tied-low-candidate-untied-root-gap +
  reachable-verifier-event-bound-for A
  trace-fri-header-tied-low-candidate-untied-query-gap)
proof –
let ?M = checked-staged-security-experiment-with-data-state A
let ?untied =
  staged-security-with-data-state-verifier-event
  trace-fri-header-tied-low-candidate-untied-reachable-witness-gap
let ?hash =
  staged-security-with-data-state-verifier-event
  hash-map-output-collision-bad
let ?root =
  staged-security-with-data-state-verifier-event
  trace-fri-header-tied-low-candidate-untied-root-gap
let ?query =
  staged-security-with-data-state-verifier-event
  trace-fri-header-tied-low-candidate-untied-query-gap
have split:
  wp-event ?M ?untied adversary-initial-state ≤
    wp-event ?M (λout. ?hash out ∨ ?root out ∨ ?query out)
    adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono-on-support)
  (auto
    dest:
      trace-fri-header-tied-low-candidate-untied-gap-imp-hash-root-or-query
    split: option.splits prod.splits)
also have ... ≤
  wp-event ?M ?hash adversary-initial-state +
  wp-event ?M (λout. ?root out ∨ ?query out)
  adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤
  wp-event ?M ?hash adversary-initial-state +
  (wp-event ?M ?root adversary-initial-state +
  wp-event ?M ?query adversary-initial-state)
by (intro add-mono order-refl wp-event-union-bound)
also have ... ≤
  trace-fri-controlled-merkle-error-for budgets +
  (reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-root-gap +
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-query-gap)
by (intro add-mono hash root query)
finally show ?thesis .
qed

```

have *transfer*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-header-tied-candidate-transfer-gap)
adversary-initial-state $\leq$
reachable-verifier-event-bound-for A
trace-fri-bad-with-header-tied-partial-candidate +
 (*trace-fri-controlled-merkle-error-for budgets* +
 (*reachable-verifier-event-bound-for A*
trace-fri-header-tied-low-candidate-untied-root-gap +
reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-untied-query-gap)))
by (*rule*
checked-staged-security-trace-fri-header-tied-candidate-transfer-gap-bound-from-staged-header-and-untied
 [*OF header untied*])

have *raw*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
trace-fri-zero-round-final-obstruction)
adversary-initial-state $\leq$
trace-fri-zero-round-agreement-set-target-error-for budgets +
trace-fri-controlled-merkle-error-for budgets +
 (*reachable-verifier-event-bound-for A*
trace-fri-bad-with-header-tied-partial-candidate +
 (*trace-fri-controlled-merkle-error-for budgets* +
 (*reachable-verifier-event-bound-for A*
trace-fri-header-tied-low-candidate-untied-root-gap +
reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-untied-query-gap)))) +
reachable-verifier-event-bound-for A
trace-fri-bad-with-header-tied-partial-candidate
by (*rule*
checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-staged-checked-merkle-transfer-header
 [*OF checked merkle transfer header*])

show *?thesis*

by (*rule order-trans*[*OF raw*]) (*simp add: algebra-simps*)

qed

lemma *checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-staged-sampled-conflict-and*

assumes *sampled-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-verifier-tied-sampled-layer-chain)
adversary-initial-state $\leq R$

and *conflict-bound*:

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
composition-fri-verifier-tied-sampled-assignment-conflict)
adversary-initial-state $\leq C$

```

and zero-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-verifier-tied-zero-round-final-obstruction)
    adversary-initial-state  $\leq Z$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-verifier-tied-partial-candidate)
    adversary-initial-state  $\leq R + (C + Z)$ 
proof –
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?sampled =
    staged-security-with-data-state-verifier-event
      composition-fri-bad-with-verifier-tied-sampled-layer-chain
  let ?missing =
    staged-security-with-data-state-verifier-event
      composition-fri-verifier-tied-missing-sampled-layer-chain
  let ?conflict =
    staged-security-with-data-state-verifier-event
      composition-fri-verifier-tied-sampled-assignment-conflict
  let ?zero =
    staged-security-with-data-state-verifier-event
      composition-fri-verifier-tied-zero-round-final-obstruction
  have missing-split:
    wp-event ?M ?missing adversary-initial-state  $\leq$ 
      wp-event ?M ( $\lambda$ out. ?conflict out  $\vee$  ?zero out)
      adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support:
      out  $\in$  set-dist (execute ?M adversary-initial-state)
      and missing: ?missing out
    show ?conflict out  $\vee$  ?zero out
    proof (cases out)
      case None
      then show ?thesis
        using missing unfolding staged-security-with-data-state-verifier-event-def
        by simp
    next
      case (Some out-data)
      then obtain data attacker-state result final-state where out-eq:
        out = Some (((data, attacker-state), result), final-state)
      by (cases out-data, auto split: prod.splits)
      let ?s =
        verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)
      from checked-staged-security-experiment-with-data-state-outcomeE
        [OF support[unfolded out-eq]]

```



```

have verify-support:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
have missing-local:
  composition-fri-verifier-tied-missing-sampled-layer-chain ?s
  (Some (result, final-state))
  using missing unfolding out-eq
  staged-security-with-data-state-verifier-event-def
  by simp
have obstruction:
  composition-fri-verifier-tied-sampled-layer-assignment-obstruction
  ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-missing-sampled-imp-assignment-obstruction
    [OF missing-local])
then have
  composition-fri-verifier-tied-sampled-assignment-conflict ?s
  (Some (result, final-state)) ∨
  composition-fri-verifier-tied-zero-round-final-obstruction ?s
  (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-assignment-obstruction-imp-conflict-or-zero-on-support
    [OF verify-support])
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
have missing-bound:
  wp-event ?M ?missing adversary-initial-state ≤ C + Z
proof –
  have wp-event ?M ?missing adversary-initial-state ≤
    wp-event ?M (λout. ?conflict out ∨ ?zero out)
    adversary-initial-state
    by (rule missing-split)
  also have ... ≤
    wp-event ?M ?conflict adversary-initial-state +
    wp-event ?M ?zero adversary-initial-state
    by (rule wp-event-union-bound)
  also have ... ≤ C + Z
    by (rule add-mono[OF conflict-bound zero-bound])
  finally show ?thesis .
qed
have split:
  wp-event ?M
  (staged-security-with-data-state-verifier-event
    composition-fri-bad-with-verifier-tied-partial-candidate)
  adversary-initial-state ≤
  wp-event ?M (λout. ?sampled out ∨ ?missing out)

```

```

    adversary-initial-state
unfolding composition-fri-verifier-tied-missing-sampled-layer-chain-def
by (rule wp-event-mono-on-support)
    (auto simp: staged-security-with-data-state-verifier-event-def
      split: option.splits prod.splits)
also have ...  $\leq$ 
    wp-event ?M ?sampled adversary-initial-state +
    wp-event ?M ?missing adversary-initial-state
by (rule wp-event-union-bound)
also have ...  $\leq R + (C + Z)$ 
by (rule add-mono[OF sampled-bound missing-bound])
finally show ?thesis .
qed

lemma checked-staged-security-partial-merkle-inconsistency-bad-bound-by-composition-one-step-controlled-merkle-err
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      partial-merkle-inconsistency-bad)
    adversary-initial-state  $\leq$ 
    composition-one-step-controlled-merkle-error-for budgets
unfolding composition-one-step-controlled-merkle-error-for-def
by (rule
    checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
    [OF wf controlled])

lemma checked-staged-security-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap)
    adversary-initial-state  $\leq (0::prob)$ 
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show  $\bigwedge s. \neg$ 
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap s None
unfolding
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-def
by simp
next
fix data attacker-state
assume Some (data, attacker-state)  $\in$ 
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
    (composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))

```

```

    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  ≤ 0
  by (rule
      wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
qed

lemma checked-staged-security-composition-fri-verifier-tied-next-value-replay-gap-bound-by-controlled-merkle:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-verifier-tied-next-value-replay-gap)
      adversary-initial-state ≤
      composition-one-step-controlled-merkle-error-for budgets + 0
proof -
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?merkle =
    staged-security-with-data-state-verifier-event
    partial-merkle-inconsistency-bad
  let ?slot =
    staged-security-with-data-state-verifier-event
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
  have split:
    wp-event ?M
      (staged-security-with-data-state-verifier-event
        composition-fri-verifier-tied-next-value-replay-gap)
      adversary-initial-state ≤
      wp-event ?M (λout. ?merkle out ∨ ?slot out)
      adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈ set-dist (execute ?M adversary-initial-state)
  and gap:
    staged-security-with-data-state-verifier-event
    composition-fri-verifier-tied-next-value-replay-gap out
  show ?merkle out ∨ ?slot out
  proof (cases out)
  case None
  then show ?thesis
    using gap unfolding staged-security-with-data-state-verifier-event-def
    by simp
  next
  case (Some out-data)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
  by (cases out-data, auto split: prod.splits)

```

```

let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support[unfolded out-eq]]
have verify-support:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
have gap-local:
  composition-fri-verifier-tied-next-value-replay-gap ?s
  (Some (result, final-state))
using gap unfolding out-eq
  staged-security-with-data-state-verifier-event-def
by simp
have
  partial-merkle-inconsistency-bad ?s
  (Some (result, final-state)) ∨
  composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
  ?s (Some (result, final-state))
by (rule
  composition-fri-verifier-tied-next-value-replay-gap-imp-merkle-or-slot-on-support
  [OF verify-support gap-local])
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
also have ... ≤
  wp-event ?M ?merkle adversary-initial-state +
  wp-event ?M ?slot adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤
  composition-one-step-controlled-merkle-error-for budgets + 0
by (intro add-mono
  checked-staged-security-partial-merkle-inconsistency-bad-bound-by-composition-one-step-controlled-merkle-error
  [OF wf controlled]
  checked-staged-security-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-verifier-tied-successor-replay-gap-bound-by-controlled-merkle:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
  composition-fri-verifier-tied-successor-replay-gap)
  adversary-initial-state ≤
  composition-one-step-controlled-merkle-error-for budgets + 0

```

```

proof –
  let ?M = checked-staged-security-experiment-with-data-state A
  let ?merkle =
    staged-security-with-data-state-verifier-event
    partial-merkle-inconsistency-bad
  let ?slot =
    staged-security-with-data-state-verifier-event
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
  have split:
    wp-event ?M
    (staged-security-with-data-state-verifier-event
     composition-fri-verifier-tied-successor-replay-gap)
    adversary-initial-state ≤
    wp-event ?M ( $\lambda out. ?merkle\ out \vee ?slot\ out$ )
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out ∈ set-dist (execute ?M adversary-initial-state)
  and gap:
    staged-security-with-data-state-verifier-event
    composition-fri-verifier-tied-successor-replay-gap out
  show ?merkle out ∨ ?slot out
  proof (cases out)
  case None
  then show ?thesis
    using gap unfolding staged-security-with-data-state-verifier-event-def
    by simp
  next
  case (Some out-data)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases out-data, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  from checked-staged-security-experiment-with-data-state-outcomeE
    [OF support[unfolded out-eq]]
  have verify-support:
    Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
    by blast
  have gap-local:
    composition-fri-verifier-tied-successor-replay-gap ?s
    (Some (result, final-state))
    using gap unfolding out-eq
    staged-security-with-data-state-verifier-event-def
    by simp
  have
    partial-merkle-inconsistency-bad ?s

```

```

    (Some (result, final-state)) ∨
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-successor-replay-gap-imp-merkle-or-slot-on-support
    [OF verify-support gap-local])
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
qed
qed
also have ... ≤
  wp-event ?M ?merkle adversary-initial-state +
  wp-event ?M ?slot adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤
  composition-one-step-controlled-merkle-error-for budgets + 0
  by (intro add-mono
    checked-staged-security-partial-merkle-inconsistency-bad-bound-by-composition-one-step-controlled-merkle-error
    [OF wf controlled]
    checked-staged-security-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
  finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-verifier-tied-final-value-selected-step-missing-gap-zero:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-fri-verifier-tied-final-value-selected-step-missing-gap)
  adversary-initial-state ≤ (0::prob)
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\bigwedge s. \neg$ 
    composition-fri-verifier-tied-final-value-selected-step-missing-gap s None
  unfolding
    composition-fri-verifier-tied-final-value-selected-step-missing-gap-def
    composition-fri-verifier-tied-final-value-authenticated-replay-gap-def
    accepted-fri-opening-transcript-def
  by simp
next
  fix data attacker-state
  assume Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  show wp-event verify-monad
    (composition-fri-verifier-tied-final-value-selected-step-missing-gap
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))

```

≤ 0
by (rule
wp-composition-fri-verifier-tied-final-value-selected-step-missing-gap-zero)
qed

lemma *checked-staged-security-composition-fri-verifier-tied-final-value-replay-gap-bound-by-controlled-merkle:*
assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-verifier-tied-final-value-replay-gap)
adversary-initial-state $\leq$
composition-one-step-controlled-merkle-error-for budgets + 0 + 0

proof –
let *?M = checked-staged-security-experiment-with-data-state A*
let *?merkle =*
staged-security-with-data-state-verifier-event
partial-merkle-inconsistency-bad
let *?slot =*
staged-security-with-data-state-verifier-event
composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
let *?selected =*
staged-security-with-data-state-verifier-event
composition-fri-verifier-tied-final-value-selected-step-missing-gap
have *split:*
wp-event ?M
(staged-security-with-data-state-verifier-event
composition-fri-verifier-tied-final-value-replay-gap)
adversary-initial-state $\leq$
wp-event ?M
(λ out. ?merkle out $\vee$?slot out $\vee$?selected out)
adversary-initial-state

proof (rule *wp-event-mono-on-support*)
fix *out*
assume *support:*
out $\in$ set-dist (execute ?M adversary-initial-state)
and *gap:*
staged-security-with-data-state-verifier-event
composition-fri-verifier-tied-final-value-replay-gap out
show *?merkle out $\vee$?slot out $\vee$?selected out*
proof (cases *out*)
case *None*
then show *?thesis*
using *gap unfolding staged-security-with-data-state-verifier-event-def*
by *simp*
next
case (*Some out-data*)
then obtain *data attacker-state result final-state* **where** *out-eq:*

```

    out = Some (((data, attacker-state), result), final-state)
  by (cases out-data, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support[unfolded out-eq]]
have verify-support:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
have gap-local:
  composition-fri-verifier-tied-final-value-replay-gap ?s
  (Some (result, final-state))
using gap unfolding out-eq
  staged-security-with-data-state-verifier-event-def
  by simp
have
  partial-merkle-inconsistency-bad ?s
  (Some (result, final-state)) ∨
  composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
  ?s (Some (result, final-state)) ∨
  composition-fri-verifier-tied-final-value-authenticated-replay-gap
  ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-final-value-replay-gap-imp-merkle-or-slot-or-authenticated-on-support
    [OF verify-support gap-local])
then show ?thesis
proof
  assume partial-merkle-inconsistency-bad ?s
  (Some (result, final-state))
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
next
  assume tail:
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
    ?s (Some (result, final-state)) ∨
    composition-fri-verifier-tied-final-value-authenticated-replay-gap
    ?s (Some (result, final-state))
  then show ?thesis
  proof
    assume
      composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
      ?s (Some (result, final-state))
    then show ?thesis
      unfolding out-eq staged-security-with-data-state-verifier-event-def
      by simp
  next
    assume auth:

```



```

    composition-fri-verifier-tied-final-value-authenticated-replay-gap
    ?s (Some (result, final-state))
  have selected:
    composition-fri-verifier-tied-final-value-selected-step-missing-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-final-value-authenticated-replay-gap-imp-selected-step-missing
    [OF auth])
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
qed
qed
qed
qed
also have ... ≤
  wp-event ?M ?merkle adversary-initial-state +
  wp-event ?M ?slot adversary-initial-state +
  wp-event ?M ?selected adversary-initial-state
proof -
  have wp-event ?M
    (λout. ?merkle out ∨ ?slot out ∨ ?selected out)
    adversary-initial-state ≤
    wp-event ?M ?merkle adversary-initial-state +
    wp-event ?M (λout. ?slot out ∨ ?selected out)
    adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤
    wp-event ?M ?merkle adversary-initial-state +
    (wp-event ?M ?slot adversary-initial-state +
    wp-event ?M ?selected adversary-initial-state)
  by (intro add-mono order-refl wp-event-union-bound)
  finally show ?thesis
    by (simp add: add.assoc)
qed
also have ... ≤
  composition-one-step-controlled-merkle-error-for budgets + 0 + 0
  by (intro add-mono
    checked-staged-security-partial-merkle-inconsistency-bad-bound-by-composition-one-step-controlled-merkle-error
    [OF wf controlled]
    checked-staged-security-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero
    checked-staged-security-composition-fri-verifier-tied-final-value-selected-step-missing-gap-zero)
  finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    composition-fri-verifier-tied-recorded-base-chunk-auth-missing)

```

```

    adversary-initial-state ≤ (0::prob)
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\bigwedge s. \neg$ 
    composition-fri-verifier-tied-recorded-base-chunk-auth-missing s None
  unfolding
    composition-fri-verifier-tied-recorded-base-chunk-auth-missing-def
    accepted-fri-opening-transcript-def
  by simp
next
  fix data attacker-state
  assume Some (data, attacker-state) ∈
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  show wp-event verify-monad
    (composition-fri-verifier-tied-recorded-base-chunk-auth-missing
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    ≤ 0
  by (rule
    wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero)
qed

```

lemma checked-staged-security-composition-fri-verifier-tied-sampled-assignment-conflict-bound-by-controlled-merkle:

assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-verifier-tied-sampled-assignment-conflict)
 adversary-initial-state ≤
 (((composition-one-step-controlled-merkle-error-for budgets + 0) +
 (composition-one-step-controlled-merkle-error-for budgets + 0) +
 (composition-one-step-controlled-merkle-error-for budgets + 0) +
 (composition-one-step-controlled-merkle-error-for budgets + 0 + 0) +
 composition-one-step-controlled-merkle-error-for budgets) + 0)

proof –

let ?M = checked-staged-security-experiment-with-data-state A
let ?conflict =
 staged-security-with-data-state-verifier-event
 composition-fri-verifier-tied-sampled-assignment-conflict
let ?merkle =
 staged-security-with-data-state-verifier-event
 partial-merkle-inconsistency-bad
let ?slot =
 staged-security-with-data-state-verifier-event
 composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap

```

let ?next =
  staged-security-with-data-state-verifier-event
  composition-fri-verifier-tied-next-value-replay-gap
let ?successor =
  staged-security-with-data-state-verifier-event
  composition-fri-verifier-tied-successor-replay-gap
let ?final =
  staged-security-with-data-state-verifier-event
  composition-fri-verifier-tied-final-value-replay-gap
have split:
  wp-event ?M ?conflict adversary-initial-state ≤
  wp-event ?M
  (λout.
    ?merkle out ∨ ?slot out ∨ ?next out ∨
    ?successor out ∨ ?final out ∨ ?merkle out ∨ ?slot out)
  adversary-initial-state
proof (rule wp-event-mono-on-support)
fix out
assume support:
  out ∈ set-dist (execute ?M adversary-initial-state)
  and conflict: ?conflict out
show ?merkle out ∨ ?slot out ∨ ?next out ∨
  ?successor out ∨ ?final out ∨ ?merkle out ∨ ?slot out
proof (cases out)
case None
then show ?thesis
  using conflict unfolding staged-security-with-data-state-verifier-event-def
  by simp
next
case (Some out-data)
then obtain data attacker-state result final-state where out-eq:
  out = Some (((data, attacker-state), result), final-state)
  by (cases out-data, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support[unfolded out-eq]]
have verify-support:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
have conflict-local:
  composition-fri-verifier-tied-sampled-assignment-conflict ?s
  (Some (result, final-state))
  using conflict unfolding out-eq
  staged-security-with-data-state-verifier-event-def
  by simp
have local-split:
  partial-merkle-inconsistency-bad ?s

```

(Some (result, final-state)) ∨
 composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
 ?s (Some (result, final-state)) ∨
 composition-fri-verifier-tied-next-value-replay-gap
 ?s (Some (result, final-state)) ∨
 composition-fri-verifier-tied-successor-replay-gap
 ?s (Some (result, final-state)) ∨
 composition-fri-verifier-tied-final-value-replay-gap
 ?s (Some (result, final-state)) ∨
 partial-merkle-inconsistency-bad ?s
 (Some (result, final-state)) ∨
 composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
 ?s (Some (result, final-state))
proof –
have *auth-or-gap*:
 composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
 ?s (Some (result, final-state)) ∨
 composition-fri-verifier-tied-assignment-auth-gap
 ?s (Some (result, final-state))
by (rule
 composition-fri-verifier-tied-assignment-imp-authenticated-or-auth-gap
 [OF conflict-local])
then show ?thesis
proof
assume *auth*:
 composition-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
 ?s (Some (result, final-state))
have *without-or-same*:
 composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
 ?s (Some (result, final-state)) ∨
 composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
 ?s (Some (result, final-state))
by (rule
 composition-fri-verifier-tied-authenticated-conflict-imp-without-same-or-same-layer
 [OF auth])
then show ?thesis
proof
assume *without*:
 composition-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
 ?s (Some (result, final-state))
then have *branches*:
 composition-fri-verifier-tied-sampled-base-opening-conflict
 ?s (Some (result, final-state)) ∨
 composition-fri-verifier-tied-sampled-next-value-conflict
 ?s (Some (result, final-state)) ∨
 composition-fri-verifier-tied-sampled-successor-opening-conflict
 ?s (Some (result, final-state)) ∨
 composition-fri-verifier-tied-sampled-final-value-conflict
 ?s (Some (result, final-state))

```

using composition-fri-verifier-tied-without-same-iff-branches
by blast
then show ?thesis
proof
  assume base:
    composition-fri-verifier-tied-sampled-base-opening-conflict
    ?s (Some (result, final-state))
  have base-split:
    composition-fri-verifier-tied-authenticated-base-opening-conflict
    ?s (Some (result, final-state))  $\vee$ 
    composition-fri-verifier-tied-base-chunk-auth-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-sampled-base-imp-authenticated-or-auth-gap
    [OF base])
  then show ?thesis
proof
  assume auth-base:
    composition-fri-verifier-tied-authenticated-base-opening-conflict
    ?s (Some (result, final-state))
  then have
    partial-merkle-inconsistency-bad ?s
    (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-authenticated-base-conflict-imp-partial-merkle)
  then show ?thesis by simp
next
  assume gap:
    composition-fri-verifier-tied-base-chunk-auth-gap
    ?s (Some (result, final-state))
  then have missing:
    composition-fri-verifier-tied-recorded-base-chunk-auth-missing
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-base-chunk-auth-gap-imp-recorded-missing)
  then have
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-recorded-base-chunk-auth-missing-imp-slot-recorded-chunk-gap)
  then show ?thesis by simp
qed
next
  assume rest:
    composition-fri-verifier-tied-sampled-next-value-conflict
    ?s (Some (result, final-state))  $\vee$ 
    composition-fri-verifier-tied-sampled-successor-opening-conflict
    ?s (Some (result, final-state))  $\vee$ 
    composition-fri-verifier-tied-sampled-final-value-conflict

```

```

      ?s (Some (result, final-state))
then show ?thesis
proof
  assume next-conflict:
    composition-fri-verifier-tied-sampled-next-value-conflict
    ?s (Some (result, final-state))
  then have
    composition-fri-verifier-tied-next-value-replay-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-sampled-next-value-imp-replay-gap)
  then show ?thesis by simp
next
assume tail:
  composition-fri-verifier-tied-sampled-successor-opening-conflict
  ?s (Some (result, final-state)) ∨
  composition-fri-verifier-tied-sampled-final-value-conflict
  ?s (Some (result, final-state))
then show ?thesis
proof
  assume successor:
    composition-fri-verifier-tied-sampled-successor-opening-conflict
    ?s (Some (result, final-state))
  then have
    composition-fri-verifier-tied-successor-replay-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-sampled-successor-imp-replay-gap)
  then show ?thesis by simp
next
assume final:
  composition-fri-verifier-tied-sampled-final-value-conflict
  ?s (Some (result, final-state))
  then have
    composition-fri-verifier-tied-final-value-replay-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-sampled-final-value-imp-replay-gap)
  then show ?thesis by simp
qed
qed
qed
next
assume same:
  composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
  ?s (Some (result, final-state))
then have
  partial-merkle-inconsistency-bad ?s
  (Some (result, final-state))

```

```

    by (rule
      composition-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-imp-partial-merkle)
  then show ?thesis by simp
qed
next
  assume gap:
    composition-fri-verifier-tied-assignment-auth-gap
    ?s (Some (result, final-state))
  have same-gap:
    composition-fri-verifier-tied-same-layer-auth-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-assignment-auth-gap-imp-same-layer-auth-gap
    [OF gap])
  have recorded-gap:
    composition-fri-verifier-tied-same-layer-recorded-auth-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-same-layer-auth-gap-imp-recorded-auth-gap
    [OF same-gap])
  have authenticated-gap:
    composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-same-layer-recorded-auth-gap-imp-authenticated-recorded-auth-gap-on-support
    [OF verify-support recorded-gap])
  have
    composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap
    ?s (Some (result, final-state))
  by (rule
    composition-fri-verifier-tied-same-layer-authenticated-recorded-auth-gap-imp-slot-recorded-chunk-gap
    [OF authenticated-gap])
  then show ?thesis by simp
qed
qed
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
have next-bound:
  wp-event ?M ?next adversary-initial-state ≤
    composition-one-step-controlled-merkle-error-for budgets
  using
    checked-staged-security-composition-fri-verifier-tied-next-value-replay-gap-bound-by-controlled-merkle
    [OF wf controlled]
  by simp
have successor-bound:
  wp-event ?M ?successor adversary-initial-state ≤

```

```

    composition-one-step-controlled-merkle-error-for budgets
using
    checked-staged-security-composition-fri-verifier-tied-successor-replay-gap-bound-by-controlled-merkle
    [OF wf controlled]
by simp
have final-bound:
    wp-event ?M ?final adversary-initial-state ≤
    composition-one-step-controlled-merkle-error-for budgets
using
    checked-staged-security-composition-fri-verifier-tied-final-value-replay-gap-bound-by-controlled-merkle
    [OF wf controlled]
by simp
have split-bound:
    wp-event ?M
    (λout.
      ?merkle out ∨ ?slot out ∨ ?next out ∨
      ?successor out ∨ ?final out ∨ ?merkle out ∨ ?slot out)
    adversary-initial-state ≤
    wp-event ?M ?merkle adversary-initial-state +
    (wp-event ?M ?slot adversary-initial-state +
    (wp-event ?M ?next adversary-initial-state +
    (wp-event ?M ?successor adversary-initial-state +
    (wp-event ?M ?final adversary-initial-state +
    wp-event ?M ?merkle adversary-initial-state))))))
proof –
    let ?nodup =
    λout.
      ?merkle out ∨ ?slot out ∨ ?next out ∨
      ?successor out ∨ ?final out ∨ ?merkle out
    have dedup:
    wp-event ?M
    (λout.
      ?merkle out ∨ ?slot out ∨ ?next out ∨
      ?successor out ∨ ?final out ∨ ?merkle out ∨ ?slot out)
    adversary-initial-state ≤
    wp-event ?M ?nodup adversary-initial-state
    by (rule wp-event-mono) auto
    have union-bound:
    wp-event ?M ?nodup adversary-initial-state ≤
    wp-event ?M ?merkle adversary-initial-state +
    (wp-event ?M ?slot adversary-initial-state +
    (wp-event ?M ?next adversary-initial-state +
    (wp-event ?M ?successor adversary-initial-state +
    (wp-event ?M ?final adversary-initial-state +
    wp-event ?M ?merkle adversary-initial-state))))))
    proof –
    let ?tail-final = λout. ?final out ∨ ?merkle out
    let ?tail-successor = λout. ?successor out ∨ ?tail-final out
    let ?tail-next = λout. ?next out ∨ ?tail-successor out

```



```

let ?tail-slot = λout. ?slot out ∨ ?tail-next out
have final-bound':
  wp-event ?M ?tail-final adversary-initial-state ≤
    wp-event ?M ?final adversary-initial-state +
    wp-event ?M ?merkle adversary-initial-state
  by (rule wp-event-union-bound)
have successor-bound':
  wp-event ?M ?tail-successor adversary-initial-state ≤
    wp-event ?M ?successor adversary-initial-state +
    (wp-event ?M ?final adversary-initial-state +
     wp-event ?M ?merkle adversary-initial-state)
proof –
  have wp-event ?M ?tail-successor adversary-initial-state ≤
    wp-event ?M ?successor adversary-initial-state +
    wp-event ?M ?tail-final adversary-initial-state
    by (rule wp-event-union-bound)
  also have ... ≤
    wp-event ?M ?successor adversary-initial-state +
    (wp-event ?M ?final adversary-initial-state +
     wp-event ?M ?merkle adversary-initial-state)
    by (intro add-mono order-refl final-bound')
  finally show ?thesis .
qed
have next-bound':
  wp-event ?M ?tail-next adversary-initial-state ≤
    wp-event ?M ?next adversary-initial-state +
    (wp-event ?M ?successor adversary-initial-state +
     (wp-event ?M ?final adversary-initial-state +
      wp-event ?M ?merkle adversary-initial-state))
proof –
  have wp-event ?M ?tail-next adversary-initial-state ≤
    wp-event ?M ?next adversary-initial-state +
    wp-event ?M ?tail-successor adversary-initial-state
    by (rule wp-event-union-bound)
  also have ... ≤
    wp-event ?M ?next adversary-initial-state +
    (wp-event ?M ?successor adversary-initial-state +
     (wp-event ?M ?final adversary-initial-state +
      wp-event ?M ?merkle adversary-initial-state))
    by (intro add-mono order-refl successor-bound')
  finally show ?thesis .
qed
have slot-bound':
  wp-event ?M ?tail-slot adversary-initial-state ≤
    wp-event ?M ?slot adversary-initial-state +
    (wp-event ?M ?next adversary-initial-state +
     (wp-event ?M ?successor adversary-initial-state +
      (wp-event ?M ?final adversary-initial-state +
       wp-event ?M ?merkle adversary-initial-state))))

```

```

proof –
  have  $wp\text{-event } ?M \text{ ?tail-slot adversary-initial-state} \leq$ 
     $wp\text{-event } ?M \text{ ?slot adversary-initial-state} +$ 
     $wp\text{-event } ?M \text{ ?tail-next adversary-initial-state}$ 
    by (rule wp-event-union-bound)
  also have  $\dots \leq$ 
     $wp\text{-event } ?M \text{ ?slot adversary-initial-state} +$ 
     $(wp\text{-event } ?M \text{ ?next adversary-initial-state} +$ 
     $(wp\text{-event } ?M \text{ ?successor adversary-initial-state} +$ 
     $(wp\text{-event } ?M \text{ ?final adversary-initial-state} +$ 
     $wp\text{-event } ?M \text{ ?merkle adversary-initial-state})))$ 
    by (intro add-mono order-refl next-bound)
  finally show ?thesis .
qed
have  $wp\text{-event } ?M \text{ ?nodup adversary-initial-state} \leq$ 
   $wp\text{-event } ?M \text{ ?merkle adversary-initial-state} +$ 
   $wp\text{-event } ?M \text{ ?tail-slot adversary-initial-state}$ 
  by (rule wp-event-union-bound)
also have  $\dots \leq$ 
   $wp\text{-event } ?M \text{ ?merkle adversary-initial-state} +$ 
   $(wp\text{-event } ?M \text{ ?slot adversary-initial-state} +$ 
   $(wp\text{-event } ?M \text{ ?next adversary-initial-state} +$ 
   $(wp\text{-event } ?M \text{ ?successor adversary-initial-state} +$ 
   $(wp\text{-event } ?M \text{ ?final adversary-initial-state} +$ 
   $wp\text{-event } ?M \text{ ?merkle adversary-initial-state}))))$ 
  by (intro add-mono order-refl slot-bound)
finally show ?thesis .
qed
show ?thesis
  by (rule order-trans[OF dedup union-bound])
qed
have split-sum-bound:
   $wp\text{-event } ?M \text{ ?merkle adversary-initial-state} +$ 
   $(wp\text{-event } ?M \text{ ?slot adversary-initial-state} +$ 
   $(wp\text{-event } ?M \text{ ?next adversary-initial-state} +$ 
   $(wp\text{-event } ?M \text{ ?successor adversary-initial-state} +$ 
   $(wp\text{-event } ?M \text{ ?final adversary-initial-state} +$ 
   $wp\text{-event } ?M \text{ ?merkle adversary-initial-state})))) \leq$ 
   $composition\text{-one-step-controlled-merkle-error-for budgets} +$ 
   $(0 +$ 
   $(composition\text{-one-step-controlled-merkle-error-for budgets} +$ 
   $(composition\text{-one-step-controlled-merkle-error-for budgets} +$ 
   $(composition\text{-one-step-controlled-merkle-error-for budgets} +$ 
   $composition\text{-one-step-controlled-merkle-error-for budgets}))))$ 
  by (intro add-mono
    checked-staged-security-partial-merkle-inconsistency-bad-bound-by-composition-one-step-controlled-merkle-error
    [OF wf controlled]
    checked-staged-security-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero
    next-bound successor-bound final-bound)

```

```

have wp-event ?M ?conflict adversary-initial-state ≤
  wp-event ?M
  (λout.
    ?merkle out ∨ ?slot out ∨ ?next out ∨
    ?successor out ∨ ?final out ∨ ?merkle out ∨ ?slot out)
  adversary-initial-state
  by (rule split)
also have ... ≤
  wp-event ?M ?merkle adversary-initial-state +
  (wp-event ?M ?slot adversary-initial-state +
  (wp-event ?M ?next adversary-initial-state +
  (wp-event ?M ?successor adversary-initial-state +
  (wp-event ?M ?final adversary-initial-state +
  wp-event ?M ?merkle adversary-initial-state))))
  by (rule split-bound)
also have ... ≤
  composition-one-step-controlled-merkle-error-for budgets +
  (0 +
  (composition-one-step-controlled-merkle-error-for budgets +
  (composition-one-step-controlled-merkle-error-for budgets +
  (composition-one-step-controlled-merkle-error-for budgets +
  composition-one-step-controlled-merkle-error-for budgets))))
  by (rule split-sum-bound)
finally show ?thesis
  by (simp add: add.assoc)
qed

lemma composition-one-step-verifier-tied-controlled-error-for-bound:
fixes comp-bad :: 'f ⇒ nat ⇒ 'f list ⇒ 'f set
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and comp-subset:
  ∧ dg i prefix.
    i < fri-round-count-for-degree-bound (to-nat dg) ⇒
    length prefix = i ⇒
    comp-bad dg i prefix ⊆
    fri-one-step-disagreement-challenges
    (composition-fri-canonical-layer-len dg i) (comp-pw dg i)
    (comp-layer dg i prefix) (comp-claimed dg i prefix)
    (comp-dom dg i prefix)
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
  composition-fri-bad-with-verifier-tied-partial-candidate)
  adversary-initial-state
  ≤ composition-one-step-verifier-tied-controlled-error-for budgets
  comp-bad
proof —
have sampled-query:

```

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   composition-fri-sampled-query-bad-candidate)
  adversary-initial-state
≤ composition-one-step-sampled-error-for budgets comp-bad
by (rule composition-one-step-sampled-error-for-bound
    [OF wf controlled comp-subset])
have sampled:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-fri-bad-with-verifier-tied-sampled-layer-chain)
    adversary-initial-state
≤ composition-one-step-sampled-error-for budgets comp-bad
by (rule
    checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-sampled-query
    [OF sampled-query])
have conflict:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-fri-verifier-tied-sampled-assignment-conflict)
    adversary-initial-state ≤
    (((composition-one-step-controlled-merkle-error-for budgets + 0) +
      (composition-one-step-controlled-merkle-error-for budgets + 0) +
      (composition-one-step-controlled-merkle-error-for budgets + 0) +
      (composition-one-step-controlled-merkle-error-for budgets + 0 + 0) +
      composition-one-step-controlled-merkle-error-for budgets) + 0)
  by (rule
      checked-staged-security-composition-fri-verifier-tied-sampled-assignment-conflict-bound-by-controlled-merkle
      [OF wf controlled])
have zero:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-fri-verifier-tied-zero-round-final-obstruction)
    adversary-initial-state ≤ (0::prob)
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show  $\bigwedge s. \neg$ 
  composition-fri-verifier-tied-zero-round-final-obstruction s None
unfolding composition-fri-verifier-tied-zero-round-final-obstruction-def
  accepted-fri-opening-transcript-def
by simp
next
fix data attacker-state
assume support: Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)
   adversary-initial-state)
show wp-event verify-monad
  (composition-fri-verifier-tied-zero-round-final-obstruction
   (verifier-state-from-adversary attacker-state)

```

```

      (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
    ≤ 0
  by (subst
      wp-composition-fri-verifier-tied-zero-round-final-obstruction-zero)
    simp
qed
have branch:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      composition-fri-bad-with-verifier-tied-partial-candidate)
  adversary-initial-state ≤
  composition-one-step-sampled-error-for budgets comp-bad +
  (((composition-one-step-controlled-merkle-error-for budgets + 0) +
    (composition-one-step-controlled-merkle-error-for budgets + 0) +
    (composition-one-step-controlled-merkle-error-for budgets + 0) +
    (composition-one-step-controlled-merkle-error-for budgets + 0 + 0) +
    composition-one-step-controlled-merkle-error-for budgets) + 0) + 0)
  by (rule
      checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-staged-sampled-conflict-and-zero
      [OF sampled conflict zero])
show ?thesis
  using branch
  by (simp add: composition-one-step-verifier-tied-controlled-error-for-def)
qed

lemma composition-one-step-verifier-tied-error-for-bound:
  fixes comp-bad :: 'f ⇒ nat ⇒ 'f list ⇒ 'f set
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and comp-subset:
    ∧ dg i prefix.
      i < fri-round-count-for-degree-bound (to-nat dg) ⇒
      length prefix = i ⇒
      comp-bad dg i prefix ⊆
      fri-one-step-disagreement-challenges
        (composition-fri-canonical-layer-len dg i) (comp-pw dg i)
        (comp-layer dg i prefix) (comp-claimed dg i prefix)
        (comp-dom dg i prefix)
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        composition-fri-bad-with-verifier-tied-partial-candidate)
      adversary-initial-state
    ≤ composition-one-step-verifier-tied-error-for budgets A comp-bad
proof -
  have sampled:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (staged-security-with-data-state-verifier-event
     composition-fri-sampled-query-bad-candidate)
    adversary-initial-state
  ≤ composition-one-step-sampled-error-for budgets comp-bad
  by (rule composition-one-step-sampled-error-for-bound
      [OF wf controlled comp-subset])
  have merkle-bound:
    ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
                      adversary-initial-state) ⇒
    wp-event verify-monad
      (partial-merkle-inconsistency-bad
       (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
      (verifier-state-from-adversary attacker-state
       (staged-proof-transcript data))
    ≤ composition-one-step-reachable-merkle-error-for A
  unfolding composition-one-step-reachable-merkle-error-for-def
  by (rule reachable-verifier-event-bound-for-ge, assumption)
  have branch:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
       composition-fri-bad-with-verifier-tied-partial-candidate)
      adversary-initial-state
    ≤ composition-one-step-sampled-error-for budgets comp-bad +
      (((composition-one-step-reachable-merkle-error-for A + 0) +
        (composition-one-step-reachable-merkle-error-for A + 0) +
        (composition-one-step-reachable-merkle-error-for A + 0) +
        (composition-one-step-reachable-merkle-error-for A + 0 + 0) +
        composition-one-step-reachable-merkle-error-for A) + 0 + 0)
  apply (rule
    checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-sampled-query-and-residual-gap
    [OF sampled])
  apply (rule wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
  apply (rule merkle-bound)
  apply assumption
  apply (rule wp-composition-fri-verifier-tied-recorded-base-chunk-auth-missing-zero)
  apply (rule wp-composition-fri-verifier-tied-next-value-replay-gap-bound-from-merkle-and-slot)
  apply (rule merkle-bound)
  apply assumption
  apply (rule wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
  apply (rule wp-composition-fri-verifier-tied-successor-replay-gap-bound-from-merkle-and-slot)
  apply (rule merkle-bound)
  apply assumption
  apply (rule wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
  apply (rule wp-composition-fri-verifier-tied-final-value-replay-gap-bound-from-merkle-slot-and-selected-step)
  apply (rule merkle-bound)
  apply assumption

```

```

apply (rule wp-composition-fri-verifier-tied-authenticated-slot-recorded-chunk-gap-zero)
apply (rule wp-composition-fri-verifier-tied-final-value-selected-step-missing-gap-zero)
apply (subst wp-composition-fri-verifier-tied-zero-round-final-obstruction-zero)
apply simp
done
show ?thesis
  using branch
  by (simp add: composition-one-step-verifier-tied-error-for-def)
qed

```

```

lemma trace-empty-one-step-error-for-bound:
fixes empty-bad :: nat  $\Rightarrow$  'f list  $\Rightarrow$  'f set
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
and empty-subset:
   $\bigwedge i$  prefix.
     $i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \implies$ 
     $\text{length prefix} = i \implies$ 
     $\text{empty-bad } i \text{ prefix} \subseteq$ 
    fri-one-step-disagreement-challenges
    (trace-fri-canonical-layer-len i) (empty-pw i)
    (empty-layer i prefix) (empty-claimed i prefix)
    (empty-dom i prefix)
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state
   $\leq$  trace-empty-one-step-error-for budgets A empty-bad
proof -
have sampled:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-query-bad-candidate)
    adversary-initial-state
   $\leq$  trace-one-step-sampled-error-for budgets empty-bad
by (rule trace-one-step-sampled-error-for-bound
  [OF wf controlled empty-subset])
have conflict-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-assignment-conflict)
    adversary-initial-state
   $\leq$  reachable-verifier-event-bound-for A
    trace-fri-sampled-assignment-conflict
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show  $\bigwedge s. \neg \text{trace-fri-sampled-assignment-conflict } s \text{ None}$ 
unfolding trace-fri-sampled-assignment-conflict-def
  trace-fri-partial-candidate-opening-evidence-def

```

```

    accepted-fri-opening-transcript-def
  by simp
next
fix data attacker-state
assume support: Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)
    adversary-initial-state)
show wp-event verify-monad
  (trace-fri-sampled-assignment-conflict
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
≤ reachable-verifier-event-bound-for A
  trace-fri-sampled-assignment-conflict
  by (rule reachable-verifier-event-bound-for-ge[OF support])
qed
have zero-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-zero-round-final-obstruction)
    adversary-initial-state
≤ trace-fri-zero-round-agreement-set-target-error-for budgets +
  trace-fri-controlled-merkle-error-for budgets +
  reachable-verifier-event-bound-for A
  trace-fri-bad-with-header-tied-partial-candidate +
  (trace-fri-controlled-merkle-error-for budgets +
    (reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-untied-root-gap +
      reachable-verifier-event-bound-for A
        trace-fri-header-tied-low-candidate-untied-query-gap)) +
  reachable-verifier-event-bound-for A
  trace-fri-bad-with-header-tied-partial-candidate
  by (rule checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-agreement-sets
    [OF wf controlled])
have branch:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state
≤ trace-one-step-sampled-error-for budgets empty-bad +
  (reachable-verifier-event-bound-for A
    trace-fri-sampled-assignment-conflict +
    trace-fri-zero-round-agreement-set-target-error-for budgets +
    trace-fri-controlled-merkle-error-for budgets +
    reachable-verifier-event-bound-for A
      trace-fri-bad-with-header-tied-partial-candidate +
      (trace-fri-controlled-merkle-error-for budgets +

```



```

    (reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-untied-root-gap +
      reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-untied-query-gap)) +
    reachable-verifier-event-bound-for A
    trace-fri-bad-with-header-tied-partial-candidate)
proof -
  have raw:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state
  ≤ trace-one-step-sampled-error-for budgets empty-bad +
    (reachable-verifier-event-bound-for A
      trace-fri-sampled-assignment-conflict +
      (trace-fri-zero-round-agreement-set-target-error-for budgets +
        trace-fri-controlled-merkle-error-for budgets +
        reachable-verifier-event-bound-for A
          trace-fri-bad-with-header-tied-partial-candidate +
          (trace-fri-controlled-merkle-error-for budgets +
            (reachable-verifier-event-bound-for A
              trace-fri-header-tied-low-candidate-untied-root-gap +
              reachable-verifier-event-bound-for A
                trace-fri-header-tied-low-candidate-untied-query-gap))) +
          reachable-verifier-event-bound-for A
            trace-fri-bad-with-header-tied-partial-candidate))
    by (rule
      checked-staged-security-trace-fri-empty-header-bound-from-staged-sampled-conflict-and-zero
      [OF sampled conflict-bound zero-bound])
  show ?thesis
    by (rule order-trans[OF raw]) (simp add: algebra-simps)
qed
show ?thesis
  using branch
  by (simp add: trace-empty-one-step-error-for-def
    trace-fri-empty-small-conflict-zero-error-for-def)
qed

lemma trace-header-one-step-error-for-bound:
  fixes trace-bad :: nat ⇒ 'f list ⇒ 'f set
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and trace-subset:
    ∧ i prefix.
    i < fri-round-count-for-degree-bound (clength - 1) ⇒
    length prefix = i ⇒
    trace-bad i prefix ⊆
    fri-one-step-disagreement-challenges
    (trace-fri-canonical-layer-len i) (empty-pw i)

```

(empty-layer i prefix) (empty-claimed i prefix)
 (empty-dom i prefix)

shows

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-partial-candidate)
 adversary-initial-state
 $\leq$ trace-header-one-step-error-for budgets A trace-bad

proof –

have sampled-query:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-sampled-query-bad-candidate)
 adversary-initial-state
 $\leq$ trace-one-step-sampled-error-for budgets trace-bad

by (rule trace-one-step-sampled-error-for-bound
 [OF wf controlled trace-subset])

have sampled-plain:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-sampled-layer-chain)
 adversary-initial-state
 $\leq$ trace-one-step-sampled-error-for budgets trace-bad

by (rule
 checked-staged-security-trace-fri-sampled-layer-chain-bound-from-sampled-query
 [OF sampled-query])

have sampled-header:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-sampled-layer-chain)
 adversary-initial-state
 $\leq$ trace-one-step-sampled-error-for budgets trace-bad

proof (rule order-trans[OF - sampled-plain])

show wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-sampled-layer-chain)
 adversary-initial-state
 $\leq$ wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-sampled-layer-chain)
 adversary-initial-state

by (rule wp-event-mono)
 (auto simp: staged-security-with-data-state-verifier-event-def
 dest: trace-fri-header-tied-sampled-layer-chain-imp-sampled-layer-chain
 split: option.splits prod.splits)

qed

have conflict-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event

```

    trace-fri-header-tied-sampled-assignment-conflict)
  adversary-initial-state
≤ trace-fri-header-replay-conflict-error-for A
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
let ?M =
  reachable-verifier-event-bound-for A partial-merkle-inconsistency-bad
let ?H =
  reachable-verifier-event-bound-for A
    trace-fri-header-tied-recorded-sibling-candidate-missing
have merkle-bound:
  wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
    ≤ ?M
by (rule reachable-verifier-event-bound-for-ge[OF support])
have recorded-missing-bound:
  wp-event verify-monad
    (trace-fri-header-tied-recorded-sibling-candidate-missing ?s) ?s
    ≤ ?H
by (rule reachable-verifier-event-bound-for-ge[OF support])
have head-neq:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-neq-gap ?s) ?s ≤ ?M
by (rule
  wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle)
  (rule merkle-bound)
have head-mismatch:
  wp-event verify-monad
    (trace-fri-header-tied-base-head-value-mismatch-gap ?s) ?s
    ≤ ?M
by (rule
  wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
  [OF head-neq])
have alignment:
  wp-event verify-monad
    (trace-fri-header-tied-base-value-alignment-gap ?s) ?s ≤ ?M
by (rule
  wp-trace-fri-header-tied-base-value-alignment-gap-bound-from-head-value-mismatch-gap
  [OF head-mismatch])
have next-bound:
  wp-event verify-monad
    (trace-fri-header-tied-next-value-replay-gap ?s) ?s ≤ ?M + 0
by (rule

```

$wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}next\text{-}value\text{-}replay\text{-}gap\text{-}bound\text{-}from\text{-}merkle\text{-}and\text{-}slot)$
(rule merkle-bound,
rule $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}authenticated\text{-}slot\text{-}recorded\text{-}chunk\text{-}gap\text{-}zero$)
have *successor*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}successor\text{-}replay\text{-}gap\ ?s)\ ?s \leq ?M + 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}successor\text{-}replay\text{-}gap\text{-}bound\text{-}from\text{-}merkle\text{-}and\text{-}slot)$
(rule merkle-bound,
rule $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}authenticated\text{-}slot\text{-}recorded\text{-}chunk\text{-}gap\text{-}zero$)
have *final*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\ ?s)\ ?s \leq ?M + 0 + 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\text{-}bound\text{-}from\text{-}merkle\text{-}slot\text{-}and\text{-}selected\text{-}step)$
(rule merkle-bound,
rule $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}authenticated\text{-}slot\text{-}recorded\text{-}chunk\text{-}gap\text{-}zero$,
rule $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}final\text{-}value\text{-}selected\text{-}step\text{-}missing\text{-}gap\text{-}zero$)
have *slot*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}authenticated\text{-}slot\text{-}recorded\text{-}chunk\text{-}gap\ ?s)\ ?s$
 ≤ 0
by (rule $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}authenticated\text{-}slot\text{-}recorded\text{-}chunk\text{-}gap\text{-}zero$)
have *authenticated-recorded-auth*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}authenticated\text{-}recorded\text{-}auth\text{-}gap\ ?s)$
 $?s \leq 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}authenticated\text{-}recorded\text{-}auth\text{-}gap\text{-}bound\text{-}from\text{-}slot\text{-}recorded\text{-}chunk\text{-}gap$
 $[OF\ slot])$
have *recorded-auth*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}recorded\text{-}auth\text{-}gap\ ?s)\ ?s \leq 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}recorded\text{-}auth\text{-}gap\text{-}bound\text{-}from\text{-}authenticated\text{-}recorded\text{-}auth\text{-}gap$
 $[OF\ authenticated\text{-}recorded\text{-}auth])$
have *same-auth*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}auth\text{-}gap\ ?s)\ ?s \leq 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}auth\text{-}gap\text{-}bound\text{-}from\text{-}recorded\text{-}auth\text{-}gap$
 $[OF\ recorded\text{-}auth])$
have *auth-gap*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}assignment\text{-}auth\text{-}gap\ ?s)\ ?s \leq 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}assignment\text{-}auth\text{-}gap\text{-}bound\text{-}from\text{-}same\text{-}layer\text{-}auth\text{-}gap$
 $[OF\ same\text{-}auth])$
have *complement*:

```

wp-event verify-monad
  (λout.
    trace-fri-header-tied-sampled-assignment-conflict ?s out ∧
    ¬ trace-fri-header-tied-recorded-sibling-candidate-missing ?s out)
  ?s ≤
  (((?M + ?M) + (?M + 0) + (?M + 0) + (?M + 0 + 0)) + ?M) + 0)
by (rule
  wp-trace-fri-header-tied-sampled-assignment-conflict-without-recorded-missing-bound-from-auth-gap-and-replay-
  (rule auth-gap, rule merkle-bound, rule alignment,
  rule next-bound, rule successor, rule final)
show wp-event verify-monad
  (trace-fri-header-tied-sampled-assignment-conflict ?s) ?s
  ≤ trace-fri-header-replay-conflict-error-for A
unfolding trace-fri-header-replay-conflict-error-for-def Let-def
by (rule
  wp-trace-fri-header-tied-sampled-assignment-conflict-bound-from-recorded-missing-and-complement)
  (rule recorded-missing-bound, rule complement)
next
fix s
show ¬ trace-fri-header-tied-sampled-assignment-conflict s None
unfolding trace-fri-header-tied-sampled-assignment-conflict-def
  accepted-fri-opening-transcript-def
by simp
qed
have zero-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
  trace-fri-header-tied-zero-round-final-obstruction)
  adversary-initial-state
  ≤ trace-fri-zero-round-agreement-set-target-error-for budgets +
  trace-fri-controlled-merkle-error-for budgets
proof –
have checked:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
  trace-fri-zero-round-checked-final-obstruction)
  adversary-initial-state
  ≤ trace-fri-zero-round-agreement-set-target-error-for budgets
by (rule checked-staged-security-trace-fri-zero-round-checked-final-obstruction-bound-from-agreement-sets
  [OF wf controlled])
have merkle:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
  partial-merkle-inconsistency-bad)
  adversary-initial-state
  ≤ trace-fri-controlled-merkle-error-for budgets
unfolding trace-fri-controlled-merkle-error-for-def
by (rule
  checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled

```

```

    [OF wf controlled])
have split:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-zero-round-final-obstruction)
    adversary-initial-state
  ≤ wp-event (checked-staged-security-experiment-with-data-state A)
    (λout.
      staged-security-with-data-state-verifier-event
        trace-fri-zero-round-checked-final-obstruction out ∨
      staged-security-with-data-state-verifier-event
        partial-merkle-inconsistency-bad out)
    adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono-on-support)
  (auto dest:
    trace-fri-header-tied-zero-round-final-obstruction-imp-checked-or-merkle
    split: option.splits prod.splits)
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-zero-round-checked-final-obstruction)
    adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      partial-merkle-inconsistency-bad)
    adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤
  trace-fri-zero-round-agreement-set-target-error-for budgets +
  trace-fri-controlled-merkle-error-for budgets
by (rule add-mono[OF checked merkle])
finally show ?thesis .
qed
have branch:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state
  ≤ trace-one-step-sampled-error-for budgets trace-bad +
    (trace-fri-header-replay-conflict-error-for A +
      (trace-fri-zero-round-agreement-set-target-error-for budgets +
        trace-fri-controlled-merkle-error-for budgets))
by (rule
  checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-conflict-and-staged-zero
  [OF sampled-header conflict-bound zero-bound])
show ?thesis
using branch
by (simp add: trace-header-one-step-error-for-def

```

trace-fri-header-replay-conflict-zero-error-for-def)
qed

definition *trace-empty-header-refined-zero-error-for*
where

trace-empty-header-refined-zero-error-for budgets A empty-bad =
(trace-fri-zero-round-agreement-set-target-error-for budgets +
trace-fri-controlled-merkle-error-for budgets +
(trace-header-one-step-error-for budgets A empty-bad +
(trace-fri-controlled-merkle-error-for budgets +
(reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-untied-root-gap +
reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-untied-query-gap))) +
trace-header-one-step-error-for budgets A empty-bad)

definition *trace-empty-header-refined-one-step-error-for*
where

trace-empty-header-refined-one-step-error-for budgets A empty-bad =
trace-one-step-sampled-error-for budgets empty-bad +
(reachable-verifier-event-bound-for A
trace-fri-sampled-assignment-conflict +
trace-empty-header-refined-zero-error-for budgets A empty-bad)

definition *concrete-trace-empty-header-refined-one-step-error-for*
where

concrete-trace-empty-header-refined-one-step-error-for budgets A =
trace-empty-header-refined-one-step-error-for budgets A
trace-one-step-split-bad

lemma *checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-agreement-sets-and-header:*
fixes *H :: prob*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

and *header-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state ≤ H

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-zero-round-final-obstruction)
adversary-initial-state ≤
trace-fri-zero-round-agreement-set-target-error-for budgets +
trace-fri-controlled-merkle-error-for budgets +
(H +
(trace-fri-controlled-merkle-error-for budgets +
(reachable-verifier-event-bound-for A

```

    trace-fri-header-tied-low-candidate-untied-root-gap +
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-query-gap))) +
  H
proof -
  have checked:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-zero-round-checked-final-obstruction)
    adversary-initial-state ≤
    trace-fri-zero-round-agreement-set-target-error-for budgets
  by (rule
    checked-staged-security-trace-fri-zero-round-checked-final-obstruction-bound-from-agreement-sets
    [OF wf controlled])
  have merkle:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        partial-merkle-inconsistency-bad)
    adversary-initial-state ≤
    trace-fri-controlled-merkle-error-for budgets
  unfolding trace-fri-controlled-merkle-error-for-def
  by (rule
    checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
    [OF wf controlled])
  have hash:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        hash-map-output-collision-bad)
    adversary-initial-state ≤
    trace-fri-controlled-merkle-error-for budgets
  unfolding trace-fri-controlled-merkle-error-for-def
  by (rule
    checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
    [OF wf controlled])
  have root:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-low-candidate-untied-root-gap)
    adversary-initial-state ≤
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-root-gap
  proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\bigwedge s. \neg \text{trace-fri-header-tied-low-candidate-untied-root-gap } s \text{ None}$ 
  unfolding trace-fri-header-tied-low-candidate-untied-root-gap-def
    trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
    accepted-fri-opening-transcript-def
  by simp
next
fix data attacker-state

```



```

assume support: Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)
    adversary-initial-state)
show wp-event verify-monad
  (trace-fri-header-tied-low-candidate-untied-root-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  ≤ reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-root-gap
  by (rule reachable-verifier-event-bound-for-ge[OF support])
qed
have query:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-low-candidate-untied-query-gap)
  adversary-initial-state ≤
  reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-query-gap
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show  $\bigwedge s. \neg \text{trace-fri-header-tied-low-candidate-untied-query-gap } s \text{ None}$ 
  unfolding trace-fri-header-tied-low-candidate-untied-query-gap-def
    trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
    accepted-fri-opening-transcript-def
  by simp
next
fix data attacker-state
assume support: Some (data, attacker-state) ∈
  set-dist
  (execute (checked-staged-transcript-program A)
    adversary-initial-state)
show wp-event verify-monad
  (trace-fri-header-tied-low-candidate-untied-query-gap
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
  ≤ reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-query-gap
  by (rule reachable-verifier-event-bound-for-ge[OF support])
qed
let ?M = checked-staged-security-experiment-with-data-state A
let ?untied =
  staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap
let ?hash =
  staged-security-with-data-state-verifier-event

```

```

    hash-map-output-collision-bad
  let ?root =
    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-untied-root-gap
  let ?query =
    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-untied-query-gap
  have untied:
    wp-event ?M ?untied adversary-initial-state ≤
      trace-fri-controlled-merkle-error-for budgets +
      (reachable-verifier-event-bound-for A
        trace-fri-header-tied-low-candidate-untied-root-gap +
        reachable-verifier-event-bound-for A
        trace-fri-header-tied-low-candidate-untied-query-gap)
  proof -
    have wp-event ?M ?untied adversary-initial-state ≤
      wp-event ?M (λout. ?hash out ∨ ?root out ∨ ?query out)
      adversary-initial-state
    unfolding staged-security-with-data-state-verifier-event-def
    by (rule wp-event-mono-on-support)
      (auto
        dest:
          trace-fri-header-tied-low-candidate-untied-gap-imp-hash-root-or-query
          split: option.splits prod.splits)
    also have ... ≤
      wp-event ?M ?hash adversary-initial-state +
      wp-event ?M (λout. ?root out ∨ ?query out)
      adversary-initial-state
    by (rule wp-event-union-bound)
    also have ... ≤
      wp-event ?M ?hash adversary-initial-state +
      (wp-event ?M ?root adversary-initial-state +
        wp-event ?M ?query adversary-initial-state)
    by (intro add-mono order-refl wp-event-union-bound)
    also have ... ≤
      trace-fri-controlled-merkle-error-for budgets +
      (reachable-verifier-event-bound-for A
        trace-fri-header-tied-low-candidate-untied-root-gap +
        reachable-verifier-event-bound-for A
        trace-fri-header-tied-low-candidate-untied-query-gap)
    by (intro add-mono hash root query)
    finally show ?thesis .
  qed
  have transfer:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-candidate-transfer-gap)
    adversary-initial-state ≤
      H +

```

```

    (trace-fri-controlled-merkle-error-for budgets +
    (reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-root-gap +
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-query-gap))
  by (rule
    checked-staged-security-trace-fri-header-tied-candidate-transfer-gap-bound-from-staged-header-and-untied
    [OF header-bound untied])
show ?thesis
  by (rule
    checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-staged-checked-merkle-transfer-header
    [OF checked merkle transfer header-bound])
qed

lemma trace-empty-header-refined-one-step-error-for-bound:
  fixes empty-bad :: nat  $\Rightarrow$  'f list  $\Rightarrow$  'f set
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and empty-subset:
     $\bigwedge i$  prefix.
       $i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \Rightarrow$ 
       $\text{length prefix} = i \Rightarrow$ 
       $\text{empty-bad } i \text{ prefix} \subseteq$ 
      fri-one-step-disagreement-challenges
      (trace-fri-canonical-layer-len i) (empty-pw i)
      (empty-layer i prefix) (empty-claimed i prefix)
      (empty-dom i prefix)
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state
     $\leq$  trace-empty-header-refined-one-step-error-for budgets A empty-bad
proof -
  have sampled:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    trace-fri-sampled-query-bad-candidate)
    adversary-initial-state
     $\leq$  trace-one-step-sampled-error-for budgets empty-bad
  by (rule trace-one-step-sampled-error-for-bound
    [OF wf controlled empty-subset])
  have conflict-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
    trace-fri-sampled-assignment-conflict)
    adversary-initial-state
     $\leq$  reachable-verifier-event-bound-for A
    trace-fri-sampled-assignment-conflict

```

```

proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show  $\bigwedge s. \neg \text{trace-fri-sampled-assignment-conflict } s \text{ None}$ 
    unfolding trace-fri-sampled-assignment-conflict-def
      trace-fri-partial-candidate-opening-evidence-def
      accepted-fri-opening-transcript-def
    by simp
next
  fix data attacker-state
  assume support: Some (data, attacker-state)  $\in$ 
    set-dist
    (execute (checked-staged-transcript-program A)
      adversary-initial-state)
  show wp-event verify-monad
    (trace-fri-sampled-assignment-conflict
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
   $\leq$  reachable-verifier-event-bound-for A
    trace-fri-sampled-assignment-conflict
  by (rule reachable-verifier-event-bound-for-ge[OF support])
qed
have header-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state
   $\leq$  trace-header-one-step-error-for budgets A empty-bad
  by (rule trace-header-one-step-error-for-bound
    [OF wf controlled empty-subset])
have zero-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-zero-round-final-obstruction)
    adversary-initial-state
   $\leq$  trace-empty-header-refined-zero-error-for budgets A empty-bad
  unfolding trace-empty-header-refined-zero-error-for-def
  by (rule
    checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-agreement-sets-and-header
      [OF wf controlled header-bound])
have branch:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state
   $\leq$  trace-one-step-sampled-error-for budgets empty-bad +
    (reachable-verifier-event-bound-for A
      trace-fri-sampled-assignment-conflict +
      trace-empty-header-refined-zero-error-for budgets A empty-bad)

```

by (rule
 checked-staged-security-trace-fri-empty-header-bound-from-staged-sampled-conflict-and-zero
 [*OF sampled conflict-bound zero-bound*])
 show ?thesis
 using branch
 by (simp add: *trace-empty-header-refined-one-step-error-for-def*)
 qed

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-one-step-fri-components*

fixes *trace-bad empty-bad* :: *nat* $\Rightarrow$ '*f list* $\Rightarrow$ '*f set*

and *comp-bad* :: '*f* $\Rightarrow$ *nat* $\Rightarrow$ '*f list* $\Rightarrow$ '*f set*

assumes *false-statement*: $\neg$ *exists-valid-trace*

and *wf*: *staged-budget-wellformed budgets*

and *controlled*: *staged-adversary-controlled budgets A*

and *trace-subset*:

$\bigwedge i$ *prefix*.

i < *fri-round-count-for-degree-bound* (*clength* - 1) $\Rightarrow$

length prefix = *i* $\Rightarrow$

trace-bad i prefix $\subseteq$

fri-one-step-disagreement-challenges

(*trace-fri-canonical-layer-len i*) (*empty-pw i*)

(*empty-layer i prefix*) (*empty-claimed i prefix*)

(*empty-dom i prefix*)

and *comp-subset*:

$\bigwedge dg$ *i prefix*.

i < *fri-round-count-for-degree-bound* (*to-nat dg*) $\Rightarrow$

length prefix = *i* $\Rightarrow$

comp-bad dg i prefix $\subseteq$

fri-one-step-disagreement-challenges

(*composition-fri-canonical-layer-len dg i*) (*comp-pw dg i*)

(*comp-layer dg i prefix*) (*comp-claimed dg i prefix*)

(*comp-dom dg i prefix*)

and *empty-subset*:

$\bigwedge i$ *prefix*.

i < *fri-round-count-for-degree-bound* (*clength* - 1) $\Rightarrow$

length prefix = *i* $\Rightarrow$

empty-bad i prefix $\subseteq$

fri-one-step-disagreement-challenges

(*trace-fri-canonical-layer-len i*) (*empty-pw i*)

(*empty-layer i prefix*) (*empty-claimed i prefix*)

(*empty-dom i prefix*)

shows *checked-staged-adversary-acceptance-probability A* $\leq$

hash-collision-budget-value 0

(*staged-attacker-query-budget budgets* +

staged-challenge-query-budget + *verifier-hash-query-budget*) +

(*composition-error-bound* +

staged-phase-relation-error size

(*staged-alpha-search-queries budgets 0*) +

hash-collision-budget-value 0

(staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds}$.
 active-route-current-prefix-authenticated-error-for A i +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 trace-header-one-step-error-for budgets A trace-bad +
 composition-one-step-verifier-tied-controlled-error-for budgets
 comp-bad +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 trace-empty-header-refined-one-step-error-for budgets A empty-bad +
 active-route-current-empty-query-error-for A)
proof –
have trace-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-partial-candidate)
 adversary-initial-state
 $\leq$ trace-header-one-step-error-for budgets A trace-bad
by (rule trace-header-one-step-error-for-bound
 [OF wf controlled trace-subset])
have comp-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state
 $\leq$ composition-one-step-verifier-tied-controlled-error-for budgets
 comp-bad
by (rule composition-one-step-verifier-tied-controlled-error-for-bound
 [OF wf controlled comp-subset])
have empty-bound:
 wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-empty-composition-header-candidates)
 adversary-initial-state
 $\leq$ trace-empty-header-refined-one-step-error-for budgets A empty-bad

by (rule trace-empty-header-refined-one-step-error-for-bound
 [OF wf controlled empty-subset])
 show ?thesis
 by (rule
 stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-bounds
 [OF false-statement wf controlled trace-bound comp-bound empty-bound])
 qed

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-one-step-endpoint:
 assumes false-statement: $\neg$ exists-valid-trace
 and wf: staged-budget-wellformed budgets
 and controlled: staged-adversary-controlled budgets A
 shows checked-staged-adversary-acceptance-probability $A \leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds.}$
 active-route-current-prefix-authenticated-error-for A i +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 concrete-trace-header-one-step-error-for budgets A +
 concrete-composition-one-step-error-for budgets A +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 concrete-trace-empty-header-refined-one-step-error-for budgets A +
 active-route-current-empty-query-error-for A)
 unfolding concrete-trace-header-one-step-error-for-def
 concrete-composition-one-step-error-for-def
 concrete-trace-empty-header-refined-one-step-error-for-def
 by (rule
 stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-one-step-fri-components

[where $\text{trace-bad} = \text{trace-one-step-split-bad}$
 and $\text{comp-bad} = \text{composition-one-step-split-bad}$
 and $\text{empty-bad} = \text{trace-one-step-split-bad}$
 and $\text{empty-pw} = \text{trace-one-step-split-pw}$
 and $\text{empty-layer} = \text{trace-one-step-split-layer}$
 and $\text{empty-claimed} = \text{trace-one-step-split-claimed}$
 and $\text{empty-dom} = \text{trace-one-step-split-domain}$
 and $\text{comp-pw} = \text{composition-one-step-split-pw}$
 and $\text{comp-layer} = \text{composition-one-step-split-layer}$
 and $\text{comp-claimed} = \text{composition-one-step-split-claimed}$
 and $\text{comp-dom} = \text{composition-one-step-split-domain}$,
 OF false-statement wf controlled])
 (simp-all add:
 $\text{trace-one-step-split-bad-def}$
 $\text{composition-one-step-split-bad-def}$
 $\text{trace-one-step-disagreement-bad-for-def}$
 $\text{composition-one-step-disagreement-bad-for-def}$)

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-sharpened:*

assumes false-statement: $\neg \text{exists-valid-trace}$
 and wf: staged-budget-wellformed budgets
 and controlled: staged-adversary-controlled budgets A
 shows checked-staged-adversary-acceptance-probability A $\leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds}$.
 active-route-current-prefix-authenticated-error-for A i +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 sharpened-trace-fri-header-error-for budgets A +
 sharpened-composition-fri-error-for budgets A +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 sharpened-trace-fri-empty-error-for budgets A +
 active-route-current-empty-query-error-for A)
 by (rule
 stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-bounds
 [OF false-statement wf controlled
 sharpened-trace-fri-header-error-for-bound[OF wf controlled]
 sharpened-composition-fri-error-for-bound[OF wf controlled]
 sharpened-trace-fri-empty-error-for-bound[OF wf controlled]])

theorem stark-soundness-concrete-fri-sharpened-endpoint:

assumes false-statement: $\neg$ exists-valid-trace

and wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

shows checked-staged-adversary-acceptance-probability A $\leq$

hash-collision-budget-value 0

(staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +

(composition-error-bound +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +

hash-collision-budget-value 0

(staged-alpha-search-queries budgets 0) +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +

(active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +

($\sum$ i < rounds.

active-route-current-prefix-authenticated-error-for A i +

(active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +

(active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +

concrete-trace-header-one-step-error-for budgets A +

concrete-composition-one-step-error-for budgets A +

(composition-error-bound +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +

hash-collision-budget-value 0

(staged-alpha-search-queries budgets 0) +

staged-phase-relation-error size

(staged-alpha-search-queries budgets 0) +

(active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +

concrete-trace-empty-header-refined-one-step-error-for budgets A +

active-route-current-empty-query-error-for A)

by (rule

stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-one-step-endpoint
[OF false-statement wf controlled])

end

hide-fact

soundness.stark-soundness-concrete-fri-sharpened-endpoint
soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-sharpened

end

theory *Soundness-FRI-Canonical-Full-Cover-Failure*

imports *Soundness-FRI-Sampled-Challenge-Cover*

begin

Canonical full-cover failure events for sampled-query FRI.

The broad sampled full-cover failure predicates quantify over arbitrary domain witnesses. The sampled-query split, however, produces the failure on the canonical FRI domains. This layer names that narrower event so later bounds do not have to prove facts about unused arbitrary domain entries.

context *soundness*

begin

definition *trace-fri-sampled-canonical-full-cover-failure*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-sampled-canonical-full-cover-failure s bad out $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table layers.}$
 $\text{trace-fri-partial-candidate-opening-evidence s out trace-roots}$
 $\text{trace-bs trace-final dg composition-roots composition-bs}$
 $\text{composition-final fri-query-idxs trace-round-layers}$
 $\text{composition-round-layers fr candidate-query-idxs trace-openings}$
 $\text{trace-table} \wedge$
 $\text{generic-fri-sampled-query-candidate-evidence trace-table-low-degree}$
 $(\text{Not} \circ \text{trace-table-low-degree}) \text{ trace-table (clength} - 1)$
 $\text{trace-roots trace-bs trace-final fri-query-idxs trace-round-layers}$
 $\text{layers} \wedge$
 $\text{generic-fri-sampled-full-cover-failure trace-table-low-degree}$
 $\text{trace-table (clength} - 1) \text{ trace-roots trace-bs trace-final}$
 $\text{fri-query-idxs trace-round-layers}$
 $(\text{fri-canonical-domains (length trace-bs)) layers}$
 $(\text{bad trace-table}))$

definition *composition-fri-sampled-canonical-full-cover-failure*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $('f \Rightarrow 'f \text{ list} \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}) \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$
where
composition-fri-sampled-canonical-full-cover-failure *s* *bad out*
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg}$
 $\text{ opening-composition-roots composition-bs composition-final}$
 $\text{ fri-query-idxs trace-round-layers composition-round-layers fr}$
 $\text{ f-fri-roots f-final as dg composition-fri-roots final}$
 $\text{ trace-query-idxs trace-openings composition-query-idxs}$
 $\text{ composition-openings trace-table composition-table layers.}$
 $\text{ composition-fri-partial-candidate-opening-evidence } s \text{ out trace-roots}$
 $\text{ trace-bs trace-final fri-dg opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr f-fri-roots f-final as dg}$
 $\text{ composition-fri-roots final trace-query-idxs trace-openings}$
 $\text{ composition-query-idxs composition-openings trace-table}$
 $\text{ composition-table} \wedge$
 $\text{ generic-fri-sampled-query-candidate-evidence}$
 $(\text{ composition-table-low-degree } (\text{to-nat fri-dg}))$
 $(\text{Not} \circ \text{ composition-table-low-degree } \text{maxDegree}) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs composition-round-layers layers} \wedge$
 $\text{ generic-fri-sampled-full-cover-failure}$
 $(\text{ composition-table-low-degree } (\text{to-nat fri-dg})) \text{ composition-table}$
 $(\text{to-nat fri-dg}) \text{ opening-composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs composition-round-layers}$
 $(\text{fri-canonical-domains } (\text{length composition-bs})) \text{ layers}$
 $(\text{bad fri-dg composition-table}))$

lemma *trace-fri-sampled-canonical-full-cover-failure-not-None[simp]:*

$\neg \text{ trace-fri-sampled-canonical-full-cover-failure } s \text{ bad None}$

unfolding *trace-fri-sampled-canonical-full-cover-failure-def*

trace-fri-partial-candidate-opening-evidence-def

accepted-fri-opening-transcript-def

by *simp*

lemma *composition-fri-sampled-canonical-full-cover-failure-not-None[simp]:*

$\neg \text{ composition-fri-sampled-canonical-full-cover-failure } s \text{ bad None}$

unfolding *composition-fri-sampled-canonical-full-cover-failure-def*

composition-fri-partial-candidate-opening-evidence-def

accepted-fri-opening-transcript-def

by *simp*

lemma *trace-fri-sampled-canonical-full-cover-failure-imp-broad:*

assumes *trace-fri-sampled-canonical-full-cover-failure* *s* *bad out*

shows *trace-fri-sampled-full-cover-failure* *s* *bad out*

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table layers*

where *evidence*:

*trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table*

and *candidate*:

*generic-fri-sampled-query-candidate-evidence trace-table-low-degree
(Not $\circ$ trace-table-low-degree) trace-table (clength $- 1$)
trace-roots trace-bs trace-final fri-query-idxs trace-round-layers
layers*

and *failure*:

*generic-fri-sampled-full-cover-failure trace-table-low-degree
trace-table (clength $- 1$) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers
(fri-canonical-domains (length trace-bs)) layers
(bad trace-table)*

unfolding *trace-fri-sampled-canonical-full-cover-failure-def* **by** *blast*

have *canonical*:

*generic-fri-canonical-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength $- 1$) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers layers*

by (rule *generic-fri-sampled-query-candidate-evidenceD(1)[OF candidate]*)

have *sampled*:

*generic-fri-sampled-layer-chain-evidence trace-table-low-degree
trace-table (clength $- 1$) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers
(fri-canonical-domains (length trace-bs)) layers*

by (rule *generic-fri-canonical-sampled-layer-chain-evidenceD*
[OF *canonical*])

show *?thesis*

unfolding *trace-fri-sampled-full-cover-failure-def*

using *evidence sampled failure* **by** *blast*

qed

lemma *composition-fri-sampled-canonical-full-cover-failure-imp-broad*:

assumes *composition-fri-sampled-canonical-full-cover-failure s bad out*

shows *composition-fri-sampled-full-cover-failure s bad out*

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs*

composition-openings trace-table composition-table layers
where *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table
and *candidate*:
generic-fri-sampled-query-candidate-evidence
(composition-table-low-degree (to-nat fri-dg))
(Not o composition-table-low-degree maxDegree) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers layers
and *failure*:
generic-fri-sampled-full-cover-failure
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers
(fri-canonical-domains (length composition-bs)) layers
(bad fri-dg composition-table)
unfolding *composition-fri-sampled-canonical-full-cover-failure-def*
by *meson*
have *canonical*:
generic-fri-canonical-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers layers
by (rule *generic-fri-sampled-query-candidate-evidenceD*(1)[*OF candidate*])
have *sampled*:
generic-fri-sampled-layer-chain-evidence
(composition-table-low-degree (to-nat fri-dg)) composition-table
(to-nat fri-dg) opening-composition-roots composition-bs
composition-final fri-query-idxs composition-round-layers
(fri-canonical-domains (length composition-bs)) layers
by (rule *generic-fri-canonical-sampled-layer-chain-evidenceD*
[*OF canonical*])
show *?thesis*
unfolding *composition-fri-sampled-full-cover-failure-def*
using *evidence sampled failure by meson*
qed

lemma *generic-fri-canonical-sampled-layer-chain-next-idx-current-bound*:
assumes *chain*:
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges)) layers
and *round-bound*: *round-idx < length query-idxs*

```

    and layer-bound: layer-idx < length challenges
  shows
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
      length (fri-canonical-domain-at layer-idx)
  proof -
    have fold:
      fri-sampled-table-fold (challenges ! layer-idx)
        (fri-evidence-layer-len roots layer-idx)
        (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
        (2 ^ layer-idx)
        (fri-canonical-domains (length challenges) ! layer-idx)
        (layers ! layer-idx)
        (round-layers ! round-idx ! layer-idx)
        (layers ! Suc layer-idx !
          fri-evidence-next-idx roots query-idxs round-idx layer-idx)
    by (rule generic-fri-sampled-layer-chain-evidence-sample(2)
      [OF chain round-bound layer-bound])
  from fold obtain xp xp-path xn xn-path where match:
    fri-opening-matches-table
      (fri-evidence-layer-len roots layer-idx)
      (fri-evidence-layer-idx roots query-idxs round-idx layer-idx)
      (layers ! layer-idx) xp xn
    by (elim fri-sampled-table-foldE)
  have partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
      roots challenges final-value query-idxs round-layers
    by (rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])
  have roots-len: length challenges = length roots
    by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
  have roots-bound: layer-idx < length roots
    using layer-bound roots-len by simp
  have len-eq:
    fri-evidence-layer-len roots layer-idx =
      length (fri-canonical-domain-at layer-idx)
    unfolding fri-evidence-layer-len-def fri-canonical-domain-at-length
    using fri-layer-lengths-nth-div[OF roots-bound, of clength * scale]
    by simp
  have raw-bound:
    fri-evidence-layer-idx roots query-idxs round-idx layer-idx <
      length (fri-canonical-domain-at layer-idx)
    using fri-opening-matches-tableD(1)[OF match] unfolding len-eq .
  show ?thesis
  proof (cases fri-evidence-layer-len roots layer-idx div 2 = 0)
    case True
    then show ?thesis
      using raw-bound unfolding fri-evidence-next-idx-def by simp
  next
    case False
    have half-bound:

```

```

    fri-evidence-layer-len roots layer-idx div 2 ≤
      length (fri-canonical-domain-at layer-idx)
  unfolding len-eq by simp
have
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    fri-evidence-layer-len roots layer-idx div 2
  unfolding fri-evidence-next-idx-def
  by (rule mod-less-divisor) (use False in simp)
then show ?thesis
  using half-bound by linarith
qed
qed

lemma generic-fri-canonical-sampled-layer-chain-next-idx-successor-bound:
  assumes chain:
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
      degree-bound roots challenges final-value query-idxs round-layers
      (fri-canonical-domains (length challenges)) layers
  and no-shape:
    ¬ generic-fri-full-cover-shape-failure roots challenges
      (fri-canonical-domains (length challenges)) layers
  and round-bound: round-idx < length query-idxs
  and layer-bound: layer-idx < length roots
  shows
    fri-evidence-next-idx roots query-idxs round-idx layer-idx <
      length (fri-canonical-domains (length challenges) ! Suc layer-idx)
proof -
  have partial:
    generic-fri-partial-evidence low-degree candidate-table degree-bound
      roots challenges final-value query-idxs round-layers
    by (rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain])
  have roots-len: length challenges = length roots
    by (rule generic-fri-partial-evidence-shapes(2)[OF partial])
  have layer-challenge-bound: layer-idx < length challenges
    using layer-bound roots-len by simp
  have no-arith:
    ¬ generic-fri-full-cover-arithmetic-shape-failure roots challenges
    using no-shape
  unfolding generic-fri-full-cover-shape-failure-def
    generic-fri-full-cover-arithmetic-shape-failure-def
  by blast
  have even-len:
    2 dvd fri-evidence-layer-len roots layer-idx
    using no-arith layer-challenge-bound
  unfolding generic-fri-full-cover-arithmetic-shape-failure-def by blast
  have roots-bound: layer-idx < length roots
    by (rule layer-bound)
  have len-eq:
    fri-evidence-layer-len roots layer-idx =

```

```

    length (fri-canonical-domain-at layer-idx)
  unfolding fri-evidence-layer-len-def fri-canonical-domain-at-length
  using fri-layer-lengths-nth-div[OF roots-bound, of clength * scale]
  by simp
have next-dom-eq:
  fri-canonical-domains (length challenges) ! Suc layer-idx =
    fri-canonical-domain-at (Suc layer-idx)
  by (rule fri-canonical-domains-nth)
    (use layer-challenge-bound in simp)
have next-len:
  length (fri-canonical-domains (length challenges) ! Suc layer-idx) =
    fri-evidence-layer-len roots layer-idx div 2
proof -
  have length (fri-canonical-domains (length challenges) ! Suc layer-idx) =
    (clength * scale) div 2 ^ Suc layer-idx
    unfolding next-dom-eq fri-canonical-domain-at-length by simp
  also have ... = ((clength * scale) div 2 ^ layer-idx) div 2
    by (metis div-mult2-eq mult.commute power-Suc)
  also have ... = fri-evidence-layer-len roots layer-idx div 2
    unfolding fri-evidence-layer-len-def
    using fri-layer-lengths-nth-div[OF roots-bound, of clength * scale]
    by simp
  finally show ?thesis .
qed
have raw-current:
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    length (fri-canonical-domain-at layer-idx)
  by (rule generic-fri-canonical-sampled-layer-chain-next-idx-current-bound
    [OF chain round-bound layer-challenge-bound])
have half-pos:
  0 < fri-evidence-layer-len roots layer-idx div 2
  using raw-current even-len unfolding len-eq by auto
have
  fri-evidence-next-idx roots query-idxs round-idx layer-idx <
    fri-evidence-layer-len roots layer-idx div 2
  unfolding fri-evidence-next-idx-def
  by (rule mod-less-divisor[OF half-pos])
then show ?thesis
  unfolding next-len .
qed

lemma generic-fri-canonical-sampled-domain-failure-imp-shape:
  assumes chain:
    generic-fri-sampled-layer-chain-evidence low-degree candidate-table
    degree-bound roots challenges final-value query-idxs round-layers
    (fri-canonical-domains (length challenges)) layers
  and sampled-domain:
    generic-fri-full-cover-sampled-domain-failure roots query-idxs
    (fri-canonical-domains (length challenges))

```


shows
generic-fri-full-cover-shape-failure roots challenges
(fri-canonical-domains (length challenges)) layers
proof (rule *ccontr*)
assume *no-shape*:
 $\neg$ *generic-fri-full-cover-shape-failure roots challenges*
(fri-canonical-domains (length challenges)) layers
have *partial*:
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers
by (rule *generic-fri-sampled-layer-chain-evidenceD(1)[OF chain]*)
have *roots-len*: *length challenges = length roots*
by (rule *generic-fri-partial-evidence-shapes(2)[OF partial]*)
have *roots-bound*: *length roots $\leq$ length challenges*
using *roots-len* **by** *simp*
have *no-sampled*:
 $\neg$ *generic-fri-full-cover-sampled-domain-failure roots query-idxs*
(fri-canonical-domains (length challenges))
proof (rule *fri-canonical-domains-no-sampled-domain-failure*
[OF roots-bound])
fix *round-idx layer-idx*
assume *round-bound*: *round-idx < length query-idxs*
and *layer-bound*: *layer-idx < length roots*
show *fri-evidence-next-idx roots query-idxs round-idx layer-idx <*
length (fri-canonical-domain-at layer-idx)
by (rule *generic-fri-canonical-sampled-layer-chain-next-idx-current-bound*
[OF chain round-bound])
(use layer-bound roots-len in simp)
next
fix *round-idx layer-idx*
assume *round-bound*: *round-idx < length query-idxs*
and *layer-bound*: *layer-idx < length roots*
show *fri-evidence-next-idx roots query-idxs round-idx layer-idx <*
length (fri-canonical-domains (length challenges) ! Suc layer-idx)
by (rule
generic-fri-canonical-sampled-layer-chain-next-idx-successor-bound
[OF chain no-shape round-bound layer-bound])
qed
show *False*
using *sampled-domain no-sampled* **by** *contradiction*
qed

lemma *generic-fri-canonical-domain-failure-imp-index-or-shape*:
assumes *chain*:
generic-fri-sampled-layer-chain-evidence low-degree candidate-table
degree-bound roots challenges final-value query-idxs round-layers
(fri-canonical-domains (length challenges)) layers
and *domain*:
generic-fri-full-cover-domain-failure roots challenges

(*fri-canonical-domains* (*length challenges*))
shows
generic-fri-full-cover-index-failure roots query-idxs $\vee$
generic-fri-full-cover-shape-failure roots challenges
 (*fri-canonical-domains* (*length challenges*)) *layers*
proof (*cases generic-fri-full-cover-index-failure roots query-idxs*)
case *True*
then show ?thesis **by** *simp*
next
case *no-index: False*
have *partial:*
generic-fri-partial-evidence low-degree candidate-table degree-bound
roots challenges final-value query-idxs round-layers
by (*rule generic-fri-sampled-layer-chain-evidenceD(1)[OF chain]*)
have *roots-len: length challenges = length roots*
by (*rule generic-fri-partial-evidence-shapes(2)[OF partial]*)
have *sampled-domain:*
generic-fri-full-cover-sampled-domain-failure roots query-idxs
 (*fri-canonical-domains* (*length challenges*))
using *generic-fri-full-cover-domain-failure-imp-index-or-sampled-domain*
 [*OF roots-len domain, of query-idxs*] *no-index*
by *blast*
then have *shape:*
generic-fri-full-cover-shape-failure roots challenges
 (*fri-canonical-domains* (*length challenges*)) *layers*
by (*rule generic-fri-canonical-sampled-domain-failure-imp-shape*
 [*OF chain*])
then show ?thesis **by** *simp*
qed

lemma *trace-fri-sampled-canonical-full-cover-failure-split-no-domain:*
assumes *trace-fri-sampled-canonical-full-cover-failure s bad out*
shows
trace-fri-sampled-full-cover-proximity-failure s bad out $\vee$
trace-fri-sampled-full-cover-index-failure s out $\vee$
trace-fri-sampled-full-cover-shape-failure s out $\vee$
trace-fri-sampled-full-cover-start-low-failure s out
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table layers
where *evidence:*
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr candidate-query-idxs trace-openings
trace-table
and *candidate:*

```

    generic-fri-sampled-query-candidate-evidence trace-table-low-degree
      (Not o trace-table-low-degree) trace-table (clength - 1)
      trace-roots trace-bs trace-final fri-query-idxs trace-round-layers
      layers
  and failure:
    generic-fri-sampled-full-cover-failure trace-table-low-degree
      trace-table (clength - 1) trace-roots trace-bs trace-final
      fri-query-idxs trace-round-layers
      (fri-canonical-domains (length trace-bs)) layers
      (bad trace-table)
  unfolding trace-fri-sampled-canonical-full-cover-failure-def by blast
have canonical:
  generic-fri-canonical-sampled-layer-chain-evidence
    trace-table-low-degree trace-table (clength - 1) trace-roots
    trace-bs trace-final fri-query-idxs trace-round-layers layers
  by (rule generic-fri-sampled-query-candidate-evidenceD(1)
    [OF candidate])
have chain:
  generic-fri-sampled-layer-chain-evidence trace-table-low-degree
    trace-table (clength - 1) trace-roots trace-bs trace-final
    fri-query-idxs trace-round-layers
    (fri-canonical-domains (length trace-bs)) layers
  by (rule generic-fri-canonical-sampled-layer-chain-evidenceD
    [OF canonical])
have cases:
  generic-fri-full-cover-proximity-failure (clength - 1) trace-bs
    (fri-canonical-domains (length trace-bs)) layers
    (bad trace-table) ∨
  generic-fri-full-cover-index-failure trace-roots fri-query-idxs ∨
  generic-fri-full-cover-shape-failure trace-roots trace-bs
    (fri-canonical-domains (length trace-bs)) layers ∨
  generic-fri-full-cover-domain-failure trace-roots trace-bs
    (fri-canonical-domains (length trace-bs)) ∨
  generic-fri-full-cover-start-low-failure (clength - 1)
    trace-table ∨
  generic-fri-full-cover-final-low-failure (clength - 1) trace-bs
    (fri-canonical-domains (length trace-bs)) layers
  by (rule generic-fri-sampled-full-cover-failure-cases[OF failure])
from cases show ?thesis
proof
  assume proximity:
    generic-fri-full-cover-proximity-failure (clength - 1) trace-bs
      (fri-canonical-domains (length trace-bs)) layers
      (bad trace-table)
  then show ?thesis
    unfolding trace-fri-sampled-full-cover-proximity-failure-def
    using evidence chain by blast
next
assume rest:

```

```

generic-fri-full-cover-index-failure trace-roots fri-query-idxs ∨
generic-fri-full-cover-shape-failure trace-roots trace-bs
  (fri-canonical-domains (length trace-bs)) layers ∨
generic-fri-full-cover-domain-failure trace-roots trace-bs
  (fri-canonical-domains (length trace-bs)) ∨
generic-fri-full-cover-start-low-failure (clength - 1)
  trace-table ∨
generic-fri-full-cover-final-low-failure (clength - 1) trace-bs
  (fri-canonical-domains (length trace-bs)) layers
then show ?thesis
proof
  assume index:
    generic-fri-full-cover-index-failure trace-roots fri-query-idxs
  then show ?thesis
    unfolding trace-fri-sampled-full-cover-index-failure-def
    using evidence chain by blast
next
  assume rest':
    generic-fri-full-cover-shape-failure trace-roots trace-bs
      (fri-canonical-domains (length trace-bs)) layers ∨
    generic-fri-full-cover-domain-failure trace-roots trace-bs
      (fri-canonical-domains (length trace-bs)) ∨
    generic-fri-full-cover-start-low-failure (clength - 1)
      trace-table ∨
    generic-fri-full-cover-final-low-failure (clength - 1) trace-bs
      (fri-canonical-domains (length trace-bs)) layers
  then show ?thesis
  proof
    assume shape:
      generic-fri-full-cover-shape-failure trace-roots trace-bs
        (fri-canonical-domains (length trace-bs)) layers
    then show ?thesis
      unfolding trace-fri-sampled-full-cover-shape-failure-def
      using evidence chain by blast
  next
    assume rest'':
      generic-fri-full-cover-domain-failure trace-roots trace-bs
        (fri-canonical-domains (length trace-bs)) ∨
      generic-fri-full-cover-start-low-failure (clength - 1)
        trace-table ∨
      generic-fri-full-cover-final-low-failure (clength - 1)
        trace-bs (fri-canonical-domains (length trace-bs)) layers
    then show ?thesis
    proof
      assume domain:
        generic-fri-full-cover-domain-failure trace-roots trace-bs
          (fri-canonical-domains (length trace-bs))
      have index-or-shape:
        generic-fri-full-cover-index-failure trace-roots

```

```

    fri-query-idxs  $\vee$ 
    generic-fri-full-cover-shape-failure trace-roots trace-bs
    (fri-canonical-domains (length trace-bs)) layers
  by (rule generic-fri-canonical-domain-failure-imp-index-or-shape
    [OF chain domain])
then show ?thesis
proof
  assume generic-fri-full-cover-index-failure trace-roots
    fri-query-idxs
  then show ?thesis
    unfolding trace-fri-sampled-full-cover-index-failure-def
    using evidence chain by blast
next
  assume generic-fri-full-cover-shape-failure trace-roots
    trace-bs (fri-canonical-domains (length trace-bs)) layers
  then show ?thesis
    unfolding trace-fri-sampled-full-cover-shape-failure-def
    using evidence chain by blast
qed
next
  assume rest''':
    generic-fri-full-cover-start-low-failure (clength - 1)
    trace-table  $\vee$ 
    generic-fri-full-cover-final-low-failure (clength - 1)
    trace-bs (fri-canonical-domains (length trace-bs)) layers
  then show ?thesis
  proof
    assume start:
      generic-fri-full-cover-start-low-failure (clength - 1)
      trace-table
    then show ?thesis
      unfolding trace-fri-sampled-full-cover-start-low-failure-def
      using evidence chain by blast
  next
    assume final:
      generic-fri-full-cover-final-low-failure (clength - 1)
      trace-bs (fri-canonical-domains (length trace-bs)) layers
    have final-event:
      trace-fri-sampled-full-cover-final-low-failure s out
    unfolding trace-fri-sampled-full-cover-final-low-failure-def
    using evidence chain final by blast
    then have shape-event:
      trace-fri-sampled-full-cover-shape-failure s out
    by (rule trace-fri-sampled-full-cover-final-low-imp-shape)
    then show ?thesis by simp
  qed
qed
qed
qed

```

qed
qed

lemma *composition-fri-sampled-canonical-full-cover-failure-split-no-domain:*

assumes *composition-fri-sampled-canonical-full-cover-failure s bad out*

shows

composition-fri-sampled-full-cover-proximity-failure s bad out $\vee$

composition-fri-sampled-full-cover-index-failure s out $\vee$

composition-fri-sampled-full-cover-shape-failure s out $\vee$

composition-fri-sampled-full-cover-start-low-failure s out

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*

opening-composition-roots composition-bs composition-final

fri-query-idxs trace-round-layers composition-round-layers fr

f-fri-roots f-final as dg composition-fri-roots final

trace-query-idxs trace-openings composition-query-idxs

composition-openings trace-table composition-table layers

where *evidence:*

composition-fri-partial-candidate-opening-evidence s out trace-roots

trace-bs trace-final fri-dg opening-composition-roots composition-bs

composition-final fri-query-idxs trace-round-layers

composition-round-layers fr f-fri-roots f-final as dg

composition-fri-roots final trace-query-idxs trace-openings

composition-query-idxs composition-openings trace-table

composition-table

and *candidate:*

generic-fri-sampled-query-candidate-evidence

(composition-table-low-degree (to-nat fri-dg))

(Not $\circ$ composition-table-low-degree maxDegree) composition-table

(to-nat fri-dg) opening-composition-roots composition-bs

composition-final fri-query-idxs composition-round-layers layers

and *failure:*

generic-fri-sampled-full-cover-failure

(composition-table-low-degree (to-nat fri-dg)) composition-table

(to-nat fri-dg) opening-composition-roots composition-bs

composition-final fri-query-idxs composition-round-layers

(fri-canonical-domains (length composition-bs)) layers

(bad fri-dg composition-table)

unfolding *composition-fri-sampled-canonical-full-cover-failure-def*

by *meson*

have *canonical:*

generic-fri-canonical-sampled-layer-chain-evidence

(composition-table-low-degree (to-nat fri-dg)) composition-table

(to-nat fri-dg) opening-composition-roots composition-bs

composition-final fri-query-idxs composition-round-layers layers

by *(rule generic-fri-sampled-query-candidate-evidenceD(1)*

[OF candidate])

have *chain:*

generic-fri-sampled-layer-chain-evidence

```

    (composition-table-low-degree (to-nat fri-dg)) composition-table
    (to-nat fri-dg) opening-composition-roots composition-bs
    composition-final fri-query-idxs composition-round-layers
    (fri-canonical-domains (length composition-bs)) layers
  by (rule generic-fri-canonical-sampled-layer-chain-evidenceD
      [OF canonical])
have cases:
  generic-fri-full-cover-proximity-failure (to-nat fri-dg)
    composition-bs (fri-canonical-domains (length composition-bs))
    layers (bad fri-dg composition-table) ∨
  generic-fri-full-cover-index-failure opening-composition-roots
    fri-query-idxs ∨
  generic-fri-full-cover-shape-failure opening-composition-roots
    composition-bs (fri-canonical-domains (length composition-bs))
    layers ∨
  generic-fri-full-cover-domain-failure opening-composition-roots
    composition-bs (fri-canonical-domains (length composition-bs)) ∨
  generic-fri-full-cover-start-low-failure (to-nat fri-dg)
    composition-table ∨
  generic-fri-full-cover-final-low-failure (to-nat fri-dg) composition-bs
    (fri-canonical-domains (length composition-bs)) layers
  by (rule generic-fri-sampled-full-cover-failure-cases[OF failure])
from cases show ?thesis
proof
  assume proximity:
    generic-fri-full-cover-proximity-failure (to-nat fri-dg)
      composition-bs (fri-canonical-domains (length composition-bs))
      layers (bad fri-dg composition-table)
  then show ?thesis
    unfolding composition-fri-sampled-full-cover-proximity-failure-def
    using evidence chain by meson
next
assume rest:
  generic-fri-full-cover-index-failure opening-composition-roots
    fri-query-idxs ∨
  generic-fri-full-cover-shape-failure opening-composition-roots
    composition-bs (fri-canonical-domains (length composition-bs))
    layers ∨
  generic-fri-full-cover-domain-failure opening-composition-roots
    composition-bs (fri-canonical-domains (length composition-bs)) ∨
  generic-fri-full-cover-start-low-failure (to-nat fri-dg)
    composition-table ∨
  generic-fri-full-cover-final-low-failure (to-nat fri-dg)
    composition-bs (fri-canonical-domains (length composition-bs))
    layers
then show ?thesis
proof
  assume index:
    generic-fri-full-cover-index-failure opening-composition-roots

```

```

    fri-query-idxs
  then show ?thesis
    unfolding composition-fri-sampled-full-cover-index-failure-def
    using evidence chain by meson
next
assume rest':
  generic-fri-full-cover-shape-failure opening-composition-roots
    composition-bs (fri-canonical-domains (length composition-bs))
    layers ∨
  generic-fri-full-cover-domain-failure opening-composition-roots
    composition-bs (fri-canonical-domains (length composition-bs)) ∨
  generic-fri-full-cover-start-low-failure (to-nat fri-dg)
    composition-table ∨
  generic-fri-full-cover-final-low-failure (to-nat fri-dg)
    composition-bs (fri-canonical-domains (length composition-bs))
    layers
  then show ?thesis
  proof
    assume shape:
      generic-fri-full-cover-shape-failure opening-composition-roots
        composition-bs (fri-canonical-domains (length composition-bs))
        layers
    then show ?thesis
      unfolding composition-fri-sampled-full-cover-shape-failure-def
      using evidence chain by meson
  next
  assume rest'':
    generic-fri-full-cover-domain-failure opening-composition-roots
      composition-bs (fri-canonical-domains (length composition-bs)) ∨
    generic-fri-full-cover-start-low-failure (to-nat fri-dg)
      composition-table ∨
    generic-fri-full-cover-final-low-failure (to-nat fri-dg)
      composition-bs (fri-canonical-domains (length composition-bs))
      layers
    then show ?thesis
    proof
      assume domain:
        generic-fri-full-cover-domain-failure opening-composition-roots
          composition-bs (fri-canonical-domains (length composition-bs))
      have index-or-shape:
        generic-fri-full-cover-index-failure opening-composition-roots
          fri-query-idxs ∨
        generic-fri-full-cover-shape-failure opening-composition-roots
          composition-bs (fri-canonical-domains (length composition-bs))
          layers
      by (rule generic-fri-canonical-domain-failure-imp-index-or-shape
        [OF chain domain])
    then show ?thesis
    proof

```



```

    assume generic-fri-full-cover-index-failure
      opening-composition-roots fri-query-idxs
    then show ?thesis
      unfolding composition-fri-sampled-full-cover-index-failure-def
      using evidence chain by meson
  next
    assume generic-fri-full-cover-shape-failure
      opening-composition-roots composition-bs
      (fri-canonical-domains (length composition-bs)) layers
    then show ?thesis
      unfolding composition-fri-sampled-full-cover-shape-failure-def
      using evidence chain by meson
  qed
next
  assume rest''':
    generic-fri-full-cover-start-low-failure (to-nat fri-dg)
    composition-table ∨
    generic-fri-full-cover-final-low-failure (to-nat fri-dg)
    composition-bs (fri-canonical-domains (length composition-bs))
    layers
  then show ?thesis
  proof
    assume start:
      generic-fri-full-cover-start-low-failure (to-nat fri-dg)
      composition-table
    then show ?thesis
      unfolding
        composition-fri-sampled-full-cover-start-low-failure-def
      using evidence chain by meson
  next
    assume final:
      generic-fri-full-cover-final-low-failure (to-nat fri-dg)
      composition-bs (fri-canonical-domains (length composition-bs))
      layers
    have final-event:
      composition-fri-sampled-full-cover-final-low-failure s out
    unfolding
      composition-fri-sampled-full-cover-final-low-failure-def
    using evidence chain final by meson
    then have shape-event:
      composition-fri-sampled-full-cover-shape-failure s out
      by (rule composition-fri-sampled-full-cover-final-low-imp-shape)
    then show ?thesis by simp
  qed
qed
qed
qed
qed
qed
qed

```

lemma *wp-trace-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases:*

assumes *proximity:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$

and *index:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-index-failure } s) \ s \leq I$

and *shape:*

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-shape-failure } s) \ s \leq Sh$

shows

wp-event verify-monad

$(\text{trace-fri-sampled-canonical-full-cover-failure } s \text{ bad}) \ s \leq P + I + Sh$

proof –

have *wp-event verify-monad*

$(\text{trace-fri-sampled-canonical-full-cover-failure } s \text{ bad}) \ s$

$\leq \text{wp-event verify-monad}$

$(\lambda out.$

$\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad out} \vee$

$\text{trace-fri-sampled-full-cover-index-failure } s \text{ out} \vee$

$\text{trace-fri-sampled-full-cover-shape-failure } s \text{ out} \vee$

$\text{trace-fri-sampled-full-cover-start-low-failure } s \text{ out}) \ s$

by $(\text{rule wp-event-mono})$

$(\text{rule trace-fri-sampled-canonical-full-cover-failure-split-no-domain})$

also have $\dots \leq$

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s +$

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-index-failure } s) \ s +$

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-shape-failure } s) \ s +$

wp-event verify-monad

$(\text{trace-fri-sampled-full-cover-start-low-failure } s) \ s$

by $(\text{rule wp-event-union-bound4})$

also have $\dots \leq P + I + Sh + 0$

using *proximity index shape*

wp-trace-fri-sampled-full-cover-start-low-failure-zero[of *s*]

by (intro add-mono)

finally show *?thesis*

by *simp*

qed

lemma *wp-composition-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases:*

assumes *proximity:*

wp-event verify-monad

$(\text{composition-fri-sampled-full-cover-proximity-failure } s \text{ bad}) \ s \leq P$

and *index:*

wp-event verify-monad
 (composition-fri-sampled-full-cover-index-failure s) $s \leq I$
and shape:
wp-event verify-monad
 (composition-fri-sampled-full-cover-shape-failure s) $s \leq Sh$
shows
wp-event verify-monad
 (composition-fri-sampled-canonical-full-cover-failure s bad) s
 $\leq P + I + Sh$
proof –
have *wp-event verify-monad*
 (composition-fri-sampled-canonical-full-cover-failure s bad) s
 $\leq$ *wp-event verify-monad*
 ($\lambda out.$
 composition-fri-sampled-full-cover-proximity-failure s bad out $\vee$
 composition-fri-sampled-full-cover-index-failure s out $\vee$
 composition-fri-sampled-full-cover-shape-failure s out $\vee$
 composition-fri-sampled-full-cover-start-low-failure s out) s
by (rule *wp-event-mono*)
 (rule
 composition-fri-sampled-canonical-full-cover-failure-split-no-domain)
also have ... $\leq$
wp-event verify-monad
 (composition-fri-sampled-full-cover-proximity-failure s bad) s +
wp-event verify-monad
 (composition-fri-sampled-full-cover-index-failure s) s +
wp-event verify-monad
 (composition-fri-sampled-full-cover-shape-failure s) s +
wp-event verify-monad
 (composition-fri-sampled-full-cover-start-low-failure s) s
by (rule *wp-event-union-bound4*)
also have ... $\leq P + I + Sh + 0$
using *proximity index shape*
wp-composition-fri-sampled-full-cover-start-low-failure-zero[of s]
by (*intro add-mono*)
finally show *?thesis*
by *simp*
qed

lemma *wp-trace-fri-sampled-canonical-full-cover-failure-bound-from-witness-shape:*
assumes *proximity:*
wp-event verify-monad
 (trace-fri-sampled-full-cover-proximity-failure s bad) $s \leq P$
and *index:*
wp-event verify-monad
 (trace-fri-sampled-full-cover-index-failure s) $s \leq I$
and *witness:*
wp-event verify-monad
 (trace-fri-sampled-full-cover-witness-length-failure s) $s \leq W$

shows

wp-event verify-monad
(*trace-fri-sampled-canonical-full-cover-failure s bad*) *s*
 $\leq P + I + W$

by (*rule*

wp-trace-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases
[*OF proximity index*])
(*rule wp-trace-fri-sampled-full-cover-shape-bound-from-witness-length*
[*OF witness*])

lemma *wp-composition-fri-sampled-canonical-full-cover-failure-bound-from-witness-shape:*

assumes *proximity:*

wp-event verify-monad
(*composition-fri-sampled-full-cover-proximity-failure s bad*) *s* $\leq P$

and *index:*

wp-event verify-monad
(*composition-fri-sampled-full-cover-index-failure s*) *s* $\leq I$

and *witness:*

wp-event verify-monad
(*composition-fri-sampled-full-cover-witness-length-failure s*) *s*
 $\leq W$

shows

wp-event verify-monad
(*composition-fri-sampled-canonical-full-cover-failure s bad*) *s*
 $\leq P + I + W$

by (*rule*

wp-composition-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases
[*OF proximity index*])
(*rule*
wp-composition-fri-sampled-full-cover-shape-bound-from-witness-length
[*OF witness*])

lemma *checked-staged-security-trace-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases:*

assumes *proximity:*

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒

wp-event verify-monad
(*trace-fri-sampled-full-cover-proximity-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) *bad*)
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq P$

and *index:*

$\bigwedge$ *data attacker-state.*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state) ⇒

```

wp-event verify-monad
  (trace-fri-sampled-full-cover-index-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ I
and shape:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
wp-event verify-monad
  (trace-fri-sampled-full-cover-shape-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)) ≤ Sh
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. trace-fri-sampled-canonical-full-cover-failure s bad))
    adversary-initial-state ≤ P + I + Sh
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show
  ∧ s. ¬ trace-fri-sampled-canonical-full-cover-failure s bad None
by simp
next
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
  (trace-fri-sampled-canonical-full-cover-failure ?s bad) ?s
  ≤ P + I + Sh
by (rule
  wp-trace-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases)
(rule proximity[OF support], rule index[OF support],
  rule shape[OF support])
qed

lemma checked-staged-security-trace-fri-sampled-canonical-full-cover-failure-bound-from-witness-shape:
  assumes proximity:
    ∧ data attacker-state.
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)

```

```

    adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (trace-fri-sampled-full-cover-proximity-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad)
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq P$ 
and index:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (trace-fri-sampled-full-cover-index-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq I$ 
and witness:
 $\wedge$ data attacker-state.
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)  $\implies$ 
wp-event verify-monad
  (trace-fri-sampled-full-cover-witness-length-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))  $\leq W$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s$ . trace-fri-sampled-canonical-full-cover-failure s bad))
    adversary-initial-state  $\leq P + I + W$ 
proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
show
   $\wedge s$ .  $\neg$  trace-fri-sampled-canonical-full-cover-failure s bad None
by simp
next
fix data attacker-state
assume support:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
  (trace-fri-sampled-canonical-full-cover-failure ?s bad) ?s

```

$\leq P + I + W$
by (rule
wp-trace-fri-sampled-canonical-full-cover-failure-bound-from-witness-shape)
(rule *proximity*[*OF support*], rule *index*[*OF support*],
rule *witness*[*OF support*])
qed

lemma *checked-staged-security-composition-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases*:
assumes *proximity*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-sampled-full-cover-proximity-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) *bad*)
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq P$
and *index*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-sampled-full-cover-index-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq I$
and *shape*:
 $\wedge$ *data attacker-state*.
Some (*data*, *attacker-state*) $\in$
set-dist (*execute* (*checked-staged-transcript-program A*)
adversary-initial-state) $\implies$
wp-event verify-monad
(*composition-fri-sampled-full-cover-shape-failure*
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)))
(*verifier-state-from-adversary attacker-state*
(*staged-proof-transcript data*)) $\leq Sh$
shows
wp-event (*checked-staged-security-experiment-with-data-state A*)
(*staged-security-with-data-state-verifier-event*
($\lambda s.$ *composition-fri-sampled-canonical-full-cover-failure s bad*))
adversary-initial-state $\leq P + I + Sh$
proof (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
show
 $\wedge s. \neg$ *composition-fri-sampled-canonical-full-cover-failure s bad*

```

    None
  by simp
next
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
let ?s =
  verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
show wp-event verify-monad
  (composition-fri-sampled-canonical-full-cover-failure ?s bad) ?s
  ≤ P + I + Sh
by (rule
  wp-composition-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases)
  (rule proximity[OF support], rule index[OF support],
  rule shape[OF support])
qed

```

lemma *checked-staged-security-composition-fri-sampled-canonical-full-cover-failure-bound-from-witness-shape:*

```

assumes proximity:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  wp-event verify-monad
    (composition-fri-sampled-full-cover-proximity-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)) bad)
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ P
and index:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  wp-event verify-monad
    (composition-fri-sampled-full-cover-index-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ I
and witness:
  ∧ data attacker-state.
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state) ⇒
  wp-event verify-monad

```



```

    (composition-fri-sampled-full-cover-witness-length-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) ≤ W
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        (λs. composition-fri-sampled-canonical-full-cover-failure s bad))
    adversary-initial-state ≤ P + I + W
  proof (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  show
    ∧s. ¬ composition-fri-sampled-canonical-full-cover-failure s bad
    None
  by simp
next
  fix data attacker-state
  assume support:
    Some (data, attacker-state) ∈
      set-dist (execute (checked-staged-transcript-program A)
        adversary-initial-state)
  let ?s =
    verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)
  show wp-event verify-monad
    (composition-fri-sampled-canonical-full-cover-failure ?s bad) ?s
    ≤ P + I + W
  by (rule
    wp-composition-fri-sampled-canonical-full-cover-failure-bound-from-witness-shape)
    (rule proximity[OF support], rule index[OF support],
    rule witness[OF support])
qed

lemma trace-fri-sampled-query-bad-candidate-imp-no-full-cover-or-canonical-failure:
  assumes sampled: trace-fri-sampled-query-bad-candidate s out
  shows
    trace-fri-sampled-query-no-full-cover-failure-candidate s bad out ∨
    trace-fri-sampled-canonical-full-cover-failure s bad out
  proof -
    from sampled obtain trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final fri-query-idxs trace-round-layers
      composition-round-layers fr candidate-query-idxs trace-openings
      trace-table layers
    where evidence:
      trace-fri-partial-candidate-opening-evidence s out trace-roots
      trace-bs trace-final dg composition-roots composition-bs
      composition-final fri-query-idxs trace-round-layers
      composition-round-layers fr candidate-query-idxs trace-openings
      trace-table
  
```

and *candidate*:
generic-fri-sampled-query-candidate-evidence trace-table-low-degree
(Not $\circ$ trace-table-low-degree) trace-table (clength $- 1$)
trace-roots trace-bs trace-final fri-query-idxs trace-round-layers
layers
unfolding *trace-fri-sampled-query-bad-candidate-def* **by** *blast*
show *?thesis*
proof (*cases*
generic-fri-sampled-full-cover-failure trace-table-low-degree
trace-table (clength $- 1$) trace-roots trace-bs trace-final
fri-query-idxs trace-round-layers
(fri-canonical-domains (length trace-bs)) layers
(bad trace-table))
case *False*
then have
trace-fri-sampled-query-no-full-cover-failure-candidate s bad out
unfolding *trace-fri-sampled-query-no-full-cover-failure-candidate-def*
using *evidence candidate* **by** *blast*
then show *?thesis* **by** *simp*
next
case *True*
then have
trace-fri-sampled-canonical-full-cover-failure s bad out
unfolding *trace-fri-sampled-canonical-full-cover-failure-def*
using *evidence candidate* **by** *blast*
then show *?thesis* **by** *simp*
qed
qed

lemma *composition-fri-sampled-query-bad-candidate-imp-no-full-cover-or-canonical-failure*:
assumes *sampled: composition-fri-sampled-query-bad-candidate s out*
shows
composition-fri-sampled-query-no-full-cover-failure-candidate s bad out $\vee$
composition-fri-sampled-canonical-full-cover-failure s bad out
proof —
from *sampled* **obtain** *trace-roots trace-bs trace-final fri-dg*
opening-composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers fr
f-fri-roots f-final as dg composition-fri-roots final
trace-query-idxs trace-openings composition-query-idxs
composition-openings trace-table composition-table layers
where *evidence*:
composition-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final fri-dg opening-composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr f-fri-roots f-final as dg
composition-fri-roots final trace-query-idxs trace-openings
composition-query-idxs composition-openings trace-table
composition-table

and *candidate*:
generic-fri-sampled-query-candidate-evidence
 (*composition-table-low-degree* (*to-nat fri-dg*))
 (*Not* $\circ$ *composition-table-low-degree maxDegree*) *composition-table*
 (*to-nat fri-dg*) *opening-composition-roots composition-bs*
composition-final fri-query-idxs composition-round-layers layers
unfolding *composition-fri-sampled-query-bad-candidate-def* **by** *meson*
show *?thesis*
proof (*cases*
generic-fri-sampled-full-cover-failure
 (*composition-table-low-degree* (*to-nat fri-dg*)) *composition-table*
 (*to-nat fri-dg*) *opening-composition-roots composition-bs*
composition-final fri-query-idxs composition-round-layers
 (*fri-canonical-domains* (*length composition-bs*)) *layers*
 (*bad fri-dg composition-table*))
case *False*
then have
composition-fri-sampled-query-no-full-cover-failure-candidate s bad out
unfolding
composition-fri-sampled-query-no-full-cover-failure-candidate-def
using *evidence candidate* **by** *meson*
then show *?thesis* **by** *simp*
next
case *True*
then have
composition-fri-sampled-canonical-full-cover-failure s bad out
unfolding *composition-fri-sampled-canonical-full-cover-failure-def*
using *evidence candidate* **by** *meson*
then show *?thesis* **by** *simp*
qed
qed

lemma *wp-trace-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-canonical-failure*:
assumes *no-failure-bound*:
wp-event verify-monad
 (*trace-fri-sampled-query-no-full-cover-failure-candidate s bad*) *s*
 $\leq N$
and *failure-bound*:
wp-event verify-monad
 (*trace-fri-sampled-canonical-full-cover-failure s bad*) *s* $\leq F$
shows
wp-event verify-monad (*trace-fri-sampled-query-bad-candidate s*) *s*
 $\leq N + F$
proof –
have *wp-event verify-monad* (*trace-fri-sampled-query-bad-candidate s*) *s*
 $\leq$ *wp-event verify-monad*
 ($\lambda out.$
trace-fri-sampled-query-no-full-cover-failure-candidate s bad out $\vee$
trace-fri-sampled-canonical-full-cover-failure s bad out) *s*

by (rule wp-event-mono)
 (rule
 trace-fri-sampled-query-bad-candidate-imp-no-full-cover-or-canonical-failure)
 also have ... $\leq$
 wp-event verify-monad
 (trace-fri-sampled-query-no-full-cover-failure-candidate s bad) s +
 wp-event verify-monad
 (trace-fri-sampled-canonical-full-cover-failure s bad) s
 by (rule wp-event-union-bound)
 also have ... $\leq N + F$
 by (rule add-mono[OF no-failure-bound failure-bound])
 finally show ?thesis .
 qed

lemma wp-composition-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-canonical-failure:
 assumes no-failure-bound:
 wp-event verify-monad
 (composition-fri-sampled-query-no-full-cover-failure-candidate s bad) s
 $\leq N$
 and failure-bound:
 wp-event verify-monad
 (composition-fri-sampled-canonical-full-cover-failure s bad) s $\leq F$
 shows
 wp-event verify-monad (composition-fri-sampled-query-bad-candidate s) s
 $\leq N + F$
 proof –
 have wp-event verify-monad (composition-fri-sampled-query-bad-candidate s) s
 $\leq$ wp-event verify-monad
 (λ out.
 composition-fri-sampled-query-no-full-cover-failure-candidate s bad
 out $\vee$
 composition-fri-sampled-canonical-full-cover-failure s bad out) s
 by (rule wp-event-mono)
 (rule
 composition-fri-sampled-query-bad-candidate-imp-no-full-cover-or-canonical-failure)
 also have ... $\leq$
 wp-event verify-monad
 (composition-fri-sampled-query-no-full-cover-failure-candidate s bad)
 s +
 wp-event verify-monad
 (composition-fri-sampled-canonical-full-cover-failure s bad) s
 by (rule wp-event-union-bound)
 also have ... $\leq N + F$
 by (rule add-mono[OF no-failure-bound failure-bound])
 finally show ?thesis .
 qed

lemma checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-canonical-failure:
 assumes no-failure-bound:

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. trace-fri-sampled-query-no-full-cover-failure-candidate s bad))
adversary-initial-state ≤ N
and failure-bound:
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. trace-fri-sampled-canonical-full-cover-failure s bad))
adversary-initial-state ≤ F
shows
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-sampled-query-bad-candidate)
adversary-initial-state ≤ N + F
proof –
have wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-sampled-query-bad-candidate)
adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (λout.
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-sampled-query-no-full-cover-failure-candidate s
        bad) out ∨
    staged-security-with-data-state-verifier-event
      (λs. trace-fri-sampled-canonical-full-cover-failure s bad) out)
adversary-initial-state
by (rule wp-event-mono)
  (auto simp: staged-security-with-data-state-verifier-event-def
    dest:
      trace-fri-sampled-query-bad-candidate-imp-no-full-cover-or-canonical-failure
      split: option.splits prod.splits)
also have ... ≤
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. trace-fri-sampled-query-no-full-cover-failure-candidate s bad))
adversary-initial-state +
wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    (λs. trace-fri-sampled-canonical-full-cover-failure s bad))
adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤ N + F
by (rule add-mono[OF no-failure-bound failure-bound])
finally show ?thesis .
qed

lemma checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-canonical-f
assumes no-failure-bound:

```

$wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 ($\lambda s.$
 $composition\text{-}fri\text{-}sampled\text{-}query\text{-}no\text{-}full\text{-}cover\text{-}failure\text{-}candidate$ s
 bad))
 $adversary\text{-}initial\text{-}state \leq N$
and $failure\text{-}bound$:
 $wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 ($\lambda s.$ $composition\text{-}fri\text{-}sampled\text{-}canonical\text{-}full\text{-}cover\text{-}failure$ s bad))
 $adversary\text{-}initial\text{-}state \leq F$
shows
 $wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $composition\text{-}fri\text{-}sampled\text{-}query\text{-}bad\text{-}candidate$)
 $adversary\text{-}initial\text{-}state \leq N + F$
proof –
have $wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 $composition\text{-}fri\text{-}sampled\text{-}query\text{-}bad\text{-}candidate$)
 $adversary\text{-}initial\text{-}state \leq$
 $wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($\lambda out.$
 $staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 ($\lambda s.$
 $composition\text{-}fri\text{-}sampled\text{-}query\text{-}no\text{-}full\text{-}cover\text{-}failure\text{-}candidate$ s
 bad) $out \vee$
 $staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 ($\lambda s.$ $composition\text{-}fri\text{-}sampled\text{-}canonical\text{-}full\text{-}cover\text{-}failure$ s bad)
 out)
 $adversary\text{-}initial\text{-}state$
by ($rule$ $wp\text{-}event\text{-}mono$)
 ($auto$ $simp$: $staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event\text{-}def$
 $dest$:
 $composition\text{-}fri\text{-}sampled\text{-}query\text{-}bad\text{-}candidate\text{-}imp\text{-}no\text{-}full\text{-}cover\text{-}or\text{-}canonical\text{-}failure$
 $split$: $option.splits$ $prod.splits$)
also have $\dots \leq$
 $wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 ($\lambda s.$
 $composition\text{-}fri\text{-}sampled\text{-}query\text{-}no\text{-}full\text{-}cover\text{-}failure\text{-}candidate$ s
 bad))
 $adversary\text{-}initial\text{-}state +$
 $wp\text{-}event$ ($checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state$ A)
 ($staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event$
 ($\lambda s.$ $composition\text{-}fri\text{-}sampled\text{-}canonical\text{-}full\text{-}cover\text{-}failure$ s bad))
 $adversary\text{-}initial\text{-}state$
by ($rule$ $wp\text{-}event\text{-}union\text{-}bound$)
also have $\dots \leq N + F$

```

    by (rule add-mono[OF no-failure-bound failure-bound])
  finally show ?thesis .
qed

end

end

```

```

theory Soundness-FRI-Trace-Transcript-Consistent
imports Soundness-FRI-Trace-Candidate-Blocker
begin

```

Trace-only verifier-consumed FRI evidence.

The broad reachable trace FRI predicate is root-parametric: it accepts any authenticated trace root present in the final verifier state. This layer names the verifier-consumed alternative, where the trace root and query indices are those appearing in the accepted FRI opening transcript. It does not change the protocol or the public theorem interface.

```

context soundness
begin

```

definition *accepted-with-partial-trace-openings-verifier-consumed*

```

:: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
  (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$ 
  'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  'f  $\Rightarrow$ 
  'f list  $\Rightarrow$  'f list  $\Rightarrow$  'f  $\Rightarrow$  nat list  $\Rightarrow$ 
  'f list list list  $\Rightarrow$  'f list list list  $\Rightarrow$ 
  'f  $\Rightarrow$  'f list  $\Rightarrow$ 
  'f authenticated-opening list list  $\Rightarrow$  bool

```

where

```

accepted-with-partial-trace-openings-verifier-consumed s out
  trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs trace-round-layers
  composition-round-layers fr as trace-openings  $\longleftrightarrow$ 
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
  dg composition-roots composition-bs composition-final query-idxs
  trace-round-layers composition-round-layers  $\wedge$ 
( $\exists$  rest.
  verifier-header-transcript s fr trace-roots trace-final as dg
  composition-roots composition-final rest)  $\wedge$ 
accepted-with-partial-trace-openings s out fr query-idxs trace-openings

```

lemma *accepted-with-partial-trace-openings-verifier-consumedD:*

assumes

```

accepted-with-partial-trace-openings-verifier-consumed s out
  trace-roots trace-bs trace-final dg composition-roots composition-bs
  composition-final query-idxs trace-round-layers
  composition-round-layers fr as trace-openings

```

shows

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
dg composition-roots composition-bs composition-final query-idxs
trace-round-layers composition-round-layers
 $\exists$ *rest. verifier-header-transcript s fr trace-roots trace-final as dg*
composition-roots composition-final rest
accepted-with-partial-trace-openings s out fr query-idxs trace-openings
using *assms*
unfolding *accepted-with-partial-trace-openings-verifier-consumed-def*
by *simp-all*

definition *trace-fri-bad-with-verifier-consumed-partial-candidate*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-bad-with-verifier-consumed-partial-candidate s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings trace-table.}$
accepted-with-partial-trace-openings-verifier-consumed s out
 $\text{ trace-roots trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings} \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\neg \text{ trace-table-low-degree trace-table})$

lemma *trace-fri-bad-with-verifier-consumed-imp-header-tied:*

assumes *trace-fri-bad-with-verifier-consumed-partial-candidate s out*

shows *trace-fri-bad-with-header-tied-partial-candidate s out*

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings trace-table}$ **where**

consumed:

accepted-with-partial-trace-openings-verifier-consumed s out
 $\text{ trace-roots trace-bs trace-final dg composition-roots composition-bs}$
 $\text{ composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings}$

and *candidate:*

$\text{ partial-trace-table-candidate trace-table trace-openings}$

and *not-low:* $\neg \text{ trace-table-low-degree trace-table}$

unfolding *trace-fri-bad-with-verifier-consumed-partial-candidate-def*

by *blast*

have *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
 $\text{ dg composition-roots composition-bs composition-final fri-query-idxs}$
 $\text{ trace-round-layers composition-round-layers}$

by (*rule* *accepted-with-partial-trace-openings-verifier-consumedD(1)*
 $[OF \text{ consumed}]$)

from *accepted-with-partial-trace-openings-verifier-consumedD(2)[OF consumed]*
obtain *rest* **where** *header:*
 verifier-header-transcript s fr trace-roots trace-final as dg
 composition-roots composition-final rest
by *blast*
have *partial:*
 accepted-with-partial-trace-openings s out fr fri-query-idxs
 trace-openings
by (rule *accepted-with-partial-trace-openings-verifier-consumedD(3)*
 [*OF consumed*])
show *?thesis*
 unfolding *trace-fri-bad-with-header-tied-partial-candidate-def*
by (intro *exI conjI*)
 (rule *fri-openings*, rule *header*, rule *partial*, rule *candidate*,
 rule *not-low*)
qed

lemma *trace-fri-bad-with-header-tied-imp-verifier-consumed:*
 assumes *trace-fri-bad-with-header-tied-partial-candidate s out*
 shows *trace-fri-bad-with-verifier-consumed-partial-candidate s out*
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr as rest trace-openings trace-table **where**
 fri-openings:
 accepted-fri-opening-transcript s out trace-roots trace-bs
 trace-final dg composition-roots composition-bs composition-final
 fri-query-idxs trace-round-layers composition-round-layers
and *header:*
 verifier-header-transcript s fr trace-roots trace-final as dg
 composition-roots composition-final rest
and *partial:*
 accepted-with-partial-trace-openings s out fr fri-query-idxs
 trace-openings
and *candidate:*
 partial-trace-table-candidate trace-table trace-openings
and *not-low: $\neg$ trace-table-low-degree trace-table*
unfolding *trace-fri-bad-with-header-tied-partial-candidate-def*
by *blast*
have *consumed:*
 accepted-with-partial-trace-openings-verifier-consumed s out
 trace-roots trace-bs trace-final dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr as trace-openings
unfolding *accepted-with-partial-trace-openings-verifier-consumed-def*
by (intro *conjI exI*)
 (rule *fri-openings*, rule *header*, rule *partial*)
show *?thesis*
 unfolding *trace-fri-bad-with-verifier-consumed-partial-candidate-def*

by (intro exI conjI)
 (rule consumed, rule candidate, rule not-low)
 qed

lemma *trace-fri-bad-with-verifier-consumed-iff-header-tied:*
trace-fri-bad-with-verifier-consumed-partial-candidate s out
 $\longleftrightarrow$ *trace-fri-bad-with-header-tied-partial-candidate s out*
proof
 assume *trace-fri-bad-with-verifier-consumed-partial-candidate s out*
 then show *trace-fri-bad-with-header-tied-partial-candidate s out*
 by (rule *trace-fri-bad-with-verifier-consumed-imp-header-tied*)
 next
 assume *trace-fri-bad-with-header-tied-partial-candidate s out*
 then show *trace-fri-bad-with-verifier-consumed-partial-candidate s out*
 by (rule *trace-fri-bad-with-header-tied-imp-verifier-consumed*)
 qed

lemma *trace-fri-header-tied-reduction-verifier-consumed-bound:*
 assumes *trace-fri-header-tied-reduction s*
 shows
wp-event verify-mono
 (*trace-fri-bad-with-verifier-consumed-partial-candidate s*) $s \leq$
trace-fri-error
proof –
 have *mono:*
wp-event verify-mono
 (*trace-fri-bad-with-verifier-consumed-partial-candidate s*) $s \leq$
wp-event verify-mono
 (*trace-fri-bad-with-header-tied-partial-candidate s*) s
 by (rule *wp-event-mono*)
 (rule *trace-fri-bad-with-verifier-consumed-imp-header-tied*)
 also have $\dots \leq$ *trace-fri-error*
 by (rule *trace-fri-header-tied-reductionD[OF assms]*)
 finally show ?thesis .
 qed

lemma *trace-fri-verifier-consumed-reduction-staged-bound:*
 assumes *reductions:*
 $\bigwedge$ *data attacker-state.*
 Some (*data, attacker-state*) $\in$
 set-dist (*execute (checked-staged-transcript-program A)*)
adversary-initial-state $\implies$
trace-fri-header-tied-reduction
 (*verifier-state-from-adversary attacker-state*)
 (*staged-proof-transcript data*))
 shows
wp-event (checked-staged-security-experiment-with-data-state A)
 (*staged-security-with-data-state-verifier-event*)
 (*trace-fri-bad-with-verifier-consumed-partial-candidate*)

```

    adversary-initial-state ≤ trace-fri-error
proof –
  have mono:
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-verifier-consumed-partial-candidate)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume consumed:
    staged-security-with-data-state-verifier-event
    trace-fri-bad-with-verifier-consumed-partial-candidate out
  then show
    staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-partial-candidate out
proof (cases out)
  case None
  then show ?thesis
    using consumed
    unfolding staged-security-with-data-state-verifier-event-def
    by simp
next
  case (Some payload)
  obtain data attacker-state result final-state where payload:
    payload = (((data, attacker-state), result), final-state)
    by (cases payload) auto
  have consumed':
    trace-fri-bad-with-verifier-consumed-partial-candidate
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result, final-state))
    using consumed Some payload
    unfolding staged-security-with-data-state-verifier-event-def
    by simp
  have header:
    trace-fri-bad-with-header-tied-partial-candidate
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data))
      (Some (result, final-state))
    by (rule trace-fri-bad-with-verifier-consumed-imp-header-tied
      [OF consumed'])
  show ?thesis
    using Some payload header
    unfolding staged-security-with-data-state-verifier-event-def
    by simp

```

```

    qed
  qed
  also have ...  $\leq$  trace-fri-error
    by (rule trace-fri-header-tied-reduction-staged-bound[OF reductions])
  finally show ?thesis .
qed

end

end

```

```

theory Soundness-FRI-Trace-Empty-Sampled-Assignment-Split
imports
  Soundness-Public-Route-Concrete-FRI-Sharpened
  Soundness-FRI-Canonical-Full-Cover-Failure
  Soundness-FRI-Trace-Transcript-Consistent
  Soundness-Conceptual-Query-Empty-Current
begin

```

First split for the remaining plain trace empty-branch sampled-assignment conflict. The plain event is intentionally broader than the low-degree sampled-query bad-candidate event: it does not require the conflicting candidate table to be non-low-degree. This layer separates the aligned verifier-tied part from the residual that still needs a Merkle/replay explanation.

```

context soundness
begin

```

```

lemma prob-le-refl:  $(x::prob) \leq x$ 
  by transfer simp

```

```

definition trace-fri-sampled-assignment-verifier-untied-gap
  :: ('f, 'a) protocol-channel-scheme  $\Rightarrow$ 
    (unit list  $\times$  ('f, 'a) protocol-channel-scheme) option  $\Rightarrow$  bool
where
  trace-fri-sampled-assignment-verifier-untied-gap s out  $\longleftrightarrow$ 
    trace-fri-sampled-assignment-conflict s out  $\wedge$ 
     $\neg$  trace-fri-verifier-tied-sampled-assignment-conflict s out

```

```

lemma trace-fri-sampled-assignment-conflict-imp-verifier-tied-or-untied:
  assumes trace-fri-sampled-assignment-conflict s out
  shows
    trace-fri-verifier-tied-sampled-assignment-conflict s out  $\vee$ 
    trace-fri-sampled-assignment-verifier-untied-gap s out
  using assms
  unfolding trace-fri-sampled-assignment-verifier-untied-gap-def by blast

```

```

lemma wp-trace-fri-sampled-assignment-conflict-bound-from-verifier-tied-and-untied:
  assumes verifier-tied-bound:

```

$wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-verifier-tied-sampled-assignment-conflict } s) \ s \leq V$
and $untied\text{-bound}$:
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-assignment-verifier-untied-gap } s) \ s \leq U$
shows
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-assignment-conflict } s) \ s \leq V + U$
proof –
have $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-assignment-conflict } s) \ s \leq$
 $wp\text{-event } verify\text{-monad}$
 $(\lambda out.$
 $trace\text{-fri-verifier-tied-sampled-assignment-conflict } s \ out \ \vee$
 $trace\text{-fri-sampled-assignment-verifier-untied-gap } s \ out) \ s$
by $(rule \ wp\text{-event-mono})$
 $(rule \ trace\text{-fri-sampled-assignment-conflict-imp-verifier-tied-or-untied})$
also have $\dots \leq$
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-verifier-tied-sampled-assignment-conflict } s) \ s +$
 $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri-sampled-assignment-verifier-untied-gap } s) \ s$
by $(rule \ wp\text{-event-union-bound})$
also have $\dots \leq V + U$
by $(rule \ add\text{-mono}[OF \ verifier\text{-tied-bound} \ untied\text{-bound}])$
finally show $?thesis$.
qed

lemma $checked\text{-staged-security-trace-fri-sampled-assignment-conflict-bound-from-verifier-tied-and-untied}$:
assumes $verifier\text{-tied-bound}$:
 $wp\text{-event } (checked\text{-staged-security-experiment-with-data-state } A)$
 $(staged\text{-security-with-data-state-verifier-event}$
 $trace\text{-fri-verifier-tied-sampled-assignment-conflict})$
 $adversary\text{-initial-state} \leq V$
and $untied\text{-bound}$:
 $wp\text{-event } (checked\text{-staged-security-experiment-with-data-state } A)$
 $(staged\text{-security-with-data-state-verifier-event}$
 $trace\text{-fri-sampled-assignment-verifier-untied-gap})$
 $adversary\text{-initial-state} \leq U$
shows
 $wp\text{-event } (checked\text{-staged-security-experiment-with-data-state } A)$
 $(staged\text{-security-with-data-state-verifier-event}$
 $trace\text{-fri-sampled-assignment-conflict})$
 $adversary\text{-initial-state} \leq V + U$
proof –
let $?M = checked\text{-staged-security-experiment-with-data-state } A$
let $?plain =$
 $staged\text{-security-with-data-state-verifier-event}$
 $trace\text{-fri-sampled-assignment-conflict}$

```

let ?tied =
  staged-security-with-data-state-verifier-event
  trace-fri-verifier-tied-sampled-assignment-conflict
let ?untied =
  staged-security-with-data-state-verifier-event
  trace-fri-sampled-assignment-verifier-untied-gap
have wp-event ?M ?plain adversary-initial-state ≤
  wp-event ?M (λout. ?tied out ∨ ?untied out)
  adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono)
  (auto dest:
    trace-fri-sampled-assignment-conflict-imp-verifier-tied-or-untied
    split: option.splits prod.splits)
also have ... ≤
  wp-event ?M ?tied adversary-initial-state +
  wp-event ?M ?untied adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤ V + U
by (rule add-mono[OF verifier-tied-bound untied-bound])
finally show ?thesis .
qed

definition trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
  :: ('f, 'a) protocol-channel-scheme ⇒
    (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool
where
  trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
    s out ⟷
    (∃ trace-roots trace-bs trace-final dg composition-roots
      composition-bs composition-final fri-query-idxs trace-round-layers
      composition-round-layers fr trace-openings trace-table.
      accepted-fri-opening-transcript s out trace-roots trace-bs
        trace-final dg composition-roots composition-bs composition-final
          fri-query-idxs trace-round-layers composition-round-layers ∧
      accepted-with-partial-trace-openings s out fr fri-query-idxs
        trace-openings ∧
      partial-trace-table-candidate trace-table trace-openings ∧
      ¬ trace-table-low-degree trace-table ∧
      generic-fri-sampled-assignment-conflict-without-same-layer
        trace-table trace-roots trace-bs trace-final fri-query-idxs
          trace-round-layers)

definition trace-fri-verifier-tied-same-layer-auth-gap
  :: ('f, 'a) protocol-channel-scheme ⇒
    (unit list × ('f, 'a) protocol-channel-scheme) option ⇒ bool
where
  trace-fri-verifier-tied-same-layer-auth-gap s out ⟷
    (∃ trace-roots trace-bs trace-final dg composition-roots

```

composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr trace-openings trace-table result
final-state.
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers $\wedge$
out = Some (result, final-state) $\wedge$
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
 $\neg$ *trace-table-low-degree trace-table* $\wedge$
generic-fri-sampled-same-layer-opening-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers $\wedge$
 $\neg$ *generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks*
trace-roots trace-bs fri-query-idxs trace-round-layers final-state)

lemma *trace-fri-verifier-tied-conflict-imp-authenticated-or-auth-gap:*

assumes *trace-fri-verifier-tied-sampled-assignment-conflict s out*

shows

trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
s out $\vee$

trace-fri-verifier-tied-same-layer-auth-gap s out

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr trace-openings trace-table

where *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers

and *partial:*

accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings

and *cand:* *partial-trace-table-candidate trace-table trace-openings*

and *not-low:* $\neg$ *trace-table-low-degree trace-table*

and *conflict:*

generic-fri-sampled-assignment-conflict trace-table trace-roots
trace-bs trace-final fri-query-idxs trace-round-layers

unfolding *trace-fri-verifier-tied-sampled-assignment-conflict-def*

by *blast*

from *fri-openings* **obtain** *result final-state* **where** *out-eq:*

out = Some (result, final-state)

unfolding *accepted-fri-opening-transcript-def* **by** *blast*

have *split:*

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers final-state $\vee$

(generic-fri-sampled-same-layer-opening-conflict trace-roots trace-bs
fri-query-idxs trace-round-layers $\wedge$

```

    ¬ generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
      trace-roots trace-bs fri-query-idxs trace-round-layers final-state)
  by (rule
    generic-fri-sampled-assignment-conflict-imp-authenticated-or-same-layer-auth-gap
    [OF conflict])
  then show ?thesis
  proof
    assume auth:
      generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
        trace-table trace-roots trace-bs trace-final fri-query-idxs
        trace-round-layers final-state
    have trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
      s out
    unfolding
      trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
    by (intro exI conjI)
      (rule fri-openings, rule out-eq, rule partial, rule cand,
       rule not-low, rule auth)
    then show ?thesis by simp
  next
    assume gap:
      generic-fri-sampled-same-layer-opening-conflict trace-roots trace-bs
        fri-query-idxs trace-round-layers ∧
      ¬ generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
        trace-roots trace-bs fri-query-idxs trace-round-layers final-state
    have trace-fri-verifier-tied-same-layer-auth-gap s out
    unfolding trace-fri-verifier-tied-same-layer-auth-gap-def
    by (intro exI conjI)
      (rule fri-openings, rule out-eq, rule partial, rule cand,
       rule not-low, use gap in simp-all)
    then show ?thesis by simp
  qed
qed

lemma wp-trace-fri-verifier-tied-conflict-bound-from-authenticated-and-auth-gap:
  fixes A G :: prob
  assumes auth-bound:
    wp-event verify-monad
      (trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s)
    s ≤ A
  and gap-bound:
    wp-event verify-monad
      (trace-fri-verifier-tied-same-layer-auth-gap s) s ≤ G
  shows
    wp-event verify-monad
      (trace-fri-verifier-tied-sampled-assignment-conflict s) s ≤ A + G
  proof -
    have wp-event verify-monad
      (trace-fri-verifier-tied-sampled-assignment-conflict s) s ≤

```



```

wp-event verify-monad
  (λout.
    trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
      s out ∨
    trace-fri-verifier-tied-same-layer-auth-gap s out) s
by (rule wp-event-mono)
  (rule trace-fri-verifier-tied-conflict-imp-authenticated-or-auth-gap)
also have ... ≤
  wp-event verify-monad
    (trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s)
    s +
  wp-event verify-monad (trace-fri-verifier-tied-same-layer-auth-gap s) s
by (rule wp-event-union-bound)
also have ... ≤ A + G
  by (rule add-mono[OF auth-bound gap-bound])
finally show ?thesis .
qed

```

lemma *trace-fri-verifier-tied-authenticated-conflict-imp-without-same-or-same-layer:*

assumes

trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
s out

shows

trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
s out ∨
trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
s out

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr trace-openings trace-table result
final-state

where *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers

and *out-eq:* *out = Some (result, final-state)*

and *partial:*

accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings

and *cand:* *partial-trace-table-candidate trace-table trace-openings*

and *not-low:* $\neg$ *trace-table-low-degree trace-table*

and *conflict:*

generic-fri-sampled-assignment-conflict-with-authenticated-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers final-state

unfolding

trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-def

by *blast*

have *split*:
generic-fri-sampled-assignment-conflict-without-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers $\vee$
generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
trace-roots trace-bs fri-query-idxs trace-round-layers final-state
by (rule
generic-fri-sampled-assignment-conflict-with-authenticated-same-layer-split
 $[OF\ conflict]$)
then show *?thesis*
proof
assume *without*:
generic-fri-sampled-assignment-conflict-without-same-layer
trace-table trace-roots trace-bs trace-final fri-query-idxs
trace-round-layers
have *trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer*
s out
unfolding
trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def
by (intro *exI conjI*)
(rule *fri-openings*, rule *partial*, rule *cand*, rule *not-low*,
rule *without*)
then show *?thesis* **by** *simp*
next
assume *same*:
generic-fri-sampled-same-layer-opening-conflict-with-authenticated-chunks
trace-roots trace-bs fri-query-idxs trace-round-layers final-state
have *trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks*
s out
unfolding
trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-def
by (intro *exI conjI*)
(rule *fri-openings*, rule *out-eq*, rule *partial*, rule *cand*,
rule *not-low*, rule *same*)
then show *?thesis* **by** *simp*
qed
qed

lemma *wp-trace-fri-verifier-tied-authenticated-conflict-bound-from-without-same-and-merkle*:
assumes *without-bound*:
wp-event verify-monad
(*trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s*)
 $s \leq W$
and *merkle-bound*:
wp-event verify-monad (partial-merkle-inconsistency-bad s) $s \leq M$
shows
wp-event verify-monad
(*trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s*)
 $s \leq W + M$

proof –
have *wp-event verify-monad*
 (*trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer s*)
 $s \leq$
wp-event verify-monad
 ($\lambda out.$
 trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
 s out $\vee$
 trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
 s out) *s*
by (*rule wp-event-mono*)
 (*rule*
 trace-fri-verifier-tied-authenticated-conflict-imp-without-same-or-same-layer)
also have $\dots \leq$
wp-event verify-monad
 (*trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer s*)
 $s +$
wp-event verify-monad
 (*trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks s*)
 s
by (*rule wp-event-union-bound*)
also have $\dots \leq W + M$
by (*rule add-mono*[*OF without-bound*
 wp-trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-bound-from-partial-merkle
 [*OF merkle-bound*]])
finally show *?thesis* .
qed

lemma *checked-staged-security-trace-fri-verifier-tied-conflict-bound-from-authenticated-and-auth-gap:*

assumes *auth-bound:*

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer)
adversary-initial-state $\leq$ *Auth*

and *gap-bound:*

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-verifier-tied-same-layer-auth-gap)
adversary-initial-state $\leq$ *Gap*

shows

wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 trace-fri-verifier-tied-sampled-assignment-conflict)
adversary-initial-state $\leq$ *Auth* + *Gap*

proof –

let *?M* = *checked-staged-security-experiment-with-data-state A*

let *?conflict* =

staged-security-with-data-state-verifier-event
trace-fri-verifier-tied-sampled-assignment-conflict

```

let ?auth =
  staged-security-with-data-state-verifier-event
  trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
let ?gap =
  staged-security-with-data-state-verifier-event
  trace-fri-verifier-tied-same-layer-auth-gap
have wp-event ?M ?conflict adversary-initial-state ≤
  wp-event ?M (λout. ?auth out ∨ ?gap out) adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono)
  (auto dest:
    trace-fri-verifier-tied-conflict-imp-authenticated-or-auth-gap
    split: option.splits prod.splits)
also have ... ≤
  wp-event ?M ?auth adversary-initial-state +
  wp-event ?M ?gap adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤ Auth + Gap
by (rule add-mono[OF auth-bound gap-bound])
finally show ?thesis .
qed

```

lemma *checked-staged-security-trace-fri-verifier-tied-authenticated-conflict-bound-from-without-same-and-merkle:*

assumes *without-bound:*

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer)
adversary-initial-state ≤ W

```

and *merkle-bound:*

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    partial-merkle-inconsistency-bad)
adversary-initial-state ≤ M

```

shows

```

wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
    trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer)
adversary-initial-state ≤ W + M

```

proof –

let ?M = *checked-staged-security-experiment-with-data-state A*

let ?auth =

```

  staged-security-with-data-state-verifier-event
  trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer

```

let ?without =

```

  staged-security-with-data-state-verifier-event
  trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer

```

let ?merkle =

```

  staged-security-with-data-state-verifier-event
  partial-merkle-inconsistency-bad

```

```

have wp-event ?M ?auth adversary-initial-state ≤
  wp-event ?M (λout. ?without out ∨ ?merkle out)
  adversary-initial-state
proof (rule wp-event-mono)
  fix out
  assume auth: ?auth out
  show ?without out ∨ ?merkle out
  proof (cases out)
    case None
    then show ?thesis
      using auth
      unfolding staged-security-with-data-state-verifier-event-def
        trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer-def
      by simp
  next
  case (Some packed)
  then obtain data attacker-state result final-state where out-eq:
    out = Some (((data, attacker-state), result), final-state)
    by (cases packed, auto split: prod.splits)
  let ?s =
    verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)
  have local-auth:
    trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer
    ?s (Some (result, final-state))
    using auth
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
  have split:
    trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
    ?s (Some (result, final-state)) ∨
    trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
    ?s (Some (result, final-state))
    by (rule
      trace-fri-verifier-tied-authenticated-conflict-imp-without-same-or-same-layer
      [OF local-auth])
  then show ?thesis
  proof
    assume without:
      trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
      ?s (Some (result, final-state))
    then show ?thesis
      unfolding out-eq staged-security-with-data-state-verifier-event-def
      by simp
  next
  assume same:
    trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks
    ?s (Some (result, final-state))
  have partial-merkle-inconsistency-bad ?s

```

```

    (Some (result, final-state))
  by (rule
    trace-fri-verifier-tied-same-layer-conflict-with-authenticated-chunks-imp-partial-merkle
    [OF same])
  then show ?thesis
    unfolding out-eq staged-security-with-data-state-verifier-event-def
    by simp
qed
qed
qed
also have ... ≤
  wp-event ?M ?without adversary-initial-state +
  wp-event ?M ?merkle adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤ W + M
  by (rule add-mono[OF without-bound merkle-bound])
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-verifier-tied-conflict-bound-from-without-same-merkle-and-auth-gap:
  fixes W M G :: prob
  assumes without-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer)
    adversary-initial-state ≤ W
  and merkle-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      partial-merkle-inconsistency-bad)
    adversary-initial-state ≤ M
  and gap-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-verifier-tied-same-layer-auth-gap)
    adversary-initial-state ≤ G
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-verifier-tied-sampled-assignment-conflict)
    adversary-initial-state ≤ (W + M) + G
proof (rule
  checked-staged-security-trace-fri-verifier-tied-conflict-bound-from-authenticated-and-auth-gap)
  show wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-verifier-tied-sampled-assignment-conflict-with-authenticated-same-layer)
    adversary-initial-state ≤ W + M
  by (rule
    checked-staged-security-trace-fri-verifier-tied-authenticated-conflict-bound-from-without-same-and-merkle

```

$[OF \text{ without-bound merkle-bound}]$
show $wp\text{-event}$ ($checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state}$ A)
 ($staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event}$
 $trace\text{-fri}\text{-verifier}\text{-tied}\text{-same}\text{-layer}\text{-auth}\text{-gap}$)
 $adversary\text{-initial}\text{-state} \leq G$
by ($rule$ $gap\text{-bound}$)
qed

definition $trace\text{-fri}\text{-empty}\text{-sampled}\text{-assignment}\text{-structural}\text{-error}\text{-for}$
where

$trace\text{-fri}\text{-empty}\text{-sampled}\text{-assignment}\text{-structural}\text{-error}\text{-for}$ budgets $A =$
 ($(reachable\text{-verifier}\text{-event}\text{-bound}\text{-for}$ A
 $trace\text{-fri}\text{-verifier}\text{-tied}\text{-sampled}\text{-assignment}\text{-conflict}\text{-without}\text{-same}\text{-layer} +$
 $trace\text{-fri}\text{-controlled}\text{-merkle}\text{-error}\text{-for}$ budgets) $+$
 $reachable\text{-verifier}\text{-event}\text{-bound}\text{-for}$ A
 $trace\text{-fri}\text{-verifier}\text{-tied}\text{-same}\text{-layer}\text{-auth}\text{-gap}$) $+$
 $reachable\text{-verifier}\text{-event}\text{-bound}\text{-for}$ A
 $trace\text{-fri}\text{-sampled}\text{-assignment}\text{-verifier}\text{-untied}\text{-gap}$

definition $trace\text{-empty}\text{-header}\text{-openings}\text{-zero}\text{-error}\text{-for}$
where

$trace\text{-empty}\text{-header}\text{-openings}\text{-zero}\text{-error}\text{-for}$ budgets A $empty\text{-bad} =$
 ($trace\text{-fri}\text{-zero}\text{-round}\text{-agreement}\text{-set}\text{-target}\text{-error}\text{-for}$ budgets $+$
 $trace\text{-fri}\text{-controlled}\text{-merkle}\text{-error}\text{-for}$ budgets $+$
 ($trace\text{-header}\text{-one}\text{-step}\text{-error}\text{-for}$ budgets A $empty\text{-bad} +$
 $reachable\text{-verifier}\text{-event}\text{-bound}\text{-for}$ A
 $trace\text{-fri}\text{-header}\text{-tied}\text{-low}\text{-candidate}\text{-untied}\text{-openings}\text{-gap}$) $+$
 $trace\text{-header}\text{-one}\text{-step}\text{-error}\text{-for}$ budgets A $empty\text{-bad}$)

definition $trace\text{-fri}\text{-header}\text{-tied}\text{-low}\text{-candidate}\text{-nonconsumed}\text{-openings}\text{-gap}$

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(unit \text{ list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow bool$

where

$trace\text{-fri}\text{-header}\text{-tied}\text{-low}\text{-candidate}\text{-nonconsumed}\text{-openings}\text{-gap}$ s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots } trace\text{-bs } trace\text{-final } fri\text{-dg } composition\text{-roots}$
 $composition\text{-bs } composition\text{-final } (fri\text{-query}\text{-idxs} :: nat \text{ list})$
 $trace\text{-round}\text{-layers } composition\text{-round}\text{-layers } (fr :: 'f) \text{ as}$
 $trace\text{-openings } trace\text{-table } (fr0 :: 'f) (query\text{-idxs}0 :: nat \text{ list})$
 $trace\text{-openings}0 \text{ trace}\text{-table}0.$
 $trace\text{-fri}\text{-reachable}\text{-header}\text{-tie}\text{-gap}$ s $out \wedge$
 $accepted\text{-with}\text{-partial}\text{-trace}\text{-openings}\text{-verifier}\text{-consumed}$ s out
 $trace\text{-roots } trace\text{-bs } trace\text{-final } fri\text{-dg } composition\text{-roots}$
 $composition\text{-bs } composition\text{-final } fri\text{-query}\text{-idxs } trace\text{-round}\text{-layers}$
 $composition\text{-round}\text{-layers } fr \text{ as } trace\text{-openings} \wedge$
 $partial\text{-trace}\text{-table}\text{-candidate}$ $trace\text{-table}$ $trace\text{-openings} \wedge$
 $trace\text{-table}\text{-low}\text{-degree}$ $trace\text{-table} \wedge$
 $trace\text{-fri}\text{-partial}\text{-candidate}\text{-evidence}$ s out $fr0$ $query\text{-idxs}0$
 $trace\text{-openings}0 \text{ trace}\text{-table}0 \wedge$

$(fr0 \neq fr \vee query_idxs0 \neq fri_query_idxs) \wedge$
 $trace_openings0 \neq trace_openings)$

definition *trace-fri-header-tied-low-candidate-nonconsumed-root-gap*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(unit\ list \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow bool$

where

trace-fri-header-tied-low-candidate-nonconsumed-root-gap $s\ out$
 $\longleftrightarrow$
 $(\exists\ trace_roots\ trace_bs\ trace_final\ fri_dg\ composition_roots$
 $\quad composition_bs\ composition_final\ (fri_query_idxs :: nat\ list)$
 $\quad trace_round_layers\ composition_round_layers\ (fr :: 'f)\ as$
 $\quad trace_openings\ trace_table\ (fr0 :: 'f)\ (query_idxs0 :: nat\ list)$
 $\quad trace_openings0\ trace_table0.$
 $trace_fri_reachable_header_tie_gap\ s\ out \wedge$
 $accepted_with_partial_trace_openings_verifier_consumed\ s\ out$
 $\quad trace_roots\ trace_bs\ trace_final\ fri_dg\ composition_roots$
 $\quad composition_bs\ composition_final\ fri_query_idxs\ trace_round_layers$
 $\quad composition_round_layers\ fr\ as\ trace_openings \wedge$
 $partial_trace_table_candidate\ trace_table\ trace_openings \wedge$
 $trace_table_low_degree\ trace_table \wedge$
 $trace_fri_partial_candidate_evidence\ s\ out\ fr0\ query_idxs0$
 $\quad trace_openings0\ trace_table0 \wedge$
 $fr0 \neq fr \wedge$
 $trace_openings0 \neq trace_openings)$

definition *trace-fri-header-tied-low-candidate-nonconsumed-query-gap*

$:: (f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(unit\ list \times (f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow bool$

where

trace-fri-header-tied-low-candidate-nonconsumed-query-gap $s\ out$
 $\longleftrightarrow$
 $(\exists\ trace_roots\ trace_bs\ trace_final\ fri_dg\ composition_roots$
 $\quad composition_bs\ composition_final\ (fri_query_idxs :: nat\ list)$
 $\quad trace_round_layers\ composition_round_layers\ (fr :: 'f)\ as$
 $\quad trace_openings\ trace_table\ (fr0 :: 'f)\ (query_idxs0 :: nat\ list)$
 $\quad trace_openings0\ trace_table0.$
 $trace_fri_reachable_header_tie_gap\ s\ out \wedge$
 $accepted_with_partial_trace_openings_verifier_consumed\ s\ out$
 $\quad trace_roots\ trace_bs\ trace_final\ fri_dg\ composition_roots$
 $\quad composition_bs\ composition_final\ fri_query_idxs\ trace_round_layers$
 $\quad composition_round_layers\ fr\ as\ trace_openings \wedge$
 $partial_trace_table_candidate\ trace_table\ trace_openings \wedge$
 $trace_table_low_degree\ trace_table \wedge$
 $trace_fri_partial_candidate_evidence\ s\ out\ fr0\ query_idxs0$
 $\quad trace_openings0\ trace_table0 \wedge$
 $query_idxs0 \neq fri_query_idxs \wedge$
 $trace_openings0 \neq trace_openings)$

definition *trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap*
 $:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final (fri-query-idxs :: nat list)}$
 $\text{ trace-round-layers composition-round-layers (fr :: 'f) as}$
 $\text{ trace-openings trace-table (fr0 :: 'f) (query-idxs0 :: nat list)}$
 $\text{ trace-openings0 trace-table0.}$
 $\text{ trace-fri-reachable-header-tie-gap s out} \wedge$
 $\text{ accepted-with-partial-trace-openings-verifier-consumed s out}$
 $\text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings} \wedge$
 $\text{ partial-trace-table-candidate trace-table trace-openings} \wedge$
 $\text{ trace-table-low-degree trace-table} \wedge$
 $\text{ trace-fri-partial-candidate-evidence s out fr0 query-idxs0}$
 $\text{ trace-openings0 trace-table0} \wedge$
 $\text{ fr0} \neq \text{ fr} \wedge$
 $\text{ query-idxs0} = \text{ fri-query-idxs} \wedge$
 $\text{ trace-openings0} \neq \text{ trace-openings})$

definition *trace-empty-header-nonnormalized-zero-error-for*

where

trace-empty-header-nonnormalized-zero-error-for budgets A empty-bad =
 $(\text{trace-fri-zero-round-agreement-set-target-error-for budgets} +$
 $\text{ trace-fri-controlled-merkle-error-for budgets} +$
 $(\text{trace-fri-controlled-merkle-error-for budgets} +$
 $((\text{reachable-verifier-event-bound-for A}$
 $\text{ trace-fri-header-tied-low-candidate-nonconsumed-query-gap} +$
 $\text{ reachable-verifier-event-bound-for A}$
 $\text{ trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap}) +$
 $\text{ reachable-verifier-event-bound-for A}$
 $\text{ trace-fri-header-tied-low-candidate-nonconsumed-query-gap})) +$
 $\text{ trace-header-one-step-error-for budgets A empty-bad})$

definition *trace-fri-zero-round-verifier-consumed-low-candidate-gap*

$:: ('f, 'a) \text{ protocol-channel-scheme} \Rightarrow$
 $(\text{unit list} \times ('f, 'a) \text{ protocol-channel-scheme}) \text{ option} \Rightarrow \text{bool}$

where

trace-fri-zero-round-verifier-consumed-low-candidate-gap s out
 $\longleftrightarrow$
 $(\exists \text{ trace-roots trace-bs trace-final fri-dg composition-roots}$
 $\text{ composition-bs composition-final fri-query-idxs trace-round-layers}$
 $\text{ composition-round-layers fr as trace-openings trace-table.}$
 $\text{ trace-fri-zero-round-final-obstruction s out} \wedge$
 $\text{ trace-fri-reachable-header-tie-gap s out} \wedge$

accepted-with-partial-trace-openings-verifier-consumed s out
trace-roots trace-bs trace-final fri-dg composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings $\wedge$
length trace-bs = 0 $\wedge$
partial-trace-table-candidate trace-table trace-openings $\wedge$
trace-table-low-degree trace-table)

lemma *trace-fri-header-tied-low-candidate-nonconsumed-root-gapE:*
assumes *trace-fri-header-tied-low-candidate-nonconsumed-root-gap s out*
obtains *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings trace-table
fr0 query-idxs0 trace-openings0 trace-table0 **where**
trace-fri-reachable-header-tie-gap s out
accepted-with-partial-trace-openings-verifier-consumed s out
trace-roots trace-bs trace-final fri-dg composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings
partial-trace-table-candidate trace-table trace-openings
trace-table-low-degree trace-table
trace-fri-partial-candidate-evidence s out fr0 query-idxs0
trace-openings0 trace-table0
fr0 $\neq$ fr
trace-openings0 $\neq$ trace-openings
using *assms*
unfolding *trace-fri-header-tied-low-candidate-nonconsumed-root-gap-def*
by *blast*

lemma *trace-fri-header-tied-low-candidate-nonconsumed-query-gapE:*
assumes *trace-fri-header-tied-low-candidate-nonconsumed-query-gap s out*
obtains *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings trace-table
fr0 query-idxs0 trace-openings0 trace-table0 **where**
trace-fri-reachable-header-tie-gap s out
accepted-with-partial-trace-openings-verifier-consumed s out
trace-roots trace-bs trace-final fri-dg composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings
partial-trace-table-candidate trace-table trace-openings
trace-table-low-degree trace-table
trace-fri-partial-candidate-evidence s out fr0 query-idxs0
trace-openings0 trace-table0
query-idxs0 $\neq$ fri-query-idxs
trace-openings0 $\neq$ trace-openings
using *assms*
unfolding *trace-fri-header-tied-low-candidate-nonconsumed-query-gap-def*
by *blast*

lemma *trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gapE*:

assumes

trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap s out

obtains *trace-roots trace-bs trace-final fri-dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as trace-openings trace-table

fr0 query-idxs0 trace-openings0 trace-table0 **where**

trace-fri-reachable-header-tie-gap s out

accepted-with-partial-trace-openings-verifier-consumed s out

trace-roots trace-bs trace-final fri-dg composition-roots

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as trace-openings

partial-trace-table-candidate trace-table trace-openings

trace-table-low-degree trace-table

trace-fri-partial-candidate-evidence s out fr0 query-idxs0

trace-openings0 trace-table0

fr0 ≠ fr

query-idxs0 = fri-query-idxs

trace-openings0 ≠ trace-openings

using *assms*

unfolding

trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap-def

by *blast*

lemma *trace-fri-zero-round-verifier-consumed-low-candidate-gapE*:

assumes

trace-fri-zero-round-verifier-consumed-low-candidate-gap s out

obtains *trace-roots trace-bs trace-final fri-dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as trace-openings trace-table **where**

trace-fri-zero-round-final-obstruction s out

trace-fri-reachable-header-tie-gap s out

accepted-with-partial-trace-openings-verifier-consumed s out

trace-roots trace-bs trace-final fri-dg composition-roots

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as trace-openings

length trace-bs = 0

partial-trace-table-candidate trace-table trace-openings

trace-table-low-degree trace-table

using *assms*

unfolding *trace-fri-zero-round-verifier-consumed-low-candidate-gap-def*

by *blast*

lemma *trace-fri-zero-round-verifier-consumed-low-candidate-gap-imp-candidate-available*:

assumes

trace-fri-zero-round-verifier-consumed-low-candidate-gap s out

shows *trace-fri-header-tied-candidate-available-gap s out*

proof (rule *trace-fri-zero-round-verifier-consumed-low-candidate-gapE*)

$[OF\ assms]$
fix *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings trace-table
assume *gap: trace-fri-reachable-header-tie-gap s out*
and *consumed:*
accepted-with-partial-trace-openings-verifier-consumed s out
trace-roots trace-bs trace-final fri-dg composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings
and *candidate:*
partial-trace-table-candidate trace-table trace-openings
have *fri-openings:*
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
by (rule *accepted-with-partial-trace-openings-verifier-consumedD(1)*
 $[OF\ consumed]$)
from *accepted-with-partial-trace-openings-verifier-consumedD(2)*
 $[OF\ consumed]$
obtain *rest where header:*
verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest
by *blast*
have *partial:*
accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings
by (rule *accepted-with-partial-trace-openings-verifier-consumedD(3)*
 $[OF\ consumed]$)
show *trace-fri-header-tied-candidate-available-gap s out*
unfolding *trace-fri-header-tied-candidate-available-gap-def*
by (intro *exI conjI*)
(rule *gap*, rule *fri-openings*, rule *header*, rule *partial*,
rule *candidate*)
qed

lemma *trace-fri-zero-round-verifier-consumed-low-candidate-gap-imp-untied-openings:*
assumes
trace-fri-zero-round-verifier-consumed-low-candidate-gap s out
shows *trace-fri-header-tied-low-candidate-untied-openings-gap s out*
proof (rule *trace-fri-zero-round-verifier-consumed-low-candidate-gapE*
 $[OF\ assms]$)
fix *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings trace-table
assume *gap: trace-fri-reachable-header-tie-gap s out*
and *consumed:*
accepted-with-partial-trace-openings-verifier-consumed s out
trace-roots trace-bs trace-final fri-dg composition-roots

```

    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings
and candidate:
    partial-trace-table-candidate trace-table trace-openings
and low: trace-table-low-degree trace-table
have fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    fri-dg composition-roots composition-bs composition-final
    fri-query-idxs trace-round-layers composition-round-layers
by (rule accepted-with-partial-trace-openings-verifier-consumedD(1)
    [OF consumed])
from accepted-with-partial-trace-openings-verifier-consumedD(2)
    [OF consumed]
obtain rest where header:
    verifier-header-transcript s fr trace-roots trace-final as fri-dg
    composition-roots composition-final rest
by blast
have partial:
    accepted-with-partial-trace-openings s out fr fri-query-idxs
    trace-openings
by (rule accepted-with-partial-trace-openings-verifier-consumedD(3)
    [OF consumed])
have low-gap: trace-fri-header-tied-low-candidate-gap s out
unfolding trace-fri-header-tied-low-candidate-gap-def
by (intro exI conjI)
    (rule gap, rule fri-openings, rule header, rule partial,
    rule candidate, rule low)
have untied:
    trace-fri-header-tied-low-candidate-untied-reachable-witness-gap s out
by (rule trace-fri-header-tied-low-candidate-gap-imp-untied-reachable-witness
    [OF low-gap])
show trace-fri-header-tied-low-candidate-untied-openings-gap s out
by (rule trace-fri-header-tied-low-candidate-untied-gap-imp-openings
    [OF untied])
qed

lemma accepted-with-partial-trace-openings-query-index-list-space:
assumes partial:
    accepted-with-partial-trace-openings s out fr query-idxs trace-openings
shows query-idxs  $\in$  fri-query-index-list-space
proof –
have len: length query-idxs = rounds
by (rule accepted-with-partial-trace-openings-shapes(1)[OF partial])
have subset: set query-idxs  $\subseteq$  query-sample-space
proof
fix idx
assume idx-in: idx  $\in$  set query-idxs
then obtain i where i-bound: i < length query-idxs
and idx-eq: query-idxs ! i = idx

```

```

    unfolding in-set-conv-nth by blast
  have i-round:  $i < \text{rounds}$ 
    using i-bound len by simp
  obtain result final-state where out-eq:  $\text{out} = \text{Some}(\text{result}, \text{final-state})$ 
  and auth:
     $\bigwedge j. j < \text{rounds} \implies$ 
      partial-authenticated-table fr (scale * clength)
      (trace-openings ! j) final-state
    using partial unfolding accepted-with-partial-trace-openings-def by blast
  have indices:
    map opening-index (trace-openings ! i) = powers-scaled idx
    using accepted-with-partial-trace-openings-shapes(3)[OF partial i-round]
    idx-eq by simp
  show  $\text{idx} \in \text{query-sample-space}$ 
    by (rule partial-authenticated-trace-openings-query-index-in-sample-space
      [OF indices auth [OF i-round]])
qed
show ?thesis
  unfolding fri-query-index-list-space-def
  using len subset by simp
qed

lemma trace-fri-header-tied-low-candidate-nonconsumed-root-gap-shapes:
  assumes trace-fri-header-tied-low-candidate-nonconsumed-root-gap s out
  obtains fri-query-idxs query-idxs0 where
    fri-query-idxs  $\in \text{fri-query-index-list-space}$ 
    query-idxs0  $\in \text{fri-query-index-list-space}$ 
proof -
  from assms obtain trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings trace-table
    fr0 query-idxs0 trace-openings0 trace-table0 where
    consumed:
      accepted-with-partial-trace-openings-verifier-consumed s out
      trace-roots trace-bs trace-final fri-dg composition-roots
      composition-bs composition-final fri-query-idxs trace-round-layers
      composition-round-layers fr as trace-openings
    and evidence0:
      trace-fri-partial-candidate-evidence s out fr0 query-idxs0
      trace-openings0 trace-table0
    and mismatch:  $\text{fr0} \neq \text{fr}$ 
    by (rule trace-fri-header-tied-low-candidate-nonconsumed-root-gapE)
  have fri-openings:
    accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
    fri-dg composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers
    by (rule accepted-with-partial-trace-openings-verifier-consumedD(1)
      [OF consumed])
  have verifier-space:

```

fri-query-idxs $\in$ *fri-query-index-list-space*
by (rule *accepted-fri-opening-transcript-query-index-list-space*
[*OF fri-openings*])
have *alternate-space*: *query-idxs0* $\in$ *fri-query-index-list-space*
by (rule *accepted-with-partial-trace-openings-query-index-list-space*)
(rule *trace-fri-partial-candidate-evidenceD(1)*[*OF evidence0*])
show ?thesis
by (rule *that*[*OF verifier-space alternate-space*])
qed

lemma *trace-fri-header-tied-low-candidate-nonconsumed-query-gap-shapes*:
assumes *trace-fri-header-tied-low-candidate-nonconsumed-query-gap s out*
obtains *fri-query-idxs query-idxs0* **where**
fri-query-idxs $\in$ *fri-query-index-list-space*
query-idxs0 $\in$ *fri-query-index-list-space*
query-idxs0 $\neq$ *fri-query-idxs*
proof –
from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings trace-table
fr0 query-idxs0 trace-openings0 trace-table0 **where**
consumed:
accepted-with-partial-trace-openings-verifier-consumed s out
trace-roots trace-bs trace-final fri-dg composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings
and *evidence0*:
trace-fri-partial-candidate-evidence s out fr0 query-idxs0
trace-openings0 trace-table0
and *mismatch*: *query-idxs0* $\neq$ *fri-query-idxs*
by (rule *trace-fri-header-tied-low-candidate-nonconsumed-query-gapE*)
have *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs trace-final
fri-dg composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers
by (rule *accepted-with-partial-trace-openings-verifier-consumedD(1)*
[*OF consumed*])
have *verifier-space*:
fri-query-idxs $\in$ *fri-query-index-list-space*
by (rule *accepted-fri-opening-transcript-query-index-list-space*
[*OF fri-openings*])
have *alternate-space*: *query-idxs0* $\in$ *fri-query-index-list-space*
by (rule *accepted-with-partial-trace-openings-query-index-list-space*)
(rule *trace-fri-partial-candidate-evidenceD(1)*[*OF evidence0*])
show ?thesis
by (rule *that*[*of fri-query-idxs query-idxs0,*
OF verifier-space alternate-space mismatch])
qed

lemma *trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap-shapes:*

assumes

trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap s out

obtains *fri-query-idxs query-idxs0* **where**

fri-query-idxs ∈ fri-query-index-list-space

query-idxs0 ∈ fri-query-index-list-space

query-idxs0 = fri-query-idxs

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as trace-openings trace-table

fr0 query-idxs0 trace-openings0 trace-table0 **where**

consumed:

accepted-with-partial-trace-openings-verifier-consumed s out

trace-roots trace-bs trace-final fri-dg composition-roots

composition-bs composition-final fri-query-idxs trace-round-layers

composition-round-layers fr as trace-openings

and *evidence0:*

trace-fri-partial-candidate-evidence s out fr0 query-idxs0

trace-openings0 trace-table0

and *mismatch: fr0 ≠ fr*

and *query-eq: query-idxs0 = fri-query-idxs*

by (rule *trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gapE*)

have *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs trace-final

fri-dg composition-roots composition-bs composition-final fri-query-idxs

trace-round-layers composition-round-layers

by (rule *accepted-with-partial-trace-openings-verifier-consumedD(1)*)

[*OF consumed*])

have *verifier-space:*

fri-query-idxs ∈ fri-query-index-list-space

by (rule *accepted-fri-opening-transcript-query-index-list-space*)

[*OF fri-openings*])

have *alternate-space: query-idxs0 ∈ fri-query-index-list-space*

by (rule *accepted-with-partial-trace-openings-query-index-list-space*)

(rule *trace-fri-partial-candidate-evidenceD(1)*[*OF evidence0*])

show *?thesis*

by (rule *that*[*OF verifier-space alternate-space query-eq*])

qed

lemma *trace-fri-header-tied-low-candidate-nonconsumed-root-gap-imp-query-or-irrelevant-root:*

assumes *trace-fri-header-tied-low-candidate-nonconsumed-root-gap s out*

shows

trace-fri-header-tied-low-candidate-nonconsumed-query-gap s out ∨

trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap

s out

proof –

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg composition-roots*

composition-bs composition-final fri-query-idxs trace-round-layers


```

    composition-round-layers fr as trace-openings trace-table
    fr0 query-idxs0 trace-openings0 trace-table0 where
gap: trace-fri-reachable-header-tie-gap s out
and consumed:
    accepted-with-partial-trace-openings-verifier-consumed s out
    trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings
and candidate: partial-trace-table-candidate trace-table trace-openings
and low: trace-table-low-degree trace-table
and evidence0:
    trace-fri-partial-candidate-evidence s out fr0 query-idxs0
    trace-openings0 trace-table0
and root-mismatch: fr0  $\neq$  fr
and openings-mismatch: trace-openings0  $\neq$  trace-openings
by (rule trace-fri-header-tied-low-candidate-nonconsumed-root-gapE)
show ?thesis
proof (cases query-idxs0  $\neq$  fri-query-idxs)
  case True
    have trace-fri-header-tied-low-candidate-nonconsumed-query-gap s out
    unfolding trace-fri-header-tied-low-candidate-nonconsumed-query-gap-def
    by (intro exI conjI)
      (rule gap, rule consumed, rule candidate, rule low, rule evidence0,
       rule True, rule openings-mismatch)
    then show ?thesis by simp
  next
  case False
    then have query-eq: query-idxs0 = fri-query-idxs
    by simp
    have trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap
      s out
    unfolding
      trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap-def
    by (intro exI conjI)
      (rule gap, rule consumed, rule candidate, rule low, rule evidence0,
       rule root-mismatch, rule query-eq, rule openings-mismatch)
    then show ?thesis by simp
qed
qed

```

lemma checked-staged-security-trace-fri-header-tied-low-candidate-nonconsumed-root-gap-bound-from-query-and-irrel

assumes query-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-header-tied-low-candidate-nonconsumed-query-gap)
 adversary-initial-state $\leq$ Q

and irrelevant-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event

$trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap)$
 $adversary-initial-state \leq I$
shows
 $wp-event \ (checked-staged-security-experiment-with-data-state \ A)$
 $(staged-security-with-data-state-verifier-event$
 $\quad trace-fri-header-tied-low-candidate-nonconsumed-root-gap)$
 $adversary-initial-state \leq Q + I$
proof –
let $?M = checked-staged-security-experiment-with-data-state \ A$
let $?root =$
 $\quad staged-security-with-data-state-verifier-event$
 $\quad \quad trace-fri-header-tied-low-candidate-nonconsumed-root-gap$
let $?query =$
 $\quad staged-security-with-data-state-verifier-event$
 $\quad \quad trace-fri-header-tied-low-candidate-nonconsumed-query-gap$
let $?irrelevant =$
 $\quad staged-security-with-data-state-verifier-event$
 $\quad \quad trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap$
have $wp-event \ ?M \ ?root \ adversary-initial-state \leq$
 $\quad wp-event \ ?M \ (\lambda out. \ ?query \ out \ \vee \ ?irrelevant \ out)$
 $\quad \quad adversary-initial-state$
unfolding $staged-security-with-data-state-verifier-event-def$
by $(rule \ wp-event-mono-on-support)$
 $(auto \ dest:$
 $\quad trace-fri-header-tied-low-candidate-nonconsumed-root-gap-imp-query-or-irrelevant-root$
 $\quad \quad split: \ option.splits \ prod.splits)$
also have $\dots \leq$
 $\quad wp-event \ ?M \ ?query \ adversary-initial-state +$
 $\quad wp-event \ ?M \ ?irrelevant \ adversary-initial-state$
by $(rule \ wp-event-union-bound)$
also have $\dots \leq Q + I$
by $(rule \ add-mono[OF \ query-bound \ irrelevant-bound])$
finally show $?thesis \ .$
qed

lemma $trace-fri-header-tied-low-candidate-nonconsumed-openings-gap-imp-root-or-query:$
assumes $trace-fri-header-tied-low-candidate-nonconsumed-openings-gap \ s \ out$
shows

$trace-fri-header-tied-low-candidate-nonconsumed-root-gap \ s \ out \ \vee$
 $trace-fri-header-tied-low-candidate-nonconsumed-query-gap \ s \ out$

proof –

from $assms$ **obtain** $trace-roots \ trace-bs \ trace-final \ fri-dg \ composition-roots$
 $\quad composition-bs \ composition-final \ fri-query-idxs$
 $\quad trace-round-layers \ composition-round-layers \ fr \ as$
 $\quad trace-openings \ trace-table \ fr0 \ query-idxs0 \ trace-openings0 \ trace-table0$

where

$gap: trace-fri-reachable-header-tie-gap \ s \ out$

and $consumed:$

$\quad accepted-with-partial-trace-openings-verifier-consumed \ s \ out$

```

    trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings
and candidate: partial-trace-table-candidate trace-table trace-openings
and low: trace-table-low-degree trace-table
and evidence0:
    trace-fri-partial-candidate-evidence s out fr0 query-idxs0
    trace-openings0 trace-table0
and mismatch: fr0  $\neq$  fr  $\vee$  query-idxs0  $\neq$  fri-query-idxs
and openings-mismatch: trace-openings0  $\neq$  trace-openings
unfolding trace-fri-header-tied-low-candidate-nonconsumed-openings-gap-def
by blast
from mismatch show ?thesis
proof
  assume root-mismatch: fr0  $\neq$  fr
  have trace-fri-header-tied-low-candidate-nonconsumed-root-gap s out
    unfolding trace-fri-header-tied-low-candidate-nonconsumed-root-gap-def
    apply (rule exI[where x=trace-roots])
    apply (rule exI[where x=trace-bs])
    apply (rule exI[where x=trace-final])
    apply (rule exI[where x=fri-dg])
    apply (rule exI[where x=composition-roots])
    apply (rule exI[where x=composition-bs])
    apply (rule exI[where x=composition-final])
    apply (rule exI[where x=fri-query-idxs])
    apply (rule exI[where x=trace-round-layers])
    apply (rule exI[where x=composition-round-layers])
    apply (rule exI[where x=fr])
    apply (rule exI[where x=as])
    apply (rule exI[where x=trace-openings])
    apply (rule exI[where x=trace-table])
    apply (rule exI[where x=fr0])
    apply (rule exI[where x=query-idxs0])
    apply (rule exI[where x=trace-openings0])
    apply (rule exI[where x=trace-table0])
    using gap consumed candidate low evidence0 root-mismatch openings-mismatch
    by blast
  then show ?thesis by simp
next
  assume query-mismatch: query-idxs0  $\neq$  fri-query-idxs
  have trace-fri-header-tied-low-candidate-nonconsumed-query-gap s out
    unfolding trace-fri-header-tied-low-candidate-nonconsumed-query-gap-def
    apply (rule exI[where x=trace-roots])
    apply (rule exI[where x=trace-bs])
    apply (rule exI[where x=trace-final])
    apply (rule exI[where x=fri-dg])
    apply (rule exI[where x=composition-roots])
    apply (rule exI[where x=composition-bs])
    apply (rule exI[where x=composition-final])

```

```

apply (rule exI[where  $x=fri\text{-}query\text{-}idxs$ ])
apply (rule exI[where  $x=trace\text{-}round\text{-}layers$ ])
apply (rule exI[where  $x=composition\text{-}round\text{-}layers$ ])
apply (rule exI[where  $x=fr$ ])
apply (rule exI[where  $x=as$ ])
apply (rule exI[where  $x=trace\text{-}openings$ ])
apply (rule exI[where  $x=trace\text{-}table$ ])
apply (rule exI[where  $x=fr0$ ])
apply (rule exI[where  $x=query\text{-}idxs0$ ])
apply (rule exI[where  $x=trace\text{-}openings0$ ])
apply (rule exI[where  $x=trace\text{-}table0$ ])
using gap consumed candidate low evidence0 query-mismatch openings-mismatch
by blast
then show ?thesis by simp
qed
qed

lemma trace-fri-header-tied-low-candidate-nonconsumed-root-gap-imp-untied-root-gap:
  assumes trace-fri-header-tied-low-candidate-nonconsumed-root-gap s out
  shows trace-fri-header-tied-low-candidate-untied-root-gap s out
proof –
  from assms obtain trace-roots trace-bs trace-final fri-dg
    composition-roots composition-bs composition-final fri-query-idxs
    trace-round-layers composition-round-layers fr as trace-openings
    trace-table fr0 query-idxs0 trace-openings0 trace-table0
  where gap: trace-fri-reachable-header-tie-gap s out
  and consumed:
    accepted-with-partial-trace-openings-verifier-consumed s out
    trace-roots trace-bs trace-final fri-dg composition-roots
    composition-bs composition-final fri-query-idxs trace-round-layers
    composition-round-layers fr as trace-openings
  and candidate: partial-trace-table-candidate trace-table trace-openings
  and low: trace-table-low-degree trace-table
  and evidence0:
    trace-fri-partial-candidate-evidence s out fr0 query-idxs0
    trace-openings0 trace-table0
  and root-mismatch: fr0 ≠ fr
  by (rule trace-fri-header-tied-low-candidate-nonconsumed-root-gapE)
have transcript:
  accepted-fri-opening-transcript s out trace-roots trace-bs
  trace-final fri-dg composition-roots composition-bs
  composition-final fri-query-idxs trace-round-layers
  composition-round-layers
  and header-ex:
   $\exists$  rest. verifier-header-transcript s fr trace-roots trace-final as
  fri-dg composition-roots composition-final rest
  and accepted:
  accepted-with-partial-trace-openings s out fr fri-query-idxs
  trace-openings

```

by (rule *accepted-with-partial-trace-openings-verifier-consumedD* [OF *consumed*]))+
from *header-ex* **obtain** *rest* **where** *header*:
 verifier-header-transcript s fr trace-roots trace-final as fri-dg
 composition-roots composition-final rest
 by *blast*
have *accepted0*:
 accepted-with-partial-trace-openings s out fr0 query-idxs0
 trace-openings0
and *candidate0*: *partial-trace-table-candidate trace-table0 trace-openings0*
and *not-low0*: $\neg$ *trace-table-low-degree trace-table0*
 by (rule *trace-fri-partial-candidate-evidenceD*[OF *evidence0*]))+
show *?thesis*
 unfolding *trace-fri-header-tied-low-candidate-untied-root-gap-def*
 by (rule *trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI* [OF *gap transcript header accepted candidate low accepted0 candidate0 not-low0*]))
 (*simp add: root-mismatch*)
qed

lemma *trace-fri-header-tied-low-candidate-nonconsumed-query-gap-imp-untied-query-gap*:
 assumes *trace-fri-header-tied-low-candidate-nonconsumed-query-gap s out*
 shows *trace-fri-header-tied-low-candidate-untied-query-gap s out*
proof –
 from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
 composition-roots composition-bs composition-final fri-query-idxs
 trace-round-layers composition-round-layers fr as trace-openings
 trace-table fr0 query-idxs0 trace-openings0 trace-table0
 where *gap*: *trace-fri-reachable-header-tie-gap s out*
 and *consumed*:
 accepted-with-partial-trace-openings-verifier-consumed s out
 trace-roots trace-bs trace-final fri-dg composition-roots
 composition-bs composition-final fri-query-idxs trace-round-layers
 composition-round-layers fr as trace-openings
 and *candidate*: *partial-trace-table-candidate trace-table trace-openings*
 and *low*: *trace-table-low-degree trace-table*
 and *evidence0*:
 trace-fri-partial-candidate-evidence s out fr0 query-idxs0
 trace-openings0 trace-table0
 and *query-mismatch*: *query-idxs0* $\neq$ *fri-query-idxs*
 by (rule *trace-fri-header-tied-low-candidate-nonconsumed-query-gapE*)
have *transcript*:
 accepted-fri-opening-transcript s out trace-roots trace-bs
 trace-final fri-dg composition-roots composition-bs
 composition-final fri-query-idxs trace-round-layers
 composition-round-layers
and *header-ex*:
 $\exists$ *rest*. *verifier-header-transcript s fr trace-roots trace-final as*
 fri-dg composition-roots composition-final rest

and *accepted*:
accepted-with-partial-trace-openings *s* out fr *fri-query-idxs*
trace-openings
by (rule *accepted-with-partial-trace-openings-verifier-consumedD*
 $[OF\ consumed]$)+
from *header-ex* **obtain** *rest* **where** *header*:
verifier-header-transcript *s* fr *trace-roots* *trace-final* as *fri-dg*
composition-roots *composition-final* *rest*
by *blast*
have *accepted0*:
accepted-with-partial-trace-openings *s* out fr0 *query-idxs0*
trace-openings0
and *candidate0*: *partial-trace-table-candidate* *trace-table0* *trace-openings0*
and *not-low0*: $\neg$ *trace-table-low-degree* *trace-table0*
by (rule *trace-fri-partial-candidate-evidenceD* $[OF\ evidence0]$)+
show *?thesis*
unfolding *trace-fri-header-tied-low-candidate-untied-query-gap-def*
by (rule *trace-fri-header-tied-low-candidate-untied-reachable-witness-caseI*
 $[OF\ gap\ transcript\ header\ accepted\ candidate\ low\ accepted0\ candidate0\ not-low0]$)
(simp add: query-mismatch)
qed

lemma *checked-staged-security-trace-fri-header-tied-low-candidate-nonconsumed-openings-gap-bound-from-root-query*

assumes *root-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-low-candidate-nonconsumed-root-gap)
adversary-initial-state $\leq R$

and *query-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-low-candidate-nonconsumed-query-gap)
adversary-initial-state $\leq Q$

shows

wp-event (*checked-staged-security-experiment-with-data-state* *A*)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-low-candidate-nonconsumed-openings-gap)
adversary-initial-state $\leq R + Q$

proof –

let *?M* = *checked-staged-security-experiment-with-data-state* *A*

let *?nonconsumed* =

staged-security-with-data-state-verifier-event
trace-fri-header-tied-low-candidate-nonconsumed-openings-gap

let *?root* =

staged-security-with-data-state-verifier-event
trace-fri-header-tied-low-candidate-nonconsumed-root-gap

let *?query* =

staged-security-with-data-state-verifier-event

trace-fri-header-tied-low-candidate-nonconsumed-query-gap
have *split*:
wp-event ?M ?nonconsumed adversary-initial-state $\leq$
wp-event ?M (λ out. ?root out $\vee$?query out) adversary-initial-state
unfolding *staged-security-with-data-state-verifier-event-def*
by (rule *wp-event-mono-on-support*)
(auto dest:
trace-fri-header-tied-low-candidate-nonconsumed-openings-gap-imp-root-or-query
split: option.splits prod.splits)
also have ... $\leq$
wp-event ?M ?root adversary-initial-state +
wp-event ?M ?query adversary-initial-state
by (rule *wp-event-union-bound*)
also have ... $\leq R + Q$
by (rule *add-mono[OF root-bound query-bound]*)
finally show *?thesis* .
qed

lemma *trace-fri-zero-round-final-obstruction-imp-checked-merkle-low-consumed-or-header*:
assumes *support: out $\in$ set-dist (execute verify-monad s)*
and *obstruction: trace-fri-zero-round-final-obstruction s out*
shows
trace-fri-zero-round-checked-final-obstruction s out $\vee$
partial-merkle-inconsistency-bad s out $\vee$
trace-fri-zero-round-verifier-consumed-low-candidate-gap s out $\vee$
trace-fri-bad-with-header-tied-partial-candidate s out
proof –
from *trace-fri-zero-round-final-obstructionE[OF obstruction]*
obtain *trace-roots trace-bs trace-final dg composition-roots*
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr0 candidate-query-idxs trace-openings0
trace-table0 **where** *evidence:*
trace-fri-partial-candidate-opening-evidence s out trace-roots
trace-bs trace-final dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers fr0 candidate-query-idxs trace-openings0
trace-table0
and *rounds-empty: length trace-bs = 0*
by (rule *trace-fri-zero-round-final-obstructionE[OF obstruction]*)
have *fri-openings*:
accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final dg composition-roots composition-bs composition-final
fri-query-idxs trace-round-layers composition-round-layers
using *trace-fri-partial-candidate-opening-evidenceD(1)[OF evidence]* .
from *fri-openings* **obtain** *result final-state* **where** *out-eq:*
out = Some (result, final-state)
unfolding *accepted-fri-opening-transcript-def* **by** *blast*
from *accepted-fri-opening-transcript-headerE[OF fri-openings]*
obtain *fr as rest* **where** *header:*

```

    verifier-header-transcript s fr trace-roots trace-final as dg
    composition-roots composition-final rest
  by blast
show ?thesis
proof (cases partial-merkle-inconsistency-bad s out)
  case True
  then show ?thesis by simp
next
case no-merkle: False
from accepted-fri-opening-transcript-trace-openings-for-query-indices
  [OF fri-openings header out-eq]
obtain trace-openings where partial:
  accepted-with-partial-trace-openings s out fr fri-query-idxs
  trace-openings
by blast
obtain trace-table where candidate:
  partial-trace-table-candidate trace-table trace-openings
using accepted-with-partial-trace-openings-candidate-if-no-partial-merkle-bad
  [OF partial[unfolded out-eq] no-merkle[unfolded out-eq]]
by blast
show ?thesis
proof (cases trace-table-low-degree trace-table)
  case True
  then show ?thesis
proof (cases trace-fri-bad-with-header-tied-partial-candidate s out)
  case True
  then show ?thesis by simp
next
case no-header: False
have reachable:
  trace-fri-bad-with-reachable-partial-candidate s out
  using trace-fri-zero-round-final-obstruction-imp-header-tied-or-gap
    [OF obstruction] no-header
  unfolding trace-fri-reachable-header-tie-gap-def by simp
have gap: trace-fri-reachable-header-tie-gap s out
  unfolding trace-fri-reachable-header-tie-gap-def
  using reachable no-header by simp
have consumed:
  accepted-with-partial-trace-openings-verifier-consumed s out
  trace-roots trace-bs trace-final dg composition-roots
  composition-bs composition-final fri-query-idxs
  trace-round-layers composition-round-layers fr as
  trace-openings
  unfolding accepted-with-partial-trace-openings-verifier-consumed-def
  by (intro conjI exI)
  (rule fri-openings, rule header, rule partial)
have low-gap:
  trace-fri-zero-round-verifier-consumed-low-candidate-gap s out
  unfolding

```



```

    trace-fri-zero-round-verifier-consumed-low-candidate-gap-def
  by (intro exI conjI)
    (rule obstruction, rule gap, rule consumed, rule rounds-empty,
     rule candidate, rule True)
  then show ?thesis by simp
qed
next
case not-low: False
have not-final:
   $\neg$  fri-final-constant-consistent trace-table trace-final
proof
  assume final:
    fri-final-constant-consistent trace-table trace-final
  have low:
    trace-table-low-degree trace-table
  by (rule
    trace-zero-round-candidate-low-degree-if-final-consistent
    [OF rounds-empty fri-openings candidate final])
  then show False
    using not-low by contradiction
qed
have header-zero:
  trace-fri-header-tied-zero-round-final-obstruction s out
  unfolding trace-fri-header-tied-zero-round-final-obstruction-def
  by (intro exI conjI)
    (rule fri-openings, rule header, rule partial, rule candidate,
     rule not-low, rule rounds-empty, rule not-final)
then show ?thesis
  using
    trace-fri-header-tied-zero-round-final-obstruction-imp-checked-or-merkle
  by blast
qed
qed
qed
lemma checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-staged-checked-merkle-low-consumed
  fixes Q M L H :: prob
  assumes checked-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-zero-round-checked-final-obstruction)
    adversary-initial-state  $\leq$  Q
  and merkle-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     partial-merkle-inconsistency-bad)
    adversary-initial-state  $\leq$  M
  and low-consumed-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (staged-security-with-data-state-verifier-event
     trace-fri-zero-round-verifier-consumed-low-candidate-gap)
    adversary-initial-state  $\leq L$ 
and header-bound:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state  $\leq H$ 
shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-zero-round-final-obstruction)
    adversary-initial-state  $\leq Q + M + L + H$ 
proof –
    let ?M = checked-staged-security-experiment-with-data-state A
    let ?zero =
        staged-security-with-data-state-verifier-event
        trace-fri-zero-round-final-obstruction
    let ?checked =
        staged-security-with-data-state-verifier-event
        trace-fri-zero-round-checked-final-obstruction
    let ?merkle =
        staged-security-with-data-state-verifier-event
        partial-merkle-inconsistency-bad
    let ?low =
        staged-security-with-data-state-verifier-event
        trace-fri-zero-round-verifier-consumed-low-candidate-gap
    let ?header =
        staged-security-with-data-state-verifier-event
        trace-fri-bad-with-header-tied-partial-candidate
    have wp-event ?M ?zero adversary-initial-state  $\leq$ 
        wp-event ?M
        ( $\lambda$ out. ?checked out  $\vee$  ?merkle out  $\vee$  ?low out  $\vee$ 
         ?header out)
        adversary-initial-state
    proof (rule wp-event-mono-on-support)
    fix out
    assume support:
        out  $\in$  set-dist (execute ?M adversary-initial-state)
    and zero: ?zero out
    show ?checked out  $\vee$  ?merkle out  $\vee$  ?low out  $\vee$  ?header out
    proof (cases out)
    case None
    then show ?thesis
        using zero unfolding staged-security-with-data-state-verifier-event-def
        by simp
    next
    case (Some out-data)
    then obtain data attacker-state result final-state where out-eq:

```

```

    out = Some (((data, attacker-state), result), final-state)
  by (cases out-data, auto split: prod.splits)
let ?s =
  verifier-state-from-adversary attacker-state
  (staged-proof-transcript data)
from checked-staged-security-experiment-with-data-state-outcomeE
  [OF support[unfolded out-eq]]
have verify-support:
  Some (result, final-state) ∈ set-dist (execute verify-monad ?s)
  by blast
have zero-local:
  trace-fri-zero-round-final-obstruction ?s
  (Some (result, final-state))
using zero unfolding out-eq
  staged-security-with-data-state-verifier-event-def
  by simp
have
  trace-fri-zero-round-checked-final-obstruction ?s
  (Some (result, final-state)) ∨
  partial-merkle-inconsistency-bad ?s
  (Some (result, final-state)) ∨
  trace-fri-zero-round-verifier-consumed-low-candidate-gap ?s
  (Some (result, final-state)) ∨
  trace-fri-bad-with-header-tied-partial-candidate ?s
  (Some (result, final-state))
  by (rule
    trace-fri-zero-round-final-obstruction-imp-checked-merkle-low-consumed-or-header
    [OF verify-support zero-local])
then show ?thesis
  unfolding out-eq staged-security-with-data-state-verifier-event-def
  by simp
qed
qed
also have ... ≤
  wp-event ?M ?checked adversary-initial-state +
  wp-event ?M ?merkle adversary-initial-state +
  wp-event ?M ?low adversary-initial-state +
  wp-event ?M ?header adversary-initial-state
  by (rule wp-event-union-bound4)
also have ... ≤ Q + M + L + H
  by (intro add-mono checked-bound merkle-bound low-consumed-bound
    header-bound)
finally show ?thesis .
qed

lemma trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-imp-nonconsumed:
  assumes
    trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap
    s out

```

shows

trace-fri-header-tied-low-candidate-nonconsumed-openings-gap s out

proof —

from *assms* **obtain** *trace-roots trace-bs trace-final fri-dg*
composition-roots composition-bs composition-final fri-query-idxs
trace-round-layers composition-round-layers fr as rest trace-openings
trace-table fr0 query-idxs0 trace-openings0 trace-table0 **where**

gap: trace-fri-reachable-header-tie-gap s out

and *fri-openings:*

accepted-fri-opening-transcript s out trace-roots trace-bs
trace-final fri-dg composition-roots composition-bs
composition-final fri-query-idxs trace-round-layers
composition-round-layers

and *header:*

verifier-header-transcript s fr trace-roots trace-final as fri-dg
composition-roots composition-final rest

and *partial:*

accepted-with-partial-trace-openings s out fr fri-query-idxs
trace-openings

and *candidate:*

partial-trace-table-candidate trace-table trace-openings

and *low: trace-table-low-degree trace-table*

and *partial0:*

accepted-with-partial-trace-openings s out fr0 query-idxs0
trace-openings0

and *candidate0:*

partial-trace-table-candidate trace-table0 trace-openings0

and *not-low0: $\neg$ trace-table-low-degree trace-table0*

and *mismatch:*

fr0 $\neq$ fr $\vee$ query-idxs0 $\neq$ fri-query-idxs

and *openings-mismatch:*

trace-openings0 $\neq$ trace-openings

unfolding

trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-def
trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def

by *blast*

have *consumed:*

accepted-with-partial-trace-openings-verifier-consumed s out
trace-roots trace-bs trace-final fri-dg composition-roots
composition-bs composition-final fri-query-idxs trace-round-layers
composition-round-layers fr as trace-openings

unfolding *accepted-with-partial-trace-openings-verifier-consumed-def*

by (*intro conjI exI*)

(*rule fri-openings, rule header, rule partial*)

have *evidence0:*

trace-fri-partial-candidate-evidence s out fr0 query-idxs0
trace-openings0 trace-table0

unfolding *trace-fri-partial-candidate-evidence-def*

by (*intro conjI*)

```

    (rule partial0, rule candidate0, rule not-low0)
show ?thesis
  unfolding trace-fri-header-tied-low-candidate-nonconsumed-openings-gap-def
  by (intro exI conjI)
    (rule gap, rule consumed, rule candidate, rule low, rule evidence0,
     rule mismatch, rule openings-mismatch)
qed

lemma checked-staged-security-trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-bound-from-n
assumes nonconsumed-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-header-tied-low-candidate-nonconsumed-openings-gap)
  adversary-initial-state  $\leq N$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap)
  adversary-initial-state  $\leq N$ 
proof (rule order-trans[OF - nonconsumed-bound])
  show wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap)
  adversary-initial-state  $\leq$ 
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-header-tied-low-candidate-nonconsumed-openings-gap)
  adversary-initial-state
  unfolding staged-security-with-data-state-verifier-event-def
  by (rule wp-event-mono-on-support)
    (auto dest:
     trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-imp-nonconsumed
     split: option.splits prod.splits)
qed

lemma checked-staged-security-trace-fri-sampled-assignment-conflict-bound-structural:
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-sampled-assignment-conflict)
  adversary-initial-state  $\leq$ 
  trace-fri-empty-sampled-assignment-structural-error-for budgets A
proof –
have without:
  wp-event (checked-staged-security-experiment-with-data-state A)
  (staged-security-with-data-state-verifier-event
   trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer)

```

```

    adversary-initial-state ≤
    reachable-verifier-event-bound-for A
    trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer
by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
    (simp add:
      trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer-def
      accepted-fri-opening-transcript-def,
      rule reachable-verifier-event-bound-for-ge)
have merkle:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     partial-merkle-inconsistency-bad)
  adversary-initial-state ≤
  trace-fri-controlled-merkle-error-for budgets
unfolding trace-fri-controlled-merkle-error-for-def
by (rule
      checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
      [OF wf controlled])
have gap:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-verifier-tied-same-layer-auth-gap)
  adversary-initial-state ≤
  reachable-verifier-event-bound-for A
  trace-fri-verifier-tied-same-layer-auth-gap
by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
    (simp add: trace-fri-verifier-tied-same-layer-auth-gap-def
      accepted-fri-opening-transcript-def,
      rule reachable-verifier-event-bound-for-ge)
have tied:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-verifier-tied-sampled-assignment-conflict)
  adversary-initial-state ≤
  (reachable-verifier-event-bound-for A
   trace-fri-verifier-tied-sampled-assignment-conflict-without-same-layer +
   trace-fri-controlled-merkle-error-for budgets) +
  reachable-verifier-event-bound-for A
  trace-fri-verifier-tied-same-layer-auth-gap
by (rule
      checked-staged-security-trace-fri-verifier-tied-conflict-bound-from-without-same-merkle-and-auth-gap
      [OF without merkle gap])
have untied:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-sampled-assignment-verifier-untied-gap)
  adversary-initial-state ≤
  reachable-verifier-event-bound-for A
  trace-fri-sampled-assignment-verifier-untied-gap

```

```

by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
  (simp add: trace-fri-sampled-assignment-verifier-untied-gap-def
    trace-fri-sampled-assignment-conflict-def
    trace-fri-partial-candidate-opening-evidence-def
    accepted-fri-opening-transcript-def,
    rule reachable-verifier-event-bound-for-ge)
show ?thesis
unfolding trace-fri-empty-sampled-assignment-structural-error-for-def
by (rule
  checked-staged-security-trace-fri-sampled-assignment-conflict-bound-from-verifier-tied-and-untied
  [OF tied untied])
qed

lemma checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-agreement-sets-header-and-openings
fixes  $H :: \text{prob}$ 
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets  $A$ 
and header-bound:
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
  (staged-security-with-data-state-verifier-event
    trace-fri-bad-with-header-tied-partial-candidate)
  adversary-initial-state  $\leq H$ 
shows
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
  (staged-security-with-data-state-verifier-event
    trace-fri-zero-round-final-obstruction)
  adversary-initial-state  $\leq$ 
  trace-fri-zero-round-agreement-set-target-error-for budgets +
  trace-fri-controlled-merkle-error-for budgets +
  ( $H$  +
    reachable-verifier-event-bound-for  $A$ 
    trace-fri-header-tied-low-candidate-untied-openings-gap) +
   $H$ 
proof –
have checked:
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
  (staged-security-with-data-state-verifier-event
    trace-fri-zero-round-checked-final-obstruction)
  adversary-initial-state  $\leq$ 
  trace-fri-zero-round-agreement-set-target-error-for budgets
by (rule
  checked-staged-security-trace-fri-zero-round-checked-final-obstruction-bound-from-agreement-sets
  [OF wf controlled])
have merkle:
  wp-event (checked-staged-security-experiment-with-data-state  $A$ )
  (staged-security-with-data-state-verifier-event
    partial-merkle-inconsistency-bad)
  adversary-initial-state  $\leq$ 
  trace-fri-controlled-merkle-error-for budgets

```

unfolding *trace-fri-controlled-merkle-error-for-def*
by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
[OF wf controlled])
have *openings*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-low-candidate-untied-openings-gap)
adversary-initial-state ≤
reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-untied-openings-gap
proof (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
show $\bigwedge s. \neg$
trace-fri-header-tied-low-candidate-untied-openings-gap s None
by (simp add: *trace-fri-header-tied-low-candidate-untied-openings-gap-def*
trace-fri-header-tied-low-candidate-untied-reachable-witness-case-def
accepted-fri-opening-transcript-def)
next
fix *data attacker-state*
assume *support*:
Some (data, attacker-state) ∈
set-dist
(execute (checked-staged-transcript-program A)
adversary-initial-state)
show *wp-event verify-monad*
(trace-fri-header-tied-low-candidate-untied-openings-gap
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
(verifier-state-from-adversary attacker-state
(staged-proof-transcript data)))
≤ reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-untied-openings-gap
by (rule *reachable-verifier-event-bound-for-ge[OF support]*)
qed
have *untied*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-low-candidate-untied-reachable-witness-gap)
adversary-initial-state ≤
reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-untied-openings-gap
proof –
have *wp-event (checked-staged-security-experiment-with-data-state A)*
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-low-candidate-untied-reachable-witness-gap)
adversary-initial-state ≤
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-low-candidate-untied-openings-gap)


```

    adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono-on-support)
    (auto dest: trace-fri-header-tied-low-candidate-untied-gap-imp-openings
      split: option.splits prod.splits)
also have ... ≤
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-openings-gap
by (rule openings)
finally show ?thesis .
qed
have transfer:
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-candidate-transfer-gap)
    adversary-initial-state ≤
    H +
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-untied-openings-gap
by (rule
    checked-staged-security-trace-fri-header-tied-candidate-transfer-gap-bound-from-staged-header-and-untied
    [OF header-bound untied])
show ?thesis
by (rule
    checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-staged-checked-merkle-transfer-header
    [OF checked merkle transfer header-bound])
qed

lemma checked-staged-security-trace-fri-header-tied-low-candidate-untied-openings-gap-bound-from-hash-and-nonnor
assumes wf: staged-budget-wellformed budgets
and controlled: staged-adversary-controlled budgets A
shows
    wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-header-tied-low-candidate-untied-openings-gap)
    adversary-initial-state ≤
    trace-fri-controlled-merkle-error-for budgets +
    ((reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap +
      reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap) +
      reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap)
proof –
let ?M = checked-staged-security-experiment-with-data-state A
let ?openings =
    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-untied-openings-gap
let ?hash =

```

```

    staged-security-with-data-state-verifier-event
    hash-map-output-collision-bad
let ?nonnormal =
    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap
let ?nonconsumed =
    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-nonconsumed-openings-gap
let ?root =
    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-nonconsumed-root-gap
let ?query =
    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-nonconsumed-query-gap
let ?irrelevant =
    staged-security-with-data-state-verifier-event
    trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap
have hash:
    wp-event ?M ?hash adversary-initial-state ≤
    trace-fri-controlled-merkle-error-for budgets
unfolding trace-fri-controlled-merkle-error-for-def
by (rule
    checked-staged-security-with-data-state-hash-map-output-collision-bad-bound-controlled
    [OF wf controlled])
have root:
    wp-event ?M ?root adversary-initial-state ≤
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-nonconsumed-root-gap
by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
    (simp add: trace-fri-header-tied-low-candidate-nonconsumed-root-gap-def
    accepted-with-partial-trace-openings-verifier-consumed-def
    trace-fri-partial-candidate-evidence-def
    accepted-fri-opening-transcript-def,
    rule reachable-verifier-event-bound-for-ge)
have query:
    wp-event ?M ?query adversary-initial-state ≤
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-nonconsumed-query-gap
by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)
    (simp add: trace-fri-header-tied-low-candidate-nonconsumed-query-gap-def
    accepted-with-partial-trace-openings-verifier-consumed-def
    trace-fri-partial-candidate-evidence-def
    accepted-fri-opening-transcript-def,
    rule reachable-verifier-event-bound-for-ge)
have irrelevant:
    wp-event ?M ?irrelevant adversary-initial-state ≤
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap
by (rule checked-staged-security-with-data-state-verifier-event-bound-from-cont)

```

```

(simp add:
  trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap-def
  accepted-with-partial-trace-openings-verifier-consumed-def
  trace-fri-partial-candidate-evidence-def
  accepted-fri-opening-transcript-def,
  rule reachable-verifier-event-bound-for-ge)
have root-refined:
  wp-event ?M ?root adversary-initial-state ≤
    reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap +
    reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap
by (rule
  checked-staged-security-trace-fri-header-tied-low-candidate-nonconsumed-root-gap-bound-from-query-and-irrelevant
    [OF query irrelevant])
have nonconsumed-root-query:
  wp-event ?M ?nonconsumed adversary-initial-state ≤
    (reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap +
    reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap) +
    reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap
by (rule
  checked-staged-security-trace-fri-header-tied-low-candidate-nonconsumed-openings-gap-bound-from-root-query
    [OF root-refined query])
have nonnormal-root-query:
  wp-event ?M ?nonnormal adversary-initial-state ≤
    (reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap +
    reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap) +
    reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap
by (rule
  checked-staged-security-trace-fri-header-tied-low-candidate-untied-nonnormalized-openings-gap-bound-from-nonconsumed-root-query
    [OF nonconsumed-root-query])
have wp-event ?M ?openings adversary-initial-state ≤
  wp-event ?M (λout. ?hash out ∨ ?nonnormal out)
    adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono-on-support)
  (auto
    dest:
      trace-fri-header-tied-low-candidate-untied-openings-gap-imp-normalized-or-nonnormalized
      trace-fri-header-tied-low-candidate-untied-normalized-openings-gap-imp-hash-collision
    split: option.splits prod.splits)
also have ... ≤
  wp-event ?M ?hash adversary-initial-state +

```

```

    wp-event ?M ?nonnormal adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤
  trace-fri-controlled-merkle-error-for budgets +
  ((reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-nonconsumed-query-gap +
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap) +
    reachable-verifier-event-bound-for A
    trace-fri-header-tied-low-candidate-nonconsumed-query-gap)
  by (rule add-mono[OF hash nonnormal-root-query])
finally show ?thesis .
qed

lemma checked-staged-security-trace-fri-zero-round-verifier-consumed-low-candidate-gap-bound-from-hash-and-nonnormal
  assumes wf: staged-budget-wellformed budgets
    and controlled: staged-adversary-controlled budgets A
  shows
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-zero-round-verifier-consumed-low-candidate-gap)
    adversary-initial-state ≤
    trace-fri-controlled-merkle-error-for budgets +
    ((reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap +
      reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap) +
      reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap)
  proof -
    have wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-zero-round-verifier-consumed-low-candidate-gap)
    adversary-initial-state ≤
    wp-event (checked-staged-security-experiment-with-data-state A)
      (staged-security-with-data-state-verifier-event
        trace-fri-header-tied-low-candidate-untied-openings-gap)
    adversary-initial-state
    unfolding staged-security-with-data-state-verifier-event-def
  by (rule wp-event-mono-on-support)
    (auto
      dest:
        trace-fri-zero-round-verifier-consumed-low-candidate-gap-imp-untied-openings
        split: option.splits prod.splits)
  also have ... ≤
    trace-fri-controlled-merkle-error-for budgets +
    ((reachable-verifier-event-bound-for A
      trace-fri-header-tied-low-candidate-nonconsumed-query-gap +
      reachable-verifier-event-bound-for A

```

$\text{trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap}) +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-low-candidate-nonconsumed-query-gap})$
by (rule
 $\text{checked-staged-security-trace-fri-header-tied-low-candidate-untied-openings-gap-bound-from-hash-and-nonnormal}$
 $[OF \text{ } wf \text{ controlled}]$)
finally show $?thesis$.
qed

lemma $\text{checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-agreement-sets-header-and-nonnormal}$
fixes $H :: prob$

assumes wf : $\text{staged-budget-wellformed budgets}$

and controlled : $\text{staged-adversary-controlled budgets } A$

and header-bound :

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$
 $\text{adversary-initial-state} \leq H$

shows

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-zero-round-final-obstruction})$
 $\text{adversary-initial-state} \leq$
 $\text{trace-fri-zero-round-agreement-set-target-error-for budgets} +$
 $\text{trace-fri-controlled-merkle-error-for budgets} +$
 $(\text{trace-fri-controlled-merkle-error-for budgets} +$
 $((\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-low-candidate-nonconsumed-query-gap} +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap}) +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-low-candidate-nonconsumed-query-gap})) +$
 H

proof –

have checked :

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-zero-round-checked-final-obstruction})$
 $\text{adversary-initial-state} \leq$
 $\text{trace-fri-zero-round-agreement-set-target-error-for budgets}$

by (rule

$\text{checked-staged-security-trace-fri-zero-round-checked-final-obstruction-bound-from-agreement-sets}$
 $[OF \text{ } wf \text{ controlled}]$)

have merkle :

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{partial-merkle-inconsistency-bad})$
 $\text{adversary-initial-state} \leq$
 $\text{trace-fri-controlled-merkle-error-for budgets}$

unfolding *trace-fri-controlled-merkle-error-for-def*
by (rule
checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
[OF wf controlled])
have *openings*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-zero-round-verifier-consumed-low-candidate-gap)
adversary-initial-state ≤
trace-fri-controlled-merkle-error-for budgets +
((reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-nonconsumed-query-gap +
reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap) +
reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-nonconsumed-query-gap)
by (rule
checked-staged-security-trace-fri-zero-round-verifier-consumed-low-candidate-gap-bound-from-hash-and-nonnorm
[OF wf controlled])
have *zero*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-zero-round-final-obstruction)
adversary-initial-state ≤
trace-fri-zero-round-agreement-set-target-error-for budgets +
trace-fri-controlled-merkle-error-for budgets +
(trace-fri-controlled-merkle-error-for budgets +
((reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-nonconsumed-query-gap +
reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap) +
reachable-verifier-event-bound-for A
trace-fri-header-tied-low-candidate-nonconsumed-query-gap))) +
H
by (rule
checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-staged-checked-merkle-low-consumed-he
[OF checked merkle openings header-bound])
show *?thesis*
by (rule *zero*)
qed

definition *trace-empty-header-sampled-split-one-step-error-for*
where

trace-empty-header-sampled-split-one-step-error-for budgets A empty-bad =
trace-one-step-sampled-error-for budgets empty-bad +
(trace-fri-empty-sampled-assignment-structural-error-for budgets A +
trace-empty-header-openings-zero-error-for budgets A empty-bad)

definition *concrete-trace-empty-header-sampled-split-one-step-error-for*

where

concrete-trace-empty-header-sampled-split-one-step-error-for budgets $A =$
trace-empty-header-sampled-split-one-step-error-for budgets A
trace-one-step-split-bad

lemma *trace-empty-header-sampled-split-one-step-error-for-bound*:

fixes *empty-bad* :: nat $\Rightarrow$ 'f list $\Rightarrow$ 'f set

assumes *wf*: *staged-budget-wellformed* budgets

and *controlled*: *staged-adversary-controlled* budgets A

and *empty-subset*:

$\bigwedge i$ *prefix*.

$i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \Rightarrow$

$\text{length } \text{prefix} = i \Rightarrow$

$\text{empty-bad } i \text{ prefix} \subseteq$

fri-one-step-disagreement-challenges

$(\text{trace-fri-canonical-layer-len } i) (\text{empty-pw } i)$

$(\text{empty-layer } i \text{ prefix}) (\text{empty-claimed } i \text{ prefix})$

$(\text{empty-dom } i \text{ prefix})$

shows

wp-event (*checked-staged-security-experiment-with-data-state* A)

$(\text{staged-security-with-data-state-verifier-event}$

$\text{trace-fri-bad-with-empty-composition-header-candidates})$

adversary-initial-state

$\leq \text{trace-empty-header-sampled-split-one-step-error-for budgets } A \text{ empty-bad}$

proof —

have *sampled*:

wp-event (*checked-staged-security-experiment-with-data-state* A)

$(\text{staged-security-with-data-state-verifier-event}$

$\text{trace-fri-sampled-query-bad-candidate})$

adversary-initial-state

$\leq \text{trace-one-step-sampled-error-for budgets empty-bad}$

by (rule *trace-one-step-sampled-error-for-bound*

[OF *wf controlled empty-subset*])

have *conflict-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* A)

$(\text{staged-security-with-data-state-verifier-event}$

$\text{trace-fri-sampled-assignment-conflict})$

adversary-initial-state

$\leq \text{trace-fri-empty-sampled-assignment-structural-error-for budgets } A$

by (rule

checked-staged-security-trace-fri-sampled-assignment-conflict-bound-structural

[OF *wf controlled*])

have *header-bound*:

wp-event (*checked-staged-security-experiment-with-data-state* A)

$(\text{staged-security-with-data-state-verifier-event}$

$\text{trace-fri-bad-with-header-tied-partial-candidate})$

adversary-initial-state

$\leq \text{trace-header-one-step-error-for budgets } A \text{ empty-bad}$

by (rule *trace-header-one-step-error-for-bound*

$[OF \text{ wf controlled empty-subset}]$
have zero-bound:
 $wp\text{-event } (checked\text{-staged-security-experiment-with-data-state } A)$
 $(staged\text{-security-with-data-state-verifier-event}$
 $\text{trace-fri-zero-round-final-obstruction})$
 $adversary\text{-initial-state}$
 $\leq \text{trace-empty-header-openings-zero-error-for budgets } A \text{ empty-bad}$
unfolding $\text{trace-empty-header-openings-zero-error-for-def}$
by (rule
 $checked\text{-staged-security-trace-fri-zero-round-final-obstruction-bound-from-agreement-sets-header-and-openings}$
 $[OF \text{ wf controlled header-bound}]$)
have branch:
 $wp\text{-event } (checked\text{-staged-security-experiment-with-data-state } A)$
 $(staged\text{-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $adversary\text{-initial-state}$
 $\leq \text{trace-one-step-sampled-error-for budgets empty-bad} +$
 $(\text{trace-fri-empty-sampled-assignment-structural-error-for budgets } A +$
 $\text{trace-empty-header-openings-zero-error-for budgets } A \text{ empty-bad})$
by (rule
 $checked\text{-staged-security-trace-fri-empty-header-bound-from-staged-sampled-conflict-and-zero}$
 $[OF \text{ sampled conflict-bound zero-bound}]$)
show ?thesis
using branch
by (simp add: $\text{trace-empty-header-sampled-split-one-step-error-for-def}$)
qed

theorem stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-sampled-split-one-step-fr
fixes $\text{trace-bad empty-bad} :: \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}$
and $\text{comp-bad} :: 'f \Rightarrow \text{nat} \Rightarrow 'f \text{ list} \Rightarrow 'f \text{ set}$
assumes $\text{false-statement}: \neg \text{exists-valid-trace}$
and $\text{wf}: \text{staged-budget-wellformed budgets}$
and $\text{controlled}: \text{staged-adversary-controlled budgets } A$
and trace-subset :
 $\bigwedge i \text{ prefix.}$
 $i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \implies$
 $\text{length prefix} = i \implies$
 $\text{trace-bad } i \text{ prefix} \subseteq$
 $\text{fri-one-step-disagreement-challenges}$
 $(\text{trace-fri-canonical-layer-len } i) (\text{empty-pw } i)$
 $(\text{empty-layer } i \text{ prefix}) (\text{empty-claimed } i \text{ prefix})$
 $(\text{empty-dom } i \text{ prefix})$
and comp-subset :
 $\bigwedge dg \text{ i prefix.}$
 $i < \text{fri-round-count-for-degree-bound } (\text{to-nat } dg) \implies$
 $\text{length prefix} = i \implies$
 $\text{comp-bad } dg \text{ i prefix} \subseteq$
 $\text{fri-one-step-disagreement-challenges}$
 $(\text{composition-fri-canonical-layer-len } dg \text{ i}) (\text{comp-pw } dg \text{ i})$

(comp-layer dg i prefix) (comp-claimed dg i prefix)
 (comp-dom dg i prefix)
and empty-subset:
 $\bigwedge i$ prefix.
 $i < \text{fri-round-count-for-degree-bound } (\text{clength} - 1) \implies$
 $\text{length prefix} = i \implies$
 $\text{empty-bad } i \text{ prefix} \subseteq$
 $\text{fri-one-step-disagreement-challenges}$
 $(\text{trace-fri-canonical-layer-len } i) (\text{empty-pw } i)$
 $(\text{empty-layer } i \text{ prefix}) (\text{empty-claimed } i \text{ prefix})$
 $(\text{empty-dom } i \text{ prefix})$
shows checked-staged-adversary-acceptance-probability $A \leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound})) +$
 $(\sum i < \text{rounds.}$
 $\text{active-route-current-prefix-authenticated-error-for } A \ i +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound}) +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound})) +$
 $\text{trace-header-one-step-error-for budgets } A \text{ trace-bad} +$
 $\text{composition-one-step-verifier-tied-controlled-error-for budgets}$
 $\text{comp-bad} +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound}) +$
 $\text{trace-empty-header-sampled-split-one-step-error-for budgets } A \text{ empty-bad} +$
 $\text{active-route-current-empty-query-error-for } A)$
proof –
have trace-bound:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$

adversary-initial-state
 $\leq$ *trace-header-one-step-error-for budgets A trace-bad*
by (rule *trace-header-one-step-error-for-bound*
[*OF wf controlled trace-subset*])
have *comp-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(*staged-security-with-data-state-verifier-event*
composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state
 $\leq$ *composition-one-step-verifier-tied-controlled-error-for budgets*
comp-bad
by (rule *composition-one-step-verifier-tied-controlled-error-for-bound*
[*OF wf controlled comp-subset*])
have *empty-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(*staged-security-with-data-state-verifier-event*
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state
 $\leq$ *trace-empty-header-sampled-split-one-step-error-for budgets A empty-bad*
by (rule *trace-empty-header-sampled-split-one-step-error-for-bound*
[*OF wf controlled empty-subset*])
show *?thesis*
by (rule
stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-bounds
[*OF false-statement wf controlled trace-bound comp-bound empty-bound*])
qed

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-sampled-split-endpoint*:

assumes *false-statement*: $\neg$ *exists-valid-trace*
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0
(*staged-attacker-query-budget budgets* +
staged-challenge-query-budget + *verifier-hash-query-budget*) +
(*composition-error-bound* +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +
hash-collision-budget-value 0
(*staged-alpha-search-queries budgets 0*) +
staged-phase-relation-error size
(*staged-alpha-search-queries budgets 0*) +
(*active-route-transcript-pre-error-for A* +
staged-concrete-transcript-target-error-bound)) +
($\sum$ *i* < *rounds*.
active-route-current-prefix-authenticated-error-for A i +
(*active-route-transcript-pre-error-for A* +
staged-concrete-transcript-target-error-bound) +
(*active-route-transcript-pre-error-for A* +

staged-concrete-transcript-target-error-bound)) +
concrete-trace-header-one-step-error-for budgets *A* +
concrete-composition-one-step-error-for budgets *A* +
(*composition-error-bound* +
staged-phase-relation-error size
(*staged-alpha-search-queries* budgets 0) +
hash-collision-budget-value 0
(*staged-alpha-search-queries* budgets 0) +
staged-phase-relation-error size
(*staged-alpha-search-queries* budgets 0) +
(*active-route-transcript-pre-error-for* *A* +
staged-concrete-transcript-target-error-bound) +
concrete-trace-empty-header-sampled-split-one-step-error-for budgets *A* +
active-route-current-empty-query-error-for *A*)
unfolding *concrete-trace-header-one-step-error-for-def*
concrete-composition-one-step-error-for-def
concrete-trace-empty-header-sampled-split-one-step-error-for-def
by (rule
stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-sampled-split-one-step-fri-comp
[**where** *trace-bad* = *trace-one-step-split-bad*
and *comp-bad* = *composition-one-step-split-bad*
and *empty-bad* = *trace-one-step-split-bad*
and *empty-pw* = *trace-one-step-split-pw*
and *empty-layer* = *trace-one-step-split-layer*
and *empty-claimed* = *trace-one-step-split-claimed*
and *empty-dom* = *trace-one-step-split-domain*
and *comp-pw* = *composition-one-step-split-pw*
and *comp-layer* = *composition-one-step-split-layer*
and *comp-claimed* = *composition-one-step-split-claimed*
and *comp-dom* = *composition-one-step-split-domain*,
OF false-statement wf controlled])
(*simp-all add*:
trace-one-step-split-bad-def
composition-one-step-split-bad-def
trace-one-step-disagreement-bad-for-def
composition-one-step-disagreement-bad-for-def)

definition *active-route-current-prefix-conceptual-residual-error-for*
where

active-route-current-prefix-conceptual-residual-error-for
(*A* :: 'f *staged-adversary*) (*i* :: nat) =
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state* *A* *i*)
checked-staged-security-with-query-prefix-conceptual-context-residual-at
adversary-initial-state

definition *active-route-current-prefix-conceptual-hit-residual-error-for*
where

active-route-current-prefix-conceptual-hit-residual-error-for

```

(A :: 'f staged-adversary) (i :: nat) =
wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  (λout.
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
      i out ∧
    checked-staged-security-with-query-prefix-conceptual-context-residual-at
      out)
  adversary-initial-state

```

definition *active-route-current-prefix-conceptual-bound-for*
where

```

active-route-current-prefix-conceptual-bound-for
  (budgets :: staged-budgets) (A :: 'f staged-adversary)
  (i :: nat) =
staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +
query-error-bound +
active-route-current-prefix-conceptual-hit-residual-error-for A i

```

lemma *active-route-current-prefix-authenticated-error-for-le-conceptual-bound:*

assumes *wf*: staged-budget-wellformed budgets
and *controlled*: staged-adversary-controlled budgets A
and *i-bound*: i < rounds

shows

```

active-route-current-prefix-authenticated-error-for A i ≤
active-route-current-prefix-conceptual-bound-for budgets A i

```

proof –

let ?M = checked-staged-security-experiment-with-query-prefix-data-state A i

let ?Hit =

```

checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
  i

```

let ?Conceptual =

```

checked-staged-security-with-query-prefix-conceptual-target-hit

```

let ?Residual =

```

λout. ?Hit out ∧
checked-staged-security-with-query-prefix-conceptual-context-residual-at
  out

```

have *event-le*:

```

wp-event ?M ?Hit adversary-initial-state ≤
wp-event ?M (λout. ?Conceptual out ∨ ?Residual out)
adversary-initial-state

```

proof (rule wp-event-mono-on-support)

fix out

assume *support*: out ∈ set-dist (execute ?M adversary-initial-state)

and *hit*: ?Hit out

from checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-imp-conceptual-or-residual-on-support

[OF wf controlled i-bound support hit]

show ?Conceptual out ∨ ?Residual out

```

    using hit by blast
qed
have conceptual-bound:
  wp-event ?M ?Conceptual adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
  by (rule
    checked-staged-security-with-query-prefix-conceptual-target-hit-bound
    [OF wf controlled i-bound])
have union-bound:
  wp-event ?M (λout. ?Conceptual out ∨ ?Residual out)
    adversary-initial-state ≤
    wp-event ?M ?Conceptual adversary-initial-state +
    wp-event ?M ?Residual adversary-initial-state
  by (rule wp-event-union-bound)
have residual-bound:
  wp-event ?M ?Residual adversary-initial-state ≤
    active-route-current-prefix-conceptual-hit-residual-error-for A i
  by (simp add:
    active-route-current-prefix-conceptual-hit-residual-error-for-def)
have sum-bound:
  wp-event ?M ?Conceptual adversary-initial-state +
  wp-event ?M ?Residual adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound +
    active-route-current-prefix-conceptual-hit-residual-error-for A i
  by (intro add-mono conceptual-bound residual-bound)
show ?thesis
  unfolding active-route-current-prefix-authenticated-error-for-def
    active-route-current-prefix-conceptual-bound-for-def
  by (rule order-trans[OF event-le])
  (rule order-trans[OF union-bound sum-bound])
qed

definition active-route-current-empty-conceptual-residual-error-for
  :: 'f itself ⇒ 'f staged-adversary ⇒ prob
where
  active-route-current-empty-conceptual-residual-error-for
    (- :: 'f itself)
    (A :: 'f staged-adversary) =
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A 0)
    (checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
      (0 :: nat))
    adversary-initial-state

definition active-route-current-empty-conceptual-bound-for
where

```

active-route-current-empty-conceptual-bound-for
 (budgets :: staged-budgets) (A :: 'f staged-adversary) =
staged-phase-relation-error size
 (staged-query-search-queries budgets 0 + 1) +
query-error-bound +
active-route-current-empty-conceptual-residual-error-for TYPE('f) A +
 (active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound)

lemma *active-route-current-empty-query-error-for-le-conceptual-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows

active-route-current-empty-query-error-for A ≤

active-route-current-empty-conceptual-bound-for budgets A

proof –

have *i-bound: 0 < rounds*

using *rounds-positive* **by** *simp*

have *residual-bound:*

wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A 0)

(checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
(0 :: nat))

adversary-initial-state ≤

active-route-current-empty-conceptual-residual-error-for TYPE('f) A

unfolding *active-route-current-empty-conceptual-residual-error-for-def*

by (rule *wp-event-mono*) *simp*

have *pre-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-transcript-pre-hit

adversary-initial-state ≤ active-route-transcript-pre-error-for A

by (*simp add: active-route-transcript-pre-error-for-def*)

have *new-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)

staged-security-with-data-state-transcript-new-hit

adversary-initial-state ≤ staged-concrete-transcript-target-error-bound

by (rule *checked-staged-security-with-data-state-transcript-new-hit-bound*
[OF *wf controlled*])

show *?thesis*

unfolding *active-route-current-empty-query-error-for-def*

active-route-current-empty-conceptual-bound-for-def

by (rule

checked-staged-security-with-data-state-current-empty-query-opening-hit-at-bound-from-conceptual-empty-residual
[OF *wf controlled i-bound residual-bound pre-bound new-bound*])

qed

definition *trace-no-full-cover-sampled-query-error-for*

where

trace-no-full-cover-sampled-query-error-for budgets bad =

$(\sum \text{challenges} \in \text{fri-challenge-space } (\text{ceil-log clength}).$
 $(1 / \text{nnreal } (\text{CARD}(f) \wedge \text{ceil-log clength})) *$
 $(\text{nnreal}$
 $(\text{card}$
 $(\text{trace-fri-sampled-query-no-full-cover-failure-query-fiber}$
 $\text{bad challenges})) *$
 $(1 / \text{nnreal } (\text{card query-sample-space})) \wedge \text{rounds})) +$
 $\text{hash-relation-budget-value}$
 $(\text{query-index-row-list-relation-fiber-bound}$
 $(\bigcup \text{challenges} \in \text{fri-challenge-space } (\text{ceil-log clength}).$
 $\text{trace-fri-sampled-query-no-full-cover-failure-query-fiber}$
 $\text{bad challenges}))$
 $(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget}) +$
 $\text{hash-relation-budget-value}$
 $(\text{card } (\text{fri-challenge-space } (\text{ceil-log clength})) * \text{ceil-log clength})$
 $(\text{staged-attacker-query-budget budgets} + \text{staged-challenge-query-budget})$

definition *trace-canonical-full-cover-shape-failure-error-for*
where

$\text{trace-canonical-full-cover-shape-failure-error-for } A \text{ bad} =$
 $\text{reachable-verifier-event-bound-for } A$
 $(\lambda s. \text{trace-fri-sampled-full-cover-proximity-failure } s \text{ bad}) +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-sampled-full-cover-index-failure} +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-sampled-full-cover-shape-failure}$

definition *trace-restricted-sampled-query-error-for*
where

$\text{trace-restricted-sampled-query-error-for budgets } A \text{ bad} =$
 $\text{trace-no-full-cover-sampled-query-error-for budgets bad} +$
 $\text{trace-canonical-full-cover-shape-failure-error-for } A \text{ bad}$

lemma *trace-restricted-sampled-query-error-for-bound:*

assumes *wf: staged-budget-wellformed budgets*
and *controlled: staged-adversary-controlled budgets A*
shows

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-sampled-query-bad-candidate})$
 $\text{adversary-initial-state}$
 $\leq \text{trace-restricted-sampled-query-error-for budgets } A \text{ bad}$

proof –

have *no-full:*
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $(\lambda s. \text{trace-fri-sampled-query-no-full-cover-failure-candidate } s \text{ bad}))$
 $\text{adversary-initial-state}$
 $\leq \text{trace-no-full-cover-sampled-query-error-for budgets bad}$

```

unfolding trace-no-full-cover-sampled-query-error-for-def
by (rule
  checked-staged-security-trace-fri-sampled-query-no-full-cover-failure-candidate-bound-from-restricted-pairs
  [OF wf controlled])
have failure:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      ( $\lambda s$ . trace-fri-sampled-canonical-full-cover-failure s bad))
    adversary-initial-state
   $\leq$  trace-canonical-full-cover-shape-failure-error-for A bad
unfolding trace-canonical-full-cover-shape-failure-error-for-def
proof (rule
  checked-staged-security-trace-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases)
fix data attacker-state
assume support:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (trace-fri-sampled-full-cover-proximity-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad)
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
   $\leq$  reachable-verifier-event-bound-for A
  ( $\lambda s$ . trace-fri-sampled-full-cover-proximity-failure s bad)
by (rule reachable-verifier-event-bound-for-ge[OF support])
next
fix data attacker-state
assume support:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (trace-fri-sampled-full-cover-index-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
   $\leq$  reachable-verifier-event-bound-for A
  trace-fri-sampled-full-cover-index-failure
by (rule reachable-verifier-event-bound-for-ge[OF support])
next
fix data attacker-state
assume support:
  Some (data, attacker-state)  $\in$ 
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad

```



```

    (trace-fri-sampled-full-cover-shape-failure
      (verifier-state-from-adversary attacker-state
        (staged-proof-transcript data)))
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))
  ≤ reachable-verifier-event-bound-for A
    trace-fri-sampled-full-cover-shape-failure
  by (rule reachable-verifier-event-bound-for-ge[OF support])
qed
have sampled:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      trace-fri-sampled-query-bad-candidate)
    adversary-initial-state
  ≤ trace-no-full-cover-sampled-query-error-for budgets bad +
    trace-canonical-full-cover-shape-failure-error-for A bad
  by (rule
    checked-staged-security-trace-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-canonical-failure
      [OF no-full failure])
show ?thesis
  using sampled
  by (simp add: trace-restricted-sampled-query-error-for-def)
qed

```

definition *concrete-sampled-split-stark-soundness-bound-for*
where

```

concrete-sampled-split-stark-soundness-bound-for budgets A =
  hash-collision-budget-value 0
  (staged-attacker-query-budget budgets +
    staged-challenge-query-budget + verifier-hash-query-budget) +
  (composition-error-bound +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
    (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
    (staged-alpha-search-queries budgets 0) +
    (active-route-transcript-pre-error-for A +
      staged-concrete-transcript-target-error-bound)) +
  (∑ i < rounds.
    active-route-current-prefix-conceptual-bound-for budgets A i +
    (active-route-transcript-pre-error-for A +
      staged-concrete-transcript-target-error-bound) +
    (active-route-transcript-pre-error-for A +
      staged-concrete-transcript-target-error-bound)) +
  concrete-trace-header-one-step-error-for budgets A +
  concrete-composition-one-step-error-for budgets A +
  (composition-error-bound +
    staged-phase-relation-error size

```

(staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 concrete-trace-empty-header-sampled-split-one-step-error-for budgets A +
 active-route-current-empty-conceptual-bound-for budgets A)

theorem stark-soundness-trace-empty-sampled-split-endpoint:

assumes false-statement: $\neg$ exists-valid-trace

and wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

shows checked-staged-adversary-acceptance-probability A $\leq$
 concrete-sampled-split-stark-soundness-bound-for budgets A

proof –

have old-bound:

checked-staged-adversary-acceptance-probability A $\leq$
 hash-collision-budget-value 0
 (staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds.}$
 active-route-current-prefix-authenticated-error-for A i +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 concrete-trace-header-one-step-error-for budgets A +
 concrete-composition-one-step-error-for budgets A +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 concrete-trace-empty-header-sampled-split-one-step-error-for budgets A +

$\text{active-route-current-empty-query-error-for } A)$
by (rule
 $\text{stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-sampled-split-endpoint}$
 $[OF \text{ false-statement wf controlled}]$)
have prefix-terms:
 $(\sum i < \text{rounds.}$
 $\text{active-route-current-prefix-authenticated-error-for } A \ i +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound}) +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound}))$
 $\leq$
 $(\sum i < \text{rounds.}$
 $\text{active-route-current-prefix-conceptual-bound-for budgets } A \ i +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound}) +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound}))$
by (intro sum-mono add-mono order-refl
 $\text{active-route-current-prefix-authenticated-error-for-le-conceptual-bound}$
 $[OF \text{ wf controlled}]$)
 simp
have empty-term:
 $\text{active-route-current-empty-query-error-for } A \leq$
 $\text{active-route-current-empty-conceptual-bound-for budgets } A$
by (rule $\text{active-route-current-empty-query-error-for-le-conceptual-bound}$
 $[OF \text{ wf controlled}]$)
have old-le-new:
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $(\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound})) +$
 $(\sum i < \text{rounds.}$
 $\text{active-route-current-prefix-authenticated-error-for } A \ i +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound}) +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\text{staged-concrete-transcript-target-error-bound})) +$
 $\text{concrete-trace-header-one-step-error-for budgets } A +$
 $\text{concrete-composition-one-step-error-for budgets } A +$
 $(\text{composition-error-bound} +$

```

    staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
    hash-collision-budget-value 0
      (staged-alpha-search-queries budgets 0) +
    staged-phase-relation-error size
      (staged-alpha-search-queries budgets 0) +
    (active-route-transcript-pre-error-for A +
      staged-concrete-transcript-target-error-bound) +
    concrete-trace-empty-header-sampled-split-one-step-error-for budgets A +
    active-route-current-empty-query-error-for A)
  ≤ concrete-sampled-split-stark-soundness-bound-for budgets A
unfolding concrete-sampled-split-stark-soundness-bound-for-def
by (intro add-mono order-refl prefix-terms empty-term)
show ?thesis
by (rule order-trans[OF old-bound old-le-new])
qed

end

hide-fact
  soundness.stark-soundness-trace-empty-sampled-split-endpoint
  soundness.stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-sampled-split-endpoint

end

theory Soundness-FRI-Trace-Restricted-Query-Endpoint
  imports Soundness-FRI-Trace-Empty-Sampled-Assignment-Split
begin

Public entry point for the staged STARK soundness theorem. The theorem in this theory combines the staged adversary experiment, Merkle authentication reductions, query bounds, composition randomization bounds, and the FRI query/challenge accounting into one explicit probability bound.

The bound uses restricted trace and composition product-accounting terms and the proved zero-round trace empty-header estimates from the preceding FRI theories.

context soundness
begin

definition trace-header-restricted-one-step-error-for
where
  trace-header-restricted-one-step-error-for budgets A trace-bad =
    trace-restricted-sampled-query-error-for budgets A trace-bad +
    trace-fri-header-replay-conflict-zero-error-for budgets A

definition concrete-trace-header-restricted-one-step-error-for
where
  concrete-trace-header-restricted-one-step-error-for budgets A =

```

trace-header-restricted-one-step-error-for budgets A
 $(\lambda\cdot \text{trace-one-step-split-bad})$

lemma *trace-fri-header-tied-sampled-assignment-conflict-replay-bound:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-sampled-assignment-conflict)
adversary-initial-state

$\leq \text{trace-fri-header-replay-conflict-error-for } A$

proof (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)

fix *data attacker-state*

assume *support:*

Some (data, attacker-state) ∈
set-dist (execute (checked-staged-transcript-program A)
adversary-initial-state)

let $?s =$

verifier-state-from-adversary attacker-state
(staged-proof-transcript data)

let $?M =$

reachable-verifier-event-bound-for A partial-merkle-inconsistency-bad

let $?H =$

reachable-verifier-event-bound-for A
trace-fri-header-tied-recorded-sibling-candidate-missing

have *merkle-bound:*

wp-event verify-monad (partial-merkle-inconsistency-bad ?s) ?s
 $\leq ?M$

by (rule *reachable-verifier-event-bound-for-ge[OF support]*)

have *recorded-missing-bound:*

wp-event verify-monad
(trace-fri-header-tied-recorded-sibling-candidate-missing ?s) ?s
 $\leq ?H$

by (rule *reachable-verifier-event-bound-for-ge[OF support]*)

have *head-neq:*

wp-event verify-monad
(trace-fri-header-tied-base-head-value-neq-gap ?s) ?s $\leq ?M$

by (rule

wp-trace-fri-header-tied-base-head-value-neq-gap-bound-from-partial-merkle)
 (rule *merkle-bound*)

have *head-mismatch:*

wp-event verify-monad
(trace-fri-header-tied-base-head-value-mismatch-gap ?s) ?s
 $\leq ?M$

by (rule

wp-trace-fri-header-tied-base-head-value-mismatch-gap-bound-from-neq-gap
 $[OF \text{ head-neq}]$)

have *alignment:*

wp-event verify-monad
(trace-fri-header-tied-base-value-alignment-gap ?s) ?s $\leq ?M$

by (rule

$wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}base\text{-}value\text{-}alignment\text{-}gap\text{-}bound\text{-}from\text{-}head\text{-}value\text{-}mismatch\text{-}gap$
 $[OF\ head\text{-}mismatch])$
have *next-bound*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}next\text{-}value\text{-}replay\text{-}gap\ ?s)\ ?s \leq ?M + 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}next\text{-}value\text{-}replay\text{-}gap\text{-}bound\text{-}from\text{-}merkle\text{-}and\text{-}slot)$
(rule *merkle-bound*,
rule *wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero*)
have *successor*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}successor\text{-}replay\text{-}gap\ ?s)\ ?s \leq ?M + 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}successor\text{-}replay\text{-}gap\text{-}bound\text{-}from\text{-}merkle\text{-}and\text{-}slot)$
(rule *merkle-bound*,
rule *wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero*)
have *final*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\ ?s)\ ?s \leq ?M + 0 + 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}final\text{-}value\text{-}replay\text{-}gap\text{-}bound\text{-}from\text{-}merkle\text{-}slot\text{-}and\text{-}selected\text{-}step)$
(rule *merkle-bound*,
rule *wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero*,
rule *wp-trace-fri-header-tied-final-value-selected-step-missing-gap-zero*)
have *slot*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}authenticated\text{-}slot\text{-}recorded\text{-}chunk\text{-}gap\ ?s)\ ?s$
 ≤ 0
by (rule *wp-trace-fri-header-tied-authenticated-slot-recorded-chunk-gap-zero*)
have *authenticated-recorded-auth*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}authenticated\text{-}recorded\text{-}auth\text{-}gap\ ?s)$
 $?s \leq 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}authenticated\text{-}recorded\text{-}auth\text{-}gap\text{-}bound\text{-}from\text{-}slot\text{-}recorded\text{-}chunk\text{-}gap$
 $[OF\ slot])$
have *recorded-auth*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}recorded\text{-}auth\text{-}gap\ ?s)\ ?s \leq 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}recorded\text{-}auth\text{-}gap\text{-}bound\text{-}from\text{-}authenticated\text{-}recorded\text{-}auth\text{-}gap$
 $[OF\ authenticated\text{-}recorded\text{-}auth])$
have *same-auth*:
 $wp\text{-}event\ verify\text{-}monad$
 $(trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}auth\text{-}gap\ ?s)\ ?s \leq 0$
by (rule
 $wp\text{-}trace\text{-}fri\text{-}header\text{-}tied\text{-}same\text{-}layer\text{-}auth\text{-}gap\text{-}bound\text{-}from\text{-}recorded\text{-}auth\text{-}gap$
 $[OF\ recorded\text{-}auth])$
have *auth-gap*:

$wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri}\text{-header}\text{-tied}\text{-assignment}\text{-auth}\text{-gap } ?s) \text{ } ?s \leq 0$
by (rule
 $wp\text{-trace}\text{-fri}\text{-header}\text{-tied}\text{-assignment}\text{-auth}\text{-gap}\text{-bound}\text{-from}\text{-same}\text{-layer}\text{-auth}\text{-gap}$
 $[OF \text{ same}\text{-auth}]$)
have complement:
 $wp\text{-event } verify\text{-monad}$
 $(\lambda out.$
 $trace\text{-fri}\text{-header}\text{-tied}\text{-sampled}\text{-assignment}\text{-conflict } ?s \text{ } out \wedge$
 $\neg trace\text{-fri}\text{-header}\text{-tied}\text{-recorded}\text{-sibling}\text{-candidate}\text{-missing } ?s \text{ } out)$
 $?s \leq$
 $(((((?M + ?M) + (?M + 0) + (?M + 0) + (?M + 0 + 0)) + ?M) + 0))$
by (rule
 $wp\text{-trace}\text{-fri}\text{-header}\text{-tied}\text{-sampled}\text{-assignment}\text{-conflict}\text{-without}\text{-recorded}\text{-missing}\text{-bound}\text{-from}\text{-auth}\text{-gap}\text{-and}\text{-replay}\text{-g}$
 $(rule \text{ auth}\text{-gap}, rule \text{ merkle}\text{-bound}, rule \text{ alignment},$
 $rule \text{ next}\text{-bound}, rule \text{ successor}, rule \text{ final})$
show $wp\text{-event } verify\text{-monad}$
 $(trace\text{-fri}\text{-header}\text{-tied}\text{-sampled}\text{-assignment}\text{-conflict } ?s) \text{ } ?s$
 $\leq trace\text{-fri}\text{-header}\text{-replay}\text{-conflict}\text{-error}\text{-for } A$
unfolding $trace\text{-fri}\text{-header}\text{-replay}\text{-conflict}\text{-error}\text{-for}\text{-def}$ $Let\text{-def}$
by (rule
 $wp\text{-trace}\text{-fri}\text{-header}\text{-tied}\text{-sampled}\text{-assignment}\text{-conflict}\text{-bound}\text{-from}\text{-recorded}\text{-missing}\text{-and}\text{-complement}$
 $(rule \text{ recorded}\text{-missing}\text{-bound}, rule \text{ complement})$
next
fix s
show $\neg trace\text{-fri}\text{-header}\text{-tied}\text{-sampled}\text{-assignment}\text{-conflict } s \text{ } None$
unfolding $trace\text{-fri}\text{-header}\text{-tied}\text{-sampled}\text{-assignment}\text{-conflict}\text{-def}$
 $accepted\text{-fri}\text{-opening}\text{-transcript}\text{-def}$
by $simp$
qed

lemma $trace\text{-fri}\text{-header}\text{-tied}\text{-zero}\text{-round}\text{-final}\text{-obstruction}\text{-agreement}\text{-bound}$:
assumes wf : $staged\text{-budget}\text{-wellformed}$ $budgets$
and $controlled$: $staged\text{-adversary}\text{-controlled}$ $budgets$ A
shows
 $wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A)$
 $(staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event}$
 $trace\text{-fri}\text{-header}\text{-tied}\text{-zero}\text{-round}\text{-final}\text{-obstruction})$
 $adversary\text{-initial}\text{-state}$
 $\leq trace\text{-fri}\text{-zero}\text{-round}\text{-agreement}\text{-set}\text{-target}\text{-error}\text{-for } budgets +$
 $trace\text{-fri}\text{-controlled}\text{-merkle}\text{-error}\text{-for } budgets$
proof –
have $checked$:
 $wp\text{-event } (checked\text{-staged}\text{-security}\text{-experiment}\text{-with}\text{-data}\text{-state } A)$
 $(staged\text{-security}\text{-with}\text{-data}\text{-state}\text{-verifier}\text{-event}$
 $trace\text{-fri}\text{-zero}\text{-round}\text{-checked}\text{-final}\text{-obstruction})$
 $adversary\text{-initial}\text{-state}$
 $\leq trace\text{-fri}\text{-zero}\text{-round}\text{-agreement}\text{-set}\text{-target}\text{-error}\text{-for } budgets$
by (rule

```

    checked-staged-security-trace-fri-zero-round-checked-final-obstruction-bound-from-agreement-sets
    [OF wf controlled])
have merkle:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     partial-merkle-inconsistency-bad)
  adversary-initial-state
  ≤ trace-fri-controlled-merkle-error-for budgets
unfolding trace-fri-controlled-merkle-error-for-def
by (rule
    checked-staged-security-with-data-state-partial-merkle-inconsistency-bad-bound-controlled
    [OF wf controlled])
have split:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-header-tied-zero-round-final-obstruction)
  adversary-initial-state
  ≤ wp-event (checked-staged-security-experiment-with-data-state A)
    (λout.
      staged-security-with-data-state-verifier-event
      trace-fri-zero-round-checked-final-obstruction out ∨
      staged-security-with-data-state-verifier-event
      partial-merkle-inconsistency-bad out)
  adversary-initial-state
unfolding staged-security-with-data-state-verifier-event-def
by (rule wp-event-mono-on-support)
    (auto dest:
      trace-fri-header-tied-zero-round-final-obstruction-imp-checked-or-merkle
      split: option.splits prod.splits)
also have ... ≤
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-zero-round-checked-final-obstruction)
  adversary-initial-state +
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     partial-merkle-inconsistency-bad)
  adversary-initial-state
by (rule wp-event-union-bound)
also have ... ≤
  trace-fri-zero-round-agreement-set-target-error-for budgets +
  trace-fri-controlled-merkle-error-for budgets
by (rule add-mono[OF checked merkle])
finally show ?thesis .
qed

lemma trace-header-restricted-one-step-error-for-bound:
  fixes trace-bad :: 'f list ⇒ nat ⇒ 'f list ⇒ 'f set
  assumes wf: staged-budget-wellformed budgets

```


and *controlled: staged-adversary-controlled budgets A*
shows
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-partial-candidate)
adversary-initial-state
 $\leq$ *trace-header-restricted-one-step-error-for budgets A trace-bad*
proof –
have *sampled-query:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-sampled-query-bad-candidate)
adversary-initial-state
 $\leq$ *trace-restricted-sampled-query-error-for budgets A trace-bad*
by (rule *trace-restricted-sampled-query-error-for-bound[OF wf controlled]*)
have *sampled-plain:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-sampled-layer-chain)
adversary-initial-state
 $\leq$ *trace-restricted-sampled-query-error-for budgets A trace-bad*
by (rule
checked-staged-security-trace-fri-sampled-layer-chain-bound-from-sampled-query
[OF sampled-query])
have *sampled-header:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-sampled-layer-chain)
adversary-initial-state
 $\leq$ *trace-restricted-sampled-query-error-for budgets A trace-bad*
proof (rule *order-trans[OF - sampled-plain]*)
show *wp-event (checked-staged-security-experiment-with-data-state A)*
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-header-tied-sampled-layer-chain)
adversary-initial-state
 $\leq$ *wp-event (checked-staged-security-experiment-with-data-state A)*
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-sampled-layer-chain)
adversary-initial-state
by (rule *wp-event-mono*)
(auto simp: staged-security-with-data-state-verifier-event-def
dest: trace-fri-header-tied-sampled-layer-chain-imp-sampled-layer-chain
split: option.splits prod.splits)
qed
have *conflict-bound:*
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-header-tied-sampled-assignment-conflict)
adversary-initial-state

$\leq \text{trace-fri-header-replay-conflict-error-for } A$
by (rule *trace-fri-header-tied-sampled-assignment-conflict-replay-bound*)
have *zero-bound*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-header-tied-zero-round-final-obstruction})$
 $\text{adversary-initial-state}$
 $\leq \text{trace-fri-zero-round-agreement-set-target-error-for budgets} +$
 $\text{trace-fri-controlled-merkle-error-for budgets}$
by (rule
 $\text{trace-fri-header-tied-zero-round-final-obstruction-agreement-bound}$
 $[OF \text{ wf controlled}]$)
have *branch*:
 $\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-header-tied-partial-candidate})$
 $\text{adversary-initial-state}$
 $\leq \text{trace-restricted-sampled-query-error-for budgets } A \text{ trace-bad} +$
 $(\text{trace-fri-header-replay-conflict-error-for } A +$
 $(\text{trace-fri-zero-round-agreement-set-target-error-for budgets} +$
 $\text{trace-fri-controlled-merkle-error-for budgets}))$
by (rule
 $\text{checked-staged-security-trace-fri-header-tied-partial-candidate-bound-from-sampled-conflict-and-staged-zero}$
 $[OF \text{ sampled-header conflict-bound zero-bound}]$)
show *?thesis*
using *branch*
by (*simp add*: *trace-header-restricted-one-step-error-for-def*
trace-fri-header-replay-conflict-zero-error-for-def)
qed

definition *trace-empty-header-restricted-openings-zero-error-for*

where

$\text{trace-empty-header-restricted-openings-zero-error-for budgets } A \text{ empty-bad} =$
 $(\text{trace-fri-zero-round-agreement-set-target-error-for budgets} +$
 $\text{trace-fri-controlled-merkle-error-for budgets} +$
 $(\text{trace-header-restricted-one-step-error-for budgets } A (\lambda-. \text{empty-bad}) +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-low-candidate-untied-openings-gap}) +$
 $\text{trace-header-restricted-one-step-error-for budgets } A (\lambda-. \text{empty-bad}))$

definition *trace-empty-header-restricted-nonnormalized-zero-error-for*

where

$\text{trace-empty-header-restricted-nonnormalized-zero-error-for budgets } A \text{ empty-bad} =$
 $(\text{trace-fri-zero-round-agreement-set-target-error-for budgets} +$
 $\text{trace-fri-controlled-merkle-error-for budgets} +$
 $(\text{trace-fri-controlled-merkle-error-for budgets} +$
 $((\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-low-candidate-nonconsumed-query-gap} +$
 $\text{reachable-verifier-event-bound-for } A$

$\text{trace-fri-header-tied-low-candidate-nonconsumed-irrelevant-root-gap}) +$
 $\text{reachable-verifier-event-bound-for } A$
 $\text{trace-fri-header-tied-low-candidate-nonconsumed-query-gap})) +$
 $\text{trace-header-restricted-one-step-error-for budgets } A (\lambda\cdot. \text{empty-bad}))$

definition *trace-empty-header-restricted-sampled-split-one-step-error-for*
where

$\text{trace-empty-header-restricted-sampled-split-one-step-error-for budgets } A \text{ empty-bad} =$
 $\text{trace-restricted-sampled-query-error-for budgets } A (\lambda\cdot. \text{empty-bad}) +$
 $(\text{trace-fri-empty-sampled-assignment-structural-error-for budgets } A +$
 $\min (\text{trace-empty-header-restricted-openings-zero-error-for budgets } A \text{ empty-bad})$
 $(\text{trace-empty-header-restricted-nonnormalized-zero-error-for budgets } A \text{ empty-bad}))$

definition *concrete-trace-empty-header-restricted-sampled-split-one-step-error-for*
where

$\text{concrete-trace-empty-header-restricted-sampled-split-one-step-error-for budgets } A =$
 $\text{trace-empty-header-restricted-sampled-split-one-step-error-for budgets } A$
 $\text{trace-one-step-split-bad}$

lemma *trace-empty-header-restricted-sampled-split-one-step-error-for-bound:*

fixes *empty-bad* :: nat $\Rightarrow$ 'f list $\Rightarrow$ 'f set

assumes *wf*: staged-budget-wellformed budgets

and *controlled*: staged-adversary-controlled budgets *A*

shows

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-bad-with-empty-composition-header-candidates})$
 $\text{adversary-initial-state}$
 $\leq \text{trace-empty-header-restricted-sampled-split-one-step-error-for}$
 $\text{budgets } A \text{ empty-bad}$

proof —

have *sampler*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-sampled-query-bad-candidate})$
 $\text{adversary-initial-state}$
 $\leq \text{trace-restricted-sampled-query-error-for budgets } A (\lambda\cdot. \text{empty-bad})$
by (rule *trace-restricted-sampled-query-error-for-bound*[OF *wf controlled*])

have *conflict-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$
 $(\text{staged-security-with-data-state-verifier-event}$
 $\text{trace-fri-sampled-assignment-conflict})$
 $\text{adversary-initial-state}$
 $\leq \text{trace-fri-empty-sampled-assignment-structural-error-for budgets } A$
by (rule
 $\text{checked-staged-security-trace-fri-sampled-assignment-conflict-bound-structural}$
 $[\text{OF } wf \text{ controlled}])$

have *header-bound*:

$\text{wp-event } (\text{checked-staged-security-experiment-with-data-state } A)$

```

    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-header-tied-partial-candidate)
    adversary-initial-state
  ≤ trace-header-restricted-one-step-error-for budgets A (λ-. empty-bad)
by (rule trace-header-restricted-one-step-error-for-bound
     [OF wf controlled])
have zero-openings-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-zero-round-final-obstruction)
    adversary-initial-state
  ≤ trace-empty-header-restricted-openings-zero-error-for budgets A empty-bad
unfolding trace-empty-header-restricted-openings-zero-error-for-def
by (rule
     checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-agreement-sets-header-and-openings
     [OF wf controlled header-bound])
have zero-nonnormalized-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-zero-round-final-obstruction)
    adversary-initial-state
  ≤ trace-empty-header-restricted-nonnormalized-zero-error-for budgets A empty-bad
unfolding trace-empty-header-restricted-nonnormalized-zero-error-for-def
by (rule
     checked-staged-security-trace-fri-zero-round-final-obstruction-bound-from-agreement-sets-header-and-nonnormalized
     [OF wf controlled header-bound])
have zero-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-zero-round-final-obstruction)
    adversary-initial-state
  ≤ min (trace-empty-header-restricted-openings-zero-error-for budgets A empty-bad)
        (trace-empty-header-restricted-nonnormalized-zero-error-for budgets A empty-bad)
using zero-openings-bound zero-nonnormalized-bound by simp
have branch:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     trace-fri-bad-with-empty-composition-header-candidates)
    adversary-initial-state
  ≤ trace-restricted-sampled-query-error-for budgets A (λ-. empty-bad) +
    (trace-fri-empty-sampled-assignment-structural-error-for budgets A +
     min (trace-empty-header-restricted-openings-zero-error-for budgets A empty-bad)
         (trace-empty-header-restricted-nonnormalized-zero-error-for budgets A empty-bad))
by (rule
     checked-staged-security-trace-fri-empty-header-bound-from-staged-sampled-conflict-and-zero
     [OF sampled conflict-bound zero-bound])
show ?thesis
using branch
by (simp add:

```

trace-empty-header-restricted-sampled-split-one-step-error-for-def)

qed

definition *composition-no-full-cover-sampled-query-error-for*
where
composition-no-full-cover-sampled-query-error-for budgets *bad* =

$$\begin{aligned}
& (\sum dg \in UNIV. \\
& \quad \sum challenges \in fri\text{-}challenge\text{-}space \ (ceil\text{-}log \ (to\text{-}nat \ dg + 1)). \\
& \quad (1 / nnreal \ (CARD('f) \wedge ceil\text{-}log \ (to\text{-}nat \ dg + 1))) * \\
& \quad (nnreal \\
& \quad \quad (card \\
& \quad \quad \quad (composition\text{-}fri\text{-}sampled\text{-}query\text{-}no\text{-}full\text{-}cover\text{-}failure\text{-}query\text{-}fiber \\
& \quad \quad \quad \quad bad \ dg \ challenges)) * \\
& \quad \quad (1 / nnreal \ (card \ query\text{-}sample\text{-}space)) \wedge rounds)) + \\
& hash\text{-}relation\text{-}budget\text{-}value \\
& (query\text{-}index\text{-}raw\text{-}list\text{-}relation\text{-}fiber\text{-}bound \\
& \quad (\bigcup dg \in UNIV. \\
& \quad \quad \bigcup challenges \in fri\text{-}challenge\text{-}space \ (ceil\text{-}log \ (to\text{-}nat \ dg + 1)). \\
& \quad \quad composition\text{-}fri\text{-}sampled\text{-}query\text{-}no\text{-}full\text{-}cover\text{-}failure\text{-}query\text{-}fiber \\
& \quad \quad \quad bad \ dg \ challenges)) \\
& \quad (staged\text{-}attacker\text{-}query\text{-}budget \ budgets + staged\text{-}challenge\text{-}query\text{-}budget) + \\
& hash\text{-}relation\text{-}budget\text{-}value \\
& (\sum dg \in (UNIV :: 'f \ set). \\
& \quad card \ (fri\text{-}challenge\text{-}space \ (ceil\text{-}log \ (to\text{-}nat \ dg + 1))) * \\
& \quad ceil\text{-}log \ (maxDegree + 1)) \\
& \quad (staged\text{-}attacker\text{-}query\text{-}budget \ budgets + staged\text{-}challenge\text{-}query\text{-}budget)
\end{aligned}$$

definition *composition-canonical-full-cover-shape-failure-error-for*
where
composition-canonical-full-cover-shape-failure-error-for *A* *bad* =

$$\begin{aligned}
& reachable\text{-}verifier\text{-}event\text{-}bound\text{-}for \ A \\
& \quad (\lambda s. composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}proximity\text{-}failure \ s \ bad) + \\
& reachable\text{-}verifier\text{-}event\text{-}bound\text{-}for \ A \\
& \quad composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}index\text{-}failure + \\
& reachable\text{-}verifier\text{-}event\text{-}bound\text{-}for \ A \\
& \quad composition\text{-}fri\text{-}sampled\text{-}full\text{-}cover\text{-}shape\text{-}failure
\end{aligned}$$

definition *composition-restricted-sampled-query-error-for*
where
composition-restricted-sampled-query-error-for budgets *A* *bad* =

$$\begin{aligned}
& composition\text{-}no\text{-}full\text{-}cover\text{-}sampled\text{-}query\text{-}error\text{-}for \ budgets \ bad + \\
& composition\text{-}canonical\text{-}full\text{-}cover\text{-}shape\text{-}failure\text{-}error\text{-}for \ A \ bad
\end{aligned}$$

lemma *composition-restricted-sampled-query-error-for-bound:*
assumes *wf*: *staged-budget-wellformed* budgets
and *controlled*: *staged-adversary-controlled* budgets *A*
shows

$$\begin{aligned}
& wp\text{-}event \ (checked\text{-}staged\text{-}security\text{-}experiment\text{-}with\text{-}data\text{-}state \ A) \\
& \quad (staged\text{-}security\text{-}with\text{-}data\text{-}state\text{-}verifier\text{-}event
\end{aligned}$$

```

    composition-fri-sampled-query-bad-candidate)
  adversary-initial-state
≤ composition-restricted-sampled-query-error-for budgets A bad
proof –
have no-full:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs.
        composition-fri-sampled-query-no-full-cover-failure-candidate s
          bad))
    adversary-initial-state
≤ composition-no-full-cover-sampled-query-error-for budgets bad
unfolding composition-no-full-cover-sampled-query-error-for-def
by (rule
  checked-staged-security-composition-fri-sampled-query-no-full-cover-failure-candidate-bound-from-restricted-pairs
    [OF wf controlled])
have failure:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
      (λs. composition-fri-sampled-canonical-full-cover-failure s bad))
    adversary-initial-state
≤ composition-canonical-full-cover-shape-failure-error-for A bad
unfolding composition-canonical-full-cover-shape-failure-error-for-def
proof (rule
  checked-staged-security-composition-fri-sampled-canonical-full-cover-failure-bound-from-shape-subcases)
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (composition-fri-sampled-full-cover-proximity-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data)) bad)
  (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data))
≤ reachable-verifier-event-bound-for A
  (λs. composition-fri-sampled-full-cover-proximity-failure s bad)
by (rule reachable-verifier-event-bound-for-ge[OF support])
next
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
      adversary-initial-state)
show wp-event verify-monad
  (composition-fri-sampled-full-cover-index-failure
    (verifier-state-from-adversary attacker-state
      (staged-proof-transcript data))))

```

```

    (verifier-state-from-adversary attacker-state
     (staged-proof-transcript data))
  ≤ reachable-verifier-event-bound-for A
    composition-fri-sampled-full-cover-index-failure
  by (rule reachable-verifier-event-bound-for-ge[OF support])
next
fix data attacker-state
assume support:
  Some (data, attacker-state) ∈
    set-dist (execute (checked-staged-transcript-program A)
                  adversary-initial-state)
show wp-event verify-monad
  (composition-fri-sampled-full-cover-shape-failure
   (verifier-state-from-adversary attacker-state
    (staged-proof-transcript data)))
  (verifier-state-from-adversary attacker-state
   (staged-proof-transcript data))
≤ reachable-verifier-event-bound-for A
  composition-fri-sampled-full-cover-shape-failure
by (rule reachable-verifier-event-bound-for-ge[OF support])
qed
have sampled:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (staged-security-with-data-state-verifier-event
     composition-fri-sampled-query-bad-candidate)
    adversary-initial-state
≤ composition-no-full-cover-sampled-query-error-for budgets bad +
  composition-canonical-full-cover-shape-failure-error-for A bad
by (rule
    checked-staged-security-composition-fri-sampled-query-bad-candidate-bound-from-no-full-cover-or-canonical-failure
    [OF no-full failure])
show ?thesis
  using sampled
  by (simp add: composition-restricted-sampled-query-error-for-def)
qed

definition composition-restricted-verifier-tied-controlled-error-for
where
  composition-restricted-verifier-tied-controlled-error-for budgets A comp-bad =
    composition-restricted-sampled-query-error-for budgets A comp-bad +
    (((composition-one-step-controlled-merkle-error-for budgets + 0) +
      (composition-one-step-controlled-merkle-error-for budgets + 0) +
      (composition-one-step-controlled-merkle-error-for budgets + 0) +
      (composition-one-step-controlled-merkle-error-for budgets + 0 + 0) +
      composition-one-step-controlled-merkle-error-for budgets) + 0 + 0)

definition concrete-composition-restricted-one-step-error-for
where
  concrete-composition-restricted-one-step-error-for budgets A =

```

composition-restricted-verifier-tied-controlled-error-for budgets A
 $(\lambda dg \text{ -. } \text{composition-one-step-split-bad } dg)$

lemma *composition-restricted-verifier-tied-controlled-error-for-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state
 $\leq$ *composition-restricted-verifier-tied-controlled-error-for budgets A*
comp-bad

proof —

have *sampled-query:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-sampled-query-bad-candidate)
adversary-initial-state
 $\leq$ *composition-restricted-sampled-query-error-for budgets A comp-bad*
by (rule *composition-restricted-sampled-query-error-for-bound*
 $[OF \text{ wf controlled}]$)

have *sampled:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-bad-with-verifier-tied-sampled-layer-chain)
adversary-initial-state
 $\leq$ *composition-restricted-sampled-query-error-for budgets A comp-bad*
by (rule
checked-staged-security-composition-fri-verifier-tied-sampled-layer-chain-bound-from-sampled-query
 $[OF \text{ sampled-query}]$)

have *conflict:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-verifier-tied-sampled-assignment-conflict)
adversary-initial-state $\leq$
 $((\text{composition-one-step-controlled-merkle-error-for budgets} + 0) +$
 $(\text{composition-one-step-controlled-merkle-error-for budgets} + 0) +$
 $(\text{composition-one-step-controlled-merkle-error-for budgets} + 0) +$
 $(\text{composition-one-step-controlled-merkle-error-for budgets} + 0 + 0) +$
 $\text{composition-one-step-controlled-merkle-error-for budgets}) + 0)$

by (rule

checked-staged-security-composition-fri-verifier-tied-sampled-assignment-conflict-bound-by-controlled-merkle
 $[OF \text{ wf controlled}]$)

have *zero:*

wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
composition-fri-verifier-tied-zero-round-final-obstruction)
adversary-initial-state $\leq (0::\text{prob})$

by (rule *checked-staged-security-with-data-state-verifier-event-bound-from-cont*)
 (simp-all add:
 composition-fri-verifier-tied-zero-round-final-obstruction-false
 wp-composition-fri-verifier-tied-zero-round-final-obstruction-zero)
have *branch*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
 (*staged-security-with-data-state-verifier-event*
 composition-fri-bad-with-verifier-tied-partial-candidate)
adversary-initial-state ≤
composition-restricted-sampled-query-error-for budgets A comp-bad +
 (((*composition-one-step-controlled-merkle-error-for budgets* + 0) +
 (*composition-one-step-controlled-merkle-error-for budgets* + 0) +
 (*composition-one-step-controlled-merkle-error-for budgets* + 0) +
 (*composition-one-step-controlled-merkle-error-for budgets* + 0 + 0) +
 composition-one-step-controlled-merkle-error-for budgets) + 0) + 0)
by (rule
 checked-staged-security-composition-fri-verifier-tied-partial-candidate-bound-from-staged-sampled-conflict-and-zero
 [*OF sampled conflict zero*])
show ?thesis
using *branch*
by (simp add:
 composition-restricted-verifier-tied-controlled-error-for-def)
qed

definition *active-route-current-prefix-conceptual-prefix-collision-hit-at*
where

active-route-current-prefix-conceptual-prefix-collision-hit-at i out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
i out ∧
 (case out of
 None ⇒ False
 | Some (((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*), *final-state*) ⇒
 hash-map-output-collision prefix-state)

definition *active-route-current-prefix-conceptual-algebraic-residual-hit-at*
where

active-route-current-prefix-conceptual-algebraic-residual-hit-at i out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
i out ∧
 (case out of
 None ⇒ False
 | Some (((*prefix*, *prefix-state*), *raw*), *raw-state*),
 ((*data*, *attacker-state*), *result*), *final-state*) ⇒
 ¬ *trace-table-low-degree*
 (*query-prefix-trace-conceptual-table prefix prefix-state*) ∨
 ¬ *composition-table-low-degree maxDegree*)

$(\text{query-prefix-composition-conceptual-table } \text{prefix } \text{prefix-state}) \vee$
 $\text{all-queries-consistent}$
 $(\text{query-prefix-trace-conceptual-table } \text{prefix } \text{prefix-state})$
 $(\text{query-prefix-composition-conceptual-table } \text{prefix } \text{prefix-state})$
 $(\text{sqp-alphas } \text{prefix})$

definition *active-route-current-prefix-conceptual-algebraic-residual-error-for*
where

$\text{active-route-current-prefix-conceptual-algebraic-residual-error-for } A \ i =$
 wp-event
 $(\text{checked-staged-security-experiment-with-query-prefix-data-state } A \ i)$
 $(\text{active-route-current-prefix-conceptual-algebraic-residual-hit-at } i)$
 $\text{adversary-initial-state}$

definition *active-route-current-prefix-conceptual-trace-low-degree-bad-at*
where

$\text{active-route-current-prefix-conceptual-trace-low-degree-bad-at } i \ \text{out}$
 $\longleftrightarrow$
 $\text{active-route-current-prefix-conceptual-algebraic-residual-hit-at } i \ \text{out} \wedge$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state}) \Rightarrow$
 $\neg \text{trace-table-low-degree}$
 $(\text{query-prefix-trace-conceptual-table } \text{prefix } \text{prefix-state}))$

definition *active-route-current-prefix-conceptual-composition-low-degree-bad-at*
where

$\text{active-route-current-prefix-conceptual-composition-low-degree-bad-at } i \ \text{out}$
 $\longleftrightarrow$
 $\text{active-route-current-prefix-conceptual-algebraic-residual-hit-at } i \ \text{out} \wedge$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state}) \Rightarrow$
 $\neg \text{composition-table-low-degree } \text{maxDegree}$
 $(\text{query-prefix-composition-conceptual-table } \text{prefix } \text{prefix-state}))$

definition *active-route-current-prefix-conceptual-alpha-bad-set-hit-at*
where

$\text{active-route-current-prefix-conceptual-alpha-bad-set-hit-at } i \ \text{out}$
 $\longleftrightarrow$
 $\text{active-route-current-prefix-conceptual-algebraic-residual-hit-at } i \ \text{out} \wedge$
 $(\text{case out of}$
 $\text{None} \Rightarrow \text{False}$
 $| \text{Some } (((((\text{prefix}, \text{prefix-state}), \text{raw}), \text{raw-state}),$
 $((\text{data}, \text{attacker-state}), \text{result})), \text{final-state}) \Rightarrow$
 $\text{sqp-alphas } \text{prefix} \in$
 $\text{composition-trace-bad-alpha-space}$

(*query-prefix-trace-conceptual-table prefix prefix-state*))

definition *active-route-current-prefix-conceptual-trace-low-degree-error-for*
where

active-route-current-prefix-conceptual-trace-low-degree-error-for A i =
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*active-route-current-prefix-conceptual-trace-low-degree-bad-at i*)
adversary-initial-state

definition *active-route-current-prefix-conceptual-composition-low-degree-error-for*
where

active-route-current-prefix-conceptual-composition-low-degree-error-for A i =
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*active-route-current-prefix-conceptual-composition-low-degree-bad-at i*)
adversary-initial-state

definition *active-route-current-prefix-conceptual-alpha-bad-set-error-for*
where

active-route-current-prefix-conceptual-alpha-bad-set-error-for A i =
wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state A i*)
 (*active-route-current-prefix-conceptual-alpha-bad-set-hit-at i*)
adversary-initial-state

lemma *active-route-current-prefix-conceptual-alpha-bad-set-hit-at-imp-data-state-alpha-prefix-bad-set-hit-on-support:*

assumes *support:*

*Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) ∈*
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A i)
adversary-initial-state)

and *candidate:*

∧ alpha-prefix alpha-prefix-state.
alpha-prefix =
(sqp-trace-root prefix, sqp-trace-fri-roots prefix,
sqp-trace-fri-challenges prefix, sqp-trace-final prefix) ⇒
Some (alpha-prefix, alpha-prefix-state) ∈
set-dist
(execute (staged-alpha-prefix-program A)
adversary-initial-state) ⇒
query-prefix-trace-conceptual-table prefix prefix-state ∈
alpha-prefix-trace-table-candidates alpha-prefix-state
(sqp-trace-root prefix)

and *hit:*

active-route-current-prefix-conceptual-alpha-bad-set-hit-at i
(Some (((((prefix, prefix-state), raw), raw-state),

```

      ((data, attacker-state), result)), final-state))
shows
  checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
  composition-trace-bad-alpha-space
  (Some (((data, attacker-state), result), final-state))
proof –
  obtain alpha-prefix alpha-prefix-state where alpha-prefix-eq:
    alpha-prefix =
      (sqp-trace-root prefix, sqp-trace-fri-roots prefix,
       sqp-trace-fri-challenges prefix, sqp-trace-final prefix)
  and alpha-support:
    Some (alpha-prefix, alpha-prefix-state) ∈
      set-dist
        (execute (staged-alpha-prefix-program A) adversary-initial-state)
  by (rule checked-staged-security-with-query-prefix-data-state-alpha-prefix-support
      [OF support])
  have table-candidate:
    query-prefix-trace-conceptual-table prefix prefix-state ∈
      alpha-prefix-trace-table-candidates alpha-prefix-state
      (sqp-trace-root prefix)
  by (rule candidate[OF alpha-prefix-eq alpha-support])
  have alphas-eq: staged-alphas data = sqp-alphas prefix
  by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq
      [OF support])
  have alpha-bad:
    sqp-alphas prefix ∈
      composition-trace-bad-alpha-space
      (query-prefix-trace-conceptual-table prefix prefix-state)
  using hit
  unfolding active-route-current-prefix-conceptual-alpha-bad-set-hit-at-def
  by simp
  have staged-alpha-bad:
    staged-alphas data ∈
      composition-trace-bad-alpha-space
      (query-prefix-trace-conceptual-table prefix prefix-state)
  using alpha-bad alphas-eq by simp
  have staged-alpha-in-space: staged-alphas data ∈ alpha-space
  using staged-alpha-bad composition-trace-bad-alpha-space-subset-alpha-space
  by blast
  have union-hit:
    staged-alphas data ∈
      alpha-prefix-union-bad-sets alpha-prefix-state
      composition-trace-bad-alpha-space (staged-trace-root data)
  proof –
  have root-eq: staged-trace-root data = sqp-trace-root prefix
  using support
  unfolding
    checked-staged-security-experiment-with-query-prefix-data-state-def
    checked-staged-after-query-prefix-receive-with-verifier-def

```

```

    checked-staged-after-query-prefix-receive-def
  by (auto elim!: set-dist-bindE split: prod.splits)
have table-candidate':
  query-prefix-trace-conceptual-table prefix prefix-state ∈
    alpha-prefix-trace-table-candidates alpha-prefix-state
    (staged-trace-root data)
  using table-candidate root-eq by simp
show ?thesis
  unfolding alpha-prefix-union-bad-sets-def
  using table-candidate' staged-alpha-bad staged-alpha-in-space
  by auto
qed
have prefix-components:
  staged-trace-root data = fst alpha-prefix ∧
  staged-trace-fri-roots data = fst (snd alpha-prefix) ∧
  staged-trace-fri-challenges data = fst (snd (snd alpha-prefix)) ∧
  staged-trace-final data = snd (snd (snd alpha-prefix))
  using support alpha-prefix-eq
  unfolding
    checked-staged-security-experiment-with-query-prefix-data-state-def
    checked-staged-after-query-prefix-receive-with-verifier-def
    checked-staged-after-query-prefix-receive-def
  by (auto elim!: set-dist-bindE split: prod.splits)
show ?thesis
  unfolding checked-staged-security-with-data-state-alpha-prefix-bad-set-hit-def
  using alpha-support prefix-components union-hit
  by auto
qed

lemma active-route-current-prefix-conceptual-alpha-bad-set-error-for-bound-from-data-state-alpha-prefix:
  assumes i-bound: i < rounds
  and candidate:
    ∧ prefix prefix-state raw raw-state data attacker-state result
      final-state alpha-prefix alpha-prefix-state.
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state) ∈
      set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state) ⇒
    alpha-prefix =
      (sqp-trace-root prefix, sqp-trace-fri-roots prefix,
        sqp-trace-fri-challenges prefix, sqp-trace-final prefix) ⇒
    Some (alpha-prefix, alpha-prefix-state) ∈
      set-dist
      (execute (staged-alpha-prefix-program A)
        adversary-initial-state) ⇒
    query-prefix-trace-conceptual-table prefix prefix-state ∈
      alpha-prefix-trace-table-candidates alpha-prefix-state

```

```

      (sqp-trace-root prefix)
and data-state-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)
    (checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
      composition-trace-bad-alpha-space)
  adversary-initial-state  $\leq C$ 
shows
  active-route-current-prefix-conceptual-alpha-bad-set-error-for A  $i \leq C$ 
proof –
  let ?Data =
    checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
    composition-trace-bad-alpha-space
  let ?Projected =
     $\lambda out. \text{case out of}$ 
      None  $\Rightarrow$  ?Data None
      | Some (packed, t)  $\Rightarrow$  ?Data (Some (snd packed, t))
  let ?E = active-route-current-prefix-conceptual-alpha-bad-set-hit-at i
  have event-le:
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      ?E adversary-initial-state  $\leq$ 
    wp-event
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      ?Projected adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support:
    out  $\in$ 
      set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
  and hit: ?E out
  show ?Projected out
  proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding active-route-current-prefix-conceptual-alpha-bad-set-hit-at-def
    by simp
  next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
        ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  have support':

```

```

    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A i)
      adversary-initial-state)
  using support unfolding out-eq .
have hit':
  active-route-current-prefix-conceptual-alpha-bad-set-hit-at i
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  using hit unfolding out-eq .
have data-hit:
  ?Data (Some (((data, attacker-state), result), final-state))
  by (rule
    active-route-current-prefix-conceptual-alpha-bad-set-hit-at-imp-data-state-alpha-prefix-bad-set-hit-on-support
    [OF support' - hit'])
  (rule candidate[OF support'])
show ?thesis
  unfolding out-eq using data-hit by simp
qed
qed
have projection:
  wp-event
  (checked-staged-security-experiment-with-query-prefix-data-state A i)
  ?Projected adversary-initial-state =
  wp-event (checked-staged-security-experiment-with-data-state A)
  ?Data adversary-initial-state
  using checked-staged-security-experiment-with-query-prefix-data-state-projection-event
  [OF i-bound, of A ?Data]
  by simp
show ?thesis
  unfolding active-route-current-prefix-conceptual-alpha-bad-set-error-for-def
  by (rule order-trans[OF event-le])
  (simp add: projection data-state-bound)
qed

definition active-route-current-prefix-split-algebraic-residual-bound-for
where
  active-route-current-prefix-split-algebraic-residual-bound-for A i =
  active-route-current-prefix-conceptual-trace-low-degree-error-for A i +
  active-route-current-prefix-conceptual-composition-low-degree-error-for A i +
  active-route-current-prefix-conceptual-alpha-bad-set-error-for A i

definition active-route-current-prefix-split-conceptual-bound-for
where
  active-route-current-prefix-split-conceptual-bound-for budgets A i =
  staged-phase-relation-error size
  (staged-query-search-queries budgets i + 1) +

```

query-error-bound +
hash-collision-budget-value 0
 (*staged-query-search-queries* budgets $i + 1$) +
active-route-current-prefix-split-algebraic-residual-bound-for $A\ i$

lemma *active-route-current-prefix-conceptual-prefix-collision-hit-at-bound*:

assumes *wf*: *staged-budget-wellformed* budgets
and *controlled*: *staged-adversary-controlled* budgets A
and *i-bound*: $i < \text{rounds}$

shows

wp-event
 (*checked-staged-security-experiment-with-query-prefix-data-state* $A\ i$)
 (*active-route-current-prefix-conceptual-prefix-collision-hit-at* i)
adversary-initial-state $\leq$
hash-collision-budget-value 0
 (*staged-query-search-queries* budgets $i + 1$)

proof –

have *i-le*: $i \leq \text{rounds}$

using *i-bound* **by** *simp*

have *prefix-bound*:

wp-event (*checked-staged-query-prefix-receive-with-state* $A\ i$)
checked-staged-query-prefix-hash-map-output-collision
adversary-initial-state $\leq$
hash-collision-budget-value 0
 (*staged-query-search-queries* budgets $i + 1$)

by (rule *checked-staged-query-prefix-hash-map-output-collision-bound*
 [OF *wf controlled i-le*])

show *?thesis*

proof (rule

checked-staged-security-experiment-with-query-prefix-data-state-bound-by-prefix
 [OF *prefix-bound*])

show

active-route-current-prefix-conceptual-prefix-collision-hit-at i None
 $\implies$ *checked-staged-query-prefix-hash-map-output-collision* None

unfolding

active-route-current-prefix-conceptual-prefix-collision-hit-at-def
checked-staged-query-prefix-hash-map-output-collision-def

by *simp*

next

fix *prefix-out* $t\ out$

assume

Some (*prefix-out*, t) $\in$
set-dist
 (*execute* (*checked-staged-query-prefix-receive-with-state* $A\ i$)
adversary-initial-state)

and *cont*:

out $\in$
set-dist
 (*execute*


```

    (get  $\gg$ 
      ( $\lambda$ prefix-state.
        checked-staged-after-query-prefix-receive-with-verifier A i
          prefix-out  $\gg$ 
            ( $\lambda$ result. return ((prefix-out, prefix-state), result))))
    t)
  and event:
    active-route-current-prefix-conceptual-prefix-collision-hit-at i out
  show
    checked-staged-query-prefix-hash-map-output-collision
      (Some (prefix-out, t))
  using cont event
  unfolding
    active-route-current-prefix-conceptual-prefix-collision-hit-at-def
    checked-staged-query-prefix-hash-map-output-collision-def
  by (auto elim!: set-dist-bindE split: option.splits prod.splits)
qed
qed

lemma active-route-current-prefix-conceptual-hit-residual-split:
assumes
  checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
    i out
and
  checked-staged-security-with-query-prefix-conceptual-context-residual-at
    out
shows
  active-route-current-prefix-conceptual-prefix-collision-hit-at i out  $\vee$ 
  active-route-current-prefix-conceptual-algebraic-residual-hit-at i out
using asms
unfolding
  active-route-current-prefix-conceptual-prefix-collision-hit-at-def
  active-route-current-prefix-conceptual-algebraic-residual-hit-at-def
  checked-staged-security-with-query-prefix-conceptual-context-residual-at-def
by (cases out) (auto split: prod.splits)

lemma active-route-current-prefix-conceptual-algebraic-residual-split-on-support:
assumes false-statement:  $\neg$  exists-valid-trace
and support:
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state)  $\in$ 
    set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A i)
        adversary-initial-state)
and hit:
  active-route-current-prefix-conceptual-algebraic-residual-hit-at i
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))

```

shows

active-route-current-prefix-conceptual-trace-low-degree-bad-at i
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state)) $\vee$
active-route-current-prefix-conceptual-composition-low-degree-bad-at i
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state)) $\vee$
active-route-current-prefix-conceptual-alpha-bad-set-hit-at i
 (Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state))

proof –

let ?trace = *query-prefix-trace-conceptual-table* prefix prefix-state

let ?composition =

query-prefix-composition-conceptual-table prefix prefix-state

from hit **have** residual:

$\neg$ *trace-table-low-degree* ?trace $\vee$

$\neg$ *composition-table-low-degree* maxDegree ?composition $\vee$

all-queries-consistent ?trace ?composition (*sqp-alphas* prefix)

unfolding

active-route-current-prefix-conceptual-algebraic-residual-hit-at-def

by *simp*

show ?thesis

proof (*cases* *trace-table-low-degree* ?trace)

case False

then show ?thesis

using hit

unfolding *active-route-current-prefix-conceptual-trace-low-degree-bad-at-def*

by *simp*

next

case *trace-low*: True

show ?thesis

proof (*cases* *composition-table-low-degree* maxDegree ?composition)

case False

then show ?thesis

using hit

unfolding

active-route-current-prefix-conceptual-composition-low-degree-bad-at-def

by *simp*

next

case *composition-low*: True

from residual *trace-low* *composition-low* **have** *all-queries*:

all-queries-consistent ?trace ?composition (*sqp-alphas* prefix)

by *simp*

have *prefix-support*:

Some ((prefix, prefix-state), prefix-state) $\in$

set-dist

(*execute* (*checked-staged-query-prefix-with-state* A i)

adversary-initial-state)

by (*rule*)

```

    checked-staged-security-with-query-prefix-data-state-prefix-support
    [OF support])
  have alpha-len: length (sqp-alphas prefix) = length spec
    by (rule checked-staged-query-prefix-with-state-alphas-length
        [OF prefix-support])
  have alpha-bad:
    sqp-alphas prefix ∈ composition-trace-bad-alpha-space ?trace
    by (rule
        low-degree-all-queries-consistent-imp-composition-trace-bad-alpha-space
        [OF false-statement alpha-len trace-low composition-low
            all-queries])
  then show ?thesis
    using hit
    unfolding
      active-route-current-prefix-conceptual-alpha-bad-set-hit-at-def
    by simp
qed
qed
qed

lemma active-route-current-prefix-conceptual-algebraic-residual-bound-from-split:
  assumes false-statement: ¬ exists-valid-trace
  shows
    active-route-current-prefix-conceptual-algebraic-residual-error-for A i
    ≤ active-route-current-prefix-split-algebraic-residual-bound-for A i
proof -
  let ?M = checked-staged-security-experiment-with-query-prefix-data-state A i
  let ?Algebraic =
    active-route-current-prefix-conceptual-algebraic-residual-hit-at i
  let ?Trace =
    active-route-current-prefix-conceptual-trace-low-degree-bad-at i
  let ?Composition =
    active-route-current-prefix-conceptual-composition-low-degree-bad-at i
  let ?Alpha =
    active-route-current-prefix-conceptual-alpha-bad-set-hit-at i
  have event-le:
    wp-event ?M ?Algebraic adversary-initial-state ≤
    wp-event ?M (λout. ?Trace out ∨ ?Composition out ∨ ?Alpha out)
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support: out ∈ set-dist (execute ?M adversary-initial-state)
    and hit: ?Algebraic out
    show ?Trace out ∨ ?Composition out ∨ ?Alpha out
  proof (cases out)
    case None
    then show ?thesis
      using hit
      unfolding

```

```

    active-route-current-prefix-conceptual-algebraic-residual-hit-at-def
  by simp
next
case (Some packed)
then obtain prefix prefix-state raw raw-state data attacker-state result
  final-state where out-eq:
  out =
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
have support':
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist (execute ?M adversary-initial-state)
  using support unfolding out-eq .
have hit':
  ?Algebraic
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  using hit unfolding out-eq .
show ?thesis
  unfolding out-eq
  by (rule
    active-route-current-prefix-conceptual-algebraic-residual-split-on-support
    [OF false-statement support' hit'])
qed
qed
have union-bound:
  wp-event ?M (λout. ?Trace out ∨ ?Composition out ∨ ?Alpha out)
  adversary-initial-state ≤
  wp-event ?M ?Trace adversary-initial-state +
  wp-event ?M ?Composition adversary-initial-state +
  wp-event ?M ?Alpha adversary-initial-state
proof -
  have wp-event ?M
    (λout. ?Trace out ∨ ?Composition out ∨ ?Alpha out)
    adversary-initial-state ≤
    wp-event ?M ?Trace adversary-initial-state +
    wp-event ?M (λout. ?Composition out ∨ ?Alpha out)
    adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤
    wp-event ?M ?Trace adversary-initial-state +
    (wp-event ?M ?Composition adversary-initial-state +
    wp-event ?M ?Alpha adversary-initial-state)
  by (intro add-mono order-refl wp-event-union-bound)
  finally show ?thesis
    by (simp add: add.assoc)
qed

```

```

show ?thesis
  unfolding
    active-route-current-prefix-conceptual-algebraic-residual-error-for-def
    active-route-current-prefix-split-algebraic-residual-bound-for-def
    active-route-current-prefix-conceptual-trace-low-degree-error-for-def
    active-route-current-prefix-conceptual-composition-low-degree-error-for-def
    active-route-current-prefix-conceptual-alpha-bad-set-error-for-def
  by (rule order-trans[OF event-le union-bound])
qed

lemma active-route-current-prefix-authenticated-error-for-le-split-conceptual-bound:
  assumes wf: staged-budget-wellformed budgets
  and controlled: staged-adversary-controlled budgets A
  and i-bound:  $i < \text{rounds}$ 
  and false-statement:  $\neg \text{exists-valid-trace}$ 
  shows
    active-route-current-prefix-authenticated-error-for A i  $\leq$ 
    active-route-current-prefix-split-conceptual-bound-for budgets A i
proof -
  let ?M = checked-staged-security-experiment-with-query-prefix-data-state A i
  let ?Hit =
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-at
    i
  let ?Conceptual =
    checked-staged-security-with-query-prefix-conceptual-target-hit
  let ?Collision =
    active-route-current-prefix-conceptual-prefix-collision-hit-at i
  let ?Algebraic =
    active-route-current-prefix-conceptual-algebraic-residual-hit-at i
  have event-le:
    wp-event ?M ?Hit adversary-initial-state  $\leq$ 
    wp-event ?M
    ( $\lambda \text{out. ?Conceptual out} \vee \text{?Collision out} \vee \text{?Algebraic out}$ )
    adversary-initial-state
  proof (rule wp-event-mono-on-support)
  fix out
  assume support: out  $\in \text{set-dist} (\text{execute } ?M \text{ adversary-initial-state})$ 
  and hit: ?Hit out
  have conceptual-or-residual:
    ?Conceptual out  $\vee$ 
    checked-staged-security-with-query-prefix-conceptual-context-residual-at
    out
  by (rule
    checked-staged-security-with-query-prefix-current-prefix-authenticated-hit-imp-conceptual-or-residual-on-support
    [where budgets=budgets and A=A and i=i and out=out,
      OF wf controlled i-bound support hit])
  show ?Conceptual out  $\vee$  ?Collision out  $\vee$  ?Algebraic out
  using conceptual-or-residual
  proof (elim disjE)

```

```

assume conceptual: ?Conceptual out
then show ?thesis by simp
next
assume residual:
  checked-staged-security-with-query-prefix-conceptual-context-residual-at
  out
have ?Collision out  $\vee$  ?Algebraic out
  by (rule active-route-current-prefix-conceptual-hit-residual-split
    [OF hit residual])
then show ?thesis by simp
qed
qed
have conceptual-bound:
  wp-event ?M ?Conceptual adversary-initial-state  $\leq$ 
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) + query-error-bound
by (rule
  checked-staged-security-with-query-prefix-conceptual-target-hit-bound
  [OF wf controlled i-bound])
have collision-bound:
  wp-event ?M ?Collision adversary-initial-state  $\leq$ 
    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1)
by (rule
  active-route-current-prefix-conceptual-prefix-collision-hit-at-bound
  [OF wf controlled i-bound])
have algebraic-bound0:
  active-route-current-prefix-conceptual-algebraic-residual-error-for A i  $\leq$ 
    active-route-current-prefix-split-algebraic-residual-bound-for A i
by (rule
  active-route-current-prefix-conceptual-algebraic-residual-bound-from-split
  [OF false-statement])
have algebraic-bound:
  wp-event ?M ?Algebraic adversary-initial-state  $\leq$ 
    active-route-current-prefix-split-algebraic-residual-bound-for A i
using algebraic-bound0
unfolding active-route-current-prefix-conceptual-algebraic-residual-error-for-def
by simp
have union-bound:
  wp-event ?M
    ( $\lambda out. ?Conceptual out \vee ?Collision out \vee ?Algebraic out$ )
    adversary-initial-state  $\leq$ 
    wp-event ?M ?Conceptual adversary-initial-state +
    wp-event ?M ?Collision adversary-initial-state +
    wp-event ?M ?Algebraic adversary-initial-state
proof –
have
  wp-event ?M
    ( $\lambda out. ?Conceptual out \vee ?Collision out \vee ?Algebraic out$ )

```

```

    adversary-initial-state ≤
    wp-event ?M ?Conceptual adversary-initial-state +
    wp-event ?M (λout. ?Collision out ∨ ?Algebraic out)
    adversary-initial-state
  by (rule wp-event-union-bound)
also have ... ≤
    wp-event ?M ?Conceptual adversary-initial-state +
    (wp-event ?M ?Collision adversary-initial-state +
    wp-event ?M ?Algebraic adversary-initial-state)
  by (intro add-mono order-refl wp-event-union-bound)
finally show ?thesis
  by (simp add: add.assoc)
qed
have sum-bound:
    wp-event ?M ?Conceptual adversary-initial-state +
    wp-event ?M ?Collision adversary-initial-state +
    wp-event ?M ?Algebraic adversary-initial-state ≤
    staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound +
    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1) +
    active-route-current-prefix-split-algebraic-residual-bound-for A i
proof -
  have tail-bound:
    wp-event ?M ?Collision adversary-initial-state +
    wp-event ?M ?Algebraic adversary-initial-state ≤
    hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1) +
    active-route-current-prefix-split-algebraic-residual-bound-for A i
  by (intro add-mono collision-bound algebraic-bound)
  have
    wp-event ?M ?Conceptual adversary-initial-state +
    (wp-event ?M ?Collision adversary-initial-state +
    wp-event ?M ?Algebraic adversary-initial-state) ≤
    (staged-phase-relation-error size
    (staged-query-search-queries budgets i + 1) +
    query-error-bound) +
    (hash-collision-budget-value 0
    (staged-query-search-queries budgets i + 1) +
    active-route-current-prefix-split-algebraic-residual-bound-for A i)
  by (intro add-mono conceptual-bound tail-bound)
  then show ?thesis
    by (simp add: add.assoc)
qed
show ?thesis
  unfolding active-route-current-prefix-authenticated-error-for-def
    active-route-current-prefix-split-conceptual-bound-for-def
  by (rule order-trans[OF event-le])

```

(rule order-trans[OF union-bound sum-bound])
qed

definition *active-route-current-empty-conceptual-prefix-collision-hit-at*
where

active-route-current-empty-conceptual-prefix-collision-hit-at
 (- :: 'f itself) out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
 0 out $\wedge$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 hash-map-output-collision prefix-state)

definition *active-route-current-empty-conceptual-trace-low-degree-bad-at*
where

active-route-current-empty-conceptual-trace-low-degree-bad-at
 (- :: 'f itself) out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
 0 out $\wedge$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 $\neg$ trace-table-low-degree
 (query-prefix-trace-conceptual-table prefix prefix-state))

definition *active-route-current-empty-conceptual-alpha-bad-set-hit-at*
where

active-route-current-empty-conceptual-alpha-bad-set-hit-at
 (- :: 'f itself) out
 $\longleftrightarrow$
checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
 0 out $\wedge$
 (case out of
 None $\Rightarrow$ False
 | Some (((((prefix, prefix-state), raw), raw-state),
 ((data, attacker-state), result)), final-state) $\Rightarrow$
 sqp-alphas prefix $\in$
 composition-trace-bad-alpha-space
 (query-prefix-trace-conceptual-table prefix prefix-state))

definition *active-route-current-empty-conceptual-prefix-collision-error-for*
where

active-route-current-empty-conceptual-prefix-collision-error-for A =
 wp-event

(checked-staged-security-experiment-with-query-prefix-data-state A 0)
(active-route-current-empty-conceptual-prefix-collision-hit-at
TYPE('f))
adversary-initial-state

definition *active-route-current-empty-conceptual-trace-low-degree-error-for*
where

active-route-current-empty-conceptual-trace-low-degree-error-for A =
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A 0)
(active-route-current-empty-conceptual-trace-low-degree-bad-at
TYPE('f))
adversary-initial-state

definition *active-route-current-empty-conceptual-alpha-bad-set-error-for*
where

active-route-current-empty-conceptual-alpha-bad-set-error-for A =
wp-event
(checked-staged-security-experiment-with-query-prefix-data-state A 0)
(active-route-current-empty-conceptual-alpha-bad-set-hit-at
TYPE('f))
adversary-initial-state

lemma *active-route-current-empty-conceptual-alpha-bad-set-hit-at-imp-data-state-alpha-prefix-bad-set-hit-on-support:*

assumes *support:*

Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state) ∈
set-dist
(execute
(checked-staged-security-experiment-with-query-prefix-data-state A 0)
adversary-initial-state)

and *candidate:*

∧ alpha-prefix alpha-prefix-state.
alpha-prefix =
(sqp-trace-root prefix, sqp-trace-fri-roots prefix,
sqp-trace-fri-challenges prefix, sqp-trace-final prefix) ⇒
Some (alpha-prefix, alpha-prefix-state) ∈
set-dist
(execute (staged-alpha-prefix-program A)
adversary-initial-state) ⇒
query-prefix-trace-conceptual-table prefix prefix-state ∈
alpha-prefix-trace-table-candidates alpha-prefix-state
(sqp-trace-root prefix)

and *hit:*

active-route-current-empty-conceptual-alpha-bad-set-hit-at
TYPE('f)
(Some (((((prefix, prefix-state), raw), raw-state),
((data, attacker-state), result)), final-state))

shows

```

checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
composition-trace-bad-alpha-space
(Some (((data, attacker-state), result), final-state))
proof –
obtain alpha-prefix alpha-prefix-state where alpha-prefix-eq:
  alpha-prefix =
    (sqp-trace-root prefix, sqp-trace-fri-roots prefix,
     sqp-trace-fri-challenges prefix, sqp-trace-final prefix)
and alpha-support:
  Some (alpha-prefix, alpha-prefix-state) ∈
    set-dist
    (execute (staged-alpha-prefix-program A) adversary-initial-state)
by (rule checked-staged-security-with-query-prefix-data-state-alpha-prefix-support
     [OF support])
have table-candidate:
  query-prefix-trace-conceptual-table prefix prefix-state ∈
    alpha-prefix-trace-table-candidates alpha-prefix-state
    (sqp-trace-root prefix)
by (rule candidate[OF alpha-prefix-eq alpha-support])
have alphas-eq: staged-alphas data = sqp-alphas prefix
by (rule checked-staged-security-with-query-prefix-data-state-alphas-eq
     [OF support])
have alpha-bad:
  sqp-alphas prefix ∈
    composition-trace-bad-alpha-space
    (query-prefix-trace-conceptual-table prefix prefix-state)
using hit
unfolding active-route-current-empty-conceptual-alpha-bad-set-hit-at-def
by simp
have staged-alpha-bad:
  staged-alphas data ∈
    composition-trace-bad-alpha-space
    (query-prefix-trace-conceptual-table prefix prefix-state)
using alpha-bad alphas-eq by simp
have staged-alpha-in-space: staged-alphas data ∈ alpha-space
using staged-alpha-bad composition-trace-bad-alpha-space-subset-alpha-space
by blast
have root-eq: staged-trace-root data = sqp-trace-root prefix
using support
unfolding
  checked-staged-security-experiment-with-query-prefix-data-state-def
  checked-staged-after-query-prefix-receive-with-verifier-def
  checked-staged-after-query-prefix-receive-def
by (auto elim!: set-dist-bindE split: prod.splits)
have union-hit:
  staged-alphas data ∈
    alpha-prefix-union-bad-sets alpha-prefix-state
    composition-trace-bad-alpha-space (staged-trace-root data)
proof –

```

```

have table-candidate':
  query-prefix-trace-conceptual-table prefix prefix-state ∈
    alpha-prefix-trace-table-candidates alpha-prefix-state
    (staged-trace-root data)
  using table-candidate root-eq by simp
show ?thesis
unfolding alpha-prefix-union-bad-sets-def
  using table-candidate' staged-alpha-bad staged-alpha-in-space
  by auto
qed
have prefix-components:
  staged-trace-root data = fst alpha-prefix ∧
  staged-trace-fri-roots data = fst (snd alpha-prefix) ∧
  staged-trace-fri-challenges data = fst (snd (snd alpha-prefix)) ∧
  staged-trace-final data = snd (snd (snd alpha-prefix))
using support alpha-prefix-eq
unfolding
  checked-staged-security-experiment-with-query-prefix-data-state-def
  checked-staged-after-query-prefix-receive-with-verifier-def
  checked-staged-after-query-prefix-receive-def
by (auto elim!: set-dist-bindE split: prod.splits)
show ?thesis
unfolding checked-staged-security-with-data-state-alpha-prefix-bad-set-hit-def
using alpha-support prefix-components union-hit
by auto
qed

lemma active-route-current-empty-conceptual-alpha-bad-set-error-for-bound-from-data-state-alpha-prefix:
assumes candidate:
  ∧prefix prefix-state raw raw-state data attacker-state result
  final-state alpha-prefix alpha-prefix-state.
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
    set-dist
    (execute
      (checked-staged-security-experiment-with-query-prefix-data-state A 0)
      adversary-initial-state) ⇒
  alpha-prefix =
    (sqp-trace-root prefix, sqp-trace-fri-roots prefix,
      sqp-trace-fri-challenges prefix, sqp-trace-final prefix) ⇒
  Some (alpha-prefix, alpha-prefix-state) ∈
    set-dist
    (execute (staged-alpha-prefix-program A)
      adversary-initial-state) ⇒
  query-prefix-trace-conceptual-table prefix prefix-state ∈
    alpha-prefix-trace-table-candidates alpha-prefix-state
    (sqp-trace-root prefix)
and data-state-bound:
  wp-event (checked-staged-security-experiment-with-data-state A)

```

```

    (checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
     composition-trace-bad-alpha-space)
    adversary-initial-state ≤ C
  shows
    active-route-current-empty-conceptual-alpha-bad-set-error-for A ≤ C
proof -
  let ?Data =
    checked-staged-security-with-data-state-alpha-prefix-bad-set-hit A
    composition-trace-bad-alpha-space
  let ?Projected =
    λout. case out of
      None ⇒ ?Data None
    | Some (packed, t) ⇒ ?Data (Some (snd packed, t))
  let ?E =
    active-route-current-empty-conceptual-alpha-bad-set-hit-at
    TYPE('f)
  have event-le:
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A 0)
    ?E adversary-initial-state ≤
    wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A 0)
    ?Projected adversary-initial-state
  proof (rule wp-event-mono-on-support)
    fix out
    assume support:
      out ∈
      set-dist
      (execute
       (checked-staged-security-experiment-with-query-prefix-data-state A 0)
       adversary-initial-state)
    and hit: ?E out
  show ?Projected out
proof (cases out)
  case None
  then show ?thesis
    using hit
    unfolding active-route-current-empty-conceptual-alpha-bad-set-hit-at-def
    by simp
next
  case (Some packed)
  then obtain prefix prefix-state raw raw-state data attacker-state result
    final-state where out-eq:
    out =
      Some (((((prefix, prefix-state), raw), raw-state),
              ((data, attacker-state), result)), final-state)
    by (cases packed) (auto split: prod.splits)
  have support':
    Some (((((prefix, prefix-state), raw), raw-state),

```

```

      ((data, attacker-state), result)), final-state) ∈
set-dist
  (execute
    (checked-staged-security-experiment-with-query-prefix-data-state A 0)
    adversary-initial-state)
using support unfolding out-eq .
have hit':
  active-route-current-empty-conceptual-alpha-bad-set-hit-at
    TYPE('f)
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  using hit unfolding out-eq .
have data-hit:
  ?Data (Some (((data, attacker-state), result), final-state))
  by (rule
    active-route-current-empty-conceptual-alpha-bad-set-hit-at-imp-data-state-alpha-prefix-bad-set-hit-on-support
      [OF support' - hit'])
    (rule candidate[OF support'])
show ?thesis
  unfolding out-eq using data-hit by simp
qed
qed
have projection:
  wp-event
    (checked-staged-security-experiment-with-query-prefix-data-state A 0)
    ?Projected adversary-initial-state =
  wp-event (checked-staged-security-experiment-with-data-state A)
    ?Data adversary-initial-state
using checked-staged-security-experiment-with-query-prefix-data-state-projection-event
  [OF rounds-positive, of A ?Data]
by simp
show ?thesis
  unfolding active-route-current-empty-conceptual-alpha-bad-set-error-for-def
  by (rule order-trans[OF event-le])
    (simp add: projection data-state-bound)
qed

definition active-route-current-empty-split-residual-bound-for
where
  active-route-current-empty-split-residual-bound-for A =
    active-route-current-empty-conceptual-prefix-collision-error-for A +
    active-route-current-empty-conceptual-trace-low-degree-error-for A +
    active-route-current-empty-conceptual-alpha-bad-set-error-for A

definition active-route-current-empty-split-conceptual-bound-for
where
  active-route-current-empty-split-conceptual-bound-for budgets A =
    staged-phase-relation-error size
    (staged-query-search-queries budgets 0 + 1) +

```

query-error-bound +
active-route-current-empty-split-residual-bound-for A +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound)

lemma *active-route-current-empty-conceptual-prefix-collision-hit-at-bound:*

assumes *wf: staged-budget-wellformed budgets*

and *controlled: staged-adversary-controlled budgets A*

shows

active-route-current-empty-conceptual-prefix-collision-error-for A ≤
hash-collision-budget-value 0
(staged-query-search-queries budgets 0 + 1)

proof –

have *i-le: 0 ≤ rounds*

by *simp*

have *prefix-bound:*

wp-event (checked-staged-query-prefix-receive-with-state A 0)
checked-staged-query-prefix-hash-map-output-collision
adversary-initial-state ≤
hash-collision-budget-value 0
(staged-query-search-queries budgets 0 + 1)

by (rule *checked-staged-query-prefix-hash-map-output-collision-bound*
[OF wf controlled i-le])

show *?thesis*

unfolding *active-route-current-empty-conceptual-prefix-collision-error-for-def*

proof (rule

checked-staged-security-experiment-with-query-prefix-data-state-bound-by-prefix
[OF prefix-bound])

show

active-route-current-empty-conceptual-prefix-collision-hit-at
TYPE('f) None
 $\implies$ *checked-staged-query-prefix-hash-map-output-collision None*

unfolding

active-route-current-empty-conceptual-prefix-collision-hit-at-def
checked-staged-query-prefix-hash-map-output-collision-def

by *simp*

next

fix *prefix-out t out*

assume

Some (prefix-out, t) ∈
set-dist
(execute (checked-staged-query-prefix-receive-with-state A 0)
adversary-initial-state)

and *cont:*

out ∈

set-dist

(execute

(get $\gg$

(λprefix-state.

```

      checked-staged-after-query-prefix-receive-with-verifier A 0
      prefix-out  $\gg$ 
      ( $\lambda$ result. return ((prefix-out, prefix-state), result))))
    t)
  and event:
    active-route-current-empty-conceptual-prefix-collision-hit-at
      TYPE('f) out
  show
    checked-staged-query-prefix-hash-map-output-collision
      (Some (prefix-out, t))
  using cont event
  unfolding
    active-route-current-empty-conceptual-prefix-collision-hit-at-def
    checked-staged-query-prefix-hash-map-output-collision-def
  by (auto elim!: set-dist-bindE split: option.splits prod.splits)
qed
qed

lemma active-route-current-empty-conceptual-residual-split-on-support:
  assumes false-statement:  $\neg$  exists-valid-trace
  and support:
    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)  $\in$ 
    set-dist
      (execute
        (checked-staged-security-experiment-with-query-prefix-data-state A 0)
        adversary-initial-state)
  and hit:
    checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
      0
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  shows
    active-route-current-empty-conceptual-prefix-collision-hit-at
      TYPE('f)
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))  $\vee$ 
    active-route-current-empty-conceptual-trace-low-degree-bad-at
      TYPE('f)
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))  $\vee$ 
    active-route-current-empty-conceptual-alpha-bad-set-hit-at
      TYPE('f)
    (Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state))
  proof -
    let ?trace = query-prefix-trace-conceptual-table prefix prefix-state
    let ?composition = replicate (scale * clength) (spp-composition-final prefix)
    from hit have residual:

```

```

hash-map-output-collision prefix-state  $\vee$ 
 $\neg$  trace-table-low-degree ?trace  $\vee$ 
all-queries-consistent ?trace ?composition (sqp-alphas prefix)
unfolding
  checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at-def
  checked-staged-security-with-query-prefix-conceptual-empty-context-residual-at-def
by simp
show ?thesis
proof (cases hash-map-output-collision prefix-state)
  case True
  then show ?thesis
    using hit
    unfolding
      active-route-current-empty-conceptual-prefix-collision-hit-at-def
    by simp
  next
  case clean: False
  show ?thesis
  proof (cases trace-table-low-degree ?trace)
    case False
    then show ?thesis
      using hit
      unfolding
        active-route-current-empty-conceptual-trace-low-degree-bad-at-def
      by simp
  next
  case trace-low: True
  from residual clean trace-low have all-queries:
    all-queries-consistent ?trace ?composition (sqp-alphas prefix)
  by simp
  have prefix-support:
    Some ((prefix, prefix-state), prefix-state)  $\in$ 
      set-dist
        (execute (checked-staged-query-prefix-with-state A 0)
          adversary-initial-state)
  by (rule
    checked-staged-security-with-query-prefix-data-state-prefix-support
    [OF support])
  have alpha-len: length (sqp-alphas prefix) = length spec
  by (rule checked-staged-query-prefix-with-state-alphas-length
    [OF prefix-support])
  have alpha-bad:
    sqp-alphas prefix  $\in$  composition-trace-bad-alpha-space ?trace
  by (rule
    low-degree-all-queries-consistent-imp-composition-trace-bad-alpha-space
    [OF false-statement alpha-len trace-low
      constant-composition-table-low-degree all-queries])
  then show ?thesis
    using hit

```



```

    unfolding
      active-route-current-empty-conceptual-alpha-bad-set-hit-at-def
    by simp
  qed
  qed
  qed

lemma active-route-current-empty-conceptual-residual-bound-from-split:
  assumes false-statement:  $\neg$  exists-valid-trace
  shows
    active-route-current-empty-conceptual-residual-error-for TYPE('f) A  $\leq$ 
      active-route-current-empty-split-residual-bound-for A
  proof -
    let ?M = checked-staged-security-experiment-with-query-prefix-data-state A 0
    let ?Residual =
       $\lambda$ out.
        checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at
          0 out
    let ?Collision =
      active-route-current-empty-conceptual-prefix-collision-hit-at
        TYPE('f)
    let ?Trace =
      active-route-current-empty-conceptual-trace-low-degree-bad-at
        TYPE('f)
    let ?Alpha =
      active-route-current-empty-conceptual-alpha-bad-set-hit-at
        TYPE('f)
    have event-le:
      wp-event ?M ?Residual adversary-initial-state  $\leq$ 
      wp-event ?M ( $\lambda$ out. ?Collision out  $\vee$  ?Trace out  $\vee$  ?Alpha out)
      adversary-initial-state
    proof (rule wp-event-mono-on-support)
      fix out
      assume support: out  $\in$  set-dist (execute ?M adversary-initial-state)
      and hit: ?Residual out
      show ?Collision out  $\vee$  ?Trace out  $\vee$  ?Alpha out
      proof (cases out)
        case None
        then show ?thesis
          using hit
          unfolding
            checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at-def
            checked-staged-security-with-query-prefix-conceptual-empty-context-residual-at-def
          by simp
      next
        case (Some packed)
        then obtain prefix prefix-state raw raw-state data attacker-state result
          final-state where out-eq:
            out =

```

```

    Some (((((prefix, prefix-state), raw), raw-state),
      ((data, attacker-state), result)), final-state)
  by (cases packed) (auto split: prod.splits)
have support':
  Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state) ∈
  set-dist (execute ?M adversary-initial-state)
  using support unfolding out-eq .
have hit':
  ?Residual
  (Some (((((prefix, prefix-state), raw), raw-state),
    ((data, attacker-state), result)), final-state))
  using hit unfolding out-eq .
show ?thesis
  unfolding out-eq
  by (rule
    active-route-current-empty-conceptual-residual-split-on-support
    [OF false-statement support' hit'])
qed
qed
have union-bound:
  wp-event ?M (λout. ?Collision out ∨ ?Trace out ∨ ?Alpha out)
  adversary-initial-state ≤
  wp-event ?M ?Collision adversary-initial-state +
  wp-event ?M ?Trace adversary-initial-state +
  wp-event ?M ?Alpha adversary-initial-state
proof -
  have
    wp-event ?M
    (λout. ?Collision out ∨ ?Trace out ∨ ?Alpha out)
    adversary-initial-state ≤
    wp-event ?M ?Collision adversary-initial-state +
    wp-event ?M (λout. ?Trace out ∨ ?Alpha out)
    adversary-initial-state
  by (rule wp-event-union-bound)
  also have ... ≤
    wp-event ?M ?Collision adversary-initial-state +
    (wp-event ?M ?Trace adversary-initial-state +
    wp-event ?M ?Alpha adversary-initial-state)
  by (intro add-mono order-refl wp-event-union-bound)
  finally show ?thesis
    by (simp add: add.assoc)
qed
show ?thesis
  unfolding
    active-route-current-empty-conceptual-residual-error-for-def
    active-route-current-empty-split-residual-bound-for-def
    active-route-current-empty-conceptual-prefix-collision-error-for-def
    active-route-current-empty-conceptual-trace-low-degree-error-for-def

```

$\text{active-route-current-empty-conceptual-alpha-bad-set-error-for-def}$
by (rule order-trans[*OF event-le union-bound*])
qed

lemma *active-route-current-empty-query-error-for-le-split-conceptual-bound*:
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
shows
 $\text{active-route-current-empty-query-error-for } A \leq$
 $\text{active-route-current-empty-split-conceptual-bound-for budgets } A$
proof –
have *i-bound*: $0 < \text{rounds}$
using *rounds-positive* **by** *simp*
have *residual-bound*:
wp-event
(*checked-staged-security-experiment-with-query-prefix-data-state A 0*)
(*checked-staged-security-with-query-prefix-conceptual-empty-context-residual-hit-at 0*)
adversary-initial-state $\leq$
active-route-current-empty-split-residual-bound-for A
using *active-route-current-empty-conceptual-residual-bound-from-split*
[*OF false-statement*]
unfolding *active-route-current-empty-conceptual-residual-error-for-def* .
have *pre-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-transcript-pre-hit
adversary-initial-state $\leq$ *active-route-transcript-pre-error-for A*
by (*simp add: active-route-transcript-pre-error-for-def*)
have *new-bound*:
wp-event (*checked-staged-security-experiment-with-data-state A*)
staged-security-with-data-state-transcript-new-hit
adversary-initial-state $\leq$ *staged-concrete-transcript-target-error-bound*
by (rule *checked-staged-security-with-data-state-transcript-new-hit-bound*
[*OF wf controlled*])
show ?thesis
unfolding *active-route-current-empty-query-error-for-def*
active-route-current-empty-split-conceptual-bound-for-def
by (rule
checked-staged-security-with-data-state-current-empty-query-opening-hit-at-bound-from-conceptual-empty-residual
[*OF wf controlled i-bound residual-bound pre-bound new-bound*])
qed

theorem *stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-restricted-trace-endpoint*:
assumes *false-statement*: $\neg \text{exists-valid-trace}$
and *wf*: *staged-budget-wellformed budgets*
and *controlled*: *staged-adversary-controlled budgets A*
shows *checked-staged-adversary-acceptance-probability A* $\leq$
hash-collision-budget-value 0

(staged-attacker-query-budget budgets +
 staged-challenge-query-budget + verifier-hash-query-budget) +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds}$.
 active-route-current-prefix-split-conceptual-bound-for budgets A i +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 concrete-trace-header-restricted-one-step-error-for budgets A +
 concrete-composition-restricted-one-step-error-for budgets A +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 concrete-trace-empty-header-restricted-sampled-split-one-step-error-for budgets A +
 active-route-current-empty-split-conceptual-bound-for budgets A)

proof –

have trace-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 trace-fri-bad-with-header-tied-partial-candidate)
 adversary-initial-state
 $\leq$ concrete-trace-header-restricted-one-step-error-for budgets A

unfolding concrete-trace-header-restricted-one-step-error-for-def

by (rule trace-header-restricted-one-step-error-for-bound
 [OF wf controlled])

have comp-bound:

wp-event (checked-staged-security-experiment-with-data-state A)
 (staged-security-with-data-state-verifier-event
 composition-fri-bad-with-verifier-tied-partial-candidate)
 adversary-initial-state
 $\leq$ concrete-composition-restricted-one-step-error-for budgets A

unfolding concrete-composition-restricted-one-step-error-for-def

by (rule composition-restricted-verifier-tied-controlled-error-for-bound
 [OF wf controlled])

have *empty-bound*:
wp-event (checked-staged-security-experiment-with-data-state A)
(staged-security-with-data-state-verifier-event
trace-fri-bad-with-empty-composition-header-candidates)
adversary-initial-state
 $\leq$ *concrete-trace-empty-header-restricted-sampled-split-one-step-error-for*
budgets A
unfolding
concrete-trace-empty-header-restricted-sampled-split-one-step-error-for-def
by (*rule trace-empty-header-restricted-sampled-split-one-step-error-for-bound*
[OF wf controlled])
have *route*:
checked-staged-adversary-acceptance-probability A $\leq$
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound)) +
 $(\sum i < \text{rounds}.$
active-route-current-prefix-authenticated-error-for A i +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound) +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound)) +
concrete-trace-header-restricted-one-step-error-for budgets A +
concrete-composition-restricted-one-step-error-for budgets A +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound) +
concrete-trace-empty-header-restricted-sampled-split-one-step-error-for budgets A +
active-route-current-empty-query-error-for A)
by (*rule*
stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-from-active-fri-bounds
[OF false-statement wf controlled trace-bound comp-bound empty-bound])
have *prefix-terms*:
 $(\sum i < \text{rounds}.$

active-route-current-prefix-authenticated-error-for A i +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound) +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound))
 $\leq$
 $(\sum i < \text{rounds}.$
active-route-current-prefix-split-conceptual-bound-for budgets A i +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound) +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound))
by (intro sum-mono add-mono order-refl
active-route-current-prefix-authenticated-error-for-le-split-conceptual-bound
[OF wf controlled - false-statement])
simp
have *empty-term:*
active-route-current-empty-query-error-for A ≤
active-route-current-empty-split-conceptual-bound-for budgets A
by (rule *active-route-current-empty-query-error-for-le-split-conceptual-bound*
[OF false-statement wf controlled])
have *route-le:*
hash-collision-budget-value 0
(staged-attacker-query-budget budgets +
staged-challenge-query-budget + verifier-hash-query-budget) +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound)) +
 $(\sum i < \text{rounds}.$
active-route-current-prefix-authenticated-error-for A i +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound) +
(active-route-transcript-pre-error-for A +
staged-concrete-transcript-target-error-bound)) +
concrete-trace-header-restricted-one-step-error-for budgets A +
concrete-composition-restricted-one-step-error-for budgets A +
(composition-error-bound +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +
hash-collision-budget-value 0
(staged-alpha-search-queries budgets 0) +
staged-phase-relation-error size
(staged-alpha-search-queries budgets 0) +

$(\text{active-route-transcript-pre-error-for } A +$
 $\quad \text{staged-concrete-transcript-target-error-bound}) +$
 $\text{concrete-trace-empty-header-restricted-sampled-split-one-step-error-for budgets } A +$
 $\text{active-route-current-empty-query-error-for } A)$
 $\leq$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\quad \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\quad \text{staged-concrete-transcript-target-error-bound})) +$
 $(\sum_{i < \text{rounds.}}$
 $\quad \text{active-route-current-prefix-split-conceptual-bound-for budgets } A \ i +$
 $\quad (\text{active-route-transcript-pre-error-for } A +$
 $\quad \quad \text{staged-concrete-transcript-target-error-bound}) +$
 $\quad (\text{active-route-transcript-pre-error-for } A +$
 $\quad \quad \text{staged-concrete-transcript-target-error-bound})) +$
 $\text{concrete-trace-header-restricted-one-step-error-for budgets } A +$
 $\text{concrete-composition-restricted-one-step-error-for budgets } A +$
 $(\text{composition-error-bound} +$
 $\quad \text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\text{hash-collision-budget-value } 0$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $\text{staged-phase-relation-error size}$
 $\quad (\text{staged-alpha-search-queries budgets } 0) +$
 $(\text{active-route-transcript-pre-error-for } A +$
 $\quad \text{staged-concrete-transcript-target-error-bound}) +$
 $\text{concrete-trace-empty-header-restricted-sampled-split-one-step-error-for budgets } A +$
 $\text{active-route-current-empty-split-conceptual-bound-for budgets } A)$
by (intro add-mono order-refl prefix-terms empty-term)
show ?thesis
by (rule order-trans[OF route route-le])
qed

definition concrete-restricted-trace-stark-soundness-bound-for
where

$\text{concrete-restricted-trace-stark-soundness-bound-for budgets } A =$
 $\text{hash-collision-budget-value } 0$
 $(\text{staged-attacker-query-budget budgets} +$
 $\quad \text{staged-challenge-query-budget} + \text{verifier-hash-query-budget}) +$
 $(\text{composition-error-bound} +$
 $\quad \text{staged-phase-relation-error size}$

(staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 ($\sum i < \text{rounds}$.
 active-route-current-prefix-split-conceptual-bound-for budgets A i +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound)) +
 concrete-trace-header-restricted-one-step-error-for budgets A +
 concrete-composition-restricted-one-step-error-for budgets A +
 (composition-error-bound +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 hash-collision-budget-value 0
 (staged-alpha-search-queries budgets 0) +
 staged-phase-relation-error size
 (staged-alpha-search-queries budgets 0) +
 (active-route-transcript-pre-error-for A +
 staged-concrete-transcript-target-error-bound) +
 concrete-trace-empty-header-restricted-sampled-split-one-step-error-for budgets A +
 active-route-current-empty-split-conceptual-bound-for budgets A)

theorem stark-soundness:

assumes false-statement: $\neg$ exists-valid-trace

and wf: staged-budget-wellformed budgets

and controlled: staged-adversary-controlled budgets A

shows checked-staged-adversary-acceptance-probability A $\leq$
 concrete-restricted-trace-stark-soundness-bound-for budgets A

unfolding concrete-restricted-trace-stark-soundness-bound-for-def

by (rule

stark-soundness-from-current-public-prefix-current-query-relevant-drift-route-restricted-trace-endpoint
 [OF false-statement wf controlled])

end

end

theory Stark-Document

imports

Soundness-FRI-Trace-Restricted-Query-Endpoint

Stark-Completeness.Completeness

Stark-Examples.Square-Sequence

begin

Document aggregation theory for the generated Isabelle proof document. The main proof entry points remain the completeness theorem, the executable example, and the public soundness endpoint imported above.

end